Étale Coverings and the Fundamental Group

(SGA 1)

English translation

Complete volume

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Source and status note

This volume is an English translation of the complete French TEX source in arXiv `math/0206203v2`. Exposé VII does not exist in the source; the numbering therefore proceeds directly from Exposé VI to Exposé VIII.

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The mathematical notation, theorem hierarchy, labels, references, diagrams, and active correction branches of that source are retained. This is a translation, not a critical edition or a certification of the mathematics.

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CHAPTER I

Étale morphisms

French source p. 1

To simplify the exposition, we assume that all preschemes under consideration are locally noetherian, at least after no. I.2.

I.1. Notions of differential calculus

Let X be a prescheme over Y, and let $\Delta_{X/Y}$, or simply Δ, be the diagonal morphism $X \to X \times_Y X$. It is an immersion, hence a closed immersion of X into an open subset V of $X \times_Y X$. Let $\mathcal{I}_X$ be the ideal of the closed subprescheme corresponding to the diagonal in V (N.B. if one wishes to proceed intrinsically, without assuming that X is separated over Y—a hypothesis that would be farcical—one should consider the set-theoretic inverse image of $\mathcal{O}_{X\times X}$ in X, and denote by $\mathcal{I}_X$ the augmentation ideal in the latter...). The sheaf $\mathcal{I}_X/\mathcal{I}_X^2$ may be regarded as a quasi-coherent sheaf on X; we denote it by $\Omega^1_{X/Y}$. It is of finite type if $X \to Y$ is of finite type. It behaves well under base change $Y' \to Y$. We also introduce the sheaves $\mathcal{O}_{X\times_Y X}/\mathcal{I}_X^{n+1} = \mathcal{P}^n_{X/Y}$; these are sheaves of *rings* on X, making X into a prescheme that may be denoted by $\Delta^n_{X/Y}$ and called the *n-th infinitesimal neighborhood of X/Y*. The resulting sorites is entirely trivial, although rather long[1]; it would be prudent to speak of it only when there is something useful to say about it, in connection with smooth morphisms.

I.2. Quasi-finite morphisms

Proposition I.2.1. *Let $A \to B$ be a local homomorphism (N.B. the rings are now noetherian), and let $\mathfrak{m}$ be the maximal ideal of A. The following conditions are equivalent:*

(i) *$B/\mathfrak{m}B$ is finite-dimensional over $k = A/\mathfrak{m}$.*
(ii) *$\mathfrak{m}B$ is an ideal of definition, and $B/\mathfrak{r}(B) = \kappa(B)$ is an extension of $k = \kappa(A)$.*
(iii) *The completion $\widehat{B}$ is finite over the completion $\widehat{A}$ of A.*

French source p. 2

One then says that B *is quasi-finite* over A. A morphism $f\colon X \to Y$ is said to be quasi-finite at x (or the Y-prescheme f is said to be quasi-finite at x) if $\mathcal{O}_x$ is quasi-finite over $\mathcal{O}_{f(x)}$. This is also equivalent to saying that x is *isolated in its fiber* $f^{-1}(f(x))$.[2] A morphism is said to be quasi-finite if it is so at every point.[3]

Corollary I.2.2. *If A is complete, quasi-finite is equivalent to finite.*

One could give the usual chain of implications (i), (ii), (iii), (iv), (v) for quasi-finite morphisms, but this does not seem indispensable here.

I.3. Unramified or net morphisms

Proposition I.3.1. *Let $f\colon X \to Y$ be a morphism of finite type, let $x \in X$, and put $y = f(x)$. The following conditions are equivalent:*

[1] Cf. EGA IV 16.3.

[2] The original printing (Exposé I, p. 2) and both French arXiv renderings print $f^{-1}(x)$. Since $x \in X$ for $f\colon X \to Y$, the type-correct fiber through x is $f^{-1}(f(x))$, which is used here with the source defect explicitly disclosed.

[3] In EGA II 6.2.3 one assumes in addition that f is of finite type.

(i) *$\mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x$ is a finite separable extension of $\kappa(y)$.*
(ii) *$\Omega^1_{X/Y}$ vanishes at x.*
(iii) *The diagonal morphism $\Delta_{X/Y}$ is an open immersion in a neighborhood of x.*

For the implication (i)⇒(ii), Nakayama's lemma immediately reduces us to the case $Y = \mathrm{Spec}(k)$, $X = \mathrm{Spec}(k')$, where the assertion is well known and, moreover, immediate from the definition of separability. The implication (ii)⇒(iii) follows from a pleasant and easy characterization of open immersions using Krull. For (iii)⇒(i), one is again reduced to the case $Y = \mathrm{Spec}(k)$, with the diagonal morphism an open immersion everywhere. One must then prove that X is finite, with a separable coordinate ring over k; for this one reduces to the case where k is algebraically closed. But then every closed point of X is isolated (it is precisely the inverse image of the diagonal under the morphism $X \to X \times_k X$ defined by x), and hence X is finite. One may therefore suppose that X consists of a single point, with ring A. Thus $A \otimes_k A \to A$ is an isomorphism, whence $A = k$, as required.

Definition I.3.2. (a) *One then says that f is* net*, or equivalently* unramified*, at x, or that X is net, or unramified, at x over Y.*
(b) *Let $A \to B$ be a local homomorphism. It is said to be* net*, or* unramified*, and B is said to be a net, or unramified, local algebra over A, if $B/\mathfrak{r}(A)B$ is a finite separable extension of $A/\mathfrak{r}(A)$; that is, if $\mathfrak{r}(A)B = \mathfrak{r}(B)$ and $\kappa(B)$ is a separable extension of $\kappa(A)$.*[4]

Remarks. The fact that B is unramified over A can already be checked on the completions of A and B. Unramified implies quasi-finite. French source p. 3

Corollary I.3.3. *The set of points at which f is unramified is open.*

Corollary I.3.4. *Let X' and X be two preschemes of finite type over Y, and let $g\colon X' \to X$ be a Y-morphism. If X is unramified over Y, the graph morphism*

$$\Gamma_g\colon X' \longrightarrow X' \times_Y X$$

is an open immersion.[5]

Indeed, it is the inverse image of the diagonal morphism $X \to X \times_Y X$ under

$$g \times_Y \mathrm{id}_X\colon X' \times_Y X \longrightarrow X \times_Y X.$$

One may also introduce the annihilator ideal $\mathfrak{d}_{X/Y}$ of $\Omega^1_{X/Y}$, called the *different ideal* of X/Y. It defines a closed subprescheme of X whose underlying set is the set of points at which X/Y is ramified, that is, not unramified.

Proposition I.3.5. (i) *An immersion is unramified.*
(ii) *The composite of two unramified morphisms is unramified.*
(iii) *A base change of an unramified morphism is unramified.*

This may be seen equally well from criterion (ii) or criterion (iii) (the latter seems more amusing to me). One can of course also be more precise by giving pointwise statements; this is more general only in appearance (except in the setting of Definition I.3.2(b)), and tedious. As usual, one obtains the following corollaries.

Corollaries I.3.6. (iv) *The Cartesian product of two unramified morphisms is unramified.*
(v) *If gf is unramified, then f is unramified.*

[4] Cf. the regrets in Exposé III, no. 1.2.

[5] The original printing (Exposé I, p. 3) and the French TeX give $X \times_Y X$ as the graph target and later write $\mathrm{id}_{X'}$ in the pullback map. The typed fiber products force $X' \times_Y X$ and id_X, respectively; those type-correct readings are used here with both source defects disclosed.

(vi) *If f is unramified, then f_{red} is unramified.*

PROPOSITION I.3.7. *Let $A \to B$ be a local homomorphism. Suppose that the residue-field extension $\kappa(B)/\kappa(A)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is unramified if and only if $\widehat{B}$, as an $\widehat{A}$-algebra, is a quotient of $\widehat{A}$.*

Remarks.

- When the residue-field extension is not assumed to be trivial, one may reduce to that case by making a suitable finite flat extension over A that kills the extension in question.
- Take the example in which A is the local ring of an ordinary double point of a curve and B is the local ring at a point of the normalization. Then $A \subset B$, and B is unramified over A with trivial residue-field extension; the map $\widehat{A} \to \widehat{B}$ is surjective but *not injective*. We shall therefore strengthen the notion of unramifiedness.

French source p. 4

I.4. Étale morphisms. Étale coverings

We shall take for granted whatever is needed concerning flat morphisms; these facts will be proved later, if necessary[6].

DEFINITION I.4.1. (a) *Let $f\colon X \to Y$ be a morphism of finite type. One says that f is* étale *at x if f is flat at x and net (equivalently, unramified) at x. One says that f is étale if it is so at every point. One says that X is étale at x over Y, or that it is a Y-prescheme étale at x, and so on.*

(b) *Let $f\colon A \to B$ be a local homomorphism. One says that f is étale, or that B is étale over A, if B is flat and unramified over A.*[7]

PROPOSITION I.4.2. *For B/A to be étale, it is necessary and sufficient that $\widehat{B}/\widehat{A}$ be étale.*

Indeed, this holds separately for "unramified" and for "flat."

COROLLARY I.4.3. *Let $f\colon X \to Y$ be of finite type, and let $x \in X$. Whether f is étale at x depends only on the local homomorphism $\mathcal{O}_{f(x)} \to \mathcal{O}_x$, and even only on the corresponding homomorphism between the completions.*

COROLLARY I.4.4. *Suppose that the residue-field extension $\kappa(A) \to \kappa(B)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is étale if and only if $\widehat{A} \to \widehat{B}$ is an isomorphism.*

This follows by combining flatness with Proposition I.3.7.

PROPOSITION I.4.5. *Let $f\colon X \to Y$ be a morphism of finite type. Then the set of points at which it is étale is open.*

Indeed, this holds separately for "unramified" and for "flat."

This proposition shows that one may dispense with "pointwise" statements when studying morphisms of finite type that are étale at some point.

PROPOSITION I.4.6. (i) *An open immersion is étale.*

(ii) *The composite of two étale morphisms is étale.*

(iii) *A base change of an étale morphism is étale.*

French source p. 5

Indeed, (i) is trivial, and for (ii) and (iii) it is enough to note that the assertions hold for "unramified" and for "flat." To tell the truth, there are also corresponding statements for local homomorphisms (without a finiteness condition); in any event, they will have to appear in the *multiplodoque*, beginning with the unramified case.

[6] Cf. Exposé IV.

[7] Cf. the regrets in Exposé III, no. 1.2.

COROLLARY I.4.7. *The Cartesian product of two étale morphisms is likewise étale.*

COROLLARY I.4.8. *Let X and X' be of finite type over Y, and let $g\colon X \to X'$ be a Y-morphism. If X' is unramified over Y and X is étale over Y, then g is étale.*

Indeed, g is the composite of the graph morphism

$$\Gamma_g\colon X \longrightarrow X \times_Y X',$$

which is an open immersion by Corollary I.3.4, and the projection $X \times_Y X' \to X'$, which is étale because it is obtained from the étale morphism $X \to Y$ by the base change $X' \to Y$.

DEFINITION I.4.9. *An* étale covering *(respectively, a* net covering*, also called an unramified covering) of Y is a Y-scheme X that is finite over Y and étale (respectively, unramified) over Y.*

The first condition means that X is defined by a coherent sheaf of algebras $\mathcal{B}$ on Y. The second then means that $\mathcal{B}$ is locally free over Y (with no corresponding local-freeness requirement in the net case), and, moreover, that for every $y \in Y$ the fiber

$$\mathcal{B}(y) = \mathcal{B}_y \otimes_{\mathcal{O}_y} \kappa(y)$$

is a separable algebra over $\kappa(y)$, that is, a finite product of finite separable field extensions of $\kappa(y)$.

PROPOSITION I.4.10. *Let X be a flat covering of Y of degree n (the definition of this term deserved to appear in Definition I.4.9), defined by a coherent locally free sheaf of algebras $\mathcal{B}$. In the usual way one defines the trace homomorphism $\mathcal{B} \to \mathcal{A}$, which is a homomorphism of $\mathcal{A}$-modules, where $\mathcal{A} = \mathcal{O}_Y$. For X to be étale, it is necessary and sufficient that the corresponding trace pairing $(x, y) \longmapsto \mathrm{Tr}_{\mathcal{B}/\mathcal{A}}(xy)$ induce an isomorphism $\mathcal{B} \xrightarrow{\sim} \mathcal{B}^\vee$,[8] or, equivalently, that the* discriminant section

$$d_{X/Y} = d_{\mathcal{B}/\mathcal{A}} \in \Gamma\Big(Y, \bigwedge^n \mathcal{B}^\vee \otimes_{\mathcal{A}} \bigwedge^n \mathcal{B}^\vee\Big)$$

be invertible, or, finally, that the discriminant ideal defined by this section be the unit ideal.

Indeed, one reduces to the case $Y = \mathrm{Spec}(k)$; the assertion is then the well-known separability criterion, and becomes immediate after passing to an algebraic closure of k. French source p. 6

Remark. There will be a less trivial statement below, when one does not assume a priori that X is flat over Y, but instead imposes a normality hypothesis.

I.5. The fundamental property of étale morphisms

THEOREM I.5.1. *Let $f\colon X \to Y$ be a morphism of finite type. For f to be an open immersion, it is necessary and sufficient that it be* étale and radicial.

Recall that radicial means injective, with radicial residue-field extensions; one may also recall that this means that the morphism remains injective after every base change. Necessity is trivial; it remains to prove sufficiency. We shall give two different proofs, the first shorter and the second more elementary.

1) A flat morphism is open, so (replacing Y by $f(X)$) we may suppose that f is a *homeomorphism* onto Y. After every base change, f remains flat, radicial, and surjective, hence a homeomorphism and therefore closed. Thus f is *proper*. Hence f is *finite* (reference: Chevalley's theorem) and is defined by a coherent sheaf $\mathcal{B}$ of algebras. The sheaf $\mathcal{B}$ is locally free and, by the hypothesis, has rank 1 everywhere. Thus $X = Y$, as required.

[8] The French TeX and the original printing instead give $\mathcal{B}$ as the target. An $\mathcal{A}$-bilinear form induces a map $\mathcal{B} \to \mathcal{B}^\vee$; the dual is therefore the type-correct target, used here with the source-original defect disclosed.

2) One may suppose Y and X *affine*. Moreover, one easily reduces to proving the following: if $Y = \mathrm{Spec}(A)$, with A local, and if $f^{-1}(y)$ is nonempty, where y is the closed point of Y, then $X = Y$. Indeed, this will imply that every $y \in f(X)$ has an open neighborhood U such that $X|U = U$. We shall have $X = \mathrm{Spec}(B)$, and we want to prove $A = B$. For this, it is enough to prove the analogous assertion after replacing A by $\widehat{A}$ and B by $B \otimes_A \widehat{A}$, taking into account that $\widehat{A}$ is faithfully flat over A. We may therefore suppose that A is *complete*. Let x be the point above y. By Corollary I.2.2, $\mathcal{O}_x$ is finite over A, and hence, being flat and radicial over A, is A itself. Thus

$$X = Y \amalg X'$$

as a disjoint union. Since X is radicial over Y, X' is empty. This completes the proof.

COROLLARY I.5.2. *Let $f\colon X \to Y$ be both a* closed immersion *and* étale. *If X is connected, f is an isomorphism from X onto a connected component of Y.*

Indeed, f is also an open immersion. It follows that: French source p. 7

COROLLARY I.5.3. *Let X be an unramified Y-scheme, with Y connected. Then every section of X over Y is an isomorphism from Y onto a connected component of X. There is therefore a one-to-one correspondence between the set of such sections and the set of connected components X_i of X for which the projection $X_i \to Y$ is an isomorphism (or, equivalently by Theorem I.5.1, is surjective, étale, and radicial).*[9] *In particular, a section is determined by its value at one point.*

Only the first assertion requires proof. By Corollary I.5.2, it is enough to observe that a section is a closed immersion (because X is separated over Y) and is étale by Corollary I.4.8.

COROLLARY I.5.4. *Let X and Y be preschemes over S, with X unramified and separated over S and Y connected. Let $f, g\colon Y \to X$ be two S-morphisms, and let $y \in Y$. Suppose that $f(y) = g(y) = x$ and that the homomorphisms of residue fields $\kappa(x) \to \kappa(y)$ induced by f and g are identical ("f and g coincide geometrically at y"). Then f and g are identical.*

This follows from Corollary I.5.3 by reducing to the case $Y = S$, replacing X by $X \times_S Y$.

Here is a particularly important variant of Corollary I.5.3.

THEOREM I.5.5. *Let S be a prescheme, let X and Y be two S-preschemes, and let S_0 be a closed subprescheme of S having the same underlying space as S. Put*

$$X_0 = X \times_S S_0 \qquad \text{and} \qquad Y_0 = Y \times_S S_0,$$

the "restrictions" of X and Y to S_0. Suppose that X is étale over S. Then the natural map

$$\mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_{S_0}(Y_0, X_0)$$

is bijective.

We are again reduced to the case $Y = S$; the assertion then follows from the "topological" description of the sections of X/Y given in Corollary I.5.3.

[9] The French TeX and the original printing give only "surjective and radicial" and cite Theorem I.5.1, which also requires étaleness. Without it the claim is false: for example, $\mathrm{Spec}(k) \to \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$ is surjective on points, radicial, unramified, separated, and of finite type, but is not an isomorphism. The logically correct criterion is used here with the source defect disclosed.

Scholium. This result includes both a *uniqueness* assertion and an *existence* assertion for *morphisms*. It may also be expressed, when X and Y are both taken to be étale over S, by saying that the functor

$$X \longmapsto X_0$$

from the category of étale S-schemes to the category of étale S_0-schemes is *fully faithful*; that is, it establishes an *equivalence* of the first category with a *full subcategory* of the second. We shall see below that it is actually an equivalence between the first and the second (which will be an *existence theorem for étale S-schemes*).

French source p. 8

The following form of Theorem I.5.5, apparently more general, is often convenient.

Corollary I.5.6 ("Extension theorem for liftings"). *Consider a commutative diagram of morphisms*

$$\begin{array}{ccc} X & \longleftarrow & Y_0 \\ \downarrow & & \downarrow \\ S & \longleftarrow & Y \end{array}$$

in which $X \to S$ is étale and $Y_0 \to Y$ is a bijective closed immersion. Then there is a unique morphism $Y \to X$ that makes the two corresponding triangles commute.

Indeed, replacing S by Y and X by $X \times_S Y$, we are reduced to the case $Y = S$; this is then the special case of Theorem I.5.5 with $Y = S$.

We should also note the following immediate consequence of Theorem I.5.1 (which we did not state as Corollary 1, so as not to interrupt the line of ideas developed after Theorem I.5.1):

Proposition I.5.7. *Let X and X' be two preschemes of finite type and flat over Y, and let $g\colon X \to X'$ be a Y-morphism. For g to be an open immersion (respectively, an isomorphism), it is necessary and sufficient that, for every $y \in Y$, the induced morphism on the fibers*

$$g \otimes_Y \kappa(y)\colon X \otimes_Y \kappa(y) \longrightarrow X' \otimes_Y \kappa(y)$$

be so.

It is enough to prove sufficiency. Since the assertion holds for surjectivity, we are reduced to the case of an open immersion. By Theorem I.5.1, one must verify that g is *radicial*, which is immediate, and that it is *étale*, which follows from Corollary I.5.9 below.

Corollary I.5.8. *(This should be moved to no. I.3.) Let X and X' be two preschemes over Y, let $g\colon X \to X'$ be a Y-morphism, let $x \in X$, and let y be its image in Y. For g to be quasi-finite (respectively, unramified) at x, it is necessary and sufficient that the same be true of $g \otimes_Y \kappa(y)$.*

Indeed, the two algebras over $\kappa(g(x))$ that one must examine to verify that the morphism is quasi-finite (respectively, unramified) at x are the same for g and for $g \otimes_Y \kappa(y)$.

French source p. 9

Corollary I.5.9. *With the notation of Corollary I.5.8, suppose that X and X' are flat and of finite type over Y. For g to be flat (respectively, étale) at x, it is necessary and sufficient that $g \otimes_Y \kappa(y)$ be so.*

For "flat," the statement is included only as a reminder: it is one of the fundamental criteria for flatness.[10] For "étale," the assertion follows from this, taking Corollary I.5.8 into account.

[10] Cf. Exposé IV, no. 5.9.

I.6. Application to étale extensions of complete local rings

This number is a special case of results on formal preschemes that will have to appear in the *multiplodoque*. Nevertheless, here one can get by more cheaply, that is, without the explicit local determination of étale morphisms in no. I.7 (using the Main Theorem). This is perhaps a sufficient reason to keep the present number (even in the *multiplodoque*) in this place.

THEOREM I.6.1. *Let A be a complete local ring (noetherian, of course), with residue field k. For every A-algebra B, put*

$$R(B) = B \otimes_A k,$$

regarded as a k-algebra; it thus depends functorially on B. Then R defines an equivalence *between the category of A-algebras* finite and étale over A *and the category of algebras* of finite rank and separable *over k.*

First, the functor in question is fully faithful, as follows from the more general fact:

COROLLARY I.6.2. *Let B and B' be two A-algebras finite over A. If B is étale over A, then the canonical map*

$$\mathrm{Hom}_{A\text{-}alg}(B, B') \longrightarrow \mathrm{Hom}_{k\text{-}alg}(R(B), R(B'))$$

is bijective.

We are reduced to the case in which A is artinian (by replacing A with $A/\mathfrak{m}^n$); this is then a special case of Theorem I.5.5.

It remains to prove that for every finite separable k-algebra L (why not say "étale"? It is shorter), there is an algebra B, étale over A, such that $R(B)$ is isomorphic to L. We may suppose that L is a separable extension of k; as such, it has a generator x, that is, it is isomorphic to an algebra $k[t]/Fk[t]$, where $F \in k[t]$ is a monic polynomial. Lift F to a monic polynomial $F_1 \in A[t]$, and take

$$B = A[t]/F_1 A[t].$$

I.7. Local construction of unramified and étale morphisms

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PROPOSITION I.7.1. *Let A be a noetherian ring, B a finite A-algebra, u a generator of B over A, and $F \in A[t]$ such that $F(u) = 0$ (we do not assume that F is monic). Put $u' = F'(u)$, where F' is the derivative of F. Let $\mathfrak{q}$ be a prime ideal of B that does not contain u', and let $\mathfrak{p}$ be its contraction to A. Then $B_\mathfrak{q}$ is unramified over $A_\mathfrak{p}$.*

In other words, put $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and $X_{u'} = \mathrm{Spec}(B_{u'})$. Then $X_{u'}$ is unramified over Y. The assertion follows from the following, more precise one.

COROLLARY I.7.2. *The different ideal of B/A contains $u'B$, and is equal to it if the natural homomorphism*

$$A[t]/FA[t] \longrightarrow B,$$

which sends t to u, is an isomorphism.

Let J be the kernel of the homomorphism $C = A[t] \to B$. This kernel contains $FA[t]$, and is equal to it in the second case considered in Corollary I.7.2. Since the homomorphism is surjective, $\Omega^1_{B/A}$ is identified with the quotient of $\Omega^1_{C/A}$ by the submodule generated by $J\Omega^1_{C/A}$ and $d(J)$ (the definition of the homomorphism d, and the computation of Ω^1 for a polynomial algebra, should have been made explicit in no. I.1). Identifying $\Omega^1_{C/A}$ with C by means of the basis dt, one obtains

$$B/(B \cdot J'),$$

where J' is the set of images in B of the derivatives of the polynomials $G \in J$; thus the different ideal is generated by J' (and it is enough to take polynomials G that generate J). Since $F \in J$, respectively since F is a generator of J, the proof is complete. (N.B. Corollary I.7.2 should be a proposition and Proposition I.7.1 a corollary.) We obtain:

COROLLARY I.7.3. *Under the hypotheses of Proposition I.7.1, suppose that F is monic and that $A[t]/FA[t] \to B$ is an isomorphism. Then $B_{\mathfrak{q}}$ is étale over $A_{\mathfrak{p}}$ if and only if $\mathfrak{q}$ does not contain u'.*

Indeed, since B is flat over A, being étale is equivalent to being unramified, and Corollary I.7.2 applies.

COROLLARY I.7.4. *Under the hypotheses of Corollary I.7.3, B is étale over A if and only if u' is invertible, or, equivalently, if and only if the ideal generated by F and F' in $A[t]$ is the unit ideal.*

The last criterion follows from the first and Nakayama's lemma (in B).

A monic polynomial $F \in A[t]$ having the property stated in Corollary I.7.4 is called a *separable polynomial*. If F is not monic, one should at least require the coefficient of its leading term to be invertible; when A is a field, this recovers the usual definition.

French source p. 11

COROLLARY I.7.5. *Let B be a* finite *algebra over the* local *ring A. Suppose that $K(A)$ is infinite or that B is local. Let n be the rank of $L = B \otimes_A K(A)$ over $K(A) = k$. Then B is unramified (respectively, étale) over A if and only if B is isomorphic to a quotient of (respectively, isomorphic to)*

$$A[t]/FA[t],$$

where F is a monic separable polynomial that may be assumed (respectively, is necessarily) to have degree n.

Only necessity has to be proved. Suppose that B is unramified over A, so that L is separable over k. The hypothesis then implies that L/k has a generator ξ, and hence the ξ^i, $0 \leq i < n$, form a basis of L over k. Let $u \in B$ lift ξ. By Nakayama's lemma, the u^i, $0 \leq i < n$, generate the A-module B (respectively, form a basis of it). In particular, one can find a monic polynomial $F \in A[t]$ such that $F(u) = 0$, and B is isomorphic to a quotient of (respectively, isomorphic to) $A[t]/FA[t]$. Finally, by Corollary I.7.4 applied to L/k, F and F' generate $A[t]$ modulo $\mathfrak{m}A[t]$. Hence, by Nakayama's lemma in $A[t]/FA[t]$, F and F' generate $A[t]$.

THEOREM I.7.6. *Let A be a local ring, and let $A \to \mathcal{O}$ be a local homomorphism such that $\mathcal{O}$ is isomorphic to a localization of an algebra of finite type over A. Suppose that $\mathcal{O}$ is unramified over A. Then there exist an A-algebra B integral over A, a maximal ideal $\mathfrak{n}$ of B, a generator u of B over A, and a monic polynomial $F \in A[t]$, such that*

$$F'(u) \notin \mathfrak{n} \qquad \text{and} \qquad \mathcal{O} \simeq B_{\mathfrak{n}}$$

as A-algebras. If $\mathcal{O}$ is étale over A, one may take $B = A[t]/FA[t]$.

(Of course, the conditions displayed here are also sufficient...)

Let us first point out the pleasant corollaries.

COROLLARY I.7.7. *The algebra $\mathcal{O}$ is unramified over A if and only if it is isomorphic to a quotient of an algebra of the same form that is étale over A.*

Indeed, take $\mathcal{O}' = B'_{\mathfrak{n}'}$, where $B' = A[t]/FA[t]$ and $\mathfrak{n}'$ is the inverse image of $\mathfrak{n}$ in B'.

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COROLLARY I.7.8. *Let $f\colon X \to Y$ be a morphism of finite type and let $x \in X$. Then f is unramified at x if and only if there is an open neighborhood U of x such that $f|_U$ factors as*

$$U \longrightarrow X' \longrightarrow Y,$$

where the first arrow is a closed immersion and the second is an étale morphism.

This is a straightforward translation of Corollary I.7.7.

Let us show how the statement of Theorem I.7.6 follows from the main assertion. By Corollary I.7.7, there is a surjective homomorphism $\mathcal{O}' \to \mathcal{O}$, where $\mathcal{O}'$ has the desired form.[11] If $\mathcal{O}$ is étale over A, then both $\mathcal{O}'$ and $\mathcal{O}$ are étale over A; the morphism $\mathcal{O}' \to \mathcal{O}$ is therefore étale by Corollary I.4.8, and hence is an isomorphism.

Proof of Theorem I.7.6. This repeats a proof from the Chevalley seminar. By the *Main Theorem*, one has $\mathcal{O} = B_{\mathfrak{n}}$, where B is a finite algebra over A and $\mathfrak{n}$ is a maximal ideal of B. Then $B/\mathfrak{n} = K(\mathcal{O})$ is a separable, hence monogenic, extension of k. If $\mathfrak{n}_i$, $1 \leq i \leq r$, are the maximal ideals of B distinct from $\mathfrak{n}$, there is therefore an element $u \in B$ that belongs to every $\mathfrak{n}_i$ and whose image in $B/\mathfrak{n}$ generates that algebra. Moreover,

$$B/\mathfrak{n} = B_{\mathfrak{n}}/\mathfrak{n}B_{\mathfrak{n}} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}},$$

where $\mathfrak{m}$ is the maximal ideal of A. Let us admit for the moment the following lemma.

LEMMA I.7.9. *Let A be a local ring, B a finite A-algebra, and $\mathfrak{n}$ a maximal ideal of B. Let $u \in B$ be an element whose image in $B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$ generates that algebra over $k = A/\mathfrak{m}$, and which belongs to every maximal ideal of B distinct from $\mathfrak{n}$. Put*

$$B' = A[u], \qquad \mathfrak{n}' = \mathfrak{n} \cap B'.$$

Then the canonical homomorphism

$$B'_{\mathfrak{n}'} \longrightarrow B_{\mathfrak{n}}$$

is an isomorphism.[12]

LEMMA I.7.10. *(This should have appeared as a corollary to Proposition I.7.1, before Corollary I.7.5, which it implies.) Let B be a finite A-algebra generated by an element u, and let $\mathfrak{n}$ be a maximal ideal of B such that $B_{\mathfrak{n}}$ is unramified over A. Then there is a monic polynomial $F \in A[t]$ such that*

$$F(u) = 0 \qquad \textit{and} \qquad F'(u) \notin \mathfrak{n}.$$

Indeed, let n be the rank of the k-algebra $L = B \otimes_A k$. By Nakayama's lemma there is a monic polynomial $F \in A[t]$ of degree n such that $F(u) = 0$. Let f be the reduction of F modulo $\mathfrak{m}$. Then L is isomorphic, as a k-algebra, to $k[t]/fk[t]$. Hence, by Corollary I.7.3, $f'(\xi)$ does not belong to the maximal ideal of L corresponding to $\mathfrak{n}$, where ξ denotes the image of t in L, that is, the image of u in L. Since $f'(\xi)$ is the image of $F'(u)$, the proof is complete.

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Theorem I.7.6 now follows by combining Lemmas I.7.9 and I.7.10. It remains to prove Lemma I.7.9. Put $S' = B' \setminus \mathfrak{n}'$, so that $B'S'^{-1} = B'_{\mathfrak{n}'}$. We have the diagram

$$\begin{array}{ccccc} B & \longrightarrow & BS'^{-1} & \longrightarrow & BS^{-1} = B_{\mathfrak{n}} \\ \uparrow & & \uparrow & & \\ B' & \longrightarrow & B'S'^{-1} = B'_{\mathfrak{n}'} & & \end{array}$$

where, similarly, $S = B \setminus \mathfrak{n}$, so that $BS^{-1} = B_{\mathfrak{n}}$. There is therefore a natural homomorphism

$$BS'^{-1} \longrightarrow BS^{-1} = B_{\mathfrak{n}}.$$

[11] The French TeX and the original printing say literally "the jargon of Theorem I.7.6," write the unprimed $\mathcal{O}$ after "where," and move directly to "since $\mathcal{O}'$ and $\mathcal{O}$ are étale." Corollary I.7.7 constructs the primed algebra with the desired étale presentation, while the latter sentence concerns the theorem's final case, in which $\mathcal{O}$ is additionally assumed étale. Both points are made explicit here.

[12] The French TeX and the original printing give $B' = B[u]$ and $\mathfrak{n}' = \mathfrak{n}B'$. Since $u \in B$, the first formula would make $B' = B$, while the proof uses B' as the finite A-subalgebra generated by u and uses $\mathfrak{n}'$ as the contraction of $\mathfrak{n}$. The type-correct formulas required by the proof are used here with the source reading disclosed.

We prove that it is an isomorphism. Equivalently, the elements of S must be invertible in BS'^{-1}; equivalently, every maximal ideal $\mathfrak{p}$ of the latter ring must be disjoint from S, that is, must induce $\mathfrak{n}$ on B. Since BS'^{-1} is finite over $B'S'^{-1} = B'_{\mathfrak{n}'}$, the ideal $\mathfrak{p}$ contracts to the unique maximal ideal $\mathfrak{n}'B'_{\mathfrak{n}'}$ of $B'_{\mathfrak{n}'}$,[13] and hence to the maximal ideal $\mathfrak{n}'$ of B'. Since B is finite over B', the ideal $\mathfrak{q}$ of B induced by $\mathfrak{p}$, lying over $\mathfrak{n}'$, is necessarily maximal. It does not contain u, and is therefore equal to $\mathfrak{n}$. Here we used the fact that u belongs to every maximal ideal of B distinct from $\mathfrak{n}$.

It remains to prove that $BS'^{-1} = B'S'^{-1}$. Since the former is finite over the latter, Nakayama's lemma reduces us to proving equality modulo $\mathfrak{n}'BS'^{-1}$, and it is enough, a fortiori, to prove equality modulo $\mathfrak{m}BS'^{-1}$. But

$$BS'^{-1}/\mathfrak{m}BS'^{-1} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$$

is generated over k by u (here we use the other property of u). Thus the image of B', and a fortiori of $B'S'^{-1}$, in this quotient is the whole ring, since it is a subring containing k and the image of u.

Remark *. It should be possible to state Theorem I.7.6 for an algebra $\mathcal{O}$ that is only semilocal, so as to subsume Corollary I.7.5 as well. One would assume that $\mathcal{O}/\mathfrak{m}\mathcal{O}$ is a *monogenic* k-algebra. One could then find $u \in B$ whose image in $B/\mathfrak{m}B$ is a generator and which belongs to every maximal ideal of B not arising from $\mathcal{O}$. Lemmas I.7.9 and I.7.10 should adapt without difficulty. More generally, ...

I.8. Infinitesimal lifting of étale schemes. Application to formal schemes

PROPOSITION I.8.1. *Let Y be a prescheme, let Y_0 be a subprescheme, let X_0 be an étale Y_0-scheme, and let x be a point of X_0. Then there exist an étale Y-scheme X, a neighborhood U_0 of x in X_0, and a Y_0-isomorphism*

$$U_0 \xrightarrow{\sim} X \times_Y Y_0.$$

Indeed, let y be the image of x in Y_0. Applying Theorem I.7.6 to the étale local homomorphism $A_0 \to B_0$ of the local rings at y and x in Y_0 and X_0, respectively, one obtains an isomorphism

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$$B_0 = (C_0)_{\mathfrak{n}_0}, \qquad C_0 = A_0[t]/F_0A_0[t],$$

where F_0 is monic and $\mathfrak{n}_0$ is a maximal ideal of C_0 that does not contain the class of $F_0'(t)$ in C_0. Let A be the local ring at y in Y, and let $F \in A[t]$ be a monic polynomial whose image under the surjective homomorphism $A \to A_0$ is F_0 (one lifts the coefficients of F_0). Finally, put

$$C = A[t]/FA[t],$$

and let $\mathfrak{n}$ be the inverse image of $\mathfrak{n}_0$ under the natural epimorphism

$$C \longrightarrow C \otimes_A A_0 = C_0.$$

Put

$$B = C_{\mathfrak{n}}.$$

It follows immediately from the construction and Proposition I.7.1 that B is étale over A, and there is an isomorphism

$$B \otimes_A A_0 \xrightarrow{\sim} B_0.$$

[13] The French TeX and the original printing write $\mathfrak{n}'B_{\mathfrak{n}'}$ here. The ring in the preceding clause is $B'_{\mathfrak{n}'}$, so the omitted prime on B is restored explicitly.

[14] We know from EGA, Chapter I, that there exist a Y-scheme X of finite type and a point $z \in X$ above y such that $\mathcal{O}_z$ is A-isomorphic to B.[15] Since the latter is étale over $A = \mathcal{O}_y$, by taking X small enough we may assume that X is étale over Y. Put $X'_0 = X \times_Y Y_0$. The local ring at z in X'_0 is then identified with

$$\mathcal{O}_z \otimes_A A_0 = B \otimes_A A_0,$$

and is consequently isomorphic to B_0. This isomorphism is induced by an isomorphism from a neighborhood U_0 of x in X_0 onto a neighborhood of z in X'_0 (loc. cit.); by taking X small enough, we may assume that the latter neighborhood is all of X'_0. This proves the proposition.

Corollary I.8.2. *There is an analogous statement for étale* coverings*, provided the residue field $\kappa(y)$ is infinite.*

The proof is the same, with Corollary I.7.5 in place of Theorem I.7.6.

Theorem I.8.3. *The functor considered in Theorem I.5.5 is an* equivalence *of* categories*.*

By Theorem I.5.5, it remains to show that every étale S_0-scheme X_0 is isomorphic to an S_0-scheme $X \times_S S_0$, where X is an étale S-scheme. The underlying topological space of X must necessarily be identical with that of X_0, with X_0 moreover identified with a closed subprescheme of X. The problem is therefore equivalent to the following one: on the topological space $|X_0|$ underlying X_0, find a sheaf of algebras $\mathcal{O}_X$ over $f_0^*(\mathcal{O}_S)$ (where $f_0 \colon X_0 \to S_0$ is here regarded as a continuous map of the underlying spaces) that makes $|X_0|$ into an étale S-prescheme X, together with a homomorphism of algebras

$$\mathcal{O}_X \longrightarrow \mathcal{O}_{X_0}$$

compatible with the homomorphism

$$f_0^*(\mathcal{O}_S) \longrightarrow f_0^*(\mathcal{O}_{S_0})$$

on the scalar sheaves, and inducing an isomorphism

$$\mathcal{O}_X \otimes_{f_0^*(\mathcal{O}_S)} f_0^*(\mathcal{O}_{S_0}) \xrightarrow{\sim} \mathcal{O}_{X_0}.$$

Then X will be an étale S-prescheme whose reduction is X_0, and hence will be separated over S, since X_0 is separated over S_0; thus X answers the question. Moreover, if (U_i) is an open covering of X_0, and if a solution to the problem has been found on each U_i, it follows from the uniqueness theorem I.5.5 that these solutions glue (that is, the sheaves of algebras that define them, equipped with their augmentation homomorphisms, glue). One checks at once that the ringed space over S constructed in this way is an étale S-prescheme X, equipped with an isomorphism

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$$X_0 \xrightarrow{\sim} X \times_S S_0.$$

It is therefore enough to find a solution locally, and this is provided by Proposition I.8.1.

Corollary I.8.4. *Let S be a locally noetherian formal prescheme, equipped with an ideal of definition $\mathcal{J}$,*[16] *and let*

$$S_0 = (|S|, \mathcal{O}_S/\mathcal{J})$$

[14] At this point both the French TeX and the original printing give $B \otimes_A A_0 = A_0$. Since $B = C_{\mathfrak{n}}$ and $C \otimes_A A_0 = C_0$, base change gives $(C_0)_{\mathfrak{n}_0} = B_0$. The type-correct target B_0 is used here with the shared source defect disclosed.

[15] The French TeX and the original printing have C here. The following equality with $B \otimes_A A_0$, and the appeal to the étaleness of the localized algebra, require $B = C_{\mathfrak{n}}$; B is therefore restored explicitly.

[16] The French TeX and the original printing alternate J and $\mathcal{J}$ for this ideal of definition and its powers. Since the quotient sheaves all use the same ideal sheaf, $\mathcal{J}$ is used consistently here.

be the corresponding ordinary prescheme. Then the functor

$$\mathfrak{X} \longmapsto \mathfrak{X} \times_S S_0$$

from the category of étale coverings of S to the category of étale coverings of S_0 is an equivalence of categories.

Of course, an étale covering of a *formal* prescheme S means a covering of S, that is, a formal prescheme over S defined by a coherent sheaf of algebras $\mathcal{B}$, such that $\mathcal{B}$ is *locally free* and the residual fibers

$$\mathcal{B}_s \otimes_{\mathcal{O}_s} \kappa(s)$$

of $\mathcal{B}$ are *separable* algebras over $\kappa(s)$. If S_n denotes the ordinary prescheme

$$S_n = (|S|, \mathcal{O}_S/\mathcal{J}^{n+1}),$$

then giving a coherent sheaf of algebras $\mathcal{B}$ on S is equivalent to giving a sequence of coherent sheaves of algebras $\mathcal{B}_n$ on the S_n, equipped with a transitive system of homomorphisms

$$\mathcal{B}_m \longrightarrow \mathcal{B}_n \qquad (m \geq n)$$

that define isomorphisms

$$\mathcal{B}_m \otimes_{\mathcal{O}_{S_m}} \mathcal{O}_{S_n} \xrightarrow{\sim} \mathcal{B}_n.$$

It is immediate that $\mathcal{B}$ is locally free if and only if all the $\mathcal{B}_n$ on the S_n are locally free, and that the separability condition holds if and only if it holds for $\mathcal{B}_0$, or, equivalently, for all the $\mathcal{B}_n$. Thus $\mathcal{B}$ is étale over S if and only if the $\mathcal{B}_n$ are étale over the S_n. In view of this, Corollary I.8.4 follows at once from Theorem I.8.3.

Remark *. It was not necessary in Corollary I.8.4 to restrict oneself to the case of *coverings*. This is, however, the only case used for the moment.

I.9. Permanence properties

Let $A \to B$ be a local étale homomorphism. We examine here several cases in which a given property of A implies the same property for B, or conversely. A number of such assertions already follow from the simple fact that B is *quasi-finite* and *flat* over A, and we shall confine ourselves to "recalling" a few of them: *A and B have the same Krull dimension and the same depth* (Serre's "cohomological codimension", in the terminology still in use). It follows, for example, that *A is Cohen–Macaulay if and only if B is.* Moreover, for every prime ideal $\mathfrak{q}$ of B whose contraction to A is $\mathfrak{p}$, the ring $B_{\mathfrak{q}}$ is again quasi-finite and flat over $A_{\mathfrak{p}}$, provided that B is a localization of an A-algebra of finite type. This follows from the fact that, for a morphism of finite type, the locus where it is quasi-finite, respectively flat, is open. Furthermore, *every* prime ideal $\mathfrak{p}$ of A is the contraction of a prime ideal $\mathfrak{q}$ of B, since B is *faithfully* flat over A. It follows, for example, that *$\mathfrak{p}$ and $\mathfrak{q}$ have the same height*, and also that *A has no embedded prime ideal if and only if B has none.*

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We shall therefore restrict ourselves to assertions that are more specific to étale morphisms.

Proposition I.9.1. *Let $A \to B$ be a local étale homomorphism. Then A is regular if and only if B is regular.*

Indeed, let k be the residue field of A, and let L be that of B. Since B is flat over A and $L = B \otimes_A k$, that is, $\mathfrak{n} = \mathfrak{m}B$, where $\mathfrak{m}$ and $\mathfrak{n}$ are the maximal ideals of A and B, respectively, the $\mathfrak{m}$-adic filtration on B is identical to its $\mathfrak{n}$-adic filtration, and hence

$$\mathrm{gr}^*(B) = \mathrm{gr}^*(A) \otimes_k L.$$

It follows that $\mathrm{gr}^*(B)$ is a polynomial algebra over L if and only if $\mathrm{gr}^*(A)$ is a polynomial algebra over k. This proves the proposition. (N.B. We did not use the fact that L/k is separable.)

Corollary I.9.2. *Let $f\colon X \to Y$ be an étale morphism. If Y is regular, then X is regular; the converse holds if f is surjective.*

Editorial note. The original printing assigns the number I.9.2 both to the preceding corollary and to the following proposition; the corrected French source records that duplicate numbering, which is retained here.

Proposition I.9.2. *Let $f\colon X \to Y$ be an étale morphism. If Y is reduced, then so is X; the converse holds if f is surjective.*

This is equivalent to the following corollary.

Corollary I.9.3. *Let $f\colon A \to B$ be a local étale homomorphism, with B isomorphic to a localization of an A-algebra of finite type. Then A is reduced if and only if B is reduced.*

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If B is reduced, then A is reduced, since $A \to B$ is injective (because B is faithfully flat over A). Conversely, suppose that A is reduced, and let $\mathfrak{p}_i$ be the minimal prime ideals of A. By hypothesis, the natural map $A \to \prod_i A/\mathfrak{p}_i$ is injective. Tensoring with the flat A-module B shows that $B \to \prod_i B/\mathfrak{p}_i B$ is injective, so it is enough to prove that the rings $B/\mathfrak{p}_i B$ are reduced. Since $B/\mathfrak{p}_i B$ is étale over $A/\mathfrak{p}_i$, we are reduced to the case where A is an integral domain.

Let K be its field of fractions. Since $A \to K$ is injective, so is $B \to B \otimes_A K$, because B is flat over A. It therefore remains to prove that the latter ring is reduced. Now B, being a localization of an A-algebra of finite type, is the local ring at a point x of a scheme $X = \mathrm{Spec}(C)$ of finite type and étale over $Y = \mathrm{Spec}(A)$. Consequently $B \otimes_A K$ is a localization, with respect to a suitable multiplicatively closed set, of the ring $C \otimes_A K$

of $X \otimes_A K$. Since $X \otimes_A K$ is étale over K, its ring is a finite product of fields that are finite separable extensions of K; the same is therefore true of $B \otimes_A K$. This proves the corollary.

COROLLARY I.9.4. *Let $f\colon A \to B$ be a local étale homomorphism, and suppose that A is analytically reduced (that is, the completion $\widehat{A}$ of A has no nilpotent elements). Then B is analytically reduced, and a fortiori reduced.*

Indeed, $\widehat{B}$[17] is *finite* and étale over $\widehat{A}$, so Corollary I.9.3 applies.

THEOREM I.9.5. *Let $f\colon A \to B$ be a local homomorphism, with B isomorphic to a localization of an A-algebra of finite type. Then:*

(i) *If f is étale, then A is normal if and only if B is normal.*
(ii) *If A is normal, then f is étale if and only if f is injective and unramified (and in that case B is normal by* (i)*).*

We shall give two different proofs of (i). The first uses certain properties of quasi-finite flat morphisms (recalled at the beginning of this number) without using Theorem I.7.6 (and hence the Main Theorem); in the second proof the roles are reversed. Finally, for (ii), it seems that the Main Theorem is needed in any case.

First proof. We use the following necessary and sufficient condition for a noetherian local ring A of nonzero dimension to be normal. French source p. 18

SERRE'S CRITERION. (i) *For every prime ideal $\mathfrak{p}$ of A of height 1, $A_\mathfrak{p}$ is normal (or, equivalently, regular).*
(ii) *For every prime ideal $\mathfrak{p}$ of A of height ≥ 2, one has* $\operatorname{depth} A_\mathfrak{p} \geq 2$.[18]

We shall admit this criterion here; it is supposed to appear in the section on flat morphisms. Its principal advantage is that it does not assume a priori that A is reduced, nor a fortiori that it is an integral domain. Here we may already assume $\dim A = \dim B \neq 0$.

By the facts recalled at the beginning of this number, the prime ideals $\mathfrak{p}$ of A of height 1 (respectively, height ≥ 2) are exactly the contractions to A of the prime ideals $\mathfrak{q}$ of B of height 1 (respectively, height ≥ 2). Finally, if $\mathfrak{p}$ and $\mathfrak{q}$ correspond, then $B_\mathfrak{q}$ is étale over $A_\mathfrak{p}$, so it has the same depth as $A_\mathfrak{p}$, and it is regular if and only if $A_\mathfrak{p}$ is regular (Proposition I.9.1). Applying Serre's criterion, we find that A is normal if and only if B is normal.

Second proof. Suppose that B is normal, and let L be its field of fractions and K that of A (the ring A is an integral domain because B is). We saw in the proof of Corollary I.9.3 that $B \otimes_A K$ is a finite product of fields. Since it is contained in L, it is a field, and since it contains B, it is L. An element of K that is integral over A is integral over B, so it belongs to B, since B is normal. It therefore belongs to A, because $B \cap K = A$ (as follows from the faithful flatness of B over A).

Now suppose that A is normal; we prove that B is normal. By Theorem I.7.6, we have $B = B'_\mathfrak{n}$, where

$$B' = A[t]/FA[t],$$

with F and $\mathfrak{n}$ as in Theorem I.7.6. Thus $L = B \otimes_A K$ is a localization of

$$B' \otimes_A K = K[t]/FK[t]$$

and is a product of fields, each a finite separable extension of K. This product, as happens whenever an artinian ring—here $B'_K = B' \otimes_A K$—is localized at a multiplicatively closed set, is a direct factor of B'_K. It therefore corresponds to a factorization $F = F_1F_2$ in $K[t]$, and the generator of L corresponding to t is already annihilated by F_1.

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[17] The original printing has B here; the corrected French TEX restores $\widehat{B}$.

[18] Cf. EGA IV 5.8.6.

But *because A is normal, the polynomials F_i belong to $A[t]$* (when they are taken to be monic). The map $B \to L = B \otimes_A K$ is injective, since $A \to K$ is injective and B is flat over A. Consequently, we already have $F_1(u) = 0$, where u is the class of t in L. If F has been chosen of minimal degree, it follows that $F_2 = 1$. (N.B. one has

$$F'(u) = F_1'(u)F_2(u) + F_1(u)F_2'(u) = F_1'(u)F_2(u),$$

because $F_1(u) = 0$; hence $F_1'(u) \neq 0$, since $F'(u) \neq 0$.)

Thus

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$$(*) \qquad L = B \otimes_A K = K[t]/FK[t].$$

Consequently, F is separable over K, though of course not necessarily separable over A.[19] (N.B. for the moment, we have essentially shown only that in Theorem I.7.6 one may choose F and $\mathfrak{n}$ in such a way that, with the notation used here, the map

$$B' \longrightarrow B'_{\mathfrak{n}} = B$$

is *injective*. We used the normality of A for this; I do not know whether the assertion remains true without the normality hypothesis.)

We now recall the well-known lemma extracted from Serre's course of last year.

LEMMA I.9.6. *Let K be a ring, let $F \in K[t]$ be a monic separable polynomial, let $L = K[t]/FK[t]$, and let u be the class of t in L (so that $F'(u)$ is a unit of L). Then, with $n = \deg F$,*

$$\mathrm{Tr}_{L/K}\left(\frac{u^i}{F'(u)}\right) = 0 \qquad \text{if } 0 \leq i < n-1,$$

$$\mathrm{Tr}_{L/K}\left(\frac{u^{n-1}}{F'(u)}\right) = 1.$$

COROLLARY I.9.7. *The determinant of the matrix*

$$\left(\mathrm{Tr}_{L/K}\left(\frac{u^j u^i}{F'(u)}\right)\right)_{0 \leq i,j \leq n-1}$$

is equal to $(-1)^{n(n-1)/2}$, and hence is a unit in every subring A of K.[20]

COROLLARY I.9.8. *Let A be a subring of K, and suppose that $F \in A[t]$.*[21] *Let V be the A-submodule of L generated by the elements u^i, $0 \leq i \leq n-1$, and let V' be the A-submodule of L consisting of the elements $x \in L$ such that*

$$\mathrm{Tr}_{L/K}(xy) \in A$$

for every $y \in V$ (that is, for $y = u^i$, $0 \leq i \leq n-1$). Then V' has as an A-basis the elements

$$\frac{u^i}{F'(u)}, \qquad 0 \leq i \leq n-1.$$

[19] The corrected French TEX reads *par nécessairement*, a grammatical regression; the original printing reads *pas nécessairement*. Since $F \in A[t]$ was already fixed, the contrast concerns separability over K and over A, not membership in the two polynomial rings.

[20] The French TEX and the original printing omit $\mathrm{Tr}_{L/K}$ from these matrix entries. Read literally, the printed matrix is an outer-product matrix of rank at most one, so for $n > 1$ its determinant is zero, contrary to the stated unit. Lemma I.9.6 and Corollary I.9.8 determine the intended trace-pairing matrix; the trace operator is restored here with the source defect disclosed. The original printing also gives the exponent n; the active French source uses $n(n-1)/2$, and its comment records Serre's correction of 27 November 2002.

[21] The French TEX and the original printing omit the hypothesis $F \in A[t]$ from this corollary. It is needed for the asserted basis. For example, take $A = \mathbb{Z}$, $K = \mathbb{Q}$, and $F = t(t - \frac{1}{2})$. Then $L \simeq \mathbb{Q} \times \mathbb{Q}$ and $u/F'(u) = (0, 1)$, but pairing with u has trace $1/2 \notin \mathbb{Z}$. Corollary I.9.9 imposes the missing coefficient hypothesis, so the application below is unaffected.

COROLLARY I.9.9. *Suppose that K is the field of fractions of a normal integral domain A, and that the coefficients of F belong to A. With the notation of Corollary I.9.8, the module V' then contains the integral closure A' of A in L. Consequently,*

$$A' \subseteq \frac{A[u]}{F'(u)} = F'(u)^{-1}A[u] \subseteq A[u][F'(u)^{-1}].$$

Apply the last corollary to the situation obtained in the proof. Since $F'(u)$ is a unit in B, and B contains $A[u]$, the ring B contains A'. By the Main Theorem (or directly from $B = A[u]_{\mathfrak{n}}$), B is a localization of A'. Since A' is normal, so is B.

Proof of (ii). We proceed as in the preceding proof to show that, in Theorem I.7.6, F can be chosen so that (*) still holds. The only a priori obstacle is that, since B is no longer assumed flat over A, we can no longer assert that $B \to L$ is injective; hence, a priori, the argument applies only to the image B_1 of B under that homomorphism. It follows at once that B_1 is flat over A (being a localization of an A-algebra that is free as an A-module). By Corollary I.4.8, the morphism $B \to B_1$ is étale, hence an isomorphism, which completes the proof. French source p. 20

(From the editorial point of view, the last two proofs should be interchanged, and the formal calculations of the lemma and its corollaries should be put in a separate number.)

COROLLARY I.9.10. *Let $f\colon X \to Y$ be an étale morphism. If Y is normal, then X is normal; the converse holds if f is surjective.*

COROLLARY I.9.11. *Let $f\colon X \to Y$ be a dominant morphism, with Y normal and X connected. If f is unramified, then f is étale; hence X is normal and therefore (being connected) irreducible.*

Let U be the set of points at which f is étale. It is open, and it suffices to show that it is also closed and nonempty. The set U contains the inverse image of the generic point of Y (for an algebra over a field, unramified is equivalent to étale); this inverse image is nonempty because X dominates Y. If x lies in the closure of U, then it lies in the closure of an irreducible component U_i of U, hence on an irreducible component $X_i \overset{v}{=} \overline{U_i}$[22] of X that meets U, and consequently dominates Y (for every irreducible component of U, being flat over Y, dominates Y). Let $y = f(x)$ be the image of x in Y. Then the homomorphism $\mathcal{O}_y \to \mathcal{O}_x$ is *injective* (since $\mathcal{O}_y$ is an integral domain). Since $\mathcal{O}_y$ is normal and $\mathcal{O}_y \to \mathcal{O}_x$ is unramified, Theorem I.9.5(ii) shows that this local homomorphism is étale. Thus $x \in U$, so U is closed. Since U is nonempty and X is connected, $U = X$, which proves the corollary.

COROLLARY I.9.12. *Let $f\colon X \to Y$ be a dominant morphism of finite type, with Y normal and X irreducible. Then the set of points at which f is étale coincides with the complement of the support of $\Omega^1_{X/Y}$, that is, with the complement of the closed subprescheme of X defined by the different ideal $\mathfrak{d}_{X/Y}$.*

(This is the "less trivial" statement alluded to in the remark in no. I.4.) French source p. 21

Remark *. One should beware of assuming that a connected étale covering of an irreducible scheme is itself irreducible when the base is not assumed to be normal. This question will be studied in no. I.11.

I.10. Étale coverings of a normal scheme

PROPOSITION I.10.1. *Let X be a prescheme that is étale and separated over a connected normal prescheme Y with function field K. Then the connected components X_i of X*

[22] The French TEX and the original printing place the superscript v above the equality sign. A nonprinting source comment tentatively suggests that it is a definition marker; we retain the printed symbol rather than assign it an unverified expansion.

are integral, their function fields K_i are finite separable extensions of K, and each X_i is identified with a nonempty open subset of the normalization of X in K_i (and hence X is identified with a dense open subset of the normalization of Y in $R(X) = L = \prod K_i$).

Editorial note. The corrected French authority TeX, the original printing, and the corrected re-typeset witness all read X in the first normalization clause, and the proof repeats X. The parenthetical conclusion and the Main Theorem argument appear to require Y. We retain the attested reading and flag the apparent source oddity.

By Corollary I.9.10, X is normal; a fortiori its local rings are integral domains, so the connected components of X are irreducible. Since X_i is normal, finite, and dominant over Y, it follows from a special case (an almost trivial one, moreover) of the Main Theorem that X_i is an open subset of the normalization of X in the field K_i of X.

Editorial note on the proof. The French authority and both checked print witnesses read "finite and dominant," "normalization of X," and "the field K_i of X." In context, the Main Theorem argument appears to require "quasi-finite and dominant"; the normalization base appears to be Y, as already noted after the statement; and K_i was defined as the function field of X_i. We retain all three attested readings and flag these tensions.

Corollary I.10.2. *Under the hypotheses of Proposition I.10.1, X is finite over Y (i.e. an étale covering of Y) if and only if X is isomorphic to the normalization Y' of Y in $L = R(X)$ (the ring of rational functions on X).*

Indeed, we know that this normalization is finite over Y (Y being normal and R/K separable); conversely, if X is finite over Y, then it is finite over Y', so its image in Y' is closed; on the other hand, that image is dense.

Editorial note on the proof. The French authority and both checked print witnesses read R/K. In the immediate context, Proposition I.10.1 gives $R(X) = L = \prod K_i$, Corollary I.10.2 gives $L = R(X)$, and the source later writes $K = R(Y)$. Thus L/K (equivalently $R(X)/K$) appears intended. We retain the attested R/K reading above and flag the tension.

An algebra L of finite rank over K will be said to be *unramified over X* (or simply unramified over K, if X is understood) if L is a separable algebra over K, that is, a direct product of separable extensions K_i, and if the normalization Y' of Y in L (the disjoint union of the normalizations of Y in the K_i) is unramified (that is, étale by Corollary I.9.11) over Y. Thus:

Editorial note. The corrected French authority TeX, the original printing, and the corrected re-typeset witness all read X in the term being defined and in the phrase "if X is understood." The defining criterion, the identity $K = R(Y)$, and Corollary I.10.3 instead use Y. Thus Y appears intended in those two places. We retain the attested X reading and flag the apparent source oddity.

Corollary I.10.3. *For every X finite over Y such that every irreducible component of X dominates Y, let $R(X)$ denote the ring of rational functions on X (the product of the local rings at the generic points of the irreducible components of X), so that $X \longmapsto R(X)$ is a functor with values in algebras of finite rank over $K = R(Y)$. This functor establishes an equivalence between the category of connected étale coverings of Y and the category of extensions L of K unramified over Y.*

French source p. 22

The inverse functor is the normalization functor.

Suppose that Y is affine, say $Y = \mathrm{Spec}(A)$, where A is normal with field of fractions K. Let L be a finite K-algebra that is a product of fields. By definition, the normalization Y' of Y in L is then isomorphic to $\mathrm{Spec}(A')$, where A' is the normalization of A in L. To say that L is unramified over Y means that A' is unramified over A, or equivalently that it is étale over A. If A is local, this is equivalent to saying that the local rings $A'_{\mathfrak{n}}$, where $\mathfrak{n}$ ranges over the finite set of maximal ideals of A', that is, its prime ideals lying over the

maximal ideal $\mathfrak{m}$ of A, are unramified (that is, étale) over the local ring A. Finally, let us also note that the discriminant criterion (Proposition I.4.10) can be applied in this situation. More generally, a variant of that criterion should be stated as follows, without a preliminary flatness condition when X dominates Y, with Y nevertheless assumed locally integral: the maps $A \longrightarrow B$ and $B \longrightarrow B \otimes_A K = L$ are injective—then $\mathrm{Tr}_{L/K}$ is defined—and $\mathrm{Tr}_{L/K}(xy)$ induces a *fundamental bilinear form* $B \times B \longrightarrow A$; that is, there exist elements $x_i \in B$ ($1 \leq i \leq n$, where $n = \mathrm{rank}_K L$) such that $\mathrm{Tr}_{L/K}(x_i x_j) \in A$ for all i, j, and $\det\Big(\big(\mathrm{Tr}_{L/K}(x_i x_j)\big)_{1 \leq i,j \leq n}\Big)$ is invertible in A.

Editorial note. The corrected French TeX capitalizes the conditional, restores $\mathfrak{n}$ in its second occurrence, deletes a stray X, and repairs the original malformed determinant phrase. We follow that corrected branch. The explicit matrix indices make visible the i, j-range already quantified in the sentence; they are not a new mathematical condition.

The sorites (I.4.6) immediately implies the sorites of unramifiedness in the classical setting:

Proposition I.10.4. *Let Y be a normal integral prescheme with function field K.* (i) *K is unramified over Y.* (ii) *If L is an extension of K unramified over Y, if Y' is a normal prescheme with function field L that dominates Y (for example, the normalization of Y in L), and if M is an extension of L unramified over Y', then M/K is unramified over X (*transitivity *of unramifiedness).* (iii) *Let Y' be a normal integral prescheme dominating Y, with function field extension K'/K; if L is an extension of K unramified over Y, then $L \otimes_K K'$ is an extension of K' unramified over Y' (the* translation *property).*

Editorial note. The corrected French TeX changes the original plural opening to the singular and supplies the full stop ending item (i); the target follows those corrections. The corrected French authority TeX, the original printing, and the corrected re-typeset witness all read X in item (ii). Proposition I.10.4 introduces Y, K, L, Y', M, K', but no X; the defining criterion preceding Corollary I.10.3 and the transitivity analogue of Proposition I.4.6 indicate that Y is intended. We retain the attested X reading and flag the apparent source-notation defect.

French source p. 23

Moreover:

Corollary I.10.5. *Under the hypotheses of Proposition I.10.4*(iii), *if $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(A')$, then $\bar{A}'$,*[23] *the coordinate ring of the normalization of Y' in $L' = L \otimes_K K'$, is identified with $\bar{A} \otimes_A A'$, where $\bar{A}$ is the normalization of A in L.*

Usually, people (who are reluctant to consider nonintegral rings, even when they are direct products of fields) state the translation property in the following (weaker) form:

Corollary I.10.6. *Under the hypotheses of Proposition I.10.4*(iii), *let L_1 be a* composite extension *of L/K (unramified over Y) and of K'/K. Then L_1/K' is unramified over Y'. If $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(A')$, one also has*

$$\overline{A'} = A[\overline{A}, A'].$$

That is, the ring $\overline{A'}$, the normalization of A' in L_1, is the A-algebra generated by A' and by the normalization $\overline{A}$ of A in L.

This last assertion is false without the unramifiedness hypothesis, even for composite extensions of number fields ...

To conclude this section, we shall give the interpretation of the notion of an étale covering that corresponds to its intuitive picture: there should be the "maximum number" of points above the point $y \in Y$ under consideration, and in particular there should not be "several points merged together" above y. To prove the results in this direction in the

[23] The original printing has the unbarred A' at this point; the corrected French TEX replaces it with $\bar{A}'$, and the target follows that corrected branch.

desired generality, we shall admit Proposition I.10.7 below (whose proof will appear in the *multiplodoque*, Chapter IV, no. 15, and uses Chevalley's technique of constructible sets and a little descent theory ...).

A morphism of finite type $f\colon X \to Y$ is called *universally open* if, for every base extension $Y' \to Y$ (with Y' locally noetherian), the morphism

$$f'\colon X' = X \times_Y Y' \longrightarrow Y'$$

is open, that is, carries open sets to open sets. It is enough to consider the case in which Y' is of finite type over Y (and even the case in which Y' is of the form $Y[t_1, \ldots, t_r]$, where the t_i are indeterminates).

A universally open morphism is *a fortiori* open (the converse is false). On the other hand, if f is open and X and Y are irreducible, then all the components of all the fibers of f have the same dimension, namely the dimension of the generic fiber $f^{-1}(z)$, where z is the generic point of Y. Finally, if Y is normal, this last condition already implies that f is universally open (Chevalley's theorem). It follows, for example, that if $f\colon X \to Y$ is quasi-finite, with Y normal and irreducible, then f is universally open (equivalently, open) if and only if every irreducible component of X dominates Y. Recall also that a flat morphism of finite type is open, and hence universally open. With these preliminaries established, let us "recall" the following proposition.

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PROPOSITION I.10.7. *Let $f\colon X \to Y$ be a quasi-finite, separated, universally open morphism. For every $y \in Y$, let $n(y)$ be the "geometric number of points of the fiber $f^{-1}(y)$", equal to the sum of the separable degrees of the residue-field extensions $\kappa(x)/\kappa(y)$, for $x \in f^{-1}(y)$. Then the function $y \longmapsto n(y)$ on Y is upper semicontinuous. If it is constant in a neighborhood of y (that is, if $n(y) = n(z_i)$, where the z_i are the generic points of the irreducible components of Y that contain y), then there must exist a neighborhood U of y such that $X|_U$ is finite over U.* [24]

COROLLARY I.10.8. *If $y \longmapsto n(y)$ is constant and Y is geometrically unibranch,*[25] *then the irreducible components of X are disjoint.*

PROPOSITION I.10.9. *Let $f\colon X \to Y$ be an* étale*, separated morphism. With the notation of Proposition I.10.7, the function $y \longmapsto n(y)$ is upper semicontinuous. For it to be constant in a neighborhood of y (that is, for $n(y) = n(z_i)$, where the z_i are the generic points of the irreducible components of Y that contain y), it is necessary and sufficient that there exist an open neighborhood U of y such that $X|_U$ is finite over U, that is, is an* étale covering *of U.*

COROLLARY I.10.10. *For a separated étale morphism $f\colon X \to Y$, with Y connected, to be finite (that is, to make X an* étale covering *of Y), it is necessary and sufficient that all the fibers of f have the same geometric number of points.*

In Proposition I.10.7 and its corollary there was no normality hypothesis on Y. If such a hypothesis is imposed, one obtains the stronger statement below (which is most often taken as the definition of the unramifiedness of a covering).

French source p. 25

THEOREM I.10.11. *Let $f\colon X \to Y$ be a* quasi-finite*, separated morphism. Suppose that Y is* irreducible*, that every component of X dominates Y, and that X is reduced (that is, $\mathcal{O}_X$ has no nilpotent elements). Let n be the degree of X over Y (the sum of the degrees, over the field K of Y, of the fields K_i of the irreducible components X_i of X). Let y be a normal point of Y. Then the geometric number $n(y)$ of points of X above y is at most n, and equality holds if and only if there exists an open neighborhood U of y such that $X|_U$ is an* étale covering *of U.*

[24] Cf. EGA IV, 15.5.1.

[25] For the definition, see no. I.11 below.

The implication from the existence of such an étale neighborhood to $n(y) = n$ is trivial. Let us prove the converse.[26] Let z be the generic point of Y. We have

$$n(z) = \text{the sum of the separable degrees of the } K_i/K \leq n,$$

and Proposition I.10.7 gives $n(y) \leq n(z) \leq n$. Equality implies that $X|_U$ is finite over U for a suitable neighborhood U of y. We may therefore suppose that X is finite over Y and that the function $n(y')$ on Y is constant. By Corollary I.10.8, X is then the disjoint union of its irreducible components; to prove that it is unramified at y, we are reduced to the case in which X is irreducible, hence integral. Finally, we may suppose that $Y = \mathrm{Spec}(\mathcal{O}_y)$. The theorem then reduces to the following classical statement.

Corollary I.10.12. *Let A be a normal local ring (noetherian, as always) with fraction field K; let L be a finite extension of K, of degree n and separable degree n_s; and let B be a subring of L, finite over A, with fraction field L. Let $\mathfrak{m}$ be the maximal ideal of A, and let n' be the separable degree of $B/\mathfrak{m}B$ over $A/\mathfrak{m}A = k$ (that is, the sum of the separable degrees of the residue-field extensions of this ring). Then $n' \leq n_s$, and* a fortiori *$n' \leq n$. Equality in the latter inequality holds if and only if B is unramified (equivalently, étale) over A.*

It remains only to show that $n' = n$ implies that B is étale over A. Recall the proof when k is infinite. It is enough to show that $R = B/\mathfrak{m}B$ is separable over k. If it were not, a familiar lemma would imply that there is an element $a \in R$ whose minimal polynomial over k has degree $> n'$. This element is the image of an element $x \in B$, whose minimal polynomial over K (as an element of L) has degree at most n. Moreover, the coefficients of this last polynomial lie in A, since A is normal. Reduction modulo $\mathfrak{m}$ therefore gives a monic polynomial $F \in k[t]$, of degree at most $n = n'$, such that $F(a) = 0$, a contradiction. French source p. 26

In the general case (where k may be finite), return to geometric language and put $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$. Consider $Y' = \mathrm{Spec}(A[t])$, which is faithfully flat over Y, and the generic point y' of the fiber $\mathrm{Spec}(k[t])$ of Y' over y. Then X is unramified over Y at y if and only if

$$X' = X \times_Y Y' = \mathrm{Spec}(B[t])$$

is unramified at y' over Y', as one sees immediately. By the choice of y', its residue field is $k(t)$, hence infinite. Since y' is a normal point of Y', this reduces the assertion to the preceding case.

I.11. Some complements

We have already said that a connected étale covering of an integral scheme need not be integral. Here are two examples of this fact.

a) Let C be an algebraic curve with an ordinary double point x, let C' be its normalization, and let a and b be the two points of C' above x. Let C'_i $(i = 1, 2)$ be two copies of C', and let a_i and b_i be the points of C'_i corresponding respectively to a and b. In the disjoint union $C'_1 \amalg C'_2$, identify a_1 with b_2, and a_2 with b_1. (The reader is left to make the identification process precise; it will be explained in Chapter VI of the *multiplodoque*, but, for curves over an algebraically closed field, it is treated in Serre's book on algebraic curves.) The result is a *connected* and *reducible* curve C'', which is an étale covering of C of degree 2. More generally, the reader will verify that the connected "Galois" étale coverings C'' of C for which the inverse image $C'' \times_C C'$ is a *trivial* covering of C' (that is, isomorphic to a disjoint union of some number of copies of C') are "cyclic" of degree n, and that, for every integer $n > 0$, one can construct a connected cyclic étale covering of degree n. In the language of the fundamental group, which will be developed later, this

[26] The French text names these two directions in the opposite order; the argument itself starts from $n(y) = n$ and proves the existence of the étale neighborhood, as translated here.

means that the quotient of $\pi_1(C)$ by the closed normal subgroup generated by the image of

$$\pi_1(C') \longrightarrow \pi_1(C)$$

(the homomorphism induced by the projection) is isomorphic to the profinite completion of $\mathbb{Z}$. More precisely, one should be able to show that the fundamental group of C is isomorphic to the topological free product of the fundamental group of C' with the profinite completion of $\mathbb{Z}$. Notice that questions of this kind gave rise to "descent theory" for schemes.

b) Let A be a complete local integral domain. Its normalization A' is known to be finite over A (Nagata); it is therefore a complete semilocal ring, and hence local because it is an integral domain. Suppose that the residue-field extension L/k thereby defined is not radicial (in the contrary case, A will be called geometrically unibranch; see below). This is the case, for example, for the ring

French source p. 27

$$\mathbb{R}[[s,t]]/(s^2+t^2)\mathbb{R}[[s,t]],$$

where $\mathbb{R}$ is the field of real numbers. Let k' be a finite Galois extension of k such that $L \otimes_k k'$ splits, and let B be a finite étale algebra over A corresponding to the residue-field extension k' (recall that B is essentially unique). Then

$$B' = A' \otimes_A B$$

over B has residue algebra $L \otimes_k k'$, which is not local. Thus B' is not a local ring and, being complete, has zero divisors. But B' is contained in the total ring of fractions of B: indeed, it is free over A', hence torsion-free over A', hence torsion-free over A, and is consequently contained in

$$B' \otimes_A K = B'_{(K)} = A'_{(K)} \otimes_K B_{(K)} = B_{(K)},$$

since $A'_{(K)} = K$. It follows that B is not an integral domain. In the case of the ring

$$\mathbb{R}[s,t]/(s^2+t^2)\mathbb{R}[s,t],$$

taking $k'/k = \mathbb{C}/\mathbb{R}$, the ring B obtained is the local ring, at their point of intersection, of two intersecting lines in the plane.

Notice also that, if there exists a connected étale covering X of an integral Y that is not irreducible, then every irreducible component of X gives an example of an unramified covering X' of Y, dominating Y, that is not étale over Y. Thus, in example a), C' is unramified over C without being étale at the two points a and b. This can also be seen directly by inspecting the completions of the local rings at x and a: from the "formal" point of view, C' at a is identified with a closed subscheme of C at x, namely one of the two "branches" of C passing through x.

In a) and b), the failure of the conclusions of Proposition I.9.5(i) and (ii) is seen to be directly connected with the fact that a point of Y "splits" into *distinct* points of the normalization. In b), the fact that the residue-field extension is not radicial is to be interpreted geometrically in this way. More precisely, we shall say that a local integral domain A is *geometrically unibranch* if its normalization has only one maximal ideal and the corresponding residue-field extension is radicial. A point y of an integral prescheme is said to be geometrically unibranch if its local ring is so. Examples are a normal point, an ordinary cusp of a curve, and so on. It seems that, if Y has a point that is not unibranch, there always exists a connected, nonirreducible étale covering of Y; this is at least what we have shown in case b), when Y is the spectrum of a complete local ring. On the other hand, one can show that *if every point of Y is geometrically unibranch, then every connected unramified Y-prescheme dominating Y is étale* and irreducible. The proof

French source p. 28

repeats that of Proposition I.9.5, using the following generalization of Theorem I.8.3, which will be proved later by means of descent theory:[27]

Let $Y' \to Y$ be a finite, radicial, surjective morphism (that is, what might be called a "universal homeomorphism"). Consider the functor

$$X \longmapsto X \times_Y Y' = X'$$

from Y-preschemes to Y'-preschemes. This functor induces an equivalence between the category of étale Y-schemes and the category of étale Y'-schemes. For example, this result may be applied when Y' is the normalization of Y, with Y assumed unibranch (and Y' finite over Y, as is true in all cases encountered in practice), or when a Y'' is "sandwiched" between Y and its normalization; in the latter case the normalization need no longer be finite over Y.

[27]Cf. IX 4.10. For a more direct proof, cf. EGA IV, 18.10.3, using a variant of Proposition I.9.5 for geometrically unibranch local rings.

CHAPTER II

Smooth morphisms: generalities and differential properties

French source p. 29

References to Exposé I are indicated by I. Recall that all rings are noetherian and all preschemes locally noetherian.

II.1. Generalities

Let Y be a prescheme, and let $t_1, \dots, t_n$ be indeterminates. We write

$$Y[t_1, \dots, t_n] = Y \otimes_{\mathbb{Z}} \mathbb{Z}[t_1, \dots, t_n]. \tag{II.1.1}$$

Thus $Y[t_1, \dots, t_n]$ is a Y-scheme, affine over Y, defined by the quasi-coherent sheaf of algebras $\mathcal{O}_Y[t_1, \dots, t_n]$. Giving a section of this prescheme over Y is therefore equivalent to giving n sections of $\mathcal{O}_Y$, which correspond to the images of the t_i under the associated homomorphism. If Y' lies over Y, then

$$Y[t_1, \dots, t_n] \times_Y Y' = Y'[t_1, \dots, t_n]. \tag{II.1.2}$$

Consequently, giving a Y-morphism from Y' to $Y[t_1, \dots, t_n]$ is equivalent to giving n sections of $\mathcal{O}_{Y'}$. We also have

$$\big(Y[t_1, \dots, t_n]\big)[t_{n+1}, \dots, t_m] = Y[t_1, \dots, t_m], \tag{II.1.3}$$

by the analogous formula for polynomial rings over $\mathbb{Z}$. Formula (II.1.2) shows that $Y[t_1, \dots, t_n]$ varies functorially with Y.

The Y-scheme $Y[t_1, \dots, t_n]$ is of finite type and flat over Y.

Definition II.1.1. *Let $f\colon X \to Y$ be a morphism, making X a Y-prescheme. We say that f is* smooth[1] *at $x \in X$, or that X is* smooth over Y at x, *if there exist an integer $n \geq 0$, an open neighborhood U of x, and an étale Y-morphism from U to $Y[t_1, \dots, t_n]$. We say that f is* smooth *(respectively, that X is* smooth over Y*) if it is smooth at every point of X.*

An algebra B over a ring A is said to be smooth at a prime ideal $\mathfrak{p}$ of B if $\operatorname{Spec}(B)$ *is smooth over* $\operatorname{Spec}(A)$ *at $\mathfrak{p}$; B is said to be smooth over A if it is smooth over A at every prime ideal $\mathfrak{p}$ of B. Finally, a local homomorphism $A \to B$ of local rings is said to be smooth (or B is said to be smooth over A)* [2] *if B is a localization of an algebra B_1 of finite type and smooth over A.*

The notion that X is smooth over Y is local on X and on Y. If X is smooth over Y, then it is locally of finite type over Y.

Proposition II.1.1. *The set of points x of X at which f is smooth is open.*

This is immediate from the definition.

Corollary II.1.2. *If B is smooth over A at $\mathfrak{p}$, then it is smooth over A at every prime ideal $\mathfrak{q}$ of B contained in $\mathfrak{p}$.*

Proposition II.1.1 also implies that the last two definitions in Definition II.1.1 agree wherever both apply.

[1] Former terminology: f is *simple* at x, or x is a *simple* point for f. This terminology caused confusion in various contexts (simple algebras, simple groups) and had to be abandoned.

[2] It is preferable in this case, as in EGA IV, 18.6.1, to say that B is *essentially smooth* over A.

PROPOSITION II.1.3. (i) *An étale morphism, and in particular an open immersion or an identity morphism, is smooth.*

(ii) *A base extension of a smooth morphism is smooth.*

(iii) *The composite of two smooth morphisms is smooth.*

Assertion (i) is immediate from the definition. More precisely, we have the following.

COROLLARY II.1.4.

$$\textit{étale} = \textit{quasi-finite} + \textit{smooth}.$$

Assertion (ii) follows at once from the analogous fact for étale morphisms (I I.4.6) and for the projections $Y[t_1, \dots, t_n] \to Y$; see (II.1.2). Assertion (iii) follows formally because it holds separately for étale morphisms (I I.4.6) and for projections of the form $Y[t_1, \dots, t_n] \to Y$, by (II.1.3), together with the two facts just cited for (ii). Indeed, suppose that Y is smooth over Z and that X is smooth over Y; we prove that X is smooth over Z. We may suppose that Y is étale over $Z[t_1, \dots, t_n]$, and that X is étale over $Y[s_1, \dots, s_m]$. The first hypothesis implies that $Y[s_1, \dots, s_m]$ is étale over

$$(Z[t_1, \dots, t_n])[s_1, \dots, s_m] = Z[t_1, \dots, t_n, s_1, \dots, s_m].$$

It follows that X is étale over $Z[t_1, \dots, t_n, s_1, \dots, s_m]$, as required.

Remark II.1.5. The integer n occurring in Definition II.1.1 is indeed determined: it is immediately seen to be the dimension of the local ring of x in its fiber $f^{-1}(f(x))$. It is called the *relative dimension* of X over Y. Relative dimension is additive under composition of morphisms.

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II.2. Some criteria for smoothness of a morphism

THEOREM II.2.1. *Let $f\colon X \to Y$ be a morphism locally of finite type, let $x \in X$, and put $y = f(x)$. Then f is smooth at x if and only if* (a) *f is flat at x, and* (b) *the fiber $f^{-1}(y)$ is smooth over $\kappa(y)$ at x.*

Since the composite of two flat morphisms is flat, and $Y[t_1, \dots, t_n] \to Y$ is flat, every smooth morphism is flat. Together with Proposition II.1.3(ii), this proves the necessity of the two conditions. Conversely, suppose that (a) and (b) hold. Let V be an affine neighborhood of y, with ring A, and let U be an affine neighborhood of x above V, with ring B. After shrinking U, condition (b) allows us to suppose that there is an étale $\kappa(y)$-morphism

$$g\colon U|_{f^{-1}(y)} \longrightarrow \operatorname{Spec} k[t_1, \dots, t_n], \qquad k = \kappa(y),$$

defined by n sections g_i of the structure sheaf of $U|_{f^{-1}(y)}$. We may plainly suppose that the g_i, which a priori are elements of

$$B \otimes_A k = BS^{-1}, \qquad S = A - \mathfrak{p},$$

where $\mathfrak{p}$ is the prime ideal of A corresponding to y, come from sections of the structure sheaf of U. Thus g is induced by a morphism, still denoted by g,

$$g\colon U \longrightarrow Y[t_1, \dots, t_n],$$

after multiplying all the g_i, if necessary, by the same nonzero element of k. Now U is flat over Y by (a), as is $Y[t_1, \dots, t_n]$; moreover, g induces an étale morphism between the fibers over y. Therefore g is étale at x, by I I.5.8, as required.

COROLLARY II.2.2. *Let S be a prescheme, and let $f\colon X \to Y$ be an S-morphism of finite type, where Y is of finite type and flat over S. Let $x \in X$, and let s be its image in S. Then f is smooth at x if and only if X is flat (or: smooth) over S at x, and the induced morphism on the fibers over s,*

$$f_s\colon X_s \longrightarrow Y_s,$$

is smooth at x.

Only sufficiency requires proof; it follows from the criterion in Theorem II.2.1, together with the flatness criterion I I.5.9.

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To state the next result, "recall" that a morphism $f\colon X \to Y$ locally of finite type is said to be *equidimensional* at $x \in X$ if, on writing $y = f(x)$, there is an open neighborhood U of x, every component of which dominates a component of Y, such that, for every $y' \in Y$, all the irreducible components of $f^{-1}(y') \cap U$ have one and the same dimension, independent of y'. In this condition it is enough to take for y' the generic points of the irreducible components of Y passing through y, *and* the point y itself. If, for example, X and Y are integral and f is dominant, the condition says that the components of $f^{-1}(y)$ passing through x have the "correct" dimension, that is, the dimension of the generic fiber. Recall that their dimension is always at least that of the generic fiber.

Suppose that f is equidimensional at x, that the dimension of its fiber at x is n, and that

$$g\colon U \longrightarrow Y' = Y[t_1, \ldots, t_n]$$

is a Y-morphism from a neighborhood U of x, inducing on the fibers over y a morphism that is quasi-finite at x. Equivalently, suppose that g itself is quasi-finite at x. One then shows that every irreducible component of U passing through x dominates an irreducible component of Y'. Moreover, by the "normalization lemma", such a g always exists. Conversely, if there is a quasi-finite Y-morphism g from an open neighborhood U of x to a Y-scheme of the form $Y' = Y[t_1, \ldots, t_n]$, such that every component of U passing through x dominates a component of Y', then f is equidimensional at x. With this understood, we have:

PROPOSITION II.2.3. *Let $f\colon X \to Y$ be a morphism locally of finite type, let $x \in X$, and put $y = f(x)$. Suppose that $\mathcal{O}_y$ is normal. Then f is smooth at x if and only if f is equidimensional at x and the fiber $f^{-1}(y)$ is smooth over $\kappa(y)$ at x.*

It follows at once from the definition that a smooth morphism is equidimensional. Notice, however, that a flat morphism of finite type need not be equidimensional at x, even if its fiber at x is irreducible. Let us prove the converse. Since $f^{-1}(y)$ is smooth over $\kappa(y)$ at x, we may suppose, after replacing X by a suitable neighborhood of x if necessary, that there is a Y-morphism

$$g\colon X \longrightarrow Y[t_1, \ldots, t_n] = Y'$$

which induces an étale morphism on the fibers over y, and is therefore quasi-finite at x. Thus g is unramified, and, since f is equidimensional at x, the irreducible components of X passing through x each dominate an irreducible component of Y'. Consequently, the homomorphism

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$$\mathcal{O}_{y'} \longrightarrow \mathcal{O}_x, \qquad y' = g(x),$$

induced by g is injective. This homomorphism is also unramified, and $\mathcal{O}_{y'}$ is normal because it is a localization of $\mathcal{O}_y[t_1, \ldots, t_n]$, which is normal since $\mathcal{O}_y$ is normal. Hence $\mathcal{O}_{y'} \to \mathcal{O}_x$ is étale, by I I.9.5(ii).

Remarks II.2.4. The preceding statement remains valid if the hypothesis that $\mathcal{O}_y$ is normal is replaced by the weaker hypothesis that Y is *geometrically unibranch* at y; see I I.11. Indeed, I I.9.5 remains valid under this hypothesis.

Let us also take this opportunity to point out that, if the residue field of a local integral domain A is algebraically closed, then analytic irreducibility—that is, the condition that $\widehat{A}$ be an integral domain—implies geometric unibranchedness. The converse also holds in every category of "good rings", more precisely in a category of rings stable under the usual operations and in which the completion of a normal local ring is normal. By Zariski's "theorem of analytic normality", this condition is satisfied in the category of affine algebras and their localizations.[3]

[3]Cf. EGA IV, 7.8.

Finally, "recall" in the present context the following result of Hironaka,[4] which can sometimes be used to ensure that $f^{-1}(y)$ is a reduced scheme, that is, that it coincides with what many algebraic geometers would loosely regard as the fiber of f above y without multiplicities, namely $f^{-1}(y)_{\mathrm{red}}$.

PROPOSITION II.2.5. *Let $f\colon X \to Y$ be a dominant morphism of finite type between reduced preschemes, and let $y \in Y$ be a point such that $\mathcal{O}_y$ is regular. Suppose that every component of $f^{-1}(y)$ has multiplicity 1 (in the sense defined below), and that $f^{-1}(y)_{\mathrm{red}}$ is normal. Then $f^{-1}(y)$ is reduced, hence normal; X is normal at every point of $f^{-1}(y)$; and X is flat over Y at every point of $f^{-1}(y)$.*

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A component Z of $f^{-1}(y)$ is said to have *multiplicity* 1 if, with x denoting the generic point of Z, the following conditions hold:

(i) $\dim \mathcal{O}_x = \dim \mathcal{O}_y$, that is, Z is not an "excess component" or, equivalently, is not of "excessively large dimension";

(ii) the maximal ideal of $\mathcal{O}_x$ is generated by the maximal ideal of $\mathcal{O}_y$, which a priori, by the choice of x, generates an ideal of definition of $\mathcal{O}_x$.

Taking Proposition II.2.3, or Theorem II.2.1, into account, we obtain:

COROLLARY II.2.6. *Let $f\colon X \to Y$ be a dominant morphism of finite type between reduced preschemes, and let $y \in Y$ be a point such that $\mathcal{O}_y$ is regular. Then f is smooth at all points of X above y if and only if the components of $f^{-1}(y)$ have multiplicity 1 and $f^{-1}(y)_{\mathrm{red}}$ is smooth over $\kappa(y)$.*

In the past, this situation was considered especially when Y was the spectrum of a discrete valuation ring A, and was commonly described in such terms as: "if the reduction of X with respect to the given valuation is nice"... . Moreover, X then denoted a closed subscheme (if one may say so) of some $\mathbb{P}^n_K$, where K is the fraction field of A; and, for want of adequate language, the more intrinsic role of an object "defined over A" rather than merely over K was scarcely apparent.

II.3. Permanence properties

PROPOSITION II.3.1. *Let $f\colon X \to Y$ be a morphism, let $x \in X$, and put $y = f(x)$. Suppose that f is smooth at x. Then $\mathcal{O}_x$ is reduced (respectively, regular; respectively, normal) if and only if $\mathcal{O}_y$ has the corresponding property.*

This statement is known when X is of the form $Y[t_1, \dots, t_n]$, and it was proved in I, no. I.9, for an étale morphism. The general case follows at once from Definition II.1.1.

We shall not detail here the other permanence properties, which already follow either from flatness alone or from the fact that X is locally quasi-finite and flat over a Y-prescheme of the form $Y[t_1, \dots, t_n]$, that is, as we shall say, from the fact that X is Cohen–Macaulay over Y. We point out only that this latter fact implies

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$$\begin{aligned} \dim \mathcal{O}_x &= \dim \mathcal{O}_y + n - d, \\ \operatorname{depth} \mathcal{O}_x &= \operatorname{depth} \mathcal{O}_y + n - d, \end{aligned} \tag{II.3.1}$$

where n is the dimension of the fiber of f at x, and d is the transcendence degree of $\kappa(x)$ over $\kappa(y)$. Thus, on putting

$$\operatorname{codepth} = \dim - \operatorname{depth},$$

we obtain[5]

$$\operatorname{codepth} \mathcal{O}_x = \operatorname{codepth} \mathcal{O}_y. \tag{II.3.2}$$

It follows, for example, that $\mathcal{O}_x$ is Cohen–Macaulay (respectively, has no embedded components) if and only if the same is true of $\mathcal{O}_y$.

[4] Cf. EGA IV, 5.12.10.

[5] For these formulas, cf. EGA IV, 6.1 and 6.3.

II.4. Differential properties of smooth morphisms

For simplicity, we shall for the most part restrict ourselves to first-order differential calculus, giving only brief indications for higher orders, where the results are just as simple.

For the definition of the sheaf $\Omega^1_{X/Y}$ of first differentials of a Y-prescheme X, see I, no. I.1. Suppose that X and Y are S-preschemes and that the morphism $f\colon X \to Y$ is an S-morphism. Then f defines a homomorphism of sheaves of modules, compatible with f,

$$f^*\colon \Omega^1_{Y/S} \longrightarrow \Omega^1_{X/S}. \tag{II.4.1}$$

In other words, $\Omega^1_{X/S}$ is *contravariant* in the S-prescheme X. Equivalently, (II.4.1) gives a homomorphism of modules on X,

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}, \tag{4.1bis}$$

also denoted by f^*, for lack of a better notation. It belongs to the canonical exact sequence

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0. \tag{II.4.2}$$

All these homomorphisms are defined by the requirements that they be local, which reduces their definition to the affine case, and that they commute with the differential operators d. The exactness of (II.4.2) is classical and immediate. In the affine case, for a homomorphism $B \to C$ of A-algebras, it becomes the exact sequence

$$\Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{C/A} \longrightarrow \Omega^1_{C/B} \longrightarrow 0. \tag{4.2bis}$$

Lemma II.4.1. *Let $f\colon X \to Y$ be a morphism of S-preschemes. If f is unramified (respectively, étale), then*

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$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is surjective (respectively, an isomorphism). The converse in the unramified case is true if f is assumed locally of finite type.

The unramified assertion follows from the exact sequence (II.4.2) and I I.3.1; it may also be seen directly as part of the étale case. For a direct proof of the latter, consider the diagram

$$\begin{array}{ccccc} X & \xrightarrow{\Delta_{X/Y}} & X \times_Y X & \longrightarrow & X \times_S X \\ & & \downarrow & & \downarrow \\ & & Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y. \end{array}$$

Here $X \times_Y X$ is the fiber product of Y and $X \times_S X$ over $Y \times_S Y$. Since f is unramified, $X \to X \times_Y X$ is an open immersion. Hence the conormal sheaf of the composite immersion $\Delta_{X/S}$ is isomorphic to the inverse image on X of the conormal sheaf of $X \times_Y X \to X \times_S X$. In the étale case, $X \to Y$ is flat, so $X \times_S X \to Y \times_S Y$ is flat. The conormal sheaf of $X \times_Y X \to X \times_S X$ is therefore the inverse image of the conormal sheaf of $Y \to Y \times_S Y$, that is, the inverse image of $\Omega^1_{Y/S}$. The assertion follows.

Lemma II.4.2. *Let $X = Y[t_1, \ldots, t_n]$, where Y is an S-prescheme. Then the sequence of canonical homomorphisms*

$$0 \longrightarrow f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact, and $\Omega^1_{X/Y}$ is free with basis $\mathrm{d}_{X/Y}t_1, \ldots, \mathrm{d}_{X/Y}t_n$.

The verification, which is purely affine, is immediate. Notice that the exactness at the right was already known from (II.4.2).

Combining these two statements with Definition II.1.1, we obtain:

Theorem II.4.3. *Let $f\colon X \to Y$ be a smooth morphism of S-preschemes. Then:*

(i) *the sequence of canonical homomorphisms*

$$0 \longrightarrow f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact;

(ii) $\Omega^1_{X/Y}$ *is locally free, and its rank* n *at* x *is equal to the relative dimension of* f *at* x.

COROLLARY II.4.4. *The homomorphism*

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is injective, and its image in $\Omega^1_{X/S}$ *is locally a direct summand.*

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Let $u\colon F \to G$ be a homomorphism of modules on the prescheme X. It is said to be *universally injective* at $x \in X$ if the homomorphism $F_x \to G_x$ of $\mathcal{O}_x$-modules is injective and remains so after tensoring with every $\mathcal{O}_x$-algebra, or, equivalently, with every $\mathcal{O}_x$-module. It is enough, for example, that there exist an open neighborhood U of x on which u identifies $F|_U$ with a direct summand of $G|_U$. This condition is also necessary when F and G are free of finite type in a neighborhood of x. More precisely, in that case the following conditions are equivalent:

(i) u is injective at x, and $\mathrm{Coker}(u)$ is free at x;

(ii) there exists an open neighborhood U of x on which u identifies $F|_U$ with a direct summand of $G|_U$;

(iii) u is universally injective at x;

(iv) the homomorphism on the fibers induced by u,

$$F_x \otimes_{\mathcal{O}_x} \kappa(x) \longrightarrow G_x \otimes_{\mathcal{O}_x} \kappa(x),$$

is injective;

(v) the transpose homomorphism $G^\vee \to F^\vee$ is surjective at x, or, equivalently, in a neighborhood of x.

The implications form a circle: (iv) implies (v) by Nakayama's lemma, while (v) implies (i) because a locally free quotient sheaf is necessarily a direct summand. Geometrically, the situation means that u corresponds to an isomorphism of the vector bundle whose sheaf of sections is F onto a vector subbundle of the analogous vector bundle defined by G. It is, of course, not enough for this that $F \to G$ merely be injective.

COROLLARY II.4.5. *Let* $f\colon X \to Y$ *be a morphism of* S*-preschemes, locally of finite type. Let* $x \in X$*, put* $y = f(x)$*, and let* s *be the common image of* x *and* y *in* S*. Suppose that* Y *is smooth over* S *at* y*. Then the following conditions are equivalent:*

(i) f *is smooth at* x*;*

(ii) X *is smooth over* S *at* x*, and*

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is universally injective at x*; that is, it is injective at* x*, and its cokernel* $\Omega^1_{X/Y}$ *is free at* x.

Necessity follows from Proposition II.1.3(iii) and Theorem II.4.3(i), (ii). Let us prove sufficiency. Since the elements $\mathrm{d}g$, $g \in \mathcal{O}_x$, generate $(\Omega^1_{X/Y})_x$, we can choose g_i, $1 \le i \le n$, such that the images of the $\mathrm{d}g_i$ form a basis of this module. After shrinking X, we may suppose that the g_i come from sections of $\mathcal{O}_X$, and hence define a Y-morphism

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$$g\colon X \longrightarrow Y' = Y[t_1, \ldots, t_n].$$

Using the hypothesis and Lemma II.4.2, one sees immediately that the corresponding homomorphism

$$g^*(\Omega^1_{Y'/S}) \longrightarrow \Omega^1_{X/S}$$

is an isomorphism at x. We are thus reduced to proving the following corollary.

COROLLARY II.4.6. *Let $f\colon X \to Y$ be a morphism between S-preschemes that are smooth over S. Then f is étale at $x \in X$ if and only if*

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is an isomorphism at x.

Necessity is known from Lemma II.4.1, and the same lemma shows that the stated condition implies that f is unramified at x. By Corollary II.2.2, we are reduced to the case $S = \mathrm{Spec}(k)$. Since Y is smooth over k, it is regular, hence in particular normal. By I I.9.5(ii), it remains only to prove that $\mathcal{O}_y \to \mathcal{O}_x$ is injective, or, equivalently, that $\mathcal{O}_y$ and $\mathcal{O}_x$ have the same dimension. These dimensions are respectively the ranks of $\Omega^1_{Y/k}$ and $\Omega^1_{X/k}$ at y and x, and are therefore equal by hypothesis.

Remarks II.4.7. When X and Y are smooth over S, the smoothness criterion in Corollary II.4.5(ii) for $f\colon X \to Y$ can also be stated by saying that, for every $x \in X$, the *tangent* map of f at x, relative to the base S, is surjective. This tangent map is the transpose of the homomorphism between the finite-dimensional $\kappa(x)$-vector spaces obtained as the fibers at x of $f^*(\Omega^1_{Y/S})$ and $\Omega^1_{X/S}$. This is a familiar hypothesis, especially among those who work with analytic spaces. Their usual nonsingularity hypothesis, which says that X and Y are "smooth over $\mathbb{C}$"; see no. II.5, seems to be due only to the fear that singular points of algebraic varieties or analytic spaces still inspire in many geometers.

We record the following special case of Corollary II.4.6.

COROLLARY II.4.8. *Let X be an S-prescheme, and let*

$$g\colon X \longrightarrow S[t_1, \ldots, t_n]$$

be an S-morphism defined by sections g_i, $1 \leq i \leq n$, of $\mathcal{O}_X$. Let $x \in X$ be a point at which X is smooth over S. Then g is étale at x if and only if the $\mathrm{d}g_i$, $1 \leq i \leq n$, form a basis of $\Omega^1_{X/S}$ at x. Equivalently, their images form a basis of the $\kappa(x)$-vector space

$$\Omega^1_{X/S}(x) = (\Omega^1_{X/S})_x \otimes_{\mathcal{O}_x} \kappa(x).$$

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Let X be a prescheme, and let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$. Then $\mathcal{J}/\mathcal{J}^2$ may be regarded as a coherent sheaf on Y; it is the *conormal sheaf* of Y in X. If X is an S-prescheme, there is a canonical exact sequence of quasi-coherent sheaves on Y:

$$\mathcal{J}/\mathcal{J}^2 \xrightarrow{\mathrm{d}} \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow 0. \tag{II.4.3}$$

The right-hand part is simply (II.4.2), with the roles of X and Y interchanged and with $\Omega^1_{Y/X} = 0$. The homomorphism

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is induced, on passing to quotients, by the generally nonlinear map $g \mapsto \mathrm{d}g$. The exactness of (II.4.3) is classical and immediate. In the affine case, for a surjective homomorphism $B \to C$ of A-algebras with kernel J, it becomes

$$J/J^2 \longrightarrow \Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{C/A} \longrightarrow 0, \qquad C = B/J. \tag{4.3bis}$$

This is the exact sequence already used implicitly in the proof of I I.7.2.

PROPOSITION II.4.9. *Let X be an S-prescheme, let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in X$. Let g_i, $1 \leq i \leq n$, be sections of $\mathcal{O}_X$ defining an S-morphism*

$$g\colon X \longrightarrow S[t_1, \ldots, t_n] = X',$$

and let p be an integer with $0 \leq p \leq n$. Suppose that X is smooth over S at x. *Then the following conditions are equivalent:*

(i) *There is an open neighborhood X_1 of x in X such that $g|_{X_1}$ is* étale *and $Y_1 = Y \cap X_1$, the trace of Y on X_1, is the* inverse image *of the closed subprescheme*

$$Y' = S[t_{p+1}, \ldots, t_n] \quad \textit{of} \quad X' = S[t_1, \ldots, t_n].$$

Equivalently, the g_i, $1 \leq i \leq p$, generate $\mathcal{J}|_{X_1}$. Thus the diagram

$$\begin{array}{ccc} Y_1 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \textit{ étale} \\ Y' = S[t_{p+1}, \ldots, t_n] & \longrightarrow & X' = S[t_1, \ldots, t_n] \end{array}$$

is cartesian.

(ii) *Y is* smooth over S at x; *the germs of the g_i, $1 \leq i \leq p$, lie in $\mathcal{J}_x$; the elements $\mathrm{d}g_i(x)$, $1 \leq i \leq n$, form a* basis *of the $\kappa(x)$-vector space $\Omega^1_{X/S}(x)$; and the $\mathrm{d}g'_i(x)$, $p+1 \leq i \leq n$, form a* basis *of $\Omega^1_{Y/S}(x)$, where g'_i denotes the restriction of g_i to Y and all differentials are relative to S.* French source p. 40

(iii) *The g_i, $1 \leq i \leq p$, form a* system of generators *of $\mathcal{J}_x$, and the $\mathrm{d}g_i(x)$, $1 \leq i \leq n$, form a* basis *of $\Omega^1_{X/S}(x)$ over $\kappa(x)$.*

(iv) *Y is* smooth over S at x; *the g_i, $1 \leq i \leq p$, form a* minimal system of generators *of $\mathcal{J}_x$; and the $\mathrm{d}g'_i(x)$, $p+1 \leq i \leq n$, form a* basis *of $\Omega^1_{Y/S}(x)$ over $\kappa(x)$.*

Moreover, under these conditions, $\mathcal{J}/\mathcal{J}^2$ is free over Y at x, with the classes of the g_i, $1 \leq i \leq p$, as a basis at x, and the canonical homomorphism

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is universally injective at x.

Remark *. It follows that p is determined by the other conditions: it is the *rank* of the free Y-module $\mathcal{J}/\mathcal{J}^2$ at x, the *minimum number of generators* of $\mathcal{J}_x$ on X, and also the integer characterized by the fact that the relative dimension of Y over S at x is $n-p$.

Proof of Proposition II.4.9. Suppose first that (i) holds. By I I.4.6(iii), Y_1 is étale over Y'; by definition it is therefore smooth over S at x, of relative dimension $n-p$, and the same is true of Y. Corollary II.4.8, applied to g and to its restriction, shows that the $\mathrm{d}g_i$, $1 \leq i \leq n$, form a basis of $\Omega^1_{X/S}$ at x, and that the $\mathrm{d}g'_i$, $p+1 \leq i \leq n$, form a basis of $\Omega^1_{Y/S}$ at x. The exact sequence (II.4.3) then shows that the classes of the g_i, $1 \leq i \leq p$, are linearly independent in the Y-module $\mathcal{J}/\mathcal{J}^2$ at x. Since these g_i generate $\mathcal{J}_x$, their classes modulo $\mathcal{J}_x^2$ form a basis of $\mathcal{J}/\mathcal{J}^2$ at x. Consequently, the g_i, $1 \leq i \leq p$, form a minimal system of generators of $\mathcal{J}_x$, and the homomorphism

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

in (II.4.3) is universally injective at x: it carries a basis of a module free at x to part of a basis of another such module, both being Y-modules. This proves that (i) implies (ii), (iii), and (iv), as well as the final assertions of Proposition II.4.9.

Condition (iii) implies (i) by Corollary II.4.8.

Condition (ii) also implies (i). Indeed, the requirement that the germs of $g_1, \ldots, g_p$ lie in $\mathcal{J}_x$ means that, after replacing X by an open neighborhood of x, the map g induces a morphism $h\colon Y \to Y'$. By Corollary II.4.8, the two basis requirements mean that g and h are étale at x. Let Y'' be the inverse image of Y' under g. Then Y is a closed subprescheme of Y'', and Y'' is étale over Y' at x, by I I.4.6(iii), because g is étale at x. The immersion $Y \to Y''$ is therefore itself étale, by I I.4.8, and hence is an open immersion, by I I.5.8 or I I.5.2. Replacing X once more by a suitable open neighborhood X_1 of x, we obtain (i). French source p. 41

The preceding argument establishes the equivalence of (i), (ii), and (iii), and shows that they imply (iv). It remains to prove that (iv) implies (ii). Since $\Omega^1_{X/S}$ is free at x, this will be immediate once we know that the smoothness of Y over S at x implies that $\mathcal{J}/\mathcal{J}^2$ is free over Y at x, and that

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is universally injective at x. This assertion is included in the following theorem.

THEOREM II.4.10. *Let X be an S-prescheme smooth over S, let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in X$. The following conditions are equivalent:*

(i) *Y is* smooth over S at x.

(ii) *There exist an open neighborhood X_1 of x in X, an étale S-morphism*

$$g\colon X_1 \longrightarrow X' = S[t_1, \dots, t_n],$$

and an integer p, such that $Y_1 = Y \cap X_1$ is the inverse image *under g of the closed subprescheme*

$$Y' = S[t_{p+1}, \dots, t_n] \quad \textit{of} \quad X' = S[t_1, \dots, t_n].$$

(iii) *There exist generators g_i, $1 \le i \le p$, of $\mathcal{J}_x$ such that the $\mathrm{d}g_i$ form part of a basis of $\Omega^1_{X/S}$ at x, or equivalently such that the $\mathrm{d}g_i(x)$ are linearly independent over $\kappa(x)$ in $\Omega^1_{X/S}(x)$.*

(iv) *The sheaf $\mathcal{J}/\mathcal{J}^2$ is free over Y at x, and the canonical homomorphism*

$$\mathrm{d}\colon \mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is universally injective at x. Equivalently, the sequence of canonical homomorphisms

$$0 \longrightarrow \mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow 0$$

is exact at x, and $\Omega^1_{Y/S}$ is locally free at x.

Proof. We already know from the preceding argument that (ii) implies (i), (iii), and (iv). Let us prove that (i) implies (ii), which will at the same time complete the proof of Proposition II.4.9. By Theorem II.4.3(ii), the last two terms of the exact sequence (II.4.3) are locally free Y-modules at x. The images of the elements $\mathrm{d}g$, $g \in \mathcal{O}_X$, generate $\Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$ at x, and hence their images generate $\Omega^1_{Y/S}$ there. We can therefore choose g_i, $p+1 \le i \le n$, in $\mathcal{O}_X$ such that the $\mathrm{d}g'_i$ form a basis of $\Omega^1_{Y/S}$. By the exactness of (II.4.3), we can then complete the $\mathrm{d}g_i$, $p+1 \le i \le n$, to a basis of the middle term by elements of the form $\mathrm{d}g_i$, $1 \le i \le p$, with $g_i \in \mathcal{J}_x$. The g_i come from sections of $\mathcal{O}_X$ on a neighborhood of x, which we may take to be X. We are now in condition (ii) of Proposition II.4.9, and we have proved that it implies condition (i) there. This gives condition (ii) of the present theorem.

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The implication (iii) $\Rightarrow$ (ii) in the present theorem follows immediately from the implication (iii) $\Rightarrow$ (i) in Proposition II.4.9. Thus (i), (ii), and (iii) are equivalent and imply (iv). Finally, (iv) implies (iii): choose elements $g_i \in \mathcal{J}_x$ whose classes form a basis of $\mathcal{J}_x/\mathcal{J}_x^2$. They generate $\mathcal{J}_x$ by Nakayama's lemma, and universal injectivity says that their differentials form part of a basis.

The preceding proof also gives the following result.

COROLLARY II.4.11. *Let X be an S-prescheme, let Y be a closed subprescheme defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in Y$. Suppose that both X and Y are* smooth over S at x. *Let g_i, $1 \le i \le p$, be sections of $\mathcal{J}$. The following conditions are equivalent:*

(i) *The g_i* generate *$\mathcal{J}_x$, and the $\mathrm{d}g_i(x)$ are* linearly independent *over $\kappa(x)$ in $\Omega^1_{X/S}(x)$.*

(ii) *The classes of the g_i modulo $\mathcal{J}^2$ form a basis of $\mathcal{J}/\mathcal{J}^2$ at x.*
(iii) *The g_i form a minimal system of generators of $\mathcal{J}_x$.*
(iv) *There exist further sections g_i, $p+1 \leq i \leq n$, of $\mathcal{O}_X$ on a neighborhood X_1 of x in X, such that, together with the preceding sections, they define an étale morphism*

$$g\colon X_1 \longrightarrow X' = S[t_1, \ldots, t_n]$$

for which $Y_1 = Y \cap X_1$ is the inverse image *of the closed subprescheme*

$$Y' = S[t_{p+1}, \ldots, t_n] \quad \textit{of} \quad X' = S[t_1, \ldots, t_n].$$

In particular:

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Corollary II.4.12. *Let X be an S-prescheme, let F be a section of $\mathcal{O}_X$, let Y be its zero subprescheme, defined by the coherent ideal $F\mathcal{O}_X$, and let $x \in Y$. Suppose that X is smooth over S at x. Then Y is smooth over S at x if and only if either F vanishes in a neighborhood of x, or $\mathrm{d}F(x) \neq 0$, where $\mathrm{d}F(x)$ denotes the image of $\mathrm{d}F$ in the $\kappa(x)$-vector space $\Omega^1_{X/S}(x)$.*

The condition is sufficient by criterion (iii) of Theorem II.4.10. It is necessary because, as $\mathcal{J}$ is generated by one element, $\mathcal{J}/\mathcal{J}^2$ at x must first be free of rank at most 1. If its rank is 0, that is, if $\mathcal{J}/\mathcal{J}^2 = 0$ at x, then $\mathcal{J} = 0$ at x by Nakayama's lemma, so F vanishes in a neighborhood of x. If its rank is 1, then F is a minimal system of generators of $\mathcal{J}$ at x, and the conclusion follows from the equivalence of (i) and (iii) in Corollary II.4.11.

Corollary II.4.13. *Let Y be an S-prescheme locally of finite type, let S' be a flat S-prescheme, put $Y' = Y \times_S S'$, let $x' \in Y'$, and let $x \in Y$ be its canonical image. Then Y is smooth over S at x if and only if Y' is smooth over S' at x'. In particular, if $S' \to S$ is flat and surjective, then Y is smooth over S if and only if Y' is smooth over S'.*

Only sufficiency remains to be proved; necessity was established in Proposition II.1.3(ii). After replacing Y by a suitable neighborhood of x, and Y' by its inverse image, we may suppose that Y is affine and of finite type over the affine prescheme S. Thus Y is isomorphic to a closed subprescheme of

$$X = S[t_1, \ldots, t_n].$$

It follows that Y' is a closed subprescheme of $X' = X \times_S S'$. Since X is smooth over S, the prescheme X' is smooth over S', and the smoothness criteria in Theorem II.4.10 apply. Criterion (iv) gives the result immediately.

Remarks II.4.14. Criterion (iii) of Theorem II.4.10 deserves to be called the *Jacobian criterion for smoothness*. It makes it possible, in theory, to determine whether a given S-prescheme Y is smooth over S at a point x of Y, since there always exists a neighborhood of Y isomorphic to a subprescheme of a smooth S-prescheme X, for example $X = S[t_1, \ldots, t_n]$. Moreover, it is for $X = S[t_1, \ldots, t_n]$, $S = \mathrm{Spec}(()A)$, that the Jacobian criterion is usually stated (of course, in the classical case considered by Zariski, A was a 44
field). We leave it to the reader to give the statement for an ideal J of $A[t_1, \ldots, t_n]$ and a prime ideal containing it, to which one is thus led. Let us note that it now seems clear (especially since Nagata succeeded, by nondifferential methods, in generalizing Zariski's theorem that the set of regular points of an algebraic scheme is open) that the Jacobian criterion is of scarcely any interest except in the form in which we give it here (i.e. using exclusively *relative* differentials and not *absolute* differentials, i.e. differentials relative to the ring of absolute constants $\mathbb{Z}$). As is so often the case, the use of differentials is more convenient here than that of derivations. Finally, let us note that if Y is smooth over S at x, of relative dimension n, then there exists an open neighborhood of x in Y isomorphic to a subprescheme of $X = S[t_1, \ldots, t_n]$ with $n = m + 1$, as follows from the definition and from (I I.7.6).

Let A be a noetherian ring, let x_i $(1 \leq i \leq n)$ be elements of A, and let J be the ideal generated by the x_i. The x_i are said to form a *regular system of generators* of J if the canonical surjective homomorphism

$$(A/J)[t_1, \ldots, t_n] \to \mathrm{gr}^J(A)$$

defined by the x_i (where the second member denotes the graded ring associated with A filtered by the powers of J) is an *isomorphism*. This condition also means that

(i) The canonical surjective homomorphism

$$S_{A/J}(J/J^2) \to \mathrm{gr}^J(A)$$

(where the first member denotes the symmetric algebra of the A/J-module J/J^2) is an isomorphism, and

(ii) J/J^2 is free and has as a basis the classes of the $x_i \mod J^2$.

In this form, we see that if $J \neq A$, the x_i form a *minimal system of generators* of J, and that *every other minimal system of generators* of J *is a regular system of generators* (N.B. "minimal"is taken in the strict sense: minimum number of elements, which is equivalent to the sense: minimal under inclusion only if A is local); on the other hand, if $J = A$, every system of generators of J is regular.

The regularity condition for a system of generators of an ideal is stable under localization by any multiplicatively closed set; on the other hand, it is immediately seen that in order for (x_i) to be a minimal system of generators of J, it already suffices that for every *maximal ideal* $\mathfrak{m}$ *containing* J, the x_i define a regular system of generators of $JA_{\mathfrak{m}}$ in $A_{\mathfrak{m}}$. We are thus reduced to the case where A is a local ring with maximal ideal $\mathfrak{m}$, and where the x_i lie in $\mathfrak{m}$. *Then the x_i form a regular system of generators of J if and only if they form an A-sequence in the sense of Serre*[6], i.e. if for every i such that $1 \leq i \leq n$, x_i is a non-zero-divisor of 0 in $A/(x_1, \ldots, x_{i-1})A$. 45

Finally, in the case where A is an algebra over a ring B, and where A/J is isomorphic as a B-algebra to B (so that J is the kernel of a homomorphism of B-algebras $A \to B$), the x_i form a regular system of generators of J if and only if the canonical homomorphism

$$B[[t_1, \ldots, t_n]] \to \hat{A}$$

defined by the x_i (where the second member denotes the separated completion $\varprojlim A/J^{n+1}$ of A for the topology defined by the powers of J) is an *isomorphism* (it is in any case *surjective*).

All these facts are well known, and doubtless appear, up to minor differences, in Serre's course on commutative algebra written up by Gabriel. (Where one finds N other characterizations of A-sequences, in the case where A is a local ring).

Let J be an ideal in a noetherian ring A. We shall say that J is a *regular ideal* if for every prime ideal $\mathfrak{p}$ of A, $JA_{\mathfrak{p}}$ admits a regular system of generators. It is plainly enough to verify this for $\mathfrak{p} \supset J$, and one may moreover restrict oneself to maximal $\mathfrak{p}$. More generally, let $\mathcal{J}$ be an Ideal on a locally noetherian prescheme X; one says that $\mathcal{J}$ is a *regular Ideal* if for every $x \in X$, $\mathcal{J}_x$ is an ideal of $\mathcal{O}_x$ which admits a regular system of generators. This is equivalent to the conjunction of the following two conditions:

(a) The canonical surjective homomorphism

$$S_{\mathcal{O}_X/\mathcal{J}}(\mathcal{J}/\mathcal{J}^2) \to \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$$

is an isomorphism, and

(b) The sheaf of $\mathcal{O}_X/\mathcal{J}$-Modules $\mathcal{J}/\mathcal{J}^2$ is locally free.

One also says then that the subprescheme Y of X defined by $\mathcal{J}$ (thus such that $\mathcal{O}_Y$, extended by 0, is isomorphic to $\mathcal{O}_X/\mathcal{J}$) is *regularly immersed* in X, and one defines in the same way (in the evident manner) the notion of a morphism that is 46

[6] We shall now rather say "A-regular sequence", cf. EGA 0_{IV} 15.1.7 and 15.1.11.

a *regular immersion*, (respectively*regular at a point* x), an immersion morphism $Y \to X$ identifying Y (respectivelya suitable neighborhood of x) with a regularly immersed closed subprescheme of an open subset of X. (One must not say: regular subprescheme, for that would mean that the local rings of Y are regular). Finally, sections x_i of $\mathcal{J}$ are called a *regular system of generators* if for every $x \in X$, the corresponding elements of $\mathcal{O}_x$ form a regular system of generators of $\mathcal{J}_x$ (terminology compatible with that introduced for generators of an ideal of a ring). This also means that the canonical surjective homomorphism

$$\mathcal{O}_Y[t_1, \ldots, t_n] \to \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$$

defined by the x_i is an isomorphism. If it is known in advance that the Ideal $\mathcal{J}$ is regular, this means simply that at every point x *of* Y, the x_i define a *basis* of $\mathcal{J}/\mathcal{J}^2$ over $\mathcal{O}_{Y,x}$. (N.B. this condition is empty if Y is empty). Thus, in order that $\mathcal{J}$ admit a regular system of generators, it is necessary and sufficient that $\mathcal{J}$ be regular, and that the $\mathcal{O}_Y$-Module $\mathcal{J}/\mathcal{J}^2$ be globally free (and not merely locally free), i.e. that the canonical homomorphism $S_{\mathcal{O}_Y}(\mathcal{J}/\mathcal{J}^2) \to \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$ be surjective, and that the $\mathcal{O}_Y$-Module $\mathcal{J}/\mathcal{J}^2$ be globally free.

An *augmented ring is said to be regular* if the augmentation ideal is regular. Thus, if A is a local ring, regarded as augmented over its residue field k, then A is a regular local ring if and only if it is a regular augmented ring.

(To tell the truth, it seems that it was unnecessary to begin by giving the preliminary chain of reasoning for rings; it is advantageous to begin with sheaves at once. If one wants something in the noetherian case, it is the definition adopted here — a priori less restrictive than that by Serre's A-sequences — which seems preferable for the needs of differential calculus. Of course, to do things properly, one would also have to develop at least part of the theory of smooth morphisms in the nonnoetherian setting[7], probably by starting from the Jacobian criterion, so as to obtain, if possible, all the essential formal 47
properties of smooth morphisms and of ètale morphisms, i.e. smooth and quasi-finite morphisms; only the converses making use of noetherian hypotheses).

After these long terminological preliminaries, a little theorem:

THEOREM II.4.15. *Let X be an S-prescheme locally of finite type, let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in X$. Suppose now that Y is* smooth over S at x, *with no assumption on X. Then the following conditions are equivalent:*

(i) *X is smooth over S at x;*
(ii) *the immersion $i\colon Y \to X$ is regular at x, that is, $\mathcal{J}_x$ is a regular ideal of $\mathcal{O}_x$.*

COROLLARY II.4.16. *Suppose that Y is smooth over S. Then X is smooth over S in a neighborhood of Y, that is, at every point of Y, if and only if Y is regularly embedded in X, or equivalently the immersion $i\colon Y \to X$ is regular.*

Proof of Theorem II.4.15. Suppose first that (i) holds. Apply criterion (ii) of Theorem II.4.10. Since the resulting morphism $g\colon X_1 \to X'$ is flat, regular immersions are preserved under its inverse image. It is therefore enough to prove that

$$Y' = S[t_{p+1}, \ldots, t_n]$$

is regularly embedded in $X' = S[t_1, \ldots, t_n]$, which is immediate: $t_1, \ldots, t_p$ form a regular system of generators of the ideal defining Y' in X'.

Conversely, suppose that (ii) holds. Let g_i, $1 \leq i \leq p$, be a regular system of generators of $\mathcal{J}_x$, and choose elements g_i, $p+1 \leq i \leq n$, of $\mathcal{O}_{X,x}$ whose images g'_i in $\mathcal{O}_{Y,x}$ define an étale morphism

$$Y_1 \longrightarrow Y' = S[t_{p+1}, \ldots, t_n]$$

[7]As is said in the foreword, this has now been done, cf. EGA IV 17, 18

from a neighborhood Y_1 of x in Y. The g_i, $1 \leq i \leq n$, come from sections of $\mathcal{O}_X$ on a neighborhood X_1 of x. We may suppose that $X_1 = X$ and $Y_1 = Y$. They define a morphism

$$g\colon X \longrightarrow X' = S[t_1, \ldots, t_n],$$

and it remains only to prove that g is étale at x.

After shrinking X, we may suppose that $g_1, \ldots, g_p$ form a regular system of generators of $\mathcal{J}$ throughout X. In particular, they generate $\mathcal{J}$, so the closed subprescheme Y of X is the inverse image under g of the closed subprescheme Y' of X'. Put $x' = g(x)$. The fiber of $X \to X'$ over x' is then the same as the fiber of $Y \to Y'$ over x', and is therefore étale over $\kappa(x')$. Thus g is unramified at x; it remains to prove that g is flat there. French source p. 48

Let $\mathcal{J}'_{x'}$ be the ideal of Y' in $\mathcal{O}_{X',x'}$. The associated graded ring of $\mathcal{O}_{X',x'}$, filtered by the powers of $\mathcal{J}'_{x'}$, is free over $\mathcal{O}_{Y',x'}$ in every degree. The associated graded ring of $\mathcal{O}_{X,x}$, filtered by the powers of $\mathcal{J}_x = \mathcal{J}'_{x'}\mathcal{O}_{X,x}$, is canonically isomorphic to the tensor product of the former graded ring with $\mathcal{O}_{Y,x}$; indeed, the two graded rings are polynomial rings in p indeterminates over $\mathcal{O}_{Y',x'}$ and $\mathcal{O}_{Y,x}$, respectively. Finally,

$$\mathcal{O}_{X,x}/\mathcal{J}_x = \mathcal{O}_{Y,x}$$

is flat over

$$\mathcal{O}_{X',x'}/\mathcal{J}'_{x'} = \mathcal{O}_{Y',x'}.$$

A general flatness criterion now applies: for a local homomorphism of noetherian local rings $A' \to A$, with an ideal $J' \neq A'$ such that the associated graded ring is free over A'/J' in every degree, these conditions imply flatness. Hence X is flat over X' at x, as required.

Corollary II.4.17. *Let X be a prescheme locally of finite type over Y, let i be a section of X over Y, let $y \in Y$, and put $x = i(y)$. Let $\mathcal{J}$ be the sheaf of ideals on X defined by the subprescheme $i(Y)$, which we assume closed in order to simplify the statement, a condition automatically satisfied when X is a scheme. The following conditions are equivalent:*

(i) *X is smooth over Y at x;*

(ii) *i is a regular immersion at y;*

(iii) *the completion of the $\mathcal{O}_y$-algebra $\mathcal{O}_x$ for the topology defined by the powers of $\mathcal{J}_x$ is isomorphic to a formal power-series algebra*

$$\mathcal{O}_y[[t_1, \ldots, t_n]];$$

(iii bis) *there is an open neighborhood U of y such that the sheaf of $\mathcal{O}_Y$-algebras*

$$\varprojlim_m i^*(\mathcal{O}_X/\mathcal{J}^{m+1})$$

over U is isomorphic to a sheaf of the form $\mathcal{O}_Y[[t_1, \ldots, t_n]]$;

(iv) *there exist an open neighborhood U of y, an open neighborhood V of x, and a Y-morphism*

$$g\colon V \longrightarrow U[t_1, \ldots, t_n]$$

such that g is étale, i induces a section of V over U, and g carries this section to the zero section of $U[t_1, \ldots, t_n]$ over U.

The equivalence of (i) and (ii) is the special case $Y = S$ of Theorem II.4.15. The equivalence of (ii) and (iii), and morally that of (ii) and (iii bis), was noted in the preceding "reminders". The equivalence of (i) and (iv) follows readily from the equivalence of (i) and (ii) in Theorem II.4.10. French source p. 49

Corollary II.4.18. *Let X be a prescheme smooth over S. Then the diagonal morphism*

$$\Delta_{X/S}\colon X \longrightarrow X \times_S X$$

is a regular immersion*; equivalently, X is* differentially smooth *over S.*

Indeed, this is a special case of Corollary II.4.16, since both X and $X \times_S X$ are smooth over S.

Remarks II.4.18. Recall from I I.1 that, if X is a prescheme over S, one introduces on X the quasi-coherent sheaves of algebras

$$\mathcal{P}^n_{X/S} = \mathcal{O}_{X\times_S X}/\mathcal{I}_X^{n+1},$$

where $\mathcal{I}_X$ is the sheaf of ideals defining the diagonal in $X \times_S X$. It is regarded as a sheaf of $\mathcal{O}_X$-algebras by means of the first projection $\mathrm{pr}_1 \colon X \times_S X \to X$. The $\mathcal{P}^n_{X/S}$ form a projective system of algebras on X. Its projective limit, denoted by $\mathcal{P}^\infty_{X/S}$, is the structure sheaf of the formal completion of $X \times_S X$ along the diagonal, where we now suppose that X is locally of finite type over S, so that the $\mathcal{P}^n_{X/S}$ are coherent.

To say that X is differentially smooth over S, that is, that the diagonal $\Delta_{X/S}$ is a regular immersion, is also to say that $\mathcal{P}^\infty_{X/S}$ is regular as a sheaf of algebras augmented toward $\mathcal{O}_X$. Equivalently, $\Omega^1_{X/S}$ is locally free and the canonical surjective homomorphism

$$\mathrm{Sym}_{\mathcal{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathrm{gr}_*(\mathcal{P}^\infty_{X/S})$$

is an isomorphism. Equivalently again, every point of X has an open neighborhood on which the augmented sheaf of algebras $\mathcal{P}^\infty_{X/S}$ is isomorphic to a sheaf $\mathcal{O}_X[[t_1, \ldots, t_n]]$.

Let s be a section of X over S, and let $\mathcal{J}$ be the sheaf of ideals on X that it defines; for simplicity, suppose that $s(S)$ is closed. There are then canonical isomorphisms of augmented $\mathcal{O}_X$-algebras French source p. 50

$$s^*(\mathcal{P}^n_{X/S}) = \mathcal{O}_X/\mathcal{J}^{n+1}, \qquad s^*(\mathcal{P}^\infty_{X/S}) = \varprojlim_n \mathcal{O}_X/\mathcal{J}^{n+1}. \tag{II.4.4}$$

These isomorphisms are functorial under base change in the evident sense and, taking this fact into account, give another characterization of the sheaves of algebras $\mathcal{P}^n_{X/S}$ on S. If, for example, $S = \mathrm{Spec}(k)$, where k is a field, then giving a section s of X over S is equivalent to giving a k-rational point $x \in X$, and the preceding formulas say that there is an isomorphism of k-algebras

$$\mathcal{P}^n_{X/S}(x) = \mathcal{O}_x/\mathfrak{m}_x^{n+1}. \tag{II.4.5}$$

This justifies the name *sheaf of principal parts of order n on X relative to S* for $\mathcal{P}^n_{X/S}$.

Formula (II.4.4) also shows that *if X is differentially smooth over S at every point of $s(S)$, then X is smooth over S at every point of $s(S)$*, by Corollary II.4.17; the converse is also true, by Corollary II.4.18. Together with Corollary II.4.13, this readily implies that, if X is an S-prescheme locally of finite type, then *X is smooth over S if and only if it is flat and differentially smooth over S*. Notice that the flatness hypothesis is essential, as one sees by taking X to be a closed subprescheme of S.

Recall also that $\mathcal{P}^n_{X/S}$ has a *second algebra structure*, obtained from the second projection $\mathrm{pr}_2 \colon X \times_S X \to X$. This structure is obtained from the first by means of the *canonical involution* of the sheaf of rings $\mathcal{P}^n_{X/S}$ induced by the symmetry automorphism of $X \times_S X$. We denote by $d^n_{X/S}$, or simply by d^n, the homomorphism of sheaves of rings

$$d^n_{X/S} \colon \mathcal{O}_X \longrightarrow \mathcal{P}^n_{X/S} \tag{II.4.6}$$

corresponding to this second algebra structure. Under the isomorphism (II.4.4), this homomorphism carries a section f of $\mathcal{O}_X$ to a section $d^n(f)$ of $\mathcal{P}^n_{X/S}$ whose inverse image under a section s of X over S is identified with the canonical image of f in $\Gamma(X, \mathcal{O}_X/\mathcal{J}^{n+1})$. This justifies the name *system of principal parts of order n of f* for $d^n f$, especially when $S = \mathrm{Spec}(k)$, as in (II.4.5). French source p. 51

Finally, the homomorphism (II.4.6) may be regarded as the *universal differential operator of order at most n* on $\mathcal{O}_X$, relative to the constant prescheme S. [8] Here a

[8] For everything concerning this paragraph, see EGA IV, 16.8–16.12.

differential operator of order at most n from $\mathcal{O}_X$ to a module F means a homomorphism of sheaves D that factors as

$$D\colon \mathcal{O}_X \xrightarrow{d^n} \mathcal{P}^n_{X/S} \xrightarrow{u} F,$$

where u is a homomorphism of $\mathcal{O}_X$-modules, uniquely determined by D. This agrees with the intuitive recursive definition: D has order at most n if, for every section g of $\mathcal{O}_X$ on an open set $U \subseteq X$, the map

$$f \longmapsto D(fg) - gD(f)$$

is a differential operator of order at most $n-1$ on U.

It follows that, *if X is differentially smooth over S, the sheaf of rings of differential operators of all orders has the familiar simple structure* from differential calculus on differentiable manifolds. In particular, it has locally a basis as an $\mathcal{O}_X$-module formed by the *divided powers* in mutually commuting operators $\delta/\delta x_i$, $1 \leq i \leq n$. If $\mathcal{O}_S$ is a sheaf of $\mathbb{Q}$-algebras, where $\mathbb{Q}$ is the field of rational numbers, it is enough to take ordinary polynomials in the $\delta/\delta x_i$. In this case, exceptionally, local freeness of $\Omega^1_{X/S}$ already suffices for X to be differentially smooth over S.

Remark II.4.19. The terminology "regular immersion", "regular ideal", and so forth, introduced in this section met with fairly strong and general opposition from Chevalley and Serre. The terms "Cohen–Macaulay ideal", "Macaulay ideal", and "Macaulayan ideal" were proposed; morally, this would also require the adoption of "Cohen–Macaulay immersion" or "Macaulay immersion". That terminology, however, conflicts with another already used in forthcoming versions of the *multiplodoque*, where a morphism of finite type is said to be "Cohen–Macaulay" at a point if it is flat there and the local ring of the fiber through that point is Cohen–Macaulay. Until a satisfactory solution is found, we shall retain, with every reservation, the terminology introduced in this section.[9]

II.5. Case of a base field

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Proposition II.5.1. *Let k be a field, let X be a prescheme of finite type over k, let $x \in X$, and let n be the dimension of X at x. Let*

$$f\colon X \longrightarrow \operatorname{Spec}(k[t_1, \ldots, t_n]) = Y$$

be the morphism defined by elements $f_i \in \Gamma(X, \mathcal{O}_X)$. The following conditions are equivalent (and imply that X is smooth over k at x, and hence, by Proposition II.3.1, regular at x):

(i) *f is étale at x;*
(ii) *the $\mathrm{d}f_i$ form a basis of $\Omega^1_{X/k}$ at x;*
(iii) *the $\mathrm{d}f_i$ generate $\Omega^1_{X/k}$ at x.*

Since (i) implies that X is smooth over k at x, the implication (i)⇒(ii) is a special case of Corollary II.4.8, and (ii)⇒(iii) is immediate. It remains to prove (iii)⇒(i). If (iii) holds, Lemma II.4.1 shows that f is unramified at x. After replacing X by an open neighborhood of x, it is therefore quasi-finite, and hence dominant by dimension considerations. Since Y is regular, it follows from I I.9.5(ii), or from I I.9.11, that f is étale.

Corollary II.5.2. *Under the hypotheses preceding the equivalent conditions in Proposition II.5.1, suppose that $\kappa(x)$ is a* finite separable *extension of k, and that $f_i \in \mathfrak{m}_x$ for $1 \leq i \leq n$. Then the preceding conditions are also equivalent to:*

(iv) *the f_i form a system of generators of $\mathfrak{m}_x$; equivalently, their classes modulo $\mathfrak{m}_x^2$ form a basis of $\mathfrak{m}_x/\mathfrak{m}_x^2$ over $\kappa(x)$.*

[9] This is the terminology adopted in EGA 0_{IV}, 15.1.7.

Indeed, (iv) implies (iii) by the exact sequence

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega^1_{\mathcal{O}_x/k} \longrightarrow \Omega^1_{\kappa(x)/k} \longrightarrow 0, \tag{II.5.1}$$

since $\Omega^1_{\kappa(x)/k} = 0$, because $\kappa(x)$ is étale over k. Conversely, (ii) implies (iv): both X and $\mathrm{Spec}(\kappa(x))$ are smooth over k at x, so by Theorem II.4.10(iv) the preceding exact sequence may be preceded by 0.

Corollary II.5.3. *Let x be a point of a prescheme X of finite type over k. If X is smooth over k at x, then $\mathcal{O}_x$ is regular. Conversely, if $\kappa(x)$ is a finite separable extension of k and $\mathcal{O}_x$ is regular, then X is smooth over k at x.*

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Indeed, the converse follows from Corollary II.5.2, by taking a regular system (f_i) of generators of $\mathfrak{m}_x$. (Instead of Corollary II.5.2, one may also invoke Theorem II.4.15.) We conclude:

Proposition II.5.4. *Let X be a prescheme of finite type over k. If X is smooth over k, then it is regular. Conversely, if k is perfect and X is regular, then X is smooth over k.*

For the converse, Corollary II.5.3 shows that X is smooth over k at every closed point, hence everywhere, since the locus on which it is smooth is open.

Theorem II.5.5. *Let X be a prescheme of finite type over k, let $x \in X$, let n be the dimension of X at x, and let k' be a perfect field extending k. The following conditions are equivalent:*

(i) *X is smooth over k at x;*
(ii) *$\Omega^1_{X/k}$ is locally free of rank n at x;*
(ii bis) *$\Omega^1_{X/k}$ is generated by n elements at x;*
(iii) *X is differentially smooth over k at x;*
(iv) *there exists an open neighborhood U of x such that $U \otimes_k k'$ is regular; that is, all its local rings are regular.*

Theorem II.4.3(ii) gives (i)⇒(ii), the implication (ii)⇒(ii bis) is immediate, and Proposition II.5.1 gives (ii bis)⇒(i). Since X is flat over k, Corollary II.4.18 gives (i)⇔(iii). Condition (i) implies (iv), because smoothness is preserved under base extension and implies regularity. Conversely, if (iv) holds, then Proposition II.5.4 shows that $U \otimes_k k'$ is smooth over k', and hence U is smooth over k by Corollary II.4.13.

Taking x to be the generic point of X, with X assumed irreducible, we obtain:

Corollary II.5.6. *Let K be a local Artinian ring which is a localization of a finitely generated algebra over the field k (for example, K may be a finitely generated extension of k), and let n be the transcendence degree over k of its residue field. The following conditions are equivalent:*

(i) *K is a finite separable extension of a purely transcendental extension $k(t_1, \ldots, t_n)$ of k.*

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(ii) *$\Omega^1_{K/k}$ is a free K-module of rank n.*
(ii bis) *$\Omega^1_{K/k}$ is a K-module generated by n elements.*
(iii) *The completion O' of $K \otimes_k K$, for the topology defined by the powers of the augmentation ideal of $K \otimes_k K \to K$, is a "regular" augmented K-algebra, that is, isomorphic to a formal power-series algebra over K. (If K is a field, this is equivalent to saying that O' is a regular local ring.)*
(iv) *K is a separable extension of k.*

Indeed, K may always be regarded as the local ring at the generic point of an irreducible scheme X of finite type over k, and the conditions in question are the conditions bearing

the same names in Theorem II.5.5, taking for k' in (iv) an algebraically closed extension of k. Only the implication

$$K \text{ separable over } k \quad \Longrightarrow \quad X \text{ smooth over } k \text{ at } x$$

requires proof. By Corollary II.4.13, we are immediately reduced to the case where the base field is k', hence algebraically closed; there is then a k-rational point a of X. By Proposition II.5.4, X is smooth over k at a, and is therefore all the more smooth over k at the generic point x. [10]

When K is a finitely generated extension of k, the equivalence of (i), (ii), (ii bis), and (iv) is well known. Notice, however, that we have used none of these already known equivalences. Of course, Proposition II.5.1 contains as a special case the familiar fact that a sequence of elements x_i, $1 \leq i \leq n$, is a separating transcendence basis of K over k if and only if the $\mathrm{d}x_i$ form a basis of the K-module $\Omega^1_{K/k}$.

COROLLARY II.5.7. *Let X be a prescheme of finite type over a field k. Then X is smooth over k if and only if $\Omega^1_{X/k}$ is locally free and the local rings at the generic points of the irreducible components of X are separable extensions of k. (The latter condition is automatically satisfied if k is perfect and X is reduced.)*

We may assume that X is connected. Let n be the rank of $\Omega^1_{X/k}$, assumed locally free. By the hypothesis and Corollary II.5.6, n is also the transcendence degree of the extensions of k defined by the local rings at the generic points of X. Thus all the irreducible components of X have dimension n, and the conclusion follows from Theorem II.5.5. French source p. 55

Care is needed here. If K is a finite extension of k, not necessarily separable, then $\Omega^1_{K/k}$ is a free k-module. Thus, putting $X = \mathrm{Spec}(K)$, the sheaf $\Omega^1_{X/k}$ is locally free and X is reduced, although X need not be smooth over k. Extending scalars to the algebraic closure of k then gives a similar example in which k is algebraically closed, but X is no longer reduced.

COROLLARY II.5.8. *Let X be a prescheme of finite type over the field k, let $x \in X$, let n be the dimension of X at x, and let $p = \dim \mathcal{O}_x$, that is, the codimension in X of the closure Y of x. Thus $n - p$ is the transcendence degree of $\kappa(x)$ over k. Let f_i, $1 \leq i \leq n$, be elements of $\mathcal{O}_x$, with $f_i \in \mathfrak{m}_x$ for $1 \leq i \leq p$. The following conditions are equivalent:*

(i) *The germ at x of the morphism*

$$X \longrightarrow \mathrm{Spec}(k[t_1, \ldots, t_n])$$

defined by the f_i is étale at x.

(ii) *The elements f_i, $1 \leq i \leq p$, generate $\mathfrak{m}_x$, that is, they form a regular system of parameters of $\mathcal{O}_x$, and the classes in $\kappa(x)$ of the f_j, $p+1 \leq j \leq n$, form a separating transcendence basis. In other words, the $\mathrm{d}\overline{f}_j$, $p+1 \leq j \leq n$, form a basis of $\Omega^1_{\kappa(x)/k}$, or equivalently generate that module.*

Suppose first that (i) holds. By Corollary II.4.8, the $\mathrm{d}f_i(x)$ form a basis of $\Omega^1_{X/k}(x)$; hence their images $\mathrm{d}\overline{f}_i(x)$ generate $\Omega^1_{\kappa(x)/k}$ over $\kappa(x)$. Since $\overline{f}_i = 0$ for $1 \leq i \leq p$, it is enough to retain the $\mathrm{d}\overline{f}_i(x)$ with $p+1 \leq i \leq n$. The transcendence degree of $\kappa(x)$ over k is $n - p$, so Corollary II.5.6, applied to $K = \kappa(x)$, shows that Y is smooth over k at its generic point x, and that the $\mathrm{d}\overline{f}_i(x)$, $p+1 \leq i \leq n$, form a basis of $\Omega^1_{\kappa(x)/k}$ over $\kappa(x)$. Condition (ii) of Proposition II.4.9 is therefore satisfied, and so is condition (iii); in particular the f_i, $1 \leq i \leq p$, generate $\mathfrak{m}_x$. Since $\mathcal{O}_x$ has dimension p, they form a regular system of parameters at x. This proves (ii). French source p. 56

Conversely, suppose that (ii) holds. The exact sequence (II.5.1) shows that the $\mathrm{d}f_i(x)$ generate $\Omega^1_{X/k}(x)$, and (i) follows from Proposition II.5.1.

[10] See the Errata at the end of the present Exposé II (p. 40).

COROLLARY II.5.9. *Let X be a prescheme of finite type over the field k, let $x \in X$, let n be the dimension of X at x, and let $p = \dim \mathcal{O}_x$, that is, the codimension of the closure Y of x in X. Thus $n-p$ is the transcendence degree of $\kappa(x)$ over k. The following conditions are equivalent:*

(i) *$\mathcal{O}_x$ is regular and $\kappa(x)$ is a separable extension of k.*

(ii) *X is smooth over k at x, and the canonical homomorphism*

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega^1_{\mathcal{O}_x/k} \otimes_{\mathcal{O}_x} \kappa(x) = \Omega^1_{X/k}(x)$$

is injective.

(iii) *There exist $f_i \in \mathcal{O}_x$, $1 \leq i \leq n$, with $f_i \in \mathfrak{m}_x$ for $1 \leq i \leq p$, such that the germ at x of the morphism from X to $\operatorname{Spec}(k[t_1, \ldots, t_n])$ defined by the f_i is étale at x. Equivalently, by Proposition II.5.1, the $\mathrm{d}f_i(x)$ generate $\Omega^1_{X/k}(x)$.*

(iv) *There exist $f_i \in \mathcal{O}_x$, $1 \leq i \leq n$, such that the f_i, $1 \leq i \leq p$, generate $\mathfrak{m}_x$, and the $\mathrm{d}f_j(x)$, $p+1 \leq j \leq n$, generate $\Omega^1_{\kappa(x)/k}$ over $\kappa(x)$.*

The equivalence of (iii) and (iv) follows from Corollary II.5.8. By Proposition II.4.9, these conditions are also equivalent to X being smooth over k at x together with condition (ii) of Theorem II.4.10. They are therefore equivalent to X being smooth over k at x together with condition (iv) of that theorem, which is condition (ii) above. Equally, they are equivalent to X being smooth over k at x together with condition (i) of Theorem II.4.10; here the latter simply means that $\kappa(x)$ is separable over k. This implies condition (i) above. To prove the converse it remains only to establish the following corollary.

COROLLARY II.5.10. *Let x be a point of a prescheme X of finite type over the field k, and suppose that $\kappa(x)$ is separable over k. Then X is smooth over k at x if and only if it is regular at x, that is, if and only if the local ring $\mathcal{O}_x$ is regular.*

Indeed, if X is regular at x, one can plainly find elements $f_i \in \mathcal{O}_x$, $1 \leq i \leq n$, satisfying condition II.5.9(iv).

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Errata. In the proof of Corollary II.5.6 above, we used the fact that a nonempty reduced scheme of finite type over an algebraically closed field has at least one regular, hence smooth, point. This fact is usually proved by differential methods, using Zariski's theorem that the set of regular points of X is open. To avoid a circular argument, one must prove the following directly. Let K/k be a finitely generated separable extension, and suppose that $f_i \in K$, $1 \leq i \leq n$, are such that the $\mathrm{d}_{K/k} f_i$ form a basis of $\Omega^1_{K/k}$. Then n is the transcendence degree of K over k; equivalently, the f_i are algebraically independent.

The usual proof uses Mac Lane's criterion; see Bourbaki, *Algèbre*, Chapter V, §9, Theorem 2. Choose a nonzero polynomial $g \in k[t_1, \ldots, t_n]$ of minimal degree such that $g(f_1, \ldots, f_n) = 0$. Then

$$\sum_i \frac{\partial g}{\partial t_i}(f_1, \ldots, f_n)\, \mathrm{d}f_i = 0.$$

Since the $\mathrm{d}f_i$ form a basis of $\Omega^1_{K/k}$, each $\partial g/\partial t_i$ vanishes at $(f_1, \ldots, f_n)$, and hence is the zero polynomial by the minimality of g. In characteristic zero this gives $g = 0$. In characteristic $p \neq 0$, one has $g = h(t_1^p, \ldots, t_n^p)$. Mac Lane's criterion then shows that $h \in k[t_1, \ldots, t_n]$ also vanishes at $(f_1, \ldots, f_n)$, and the minimality of g again gives $g = 0$.

CHAPTER III

Smooth morphisms: lifting properties

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III.1. Formally smooth homomorphisms

In Exposé II we confined ourselves to homomorphisms of finite type and, consequently, in the case of local homomorphisms $A \to B$ of local rings, to the case where B is isomorphic to a localization of a finitely generated A-algebra. This case is insufficient for various applications, especially in formal or analytic geometry. For example, the formal power-series ring $B = A[[t_1, \ldots, t_n]]$ has, from the viewpoint of formal geometry, the properties of a smooth algebra over A. Likewise, in analytic geometry, the local ring at a point (y, z) of a product $Y \times \mathbb{C}^n$, regarded as an algebra over the local ring at y, has these properties. Indeed, its completion is isomorphic to the algebra of formal power series in n indeterminates over the completion of the base ring $\mathcal{O}_y$. This leads to the following definition.

Definition III.1.1. *Let $u\colon A \to B$ be a local homomorphism of local rings, which, as always, are assumed Noetherian. Suppose that the residue-field extension $\kappa(B)/\kappa(A)$ is finite. We say that u is a* formally smooth homomorphism*, or that the algebra B is* formally smooth over A*, if there exists a finite local $\overline{A}$-algebra A', free over $\overline{A}$, such that the local components of the semilocal ring*

$$B' = \overline{B} \otimes_{\overline{A}} A'$$

are A'-isomorphic to formal power-series algebras over A'. [1]

Here $\overline{A}$ and $\overline{B}$ denote the completions of A and B. Since B' is finite and free over $\overline{B}$, it is indeed a semilocal ring, a direct product of complete local rings. Each of these is again a free $\overline{B}$-module, and therefore has the same dimension as $\overline{B}$, hence as B. It follows that the number of variables t_i in the formal power-series rings of Definition III.1.1 is

$$\dim \overline{B} - \dim \overline{A} = \dim B - \dim A,$$

and in particular is independent of the chosen local component. It is immediately seen to be also the dimension of the ring $B \otimes_A k = B/\mathfrak{m}B$, where $k = A/\mathfrak{m}$ is the residue field of A. We call it the *relative dimension of B over A*.

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Remarks III.1.2. Definition III.1.1 plainly depends only on the homomorphism of completions $\overline{A} \to \overline{B}$ induced by $A \to B$, which to some extent justifies the terminology. We here retract the terminology of I I.3.2(b) and I I.4.1(b), which risks being misleading, and prefer to say "formally unramified" and "formally étale" in the cases treated by those definitions. We reserve "unramified" and "étale" for the case where B is a localization of a finitely generated A-algebra. [2]

The reader will verify at once that "formally étale" is equivalent to "formally smooth and quasi-finite". Finally, there is a reasonable definition of "formally smooth" without any preliminary hypothesis on the residue extension $\kappa(B)/\kappa(A)$, assumed finite here. Among other cases, it includes local homomorphisms $A \to B$ for which B is flat over A and $B/\mathfrak{m}B$ is a separable extension of $A/\mathfrak{m} = k$, not necessarily of finite type. For

[1] For a more general and more conceptual definition, motivated by Theorem III.2.1 below, see EGA 0_{IV}, 19.3.1.

[2] Or, better, "essentially unramified" and "essentially étale"; compare EGA IV, 18.6.1.

example, a Cohen p-ring is formally smooth over the ring of p-adic integers. In this general setting, the lifting property for homomorphisms (compare Theorem III.2.1) should be taken as the definition. For the applications we have in mind, the case treated in Definition III.1.1 will suffice. Henceforth, "formally smooth" will tacitly mean "with finite residue extension".

LEMMA III.1.3. *If B is formally smooth over A, then B is flat over A.*

Since flatness is invariant under completion, we may suppose that A and B are complete. Since flatness is also invariant under a flat local, hence faithfully flat, extension of the base ring, Definition III.1.1 reduces us to the case where B is a formal power-series algebra over A. As an A-module, B is then isomorphic to a product of A-modules isomorphic to A, and is therefore A-flat, since the base ring A is Noetherian.

We now place ourselves under the hypotheses of Definition III.1.1. Since the residue extensions of the local components of B' over A' are trivial, $L \otimes_k k'$ is an Artinian k'-algebra whose local components have trivial residue extensions, where L, k, and k' are the residue fields of B, A, and A', respectively. This condition, necessary for the finite free extension A' to satisfy the requirement in Definition III.1.1, is also sufficient, as follows immediately from Proposition III.1.4(i) and Proposition III.1.5 below.

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PROPOSITION III.1.4. *Let $A \to B$ be a local homomorphism of local rings with finite residue extension, and let A' be a finite local A-algebra. Then $B' = B \otimes_A A'$ is finite over B, and is therefore a Noetherian semilocal ring.*

(i) *If B is formally smooth over A, then the localizations of B' at its maximal ideals are formally smooth over A'.*

(ii) *The converse holds if A' is free over A.*

We reduce at once to the case in which A and B are complete.

For (i), let A'' be a finite free local extension of A such that the local components of $B'' = B \otimes_A A''$ are formal power-series algebras over A''. Make the scalar extension

$$A'' \longrightarrow A'' \otimes_A A' \longrightarrow A''',$$

where A''' is a local component of $A'' \otimes_A A'$. The local components of

$$B'' \otimes_{A''} A''' = B \otimes_A A'''$$

are then formal power-series algebras over A'''. We also have

$$B \otimes_A A''' = (B \otimes_A A') \otimes_{A'} A''' = B' \otimes_{A'} A'''.$$

Moreover, since A'' is free over A, the algebra $A'' \otimes_A A'$ is free over A'; its direct factor A''' is therefore free over the local ring A'. This proves that the local components of B' are formally smooth over A'.

For (ii), let A'' be a finite free local A'-algebra such that the local components of

$$B' \otimes_{A'} A'' = B \otimes_A A''$$

are formal power-series algebras over A''. Since A' is free over A, the algebra A'' is also free over A, and hence B is formally smooth over A.

PROPOSITION III.1.5. *Let $A \to B$ be a local homomorphism of local rings with* trivial *residue extension. Then B is formally smooth over A if and only if $\overline{B}$ is isomorphic to a formal power-series algebra over $\overline{A}$.*

Only necessity remains to be proved, and we may suppose that A and B are complete. Let $\mathfrak{m}$ and $\mathfrak{n}$ be the maximal ideals of A and B, respectively, and choose elements $t_1, \ldots, t_n \in \mathfrak{n}$ whose classes form a basis of the vector space

$$(\mathfrak{n}/\mathfrak{n}^2)/\operatorname{Im}(\mathfrak{m}/\mathfrak{m}^2) = \mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B).$$

These elements define a local homomorphism of A-algebras

$$B_1 = A[[t_1, \ldots, t_n]] \longrightarrow B.$$

We shall prove that it is an isomorphism. It is enough to prove, for every power $\mathfrak{m}^q$, that reduction modulo $\mathfrak{m}^q$ gives an isomorphism French source p. 61

$$B_1/\mathfrak{m}^q B_1 \longrightarrow B/\mathfrak{m}^q B,$$

since B_1 and B are the projective limits of the corresponding rings modulo $\mathfrak{m}^q$, as q varies. The A-modules B and B_1 are flat. Consequently, their associated graded rings for the $\mathfrak{m}$-adic filtration are obtained by tensoring $B_1/\mathfrak{m}B_1$, respectively $B/\mathfrak{m}B$, over $k = A/\mathfrak{m}$ with $\mathrm{gr}(A)$. We are thus reduced to showing that

$$B_1/\mathfrak{m}B_1 \longrightarrow B/\mathfrak{m}B$$

is an isomorphism. In view of Lemma III.1.3, this reduces us to the case in which A is a field k.

On the other hand, let A' be a finite free local A-algebra such that $B \otimes_A A'$ is a formal power-series algebra over A'. (This algebra is local, since the residue extension of B over A is trivial.) To prove that $B_1 \to B$ is an isomorphism, it is enough to prove that

$$B_1 \otimes_A A' \longrightarrow B \otimes_A A'$$

is an isomorphism. This reduces us to the case in which B is already a formal power-series algebra. This reduction should in fact have been made before reducing to the case of a base field. But then B is a regular local ring with coefficient field k, and it is well known—and follows immediately by considering the associated graded rings for the $\mathfrak{n}_1$-adic and $\mathfrak{n}$-adic filtrations on B_1 and B, where $\mathfrak{n}_1$ is the maximal ideal of B_1—that $B_1 \to B$ is an isomorphism. This completes the proof.

COROLLARY III.1.6. *If B is formally smooth over A, then there exists a finite local A-algebra A' such that the local components of*

$$\overline{B} \otimes_{\overline{A}} \overline{A'} = \overline{B \otimes_A A'}$$

are isomorphic to formal power-series algebras over $\overline{A'}$.

Indeed, let L/k be the residue extension of B/A, and choose a finite extension k'/k such that the residue extensions of the local components of the k'-algebra $L \otimes_k k'$ are trivial. Let A' be a finite free A-algebra such that $A'/\mathfrak{m}A' = k'$. Such an algebra exists: one may reduce step by step to the case in which k'/k is generated by one element, and then lift to A the coefficients of the minimal polynomial of a generator of k' over k. The algebra A' is local. At each maximal ideal of $B \otimes_A A'$, the residue extension over the residue field k' of A' is trivial, and Proposition III.1.5 completes the proof.

COROLLARY III.1.7. *Let $A \to B$ be a local homomorphism of local rings. Then B is formally smooth over A if and only if B is flat over A and $B/\mathfrak{m}B$ is formally smooth over $k = A/\mathfrak{m}$.*

After making a suitable finite free local extension A' of A and using Proposition III.1.4(ii), we reduce to the case in which the residue extension of B/A is trivial. Proposition III.1.4(i) and Lemma III.1.3 show that the stated conditions are necessary. For sufficiency, it is enough to observe that, under the present hypotheses, the proof of Proposition III.1.5 shows that B is a formal power-series algebra over A, after replacing A and B by their completions, as we may. French source p. 62

Remark III.1.8. It would not be difficult to develop, for formally smooth homomorphisms, analogues of all the properties of smooth morphisms studied in Exposé II. For differential properties, however, this requires modifying the usual definition of Kähler differentials (cf. I I.1), with completed tensor products replacing ordinary tensor products. We shall merely gesture toward these abysses here; the preceding results suffice for our purposes.

It remains to connect formal smoothness with the notion of smoothness developed in Exposé II, which we have not yet used at all:

PROPOSITION III.1.9. *Let $A \to B$ be a local homomorphism, where B is a localization of an A-algebra of finite type. Then B is smooth over A if and only if it is formally smooth over A.*

Using Corollary III.1.7 and Theorem II.2.1, we reduce to the case in which A is a field k.

Using Proposition III.1.4(ii) and Corollary II.4.13, a suitable extension k'/k reduces us to the case in which the residue extension of B/k is trivial. By Corollary II.5.2, respectively Proposition III.1.5, the algebra B is smooth over k, respectively formally smooth over k, if and only if B is a regular local ring, respectively its completion is a formal power-series algebra over k. These two conditions are well known to be equivalent when the residue extension is trivial.

III.2. The characteristic lifting property of formally smooth homomorphisms

THEOREM III.2.1. *Let $A \to B$ be a local homomorphism of local rings inducing a finite residue-field extension. The following conditions are equivalent:*

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(i) *B is formally smooth over A.*

(ii) *For every local homomorphism $A \to C$, where C is a complete local ring, every ideal $J \subseteq \mathfrak{r}(C)$, and every local A-homomorphism $B \to C/J$, there exists an A-homomorphism $B \to C$, necessarily local, which lifts it.*

(iii) *For every A-algebra C, not necessarily Noetherian, every nilpotent ideal J of C, and every continuous A-homomorphism $B \to C/J$, that is, one which vanishes on some power of $\mathfrak{r}(B)$, there exists an A-homomorphism $B \to C$ lifting it; this lift is necessarily continuous as well.*

(iv) *The same lifting assertion as in* (ii) *and* (iii)*, with C restricted to be a local Artinian ring finite over A.*

(iv bis) *The same as* (iv)*, with the additional requirement $J^2 = 0$.*

Remark. In the remainder of this Exposé we shall use only the implication (iv)⇒(i), or (iv bis)⇒(i). The direct implication (i)⇒(ii) will be proved by another method in the next section when B is a localization of a finitely generated A-algebra. Recall that in the "good" theory of Cohen theorems,[3] property (ii) or (iii) becomes the definition of formally smooth homomorphisms, whereas Definition III.1.1 becomes a characteristic property valid only when the residue extension is finite.

Notice that neither property (ii) nor property (iii) is more general than the other. One could give an equivalent property encompassing both by taking a linearly topologized ring C, separated and complete, a closed topologically nilpotent ideal of C, and a continuous homomorphism $A \to C$, thereby making C a topological A-algebra. We leave this modification to the reader.

Proof of Theorem III.2.1. We shall prove

$$\text{(i)} \Longrightarrow \text{(iii)} \Longrightarrow \text{(ii)}$$

[3] See EGA 0_{IV}, 19.3 and 19.8.

and then (iv) ⇒ (i). Since (ii) ⇒ (iv) is trivial, and the equivalence of (iv) and (iv bis) follows by immediate induction on an integer n such that $J^n = 0$, this will complete the proof.

(i)⇒(iii). An immediate induction reduces us to the case $J^2 = 0$. Continuity of the given map, together with $J^2 = 0$, implies that some power of the maximal ideal $\mathfrak{m}$ of A acts trivially on C. After quotienting by that power of $\mathfrak{m}$, and observing from Proposition III.1.4(i) that $B/\mathfrak{m}^q B$ remains formally smooth over $A/\mathfrak{m}^q$, we may suppose that A is Artinian. By Lemma III.1.3, B is flat over A, and hence is free as an A-module because A is Artinian. There is therefore an A-linear map French source p. 64

$$w\colon B \longrightarrow C$$

lifting the given homomorphism $u\colon B \to C/J$. Put

$$f(x,y) = w(xy) - w(x)w(y) \qquad (x, y \in B).$$

Then $f(x,y) \in J$, so f is an A-bilinear map from $B \times B$ to J. A lift $v\colon B \to C$ of u is an algebra homomorphism if and only if there is an A-linear map $g\colon B \to J$ such that $v = w + g$ and

$$g(1) = 1 - w(1),$$
$$g(xy) - u(x)g(y) - u(y)g(x) = -f(x,y) \qquad (x, y \in B).$$

Here the action of C/J on J is well defined because $J^2 = 0$.

This is a system of linear equations in $\mathrm{Hom}_A(B, J)$, with right-hand sides in J. It has a solution if and only if the corresponding system in

$$\mathrm{Hom}_A(B, J) \otimes_A A', \qquad J' = J \otimes_A A',$$

has a solution, where A' is any faithfully flat A-algebra. Choose a finite free local A-algebra A' such that $B' = B \otimes_A A'$ is a formal power-series algebra over A'; we may suppose A and B complete in this argument. Since A' is finite free over A,

$$\mathrm{Hom}_A(B, J) \otimes_A A' = \mathrm{Hom}_{A'}(B', J').$$

The resulting system in $\mathrm{Hom}_{A'}(B', J')$ is exactly the one determining the A'-algebra homomorphisms

$$B' \longrightarrow C' = C \otimes_A A'$$

which lift the scalar extension $u'\colon B' \to C'/J'$ of u, by correcting the scalar extension $w'\colon B' \to C'$ of w with an A'-linear map $g'\colon B' \to J'$. Notice that B generates B' as an A'-module.

We are thus reduced to proving (iii) when

$$B = A[[t_1, \ldots, t_n]]$$

is a formal power-series algebra over A. Lift the images of the t_i in C/J arbitrarily to elements $z_i \in C$. Since the z_i mod J are nilpotent, by continuity of $u\colon B \to C/J$, and J itself is nilpotent, the z_i are nilpotent. They therefore define a continuous homomorphism of topological A-algebras from B to the discrete ring C, which plainly lifts u.

(iii)⇒(ii). Let $\mathfrak{n}$ be the maximal ideal of C, and, for each integer $q > 0$, set French source p. 65

$$C_q = C/\mathfrak{n}^q, \qquad J_q = (J + \mathfrak{n}^q)/\mathfrak{n}^q.$$

Thus C_q/J_q is naturally a quotient algebra of C/J. Moreover, the composite homomorphism

$$u_q\colon B \longrightarrow C/J \longrightarrow C_q/J_q$$

is continuous when C_q/J_q is given the discrete topology, and J_q is a nilpotent ideal of C_q. We construct successively A-homomorphisms

$$v_q\colon B \longrightarrow C_q$$

such that (a) v_q lifts u_q, and (b) v_q lifts v_{q-1}. The induction can be carried out as follows. The maps

$$u_q\colon B \longrightarrow C/(J+\mathfrak{n}^q), \qquad v_{q-1}\colon B \longrightarrow C/\mathfrak{n}^{q-1}$$

induce the same homomorphism

$$B \longrightarrow C/\big((J+\mathfrak{n}^q)+\mathfrak{n}^{q-1}\big) = C/(J+\mathfrak{n}^{q-1}) = C_{q-1}/J_{q-1},$$

namely u_{q-1}. They therefore determine a homomorphism

$$B \longrightarrow C/J'_q, \qquad J'_q = (J+\mathfrak{n}^q)\cap\mathfrak{n}^{q-1} \supseteq \mathfrak{n}^q,$$

from which both are obtained by reduction. It remains only to lift this homomorphism from B to the quotient of C_q by $J'_q/\mathfrak{n}^q$. This ideal is contained in J_q, hence is nilpotent, so the required lift exists by hypothesis (iii).

The maps v_q now determine a homomorphism from B to the inverse limit C of the rings C_q. Since J is closed, it is the inverse limit of the ideals J_q; consequently this homomorphism lifts u.

(iv)⇒(i). First observe that if condition (iv) holds, then it also holds over A' for every local component of $B\otimes_A A'$, whenever A' is a finite local A-algebra. Choose A' free over A and such that the residue extensions of the local components of $B' = B\otimes_A A'$ over A' are trivial. Proposition III.1.4 (ii) then reduces us to the case in which the residue extension of B over A is trivial. We shall prove the following slightly more precise result.

COROLLARY III.2.2. *Under the hypotheses of Theorem III.2.1, suppose in addition that the residue extension of B over A is trivial. Then the equivalent conditions of Theorem III.2.1 are also equivalent to each of the following two conditions (with A and B assumed complete in* (v)*):*

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(iv ter) *The same assertion as in* (iv)*, but with the finite local Artinian A-algebra C required to have trivial residue extension (and, if desired, with $J^2=0$).*

(v) *There is a local A-homomorphism, where*

$$n = \dim\big(\mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B)\big),$$

$$u\colon B \longrightarrow B_1 = A[[t_1,\dots,t_n]],$$

which induces an isomorphism

$$\mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B) \xrightarrow{\sim} \mathfrak{n}_1/(\mathfrak{n}_1^2+\mathfrak{m}B_1),$$

where $\mathfrak{n}$ and $\mathfrak{n}_1$ are the maximal ideals of B and B_1, respectively, and $\mathfrak{m}$ is the maximal ideal of A.

Proof. Since (iv bis) plainly implies (iv ter)—even without using the square-zero proviso—it is enough to prove

$$\text{(iv ter)} \Rightarrow \text{(v)} \Rightarrow \text{(i)}.$$

(iv ter)⇒(v). Choose a basis $a_1,\dots,a_n$ of $\mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B)$. This determines a local homomorphism of A-algebras

$$B \longrightarrow B_1/(\mathfrak{n}_1^2+\mathfrak{m}B_1) = k[t_1,\dots,t_n]/(t_1,\dots,t_n)^2.$$

By (iv ter) it can be lifted successively to homomorphisms of A-algebras from B to $B_1/\mathfrak{n}_1^2, B_1/\mathfrak{n}_1^3,\dots$. Passing to the inverse limit gives a homomorphism $B\to B_1$ having the required property.

(v)⇒(i). In the commutative diagram

$$\begin{array}{ccccccc}
\mathfrak{m}/\mathfrak{m}^2 & \longrightarrow & \mathfrak{n}/\mathfrak{n}^2 & \longrightarrow & \mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B) & \longrightarrow & 0 \\
\downarrow & & \downarrow & & \downarrow & & \\
\mathfrak{m}/\mathfrak{m}^2 & \longrightarrow & \mathfrak{n}_1/\mathfrak{n}_1^2 & \longrightarrow & \mathfrak{n}_1/(\mathfrak{n}_1^2+\mathfrak{m}B_1) & \longrightarrow & 0.
\end{array}$$

the rows are exact and the two outer vertical arrows are surjective. The middle vertical arrow is therefore surjective as well; since B is complete, it follows that $u\colon B \to B_1$ is surjective. Choose elements $x_i \in B$, $1 \leq i \leq n$, lifting the t_i. They define an A-algebra homomorphism $B_1 \to B$, which is surjective for the same reason as u, and whose composite with u is the identity by construction. Thus $B_1 \to B$ is also injective and hence is an isomorphism. We have therefore also proved the following.

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COROLLARY III.2.3. *Under the hypotheses of Corollary III.2.2*(v)*, the map u is necessarily an isomorphism.*

This completes the proof that B is formally smooth over A. At the same time we have recovered Proposition III.1.5, although there is little merit in doing so.

III.3. Local infinitesimal lifting of morphisms into a smooth S-scheme

THEOREM III.3.1. *Let $f\colon X \to Y$ be a morphism locally of finite type. The following conditions are equivalent:*

(i) *f is smooth.*

(ii) *For every prescheme Y' over Y, every closed subprescheme $Y'_0 \subseteq Y'$ having the same underlying space as Y', every Y-morphism $g_0\colon Y'_0 \to X$, and every $z \in Y'_0$, there exist an open neighborhood U of z in Y' and a Y-morphism $g\colon U \to X$ extending $g_0|_{Y'_0 \cap U}$.*

(ii bis) *For Y', Y'_0, and z as in* (ii)*, put*

$$X' = X \times_Y Y', \qquad X'_0 = X \times_Y Y'_0.$$

Every section of X'_0 over Y'_0 extends to a section of X' over an open neighborhood U of z.

(iii) *For every Y-scheme Y' which is the spectrum of a local Artinian ring finite over some $\mathcal{O}_y$, $y \in Y$, every nonempty closed subprescheme $Y'_0 \subseteq Y'$, and every Y-morphism $g_0\colon Y'_0 \to X$, there exists a Y-morphism $g\colon Y' \to X$ extending g_0.*

(iii bis) *For Y' and Y'_0 as in* (iii)*, put*

$$X' = X \times_Y Y', \qquad X'_0 = X \times_Y Y'_0.$$

Every section of X'_0 over Y'_0 extends to a section of X' over Y'.

Proof. The equivalence of (ii) and (ii bis), and that of (iii) and (iii bis), are immediate, as is the implication (ii) ⇒ (iii). It remains to prove

$$\text{(i)} \Rightarrow \text{(ii)} \qquad \text{and} \qquad \text{(iii)} \Rightarrow \text{(i)}.$$

(i)⇒(ii). Put $x = g_0(z)$. By replacing X with a suitable open neighborhood of x, and Y' with the prescheme induced on the inverse image of that open set under g_0, we may suppose that X is étale over $Y[t_1, \ldots, t_n]$. The composite Y-morphism

$$Y'_0 \longrightarrow X \longrightarrow Y[t_1, \ldots, t_n]$$

is defined by n sections of $\mathcal{O}_{Y'_0}$. Near z, these extend to sections of $\mathcal{O}_{Y'}$, so after shrinking we may suppose that the displayed morphism extends to a Y-morphism

$$Y' \longrightarrow Y[t_1, \ldots, t_n].$$

By I I.5.6, there is then a unique Y-morphism $g\colon Y' \to X$ lifting this morphism and at the same time extending g_0.

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(iii)⇒(i). Since the locus where f is smooth is open, it is enough to show that it contains every $x \in X$ which is closed in its fiber. Put $y = f(x)$. Then $\mathcal{O}_x$ is an $\mathcal{O}_y$-algebra which is a localization of a finitely generated algebra and has finite residue extension. Hypothesis (iii) implies that every homomorphism from $\mathcal{O}_x$ to A/J, where A is a local Artinian algebra finite over $\mathcal{O}_y$ and J is an ideal contained in its radical, lifts to a homomorphism $\mathcal{O}_x \to A$. Here we use the fact that a morphism $\mathrm{Spec}(B) \to X$, with B local and the closed point mapping to x, is determined bijectively by a local homomorphism $\mathcal{O}_x \to B$. Theorem III.2.1 therefore shows that $\mathcal{O}_x$ is formally smooth over $\mathcal{O}_y$, and hence smooth over $\mathcal{O}_y$ by Proposition III.1.9.

COROLLARY III.3.2. *Let $f\colon X \to Y$ be as in Theorem III.3.1. The following conditions are equivalent:*

(i) *f is étale.*
(ii) *Condition* (ii) *of Theorem III.3.1 holds, with the extension g of g_0 to U unique.*
(iii) *Condition* (iii) *of Theorem III.3.1 holds, with g unique.*

It is enough to observe in the proof of (i) ⇒ (ii) above that uniqueness, when Y_0' is not identical to Y' near z, is possible only if $n = 0$, a condition already known to be sufficient.

COROLLARY III.3.3. *Let X be a prescheme locally of finite type over a complete local ring A, let y be the closed point of $Y = \mathrm{Spec}(A)$, and let $x \in f^{-1}(y)$ be rational over $\kappa(y)$. If X is smooth over A at x, then there exists a section s of X over Y passing through x, that is, satisfying $s(y) = x$.*

In particular, if X is smooth over A, then the natural map

$$\Gamma(X/Y) \longrightarrow \Gamma(X \otimes_A k/k)$$

from the sections of X over Y to the set of k-rational points of the fiber $f^{-1}(y) = X \otimes_A k$ is surjective. This fact was especially well known and used when A is a discrete valuation ring and X is proper over A, in fact projective. In that case the sections of X over Y, that is, the A-valued points of X, also identify with the rational sections, namely with the K-valued points of

$$X \otimes_A K = X_K,$$

which is a proper smooth scheme over the fraction field K of A.

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III.4. Local infinitesimal lifting of smooth S-schemes

THEOREM III.4.1. *Let Y be a locally Noetherian prescheme, let Y_0 be a closed subprescheme with the same underlying space, let X_0 be a smooth Y_0-prescheme, and let $x \in X_0$. There exist an open neighborhood U_0 of x, a prescheme X smooth over Y, and a Y_0-isomorphism*

$$h\colon U_0 \xrightarrow{\sim} X \times_Y Y_0.$$

Moreover, if (U_0', X', h') is another solution of this problem, then it is isomorphic to the first in a neighborhood of x.

We leave it to the reader to make the last assertion precise. For a fixed U_0, a solution of the problem amounts to giving on U_0 a sheaf of algebras $\mathcal{B}$ over $f_0^{-1}(\mathcal{O}_Y)$, where f_0 is the continuous map underlying the structural morphism $U_0 \to Y_0$, together with a ring homomorphism

$$\mathcal{B} \longrightarrow \mathcal{O}_{U_0}$$

compatible with $f_0^{-1}(\mathcal{O}_Y) \to f_0^{-1}(\mathcal{O}_{Y_0})$, such that:

(a) this homomorphism induces an isomorphism

$$\mathcal{B} \otimes_{f_0^{-1}(\mathcal{O}_Y)} f_0^{-1}(\mathcal{O}_{Y_0}) \xrightarrow{\sim} \mathcal{O}_{U_0};$$

(b) U_0, equipped with $\mathcal{B}$, becomes a smooth Y-prescheme.

In this formulation, the precise meaning of the local uniqueness assertion becomes particularly clear.

Proof. We may first suppose that X_0 is étale over

$$Y_0[t_1, \dots, t_n] = Y_0'.$$

The latter may be regarded as a closed subprescheme, with the same underlying space, of $Y' = Y[t_1, \dots, t_n]$. By I I.8.3, there exist a prescheme X étale over Y' and a Y_0'-isomorphism

$$X \times_{Y'} Y_0' \xrightarrow{\sim} X_0.$$

This proves existence. For uniqueness, use the lifting property in Theorem III.3.1(ii) for smooth morphisms, together with the following lemma. French source p. 70

LEMMA III.4.2. *Let Y be a prescheme, let Y_0 be a closed subprescheme defined by a locally nilpotent sheaf of ideals $\mathcal{J}$, let X and X' be Y-preschemes, and let $u\colon X \to X'$ be a Y-morphism. Suppose that X is flat over Y. Then u is an isomorphism if and only if*

$$u_0\colon X \times_Y Y_0 \longrightarrow X' \times_Y Y_0$$

is an isomorphism.

This follows easily by passing to the affine case and considering the associated graded objects. Notice also that the analogous statement obtained by replacing "isomorphism" with "closed immersion" is valid without the flatness hypothesis.

Remark III.4.3. It is essential to observe that the local lifting X obtained in Theorem III.4.1 is *not canonical*. In other words, the local isomorphism between two solutions is not unique: in general there are nontrivial Y-automorphisms of X inducing the identity on the closed subprescheme $X_0 = X \times_Y Y_0$. Consequently, in constructing global infinitesimal liftings of smooth preschemes one should expect a cohomological obstruction, which will be made precise below in Section III.6.

III.5. Global infinitesimal lifting of morphisms

Let T be a topological space, let $\mathcal{G}$ be a sheaf of groups on T, and let $\mathcal{P}$ be a sheaf of sets on T on which $\mathcal{G}$ acts, on the right for definiteness. We say that $\mathcal{P}$ is *formally principal homogeneous* under $\mathcal{G}$ if the familiar homomorphism

$$\mathcal{G} \times \mathcal{P} \longrightarrow \mathcal{P} \times \mathcal{P}$$

of sheaves of sets, induced by the action of $\mathcal{G}$ on $\mathcal{P}$, is an *isomorphism*. Equivalently, for every $x \in T$, the stalk $\mathcal{P}_x$ is either empty or a principal homogeneous space under the ordinary group $\mathcal{G}_x$. Equivalently again, for every open subset $U \subseteq T$, the set $\mathcal{P}(U)$ is either empty or a principal homogeneous space under the ordinary group $\mathcal{G}(U)$.

We say that $\mathcal{P}$ is a *principal homogeneous sheaf* under $\mathcal{G}$ if it is formally principal homogeneous and all its stalks $\mathcal{P}_x$ are nonempty; in other words, if every $\mathcal{P}_x$ is a nonempty principal homogeneous space under $\mathcal{G}_x$. [4] Recall that the set of isomorphism classes of principal homogeneous sheaves under $\mathcal{G}$ is identified with the cohomology set $H^1(T, \mathcal{G})$. When $\mathcal{G}$ is commutative, this is the usual cohomology group of T with coefficients in $\mathcal{G}$. Thus every principal homogeneous sheaf $\mathcal{P}$ has a characteristic class French source p. 71

$$c(\mathcal{P}) \in H^1(T, \mathcal{G}).$$

This class vanishes if and only if $\mathcal{P}$ is trivial—that is, isomorphic to $\mathcal{G}$, acting on itself by right translations—or, equivalently, if and only if $\mathcal{P}$ has a section.

[4] It seems preferable to adopt the shorter and more expressive term "*torsor under* $\mathcal{G}$", introduced in the thesis of J. Giraud.

PROPOSITION III.5.1. *Let S be a prescheme, let X and Y be preschemes over S, and let Y_0 be a closed subprescheme of Y, defined by an ideal sheaf $\mathcal{J}$ on Y with $\mathcal{J}^2 = 0$. Let $g_0 \colon Y_0 \to X$ be an S-morphism, and let $\mathcal{P}(g_0)$ be the sheaf on Y whose sections on an open subset U are the S-morphisms $g \colon U \to X$ extending $g_0|_{U\cap Y_0}$. Then $\mathcal{P}(g_0)$ is naturally a* formally principal homogeneous sheaf *under the commutative sheaf of groups*

$$\mathcal{G} = \mathcal{H}om_{\mathcal{O}_{Y_0}}(g_0^*\Omega^1_{X/S}, \mathcal{J}).$$

Put $\mathcal{P} = \mathcal{P}(g_0)$. For every open subset U of Y, we must define a map

$$\mathcal{P}(U) \times \mathcal{G}(U) \longrightarrow \mathcal{P}(U)$$

such that: (a) for fixed $g \in \mathcal{P}(U)$, the map $s \mapsto gs$ from $\mathcal{G}(U)$ to $\mathcal{P}(U)$ is bijective; (b) $\mathcal{P}(U)$ becomes a right $\mathcal{G}(U)$-set; and (c) these maps are compatible with restriction to every open subset $V \subseteq U$. Condition (c) will be immediate, so for simplicity we may take $U = Y$. We leave condition (b), which is local in nature, to the reader. It remains, for a given $g \in \mathcal{P}(Y)$, to define a natural bijection from $\mathcal{G}(Y)$ onto $\mathcal{P}(Y)$.

We are therefore given an S-morphism $g \colon Y \to X$, and seek a canonical bijection

$$(*) \qquad \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*\Omega^1_{X/S}, \mathcal{J}) \xrightarrow{\sim} \mathcal{P}(Y),$$

where $\mathcal{P}(Y)$ is the set of S-morphisms $g' \colon Y \to X$ that induce on Y_0 the same morphism $g_0 \colon Y_0 \to X$ as g. Giving such a g' is equivalent to giving an S-morphism $h \colon Y \to X \times_S X$ such that

$$\mathrm{pr}_1 \circ h = g, \qquad h \circ i = (g_0, g_0),$$

where $\mathrm{pr}_1 \colon X \times_S X \to X$ is the first projection, $i \colon Y_0 \to Y$ is the canonical immersion, and $(g_0, g_0) = \Delta_{X/S} g_0 \colon Y_0 \to X \times_S X$ has components g_0, g_0. This is summarized by the diagram French source p. 72

$$\begin{array}{ccc} Y_0 & \xrightarrow{h_0=(g_0,g_0)} & X \times_S X \\ {\scriptstyle i}\big\downarrow & \nearrow_h & \big\downarrow{\scriptstyle \mathrm{pr}_1} \\ Y & \xrightarrow{g} & X. \end{array}$$

The morphism h_0 factors through the diagonal immersion $\Delta_{X/S}$. Since Y lies in the first infinitesimal neighborhood of Y_0, that is, since $\mathcal{J}^2 = 0$, every such h necessarily factors, uniquely, through the first infinitesimal neighborhood of the diagonal. As an X-prescheme via pr_1, this neighborhood is identified with the relative spectrum X' of the sheaf of algebras

$$\mathcal{O}_X \oplus \Omega^1_{X/S},$$

where the second summand is a square-zero ideal; the diagonal morphism $X \to X'$ corresponds to the canonical augmentation of this algebra sheaf.

Put

$$Y' = X' \times_X Y, \qquad Y'_0 = Y' \times_Y Y_0 = X' \times_X Y_0.$$

The desired morphisms h are then in bijection with the sections u of Y' over Y that extend a fixed section u_0 of Y'_0 over Y_0. More explicitly, Y' is the relative spectrum over Y of

$$\mathcal{A} = g^*(\mathcal{O}_X \oplus \Omega^1_{X/S}) = \mathcal{O}_Y \oplus g^*\Omega^1_{X/S},$$

and Y'_0 is the relative spectrum over Y_0 of

$$\mathcal{A}_0 = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} = \mathcal{O}_{Y_0} \oplus g_0^*\Omega^1_{X/S}.$$

The section u_0 is defined by the canonical augmentation $\mathcal{A}_0 \to \mathcal{O}_{Y_0}$. Hence $\mathcal{P}(Y)$ is identified with the algebra homomorphisms $\mathcal{A} \to \mathcal{O}_Y$ inducing that augmentation on $\mathcal{A}_0$.

For brevity put $\mathcal{M} = g^*\Omega^1_{X/S}$ and $\mathcal{M}_0 = \mathcal{M} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}$. Such an algebra homomorphism is determined by an $\mathcal{O}_Y$-linear homomorphism $\mathcal{M} \to \mathcal{O}_Y$. The condition on $\mathcal{A}_0$ says

precisely that its image lies in $\mathcal{J}$; conversely, because $\mathcal{J}^2 = 0$, every $\mathcal{O}_Y$-linear map $\mathcal{M} \to \mathcal{J}$ defines such an algebra homomorphism. We therefore obtain

$$\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{J}) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{M}_0, \mathcal{J}),$$

because the action of $\mathcal{O}_Y$ on $\mathcal{J}$ factors through $\mathcal{O}_{Y_0}$. This is precisely the canonical bijection (*).

Taking into account the implication (i) ⇒ (iii) in Theorem III.3.1, we obtain:

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COROLLARY III.5.2. *If X is smooth over S, at least at the points of $g_0(Y_0)$, then $\mathcal{P}$ is in fact a* principal homogeneous sheaf *under the commutative sheaf of groups $\mathcal{G}$. In this case it may also be written*

$$\mathcal{G} = g_0^*(\mathfrak{g}_{X/S}) \otimes_{\mathcal{O}_{Y_0}} \mathcal{J},$$

where $\mathfrak{g}_{X/S} = (\Omega^1_{X/S})^\vee$ is the sheaf on X dual to $\Omega^1_{X/S}$, that is, the tangent sheaf*, or* sheaf of derivations*, of X relative to S. The displayed formula follows from the fact that $\Omega^1_{X/S}$ is then locally free of finite rank.*

In particular, this principal homogeneous sheaf determines a cohomology class in $H^1(Y_0, \mathcal{G})$. Its vanishing is necessary and sufficient for the existence of an S-morphism g extending g_0. If such an extension exists, the set of all possible extensions is a homogeneous space under the group $H^0(Y_0, \mathcal{G})$.

In applications of the methods of formal geometry, the situation is most often the following. Let X and Y be S-preschemes, let $\mathcal{I}$ be a coherent ideal sheaf on S, let S_n denote the closed subprescheme of S defined by $\mathcal{I}^{n+1}$, and put

$$X_n = X \times_S S_n, \qquad Y_n = Y \times_S S_n.$$

Suppose that an S_n-morphism

$$g_n \colon Y_n \longrightarrow X_n$$

is given. Equivalently, this is an S-morphism $Y_n \to X$, or an S_{n+1}-morphism $Y_n \to X_{n+1}$, since any such morphism necessarily induces $Y_n \to X_n$. We seek to extend it to an S_{n+1}-morphism

$$g_{n+1} \colon Y_{n+1} \longrightarrow X_{n+1}.$$

If this process can be continued indefinitely, it yields a morphism $\widehat{Y} \to \widehat{X}$ between the formal preschemes obtained by completing Y and X with respect to the ideals $\mathcal{I}\mathcal{O}_Y$ and $\mathcal{I}\mathcal{O}_X$, respectively.

Apply Proposition III.5.1 with

$$(S, X, Y, Y_0, g_0) = (S_{n+1}, X_{n+1}, Y_{n+1}, Y_n, g_n).$$

The sheaf $\mathcal{G}$ is now the sheaf of homomorphisms from

$$g_n^* \Omega^1_{X_{n+1}/S_{n+1}}$$

to

$$\mathcal{J} = \mathcal{I}^{n+1}\mathcal{O}_Y / \mathcal{I}^{n+2}\mathcal{O}_Y.$$

Since $\mathcal{J}$ is annihilated by $\mathcal{I}\mathcal{O}_Y$, the first sheaf may be replaced by its restriction to $Y_0 = Y \times_S S_0$. Let $h_0 \colon Y_0 \to X_{n+1}$ be the composite $Y_0 \to Y_n \xrightarrow{g_n} X_n \to X_{n+1}$, and let $g_0 \colon Y_0 \to X_0$ be the morphism induced by g_n. Equivalently, h_0 is the composite $Y_0 \xrightarrow{g_0} X_0 \to X_{n+1}$. The restricted sheaf is $h_0^* \Omega^1_{X_{n+1}/S_{n+1}}$. Since the inverse image of $\Omega^1_{X_{n+1}/S_{n+1}}$ on

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$$X_0 = X_{n+1} \times_{S_{n+1}} S_0$$

is $\Omega^1_{X_0/S_0}$, we may write

$$\mathcal{G} = \mathcal{H}om_{\mathcal{O}_{Y_0}} \left(g_0^* \Omega^1_{X_0/S_0}, \mathcal{I}^{n+1}\mathcal{O}_Y / \mathcal{I}^{n+2}\mathcal{O}_Y \right).$$

Thus we obtain the following corollary.

COROLLARY III.5.3. *Let $S, X, Y, \mathcal{I}, g_n$ be as above. Let $\mathcal{P}(g_n)$ be the sheaf on Y whose sections over an open set U are the extensions g_{n+1} of g_n to an S_{n+1}-morphism $Y_{n+1} \to X_{n+1}$. Then $\mathcal{P}(g_n)$ is a pseudo-torsor under the sheaf of groups*

$$\mathcal{G} = \mathcal{H}om_{\mathcal{O}_{Y_0}}\left(g_0^*\Omega^1_{X_0/S_0}, \mathrm{gr}^{n+1}_{\mathcal{I}\mathcal{O}_Y}(\mathcal{O}_Y)\right).$$

In particular:

COROLLARY III.5.4. *If, in addition, X is smooth over S, at least at the points of $g_0(Y_0)$, then $\mathcal{P}(g_n)$ is a torsor. It therefore defines an obstruction class in $H^1(Y_0, \mathcal{G})$, whose vanishing is necessary and sufficient for the existence of a global extension g_{n+1} of g_n. If such an extension exists, the set of all global extensions is a principal homogeneous space under $H^0(Y_0, \mathcal{G})$. In this case the sheaf $\mathcal{G}$ may also be written*

$$\mathcal{G} = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes_{\mathcal{O}_{Y_0}} \mathrm{gr}^{n+1}_{\mathcal{I}\mathcal{O}_Y}(\mathcal{O}_Y).$$

Proceeding step by step, we see that if all the $H^1(Y_0, \mathcal{G}_n)$ vanish, where

$$\mathcal{G}_n = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes \mathrm{gr}^n_{\mathcal{I}\mathcal{O}_Y}(\mathcal{O}_Y),$$

then, starting with any g_k, one may successively extend it to $g_{k+1}, \ldots$. In particular, if $\mathcal{I}$ is nilpotent, an extension of g_k to Y exists. The required H^1-vanishing holds in particular when Y_0 is affine. Thus:

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COROLLARY III.5.5. *In Theorem III.3.1, one obtains another equivalent necessary and sufficient condition by restricting the Y' occurring in* (ii)*, or in* (ii bis)*, to be affine and requiring a global extension g of g_0 over all of Y'.*

Notice that the proof of Theorem III.3.1 could not have given this result directly.

An important case is that in which Y is flat over S. Then

$$\mathrm{gr}^n(\mathcal{O}_Y) = \mathrm{gr}^n(\mathcal{O}_S) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}.$$

If, in addition, the $\mathrm{gr}^n(\mathcal{O}_S)$ are locally free over S, then

$$\mathcal{G}_n = \mathcal{H}om_{\mathcal{O}_{Y_0}}(g_0^*\Omega^1_{X_0/S_0}, \mathcal{O}_{Y_0}) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}^n(\mathcal{O}_S).$$

Equivalently, if $\Omega^1_{X_0/S_0}$ is also locally free, for example if X is smooth over S,

$$\mathcal{G}_n = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}^n(\mathcal{O}_S).$$

For example, suppose that S is affine with coordinate ring A, and that $\mathcal{I}$ is defined by an ideal $I \subseteq A$. Then, for every i,

$$H^i(Y_0, \mathcal{G}_n) = H^i(Y_0, \mathcal{G}_0) \otimes_A \mathrm{gr}^n_I(A).$$

Indeed, the question is local on S_0, and reduces to tensoring with a free module. In this case, the vanishing of $H^1(Y_0, \mathcal{G}_0)$ implies that every obstruction to the successive extensions of g_n vanishes. We obtain:

COROLLARY III.5.6. *Let $(S, X, Y, \mathcal{I}, g_n)$ be as above. Suppose in addition that X is smooth over S, that Y is flat over S, that S is affine, and that*

$$\mathrm{gr}^n(\mathcal{O}_S) = \mathcal{I}^n/\mathcal{I}^{n+1}$$

is locally free for every n. Then the obstruction to constructing g_{n+1} lies in

$$H^1(Y_0, \mathcal{G}_0) \otimes_A \mathrm{gr}^{n+1}_I(A),$$

where A is the ring of S, I is the ideal defining $\mathcal{I}$, and

$$\mathcal{G}_0 = g_0^*(\mathfrak{g}_{X_0/S_0}).$$

If $H^1(Y_0, \mathcal{G}_0) = 0$, then g_n extends to an $\widehat{S}$-morphism

$$\widehat{g}\colon \widehat{Y} \longrightarrow \widehat{X}.$$

The same result plainly remains valid if, instead of ordinary S-preschemes X and Y, one starts with adic formal $\widehat{S}$-preschemes $\mathfrak{X}$ and $\mathfrak{Y}$. For example, it can be used to show that certain formal schemes proper over a complete local ring are in fact algebraic. Proceeding as in Lemma III.4.2, one obtains:

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COROLLARY III.5.7. *Under the hypotheses of Corollary III.5.6, if g_0 is an isomorphism, then so is $\widehat{g}$.*

The same statement holds for closed immersions.

PROPOSITION III.5.8. *Let A be a complete local ring with maximal ideal $\mathfrak{m}$ and residue field k. Let $\mathfrak{X}$ and $\mathfrak{Y}$ be two $\mathfrak{m}$-adic formal preschemes over A, flat over A; that is, for every n, X_n and Y_n are flat over $A_n = A/\mathfrak{m}^{n+1}$. Suppose that*

$$X_0 = \mathfrak{X} \otimes_A k$$

is smooth over k and that

$$H^1(X_0, \mathfrak{g}_{X_0/k}) = 0.$$

Then every k-isomorphism $Y_0 \xrightarrow{\sim} X_0$ extends to an A-isomorphism $\mathfrak{Y} \xrightarrow{\sim} \mathfrak{X}$. This extension is unique if, in addition,

$$H^0(X_0, \mathfrak{g}_{X_0/k}) = 0.$$

This gives, in particular, a uniqueness result for a smooth formal prescheme over A reducing to a given prescheme X_0, provided $H^1(X_0, \mathfrak{g}_{X_0/k}) = 0$. Furthermore, if $\mathfrak{X}$ and $\mathfrak{Y}$ arise as the formal completions of ordinary schemes X and Y proper over A, then the existence theorem for sheaves in formal geometry (see Bourbaki Seminar, no. 182, and EGA III, 5.4.1) gives a bijection between the A-isomorphisms $Y \xrightarrow{\sim} X$ and the A-isomorphisms of their formal completions. Hence:

COROLLARY III.5.9. *Proposition III.5.8 remains valid when $\mathfrak{X}$ and $\mathfrak{Y}$ are replaced by ordinary A-schemes X and Y proper over A.*

Finally, suppose that $\mathfrak{X}$ is a formal scheme proper over A, while $\mathfrak{Y} = \widehat{Y}$ for an ordinary scheme Y proper over A. Proposition III.5.8 gives sufficient conditions for an isomorphism $\mathfrak{X} \xrightarrow{\sim} \widehat{Y}$ to exist, and hence for $\mathfrak{X}$ to be algebraic, that is, isomorphic to $\widehat{X}$ for an ordinary scheme X proper over A; this X is then canonically determined. This applies in particular when $X_0 = \mathbb{P}^r_k$, and more generally when X_0 is a Severi–Brauer scheme, that is, becomes isomorphic to projective space over the algebraic closure of k. Every formal scheme proper and flat over A with special fiber $\mathbb{P}^r_k$ is algebraizable; more precisely, it is isomorphic to the $\mathfrak{m}$-adic formal completion of $\mathbb{P}^r_A$. Consequently, by the existence theorem, every ordinary scheme proper over a complete local ring A, with special fiber $\mathbb{P}^r_k$, is isomorphic to $\mathbb{P}^r_A$.

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Using descent theory, one can prove that when A is not complete, X becomes isomorphic to $\mathbb{P}^r$ after a finite étale base extension $A \to A'$. In this form the result remains valid when the special fiber is a Severi–Brauer scheme.

III.6. Global infinitesimal lifting of smooth S-schemes

Under the hypotheses of Theorem III.4.1, we ask whether there is a prescheme X smooth over Y such that $X \times_Y Y_0$ is Y_0-isomorphic to X_0, knowing that such a scheme exists locally on X_0. Returning to the step-by-step construction, replace Y by S. Let S_0 be a closed subprescheme of S defined by a sheaf of ideals $\mathcal{I}$, no longer necessarily locally nilpotent, and let S_n be the closed subprescheme of S defined by $\mathcal{I}^{n+1}$. Suppose that a prescheme X_n, smooth over S_n, has been given. We seek an S_{n+1}-prescheme X_{n+1} reducing to X_n, that is, equipped with an isomorphism

$$X_n \xrightarrow{\sim} X_{n+1} \times_{S_{n+1}} S_n,$$

and smooth over S_{n+1}, or equivalently flat over S_{n+1} by II II.2.1.

As observed in Section III.4, such data amount to a sheaf of algebras $\mathcal{B}$ over $f^{-1}(\mathcal{O}_{S_{n+1}})$, where f is the continuous map underlying the structural morphism $X_n \to S_n$, together with an augmentation $\mathcal{B} \to \mathcal{O}_{X_n}$ compatible with

$$f^{-1}(\mathcal{O}_{S_{n+1}}) \longrightarrow f^{-1}(\mathcal{O}_{S_n}),$$

and satisfying the two conditions (a) and (b) already given there. We need only recall that they are local on the underlying topological space of X_n. Theorem III.4.1 gives a solution locally, unique locally up to a nonunique isomorphism. We first make this last point precise.

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PROPOSITION III.6.1. *Let X_{n+1} over S_{n+1} reduce to X_n over S_n. The sheaf, on the underlying topological space of X_n, equivalently of X_0, of S_{n+1}-automorphisms of X_{n+1} which induce the identity on X_n is canonically isomorphic, as a sheaf of groups, to*

$$\mathcal{G} = \mathfrak{g}_{X_0/S_0} \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_S).$$

Indeed, by Corollary III.5.4 and Lemma III.4.2, this automorphism sheaf is a torsor under $\mathcal{G}$. It has the distinguished section given by the identity automorphism of X_{n+1}, and hence identifies with $\mathcal{G}$ as a sheaf of sets. Compatibility with the group structures is immediate. It is also a special case of a more general compatibility, omitted here, between the torsor structures of Proposition III.5.1 and Corollary III.5.3 and the composition of morphisms.

In particular, the sheaf on X_0 of germs of automorphisms of X_{n+1}, with the structures just described, is commutative. Thus, if X'_{n+1} is another solution and is isomorphic to X_{n+1} over an open set $U \subseteq X_0$, the isomorphism

$$\mathcal{A}ut(X_{n+1})|_U \xrightarrow{\sim} \mathcal{A}ut(X'_{n+1})|_U$$

obtained by transport of structure through an isomorphism $X_{n+1}|_U \xrightarrow{\sim} X'_{n+1}|_U$ does not depend on the chosen isomorphism. Under the identifications of Proposition III.6.1, it is simply the identity of $\mathcal{G}$.

We deduce:

COROLLARY III.6.2. *Let X_{n+1} and X'_{n+1} be smooth over S_{n+1} and reduce to X_n. Then the sheaf, on the underlying space of X_0, of S_{n+1}-isomorphisms*

$$X_{n+1} \xrightarrow{\sim} X'_{n+1}$$

which induce the identity on X_n is naturally a torsor under $\mathcal{G}$.

This says precisely that X_{n+1} and X'_{n+1} are locally isomorphic and that the sheaf of germs of automorphisms of either one is $\mathcal{G}$.

By Theorem III.4.1, one can choose an open cover (U_i) of X_n, which may be assumed affine, and for each i a smooth scheme X^i over S_{n+1} reducing to U_i. For simplicity suppose that X_n is separated. The intersections $U_{ij} = U_i \cap U_j$ are then affine. Since $H^1(U_{ij}, \mathcal{G}) = 0$ for the quasi-coherent sheaf $\mathcal{G}$, Corollary III.6.2 shows that $X^i|_{U_{ij}}$ and $X^j|_{U_{ij}}$ are isomorphic. Choose an isomorphism

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$$f_{ji} \colon X^i|_{U_{ij}} \xrightarrow{\sim} X^j|_{U_{ij}}.$$

It is determined up to a section of $\mathcal{G}$ over U_{ij}. For every triple of indices put

$$f_{ji}^{(k)} = f_{ji}|_{U_{ijk}}, \qquad U_{ijk} = U_i \cap U_j \cap U_k.$$

If

$$f_{kj}^{(i)} f_{ji}^{(k)} = f_{ki}^{(j)}, \tag{1}$$

then the X^i glue through the f_{ji}, and hence define the desired solution $X = X_{n+1}$. More generally, a solution exists if the f_{ji} can be modified to

$$f'_{ji} = f_{ji} g_{ji}, \qquad g_{ji} \in \Gamma(U_{ij}, \mathcal{G}), \tag{2}$$

so that the f'_{ji} satisfy the preceding cocycle condition. This sufficient condition is also necessary. Indeed, any solution X is locally isomorphic to X^i; choosing isomorphisms

$$f_i \colon X|_{U_i} \xrightarrow{\sim} X^i$$

gives

$$f'_{ji} = (f_j|_{U_{ij}})(f_i|_{U_{ij}})^{-1} \colon X^i|_{U_{ij}} \xrightarrow{\sim} X^j|_{U_{ij}},$$

which satisfy the gluing condition.

Now set

$$f_{ijk} = (f^{(j)}_{ki})^{-1} f^{(i)}_{kj} f^{(k)}_{ji}. \tag{3}$$

This is an automorphism of $X^i|_{U_{ijk}}$, which we identify with a section of $\mathcal{G}$ by Proposition III.6.1. A direct formal calculation, left to the reader, shows that $f = (f_{ijk})$ is a Čech 2-cocycle of the open cover $\mathcal{U} = (U_i)$ with coefficients in $\mathcal{G}$. The same calculation shows that, after the modification (2), the gluing condition (1) for the f'_{ij} is equivalent to

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$$f = dg, \tag{4}$$

where $g = (g_{ij})$ is regarded as a Čech 1-cochain of $\mathcal{U}$ with coefficients in $\mathcal{G}$. Thus the necessary and sufficient condition for a solution to exist is the vanishing of the cohomology class in $H^2(\mathcal{U}, \mathcal{G})$ defined by the cocycle (3).

Since $\mathcal{U}$ is an affine open cover of the scheme X_0, $H^2(\mathcal{U}, \mathcal{G})$ identifies with $H^2(X_0, \mathcal{G})$. The resulting class in $H^2(X_0, \mathcal{G})$ is plainly independent of the chosen affine cover. We call it the obstruction class to extending X_n to a scheme X_{n+1} smooth over S_{n+1}.

Suppose this obstruction vanishes. The argument just given shows that every solution $X = X_{n+1}$ is isomorphic to one obtained by gluing with isomorphisms f'_{ji}, which may be assumed of the form (2). The gluing condition is then (4). The set of admissible g is a principal homogeneous space under the group $Z^1(\mathcal{U}, \mathcal{G})$ of Čech 1-cocycles. Moreover, two cochains g and g', with $dg = dg' = f$, define isomorphic solutions if and only if the cocycle $g - g'$ is of the form dh, where $h = (h_i) \in C^0(\mathcal{U}, \mathcal{G})$. We therefore obtain:

THEOREM III.6.3. *Let $(S, \mathcal{I}, X_n)$ be as above, and suppose that X_n is separated.*[5] *There is a canonically defined obstruction class in $H^2(X_0, \mathcal{G})$, where $\mathcal{G}$ is defined in Proposition III.6.1. Its vanishing is necessary and sufficient for the existence of a scheme X_{n+1}, smooth over S_{n+1}, which reduces to X_n. If the obstruction vanishes, the set of isomorphism classes, with the isomorphisms inducing the identity on X_n, of S_{n+1}-preschemes X_{n+1} reducing to X_n is naturally a principal homogeneous space under $H^1(X_0, \mathcal{G})$.*

REMARKS III.6.4. From Proposition III.6.1 onward, the arguments are purely formal and are better expressed in the language of local categories, or even general fibered categories. In that general setting, one obtains an obstruction class to the existence of a global object of a category in which objects exist locally, any two objects are locally isomorphic, and the automorphism group of every object is commutative. As a special case it contains the second connecting homomorphism in an exact sequence of sheaves of groups which need not be commutative, studied for example by Grothendieck in Kansas or Tôhoku. The elementary cocycle calculation given here should therefore be regarded as a makeshift necessitated by the absence of a satisfactory reference.

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[5] This hypothesis is in fact unnecessary, and the cocycle calculations above can be avoided. See J. Giraud, *Cohomologie non abélienne*, forthcoming from Springer-Verlag in 1971, and compare Remarks III.6.4.

III.6.5. In Theorem III.6.3, the principal homogeneous space under $H^1(X_0, \mathcal{G})$ generally has no distinguished element. After localizing on S, this is reflected by a torsor on S_0 with structural group $R^1f_*(\mathcal{G})$, which need not be trivial and thus defines a cohomology class in

$$H^1(S_0, R^1f_*(\mathcal{G}))$$

which need not vanish. This is the situation when the class $d \in H^2(X_0, \mathcal{G})$, although not zero, vanishes locally over S; equivalently, it defines the zero section of $R^2f_*(\mathcal{G})$, or the zero element of $H^0(S_0, R^2f_*(\mathcal{G}))$.

III.6.6. At present almost nothing is known about the general algebraic mechanism of the cohomology classes introduced in this section and the preceding one, and nothing precise can be said even in the simplest special cases, such as abelian schemes over Artinian rings. [6] It is to be hoped that someone will work out this particularly interesting question in detail. It is closely connected, in particular, with the moduli theory of algebraic structures.

COROLLARY III.6.7. *If $H^2(X_0, \mathcal{G}) = 0$, then an X_{n+1} exists; it is unique up to isomorphism if, in addition, $H^1(X_0, \mathcal{G}) = 0$.*

Proceeding step by step, and recalling that an affine scheme is acyclic for a quasi-coherent sheaf, we obtain:

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COROLLARY III.6.8. *Under the hypotheses of Theorem III.4.1, if X_0 is affine, then there exists a prescheme X, smooth over Y, reducing to X_0; it is unique up to a nonunique isomorphism.*

Notice that the direct proof of Theorem III.4.1 could not have given this result.

COROLLARY III.6.9. *Under the hypotheses of Theorem III.6.3, suppose that S is affine with ring A, that $\mathcal{I}$ is defined by an ideal $I \subseteq A$, and that*

$$\mathrm{gr}^n_{\mathcal{I}}(\mathcal{O}_S) = \mathcal{I}^n/\mathcal{I}^{n+1}$$

is locally free for every n. Then

$$H^i(X_0, \mathcal{G}) = H^i(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^{n+1}(A), \qquad \mathcal{G}_0 = \mathfrak{g}_{X_0/S_0}.$$

Thus the obstruction class to extending X_n lies in

$$H^2(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^{n+1}(A).$$

If it vanishes, the set of isomorphism classes of solutions is a principal homogeneous space under

$$H^1(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^{n+1}(A).$$

In particular:

COROLLARY III.6.10. *Under the hypotheses of Corollary III.6.9, suppose that*

$$H^2(X_0, \mathfrak{g}_{X_0/S_0}) = 0.$$

Then there exists an I-adic formal scheme $\mathfrak{X}$ over the $\mathcal{I}$-adic formal completion $\widehat{S}$ of S, smooth over $\widehat{S}$ in the sense that each $\mathfrak{X}_p$ is smooth over S_p, and reducing to X_n; that is, equipped with an isomorphism

$$X_n \xrightarrow{\sim} \mathfrak{X} \times_S S_n.$$

If, in addition,

$$H^1(X_0, \mathfrak{g}_{X_0/S_0}) = 0,$$

then such a formal scheme $\mathfrak{X}$ is unique up to isomorphism.

[6] It is now known that the obstruction always vanishes in this case. [Added in 2003 (MR): see F. Oort, "Finite group schemes, local moduli for abelian varieties and lifting problems," *Algebraic Geometry, Oslo 1970*, Wolters-Noordhoff, 1972, pp. 223–254.]

Indeed, one constructs $X_{n+1}, X_{n+2}, \ldots$ successively and passes to the limit to obtain $\mathfrak{X}$. The uniqueness assertion was already proved in the preceding paragraph.

III.7. Application to the construction of formal and ordinary schemes smooth over a complete local ring A

The results of the preceding section sometimes prove the existence of an $\mathfrak{m}$-adic formal scheme over such a ring, reducing to a given smooth scheme X_0 over k. We distinguish two cases. French source p. 83

(a) Equal characteristics. This includes, in particular, the case where k has characteristic zero. There is then a coefficient field in A, that is, a subfield $k' \subseteq A$ such that $A \to k$ induces an isomorphism $k' \xrightarrow{\sim} k$. In fact, there is then an ordinary scheme smooth over A reducing to X_0, namely

$$X = X_0 \otimes_k A,$$

where A is regarded as a k-algebra through the section $k \to k' \to A$ defined by k'.

This construction is not natural. Already for the dual-number algebra $A = k[t]/(t^2)$, a different lifting homomorphism $k \to A$, defined in this case by an absolute derivation of k into itself, gives an A-scheme X' which in general is not isomorphic to X when

$$H^1(X_0, \mathfrak{g}_{X_0/k}) \neq 0.$$

It would be interesting, when k has characteristic zero, or is imperfect of characteristic $p > 0$, to determine which smooth A-schemes arise in this way and when two homomorphisms $k \to A$ give isomorphic A-schemes.

Nevertheless, the existence of k' implies that the first obstruction to lifting X_0, which lies in

$$H^2(X_0, \mathfrak{g}_{X_0/k}) \otimes_k \mathfrak{m}/\mathfrak{m}^2,$$

must vanish. After X_0 has been lifted to an X_1 smooth over $A/\mathfrak{m}^2$, however, the new obstruction to constructing X_2 need not vanish. It depends on a variable element in a certain principal homogeneous space under

$$H^1(X_0, \mathcal{G}_0) \otimes \mathfrak{m}/\mathfrak{m}^2$$

and lies in

$$H^2(X_0, \mathcal{G}_0) \otimes \mathfrak{m}^2/\mathfrak{m}^3.$$

The situation deserves a detailed study. [7]

(b) Unequal characteristics. In this case essentially nothing is known, except when by good fortune $H^2(X_0, \mathfrak{g}_{X_0/k}) = 0$, in which case an $\mathfrak{m}$-adic formal scheme smooth over A and reducing to X_0 can be constructed. Even for $A = \mathbb{Z}/p^2\mathbb{Z}$ and X_0 an abelian scheme of dimension 2, it was not known whether X_0 could be lifted to an $X = X_1$ smooth over A. [8] Nor was there an example proved not to arise from an ordinary scheme smooth over A; one expected such examples among projective surfaces. [9]

By Cohen's theorem there is a Cohen p-ring B with residue field k and a homomorphism $B \to A$ inducing the identity on residue fields. The strongest lifting result is therefore obtained by taking A itself to be a Cohen p-ring: if an ordinary or formal solution exists over such a ring, one exists over every complete local ring with residue field k. For a French source p. 84

[7] It is presumably described by the Kodaira–Spencer bracket; see the Cartan Seminar 1960/61, Exposé 4.

[8] This has now been proved; see the note to Section III.6.6.

[9] Such an example was subsequently constructed by J.-P. Serre, *Proc. Nat. Acad. Sci. USA* **47** (1961), no. 1, pp. 108–109, at least in certain dimensions. D. Mumford gave an unpublished example with an algebraic surface.

Cohen p-ring, $\mathfrak{m}/\mathfrak{m}^2$ identifies canonically with k. Hence, for every smooth scheme X_0 over a field k of characteristic $p > 0$, there is a cohomology class in

$$H^2(X_0, \mathfrak{g}_{X_0/k})$$

which is the first obstruction to lifting X_0 to a smooth scheme over a Cohen p-ring. It was not known whether this class could be nonzero.[10]

Even if the X_n reducing to X_0 can be constructed successively, this generally yields only a formal scheme $\mathfrak{X}$, smooth over A, reducing to X_0. When X_0 is proper over k, one must still decide whether $\mathfrak{X}$ is algebraizable, so as to obtain an ordinary scheme proper and smooth over A reducing to X_0. The only known criterion, stated in the Bourbaki Seminar and in EGA III, 4.7.1, is the following. Suppose that $\mathfrak{X}$ is proper over A, and that $\mathcal{L}$ is an invertible sheaf on $\mathfrak{X}$ whose restriction $\mathcal{L}_0$ to X_0 is ample, meaning that some positive tensor power $\mathcal{L}_0^{\otimes n}$ comes from a projective immersion of X_0. Then there exist a scheme X projective over A and an ample invertible sheaf on X whose $\mathfrak{m}$-adic completion is $(\mathfrak{X}, \mathcal{L})$.

This reduces us to extending a locally free sheaf $\mathcal{E}_0$ on X_0, chosen invertible and ample for the present purpose, to a locally free sheaf $\mathcal{E}$ on $\mathfrak{X}$. One constructs locally free sheaves $\mathcal{E}_n$ on the X_n successively. The discussion is entirely analogous to Section III.6; compare Remarks III.6.4. The essential role is played by the sheaf of automorphisms of an $\mathcal{E}_{n+1}$ which induce the identity on $\mathcal{E}_n$. This sheaf identifies with

$$(*) \qquad \mathcal{G} = \mathcal{H}om_{\mathcal{O}_{X_0}}\left(\mathcal{E}_0, \mathcal{E}_0 \otimes \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_X)\right) = \mathcal{E}nd_{\mathcal{O}_{X_0}}(\mathcal{E}_0) \otimes \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_X),$$

which is again a sheaf of commutative groups. We obtain:

Proposition III.7.1. *Let S be a prescheme equipped with a quasi-coherent sheaf of ideals $\mathcal{I}$, let X be a prescheme over S, let S_n be the closed subprescheme of S defined by $\mathcal{I}^{n+1}$, and put $X_n = X \times_S S_n$ for every n. Let $\mathcal{E}_n$ be a locally free sheaf on X_n. The obstruction to extending it to a locally free sheaf $\mathcal{E}_{n+1}$ on X_{n+1} is a canonical class in $H^2(X_0, \mathcal{G})$, where $\mathcal{G}$ is the quasi-coherent sheaf in (*). Its vanishing is necessary and sufficient for such an extension to exist. If it vanishes, the set of isomorphism classes, with isomorphisms inducing the identity on $\mathcal{E}_n$, of extensions $\mathcal{E}_{n+1}$ is a principal homogeneous space under $H^1(X_0, \mathcal{G})$.*

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This proposition has the usual corollaries. We record only that if X is flat over S, then

$$\mathcal{G} = \mathcal{E}nd_{\mathcal{O}_{X_0}}(\mathcal{E}_0) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_S).$$

Consequently, if S is affine with ring A and the $\mathcal{I}^n/\mathcal{I}^{n+1}$ are locally free, the condition

$$H^2(X_0, \mathcal{G}_0) = 0, \qquad \mathcal{G}_0 = \mathcal{E}nd_{\mathcal{O}_{X_0}}(\mathcal{E}_0),$$

is sufficient for $\mathcal{E}_{n+1}$ to exist, and hence, step by step, for all successive extensions $\mathcal{E}_m$, $m = n, n+1, \ldots$, to exist.

Returning to the original situation, we obtain:

Proposition III.7.2. *Let A be a complete local ring, and let $\mathfrak{X}$ be a formal scheme proper and flat over A. Suppose that X_0 is projective and*

$$H^2(X_0, \mathcal{O}_{X_0}) = 0.$$

Then there exists a scheme X projective over A whose $\mathfrak{m}$-adic formal completion is isomorphic to $\mathfrak{X}$.

Combining this with Corollary III.6.10, we obtain:

[10] It can be nonzero, as noted in Footnote 9.

THEOREM III.7.3. *Let A be a complete local ring with residue field k, and let X_0 be a smooth projective scheme over k such that*

$$H^2(X_0, \mathfrak{g}_{X_0/k}) = H^2(X_0, \mathcal{O}_{X_0}) = 0.$$

Then there exists a smooth projective scheme X over A reducing to X_0.

More generally, if X_n is smooth over $A_n = A/\mathfrak{m}^{n+1}$ and reduces to X_0, then there exist a scheme X smooth and proper over A and an isomorphism

$$X \otimes_A A_n \xrightarrow{\sim} X_n.$$

COROLLARY III.7.4. *Every smooth proper curve over k arises by reduction from a smooth proper curve over A.*

This result, together with the existence theorem for sheaves in formal geometry, will be the essential tool for studying the fundamental group of X_0 by transcendental methods.

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CHAPTER IV

Flat morphisms

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We give here chiefly the properties of flatness used in the preceding exposés. A more detailed study will appear in Chapter IV of the forthcoming *Éléments de Géométrie Algébrique*[1], where the following situation is treated systematically: X is locally of finite type over a locally Noetherian prescheme Y, and $\mathcal{F}$ is a coherent sheaf on X, flat over Y; one seeks relations among properties of Y, properties of $\mathcal{F}$, and properties of the coherent sheaves induced by $\mathcal{F}$ on the fibers of $X \to Y$, especially with respect to dimension, cohomological dimension, and depth. In particular, this yields a systematic way of obtaining theorems of Seidenberg or Bertini type for hyperplane sections.

The essential result for applying flatness methods in this setting is the following, to be proved below: if Y is integral, X is of finite type over Y, and $\mathcal{F}$ is coherent on X, then there exists a nonempty open subset $U \subseteq Y$ such that $\mathcal{F}$ is flat over Y at every point of X lying over U.

A second, and perhaps still more important, way in which flatness enters algebraic geometry is through descent theory. See, for example, Grothendieck's two Bourbaki Seminar talks on the subject, and, for a more detailed account, Exposés VIII and IX below. Flatness thus appears to be one of the central technical notions of algebraic geometry.

Recall that Serre introduced the notions of flatness and faithful flatness in GAGA. The material of Sections IV.1 and IV.2 below may also be found in Bourbaki's *Algèbre commutative*, which, as its title indicates, does not restrict itself to commutative base rings.[2]

Unlike the preceding exposés, we do not assume that the rings under consideration are necessarily Noetherian.

IV.1. Elementary facts about flat modules

An A-module M is called *flat*, or A-flat when the ring must be specified, if the functor

$$T_M : N \longmapsto M \otimes_A N,$$

which is always right exact, is exact; equivalently, if it takes monomorphisms to monomorphisms. This is equivalent to the vanishing of the first left-derived functor, or of all the left-derived functors: French source p. 88

$$\mathrm{Tor}_1^A(M, N) = 0 \qquad \text{for every } N,$$

or, respectively,

$$\mathrm{Tor}_i^A(M, N) = 0 \qquad \text{for every } i > 0 \text{ and every } N.$$

Since the Tor_i commute with filtered direct limits, it is enough to verify these conditions for finitely generated N. Using a filtration of N with cyclic quotients, it is enough even to require

$$\mathrm{Tor}_1^A(M, N) = 0$$

[1] See EGA IV, 11 and 12.

[2] N. Bourbaki, *Algèbre commutative*, Chapter I (Modules plats), Act. Sci. Ind. 1290, Hermann, Paris, 1961.

when N is cyclic, hence of the form A/I for an ideal $I \subseteq A$. Moreover,

$$\mathrm{Tor}_1^A(M, A/I) = 0 \quad \Longleftrightarrow \quad I \otimes_A M \longrightarrow M = A \otimes_A M \text{ is injective.}$$

This follows from the exact Tor-sequence, since $\mathrm{Tor}_1^A(M, A) = 0$. Thus M is flat if and only if, for every ideal I, the natural homomorphism

$$I \otimes_A M \longrightarrow IM$$

is an isomorphism. It is enough to verify this for finitely generated ideals I; a fortiori it is enough to verify that $M \otimes_A -$ is exact on finitely generated modules.

As for every exact functor T, if $T(N')$ is identified with a subobject of $T(N)$ whenever $N' \subseteq N$, then

$$T(N' \cap N'') = T(N') \cap T(N''), \qquad T(N' + N'') = T(N') + T(N'')$$

for any two subobjects N' and N'' of N.

A direct sum of flat modules is flat, and a direct summand of a flat module is flat. Since A is flat over itself, every free module, and hence every projective module, is flat. The tensor product of two flat modules is flat. If M is flat over A, then $M \otimes_A B$ is flat over B after every base change $A \to B$. These assertions follow from associativity of the tensor product and from the fact that a composite of exact functors is exact. Likewise, if M is flat over B and B is flat over A, then M is flat over A.

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The exact Tor-sequence, together with the symmetry of Tor, gives:

PROPOSITION IV.1.1. *Let*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be an exact sequence of A-modules, with M'' flat. Then:

(i) *the sequence remains exact after tensoring with any A-module N;*
(ii) *M is flat if and only if M' is flat.*

Thus, from the viewpoint of tensor products, flat modules are as well behaved as free or projective modules. In particular, the exact sequence in Proposition IV.1.1 behaves as well as if it split.

Let S be a multiplicatively closed subset of A. Then $S^{-1}A$ is flat over A, since

$$S^{-1}A \otimes_A N = S^{-1}N$$

is an exact functor of N. If M is A-flat, then $S^{-1}M = S^{-1}A \otimes_A M$ is $S^{-1}A$-flat. Conversely, if $M \to S^{-1}M$ is an isomorphism, equivalently if every $s \in S$ acts bijectively on M, then $S^{-1}A$-flatness of M implies A-flatness by transitivity.

More generally, the study of the flatness over Y of a quasi-coherent sheaf $\mathcal{F}$ on a prescheme $X \to Y$ leads to the following two-ring situation.

PROPOSITION IV.1.2. *Let $A \to B$ be a ring homomorphism, let M be a B-module, and let T be a multiplicatively closed subset of B.*

(i) *If M is A-flat, then $T^{-1}M$ is A-flat. Consequently, it is also $S^{-1}A$-flat for every multiplicatively closed subset $S \subseteq A$ whose image lies in T.*
(ii) *Conversely, suppose that for every maximal ideal $\mathfrak{n}$ of B, $M_\mathfrak{n}$ is flat over A, or equivalently over $A_\mathfrak{m}$, where $\mathfrak{m}$ is the inverse image of $\mathfrak{n}$ in A. Then M is A-flat.*

Indeed, functorially in the A-module N,

$$T^{-1}M \otimes_A N = T^{-1}(M \otimes_A N).$$

Both sides are naturally isomorphic, by associativity, to

$$T^{-1}B \otimes_B M \otimes_B (N \otimes_A B).$$

Thus exactness of $M \otimes_A N$ in N implies that of $T^{-1}M \otimes_A N$, proving (i). It also proves (ii), since exactness of a sequence of B-modules may be checked after localization at every maximal ideal of B.

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PROPOSITION IV.1.3. (i) *Let M be a flat A-module. If $x \in A$ is not a zero divisor in A, then it is not a zero divisor on M. In particular, if A is an integral domain, then M is torsion-free.*

(ii) *Suppose that A is an integral domain and that $A_{\mathfrak{m}}$ is a principal ideal domain for every maximal ideal $\mathfrak{m}$ of A, for example when A is a Dedekind domain or a principal ideal domain. An A-module M is flat if and only if it is torsion-free.*

For (i), multiplication by x on M is obtained by tensoring multiplication by x on A with M. For (ii), Proposition IV.1.2(ii) reduces us to the case where A is a principal ideal domain. If M is torsion-free, then for every ideal $I = (x)$, tensoring the injection $I \to A$ with M remains injective, since x is not a zero divisor on M. This is precisely the flatness criterion above.

IV.2. Faithfully flat modules

A functor F from one category to another is called *faithful* if, for every pair of objects X, Y, the map

$$\mathrm{Hom}(X, Y) \longrightarrow \mathrm{Hom}(F(X), F(Y))$$

is injective. For an additive functor between additive categories, this is equivalent to saying that $F(u) = 0$ implies $u = 0$, and it implies that $F(X) = 0$ only if $X = 0$.

For a functor between abelian categories to be faithful and exact, it is necessary and sufficient that a sequence

$$M' \longrightarrow M \longrightarrow M''$$

be exact if and only if its image

$$F(M') \longrightarrow F(M) \longrightarrow F(M'')$$

is exact. Equivalently, F must be exact and $F(X) = 0$ must imply $X = 0$.

Suppose that a family (M_i) of nonzero objects has the property that every nonzero object has a subobject admitting a quotient isomorphic to some M_i. Then F is faithful and exact if and only if F is exact and $F(M_i) \neq 0$ for every i. In the category of modules over a ring A, one may take the family of modules $A/\mathfrak{m}$, where $\mathfrak{m}$ runs through the maximal ideals of A. Indeed, every nonzero module has a nonzero cyclic submodule, isomorphic to A/I for some proper ideal I, and by Krull's theorem this has a quotient $A/\mathfrak{m}$. Hence:

PROPOSITION IV.2.1. *For an A-module M, the following conditions are equivalent:*

(i) *The functor $M \otimes_A -$ is faithful and exact.*

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(i bis) *M is flat, and $M \otimes_A N = 0$ implies $N = 0$.*

(i ter) *M is flat, and*

$$M \otimes_A A/\mathfrak{m} \neq 0$$

for every maximal ideal $\mathfrak{m}$ of A.

(ii) *For every sequence of homomorphisms $N' \to N \to N''$, the sequence obtained by tensoring with M is exact if and only if the original sequence is exact.*

In this case M is called a *faithfully flat A-module*. In particular, if M is faithfully flat, a homomorphism $N \to N'$ is a monomorphism, epimorphism, or isomorphism if and only if the homomorphism obtained by tensoring with M has the same property. A faithfully flat module is faithful, since multiplication by f on M is obtained by tensoring multiplication by f on A with M.

The usual transitivity properties follow as in Section IV.1. The tensor product of two faithfully flat modules is faithfully flat. If M is faithfully flat over A, then $M \otimes_A B$ is

faithfully flat over B after every base extension $A \to B$. If B is a faithfully flat A-algebra and M is faithfully flat as a B-module, then M is faithfully flat over A.

COROLLARY IV.2.2. *Let $A \to B$ be a local homomorphism of local rings, and let M be a finitely generated B-module. Then M is faithfully flat over A if and only if it is flat over A and nonzero.*

This follows from criterion (i ter) and Nakayama's lemma. In particular, B is flat over A if and only if it is faithfully flat over A.

PROPOSITION IV.2.3. *Let $A \to B$ be a ring homomorphism, and let M be a B-module which is faithfully flat over A. For every prime ideal $\mathfrak{p}$ of A, there exists a prime ideal $\mathfrak{q}$ of B whose contraction to A is $\mathfrak{p}$.*

After quotienting by $\mathfrak{p}$, we reduce to $\mathfrak{p} = 0$. Localizing at the prime ideal 0, we reduce to the case where A is a field. Since M is faithfully flat over A, it is nonzero, and hence $B \neq 0$. The ring B has a prime ideal, which necessarily contracts to the unique prime ideal of A. Geometrically, the existence on $X = \mathrm{Spec}(B)$ of a quasi-coherent sheaf which is faithfully flat relative to A implies that

$$X \longrightarrow Y = \mathrm{Spec}(A)$$

is surjective.

COROLLARY IV.2.4. *Suppose that M is flat over A, finitely generated over B, and has full support:*

$$\mathrm{Supp}\, M = \mathrm{Spec}(B).$$

For every prime ideal $\mathfrak{p}$ of A, each prime ideal $\mathfrak{q}$ of B minimal among those containing $\mathfrak{p}B$ contracts to $\mathfrak{p}$.

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All the hypotheses are preserved after quotienting, so once again we reduce to $\mathfrak{p} = 0$, with A an integral domain. We are reduced to:

COROLLARY IV.2.5. *Let M be as above. Every minimal prime $\mathfrak{q}$ of B contracts to a minimal prime $\mathfrak{p}$ of A.*

Indeed, after localizing at $\mathfrak{p}$ and $\mathfrak{q}$, it is enough to prove the following. Let A and B be local rings, let $A \to B$ be local, let M be a nonzero B-module flat over A, and suppose that B has dimension zero. Then A has dimension zero. By Corollary IV.2.2 and Proposition IV.2.3, every prime ideal of A is induced by a prime ideal of B, hence by the maximal ideal of B, and is therefore the maximal ideal of A.

Geometrically, Corollary IV.2.5 says that every irreducible component of $X = \mathrm{Spec}(B)$ dominates an irreducible component of $Y = \mathrm{Spec}(A)$, provided there is a finitely generated quasi-coherent sheaf on X, with support X, which is flat relative to Y.

Notice that Corollary IV.2.4 did not require M to be faithfully flat over A. In that case, however, there is no guarantee that a prime ideal containing $\mathfrak{p}B$ exists.

PROPOSITION IV.2.6. *Let $i\colon A \to B$ be a ring homomorphism. The following conditions are equivalent:*

(i) *B is faithfully flat as an A-module.*
(ii) *B is flat over A, and $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is surjective.*
(ii bis) *B is flat over A, and every maximal ideal of A is the contraction of a prime ideal of B.*
(iii) *i is injective and $\mathrm{Coker}\, i$ is a flat A-module.*
(iv) *The functor*

$$M \longmapsto M_{(B)} = M \otimes_A B$$

is exact, and the canonical natural homomorphism $M \to M_{(B)}$ is injective.

(iv bis) *For every ideal $I \subseteq A$, the map*

$$I \otimes_A B \longrightarrow IB$$

is an isomorphism, and the inverse image of IB in A is I.

The implication (i) ⇒ (ii) follows from Proposition IV.2.3; (ii) ⇒ (ii bis) is immediate; and (ii bis) ⇒ (i) follows from criterion (i ter) of Proposition IV.2.1. Proposition IV.1.1 gives (iii) ⇒ (iv), and (iv) ⇒ (iv bis) is immediate. Condition (iv bis) implies (i) by the ideal criterion for flatness in Section IV.1 and criterion (i ter) of Proposition IV.2.1. The implication (iv) ⇒ (iii) follows from the converse form of Proposition IV.1.1.

Finally, (i) implies (iv) as follows. If N is the kernel of

$$M \longrightarrow M \otimes_A B = T(M),$$

then exactness of T makes the map $N \to T(N)$ zero. Hence $T(N) = N \otimes_A B = 0$, and faithful flatness gives $N = 0$.

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IV.3. Relations with completion

Let A be a Noetherian ring, let I be an ideal of A, and let $\widehat{A}$ be the separated completion of A for the I-adic topology. For every A-module M, let $\widehat{M}$ be its I-adic completion. It is an $\widehat{A}$-module, and hence there is a canonical homomorphism

$$M \otimes_A \widehat{A} \longrightarrow \widehat{M}.$$

On finitely generated modules, the functor $M \mapsto \widehat{M}$ is exact. This follows readily from Krull's theorem: if $N \subseteq M$, the I-adic topology of N is the topology induced from M. Since $M \mapsto M \otimes_A \widehat{A}$ is right exact, a finite free presentation $L \to L' \to M$ shows that the displayed natural homomorphism is an isomorphism, completion also being right exact. Consequently $M \mapsto M \otimes_A \widehat{A}$ is exact on finitely generated modules. Hence:

Proposition IV.3.1. *Let A be a Noetherian ring and I an ideal of A. The separated I-adic completion $\widehat{A}$ of A is flat over A.*

Corollary IV.3.2. *The ring $\widehat{A}$ is faithfully flat over A if and only if I is contained in the Jacobson radical of A.*

Apply criterion (i ter) of Proposition IV.2.1.

These statements summarize everything that linear algebra alone says about the relation between A and $\widehat{A}$. Corollary IV.3.2 is used especially when A is a Noetherian local ring and I is contained in its maximal ideal $\mathfrak{m}$, most often with $I = \mathfrak{m}$.

IV.4. Relations with free modules

Proposition IV.4.1. *Let A be a ring, I an ideal of A, and M an A-module. Suppose that one of the following hypotheses holds:*

(a) *I is nilpotent;*
(b) *A is Noetherian, I is contained in the Jacobson radical of A, and M is finitely generated.*

Then M is free over A if and only if $M \otimes_A A/I$ is free over A/I and

$$\operatorname{Tor}_1^A(M, A/I) = 0.$$

Necessity is clear; we prove sufficiency. Choose elements (e_i) of M whose images form a basis of

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$$M \otimes_A A/I = M/IM$$

over A/I; the family is finite in case (b). Let L be the free A-module on the same index set. The resulting homomorphism $L \to M$ becomes an isomorphism after tensoring

with A/I. If Q is its cokernel, then $Q/IQ = 0$, and Nakayama's lemma, valid under either hypothesis, gives $Q = 0$. Thus $L \to M$ is surjective. Let R be its kernel:

$$0 \longrightarrow R \longrightarrow L \longrightarrow M \longrightarrow 0.$$

Since $\mathrm{Tor}_1^A(M, A/I) = 0$, tensoring with A/I gives an exact sequence

$$0 \longrightarrow R/IR \longrightarrow L/IL \longrightarrow M/IM \longrightarrow 0.$$

Hence $R/IR = 0$, and Nakayama again gives $R = 0$. In case (b), note that R is finitely generated because A is Noetherian.

COROLLARY IV.4.2. *The condition* $\mathrm{Tor}_1^A(M, A/I) = 0$ *may be replaced by the condition that the canonical surjection*

$$(*) \qquad \mathrm{gr}_I^0(M) \otimes_{A/I} \mathrm{gr}_I(A) \longrightarrow \mathrm{gr}_I(M)$$

is an isomorphism.

The condition plainly holds when M is free. Conversely, suppose that $M \otimes_A A/I$ is free over A/I and that (*) is an isomorphism. Construct $L \to M$ as in the proof above. The hypothesis shows that this homomorphism induces an isomorphism on associated graded modules. Its kernel is therefore contained in $\bigcap_n I^n L$, which is zero: trivially in case (a), and by the standard separation theorem in case (b).

COROLLARY IV.4.3. *Suppose that A/I is a field. The following conditions on M are equivalent:*

(i) *M is free;*
(ii) *M is projective;*
(iii) *M is flat;*
(iv) $\mathrm{Tor}_1^A(M, A/I) = 0$*;*
(v) *the canonical homomorphism* (*) *is an isomorphism.*

Indeed, $M \otimes_A A/I$ is automatically free in this case. The corollary applies in either of the following situations:

(a) M is an arbitrary module over a local ring A whose maximal ideal I is nilpotent, for example an Artinian local ring;
(b) M is a finitely generated module over a Noetherian local ring A.

For reference, we recall:

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COROLLARY IV.4.4. *Let A be a Noetherian local integral domain with maximal ideal $\mathfrak{m} = I$, residue field $k = A/I$, and fraction field K. Let M be a finitely generated A-module. Conditions* (i)–(v) *above are also equivalent to:*

(vi) *The vector spaces $M \otimes_A K$ and $M \otimes_A k$ have the same dimension; equivalently, the rank of M over A equals the minimum number of generators of the A-module M.*

The proof is immediate. The reader may generalize it to the case where A is only assumed reduced: one must then require the ranks of M at all minimal prime ideals of A to equal the dimension of the vector space $M \otimes_A k$.

IV.5. Local criteria for flatness

PROPOSITION IV.5.1. *Let A be a ring equipped with an ideal I, and let M be an A-module. Suppose that*

$$\mathrm{Tor}_1^A(M, A/I^n) = 0 \qquad \textit{for every } n > 0.$$

Then the canonical surjective homomorphism

$$(*) \qquad \mathrm{gr}_I^0(M) \otimes_{A/I} \mathrm{gr}_I(A) \longrightarrow \mathrm{gr}_I(M)$$

is an isomorphism. The converse holds if I is nilpotent.

The hypothesis means that the homomorphisms

$$I^n \otimes_A M \longrightarrow I^n M$$

are isomorphisms. It follows immediately that the homomorphisms

$$(I^n/I^{n+1}) \otimes_A M \longrightarrow I^n M/I^{n+1}M$$

are isomorphisms. Conversely, suppose this last condition holds and that I is nilpotent. We prove that $\operatorname{Tor}_1^A(M, A/I^n) = 0$ for every n. This is true for large n; proceed by descending induction, assuming it for $n+1$. There is a commutative diagram

$$\begin{array}{ccccccccc} & & M \otimes_A I^{n+1} & \longrightarrow & M \otimes_A I^n & \longrightarrow & M \otimes_A (I^n/I^{n+1}) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & MI^{n+1} & \longrightarrow & MI^n & \longrightarrow & MI^n/MI^{n+1} & \longrightarrow & 0, \end{array}$$

whose rows are exact. By hypothesis, the last vertical arrow is an isomorphism, while the induction hypothesis says that the first vertical arrow is an isomorphism. The middle vertical arrow is therefore also an isomorphism, completing the proof. French source p. 96

The following proposition was isolated by Serre during the seminar; it allows substantial simplifications in this section.

PROPOSITION IV.5.2. *Let $A \to B$ be a ring homomorphism and let M be an A-module. The following conditions are equivalent:*

(i) *for every B-module N, $\operatorname{Tor}_1^A(M, N) = 0$;*
(ii) *$\operatorname{Tor}_1^A(M, B) = 0$, and $M_{(B)} = M \otimes_A B$ is flat over B.*

There is a functorial isomorphism

$$M \otimes_A N = (M \otimes_A B) \otimes_B N.$$

Regarding the left-hand side as a functor of M, this expresses it as the composite of the two functors

$$M \longmapsto M \otimes_A B, \qquad P \longmapsto P \otimes_B N.$$

The first carries free A-modules to free B-modules, and hence projective modules to projective modules. The spectral sequence for composite functors therefore gives

$$E^2_{p,q} = \operatorname{Tor}_p^B(\operatorname{Tor}_q^A(M, B), N) \Longrightarrow \operatorname{Tor}_{p+q}^A(M, N),$$

and its low-degree terms give an exact sequence

$$\operatorname{Tor}_1^A(M, B) \otimes_B N \longrightarrow \operatorname{Tor}_1^A(M, N) \longrightarrow \operatorname{Tor}_1^B(M \otimes_A B, N) \longrightarrow 0.$$

If (i) holds, then $\operatorname{Tor}_1^A(M, B) = 0$, and the exact sequence shows that $\operatorname{Tor}_1^B(M \otimes_A B, N) = 0$ for every N. Thus $M \otimes_A B$ is flat over B, proving (ii). Conversely, if (ii) holds, the two terms surrounding $\operatorname{Tor}_1^A(M, N)$ vanish, proving (i).

COROLLARY IV.5.3. *Suppose that $B = A/I$. Then the preceding conditions are also equivalent to:*

(iii) *$\operatorname{Tor}_1^A(M, N) = 0$ for every A-module N annihilated by a power of I.*

Indeed, condition (i) says this when N is annihilated by I. Condition (iii) follows by applying this assertion successively to the modules $I^n N/I^{n+1}N$.

COROLLARY IV.5.4. *Under the hypotheses of Corollary IV.5.3, the conditions under consideration imply that the functorial homomorphism* French source p. 97

$$(*) \qquad \operatorname{gr}_I^0(M) \otimes_{A/I} \operatorname{gr}_I(A) \longrightarrow \operatorname{gr}_I(M)$$

is an isomorphism, and that $M \otimes_A A/I$ is flat over A/I.

Apply condition (iii) and Proposition IV.5.1. Using the converse in Proposition IV.5.1 when I is nilpotent gives:

COROLLARY IV.5.5. *Let A be a ring equipped with a nilpotent ideal I, and let M be an A-module. The following conditions are equivalent:*

(i) *M is flat over A;*
(ii) *$M \otimes_A A/I$ is flat over A/I, and $\mathrm{Tor}_1^A(M, A/I) = 0$;*
(iii) *$M \otimes_A A/I$ is flat over A/I, and the canonical homomorphism (*) on associated graded modules is an isomorphism.*

These are, respectively, condition (iii) and condition (ii) above, and the conditions of Corollary IV.5.4.

Now drop the assumption that I is nilpotent. A priori, in Corollary IV.5.5 we retain only the implications

$$\text{(i)} \Rightarrow \text{(ii)} \Rightarrow \text{(iii)}.$$

Since condition (iii) remains stable after quotienting by a power of I, Corollary IV.5.5 shows that it implies:

(iv) *for every $n > 0$, $M \otimes_A A/I^n$ is flat over A/I^n.*

We seek conditions under which (iv) implies (i), that is, under which M is flat over A. It is enough that A be Noetherian and that M satisfy the following finiteness condition: *for every finitely generated A-module N, the module $M \otimes_A N$ is separated for the I-adic topology.* It would be enough to check this when N is a finitely generated ideal of A.

Indeed, suppose these conditions hold, and let $N' \to N$ be an injective homomorphism of finitely generated modules. We prove that

$$M \otimes_A N' \longrightarrow M \otimes_A N$$

is injective. It is enough to show that its kernel is contained in every

$$I^n(M \otimes_A N') = \mathrm{Im}(M \otimes_A I^n N' \longrightarrow M \otimes_A N'),$$

or, equivalently, in every

$$\mathrm{Im}(M \otimes_A V_n' \longrightarrow M \otimes_A N') = \mathrm{Ker}(M \otimes_A N' \longrightarrow M \otimes_A (N'/V_n')),$$

where (V_n') runs through a countable fundamental system of neighborhoods of zero in N', equipped with its I-adic topology. By Krull's theorem, the I-adic topology of N' is induced by that of N, so we may take

$$V_n' = N' \cap I^n N.$$

Consider the commutative diagram

$$\begin{array}{ccc} M \otimes_A N' & \longrightarrow & M \otimes_A (N'/V_n') \\ \downarrow & & \downarrow \\ M \otimes_A N & \longrightarrow & M \otimes_A (N/I^n N). \end{array}$$

Since N'/V_n' and N/I^nN are annihilated by I^n, the right-hand vertical homomorphism is obtained from the *injective* homomorphism French source p. 98

$$N'/V_n' \longrightarrow N/I^n N$$

by tensoring over A/I^n with the *flat* (A/I^n)-module $M \otimes_A A/I^n$. It is therefore injective. Hence the kernel of $M \otimes_A N' \to M \otimes_A N$ is contained in the kernel of

$$M \otimes_A N' \longrightarrow M \otimes_A (N'/V_n'),$$

as required.

The finiteness condition just imposed on M holds in particular if M is a finitely generated module over a Noetherian A-algebra B such that IB is contained in the Jacobson radical of B. Indeed, for every finitely generated A-module N, the module $M \otimes_A N$ is finitely generated over B, and is therefore separated by Krull's theorem for the I-adic topology, which is its (IB)-adic topology. We have proved:

THEOREM IV.5.6. *Let $A \to B$ be a homomorphism of Noetherian rings, let I be an ideal of A such that IB is contained in the Jacobson radical of B, and let M be a finitely generated B-module. The following conditions are equivalent:*

(i) *M is flat over A;*
(ii) *$M \otimes_A A/I$ is flat over A/I, and $\operatorname{Tor}_1^A(M, A/I) = 0$;*
(iii) *$M \otimes_A A/I$ is flat over A/I, and the canonical homomorphism*

$$\operatorname{gr}_I^0(M) \otimes_{A/I} \operatorname{gr}_I(A) \longrightarrow \operatorname{gr}_I(M)$$

is an isomorphism;
(iv) *for every $n > 0$, $M \otimes_A A/I^n$ is flat over A/I^n.*

This result is used chiefly when A and B are Noetherian *local* rings, $A \to B$ is a local homomorphism, and I is an ideal of A contained in its maximal ideal; Theorem IV.5.6 may in fact be reduced immediately to this case. An interesting special case is that in which A/I is a field, that is, I is maximal. The condition that $M \otimes_A A/I$ be flat over A/I is then automatic. Moreover, the rings A/I^n are Artinian local rings, so condition (iv) says that the modules $M \otimes_A A/I^n$ are *free* over A/I^n.

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COROLLARY IV.5.7. *Let $A \to B$ be a local homomorphism of Noetherian local rings, and let $u\colon M' \to M$ be a homomorphism of finitely generated B-modules. Suppose that M is flat over A. The following conditions are equivalent:*

(i) *u is injective and $\operatorname{Coker}(u)$ is flat over A;*
(ii) *the homomorphism*

$$u \otimes_A k\colon M' \otimes_A k \longrightarrow M \otimes_A k$$

is injective, where k is the residue field of A.

The implication (i)⇒(ii) follows from Proposition IV.1.1. Conversely, u is first of all injective: it is enough to check this on associated graded modules, where it follows from a commutative square that we leave to the reader. Put $M'' = \operatorname{Coker}(u)$. We then have an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

The long exact sequence of Tor, together with hypothesis (ii) and $\operatorname{Tor}_1^A(M, k) = 0$, gives $\operatorname{Tor}_1^A(M'', k) = 0$. Theorem IV.5.6 therefore shows that M'' is flat over A.

COROLLARY IV.5.8. *Under the hypotheses of Theorem IV.5.6, let J be an ideal of B containing IB and contained in the Jacobson radical. Let $\widehat{A}$ be the I-adic completion of A, and let $\widehat{B}$ and $\widehat{M}$ be the J-adic completions of B and M, respectively. Then M is flat over A if and only if $\widehat{M}$ is flat over $\widehat{A}$.*

The sufficiency already follows readily from Corollary IV.3.2. Apply criterion (iii) of Theorem IV.5.6 to the situations

$$(A, B, I, M) \qquad \text{and} \qquad (\widehat{A}, \widehat{B}, I\widehat{A}, \widehat{M}).$$

The resulting conditions in the two cases are equivalent by Corollary IV.3.2.

COROLLARY IV.5.9. *Let $A \to B \to C$ be local homomorphisms of Noetherian local rings, and let M be a finitely generated C-module. (The ring C occurs only so that a finiteness condition can be imposed on M.) Suppose that B is flat over A, and let k be the residue field of A. The following conditions are equivalent:*

(i) *M is flat over B;*
(ii) *M is flat over A, and $M \otimes_A k$ is flat over $B \otimes_A k$.*

The implication (i)⇒(ii) is immediate. For the converse, apply criterion (iii) of Theorem IV.5.6 to

$$(B, C, I = \mathfrak{m}B, M),$$

where $\mathfrak{m}$ is the maximal ideal of A. Since

$$M \otimes_B (B/I) = M \otimes_B (B \otimes_A k) = M \otimes_A k,$$

the first condition of that criterion says precisely that $M \otimes_A k$ is flat over $B \otimes_A k$, as required. The second condition is satisfied because M and B are flat over A, by associativity of the tensor product. French source p. 100

Of course, using Corollary IV.5.5 instead of Theorem IV.5.6 gives an analogous statement without Noetherian or finiteness hypotheses, provided instead that the ideal $\mathfrak{m}$ of A is nilpotent. The fact that $\mathfrak{m}$ was taken maximal did not enter the argument; nevertheless, in a sense, the case of maximal $\mathfrak{m}$ is the best possible one.

IV.6. Flat morphisms and open subsets

We first recall some results on constructible sets, also proved in circulating notes from the Dieudonné–Rosenlicht Seminar on schemes. [3]

Following Chevalley, a subset of a topological space is called *constructible* if it is a finite union of locally closed subsets.

LEMMA IV.6.1. *Let X be a Noetherian topological space and $Z \subseteq X$. Then Z is constructible if and only if, for every irreducible closed subset $Y \subseteq X$, the intersection $Z \cap Y$ is either not dense in Y or contains a nonempty open subset of Y.*

Using a familiar lemma from commutative algebra, one deduces:

LEMMA IV.6.2 (Chevalley). *Let $f\colon X \to Y$ be a morphism of finite type between preschemes, with Y Noetherian. Then $f(X)$ is constructible.*

LEMMA IV.6.3. *Let X be a Noetherian topological space in which every irreducible closed subset has a generic point, let $U \subseteq X$ be constructible, and let $x \in X$. Then U is a neighborhood of x if and only if every generalization y of x, that is, every y such that $x \in \overline{\{y\}}$, belongs to U.*

In particular:

COROLLARY IV.6.4. *Let X be a Noetherian topological space in which every irreducible closed subset has a generic point. A subset $U \subseteq X$ is open if and only if it satisfies both of the following conditions:*

(a) *U contains every generalization of each of its points;*
(b) *if $x \in U$, then $U \cap \overline{\{x\}}$ contains a nonempty open subset of $\overline{\{x\}}$.*

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Condition (b) and Lemma IV.6.1 show that U is constructible, and Lemma IV.6.3 then shows that U is a neighborhood of each of its points.

COROLLARY IV.6.5. *Let $f\colon X \to Y$ be a morphism of finite type between preschemes, with Y locally Noetherian. Let $x \in X$ and $y = f(x)$. Then f maps every neighborhood of x onto a neighborhood of y if and only if, for every generalization y' of y, there is a generalization x' of x such that $f(x') = y'$.*

We may suppose that X and Y are affine, hence Noetherian. The condition is sufficient because it is enough to show that $f(X)$ is a neighborhood of y. The image $f(X)$ is constructible by Lemma IV.6.2, and Lemma IV.6.3 applies.

Conversely, let $Y' = \overline{\{y'\}}$, and let F be the union of the irreducible components of $f^{-1}(Y')$ which do not contain x. Then $X - F$ is an open neighborhood of x, so its image

[3] See EGA 0_{III}, 9, and EGA IV, 1.8 and 1.10.

is a neighborhood of y, and in particular contains y'. Choose $x'_1 \in X - F$ with $f(x'_1) = y'$. An irreducible component of $f^{-1}(Y')$ containing x'_1 must also contain x, for otherwise it would lie in F. Let x' be its generic point. Then x' is a generalization of x, and $f(x')$ is a generalization of $f(x'_1) = y'$ lying in Y'; hence $f(x') = y'$.

THEOREM IV.6.6. *Let $f\colon X \to Y$ be a morphism locally of finite type, with Y locally Noetherian, and let F be a coherent sheaf on X, with support X, flat relative to Y. Then f is an open morphism; that is, it maps open sets to open sets.*

It is enough to verify the criterion of Corollary IV.6.5 at each $x \in X$. Generalizations of x correspond to prime ideals of $\mathcal{O}_x$, and generalizations of $y = f(x)$ correspond to prime ideals of $\mathcal{O}_y$. We must therefore show that every prime ideal of $\mathcal{O}_y$ is the contraction of a prime ideal of $\mathcal{O}_x$. The stalk F_x is a nonzero finitely generated $\mathcal{O}_x$-module flat over $\mathcal{O}_y$, hence faithfully flat over $\mathcal{O}_y$ by Corollary IV.2.2. Proposition IV.2.3 now finishes the proof.

Remarks. Since flatness is preserved under base change, the preceding theorem also shows that f is universally open. If Y is integral and X is of finite type over Y, it was not known whether the restriction of f to each irreducible component X_i of X must be open, or even merely equidimensional. [4] Here equidimensional means that all components of all fibers have the same dimension; one knows only that X_i dominates Y. The question is related to the following one. Let $A \to B$ be a local homomorphism of Noetherian local rings, with B flat over A and $\mathfrak{m}B$ an ideal of definition of B, which in particular implies $\dim B = \dim A$. Is it true that

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$$\dim B/\mathfrak{p}_i = \dim B$$

for every minimal prime ideal $\mathfrak{p}_i$ of B?

We merely note that the answer to the first question is negative if the flatness hypothesis in Theorem IV.6.6 is replaced by the assumption that f is universally open.

LEMMA IV.6.7. *Let A be a Noetherian integral domain, let B be a finitely generated A-algebra, and let M be a finitely generated B-module. There exists a nonzero $f \in A$ such that M_f is free, and hence flat, over A_f.*

Let K be the fraction field of A. Then $B \otimes_A K$ is a finitely generated K-algebra and $M \otimes_A K$ is a finitely generated module over it. Let n be the dimension of the support of this module; we argue by induction on n. If $n < 0$, that is, if $M \otimes_A K = 0$, choose finitely many generators of M over B. Some nonzero $f \in A$ annihilates all of them, hence annihilates M, so $M_f = 0$.

Suppose $n \geq 0$. The B-module M has a prime filtration whose successive quotients are $B/\mathfrak{p}_i$. Since an extension of free modules is free, we reduce to $M = B/\mathfrak{p}$, or, equivalently, to $M = B$ with B an integral A-algebra. Applying Noether normalization to $B \otimes_A K$, after inverting a suitable nonzero element of A we may suppose that B is integral over

$$C = A[t_1, \ldots, t_n].$$

It is then a finitely generated torsion-free C-module. If m is its rank, there is an exact sequence of C-modules

$$0 \longrightarrow C^m \longrightarrow B \longrightarrow M' \longrightarrow 0$$

with M' a torsion C-module. The Krull dimension of the $C \otimes_A K$-module $M' \otimes_A K$ is strictly smaller than the dimension n of $C \otimes_A K$. By induction, after inverting another suitable nonzero $f \in A$, M' is free over A. The module C^m is also free over A, so the sequence splits over A and B is free over A.

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[4] The answer to the second question is affirmative and to the first negative, even when f is étale; see EGA IV, 12.1.1.5, and EGA Errata IV, 33.

LEMMA IV.6.8. *Let A be Noetherian, let B be a finitely generated A-algebra, let M be a finitely generated B-module, and let $\mathfrak{p}$ be a prime ideal of B, with contraction $\mathfrak{q}$ in A. Suppose that $M_\mathfrak{p}$ is flat over $A_\mathfrak{q}$, equivalently over A. Then there exists $g \in B - \mathfrak{p}$ such that:*

(a) *$(M/\mathfrak{q}M)_g$ is flat over $A/\mathfrak{q}$;*
(b) *$\operatorname{Tor}_1^A(M, A/\mathfrak{q})_g = 0$.*

Apply Lemma IV.6.7 to

$$(A/\mathfrak{q},\ B/\mathfrak{q}B,\ M/\mathfrak{q}M).$$

There is first an $f \in A - \mathfrak{q}$ such that $(M/\mathfrak{q}M)_f$ is flat over $A/\mathfrak{q}$. Since $M_\mathfrak{p}$ is flat over A,

$$\operatorname{Tor}_1^A(M, A/\mathfrak{q})_\mathfrak{p} = \operatorname{Tor}_1^A(M_\mathfrak{p}, A/\mathfrak{q}) = 0.$$

The left-hand module is finitely generated over B, so there exists $g \in B - \mathfrak{p}$ for which (b) holds. Replacing g by gf, we may arrange that (a) holds as well.

COROLLARY IV.6.9. *With the notation of Lemma IV.6.8, let g be as there. For every prime ideal $\mathfrak{p}'$ of B containing $\mathfrak{p}$ and not containing g, the module $M_{\mathfrak{p}'}$ is flat over A, equivalently over $A_{\mathfrak{q}'}$, where $\mathfrak{q}'$ is the contraction of $\mathfrak{p}'$ to A.*

Apply criterion (ii) of Theorem IV.5.6 to

$$(A,\ B_{\mathfrak{p}'},\ \mathfrak{q},\ M_{\mathfrak{p}'}),$$

using localization of Tor.

THEOREM IV.6.10. *Let $f\colon X \to Y$ be a morphism of finite type, with Y locally Noetherian, and let F be a coherent sheaf on X. The set*

$$U = \{x \in X \mid F_x \text{ is flat over } \mathcal{O}_{f(x)}\}$$

is open.

Proof. We may suppose that X and Y are affine, with rings B and A, and that F is defined by a finitely generated B-module M. Apply Corollary IV.6.4. Condition (a) follows immediately from Proposition IV.1.2(i), and condition (b) is exactly Lemma IV.6.8 and Corollary IV.6.9.

For many purposes the following weaker form of Theorem IV.6.10 suffices. It already follows from Lemma IV.6.7, and therefore requires neither constructible sets nor Theorem IV.5.6.

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COROLLARY IV.6.11. *Under the hypotheses of Theorem IV.6.10, suppose that Y is integral. There exists a nonempty open subset $V \subseteq Y$ such that F is flat relative to Y at every point of $f^{-1}(V)$.*

Indeed, the open set U contains the fiber over the generic point of Y, since its local ring is a field. As X is of finite type over Y, U therefore contains an open set of the form $f^{-1}(V)$.

Corollary IV.6.11 also readily implies the following statement when Y is Noetherian but not necessarily integral. There exists a partition of Y into locally closed subsets Y_i such that, after giving Y_i its induced reduced structure, the restriction of F to $X_i = X \times_Y Y_i$ is flat relative to Y_i.

CHAPTER V

The fundamental group: generalities

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V.0. Introduction

The present Seminar is the continuation of the 1960 Seminar. We shall refer to the latter by abbreviations such as I I.9.7, meaning: Seminar on Algebraic Geometry, Exposé I, no. 9.7. The numbers of the 1961 exposés will follow those of 1960. We shall refer to the *Éléments de Géométrie Algébrique* of Dieudonné and Grothendieck by abbreviations such as EGA I, 8.7.3.

The present exposé summarizes, with slight additions, the final exposés of 1960, which had not been written up.

As in 1961, we shall as a rule restrict ourselves to locally Noetherian preschemes, although this restriction is often inessential. In Exposé VI we shall assume the theory of faithfully flat descent, summarized in Bourbaki Seminar no. 190. If appropriate, we shall give a more detailed account in a later exposé, [1] once the reader has had occasion to appreciate the usefulness of this technique in the theory of the fundamental group.

V.1. Preschemes with a finite group of operators; quotient preschemes

Let X be a prescheme and let a finite group G act on X by automorphisms, on the right for definiteness. If X is affine with ring A, then G acts on A by automorphisms on the left.

For every prescheme Z, the group G acts on the left on the set $\mathrm{Hom}(X, Z)$. We may therefore consider the set

$$\mathrm{Hom}(X, Z)^G$$

of G-invariant morphisms. It depends functorially on Z, and we may ask whether this functor is representable, that is, isomorphic to a functor $Z \mapsto \mathrm{Hom}(Y, Z)$. This means finding a prescheme Y and a G-invariant morphism

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$$p\colon X \longrightarrow Y$$

such that, for every Z, the corresponding map $g \mapsto gp$,

$$\mathrm{Hom}(Y, Z) \longrightarrow \mathrm{Hom}(X, Z)^G,$$

is bijective. The pair (Y, p) is then called a *quotient prescheme* of X by G; it is determined up to unique isomorphism.

Proposition V.1.1. *Let A be a ring on which the finite group G acts on the left, let $B = A^G$ be the subring of invariants, let $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$, and let $p\colon X \to Y$ be the canonical morphism, which is plainly G-invariant. Then:*

(i) *A is integral over B; equivalently, p is an integral morphism.*

(ii) *The morphism p is surjective, its fibers are the G-orbits, and the topology of Y is the quotient topology of that of X.*

(iii) *Let $x \in X$, let $y = p(x)$, and let G_x be the stabilizer of x. Then $\kappa(x)$ is a quasi-Galois algebraic extension of $\kappa(y)$, and the canonical map*

$$G_x \longrightarrow \mathrm{Gal}(\kappa(x)/\kappa(y))$$

[1] See Exposés VI and VIII.

to the group of $\kappa(y)$-automorphisms of $\kappa(x)$ is surjective.

(iv) *The pair (Y, p) is a quotient prescheme of X by G.*

Statements (i), (ii), and (iii) are well-known in commutative algebra, [2] and are recalled only for reference, except for the assertion about the topology. The latter follows from this general fact, an easy consequence of the Cohen–Seidenberg theorem: an integral morphism is closed, that is, it maps closed sets to closed sets. We note at once:

COROLLARY V.1.2. *Under the preceding hypotheses, the natural homomorphism*

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_X)^G$$

is an isomorphism.

This follows immediately from the formula

$$(S^{-1}A)^G = S^{-1}(A^G),$$

valid for every multiplicatively closed subset S of $B = A^G$. The formula has a module version, and more generally holds after a flat base change $A \to A'$. Apply it when S is generated by one element $f \in B$.

Assertion (ii) and Corollary V.1.2 readily imply (iv). More generally:

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PROPOSITION V.1.3. *Let X be a prescheme with a finite group G of automorphisms, and let $p\colon X \to Y$ be an invariant affine morphism such that*

$$\mathcal{O}_Y \xrightarrow{\sim} p_*(\mathcal{O}_X)^G.$$

Then conclusions (i), (ii), (iii), *and* (iv) *of Proposition V.1.1 still hold.*

For (i), (ii), and (iii), we may suppose that Y, and hence X, is affine. If B and A are their rings, the hypothesis implies $B = A^G$, so Proposition V.1.1 applies. For (iv), use (ii) together with $\mathcal{O}_Y = p_*(\mathcal{O}_X)^G$.

COROLLARY V.1.4. *Under the hypotheses of Proposition V.1.3, for every open subset $U \subseteq Y$, the prescheme U is a quotient of $X|_U = p^{-1}(U)$ by G.*

Indeed, the morphism $p^{-1}(U) \to U$ induced by p satisfies the same hypotheses as p.

Now suppose that X is a Z-prescheme and that the operations of G are Z-automorphisms. By (iv), Y is then a Z-prescheme. Thus:

COROLLARY V.1.5. *The prescheme X is affine, respectively separated, over Z if and only if Y is. If X is of finite type over Z, then it is finite over Y. If, in addition, Z is locally Noetherian, then Y is of finite type over Z.*

Since X is affine, and hence separated, over Y, it is affine, respectively separated, over Z whenever Y is. Conversely, suppose first that X is affine over Z, and let us prove that Y is. By Corollary V.1.4, we may suppose that Z is affine. We are then reduced to proving that Y is affine when X is affine, and this follows from the explicit description $Y = \operatorname{Spec}(A^G)$ in Proposition V.1.1.

Likewise, since $p\colon X \to Y$ is integral, hence universally closed, and surjective, it follows that Y is separated over Z whenever X is. Indeed, in the diagram

$$\begin{array}{ccc} X \times_Z X & \xrightarrow{p \times_Z p} & Y \times_Z Y \\ \uparrow{\scriptstyle \Delta_{X/Z}} & & \uparrow{\scriptstyle \Delta_{Y/Z}} \\ X & \xrightarrow{p} & Y, \end{array}$$

the morphism $X \times_Z X \to Y \times_Z Y$ is closed. It therefore maps the closed diagonal of X to a closed subset of $Y \times_Z Y$, and this image is precisely the diagonal of Y, because p is surjective.

[2] See N. Bourbaki, *Commutative Algebra*, Chapter V, §§1–2, Theorem 2.

If X is of finite type over Z, it is a fortiori of finite type over Y, and is therefore finite over Y, since it is already integral over Y. Suppose further that Z is locally Noetherian; we prove that Y is of finite type over Z. By Corollary V.1.4, we may suppose that Z is affine. The topological space X is quasi-compact and $p\colon X \to Y$ is surjective, so Y is also quasi-compact and hence is a finite union of affine open subsets. Using Corollary V.1.4 again, we reduce to the case in which Y and X are affine. The ring A of X is then a finitely generated algebra over the Noetherian ring C of Z. It is known that $B = A^G$ is likewise a finitely generated C-algebra. Indeed, A is integral, hence finite, over a C-subalgebra $B' \subseteq B$ of finite type; since B' is Noetherian, B is also finite over B', and hence of finite type over C.

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Corollary V.1.6. *The prescheme X is affine, respectively a scheme, if and only if Y is.*

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Definition V.1.7. *Let X be a prescheme on which a finite group G acts on the right. The action of G is called* admissible *if there exists a morphism $p\colon X \to Y$ having the properties of Proposition V.1.3. This implies that X/G exists and is isomorphic to Y.*

Proposition V.1.8. *Let X be a prescheme on which the finite group G acts on the right. The action is admissible if and only if X is a union of G-invariant affine open subsets, or equivalently, if and only if every G-orbit in X is contained in an affine open subset.*

The first condition plainly implies the second, and the second in turn implies the first. Indeed, let T be a G-orbit and let U be an affine open subset containing it. The intersection of the translates gU, for $g \in G$, is a G-stable open subset U' containing T and contained in the affine open U. In an affine scheme, every finite subset has a fundamental system of affine open neighborhoods, so T has an affine open neighborhood $V \subseteq U'$. Its translates gV are affine and contained in the separated open subset U'. Their intersection is therefore an affine, G-invariant open subset U'' containing T.

Now the condition in Proposition V.1.8 is necessary: if a quotient $p\colon X \to Y$ as in Proposition V.1.3 exists, take the inverse images of an affine open cover of Y. It is also sufficient. Indeed, Proposition V.1.1 constructs the quotients $Y_i = X_i/G$ of the invariant affine opens X_i. In each Y_i, the image of $X_i \cap X_j$ is an open subset Y_{ij}, identified with X_{ij}/G by Corollary V.1.4. We thereby obtain isomorphisms $Y_{ij} \xrightarrow{\sim} Y_{ji}$, with which the Y_i can be glued to construct Y. Serre instead constructs directly the quotient topological space $Y = X/G$, equips it with the sheaf $p_*(\mathcal{O}_X)^G$, and verifies that Y becomes a prescheme satisfying the conditions of Proposition V.1.3.

Corollary V.1.7. *If the action of G on X is admissible, then the action of every subgroup $H \subseteq G$ is admissible as well; in particular, X/H exists.*

This can also be checked directly in the situation of Proposition V.1.3. We may always suppose that X is affine over some Z and that the elements of G act by Z-automorphisms, for example by taking $Z = Y$. Indeed:

Corollary V.1.8. *Suppose that X is affine over Z and that the operations of G are Z-automorphisms. Then the action of G on X is admissible. If X is defined by a quasi-coherent sheaf of algebras $\mathcal{A}$, then Y is defined by the sheaf of invariant algebras $\mathcal{A}^G$.*

Proposition V.1.9. *Suppose that G acts admissibly on X, that $X/G = Y$ is a Z-prescheme, and let $Z' \to Z$ be a base-change morphism. Put*

$$X' = X \times_Z Z', \qquad Y' = Y \times_Z Z'.$$

The group G acts on X' by transport of structure, and the morphism $p'\colon X' \to Y'$ is invariant. If Z' is flat over Z, then p' again satisfies the hypotheses of Proposition V.1.3;

that is,

$$\mathcal{O}_{Y'} \xrightarrow{\sim} p'_*(\mathcal{O}_{X'})^G.$$

Thus G acts admissibly on X', and

$$(X/G) \times_Z Z' \xrightarrow{\sim} (X \times_Z Z')/G.$$

We may plainly take $Z = Y$, and are then reduced to the case in which Y and Y' are affine. We must show the following. Let B be the invariant subring for the action of G on A, and let B' be a flat B-algebra. Then B' is the invariant subring of $A' = A \otimes_B B'$. This is immediate, since the exact sequence

$$0 \longrightarrow B \xrightarrow{i} A \xrightarrow{j} A^{(G)},$$

where the last term denotes a power of A and $j(x) = (s \cdot x - x)_{s \in G}$, remains exact after tensoring with B'.

The flatness hypothesis is essential. In particular, if Y' is a closed subscheme of Y, even a closed point, and X' is its inverse image in X, then in general Y' is *not* identified with X'/G. We shall see that it is so nevertheless when X is étale over Y.

We conclude with a convenient, if elementary, formalism. Let Y be a prescheme. Since coproducts exist in the category of preschemes, for every set E we may form the sum of a family $(Y_i)_{i \in E}$ of preschemes all equal to Y. We denote this prescheme by $Y \times E$. It is characterized by the formula

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$$(*) \qquad \mathrm{Hom}(Y \times E, Z) = \mathrm{Hom}\big(E, \mathrm{Hom}(Y, Z)\big).$$

Here the second Hom on the right means, of course, the set of maps from E to the set $\mathrm{Hom}(Y, Z)$. There is a canonical morphism

$$Y \times E \longrightarrow Y,$$

making $Y \times E$ a prescheme over Y. Since fiber products commute with coproducts in the category of preschemes, if Y is a Z-prescheme and $Z' \to Z$ is a base change, then

$$(Y \times E) \times_Z Z' = (Y \times_Z Z') \times E.$$

This formula is especially useful when $Z = Y$. The definition also immediately gives

$$(Y \times E) \times F = Y \times (E \times F) = (Y \times E) \times_Y (Y \times F),$$

the last equality also following from the compatibility with fiber products just noted.

For fixed Y, the construction $E \mapsto Y \times E$ is a functor from sets to preschemes over Y. By the preceding formula it commutes with finite products. Thus, for example, every ordinary group G gives rise to a group prescheme $Y \times G$ over Y, which is finite over Y when G is finite. More generally the functor is "left exact," although we shall not use this here. It plainly commutes with coproducts and is also "right exact," as follows at once from the defining formula (*). In particular, if a finite group G acts on the right on a set E, then it acts on the right on $Y \times E$, and

$$(Y \times E)/G = Y \times (E/G).$$

The quotient on the left in fact satisfies the conditions of Proposition V.1.3, as is immediate.

V.2. Decomposition and inertia groups. The étale case

Let G a finite group acting on the right on the prescheme X. If $x \in X$, the stabilizer $G_d(x)$ of x is called the *decomposition group of x*. This group acts canonically (on the left) on the residue field $\kappa(x)$, and the set of elements of $G_d(x)$ which act trivially is called the *inertia group* of x, denoted $G_i(x)$. 111

Suppose that G acts admissibly on X and that Y is a prescheme over a prescheme Z. Fix $z \in Z$, and an algebraically closed extension Ω of $\kappa(z)$ having transcendence degree

greater than that of the $\kappa(x)/\kappa(z)$, where x is a point of X above z. We may regard $\mathrm{Spec}(()\Omega)$ as a Z-scheme, and the Ω-valued points of X correspond to the homomorphisms of $\kappa(z)$-algebras $\kappa(x) \to \Omega$, where x is a point of X above z. Since Ω was chosen sufficiently large, every point x of X above z is the underlying point of an Ω-valued point of X. If $X(\Omega)$ and $Y(\Omega)$ denote, respectively, the sets of Ω-valued points of X and Y, there is a natural map

$$X(\Omega) \to Y(\Omega),$$

and, moreover, G acts on $X(\Omega)$ and the preceding map is G-invariant. With this understood, conclusions (ii) and (iii) of V.1.3 may also be interpreted as follows: *the preceding map is surjective and identifies $Y(\Omega)$ with the quotient $X(\Omega)/G$. Moreover, if x is the underlying point of $a \in X(\Omega)$, then the stabilizer of a in G is precisely the inertia group $G_i(x)$.* All this is in fact true without assuming that Ω is "sufficiently large"; the latter hypothesis serves only to ensure that the inertia group of every point of X above z can be characterized as a "geometric" stabilizer. We immediately deduce, for example:

PROPOSITION V.2.1. *Make a base extension $Z' \to Z$, and hence put $X' = X \times_Z Z'$. Let x' be a point of X' and x its image in X. Then $G_i(x) = G_i(x')$.*

In the preceding considerations, it is enough to take for Ω a sufficiently large extension of $\kappa(z')$ (where z, z' are the images of x, x' in Z, Z', respectively).

PROPOSITION V.2.2. *Under the conditions of V.1.3, suppose that Y is locally Noetherian and that X is finite over Y. Let H be a subgroup of G, put $X' = X/H$ (cf. V.1.7), and let $x \in X$, with image x' in X' and image y in Y.* 112

(i) *If $H \supset G_d(x)$, then the homomorphism $\mathcal{O}_y \to \mathcal{O}_{x'}$ induces an isomorphism on completions.*

(ii) *If $H \supset G_i(x)$, then the homomorphism $\mathcal{O}_y \to \mathcal{O}_{x'}$ is étale, i.e. X' is étale over Y at x'.*

Let $Y_1 = \mathrm{Spec}(()\widehat{\mathcal{O}_y})$, and make the base change $Y_1 \to Y$. We obtain $X_1 = X \times_Y Y_1$, finite over Y_1, on which G acts, with quotient Y_1 by V.1.9. Let y_1 be the unique point of Y_1 above y. Since $\kappa(y) = \kappa(y_1)$, the fiber of X at y is isomorphic to that of X_1 at y_1; hence there is a unique point x_1 of X_1 above x. Moreover, by V.1.9, we have $X_1/H = X_1' = X' \times_Y Y_1$. Let x_1' be the image of x_1 in X_1'. It lies above x', and one checks easily (since X' is of finite type over Y) that the homomorphism $\mathcal{O}_{x'} \to \mathcal{O}_{x_1'}$ induces an isomorphism on completions. We are therefore reduced to the case where Y is the spectrum of a complete local ring B, and hence X is the spectrum of a finite B-algebra A, which is a product of finitely many local rings A_x corresponding to the points x_i of X above Y. If A_0 corresponds to $x = x_0$, then A identifies with the ring $\mathrm{Hom}_{G_d}(G, A_0)$ of functions $f\colon G \to A_0$ such that $f(st) = sf(t)$ for $s \in G_d$, the operations of G on these functions being defined by $(uf)(t) = f(tu)$. Thus, if H is any subgroup of G, then A^H is the ring of functions $f\colon G \to A_0$ such that

$$f(stu) = sf(t) \qquad \text{for } s \in G_d,\ u \in H$$

It is therefore a semilocal ring whose local components correspond to the double cosets $G_d a H$ in G; to the double coset defined by $a \in G$ corresponds (by means of the map $f \longmapsto f(a)$) the subring $A_0^{H(a)}$ of A_0, where $H(a) = G_d \cap aHa^{-1}$. Moreover, the local component of A^H corresponding to the image x' of x is also the one corresponding to the double coset $G_d H$ of the identity element; its local component is therefore $A_0^{G_d \cap H}$. Thus, if $G_d \subset H$, we obtain $A_0^{G_d} = A^G = B$, which proves (i). To prove (ii), we may, by passing to a suitable finite flat extension of A and using V.2.1, reduce to the case where the residue-field extension $\kappa(x)/\kappa(y)$ is trivial. But then $G_i(x) = G_d(x)$, and we are reduced to the preceding case.

COROLLARY V.2.3. *Under the conditions of V.2.2, suppose that $G_i(x) = (e)$. Then* 113
X is étale over Y at x. Consequently, if $G_i(x) = (e)$ for every $x \in X$, then $X \to Y$ is an étale morphism.

There is a partial converse:

COROLLARY V.2.4. *Suppose that X is connected and that the action of G on X is faithful. The morphism $p\colon X \to Y = X/G$ is étale if and only if the inertia groups of the points of X are reduced to the identity element. If this is so, G identifies with the group of all Y-automorphisms of the Y-scheme X.*

In view of V.2.3, we may suppose that X is étale over Y. But if an $s \in G$ belongs to some $G_i(x)$, it then follows from I I.5.4 that s acts trivially on G, and hence is the identity element since G is faithful; this proves the first assertion. Let u be a Y-automorphism of X, and let $x \in X$. By Proposition V.1.3, there exists an $s \in G$ such that $s(x) = u(x)$ and inducing the same residue homomorphism $\kappa(x)/\kappa(x')$ as u. By loc. cit. , we therefore have $s = u$, which completes the proof.

REMARK V.2.5. The hypothesis that G acts faithfully is clearly not superfluous in Corollary V.2.4. The same is true of the hypothesis that X is connected, as is seen, for example, by taking $X = Y \times E$, where E is a finite set, and taking G to be the group of permutations of E: G acts with considerable inertia, yet $(Y \times E)/G = Y \times (E/G) = Y$, and X is étale over Y. Taking for G a group strictly smaller than the symmetric group of E but acting transitively on E, one sees that there will also be Y-automorphisms of X which do not come from G.

The standard example of a group G acting without inertia is that of $Y \times G$, on which G acts through its right-translation action on the factor G: a Y-prescheme X with right operator group G is said to be *trivial* if it is isomorphic to $Y \times G$.

The following considerations are useful for relating preschemes with finite operator groups to the notion of a principal bundle in a category (a relation which, although we shall not need it in the rest of the seminar, is important in other contexts). We fix a base prescheme Y and work in the category of Y-preschemes. If G is a finite group, we shall 114
write $G_Y = Y \times G$ for short; this is thus a finite group scheme over Y (cf. no. 1), and if X is a Y-prescheme, then

$$X \times_Y G_Y = X \times G$$

(same reference). Giving a Y-morphism $X \times_Y G_Y \to X$ is therefore equivalent to giving a Y-morphism $X \times G \to X$, i.e. to giving, for each $g \in G$, a Y-morphism $T_g\colon X \to X$. One checks at once that the data of the T_g define on X the structure of a prescheme with right operator group G (i.e. $T_{gg'} = T_{g'}T_g$, $T_e = \mathrm{id}_x$) if and only if the corresponding Y-morphism $X \times_Y G_Y \to X$ defines on X the structure of a Y-prescheme with operator Y-group scheme (in the general sense of objects with a $\mathcal{C}$-operator group in a category $\mathcal{C}$). Suppose this is so. Recall that X is said to be *formally principal homogeneous* under G_Y[3] if the canonical morphism

$$X \times_Y G_Y \to X \times_Y X$$

whose components are respectively pr_1 and the multiplication morphism $\pi\colon X \times_Y G_Y \to X$, is an isomorphism. In the present case, identifying the first member with $X \times G$, the morphism in question is the one which associates to each $g \in G$ the morphism

$$(\mathrm{id}_X, T_g) = (\mathrm{id}_X \times_Y T_g)\Delta_{X/Y}\colon X \to X \times_Y X$$

Consequently, to say that X is formally principal homogeneous under G_Y also means that $X \times_Y X$ is isomorphic to the disjoint sum of the transforms of the diagonal by the elements (e, g) of $G \times G$ (acting on $X \times_Y X$ in the evident way), where e denotes the

[3]One now says instead that X is a pseudo-torsor under G_Y.

identity element of G. If one does not wish to distinguish left from right, and wants a formula which remains applicable to a product of more than two factors identical to X, the condition may be formulated by saying that the canonical morphism

$$X \times_G (G \times G) \to X \times_Y X$$

obtained by assigning to the pair (g, g') the morphism

$$(T_g, T_{g'}) = (T_g \times_Y T_{g'})\Delta_{X/Y} \colon X \to X \times_Y X$$

and making G act on the left on $G \times G$ by the diagonal homomorphism,

$$s(g, g') = (sg, sg'),$$

is an *isomorphism*.

The notion of a *principal homogeneous space* is obtained from that of a formally 115
principal homogeneous space by adding a further axiom ensuring that the "quotient" of X by G_Y exists and is precisely the right unit object of the category, here Y. This axiom may vary with the context, and is often most conveniently stated explicitly (in the yoga of "descent") by requiring that the object with operators become "trivial", i.e. isomorphic to the product $X \times_Y G_Y$ (in the present case $X \times G$), after a suitable base change of a specified type (in such a way, in practice, as to permit the descent technique; cf. Grothendieck, Technique de descente et théorèmes d'existence en Géométrie Algébrique, Sém. Bourbaki no. 190, pages 26 to 28)[4]. In this vein, we record here the characterization of principal homogeneous bundles with group G (in the sense of loc. cit.):

Proposition V.2.6. *Let Y be a locally Noetherian prescheme and let X be a Y-prescheme with a finite group G of operators acting on the right. The following conditions are equivalent:*

(i) *X is finite over Y, $Y = X/G$, and the inertia groups of the points of X are reduced to the identity.*

(ii) *There exists a faithfully flat and quasi-compact base change $Y_1 \to Y$ such that $X_1 = X \times_Y Y_1$ is a trivial Y_1-prescheme with operators, i.e. isomorphic to $Y_1 \times G$.*

(ii bis) *As in (ii), but with $Y_1 \to Y$ finite, étale, and surjective.*

(iii) *X is formally principal homogeneous under G_Y, and is faithfully flat and quasi-compact over Y.*

Proof (i)$\Longrightarrow$(ii bis). Take $Y_1 = X$, noting that $X \to Y$ is indeed finite, étale by V.2.3, and surjective. Let us show that X_1 is then trivial over Y_1, which will follow from the

Corollary V.2.7. *If* (i) *holds and X admits a section over Y, then X is a trivial space with operators.*

Indeed, this section makes it possible to define a G-morphism $X \times G \to X$, which is surjective because G is transitive on the fibers of X, injective because G acts without inertia, and finally a local isomorphism by I I.5.3, since X is étale over Y. It is therefore an isomorphism.

(ii bis) trivially implies (ii), which implies (i), since the ingredients of (i) are "invariant" 116
under faithfully flat quasi-compact base extension (cf. the Bourbaki Seminar cited above for "finite"; for inertia groups apply V.2.1, and for $Y = X/G$ use a converse to V.1.9 in the case of a *faithfully flat* base change, which we had forgotten to state explicitly).

In proving (i)$\Longrightarrow$(ii bis), we proved (i)$\Longrightarrow$(iii) along the way. Finally, (iii)$\Longrightarrow$(ii), because the first hypothesis in (iii) means precisely that X becomes trivial after making the base change $Y_1 = X$; hence (ii), since X is faithfully flat and quasi-compact over Y.

Definition V.2.8. *A Y-prescheme X with right operator group G satisfying the equivalent conditions of V.2.6 is called a* principal covering of Y, with Galois group G.

[4]Cf. Exp. VIII for the theory of flat descent.

V.3. Automorphisms and morphisms of étale coverings

PROPOSITION V.3.1. *Let X be separated, étale, and of finite type over a locally noetherian prescheme Y, and let G be a finite group acting on X by Y-automorphisms. Then the action of G is admissible, and the quotient prescheme X/G is étale over Y.*

Although X is not assumed finite over Y, it is quasi-projective over Y, so the existence of X/G follows from V.1.8. Let us first prove:

COROLLARY V.3.2. *The morphism $X \to X/G$ is étale.*

We may clearly suppose first that G acts transitively on the set of connected components of X, and then, by considering the stabilizer of one connected component, that X itself is connected. Finally, we may suppose that G acts faithfully. But then, just as in V.2.4, one sees that G acts without inertia; hence V.2.3 implies that $X \to X/G$ is étale. The conclusion follows from:

LEMMA V.3.3 (a regret concerning Exposé I). *Let $X \to X' \to Y$ be morphisms of finite type, let x be a point of X, and let x' and y be its images. Suppose that Y is locally noetherian. If two of the three morphisms in question are étale at the indicated points, then so is the third.*

It remains only to consider the case in which $X \to X'$ and $X \to Y$ are étale at x, and to prove that $X' \to Y$ is étale at x' (which is the case needed for V.3.1). After making 117
a suitable flat extension of the base Y, we reduce to the case in which the residue-field extension $\kappa(x)/\kappa(y)$ is trivial. Consider the homomorphisms $\mathcal{O}_y \to \mathcal{O}_{x'} \to \mathcal{O}_x$, together with the homomorphisms obtained by passing to completions. The hypothesis says that

$$\widehat{\mathcal{O}}_y \longrightarrow \widehat{\mathcal{O}}_x \quad \text{and} \quad \widehat{\mathcal{O}}_{x'} \longrightarrow \widehat{\mathcal{O}}_x$$

are isomorphisms. It follows at once that $\widehat{\mathcal{O}}_y \to \widehat{\mathcal{O}}_{x'}$ is also an isomorphism, which proves the lemma.

COROLLARY V.3.4. *If X is finite and étale over Y, then X/G is finite and étale over Y.*

PROPOSITION V.3.5. *Let X and X' be two étale coverings of Y. Then every Y-morphism $f\colon X \to X'$ factors as the composite of a surjective étale morphism $X \to X''$ and the canonical immersion $X'' \to X'$ of a subset X'' of X' that is both open and closed.*

It is known (I I.4.8) that f is étale, and hence an open morphism. On the other hand, since X is finite over Y, the morphism f is closed. Thus $f(X) = X''$ is both open and closed in X', and the proof is complete. (N.B. it would have been enough for X', instead of being an étale covering, to be unramified over Y.)

COROLLARY V.3.6. *With the preceding notation, $X \to X''$ is a strict epimorphism in the category of preschemes, and $X'' \to X'$ is a monomorphism (indeed, a strict monomorphism) in the category of preschemes.*

By definition, the first assertion means that the sequence of morphisms

$$X \times_{X''} X \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} X \longrightarrow X''$$

is exact. This follows from the fact that $X \to X''$ is finite and faithfully flat, as is easily seen (cf. Grothendieck, loc. cit.). The dual assertion for $X'' \to X'$ is still more immediate.

Corollary V.3.6 will be useful in the theory of the fundamental group in the next section. Those who dislike the notion of strict epimorphism may replace Corollary V.3.6 by whatever variant they prefer. Let us merely take this opportunity to point out that a factorization $f = f'f''$, with f'' a strict epimorphism and f' a monomorphism, 118

is necessarily unique up to unique isomorphism (in any category). At the same time, however, there may exist a factorization $f = f_1 f_2$ having the dual properties: f_2 is an epimorphism and f_1 a strict monomorphism (again unique up to unique isomorphism), which is not isomorphic to the preceding one. It is enough, for example, to take the category of topological vector spaces (Hausdorff, if desired) and a morphism $u\colon X \to X'$ such that $u(X)$ is not closed.

PROPOSITION V.3.7. *Let Y be a connected locally noetherian prescheme, let y be a point of Y, and let Ω be an algebraically closed extension of $\kappa(y)$. For every X over Y, denote by $X(\Omega)$ the set of points of X with values in Ω. Let X and X' be étale coverings of Y, and let $u\colon X \to X'$ be a Y-morphism such that the corresponding map*

$$X(\Omega) \longrightarrow X'(\Omega)$$

is a bijection. Then u is an isomorphism.

We reduce immediately to the case in which X' is connected. Since $X \to X'$ is finite and étale, the geometric number of points in a fiber of $X \to X'$ is constant, and it is equal to 1 if and only if the morphism in question is an isomorphism. But this number is also the number of elements in a fiber of $X(\Omega) \to X'(\Omega)$, and the conclusion follows.

V.4. Axiomatic conditions for a Galois theory

Let $\mathcal{C}$ be a category and F a covariant functor from $\mathcal{C}$ to the category of finite sets. Assume that the following conditions are satisfied:

(G 1) $\mathcal{C}$ has a terminal object[5], and the fiber product of two objects over a third object in $\mathcal{C}$ exists (this axiom may also be stated by saying that finite inverse limits exist in $\mathcal{C}$).

(G 2) Finite coproducts exist in $\mathcal{C}$ (and hence so does an initial object $\emptyset_{\mathcal{C}}$, playing the role of the empty set), as does the quotient of an object of $\mathcal{C}$ by a finite group of automorphisms.

(G 3) Let $u\colon X \to Y$ be a morphism in $\mathcal{C}$. Then u factors as a composite $X \xrightarrow{u'} Y' \xrightarrow{u''} Y$, where u' is a *strict* epimorphism and u'' is a monomorphism which is an isomorphism onto a direct summand of Y.

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(G 4) The functor F is left exact (i.e. it sends a terminal object to a terminal object and commutes with fiber products).

(G 5) F commutes with finite coproducts, sends strict epimorphisms to epimorphisms, and commutes with passage to the quotient by a finite group of automorphisms.

(G 6) Let $u\colon X \to Y$ be a morphism in $\mathcal{C}$ such that $F(u)$ is an isomorphism. Then u is an isomorphism.

Our aim is to construct a topological group π, an inverse limit of finite groups, and an equivalence between the category $\mathcal{C}$ and the category $\mathcal{C}(\pi)$ *of finite sets on which π acts continuously* (i.e. in such a way that the stabilizer of a point is an open subgroup, or, equivalently, that there is a discrete quotient group which already acts on the set in question), the equivalence constructed sending the given functor F to the evident inclusion functor from $\mathcal{C}(\pi)$ into the category of finite sets. We note at once that the category $\mathcal{C}(\pi)$ constructed from a topological group π, together with the preceding inclusion functor, does indeed satisfy conditions (G 1) through (G 6).

We proceed in several steps.

a) *Let $u\colon X \to Y$ be a morphism in $\mathcal{C}$. Then u is a monomorphism if and only if $F(u)$ is a monomorphism.* (This uses (G 1), (G 4), and (G 6).)

Indeed, to say that u is a monomorphism means that the projection $\mathrm{pr}_1\colon X \times_Y X \to X$ is an isomorphism.

[5]Recall that an object e of $\mathcal{C}$ is called a *terminal object* if, for every X in $\mathcal{C}$, $\mathrm{Hom}(X, e)$ has exactly one element. An *initial object* of $\mathcal{C}$ is defined dually.

b) *Every object X of $\mathcal{C}$ is Artinian.*

Indeed, if $X' \to X'' \to X$ are monomorphisms such that $F(X')$ and $F(X'')$ have the same image in $F(X)$, then by a), $F(X') \to F(X'')$ is an isomorphism, and hence $X' \to X''$ is an isomorphism by (G 6).

c) *The functor F is strictly pro-representable.* (cf. Grothendieck, Technique de descente et théorèmes d'existence en Géométrie Algébrique, II, Séminaire Bourbaki 195, February 1960.)

Indeed, by loc. cit. prop. V.3.1, this follows from b) and (G 4). Thus one can find an inverse system, indexed by a directed ordered set I,

$$P = (P_i)_{i \in I}$$

in $\mathcal{C}$, regarded as a pro-object of $\mathcal{C}$, and a functorial isomorphism 120

$$(*) \qquad F(X) = \mathrm{Hom}_{\mathrm{Pro}(\mathcal{C})}(P, X) = \varinjlim_i \mathrm{Hom}_{\mathcal{C}}(P_i, X).$$

This isomorphism is realized by an element

$$\varphi \in \varprojlim_i F(P_i) = F(P).$$

One may moreover assume that the transition homomorphisms $\varphi_{ji} \colon P_i \to P_j$ $(i \geq j)$ are *epimorphisms*, and that *every epimorphism $P_i \to P'$* is equivalent to an epimorphism $P_i \to P_j$ for a suitable $j \leq i$ (which determines the inverse system P in an essentially unique way).

An object $X \in \mathcal{C}$ is called *connected* if it is not isomorphic to the coproduct of two objects of $\mathcal{C}$ neither of which is isomorphic to the initial object $\emptyset_{\mathcal{C}}$.

d) *The P_i are connected and are not isomorphic to $\emptyset_{\mathcal{C}}$.*

If X is an initial object, then $F(X) = \emptyset$ by (G 5), applied to the coproduct of an empty family, and the converse follows from (G 6). Thus if X' is an object of $\mathcal{C}$ which is not initial, i.e. such that $F(X') \neq \emptyset$, there is no morphism from X' to X. Consequently, if some P_i is initial, then i is a greatest element of the directed ordered index set I, and formula $(*)$ would say that $F(X) = \mathrm{Hom}(P_i, X)$ is a singleton for every X, which is absurd since $F(\emptyset_{\mathcal{C}}) = \emptyset$. The P_i are therefore not isomorphic to $\emptyset_{\mathcal{C}}$.

Suppose that $P_i = A \amalg B$. By (G 5) this gives $F(P_i) = F(A) \amalg F(B)$. In particular, the element a_i of $F(P_i)$ corresponding under $(*)$ to the identity homomorphism $P_i \to P_i$ belongs to $F(A) \amalg F(B)$, say to $F(A)$. This means that there is a $j \geq i$ such that $\varphi_{ij} \colon P_j \to P_i$ factors as $P_j \to A \to P_i = A \amalg B$, where the second arrow is the canonical morphism. Hence $F(P_j) \to F(P_i)$ factors as $F(P_j) \to F(A) \to F(P_i) = F(A) \amalg F(B)$; since $F(P_j) \to F(P_i)$ is surjective by (G 5), it follows that $F(B) = \emptyset$, and hence that B is isomorphic to $\emptyset_{\mathcal{C}}$. 121

e) *Every morphism $u \colon X \to Y$ in $\mathcal{C}$, with X not isomorphic to $\emptyset_{\mathcal{C}}$ and Y connected, is a strict epimorphism. Every endomorphism of a connected object is an automorphism.*

Consider the factorization of u in (G 3). Since $X \neq \emptyset_{\mathcal{C}}$, it follows from (G 6) that $F(X) \neq \emptyset$, hence $F(Y') \neq \emptyset$, hence $Y' \neq \emptyset_{\mathcal{C}}$. Since Y is connected, Y' is therefore identified with Y, and thus u is a strict epimorphism. Suppose that u is an endomorphism of the connected object X; let us show that it is an automorphism. We may assume that X is not isomorphic to $\emptyset_{\mathcal{C}}$, so by what precedes u is a strict epimorphism. Hence $F(u)$ is an epimorphism by (G 5), and since $F(X)$ is a finite set, $F(u)$ is bijective. Thus u is an automorphism by (G 6).

In particular, *every endomorphism of a P_i is an automorphism.*

f) *The following conditions on a P_i are equivalent*: (i) The natural injective map $\mathrm{Hom}(P_i, P_i) \to \mathrm{Hom}(P, P_i) \simeq F(P_i)$ is also surjective, i.e. for every $u \colon P \to P_i$

there is a $v\colon P_i \to P_i$ such that $u = v\varphi_i$ (where φ_i is the canonical homomorphism $P \to P_i$). (ii) The group $\mathrm{Aut}(P_i)$ acts transitively on $F(P_i)$. (iii) The group $\mathrm{Aut}(P_i)$ acts simply transitively on $F(P_i)$.

Upon identifying $\mathrm{Hom}(P, P_i)$ with $F(P_i)$, the map in (i) is precisely $v \longmapsto F(v)(\varphi_i)$. The equivalence of the three conditions then follows from the fact that $\mathrm{Hom}(P_i, P_i) = \mathrm{Aut}(P_i)$ and that the preceding map is already injective.

A P_i satisfying the equivalent conditions (i), (ii), (iii) of f) is called *Galois*.

g) *For every X in $\mathcal{C}$, there is a Galois P_i such that every $u \in \mathrm{Hom}(P, X)$ factors as $P \xrightarrow{\varphi_i} P_i \longrightarrow X$.*

Let $J = \mathrm{Hom}(P, X) = F(X)$. This is a finite set, so there is a P_j such that every $u\colon P \to X$ factors as $P \xrightarrow{j} P_j \longrightarrow X$, or, equivalently, such that the natural morphism

$$P \to X^J \qquad (J = \mathrm{Hom}(P, X))$$

factors as

$$P \xrightarrow{\varphi_j} P_j \to X^J.$$

By (G 3), the morphism $P_j \to X^J$ factors as the composite of a strict epimorphism 122
and a monomorphism, where the former may be taken to be of the form $\varphi_{ij}\colon P_j \to P_i$. We are thus reduced to proving that P_i is Galois. Let k be an index $\geq j$ such that every morphism $P \to P_i$ factors through $P \xrightarrow{\varphi_k} P_k \longrightarrow P_i$. Note that the natural morphism $P_k \to X^J$ still factors as the composite

$$P_k \xrightarrow{\varphi_{ik}} P_i \xrightarrow{U} X^J,$$

where the first arrow is a strict epimorphism by e), and the second is a monomorphism. We wish to show that, for a given morphism $\psi\colon P_k \to P_i$, there is an endomorphism v of P_i such that $\psi = v\varphi_{ik}$. For every $u \in \mathrm{Hom}(P_i, X)$, consider $u\psi \in \mathrm{Hom}(P_k, X)$. It is of the form $u'\varphi_{ik}$ for a uniquely determined $u' \in \mathrm{Hom}(P_i, X)$. The map $u \longmapsto u'$ from J to J thus defined by ψ is injective, since ψ is an epimorphism by e); it is therefore bijective because J is finite. The bijection $u \longmapsto u'$ from J to J thus defines an isomorphism $\alpha\colon X^J \xrightarrow{\sim} X^J$ making the diagram

$$\begin{array}{ccccc} P_k & \xrightarrow{\varphi_{ik}} & P_i & \xrightarrow{U} & X^J \\ \| & & & & \simeq \downarrow \alpha \\ P_k & \xrightarrow{\psi} & P_i & \xrightarrow{U} & X^J \end{array}$$

commutative. By the uniqueness properties of the factorization of a morphism as a composite of a strict epimorphism and a monomorphism, it follows (since ψ too is a strict epimorphism by e)) that one can find a morphism $v\colon P_i \to P_i$ which makes the diagram commutative, as required.

It follows in particular that *the Galois P_i form a cofinal subsystem of the system (P_j)*. Since for a Galois object P_i one has

$$\mathrm{Hom}(P, P_i) = \mathrm{Hom}(P_i, P_i) = \mathrm{Aut}(P_i),$$

passage to the limit therefore gives

$$\mathrm{Hom}(P, P) = \varprojlim_i \mathrm{Hom}(P, P_i) = \varprojlim_i \mathrm{Hom}(P_i, P_i) = \varprojlim_i \mathrm{Aut}(P_i),$$

where the inverse limit is taken over the Galois P_i. Moreover, under the 123
identification $\mathrm{Hom}(P, P_i) = F(P_i)$, and taking account of the fact that F sends epimorphisms to epimorphisms, one sees that the transition homomorphisms in the preceding inverse system are surjective. From all this we conclude:

h) *One has*

$$\mathrm{Hom}(P,P) = \mathrm{Aut}(P) = \varprojlim_i F(P_i) = \varprojlim_i \mathrm{Aut}(P_i),$$

where the inverse limit is taken over the Galois P_i.

In particular, $\mathrm{Aut}(P)$ appears as the inverse limit of an inverse system of finite groups (with surjective transition homomorphisms); we endow it with the inverse-limit topology of the discrete topologies. *We denote by* π *and call the fundamental group* (of $\mathcal{C}$ equipped with F) *the group opposite to* $\mathrm{Aut}(P)$. Thus this group acts *on the right* on P; it is the inverse limit of finite groups π_i acting on the right on the Galois P_i, where π_i is the group opposite to $\mathrm{Aut}(P_i)$.

In view of the functorial isomorphism

$$F(X) = \mathrm{Hom}(P,X)$$

and the definition of π, we therefore see that π acts *on the left* on $F(X)$, and moreover acts continuously by g) (for, with the notation of g), it is in fact π_i that acts on $F(X)$). It is immediate that, for every morphism $u\colon X \to Y$ in $\mathcal{C}$, the morphism $F(u)\colon F(X) \to F(Y)$ is compatible with the actions of π. *Henceforth one may therefore regard* F *as a covariant functor*

$$F\colon \mathcal{C} \to \mathcal{C}(\pi),$$

where $\mathcal{C}(\pi)$ *is the category of finite sets on which* π *acts continuously on the left.*

We now define a functor in the opposite direction:

$$G\colon \mathcal{C} \longleftarrow \mathcal{C}(\pi)$$

by the formula

$$G(E) = P \times_\pi E,$$

where $P \times_\pi E$ is defined as a solution of the universal problem summarized by

$$\mathrm{Hom}_{\mathcal{C}}(P \times_\pi E, X) \xrightarrow{\sim} \mathrm{Hom}_\pi(E, \mathrm{Hom}(P,X))$$

(where, in the right-hand side, $\mathrm{Hom}(P,X) = F(X)$ is regarded as a set on which π acts on the left). We must prove the existence of the object $P \times_\pi E$. 124

i) *Let* Q *be an object of* $\mathcal{C}$ *on which a finite group* G *acts on the right, and let* E *be a finite set on which* G *acts on the left. Then* $Q \times_G E$ *exists, and the canonical map*

$$F(Q) \times_G E \to F(Q \times_G E)$$

is an isomorphism.

Since finite coproducts exist in $\mathcal{C}$ by (G 2), and F commutes with them by (G 5), one is immediately reduced to the case in which G acts transitively on E: indeed, if the E_j are the orbits of G in E, then

$$Q \times_G E = \coprod_j Q \times_G E_j.$$

Let $a \in E$, and let H be its stabilizer. It follows at once from the definition that $Q \times_G E$ is identified with Q/H. Existence thus follows from (G 2), and the compatibility property for F follows from (G 5).

j) *Let* E *be an object of* $\mathcal{C}(\pi)$, *and let* P_i *be Galois and such that* π_i *already acts on* E. *Then* $P_i \times_{\pi_i} E$ *exists and there is a canonical isomorphism*

$$E \xrightarrow{\sim} F(P_i \times_{\pi_i} E).$$

If $j \geq i$ *is such that* P_j *is Galois, then the canonical homomorphism* $P_j \times_{\pi_j} E \to P_i \times_{\pi_i} E$ *is an isomorphism.*

The first assertion is a special case of i), taking account of the fact that π_i acts simply transitively on $F(P_i)$, which is equipped with the marked point φ_i;

this gives an isomorphism $F(P_i) \times_{\pi_i} E \simeq E$. For the second assertion one may, for example, use (G 6).

For every j, let $\mathcal{C}_j$ be the full subcategory of $\mathcal{C}$ formed by those X for which $\mathrm{Hom}(P_j, X) \to \mathrm{Hom}(P, X) \simeq F(X)$ is bijective. We know from g) that $\mathcal{C}$ *is the filtered union of the* $\mathcal{C}_j$. Thus, for $X \in \mathcal{C}_j$, one has

$$\begin{aligned}\mathrm{Hom}_\pi(E, \mathrm{Hom}(P, X)) &\simeq \mathrm{Hom}_\pi(E, \mathrm{Hom}(P_j, X)) \simeq \mathrm{Hom}_{\pi_j}(E, \mathrm{Hom}(P_j, X)) \\ &\simeq \mathrm{Hom}(P_j \times_{\pi_j} E, X),\end{aligned}$$

and, in view of the last assertion in j), one obtains an isomorphism functorial in the object X of $\mathcal{C}_j$: 125

$$\mathrm{Hom}_\pi(E, \mathrm{Hom}(P, X)) \simeq \mathrm{Hom}(P_i \times_{\pi_i} E, X).$$

Since this is true for every j, and since these functorial isomorphisms, as j varies, induce one another, we conclude:

k) *Under the conditions of* j), *the composite of the canonical morphisms*

$$E \to \mathrm{Hom}(P_i, P_i \times_{\pi_i} E) \to \mathrm{Hom}(P, P_i \times_{\pi_i} E)$$

makes $P_i \times_{\pi_i} E$ *a solution of the universal problem defining* $P \times_\pi E$, i.e. *the latter exists and there is an isomorphism*

$$P \times_\pi E \xrightarrow{\sim} P_i \times_{\pi_i} E.$$

This completes the construction of the functor $G(E)$. On the other hand, there is a functorial homomorphism

$$\alpha \colon \mathrm{id}_{\mathcal{C}(\pi)} \to FG,$$

that is, a homomorphism functorial in the object E of $\mathcal{C}(\pi)$:

$$\alpha(E) \colon E \to FG(E) = F(P \times_\pi E),$$

namely the composite of the canonical morphisms

$$E \to F(P) \times_\pi E \to F(P \times_\pi E)$$

(where the first comes from the marked point $\varphi \in F(P)$). Combining j) and k), one finds:

l) *The homomorphism* α *is an isomorphism.*

Similarly, one defines a functorial homomorphism

$$\beta \colon GF \to \mathrm{id}_{\mathcal{C}},$$

that is, a homomorphism functorial in the object X of $\mathcal{C}$:

$$\beta(X) \colon P \times_\pi F(X) \to X,$$

as the morphism associated with the π-homomorphism 126

$$F(X) \to \mathrm{Hom}(P, X)$$

inverse to the canonical isomorphism $\mathrm{Hom}(P, X) \xrightarrow{\sim} F(X)$.

m) *The composites*

$$F(X) \xrightarrow{\alpha(F(X))} FGF(X) \xrightarrow{F(\beta(X))} F(X)$$

$$G(E) \xrightarrow{G(\alpha(E))} GFG(E) \xrightarrow{\beta(G(E))} G(E)$$

are the identity isomorphisms.

A trotting donkey.

In view of l), it follows that:

n) *The homomorphism* β *is an isomorphism.*

We have thus obtained the promised result:

THEOREM V.4.1. *Let $\mathcal{C}$ be a category satisfying conditions* (G 1), (G 2), *and* (G 3) *at the beginning of this section, and let F be a covariant functor from $\mathcal{C}$ to the category of finite sets, satisfying conditions* (G 4), (G 5), *and* (G 6). *Then the preceding canonical constructions define equivalences of categories $F\colon \mathcal{C} \to \mathcal{C}(\pi)$ and $G\colon \mathcal{C}(\pi) \to \mathcal{C}$ which are quasi-inverse to one another. More precisely, there is a pro-object P of $\mathcal{C}$ and a functorial isomorphism $F(X) \xleftarrow{\sim} \mathrm{Hom}(P, X)$; π is the group opposite to the automorphism group of P, endowed with a suitable topology, so that π acts continuously on the sets $\mathrm{Hom}(P, X) \simeq F(X)$. Finally, G is given by $G(E) \simeq P \times_\pi E$.*

REMARKS V.4.2. The statement of conditions (G 1) through (G 6) becomes simpler and more pleasing if (G 2) and (G 5) are replaced respectively by:

(G' 2) Finite direct limits exist in $\mathcal{C}$.

(G' 5) The functor F is right exact (i.e. it commutes with finite direct limits).

These conditions appear stronger than (G 2) and (G 5), but it follows at once from the structure theorem V.4.1 that they are implied by (G 1) through (G 6). Note, however, that in the cases of interest to us, checking (G 2) and (G 5) does in fact seem simpler than checking (G' 2) and (G' 5). I do not know whether, in condition (G 3), the requirement that u'' be an isomorphism onto a direct summand of Y could be omitted.

V.5. Galois categories

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DEFINITION V.5.1. *A* Galois category *is a category $\mathcal{C}$ equivalent to a category $\mathcal{C}(\pi)$, where π is a compact group that is a projective limit of finite groups (i.e. is totally disconnected).*

For the definition of $\mathcal{C}(\pi)$, see the beginning of no. V.4. By Theorem V.4.1, $\mathcal{C}$ is Galois if and only if it satisfies conditions (G 1) through (G 3), and there exists a functor F from $\mathcal{C}$ to the category of finite sets satisfying conditions (G 4) through (G 6), that is, one which is exact *and* conservative, *in general terminology. Such a functor is called a* fundamental functor *of the Galois category $\mathcal{C}$*[6]*; it is pro-representable by a pro-object which we denote by P_F. A pro-object P for which the associated functor F is fundamental is called a* fundamental pro-object. *In this way, the category of fundamental functors on $\mathcal{C}$ is anti-equivalent to the category of fundamental pro-objects. If F and P correspond, the group* $\mathrm{Aut}\, F$ *is therefore isomorphic to the opposite of the group* $\mathrm{Aut}\, P$*; hence the group denoted by π in the preceding section is none other than* $\mathrm{Aut}\, P$. *Recall that in the preceding section, starting from a* given *fundamental functor F, we constructed an equivalence of $\mathcal{C}$ with $\mathcal{C}(\pi)$, where $\pi = \mathrm{Aut}(F)$, which carries F to the canonical functor from $\mathcal{C}(\pi)_x$ to the category of finite sets. In the standard case $\mathcal{C} = \mathcal{C}(\pi)$, with F the canonical functor, the fundamental pro-object associated with F is precisely the projective system of discrete quotients π_i of π.*

It may be useful to describe explicitly the category of pro-objects of $\mathcal{C}(\pi)$. We obtain:

PROPOSITION V.5.2. *The category* Pro-$\mathcal{C}(\pi)$ *is canonically equivalent to the category $\mathcal{C}'(\pi)$ of spaces, equipped with a topological group of operators π, which are compact and totally disconnected.*

The latter category contains $\mathcal{C}(\pi)$ as a full subcategory, corresponding to compact discrete spaces with operators, and projective limits exist in it. There is therefore a canonical functor

$$g\colon \text{Pro-}\mathcal{C}(\pi) \to \mathcal{C}'(\pi),$$

which associates to a projective system $Q = (Q_i)$ the object $X = \varprojlim_i Q_i$ of $\mathcal{C}'(\pi)$. To define a functor in the opposite direction, it is enough to define a contravariant functor

[6] It seems preferable to adopt the more expressive term "fiber functor."

from $\mathcal{C}'(\pi)$ to the category of left-exact functors $\mathcal{C} \to \mathbf{Ens}$. For every $X \in \mathcal{C}'(\pi)$, take

$$h(X)(E) = \mathrm{Hom}(X, E)$$

(where Hom is taken in $\mathcal{C}'(\pi)$). It follows immediately from the definitions that h and g 128
are adjoint, and that hg is canonically isomorphic to the identity functor of Pro-$\mathcal{C}(\pi)$. To prove that g and h are quasi-inverse, it remains to show that every object of $\mathcal{C}'(\pi)$ is isomorphic to an object of the form $g(Q)$, with $Q \in \text{Pro-}\mathcal{C}(\pi)$. In other words, *every compact, totally disconnected space X equipped with a topological group of operators π is isomorphic to a projective limit of finite discrete spaces with operators.*

As a topological space without operators, X is the projective limit of its finite discrete quotients. We are therefore reduced to showing that among these quotients there is a cofinal system invariant under π. For a given quotient X', it is enough to show that the set of its transforms under the operations of π is finite; one then takes the supremum of these transforms, which is an invariant quotient dominating X'. Equivalently, it is enough to find an open normal subgroup π' of π whose elements act trivially on X'. The quotient X' amounts to a finite partition of X into open subsets X_i. By continuity and compactness of π, there exists a neighborhood V of the identity of π such that $s \in V$ implies $s \cdot X_i \subset X_i$ for every i; thus s acts trivially on X'. The open normal subgroups of π form a fundamental system of neighborhoods of the identity, which completes the proof.

Notice, more simply, that the category $\mathrm{Ind}\,\mathcal{C}(\pi)$ is canonically equivalent to the category of sets on which π acts continuously. We shall not need this fact here.

Proposition V.5.3. *Let $\mathcal{C}$ be a Galois category, let F be a fundamental functor on $\mathcal{C}$, and let $P = (P_i)$ be the associated pro-object, normalized in the usual way. Let $X \in \mathcal{C}$; then X is connected if and only if π acts transitively on $E = F(X)$.*

We reduce to the standard case $\mathcal{C} = \mathcal{C}(\pi)$, with F the canonical functor, where the assertion is immediate.

Corollary V.5.4. *The following conditions on X are equivalent:* (i) *X is connected and $\not\simeq \emptyset_{\mathcal{C}}$;* (ii) *the group π acts transitively on $E = F(X)$, and $F(X) \neq \emptyset$;* (iii) *X is isomorphic to some P_i.*

The equivalence of (i) and (iii) also follows readily from no. V.4, e). 129

Proposition V.5.5. *Let $Q = (Q_i)_{i \in I}$ be a pro-object of $\mathcal{C}$, normalized in the usual way, and let G be the corresponding functor $G(X) = \mathrm{Hom}(Q, X)$ from $\mathcal{C}$ (to* **Ens***). The following conditions are equivalent:*

(i) *G commutes with finite direct sums;*
(ii) *G commutes with the direct sum of two objects;*
(iii) *the Q_i are connected and $\not\simeq \emptyset_{\mathcal{C}}$;*
(iv) *Q is isomorphic to π/H, where H is a closed subgroup of π;*
(v) *the functor G is isomorphic to the functor $E \longmapsto E^H$, the set of H-invariants, defined by a closed subgroup H of π.*

N.B. In conditions (iv) and (v), a fundamental functor is understood to have been chosen, thereby identifying $\mathcal{C}$ with the category $\mathcal{C}(\pi)$.

Proof. We may suppose that $\mathcal{C} = \mathcal{C}(\pi)$. The implication (i)$\Longrightarrow$(ii) is immediate, and (ii)$\Longrightarrow$(iii) is proved as in property d) of no. V.4. Let us prove (iii)$\Longrightarrow$(iv). The set $\varprojlim_i Q_i$ is nonempty, being the projective limit of nonempty finite sets. Let $a \in \varprojlim_i Q_i$. It defines a homomorphism of spaces with operators

$$\pi \to Q$$

which is *surjective*: for every i, the composite $\pi \to Q \to Q_i$ is surjective, since π acts transitively on Q_i by Proposition V.5.3. If H is the stabilizer of a, we obtain an isomorphism $\pi/H \xrightarrow{\sim} Q$. The implications (iv)$\Longrightarrow$(v) and (v)$\Longrightarrow$(i) are again immediate.

PROPOSITION V.5.6. *Let $\mathcal{C}$ be a Galois category, let P be a fundamental pro-object of $\mathcal{C}$, and let F be the associated fundamental functor. Let $P' = (P'_i)_{i\in I}$ be a pro-object of $\mathcal{C}$ in normal form, and let F' be the associated functor $F'(X) = \mathrm{Hom}(P', X)$ from $\mathcal{C}$ to* **Ens**. *The following conditions are equivalent:*

(i) *$P' \simeq P$, or equivalently $F' \simeq F$;*
(ii) *P' is fundamental, or equivalently F' is fundamental;*
(iii) *F' carries a direct sum of two objects to their direct sum, and $X \not\simeq \emptyset_{\mathcal{C}}$ implies $F'(X) \neq \emptyset$;*
(iv) *the connected objects of $\mathcal{C}$ distinct from $\emptyset_{\mathcal{C}}$ are exactly those isomorphic to some P'_i.*

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We have trivially (i)$\Longrightarrow$(iii) and (i)$\Longrightarrow$(ii); moreover, (ii)$\Longrightarrow$(iv) by Corollary V.5.4, applied to P' in place of P. Furthermore, either (iii) or (iv), by Proposition V.5.5, implies that P' is of the form π/H, where H is a closed subgroup of π. In case (iii), for every open normal subgroup π' of π there is a π-homomorphism $P' = \pi/H \to \pi/\pi'$. Hence $H \subset \pi'$, so $H = (0)$, and consequently (i) holds. This proves the assertion.

COROLLARY V.5.7. *Let $\mathcal{C}$ be a Galois category. Its fundamental pro-objects are isomorphic, and its fundamental functors are isomorphic.*

In other words, *the category of fundamental functors is a connected groupoid* Γ, which may be called the *fundamental groupoid* of the Galois category $\mathcal{C}$. If $\mathcal{C} = \mathcal{C}(\pi)$, the automorphism group of an object of the fundamental groupoid is isomorphic to π, this isomorphism being well determined up to an inner automorphism. (Recall that a *groupoid* is a category in which every morphism is an isomorphism, and a *connected* groupoid is a groupoid all of whose objects are isomorphic.) The fundamental pro-objects of $\mathcal{C}$ form a connected groupoid equivalent to the *opposite* of the fundamental groupoid. If F, F' are two fundamental functors, associated with fundamental pro-objects P, P', then $\mathrm{Hom}(F, F') = \mathrm{Isom}(F, F')$ is sometimes denoted by $\pi_{F',F}$ and plays the role of a "set of *homotopy classes of paths* from F to F'". In particular, $\pi_{F,F} = \pi_F$ is precisely the *fundamental group of $\mathcal{C}$ at F* constructed in the preceding section. The pro-object P associated with F plays the role of a *universal covering at F* of the final object $e_{\mathcal{C}}$ of $\mathcal{C}$.

It may be convenient to have a description of $\mathcal{C}$, up to equivalence, in terms of its fundamental groupoid Γ, without choosing a particular object F of the latter. To every object X of $\mathcal{C}$ is associated a functor E_X on the fundamental groupoid, defined by

$$E_X(F) = F(X),$$

with values in **Ens**. Such a functor is known in topology as a "local system" on the groupoid. The set $F(X) = E_X(F)$ may be called the *fiber* of X at F, and E_X the fiber functor associated with X. The functor E_X has the following property: *for every F,* 131
$E_X(F)$ is a finite set on which the topological group $\pi_F = \mathrm{Aut}(F)$ acts continuously. For a given covariant functor ξ from the fundamental groupoid to **Ens**, the preceding condition is equivalent to the same condition for any one fixed F. With this understood:

PROPOSITION V.5.8. *The functor $X \longmapsto E_X$ is an equivalence from the category $\mathcal{C}$ to the category of covariant functors from the fundamental groupoid Γ of $\mathcal{C}$ to* **Ens** *which satisfy the condition highlighted above.*

Indeed, let F_0 be an object of the fundamental groupoid and put $\pi_0 = \mathrm{Aut}(F_0)$. The functor $\xi \longmapsto \xi(F_0)$ is an equivalence from the second category in Proposition V.5.8 to the category $\mathcal{C}(\pi_0)$, as is seen immediately. On the other hand, its composite with the functor $X \longmapsto E_X$ is the natural equivalence $\mathcal{C} \to \mathcal{C}(\pi_0)$. It follows that $X \longmapsto E_X$ itself is an equivalence.

COROLLARY V.5.9. *The category* Pro-$\mathcal{C}$ *is canonically equivalent to the category of covariant functors ξ from the fundamental groupoid Γ to the category of topological spaces*

satisfying the following condition: for every object F of Γ, $\xi(F)$ is a compact, totally disconnected space equipped with a topological group of operators π_F.

Here again, it is enough to verify this condition on ξ for a single F. The proof is the same as that of Proposition V.5.8.

REMARKS V.5.10. Let $(F_s)_{s\in S}$ be a family of objects of the fundamental groupoid Γ. For $s, s' \in S$, put

$$\mathrm{Hom}(s, s') = \mathrm{Hom}(F_s, F_{s'}).$$

The set S thereby becomes a connected groupoid, and the map $s \longmapsto F_s$ becomes a fully faithful functor $f\colon S \to \Gamma$. Consider the functor $X \longmapsto E_X \circ f$ from $\mathcal{C}$ to the functor category $\mathrm{Hom}(S, \mathbf{Ens})$. This gives a variant of Proposition V.5.8, and of Corollary V.5.9, with Γ replaced by S. The resulting statement reduces to Theorem V.4.1 when S consists of one point, and is precisely Proposition V.5.8 itself when S is the set of objects of Γ.

We shall use Corollary V.5.9 to define a canonical pro-object of $\mathcal{C}$. To this end, consider the functor from Γ to the category of topological spaces and indeed of topological 132
groups:

$$f\colon F \longmapsto \mathrm{Aut}(F) = \pi_F.$$

This functor satisfies the condition in Proposition V.5.8. The space with operators $f(F)$ under π_F is simply π_F itself, regarded as a space with operators under itself by inner automorphisms. Thus f corresponds to a pro-object of $\mathcal{C}$ determined up to unique isomorphism. It is in fact a pro-group in $\mathcal{C}$, called the *fundamental pro-group of* $\mathcal{C}$, and plays the role of a local system of fundamental groups. Thus it is a pro-group Π of $\mathcal{C}$ defined by the existence of an isomorphism, functorial in F,

$$F(\Pi) \simeq \pi_F.$$

If X is any pro-object of $\mathcal{C}$, there is a canonical morphism

$$\Pi \times X \to X$$

which makes X an object with a left group of operators Π in Pro-$\mathcal{C}$. It is enough to observe that, for variable F, there is a canonical map

$$\Pi(F) \times X(F) \to X(F),$$

that is,

$$\mathrm{Aut}(F) \times E_X(F) \to E_X(F), \quad \text{or} \quad \pi_F \times F(X) \to F(X),$$

which is functorial in F. It is also functorial in X, so for every morphism $X \to Y$ of pro-objects the corresponding diagram

$$\begin{array}{ccc} \Pi \times X & \longrightarrow & X \\ \downarrow & & \downarrow \\ \Pi \times Y & \longrightarrow & Y \end{array}$$

is commutative.

REMARKS V.5.11. One must not confuse a fundamental pro-object P, which has no group structure and is connected, with the fundamental pro-group Π, which is a pro-*group* and is generally not connected. More precisely, Π is connected if and only if π_F, acting on itself by inner automorphisms, acts transitively; that is, if and only if π consists only of the identity, or equivalently if $\mathcal{C}$ is equivalent to the category of finite sets. Another essential difference is that Π is determined up to unique isomorphism, whereas P is determined 133
only up to nonunique isomorphism.

Let E be a finite set and consider the constant functor on the groupoid Γ with value E. By Proposition V.5.8, it defines an object of $\mathcal{C}$, denoted by $E_{\mathcal{C}}$, which may also be interpreted as the direct sum of E copies of the final object $e_{\mathcal{C}}$ of $\mathcal{C}$. We may regard

$E_{\mathcal{C}}$ as a functor of E, from the category of finite sets to $\mathcal{C}$. This functor is *exact*, and hence carries finite groups to $\mathcal{C}$-groups, etc. If an object X of $\mathcal{C}$ carries a right action of a finite group G, we may therefore regard X as an object of $\mathcal{C}$ with a right group of operators, the $\mathcal{C}$-group $G_{\mathcal{C}}$. Extending the general terminology for objects with $\mathcal{C}$-groups of operators, we say that X is *formally principal homogeneous* under G if it is formally principal homogeneous under $G_{\mathcal{C}}$, that is, if the canonical morphism

$$X \times G_{\mathcal{C}} \to X \times X$$

induced by the right action of $G_{\mathcal{C}}$ on X is an isomorphism. We say that X is *principal homogeneous* under G if it is so under $G_{\mathcal{C}}$, that is, if it is formally principal homogeneous and, in addition, the quotient $X/G = X/G_{\mathcal{C}}$ is $e_{\mathcal{C}}$.

Fix a fundamental functor, and hence an equivalence of $\mathcal{C}$ with a category $\mathcal{C}(\pi)$. The object X corresponds to a set $E = F(X)$ on which π acts continuously on the left. Giving a right action of G on X is then equivalent to giving a right action of G on the set E which commutes with the action of π. It follows at once that X is principal homogeneous under G if and only if E is a principal homogeneous space under G, that is, if and only if G acts simply transitively on E. Moreover, X is formally principal homogeneous $if and only if$ E is principal homogeneous *or* empty. Comparing with Proposition V.5.3, we see that if X is principal homogeneous under G *and connected*, then the given homomorphism from G to the group opposite to $\mathrm{Aut}(X)$ is an *isomorphism*. Furthermore, an object X of $\mathcal{C}$ is connected and principal homogeneous under the group opposite to $\mathrm{Aut}(X)$ if and only if, with the notation of no. V.4, it is isomorphic to a *Galois* P_i. In the standard case $\mathcal{C} = \mathcal{C}(\pi)$, this means that X is isomorphic to a quotient of π by a normal subgroup.

Continue to fix a fundamental functor F. The data of an object X principal homo- 134
geneous under a finite group G acting on the right, together with a point $a \in F(X)$, are equivalent to the data of a homomorphism from π to G. Indeed, to such a homomorphism we associate the set $E = G$, letting π act on the left through the given homomorphism $\pi \to G$ and the left translations of G, and letting G act on the right by right translations; the marked point a of E is the identity element of G. By the preceding discussion, this gives, in an essentially unique way, every triple (X, G, a) with the properties considered above, since a pointed principal homogeneous space under a group G is identified with G itself. In this way we obtain a direct geometric interpretation of the functor $G \longmapsto \mathrm{Hom}(\pi, G)$ from the category of finite groups to **Ens**. This functor is pro-representable by π, and its consideration would therefore provide another construction of the group π associated with F.

V.6. Exact functors from one Galois category to another

PROPOSITION V.6.1. *Let $\mathcal{C}$ and $\mathcal{C}'$ be two Galois categories, let $H\colon \mathcal{C} \to \mathcal{C}'$ be a covariant functor, let F' be a fundamental functor on $\mathcal{C}'$, and set $F = F' \circ H$. The following conditions are equivalent:*

(i) *H is* exact, *i.e. both left exact and right exact.*
(ii) *H is left exact, carries finite sums to finite sums, and epimorphisms to epimorphisms (or, equivalently, carries objects not isomorphic to $\emptyset_{\mathcal{C}}$ to objects not isomorphic to $\emptyset_{\mathcal{C}'}$).*
(iii) *F is a fundamental functor on $\mathcal{C}$.*

The implication (i)$\Longrightarrow$(ii) is a general categorical fact. Moreover, the first form of (ii) implies the second. Indeed, if X is an object of $\mathcal{C}$, then $X \not\approx \emptyset_{\mathcal{C}}$ if and only if the morphism $X \to e_{\mathcal{C}}$ is an epimorphism; note also that, since F is assumed left exact, it carries $e_{\mathcal{C}}$ to $e_{\mathcal{C}'}$. The second form of (ii) implies (iii), since F, being left exact and therefore pro-representable, satisfies criterion (iii) of V.5.6. Finally, (iii) implies (i), because F is exact and conservative (i.e. it satisfies axiom (G 6) of no. V.4).

Let Γ now be the fundamental groupoid of $\mathcal{C}$, and Γ' that of $\mathcal{C}'$. Thus, if H is exact, 135

then $F' \longmapsto F' \circ H$ is a functor from the groupoid Γ' to the groupoid Γ, which we denote by ${}^tH$:

$$ {}^tH(F')(X) = F'(H(X)). $$

Using the notation $F(X) = E_X(F)$ introduced in no. V.6, this may also be written

$$ E_{H(X)}(F') = E_X({}^tH(F')). $$

This last formula, together with V.5.8 or V.4.1, shows that the exact functor H is determined up to unique isomorphism once the corresponding functor ${}^tH$ is known. Fix F', and set $F = {}^tH(F')$. Then ${}^tH$ defines a homomorphism from $\pi_{F'} = \mathrm{Aut}(F')$ to $\pi_F = \mathrm{Aut}(F)$:

$$ {}^tH\colon \pi_{F'} \longrightarrow \pi_F \qquad (F = {}^tH(F') = F' \circ H). $$

The preceding formula also shows, in view of V.5.8, that this homomorphism has the following property: for every finite set E on which π_F acts continuously, the group $\pi_{F'}$ also acts *continuously* through the preceding homomorphism $\pi_{F'} \to \pi_F$. Applying this to the quotients of π_F by its open normal subgroups, one sees that the preceding condition also means that the homomorphism in question is continuous.

Conversely, let F be an object of Γ, let F' be an object of Γ', and let

$$ u\colon \pi_{F'} \longrightarrow \pi_F $$

be a continuous homomorphism. It determines a manifestly exact functor from $\mathcal{C}(\pi)$ to $\mathcal{C}(\pi')$, and hence, by V.4.1, an exact functor $H\colon \mathcal{C} \to \mathcal{C}'$ for which ${}^tH\colon \pi_{F'} \to \pi_F$ is precisely u. Instead of starting with a group homomorphism, one may start with a *functor*

$$ U\colon \Gamma' \longrightarrow \Gamma $$

such that, for *every* $F' \in \Gamma'$ (or for *one* $F' \in \Gamma'$, which amounts to the same thing), the corresponding homomorphism $\pi_{F'} \to \pi_F$ is continuous. Every such functor is isomorphic to one of the form ${}^tH$, where $H\colon \mathcal{C} \to \mathcal{C}'$ is an exact functor determined up to unique isomorphism. Thus:

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Corollary V.6.2. *A functor $H\colon \mathcal{C} \to \mathcal{C}'$ between Galois categories is exact if and only if there exist equivalences $\mathcal{C}(\pi) \to \mathcal{C}$ and $\mathcal{C}' \to \mathcal{C}(\pi')$ which carry H to the functor $\mathcal{C}(\pi) \to \mathcal{C}(\pi')$ associated with a homomorphism of topological groups $\pi' \to \pi$.*

Corollary V.6.3. *Let $\mathcal{C}$ and $\mathcal{C}'$ be two Galois categories, with fundamental groupoids Γ and Γ'. Then the category of exact functors from $\mathcal{C}$ to $\mathcal{C}'$ is equivalent to the category of functors $U\colon \Gamma' \to \Gamma$ having the following property: for every F' in Γ' (or for one F' in Γ', which amounts to the same thing), if $F = U(F')$, then the homomorphism*

$$ \pi_{F'} = \mathrm{Aut}(F') \longrightarrow \pi_F = \mathrm{Aut}(F) $$

defined by U is continuous.

Consider the fundamental pro-group Π of $\mathcal{C}$. An exact functor H carries it to a pro-group $H(\Pi)$ in $\mathcal{C}'$. We shall define a homomorphism

$$ \Pi' \longrightarrow H(\Pi), $$

where Π' is the fundamental pro-group of $\mathcal{C}'$, by requiring that, for every object F' of Γ', the corresponding homomorphism

$$ F'(\Pi') = \pi_{F'} \longrightarrow F'(H(\Pi)) = \pi_F \qquad (\text{where } F = F' \circ H = {}^tH(F')) $$

be the natural homomorphism

$$ \mathrm{Aut}(F') \longrightarrow \mathrm{Aut}(F' \circ H). $$

Since the latter is functorial in F', it does indeed define, by V.5.8, a homomorphism of pro-objects, and in fact of pro-groups, in $\mathcal{C}'$. This homomorphism is said to be *associated* with the functor H.

Now let H' be a second exact functor, from the Galois category $\mathcal{C}'$ to a Galois category $\mathcal{C}''$. It is immediate that

$$^t(H'H) = {}^tH\,{}^tH'$$

(N.B. this is an identity of functors, not merely a canonical isomorphism). There is an analogous transitivity property for the associated homomorphisms of fundamental pro-groups.

We now interpret the properties of the exact functor H in terms of the corresponding 137
homomorphism

$$u\colon \pi_{F'} \longrightarrow \pi_F \qquad (\text{where } F = F' \circ H).$$

It is convenient to introduce the notion of a *pointed object* of the Galois category $\mathcal{C}$, equipped with its fundamental functor F. By definition, this is an object X of $\mathcal{C}$ together with an element a of $F(X)$. It may therefore be viewed as a finite set on which π_F acts continuously on the left, together with a point a. Consequently, by V.5.3, the *connected* pointed objects of $\mathcal{C}$ are identified with the open subgroups of π_F. If U and V are two such subgroups, corresponding to connected pointed objects X and Y of $\mathcal{C}$, then a pointed homomorphism from X to Y exists if and only if $U \subset V$, and it is then unique.

Of course, the functor H carries pointed objects to pointed objects, since $F = F' \circ H$. Also note that a closed subgroup of a group such as π_F is the intersection of the open subgroups containing it. Thus, if M and N are two closed subgroups, then $M \subset N$ if and only if every open subgroup containing N also contains M. These remarks easily yield the following results:

Proposition V.6.4. *Let X be a connected pointed object of $\mathcal{C}$, associated with an open subgroup U of π_F. Then U contains $u(\pi_{F'})$ (respectively, the closed normal subgroup generated by $u(\pi_{F'})$) if and only if $H(X)$ admits a pointed section (respectively, is completely decomposed).*

By a *section* of an object X of a Galois category $\mathcal{C}$ we mean, implicitly over the final object, a morphism from the final object $e_{\mathcal{C}}$ to X. Equivalently, it is an element a of $F(X)$ fixed by π_F. If X is pointed, this is called a *pointed section* when it is compatible with the pointed structures on X and $e_{\mathcal{C}}$, i.e. when a is precisely the distinguished element of $F(X)$. Such a section is therefore unique, and exists if and only if the distinguished element of $F(X)$ is fixed by π_F. Finally, an object of a Galois category is said to be *completely decomposed* if it is isomorphic to a sum of final objects, i.e. if π_F acts trivially on $F(X)$. For a pointed X, this condition is clearly stronger than the existence of a pointed section. Proposition V.6.4 follows immediately from the preceding definitions and remarks.

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Corollary V.6.5. *The homomorphism u is trivial if and only if $H(X)$ is completely decomposed for every object X of $\mathcal{C}$.*

Proposition V.6.6. *Let X' be a connected pointed object of $\mathcal{C}'$, associated with an open subgroup U' of $\pi_{F'}$. Then U' contains* $\operatorname{Ker} u$ *if and only if there exist a connected pointed object X of $\mathcal{C}$ and a pointed homomorphism from the distinguished connected component X'_0 of $H(X)$ to X'. Equivalently, X' is isomorphic, as a pointed object, to a quotient of the distinguished connected component of the inverse image of a pointed object of $\mathcal{C}$. If u is surjective, the preceding condition is also equivalent to the following: X' is isomorphic to an object $H(X)$, where X is a pointed object of $\mathcal{C}$.*

(The *distinguished connected component* of a pointed object X of a Galois category $\mathcal{C}$ is the unique connected pointed subobject of X. By V.5.3, it corresponds to the orbit under π_F of the distinguished point of $F(X)$.) Since the fact that U' contains $\operatorname{Ker} u$ does not depend on the chosen pointing of X' (another pointing replaces U' by a conjugate subgroup), we obtain:

COROLLARY V.6.7. *The subgroup U' contains* $\operatorname{Ker} u$ *if and only if there exist an object X of $\mathcal{C}$, which may be assumed connected, and a morphism from a connected component of $H(X)$ to X'. If u is surjective, this also says that X' is isomorphic to an object of the form $H(X)$.*

COROLLARY V.6.8. *The homomorphism u is injective if and only if, for every object X' of $\mathcal{C}'$, there exist an object X of $\mathcal{C}$ and a homomorphism from a connected component of $H(X)$ to X'.*

PROPOSITION V.6.9. *The following conditions are equivalent:*

(i) *The homomorphism $u\colon \pi_{F'} \to \pi_F$ is surjective.*
(ii) *For every connected object X of $\mathcal{C}$, the object $H(X)$ is connected.*
(iii) *The functor H is fully faithful.*

The last condition means that, for any two objects X, Y of $\mathcal{C}$, the natural map

$$\operatorname{Hom}(X, Y) \longrightarrow \operatorname{Hom}(H(X), H(Y))$$

is bijective.

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COROLLARY V.6.10. *The homomorphism u is an isomorphism if and only if H is an equivalence of categories, or equivalently if the following two conditions hold:*

a) *for every connected object X of $\mathcal{C}$, the object $H(X)$ is connected;*
b) *every object of $\mathcal{C}'$ is isomorphic to an object of the form $H(X)$.*

PROPOSITION V.6.11. *Let $H\colon \mathcal{C} \to \mathcal{C}'$ and $H'\colon \mathcal{C}' \to \mathcal{C}''$ be exact functors between Galois categories. Let F'' be a fundamental functor on $\mathcal{C}''$, set $F' = F''H'$ and $F = F'H$, and consider the associated homomorphisms*

$$u'\colon \pi_{F''} \longrightarrow \pi_{F'} \qquad \text{and} \qquad u\colon \pi_{F'} \longrightarrow \pi_F.$$

The inclusion $\operatorname{Ker} u \supset \Im u'$, *or equivalently the triviality of uu', holds if and only if $H'(H(X))$ is completely decomposed for every object X of $\mathcal{C}$. The inclusion* $\operatorname{Ker} u \subset \Im u'$ *holds if and only if, for every connected pointed object X' of $\mathcal{C}'$ such that $H'(X')$ admits a pointed section, there exist an object X of $\mathcal{C}$ and a homomorphism from a connected component of $H(X)$ to X'.*

The first assertion follows from the last assertion of V.6.4. The second follows by combining V.6.4 and V.6.6.

REMARKS V.6.12. Under the conditions of V.6.8, it is not true in general that X' is isomorphic to an object of the form $H(X)$. One can show that every connected object, and hence every object, of $\mathcal{C}'$ is isomorphic to an object of the form $H(X)$ if and only if u is an isomorphism from $\pi_{F'}$ onto a *direct-factor* subgroup of π_F. In practice, however, one constructs directly a right inverse $\pi_F \to \pi_{F'}$ of u, using a suitable exact functor from $\mathcal{C}'$ to $\mathcal{C}$.

PROPOSITION V.6.13. *Let $\mathcal{C}$ be a Galois category equipped with a fundamental functor F, let S be a connected object of $\mathcal{C}$, and let $\mathcal{C}'$ be the category of objects of $\mathcal{C}$ over S. Then $\mathcal{C}'$ is a Galois category, and the functor*

$$X \longmapsto H(X) = X \times S$$

from $\mathcal{C}$ to $\mathcal{C}'$ is exact. Let $a \in F(S)$, and let F' be the functor from $\mathcal{C}'$ to the category of finite sets defined by

$$F'(X') = \text{the inverse image of } a \text{ under } F(X') \longrightarrow F(S).$$

Then there is an isomorphism $F \cong F' \circ H$, and the corresponding homomorphism

$$u\colon \pi_{F'} \longrightarrow \pi_F$$

is an isomorphism from $\pi_{F'}$ onto the open subgroup U of π_F stabilizing the distinguished 140

element a of $F(S)$.

The proof is left to the reader.

V.7. The case of preschemes

Let S be a connected locally noetherian prescheme, and let

$$a\colon \operatorname{Spec}(()\Omega) \longrightarrow S$$

be a geometric point of S, with values in an algebraically closed field Ω. Set

$$\mathcal{C} = \text{the category of étale coverings of } S,$$

and, for an object X of $\mathcal{C}$, i.e. an étale covering X of S, set

$$F(X) = \text{the set of geometric points of } X \text{ lying over } a.$$

Thus F becomes a functor from $\mathcal{C}$ to the category of finite sets. Properties (G 1) through (G 6) hold: (G 1) is included among the standard consequences in I I.4.6; (G 2) follows from V.3.4; (G 3) from V.3.5; (G 4) is immediate from the definition; (G 5) follows from V.3.5 and the beginning of no. V.2; and (G 6) is proved in V.3.7. We may therefore apply the results of no. V.4, V.5, and V.6.

In particular, this defines a pro-object P of $\mathcal{C}$ representing F, called the *universal covering of S at the point a*, and a topological group $\pi = \operatorname{Aut}(F) = \operatorname{Aut}^0 P$, called the *fundamental group of S at a*, and denoted by $\pi_1(S, a)$. The functor F then defines an equivalence between $\mathcal{C}$ and the category of finite sets on which $\pi = \pi_1(S, a)$ acts continuously. This equivalence interprets the usual finite projective- and inductive-limit operations on coverings—products, fiber products, sums, passage to quotients, etc.—in terms of the analogous operations in $\mathcal{C}(\pi)$, i.e. the evident operations on finite sets with a π-action. Since the topological connected components of an étale covering are themselves étale coverings, note also that *an object X of $\mathcal{C}$ is connected in $\mathcal{C}$ if and only if it is topologically connected.* By V.5.3, this says that π_1 acts transitively on $F(X)$. 141

For an object X of $\mathcal{C}$ to be faithfully flat and quasi-compact over S—it is already flat and quasi-compact over S—it is necessary and sufficient that $X \to S$ be surjective, i.e. an epimorphism in $\mathcal{C}$, or equivalently that $X \neq \emptyset$. Criterion V.2.6 (iii) therefore shows that *X is a principal covering of S with group G if and only if it is a principal homogeneous space under G in the category $\mathcal{C}$*, in the sense defined in no. V.5.

If a' is another geometric point of S, corresponding to an algebraically closed field Ω', which may differ from Ω and may even have a different characteristic, then it defines another fiber functor $F' = F_{a'}$ from $\mathcal{C}$ to the category of finite sets. This functor is again exact, and hence is isomorphic to $F = F_a$. Consequently, the fundamental groups $\pi_1(S; a)$ for varying a are isomorphic to one another.

Let $\pi_1(S; a, a')$ denote the set of isomorphisms—or, equivalently, the set of homomorphisms—from F_a to $F_{a'}$ between the associated fiber functors. We thereby obtain a *groupoid* whose objects are the geometric points of S, the fundamental groups being the automorphism groups of the objects of this groupoid. The set $\pi_1(S; a', a)$ may be called the *set of homotopy classes of paths from a to a'*. These classes compose in the evident way.

Finally, one may define a pro-group Π_1^S in $\mathcal{C}$, called the *fundamental pro-group of S* or the *local system of fundamental groups on S*, characterized up to unique isomorphism by the existence of an isomorphism, functorial in the geometric point a of S,

$$F_a(\Pi_1^S) = \pi_1(S; a)$$

(cf. Remark V.5.10). In particular, if s is an ordinary point of S, the fiber of Π_1^S at s is a pro-group over $\kappa(s)$, a projective limit of finite étale groups over $\kappa(s)$. One might call this pro-group the *fundamental group of S at the ordinary point s* and denote it by $\pi_1(S, s)$. By definition, its points with values in an algebraically closed extension Ω of $\kappa(s)$ are the elements of $\pi_1(S; a)$, where a is the geometric point of S defined by that extension.

In particular, by taking S to be the spectrum of a field, every field k has canonically and functorially associated with it a pro-group over k, which one might denote by $\pi_1(k)$. It is a projective limit of finite étale groups over k, and its points in an algebraically closed extension Ω of k identify with the elements of the topological Galois group of $\bar{k}/k$, 142
where $\bar{k}$ is the separable closure of k in Ω (cf. V.8.1). This group $\pi_1(k)$ does not yet seem to have attracted the attention of algebraists.

Now let

$$f\colon S' \longrightarrow S$$

be a morphism from one connected locally noetherian prescheme to another, let a' be a geometric point of S', and let $a = f(a')$ be its direct image in S. The inverse-image functor induces a functor from the category $\mathcal{C}(S)$ of étale coverings of S to the category $\mathcal{C}(S')$ of étale coverings of S':

$$f^\bullet\colon \mathcal{C}(S) \longrightarrow \mathcal{C}(S').$$

There is an isomorphism of functors

$$F_a \cong F_{a'} \circ f^\bullet,$$

so $f^\bullet$ is an *exact* functor to which the results of no. V.6 apply. In particular, there is a canonical homomorphism

$$u = \pi_1(f; a')\colon \pi_1(S', a') \longrightarrow \pi_1(S; a) \qquad (a = f(a')).$$

This homomorphism recovers the inverse-image functor as restriction of the acting groups. Thus the properties of $f^\bullet$ are expressed simply in terms of the properties of the associated group homomorphism, as explained in no. V.6. In particular, if S' is an étale covering of S, then u is an isomorphism from $\pi_1(S', a')$ onto the open subgroup of $\pi_1(S, a)$ defining the pointed connected étale covering S' of S, i.e. the stabilizer U of $a' \in F_a(S')$ in $\pi_1(S; a)$.

To interpret the homomorphisms $\pi_1(f; a')$ as the geometric point a' varies, one must, in accordance with no. V.6, consider a homomorphism

$$\Pi_1(f)\colon \Pi_1^{S'} \longrightarrow f^\bullet(\Pi_1^S)$$

of pro-groups over S', and then take the corresponding homomorphism on geometric fibers.

V.8. The case of a normal base prescheme

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Proposition V.8.1. *Let S be the spectrum of a field k, and let Ω be an algebraically closed extension of k, defining a geometric point a of S with values in Ω. Let $\bar{k}$ be the separable closure of k in Ω. Then there is a canonical isomorphism from $\pi_1(S, a)$ onto the topological Galois group of $\bar{k}/k$.*

Let k' be the algebraic closure of k in Ω; it defines a geometric point b of S with values in k'. The natural homomorphism of functors $F_b \to F_a$ is plainly an isomorphism, because every k-homomorphism from a finite separable extension of k into Ω necessarily has its image in $\bar{k}$, and hence in k'. The group π' of k-automorphisms of k' acts naturally on F_b, giving a homomorphism

$$\pi' \longrightarrow \mathrm{Aut}(F_b) \xrightarrow{\sim} \mathrm{Aut}(F_a) = \pi_1(S; a).$$

It is well known that the natural homomorphism from π' to the group π of automorphisms of $\bar{k}/k$ is an isomorphism. We thus obtain a canonical homomorphism $\pi \to \pi_1(S; a)$, and it remains to show that it is an isomorphism. It is injective because an element of its kernel is an automorphism of $\bar{k}/k$ inducing the identity on every finite separable subextension, and is therefore trivial. It is surjective because, if X is a *connected* étale covering of S, hence defined by a finite separable *extension* L/k, then π acts transitively on the set of k-homomorphisms from L into k', as is well known.

PROPOSITION V.8.2. *Let S be a connected, locally noetherian, normal prescheme, let $K = \kappa(s)$ be its function field, equal to the residue field at its generic point s, and let Ω be an algebraically closed extension of K, defining a geometric point a' of $S' = \mathrm{Spec}(()K)$ and a geometric point a of S. Then the homomorphism*

$$\pi_1(S'; a') \longrightarrow \pi_1(S; a)$$

is surjective. If the first group is identified with the Galois group of the separable closure $\bar{K}$ of K in Ω (cf. V.8.1), then the kernel of the preceding homomorphism corresponds under Galois theory to the subextension of $\bar{K}/K$ obtained by composing the finite extensions of K in Ω that are unramified over S.

The first assertion says that the inverse image on S' of a connected étale covering X of S is connected, i.e. that X is integral; this is precisely I I.10.1. The kernel of the preceding homomorphism consists of those automorphisms of $\bar{K}/K$ that induce the identity on the sets $F_a(X)$, where the étale covering X of S may be assumed connected. 144
This means exactly that the automorphism induces the identity on the finite subextensions of $\bar{K}/K$ that are unramified over S, proving the last assertion.

REMARK. Through this interpretation of the fundamental group of the normal prescheme S in terms of ordinary Galois theory, the definition in this case has long been known.

V.9. The case of disconnected preschemes: multi-Galois categories

Let S be a locally noetherian prescheme, and let $(S_i)_{i\in I}$ be its connected components. The category $\mathcal{C}(S)$ of étale coverings of S is equivalent to the product category of the $\mathcal{C}(S_i)$, which may be interpreted in terms of the fundamental groups of the S_i once a geometric point has been chosen in each S_i. In applying descent theory to étale morphisms, it is sometimes inconvenient to choose a geometric point in every S_i. It is then preferable to use the natural generalization of V.5.8, interpreting $\mathcal{C}(S)$ as a category of functors on the groupoid of geometric points of S, regarded as the sum of the groupoids corresponding to the connected components of S. The functors in question take values in the category of finite sets and satisfy the continuity condition analogous to that used in V.5.8.

In practice, one has a family $(a_t)_{t\in E}$ of geometric points of S such that every connected component S_i of S contains at least one of them. As in V.5.10, one may then replace the groupoid of all geometric points of S by the analogous groupoid whose underlying set is E. These considerations should of course be incorporated into general definitions concerning categories equivalent to product categories of categories of the form $\mathcal{C}(\pi)$, which may be called *multi-Galois categories*. We leave the details to the reader.

CHAPTER VI

Fibered categories and descent

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VI.0. Introduction

Contrary to what was announced in the introduction to the preceding exposé, it has proved impossible to carry out descent in the category of preschemes, even in special cases, without first developing with sufficient care the language of descent in general categories.

The notion of "descent" provides the general framework for every process of "gluing" objects, and consequently for gluing categories. The most classical gluing situation begins with a topological space X and an open cover (X_i). Suppose that for every i we are given a fibered space E_i over X_i, and for every pair (i,j) an isomorphism

$$f_{ji}\colon E_i|_{X_{ij}} \xrightarrow{\sim} E_j|_{X_{ij}}, \qquad X_{ij} = X_i \cap X_j,$$

satisfying the familiar transitivity condition, written in abbreviated form as $f_{kj}f_{ji} = f_{ki}$. Then there exists a fibered space E over X, determined up to isomorphism by the existence of isomorphisms

$$f_i\colon E|_{X_i} \xrightarrow{\sim} E_i$$

satisfying $f_{ji} = f_j f_i^{-1}$, with the usual abuse of notation.

Let $X' = \coprod_i X_i$, which is a fibered space over X, that is, is equipped with a continuous map $X' \to X$. The family (E_i) can be described more concisely as a fibered space E' over X'. The family of maps f_{ji} becomes an isomorphism between the two pullbacks E_1'' and E_2'' of E' to

$$X'' = X' \times_X X'$$

by the two canonical projections. The gluing condition can then be written as an identity between isomorphisms of the fibered spaces E_1''' and E_3''' over the triple fiber product

$$X''' = X' \times_X X' \times_X X',$$

where E_i''' is the pullback of E' to X''' by the i-th canonical projection.

The construction of E from E' and the gluing isomorphism is a typical descent process. Conversely, starting with a fibered space E over X, one says that E is *locally trivial with fiber* F if there is an open cover (X_i) of X such that every $E|_{X_i}$ is isomorphic to $F \times X_i$. Equivalently, the pullback E' of E to $X' = \coprod_i X_i$ is isomorphic to $X' \times F$.

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Thus both the gluing of objects and the localization of a property are connected with the study of certain kinds of base change $X' \to X$. In algebraic geometry, many other kinds of base change, notably faithfully flat morphisms $X' \to X$, must be regarded as providing a localization process for preschemes, or other objects, over X. This kind of localization is used just as much as ordinary topological localization, of which the latter is in fact a special case. The same is true, to a lesser extent, in analytic geometry.

Most proofs, being reduced to verifications, are omitted or only sketched. When useful, we spell out the less evident diagrams arising in a proof.

VI.1. Universes, categories, and equivalences of categories

To avoid certain logical difficulties, we shall use the notion of a *universe*: a set large enough that the usual operations of set theory do not lead outside it. A "universe axiom" guarantees that every object belongs to some universe. For details, see the book in

preparation by C. Chevalley and the lecturer. [1] Thus **Set** denotes not the category of all sets, a notion which has no meaning, but the category of sets belonging to a fixed universe, which will not be indicated in the notation. Likewise, **Cat** denotes the category of categories belonging to that universe; by definition, the morphisms from an object X of **Cat** to another object Y are the functors from X to Y.

If $\mathcal{C}$ is a category, we denote by $\mathrm{Ob}(\mathcal{C})$ its *set of objects*, and by $\mathrm{Fl}(\mathcal{C})$ its *set of arrows*, or morphisms. We shall therefore write $X \in \mathrm{Ob}(\mathcal{C})$, and avoid the customary abuse $X \in \mathcal{C}$. If $\mathcal{C}$ and $\mathcal{C}'$ are categories, a *functor* from $\mathcal{C}$ to $\mathcal{C}'$ will always mean what is commonly called a *covariant* functor. Its specification includes its source and target categories $\mathcal{C}$ and $\mathcal{C}'$. The functors from $\mathcal{C}$ to $\mathcal{C}'$ form a set $\mathrm{Hom}(\mathcal{C},\mathcal{C}')$, which is the set of objects of the functor category $\mathrm{Fun}(\mathcal{C},\mathcal{C}')$. By definition, a *contravariant functor* from $\mathcal{C}$ to $\mathcal{C}'$ is a functor from the opposite category $\mathcal{C}^\circ$ to $\mathcal{C}'$.

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We shall use the notions of projective and inductive limits of a functor $F\colon \mathcal{I} \to \mathcal{C}$, and in particular their most common special cases: Cartesian products and fiber products, the dual notions of coproducts and amalgamated sums, and the usual formal properties of these operations.

For example, projective limits exist in the category **Cat** introduced above, provided the indexing categories $\mathcal{I}$ belong to the chosen universe. The set of objects, respectively the set of arrows, of a projective-limit category $\mathcal{C}$ of categories $\mathcal{C}_i$ is obtained by taking the projective limit of the sets of objects, respectively arrows, of the categories $\mathcal{C}_i$. The best-known case is the product of a family of categories. We shall constantly use below the fiber product of two categories over a third.

For matters concerning categories and functors, pending the book already mentioned, see the reference cited as VI.1. It is necessarily very incomplete, even concerning the generalities sketched in this section.

We take this opportunity to make explicit the notion of equivalence of categories, which is not presented satisfactorily in that reference. A functor

$$F\colon \mathcal{C} \longrightarrow \mathcal{C}'$$

is called *faithful*, respectively *fully faithful*, if for every pair of objects S, T of $\mathcal{C}$, the map

$$\mathrm{Hom}(S,T) \longrightarrow \mathrm{Hom}(F(S),F(T)), \qquad u \longmapsto F(u),$$

is injective, respectively bijective. The functor F is called an *equivalence of categories* if it is fully faithful and every object S' of $\mathcal{C}'$ is isomorphic to an object of the form $F(S)$.

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Equivalently, there is a functor

$$G\colon \mathcal{C}' \longrightarrow \mathcal{C}$$

quasi-inverse to F, with GF isomorphic to $\mathrm{id}_{\mathcal{C}}$ and FG isomorphic to $\mathrm{id}_{\mathcal{C}'}$. The data of a functor $G\colon \mathcal{C}' \to \mathcal{C}$ and an isomorphism

$$\varphi\colon FG \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}'}$$

amount to choosing, for every $S' \in \mathrm{Ob}(\mathcal{C}')$, an object $S = G(S')$ of $\mathcal{C}$ and an isomorphism $u = \varphi(S')\colon F(S) \xrightarrow{\sim} S'$, compatibly with morphisms. More precisely, there is a unique functor $\mathcal{C}' \to \mathcal{C}$ having the prescribed map $S' \mapsto G(S')$ on objects and for which $S' \mapsto \varphi(S')$ is a natural isomorphism $FG \to \mathrm{id}_{\mathcal{C}'}$.

Finally, if G is quasi-inverse to F, and if isomorphisms

$$\varphi\colon FG \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}'}, \qquad \psi\colon GF \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}}$$

are chosen, then the two compatibility conditions on φ and ψ stated in VI.1, I.1.2 are equivalent. For every chosen φ, there is a unique ψ for which those conditions hold.

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[1] The eventual authors are C. Chevalley and P. Gabriel. The book is due to appear in the year 3000. In the meantime, see also SGA 4, Exposé I.

VI.2. Categories over another category

Let $\mathcal{E}$ be a category belonging to $\mathfrak{U}$, hence an object of **Cat**. We may therefore consider the category

$$\mathbf{Cat}_{/\mathcal{E}}$$

of "objects of **Cat** over $\mathcal{E}$." An object of this category is a functor

$$p\colon \mathcal{F} \longrightarrow \mathcal{E}.$$

The category $\mathcal{F}$, equipped with such a functor, is also called a *category over* $\mathcal{E}$, or an *$\mathcal{E}$-category*. Accordingly, an $\mathcal{E}$-functor from a category $\mathcal{F}$ over $\mathcal{E}$ to a category $\mathcal{G}$ over $\mathcal{E}$ is a functor

$$f\colon \mathcal{F} \longrightarrow \mathcal{G}$$

such that

$$qf = p,$$

where p and q are the projection functors for $\mathcal{F}$ and $\mathcal{G}$, respectively. The set of $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$ is thus in bijection with the set of arrows from $\mathcal{F}$ to $\mathcal{G}$ in $\mathbf{Cat}_{/\mathcal{E}}$. This is not literally an identity, since the datum of f alone does not determine $\mathcal{F}$ and $\mathcal{G}$ as categories over $\mathcal{E}$; nevertheless, as in every category $\mathcal{C}_{/S}$, we shall routinely make the harmless abuse of identifying $\mathcal{E}$-functors in the above sense with arrows in $\mathbf{Cat}_{/\mathcal{E}}$.

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We denote by

$$\mathrm{Hom}_{\mathcal{E}}(\mathcal{F}, \mathcal{G})$$

the set of $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$. Of course, a composite of $\mathcal{E}$-functors is an $\mathcal{E}$-functor; by definition, the composition in question corresponds to composition of arrows in $\mathbf{Cat}_{/\mathcal{E}}$.

Now consider two $\mathcal{E}$-functors

$$f, g\colon \mathcal{F} \longrightarrow \mathcal{G}$$

and a natural transformation

$$u\colon f \longrightarrow g.$$

We call u an *$\mathcal{E}$-homomorphism*, or a *homomorphism of $\mathcal{E}$-functors*, if, for every $\xi \in \mathrm{Ob}(\mathcal{F})$,

$$q\big(u(\xi)\big) = \mathrm{id}_{p(\xi)}\,.$$

In words, put

$$S = p(\xi) = qf(\xi) = qg(\xi) \in \mathrm{Ob}(\mathcal{E}).$$

Then the morphism

$$u(\xi)\colon f(\xi) \longrightarrow g(\xi)$$

in $\mathcal{G}$ is an id_S-morphism. More generally, for every morphism $\alpha\colon T \to S$ in $\mathcal{E}$, and every category $\mathcal{G}$ over $\mathcal{E}$, a morphism v in $\mathcal{G}$ is called an *α-morphism* if $q(v) = \alpha$, where $q\colon \mathcal{G} \to \mathcal{E}$ is the projection functor.

If $h\colon \mathcal{F} \to \mathcal{G}$ is a third $\mathcal{E}$-functor and $v\colon g \to h$ is an $\mathcal{E}$-homomorphism, then vu is again an $\mathcal{E}$-homomorphism. Thus the $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$, together with their $\mathcal{E}$-homomorphisms, form a subcategory of the category $\mathrm{Fun}(\mathcal{F}, \mathcal{G})$ of all functors from $\mathcal{F}$ to $\mathcal{G}$. We call it the *category of $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$* and denote it by

$$\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}).$$

It is also the kernel subcategory of the pair of functors

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$$\mathrm{Fun}(\mathcal{F}, \mathcal{G}) \underset{S}{\overset{R}{\rightrightarrows}} \mathrm{Fun}(\mathcal{F}, \mathcal{E}),$$

where R is the constant functor defined by the object $p \in \mathrm{Fun}(\mathcal{F}, \mathcal{E})$, while S is the functor $f \mapsto q \circ f$ defined by $q\colon \mathcal{G} \to \mathcal{E}$.

To finish these generalities, it remains to define the natural pairings among the categories $\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ by composition of $\mathcal{E}$-functors. In other words, for three categories $\mathcal{F},\mathcal{G},\mathcal{H}$ over $\mathcal{E}$, we want to define a "composition functor"

$$\text{(i)}\qquad \mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})\times\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{G},\mathcal{H})\longrightarrow\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{H})$$

which, on objects, induces the composition map $(f,g)\mapsto gf$ for $\mathcal{E}$-functors $f\colon\mathcal{F}\to\mathcal{G}$ and $g\colon\mathcal{G}\to\mathcal{H}$.

For this purpose, recall the canonical functor

$$\text{(ii)}\qquad \mathrm{Fun}(\mathcal{F},\mathcal{G})\times\mathrm{Fun}(\mathcal{G},\mathcal{H})\longrightarrow\mathrm{Fun}(\mathcal{F},\mathcal{H}),$$

which, on objects, is just the composition map $(f,g)\mapsto gf$. It sends an arrow (u,v), where

$$u\colon f\to f',\qquad v\colon g\to g'$$

are arrows in $\mathrm{Fun}(\mathcal{F},\mathcal{G})$ and $\mathrm{Fun}(\mathcal{G},\mathcal{H})$, respectively, to the arrow

$$v*u\colon gf\longrightarrow g'f'$$

defined by

$$(v*u)(\xi)=v(f'(\xi))\circ g(u(\xi))=g'(u(\xi))\circ v(f(\xi)).$$

It is well known that this does define a natural transformation from gf to $g'f'$, and that, as f,g,u,v vary, it defines the functor in (ii); that is, French source p. 151

$$\text{(I)}\qquad \mathrm{id}_g*\mathrm{id}_f=\mathrm{id}_{gf},$$

and

$$\text{(II)}\qquad (v'*u')\circ(v*u)=(v'\circ v)*(u'\circ u).$$

Recall also the associativity formula for the canonical pairings in (ii). It is expressed both by the associativity $(hg)f=h(gf)$ of composition of functors and by

$$\text{(III)}\qquad (w*v)*u=w*(v*u)$$

for composition products of natural transformations. Here $u\colon f\to f'$ and $v\colon g\to g'$ are as above, and one is also given a natural transformation $w\colon h\to h'$ between functors $h,h'\colon\mathcal{H}\to\mathcal{K}$.

We now claim that, when $\mathcal{F}$ and $\mathcal{G}$ are $\mathcal{E}$-categories, the canonical composition functor in (ii) induces the functor in (i). Since we already know that the composite of two $\mathcal{E}$-functors is an $\mathcal{E}$-functor, this amounts to saying that if $u\colon f\to f'$ and $v\colon g\to g'$ are homomorphisms of $\mathcal{E}$-functors, then $v*u\colon gf\to g'f'$ is again a homomorphism of $\mathcal{E}$-functors. This follows immediately from the definitions. Because the pairings in (i) are induced by those in (ii), they satisfy the same associativity property, expressed by the formulas $(hg)f=h(gf)$ and $(w*v)*u=w*(v*u)$ for $\mathcal{E}$-functors and homomorphisms of $\mathcal{E}$-functors.

To complete the collection of formulas (I)–(III), recall also

$$\text{(IV)}\qquad v*\mathrm{id}_{\mathcal{F}}=v,\qquad \mathrm{id}_{\mathcal{G}}*u=u.$$

For brevity, one writes $v*f$ or $u*g$ instead of $v*u$ when u, respectively v, is the identity automorphism of f, respectively g.

It follows from the definition of the pairings in (i) that $\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ is a functor in $\mathcal{F}$ and $\mathcal{G}$, from the product category French source p. 152

$$\mathbf{Cat}^{\mathrm{op}}_{/\mathcal{E}}\times\mathbf{Cat}_{/\mathcal{E}}\quad\text{to}\quad\mathbf{Cat}.$$

Indeed, if $g\colon\mathcal{G}\to\mathcal{G}_1$ is an $\mathcal{E}$-functor, hence an object of $\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{G},\mathcal{G}_1)$, then, on putting $\mathcal{H}=\mathcal{G}_1$ in (i), it gives a functor

$$g_*\colon\ \mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})\longrightarrow\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G}_1).$$

Similarly, an $\mathcal{E}$-functor $f\colon \mathcal{F}_1 \to \mathcal{F}$ defines a functor

$$f^*\colon \mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G}) \longrightarrow \mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F}_1,\mathcal{G}).$$

For short, these functors are also denoted by the symbols $f \mapsto g\circ f$ and $g \mapsto g\circ f$, respectively; strictly speaking, those symbols describe only the corresponding maps on sets of objects. The associativity property established above shows that this does indeed give the announced functor

$$\mathbf{Cat}^{\mathrm{op}}_{/\mathcal{E}} \times \mathbf{Cat}_{/\mathcal{E}} \longrightarrow \mathbf{Cat}.$$

VI.3. Base change in categories over $\mathcal{E}$

Since inverse limits (with respect to categories $\mathcal{I}$ that are elements of $\mathfrak{U}$) exist in **Cat**, the same is true in $\mathbf{Cat}_{/\mathcal{E}}$; in particular, Cartesian products exist there, and are interpreted as fiber products in **Cat**. In accordance with the general notation, if $\mathcal{F}$ and $\mathcal{G}$ are categories over $\mathcal{E}$, we denote by

$$\mathcal{F} \times_{\mathcal{E}} \mathcal{G}$$

their product in $\mathbf{Cat}_{/\mathcal{E}}$, that is, their fiber product over $\mathcal{E}$ in **Cat**, regarded as a category 153
over $\mathcal{E}$. Thus $\mathcal{F}\times_{\mathcal{E}}\mathcal{G}$ is equipped with two $\mathcal{E}$-functors pr_1 and pr_2, which define, for every category $\mathcal{H}$ over $\mathcal{E}$, a bijection

$$\mathrm{Hom}_{\mathcal{E}}(\mathcal{H},\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{H},\mathcal{F}) \times \mathrm{Hom}_{\mathcal{E}}(\mathcal{H},\mathcal{G}).$$

Moreover, this bijection is obtained from an isomorphism of categories

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{F}) \times \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{G})$$

by taking the sets of objects on both sides, where the displayed functor has as its components the functors $h \longmapsto \mathrm{pr}_1\circ h$ and $h \longmapsto \mathrm{pr}_2\circ h$ from the left-hand side to the two factors on the right. We leave it to the reader to verify that this indeed gives an isomorphism (the analogous fact holds, more generally, whenever one has an inverse limit of categories—and not only in the case of a fiber product).

Recall also that (as was stated in no. VI.1):

$$\begin{aligned}\mathrm{Ob}(\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) &= \mathrm{Ob}(\mathcal{F}) \times_{\mathrm{Ob}(\mathcal{E})} \mathrm{Ob}(\mathcal{G})\\ \mathrm{Fl}(\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) &= \mathrm{Fl}(\mathcal{F}) \times_{\mathrm{Fl}(\mathcal{E})} \mathrm{Fl}(\mathcal{G}),\end{aligned}$$

the composition of arrows being performed componentwise.

In what follows, we consider a functor

$$\lambda\colon \mathcal{E}' \to \mathcal{E}$$

and, for every category $\mathcal{F}$ over $\mathcal{E}$, regard $\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'$ as a category over $\mathcal{E}'$ by means of pr_2; in other words, we interpret the "fiber product"operation as a *"base change"* operation, and call the functor $\lambda\colon \mathcal{E}' \to \mathcal{E}$ a *"base-change functor"*. In accordance with the familiar general facts, this yields a functor, called the *base-change functor* for λ: 154

$$\lambda^*\colon \mathbf{Cat}_{/\mathcal{E}} \to \mathbf{Cat}_{/\mathcal{E}'},$$

(adjoint to the "restriction of base"functor which, to every category $\mathcal{F}'$ over $\mathcal{E}'$ with projection functor p', assigns $\mathcal{F}'$, regarded as a category over $\mathcal{E}$ through the functor $p = \lambda p'$). As is well known in the general case of a base-change functor in a category, the base-change functor "commutes with inverse limits", and in particular "transforms"fiber products over $\mathcal{E}$ into fiber products over $\mathcal{E}'$.

Let $\mathcal{F}$ and $\mathcal{G}$ be two categories over $\mathcal{E}$. We shall define a *canonical isomorphism*:

(i) $\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}',\mathcal{G}') \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}',\mathcal{G})$

where $\mathcal{F}' = \mathcal{F}\times_{\mathcal{E}}\mathcal{E}'$, $\mathcal{G}' = \mathcal{G}\times_{\mathcal{E}}\mathcal{E}'$.

For this, consider the functor

$$\mathrm{pr}_1 \colon \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}' \to \mathcal{G},$$

and define (i) by

$$F \longmapsto \mathrm{pr}_1 \circ F,$$

which a priori denotes a functor

(ii) $$\mathbf{Hom}(\mathcal{F}', \mathcal{G}') \to \mathbf{Hom}(\mathcal{F}', \mathcal{G})$$

It therefore remains only to verify that the latter induces a functor on the subcategories in (i), and that this functor is an isomorphism. That (ii) induces a bijection

$$\mathrm{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$$

is the characteristic property of the base-change functor. It therefore remains to prove 155
that if F, G are $\mathcal{E}'$-functors $\mathcal{F}' \to \mathcal{G}'$, then *the map*

$$u \longmapsto \mathrm{pr}_1 \circ u$$

induces a bijection

$$\mathrm{Hom}_{\mathcal{E}'}(F, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathrm{pr}_1 \circ F, \mathrm{pr}_1 \circ G).$$

The verification of this fact is immediate and is left to the reader.

It follows from this isomorphism (i), and from the end of the preceding subsection, that

$$\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}')$$

may be regarded as a functor in $\mathcal{E}', \mathcal{F}, \mathcal{G}$, *from the category* $\mathbf{Cat}^{\circ}_{/\mathcal{E}} \times \mathbf{Cat}^{\circ}_{/\mathcal{E}} \times \mathbf{Cat}_{/\mathcal{E}}$ *to the category* **Cat**, isomorphic to the functor defined by the expression $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$. In particular, for fixed $\mathcal{F}$, $\mathcal{G}$, one obtains a functor in $\mathcal{E}'$, $\mathcal{E}' \to \mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F}', \mathcal{G}') = \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}')$, and in particular the projection $\mathcal{E}$-functor $\lambda \colon \mathcal{E}' \to \mathcal{E}$ defines a morphism i.e. a functor

$$\lambda^*_{\mathcal{F},\mathcal{G}} \colon \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \to \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}')$$

which we shall make explicit. On the sets of objects of the two sides, this is the map

$$f \longmapsto f' = f \times_{\mathcal{E}} \mathcal{E}'$$

which expresses the functorial dependence of $\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$ on the object $\mathcal{F}$ over $\mathcal{E}$. On the other hand, consider two $\mathcal{E}$-functors

$$f, g \colon \mathcal{F} \to \mathcal{G}$$

and a homomorphism of $\mathcal{E}$-functors

$$u \colon f \to g,$$

we shall make explicit the corresponding homomorphism of $\mathcal{E}'$-functors: 156

$$u' \colon f' \to g'.$$

For every

$$\xi' = (\xi, S') \in \mathrm{Ob}(\mathcal{F}')$$

with

$$\xi \in \mathrm{Ob}(\mathcal{F}), \quad S' \in \mathrm{Ob}(\mathcal{E}'), \quad p(\xi) = \lambda(S') = S$$

the morphism

$$u'(\xi') \colon f'(\xi') = (f(\xi), S') \to g'(\xi') = (g(\xi), S') \quad \text{in } \mathcal{G}'$$

is defined by the formula

$$u'(\xi') = (u(\xi), \mathrm{id}_{S'})$$

(which is indeed an S'-morphism in $\mathcal{G}'$, since $q(u(\xi)) = \lambda(\mathrm{id}_{S'}) = \mathrm{id}_S$).

Now consider an arbitrary $\mathcal{E}$-functor

$$\lambda' \colon \mathcal{E}'' \to \mathcal{E}'$$

and the corresponding functor

$$\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}') \to \mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}''),$$

I claim that this functor is none other than the functor obtained by the preceding procedure, starting from $\mathcal{F}'$ and $\mathcal{G}'$ over $\mathcal{E}'$ and regarding $\mathcal{E}''$ as a category over $\mathcal{E}'$, taking into account the isomorphisms of *"transitivity of base change"*:

$$\mathcal{F}' \times_{\mathcal{E}'} \mathcal{E}'' \xrightarrow{\sim} \mathcal{F}'' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'' \quad \text{and} \quad \mathcal{G}' \times_{\mathcal{E}'} \mathcal{E}'' \xrightarrow{\sim} \mathcal{G}'' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'',$$

which induce a canonical isomorphism

$$\mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F}' \times_{\mathcal{E}'} \mathcal{E}'', \mathcal{G}' \times_{\mathcal{E}'} \mathcal{E}'') \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'')$$

The verification of this compatibility is immediate and is left to the reader. 157

The functors just defined are compatible with the pairings defined in the preceding subsection, in the following precise sense: if $\mathcal{F}$, $\mathcal{G}$, and $\mathcal{H}$ are categories over $\mathcal{E}$ and if we set

$$\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \quad \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}', \quad \mathcal{H}' = \mathcal{H} \times_{\mathcal{E}} \mathcal{E}',$$

then the following diagram of functors is commutative:

$$\begin{array}{ccc} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G}) \times \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G},\mathcal{H}) & \longrightarrow & \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{H}) \\ \downarrow{\scriptstyle \lambda^*_{\mathcal{F},\mathcal{G}} \times \lambda^*_{\mathcal{G},\mathcal{H}}} & & \downarrow{\scriptstyle \lambda^*_{\mathcal{F},\mathcal{H}}} \\ \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}',\mathcal{G}') \times \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{G}',\mathcal{H}') & \longrightarrow & \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}',\mathcal{H}') \end{array}$$

where the horizontal arrows are the composition functors defined in the preceding subsection. This commutativity is expressed by the formulas

$$(gf)' = g'f'$$

for $f \in \mathrm{Hom}_{\mathcal{E}}(\mathcal{F},\mathcal{G})$, $g \in \mathrm{Hom}_{\mathcal{E}}(\mathcal{G},\mathcal{H})$ (a formula that simply expresses the functoriality of base change), and by the formula

$$(v * u)' = v' * u'$$

when $u \colon f \to f_1$ is an arrow of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ and $v \colon g \to g_1$ is an arrow of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G},\mathcal{H})$. The verification of this formula follows readily from the definitions.

In what follows, we shall be principally interested in $\mathbf{Hom}_{\mathcal{E}}(\mathcal{F},\mathcal{G})$ (and certain noteworthy subcategories thereof) when $\mathcal{F} = \mathcal{E}$, and for this reason we introduce special notation:

$$\mathbf{\Gamma}(\mathcal{G}/\mathcal{E}) = \mathbf{Hom}_{\mathcal{E}}(\mathcal{E},\mathcal{G}), \quad \mathbf{\Gamma}(\mathcal{G}/\mathcal{E}) = \mathrm{Ob}(\mathbf{\Gamma}(\mathcal{G}/\mathcal{E})) = \mathrm{Hom}_{\mathcal{E}}(\mathcal{E},\mathcal{G}).$$

Remarks. When $\mathcal{E}$ is a one-point category, i.e. when $\mathrm{Ob}(\mathcal{E})$ and $\mathrm{Fl}(\mathcal{E})$ are each reduced to a single element, which also means that $\mathcal{E}$ is a final object of the category **Cat**, the datum of a category over $\mathcal{E}$ is equivalent to the datum of a category, plain and simple (since there is a unique functor from $\mathcal{F}$ to $\mathcal{E}$). More precisely, $\mathbf{Cat}_{/\mathcal{E}}$ is then isomorphic to **Cat**. Moreover, the categories $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ are then precisely the categories $\mathbf{Hom}(\mathcal{F},\mathcal{G})$. Recall that the fundamental formula 158

$$\mathrm{Hom}(\mathcal{H}, \mathbf{Hom}(\mathcal{F},\mathcal{G})) \xrightarrow{\sim} \mathrm{Hom}(\mathcal{F} \times \mathcal{H}, \mathcal{G})$$

(an isomorphism functorial in the three arguments occurring therein) allows $\mathbf{Hom}(\mathcal{F},\mathcal{G})$ to be interpreted axiomatically, in terms internal to the category **Cat**, so that the familiar formalism of the categories **Hom** appears as a special case of a formalism valid in categories such as **Cat**, in which "**Hom** objects" (defined by the preceding formula) exist. There is

an analogous interpretation of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ when $\mathcal{E}$ is again arbitrary, by means of the formula

$$\mathrm{Hom}(\mathcal{H},\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{F}\times\mathcal{H},\mathcal{G})$$

(an isomorphism functorial in the three arguments). Thus the formal properties set out in no. VI.2, VI.3 are special cases of more general results, valid in categories in which the objects $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ (when $\mathcal{F}$ and $\mathcal{G}$ are two objects of the category over a third object $\mathcal{E}$) exist.

VI.4. Fiber categories; equivalence of $\mathcal{E}$-categories

Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let $S\in\mathrm{Ob}(\mathcal{E})$. The *fiber category of $\mathcal{F}$ at S* is the subcategory $\mathcal{F}_S$ of $\mathcal{F}$ that is the inverse image of the one-point subcategory of $\mathcal{E}$ defined by S. Thus the objects of $\mathcal{F}_S$ are the objects ξ of $\mathcal{F}$ such that $p(\xi)=S$, and its morphisms are the morphisms u of $\mathcal{F}$ such that $p(u)=\mathrm{id}_S$, i.e. the S-morphisms in $\mathcal{F}$. 159

Of course, $\mathcal{F}_S$ is canonically isomorphic to the fiber product $\mathcal{F}\times_{\mathcal{E}}\{S\}$, where $\{S\}$ denotes the one-point subcategory of $\mathcal{E}$ defined by S, equipped with its inclusion functor into $\mathcal{E}$. It follows, in view of the transitivity of base change, that if one makes the base change $\lambda\colon\mathcal{E}'\to\mathcal{E}$, then for every $S'\in\mathrm{Ob}(\mathcal{E}')$, *the projection*

$$\mathrm{pr}_1\colon\mathcal{F}'=\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'\longrightarrow\mathcal{F}$$

induces an isomorphism

$$\mathcal{F}'_{S'}\longrightarrow\mathcal{F}_S\qquad(\text{where } S=\lambda(S')).$$

Proposition VI.4.1. *Let $f\colon\mathcal{F}\to\mathcal{G}$ be an $\mathcal{E}$-functor. If f is fully faithful, then for every base change $\mathcal{E}'\to\mathcal{E}$, the corresponding functor*

$$f'\colon\mathcal{F}'=\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'\longrightarrow\mathcal{G}'=\mathcal{G}\times_{\mathcal{E}}\mathcal{E}'$$

is fully faithful.

The verification is immediate. More generally, one can show that every projective limit of fully faithful functors (here f and the identity functors in $\mathcal{E},\mathcal{E}'$) is a fully faithful functor.

The analogue of Proposition VI.4.1 in which where "fully faithful" is replaced by "equivalence of categories" is false, already when $\mathcal{G}=\mathcal{E}$. Nevertheless:

Proposition VI.4.2. *Let $f\colon\mathcal{F}\to\mathcal{G}$ be an $\mathcal{E}$-functor. The following conditions are equivalent:*

(i) *There exist an $\mathcal{E}$-functor $g\colon\mathcal{G}\to\mathcal{F}$ and $\mathcal{E}$-isomorphisms*

$$gf\xrightarrow{\sim}\mathrm{id}_{\mathcal{F}},\qquad fg\xrightarrow{\sim}\mathrm{id}_{\mathcal{G}}.$$

(ii) *For every category $\mathcal{E}'$ over $\mathcal{E}$, the functor*

$$f'=f\times_{\mathcal{E}}\mathcal{E}'\colon\mathcal{F}'=\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'\longrightarrow\mathcal{G}'=\mathcal{G}\times_{\mathcal{E}}\mathcal{E}'$$

is an equivalence of categories. 160

(iii) *f is an equivalence of categories, and for every $S\in\mathrm{Ob}(\mathcal{E})$, the functor*

$$f_S\colon\mathcal{F}_S\longrightarrow\mathcal{G}_S$$

induced by f is an equivalence of categories.

(iii bis) *f is fully faithful, and for every $S\in\mathrm{Ob}(\mathcal{E})$ and every $\eta\in\mathrm{Ob}(\mathcal{G}_S)$, there exist $\xi\in\mathrm{Ob}(\mathcal{F}_S)$ and an S-isomorphism $u\colon f(\xi)\to\eta$.*

Proof. Condition (i) plainly implies that f is an equivalence of categories, a notion defined by the same condition except that the functor isomorphisms are not required to be $\mathcal{E}$-morphisms. Moreover, the functoriality established in the preceding section shows that condition (i) is preserved after a base change $\mathcal{E}' \to \mathcal{E}$. Thus (i)$\Longrightarrow$(ii). Clearly (ii)$\Longrightarrow$(iii), since it is enough to take $\mathcal{E}' = \mathcal{E}$ and $\mathcal{E}' = \{S\}$. The implication (iii)$\Longrightarrow$(iii bis) is still more immediate, so it remains to prove (iii bis)$\Longrightarrow$(i).

For every $\eta \in \mathrm{Ob}(\mathcal{G})$, choose $g(\eta) \in \mathrm{Ob}(\mathcal{F})$ and an isomorphism

$$u(\eta)\colon f(g(\eta)) \longrightarrow \eta$$

such that $q(u(\eta)) = \mathrm{id}_S$, where $S = q(\eta)$. This is possible by the second condition in (iii bis). As is well known and immediate, the full faithfulness of f implies that g can be made, in a unique way, into a functor from $\mathcal{G}$ to $\mathcal{F}$ so that the $u(\eta)$ define a functorial homomorphism, hence an isomorphism,

$$u\colon fg \xrightarrow{\sim} \mathrm{id}_{\mathcal{G}}.$$

Moreover, by construction, g is an $\mathcal{E}$-functor and u an $\mathcal{E}$-homomorphism. The preceding data then determine a functorial isomorphism

$$v\colon gf \longrightarrow \mathrm{id}_{\mathcal{F}},$$

defined by the condition $f * v = u * f$, and one sees at once that this, too, is an $\mathcal{E}$-morphism. This proves the proposition.

Definition VI.4.3. *When the preceding conditions hold, f is called an* equivalence of categories over $\mathcal{E}$, *or an* $\mathcal{E}$-equivalence.

Corollary VI.4.4. *Suppose that the projection functor $p\colon \mathcal{F} \to \mathcal{E}$ is transportable, i.e. that for every isomorphism $\alpha\colon T \to S$ in $\mathcal{E}$ and every object ξ in $\mathcal{F}_T$, there exists an isomorphism u in $\mathcal{F}$ with source ξ such that $p(u) = \alpha$. Then every $\mathcal{E}$-functor $f\colon \mathcal{F} \to \mathcal{G}$ that is an equivalence of categories is an $\mathcal{E}$-equivalence.*

This follows from criterion (iii bis). 161

Corollary VI.4.5. *Let $f\colon \mathcal{F} \to \mathcal{G}$ be an $\mathcal{E}$-equivalence. Then, for every category $\mathcal{H}$ over $\mathcal{E}$, the corresponding functors*

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G}, \mathcal{H}) \longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{H}),$$
$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H}, \mathcal{F}) \longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H}, \mathcal{G})$$

(cf. no. VI.2) are equivalences of categories.

This follows from criterion (i) by the usual argument.

VI.5. Cartesian morphisms, inverse images, Cartesian functors

Let $\mathcal{F}$ be a category over $\mathcal{E}$, with projection functor p.

Definition VI.5.1. *Consider a morphism*

$$\alpha\colon \eta \longrightarrow \xi$$

in $\mathcal{F}$, and let

$$S = p(\xi), \qquad T = p(\eta), \qquad f = p(\alpha).$$

The morphism α is called a Cartesian morphism *if, for every $\eta' \in \mathrm{Ob}(\mathcal{F}_T)$ and every f-morphism $u\colon \eta' \to \xi$, there exists a unique T-morphism $\overline{u}\colon \eta' \to \eta$ such that $u = \alpha \circ \overline{u}$.*

Thus, for every $\eta' \in \mathrm{Ob}(\mathcal{F}_T)$, the map $v \longmapsto \alpha \circ v$,

(i) $$\mathrm{Hom}_T(\eta', \eta) \longrightarrow \mathrm{Hom}_f(\eta', \xi),$$

is bijective. Equivalently, the pair (η, α) *represents the functor in η'* $\mathcal{F}_T^{\circ} \to$ **Ens** appearing on the right.

If, for a given morphism $f\colon T \to S$ in $\mathcal{E}$ and a given $\xi \in \mathrm{Ob}(\mathcal{F}_S)$, such a pair (η, α) exists, i.e. if there is a Cartesian morphism α in $\mathcal{F}$ with target ξ and $p(\alpha) = f$, then η is 162
determined in $\mathcal{F}_T$ up to unique isomorphism. One then says that *the inverse image of ξ under f exists*, and an object η of $\mathcal{F}_T$, together with a Cartesian f-morphism $\alpha\colon \eta \to \xi$, is called *an inverse image of ξ under f*. Often such an inverse image is chosen whenever it exists, with $\mathcal{F}$ fixed. It is then denoted by $f^*_{\mathcal{F}}(\xi)$, or simply $f^*(\xi)$, or $\xi \times_S T$ when these notations cause no confusion. The canonical morphism $\alpha\colon \eta \to \xi$ will be denoted below by $\alpha_f(\xi)$.

If the inverse image of every $\xi \in \mathrm{Ob}(\mathcal{F}_S)$ under f exists, one also says that *the inverse-image functor under f in $\mathcal{F}$ exists*; then $f^*(\xi)$ becomes a covariant functor in ξ, from $\mathcal{F}_S$ to $\mathcal{F}_T$. This follows because the right-hand side of (i) depends covariantly on ξ, or more precisely defines a functor from $\mathcal{F}_T^\circ \times \mathcal{F}_S$ to **Ens**.

This functorial dependence of $f^*(\xi)$ is explicit as follows. Consider Cartesian f-morphisms

$$\alpha\colon \eta \longrightarrow \xi, \qquad \alpha'\colon \eta' \longrightarrow \xi',$$

and an S-morphism $\lambda\colon \xi \to \xi'$. There exists a unique T-morphism $\mu\colon \eta \to \eta'$ such that

$$\alpha'\mu = \lambda\alpha,$$

as follows from the fact that α' is Cartesian.

We also record the following immediate fact. Consider a commutative diagram

$$\begin{array}{ccc} \xi & \xleftarrow{\ \alpha\ } & \eta \\ {\scriptstyle\lambda}\downarrow & & \downarrow{\scriptstyle\mu} \\ \xi' & \xleftarrow{\ \alpha'\ } & \eta' \end{array}$$

in $\mathcal{F}$, where α and α' are f-morphisms, λ is an S-isomorphism, and μ a T-isomorphism. 163
Then *α is Cartesian if and only if α' is Cartesian.*

Definition VI.5.2. *An $\mathcal{E}$-functor $F\colon \mathcal{F} \to \mathcal{G}$ is called a* Cartesian functor *if it carries Cartesian morphisms to Cartesian morphisms. Denote by $\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}, \mathcal{G})$ the full subcategory of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ formed by the Cartesian functors.*

For example, regard $\mathcal{E}$ as a category over itself by means of the identity functor. Every morphism of $\mathcal{E}$ is Cartesian, so a Cartesian functor from $\mathcal{E}$ to $\mathcal{F}$ is a section functor $F\colon \mathcal{E} \to \mathcal{F}$ that carries every morphism of $\mathcal{E}$ to a Cartesian morphism. Such a functor is called a Cartesian section *of $\mathcal{F}$ over $\mathcal{E}$.*

Proposition VI.5.3. (i) *A functor $F\colon \mathcal{F} \to \mathcal{G}$ that is an $\mathcal{E}$-equivalence is Cartesian.*

(ii) *Let $F, G\colon \mathcal{F} \to \mathcal{G}$ be isomorphic $\mathcal{E}$-functors. If one is Cartesian, then so is the other.*

(iii) *The composite of two Cartesian functors $\mathcal{F} \to \mathcal{G}$ and $\mathcal{G} \to \mathcal{H}$ is Cartesian.*

Assertion (iii) is immediate from the definition, (ii) follows from the remark preceding VI.5.2, and (i) follows readily from the definition and criterion VI.4.2 (iii). More precisely, a morphism α in $\mathcal{F}$ is Cartesian if and only if $F(\alpha)$ is Cartesian.

Corollary VI.5.4. *Let $F\colon \mathcal{F} \to \mathcal{G}$ be an $\mathcal{E}$-equivalence. Then, for every category $\mathcal{H}$ over $\mathcal{E}$, the corresponding functors $G \longmapsto G \circ F$ and $G \longmapsto F \circ G$ induce equivalences of categories*

$$\begin{aligned} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{H}) &\xrightarrow{\approx} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}, \mathcal{H}), \\ \mathbf{Hom}_{\mathrm{cart}}(\mathcal{H}, \mathcal{F}) &\xrightarrow{\approx} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{H}, \mathcal{G}). \end{aligned}$$

This follows in the usual way from criterion VI.4.2 (i) and VI.5.3 (i), (ii), and (iii). 164

More precisely, *an $\mathcal{E}$-functor $G\colon \mathcal{G} \to \mathcal{H}$ is Cartesian if and only if $G \circ F$ is Cartesian*; likewise, *an $\mathcal{E}$-functor $G\colon \mathcal{H} \to \mathcal{F}$ is Cartesian if and only if $F \circ G$ is Cartesian.*

It follows from VI.5.3 (iii) that, if one considers the subcategory $\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ of $\mathbf{Cat}_{/\mathcal{E}}$ having the same objects as $\mathbf{Cat}_{/\mathcal{E}}$ and Cartesian functors as its morphisms, then, as in no. VI.2, there are composition pairings

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{G}) \times \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G},\mathcal{H}) \longrightarrow \mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{H})$$

induced by those of no. VI.2. These allow $\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{G})$ to be regarded as a functor in $\mathcal{F},\mathcal{G}$, from $(\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}})^{\circ} \times \mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ to $\mathbf{Cat}$. We shall chiefly need this remark when $\mathcal{F}=\mathcal{G}$.

Definition VI.5.5. *Let $\mathcal{F}$ be a category over $\mathcal{E}$. Denote by*

$$\varprojlim \mathcal{F}/\mathcal{E}$$

the category of Cartesian $\mathcal{E}$-functors $\mathcal{E} \to \mathcal{F}$, i.e. of Cartesian sections of $\mathcal{F}$ over $\mathcal{E}$.

By what has just been said, $\varprojlim \mathcal{F}/\mathcal{E}$ is a functor in $\mathcal{F}$, from the category $\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ to the category $\mathbf{Cat}$.

We shall see below the relations between this operation $\varprojlim$ and the notion of a projective limit of categories, as well as many examples.

VI.6. Fibered categories and prefibered categories. Products and base change therein

Definition VI.6.1. *A category $\mathcal{F}$ over $\mathcal{E}$ is called a* fibered category *(and the functor $\mathcal{F} \to \mathcal{E}$ is then said to be* fibered*) if it satisfies the following two axioms:*

- Fib$_{\mathrm{I}}$ *For every morphism $f\colon T \to S$ in $\mathcal{E}$, the inverse-image functor by f in $\mathcal{F}$ exists.* 165
- Fib$_{\mathrm{II}}$ *The composite of two Cartesian morphisms is Cartesian.*

A category $\mathcal{F}$ over $\mathcal{E}$ satisfying condition Fib$_{\mathrm{I}}$ *is called a* prefibered category over $\mathcal{E}$.

If $\mathcal{F}$ is a fibered category (respectivelyprefibered category) over $\mathcal{E}$, a subcategory $\mathcal{G}$ of $\mathcal{F}$ is called a *fibered subcategory* (respectivelya *prefibered subcategory*) if it is a fibered category (respectivelyprefibered category) over $\mathcal{E}$ and, moreover, the inclusion functor is Cartesian. If, for example, $\mathcal{G}$ is a *full* subcategory of $\mathcal{F}$, one sees that this means that, for every morphism $f\colon T \to S$ in $\mathcal{E}$ and every $\xi \in \mathrm{Ob}(\mathcal{G}_S)$, $f^*_{\mathcal{F}}(\xi)$ is T-isomorphic to an object of $\mathcal{G}_T$. Another interesting case is the following: with $\mathcal{F}$ a fibered category over $\mathcal{E}$, consider the subcategory $\mathcal{G}$ of $\mathcal{F}$ having the same objects and whose morphisms are the *Cartesian* morphisms of $\mathcal{F}$; in particular, the morphisms of $\mathcal{G}_S$ are the isomorphisms of $\mathcal{F}_S$. One sees at once that this is indeed a fibered subcategory of $\mathcal{F}$, since in the bijection

$$\mathrm{Hom}_T(\eta',\eta) \xrightarrow{\sim} \mathrm{Hom}_f(\eta',\xi)$$

associated with a Cartesian f-morphism α in $\mathcal{F}$, the T-isomorphisms on the left correspond to the Cartesian morphisms on the right. By definition, the Cartesian sections $\mathcal{E} \to \mathcal{F}$ correspond bijectively to arbitrary $\mathcal{E}$-functors $\mathcal{E} \to \mathcal{G}$ (but note that the natural functor

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{E},\mathcal{G}) \to \mathbf{Hom}_{\mathrm{cart}}(\mathcal{E},\mathcal{F}) = \varprojlim(\mathcal{F}/\mathcal{E})$$

is faithful, but in general is not fully faithful, i.e. is not an isomorphism).

Remarks. Let $\mathcal{F}$ be a category over $\mathcal{E}$. The following conditions are equivalent: (i) Every morphism of $\mathcal{F}$ is Cartesian; (ii) $\mathcal{F}$ is a fibered category over $\mathcal{E}$, and the $\mathcal{F}_S$ are groupoids (i.e. every morphism in $\mathcal{F}_S$ is an isomorphism). In this case $\mathcal{F}$ is said to be a category *fibered in groupoids* over $\mathcal{E}$. These are the categories encountered especially 166
in "moduli theory". If $\mathcal{E}$ is a groupoid, one shows that conditions (i) and (ii) are also equivalent to the following: (iii) $\mathcal{F}$ is a groupoid, and the projection functor $p\colon \mathcal{F} \to \mathcal{E}$ is transportable (cf. VI.4.4). For example, if $\mathcal{E}$ and $\mathcal{F}$ are groupoids such that $\mathrm{Ob}(\mathcal{E})$ and $\mathrm{Ob}(\mathcal{F})$ are each reduced to one point, so that $\mathcal{E}$ and $\mathcal{F}$ are defined, up to isomorphism, by groups E and F, and the functor $p\colon \mathcal{F} \to \mathcal{E}$ is defined by a group homomorphism

$p\colon F \to E$, then $\mathcal{F}$ is fibered over $\mathcal{E}$ if and only if p is surjective, i.e. if p defines an *extension* of the group E by the group $G = \operatorname{Ker} p$.

PROPOSITION VI.6.2. *Let $F\colon \mathcal{F} \to \mathcal{G}$ be an $\mathcal{E}$-equivalence. Then $\mathcal{F}$ is a fibered (respectively, prefibered) category over $\mathcal{E}$ if and only if $\mathcal{G}$ is.*

This follows readily from the definitions and from the observation made above that a morphism α in $\mathcal{F}$ is Cartesian if and only if $F(\alpha)$ is Cartesian.

PROPOSITION VI.6.3. *Let $\mathcal{F}_1, \mathcal{F}_2$ be two categories over $\mathcal{E}$, and let $\alpha = (\alpha_1, \alpha_2)$ be a morphism in $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$. Then α is Cartesian if and only if both its components are Cartesian.*

Let ξ_i be the target and η_i the source of α_i, and let $f\colon T \to S$ be the morphism of $\mathcal{E}$ for which α_1 and α_2 are f-morphisms. For every $\eta' = (\eta'_1, \eta'_2)$ in $\mathcal{F}_T$, there is a commutative diagram

$$\begin{array}{ccc} \operatorname{Hom}_T(\eta', \eta) & \longrightarrow & \operatorname{Hom}_f(\eta', \xi) \\ \downarrow & & \downarrow \\ \operatorname{Hom}_T(\eta'_1, \eta_1) \times \operatorname{Hom}_T(\eta'_2, \eta_2) & \longrightarrow & \operatorname{Hom}_f(\eta'_1, \xi_1) \times \operatorname{Hom}_f(\eta'_2, \xi_2) \end{array}$$

whose vertical arrows are bijections. Thus, if either horizontal arrow is a bijection, then so is the other. This already shows that if α_1 and α_2 are Cartesian, so that the lower horizontal arrow is bijective, then α is Cartesian. Conversely, in the preceding diagram first put $\eta'_2 = \eta_2$, so that $\operatorname{Hom}_T(\eta'_2, \eta_2) \neq \emptyset$; this proves that α_1 is Cartesian. Then put $\eta'_1 = \eta_1$, proving that α_2 is Cartesian.

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COROLLARY VI.6.4. *Let $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$, and let $F = (F_1, F_2)$ be an $\mathcal{E}$-functor $\mathcal{G} \to \mathcal{F}$. Then F is Cartesian if and only if F_1 and F_2 are Cartesian. This gives an isomorphism of categories*

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2) \xrightarrow{\sim} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_1) \times \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_2),$$

and, in particular, by taking $\mathcal{G} = \mathcal{E}$, an isomorphism of categories

$$\varprojlim(\mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2/\mathcal{E}) \xrightarrow{\sim} \varprojlim(\mathcal{F}_1/\mathcal{E}) \times \varprojlim(\mathcal{F}_2/\mathcal{E}).$$

COROLLARY VI.6.5. *Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two fibered (respectively, prefibered) categories over $\mathcal{E}$. Then their fiber product $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$ is a fibered (respectively, prefibered) category over $\mathcal{E}$.*

These results extend to the fiber product of an arbitrary family of categories over $\mathcal{E}$.

PROPOSITION VI.6.6. *Let $\mathcal{F}$ be a category over $\mathcal{E}$, with projection functor p, and let $\lambda\colon \mathcal{E}' \to \mathcal{E}$ be a functor. Regard*

$$\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$$

as a category over $\mathcal{E}'$ by means of the projection functor $p' = p \times_{\mathcal{E}} \mathrm{id}_{\mathcal{E}'}$. Let α' be a morphism of $\mathcal{F}'$. Then α' is Cartesian if and only if its image α in $\mathcal{F}$ is Cartesian.

The proof is immediate and is left to the reader.

COROLLARY VI.6.7. *For every Cartesian functor $F\colon \mathcal{F} \to \mathcal{G}$ between categories over $\mathcal{E}$, the functor*

$$F' = F \times_{\mathcal{E}} \mathcal{E}'\colon \mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}' \longrightarrow \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'$$

is Cartesian.

Consequently, the functor

$$\mathbf{Hom}_{\mathcal{E}}(\mathcal{F},\mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathcal{E}'}(\mathcal{F}',\mathcal{G}')$$

considered in no. VI.3 induces a functor

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}',\mathcal{G}').$$

In other words, for fixed $\mathcal{F},\mathcal{G}$, one may regard

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}',\mathcal{G}\times_{\mathcal{E}}\mathcal{E}')$$

as a functor in $\mathcal{E}'$, from the category $\mathbf{Cat}_{/\mathcal{E}}{}^{\circ}$ *to* **Cat**. If $\mathcal{F},\mathcal{G}$ are also allowed to vary, one 168
obtains a functor from

$$\mathbf{Cat}_{/\mathcal{E}}{}^{\circ}\times(\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}})^{\circ}\times\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$$

to **Cat**.

Taking into account the isomorphism

$$\mathrm{Hom}_{\mathcal{E}'}(\mathcal{F}',\mathcal{G}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}',\mathcal{G})$$

considered in no. VI.3, the Cartesian $\mathcal{E}'$-functors from $\mathcal{F}'$ to $\mathcal{G}'$ correspond to the $\mathcal{E}$-functors

$$\mathcal{F}\times_{\mathcal{E}}\mathcal{E}' \longrightarrow \mathcal{G}$$

that carry every morphism whose first projection is a Cartesian morphism of $\mathcal{F}$ to a Cartesian morphism of $\mathcal{G}$. Taking $\mathcal{F}=\mathcal{E}$ and changing notation gives:

Corollary VI.6.8. $\varprojlim(\mathcal{F}'/\mathcal{E}')$ *is isomorphic to the full subcategory of* $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{E}',\mathcal{F})$ *formed by those $\mathcal{E}$-functors $\mathcal{E}'\to\mathcal{F}$ that carry arbitrary morphisms to Cartesian morphisms. In particular, if $\mathcal{F}$ is a fibered category and $\widetilde{\mathcal{F}}$ is the subcategory of $\mathcal{F}$ whose morphisms are the Cartesian morphisms of $\mathcal{F}$, then there is a bijection*

$$\mathrm{Ob}\varprojlim(\mathcal{F}'/\mathcal{E}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}/-}(\mathcal{E}',\widetilde{\mathcal{F}}).$$

This specifies how the expression

$$\varprojlim(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'/\mathcal{E}')$$

is to be regarded as a functor in $\mathcal{E}'$ and $\mathcal{F}$, from $\mathbf{Cat}_{/\mathcal{E}}{}^{\circ}\times\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ to **Cat**. We shall later obtain a fuller functorial dependence on $\mathcal{E}'$ when $\mathcal{F}$ is required to be a fibered category.

Corollary VI.6.9. *Let $\mathcal{F}$ be a fibered (respectively, prefibered) category over $\mathcal{E}$. Then $\mathcal{F}'=\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'$ is a fibered (respectively, prefibered) category over $\mathcal{E}'$.*

Proposition VI.6.10. *Let $\mathcal{F},\mathcal{G}$ be prefibered categories over $\mathcal{E}$, and let F be a Cartesian $\mathcal{E}$-functor from $\mathcal{F}$ to $\mathcal{G}$. Then F is faithful (respectively fully faithful, respectively* 169
an $\mathcal{E}$-equivalence) if and only if, for every $S\in\mathrm{Ob}(\mathcal{E})$, the induced functor

$$F_S\colon \mathcal{F}_S \longrightarrow \mathcal{G}_S$$

is faithful (respectively fully faithful, respectively an equivalence).

The proof is immediate from the definitions.

To conclude this section, we give some properties of fibered categories that use axiom $\mathrm{Fib}_{\mathrm{II}}$.

Proposition VI.6.11. *Let $\mathcal{F}$ be a prefibered category over $\mathcal{E}$. Then $\mathcal{F}$ is fibered if and only if it satisfies the following condition:*

$\mathrm{Fib}'_{\mathrm{II}}$*: Let $\alpha\colon\eta\to\xi$ be a Cartesian morphism in $\mathcal{F}$ lying over a morphism $f\colon T\to S$ of $\mathcal{E}$. For every morphism $g\colon U\to T$ in $\mathcal{E}$, and every $\zeta\in\mathrm{Ob}(\mathcal{F}_U)$, the map $u\longmapsto\alpha\circ u$,*

$$\mathrm{Hom}_g(\zeta,\eta) \longrightarrow \mathrm{Hom}_{fg}(\zeta,\xi),$$

is bijective.

In other words, in a category *fibered* over $\mathcal{E}$, Cartesian diagrams are characterized by a property that is a priori stronger than the one in the definition (obtained by taking $g = \mathrm{id}_T$ in the preceding statement).

COROLLARY VI.6.12. *Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let α be a morphism in $\mathcal{F}$. If α is an isomorphism, then $p(\alpha) = f$ is an isomorphism and α is Cartesian. The converse holds if $\mathcal{F}$ is fibered over $\mathcal{E}$.*

Indeed, if α is an isomorphism, then $f = p(\alpha)$ plainly is as well. For every $\eta' \in \mathrm{Ob}(\mathcal{F}_T)$, the map $u \longmapsto \alpha \circ u$,

$$\mathrm{Hom}(\eta', \eta) \longrightarrow \mathrm{Hom}(\eta', \xi),$$

is bijective. Since f is an isomorphism, an element on the left is a T-morphism if and only if its image on the right is an f-morphism. We therefore obtain a bijection

$$\mathrm{Hom}_T(\eta', \eta) \longrightarrow \mathrm{Hom}_f(\eta', \xi),$$

which proves the first assertion. 170

Conversely, suppose that f is an isomorphism and that α satisfies condition $\mathrm{Fib}'_{\mathrm{II}}$, which, when $\mathcal{F}$ is fibered over $\mathcal{E}$, means that α is Cartesian. Then one sees at once that, for every $\zeta \in \mathrm{Ob}(\mathcal{F})$, the map

$$\mathrm{Hom}(\zeta, \eta) \longrightarrow \mathrm{Hom}(\zeta, \xi), \qquad u \longmapsto \alpha \circ u,$$

is bijective. Thus α is an isomorphism.

COROLLARY VI.6.13. *Let $\alpha\colon \eta \to \xi$ and $\beta\colon \zeta \to \eta$ be two composable morphisms in the category $\mathcal{F}$ fibered over $\mathcal{E}$. If α is Cartesian, then β is Cartesian if and only if $\alpha\beta$ is Cartesian.*

Use the strengthened form of the definition of Cartesian morphisms given in VI.6.11.

VI.7. Cloven categories over $\mathcal{E}$

DEFINITION VI.7.1. *Let $\mathcal{F}$ be a category over $\mathcal{E}$. A* cleavage *of $\mathcal{F}$ over $\mathcal{E}$ is a function that assigns to every $f \in \mathrm{Fl}(\mathcal{E})$ an inverse-image functor f^* for f in $\mathcal{F}$. The cleavage is called* normalized *if $f = \mathrm{id}_S$ implies $f^* = \mathrm{id}_{\mathcal{F}_S}$. A* cloven category *(respectively a* normalized cloven category*) is a category $\mathcal{F}$ over $\mathcal{E}$ equipped with a cleavage (respectively a normalized cleavage).*

It is clear that $\mathcal{F}$ admits a cleavage if and only if it is prefibered over $\mathcal{E}$, and in that case it admits a normalized cleavage. The set of cleavages on $\mathcal{F}$ is in bijection with the set of subsets $K \subset \mathrm{Fl}(\mathcal{F})$ satisfying the following conditions:

(a) Every $\alpha \in K$ is cartesian.
(b) For every morphism $f\colon T \to S$ in $\mathcal{E}$ and every $\xi \in \mathrm{Ob}(\mathcal{F}_S)$, there is a unique f-morphism in K with target ξ.

The cleavage defined by K is normalized if and only if K also satisfies

(c) The identity morphisms of $\mathcal{F}$ belong to K.

The elements of K will be called the *transport morphisms* for the cleavage in question. French source p. 171

The notion of an isomorphism of cloven categories over $\mathcal{E}$ is clear. More generally, one may define morphisms of cloven $\mathcal{E}$-categories to be the $\mathcal{E}$-functors $\mathcal{F} \to \mathcal{G}$ that send transport morphisms to transport morphisms; in particular, these are cartesian functors. In this way the cloven categories over $\mathcal{E}$ are the objects of a category, the *category of cloven categories over $\mathcal{E}$*. The reader may spell out the existence of products: if a category over $\mathcal{E}$ is the product of categories $\mathcal{F}_i$ over $\mathcal{E}$, each supplied with a cleavage, then their product has the corresponding natural cleavage. We likewise leave the notion of base change for cloven categories to the reader.

We denote by $\alpha_f(\xi)$ the canonical morphism

$$\alpha_f(\xi)\colon f^*(\xi) \longrightarrow \xi.$$

As already noted, it is functorial in ξ; that is, there is a natural transformation

$$\alpha_f\colon i_T f^* \longrightarrow i_S,$$

where, for every $S \in \mathrm{Ob}(\mathcal{E})$, $i_S\colon \mathcal{F}_S \to \mathcal{F}$ denotes the inclusion functor.

Now consider morphisms

$$f\colon T \longrightarrow S, \qquad g\colon U \longrightarrow T$$

in $\mathcal{E}$, and let $\xi \in \mathrm{Ob}(\mathcal{F}_S)$. There is a unique U-morphism

$$c_{f,g}(\xi)\colon g^*f^*(\xi) \longrightarrow (fg)^*(\xi)$$

making the diagram

$$\begin{array}{ccc} f^*(\xi) & \xleftarrow{\alpha_g(f^*(\xi))} & g^*f^*(\xi) \\ {\scriptstyle \alpha_f(\xi)}\downarrow & & \downarrow{\scriptstyle c_{f,g}(\xi)} \\ \xi & \xleftarrow[\alpha_{fg}(\xi)]{} & (fg)^*(\xi) \end{array}$$

commutative, by the definition of $(fg)^*(\xi)$. As ξ varies, this morphism is functorial; in other words, there is a natural transformation

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$$c_{f,g}\colon g^*f^* \longrightarrow (fg)^*$$

between functors $\mathcal{F}_S \to \mathcal{F}_U$.

Proposition VI.7.2. *A cloven category $\mathcal{F}$ over $\mathcal{E}$ is fibered if and only if all the transformations $c_{f,g}$ are isomorphisms.*

Taking f to be an isomorphism and g its inverse, and considering $c_{f,g}$ and $c_{g,f}$, gives the following.

Corollary VI.7.3. *If $\mathcal{F}$ is a cloven category fibered over $\mathcal{E}$, then, for every isomorphism $f\colon T \to S$ in $\mathcal{E}$, the functor*

$$f^*\colon \mathcal{F}_S \longrightarrow \mathcal{F}_T$$

is an equivalence of categories.

Proposition VI.7.4. *Let $\mathcal{F}$ be a cloven category over $\mathcal{E}$. Then*

$$\text{A)} \qquad \begin{cases} c_{f,\mathrm{id}_T}(\xi) = \alpha_{\mathrm{id}_T}(f^*(\xi)), \\ c_{\mathrm{id}_S,f}(\xi) = f^*(\alpha_{\mathrm{id}_S}(\xi)), \end{cases}$$

and

$$\text{B)} \qquad c_{f,gh}(\xi) \circ c_{g,h}(f^*(\xi)) = c_{fg,h}(\xi) \circ h^*(c_{f,g}(\xi)).$$

Here f, g, h are successive morphisms

$$V \longrightarrow U \longrightarrow T \longrightarrow S$$

and ξ is an object of $\mathcal{F}_S$.

For a normalized cleavage, the first two relations take the simpler form

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$$\text{(A')} \qquad c_{f,\mathrm{id}_T} = \mathrm{id}_{f^*}, \qquad c_{\mathrm{id}_S,f} = \mathrm{id}_{f^*}.$$

The third relation is visualized by the commutative diagram

(D)
$$\begin{array}{ccc} h^*g^*f^*(\xi) & \xrightarrow{c_{g,h}(f^*(\xi))} & (gh)^*f^*(\xi) \\ {\scriptstyle h^*(c_{f,g}(\xi))}\downarrow & & \downarrow{\scriptstyle c_{f,gh}(\xi)} \\ h^*(fg)^*(\xi) & \xrightarrow[c_{fg,h}(\xi)]{} & (fgh)^*(\xi). \end{array}$$

For fibered categories, where the $c_{f,g}$ are isomorphisms, this commutativity says intuitively that successive use of isomorphisms of the form $c_{f,g}$ does not lead to "contradictory identifications." The same formula, written without the argument ξ and using the convolution product of natural transformations, is

$$c_{fg,h} \circ (h^* * c_{f,g}) = c_{f,gh} \circ (c_{g,h} * f^*).$$

The first two formulas in VI.7.4 are immediate; we sketch the proof of the third. In addition to square (D), consider the square of natural transformations

(D')
$$\begin{array}{ccc} g^*f^*(\xi) & \xrightarrow{\alpha_g(f^*(\xi))} & f^*(\xi) \\ {\scriptstyle c_{f,g}(\xi)}\downarrow & & \downarrow{\scriptstyle \alpha_f(\xi)} \\ (fg)^*(\xi) & \xrightarrow[\alpha_{fg}(\xi)]{} & \xi, \end{array}$$

which is commutative by the definition of $c_{f,g}(\xi)$. Join the vertices of (D) to the corresponding vertices of (D') by the natural transformations

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$$\begin{array}{ll} \alpha_h(g^*f^*(\xi)), & \alpha_{gh}(f^*(\xi)), \\ \alpha_h((fg)^*(\xi)), & \alpha_{fgh}(\xi). \end{array}$$

The four lateral faces of the resulting cube are also commutative. For the left face this follows because the left column of (D) is obtained from the left column of (D') by applying h^*, and because α_h is natural. For the other three faces it is simply the definition of the c-operations on the remaining three sides of (D). Thus the five faces other than the top face commute. It follows that the two (fgh)-morphisms

$$h^*g^*f^*(\xi) \longrightarrow (fgh)^*(\xi)$$

defined by (D) have the same composite with $\alpha_{fgh}(\xi)\colon (fgh)^*(\xi) \to \xi$; hence they are equal by the definition of $(fgh)^*$.

For what follows, restrict attention to *normalized* cloven categories. Such a category gives rise to the following data:

(a) A map $S \mapsto \mathcal{F}_S$ from $\mathrm{Ob}(\mathcal{E})$ to **Cat**.
(b) A map $f \mapsto f^*$ assigning to every $f \in \mathrm{Fl}(\mathcal{E})$, with source T and target S, a functor $f^*\colon \mathcal{F}_S \to \mathcal{F}_T$.
(c) A map $(f, g) \mapsto c_{f,g}$ assigning to every composable pair of arrows (f, g) of $\mathcal{E}$ a natural transformation $c_{f,g}\colon g^*f^* \to (fg)^*$.

These data satisfy conditions (A') and (B) above. If we had not restricted to normalized cleavages, one further datum would be needed: a map $S \mapsto \alpha_S$ assigning to every object S of $\mathcal{E}$ a natural transformation

$$\alpha_S\colon (\mathrm{id}_S)^* \longrightarrow \mathrm{id}_{\mathcal{F}_S};$$

condition (A') would then be replaced by condition (A).

We shall now show how the preceding data reconstruct, up to unique isomorphism, the normalized cloven category $\mathcal{F}$ over $\mathcal{E}$.

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VI.8. Cloven category defined by a pseudofunctor $\mathcal{E}^\circ \to \mathbf{Cat}$

For brevity, call a *pseudofunctor* from $\mathcal{E}^\circ$ to **Cat**—more precisely, a normalized pseudofunctor—a set of data a), b), c) as above satisfying conditions A') and B). In the preceding section we associated a pseudofunctor $\mathcal{E}^\circ \to \mathbf{Cat}$ with a normalized cloven category over $\mathcal{E}$; we now describe the inverse construction. We leave to the reader the verification of most of the details, as well as the fact that these constructions are indeed inverse to one another. More precisely, one should regard the pseudofunctors $\mathcal{E}^\circ \to \mathbf{Cat}$ as the objects of a new category and show that our constructions give mutually quasi-inverse equivalences between this category and the category of cloven categories over $\mathcal{E}$ defined in the preceding section.

Set

$$\mathcal{F}_\circ = \coprod_{S \in \mathrm{Ob}(\mathcal{E})} \mathrm{Ob}(\mathcal{F}(S)),$$

the disjoint union of the sets $\mathrm{Ob}(\mathcal{F}(S))$. (N.B. Here we write $\mathcal{F}(S)$, rather than $\mathcal{F}_S$, for the value of the given pseudofunctor at the object S of $\mathcal{E}$, in order to avoid notational confusion below.) There is an evident map

$$p_\circ \colon \mathcal{F}_\circ \longrightarrow \mathrm{Ob}(\mathcal{E}).$$

Let

$$\overline{\xi} = (S, \xi), \qquad \overline{\eta} = (T, \eta) \qquad \text{with } \xi \in \mathrm{Ob}(\mathcal{F}(S)),\ \eta \in \mathrm{Ob}(\mathcal{F}(T))$$

be two elements of $\mathcal{F}_\circ$, and let $f \in \mathrm{Hom}(T, S)$. Define

$$h_f(\overline{\eta}, \overline{\xi}) = \mathrm{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)).$$

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If, in addition, $g \colon U \to T$ is a morphism in $\mathcal{E}$ and $\zeta \in \mathrm{Ob}(\mathcal{F}(U))$, define a map, denoted $(u, v) \longmapsto u \circ v$,

$$h_f(\overline{\eta}, \overline{\xi}) \times h_g(\overline{\zeta}, \overline{\eta}) \longrightarrow h_{fg}(\overline{\zeta}, \overline{\xi}),$$

i.e. a map

$$\mathrm{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)) \times \mathrm{Hom}_{\mathcal{F}(U)}(\zeta, g^*(\eta)) \longrightarrow \mathrm{Hom}_{\mathcal{F}(U)}(\zeta, (fg)^*(\xi)),$$

by the formula

$$u \circ v = c_{f,g}(\xi) \cdot g^*(u) \cdot v.$$

Thus $u \circ v$ is the composite

$$\zeta \xrightarrow{v} g^*(\eta) \xrightarrow{g^*(u)} g^* f^*(\xi) \xrightarrow{c_{f,g}(\xi)} (fg)^*(\xi).$$

Also set

$$h(\overline{\eta}, \overline{\xi}) = \coprod_{f \in \mathrm{Hom}(T,S)} h_f(\overline{\eta}, \overline{\xi}).$$

The preceding pairings then define pairings

$$h(\overline{\eta}, \overline{\xi}) \times h(\overline{\zeta}, \overline{\eta}) \longrightarrow h(\overline{\zeta}, \overline{\xi}),$$

while the definition of $h(\overline{\eta}, \overline{\xi})$ gives an evident map

$$p_{\overline{\eta},\overline{\xi}} \colon h(\overline{\eta}, \overline{\xi}) \longrightarrow \mathrm{Hom}(T, S).$$

One now verifies the following points:

1) Composition between elements of $h(\overline{\eta}, \overline{\xi})$ is *associative*. 177
2) For every $\overline{\xi} = (S, \xi)$ in $\mathcal{F}_\circ$, consider the identity element of

$$h_{\mathrm{id}_S}(\overline{\xi}, \overline{\xi}) = \mathrm{Hom}_{\mathcal{F}(S)}(\mathrm{id}_S^*(\xi), \xi) = \mathrm{Hom}_{\mathcal{F}(S)}(\xi, \xi),$$

and its image in $h(\overline{\xi}, \overline{\xi})$. This element is a left and right *identity* for composition between elements of the $h(\overline{\eta}, \overline{\xi})$.

This already shows that one obtains a category $\mathcal{F}$ by setting

$$\mathrm{Ob}(\mathcal{F}) = \mathcal{F}_\circ, \qquad \mathrm{Fl}(\mathcal{F}) = \coprod_{\overline{\xi},\overline{\eta}\in\mathcal{F}_\circ} h(\overline{\eta},\overline{\xi}).$$

(N.B. One cannot simply take $\mathrm{Fl}(\mathcal{F})$ to be the *union* of the sets $h(\overline{\eta},\overline{\xi})$, since these sets need not be disjoint.) Moreover:

3) The maps

$$p_\circ\colon \mathrm{Ob}(\mathcal{F}) \to \mathrm{Ob}(\mathcal{E}), \qquad p_1 = (p_{\overline{\eta},\overline{\xi}})\colon \mathrm{Fl}(\mathcal{F}) \to \mathrm{Fl}(\mathcal{E})$$

define a *functor* $p\colon \mathcal{F} \to \mathcal{E}$. Thus $\mathcal{F}$ becomes a category over $\mathcal{E}$, and the evident map

$$h_f(\overline{\eta},\overline{\xi}) \longrightarrow \mathrm{Hom}(\overline{\eta},\overline{\xi})$$

induces a *bijection*

$$h_f(\overline{\eta},\overline{\xi}) \xrightarrow{\sim} \mathrm{Hom}_f(\overline{\eta},\overline{\xi}).$$

4) The evident maps

$$\mathrm{Ob}(\mathcal{F}(S)) \longrightarrow \mathcal{F}_\circ = \mathrm{Ob}(\mathcal{F}), \qquad \mathrm{Fl}(\mathcal{F}(S)) \longrightarrow \mathrm{Fl}(\mathcal{F}),$$

where the second is defined by the evident maps

$$\mathrm{Hom}_{\mathcal{F}(S)}(\xi,\xi') = h_{\mathrm{id}_S}(\overline{\xi},\overline{\xi'}) \longrightarrow \mathrm{Hom}(\overline{\xi},\overline{\xi'}),$$

define an *isomorphism*

$$i_S\colon \mathcal{F}(S) \xrightarrow{\sim} \mathcal{F}_S.$$

5) For every object $\overline{\xi} = (S,\xi)$ of $\mathcal{F}$, and every morphism $f\colon T \to S$ of $\mathcal{E}$, consider the element $\overline{\eta} = (T,\eta)$ of $\mathcal{F}_T$, where $\eta = f^*(\xi)$, and the element $\alpha_f(\xi)$ of $\mathrm{Hom}(\overline{\eta},\overline{\xi})$ that is the image of $\mathrm{id}_{f^*(\xi)}$ under the map 178

$$\mathrm{Hom}_{\mathcal{F}(T)}(f^*(\xi), f^*(\xi)) = h_f(\overline{\eta},\overline{\xi}) \longrightarrow \mathrm{Hom}_f(\overline{\eta},\overline{\xi}).$$

This element is Cartesian, and is the identity of $\overline{\xi}$ *when* $f = \mathrm{id}_S$. In other words, the collection of the $\alpha_f(\xi)$ defines a *normalized cleavage of* $\mathcal{F}$ *over* $\mathcal{E}$. Moreover, by construction, the diagram of functors

$$\begin{array}{ccc} \mathcal{F}(S) & \xrightarrow{f^*} & \mathcal{F}(T) \\ {\scriptstyle i_S}\downarrow & & \downarrow{\scriptstyle i_T} \\ \mathcal{F}_S & \xrightarrow{f^*_{\mathcal{F}}} & \mathcal{F}_T \end{array}$$

is commutative, where $f^*_{\mathcal{F}}$ is the inverse-image functor under f relative to the cleavage just defined on $\mathcal{F}$. Finally:

6) Under the isomorphisms i_S, the homomorphisms $c_{f,g}$ supplied with the pseudofunctor are carried to the functorial homomorphisms $c_{f,g}$ associated with the cleavage of $\mathcal{F}$.

We give only the verification of 1), which is, if anything, less immediate than the others. It suffices to prove associativity of composition between elements of sets of the form $h_f(\overline{\eta},\overline{\xi})$. Consider morphisms in $\mathcal{E}$

$$S \xleftarrow{f} T \xleftarrow{g} U \xleftarrow{h} V,$$

objects

$$\xi,\ \eta,\ \zeta,\ \tau$$

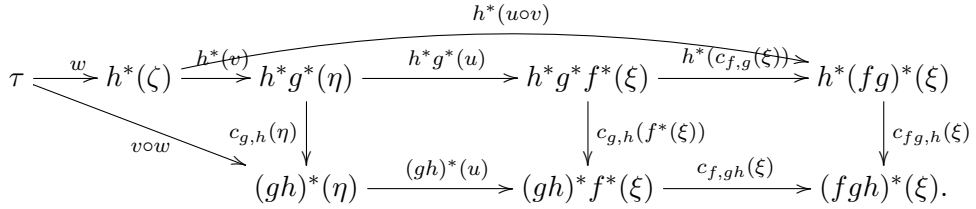
in $\mathcal{F}(S), \mathcal{F}(T), \mathcal{F}(U), \mathcal{F}(V)$, respectively, and elements

$$\begin{aligned} u \in h_f(\overline{\eta},\overline{\xi}) &= \mathrm{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)), \\ v \in h_g(\overline{\zeta},\overline{\eta}) &= \mathrm{Hom}_{\mathcal{F}(U)}(\zeta, g^*(\eta)), \\ w \in h_h(\overline{\tau},\overline{\zeta}) &= \mathrm{Hom}_{\mathcal{F}(V)}(\tau, h^*(\zeta)). \end{aligned}$$

We must prove 179

$$(u \circ v) \circ w = u \circ (v \circ w),$$

an equality in $\mathrm{Hom}_{\mathcal{F}(V)}(\tau, (fgh)^*(\xi))$. By the definitions, the two sides are obtained by composing along the upper and lower boundaries, respectively, of the following diagram:



The middle square is commutative because $c_{g,h}$ is a functorial homomorphism, and the square on the right is commutative by condition B) for a pseudofunctor. This proves the assertion.

Of course, when the pseudofunctor in question already comes from a normalized cloven category $\mathcal{F}'$ over $\mathcal{E}$, one must still specify the natural isomorphism between $\mathcal{F}'$ and $\mathcal{F}$. We leave the details to the reader.

We also leave to the reader the interpretation, in terms of pseudofunctors, of the inverse image of a cloven category $\mathcal{F}$ over $\mathcal{E}$ under a base-change functor $\mathcal{E}' \to \mathcal{E}$.

VI.9. Example: a cloven category defined by a functor $\mathcal{E}^\circ \to \mathbf{Cat}$; split categories over $\mathcal{E}$

Suppose that a functor

$$\varphi\colon \mathcal{E}^\circ \to \mathbf{Cat}$$

is given. It defines a pseudofunctor by setting

$$\mathcal{F}(S) = \varphi(S), \qquad f^* = \varphi(f), \qquad c_{f,g} = \mathrm{id}_{(fg)^*}.$$

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The construction of the preceding section therefore gives a cloven category $\mathcal{F}$ over $\mathcal{E}$, called the category associated with the functor φ. A cloven category over $\mathcal{E}$ is isomorphic to a cloven category defined by a functor $\varphi\colon \mathcal{E}^\circ \to \mathbf{Cat}$ if and only if it satisfies

$$(fg)^* = g^*f^*, \qquad c_{f,g} = \mathrm{id}_{(fg)^*}.$$

In terms of the set K of transport morphisms, this simply says that *the composite of two transport morphisms is a transport morphism.* A cleavage of a category $\mathcal{F}$ over $\mathcal{E}$ satisfying this condition is called a *splitting* of $\mathcal{F}$ over $\mathcal{E}$, and a category $\mathcal{F}$ over $\mathcal{E}$ equipped with a splitting is called a *split category over* $\mathcal{E}$. This is a special case of the notion of a cloven category. The category of split categories over $\mathcal{E}$ is therefore equivalent to $\mathbf{Hom}(\mathcal{E}^\circ, \mathbf{Cat})$. Note that a split category over $\mathcal{E}$ is, in particular, a cloven category over $\mathcal{E}$.

If $\mathcal{F}$ is a fibered category over $\mathcal{E}$, a splitting of $\mathcal{F}$ need not exist. Suppose, for example, that $\mathrm{Ob}(\mathcal{E})$ and $\mathrm{Ob}(\mathcal{F})$ each consist of one element and that the endomorphisms of those elements form groups E and F, respectively. The projection functor p is then given by a group homomorphism $p\colon F \to E$, which is surjective because p is fibered. One immediately verifies that the set of cleavages of $\mathcal{F}$ over $\mathcal{E}$ is in bijection with the set of maps

$$s\colon E \longrightarrow F \qquad \text{such that} \qquad ps = \mathrm{id}_E;$$

these are the systems of representatives for the cosets modulo the subgroup $G = \mathrm{Ker}\, p$ of the surjective homomorphism $p\colon F \to E$. A cleavage is a splitting if and only if s is a group homomorphism. Thus a splitting exists precisely when the extension F of E by G is split; when G is commutative, this is expressed by the vanishing of a certain cohomology class in $H^2(E, G)$, where G is regarded as a group on which E acts.

Suppose, however, that $\mathcal{F}$ is a fibered category over $\mathcal{E}$ whose fiber categories $\mathcal{F}_S$ are 181
rigid, i.e. every object of $\mathcal{F}_S$ has trivial automorphism group. It is then easy to prove

that $\mathcal{F}$ admits a splitting over $\mathcal{E}$. First, the existence of a splitting is unchanged if $\mathcal{F}$ is replaced by an $\mathcal{E}$-equivalent category. We may therefore reduce to the case in which the $\mathcal{F}_S$ are both rigid and *reduced*, meaning that two isomorphic objects in $\mathcal{F}_S$ are equal. If G is a rigid reduced category, every isomorphism between two functors $H \to G$, with H arbitrary, is an identity. Consequently, if $\mathcal{F}$ is fibered over $\mathcal{E}$ and its fiber categories are rigid and reduced, then $\mathcal{F}$ has a *unique* cleavage, which is necessarily a splitting. Thus $\mathcal{F}$ is isomorphic to the category defined by a functor

$$\varphi\colon \mathcal{E}^\circ \longrightarrow \mathbf{Cat}$$

whose $\varphi(S)$ are rigid discrete categories, and the functor φ is determined up to isomorphism.

VI.10. Cofibered categories and bifibered categories

Consider a category $\mathcal{F}$ over $\mathcal{E}$, with projection functor

$$p\colon \mathcal{F} \longrightarrow \mathcal{E}.$$

It defines a category $\mathcal{F}^\circ$ over $\mathcal{E}^\circ$, with projection functor

$$p^\circ\colon \mathcal{F}^\circ \longrightarrow \mathcal{E}^\circ.$$

A morphism $\alpha\colon \eta \to \xi$ in $\mathcal{F}$ is called *cocartesian* if it is Cartesian for $\mathcal{F}^\circ$ over $\mathcal{E}^\circ$. Explicitly, this means that, for every object ξ' of $\mathcal{F}_S$, the map $u \longmapsto u \circ \alpha$,

$$\mathrm{Hom}_S(\xi, \xi') \longrightarrow \mathrm{Hom}_f(\eta, \xi'),$$

is bijective. The pair (ξ, α) is then also called a *direct image* of η under f, in the category $\mathcal{F}$ over $\mathcal{E}$. If such a direct image exists for every η in $\mathcal{F}_T$, the direct-image functor under f is said to exist; once chosen, it is denoted by $f_*^{\mathcal{F}}$, or simply f_*. It is characterized by an 182
isomorphism of bifunctors on $\mathcal{F}_T^\circ \times \mathcal{F}_S$:

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_f(\eta, \xi).$$

If f_* exists, then f^* exists if and only if f_* admits a right adjoint, i.e. if and only if there exist a functor

$$f^*\colon \mathcal{F}_S \longrightarrow \mathcal{F}_T$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_T(\eta, f^*(\xi)).$$

Let $g\colon U \to T$ be another morphism in $\mathcal{E}$, and suppose that the inverse and direct images under f, g, and fg exist. Consider the functorial homomorphisms

$$\begin{array}{rcl} c^{f,g}\colon f_*g_* & \longleftarrow & (fg)_*, \\ c_{f,g}\colon g^*f^* & \longrightarrow & (fg)^*. \end{array}$$

If f_*g_* and g^*f^* are regarded as a pair of adjoint functors, as are $(fg)_*$ and $(fg)^*$, then these two homomorphisms are adjoint to one another. Hence one is an isomorphism if and only if the other is. In particular:

Proposition VI.10.1. *Suppose that the category $\mathcal{F}$ over $\mathcal{E}$ is both prefibered and coprefibered. Then it is fibered if and only if it is cofibered.*

Of course, $\mathcal{F}$ is called coprefibered, respectively cofibered, over $\mathcal{E}$ if $\mathcal{F}^\circ$ is prefibered, respectively fibered, over $\mathcal{E}^\circ$. We call $\mathcal{F}$ *bifibered over* $\mathcal{E}$ if it is both fibered and cofibered over $\mathcal{E}$.

VI.11. Miscellaneous examples

a) **Arrow categories of $\mathcal{E}$.** Let $\mathcal{E}$ be a category. Denote by $\mathbf{\Delta^1}$ the category associated with the totally ordered two-element set $[0,1]$; it thus has two objects, 183
0 and 1, and, in addition to the two identity morphisms, an arrow $(0,1)$ with source 0 and target 1. Let

$$\mathbf{Fl}(\mathcal{E}) = \mathbf{Hom}(\mathbf{\Delta^1}, \mathcal{E})$$

which is called the *arrow category of* $\mathcal{E}$. The object 1 of $\mathbf{\Delta^1}$ defines a canonical functor, called the *target functor*,

$$\mathbf{Fl}(\mathcal{E}) \to \mathcal{E}$$

(the functor defined by the object 0 of $\mathbf{\Delta^1}$ is called the *source functor*). For every object S of $\mathcal{E}$, the fiber category $\mathbf{Fl}(\mathcal{E})_S$ is canonically isomorphic to the category $\mathcal{E}_{/S}$ of objects of $\mathcal{E}$ over S.

Consider a morphism $f\colon T \to S$ in $\mathcal{E}$. It corresponds to a canonical functor

$$f_*\colon \mathcal{E}_{/T} = \mathcal{F}_T \to \mathcal{E}_{/S} = \mathcal{F}_S$$

and a functorial isomorphism

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_f(\eta, \xi)$$

which thus makes f_* a direct-image functor for f in $\mathcal{F}$. Moreover, here one has

$$(\mathrm{id}_S)_* = \mathrm{id}_{\mathcal{F}_S}, \quad (fg)_* = f_* g_*, \quad c^{f,g} = \mathrm{id}_{(fg)},$$

i.e. $\mathcal{F}$ is equipped with a cosplitting over $\mathcal{E}$. A fortiori, $\mathcal{F}$ is cofibered over $\mathcal{E}$. Note now that the set of morphisms in $\mathcal{F}$ is in bijection with the set of commutative square diagrams in $\mathcal{E}$:

$$\begin{array}{ccc} X & \xleftarrow{f'} & Y \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ S & \xleftarrow{f} & T \end{array}$$

By definition, the morphism in question is Cartesian if the square is Cartesian 184
in $\mathcal{E}$, i.e. if it makes Y a fiber product of X and T over S. The inverse-image functor f^* therefore exists if and only if, for every object X over S, the fiber product $X \times_S T$ exists. It follows from VI.10.1 that if the product of two objects over a third always exists in $\mathcal{E}$, i.e. if $\mathcal{F}$ is prefibered over $\mathcal{E}$, then $\mathcal{F}$ is in fact fibered over $\mathcal{E}$.

b) **Category of presheaves or sheaves on varying spaces.**

Let $\mathcal{E} = \mathbf{Top}$ be the category of topological spaces. If T is a topological space, we denote by $\mathcal{U}(T)$ the category of open subsets of T, whose morphisms are the inclusion maps. If $\mathcal{C}$ is a category, a functor $\mathcal{U}(T)^\circ \to \mathcal{C}$ is called a *presheaf* on T with values in $\mathcal{C}$, and a *sheaf* if it satisfies a left-exactness condition that we shall not repeat here. The *category $\mathcal{P}(T)$ of presheaves on T with values in $\mathcal{C}$* is, by definition, the category $\mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C})$, and the category $\mathcal{F}(T)$ of sheaves on T with values in $\mathcal{C}$ is the full subcategory whose objects are those objects of $\mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C})$ that are sheaves. If $f\colon T \to S$ is a morphism in $\mathcal{E}$, i.e. a continuous map of topological spaces, the order-preserving map $U \longmapsto f^{-1}(U)$ gives a functor $\mathcal{U}(S) \to \mathcal{U}(T)$, hence a functor

$$f_*\colon \mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C}) \to \mathbf{Hom}(\mathcal{U}(S)^\circ, \mathcal{C}),$$

called the *direct-image functor for presheaves under f*. It is immediate that the direct image of a sheaf is a sheaf. Thus the functor

$$f_*\colon \mathcal{P}(T) \to \mathcal{P}(S)$$

induces a functor, also denoted

$$f_*\colon \mathcal{F}(T) \to \mathcal{F}(S).$$

Moreover, one checks trivially, by associativity of the composition of functors, that for a second continuous map $g\colon U \to T$ one has

$$(gf)_* = g_* f_*, \qquad \text{and likewise} \qquad (\mathrm{id}_S)_* = \mathrm{id}_{\mathcal{P}(S)}.$$

In this way we have obtained a functor

$$S \longmapsto \mathcal{P}(S),$$

respectively

$$S \longmapsto \mathcal{F}(S),$$

from $\mathcal{E}$ to **Cat**. In fact, we are interested in the corresponding functor 185

$$S \longmapsto \mathcal{P}(S)^\circ, \qquad \text{respectively} \qquad S \longmapsto \mathcal{F}(S)^\circ.$$

It defines a cofibered, and indeed cosplit, category over the category of topological spaces, called the *cofibered category of presheaves* (respectively *sheaves*) *with values in* $\mathcal{C}$ (understood: on varying spaces). Spelling out the construction of no. VI.8, one sees that a morphism from a presheaf B on T to a presheaf A on S is a pair (f, u) consisting of a continuous map from T to S and a morphism $u\colon A \to f_*(B)$ in the category $\mathcal{P}(S)$. This description is equally valid for morphisms of sheaves, since $\mathcal{F}$ is a full subcategory of $\mathcal{P}$.

In the most important cases, the categories $\mathcal{P}$ and $\mathcal{F}$ over $\mathcal{E}$ are also fibered categories; that is, for every continuous map, the direct-image functors $\mathcal{P}(T) \to \mathcal{P}(S)$ and $\mathcal{F}(T) \to \mathcal{F}(S)$ have an adjoint functor, then denoted f^* and called the inverse-image functor for presheaves, respectivelythe inverse-image functor for sheaves, under the continuous map f. This functor exists, for example, if $\mathcal{C} = \mathbf{Ens}$. One can show that $f^*\colon \mathcal{P}(S) \to \mathcal{P}(T)$ exists whenever inductive limits (relative to diagrams in the Universe under consideration) exist in $\mathcal{C}$. The question is less easy for $\mathcal{F}$. Indeed, even when $\mathcal{C} = \mathbf{Ens}$, the inverse image of a presheaf that is a sheaf is not in general a sheaf; in other words, the inverse-image functor for sheaves is not isomorphic to the functor induced by the inverse-image functor for presheaves (despite the common notation f^*). Thus $\mathcal{F}$ is a cofibered subcategory of $\mathcal{P}$, but not a fibered subcategory; that is, *the inclusion functor $\mathcal{F} \to \mathcal{P}$ is not fibrant.*

The cofibered category $\mathcal{P}$ can be deduced from a more general cofibered category—or rather, fibered category—constructed as follows. For every category $\mathcal{U}$ (in the fixed Universe), put

$$\mathcal{P}(\mathcal{U}) = \mathbf{Hom}(\mathcal{U}, \mathcal{C}),$$

and note that $\mathcal{U} \longmapsto \mathcal{P}(\mathcal{U})$ is naturally a contravariant functor in $\mathcal{U}$, from **Cat** 186
to **Cat**. It therefore defines a split category over $\mathcal{E} = \mathbf{Cat}$, which we denote $\mathbf{Cat}_{//\mathcal{C}}$. The objects of this category are pairs $(\mathcal{U}, p)$ consisting of a category $\mathcal{U}$ and a functor $p\colon \mathcal{U} \to \mathcal{C}$, and a morphism from $(\mathcal{U}, p)$ to $(\mathcal{V}, q)$ is essentially a pair (f, u), where f is a functor $\mathcal{U} \to \mathcal{V}$ and u is a homomorphism of functors $u\colon p \to qf$. We leave it to the reader to spell out the composition of morphisms in $\mathbf{Cat}_{//\mathcal{C}}$. The projection functor

$$\mathcal{F} = \mathbf{Cat}_{//\mathcal{C}} \to \mathcal{E} = \mathbf{Cat}$$

assigns to the pair $(\mathcal{U}, p)$ the object $\mathcal{U}$; the fiber category at $\mathcal{U}$ is the category $\mathbf{Hom}(\mathcal{U}, \mathcal{C})$, up to isomorphism. When inductive limits exist in $\mathcal{C}$, one easily shows that the fibered category $\mathbf{Cat}_{//\mathcal{C}}$ over **Cat** is also cofibered over **Cat**; that is, one can define the notion of the *direct image of a functor* $p\colon \mathcal{U} \to \mathcal{C}$ under a

functor $f\colon \mathcal{U} \to \mathcal{V}$. The category of presheaves is obtained from the preceding fibered category by the base change

$$\mathbf{Top}^\circ \to \mathbf{Cat}$$

(the functor $S \longmapsto \mathcal{U}(S)$ defined above). This gives a fibered category over $\mathbf{Top}^\circ$, and on passing to the opposite category one obtains the cofibered category $\mathcal{P}$ of presheaves over **Top**. The notion of inverse image of a functor corresponds to that of direct image of a presheaf, and the notion of direct image of a functor to that of inverse image of a presheaf.

c) **Objects with operators over an object with operators.**

Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let S be an object of $\mathcal{E}$ on which a group G acts, on the left to fix ideas. This object with operators can be interpreted as corresponding to a functor $\lambda\colon \mathcal{E}' \to \mathcal{E}$ from the category $\mathcal{E}'$ defined by G (having a single object and G as its group of endomorphisms) to the category $\mathcal{E}$. By base change it therefore defines a category $\mathcal{F}'$ over $\mathcal{E}'$, which is fibered, respectivelycofibered, when $\mathcal{F}$ is so over $\mathcal{E}$. A section of $\mathcal{E}'$ over $\mathcal{F}'$ (necessarily 187
Cartesian, since $\mathcal{E}'$ is a groupoid and every isomorphism in $\mathcal{F}'$ is Cartesian by VI.6.12) can also be interpreted as an $\mathcal{E}$-functor $\mathcal{E}' \to \mathcal{F}$ over λ, or as an object with operators ξ in $\mathcal{F}$ "over" the object with operators S.

d) **Pairs of quasi-inverse adjoint functors; autodualities.**

When the base category $\mathcal{E}$ is reduced to two objects a, b and, besides the identity arrows, two mutually inverse isomorphisms $f\colon a \to b$ and $g\colon b \to a$ (that is, $\mathcal{E}$ is a rigid connected groupoid with two objects), a normalized cloven category over $\mathcal{E}$ is essentially the same thing as a system consisting of two categories $\mathcal{F}_a$ and $\mathcal{F}_b$ and a *pair of adjoint functors*

$$G\colon \mathcal{F}_a \to \mathcal{F}_b, \qquad F\colon \mathcal{F}_b \to \mathcal{F}_a,$$

which are equivalences of categories (and hence quasi-inverse to one another). One takes for $\mathcal{F}_a$ and $\mathcal{F}_b$ the fiber categories of $\mathcal{F}$, for F and G the functors f^* and g^*, and the two isomorphisms

$$u\colon FG \xrightarrow{\sim} \mathrm{id}_{\mathcal{F}_a}, \qquad v\colon GF \xrightarrow{\sim} \mathrm{id}_{\mathcal{F}_b}$$

are $c_{g,f}$ and $c_{f,g}$. The two usual compatibility conditions between u and v are nothing other than condition VI.7.4 B) for the composites fgf and gfg. It is easy to show that these conditions suffice to imply that one indeed has a pseudofunctor $\mathcal{E}^\circ \to \mathbf{Cat}$.

An interesting case is that in which

$$\mathcal{F}_b = \mathcal{F}_a^\circ, \qquad G = F^\circ, \qquad v = u^\circ.$$

An *autoduality* in a category $\mathcal{C}$ means the data of a functor $D\colon \mathcal{C} \to \mathcal{C}^\circ$ and an isomorphism $u\colon DD^\circ \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}}$ such that u and the isomorphism $u^\circ\colon D^\circ D \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}^\circ}$ make (D, D°) a pair of adjoint functors (necessarily quasi-inverse to one another). This condition is written

$$D(u(x)) = u(D(x))qquadfor\ every\ x \in \mathrm{Ob}(\mathcal{C}).$$

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e) **Categories over a discrete category $\mathcal{E}$.** One says that $\mathcal{E}$ is a *discrete category* if every arrow in it is an identity arrow, so that $\mathcal{E}$ is determined up to unique isomorphism by the set $I = \mathrm{Ob}(\mathcal{E})$. The data of a category $\mathcal{F}$ over $\mathcal{E}$ is therefore equivalent (up to unique isomorphism) to the data of a family of categories $\mathcal{F}_i$ $(i \in I)$, the fiber categories. Every category $\mathcal{F}$ over $\mathcal{E}$ is fibered, every $\mathcal{E}$-functor $\mathcal{F} \to \mathcal{G}$ is Cartesian, and there is a canonical isomorphism

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \prod_i \mathbf{Hom}(\mathcal{F}_i, \mathcal{G}_i).$$

In particular, one obtains

$$\mathbf{\Gamma}(\mathcal{F}/\mathcal{E}) = \varprojlim \mathcal{F}/\mathcal{E} \xrightarrow{\sim} \prod_i \mathcal{F}_i.$$

f) **Suppose that $\mathcal{E}$ has exactly two objects S and T,** and, besides the **identity morphisms, one morphism** $f\colon T \to S$. Then a category $\mathcal{F}$ over $\mathcal{E}$ is determined, up to unique $\mathcal{E}$-isomorphism, by two categories $\mathcal{F}_S$ and $\mathcal{F}_T$ and a bifunctor $H(\eta, \xi)$ on $\mathcal{F}_T^\circ \times \mathcal{F}_S$ with values in **Ens**. Indeed, if $\mathcal{F}$ is a category over $\mathcal{E}$, one associates with it the two fiber categories $\mathcal{F}_S$ and $\mathcal{F}_T$ and the bifunctor $H(\eta, \xi) = \mathrm{Hom}_f(\eta, \xi)$. We leave it to the reader to spell out the construction in the opposite direction. For the category in question to be fibered (or prefibered, which amounts to the same thing), it is necessary and sufficient that the functor H be representable with respect to the argument ξ. For it to be cofibered, it is necessary and sufficient that H be representable with respect to the argument η.

g) Let $\mathcal{F} = \mathcal{C} \times \mathcal{E}$, regarded as a category over $\mathcal{E}$ by means of pr_2. Then $\mathcal{F}$ is fibered and cofibered over $\mathcal{E}$, and is even endowed with a canonical splitting and cosplitting, corresponding to the constant functor on $\mathcal{E}$, respectivelyon $\mathcal{E}^\circ$, with values in **Cat** and value $\mathcal{C}$. One has

$$\mathbf{\Gamma}(\mathcal{F}/\mathcal{E}) \simeq \mathbf{Hom}(\mathcal{E}, \mathcal{C}),$$

and $\varprojlim \mathcal{F}/\mathcal{E}$ corresponds to the full subcategory formed by the functors $F\colon \mathcal{E} \to \mathcal{C}$ 189
that send arbitrary morphisms to isomorphisms.

VI.12. Functors on a cloven category

Let $\mathcal{F}$ be a normalized cloven category over $\mathcal{E}$. For every object S of $\mathcal{E}$, denote by

$$i_S\colon \mathcal{F}_S \longrightarrow \mathcal{F}$$

the inclusion functor. Thus, for every morphism $f\colon T \to S$ in $\mathcal{E}$, there is a functorial homomorphism

$$\alpha_f\colon i_T f^* \longrightarrow i_S,$$

where $f^*\colon \mathcal{F}_S \to \mathcal{F}_T$ is the base-change functor defined by the cleavage. Now let

$$F\colon \mathcal{F} \longrightarrow \mathcal{C}$$

be a functor from $\mathcal{F}$ to a category $\mathcal{C}$. For every $S \in \mathrm{Ob}(\mathcal{E})$, set

$$F_S = F \circ i_S\colon \mathcal{F}_S \longrightarrow \mathcal{C},$$

and, for every $f\colon T \to S$ in $\mathcal{E}$, set

$$\varphi_f = F * \alpha_f\colon F_T f^* \longrightarrow F_S.$$

Thus every functor $F\colon \mathcal{F} \to \mathcal{C}$ determines a family (F_S) of functors $\mathcal{F}_S \to \mathcal{C}$ and a family (φ_f) of functorial homomorphisms $F_T f^* \to F_S$. These families satisfy the following conditions:

a) $\qquad \varphi_{\mathrm{id}_S} = \mathrm{id}_{F_S}.$

For morphisms $f\colon T \to S$ and $g\colon U \to T$ in $\mathcal{E}$, the following square of functorial homomorphisms is commutative:

(b))
$$\begin{array}{ccc} F_U g^* f^* & \xrightarrow{F_U * c_{f,g}} & F_U (fg)^* \\ \downarrow{\scriptstyle \varphi_g * f^*} & & \downarrow{\scriptstyle \varphi_{fg}} \\ F_T f^* & \xrightarrow{\varphi_f} & F_S. \end{array}$$

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The first relation is immediate, and the second follows by applying the functor F to the commutative diagram

$$\begin{array}{ccc} g^*f^*(\xi) & \xrightarrow{c_{f,g}(\xi)} & (fg)^*(\xi) \\ {\scriptstyle \alpha_g(f^*(\xi))}\downarrow & & \downarrow{\scriptstyle \alpha_{fg}(\xi)} \\ f^*(\xi) & \xrightarrow{\alpha_f(\xi)} & \xi \end{array}$$

for a varying object ξ in $\mathcal{F}_S$.

If $G\colon \mathcal{F} \to \mathcal{C}$ is a second functor, giving rise to functors $G_S\colon \mathcal{F}_S \to \mathcal{C}$ and functorial homomorphisms $\psi_f\colon G_T f^* \to G_S$, and if $u\colon F \to G$ is a functorial homomorphism, then the homomorphisms $u * i_S$ give

$$u_S\colon F_S \longrightarrow G_S.$$

For every morphism $f\colon T \to S$ in $\mathcal{E}$, the squares

$$\text{(c))} \qquad \begin{array}{ccc} F_T f^* & \xrightarrow{\varphi_f} & F_S \\ {\scriptstyle u_T * f^*}\downarrow & & \downarrow{\scriptstyle u_S} \\ G_T f^* & \xrightarrow{\psi_f} & G_S \end{array}$$

are commutative.

Proposition VI.12.1. *Let $\mathcal{H}(\mathcal{F},\mathcal{C})$ be the category whose objects are pairs consisting of a family (F_S), indexed by $S \in \mathrm{Ob}(\mathcal{E})$, of functors $\mathcal{F}_S \to \mathcal{C}$, and a family (φ_f), indexed by $f \in \mathrm{Fl}(\mathcal{E})$, of functorial homomorphisms $F_T f^* \to F_S$, satisfying conditions* a) and b). *Its morphisms are the families (u_S), indexed by $S \in \mathrm{Ob}(\mathcal{E})$, of homomorphisms $F_S \to G_S$* 191 *satisfying commutativity condition* c) *above; composition is defined by composing the homomorphisms of functors $\mathcal{F}_S \to \mathcal{C}$. Then the two constructions described above define an* isomorphism

$$K\colon \mathbf{Hom}(\mathcal{F},\mathcal{C}) \xrightarrow{\sim} \mathcal{H}(\mathcal{F},\mathcal{C}).$$

It is immediate that these constructions define a functor from the first category to the second. This functor is fully faithful. For given F, G, the map

$$\mathrm{Hom}(F,G) \longrightarrow \mathrm{Hom}(K(F),K(G))$$

is plainly injective. To prove surjectivity, note that condition c) expresses the functoriality of the maps

$$u(\xi) = u_S(\xi)\colon F(\xi) = F_S(\xi) \longrightarrow G(\xi) = G_S(\xi)$$

for homomorphisms of the form $\alpha_f(\xi)$ in $\mathcal{F}$. Functoriality also holds on every fiber category, i.e. for the morphisms in $\mathcal{F}$ that are T-morphisms, with $T \in \mathrm{Ob}(\mathcal{E})$. It therefore holds for every morphism in $\mathcal{F}$, since every f-morphism, with $f\colon T \to S$ in $\mathcal{E}$, is uniquely the composite of a morphism $\alpha_f(\xi)$ and a T-morphism.

It remains to show that K is bijective on objects. The preceding argument already shows that K is injective on objects. For surjectivity, start with a system $(F_S), (\varphi_f)$ satisfying a) and b), and define a map $\mathrm{Ob}(\mathcal{F}) \to \mathrm{Ob}(\mathcal{C})$ by

$$F(\xi) = F_S(\xi) \qquad \text{for } \xi \in \mathrm{Ob}(\mathcal{F}_S) \subset \mathrm{Ob}(\mathcal{F}),$$

and a map $\mathrm{Fl}(\mathcal{F}) \to \mathrm{Fl}(\mathcal{C})$ by

$$F(\alpha_f(\xi)u') = \varphi_f(\xi)\, F_T(u')$$

for every morphism $f\colon T \to S$ in $\mathcal{E}$, every object ξ of $\mathcal{F}_S$, and every T-morphism u' with target $f^*(\xi)$. These maps define a *functor* $F\colon \mathcal{F} \to \mathcal{C}$. Indeed, $F(\mathrm{id}_\xi) = \mathrm{id}_{F(\xi)}$ is immediate. It remains to prove multiplicativity $F(uv) = F(u)F(v)$ for an f-morphism

$u\colon \eta \to \xi$ and a g-morphism $v\colon \zeta \to \eta$, where $f\colon T \to S$ and $g\colon U \to T$ are morphisms of $\mathcal{E}$. Setting $w = uv$, we have

$$u = \alpha_f(\xi)u', \qquad v = \alpha_g(\eta)v', \qquad w = \alpha_{fg}(\xi)w',$$

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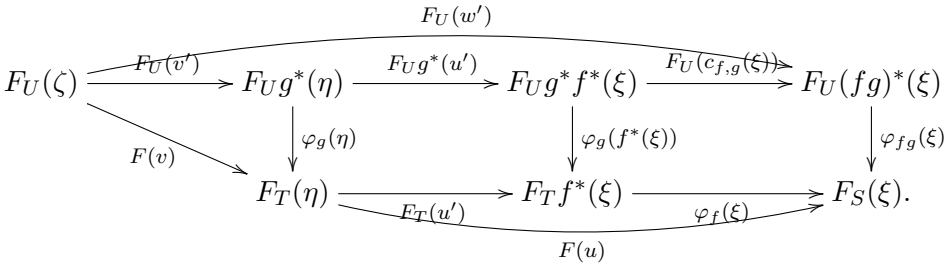
with

$$w' = c_{f,g}(\xi)g^*(u')v' \qquad (cf.\ no.\ VI.8).$$

With this notation, we must prove commutativity along the outer boundary of the following diagram:



The left triangle is commutative by the definition of $F(v)$. The middle square is commutative because it is obtained from the homomorphism $u'\colon \eta \to f^*(\xi)$ by the functorial homomorphism φ_g, and the square on the right is commutative by condition b). The desired conclusion follows.

Now suppose that $\mathcal{C}$ is itself a normalized cloven category over $\mathcal{E}$, which we shall henceforth call $\mathcal{G}$, and that we are interested in the $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$. Every such functor F induces functors

$$F_S\colon \mathcal{F}_S \longrightarrow \mathcal{G}_S$$

on the fiber categories. Moreover, for every morphism $f\colon T \to S$ in $\mathcal{E}$ and every object ξ of $\mathcal{F}_S$, the f-morphism $F(\alpha_f(\xi))$ factors uniquely through a T-morphism

$$\varphi_f(\xi)\colon F_T(f^*_{\mathcal{F}}(\xi)) \longrightarrow f^*_{\mathcal{G}}(F_S(\xi)).$$

Here the subscript $\mathcal{F}$ or $\mathcal{G}$ indicates the cloven category whose inverse-image functor is being used. We thereby obtain a functorial homomorphism between functors from $\mathcal{F}_S$ to $\mathcal{G}_T$:

$$\varphi_f\colon F_T f^*_{\mathcal{F}} \longrightarrow f^*_{\mathcal{G}} F_S.$$

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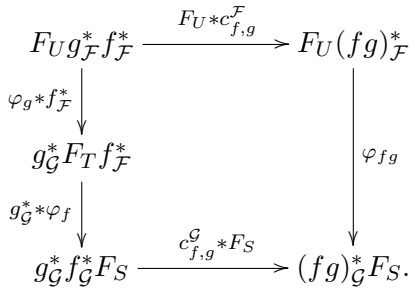
The two systems (F_S) and (φ_f) satisfy the following conditions:

a') $\qquad \varphi_{\mathrm{id}_S} = \mathrm{id}_{F_S}.$

For morphisms $f\colon T \to S$ and $g\colon U \to T$ in $\mathcal{E}$, the following diagram of functorial homomorphisms is commutative:

(b'))



We leave to the reader the verification, as well as the statement and proof of the analogue of Proposition VI.12.1. This analogue shows that the set of $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$ is in bijection with the set of systems $(F_S), (\varphi_f)$ satisfying conditions a') and b') above. Under this correspondence, the Cartesian functors are precisely those for which the homomorphisms φ_f are isomorphisms.

REMARK. It is usually preferable, of course, to reason directly with fibered categories without using explicit cleavages. In particular, this avoids having to use a cumbersome interpretation such as the one above for the simple notion of an $\mathcal{E}$-functor or a Cartesian $\mathcal{E}$-functor. It was in order to avoid unbearable heaviness and to obtain more intrinsic statements that we had to abandon the approach taken in [**2**], starting from the notion of a 194
cloven category, called there a "fibered category", which now takes second place behind the notion of a fibered category. It is moreover likely that, contrary to the still predominant usage inherited from older habits of thought, universal problems will eventually prove more conveniently handled by not emphasizing *one* solution chosen once and for all, but by placing all solutions on an equal footing.

Bibliography

[1] A. Grothendieck, *Sur quelques points d'algèbre homologique*, Tôhoku Math. J. **9** (1957), pp. 119–221.
[2] A. Grothendieck, *Technique de descente et Théorèmes d'existence, I*, Séminaire Bourbaki 190, December 1959.

CHAPTER VIII

Faithfully flat descent

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VIII.1. Descent of quasi-coherent modules

Let **Sch** denote the category of preschemes. Proceeding as in VI VI.11.b, one finds that the category of pairs (X, F) consisting of a prescheme X and a module F on X (with morphisms defined as in loc. cit. by means of the direct image of a module under a morphism of ringed spaces) may be regarded as a fibered category over **Sch**. The base-change functor associated with a morphism $f\colon X \to Y$ in **Sch** is the inverse-image functor for modules under f. Notice that the fiber category at $X \in \mathrm{Ob}(\mathbf{Sch})$ of the preceding fibered category is the category *opposite* to the category of modules on X.

Since the inverse image of a quasi-coherent module is quasi-coherent, the full subcategory of the category of pairs (X, F) formed by those pairs for which F is quasi-coherent is a fibered subcategory of the preceding fibered category. (By contrast, without hypotheses on f, the direct image of a quasi-coherent module is not in general a quasi-coherent module.) We shall simply call this the *fibered category of quasi-coherent modules on preschemes*.

Recall also that a morphism $f\colon X \to Y$ of ringed spaces is called *faithfully flat* if it is *flat* (that is, for every $x \in X$, $\mathcal{O}_{X,x}$ is a flat module over $\mathcal{O}_{Y,f(x)}$, (IV)) and *surjective*. The morphism f is called *quasi-compact* if the inverse image under f of every quasi-compact subset is quasi-compact. When f is a morphism of preschemes, this also means that the inverse image under f of every affine open subset of Y is a *finite* union of affine open subsets of X.

THEOREM VIII.1.1. *Let $\mathcal{F}$ be the fibered category of quasi-coherent modules on preschemes. Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes. Then g is a morphism of effective $\mathcal{F}$-descent.* 196

Recall[1] that this means two things.

COROLLARY VIII.1.2 (Descent of homomorphisms of modules). *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes, let F and G be two quasi-coherent modules on S, let F' and G' be their inverse images on S', and finally let F'' and G'' be their inverse images on $S'' = S' \times_S S'$. Consider the diagram of maps of sets defined by the base-change functors for g, p_1, and p_2, where $p_1, p_2\colon S' \times_S S' \rightrightarrows S'$ are the two projections:*

$$\mathrm{Hom}_S(F, G) \longrightarrow \mathrm{Hom}_{S'}(F', G') \rightrightarrows \mathrm{Hom}_{S''}(F'', G'').$$

This diagram is exact; that is, it identifies the first set bijectively with the equalizer of the two displayed maps from the second set to the third.

In other words, the base-change functor for g, $F \longmapsto F'$, defines a *fully faithful* functor from the category of quasi-coherent modules on S to the category of quasi-coherent modules on S' endowed with descent data relative to g. Moreover:

[1] We shall admit here the general theory of descent set out in detail in the article of J. GIRAUD cited in the footnote to the Warning, a work that we shall cite below as [**D**]. Cf. also [**2**] for a succinct account.

COROLLARY VIII.1.3 (Descent of modules). *For every quasi-coherent module F' on S', every descent datum on F' relative to g is* effective*; that is, F', together with its descent datum, is isomorphic to the inverse image under g of a quasi-coherent module on S (determined up to unique isomorphism by VIII.1.2).*

In other words, the preceding fully faithful functor is in fact an *equivalence*. In practical terms, giving a quasi-coherent module on S amounts to the same thing as giving a quasi-coherent module on S' endowed with descent data relative to g.

Proof of VIII.1.1. Let T first be an S-prescheme that is S-isomorphic to the disjoint 197
union of a family of open subpreschemes S_i induced from S and covering S. It is then evident that the structural morphism $T \to S$ is a morphism of effective $\mathcal{F}$-descent. This means precisely that giving a quasi-coherent module F on S is equivalent to giving quasi-coherent modules F_i on the S_i, together with gluing isomorphisms

$$\varphi_{ji} \colon F_i|S_i \cap S_j \longrightarrow F_j|S_i \cap S_j$$

satisfying the familiar cocycle condition. By VII, 8, it follows that, to verify that $g \colon S' \to S$ is a morphism of effective $\mathcal{F}$-descent, it suffices to verify this for the morphism

$$g_T \colon T' = T \times_S S' \longrightarrow T$$

obtained from g by the base change $T \to S$. Notice that the hypothesis on $T \to S$ is stable under arbitrary base change, so that $T \to S$ is in fact a *universal* morphism of effective $\mathcal{F}$-descent. Taking the S_i to be affine open subsets covering S, we are therefore reduced to the case where S is affine.

Then S' is a finite union of affine open subsets. Taking their disjoint union as an S-scheme gives an affine S-scheme S_1 and a flat surjective S-morphism $S_1 \to S'$. Thus S_1 is also faithfully flat over S. If, therefore, one proves that a faithfully flat affine morphism is a morphism of effective $\mathcal{F}$-descent, hence a strict universal morphism of $\mathcal{F}$-descent (the hypothesis being stable under base change), it follows in particular that the structural morphism $S_1 \to S$ is a strict universal morphism of $\mathcal{F}$-descent. Since there is an S-morphism $S_1 \to S'$, it then follows from [**D**] that $g \colon S' \to S$ is indeed a strict morphism of $\mathcal{F}$-descent.

We are thus reduced to the case in which g is affine. As we have seen, we may then also suppose that S is affine; hence *we may suppose that S and S' are affine.* In this case, VIII.1.2 is equivalent to the following lemma.

LEMMA VIII.1.4. *Let A be a ring, let A' be a faithfully flat A-algebra, let M and N be two A-modules, let M' and N' be the A'-modules obtained by the change of rings $A \to A'$, and let M'' and N'' be the $A'' = A' \otimes_A A'$-modules obtained by the change of rings $A \to A''$. Then the sequence of maps of sets*

$$\mathrm{Hom}_A(M, N) \rightarrow \mathrm{Hom}_{A'}(M', N') \rightrightarrows \mathrm{Hom}_{A''}(M'', N'')$$

is exact.

Since the homomorphism $N \to N'$ is injective (because A' is faithfully flat over A), the first arrow is injective. It remains to prove that if an A'-homomorphism $u' \colon M' \to N'$ is compatible with the descent data, then it comes from an A-homomorphism $u \colon M \to N$. This simply means that u' maps the subset M of M' into the subset N of N'. The induced map $u \colon M \to N$ is then automatically A-linear because u' is A'-linear, and one sees similarly that u' is necessarily equal to $u \otimes_A A'$.

Now, if $x \in M$, then $u'(x)$ belongs to the equalizer of the pair of maps $N' \rightrightarrows N''$. Thus, to prove VIII.1.4, we are reduced to the following special case (corresponding to $M = A$):

COROLLARY VIII.1.5. *Let N be an A-module. Then the sequence of maps of sets*

$$N \rightarrow N' \rightrightarrows N''$$

is exact.

Let A_1 be a faithfully flat A-algebra. To prove exactness of the displayed sequence, it is enough to prove exactness after the base change $A \to A_1$. The resulting sequence is plainly the sequence associated with the A_1-module

$$N_1 = N \otimes_A A_1$$

and the A_1-algebra $A'_1 = A_1 \otimes_A A'$. It is therefore enough to find a faithfully flat A-algebra A_1 for which

$$\mathrm{Spec}(()A'_1) \longrightarrow \mathrm{Spec}(()A_1)$$

is a strict $\mathcal{F}$-descent morphism. We may take $A_1 = A'$: the preceding morphism then has a right inverse, and hence, by [**D**], is an effective descent morphism for every category fibered over **Sch**.

It remains to show that if N' is an A'-module equipped with a descent datum relative to $A \to A'$, i.e. with an isomorphism

$$\varphi\colon N'_1 \xrightarrow{\sim} N'_2$$

between the two modules obtained from N' by the two base changes

$$A' \rightrightarrows A' \otimes_A A',$$

then N', together with its descent datum, is isomorphic to a module of the form $N \otimes_A A'$. 199
In view of VIII.1.5, this statement is readily seen to be equivalent to the following:

Lemma VIII.1.6. *Let N' be an A'-module equipped with a descent datum relative to $A \to A'$, where A' is an A-algebra. Let N be the A-submodule of N' formed by the elements x such that*

$$\varphi(x \otimes_{A'} 1_{A'}) = 1_{A'} \otimes_{A'} x,$$

and consider the canonical homomorphism

$$N \otimes_A A' \longrightarrow N',$$

which is compatible with the descent data. If A' is faithfully flat over A, this homomorphism is an isomorphism.

Let us prove the lemma. Again let A_1 be a faithfully flat A-algebra. To show that the homomorphism in question is an isomorphism, it is enough to show that it becomes one after the base change $A \to A_1$. Using the flatness of A_1 over A, one sees that the resulting homomorphism is precisely the one obtained directly from the $A'_1 = A' \otimes_A A_1$-module $N' \otimes_A A_1$, equipped with the descent datum relative to $A_1 \to A'_1$ obtained canonically by base change from the descent datum on N'. Thus it is enough to choose a faithfully flat A-algebra A_1 such that $\mathrm{Spec}(()A'_1) \to \mathrm{Spec}(()A_1)$ is an effective $\mathcal{F}$-descent morphism. As above, take $A_1 = A'$. This proves VIII.1.6, and hence also VIII.1.1.

Corollary VIII.1.7 (Descent of sections of modules). *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes. For every quasi-coherent module G on S, let G' and G'' be its inverse images on S' and $S'' = S' \times_S S'$, respectively. Then the diagram of homomorphisms of modules on S*

$$G \rightarrowtail g_*(G') \rightrightarrows h_*(G'')$$

where $h\colon S'' \to S$ is the structural morphism, is exact.

Indeed, this means that, for every open subset U of S, the corresponding diagram of 200
sections over U is exact. We may plainly assume $U = S$, and the asserted exactness is then the special case of VIII.1.2 obtained by taking $F = \mathcal{O}_S$.

Since the inverse-image functor on modules is right exact, VIII.1.1 formally implies:

COROLLARY VIII.1.8 (Descent of quotient modules). *With the notation of VIII.1.7, let* $\mathrm{Quot}(F)$*, for every quasi-coherent module* F *on a prescheme, denote the set of quasi-coherent quotient modules of* F*. Then the diagram of maps of sets*

$$\mathrm{Quot}(G) \longrightarrow \mathrm{Quot}(G') \rightrightarrows \mathrm{Quot}(G'')$$

is exact.

(The same statement plainly holds with submodules in place of quotient modules, since the two are in bijection.) Taking $G = \mathcal{O}_S$, we obtain:

COROLLARY VIII.1.9 (Descent of closed subpreschemes). *For every prescheme* X*, let* $H(X)$ *denote the set of closed subpreschemes of* X*. With this notation, and under the hypotheses of VIII.1.7, the diagram of maps of sets*

$$H(S) \longrightarrow H(S') \rightrightarrows H(S'')$$

is exact.

Theorem VIII.1.1 should be supplemented by the following result:

PROPOSITION VIII.1.10 (Descent of properties of modules). *Let* $g\colon S' \to S$ *be faithfully flat and quasi-compact, and let* F *be a quasi-coherent module on* S*. Then* F *is of finite type, respectively of finite presentation, respectively finite locally free, if and only if its inverse image* F' *on* S' *has the corresponding property.*

Only sufficiency requires proof. We may plainly suppose that S is affine. Replacing 201
S' by a sum of affine open subsets covering S', we reduce to the case in which S' is also affine. The assertion is then equivalent to the following:

COROLLARY VIII.1.11. *Let* A *be a ring, let* A' *be a faithfully flat* A*-algebra, let* M *be an* A*-module, and let* $M' = M \otimes_A A'$*. Then* M *is of finite type, respectively of finite presentation, respectively finite locally free, if and only if* M' *has the corresponding property.*

Indeed, write $M = \varinjlim_i M_i$, where the M_i are the finite-type submodules of M. Then $M' = \varinjlim_i M'_i$. If M' is of finite type, it equals one of the M'_i; faithful flatness then gives $M = M_i$, so M is of finite type. Consequently, there is an exact sequence

$$0 \longrightarrow R \longrightarrow L \longrightarrow M \longrightarrow 0$$

with L finite free, and hence an exact sequence

$$0 \longrightarrow R' \longrightarrow L' \longrightarrow M' \longrightarrow 0$$

with L' finite free. If M' is finitely presented, then R' is of finite type; by what precedes, R is of finite type, and therefore M is finitely presented. Finally, M is finite locally free precisely when it is finitely presented and flat (cf. IV in the noetherian case; the general case is left to the reader). Each of these properties descends, and so does their conjunction. This completes the proof.

REMARK VIII.1.12. Combining VIII.1.1 and VIII.1.10 shows that VIII.1.1 remains valid if the fibered category $\mathcal{F}$ is replaced by the fibered subcategory consisting of quasi-coherent modules of finite type, respectively of finite presentation, respectively finite locally free, respectively locally free of a specified rank n.

VIII.2. Descent of affine preschemes over another

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Since the inverse-image functor for modules is compatible with tensor products and other tensor operations, Theorem VIII.1.1 implies various variants obtained by considering, in place of a single quasi-coherent module, a quasi-coherent module or a system of quasi-coherent modules endowed with various additional structures expressed by means of tensor operations. For example, giving three quasi-coherent modules F, G, and H on S and a pairing

$$F \otimes G \longrightarrow H$$

is equivalent to giving three quasi-coherent modules F', G', and H' on S', endowed with descent data relative to $g\colon S' \to S$, together with a pairing

$$F' \otimes G' \longrightarrow H'$$

"compatible" with those descent data, in the evident sense of the term. For example, if $F = G = H$, one sees that giving a quasi-coherent module F on S endowed with an algebra law (which for the moment we do not require to satisfy any axiom of associativity, commutativity, or the existence of a unit section) is equivalent to giving the same data on S', together with descent data. Using the results of the preceding section, one sees at once that F satisfies any one of the usual axioms just mentioned if and only if F' does.

For example, giving a quasi-coherent algebra $\mathcal{A}$ on S (by which from now on we mean associative, commutative, and equipped with a unit section) is equivalent to giving a quasi-coherent algebra $\mathcal{A}'$ on S' endowed with descent data relative to $g\colon S' \to S$. Recalling the equivalence between the opposite category of quasi-coherent algebras on S and the category of affine S-preschemes over S (EGA II, §1), one obtains at once:

THEOREM VIII.2.1. *Let $\mathcal{F}'$ be the fibered category of affine morphisms of preschemes $f\colon X \to S$, regarded as a fibered subcategory of the fibered category of arrows in the category* 203 **Sch** *of preschemes* (VI VI.11.a). *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes. Then g is a morphism of effective $\mathcal{F}'$-descent.*

VIII.3. Descent of set-theoretic properties and finiteness properties of morphisms[2]

PROPOSITION VIII.3.1. *Let $f\colon X \to Y$ be an S-morphism, let $g\colon S' \to S$ be a surjective morphism, and let*

$$f'\colon X' = X \times_S S' \longrightarrow Y' = Y \times_S S'$$

be the morphism obtained from f by base change along $g\colon S' \to S$. The morphism f is surjective (respectivelyradicial) if and only if f' is so.

Notice that f' can also be obtained by the base change $Y' \to Y$, which is itself surjective because it is obtained from the surjective morphism $g\colon S' \to S$. On the other hand, for every $y \in Y$ and every $y' \in Y'$ lying over y, there is an isomorphism

$$X'_{y'} \xleftarrow{\sim} X_y \otimes_{\kappa(y)} \kappa(y'),$$

where X_y denotes the fiber of X at y, and $X'_{y'}$ that of X' at y'. It follows that X_y is nonempty (respectivelyhas at most one point, and that point corresponds to a radicial residue-field extension) if and only if $X'_{y'}$ has the same property. This proves VIII.3.1.

COROLLARY VIII.3.2. *Under the hypotheses of VIII.3.1, if f' is injective (respectively-bijective), then f is likewise.*

Indeed, if $X'_{y'}$ has at most one point (respectivelyexactly one point), the same is true of X_y. This follows because the morphism $X'_{y'} \to X_y$ is surjective, being obtained from the surjective morphism $\mathrm{Spec}(\kappa(y')) \to \mathrm{Spec}(\kappa(y))$.

[2]For other results of the kind treated in Sections 3 and 4, cf. EGA IV 2.3, 2.6, and 2.7.

PROPOSITION VIII.3.3. *With the notation of VIII.3.1, suppose that $g\colon S' \to S$ is surjective and quasi-compact (respectivelyfaithfully flat and quasi-compact). The morphism f is quasi-compact (respectivelyof finite type) if and only if f' is so.*

Only the sufficiency needs to be proved. We may clearly suppose $S = Y$, since the hypothesis on $g\colon S' \to S$ is preserved for $Y' \to Y$. We may also suppose that Y is affine. Then Y' is quasi-compact, and therefore X' is quasi-compact, since f' is so by hypothesis. Let $(X_i)_{i\in I}$ be a family of affine open subsets of X covering X. The X'_i are open subsets covering X', so a finite subfamily covers X'. Since $X' \to X$ is surjective, the corresponding X_i already cover X. Thus X is quasi-compact, that is, f is quasi-compact.

Now suppose that f' is of finite type, and let us prove that f is of finite type, assuming g faithfully flat. Replacing Y' by the disjoint union of a family of affine open subsets covering it, we may suppose that Y' is affine. Finally, by the preceding argument X is covered by finitely many affine open subsets X_i; it remains to show that they are of finite type over Y, knowing that X'_i is of finite type over Y'. This reduces us to the following result.

COROLLARY VIII.3.4. *Let B be an A-algebra, let A' be a faithfully flat A-algebra, and let $B' = B \otimes_A A'$ be the A'-algebra obtained from B by change of rings. Then B is of finite type if and only if B' is of finite type.*

Again, only the sufficiency needs proof. We have $B = \varinjlim_i B_i$, where the B_i are the finite-type subalgebras of B. Hence $B' = \varinjlim_i B'_i$, and if B' is of finite type over A', then B' is equal to one of the B'_i. Faithful flatness then gives $B = B_i$, so B is of finite type.

COROLLARY VIII.3.5. *Suppose again that the base-change morphism $g\colon S' \to S$ is faithfully flat and quasi-compact. The morphism f is quasi-finite if and only if f' is quasi-finite.*

Indeed, the property of being quasi-finite is, by definition, the conjunction of being of finite type and having finite fibers. Each of these properties descends under g: the first by VIII.3.3, and the second by the argument of VIII.3.1, which uses only the surjectivity of g.

REMARKS VIII.3.6. Let A be a ring and X an A-prescheme. One sees easily that the 205
following conditions are equivalent:

(i) There exist a noetherian ring A_0 (which, if desired, may be assumed to be a finite-type subring of A), an A_0-prescheme X_0 of finite type, a homomorphism $A_0 \to A$, and an A-isomorphism

$$X \xrightarrow{\sim} X_0 \times_{A_0} A.$$

(ii) The diagonal morphism $X \to X \times_{\mathrm{Spec}(A)} X$ is quasi-compact (a vacuous condition if X is separated over A), and X is the union of finitely many affine open subsets X_i whose rings B_i are algebras of finite presentation over A; that is, quotients of polynomial algebras in finitely many indeterminates by ideals of finite type.

If X is itself affine, with ring B, these conditions simply mean that B is an algebra of finite presentation over A.

A morphism $f\colon X \to Y$ is called a *morphism of finite presentation*, and X is also said to be of finite presentation over Y, if Y is the union of affine open subsets Y_i such that $X|Y_i$, as a Y_i-prescheme, satisfies the preceding equivalent conditions. The same is then true of $X|Y'$ for *every* affine open subset Y' of Y. This property is stable under base change, and the composite of two morphisms of finite presentation is again of finite presentation.

With these notions in place, condition (ii), by the same argument as in VIII.1.10, shows that Proposition VIII.3.3 remains valid when "of finite type" is replaced by "of finite presentation."

VIII.4. Descent of topological properties

THEOREM VIII.4.1. *Let $g\colon Y' \to Y$ be a morphism, and let Z be a subset of Y. Suppose that g is flat and that there exists a quasi-compact morphism $f\colon X \to Y$ such that $Z = f(X)$ (N.B. if Y is noetherian, this last condition is implied by "Z being constructible"). Then*

$$g^{-1}(\overline{Z}) = \overline{g^{-1}(Z)}.$$

We may assume Y affine, and then Y' affine. Since Y is affine, X is a finite union 206
of affine open subsets X_i, and, replacing X by the disjoint union of the X_i, we may also assume X affine. Let A, A', B be the rings of Y, Y', X, let $B' = B \otimes_A A'$ be that of $X' = X \times_Y Y'$, let I be the kernel of $A \to B$, and let I' be the kernel of $A' \to B'$. Thus the closed subsets of Y and Y' defined by these ideals are respectively the closure of $Z = f(X)$ and the closure of $Z' = f'(X') = g^{-1}(Z)$. We wish to show that the latter equals $g^{-1}(\overline{Z})$; this follows from $I' = IA'$, which in turn follows from the flatness of A' over A.

COROLLARY VIII.4.2. *Let $g\colon Y' \to Y$ be a flat and quasi-compact morphism, and let Z' be a closed subset of Y' saturated for the set-theoretic equivalence relation defined by g. Then $Z' = g^{-1}(\overline{g(Z')})$.*

Indeed, $Z' = g^{-1}(Z)$, with $Z = g(Z')$. We may then apply VIII.4.1, noting that the condition imposed on Z in VIII.4.1 is indeed satisfied by taking for X the prescheme Z' equipped with the reduced structure induced by Y'. (The fact that g is quasi-compact then ensures that the induced morphism $f\colon Z' \to Y$ is quasi-compact.)

Statement VIII.4.2 also means that *the topology on $g(Y')$ induced by Y is the quotient of that on Y'*. In particular:

COROLLARY VIII.4.3. *Let $g\colon Y' \to Y$ be a faithfully flat and quasi-compact morphism. Then g makes Y a quotient topological space of Y', i.e. for a subset Z of Y, Z is closed (respectivelyopen) if and only if $Z' = g^{-1}(Z)$ is so.*

Recall now that two elements a, b of Y' have the same image in Y if and only if they are of the form $p_1(c), p_2(c)$ for a suitable element c of $Y'' = Y' \times_Y Y'$. It follows that if g is surjective, there is an *exact* diagram of sets

$$\mathcal{P}(Y) \longrightarrow \mathcal{P}(Y') \rightrightarrows \mathcal{P}(Y'') \ ,$$

where, for every set E, $\mathcal{P}(E)$ denotes its power set. With this notation, VIII.4.3 may also be interpreted as follows:

COROLLARY VIII.4.4 (Descent of open respectivelyclosed subsets). *Let $g\colon Y' \to Y$ be* 207
as in VIII.4.3. For every prescheme X, let $\mathrm{Ouv}(X)$ *respectively*$\mathrm{Fer}(X)$ *be the set of its open subsets respectivelythe set of its closed subsets. Then the following are* exact *diagrams of maps of sets (deduced from g and the two projections of $Y'' = Y' \times_Y Y'$):*

$$\mathrm{Ouv}(Y) \longrightarrow \mathrm{Ouv}(Y') \rightrightarrows \mathrm{Ouv}(Y'')$$

$$\mathrm{Fer}(Y) \longrightarrow \mathrm{Fer}(Y') \rightrightarrows \mathrm{Fer}(Y'')$$

The following supplements VIII.4.3:

COROLLARY VIII.4.5. *Let $g\colon Y' \to Y$ be as in VIII.4.3, and let Z be a subset of Y for which there exists a quasi-compact morphism $f\colon X \to Y$ with image Z (for example, Z constructible and Y noetherian). Then Z is a locally closed subset of Y if and only if $Z' = g^{-1}(Z)$ is a locally closed subset of Y'.*

It suffices to prove the "if"direction. Let Y_1 be the closed subprescheme of Y given by the closure of Z equipped with the induced reduced structure, and let $Y_1' = Y_1 \times_Y Y'$ be the closed subprescheme of Y' that is the inverse image of Y_1. Its underlying set is the inverse image $g^{-1}(Y_1) = g^{-1}(\overline{Z})$, and hence equals $\overline{Z'}$ by VIII.4.1. Since Z' is locally closed in Y', it is open in $\overline{Z'}$, and hence open in Y_1'. The latter is faithfully flat and quasi-compact over Y_1, so by VIII.4.3 we conclude that Z is open in Y_1, i.e. in $\overline{Z}$, which means that Z is locally closed.

COROLLARY VIII.4.6. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism, let $f\colon X \to Y$ be an S-morphism, and let $f'\colon X' \to Y'$ be the S'-morphism obtained from it by base change. Suppose that f' is an open map (respectivelya closed map, respectivelyquasi-compact and a homeomorphism into, respectivelya homeomorphism onto); then f has the same property.*

Since Y' is faithfully flat and quasi-compact over Y, we may assume $Y = S$. Let Q be a subset of X. Then (denoting by h the projection morphism $X' \to X$) one has: 208

$$g^{-1}(f(Q)) = f'(h^{-1}(Q)).$$

If Q is open (respectivelyclosed), the same is true of $h^{-1}(Q)$, and hence also of $f'(h^{-1}(Q))$ if f' is assumed to be an open (respectivelyclosed) map; the same is therefore true of $f(Q)$ by the preceding formula and VIII.4.3. This proves the first two assertions in VIII.4.6. It remains to consider the case in which f' is a homeomorphism into, and then to prove that f is a homeomorphism into. (The case of a homeomorphism onto will then follow from VIII.3.1.) By VIII.3.2 f is injective; it remains to prove that the map $X \to f(X)$ is open. We already know that f is quasi-compact by VIII.3.3. It suffices to prove that, for every closed subset Z of X, one has $Z = f^{-1}(\overline{f(Z)})$. This is equivalent to the analogous formula for the inverse images under the surjective map $h\colon X' \to X$, i.e. to

$$Z' = f'^{-1}(g^{-1}(\overline{f(Z)})),$$

where $Z' = h^{-1}(Z)$. But by VIII.4.1, applied to the subset $f(Z)$ of Y, one has $g^{-1}(\overline{f(Z)}) = \overline{g^{-1}(f(Z))}$, and the formula to be proved is equivalent to

$$Z' = f'^{-1}(\overline{f'(Z')}),$$

which follows from the hypothesis that f' is a homeomorphism into.

N.B. In this last argument, already supposing that f is quasi-compact, we did not use the quasi-compactness of g, but only that g is faithfully flat. Thus it is under this hypothesis that one may descend the property "homeomorphism into", or "homeomorphism onto", or, by the preceding argument, the property "f' is quasi-compact and makes $f'(X')$ a quotient topological space of X'".

We shall say that a morphism $f\colon X \to Y$ of preschemes is *universally open* (respectively*universally closed*, respectively*universally bicontinuous*, etc.) if for every base change $Y' \to Y$, $f'\colon X' \to Y'$ is an open morphism (respectivelyclosed, respectivelya homeomorphism onto its image space). We then deduce from VIII.4.6:

COROLLARY VIII.4.7. *Under the conditions of VIII.4.6, f is universally open (respectivelyuniversally closed, respectivelya universal homeomorphism into, respectivelya universal homeomorphism) if and only if f' is so.* 209

COROLLARY VIII.4.8. *Under the conditions of VIII.4.6, f is separated (respectivelyproper) if and only if f' is so.*

To say that f is separated means that the diagonal morphism $X \to X \times_Y X$ is closed, or equivalently universally closed, and the first assertion of VIII.4.8 therefore follows from VIII.4.7. To say that f is proper means that f satisfies the conditions a) f is of finite type, b) f is separated, c) f is universally closed. Condition a) does descend by VIII.3.3, b) does as well by what we have just seen, and finally c) does by VIII.4.7.

REMARKS VIII.4.9. Recall that when $g\colon Y' \to Y$ is a flat morphism of finite type, with Y locally noetherian, then g is an open morphism (VI IV.6.6), which is a more precise result than VIII.4.3. Note, however, that if f is a faithfully flat and quasi-compact morphism of noetherian preschemes, then f is not in general an open morphism. For example, let Y be an irreducible scheme whose generic point y is not open (for example, an algebraic curve), and take for Y' the disjoint-union scheme $Y \amalg \operatorname{Spec}(()\kappa(y))$. Then the image of the open subset $\operatorname{Spec}(()\kappa(y))$ under the structure morphism $Y' \to Y$ is not an open subset of Y. The reader will also note that various statements of the present exposé become false if the hypothesis that the faithfully flat morphism under consideration is also quasi-compact is dropped. The typical counterexample is obtained by taking for Y' the disjoint-union scheme of the spectra of the local rings at the points of Y. For example, again taking Y to be an irreducible algebraic curve and Z to be the subset of Y consisting of the generic point, its inverse image in Y' is open although Z is not open.

. Various statements of the present exposé remain valid if the hypothesis that Y' is 210
flat over Y is replaced by the following: there exists a module of finite type F on Y', with support Y', that is flat relative to Y; the hypothesis of faithful flatness is then replaced by the preceding one together with the hypothesis that $Y' \to Y$ is surjective. This applies to the first two assertions in VIII.1.10, to VIII.3.3, VIII.3.5, VIII.4.1, and consequently to all the results of the present section.

VIII.5. Descent of morphisms of preschemes

PROPOSITION VIII.5.1. *Let $g\colon S' \to S$ be a morphism of preschemes.*

a) *Suppose that g is surjective, and that the homomorphism*

$$g^*\colon \mathcal{O}_S \to g_*(\mathcal{O}_{S'})$$

is injective. Then g is an epimorphism in the category of preschemes, and even in the category of ringed spaces.

b) *Suppose that g is surjective and makes S a quotient topological space of S'. Let $S'' = S' \times_S S'$ and let $h\colon S'' \to S$ be the structural morphism; consider the canonical diagram of homomorphisms:*

$$\mathcal{O}_S \rightarrowtail g_*(\mathcal{O}_{S'}) \rightrightarrows h_*(\mathcal{O}_{S''})\ ,$$

and suppose that this diagram is exact*. Then g is an effective epimorphism in the category of preschemes (and also in the category of ringed spaces), i.e. the diagram*

$$S \leftarrow S' \leftleftarrows S''$$

is exact.

Proof. a) We must show that a morphism of ringed spaces $f\colon S \to Z$ is determined once fg is known. Since g is surjective, the underlying set map f_0 of f is known; it remains to determine the homomorphism of sheaves of rings $\mathcal{O}_Z \to \mathcal{O}_S$, or, what amounts to the same thing, the homomorphism of sheaves of rings 211

$$u\colon f_0^{-1}(\mathcal{O}_Z) \to \mathcal{O}_S$$

defined by f. We already know the homomorphism

$$(fg)_0^{-1}(\mathcal{O}_Z) = g_0^{-1}(f_0^{-1}(\mathcal{O}_Z)) \to \mathcal{O}_{S'}$$

defined by fg, or, what amounts to the same thing, we have a homomorphism

$$f_0^{-1}(\mathcal{O}_Z) \to g_{0*}(\mathcal{O}_{S'}) = g_*(\mathcal{O}_{S'}).$$

One immediately sees that the latter is none other than the composite of $g^*\colon \mathcal{O}_S \to g_*(\mathcal{O}_{S'})$ with u, and since g^* is injective, u is known when g^*u is known. [N.B. we have obviously not used the fact that $g\colon S' \to S$ is a morphism of preschemes; the statement would hold

for an arbitrary morphism of ringed spaces; the same remark applies to b), both in the category of ringed spaces and in the category of locally ringed spaces. Note also that if g is a morphism of preschemes, not necessarily surjective, but such that $g^*\colon \mathcal{O}_S \to g_*(\mathcal{O}_{S'})$ is injective, then for two morphisms f_1, f_2 from S to a *scheme* Z such that $f_1 g = f_2 g$, one has $f_1 = f_2$; indeed, if I is the Ideal on S that defines the subprescheme of S where f_1, f_2 coincide (the inverse image of the diagonal subprescheme of $Z \times Z$ under (f_1, f_2)), one sees that I is contained in $\mathrm{Ker}(g^*)$].

b) We must show that for every ringed space Z, the following diagram of maps

$$\mathrm{Hom}(S, Z) \longrightarrow \mathrm{Hom}(S', Z) \rightrightarrows \mathrm{Hom}(S'', Z)$$

is exact, and that the same holds when Z is a locally ringed space and one restricts to morphisms of locally ringed spaces. Since we already know from a) that the first map is injective, it remains to see that if $f'\colon S' \to Z$ is a morphism of ringed spaces such that $f'p_1 = f'p_2$, then f' is of the form fg, where $f\colon S \to Z$ is a morphism of ringed 212
spaces. Since g is surjective, it is then evident that if f' is a morphism of locally ringed spaces, the same will hold for f.

It follows from the hypothesis on f' that the underlying set map f'_0 is constant on the fibers of the map g_0; hence, since the latter is surjective, f'_0 factors uniquely as $f'_0 = f_0 g_0$, where $f_0\colon S \to Z$ is a map, necessarily continuous since g_0 identifies S with a quotient topological space of S'. Now consider the homomorphism

$$f_0^{-1}(\mathcal{O}_Z) \to g_*(\mathcal{O}_{S'})$$

deduced from the homomorphism $(f_0 g_0)^{-1}(\mathcal{O}_Z) \to \mathcal{O}_{S'}$ corresponding to f'. The hypothesis $f'p_1 = f'p_2$ is then interpreted as saying that the composites of the preceding homomorphism with the two homomorphisms $g_*(\mathcal{O}_{S'}) \rightrightarrows h_*(\mathcal{O}_{S''})$ are the same; hence, by hypothesis b), it factors through a morphism

$$f_0^{-1}(\mathcal{O}_Z) \to \mathcal{O}_S.$$

The latter defines a morphism of ringed spaces $f\colon S \to Z$, which is the desired morphism.

THEOREM VIII.5.2. *Let $\mathcal{F}$ be the fibered category of arrows in the category* **Sch** *of preschemes* (VI VI.11.a). *Then every faithfully flat and quasi-compact morphism $g\colon S' \to S$ is a morphism of $\mathcal{F}$-descent (or, as one also says, a descent morphism in* **Sch***).*

This means the following: let $S'' = S' \times_S S'$, and for two preschemes X, Y over S, consider their inverse images X', Y' over S' and their inverse images X'', Y'' over S'', giving a diagram of maps

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S'}(X', Y') \rightrightarrows \mathrm{Hom}_{S''}(X'', Y'')\ ;$$

with these notations, VIII.5.2 means that this diagram is exact. It should be noted that it is not true in general that g is a morphism of effective descent, i.e. that for every 213
prescheme X' over S', every descent datum on X' relative to $g\colon S' \to S$ is effective. The question of effectivity, often delicate, will be examined in no. VIII.7.

We have seen in [**D**] (taking into account that fiber products exist in **Sch**) that statement VIII.5.2 is equivalent to the following:

COROLLARY VIII.5.3. *A faithfully flat and quasi-compact morphism of preschemes is a universal effective epimorphism.*

Since a faithfully flat and quasi-compact morphism remains so under every base change, we are reduced to proving that it is an effective epimorphism. We then apply the criterion VIII.5.1 b), which gives the desired result, taking VIII.4.3 and VIII.1.7 into account.

COROLLARY VIII.5.4. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism, let $f\colon X \to Y$ be an S-morphism, and let $f'\colon X' \to Y'$ be the S'-morphism deduced from it by the base change $S' \to S$. Then f is an isomorphism if and only if f' is.*

Indeed, if f' is an isomorphism, it is also an isomorphism for the natural descent structures on X', Y', and since the functor $X \longmapsto X'$ from $\mathbf{Sch}_{/S}$ to the category of preschemes over S' endowed with a descent datum relative to g is fully faithful by VIII.5.2, the desired conclusion follows.

COROLLARY VIII.5.5. *Under the conditions of VIII.5.4, f is a closed immersion (respectivelyan open immersion, respectivelya quasi-compact immersion) if and only if f' is.*

As usual, we may suppose $Y = S$, and only the "if" direction has to be proved. Note that the fact that X'/Y' is endowed with a descent datum relative to $g\colon Y' \to Y$, and that the structural morphism $f'\colon X' \to Y'$ is an immersion, hence a monomorphism, implies that the two subobjects of Y'' obtained as inverse images of X'/Y' under either projection 214
from S'' to S' are the same. If f' is a closed immersion, it follows by virtue of VIII.1.9 that there exists a closed subprescheme X_1 of Y whose inverse image under $g\colon Y' \to Y$ is X'. Thus, by the uniqueness of the solution to a descent problem relative to a morphism of $\mathcal{F}$-descent, it follows that X_1 is Y-isomorphic to X, so $f\colon X \to Y$ is a closed immersion. We proceed in the same way for an open immersion, using VIII.4.4. Finally, if f' is a quasi-compact immersion, f is quasi-compact by virtue of VIII.3.3, so we may apply the criterion VIII.4.5 to the subset $f(X)$ of Y, which proves that $f(X)$ is locally closed since its inverse image $f'(X')$ in Y' is so. Replacing Y by an open subset in which $f(X)$ is closed, we are reduced to the case where f' is a closed immersion, so f is one by what precedes.

COROLLARY VIII.5.6. *Under the conditions of VIII.5.4, f is affine if and only if f' is.*

We proceed as in VIII.5.5, using VIII.2.1 (One may also use Serre's cohomological criterion [EGA II 5.2], which proves VIII.5.6 without using descent techniques).

COROLLARY VIII.5.7. *Under the conditions of VIII.5.4, f is integral (respectivelyfinite, respectivelyfinite and locally free) if and only if f' is.*

Only the "if" direction has to be proved, and as usual we may suppose $Y = S$, with Y affine and Y' affine. Since the hypothesis implies that f' is affine, the same is true of f by VIII.5.6, so X and consequently X' are affine. Let $A, A', B, B' = B \otimes_A A'$ be the rings of Y, Y', X, X'. We have $B = \varinjlim_i B_i$, where B_i ranges over the sub-A-algebras of B that are of finite type over A, whence $B' = \varinjlim_i B'_i$, where the B'_i are finite-type subalgebras of the A'-algebra B'. If B' is integral over A', the B'_i are modules of finite type over A'; since A' is faithfully flat over A, the B_i are modules of finite type over A, i.e. B is integral over A. One sees in the same way that if B' is finite over A', B is finite over A. The same conclusion holds for "locally free of finite type", cf. VIII.1.11.

COROLLARY VIII.5.8. *Under the conditions of VIII.5.4, suppose f quasi-compact and* 215
let $\mathcal{L}$ be an invertible Module on X, and $\mathcal{L}'$ its inverse image on X'. Then $\mathcal{L}$ is ample (respectivelyvery ample) relative to f if and only if $\mathcal{L}'$ is ample (respectivelyvery ample) relative to f'.

Only the "if" direction has to be proved. The hypothesis on $\mathcal{L}$ implies in any case that f' is separated, hence f is separated by VIII.4.8; and since f is quasi-compact and $g\colon Y' \to Y$ is flat, the computation of direct images by affine coverings shows that there are isomorphisms

$$g^*(f_*(\mathcal{L}^{\otimes n})) \xrightarrow{\sim} f'_*(\mathcal{L}'^{\otimes n})$$

for every integer n, and consequently there is an isomorphism

$$g^*(\mathcal{S}) \xrightarrow{\sim} \mathcal{S}',$$

where $\mathcal{S}$ (respectively$\mathcal{S}'$) denotes the quasi-coherent graded Algebra on Y (respectivelyon Y') that is the direct sum of the $f_*(\mathcal{L}^{\otimes n})$ (respectivelythe $f'_*(\mathcal{L}'^{\otimes n})$) for $n \geq 0$. Note that for every $n \geq 0$, the cokernel of the canonical homomorphism $f'^*(\mathcal{S}'_n) \to \mathcal{L}'^{\otimes n}$ is the inverse image under $X' \to X$ of the cokernel of $f^*(\mathcal{S}_n) \to \mathcal{L}^{\otimes n}$, and hence its support Z'_n is the inverse image of the support Z_n. If $\mathcal{L}'$ is ample, the intersection of the Z'_n is empty; hence, since $X' \to X$ is surjective, the intersection of the Z_n is empty, i.e. there is a canonical morphism

$$j\colon X \to \mathrm{Proj}(\mathcal{S})$$

(EGA II 3). Moreover, the analogous morphism

$$j'\colon X' \to \mathrm{Proj}(\mathcal{S}')$$

is none other than the one deduced from the preceding morphism by the base change $Y' \to Y$ (loc. cit.). This being so, to say that $\mathcal{L}'$ is ample relative to f' means that j' is an immersion, necessarily quasi-compact since f' is quasi-compact. Thus, by virtue of VIII.5.5, j is an immersion, i.e. $\mathcal{L}$ is ample relative to f. — We proceed in an entirely analogous manner in the case of "very ample", restricting the above to $n = 1$, and replacing the 216
consideration of $\mathrm{Proj}(\mathcal{S})$ by that of the projective bundle $\mathcal{P}(\mathcal{S}_1)$ associated with $\mathcal{S}_1$.

Recall (EGA II 5.1.1) that a quasi-compact morphism f is called *quasi-affine* if for every affine open subset U of Y, $f^{-1}(U)$ is a prescheme isomorphic to an open subscheme of an affine scheme. It is shown (loc. cit.) that this is equivalent to saying that $\mathcal{O}_X$ is ample (or also: very ample) relative to f. Thus VIII.5.8 implies:

Corollary VIII.5.9. *Under the conditions of VIII.5.4, and assuming f quasi-compact, f is quasi-affine if and only if f' is.*

Remarks VIII.5.10. Hironaka's example of a nonprojective variety shows that one can have a proper morphism $f\colon X \to Y$ of nonsingular algebraic varieties (with Y projective), such that Y is the union of two open subsets Y_i for which $X_i = X \times_Y Y_i$ is projective over Y_i, while f is not projective. Thus, putting $Y' = Y_1 \amalg Y_2$, Y' is faithfully flat and quasi-compact (and even quasi-finite) over Y, and $f'\colon X' \to Y'$ is projective, but f is not projective. One must therefore take care: in order to apply VIII.5.8, and to deduce from the fact that f' is projective the same conclusion for f, one must already have on X' an invertible Module $\mathcal{L}'$ ample for f', *endowed with a descent datum relative to $X' \to X$* (which allows $\mathcal{L}'$ to be regarded as the inverse image of an invertible Module $\mathcal{L}$ on X, which will then be ample for f by virtue of VIII.5.8). When $g\colon S' \to S$ is finite and locally free, however, see VIII.7.7.

VIII.6. Application to finite and quasi-finite morphisms[3]

We shall prove the following two theorems:

Theorem VIII.6.1. *Let $f\colon X \to Y$ be a morphism that is* proper with finite fibers, *with Y locally Noetherian. Then f is finite.*

Theorem VIII.6.2. *Let $f\colon X \to Y$ be a* quasi-finite *and* separated *morphism, with Y locally Noetherian. Then f is quasi-affine, and a fortiori quasi-projective.*

Remarks VIII.6.3. The theorem VIII.6.1 is well known, and is due to Chevalley in 217
the case of algebraic varieties; a simple proof may also be found in [EGA III 4], using the "theorem on holomorphic functions". The proof given here does not use the latter theorem, but instead uses descent theory; we give it as a bonus to the reader, since one

[3]Cf. EGA IV 18.12 for generalizations to preschemes that are not necessarily locally Noetherian

obtains it "for free"at the same time as the proof of VIII.6.2. Recall also ([EGA III 4] or [**1**]) that the global form of Zariski's "Main Theorem", deduced from the "theorem on holomorphic functions", asserts that if $f\colon X \to Y$ is quasi-finite and *quasi-projective*, with Y Noetherian, then X is Y-isomorphic to an open subprescheme of a Y-prescheme *finite* Z. The conjunction of the "Main Theorem"and of VIII.6.2 may therefore be stated as follows:

COROLLARY VIII.6.4. *Let $f\colon X \to Y$ be a quasi-finite and separated morphism, with Y Noetherian. Then X is Y-isomorphic to an open subprescheme of a finite Y-prescheme Z.*

Another interesting consequence of VIII.6.2 for descent theory will be given along with VIII.7.9.

Proof of VIII.6.1 and VIII.6.2. We shall use the following fact, whose proof is easy[4]:

LEMMA VIII.6.5. *Let X be a prescheme of finite type over a locally Noetherian prescheme Y, and let $y \in Y$. In order that there exist an open neighborhood U of y such that $X|U$ is finite (respectivelyquasi-affine, respectively ...) over U, it is necessary and sufficient that $X \times_Y \mathrm{Spec}(()\mathcal{O}_y)$ be finite (respectivelyquasi-affine, respectively ...) over $\mathrm{Spec}(()\mathcal{O}_y)$.*

Since, on the other hand, the property for $f\colon X \to Y$ of being finite, respectivelyquasi-affine, is local on Y, we are reduced, in order to prove VIII.6.1 and VIII.6.2, to the case where Y is the spectrum of a local ring, and hence in particular is of finite dimension. (N.B. by the *dimension of a prescheme Y* we mean the supremum of the Krull dimensions of its local rings). We proceed by induction on

$$n = \dim(Y),$$

the assertion being trivial for $n < 0$. We may therefore suppose $n \geq 0$, and the assertion 218
proved for dimensions $n' < n$. We may again suppose that Y is the spectrum of a Noetherian local ring A, of dimension n. Note that the hypotheses made in VIII.6.1 and VIII.6.2 are stable under base change (this was already used in the initial reduction), so they will remain true after the base change $\mathrm{Spec}(()\widehat{A}) \to \mathrm{Spec}(()A)$. Since the latter is faithfully flat and quasi-compact, the statements VIII.5.7 and VIII.5.9 reduce us to the case where A is moreover complete. Using then the fact that every Noetherian local ring B over A that is quasi-finite over A is finite over A, and the fact that X is separated over Y and the fiber at y consists of isolated points, we obtain a decomposition

$$X = X' \amalg X''$$

where X' is *finite* over Y, and where the fiber of X'' at y is empty. If X is proper over Y, then so is X'', hence its image in Y is closed, and since it does not contain y, it is empty; thus $X'' = \emptyset$ and hence $X = X'$, which shows that X is finite over Y and proves VIII.6.1 (N.B. the induction hypothesis is unnecessary here). If X is quasi-finite over Y, so is X'', but X'' in fact lies over the open subset $Y - \{y\}$ of Y, *which has dimension $< n$*. By the induction hypothesis, X'' is quasi-affine over $Y - \{y\}$, hence also over Y; the same is plainly true of X', and hence also of their sum X, which proves VIII.6.2.

REMARK VIII.6.6. Theorems VIII.6.1 and VIII.6.2 remain valid if Y is no longer assumed locally Noetherian, provided one specifies that f is assumed to be of finite presentation (cf. VIII.3.6). Indeed, we may still suppose Y affine, and then one verifies without difficulty that the situation $f\colon X \to Y$ is deduced, by a base change $Y \to Y_0$, from a situation $f_0\colon X_0 \to Y_0$ satisfying the same hypotheses as f, with Y_0 *Noetherian*. Therefore, by VIII.6.1 respectivelyVIII.6.2, f_0 is finite respectivelyquasi-affine, and hence so is f. This kind of argument is often useful for dispensing with Noetherian hypotheses, (which always end up being troublesome in applications). 219

[4]Cf. EGA IV 8

VIII.7. Criteria for the effectivity of a descent datum

As usual, consider a morphism of preschemes

$$g\colon S' \longrightarrow S$$

and an S'-prescheme X'. In accordance with the general facts in VII, 9[5], giving a descent datum on X' relative to g is equivalent to giving an equivalence pair [**3**]

$$q_1, q_2\colon X'' \rightrightarrows X'$$

such that the structural morphism $X' \to S'$ is compatible with this pair and the equivalence pair

$$p_1, p_2\colon S'' = S' \times_S S' \rightrightarrows S'$$

defined by g, and such that the two squares (or either one of them, which amounts to the same thing by symmetry) extracted from the corresponding diagram

$$\begin{array}{ccc} X' & \leftleftarrows & X'' \\ \downarrow & & \downarrow \\ S' & \leftleftarrows & S'' \end{array}$$

(using either p_1, q_1 or p_2, q_2) are *cartesian*. A solution to the descent problem posed by this descent datum, i.e. an object X over S endowed with an isomorphism $X \times_S S' \xleftarrow{\sim} X'$ compatible with the descent data, is equivalent to a *cartesian* square

$$\begin{array}{ccc} X & \xleftarrow{h} & X' \\ \downarrow & & \downarrow \\ S & \xleftarrow{g} & S' \end{array}$$

satisfying $hq_1 = hq_2$.

Since the class of faithfully flat and quasi-compact morphisms is stable under base change, and since a faithfully flat and quasi-compact morphism is an effective epimorphism by VIII.5.3, the general theory [**D**] gives: 220

Proposition VIII.7.1. *Suppose that $g\colon S' \to S$ is faithfully flat and quasi-compact. A descent datum on X' relative to g is effective if and only if the equivalence relation $R = (q_1, q_2)$ that it defines is effective (i.e. the quotient X'/R exists and X'' becomes the fiber square of X' over X'/R), and the canonical morphism $X' \to X'/R$ is faithfully flat and quasi-compact.*

Thus the question of the effectivity of a descent datum is a special case of the question of the effectivity of an equivalence graph, and various effectivity criteria given in this section may be obtained in this way. Nevertheless, in the context of descent one has Theorem VIII.2.1, which implies that *if X' is affine over S', every descent datum on X' relative to g is effective.* This statement has no analogue for passage to the quotient by a general flat equivalence graph. All the effectivity criteria given here may also be regarded as consequences of that theorem.

Let U' be a subprescheme of X' (or, more generally, a subobject of X' in the category **Sch**). We say that U' is *stable under the descent datum* on X' if U' can be endowed with a descent datum relative to g for which the immersion $U' \to X'$ is compatible with the descent data. This also means that the inverse images of U' in X'' under q_1 and q_2 are the same (or, as one also says, that U' is *stable under the equivalence relation R*). The descent datum in question on U' is then plainly unique and is called the *induced descent datum* from that on X'. With this terminology:

[5]This section does not exist; see Exposé 212 of the Séminaire Bourbaki (1962).

PROPOSITION VIII.7.2. *Let (X'_i) be a covering of X' by open subsets X'_i stable under the descent datum. The descent datum on X' is effective if and only if the induced descent data on all the X'_i are effective.* 221

This is, for example, an immediate consequence of VIII.7.1; the details of the proof are left to the reader.

COROLLARY VIII.7.3. *Let (S_i) be an open covering of S, and for every i let S'_i and X'_i be deduced from S' and X' by the base change $S_i \to S$. The descent datum on X' is effective if and only if, for every i, the descent datum on X'_i relative to $g_i\colon S'_i \to S_i$ is effective.*

In practice this criterion always reduces us to the case in which S is affine. When S' is affine as well, as is most often the case in applications, we have:

COROLLARY VIII.7.4. *Suppose that S and S' are affine. The descent datum on X' is effective if and only if X' is the union of affine open subsets X'_i stable under the descent datum.*

Sufficiency follows from VIII.7.2 and the fact that, if X'_i is affine, then it is affine over S', so that VIII.2.1 applies. For necessity, observe that if X' comes from X, and if X is covered by affine open subsets X_i, then the $X'_i = X_i \times_S S'$ are affine open subsets stable under the descent datum and covering X'.

COROLLARY VIII.7.5. *Let $g\colon S' \to S$ be a faithfully flat, quasi-compact, and* radicial *morphism. Then g is an* effective descent *morphism, i.e. for every X' over S', every descent datum on X' relative to $g\colon S' \to S$ is effective.*

Indeed, by VIII.7.3 we may suppose that S is affine. Since S' is radicial over S, and hence separated over S, it is separated. Moreover, for every $x' \in X'$, the fiber

$$R(x') = q_2(q_1^{-1}(x'))$$

of the set-theoretic equivalence relation defined by R consists of a single point. Indeed, 222
since g is radicial, the same holds for p_1, p_2, which are deduced from g by the base change $S' \to S$, and hence also for q_1, q_2, which are deduced from the former by the base change $X' \to S''$. Therefore *every open subset of X' is stable* under the descent datum. Cover X' by affine open subsets X'_i. They are affine over S', since S' is separated, so the induced descent datum is effective by VIII.2.1. We conclude by VIII.7.2.

Observe that VIII.7.5 gives the only known instance of an effective descent morphism in the category of preschemes, and it is probably the only instance, even if one restricts to noetherian schemes or to schemes of finite type over a field.

When S is assumed locally noetherian and S' of finite type over S, statement VIII.7.5 is also a special case of the following result (which generalizes Weil's Galois descent and Cartier's inseparable descent):

COROLLARY VIII.7.6. *Let $g\colon S' \to S$ be a finite locally free morphism (i.e. defined by an algebra on S which, as a module, is finite locally free) and suppose that g is surjective (so g is faithfully flat and quasi-compact, and hence a descent morphism). Let X' be an S'-prescheme endowed with a descent datum. This datum is effective if and only if, for every $x' \in X'$, the fiber $R(x') = q_2(q_1^{-1}(x'))$ is contained in an affine open subset (a condition automatically satisfied if X' is quasi-projective over S').*

The parenthetical observation follows from the fact that, if s is the point of S below x', then $R(x')$ is finite and contained in the fiber X'_s, while, since X' is quasi-projective over S' and S' is finite over S, X' is quasi-projective over S; this implies that a fiber of X' over S is contained in an affine open subset.

Since every finite subset of an affine scheme has a fundamental system of affine neighborhoods, the hypothesis is preserved when we restrict over an affine open subset

of S. By VIII.7.3, this reduces us to the case in which S is affine. By VIII.7.4, it is enough to show that x' is contained in an affine open subset *stable* under the descent 223
datum. Indeed, let U be an affine open subset containing $R(x')$. Then the saturation

$$R(X' - U) = q_2(q_1^{-1}(X' - U))$$

does not meet $R(x')$. Moreover, since q_2 is finite (because g, and hence p_2, is finite), and therefore closed, the right-hand side is a closed subset of X'. Let U' be its complement in X'. Thus U' is a *saturated* open subset, and

$$R(x') \subset U' \subset U,$$

where U is affine, although U' need not be affine a priori. Since the finite subset $R(x')$ of the affine scheme U has a fundamental system of affine neighborhoods of the form U_f, one sees, by replacing f by its restriction to U', that there is a section f of $\mathcal{O}_U$ such that

$$R(x') \subset U'_f, \qquad U'_f \text{ is affine.}$$

Set $U'' = q_1^{-1}(U') = q_2^{-1}(U')$, continue to write q_1, q_2 for the induced morphisms $U'' \to U'$, and consider

$$f' = \mathrm{N}_{q_2}(q_1^*(f)),$$

where N_{q_2} denotes the *norm* relative to the finite locally free morphism $q_2 \colon U'' \to U'$. Compatibility of the formation of the norm with base change readily implies that f' is an *invariant* section:

$$q_1^*(f') = q_2^*(f').$$

It follows that $U'_{f'}$ is a saturated open subset of U'. More precisely, writing $Z(f')$ for the zero set of a section f', the properties of norms give

$$Z(f') = q_2(Z(q_1^*(f))) = q_2(q_1^*(Z(f))) = R(U' - U'_f).$$

Consequently, $U'_{f'} = U' - Z(f')$ is saturated, contains $R(x')$, and is contained in U'_f. Since 224
the latter is affine, $U'_{f'}$ is affine as well (it is equal to $(U'_f)_{f''}$, with $f'' = f'|_{U'_f}$). It is therefore a saturated affine open subset containing $R(x')$, and hence x'. This completes the proof.

The same argument applies whenever one has an equivalence relation (or even merely a pre-equivalence relation, cf. [**3**]) on a prescheme X' which is finite and locally free. Moreover, VIII.7.6 is itself a special case of the analogous result for finite locally free pre-equivalence relations; see loc. cit. The same observation applies to VIII.7.7 below.

Alternatively, once a saturated quasi-affine open subset U' containing x' has been obtained, one may invoke VIII.7.9 and VIII.7.2, thereby avoiding the use of norms.

Observe also that under the hypotheses of VIII.7.6, if the descent datum on X' is effective and X' comes from X over S, then the morphism $X' \to X$ is finite, locally free, and surjective, since it is deduced from g by the base change $X \to S$. It follows (EGA II, 6.6.4) that if X' is quasi-projective over S', and hence over S, then X is quasi-projective over S. A relatively ample invertible sheaf on X is obtained by taking the *norm* of an invertible sheaf on X' that is relatively ample over S, or equivalently over S'. Thus we obtain:

Corollary VIII.7.7. *A finite locally free and surjective morphism $g \colon S' \to S$ is an effective descent morphism for the fibered category of preschemes quasi-projective over other preschemes. In other words, for every X' quasi-projective over S', every descent datum on X' relative to g is effective, and the descended S-prescheme X is quasi-projective over S.*

Proposition VIII.7.8. *Let $g \colon S' \to S$ be faithfully flat and quasi-compact. Then g is an effective descent morphism for the fibered category of preschemes Z quasi-compact over a prescheme T, equipped with an invertible sheaf ample relative to T. In particular, let X'* 225

be an S'-prescheme endowed with a descent datum relative to $g\colon S' \to S$, and let $\mathcal{L}'$ be an invertible sheaf on X', ample relative to S', likewise endowed with a descent datum relative to the given datum on X' (i.e. endowed with an isomorphism between $q_1^*(\mathcal{L}')$ and $q_2^*(\mathcal{L}')$ satisfying the usual transitivity condition). Then the descent datum on X' is effective, and the invertible sheaf $\mathcal{L}$ on the descended prescheme X, obtained from $\mathcal{L}'$ by descent, is ample relative to S.

The proof is entirely analogous to that of VIII.5.8. On the quasi-coherent graded algebra $\mathcal{S}'$ over S' defined by $\mathcal{L}'$, there is a descent datum. By VIII.1.1, this allows one to construct a quasi-coherent graded algebra $\mathcal{S}$ over S, and hence an S-prescheme $P = \mathrm{Proj}(\mathcal{S})$, such that $P' = \mathrm{Proj}(\mathcal{S}')$, together with its descent datum, is identified with $P \times_S S'$. By hypothesis, X' is identified with an open subset of P', necessarily stable under the descent datum on P'. The descent datum on X' is therefore effective as well, and the descended prescheme is obtained as an open subset of P. The details are left to the reader. In particular, taking $\mathcal{L}' = \mathcal{O}_{X'}$, we find:

Corollary VIII.7.9. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism, and let X' be a prescheme* quasi-affine *over S'. Then every descent datum on X' relative to g is effective, and the descended prescheme X is quasi-affine over S.*

By VIII.6.2, this result applies in particular when S' is locally noetherian and X' is quasi-finite and separated over S', and more generally when S' is arbitrary and X' is of finite presentation, quasi-finite, and separated over S' (cf. VIII.6.6).

Remarks VIII.7.10. The results given in this section exhaust the currently known effectivity criteria, and probably even all the useful criteria that exist[6]. The following counterexamples support this assertion:

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(i) If S is the spectrum of a field and S' is the spectrum of a quadratic Galois extension, there exists an X' over S', proper and smooth over S', of dimension 3, endowed with a descent datum which is not effective (Serre).

(ii) There exist an S which is the spectrum of a regular local ring of dimension 3 (if desired, the local ring of an algebraic scheme over a field of prescribed characteristic), a principal covering T of S with group $\mathbb{Z}/2\mathbb{Z}$, and the following property. If t denotes one of the points of T above the closed point s of S, and $S' = T - \{t\}$, there exists a regular X', *projective* over S', endowed with a descent datum relative to $g\colon S' \to S$, which is not effective.

These constructions use Hironaka's example of nonprojective varieties. For (i), it is enough to use the fact that there exists over k a proper smooth scheme X_0 of dimension 3 on which $G = \mathbb{Z}/2\mathbb{Z}$ acts without inertia, and in which there are two rational points a, b, over S, conjugate under G, which are not contained in an affine open subset. Set $X' = X_0 \times_k k'$, and let G act on X' by its actions on both factors. This gives a descent datum on X' relative to $g\colon \mathrm{Spec}(()k') \to \mathrm{Spec}(()k)$. Above a, respectively b, there is exactly one point a', respectively b', with quadratic residue-field extension, and a' and b' are conjugate under G, since $X' \to X_0$ is compatible with the actions of G. The points a' and b' cannot be contained in an affine open subset U'. Indeed, otherwise

$$U = X_0 - \mathrm{Im}(X' - U')$$

would be an open subset of X_0 containing a, b, whose inverse image in X' would be contained in U', and hence would be quasi-affine. It would follow that U is quasi-affine, and thus that a, b have an affine neighborhood in U.

[6]This opinion proved partly mistaken; see, for example, J.P. Murre, Sém. Bourbaki 294 (Appendix), May 1965, and special results, notably those of Néron and Raynaud. For the descent of group schemes, cf. M. Raynaud, thesis (1968).

For (ii), one uses the fact that in Hironaka's example X_0 is obtained as a prescheme proper over a projective k-scheme Y, smooth over k. The morphism $f\colon X_0 \to Y$ is in fact birational, but this is irrelevant. The group G also acts on Y compatibly with its action on X_0. Finally, on setting $S' = Y - f(b)$ and $X' = X_0|S'$, the prescheme X' is projective over S'. The prescheme X_0 is then endowed with a natural descent datum relative to the canonical morphism $Y \to S = Y/G$, by virtue of the action of G on X_0 compatible with 227
its action on Y. This descent datum is not effective, since a, b are not contained in an affine open subset. The induced descent datum on X' relative to $g\colon S' \to S$ is then not effective either, as is readily verified.

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CHAPTER IX

Descent of étale morphisms. Application to the fundamental group

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IX.1. Review of étale morphisms

We shall review here the properties of étale morphisms (developed in I) that we shall use, taking this opportunity to remove the superfluous noetherian hypotheses from the theory. The reader should note that even if one is interested only in noetherian schemes, the technique of descent leads one to introduce nonnoetherian schemes (such as $\mathrm{Spec}(()\widehat{A} \otimes_A \widehat{A})$, where A is a noetherian local ring). In order to use the language of fibered categories, it is therefore important to define notions such as étale morphisms, etc., without introducing noetherian restrictions. A reader reluctant to verify or accept that the statements below hold without noetherian hypotheses may simply take them under the noetherian hypotheses of I, provided that the same noetherian hypotheses are inserted into the statements of the following sections, and that Definition IX.1.1 below is used for the nonnoetherian schemes that arise in the arguments.

Definition IX.1.1. *Let $f\colon X \to S$ be a morphism of preschemes and x a point of X. We say that f is* étale at *x, or that X is étale over S at x, if there exist an affine open neighborhood U of $s = f(x)$, an affine open neighborhood V of x above U, an affine* noetherian *scheme U_0, an affine U_0-scheme V_0 which is* étale (I)*, a morphism $U \to U_0$, and a U-isomorphism*

$$V \xrightarrow{\sim} V_0 \times_{U_0} U.$$

When S is locally noetherian, this terminology agrees with that of loc. cit. Likewise, 229
we say that *f is étale*, or that *X is étale over S*, if f is étale at every point x of X. With these definitions, the propositions below reduce without difficulty to the noetherian case, where they are proved in I, no. I.4, I.5, and I.7. For details, the reader may consult EGA IV[1].

Remarks IX.1.2. If f is étale at x, then f is *of finite presentation at x* (VIII VIII.3.5), the local ring of x in the fiber $f^{-1}(s)$ is a *finite separable extension* of $\kappa(s)$, and f is *flat at x*. One can prove the converse, so Definition IX.1.1 is the same as in the case where S is locally noetherian, except that the condition *of finite type at x* must be replaced by *of finite presentation at x*. Since this result is delicate to prove, we have not chosen it here as the definition of an étale morphism; it does not lend itself directly to proving the properties that follow.

First, we have trivially:

Proposition IX.1.3. *If $f\colon X \to S$ is étale, then every morphism $f'\colon X' \to S'$ deduced from it by a base change $S' \to S$ is likewise étale.*

Thus one may say that étale morphisms form a *fibered subcategory* of the category of arrows in **Sch** (cf. VI VI.11 a)). The purpose of the present exposé is to study the exactness properties of this category fibered over **Sch**.

[1]More precisely, EGA IV, 17 and 18.

PROPOSITION IX.1.4. *Let $f\colon X \to S$ be a morphism of preschemes. It is an open immersion if and only if it is étale and radicial.*

Cf. I I.5.1. It follows that if X is étale over S, every section of X over S is an open immersion. Applying IX.1.4 once more gives:

COROLLARY IX.1.5. *Let X be an étale S-prescheme. There is a bijection between the set of sections of X over S and the set of open subsets Γ of X for which the morphism* 230
$\Gamma \to S$ induced by the structural morphism is radicial *and* surjective.

If X is separated over S, then Γ is both open and closed in X, but this is immaterial. After an evident base change, IX.1.5 may be put in the following apparently more general form:

COROLLARY IX.1.6. *Let X and Y be two S-preschemes, with Y étale over S. The map $f \longmapsto \Gamma_f$, which assigns to every S-morphism $f\colon X \to Y$ the underlying subset of the graph of f in $X \times_S Y$, is a bijection from $\mathrm{Hom}_S(X, Y)$ onto the set of open subsets Γ of $X \times_S Y$ such that the morphism $\Gamma \to X$ induced by pr_1 is* radicial *and* surjective.

PROPOSITION IX.1.7. *Let S_0 be the subprescheme of S defined by a quasi-coherent nil ideal, i.e. such that S_0 has the same underlying set as S. Then the functor*

$$X \longmapsto X \times_S S_0$$

from the category of preschemes étale over S to the category of preschemes étale over S_0 is an equivalence of categories.

The fact that this functor is fully faithful is an immediate consequence of IX.1.6. Its essential surjectivity is contained in I I.8.3. Observe that under the preceding equivalence, X is of finite type, i.e. quasi-finite, over S (respectively finite, i.e. an étale covering of S) if and only if X_0 satisfies the analogous condition over S_0. The same observation applies to separatedness. These facts are immediate, and are also contained in IX.2.4 below.

COROLLARY IX.1.8. *Let A be a complete noetherian local ring with residue field k. Then the functor*

$$B \longmapsto B \otimes_A k$$

is an equivalence from the category of finite étale algebras over A to the category of finite étale algebras over k (i.e. finite products of finite separable extensions of k).

PROPOSITION IX.1.9. *A prescheme X is an* étale covering *of S, i.e. finite and étale over S, if and only if X is S-isomorphic to the spectrum of an algebra $\mathcal{A}$ on S which is finite locally free as a module and such that, for every $s \in S$, the algebra $\mathcal{A}_s \otimes_{\mathcal{O}_s} \kappa(s)$ is* 231
separable over $\kappa(s)$, and hence in this case is a finite product of finite separable extensions of $\kappa(s)$.

Finally, the following result is less elementary, being the conjunction of I I.8.4 and the *existence theorem for sheaves in algebraic geometry* (EGA III, 5; cf. also [**1**], Th. 3).

THEOREM IX.1.10. *Let S be the spectrum of a complete noetherian local ring, let X be a proper S-scheme, and let X_0 be the fiber of X over the closed point of S, so that X_0 is a closed subscheme of X. Then the restriction functor*

$$X' \longmapsto X' \times_X X_0$$

is an equivalence from the category of étale coverings of X to the category of étale coverings of X_0.

IX.2. Submersive and universally submersive morphisms

DEFINITION IX.2.1. *A morphism $g\colon S' \to S$ of preschemes is called* submersive *if it is surjective and makes S a quotient topological space of S' (i.e. every subset U of S for which $g^{-1}(U)$ is open is itself open). We say that g is* universally submersive *if, for every morphism $T \to S$, the morphism $g'\colon T' = S' \times_S T \to T$ deduced from g by base change is submersive.*

It is immediate that the composite of two submersive (respectively universally submersive) morphisms is submersive (respectively universally submersive), and that a base change of a universally submersive morphism is universally submersive, as the definition was made precisely for this purpose. If fg is submersive (respectively universally submersive), then f is so.

EXAMPLES IX.2.2. *a) A surjective morphism which is open or closed is submersive. Consequently, a surjective universally closed or universally open morphism is universally submersive. For example,* a proper surjective morphism is universally submersive*. Moreover,* a faithfully flat and quasi-compact morphism is universally submersive *(VIII VIII.4.3). These will be the two most important cases for us.*

The arguments of VIII VIII.4.3 may be applied to a submersive or universally sub- 232
mersive morphism $g\colon S' \to S$. In particular, one obtains:

PROPOSITION IX.2.3. *Suppose that $g\colon S' \to S$ is submersive. Then the following diagram of maps is exact:*

$$\mathrm{Ouv}(S) \longrightarrow \mathrm{Ouv}(S') \rightrightarrows \mathrm{Ouv}(S''),$$

where $S'' = S' \times_S S'$, and where $\mathrm{Ouv}(X)$ denotes the set of open subsets of the prescheme X.

PROPOSITION IX.2.4. *Let $g\colon S' \to S$ be universally submersive, let $f\colon X \to Y$ be an S-morphism, and let $f'\colon X' \to Y'$ be the S'-morphism deduced from f by base change. For f to be open (respectively closed), it is sufficient that f' be so. The morphism f is universally open, respectively universally closed, respectively separated, if and only if f' is so. If, in addition, g is quasi-compact and f is locally of finite type, then f is proper if and only if f' is proper.*

For the last assertion, observe that if f' is proper, and hence quasi-compact, then f is quasi-compact (VIII VIII.3.3), and therefore of finite type since it is locally of finite type. Moreover, by what precedes, it is separated and universally closed, and hence proper.

PROPOSITION IX.2.5. *Let S' be a prescheme of finite type over the spectrum S of a complete noetherian local ring. Suppose that the fiber over the closed point s of S is finite, so that the local rings of S' at the points s' of this fiber are finite over $A = \mathcal{O}_s$. Let S'' be the scheme which is the disjoint sum of the spectra of the rings $\mathcal{O}_{S',s'}$ in question, regarded as a finite S-scheme. Then $g\colon S' \to S$ is universally submersive if and only if the structural morphism $S'' \to S'$ is surjective.*

Since there is a natural S-morphism $S'' \to S'$, and a finite surjective morphism is universally submersive by IX.2.2, the stated condition is sufficient. We therefore show that if $S'' \to S'$ is not surjective, then g is not universally submersive. Let t be a point of S not in the image of S''. There is then an S-scheme T, the spectrum of a discrete valuation ring, whose image in S is $\{s, t\}$. The image of S'' in S' is open, since the 233
morphism $S'' \to S'$ is a local isomorphism; moreover, this image contains S'_s and does not meet S'_t. Its inverse image in $T' = S' \times_S T$ is therefore *open*, and is precisely the inverse image of the closed point of T. This shows that $T' \to T$ is not submersive, and hence that $S' \to S$ is not universally submersive.

REMARK IX.2.6. Using the criterion IV IV.6.3 for a constructible subset of a noetherian space to be open, one readily obtains the following valuative criterion for a morphism $g\colon S' \to S$ *of finite type*, with S locally noetherian, to be universally submersive: it is necessary and sufficient that, for every S-scheme T which is the spectrum of a discrete valuation ring, on setting $T' = S' \times_S T$, the inverse image in T' of the closed point of T is not open.

IX.3. Descent of étale morphisms of preschemes

PROPOSITION IX.3.1. *Let $g\colon S' \to S$ be a* surjective *morphism of preschemes, let X and Y be two preschemes over S, and let X' and Y' be their inverse images over S'. If Y is unramified over S, then the canonical map*

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S'}(X', Y')$$

is injective.

Indeed, by IX.1.6, an S-morphism $f\colon X \to Y$ is determined by the underlying set of its graph Γ, which is a subset of $Z = X \times_S Y$. Since

$$Z' = Z \times_S S' = X' \times_{S'} Y' \longrightarrow Z$$

is surjective, because $S' \to S$ is surjective, the subset Γ is determined by its inverse image in $X' \times_{S'} Y'$, which is precisely the underlying set of the graph of f'. This proves the assertion.

A subset Γ of Z is the graph of an S-morphism $f\colon X \to Y$ if and only if it is open and the morphism $\Gamma \to X$ induced by pr_1 is radicial and surjective; see IX.1. When the 234
first property holds, the second holds if and only if the inverse image Γ' of Γ in Z' satisfies the same condition (VIII VIII.3.1). Finally, if $Z' \to Z$ is known to be submersive, as is in particular the case when $S' \to S$ is universally submersive, then Γ is open if and only if Γ' is open.

Thus $\mathrm{Hom}_S(X, Y)$ is in bijection with the set of open subsets Γ' of Z' for which the projection $\mathrm{pr}_1\colon \Gamma' \to X'$ is radicial and surjective (i.e. those corresponding to an S'-morphism $f'\colon X' \to Y'$), and which are saturated for the equivalence relation defined by $Z' \to Z$. The latter condition says that their two inverse images in

$$Z'' = Z' \times_Z Z' = Z \times_S S'', \qquad S'' = S' \times_S S',$$

under the two projections are equal. These inverse images are the graphs of the two S''-morphisms $X'' \to Y''$ deduced from f' by base change along the two projections $S'' \to S'$. We have therefore proved:

PROPOSITION IX.3.2. *Let $g\colon S' \to S$ be a* universally submersive *morphism of preschemes, let $S'' = S' \times_S S'$, let X and Y be two S-preschemes, let X' and Y' be their inverse images over S', and let X'' and Y'' be their inverse images over S''. If Y is étale over S, then the following canonical diagram of maps is exact:*

$$\mathrm{Hom}_S(X, Y) \rightarrowtail \mathrm{Hom}_{S'}(X', Y') \rightrightarrows \mathrm{Hom}_{S''}(X'', Y'').$$

Taking X and Y étale over S, one obtains the following statement, which moreover recovers IX.3.2, even after restricting to $X = S$, a case to which one may always reduce in IX.3.2 by the base change $X \to S$:

COROLLARY IX.3.3. *A universally submersive morphism of preschemes is a descent morphism for the fibered category of preschemes étale over other preschemes.*

I do not know whether it is necessarily an *effective* descent morphism for the fibered category in question, even under the additional hypotheses that S is noetherian and g is of finite type, and even if one restricts to étale coverings. In the next section we shall nevertheless give useful effectivity criteria.

COROLLARY IX.3.4. *Let $g\colon S' \to S$ be a universally submersive morphism whose fibers* 235
$g^{-1}(s)$ are geometrically connected, *i.e. such that for every extension $K/\kappa(s)$, the scheme $g^{-1}(s) \otimes_{\kappa(s)} K$ is connected. Then S' is connected whenever S is connected. The functor, defined by g, from the category of preschemes étale over S to the category of preschemes étale over S' is fully faithful.*

A subset of S' which is both open and closed is saturated for the set-theoretic equivalence relation defined by g, since the fibers are connected. It is therefore the inverse image of a subset of S, which is necessarily open and closed because g is submersive. Thus if S is connected, so is S'.

This also implies the following fact: if f and g are two morphisms with universally connected fibers, and f is universally submersive, then the composite fg has universally connected fibers. If S'_1 and S'_2 over S have universally connected fibers, then so does $S'_1 \times_S S'_2$. In particular, under the hypotheses of IX.3.4, S'' has universally connected fibers over S.

Now let X and Y be étale over S, and let u' be an S'-morphism from X' to Y'. We prove that it is compatible with the descent data, which gives the desired conclusion by IX.3.3. Let u''_1 and u''_2 be the two S''-morphisms $X'' \to Y''$ deduced from u'. The subprescheme of S'' on which u''_1 and u''_2 coincide is an induced open subprescheme, closed on each fiber, since it is the inverse image of the diagonal prescheme of Y'' over S'' [2]. It is therefore the inverse image of a subset of S. Since it contains the diagonal in S'', it is all of S''. Thus $u''_1 = u''_2$, as required.

IX.4. Descent of étale preschemes: effectivity criteria

PROPOSITION IX.4.1. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism. Then g is an effective descent morphism for the fibered category of preschemes that are étale, separated, and of finite type over other preschemes.*

Indeed, it is a descent morphism for the fibered category in question, by IX.3.3 or VIII VIII.5.2, as one prefers. It remains to show that if X' is étale, separated, and of finite type over S', and endowed with a descent datum relative to $g\colon S' \to S$, the latter is effective in the fibered category in question. But one sees easily that if X is a prescheme over S, then it is étale over S if and only if it is étale over S' (by Definition IX.1.1 and loc. 236
cit. VIII.3.6). Thus it is étale, separated, and of finite type over S if and only if X' is so over S', cf. for example IX.2.4. It therefore suffices to ensure the effectivity of the descent datum on X for the fibered category of arrows in **Sch**. But X' is quasi-affine over S' by VIII VIII.6.2 and VIII.6.6. We may then conclude by using VIII VIII.7.9. The reader will moreover note that the proof requires less if one restricts oneself to preschemes that are étale and *finite* over other preschemes, since one may then invoke VIII VIII.2.1 directly.

COROLLARY IX.4.2. *Let $g\colon S' \to S$ be a universally submersive morphism, let X' be an S'-prescheme that is étale, separated, and of finite type, endowed with a descent datum relative to g, let $S_1 \to S$ be a faithfully flat and quasi-compact morphism, and let S'_1 and X'_1 be deduced from S' and X' by base change, so that $S'_1 \to S_1$ is universally submersive, while X'_1 is étale, separated, and of finite type over S'_1, and is endowed with a descent datum relative to $g_1\colon S'_1 \to S_1$. The descent datum on X' is effective if and only if the descent datum on X'_1 is effective.*

This follows from descent theory in categories [**D**], in view of IX.4.1 and IX.3.3.

One proves similarly:

COROLLARY IX.4.3. *Let $g\colon S' \to S$ be a universally submersive morphism, let X' be an S'-prescheme that is étale and endowed with a descent datum relative to g, and let (S_i) be a covering of S by open subsets. The descent datum is effective if and only if,*

[2] Observe that the fibers of S' over S are separated!

for every i, the corresponding descent datum on $X'_i = X \times_S S_i$, relative to the morphism $g_i \colon S'_i = S' \times_S S_i \to S_i$, is effective.

This last result leads to the following local effectivity criterion:

PROPOSITION IX.4.4. *Let $g \colon S' \to S$ be a morphism of finite presentation* VIII *VIII.3.6 and universally submersive, let X' be a prescheme that is étale and of finite presentation* 237 *over S', endowed with a descent datum relative to g, and finally let a be a point of S. In order that there exist an open neighborhood U of a such that the corresponding descent datum on $X'_U = X' \times_S U$ relative to the morphism*

$$g_U \colon S'_U = S' \times_S U \to S_U = U$$

is effective, it is necessary and it is sufficient that the corresponding descent datum on $X'_a = X' \times_S \operatorname{Spec}(()\mathcal{O}_a)$, relative to the morphism

$$g_a \colon S'_a = S' \times_S \operatorname{Spec}(()\mathcal{O}_a) \to S_a = \operatorname{Spec}(()\mathcal{O}_a),$$

be effective.

Necessity being trivial, let us prove sufficiency. We thus have an étale prescheme of finite type X_a over S_a, and an isomorphism

$$(*) \qquad X'_a \xrightarrow{\sim} X_a \times_{S_a} S'_a$$

compatible with the descent data. In accordance with a general elementary limit argument for preschemes defined over a direct limit of rings (here the rings A_f, where A is the ring of an affine open neighborhood of a, and where f ranges over the elements of A that do not belong to the prime ideal corresponding to a), one can find an open neighborhood U of a, an étale prescheme of finite type X_U over $U = S_U$, and an S_a-isomorphism $X_a \xrightarrow{\sim} X_U \times_{S_U} S_a$. Moreover, on taking U sufficiently small, one may then suppose that the isomorphism $(*)$ comes from an isomorphism:

$$X'_U \xrightarrow{\sim} X_U \times_{S_U} S'_U;$$

the latter might not be compatible with the descent data; nevertheless, after shrinking U, it will be compatible with the descent data. This completes the proof.

COROLLARY IX.4.5. *Under the conditions of IX.4.4, the descent datum on X' is effective if and only if, for every $a \in S$, the corresponding descent datum on X'_a, relative to the morphism $S'_a = S' \times_S \operatorname{Spec}(()\mathcal{O}_a) \to \operatorname{Spec}(()\mathcal{O}_a)$, is effective. When S is locally Noetherian and X' is separated over S', one may in the preceding criterion also replace $\mathcal{O}_a$ by its completion.*

The first assertion follows from IX.4.4 and IX.4.3; the second is then a consequence 238 of IX.4.2. Using IX.4.2 once more and the fact that for every Noetherian local ring A, one can find a complete Noetherian local ring B, and a local homomorphism $A \to B$, such that B is flat over A and $B/\mathfrak{m}B$ is a prescribed extension of the residue field $k = A/\mathfrak{m}$ of A, one obtains:

COROLLARY IX.4.6. *Under the conditions of IX.4.4, suppose in addition that X' is separated over S', and that S is locally Noetherian. The descent datum on X' is effective if and only if, for every prescheme S_1 over S that is the spectrum of a complete local ring with algebraically closed residue field, the corresponding descent datum on $X'_1 = X' \times_S S_1$, relative to the morphism $g_1 \colon S'_1 \to S_1$, is effective.*

THEOREM IX.4.7. *Let $g \colon S' \to S$ be a finite and surjective morphism of finite presentation (the latter hypothesis being a consequence of the others if S is locally Noetherian)*[3]*. Then g is an effective descent morphism for the fibered category of preschemes that are étale, separated, and of finite type over other preschemes.*

[3] One can show that it in fact suffices for g to be an *integral* morphism, by reducing to the case in the text through a limit process in the style of EGA IV 8.

We must show that if X' is étale, separated, and of finite type over S', and endowed with a descent datum relative to g, then this datum is effective. Using IX.4.3, one reduces easily to the case where S is Noetherian. By IX.4.5, we may therefore suppose that S is the spectrum of a Noetherian local ring, and hence in particular that

$$\dim S = n < +\infty.$$

We then argue by induction on $\dim S = n$, the assertion being trivial for $n < 0$. Suppose therefore that $n \geq 0$ and that the theorem has been proved for dimensions $n' < n$. By IX.4.6, we are reduced to the case where S is the spectrum of a complete local ring, so S' is a finite union of spectra of complete local rings. We therefore have

$$X' = X'_1 \sqcup X'_2$$

where X'_1 is *finite* over S', and where X'_2 has no point above any of the closed points of S'. 239
Consider the morphisms

$$q_1, q_2 \colon X'' \rightrightarrows X'$$

corresponding to the descent datum, compatible with $p_1, p_2 \colon S'' \rightrightarrows S'$. One sees at once that

$$X'' = q_i^{-1}(X'_1) \sqcup q_i^{-1}(X'_2) \quad i = 1, 2$$

is the analogous canonical decomposition of X'' over S'', which implies $q_1^{-1}(X'_1) = q_2^{-1}(X'_1)$, and consequently X'_1 and X'_2 are endowed with induced descent data. Now let T be the open subset of S complementary to its closed point; then $T' = S' \times_S T$ is the part of S' complementary to the set of closed points, and X'_2, which lies entirely over T', is endowed with a descent datum relative to the morphism $T' \to T$ induced by g. Since the latter is finite and surjective, and $\dim T < \dim S = n$, this descent datum is effective by the induction hypothesis. Thus one sees that it suffices to prove that the descent datum on X'_1 is effective, so we may now suppose that X' is étale and *finite* over S'. (N.B. the inductive argument is unnecessary if one restricts to étale coverings in the statement IX.4.7). Let S_0 then be the spectrum of the residue field of A, let $S'_0 = S' \times_S S_0$, and define S''_0, S'''_0 similarly from the fibered squares and cubes S'' and S''' of S' over S. By IX.1.8, the morphisms $S_0 \to S$, $S'_0 \to S'$, etc. induce equivalences for the categories of étale coverings of S and S_0, on the one hand, and of S' and S'_0 on the other hand, etc. By the standard consequences of descent theory in categories [**D**], it follows that $g \colon S' \to S$ is an effective descent morphism for the fibered category of étale coverings if and only if the same is true of $g_0 \colon S'_0 \to S_0$. But this is indeed the case, for example as a special case of IX.4.1. This completes the proof.

Corollary IX.4.8. *The conclusion of IX.4.7 remains valid if one assumes only that $S' \to S$ is universally submersive, of finite type, and quasi-finite, provided S is assumed locally Noetherian.*

Indeed, by IX.4.6, we may suppose that S is the spectrum of a complete Noetherian 240
local ring. Then, by IX.2.5, there exists a finite and surjective morphism $S_1 \to S$, and an S-morphism $S_1 \to S'$. Since $S_1 \to S$ is a universal strict descent morphism for the fibered category under consideration, by IX.4.7, and $S' \to S$ is a universal descent morphism for that category, IX.4.8 follows from the general formal consequences [**D**].

Corollary IX.4.9. *Let $g \colon S' \to S$ be a morphism of finite type, surjective and universally open, with S locally Noetherian. Then g is an effective descent morphism for the fibered category of preschemes that are étale, separated, and of finite type over other preschemes.*

Proceeding as in IX.4.7, we reduce to the case where S is the spectrum of a complete Noetherian local ring A. Let A_1 be a finite algebra over A, with spectrum S_1, such that $S_1 \to S$ is finite and *surjective*, and hence a universal effective descent morphism for

the fibered category under consideration, by IX.4.7. It then follows from the general theorems [**D**] that g is an effective descent morphism for the fibered category under consideration if and only if the corresponding morphism $g_1 : S'_1 = S' \times_S S_1 \to S_1$ is so. Since the latter satisfies the same hypotheses as g, we are reduced to proving IX.4.9 with S_1 in place of S. Taking first for A_1 the direct product of the $A/\mathfrak{p}_i$, for the minimal prime ideals $\mathfrak{p}_i$ of A, we are reduced to the case where A is an *integral domain*. One then shows[4] that there exists an integral subscheme S_1 of S', quasi-finite over S and dominating S, passing through a point of the fiber of S' over the closed point y of S (using the fact that S' is universally open and of finite type over the integral Noetherian local scheme S, and $S'_y \neq \emptyset$). Since A is complete, S_1 is finite over S, and since it dominates S, the morphism $S_1 \to S$ is surjective. Replacing S once more by S_1, we are reduced to the case where S' has a section over S, in which case the statement is trivial.

THEOREM IX.4.10. *Let $g : S' \to S$ be a finite, radical, and surjective morphism of finite presentation (the latter condition being superfluous if S is locally Noetherian[5]). Then the inverse-image functor induces an equivalence from the category of preschemes* 241
that are étale over S to the category of preschemes that are étale over S'.

Since the diagonal morphisms from S' into $S' \times_S S'$ and $S' \times_S S' \times_S S'$ are surjective immersions, and hence by IX.1.9 induce equivalences from the categories of preschemes that are étale over $S' \times_S S'$ respectively$S' \times_S S' \times_S S'$ to the category of preschemes that are étale over S', it follows from the standard consequences of descent [**D**] that every X' étale over S' is endowed with one and only one descent datum relative to $g : S' \to S$. Thus IX.3.3 implies that the inverse-image functor under g, from the category of preschemes that are étale over S to the category of preschemes that are étale over S', is *fully faithful*. It remains to show that it is essentially surjective, i.e. that every X' étale over S' is isomorphic to the inverse image of an X étale over S. Since the question is plainly local on S *and on* X', we may suppose that S, S', X' are affine. But then X' is separated and of finite type over S', and we may apply the effectivity criterion IX.4.7.

COROLLARY IX.4.11. *The conclusion of IX.4.9 remains valid if the hypothesis on g is replaced by: g is faithfully flat, quasi-compact, and radicial.*

The same proof applies, invoking IX.4.1 instead of IX.4.7.

Note that the proof of IX.4.7 is "elementary"in that it does not use the theorems on finiteness and comparison for proper morphisms (EGA III 3, 4, 5). The same is not true of the following result:

THEOREM IX.4.12. *Let $g : S' \to S$ be a proper and surjective morphism of finite presentation (the latter hypothesis being a consequence of the first if S is locally Noetherian). Then g is an effective descent morphism for the fibered category of étale coverings of preschemes.*

By IX.3.3 and IX.2.2, we are reduced to proving that for every étale covering X' over S', endowed with a descent datum relative to $g : S' \to S$, this descent datum is effective. Using IX.4.3, we reduce easily to the case where S is Noetherian, and using IX.4.6, we 242
may therefore suppose that S is the spectrum of a *complete* Noetherian local ring A. Introduce S'' and S''' as usual, let S_0 be the spectrum of the residue field of A, and let S'_0, S''_0, S'''_0 be deduced from S', S'', S''' by the base change $S_0 \to S$, i.e. the fibers of S', S'', S''' at the closed point of S. By IX.1.10, the morphisms $S_0 \to S$, $S'_0 \to S'$, etc. induce equivalences from the category of étale coverings over the target scheme to the category of étale coverings over the source scheme. Consequently, $g : S' \to S$ is a strict

[4]Cf. EGA IV 14.3.13 and 14.5.4.

[5]It even suffices for g to be integral, radicial, and surjective, as follows from an easy reduction to the case in the text, in the style of EGA IV 8, cf. SGA 4 VIII 1.1.

descent morphism for the fibered category of étale coverings of preschemes if and only if $g_0\colon S'_0 \to S_0$ is so, which is indeed the case by IX.4.1. This completes the proof of IX.4.12. (In this argument, from IX.1.10 we needed only the fact that the functor considered in IX.1.10 is *fully faithful*, which does *not* use the existence theorem for coherent sheaves in algebraic geometry.)

IX.5. Translation in terms of the fundamental group

Let

$$g\colon S' \to S$$

be an *effective descent morphism* for the fibered category of *étale coverings* of preschemes, for example a proper, surjective morphism of finite presentation (IX.4.12), or a faithfully flat and quasi-compact morphism. Introducing S'', S''' as usual, and denoting by $\mathcal{C}$, $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$ the categories of étale coverings of S, S', S'', and S''', respectively, we thus have a 2-exact diagram of categories

$$(*) \qquad \mathcal{C} \xrightarrow{p^*} \mathcal{C}' \underset{}{\overset{p_1^*,p_2^*}{\rightrightarrows}} \mathcal{C}'' \underset{}{\overset{p_{21}^*,p_{32}^*,p_{31}^*}{\substack{\rightarrow\\ \rightarrow\\ \rightarrow}}} \mathcal{C}'''$$

corresponding to the diagram

$$S \xleftarrow{p} S' \underset{}{\overset{p_1,p_2}{\rightrightarrows}} S'' \underset{}{\overset{p_{21},p_{32},p_{31}}{\substack{\leftarrow\\ \leftarrow\\ \leftarrow}}} S'''.$$

Suppose that the preschemes S, S', S'', and S''' are disjoint unions of connected preschemes, 243
as will be the case in particular if they are locally connected preschemes, and a fortiori if they are locally noetherian (for example, if S' is of finite type over the locally noetherian prescheme S). The categories $\mathcal{C}$, $\mathcal{C}'$ ... in $(*)$ are then multigalois categories V V.9, each described by a collection of compact totally disconnected topological groups, namely the fundamental groups of the connected components of the preschemes S, S', S'', and S'''. For simplicity, we suppose that S is connected, and we shall then give a procedure for computing its fundamental group in terms of the fibered category formed from $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$, suitably made explicit by means of the fundamental groups describing these categories. The reader will note that the procedure outlined is in fact valid in the general framework of multigalois categories (which need not arise from given preschemes S, S', S'', and S'''). It is moreover the analogue of the well-known procedure for computing the fundamental group of a topological space S that is a locally finite union of closed subspaces S_i (or an arbitrary union of open subspaces S_i), by means of the fundamental groups of the components of the S_i and of the components of the $S_i \cap S_j$. Of course, the analogous situation for preschemes does fall within the general framework of descent, by introducing the prescheme S' that is the disjoint union of the S_i and the canonical morphism $g\colon S' \to S$.

Set

$$E' = \pi_0(S'), \quad E'' = \pi_0(S''), \quad E''' = \pi_0(S'''),$$

where π_0 denotes the "set of connected components" functor. Since the fiber products of S' over S form a simplicial object of **Sch**, it is transformed by the functor π_0 into a simplicial set of which E', E'', and E''' are the components in dimensions 0, 1, and 2. We shall have to use the simplicial maps

$$q_i = \pi_0(p_i), \quad (i = 1, 2) \quad \text{and} \quad q_{ij} = \pi_0(p_{ij}), \quad (i, j) = (2, 1), (3, 2), (3, 1),$$

displayed in the diagram

$$(1) \qquad E' \underset{}{\overset{q_1,q_2}{\rightrightarrows}} E'' \underset{}{\overset{q_{21},q_{32},q_{31}}{\substack{\leftarrow\\ \leftarrow\\ \leftarrow}}} E'''.$$

The elements of E' will be denoted with one prime, as in s', while those of E'' respectively E''' 244
will be denoted with $''$ respectively with $'''$. The fact that S is connected is expressed by

$\pi_0(K) = 0$, where K is the simplicial set defined by $g\colon S' \to S$, or equivalently by the fact that the equivalence relation on E' generated by the pair of maps (q_1, q_2) is transitive.

We shall choose once and for all an element s'_0 in E', and, for every s' in E', an element $\overline{s'} \in E''$ such that[6]

$$q_1(\overline{s'}) = s'_0, \quad q_2(\overline{s'}) = s',$$

thus exhibiting the connectedness of S. For every $s' \in E'$, choose a geometric point $\underline{s}'$ in the connected component s' of S'; this point will in fact enter through the corresponding fiber functor $F'_{s'}$ on the multigalois category $\mathcal{C}'$. The automorphism group of this functor, i.e. the fundamental group of S' at $\underline{s}'$, will be denoted $\pi_{s'}$. We similarly choose $\underline{s}''$ and $\underline{s}'''$, hence functors $F''_{s''}$ and $F'''_{s'''}$, and thus fundamental groups $\pi_{s''}$ and $\pi_{s'''}$. Hence

$$\pi_{s'} = \pi_1(S', \underline{s}'), \quad \pi_{s''} = \pi_1(S'', \underline{s}''), \quad \pi_{s'''} = \pi_1(S''', \underline{s}''').$$

For every $s'' \in E''$, $p_1(\underline{s}'')$ lies in the same connected component as $\underline{q_1(s'')}$, so there exists an isomorphism of functors $F''_{s''} \circ p_1^* \xrightarrow{\sim} F'_{s'}$ (i.e. a "path class" from $p_1(\underline{s}'')$ to $\underline{q_1(s'')}$). This observation applies likewise to q_2 and to the q_{ij}. Choose all these path classes:

$$F''_{s''} \circ p_i^* \xrightarrow{\sim} F'_{q_i(s'')}, \quad F'''_{s'''} \circ p_{ij}^* \xrightarrow{\sim} F'''_{q_{ij}(s''')},$$

(for $i = 1, 2$ and $(i, j) = (2, 1), (3, 2), (3, 1)$). In particular, this yields group homomor- 245
phisms:

$$q_i^{s''} : \pi_{s''} \to \pi_{q_i(s'')}, \quad q_{ij}^{s'''} : \pi_{s'''} \to \pi_{q_{ij}(s''')}, \tag{2}$$

(with the same values of i and (i, j)). Finally, recall that the structure of a cloven category with fibers $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$ also includes isomorphisms of functors:

$$p_{21}^* p_1^* \xrightarrow{\sim} p_{31}^* p_1^*, \quad p_{21}^* p_2^* \xrightarrow{\sim} p_{32}^* p_1^*, \quad p_{31}^* p_2^* \xrightarrow{\sim} p_{32}^* p_2^*,$$

deduced from isomorphisms of the two sides, respectively, with the u_i^* $(i = 1, 2, 3)$, where the u_i are the three projections from S''' to S'. When these data are made explicit, one obtains, for every s''', a well-determined element

$$a_i^{s'''} \in \pi_{v_i(s''')}, \tag{3}$$

(where the v_i, $i = 1, 2, 3$, are the three maps $E''' \to E'$ defined by $v_i = \pi_0(u_i)$), which moreover satisfies the conditions:

$$q_1^{s''_1} q_{21}^{s'''} = \mathrm{int}(a_1^{s'''}) q_1^{s''_2} q_{31}^{s'''} \qquad (s''_1 = q_{21}(s'''),\ s''_2 = q_{31}(s''')),$$

and the two analogous conditions involving a_2 and a_3. The reader will moreover note that the data (1), (2), and (3) allow one to reconstruct, up to an equivalence of fibered categories, the fibered category under consideration with fibers $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$. They must therefore, in principle, allow one to reconstruct $\mathcal{C}$ up to equivalence, and hence its fundamental group up to isomorphism. In fact, we shall determine the fundamental group at the geometric point $p(\underline{s}'_0)$ of S, i.e. the automorphism group of $F'_{s'_0} \circ p^*$.

Note that the datum of an object X' of $\mathcal{C}'$ is essentially equivalent to the datum of finite sets $X'_{s'}$ $(s' \in E')$ on which the $\pi_{s'}$ act continuously. A gluing datum on such an 246
object then amounts to the datum, for every $s'' \in E''$, of a bijection

$$\varphi_{s''} : X'_{q_1(s'')} \xrightarrow{\sim} X'_{q_2(s'')}$$

[6]Care must be taken here: the element $\overline{s'}$, whose existence is implicitly assumed, does not exist in every case. Consequently, Theorem IX.5.1, as stated, does not apply in every case. It is not difficult, however, by following the ideas of the written text, to modify the statement of this theorem so that it gives a computational method applicable in every case. In particular, the corollaries of the said theorem remain valid as stated.

compatible with the actions of $\pi_{s''}$, acting on the two sides through the homomorphisms $q_i^{s''}: \pi_{s''} \to \pi_{q_i(s'')}$. Taking first the s'' of the form $\overline{s'}$, one sees that such a datum defines bijections

$$\psi_{s'}: X'_{s'_0} = F'_0(X') \to X'_{s'}$$

which allow the $X'_{s'}$ to be identified with the same set $F'_0(X') = X'_{s'_0}$, on which all the groups $\pi_{s'}$ will henceforth act. With this understood, the bijections $\varphi_{s''}$ correspond to bijections

$$g_{s''}: F'_0(X') \xrightarrow{\sim} F'_0(X'),$$

subject, on the one hand, to the commutation relations with $\pi_{s''}$:

a) $$g_{s''} q_1^{s''}(g'') = q_2^{s''}(g'') g_{s''} \qquad (s'' \in E'',\ g'' \in \pi_{s''}),$$

and, on the other hand, to the relations

b) $$g_{\overline{s'}} = g_{\overline{s'_0}} \qquad (s' \in E'),$$

which express the manner in which we identified the $X_{s'}$ with one another. On making explicit the condition for such a gluing datum to be a descent datum, one obtains the relations

c) $$a_3^{s'''} g_{q_{31}(s''')} a_1^{s'''} = g_{q_{32}(s''')} a_2^{s'''} g_{q_{21}(s''')} \qquad (s''' \in E''').$$

This gives an equivalence between the category of objects of $\mathcal{C}'$ equipped with a descent datum and the category of finite sets on which the groups $\pi_{s'}$ act continuously, 247
equipped in addition with bijections $g_{s''}$ satisfying the relations A), B), and C). Let G be the group generated by the groups $\pi_{s'}$ and the new generators $g_{s''}$, subject to the relations A), B), and C), and let π be the inverse-limit group of the quotients of G by the subgroups of finite index whose inverse images in the groups $\pi_{s'}$ are open subgroups. One also says that π is the *group of Galois type generated by the $\pi_{s'}$ and the $g_{s''}$, subject to the relations A), B), and C).* It is immediately seen that the category under consideration is also equivalent to the category of finite sets on which the topological group π acts continuously. This proves the following statement:

THEOREM IX.5.1. *Let $g: S' \to S$ be a morphism of preschemes which is an effective descent morphism for the fibered category of étale coverings of preschemes (cf. IX.4.9 and IX.4.12). Suppose that S is connected, and that S', its fiber square S'', and its fiber cube S''' are disjoint unions of connected preschemes (as is the case, for example, if S' is of finite type over a locally noetherian connected S). Choose as above: a geometric point in every connected component of S', S'', and S''', certain path classes, an $s'_0 \in E'$, and, for every $s' \in E'$, an $s'' \in E''$ whose two images in E' are s'_0 and s'. (E', E'', and E''' denote respectively the sets of connected components of S', S'', and S'''.) Then the fundamental group of S at the geometric point that is the image of s'_0 is canonically isomorphic to the group of Galois type generated by the $\pi_{s'} = \pi_1(S', \underline{s}')$ ($s' \in E'$) and generators $g_{s''}$ ($s'' \in E''$), subject to the above relations A), B), and C), which involve the elements of the groups $\pi_{s''} = \pi_1(S'', \underline{s}'')$, and the elements $a_i^{s'''}$ ($i = 1, 2, 3$, $s''' \in E'''$) introduced above.*

COROLLARY IX.5.2. *Suppose that S' and S'' have only finitely many connected components, and that the fundamental groups of the connected components of S' are topologically finitely generated. Then the fundamental group of S is topologically finitely generated.*

Thus, we shall prove later that the fundamental group of a normal projective scheme 248
over an algebraically closed field is topologically finitely generated. Using Chow's lemma and the normalization of algebraic schemes, it will follow that the same result holds for every proper scheme over an algebraically closed field.

COROLLARY IX.5.3. *Suppose that S', S'', and S''' have only finitely many connected components, that the fundamental groups of the connected components of S' are topologically* finitely presented*, and that the fundamental groups of the connected components of S'' are topologically* finitely generated*. Then the fundamental group of S is topologically* finitely presented*.*

Note that IX.4.9 (restricted to étale *coverings*) may be expressed by saying that *a finite radicial surjective morphism of noetherian preschemes induces an isomorphism of fundamental groups*; figuratively speaking, one may therefore say that the fundamental group is a *topological invariant* for preschemes. More generally, using IX.5.1, one can make explicit the effect on the fundamental group of operations on preschemes, such as the "pinching"of a prescheme along a finite set of points, which has a simple topological meaning. For example, one obtains:

COROLLARY IX.5.4. *Let $g\colon S' \to S$ be a finite morphism of finite presentation, and let T be a discrete subset of S. For every $s \in S$, let $n(s)$ be the "geometric number of points"in the fiber $g^{-1}(s)$ (which may also be described as the separable degree of $g^{-1}(s)$ over $\kappa(s)$, the sum of the separable degrees of its residue extensions). Suppose that $n(s) = 1$ for $s \in S - T$. For every $s \in T$, let K_s be an algebraically closed extension of $\kappa(s)$, let I_s be the set of geometric points of S' with values in K_s (a set with $n(s)$ elements), let I'_s be the complement of a chosen point of I_s, and finally let I' be the union of the I'_s. Suppose that S' is connected. Then the fundamental group of S is isomorphic to the group of Galois type generated by the fundamental group of S' and generators g_i ($i \in I'$), subject to no additional relations.*

The details of the proof are left to the reader; the resulting statement is merely the 249
translation into the language of group theory of the fact that the category C of étale coverings of S is equivalent to the category of étale coverings X' of S', *equipped*, for every $s \in T$, with a transitive system of bijections between the $n(s)$ fibers of X' at the points of $g^{-1}(s)$ with values in K_s. (In this intrinsic form, of course, it is no longer necessary to suppose that S' is connected.)

EXAMPLE IX.5.5. *It is easy to prove that the rational curve $\mathbb{P}^1_k$ over an algebraically closed field k is simply connected*[7]*. Hence the fundamental group of a complete rational curve having exactly one double point, with n analytic branches, is the free group of Galois type generated by $n-1$ generators. For example, in the case of an ordinary double point, one obtains the fundamental group $\hat{\mathbb{Z}}$, as announced in (I I.11 a)). By contrast, the presence of a cusp (which is a "geometrically unibranch"point) has no effect on the fundamental group.*

COROLLARY IX.5.6. *Let $g\colon S' \to S$ be a universally submersive morphism of preschemes with geometrically connected fibers, where S is connected. Then S' is connected, and, on choosing a geometric point s' of S' and denoting its image in S by s, the homomorphism*

$$\pi_1(S', s') \to \pi_1(S, s)$$

is surjective*. If g is an effective descent morphism for the fibered category of étale coverings of preschemes (cf.* IX.4.12*), introduce the geometric point $s'' = \mathrm{diag}(s')$ of $S'' = S' \times_S S'$ and the two homomorphisms*

$$p_{1*}, p_{2*}\colon \pi_1(S'', s'') \to \pi_1(S', s')$$

induced by the two projections. Then $\pi_1(S, s)$ is isomorphic to the cokernel of this pair of homomorphisms in the category of groups of Galois type, i.e. to the quotient of $\pi_1(S', s')$ by the closed normal subgroup generated by the elements of the form $p_{1*}(g'')p_{2*}(g'')^{-1}$, 250
with $g'' \in \pi_1(S'', s'')$.

[7] Cf. Exp. XI XI.1.1.

Indeed, by IX.3.4, the functor $X \longmapsto X \times_S S'$ from étale coverings of S to étale coverings of S' is fully faithful, which is equivalent to the homomorphism on fundamental groups being an epimorphism (V V.6.9). The last assertion follows immediately from the description in IX.5.1.

Remark IX.5.7. It is not known at present whether the fundamental group of a proper scheme over an algebraically closed field k is topologically finitely presented[8]. Using IX.5.3, a well-known hyperplane-section technique, and the resolution of singularities of normal surfaces, one is reduced to the case of a *smooth surface over* k. This at least makes it possible to show, by transcendental methods, that the answer is affirmative in characteristic 0 (without having to assume the triangulability of singular algebraic varieties). In characteristic $p > 0$, the principal difficulty seems to lie in the case of curves, whose fundamental group is known only to be a quotient of the group arising in the classical case (cf. the following exposé), while the kernel by which one takes the quotient is very poorly understood.

Remark IX.5.8. One could make explicit other special cases besides IX.5.4 and IX.5.6 in which IX.5.1 takes a particularly simple form. An interesting case is that in which S is the quotient of S' by a finite group of automorphisms Γ. *Then the category of étale coverings of S' is equivalent to the category of étale coverings X' of S' on which the group Γ acts compatibly with its action on S', in such a way that for every $s' \in S'$ and every $g \in \Gamma_{s'}$* (where $\Gamma_{s'}$ denotes *the inertia group* of s' in Γ), *g acts trivially on the fiber $X'_{s'}$*. If S' is connected, this statement is interpreted as follows. Let C'_0 be the category of étale coverings of S' on which Γ acts compatibly with its action on S' (but without necessarily satisfying the preceding condition on the inertia groups of the points of S'). One sees easily that this is a Galois category (V V.5), and that for every geometric point a' of S', 251
the fiber functor $X' \longmapsto X'_{a'}$ on C'_0 is a fundamental functor. Let $\pi_1(S', \Gamma; a') = G$ be the group of automorphisms of this functor, endowed with its usual topology. There is then a canonical exact sequence

$$e \to \pi_1(S', a') \to G \to \Gamma \to e$$

(a special case of (V V.6.13), where one takes for S the trivial covering $S' \times \Gamma$ of S' defined by Γ, on which Γ is made to act in the evident way). Moreover, for every geometric point b' of S', there is an isomorphism $\pi_1(S', \Gamma; b') \to G = \pi_1(S', \Gamma; a')$, defined up to an inner automorphism arising from $\pi_1(S', a')$; and since $\Gamma_{b'}$ maps in the evident way into the first group, one obtains a homomorphism

$$u_{b'} : \Gamma_{b'} \to G,$$

defined up to an inner automorphism (arising from $\pi_1(S', a')$), whose composite with the canonical homomorphism $G \to \Gamma$ is the canonical inclusion $\Gamma_{b'} \to \Gamma$. With this understood, *the fundamental group $\pi_1(S, a)$* is *canonically isomorphic to the quotient of $G = \pi_1(S', \Gamma; a')$ by the closed normal subgroup generated by the images of the homomorphisms $\Gamma_{b'} \to G$*. In particular, the image of $\pi_1(S', a')$ in $\pi_1(S, a)$ is a normal subgroup, and the corresponding quotient is isomorphic to a quotient of Γ. One can moreover reduce the number of "relations"introduced by considering, for every $g \in \Gamma$, $g \neq e$, the coincidence subprescheme S'_g of the automorphisms id_S and g of S, choosing a geometric point $b'_{g,i}$ in each connected component of S'_g, and then choosing one of the corresponding homomorphisms $\pi_1(S', \Gamma; b'_{g,i}) \to G$, thereby obtaining lifts $\bar{g}_i$ of g in G. It then suffices to take the quotient of G by the closed normal subgroup of G generated by the $\bar{g}_i$.

When a' is fixed by Γ, one sees easily that Γ acts naturally on $\pi_1(S', a')$, and G identifies with the corresponding semidirect product. Identifying Γ with a subgroup of G,

[8] This seems very unlikely in the case of smooth curves of genus $g \geq 2$ in characteristic $p > 0$. On the other hand, when π_1 is replaced by its largest prime-to-p quotient, it appears that well-known techniques give an affirmative answer, even without the properness hypothesis. Cf. forthcoming work of J.P. Murre.

one sees that among the relations introduced above, on taking $b' = a'$, one finds "$g = e$"for 252
$g \in I$. Thus *if S' has a geometric point a' fixed by I* (i.e. a point s' whose inertia group is Γ), *then $\pi_1(S, a)$ is a quotient group of the Galois-type quotient group of $\pi_1(S', a')$ obtained by* "making trivial" *the actions of Γ on $\pi_1(S', a')$; and it is even isomorphic to this latter group if one assumes that for every $g \in G$, the inertia locus S'_g is connected, and therefore contains the underlying point of a'.* Indeed, this last assertion is contained in the second description given above of the relations to be imposed on G.

This last result applies in particular if one takes for S' the Cartesian power X^n of a connected prescheme over an algebraically closed field, and for Γ the symmetric group $\Gamma = \mathfrak{S}_n$, acting in the usual way, so that S is the symmetric power of degree n of X. Taking for a' a geometric point whose underlying point lies on the diagonal, one is under the preceding conditions, since all the inertia loci S'_g contain the diagonal. Using the fact, proved in the following exposé, that if X is proper and connected over k, the fundamental group of X^n identifies with $\pi_1(X)^n$, one obtains the following amusing result: *If X is proper and connected over an algebraically closed field k, the fundamental group of its symmetric power of degree n, $n \geq 2$, is isomorphic to the fundamental group of X made abelian.* (I do not know whether the analogous fact in algebraic topology is known; it should be possible to establish it by the same descent method.) Take, for example, the rational curve $X = \mathbb{P}^1_k$; one obtains yet another proof of the fact that $\mathbb{P}^r_k$ is simply connected, using the fact that $\mathbb{P}^1_k$ is so. Now take for X a nonsingular curve over k, and let $n \geq 2g - 1$, so that $\mathrm{Sym}^n(X)$ is fibered over the Jacobian J, with projective spaces as fibers, and hence (as will be seen using the results of the next two exposés) has the same fundamental group as J. One thereby recovers, without dévissage, the well-known fact that *the fundamental group of the Jacobian of X is isomorphic to the fundamental group of X made abelian.*

IX.6. A fundamental exact sequence. Descent by morphisms with relatively connected fibers

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THEOREM IX.6.1. *Let S be the spectrum of an Artinian ring A with residue field k, let $\overline{k}$ be an algebraic closure of k, let X be an S-prescheme, set*

$$X_0 = X \otimes_A k, \qquad \overline{X}_0 = X \otimes_A \overline{k},$$

and let $\overline{a}$ be a geometric point of $\overline{X}_0$, with image a in X and image b in S. Suppose that X_0 is quasi-compact and geometrically connected *over k. (If X is proper over S, this means that $\mathrm{H}^0(X_0, \mathcal{O}_{X_0})$ is a* local *Artinian ring whose residue field is* radicial *over k.) Then the sequence of canonical homomorphisms*

$$e \longmapsto \pi_1(\overline{X}_0, \overline{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(S, b) \longrightarrow e$$

is exact, and

$$\pi_1(S, b) \overset{\sim}{\leftarrow} \pi_1(k, \overline{k}) = \textit{the Galois group of } \overline{k} \textit{ over } k.$$

Since fundamental groups are unchanged on killing nilpotent elements, we may suppose that $A = k$, which already makes the last isomorphism evident. Let k' be the separable closure of k in $\overline{k}$, consider $X' = X \otimes_k k'$, and let a' be the image of $\overline{a}$ in X'. There is a sequence of canonical homomorphisms

$$e \longmapsto \pi_1(\overline{X}_0, \overline{a}) \longrightarrow \pi_1(X', a') \longrightarrow \pi_1(S', b') \longrightarrow e,$$

where $S' = \mathrm{Spec}(()k')$. The diagram

$$\begin{array}{ccccc} S & \longleftarrow & X & \longleftarrow & \overline{X}_0 \\ \uparrow & & \uparrow & & \| \\ S' & \longleftarrow & X' & \longleftarrow & \overline{X}_0 \end{array}$$

gives a canonical homomorphism from this sequence to the corresponding sequence for X/k. By IX.4.11, this homomorphism of sequences of groups is an isomorphism. We are therefore reduced to proving that the second sequence is exact, i.e. we may assume that k is *perfect*.

Let k_i run through the finite Galois subextensions of k in $\overline{k}$, set $X_i = X \otimes_k k_i$, and let a_i be the image of $\overline{a}$ in X_i. We leave it to the reader to verify that the natural homomorphism

$$\pi_1(\overline{X}_0, \overline{a}) \xrightarrow{\sim} \varprojlim_i \pi_1(X_i, a_i)$$

is an isomorphism. This simply means that an étale covering of $\overline{X}_0$ comes from an étale 254
covering of some X_i, and that the latter is essentially unique after passing to an X_j with $j \geq i$.

Let π_i be the Galois group of k_i over k, i.e. the group opposite to the group of S-automorphisms of $S_i = \mathrm{Spec}(()k_i)$. Since the functor

$$S' \longmapsto X \times_S S'$$

from étale coverings of S to étale coverings of X is fully faithful by (IX.3.4), it follows that π_i is also isomorphic to the group opposite to the group of X-automorphisms of the connected principal covering X_i of X. Consequently, V V.6.13 gives an exact sequence

$$e \longmapsto \pi_1(X_i, a_i) \longrightarrow \pi_1(X, a) \longrightarrow \pi_i \longrightarrow e.$$

Passing to the inverse limit over i in these exact sequences gives an exact sequence, since we are in the category of groups of Galois type. This is precisely the sequence considered in IX.6.1, and the proof is complete.

The geometric translation of exactness on the right in IX.6.1 is the following:

Corollary IX.6.2. *With the preceding notation, let X' be an étale covering of X, and let $\overline{X'}_0$ be the corresponding étale covering of $\overline{X}_0$. The following conditions are equivalent:*

(i) *There exist an S' étale over S and an X-isomorphism $X' \xrightarrow{\sim} X \times_S S'$. In that case S' is determined up to unique isomorphism by IX.3.4.*

(ii) *The covering $\overline{X'}_0$ is completely split over $\overline{X}_0$.*

If X' is connected, these conditions are also equivalent to:

(ii bis) *The covering $\overline{X'}_0$ has a section over $\overline{X}_0$.*

The last qualification is essential. The equivalence of (i) and (ii) means only that $\pi_1(S, b)$ is the quotient of $\pi_1(X, a)$ by the closed normal subgroup generated by the image 255
of $\pi_1(\overline{X}_0, \overline{a})$, and not by that image itself. Under the preceding hypotheses, we shall say that X' is a *geometrically trivial* covering of X.

Remark IX.6.3. In IX.6.1, one cannot replace $\overline{k}$ by an arbitrary algebraically closed extension of k, even if k is already assumed algebraically closed. In other words, it is not true in general that, if X is a connected algebraic scheme over an algebraically closed field k, its fundamental group is unchanged when k is replaced by an algebraically closed extension. This already fails, for example, in characteristic $p > 0$ for the affine line over k, because of phenomena of *higher ramification* at the point at infinity, which give the fundamental group a *continuous* structure. In the next exposé we shall see, however, that such phenomena cannot occur if X is *proper* over k. We shall also prove by transcendental methods that the same holds when k has characteristic zero.

Corollary IX.6.4. *Suppose that a is localized at a point $x \in X$ rational over k, or more generally with residue field radicial over k. Then the exact sequence IX.6.1 splits.*

We may suppose that $S = \mathrm{Spec}(()k)$. If x is rational over k, it corresponds to a section $S \to X$ of X over S taking b to a. This section defines a homomorphism $\pi_1(S, b) \to \pi_1(X, a)$, which is the required splitting. If $\kappa(x)$ is radicial over k, we reduce to the preceding case by the base extension $\mathrm{Spec}(()\kappa(x)) \to \mathrm{Spec}(()k)$.

THEOREM IX.6.5. *Let $f\colon X \to S$ be a proper surjective morphism of finite presentation with geometrically connected fibers, let X' be a prescheme of finite presentation and proper over X, let s be a point of S, let $F = X_s$ be the fiber of X at s, and let F'_1 be a connected component of the fiber $F' = X'_s$ of X' at s. There exist an open neighborhood X'_1 of F'_1 in X', an S-scheme S'_1 étale over S, and an X-isomorphism*

$$X'_1 \xrightarrow{\sim} S'_1 \times_S X$$

if and only if X' is étale over X at every point of F'_1, and F'_1 is a geometrically trivial covering of F.

The necessity of the condition is trivial, so it remains to prove sufficiency. We reduce 256
readily to the case where S is noetherian. Consider the Stein factorization $X \to T \to S$ of f, where T is the spectrum of the algebra $f_*(\mathcal{O}_X)$ on S. Since the fibers of X over S are geometrically connected and f is surjective, the morphism $T \to S$ is finite, surjective, and radicial. Hence, by (IX.4.10), every T' étale over T is the inverse image of an S' étale over S. This reduces us to proving IX.6.5 after replacing S by T, i.e. to the case where $f_*(\mathcal{O}_X) = \mathcal{O}_S$. Now consider the Stein factorization

$$X' \longrightarrow S' \longrightarrow S$$

of the proper morphism $h\colon X' \to S$, where S' is the spectrum of the algebra $h_*(\mathcal{O}_{X'})$. The morphisms $X' \to X$ and $X' \to S'$ define a canonical morphism

$$X' \longrightarrow X \times_S S',$$

and our assertion is contained in the following result.

COROLLARY IX.6.6. *Let $f\colon X \to S$ be a proper morphism of locally noetherian preschemes such that $f_*(\mathcal{O}_X) = \mathcal{O}_S$, and let X' be a prescheme proper over X. Consider the* Stein *factorization*

$$X' \longrightarrow S' \longrightarrow S$$

of $X' \to S$, and the canonical morphism $X' \to X \times_S S'$. Let s be a point of S, and let s' be a point of S' above s, corresponding to a connected component F'_1 of the fiber X'_s. The morphism $X' \to X \times_S S'$ is an isomorphism over an open neighborhood U' of s' which is étale over S if and only if X' is étale over X at the points of F'_1, and F'_1 is a geometrically trivial covering of the fiber $F = X_s$.

Necessity is again trivial, so it remains to prove sufficiency. The conclusion also says that

a) the morphism deduced from $X' \to X \times_S S'$ by the base change $\mathrm{Spec}(()\widehat{\mathcal{O}}_{s'}) \to S'$ is an isomorphism;
b) S' is étale over S at s', i.e. $\widehat{\mathcal{O}}_{s'}$ is étale over $\widehat{\mathcal{O}}_s$.

In this form, the conclusion is invariant under the base change $\mathrm{Spec}(()\widehat{\mathcal{O}}_s) \to S$. Since the hypotheses are also stable under this base change, we may suppose that S is the spectrum of a complete noetherian local ring. We may moreover plainly suppose that X' is connected, which here implies that

$$S' = \mathrm{Spec}(()\mathcal{O}_{s'}), \qquad F' = F'_1.$$

The set of points of X' at which X' is étale over X is open and contains the fiber 257
$X'_{s'} = F'$. Since X' is proper over S, it follows that X' is étale over X. On $F = X_s$, it induces an étale covering isomorphic to

$$F \otimes_{\kappa(s)} L,$$

where L is étale over $\kappa(s)$. It follows from IX.1.10 that this covering is isomorphic to one of the form $X \times_S T$, with T étale over S. Here again, it is enough to use the full faithfulness of the functor in IX.1.10; this follows from the fact that a formal isomorphism of coherent sheaves on X comes from an isomorphism of those sheaves.

Thus, if T is defined by the finite A-algebra B, then X' is identified with the spectrum of the algebra $\mathcal{O}_X \otimes_A B$ on X. Since $f_*(\mathcal{O}_X) = \mathcal{O}_S$, it follows at once that $h_*(\mathcal{O}_{X'})$ is defined by B. Consequently, the canonical homomorphism $X' \to X \times_S S'$ is precisely the isomorphism $X' \xrightarrow{\sim} X \times_S T$ under consideration. This completes the proof.

COROLLARY IX.6.7. *Under the hypotheses of IX.6.5, there exist a prescheme S' étale over S and an X-isomorphism*

$$X' \xrightarrow{\sim} X \times_S S'$$

if and only if X' is étale over X and, for every $s \in S$, X'_s is a geometrically trivial covering of X_s.

Indeed, if this is so, then X' is the union of open subsets X'_i isomorphic to inverse images of S'_i étale over S. It is readily seen that these S'_i glue to an S' étale over S, yielding an isomorphism $X' \xrightarrow{\sim} X \times_S S'$. For example, one may say that the X'_i carry descent data relative to $X \to S$, and that these necessarily glue to a descent datum on all of X' relative to $X \to S$. Since the latter is effective on the X'_i, it follows readily, by a sorites omitted from Section IX.4, that it is effective. We may also state IX.6.7 in the following form:

COROLLARY IX.6.8. *Let $f\colon X \to S$ be a proper surjective morphism of finite presentation with geometrically connected fibers. Then f is an effective descent morphism for the fibered category of preschemes finite and étale over other preschemes. The functor*

$$S' \longmapsto X \times_S S'$$

induces an equivalence from the category of preschemes finite and étale over S to the category of preschemes finite and étale over X which induce a geometrically trivial covering 258
on every fiber X_s.

REMARK IX.6.9. Let $f\colon X \to S$ be proper and surjective, with S locally noetherian. Then f factors as a morphism $X \to S'$ satisfying the hypothesis of IX.6.8, followed by a finite surjective morphism $S' \to S$ to which IX.4.7 applies. Thus f is the composite of two morphisms which are *universal effective descent morphisms* for the fibered category of preschemes finite and étale over other preschemes. It follows that f itself is a universal effective descent morphism for the fibered category in question. This recovers IX.4.12 by a different method.

REMARK IX.6.10. The conclusion of IX.6.7 no longer holds if the hypothesis that f is proper is replaced by the following: X is of finite type over S and has a section over S. Under this replacement, f is universally submersive and is a descent morphism for the fibered category of preschemes étale over other preschemes, but the conclusion fails even when S is the spectrum of a discrete valuation ring and X' is an étale covering of X.

To see this, start with a Z proper over S whose generic fiber is a nonsingular rational curve and whose special fiber Z_0 consists of two intersecting lines. For example, if t is a uniformizer of the valuation ring A, take the closed subscheme Z of $\mathbb{P}^2_A$ defined by the homogeneous equation

$$x^2 + y^2 + tz^2 = 0$$

in homogeneous coordinates x, y, z. Let X be the complement of the singular point a of Z_0 in the union $Z \cup \mathbb{P}^2_k$. The fibers of X are $\mathbb{P}^1_k$ and $\mathbb{P}^2_k - a$, and are therefore geometrically simply connected (i.e. every étale covering of such a fiber is geometrically trivial).

Nevertheless, arguing as in Section IX.4, one readily constructs étale coverings of X 259
which do not come from étale coverings of S, by gluing trivial coverings of $Z - a$ and $\mathbb{P}^2_k - a$. It is possible, on the other hand, that the conclusion of IX.6.7 remains valid if properness is replaced by the assumption that X is *universally open* and of finite presentation over S [9]. This is at least true if the fibers of X over S are assumed geometrically irreducible,

[9] This has now been proved with g merely universally open and surjective; cf. SGA 4, XV 1.15.

and not merely geometrically connected. We mention only that this question may be reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field.

The interpretation of IX.6.7 in terms of the fundamental group is the following:

COROLLARY IX.6.11. *Let $f\colon X \to S$ be a proper surjective morphism of finite presentation with geometrically connected fibers. Suppose that X, and hence S, is connected. Let a be a geometric point of X and b its image in S. For every $s \in S$, choose an algebraic closure $\overline{\kappa(s)}$ of $\kappa(s)$, a geometric point a_s of X_s with values in this extension, and a path class from a_s to a. This gives a homomorphism*

$$\pi_1(\overline{X}_s, a_s) \longrightarrow \pi_1(X, a), \qquad \overline{X}_s = X_s \otimes_{\kappa(s)} \overline{\kappa(s)}.$$

Then the homomorphism $\pi_1(X, a) \to \pi_1(S, b)$ is surjective, and its kernel is the closed normal subgroup of $\pi_1(X, a)$ generated by the images of the groups $\pi_1(\overline{X}_s, a_s)$.

REMARK IX.6.12. Under the hypotheses of IX.6.7, with S noetherian, it is readily seen that the set of points $s \in S$ for which X'_s is geometrically trivial over X_s is constructible. If X' is proper over X, it is even open, as follows from IX.6.6. Consequently, if S is a Jacobson prescheme (for example, of finite type over a field), or if X' is proper over X, 260
it is enough, in verifying the hypotheses of IX.6.7, to consider the *closed* points s of S. Likewise, in IX.6.11, it is then enough to use the groups $\pi_1(\overline{X}_s, a_s)$ for closed points s of S.

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CHAPTER X

Specialization theory of the fundamental group

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In the present exposé we restrict ourselves to studying the fundamental group of geometric fibers in a *proper* morphism, i.e. the fundamental group of a varying *proper* algebraic scheme. In a later exposé we shall extend the technique used here to étale coverings *tamely ramified* at infinity. This will give, for example, a solution of the *three-point problem* for Galois coverings of order prime to the characteristic (i.e. a determination of the Galois coverings of the line $\mathbb{P}^1_k$ ramified at no more than three specified points and tamely ramified at those points), together with its evident variants.

X.1. The homotopy exact sequence for a proper and separable morphism

Definition X.1.1. *A prescheme X over a field k is called* separable*, or* separable over k*, if $X \otimes_k K$ is reduced for every extension K of k. If $f\colon X \to Y$ is a morphism of preschemes, we say that f is* separable*, or that X* is separable over Y*, if X is flat over Y and, for every $y \in Y$, the fiber $X \otimes_Y \kappa(y)$ is separable over $\kappa(y)$.*

For a prescheme X over a field k, separability also means that X is *reduced* and that the fields $\kappa(x)$, for x the generic point of an irreducible component of X, are separable extensions of k. Thus, if k is perfect, this is equivalent to saying that X is reduced. Observe that if X is separable over Y, then after every base change $Y' \to Y$, $X' = X \times_Y Y'$ is separable over Y'. Under suitable finiteness hypotheses one can also prove that the composite of two separable morphisms is separable. We shall need this only in the 262
following form: *if X is separable over Y, and X' is étale over X, then X' is separable over Y*. This is an immediate consequence of the definitions and I I.9.2. The hypothesis of a separable morphism will otherwise enter through the following proposition:

Proposition X.1.2. *Let $f\colon X \to Y$ be a proper and separable morphism, with Y locally noetherian, and consider its* Stein *factorization*

$$X \xrightarrow{f'} Y' \to Y,$$

where $f'_*(\mathcal{O}_X) = \mathcal{O}_{Y'}$, and where Y', finite over Y, is isomorphic to the spectrum of the algebra $f_*(\mathcal{O}_X)$. Then Y' is an étale covering *of Y.*

This proposition appears in EGA III, 7[1]. Let us indicate the principle of the proof. One readily reduces to the case where Y is the spectrum of a complete local ring A. After making a suitable finite flat extension of A, corresponding to a suitable residue-field extension, one may suppose that the connected components of the fiber over the closed point y are geometrically connected. This also means that $\mathrm{H}^0(X_y, \mathcal{O}_{X_y})$ decomposes as a product of fields equal to $k = \kappa(y)$. We may then suppose X connected, in which case

$$\mathrm{H}^0(X_y, \mathcal{O}_{X_y}) = k.$$

Thus the homomorphism

$$A \longrightarrow \mathrm{H}^0(X_y, \mathcal{O}_{X_y})$$

[1] Cf. EGA III, 7.8.10 (i).

is *surjective*. A general proposition of Künneth type then shows that $f_*(\mathcal{O}_X)$ is defined by an A-module B which is free over A, and that

$$B/\mathfrak{m}B \longrightarrow \mathrm{H}^0(X_y, \mathcal{O}_{X_y}) = k$$

is bijective. In the present case B is therefore an étale algebra over A, which completes the proof.

THEOREM X.1.3. *Let $f\colon X \to Y$ be a proper and separable morphism, with Y locally noetherian and connected, and suppose that $f_*(\mathcal{O}_X) = \mathcal{O}_Y$. This implies that the fibers of X over Y are geometrically connected, and the converse follows from X.1.2. Let y be a point of Y, let $\overline{\kappa(y)}$ be an algebraic closure of $\kappa(y)$, and put*

$$\overline{X}_y = X_y \otimes_{\kappa(y)} \overline{\kappa(y)}.$$

Finally, let X' be a connected *étale covering of X, and put*

$$\overline{X}'_y = X'_y \otimes_{\kappa(y)} \overline{\kappa(y)}.$$

There exist an étale covering Y' of Y and an X-isomorphism 263

$$X' \xrightarrow{\sim} X \times_Y Y'$$

if and only if $\overline{X}'_y$ has a section over $\overline{X}_y$.

Set $Y' = \mathrm{Spec}(())h_*(\mathcal{O}_{X'}))$, where $h\colon X' \to Y$ is the composite $X' \to X \to Y$. It is enough to prove that the canonical Y-morphism

$$X' \longrightarrow X \times_Y Y'$$

is an *isomorphism*, and that Y' is étale over Y. By X.1.2, we already know that Y' is étale over Y. Consequently $X \times_Y Y'$ is étale over X, and hence $X' \to X \times_Y Y'$ is likewise étale (I I.4.8). Moreover, Y' is connected, as the image of the connected prescheme X', so $X \times_Y Y'$ is connected because X has connected fibers over Y (IX IX.3.4 and V V.6.9 (iii)). Thus, to prove that $X' \to X \times_Y Y'$ is an isomorphism, it is enough to show that its degree at *one* point of $X \times_Y Y'$ equals 1. This follows readily from the hypothesis that $\overline{X}'_y$ has a section over $\overline{X}_y$, either by using IX IX.6.6, or more simply by observing that it is enough to prove the existence of such a point in $X \times_Y Y'$ after the base change $\mathrm{Spec}(()\overline{\kappa(y)}) \to Y$, where it is evident. This completes the proof of X.1.3.

Taking account of IX IX.3.4 and the dictionary V V.6.9 and V V.6.11, we may put X.1.3 in the following equivalent form:

COROLLARY X.1.4. *With the preceding notation for $f\colon X \to Y$ and $\overline{X}_y$, let $\overline{a}$ be a geometric point of $\overline{X}_y$, let a be its image in X, and let b be its image in Y. Then the following sequence of group homomorphisms is* exact*:*

$$\pi_1(\overline{X}_y, \overline{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Y, b) \longrightarrow e.$$

REMARKS X.1.5. Observe that the proof of X.1.3 uses X.1.2 in an essential way, and hence the *first comparison theorem* in algebraic-formal geometry. By contrast, the descent theory of Exposé IX entered only through IX IX.3.4, for which a direct proof is easy in 264
the case of a *proper* morphism $f\colon X \to Y$ such that $f_*(\mathcal{O}_X) = \mathcal{O}_Y$. Indeed, let Y' be étale over Y, and suppose that $X' = X \times_Y Y'$ is the disjoint sum of two nonempty open subsets. We prove that the same is true of Y'. We may write $Y' = \mathrm{Spec}(()\mathcal{A})$, and hence $X' = \mathrm{Spec}(()\mathcal{B})$, with

$$\mathcal{B} = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_X.$$

The decomposition of X' as a disjoint sum corresponds to a decomposition of $\mathcal{B}$ as a product of two nonzero algebras $\mathcal{B}_1$ and $\mathcal{B}_2$. Since $f_*(\mathcal{O}_X) = \mathcal{O}_Y$, it follows readily that $f_*(\mathcal{B}) = \mathcal{A}$. Thus $\mathcal{A}$ is a product of the two algebras $f_*(\mathcal{B}_1)$ and $f_*(\mathcal{B}_2)$, which are likewise nonzero because their unit sections are nonzero. This proves the assertion.

. Continue to suppose that f is proper and separable, but make no further assumption on $f_*(\mathcal{O}_X)$. The latter corresponds to a well-determined étale covering Y' of Y, pointed above b by the image b' of a. Applying X.1.4 to the canonical morphism $X \to Y'$, and assuming f surjective, the exact sequence X.1.4 is replaced by the following sequence, analogous to the homotopy exact sequence of fiber spaces in algebraic topology:

$$\pi_1(\overline{X}_y, \overline{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Y, b) \longrightarrow \pi_0(\overline{X}_y, \overline{a}) \longrightarrow \pi_0(X, a) \longrightarrow \pi_0(Y, b) \longrightarrow e.$$

Of course, in X.1.4 one cannot in general assert that the homomorphism

$$\pi_1(\overline{X}_y, \overline{a}) \longrightarrow \pi_1(X, a)$$

is injective. In algebraic topology its kernel is the image of $\pi_2(Y, b)$, and in algebraic geometry as well one ought to introduce homotopy groups in every dimension, together with the full homotopy exact sequence for a proper morphism satisfying suitable hypotheses, for example being smooth). At present no result of this kind is available, apart from a reasonable, if not definitive, definition of the higher homotopy groups.

COROLLARY X.1.6. *Let k be an algebraically closed field, and let X and Y be two connected preschemes over k, with X proper over k and Y locally noetherian. Let a* 265
be a geometric point of X and b a geometric point of Y, both with values in the same algebraically closed extension K of k. Let $c = (a, b)$ be the corresponding geometric point of $X \times_k Y$. Then the homomorphism

$$\pi_1(X \times_k Y, c) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b)$$

induced by the two projections $X \times_k Y \to X$ and $X \times_k Y \to Y$ is an isomorphism.

First suppose that $K = k$. Set $Z = X \times_k Y$, consider the projection $f\colon Z \to Y$ and the underlying point y of the geometric point b of Y, and apply X.1.4. For this purpose, observe that after replacing X by X_{red}, which does not change the fundamental groups in question, we may assume that X is reduced, and hence separable over k. Thus Z is separable over Y, and plainly has geometrically connected fibers, since X is connected. The geometric fiber of Z at b is canonically isomorphic to $X \otimes_k K = X$.

Moreover, since the composite of the morphisms $X \to Z \to X$ is the identity, the homomorphism $\pi_1(X, a) \to \pi_1(Z, c)$ is injective, and X.1.4 gives an exact sequence

$$e \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Z, c) \longrightarrow \pi_1(Y, b) \longrightarrow e.$$

There is also the canonical exact sequence

$$e \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b) \longrightarrow \pi_1(Y, b) \longrightarrow e,$$

in which the displayed homomorphisms are the canonical inclusion and projection. Finally, the identity homomorphisms on the outer terms together with the canonical homomorphism

$$\pi_1(Z, c) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b)$$

on the middle terms define a homomorphism from the first exact sequence to the second. The resulting diagram is plainly commutative. Since the homomorphisms on the outer terms are isomorphisms, so is the homomorphism on the middle terms. This proves X.1.6 in this case.

When K is no longer assumed equal to k, the argument gives only an isomorphism

$$\pi_1(Z, c) \longrightarrow \pi_1(X \otimes_k K, a) \times \pi_1(Y, b),$$

and X.1.6 is then equivalent to the following special case: 266

COROLLARY X.1.7. *Let X be a proper connected scheme over an algebraically closed field k, let k' be an algebraically closed extension of k, let a' be a geometric point of $X \otimes_k k'$, and let a be its image in X. Then the canonical homomorphism*

$$\pi_1(X \otimes_k k', a') \longrightarrow \pi_1(X, a)$$

is an isomorphism.

Surjectivity of this homomorphism is equivalent to the assertion that, if X' is a connected étale covering of X, then $X' \otimes_k k'$ is connected as well. This follows at once from the fact that k is algebraically closed, and is also a special case of IX IX.3.4. The properness hypothesis on X has not yet been used. Injectivity of the homomorphism in question means the following: *every étale covering of $X \otimes_k k'$ is isomorphic to the inverse image of an étale covering of X.*

By an essentially formal argument, one can find a k-subalgebra A of k', of finite type over k, and an étale covering of $X \otimes_k A$ whose inverse image over $X \otimes_k k'$ is isomorphic to the given covering. Let $Y = \mathrm{Spec}(()A)$. This is an integral k-scheme of finite type, and hence has rational points over k. Apply X.1.6 to the fundamental group of $X \times Y$ at a point (a, b) rational over k. Every connected étale covering of $X \times Y$ is then isomorphic to a quotient of a covering $X' \times Y'$, where X' and Y' are Galois étale coverings of X and Y, with groups G and G', by a subgroup H of $G \times G'$. It follows that the inverse image of this covering of $X \times Y$ over $X \times Y'$ is isomorphic to a covering of the form $X'_1 \times Y'$, where X'_1 is an étale covering of X.

Let L be the function field of Y, equal to the fraction field of A inside k'. The étale covering of $X \otimes_k L$ induced by the given covering of $X \times_k Y$ has the following property: there is a finite separable extension L'/L such that its inverse image over $X \otimes_k L'$ is isomorphic to $X'_1 \otimes_k L'$. Since k' is algebraically closed, we may suppose that L' is contained in k'. This proves that the given étale covering of $X \otimes_k k'$ is isomorphic to $X'_1 \otimes_k k'$, as required.

The explicit description just given for étale coverings of a product $X \times_k Y$ immediately 267
implies:

Corollary X.1.8. *Let k be an algebraically closed field, let X and Y be two locally noetherian preschemes over k, let $Z = X \times_k Y$, and let Z' be an étale covering of Z. For every point $y \in Y$ rational over k, let $i_y \colon \mathrm{Spec}(()k) \to Y$ be the canonically associated morphism, let $j_y = \mathrm{id}_X \times_k i_y \colon X \to Z$, and let X'_y be the étale covering of X obtained from Z' by inverse image under j_y. Suppose that Y is connected, and that X or Y is proper over k. Then all the coverings X'_y of X are isomorphic.*

Figuratively, one may say that *a family of étale coverings of X, parametrized by a connected prescheme Y, is constant if X, or the parameter prescheme Y, is proper over k.*

Remarks X.1.9. Corollaries X.1.6 through X.1.8 are due to Lang–Serre [**2**] in the case of normal algebraic schemes. Their work was the initial motivation for the theory of the fundamental group developed in this Seminar. As those authors observed, these results become false when the properness hypothesis is dropped, at least in characteristic $p > 0$. For example, take $X = \mathrm{Spec}(()k[t])$, the affine line. The coverings of X parametrized by the affine line $Y = \mathrm{Spec}(()k[s])$, defined by the equations

$$x^p - x = st,$$

are étale and pairwise nonisomorphic. This contradicts X.1.8, and a fortiori X.1.6. Likewise, if s is regarded as an element transcendental over k in an algebraically closed extension K of k, one obtains an étale covering X' of $X \otimes_k K$ which does not come from an étale covering of X.

X.2. Application of the existence theorem for sheaves: a semicontinuity theorem for the fundamental groups of the fibers of a proper and separable morphism

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Theorem X.2.1. *Let Y be the spectrum of a* complete *noetherian local ring A, with residue field k, let X be a proper Y-scheme, let $X_0 = X \otimes_A k$, let a_0 be a geometric*

point of X_0, and let a be the corresponding geometric point of X. Then the canonical homomorphism

$$\pi_1(X_0, a_0) \longrightarrow \pi_1(X, a)$$

is an isomorphism.

This is merely the translation into the language of the fundamental group of the result recalled in IX IX.1.10. It is here that the existence theorem for sheaves in algebraic-formal geometry enters the theory of the fundamental group in an essential way.

Now introduce an algebraic closure $\overline{k}$ of the residue field k, and the geometric fiber

$$\overline{X}_0 = X_0 \otimes_k \overline{k}.$$

We have the exact sequence (IX IX.6.1)

$$e \to \pi_1(\overline{X}_0, \overline{a}) \to \pi_1(X_0, a_0) \to \pi_1(k, \overline{k}) \to e.$$

On the other hand, there is the isomorphism X.2.1, together with the analogous and more elementary isomorphism

$$\pi_1(k, \overline{k}) \longrightarrow \pi_1(Y, b),$$

where b is the image of a in Y. Thus we obtain:

Corollary X.2.2. *With the preceding notation, suppose that $\overline{X}_0 = X_0 \otimes_k \overline{k}$ is connected, and let $\overline{a}_0$ be a geometric point of $\overline{X}_0$, with image a_0 in X, and let b_0 be its image in Y. Then the following sequence of canonical homomorphisms*

$$e \to \pi_1(\overline{X}_0, \overline{a}) \to \pi_1(X, a_0) \to \pi_1(Y, b_0) \to e$$

is exact.

Compare this sequence with the exact sequence X.1.4, but note that a) no flatness or fiberwise separability hypothesis was required for $X \to Y$; and b) one has the important additional fact that *the homomorphism $\pi_1(\overline{X}_0, \overline{a}_0) \to \pi_1(X, a_0)$ is injective.*

This last fact will allow us to compare the fundamental groups of the other geometric 269
fibers of X over Y with that of $\overline{X}_0$. Let y_1 be an arbitrary point of Y, let X_1 be its fiber, and let $\overline{X}_1$ be its geometric fiber with respect to a chosen algebraically closed extension of $\kappa(y_1)$. Let $\overline{a}_1$ be a geometric point of $\overline{X}_1$, let a_1 be its image in X, and let b_1 be its image in Y. Choose a path class from a_1 to a_0, and hence a path class from b_1 to b_0. This gives a commutative diagram of homomorphisms

$$\begin{array}{ccccccc} & & \pi_1(\overline{X}_1, \overline{a}_1) & \longrightarrow & \pi_1(X, a_1) & \longrightarrow & \pi_1(Y, b_1) \longrightarrow e \\ & & & & \downarrow & & \downarrow \\ e \longrightarrow & \pi_1(\overline{X}_0, \overline{a}_0) & \longrightarrow & \pi_1(X, a_0) & \longrightarrow & \pi_1(Y, b_0) & \longrightarrow e, \end{array}$$

in which the two displayed vertical arrows are isomorphisms. Since the second row is exact, we obtain a canonical homomorphism, which we shall call the *specialization homomorphism for the fundamental group*:

$$\pi_1(\overline{X}_1, \overline{a}_1) \longrightarrow \pi_1(\overline{X}_0, \overline{a}_0).$$

It depends only on the chosen path class from a_1 to a_0, and is therefore *defined up to an inner automorphism* of $\pi_1(X, a_0)$. When the first row above is exact as well, it follows at once that the specialization homomorphism is surjective. Taking X.1.4 into account, we therefore obtain:

Corollary X.2.3. *Under the hypotheses of X.2.1, suppose in addition that $f\colon X \to Y$ is* separable (X.1.1) *and that $\overline{X}_0$ is connected. Then $f_*(\mathcal{O}_X) = \mathcal{O}_Y$ by X.1.2. For every geometric fiber $\overline{X}_1$ of X over Y, endowed with a geometric point $\overline{a}_1$, the specialization homomorphism defined above is* surjective.

This is a *semicontinuity* result for the fundamental group, which does not yet seem to have an analogue in algebraic topology. It may moreover be stated under more general hypotheses:

COROLLARY X.2.4. *Let $f\colon X \to Y$ be a proper morphism with geometrically connected* 270
fibers, with Y locally noetherian. Let y_0, y_1 be two points of Y such that $y_0 \in \overline{\{y_1\}}$, and let $\overline{X}_0, \overline{X}_1$ be the geometric fibers of X corresponding to chosen algebraically closed extensions of $\kappa(y_0)$ and $\kappa(y_1)$. Let $\overline{a}_0$, respectively $\overline{a}_1$, be a geometric point of $\overline{X}_0$, respectively $\overline{X}_1$. Then one can naturally define a specialization homomorphism

$$\pi_1(\overline{X}_1, \overline{a}_1) \longrightarrow \pi_1(\overline{X}_0, \overline{a}_0),$$

well-defined up to inner automorphism. This homomorphism is surjective *if f is separable* (X.1.1).

First, it follows from X.1.7 that X.2.4 is essentially independent of the chosen algebraically closed extensions of the residue fields $\kappa(y_0)$ and $\kappa(y_1)$. We may therefore replace Y by a prescheme Y' over Y having a point y'_0 above y_0, respectively a point y'_1 above y_1. Take for Y' the spectrum of the completion of the local ring of Y at y_0, and apply X.2.3.

REMARKS X.2.5. The final conclusion of X.2.4, concerning surjectivity of the specialization homomorphism, and a fortiori the results X.1.3 and X.1.4 from which it follows, becomes false if $f\colon X \to Y$ is no longer assumed separable, even for projective schemes over an algebraically closed field of characteristic 0. We shall see examples below both when f is flat but has a nonseparable fiber, although X and Y are smooth over k, and when the fibers of f are separable but f is not flat, for example when $f\colon X \to Y$ is a birational morphism of normal integral schemes; see XI XI.3. In these examples, the fundamental group of the geometric generic fiber may be trivial while that of a suitable geometric special fiber is not.

Moreover, even if $f\colon X \to Y$ is proper and separable as in X.2.4, the specialization homomorphism often fails to be an isomorphism. For example, it is easy to give cases in 271
which $\overline{X}_1$ is a nonsingular elliptic curve, so its fundamental group is commutative and its ℓ-primary component, for a prime ℓ different from the characteristic, is isomorphic to $\mathbb{Z}_\ell^2$; see XI. Meanwhile $\overline{X}_0$ may consist of two nonsingular rational curves meeting in two points, or of two rational curves tangent at one point, or of a rational curve having a singular point which is a cusp. For the complete classification of degenerate elliptic curves, see the recent work of Kodaira [**1**] and Néron. In these three cases the fundamental group of $\overline{X}_0$ is respectively $\widehat{\mathbb{Z}}$, e, and e, and is therefore *strictly smaller* than that of $\overline{X}_1$.

We shall nevertheless obtain below, when f is *smooth*, a bound on the kernel of the specialization homomorphism which implies in particular that, if $\kappa(y_0)$ has characteristic 0, the specialization homomorphism is an isomorphism. Even for a smooth morphism, however, if $\kappa(y_0)$ has positive characteristic, the specialization homomorphism need not be an isomorphism, as one sees for example when X is an abelian scheme over Y, even of relative dimension 1, cf. XI XI.2. A satisfactory theory of specialization of the fundamental group must take account of the *continuous component* of the *true* fundamental group, corresponding to the classification of principal coverings whose structural groups are infinitesimal groups. One might then expect that the true fundamental groups of the geometric fibers of a smooth proper morphism $f\colon X \to Y$ form a well-behaved local system on X, an inverse limit of finite flat group schemes over X[2]. We shall return later to this point of view. Our present aim, by contrast, is to push as far as possible the phenomena common to the topological and scheme-theoretic theories of the fundamental group.

[2]This extremely attractive conjecture is unfortunately disproved by an unpublished example of M. Artin, already when the fibers of f are algebraic curves of a fixed genus $g \geq 2$.

Now let X_0 be a proper smooth connected curve of genus g over an algebraically closed field k. If k has characteristic zero, its fundamental group may be determined by transcendental methods as follows. We know that X_0 is obtained by base extension from a curve defined over an algebraically closed extension of finite transcendence degree of 272
the prime field $\mathbb{Q}$. By X.1.7, we may suppose that k itself has finite transcendence degree over $\mathbb{Q}$. We may therefore regard k as a subfield of the complex numbers $\mathbb{C}$, and a further application of X.1.7 allows us to suppose $k = \mathbb{C}$. It is then not difficult to verify that the fundamental group of X_0 is isomorphic to the profinite completion of the fundamental group of the associated topological space $\tilde{X}_0$, a compact oriented surface of genus g, with respect to the topology defined by subgroups of finite index[3]. It is classical that the topological fundamental group is generated by $2g$ generators s_i, t_i, $1 \leq i \leq g$, subject to the single relation

$$(s_1 t_1 s_1^{-1} t_1^{-1}) \cdots (s_g t_g s_g^{-1} t_g^{-1}) = 1.$$

Thus the fundamental group of X_0 has $2g$ *topological* generators s_i, t_i, $1 \leq i \leq g$, satisfying the preceding relation.

Now suppose that k has characteristic $p > 0$. Let A be the ring of Witt vectors constructed from k, and let K be an algebraically closed extension of its fraction field. We saw in III III.7.4 that there exists a scheme X over $Y = \mathrm{Spec}(()A)$, proper and smooth over Y, whose reduction is X_0. Applying X.2.3, we obtain a *surjective* homomorphism

$$\pi_1(X_1) \longrightarrow \pi_1(X_0),$$

where $X_1 = X \otimes_A K$. It is immediate[4] that X_1 is smooth over K, connected by (X.1.2), and of dimension 1, and that its genus equals g, by invariance of the Euler–Poincaré characteristic; see EGA III, 7. Since K has characteristic 0, the preceding result applies. We have therefore proved, by *transcendental methods*:

Theorem X.2.6. *Let X_0 be a smooth proper connected algebraic curve over an algebraically closed field k, and let g be its genus. Then $\pi_1(X_0)$ has a system of $2g$* 273
topological generators subject to the relation displayed above. If k has characteristic 0, $\pi_1(X_0)$ is moreover the free group of Galois type on those generators subject to that relation.

Remarks X.2.7. At present, to the author's knowledge, there is no purely algebraic proof of the preceding result, except in genera 0 and 1. To begin with, it is not at all clear how to distinguish $2g$ elements of $\pi_1(X_0)$ which one might then expect to form a system of topological generators. In this respect, the situation of the rational line with n points removed, and the study of its coverings tamely ramified at those points, is more congenial: the ramification groups at the n points provide n elements of the fundamental group under study, and one proves that they generate it topologically, as we shall see later[5]. Yet even in this particularly concrete case no purely algebraic proof seems to be known. Such a proof would plainly be extremely interesting. The only fact about the fundamental group of a curve that appears to admit a purely algebraic proof, apart from the weak finiteness theorem X.2.11 below, proved algebraically by Lang–Serre [**2**], is the determination of its abelianization by means of the Jacobian, mentioned in the last line of IX IX.5.8.

. The final assertion of X.2.6 no longer holds in characteristic $p > 0$, as one sees already in the case of elliptic curves. As we have already pointed out, we do not know whether the fundamental group of X_0 is topologically finitely presented; this seems wholly unlikely.

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[3]This deduction was explained in one of the oral exposés which were not written up.

[4]Cf. EGA IV, 12.2.

[5]Cf. Exposé XII. Even these elements are really determined only up to conjugacy, and one must make a judicious *simultaneous* choice of representatives in their conjugacy classes.

THEOREM X.2.8. *Let k be an algebraically closed field, and let X be a proper connected scheme over k. Then the fundamental group of X is topologically finitely generated.*

We proceed by induction on $n = \dim X$, the assertion being trivial for $n \leq 0$. Thus 274
suppose that $n > 0$, and that the theorem has been proved in dimensions $n' < n$. By Chow's lemma (EGA II 5.6.2), there exists a projective scheme X' over k and a surjective morphism $X' \to X$. We may clearly suppose that X' is reduced and, on passing to its normalization, normal. By descent theory, it suffices to prove that the fundamental groups of the connected components of X' are topologically finitely generated (IX IX.5.2). We are thus reduced to the case in which X is *projective* and *normal*. If $n = 1$, it is enough to apply X.2.6. If $n \geq 2$, consider a projective immersion $X \to \mathbb{P}^r_k$ and a hyperplane section $Y = X \cdot H$ (equipped with the induced reduced structure), such that $Y \neq X$, i.e. $H \not\supset X$. We then have $\dim Y < n$, and, taking the induction hypothesis into account, it suffices to prove that $\pi_1(Y) \to \pi_1(X)$ is *surjective.* More generally:

LEMMA X.2.9. *Let X be a proper prescheme over an algebraically closed field k, and let $g\colon X \to \mathbb{P}^r_k$ be a morphism. Suppose that X is irreducible and normal and that $\dim g(X) \geq 2$. Let H be a hyperplane of $\mathbb{P}^r_k$ and let $Y = X \times_{\mathbb{P}^r_k} H$. Then Y is connected, and the homomorphism $\pi_1(Y) \to \pi_1(X)$ is surjective.*

These assertions in fact follow from the following one:

COROLLARY X.2.10. *Under the preceding conditions, let X' be a connected étale covering of X, and let $Y' = X' \times_X Y = X' \times_{\mathbb{P}^r_k} H$ be the induced covering of Y. Then Y' is connected.*

Since X is normal, X' is normal; hence, being connected, X' is irreducible. Moreover, its image in $\mathbb{P}^r_k$ has dimension ≥ 2. A well-known lemma due to Zariski (called the "*Bertini theorem*") therefore implies that if H'_1 is the generic hyperplane in $\mathbb{P}^r_k$, defined over an extension K of k, then $X' \times_{\mathbb{P}^r} H_1$ is universally irreducible, hence universally connected over K. Zariski's connectedness theorem (EGA III 4) then implies that for *every* hyperplane H (defined over an arbitrary extension of k), $X' \times_{\mathbb{P}^r} H$ is geometrically connected. This completes the proof of X.2.10, and hence of X.2.8.

COROLLARY X.2.11 (Lang-Serre). *Under the conditions of X.2.8, for every finite* 275
group G, the set of isomorphism classes of principal coverings of X with group G is finite.

REMARK X.2.12. Under the conditions of X.2.9, when $\dim g(X) \geq 3$ (at least when g is an immersion and X is regular), we shall prove a more precise result, known in algebraic geometry as the "*Lefschetz theorem*": $\pi_1(Y) \to \pi_1(X)$ *is an isomorphism.*[6] In the classical cases there are analogous statements for homology groups and higher homotopy groups, which sooner or later will have to be encompassed within abstract algebraic geometry. Even for Hodge cohomology $\mathrm{H}^p(X, \Omega^q)$, the question does not seem to have been studied yet; moreover, it is hardly likely that for the latter the Lefschetz theorems remain unchanged in characteristic $p > 0$.

REMARK X.2.13. Let R be a complete discrete valuation ring with algebraically closed residue field κ of characteristic $p > 0$ and fraction field K, and let Y be a proper smooth connected curve of genus g over R. Thus there is a surjective specialization morphism $\mathrm{sp} : \pi_1(Y_{\overline{K}}) \to \pi_1(Y_\kappa)$. We have already noted that if K has characteristic 0, sp is not an isomorphism as soon as $g \geq 1$. Suppose that K has characteristic p, so that R is isomorphic to the formal power series ring $\kappa[[T]]$. In equal characteristic $p > 0$, the fundamental group is not determined by the genus g, as one sees already from elliptic curves, which may be ordinary or supersingular. We mention the recent result of A. Tamagawa (not yet published). If G is a profinite group, we denote by G^{res} the quotient

[6] Cf. the SGA 2 seminar (1962), which followed this one, Theorem X 3.10.

profinite group of G given by the inverse limit of the finite solvable topological quotients of G.

THEOREM (A. Tamagawa). *Suppose that $g \geq 2$, that the special fiber Y_κ is definable over a finite field, and that the morphism* $\mathrm{sp}^{\mathrm{res}} : \pi_1(Y_{\overline{K}})^{\mathrm{res}} \to \pi_1(Y_\kappa)^{\mathrm{res}}$ *induced by* sp *upon passing to the quotient is bijective. Then the curve Y is constant over R.*

Note that the Galois group of $\overline{K}/K$ is solvable. The preceding statement may also be expressed as follows: suppose that every finite étale covering $X_K \to Y_K$ of the generic fiber Y_K, Galois with solvable Galois group, has potentially good reduction (that is, extends to a finite étale covering of Y after possibly replacing R by its normalization in a finite extension of K). Then Y is constant over R.

X.3. Application of the purity theorem: continuity theorem for the fundamental groups of the fibers of a proper and smooth morphism

Recall without proof[7] the

PURITY THEOREM X.3.1 (Zariski-Nagata). *Let $f\colon X \to Y$ be a quasi-finite and dominant morphism of integral preschemes, with X normal and Y regular and locally Noetherian, and let Z be the set of points of X where f is not étale, i.e. where f is ramified (these are equivalent,* I I.9.5 (ii)*). If $Z \neq X$, then Z has codimension* 1 *in X at all its points, i.e. for every irreducible component Z' of Z with generic point z, the Krull dimension of $\mathcal{O}_{X,z}$ is equal to* 1.

Recall that a prescheme is called *normal* respectively*regular* if its local rings are normal respectivelyregular, and that the relation $Z \neq X$ also means that the finite extension $R(Z)/R(X)$ (where R denotes the field of rational functions) is *separable*. Passing to the generic point z of a component Z' of Z, and localizing at the point y of Y below z, one obtains the equivalent statement: 276

COROLLARY X.3.2. *Let A be a regular Noetherian local ring, and let $A \to B$ be an injective local homomorphism such that B is normal, a localization of an algebra of finite type over A, and* quasi-finite *over A; suppose in addition that* $\dim A (= \dim B) \geq 2$*, and that for every prime ideal $\mathfrak{p}$ of B distinct from the maximal ideal, B is étale over A at $\mathfrak{p}$, i.e. $B_\mathfrak{p}$ is étale over $A_\mathfrak{q}$ (where $\mathfrak{q} = A \cap \mathfrak{p}$). Then B is étale over A.*

Moreover, it is not difficult to reduce this last statement to the case where A is a *complete* local ring, and hence where B is *finite* over A. Zariski [**5**] gives a simple proof of this result, valid in the equal-characteristic case; the general case is due to Nagata [**3**], who relies on a delicate result of Chow; the latter has not been verified by any of the participants in the Seminar, and ought to be the subject of a later lecture. We mention here only the very simple proof in the special case where $\dim A = 2$, which is sufficient for the most important application we shall make of it in the present section. Since B is normal, it is a B-module of depth (older terminology: cohomological codimension) ≥ 2, and hence it is an A-module of depth ≥ 2; since A is regular of dimension 2, it follows that B is a *free module* over A[8]. It then follows from I I.4.10 that the set of prime ideals $\mathfrak{q}$ of A at which B is ramified over A is the subset of $\mathrm{Spec}(()A)$ defined by a principal ideal (generated by the discriminant of a basis of B over A), and is therefore empty if it is contained in the closed point of $\mathrm{Spec}(()A)$, which proves X.3.2 when $\dim A = 2$.

We shall chiefly use X.3.1 in the equivalent form:

COROLLARY X.3.3. *Let X be a regular locally Noetherian prescheme, and let U be an open subset of X complementary to a closed subset Z of X of codimension ≥ 2. Then*

[7] For a proof, cf. SGA 2 X X.3.4.

[8] Cf. EGA 0_{IV} 17.3.4

the functor $X' \longmapsto X' \times_X U$ from the category of étale coverings of X to the category of étale coverings of U is an equivalence of categories; in particular, if a is a geometric point 277
of U, the canonical homomorphism $\pi_1(U,a) \to \pi_1(X,a)$ is an isomorphism.

The last assertion is plainly a consequence of the first, and for the latter we may plainly suppose that X is connected, and hence irreducible. The normality of X already implies that the functor $X' \longmapsto X' \times_X U$ from the category of locally free coverings (not necessarily étale) of X' to the category of coverings of U is fully faithful, since the functor $\mathcal{E} \longmapsto \mathcal{E}|U$ from the category of locally free Modules on X to the category of locally free Modules on U is so. It therefore remains to prove that for every *étale* covering U' of U, there exists an étale covering X' of X (necessarily unique by what precedes) such that U' is isomorphic to $X' \times_X U$. We may plainly suppose that U' is connected, and hence irreducible since (U being normal) U' is normal. Let K be the field of rational functions on X, or on U (it is the same), and let K' be that of U'; then U' identifies with the normalization of U in K' I I.10.3. Let X' be the normalization of X in K' (EGA II 6.3); then $X' = X \times_X U \cong U'$, while X' is normal and integral, and the structural morphism $f\colon X' \to X$ is *finite* and dominant (since X is normal and K'/K is a finite separable extension). It is étale on $U' = f^{-1}(U) = X' = f^{-1}(Z)$, and since Z has codimension ≥ 2 in X, $f^{-1}(Z)$ has codimension ≥ 2 in X'. It then follows from X.3.1 that X' is étale over X, which completes the proof.

Now let $f\colon X \to Y$ be a rational map from a regular locally Noetherian prescheme X to a prescheme Y, and suppose that f is defined on an open subset U complementary to a closed subset of codimension ≥ 2. Then X.3.3 yields a functor, defined up to isomorphism, from the category of étale coverings of Y to the category of étale coverings of X, and hence, for every geometric point a of U, with image b in Y, a canonical homomorphism

$$\pi_1(X,a) \to \pi_1(Y,b)$$

(deduced from the canonical homomorphism $\pi_1(U,a) \to \pi_1(Y,b)$ by means of the isomor- 278
phism $\pi_1(U,a) \xrightarrow{\sim} \pi_1(X,a)$). When f is a dominant morphism, with X and Y integral with function fields K and L, so that K is an extension of L, and with Y normal, these correspondences may be made more precise in terms of field extensions by noting that for every finite extension L' of L that is unramified over Y, the algebra $K' = L' \otimes_L K$ over K is unramified over X.

In particular, these observations show that the fundamental group of connected regular locally Noetherian preschemes, pointed by geometric points located in codimension ≤ 1, is a *functor* when the morphisms in this category are taken to be dominant rational maps defined on the complements of closed subsets of codimension ≥ 2. Recalling, for example, that a rational map from a normal scheme over a field k to a proper scheme over k is defined on the complement of a subset of codimension ≥ 2, one obtains:

Corollary X.3.4 (Birational invariance of the fundamental group). *Let k be a field, let X and Y be two regular schemes proper over k, let $f\colon X \to Y$ be a birational map from X to Y, and let Ω be an algebraically closed extension of the function field K of X, making it possible to define the fundamental group of X and the fundamental group of Y. These groups are then canonically isomorphic.*

This also means that if a finite extension K' of K is unramified over one proper nonsingular "model"X of K, then it is unramified over every other proper nonsingular model.

Remark X.3.5. For other applications of the purity theorem, see the works of Abhyankar presented in [**4**], inspired by the results of Zariski [**6**, chap. VIII], proved by topological methods. The latter are far from having been assimilated by "abstract"algebraic geometry and deserve further effort.

We shall need a few elementary facts from ramification theory. Let V be a discrete 279
valuation ring with fraction field K and residue field k, let L be a Galois extension of K with group G, and let V' be the normalization of V in L, which is a free module of rank $n = [L\colon K]$ over V; let $\mathfrak{m}'$ be a maximal ideal of V', let G_d be the subgroup of G consisting of the elements that leave $\mathfrak{m}'$ invariant, so that G_d acts on the residue extension $k' = V'/\mathfrak{m}'$ of k, and let G_i be the subgroup of elements of G_d that act trivially (recall that G_d and G_i are called, respectively, the *decomposition* and *inertia* subgroups of G). One says that L is *tamely ramified* over V if $n_i = [G_i\colon e]$ is prime to the characteristic p of k (a condition that is always satisfied if k has characteristic 0). It is well known that G_i then embeds canonically into the group k'^*, and is therefore isomorphic to the group of n_i-th roots of unity in k'^*; this implies in particular that G_i *is cyclic*. The standard example of this situation is obtained by setting $L = K[t]/(t^n - u)$, where u is a uniformizer of V and n is an integer prime to p: if K contains the n-th roots of unity, L is a totally ramified Galois extension of K, with Galois group $G = G_i$ isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Lemma X.3.6 (Abhyankar's lemma). *Let V be a discrete valuation ring with fraction field K, let L and K' be two Galois extensions of K that are* tamely ramified *over V, and let n and m be the orders of the corresponding inertia groups , Let L' be a composite extension of L and K' over K. If m is a multiple of n, then L' is unramified over the localizations of the integral closure V' of V in K'.*

Indeed, let W' be the normalization of V' in L', let $\mathfrak{m}'$ be a maximal ideal of V', and let $\mathfrak{n}'$ be a maximal ideal of W' lying above $\mathfrak{m}'$, let $\mathfrak{n}$ be the maximal ideal it induces on the normalization W of V in L, let $G,\ H,\ M$ be the Galois groups of $L,\ K',\ L'$ over K, and let $G_i,\ H_i,\ L'_i$ be the inertia groups corresponding to the chosen maximal ideals. Then M embeds into the product $G \times H$ and M_i into the product $G_i \times H_i$, in such a way that the projections $M \to G$ and $M \to H$, $M_i \to G_i$ and $M_i \to H_i$ are surjective (the 280
intermediate-field argument). It already follows, since G_i and H_i are by hypothesis cyclic of orders m and n prime to p, that M_i has order prime to p, and hence is cyclic; and since m is a multiple of n, the elements of $G_i \times H_i$ have trivial m-th power, so M_i has order dividing m, and therefore has order equal to m since $M_i \to H_i$ is surjective. This last homomorphism is therefore also injective. But its kernel is the inertia group of $\mathfrak{n}'$ over $\mathfrak{m}'$, which proves that L' is unramified over K' at $\mathfrak{n}'$. This proves the lemma.

Let us now place ourselves under the conditions of X.2.4, where we have a specialization homomorphism

$$\pi_1(\overline{X}_1, \overline{a}_1) \to \pi_1(\overline{X}_0, \overline{a}_0)$$

that is *surjective*, relative to a proper and separable morphism $f\colon X \to Y$. We wish to determine the kernel of this homomorphism. Proceeding as in the proof of X.2.4, we see that for this question we may always suppose that Y is the spectrum of a *complete discrete valuation ring V with algebraically closed residue field* (since one can always find such a ring and a morphism from its spectrum Y' to Y whose image is $\{y_0, y_1\}$). Then $X_0 = \overline{X}_0$, $\kappa(y_0) = k =$ the residue field of V, and $\kappa(y_1) = K =$ the fraction field of V. Let K_s be the separable closure of K, let $\overline{K}$ be its algebraic closure, and for every subring W of $\overline{K}$ containing V, put $X_W = X \otimes_V W$. In particular, we have

$$X_V = X, \qquad X_K = X_1, \qquad X_{\overline{K}} = \overline{X}_1.$$

Moreover, the canonical morphism $\overline{X}_1 = X_{\overline{K}} \to X_{K_s}$ induces an isomorphism on fundamental groups (IX IX.4.11); hence, in view of the isomorphism X.2.1 $\pi_1(X_0) \to \pi_1(X)$, we are reduced to studying the surjective homomorphism

$$\pi_1(X_{K_s}) \to \pi_1(X)$$

associated with the canonical morphism $X_{K_s} \to X$. Determining the kernel of the latter 281
amounts to solving the following problem: *given a connected principal covering Z_{K_s} of*

X_{K_s} with group G (thus associated with a homomorphism from $\pi_1(X_{K_s})$ to G), *determine under what conditions it is isomorphic to the inverse image of a principal covering Z of X with group G.*

First note that K_s is the increasing filtered union of its finite subextensions K' over K, and consequently Z_{K_s} is isomorphic to the inverse image of a principal covering $Z_{K'}$ of $X_{K'}$ for a suitable K' (note, however, that for fixed K', $Z_{K'}$ is not uniquely determined). To say that Z_{K_s} is isomorphic to the inverse image of a principal covering Z of X means that there exists a finite subextension $K'' \supset K'$ of K_s such that $Z_{K''} = Z_{K'} \otimes_{K'} K''$ is isomorphic to $Z \otimes_V K''$. For a finite subextension K' of K_s, now denote by V' the normalization of V in K'; this is a complete discrete valuation ring with residue field k. Thus the canonical morphism $X_{V'} \to X_V$ induces an isomorphism on the fibers over the closed points of $Y = \mathrm{Spec}(()V)$ and $Y' = \mathrm{Spec}(()V')$, and it then follows from X.2.1, applied to X_V and $X_{V'}$, that the induced homomorphism on fundamental groups $\pi_1(X_{V'}) \to \pi_1(X_V)$ is an isomorphism, or equivalently that every principal covering of $X_{V'}$ is the inverse image of a principal covering of X_V determined up to isomorphism. This therefore implies the

LEMMA X.3.7. *Let $Z_{K'}$ be a connected principal covering of $X_{K'}$ with group G, and let Z_K be its inverse image over X_{K_s}. This latter covering is isomorphic to the inverse image of a principal covering Z of X if and only if there exists a finite extension $K'' \supset K'$ of K contained in K_s such that the principal covering $Z_{K''}$ of $X_{K''}$ is induced by a principal covering of $X_{V''}$.*

Suppose in particular that the $X_{V''}$ are normal (it suffices for this, for example, that X_0 be normal, and a fortiori that X_0 be simple, cf. I I.9.1). Since they are connected, 282
they are therefore irreducible. Let L be the field of rational functions of X and X_K, let L' be that of $X_{V'}$ and $X_{K'}$, and let L'' be that of $X_{V''}$ and $X_{K''}$. Then under the conditions of X.3.7, $Z_{K'}$ defines a finite separable extension R' of L', and $Z_{K''}$ defines the extension $R'' = R' \otimes_{L'} L'' = R' \otimes_{K'} K''$. The condition considered in X.3.7 thus also means that there exists a finite separable extension K'' of K' such that $R'' = R' \otimes_{K'} K''$ is *unramified* over the normal scheme $X_{V''}$ with function field $L'' = L' \otimes_{K'} K''$, and not only over the open subset $X_{K''}$ of $X_{V''}$.

From now on, suppose that $f\colon X \to Y$ is a *smooth* morphism, so the morphisms $X_{V'} \to \mathrm{Spec}(()V')$ are smooth, and hence the schemes $X_{V'}$ are *regular*. Note that the fiber of the closed point of $\mathrm{Spec}(()V')$ in $X_{V'}$ is irreducible and of codimension 1. Let $\mathfrak{o}'$ be its local ring; it is therefore a discrete valuation ring with fraction field L' and residue field isomorphic to the field of rational functions of X_0, and thus of the same characteristic as k. Define $\mathfrak{o}''$ similarly in L''; it is plainly the normalization of $\mathfrak{o}'$ in L''. It then follows from the purity theorem X.3.1 or X.3.3 that R'' is unramified over $X_{V''}$ if and only if R'' is unramified over $\mathfrak{o}''$, the normalization of $\mathfrak{o}'$ in L''.

Now note that if u' is a uniformizer of V', it is also a uniformizer of $\mathfrak{o}'$. Thus, if n is an integer prime to the characteristic p of k, and if we take $K'' = K'[t]/(t^n - u)$, then K'' is a finite Galois extension of K', and L'' is isomorphic to $L'[t]/(t^n - u')$, and hence is tamely ramified over $\mathfrak{o}'$ with inertia group of order n. Now suppose that *G has order prime to p;* this implies that R' is tamely ramified over $\mathfrak{o}'$, and choose for n a multiple, prime to p, of the order of the inertia group of R' over $\mathfrak{o}'$ (for example, $n = [G\colon e]$). Applying the ABHYANKAR's lemma X.3.6, we see that the condition considered in X.3.7 is satisfied.

This proves the following theorem:

THEOREM X.3.8. *Let $f\colon X \to Y$ be a proper and smooth morphism with geometrically* 283
connected fibers, with Y locally Noetherian, and let y_0 and y_1 be two points of Y such that $y_0 \in \overline{\{y_1\}}$; let $\overline{X}_0$ and $\overline{X}_1$ be the corresponding geometric fibers, and consider the specialization homomorphism (X.2.4) $\pi_1(\overline{X}_1) \to \pi_1(\overline{X}_0)$. *This homomorphism is surjective, and every continuous homomorphism from $\pi_1(\overline{X}_1)$ to a finite group G of order prime to the characteristic p of $\kappa(y_0)$ comes from a homomorphism from $\pi_1(\overline{X}_0)$ to G.*

In other words:

COROLLARY X.3.9. *If $\kappa(y_0)$ has characteristic zero, then the specialization homomorphism is an isomorphism. If $\kappa(y_0)$ has characteristic $p > 0$, then the kernel of the specialization homomorphism is contained in the intersection of the kernels of the continuous homomorphisms from $\pi_1(\overline{X}_1)$ to finite groups of order prime to p (or equivalently, in the closed normal subgroup generated by a Sylow p-subgroup of the group of Galois type $\pi_1(\overline{X}_1)$); thus, if $\pi_1(\overline{X}_1)^{(p)}$ denotes the quotient group of $\pi_1(\overline{X}_1)$ by the preceding closed subgroup, and if $\pi_1(\overline{X}_0)^{(p)}$ is defined similarly, then the specialization homomorphism induces an* isomorphism

$$\pi_1(\overline{X}_1)^{(p)} \xrightarrow{\sim} \pi_1(\overline{X}_0)^{(p)}$$

Note that the proof of X.3.8 is purely algebraic. Proceeding as in X.2.6, we conclude *by transcendental methods:*

COROLLARY X.3.10. *Let X_0 be a proper, smooth, and connected curve of genus g over an algebraically closed field of characteristic p. With the notation introduced in X.3.9, the group $\pi_1(X_0)^{(p)}$ is isomorphic to $\Gamma^{(p)}$, where Γ is the group of Galois type generated by elements $s_i,\ t_i$ $(1 \leq i \leq g)$ subject to the relation*

$$(s_1 t_1 s_1^{-1} t_1^{-1}) \cdots (s_g t_g s_g^{-1} t_g^{-1}) = 1$$

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REMARKS X.3.11. When $\kappa(y_0)$ has characteristic zero, the result X.3.9 is well known by transcendental methods. Note that the proof of X.3.10 invokes the purity theorem in the mixed-characteristic case, but only for rings of dimension 2, where the proof of that theorem is easy and was recalled in the text.

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CHAPTER XI

Examples and complements

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XI.1. Projective spaces, unirational varieties

PROPOSITION XI.1.1. *Let k be an algebraically closed field, and let $X = \mathbb{P}^r_k$ be projective space of dimension r over k. Then X is* simply connected, *i.e. $\pi_1(X) = 0$.*

For $r = 0$, this is trivial. If $r = 1$, we must show that if X' is a nonempty connected étale covering of $X = \mathbb{P}^1_k$, then $X' \xrightarrow{\sim} X$. The genus formula gives, in this case, if g and g' are the genera of X and X':

$$1 - g' = d(1 - g),$$

where d is the degree of X' over X. Since $g = 0$, we therefore have $1 - g' = d$, which requires $d = 1$ since $g' \geq 0$, and this proves $X' \xrightarrow{\sim} X$. When $r \geq 2$, we proceed by induction on r, assuming that $\mathbb{P}^{r'}$ is simply connected for $r' < r$. Applying this to a hyperplane in $\mathbb{P}^r$ and using (X X.2.9), it indeed follows that $\mathbb{P}^r$ is simply connected. Another proof: by (X X.1.6), we have $\pi_1(\mathbb{P}^1 \times \cdots \times \mathbb{P}^1) = \pi_1(\mathbb{P}^1) \times \cdots \times \pi_1(\mathbb{P}^1)$, and hence $(\mathbb{P}^1)^r$ is simply connected since $\mathbb{P}^1$ is, so $\mathbb{P}^r$ is simply connected by the birational invariance of the fundamental group (X X.3.4). This proof shows more generally:

COROLLARY XI.1.2. *Let X be a proper normal scheme over an algebraically closed field k; if X is a rational variety, i.e. it is integral and its function field is a purely transcendental extension of k, then X is simply connected.*

This result applies in particular to Grassmann varieties and, more generally, to 286
varieties G/H, where G is a connected linear group over k and H is an algebraic subgroup containing a Borel subgroup of G.

Recall that a *unirational variety over k* is a proper integral scheme over k whose function field K is contained in a purely transcendental extension K' of k that is finite over K (i.e. has the same transcendence degree over k as K), i.e. if there exists a dominant rational map $f\colon \mathbb{P}^r_k \to X$, with $r = \dim X$. If X is normal, it follows from the considerations preceding X X.3.4 that, for every connected étale covering X' of X, with function field L/K, the algebra $L \otimes_K K'$ over K' is unramified on the model $\mathbb{P}^r$, and is therefore completely split by X.1.1; this shows that L is K-isomorphic to a subextension of K'/K. Taking V V.8.2 into account, this therefore proves:

COROLLARY XI.1.3. *The fundamental group of a normal unirational variety over an algebraically closed field is finite.*

(N.B. Note that in the definition of "unirational", it was not necessary for K'/K to be finite).

REMARKS XI.1.4. Of course, the results of this section are well known. On the other hand, J-P. Serre showed [**10**] that when X is a smooth projective unirational variety over an algebraically closed field of *characteristic zero, X* is simply connected. His proof is transcendental in that it uses the Hodge symmetry theorem.

[Added in 2003 (MR) Let us restrict ourselves to the case of an algebraically closed field k of characteristic $p \geq 0$. In characteristic $p > 0$, the definition of a unirational k-variety given above is that of a weakly unirational k-variety, as opposed to that of

a strongly unirational k-variety, where one assumes in addition that K' is a separable extension of K.

In dimension 2, there exist smooth projective weakly unirational surfaces whose fundamental group (finite by 1.3) is nontrivial, and which are therefore not rational (T. Shioda, On unirationality of supersingular surfaces, Math. Ann. **225** (1977), p. 155–159.). On the other hand, a strongly unirational surface is rational: this follows from Castelnuovo's rationality criterion, extended to characteristic $p > 0$ by O. Zariski (cf. J-P. Serre, Sém. Bourbaki no. 146, 1957 and Œuvres–Collected papers, vol. 1, p. 467).

Over the field $\mathbb{C}$ of complex numbers, examples are known of smooth projective varieties of dimension ≥ 3 that are unirational but not rational (cf. P. Deligne, Variétés unirationnelles non rationnelles [d'après M. Artin et D. Mumford], Sém. Bourbaki no. 402, 1971-72, Lecture Notes vol. 317). This is the case, in particular, of a smooth cubic hypersurface in projective space $\mathbb{P}^4$. Some of these examples extend to characteristic > 0 to give strongly unirational nonrational varieties (cf. J.P. Murre, Reduction of the proof of the non-rationality of a non singular cubic threefold to a result of Mumford, Compositio Math. **27** (1973), p. 63–82).

Let V be a normal integral projective k-variety. We say that V is rationally connected if there exist an integral k-scheme T of finite type and a k-morphism $F : T \times \mathbb{P}^1 \to V$ such that the morphism

$$\begin{aligned} T \times \mathbb{P}^1 \times \mathbb{P}^1 &\to V \times V \\ (t, u, u') &\longmapsto (F(t, u), F(t, u')) \end{aligned} \tag{1}$$

is dominant. In particular, a rational curve passes through any two sufficiently general rational points of V. The terminology is justified by the fact that if V is rationally connected, two rational points can be joined by a finite chain of rational curves. If k has characteristic $p > 0$, we strengthen the preceding definition by requiring the map (1) to be generically smooth. This gives the notion of a separably rationally connected variety. Thus unirational varieties are rationally connected, and in characteristic $p > 0$, strongly unirational varieties are separably rationally connected. J. Kollár showed that separably rationally connected varieties have trivial algebraic fundamental group (cf. O. Debarre, Variétés rationnellement connexes [d'après T. Graber, J. Harris, J. Starr et A.J. de Jong], Sém. Bourbaki no. 906, 2001-2002).**]**

XI.2. Abelian varieties

Let k be an algebraically closed field, let A be an abelian variety over k, i.e. a group scheme over k that is proper over k, smooth over k, and connected, and finally let G be a commutative group scheme of finite type over k. Denote by $\mathrm{Ext}(A, G)$ the group of classes of commutative extensions of A by G, and by $\mathrm{H}^1(A, G)$ the group of classes of principal bundles over A with group G (compare no. XI.4 below), and consider the canonical homomorphism

$$\mathrm{Ext}(A, G) \to \mathrm{H}^1(A, G).$$

An argument of Serre [**5**, chap. VII, th. 5] shows that this is an injective homomorphism 287
whose image is the set of "primitive elements"of $\mathrm{H}^1(A, G)$, i.e. the elements ξ for which

$$\pi^*(\xi) = \mathrm{pr}_1^*(\xi) + \mathrm{pr}_2^*(\xi),$$

where the pr_i are the two projections from $A \times A$ onto A, and $\pi\colon A \times A \to A$ is the composition law of A (N.B. Serre states his theorem only for G linear and connected, and of course smooth over k, but, on simplifying the first part of his argument, one sees that these restrictions are unnecessary: it is enough to note that every morphism from A into a group scheme E of finite type over k which takes the identity to the identity is a group homomorphism, and to apply this to the sections over A of an extension E of A by G).

We shall apply this result when G is a finite separable group scheme over k, i.e. an ordinary finite group, assumed commutative. Using then $\pi_1(A \times A) \xrightarrow{\sim} \pi_1(A) \times \pi_1(A)$ (X X.1.6) and interpreting $\mathrm{H}^1(X, G)$ as $\mathrm{Hom}(\pi_1(X), G)$ for every algebraic scheme X, in particular for $X = A$ or $X = A \times A$, one sees that every class in $\mathrm{H}^1(A, G)$ is primitive, and hence there is an isomorphism

$$\mathrm{Ext}(A, G) \xrightarrow{\sim} \mathrm{H}^1(A, G),$$

in other words, *every principal covering of A with commutative structural group G, pointed above the origin of A, is endowed in a unique way with the structure of an algebraic group having the marked point as its origin and such that $A' \to A$ is a homomorphism of algebraic groups.* In particular, if A' is connected, it too is an abelian variety, isogenous to A.

On the other hand, since the functor $X \longmapsto \pi_1(X)$ from pointed algebraic schemes X to groups commutes with products (IX IX.1.7), it carries a group object in the first category to a group object in the category of groups, i.e. to a *commutative* group. Thus, *if A is an abelian variety, $\pi_1(A)$ is a commutative group.* Therefore, to determine $\pi_1(A)$, it 288
suffices to know the functor $G \longmapsto \mathrm{H}^1(A, G) = \mathrm{Hom}(\pi_1(A), G)$ as G ranges over the finite *commutative* groups. Finally, recall that for every integer $n > 0$, the multiplication-by-n homomorphism on A,

$$A \xrightarrow{n} A$$

is surjective and hence has finite kernel, i.e. it is an isogeny, and it follows that every isogeny $A' \to A$ is a quotient of an isogeny of the preceding type. From this and standard arguments (cf. for example [**6**]), one obtains:

THEOREM XI.2.1 (Serre-Lang). *Let A be an abelian variety over an algebraically closed field k. For every integer $n > 0$, consider the ordinary finite group K_n underlying the kernel ${}_nA$ of multiplication by n on A, and, for every prime number ℓ, set*

$$T_\ell(A) = \varprojlim_r K_{\ell^r}$$

and

$$T_\cdot(A) = \prod_\ell T_\ell(A) = \varprojlim_n K_n$$

(where, if m is a multiple of n, $m = ns$, the map from K_m to K_n is multiplication by s). Then the group $\pi_1(A)$ is canonically isomorphic to $T_\cdot(A)$; consequently, for every prime number ℓ, the ℓ-primary component of $\pi_1(A)$ is canonically isomorphic to $T_\ell(A)$.

These isomorphisms are functorial as A varies. The module $T_\ell(A)$ is called the *ℓ-adic Tate module* of the abelian variety A. It is an additive functor in A; in particular, it gives rise to a representation of the ring $\mathrm{Hom}(A, A)$ of endomorphisms of A on $T_\ell(A)$, called the *ℓ-adic Weil representation*, which plays an important role in the theory of abelian varieties (cf. for example [**4**, chap VII]). Theorem XI.2.1 interprets it as the natural representation on the *ℓ-adic homology group* of A, $\mathrm{H}_1(A, \mathbb{Z}_\ell) = \pi_1(A)_\ell$, which is clearly more satisfactory 289
a priori, especially from the point of view of the Lefschetz formula, [**4**, Chap V]. Let us recall here Weil's results on the structure of $T_\ell(A)$:

a) If n is prime to $\mathrm{char}(k)$, then K_n is a free module of rank $2 \dim A$ over $\mathbb{Z}/n\mathbb{Z}$; hence, if ℓ is a prime number $\neq \mathrm{char}(k)$, $T_\ell(A)$ is a free module of rank $2 \dim A$ over the ring $\mathbb{Z}_\ell$ of ℓ-adic integers;
b) If n is a power of $\mathrm{char}(k) = p$, then K_n is a free module of rank $\nu \leq \dim A$ over $\mathbb{Z}/n\mathbb{Z}$, with ν independent of n; hence $T_p(A)$ is a free module of rank $\nu \leq \dim A$ over the ring $\mathbb{Z}_p$ of p-adic integers.

This shows that, in the theory of the fundamental group developed here, the fundamental group of a varying abelian variety does not vary regularly with the parameter: its p-primary component may drop abruptly at values of the parameter t corresponding to

residual characteristic p. The best-known case of this phenomenon is that of elliptic curves. Note, however, that, whatever n may be (whether or not it is prime to the characteristic), the actual kernel ${}_nA$ in A of multiplication by n is a finite group scheme over k of degree n^{2g}, where $g = \dim A$, and it is nonseparable over k if n is a multiple of $p = \mathrm{char}(k)$. Moreover, when A varies in a family of abelian varieties, i.e. when there is an abelian scheme A over a base scheme S, one proves more generally that ${}_nA$ is a finite flat group scheme over S, of degree n^{2g} over S; that is, provided one takes the infinitesimal parts of the kernels ${}_nA$ into account, they behave regularly for every n. This suggests that the "true"fundamental group of an abelian variety A is the pro-algebraic group (the formal inverse limit of finite groups over k) $\varprojlim_n {}_nA$, where by the "true fundamental group"of an algebraic scheme X one means the pro-group that classifies the principal coverings of X whose structural group is an arbitrary finite group G over k (not necessarily separable over k). In this way, for example, the representations of $\mathrm{Hom}(A, A)$ on the p-primary component of the true fundamental group of A recover the Weil characteristic polynomial, defined by Weil using the primes $\ell \neq p$, more naturally than does Serre's construction [**8**].

XI.3. Projective cones, an example of Zariski

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Continue to assume, for simplicity, that k is an algebraically closed field, and let V be a connected projective k-scheme, a closed subscheme of $\mathbb{P}^r_k$, which may, if desired, be assumed nonsingular. Let $Y = \hat{C}$ be the projective cone over V, let y_0 be its vertex, let $X = \hat{C}_V$ be the usual projective completion of the vector bundle $C_V = \mathbf{V}(\mathcal{O}_V(1))$ associated with $\mathcal{O}_V(1)$, and finally let

$$f\colon X \to Y$$

be the canonical morphism, contracting the zero section X_0 of C_V in X to a point (EGA II 8.6.4). Since X is a locally trivial bundle over V, with fibers $\mathbb{P}^1$ and hence simply connected fibers, the morphism $p\colon X \to V$ induces by (X X.1.4) an isomorphism:

$$\pi_1(X) \xrightarrow{\sim} \pi_1(V).$$

Since p induces an isomorphism $X_0 \to V$, we conclude that *an étale covering of X is completely split if and only if its restriction to X_0 is completely split.* Now, for every étale covering Y' of Y, the inverse image $X' = X \times_Y Y'$ is an étale covering of X that is completely split over the fiber X_0, and hence is trivial. Since the homomorphism $\pi_1(X) \to \pi_1(Y)$ is surjective (IX IX.3.4), we conclude that

$$\pi_1(Y) = (e)$$

in other words, *every projective cone is simply connected.* (N.B. in characteristic 0, the same result holds when Y is taken to be the affine cone).

Now suppose that V is regular, i.e. smooth over k; then X is regular, and for a suitable projective embedding of V, one obtains a *normal* projective cone Y. If V is not simply connected, and hence X is not simply connected, let X' be a nontrivial connected étale covering of X. Since the fibers of X over points $y \in Y$ distinct from y_0 are reduced to a point, we see that the restriction of X' to these fibers (in particular, to the generic fiber) is trivial; nevertheless, X is not obtained as the inverse image of an étale covering 291
of Y, since Y is simply connected and X' would be completely split. This shows that (X X.1.3 and X.1.4) become false if the hypothesis that f is separable is replaced by the weaker hypothesis that its fibers are separable algebraic schemes (or even smooth ones) over the $\kappa(s)$. Likewise, note that the fundamental groups of the geometric fibers $\overline{X}_y$ for $y \neq y_0$ are obviously reduced to (e), since these fibers are reduced to a point, whereas $\pi_1(X_0) \neq e$; thus the semicontinuity theorem (X X.2.4) likewise fails for f.

Finally, let us give the example, pointed out by Zariski, in which these same theorems fail when the hypothesis that f is separable is replaced by the hypothesis that f is flat.

Let $f\colon X \to Y$ be a morphism from a nonsingular projective surface to the rational line $Y = \mathbb{P}^1$, such that $K = k(X)$ is a "regular"extension of $k(f)$, i.e. primary and separable (i.e. the geometric generic fiber is connected and separable), and such that the divisor $(f) = X_0 - X_\infty$ is an n-fold multiple of a divisor (where n is an integer prime to the characteristic). It is possible to construct such examples in every characteristic. Let X' be the normalization of X in $K(f^{1/n})$, where $K = k(X)$ is the function field of X. The hypothesis on (f) implies that X' is étale over X. Let Y' be the normalization of Y in $k(t)(t^{1/n})$; it is ramified over Y exactly at the points $t = 0$ and $t = \infty$, and the restriction $X'|f^{-1}(U)$ is isomorphic to the inverse image of $Y'|U$. In particular, the restriction of X' to the *geometric* generic fiber of X over Y is completely split. Nevertheless, X' is not isomorphic to the inverse image of an étale covering of Y, since one sees at once that the latter would necessarily be Y', which is absurd because Y' is ramified over Y [1].

Here is (following Serre) a simple way to realize the conditions of this example, inspired by [**5**, no. 20]: take n to be a prime number ≥ 5, distinct from the characteristic, and let $G = \mathbb{Z}/n\mathbb{Z}$ act on $\underline{k}^{4}$ [*] by multiplying the coordinates by four distinct characters of G (which is possible since $n \geq 5$). Then G acts on projective space $\mathbb{P}^3_k$, and the only points fixed by G are the four points corresponding to the coordinate axes. The surface X' with equation $x^n + y^n + z^n + t^n = 0$ is smooth over k (Jacobian criterion), and contains none 292
of the fixed points; thus, since G has prime order, it acts on X "without fixed points", i.e. X is a principal covering of $X = X'/G$ with group G. Let $g = x/y$ in $k(X') = K'$; it is a Kummer generator of K' over $K = k(X)$ if the chosen characters were χ^i, $i = 0, 1, 2, 3$, with χ a primitive character. Let f be its n-th power, which is an element of K. We see at once that K' is a regular extension of $k(g)$; this follows from the fact that the plane curve with homogeneous equation in U, T, Z, $T^n + Z^n + (1 + g^n)U^n = 0$, is smooth over $k(g)$ (Jacobian criterion), and from the fact that every plane curve is known to be connected. On the other hand, $k(f) = K \cap k(g)$, since the right-hand side is an extension of $k(f)$ contained in $k(g)$, an extension of prime degree, and is different from $k(g)$ (since $g \notin K$). This implies that K is a regular extension of $k(f)$. Finally, the divisor of f on X is an n-fold multiple of a divisor, since its inverse image on X' is the divisor of g^n, and hence an n-fold multiple, and one can descend because X' is étale over X. We would be done if the rational map $f\colon X \to \mathbb{P}^1$ were a morphism, that is, if the divisors of the zeros and poles of f did not meet. In fact, one checks easily (again by looking on X') that the two divisors in question are the multiples by n of two smooth curves over k, meeting transversely at a point a. If we now replace X by the scheme obtained by blowing up a, the preceding conditions ($\mathrm{div}(f)$ divisible by n, and $k(X_1) = k(X)$ a regular extension of $k(X)$) remain satisfied; moreover, f is a *morphism* $X_1 \to \mathbb{P}^1$, so the required conditions hold.

XI.4. The cohomology exact sequence

Let S be a prescheme. The category $\mathbf{Sch}_{/S}$ of preschemes over S is then determined, as is the notion of a group in this category; such a group will also be called a *group prescheme over S*, or simply an *S-group*. To simplify the exposition and fix ideas, we shall most often restrict ourselves below to groups which are *affine* and *flat* over S [2]. This will suffice for the applications we have in mind, although of course many cases arise in which neither hypothesis holds.

Let G be such an S-group, and let P be a prescheme over S on which G acts on the 293

[1] One may observe, from the viewpoint of the "étale topology"(SGA 4 VII), that in this example $\mathrm{R}^1(f_{\text{ét}})_*(\mathbb{Z}/n\mathbb{Z})$ is "not separated"over S.

[*] **[**Added in 2003: $\underline{k}^4$ denotes affine space of dimension 4 over k.**]**

[2] In fact, for what follows, quasi-affine instead of affine would suffice; see footnote (5), p. 183.

right. In particular, there is a morphism

$$\pi\colon P \times_S G \longrightarrow P$$

in the category $\mathbf{Sch}_{/S}$, satisfying the familiar axioms. We say that P is *formally principal homogeneous under* G if the morphism

$$P \times_S G \longrightarrow P \times_S P$$

whose components are pr_1 and π is an isomorphism. Equivalently, for every object S' of $\mathbf{Sch}_{/S}$, the set

$$P(S') = \mathrm{Hom}_S(S', P),$$

regarded as a set with operator group

$$G(S') = \mathrm{Hom}_S(S', G),$$

is either empty or principal homogeneous, i.e. either empty or isomorphic to $G(S')$ with $G(S')$ acting by right translations.

We say that P is *trivial* if it is isomorphic to G, on which G acts by right translations, or equivalently if each of the operator sets $P(S')$ under $G(S')$ is trivial. One verifies, for example by the standard procedure of passing to the set-theoretic case, that *P is trivial if and only if it is formally principal homogeneous and has a section over S*. The latter condition may be stated categorically by saying that P has a section over the final object $e = S$ of $\mathbf{Sch}_{/S}$, i.e. that there is a morphism from e to P.

To define a principal homogeneous bundle P under G, a notion stronger than that of a formally principal homogeneous bundle, one must first specify in $\mathbf{Sch}_{/S}$ a class of morphisms to be used for descent and to play the role of localization morphisms that trivialize bundles. The most suitable choice varies with the context, and no one choice contains all the others[3]. Here it will be convenient to adopt the following definition:

Definition XI.4.1. *Let G be an S-group. A* principal homogeneous bundle *on the right under G is an S-prescheme P with a right action of the S-group G, such that there exist a covering of S by open subsets U_i, and for every i a faithfully flat and quasi-compact base-change morphism $S'_i \to U_i$, for which $P' = P \times_S S'$ is a trivial prescheme with operators under $G' = G \times_S S'$, where S' is the S-prescheme given by the disjoint sum of the S'_i.*

Observe that the base-change functor $X \longmapsto X' = X \times_S S'$, being left exact, sends 294
groups to groups and objects with operator groups to objects with operator groups. Definition XI.4.1 is stable under base change. We also have:

Proposition XI.4.2. *Let G be an S-group, flat and quasi-compact over S, and let P be an S-prescheme on which G acts on the right. The following conditions are equivalent:*

(i) *P is a principal homogeneous bundle under G;*
(ii) *P is formally principal homogeneous under G, and the structural morphism $P \to S$ is faithfully flat and quasi-compact.*

If P is principal homogeneous under G, then, with the notation of XI.4.1, P' is faithfully flat and quasi-compact over S', since G' is so and P' is S'-isomorphic to G'. Hence P has the same properties over S. For surjectivity and quasi-compactness, see VIII VIII.3.1; flatness was omitted from the sorites of Exposé VIII. Conversely, if (ii) holds, take the base change $S' = P$, which is faithfully flat and quasi-compact over S. Then P' is formally principal homogeneous over S', since P is so over S and the base-change functor is left exact. Moreover, P' has a section over S', namely the diagonal section. It is therefore a trivial principal bundle, which completes the proof.

[3] See SGA 3, IV, especially §4.

COROLLARY XI.4.3. *If G is affine and flat over S, every principal homogeneous bundle P under G is affine and flat over S.*

Indeed, it becomes so after a faithfully flat and quasi-compact extension of the base, and VIII VIII.5.6 applies.

The usefulness of Definition XI.4.1 for S-groups *flat* and *affine* over S rests on VIII VIII.2.1: the morphisms $S' \to S$ considered in XI.4.1 are effective descent morphisms for the category of preschemes affine over other preschemes. Thanks to this fact, the verification of the statements outlined below is essentially categorical[4].

Let E be an S-prescheme on which the S-group G acts on the left, and let P be a principal homogeneous bundle on the right under G. We wish to define an associated bundle $E^{(P)}$, locally isomorphic to E. Let G act on the right on $P \times_S E$ by 295

$$(x, y) \longmapsto (xg, g^{-1}y),$$

the formula describing such actions in the set-theoretic context, and extended to categories by the standard procedure. Subject to its existence, set

$$E^{(P)} = (P \times_S E)/G.$$

Then $P \times_S E$ is a prescheme over $T = E^{(P)}$ with right operator group $G_T = G \times_S T$. To work comfortably, we would like $P \times_S E$ to be, in addition, a principal homogeneous bundle over T with group G_T.

To verify the existence of $E^{(P)}$ and this additional property, take the S' of Definition XI.4.1 and pass to the inverse image over S' of the original situation. Since P' is trivial, i.e. isomorphic to G'_d, one sees immediately that $E'^{(P')}$ exists and has the required exactness property. Indeed, $E' \times_{S'} P'$ is G'-isomorphic to $E' \times_{S'} G'$, so $E'^{(P')}$ is isomorphic to E'.

Moreover, formation of the associated bundle in the case of a trivial object with operators commutes with every base extension. Applying this to the various base extensions

$$S'' \rightrightarrows S' \qquad \text{and} \qquad S''' \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; S',$$

where S'' and S''' are the double and triple fiber products of S' over S, shows that $E'^{(P')}$ *is endowed with a descent datum relative to $S' \to S$, and that $E^{(P)}$ exists with the required properties if and only if this descent datum is effective.* In that case $E^{(P)}$ is, of course, precisely the descended object. Here one uses the fact that $S' \to S$ is a descent morphism in the category of S-preschemes; see VIII VIII.5.2. It follows that *the associated bundle exists if E is affine over S.*

We shall apply this construction when an S-group homomorphism $G \to H$ is given, taking for E the S-prescheme H with the left action of G induced by the given morphism. The group H acts on itself on the right in a way that commutes with the action of G on H, and formation of the associated bundle, whenever it exists over S, commutes with base extension. It follows readily that H acts on the right on $P^{(H)}$, which is therefore a principal homogeneous bundle under H in the sense of XI.4.1, and is in fact trivialized by 296 the same morphism $S' \to S$ as P. In particular, *to every principal homogeneous bundle P under G and every homomorphism of S-groups $G \to H$, with H affine over S, there is functorially associated a principal homogeneous bundle with group H.* This construction is functorial in $(G \to H)$ and compatible with arbitrary base changes $T \to S$.

DEFINITION XI.4.4. *Let G be an S-prescheme. We denote by $\mathrm{H}^0(S, G)$ the set of sections of G over S, regarded as a group when G is an S-group. In that case, $\mathrm{H}^1(S, G)$*

[4]Cf. loc. cit. in footnote (3), p. 181.

denotes the set of isomorphism classes of principal homogeneous bundles over S with group G, endowed with the distinguished point corresponding to trivial bundles[5].

Thus $\mathrm{H}^0(S,G)$ is a functor in the S-prescheme G, with values in the category of sets. This functor is left exact, and a fortiori commutes with finite products; it therefore sends groups to groups and commutative groups to commutative groups. Similarly, by formation of associated bundles, $\mathrm{H}^1(S,G)$ is a functor in the *affine* S-group G, with values in the category of pointed sets, and one readily verifies that it commutes with finite products. In particular, it sends groups in the category of affine S-groups, i.e. *commutative affine S-groups*, to groups in the category of pointed sets, and indeed to commutative groups, since the group objects in the first category are commutative. Thus, *if G is a commutative affine S-group, then $\mathrm{H}^1(S,G)$ is a commutative group*, and a homomorphism $G \to H$ of commutative affine S-groups gives rise to a group homomorphism

$$\mathrm{H}^1(S,G) \longrightarrow \mathrm{H}^1(S,H).$$

For simplicity, in what follows we restrict ourselves to *affine commutative* S-groups. Let

$$0 \longrightarrow G' \xrightarrow{u} G \xrightarrow{v} G'' \longrightarrow 0$$

be a sequence of morphisms of such groups. *We shall say that this sequence is exact if* $vu = 0$, which allows G to be regarded as a prescheme over G'' with right operator group $G'_{G''}$, *and if G is a principal homogeneous bundle over G'' with group $G'_{G''} = G' \times_S G''$.* 297
This implies in particular that $u\colon G' \to G$ is a kernel of v, and hence that the sequence

$$0 \longrightarrow \mathrm{H}^0(S,G') \longrightarrow \mathrm{H}^0(S,G) \longrightarrow \mathrm{H}^0(S,G'')$$

is exact. It also allows one to define a map

$$\partial\colon \mathrm{H}^0(S,G'') \longrightarrow \mathrm{H}^1(S,G')$$

as follows. To every section of G'' over S, i.e. every S-morphism $f\colon S \to G''$, associate the principal homogeneous bundle P_f over S, with group $G' \simeq f^*(G'_{G''})$, obtained by inverse image of the principal homogeneous bundle G over G''. As an S-prescheme, this is simply the inverse image under $v\colon G \to G''$ of the subprescheme which is the image of S under the immersion f; the action of G' on P_f is induced by the right action of G' on G.

We also leave to the reader the verification of the following proposition, which presents no difficulty apart from its formulation:

Proposition XI.4.5. *The map $\partial\colon \mathrm{H}^0(S,G'') \to \mathrm{H}^1(S,G')$ is a group homomorphism. The following sequence of homomorphisms is exact:*

$$0 \longrightarrow \mathrm{H}^0(S,G') \xrightarrow{u^0} \mathrm{H}^0(S,G) \xrightarrow{v^0} \mathrm{H}^0(S,G'') \xrightarrow{\partial} \mathrm{H}^1(S,G') \xrightarrow{u^1} \mathrm{H}^1(S,G) \xrightarrow{v^1} \mathrm{H}^1(S,G'').$$

All the homomorphisms other than ∂ arise from the functoriality of H^0, respectively H^1.

Remarks XI.4.6. The point of view presented here for the study of principal homogeneous bundles is visibly inspired by Serre [**7**], which the reader is strongly advised to consult. For a formalism that also applies to structural S-groups quasi-projective over S, so as in particular to include projective abelian schemes, it is preferable to modify XI.4.1 by requiring S' to be a disjoint sum of preschemes S'_i which are finite and locally free over open subsets S_i covering S. The preceding developments, including in particular XI.4.5, then remain valid after replacing the affine hypothesis everywhere by the quasi-projective hypothesis, and interpreting accordingly the definition given above of an exact sequence of S-groups. Indeed, one need only replace the reference to VIII VIII.2.1 by VIII VIII.7.7: 298

[5]This notation is compatible with the general cohomological notation (SGA 4, V) only when descent effectivity criteria are available. These are hardly assured unless G is affine, or merely quasi-affine; see VIII VIII.7.9.

the morphisms $S' \to S$ used here are effective descent morphisms for the fibered category of preschemes quasi-projective over other preschemes. One should note, however, that this second notion of principal homogeneous bundle is more restrictive than the first one in XI.4.1.

. An even more restrictive notion of principal homogeneous bundle is obtained by requiring S to be covered by open subsets S_i such that, for every i, $P|S_i$ is a trivial bundle with operators under $G|S_i$. We then say that P is a *locally trivial principal homogeneous bundle*. For fixed G, the classes of these bundles form a subset of $\mathrm{H}^1(S, G)$, in bijection with $\mathrm{H}^1(S, \mathcal{O}_S(G))$, where $\mathcal{O}_S(G)$ is the sheaf, in the ordinary sense, of sections of G over S; see [**2**]. For these H^1 sets still to give rise to the cohomology exact sequence XI.4.5, one must plainly suppose that

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$

is exact in the appropriate sense for this new context: G must be a locally trivial bundle over G'' with group $G'_{G''}$. Equivalently, $u\colon G' \to G$ is a kernel of $v\colon G \to G''$, and G locally admits a section over G''.

. It is plainly very desirable to continue the exact sequence XI.4.5 by introducing the higher cohomology groups $\mathrm{H}^i(S, G)$. This is possible in the framework of *Weil cohomology*. Consider the category $\mathcal{B}$ of preschemes quasi-compact over S, equipped with the class $\mathcal{M}$ of faithfully flat and quasi-compact morphisms, called localizing morphisms. An abelian *Weil sheaf* on S, or more precisely on $(\mathcal{B}, \mathcal{M})$, is a contravariant functor $\mathcal{F}$ from $\mathcal{B}$ to the category of abelian groups, sending sums to products and every sequence

$$T'' = T' \times_T T' \overset{\mathrm{pr}_1, \mathrm{pr}_2}{\rightrightarrows} T' \overset{f}{\longrightarrow} T\ , \qquad f \in \mathcal{M},$$

to an *exact* diagram of sets

$$\mathcal{F}(T) \longrightarrow \mathcal{F}(T') \rightrightarrows \mathcal{F}(T'').$$

Weil sheaves form an abelian category with exact direct limits and a generator, and hence with enough injective objects [**1**]. The right derived functors of

$$\Gamma(\mathcal{F}) = \mathcal{F}(S)$$

are denoted by $\mathrm{H}^i(S, \mathcal{F})$. Moreover, every commutative S-group plainly defines a Weil sheaf (VIII VIII.5.2), whose H^0 and H^1 are precisely $\mathrm{H}^0(S, G)$ and $\mathrm{H}^1(S, G)$. This gives a reasonable definition of the remaining groups $\mathrm{H}^i(S, G)$. One also proves that an exact 299
sequence of S-groups defines an exact sequence of Weil sheaves, recovering and extending the exact sequence XI.4.5[6].

. It would be appropriate to develop the noncommutative variants of XI.4.5, as in [**2**]. For a systematic development, in the proper framework, of the various cohomological notions outlined in this section, we refer to a work then in preparation by J. Giraud[7].

XI.5. Special cases of principal bundles

Suppose now that S is connected and endowed with a geometric point a. Its fundamental group $\pi_1(S, a)$ classifies the étale coverings of S: the category of étale coverings of S is equivalent to the category of finite sets on which π_1 acts continuously. It follows that a finite étale group scheme G over S is determined essentially by an ordinary finite group $\mathcal{G}$, on which π_1 acts continuously by group automorphisms. An étale covering P of S on which G acts on the right is determined essentially by a finite set $\mathcal{P}$ on which

[6]For a systematic study of this point of view, cf. SGA 4, I through IX.

[7]Cf. J. Giraud, *Cohomologie non abélienne*, Springer-Verlag, 1971.

π_1 acts continuously on the left, and on which $\mathcal{G}$ acts on the right compatibly with the action of π_1:

$$s(p \cdot g) = (sp) \cdot (sg) \qquad \text{for } s \in \pi_1,\ p \in \mathcal{P},\ g \in \mathcal{G}.$$

One verifies that P is a principal homogeneous bundle in the sense of XI.4.1 if and only if $\mathcal{P}$ is a principal homogeneous set under $\mathcal{G}$, for example by using the criterion XI.4.2. In other words, *the category of principal homogeneous bundles over S with group G is equivalent to the category of principal homogeneous bundles with group G in the category of finite sets on which π_1 acts continuously.* In particular, one obtains a canonical bijection, functorial in G:

$$(*) \qquad \mathrm{H}^1(S, G) \xrightarrow{\sim} \mathrm{H}^1(\pi_1, \mathcal{G}).$$

Here the right-hand side denotes the set of isomorphism classes of principal homogeneous 300
bundles under $\mathcal{G}$ in the category of finite sets on which π_1 acts; there is no need to specify continuity. As is well known, this set is the quotient of the set $Z^1(\pi_1, \mathcal{G})$ of 1-cocycles $\varphi\colon \pi_1 \to \mathcal{G}$, satisfying

$$\varphi(1) = 1, \qquad \varphi(st) = \varphi(s)(s \cdot \varphi(t)),$$

by the natural action of $\mathcal{G}$.

An important case is that in which π_1 acts trivially on $\mathcal{G}$, i.e. when G is a completely split covering of S, isomorphic to the disjoint sum of $\mathcal{G}$ copies of S. One then also writes $\mathrm{H}^1(S, \mathcal{G})$ in place of $\mathrm{H}^1(S, G)$, and the bijection $(*)$ identifies this set with

$$\mathrm{H}^1(\pi_1, \mathcal{G}) = \mathrm{Hom}(\pi_1, \mathcal{G})/\{\text{inner automorphisms of } \mathcal{G}\}.$$

In this case, a principal homogeneous bundle over S with group G is nothing other than a *principal covering* of S with group $\mathcal{G}$ (V V.2.7). The preceding bijection is the one deduced from the correspondence between principal coverings of S with group $\mathcal{G}$, *pointed* above a, and continuous homomorphisms from $\pi_1(S, a)$ to $\mathcal{G}$; see the end of V, no. V.5.

The usefulness of connecting the theory of étale coverings with that of principal bundles, already implicit in A. Weil, *Généralisation des Fonctions Abéliennes*, and made explicit by S. Lang in his geometric class-field theory; see Serre [**5**], stems from the following fact. Every S-group finite and étale over S can be embedded into an S-group H, affine and smooth over S, with connected fibers, and commutative when G is commutative. The exact sequence XI.4.5, together where necessary with its noncommutative variants, then allows the discrete classification of principal coverings with group G to be studied through the continuous classification of principal bundles with group H, and conversely. For the idea of the general construction of the embedding $G \to H$, which appears to be used rather little in practice, see [**5**, VI 2.8]. In the next section we shall develop only two important classical special cases. We shall need the following auxiliary result:

Proposition XI.5.1. *Let S be a prescheme, and let G be an S-group isomorphic* 301
to $\mathbf{GL}(n)_S$, for example $\mathbf{G}_{m\,S}$, or to $\mathbf{G}_{a\,S}$. Then every principal homogeneous bundle under G is locally trivial.

Here $\mathbf{GL}(n)_S$, with $n \geq 0$, denotes the S-group representing the contravariant functor

$$T \longmapsto \mathbf{GL}(n, \Gamma(T, \mathcal{O}_T))$$

in the S-prescheme T. In particular, the multiplicative group $\mathbf{G}_{m\,S}$ represents the contravariant functor

$$T \longmapsto \Gamma(T, \mathcal{O}_T^*),$$

and hence, as a prescheme over S, is isomorphic to

$$\mathrm{Spec}(\mathcal{O}_S[t, t^{-1}]),$$

where t is an indeterminate. Similarly, $\mathbf{G}_{a\,S}$ represents the contravariant functor

$$T \longmapsto \Gamma(T, \mathcal{O}_T),$$

and is therefore isomorphic, as an S-prescheme, to $\mathrm{Spec}(()\mathcal{O}_S[t])$. By dévissage, XI.5.1 recovers Rosenlicht's local-triviality result for the case in which G has a composition series whose successive factors are groups of the kind considered here. For a finer study of local triviality questions for principal homogeneous bundles, see [**7**] and [**3**].

The first assertion is proved by observing that

$$G(T) = \mathrm{Aut}(\mathcal{O}_T^n),$$

and that the morphisms $S' \to S$ occurring in XI.4.1, i.e. the faithfully flat and quasi-compact morphisms, are effective descent morphisms for the fibered category of modules locally isomorphic to $\mathcal{O}_T^n$, i.e. locally free of rank n (VIII VIII.1.12). The second assertion is proved similarly, noting that in this case

$$G(T) = \mathrm{Aut}(\mathcal{E}_T),$$

where $\mathcal{E}_T$ is the *trivial extension* of $\mathcal{O}_T$ by $\mathcal{O}_T$, and where the automorphisms must of course preserve the extension structure. The morphisms $S' \to S$ in XI.4.1 are effective descent morphisms for the fibered category of extensions of $\mathcal{O}_T$ by $\mathcal{O}_T$, as follows readily from VIII VIII.1.1, and such extensions are automatically locally trivial.

Remark XI.5.2. The same type of proof applies to the symplectic group $\mathbf{Sp}(2n)_S$, since a nondegenerate alternating form on a module locally isomorphic to $\mathcal{O}_S^{2n}$, i.e. a form defining an isomorphism of that module with its dual, is locally isomorphic to the standard form. The corresponding result for the orthogonal group is false, however, already when S is the spectrum of a field, since there may be quadratic forms over a field which are not isomorphic to the standard form. In fact, it is proved essentially in [**3**] that 302
the groups $\mathbf{GL}$, $\mathbf{Sp}$, $\mathbf{G}_a$, and those that can be dévisséd into such groups are, up to minor qualifications, the only ones for which a local-triviality result of the kind considered here holds.

Corollary XI.5.3. *There are canonical bijections*

$$\mathrm{H}^1(S, \mathbf{GL}(n)_S) \xleftarrow{\sim} \mathrm{H}^1(S, \mathbf{GL}(n, \mathcal{O}_S)),$$

in particular

$$\mathrm{H}^1(S, \mathbf{G}_{m\,S}) \xleftarrow{\sim} \mathrm{H}^1(S, \mathcal{O}_S^*),$$

and

$$\mathrm{H}^1(S, \mathbf{G}_{a\,S}) \xleftarrow{\sim} \mathrm{H}^1(S, \mathcal{O}_S),$$

where the right-hand sides denote the cohomology of the topological space S with coefficients in ordinary sheaves.

In particular, $\mathrm{H}^1(S, \mathbf{GL}(n)_S)$ is identified with the set of isomorphism classes of locally free modules of rank n on S, and $\mathrm{H}^1(S, \mathbf{G}_{a\,S})$ with the set of extension classes of the module $\mathcal{O}_S$ by itself.

XI.6. Application to principal coverings: Kummer and Artin–Schreier theories

Proposition XI.6.1. *Let S be a prescheme, let $n > 0$ be an integer, let*

$$u_n \colon \mathbf{G}_{m\,S} \longrightarrow \mathbf{G}_{m\,S}$$

be the n-th power homomorphism, and let $\boldsymbol{\mu}_{n\,S}$ be its kernel. Then $\boldsymbol{\mu}_{n\,S}$ is finite and locally free of rank n over S, and is étale over S if and only if the characteristic of every $s \in S$ is prime to n. The sequence of homomorphisms

$$0 \longrightarrow \boldsymbol{\mu}_{n\,S} \longrightarrow \mathbf{G}_{m\,S} \xrightarrow{u_n} \mathbf{G}_{m\,S} \longrightarrow 0$$

is exact in the sense of Section XI.4. It will be called the Kummer exact sequence *over S, relative to the integer n.*

We have 303

$$\mathbf{G}_m = \mathrm{Spec}()O_S[t, t^{-1}],$$

and u_n corresponds to the homomorphism on affine $\mathcal{O}_S$-algebras given by

$$u_n(t) = t^n.$$

The unit section of $\mathbf{G}_{m\,S}$, on the other hand, corresponds to the augmentation homomorphism of $\mathcal{O}_S$-algebras given by

$$\varepsilon(t) = 1,$$

whose kernel is the principal ideal $(t-1)$. Its image under u_n is therefore the principal ideal $(1-t^n)$, and hence

$$\boldsymbol{\mu}_{n\,S} = \mathrm{Spec}()(\mathcal{O}_S[t]/(1-t^n)).$$

This shows in particular that $\boldsymbol{\mu}_{n\,S}$ is finite over S and is defined by an algebra over S which is free of rank n, with basis t^i, $0 \leq i \leq n-1$.

For this group to be étale at $s \in S$, it is necessary and sufficient that the reduced algebra $k[t]/(1-t^n)$, where $k = \kappa(s)$, obtained by formally adjoining the n-th roots of unity to k, be separable over k. Equivalently, the roots of $1-t^n$ in an algebraic closure of k must all be distinct, which is the same as n being prime to the characteristic.

Finally, by the criterion XI.4.2, exactness of the sequence in XI.6.1 reduces to proving that u_n is faithfully flat. We may suppose S affine with ring A, so $\mathbf{G}_{m\,S}$ is affine with ring $B = A[t, t^{-1}]$. It is enough to check that u_n makes B a free module of rank n over B, or equivalently that u_n is injective and that

$$A[t, t^{-1}]$$

is a free module of rank n over $A[t^n, t^{-n}]$. The elements t^i, $0 \leq i \leq n-1$, plainly form a basis, completing the proof.

Definition XI.6.2. *The group $\boldsymbol{\mu}_{n\,S}$ is called the* Kummer group of rank n over S, *and a* Kummer principal covering of rank n over S *is a principal homogeneous bundle* 304 *over S whose group is the Kummer group of rank n.*

The set of these coverings is a group, denoted $\mathrm{H}^1(S, \boldsymbol{\mu}_{n\,S})$, or simply $\mathrm{H}^1(S, \boldsymbol{\mu}_n)$. Formation of the Kummer group of rank n over S is compatible with base extension, so $\boldsymbol{\mu}_{n\,S}$ comes by base extension from the *absolute Kummer group* $\boldsymbol{\mu}_n$ over $\mathrm{Spec}(()\mathbb{Z})$.

Write $(\mathbb{Z}/n\mathbb{Z})_S$ for the S-group defined by the ordinary finite group $\mathbb{Z}/n\mathbb{Z}$. If G is any S-group, the homomorphisms of S-groups

$$u\colon (\mathbb{Z}/n\mathbb{Z})_S \longrightarrow G$$

are in bijection, compatibly with base change, with the sections of G over S whose n-th power is the unit section. The correspondence sends u to the image under u of the section of $(\mathbb{Z}/n\mathbb{Z})_S$ over S defined by the generator $1 \bmod n\mathbb{Z}$. Thus:

Corollary XI.6.3. *If $\boldsymbol{\mu}_{n\,S}$ is étale over S, there is a bijection between isomorphisms of S-groups*

$$(\mathbb{Z}/n\mathbb{Z})_S \xrightarrow{\sim} \boldsymbol{\mu}_{n\,S}$$

and sections of $\mathcal{O}_S^*$ having exact order n on every connected component of S. Such a section is called a primitive n-th root of unity over S. *Consequently, $\boldsymbol{\mu}_{n\,S}$ is isomorphic as an S-group to $(\mathbb{Z}/n\mathbb{Z})_S$ if and only if it is étale over S, i.e. the residue characteristics of S are prime to n, and there exists a primitive n-th root of unity over S.*

This explains the role in classical Kummer theory of the hypotheses that the base field, playing the role of S, have characteristic prime to n and contain the n-th roots of unity, together with the choice of a primitive n-th root of unity. Once the language of schemes is available, these hypotheses need no longer encumber the theory: one should reason directly with $\boldsymbol{\mu}_n$, rather than with $\mathbb{Z}/n\mathbb{Z}$. Combining XI.6.1, XI.4.5, and XI.5.3

gives the following general relation between the theory of Kummer principal coverings and that of Picard groups:

PROPOSITION XI.6.4. *Let S be a prescheme. There is a canonical exact sequence* 305

$$0 \to \mathrm{H}^0(S, \boldsymbol{\mu}_n) \to \mathrm{H}^0(S, \mathcal{O}_S^*) \to \mathrm{H}^0(S, \mathcal{O}_S^*) \to \mathrm{H}^1(S, \boldsymbol{\mu}_n) \to \mathrm{H}^1(S, \mathcal{O}_S^*) \to \mathrm{H}^1(S, \mathcal{O}_S^*).$$

On setting $\mathrm{H}^1(S, \mathcal{O}_S^*) = \mathrm{Pic}(S)$, and, for every abelian group A, denoting the kernel and cokernel of multiplication by n on A by ${}_nA$ and A_n, respectively, one obtains the exact sequence

$$0 \longrightarrow \mathrm{H}^0(S, \mathcal{O}_S^*)_n \longrightarrow \mathrm{H}^1(S, \boldsymbol{\mu}_n) \longrightarrow {}_n\mathrm{Pic}(S) \longrightarrow 0.$$

We spell out two important cases in which one or the other outer term of this exact sequence vanishes:

COROLLARY XI.6.5. *Suppose that ${}_n\mathrm{Pic}(S) = 0$, for example if S is the spectrum of a local ring or of a factorial ring, and set*

$$A = \mathrm{H}^0(S, \mathcal{O}_S).$$

Then there is a canonical isomorphism

$$\mathrm{H}^1(S, \boldsymbol{\mu}_n) \xrightarrow{\sim} A^*/A^{*n}.$$

This is essentially the classical statement of Kummer theory when S is the spectrum of a field.

COROLLARY XI.6.6. *Suppose that every element of $\mathrm{H}^0(S, \mathcal{O}_S)$ is an n-th power, for example if $\mathrm{H}^0(S, \mathcal{O}_S)$ is a product of algebraically closed fields, or if S is reduced and proper over an algebraically closed field k. Then there is a canonical isomorphism*

$$\mathrm{H}^1(S, \boldsymbol{\mu}_n) \xrightarrow{\sim} {}_n\mathrm{Pic}(S).$$

In particular, when S is proper and connected over an algebraically closed field k, this relates the fundamental group of S to the finite-order points of the Picard scheme P of S over k. Thus, for n prime to the characteristic, one has an isomorphism

$$\mathrm{Hom}(\pi_1(S), \mathbb{Z}/n\mathbb{Z}) \simeq {}_nP(k),$$

which is often used in algebraic geometry. If the connected component P^0 of P is a 306
complete group scheme of dimension g, the results recalled in Section XI.2, together with finiteness of the Néron–Severi torsion group, show that for every prime ℓ different from the characteristic, the ℓ-primary component of the abelianized fundamental group $\pi_1(S)$ is a finite-type module of rank $2g$ over the ring $\mathbb{Z}_\ell$ of ℓ-adic integers. It is moreover free for all but at most finitely many values of ℓ.

As Serre observed, this can be used under suitable hypotheses to prove that, when X is flat and projective over connected S, the Picard schemes of the fibers of X all have the same dimension, by applying the semicontinuity theorem X X.2.3. Serre's argument applies whenever the Picard scheme of X over S exists and the connected Picard schemes of the fibers of X over S are proper group schemes. This holds, for example, when the geometric fibers of X over S are normal, with X still flat and projective over S, and in particular when X is smooth and projective over S.

Now let p be a prime, and suppose that S is a prescheme of characteristic p, i.e. $p\mathcal{O}_S = 0$. The p-th power map on $\mathcal{O}_S$ is additive, and the corresponding morphism, obtained by replacing S by a variable T over S,

$$F\colon \mathbf{G}_{a\,S} \longrightarrow \mathbf{G}_{a\,S},$$

is therefore a homomorphism of S-groups, called the *Frobenius homomorphism*. Such a morphism is defined for every S-prescheme G obtained by base extension from a prescheme

G_0 over the prime field $\mathbb{Z}/p\mathbb{Z}$, and it is a group homomorphism if G_0 is a group prescheme. We set

$$\wp = \mathrm{id} - F\colon \mathbf{G}_{a\,S} \longrightarrow \mathbf{G}_{a\,S}.$$

Consider, on the other hand, the S-group $(\mathbb{Z}/p\mathbb{Z})_S$ defined by the ordinary finite group $\mathbb{Z}/p\mathbb{Z}$. We have said that, for every S-group G, the homomorphisms of S-groups from $(\mathbb{Z}/p\mathbb{Z})_S$ to G correspond bijectively to the sections of G over S whose p-th power is the identity section. When $G = \mathbf{G}_{a\,S}$, they therefore correspond to arbitrary sections 307
of G over S. Taking in particular the section of $\mathbf{G}_{a\,S}$ over S corresponding to the unit section of the sheaf of rings $\mathcal{O}_S$, we obtain a homomorphism of S-groups

$$i\colon (\mathbb{Z}/p\mathbb{Z})_S \to \mathbf{G}_{a\,S}$$

Proposition XI.6.7. *The sequence of homomorphisms of S-groups*

$$0 \to (\mathbb{Z}/p\mathbb{Z})_S \to \mathbf{G}_{a\,S} \to \mathbf{G}_{a\,S} \to 0$$

is exact (in the sense of no. XI.4). (It is called the Artin–Schreier exact sequence *over S.)*

It is enough to prove this over the prime field $k = \mathbb{Z}/p\mathbb{Z}$. It is enough to observe that the homomorphism $\wp^*\colon k[t] \to k[t]$ defined by $\wp^*(t) = t - t^p$ makes $k[t]$ a free module of rank p over $k[t]$: more precisely, $k[t]$ is a free module over $k[s]$, where $s = t - t^p$, with basis given by the t^i $(0 \leq i \leq p-1)$.

Using XI.4.5 and XI.5.3, we conclude:

Proposition XI.6.8. *There is a canonical exact sequence*

$$0 \to \mathrm{H}^0(S, \mathbb{Z}/p\mathbb{Z}) \to \mathrm{H}^0(S, \mathcal{O}_S) \to \mathrm{H}^0(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \\ \to \mathrm{H}^1(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathcal{O}_S),$$

and hence an exact sequence

$$0 \to \mathrm{H}^0(S, \mathcal{O}_S)/\wp\mathrm{H}^0(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \to \mathrm{H}^1(S, \mathcal{O}_S)^F \to 0,$$

(where the superscript F in the last term denotes the subgroup of invariants under the endomorphism F, equal to the kernel of $\wp = \mathrm{id} - F$).

Let us spell out two extreme cases:

Corollary XI.6.9. *Suppose that $\mathrm{H}^1(S, \mathcal{O}_S)^F = 0$, for example that S is an affine scheme. Then, putting $A = \mathrm{H}^0(S, \mathcal{O}_S)$, there is a canonical isomorphism*

$$\mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \xrightarrow{\sim} A/\wp A.$$

This is *Artin–Schreier theory* in its classical form, at least when S is the spectrum of a field.

Corollary XI.6.10. *Suppose that $\wp\mathrm{H}^0(S, \mathcal{O}_S) = \mathrm{H}^0(S, \mathcal{O}_S)$, for example that $\mathrm{H}^0(S, \mathcal{O}_S)$* 308
is a product of algebraically closed fields, or that S is proper over an algebraically closed field. Then there is a canonical isomorphism

$$\mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \xrightarrow{\sim} \mathrm{H}^1(S, \mathcal{O}_S)^F.$$

Remarks XI.6.11. The last statement is due to J-P. Serre [**9**]. It is also possible to develop an analogous theory for the structure group $\mathbb{Z}/p^n\mathbb{Z}$, for arbitrary n, by using the Witt group scheme $\mathbf{W}_n$ in place of $\mathbf{G}_a$, cf. loc. cit. Notice that in characteristic $p > 0$, Kummer theory no longer gives information about principal coverings of order p, since $\boldsymbol{\mu}_p$ is then an *infinitesimal* group, i.e. radicial over the base, and hence has no direct connection with $\mathbb{Z}/p\mathbb{Z}$. Thus, at first sight, the theory of these coverings can no longer be handled (when, for definiteness, S is a proper scheme over an algebraically closed field) by the theory of the Picard scheme as in XI.6.6. Nevertheless, recalling that

the Zariski tangent space at the origin of $\mathbf{Pic}_{S/K}$[8] is identified with $\mathrm{H}^1(S, \mathcal{O}_S)$, we see that *knowledge of the group scheme* ${}_p\mathbf{Pic}_{S/k}$*, the kernel of multiplication by* p *in* $\mathbf{Pic}_{S/k}$*, determines* $\mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z})$ *as well as* $\mathrm{H}^1(S, \boldsymbol{\mu}_p)$*; note that it also determines* $\mathrm{H}^1(S, \boldsymbol{\alpha}_p)$. Here $\boldsymbol{\alpha}_p$ denotes the infinitesimal group scheme over the prime field that is the kernel of $F\colon \mathbf{G}_a \to \mathbf{G}_a$ (it may also be described as the spectrum of the restricted enveloping algebra of the trivial one-dimensional p-Lie algebra): indeed, the exact sequence XI.4.5 gives in this case

$$\mathrm{H}^1(S, \boldsymbol{\alpha}_p) \simeq \mathrm{Ker}(F\colon \mathrm{H}^1(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathcal{O}_S)),$$

and, more generally, if $\boldsymbol{\alpha}_{p^n}$ denotes the kernel in $\mathbf{G}_a$ of the n-th iterate of F, then

$$\mathrm{H}^1(S, \boldsymbol{\alpha}_{p^n}) \simeq \mathrm{Ker}(F^n\colon \mathrm{H}^1(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathcal{O}_S)).$$

In fact, knowledge of ${}_p\mathbf{Pic}_{S/k}$ is equivalent to knowledge of $\mathrm{H}^1(S, G)$ for every finite commutative algebraic group G annihilated by p. More generally, knowledge of ${}_{p^n}\mathbf{Pic}_{S/k}$ 309
is equivalent to knowledge of $\mathrm{H}^1(S, G)$ for every finite commutative algebraic group G annihilated by p^n, by virtue of the following theorem, which in the situation under consideration encompasses both Kummer theory and Artin–Schreier theory:

Let G be a finite algebraic group over k, and let $D(G) = \mathbf{Hom}_{k\text{-groups}}(G, \mathbf{G}_m)$ be its *Cartier dual* (whose affine algebra is carried by the vector space dual to the affine algebra of G, i.e. by the hyperalgebra of G in the sense of Dieudonné–Cartier). Then there is a canonical isomorphism

$$(*) \qquad \mathrm{H}^1(S, G) \simeq \mathrm{Hom}_{k\text{-groups}}(D(G), \mathbf{Pic}_{S/k}).$$

(N.B. S is a proper scheme over the algebraically closed field k, with $\mathrm{H}^0(S, \mathcal{O}_S) = k$.) This formula can also be expressed by saying that the *true fundamental group* of S alluded to in no. XI.2, after abelianization, is isomorphic to the projective limit of the $D(P_i)$, where P_i ranges over the *finite* algebraic subgroups of $\mathbf{Pic}_{S/k}$. We denote it by $T^\bullet(\mathbf{Pic}_{X/k})$. When S is an abelian variety, we saw in XI.2.1 that this group is also isomorphic to the *true* Tate module $T_\bullet(S) = \varprojlim {}_nS$, and the isomorphism $(*)$ then takes the more striking form

$$\mathrm{Ext}^1(A, G) \simeq \mathrm{Hom}(D(G), B),$$

where A is an abelian variety, B its dual, and G a finite algebraic group over k. The results just indicated can moreover be generalized to the case where k is replaced by an arbitrary base prescheme, and to coefficient groups G other than finite groups.

[8] For the definition of $\mathbf{Pic}_{S/K}$, cf. A. Grothendieck, Sém. Bourbaki no. 232 (February 1962).

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CHAPTER XII

Algebraic geometry and analytic geometry

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Mme M. RAYNAUD[1]

Proceeding as in [**10**], one associates with every scheme X locally of finite type over the field of complex numbers $\mathbb{C}$ an analytic space X^{an} whose underlying set is $X(\mathbb{C})$.

In no. XII.2 and XII.3 of this exposé, we give a "dictionary"between the usual properties of X and X^{an}, and between the properties of a morphism $f\colon X \to Y$ and those of the associated morphism $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$.

We then show that the comparison theorems for coherent sheaves on X and X^{an}, established in [**10**, no. 12] for a projective variety, remain valid when X is a proper scheme.

Finally, in no. XII.5 we prove the equivalence between the category of finite étale coverings of X and the category of finite étale coverings of X^{an}. As a bonus for the reader, we give a new proof of the Grauert-Remmert theorem [**6**], using resolution of singularities [**8**].

XII.1. Analytic space associated with a scheme

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Let X be a scheme locally of finite type over $\mathbb{C}$. Let Φ be the functor from the category of analytic spaces [**4**, no. 9] to the category of sets which assigns to an analytic space $\mathcal{X}$ the set of morphisms of ringed spaces in $\mathbb{C}$-algebras $\mathrm{Hom}_{\mathbb{C}}(\mathcal{X}, X)$. We have the following theorem:

THEOREM AND DEFINITION XII.1.1. *The functor Φ is representable by an analytic space X^{an} and a morphism $\varphi\colon X^{\mathrm{an}} \to X$. We say that X^{an} is the analytic space associated with X.*

If $|X^{\mathrm{an}}|$ is the underlying set of X^{an}, φ induces a bijection from $|X^{\mathrm{an}}|$ onto the set $X(\mathbb{C})$ of points of X with values in $\mathbb{C}$. Moreover, for every point x of X^{an}, the morphism

$$\varphi_x\colon \mathcal{O}_{X,\varphi(x)} \to \mathcal{O}_{X^{\mathrm{an}},x},$$

which is necessarily local, yields upon passage to the completions an isomorphism

$$\widehat{\varphi}_x\colon \widehat{\mathcal{O}}_{X,\varphi(x)} \xrightarrow{\sim} \widehat{\mathcal{O}}_{X^{\mathrm{an}},x},$$

In particular, the morphism φ is flat.

Note that the fact that φ induces a bijection from X^{an} onto $X(\mathbb{C})$ follows from the universal property of X^{an}. We also have the following assertions:

a) If the theorem is true for a scheme Y, it is also true for every subscheme X of Y. Suppose first that X is an open subscheme of Y; if $\psi\colon Y^{\mathrm{an}} \to Y$ is the canonical morphism, $\psi^{-1}(X)$ is an open subset of Y^{an}, which we endow with 313
the analytic-space structure induced by that of Y^{an}. Since every morphism from an analytic space $\mathcal{X}$ to X factors through Y^{an} by the universal property of the latter, and therefore through X^{an}, which is the fiber product $Y^{\mathrm{an}} \times_Y X$, X^{an} is the analytic space associated with X. Finally, the assertion concerning the φ_x is clear.

[1]Based on unpublished notes by A. Grothendieck.

It remains only to consider the case where X is a closed subscheme of Y. Let I be the coherent $\mathcal{O}_Y$-ideal defining X; then $I \cdot \mathcal{O}_{Y^{\mathrm{an}}}$ is a coherent sheaf of ideals of $\mathcal{O}_{Y^{\mathrm{an}}}$ which defines a closed analytic subspace X^{an} of Y^{an}; as in the case of an open subscheme, one sees that X^{an} is the analytic space associated with X. Let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. For every point x of X^{an}, the morphism φ_x is none other than the morphism

$$\mathcal{O}_{Y,\psi(x)}/I_{\psi(x)} \to \mathcal{O}_{Y^{\mathrm{an}},x}/I_{\psi(x)} \cdot \mathcal{O}_{Y^{\mathrm{an}},x}$$

induced by ψ_x; its completion

$$\widehat{\varphi}_x\colon \widehat{\mathcal{O}}_{Y,\psi(x)}/I_{\psi(x)} \cdot \widehat{\mathcal{O}}_{Y,\psi(x)} \to \widehat{\mathcal{O}}_{Y^{\mathrm{an}},x}/I_{\psi(x)} \cdot \widehat{\mathcal{O}}_{Y^{\mathrm{an}},x}$$

is an isomorphism since $\widehat{\psi}_x$ is one, which proves a).

b) If we have two $\mathbb{C}$-schemes X_1, X_2 such that X_1^{an} and X_2^{an} exist, then $(X_1 \times X_2)^{\mathrm{an}}$ exists as well. Indeed, let $\varphi_1\colon X_1^{\mathrm{an}} \to X_1$, $\varphi_2\colon X_2^{\mathrm{an}} \to X_2$ be the canonical morphisms, and let p_1, p_2 be the two projections of $X_1^{\mathrm{an}} \times X_2^{\mathrm{an}}$. It follows formally from EGA I 1.8.1 that $X_1 \times X_2$ is the product of X_1 and X_2 in the category of locally ringed spaces; consequently, the morphisms $\varphi_1 \cdot p_1$ and $\varphi_2 \cdot p_2$ define a morphism $\varphi\colon X_1^{\mathrm{an}} \times X_2^{\mathrm{an}} \to X_1 \times X_2$, and the pair $(X_1^{\mathrm{an}} \times X_2^{\mathrm{an}}, \varphi)$ 314
represents the functor $\mathcal{X} \longmapsto \mathrm{Hom}_{\mathbb{C}}(\mathcal{X}, X_1 \times X_2)$.

c) If $\mathcal{E}^1$ denotes affine space of dimension 1, i.e. the topological space $\mathbb{C}$ endowed with the sheaf of holomorphic functions, the functor* $\mathcal{X} \longmapsto \mathrm{Hom}_{\mathbb{C}}(\mathcal{X}, E^1_{\mathbb{C}})$ is representable by $\mathcal{E}^1$, the canonical morphism $\varphi\colon \mathcal{E}^1 \to E^1_{\mathbb{C}}$ being the evident morphism. Indeed, to give a morphism from an analytic space $\mathcal{X}$ to $E^1_{\mathbb{C}}$ is equivalent to giving an element of $\Gamma(\mathcal{X}, \mathcal{O}_{\mathcal{X}})$, which is also equivalent to giving a morphism from $\mathcal{X}$ to $\mathcal{E}^1$. We plainly have a bijection $|\mathcal{E}^1| \simeq E^1(\mathbb{C})$, and, for every point $x \in \mathcal{E}^1$, the morphism $\widehat{\varphi}_x$ is none other than the identity morphism of a ring of formal power series in one variable over $\mathbb{C}$.

It follows from b) and c) that the theorem is true for affine space $E^n_{\mathbb{C}}$, $n \geq 0$. Using a), one sees that it is also true for every affine scheme X locally of finite type over $\mathbb{C}$. If X is no longer assumed affine and if (X_i) is a covering of X by affine open subsets, it follows from the universal property and from a) that the X_i^{an} glue together and thus define the analytic space X^{an} associated with X.

. Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type. If $\varphi\colon X^{\mathrm{an}} \to X$ and $\psi\colon Y^{\mathrm{an}} \to Y$ are the canonical morphisms, it follows from the universal property of Y^{an} that there exists a unique morphism $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$ such that the diagram

$$\begin{array}{ccc} X^{\mathrm{an}} & \longrightarrow & X \\ {\scriptstyle f^{\mathrm{an}}}\downarrow & & \downarrow{\scriptstyle f} \\ Y^{\mathrm{an}} & \longrightarrow & Y \end{array}$$

is commutative. We have therefore defined a functor Φ from the category of $\mathbb{C}$-schemes 315
locally of finite type to the category of analytic spaces.

The functor Φ commutes with finite inverse limits. It suffices indeed to see that Φ commutes with fiber products. Now, if X, Y, Z are schemes locally of finite type over $\mathbb{C}$, it follows from the fact that $X \times_Z Y$ is the fiber product of X and Y over Z in the category of locally ringed spaces that $X^{\mathrm{an}} \times_{Z^{\mathrm{an}}} Y^{\mathrm{an}}$ satisfies the universal property characterizing $(X \times_Z Y)^{\mathrm{an}}$.

*[Added in 2003: $E^1_{\mathbb{C}}$ denotes the (algebraic) affine line over $\mathbb{C}$.]

. Let X be a $\mathbb{C}$-scheme locally of finite type, let X^{an} be the associated analytic space, and let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. If F is an $\mathcal{O}_X$-module, the inverse image $\varphi^*F = F^{\mathrm{an}}$ is a sheaf of modules over $\mathcal{O}_{X^{\mathrm{an}}}$. This defines a functor from the category of $\mathcal{O}_X$-modules to the category of modules on X^{an}. This functor commutes with inductive limits (EGA 0 4.3.2). Since the sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ is coherent [**4**, no. 18 §2 th. 2], it sends coherent sheaves to coherent sheaves (EGA 0 5.3.11). Moreover:

Proposition XII.1.2. *The functor which assigns to an $\mathcal{O}_{X^{\mathrm{an}}}$-module F its inverse image F^{an} on X^{an} is exact, faithful, and conservative.*

Exactness follows from the fact that the morphism $\varphi\colon X^{\mathrm{an}} \to X$ is flat (XIII.1.1). Let us prove that the functor $F \longmapsto F^{\mathrm{an}}$ is faithful. In view of exactness, it suffices to show that if F^{an} is zero, then so is F. But, for every point x of X^{an}, we then have $F_{\varphi(x)} \otimes_{\mathcal{O}_{X,\varphi(x)}} \mathcal{O}_{X^{\mathrm{an}},x} = 0$. Since the morphism $\mathcal{O}_{X,\varphi(x)} \to \mathcal{O}_{X^{\mathrm{an}},x}$ is faithfully flat, we have $F_{\varphi(x)} = 0$ for every closed point $\varphi(x)$ of X, and, since X is Jacobson (EGA IV 10.4.8), this implies that F is zero.

The fact that the functor $F \longmapsto F^{\mathrm{an}}$ is conservative follows formally from exactness 316
and faithfulness.

XII.2. Comparison of the properties of a scheme and of its associated analytic space

Proposition XII.2.1. *Let X be a $\mathbb{C}$-scheme locally of finite type, let X^{an} be the associated analytic space, and let n be an integer. Consider the property P of being*

(i) *nonempty*
(i') *discrete*
(ii) *Cohen–Macaulay*
(iii) (S_n)
(iv) *regular*
(v) (R_n)
(vi) *normal*
(vii) *reduced*
(viii) *of dimension n.*

Then X has property P if and only if X^{an} does.

Let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. (i) follows from the equality $|X^{\mathrm{an}}| = X(\mathbb{C})$ (XIII.1.1) and from the fact that X is Jacobson (EGA IV 10.4.8). To say that X (respectivelyX^{an}) is discrete is equivalent to saying that $\dim X = 0$ (respectively$\dim X^{\mathrm{an}} = 0$, by [**4**, no. 19 §4 cor. 6]); (i') therefore follows from (viii).

Let P be one of the properties (ii) through (vii). The scheme X has property P if and only if P holds at every closed point of X; indeed, since X is excellent (EGA IV 7.8.6 (iii)), the set of points at which X has property P is open (loc. cit.), and if this open set 317
contains all the closed points, it is all of X. Thus, to say that X (respectivelyX^{an}) has property P is equivalent to saying that, for every point x of X^{an}, the local ring $\mathcal{O}_{X,\varphi(x)}$ (respectively$\mathcal{O}_{X^{\mathrm{an}},x}$) has property P. Since whether an excellent local ring has property P can be checked after passing to its completion, the proposition follows from the isomorphisms $\widehat{\mathcal{O}}_{X,\varphi(x)} \xrightarrow{\sim} \widehat{\mathcal{O}}_{X^{\mathrm{an}},x}$ in cases (ii) through (vii). The same is true in case (viii), in view of the equalities

$$\dim X = \sup_x \dim \mathcal{O}_{X,\varphi(x)} \qquad \dim X^{\mathrm{an}} = \sup_x \dim \mathcal{O}_{X^{\mathrm{an}},x}$$

where $x \in X^{\mathrm{an}}$. This completes the proof.

Proposition XII.2.2. *Let X be a $\mathbb{C}$-scheme locally of finite type, let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism, and let T be a locally constructible subset of X. Then*

$$\varphi^{-1}(\overline{T}) = \overline{\varphi^{-1}(T)}.$$

We may assume that T is a dense open subset of X. Let H be the reduced closed subscheme of X whose underlying space is $X - T$; the associated analytic space H^{an} is a closed analytic subspace of X^{an} whose underlying space is $X^{\mathrm{an}} - \varphi^{-1}(T)$. We must show that every point x of H^{an} belongs to $\overline{\varphi^{-1}(T)}$. At such a point x, the germ (X^{an}, x) contains the subgerm (H^{an}, x), and the latter is defined by a non-nilpotent ideal of $\mathcal{O}_{X^{\mathrm{an}},x}$. It then follows from the Nullstellensatz [**4**, no. 19 §4 cor. 3] that every open neighborhood of x contains points of X^{an} which do not belong to H^{an}, proving that $x \in \overline{\varphi^{-1}(T)}$.

Corollary XII.2.3. *Let X be a $\mathbb{C}$-scheme locally of finite type, let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism, and let T be a locally constructible subset of X. The subset T is open (respectivelyclosed, respectivelydense) if and only if $\varphi^{-1}(T)$ has the corresponding property.* 318

The corollary follows from XII.2.2 and from the fact that, since X is a Jacobson scheme (EGA IV 10.4.8), two locally constructible subsets of X having the same trace on the very dense subset $X(\mathbb{C})$ are equal.

Proposition XII.2.4. *Let X be a $\mathbb{C}$-scheme locally of finite type. Then X is connected (respectivelyirreducible) if and only if the same is true of X^{an}.*

Suppose that X^{an} is connected (respectivelyirreducible). The image $X(\mathbb{C})$ of X^{an} in X is then connected (respectivelyirreducible). It follows that X is connected (respectivelyirreducible), since the closed subsets of X and of $X(\mathbb{C})$ correspond bijectively (EGA IV 10.1.2).

Conversely, suppose that X is connected (respectivelyirreducible), and let us show that the same is true of X^{an}. We may reduce to the case where X is irreducible. Indeed, suppose that X is connected. Given a point x of X, the set of points $y \in X$ for which there exists a finite sequence of irreducible closed subschemes $X_1, \ldots, X_n$ of X, with $x \in X_1$, $y \in X_n$, $X_i \cap X_{i+1} \neq \emptyset$ for $1 \leq i \leq n-1$, is both open and closed, and hence is all of X. For a sequence $X_1, \ldots, X_n$ as above, we also have $X_i^{\mathrm{an}} \cap X_{i+1}^{\mathrm{an}} \neq \emptyset$ for $1 \leq i \leq n-1$; thus, once it is known that the X_i^{an} are connected, it follows that X^{an} is connected as well.

We now suppose that X is irreducible. We may further assume that X is affine. Indeed, if $(U_i)_{i \in I}$ is a covering of X by affine open subsets, any two of these open subsets have nonempty intersection, and the same property therefore holds for the covering $(U_i^{\mathrm{an}})_{i \in I}$ of X^{an}; once it is known that the U_i^{an} are irreducible, it follows that X^{an} is irreducible as well. 319

We may further assume that X is normal. Indeed, let $\tilde{X}$ be the normalization of X; since the morphism $\tilde{X} \to X$ is surjective, the same is true of $\tilde{X}^{\mathrm{an}} \to X^{\mathrm{an}}$. This shows that if $\tilde{X}^{\mathrm{an}}$ is irreducible, then X^{an} is irreducible as well.

We now suppose that X is affine and normal. Since the local rings of X^{an} are integral domains, saying that X^{an} is irreducible is equivalent to saying that it is connected. Indeed, if $\mathcal{F}$ is a closed analytic subset of X^{an}, the set of points x of X^{an} at which $\operatorname{codim}_x(\mathcal{F}, X^{\mathrm{an}}) = 0$ is a closed analytic subset of X^{an} [**4**, no. 20 A Cor. 1] that is also open; if X^{an} is connected, this shows that if $\mathcal{F} \neq X^{\mathrm{an}}$, then $\mathcal{F}$ is nowhere dense, and hence that X^{an} is irreducible. We are thus reduced to showing that X^{an} is connected.

Let

$$i\colon X \to P$$

be a compactification of X, where P is a normal projective $\mathbb{C}$-scheme and i is a dominant open immersion. It then follows from [**10**, no. 12 th. 1] that P^{an} is connected. Since X^{an} is obtained by removing from P^{an} a nowhere dense closed analytic subset, it follows from XII.2.5 below that X^{an} is connected as well.

Lemma XII.2.5. *Let $\mathcal{P}$ be a connected normal analytic space and $\mathcal{Y}$ a nowhere dense closed analytic subset. Then $\mathcal{X} = \mathcal{P} - \mathcal{Y}$ is connected.* 320

When $\mathcal{Y}$ has codimension ≥ 2, the proposition follows from [**11**, no. 3 prop. 4]. In the general case, after removing from $\mathcal{P}$ a closed analytic subset of codimension ≥ 2, we may assume that $\mathcal{P}$ and $\mathcal{Y}$ (regarded as a reduced analytic subspace of $\mathcal{P}$) are regular. By the implicit function theorem, every point y of $\mathcal{Y}$ has a neighborhood $\mathcal{U}$ isomorphic to a ball in an affine space $\mathcal{E}^n$, in such a way that $\mathcal{U} \cap \mathcal{Y}$ is defined by the vanishing of a certain number of coordinate functions. This shows that $\mathcal{U} - \mathcal{U} \cap \mathcal{Y}$ is connected, and therefore so is $\mathcal{X}$.

COROLLARY XII.2.6. *Let X be a $\mathbb{C}$-scheme locally of finite type; the morphism*

$$\pi_0(X^{\mathrm{an}}) \to \pi_0(X)$$

induced by the canonical morphism $X^{\mathrm{an}} \to X$ is bijective.

XII.3. Comparison of properties of morphisms

PROPOSITION XII.3.1. *Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type, and let $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$ be the morphism induced by f on the associated analytic spaces. Let P be the property of being*

(i) *flat*
(ii) *net (i.e. unramified)*
(iii) *étale*
(iv) *smooth*
(v) *normal*
(vi) *reduced*
(vii) *injective*
(viii) *separated*
(ix) *an isomorphism*
(x) *a monomorphism*
(xi) *an open immersion.*

Then f has property P if and only if f^{an} does.

Denote by $\varphi\colon X^{\mathrm{an}} \to X$ and $\psi\colon Y^{\mathrm{an}} \to Y$ the canonical morphisms. Let x be a point of X^{an}, and put $y = f^{\mathrm{an}}(x)$. The morphisms $\mathcal{O}_{Y^{\mathrm{an}},y} \to \mathcal{O}_{X^{\mathrm{an}},x}$ and $\mathcal{O}_{Y,\psi(y)} \to \mathcal{O}_{X,\varphi(x)}$ induced by f and f^{an} yield the same morphism after passage to completions (XII.1.1). By [**2**, ch. 3 §5 prop. 4] (respectivelyEGA IV 17.4.4), saying that f^{an} satisfies property (i) (respectively(ii)) is therefore equivalent to saying that f satisfies (i) (respectively(ii)) at every closed point of X. Since the set of points of X at which (i) (respectively(ii)) holds is open (EGA IV 11.1.1 and I 3.3), this proves (i) and (ii), and hence also (iii).

Let P be property (iv) (respectively(v), respectively(vi)). In view of XII.2.1 ((v), (vi), (vii)), the geometric fibers of f^{an} at the various points y of Y^{an} are regular (respectively-normal, respectivelyreduced) if and only if the same holds for the geometric fibers of f at the various closed points $\psi(y)$ of Y. The cases (iv) (respectively(v), respectively(vi)) then follow from (i) and from the fact that the set of points of Y where the geometric fibers of f are regular is open (EGA IV 12.1.7).

(vii). If f is injective, then so is f^{an}. Conversely, suppose that f^{an} is injective, and let us show that f is as well. We may suppose that f is of finite type. Since f^{an} is injective, 322
the fibers of f over the closed points of Y are radicial; since the set of points of Y whose fiber is radicial is locally constructible (EGA IV 9.6.1), and since Y is a Jacobson scheme, all the fibers of f are radicial, and hence the morphism is injective.

(viii). Let $\Delta\colon X \to X \times_Y X$ and $\Delta^{\mathrm{an}}\colon X^{\mathrm{an}} \to X^{\mathrm{an}} \times_{Y^{\mathrm{an}}} X^{\mathrm{an}}$ be the diagonal immersions, and let $\Theta\colon X^{\mathrm{an}} \times_{Y^{\mathrm{an}}} X^{\mathrm{an}} \to X \times_Y X$ be the canonical morphism. By XII.2.3, saying that $\Delta(X)$ is closed in $X \times_Y X$ is equivalent to saying that $\Delta^{\mathrm{an}}(X^{\mathrm{an}})$ is closed in $X^{\mathrm{an}} \times_{Y^{\mathrm{an}}} X^{\mathrm{an}}$.

Since an open immersion is nothing but an étale injective morphism (EGA IV 17.9.1 and [**4**, no. 13 §1]), (xi) follows from (iii) and (vii). Since an isomorphism is the same thing

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as a surjective open immersion, (ix) follows from (xi) and (XII.3.2 (i)) below. To say that f is a monomorphism is equivalent to saying that the diagonal morphism $\Delta\colon X \to X \times_Y X$ is an isomorphism, so (x) follows from (ix).

PROPOSITION XII.3.2. *Let X and Y be two $\mathbb{C}$-schemes locally of finite type, let $f\colon X \to Y$ be a morphism of finite type, and let $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$ be the morphism induced by f on the associated analytic spaces. Let P be the property of being*

(i) *surjective*
(ii) *dominant*
(iii) *a closed immersion*
(iv) *an immersion*
(v) *proper*[2]
(vi) *finite.*

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Then f has property P if and only if f^{an} does.

Let $\varphi\colon X^{\mathrm{an}} \to X$ and $\psi\colon Y^{\mathrm{an}} \to Y$ be the canonical morphisms.

(i). If f is surjective, then for every point y of Y^{an}, $f^{-1}(\psi(y))$ is a nonempty closed subset of X; it therefore contains at least one closed point, which proves that f^{an} is surjective. Conversely, if f^{an} is surjective, $f(X)$ is a locally constructible subset of Y (EGA IV 1.8.4) containing all the closed points of Y; hence $f(X) = Y$.

(ii) follows from XII.2.2.

(iii). If f is a closed immersion, then so is f^{an} by XII.1.1 a). Conversely, if f^{an} is a closed immersion, then so is f by XII.3.1 (x) and XII.3.2 (v), since this is equivalent to saying that f is a proper monomorphism (EGA IV 8.11.5).

(iv). It is clear that if f is an immersion, then so is f^{an}. Conversely, suppose that f^{an} is an immersion, and let T be the image of X in Y, and $\overline{T}$ the scheme-theoretic closure of f. We have a factorization of f,

$$X \xrightarrow{i} \overline{T} \xrightarrow{j} J,$$

where j is a closed immersion, i is the canonical morphism, and from this we deduce the following factorization of f^{an}:

$$X^{\mathrm{an}} \xrightarrow{i^{\mathrm{an}}} \overline{T}^{\mathrm{an}} \xrightarrow{j^{\mathrm{an}}} Y^{\mathrm{an}} .$$

Since $T = f(X)$ is a locally constructible subset of Y (EGA IV 1.8.4), we have, by XII.2.2, 324
$\overline{T}^{\mathrm{an}} = \overline{f^{\mathrm{an}}(X^{\mathrm{an}})}$. It follows that $i^{\mathrm{an}}(X^{\mathrm{an}})$ is open in $\overline{T}^{\mathrm{an}}$, and hence that $i(X)$ is open in $\overline{T}$. Consider the canonical factorization of i:

$$X \xrightarrow{i_1} i(X) \xrightarrow{i_2} \overline{T}.$$

The morphism i_1^{an} is a proper monomorphism, and hence so is i_1 by XII.3.2 (v) and XII.3.1 (x); this proves that i_1, and hence also f, is an immersion.

(v). Suppose that f is proper, and let us show that f^{an} is as well. Since properness of f^{an} is local on Y^{an}, we may suppose that Y is affine. By Chow's lemma (EGA II 5.6.1), one can find a projective Y-scheme X' and a projective surjective morphism

$$g\colon X' \to X.$$

The morphism $(fg)^{\mathrm{an}} = f^{\mathrm{an}}g^{\mathrm{an}}$ is projective and hence proper, g^{an} is surjective, and it follows from [**1**, ch. 1 §10] that f^{an} is proper.

Conversely, suppose that f^{an} is proper, and let us show that f is as well. By XII.3.1 (viii), f is separated. It remains to prove that f is universally closed, and it even

[2]We shall say that a morphism of analytic spaces is proper if it is proper in the sense of [**1**, ch. 1 §10 no. 1] and if it is separated.

suffices to show that f is closed; indeed, for every Y-scheme Y' locally of finite type, the morphism

$$f_{(Y')} = h\colon X \times_Y Y' \to Y'$$

will also be closed, since h^{an} is proper. Let T be a closed subset of X; $f(T)$ is a locally constructible set, and we have

$$f^{\mathrm{an}}(\varphi^{-1}(T)) = \psi^{-1}(f(T)).$$

Since f^{an} is proper, $\psi^{-1}(f(T))$ is a closed subset of Y^{an}, and it therefore follows 325
from XII.2.2 that

$$\psi^{-1}(\overline{f(T)}) = \psi^{-1}(f(T)).$$

This implies that $f(T) = \overline{f(T)}$, i.e. that f is closed and hence that f is proper.

(vi). To say that a morphism is finite is equivalent to saying that it is proper with finite fibers (EGA III 4.4.2 and [**4**, no. 19 §5]). Since the set of points where the fibers of f are finite is locally constructible (EGA IV 9.7.9), the fibers of f are finite if and only if the fibers of f^{an} are finite; (vi) therefore follows from (v).

Remark XII.3.3. a) Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type. The fact that f^{an} is a local isomorphism does not imply that the same is true of f. Indeed, if f is étale, then f^{an} is étale and hence a local isomorphism [**4**, no. 13 §1], but f need not be one.

b) The statement XII.3.2 is not true if f is not assumed to be of finite type. For example, f^{an} can be a closed immersion without f being one. Indeed, it suffices to take X to be the disjoint union of $\mathbb{Z}$ copies of $\mathrm{Spec}(\mathbb{C})$, Y to be the affine line, and f to be the morphism obtained by sending the points of X to distinct points of Y forming a discrete subset.

XII.4. Cohomological comparison and existence theorems

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The purpose of this section is to reprove the results of [**3**, no. 2 th. 5 and th. 6]; the latter generalize the theorems established in [**10**, no. 12] for projective X to the case of a proper scheme, and extend them to the relative case. More general results concerning proper relative schemes over an analytic space are proved in [**7**, ch. VIII no. 3].

Recall that the Čech cohomology used in [**10**, no. 12] coincides with the usual cohomology in both the algebraic and the analytic cases (EGA III 1.4.1. and [**5**, II 5.10]).

. Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type, and consider the commutative diagram

$$\begin{array}{ccc} X^{\mathrm{an}} & \xrightarrow{\varphi} & X \\ {\scriptstyle f^{\mathrm{an}}}\downarrow & & \downarrow{\scriptstyle f} \\ Y^{\mathrm{an}} & \xrightarrow{\psi} & Y. \end{array}$$

If F is an $\mathcal{O}_X$-Module, then, for every integer $p \geq 0$, there are morphisms

$$\mathrm{R}^p f_* F \xrightarrow{i} \mathrm{R}^p f_*(\varphi_* F^{\mathrm{an}}) \xrightarrow{j} \mathrm{R}^p (f\cdot\varphi)_* F^{\mathrm{an}} \xrightarrow{k} \psi_*(\mathrm{R}^p f^{\mathrm{an}}_* F^{\mathrm{an}}),$$

where i is induced by the canonical morphism $F \to \varphi_* F^{\mathrm{an}}$, and j, k are "edge homomorphisms" of Leray spectral sequences. The composite $k.j.i$ gives rise to a canonical morphism

$$\theta_p\colon (\mathrm{R}^p f_* F)^{\mathrm{an}} \to \mathrm{R}^p f^{\mathrm{an}}_*(F^{\mathrm{an}}) \tag{4.1.1}$$

Theorem XII.4.1. *Let $f\colon X \to Y$ be a proper morphism of $\mathbb{C}$-schemes locally of finite type, and let F be a coherent $\mathcal{O}_X$-Module. Then, for every integer $p \geq 0$, the morphism* (4.1.1) 327

$$\theta_p\colon (\mathrm{R}^p f_* F)^{\mathrm{an}} \to \mathrm{R}^p f^{\mathrm{an}}_*(F^{\mathrm{an}})$$

is an isomorphism.

1) *The case where f is projective.* The proof is analogous to that of [**10**, no. 13]. Let us recall it briefly. One reduces to the case where X is a projective space of the form $\mathbb{P}^r_Y$ over Y. Let $\mathcal{Y} = Y^{\mathrm{an}}$, $\mathcal{P} = \mathbb{P}^r_{\mathcal{Y}}$; one first proves that

$$f_*^{\mathrm{an}}\mathcal{O}_{\mathcal{P}} = \mathcal{O}_{\mathcal{Y}}, \qquad \mathrm{R}^p f_*^{\mathrm{an}}(\mathcal{O}_{\mathcal{P}}) = 0 \qquad \text{for } p > 0$$

Indeed, to verify the preceding relations, one may reduce to the case where $\mathcal{Y}$ is a ball $\mathcal{B}$ in an affine space $\mathcal{E}^n$. Consider the "standard covering" $\{\mathcal{U}_i\}$ of $\mathcal{P}$ by $r+1$ open subsets isomorphic to $\mathcal{B} \times \mathcal{E}^r$. Since these open subsets are Stein, for every integer $p \geq 0$ there are isomorphisms

$$\mathrm{H}^p(\{\mathcal{U}_i\}, \mathcal{O}_{\mathcal{P}}) \xrightarrow{\sim} \mathrm{H}^p(\mathcal{P}, \mathcal{O}_{\mathcal{P}})$$

One can then express the sections of the structure sheaf $\mathcal{O}_{\mathcal{P}}$ over the open subsets $\mathcal{U}_i$ and over their intersections in terms of Laurent series; an easy computation shows that

$$\mathrm{H}^0(\mathcal{P}, \mathcal{O}_{\mathcal{P}}) \xrightarrow{\sim} \mathrm{H}^0(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}}), \qquad \mathrm{H}^p(\mathcal{P}, \mathcal{O}_{\mathcal{P}}) = 0 \qquad \text{for } p > 0$$

The proof is then completed by repeating the argument of [**10**, no. 12 lemma 5], with the cohomology groups replaced by the cohomology sheaves.

2) *The case where f is proper.* We use EGA III 3.1.2 to reduce to the projective case. 328
Let $\mathcal{K}$ be the category of coherent $\mathcal{O}_X$-Modules for which θ_p is an isomorphism for every $p \geq 0$. It suffices to prove that, for every exact sequence $0 \to F' \to F \to F'' \to 0$ two of whose terms belong to $\mathcal{K}$, the same is true of the third; that a direct summand of an object of $\mathcal{K}$ belongs to $\mathcal{K}$; and that, for every point x of X, one can find an object F of $\mathcal{K}$ such that $F_x \neq 0$.

The first condition follows by applying the five lemma to the following commutative diagram, whose rows are exact:

$$\begin{array}{ccccccccc}
\longrightarrow & (\mathrm{R}^p f_* F')^{\mathrm{an}} & \longrightarrow & (\mathrm{R}^p f_* F)^{\mathrm{an}} & \longrightarrow & (\mathrm{R}^p f_* F'')^{\mathrm{an}} & \longrightarrow & (\mathrm{R}^{p+1} f_* F')^{\mathrm{an}} & \longrightarrow \\
 & \downarrow & & \downarrow & & \downarrow & & \downarrow & \\
\longrightarrow & \mathrm{R}^p f_*^{\mathrm{an}} F'^{\mathrm{an}} & \longrightarrow & \mathrm{R}^p f_*^{\mathrm{an}} F^{\mathrm{an}} & \longrightarrow & \mathrm{R}^p f_*^{\mathrm{an}} F''^{\mathrm{an}} & \longrightarrow & \mathrm{R}^{p+1} f_*^{\mathrm{an}} F'^{\mathrm{an}} & \longrightarrow ,
\end{array}$$

and the second condition is verified similarly.

To verify the third condition, one may restrict to the case where X is an irreducible scheme with generic point x. We could have assumed from the outset that Y is noetherian. By Chow's lemma (EGA II 5.6.1), one can find a projective Y-scheme X' and a projective surjective morphism $g\colon X' \to X$. Moreover, there is an integer n such that $\mathrm{R}^p g_*(\mathcal{O}_{X'}(n)) = 0$ for every $p > 0$ and the canonical morphism $g^* g_*(\mathcal{O}_{X'}(n)) \to \mathcal{O}_{X'}(n)$ is surjective (EGA III 2.2.1). If we put $F = g_*(\mathcal{O}_{X'}(n))$, then the sheaf F has the required property. Indeed, $F_x \neq 0$; moreover, since the Leray spectral sequence

$$\mathrm{R}^p f_*(\mathrm{R}^q g_*(\mathcal{O}_{X'}(n))) \Longrightarrow \mathrm{R}^{p+q}(f \cdot g)_*(\mathcal{O}_{X'}(n))$$

degenerates, there is an isomorphism

$$\mathrm{R}^p f_* F \xrightarrow{\sim} \mathrm{R}^p(f \cdot g)_*(\mathcal{O}_{X'}(n))$$

As in the algebraic case, there is a canonical isomorphism 329

$$\mathrm{R}^p f_*^{\mathrm{an}} F^{\mathrm{an}} \xrightarrow{\sim} \mathrm{R}^p(f \cdot g)_*^{\mathrm{an}}(\mathcal{O}_{X'}(n)^{\mathrm{an}}),$$

and the diagram

$$\begin{array}{ccc}
(\mathrm{R}^p f_* F)^{\mathrm{an}} & \xrightarrow{\sim} & (\mathrm{R}^p(f \cdot g)_*(\mathcal{O}_{X'}(n)))^{\mathrm{an}} \\
{\scriptstyle\theta_p}\downarrow & & \downarrow{\scriptstyle\psi_p} \\
\mathrm{R}^p f_*^{\mathrm{an}} F^{\mathrm{an}} & \xrightarrow{\sim} & \mathrm{R}^p(f \cdot g)_*^{\mathrm{an}}(\mathcal{O}_{X'}(n)^{\mathrm{an}})
\end{array}$$

is commutative. By 1), ψ_p is an isomorphism; hence so is θ_p, which completes the proof.

COROLLARY XII.4.2. *Let X be a proper $\mathbb{C}$-scheme, and let F be a coherent $\mathcal{O}_X$-Module. Then, for every integer $p \geq 0$, the canonical morphism*

$$\mathrm{H}^p(X, F) \to \mathrm{H}^p(X^{\mathrm{an}}, F^{\mathrm{an}})$$

is an isomorphism.

THEOREM XII.4.3. *Let X be a proper $\mathbb{C}$-scheme. The functor which assigns to every coherent $\mathcal{O}_X$-Module F its inverse image F^{an} on X^{an} is an equivalence of categories.*

1) *The functor is fully faithful.* Indeed, let F and G be two coherent $\mathcal{O}_X$-Modules. The canonical morphism

$$\mathrm{Hom}_{\mathcal{O}_X}(F, G) \to \mathrm{Hom}_{\mathcal{O}_{X^{\mathrm{an}}}}(F^{\mathrm{an}}, G^{\mathrm{an}})$$

is identified with the canonical morphism

$$\mathrm{H}^0(X, \mathbf{Hom}_{\mathcal{O}_X}(F, G)) \to \mathrm{H}^0(X^{\mathrm{an}}, \mathbf{Hom}_{\mathcal{O}_X}(F, G))$$

(EGA 0_{I} 6.7.6). Since $\mathbf{Hom}_{\mathcal{O}_X}(F, G)$ is coherent, it follows from XII.4.2 that this morphism 330
is bijective.

2) *The functor is essentially surjective.* When X is projective, the assertion follows from [**10**, no. 12 th. 3]. The general case reduces to the preceding one by using Chow's lemma (EGA II 5.6.1). Indeed, let X' be a projective $\mathbb{C}$-scheme, let $f\colon X' \to X$ be a projective surjective morphism, and let U be a dense open subset of X such that f induces an isomorphism $f^{-1}(U) \simeq U$. We argue by noetherian induction on X; we may therefore suppose that, for every coherent sheaf $\mathcal{G}$ on X^{an} for which there exists a closed subset Y of X, distinct from X, satisfying $Y^{\mathrm{an}} \supset \mathrm{Supp}\,\mathcal{G}$, there is a coherent sheaf G on X such that $G^{\mathrm{an}} \simeq \mathcal{G}$.

Let $\mathcal{F}$ be a coherent sheaf of $\mathcal{O}_{X^{\mathrm{an}}}$-modules, and let $\mathcal{K}$ and $\mathcal{L}$ be the coherent sheaves defined by the condition that the sequence

$$0 \to \mathcal{K} \to \mathcal{F} \to f_*^{\mathrm{an}} f^{\mathrm{an}*}\mathcal{F} \to \mathcal{L} \to 0$$

be exact. Since X' is projective, there is a coherent $\mathcal{O}_{X'}$-Module F' such that $F'^{\mathrm{an}} \simeq f^{\mathrm{an}*}\mathcal{F}$; it then follows from XII.4.1 that there is an isomorphism $(f_*F')^{\mathrm{an}} \simeq f_*^{\mathrm{an}} f^{\mathrm{an}*}\mathcal{F}$. Since $\mathcal{K}|U^{\mathrm{an}}$ and $\mathcal{L}|U^{\mathrm{an}}$ vanish, there are coherent $\mathcal{O}_X$-Modules K and L such that $K^{\mathrm{an}} \simeq \mathcal{K}$ and $L^{\mathrm{an}} \simeq \mathcal{L}$. By 1), the morphism $f_*^{\mathrm{an}} f^{\mathrm{an}*}\mathcal{F} \to \mathcal{L}$ comes from a unique morphism $f_*F' \to L$; let $I = \mathrm{Ker}(f_*F' \to L)$. The sheaf $\mathcal{F}$ is then an extension of I^{an} by K^{an}, and it remains to see that this extension is obtained by inverse image from an extension of I 331
by K. It therefore suffices to prove that the canonical morphism

$$(*) \qquad \mathrm{Ext}^q_{\mathcal{O}_X}(I, K)^{\mathrm{an}} \xrightarrow{\sim} \mathrm{Ext}^q_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}}) \quad q \neq 1$$

is bijective. But there are isomorphisms $\mathbf{Ext}^q_{\mathcal{O}_X}(I, K)^{\mathrm{an}} \xrightarrow{\sim} \mathbf{Ext}^q_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}})$ for every integer $q \geq 0$ (EGA 0_{III} 12.3.5), and a morphism of spectral sequences

$$\begin{array}{ccc} \mathrm{H}^p(X, \mathbf{Ext}^q_{\mathcal{O}_X}(I, K)) & \Longrightarrow & \mathrm{Ext}^{p+q}_{\mathcal{O}_X}(I, K) \\ \downarrow & & \downarrow \\ \mathrm{H}^p(X^{\mathrm{an}}, \mathbf{Ext}^q_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}})) & \Longrightarrow & \mathrm{Ext}^{p+q}_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}}). \end{array}$$

This morphism is an isomorphism because, by XII.4.2, it is an isomorphism on the terms E_2^{pq}, and this proves the bijectivity of $(*)$.

COROLLARY XII.4.4. *The functor which assigns to every proper $\mathbb{C}$-scheme X the analytic space X^{an} is fully faithful.*

We must show that, if X and Y are two proper $\mathbb{C}$-schemes, the canonical map

$$\mathrm{Hom}_{\mathbb{C}}(X, Y) \to \mathrm{Hom}(X^{\mathrm{an}}, Y^{\mathrm{an}})$$

is bijective. Now, to give a morphism from X to Y (respectivelyfrom X^{an} to Y^{an}) is equivalent to giving its graph, i.e. a closed subscheme Z of $X \times Y$ (respectivelya closed analytic subspace $\mathcal{Z}$ of $X^{\mathrm{an}} \times Y^{\mathrm{an}}$), such that the restriction of the first projection $X \times Y \to X$ to Z (respectivelyof $X^{\mathrm{an}} \times Y^{\mathrm{an}} \to X^{\mathrm{an}}$ to $\mathcal{Z}$) is an isomorphism. Since giving 332
a closed subscheme of $X \times Y$ (respectivelya closed analytic subspace of $X^{\mathrm{an}} \times Y^{\mathrm{an}}$) is equivalent to giving a coherent sheaf of ideals in $\mathcal{O}_{X\times Y}$ (respectivelyin $\mathcal{O}_{X^{\mathrm{an}}\times Y^{\mathrm{an}}}$), the corollary follows from XII.4.3.

Corollary XII.4.5. *Let X be a proper $\mathbb{C}$-scheme. The functor which assigns to every scheme X' finite (respectivelyfinite étale) over X the analytic space X'^{an} is an equivalence from the category of schemes finite (respectivelyfinite étale) over X to the category of analytic spaces finite (respectivelyfinite étale) over X^{an}.*

Indeed, to give a finite morphism $X' \to X$ (respectively$X'^{\mathrm{an}} \to X^{\mathrm{an}}$) is equivalent to giving a coherent sheaf of algebras over $\mathcal{O}_X$ (respectivelyover $\mathcal{O}_{X^{\mathrm{an}}}$) [**4**, no. 19 §5 th. 2]. The corollary therefore follows from XII.4.3 in the unparenthesized case, and the parenthesized case follows from it in view of XII.3.1 (iii).

XII.5. Comparison theorems for étale coverings

. Let us make precise the notion of a finite covering of an analytic space. If $\mathcal{X}$ is an analytic space, an analytic space $\mathcal{X}'$ finite over $\mathcal{X}$ is said to be a finite covering of $\mathcal{X}$ if every irreducible component of $\mathcal{X}'$ dominates an irreducible component of $\mathcal{X}$.

Theorem XII.5.1 ("Riemann existence theorem"). *Let X be a $\mathbb{C}$-scheme locally of finite type, and let X^{an} be the analytic space associated with X. The functor Ψ which assigns to every finite étale covering X' of X the space X'^{an} is an equivalence from the* 333
category of finite étale coverings of X to the category of finite étale coverings of X^{an}.

1) *The functor Ψ is fully faithful.* Let X' and X'' be two finite étale coverings of X, and let us prove that the canonical map

$$(*) \qquad \mathrm{Hom}_X(X', X'') \to \mathrm{Hom}_{X^{\mathrm{an}}}(X'^{\mathrm{an}}, X''^{\mathrm{an}})$$

is bijective. We may suppose that X' is connected. To give an X-morphism from X' to X'' is equivalent to giving a connected component X_i of $X' \times_X X''$ such that the morphism $X_i \to X'$ induced by the first projection is an isomorphism. Since the connected components of $X' \times_X X''$ correspond bijectively to the connected components of $X'^{\mathrm{an}} \times_{X^{\mathrm{an}}} X''^{\mathrm{an}}$ (XII.2.6), and since a morphism $X_i \to X'$ is an isomorphism if and only if $X_i^{\mathrm{an}} \to X'^{\mathrm{an}}$ is an isomorphism, this proves the bijectivity of $(*)$.

2) *The functor Ψ is essentially surjective.* Let $\mathcal{X}'$ be a finite étale covering of X^{an}, and let us prove that there exists an étale covering X' of X together with an isomorphism $X'^{\mathrm{an}} \xrightarrow{\sim} \mathcal{X}'$. In view of 1), the question is local on X, and we may therefore suppose that X is affine.

a) Reduction to the case where X is normal. We may suppose that X is reduced. Indeed, suppose that the theorem has been proved for X_{red}. The functor which assigns to a finite étale covering X' of X the finite étale covering $X'^{\mathrm{an}}_{\mathrm{red}}$ of $X^{\mathrm{an}}_{\mathrm{red}}$ is then an equivalence. Since it is obtained by composing Ψ with the functor Θ which assigns to a finite étale covering of X^{an} its inverse image over $X^{\mathrm{an}}_{\mathrm{red}}$, and since Θ is fully faithful, this shows that Ψ is an equivalence of categories.

We may suppose that X is normal. Indeed, let $\widetilde{X}$ be the normalization of X, and let 334
$p\colon \widetilde{X} \to X$ be the canonical morphism. Since p is finite, p is an effective descent morphism for the category of étale coverings (IX IX.4.7). Supposing that the theorem has been

proved for $\widetilde{X}$, put $\widetilde{\mathcal{X}'} = \mathcal{X}' \times_{X^{\mathrm{an}}} \widetilde{X}^{\mathrm{an}}$. There then exist an étale covering $\widetilde{X}'$ of $\widetilde{X}$ and an isomorphism $\widetilde{X}'^{\mathrm{an}} \simeq \widetilde{\mathcal{X}'}$. It follows from 1) that the natural descent datum on $\widetilde{\mathcal{X}'}$ lifts to a descent datum on $\widetilde{X}'$ relative to $\widetilde{X} \to X$; this proves the existence of an étale covering X' of X together with an isomorphism $i\colon X'^{\mathrm{an}} \times_{X^{\mathrm{an}}} \widetilde{X}^{\mathrm{an}} \simeq \widetilde{\mathcal{X}'}$ whose inverse images under the two projections from $\widetilde{X}^{\mathrm{an}} \times_{X^{\mathrm{an}}} \widetilde{X}^{\mathrm{an}}$ are the same. By IX IX.3.2, whose proof is valid in the analytic case, the morphism $\widetilde{X}^{\mathrm{an}} \to X^{\mathrm{an}}$ is a descent morphism for the category of étale coverings, and consequently i comes from an isomorphism $X'^{\mathrm{an}} \simeq \mathcal{X}'$.

b) Reduction to the case where X is regular. Let U be the open subset of regular points of X, and let $i\colon U \to X$ and $i^{\mathrm{an}}\colon U^{\mathrm{an}} \to X^{\mathrm{an}}$ be the canonical morphisms; since X is normal, we have $\operatorname{codim}(X - U, X) \geq 2$. Suppose that there exists an étale covering U' of U such that $U'^{\mathrm{an}} \simeq \mathcal{X}'|U^{\mathrm{an}}$, and let us show that U' then extends to an étale covering X' of X such that $X'^{\mathrm{an}} \simeq \mathcal{X}'$. It suffices to see that U' extends to an étale covering X' of X; indeed, we will then have an isomorphism $X'^{\mathrm{an}}|U^{\mathrm{an}} \simeq \mathcal{X}'|U^{\mathrm{an}}$. But if $\mathcal{F}$ and $\mathcal{G}$ are the coherent sheaves of algebras over $\mathcal{O}_{X^{\mathrm{an}}}$ defining $\mathcal{X}'$ and X'^{an} respectively, the fact that X is normal and that $\operatorname{codim}(X - U, X) \geq 2$ implies that the canonical morphisms

$$\mathcal{F} \to i^{\mathrm{an}}_*(\mathcal{F}|U^{\mathrm{an}}) \qquad \mathcal{G} \to i^{\mathrm{an}}_*(\mathcal{G}|U^{\mathrm{an}})$$

are isomorphisms [**11**, no. 3 prop. 4]. It follows that $\mathcal{F}$ and $\mathcal{G}$, and hence also X'^{an} and 335
$\mathcal{X}'$, are isomorphic.

Let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. Since the problem of extending U' to X is local on X, it is enough to prove that, for every point y of $X^{\mathrm{an}} - U^{\mathrm{an}}$, the étale covering $U'_{\varphi(y)} = U' \times_X \operatorname{Spec}()O_{X,\varphi(y)}$ of $U_{\varphi(y)} = U \times_X \operatorname{Spec}()O_{X,\varphi(y)}$ extends to $\operatorname{Spec}()O_{X,\varphi(y)}$. Let H be the coherent $\mathcal{O}_U$-algebra defining U'. The canonical morphism

$$\alpha\colon (i_*H)^{\mathrm{an}} \to i^{\mathrm{an}}_*(H^{\mathrm{an}}) = \mathcal{F}$$

defines a morphism of sheaves of modules on $\operatorname{Spec}()O_{X^{\mathrm{an}},y}$:

$$\alpha_y\colon (i_*H)^{\mathrm{an}}_y \to \mathcal{F}_y,$$

whose restriction to $U_y = U_{\varphi(y)} \times_{\operatorname{Spec}()O_{X,\varphi(y)}} \operatorname{Spec}()O_{X^{\mathrm{an}},y}$ is an isomorphism. But this proves that $H|U_y$ is trivial, and hence that $U'_{\varphi(y)}$ extends to $\operatorname{Spec}()O_{X,\varphi(y)}$.

c) The case where X is affine and regular. Let

$$j\colon X \to P$$

be a compactification of X, where P is a projective $\mathbb{C}$-scheme and j is a dominant open immersion. By the resolution of singularities theorem [**8**], one can find a regular scheme R and a projective morphism $r\colon R \to P$ such that r induces an isomorphism $r^{-1}(X) \simeq X$ and $r^{-1}(X)$ is the complement in R of a divisor with normal crossings. Let 336

$$k\colon X \to R$$

be the canonical immersion. We shall show that there exists a finite normal covering (XII.5) $\mathcal{R}'$ of R^{an} extending the étale covering X'^{an}. By Proposition XII.5.3 below, such a covering is unique; the problem of extending X'^{an} is therefore local on R^{an} in a neighborhood of $R^{\mathrm{an}} - X^{\mathrm{an}}$. Now every point of $R^{\mathrm{an}} - X^{\mathrm{an}}$ has an open neighborhood $\mathcal{V}$ isomorphic to a ball in an affine space $\mathcal{E}^n$ such that $\mathcal{V} - \mathcal{V} \cap X^{\mathrm{an}}$ is defined by the vanishing of the first p coordinate functions $z_1, \ldots, z_p$, where $0 \leq p \leq n$. The fundamental group of $\mathcal{U} = \mathcal{V} \cap X^{\mathrm{an}}$ is isomorphic to $\mathbb{Z}^p$, and every étale covering of $\mathcal{U}$ is a quotient of a covering of the form

$$\mathcal{U}'' = \mathcal{U}[T_1, \ldots, T_p]/(T_1^{n_1} - z_1, \ldots, T_p^{n_p} - z_p),$$

where the n_i are integers > 0, by a subgroup H of the Galois group $\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_p\mathbb{Z}$ of $\mathcal{U}''$. Now $\mathcal{U}''$ extends to the regular covering

$$\mathcal{V}'' = \mathcal{V}[T_1, \ldots, T_p]/(T_1^{n_1} - z_1, \ldots, T_p^{n_p} - z_p),$$

of $\mathcal{V}$ on which H acts, and the quotient of $\mathcal{V}''$ by H is the desired extension.

The proof is then completed by XII.4.5. The covering $\mathcal{R}'$ comes from a finite covering R' of R; the restriction of R' to X is a covering X' of X such that $X'^{\mathrm{an}} \simeq \mathcal{X}'$, and by XII.3.1(iii), X' is an étale covering of X.

COROLLARY XII.5.2. *Let X be a connected $\mathbb{C}$-scheme locally of finite type, let* 337
$\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism, and let x be a point of X^{an}. Let $\pi_1(X^{\mathrm{an}}, x)$ be the fundamental group of the topological space X^{an} at x, and let $\pi_1(X, \varphi(x))$ be the fundamental group of the scheme X at $\varphi(x)$ (V V.7). *Then $\pi_1(X, \varphi(x))$ is canonically isomorphic to the completion of $\pi_1(X^{\mathrm{an}}, x)$ with respect to the topology of subgroups of finite index.*

Indeed, let $\mathcal{C}$ be the category of finite étale coverings of X^{an}, let F be the functor from $\mathcal{C}$ to **Ens** which assigns to every finite étale covering $\mathcal{X}'$ of X^{an} the set of points of $\mathcal{X}'$ above x, and let $\widehat{\pi}_1(X^{\mathrm{an}}, x)$ be the profinite group associated with $\mathcal{C}$ and F as in V V.4. Since every finite étale covering of X^{an} is a quotient of the universal covering by a subgroup of finite index, $\widehat{\pi}_1(X^{\mathrm{an}}, x)$ is precisely the completion of $\pi_1(X^{\mathrm{an}}, x)$ for the topology of subgroups of finite index. The corollary therefore follows from XII.5.1 and V V.6.10.

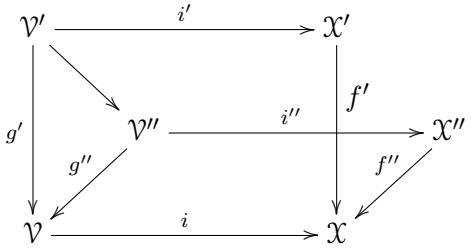
PROPOSITION XII.5.3. *Let $\mathcal{X}$ be a normal analytic space, and let $\mathcal{Y}$ be a closed analytic subset such that $\mathcal{U} = \mathcal{X} - \mathcal{Y}$ is dense in $\mathcal{X}$. Then the functor which assigns to every finite normal covering* (XII.5) *$\mathcal{X}'$ of $\mathcal{X}$ its restriction to $\mathcal{U}$ is fully faithful.*

Let $\mathcal{X}'$ and $\mathcal{X}''$ be two finite normal coverings of $\mathcal{X}$. We must show that the canonical map

$$\mathrm{Hom}_{\mathcal{X}}(\mathcal{X}', \mathcal{X}'') \to \mathrm{Hom}_{\mathcal{U}}(\mathcal{X}'|\mathcal{U}, \mathcal{X}''|\mathcal{U})$$

is bijective. Let u, v be two $\mathcal{X}$-morphisms from $\mathcal{X}'$ to $\mathcal{X}''$ whose restrictions to $\mathcal{U}$ are the same, and let us prove that $u = v$. The morphisms u and v coincide on the dense open 338
subset $\mathcal{U} \times_{\mathcal{X}} \mathcal{X}'$, and hence on the underlying topological spaces. By [**4**, no. 19 §4 cor. 5], this proves that $u = v$.

Now let u be a $\mathcal{U}$-morphism from $\mathcal{X}'|\mathcal{U}$ to $\mathcal{X}''|\mathcal{U}$, and let us show that it extends to all of $\mathcal{X}'$. We may suppose that $\mathcal{X}'$ is regular. Indeed, since $\mathcal{X}'$ is normal, one can find an open subset $\mathcal{V}$ of $\mathcal{X}$ whose complement is an analytic subset of codimension ≥ 2, such that $\mathcal{X}' \times_{\mathcal{X}} \mathcal{V} = \mathcal{V}'$ is regular. Let $\mathcal{V}'' = \mathcal{X}'' \times_{\mathcal{X}} \mathcal{V}$, and suppose the proposition proved for $\mathcal{V}$. Consider the commutative diagram



The morphism u determines a morphism of $\mathcal{O}_{\mathcal{V}}$-algebras $g''_*\mathcal{O}_{\mathcal{V}''} \to g'_*\mathcal{O}_{\mathcal{V}'}$, from which one deduces a morphism

$$i_*g''_*\mathcal{O}_{\mathcal{V}''} \to i_*g'_*\mathcal{O}_{\mathcal{V}'}.$$

In view of the isomorphisms $i'_*\mathcal{O}_{\mathcal{V}'} \simeq \mathcal{O}_{\mathcal{X}'}$, $i''_*\mathcal{O}_{\mathcal{V}''} \simeq \mathcal{O}_{\mathcal{X}''}$ [**11**, no. 3 prop. 4], one deduces a morphism of $\mathcal{O}_{\mathcal{X}}$-algebras

$$f''_*\mathcal{O}_{\mathcal{X}''} \to f'_*\mathcal{O}_{\mathcal{X}'},$$

and hence the desired morphism $\mathcal{X}' \to \mathcal{X}''$.

We henceforth suppose that $\mathcal{X}'$ is regular. Let $\mathcal{U}' = \mathcal{U} \times_{\mathcal{X}} \mathcal{X}'$, $\mathcal{Y}' = \mathcal{X}' - \mathcal{U}'$. We 339
regard $\mathcal{Y}'$ as a reduced analytic subspace of $\mathcal{X}'$; if $\mathcal{Y}'_1$ is the singular locus of $\mathcal{Y}'$, then $\dim \mathcal{Y}'_1 < \dim \mathcal{Y}'$ [**4**, no. 20 D. Th. 3]. Thus, by induction on the dimension of $\mathcal{Y}'$, we may suppose that $\mathcal{Y}'$ is smooth. Since it suffices to extend u to an open neighborhood of each

point of $\mathcal{Y}'$, the implicit function theorem allows us to suppose that $\mathcal{X}'$ is a ball in an affine space $\mathcal{E}^n$ and that $\mathcal{Y}'$ is the closed subset defined by the vanishing of the first p coordinate functions $z_1, \ldots, z_p$, with $0 \leq p \leq n$.

The morphism u determines a section s of $p\colon \mathcal{X}' \times_{\mathcal{X}} \mathcal{X}'' \to \mathcal{X}'$ over $\mathcal{U}'$; after restricting $\mathcal{X}'$ if necessary, we may suppose that $p_*(\mathcal{O}_{\mathcal{X}'\times_{\mathcal{X}}\mathcal{X}''})$ is generated by elements $x_1, \ldots, x_q$ of $\Gamma(\mathcal{X}', p_*\mathcal{O}_{\mathcal{X}'\times_{\mathcal{X}}\mathcal{X}''})$; let $u_1, \ldots, u_q \in \Gamma(\mathcal{U}', \mathcal{O}_{\mathcal{X}'})$ be the images under s of $x_1|\mathcal{U}', \ldots, x_q|\mathcal{U}'$. To say that s extends to $\mathcal{X}'$ is equivalent to saying that $u_1, \ldots, u_q$ extend to sections of $\Gamma(\mathcal{X}', \mathcal{O}_{\mathcal{X}'})$. But since f is finite, each u_i is a Laurent series in $z_1, \ldots, z_p$, with coefficients that are convergent power series in $z_{p+1}, \ldots, z_n$, and satisfies integral-dependence relations. It follows that u_i is bounded, and hence is a convergent power series in $z_1, \ldots, z_n$; consequently it extends to $\mathcal{X}'$.

One may ask whether the functor introduced in XII.5.3 is an equivalence of categories. The GRAUERT-REMMERT theorem [**6**] answers this question; below we give a proof using resolution of singularities. One could also have used the GRAUERT-REMMERT theorem to prove XII.5.1; this is what was done before [**8**] became available.

THEOREM XII.5.4 (GRAUERT-REMMERT theorem). *Let $\mathcal{X}$ be a normal analytic space,* 340
and let $\mathcal{Y}$ be a closed analytic subset such that $\mathcal{U} = \mathcal{X} - \mathcal{Y}$ is dense in $\mathcal{X}$. Let $\mathcal{U}'$ be a finite normal covering of $\mathcal{U}$; suppose that there exists a nowhere dense closed analytic subset $\mathcal{S}$ of $\mathcal{X}$ such that the restriction of $\mathcal{U}'$ to $\mathcal{U} - \mathcal{U} \cap \mathcal{S}$ is étale. Then there exists a finite normal covering $\mathcal{X}'$ of $\mathcal{X}$ which extends $\mathcal{U}'$, and $\mathcal{X}'$ is unique up to isomorphism.

Uniqueness follows from XII.5.3. The problem of extending $\mathcal{U}'$ is therefore local on $\mathcal{X}$. We may suppose that $\mathcal{U}$ is regular and that $\mathcal{U}'$ is étale over $\mathcal{U}$. Indeed, the set of regular points of $\mathcal{U}$ is a dense open subset $\mathcal{V}$ of $\mathcal{X}$ whose complement is an analytic subset [**4**, no. 20 D th. 2], and it suffices to replace $\mathcal{U}$ by the open subset $\mathcal{V} - \mathcal{V} \cap \mathcal{S}$.

Let y be a point of $\mathcal{X} - \mathcal{U}$, and let us show that $\mathcal{U}'$ can be extended to a neighborhood of y. After restricting $\mathcal{X}$ to an open neighborhood of y, it follows from the resolution of singularities theorem [**8**] that one can find a regular analytic space $\mathcal{X}_1$ and a projective morphism $f\colon \mathcal{X}_1 \to \mathcal{X}$ which, upon restriction to $\mathcal{U}$, induces an isomorphism $\mathcal{U}_1 = f^{-1}(\mathcal{U}) \simeq \mathcal{U}$, such that $\mathcal{U}_1$ is the complement in $\mathcal{X}_1$ of a normal crossings divisor. Let us show that $\mathcal{U}'$ extends to a normal finite covering of $\mathcal{X}_1$. Since the question is local on $\mathcal{X}_1$, we may suppose that $\mathcal{X}_1$ is a ball in an affine space $\mathcal{E}^n$ and that $\mathcal{X}_1 - \mathcal{U}_1$ is defined by the vanishing of the first p coordinate functions $z_1, \ldots, z_p$, with $0 \leq p \leq n$. The étale covering $\mathcal{U}'$ of $\mathcal{U}_1$ is a quotient of a covering of the form

$$\mathcal{U}_2 = \mathcal{U}_1[T_1, \ldots, T_p] / \left(T_1^{n_1} - z_1, \ldots, T_p^{n_p} - z_p\right)$$

by a subgroup H of the Galois group of $\mathcal{U}_2$. The covering $\mathcal{U}_2$ extends to the covering 341

$$\mathcal{X}_2 = \mathcal{X}_1[T_1, \ldots, T_p] / \left(T_1^{n_1} - z_1, \ldots, T_p^{n_p} - z_p\right)$$

of $\mathcal{X}_1$, on which H acts, and $\mathcal{X}_2/H$ extends $\mathcal{U}'$ to $\mathcal{X}_1$.

Let $\mathcal{X}_1'$ denote the finite normal covering of $\mathcal{X}_1$ which extends $\mathcal{U}'$, and let $\mathcal{F}_1$ denote the coherent $\mathcal{O}_{\mathcal{X}_1}$-algebra defined by $\mathcal{F}_1$. By the GRAUERT-REMMERT finiteness theorem [**4**, no. 15 th. 1.1], $f_*\mathcal{F}_1$ is a coherent $\mathcal{O}_{\mathcal{X}}$-algebra. It therefore corresponds to a finite covering $\mathcal{X}'$ of $\mathcal{X}$, which is moreover normal since $\mathcal{X}_1'$ is normal, and $\mathcal{X}'$ is the desired extension of $\mathcal{U}'$.

REMARK XII.5.5. In the statement XII.5.4, the hypothesis on the locus of points where the morphism $\mathcal{U}' \to \mathcal{U}$ is not étale cannot be omitted. For example, let $\mathcal{X}$ be the unit disk in the complex plane, let $\mathcal{U}$ be the complement of the origin in $\mathcal{X}$, and let $\mathcal{U}' = \mathcal{U}[T]/(T^2 - \sin 1/z)$, where z is the coordinate function on $\mathcal{X}$. Then $\mathcal{U}'$ is a normal finite covering of $\mathcal{U}$ which does not extend to $\mathcal{X}$. Indeed, suppose that $\mathcal{U}'$ extends to a finite covering $\mathcal{X}'$ of $\mathcal{X}$; the locus of points of $\mathcal{X}$ where the morphism $\mathcal{X}' \to \mathcal{X}$ is not étale would then be a closed analytic subset containing all points z such that $\sin 1/z = 0$, which is absurd.

The hypothesis on the singular locus of the morphism $\mathcal{U}' \to \mathcal{U}$ can, however, be omitted when $\operatorname{codim}(\mathcal{X} - \mathcal{U}, \mathcal{X}) \geq 2$. Indeed, we may suppose that $\mathcal{U}$ is regular. The locus of points of $\mathcal{U}$ where $\mathcal{U}' \to \mathcal{U}$ is not étale is a divisor of $\mathcal{U}$, and it follows from the REMMERT-STEIN theorem [**9**, th. 3] that it is the trace on $\mathcal{U}$ of a divisor of $\mathcal{X}$. Now, in 342
this case, if $\mathcal{A}$ is a coherent $\mathcal{O}_{\mathcal{U}}$-algebra such that $\mathcal{U}' = \operatorname{Spec}(\mathrm{an})(\mathcal{A})$, and if $i : \mathcal{U} \to \mathcal{X}$ is the canonical morphism, it suffices to take $\mathcal{X}' = \operatorname{Spec}(\mathrm{an})(i_*\mathcal{A})$; indeed, we know that $i_*\mathcal{A}$ is coherent [**11**, no. 1 th. 1].

REMARK. **[**Added in 2003 (MR): There exist nontrivial finitely presented groups G which have no finite-index subgroups distinct from G, for example G. Higman's group (cf. J-P. Serre, Arbres et amalgames, Astérisque N°46, prop. 6, ch. I, §1). Consequently, if such a group is realized as the topological fundamental group of a scheme V over $\mathbb{C}$, say a smooth projective scheme, the topological space V_{top} underlying V is not simply connected, but V is algebraically simply connected. At present, no such V is known. Let us nevertheless mention that D. Toledo constructed smooth projective schemes V over $\mathbb{C}$ whose topological fundamental group is not Hausdorff for the topology of finite-index subgroups (the natural morphism $\pi_1(V_{\mathrm{top}}) \to \pi_1(V_{\mathrm{alg}})$ is not injective). [D. Toledo, Projective varieties with non-residually finite fundamental group, Publ. Math. IHES **77** (1993), p. 103–119]. **]**

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CHAPTER XIII

Cohomological properness for sheaves of sets and sheaves of noncommutative groups

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by Mme M. Raynaud[1]

The purpose of this exposé is to use étale cohomology to generalize certain results of IX and X. It also shows how the results of SGA 5 II that still make sense for sheaves of noncommutative groups can be extended to such sheaves. We assume familiarity with the notions of étale cohomology presented in SGA 4.

The main result (XIII.2.4) gives an important example of a nonproper morphism $f: U \to S$ that is *cohomologically proper in dimension* ≤ 1, that is, such that for certain sheaves of groups F on U (for the étale topology), the formation of f_*F and $\mathrm{R}^1 f_* F$ commutes with every base change $S' \to S$. This property is indeed satisfied by the open subset U of a scheme X proper over S that is the complement of a divisor D with normal crossings relative to S, at least if F is required to be finite constant, of order prime to the residue characteristics of S. If F is no longer assumed to have order prime to the residue characteristics of S, there is an analogous result with $\mathrm{R}^1 f_* F$ replaced by the subsheaf $\mathrm{R}^1_{\mathrm{t}} f_* F$ obtained by considering only torsors under F that are *tamely ramified on X relative to S*. In particular, this makes it possible to show that the tamely ramified fundamental group of a proper smooth algebraic curve over a separably closed field, with finitely many closed points removed, is topologically of finite type (XIII.2.11).

The no. XIII.4 is devoted to the homotopy exact sequence and the Künneth formula. 345

Finally, an appendix gives useful variants of Abhyankar's lemma proved in X.X.3.6.

XIII.0. Review of the theory of stacks

We shall use below the theory of stacks presented in [**1**] and [**2**]. We restrict ourselves to the case of the étale site of a scheme. Given a scheme X, denote by $X_{\text{ét}}$ the étale site of X. Recall that a stack F on X is a category fibered over $X_{\text{ét}}$ such that, for every scheme X' étale over X and every pair of objects x, y of the fiber $F_{X'}$, the presheaf $\mathbf{Hom}_{X'}(x, y)$ is a sheaf, and such that, for every surjective étale morphism $X'' \to X'$, every object of $F_{X''}$ equipped with descent data relative to $X'' \to X'$ is the inverse image of an object of $F_{X'}$.

We denote by $F(X')$ the category of cartesian sections of F/X'. More generally, if $\mathbf{Sch}_X$ is the category of schemes over X, endowed with the étale topology, the stack F can be extended to a stack $\mathcal{F}$ on $\mathbf{Sch}_X$, and for every morphism $f: X' \to X$, we again denote by $F(X')$ the category of cartesian sections of this stack $\mathcal{F}$ over X'.

A gerbe is a stack such that, for every scheme X' étale over X and every pair of objects x, y of $F_{X'}$, every morphism from x to y is an isomorphism, x and y are locally isomorphic, and the collection of objects X' of $X_{\text{ét}}$ for which $F_{X'}$ is nonempty forms a covering of $X_{\text{ét}}$. For example, the stack of torsors under a sheaf of groups is a gerbe that,

[1] Based on unpublished notes by A. Grothendieck.

moreover, has a cartesian section. Conversely, a gerbe that has a section, i.e. for which there exists an object x of F_X, is equivalent to the stack of torsors under the sheaf of 346
groups $\mathbf{Aut}_X(x)$.

There is an evident notion of a subgerbe and of a maximal subgerbe of a stack F. Given a cartesian section x of $F(X)$, there is a unique maximal subgerbe G_x of F through which x factors. We call G_x the subgerbe generated by x; it is, by definition, a trivial gerbe. The presheaf SF defined by

$$SF(X') = \{\text{maximal subgerbes of } F|X'\}$$

is a sheaf, called the sheaf of maximal subgerbes of F. Let O be the presheaf defined by

$$O(X') = \{\text{classes of objects of } F_{X'} \text{ modulo isomorphism}\}.$$

By associating with every object x of $F_{X'}$ the maximal subgerbe of $F|X'$ generated by x, we obtain a morphism

$$O \to SF;$$

by [**2**, III 2.1.4], this morphism makes SF a sheaf associated with O.

A stack F is said to be *constructible* (*respectivelyind-*$\mathbb{L}$*-finite*, where $\mathbb{L}$ is a set of prime numbers) if, for every scheme X' étale over X and every object x of $F_{X'}$, the sheaf $\mathbf{Aut}_{X'}(x)$ has the corresponding property [**2**, VII 2.2.1]. A stack is said to be 1-constructible if it is constructible and its sheaf of maximal subgerbes is constructible.

XIII.1. Cohomological properness

. Let S be a scheme and $f\colon X \to Y$ a morphism of S-schemes. If S' is an S-scheme, consider the following diagram, all of whose squares are cartesian: 347

(XIII.1.-1.1)
$$\begin{array}{ccc} X & \xleftarrow{h} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{g} & Y' \\ \downarrow & & \downarrow \\ S & \longleftarrow & S'. \end{array}$$

If Y_1 is a scheme étale over Y, put $X_1 = X \times_Y Y_1$, $Y_1' = Y' \times_Y Y_1$, and consider the cartesian square

(XIII.1.-1.2)
$$\begin{array}{ccc} X_1 & \xleftarrow{h_1} & X_1' \\ {\scriptstyle f_1}\downarrow & & \downarrow{\scriptstyle f_1'} \\ Y_1 & \xleftarrow[g_1]{} & Y_1'. \end{array}$$

Definition XIII.1.1. *Let F be a stack on X. We say that (F, f) is* cohomologically proper relative to S in dimension $\leqslant -1$ *(respectivelyin dimension $\leqslant 0$, respectivelyin dimension $\leqslant 1$) if, for every S-scheme S', the canonical functor (defined in the evident way by the universal property of the inverse image of stacks)*

$$g^* f_* F \to f'_* h^* F \qquad \textit{(cf. 1.0.1)}$$

is faithful (respectivelyfully faithful, respectivelyan equivalence of categories).

If there is no possible ambiguity about S, in particular if $S = Y$, we say cohomologically proper instead of cohomologically proper relative to S.

. Let F be a sheaf of sets on X, and let Φ be the associated stack in discrete categories, i.e. the stack whose fiber over every scheme X_1 étale over X is the discrete category with object set $F(X_1)$. We say that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$) if (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$).

The canonical morphism 348

$$g^* f_* F \to f'_* h^* F \tag{XIII.1.-1.1}$$

induces, upon passing to the associated stacks in discrete categories, the canonical morphism

$$g^* f_* \Phi \to f'_* h^* \Phi.$$

Consequently, saying that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$) is equivalent to saying that, for every S-scheme S', the morphism (XIII.1.-1.1) is injective (respectivelybijective).

. Let F be a sheaf of groups on X, and let Φ be the stack of torsors on X with group F [**1**, II 2.3.2]. We say that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) if (Φ, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$). The cohomological properness condition can be made explicit as follows.

Proposition XIII.1.2. *Use the notation of* (XIII.1.-1.1) *and* (XIII.1.-1.2)*. Let F be a sheaf of groups on X. Denote by F' (respectivelyF_1, respectivelyF'_1, etc.) the inverse image of F on X' (respectivelyon X_1, respectivelyon X'_1, etc.). Then the following conditions are equivalent:*

(i) *(F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$).*

(ii) *For every morphism $S' \to S$, every scheme Y_1 étale over Y, and every torsor P on X_1 with group F_1, if ${}^P F_1$ denotes the twist of F_1 by P* [**1**, II 4.1.2.3]*, the canonical morphism*

$$a_0 \colon g_1^*(f_{1*}({}^P F_1)) \to f'_{1*}({}^{P'} F'_1)$$

is injective (respectivelya$_0$ is bijective and the canonical morphism

$$a_1 \colon g^*(\mathrm{R}^1 f_* F) \to \mathrm{R}^1 f'_* F'$$

is injective, respectivelya$_0$ and a_1 are bijective). 349

(ii bis) *For every morphism $S' \to S$, every scheme Y_1 étale over Y, every torsor P on X_1 with group F_1, and every torsor R under ${}^P F_1$, the canonical morphism*

$$\alpha_0 \colon g_1^*(f_{1*} R) \to f'_{1*} R'$$

is injective (respectivelyα$_0$ is bijective, respectivelythe morphisms α_0 and

$$\alpha_1 \colon g_1^*(\mathrm{R}^1 f_{1*}({}^P F_1)) \to \mathrm{R}^1 f'_{1*}({}^{P'} F'_1)$$

are bijective).

Proof. (i)$\Longrightarrow$(ii bis). By [**1**, II 4.2.5], every torsor R with group ${}^P F_1$ is of the form $R = Q \overset{F_1}{\wedge} P^\circ$, where Q is a torsor with group F_1 and P° is the opposite of P. We then have $R' \simeq Q' \overset{F'_1}{\wedge} P'^\circ$. Let Φ be the stack of torsors under F, and let x, y (respectivelyx', y') be the objects of the fiber category $(g^* f_* \Phi)_{Y'_1}$ (respectively$(f'_* \Phi')_{Y'_1}$) associated with P, Q (respectivelyP', Q'). We have

$$Q \overset{F_1}{\wedge} P^\circ \simeq \mathbf{Hom}_{F_1}(P, Q),$$

and it follows that there are canonical isomorphisms

$$\mathbf{Hom}_{Y_1'}(x,y) \simeq g_1^* f_{1*}(Q \overset{F_1}{\wedge} P^\circ), \quad \mathbf{Hom}_{Y_1'}(x',y') \simeq f_{1*}'(Q' \overset{F_1'}{\wedge} P'^\circ).$$

Consequently, the morphism α_0 is identified with the morphism

$$\mathbf{Hom}_{Y_1'}(x,y) \to \mathbf{Hom}_{Y_1'}(x',y'),$$

from which it follows that if (F,f) is cohomologically proper relative to S in dimension $\leqslant -1$, then α_0 is injective, and that if (F,f) is cohomologically proper in dimension $\leqslant 0$, then α_0 is bijective.

Now suppose that (F,f) is cohomologically proper relative to S in dimension $\leqslant 1$, i.e. that the canonical morphism 350

$$\varphi\colon g^* f_* \Phi \to f_*' \Phi'$$

is an equivalence. Let G be the sheaf of maximal subgerbes of the stack $f_*\Phi$ [**1**, III 2.1.8]; then there is an isomorphism $G \simeq \mathrm{R}^1 f_* F$. Since g^*G is the sheaf of maximal subgerbes of $g^* f_* \Phi$ [**2**, III 2.1.5.5], the morphism α_1 is obtained from $\varphi | Y_1'$ by taking the sheaves of maximal subgerbes, and is therefore an isomorphism.

(ii bis)$\Longrightarrow$(ii). It is enough to show that if the morphisms α_0 are bijective, then the morphisms a_1 are injective. Let Y_1' be a scheme étale over Y', and let s and t be two elements of $g^*(\mathrm{R}^1 f_* F)(Y_1')$ having the same image in $\mathrm{R}^1 f_*' F'(Y_1')$; we shall show that $s = t$. The assertion is local for the étale topology of Y_1', and, in view of the definition of the inverse image $g^*(\mathrm{R}^1 f_* F)$, we may suppose that Y_1' is the inverse image of a scheme Y_1 étale over Y, and that s and t come from torsors P and Q on X_1. The hypothesis on s and t then means that the inverse images P' and Q' of P and Q on X_1' are locally isomorphic for the étale topology of Y_1'. If we put $R = \mathbf{Hom}_{F_1}(P,Q)$, the fact that the morphism

$$g_1^* f_{1*} R \to f_{1*}' R'$$

is bijective proves that P and Q are locally isomorphic for the étale topology of Y_1, and hence that $s = t$.

(ii)$\Longrightarrow$(i). To prove that φ is faithful (respectivelyfully faithful), it is enough to show that if Y_1 is a scheme étale over Y, if P, Q are two torsors on X_1 with group F_1, and if x, y (respectivelyx', y') are the objects of $(g^* f_* \Phi)_{Y_1'}$ (respectively$(f_*' \Phi')_{Y_1'}$) associated with P, Q (respectivelyP', Q'), then the morphism

$$a\colon \mathrm{Hom}(x,y) \to \mathrm{Hom}(x',y')$$

is injective (respectivelybijective). But a is identified with the canonical morphism

$$\mathrm{H}^0(Y_1', g_1^* f_{1*}(Q \overset{F_1}{\wedge} P^\circ)) \to \mathrm{H}^0(Y_1', f_{1*}'(Q' \overset{F_1'}{\wedge} P'^\circ)).$$

If $\mathrm{Hom}(x,y) \neq \emptyset$, then $Q \overset{F_1}{\wedge} P^\circ$ is a torsor under ${}^P F_1$ that is locally trivial over Y_1. It 351
follows that $f_{1*}(Q \overset{F_1}{\wedge} P^\circ)$ is a torsor under $f_{1*}({}^P F_1)$, and that $g_1^* f_{1*}(Q \overset{F_1}{\wedge} P^\circ)$ is a trivial torsor. The morphism a is then identified with the canonical morphism

$$\mathrm{H}^0(Y_1, g_1^* f_{1*}({}^P F_1)) \to \mathrm{H}^0(Y_1', f_{1*}'({}^{P'} F_1')).$$

The same holds if $\mathrm{Hom}(x',y') \neq \emptyset$ and a_1 is injective, for then $Q' \overset{F_1'}{\wedge} P'^\circ$ is trivial, and the injectivity of a_1 implies that P and Q are locally isomorphic over Y_1. We conclude that if a_0 is injective (respectivelyif a_0 is bijective and a_1 injective), then φ is faithful (respectivelyfully faithful).

It remains to show that if a_0 and a_1 are bijective, the functor φ is essentially surjective. Let Y'' be a scheme étale over Y', put $X'' = X' \times_{Y'} Y''$, and let P'' be a torsor on X'' with group $F'' = F' | X''$. We shall show that there is an element x of $(g^* f_* \Phi)_{Y''}$ whose image in $(f_*' \Phi')_{Y''}$ is isomorphic to P''. Let p'' be the class of P''. Since a_1 is surjective,

we can find a surjective étale morphism $Y_1'' \to Y''$, an étale morphism $Y_1 \to Y$ such that there is a morphism $Y_1'' \to Y_1'$, and a torsor P_1 on X_1 with group F_1, whose inverse image P_1'' on X_1'' is isomorphic to the inverse image of P''. Using the fact that φ is fully faithful, we see that the object x_1 of $(g^* f_* \Phi)_{Y_1''}$ corresponding to P_1'' is equipped with descent data relative to $Y_1'' \to Y''$, and hence comes from an element x of $(g^* f_* \Phi)_{Y''}$. Since the image of x in $(f'_* \Phi')_{Y''}$ is P'', this proves that φ is essentially surjective and completes the proof.

EXAMPLE XIII.1.3. *Let $f\colon X \to Y$ be a proper morphism. It follows from* [**2**, VII 2.2.2] *that, for every ind-finite stack F on X, the pair (F, f) is cohomologically proper (relative to Y) in dimension $\leqslant 1$. In particular, for every sheaf of sets (respectivelyevery sheaf of groups, respectivelyevery ind-finite sheaf of groups) F on X, (F, f) is cohomologically proper in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 0$, respectivelyin dimension $\leqslant 1$).* 352

REMARKS XIII.1.4. a) Let F be a sheaf of groups on X such that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$). If F is regarded as a sheaf of sets, (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$), but the converse is false.

For example, let Y be the spectrum of a strictly local discrete valuation ring, with closed point t and generic point s; let $f\colon X \to Y$ be a nonempty scheme over Y whose closed fiber is empty; let F be a nontrivial constant sheaf of groups on X; and let P be a torsor under F such that $\mathrm{H}^0(X_s, {}^P F|X_s) = 1$. Then $({}^P F, f)$ is cohomologically proper relative to Y in dimension $\leqslant -1$ when ${}^P F$ is regarded as a sheaf of sets. If ${}^P F$ is regarded as a sheaf of groups, there is an isomorphism ${}^{P^\circ}({}^P F) \simeq F$; since the canonical morphism

$$\mathrm{H}^0(X, F) \to \mathrm{H}^0(X_t, F|X_t) = 1$$

is not injective, this proves that $({}^P F, f)$ is not cohomologically proper relative to Y in dimension $\leqslant -1$.

b) Suppose that f is coherent (i.e. quasi-compact and quasi-separated). Let F be a stack on X. For every geometric point $\overline{y}$ of Y', denote by $\overline{Y}$ (respectively$\overline{Y'}$) the strict localization of Y (respectivelyY') at $\overline{y}$, and put $\overline{X} = X \times_Y \overline{Y}$, $\overline{X'} = X' \times_{Y'} \overline{Y'}$, etc. The pair (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) if and only if, for every S-scheme S' and every geometric point $\overline{y}$ of Y', the canonical functor

$$\overline{F}(\overline{X}) \to \overline{F'}(\overline{X'})$$

is faithful (respectivelyfully faithful, respectivelyan equivalence).

Indeed, if S' is an S-scheme, the functor 353

$$g^* f_* F \to f'_* F'$$

is faithful (respectivelyfully faithful, respectivelyan equivalence) if and only if the same is true of the functor induced on the fibers at the various geometric points $\overline{y}'$ of $\overline{Y'}$ [**2**, III 2.1.5.9]. The assertion therefore follows from the calculation of the geometric fibers of the direct image of a stack under a coherent morphism [**2**, VII 2.1.5].

c) Let F be a stack on X. The property that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) is local on Y for the étale topology.

Let S' be an S-scheme, and let F' be the inverse image of F on X' (cf. (XIII.1.-1.1)). If (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, then the same is true of (F', f'). But if (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$), the same need not be true of (F', f').

For example, let S' be a discrete valuation ring, let $f'\colon E_{S'} \to S'$ be affine space over S', let x be a closed point of $E_{S'}$ over the generic point of S', and let F' be the sheaf of sets on $E_{S'}$ whose restriction to $E_{S'} - \{x\}$ is the one-element constant sheaf and whose fiber at a geometric point over x has two elements. Then (F', f') is not cohomologically

proper relative to S' in dimension $\leqslant -1$. Let $S = S'[Z]$, let $f\colon E_S \to S$ be affine space over S, and let T be a closed subset of $X = E_S$ that does not meet the closed subset $Z = 0$ and such that $f(T)$ contains the generic point of S. Let G be the inverse image of F' on X, and let F be the sheaf on X obtained by extending $G|X - T$ by the empty sheaf. Then (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$, but this is no longer true after the base change $S' \to S$ defined by $Z = 0$.

d) Let F be a stack on X such that (F, f) is cohomologically proper relative to Y in 354
dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$). Then, for every geometric point $\overline{y}$ of Y, the canonical functor

$$(f_*F)_{\overline{y}} \to F(X_{\overline{y}})$$

is faithful (respectivelyfully faithful, respectivelyan equivalence of categories).

PROPOSITION XIII.1.5. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be two S-morphisms, and let Φ be a stack on X.*

1) *Suppose that (Φ, f) and $(f_*\Phi, g)$ are cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$). Then the same is true of (Φ, gf).*
2) *Suppose that (Φ, gf) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelythat (Φ, gf) is cohomologically proper relative to S in dimension $\leqslant 0$ and (Φ, f) is cohomologically proper relative to S in dimension $\leqslant -1$, respectivelythat (Φ, gf) is cohomologically proper relative to S in dimension $\leqslant 1$ and (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 0$). Then $(f_*\Phi, g)$ is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$, respectivelyin dimension $\leqslant 1$).*

For every S-scheme S', consider the following diagram, all of whose squares are cartesian:

$$\begin{array}{ccccccc} X' & \xrightarrow{f'} & Y' & \xrightarrow{g'} & Z' & \longrightarrow & S' \\ {\scriptstyle h}\downarrow & & {\scriptstyle k}\downarrow & & {\scriptstyle m}\downarrow & & \downarrow \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & S. \end{array} \tag{XIII.1.-1.1}$$

The canonical morphism

$$m^*(g_*f_*\Phi) \to g'_*f'_*(h^*\Phi)$$

is identified with the composite of the canonical morphisms 355

$$m^*(g_*f_*\Phi) \xrightarrow{i} g'_*(k^*f_*\Phi) \xrightarrow{j} g'_*f'_*(h^*\Phi).$$

1) The hypothesis implies that i and j are faithful (respectivelyfully faithful, respectivelyequivalences); hence the same is true of ji.
2) The hypothesis implies that ji is faithful (respectivelythat ji is fully faithful and j is faithful, respectivelythat ji is an equivalence and j is fully faithful); it follows that i is faithful (respectivelyfully faithful, respectivelyan equivalence).

COROLLARY XIII.1.6. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be two S-morphisms, and let F be a sheaf of groups on X. Suppose that (F, gf) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelythat (F, gf) is cohomologically proper relative to S in dimension $\leqslant 0$ and that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$). Then (f_*F, g) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$).*

Retain the notation of (XIII.1.-1.1), and, for every scheme Y_1 étale over Y, denote by f_1, F_1 the respective inverse images of f, F under the morphism $Y_1 \to Y$. Let Φ be

the stack of torsors under F, and let Ψ be the stack of torsors under f_*F. There is a canonical functor

$$\varphi\colon \Psi \to f_*\Phi,$$

obtained by associating with every scheme Y_1 étale over Y and every torsor P on Y_1 with group $f_{1*}F_1$, the torsor $\widetilde{P}$ on X_1 obtained from f_1^*P by extension of structure group along $f_1^*f_{1*}F_1 \to F_1$. The functor φ is fully faithful. Indeed, if P and Q are two torsors on Y_1 with group $f_{1*}F_1$, there is a canonical morphism

$$\mathbf{Isom}_{f_{1*}F_1}(P,Q) \to f_{1*}(\mathbf{Isom}_{F_1}(\widetilde{P},\widetilde{Q}))$$

which is an isomorphism because this is so locally. It follows that the canonical morphism 356

$$\mathrm{Isom}_{f_{1*}F_1}(P,Q) \to \mathrm{Isom}_{F_1}(\widetilde{P},\widetilde{Q})$$

is an isomorphism, and hence that φ is fully faithful.

There is a commutative diagram

$$\begin{array}{ccc} g'_*k^*\Psi & \xrightarrow{\ i\ } & m^*(g_*\Psi) \\ {\scriptstyle g'_*k^*(\varphi)}\downarrow & & \downarrow{\scriptstyle m^*g_*(\varphi)} \\ g'_*k^*(f_*\Phi) & \xrightarrow{\ j\ } & m^*(g_*f_*\Phi), \end{array}$$

where i and j are the base-change morphisms. It follows from XIII.1.5 2) that j is faithful (respectivelyfully faithful). Since $g'_*k^*(\varphi)$ and $m^*g_*(\varphi)$ are fully faithful, the diagram above shows that i is faithful (respectivelyfully faithful).

Corollary XIII.1.7. *Let $f\colon X \to Y$ be a coherent S-morphism, let $g\colon Y \to Z$ be a proper S-morphism, and let Φ be an ind-finite stack on X* [**2**, VII 2.2.1]. *Suppose that (Φ, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$); then the same is true of (Φ, gf).*

Since f is coherent, $f_*\Phi$ is an ind-finite stack (SGA 4 IX 1.6 (ii)). The corollary therefore follows from XIII.1.5 1) and XIII.1.3.

Corollary XIII.1.8. *Let $f\colon X \to Y$ be an integral S-morphism, and let $g\colon Y \to Z$ be an S-morphism. If F is a sheaf of sets on X, then (f_*F, g) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$) if and only if the same is true of (F, gf). If F is a sheaf of groups on X, then (f_*F, g) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) if and only if the same is true of (F, gf).*

The assertion for a sheaf of sets follows from XIII.1.5 and from the fact that (F, f) 357 is cohomologically proper relative to S in dimension $\leqslant 0$. Let F be a sheaf of groups on X, and let Φ be the stack of torsors under F. By SGA 4 VIII 5.8, every torsor under F is locally trivial over Y. It follows that the stack $f_*\Phi$ is equivalent to the stack of torsors under f_*F, the equivalence being obtained by associating with every scheme Y_1 étale over Y and every torsor P on $X_1 = X \times_Y Y_1$ with group $F|X_1$, the torsor f_*P with group $f_*F|Y_1$. Since (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, the corollary therefore follows from XIII.1.5.

Definitions XIII.1.9.

. Let E be a category, and consider a diagram

$$\Phi \xrightarrow{\ p\ } \Phi_1 \underset{p_2}{\overset{p_1}{\rightrightarrows}} \Phi_2\ ,$$

where Φ, Φ_1, Φ_2 are categories fibered over E, the arrows are morphisms of fibered categories, and

$$a\colon p_1p \xrightarrow{\sim} p_2p$$

is an isomorphism of functors.

We say that the diagram above is exact if the following condition is satisfied:

a) For every pair of objects x, y of Φ and every morphism $f\colon p(x) \to p(y)$ such that $p_1(f) = p_2(f)$ (p_1p and p_2p being identified by means of a), there exists a unique morphism $g\colon x \to y$ such that $p(g) = f$.

. Consider the diagram

$$\Phi \xrightarrow{p} \Phi_1 \underset{}{\overset{p_1,p_2}{\rightrightarrows}} \Phi_2 \underset{}{\overset{p_{12},p_{23},p_{31}}{\rightrightarrows}} \Phi_3 \ ,$$

where Φ, Φ_i, $1 \leqslant i \leqslant 3$, are categories fibered over E, and the arrows are morphisms of 358
fibered categories. Suppose that we are given isomorphisms of functors

$$a\colon p_1p \xrightarrow{\sim} p_2p$$

$$a_1\colon p_{31}p_2 \xrightarrow{\sim} p_{12}p_1, \quad a_2\colon p_{12}p_2 \xrightarrow{\sim} p_{23}p_1, \quad a_3\colon p_{23}p_2 \xrightarrow{\sim} p_{31}p_1$$

such that the following diagram commutes:

$$\begin{array}{ccccc} p_{23}p_1p & \xrightarrow{\mathrm{id}.a} & p_{23}p_2p & \xrightarrow{a_3.\mathrm{id}} & p_{31}p_1p \\ {\scriptstyle a_2.\mathrm{id}}\uparrow & & & & \downarrow{\scriptstyle \mathrm{id}.a} \\ p_{12}p_2p & \xleftarrow[\mathrm{id}.a]{} & p_{12}p_1p & \xleftarrow[a_1.\mathrm{id}]{} & p_{31}p_2p \end{array} \ .$$

We identify p_1p and p_2p, $p_{31}p_2$ and $p_{12}p_1$, and so on.

We say that the diagram above is exact if the following conditions are satisfied:

a) The analogue of condition a) of XIII.1.9.

b) For every object x_1 of Φ_1 and every isomorphism $u\colon p_1(x_1) \xrightarrow{\sim} p_2(x_1)$ such that

(XIII.1.-1.0.1) $$p_{23}(u)p_{31}(u) = p_{12}(u)^{-1},$$

there exists an object x of Φ and an isomorphism $i\colon p(x) \xrightarrow{\sim} x_1$ making the diagram

(XIII.1.-1.0.2) $$\begin{array}{ccc} p_1p(x) & = & p_2p(x) \\ {\scriptstyle p_1(i)}\downarrow & & \downarrow{\scriptstyle p_2(i)} \\ p_1(x_1) & \xrightarrow[u]{\sim} & p_2(x_1) \end{array}$$

commute.

. The notion of a morphism of exact diagrams of fibered categories over a category E is defined in the evident way.

. We shall use in particular the notion of an exact diagram in the case where E is a 359
site and Φ, Φ_i, $1 \leqslant i \leqslant 3$, are stacks on E.

Let $f\colon E \to E'$ be a morphism of sites and let

(XIII.1.-1.0.1) $$\Phi \xrightarrow{p} \Phi_1 \underset{}{\overset{p_1,p_2}{\rightrightarrows}} \Phi_2 \underset{}{\overset{p_{12},p_{23},p_{31}}{\rightrightarrows}} \Phi_3$$

be an exact diagram of stacks on E. Taking direct images gives a diagram

$$f_*\Phi \longrightarrow f_*\Phi_1 \rightrightarrows f_*\Phi_2 \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} f_*\Phi_3$$

which is evidently exact.

If $h\colon E'' \to E$ is a morphism of sites, one likewise has an exact diagram

$$h^*\Phi \xrightarrow{p''} h^*\Phi_1 \underset{}{\overset{p''_1,p''_2}{\rightrightarrows}} h^*\Phi_2 \underset{}{\overset{p''_{12},p''_{23},p''_{31}}{\rightrightarrows}} h^*\Phi_3$$

Let us first verify condition a) of XIII.1.9. Let F'' be an object of E'', let x'' and y'' be two objects of $(h^*\Phi)_{F''}$, let x''_1 and y''_1 be their respective images in $h^*\Phi_1$, and let x''_2, y''_2 be their images in $h^*\Phi_2$. Let $u''_1\colon x''_1 \to y''_1$ be a morphism such that $p''_1(u''_1) = p''_2(u''_1)$, and let us prove that u''_1 comes from a unique morphism $u''\colon x'' \to y''$. Since the question is

local on F'', we may suppose that we have an object F_1 of E, a morphism from F'' to the inverse image F_1'' of F_1 under h, and objects x, y of Φ_{F_1} whose inverse images on F'' are x'' and y''. Let x_1, y_1 (respectivelyx_2, y_2) be the images of x, y in Φ_1 (respectively Φ_2). We may suppose that u_1'' comes from a morphism $u_1\colon x_1 \to y_1$ such that $p_1(u_1) = p_2(u_1)$. By the exactness of (XIII.1.-1.0.1), we obtain a unique morphism $u\colon x \to y$ whose inverse image under h is the desired morphism u''.

The condition b) of XIII.1.9 is verified in the same way. Let x_1'' be an object of $(h^*\Phi_1)_{F''}$ 360
and let $u''\colon p_1''(x_1'') \to p_2''(x_1'')$ be a morphism satisfying the relation

$$p_{23}''(u'')p_{31}''(u'') = p_{12}''(u'')^{-1},$$

and let us prove that there exists an object x'' of $(h^*\Phi)_{F''}$ and an isomorphism $i''\colon p''(x'') \simeq x_1''$ making a diagram analogous to (XIII.1.-1.0.2) commute. Since the question is local on F'', we may suppose that we have an object F_1, a morphism $F'' \to F_1''$ as above, and an object x_1 of $(\Phi_1)_{F_1}$ whose inverse image in $(h^*\Phi_1)_{F''}$ is x_1''. Likewise, we may suppose that u'' comes from a morphism $u\colon p_1(x_1) \to p_2(x_1)$ satisfying (XIII.1.-1.0.2). The existence of an object x of Φ_{F_1} whose inverse image under h is an object x'' with the required property then follows from the exactness of (XIII.1.-1.0.1).

Examples XIII.1.10. 1) *Let $f\colon X_1 \to X$ be a* descent morphism *for the category of étale sheaves on variable schemes (for example, a universally submersive morphism (SGA 4 VIII 9.3)). Let $X_2 = X_1 \times_X X_1$, let $g\colon X_2 \to X$ be the canonical projection, and let F be a sheaf of sets on X. It then follows from loc. cit. that there is an exact sequence of sheaves of sets*

(XIII.1.-1.1) $$F \longrightarrow f_*f^*F \rightrightarrows g_*g^*F.$$

If Φ is the stack in discrete categories associated with F, and Φ_3 is the final stack on X, i.e. the stack all of whose fibers consist of a single object whose only morphism is the identity morphism, saying that the sequence (XIII.1.-1.1) *is exact is equivalent to saying that the diagram of stacks*

$$\Phi \longrightarrow f_*f^*\Phi \rightrightarrows g_*g^*\Phi \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} \Phi_3\ .$$

is exact.

2) *Let $f\colon X_1 \to X$ be an* effective descent morphism *for the category of étale sheaves on variable schemes (for example, a proper surjective morphism, an integral surjective morphism, or a faithfully flat morphism locally of finite presentation (SGA 4 VIII 9.4)). Let $X_2 = X_1 \times_X X_1$, let $g\colon X_2 \to X$ be the canonical projection, let $X_3 = X_1 \times_X X_1 \times_X X_1$, and let $h\colon X_3 \to X$ be the canonical morphism. Let Φ be a stack on X, let $\Phi_1 = f_*f^*\Phi$, $\Phi_2 = g_*g^*\Phi$, and $\Phi_3 = h_*h^*\Phi$. We then have an exact diagram* 361

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} \Phi_3,$$

where the arrows are the canonical morphisms associated with the projections.

Indeed, regard Φ as a stack on the category $\mathbf{Sch}_X$ of schemes over X, endowed with the étale topology. Then, by [**2**, VII 2.2.8], *Φ is also a stack for the finest topology on $\mathbf{Sch}_X$ for which the covering morphisms are the effective descent morphisms for the category of étale sheaves. The exactness of the diagram above follows immediately.*

Proposition XIII.1.11. *Let S be a scheme, and let $f\colon X \to Y$ be an S-morphism.*

1) *Let*

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2$$

be an exact diagram of stacks on X. Suppose that (Φ_1, f) is cohomologically proper relative to S in dimension $\leqslant 0$ and that (Φ_2, f) is cohomologically proper

relative to S in dimension $\leqslant -1$. Then (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 0$.

2) *Let*

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \;\substack{\rightarrow\\\rightarrow\\\rightarrow}\; \Phi_3$$

be an exact diagram of stacks on X. Suppose that (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 1$, that (Φ_2, f) is cohomologically proper relative to S in dimension $\leqslant 0$, and that (Φ_3, f) is cohomologically proper relative to S in dimension $\leqslant -1$. Then (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 1$. 362

For every S-scheme S', consider the following commutative diagram, all of whose squares are cartesian:

$$\begin{array}{ccccc} X' & \xrightarrow{f'} & Y' & \longrightarrow & S' \\ {\scriptstyle h}\downarrow & & {\scriptstyle g}\downarrow & & \downarrow \\ X & \xrightarrow{f} & Y & \longrightarrow & S \end{array}$$

Let us prove 2), the proof of 1) being analogous. Since the direct and inverse image functors take exact diagrams of stacks to exact diagrams (XIII.1), there is the following morphism of exact diagrams of stacks:

$$\begin{array}{ccccccc} g^*f_*\Phi & \xrightarrow{\pi} & g^*f_*\Phi_1 & \underset{\pi_2}{\overset{\pi_1}{\rightrightarrows}} & g^*f_*\Phi_2 & \;\substack{\rightarrow\\\rightarrow\\\rightarrow}\; & g^*f_*\Phi_3\ . \\ {\scriptstyle\varphi}\downarrow & & {\scriptstyle\varphi_1}\downarrow & & {\scriptstyle\varphi_2}\downarrow & & {\scriptstyle\varphi_3}\downarrow \\ f'_*h^*\Phi & \longrightarrow & f'_*h^*\Phi_1 & \rightrightarrows & f'_*h^*\Phi_2 & \;\substack{\rightarrow\\\rightarrow\\\rightarrow}\; & f'_*h^*\Phi_3 \end{array}$$

By hypothesis, φ_1 is an equivalence of categories, φ_2 is fully faithful, and φ_3 is faithful. It therefore follows from the preceding diagram that φ is an equivalence.

PROPOSITION XIII.1.12. *Let $f\colon X \to Y$ be an S-morphism.*

1) *Let*

$$F \longrightarrow G \rightrightarrows H$$

be an exact diagram of sheaves of sets on X. Suppose that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ and that (H, f) is cohomologically proper relative to S in dimension $\leqslant -1$. Then (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$.

2) *Let $F \to G$ be a monomorphism of sheaves of groups on X. If Y_1 is a scheme étale over Y, put $X_1 = Y_1 \times_Y X$, and denote by f_1 (respectivelyF_1, respectivelyG_1) the inverse image of f (respectively$F$, respectively$G$) over Y_1 (cf. XIII.1.-1.2). Suppose that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$) and that, for every scheme Y_1 étale over Y and every torsor Q under G_1, $(Q/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$). Then (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$).* 363

3) *Let $F \to G$ be a monomorphism of sheaves of groups on X. Suppose that (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$ and that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$. Then, for every torsor Q under G, $(Q/F, f)$ is cohomologically proper relative to S in dimension $\leqslant 0$.*

Proof. 1) Let Φ (respectivelyΦ_1, respectivelyΦ_2) be the stack in discrete categories associated with F (respectively$G$, respectively$H$), and let Φ_3 be the final stack on X. There is then an exact diagram

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \;\substack{\rightarrow\\\rightarrow\\\rightarrow}\; \Phi_3\ .$$

By hypothesis, (Φ_1, f) is cohomologically proper relative to S in dimension $\leqslant 1$, and (Φ_2, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (XIII.1). Since (Φ_3, f) is clearly cohomologically proper relative to S in dimension $\leqslant -1$, it follows from XIII.1.11 that (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 1$, i.e. that (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$.

2) Let us first show that, if (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ and if the pairs $(Q/F_1, f_1)$ are cohomologically proper relative to S in dimension $\leqslant -1$, then (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$. By XIII.1.2, it suffices to prove that, for every scheme Y_1 ètale over Y and every torsor P on X_1 with group F_1, $({}^PF_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant 0$ when ${}^PF_1$ is regarded as a sheaf of sets, and that the canonical morphism 364

$$d\colon g^*(\mathrm{R}^1 f_* F) \to \mathrm{R}^1 f'_* F'$$

is injective. The first assertion follows at once from 1), for if Q denotes the torsor obtained from P by extension of structure group along $F_1 \to G_1$, there is an isomorphism ${}^QG_1/{}^PF_1 \xrightarrow{\sim} Q/F_1$.

Let us show that d is injective. It suffices to prove that, if Y_1 is a scheme ètale over Y, and if P and $\tilde{P}$ are two torsors under F_1 whose inverse images P' and $\tilde{P}'$ on X'_1 are isomorphic, then, after replacing Y_1 by a surjective ètale extension, P and $\tilde{P}$ become isomorphic. Choose an isomorphism $p'\colon P' \xrightarrow{\sim} \tilde{P}'$. Let Q (respectively $\tilde{Q}$) be the torsor obtained from P (respectively$\tilde{P}$) by extension of structure group along $F_1 \to G_1$. The inverse images Q' (respectively$\tilde{Q}'$) of Q (respectively$\tilde{Q}$) on X'_1 are obtained from P' (respectively$\tilde{P}'$) by the extension of the structure group $F'_1 \to G'_1$; let $q'\colon Q' \xrightarrow{\sim} \tilde{Q}'$ be the isomorphism obtained in the same way from p'. Since (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$, after replacing Y_1 by a surjective ètale extension, we may suppose that q' is the inverse image of an isomorphism $q\colon Q \xrightarrow{\sim} \tilde{Q}$. Associated with the torsor P (respectively $\tilde{P}$) is a section x of Q/F_1 (respectivelya section $\tilde{x}$ of $\tilde{Q}/F_1$), and, for P and $\tilde{P}$ to be isomorphic, it is necessary and sufficient that there be an isomorphism $Q \xrightarrow{\sim} \tilde{Q}$ such that the induced isomorphism

$$e\colon \mathrm{H}^0(X_1, Q/F_1) \to \mathrm{H}^0(X_1, \tilde{Q}/F_1)$$

carries x to $\tilde{x}$. Take the isomorphism q. The sections $e(x)$ and $\tilde{x}$ of $\mathrm{H}^0(X_1, \tilde{Q}/F_1)$ have the same image in $\mathrm{H}^0(X'_1, \tilde{Q}'/F'_1)$. Since $(\tilde{Q}/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant -1$, after replacing Y_1 by a surjective ètale extension, we indeed have $e(x) = \tilde{x}$, which proves the injectivity of d. 365

To complete the proof, it remains to show that, if (G, f) is cohomologically proper relative to S in dimension $\leqslant 1$, and if, for every scheme Y_1 ètale over Y and every torsor Q on X_1 with group F_1, $(Q/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant 0$, then the morphism d is surjective. Let P' be a torsor on X'_1 with group F'_1, and let Q' be the torsor under G'_1 obtained from P' by extension of structure group. The data of P' are equivalent to those of Q' together with a section x' of $\mathrm{H}^0(X'_1, Q'/F'_1)$. It follows from the surjectivity of the morphism

$$g^*(\mathrm{R}^1 f_* G) \to \mathrm{R}^1 f'_* G'$$

that, after replacing Y_1 by a surjective ètale extension, there is a torsor Q under G_1 whose inverse image on X'_1 is isomorphic to Q'. Using the fact that $(Q/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant 0$, we may likewise suppose that there is an element x of $\mathrm{H}^0(X_1, Q/F_1)$ whose image in $\mathrm{H}^0(X'_1, Q'/F'_1)$ is x'. Together, Q and x determine a torsor P under F_1 whose inverse image on X_1 is isomorphic to P', which proves the surjectivity of d.

3) Let us show that $(Q/F, f)$ is cohomologically proper relative to S in dimension $\leqslant -1$, i.e. that, for every S-scheme S', and every scheme Y_1 ètale over Y, if x, $\tilde{x}$ are two

elements of $\mathrm{H}^0(X_1, Q_1/F_1)$ whose images x', $\tilde{x}'$ in $\mathrm{H}^0(X'_1, Q'_1/F'_1)$ are equal, then, after a surjective extension of Y_1, we have $x = \tilde{x}$. Associated with x (respectively $\tilde{x}$) is a torsor P (respectively $\tilde{P}$) under F_1 such that Q_1 is obtained from P (respectively $\tilde{P}$) by extension of structure group along $F_1 \to G_1$. It follows from the relation $x' = \tilde{x}'$ that there is an isomorphism $u' \colon P' \xrightarrow{\sim} \tilde{P}'$ such that the isomorphism induced on Q'_1 by u' is the identity. Since (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$, it follows that, after 366
a surjective ètale extension of Y_1, there is an isomorphism $u \colon P \to \tilde{P}$ lifting u'; the fact that (G, f) is cohomologically proper relative to S in dimension $\leqslant -1$ then implies that $x = \tilde{x}$.

Let us show that $(Q/F, f)$ is cohomologically proper relative to S in dimension $\leqslant 0$. Let Y'' be a scheme ètale over Y', and let x'' be an element of $\mathrm{H}^0(X'', Q''/F'')$. Associated with x'' is a torsor P'' on X'' with group F''. Since (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, one can find surjective ètale morphisms $Y''_1 \to Y''$ and $Y_1 \to Y$ such that there is a morphism $Y''_1 \to Y'_1$, and a torsor P on X_1 with group F_1 whose inverse image on X''_1 is isomorphic to the inverse image of P''. It then follows from the fact that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ that one can even choose Y''_1 and Y_1 so that the torsor obtained from P by extension of structure group along $F_1 \to G_1$ is isomorphic to Q_1; the torsor P corresponds to an element x of $\mathrm{H}^0(X_1, Q_1/F_1)$ whose image in $\mathrm{H}^0(X''_1, Q''_1/F''_1)$ is isomorphic to the inverse image of x'', which completes the proof.

Proposition XIII.1.13. *Let $f \colon X \to S$ be the structure morphism of an S-scheme, and let F be a sheaf of sets or groups on X (respectivelya sheaf of ind-$\mathbb{L}$-groups, where $\mathbb{L}$ is a set of prime numbers). Suppose that F is locally constant, that (F, f) is cohomologically proper in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$), and that f is locally 0-acyclic (respectivelylocally 1-aspherical for $\mathbb{L}$)* (SGA 4 XV 1.11). *Then, for every specialization $\overline{s}_1 \to \overline{s}_2$ of geometric points of S, the specialization morphism* (SGA 4 VIII 7.1)

$$a_0 \colon (f_* F)_{\overline{s}_2} \to (f_* F)_{\overline{s}_1}$$

is an isomorphism, and, if F is a sheaf of groups, the morphism

$$a_1 \colon (\mathrm{R}^1 f_* F)_{\overline{s}_2} \to (\mathrm{R}^1 f_* F)_{\overline{s}_1}$$

is injective (respectivelythe morphisms a_0 and a_1 are isomorphisms).

The proof is obtained by reproducing verbatim that of SGA 4 XVI 2.3, with the 367
expression "proper"replaced by the expression "cohomologically proper".

Corollary XIII.1.14. *Let $f \colon X \to S$ be a morphism, let Φ be a stack on X, and let $\mathbb{L}$ be a set of prime numbers. Suppose that, for every scheme X_1 étale over X and every pair of objects x, y of Φ_{X_1}, the sheaf $\mathbf{Hom}_{X_1}(x, y)$ is locally constant, the sheaf $\mathbf{Aut}_{X_1}(x)$ is a locally constant ind-$\mathbb{L}$-group, and the sheaf $S\Phi$ of maximal subgerbes of Φ* [**1**, III 2.1.7] *is locally constant. Suppose that (Φ, f) is cohomologically proper in dimension $\leqslant 1$ and that f is locally 1-aspherical for $\mathbb{L}$. Then, for every specialization $\overline{s}_1 \to \overline{s}_2$ of geometric points of S, the specialization morphism*

$$a \colon (f_* \Phi)_{\overline{s}_2} \to (f_* \Phi)_{\overline{s}_1}$$

is an equivalence of categories.

Let $\overline{S}_1$ (respectively$\overline{S}_2$) be the strict localization of S at $\overline{s}_1$ (respectivelythe strict localization of S at $\overline{s}_2$), and let $\overline{X}_2$, $\overline{\Phi}_2$ (respectively$\overline{X}_1$, $\overline{\Phi}_1$) be the inverse images of X_2, Φ_2 on $\overline{S}_2$ (respectivelyof X_1, Φ_1 on $\overline{S}_1$), and consider the cartesian square

$$\begin{array}{ccc} \overline{X}_1 & \xrightarrow{h} & \overline{X}_2 \\ {\scriptstyle \overline{f}_1}\downarrow & & \downarrow{\scriptstyle \overline{f}_2} \\ \overline{S}_1 & \xrightarrow{g} & \overline{S}_2 \end{array} .$$

We must show that the functor

$$\varphi\colon \overline{\Phi}_2(\overline{X}_2) \to \overline{\Phi}_1(\overline{X}_1)$$

is an equivalence. The functor φ is fully faithful; in fact, let x, y be two objects of $(\overline{\Phi}_2)_{\overline{X}_2}$; 368
the canonical morphism

$$\mathrm{Hom}_{\overline{X}_2}(x,y) \to \mathrm{Hom}_{\overline{X}_1}(\varphi(x),\varphi(y))$$

is identified with the canonical morphism

$$\mathrm{H}^0(\overline{X}_2, \mathbf{Hom}_{\overline{X}_2}(x,y)) \to \mathrm{H}^0(\overline{X}_1, h^*(\mathbf{Hom}_{\overline{X}_2}(x,y)).$$

This morphism is an isomorphism by XIII.1.13.

Let us show that φ is an equivalence. Let x_1 be an object of $\overline{\Phi}_1(\overline{X}_1)$, and let G_1 be the maximal subgerbe of $\overline{\Phi}_1$ generated by x_1. The morphism

$$\mathrm{H}^0(\overline{X}_2, S\overline{\Phi}_2) \to \mathrm{H}^0(\overline{X}_1, h^*(S\overline{\Phi}_2)) = \mathrm{H}^0(\overline{X}_1, S\overline{\Phi}_1)$$

is bijective, and hence there exists a maximal subgerbe G_2 of $\overline{\Phi}_2$ such that h^*G_2 is isomorphic to G_1. It then suffices to prove that the functor

$$G_2 \to h_*h^*G_2$$

is an equivalence. In this form, however, the question is local for the étale topology on $\overline{X}_2$. We may therefore suppose that G_2 is a gerbe of torsors under the automorphism group of an object of G_2, in which case the assertion follows from XIII.1.13.

Corollary XIII.1.15. *The hypotheses are those of XIII.1.13. Suppose in addition that F is a sheaf of sets (respectivelyof ind-$\mathbb{L}$-groups) and that f_*F (respectivelyR^1f_*F) is constructible. Then f_*F (respectivelyR^1f_*F) is locally constant.*

The corollary follows from XIII.1.13 by SGA 4 IX 2.11.

Remark XIII.1.16. Recall that the condition that f be locally 0-acyclic is satisfied if f is flat with separable fibers and X and Y are locally Noetherian (SGA 4 XV 4.1), and that the condition that f be locally 1-aspherical for $\mathbb{L}$ is satisfied if f is smooth, with $\mathbb{L}$ 369
the set of prime numbers distinct from the residual characteristics of S (SGA 4 XV 2.1).

XIII.2. A special case of cohomological properness: relative normal crossings divisors

. Let R be a discrete valuation ring with fraction field K, and let L be an étale K-algebra. Then L is the direct product of finitely many fields L_i, where L_i is an étale extension of K. If L_i' denotes the Galois extension generated by L_i in an algebraic closure of L_i, we say that L is *tamely ramified* over R if the L_i' are tamely ramified extensions in the sense of X X.3, i.e. if an inertia group I_i of $L_i'|K$ has order prime to the residue characteristic p of R.

It is known that I_i is in any case an extension of a cyclic group of order prime to p by a p-group. (This follows from [**5**, ch. IV prop. 7 cor. 4] when the residue extension of R is separable. The proof given in loc. cit. extends to the general case as follows. Retain the hypotheses and notation of loc. cit. , but without assuming the residue extension separable. Let H_i be the subgroup of the inertia group G_0 consisting of the elements s of G_0 such that $s\pi/\pi \in U^i$ for every uniformizer π of A_L. One then checks that G_0/H_1 is a group of order prime to p and that, for $i \geqslant 1$, the H_i/H_{i+1} are p-groups, which gives the asserted result.)

. When R is strictly local, there is the following simple characterization: the K-algebra L is tamely ramified over R if and only if the $[L_i : K]$ are prime to p. Moreover, if L is tamely ramified over R, the L_i are cyclic extensions of K. Indeed, when R is strictly local, I_i is equal to the Galois group of L'_i over K. As just recalled, I_i is an extension of 370
a cyclic group of order prime to p by a p-group. If L'_i is assumed tamely ramified over R, then I_i is a cyclic group of order prime to p. It follows that $[L_i : K]$ is prime to p and that $L_i = L'_i$. Conversely, if $[L_i : K]$ is prime to p, then I_i cannot contain a nontrivial normal p-subgroup; I_i is therefore a cyclic group of order prime to p, which proves that L_i is tamely ramified over R.

. Let R be a discrete valuation ring with fraction field K, let L be an étale K-algebra, let $\overline{R}$ be a strict localization of R, let $\overline{K}$ be its fraction field, and put $\overline{L} = L \otimes_K \overline{K}$. Then L is tamely ramified over R if and only if $\overline{L}$ is tamely ramified over $\overline{R}$. We may indeed reduce to the case where L is a field. Write $\overline{L} = \prod_i \overline{L}_i$, where the $\overline{L}_i$ are field extensions of $\overline{K}$. If L' is the Galois extension generated by L, and if $\overline{L'} = L' \otimes_K \overline{K}$, then $\overline{L'}$ likewise decomposes as a product of fields, $\overline{L'} = \prod_j \overline{L'_j}$ and every $\overline{L}_i$ is a subextension of at least one $\overline{L'_j}$. Since L' is a Galois extension of K, the $\overline{L'_j}$ are Galois extensions of $\overline{K}$. Suppose that L is tamely ramified over R. Since the Galois group of $\overline{L'_j}|\overline{K}$ is isomorphic to the inertia group of $L'|K$, the $\overline{L'_j}$ are also tamely ramified over $\overline{R}$, and hence so are the $\overline{L}_i$. Conversely, suppose that $\overline{L}$ is tamely ramified over $\overline{R}$. For each j, let v_j be the discrete valuation of $\overline{L'_j}$ extending the valuation of $\overline{K}$, and also denote by v_j the induced valuation on L'. As j varies, v_j ranges over the valuations of L' extending the valuation of K. Let $G = \mathrm{Gal}(L'|K)$, $H = \mathrm{Gal}(L'|L)$, and let I_j be the inertia group of $L'|K$ at v_j, and J_j the inertia group of $L'|L$ at v_j. The group I_j is an extension of a cyclic group of order prime to p by a p-group P_j. Since the $\overline{L}_i$ are tamely ramified over $\overline{R}$, I_j/J_j has order prime to p, and hence $P_j \subset J_j$. Consequently, the group H contains every P_j, and therefore also the group P generated by the P_j as j varies. But P is invariant in G, since an 371
inner automorphism of G permutes the I_j, and hence also the P_j. It follows that P is a subgroup of H normal in G, and therefore, since L' is the Galois extension generated by L, that $P = 1$. This proves that L is tamely ramified over R.

More generally, let $R \to R'$ be a morphism of discrete valuation rings such that the image of a uniformizer π of R is a uniformizer π' of R', and such that the residue extension $\kappa(R')|\kappa(R)$ is separable. Let K be the fraction field of R, K' the fraction field of R', L an étale K-algebra, and $L' = L \otimes_K K'$. Then L is tamely ramified over R if and only if L' is tamely ramified over R'. We may indeed suppose that R and R' are strictly local. By XIII.2, it is enough to prove that when L is a field, so is L'. Let $\widetilde{R}$ be the normalization of R in L, let $\widetilde{\pi}$ be a uniformizer of $\widetilde{R}$, and put $\widetilde{R}' = \widetilde{R} \otimes_R R'$. Since the extension $\kappa(\widetilde{R})|\kappa(R)$ is radicial and the extension $\kappa(R')|\kappa(R)$ is étale, $\kappa(\widetilde{R}) \otimes_{\kappa(R)} \kappa(R')$ is a field [EGA IV 4.3.2 and 4.3.5]. This proves that $\widetilde{R}'$ is a local ring, and, since π maps to π' in R', we have $\kappa(\widetilde{R}') = \widetilde{R}'/(\widetilde{\pi})$. Consequently, $\widetilde{R}'$ is a discrete valuation ring [**5**, ch. I §2 prop. 2], and hence L' is a field.

. By reducing to the strictly local case, we see that a subalgebra of a tamely ramified algebra is tamely ramified; that the tensor product of two tamely ramified algebras is tamely ramified; that a tamely ramified algebra remains so after extending the discrete valuation ring; and that an algebra which becomes tamely ramified after a tamely ramified extension is itself tamely ramified.

. Let X be an S-scheme, and let D be a divisor ≥ 0 on X. Recall (SGA 5 II 4.2) 372
that D is said to have *strict normal crossings relative to* S if there exists a finite family $(f_i)_{i\in I}$ of elements of $\Gamma(X, \mathcal{O}_X)$ such that $D = \sum_{i\in I} \mathrm{div}(f_i)$ and the following condition is satisfied:

. For every point x of $\operatorname{Supp} D$, X is smooth over S at x, and, if $I(x)$ denotes the set of $i \in I$ such that $f_i(x) = 0$, the subscheme $V((f_i)_{i\in I(x)})$ is smooth over S of codimension $\operatorname{card} I(x)$ in X.

The divisor D is said to have *normal crossings relative to S* if, locally on X for the étale topology, it has strict normal crossings.

Let D be a divisor with normal crossings relative to S. Put $Y = \operatorname{Supp} D$, $U = X - Y$, and denote by $i\colon U \to X$ the canonical immersion. For every geometric point $\overline{s}$ of S and every maximal point y of the geometric fiber $Y_{\overline{s}}$, the ring $R = \mathcal{O}_{X_{\overline{s}},y}$ is a discrete valuation ring.

In the remainder of this number, we shall use the following technical definition:

Definition XIII.2.1. *Let F be a sheaf of sets on U. We say that F is* tamely ramified *on X (*along D*)* relative to *S if, for every geometric point $\overline{s}$ of S, the following condition is satisfied:*

For every maximal point y of $Y_{\overline{s}}$, the restriction of F to the fraction field K of $\mathcal{O}_{X_{\overline{s}}}$ is representable by the spectrum of an étale K-algebra L which is tamely ramified over $\mathcal{O}_{X_{\overline{s}},y}$.

Most often, when no confusion can result, we shall omit mention of D from the terminology.

Definition XIII.2.2. *If F is a sheaf of groups on U, tamely ramified on X relative to S, we denote by* 373

$$\mathrm{H}^1_{\mathrm{t}}(U, F)$$

the subset of $\mathrm{H}^1(U, F)$ consisting of the classes of torsors under F which are tamely ramified on X relative to S.

Let

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & X \\ {\scriptstyle g}\downarrow & \swarrow{\scriptstyle f} & \\ T & & \end{array}$$

be a commutative diagram of S-schemes, with i as in XIII.2; we denote by

$$\mathrm{R}^1_{\mathrm{t}} g_* F$$

the sheaf on T associated with the presheaf $T' \longmapsto \mathrm{H}^1_{\mathrm{t}}(U', F)$, where T' ranges over the schemes étale over T and where $U' = U \times_T T'$; $\mathrm{R}^1_{\mathrm{t}} g_* F$ is a subsheaf of $\mathrm{R}^1 g_* F$.

Note that, if g is coherent, if $\overline{t}$ is a geometric point of T, if $\overline{T}$ is the strict localization of T at $\overline{t}$, and if $\overline{U} = U \times_T \overline{T}$, then there is an isomorphism

$$(\mathrm{R}^1_{\mathrm{t}} g_* F)_{\overline{t}} \simeq \mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \tag{XIII.2.-1.-1.1}$$

. Let $C_t((U, X)/S)$, or simply C_t, be the category of étale coverings of U that are tamely ramified on X relative to S. Suppose that U is connected, and let a be a geometric point of U. Let Γ_t be the functor which assigns to an étale covering U' of U tamely ramified on X relative to S the set of geometric points of U' above a. It follows from XIII.2 that the pair (C_t, Γ_t) satisfies axioms (G_1) through (G_6) of V V.4. Consequently, Γ_t is representable by a pro-object called the *tamely ramified universal covering of* (U, X) 374 *relative to S pointed at a*. The opposite of the group of U-automorphisms of the tamely ramified universal covering is called the *tamely ramified fundamental group* and is denoted by

$$\pi^{\mathrm{t}}_1((U, X)/S, a) \text{ or simply } \pi^{\mathrm{t}}_1(U, a) \text{ or even } \pi^{\mathrm{t}}_1(U).$$

It is clearly a quotient of the fundamental group $\pi_1(U, a)$ (V V.6.9).

. Let F be a sheaf of groups on U, let P be a right torsor with group F, and let Q be a left torsor with group F. Suppose that P and Q are tamely ramified on X relative to S. Then the same is true of $P \overset{F}{\wedge} Q$. Indeed, we reduce to showing that, if R is a discrete valuation ring with fraction field K, if F is a finite étale group scheme over K, and if P and Q are two torsors under F tamely ramified over R, then the same is true of $P \overset{F}{\wedge} Q$. Now $T = P \overset{F}{\wedge} Q$ is a quotient of $P \times_K Q$. If L, M, N denote the K-algebras representing T, P, Q, respectively, then L is a subalgebra of $M \otimes_K N$, and it follows from XIII.2 that L is tamely ramified over R.

It follows from the preceding that, if F is a sheaf of groups on U and if there exists a torsor with group F tamely ramified on X relative to S, then F is tamely ramified on X relative to S. Indeed, the opposite torsor P° of P is tamely ramified on X relative to S, since it is isomorphic to P as a sheaf of sets. If ${}^PF$ is the group obtained by twisting F by P, there is an isomorphism

$$F \simeq P^\circ \overset{{}^PF}{\wedge} P$$

and consequently F is tamely ramified on X relative to S.

As before, we see that, if $F \to F'$ is a morphism of sheaves of groups on U tamely ramified on X relative to S, and if P is a torsor under F tamely ramified on X relative to S, then the torsor P' deduced from P by extension of the structure group $F \to F'$ is 375
tamely ramified on X relative to S.

In particular, the canonical morphism

$$\mathrm{H}^1(U, F) \to \mathrm{H}^1(U, F')$$

induces, by restriction to $\mathrm{H}^1_{\mathrm{t}}(U, F)$, a canonical morphism

$$\mathrm{H}^1_{\mathrm{t}}(U, F) \to \mathrm{H}^1_{\mathrm{t}}(U, F')$$

. Let $S' \to S$ be a morphism, and denote by U' (respectivelyX', etc.) the inverse image of U (respectively$X$, etc.) on S'. If F is a sheaf of sets on U tamely ramified on X relative to S, it follows from Definition XIII.2.1 and from XIII.2 that F' is tamely ramified on X' relative to S'.

If now F is a sheaf of groups on U, the inverse image on S' of a torsor under F tamely ramified on X relative to S is a torsor under F' tamely ramified on X' relative to S'. In particular, there is a canonical functor

$$C_t((U, X)/S) \to C_t((U', X')/S'). \tag{XIII.2.-1.-1.1}$$

Suppose that U and U' are connected, and let a be a geometric point of U and a' a geometric point of U' above a; the preceding gives a canonical morphism

$$\pi_1^{\mathrm{t}}(U', a') \to \pi_1^{\mathrm{t}}(U, a). \tag{XIII.2.-1.-1.2}$$

If $S' \to S$ is a morphism and $h\colon T' \to T$ is the canonical projection, the morphism

$$h^*(\mathrm{R}^1 g_* F) \to \mathrm{R}^1 g'_* F'$$

restricts to a canonical morphism

$$h^*(\mathrm{R}^1_{\mathrm{t}} g_* F) \to \mathrm{R}^1_{\mathrm{t}} g'_* F' \tag{XIII.2.-1.-1.3}$$

. Let F be a sheaf of groups on U, tamely ramified on X relative to S. With the 376
notation of XIII.2.2, there are canonical exact sequences:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathrm{H}^1(X, i_* F) & \longrightarrow & \mathrm{H}^1_{\mathrm{t}}(U, F) & \longrightarrow & \mathrm{H}^0(X, \mathrm{R}^1_{\mathrm{t}} i_* F) \\ 1 & \longrightarrow & \mathrm{R}^1 f_*(i_* F) & \longrightarrow & \mathrm{R}^1_{\mathrm{t}} g_* F & \longrightarrow & f_*(\mathrm{R}^1_{\mathrm{t}} i_* F). \end{array} \tag{XIII.2.-1.-1.1}$$

The first is obtained from the exact sequence (SGA 4 III 3.2):

$$1 \to \mathrm{H}^1(X, i_* F) \to \mathrm{H}^1(U, F) \to \mathrm{H}^0(X, \mathrm{R}^1 i_* F).$$

Indeed, it suffices to show that the image of $\mathrm{H}^1(X, i_*F)$ in $\mathrm{H}^1(U, F)$ is in fact contained in $\mathrm{H}^1_{\mathrm{t}}(U, F)$ and that the image of $\mathrm{H}^1_{\mathrm{t}}(U, F)$ in $\mathrm{H}^0(X, \mathrm{R}^1 i_*F)$ is in fact contained in $\mathrm{H}^0(X, \mathrm{R}^1_{\mathrm{t}} i_*F)$. Now the inverse image on U of a torsor under i_*F is a torsor under i^*i_*F which is clearly tamely ramified on X relative to S; the same is therefore true after extension of the structure group $i^*i_*F \to F$, which proves the existence of the arrow $\mathrm{H}^1(X, i_*F) \to \mathrm{H}^1_{\mathrm{t}}(U, F)$. The fact that the image of $\mathrm{H}^1_{\mathrm{t}}(U, F)$ in $\mathrm{H}^0(X, \mathrm{R}^1 i_*F)$ is contained in $\mathrm{H}^0(X, \mathrm{R}^1_{\mathrm{t}} i_*F)$ follows at once from the definition of $\mathrm{R}^1_{\mathrm{t}} i_*F$. This proves the existence of the first exact sequence, and the second follows from it by localization.

. Retain the notation of XIII.2. We shall define a notion of a tamely ramified object of a stack Φ on U when the latter is given, locally *on* X and on S for the étale topology, as the *inverse image of a stack* Ψ *on* S.

First let G be a gerbe on U, and suppose that we are given a surjective étale morphism $S_1 \to S$, a surjective étale morphism $X_2 \to X \times_S S_1$, a trivial gerbe H on S_1, and an isomorphism

$$G|U_2 \to H|U_2,$$

where $U_1 = U \times_X X_1$, $U_2 = U \times_X X_2$. Upon choosing a trivialization of $H|X_2$, the 377
isomorphism above identifies $G|U_2$ with the stack of torsors under a sheaf of groups F. An object x of G_U is said to be tamely ramified on X relative to S if the restriction of x to U_2 is a torsor tamely ramified on X relative to S. By XIII.2, this notion does not depend on the chosen trivialization of $H|X_2$.

Now let Φ be a stack on U, and suppose that we are given a surjective étale morphism $S_1 \to S$, a surjective étale morphism $X_2 \to X \times_S S_1$, a stack Ψ on S_1, and an isomorphism

$$i\colon \Phi|U_2 \to \Psi|U_2.$$

Let x be an object of Φ_U, let G_x be the maximal subgerbe of Φ generated by x [**1**, III 2.1.7], and let $S\Phi$ be the sheaf of maximal subgerbes of Φ. The isomorphism i induces an isomorphism

$$S\Phi|U_2 \to S\Psi|U_2.$$

It follows from XIII.5.8 that, after replacing S_1 by a surjective étale extension if necessary, there is a unique maximal subgerbe H of Ψ, which may be assumed trivial, such that i defines an isomorphism

$$G_x|U_2 \to H|U_2.$$

The object x is said to be tamely ramified on X relative to S if it is so as an object of G_x equipped with the isomorphism above.

. Let Φ be a stack on U given, locally on X and S, as the inverse image of a stack on S, and let

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & X \\ {\scriptstyle g}\downarrow & \swarrow{\scriptstyle f} & \\ T & & \end{array}$$

be a diagram as in XIII.2.2. For every scheme T' étale over T, if $U' = U \times_T T'$, consider the subset $(g^{\mathrm{t}}_*\Phi)_{T'}$ of $(g_*\Phi)_{T'} = \Phi_{U'}$, consisting of the objects of $\Phi_{U'}$ that are tamely 378
ramified on X relative to S. The *tamely ramified direct image* of Φ under g, denoted

$$g^{\mathrm{t}}_*\Phi,$$

is the full subcategory of $g_*\Phi$ whose objects over a scheme T' étale over T are the elements of $(g^{\mathrm{t}}_*\Phi)_{T'}$. It is clear that $g^{\mathrm{t}}_*\Phi$ is a substack of $g_*\Phi$.

. By reducing to the case of a stack of torsors, we see that if Φ is a stack on U which, locally for the étale topology of S and X, is the inverse image of a stack on S, then the canonical morphism

$$h^*(g_*\Phi) \to g'_*\Phi'$$

restricts to a canonical morphism

$$h^*(g^{\mathrm{t}}_*\Phi) \to g'^{\mathrm{t}}_*\Phi'.$$

REMARKS XIII.2.3. a) If F is a constructible locally constant sheaf of sets on U, then, for F to be tamely ramified on X relative to S, it is enough that the condition of XIII.2.1 hold at the geometric points of S lying over the maximal points of S. To see this, we may suppose that D is a strict normal crossings divisor. The sheaf F is represented by an étale covering V of U. If $\overline{s}$ is a geometric point of S and y a maximal point of $Y_{\overline{s}}$, denote by $\overline{S}$ the strict localization of S at $\overline{s}$, by $\overline{X}$ the strict localization of X at $\overline{y}$, and put $\overline{U} = U \times_X \overline{X}$, $\overline{V} = V \times_X \overline{X}$. If the condition of XIII.2.1 holds at geometric points over the maximal points of S, it follows from XIII.5.5 below that $\overline{V}$ is a covering of $\overline{U}$ tamely ramified on $\overline{X}$ relative to $\overline{S}$. Consequently, V is an étale covering of U tamely ramified on X relative to S.

b) Let F be a sheaf of groups on U tamely ramified on X relative to S. If $\overline{s}$ is a 379
geometric point of S and y a maximal point of $Y_{\overline{s}}$, denote by K the fraction field of $\mathcal{O}_{X_{\overline{s}},y}$. Suppose that, for every point $\overline{s}$ and every point y, the K-algebra L whose spectrum represents $F|K$ has rank prime to the residue characteristic p of $\mathcal{O}_{X_{\overline{s}}}$. By an abuse of language, we shall sometimes say that F is prime to the residue characteristics of S. When this is so, every torsor P under F is tamely ramified on X relative to S.

Indeed, let $\overline{R}$ be the strict localization of $\mathcal{O}_{X_{\overline{s}},y}$ at $\overline{y}$, let $\overline{K}$ be its fraction field, and let $\overline{F}$ be the inverse image of F on $\overline{K}$. We show that F may be assumed constant. Since F is tamely ramified on X relative to S, $\overline{F}$ is represented by the spectrum of a $\overline{K}$-algebra $L = \prod L_i$, where the L_i are extensions of $\overline{K}$ of degree prime to p. We can therefore find an extension K' of $\overline{K}$ of degree prime to p such that $\overline{F}|K'$ is a constant sheaf. By XIII.2, to prove that $P|\overline{K}$ is tamely ramified over $\overline{R}$, it is enough to see that $P|K'$ is tamely ramified over the integral closure of $\overline{R}$ in K'; this reduces us to the case where $\overline{F}$ is constant. Henceforth suppose that $\overline{F}$ is constant. The $\overline{K}$-algebra H representing $P|\overline{K}$ is then a product of mutually isomorphic extensions H_i of $\overline{K}$. Since the rank of H is prime to p, the same is true of $[H_1 : \overline{K}]$, which proves that H is tamely ramified on X relative to S.

c) Let X be a regular scheme, let D be a normal crossings divisor of X (SGA 5 I 3.1.5), let $U = X - \operatorname{Supp} D$, and let F be a sheaf of sets on U. If y is a maximal point of $\operatorname{Supp} D$, denote by K the fraction field of $\mathcal{O}_{X,y}$. We say that F is *tamely ramified relative to D* if, for every maximal point y of $\operatorname{Supp} D$, $F|K$ is represented by a K-algebra tamely ramified over $\mathcal{O}_{X,y}$.

THEOREM XIII.2.4. *Let $f\colon X \to S$ be an S-scheme, let D be a divisor on X with* 380
normal crossings relative to S (XIII.2)*, put $Y = \operatorname{Supp} D$ and $U = X - Y$, and let $i\colon U \to X$ be the canonical immersion. Let F be a sheaf of sets (respectivelygroups) on U, satisfying one of the following conditions:*

a) *Locally for the étale topology* on *X and on S, F is the inverse image of a sheaf of sets (respectivelya constructible sheaf of groups) on S.*
b) *F is constructible locally constant on U and is tamely ramified on X relative to S.*

Then the following conclusions hold:

1) *(F, i) is cohomologically proper relative to S in dimension ≤ 0 (respectivelyfor every morphism $h\colon S' \to S$, if $i'\colon U' \to X'$ is the inverse image of i on S', if*

$F' = F|U'$, and if $k = h_{(X)}$, the canonical morphism

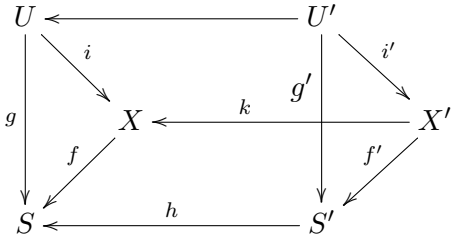
$$\Psi\colon k^*(\mathrm{R}^1_{\mathrm{t}} i_* F) \to \mathrm{R}^1_{\mathrm{t}} i'_* F'$$

is an isomorphism).

If F is a sheaf of groups prime to the residue characteristics of S (XIII.2.3.b)) (and tamely ramified on X relative to S), then (F, i) is cohomologically proper relative to S in dimension ≤ 1.

2) *If F is a constructible sheaf of sets (respectivelygroups), then i_*F (respectively$\mathrm{R}^1_{\mathrm{t}} i_* F$) is constructible.*

Proof. For every S-scheme S', consider the following diagram, all of whose squares are cartesian:



Since the question is local on X for the étale topology, we may suppose that D is a 381
strict normal crossings divisor relative to S (XIII.2); moreover, after restricting X to a neighborhood of Y if necessary, we may suppose that X is smooth over S.

Proof of XIII.2.4 1).

. *Case of a sheaf of sets satisfying* a). We may suppose that $F = g^*G$, where G is a sheaf on S. It then follows from (SGA 4 XVI 3.2) that the canonical morphism

$$f^*G \to i_*F \tag{XIII.2.-1.-1.1}$$

is an isomorphism. For every S-scheme $h\colon S' \to S$, there is similarly an isomorphism $f'^*G' \to i'_*F'$; consequently, the canonical morphism

$$\varphi\colon k^*(i_*F) \to i'_*F'$$

is identified with the natural isomorphism

$$k^* f^* G \simeq f'^* G'$$

. *Case of a sheaf of sets satisfying* b). We must show that φ is an isomorphism, and it is enough to check this at every geometric point $\overline{x}'$ of X'. Let $\overline{S}$ (respectively$\overline{X}$, respectively$\overline{S'}$, respectively$\overline{X'}$) be the strict localization of S (respectively X, respectively S', respectivelyX') at the corresponding image of $\overline{x}'$, and put $\overline{U} = U_{(\overline{X})}$, $\overline{U'} = U_{(\overline{X'})}$, etc. The morphism $\varphi_{\overline{x}}$ is identified with the canonical morphism

$$\overline{\varphi}\colon \mathrm{H}^0(\overline{U}, \overline{F}) \to \mathrm{H}^0(\overline{U'}, \overline{F'})$$

One can find a principal covering V of $\overline{U}$, of the type appearing in XIII.5.4, such that the inverse image of $\overline{F}$ on V is a constant sheaf with value C. If Π is the Galois group of V over $\overline{U}$, then Π acts on $\overline{F}|V$, and there is an isomorphism

$$\mathrm{H}^0(\overline{U}, \overline{F})) \simeq \mathrm{H}^0(V, C_V)^{\Pi} \tag{XIII.2.-1.-1.1}$$

where the right-hand side denotes the set of elements of $\mathrm{H}^0(V, C_V)$ invariant under Π. 382
Since $V' = V \times_{\overline{U}} \overline{U'}$ is a principal covering of $\overline{U'}$ with Galois group $\Pi' \simeq \Pi$, one sees that the morphism $\overline{\varphi}$ is obtained, by taking the invariants under Π, from the canonical morphism

$$\mathrm{H}^0(V, C_V) \to \mathrm{H}^0(V', C_{V'}).$$

Since V and V' are connected (XIII.5.4), this morphism, and hence also $\overline{\varphi}$, is an isomorphism.

Note that if, in addition, F is a sheaf of groups and P is a torsor on $\overline{U}$ with group $\overline{F}$, tamely ramified relative to D, it follows from the preceding proof and from XIII.2 that $({}^{P}\overline{F}, \bar{i})$ is cohomologically proper relative to S in dimension ≤ 0.

. *Case of a sheaf of groups.* To show that Ψ is an isomorphism, it suffices to prove that, for every geometric point $\overline{y}'$ of Y', the morphism

$$\Psi_{\overline{y}'} : (k^*(\mathrm{R}^1_{\mathrm{t}} i_* F))_{\overline{y}'} \to (\mathrm{R}^1_{\mathrm{t}} i'_* F')_{\overline{y}'}$$

is an isomorphism. Now, by XIII.2.2, $\Psi_{\overline{y}'}$ is identified with the canonical morphism

$$\overline{\Psi} : \mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \to \mathrm{H}^1_{\mathrm{t}}(\overline{U'}, \overline{F'}).$$

Let $\widetilde{U}$ be the tamely ramified universal covering of $\overline{U}$ (XIII.2), and let $\widetilde{F}$ be the inverse image of $\overline{F}$ on $\widetilde{U}$. It follows from XIII.5.8 in case a) and from XIII.5.5 in case b) that $\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F})$ is identified with the subset of $\mathrm{H}^1(\overline{U}, \overline{F})$ consisting of the classes of $\overline{F}$-torsors whose inverse image on $\widetilde{U}$ is trivial. On the other hand, a standard (cf. IX IX.5, p. 185) argument shows that the set of elements of $\mathrm{H}^1(\overline{U}, \overline{F})$ whose inverse image on $\widetilde{U}$ is trivial is identified with $\mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F}))$. Thus one obtains a canonical isomorphism

$$\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \xrightarrow{\sim} \mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F})). \tag{XIII.2.-1.-1.1}$$

Consequently, the morphism $\overline{\Psi}$ is identified with the canonical morphism

$$\mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F})) \to \mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U'}), \mathrm{H}^0(\widetilde{U}', \widetilde{F}')).$$

Let us show that this morphism is an isomorphism. The morphism $\pi_1^{\mathrm{t}}(\overline{U'}) \to \pi_1^{\mathrm{t}}(\overline{U})$ is an isomorphism by XIII.5.6, and the same is true of the morphism $\mathrm{H}^0(\widetilde{U}, \widetilde{F}) \to \mathrm{H}^0(\widetilde{U}', \widetilde{F}')$. Indeed, this is clear in case b), since $\widetilde{F}$ is constant and $\widetilde{U}$ and $\widetilde{U}'$ are connected. In case a), let $\overline{G}$ be a constructible sheaf of groups on $\overline{S}$ such that $\overline{F} = \overline{g}^*\overline{G}$; since the morphisms $\widetilde{U} \to \overline{S}$ and $\widetilde{U}' \to \overline{S'}$ are 0-acyclic (XIII.5.8), there are isomorphisms 383

$$\mathrm{H}^0(\widetilde{U}, \widetilde{F}) \simeq \mathrm{H}^0(\overline{S}, \overline{G}) \simeq \mathrm{H}^0(\overline{S'}, \overline{G'}) \simeq \mathrm{H}^0(\widetilde{U}', \widetilde{F}'),$$

which implies that Ψ is an isomorphism. The last assertion of XIII.2.4 1) follows from the foregoing, in view of XIII.2.3 b).

Proof of XIII.2.4 2). The case of a constructible sheaf of sets satisfying a) follows immediately from (XIII.2.-1.-1.1). Let $F = g^*G$ be a sheaf of groups satisfying a), where G is a constructible sheaf; we may suppose that S is affine. Let $(S_j)_{j\in J}$ be a finite family of reduced closed subschemes of S whose union covers S, such that the inverse image of G on S_j is a locally constant sheaf. In view of XIII.2.4 1), in order to prove that $\mathrm{R}^1_{\mathrm{t}} i_* F$ is constructible, it suffices to prove this after the base change $S_j \to S$, for every $j \in J$. We are therefore reduced to case b), where F is locally constant.

From now on, suppose that F is a sheaf of sets or of groups satisfying b). Since the question is local for the étale topology on X, we may suppose that X is of finite presentation over S, and, by passage to the limit, we may suppose that X and S are noetherian.

Let $D = \sum_{1\leq i\leq r} \operatorname{div} f_i$, where, for every point x of $\operatorname{Supp} D$, if $I(x)$ is the set of i such that $f_i(x) = 0$, the closed subscheme $V((f_i)_{i\in I(x)})$ is smooth over S of codimension $\operatorname{card} I(x)$ in X. Let $\mathcal{P}$ be the set of subsets of $[1, r]$, and, for every $I \in \mathcal{P}$, put

$$X_I = \Big(\bigcap\nolimits_{i\in I} V(f_i)\Big) \cap \Big(\bigcap\nolimits_{i\notin I} X_{f_i}\Big).$$

Let z be a point of X_I. After first passing to an étale neighborhood of z, one can find an open subset W of X containing z and a principal covering V of $U \cap W$, tamely ramified on W relative to S, of the type considered in XIII.5.7, such that the inverse image of F on V is a constant sheaf with value C. Let π be the Galois group of the covering V. For every geometric point $\overline{x}$ of X_I, one then has, by (XIII.2.-1.-1.1), 384

$$\mathrm{H}^0(\overline{U}, \overline{F}) \simeq \mathrm{H}^0(V, C_V)^{\pi}.$$

It follows that $i_*F|X_I \cap W$ is locally constant (SGA 4 IX 2.13), and hence i_*F is constructible.

Finally, let us show that if F is a locally constant sheaf of groups, then $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X_I}$ is constructible. If $\overline{x}$ is a geometric point of X, we have obtained in (XIII.2.-1.-1.1) the expression

$$\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \xrightarrow{\sim} \mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F})).$$

If p is the residue characteristic of $\overline{X}$, then by XIII.5.6, $\pi_1^{\mathrm{t}}(\overline{U}) = \prod_{l \neq p} \mathbb{Z}_l(1)^{\operatorname{card} I}$. Let $\mathbb{L}$ be the set of prime numbers that divide the order of the finite group $\mathrm{H}^0(\widetilde{U}, \widetilde{F})$, and let $K = \prod_{l \in \mathbb{L} - \{p\} \cap \mathbb{L}} \mathbb{Z}_l(1)^{\operatorname{card} I}$. It follows from [**4**, I §5 ex.2] that there is an isomorphism

$$\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \simeq \mathrm{H}^1(K, \mathrm{H}^0(\widetilde{U}, \widetilde{F})).$$

Since K is topologically finitely generated and $\mathrm{H}^0(\widetilde{U}, \widetilde{F})$ is finite, we first deduce that the fibers of the sheaf $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X_I}$ are finite. Moreover, the set $\mathbb{L}$ does not depend on the point $\overline{x}$. For every $q \in \mathbb{L}$, let $X_{I,q}$ be the closed subset of X_I defined by the equation $q = 0$, and let $X_{I'}$ be the open subset of X_I complementary to the union of the $X_{I,q}$. Then $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X_{I,q}}$ and $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X'_I}$ are locally constant; indeed, a specialization morphism between geometric points of $X_{I,q}$ (respectivelyof $X_{I'}$) induces an isomorphism on the groups K (XIII.5.7), hence also on the sets $\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F})$, and one may apply SGA 4 IX 2.13.

Corollary XIII.2.5. *Let $f\colon X \to S$ be a morphism, let D be a divisor on X with* 385
normal crossings relative to S (XIII.2)*, let $Y = \operatorname{Supp} D$, $U = X - Y$, and let $i\colon U \to X$ be the canonical immersion. Let Φ be a stack on U, and suppose that we are given surjective étale morphisms $S_1 \to S$ and $X_2 \to X \times_S S_1$, a stack Ψ on S_1, and an isomorphism $\Phi|U_2 \simeq \Psi|U_2$ (cf. XIII.2).*

Then, for every morphism $h\colon S' \to S$, if $k = h_{(X)}$, the canonical functor

$$\varphi\colon k^* i_* \Phi \to i'_* \Phi'$$

is fully faithful. If Ψ is constructible, the canonical functor

$$\psi\colon k^* i^{\mathrm{t}}_* \Phi \to i'^{\mathrm{t}}_* \Phi'$$

is an equivalence of categories.

Moreover, if the stack Ψ is constructible (respectivelyif Ψ is 1*-constructible* (XIII.0) *and S is locally noetherian), $i_*\Phi$ is constructible (respectivelyi$^{\mathrm{t}}_*\Phi$ is* 1*-constructible).*

Let us show that φ is fully faithful. It is enough to see that, for every geometric point $\overline{x}'$ of X', the same is true of the functor

$$\overline{\varphi}\colon \Phi(\overline{U}) \to \Phi'(\overline{U'})$$

(we again use the notation of XIII.2). Let a, b be two objects of $\Phi(\overline{U})$, and let a', b' be their images under $\overline{\varphi}$. Since the morphism $\overline{U} \to \overline{S}$ is locally 0-acyclic (XIII.5.8), we have an isomorphism

$$\mathrm{H}^0(\overline{U}, \overline{S\Phi}) \xrightarrow{\sim} \mathrm{H}^0(S, \overline{S\Psi}).$$

Consequently, a and b are inverse images of objects of Ψ, and the same is therefore true of $F = \mathbf{Hom}_{\overline{U}}(a, b)$. Since the sheaf $F' = \mathbf{Hom}_{\overline{U'}}(a', b')$ is the inverse image on $\overline{U'}$ of F, it follows from XIII.2 that the canonical morphism

$$\mathrm{H}^0(\overline{U}, F) \to \mathrm{H}^0(\overline{U'}, F')$$

is an isomorphism, which proves that $\overline{\varphi}$ is fully faithful.

Let us show that, for every geometric point $\overline{x}'$ of X', the functor 386

$$\overline{\psi}\colon i^{\mathrm{t}}_* \Phi(\overline{X'}) \to i'^{\mathrm{t}}_* \Phi'(\overline{X'})$$

is an equivalence. By what precedes, $\overline{\psi}$ is fully faithful. Let us show that $\overline{\psi}$ is essentially surjective. Let a' be an object of $\Phi'(\overline{U'})$ tamely ramified on X' relative to S' (XIII.2),

and let us show that it is the image of a tamely ramified object of $\Phi(\overline{U})$. It follows from XIII.2.4 1) that the canonical morphism

$$\mathrm{H}^0(\overline{U}, \overline{S\Phi}) \to \mathrm{H}^0(\overline{U'}, \overline{S\Phi}')$$

is an isomorphism. Let G' be the maximal subgerbe of Φ' generated by a'; there is then a maximal subgerbe G of $\overline{\Phi}$, the inverse image of a gerbe on $\overline{S}$, such that

$$\overline{m}^* G \simeq G',$$

where m is the morphism $\overline{U'} \to \overline{U}$. The canonical functor

$$(*) \qquad \overline{k}^* \bar{i}_*^{\mathrm{t}} G \to \bar{i}_*'^{\mathrm{t}} G'$$

is an equivalence, because G identifies with a gerbe of torsors under a constructible sheaf of groups coming from $\overline{S}$, and we can apply XIII.2.4 1). It then follows from $(*)$ that there exists an object a of $G(\overline{U})$, tamely ramified on X relative to S, whose inverse image on $\overline{U'}$ is a'. This proves that ψ is an equivalence.

If Ψ is constructible, then so is $i_*\Phi$; indeed, an object x of $i_*\Phi$ is, locally for the étale topology on S, the inverse image of an object y of Ψ; it therefore follows from XIII.2.-1.-1.1 that $\mathbf{Aut}(x)$ is the inverse image of $\mathbf{Aut}(y)$, and hence is constructible. Finally, if Ψ is 1-constructible, then so is $i_*^{\mathrm{t}}\Phi$ by XIII.6.5 below.

Corollary XIII.2.6. *The notation is that of XIII.2.4. Suppose that S has characteristic zero at every point s such that $Y_s \neq \emptyset$. Then, if $\mathcal{F}$ is a locally constant constructible sheaf of groups on U (respectivelya constructible stack on U which, locally on X and S, is the inverse image of a constructible stack on S), $(\mathcal{F}, i)$ is cohomologically proper relative to S in dimension ≤ 1.* 387

Since every constructible sheaf of sets on U is tamely ramified on X relative to S, the corollary follows from XIII.2.4 (respectivelyXIII.2.5).

Corollary XIII.2.7. *The notation is that of XIII.2.4, but in addition we are given an S-scheme T, a proper morphism $p\colon X \to T$, and we suppose that X and T are of finite presentation over S; let $q = pi$. Let $\mathcal{F}$ be a constructible sheaf of sets on U satisfying one of conditions* a) *or* b) *of XIII.2.4 (respectivelya sheaf of groups satisfying one of conditions* a), b) *of XIII.2.4, respectivelya stack on U which, locally on X and S, is the inverse image of a constructible stack G on S). Then the following conclusions hold:*

1) *$(\mathcal{F}, q)$ is cohomologically proper relative to S in dimension ≤ 0 (respectivelyfor every morphism $h\colon S' \to S$, if $m = h_{(T)}$, the canonical morphism:*

$$\Theta\colon m^*(\mathrm{R}^1_{\mathrm{t}} q_* \mathcal{F}) \to \mathrm{R}^1_{\mathrm{t}} q'_* \mathcal{F}'$$

is an isomorphism, respectivelyfor every morphism $S' \to S$, the canonical morphism

$$\xi\colon m^*(q_*^{\mathrm{t}} \mathcal{F}) \to q'^{\mathrm{t}}_* \mathcal{F}'$$

is an equivalence).

2) *The sheaf $q_*\mathcal{F}$ (respectivelythe sheaf $\mathrm{R}^1_{\mathrm{t}} q_* \mathcal{F}$, respectivelythe stack $q_*^{\mathrm{t}} \mathcal{F}$) is constructible. In the last case, if S is assumed locally noetherian and G is 1-constructible, then so is $q_*^{\mathrm{t}} \mathcal{F}$.*

The first part follows immediately from XIII.2.4, XIII.2.5, and the proof of XIII.1.7. 388
Let us prove 2). If $\mathcal{F}$ is a constructible sheaf of sets on U satisfying XIII.2.4 a) or XIII.2.4 b), it follows from XIII.2.4 2) that $i_*\mathcal{F}$ is constructible; hence so is $q_*\mathcal{F} = p_*(i_*\mathcal{F})$ (SGA 4 XIV 1.1).

Let $\mathcal{F}$ be a constructible sheaf of groups on U satisfying XIII.2.4 a) or XIII.2.4 b), and let us prove that $\mathrm{R}^1_{\mathrm{t}} q_* \mathcal{F}$ is constructible. By passage to the limit (EGA IV 8.10.5 and 17.7.8) and using 1), we may suppose that S is noetherian. Let Φ then be the stack on X

whose fiber over every scheme X' étale over X consists of the torsors on $U' = U \times_X X'$, with group $\mathcal{F}|U$, which are tamely ramified on X relative to S; thus

$$S(i_*^{\mathrm{t}}\Phi) \simeq \mathrm{R}_{\mathrm{t}}^1 i_*\mathcal{F}$$

and this sheaf is constructible by XIII.2.4 2). It follows from XIII.6.5 below that $S(p_*\Phi)$ is constructible, i.e. that $\mathrm{R}_{\mathrm{t}}^1 q_*\mathcal{F}$ is constructible.

Finally, if $\mathcal{F}$ is a stack on U which, locally on X and S, is the inverse image of a constructible stack on S, $i_*^{\mathrm{t}}\mathcal{F}$ is constructible, and hence so is $q_*^{\mathrm{t}}\mathcal{F} = p_*(i_*^{\mathrm{t}}\mathcal{F})$. If in addition S is locally noetherian and SG is constructible, $S(i_*^{\mathrm{t}}\mathcal{F})$ is constructible by 6.3; hence the same is true of $S(q_* i_*^{\mathrm{t}}\mathcal{F})$, i.e. of $S(q_*^{\mathrm{t}}\mathcal{F})$ XIII.6.4.

Corollary XIII.2.8. *Let*

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & X \\ {\scriptstyle g}\downarrow & \swarrow{\scriptstyle f} & \\ S & & \end{array}$$

be a commutative diagram of schemes in which U is the open complement in X of a divisor 389
with normal crossings relative to S, and f is a proper *morphism of finite presentation. Let $\mathbb{L}$ be a set of prime numbers. Suppose that g is locally* 0*-acyclic (respectivelylocally* 1*-aspherical for $\mathbb{L}$). Then, if $\mathcal{F}$ is a sheaf of sets on U (respectivelya sheaf of $\mathbb{L}$-groups) that is locally constant and constructible on U, and tamely ramified on X relative to S, $f_*\mathcal{F}$ (respectively$\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F}$) is locally constant constructible, and $(\mathcal{F}, f)$ is cohomologically proper relative to S in dimension ≤ 0 (respectivelythe formation of $\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F}$ commutes with every base change $S' \to S$). In the non-respective case, if $\mathcal{F}$ is a sheaf of groups, then for every specialization $\overline{s}_1 \to \overline{s}_2$ of geometric points of S, the specialization morphism*

$$(\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F})_{\overline{s}_2} \to (\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F})_{\overline{s}_1}$$

is injective.

The corollary follows immediately from XIII.2.6 and from XIII.1.13 (respectivelyfrom the analogue of XIII.1.13 for $\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F}$, which is proved as in loc. cit.).

Corollary XIII.2.9. *Let*

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & X \\ {\scriptstyle g}\downarrow & \swarrow{\scriptstyle f} & \\ S & & \end{array}$$

be a commutative diagram of schemes in which U is the open complement in X of a divisor with normal crossings relative to S, and f is a proper smooth morphism of finite presentation. Let $\mathbb{L}$ be the set of prime numbers distinct from the residue characteristics of S. Let $\mathcal{F}$ be a locally constant constructible sheaf of $\mathbb{L}$-groups on U, tamely ramified on X relative to S. Then $\mathrm{R}^1 f_*\mathcal{F}$ is locally constant constructible, and $(\mathcal{F}, f)$ is cohomologically proper relative to S in dimension ≤ 1. 390

The corollary follows from XIII.2.8 and from the fact that $\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F} = \mathrm{R}^1 f_*\mathcal{F}$ (XIII.2.3 b)).

. If U is a connected scheme, a is a geometric point of U, and $\mathbb{L}$ is a set of prime numbers, we denote by

$$\pi_1^{\mathbb{L}}(U, a) \tag{2.10.0}$$

the inverse limit of the finite quotients of $\pi_1(U, a)$ whose orders have all their prime factors in $\mathbb{L}$.

We shall define specialization homomorphisms for the fundamental group, generalizing X.X.2.

Let $g\colon U \to S$ be a coherent morphism with geometrically connected fibers (respectivelya morphism of the form $g = fi$, where $f\colon X \to S$ is a proper morphism of finite presentation and $i\colon U \to X$ is an open immersion such that U is the complement in X of a divisor with normal crossings relative to S (cf. XIII.2.8)). Let $\mathbb{L}$ be a set of prime numbers and suppose, in the first case, that for every finite constant $\mathbb{L}$-group C, (C_U, g) is cohomologically proper relative to S in dimension ≤ 1. Let $\bar{s}_1 \to \bar{s}_2$ be a specialization morphism of geometric points of S, let $\overline{S}$ be the strict localization of S at $\bar{s}_2$, and put $\overline{U} = U \times_S \overline{S}$. There is a commutative diagram

$$\begin{array}{ccccc} U_{\bar{s}_1} & \xrightarrow{h_1} & \bar{U} & \xleftarrow{h_2} & U_{\bar{s}_1} \\ \downarrow & & \downarrow{\scriptstyle \bar{g}} & & \downarrow \\ \bar{s}_1 & \longrightarrow & \overline{S} & \longleftarrow & \bar{s}_2\,. \end{array}$$

Let a_1 be a geometric point of $U_{\bar{s}_1}$, a_2 a geometric point of $U_{\bar{s}_2}$; the morphisms h_1 and h_2 define canonical homomorphisms 391

$$\pi_1\colon \pi_1^{\mathbb{L}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathbb{L}}(\bar{U}, a_1) \qquad \pi_2\colon \pi_1^{\mathbb{L}}(U_{\bar{s}_2}, a_2) \to \pi_1^{\mathbb{L}}(\bar{U}, a_2)$$
$$\text{(respectively } \pi_1\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathrm{t}}(\bar{U}, a_1) \qquad \pi_2\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_2}, a_2) \to \pi_1^{\mathrm{t}}(\bar{U}, a_2))$$

(V V.7 and XIII.2.-1.-1.2). The hypotheses of cohomological properness (respectivelyXIII.2.8) show that π_2 is an isomorphism. If we choose a path class from a_1 to a_2, we obtain an isomorphism

$$\pi_{12}\colon \pi_1^{\mathbb{L}}(\bar{U}, a_1) \xrightarrow{\sim} \pi_1^{\mathbb{L}}(\bar{U}, a_2) \quad (\text{respectively } \pi_{12}\colon \pi_1^{\mathrm{t}}(\bar{U}, a_1) \xrightarrow{\sim} \pi_1^{\mathrm{t}}(\bar{U}, a_2))$$

and hence a homomorphism $\pi = \pi_2^{-1}\pi_{12}\pi_1$

$$\pi\colon \pi_1^{\mathbb{L}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathbb{L}}(U_{\bar{s}_2}, a_2) \quad (\text{respectively } \pi\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathrm{t}}(U_{\bar{s}_2}, a_2)).$$

Changing the path class from a_1 to a_2 amounts to modifying π by an inner automorphism of $\pi_1^{\mathbb{L}}(X_{\bar{s}_2}, a_2)$ (respectivelyof $\pi_1^{\mathrm{t}}(X_{\bar{s}_2}, a_2)$). One calls any one of the homomorphisms defined above the *specialization homomorphism for the fundamental group* associated with the morphism $\bar{s}_1 \to \bar{s}_2$, and writes simply

$$\pi\colon \pi_1^{\mathbb{L}}(X_{\bar{s}_1}) \to \pi_1^{\mathbb{L}}(X_{\bar{s}_2}) \quad (\text{respectively } \pi\colon \pi_1^{\mathrm{t}}(X_{\bar{s}_1}) \to \pi_1^{\mathrm{t}}(X_{\bar{s}_2}))$$

for one of the homomorphisms defined above.

LEMMA XIII.2.10. *Let $f\colon X \to S$ be a proper morphism of finite presentation, let D be a divisor on X with normal crossings relative to S, put $Y = \operatorname{Supp} D$, $U = X - Y$, and let $i\colon U \to X$ be the canonical morphism. Let $\bar{s}_1 \to \bar{s}_2$ be a specialization morphism of geometric points of S, let y_1 be a geometric point of $Y_{\bar{s}_1}$, and let y_2 be a geometric point of $Y_{\bar{s}_2}$ such that the projection z_1 of y_1 on X is a generization of the projection z_2 of y_2. Let I_{y_1} be an inertia subgroup of $\pi_1^{\mathrm{t}}(\bar{U}_{\bar{s}_1})$ at y_1. Then the image of I_{y_1} under the specialization homomorphism*

$$\pi\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_1}) \to \pi_1^{\mathrm{t}}(U_{\bar{s}_2})$$

is an inertia subgroup of $\pi_1^{\mathrm{t}}(U_{\bar{s}_2})$ at y_2. 392

Indeed, let $\overline{X}$ (respectively$\widetilde{X}$) be the strict localization of X at y_2 (respectivelyat y_1), and put $\overline{U} = U \times_X \overline{X}$ (respectively$\widetilde{U} = U \times_X \widetilde{X}$). There is a canonical morphism $\widetilde{U} \to \overline{U}$, and it follows from XIII.1.9 that there is a commutative diagram

$$\begin{array}{ccc} \pi_1^{\mathrm{t}}\left(\widetilde{U}_{\bar{s}_1}\right) & \xrightarrow{\pi'} & \pi_1^{\mathrm{t}}\left(\overline{U}_{\bar{s}_2}\right) \\ \downarrow & & \downarrow \\ \pi_1^{\mathrm{t}}\left(U_{\bar{s}_1}\right) & \xrightarrow{\pi} & \pi_1^{\mathrm{t}}\left(U_{\bar{s}_2}\right) \end{array}$$

where π' is the composite of the canonical homomorphism $\pi_1^{\mathrm{t}}(\widetilde{U}_{\overline{s}_1}) \to \pi_1^{\mathrm{t}}(\overline{U}_{\overline{s}_1})$ and the specialization homomorphism. Since $\pi_1^{\mathrm{t}}(\widetilde{U}_{\overline{s}_1})$ (respectively$\pi_1^{\mathrm{t}}(\overline{U}_{\overline{s}_1})$) is an inertia group of $\pi_1^{\mathrm{t}}(U_{\overline{s}_1})$ at y_1 (respectivelyof $\pi_1^{\mathrm{t}}(U_{\overline{s}_2})$ at y_2), it suffices to prove that π' is surjective. This follows from the expression obtained in XIII.5.6.

COROLLARY XIII.2.11. *Let X be a connected proper smooth curve of genus g over a separably closed field k of characteristic $p \geq 0$. Let U be the open subset obtained by removing from X the n distinct closed points $a_1, \ldots, a_n$. Then the tamely ramified fundamental group $\pi_1^{\mathrm{t}}(U)$ (XIII.2) can be generated by $2g+n$ elements x_i, y_i, σ_j, with $1 \leq i \leq g$, $1 \leq j \leq n$, such that σ_j is a generator of an inertia group corresponding to a_j, and the following relation holds:*

$$(*) \qquad \prod_{1 \leq i \leq g} (x_i y_i x_i^{-1} y_i^{-1}) \cdot \prod_{1 \leq j \leq n} \sigma_j = 1.$$

For every finite group G of order prime to p, *generated by elements $\overline{x}_i, \overline{y}_i, \overline{\sigma}_j$ satisfying the relation $(*)$, there exists an étale covering of U with group G, corresponding to a homomorphism $\pi_1^{\mathrm{t}}(U) \to G$ which sends x_i, y_i, σ_j to $\overline{x}_i, \overline{y}_i, \overline{\sigma}_j$, respectively. In other words,* 393
if p' denotes the set of prime numbers distinct from p, then $\pi_1^{p'}(U)$ is the pro-p'-group generated by x_i, y_i, σ_j subject to the single relation $(*)$.

Proof. We may suppose that k is algebraically closed. Suppose first that k has characteristic zero. There then exists an algebraically closed subfield k' of k, of finite transcendence degree over $\mathbb{Q}$, such that X is obtained by extension of scalars from a proper smooth curve X' defined over k', and we may suppose that the points $a_1, \ldots, a_n$ come from rational points $a'_1, \ldots, a'_n$ of X'. Since k' has finite transcendence degree over $\mathbb{Q}$, there is an embedding of k' into the field of complex numbers $\mathbb{C}$; put $\widetilde{U} = U' \times_{k'} \mathbb{C}$. Let k'' be an algebraically closed extension of k' such that there are k'-morphisms from k and from $\mathbb{C}$ to k''. If $g' \colon U' \to k'$ is the structural morphism, and if $\mathcal{F}$ is a finite constant sheaf of groups on U'', it follows from XIII.2.9 that the specialization morphisms

$$(\mathrm{R}^1 g'_* \mathcal{F}')_{\mathbb{C}} \to (\mathrm{R}^1 g'_* \mathcal{F}')_{k''} \longleftarrow (\mathrm{R}^1 g'_* \mathcal{F}')_k$$

are isomorphisms. In terms of fundamental groups, this shows that there is an isomorphism, defined up to inner automorphism,

$$\pi_1(U) \to \pi_1(\widetilde{U}),$$

and it is clear that this isomorphism sends an inertia group relative to a point of $X' - U'$ to an inertia group relative to the same point. We may therefore suppose that $k = \mathbb{C}$. In this case, the Riemann existence theorem (XII XII.5.2) shows that the fundamental group $\pi_1(U)$ is precisely the completion, for the topology of subgroups of finite index, of the fundamental group of the analytic space associated with U. The latter can be computed by transcendental methods [**3**, ch. 7 §47]; it can be generated by $2g+n$ elements x_i, y_i, σ_j 394
such that σ_j is the image of a generator of the local fundamental group $\pi_1(D_j)$ of a small disk centered at a_j, i.e. a generator of an inertia group corresponding to the point a_j, and these elements satisfy the single relation $(*)$.

Now suppose that k has characteristic $p > 0$. There exists a complete discrete valuation ring A with residue field k and fraction field K of characteristic zero, and a connected scheme X_1, proper and smooth over $S = \mathrm{Spec}(A)$, such that $X_1 \times_S \mathrm{Spec}(k) \simeq X$ (III III.7.4). The points a_j then lift to sections s_j of X_1 over S. Let Y_{1j} be the reduced closed subscheme with underlying space $s_j(S)$, let Y_1 be the union of the Y_{1j}, put $U_1 = X_1 - Y_1$, and let $g_1 \colon U_1 \to S$ be the structural morphism. Let $\overline{K}$ be an algebraically closed extension of K, and put $\overline{U} = U_1 \times_S \overline{K}$. If C is a finite constant group, it follows from XIII.2.8 that the specialization morphism

$$(\mathrm{R}^1 g_{1*} C_{U_1})_k \to (\mathrm{R}^1 g_{1*} C_{U_1})_{\overline{K}}$$

is injective, and even bijective if C has order prime to p. In terms of fundamental groups, this means that the specialization homomorphism (XIII.1.9)

$$\pi\colon \pi_1(\overline{U}) \to \pi_1^{\mathrm{t}}(U)$$

is surjective, and that the specialization homomorphism

$$\pi_1^{p'}(\overline{U}) \to \pi_1^{p'}(U)$$

is bijective. Finally, if x_i, y_i, σ_j are generators of $\pi_1(\overline{U})$ such that σ_j is a generator of an inertia group corresponding to the point $b_j = Y_{1j}(\overline{K})$ of $\overline{X}$, then, by XIII.1.10, $\pi(\sigma_j)$ is a generator of an inertia group corresponding to a_j. This completes the proof.

Remark. [Added in 2003 (MR): Let k be an algebraically closed field of characteristic $p > 0$, let X be a connected proper smooth algebraic curve over k of genus g, and let U be an affine open subset of X, the complement of $r \geq 1$ rational points of X. We have the fundamental group $\pi_1(U)$, its quotient $\pi_1^{\mathrm{t}}(U)$, which classifies the finite étale coverings of U tamely ramified at the points of $X - U$, and the quotient $\pi_1^{p'}(U)$ of $\pi_1^{\mathrm{t}}(U)$, which classifies the Galois étale coverings of U whose Galois group has order prime to p.

In (Coverings of algebraic curves, Amer. J. Math. **79** (1957), p. 825–856), S. Abhyankar formulated a number of conjectures on the structure of the finite groups that occur as Galois groups of finite étale coverings of U.

The conjectures concerning Galois groups of connected étale coverings of U whose order is prime to p are proved and made precise in Corollary XIII.2.11. Before turning to the finite quotients of $\pi_1(U)$, let us first give some indications about the "size"of this group. Let $X = \mathrm{Spec}(()A)$. We know that $\mathrm{Hom}_{\mathrm{cont}}(\pi_1(U), \mathbb{Z}/p\mathbb{Z})$ is described by Artin–Schreier theory (Corollary. XI XI.6.9). This group is isomorphic to $A/\wp(A)$, where $\wp$ is the map from A to A that sends a to $a^p - a$. Suppose first that U is the affine line $\mathbf{A}^1$, with ring $k[T]$. Let E be the set of elements of $k[T]$ of the form $\sum_i a_i T^i$, with $a_i = 0$ whenever p divides i. It is immediate that the composite map $E \to k[T] \to k[T]/\wp k[T]$ is bijective. Hence the coefficients a_i, $(i, p) = 1$, behave like coordinates on a space parametrizing the cyclic coverings of degree p of the affine line. In particular, it follows that $\pi_1(\mathbf{A}^1)$ is not topologically finitely generated and that, if one makes an extension $k \to k'$ of algebraically closed fields, the natural homomorphism $\pi_1(\mathbf{A}^1 \times_k k') \to \pi_1(\mathbf{A}^1)$, which is surjective, is not bijective, unlike in the proper case. In general, the curve U can be realized as a finite scheme over the affine line, and we conclude that the same phenomena occur for $\pi_1(U)$ (and indeed, more generally, for $\pi_1(V)$ for every connected affine k-scheme V of finite type and dimension ≥ 1).

With this said, if G is a finite group, let $G^{(p')}$ denote the largest quotient group of G whose order is prime to p. For a finite group G to be a topological quotient of $\pi_1(U)$, it is necessary that $G^{(p')}$ be a topological quotient of $\pi_1^{p'}(U)$, a condition that can in principle be decided using Corollary XIII.2.11. In the article cited above, Abhyankar conjectured that this necessary condition is also sufficient. This conjecture was proved by M. Raynaud in the case of the affine line and by D. Harbater in the general case (Invent. Math. **116** (1994), p. 425–462 and **117**, p. 1–25). For example, for the affine line $\pi_1^{p'}(\mathbf{A}^1) = 1$, and it follows that a finite group G is the Galois group of a connected étale covering of $\mathbf{A}^1$ if and only if $G^{(p')} = 1$, that is, if and only if G is generated by its Sylow p-subgroups. Thus every finite simple group whose order is a multiple of p has the required property.]

XIII.3. Cohomological properness and generic local acyclicity

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Theorem XIII.3.1. *Let S be an irreducible scheme with generic point s, let X and Y be two S-schemes of finite presentation, and let $f\colon X \to Y$ be an S-morphism. For every S-scheme S', denote by Y', X', etc. the inverse images of Y, X, etc. under the morphism $S' \to S$. The following properties hold:*

1) a) *One can find a nonempty open subset S' of S such that, for every finite constant sheaf of sets F' on X', f'_*F' is constructible, and (F', f') is cohomologically proper relative to S' in dimension ≤ 0.*

b) *Let F be a constructible sheaf of sets on X. Then one can find a nonempty open subset S' of S (depending on F) such that f'_*F' is constructible and (F', f') is cohomologically proper relative to S' in dimension ≤ 0.*

2) *Suppose that the schemes of finite type of dimension $\leq \dim X_s$ over an algebraic closure $\overline{k}$ of $\kappa(s)$ are strongly desingularizable* (SGA 5 I 3.1.5). *Then the following additional properties hold:*

a) *One can find a nonempty open subset S' of S such that, for every finite constant sheaf of groups F' on X' whose order is prime to the residual characteristics of S, if Φ' is the stack of torsors under F', then $f'_*\Phi'$ is 1-constructible, and (F', f') is cohomologically proper relative to S' in dimension ≤ 1.*

b) *Let $\mathbb{L}$ be the set of prime numbers distinct from the residual characteristics of S, and let Φ be an ind-$\mathbb{L}$-stack that is 1-constructible on X* (XIII.0) *such that, for every scheme X_1 étale over X and every pair of objects x, x_1 of Φ_{X_1}, the sheaf $\mathbf{Hom}_{X_1}(x, x_1)$ is constructible. Suppose moreover that S is locally noetherian. Then one can find a nonempty open subset S' of S such that $f'_*\Phi'$ is 1-constructible, such that, for every pair of objects y, y_1 of a fiber $(f'_*\Phi')_{Y_1}$, $\mathbf{Hom}_{Y_1}(y, y_1)$ is constructible, and such that (Φ', f') is cohomologically proper* 396
relative to S' in dimension ≤ 1.

Proof. We may suppose that S is affine; by SGA 4 VIII 1.1, we may suppose that S is integral; finally, by passage to the limit, we may suppose that S is the spectrum of an algebra of finite type over $\mathbb{Z}$; in particular, S is then noetherian. Since the question is local on Y, we may suppose that Y is affine. Moreover, in order to prove the theorem, it suffices to prove it after a finite extension $S' \to S$, where S' is an integral scheme and where $S' \to S$ is a composite of étale morphisms and finite surjective radicial morphisms.

1) Case of constant sheaves of sets. 1) 1. Reduction to the case where X is normal over S. Let $X_{1\overline{s}}$ be the normalization of $(X_{\overline{s}})_{\mathrm{red}}$; after restricting S to a nonempty open subset and making a radicial extension of S, we may suppose that $X_{1\overline{s}}$ comes from a scheme X_1 normal over S, and that the morphism $X_{1\overline{s}} \to X_{\overline{s}}$ comes from a finite surjective morphism $p\colon X_1 \to X$ (EGA IV 8.8.2 and 9.6.1). Suppose that the theorem has been proved for fp. After restricting S to an open subset, we may suppose that, for every constant sheaf of sets F on X, (p^*F, fp) is cohomologically proper relative to S in dimension ≤ 0 and that $f_*p_*(p^*F)$ is constructible. By XIII.1.8, (p_*p^*F, f) is then cohomologically proper relative to S in dimension ≤ 0. The morphism

$$F \to p_*p^*F = G$$

is a monomorphism. Since f_*F is a subsheaf of f_*G, it already follows that f_*F is constructible (SGA 4 IX 2.9 (ii)) and that (F, f) is cohomologically proper relative to S in dimension ≤ -1.

Let $X_2 = X_1 \times_X X_1$, and let $q\colon X_2 \to X$ be the canonical morphism. By XIII.1.10 1), there is an exact sequence 397

$$F \longrightarrow G \rightrightarrows q_*q^*F\ .$$

By what we have just proved, applied to fq in place of f, after restricting S to a nonempty open subset, we may suppose that, for every constant sheaf of sets F on X, (q^*F, fq) is cohomologically proper relative to S in dimension ≤ -1, and hence that (q_*q^*F, f) is cohomologically proper relative to S in dimension ≤ -1. It then follows from XIII.1.12 1) that (F, f) is cohomologically proper relative to S in dimension ≤ 0.

1) 2. Reduction to the case where X is normal affine over S.

Let U_s be an affine open subset of X_s dense in X_s. After restricting S to a nonempty open subset, we may suppose that $U_s \to X_s$ lifts to an open immersion $i\colon U \to X$, schematically dominant relative to S (EGA IV 8.9.1). Since the morphism $X \to S$ is normal, by SGA 2 XIV 1.18 we have

$$\operatorname{prof} \text{ét}_{S-U}(X) \geq 2;$$

consequently, for every constant sheaf F on X, the canonical morphism

$$F \to i_* i^* F$$

is an isomorphism. It follows that, if the theorem is assumed proved for fi and i, then, after restricting S to a nonempty open subset, (i^*F, i) and (i^*F, fi) are cohomologically proper relative to S in dimension ≤ 0. The same is therefore true of (F, f) (XIII.1.5 2)). Moreover, $f_*F = (fi)_*(i^*F)$ is constructible, which completes the reduction.

1) 3. End of the proof.

We may suppose that S is normal (EGA IV 7.8.3). One can find a compactification of X_s: 398

$$\begin{array}{ccc} X_s & \xrightarrow{j_s} & P_s \\ {\scriptstyle f_s}\downarrow & \swarrow{\scriptstyle g_s} & \\ Y_s & & \end{array}$$

where j_s is a dominant open immersion and g_s is a proper morphism; after making a radicial extension of $\kappa(s)$ and replacing P_s by its normalization, which does not change X_s, we may suppose that P_s is geometrically normal. After restricting S to a nonempty open subset and making a surjective radicial extension, we may suppose that the diagram above comes from a diagram

$$\begin{array}{ccc} X & \xrightarrow{j} & P \\ {\scriptstyle f}\downarrow & \swarrow{\scriptstyle g} & \\ Y & & \end{array}$$

where P is a scheme normal over S, j is a schematically dominant open immersion relative to S, and g is a proper morphism (EGA IV 6.9.1, 9.9.4 and 9.6.1). For every finite constant sheaf of sets F on X with value C, j_*F is the constant sheaf with value C (SGA 4 2.14.1), and the same is true after every base change $S' \to S$; it follows that (F, j) is cohomologically proper relative to S in dimension ≤ 0. The same is therefore true of (F, f), since g is proper (XIII.1.7). Since g is proper, $f_*F = g_*C_P$ is constructible, which completes the proof of 1) a).

2) Case of a constructible sheaf of sets. Let F be a constructible sheaf of sets on X. By SGA 4 IX 2.14 (ii), one can find a finite family of morphisms $p_i\colon Z_i \to X$ and, on each Z_i, a finite constant sheaf of sets C_i, such that there is a monomorphism 399

$$j\colon F \to \prod_i p_{i*} C_i = G.$$

By 1) a), after restricting S to a nonempty open subset, we may suppose that the (C_i, fp_i) are cohomologically proper relative to S in dimension ≤ 0, and that the $f_* p_{i*} C_i$ are constructible. We already conclude that (G, f) is cohomologically proper relative to S in dimension ≤ 0 (XIII.1.8), hence that (F, f) is cohomologically proper relative to S in dimension ≤ -1, and that f_*F is constructible. Let K be the amalgamated sum $K = G \coprod_F G$; since F and G are constructible, so is K. We therefore conclude from the preceding that, after restricting S to a nonempty open subset, we may suppose that (K, f) is cohomologically proper relative to S in dimension ≤ -1. It then follows from XIII.1.12 1) that (F, f) is cohomologically proper relative to S in dimension ≤ 0.

3) Case of constant sheaves of groups. If F is a constant sheaf of groups on X, we denote by Φ the stack of torsors under F.

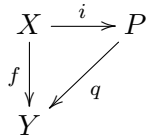
3) 1. Let us first show that, after restricting S to a nonempty open subset, for every constant sheaf of groups F on X whose order is prime to the residue characteristics of S, (F, f) is cohomologically proper relative to S in dimension ≤ 0, and $f_*\Phi$ is constructible.

For this, we reduce to the case where X is smooth over S. After making a finite extension of $\kappa(s)$, which is permissible because it can be regarded as the composite of an étale extension and a radicial extension, one can find a proper surjective morphism $p_s\colon X_{1s} \to X_s$, where X_{1s} is a scheme smooth over S of the same dimension as X_s, and, after restricting S to a nonempty open subset, we may suppose that p_s comes from a proper surjective morphism $p\colon X_1 \to X$, where X_1 is a scheme smooth over S (EGA IV 400
9.6.1 and 12.1.6). Let $X_2 = X_1 \times_X X_1$, and let $q\colon X_2 \to X$ be the canonical morphism. There is an exact diagram of stacks on X

$$\Phi \longrightarrow p_*p^*\Phi \rightrightarrows q_*q^*\Phi$$

(XIII.1.10 2)). Assuming that the theorem has been proved in the smooth case, we first see that we may suppose $f_*p_*p^*\Phi$ constructible; hence the same is true of $f_*\Phi$ (XIII.3.2 below). Moreover, by XIII.1.5 2), we may suppose $(p_*p^*\Phi, f)$ cohomologically proper relative to S in dimension ≤ 0; it follows that (Φ, f) is cohomologically proper relative to S in dimension ≤ -1. We may therefore suppose that $(q_*q^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ -1, and this implies that (Φ, f) is cohomologically proper relative to S in dimension ≤ 0 (XIII.1.11 1)).

We then reduce, as in 1) 2, to the case where X is smooth and affine over S. Let



be a compactification of X, where i is a dominant open immersion and q is a proper morphism. Since $\dim P_s = \dim X_s$, we may apply the resolution of singularities hypothesis to $P_{\bar{s}}$. After making an étale extension and a radicial extension of S, one can find a proper morphism $r\colon Z \to P$, where Z is smooth over S, $r^{-1}(X) \simeq X$, and $r^{-1}(X)$ is the complement in Z of a divisor with normal crossings relative to S. Every torsor under F is then tamely ramified on Z relative to S (XIII.2.3 b)). It therefore follows from XIII.2.7 that (F, f) is cohomologically proper relative to S in dimension ≤ 0, which proves our assertion.

3) 2. Reduction to the case where X is smooth over S. 401

After making a finite extension of $\kappa(s)$, one can find a proper surjective morphism $p_s\colon X_{1s} \to X$, where X_{1s} is a scheme smooth over s, and we may suppose that p_s comes from a proper surjective morphism $p\colon X_1 \to X$, where X_1 is smooth over S. Suppose that the theorem has been proved for fp, and let us prove it for f. Let F be a finite constant sheaf of groups on X, of order prime to the residue characteristics of S, and let Φ be the stack of torsors under F. Let $X_2 = X_1 \times_X X_1$, $X_3 = X_1 \times_X X_1 \times_X X_1$, and let $q\colon X_2 \to X$, $r\colon X_3 \to X$ be the canonical morphisms. By XIII.1.10 2), there is an exact diagram of stacks

$$\Phi \longrightarrow p_*p^*\Phi \rightrightarrows q_*q^*\Phi \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; r_*r^*\Phi\ .$$

To prove that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1, it is enough to show that the same is true of $(p_*p^*\Phi, f)$, that $(q_*q^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0, and that $(r_*r^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ -1 (XIII.1.11 2)). By 3) 1 above, we may suppose that, for every finite constant sheaf of groups F, $(q^*\Phi, fq)$, $(q^*\Phi, q)$, $(r^*\Phi, fr)$, $(r^*\Phi, r)$ are cohomologically proper relative to S in dimension ≤ 0. It then follows from XIII.1.5 2) that $(q_*q^*\Phi, f)$ and $(r_*r^*\Phi, f)$ are cohomologically proper relative to S in dimension ≤ 0. Since the theorem is assumed proved in the smooth case, $(p^*\Phi, fp)$ and $(p^*\Phi, p)$, and hence also $(p_*p^*\Phi, f)$ (XIII.1.5 2)), are cohomologically proper relative to S in dimension ≤ 1. This indeed shows that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1.

Moreover, $f_*p_*p^*\Phi$ is 1-constructible by hypothesis; by XIII.3.1, we may suppose that $f_*q_*q^*\Phi$ is constructible; it therefore follows from XIII.3.2 below that $f_*\Phi$ is 1-constructible.

3) 3. Reduction to the case where X is smooth and affine over S. 402

By 3) 2, we may suppose that X is smooth over S. Let $(U_i)_{i\in I}$ be a finite covering of X by affine open subsets, let X_1 be the disjoint union of the U_i, and let $p\colon X_1 \to X$ be the canonical morphism. Since p is an effective descent morphism for the category of étale sheaves of finite type on variable schemes, we see as in 3) 2 that, if the theorem is assumed proved for the U_i, i.e. for X_1, it is also true for X.

3) 4. The case where X is smooth and affine over S.

As in 3)1, after restricting S to a nonempty open subset and making a surjective étale extension and a surjective radicial extension of S, one can find a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\ i\ } & P \\ {\scriptstyle f}\downarrow & \swarrow{\scriptstyle q} & \\ Y & & \end{array}$$

where P is a scheme smooth over S, X is the complement in P of a divisor with normal crossings relative to S, and g is a proper morphism. If F is a constant sheaf of order prime to the residue characteristics of S, every torsor under F is tamely ramified on P relative to S (XIII.2.3 b)). The fact that (F, f) is cohomologically proper relative to S in dimension ≤ 1 and that $f_*\Phi$ is constructible then follows from XIII.2.7.

4) Proof of 2) b). 4) 1. The case where Φ is a gerbe.

One can find a surjective étale morphism of finite type $p\colon X_1 \to X$ such that $p^*\Phi$ is a trivial gerbe. By descent, as in 3) 2, we see that it is enough to prove the theorem for X_1, $X_1 \times_X X_1$, and $X_1 \times_X X_1 \times_X X_1$. We are therefore reduced to the case where Φ is the 403
gerbe of torsors under a constructible sheaf of groups F whose fibers have order prime to the residue characteristics of S.

By SGA 4 IX 2.14, one can find a finite family of finite morphisms $p_i\colon Z_i \to X$ and, for each i, a finite constant sheaf of groups C_i, of order prime to the residue characteristics

of S, such that there is a monomorphism

$$j\colon F \to \prod_i p_{i*}C_i = G.$$

Let Φ_i be the stack of torsors under C_i, and let Ψ be the stack of torsors under G. It follows from 2) a) that, after restricting S to a nonempty open subset, we may suppose that the (C_i, fp_i) are cohomologically proper relative to S in dimension ≤ 1, and that the stacks $f_*p_{i*}\Phi_i$ are 1-constructible. It then follows from XIII.1.8 that the $(p_{i*}C_i, f)$ are cohomologically proper relative to S in dimension ≤ 1; hence the same is true of (G, f). Moreover, since $p_{i*}\Phi_i$ is equivalent to the stack of torsors under the group $p_{i*}C_i$ (SGA 4 VIII 5.8), we see that $f_*\Psi$ is 1-constructible.

Since $\mathrm{R}^1 f_* G$ is constructible, one can find a sheaf representable by an étale Y-scheme of finite type T and an epimorphism

$$a\colon T \to \mathrm{R}^1 f_* G$$

(SGA 4 IX 2.7); moreover, we may suppose that the image of the identity section of $T(T)$ is defined by a torsor Q on $X \times_Y T = X_T$, with group $G|X_T$. Let $f_T\colon X_T \to T$ be the canonical morphism, and put $F_T = F|X_T$, etc. By 1) b), after restricting S to a nonempty open subset, we may suppose that $(Q/F_T, f_T)$ is cohomologically proper relative to S in dimension ≤ 0 and that $f_{T*}(Q/F_T)$ is constructible. It then follows from XIII.3.3 that $f_*\Psi$ is constructible.

Let us show that (F, f) is cohomologically proper relative to S in dimension ≤ 1. By 404
XIII.1.12 2), it is enough to prove that, for every scheme Y_1 étale over Y and every torsor Q_1 on $X_1 = X \times_Y Y_1$, if $f_1\colon X_1 \to Y_1$ is the canonical morphism, then $(Q_1/F_1, f_1)$ is cohomologically proper relative to S in dimension ≤ 0. But, by definition of T, Q_1 is, locally for the étale topology on Y_1, the inverse image of Q, which proves our reduction.

4) 2. The general case.

Using Lemma XIII.6.2, 4) 1, and 1) a), we see that, after restricting S to a nonempty open subset, we may suppose that $S(f_*\Phi)$ is constructible and $(S\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0. We can then find a sheaf representable by an étale Y-scheme of finite type T and an epimorphism

$$a\colon T \to S(f_*\Phi).$$

We again use the notation of 4) 1, and in addition put $Z = T \times_Y T$, $X_Z = X \times_Y Z$, and denote by $f_Z\colon X_Z \to Z$ the canonical morphism. We may suppose that T is chosen so that the image q of the identity section of $T(T)$ under a is defined by an object p of $(f_*\Phi)_T = \Phi_{X_T}$. Let p_1 and p_2 (respectively q_1 and q_2) be the inverse images of p (respectively q) under the two projections from Z to T. After restricting S to a nonempty open subset, we may suppose that $(\mathbf{Aut}_{X_T}(p), f_T)$ is cohomologically proper relative to S in dimension ≤ 1, that $(\mathbf{Hom}_{X_Z}(p_1, p_2), f_Z)$ is cohomologically proper relative to S in dimension ≤ 0, and that $f_{T*}(\mathbf{Aut}_{X_T}(p))$, $f_{Z*}(\mathbf{Hom}_{X_Z}(p_1, p_2))$ are constructible (a) 1) and 4) 1)).

It follows first that, for every scheme Y_1 étale over Y and every pair of objects y, y_1 of $(f_*\Phi)_{Y_1}$, the sheaf $\mathbf{Aut}_{Y_1}(y)$ (respectively$\mathbf{Hom}_{Y_1}(y, y_1)$) is constructible: locally for the étale topology of Y_1, such a sheaf is the inverse image of $f_{T*}(\mathbf{Aut}_{X_T}(p))$ (respectivelyof $f_{Z*}(\mathbf{Hom}_{X_Z}(p_1, p_2))$).

It remains to prove that (Φ, f) is cohomologically proper relative to S in dimension 405
≤ 1. For this, it is enough to show that, for every S-scheme S' and every geometric point $\overline{y}'$ of Y', if $\overline{Y}$ denotes the strict localization of Y at y', and $\overline{X} = X \times_Y \overline{Y}$, etc., then the canonical functor

$$\overline{\varphi}\colon \Phi(\overline{X}) \to \Phi'(\overline{X'})$$

is an equivalence of categories.

Let us show that $\overline{\varphi}$ is fully faithful. Let $x, y \in \Phi(\overline{X})$, and let x', y' be their images in $\Phi'(\overline{X'})$. We must show that the canonical morphism

$$(*) \qquad \mathrm{Hom}_{\overline{X}}(x, y) \to \mathrm{Hom}_{\overline{X'}}(x', y')$$

is bijective. By the definition of T, there are two morphisms from $\overline{Y}$ to T such that x and y are the inverse images of p under these two morphisms. This amounts to giving a morphism $\overline{Y} \to Z$, and hence a morphism $h\colon \overline{X} \to X_Z$ such that

$$h^*(p_1) = x \qquad h^*(p_2) = y.$$

Consequently, there is a canonical isomorphism

$$\mathbf{Hom}_{\overline{X}}(x, y) = h^*(\mathbf{Hom}_{X_Z}(p_1, p_2)).$$

But, since $(\mathbf{Hom}_{X_Z}(p_1, p_2), f_Z)$ is cohomologically proper relative to S in dimension ≤ 0, the same is true of $(\mathbf{Hom}_{\overline{X}}(x, y), \overline{f})$, which proves that the morphism $(*)$ is bijective.

Let us show that $\overline{\varphi}$ is essentially surjective. Let $x' \in \Phi'(\overline{X'})$. Since $(S\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0, the canonical morphism

$$\mathrm{H}^0(\overline{X}, \overline{S\Phi}) \to \mathrm{H}^0(\overline{X'}, \overline{S\Phi}')$$

is bijective. Let G' be the maximal subgerbe of Φ' generated by x'. There is then a maximal subgerbe G of $\overline{\Phi}$ whose inverse image on $\overline{X'}$ is G'. By 4) 1, $(G, \overline{f})$ is cohomologically proper relative to S in dimension ≤ 1. Consequently, the canonical functor

$$G(\overline{X}) \to G'(\overline{X'})$$

is an equivalence of categories. This proves the existence of an object x of $\Phi(\overline{X})$ whose image in $\Phi'(X')$ is isomorphic to x', and completes the proof of the theorem. 406

Lemma XIII.3.2. *Let S be a locally Noetherian scheme, and let*

$$\Phi \xrightarrow{p} \Phi_1 \underset{\longrightarrow}{\overset{p_1,p_2}{\rightrightarrows}} \Phi_2$$

be an exact diagram of stacks on S (XIII.1.9). *If Φ_1 is constructible, then so is Φ. If, for every scheme S' étale over S and every pair of objects x_1, y_1 of $(\Phi_1)_{S'}$, the sheaf $\mathbf{Hom}_{S'}(x_1, y_1)$ is constructible, then for every pair of objects x, y of $\Phi_{S'}$, the same is true of $\mathbf{Hom}_{S'}(x, y)$. Suppose that Φ_1 is 1-constructible (XIII.0) and that Φ_2 is constructible; then Φ is 1-constructible.*

For every scheme S' étale over S and every object x of $\Phi_{S'}$, there is a monomorphism

$$\mathbf{Aut}_{S'}(x) \to \mathbf{Aut}_{S'}(p(x)).$$

It follows from SGA 4 IX 2.9 that if Φ_1 is constructible, then so is Φ, and the second assertion of the lemma is proved in the same way.

Now suppose that Φ_1 is 1-constructible and that Φ_2 is constructible. The morphism p induces a morphism on the sheaves of maximal subgerbes,

$$\varphi\colon S\Phi \to S\Phi_1.$$

Let G be the image of $S\Phi$ under φ. By SGA 4 IX 2.9, G is a constructible sheaf. We can therefore find a sheaf represented by an étale S-scheme T of finite type, and an epimorphism

$$a\colon T \to G$$

(SGA 4 IX 2.7). Moreover, T can be chosen so that the image y of the identity section of $T(T)$ under a is defined by an object x_1 of $(\Phi_1)_T$ of the form $x_1 = p(x)$, where $x \in \Phi_T$. 407

It is enough to show that, for every point s of S, there is a nonempty open subset U of $\overline{\{s\}}$ such that $S\Phi|U$ is constructible locally constant. Let $s \in S$, let $\overline{s}$ be a geometric point over s, and let $\overline{y}_1, \ldots, \overline{y}_n$ be the elements of $G_{\overline{s}}$. By the definition of T, there are morphisms $h_i\colon \overline{s} \to T$ such that $h_i^*(y) = \overline{y}_i$. Let S' be the fiber product over S

of n schemes isomorphic to T, let $h\colon \overline{s} \to S'$ be the fiber product of the h_i, and let y_i (respectivelyx_i) be the inverse image of y (respectively$x$) under the i-th projection from S' to T. Let F_i be the subsheaf of $S\Phi|S'$ that is the inverse image of y_i, and let us show that the F_i are constructible. The sheaf F_i is a quotient of the sheaf F_i' such that, for every scheme S'' étale over S',

$$F_i'(S'') = \{\text{classes modulo isomorphism of objects } z \text{ of } \Phi_{S''} \text{ equipped with an isomorphism } i\colon p(z) \simeq p(x_i|S'')\}.$$

It is enough to show that the F_i' are constructible. Put $z_i = p_1p(x_i)$, $z_i' = p_2p(x_i)$. There is a monomorphism

$$\Psi_i\colon F_i' \to \mathbf{Isom}_{S'}(z_i, z_i'),$$

obtained by associating with every scheme S'' étale over S' and every object z of $\Phi_{S''}$ equipped with an isomorphism $i\colon p(z) \simeq p(x_i|S'')$, the isomorphism from z_i to z_i' defined by requiring the diagram

$$\begin{array}{ccc} p_1p(z) & \xrightarrow{p_1(i)} & z_i \\ {\scriptstyle j}\downarrow & & \downarrow \\ p_2p(z) & \xrightarrow{p_2(i)} & z_i' \end{array}$$

to be commutative (j is the canonical morphism associated with the exact diagram). 408

The morphism Ψ_i is injective: to say that two objects z, z', equipped with isomorphisms $i\colon p(z) \xrightarrow{\sim} p(x_i|S'')$ and $i'\colon p(z') \xrightarrow{\sim} p(x_i|S'')$, define the same element of $\mathbf{Isom}_{S''}(z_i, z_i')$ amounts to saying that $p_1(i'^{-1}i) = p_2(i'^{-1}i)$, i.e. that $i'^{-1}i$ comes from an isomorphism $z \to z'$. Since F_i' is a subsheaf of $\mathbf{Isom}_{S'}(z_i, z_i')$, it is constructible.

Since there is a nonempty open subset U of $\overline{\{s\}}$ such that $y_1|U, \ldots, y_n|U$ generate G, it follows from Lemma XIII.6.3 below that $S\Phi|U$ is constructible. Thus, after restricting U if necessary, $S\Phi|U$ is constructible locally constant.

LEMMA XIII.3.3. *Let S be a locally Noetherian scheme, let $f\colon X \to S$ be a morphism, let $F \to G$ be a monomorphism of sheaves of groups on X, let Ψ (respectivelyΨ_1) be the stack of torsors under F (respectivelyunder G), and put $\Phi = f_*\Psi$, $\Phi_1 = f_*\Psi_1$. Suppose that we are given a sheaf on S represented by an étale S-scheme T of finite type and a surjective morphism*

$$a\colon T \to S\Phi_1 \simeq \mathrm{R}^1 f_* G,$$

such that there is a torsor Q on $X_T = X \times_S T$ with group $G|X_T$, defining in $\mathrm{R}^1 f_* G(T)$ the image under a of the identity section of $T(T)$. Let $f_T\colon X_T \to T$ be the canonical morphism, and put $F_T = F|X_T$. Suppose that Φ_1 is 1-constructible and that $f_{T*}(Q/F_T)$ is constructible. Then Φ is 1-constructible.

Indeed, it is enough to repeat the proof of XIII.3.2, with the existence of the morphisms Ψ_i replaced by the isomorphisms

$$F_i' \xrightarrow{\sim} f_*(Q/F_T)|S'.$$

REMARK XIII.3.4. *Suppose that $k = \kappa(s)$ has characteristic zero. Then schemes of finite type over $\overline{k}$, of dimension $\leq \dim X$, admit strong resolution of singularities, and the proof of XIII.3.1 gives the following results:* 409

a) *There is a nonempty open subset S_1 of S such that, for every scheme S' over S_1 whose maximal points have characteristic zero and every constructible locally constant sheaf of sets F on $X' = X \times_S S'$, (F, f') is cohomologically proper relative to S' in dimension ≤ 0.*

b) *If all the residue characteristics of S are zero, there is a nonempty open subset S_1 of S such that, for every scheme S' over S_1 and every constructible locally constant sheaf of groups F on X', (F, f') is cohomologically proper relative to S' in dimension ≤ 1.*

Indeed, it is enough to repeat the proof of XIII.3.1.2) a). Proposition XIII.2.7, used in 3) 4, applies to a locally constant sheaf F, because, with the notation of 3) 4, every torsor under F is tamely ramified on P relative to S, since all the residue characteristics of S are zero.

Corollary XIII.3.5. *Let k be a field of characteristic $p \geq 0$, let p' be the set of prime numbers distinct from p, and let $f\colon X \to k$ be a coherent morphism.*

1) *For every sheaf of sets F, (F, f) is cohomologically proper in dimension ≤ 0.*
2) *Suppose that one of the following two conditions holds:*
   a) *f is of finite type, and schemes of finite type of dimension $\leq \dim X$ over an algebraic closure of k admit strong resolution of singularities.*
   b) *Schemes of finite type over an algebraic closure of k admit strong resolution of singularities.* 410

   Then, for every sheaf of ind-p'-groups F, (F, f) is cohomologically proper in dimension ≤ 1.

Let F be a sheaf of sets (respectivelyof ind-p'-groups). By SGA 4 IX 2.7.2, F can be written as a filtered inductive limit

$$F = \varinjlim F_i,$$

where the F_i are constructible sheaves of sets (respectivelyof ind-p'-groups). Since f is coherent, f_* (respectively$\mathrm{R}^1 f_*$) commutes with inductive limits (SGA 4 VII 3.3). If the (F_i, f) are known to be cohomologically proper in dimension ≤ 0 (respectivelyif every sheaf of sets is cohomologically proper in dimension ≤ 0 and the (F_i, f) are cohomologically proper in dimension ≤ 1), then the same is true of (F, f). We may therefore suppose that F is constructible.

If f is assumed to be of finite type (respectivelyto satisfy a)), the proposition follows from XIII.3.1 1) b) (respectivelyfrom XIII.3.1 2) b)). We now prove XIII.3.5 without assuming that f is of finite type. For every scheme S' over k and every geometric point $\overline{s}$ of S', denote by $\overline{k}$ (respectively$\overline{S'}$) the strict localization of k at $\overline{s}$ (respectivelyof S' at $\overline{s}$), let $\overline{X}$ be the inverse image of X on $\overline{k}$, and consider the cartesian square

$$\begin{array}{ccc} \overline{X} & \xleftarrow{g} & \overline{X'} \\ {\scriptstyle \overline{f}}\downarrow & & \downarrow{\scriptstyle \overline{f}'} \\ \overline{k} & \longleftarrow & \overline{S'} \end{array}$$

It is enough to prove that, for every S' and every $\overline{s}$, the canonical morphism

$$\mathrm{H}^0(\overline{X}, \overline{F}) \to \mathrm{H}^0(\overline{X'}, \overline{F'}) \qquad (\text{respectively} \mathrm{H}^1(\overline{X}, \overline{F}) \to \mathrm{H}^1(\overline{X'}, \overline{F'}))$$

is an isomorphism. It is enough to show the relations 411

$$(*) \qquad \overline{F} \simeq g_* g^* \overline{F} \qquad (\text{respectively} \mathrm{R}^1 g_*(g^* \overline{F}) = 0).$$

In this form, the question is local on X for the étale topology. We may therefore suppose that X is affine; by passage to the limit, we may suppose that X is of finite type over k. It is then known that (F, f) is cohomologically proper in dimension ≤ 0 (respectively≤ 1), and that the same is true when X is replaced by a scheme étale and of finite type over X. This proves $(*)$.

Theorem XIII.3.6. *Let S be an irreducible scheme with generic point s, and let $f\colon X \to S$ be a morphism of finite presentation. Suppose that schemes of finite type of*

dimension $\leqslant \dim X_s$ *over an algebraic closure* $\bar{k}$ *of* k *admit resolution of singularities* (EGA IV 7.9.1). *If* $\mathbb{L}$ *denotes the set of prime numbers distinct from the residue characteristics of* S*, then there is a nonempty open subset* S_1 *of* S *such that the morphism* $f|S_1$ *is universally locally* 1*-aspherical for* $\mathbb{L}$.

We may suppose that S is integral and X is reduced (SGA 4 VIII 1.1). By passage to the limit, we may suppose that S is Noetherian. Moreover, to prove the theorem, it is enough to prove it after a finite extension $S_1 \to S$, where S_1 is an integral scheme and $S_1 \to S$ is a composite of étale extensions and surjective radicial extensions.

First let us show that, after restricting S to a nonempty open subset if necessary, f is universally locally 0-acyclic. After a radicial extension of $\kappa(s)$, we may suppose that $(X_s)_{\mathrm{red}} \to s$ is separable. Thus, after restricting S to a nonempty open subset and making a surjective radicial extension of S, we may suppose that f is flat with geometrically separable fibers (EGA IV 12.1.1), which implies that f is universally 0-acyclic (SGA 4 XV 4.1).

Let us show that, after restricting S to a nonempty open subset if necessary, f is universally locally 1-aspherical for $\mathbb{L}$. After making a finite extension of $\kappa(s)$, which is 412
allowed because it may be regarded as a composite of an étale extension and a radicial extension, we can find a proper surjective morphism $p_s\colon Y_s \to X_s$, where Y_s is a scheme smooth over s and has the same dimension as X_s. We may suppose that p_s comes from a proper surjective morphism $p\colon Y \to X$, where Y is smooth over S (EGA IV 9.6.1 and 12.1.6). It is enough to show that, after restricting S to a nonempty open subset if necessary, for every diagram with cartesian squares

$$\begin{array}{ccc} S'' & \xleftarrow{f''} & X'' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ S' & \xleftarrow{f'} & X' \\ \downarrow & & \downarrow \\ S & \xleftarrow{f} & X, \end{array}$$

where i is étale of finite presentation, and every sheaf of ind-$\mathbb{L}$-groups F on S'', if Φ is the stack of torsors under F, then the canonical morphism

$$f'^{*} i_{*} \Phi \to j_{*} f''^{*} \Phi$$

is an equivalence. Put $Z = Y \times_X Y$ and $T = Y \times_X Y \times_X Y$. There is a natural commutative diagram

$$\begin{array}{ccccccc} S'' & \xleftarrow{f''} & X'' & \xleftarrow{p''} & Y'' & \leftleftarrows & Z'' \;\lhd\!\lhd\!\lhd\; T'' \\ {\scriptstyle i}\downarrow & & {\scriptstyle j}\downarrow & & \downarrow & & \downarrow \\ S' & \xleftarrow{f'} & X' & \xleftarrow{p'} & Y' & \leftleftarrows & Z' \;\lhd\!\lhd\!\lhd\; T' \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ S & \xleftarrow{f} & X & \xleftarrow{p} & Y & \leftleftarrows & Z' \;\lhd\!\lhd\!\lhd\; T' \end{array}$$

Let $q\colon Z \to X$ and $r\colon T \to X$ be the canonical morphisms, with q', r', q'', r'' having the 413
evident meanings. By XIII.1.10 2), there is the following essentially commutative diagram, whose rows are exact:

$$\begin{array}{ccccccc} f'^{*} i_{*} \Phi & \longrightarrow & p'_{*} p'^{*} (f'^{*} i_{*} \Phi) & \rightrightarrows & q'_{*} q'^{*} (f'^{*} i_{*} \Phi) & \rightrightarrows\!\rightarrow & r'_{*} r'^{*} (f'^{*} i_{*} \Phi) \\ {\scriptstyle a}\downarrow & & {\scriptstyle b}\downarrow & & {\scriptstyle c}\downarrow & & {\scriptstyle d}\downarrow \\ j_{*} f''^{*} \Phi & \longrightarrow & j_{*} p''_{*} p''^{*} (f''^{*} \Phi) & \rightrightarrows & j_{*} q''_{*} q''^{*} (f''^{*} \Phi) & \rightrightarrows\!\rightarrow & j_{*} r''_{*} r''^{*} (f''^{*} \Phi) \end{array}$$

Since Y is smooth over S, the morphism fp is universally locally 1-aspherical for $\mathbb{L}$ (SGA XV 2.1), and it follows from [**2**, VII 2.1.7] that b is an equivalence of categories. On the other hand, after restricting S to a nonempty open subset if necessary, we may suppose that the morphisms $Z \to S$ and $T \to S$ are universally locally 0-acyclic. It follows that the functors c and d are fully faithful, and the diagram above then shows that a is an equivalence, completing the proof.

Corollary XIII.3.7. *Let k be a field of characteristic $p \geqslant 0$, let p' be the set of prime numbers distinct from p, and let $f\colon X \to k$ be a coherent morphism. Suppose that one of the following two conditions holds:*

a) *f is of finite type, and schemes of finite type of dimension $\leqslant \dim X$ over an algebraic closure of k admit resolution of singularities.*
b) *Schemes of finite type over an algebraic closure of k admit resolution of singularities.*

Then f is universally locally 1-aspherical for p'.

Case a) follows from XIII.3.6. In case b), since the question is local on X, we may suppose that X is affine; passage to the limit (SGA 4 XV 1.3) reduces us to the case where X is of finite type over k.

Corollary XIII.3.8. *Let S be an irreducible scheme with generic point s, and let* 414
$f\colon X \to S$ be a morphism of finite presentation. Suppose that schemes of finite type of dimension $\leqslant \dim X_s$ over an algebraic closure $\overline{k}$ of $\kappa(s)$ admit strong resolution of singularities (SGA 5 I 3.1.5). *If $\mathbb{L}$ denotes the set of prime numbers distinct from the residue characteristics of S, then there is a nonempty open subset S_1 of S such that, for every specialization $\bar{s}_1 \to \bar{s}_2$ of geometric points of S_1, the specialization morphism* (XIII.2)

$$\pi_1^{\mathbb{L}}(X_{\bar{s}_1}) \to \pi_1^{\mathbb{L}}(X_{\bar{s}_2})$$

is bijective.

By XIII.3.1 and XIII.3.6, after restricting S to a nonempty open subset if necessary, we may suppose that f is locally 1-aspherical for $\mathbb{L}$, and that, for every finite constant sheaf of $\mathbb{L}$-groups F on X, (F, f) is cohomologically proper in dimension $\leqslant 1$. It then follows from XIII.1.13 that, for every specialization $\bar{s}_1 \to \bar{s}_2$ of geometric points of S_1, the specialization morphism

$$\left(\mathrm{R}^1 f_* F\right)_{\bar{s}_2} \to \left(\mathrm{R}^1 f_* F\right)_{\bar{s}_1}$$

is bijective. The corollary is precisely the translation of the preceding statement into the language of fundamental groups.

XIII.4. Exact homotopy sequences

. Let X and S be two connected schemes, let $f\colon X \to S$ be a morphism, let a be a geometric point of X, and let $\mathbb{L}$ be a set of prime numbers. Let K be the kernel of the canonical homomorphism $\pi_1(X, a) \to \pi_1(S, a)$, and let N be the smallest normal pro-subgroup of K such that K/N is the pro-$\mathbb{L}$-group $K^{\mathbb{L}}$. Then N is normal in $\pi_1(X, a)$, and we denote by

$$\pi_1'(X, a)$$

the quotient of $\pi_1(X, a)$ by N. If a is a geometric point of a geometric fiber $X_{\bar{s}}$, the 415
canonical morphisms

$$\pi_1(X_{\bar{s}}, a) \to \pi_1(X, a) \to \pi_1(S, a)$$

define canonical morphisms

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi_1'(X, a) \xrightarrow{v} \pi_1(S, a)$$

We have $vu = 0$.

PROPOSITION XIII.4.1. *Let S be a connected scheme, and let $f\colon X \to S$ be a locally* 0*-acyclic morphism* (SGA 4 XV 1.11)*; suppose moreover that f is* 0*-acyclic (which, when f is coherent, amounts to saying that the geometric fibers of f are connected* (SGA 4 XV 1.16)*). Let $\mathbb{L}$ be a set of prime numbers. If S' is a scheme étale over S, denote by X' and f' the inverse images of X and f over S'. Suppose that, for every étale covering S' of S and every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, (E, f') is cohomologically proper relative to S' in dimension $\leqslant 0$ and f'_*E is constructible. Then, if $\bar{s}$ is a geometric point of S and a is a geometric point of the fiber $X_{\bar{s}}$, the sequence of group homomorphisms*

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi'_1(X, a) \xrightarrow{v} \pi_1(S, a) \to 1 \tag{XIII.4.-1.1}$$

is exact.

This statement generalizes X.X.1.4, whose proof we shall repeat.

Let us first show that v is surjective. It suffices to show that, for every connected étale covering S' of S, X' is also connected (V V.6.9). Let C be a set having at least two elements. Since f is 0-acyclic, the canonical morphism

$$\mathrm{H}^0(S', C_{S'}) \to \mathrm{H}^0(X', C_{X'})$$

is bijective, and hence X' is connected, which proves the surjectivity of v.

By definition of $K^{\mathbb{L}}$ (XIII.4), there is an exact sequence 416

$$1 \to K^{\mathbb{L}} \to \pi'_1(X, a) \to \pi_1(S, a) \to 1$$

Let $\widetilde{S}$ be the universal covering of S, and put $\widetilde{X} = \widetilde{S} \times_S X$; the group $K^{\mathbb{L}}$ classifies the Galois coverings P whose group is an $\mathbb{L}$-group and for which there exist an étale covering S' of S and a Galois covering Q of $X' = X \times_S S'$ such that there is an isomorphism $P \simeq Q \times_{X'} \widetilde{X}$. For the sequence (XIII.4.-1.1) to be exact, it is necessary and sufficient that the canonical morphism

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \to K^{\mathbb{L}}$$

be surjective. By the interpretation of $K^{\mathbb{L}}$, this amounts to saying that, for every étale covering S' of S and every Galois covering Q of X' whose group is an $\mathbb{L}$-group, such that $P = Q \times_{X'} \widetilde{X}$ is connected, $Q|X_{\bar{s}}$ is connected. Let us show that this last condition is satisfied. Indeed, let S' be an étale covering of S, and let Q be a Galois covering of X' whose group is an $\mathbb{L}$-group F, such that $Q|X_{\bar{s}}$ is disconnected; let us show that, after possibly replacing S' by an étale covering, Q becomes disconnected. There is a subgroup G of F distinct from F and a torsor R under $G|X_{\bar{s}}$ such that $Q|X_{\bar{s}}$ is obtained by extension of structure group along $G \to F$ from R. The étale covering $E = Q/G$ of X' is such that $E|X_{\bar{s}}$ has a section. By XIII.1.15, f_*E is locally constant constructible and, after possibly replacing S' by an étale covering, we may even suppose that f_*E is constant. Since (E, f') is cohomologically proper relative to S' in dimension $\leqslant 0$ and $\mathrm{H}^0(X_{\bar{s}}, E|X_{\bar{s}})$ is nonempty, one sees that E has a section. But this proves that Q is disconnected, which completes the proof.

From XIII.4.1, we deduce the following lemma, which will be used in XIII.4.6.

LEMMA XIII.4.2. *Let S be a connected scheme, let $f\colon X \to S$ be a locally* 0*-acyclic and* 0*-acyclic morphism, and let $\mathbb{L}$ be a set of prime numbers. Suppose that, for every* 417 *finite constant sheaf of $\mathbb{L}$-groups F on X, (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, and that, for every étale covering S' of S and every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, f'_*E is constructible. Then, if $\bar{s}$ is a geometric point of S and a is a geometric point of the fiber $X_{\bar{s}}$, the sequence of group homomorphisms*

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \to \pi'_1(X, a) \to \pi_1(S, a) \to 1$$

is exact.

The hypotheses of XIII.4.1 are satisfied. Indeed, it follows from XIII.1.12 3) that, for every scheme S' étale over S and every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, (E, f') is cohomologically proper relative to S' in dimension $\leqslant 0$.

PROPOSITION XIII.4.3. *Let S be a connected scheme, let $\mathbb{L}$ be a set of prime numbers, let $f\colon X \to S$ be a 0-acyclic morphism, locally 1-aspherical for $\mathbb{L}$* (SGA 4 XV 1.11)*, and let $g\colon S \to X$ be a section of f. Let $\bar{s}$ be a geometric point of S, and let a be a geometric point of the fiber $X_{\bar{s}}$. Suppose that, for every constant sheaf of $\mathbb{L}$-groups F, (F, f) is cohomologically proper in dimension $\leqslant 1$, that the direct image under f of the stack of torsors under F is a 1-constructible stack* (XIII.0)*, and that, for every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, f'_*E is constructible. Then the sequence of group homomorphisms*

$$1 \to \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi_1'(X, a) \xrightarrow{v} \pi_1(S, a) \to 1 \tag{XIII.4.-1.1}$$

is exact.

In view of XIII.4.2, it suffices to prove the injectivity of the morphism u, i.e. to prove that, for every principal covering $\overline{Z}$ of $X_{\bar{s}}$ whose group is an $\mathbb{L}$-group C, there exist 418
an étale covering Z of X and a morphism from a connected component of $Z|X_{\bar{s}}$ to $\overline{Z}$ (V V.6.8). Thus let $\overline{Z}$ be a principal covering of $X_{\bar{s}}$ whose group is an $\mathbb{L}$-group C, and let $\bar{z}$ be its class in $\mathrm{H}^1(X_{\bar{s}}, C_{X_{\bar{s}}})$. By XIII.1.4 d), there is a canonical isomorphism

$$\left(\mathrm{R}^1 f_* C_X\right)_{\bar{s}} \xrightarrow{\sim} \mathrm{H}^1(X_{\bar{s}}, C_{X_{\bar{s}}}),$$

and by XIII.1.15, $\mathrm{R}^1 f_* C_X$ is a locally constant constructible sheaf. Thus one can find an étale covering S' of S such that $\mathrm{R}^1 f_* C_X | S'$ is constant. If $\bar{s} \to S'$ is a geometric point above the geometric point $\bar{s} \to S$, there is an element z of $\mathrm{H}^0(S', \mathrm{R}^1 f_* C_X)$ whose image in $\mathrm{H}^1(X_{\bar{s}}, C_{X_{\bar{s}}})$ is $\bar{z}$. By the lemma XIII.4.4 below, one can find an étale covering S'_1 of S' and a torsor P on X'_1 with group C whose image in $\mathrm{H}^0(S'_1, \mathrm{R}^1 f_* C)$ is equal to the restriction of z. The torsor P is represented by an étale covering Z of $X'_1 = X \times_S S'_1$ such that $Z \times_{X'_1} X_{\bar{s}}$ is isomorphic to $\overline{Z}$. If Z is regarded as an étale covering of X, there is then a morphism from $Z \times_X X_{\bar{s}}$ to $\overline{Z}$, which completes the proof.

LEMMA XIII.4.4. *Let $f\colon X \to S$ be a 0-acyclic and locally 0-acyclic morphism, and let g be a section of f. Let C be a finite constant group such that (C_X, f) is cohomologically proper in dimension $\leqslant 0$ and the direct image under f of the stack of torsors under C_X is constructible. Then, for every section z of $\mathrm{H}^0(S, \mathrm{R}^1 f_* C_X)$, one can find an étale covering S_1 of S and, if $X_1 = X \times_S S_1$, an element of $\mathrm{H}^1(X_1, C_{X_1})$ whose image under the canonical morphism*

$$\mathrm{H}^1(X_1, C_{X_1}) \to \mathrm{H}^0(S_1, \mathrm{R}^1 f_* C_{X_1})$$

is equal to the restriction of z to $\mathrm{H}^0(S_1, \mathrm{R}^1 f_* C_{X_1})$.

For every scheme S' étale over S, put $X' = X \times_S S'$, and denote by g' (respectively F', etc.) the inverse image of g (respectively F, etc.) under the morphism $S' \to S$. The 419
presheaf G on S defined by

$$G(S') = \left\{ \begin{array}{l} \text{isomorphism classes of torsors } P \text{ on } X' \text{ with group } C_{X'}, \text{ endowed with} \\ \text{an isomorphism } g'^* P \xrightarrow{\sim} C_{S'} \end{array} \right\}$$

is then a sheaf. Indeed, this follows by descent from the fact that an automorphism of a torsor P on X' is completely determined by its restriction to $g'(S')$. Moreover, there is a surjective morphism

$$G \to \mathrm{R}^1 f_* F.$$

Let z be an element of $\mathrm{H}^0(S, \mathrm{R}^1 f_* C_X)$, and let H be the subsheaf of G that is the inverse image of z. It suffices to show that H is a locally constant constructible sheaf. Since

this property is local on S, we may suppose that z comes from an element of $\mathrm{H}^1(X, C_X)$ represented by a torsor P such that g^*P is isomorphic to C_S. To give an isomorphism $i\colon g^*P \xrightarrow{\sim} C_S$ amounts to giving a global section of $\mathbf{Aut}_{C_S}(g^*P)$, and two isomorphisms i and i' define the same element of $G(X)$ if and only if ii'^{-1} is the image of an element of $\mathrm{Aut}_{C_X}(P)$. Consider the canonical injection

$$f_*\,\mathbf{Aut}_{C_X}(P) \to \mathbf{Aut}_{C_S}(g^*P) \simeq C,$$

so H is identified with the quotient of $\mathbf{Aut}_{C_S}(g^*P)$ by $f_*\,\mathbf{Aut}_{C_X}(P)$. By XIII.1.15, $f_*\,\mathbf{Aut}_{C_X}(P)$ is locally constant; hence the same is true of H, which completes the proof.

Examples XIII.4.5. *Note that, if S is a connected scheme, the hypotheses of XIII.4.1 are satisfied when the morphism f is proper, flat, and of finite presentation, with separable, connected geometric fibers, and $\mathbb{L}$ is arbitrary (cf. X X.1.3). The hypotheses of XIII.4.3 are satisfied if, in addition, f is smooth and has a section, with $\mathbb{L}$ denoting the set of prime numbers distinct from the residual characteristics of S (SGA 4 XV 2.1 and XVI 5.2).*

The hypotheses of XIII.4.1 are also satisfied if S is connected, and if there is a scheme Z that is proper, flat, and of finite presentation over S, with separable, connected 420
geometric fibers, such that X is the complement in Z of a normal crossings divisor relative to S, with $\mathbb{L}$ the set of prime numbers distinct from the residual characteristics of S (XIII.2.9). *The hypotheses of XIII.4.3 are satisfied if, in addition, f is smooth and has a section.*

. We retain the notation and hypotheses of XIII.4.3. If $\bar{s}$ is a geometric point of S and $a = g(\bar{s})$, the section g defines a morphism

$$w\colon \pi_1(S, a) \to \pi_1'(X, a),$$

such that $\pi_1'(X, a)$ is identified with the semidirect product of $\pi_1(S, a)$ by $\pi_1(X_{\bar{s}}, a)$. The profinite group $\pi_1(S, a)$ therefore acts on $\pi_1(X_{\bar{s}}, a)$. Since $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is a strict inverse limit of groups invariant under the action of $\pi_1(S, a)$, specifying $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ together with this action is equivalent to specifying a strict inverse system of finite étale group schemes over S, which we denote by

$$\pi_1^{\mathbb{L}}(X/S, g, \bar{s}) \quad \text{or simply} \quad \pi_1^{\mathbb{L}}(X/S, g).$$

The following properties then hold:

. For every finite étale group scheme G over S whose fibers are $\mathbb{L}$-groups, the set E of classes of torsors P under the pullback G_X of G to X, equipped with an isomorphism $g^*P \xrightarrow{\sim} G$, is canonically isomorphic to the set

$$\mathrm{Hom}_S(\pi_1^{\mathbb{L}}(X/S, g, \bar{s}), G) \quad \text{modulo inner automorphisms of } G.$$

. For every finite étale group scheme G over S whose fibers are $\mathbb{L}$-groups, the sheaf $\mathrm{R}^1 f_* G_X$ is canonically isomorphic to the sheaf associated with the presheaf

$$S' \longmapsto \mathrm{Hom}_{S'}(\pi_1^{\mathbb{L}}(X/S, g, \bar{s}), G) \text{ modulo inner automorphisms of } G.$$

(S' denotes a scheme étale over S).

. Let S' be a connected S-scheme , let $\bar{s}$ be a geometric point of S', and let X', g' be the 421
respective pullbacks of X, g to S'. Then $\pi_1^{\mathbb{L}}(X'/S', g', \bar{s})$ is canonically isomorphic to the pullback of $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})$ to S'. For every geometric point ξ of S, the fiber $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})_\xi$ is isomorphic to $\pi_1^{\mathbb{L}}(X_\xi)$.

Indeed, specifying G is equivalent to specifying an abstract $\mathbb{L}$-group $\mathbf{G}$ on which $\pi_1(S, a)$ acts, thereby inducing an action of $\pi_1'(X, a)$ on $\mathbf{G}$. The isomorphism defined in XIII.4 is then obtained, by restriction to the subset E, from the canonical morphism

$$\mathrm{H}^1(\pi_1'(X, a), \mathbf{G}) \to \mathrm{H}^1(\pi_1^{\mathbb{L}}(X_{\bar{s}}, a), \mathbf{G}) = \mathrm{Hom}(\pi_1^{\mathbb{L}}(X_{\bar{s}}, a), \mathbf{G})/\text{inner aut. } G,$$

with the set E mapping bijectively onto the subset of morphisms from $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ to $\mathbf{G}$ that are compatible with the action of $\pi_1(S, a)$. Assertion XIII.4 then follows from the definition of $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})$, taking into account the exact homotopy sequence (XIII.4.-1.1), and XIII.4 follows from XIII.4 and XIII.4.

Proposition XIII.4.6 (Künneth formula). *Let k be a separably closed field of characteristic $p \geq 0$, let X and Y be two connected k-schemes, let a be a geometric point of X, let b be a geometric point of Y, and let c be a geometric point of $X \times_k Y$ lying over a and b. Suppose that one of the following two conditions holds:*

a) *X is of finite type over k, and schemes of finite type over an algebraic closure $\bar{k}$ of k of dimension $\leq \dim X$ are strongly desingularizable.* (SGA 5 I 3.1.5).
b) *X is quasi-compact and quasi-separated, and every scheme of finite type over $\bar{k}$ is strongly desingularizable.*

Then, if p' is the set of prime numbers distinct from p, the morphism

$$\pi_1^{p'}(X \times_k Y, c) \to \pi_1^{p'}(X, a) \times \pi_1^{p'}(Y, b), \tag{XIII.4.-1.0}$$

induced by the homomorphisms on fundamental groups associated with the projections 422

$$X \times_k Y \to X \quad \text{and} \quad X \times_k Y \to Y,$$

is an isomorphism.

We may assume that k is algebraically closed and that X is reduced (SGA 4 VIII 1.1). Let $Z = X \times_k Y$, and let $g\colon X \to k$ and $f\colon Z \to Y$ be the canonical morphisms. The morphism g, and hence also f, is universally locally 1-aspherical for p' by XIII.3.8. Since X is connected, f is 0-acyclic (SGA 4 XV 1.16). Moreover, it follows from XIII.3.5 that, for every finite p'-group C, (C_X, g) is cohomologically proper relative to k in dimension ≤ 1. It follows that (C_Z, f) is cohomologically proper relative to Y in dimension ≤ 1 (XIII.1.4 c)) and that f_*C_Z and $\mathrm{R}^1 f_* C_Z$ are constant sheaves. Consequently, g satisfies all the hypotheses of XIII.4.2. We therefore have the exact sequence

$$\pi_1^{p'}(X_b, c) \to \pi_1^{p'}(Z, c) \to \pi_1^{p'}(Y, b) \to 1.$$

Moreover, the composite morphism

$$\pi_1^{p'}(X_b, c) \to \pi_1^{p'}(Z, c) \to \pi_1^{p'}(Y, b)$$

is an isomorphism, and we therefore have the exact sequence

$$1 \to \pi_1^{p'}(X, a) \to \pi_1^{p'}(Z, c) \to \pi_1^{p'}(Y, b) \to 1.$$

On the other hand, the morphism (XIII.4.-1.0) defines a morphism from this exact sequence to the exact sequence

$$1 \to \pi_1^{p'}(X, a) \to \pi_1^{p'}(X, a) \times \pi_1^{p'}(Y, b) \times \pi_1^{p'}(Y, b) \to 1,$$

and it follows that the morphism (XIII.4.-1.0) is an isomorphism.

. Let

$$\begin{array}{ccccc} X & \longleftarrow & X_U & \longleftarrow & X_{\bar{s}} \\ {\scriptstyle f}\downarrow & & {\scriptstyle f_U}\downarrow & & \downarrow \\ S & \longleftarrow & U & \longleftarrow & \bar{s} \end{array}$$

be a diagram whose squares are cartesian, where S is an arcwise connected scheme (SGA 4 423 IX 2.12), U is a connected open subset of S, and $\bar{s}$ is a geometric point of U. Let a be a geometric point of $X_{\bar{s}}$, and let $\mathbb{L}$ be a set of prime numbers. Let g be a section of f, and suppose that the following conditions are satisfied:

a) The morphism f is 0-acyclic and locally 0-acyclic, and, for every étale covering S' of S and every étale covering E of $X \times_S S'$ which is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, $(E, f_{(S')})$ is cohomologically proper relative to S' in dimension ≤ 0.

b) The morphism f_U is locally 1-aspherical for $\mathbb{L}$, and, for every finite constant sheaf of $\mathbb{L}$-groups F on X_U, (F, f_U) is cohomologically proper in dimension ≤ 1, and the fibers of $\mathrm{R}^1 f_{U*}F$ are finite.

We then deduce from XIII.4.1 and XIII.4.3 the following commutative diagram, whose rows are exact:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) & \longrightarrow & \pi_1'(X_U, a) & \longrightarrow & \pi_1(U, a) & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ & & \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) & \longrightarrow & \pi_1'(X, a) & \longrightarrow & \pi_1(S, a) & \longrightarrow & 1 \end{array} \tag{XIII.4.-1.0}$$

Thanks to the section g, we have morphisms

$$\pi_1(U, a) \to \pi_1'(X_U, a), \quad \pi_1(S, a) \to \pi_1'(X, a)$$

from these we deduce a morphism from the amalgamated sum of $\pi_1(S, a)$ and $\pi_1'(X_U, a)$ over $\pi_1(U, a)$ into $\pi_1'(X, a)$:

$$\varphi\colon \pi = \pi_1(S, a) \coprod_{\pi_1(U,a)} \pi_1'(X_U, a) \to \pi_1'(X, a). \tag{XIII.4.-1.1}$$

Suppose that the following condition is satisfied:

c) If $T = S - U$, then $\operatorname{prof} \text{ét}_T(S) \geq 2$ (SGA 2 XIV 1.1).

Then the functor which assigns to an étale covering of S its restriction to U is fully faithful (SGA 2 XVI 1.4). It follows that the morphism $\pi_1(U, a) \to \pi_1(S, a)$ is surjective (V V.6.9), and we deduce from the diagram (XIII.4.-1.0) that the same is true of the 424
morphism $\pi_1'(X_U, a) \to \pi_1'(X, a)$; a fortiori, φ is an epimorphism. Let

$$K = \operatorname{Ker}(\pi_1(U, a) \to \pi_1(S, a)). \tag{XIII.4.-1.2}$$

The group π of (XIII.4.-1.1) is identified with the quotient of $\pi_1'(X_U, a)$ by the closed normal subgroup generated by the image L of K in $\pi_1'(X_U, a)$. Regard $\pi_1'(X_U, a)$ as the semidirect product of $\pi_1(U, a)$ by $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$. The group K then acts by inner automorphisms on $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$, and the quotient $\pi = \pi_1'(X_U, a)/L$ is identified with the semidirect product of $\pi_1(U, a)/K = \pi_1(S, a)$ by the group $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K$ of coinvariants of $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ under K. Finally, we have an epimorphism

$$\varphi\colon \pi = \pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K \cdot \pi_1(S, a) \to \pi_1'(X, a). \tag{XIII.4.-1.3}$$

The following proposition gives conditions under which the morphism φ is an isomorphism.

Proposition XIII.4.7. *The notation is that of XIII.4. In addition to conditions* a), b), c)*, suppose that the following conditions are satisfied:*

d) *For every point t of $T = S - U$, the morphism f is locally 1-aspherical for $\mathbb{L}$ at $g(t)$.*

e) *For every point $t \in T$, every irreducible component of the fiber X_t contains $g(t)$ and, for every point x of $X_t - \{g(t)\}$ which is not maximal, one has*

$$\operatorname{prof} \operatorname{hop}_x(X) \geq 3 \quad \text{(SGA 2 XIV 1.2)}$$

and the ring $\mathcal{O}_{X,x}$ is noetherian.

Then the morphism (XIII.4.-1.3) *is an isomorphism.*

As stated above, the group π is identified with the quotient of $\pi_1'(X_U, a)$ by the closed normal subgroup L generated by the image of K (XIII.4.-1.2) in $\pi_1'(X_U, a)$. This amounts to saying that π classifies the principal coverings Z of X_U such that $g_U^{-1}(Z)$ extends to 425
an étale covering of S, and which induce on $X_{\bar{s}}$ a covering obtained by extension of the structure group from a principal covering whose group is an $\mathbb{L}$-group. To prove that φ is an isomorphism, it is enough to show that such a covering Z extends to all of X.

Let us first show that Z extends to an étale covering of an open subset containing X_U and $g(S)$. Let W be a scheme étale over X whose image contains X_U and $g(S)$, and put $W_U = W \times_S U$. Since the morphism $W \to S$ is 0-acyclic and $\operatorname{prof.ét}_T S \geq 2$, we have

$$\operatorname{prof.ét}_{(W-W_U)} W \geq 2$$

(SGA 2 XIV 1.13); consequently, if $Z|W_U$ extends to an étale covering of W, this extension is unique up to unique isomorphism. It follows that the problem of extending Z to a neighborhood of $g(S) \cup X_U$ is local for the étale topology near the points of $g(T)$. If t is a point of T, put $x = g(t)$, and denote by $\bar{X}$ (respectively $\bar{S}$) the strict localization of X at $\bar{x}$ (respectively of S at $\bar{t}$), put $\bar{U} = U \times_S \bar{S}$, $\bar{X}_U = \bar{X} \times_X X_U$, and let $\bar{g}\colon \bar{S} \to \bar{X}$ be the morphism induced by g. It is enough to show that, for every point t of T, the inverse image $\bar{Z}$ of Z on $\bar{X}_U$ extends to $\bar{X}$ or, equivalently, is trivial. But, by definition of Z, the inverse image of $\bar{Z}$ on $\bar{U}$ is trivial. To prove that $\bar{Z}$ is trivial, it is enough to show that it is of the form $\bar{f}_U^* E$, where E is a principal covering of $\bar{U}$; indeed, we will then have $E \simeq \bar{g}_U^* \bar{f}_U^* E \simeq \bar{g}_U^* \bar{Z}$, whence the result, since $\bar{g}_U^* \bar{Z}$ is trivial. Since the morphism $\bar{f}_U$ is 0-acyclic and locally 0-acyclic, to prove that $\bar{Z}$ comes from $\bar{U}$, it is enough to show that, for every algebraic geometric point over a point of $\bar{U}$, which we may suppose to be the point $\bar{s}$, $\bar{Z}|X_{\bar{s}}$ is trivial (SGA 4 XV 1.15). But $\bar{Z}|X_{\bar{s}}$, being obtained by extension of the structure group from a principal covering whose group is an $\mathbb{L}$-group, is trivial because the morphism $\bar{f}$ is 1-aspherical for $\mathbb{L}$.

We have therefore shown that there exists an open neighborhood V of $g(S) \cup X_U$ such 426
that Z extends to an étale covering Z_V of V. Let us show that Z_V extends to all of X. It is enough to see that, for every point x of $X - V$, one has

$$\operatorname{prof} \operatorname{hop}_x X \geq 3.$$

But this follows from hypothesis e) and from the fact that a point x of $X - V$ cannot be maximal in its fiber X_t: since every irreducible component of X_t contains $g(t)$, every maximal point of X_t belongs to V.

Corollary XIII.4.8. *The hypotheses are those of XIII.4.7, but suppose in addition that $\pi_1(S, a) = 1$. Then there is an isomorphism*

$$(*) \qquad \pi_1^{\mathbb{L}}(X, a) \xrightarrow{\sim} \pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K.$$

In particular, if $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is topologically finitely presented and if K acts on $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ through a finitely generated group, then $\pi_1^{\mathbb{L}}(X, a)$ is topologically finitely presented.

The isomorphism $(*)$ was proved in XIII.4.7. Suppose that $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is the quotient of the free pro-$\mathbb{L}$-group on n generators $L(x_1, \ldots, x_n)$ by the closed normal subgroup generated by the elements $y_1, \ldots, y_p$ of $L(x_1, \ldots, x_n)$, and suppose that K acts through a group generated by elements $k_1, \ldots, k_q$. If, for every $i \in [1, n]$, $j \in [1, q]$, z_{ij} denotes an element of $L(x_1, \ldots, x_n)$ lifting the element $(k_j \cdot x_i) x_i^{-1}$, then $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K$ is the quotient of $L(x_1, \ldots, x_n)$ by the closed normal subgroup generated by the elements $(y_i)_{i \in [1,p]}$, $(z_{ij})_{i \in [1,n], j \in [1,q]}$.

Remarks XIII.4.9. a) Conditions a) through e) of XIII.4 are satisfied when S is a connected normal scheme, U is a dense retrocompact open subset of S, and f is a proper morphism of finite presentation, with geometrically connected and irreducible fibers at every point t of T, f being moreover separable, smooth at the points of $X_U \cup g(T)$, $\mathbb{L}$ 427

being the set of prime numbers distinct from the residue characteristics of S, and X being regular at every point of X_t. Indeed, condition a) follows from SGA 4 XV 4.1 and 1.4; conditions b) and d) follow from XIII.1.3 and SGA 4 XV 2.1 and XVI 5.2. Finally, e) follows from SGA XIV 1.11.

b) Corollary XIII.4 applies to compute the fundamental group $\pi_1^{p'}(X)$ of a proper smooth surface X over a separably closed field k of characteristic p (p' denoting the set of prime numbers distinct from p). The method was communicated to us by J.P. Murre; it consists in reducing, by blowing up X, to the case where there is a fibration $X \to \mathbb{P}^1_k$ and an open subset U of $\mathbb{P}^1_k$ satisfying the hypotheses of XIII.4 (see SGA 7 for more details). The same method may be used more generally (loc. cit.) to prove that, if X is a connected k-scheme of finite type and if the schemes of finite type of dimension $\leq \dim X$ over an algebraic closure of k are strongly desingularizable (SGA 5 I 3.1.5), then $\pi_1^{p'}(X)$ is topologically finitely presented.

XIII.5. Appendix I: Variations on Abhyankar's lemma

This appendix contains several variants of Abhyankar's lemma.

Proposition XIII.5.1. *Let $X = \mathrm{Spec}(A)$ be a regular local scheme, and let $D = \sum_{1\leq i\leq r} \mathrm{div}\, f_i$ be a normal crossings divisor, where the f_i are elements of the maximal ideal of A belonging to a regular system of parameters. Let n_i, $1 \leq i \leq r$, be integers ≥ 0, and put*

$$X' = X[T_1, \ldots, T_r]/(T_1^{n_1} - f_1, \ldots, T_r^{n_r} - f_r),$$

$U' = U \times_X X'$. Then X' is regular, and U' is the complement in X' of the normal crossings divisor $\sum_{1\leq i\leq r} \mathrm{div}\, T_i$. If the integers n_i are prime to the residue characteristic p of X, 428
then U' is a connected étale covering of U, tamely ramified relative to D (XIII.2.3 c)).

Indeed, X' is the spectrum of a local ring A' whose maximal ideal is generated by $T_1, \ldots, T_r$. (We may suppose that $r = \dim(A)$.) Since A' is finite and flat over A, and hence has dimension r, A' is regular (EGA 0_{IV} 17.1.1), and the T_i form a regular system of parameters of A'. Suppose that the n_i are prime to p. Since all the f_i are invertible on U, the fact that U' is étale over U follows from I I.7.4. Moreover, U' is tamely ramified relative to D. Indeed, let x_i be the generic point of $V(f_i)$, let $\bar{R}$ be the strict localization of $R = \mathcal{O}_{X,x_i}$, and let $\bar{K}$ be the fraction field of $\bar{R}$. Then the $\bar{K}$-algebra representing $U'|\bar{K}$ is obtained from the field $\bar{K}[T_i]/(T_i^{n_i} - f_i)$ by an unramified extension; it is therefore tamely ramified over $\bar{R}$.

Proposition XIII.5.2 (Absolute Abhyankar lemma). *Let X be a regular local scheme, let*

$$D = \sum_{1\leq i\leq r} \mathrm{div}\, f_i$$

be a normal crossings divisor as in XIII.5.1, put $Y = \mathrm{Supp}\, D$, $U = X - Y$, and let V be an étale covering of U tamely ramified relative to D. If x_i is the generic point of the closed subset $V(f_i)$, then $\mathcal{O}_{X,x_i}$ is a discrete valuation ring with fraction field K_i, and $V|K_i = \mathrm{Spec}(() \prod_{j\in J_i} L_j)$, where the L_j are finite separable extensions of K_i. Let n_j be the order of the inertia group of a Galois extension generated by L_j, and let n_i be the least common multiple of the n_j as j ranges over J_i. If

$$X' = X[T_1, \ldots, T_r]/(T_1^{n_1} - f_1, \ldots, T_r^{n_r} - f_r),$$

$U' = U_{(X')}$, $V' = V_{(X')}$, etc., then the étale covering V' of U' extends, uniquely up to unique isomorphism, to an étale covering of X', and the n_i are prime to the residue characteristic p of X.

Uniqueness follows from the fact that X' is normal (XIII.5.1). Indeed, an étale 429
covering of X' extending V' is isomorphic to the normalization of X' in the fiber of V' at the generic point of X' (I.10.2). If $\bar{x}'$ is a geometric point of Y', denote by $\bar{X}'$ the strict localization of X' at $\bar{x}'$, and put $\bar{V}' = V'_{(\bar{X}')}$, etc. By descent, in view of uniqueness, it is enough to show that for every geometric point $\bar{x}'$ of Y', the étale covering $\bar{V}'$ of $\bar{U}'$ extends to $\bar{X}'$. Since an étale covering of an open subset of the regular scheme X' containing all points x' such that $\dim \mathcal{O}_{X',x'} \leq 1$ extends to all of X' (SGA 2 XIV 1.11), we may even restrict ourselves to the points $\bar{x}'$ lying over a maximal point of Y'. At such a point $\bar{x}'$, the fact that $\bar{V}'$ extends to an étale covering of $\bar{X}'$ follows from X.3.6.

Let us show that the n_i are prime to p. Otherwise, for example, we would have $p|n_1$. After replacing X by

$$X[T_1, \ldots, T_r]/(T_1^{n_1/p} - f_1, T_2^{n_2} - f_2, \ldots, T_r^{n_r} - f_r),$$

we reduce to the case $X' = X[T_1]/(T_1^p - f_1)$. It is enough to show that V extends to an étale covering of X, since we would then have $n_1 = 1$, contrary to the hypothesis. For this, we may suppose that X is strictly local. Let Z be the closed subscheme of X defined by $p = 0$, and let $Z_1 = Z \cap X_{f_1}$; Z_1 is a nonempty open subset of Z. By what precedes, the étale covering V' of U' extends to an étale covering W' of X'. Let W''_1 and W''_2 be the inverse images of W' under the two projections $X'' = X' \times_X X' \rightrightarrows X'$. We show that the descent isomorphism $u\colon W''_1|U'' \to W''_2|U''$ extends to an X''-morphism $W''_1 \to W''_2$, which will necessarily be descent data on W' relative to $X' \to X$. Let Z'' (respectively Z''_1) be the inverse image of Z (respectively Z_1) in X''. Since the morphism $Z'' \to Z$ is radicial, there is an isomorphism $v\colon W''_1|Z'' \to W''_2|Z''$ extending the isomorphism $u|Z''_1$. But since X is Henselian, there is a bijection

$$\mathrm{Hom}_{X''}(W''_1, W''_2) \simeq \mathrm{Hom}_{Z''}(W''_1|Z'', W''_2|Z''),$$

and hence a morphism $w\colon W''_1 \to W''_2$ lifting v. The open-and-closed subscheme of X'' 430
over which u and w coincide contains Z''_1, and therefore equals X''. Thus V extends to X.

. Retain the hypotheses and notation of XIII.5.2, and suppose in addition that X is strictly local. It follows from loc. cit. that every connected étale covering of U tamely ramified relative to D is a quotient of a tamely ramified covering of the form

$$U' = U[T_1, \ldots, T_r]/(T_1^{n_1} - f_1, \ldots, T_r^{n_r} - f_r),$$

where the n_i are integers prime to p. Let $\boldsymbol{\mu}_n$ be the group of n-th roots of unity of U. The group of U-automorphisms of U' is precisely $\boldsymbol{\mu}_{n_1} \times \cdots \times \boldsymbol{\mu}_{n_r}$, an n_i-th root of unity ξ_i acting on U' by sending T_i to $\xi_i T_i$. We therefore have the following statement:

Corollary XIII.5.3. *Let X be a regular strictly local scheme of residue characteristic $p \geq 0$, let $D = \sum_{1 \leq i \leq n} \mathrm{div}\, f_i$ be a normal crossings divisor on X, and put $U = X - \mathrm{Supp}\, D$. Let*

$$\widetilde{U} = \varprojlim_{(n_i)} U[T_1, \ldots, T_r]/(T_1^{n_1} - f_1, \ldots, T_r^{n_r} - f_r),$$

where the projective limit is taken over the filtered ordered set (under divisibility) of families of integers $n_i > 0$ prime to p. Then $\widetilde{U}$ is a universal tamely ramified covering of U. Consequently, the tamely ramified fundamental group of U is

$$\pi_1^{\mathrm{t}}(U) \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell[1]^r \qquad \textit{(canonical isomorphism)}$$

where $\mathbb{Z}_\ell[1] = \varprojlim_{n>0} \boldsymbol{\mu}_{\ell^n}$. The group $\pi_1^{\mathrm{t}}(U)$ is noncanonically isomorphic to $\prod_{\ell \neq p} \mathbb{Z}_\ell^r$.

Proposition XIII.5.4. *Let $f\colon X \to S$ be a morphism of schemes, and let $D = \sum_{1 \leq i \leq r} \mathrm{div}\, f_i$ be a divisor on X with normal crossings relative to S* (XIII.2)*, where, for* 431
every point x of $Y = \mathrm{Supp}\, D$, if $I(x) \subset [1, r]$ is the set of i such that $f_i(x) = 0$, the subscheme $V((f_i)_{i \in I(x)})$ is smooth over S, of codimension $\mathrm{card}\, I(x)$ in X. Let $U = X - Y$.

Let x be a point of Y, put $X_1 = \mathrm{Spec}()O_{X,x}$, $U_1 = U \times_X X_1$, let n_i, $i \in I(x)$, be integers, and put

$$X' = X[T_i]_{i\in I(x)}/(T_i^{n_i} - f_i).$$

Then, if x' is the point of X' over x, X' is smooth over S at x'. If the integers n_i are prime to the characteristic p of $\kappa(x)$, then $U_1' = U_1 \times_X X'$ is a connected étale covering of U_1, tamely ramified on X_1 relative to S_1 (XIII.2.1).

If $s = f(x)$, the geometric fiber $X'_{\overline{s}}$ is regular at the point x' (XIII.5.1). Since X' is flat over S in a neighborhood of Y, this proves that X' is smooth over S at x' (EGA IV 12.1.6). If the integers n_i are prime to p, U_1' is an étale covering of U_1 (I.7.4); it is tamely ramified on X_1 relative to S because this is so on the geometric fibers at every point of S (XIII.5.1). Finally, the connectedness of U_1' follows from SGA 4 XVI 3.2.

Proposition XIII.5.5 (Relative Abhyankar lemma). *Let X be an S-scheme and let D be a normal crossings divisor relative to S, as in* (XIII.5.4). *Let $Y = X - \mathrm{Supp}\, D$, $U = X - Y$, let x be a point of Y, let X_1 be the strict localization of X at a geometric point above x, let $U_1 = U \times_X X_1$, and let V_1 be an étale covering of U_1. Suppose that, for every maximal point s of S, $V_{1\overline{s}}$ is tamely ramified on $X_{1\overline{s}}$ relative to $\overline{s}$. Then there exist integers n_i prime to the characteristic p of $\kappa(x)$, with $i \in I(x)$, such that, upon putting*

$$X_1' = X_1[T_i]_{i\in I(x)}/(T_i^{n_i} - f_i),$$

$U_1' = U_1 \times_{X_1} X_1'$, etc., the étale covering V_1' of U_1' extends uniquely, up to unique isomorphism, to an étale covering of X_1'. In particular, V_1 is tamely ramified on X_1 432
relative to S.

We may assume that S is local noetherian with closed point $f(x)$. For every maximal point s of S and every $i \in I(x)$, let x_i be the generic point of the closed subset $V(f_i)$ of the fiber $X_{1\overline{s}}$. The local ring $(\mathcal{O}_{X_1,x_i})_{\mathrm{red}}$ is a discrete valuation ring with fraction field K_i, and we have $V_1|K_i = \mathrm{Spec}(()\prod_{j\in I(x_i)} L_j)$, where L_j is a finite separable extension of K; let n_j denote the order of the inertia group of a Galois extension generated by L_j, and let n_i be the l.c.m. of the n_j as s ranges over the maximal points of S and $j \in I(x_i)$.

With the n_i so chosen, we shall show that V_1' extends uniquely to an étale covering of X_1'. Uniqueness follows from the fact that, since X' is smooth over S at the points of Y, we have $\mathrm{prof}\,\text{ét}_{Y_1'}(X_1') \geqslant 2$ (SGA 4 XVI 3.2 or SGA 2 XIV 1.19). Let x_1' be a point of Y_1', let $\overline{x}_1'$ be a geometric point above x_1', and denote by $\overline{X_1'}$ the strict localization of X_1' at $\overline{x}_1'$, $\overline{U_1'} = U'_{1(\overline{X_1'})}$, etc. By descent, in view of uniqueness, it suffices to show that $\overline{V}_1'$ extends to $\overline{X_1'}$. Moreover, it is enough to take for x_1' only the maximal points of Y_1'; indeed, we shall then have an extension of V_1' over an open subset W_1' of X_1' containing the maximal points of Y_1'. Now, if $Z_1' = X_1' - W_1'$, then $\mathrm{codim}(Z'_{1s}, X'_{1s}) \geqslant 2$ if s is a maximal point of S, and $\mathrm{codim}(Z'_{1s}, X'_{1s}) \geqslant 1$, $\mathrm{prof}\,\text{ét}_s(S) \geqslant 1$ if s is a point of S that is not maximal; the fact that V_1' extends to all of X_1' then follows from (SGA 2 XIV 1.20). But at a geometric point $\overline{x}_1'$ above a maximal point of Y_1', $X'_{1\mathrm{red}}$ is the spectrum of a discrete valuation ring, and the fact that $\overline{V_1'}$ extends to $\overline{X_1'}$ then follows from X X.3.6.

Let us show that the n_i are prime to p. Indeed, if this were not so, there would be an index $i_0 \in I(x)$ such that p divides n_{i_0}. After replacing X by $X_1[T_{i_0}, T_i]_{i\in I(x)}/(T_i^{n_{i_0}/p} - f_{i_0}, T_i^{n_i} - f_i)$, we are reduced to the case $X_1' = X_1[T]/T^p - f_{i_0}$. By what precedes, the étale covering V_1' of U_1' extends to an étale covering E_1' of X_1'. Let η be the closed point 433
of S; since the morphism $X'_{1\eta} \to X_{1\eta}$ is radicial, $V_{1\eta}$ extends to an étale covering $E_{1\eta}$ of $X_{1\eta}$. It then follows, as in XIII.5.2, that E_1' is endowed with descent data relative to the morphism $X_1' \to X_1$ extending the natural descent data on $E_1'|U_1'$. It follows that V_1 extends to X_1; but this implies $n_{i_0} = 1$, contrary to the hypothesis $n_{i_0} = p$.

COROLLARY XIII.5.6. *Let X be an S-scheme, and let $D = \sum_{1\leqslant i\leqslant r} \operatorname{div} f_i$ be a normal crossings divisor relative to S, as in XIII.5.4. Let $\overline{x}$ be a geometric point of X, let $\overline{X}$ be the strict localization of X at $\overline{x}$, let $\overline{Y} = Y_{(\overline{X})}$, $\overline{U} = \overline{X} - \overline{Y}$, and let*

$$\widetilde{U} = \varprojlim_{(n_i)} \overline{U}[T_i]_{i\in I(x)}/(T_i^{n_i} - f_i),$$

where the projective limit is taken over the filtered set of families of integers $n_i > 0$ prime to the characteristic p of $\kappa(x)$. Then $\widetilde{U}$ is a universal tamely ramified covering of $\overline{U}$ relative to S. Consequently, the tamely ramified fundamental group of $\overline{U}$ is

$$\pi_1^{\mathrm{t}}(\overline{U}) \simeq \prod_{\ell\neq p} \mathbb{Z}_\ell[1]^{I(x)} \qquad \textit{(canonical isomorphism).}$$

The group $\pi_1^{\mathrm{t}}(\overline{U})$ is noncanonically isomorphic to $\prod_{\ell\neq p} \mathbb{Z}_\ell^{I(x)}$.

REMARK XIII.5.7. *Let X be an S-scheme, let $D = \sum_{1\leqslant i\leqslant r} \operatorname{div} f_i$ be a normal crossings divisor relative to S, as in XIII.5.4, and let $U = X - \operatorname{Supp} D$. For every subset $I \subset [1, r]$, let*

$$X_I = \Big(\bigcap_{i\in I} V(f_i)\Big) \cap \Big(\bigcap_{i\in \complement I} X_{f_i}\Big).$$

Let p be a prime integer or zero, and let Z be a subset of X_I all of whose points have characteristic p. Let

$$\widetilde{U}_I = \varprojlim_{(n_i)} U[T_i]_{i\in I}/(T_i^{n_i} - f_i),$$

where the projective limit is taken over the filtered set of families of integers $n_i > 0$ prime 434
to p. Then, for every geometric point $\overline{x}$ of Z, the pullback of $\widetilde{U}_I$ to $\overline{U}$ identifies with the universal tamely ramified covering of $\overline{U}$.

COROLLARY XIII.5.8. *Use the notation of XIII.5.6. Let $\overline{S}$ be the strict localization of S at $\overline{x}$, and let*

$$\overline{g}\colon \overline{U} \to \overline{S} \quad \textit{and} \quad \tilde{g}\colon \widetilde{U} \to \widetilde{S}$$

be the canonical morphisms. Then the morphisms $\overline{g}$ and $\tilde{g}$ are 0-acyclic (SGA 4 XV 1.3). *Let G be a constructible sheaf of groups on $\overline{S}$, let $F = \overline{g}^*G$, and let P be a torsor under F. Then P is tamely ramified on $\overline{X}$ relative to $\overline{S}$ if and only if its pullback $\widetilde{P}$ to $\widetilde{U}$ is trivial.*

Indeed, for every scheme $\overline{X'} = \overline{X}[T_i]_{i\in I(x)}/(T_i^{n_i} - f_i)$, where the n_i are integers > 0 prime to p, the morphism $\overline{f}'\colon \overline{X'} \to \overline{S}$ is 0-acyclic. The geometric fibers of $\overline{f}'$ at the various points of $\overline{S}$ are therefore connected, and indeed irreducible. The same is consequently true of the geometric fibers of the morphisms $\overline{g}'\colon \overline{U'} \to \overline{S}$, which proves that the $\overline{g}'$, and hence also $\tilde{g}$, are 0-acyclic (SGA 4 XV 1.16).

It is clear that a torsor P on $\overline{U}$ under F whose pullback to $\widetilde{U}$ is trivial is tamely ramified on $\overline{X}$ relative to $\overline{S}$. Conversely, suppose that P is tamely ramified on $\overline{X}$ relative to $\overline{S}$; we shall show that its pullback to $\widetilde{U}$ is trivial.

It follows from (SGA 4 IX 2.14 (ii)) that there exist a finite morphism $n\colon S_1 \to \overline{S}$, a constant sheaf of groups C on S_1, and a monomorphism $G \to n_*C$. Consider the following commutative diagram formed by Cartesian squares:

$$\begin{array}{ccccc} \widetilde{U}_1 & \longrightarrow & U_1 & \xrightarrow{g_1} & S_1 \\ {\scriptstyle r}\downarrow & & {\scriptstyle q}\downarrow & & {\scriptstyle n}\downarrow \\ \widetilde{U} & \longrightarrow & \overline{U} & \xrightarrow{\overline{g}} & \overline{S}. \end{array}$$

Let C_1 (respectively$\widetilde{C}_1$) be the pullback of C to U_1 (respectivelyto $\widetilde{U}_1$). We have a 435

commutative diagram in which i and j are isomorphisms (SGA 4 VIII 5.8):

$$(*)\qquad \begin{array}{ccc} \mathrm{H}^1(\overline{U}, q_*C_1) & \xrightarrow{i} & \mathrm{H}^1(U_1, C_1) \\ \downarrow & & \downarrow \\ \mathrm{H}^1(\widetilde{U}, r_*\widetilde{C}_1) & \xrightarrow{j} & \mathrm{H}^1(\widetilde{U}_1, \widetilde{C}_1). \end{array}$$

Let Q be the torsor under q_*C_1 obtained from P by extension of structure group along $F \to q_*C_1$. By XIII.2, Q is tamely ramified on $\overline{X}$ relative to $\overline{S}$. The torsor Q corresponds, via i, to a torsor Q_1 under C_1, and it is clear that Q_1 is tamely ramified on $X_1 = \overline{X} \times_{\overline{S}} S_1$ relative to S_1. It therefore follows from XIII.5.6 that the pullback $\widetilde{Q}_1$ of Q_1 to $\widetilde{U}_1$ is trivial, and diagram $(*)$ then shows that the pullback $\widetilde{Q}$ of Q to $\widetilde{U}$ is trivial.

Consider the following commutative diagram, whose second row is exact (SGA 4 XII 3.1):

$$(**)\qquad \begin{array}{ccccc} \mathrm{H}^0(\overline{S}, n_*C/G) & \longrightarrow & \mathrm{H}^1(\overline{S}, G) = 1 & & \\ {\scriptstyle k}\downarrow & & \downarrow & & \\ \mathrm{H}^0(\widetilde{U}, r_*\widetilde{C}_1/\widetilde{F}) & \longrightarrow & \mathrm{H}^1(\widetilde{U}, \widetilde{F}) & \longrightarrow & \mathrm{H}^1(\widetilde{U}, r_*\widetilde{C}_1). \end{array}$$

Since the morphism $\widetilde{U} \to \overline{S}$ is 0-acyclic, k is an isomorphism. It then follows from $(**)$ that $\widetilde{P}$ is trivial.

XIII.6. Appendix II: finiteness theorem for direct images of stacks

PROPOSITION XIII.6.1. *Let S be a locally noetherian scheme and let $f\colon X \to S$ be a morphism. If S' is an S-scheme, we denote by X' (respectivelyf', etc.) the pullback of X (respectivelyf, etc.) by the morphism $S' \to S$. Suppose that, for every scheme S'* 436 *étale over S and every constructible sheaf of sets F on X', f'_*F is constructible, and that, for every constructible sheaf of groups F on X', $\mathrm{R}^1 f'_*F$ is constructible. Let Φ be a 1-constructible stack on X* (XIII.0). *Then $f_*\Phi$ is 1-constructible.*

For every scheme S' étale over S and every object x of $(f_*\Phi)_{S'}$, there is an isomorphism

$$\mathbf{Aut}_{S'}(x) \simeq f'_* \mathbf{Aut}_{X'}(x),$$

where, on the right-hand side, x is regarded as an object of $\Phi_{X'}$. The hypotheses therefore imply that $f_*\Phi$ is constructible. Let $S\Phi$ be the sheaf of maximal subgerbes of Φ [**1**, III 2.1.7]. Since $f_*(S\Phi)$ is constructible, we may apply (SGA 4 IX 2.7) to it, and the fact that $f_*\Phi$ is 1-constructible then follows from the sublemma below.

LEMMA XIII.6.2. *Let S be a locally noetherian scheme, let $f\colon X \to S$ be a morphism, and let Φ be a stack on X. Suppose we are given a sheaf on S, representable by an étale S-scheme of finite type T, a surjective morphism*

$$a\colon T \to f_*(S\Phi)$$

and an object p of the fiber Φ_{X_T} (where $X_T = X \times_S T$), defining in $f_*(S\Phi)(T) = S\Phi(X_T)$ an element equal to the image q under a of the identity section of $T(T)$. Let $f_T\colon X_T \to T$ be the canonical morphism, and suppose that the sheaf $\mathrm{R}^1 f_{T*}(\mathbf{Aut}_{X_T}(p))$ is constructible; then the same holds for $S(f_*\Phi)$.

The canonical morphism $f^*f_*\Phi \to \Phi$ gives a morphism

$$S(f^*f_*\Phi) \simeq f^*(S(f_*\Phi)) \to S\Phi,$$

and hence a canonical morphism

$$\varphi\colon S(f_*\Phi) \to f_*(S\Phi).$$

Let $F = S(f_*\Phi)$, and let G be the image of F under φ; by (SGA 4 IX, 2.9), G is a 437
constructible sheaf.

It is enough to show that, for every point s of S, there exists a nonempty open neighborhood U of s such that $F|U$ is locally constant constructible. Let $s \in S$, let $\overline{s}$ be a geometric point lying over s, and let $\overline{q}_1, \ldots, \overline{q}_n$ be the elements of $G_{\overline{s}}$. By the definition of T, there exist S-morphisms $h_i \colon \overline{s} \to S'$ such that $\overline{q}_i = h_i^*(q)$. Let S' be the fiber product over S of n schemes isomorphic to T, let $\overline{s} \to S'$ be the fiber product of the h_i, let $X' = X \times_S S'$, and let q_i (respectivelyp_i) be the pullback of q (respectively$p$) by the i-th projection of S' onto T. If Ψ_i is the maximal subgerbe of $\Phi|X'$ generated by p_i, the sheaf $F_i = \mathrm{R}^1 f'_*(\mathbf{Aut}_{X'}(p_i))$ is precisely the sheaf $S(f'_*\Psi_i)$ of maximal subgerbes of $f'_*\Psi_i$. In particular, the canonical injection $\Psi_i \to \Phi|X'$ gives a morphism

$$\alpha_i \colon F_i \to F|S'.$$

We shall show that α_i is a bijection from F_i onto the inverse image of q_i in $F|S'$. For every scheme S'' étale over S', every section y of $F_i(S'')$ has image $q_i|S''$ in $F(S'')$, because, locally for the étale topology on S'', y is defined by an object x of $\Phi_{X''}$ that is isomorphic to $p_i|X''$. Conversely, if $y \in F(S'')$ has image $q_i|S''$ in $F(S'')$, then, locally for the étale topology on S'', y is defined by an object x of $\Phi_{X''}$ that is isomorphic to p_i; consequently, x is an object of $\Psi_{iX''}$ and therefore $y \in F_i(S'')$.

The proof is completed using XIII.6.3 below. Indeed, one can find an open neighborhood U' of s such that $q_1|U', \ldots, q_n|U'$ are sections of $G(U')$ and generate this sheaf. Since the $F_i|U'$ and $G|U'$ are constructible, so is $F|U'$ by XIII.6.3; after replacing U by a smaller open subset if necessary, $F|U$ is locally constant, which completes the proof.

Lemma XIII.6.3. *Let S be a locally noetherian scheme, and let $F \to G$ be a surjective* 438
morphism of sheaves of groups on S. Let q_i be a finite family of sections of G on X that generate G, and, for each i, let F_i be the subsheaf of F that is the inverse image of q_i. Then, if G and the F_i are constructible, so is F.

To prove that F is constructible, it is enough to show that, for every point s of S, there exists an open neighborhood U of s such that $F|U$ is locally constant constructible. Let s be a point of S. Since the sheaves F_i and G are constructible, one can find an open neighborhood U of s such that $F_i|U$ and $G|U$ are locally constant. We now show that $F|U$ is locally constant. By (SGA 4 IX 2.13 (i)), it is enough to see that, if $\overline{s}$ is a geometric point lying over s, $\tilde{s}$ is a geometric point of U, and $\overline{s} \to \tilde{s}$ is a specialization morphism, the canonical morphism

$$F_{\tilde{s}} \to F_{\overline{s}}$$

is bijective.

Consider the commutative diagrams

$$\begin{array}{ccccc}
(F_i)_{\tilde{s}} & \xrightarrow{\tilde{a}_i} & F_{\tilde{s}} & \xrightarrow{\tilde{a}} & G_{\tilde{s}} \\
\downarrow \wr & & \downarrow b & & \downarrow \wr \\
(F_i)_{\overline{s}} & \xrightarrow{\overline{a}_i} & F_{\overline{s}} & \xrightarrow{\overline{a}} & G_{\overline{s}}
\end{array}$$

Let $\overline{q}_i$ (respectively$\tilde{q}_i$) be the pullback of q_i in $G_{\overline{s}}$ (respectively$G_{\tilde{s}}$); the morphisms $\overline{a}$ and $\tilde{a}$ are surjective, and the morphism $\overline{a}_i$ (respectively$\tilde{a}_i$) induces a bijection from $(F_i)_{\overline{s}}$ (respectively$(F_i)_{\tilde{s}}$) onto $\overline{a}^{-1}(q_i)$ (respectivelyonto $\tilde{a}^{-1}(q_i)$). It therefore follows from the diagram above that b is an isomorphism.

Corollary XIII.6.4. *Let S be a locally noetherian scheme and let $f \colon X \to S$ be* 439
a proper morphism. If Φ is a 1-constructible stack on X, then $f_*\Phi$ is a 1-constructible stack.

The proof of XIII.6.1 also proves the following result, in view of XIII.2.4 2).

COROLLARY XIII.6.5. *Let S be a locally noetherian scheme, let $f\colon X \to S$ be a morphism, let D be a divisor on X with normal crossings relative to S* (XIII.2)*, let $Y = \operatorname{Supp} D$, let $U = X - Y$, and let $i\colon U \to X$ be the canonical immersion. Let Φ be a stack on U given, locally for the étale topology on X and S, as the pullback of a stack Ψ that is* 1*-constructible on S. Then the stack $i^{\mathrm{t}}_{*}\Phi$ is* 1*-constructible.*

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Seminar on Algebraic Geometry at Bois Marie 1962

Local Cohomology of Coherent Sheaves and Local and Global Lefschetz Theorems (SGA 2)

Alexander Grothendieck
(written up by a group of participants)
With an additional exposé by Michèle Raynaud

July 22, 2026

Abstract

This volume is a newly updated edition of the book *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux (SGA 2)*, Advanced Studies in Pure Mathematics 2, North-Holland Publishing Company, Amsterdam, 1968, by A. Grothendieck *et al.*

The monograph gives necessary and sufficient conditions for the finiteness of the local cohomology sheaves of coherent sheaves. These results yield algebraization theorems which, together with purity results also proved in the text, lead to Lefschetz-type theorems for both the fundamental group and the Picard group.

Keywords. Algebraization, Lefschetz theorem, local duality, local cohomology, depth, homotopical depth, fundamental group, Picard group.

2020 Mathematics Subject Classification. 14-02, 14B10, 14B15, 14B20, 14C22, 14J70, 32S20.

Editor's Preface

The present text is a new updated edition of the book "Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux (SGA 2)", Advanced Studies in Pure Mathematics 2, North-Holland Publishing Company, Amsterdam, 1968, by A. Grothendieck *et al.* It is the second part of the SGA project initiated by B. Edixhoven, who prepared a new edition of SGA 1. This version is intended to reproduce the original text as closely as possible, while incorporating minor typographical corrections and numbered editorial footnotes explaining the current status of questions raised in the first edition. Further details have also been supplied for some proofs. To avoid confusion, original notes retain star-based numbering, whereas new editorial notes use Arabic numbers. The page numbers of the original edition are shown in the margins.

I thank the mathematicians who carried out most of the initial typesetting in LaTeX 2_ε: L. Bayle, N. Borne, O. Brinon, J. Buresi, M. Chardin, F. Ducrot, P. Graftiaux, F. Han, P. Karwasz, L. Koelblen, D. Madore, S. Morel, D. Naie, B. Osserman, J. Riou, and V. Sécherre, as well as C. Sabbah for adapting the text to the SMF format. I also thank J.-B. Bost, P. Colmez, O. Gabber, W. Fulton, S. Kleiman, F. Orgogozo, M. Raynaud, and J.-P. Serre for their comments and advice.

Yves Laszlo, editor

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Introduction

We present here, in revised and expanded form, a photo-offset reissue of the second Algebraic Geometry Seminar of the Institut des Hautes Études Scientifiques, held in 1962 and originally issued in mimeographed form. 1

The reader is referred to the Introduction to the first of these seminars (cited below as SGA 1) for the aims pursued by these seminars and their relation to the Éléments de Géométrie Algébrique.

The text of Exposés I through XI was written up as the seminar proceeded, from my oral lectures and handwritten notes, by a group of participants comprising I. Giorgiutti, J. Giraud, Mlle M. Jaffe (now Mme M. Hakim), and A. Laudal. These notes were originally regarded as provisional and intended for very limited circulation, pending their absorption into the EGA—an absorption that has now become problematic, to say the least, as it has for the other parts of the SGA. As the warning to the original edition said, the "confidential"character of the notes was meant to excuse certain "stylistic weaknesses", doubtless more apparent in the present seminar, SGA 2, than in the others. I have tried, as far as possible, to remedy them in the present reissue by a fairly thorough revision of the original text. In particular, I have harmonized the systems used to number statements in the different Exposés, introducing everywhere the same decimal system, already used in most of the original Exposés in the present SGA 2 as well as in all the other parts of the SGA. This led me, in particular, to revise completely the numbering[1] of the statements in Exposés III through VIII (and consequently the references to those Exposés)[2]. I have also tried to remove from the original text the principal typing and syntactical errors, which were numerous and troublesome. Moreover, Mme M. Hakim kindly undertook to rewrite Exposé IV in a less telegraphic style than the original. As in the other SGA reissues, I have also added a number of footnotes, either to give additional references or to indicate the status of questions on which progress has been made since the original text was written. Finally, the seminar has been enlarged by a new Exposé, namely Exposé XIV, written by Mme Michèle Raynaud in 1967, which takes up and completes suggestions contained in the "Comments on Exposé XIII"(XIII XIII.6), written in March 1963. This Exposé treats Lefschetz-type theorems from the viewpoint of étale cohomology, using the results on étale cohomology presented in SGA 4 and SGA 5 (to appear in the same Series in Pure Mathematics)[3]; it is therefore less "elementary"in nature than the other Exposés of the present volume, which use little more than the substance of Chapters I through III of the EGA. Here is an outline of the contents of the present volume. Exposé I contains the sorites of "cohomology with support in Y", $H^*_Y(X, F)$, where Y is a closed subset of a space X, cohomology that may be interpreted as the cohomology of X *modulo* the open subset $X - Y$, and that is the abutment of a very useful "*local-to-global spectral sequence*"I 2.6, involving 2

[1] *Editor's note.* The original numbering has been preserved as far as possible, with the adverb *bis* added when ambiguous duplicates occurred here and there.

[2] It goes without saying that every reference to SGA 2 appearing in those parts of the SGA published in the Series in Pure Mathematics refers to the present volume, not to the original edition of SGA 2!

[3] *Editor's note.* In fact, these seminars were published by Springer (numbers 269, 279, 305, and 589), but unfortunately are out of print.

the *sheaves* of cohomology "with support in Y", $\underline{H}^*_Y(F)$[4]. In many questions this formalism can play a "localization"role analogous to that played by tubular neighborhoods of Y in 3
differential geometry. Exposé II studies the preceding notions for quasi-coherent sheaves on preschemes, and Exposé III relates them to the classical notion of *depth* (III 3.3).

Exposés IV and V give forms of *local duality*, which may be compared with Serre's projective duality theorem (XII XII.1.1). We note that both kinds of duality theorem are substantially generalized in Hartshorne's seminar (cited in the footnote at the end of Exp. IV) [5]

Exposés VI and VII give elementary technical notions used in Exposé VIII to prove the *finiteness theorem* (VIII VIII.2.3). This theorem gives necessary and sufficient conditions, for a coherent sheaf F on a noetherian scheme X, for the local cohomology sheaves $\underline{H}^i_Y(F)$ to be coherent for $i \leq n$ (or, equivalently, for the sheaves $\mathrm{R}^i f_*(F|X-Y)$ to be coherent for $i \leq n-1$, where $f\colon X-Y \to X$ is the inclusion). It is one of the central technical results of the seminar, and in Exposé IX we show how a theorem of this kind can be used to establish a "comparison theorem"and an "existence theorem"in formal geometry, following and generalizing the use made in (EGA III §§4 and 5) of the finiteness theorem for a proper morphism.

These last results are applied in X and XI, devoted respectively to Lefschetz-type theorems for the fundamental group and for the Picard group. These theorems compare, under certain 4
conditions, the invariants (π_1 or Pic) associated respectively with a scheme X and a subscheme Y, which plays the role of a hyperplane section, and in particular give conditions under which they are isomorphic. Roughly speaking, the hypotheses allow one to pass from Y to the formal completion of X along Y, and then to apply the results of IX in order to pass from there to an open *neighborhood* U of Y in X. To pass from U to X, one also needs information of a "purity"or "parafactoriality"type for the local rings of X at the points of $Z = X-U$, which is a finite discrete set in the cases under consideration. This explains the interaction, in the proofs of Exposés X, XI, and XII, between local and global results, particularly in certain inductions. The main results obtained in X and XI are the theorems of *local nature* X 3.4 (the *purity theorem*) and XI XI.3.15 (the *parafactoriality theorem*). Notice that these theorems are proved by cohomological techniques of an essentially global nature. In XII, using the preceding local results, one obtains global variants for projective schemes over a field, or more generally over a more or less arbitrary base scheme; typical statements include XII XII.3.5 and XII XII.3.7.

In XIII, we review some of the many problems and conjectures suggested by the results and methods of the seminar. Perhaps the most interesting concern cohomological and homotopical Lefschetz-type theorems for complex analytic spaces; cf. XIII, pages 26 and following [6]. In the context of the étale cohomology of schemes , the corresponding conjectures are proved 5
in XIV by a duality technique that should apply equally in the complex analytic case (cf. the comments in XIII, p. 25, and XIV XIV.6.4). But the corresponding homotopical statements for analytic spaces, and more particularly those involving the fundamental group, seem to require entirely new techniques (cf. XIV XIV.6.4).

I am happy to thank all those who, in various capacities, helped bring the present volume

[4] *Editor's note.* The underlined notation has been retained for sheafified versions of functors, since the calligraphic analogue of Γ is not clear.

[5] *Editor's note.* Hartshorne's book contains sign errors and, more importantly, does not actually prove the compatibility of the trace with base change. Conrad completely redid this work and proved this crucial and highly nontrivial compatibility (*Grothendieck duality and base change*, Lect. Notes in Math., vol. 1750, Springer-Verlag, Berlin, 2000). Unfortunately, errors remain (cf. two preprints: Conrad B., "Clarifications and corrections to "*Grothendieck duality and base change*"" and "An addendum to Chapter 5 of "*Grothendieck duality and base change*""). For a more concrete aspect, with particular attention to the notion of residue, see Lipman's work, especially Lipman J., *Dualizing sheaves, differentials and residues on algebraic varieties*, Astérisque, vol. 117, Société mathématique de France, 1984. A categorical proof of the duality theorem, based on Brown's representability theorem, was obtained by Neeman (Neeman A., "The Grothendieck duality theorem via Bousfield's techniques and Brown representability", *J. Amer. Math. Soc.* **9** (1996), no. 1, p. 205–236).

[6] *Editor's note.* Essentially all the conjectures stated in XIII and XIV have now been proved; see the footnotes in those sections for references and comments.

to publication, including the collaborators already named in this Introduction. In particular, I wish to thank Mlle Chardon for the good grace with which she carried out the thankless task of physically preparing the final manuscript for photo-offset reproduction.

Bures-sur-Yvette, April 1968

A. Grothendieck.

Exposé I

Global and Local Cohomological Invariants Relative to a Closed Subspace

1. The functors Γ_Z and $\underline{\Gamma}_Z$

Let X be a topological space, and let $\underline{\mathbf{C}}_X$ be the category of sheaves of abelian groups on X. Let Φ be a family of supports in the sense of Cartan. The functor Γ_Φ on $\underline{\mathbf{C}}_X$ is defined by Original French p. 6

$$\Gamma_\Phi(F) = \{f \in \Gamma(F) \mid \mathrm{supp}(f) \in \Phi\}. \tag{1}$$

If Z is a closed subset of X, by an abuse of notation we denote by Γ_Z the functor Γ_Φ, where Φ is the family of closed subsets of X contained in Z. Thus

$$\Gamma_Z(F) = \{f \in \Gamma(F) \mid \mathrm{supp}(f) \subset Z\}. \tag{2}$$

We wish to extend this definition to the case in which Z is a *locally closed* subset of X, hence closed in a suitable open subset V of X. In that case we set

$$\Gamma_Z(F) = \Gamma_Z(F|_V). \tag{3}$$

We must verify that $\Gamma_Z(F)$ "does not depend" on the chosen open set. It is enough to show that, if V' is open and $V \supset V' \supset Z$, then the map $\rho^V_{V'} : F(V) \to F(V')$ maps $\Gamma_Z(F|_V)$ isomorphically onto $\Gamma_Z(F|_{V'})$. But

$$\Gamma_Z(F|_V) = \ker \rho^V_{V-Z}. \tag{4}$$

Consequently, if $f \in \Gamma_Z(F|_V)$ and $\rho^V_{V'}(f) = \rho^V_{V-Z}(f) = 0$, then $f = 0$, since $(V', V - Z)$ is a covering of V. Likewise, if $f' \in \Gamma_Z(F|_{V'})$, then $f' \in F(V')$ and $0 \in F(V - Z)$ define an element $f \in F(V)$ such that $\rho^V_{V'}(f) = f'$ and $f \in \Gamma_Z(F|_V)$. Thus $\rho^V_{V'}$ induces an isomorphism Original French p. 7

$$\Gamma_Z(F|_V) \longrightarrow \Gamma_Z(F|_{V'}).$$

Note that every open subset W of Z is induced by an open subset U of X in which W is closed. It follows that $W \mapsto \Gamma_W(F)$ defines a presheaf on Z; one verifies that it is a sheaf, denoted by $i^!(F)$, where $i : Z \to X$ is the canonical immersion. We obtain

$$\Gamma_Z(F) = \Gamma(i^!(F)). \tag{5}$$

The sheaf $i^!(F)$ is a subsheaf of $i^*(F)$. Indeed, the canonical homomorphism

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

is injective on $\Gamma_{U\cap Z}(F|_U) \subset \Gamma(F|_U)$. In summary, we have the following result.

Proposition 1.1. *There exists a unique subsheaf $i^!(F)$ of $i^*(F)$ such that, for every open subset U of X for which $U \cap Z$ is closed in U, the map*

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

induces an isomorphism

$$\Gamma_{U\cap Z}(F|_U) \longrightarrow \Gamma(U \cap Z, i^!(F)).$$

If Z is open, then simply

$$i^!(F) = i^*(F) = F|_Z, \qquad \Gamma_Z(F) = \Gamma(Z, F). \tag{6}$$

Suppose again that Z is arbitrary. As U ranges through the open subsets of X, the assignment

$$U \longmapsto \Gamma_{U\cap Z}(F|_U) = \Gamma(U \cap Z, i^!(F))$$

is a sheaf on X, which we denote by $\underline{\Gamma}_Z(F)$. More precisely, the preceding formula, which expresses the fact that $i^!$ commutes with restriction to open subsets, gives an isomorphism

$$\underline{\Gamma}_Z(F) = i_*(i^!(F)). \tag{7}$$

By definition, for every open subset U of X,

Original French p. 8

$$\Gamma(U, \underline{\Gamma}_Z(F)) = \Gamma_{U\cap Z}(F|_U). \tag{8}$$

Here there is a characteristic difference between the case in which Z is closed and the case in which Z is open. In the first case, formula (8) shows that $\underline{\Gamma}_Z(F)$ may be regarded as a subsheaf of F; hence there is a *canonical immersion*

$$\underline{\Gamma}_Z(F) \hookrightarrow F. \tag{8'}$$

When Z is open, on the other hand, equation (6) shows that the right-hand side of equation (8) is $\Gamma(U \cap Z, F)$ and therefore receives a homomorphism from $\Gamma(U, F)$. Thus there is a *canonical homomorphism* in the opposite direction:[1]

$$F \longrightarrow \underline{\Gamma}_Z(F). \tag{8''}$$

This is none other than the canonical homomorphism

$$F \longrightarrow i_*i^*(F),$$

in view of the isomorphism

$$\underline{\Gamma}_Z(F) \simeq i_*i^*(F) \tag{6 bis}$$

deduced from equations (6) and (7).

For variable F, the objects $\Gamma_Z(F)$, $\underline{\Gamma}_Z(F)$, and $i^!(F)$ may of course be regarded as functors in F, with values, respectively, in the category of abelian groups, the category of sheaves of abelian groups on X, and the category of sheaves of abelian groups on Z. It is sometimes convenient to interpret the functor

$$i^! : \underline{\mathbf{C}}_X \longrightarrow \underline{\mathbf{C}}_Z$$

as the right adjoint of the familiar functor

$$i_! : \underline{\mathbf{C}}_Z \longrightarrow \underline{\mathbf{C}}_X$$

defined by the following proposition.

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Proposition 1.2. *Let G be a sheaf of abelian groups on Z. There exists a unique subsheaf of $i_*(G)$, denoted $i_!(G)$, such that for every open subset U of X the (identity) isomorphism*

$$\Gamma(U \cap Z, G) = \Gamma(U, i_*(G))$$

induces an isomorphism

$$\Gamma_{\Phi_{U\cap Z,U}}(U \cap Z, G) = \Gamma(U, i_!(G)),$$

where $\Phi_{U\cap Z,U}$ denotes the family of subsets of $U \cap Z$ that are closed in U.

The verification reduces to observing that the left-hand side is a sheaf as U varies: namely, for a section of $i_*(G)$ on U, viewed as a section of G on $U \cap Z$, the property of having support closed *in* U is *local on* U. The sheaf $i_!(G)$ just defined is also known as the *sheaf obtained from G by extension by zero* outside Z; see Godement [Godement]. In particular, if Z is closed, then

$$i_!(G) = i_*(G). \tag{9}$$

In general, however, the canonical injection $i_!(G) \to i_*(G)$ is not an isomorphism, as is already well known when Z is open. Evidently $i_!(G)$ depends functorially on G (and is in fact an exact functor of G). We therefore have:

Proposition 1.3. *There is an isomorphism of bifunctors in G and F, where G is a sheaf of abelian groups on Z and F is a sheaf of abelian groups on X:*

$$\mathrm{Hom}(i_!(G), F) = \mathrm{Hom}(G, i^!(F)). \tag{10}$$

To define such an isomorphism is equivalent to defining functorial homomorphisms

$$i_!i^!(F) \longrightarrow F, \qquad G \longrightarrow i^!i_!(G),$$

satisfying the familiar compatibility conditions (see, for example, Shih's exposé in the Cartan seminar on cohomological operations).

Recalling that $i_!$ is exact, and hence takes monomorphisms to monomorphisms, we conclude:

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Corollary 1.4. *If F is injective, then $i^!(F)$ is injective; consequently $\underline{\Gamma}_Z(F) = i_*i^!(F)$ is also injective.*

Replacing X by a variable open subset U of X, Proposition 1.3 also gives:

Corollary 1.5. *There is an isomorphism, functorial in F and G,*

$$\underline{\mathrm{Hom}}(i_!(G), F) = i_*(\underline{\mathrm{Hom}}(G, i^!(F))). \tag{11}$$

Taking G to be the constant sheaf on Z defined by $\mathbf{Z}$, denoted $\mathbf{Z}_Z$, Proposition 1.3 and Corollary 1.5 specialize as follows.

Corollary 1.6. *There are isomorphisms, functorial in F,*

$$\begin{aligned} \Gamma_Z(F) &= \mathrm{Hom}(\mathbf{Z}_{Z,X}, F), \\ \underline{\Gamma}_Z(F) &= \underline{\mathrm{Hom}}(\mathbf{Z}_{Z,X}, F), \end{aligned} \tag{12}$$

where $\mathbf{Z}_{Z,X} = i_!(\mathbf{Z}_Z)$ is the sheaf of abelian groups on X obtained from the constant sheaf $\mathbf{Z}_Z$ on Z by extension by zero outside Z.

[1] The original reference was (8).

Remark 1.7. Suppose that X is a ringed space, equip Z with the sheaf of rings $\mathcal{O}_Z = i^{-1}(\mathcal{O}_X)$, and now let $\underline{\mathrm{C}}_X$ and $\underline{\mathrm{C}}_Z$ denote the categories of modules on X and Z, respectively. The preceding considerations extend word for word, with F a module on X and G a module on Z, and with statements 1.3–1.6 interpreted accordingly.

To conclude these generalities, let us examine what happens when the locally closed subset Z is changed. Let $Z' \subset Z$ be another locally closed subset, and let

$$j : Z' \longrightarrow Z, \qquad i' : Z' \longrightarrow X, \qquad i' = ij$$

be the canonical inclusions. There are functorial isomorphisms

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$$(ij)^! = j^! i^!, \qquad (ij)_! = i_! j_!. \tag{13}$$

The first isomorphism in (13) defines a functorial isomorphism

$$\Gamma_{Z'}(F) = \Gamma(Z', (ij)^!(F)) \simeq \Gamma(Z', j^!(i^!(F))) = \Gamma_{Z'}(i^!(F)). \tag{14}$$

Suppose now that Z' is *closed* in Z, and let

$$Z'' = Z - Z'$$

be its complement in Z. This is open in Z and hence locally closed in X. Applying the canonical injection $(8')$ to $i^!(F)$ on Z with the subset Z', and using (14), gives an injective functorial canonical homomorphism

$$\Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F). \tag{15}$$

If Z' is replaced by Z'' in (14) and equation $(8'')$ is used, one obtains a functorial canonical homomorphism

$$\Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F). \tag{15'}$$

Proposition 1.8. *Under the preceding hypotheses, the sequence of functorial homomorphisms*

$$0 \longrightarrow \Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F) \tag{16}$$

is exact. If F is flasque, the sequence remains exact after a zero is added on the right.

Proof. Replace X by an open subset V in which Z is closed. This reduces the assertion to the case in which Z, and hence Z', is closed. Then Z'' is closed in the open subset $X - Z'$, and there is a canonical injection

$$\Gamma_{Z''}(F) \longrightarrow \Gamma(X - Z', F).$$

The exactness of (16) then says simply that the sections of F with support in Z' are precisely those whose restriction to $X - Z'$ is zero.

If F is flasque, every element of $\Gamma_{Z''}(F)$, regarded as a section of F on $X - Z'$, extends to a section of F on X. The latter section plainly has support in Z, proving that the last homomorphism in (16) is then surjective. □

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Corollary 1.9. *There is a functorial exact sequence*

$$0 \longrightarrow \underline{\Gamma}_{Z'}(F) \longrightarrow \underline{\Gamma}_Z(F) \longrightarrow \underline{\Gamma}_{Z''}(F), \tag{16 bis}$$

and, if F is flasque, the sequence remains exact after a zero is added on the right.

Proposition 1.8 may be interpreted, by means of Corollary 1.6, as a statement about the functors Hom and $\underline{\mathrm{Hom}}$, as follows. First note that if G is a sheaf of abelian groups on Z,

inducing the sheaves $j^*(G)$ and $k^*(G)$ on Z' and Z'', respectively, where $j : Z' \to Z$ and $k : Z'' \to Z$ are the canonical injections, then there is a canonical exact sequence of sheaves on X:

$$0 \longrightarrow k^*(G)_X \longrightarrow G_X \longrightarrow j^*(G)_X \longrightarrow 0. \tag{17}$$

Here, to simplify notation, the subscript X denotes the sheaf on X obtained by extension by zero outside the space on which the sheaf in question is defined. Sequence (17) generalizes a familiar exact sequence for $Z = X$ (see Godement [Godement]); it follows from that sequence by writing it on Z and applying the functor $i_!$. Taking $G = \mathbf{Z}_Z$, we obtain in particular:

Proposition 1.10. *Under the preceding hypotheses, there is an exact sequence of sheaves of abelian groups on X:*

$$0 \longrightarrow \mathbf{Z}_{Z'',X} \longrightarrow \mathbf{Z}_{Z,X} \longrightarrow \mathbf{Z}_{Z',X} \longrightarrow 0. \tag{18}$$

The two exact sequences in Proposition 1.8 and Corollary 1.9 are exactly the sequences obtained from (18) *by applying, respectively, the functors* $\mathrm{Hom}(-, F)$ *and* $\underline{\mathrm{Hom}}(-, F)$.

This also gives an immediate proof that sequences (16) and (16 bis) remain exact after a zero is added on the right, provided that F is *injective*.

2. The functors $\mathrm{H}^*_Z(X, F)$ and $\underline{\mathrm{H}}^*_Z(F)$

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Definition 2.1. *The derived functors, in the variable F, of the functors $\Gamma_Z(F)$ and $\underline{\Gamma}_Z(F)$ are denoted by $\mathrm{H}^*_Z(X, F)$ and $\underline{\mathrm{H}}^*_Z(F)$, respectively.*

They are cohomological functors with values, respectively, in the category of abelian groups and in the category of sheaves of abelian groups on X. When Z is closed, $\mathrm{H}^*_Z(X, F)$ is, by definition, simply $\mathrm{H}^*_\Phi(X, F)$, where Φ denotes the family of closed subsets of X contained in Z. When Z is open, we shall see that $\mathrm{H}^*_Z(X, F)$ is simply $\mathrm{H}^*(Z, F) = \mathrm{H}^*(Z, F|_Z)$, by the following proposition.

Proposition 2.2 (Excision theorem). *Let V be an open subset of X containing Z. There is an isomorphism of cohomological functors in F:*

$$\mathrm{H}^*_Z(X, F) \longrightarrow \mathrm{H}^*_Z(V, F|_V). \tag{19}$$

Indeed, there is a functorial isomorphism $\Gamma^X_Z \simeq \Gamma^V_Z j^!$, where $j\colon V \to X$ is the inclusion and $j^!$ is therefore the restriction functor (cf. equation (14)). This last functor is exact and, by Corollary 1.4, takes injectives to injectives; the isomorphism (19) follows immediately.

When Z is open, we may take $V = Z$, and obtain:

Corollary 2.3. *Suppose that Z is open. There is an isomorphism of cohomological functors*

$$\mathrm{H}^*_Z(X, F) = \mathrm{H}^*(Z, F). \tag{20}$$

The isomorphisms of Corollary 1.6 and the definitions give the following (cf. [Tôhoku]).

Proposition 2.3 bis. *There are isomorphisms of cohomological functors*

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$$\mathrm{H}^*_Z(X, F) \simeq \mathrm{Ext}^*(X; \mathbf{Z}_{Z,X}, F), \tag{21}$$

$$\underline{\mathrm{H}}^*_Z(F) \simeq \underline{\mathrm{Ext}}^*(\mathbf{Z}_{Z,X}, F). \tag{21 bis}$$

We may therefore apply the results of [Tôhoku] on the Ext of modules. First, we record the following interpretation of the sheaves $\underline{\mathrm{H}}^*_Z(F)$ in terms of the global groups $\mathrm{H}^*_Z(X, F)$.

Corollary 2.4. $\underline{\mathrm{H}}^*_Z(F)$ *is canonically isomorphic to the sheaf associated with the presheaf*

$$U \longmapsto \mathrm{H}^*_{Z\cap U}(U, F|_U).$$

In particular, using Corollary 2.3, we obtain:

Corollary 2.5. *Suppose that Z is open. There is an isomorphism of cohomological functors*

$$\underline{\mathrm{H}}^*_Z(F) = \mathrm{R}^* i_* i^*(F), \tag{22}$$

where $i\colon Z \to X$ is the inclusion.

The Ext spectral sequence gives the following important spectral sequence.

Theorem 2.6. *There is a spectral sequence, functorial in F, abutting to $\mathrm{H}^*_Z(X, F)$ and having initial term*

$$E_2^{p,q}(F) = \mathrm{H}^p(X, \underline{\mathrm{H}}^q_Z(F)). \tag{23}$$

Remarks 2.7. It follows immediately from Corollary 2.4 that the sheaves $\underline{\mathrm{H}}^q_Z(F)$ vanish on $X - \overline{Z}$, and also vanish in the interior of Z when $q \neq 0$; hence, for such q, $\underline{\mathrm{H}}^q_Z(F)$ is supported on the boundary of Z.

Consequently, the right-hand side of (23) may be interpreted as a cohomology group on $\overline{Z}$. We shall use Theorem 2.6 when Z is closed in X; in that case the right-hand side of (23) may be interpreted as a cohomology group computed on Z:

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$$E_2^{p,q}(F) = \mathrm{H}^p(Z, \underline{\mathrm{H}}^q_Z(F)). \tag{23 bis}$$

The corresponding equation was numbered (23) in the original edition.

When Z is open, the spectral sequence of Theorem 2.6 is simply the Leray spectral sequence for the continuous map $i\colon Z \to X$, in view of the interpretation in Corollary 2.5 used to compute its initial term.

We return to the exact sequence (18). It gives rise to an exact Ext sequence (cf. [Tôhoku]).

Theorem 2.8. *Let Z be a locally closed subset of X, let Z' be a closed subset of Z, and put $Z'' = Z - Z'$. There is a long exact sequence, functorial in F:*

$$\begin{aligned} 0 \longrightarrow \mathrm{H}^0_{Z'}(X,F) \longrightarrow \mathrm{H}^0_Z(X,F) \longrightarrow \mathrm{H}^0_{Z''}(X,F) \xrightarrow{\partial} \mathrm{H}^1_{Z'}(X,F) \longrightarrow \mathrm{H}^1_Z(X,F) \longrightarrow \cdots \\ \cdots \longrightarrow \mathrm{H}^i_{Z'}(X,F) \longrightarrow \mathrm{H}^i_Z(X,F) \longrightarrow \mathrm{H}^i_{Z''}(X,F) \xrightarrow{\partial} \mathrm{H}^{i+1}_{Z'}(X,F) \longrightarrow \cdots . \end{aligned} \tag{24}$$

Recall how this exact sequence is obtained. Let $C(F)$ be an injective resolution of F. Then the exact sequence (18) gives the exact sequence

$$0 \longrightarrow \Gamma_{Z'}(C(F)) \longrightarrow \Gamma_Z(C(F)) \longrightarrow \Gamma_{Z''}(C(F)) \longrightarrow 0. \tag{25}$$

This is precisely the sequence defined in Proposition 1.8. Its cohomology long exact sequence is (24).

The most important case for us is the one in which Z is closed (and we may always reduce to it by replacing X with an open subset V in which Z is closed). Then Z' is closed, Z'' is closed in the open subset $X - Z'$, and we may write

$$\mathrm{H}^i_{Z''}(X, F) = \mathrm{H}^i_{Z''}(X - Z', F|_{X-Z'}). \tag{26}$$

This permits us to write the exact sequence (24) in terms of cohomology with support in a given closed subset. The most common case is $Z = X$. Writing $Z' = A$ for simplicity, we obtain:

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Corollary 2.9. *Let A be a closed subset of X. There is a long exact sequence, functorial in F:*

$$\begin{aligned} 0 \longrightarrow \mathrm{H}^0_A(X,F) \longrightarrow \mathrm{H}^0(X,F) \longrightarrow \mathrm{H}^0(X-A,F) \xrightarrow{\partial} \mathrm{H}^1_A(X,F) \longrightarrow \cdots \\ \cdots \longrightarrow \mathrm{H}^i_A(X,F) \longrightarrow \mathrm{H}^i(X,F) \longrightarrow \mathrm{H}^i(X-A,F) \xrightarrow{\partial} \mathrm{H}^{i+1}_A(X,F) \longrightarrow \cdots . \end{aligned} \tag{27}$$

This exact sequence shows that $\mathrm{H}^i_A(X,F)$ plays the role of a relative cohomology group of X modulo $X-A$, with coefficients in F. It is in this capacity that it arises naturally in applications. By "sheafifying" (24) and (27), or by arguing directly, and taking into account that the sheaf associated with $U \mapsto \mathrm{H}^i(U,F)$ is zero for $i>0$, we obtain:

Corollary 2.10. *Under the hypotheses of Theorem 2.8 there is a long exact sequence, functorial in F:*

$$\cdots \longrightarrow \underline{\mathrm{H}}^i_{Z'}(F) \longrightarrow \underline{\mathrm{H}}^i_Z(F) \longrightarrow \underline{\mathrm{H}}^i_{Z''}(F) \xrightarrow{\partial} \underline{\mathrm{H}}^{i+1}_{Z'}(F) \longrightarrow \cdots . \tag{24 bis}$$

Corollary 2.11. *Let A be a closed subset of X. There is an exact sequence, functorial in F:*

$$0 \longrightarrow \underline{\mathrm{H}}^0_A(F) \longrightarrow F \longrightarrow f_*(F|_{X-A}) \xrightarrow{\partial} \underline{\mathrm{H}}^1_A(F) \longrightarrow 0, \tag{28}$$

and, for $i \geq 2$, canonical isomorphisms

$$\underline{\mathrm{H}}^i_A(F) = \underline{\mathrm{H}}^{i-1}_{X-A}(F) = \mathrm{R}^{i-1}f_*(F|_{X-A}), \tag{29}$$

where $f\colon X-A \to X$ is the inclusion.

Thus $\underline{\mathrm{H}}^0_A(F)$ and $\underline{\mathrm{H}}^1_A(F)$ are defined, respectively, as the kernel and cokernel of the canonical homomorphism

$$F \longrightarrow f_*f^*(F) = f_*(F|_{X-A}),$$

and $\underline{\mathrm{H}}^i_A(F)$ for $i \geq 2$ is expressed in terms of the derived functors of f_*.

Corollary 2.12. *Let F be a sheaf of abelian groups on X. If F is flasque, then for every locally closed subset Z of X and every integer $i \neq 0$,*

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$$\mathrm{H}^i_Z(X,F) = 0, \qquad \underline{\mathrm{H}}^i_Z(F) = 0.$$

Conversely, if $\mathrm{H}^1_Z(X,F) = 0$ for every closed subset Z of X, then F is flasque.

Suppose that F is flasque. It induces a flasque sheaf on every open subset. To prove that $\mathrm{H}^i_Z(X,F) = 0$ for $i>0$, we may therefore assume that Z is closed; the assertion then follows from the exact sequence (27). For every locally closed Z, "sheafifying"—that is, applying Corollary 2.4—gives $\underline{\mathrm{H}}^i_Z(F) = 0$ for $i>0$. Conversely, suppose that $\mathrm{H}^1_Z(X,F) = 0$ for every closed Z. Then (27) shows that $\mathrm{H}^0(X,F) \to \mathrm{H}^0(X-Z,F)$ is surjective for every such Z, which means precisely that F is flasque.

Combining Theorems 2.6 and 2.8, we deduce:

Proposition 2.13. *Let F be a sheaf of abelian groups on X, let Z be a closed subset of X, put $U = X-Z$, and let N be an integer. The following conditions are equivalent:*

(i) $\underline{\mathrm{H}}^i_Z(F) = 0$ *for $i \leq N$.*

(ii) *For every open subset V of X, the canonical homomorphism*

$$\mathrm{H}^i(V,F) \longrightarrow \mathrm{H}^i(V \cap U, F)$$

is

a) *bijective for* $i < N$*;*

b) *injective for* $i = N$.

If $N > 0$*, it is enough in* (ii) *to require* a).

To prove (i) $\Rightarrow$ (ii), the local nature of the sheaves $\underline{\mathrm{H}}^i_Z(F)$ reduces us to proving the following.

Corollary 2.14. *If condition* (i) *of Proposition 2.13 holds, then*

$$\mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F)$$

is bijective for $i < N$ *and injective for* $i = N$.

Indeed, by the exact sequence (27), this is equivalent to $\mathrm{H}^i_Z(X, F) = 0$ for $i \leq N$, and this relation is an immediate consequence of the spectral sequence (23 bis). Original French p. 18

Conversely, condition (ii) of Proposition 2.13 says that, for every open subset V of X,

$$\mathrm{H}^i_{Z\cap V}(V, F|_V) = 0 \qquad \text{for } i \leq N,$$

which implies (i) by Corollary 2.4. Moreover, if $N > 0$, condition b) is superfluous. Indeed, if $N = 1$, condition a) together with (28) gives $\underline{\mathrm{H}}^i_Z(F) = 0$ for $i \leq N$. If $N > 1$, condition a) for $i = N - 1 > 0$, together with (29), gives $\underline{\mathrm{H}}^i_Z(F) = 0$ for $i \leq N$.

Taking Corollary 2.11 into account, this also proves condition (i) of Proposition 2.13.

Remark. Let $Y \to X$ be a closed immersion, and suppose that locally it has the form $\{0\} \times Y \subset R^n \times Y$. Suppose that F is a locally constant sheaf on X. Then

$$\underline{\mathrm{H}}^i_Y(F) \simeq \begin{cases} 0, & i \neq n, \\ F \otimes \underline{\mathrm{T}}_{Y,X}, & i = n, \end{cases} \qquad \underline{\mathrm{T}}_{Y,X} \simeq \underline{\mathrm{H}}^n_Y(\mathbf{Z}_X), \tag{30}$$

where $\underline{\mathrm{T}}_{Y,X}$ is the extension to X of a sheaf on Y locally isomorphic to $\mathbf{Z}_Y$; it is called the *normal orientation sheaf* of Y in X.

Using the spectral sequence (23 bis), we obtain in this case

$$\mathrm{H}^i_Y(X, F) \simeq \mathrm{H}^{i-n}(Y, F \otimes \underline{\mathrm{T}}_{Y,X}), \tag{31}$$

and recover the *Gysin homomorphism*

$$\mathrm{H}^j(Y, F \otimes \underline{\mathrm{T}}_{Y,X}) \longrightarrow \mathrm{H}^{j+n}(X, F). \tag{32}$$

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Exposé II

Application to Quasi-coherent Sheaves on Preschemes

Proposition 1. *Let X be a prescheme and let Z be a locally closed subset of the form $Z = U - V$, where U and V are open subsets of X such that $V \subset U$ and the canonical immersions $U \to X$ and $V \to X$ are quasi-compact. Then, for every quasi-coherent $\mathcal{O}_X$-module F, the sheaves $\underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent.*

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By Exposé I, equation (24), there is an exact sequence of relative cohomology sheaves

$$\longrightarrow \underline{\mathrm{H}}^{i-1}_U(F) \longrightarrow \underline{\mathrm{H}}^i_V(F) \longrightarrow \underline{\mathrm{H}}^i_Z(F) \longrightarrow \underline{\mathrm{H}}^i_U(F) \longrightarrow \underline{\mathrm{H}}^{i+1}_V(F) \longrightarrow .$$

By EGA III, 1.4.17, for the sheaves $\underline{\mathrm{H}}^i_Z(F)$ to be quasi-coherent it therefore suffices that the sheaves $\underline{\mathrm{H}}^i_U(F)$ and $\underline{\mathrm{H}}^i_V(F)$ be quasi-coherent. We may thus suppose that Z is open and that the canonical immersion $j\colon Z \to X$ is quasi-compact.

Since Z is open, Exposé I, equation (22), gives a canonical isomorphism

$$\underline{\mathrm{H}}^i_Z(F) \simeq \mathrm{R}^i j_*(F|_Z).$$

But j is separated (EGA I, 5.5.1) and quasi-compact; hence, by EGA III, 1.4.10, the sheaves $\mathrm{R}^i j_*(F|_Z) = \underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent. This completes the proof.

Corollary 2. *Let Z be a closed subset of X such that the canonical immersion $X - Z \to X$ is quasi-compact. Then the $\mathcal{O}_X$-modules $\underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent.*

Corollary 3. *If X is locally noetherian, then, for every locally closed subset Z of X and every quasi-coherent $\mathcal{O}_X$-module F, the sheaves $\underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent.*

This follows immediately from Corollary 2 and EGA I, 6.6.4.

Corollary 4. *Suppose that X is the spectrum of a ring A. Let U be a quasi-compact open subset of X, put $Y = X - U$, and let F be a quasi-coherent $\mathcal{O}_X$-module. There is an isomorphism of cohomological functors in F:*

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$$\underline{\mathrm{H}}^i_Y(F) = \widetilde{(\mathrm{H}^i_Y(X, F))}. \tag{4.1}$$

There is moreover an exact sequence, functorial in F,

$$0 \longrightarrow \mathrm{H}^0_Y(X, F) \longrightarrow \mathrm{H}^0(X, F) \longrightarrow \mathrm{H}^0(U, F) \longrightarrow \mathrm{H}^1_Y(X, F) \longrightarrow 0, \tag{4.2}$$

and there are isomorphisms, functorial in F,

$$\mathrm{H}^i_Y(X, F) \simeq \mathrm{H}^{i-1}(U, F), \qquad i \geq 2. \tag{4.3}$$

By Corollary 2, the sheaves $\underline{\mathrm{H}}^i_Y(F)$ are quasi-coherent. Since X is affine, it follows that $\mathrm{H}^p(X, \underline{\mathrm{H}}^i_Y(F)) = 0$ for $p > 0$. The spectral sequence of Exposé I, equation (23), therefore degenerates, and

$$\mathrm{H}^i_Y(X, F) = \Gamma(\underline{\mathrm{H}}^i_Y(F)).$$

Equation (4.1) now follows from EGA I, 1.1.3.7; equations (4.2) and (4.3) follow from the cohomology exact sequence of Exposé I, equation (27), and from the vanishing $\mathrm{H}^i(X, F) = 0$ for $i > 0$, since X is affine.

Under the hypotheses of Corollary 4, U is a finite union of affine open subsets X_f. Consequently, one can find an ideal I, generated by finitely many elements f_α and defining Y; write $\mathbf{f} = (f_\alpha)$. With the notation of EGA III, 1,[1] we have the following result.

Proposition 5. *Suppose that X is the spectrum of a ring A. Let $\mathbf{f} = (f_\alpha)$ be a finite family of elements of A, let Y be the closed subset of X that they define, let M be an A-module, and let F be the associated sheaf. There are isomorphisms of ∂-functors in M:*

$$\mathrm{H}^i((\mathbf{f}), M) \simeq \mathrm{H}^i_Y(X, F). \tag{5.1}$$

We shall also write $\mathrm{H}^i_J(M) = \mathrm{H}^i_Y(X, F)$ when Y is the closed subset of $X = \operatorname{Spec} A$ defined by an ideal J of A.

For $i = 0$ and $i = 1$, use the exact sequences (4.2) and EGA III, 1.4.3.2; for $i \geq 2$, use (4.3) and EGA III, 1.4.3.1. This gives isomorphisms functorial in M. One verifies that, up to a sign depending only on i, they are compatible with the boundary operator; hence the isomorphism of ∂-functors in (5.1) exists.

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Now let X be a prescheme, let Y be a closed subset of X, and let $f\colon Y \to X$ be the inclusion. Let I be a quasi-coherent ideal defining Y in X, and let F be a sheaf on X.

We have seen that there are isomorphisms of ∂-functors in F

$$\operatorname{Ext}^i_{\mathcal{O}_X}(X; f_* f^{-1}(\mathcal{O}_X), F) \longrightarrow \mathrm{H}^i_Y(X, F), \tag{$*$}$$

$$\underline{\operatorname{Ext}}^i_{\mathcal{O}_X}(f_* f^{-1}(\mathcal{O}_X), F) \longrightarrow \underline{\mathrm{H}}^i_Y(F). \tag{$**$}$$

Let n, m be integers such that $m \geq n \geq 0$. Denote by $i_{n,m}$ the canonical map

$$\mathcal{O}_{Y_m} = \mathcal{O}_X / I^{m+1} \longrightarrow \mathcal{O}_X / I^{n+1} = \mathcal{O}_{Y_n},$$

and by j_n the map $f_* f^{-1}(\mathcal{O}_X) \to \mathcal{O}_{Y_n}$. The pairs $(\mathcal{O}_{Y_n}, i_{n,m})$ form a projective system, and the maps j_n are compatible with the $i_{n,m}$.

Applying the functor $\operatorname{Ext}^i_{\mathcal{O}_X}(X; \cdot, F)$ yields a morphism

$$\varphi'\colon \varinjlim_n \operatorname{Ext}^i_{\mathcal{O}_X}(X; \mathcal{O}_{Y_n}, F) \longrightarrow \operatorname{Ext}^i_{\mathcal{O}_X}(X; f_* f^{-1}(\mathcal{O}_X), F).$$

It is readily seen to be a morphism of cohomological functors in F. Consequently, the morphism

$$\varphi\colon \varinjlim_n \operatorname{Ext}^i_{\mathcal{O}_X}(X; \mathcal{O}_{Y_n}, F) \longrightarrow \mathrm{H}^i_Y(X, F),$$

obtained by composing φ' with $(*)$, is itself a morphism of cohomological functors in F.

We define similarly

$$\underline{\varphi}\colon \varinjlim_n \underline{\operatorname{Ext}}^i_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F) \longrightarrow \underline{\mathrm{H}}^i_Y(F).$$

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[1] Recall that $\mathrm{H}^\bullet(\mathbf{f}, M)$ is the Koszul cohomology $\mathrm{H}^\bullet(\operatorname{Hom}(K_\bullet(\mathbf{f}), M))$ of $\mathbf{f}$ with values in M (EGA III, 1.1.2), and that $\mathrm{H}^\bullet((\mathbf{f}), M)$ is the direct limit (ibid., 1.1.6.5) $\varinjlim_n \mathrm{H}^\bullet(\mathbf{f}^n, M)$, whose transition morphisms are induced by the natural morphisms $K_\bullet(\mathbf{f}^{n+1}) \to K_\bullet(\mathbf{f}^n)$ (ibid., 1.1.6).

Our goal is the following theorem.

Theorem 6. a) *Let X be a locally noetherian prescheme, let Y be a closed subset of X defined by a coherent ideal I, and let F be a quasi-coherent $\mathcal{O}_X$-module. Then $\boldsymbol{\varphi}$ is an isomorphism.*
b) *If X is noetherian, then φ is an isomorphism.*

Theorem 6 will follow from Theorem 6(a) and the following lemma.

Lemma 7. *If the underlying topological space of X is noetherian and $\boldsymbol{\varphi}$ is an isomorphism, then φ is also an isomorphism.*

We first prove Lemma 7. There is a spectral sequence

$$\mathrm{H}^p(X, \underline{\mathrm{H}}^q_Y(F)) \Longrightarrow \mathrm{H}^*_Y(X, F). \tag{7.1}$$

There is, on the other hand, a direct system of spectral sequences

$$\mathrm{H}^p(X, \underline{\mathrm{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)) \Longrightarrow \mathrm{Ext}^*_{\mathcal{O}_X}(X; \mathcal{O}_{Y_n}, F). \tag{7.2$_n$}$$

It follows from the definitions of φ and $\underline{\varphi}$ that these morphisms are associated with a homomorphism Φ of spectral sequences, from the direct limit of (7.2$_n$) to (7.1). If the underlying space of X is noetherian, then by Godement, 4.12.1,[2]

$$\varinjlim_n \mathrm{H}^p(X, \underline{\mathrm{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)) \xrightarrow{\sim} \mathrm{H}^p\left(X, \varinjlim_n \underline{\mathrm{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)\right).$$

Thus Φ_2 may be written as a morphism

$$\mathrm{H}^p\left(X, \varinjlim_n \underline{\mathrm{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)\right) \longrightarrow \mathrm{H}^p(X, \underline{\mathrm{H}}^q_Y(F)),$$

which is precisely the morphism deduced from $\underline{\varphi}$.

If $\underline{\varphi}$ is an isomorphism, then so is Φ_2, and hence so is φ by EGA 0_{III}, 11.1.5. This proves Lemma 7.

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We now prove Theorem 6(a). This is a local question on X. By Corollary 4 and EGA I, 1.3.9 and 1.3.12, we may suppose that X is the spectrum of a ring A. Write M for the A-module whose associated sheaf is F. It is therefore enough to prove that, under the hypotheses of Theorem 6(a), the canonical homomorphism

$$\varinjlim_n \mathrm{Ext}^i_A(A/I^n, M) \longrightarrow \mathrm{H}^i_Y(X, M) \tag{7.3}$$

is an isomorphism.

Let f_α be finitely many elements of A that generate I, and put $\mathbf{f} = (f_\alpha)$. The sequence of ideals $(\mathbf{f}^n)$ is decreasing and cofinal with the sequence of ideals I^n, so that (7.3) is equivalent to a morphism of ∂-functors in M:

$$\varinjlim_n \mathrm{Ext}^i_A(A/(\mathbf{f}^n), M) \longrightarrow \mathrm{H}^i_Y(X, M). \tag{7.4}$$

On the other hand, there are canonical isomorphisms

$$\begin{aligned} \varinjlim_n \mathrm{Hom}_A(A/(\mathbf{f}^n), M) &\simeq \varinjlim_n (m \in M \mid (\mathbf{f}^n)m = 0) \\ &\simeq \mathrm{H}^0((\mathbf{f}), M). \end{aligned} \tag{7.5}$$

[2] Cf. the first bibliographic reference at the end of Exposé I.

Since $\varinjlim_n \mathrm{Ext}^i_A(A/(\mathbf{f}^n), M)$ is a universal ∂-functor in M, there is a unique morphism of ∂-functors in M

$$\varinjlim_n \mathrm{Ext}^i_A(A/(\mathbf{f}^n), M) \longrightarrow \mathrm{H}^i((\mathbf{f}), M) \tag{7.6}$$

that agrees in degree zero with (7.5).

The comparison obtained from (7.3) and (5.1) is a morphism of ∂-functors in M that agrees with (7.6) in degree zero; it therefore agrees with (7.6) in every degree. Theorem 6(a) is consequently an immediate consequence of the following lemma.

Lemma 8. *Let A be a noetherian ring, let I be an ideal generated by a finite family $\mathbf{f} = (f_\alpha)$ of elements, and let M be an A-module. Then the homomorphisms* (7.6) *are isomorphisms.*

Lemma 9. *Let A be a ring, let $\mathbf{f} = (f_\alpha)$ be a finite family of elements of A, let I be the ideal generated by $\mathbf{f}$, and let $i > 0$ be an integer. The following conditions are equivalent:*

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a) *The homomorphism* (7.6) *is an isomorphism for every M.*

b) $\mathrm{H}^i((\mathbf{f}), M) = 0$ *for every injective M.*

c) *The projective system $(\mathrm{H}_i(\mathbf{f}^n, A)) = \mathrm{H}_{i,n}$ is essentially zero; that is, for every n there exists $n' > n$ such that $\mathrm{H}_{i,n'} \to \mathrm{H}_{i,n}$ is zero.*

Condition a) trivially implies b).

Condition b) implies a). Indeed, b) implies that $M \mapsto \mathrm{H}^i((\mathbf{f}), M)$ is a universal cohomological functor, so (7.6) is a morphism of universal cohomological functors. It is an isomorphism in degree zero and hence in every degree.

Condition c) implies b). Indeed, if M is injective, then for every n

$$\mathrm{H}^i(\mathbf{f}^n, M) = \mathrm{Hom}(\mathrm{H}_i(\mathbf{f}^n, A), M) = \mathrm{Hom}(\mathrm{H}_{i,n}, M).$$

Thus c) implies that, for every i, the direct system $(\mathrm{H}^i(\mathbf{f}^n, M))_{n\in\mathbf{Z}}$ is essentially zero, whence b).

Condition b) implies c). Indeed, fix $n > 0$, and let j be a monomorphism from $\mathrm{H}_{i,n}$ into an injective module M. For $n' \geq n$, let $j_{n'} \in \mathrm{Hom}(\mathrm{H}_{i,n'}, M)$ be the composite of j with the transition homomorphism $t_{n',n}\colon \mathrm{H}_{i,n'} \to \mathrm{H}_{i,n}$. The $j_{n'}$ define an element of $\mathrm{H}^i((\mathbf{f}), M)$, which is zero by hypothesis. There is therefore an n_0 such that $j_{n'} = 0$ if $n' > n_0$. Since j is a monomorphism, $j_{n'} = 0$ implies $t_{n',n} = 0$, which proves c).

Corollary 10. *Suppose that the underlying space of $X = \mathrm{Spec}(A)$ is noetherian. The preceding conditions hold for every finite family of elements of A and every $i > 0$ (or, equivalently, for $i = 1$) if and only if, for every injective A-module M, the sheaf F associated with M is flasque.*

The condition is necessary. Indeed, let $\mathbf{f} = (f_\alpha)$ be a finite family of elements of A, let Y be the closed subset defined by $\mathbf{f}$, and put $U = X - Y$. We then have the exact sequence

$$\mathrm{H}^0(X, F) \longrightarrow \mathrm{H}^0(U, F) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0,$$

and Lemma 9(b) shows that $\mathrm{H}^0(X, F) \to \mathrm{H}^0(U, F)$ is surjective.

The condition is sufficient by (5.1) and the fact that, for every closed subset Y of X and every flasque sheaf F on X, $\mathrm{H}^i_Y(X, F) = 0$ for $i > 0$.

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Lemma 11. *Under the hypotheses of Lemma 9, for every noetherian A-module N and every $i > 0$, the projective system $(\mathrm{H}_{i,n}(N))_{n\in\mathbf{Z}}$, where $\mathrm{H}_{i,n}(N) = \mathrm{H}_i(\mathbf{f}^n, N)$, is essentially zero.*

Proof. We argue by induction on the number m of elements of $\mathbf{f}$.

If $m = 1$, the family $\mathbf{f}$ consists of a single element, say f. Then $\mathrm{H}_{i,n}(N) = 0$ for $i > 1$, and $\mathrm{H}_{1,n}(N)$ is canonically isomorphic to the annihilator $N(n)$ of f^n in N. For $n' \geq n$, the transition homomorphism $N(n') \to N(n)$ is multiplication by $f^{n'-n}$. The $N(n)$ form an increasing sequence of submodules of N. Since N is noetherian, there is an n_0 such that $N(n) = N(n_0)$ for $n \geq n_0$. Hence all the $N(n)$ are annihilated by f^{n_0}, and every transition homomorphism $N(n') \to N(n)$ is zero when $n' \geq n + n_0$. The lemma is thus proved for $m = 1$.

Now suppose that $m > 1$ and that the lemma has been proved for all integers $m' < m$. Put $\mathbf{g} = (f_1, \ldots, f_{m-1})$ and $\mathbf{h} = f_m$.

For every $n > 0$, EGA III, 1.1.4.1 gives an exact sequence

$$0 \longrightarrow \mathrm{H}_0(\mathbf{h}^n, \mathrm{H}_i(\mathbf{g}^n, N)) \longrightarrow \mathrm{H}_i(\mathbf{f}^n, N) \longrightarrow \mathrm{H}_1(\mathbf{h}^n, \mathrm{H}_{i-1}(\mathbf{g}^n, N)) \longrightarrow 0,$$

and, as n varies, a projective system of exact sequences. The induction hypothesis shows that, for $i > 0$, the modules $\mathrm{H}_i(\mathbf{g}^n, N)$ form an essentially zero projective system. The same is therefore true of $\mathrm{H}_0(\mathbf{h}^n, \mathrm{H}_i(\mathbf{g}^n, N))$, which is identified with a quotient of $\mathrm{H}_i(\mathbf{g}^n, N)$. For the terms on the right, factor the transition morphisms from n' to n as

$$\begin{aligned} \mathrm{H}_1(\mathbf{h}^{n'}, \mathrm{H}_{i-1}(\mathbf{g}^{n'}, N)) &\longrightarrow \mathrm{H}_1(\mathbf{h}^{n'}, \mathrm{H}_{i-1}(\mathbf{g}^n, N)) \\ &\longrightarrow \mathrm{H}_1(\mathbf{h}^n, \mathrm{H}_{i-1}(\mathbf{g}^n, N)). \end{aligned}$$

Since $\mathrm{H}_{i-1}(\mathbf{g}^n, N)$ is a noetherian module, the case $m = 1$ shows that, for fixed n, there exists $n' > n$ such that the second arrow is zero. One therefore sees that, in this projective system of exact sequences, the two outer projective systems are essentially zero. The middle projective system is therefore essentially zero as well. □

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We have thus proved Lemma 11, hence Lemma 8, and consequently Theorem 6.

Remark. Theorem 6 can also be obtained by proving the condition of Corollary 10 with the aid of the structure theorems for injective modules over a noetherian ring (Matlis, Gabriel).

Exposé III

Cohomological Invariants and Depth

1. Review

We shall state a few definitions and results which the reader may find, for example, in Chapter I of the course given by J.-P. Serre at the Collège de France in 1957–58.[1]

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Definition 1.1. *Let A be a ring (commutative with identity, as throughout what follows), and let M be an A-module (unital, as throughout what follows). We call:*

- *the* annihilator of M, *denoted* $\operatorname{Ann} M$, *the set of $a \in A$ such that $am = 0$ for every $m \in M$;*
- *the* support of M, *denoted* $\operatorname{Supp} M$, *the set of prime ideals $\mathfrak{p}$ of A such that the localization $M_{\mathfrak{p}}$ is nonzero;*
- *the "*assassin of M,*" or the "*set of prime ideals associated with M,*" denoted* $\operatorname{Ass} M$, *the set of prime ideals $\mathfrak{p}$ for which there exists a nonzero element of M whose annihilator is $\mathfrak{p}$.*

If $\mathfrak{a}$ is an ideal of A, we denote by $\mathfrak{r}(\mathfrak{a})$ the radical of $\mathfrak{a}$ in A, that is, the set of elements of A some power of which lies in $\mathfrak{a}$.

The following results hold under the assumption that *A is noetherian and M is finitely generated.*

Proposition 1.1.

(i) $\operatorname{Ass} M$ *is a finite set.*

(ii) *An element of A annihilates a nonzero element of M if and only if it belongs to one of the prime ideals associated with M.*

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(iii) *The radical of the annihilator of M, $\mathfrak{r}(\operatorname{Ann} M)$, is the intersection of the prime ideals associated with M that are minimal with respect to inclusion in* $\operatorname{Ass} M$.

Proposition 1.2. *Let $\mathfrak{p}$ be a prime ideal of A. The following assertions are equivalent:*

(i) $\mathfrak{p} \in \operatorname{Supp} M$.

[1] *Editorial note.* The reissue of Serre's text (J.-P. Serre, *Algèbre locale. Multiplicités*, course at the Collège de France, 1957–58, written up by Pierre Gabriel, second edition, Lecture Notes in Mathematics, vol. 11, Springer-Verlag, 1965) no longer contains proofs of these statements. One may consult N. Bourbaki, *Algèbre commutative*, Masson, as Serre himself suggests.

(ii) *There exists* $\mathfrak{q} \in \operatorname{Ass} M$ *such that* $\mathfrak{q} \subset \mathfrak{p}$.

(iii) $\mathfrak{p} \supset \operatorname{Ann} M$.

(iii bis) $\mathfrak{p} \supset \mathfrak{r}(\operatorname{Ann} M)$.

Proposition 1.3. *Let N be a finitely generated A-module. Then*

$$\operatorname{Ass} \operatorname{Hom}_A(N, M) = \operatorname{Supp} N \cap \operatorname{Ass} M.$$

2. Depth

Throughout this section, A denotes a commutative ring, I an ideal of A, and M and N two A-modules. We denote by X the prime spectrum of A (its structure sheaf will not be used in this section), and by Y the closed set defined by I, namely $Y = \operatorname{Supp}(A/I) = \{\mathfrak{p} \in X \mid \mathfrak{p} \supset I\}$. Original French p. 28

Lemma 2.1. *Suppose that A is noetherian, that M and N are finitely generated, and that* $\operatorname{Supp} N = Y$. *Then the following assertions are equivalent:*

(i) $\operatorname{Hom}_A(N, M) = 0$.

(ii) $\operatorname{Supp} N \cap \operatorname{Ass} M = \varnothing$.

(iii) *The ideal I is not a zero divisor on M, meaning that, for every $m \in M$, the equality $Im = 0$ implies $m = 0$.*

(iv) *There exists an M-regular element in I. (An element $a \in A$ is called M-regular if multiplication by a on M is injective.)*

(v) *For every $\mathfrak{p} \in Y$, the maximal ideal $\mathfrak{p}A_\mathfrak{p}$ of the local ring $A_\mathfrak{p}$ is not associated with $M_\mathfrak{p}$; in symbols, $\mathfrak{p}A_\mathfrak{p} \notin \operatorname{Ass} M_\mathfrak{p}$.*

Proof. The equivalence (i)⇔(ii) follows because $\operatorname{Ass} \operatorname{Hom}_A(N, M) = \varnothing$ is equivalent to (ii) by Proposition 1.3, and to (i) by an immediate consequence of Proposition 1.2.

The implication (iii)⇒(ii) follows by contradiction. If $\mathfrak{p} \in \operatorname{Supp} N \cap \operatorname{Ass} M$, then $\mathfrak{p} \supset I$ and there is an $m \in M$ whose annihilator is $\mathfrak{p}$; hence $Im \subset \mathfrak{p}m = 0$, contrary to (iii). Original French p. 29

The implication (iv)⇒(iii) is immediate.

Conditions (ii) and (iv) are equivalent because $\operatorname{Supp} N = Y$, so (ii) says that I is contained in no prime ideal associated with M. Since those associated ideals are prime and finite in number, this is equivalent to saying that I is not contained in their union. By Proposition 1.1(ii), that union is precisely the set of elements of A which are not M-regular.

To prove (i)⇒(v), let $\mathfrak{p} \in Y$. From $\operatorname{Hom}_A(N, M) = 0$ and the formula

$$\big(\operatorname{Hom}_A(N, M)\big)_\mathfrak{p} = \operatorname{Hom}_{A_\mathfrak{p}}(N_\mathfrak{p}, M_\mathfrak{p})$$

one obtains $\operatorname{Hom}_{A_\mathfrak{p}}(N_\mathfrak{p}, M_\mathfrak{p}) = 0$. Thus, by Proposition 1.3,

$$\operatorname{Supp} N_\mathfrak{p} \cap \operatorname{Ass} M_\mathfrak{p} = \varnothing.$$

But $\mathfrak{p}A_\mathfrak{p} \in \operatorname{Supp} N_\mathfrak{p}$, so $\mathfrak{p}A_\mathfrak{p} \notin \operatorname{Ass} M_\mathfrak{p}$.

Finally, suppose (v), and let $\mathfrak{p} \in \operatorname{Ass} M$. There is an $m \in M$ with annihilator $\mathfrak{p}$. Its canonical image in $M_\mathfrak{p}$ is nonzero, and its annihilator contains $\mathfrak{p}A_\mathfrak{p}$ and hence equals it. Thus $\mathfrak{p}A_\mathfrak{p} \in \operatorname{Ass} M_\mathfrak{p}$, so (v) gives $\mathfrak{p} \notin Y$. This proves (i). □

We shall now work with these conditions after replacing the functor Hom by its derived functors.

Theorem 2.2. *Let A be a commutative ring, I an ideal of A, M an A-module, and n an integer.*

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(a) *If there is a sequence $f_1, \dots, f_{n+1}$ of elements of I which is M-regular (that is, f_1 is M-regular and f_{i+1} is regular on $M/(f_1, \dots, f_i)M$ for $i \leq n$), then, for every A-module N annihilated by a power of I,*

$$\operatorname{Ext}^i_A(N, M) = 0 \qquad \text{for } i \leq n.$$

(b) *If, in addition, A is noetherian, M is finitely generated, and there is a finitely generated A-module N with $\operatorname{Supp} N = V(I)$ and $\operatorname{Ext}^i_A(N, M) = 0$ for $i \leq n$, then there exists an M-regular sequence $f_1, \dots, f_{n+1}$ of elements of I.*

We first prove (a), by induction. If $n < 0$, the assertion is empty. Suppose $n \geq 0$ and that (a) has been proved for every $n' < n$. By hypothesis there is an M-regular element $f_1 \in I$. Denote by f_1^i multiplication by f_1 on $\operatorname{Ext}^i_A(N, M)$ and by f_1^M multiplication by f_1 on M. The sequence

$$0 \longrightarrow M \xrightarrow{f_1^M} M \longrightarrow M/f_1M \longrightarrow 0 \tag{III.1}$$

is exact, and therefore so is

$$\operatorname{Ext}^{i-1}_A(N, M/f_1M) \xrightarrow{\delta} \operatorname{Ext}^i_A(N, M) \xrightarrow{f_1^i} \operatorname{Ext}^i_A(N, M).$$

By hypothesis, $I^r N = 0$ for some r, so f_1^0 is nilpotent. Since Ext^i is a universal cohomological functor, the same is true of f_1^i for every i. Moreover, M/f_1M has a regular sequence of length n. The induction hypothesis therefore gives

$$\operatorname{Ext}^{i-1}_A(N, M/f_1M) = 0 \qquad \text{for } i \leq n.$$

Consequently, for $i \leq n$, f_1^i is both injective and nilpotent, so $\operatorname{Ext}^i_A(N, M) = 0$.

We now prove (b), again by induction. If $n < 0$, the assertion is empty. If $n = 0$, it follows from Lemma 2.1, (i)⇒(iv). If $n > 0$, the case $n = 0$ gives an M-regular element $f_1 \in I$. From (III.1) one obtains the exact sequence

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$$\operatorname{Ext}^{i-1}_A(N, M) \longrightarrow \operatorname{Ext}^{i-1}_A(N, M/f_1M) \longrightarrow \operatorname{Ext}^i_A(N, M). \tag{III.2}$$

The hypotheses of (b) therefore hold for M/f_1M and $n - 1$. By induction there is a regular sequence of n elements of I on M/f_1M, and hence an M-regular sequence of $n + 1$ elements of I beginning with f_1.

This theorem suggests the following generalization of the classical definition of the depth of a finitely generated module over a noetherian ring.

Definition 2.3. *Let A be a commutative ring with identity, M an A-module, and I an ideal of A. The* I-depth of M, *denoted* $\operatorname{prof}_I M$, *is the supremum in $\mathbb{N} \cup \{+\infty\}$ of the natural numbers n such that, for every finitely generated A-module N annihilated by a power of I,*

$$\operatorname{Ext}^i_A(N, M) = 0 \qquad \text{for every } i < n.$$

It follows from the preceding theorem that, if n is the supremum of the lengths of M-regular sequences of elements of I, then $n \leq \operatorname{prof}_I M$. More precisely:

Proposition 2.4. *Let A be a commutative ring, I an ideal of A, M an A-module, and $n \in \mathbb{N}$. Consider the assertions:*

(1) $n \leq \operatorname{prof}_I M$.

(2) *For every finitely generated A-module N annihilated by a power of I, one has*

$$\operatorname{Ext}^i_A(N, M) = 0 \qquad \text{for } i < n.$$

(3) *There exists a finitely generated A-module N such that* $\operatorname{Supp} N = V(I)$*, and such that* $\operatorname{Ext}^i_A(N, M) = 0$ *whenever* $i < n$. Original French p. 32

(4) *There is an M-regular sequence of length n consisting of elements of I.*

The following logical implications hold:

$$\begin{array}{ccccc} (1) & \Longleftrightarrow & (2) & \Longleftarrow & (4) \\ & & \Downarrow & & \\ & & (3) & & \end{array}$$

If A is noetherian and M is finitely generated, all four conditions are equivalent.

Proof. Conditions (1) and (2) are equivalent by definition, and (2)⇒(3) follows by taking $N = A/I$. The implication (4)⇒(2) is Theorem 2.2(a). Finally, when A is noetherian and M is finitely generated, (3)⇒(4) is Theorem 2.2(b). □

From now until the end of the section, we assume that A is noetherian and M is finitely generated.

Corollary 2.5. *If $f \in I$ is M-regular, then*

$$\operatorname{prof}_I M = \operatorname{prof}_I(M/fM) + 1.$$

Indeed, if $n \leq \operatorname{prof}_I(M/fM)$, there is an (M/fM)-regular sequence $f_1, \ldots, f_n$ of elements of I. Hence $f, f_1, \ldots, f_n$ is M-regular, so $n+1 \leq \operatorname{prof}_I M$ and $\operatorname{prof}_I M \geq \operatorname{prof}_I(M/fM) + 1$. Conversely, for every finitely generated A-module N annihilated by a power of I, the exact sequence (III.2) shows that if $i < \operatorname{prof}_I M$, then $\operatorname{Ext}^{i-1}_A(N, M/fM) = 0$. Thus $\operatorname{prof}_I M - 1 \leq \operatorname{prof}_I(M/fM)$.

Corollary 2.6. *Every finite M-regular sequence consisting of elements of I can be extended to a maximal M-regular sequence, whose length is necessarily the I-depth of M.*

Remark 2.7. *One can scarcely refrain from saying that the greater the depth of an A-module, the more beautiful it is. A module whose support does not meet $V(I)$ is among the most beautiful: indeed,* $\operatorname{prof}_I M$ *is finite if and only if* $\operatorname{Supp} M \cap V(I) \neq \varnothing$. Original French p. 33

Remark 2.8. *If A is a semi-local ring, let $\mathfrak{r}(A)$ be its radical and $k = A/\mathfrak{r}(A)$ its residue ring. The relevant notion of depth is obtained by taking $I = \mathfrak{r}(A)$; accordingly, we write simply* $\operatorname{prof} M$ *for the $\mathfrak{r}(A)$-depth of an A-module M. This is the notion of "homological codimension" (cf. Serre,* op. cit.*), formerly denoted* $\operatorname{codh}_A M$*, defined as the infimum of the integers i such that* $\operatorname{Ext}^i_A(k, M) \neq 0$. *Indeed,* $\operatorname{Supp} k = V(\mathfrak{r}(A))$.

Proposition 2.9. *If A is noetherian and M is finitely generated, then*

$$\operatorname{prof}_I M = \inf_{\mathfrak{p} \in V(I)} \operatorname{prof} M_{\mathfrak{p}}.$$

Corollary 2.10. *If A is a noetherian semi-local ring and M is a finitely generated A-module, then*

$$\operatorname{prof} M = \inf_{\mathfrak{m}} \operatorname{prof} M_{\mathfrak{m}},$$

where $\mathfrak{m}$ *ranges over the maximal ideals of* A.

The corollary follows immediately from Proposition 2.9, since the prime ideals containing the radical are precisely the maximal ideals.

We now prove Proposition 2.9. Let $f \in I$. If f is M-regular and $\mathfrak{p} \supset I$, then the image g of f in $A_\mathfrak{p}$ belongs to the maximal ideal $\mathfrak{p}A_\mathfrak{p}$ and is $M_\mathfrak{p}$-regular, as follows from the exact sequence

$$0 \longrightarrow M_\mathfrak{p} \xrightarrow{g'} M_\mathfrak{p} \longrightarrow (M/fM)_\mathfrak{p} \longrightarrow 0, \tag{III.3}$$

where g' denotes multiplication by g on $M_\mathfrak{p}$. This also shows that $(M/fM)_\mathfrak{p} \simeq M_\mathfrak{p}/gM_\mathfrak{p}$. Applying Corollary 2.5 to M and to $M_\mathfrak{p}$ gives, by induction, $\operatorname{prof}_I M \leq \nu(M)$, where

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$$\nu(M) = \inf_{\mathfrak{p} \in V(I)} \operatorname{prof} M_\mathfrak{p}.$$

More precisely, induction gives $\nu(M) = \nu(M/fM) + 1$ whenever f is M-regular. It remains to prove that, if $\nu(M) \geq 1$, there is an M-regular element in I. Applying Lemma 2.1 to $M_\mathfrak{p}$, $A_\mathfrak{p}$, and $\mathfrak{p}A_\mathfrak{p}$ for every $\mathfrak{p} \in V(I)$ shows that $\mathfrak{p}A_\mathfrak{p} \notin \operatorname{Ass} M_\mathfrak{p}$. Applying the same lemma to A, M, and I gives the conclusion.

Proposition 2.11. *Let* $u : A \to B$ *be a homomorphism of noetherian rings, let* I *be an ideal of* A*, and let* M *be a finitely generated* A*-module. Put* $I_B = I \otimes_A B$ *and* $M_B = M \otimes_A B$*. If* B *is flat over* A*, then*

$$\operatorname{prof}_{I_B} M_B \geq \operatorname{prof}_I M;$$

if B *is faithfully flat over* A*, equality holds.*

Indeed, put $N = A/I$. Flatness gives $N \otimes_A B = B/I_B$; set $N_B = N \otimes_A B$. Again by flatness and the noetherian hypotheses,

$$\operatorname{Ext}^i_B(N_B, M_B) = \operatorname{Ext}^i_A(N, M) \otimes_A B.$$

Thus $\operatorname{Ext}^i_A(N, M) = 0$ implies $\operatorname{Ext}^i_B(N_B, M_B) = 0$, and the converse holds when B is faithfully flat over A.

3. Depth and Topological Properties

Lemma 3.1. *Let* X *be a topological space,* Y *a closed subspace, and* F *a sheaf of abelian groups on* X*. Put* $U = X - Y$*. If* n *is an integer, the following conditions are equivalent:*

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(i) $\mathcal{H}^i_Y(X, F) = 0$ *for* $i < n$.

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(ii) *For every open subset* V *of* X*, the group homomorphism*

$$\mathrm{H}^i(V, F) \longrightarrow \mathrm{H}^i(V \cap U, F)$$

is bijective for $i < n - 1$ *and injective for* $i = n - 1$.

(iii) *For every open subset* V *of* X*,*

$$\mathrm{H}^i_{Y \cap V}(V, F|_V) = 0 \qquad \text{for } i < n.$$

Proof. First, (ii) and (iii) are equivalent. Indeed, let V be an open subset of X, and put $X' = V$, $Y' = Y \cap V$, $F' = F|_V$, and $U' = X' - Y'$. Since Y' is closed in X', there is an exact sequence

$$\mathrm{H}^i_{Y'}(X', F') \longrightarrow \mathrm{H}^i(X', F') \xrightarrow{\rho_i} \mathrm{H}^i(U', F') \longrightarrow \mathrm{H}^{i+1}_{Y'}(X', F').$$

If the two outer terms vanish, ρ_i is bijective; if the term on the left vanishes, ρ_i is injective. Thus (iii) implies (ii). Conversely, if $i < n$, then $\mathrm{H}^i_{Y'}(X', F')$ vanishes because ρ_i is injective and ρ_{i-1} is surjective.

Next, (i) implies (iii). The local-to-global spectral sequence gives

$$\mathrm{H}^p(X, \mathcal{H}^q_Y(X, F)) \Longrightarrow \mathrm{H}^{p+q}_Y(X, F).$$

By hypothesis, $\mathcal{H}^q_Y(X, F) = 0$ for $q < n$; hence $\mathrm{H}^{p+q}_Y(X, F) = 0$ whenever $p + q < n$.

Finally, (iii) implies (i), since (iii) says that the presheaf

$$V \longmapsto \mathrm{H}^i_{Y\cap V}(V, F|_V)$$

is zero. Its associated sheaf is therefore zero; because Y is closed, that associated sheaf is $\mathcal{H}^i_Y(X, F)$. □

Remark 3.2. *The equivalence of (i) and (ii) was proved in Proposition I.2.13. As was observed there, if $n \geq 2$, the condition that ρ_{n-1} be injective may be omitted.*[2]

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Proposition 3.3. *Let X be a locally noetherian prescheme, Y a closed subprescheme of X, and F a coherent $\mathcal{O}_X$-module. The conditions of Lemma 3.1 are equivalent to each of the following conditions:*

(iv) *For every $x \in Y$, one has* $\operatorname{prof} F_x \geq n$.

(v) *For every coherent $\mathcal{O}_X$-module G on X whose support is contained in Y,*

$$\mathcal{E}xt^i_{\mathcal{O}_X}(G, F) = 0 \qquad \text{for } i < n.$$

(vi) *There exists a coherent $\mathcal{O}_X$-module G whose support equals Y and such that*

$$\mathcal{E}xt^i_{\mathcal{O}_X}(G, F) = 0 \qquad \text{for } i < n.$$

If X is affine, Proposition 2.4 supplies what is needed to establish the equivalence of the three conditions in Proposition 3.3. These conditions are local except for (v)⇒(vi); for that implication one may take $G = \mathcal{O}_Y$ and invoke Proposition 2.4 again. It therefore suffices to prove (i)⇒(vi) and (v)⇒(i).

Let J be the ideal of Y; it is a coherent sheaf of ideals. Put $\mathcal{O}_{Y_m} = \mathcal{O}_X/J^{m+1}$. This is a coherent $\mathcal{O}_X$-module whose support equals Y, and Theorem II.6(b) gives

$$\mathcal{H}^i_Y(X, F) = \varinjlim_m \mathcal{E}xt^i_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F).$$

Thus (v) implies (i). Moreover, the transition morphisms in the projective system of the $\mathcal{O}_{Y_m}$ are epimorphisms.

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If the functor $\mathcal{E}xt^i$ is left exact in its first argument—at least when that argument lies in the category of coherent $\mathcal{O}_X$-modules with support contained in Y—then the transition morphisms of the inductive system obtained by applying $\mathcal{E}xt^i$ to the $\mathcal{O}_{Y_m}$ are injective. Condition (i) makes the limit zero, and therefore makes $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F)$ zero for every m.

We argue by induction. The assertion is trivial for $n < 0$. Suppose that (i)⇒(vi) has been proved in degrees below q. Then (i)⇒(v), so the long exact $\mathcal{E}xt$ sequence shows that $\mathcal{E}xt^q$ is left exact in its first argument. Hence $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F) = 0$ for every m. Thus (i)⇒(vi) holds through degree q. □

Example 3.4. *Let A be a noetherian local ring, $\mathfrak{m}$ its maximal ideal, M a finitely generated A-module, and n an integer. Put $X = \operatorname{Spec}(A)$, $Y = \{\mathfrak{m}\}$, and $U = X - Y$. Let F be the sheaf associated with M. The following conditions are equivalent:*

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[2] The original edition repeated a proof here, but that proof was not entirely correct.

(1) $\operatorname{prof} M \geq n$.

(2) *The natural homomorphism*

$$\mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F)$$

is injective if $i = n-1$ *and bijective if* $i < n-1$.

(3) $\operatorname{Ext}^i_A(k, M) = 0$ *if* $i < n$, *where* $k = A/\mathfrak{m}$.

(4) $\mathrm{H}^i_Y(X, F) = 0$ *if* $i < n$.

Taking Remark 3.2 into account, one obtains the following corollary.

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Corollary 3.5. *Let* X *be a locally noetherian prescheme,* Y *a closed subprescheme of* X, *and* F *a coherent* $\mathcal{O}_X$*-module. The following conditions are equivalent:*

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(1) *For every* $x \in Y$, $\operatorname{prof} F_x \geq 2$.

(2) *For every open subset* V *of* X, *the natural homomorphism*

$$\mathrm{H}^0(V, F) \longrightarrow \mathrm{H}^0(V \cap (X - Y), F)$$

is bijective.

Theorem 3.6 (Hartshorne). *Let* X *be a locally noetherian prescheme and* Y *a closed subprescheme of* X. *Suppose that* $\operatorname{prof} \mathcal{O}_{X,x} \geq 2$ *for every* $x \in Y$. *Then the natural map*

$$\pi_0(X) \longrightarrow \pi_0(X - Y)$$

is bijective.

Proof. Since X is locally noetherian, it is locally connected. It therefore suffices to prove that X is connected if and only if $X - Y$ is connected. A locally ringed space $(X, \mathcal{O}_X)$ is connected if and only if $\mathrm{H}^0(X, \mathcal{O}_X)$ is not a direct product of two nonzero rings. By Corollary 3.5, applied with $F = \mathcal{O}_X$, the hypothesis implies that

$$\mathrm{H}^0(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^0(X - Y, \mathcal{O}_X)$$

is an isomorphism, and the conclusion follows. □

Corollary 3.7. *Let* X *be a locally noetherian prescheme. Let* d *be an integer such that* $\dim \mathcal{O}_{X,x} \geq d$ *implies* $\operatorname{prof} \mathcal{O}_{X,x} \geq 2$. *If* X *is connected, then it is connected in codimension* $d-1$*: namely, if* X' *and* X'' *are two irreducible components of* X, *there is a sequence of irreducible components of* X,

$$X' = X_0, X_1, \ldots, X_n = X'',$$

such that, for every i *with* $0 \leq i < n$, *the codimension in* X *of* $X_i \cap X_{i+1}$ *is at most* $d-1$.

First note that if X is Cohen–Macaulay, then $d = 2$ has the property just invoked. Recall also that the codimension of Y in X is defined as the infimum of the dimensions of the local rings of X at the points of Y.

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Proof (reduction). Let $\mathcal{F}$ be the collection of closed subsets of X whose codimension in X is at least d. This is an antifilter of closed subsets of X. A closed subset Y belongs to $\mathcal{F}$ if and only if, for every $y \in Y$, there is an open neighborhood V of y such that $Y \cap V$ has codimension at

least d in V. Finally, if X is connected and $Y \in \mathcal{F}$, then $X - Y$ is connected by Hartshorne's theorem. The corollary therefore follows from the next lemma, which is purely topological.

Lemma 3.8. *Let X be a connected, locally noetherian topological space, and let $\mathcal{F}$ be an antifilter of closed subsets of X. Suppose that every closed subset $Y \subset X$ which belongs locally to $\mathcal{F}$—that is, for every $x \in X$ there are an open neighborhood V of x in X and a $Y' \in \mathcal{F}$ such that $V \cap Y = V \cap Y'$—belongs to $\mathcal{F}$. The following conditions are equivalent:*

(i) *For every $Y \in \mathcal{F}$, the complement $X - Y$ is connected.*

(ii) *If X' and X'' are distinct irreducible components, there is a sequence $X_0, X_1, \ldots, X_n$ of irreducible components of X such that $X' = X_0$, $X'' = X_n$, and*

$$X_i \cap X_{i+1} \notin \mathcal{F} \qquad (0 \leq i < n).$$

Proof. Assume (ii), and let $Y \in \mathcal{F}$. We must prove that $U = X - Y$ is connected. If U' and U'' are two irreducible components of U, there are irreducible components X' and X'' of X such that $X' \cap U = U'$ and $X'' \cap U = U''$. Choose a sequence $X_0, \ldots, X_n$ having the property in (ii), and put $U_i = X_i \cap U$ for $0 \leq i \leq n$. The U_i are irreducible components of U, and $U_i \cap U_{i+1}$ is nonempty for $0 \leq i < n$. Otherwise $X_i \cap X_{i+1} \subset Y$ would belong to $\mathcal{F}$, contrary to the choice of the X_i. Hence U is connected. Original French p. 40

Conversely, assume (i). Let

$$Y = \bigcup_{X', X''} (X' \cap X''),$$

where the union is over distinct irreducible components X', X'' of X for which $X' \cap X'' \in \mathcal{F}$. This family of intersections is locally finite because X is locally noetherian; its members are closed, so Y is closed. Moreover, Y belongs locally to $\mathcal{F}$, and hence $Y \in \mathcal{F}$. Thus $U = X - Y$ is connected.

Let X' and X'' be distinct irreducible components of X, and let U' and U'' be their traces on U; by the construction of Y, these traces are nonempty. They are irreducible components of U. Since U is connected and locally noetherian, there is a sequence $U_0, \ldots, U_n$ of irreducible components of U such that $U_0 = U'$, $U_n = U''$, and, for $0 \leq i < n$,

$$U_i \cap U_{i+1} \neq \varnothing \quad \text{and} \quad U_i \cap U_{i+1} \neq U_i.$$

Let $X_0, \ldots, X_n$ be the irreducible components of X satisfying $X_i \cap U = U_i$. If $X_i \cap X_{i+1} \in \mathcal{F}$, then by the construction of Y one would have either $U_i \cap U_{i+1} = \varnothing$ or $U_i = U_{i+1}$, contradicting the choice of the U_i. □

Corollary 3.9. *Let A be a noetherian local ring. Suppose that, for every prime ideal $\mathfrak{p}$ of A,*

$$(\dim A_{\mathfrak{p}} \geq 2) \Longrightarrow (\operatorname{prof} A_{\mathfrak{p}} \geq 2).$$

Suppose also that A satisfies the chain condition.[3] *Then every minimal prime ideal $\mathfrak{p}$ of A satisfies* $\dim(A/\mathfrak{p}) = \dim A$. *Equivalently, all irreducible components of* $\operatorname{Spec} A$ *have the same dimension, namely* $\dim A$.

Indeed, put $X = \operatorname{Spec} A$. If X' and X'' are two irreducible components of X, Corollary 3.7 joins them by a chain with the stated properties. It is therefore enough to prove that two successive components have the same dimension, which follows from the second hypothesis.

Example 3.10. *Let X be the union of two complementary linear subspaces of respective* Original French p. 41

[3] Cf. EGA 0_{IV}, 14.3.2.

dimensions 2 *and* 3 *in a vector space of dimension* 5. *More precisely, let* $X = \operatorname{Spec} A$, *where*

$$A = B/(\mathfrak{p} \cap \mathfrak{q}), \qquad B = k[X_1, \ldots, X_5],$$

$\mathfrak{p} = (X_1, X_2, X_3)$, *and* $\mathfrak{q} = (X_4, X_5)$. *Removing the intersection point* x *of the two linear subspaces disconnects* X. *Consequently, the depth of* $\mathcal{O}_{X,x}$ *is* 1, *since it cannot be at least* 2 *by Theorem 3.6. The failure of the preceding corollary's equidimensionality conclusion gives another reason.*

More generally, let X *be the union of two linear subspaces of dimensions* $p, q \geq 2$ *in a vector space of dimension* $p + q$. *Under no embedding of* X *into a regular scheme is* X *even set-theoretically a complete intersection at the origin. Indeed, after possibly modifying its scheme structure without changing the underlying topological space near the origin,* X *would then be Cohen–Macaulay and hence of depth at least* 2 *at the origin, which it is not.*

Remark 3.11. *Let* X *be a locally noetherian prescheme,* Y *a closed subprescheme of* X, *and* F *an* $\mathcal{O}_X$*-module. Depth is a purely topological notion, expressed in terms of the vanishing of* $\mathcal{H}^i_Y(X, F)$ *for* $i < n$. *One also wishes to study these sheaves for a fixed* i, *or for* $i > n$. *The following result addresses this question.*

Lemma 3.12. *Let* m *be an integer. The equality* $\mathcal{H}^i_Y(X, F) = 0$ *for every* $i > m$ *and every coherent* F *holds if and only if it holds for* $F = \mathcal{O}_X$.

By passage to a direct limit, it then holds for every quasi-coherent sheaf. For example, if Y can be locally described by m equations—or, as one says, if Y is locally a set-theoretic complete intersection (as happens, for example, when X and Y are nonsingular)—the computation of $\mathcal{H}^i_Y(X, F)$ by the Koszul complex shows that these sheaves vanish for $i > m$. This fact was used implicitly in Example 3.10. The cohomological condition is nevertheless insufficient, as the following example shows.

Example 3.13. *Let* $X = \operatorname{Spec} A$, *where* A *is a normal noetherian local ring of dimension* 2, *and let* Y *be a curve in* X. *The complement* $X - Y$ *can be shown to be affine. Hence, for* $i > 1$, *the supported-cohomology exact sequence gives*

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$$\mathrm{H}^i_Y(X, \mathcal{O}_X) \simeq \mathrm{H}^{i-1}(X - Y, \mathcal{O}_X) = 0.$$

Consequently the corresponding local-cohomology sheaf $\mathcal{H}^i_Y(\mathcal{O}_X)$ *vanishes. Nevertheless, one can construct a curve which is not described by a single equation.*

We shall seek[4] conditions ensuring that $\mathcal{H}^i_Y(X, F)$ is coherent for a fixed i. In general it is not: for example, consider $\mathrm{H}^n_{\mathfrak{m}}(A)$ when A is a noetherian local ring of dimension $n > 0$. If A is a discrete valuation ring with fraction field K, then

$$\mathrm{H}^1_{\mathfrak{m}}(A) \simeq K/A,$$

which is not a finitely generated A-module.

To illuminate the question, it is equivalent to the following problem. Let $f : U \to X$ be an open immersion and G a coherent sheaf on U. Find criteria ensuring that the higher direct images $\mathrm{R}^i f_* G$ are coherent sheaves on X for a fixed i. Such conditions are necessary for the use of formal geometry encountered in Exposé IX and the subsequent exposés.

[4] Cf. Exposé VIII.

Exposé IV

Dualizing Modules and Functors

1. Generalities on Functors of Modules

Fix a commutative noetherian ring A. Let $\mathcal{C}$ be the category of finitely generated A-modules, $\mathcal{C}'$ the category of all A-modules, and **Ab** the category of abelian groups. This section studies certain properties of additive functors

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$$T : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{Ab}.$$

For $M \in \mathrm{Ob}\,\mathcal{C}$, the group $T(M)$ has a canonical A-module structure. If $f_M : M \to M$ denotes multiplication by $f \in A$, then f acts on $T(M)$ as $T(f_M)$. In other words, T factors as

$$\mathcal{C}^{\mathrm{op}} \xrightarrow{T_0} \mathcal{C}' \longrightarrow \mathbf{Ab},$$

where the second arrow is the forgetful functor. We henceforth regard $T(M)$ as endowed with this A-module structure.

Compose the isomorphism $M \xrightarrow{\sim} \mathrm{Hom}_A(A, M)$ with the map

$$\mathrm{Hom}_A(A, M) \longrightarrow \mathrm{Hom}_A(T(M), T(A))$$

induced by T. This gives the following mutually adjoint forms of the same pairing:

$$\begin{aligned} M &\longrightarrow \mathrm{Hom}_A(T(M), T(A)), \\ M \times T(M) &\longrightarrow T(A), \\ T(M) &\longrightarrow \mathrm{Hom}_A(M, T(A)). \end{aligned}$$

They define a natural morphism of contravariant functors

$$\varphi_T : T \longrightarrow \mathrm{Hom}_A(-, T(A)).$$

Proposition 1.1. *The following two properties are equivalent:*

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i) *φ_T is an isomorphism of functors.*

ii) *T is left exact.*

The implication (i)⇒(ii) is immediate. For the converse, use the following fact: if $u : F \to F'$ is a morphism between additive left-exact functors from $\mathcal{C}^{\mathrm{op}}$ to **Ab** and $u(A)$ is an isomorphism, then u is an isomorphism. Here one uses that A is noetherian, so every finitely generated A-module is finitely presented.

Remark 1.2. *In particular, the representable functors $T : \mathcal{C}'^{\mathrm{op}} \to \mathbf{Ab}$ are precisely those which commute with arbitrary inverse limits in $\mathcal{C}'^{\mathrm{op}}$ (indexed by a preordered set, not necessarily filtered).*

Let $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Ab})_{\mathrm{lex}}$ denote the full subcategory of additive functors that are left exact. We have proved an equivalence of categories

$$\mathcal{C}' \xrightarrow{\sim} \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Ab})_{\mathrm{lex}},$$

with mutually quasi-inverse functors

$$H \longmapsto \mathrm{Hom}_A(-, H), \qquad T \longmapsto T(A).$$

Let J be an ideal of A, put $Y = V(J) \subset \mathrm{Spec}\, A$, and let $\mathcal{C}_Y$ be the full subcategory of $\mathcal{C}$ consisting of finitely generated A-modules M with $\mathrm{Supp}\, M \subset Y$. Then

$$\mathcal{C}_Y = \bigcup_n \mathcal{C}^{(n)},$$

where $\mathcal{C}^{(n)}$ is the full subcategory of modules satisfying $J^n M = 0$.

Proposition 1.3. *With this notation, let $T : \mathcal{C}_Y^{\mathrm{op}} \to \mathbf{Ab}$ be an additive functor, and put*

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$$H = \varinjlim_n T(A/J^n).$$

There is a natural morphism

$$\varphi_T : T \longrightarrow \mathrm{Hom}_A(-, H),$$

and the following conditions are equivalent:

i) *φ_T is an isomorphism.*

ii) *T is left exact.*

Proof. First define φ_T. For $M \in \mathrm{Ob}\, \mathcal{C}_Y$, choose n with $J^n M = 0$. Then M is an A/J^n-module. If T_n is the restriction of T to $\mathcal{C}^{(n)\mathrm{op}}$ and $H_n = T(A/J^n)$, the construction of Proposition 1.1 gives

$$T_n \longrightarrow \mathrm{Hom}_A(-, H_n).$$

Composing with $H_n \to H$ gives, for every such M,

$$T(M) = T_n(M) \longrightarrow \mathrm{Hom}_A(M, H),$$

and these maps define φ_T.

It is clear that (i) implies (ii). Conversely, suppose T is left exact and let $M \in \mathrm{Ob}\, \mathcal{C}^{(n)}$. Proposition 1.1 applied over A/J^n gives

$$T_n(M) \xrightarrow{\sim} \mathrm{Hom}_A(M, H_n).$$

For every $n' \geq n$, Proposition 1.1 applied over $A/J^{n'}$ gives a compatible isomorphism with $H_{n'}$. Hence

$$T(M) \simeq \varinjlim_{n' \geq n} \mathrm{Hom}_A(M, H_{n'}).$$

Because this is a filtered direct limit and M is finitely presented,

$$\varinjlim_{n' \geq n} \mathrm{Hom}_A(M, H_{n'}) \xrightarrow{\sim} \mathrm{Hom}_A\left(M, \varinjlim_{n' \geq n} H_{n'}\right) = \mathrm{Hom}_A(M, H).$$

Consequently, if $\mathcal{C}'_Y$ denotes the category of all A-modules supported in Y, without a finite-generation condition, there is a natural equivalence

$$\mathcal{C}'_Y \xrightarrow{\sim} \mathrm{Fun}(\mathcal{C}_Y^{\mathrm{op}}, \mathbf{Ab})_{\mathrm{lex}}.$$

□

Application

Retain the preceding notation, and let

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$$T^* : \mathcal{C}_Y^{\mathrm{op}} \longrightarrow \mathbf{Ab}$$

be an exact ∂-functor. For every $i \in \mathbb{Z}$, put

$$H_n^i = T^i(A/J^n), \qquad H^i = \varinjlim_n H_n^i.$$

Theorem 1.4. *Let $n \in \mathbb{Z}$. Suppose there is an $i_0 \in \mathbb{Z}$ such that $T^i = 0$ for every $i < i_0$. Then the following three conditions are equivalent:*

i) *$T^i = 0$ for every $i < n$.*

ii) *$H^i = 0$ for every $i < n$.*

iii) *There exists an $M_0 \in \mathcal{C}_Y$ such that $\operatorname{Supp} M_0 = Y$ and $T^i(M_0) = 0$ for every $i < n$.*

Proof. Clearly (i) implies (ii) and (iii), taking $M_0 = A/J$. We prove (ii)⇒(i) by induction on n. It is true for $n = i_0$; suppose it has been proved through n. If $H^i = 0$ for every $i < n+1$, the induction hypothesis gives $T^i = 0$ for $i < n$. The vanishing of T^{n-1} makes T^n left exact, so Proposition 1.3 gives

$$T^n \xrightarrow{\sim} \operatorname{Hom}_A(-, H^n) = 0.$$

It remains to prove (iii)⇒(ii), again by induction. The assertion is true for $n = i_0$. Suppose it has been proved through n, and let $M_0 \in \mathcal{C}_Y$ satisfy $\operatorname{Supp} M_0 = Y$ and $T^i(M_0) = 0$ for every $i < n+1$. The induction hypothesis gives $H^i = 0$ for $i < n$; it remains to show $H^n = 0$. The preceding implication gives $T^{n-1} = 0$, and therefore

$$T^n \xrightarrow{\sim} \operatorname{Hom}_A(-, H^n).$$

Moreover,

$$\operatorname{Ass} \operatorname{Hom}_A(M_0, H^n) = \operatorname{Supp} M_0 \cap \operatorname{Ass} H^n = \operatorname{Ass} H^n,$$

because

$$\operatorname{Ass} H^n \subset \operatorname{Supp} H^n \subset Y = \operatorname{Supp} M_0.$$

Thus

$$T^n(M_0) = 0 \quad \Longleftrightarrow \quad \operatorname{Ass} H^n = \varnothing \quad \Longleftrightarrow \quad H^n = 0,$$

which completes the proof. □

2. Characterization of Exact Functors

The ring A remains commutative and noetherian. Retain the notation of Proposition 1.3:

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$$Y = V(J), \qquad T : \mathcal{C}_Y^{\mathrm{op}} \longrightarrow \mathbf{Ab}, \qquad H = \varinjlim_n T(A/J^n).$$

Assume that T is left exact, so that

$$T(M) \xrightarrow{\sim} \operatorname{Hom}_A(M, H).$$

Proposition 2.1. *The following properties are equivalent:*

i) *T is exact.*

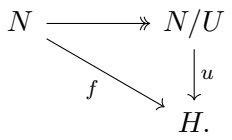
ii) *H is injective in $\mathcal{C}'$, the category of all A-modules.*

Proof. Only (i)⇒(ii) needs proof. Suppose the restriction of $\mathrm{Hom}_A(-, H)$ to $\mathcal{C}_Y^{\mathrm{op}}$ is exact. Since A is noetherian, it is enough to prove the following extension property: whenever $N \subset M$ are finitely generated A-modules, every homomorphism $f : N \to H$ extends to a homomorphism $\bar{f} : M \to H$.

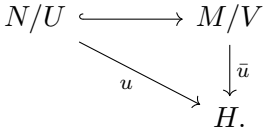
By the definition of H and finite generation of N, there is an integer n such that $J^n f(N) = 0$. Give M and N their J-adic topologies. By Krull's theorem, the J-adic topology on N is equivalent to the topology induced from the J-adic topology on M. Hence there is a $V = J^k M$ such that

$$U = V \cap N \subset J^n N.$$

Thus $U \subset \ker f$, and f factors as



Both N/U and M/V belong to $\mathcal{C}_Y$. Exactness therefore extends u to a map $\bar{u} : M/V \to H$:



The composite $M \to M/V \xrightarrow{\bar{u}} H$ is the required extension $\bar{f}$. □

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Retain the notation $Y = V(J)$ and let $\mathcal{C}_Y$ denote the category of finitely generated A-modules whose support is contained in Y.

Corollary 2.2. *Let K be an injective A-module. Then the submodule*

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$$H^0_J(K) = \{x \in K : J^n x = 0 \text{ for some } n \geq 0\}$$

of elements annihilated by a suitable power of J is injective.

Proof. By Proposition 2.1, it is enough to verify that the restriction to $\mathcal{C}_Y^{\mathrm{op}}$ of $\mathrm{Hom}_A(-, H^0_J(K))$ is exact. Let $M \in \mathcal{C}_Y$. There is an integer k such that $J^k M = 0$, and the inclusion

$$\mathrm{Hom}_A(M, H^0_J(K)) \longrightarrow \mathrm{Hom}_A(M, K)$$

is an isomorphism: every map $M \to K$ has image annihilated by J^k and therefore lands in $H^0_J(K)$. The assertion follows because $\mathrm{Hom}_A(-, K)$ is exact. □

3. The Case Where T Is Left Exact and Every $T(M)$ Is Finitely Generated

As above, let

$$T : \mathcal{C}_Y^{\mathrm{op}} \longrightarrow \mathbf{Ab}.$$

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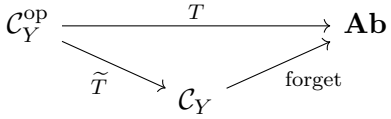
From now on, assume that T is left exact and factors through $\mathcal{C}_Y$:



Thus $T(T(M)) = (T \circ T)(M)$ is defined. Under the representation $T(M) \simeq \mathrm{Hom}_A(M, H)$, the canonical evaluation map

$$M \longrightarrow \mathrm{Hom}_A(\mathrm{Hom}_A(M, H), H)$$

defines a natural morphism

$$\eta_M : M \longrightarrow (T \circ T)(M).$$

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Proposition 3.1. *Assume, as before, that A is noetherian, and assume in addition that A/J is artinian. The following conditions are equivalent:*

i) *T is left exact and, for every $M \in \mathcal{C}_Y$, the module $T(M)$ is finitely generated and $\eta_M : M \to (T \circ T)(M)$ is an isomorphism.*

ii) *T is exact and, for every residue field $k = A/\mathfrak{m}$ of a maximal ideal $\mathfrak{m} \supset J$, one has $T(k) \simeq k$.*

iii) *There is an isomorphism $T \simeq \mathrm{Hom}_A(-, H)$ with H injective and, for every k as in* (ii), *one has $\mathrm{Hom}_A(k, H) \simeq k$.*

iv) *T is exact and $\mathrm{length}_A T(M) = \mathrm{length}_A M$ for every $M \in \mathcal{C}_Y$.*

Proof. The equivalence of *(ii)* and *(iii)* follows from Proposition 2.1.

(ii)⇒(iv). If $M \in \mathcal{C}_Y$, then M is an A/J^n-module for some n; since A/J^n is artinian, M has finite length. We argue by induction on $\mathrm{length}_A M$. When $\mathrm{length}_A M = 1$, the module M is one of the residue fields occurring in *(ii)*, so the assertion holds. If $\mathrm{length}_A M > 1$, choose a nonzero proper submodule $M' \subset M$ and put $M'' = M/M'$. Exactness of T turns

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

into

$$0 \longrightarrow T(M'') \longrightarrow T(M) \longrightarrow T(M') \longrightarrow 0.$$

Both end terms have smaller length, and therefore

$$\begin{aligned} \mathrm{length}_A T(M) &= \mathrm{length}_A T(M') + \mathrm{length}_A T(M'') \\ &= \mathrm{length}_A M' + \mathrm{length}_A M'' = \mathrm{length}_A M. \end{aligned}$$

(ii)⇒(i). By *(iv)*, $T(M)$ has finite length and hence is finitely generated. It remains to prove that every η_M is an isomorphism. Again use induction on $\mathrm{length}_A M$. For a residue field, $T(k)$ and $(T \circ T)(k)$ both have length one; evaluation at $1 \in k$ shows that η_k is nonzero, hence it is an isomorphism. For an exact sequence as above, naturality gives the commutative diagram with exact rows

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$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \eta_{M'}} & & \downarrow{\scriptstyle \eta_M} & & \downarrow{\scriptstyle \eta_{M''}} & & \\ 0 & \longrightarrow & (T\circ T)(M') & \longrightarrow & (T\circ T)(M) & \longrightarrow & (T\circ T)(M'') & \longrightarrow & 0. \end{array}$$

The outer vertical arrows are isomorphisms by induction, hence so is η_M.

(i)⇒(ii). Let

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be exact in $\mathcal{C}_Y$, and let Q be the cokernel of $T(M) \to T(M')$. Left exactness gives the exact sequence

$$0 \longrightarrow T(M'') \longrightarrow T(M) \longrightarrow T(M') \longrightarrow Q \longrightarrow 0.$$

Applying T at the left end and using naturality of evaluation yields

$$\begin{array}{ccccc} 0 \longrightarrow T(Q) & \longrightarrow & (T\circ T)(M') & \longrightarrow & (T\circ T)(M) \\ & & {\scriptstyle \eta_{M'}}\uparrow{\scriptstyle \sim} & & {\scriptstyle \sim}\uparrow{\scriptstyle \eta_M} \\ & & M' & \hookrightarrow & M. \end{array}$$

Thus $T(Q) = 0$, and condition *(i)* gives $Q \simeq (T \circ T)(Q) = 0$. Therefore $T(M) \to T(M')$ is surjective and T is exact.

Now let $k = A/\mathfrak{m}$ with $\mathfrak{m} \supset J$. The A-module $T(k)$ is a finite-dimensional k-vector space. It is nonzero because $k \simeq (T \circ T)(k)$. Choose a decomposition $T(k) \simeq k \oplus V$. Additivity and biduality give

$$\begin{aligned}(T \circ T)(k) &\simeq T(k) \oplus T(V) \\ &\simeq k \oplus V \oplus T(V) \simeq k,\end{aligned}$$

so comparison of finite lengths forces $V = 0$. Hence $T(k) \simeq k$.

(iv)⇒(iii). Proposition 2.1 supplies an injective module H and an isomorphism

$$T \simeq \operatorname{Hom}_A(-, H).$$

It remains only to check the residue fields. For $k = A/\mathfrak{m}$ as above, $T(k)$ has length one, hence is isomorphic to a residue field $k' = A/\mathfrak{m}'$. Moreover

$$\operatorname{Supp} k' = \operatorname{Supp} \operatorname{Hom}_A(k, H) \subseteq \operatorname{Supp} k = \{\mathfrak{m}\}.$$

Thus $\mathfrak{m}' = \mathfrak{m}$ and $k' \simeq k$, as required. □

Remark 3.2. *One can show that condition* (iv) *is equivalent to the condition*

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(iv)′ *For every $M \in \operatorname{Ob} \mathcal{C}_Y$, one has*

$$\operatorname{length} T(M) = \operatorname{length} M.$$

4. Dualizing Module. Dualizing Functor

Definition 4.1. *Let A be a noetherian local ring and let $\mathfrak{m}$ be its maximal ideal. A* dualizing functor *for A is a functor*

$$T : \mathcal{C}_{\mathfrak{m}}^{\mathrm{op}} \longrightarrow \mathbf{Ab},$$

where $\mathcal{C}_{\mathfrak{m}}$ denotes $\mathcal{C}_Y$ when $Y = V(\mathfrak{m})$, that satisfies the equivalent conditions of Proposition 3.1. An A-module I is said to be dualizing for A *if the functor*

$$M \longmapsto \operatorname{Hom}_A(M, I)$$

is dualizing.

Definition 4.1 can be generalized without assuming that A is local.

Definition 4.2. *Let A be a noetherian ring, and let $\overline{\mathcal{C}}$ be the full subcategory of $\mathcal{C}$ consisting of the A-modules of finite length. A* dualizing functor *is an exact A-linear functor*

$$T : \overline{\mathcal{C}}^{\mathrm{op}} \longrightarrow \overline{\mathcal{C}}$$

such that the morphism of functors

$$\mathrm{id} \longrightarrow T \circ T$$

is an isomorphism.

We shall prove an existence theorem and also prove that the module I representing such a functor is locally artinian. We shall further show that, for every maximal ideal $\mathfrak{m}$ of A, the $\mathfrak{m}$-primary component of the socle of I has length one.

Proposition 4.3. *Let A and B be noetherian local rings with maximal ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$, and suppose that B is a finite A-algebra. If I is a dualizing module for A, then* $\mathrm{Hom}_A(B, I)$ *is a dualizing module for B.*

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Proof. Let

$$R : \mathcal{C}_{\mathfrak{m}_B} \longrightarrow \mathcal{C}_{\mathfrak{m}_A}$$

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be the restriction-of-scalars functor; it is exact. Let T be a dualizing functor for A,

$$T : \mathcal{C}_{\mathfrak{m}_A}^{\mathrm{op}} \longrightarrow \mathbf{Ab}.$$

It is exact, and for every $M \in \mathrm{Ob}\,\mathcal{C}_{\mathfrak{m}_A}$ the natural morphism

$$M \longrightarrow (T \circ T)(M)$$

is an isomorphism. Consequently, the contravariant composite $T \circ R$ is a dualizing functor for B. If I represents T, the classical formula

$$\mathrm{Hom}_A(M, I) \simeq \mathrm{Hom}_B\big(M, \mathrm{Hom}_A(B, I)\big),$$

valid for every B-module M, shows that $\mathrm{Hom}_A(B, I)$ is a dualizing module for B. □

Corollary 4.4. *Let A be a noetherian local ring, let $\mathfrak{a}$ be an ideal of A, and put $B = A/\mathfrak{a}$. If I is a dualizing module for A, then the annihilator of $\mathfrak{a}$ in I is a dualizing module for B.*

Lemma 4.5. *Let A be a noetherian local ring and let I be a locally artinian A-module. There is a canonical isomorphism*

$$I \xrightarrow{\sim} \widehat{I} = I \otimes_A \widehat{A}.$$

Proof. Let I_n be the annihilator of $\mathfrak{m}^n$ in I, where $\mathfrak{m}$ is the maximal ideal of A. To say that I is locally artinian means that I is the direct limit of the I_n and that each I_n has finite length. Since tensor product commutes with direct limits, we are reduced to the case in which I is artinian. In that case some power $\mathfrak{m}^k$ of the maximal ideal annihilates I. Thus, for $p \geq k$,

$$I \xrightarrow{\sim} I \otimes_A A/\mathfrak{m}^p,$$

and hence

$$I \xrightarrow{\sim} I \otimes_A \widehat{A},$$

because A is noetherian and I is finitely generated.

It follows that restriction of scalars from $\widehat{A}$ to A and extension of scalars induce mutually quasi-inverse *equivalences* between the category of locally artinian $\widehat{A}$-modules and the category of locally artinian A-modules. □

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Proposition 4.6. *Let A be a noetherian local ring, let $\widehat{A}$ be its completion, let I be a dualizing module for A (respectively, for $\widehat{A}$), and let J be the completed module*[1] *of I (respectively, the A-module obtained by restriction of scalars). Then J is a dualizing module for $\widehat{A}$ (respectively, for A). Moreover, the underlying abelian groups of I and J are isomorphic.*

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[1] Here "the completed module" means $\widehat{I} = I \otimes_A \widehat{A}$ (cf. Lemma 4.5): this is I endowed with its canonical $\widehat{A}$-module structure, not its $\mathfrak{m}$-adic completion. For example, if p is prime and $A = \widehat{A} = \mathbb{Z}_p$ is the ring of p-adic integers, then the injective envelope of $k = \mathbb{Z}/p\mathbb{Z}$ is the discrete $\mathbb{Z}_p$-module $\mathbb{Q}_p/\mathbb{Z}_p$, whose p-adic completion is zero.

Proof. One merely observes that the equivalence between the category of locally artinian A-modules and the category of locally artinian $\widehat{A}$-modules induces an isomorphism between the bifunctors $\mathrm{Hom}_A(-,-)$ and $\mathrm{Hom}_{\widehat{A}}(-,-)$, and that the characterization of a dualizing functor or a dualizing module involves only these bifunctors. □

Theorem 4.7. *Let A be a noetherian local ring.*

a) *A dualizing module I always exists.*

b) *Any two dualizing modules are isomorphic (by a noncanonical isomorphism).*

c) *An A-module I is dualizing if and only if it is an injective envelope of the residue field k of A.*

Remark 4.8. *Proposition 4.6 reduces the question to the case of a complete noetherian local ring. By a structure theorem of Cohen,*[2] *such a ring is a quotient of a regular ring. Proposition 4.3 then permits us to assume that A is regular. As we shall see later, this observation gives an explicit calculation of the dualizing module.*[3] *We shall nevertheless prove Theorem 4.7 by other means.*

Recollections

Before proving the theorem, we recall several facts about injective envelopes; see Gabriel, *Des Catégories Abéliennes*, thesis, Paris, 1961, Chapter II, §5.

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Let $\mathcal{C}$ be an abelian category in which direct limits exist and are exact[4] (for example, the category of modules). Every object M embeds in an injective object, and an *injective envelope* of M means an injective object containing M that is minimal among such objects. The following properties hold:

i) Every object M has an injective envelope I.

ii) If I and J are two injective envelopes of M, there is an isomorphism $I \simeq J$ (generally not unique) that induces the identity on M.

iii) The object I is an *essential extension* of M: if $P \subset I$ and $P \cap M = \{0\}$, then $P = \{0\}$. Conversely, if I is injective and is an essential extension of M, then I is an injective envelope of M.

Granting these facts, it plainly suffices to prove part *c)* of Theorem 4.7.

Proof. Suppose first that I is a dualizing module for A. Then I is injective and $\mathrm{Hom}_A(k, I) \simeq k$. Composing this isomorphism with the inclusion

$$\mathrm{Hom}_A(k, I) \hookrightarrow \mathrm{Hom}_A(A, I) \simeq I$$

gives an inclusion

$$k \hookrightarrow I.$$

[2] See I. S. Cohen, "On the structure and ideal theory of complete local rings," *Trans. Amer. Math. Soc.* **59** (1946), 54–106.

[3] This was the method followed by Grothendieck in 1957. The method by injective envelopes used below appears to be due to K. Morita, "Duality for modules and its applications to the theory of rings with minimum conditions," *Sc. Rep. Tokyo Kyoiku Daigaku* **6** (1958/59), 83–142. Morita's work is independent of Grothendieck's, considerably predates this seminar, and is not restricted to commutative base rings.

[4] More precisely, small filtered direct limits are assumed exact; one should also assume the existence of a generator. See [Tôhoku]. For the category of modules, which suffices here, one may instead consult Chapter 10 of Bourbaki's *Algèbre*.

We show that I is an injective envelope of k. Let J be an injective module such that

$$k \subset J \subset I.$$

Since J is injective, there is an injective A-submodule J' of I such that $I = J \oplus J'$. We claim that $\mathrm{Hom}_A(k, J') = 0$. Indeed,

$$\mathrm{Hom}_A(k, I) \simeq \mathrm{Hom}_A(k, J) \oplus \mathrm{Hom}_A(k, J').$$

The first summand is a nonzero vector subspace of $\mathrm{Hom}_A(k, I) \simeq k$ (it contains the inclusion $k \subset J$). Consequently $\mathrm{Hom}_A(k, J) \simeq k$ and $\mathrm{Hom}_A(k, J') = 0$. Original French p. 55

Induction on length now gives $\mathrm{Hom}_A(M, J') = 0$ for every finite-length A-module M. The module I is the direct limit of the modules $\mathrm{Hom}_A(A/\mathfrak{m}^n, I)$ (cf. Proposition 1.3), which have finite length by hypothesis. Hence the projection $I \to J'$ vanishes, and therefore $J' = 0$.

Conversely, let I be an injective envelope of k. By Propositions 2.1 and 3.1*(ii)*, to prove that I is dualizing it suffices to show that $V = \mathrm{Hom}_A(k, I)$ is isomorphic to k. There are inclusions

$$k \subset V \subset I.$$

As a k-vector space, V decomposes as $k \oplus V'$, where V' is a vector subspace of I with $V' \cap k = 0$. Since I is an essential extension of k, we have $V' = 0$, and thus $V = k$. □

Corollary 4.9. *Let A be a noetherian local ring. Every dualizing module for A is locally artinian.*

Proof. Let I be a dualizing module; it is an injective envelope of k. Using the notation and result of Corollary 2.2, we have

$$k \subset H^0_{\mathfrak{m}}(I) \subset I,$$

and $H^0_{\mathfrak{m}}(I)$ is injective. It follows that $I = H^0_{\mathfrak{m}}(I)$, and hence that I is locally artinian.[5] □

5. Consequences of the theory of dualizing modules

The functor Original French p. 56

$$T = \mathrm{Hom}_A(-, I) \colon \mathcal{C}_{\mathfrak{m}} \longrightarrow \mathcal{C}_{\mathfrak{m}}$$

is an anti-equivalence. Indeed, $T \circ T$ is isomorphic to the identity functor, and the argument is formal from that point on.

One deduces the usual properties of the notion of orthogonality:

Let $M^* = \mathrm{Hom}_A(M, I) = T(M)$, and let $N \subset M$ be a submodule. The *orthogonal of* N is the submodule N' of M^* consisting of those elements of M^* that vanish on N. In this way one obtains a bijection between the set of submodules of M and the set of submodules of M^*, reversing the order relation.

In particular:

- $\mathrm{length}_M N = \mathrm{colength}_{M^*} N'$.
- Cyclic modules, that is, those for which $M/\mathfrak{m}M$ has dimension 0 or 1, correspond under the duality to modules whose socle has length 0 or 1.
- If A is artinian, the ideals of A correspond to the submodules of I.

[5] As already observed, one may alternatively note directly that I is the direct limit of the modules $\mathrm{Hom}_A(A/\mathfrak{m}^n, I)$.

And so on.

Let A be a noetherian local ring. Write $\mathrm{D}A$ for the category of A-modules M such that, for every $n \in \mathbb{N}$,

$$M^{(n)} = M/\mathfrak{m}^{n+1}M$$

has finite length and such that

$$M = \varprojlim_n M^{(n)},$$

and let $\widehat{A}$ be the completion of A. Restriction of scalars and completion are quasi-inverse equivalences between $\mathrm{D}A$ and $\mathrm{D}\widehat{A}$; up to isomorphism, they are compatible with taking the underlying abelian groups of the modules in question. Write $\mathrm{C}A$ for the category of locally artinian A-modules with finite-dimensional socle.

Proposition 5.1. *Let A be a noetherian local ring, and let I be a dualizing module for A. The functors*

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$$\mathrm{Hom}_A(-, I)\colon (\mathrm{C}A)^{\mathrm{op}} \longrightarrow \mathrm{D}A$$

and

$$\mathrm{Hom}_{\widehat{A}}(-, I)\colon \mathrm{D}A \longrightarrow (\mathrm{C}A)^{\mathrm{op}}$$

are equivalences of categories, quasi-inverse to one another.

Moreover, transporting these functors by the equivalences of categories between $\mathrm{D}A$ and $\mathrm{D}\widehat{A}$ on the one hand, and between $\mathrm{C}A$ and $\mathrm{C}\widehat{A}$ on the other, yields the functor $\mathrm{Hom}_{\widehat{A}}(-, I)$.

Proof. Let $X \in \mathrm{Ob}(\mathrm{C}A)$. By definition,

$$X = \varinjlim_{k\in\mathbb{N}} X_k, \qquad X_k = \mathrm{Hom}_A(A/\mathfrak{m}^{k+1}, X),$$

and therefore

$$\mathrm{Hom}_A(X, I) = \varprojlim_k \mathrm{Hom}_A(X_k, I).$$

Thus

$$Y = \mathrm{Hom}_A(X, I) = \varprojlim_k \mathrm{Hom}_A(X_k, I)$$

is a finitely generated $\widehat{A}$-module, as follows from *EGA* 0_{I}, 7.2.9. In this connection, observe that $\mathrm{D}\widehat{A}$ is also the category of finitely generated $\widehat{A}$-modules, or, equivalently, that $\mathrm{D}A$ is the category of A-modules that are complete and finitely generated over $\widehat{A}$.

Now let Y be such a module, and let $f\colon Y \to I$ be a $\widehat{A}$-homomorphism. The image of f is a finitely generated submodule, hence is annihilated by $\mathfrak{m}^k$ for some k: indeed, every $x \in I$ is annihilated by some power of $\mathfrak{m}$. Consequently f factors through $Y/\mathfrak{m}^k Y$, and it follows that

$$\begin{aligned} \mathrm{Hom}_{\widehat{A}}(Y, I) &= \varinjlim_k \mathrm{Hom}_{\widehat{A}}(Y^{(k)}, I), \qquad \text{where } Y^{(k)} = Y/\mathfrak{m}^{k+1}Y, \\ &= \varinjlim_k (Y^{(k)})^*. \end{aligned}$$

This object belongs to $\mathrm{Ob}(\mathrm{C}A)$. It follows immediately that the two functors in the statement are quasi-inverse to one another. □

It follows from the preceding discussion that replacing A by $\widehat{A}$ changes neither the categories or functors under consideration nor the underlying abelian groups of the modules involved. Proposition 5.1 may therefore be restated as follows.

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The restriction of $\mathrm{Hom}_{\widehat{A}}(-, I)$ to the category of finitely generated $\widehat{A}$-modules takes its values in the category of locally artinian $\widehat{A}$-modules having finite-dimensional socle. It

admits a quasi-inverse, namely the restriction of $\mathrm{Hom}_{\widehat{A}}(-, I)$ in the opposite direction. On the intersection of these two categories the two functors coincide (of course) and define an autoduality of the category of finite-length $\widehat{A}$-modules.

Example 5.2 (Macaulay). *Let A be a local ring with residue field k. Let k_0 be a subfield of A such that k is finite over k_0, and put $[k : k_0] = d$. Every finite-length A-module may be regarded as a finite-dimensional k_0-vector space of dimension*

$$d \ \mathrm{length}(M).$$

The functor

$$T(M) = \mathrm{Hom}_{k_0}(M, k_0)$$

is exact and preserves length, hence is dualizing for A. Its associated dualizing module is

$$A' = \varinjlim_n \mathrm{Hom}_{k_0}(A/\mathfrak{m}^n, k_0),$$

the topological dual of A endowed with the $\mathfrak{m}$-adic topology.

Example 5.3. *Let A be a regular noetherian local ring of dimension n, let $\mathfrak{m}$ be its maximal ideal, and let k be its residue field. There is a regular system of parameters $(x_1, x_2, \ldots, x_n)$ that generates $\mathfrak{m}$ and is an A-regular sequence. The Koszul complex therefore gives*

$$\mathrm{Ext}^i_A(k, A) = 0 \quad \text{if } i \neq n,$$
$$\mathrm{Ext}^n_A(k, A) \simeq k.$$

Since the depth of A is n, one has $\mathrm{Ext}^i_A(M, A) = 0$ for $i < n$ whenever M is annihilated by a power of $\mathfrak{m}$. One also has $\mathrm{Ext}^i_A(M, A) = 0$ for $i > n$, because the global cohomological dimension of A equals n. Thus $\mathrm{Ext}^n_A(-, A)$ is exact, and $\mathrm{Ext}^n_A(k, A) \simeq k$. It follows that:

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Proposition 5.4. *If A is a regular noetherian local ring of dimension n, the functor*

$$M \longmapsto \mathrm{Ext}^n_A(M, A)$$

is dualizing. The associated dualizing module is

$$I = \varinjlim_r \mathrm{Ext}^n_A(A/\mathfrak{m}^r, A).$$

It is isomorphic to $H^n_{\mathfrak{m}}(A)$ (Exposé II, Theorem 6).[6]

Remark 5.5. *If A satisfies the hypotheses of both preceding examples, the two resulting dualizing modules are isomorphic. Suppose, for example, that A is regular of dimension n, complete, and of equal characteristic. There is then a coefficient field, say K. After choosing a system of parameters $(x_1, \ldots, x_n)$ of A, one can construct an isomorphism*

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$$A \simeq K[[T_1, \ldots, T_n]],$$

and hence, as we shall see, an explicit isomorphism between the two dualizing modules

$$v : H^n_{\mathfrak{m}}(A) \longrightarrow A'.$$

[6] For a ring A, an ideal $J \subset A$, an A-module M, and $i \in \mathbb{Z}$, set $H^i_J(M) = H^i_Y(X, \widetilde{M})$, where $X = \mathrm{Spec}(A)$ and $Y = V(J)$.

This isomorphism has an intrinsic interpretation in terms of the module

$$\Omega^n = \Omega^n(A/K)$$

of completed relative differentials of top degree. Indeed, Ω^n has a basis consisting of

$$dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n.$$

There is therefore an isomorphism

$$u : H^n_{\mathfrak{m}}(\Omega^n) \longrightarrow H^n_{\mathfrak{m}}(A).$$

A remarkable fact is that the composite

$$vu = w : H^n_{\mathfrak{m}}(\Omega^n) \longrightarrow A'$$

is independent of the chosen system of parameters and commutes with change of the base field.

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To construct v, compute $H^n_{\mathfrak{m}}(A)$ from the Koszul complex associated with the x_i. One obtains

$$H^n_{\mathfrak{m}}(A) = \varinjlim_r A/(x_1^r, \ldots, x_n^r).$$

The transition morphisms are as follows. Put

$$I_r = A/(x_1^r, \ldots, x_n^r),$$

and let $e^r_{a_1,\ldots,a_n}$ be the image of $x_1^{a_1}x_2^{a_2}\cdots x_n^{a_n}$ in I_r. The elements $e^r_{a_1,\ldots,a_n}$, for $0 \leq a_i < r$, form a basis of I_r.

For an integer s, the transition morphism

$$t_{r,r+s} : I_r \longrightarrow I_{r+s}$$

is multiplication by $x_1^s x_2^s \cdots x_n^s$; consequently,

$$t_{r,r+s}(e^r_{a_1,\ldots,a_n}) = e^{r+s}_{a_1+s,\ldots,a_n+s}.$$

Giving an A-homomorphism $w : M \to A'$ is equivalent to giving a K-linear form $w' : M \to K$ that is continuous on every finitely generated submodule. For $M = H^n_{\mathfrak{m}}(\Omega^n)$, defining w is therefore equivalent to defining a linear form

$$\rho : H^n_{\mathfrak{m}}(\Omega^n) \longrightarrow K,$$

called the residue form.[7] *To construct ρ, it is enough to define compatible forms $\rho_r : I_r \to K$ by*

$$\rho_r(e^r_{a_1,\ldots,a_n}) = \begin{cases} 1, & \text{if } a_i = r-1 \text{ for } 1 \leq i \leq n, \\ 0, & \text{otherwise.} \end{cases}$$

[7] For a fuller treatment of residues, see R. Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics 20, Springer, 1966.

Exposé V

Local Duality and Structure of the $H^i(M)$

1. Complexes of homomorphisms

1.1

Let $F^\bullet$ and $G^\bullet$ be two graded modules. Write

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$$\mathrm{Hom}^\bullet(F^\bullet, G^\bullet) \tag{1}$$

for the graded module of homomorphisms of graded modules from $F^\bullet$ to $G^\bullet$. Thus

$$\mathrm{Hom}^s(F^\bullet, G^\bullet) = \prod_k \mathrm{Hom}(F^k, G^{k+s}). \tag{2}$$

Suppose that $F^\bullet$ (respectively $G^\bullet$) is a complex, and let d_1 (respectively d_2) be its differential. For $h \in \mathrm{Hom}^s(F^\bullet, G^\bullet)$, set[1]

$$d(h) = h \circ d_1 + (-1)^{s+1} d_2 \circ h. \tag{3}$$

One checks immediately that $d \circ d = 0$; hence $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$, equipped with d, is a complex. The cohomology group of this complex is denoted by

$$\underline{\mathrm{H}}^\bullet(F^\bullet, G^\bullet). \tag{4}$$

If $G^\bullet$ is injective in every degree, then

$$F^\bullet \longmapsto \underline{\mathrm{H}}^\bullet(F^\bullet, G^\bullet)$$

is an exact ∂-functor. Likewise, for arbitrary $F^\bullet$,

$$G^\bullet \longmapsto \underline{\mathrm{H}}^\bullet(F^\bullet, G^\bullet)$$

is an exact δ-functor on the category of complexes $G^\bullet$ that are injective in every degree.

Remark 1.2. The cycles of $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ that commute or anticommute with the differentials, according to the degree. The boundaries of $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ that are homotopic to zero.

[1] *Editor's note.* The original sign convention was different, but it is incompatible with the convention of Exposé VIII. The latter seems more reasonable because in that case the degree-zero cohomology is the set of homotopy classes of morphisms from $F^\bullet$ to $G^\bullet$. The calculations below have been modified accordingly.

Let A be a ring, let M (respectively N) be an A-module, and let $R(M)$ (respectively $R(N)$) be an injective resolution of M (respectively N). There is then a canonical isomorphism[2]

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$$\underline{\mathrm{H}}^s(R(M), R(N)) \simeq \mathrm{Ext}^s(M, N). \tag{1.3}$$

Indeed, let $i\colon M \to R(M)$ be the canonical augmentation, and let $h \in \mathrm{Hom}^s(R(M), R(N))$. Denote by t_s the map

$$h \longmapsto h_0 \circ i$$

from $\mathrm{Hom}^s(R(M), R(N))$ to $\mathrm{Hom}(M, R(N)^s)$. The family $(t_s)_{s\geq 0}$ defines a homomorphism of (ordinary) complexes[3]

$$t\colon \mathrm{Hom}^\bullet(R(M), R(N)) \longrightarrow \mathrm{Hom}^\bullet(M, R(N));$$

that is,

$$(dh)_0 \circ i = d_2 \circ h_0 \circ i.$$

One checks easily that, on passing to cohomology, t induces an isomorphism. It follows in particular that

$$\underline{\mathrm{H}}^\bullet(R(M), R(N))$$

does not "depend" on the chosen injective resolution $R(M)$ (respectively $R(N)$) of M (respectively N).

To every exact sequence of A-modules

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$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \tag{5}$$

one associates an exact sequence of injective resolutions

$$0 \longrightarrow R(M') \longrightarrow R(M) \longrightarrow R(M'') \longrightarrow 0. \tag{6}$$

One checks that the isomorphism *(1.3)* commutes with the homomorphisms

$$\underline{\mathrm{H}}^s(R(M'), R(N)) \longrightarrow \underline{\mathrm{H}}^{s+1}(R(M''), R(N)), \tag{8}$$

$$\mathrm{Ext}^s(R(M'), R(N)) \longrightarrow \mathrm{Ext}^{s+1}(R(M''), R(N)), \tag{9}$$

deduced from *(6)* and *(5)*.

Let P be a third A-module, and let $R(P)$ be an injective resolution of P. Then composition of graded morphisms gives a pairing

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$$\mathrm{Hom}^i(R(N), R(M)) \times \mathrm{Hom}^j(R(M), R(P)) \longrightarrow \mathrm{Hom}^{i+j}(R(N), R(P)), \tag{10}$$

which defines a pairing

$$\underline{\mathrm{H}}^i(R(N), R(M)) \times \underline{\mathrm{H}}^j(R(M), R(P)) \longrightarrow \underline{\mathrm{H}}^{i+j}(R(N), R(P)), \tag{11}$$

and hence a homomorphism of functors in M:

$$\underline{\mathrm{H}}^i(R(N), R(M)) \longrightarrow \mathrm{Hom}\Big(\underline{\mathrm{H}}^j(R(M), R(P)), \underline{\mathrm{H}}^{i+j}(R(N), R(P))\Big). \tag{1.4}$$

We shall see that *(1.4)* is a homomorphism of δ-functors in M. The exact sequences *(5)* and *(6)* give a commutative diagram:

[2] *Editor's note.* The strange original numbering has been preserved.

[3] Here M also denotes the complex $M[0]$ consisting of M placed in degree zero.

$$\begin{array}{ccc}
\mathrm{Hom}^i(R(N),R(M')) & \longrightarrow & \mathrm{Hom}\Big(\mathrm{Hom}^j(R(M'),R(P)),\mathrm{Hom}^{i+j}(R(N),R(P))\Big) \\
\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{Hom}(q,\mathrm{id})} \\
\mathrm{Hom}^i(R(N),R(M)) & \longrightarrow & \mathrm{Hom}\Big(\mathrm{Hom}^j(R(M),R(P)),\mathrm{Hom}^{i+j}(R(N),R(P))\Big) \\
\big\downarrow{\scriptstyle p} & & \big\downarrow \\
\mathrm{Hom}^i(R(N),R(M'')) & \longrightarrow & \mathrm{Hom}\Big(\mathrm{Hom}^j(R(M''),R(P)),\mathrm{Hom}^{i+j}(R(N),R(P))\Big).
\end{array}$$

Let $h \in \mathrm{Hom}^i(R(N),R(M''))$ (respectively $g \in \mathrm{Hom}^j(R(M'),R(P))$) be a cycle, and let $h' \in \mathrm{Hom}^i(R(N),R(M))$ (respectively $g' \in \mathrm{Hom}^j(R(M),R(P))$) be such that $p(h') = h$ (respectively $q(g') = g$). To say that *(1.4)* is a homomorphism of δ-functors in M is to say that

$$g \circ dh' - dg' \circ h \tag{12}$$

is a coboundary in $\mathrm{Hom}^\bullet(R(N),R(P))$.

Now

$$dh' = h' \circ d_1 + (-1)^{i+1} d_2 \circ h',$$
$$dg' = g' \circ d_2 + (-1)^{j+1} d_3 \circ g',$$

with the obvious notation. Thus *(12)* becomes

$$g \circ h' \circ d_1 + (-1)^{i+1} g \circ d_2 \circ h' - g' \circ d_2 \circ h - (-1)^{j+1} d_3 \circ g' \circ h.$$

On the other hand, since h and g are cycles,

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$$g \circ d_2 = (-1)^j d_3 \circ g,$$
$$d_2 \circ h = (-1)^i h \circ d_1.$$

Consequently *(12)* is finally written as

$$d\Big(g \circ h' + (-1)^{i+1} g' \circ h\Big),$$

which completes the proof.

Thus *(1.3)* and *(1.4)* give a homomorphism of δ-functors in M:

$$\mathrm{Ext}^i(N,M) \longrightarrow \mathrm{Hom}\Big(\mathrm{Ext}^j(M,P),\mathrm{Ext}^{i+j}(N,P)\Big). \tag{1.5}$$

2. The local duality theorem for a regular local ring

Let A be a regular local ring of dimension r, let $\mathfrak{m}$ be the maximal ideal of A, and let M be a finitely generated A-module. Set

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$$H^i(M) = H^i_{\mathfrak{m}}(M) \qquad (\text{so } H^i(M) = \varinjlim_k \mathrm{Ext}^i(A/\mathfrak{m}^k, M)).$$

We saw in *IV, 5.4* that $I = H^r(A)$ is a dualizing module for A; denote the associated dualizing functor by D. In *(1.5)*, put $N = A/\mathfrak{m}^k$ and $P = A$. We then obtain a homomorphism of δ-functors in M,

$$\varphi_k \colon \mathrm{Ext}^i(A/\mathfrak{m}^k, M) \longrightarrow \mathrm{Hom}\Big(\mathrm{Ext}^{r-i}(M,A),\mathrm{Ext}^r(A/\mathfrak{m}^k,A)\Big). \tag{13}$$

Passing to the direct limit over k gives a homomorphism of δ-functors

$$\varphi\colon H^i(M) \longrightarrow D\Big(\operatorname{Ext}^{r-i}(M, A)\Big). \tag{14}$$

Theorem 2.1 (Local duality theorem). The preceding functorial homomorphism φ is an isomorphism.

Proof. If $i > r$, the right-hand side of *(14)* is trivially zero, and the left-hand side is zero because Original French p. 65

$$H^i(M) = \varinjlim_k \operatorname{Ext}^i(A/\mathfrak{m}^k, M),$$

and each $\operatorname{Ext}^i(A/\mathfrak{m}^k, M)$ is zero (syzygy theorem).

If $i = r$, the preceding discussion shows that the two functors in M, $H^r(M)$ and $D(\operatorname{Hom}(M, A))$, are right exact. Since A is noetherian and M is finitely generated, it is enough to verify the isomorphism for $M = A$, where it is immediate.

To show that φ is a functorial isomorphism, it now suffices to proceed by descending induction on i and observe that every finitely generated module admits a finite presentation, while for $i < r$ both sides of *(14)* vanish when M is finitely generated and free. This is clear for the right-hand side. Since H^i commutes with finite direct sums, for the left-hand side it is enough to show that $H^i(A) = 0$ for $i < r$. This follows from *III, 3.4*, since $\operatorname{prof}(A) = r$. □

3. Application to the structure of the $H^i(M)$

Theorem 3.1. *Let A be a Noetherian local ring, let D be a dualizing functor for A, and let $M \neq 0$ be a finitely generated A-module of dimension n. Then:* Original French p. 65

(i) $H^i(M) = 0$ *if* $i < 0$ *or if* $i > n$.

(ii) $D(\widehat{H^i(M)})$ *is a finitely generated* $\widehat{A}$*-module of dimension at most* i.

(iii) $H^n(M) \neq 0$*; moreover, if A is complete, then $D(H^n(M))$ has dimension n and*

$$\operatorname{Ass}(D(H^n(M))) = \{\, \mathfrak{p} \in \operatorname{Ass}(M) \mid \dim(A/\mathfrak{p}) = n \,\}.$$

Proof. Let I be the dualizing module associated with D. We know that $\widehat{I}$ is a dualizing module for $\widehat{A}$. On the other hand,

$$\widehat{H^i(M)} = H^i(\widehat{M}),$$
$$D(\widehat{H^i(M)}) = \operatorname{Hom}(H^i(\widehat{M}), \widehat{I}),$$
$$\dim \widehat{M} = \dim M.$$

Consequently, we may suppose that A is complete. By Cohen's theorem, every complete local ring is a quotient of a regular local ring. To reduce to that case, we need the following lemma. Original French p. 66

Lemma 3.2. *Let X (respectively Y) be a ringed space, let X' (respectively Y') be a closed subspace of X (respectively Y), and let $f\colon X \to Y$ be a morphism of ringed spaces such that $f^{-1}(Y') = X'$. Let F be an $\mathcal{O}_X$-module. Denote by A (respectively B) the ring $\Gamma(\mathcal{O}_X)$ (respectively $\Gamma(\mathcal{O}_Y)$), and by $\overline{f}\colon B \to A$ the ring homomorphism corresponding to f. There is a spectral sequence of B-modules with initial term* Original French p. 66

$$E_2^{p,q} = H^p_{Y'}(Y, R^q f_*(F)), \tag{15}$$

converging to the B-module $H^\bullet_{X'}(X, F)_{[\overline{f}]}$.

Proof. Let $\mathcal{O}_{Y,Y'}$ be the sheaf $\mathcal{O}_Y|_{Y'}$, extended by zero outside Y' (see Exposé I). There is an isomorphism of B-modules

$$\operatorname{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)) \simeq \operatorname{Hom}(f^*(\mathcal{O}_{Y,Y'}), F)_{[\overline{f}]}. \tag{16}$$

Now

$$f^*(\mathcal{O}_{Y,Y'}) = \mathcal{O}_{X,X'}. \tag{17}$$

Moreover, if G is an injective $\mathcal{O}_X$-module, then $f_*(G)$ is an injective $\mathcal{O}_Y$-module, at least when f is flat; one can easily reduce to this case by replacing $\mathcal{O}_X$, and so forth, by the constant sheaves of rings $\mathbb{Z}$. Hence the spectral sequence of the composite functor

$$F \longmapsto \operatorname{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)),$$

whose initial term is

$$E_2^{p,q} = \operatorname{Ext}^p(Y; \mathcal{O}_{Y,Y'}, R^q f_*(F)),$$

converges, in view of (16) and (17), to

$$\operatorname{Ext}^\bullet(X; \mathcal{O}_{X,X'}, F)_{[\overline{f}]}.$$

The lemma then follows from Exposé I, Proposition 2.3 bis. □

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Now let $f\colon B \to A$ be a surjective homomorphism of local rings. Let

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$$f\colon \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(B)$$

be the corresponding morphism of affine schemes. Set $X = \operatorname{Spec}(A)$ (respectively $X' = \{\mathfrak{m}_A\}$) and $Y = \operatorname{Spec}(B)$ (respectively $Y' = \{\mathfrak{m}_B\}$), and let M be an A-module and $\widetilde{M}$ the corresponding $\mathcal{O}_X$-module. Since $R^q f_*(\widetilde{M}) = 0$ for $q > 0$, the spectral sequence (15) degenerates, and Lemma 3.2 gives an isomorphism of B-modules

$$H^n_{\{\mathfrak{m}_B\}}(Y, f_*(\widetilde{M})) \simeq H^n_{\{\mathfrak{m}_A\}}(X, \widetilde{M})_{[f]}, \tag{18}$$

hence an isomorphism of B-modules

$$H^n_{\mathfrak{m}_B}(M_{[f]}) \simeq H^n_{\mathfrak{m}_A}(M)_{[f]}. \tag{19}$$

On the other hand, if D_A (respectively D_B) is the dualizing functor for A (respectively B), then

$$D_A(M)_{[f]} \simeq D_B(M_{[f]}). \tag{20}$$

Finally, since there is an isomorphism of rings

$$B/\operatorname{Ann} M_{[f]} \simeq A/\operatorname{Ann} M, \tag{21}$$

it follows that the change of base rings under consideration changes nothing. We may therefore suppose that A is regular of dimension r.

By (2.1), we have

$$D(H^i(M)) = \operatorname{Ext}^{r-i}(M, A). \tag{22}$$

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We shall prove the equivalence of the following properties:

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(a) $\dim \operatorname{Ext}^j(M, A) \leq r - j$;

(b) for every $\mathfrak{p} \in X = \operatorname{Spec}(A)$ such that $\dim A_\mathfrak{p} < j$, one has $\operatorname{Ext}^j(M, A)_\mathfrak{p} = 0$;

(c) $\operatorname{codim}(\operatorname{Supp}(\operatorname{Ext}^j(M, A)), X) \geq j$.

To prove (a) $\Longrightarrow$ (b), let $\mathfrak{p} \in X$ with $\dim A_\mathfrak{p} < j$. Then $\dim A/\mathfrak{p} > r - j$, and hence, by (a),

$$\operatorname{Ann}\big(\operatorname{Ext}^j(M, A)\big) \not\subset \mathfrak{p},$$

which implies that $\operatorname{Ext}^j(M, A)_\mathfrak{p} = 0$. Let $\mathfrak{p} \in \operatorname{Supp}(\operatorname{Ext}^j(M, A))$. Then $\operatorname{Ext}^j(M, A)_\mathfrak{p} \neq 0$, so by (b) we have $\dim A_\mathfrak{p} \geq j$. Therefore

$$\operatorname{codim}\big(\operatorname{Supp}(\operatorname{Ext}^j(M, A)), X\big) = \inf\{\dim A_\mathfrak{p} \mid \mathfrak{p} \in \operatorname{Supp}(\operatorname{Ext}^j(M, A))\} \geq j,$$

that is, (b) $\Longrightarrow$ (c). Finally, (c) implies (a) trivially.
Let us now prove the theorem.

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(i) Let $x = (x_1, \ldots, x_r)$ be a system of parameters for A such that $x_i \in \operatorname{Ann} M$ for $i = 1, \ldots, r - n$. Let $K^\bullet((x^k), M)$ be the Koszul complex. It is easy to see that the map

$$K^i((x^k), M) \longrightarrow K^i((x^{k'}), M)$$

for $k < k'$ is zero if $i > n$. It follows that

$$H^i(M) = \varinjlim H^i((x^k), M) = 0$$

if $i > n$. On the other hand, it is trivial that $H^i(M) = 0$ if $i < 0$; hence (i) is proved.

(ii) Since A is regular, $\dim A_\mathfrak{p} < j$ implies that the global homological dimension of $A_\mathfrak{p}$ is strictly less than j, and hence

$$\operatorname{Ext}^j(M, A)_\mathfrak{p} = \operatorname{Ext}^j_{A_\mathfrak{p}}(M_\mathfrak{p}, A_\mathfrak{p}) = 0.$$

Thus we have proved (b), and consequently (a). Assertion (ii) then follows from (22) and (a).

(iii) There exists a $\mathfrak{p} \in \operatorname{Supp}(M)$ such that $\dim A_\mathfrak{p} = r - n$ and such that $\operatorname{Supp}(M_\mathfrak{p}) = \{\mathfrak{m}A_\mathfrak{p}\}$. Since $A_\mathfrak{p}$ is regular if A is, we have $\operatorname{depth} A_\mathfrak{p} = r - n$, and therefore

$$\operatorname{Ext}^{r-n}_A(M, A)_\mathfrak{p} = \operatorname{Ext}^{r-n}_{A_\mathfrak{p}}(M_\mathfrak{p}, A_\mathfrak{p}) \neq 0. \tag{23}$$

Taking (22) into account, this implies, on the one hand,

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$$H^n(M) \neq 0,$$

and, on the other hand,

$$\dim D\big(H^n(M)\big) \geq n,$$

hence, by (ii),

$$\dim D\big(H^n(M)\big) = n.$$

Now let $Y = \operatorname{Supp}(M)$. By (i), we know that $D\big(H^n(M')\big) = \operatorname{Ext}^{r-n}(M', A)$ is a left exact functor in M' on the category $(\mathcal{C}_Y)^\circ$. Hence there exists an A-module H and an isomorphism of functors in M':

$$\operatorname{Ext}^{r-n}(M', A) = \operatorname{Hom}(M', H).$$

Let Y_i, $i = 1, \ldots, k$, be the irreducible components of Y of maximum dimension. We shall see that the assertion $\operatorname{Ext}^{r-n}(M', A) \neq 0$ is equivalent to the assertion that there exists an i such that $\operatorname{Supp} M' \supset Y_i$. Indeed, if $\operatorname{Supp} M' \supset Y_i$, then $\dim(M') = n$, and hence $\operatorname{Ext}^{r-n}(M', A) \neq 0$.

If $\operatorname{Supp} M' \not\supset Y_i$ for every $i = 1, \dots, k$, then $\dim M' < n$, and

$$D\big(H^n(M')\big) = \operatorname{Ext}^{r-n}(M', A) = 0.$$

Since $\operatorname{Ass}\big(\operatorname{Ext}^{r-n}(M, A)\big) = \operatorname{Supp} M \cap \operatorname{Ass}(H)$, we see that the last assertion of (iii) follows from the following lemma:

Lemma 3.3. *Let $X = \operatorname{Spec}(A)$, let Y be a closed subset of X, let*

$$T\colon (\mathcal{C}_Y)^\circ \longrightarrow \mathbf{Ab}$$

be a left exact functor, and let Y_i, $i = 1, \dots, k$, be a family of irreducible components of Y such that the assertion $T(M) = 0$ is equivalent to the assertion that, for every i, $\operatorname{Supp} M \not\supset Y_i$. Then T is representable by a module H such that

$$\operatorname{Ass}(H) = \bigcup_{i=1}^{k} \{y_i\},$$

where y_i is the generic point of Y_i, for $i = 1, \dots, k$.

Proof. For $y \in Y$, construct an A-module $M(y)$ such that

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$$\operatorname{Supp}\big(M(y)\big) = \overline{\{y\}}.$$

Suppose that $y \neq y_i$ for every $i = 1, \dots, k$. Then $Y_i \not\subset \operatorname{Supp}\big(M(y)\big)$ for every $i = 1, \dots, k$, and hence $T\big(M(y)\big) = 0$. It follows that

$$\operatorname{Ass}\big(T(M(y))\big) = \operatorname{Supp}\big(M(y)\big) \cap \operatorname{Ass}(H) = \varnothing,$$

and therefore $y \notin \operatorname{Ass}(H)$. If $y = y_i$, then $Y_i \subset \operatorname{Supp}\big(M(y)\big)$, so $T\big(M(y)\big) \neq 0$, and hence

$$\operatorname{Ass}\big(T(M(y))\big) = \operatorname{Supp}\big(M(y)\big) \cap \operatorname{Ass}(H) \neq \varnothing.$$

By the first part of the proof, this implies $y \in \operatorname{Ass}(H)$, which proves the lemma. □

Example 3.4. *Let A be a noetherian ring, let $X = \operatorname{Spec}(A)$, and let Y be a closed subset of X such that $X - Y$ is affine. Then, for every irreducible component Y_α of Y, one has*

$$\operatorname{codim}(Y_\alpha, X) \leq 1.$$

Proof. Consider X as a prescheme over X. Let $y_\alpha \in Y_\alpha$ be a generic point, and consider the morphism

$$\operatorname{Spec}(\mathcal{O}_{X,y_\alpha}) \longrightarrow X.$$

The affine scheme obtained by base-changing X to $\operatorname{Spec}(\mathcal{O}_{X,y_\alpha})$ is canonically isomorphic to $\operatorname{Spec}(\mathcal{O}_{X,y_\alpha})$.

By (EGA I 3.2.7), if y_0 is the unique closed point of

$$Y_0 = \operatorname{Spec}(\mathcal{O}_{X,y_\alpha}),$$

then $Y_0 - y_0$ is affine. By (EGA III 1.3.1), one obtains

$$\mathrm{H}^i(Y_0 - y_0, \mathcal{O}_{Y_0}) = 0 \qquad \text{if } i > 0.$$

Hence, by (I 2.9),

$$\mathrm{H}^{i-1}(\mathcal{O}_{X,y_\alpha}) = \mathrm{H}^i_{\{y_0\}}(Y_0, \mathcal{O}_{Y_0}) = 0 \qquad \text{if } i \geq 2.$$

By 3.1(iii), it follows that

$$\dim \mathcal{O}_{X,y_\alpha} \leq 1,$$

and therefore

$$\operatorname{codim}(Y_\alpha, X) = \inf_{y \in Y_\alpha} \dim \mathcal{O}_{X,y} \leq 1.$$

□

Let A be a noetherian local ring, let $\mathfrak{m}$ be its maximal ideal, and let M be a finitely generated A-module. Suppose that A is a quotient of a regular local ring. Set $X = \operatorname{Spec}(A)$ and, for every $x \in X$, set $\mathfrak{m}_x = \mathfrak{m}A_x$.

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Proposition. The following two conditions are equivalent:

(a) $\mathrm{H}^i(M)$ has finite length;

(b) for every $x \in X - \{\mathfrak{m}\}$,

$$\mathrm{H}^{i-\dim \overline{\{x\}}}_{\mathfrak{m}_x}(M_x) = 0.$$

Proof. In view of (3.2), we may suppose that A is regular. By (2.1), we have

$$\mathrm{H}^i(M) = \mathrm{D}\big(\operatorname{Ext}^{r-i}(M, A)\big),$$

where $r = \dim A$. By (IV 4.7), condition (a) is equivalent[4] to

$$\operatorname{Ext}^{r-i}(M, A) \text{ has finite length.} \tag{24}$$

Condition (24) is equivalent to

$$\text{for every } x \in X - \{\mathfrak{m}\}, \qquad \operatorname{Ext}^{r-i}_A(M, A)_x = 0. \tag{25}$$

On the other hand, A_x is regular of dimension $r - \dim \overline{\{x\}}$. Hence, by (2.1),

$$\begin{aligned} \mathrm{H}^{i-\dim \overline{\{x\}}}_{\mathfrak{m}_x}(M_x) &= \mathrm{D}\Big(\operatorname{Ext}^{(r-\dim \overline{\{x\}})-(i-\dim \overline{\{x\}})}_{A_x}(M_x, A_x)\Big) \\ &= \mathrm{D}\Big(\operatorname{Ext}^{r-i}_{A_x}(M_x, A_x)\Big). \end{aligned} \tag{26}$$

Since M is finitely generated, one has

$$\operatorname{Ext}^{r-i}_A(M, A)_x = \operatorname{Ext}^{r-i}_{A_x}(M_x, A_x),$$

which proves the proposition.

Corollary. For $\mathrm{H}^i(M)$ to have finite length for $i \leq n$, it is necessary and sufficient that

$$\operatorname{depth}(M_x) > n - \dim \overline{\{x\}}$$

for every $x \in X - \{\mathfrak{m}\}$.

Proof. This follows from (3.5) and (III 3.1).

[4]Indeed, one must show that, if E is a finitely generated A-module and $\mathrm{D}(E)$ has finite length, then E has finite length. Let K (respectively Q) be the kernel (respectively cokernel) of the canonical morphism $\epsilon\colon E \to \mathrm{DD}(E)$. The canonical morphism $\gamma\colon \mathrm{D}(E) \to \mathrm{DDD}(E)$ satisfies $\mathrm{D}(\epsilon) \circ \gamma = \mathrm{id}_{\mathrm{D}(E)}$. Since $\mathrm{D}(E)$ has finite length, γ is an isomorphism, and hence so is $\mathrm{D}(\epsilon)$. Since D is exact, it follows that $\mathrm{D}(K)$ and $\mathrm{D}(Q)$ are zero. It is enough to prove that an A-module whose dual is zero is itself zero, for then $E = \mathrm{DD}(E)$ has finite length, just as $\mathrm{D}(E)$ does. Indeed, let M_0 be a finitely generated submodule of such a module M. Since D is exact, $\mathrm{D}(M_0)$ is a quotient of $\mathrm{D}(M)$ and is therefore zero. Again by exactness, $\mathrm{D}(M_0/\mathfrak{m}_A M_0) = 0$; by biduality, the finite-length module $M_0/\mathfrak{m}_A M_0$ is therefore zero. Nakayama's lemma then gives $M_0 = 0$, and finally $M = 0$.

Exposé VI

The Functors $\mathrm{Ext}^\bullet_Z(X; F, G)$ and $\mathcal{E}xt^\bullet_Z(F, G)$

1. Generalities

1.1

Let $(X, \mathcal{O}_X)$ be a ringed space and let Z be a locally closed subset of X. Let F and G be $\mathcal{O}_X$-modules. We shall denote by $\mathrm{Ext}^i_Z(X; F, G)$ (respectively $\mathcal{E}xt^i_Z(F, G)$) the ith derived functor of

$$G \longmapsto \Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G)) \quad \text{(respectively } G \longmapsto \underline{\Gamma}_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G))\text{)}.$$

Lemma 1.2. *The sheaf $\mathcal{E}xt^i_Z(F, G)$ is canonically isomorphic to the sheaf associated with the presheaf*

$$U \longmapsto \mathrm{Ext}^i_{Z\cap U}(U; F|_U, G|_U).$$

This follows from ([Tôhoku], 3.7.2) and from the fact that

$$\Gamma(U; \underline{\Gamma}_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G))) \simeq \Gamma_{Z\cap U}\Big(\mathcal{H}om_{\mathcal{O}_X|_U}(F|_U, G|_U)\Big)$$

canonically.

Theorem 1.3 (Excision theorem). *Let V be an open subset of X containing Z. Then there is an isomorphism of cohomological functors*

$$\mathrm{Ext}^\bullet_X(X; F, G) \simeq \mathrm{Ext}^\bullet_V(V; F|_V, G|_V). \tag{1.3.1}$$

Indeed, if $G^\bullet$ is an injective resolution of G, then $G^\bullet|_V$ is an injective resolution of $G|_V$. The theorem follows immediately.

1.4

Let $\mathcal{O}_{X,Z}$ be the $\mathcal{O}_X$-module defined by the following conditions ([Godement], 2.9.2):

$$\mathcal{O}_{X,Z}|_{X-Z} = 0 \qquad \text{and} \qquad \mathcal{O}_{X,Z}|_Z = \mathcal{O}_X|_Z.$$

We have seen that, for every $\mathcal{O}_X$-module H, there is a functorial isomorphism

$$\Gamma_Z(H) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, H).$$

Consequently, there are isomorphisms functorial in F and G:

$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, \mathcal{H}om_{\mathcal{O}_X}(F,G)), \tag{1.4.1}$$
$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G), \tag{1.4.2}$$
$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \mathrm{Hom}_{\mathcal{O}_X}(F, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, G)) = \mathrm{Hom}_{\mathcal{O}_X}(F, \underline{\Gamma}_Z(G)). \tag{1.4.3}$$

In particular, it follows from (1.4.2) that there is a ∂-functorial isomorphism in F and G,

$$\theta\colon \mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G) \xrightarrow{\sim} \mathrm{Ext}^i_Z(X; F, G).$$

1.5

By definition, the functor $G \longmapsto \Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G))$ is the composite of $G \longmapsto \mathcal{H}om_{\mathcal{O}_X}(F,G)$ and Γ_Z. Since Γ_Z is left exact (I 1.9), since $\mathcal{H}om_{\mathcal{O}_X}(F,G)$ is flasque when G is injective, and since Γ_Z is exact on flasque sheaves (I 2.12), it follows from ([Tôhoku], 2.4.1) that there is a spectral functor abutting to $\mathrm{Ext}^\bullet_Z(X; F, G)$ whose initial term is

$$H^p_Z(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F,G)).$$

On the other hand, it follows from (1.4.3) that $\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G))$ is the composite of $G \longmapsto \underline{\Gamma}_Z(G)$ and $H \longmapsto \mathrm{Hom}_{\mathcal{O}_X}(F,H)$. Since $\underline{\Gamma}_Z$ takes injectives to injectives (I 1.4), it follows from ([Tôhoku], 2.4.1) that there is a spectral functor abutting to $\mathrm{Ext}^\bullet_Z(X; F, G)$ whose initial term is

$$\mathrm{Ext}^p_{\mathcal{O}_X}(X; F, \mathcal{H}^q_Z(G)).$$

Finally, it follows from (1.4.2) and the Ext spectral sequence that there is a spectral functor abutting to $\mathrm{Ext}^\bullet_Z(X; F, G)$ whose initial term is $H^p(X; \mathcal{E}xt^\bullet_Z(F,G))$. Hence:

Theorem 1.6. *There are three spectral functors abutting to* $\mathrm{Ext}^\bullet_Z(X; F, G)$ *whose initial terms are respectively*

$$H^p_Z(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F,G)), \tag{1.6.1}$$
$$H^p(X, \mathcal{E}xt^q_Z(F,G)), \tag{1.6.2}$$
$$\mathrm{Ext}^p_{\mathcal{O}_X}(X; F, \mathcal{H}^q_Z(G)). \tag{1.6.3}$$

1.7

Let Z' be a closed subset of Z, and set $Z'' = Z - Z'$. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_{X,Z''} \longrightarrow \mathcal{O}_{X,Z} \longrightarrow \mathcal{O}_{X,Z'} \longrightarrow 0. \tag{1.7.1}$$

It generalizes the exact sequence of ([Godement], 2.9.3). This exact sequence splits locally; hence, for every $\mathcal{O}_X$-module F, there is another exact sequence

$$0 \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z''} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z'} \longrightarrow 0. \tag{1.7.2}$$

Now let G be an $\mathcal{O}_X$-module. Applying the functor $\mathrm{Hom}_{\mathcal{O}_X}(\bullet, G)$ to (1.7.2), one obtains from (1.4.2) and the long exact sequence of Ext the following theorem.

Theorem 1.8. *Let Z be a locally closed subset of X, let Z' be a closed subset of Z, and set $Z'' = Z - Z'$. There is an exact sequence, functorial in F and G,*

$$0 \longrightarrow \mathrm{Hom}_{Z'}(F,G) \longrightarrow \mathrm{Hom}_Z(F,G) \longrightarrow \mathrm{Hom}_{Z''}(F,G) \longrightarrow \mathrm{Ext}^1_{Z'}(F,G) \longrightarrow \cdots$$
$$\cdots \longrightarrow \mathrm{Ext}^i_Z(F,G) \longrightarrow \mathrm{Ext}^i_{Z''}(F,G) \longrightarrow \mathrm{Ext}^{i+1}_{Z'}(F,G) \longrightarrow \cdots.$$

Corollary 1.9. *Let Y be a closed subset of X, and set $U = X - Y$. There is an exact sequence, functorial in F and G,*

$$0 \longrightarrow \mathrm{Hom}_Y(F,G) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(F,G) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X|_U}(F|_U, G|_U) \longrightarrow \mathrm{Ext}^1_Y(F,G) \longrightarrow \cdots$$
$$\cdots \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X}(F,G) \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X|_U}(F|_U, G|_U) \longrightarrow \mathrm{Ext}^{i+1}_Y(F,G) \longrightarrow \cdots .$$

This corollary is an immediate consequence of Theorems (1.3) and (1.8).

2. Applications to Quasi-Coherent Sheaves on Preschemes

Proposition 2.1. *Let X be a locally Noetherian prescheme. For every locally closed subset Z of X, every coherent $\mathcal{O}_X$-module F, and every quasi-coherent $\mathcal{O}_X$-module G, the sheaves $\mathcal{E}xt^i_Z(F,G)$ are quasi-coherent.*

As in (1.6.3), one shows that the sheaves $\mathcal{E}xt^i_Z(F,G)$ are obtained as the abutment of a spectral sequence whose initial term is

$$\mathcal{E}xt^p_{\mathcal{O}_X}(F, \mathcal{H}^q_Z(G)).$$

By (II, Corollary 3), the $\mathcal{H}^q_Z(G)$ are quasi-coherent, and therefore so are the $\mathcal{E}xt^p_{\mathcal{O}_X}(F, \mathcal{H}^q_Z(G))$, since F is coherent. The proposition follows immediately.

2.2

Let Y now be a closed subprescheme of X, and let $\mathcal{I}$ be an ideal defining Y. Let m and n be integers such that $m \geq n \geq 0$. We denote by $i_{n,m}$ the canonical map

$$\mathcal{O}_{Y_m} = \mathcal{O}_X/\mathcal{I}^{m+1} \longrightarrow \mathcal{O}_X/\mathcal{I}^{n+1} = \mathcal{O}_{Y_n}$$

and by j_n the map $\mathcal{O}_{X,Y} \to \mathcal{O}_{Y_n}$. The $(\mathcal{O}_{Y_n}, i_{n,m})$ form a projective system, and the j_n are compatible with the $i_{n,m}$.

Applying the functor $\mathrm{Ext}^i_{\mathcal{O}_X}(F \otimes \bullet, G)$, we obtain a morphism

$$\varphi' \colon \varinjlim_n \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{X,Y}, G);$$

this is a morphism of cohomological functors in G. The morphism

$$\varphi \colon \varinjlim_n \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathrm{Ext}^i_Y(X; F, G),$$

which is the composite $\theta \circ \varphi'$ (cf. (1.4)), is therefore also a morphism of cohomological functors in G.

In the same way one defines

$$\underline{\varphi} \colon \varinjlim_n \mathcal{E}xt^i_{\mathcal{O}_X}(F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathcal{E}xt^i_Y(X; F, G).$$

Theorem 2.3. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X defined by a coherent ideal $\mathcal{I}$, let F be a coherent $\mathcal{O}_X$-module, and let G be a quasi-coherent $\mathcal{O}_X$-module. Then:*

a) $\underline{\varphi}$ is an isomorphism.

b) If X is Noetherian, φ is an isomorphism.

Proof of Theorem 2.3

The proof of b) is almost word for word the same as that of (II 6 b)), using the spectral sequence (1.6.2), so we do not reproduce it.

For the proof of a), by (2.1) we may assume that X is affine with ring A, that F (respectively G) is defined by an A-module M (respectively N), and that $\mathcal{I}$ is defined by an ideal I. It is enough to prove that the homomorphism

$$\varinjlim_n \mathrm{Ext}^i_A(M/I^nM, N) \longrightarrow \mathrm{Ext}^i_Y(X, F, G) \tag{2.3.1}$$

induced by φ is an isomorphism.

Indeed, for $i = 0$, the two sides of (2.3.1) may be canonically identified with the submodule of $\mathrm{Hom}_A(M, N)$ consisting of the elements of $\mathrm{Hom}_A(M, N)$ annihilated by a power of I. One then sees that the homomorphism (2.3.1) is simply the identity map.

The functor

$$N \longmapsto \varinjlim_n \mathrm{Ext}^\bullet_A(M/I^nM, N)$$

is a universal ∂-functor. We shall show that the same holds for the functor $N \longmapsto \mathrm{Ext}^\bullet_Y(M, N)$. Indeed, if N is an injective module, then by (II, 9 and 11), $\mathcal{H}^q_Y(N) = 0$ for $q \neq 0$; and by (IV.2.2), $H^0_Y(N)$ is injective.

It follows that

$$\mathrm{Ext}^p_{\mathcal{O}_X}(X; M, \mathcal{H}^q_Y(N)) = 0 \qquad \text{if } p + q \neq 0.$$

Thus, by (1.6.3), $\mathrm{Ext}^i_Y(M, N) = 0$ for $i \neq 0$ when N is injective. This completes the proof.

Bibliography

The same references as those listed at the end of Exposé I, cited respectively as [Tôhoku] and [Godement].

Exposé VII

Vanishing Criteria and Coherence Conditions for the Sheaves $\mathcal{E}xt^i_Y(F,G)$

1. Study for $i < n$

We first prove a lemma.

Lemma 1.1. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X, and let G be a quasi-coherent $\mathcal{O}_X$-module. Suppose that, for every coherent $\mathcal{O}_X$-module F whose support is contained in Y, one has*

$$\mathcal{E}xt^{n-1}(F,G) = 0.$$

Then, for every coherent $\mathcal{O}_X$-module F and every closed subset Z of X such that $Y \supset \operatorname{Supp} F \cap Z$, one has an isomorphism

$$\mathcal{E}xt^n_Z(F,G) \cong \mathcal{H}om\big(F, \mathcal{H}^n_Y(G)\big).$$

Proof of Lemma 1.1

We first observe that

$$\mathcal{E}xt^i_Z(F,G) = \mathcal{E}xt^i_{Z\cap \operatorname{Supp} F}(F,G).$$

(This is immediate; cf. Exposé VI.) We first prove the assertion for $Z = X$, so that $\operatorname{Supp} F \subset Y$. On the category of coherent $\mathcal{O}_X$-modules whose support is contained in Y, the functor

$$F \longmapsto \mathcal{E}xt^n(F,G)$$

is left exact. By (IV, 1.3), it is represented by

$$I = \varinjlim_k \mathcal{E}xt^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G),$$

where $\mathcal{I}$ is a defining ideal of Y. By (II, 6), one has

$$\mathcal{H}^n_Y(G) \cong \varinjlim_k \mathcal{E}xt^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G).$$

This proves the assertion when $Z = X$. Again by (VI, 2.3),

$$\mathcal{E}xt^n_Z(F,G) \cong \varinjlim_k \mathcal{E}xt^n(F/\mathcal{J}^{k+1}F, G),$$

where $\mathcal{J}$ is a defining ideal of Z. The support of $F/\mathcal{J}^{k+1}F$ is contained in Y whenever $Z \cap \operatorname{Supp} F \subset Y$. What we have just proved therefore gives

$$\mathcal{E}xt^n_Z(F,G) \cong \varinjlim_k \mathcal{H}om(F/\mathcal{J}^{k+1}F, \mathcal{H}^n_Y(G)).$$

It remains to show that the natural homomorphism

$$\varinjlim_k \mathcal{H}om(F/\mathcal{J}^{k+1}F, \mathcal{H}^n_Y(G)) \longrightarrow \mathcal{H}om(F, \mathcal{H}^n_Y(G))$$

is an isomorphism when $Z \cap \operatorname{Supp} F \subset Y$. Since X admits a cover by Noetherian affine open subsets, we reduce to the case where X is affine Noetherian. Then $F(X)$ is a finitely generated $\mathcal{O}_X(X)$-module and $\operatorname{Supp} \mathcal{H}^n_Y(G) \subset Y$. Hence every homomorphism

$$u : F(X) \longrightarrow \mathcal{H}^n_Y(G)(X)$$

is annihilated by a power of $\mathcal{I}$, and therefore by a power of $\mathcal{J}$. This proves the lemma.

Proposition 1.2. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X, let G be a quasi-coherent $\mathcal{O}_X$-module, and let n be an integer. For any closed subsets Z and S of X such that $Z \cap S = Y$, the following conditions are equivalent:*

(i) $\mathcal{H}^i_Y(G) = 0$ *for* $i < n$;

(ii) there exists a coherent $\mathcal{O}_X$-module F with support S such that

$$\mathcal{E}xt^i_Z(F,G) = 0 \qquad \text{for } i < n;$$

(iii) for every coherent $\mathcal{O}_X$-module F whose support is contained in S (that is, $\operatorname{Supp} F \cap Z = \operatorname{Supp} F \cap Y$*), one has*

$$\mathcal{E}xt^i_Z(F,G) = 0 \qquad \text{for } i < n;$$

(iv) for every coherent $\mathcal{O}_X$-module F, one has

$$\mathcal{E}xt^i_Y(F,G) = 0 \qquad \text{for } i < n.$$

Moreover, if these conditions hold, then for every coherent $\mathcal{O}_X$-module F and every closed subset Z' of X such that $Z' \cap \operatorname{Supp} F = Y \cap \operatorname{Supp} F$, one has isomorphisms

$$\mathcal{E}xt^n_Z(F,G) \cong \mathcal{E}xt^n_Y(F,G) \cong \mathcal{H}om(F, \mathcal{H}^n_Y(G)).$$

Proof of Proposition 1.2: induction

We argue by induction. The proposition is immediate for $n < 0$. Suppose that it has been proved for $n < q$. If one of the conditions holds for $n = q$ and for two subsets Z and S as in the statement, then the induction hypothesis gives, for every closed subset Z' of X and every coherent $\mathcal{O}_X$-module F such that $Z' \cap \operatorname{Supp} F = Y \cap \operatorname{Supp} F$, isomorphisms

$$\mathcal{E}xt^{q-1}_{Z'}(F,G) \cong \mathcal{H}om(F, \mathcal{H}^{q-1}_Y(G)) \cong \mathcal{E}xt^{q-1}_Y(F,G). \tag{1.1}$$

Consequently:

(i)⇒(iv) take $Z' = Y$ in (1.1);

(iv)⇒(iii) take $Z' = Z$ in (1.1);

(iii)⇒(ii) take $F = \mathcal{O}_S$;

(ii)⇒(i) take $Z' = Z$ in (1.1). This gives $\mathcal{H}om(F, \mathcal{H}^{q-1}_Y(G)) = 0$. Moreover,

$$\operatorname{Supp} \mathcal{H}^{q-1}_Y(G) \subset Y = Z \cap S \subset S = \operatorname{Supp} F,$$

so the following lemma applies.

Lemma 1.3. *Let X be a prescheme, let P be a coherent $\mathcal{O}_X$-module, and let H be a quasi-coherent $\mathcal{O}_X$-module such that*

$$\mathcal{H}om(P, H) = 0 \qquad \textit{and} \qquad \operatorname{Supp} P \supset \operatorname{Supp} H.$$

Then $H = 0$.

Proof of Lemma 1.3

It suffices to prove the lemma when X is affine, since the affine open subsets form a basis for the topology of X and the hypotheses are preserved under restriction to an open subset. In that case, we are reduced to a problem about A-modules, where $X = \operatorname{Spec}(A)$. We apply the formula (valid under the sole hypothesis that M is of finite type):

$$\operatorname{Ass} \operatorname{Hom}_A(P, H) = \operatorname{Supp} P \cap \operatorname{Ass} H.$$

We know that

$$\operatorname{Ass} H \subset \operatorname{Supp} H \subset \operatorname{Supp} P \qquad \text{and} \qquad \operatorname{Ass} \operatorname{Hom}_A(P, H) = \varnothing;$$

hence $\operatorname{Ass} H = \varnothing$, and therefore $H = 0$.

Close of the proof of Proposition 1.2

To complete the proof of the proposition, it remains to observe that condition (iv) permits the application of Lemma 1.1.

Corollary 1.4. *Let G be a coherent Cohen–Macaulay $\mathcal{O}_X$-module, and let $n \in \mathbb{Z}$. The conditions of Proposition 1.2 are equivalent to:*

$$\text{(v)} \qquad \operatorname{codim}(Y \cap \operatorname{Supp} G, \operatorname{Supp} G) \geq n.$$

Proof

Recall first that an $\mathcal{O}_X$-module is said to be Cohen–Macaulay if, for every $x \in X$, the stalk G_x is a Cohen–Macaulay $\mathcal{O}_{X,x}$-module; that is, for every $x \in S = \operatorname{Supp} G$,

$$\operatorname{depth} G_x = \dim G_x = \dim \mathcal{O}_{S,x}. \tag{1.2}$$

By Proposition III.3.3, condition (i) of Proposition 1.2 is equivalent to

$$\operatorname{depth}_Y G = \inf_{x \in Y} \operatorname{depth} G_x \geq n, \tag{1.3}$$

and hence also to

$$\operatorname{depth}_Y G = \inf_{x \in Y \cap S} \operatorname{depth} G_x \geq n,$$

since the depth of the zero module is infinite.

By definition,

$$\operatorname{codim}(Y \cap S, S) = \inf_{x \in S \cap Y} \dim \mathcal{O}_{S,x};$$

the conclusion follows by applying formula (1.2).

We shall now prove a result that will allow us to deduce the coherence conditions we have in view from certain vanishing criteria.

Lemma 1.5. *Let X be a locally Noetherian prescheme. Let T^* be an exact contravariant ∂-functor defined on the category of coherent $\mathcal{O}_X$-modules and taking values in the category of $\mathcal{O}_X$-modules. Let Y be a closed subset of X, and let $i \in \mathbb{Z}$. Suppose that, for every coherent $\mathcal{O}_X$-module F whose support is contained in Y, the modules T^iF and $T^{i-1}F$ are coherent. Let F be a coherent $\mathcal{O}_X$-module. Then T^iF is coherent if and only if T^iF'' is coherent, where*

$$F'' = F/\Gamma_Y(F).$$

Indeed, $F' = \Gamma_Y(F)$ is coherent because X is locally Noetherian. The long exact cohomology sequence of T^* then gives

$$T^{i-1}F' \longrightarrow T^iF'' \longrightarrow T^iF \longrightarrow T^iF',$$

where the two outer terms are coherent; the conclusion follows.

Lemma 1.6. *If F and G are coherent and $\operatorname{Supp} F \subset Y$, then $\mathcal{E}xt^i_Y(F,G)$ is coherent.*

Indeed, $\mathcal{E}xt^i_Y(F,G)$ is isomorphic to $\mathcal{E}xt^i(F,G)$. More generally, this holds on any ringed space X: if Z is a closed subset containing $Y \cap \operatorname{Supp} F$, then $\mathcal{E}xt^i_Z(F,G)$ is isomorphic to $\mathcal{E}xt^i_Y(F,G)$ (cf. Exposé VI).

Proposition 1.7. *Suppose that F and G are coherent, and put*

$$\operatorname{Supp} F = S, \qquad S' = \overline{S \cap (X - Y)}.$$

Suppose that $\operatorname{depth} G_x \geq n$ for every $x \in Y \cap S'$. Then $\mathcal{E}xt^i_Y(F,G)$ is coherent for $i < n$.

Indeed, Lemma 1.6 allows us to apply Lemma 1.5 to

$$T^*(F) = \mathcal{E}xt^*_Y(F,G).$$

Putting $F'' = F/\Gamma_Y(F)$, we see that $\operatorname{Supp} F'' = S'$. By Proposition III.3.3, the hypothesis on the depth of G ensures that $\mathcal{H}^i_{Y \cap S'}(G) = 0$ for $i < n$; Proposition 1.2 then gives $T^iF'' = 0$ for $i < n$, and the conclusion follows from Lemma 1.5.

2. Study for $i > n$

Let X be a locally Noetherian *regular* prescheme; that is, suppose that all its local rings are regular. Let Y be a closed subset of X, and let F and G be two coherent $\mathcal{O}_X$-modules. Put

$$S = \operatorname{Supp} F, \qquad S' = \overline{S \cap (X - Y)},$$

and put

$$m = \sup_{x \in Y \cap S} \dim \mathcal{O}_{X,x},$$
$$n = \sup_{x \in Y \cap S'} \dim \mathcal{O}_{X,x}.$$

Then $n \leq m$.

Proposition 2.1. *In the situation described above:*

1. $\mathcal{E}xt^i_Y(F,G) = 0$ *if* $i > m$;

2. $\mathcal{E}xt^i_Y(F,G)$ *is coherent if* $i > n$.

Proof of Proposition 2.1

First note that $\mathcal{E}xt^i_Y(F,G)$ is coherent for every i when $\operatorname{Supp} F \subset Y$. Moreover, putting $F'' = F/\Gamma_Y(F)$ as above, we see that $\operatorname{Supp} F'' = S'$. Thus assertion (2) follows from assertion (1) and Lemma 1.5.

To prove assertion (1), first observe that

$$\mathcal{E}xt^i_Y(F,G) \cong \varinjlim_k \mathcal{E}xt^i(F/\mathcal{J}^k F, G),$$

where $\mathcal{J}$ is the ideal defining Y. On the other hand, Theorem 4.2.2 of A. Grothendieck, "Sur quelques points d'algèbre homologique," *Tôhoku Mathematical Journal* **9** (1957), 119–221, shows that the $\mathcal{E}xt$ sheaves commute with the formation of stalks, at least when X is a locally Noetherian prescheme and the first argument is coherent. Since direct limits do likewise, for every $x \in X$ there are isomorphisms

$$(\mathcal{E}xt^i_Y(F,G))_x \cong \varinjlim_k \operatorname{Ext}^i_{\mathcal{O}_{X,x}}((F/\mathcal{J}^k F)_x, G_x).$$

Since

$$\operatorname{Supp} \mathcal{E}xt^i_Y(F,G) \subset S \cap Y,$$

it is enough to note that $x \in Y \cap S$ implies $\dim \mathcal{O}_{X,x} \leq m$, and hence

$$\operatorname{Ext}^i_{\mathcal{O}_{X,x}}((F/\mathcal{J}^k F)_x, G_x) = 0 \qquad \text{if } i > m,$$

because the global cohomological dimension of a regular local ring is equal to its dimension.[1]

Let X be a locally Noetherian prescheme. For every subset P of X, set

$$D(P) = \{\dim \mathcal{O}_{X,p} \mid p \in P\}.$$

Lemma 2.2

If P is the underlying space of a connected subprescheme of X, then $D(P)$ is an interval.

Indeed, let a and b belong to $D(P)$, corresponding to points p and q of P. We show that there is a sequence of points of P,

$$p = p_1, \ldots, p_n = q,$$

such that, for $1 \leq i < n$,

$$|\dim \mathcal{O}_{X,p_i} - \dim \mathcal{O}_{X,p_{i+1}}| = 1.$$

It will follow that $D(P)$ contains the interval $[a,b]$.

For this, observe that p and q can be joined by a chain of irreducible components of P in which two successive components meet. We are reduced to the case where p is the generic point of an irreducible component Q of P, and where $q \in Q$, hence $q \supset p$ as ideals of $\mathcal{O}_q$; there the assertion follows immediately from the definition of dimension.

[1] Cf. EGA 0_{IV}, 17.3.1.

Proposition 2.3

Let X be a locally Noetherian regular prescheme, let Y be a closed subset of X, and let F be a coherent $\mathcal{O}_X$-module. Put

$$P = Y \cap \overline{\operatorname{Supp} F \cap (X - Y)}.$$

Let $n \in \mathbb{Z}$, and suppose that $n \notin D(P)$. Then $\mathcal{E}xt^n_Y(F, \mathcal{O}_X)$ is coherent.

Proof

The conclusion is local, and the hypotheses are preserved under restriction to an open subset. Since P is closed, it is locally Noetherian and hence locally connected. We may therefore suppose that X is affine and Noetherian and that P is connected.

If $P = \varnothing$, the depth hypothesis of Proposition 1.7 with threshold $n+1$ is vacuous, and the conclusion follows. Assume henceforth that $P \neq \varnothing$. By Lemma 2.2, $D(P)$ is an interval. Since $n \notin D(P)$, exactly one of the following alternatives holds:[2]

$$n > d \quad \text{for every } d \in D(P), \qquad \text{or} \qquad n < d \quad \text{for every } d \in D(P).$$

In the first case, put $m = \sup D(P)$. Then $n > m$, and Proposition 2.1, applied with $G = \mathcal{O}_X$, shows that $\mathcal{E}xt^n_Y(F, \mathcal{O}_X)$ is coherent.

In the second case, for every $x \in P$ regularity gives

$$n < \dim \mathcal{O}_{X,x} = \operatorname{depth} \mathcal{O}_{X,x},$$

and hence $\operatorname{depth} \mathcal{O}_{X,x} \geq n+1$. Proposition 1.7 with threshold $n+1$ therefore gives coherence in every degree $i < n+1$, in particular in degree $i = n$.

[2]The French TeX and print write $D(P) = [a, b[$ and then treat only $n > b$ and $n < a$, leaving the allowed case $n = b$ uncovered. The print does not decide whether the intended one-character repair was a closed upper endpoint or a non-strict upper inequality. The reader states the order alternatives forced by $n \notin D(P)$ and Lemma 2.2 and therefore does not guess that glyph.

Exposé VIII

The Finiteness Theorem

1. A Biduality Spectral Sequence[1]

Let us state the result we wish to establish:

Proposition 1.1. *Let A be a Noetherian ring and let I be an ideal of A. Put $X = \operatorname{Spec}(A)$ and $Y = V(I)$. Let M be a finitely generated A-module of finite projective dimension. Let $\mathcal{F} = \widetilde{M}$ be the $\mathcal{O}_X$-module associated with M.*

1) *There is a spectral sequence*

$$\mathrm{H}_Y(X, \mathcal{F}) \Longleftarrow \operatorname{Ext}_Y^p(\operatorname{Ext}^{-q}(M, A), A).$$

2) *There is a spectral sequence*

$$\mathcal{H}_Y(X, \mathcal{F}) \Longleftarrow \mathcal{E}xt_Y^p(\mathcal{E}xt^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

Of course, 2) follows from 1) by observing that, if M and N are two finitely generated A-modules and if we put $\mathcal{F} = \widetilde{M}$ and $\mathcal{G} = \widetilde{N}$, then there are isomorphisms

$$\mathcal{H}_Y(\mathcal{F}) \simeq \widetilde{\mathrm{H}_Y(X, \mathcal{F})},$$
$$\mathcal{E}xt_Y(\mathcal{F}, \mathcal{G}) \simeq \widetilde{\operatorname{Ext}_Y(\mathcal{F}, \mathcal{G})},$$
$$\mathcal{E}xt_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \simeq \widetilde{\operatorname{Ext}_A(M, N)}.$$

Let $\mathcal{C}$ be the category of A-modules, and let $\mathbf{Ab}$ be the category of abelian groups. Let $\underline{\mathrm{F}}$ be the functor

$$\underline{\mathrm{F}} \colon \mathcal{C} \longrightarrow \mathbf{Ab},$$
$$M \longmapsto \Gamma_Y(\widetilde{M}).$$

We know from Exposé II that there is an isomorphism of ∂-functors

$$\mathrm{H}_Y^*(X, \widetilde{M}) \simeq \mathrm{R}^*\underline{\mathrm{F}}(M).$$

Furthermore, let Ext_Y^* denote the right derived functors, in the second argument, of

$$\underline{\mathrm{F}} \circ \operatorname{Hom} \colon \mathcal{C}^\circ \times \mathcal{C} \longrightarrow \mathbf{Ab}.$$

[1] A reader familiar with Verdier's language of derived categories will recognize the spectral sequence associated with a *biduality isomorphism*. Cf. SGA 6, I.

We know from Exposé VI that there is an isomorphism of ∂-functors

$$\mathrm{Ext}^*_Y(M,N) \simeq \mathrm{Ext}^*_Y(\widetilde{M},\widetilde{N}).$$

Finally, let us record the following result from Exposé VI: if C is an injective A-module and N is a finitely generated A-module, the sheaf

$$\mathcal{H}om(\widetilde{N},\widetilde{C}) \simeq \widetilde{\mathrm{Hom}(N,C)}$$

is flasque; hence

$$\mathrm{R}^1\underline{\mathrm{F}}(\mathrm{Hom}(N,C)) = 0.$$

It remains for us to prove the following result.

Lemma 1.2. — Let A be a Noetherian ring and let $\mathcal{C}$ be the category of A-modules. Let $\underline{\mathrm{F}}\colon \mathcal{C} \longrightarrow \mathbf{Ab}$ be a left exact additive functor such that, for every finitely generated A-module N and every injective A-module C, one has

$$\mathrm{R}^1\underline{\mathrm{F}}(\mathrm{Hom}(N,C)) = 0.$$

Let M be a finitely generated A-module of finite projective dimension. There is a spectral sequence

$$\mathrm{R}^*\underline{\mathrm{F}}(M) \Longleftarrow \mathrm{Ext}^p_{\underline{\mathrm{F}}}(\mathrm{Ext}^{-q}(M,A),A),$$

where $\mathrm{Ext}^p_{\underline{\mathrm{F}}}$ denotes the p-th right derived functor of $\underline{\mathrm{F}} \circ \mathrm{Hom}$.

We shall consider only complexes whose differential has degree $+1$. By the hypothesis on M, there exists a projective resolution of M of finite length,

$$u\colon L^\bullet \longrightarrow M,$$

where, moreover, the L^p are finitely generated modules and $L^p = 0$ if $p \notin [-n,0]$. On the other hand, let $v\colon M \longrightarrow I^\bullet$ be an injective resolution of M. I claim that

$$v \circ u\colon L^\bullet \longrightarrow I^\bullet \tag{1.1}$$

is an injective resolution of $L^\bullet$. We must specify what this means.

Definition 1.3. — Let $X^\bullet$ be a complex of A-modules. An *injective resolution* of $X^\bullet$ is a homomorphism of complexes

$$x\colon X^\bullet \longrightarrow CX^\bullet$$

such that CX^p is injective for every $p \in \mathbb{Z}$, and x induces an isomorphism in homology.

Proposition 1.4. — Every left-bounded complex, i.e., every complex for which there exists $q \in \mathbf{Z}$ such that $X^p = 0$ for $p < q$, admits an injective resolution. Moreover, suppose that $u\colon X^\bullet \longrightarrow Y^\bullet$ is a homomorphism of complexes (both left-bounded), and that $x\colon X^\bullet \longrightarrow CX^\bullet$ and $y\colon Y^\bullet \longrightarrow CY^\bullet$ are injective resolutions of $X^\bullet$ and $Y^\bullet$. Then there exists a homomorphism of complexes

$$Cu\colon CX^\bullet \longrightarrow CY^\bullet,$$

unique up to homotopy, such that the diagram

$$\begin{array}{ccc} X^\bullet & \xrightarrow{x} & CX^\bullet \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle Cu} \\ Y^\bullet & \xrightarrow{y} & CY^\bullet \end{array}$$

commutes up to homotopy.

The proof is left to the reader.[2]

Let us recall a notation introduced in Exposé V.

Notation. — Let $X^\bullet$ and $Y^\bullet$ be two complexes. We denote by $\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)$ the *simple complex* whose component of degree n is

$$(\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet))^n = \prod_{-p+q=n} \mathrm{Hom}(X^p, Y^q),$$

also denoted by $\mathrm{Hom}^n(X^\bullet, Y^\bullet)$, and whose differential is given by

$$\partial_n \colon \mathrm{Hom}^n(X^\bullet, Y^\bullet) \longrightarrow \mathrm{Hom}^{n+1}(X^\bullet, Y^\bullet),$$
$$\partial_n = d' + (-1)^{n+1} d'',$$

where d' and d'' are the differentials (of degree $+1$) induced by those of $X^\bullet$ and $Y^\bullet$.

Let $A^\bullet$ then be the complex defined by $A^p = 0$ if $p \neq 0$ and $A^0 = A$. Let

$$a \colon A^\bullet \longrightarrow CA^\bullet$$

be an injective resolution of $A^\bullet$. Consider the double complex

$$Q^{p,q} = \mathrm{Hom}(\mathrm{Hom}(L^{-q}, A), CA^p). \tag{1.2}$$

The first spectral sequence of the bicomplex $\underline{\mathrm{F}}Q^{\bullet\bullet}$ will give the conclusion of Lemma 1.2.

Set

$$L'^\bullet = \mathrm{Hom}^\bullet(L^\bullet, A^\bullet), \tag{1.3}$$

and

$$P^\bullet = \mathrm{Hom}^\bullet(L'^\bullet, CA^\bullet). \tag{1.4}$$

One sees easily that $P^\bullet$ is the simple complex associated with $Q^{\bullet\bullet}$. Let us compute the abutment of the spectral sequence, that is, the homology of $\underline{\mathrm{F}}P^\bullet$. To this end, using the fact that $L^\bullet$ is projective of finite type in every degree, one proves that

$$L^\bullet \cong \mathrm{Hom}^\bullet(L'^\bullet, A^\bullet).$$

From the homomorphism $a \colon A^\bullet \to CA^\bullet$, one obtains a homomorphism

$$b \colon \mathrm{Hom}^\bullet(L'^\bullet, A^\bullet) \longrightarrow \mathrm{Hom}^\bullet(L'^\bullet, CA^\bullet),$$

or equivalently a homomorphism

$$c \colon L^\bullet \longrightarrow P^\bullet. \tag{1.5}$$

This being said, it is easy to see, using the fact that $L'^\bullet$ is projective of finite type in every degree and bounded on the left, that (1.5) is an injective resolution of $L^\bullet$. Using Proposition 1.4, one concludes that $P^\bullet$ is homotopy equivalent to $I^\bullet$, where $I^\bullet$ is the injective resolution of M introduced above in (1.1). It follows that the *abutment* of the first spectral sequence of the double complex $\underline{\mathrm{F}}Q^{\bullet\bullet}$, which is $H^*(\underline{\mathrm{F}}P^\bullet)$, *is isomorphic* to $R^*\underline{\mathrm{F}}(M)$.

The initial term of the first spectral sequence of the bicomplex $\underline{\mathrm{F}}Q^{\bullet\bullet}$ is

$$E_2^{p,q} = {}'H^p({}''H^q(\underline{\mathrm{F}}Q^{\bullet\bullet})).$$

For every $p \in \mathbf{Z}$, CA^p is injective. By the hypothesis on $\underline{\mathrm{F}}$, the functor (restricted to the category of finitely generated modules)

$$N \longmapsto \underline{\mathrm{F}}\,\mathrm{Hom}(N, CA^p)$$

[2] See also H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton Mathematical Series, vol. 19, Princeton University Press, 1956.

is exact. Hence one obtains isomorphisms

$$''H^q(\underline{\mathrm{F}}\,\mathrm{Hom}(L'^{\bullet}, CA^p)) \cong \underline{\mathrm{F}}\,\mathrm{Hom}(H^{-q}(L'^{\bullet}), CA^p).$$

By the definition of $\mathrm{Ext}^*_{\underline{\mathrm{F}}}$ as the right-derived functor of $\underline{\mathrm{F}} \circ \mathrm{Hom}$, one therefore obtains isomorphisms

$$E_2^{p,q} \cong \mathrm{Ext}^p_{\underline{\mathrm{F}}}(H^{-q}(L'^{\bullet}), A).$$

Now

$$L'^{\bullet} = \mathrm{Hom}^{\bullet}(L^{\bullet}, A^{\bullet}),$$

where $L^{\bullet}$ is a projective resolution of M; hence there are isomorphisms

$$\mathrm{Ext}^q(M, A) \cong H^q(L'^{\bullet}).$$

Substituting $-q$ in this last identity gives the conclusion. □

VIII.2 The Finiteness Theorem

Theorem VIII.2.1. [(1)]. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Suppose that X is locally embeddable in a regular prescheme*[*]*. Let $i \in \mathbf{Z}$. Suppose that:*

a) *for every $x \in U = X - Y$, one has*

$$H^{i-c(x)}(\mathcal{F}_x) = 0,$$

where we have set[(2)]*:*

$$c(x) = \mathrm{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \tag{VIII.2.6}$$

Then:

b) *$\mathcal{H}^i_Y(\mathcal{F})$ is coherent.*

Corollary VIII.2.2. *Editor's note. Strictly speaking, this is a corollary of the proof that follows, not of the statement. The implication* c)⇒a) *is tautological. The converse is not, but follows from the proof. More precisely, as below, one covers X by open subsets embeddable in regular schemes; as explained below, this permits one to reduce to the case where $X = \mathrm{Spec}(A)$ is affine and regular and $\mathcal{F} = \widetilde{M}$, where M is an A-module of finite projective dimension. In this case it is shown that conditions* a) *and* c) *are equivalent to the dual conditions* a′) *and* c′)*. One then shows that* c′) *implies condition* d) *(see below), which in turn implies* a′)*. See the discussion following 2.4. Under the hypotheses of the preceding theorem, condition* a) *is equivalent to:*

c) *for every $x \in U$ such that $c(x) = 1$, one has $H^{i-1}(\mathcal{F}_x) = 0$.*

Corollary VIII.2.3. *Let X be a locally Noetherian prescheme that is locally embeddable in a regular prescheme, let Y be a closed subset of X, let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module, and let n be an integer. The following conditions are equivalent:*

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(1) *Editor's note.* For an analogous statement, but in a somewhat more general situation, see Mme Raynaud (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental," *Ann. Sci. Éc. Norm. Sup.* (4) **7** (1974), pp. 29–52, Proposition II.2.1).

[*] This condition may be generalized to the hypothesis that a "dualizing complex," in the sense defined in R. Hartshorne, *Residues and Duality* (cited in the note to Exposé IV, p. 46), exists locally on X.

(2) *Editor's note.* As in Exposé V, $H^*(\mathcal{F}_x)$ denotes the local cohomology $H^*_{\mathfrak{m}_x}(\mathcal{F}_x)$.

(i) *for every* $x \in U$, *one has* $\operatorname{prof} \mathcal{F}_x > n - c(x)$;

(ii) *for every* $x \in U$ *such that* $c(x) = 1$, *one has* $\operatorname{prof} \mathcal{F}_x \geq n$;

(iii) *for every* $i \in \mathbb{Z}$, $\mathcal{H}^i_Y(\mathcal{F})$ *is coherent if* $i \leq n$;

(iv) $R^i i_*(\mathcal{F}|_U)$ *is coherent for* $i < n$[4].

Assume these results established when X is the spectrum of a regular Noetherian ring A and when $\mathcal{F}$ is the sheaf associated with an A-module of finite projective dimension.

First observe that, if $(X_j)_{j\in J}$ is an open covering of X by open subsets embeddable in a regular scheme, each of the conditions above is equivalent to the conjunction of the analogous conditions obtained by replacing X by X_j, Y by $Y_j = Y \cap X_j$, and $\mathcal{F}$ by $\mathcal{F}|_{X_j}$. Indeed, only the conditions involving $c(x)$ can present a difficulty. Let $x \in U$. If $x \in X_j$, set

$$c_j(x) = \operatorname{codim}(X_j \cap \overline{\{x\}} \cap Y, X_j \cap \overline{\{x\}}).$$

Necessarily $c_j(x) \geq c(x)$. Let $y \in \overline{\{x\}} \cap Y$ be a point that "realizes the codimension," that is, such that $c(x) = \dim \mathcal{O}_{\overline{\{x\}},y}$, and let X_j be a member of the covering such that $y \in X_j$. Then $x \in X_j$, and hence $c_j(x) = c(x)$. It follows that condition (a) for the X_j implies condition (a) for X. At this point we have only a partial converse: condition (a) for X implies condition (a) for those X_j such that $c(x) = c_j(x)$, which suffices for our purposes.[5]

Choose a covering of X by open subsets embeddable in a regular prescheme. Applying the preceding observation, we see that we may assume that X is closed in a regular X'. The reduction to X' is then immediate.

We may therefore assume that X is regular, and even affine, by covering X with affine open subsets. That we may assume that $\mathcal{F} = \widetilde{M}$, where M has finite projective dimension, will follow from the next lemma.

Lemma VIII.2.4. *Let X be a regular Noetherian prescheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. The function that assigns to each $x \in X$ the projective dimension of $\mathcal{F}_x$ is bounded above.*

Indeed, let $x \in X$, and let U be an affine open neighborhood of x. Let $L^\bullet$ be a projective resolution of the module $\mathcal{F}(U)$, where the L^i are finitely generated. By hypothesis, the ring $\mathcal{O}_{X,x}$ is regular, so the projective dimension of $\mathcal{F}_x$ is finite; denote this integer by d. Set

$$K = \ker(L^{-d} \longrightarrow L^{-d+1}).$$

The module K_x is free, since d is the projective dimension of $\mathcal{F}_x$ ([M], Ch. VI, Prop. 2.1). By (EGA 0_{I}, 5.4.1 Errata), it follows that the $\mathcal{O}_U$-module $\widetilde{K}$ is free on a neighborhood U' of x, with $U' \subset U$. Choose $f \in \mathcal{O}_X(U)$ such that $x \in D(f) \subset U'$. We then have a projective resolution of M_f, where $M = \mathcal{F}(U)$:

$$0 \longrightarrow K_f \longrightarrow (L^{-d})_f \longrightarrow \cdots \longrightarrow M_f \longrightarrow 0.$$

This proves that the function under consideration is upper semicontinuous. Since X is quasi-compact, the conclusion follows.

[4] *Editor's note.* This condition appeared only in the body of the proof, but not in the statement of the corollary; since it is used in Section 3, it has been added.

[5] *Editor's note.* In fact, condition (a) for X implies condition (a) for every X_j, as asserted in the original text, but establishing this requires reading the proof below in detail. At this point the implication does not appear formal. For a property (a), (b), or (c), append a subscript J to denote its conjunction over the X_j. The proof *below* establishes $(c_J) \Rightarrow (a_J)$ (this is the chain of implications $(c') \Rightarrow (d) \Rightarrow (a')$). But tautologically $(a) \Rightarrow (c)$, and $(c) \Longleftrightarrow (c_J)$; hence $(a) \Rightarrow (a_J)$.

We henceforth assume that X is affine, Noetherian, and regular, and that $\mathcal{F} = \widetilde{M}$, where M is a finitely generated A-module, necessarily of finite projective dimension. We shall proceed in several steps. First, we find a condition (d), equivalent to (a), and prove that it is also equivalent to (c). Then, using the spectral sequence of the preceding subsection, we prove $(d) \Rightarrow (b)$. It remains to prove (iii) $\Rightarrow$ (ii); indeed, (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii) follows immediately from $(a) \Leftrightarrow (c) \Rightarrow (b)$.

Let $x \in U$. By hypothesis, $\mathcal{O}_{X,x}$ is a regular local ring. If D denotes the dualizing functor for $\mathcal{O}_{X,x}$, it follows from (V, 2.1) that

$$D\,H^{i-c(x)}(\mathcal{F}_x) \simeq \mathrm{Ext}^{d(x)-i}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{O}_{X,x}),$$

where

$$d(x) = \dim \mathcal{O}_{X,x} + c(x) = \dim \mathcal{O}_{X,x} + \mathrm{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \tag{2.2}$$

Since X is Noetherian and $\mathcal{F}$ is coherent,

$$D\,H^{i-c(x)}(\mathcal{F}_x) \simeq (\mathcal{E}xt^{d(x)-i}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X))_x. \tag{2.3}$$

Moreover, a module vanishes if and only if its dual vanishes (cf. the editor's note (4) on p. 54). For every $q \in \mathbb{Z}$, set

$$\begin{cases} S_q = \mathrm{Supp}\,\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X), \\ S'_q = S_q \cap U, \quad (U = X - Y), \\ Z_q = \overline{S'_q} \cap Y. \end{cases} \tag{2.4}$$

It follows from formula (2.3) that conditions (a) and (c) are respectively equivalent to:

(a′) for every $q \in \mathbb{Z}$ and every $x \in S'_q$, one has $q + i \neq d(x)$;

(c′) for every $q \in \mathbb{Z}$ and every $x \in S'_q$, if $c(x) = 1$, then $q + i \neq d(x)$.

Here is condition (d), promised above:

(d) for every $q \in \mathbb{Z}$ and every $y \in Z_q$, one has $q + i \neq \dim \mathcal{O}_{X,y}$.

These conditions are equivalent.

(a′) $\Rightarrow$ (c′) is recorded for completeness.

(d) $\Rightarrow$ (a′). Indeed, let $q \in \mathbb{Z}$ and $x \in S'_q$. Choose a point $y \in \overline{\{x\}} \cap Y$ that realizes the codimension,[6] that is, such that

$$\dim \mathcal{O}_{\overline{\{x\}},y} = \mathrm{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}) = c(x). \tag{2.5}$$

Since X is regular at y, it follows that

$$\dim \mathcal{O}_{X,y} = d(x) \qquad \text{(cf. (2.2))}. \tag{2.6}$$

But $y \in \overline{\{x\}}$, and hence $y \in Z_q$; the conclusion follows.

(c′) $\Rightarrow$ (d). Let $q \in \mathbb{Z}$ and $y \in Z_q$. Assume provisionally that there exists $x \in S'_q$ such that

$$y \in \overline{\{x\}} \qquad \text{and} \qquad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

(One also says that y is an immediate specialization of x.) It follows that $c(x) = 1$, since y realizes the codimension of $\overline{\{x\}} \cap Y$ in $\overline{\{x\}}$, while $x \notin Y$. Condition (c′) therefore gives

$$q + i \neq d(x).$$

[6] *Editor's note.* Throughout what follows, closures of points are endowed with the reduced structure.

Whence the conclusion, upon observing that $d(x) = \dim \mathcal{O}_{X,y}$ by (2.6). The assertion provisionally assumed above is the content of the following lemma.

Lemma 2.5. Let X be a locally Noetherian prescheme and Y a closed subset of X. Set $U = X - Y$, and suppose that U is dense in X. For every $y \in Y$, there exists $x \in U$ such that y is an immediate specialization of x, that is,

$$y \in \overline{\{x\}} \qquad \text{and} \qquad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

We have applied the lemma with X equal to the prescheme $\overline{S'_q}$ and Y equal to the subset $Y \cap \overline{S'_q}$.

Proof of Lemma 2.5. There exists $x \in U$ such that $y \in \overline{\{x\}}$. Choose such an x for which $r = \dim \mathcal{O}_{\overline{\{x\}},y}$ is minimal. We must prove that $r = 1$. By the choice of x, every $z \in \operatorname{Spec}(\mathcal{O}_{\overline{\{x\}},y})$ with $z \neq x$ lies in Y. Consequently, $\{x\}$ is open in $\operatorname{Spec}(\mathcal{O}_{\overline{\{x\}},y})$, and the conclusion follows.□

The second step is to deduce (b) *from* (d).

Set

$$D(Z_q) = \{\dim \mathcal{O}_{X,y} \mid y \in Z_q\}.$$

By (d), for every $q \in \mathbb{Z}$ one has $q + i \notin D(Z_q)$. Applying Proposition VII.2.3, we see that

$$\mathcal{E}xt_Y^{q+i}(\mathcal{E}xt^q(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X) \quad \text{is coherent.}$$

The initial term of the spectral sequence of the preceding subsection is

$$E_2^{p,q} = \mathcal{E}xt_Y^p(\mathcal{E}xt^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

It follows that $E_2^{p,q}$ is coherent for every $p, q \in \mathbb{Z}$ such that $p + q = i$. Now there are only finitely many pairs (p, q) such that $p + q = i$, and this spectral sequence converges to $\mathcal{H}_Y^*(\mathcal{F})$; whence the conclusion.

It remains to prove that (iii) *implies* (ii). Let

$$i\colon U \longrightarrow X$$

denote the canonical immersion of U into X. In view of the exact homology sequence associated with the closed subset Y (I, 2.11), one sees that (iii) is *equivalent to*:

(iv) $R^i i_*(\mathcal{F}|_U)$ is coherent for $i < n$.

French printed p. 94 Indeed, there is an exact sequence

$$0 \longrightarrow \mathcal{H}_Y^0(\mathcal{F}) \longrightarrow \mathcal{F} \longrightarrow i_*(\mathcal{F}|_U) \longrightarrow \mathcal{H}_Y^1(\mathcal{F}) \longrightarrow 0.$$

Now $\mathcal{H}_Y^0(\mathcal{F})$ is a quasi-coherent subsheaf of the coherent sheaf $\mathcal{F}$, and is therefore coherent. Hence $\mathcal{H}_Y^1(\mathcal{F})$ is coherent if and only if $i_*(\mathcal{F}|_U)$ is coherent. Moreover, for $p > 0$, the exact cohomology sequence associated with the closed subset Y reduces to isomorphisms

$$R^p i_*(\mathcal{F}|_U) \xrightarrow{\approx} \mathcal{H}_Y^{p+1}(\mathcal{F}).$$

We shall prove that (iv) *implies* (ii). To this end, recall (ii):

(ii) For every $x \in U$ such that $c(x) = 1$, one has

$$\operatorname{prof} \mathcal{F}_x \geq n.$$

We argue by induction on n.

If $n = 0$, both conditions are vacuous.

If $n = 1$, assume that $i_*(\mathcal{F}|_U)$ is coherent. We argue by contradiction and suppose that there exists $x \in U$ such that $c(x) = 1$ and

$$\operatorname{prof} \mathcal{F}_x = 0,$$

i.e. $x \in \operatorname{Ass} \mathcal{F}_x$. Let $y \in \overline{\{x\}} \cap Y$ be such that

$$\dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

Set

$$A = \mathcal{O}_{X,y} \qquad \text{and} \qquad X' = \operatorname{Spec}(A).$$

Perform the base change $v\colon X' \longrightarrow X$, which is flat:

$$\begin{array}{ccc} U' = X' \times_X U & \xrightarrow{v'} & U \\ {\scriptstyle i'}\downarrow & & \downarrow{\scriptstyle i} \\ X' & \xrightarrow{v} & X. \end{array} \tag{2.7}$$

The morphism i is separated, since it is an immersion, and is of finite type, since it is an open immersion and X is locally Noetherian. Since the base change is flat, EGA III, 1.4.15 gives an isomorphism

$$v^*(i_*(\mathcal{F}|_U)) \approx i'_*(v'^*(\mathcal{F}|_U)). \tag{2.8}$$

French printed p. 95 Denote by $\underline{x}$ (respectively, $\underline{y}$) the ideal of A corresponding to x (respectively, y).

Set $\mathcal{G} = v'^*(\mathcal{F}|_U)$. Then $\mathcal{G}$ is coherent and $\underline{x} \in \operatorname{Ass} \mathcal{G}$, so there exists a monomorphism

$$\mathcal{O}_{\overline{\{x\}}} \longrightarrow \mathcal{G};$$

consequently, $i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})$ is coherent. By the choice of y, $\dim(A/\underline{x}) = 1$, and therefore the support of $\mathcal{O}_{\overline{\{x\}}}$ is precisely

$$\overline{\{\underline{x}\}} = \{\underline{x}\} \cup \{\underline{y}\},$$

since $\overline{\{\underline{x}\}} = \operatorname{Spec}(A/\underline{x})$ as a scheme. It follows that

$$(\mathcal{O}_{\overline{\{x\}}}|_{U'})(U') = \operatorname{Frac}(A/\underline{x}),$$

the field of fractions of $A/\underline{x}$, and

$$i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})(X') = \operatorname{Frac}(A/\underline{x}).$$

But $\operatorname{Frac}(A/\underline{x})$ is not a finitely generated A-module, because $\underline{x}$ is different from the maximal ideal of A. This is a contradiction.

Suppose that $n > 1$, and assume the result established for every $n' < n$. By the induction hypothesis, for every $x \in U$ such that $c(x) = 1$, one has $x \notin \operatorname{Ass} \mathcal{F}_x$. Let x be such a point, and let $y \in \overline{\{x\}} \cap Y$ be an immediate specialization of x, i.e.

$$\dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

Perform the base change

$$v\colon \operatorname{Spec}(\mathcal{O}_{X,y}) \longrightarrow X,$$

retaining the notation of diagram (2.7). Applying EGA III, 1.4.15, one obtains isomorphisms

$$v^*(R^p i_*(\mathcal{F}|_U)) \simeq R^p i'_*(v'^*(\mathcal{F}|_U)), \qquad p \in \mathbb{Z}.$$

Thus we are reduced to the case in which X is the spectrum of a local ring A in which $\underline{x}$ is a prime ideal of dimension 1, i.e. $\dim(A/x) = 1$. Set

$$\mathcal{F}' = \underline{\Gamma}_Y(\mathcal{F}) \qquad \text{and} \qquad \mathcal{F}'' = \mathcal{F}/\mathcal{F}'.$$

We have $\mathcal{F}_x \simeq \mathcal{F}''_x$ and $y \notin \operatorname{Ass} \mathcal{F}''$. Moreover, $\mathcal{F}'|_U = 0$; hence the long exact sequence of the $R^p i_*$ gives isomorphisms

$$R^p i_*(\mathcal{F}|_U) \simeq R^p i_*(\mathcal{F}''|_U), \qquad p \in \mathbb{Z}.$$

French printed p. 96 Since $n > 1$, neither x nor y belongs to $\operatorname{Ass} \mathcal{F}''$. Now $\underline{x}$ and $\underline{y}$ are the only prime ideals of A containing $\underline{x}$. It follows from Exposé III, 2.1 that there exists an element $g \in \underline{x}$ that is M-regular, where we have set

$$\mathcal{F} = \widetilde{M}, \qquad M = \mathcal{F}(X).$$

Thus there is an exact sequence

$$0 \longrightarrow M \xrightarrow{g'} M \longrightarrow N \longrightarrow 0,$$

in which g' denotes multiplication by g on M. From the resulting long exact sequence one sees that

$$R^p i_*(\widetilde{N}|_U) \quad \text{is coherent for } p < n-1.$$

Therefore, by the induction hypothesis,

$$\operatorname{prof}(\widetilde{N})_x \geq n-1, \qquad \text{and hence} \qquad \operatorname{prof} \mathcal{F}_x \geq n.$$

□

3. Applications

From these results one deduces a coherence condition for the higher direct images of a coherent sheaf under a *morphism that is not proper*.

Theorem 3.1.—Let $f\colon X \to Y$ be a morphism of preschemes. Suppose that Y is *locally Noetherian* and that f is *proper*. Suppose that X is *locally embeddable in a regular prescheme*. Let $n \in \mathbb{Z}$. Let U be an open subset of X, and let $\mathcal{F}$ be a *coherent* $\mathcal{O}_U$-module. Suppose that, for every $x \in U$ such that

$$\operatorname{codim}(\overline{\{x\}} \cap (X \setminus U), \overline{\{x\}}) = 1,$$

one has $\operatorname{prof} \mathcal{F}_x \geq n$. *Then* the $\mathcal{O}_Y$-modules

$$R^p(f \circ g)_*(\mathcal{F})$$

are coherent for $p < n$, where g is the canonical immersion of U into X.

Indeed, there is a Leray spectral sequence whose abutment is $R^*(f \circ g)_*(\mathcal{F})$ and whose initial term is given by

$$E_2^{p,q} = R^p f_*(R^q g_*(\mathcal{F})).$$

Moreover, there exists a *coherent* $\mathcal{O}_X$-module $\mathcal{G}$ such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). It then follows from the preceding section that condition (iv) on p. 74 is satisfied [French printed p. 97], i.e.

that $R^q g_*(\mathcal{G}|_U)$ is coherent for $q < n$. Applying the finiteness theorem of EGA III, 3.2.1 to f and the sheaves $R^q g_*(\mathcal{F})$, one finds that $E_2^{p,q}$ is coherent for $q < n$, whence the conclusion.

Proposition 3.2.—Let X be a locally Noetherian prescheme, locally embeddable in a regular prescheme. Let U be an open subset of X, and let $i\colon U \to X$ be the canonical immersion. Let $n \in \mathbb{Z}$. Finally, let $\mathcal{F}$ be a coherent Cohen–Macaulay $\mathcal{O}_U$-module (on U!). The following conditions are equivalent:

(a) $R^p i_*(\mathcal{F})$ is coherent for $p < n$;

(b) for every irreducible component S' of the closure $\overline{S}$ of the support S of $\mathcal{F}$, one has

$$\operatorname{codim}(S' \cap (X \setminus U), S') > n.$$

Proof, opening.—Let $\mathcal{G}$ be a coherent $\mathcal{O}_X$-module such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). Applying Corollary 2.3 to $\mathcal{G}$, we find that condition (a) is equivalent to (c):

(c) for every $x \in S$, one has

$$\operatorname{prof} \mathcal{F}_x > n - c(x), \qquad c(x) = \operatorname{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}).$$

Proof, (a)⇒(b).—Indeed, let S' be an irreducible component of $\overline{S}$, and let s be its generic point. Since $\mathcal{F}$ is Cohen–Macaulay, one has $\operatorname{prof} \mathcal{F}_s = \dim \mathcal{O}_{S,s} = 0$. Moreover, $\overline{\{s\}} = S'$, and the conclusion follows.

Proof, (b)⇒(a)*, opening.*—Let $x \in S$, and let S' be an irreducible component of $\overline{S}$ such that $x \in S'$. Let s be the generic point of S'. Since $\mathcal{F}$ is Cohen–Macaulay, we know that

$$\operatorname{prof} \mathcal{F}_x = \dim \mathcal{O}_{\overline{\{s\}},x}.$$

Proof, (b)⇒(a)*, case split.*—If $c(x) = +\infty$, there is nothing to prove. Otherwise, there exists $y \in Y \cap \overline{\{x\}}$ such that

$$c(x) = \dim \mathcal{O}_{\overline{\{x\}},y}.$$

Proof, (b)⇒(a)*, conclusion.*—Now $\mathcal{O}_{X,y}$ is a quotient of a regular local ring by hypothesis; hence

$$\dim \mathcal{O}_{\overline{\{s\}},y} = \dim \mathcal{O}_{\overline{\{s\}},x} + \dim \mathcal{O}_{\overline{\{x\}},y} > n.$$

□

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Exposé IX

Algebraic Geometry and Formal Geometry

The purpose of this exposé is to generalize Theorems 3.4.2 and 4.1.5 of *EGA III* to the case of a morphism that is not proper.

1. The Comparison Theorem

Let $f\colon X \to X'$ be a morphism of preschemes that is *separated* and of *finite type*. Suppose that X' is locally noetherian. Let Y' be a closed subset of X', and let $Y = f^{-1}(Y')$. Let $\widehat{X}$ and $\widehat{X'}$ be the formal completions of X and X' along Y and Y', respectively. Let $\widehat{f}$ be the morphism induced by f upon passing to the completions.

$$\begin{array}{ccc} X & \longleftarrow & Y \\ {\scriptstyle f}\downarrow & & \downarrow \\ X' & \longleftarrow & Y' \end{array} \qquad , \qquad \begin{array}{ccc} \widehat{X} & \xrightarrow{\ j\ } & X \\ {\scriptstyle \widehat{f}}\downarrow & & \downarrow{\scriptstyle f} \\ \widehat{X'} & \xrightarrow{\ i\ } & X'. \end{array} \tag{1.1}$$

We denote by j (resp. i) the morphism from $\widehat{X}$ to X (resp. from $\widehat{X'}$ to X'). It is known that i and j are *flat*.

Let $\mathcal{I}'$ be an ideal of definition of Y', and let $\mathcal{J} = f^*(\mathcal{I}') \cdot \mathcal{O}_X$; this is an ideal of definition of Y. Hence

$$\widehat{X'} = \left(Y', \varprojlim_{k\in\mathbb{N}} \mathcal{O}_{X'}/{\mathcal{I}'}^{k+1}\right), \qquad \widehat{X} = \left(Y, \varprojlim_{k\in\mathbb{N}} \mathcal{O}_X/\mathcal{J}^{k+1}\right). \tag{1.2}$$

For every $k \in \mathbb{N}$, set

$$Y'_k = \left(Y', \mathcal{O}_{X'}/(\mathcal{I}')^{k+1}\right), \qquad Y_k = \left(Y, \mathcal{O}_X/\mathcal{J}^{k+1}\right). \tag{1.3}$$

Let F be a coherent $\mathcal{O}_X$-module. For every $k \in \mathbb{N}$, set

$$F_k = F/\mathcal{J}^{k+1}F, \qquad \widehat{F} = j^*(F) \simeq \varprojlim F_k. \tag{1.4}$$

If we set

$$R^i f_*(F)^\wedge = \varprojlim_{k\in\mathbb{N}} \left(R^i f_*(F) \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{Y'_k}\right), \qquad i \in \mathbb{Z}, \tag{1.5}$$

there is a natural homomorphism

$$r_i : i^*(R^i f_*(F)) \longrightarrow R^i f_*(F)^\wedge, \tag{1.6}$$

which is an *isomorphism* when $R^i f_*(F)$ is *coherent*.

As explained in *EGA III*, 4.1.1, there is a commutative diagram

$$\begin{array}{ccc} i^*(R^i f_*(F)) & \xrightarrow{\quad\rho_i\quad} & R^i \widehat{f}_*(\widehat{F}) \\ \Big\downarrow{\scriptstyle r_i} & & \Big\downarrow{\scriptstyle \psi_i} \\ R^i f_*(F)^\wedge & \xrightarrow{\quad\varphi_i\quad} & \varprojlim_{k\in\mathbb{N}} R^i f_*(F_k). \end{array} \tag{1.7}$$

In the cited passage one obtains a commutative diagram because $R^i f_*(F)$ is known to be coherent, and the source and target of (1.6) are identified. In our case, $R^i f_*(F)$ will be coherent only for certain values of i, for which we shall study (1.7).

Consider the graded ring

$$\mathcal{S} = \bigoplus_{k\in\mathbb{N}} \mathcal{I}'^k, \tag{1.8}$$

and the graded $\mathcal{S}$-module

$$\mathcal{H}^i = \bigoplus_{k\in\mathbb{N}} R^i f_*(\mathcal{J}^k F), \qquad i \in \mathbb{Z}, \tag{1.9}$$

whose $\mathcal{S}$-module structure is defined as follows.

The sheaf $R^i f_*(\mathcal{J}^k F)$ is associated with the presheaf that assigns to each affine open subset U' of X'

$$H^i\Big(f^{-1}(U'), \mathcal{J}^k F\Big|_{f^{-1}(U')}\Big). \tag{1.10}$$

Thus let U' be an affine open subset of X', set

$$U = f^{-1}(U'),$$

and let $x' \in \mathcal{I}'^m(U')$. Let x be the image of x' in $\mathcal{J}^m(U)$. Multiplication by x on $F|_U$ maps $\mathcal{J}^k F|_U$ into $\mathcal{J}^{k+m}F|_U$ and hence, by functoriality, gives a morphism

$$\mu^i_{x',k}(U') : H^i(U, \mathcal{J}^k F|_U) \longrightarrow H^i(U, \mathcal{J}^{k+m}F|_U), \tag{1.11}$$

defined for every $i \in \mathbb{Z}$ and every $k \in \mathbb{N}$. Upon passage to the associated sheaf, these morphisms give $\mathcal{H}^i$ its graded $\mathcal{S}$-module structure.

Theorem 1.1. Let n be an integer. Suppose that the graded $\mathcal{S}$-module $\mathcal{H}^i$ is finitely generated for $i = n-1$ and $i = n$. Then:

(0) r_n and r_{n-1} are isomorphisms, and $R^i \widehat{f}_*(\widehat{F})$ is coherent for $i = n-1$;

(1) for $i = n-1$, the maps ρ_i, φ_i, and ψ_i are topological isomorphisms. In particular, the filtration on $R^{n-1}f_*(F)$ defined by the kernels of the homomorphisms

$$R^{n-1}f_*(F) \longrightarrow R^{n-1}f_*(F_k) \tag{1.12}$$

is $\mathcal{I}'$-good;

(2) for $i = n$, the maps ρ_i, φ_i, and ψ_i are monomorphisms; moreover, the filtration on $R^n f_*(F)$ is $\mathcal{I}'$-good, and ψ_n is an isomorphism;

(3) for $i = n-2, n-1$, the projective system $R^i f_*(F_k)$ satisfies the uniform Mittag–Leffler condition; that is, there is a *fixed* integer $k \geq 0$ such that, for every $p \geq 0$ and every $p' \geq p+k$,

$$\mathrm{Im}\Big[R^i f_*(F_{p'}) \longrightarrow R^i f_*(F_p)\Big] = \mathrm{Im}\Big[R^i f_*(F_{p+k}) \longrightarrow R^i f_*(F_p)\Big].$$

Proceeding as in *EGA III*, 4.1.8, it is easy to *reduce to the case where X' is the spectrum of a noetherian ring A*. In this case, it is known that

$$R^i f_*(F) = \widetilde{H^i(X, F)} \qquad \text{(cf. (1.10))}. \tag{1.13}$$

Let I be the ideal of A such that $\widetilde{I} = \mathcal{I}'$, and set

$$S = \bigoplus_{k\in\mathbb{N}} I^k, \tag{1.14}$$

$$H^i = \bigoplus_{k\in\mathbb{N}} H^i(X, \mathcal{J}^k F), \qquad i \in \mathbb{Z}, \tag{1.15}$$

where H^i carries the graded S-module structure defined by (1.11), with $U' = X'$.

The proof is modeled on that of *EGA III*, 4.1.5; let us give a summary.

We now work with φ_i and ψ_i, to which the following module homomorphisms correspond:

$$\begin{array}{ccc} & & H^i(\widehat{X}, \widehat{F}) \\ & & \Big\downarrow{\scriptstyle \psi_i} \\ H^i(X, F)^\wedge & \xrightarrow{\varphi_i} & \varprojlim_{k\in\mathbb{N}} H^i(X, F_k). \end{array} \tag{1.16}$$

(a) Suppose only that H^i is a graded S-module of finite type. It follows that *the filtration on $H^i(X, F)$ defined by the modules*

$$R^i_k = \ker\big(H^i(X, F) \longrightarrow H^i(X, F_k)\big) \tag{1.17}$$

is I-good. Indeed, the exact cohomology sequence

$$H^i(X, \mathcal{J}^{k+1}F) \longrightarrow H^i(X, F) \longrightarrow H^i(X, F_k) \tag{1.18}$$

shows that the *graded S-module $\bigoplus_{k\in\mathbb{N}} R^i_k$ is a quotient* of the graded S-submodule

$$\bigoplus_{k\in\mathbb{N}} H^i(X, \mathcal{J}^{k+1}F)$$

of H^i, and hence *is of finite type*, since S is noetherian. This proves the first assertion.

Set

$$M^i = H^i(X, F), \qquad H^i_k = H^i(X, F_k). \tag{1.19}$$

There is a commutative diagram

$$\begin{array}{ccc} H^i(X, F)^\wedge & \xrightarrow{s_i} & \varprojlim_k (M^i/R^i_k) \\ & \searrow^{\varphi_i} & \Big\downarrow{\scriptstyle t_i} \\ & & \varprojlim_k H^i_k, \end{array} \tag{1.20}$$

in which s_i is an *isomorphism*, since the filtration of $H^i(X, F)$ is I-good. Moreover, t_i is a *monomorphism*: the functor $\varprojlim$ is left exact and, for every $k \geq 0$, the natural morphism $M^i/R^i_k \longrightarrow H^i_k$ is a monomorphism by the definition of R^i_k.

To study the surjectivity of t_i, introduce

$$Q_k^i = \operatorname{coker}\big(H^i(X,F) \longrightarrow H^i(X,F_k)\big), \tag{1.21}$$

which gives a projective system of exact sequences

$$0 \longrightarrow R_k^i \longrightarrow M^i \longrightarrow H_k^i \longrightarrow Q_k^i \longrightarrow 0. \tag{1.22}$$

Using the exact cohomology sequence

$$H^i(X,F) \longrightarrow H^i(X,F_k) \longrightarrow H^{i+1}(X,\mathcal{J}^{k+1}F), \tag{1.23}$$

we see that the *graded S-module*

$$Q^i = \bigoplus_{k\in\mathbb{N}} Q_k^i \tag{1.24}$$

is a graded S-submodule of H^{i+1}. Moreover, for every $k \geq 0$,

$$I^{k+1}Q_k^i = 0, \tag{1.25}$$

because Q_k^i is the image of H_k^i.
(b) Suppose only that H^{i+1} is of finite type, and consider t_i alone, disregarding s_i. Since S is noetherian, Q^i is of finite type; as $I^{k+1}Q_k^i = 0$, there exist integers $r \geq 0$ and $k_0 \geq 0$ such that

$$I^r Q_k^i = 0 \qquad \text{for } k \geq k_0. \tag{1.26}$$

It follows that the *projective system $(Q_k^i)_{k\in\mathbb{N}}$ is essentially zero*, and hence that the *projective system $(H_k^i)_{k\in\mathbb{N}}$ satisfies the uniform Mittag–Leffler condition*. From the exact sequence (1.22) we obtain the exact sequence

$$0 \longrightarrow M^i/R_k^i \longrightarrow H_k^i \longrightarrow Q_k^i \longrightarrow 0, \tag{1.27}$$

and therefore the exact sequence

$$0 \longrightarrow \varprojlim_k M^i/R_k^i \xrightarrow{t_i} \varprojlim_k H_k^i \longrightarrow \varprojlim_k Q_k^i. \tag{1.28}$$

But the projective system $(Q_k^i)_{k\in\mathbb{N}}$ is essentially zero, so t_i is an *isomorphism*.
(c) Let us prove that if H^i is of finite type, then ψ_i is an isomorphism. It suffices to apply *EGA* 0_{III}, 13.3.1, taking the affine open subsets as a basis of open subsets of X. This is legitimate: by (b), the projective system $(H_k^{i-1})_{k\in\mathbb{N}}$ satisfies the Mittag–Leffler condition.

The theorem follows formally from (a), (b), and (c). Observe that, in fact, at each step the proof uses finite generation of H^i for only one value of i.

Let us give some examples in which the hypothesis of Theorem 1.1 is satisfied.

Corollary 1.2. *Suppose that $\mathcal{I}'$ is generated by a section t' of $\mathcal{O}_{X'}$, and denote by t the corresponding section of $\mathcal{O}_X$. Let F be a coherent $\mathcal{O}_X$-module, and let n be an integer. Assume that:*

(i) *t is F-regular (that is, multiplication by t on F is a monomorphism);*

(ii) *$R^i f_*(F)$ is coherent for $i = n-1$ and $i = n$.*

Then the hypothesis of Theorem 1.1 is satisfied.

Indeed, multiplication by t^k defines an isomorphism $F \xrightarrow{\sim} \mathcal{J}^k F$, from which it follows that

$$H^i \simeq R^i f_*(F) \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{X'}[T], \tag{1.29}$$

where T is an indeterminate. This proves the result.

Corollary 1.3. *Suppose that $X' = \mathrm{Spec}(A)$, where A is a noetherian ring that is separated and complete for the I-adic topology. Suppose that the graded S-module H^i is of finite type for $i = n-1$ and $i = n$ (cf. (1.14) and (1.15)). Then the hypotheses of Theorem 1.1 are satisfied, and there is a commutative diagram of isomorphisms*

$$\begin{array}{ccc} H^i(X,F) & \xrightarrow{\quad\rho'_i\quad} & H^i(\widehat{X},\widehat{F}) \\ & {\scriptstyle\varphi'_i}\searrow \quad \swarrow{\scriptstyle\psi_i} & \\ & \varprojlim_k H^i(X,F_k) & \end{array} \qquad \text{for } i = n-1. \tag{1.30}$$

Editor's note (1)

In the same vein, see the article by W.-L. Chow, "Formal functions on homogeneous spaces," *Invent. Math.* **86** (1986), no. 1, pp. 115–130. The author proves the following result. Let X be an algebraic variety over a field, homogeneous under an algebraic group G, and let Z be a complete subvariety of X of dimension > 0. Suppose that Z generates X in the following sense: given $p \in Z$, let Γ_p be the set of elements of G that send p into Z. One says that Z generates X if the group generated by the identity component of Γ_p is all of G. In that case, every formal rational function on X along Z is algebraic; compare the results of Hironaka and Matsumura cited in editor's note (3), p. 138.

Along the lines of the techniques introduced by those authors, we also mention the elegant algebraization result due to D. Gieseker, "On two theorems of Griffiths about embeddings with ample normal bundle," *Amer. J. Math.* **99** (1977), no. 6, pp. 1137–1150, Theorems 4.1 and 4.2. Let X be a connected projective variety of dimension > 0, locally a complete intersection over an algebraically closed field. Suppose there are two embeddings of X into smooth projective varieties Y and W. If the formal completions of X in Y and W are equivalent, then there exists a scheme U containing X as a closed subscheme which embeds into both Y and W as an étale neighborhood of X. In other words, formal equivalence implies étale equivalence.

See also G. Faltings, "Formale Geometrie und homogene Räume," *Invent. Math.* **64** (1981), pp. 123–165.

Proof. We simply note that $H^i(X,F)$ is of finite type and hence isomorphic to its completion. One obtains (1.30) by transcribing the diagram of modules (1.7) into the category of A-modules and replacing the left-hand vertical by $H^i(X,F)$.

Proposition 1.4. *Let A be a noetherian ring. Let $t \in A$, and suppose that A is separated and complete for the (tA)-adic topology. Set*

$$X' = \mathrm{Spec}(A), \qquad Y' = V(t), \qquad I = (tA). \tag{1.31}$$

Let T be a closed subset of X', and set

$$X = X' \setminus T, \qquad Y = Y' \cap X = Y' \setminus (Y' \cap T). \tag{1.32}$$

Let F be a coherent $\mathcal{O}_X$-module. Finally, set

$$T' = \{x \in X' \mid \mathrm{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) = 1\}. \tag{1.33}$$

Suppose that

a) t is F-regular;

b) $\operatorname{prof}_{T'}(F) \geq n+1$*;*

c) A is a quotient of a regular noetherian ring.

Then, in diagram (1.30), the morphisms ρ'_i, φ'_i, and ψ_i are isomorphisms for $i < n$ and monomorphisms for $i = n$. Moreover, ψ_n is an isomorphism.

Proof. By (1.3) and (1.2), it suffices to prove that $R^i f_*(F)$ is coherent for $i \leq n$; this follows from the finiteness theorem, Exposé VIII, Theorem 2.1.

In particular:

Example 1.5. Proposition 1.4 applies when A is a local ring and t belongs to the radical $\mathfrak{r}(A)$ of A. We then take $T = \{\mathfrak{r}(A)\}$. In this case, for $n = 1$, we obtain the following statement:

If A is noetherian, separated and complete for the t-adic topology, and is a quotient of a regular ring (for example, if A is complete), if moreover t is F-regular and $\operatorname{prof} F_x \geq 2$ *for every $x \in \operatorname{Spec}(A)$ such that* $\dim(A/x) = 1$*, then the natural homomorphism*

$$\Gamma(X, F) \longrightarrow \Gamma(\widehat{X}, \widehat{F})$$

is an isomorphism.

Indeed, retaining the notation of Proposition 1.4, we have $T = \{\mathfrak{r}(A)\}$, and formula (1.33) says that

$$T' = \{x \in \operatorname{Spec}(A) \mid \dim(A/x) = 1\}.$$

IX.2 The Existence Theorem

Let us first state EGA III, 3.4.2 in a slightly more general form.

Let $f : \mathfrak{X} \to \mathfrak{X}'$ be an adic morphism(*) of formal preschemes, with $\mathfrak{X}'$ noetherian. Let $\mathcal{I}'$ be an ideal of definition of $\mathfrak{X}'$; since f is adic, $f^*\mathcal{I}' = \mathcal{J}$ is(2) an ideal of definition of $\mathfrak{X}$.

For every $n \in \mathbb{N}$, set

$$\mathfrak{X}_n = (\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathcal{J}^{n+1}); \tag{IX.2.1}$$

this is an ordinary prescheme with the same underlying topological space as $\mathfrak{X}$.

Let $\mathfrak{F}$ be a coherent $\mathcal{O}_{\mathfrak{X}}$-module. For every $k \in \mathbb{N}$, the $\mathcal{O}_{\mathfrak{X}_k}$-modules

$$F_k = \mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F} \tag{2.2}$$

are coherent. For every i, there is a homomorphism

$$\psi_i : R^i f_*(\mathfrak{F}) \longrightarrow \varprojlim_k R^i f_*(F_k), \tag{2.3}$$

induced by functoriality from the natural homomorphism

$$\mathfrak{F} \longrightarrow F_k. \tag{2.4}$$

Set

$$\mathbf{S} = \operatorname{gr}_{\mathcal{I}'} \mathcal{O}_{\mathfrak{X}'} = \bigoplus_{k \in \mathbb{N}} \mathcal{I}'^k/\mathcal{I}'^{k+1}, \tag{2.5}$$

(*) This hypothesis is not essential; cf. Exposé XII, p. 118.

(2) *Editor's note.* By the very definition; cf. EGA I, 10.12.1.

$$\mathrm{gr}_{\mathcal{J}}(\mathfrak{F}) = \bigoplus_{k\in\mathbb{N}} \mathcal{J}^k\mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F}, \tag{2.6}$$

$$\mathbf{K}^i = R^i f_*(\mathrm{gr}_{\mathcal{J}}(\mathfrak{F})) = \bigoplus_{k\in\mathbb{N}} R^i f_*(\mathcal{J}^k\mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F}). \tag{2.7}$$

It is clear that $\mathbf{K}^i$ is endowed with the structure of a graded $\mathbf{S}$-module.

Theorem 2.1. *Suppose that $\mathbf{K}^i$ is a graded $\mathbf{S}$-module of finite type for $i = n-1$, $i = n$, and $i = n+1$. Then:*

(i) $R^n f_*(\mathfrak{F})$ is coherent.

(ii) The homomorphism ψ_n in (2.3) is a topological isomorphism. The natural filtration on the right-hand side of (2.3) is $\mathcal{I}'$-good.

(iii) The projective system of the $R^n f_*(F_k)$ satisfies the uniform Mittag-Leffler condition.

Proof. The proof follows very easily from EGA 0_{III}, 13.7.7 (cf. EGA III, 3.4.2), provided that the text on page 78 is corrected as indicated in (EGA III 2, $\mathrm{Err}_{\mathrm{III}}$ 24).

Theorem 2.2[(3)]. *Let A be a noetherian adic ring, and let I be an ideal of definition of A. Let T be a closed subset of $X' = \mathrm{Spec}(A)$. Suppose that I is generated by an element $t \in A$. Retain the notation of (1.31), (1.32), and (1.33). Let $\mathfrak{F}$ be a coherent $\mathcal{O}_{\widehat{X}}$-module. Set*

$$F_0 = \mathfrak{F}/\mathcal{J}\mathfrak{F}, \tag{2.8}$$

where $\mathcal{J} = t\mathcal{O}_{\widehat{X}}$ is an ideal of definition of $\widehat{X}$. Suppose that *A is a quotient of a regular noetherian ring and that:*

(1) t is $\mathfrak{F}$-regular;

(2) $\mathrm{prof}_{T'} F_0 \geq 2$.

Then there exists a coherent $\mathcal{O}_X$-module F such that $\widehat{F} \simeq \mathfrak{F}$.

Proof. It suffices to prove that $\widehat{f}_*(\mathfrak{F})$ is a coherent $\mathcal{O}_{\widehat{X'}}$-module, where $\widehat{f}\colon \widehat{X} \to \widehat{X'}$ is the morphism of formal preschemes obtained from the inclusion of X into X' by completion with respect to t. Indeed, A is separated and complete for the t-adic topology, so there exists an A-module F' whose completion is isomorphic to $\widehat{f}_*(\mathfrak{F})$. Since X is open in X', we may take

$$F = \widetilde{F'}|_X.$$

It remains to show that Theorem 2.1 applies to the morphism of formal preschemes $\widehat{f}$ and to $\mathfrak{F}$. By hypothesis (1), for every $k \in \mathbb{N}$ there is an isomorphism

$$\mathcal{J}^k\mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F} \to \mathfrak{F}/\mathcal{J}\mathfrak{F},$$

and hence the hypothesis of Theorem 2.1 is satisfied provided that

$$R^i f_*(F_0) \text{ is coherent for } i \leq 1.$$

This follows from (2) and the finiteness theorem of Exposé VIII, Theorem 2.1. This proves the result.

It remains to specialize this statement under the assumption that A is a local ring.

Corollary 2.3. *Let A be a noetherian local ring, and let $t \in \mathfrak{r}(A)$ be an element of the radical of A. Suppose that A is separated and complete for the t-adic topology and, moreover, is a quotient of a regular ring (for example, suppose that A is a complete noetherian local ring). Set*

$$X' = \operatorname{Spec} A, \qquad T = \{\mathfrak{r}(A)\}, \tag{2.9}$$

and retain the notation of (1.31), (1.32), and (1.33). Let $\mathfrak{F}$ be an $\mathcal{O}_{\widehat{X}}$-module. Suppose that

1. *t is $\mathfrak{F}$-regular;*
2. *$\operatorname{prof}_{T'} F_0 \geq 2$, where $F_0 = \mathfrak{F}/\mathcal{J}\mathfrak{F}$ and $\mathcal{J} = t\mathcal{O}_{\widehat{X}}$.*

Then there exists a coherent $\mathcal{O}_X$-module F such that $\widehat{F} \simeq \mathfrak{F}$.

Note that here T' is the set of prime ideals $\mathfrak{p}$ of A such that

$$\dim(A/\mathfrak{p}) = 1.$$

Exposé X

Application to the Fundamental Group

Throughout this exposé, X will denote a locally noetherian prescheme, Y a closed subset of X, U a variable open neighborhood of Y in X, and $\widehat{X}$ the formal completion of X along Y (EGA I, 10.8). For every prescheme Z, let $\mathbf{Et}(Z)$ denote the category of étale coverings of Z, and let $\mathbf{L}(Z)$ denote the category of coherent locally free modules on Z.

1. Comparison of $\mathbf{Et}(\widehat{X})$ and $\mathbf{Et}(Y)$

Let $\mathcal{I}$ be an ideal of definition of Y in X. For every $n \in \mathbb{N}$, set

$$Y_n = \left(Y, (\mathcal{O}_X/\mathcal{I}^{n+1})\Big|_Y\right).$$

The Y_n form a direct system of ordinary preschemes, or also of formal preschemes when the structure sheaves are endowed with the discrete topology. We know (EGA I, 10.6.2) that $\widehat{X}$ is the direct limit, in the category of formal preschemes, of the direct system (Y_n). We also know (EGA I, 10.13) that to give a formal $\widehat{X}$-prescheme R of *finite type* is to give a direct system of ordinary Y_n-preschemes R_n of *finite type*, such that

$$R_n \simeq R_{n+1} \times_{Y_{n+1}} Y_n.$$

Moreover, for R to be an étale covering of $\widehat{X}$, it is necessary and sufficient that, for every n, R_n be an étale covering of Y_n.

That said, it is easy to see that nilpotent elements make no difference to étale coverings (SGA 1, 8.3); in other words, the base-change functor

$$\mathbf{Et}(Y_{n+1}) \longrightarrow \mathbf{Et}(Y_n)$$

is an equivalence of categories for every $n \in \mathbb{N}$. Hence:

Proposition 1.1. *With the notation introduced above, the natural functor*

$$\mathbf{Et}(\widehat{X}) \longrightarrow \mathbf{Et}(Y)$$

is an equivalence of categories (cf. SGA 1, 8.4).

2. Comparison of $\mathbf{Et}(Y)$ and $\mathbf{Et}(U)$, for varying U

We shall introduce two conditions from which the comparison theorem announced above will follow easily. Let X be a locally noetherian prescheme, and let Y be a closed subset of X. We say that the pair (X, Y) satisfies the *Lefschetz condition*, written $\mathrm{Lef}(X, Y)$, if, for every open subset U of X containing Y and every locally free coherent sheaf E on U, the natural homomorphism

$$\Gamma(U, E) \longrightarrow \Gamma(\widehat{X}, \widehat{E})$$

is an isomorphism.

We say that the pair (X, Y) satisfies the *effective Lefschetz condition*, written $\mathrm{Leff}(X, Y)$, if $\mathrm{Lef}(X, Y)$ holds and, in addition, for every locally free coherent sheaf $\mathcal{E}$ on $\widehat{X}$, there exist an open neighborhood U of Y, a locally free coherent sheaf E on U, and an isomorphism

$$\widehat{E} \simeq \mathcal{E}.$$

These conditions hold in two important examples:

Example 2.1[1]. — Let A be a noetherian ring, and let $t \in \mathfrak{r}(A)$ be an A-regular element belonging to the radical $\mathfrak{r}(A)$ of A. Assume that A is a quotient of a regular local ring and that A is complete for the t-adic topology (for example, that A is complete for the $\mathfrak{r}(A)$-adic topology). Set

$$X' = \mathrm{Spec}(A), \qquad Y' = V(t),$$

and, further, set

$$x = \mathfrak{r}(A), \qquad X = X' \setminus \{x\}, \qquad Y = Y' \setminus \{x\}.$$

Thus X is open in X' and $Y = X \cap Y'$. Then:

(i) If, for every prime ideal $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$ (i.e. for every closed point of X), we have $\mathrm{depth}\, A_{\mathfrak{p}} \geq 2$, then $\mathrm{Lef}(X, Y)$ holds;

(ii) if, moreover, for every prime ideal $\mathfrak{p}$ of A such that $t \in \mathfrak{p}$ and $\dim A/\mathfrak{p} = 1$ (i.e. for every closed point of Y), we have $\mathrm{depth}\, A_{\mathfrak{p}} \geq 3$, then $\mathrm{Leff}(X, Y)$ holds.

Let us first prove that, for every open neighborhood U of Y in X, the complement of U in X is a union of finitely many closed points (in X). Observe that U is open in X, hence in X', and therefore $Z' = X' \setminus U$ is closed. Let I be an ideal defining Z'; it is enough to prove that A/I has dimension 1. Since

$$Z' \cap Y' = \{x\},$$

the ring $A/(I + (t))$ is Artinian, and the conclusion follows from the "Hauptidealsatz."

The *first hypothesis is equivalent to*: "for every prime ideal $\mathfrak{p}$ of A with $\mathfrak{p} \neq \mathfrak{r}(A)$, one has

$$\mathrm{depth}\, A_{\mathfrak{p}} \geq 3 - \dim(A/\mathfrak{p}).$$

Indeed, A is a quotient of a regular ring, so we may apply VIII, 2.3 to the prescheme X', the closed subset $\{x\}$, and the coherent sheaf $\mathcal{O}_{X'}$, observing that $c(\mathfrak{p}) = \dim(A/\mathfrak{p})$ for $\mathfrak{p} \in X = X' \setminus \{x\}$ (since x is the closed point of X').

[1] *Editor's note.* One can slightly improve (i): see Mme Raynaud (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental," *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), pp. 29–52, Corollaries I.1.4 and I.5). Condition (ii) can be improved so as to dispense with the depth conditions along Y (see Theorem 3.3 of *loc. cit.* for a precise statement). The proof of this latter point is very technical; moreover, the article cited above gives only indications of proof and refers to an earlier, detailed version published in the *Bulletin de la Société mathématique de France*.

Let U be an open neighborhood of Y in X, and let E be a locally free $\mathcal{O}_U$-module. Set $Z = X \setminus U$, and let $u\colon U \to X$ be the canonical immersion. We shall first prove that $u_*(E)$ is a coherent $\mathcal{O}_X$-module, or, equivalently, that $\mathcal{H}^i_Z(E')$ is coherent for $i = 0, 1$, where E' is a coherent extension of E to X. For this, we apply Theorem VIII, 2.1 to the prescheme X, the closed subset Z, and the coherent sheaf E'. It is enough to verify that, for every point $p \in U$ such that $c(p) = 1$, one has $\operatorname{depth} E'_p \geq 1$, where we have set

$$c(p) = \operatorname{codim}(\overline{\{p\}} \cap Z, \overline{\{p\}}).$$

Now suppose $p \in U$ and $c(p) = 1$, and let $\mathfrak{p}$ denote the ideal of A corresponding to p. Then $\dim(A/\mathfrak{p}) = 2$, since the complement of U is a union of finitely many closed points and A is a quotient of a regular ring. Moreover, E is locally free, so for every $p \in \operatorname{Supp} E$ one has $\operatorname{depth} E_p = \operatorname{depth} \mathcal{O}_{U,p}$. Finally, if $p \in U$ and $c(p) = 1$, then

$$\operatorname{depth} E'_p = \operatorname{depth} E_p = \operatorname{depth} \mathcal{O}_{U,p} = \operatorname{depth} A_{\mathfrak{p}} \geq 3 - 2 = 1.$$

We must now prove that the natural homomorphism

$$\Gamma(U, E) \longrightarrow \Gamma(\widehat{X}, \widehat{E}) \tag{X.0.1}$$

is an isomorphism. Set $\overline{E} = u_*(E)$. The sheaf $\overline{E}$ is coherent and has depth at least 2 at every closed point of X. It follows that $R^i f_*(\overline{E})$ is coherent for $i = 0, 1$, where $f\colon X \to X'$ denotes the canonical immersion of X into $X' = \operatorname{Spec}(A)$ (Exposé VIII). Applying IX, 1.5, we conclude that

$$\Gamma(U,\overline{E}) \longrightarrow \Gamma(\widehat{X},\widehat{\overline{E}}) \tag{X.0.2}$$

is an isomorphism, since A is complete for the t-adic topology.

There is a commutative diagram

$$\begin{array}{ccc} \Gamma(X,\overline{E}) & \xrightarrow{\quad\simeq\quad} & \Gamma(U, E) \\ & {\scriptstyle\simeq}\searrow \quad \swarrow & \\ & \Gamma(\widehat{X}, \widehat{E}) & \end{array}$$

and the conclusion follows.

Now let $\mathcal{E}$ be a locally free coherent sheaf on $\widehat{X}$. If one has proved that $\mathcal{E}$ is algebraizable, that is, isomorphic to the formal completion of a coherent $\mathcal{O}_X$-module E, it is easy to see that E is locally free in a neighborhood of Y, and hence to prove $\operatorname{Leff}(X, Y)$. Let $\widehat{X'}$ be the formal spectrum of A for the t-adic topology; it is identified with the formal completion of X' along Y'. Denote by f the canonical immersion of X into X', by f' the canonical immersion of Y into Y', and by $\widehat{f}$ the morphism obtained by passing to the completions. For $\mathcal{E}$ to be algebraizable, it suffices that $\widehat{f}_*(\mathcal{E})$ be a coherent $\mathcal{O}_{\widehat{X'}}$-module, since A is complete for the t-adic topology.

Set $\mathcal{I} = t\mathcal{O}_{\widehat{X}}$; this is an ideal of definition of $\widehat{X}$.

For every $n \geq 0$, set

$$\mathcal{E}_n = \mathcal{E}/\mathcal{I}^{n+1}\mathcal{E}.$$

At every closed point $y \in Y$, the depth of $\mathcal{E}_0$ is at least two; indeed, t is an A-regular element, and therefore

$$\operatorname{depth} \mathcal{O}_{Y,y} = \operatorname{depth} \mathcal{O}_{X,y} - 1 \geq 2.$$

It follows that $\widehat{f}_*(\mathcal{E})$ is coherent (IX 2.3). Q.E.D.

Example 2.2 (This will make it possible to compare the fundamental group of a projective variety with that of a hyperplane section).

Let K be a field, and let X be a proper K-prescheme. Let $\mathcal{L}$ be an ample invertible $\mathcal{O}_X$-module. Let $t \in \Gamma(X, \mathcal{L})$ be an $\mathcal{O}_X$-regular element; this means that, for every open subset U and every isomorphism

$$u\colon \mathcal{L}|_U \xrightarrow{\sim} \mathcal{O}_U,$$

the section $u(t)$ is a non-zero-divisor in $\mathcal{O}_U$ (a condition that does not depend on u). Let $Y = V(t)$ be the subscheme of X defined by the equation $t = 0$[2]. Then:

(i) If, for every closed point x of X, one has $\operatorname{depth} \mathcal{O}_{X,x} \geq 2$, then $\mathrm{Lef}(X, Y)$ holds;

(ii) if, moreover, for every closed point $y \in Y$, one has $\operatorname{depth} \mathcal{O}_{X,y} \geq 3$, then $\mathrm{Leff}(X, Y)$ holds.

This example will be treated in detail in Exposé XII.

Let S be a prescheme. It is known (EGA II, 6.1.2) that the functor which assigns to every finite flat covering $r\colon R \to S$ the $\mathcal{O}_S$-algebra $r_*(\mathcal{O}_R)$ induces an equivalence between the category of finite flat coverings of S and the category of coherent locally free $\mathcal{O}_S$-algebras.

Let U be an open neighborhood of Y, and let $r\colon R \to U$ be a finite flat covering of U. Let $\widehat{R}$ be the finite flat covering of $\widehat{X}$ obtained from it by base change. Then

$$\widehat{r}_*(\mathcal{O}_{\widehat{R}}) \simeq \widehat{r_*(\mathcal{O}_R)}.$$

Assume now that $\mathrm{Lef}(X, Y)$ holds. This implies that, for every U, the inverse-image functor

$$\mathsf{L}(U) \longrightarrow \mathsf{L}(\widehat{X})$$

is fully faithful. Indeed, let E and F be two coherent locally free $\mathcal{O}_U$-modules; $\mathcal{H}om(E, F)$ is likewise coherent and locally free.

By hypothesis, the natural map

$$\Gamma(U, \mathcal{H}om(E, F)) \longrightarrow \Gamma(\widehat{X}, \widehat{\mathcal{H}om(E, F)})$$

is an isomorphism, which proves the assertion, since $\mathcal{H}om$ commutes with formal completion when all the modules involved are locally free. Formal completion also commutes with tensor products. It follows that the functor which assigns to every coherent locally free $\mathcal{O}_U$-algebra $\mathcal{A}$ the $\mathcal{O}_{\widehat{X}}$-algebra $\widehat{\mathcal{A}}$ is fully faithful. More precisely, if E is a coherent locally free $\mathcal{O}_U$-module, commutative $\mathcal{O}_U$-algebra structures on E correspond bijectively to commutative $\mathcal{O}_{\widehat{X}}$-algebra structures on $\widehat{E}$.

Proposition 2.3. Let X be a locally noetherian prescheme, and let Y be a closed subset of X. Let $\widehat{X}$ be the formal completion of X along Y. For every open subset U of X containing Y, let L_U (respectively P_U, respectively E_U) denote the functor that assigns to every locally free coherent $\mathcal{O}_U$-module (respectively every finite flat covering of U, respectively every étale covering of U) its pullback along $\widehat{X} \to X$.

(i) If $\mathrm{Lef}(X, Y)$ holds, then, for every open neighborhood U of Y, the functors L_U, P_U, and E_U are fully faithful.

[2] *Editor's note.* Condition (ii) is superfluous; see the editor's note to Example 2.1 (original running p. 90).

(ii) If $\mathrm{Leff}(X,Y)$ holds, then, for every locally free coherent $\mathcal{O}_{\widehat{X}}$-module $\mathcal{E}$, there are an open neighborhood U of Y and a locally free coherent $\mathcal{O}_U$-module E such that

$$L_U(E) \simeq \mathcal{E}.$$

The analogous assertions hold for finite flat coverings and for étale coverings, with P_U and E_U, respectively.

Proof. Part (i) has already been proved.

Part (ii) follows from (i) and the hypothesis, at least for L_U and P_U. Moreover, if R is an étale covering of $\widehat{X}$, there are an open neighborhood U of Y in X and a finite flat covering R' of U such that

$$\widehat{R'} \simeq R.$$

By restriction this gives a covering R'' of Y, which is étale by Proposition 1.1. Hence the restriction of R' to some neighborhood U' of Y is étale. □

Corollary 2.4. Suppose that $\mathrm{Lef}(X,Y)$ holds. A finite flat covering R of an open neighborhood U of Y is connected if and only if $R \times_U \widehat{X}$ is connected. In particular, Y is connected if and only if U is connected, and this is also equivalent to X being connected.

Proof. Indeed, a locally ringed space $(X, \mathcal{O}_X)$ is connected if and only if $\Gamma(X, \mathcal{O}_X)$ is not a direct product of two nonzero rings. But, by $\mathrm{Lef}(X,Y)$, one has

$$\Gamma\big(U, r_*\mathcal{O}_R\big) \simeq \Gamma\big(\widehat{X}, \widehat{r}_*\mathcal{O}_{\widehat{R}}\big).$$

Corollary 2.5. Suppose that $\mathrm{Lef}(X,Y)$ holds. Then, for every open neighborhood U of Y, the functor

$$\mathbf{Et}(U) \longrightarrow \mathbf{Et}(Y)$$

is fully faithful. Suppose that $\mathrm{Leff}(X,Y)$ holds. Then, for every étale covering R of Y, there are an open neighborhood U of Y and a covering R' of U such that $R' \times_U Y \simeq R$.

Corollary 2.6[(3)]. Suppose that $\mathrm{Lef}(X,Y)$ holds and that Y is connected. Then every open neighborhood U of Y is connected, and the natural homomorphism $\pi_1(Y) \longrightarrow \pi_1(U)$ is surjective. Suppose, moreover, that $\mathrm{Leff}(X,Y)$ holds. Then the natural homomorphism

$$\pi_1(Y) \longrightarrow \varprojlim_U \pi_1(U)$$

is an isomorphism. (N.B. A "base point" in Y is understood to have been chosen and is also taken as a base point in X for the definition of the fundamental groups.)

All of this follows immediately from Propositions 1.1 and 2.3.

Editor's note (3)

Combined with Proposition 3.3 and criteria XII.2.4 and XII.3.4, this yields the following relative Lefschetz theorem. Let $f\colon X \to S$ be a projective and flat morphism of connected noetherian schemes, and let D be an effective relative Cartier divisor on X that is relatively ample. If, for every $s \in S$, the depth of X_s at every closed point is ≥ 2, then D is connected and, for every open subset U of X containing D, the map

$$i_U\colon \pi_1(D) \longrightarrow \pi_1(U)$$

is surjective. If, in addition, the depth of X_s at every closed point of D_s is ≥ 3 and the local rings of X at its closed points are pure—for example, are complete intersections (cf. X, 3.4)—then i_X is an isomorphism.

Cf. J.-B. Bost, "Lefschetz theorem for Arithmetic Surfaces," *Ann. Sci. Éc. Norm. Sup. (4)* **32** (1999), pp. 241–312, Theorems 1.1 and 2.1.

When X is merely a smooth projective surface that is geometrically connected over a field, D is always connected and

$$\pi_1(D) \longrightarrow \pi_1(U)$$

is surjective (where U is an open subset containing D), provided only that D is nef and has self-intersection > 0; cf. *loc. cit.*, Theorem 2.3, and also Theorem 2.4 for surfaces assumed only normal and complete.

In the case of an arithmetic surface (normal and quasi-projective) X over a ring of integers $\mathcal{O}_K$, Bost, improving results of Y. Ihara (Y. Ihara, "Horizontal divisors on arithmetic surfaces associated with Belyĭ uniformizations," in *The Grothendieck theory of dessins d'enfants (Luminy, 1993)*, London Math. Soc. Lect. Note Series, vol. 200, Cambridge Univ. Press, Cambridge, 1994, pp. 245–254, or *loc. cit.*, Corollary 7.2), proved that if a point $P \in X(\mathcal{O}_K)$, which plays the role of the divisor D in the geometric situation, satisfies certain positivity conditions, then the map

$$\pi_1(X) \longrightarrow \pi_1(\operatorname{Spec} \mathcal{O}_K)$$

induced by the projection is an isomorphism whose inverse is the map

$$\pi_1(\operatorname{Spec} \mathcal{O}_K) \longrightarrow \pi_1(X)$$

induced by P; cf. *loc. cit.*, Theorem 1.2.

3. Comparison of $\pi_1(X)$ and $\pi_1(U)$

Definition 3.1. Let X be a prescheme and Z a closed subset of X. Set $U = X \setminus Z$. We say that the pair (X, Z) is *pure* if, for every open subset V of X, the functor

$$\begin{aligned} \mathbf{Et}(V) &\longrightarrow \mathbf{Et}(V \cap U), \\ V' &\longmapsto V' \times_V (V \cap U) \end{aligned}$$

is an equivalence of categories[*].

Definition 3.2. Let A be a noetherian local ring. Set $X = \operatorname{Spec} A$. Let $\mathfrak{r}(A)$ be the radical of A, and let $x = \mathfrak{r}(A)$ be the closed point of X. We say that A is *pure* if the pair $(X, \{x\})$ is pure.

We leave to the reader the task of not proving the following proposition:

Proposition 3.3. Let X be a locally noetherian prescheme and let Z be a closed subset of X. The pair (X, Z) is pure if and only if, for every $z \in Z$, the local ring $\mathcal{O}_{X,z}$ is pure[**].

That said, the following theorem is the main result of this section:

Theorem 3.4 (Purity theorem)[(4)]**.**

[*] For a notion that is more satisfactory in certain respects, cf. the commentary XIV, 1.6(d).

[**] Compare the noncommutative case of XIV, 1.8, whose proof is essentially the same as that of Proposition 3.3.

[(4)] N.D.E.: For the history of the methods employed, see Grothendieck's letter of October 1, 1961 to Serre, *Correspondance Grothendieck-Serre*, edited by Pierre Colmez and Jean-Pierre Serre, Documents Mathématiques, vol. 2, Société Mathématique de France, Paris, 2001.

(i) A regular noetherian local ring of dimension ≥ 2 is pure (Zariski–Nagata purity theorem).

(ii) A noetherian local ring of dimension ≥ 3 that is a complete intersection is pure.

Recall that a local ring A is said to be a *complete intersection* if and only if there exist a *regular* noetherian local ring B and a B-regular sequence $(t_1, \ldots, t_k)$ of elements of the radical $\mathfrak{r}(B)$ of B such that

$$A \simeq B/(t_1, \ldots, t_k).$$

In this connection, let us note that it would be less ambiguous to say that A is an *absolute* complete intersection, in contrast to the situation already encountered, in which X is a locally noetherian prescheme (not necessarily regular) and Y is a closed subset of X that is said to be "locally a set-theoretic complete intersection *in* X."

Let us first prove a few lemmas.

Lemma 3.5. — *Let X be a locally noetherian prescheme, and let U be an open subset of X. Set $Z = X \setminus U$. Let $i\colon U \to X$ be the canonical immersion of U into X. The following conditions are equivalent:*

(i) *For every open subset V of X, if $V' = V \cap U$, the restriction functor $F \longmapsto F|_{V'}$ from the category of locally free coherent $\mathcal{O}_V$-modules to the category of locally free coherent $\mathcal{O}_{V'}$-modules is fully faithful;*

(ii) *the natural homomorphism $\mathcal{O}_X \to i_*\mathcal{O}_U$ is an isomorphism;*

(iii) *for every $z \in Z$, one has* $\operatorname{prof} \mathcal{O}_{X,z} \geq 2$.

We have already seen (III 3.3) the equivalence of (ii) and (iii). Let us show that (ii) implies (i). Let F and G be two locally free coherent $\mathcal{O}_V$-Modules. Then $\mathcal{H}om(F, G)$ is also locally free and coherent, so

$$\mathcal{H}om(F, G) \longrightarrow i_*(\mathcal{H}om(F_{|V'}, G_{|V'}))$$

is an isomorphism, and therefore

$$\operatorname{Hom}(F, G) \simeq \operatorname{Hom}(F_{|V'}, G_{|V'}).$$

Conversely, take $F = G = \mathcal{O}_X$ and apply (i) to every open subset V of X.

Here is a useful "descent lemma":

Lemma 3.6. — *Let X be a locally noetherian prescheme, and let Z be a closed subset of X. Set $U = X \setminus Z$. Suppose that the homomorphism $\mathcal{O}_X \to i_*\mathcal{O}_U$ is an isomorphism. Let $f\colon X_1 \to X$ be a faithfully flat and quasi-compact morphism. Set $Z_1 = f^{-1}(Z)$. If the pair (X_1, Z_1) is pure, then so is (X, Z).*

Proof of Lemma 3.6. Observe that the hypothesis $\mathcal{O}_X \simeq i_*\mathcal{O}_U$ is preserved under flat base extension, since i is quasi-compact and, in this case, formation of the direct image commutes with inverse image. This hypothesis implies that the functor

$$\mathbf{Et}(V) \longrightarrow \mathbf{Et}(U \cap V)$$

defined by

$$V' \longmapsto V' \times_V (V \cap U)$$

is fully faithful, as follows from the interpretation of an étale covering in terms of a coherent locally free algebra. It remains to prove effectivity. One may, for example, form the fiber square X_2 and the fiber cube X_3 of X_1 over X, and observe that a faithfully flat and quasi-compact

morphism is a morphism of universal effective descent for the fibered category of étale coverings over the category of preschemes. The conclusion then follows formally(*).

Remark 3.7. — *We proved along the way that if $\mathcal{O}_X \to i_*\mathcal{O}_U$ is an isomorphism, then X is connected if and only if U is connected, and then $\pi_1(U) \to \pi_1(X)$ is surjective.*

Corollary 3.8. — *Let A be a noetherian local ring. Assume that* $\operatorname{depth} A \geq 2$. *If $\widehat{A}$ is pure, then A is pure.*

Proof. This follows from Lemma 3.5 and Lemma 3.6.

The following lemma is the essential point in the proof of the purity theorem:

Lemma 3.9. — *Let A be a noetherian local ring, and let $t \in \mathfrak{r}(A)$ be an A-regular element. Suppose that A is complete for the t-adic topology and is moreover a quotient of a regular local ring (for example, if A is complete). Set $B = A/tA$.*

(i) *If, for every prime ideal $\mathfrak{p}$ of A such that* $\dim A/\mathfrak{p} = 1$, *one has* $\operatorname{depth} A_\mathfrak{p} \geq 2$, *then the purity of B implies the purity of A.*

(ii) *If, for every prime ideal $\mathfrak{p}$ of A such that* $\dim A/\mathfrak{p} = 1$, *one has* $\operatorname{depth} A_\mathfrak{p} \geq 2$, *if $A_\mathfrak{p}$ is pure whenever $t \notin \mathfrak{p}$, and if*(5) $\operatorname{depth} A_\mathfrak{p} \geq 3$ *whenever $t \in \mathfrak{p}$, then the purity of A implies the purity of B.*

Let $X' = \operatorname{Spec}(A)$ and let $Y' = V(t)$, which we identify with the spectrum of B. Let $x = \mathfrak{r}(A)$, and set $X = X' \setminus \{x\}$ and $Y = Y' \setminus \{x\} = X \cap Y'$. Denote by $\widehat{X'}$ the formal spectrum of A for the t-adic topology; it is identified with the formal completion of X' along Y'.

Since A is complete for the t-adic topology, observe that $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(\widehat{X'})$ is an equivalence of categories. Likewise, $\mathbf{Et}(\widehat{X'}) \longrightarrow \mathbf{Et}(Y')$ is an equivalence by Proposition 1.1; hence $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(Y')$ is an equivalence of categories.

We prove (i). Consider the diagrams

$$\begin{array}{ccc} X' & \longleftarrow & X \\ \uparrow & & \uparrow \\ Y' & \longleftarrow & Y \end{array} \qquad\qquad \begin{array}{ccc} \mathbf{Et}(X') & \xrightarrow{a} & \mathbf{Et}(X) \\ {\scriptstyle c}\downarrow & & \downarrow{\scriptstyle b} \\ \mathbf{Et}(Y') & \xrightarrow{d} & \mathbf{Et}(Y) \end{array}$$

We have just seen that c is an equivalence; d is also an equivalence by the hypothesis that B is pure; and finally b is fully faithful, as was seen in Example 2.1; cf. Proposition 2.3(i).

We prove (ii). This time suppose that A is pure, so a is an equivalence; likewise c. Let us show that b is an equivalence. By Example 2.1, we know that $\operatorname{Leff}(X, Y)$ holds, so b is already fully faithful; let us prove that it is essentially surjective. Apply Proposition 2.3(ii), noting that if U is an open neighborhood of Y in X, then $X \setminus U$ is a union of finitely many closed points. The pair $(X, X \setminus U)$ is therefore pure by Proposition 3.3, since at any such point $\mathfrak{p}$ one has $\mathcal{O}_{X,\mathfrak{p}} = A_\mathfrak{p}$, which is pure by hypothesis. This proves the assertion.

Proof of the purity theorem. We first prove (i) by induction on the dimension. Let A be a noetherian local ring of dimension 2. Set $X' = \operatorname{Spec}(A)$, $x = \mathfrak{r}(A)$, and $X = X' \setminus \{x\}$. We have $\operatorname{depth} A = 2$. We may therefore apply Lemma 3.5 to the pair $(X', \{x\})$, and hence $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(X)$ is fully faithful.

Now let $r\colon R \longrightarrow X$ be an étale covering defined by the coherent, locally free, étale $\mathcal{O}_X$-algebra $\mathcal{A} = r_*(\mathcal{O}_R)$. Let $i\colon X \longrightarrow X'$ denote the canonical immersion. I claim that

(*) Cf. J. Giraud, *Méthode de la descente*, Mémoire no. 2 du *Bulletin de la Société Mathématique de France* (1964).

(5) N.D.E.: This last condition can be improved; cf. the editor's note (1) on p. 90.

$i_*(\mathcal{A}) = \mathcal{B}$ is a coherent $\mathcal{O}_{X'}$-algebra. Indeed, it suffices to apply the "finiteness theorem," VIII, 2.3. I also claim that this algebra has depth at least 2 at x: it is the direct image of an $\mathcal{O}_X$-module, with $X = X' \setminus \{x\}$. Since A is a regular ring of dimension 2, we have

$$\operatorname{pd} \mathcal{B} + \operatorname{depth} \mathcal{B} = \dim A = 2,$$

where $\operatorname{pd} \mathcal{B}$ denotes the projective dimension of $\mathcal{B}$. Thus $\operatorname{pd} \mathcal{B} = 0$, so $\mathcal{B}$ is projective and therefore free. Consequently, $\mathcal{B}$ defines a finite flat covering of $X' = \operatorname{Spec}(A)$. The set of points of X' at which this covering is not étale is a closed subset of X' whose defining ideal is principal, namely the discriminant ideal of $\mathcal{B}/A$. By construction this closed subset is contained in $x = \mathfrak{r}(A)$; it is therefore empty because $\dim A = 2$.

Let A now be a regular noetherian local ring with $\dim A = n \geq 3$. Suppose that (i) has been proved for rings of dimension less than n. To prove that A is pure, we may assume by Corollary 3.8 that A is complete. Choose $t \in \mathfrak{r}(A)$ whose image in $\mathfrak{r}(A)/\mathfrak{r}(A)^2$ is nonzero. Then $B = A/tA$ is a regular noetherian local ring of dimension $n-1$, and is therefore pure because $n - 1 \geq 2$. The conclusion follows from Lemma 3.9(i), which is applicable because A is complete.

Let us prove (ii). Let A be a noetherian local ring of dimension at least 3. Suppose that there exist a regular noetherian local ring B and a B-sequence $(t_1, \ldots, t_k)$ such that

$$A \simeq B/(t_1, \ldots, t_k).$$

We prove that A is pure by induction on k. If $k = 0$, this follows from (i). Suppose that $k \geq 1$ and that the result has been proved for $k' < k$. By Corollary 3.8, we may assume that A, and hence also B, is complete. Set

$$C = B/(t_1, \ldots, t_{k-1}).$$

Then $A \simeq C/t_k C$, and t_k is C-regular. By the induction hypothesis, C is pure; it therefore suffices to verify that Lemma 3.9(ii) is applicable. In the notation of that lemma, its A and B are here C and A, respectively. We have $\dim C \geq 4$; hence, for every prime ideal $\mathfrak{p}$ of C such that $\dim(C/\mathfrak{p}) = 1$, one has $\operatorname{depth} C_{\mathfrak{p}} \geq 3$. Moreover, $C_{\mathfrak{p}}$ is a complete intersection with $k' \leq k - 1$, and is therefore pure by the induction hypothesis.

Theorem 3.10. — *Let X be a locally noetherian prescheme, and let Y be a closed subset of X. Suppose that* $\operatorname{Leff}(X, Y)$ *holds (cf. Examples 2.1 and 2.2). Suppose further that, for every open neighborhood U of Y and every $x \in X \setminus U$, the local ring $\mathcal{O}_{X,x}$ is either regular of dimension ≥ 2 or a complete intersection of dimension ≥ 3. Then*

$$\pi_0(Y) \longrightarrow \pi_0(X)$$

is a bijection, and, if X is connected,

$$\pi_1(Y) \longrightarrow \pi_1(X)$$

is an isomorphism.

There is nothing more to prove. Observe that in the two cited Examples 2.1 and 2.2, the complement of U is a union of finitely many closed points; it follows that the hypothesis on the dimension of $\mathcal{O}_{X,x}$ is not farcical.

Exposé XI

Application to the Picard Group

This exposé follows the pattern of the preceding one, but this time the result of no. 1 is weaker.

Throughout this exposé, X will denote a locally noetherian prescheme; $\mathcal{I}$, a quasi-coherent ideal of $\mathcal{O}_X$ (so $Y = V(\mathcal{I})$ is a closed subset of X); U, a variable open neighborhood of Y in X; and $\widehat{X}$, the formal completion of X along Y.

For every ringed space $(Z, \mathcal{O}_Z)$, let $\mathbf{P}(Z)$ denote the category of invertible $\mathcal{O}_Z$-modules, that is, modules locally free of rank 1, and let $\mathrm{Pic}(Z)$ denote the group of isomorphism classes of invertible modules on Z.

XI.1 Comparison of $\mathrm{Pic}(\widehat{X})$ and $\mathrm{Pic}(Y)$

For every $n \in \mathbf{N}$, set

$$X_n = (Y, \mathcal{O}_X/\mathcal{I}^{n+1}) \qquad \text{and} \qquad P_n = \mathcal{I}^{n+1}/\mathcal{I}^{n+2}.$$

The sequence of sheaves of abelian groups on Y

$$0 \longrightarrow P_n \xrightarrow{u} \mathcal{O}^*_{X_{n+1}} \xrightarrow{v} \mathcal{O}^*_{X_n} \longrightarrow 1 \tag{XI.1.1}$$

is *exact*. More precisely, the group structure on P_n is the additive structure, $u(x) = 1 + x$ for every $x \in P_n$, and v is the homomorphism induced by the injection $\mathcal{I}^{n+2} \longrightarrow \mathcal{I}^{n+1}$. To see that v is surjective, observe that, for every $y \in Y$, $\mathcal{O}_{X_n,y}$ is a local ring, the quotient of $\mathcal{O}_{X_{n+1},y}$ by a nilpotent ideal; the rest is equally trivial. From (XI.1.1) we deduce an exact cohomology sequence:

$$\mathrm{H}^1(Y, P_n) \xrightarrow{u^1} \mathrm{H}^1(Y, \mathcal{O}^*_{X_{n+1}}) \xrightarrow{v^1} \mathrm{H}^1(Y, \mathcal{O}^*_{X_n}) \xrightarrow{d} \mathrm{H}^2(Y, P_n). \tag{$*$}$$

Moreover, for every $n \in \mathbf{N}$, we know how to identify $\mathrm{Pic}(X_n)$ with $\mathrm{H}^1(Y, \mathcal{O}^*_{X_n})$. Furthermore, if E is an invertible $\mathcal{O}_{X_{n+1}}$-module corresponding to a cohomology class $c(E)$, the cohomology **6**
class corresponding to the pullback of E to X_n is equal to $v^1(c(E))$. Hence the following proposition:

Proposition XI.1.1. *Retain the notation introduced above. Let $p \in \mathbf{N}$. The map*

$$\mathrm{Pic}(\widehat{X}) \longrightarrow \mathrm{Pic}(X_n)$$

has the following properties:

(i) it is injective for $n \geq p$ if $\mathrm{H}^1(Y, P_n) = 0$ for every $n \geq p$;

(ii) it is an isomorphism for $n \geq p$ if $\mathrm{H}^i(Y, P_n) = 0$ for every $n \geq p$ and $i = 1, 2$.

Of course, the exact sequence $(*)$ contains more information than the proposition above. The reader will have noticed that nothing has been said about the functor $\mathbf{P}(\widehat{X}) \longrightarrow \mathbf{P}(Y)$. Given two invertible $\mathcal{O}_{\widehat{X}}$-modules E, F, $H = \mathcal{H}om(E, F)$ is also invertible. If reduction modulo $\mathcal{I}^{n+1}$ is denoted by a subscript n, we obtain an exact sequence:

$$0 \longrightarrow H_0 \otimes P_n \longrightarrow \mathcal{H}om(E_{n+1}, F_{n+1}) \longrightarrow \mathcal{H}om(E_n, F_n) \longrightarrow 0.$$

Hence there is an exact cohomology sequence that we shall not write down and whose interpretation is evident; this observation can be used to study the functor $\mathbf{P}$.

XI.2 Comparison of $\mathrm{Pic}(X)$ and $\mathrm{Pic}(\widehat{X})$

The reader will find in Exposé X, no. 2, the proof of the following:

Proposition XI.2.1. *Suppose that* $\mathrm{Lef}(X, Y)$ *holds; then, for every open neighborhood U of Y in X, the functor*

$$\mathbf{P}(U) \to \mathbf{P}(\hat{X}) \tag{XI.2.2}$$

is fully faithful, and the map

$$\mathrm{Pic}(U) \to \mathrm{Pic}(\hat{X}) \tag{XI.2.3}$$

is therefore injective. If $\mathrm{Leff}(X, Y)$ *holds, the map (XI.2.4) is an isomorphism:* **7**

$$\varinjlim_{U} \mathrm{Pic}(U) \to \mathrm{Pic}(\hat{X}). \tag{XI.2.4}$$

Corollary XI.2.2. *Suppose that* $\mathrm{Lef}(X, Y)$ *holds and that, for every integer $n \geq p$, one has* $\mathrm{H}^1(Y, P_n) = 0$*; then, for every open subset $U \supset Y$, the maps*

$$\mathrm{Pic}(X) \to \mathrm{Pic}(U) \to \mathrm{Pic}(X_n)$$

are injective if $n \geq p$. If $\mathrm{Leff}(X, Y)$ *holds and if, moreover, for every integer $n \geq p$, one has* $\mathrm{H}^i(Y, P_n) = 0$ *for $i = 1$ and $i = 2$, then the map*

$$\varinjlim_{U} \mathrm{Pic}(U) \to \mathrm{Pic}(X_n)$$

is an isomorphism for $n \geq p$.

XI.3 Comparison of $\mathbf{P}(X)$ and $\mathbf{P}(U)$

A definition:

[. *remark]Definition 3.1*$^{(*)}$ *Let X be a prescheme and let Z be a closed subset of X. Put $U = X - Z$. We say that X is* parafactorial *at the points of Z if, for every open subset V of X, the functor $\mathbf{P}(V) \to \mathbf{P}(V \cap U)$ is an equivalence of categories. We also say that the pair (X, Z) is parafactorial.*

Recall that $\mathbf{P}(Z)$ denotes the category of locally free modules of rank 1 on Z.

Definition XI.3.2. A noetherian local ring is said to be *parafactorial* if the pair $(\mathrm{Spec}(A), \{\mathfrak{r}(A)\})$ is parafactorial.

One proves the following proposition, which shows that the notion is "pointwise":

Proposition XI.3.3. *Suppose that X is locally noetherian. The pair (X, Z) is parafactorial if and only if, for every $z \in Z$, the local ring $\mathcal{O}_{X,z}$ is parafactorial.* 8

Note that "parafactorial"includes "fully faithful". As in Lemma 3.5 of Exposé X, one proves the following:

Lemma XI.3.4. *If X is a locally noetherian prescheme and if $Z = X - U$ is a closed subset of X, the following conditions are equivalent:*

(i) for every open subset V of X, the functor $\mathbf{P}(V) \to \mathbf{P}(V \cap U)$ is fully faithful;

(ii) the homomorphism $\mathcal{O}_X \to i_*(\mathcal{O}_U)$ is an isomorphism;

(iii) for every $z \in Z$, one has $\operatorname{prof}(\mathcal{O}_{X,z}) \geq 2$.

Thus "parafactorial"means that the conditions of XI.3.4 are satisfied and that, for every open subset V of X, the homomorphism $\operatorname{Pic}(V) \to \operatorname{Pic}(V \cap U)$ is surjective. In particular, if X is the spectrum of a noetherian local ring, one obtains:

Proposition XI.3.5. *Let A be a noetherian local ring; it is parafactorial if and only if $\operatorname{prof} A \geq 2$ and $\operatorname{Pic}(X' - \{x\}) = 0$, where $X' = \operatorname{Spec}(A)$ and x is the unique closed point of X'.*

Notice that a local ring of dimension ≤ 1 is never parafactorial, since its depth is ≤ 1. Thus *factorial does not imply parafactorial*; the implication does, however, hold for noetherian local rings of dimension ≥ 2, as we shall see below.

Lemma XI.3.6. *Let X be a locally noetherian prescheme and let Z be a closed subset of X. Let $f\colon X_1 \to X$ be a faithfully flat and quasi-compact morphism. Put $Z_1 = f^{-1}(Z)$. If (X_1, Z_1) is parafactorial, then (X, Z) is parafactorial.*

First note that, if $i\colon (X - Z) \to X$ denotes the canonical immersion of $U = X - Z$ into X, the formation of the direct image under i of a quasi-coherent $\mathcal{O}_U$-module commutes with base change by f, since f is flat. It is therefore equivalent to assume the equivalent conditions of Lemma XI.3.5 for (X, Z) or for (X_1, Z_1), since f is a morphism of descent for the category of quasi-coherent sheaves. It remains to prove that, for every open subset V of X, $\operatorname{Pic}(V) \to \operatorname{Pic}(V \cap U)$ is surjective. We make the base change $V \to X$, which changes nothing (*sic*), and we are reduced to $V = X$. We then note that, if $\mathcal{L}$ is an invertible $\mathcal{O}_U$-module and if $\mathcal{L}$ admits a locally free extension, this extension is isomorphic to $i_*(\mathcal{L})$, by what we have just seen. It remains to prove that $i_*(\mathcal{L})$ is invertible. Using once again the fact that direct image under i commutes with flat base change, and that "locally free of rank 1"is a property that descends under faithfully flat and quasi-compact morphisms, we are done. 9

Corollary XI.3.7. *Let A be a noetherian local ring; if $\hat{A}$ is parafactorial, then A is parafactorial.*

One should not believe that if A is parafactorial, then $\hat{A}$ is parafactorial(*). △

Before turning to the statement of the main theorem of this no. , let us make the connection with divisor theory and the notion of a factorial ring(*).

(*) For a more detailed study of the notion of parafactoriality, and for the proof of XI.3.3, cf. EGA IV 21.13, 21.14.

(*) *Editor's note.* For a precise study of the relationship between the factoriality of A and that of its completion, see (Heitmann R., "Characterization of completions of unique factorization domains", *Trans. Amer. Math. Soc.* **337** (1993), no. 1, pp. 379–387).

(*) For the generalities that follow, cf. also EGA IV 21.

Let X be a *noetherian* and *normal* prescheme. Let $Z^1(X)$ be the free abelian group generated by the $x \in X$ such that $\dim \mathcal{O}_{X,x} = 1$. The local ring at such a point is a discrete valuation ring. We denote the corresponding normalized valuation by v_x. Let $K(X)$ be the ring of rational functions on X, and let

$$p\colon K(X)^* \to Z^1(X)$$

be the map that associates to every $f \in K(X)^*$ the codimension-one cycle:

$$(f) = \sum_{x \in X,\, \dim \mathcal{O}_{X,x}=1} v_x(f) \cdot x.$$

The image of p is denoted by $P(X)$, and its elements are called *principal divisors*(*). We set **10**

$$\mathrm{Cl}(X) = Z^1(X)/P(X).$$

Let $Z'^1(X)$ be the subgroup of $Z^1(X)$ whose elements are the *locally principal divisors*. We know that

$$\mathrm{Pic}(X) \simeq Z'^1(X)/P(X),$$

and consequently $\mathrm{Pic}(X)$ is identified with a subgroup of $\mathrm{Cl}(X)$.

Note that if U is a dense open subset of X, then $K(X) \to K(U)$ is an isomorphism, and that if $\mathrm{codim}(X - U, X) \geq 2$, i.e. if every $x \in X$ such that $\dim \mathcal{O}_{X,x} \leq 1$ belongs to U, the homomorphism $Z^1(X) \to Z^1(U)$, and consequently $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$, is also an isomorphism. Finally, if every $x \in U$ is factorial, i.e. if $\mathcal{O}_{X,x}$ is factorial, then $Z^1(U) = Z'^1(U)$, and hence $\mathrm{Pic}(U) \simeq \mathrm{Cl}(U)$.

Proposition XI.3.8. *Let X be a noetherian and normal prescheme. Let $(U_i)_{i \in I}$ be a family of open subsets of X such that:*

a) the U_i form a filter base(*)*;*

b) if we set $Y_i = X - U_i$, then $\mathrm{codim}(Y_i, X) \geq 2$ for every i,

c) if $x \in U_i$ for every $i \in I$, then $\mathcal{O}_{X,x}$ is factorial.

Then there is an isomorphism:

$$\varinjlim_{i \in I} \mathrm{Pic}(U_i) \xrightarrow{\approx} \mathrm{Cl}(X).$$

Note that b) implies that every $x \in X$ such that $\dim \mathcal{O}_{X,x} \leq 1$ belongs to U_i for every i. Thus the U_i are dense, and moreover the homomorphism $Z^1(U_i) \to Z^1(X)$ is an isomorphism, as is $K(U_i) \to K(X)$. Thus $\mathrm{Pic}(U) \subset \mathrm{Cl}(U_i) \simeq \mathrm{Cl}(X)$. To prove the desired assertion, it is therefore enough to show that every $D \in Z^1(X)$ belongs to $Z'^1(U_i)$ for a suitable i. It is enough to do this for irreducible positive "divisors". Thus let $x \in X$ be such that $\dim \mathcal{O}_{X,x} = 1$. It **11** is enough to prove that there exists an $i \in I$ such that $\overline{\{x\}}$ is locally principal at the points of U_i. Let $\mathcal{I}$ be the largest ideal of definition of the closed subset $\overline{\{x\}}$. The set of points in a neighborhood of which $\mathcal{I}$ is free is an open subset U. Now $U \supset \bigcap_{i \in I} U_i$ by c). If we set $Y = X - U$, then $Y \subset \bigcup_{i \in I} Y_i$ with $Y_i = X - U_i$; now Y is closed, and therefore has only *finitely* many generic points, so it is contained in the union of finitely many Y_i, and hence in Y_j for some $j \in I$, since the U_i form a filter base. Thus $U \supset U_j$. □

(*) In accordance with the terminology of EGA IV 21, we now prefer to reserve the term "divisors"for "locally principal divisors"or "Cartier divisors".

(*) *Editor's note.* i.e. a decreasing filtered family.

Corollary XI.3.9. *Let X be a noetherian and normal prescheme, and let Y be a closed subset of codimension ≥ 2. Suppose that, for every $p \in X - Y$, $\mathcal{O}_{X,p}$ is factorial. Then*

$$\mathrm{Pic}(X - Y) \to \mathrm{Cl}(X - Y) \to \mathrm{Cl}(X)$$

are isomorphisms.

Corollary XI.3.10. *Let A be a noetherian normal local ring. Set $X' = \mathrm{Spec}(A)$ and $x = \mathfrak{r}(A)$. For A to be* factorial, *it is necessary and sufficient that $\mathrm{Pic}(X' - \{x\}) = 0$ and that $\mathfrak{p} \in X' - \{x\}$ imply that $A_\mathfrak{p}$ is factorial.*

Indeed, for A to be factorial, it is necessary and sufficient that $\mathrm{Cl}(X') = 0$ [(*)].

Corollary XI.3.11. *Let A be a noetherian local ring of dimension ≥ 2. Set $X' = \mathrm{Spec}(A)$, and let $x = \mathfrak{r}(A)$. Set $X = X' - \{x\}$. The following conditions are equivalent:*

(i) A is factorial;

(ii) a) *for every $y \in X$, $\mathcal{O}_{X,p}$ is factorial, and*

b) *A is parafactorial, i.e.* $\mathrm{prof}\, A \geq 2$ *and* $\mathrm{Pic}(X) = 0$.

Before proving this corollary, let us state the **12**

Serre's normality criterion 3.11 [(*)]. *Let A be a noetherian local ring. For A to be normal, it is necessary and sufficient that*

(i) for every prime ideal $\mathfrak{p}$ of A such that $\dim A_\mathfrak{p} \leq 1$, $A_\mathfrak{p}$ is normal,

(ii) for every prime ideal $\mathfrak{p}$ of A such that $\dim A_\mathfrak{p} \geq 2$, one has $\mathrm{prof}\, A_\mathfrak{p} \geq 2$.

Let us prove XI.3.11.

(i) $\Longrightarrow$ (ii). Since a localization of a factorial ring is again factorial, we have (ii) a). Moreover, A is normal, and hence $\mathrm{prof}\, A \geq 2$, since $\dim A \geq 2$ (XI.3.12 (ii)). Finally, A is parafactorial; indeed, $\mathrm{Pic}(X) \simeq \mathrm{Cl}(X') = 0$ (cf. XI.3.10).

(ii) $\Longrightarrow$ (i). Let us first prove that A is *normal* by applying Serre's criterion. Since $\dim A \geq 2$, condition (i) of the criterion is among the hypotheses. Moreover, for every $\mathfrak{p} \in X$, $A_\mathfrak{p}$ is factorial, hence normal, and therefore of depth ≥ 2, at least if $\dim A_\mathfrak{p} \geq 2$. Finally, $\mathrm{prof}\, A \geq 2$ by (ii) b). It remains to apply XI.3.10.

Let us summarize the foregoing:

Proposition XI.3.13. *Let X be a locally noetherian prescheme, and let $\mathcal{I}$ be a quasi-coherent ideal of X. Let $Y = V(\mathcal{I})$. Let $p \in \mathbf{N}$. Suppose that:*

*1) one has $\mathrm{Leff}(X, Y)$ (*Exposé *X);*

2) $\mathrm{H}^i(X, \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$ if $i = 1$ or 2 and if $n \geq p$;

3) for every open neighborhood U of Y in X and every $x \in X - U$, the ring $\mathcal{O}_{X,x}$ is parafactorial.

Then, for every $n \geq p$, and every open neighborhood U of Y, the homomorphisms

$$\begin{array}{ccc} \mathrm{Pic}(X) & \longrightarrow & \mathrm{Pic}(X_n) \\ & \searrow \quad \nearrow & \\ & \mathrm{Pic}(U) & \end{array}$$

are isomorphisms.

We know some parafactorial rings: 13

Theorem XI.3.14. *(i) (Auslander–Buchsbaum)*(*) *A regular noetherian local ring is factorial (and hence parafactorial if its dimension is* ≥ 2*).*

(ii) A noetherian local ring of dimension ≥ 4 *that is a* complete intersection *is parafactorial.*

Corollary XI.3.15 (Conjecture of Samuel(*))**.** *A noetherian local ring* A *that is a complete intersection and is factorial in codimension* ≥ 3 *(i.e.* $\dim A_{\mathfrak{p}} \leq 3$ *implies that* $A_{\mathfrak{p}}$ *is factorial) is factorial.*

Proof of the corollary

We argue by induction on the dimension of A.

If $\dim A \leq 3$, then A is factorial by hypothesis.

If $\dim A > 3$, then by the induction hypothesis, noting that a localization of a complete intersection is again a complete intersection, all localizations of A other than A are factorial. By Theorem XI.3.14 (ii), A is parafactorial, and therefore factorial by XI.3.11.

Let us prove XI.3.14 (i) (following Kaplansky)(*).

Let A be a regular noetherian local ring, and set $\dim A = n$. If $n = 0$ or 1, the result is known. Suppose $n \geq 2$; we argue by induction on n, and suppose $n \geq 2$ and the theorem proved for rings of dimension $< n$. Set $X' = \mathrm{Spec}(A)$ and $X = X' - \{x\}$, where $x = \mathfrak{r}(A)$. The localizations of A other than A are regular and of dimension $< n$, and hence factorial. Moreover, $\mathrm{prof}\, A = \dim A \geq 2$. It is therefore enough to prove that $\mathrm{Pic}(X) = 0$ (cor. XI.3.11). Thus let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-Module; we know that it can be extended to a coherent $\mathcal{O}_{X'}$-Module $\mathcal{L}'$. There exists a resolution of $\mathcal{L}'$ by free $\mathcal{O}_X$-Modules:

$$0 \longleftarrow \mathcal{L}' \longleftarrow \mathcal{L}'_1 \longleftarrow \cdots \longleftarrow \mathcal{L}'_n \longleftarrow 0,$$

because the cohomological dimension of A is finite. By restriction to X' we obtain a finite 14
free resolution. It is therefore enough to prove the following lemma:

Lemma XI.3.16. *Let* $(X, \mathcal{O}_X)$ *be a ringed space, and let* $\mathcal{L}$ *be a locally free* $\mathcal{O}_X$*-Module that admits a finite resolution by* finite-type free *modules. Then* $\det(\mathcal{L}) \simeq \mathcal{O}_X$*.*

Recall that $\det(\mathcal{L})$ is defined as the highest exterior power of $\mathcal{L}$. In the case under consideration, $\det(\mathcal{L}) \simeq \mathcal{L}$, because $\mathcal{L}$ is invertible, and hence the lemma gives the conclusion. Let us prove this lemma. Let

$$0 \longleftarrow \mathcal{L}_0 \longleftarrow \mathcal{L}_1 \longleftarrow \mathcal{L}_2 \longleftarrow \cdots \longleftarrow \mathcal{L}_n \longleftarrow 0,$$

be the asserted exact sequence, where $\mathcal{L}_0 = \mathcal{L}$. Since everything is locally free, we have:

$$\bigotimes_{0 \leq i \leq n} (\det(\mathcal{L}_i))^{(-1)^i} \simeq \mathcal{O}_X$$

(*) *Editor's note.* see Bourbaki, *Algèbre commutative* VII.1.4, corollary to Theorem 2, and VII.3.2, Theorem 1

(*) Cf. EGA IV 5.8.6.

(*) *Editor's note.* compare with the following purity result, due to Gabber. Let X be the spectrum of a regular local ring A of dimension 3, let a have nonzero differential, i.e. $a \in \mathfrak{m} - \mathfrak{m}^2$, and let U be the complement of $V(a)$. Then a vector bundle on U is free (for a simple proof, see (Swan R.G., "A simple proof of Gabber's theorem on projective modules over a localized local ring" *Proc. Amer. Math. Soc.* **103** (1988), no. 4, p. 1025–1030). The rank-1 case is a special case of Theorem XI.3.14. For purity results concerning vector bundles of arbitrary rank, whether in the analytic or algebraic setting, see (Gabber O., "On purity theorems for vector bundles", *Internat. Math. Res. Notices* (2002), no. 15, p. 783–788).

(*) *Editor's note.* for a proof in the same spirit, but more elementary, see Call F. & Lyubeznik G., "A simple proof of Grothendieck's theorem on the parafactoriality of local rings", in *Commutative algebra: syzygies, multiplicities, and birational algebra (South Hadley, MA, 1992)*, Contemp. Math., vol. 159, American Mathematical Society, Providence, RI, 1994, p. 15–18.

(*) This is the proof reproduced in EGA IV 21.11.1.

and all the $\mathcal{L}_i$, $i > 0$, are free, and hence so are their determinants, and consequently so is the determinant of $\mathcal{L}_0 = \mathcal{L}$. □

It remains for us to prove part (ii) of the theorem. First, let us prove a lemma that will allow us to proceed by induction:

Lemma XI.3.17. *Let A be a noetherian local ring that is a quotient of a regular ring. Let $t \in \mathfrak{r}(A)$ be an A-regular element. Suppose that A is complete for the t-adic topology. Set $X' = \mathrm{Spec}(A)$, $Y' = V(t) \simeq \mathrm{Spec}(B)$, $B = A/tA$, $X = X' - \{x\}$, $Y = Y' - \{x\}$, and $x = \mathfrak{r}(A)$. Suppose that:*

a) for every $y \in X$ closed in X, one has $\operatorname{prof} \mathcal{O}_{X,y} \geq 2$,

b) $\operatorname{prof} A/tA \geq 3$,

then the map $\mathrm{Pic}(X) \to \mathrm{Pic}(Y)$ *is injective. In particular, if B is parafactorial, then A is also parafactorial.* **15**

We know that a) implies $\mathrm{Lef}(X, Y)$ by X 2.1. If we prove that $\mathrm{H}^1(Y, P_n) = 0$ for every $n \geq 0$, then we will know by (XI.2.2) that $\mathrm{Pic}(X) \to \mathrm{Pic}(Y)$ is injective. If, moreover, B is parafactorial, then we will know that $\mathrm{Pic}(Y) = 0$ (XI.3.5), and hence $\mathrm{Pic}(X) = 0$; moreover, $\operatorname{prof}(A) = 3 + 1 \geq 2$, since t is A-regular, and thus A will be parafactorial by XI.3.5.

Let $\mathcal{I} = \widetilde{(tA)}$ be the $\mathcal{O}_{X'}$-Module associated with the ideal tA. In no. XI.1, we set $P_n = (\mathcal{I}^{n+1}/\mathcal{I}^{n+2})_{|Y}$ for every $n \geq 0$. Since t is A-regular, $P_n \simeq \mathcal{O}_Y$. *It therefore remains for us to prove that* $\mathrm{H}^1(Y, \mathcal{O}_Y) = 0$. Now $Y = Y' - \{x\}$ is an open subset of Y', and hence we have an exact sequence (I (27)):

$$\mathrm{H}^1(Y', \mathcal{O}_{Y'}) \to \mathrm{H}^1(Y, \mathcal{O}_Y) \to \mathrm{H}^2_x(Y', \mathcal{O}_{Y'}),$$

whose right-hand term is zero by hypothesis b), and whose left-hand term is zero because Y' is affine. □

Lemma XI.3.18. *Keeping the hypotheses of XI.3.17, suppose in addition that:*

c) for every y closed in Y, one has $\operatorname{prof} \mathcal{O}_{X,y} \geq 3$,

d) $\operatorname{prof} A/tA \geq 4$ *(stronger than b)),*

e) for every y closed in X with $y \in Y$, the ring $\mathcal{O}_{X,y}$ is parafactorial.

Then the map $\mathrm{Pic}(X) \to \mathrm{Pic}(Y)$ *is an isomorphism; in particular, for A to be parafactorial, it is necessary and sufficient that B be parafactorial.*

We know (X 2.1) that a) and c) imply $\mathrm{Leff}(X, Y)$. Moreover, by the argument just given, d) implies that $\mathrm{H}^1(Y, P_n) = 0$ for every $n \geq 0$ and $i = 1$ or $i = 2$. Moreover, for every open neighborhood U of Y in X, the complement of U in X consists of finitely many closed points. By e) and Theorem XI.3.13, we deduce that $\mathrm{Pic}(X) \to \mathrm{Pic}(Y)$ is an isomorphism. On the other hand, $\operatorname{prof} A \geq \operatorname{prof} B \geq 2$; by the criterion XI.3.5, we deduce that A is parafactorial if and only if B is. **16**

Let us now prove XI.3.14 (ii). Let R be a regular noetherian local ring. Let $(t_1, \ldots, t_k)$ be an R-sequence. Set $B = R/(t_1, \ldots, t_k)$ and suppose that $\dim B \geq 4$. We must prove that B is parafactorial. We argue by induction on k. If $k = 0$, then B is regular, and hence factorial by XI.3.14 (i), and therefore parafactorial by XI.3.11. Suppose $k \geq 1$ and the theorem proved for $k' < k$. Set $A = R/(t_1, \ldots, t_{k-1})$, so that $B = A/t_k A$. We may suppose that B is complete by XI.3.7. By the induction hypothesis, A is parafactorial. Let us prove that Lemma XI.3.18 applies. We have assumed that B is complete, and hence so is A, and therefore A is complete for the t-adic topology. If $x \in X$ and x is closed in X, then A_x is a complete intersection of dimension ≥ 4, with $k' < k$. By the induction hypothesis, A_x is parafactorial, and moreover has depth ≥ 4. This gives a), c), and e). Moreover, $\dim A \geq 5$, which gives d). □

Theorem XI.3.19. *Let X be a locally noetherian prescheme, and let $\mathcal{I}$ be a coherent sheaf of ideals of X. Set $Y = V(\mathcal{I})$. Let n be an integer. Suppose that:*

(i) one has $\mathrm{Leff}(X, Y)$ *(cf. examples X 2.1 and X 2.2),*

(ii) for every $p \geq n$, one has $\mathrm{H}^i(Y, \mathcal{I}^{p+1}/\mathcal{I}^{p+2}) = 0$ for $i = 1$ and $i = 2$,

(iii) for every open subset $U \supset Y$ and every $x \in X - U$, the ring $\mathcal{O}_{X,x}$ is regular of dimension ≥ 2 or a complete intersection of dimension ≥ 4.

Then, for every open subset $U \supset Y$ and every integer $p \geq n$, the homomorphisms

$$\mathrm{Pic}(X) \to \mathrm{Pic}(U) \to \mathrm{Pic}(Y_p)$$

are isomorphisms, where Y_p denotes the prescheme $(Y, \mathcal{O}_X/\mathcal{I}^{p+1})$.

It is enough to combine XI.3.13 and XI.3.14.

Exposé XII

Applications to projective algebraic schemes

XII.1 Projective duality theorem and finiteness theorem(*)

The following theorem, essentially contained in FAC(*) (except that, at the time, Serre did not have at his disposal the language of Ext for sheaves of modules(*)), is the global analogue of the local duality theorem (Exposé IV), which was modeled on it. **17**

Theorem XII.1.1. *Let k be a field, $X = \mathbf{P}^r_k$ projective space of dimension r over k, and F a variable coherent module on X; then there is an isomorphism of ∂-functors in F:*

$$\mathrm{H}^i(X,F)' \xleftarrow{\sim} \mathrm{Ext}^{r-i}(X; F, \Omega^r_{X/k}), \tag{1}$$

where we set

$$\Omega^r_{X/k} = \mathcal{O}_{\mathbf{P}^r_k}(-r-1). \tag{2}$$

Remark. This module is of course also the module of relative differentials of degree r of X over k. In this form, the theorem remains valid if X is a proper smooth scheme over k (for the projective case, see A. Grothendieck, "*Théorèmes de dualité pour les faisceaux algébriques cohérents*", Séminaire Bourbaki, May 1957)(*). When F is locally free, one recovers Serre's duality theorem $\mathrm{H}^i(X,F)' \xrightarrow{\sim} \mathrm{H}^{r-i}(\mathcal{H}om_{\mathcal{O}_X}(F, \Omega^r_{X/k}))$. Theorem XII.1.1 (which also recovers the case of a projective X over k, as in *loc. cit.*) will suffice for our purposes.

The homomorphism (1) is deduced from the Yoneda pairing **18**

$$\mathrm{H}^i(X,F) \times \mathrm{Ext}^{r-i}(X; F, \Omega^r_{X/k}) \to \mathrm{H}^r(X, \Omega^r_{X/k}), \tag{3}$$

and from a well-known isomorphism (cf. FAC, or EGA III 2.1.12):

$$\mathrm{H}^r(X, \Omega^r_{X/k}) = \mathrm{H}^r(\mathbf{P}^r_k, \mathcal{O}_{\mathbf{P}^r_k}(-r-1)) \xrightarrow{\sim} k. \tag{4}$$

To show that (1) is an isomorphism, one proceeds as in the case of the local duality theorem, noting that $\mathrm{H}^r(X,F)$, as a functor of F, is right exact (since $\mathrm{H}^n(X,F) = 0$ for $n > r$), and

(*)The present exposé, written in January 1963, is considerably more detailed than was the oral presentation in June 1962.

(*)J.-P. Serre, "Faisceaux algébriques cohérents", *Ann. of Math.* **61** (1955), p. 197-278.

(*)*Editor's note.* Readers interested in the history of mathematics will profitably consult the letter that Grothendieck wrote to Serre on 15 December 1955 and Serre's reply of 22 December of the same year; see *Correspondance Grothendieck-Serre*, edited by Pierre Colmez and Jean-Pierre Serre, Documents Mathématiques, vol. 2, Société Mathématique de France, Paris, 2001.

(*)For a more general duality theorem, cf. the Hartshorne seminar cited at the end of Exp. IV.

that every coherent module is isomorphic to a quotient of a direct sum of modules of the form $\mathcal{O}(-m)$, with m large. By descending induction on i, this reduces us to checking the assertion for a sheaf of the form $\mathcal{O}(-m)$, where it is contained in the well-known explicit computations (FAC, or EGA III 2.1.12). We may moreover assume $-m \leq -r-1$, in which case $\mathrm{H}^i(X, \mathcal{O}(-m)) = 0$ for $i \neq r$.

Corollary XII.1.2. *For a given coherent F, and for m sufficiently large, there is a canonical isomorphism*

$$\mathrm{H}^i(X, F(-m))' \xrightarrow{\sim} \mathrm{H}^0(X, \mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \Omega^r_{X/k})(m)) \tag{5}$$

(where $'$ denotes the vector-space dual).

Indeed, on projective space X, for every pair of coherent sheaves F, G and for n sufficiently large, there is a canonical isomorphism:

$$\mathrm{Ext}^n(X; F(-m), G) \simeq \mathrm{Ext}^n(X; F, G(m)) \xrightarrow{\sim} \mathrm{H}^0(X, \mathcal{E}xt^n_{\mathcal{O}_X}(F, G)(m)), \tag{6}$$

(the isomorphism between the first two terms being trivially valid for every m), as follows from the global Ext spectral sequence

$$\mathrm{H}^p(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F, G(m))) \Longrightarrow \mathrm{Ext}^\bullet(X; F, G(m)),$$

which degenerates for m sufficiently large, by virtue of the fact that **19**

$$\mathcal{E}xt^q_{\mathcal{O}_X}(F, G(m)) \simeq \mathcal{E}xt^q_{\mathcal{O}_X}(F, G)(m),$$

and that the $\mathcal{E}xt^q_{\mathcal{O}_X}(F, G)$ are coherent sheaves. Thus (5) follows from (6) and (1).

Corollary XII.1.3. *For given i, F, the following conditions are equivalent:*

(i) $\mathrm{H}^i(X, F(-m)) = 0$ *for m large.*

(i bis) $\mathrm{H}^i(X, F(\cdot)) = \bigoplus_{m \in \mathbf{Z}} \mathrm{H}^i(X, F(-m))$ *is a finite-type S-module, where $S = k[t_o, \ldots, t_r]$.*

(ii) $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \Omega^r_{X/k}) = 0$.

(ii bis) $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \mathcal{O}_X) = 0$.

(iii) $\mathrm{H}^i_x(F_x) = 0$ *for every closed point x of X.*

(iv) $\mathrm{H}^{i+1}_x(\widetilde{F}_x) = 0$ *for every closed point x of the punctured affine cone $\widetilde{X} = \mathrm{Spec}(S) - \mathrm{Spec}(k)$ over X, where $\widetilde{F}$ denotes the inverse image of F under the canonical morphism $\widetilde{X} \to X$.*

Proof. (i) $\Longleftrightarrow$ (i bis), since the submodule of $\mathrm{H}^i(X, F(\cdot))$ that is the sum of the homogeneous components of degree $\geq \nu$ is of finite type over S (in fact, for $i \neq 0$, it is even of finite type over k), (cf. FAC or EGA III 2.2.1 and 2.3.2).

(i) $\Longleftrightarrow$ (ii) by Corollary XII.1.2.

(ii) $\Longleftrightarrow$ (ii bis), because $\Omega^r_{X/k}$ is locally isomorphic to $\mathcal{O}_X$.

(ii bis) $\Longleftrightarrow$ (iii) by the local duality theorem for $\mathcal{O}_{X,x}$ (which is a regular local ring of **20** dimension r), according to which the "dual"of $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \mathcal{O}_X)_x$ is identified with $\mathrm{H}^i_x(F_x)$ (V 2.1).

(ii bis) is equivalent to the analogous relation

$$\mathcal{E}xt^{r-i}_{\mathcal{O}_{\widetilde{X}}}(\widetilde{F}, \mathcal{O}_{\widetilde{X}}) = 0$$

(because $\widetilde{X} \to X$ is faithfully flat, so the inverse image of $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \mathcal{O}_X)$ is isomorphic to $\mathcal{E}xt^{r-i}_{\mathcal{O}_{\widetilde{X}}}(\widetilde{F}, \mathcal{O}_{\widetilde{X}})$), and this last relation is equivalent to (iv) by the local duality theorem for the local ring $\mathcal{O}_{X,x}$, which is regular of dimension $r+1$.

In particular, applying this to all $i \leq n$, we find:

Corollary XII.1.4. *Equivalent conditions for given n, F:*

(i) $\mathrm{H}^i(X, F(-m)) = 0$ *for $i \leq n$ and m large.*

(i bis) $\mathrm{H}^i(X, F(\cdot))$ *is a finite-type S-module for $i \leq n$.*

(ii) $\mathrm{prof}(F_x) > n$ *for every closed point x of X.*

(iii) $\mathrm{prof}(\widetilde{F}_x) > n+1$ *for every closed point x of $\widetilde{X}$.*

The point of Corollaries XII.1.3 and XII.1.4 is to express a global condition (i) or (i bis) in terms of local conditions, namely the vanishing of local invariants such as $\mathrm{H}^i_x(X, F_x)$ or $\mathrm{H}^i_x(X, \widetilde{F}_x)$, or an inequality involving depth. In this form, these results remain trivially valid for an arbitrary projective scheme X, and a very ample invertible sheaf $\mathcal{O}_X(1)$ on X, as one sees by means of a suitable projective immersion $X \to \mathbf{P}^r_k$. (Of course, conditions XII.1.3 (ii) and XII.1.3 (ii bis) are no longer equivalent to the others in this general case, unless, for example, one assumes that X is regular.) One can moreover generalize to the case of a projective morphism $X \to S$ as follows:

Proposition XII.1.5. *Let $f : X \to S$ be a projective morphism, with S noetherian, let $\mathcal{O}_X(1)$* **21**
be an invertible module on X that is very ample relative to S, let F be a coherent module on X, flat over S, let s be a point of S, let X_s be the fiber of X at s (regarded as a projective scheme over $k(s)$), let F_s be the sheaf induced on X_s by F, and finally let i be an integer. Suppose that, for every closed point x of X_s, one has $\mathrm{H}^i_x(F_{s,x}) = 0$ (for example, $\mathrm{prof}(F_{s,x}) > i$). Then there exists an open neighborhood U of s such that the same condition holds for every $s' \in U$. Moreover, for such a U, one has

$$\mathrm{R}^i f_*(F(-m)) = 0 \text{ for } m \text{ large,}$$

and if $\mathcal{S}$ is a quasi-coherent graded algebra on S, generated by $\mathcal{S}^1$, and defining X together with $\mathcal{O}_X(1)$ as $X = \mathrm{Proj}(\mathcal{S})$, $\mathcal{O}_X(1) = \mathrm{Proj}(\mathcal{S}(1))$, then the S-module

$$\mathrm{R}^i f_*(F(\cdot)) = \bigoplus_{m \in \mathbf{Z}} \mathrm{R}^i f_*(F(m))$$

is of finite type over U.

Embed X in an $X' = \mathbf{P}^r_S$ in such a way that $\mathcal{O}_X(1)$ is induced by $\mathcal{O}_{X'}(1)$ (this is possible provided that S is replaced by an affine neighborhood of s). Put [(*)] for every integer j, and every $t \in S$:

$$\underline{E}^j(t) = \mathcal{E}xt^j_{\mathcal{O}_{X'_t}}(F_{t'}, \mathcal{O}_{X'_t}(-r-1)) \tag{7}$$

Thus $\underline{E}^j(t)$ is a coherent module on X_t; I claim that, as t varies, the family of these modules is "constructible"in the following sense: for every $t \in S$ there exists a nonempty open subset V of the closure $\bar{t}$, endowed with the induced reduced structure, and a coherent module $\underline{E}^j(V)$ on $X_V = X \times_S V$, *flat* over V, such that, for every $t' \in V$, $\underline{E}^j(t)$ is isomorphic to the module induced by $\underline{E}^j(V)$ on X_t. To verify this assertion, putting $Z = \bar{t}$ with its induced structure, we consider the coherent modules **22**

$$\underline{E}^j(Z) = \mathcal{E}xt^j_{\mathcal{O}_{X'_Z}}(F_Z, \mathcal{O}_{X'_Z}(-r-1))$$

[(*)] The first part of XII.1.5 can also be obtained immediately by applying the purely local statement EGA IV 12.3.4 to the preceding $\underline{E}^j$, thereby bypassing most of the proof that follows.

(where the subscript Z still means that one takes the induced object over Z), and we take for V a nonempty open subset of Z such that the modules $\underline{E}^j(Z)$ are flat over V: this is possible, since one verifies immediately that $\underline{E}^j(Z) = 0$ for j outside the interval $[0, r]$, and one can apply SGA 1 IV 6.11. We then take $\underline{E}^j(V) = \underline{E}^j(Z)|_{X_V}$, and one readily verifies that it has the required property.

It follows from the preceding remark that there exists a finite partition of S formed by sets V_α of the form $V = V(t)$ as above (noetherian induction), and, applying Serre's theorem EGA III 2.2.1 to the $\underline{E}^j(V_i)$, one sees that there exists an integer m_0 such that

$$\mathrm{R}^i f_{V_\alpha *}(\underline{E}^j(V_\alpha)) = 0 \text{ for } i \neq 0, m \geq m_0, \text{ for } j,$$

from which it follows, using the flatness of $\underline{E}^j(V_\alpha)$ over V_α and straightforward Künneth-type relations (cf. EGA III par. 7), that

$$\mathrm{H}^i(X_t, \underline{E}^j(t)(m)) = 0 \text{ for } i \neq 0, m \geq m_0, \text{ for } j,$$

for every $t \in V_\alpha$, and hence for every t, since the V_α cover S. From this and the spectral sequence for global Ext it follows, by virtue of XII.1.1, as in the proof of XII.1.2, that there is an isomorphism

$$\mathrm{H}^i(X_t, F_t(-m))' \xleftarrow{\sim} \mathrm{H}^0(X_t, \underline{E}^{r-i}(t)(m)) \text{ for } m \geq m_0, \tag{8}$$

for every integer i, and every $t \in S$.

Now use the hypothesis on F_s, which can be written

$$\underline{E}^{r-i}(s) = 0, \tag{9}$$

and which, by virtue of (8), is equivalent to

$$\mathrm{H}^i(X_s, F_s(-m)) = 0 \text{ for } m \geq m_0. \tag{10}$$

Since F, and hence $F(-m)$, is flat over S, it follows from the Künneth-type relations already invoked that (for a given m) the same relation (10) holds when s is replaced by a t near s, in **23** particular by every generization t of s. By virtue of (8), we shall therefore have, for such a generization,

$$\underline{E}^{r-i}(t) = 0, \tag{11}$$

and the set of $t \in s$ for which this relation holds is plainly constructible (since it induces an open subset on each V_α); as it contains the generizations of s, it contains an open neighborhood U of s. This proves the first assertion of XII.1.5. Moreover, for $t \in U$, it follows from (11) and (8) that

$$\mathrm{H}^i(X_t, F_t(-m) = 0 \text{ for } m \geq m_0, \ t \in U, \tag{12}$$

which, by virtue of the Künneth-type relations, implies (and in fact is much stronger than)

$$\mathrm{R}^i f_*(F(-m)) = 0 \text{ on } U, \text{ for } m \geq m_0. \tag{13}$$

This proves the second assertion of XII.1.5. Finally, the last assertion follows immediately by proceeding as at the beginning of the proof of XII.1.3.

[. *remark]Remark 1.6*(*) *The proof simplifies considerably (eliminating all consideration of constructibility) when one assumes from the outset that the hypothesis made for $s \in S$ holds for every $s' \in S$. In fact, when one assumes that F_s has depth $> i$ at the closed points of X,*

(*)This remark is made more precise by footnote 95.

there is a general statement, local in nature on X*, which says that the same condition holds for all the* X_t*, provided that* X *is replaced by a suitable open neighborhood of the fiber* X_s *(in other words, a certain subset of* X*, defined by conditions on the modules induced by* F *on the fibers, is open, cf.* EGA *IV). Since* f *is proper here, this neighborhood can therefore be taken to be of the form* $f^{-1}(U)$*, which recovers the first assertion of XII.1.5 without laborious dévissage. In this general case, one can still prove by the method of loc. cit. that the first assertion of XII.1.5 (proved here globally, using the fact that* X *is projective over* S*) again follows from a purely local statement on* X *(which the reader may formulate explicitly if useful).*

XII.2 Lefschetz theory for a projective morphism: Grauert comparison theorem

It is the following theorem: **24**

Theorem XII.2.1. *Let* $f : X \to S$ *be a projective morphism, with* S *Noetherian, let* $\mathcal{O}_X(1)$ *be an invertible module on* X*, relatively ample over* S*, let* Y *be the prescheme of zeros of a section* t *of* $\mathcal{O}_X(1)$*, let* J *be the ideal defining* Y*, let* X_n *be the closed subprescheme of* X *defined by* J^{n+1}*, let* $\widehat{X}$ *be the formal completion of* X *along* Y*, let* $\widehat{f} : \widehat{X} \to S$ *be the composite morphism* $\widehat{X} \to X \to S$*, and let* F *be a coherent module on* X*, flat over* S*. Assume in addition that, for every* $s \in S$*, the module* F_s *induced on the fiber* X_s *has depth* $> n$ *at the points of that fiber, and that* t *is* F*-regular. Under these conditions:*

(i) The canonical homomorphism

$$\mathrm{R}^i f_*(F) \to \mathrm{R}^i \widehat{f}_*(\widehat{F})$$

is an isomorphism for $i < n$*, and a monomorphism for* $i = n$*.*

(ii) The canonical homomorphism

$$\mathrm{R}^i \widehat{f}_*(\widehat{F}) \to \varprojlim_m \mathrm{R}^i f_*(F_m)$$

is an isomorphism for $i \leq n$*.*

Proof. We immediately reduce to the case where S is affine, and then to proving the

Corollary XII.2.2. *Under the conditions of XII.2.1, assume in addition that* S *is affine. Then:*

(i) The canonical homomorphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(\widehat{X}, \widehat{F})$$

is an isomorphism for $i < n$*, and a monomorphism for* $i = n$*.* **25**

(ii) The canonical homomorphism

$$\mathrm{H}^i(\widehat{X}, \widehat{F}) \to \varprojlim_m \mathrm{H}^i(X_m, F_m)$$

is an isomorphism for $i \leq n$*.*

After replacing $\mathcal{O}_X(1)$ by a tensor power, and t by a power of t, we may assume that $\mathcal{O}_X(1)$ is very ample relative to S. On the other hand, since t, and hence t^m, is F-regular, multiplication by t^m, regarded as a homomorphism from $F(-m)$ to F, is injective; therefore, for every $m \geq 0$, there is an exact sequence:

$$0 \to F(-m) \xrightarrow{t^m} F \to F_m \to 0, \tag{14}$$

and hence an exact cohomology sequence

$$\mathrm{H}^i(X, F(-m)) \to \mathrm{H}^i(X, F) \to \mathrm{H}^i(X, F_m) \to \mathrm{H}^{i+1}(X, F(-m)).$$

But by XII.1.5, we have $\mathrm{H}^i(X, F(-m)) = 0$ for $i \leq n$ and m sufficiently large, which proves the

Lemma XII.2.3. *For m large, the canonical homomorphism*

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(X, F_m)$$

is bijective if $i < n$, and injective if $i = n$.

This shows that, for $i < n$, the inverse system $(\mathrm{H}^i(X_m, F_m))_{m \geq 0}$ is essentially constant, and *a fortiori* satisfies the Mittag-Leffler condition; hence (in view of $\widehat{F} = \varprojlim F_m$), we obtain (ii) from EGA 0_{III} 13.3. On the other hand, (i) follows trivially from this, taking XII.2.3 into account.

Corollary 2.4$^{(*)}$**.** *Let $f : X \to S$ be a projective flat morphism, with S locally Noetherian,* **26** *let $\mathcal{O}_X(1)$ be an invertible module on X, relatively ample over S, let t be a section of this module that is $\mathcal{O}_X$-regular, let Y be the closed subprescheme of zeros of t, and let $\widehat{X}$ be the formal completion of X along Y. Assume that, for every $s \in S$, X_s has depth ≥ 1 (resp. depth ≥ 2) at its closed points. Then, for every open neighborhood U of Y, the functor*

$$F \longmapsto \widehat{F}$$

*from the category of locally free coherent modules on U to the category of locally free coherent modules on $\widehat{X}$ is faithful (resp. fully faithful, i.e. the Lefschetz condition (*Lef*) of X 2 holds).*

For two locally free Modules F and G on U, introduce the Module

$$H = \mathcal{H}om_{\mathcal{O}_U}(F, G)$$

we are reduced to proving that the canonical homomorphism

$$\mathrm{H}^0(U, H) \to \mathrm{H}^0(\widehat{U}, \widehat{H}) \tag{15}$$

is injective (resp. bijective). Now the Modules H_t have depth ≥ 1 (resp. ≥ 2) at the closed points of X_t, and hence we can apply XII.2.1, which gives the conclusion XII.2.4 in the case where $U = X$. In the case of an arbitrary U, note that the question is local on S, so we may suppose that S is affine. Then every coherent Module on X is a quotient of a coherent locally free Module (since $\mathcal{O}_X(1)$ is an invertible relatively ample Module on X). Since the dual Module $H' = \mathcal{H}om(H, \mathcal{O}_U)$ extends to a coherent Module on X, which is therefore isomorphic to a cokernel of a homomorphism of locally free Modules on X, it follows by transposition that one can find a homomorphism

$$u' : L'^0 \to L'^1$$

of locally free Modules on X, inducing a homomorphism **27**

$$u : L^0 \to L^1$$

of locally free Modules on U, such that there is an exact sequence

$$0 \to H \to L^0 \xrightarrow{u} L^1.$$

Using the five lemma (which becomes the three lemma), and the left exactness of the functor H^0, we are reduced to proving that (15) is injective (resp. bijective) when H is replaced there by L^0, L^1, which brings us back to the case where H is induced by a locally free Module H' on X. Moreover, in the non-parenthetical case this reduction is even unnecessary, since the kernel of (15) consists in any event of the sections of H on U that vanish in some suitable open neighborhood V of Y; but the restriction homomorphism $\mathrm{H}^0(U, H) \to \mathrm{H}^0(V, H)$ is injective, since H has depth ≥ 1 at the points of every closed subset Z of X not meeting Y (cf. the lemma below). In the parenthetical case, we are reduced to proving that

$$H^0(X, H') \to \mathrm{H}^0(U, H')$$

is bijective, which follows from the fact that H' has depth ≥ 2 at all the points of a closed subset $Z = X - U$ of X not meeting Y. It therefore remains only to prove the

Lemma XII.2.5. *Let F be a coherent Module on X, flat over S, such that for every $s \in S$, F_s has depth $\geq n$ at every closed point of X_s. Then, for every closed subset Z of X not meeting Y, F has depth $\geq n$ at all the points of Z.*

Indeed, for every $x \in X$, setting $s = f(x)$, we have

$$\operatorname{prof}(F_x) \geq \operatorname{prof}(F_{s,x}), \tag{16}$$

as can be seen by lifting arbitrarily a maximal $F_{s,x}$-regular sequence of elements of $\mathfrak{r}(\mathcal{O}_{X_s,x})$, **28** which gives an F_x-regular sequence by SGA 1 IV 5.7. Now if x belongs to a Z as in Lemma XII.2.5, then x is necessarily closed in X_s; in other words, Z is *finite* over S. Indeed, Z (endowed with a structure induced from X) is projective over S as a closed subprescheme of X, which is projective, and Z is affine over S as a closed subprescheme of $X - Y$, which is affine.

Remark XII.2.6. Suppose that for every $s \in S$, the section t_s of $\mathcal{O}_{X_s}(1)$ induced by t is $\mathcal{O}_{X_s}$-regular (which implies by SGA 1 IV 5.7 that t is $\mathcal{O}_X$-regular). Then the hypotheses made are stable under extension of the base $S' \to S$ (S' locally noetherian). Thus the conclusion remains valid after every base change.

XII.3 Lefschetz theory for a projective morphism: existence theorem

Theorem 3.1(*)**.** *Let $f : X \to S$ be a projective morphism, with S noetherian, let $\mathcal{O}_X(1)$ be an invertible Module on X that is ample relative to S, let X_0 be the closed subprescheme*

(*) *Editor's note.* See Corollary I.1.4 of the article by Mme Raynaud (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental", *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52).

(*) *Editor's note.* For a version without the flatness hypothesis, see (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental", *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52, Theorem II.3.3).

of zeros of a section t of $\mathcal{O}_X(1)$, let $\widehat{X}$ be the formal completion of X along X_0, let $\mathfrak{F}$ be a coherent Module on $\widehat{X}$, and let F_0 be the Module that it induces on X_0. Assume moreover:

a) $\mathfrak{F}$ is flat over S.

b) For every $s \in S$, the section t_s induced by t on X_s is $\mathfrak{F}_s$-regular (which implies that F_0 is also flat over S, cf. SGA 1 IV 5.7*).*

c) For every $s \in S$, F_{0s} has depth ≥ 2 at the closed points of X_{0s}.

Assume moreover that S admits an ample invertible sheaf. Under these conditions, there exists a coherent Module F on X, and an isomorphism from its formal completion $\widehat{F}$ to $\mathfrak{F}$.

This statement will follow from the following one:

Corollary XII.3.2. *Under conditions A)), B)), C)) above, the following hold:*

(i) The Module $\widehat{f}_*(\mathfrak{F})$ on S is coherent; hence, for every n, the Module $\widehat{f}_*(\mathfrak{F}(n))$ on S is coherent. **29**

(ii) For n large, the canonical homomorphism $\widehat{f}^*\widehat{f}_*(\mathfrak{F}(n)) \to \mathfrak{F}(n)$ is surjective.

Assume the corollary for the moment, and let us prove XII.3.1. By virtue of the last hypothesis made in XII.3.1, we may reduce to the case $X = \mathbf{P}^r_S$, after replacing $\mathcal{O}_X(1)$ and t by a suitable power. I claim that we may further suppose that $t_s \neq 0$ for every s. Otherwise, indeed, $\mathfrak{F}_s = 0$ by b), or equivalently, by Nakayama, $F_{0s} = 0$, i.e. s does not belong to the image of $\operatorname{Supp} F_0$ under the morphism $f_0 : X_0 \to S$ induced by f. Now this image S' is open by virtue of a), b), since F_0 is flat over S, and it is clear that it suffices to prove the conclusion of XII.3.1 in the situation obtained by restricting over S', since the coherent Module F' on $X|_{S'}$ thereby obtained will be the restriction of a coherent Module F on X, which will have the required property. We may therefore suppose that, in addition to hypotheses a), b), c), the following hypotheses also hold:

a′) $\mathcal{O}_X$ is flat over S.

b′) For every $s \in S$, the section t_s is $\mathcal{O}_{X_s}$-regular.

c′) For every $s \in S$, $\mathcal{O}_{X_{0s}}$ has depth ≥ 2 at the closed points of X_{0s}. (It suffices to choose $X = \mathbf{P}^r_S$ with $r \geq 3$, which is permissible.)

Now XII.3.2 implies that one can find an epimorphism

$$\widehat{L} \to \mathfrak{F} \to 0, \tag{17}$$

where L is a Module on X of the form $f_*(G)(-n)$, with G a coherent locally free Module on S: indeed, for n large, it suffices to express the coherent Module $\widehat{f}_*(\mathfrak{F})$ on S as a quotient of such a G. On the other hand, hypotheses a), b), c) on f, t imply that $\widehat{L}$ satisfies the same conditions a), b), c) as $\mathfrak{F}$. It follows easily that the same is true of the kernel of (17), to which the same argument may therefore be applied, so that $\mathfrak{F}$ is represented as the cokernel of a homomorphism

$$\widehat{L'} \to \widehat{L}, \tag{18}$$

where L, L' are locally free Modules on X. Now, by virtue of a′), the second part of c′), and either XII.2.1 or XII.2.4, the homomorphism (18) comes from a homomorphism $L' \to L$ of Modules on X. It now suffices to take for F the cokernel of $L' \to L$, and we are done.

It remains to prove XII.3.2. This had been done in the seminar by a somewhat laborious expedient, consisting in interpreting everything in terms of cohomology on the punctured

projective cone of X relative to S, in order to reduce to Theorem 2.1. A more direct and more satisfactory method (although substantially identical) now seems to me to be the following. It consists in noting that in IX, no. IX.2 (and with the notation of that exposé), the hypothesis that the morphism $f : \mathcal{X} \to \mathcal{X}'$ is adic is not used anywhere in the proof of 2.1, via EGA 0_{III} 13.7.7; it suffices instead to suppose that $\mathcal{X}$ is also adic, and to choose two ideals of definition $\mathcal{J}$ for $\mathcal{X}'$, $\mathcal{I}$ for $\mathcal{X}$, such that $\mathcal{J}\mathcal{O}_{\mathcal{X}} \subset \mathcal{I}$, and to define $\mathcal{S} = \mathrm{gr}_{\mathcal{J}}(\mathcal{O}_{\mathcal{X}'})$, and consider $\mathrm{gr}_{\mathcal{I}}(\mathfrak{F})$. In any event, 2.1 can be applied directly to the morphism $\widehat{f} : \widehat{X} \to S$ considered in the present subsection, where one simply takes $\mathcal{J} = 0$. Thus, to verify that $\widehat{f}_*(\mathfrak{F})$ is coherent, it suffices, by virtue of *loc. cit.*, to verify that $\mathrm{R}^i f_{0*}(\mathrm{gr}_{\mathcal{I}}(\mathfrak{F}))$ is coherent on S for $i = 0, 1$; for this, note that by virtue of a) and b), the Module in question is none other than $\bigoplus_{m\geq 0} \mathrm{R}^i f_{0*}(F_0(-m))$, which is indeed coherent by virtue of hypothesis c) and XII.1.5.

This proves XII.3.2 (i). For XII.3.2 (ii), we shall need the

Lemma XII.3.3. *Under the conditions of A)), B)), C)) of XII.3.1, set* **30**

$$G_m = \widehat{f}_*(\mathfrak{F}(\cdot)_m) = \bigoplus_n \widehat{f}_*(\mathfrak{F}_m(n))$$

Then the projective system (G_m) satisfies the Mittag-Leffler condition.

We may suppose that S is affine, with ring A. Let $\mathcal{S}$ be a finitely generated graded A-algebra in positive degrees, and let $t' \in \mathcal{S}_1$, such that X embeds in $\mathrm{Proj}(\mathcal{S})$, with $\mathcal{O}_X(1)$ induced by $\mathcal{O}(1)$ and the section t the image of t'. Endow $\mathcal{S}$ with the $\mathcal{J}$-adic filtration, where $\mathcal{J} = t'\mathcal{S}$, and consider the projective system of the $\mathfrak{F}(\cdot)_m$ in the category of abelian sheaves on X_0. We are again under the preliminary conditions of EGA 0_{III} 13.7.7$^{(*)}$, and moreover $\mathrm{H}^i(X_0, \mathrm{gr}(\mathfrak{F}(\cdot))$ is a finite-type Module over $\mathrm{gr}_{\mathcal{J}}(\mathcal{S})$ for $i = 0, 1$. Indeed, since t' is $\mathfrak{F}$-regular, one sees at once that, as a Module over $(\mathcal{S}/t'\mathcal{S})[T]$ (of which $\mathrm{gr}_{\mathcal{J}}(\mathcal{S})$ is a quotient), the Module in question identifies with $\mathrm{H}^i(X_0, F_0(\cdot)) \otimes_{\mathcal{S}/t'\mathcal{S}} (\mathcal{S}/t'\mathcal{S})[T]$; but by XII.1.5, $\mathrm{H}^i(X_0, F_0(\cdot))$ is of finite type over $\mathcal{S}$, and hence over $\mathcal{S}/t'\mathcal{S}$, for $i = 0, 1$, which proves our assertion. Consequently, we are under the conditions for applying 0_{III} 13.7.7 with $n = 1$, which proves XII.3.3.

This point being established (and still supposing S affine, as we may for proving XII.3.2 (ii)), let m_0 be such that $m \geq m_0$ implies $\Im(G_m \to G_0) = \Im(G_{m_0} \to G_0)$, so that both sides are also equal to $\Im(\varprojlim G_m \to G_0) = \Im(\widehat{f}_*(\mathfrak{F}(\cdot)) \to f_*\mathfrak{F}_0(\cdot))$. Now note that for large n, $\mathfrak{F}_{m_0}(n)$ is generated by its sections, and therefore $\mathfrak{F}_0(n)$ is generated by sections that lift to $\mathfrak{F}_{m_0}$, and hence (by the choice of m_0) that lift to $\mathfrak{F}$. Thus the sections of $\mathfrak{F}$ generate $\mathfrak{F}_0$, and hence also $\mathfrak{F}$ by Nakayama. This proves XII.3.2 (ii), and therefore XII.3.1.

Corollary XII.3.4. *Let $f : X \to S$ be a projective flat morphism, with S locally noetherian, let $\mathcal{O}_X(1)$ be an invertible Module on X, relatively ample over S, and let t be a section of this Module such that for every $s \in S$, the induced section t_s on the fiber X_s is $\mathcal{O}_{X_s}$-regular; let X_0* **31**
*be the closed subprescheme of zeros of t, and let $\widehat{X}$ be the formal completion of X along X_0. Suppose that for every $s \in S$, X_{0s} has depth ≥ 2 at its closed points (i.e. X_s has depth ≥ 3 at the closed points of X_{0s}), and X_s has depth ≥ 2 at its closed points. Under these conditions, the pair (X, X_0) satisfies the effective Lefschetz condition (*Leff*) of paragraph 2 of Exposé* X*, i.e. :*

a) For every open neighborhood U of X_0 in X, the functor

$$F \longmapsto \widehat{F}$$

from the category of coherent locally free Modules on U to the category of coherent locally free Modules on $\widehat{X}$ is fully faithful.

$^{(*)}$Corrected as indicated in IX p. 74.

b) For every coherent locally free Module $\mathfrak{F}$ on $\widehat{X}$, there exist an open neighborhood U of X_0 and a coherent locally free Module F on U such that $\mathfrak{F}$ is isomorphic to $\widehat{F}$.

Indeed, a) has already been noted in XII.2.4 under weaker conditions. For b), we apply XII.3.1, which gives the conclusion, at least if S is noetherian and admits an absolutely ample invertible Module, in particular if S is affine. Indeed, if F is a coherent module on X such that $\widehat{F}$ is isomorphic to $\mathfrak{F}$, and hence locally free, it follows that F is locally free on a neighborhood U of X_0, and $F|_U$ will satisfy the required condition. But now note that by XII.2.5, for such an F, its direct image under the immersion $U \to X$ is coherent, and is moreover independent of the chosen solution (U, F) (taking account of the fact that two solutions coincide in a neighborhood of X_0, by a)). More precisely, one can find a coherent Module F on X and an isomorphism $\widehat{F} \xrightarrow{\sim} \mathfrak{F}$, such that F has depth ≥ 2 at all the points of X that are closed in their fiber, and this determines F up to unique isomorphism. By this uniqueness property, the solutions of the problem obtained over the affine open subsets of S glue together, giving a coherent F on all of X and an isomorphism $\widehat{F} \simeq \mathfrak{F}$. Restricting F to the open subset U of points at which it is free gives what was required.

Thanks to XII.2.4 and XII.3.4, in the setting of a projective algebraic scheme and a **32** "hyperplane section"thereof, one can apply the general facts established in Exposés X and XI concerning conditions (Lef) and (Leff). Thus:

Corollary XII.3.5. *Let X be a projective algebraic scheme equipped with an ample invertible module $\mathcal{O}_X(1)$, let t be a section of this module that is $\mathcal{O}_X$-regular, and let X_0 be the closed subscheme of zeros of t. Assume that X has depth ≥ 2 at its closed points (resp. and depth ≥ 3 at the closed points of X_0). Then $\pi_0(X_0) \to \pi_0(X)$ is bijective; in particular, X is connected if and only if X_0 is connected, and, after choosing a geometric base point in X_0, $\pi_1(X_0) \to \pi_1(X)$ is surjective. More generally, for every open $U \supset X_0$, the homomorphism $\pi_1(X_0) \to \pi_1(U)$ is surjective (resp. the homomorphism $\pi_1(X_0) \to \varprojlim_U \pi_1(U)$ is bijective). In the parenthesized case, if one assumes in addition that the local ring at every closed point of X not in X_0 is pure (3.2) (for example, regular, or merely a complete intersection), then $\pi_1(X_0) \to \pi_1(X)$ is an isomorphism.*

Apply 2.4 and XII.3.4. Note that, in the parenthesized case, the hypothesis that X have depth ≥ 3 at the closed points of X_0 implies that every irreducible component of X having dimension $\neq 0$ has dimension ≥ 3 (as one sees by noting that such a component necessarily meets X_0, and by considering a closed point of the intersection).

Remark. When X is normal and has dimension ≥ 2 at all its points, it has depth ≥ 2 at its closed points, and the non-parenthesized hypotheses of XII.3.5 hold. In this case there is a more elementary proof of the surjectivity of $\pi_1(X_0) \to \pi_1(X)$ using Bertini's theorem (cf. SGA 1 X.2.10). If one assumes in addition that X_0 is normal and that X has dimension ≥ 3 at all its points, then the parenthesized hypotheses of XII.3.5 hold. In this case, XII.3.5 was established by Grauert (indeed, thanks to the normality hypothesis, one can then dispense with the existence theorem XII.3.1 by various devices) . This proof of Grauert's was the **33** starting point of the "Lefschetz theory"that is the subject of the present seminar.

Corollary XII.3.6. *Let X, $\mathcal{O}_X(1)$, t, and X_0 be as in XII.3.5. Assume that X has depth ≥ 2 at its closed points, and that*

$$\mathrm{H}^i(X_0, \mathcal{O}_{X_0}(-n)) = 0$$

for $n > 0$ and for $i = 1$ (resp. for $i = 1$ and $i = 2$), which, by XII.1.4, implies that X_0 has depth ≥ 2 (resp. ≥ 3) at its closed points, i.e. that X has depth ≥ 3 (resp. ≥ 4) at the closed points of X_0. Under these conditions, for every open neighborhood U of X_0, $\mathrm{Pic}(U) \to \mathrm{Pic}(X_0)$ is injective; in particular, $\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$ is injective (resp. $\varprojlim_U \mathrm{Pic}(U) \to \mathrm{Pic}(X_0)$ is

bijective). In the parenthesized case, if one assumes in addition that the local ring of X at every closed point not in X_0 is parafactorial (XI.3.1) (for example, regular, or more generally a complete intersection), then $\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$ *is bijective.*

Apply XI XI.3.13 and XI.3.14, noting that the parenthesized hypothesis implies that the irreducible components of X having dimension $\neq 0$ have dimension ≥ 4. In particular, applying this when X is a global complete intersection of dimension ≥ 4 in projective space, one obtains:

Corollary XII.3.7. *Let X be an algebraic scheme of dimension ≥ 3 that is a complete intersection in the scheme $\mathbf{P}_k^r$. Then* $\mathrm{Pic}(X)$ *is the free group generated by the class of the sheaf $\mathcal{O}_X(1)$.*

Proceed by induction on the number of hypersurfaces whose intersection is X, applying XII.3.6 and noting that, for a complete intersection X of dimension ≥ 3, one has $\mathrm{H}^i(X, \mathcal{O}_X(n)) = 0$ for $i = 1, 2$ and every n.

Remark XII.3.8. In the case where X is a nonsingular hypersurface, XII.3.7 is due to **34**
Andreotti. The result XII.3.7 can also be expressed (when X is nonsingular) by saying that the homogeneous coordinate ring of X is factorial, and in this form it is contained in XI XI.3.14 (ii). We also note that Serre had given a proof of XII.3.7 in the nonsingular case by transcendental methods, using a specialization argument to reduce to characteristic 0, where the Lefschetz theorem is available in its classical form. Of course, the fact that the purely algebraic proof given here makes it possible to dispense with nonsingularity hypotheses in the statement of the Lefschetz theorem invites one to reconsider it in the classical case as well. Cf. the following exposé, which proposes conjectures in this direction.

In Corollaries XII.3.5 and XII.3.7, we have worked over a base field, whereas the key theorems XII.2.4 and XII.3.4 are valid over an arbitrary base. To generalize Corollaries XII.3.5 and XII.3.6 to a general S, we must give workable criteria for a point of X (flat over S) to have a "pure"resp. parafactorial local ring. This will be the subject of the following n°.

XII.4 Formal completion and normal flatness

Theorem XII.4.1. *Let X be a locally Noetherian prescheme, locally embeddable in a regular scheme, Y a closed subset of X, $U = X - Y$, X_0 a closed subprescheme of X defined by an ideal $\mathcal{J}$, $\widehat{X}$ the formal completion of X along X_0, U_0 the trace of X_0 on U, $\widehat{U}$ the formal completion of U along U_0, $i : U \to X$ and $\widehat{i} : \widehat{U} \to \widehat{X}$ the canonical immersions, and n an integer. Assume that*

a) X is normally flat along X_0 at the points of $Y \cap X_0$, i.e. at these points the Modules $\mathcal{J}^n/\mathcal{J}^{n+1}$ over X_0 are flat, i.e. locally free.

b) For every $x \in Y \cap X_0$, one has $\operatorname{prof} \mathcal{O}_{X_0,x} \geq n + 2$.

Under these conditions, the following assertions hold: **35**

1. Let F be a coherent Module on U, and suppose that:

c) For every $x \in Y - Y \cap X_0$, one has $\operatorname{prof} \mathcal{O}_{X,x} \geq n + 2$.

d) F is free at the points of U_0, and has depth $\geq n + 1$ at all the points of U where it is not free.

Then the graded Module

$$\bigoplus_{m \geq 0} \mathrm{R}^p i_*(\mathcal{J}^m F)$$

over $\bigoplus_{m \geq 0} \mathcal{J}^m$ is of finite type for $p \leq n$.

2. Let $\mathfrak{F}$ be a coherent Module on $\widehat{U}$; then the graded Module

$$\bigoplus_{m\geq 0} \mathrm{R}^p i_*(\mathcal{J}^m\mathfrak{F}/\mathcal{J}^{m+1}\mathfrak{F})$$

over $\bigoplus_{m\geq 0}\mathcal{J}^m/\mathcal{J}^{m+1} = \mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X)$ is of finite type for $p \leq n$.

Proof. 1) Let $X' = \mathrm{Spec}(\bigoplus_{m\geq 0}\mathcal{J}^m)$; the base change $f : X' \to X$ then defines $U' = X' - Y'$, X'_0, $U'_0 = X'_0U'$, and immersions $i' : U' \to X'$, $i'_0 : U'_0 \to X'_0$. Thus we have a Cartesian square

$$\begin{array}{ccc} X' & \xleftarrow{i'} & U' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xleftarrow{i} & U \end{array}$$

and we have

$$\bigoplus_{m\geq 0} \mathrm{R}^p i_*(\mathcal{J}^m F) = \mathrm{R}^p i_*\Big(\bigoplus_{m\geq 0}\mathcal{J}^m F\Big) = \mathrm{R}^p i_*(g_*(F')), \tag{19}$$

where $F' = g^*F$, so that there is indeed a canonical isomorphism **36**

$$g_*(F') \xrightarrow{\sim} \bigoplus_{m\geq 0}\mathcal{J}^m F,$$

since this is true at the points of U_0, because F is free there by d), and also at the points of U_0, because there one has $\mathcal{J}^m = \mathcal{O}_U$, (so that in both cases, $\mathcal{J}^m \otimes_{\mathcal{O}_U} F \to \mathcal{J}^m F$ is an isomorphism).

On the other hand, since f, and hence g, are affine, one has

$$\mathrm{R}^p i_*(g_*(F')) = \mathrm{R}^p(ig)_*(F') = \mathrm{R}^p(fi')_*(F') = f_*(R^p i'_*(F')) \tag{20}$$

and therefore, comparing (19) and (20), we see that assertion 1 is equivalent to the following: $R^p i'_*(F')$ is a Module of finite type, i.e. coherent, on X', for every $p \leq n$. Now, since X is locally embeddable in a regular scheme, the same is true of X', which is of finite type over X, and we may apply the coherence criterion VIII VIII.2.3 to a coherent extension F'' of F': we wish to express that $\mathcal{H}^p_Y(F'')$ is coherent for $p \leq n+1$, and this is also equivalent to saying that, for every $x' \in U'$ such that

$$\mathrm{codim}(\overline{x'} \cap Y', \overline{x'}) = 1, \tag{20 bis}$$

one has

$$\mathrm{prof}\, F'_{x'} \geq n+1. \tag{21}$$

Now this condition is satisfied at the points x' where F' is not free, because for such an x' one has $x' \notin U'_0$ by d), and therefore g is an isomorphism there; again by d), F has depth $\geq n+1$ at $g(x')$, and hence F' has depth $\geq n+1$ at x'. It is therefore enough to verify condition (21) at those $x' \in U'$ satisfying (21) at which F' is free. For this, it is enough to prove that one has **37**

$$\mathrm{prof}\, \mathcal{O}_{X',x'} \geq n+1 \tag{21 bis}$$

at these points; *a fortiori* it is enough to establish this relation at all the points x of U' satisfying (20 bis). Now, again by criterion VIII.2.3 in Exposé VIII, this is equivalent to the assertion that the Modules

$$\mathcal{H}^p_{Y'}(\mathcal{O}_{X'}) \quad \text{for } p \leq n+1$$

are coherent. In fact, we shall prove that they are even zero, or, equivalently by Exposé III, that one has

$$\mathrm{prof}\, \mathcal{O}_{X',x'} \geq n+2 \quad \text{for every } x' \in Y'. \tag{22}$$

For this, we distinguish two cases. If $x' \notin X'_0$, then f is an immersion at x', and we must verify that F has depth $\geq n+2$ at the image $x = f(x')$, which is precisely condition c). If, on the other hand, $x' \in X'_0$, i.e. $x = f(x') \in X_0$ and hence $x \in Y \cap X_0$, we apply conditions a) and b) by means of the

Lemma XII.4.2. *Let X be a locally Noetherian prescheme, X_0 a closed subprescheme of X defined by an ideal $\mathcal{J}$, $X' = \mathrm{Spec}(\bigoplus_{m\geq 0}\mathcal{J}^m)$, $X'_0 = \mathrm{Spec}(\bigoplus_{m\geq 0}\mathcal{J}^m/\mathcal{J}^{m+1}) = X' \times_X X_0$, and x a point of X_0 at which X is normally flat along X_0, i.e. such that $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X) = \bigoplus_{m\geq 0}\mathcal{J}^m/\mathcal{J}^{m+1}$ is flat there as a Module over X_0. Then, for every sequence of elements f_i $(1 \leq i \leq m)$ of $\mathcal{O}_{X,x}$ whose images in $\mathcal{O}_{X_0,x}$ form an $\mathcal{O}_{X_0,x}$-regular sequence, and every $x' \in X'$ above x, the images of the f_i in $\mathcal{O}_{X',x'}$ (respectively in $\mathcal{O}_{X'_0,x'}$) likewise form an $\mathcal{O}_{X',x'}$-regular sequence (respectively an $\mathcal{O}_{X'_0,x'}$-regular sequence); in particular, one has*

$$\mathrm{prof}\,\mathcal{O}_{X',x'} \geq \mathrm{prof}\,\mathcal{O}_{X_0,x}, \quad \mathrm{prof}\,\mathcal{O}_{X'_0,x'} \geq \mathrm{prof}\,\mathcal{O}_{X_0,x}. \tag{23}$$

To prove it, we may suppose that X is local with closed point x, and hence affine with **38**
ring $A = \mathcal{O}_{X,x}$, with $\mathcal{J}$ defined by the Ideal J; it is enough to prove that, for every sequence f_i $(1 \leq i \leq m)$ of elements of A whose images in A/J form an A/J-regular sequence, the f_i likewise form a $(\bigoplus_{m\geq 0} J^m)$-regular sequence and a $(\bigoplus_{m\geq 0} J^m/J^{m+1})$-regular sequence, i.e. for every m, it forms a J^m-regular sequence and a J^m/J^{m+1}-regular sequence. The second assertion is trivial, since J^m/J^{m+1} is a free module over A/J. The first follows by considering the J-adic filtration of J^m and noting that the sequence of the f_i is regular on the graded module associated with J^m for that filtration.

This proves XII.4.2, and consequently XII.4.1 1).

Let us prove XII.4.1 2). For this, use the Cartesian square

$$\begin{array}{ccc} X'_0 & \xleftarrow{i'_0} & U'_0 \\ {\scriptstyle f_0}\downarrow & & \downarrow{\scriptstyle g_0} \\ X_0 & \xleftarrow{i_0} & U_0 \end{array}$$

and, proceeding as at the beginning of the proof of 1), we find that

$$\bigoplus_{m\geq 0} \mathrm{R}^p i_*(\mathcal{J}^m\mathfrak{F}/\mathcal{J}^{m+1}\mathfrak{F}) \simeq f_{0*}(\mathrm{R}^p i'_{0*}(\mathfrak{F}'_0)), \tag{24}$$

where $\mathfrak{F}_0 = \mathfrak{F}/\mathcal{J}\mathfrak{F}$ and $\mathfrak{F}'_0 = g_0^*(\mathfrak{F}_0)$, (using the fact that $\mathfrak{F}$ is locally free). Thus the conclusion of 2) is equivalent to saying that, for $p \leq n$, $\mathrm{R}^p i'_{0*}(\mathfrak{F}'_0)$ is a coherent Module.

Here again, taking into account that $\mathfrak{F}'_0$ is locally free, criterion VIII VIII.2.3 allows us to reduce to proving the same assertion after replacing $\mathfrak{F}'_0$ by $\mathcal{O}_{U'_0}$, i.e. to proving that the Modules **39**

$$\mathcal{H}^p_{Y'_0}(\mathcal{O}_{X'_0}) \quad \text{for } p \leq n+1 \quad (\text{where } Y'_0 = Y' \cap X'_0 = X'_0 - U'_0)$$

are coherent. Once again, we prove that they are in fact zero, i.e., that one has

$$\mathrm{prof}\,\mathcal{O}_{X'_0,x'} \geq n+2 \quad \text{for every } x' \in Y'_0. \tag{25}$$

This does indeed follow from conditions a) and b), in view of XII.4.2. This completes the proof of XII.4.1.

Remark XII.4.3. It follows immediately, by descent, that the hypothesis that X be locally embeddable in a regular scheme can be replaced by the following weaker hypothesis: there exists a faithfully flat and quasi-compact morphism $\overline{X} \to X$ such that $\overline{X}$ is locally embeddable in a regular scheme.

Theorem XII.4.1 enables us to apply the results of Exposé IX (the comparison and existence theorems). We shall be interested in particular in the

Corollary XII.4.4. *Assume that conditions a), b), and c) of Theorem XII.4.1 hold, with $n = 1$ and $X = \mathrm{Spec}(A)$, where A is separated and complete for the J-adic topology. Then*

1. *The functor $F \longmapsto \widehat{F}$ from the category of locally free coherent modules on U to the category of locally free coherent modules on $\widehat{U}$ is fully faithful.*

2. *For every locally free coherent module $\mathfrak{F}$ on $\widehat{U}$, there exists a coherent module F on U and an isomorphism $\widehat{F} \xrightarrow{\sim} \mathfrak{F}$.*

In particular, if, for every $x \in U$ whose closure in U does not meet U_0, i.e. such that $\overline{x} \cap X_0 \subset Y$, one has $\operatorname{prof} \mathcal{O}_{U,x} \geq 2$*, then the pair (U, U_0) satisfies the effective Lefschetz condition (*Leff*) of Exposé X.*

(For the last assertion, one proceeds as in X 2.1.)

Special case of XII.4.4:

Corollary XII.4.5. *Let A be a Noetherian ring, let J be an ideal of A contained in the* **40** *radical, and let $A_0 = A/J$. Assume*

(i) $\operatorname{prof} A_0 \geq 3$.

(ii) $\operatorname{gr}_J(A)$ *is a free A_0-module.*

(iii) A is complete for the J-adic topology.

Let $X = \operatorname{Spec}(A)$, $X_0 = \operatorname{Spec}(A_0) = V(J)$, let a be the closed point of X, let $U = X - \{a\}$, $U_0 = X_0 - \{a\}$, and let $\widehat{U}$ be the formal completion of U along U_0. Then the functor $F \longmapsto \widehat{F}$ from the category of locally free coherent modules on U to the category of locally free coherent modules on $\widehat{U}$ is fully faithful. Moreover, for every locally free coherent module $\mathfrak{F}$ on $\widehat{U}$, there exists a coherent module (not necessarily locally free!) F on U, and an isomorphism $\widehat{F} \simeq \mathfrak{F}$.

Note that, thanks to XII.4.3, we did not have to assume that A is a quotient of a regular ring, because the completion of A for the $\mathfrak{r}(A)$-adic topology satisfies this condition in any case.

Proceeding as in Exposés X and XI, we deduce from XII.4.5

Corollary XII.4.6. *Under the conditions of XII.4.5, one has the following:*

a) U and U_0 are connected (III 3.1).

Choosing a geometric base point in U_0, the homomorphism

$$\pi_1(U_0) \to \pi_1(U)$$

is surjective.

b) The homomorphism

$$\operatorname{Pic}(U) \to \operatorname{Pic}(U_0)$$

is injective.

To prove b), in view of XII.4.5, this amounts to checking that every isomorphism $L'_0 \xrightarrow{\sim} L_0$ **41** lifts to an isomorphism $\widehat{L'} \xrightarrow{\sim} \widehat{L}$. For this, one lifts step by step to isomorphisms $L'_n \xrightarrow{\sim} L_n$; the obstructions lie in $\mathrm{H}^1(U_0, \mathcal{J}^n/\mathcal{J}^{n+1})$, and these modules are zero because J^n/J^{n+1} is free and $\operatorname{prof} A_0 \geq 3$.

We are now in a position to prove the

Theorem XII.4.7. *Let A be a noetherian local ring, let J be an ideal of A contained in its radical, and let $A_0 = A/J$. Suppose that*

(i) $\operatorname{prof} A_0 \geq 3$.

(ii) $\operatorname{gr}_J(A)$ *is a free module over* A_0.

Then, if A_0 *is "pure"(X 3.1) (resp. parafactorial (XI XI.3.1)), the same holds for* A.

Proof. By descent, we may suppose that we also have

(iii) A is complete for the J-adic topology.

Indeed, by (i) and (ii), we have $\operatorname{prof}(A) \geq 3$, and hence $\operatorname{prof}(\widehat{A}) \geq 3$, where $\widehat{A}$ is the completion of A for the J-adic topology, and we apply X 3.6 and XI XI.3.6. We are therefore under the conditions of XII.4.5. Since $\operatorname{prof}(A) \geq 3 \geq 2$, to say that A is parafactorial means simply that $\operatorname{Pic}(U) = 0$, and by XII.4.6 b), it is enough for this that $\operatorname{Pic}(U_0) = 0$, i.e. that A_0 be parafactorial. To prove that A is "pure"if A_0 is, we must prove that if V is an étale covering of U, defined by an algebra $\mathcal{B}$ on U, then $\mathrm{H}^0(U, \mathcal{B})$ is a finite étale algebra over A. Now, since A_0 is pure, the same holds for the A_n (which differ from it only by nilpotent elements), and therefore, for every n, $B_n = \mathrm{H}^0(U, \mathcal{B}_n)$ is an étale algebra over A/J^{n+1}; these algebras of course fit together, so that $\varprojlim B_n$ is an étale algebra over A. By XII.4.5, this algebra is none other than $\mathrm{H}^0(U, \mathcal{B})$, which proves our assertion.

Corollary XII.4.8. *Let* $f : X \to Y$ *be a flat morphism of locally noetherian preschemes,* **42**
let $x \in X$*, and let* $y = f(x)$*. Suppose that* $\mathcal{O}_{X_y,x}$ *is a "pure"(resp. parafactorial) local ring of depth* ≥ 3*. Then the same holds for* $\mathcal{O}_{X,x}$.

This is the result of the kind promised at the end of the preceding no. , for generalizing Corollaries XII.3.5 and following. Thus, using XII.3.4, we obtain the

Corollary XII.4.9. *Let* $f : X \to S$ *be a projective flat morphism, with* S *locally Noetherian, let* $\mathcal{O}_X(1)$ *be an invertible module on* X*, ample relative to* S*, let* t *be a section of* $\mathcal{O}_X(1)$ *such that, for every* $s \in S$*, the section* t_s *induced on* X_s *is* $\mathcal{O}_{X_s}$*-regular, let* X_0 *be the closed subscheme of zeros of* t*, and let* X_m *be the closed sub-scheme of zeros of* t^{m+1}*. Assume that, for every* $s \in S$*,* X_s *has depth* ≥ 3 *at all its closed points. Then:*

a) If the local rings at the closed points of $X_s - X_{0,s}$ *(*$s \in S$*) are "pure"—for example, if they are complete intersections—then the functor* $X' \longmapsto X'_0 = X' \times_X X_0$ *from the category of finite ètale coverings of* X *to the category of finite ètale coverings of* X_0 *is an equivalence of categories; in particular, after choosing a geometric base point in* X_0*, the homomorphism*

$$\pi_1(X_0) \to \pi_1(X)$$

is an isomorphism (*)

(*)*Editor's note.* Let us point out the striking connectedness result obtained subsequently by Fulton and Hansen, in the case where $S = \operatorname{Spec}(k)$ (k an algebraically closed field). Let $g : X \to \mathbf{P}^m_k \times \mathbf{P}^m_k$ be such that $\dim g(X) > m$; then the inverse image of the diagonal is connected. Among other things, this makes it possible to generalize Corollary XII.4.9 when f is the structural morphism of projective space $\mathbf{P}^m_k$ over $S = \operatorname{Spec}(k)$: more precisely, an irreducible subvariety X of $\mathbf{P}^m_k$ of dimension $> m/2$ has trivial fundamental group! (cf. Fulton W. & Hansen J., "A connectedness theorem for projective varieties, with applications to intersections and singularities of mappings", *Ann. of Math. (2)* **110** (1979), no. 1, p. 159-166). For generalizations to Grassmannians or abelian varieties, see Debarre O., "Théorèmes de connexité pour les produits d'espaces projectifs et les grassmanniennes", *Amer. J. Math.* **118** (1996), no. 6, p. 1347-1367 and "Théorèmes de connexité et variétés abéliennes", *Amer. J. Math.* **117** (1995), no. 3, p. 787–805. The triviality result for the fundamental group of X above was obtained independently by Faltings, who moreover proves that $\operatorname{Pic}(X)$ has no torsion prime to the characteristic of k, using methods of algebraization of formal vector bundles, more in the spirit of Grothendieck's techniques, cf. (Faltings G., "Algebraization of some formal vector bundles" *Ann. of Math. (2)* **110** (1979), no. 3, p. 501–514).

b) If the local rings at the closed points of $X_s - X_{0,s}$ ($s \in S$) are "parafactorial"—for example, regular, or complete intersections of dimension ≥ 4—then, for every integer m such that $\mathrm{R}^i f_{0*}(\mathcal{O}_{X_0}(-n)) = 0$ for $n > m$ and $i = 1, 2$, the map $\mathrm{Pic}(X) \to \mathrm{Pic}(X_m)$ is bijective.

Moreover, if S is Noetherian and the $X_{0,s}$ have depth ≥ 3 at their closed points, such m exist (cf. XII.1.5).

Remark XII.4.10. Under the conditions of the last assertion of XII.4.9 b), we saw in XII.1.5 **43** that there exists an m such that $n > m$ even implies $\mathrm{H}^i(X_0, \mathcal{O}_{X_0}(-n)) = 0$ for $i = 1, 2$, and indeed for $i \leq 2$). This condition is stronger than $\mathrm{R}^i f_*(\mathcal{O}_{X_0}(-n)) = 0$ for $i = 1, 2$, and it has the further advantage of being stable under base change. The same is true of the depth hypotheses made in XII.4.9, and also of a hypothesis of the type "the X_s are locally complete intersections". It follows under these conditions that XII.4.9 b) also implies that the morphism of functors

$$\mathbf{Pic}_{X/S} \to \mathbf{Pic}_{X_n/S}$$

in $\mathbf{Sch}_{/S}$ is an isomorphism, and hence so is the morphism of relative Picard schemes, when these exist:

$$\mathrm{Pic}_{X/S} \to \mathrm{Pic}_{X_n/S}\,.$$

Even when S is the spectrum of an algebraically closed field, this statement is considerably more precise than the statement that merely says that $\mathrm{Pic}(X) \to \mathrm{Pic}(X_n)$ is bijective.

One may ask whether one can always take $n = 0$ in the preceding conclusions (thus assuming that the $X_{0,s}$ have depth ≥ 3 at their closed points). When X_0 is smooth over S and the residual characteristics of S are zero, this is indeed so, by the Kodaira "vanishing theorem"(proved by transcendental methods, using a Kähler metric), which implies that, for every connected smooth projective scheme of dimension n over a field k of characteristic zero and every ample invertible module L on X, one has $\mathrm{H}^i(X, L^{-1}) = 0$ for $i \neq n$. It is not known(*) at present whether this theorem can be replaced by a generalization in characteristic $p > 0$, and whether the smoothness hypothesis can be replaced by a hypothesis of a more general nature (involving depth, or of "complete intersection"type, and so on).

XII.5 Universal finiteness conditions for a nonproper morphism

Recall for reference the **44**

Proposition XII.5.1. *Let $f : X \to S$ be a proper morphism of preschemes, with S locally noetherian, let U be an open subset of X, let $g : U \to X$ be the canonical immersion, let $h = fg : U \to S$, and let F be a Module on U. Suppose that the Modules $\mathrm{R}^i g_*(F)$ are coherent for $i \leq n$ (a hypothesis local in nature on X, which in practice is checked using the criterion VIII 2.3). Then $\mathrm{R}^i h_*(F)$ is coherent for $i \leq n$.*

(*) *Editor's note.* As Raynaud observed, the decomposition result for the de Rham complex of Deligne and Illusie readily implies the vanishing of the group $H^i(X, L^{-1})$ (with L ample on X, projective and smooth over k of characteristic $p > 0$) for $i < \inf(p, \dim(X))$, provided that X can be lifted to a flat scheme over $W_2(k)$ (cf. Deligne P. & Illusie L., "Relèvements modulo p^2 et décomposition du complexe de de Rham", *Invent. Math.* **89** (1987), no. 2, p. 247–270); this gives a purely algebraic proof of Kodaira's result for projective varieties in characteristic zero. If X cannot be lifted, it is well known that the "vanishing theorem"is false; cf. the example in (Raynaud M., "Contre-exemple au "vanishing theorem" en caractéristique $p > 0$", in *C.P. Ramanujam—a tribute*, Tata Inst. Fund. Res. Studies in Math., vol. 8, Springer, Berlin-New York, 1978, p. 273–278); see also the very elegant examples in (Haboush W. & Lauritzen N., "Varieties of unseparated flags", in *Linear algebraic groups and their representations (Los Angeles, CA, 1992)*, Contemp. Math., vol. 153, American Mathematical Society, Providence, RI, 1993, p. 35–57), simplified in (Lauritzen N. & Rao A.P., "Elementary counterexamples to Kodaira vanishing in prime characteristic", *Proc. Indian Acad. Sci. Math. Sci.* **107** (1997), no. 1, p. 21–25). On the other hand, I know of no example in which the map $\mathrm{Pic}(X_{n+1}) \to \mathrm{Pic}(X_n)$ is not surjective for $n > 1$ in positive characteristic, where X_n denotes a thickened hyperplane section of a smooth projective X as above.

This follows at once from the Leray spectral sequence

$$\mathrm{E}_2^{p,q} = \mathrm{R}^p f_*(\mathrm{R}^q g_*(F)) \Longrightarrow \mathrm{R}^* h_*(F), \tag{26}$$

and from the fact that the higher direct images under f of a coherent Module on X are coherent (EGA III 3.2.1).

Proposition XII.5.2. *Let S be a locally noetherian prescheme, let $\mathcal{S}$ be a quasi-coherent graded Algebra of finite type over S, generated by $\mathcal{S}_1$, let X be a subprescheme of* $\mathrm{Proj}(\mathcal{S})$*, let $\mathcal{O}_X(1)$ be the invertible Module on X induced by* $\mathrm{Proj}(\mathcal{S}(1))$ *and very ample relative to S, let U be an open subset of X, let $g : U \to X$ be the canonical immersion, let $h = fg : U \to S$, and let F be a quasi-coherent Module on U, giving twisted Modules $F(m) = F \otimes \mathcal{O}_X(m)$) ($m \in \mathbf{Z}$); let n be an integer and m_0 an integer. The following conditions are equivalent:*

(i) $\mathrm{R}^i g_*(F)$ *is coherent for $i \leq n$.*

(ii) $\bigoplus_{m>m_0} \mathrm{R}^i h_*(F(m))$ *is a finite-type $\mathcal{S}$-Module for $i \leq n$.*

Proof. Replacing F by $F(m)$ in the spectral sequence above, we obtain a spectral sequence of graded $\mathcal{S}$-Modules **45**

$$\mathrm{E}_2^{p,q} = \bigoplus_{m\geq m_0} \mathrm{R}^p f_*(\mathrm{R}^q g_*(F(m)) \Longrightarrow \bigoplus_{m\geq m_0} \mathrm{R}^* h_*(F(m)).$$

Since we have

$$\mathrm{R}^q g_*(F(m)) \simeq \mathrm{R}^q g_*(F)(m),$$

we see that if the $\mathrm{R}^i g_*(F)$ are coherent, then $\mathrm{E}_2^{p,q}$ is of finite type over $\mathcal{S}$ for $q \leq n$, by part a) of Lemma XII.5.3 below, which implies that the abutment is of finite type over $\mathcal{S}$ in degree $i \leq n$. This proves (i) $\Longrightarrow$ (ii). Moreover, reasoning in the abelian category of graded $\mathcal{S}$-Modules modulo the thick subcategory $\mathcal{C}$ of those that are quasi-coherent of finite type, we obtain from the preceding spectral sequence

$$\bigoplus_{m\geq m_0} \mathrm{R}^{n+1} h_*(F(m)) \simeq \bigoplus_{m\geq m_0} f_*(\mathrm{R}^{n+1} g_*(F)(m)) \mod \mathcal{C},$$

which proves that if the left-hand side is a finite-type $\mathcal{S}$-Module, then $\mathrm{R}^{n+1} g_*(F)$ is coherent, by part b) of Lemma XII.5.3. This proves the implication (ii) $\Longrightarrow$ (i) by induction on n. It remains to prove:

Lemma XII.5.3. *Let S, $\mathcal{S}$, X, f be as in XII.5.2, let G be a quasi-coherent Module on X, and let m_0 be an integer. Then*

a) If G is coherent, then for every integer i, the graded Module

$$\bigoplus_{m\geq m_0} \mathrm{R}^i f_*(G(m))$$

over $\mathcal{S}$ is of finite type.

b) Conversely, suppose that the Module $\bigoplus_{m\geq m_0} \mathrm{R}^i f_*(G(m))$ over $\mathcal{S}$ is of finite type; then G is coherent.

Proof of XII.5.3. For a), the case $i = 0$ is given in EGA III 2.3.2, and the case $i > 0$ in EGA III 2.2.1 (i)(ii), which says that the $\mathrm{R}^i f_* G(m)$ are coherent and zero for large m (if we suppose S **46** noetherian, as we may). For b), note that G is isomorphic to $\mathbf{Proj}(\bigoplus_{m\geq m_0} f_*(G(m))$ (EGA II 3.4.4 and 3.4.2), which proves that G is coherent if $\bigoplus_{m\geq m_0} f_*(G(m))$ is of finite type over $\mathcal{S}$, by *loc. cit.* 3.4.4.

Corollary XII.5.4 (S noetherian)**.** *Suppose that* $\mathrm{R}^i g_*(F)$ *is coherent for* $i \leq n$*. Then, for* $i \leq n+1$ *and large* m*, there is a canonical isomorphism:*

$$\mathrm{R}^i h_*(F(m)) \simeq f_*(\mathrm{R}^i g_*(G)(m)).$$

Indeed, the spectral sequence (26) for $F(m)$ then degenerates in degree $\leq n$, by EGA III 2.2.1 (ii), which gives the result at once (and in fact recovers the implication (ii) $\Longrightarrow$ (i) of 5.2).

Corollary XII.5.5. *Under the preliminary conditions of XII.5.2, with* S *noetherian, the following conditions are equivalent:*

(i) $\bigoplus_{m \geq m_0} h_*(F(m))$ *is of finite type over* S*, and* $\mathrm{R}^i h_*(F(m)) = 0$ *for* $0 < i \leq n$ *and large* m*.*

(ii) $g_*(F)$ *is coherent, and* $\mathrm{R}^i g_*(F) = 0$ *for* $0 < i \leq n$*.*

(ii bis) $g_*(F)$ *is coherent, and* $\operatorname{prof}_Y g_*(F) > n+1$*.*

The equivalence of (ii) and (ii bis) is contained in III 3.3. Moreover, by XII.5.2, conditions (i) and (ii) both imply that the $\mathrm{R}^i g_*(F)$ ($i \leq n$) are coherent. The equivalence of (i) and (ii) then follows from XII.5.4, taking into account the fact that for every coherent Module G on X, one has $G = 0$ if and only if $f_*(G(m)) = 0$ for large m, for example by EGA III 2.2.1 (iii).

Remark XII.5.6. The criteria XII.5.2 and XII.5.5 can be interpreted by saying that the **47**
"simultaneous finiteness condition"XII.5.2 (ii) is expressed by local regularity properties (in terms of depth by VIII VIII.2.1) of F at the points of U near $Y = X - U$, whereas the "asymptotic vanishing condition"XII.5.5 (i) is distinctly stronger in nature and is expressed by local regularity conditions on $g_*(F)$ at the points of Y itself. It would be interesting, in order to generalize Lefschetz-type theorems for projective morphisms to quasi-projective morphisms, to find local criteria on X that are necessary and sufficient for the $\mathcal{S}$-Modules $\bigoplus_{m \geq 0} \mathrm{R}^i h_*(F(m))$ for $i \leq n$ to be of finite type. When S is the spectrum of a field (and doubtless more generally if it is the spectrum of an artinian ring) and $Y = X - U$ is finite, one can show that it is necessary and sufficient that the following conditions hold:

1°) $\operatorname{prof} F_x > n$ for every closed point x of U (compare XII.1.4).

2°) $\mathrm{R}^i g_*(F)$ is coherent for $i \leq n$, or equivalently, there exists an open neighborhood V of Y such that for every closed point x of $U \cap V$, one has $\operatorname{prof} F_x > n+1$.

Proposition XII.5.7. *Let* S *be a locally Noetherian prescheme, let* $g : U \to X$ *be a morphism of preschemes of finite type over* $S^{(*)}$*, with structural morphisms* h *and* f*, let* F *be a quasi-coherent module on* U*, and let* n *be an integer. The following conditions are equivalent:*

(i) For every base change $S' \to S$*, with* S' *Noetherian, the module* $\mathrm{R}^n g'_*(F')$ *on* X' *is coherent.*

(ii) For every base change as above and every coherent ideal $\mathcal{J}$ *on* S'*, denoting by* $\mathcal{I}$ *the ideal* $\mathcal{J}\mathcal{O}_{X'}$ *on* X'*, the graded module*

$$\bigoplus_{m \geq 0} \mathrm{R}^n g'_*(\mathcal{I}^m F')$$

over $\bigoplus_{m \geq 0} \mathcal{I}^m$ *is of finite type.*

(iii) For every base change $S' \to S$, and $\mathcal{J}$ as above, the graded module **48**

$$\bigoplus_{m\geq 0} \mathrm{R}^n g'_*(\mathcal{I}^m F'/\mathcal{I}^{m+1}F')$$

over $\mathrm{gr}_I(\mathcal{O}_{X'}) = \bigoplus_{m\geq 0} \mathcal{I}^m/\mathcal{I}^{m+1}$ *is of finite type.*

Obviously, (ii) $\Longrightarrow$ (i) and (iii) $\Longrightarrow$ (i), as one sees by taking $\mathcal{I} = 0$ in conditions (ii) and (iii). The reverse implications are obtained by applying (i) to the composite base change $S'' \to S' \to S$, where S'' is equal to $\mathrm{Spec} \bigoplus_{m\geq 0} \mathcal{J}^m$ resp. $\mathrm{Spec} \bigoplus_{m\geq 0} \mathcal{J}^m/\mathcal{J}^{m+1}$.

The point of this proposition is that conditions of the form (ii) are those that occur in the "algebraic-formal comparison theorems", whereas conditions of the form (iii) occur in the "existence theorems"that complement them; cf. Exposé IX. A first interesting case is that in which $f : X \to S$ is the identity, so that one is dealing with conditions on a morphism $h : U \to S$ locally of finite type and a quasi-coherent module F on U, flat over S. To obtain sufficient conditions, we shall assume that U embeds by $g : U \to X$ as an open subprescheme of an X that is *proper* over S. Applying XII.5.1, we thus see:

Corollary XII.5.8. *Let $f : X \to S$ be a proper morphism, with S locally Noetherian, let U be an open subset of X, let $g : U \to X$ be the canonical immersion, let $h = fg : U \to S$, and let F be a quasi-coherent module on X, flat over S. Assume that, for every base change $S' \to S$, with S' locally Noetherian, $\mathrm{R}^i g'_*(F')$ is coherent on X' for $i \leq n$. Then the following hold:*

(i) For every base change $S' \to S$, with S' locally Noetherian, $\mathrm{R}^i h'_*(F')$ is coherent on S' for $i \leq n$.

(ii) For every $S' \to S$ as above, and every coherent ideal $\mathcal{J}$ on S', the graded modules **49**

$$\bigoplus_{m\geq 0} \mathrm{R}^i h'_*(\mathcal{J}^m F')$$

over $\bigoplus_{m\geq 0} \mathcal{J}^m$ are of finite type for $i \leq n$.

(iii) For every $S' \to S$ and $\mathcal{J}$ as above, the graded modules

$$\bigoplus_{m\geq 0} \mathrm{R}^i h'_*(\mathcal{J}^m F'/\mathcal{J}^{m+1}F')$$

over $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_{S'}) = \bigoplus_{m\geq 0} \mathcal{J}^m/\mathcal{J}^{m+1}$ are of finite type for $i \leq n$.

Moreover, under the conditions of (ii), and by the comparison theorem 1.1, denoting by $\widehat{S'}$ the formal completion of S' with respect to $\mathcal{J}$, and by $\widehat{U'}$ that of U' with respect to $\mathcal{J}\mathcal{O}_{U'}$, the canonical homomorphisms

$$\widehat{\mathrm{R}^i h'_*(F')} \to \mathrm{R}^i \widehat{h'}_*(\widehat{F'}) \to \varprojlim_k \mathrm{R}^i h'_*(F'_k)$$

are isomorphisms for $i \leq n-1$.

Remark XII.5.9. Assume in addition that the conditions of XII.5.8 hold with F *coherent*, and consider a base change $S' \to S$ as in XII.5.9 (i). Assume further that S' is locally embeddable in a regular scheme, or, more generally, that there exists a faithfully flat and quasi-compact morphism $S'' \to S$ such that S'' is locally embeddable in a regular scheme; this condition holds in particular if S' is local. Then the conclusion of XII.5.8 (i) and (ii) remains valid when F' is replaced there by a module G' on U' such that every point of U' has an open neighborhood on which G' is isomorphic to a module of the form F'^n. Indeed, one reduces to the case where S' itself is locally embeddable in a regular scheme, and hence so are $S'' = \mathrm{Spec}\,(\bigoplus_{m\geq 0} \mathcal{J}^m)$ and

$X'' = X' \times_{S'} S'' = X \times_S S''$, which are of finite type over it. One then applies the finiteness criterion VIII VIII.2.3 to the direct images for $i \leq n$ of G'' under the immersion $U'' \to X''$, noting that its conditions hold by hypothesis for F'' , and therefore also for G'', since they **50** are expressed in terms of depth and G'' is locally isomorphic to an F''^n. The same argument shows that, if $\mathcal{G}'$ is a coherent module on $\widehat{U'}$ (the completion of U' with respect to the ideal $\mathcal{J}\mathcal{O}_{U'}$), such that $\mathcal{G}'_0 = \mathcal{G}'/\mathcal{J}\mathcal{G}'$ is locally of the form $F_0'^n$, then the conclusion of (iii) remains valid when F' is replaced there by $\mathcal{G}'$. Using the results of Exposé IX, one thus obtains the following result:

Corollary XII.5.10. *Let $f : X \to S$ be a proper morphism, with S locally Noetherian, and let U be an open subset of X; assume that U is flat over S, and that, for every base change $S' \to S$, with S' locally Noetherian, $R^i g'_*(\mathcal{O}_{U'})$ is coherent on X' for $i = 0, 1$. Assume now that S' is of the form* Spec(A)*, where A is a Noetherian ring equipped with an ideal J such that A is separated and complete for the J-adic topology. Under these conditions:*

(i) The functor $F \longmapsto \widehat{F}$ from the category of locally free modules on U' to the category of locally free modules on $\widehat{U'}$ is fully faithful.

(ii) For every locally free module $\mathfrak{F}$ on $\widehat{U'}$, there exists a coherent module F on U' (not necessarily locally free, alas), and an isomorphism $\widehat{F} \simeq \mathfrak{F}$.

It remains only to prove (ii), by XII.5.9. But it follows from this remark and 2.1 that $\mathfrak{F}$ is induced by a coherent module $\mathfrak{G}$ on $\widehat{X'}$. By the existence theorem EGA III 5.1.4, $\mathfrak{G}$ is of the form $\widehat{F}$, where F is coherent on X, which proves the conclusion.

Remarks XII.5.11. 1°) Using XII.5.10, XII.4.7, and a suitable hypothesis saying that certain local rings of the geometric fibers of $X' \to S'$ are "pure"resp. parafactorial, one should be able to obtain statements saying that the functor $Z' \longmapsto \widehat{Z'}$ from the category of finite ètale coverings of X' to the category of finite ètale coverings of $\widehat{X'}$ (or, equivalently, of X'_0) is **51** an equivalence of categories, resp. that the functor $L \longmapsto \widehat{L}$ from the category of invertible modules on X' to the category of invertible modules on $\widehat{X'}$ is an equivalence. Using recent results of Murre, it is likely that one should be able to deduce existence theorems for Picard schemes of certain *nonproper* algebraic schemes(*). In general, the removal of purity hypotheses

(*)It is in fact enough that g be quasi-compact and quasi-separated (EGA IV1.2.1), with no condition on U, X.

(*)*Editor's note.* Of course, in the projective case one should refer to Grothendieck's existence theorems in FGA; cf. Grothendieck A., "Technique de descente et théorèmes d'existence en géométrie algébrique. VI. Les schémas de Picard: propriétés générales", in *Séminaire Bourbaki*, vol. 7, Société mathématique de France, Paris, 1995, Exp. 236, p. 221–243 and "Technique de descente et théorèmes d'existence en géométrie algébrique. V. Les schémas de Picard: théorèmes d'existence", in *Séminaire Bourbaki*, vol. 7, Société mathématique de France, Paris, 1995, Exp. 232, 143–161. The nine finiteness conjectures found there are proved in Exposés XII and XIII by Mme Raynaud and Kleiman in SGA 6. For an excellent elementary account of the subject, see Kleiman's expository article (Kleiman S., "The Picard scheme", forthcoming in Contemp. Math.). For an application of these techniques to the global generalized Jacobians of a relative smooth curve, see (Contou-Carrère C., "La jacobienne généralisée d'une courbe relative; construction et propriété universelle de factorisation", *C. R. Acad. Sci. Paris Sér. A-B* **289** (1979), no. 3, A203–A206 and "Jacobiennes généralisées globales relatives", in *The Grothendieck Festschrift, Vol. II*, Progr. Math., vol. 87, Birkhäuser, Boston, 1990, p. 69–109). See also, by the same author and in the purely local setting, the construction and study of the "local generalized Jacobian"functor ("Jacobienne locale, groupe de bivecteurs de Witt universel, et symbole modéré", *C. R. Acad. Sci. Paris Sér. I Math.* **318** (1994), no. 8, p. 743–746). Moreover, although in the case of a projective smooth morphism the connected components of the Picard scheme are proper, this is no longer so in the singular case. The problem of compactifying Picard schemes then arises naturally; this problem has been studied in detail, notably in (Altman A.B. & Kleiman S., "Compactifying the Picard scheme", *Adv. in Math.* **35** (1980), no. 1, p. 50–112, and "Compactifying the Picard scheme. II", *Amer. J. Math.* **101** (1979), no. 1, p. 10–41). The case of curves had been studied earlier (D'Souza C., "Compactification of generalised Jacobians", *Proc. Indian Acad. Sci. Sect. A Math. Sci.* **88** (1979), no. 5, p. 419–457). One even knows exactly when the compactified Jacobian of a curve is irreducible (Rego C.J., "The compactified Jacobian", *Ann. Sci. Éc. Norm. Sup. (4)* **13** (1980), no. 2, p. 211–223), the latter being the closure of the ordinary Jacobian when the curve is geometrically integral and locally planar; for a construction in families of compactified Jacobians, see (Esteves E., "Compactifying the

in various existence theorems, especially representability theorems for functors such as the Hilbert or Picard functors, by means of the techniques developed in this seminar, deserves systematic study.

2°) One may seek manageable necessary and sufficient conditions, in terms of depth, for the universal finiteness condition considered in XII.5.10 to hold. When S is the spectrum of a field, it follows easily from EGA III 1.4.15 that it is necessary and sufficient for the $\mathrm{R}^i g_*(F)$ $(i \leq n)$ to be coherent, which is indeed expressed in terms of depth by VIII VIII.2.3. In the general case, however, note that it is not enough to require the preceding condition to hold for all the fibers $U_s \subset X_s$ $(s \in S)$, even when $n = 0$. For example, take $X = S$, let S be the spectrum of a discrete valuation ring, let U be the open subset consisting only of the generic point, and let $F = \mathcal{O}_U$.

3°) Here, however, is a *sufficient* condition ensuring that the hypotheses of XII.5.10 hold: it is enough that f be flat and that, for every $s \in S$ and every $x \in Y_s = X_s - U_s$, one have

$$\operatorname{prof} \mathcal{O}_{X_s,x} \geq n + 2.$$

Indeed, taking Lemma XII.2.5 into account (cf. relation (16) following XII.2.5), it follows that $g_*(\mathcal{O}_U) \simeq \mathcal{O}_X$ and $\mathrm{R}^i g_*(\mathcal{O}_U) = 0$ for $0 < i \leq n$, and the same relations will clearly hold after every base change $S' \to S$.

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relative Jacobian over families of reduced curves", *Trans. Amer. Math. Soc.* **353** (2001), no. 8, p. 3045–3095). Since then, existence results for the Picard scheme in the proper case have advanced beyond the original edition of SGA 2; cf. (Murre J.P., "On contravariant functors from the category of pre-schemes over a field into the category of abelian groups (with an application to the Picard functor)", *Publ. Math. Inst. Hautes Études Sci.* **23** (1964), p. 5–43), and especially (Artin M., "Algebraization of formal moduli. I", in *Global Analysis (Papers in Honor of K. Kodaira)*, Univ. Tokyo Press, Tokyo, 1969, p. 21–71). See also (Raynaud M., "Spécialisation du foncteur de Picard", *Publ. Math. Inst. Hautes Études Sci.* **38** (1970), p. 27–76) for a scheme proper over a discrete valuation ring but not necessarily flat. For a discussion of more recent results, in particular Artin's, for the Picard functor of proper flat schemes, especially in the cohomologically flat case in dimension 0, see Chapter VIII of (Bosch S., Lütkebohmert W. & Raynaud M., *Néron models*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 21, Springer-Verlag, Berlin, 1990) and the references cited there. Much more recently, very precise results have been obtained for relative curves $f : X \to S$ over the spectrum S of a discrete valuation ring with perfect residue field. More precisely, one assumes that f is proper and flat, that X is regular, and that $f_*\mathcal{O}_X = \mathcal{O}_S$. On the other hand, one does not assume that f is cohomologically flat in dimension 0, i.e. one does not assume that $H^1(X, \mathcal{O})$ is torsion-free. The Picard functor is then not representable, either by a scheme or by an algebraic space, as soon as the torsion in question is nonzero. Let J be the Néron model of the generic fiber of f; it is a quotient of the Picard functor P. Raynaud then showed that the kernel of the tangent map $H^1(X, \mathcal{O}) = \mathrm{Lie}(P) \to \mathrm{Lie}(J)$ coincides with the torsion subgroup of H^1, and that the cokernel has the same length (see Theorem 3.1 of (Liu Q., Lorenzini D. & Raynaud M., "Néron models, Lie algebras, and reduction of curves of genus one", *Invent. Math.* **157** (2004), p. 455–518)). This result enables those authors to study the relation between the Birch–Swinnerton-Dyer and Artin–Tate conjectures (see Th.. 6.6 of *loc. cit.*). As for the local Picard scheme, see Boutot's thesis, cited in editor's note ((*)) on page 125.

Exposé XIII

Problems and conjectures

XIII.1 Relations between global and local results. Affine problems connected with duality

It is well known that many statements concerning a projective scheme X can be formulated **52**
in terms of statements concerning a certain graded ring, or better, a complete local ring, namely the homogeneous coordinate ring of X (i.e. the affine ring of the cone $\tilde{X}$ over X), or its completion (i.e. the completion of the local ring at the vertex of $\tilde{X}$). The value of this reformulation is that, starting from known global results, it often makes it possible to conjecture, or even prove, analogous results for complete Noetherian local rings more general than those which actually occur in the global statement, for example for local rings that are not necessarily of equal characteristic. Thus, Serre's duality theorem for projective space XII XII.1.1 suggested the useful local duality theorem V 2.1. Serre's fundamental theorem on the cohomology of coherent Modules on projective space (finiteness, asymptotic behavior for large n of $\mathrm{H}^i(X, F(n))$, cf. EGA III 2.2.1) generalizes to a structure theorem for the local invariants $\mathrm{H}^i_{\mathfrak{m}}(M)$; see V 3. Likewise, the Lefschetz theorems for the fundamental group and the Picard group ("equivalence criteria"), well known in the classical case and subsequently extended to an arbitrary base field, suggested the "local"Lefschetz theorems of Exposés X and XI. Of course, the local theorems are in turn valuable tools for obtaining global statements. For example, local duality allows a global asymptotic property XII XII.1.3 (i) to be formulated by the vanishing of certain local invariants $\mathrm{H}^i(F_x)$. More substantially, the local Lefschetz theorems, implying for example the "purity"or parafactoriality of certain complete-intersection local rings (X 3.4 and XI XI.3.14), make it possible in the global Lefschetz theorems to dispense with certain nonsingularity hypotheses, as in X 3.5, 3.6, 3.7.

Another useful generalization of theorems concerning projective schemes over a field k **53**
consists in replacing k by a general base scheme. Thus, the sequel to EGA III will give(*) a generalization in this direction of Serre duality(*); the finiteness and asymptotic-behavior theorems for $\mathrm{H}^i(X, F(n))$ were stated in EGA III 2.2.1 over a general base scheme; finally, the Lefschetz theorems can likewise be developed for a projective morphism, as we saw in XII XII.4.9, thanks to the local theorem XII XII.4.7. Of course, working over a general base scheme also leads to essentially new statements, such as the "comparison theorem"EGA III 4.15 and the existence theorem for sheaves EGA III 5.1.4 (which, as we saw elsewhere in IX, arise from the same key theorems of a cohomological nature as the Lefschetz theorems for π_1 and Pic).

It is therefore necessary to isolate theorems that simultaneously encompass the two

(*) *Editor's note.* in fact, this generalization is not found there; see the note below and the editor's note (5) on page 2

(*) cf. the Hartshorne Seminar, cited at the end of Exp. IV.

generalizations just mentioned of statements concerning projective schemes over a field. The natural objects for such a common generalization are the *Noetherian rings separated and complete for an I-adic topology.* From this point of view, their study has not yet been seriously undertaken and seems to me at present the most interesting subject in the local theory of coherent sheaves. Here is a typical problem in this direction:

Conjecture 1.1 ("Second affine finiteness theorem"[**] [(*)]). *Let M be a module of finite type over a Noetherian ring A (which may, if necessary, be assumed to be a quotient of a regular ring), and let J be an ideal of A. Prove that the modules $\mathrm{H}^i_J(M)$ are "J-cofinite", i.e. that the modules*

$$\mathrm{Hom}_A(A/J, \mathrm{H}^i_J(M))$$

are of finite type.

Recall that $\mathrm{H}^i_J(M)$ denotes the module $\mathrm{H}^i_Y(X, \tilde{M})$ (where $X = \mathrm{Spec}(A)$, $Y = V(J)$) of Exp. I, interpreted in II in terms of an inductive limit of the cohomology of Koszul complexes, **54**
or, for $i \geq 2$, the module $\mathrm{H}^{i-1}(X - Y, \tilde{M})$. To tell the truth, XIII.1.1 should be a consequence of a more precise statement, implying that the $\mathrm{H}^i_J(M)$ belong to a suitable abelian subcategory $\underline{D}_J$ of the category $\underline{C}_J$ of A-modules with support $\subset Y = V(J)$, such that $H \in \mathrm{Ob}\,\underline{D}_J$ implies that H is J-cofinite. (N.B. The category of modules H with support contained in $V(J)$ and which are J-cofinite is unfortunately not stable under passage to quotients!). The essential problem would then consist in defining $\underline{D}_J$. More precisely, the solution of problem XIII.1.1 should emerge (at least if A is a quotient of a regular ring) from a duality theory generalizing both local duality and the duality theory for projective morphisms alluded to above, and which would be of the following kind:

Conjecture 1.2 ("Affine duality"[(*)]). *Suppose that A is regular, separated, and complete for the J-adic topology. Let $C^\bullet(A)$ be an injective resolution of A.*

(i) Prove that the functor

$$D_J : L_\bullet \longmapsto \mathrm{Hom}_J(L_\bullet, C^\bullet(A))$$

from the category of complexes of A-modules that are free of finite type in every dimension and bounded above (where the morphisms are homomorphisms of complexes up to homotopy) to the category of complexes of A-modules $K^\bullet$ that are injective in every dimension and bounded above (where the homomorphisms are defined in the same way), is fully faithful.

[**]This conjecture, and Conjecture 1.2 below, are false, as was shown by R. Hartshorne, "Affine duality and cofinite modules", *Invent. Math.* **9** (1969/70), p. 145-164, Section 3.

[(*)]*Editor's note.* However, if A is complete local (respectively regular of positive characteristic) and J is the maximal ideal, the statement is true for M of finite type (respectively $M = A$), cf. (Hartshorne R., "Affine duality and cofinite modules", *Invent. Math.* **9** (1969/70), p. 145-164, Corollary 1.4) (respectively (Huneke C. & Sharp R., "Bass numbers of local cohomology modules", *Trans. Amer. Math. Soc.* **339** (1993), no. 2, p. 765–779), which moreover contains much stronger results). For completely different methods (D-modules) for treating characteristic zero, see (Lyubeznik G., "Finiteness properties of local cohomology modules (an application of D-modules to commutative algebra)", *Invent. Math.* **113** (1993), no. 1, p. 41–55); see also, by the same author, "Finiteness properties of local cohomology modules for regular local rings of mixed characteristic: the unramified case", *Comm. Algebra* **28** (2000), no. 12, p. 5867–5882, Special issue in honor of Robin Hartshorne, and "Finiteness properties of local cohomology modules: a characteristic-free approach", *J. Pure Appl. Algebra* **151** (2000), no. 1, p. 43–50. The notion of a cofinite module has since evolved under Hartshorne's leadership. One says that M is J-cofinite if its support is contained in $V(J)$ and if all the $\mathrm{Ext}^i_A(A/J, \mathrm{H}^i_J(M))$ are of finite type. On this subject, see for example (Delfino D. & Marley Th., "Cofinite modules and local cohomology", *J. Pure Appl. Algebra* **121** (1997), no. 1, p. 45–52).

[(*)]This conjecture, false as stated, was nevertheless established in a fairly similar form by R. Hartshorne, "Affine duality and cofinite modules", *Invent. Math.* **9** (1969/70), p. 145-164.

(ii) Prove that, for every $K^\bullet$ of the form $D_J(L_\bullet)$, the $\mathrm{H}^i(K^\bullet)(= \mathrm{Ext}^i_Y(X; L_\bullet, \mathcal{O}_X))$ are J-cofinite.

(iii) More precisely, prove that the $K^\bullet$ which are homotopic to a complex of the form $D_J(L_\bullet)$ can be characterized by finiteness properties of the $\mathrm{H}^i(K^\bullet)$ stronger than the one considered in (ii), for example by the property $\mathrm{H}^i(K^\bullet) \in \mathrm{Ob}\,\underline{D}_J$, where $\underline{D}_J$ is a suitable abelian category, as contemplated above. 55

Note that the problem has an affirmative solution when A is local and J is an ideal of definition (cf. Exp. IV), and also when J is the zero ideal. In these two exceptional cases, one may take for $\underline{D}_J$ simply the category of Modules with support $V(J)$ which are J-cofinite (which in the second case means simply that one takes the category of Modules of finite type over A). An affirmative solution of Conjecture XIII.1.2 in general would give one for XIII.1.1, by taking for $L_\bullet$ the dual of a finite free resolution of M. On the other hand, an affirmative solution of XIII.1.1 would give an affirmative answer to the first part of the following conjecture, which we formulate in "global"form:

Conjecture XIII.1.3. *Let $X \subset \mathbf{P}^r_k$ be a closed subscheme of the standard projective scheme which is locally a complete intersection and each of whose irreducible components has codimension $\geq s$. Let $U = \mathbf{P}^r_k - X$.*

(i) Prove that, for every coherent Module F on U, one has

$$\dim \mathrm{H}^i(U, F) < +\infty \text{ for } i \geq s.^{(*)}$$

(ii) Give an example, with X connected and regular, for which one has

$$\mathrm{H}^s(U, F) \neq 0.$$

To see that (i) is a special case of XIII.1.1, consider

$$\mathrm{H}^i(U, F(\cdot)) = \bigoplus_n \mathrm{H}^i(U, F(n)) = \mathrm{H}^i(E^{r+1} - \tilde{X}, \tilde{F})$$

as a module over the affine ring $k[t_0, \ldots, t_r]$ of the cone E^{r+1} over $\mathbf{P}^r$. This module is none other than $\mathrm{H}^{i+1}_J(M)$, where J is the ideal of the cone $\tilde{X}$ over X in E^{r+1}. On the other hand, the hypothesis on X, which implies that $\tilde{X}$ is also a complete intersection of codimension $\geq s$ at every point of E^{r+1} distinct from the origin, implies that $\mathrm{H}^{i+1}_J(M)$ is zero away from the 56 origin for $i \geq s$. Thus, if it is J-cofinite as required by XIII.1.1, it is *a fortiori* $\mathfrak{m}$-cofinite, which readily implies that it is finite-dimensional in every degree $^{(*)}$ Note moreover that Conjecture

$^{(*)}$Partie (i) of this conjecture was proved by R. Hartshorne when X is smooth cf. *Ample subvarieties of algebraic varieties*, Notes written in collaboration with C. Musili, Lect. Notes in Math, vol. 156, Springer-Verlag, Berlin-New York, 1970, Theorem III.5.2. This author also found an example for (ii), cf. R. Hartshorne, "Cohomological dimension of algebraic varieties", *Ann. of Math. (2)* **88** (1968), p. 403–450, example on page 449.

$^{(*)}$*Editor's note.* Hartshorne proved (Hartshorne R., "Cohomological dimension of algebraic varieties", *Ann. of Math. (2)* **88** (1968), p. 403–450) that the cohomology $H^{n-1}(\mathbf{P}^n_k - X, F)$ is zero for F coherent and X of positive dimension (k algebraically closed). In fact, thanks essentially to Serre duality and the Lichtenbaum theorem — the vanishing of the cohomology of coherent sheaves in the maximal dimension of irreducible noncomplete quasi-projective varieties —, one is reduced to proving that the formal completion $\hat{\mathbf{P}}^n_k$ and X have the same field of rational functions. This is the difficult point (Theorem 7.2 of *loc. cit.*); in other words, $\mathbf{P}^n_k$ is $G3$ in the terminology of (Hironaka H. & Matsumura H., "Formal functions and formal embeddings", *J. Math. Soc. Japan* **20** (1968), p. 52–82). These authors independently proved the preceding results, and in fact much more. They proved that X is universally $G3$ and calculated the field of rational functions of the formal completion of an abelian variety along a positive-dimensional subvariety. It is in this article that the now-classical conditions $G1, G2, G3$ appear for the first time.

XIII.1.3 already arises for a nonsingular irreducible curve X in $\mathbf{P}^3$: it is unknown whether in this case the $\mathrm{H}^2(\mathbf{P}^3 - X, \mathcal{O}_X(n))$ are finite-dimensional, or whether they are necessarily zero(*). It is not even known whether there exists an irreducible curve in $\mathbf{P}^3$ which is not set-theoretically the intersection of two hypersurfaces(*).

Problem XIII.1.4. *Give an affine variant of the "comparison theorem"*EGA III 4.1.5 *as a theorem on the commutation of the functors* H^i_J *with certain inverse limits.*

Finally, in the present line of thought, I had posed the following problem: let A be a *complete* regular Noetherian local ring, and K its field of fractions; prove that $\mathrm{Ext}^i_A(K, A) = 0$ for every i. An affirmative answer was given on the spot by M. Auslander; the regularity hypothesis may be replaced by the hypothesis that A is integral, and in fact $\mathrm{Ext}^i_A(K, M) = 0$ for every i whenever M is of finite type over A. This follows immediately from the following statement, due to Auslander: if A is a complete Noetherian local ring, then for every module of finite type M over A, the functors $\mathrm{Ext}^i_A(., M)$ transform inductive limits into inverse limits(*).

XIII.2 Problems related to π_0: local Bertini theorems

Let A be a complete noetherian local ring, f an element of its maximal ideal, $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(A/fA)$. The use of the local "Lefschetz"technique makes it possible to give criteria for $Y' = X' \cap Y$ (where $X' = X - \{\mathfrak{m}\}$) to be connected, in terms of hypotheses on X'. Thus, it suffices that: a) X' is connected, b) $\mathrm{prof}\, \mathcal{O}_{X',x} \geq 2$ for every closed point x of X', c) f is **57**
A-regular. Note, however, that hypotheses b) and c) are not purely topological in nature; for example, they are not invariant under replacement of X by X_{red}. In the analogous situation for a projective scheme X' over a field and a hyperplane section Y' of X', the use of the "Bertini theorem"and Zariski's "connectedness theorem"in fact gives results of a distinctly more satisfactory form, which had led me, in the oral seminar, to state a conjecture that I have since solved affirmatively. Let us therefore state here:

Theorem XIII.2.1. *Let A be a complete noetherian local ring, X its spectrum, $\mathfrak{a}$ the closed point of X, $X' = X - \{\mathfrak{a}\}$. Suppose that X satisfies the following conditions (where k denotes an integer ≥ 1):*

a_k) The irreducible components of X' have dimension $\geq k + 1$.

b_k) X' is connected in dimension $\geq k$, i.e. it cannot be disconnected by a closed subset of dimension $< k$ (cf. III 3.8).

Let m be an integer, $0 \leq m \leq k$, and let $f_1, \ldots, f_m \in \mathfrak{r}(A)$; set $B = A/\sum_i f_i A$, $Y = \mathrm{Spec}(B) = V(f_1) \cap \cdots \cap V(f_m)$, $Y' = X' \cap Y = Y - \{\mathfrak{a}\}$. Then Y satisfies conditions a_{k-m}), b_{k-m}). In particular, for every sequence of $m \leq k$ elements $f_1, \ldots, f_m$ of $\mathfrak{r}(A)$, $Y' = X' \cap V(f_1) \cap \cdots \cap V(f_m)$ is connected.

It is moreover easy to see that if the last conclusion holds (it plainly suffices to take $m = k$ there), and if the case where X is irreducible of dimension 0 or 1 is excluded, it follows that the irreducible components of X' have dimension $\geq k + 1$, and X' is connected in dimension $\geq k$, so that, in a sense, XIII.2.1 is a "best possible"result.

Let us give the principle of the proof of XIII.2.1. Only condition b_{k-m}) for Y presents a problem. For a given k, one is easily reduced to the case where X is integral, and even (by passing to the normalization, which is finite over X) to the case where X is *normal*. If $k = 1$, **58**

(*)The question has just been answered affirmatively by R. Hartshorne (Hartshorne R., "Ample vector bundles", *Publ. Math. Inst. Hautes Études Sci.* **29** (1966), p. 63–94, Theorem 8.1) and H. Hironaka.

(*)*Editor's note.* see Conjecture XII.3.5 and the corresponding note.

(*)*Editor's note.* write $K = \varinjlim_{a \neq 0} A[1/a]$ and observe that $a \neq 0$ is A-regular.

hence $\dim X' \geq 2$, then X' has depth ≥ 2 at its closed points and one may apply the result recalled at the beginning of n°, which shows that $Y' = X' \cap V(f)$ is connected. In the case $k \geq 1$, suppose the theorem proved for $k' < k$. By induction on m, one is reduced to the case $m = 1$, i.e. to verifying that, for $f_1 \in \mathfrak{r}(A)$, $X' \cap V(f_1)$ is connected in dimension $\geq k-1$. If it were not, i.e. if it were disconnected by a Z' of dimension $< k-1$, there would exist a sequence $f_2, \ldots, f_k$ such that $X' \cap V(f_1) \cap \cdots \cap V(f_k)$ is disconnected, and in this sequence one may choose $f_2 \in \mathfrak{r}(A)$ arbitrarily, subject only to the condition that it not vanish at any point of a certain finite subset F of X' (namely the set of maximal points of Z'). Moreover, one easily verifies, using the fact that X' is normal and therefore satisfies Serre's condition (S_2)(*), that there exists a finite subset F' of X' such that $f \in \mathfrak{r}(A)$, $V(f) \cap F' = \emptyset$ implies that $V(f) \cap X'$ also satisfies condition (S_2). One may then choose f_2 so that f_2 vanishes neither on F nor on F', and hence so that $X' \cap V(f_2)$ satisfies (S_2). But then, by the theorem of Hartshorne III 3.6, $X' \cap V(f_2)$ is connected in codimension 1 and thus (since every component of $X' \cap V(f_2)$ has dimension $\geq k$) it is connected in dimension $\geq k-1$. Applying the induction hypothesis to $V(f_2) = \mathrm{Spec}(A/f_2A)$, it follows that $X' \cap V(f_2) \cap V(f_1) \cap V(f_3) \cap \cdots \cap V(f_k)$ is connected, whereas it had been constructed disconnected, a contradiction.

Let us mention some interesting corollaries:

Corollary XIII.2.2. *Let $f\colon X \to Y$ be a proper morphism of locally noetherian preschemes, with Y integral, $y_0 \in Y$, and y_1 the generic point of Y. Assume that*

a) Y is unibranch at y_0, and every irreducible component of X dominates Y.

b) The irreducible components of X_1 have dimension $\geq k+1$, and X_1 is connected in dimension $\geq k$.

Then the irreducible components of X_0 have dimension $\geq k+1$, and X_0 is connected in dimension $\geq k$. **59**

Indeed, Zariski's connectedness theorem (cf. EGA III 4.3.1) implies that X_0 is connected; to show that it is not disconnected by a closed subset of dimension $< k$, one is reduced to showing that the local rings at points $x \in X_0$ such that $\dim \overline{x} < k$ have a spectrum that is not disconnected by x. But this is true without assuming either that f is proper or that Y is unibranch at y_0. To see this, one is reduced to the case where X is integral and dominates Y, and, if desired, where Y is affine of finite type over $\mathbf{Z}$, so that one is under the hypotheses of the dimension formula for $\mathcal{O}_{X,x}$ over $\mathcal{O}_{Y,y_0}$. Using the finiteness of the normalization in this case, one may even suppose X *normal*; hence, by a theorem of Nagata(*), the completion of a local ring $\mathcal{O}_{X,x}$ of X' is still normal, and therefore (if $\mathcal{O}_{X,x}$ has dimension N) $\mathrm{Spec}(\hat{\mathcal{O}}_{X,x})$ is connected in dimension $\geq N-1$. Let $n = \dim \mathcal{O}_{Y,y_0}$; then $\deg \mathrm{tr}\ k(x)/k(y) < k$ implies $\dim \mathcal{O}_{X,x} > n + (k+1) - k = n+1$, in view of $\dim X_1 \geq k+1$, and, taking a system $f_1, \ldots, f_n$ of parameters of $\mathcal{O}_{Y,y_0}$ and lifting them to elements of $\mathcal{O}_{X,x}$, one sees from XIII.2.1 that $\mathrm{Spec}(\hat{\mathcal{O}}_{X,x}/\sum f_i \hat{\mathcal{O}}_{X,x})$ is connected in dimension ≥ 1, i.e. is not disconnected by its closed point, or, equivalently, $\mathrm{Spec}(\hat{\mathcal{O}}_{X_0,x})$ is not disconnected by its closed point; *a fortiori* the same holds for $\mathrm{Spec}(\mathcal{O}_{X_0,x})$.

As in the case of the ordinary connectedness theorem, one may vary XIII.2.2 by taking geometric fibers (over algebraic closures of the residue fields), provided that one assumes Y to be *geometrically* unibranch at y_0, or (with no hypothesis other than that Y is noetherian) that f is universally open. Applying this to the case where Y is the dual scheme of a projective scheme $\mathbf{P}^r_k$ over a field, one recovers a strengthened form of the global result that had inspired XIII.2.1, namely:

(*) Cf. EGA IV 5.7.2.

(*) Cf. EGA IV 7.8.3 (i) (ii) (v).

Corollary 2.3(*). *Let X be a closed subscheme of $\mathbf{P}^r_k$ (k a field); suppose that the irreducible components of X have dimension $\geq l+1$, and that X is geometrically connected in dimension $\geq$* **60**
l. Then for every sequence $H_1, \ldots, H_m$ of m hyperplanes of $\mathbf{P}^r_k$ ($0 \leq m \leq l-1$), $X \cap H_1 \cap \cdots \cap H_m$ satisfies the same condition with $l-m$; in particular, it is geometrically connected in dimension $\geq l-1$.

One may moreover modify this statement in the obvious way for the case where one is given a proper morphism $X \to \mathbf{P}^r_k$, which is not necessarily an immersion, and an analogous extension is possible for XIII.2.1 (by considering a scheme proper over X'). These statements moreover follow formally from the statements given here, in view of the ordinary connectedness theorem, which reduces us to the case of a finite morphism.

Corollary XIII.2.4. *Let A be a complete normal noetherian local ring of dimension $\geq k+2$. Let $X = \mathrm{Spec}(A)$, $X' = X - \{\mathfrak{a}\}$, and let $f_1, \ldots, f_k$ be elements of $\mathfrak{r}(A)$; then $Y' = X' \cap V(f_1) \cap \cdots \cap V(f_k)$ is connected, and $\pi_1(Y') \to \pi_1(X')$ is surjective.*

One proceeds as in SGA 1 X 2.11.

Throughout the preceding discussion, only questions of *connectedness* were involved. In the global case, however, well-known theorems assert that for an irreducible projective variety $X \subset \mathbf{P}^r_k$, with k algebraically closed, its intersection with a sufficiently "general"hyperplane H is irreducible (and not merely connected): this is the *Bertini theorem*, proved by Zariski, which in turn implies, by Zariski's connectedness theorem, that for *every* H, $H \cap X$ is connected (although not necessarily irreducible). One may moreover proceed in the opposite direction, by proving the latter result using a Lefschetz-type technique and deducing the Bertini theorem by reducing to the case where X is normal and using the following result: for a "sufficiently general"H, $X \cap H$ is also normal. This suggests the

Conjecture 2.5(*). *Let A be a complete normal noetherian local ring. Show that there exists a nonzero $f \in \mathfrak{r}(A)$ such that $Y' = X' \cap V(f) = Y - \{\mathfrak{a}\}$ (where $Y = \mathrm{Spec}(A/fA)$) is normal* **61**
(and hence irreducible by XIII.2.1 if $\dim A \geq 3$).

Ideally, one should show that, in a suitable sense, there are even "many"elements f having the property in question; for example, that f may be chosen in an arbitrary power of the maximal ideal. Using Serre's normality criterion and the remark made above concerning Serre's property (S_2), one sees that XIII.2.5 would have an affirmative answer if the same were true for the

Conjecture 2.6(*). *Let A be a complete noetherian local ring, U an open subset of its spectrum X, and F a finite subset of $X' = X - \{\mathfrak{a}\}$. Suppose that U is regular. Prove that there exists an $f \in \mathfrak{r}(A)$ such that $V(f) \cap U$ is regular, and $V(f) \cap F = \emptyset$.*

For a result of "local Bertini"type, see Chow [2]

(*)*Editor's note.* for a very beautiful direct proof, see (Fulton W. & Lazarsfeld. R., "Connectivity and its applications in algebraic geometry", in *Algebraic geometry (Chicago, Ill., 1980)*, Lect. Notes in Math., vol. 862, Springer, Berlin-New York, 1981, p. 26–92, Theorem 2.1). Cf. also [HL], cited in the editor's note ((*)) on page 129.

(*)*Editor's note.* see the following editor's note.

(*)*Editor's note.* a proof of this conjecture can now be found in the literature, and the preceding conjecture must therefore also be regarded as proved, as indicated above. One can also find two earlier published but unfortunately unsuccessful proof attempts, by Flenner and Trivedi. See Trivedi V., "Erratum: "A local Bertini theorem in mixed characteristic"", *Comm. Algebra* **25** (1997), no. 5, p. 1685-1686. However, the editor has not verified that the proof is now complete.

XIII.3 Problems related to π_1

Here again there are many questions, suggested by the global results or by the transcendental results.

Conjecture 3.1(*). *Let A be a complete noetherian local ring with algebraically closed residue field, let $X = \mathrm{Spec}(A)$, let $X' = X - \{\mathfrak{a}\}$, where $\mathfrak{a}$ is the closed point. Suppose that the irreducible components of X have dimension ≥ 2 and that X' is connected.*

(i) Prove that $\pi_1(X')$ is topologically finitely generated.

(ii) If p is the characteristic exponent of the residue field k of A, prove that the largest topological quotient group of $\pi_1(X')$ that is "of order prime to p"is finitely presented.

For part (i), using the descent theory of SGA 1 IX 5.2 and Theorem 2.4, one is reduced to **62**
the case in which A is normal of dimension 2. In this case, a systematic method for studying the fundamental group of X', introduced by Mumford [5] in the transcendental setting, consists in *desingularizing* X, i.e. considering a projective birational morphism $Z \to X$, with Z integral and regular, which induces an isomorphism $Z' = Z|X' \to X'$. It is plausible that such a Z always exists; in any case this is what Abhyankar's method [1] proves in the case of "equal characteristic"(*). Let C be the fiber over the closed point of X under $Z \to X$; it is an algebraic curve over the residue field k, connected by the connectedness theorem. The solution of XIII.3.1 then seems to be related to the following problem.

Problem XIII.3.2. *With the preceding notation, relate $\pi_1(X')$ to the topological invariants of C, in particular to its fundamental group, so as to exhibit the topological finite generation of $\pi_1(X')$, using for example* SGA I, *Theorem* X 2.6.

Another method would be to regard A as a finite algebra over a complete *regular* local ring B of dimension 2, ramified along a curve C contained in $\mathrm{Spec}(B) = Y$. This leads to the following problem.

Problem XIII.3.3. *Let A be a complete regular local ring of dimension 2, with algebraically closed residue field k; let X be its spectrum, and let C be a closed subset of X of dimension 1. Define local invariants of the embedded curve C that have a meaning independent of the residue characteristic and whose knowledge makes it possible to calculate the fundamental group of $X - C$ by generators and relations when k has characteristic zero. Prove that when k has characteristic $p > 0$, the "tame"fundamental group of $X - C$ is a quotient of the preceding one, and that the two fundamental groups (in characteristic 0 and in characteristic $p > 0$) have the same maximal quotient of order prime to p.*

Of course, XIII.3.3 shows that in XIII.3.1 one should also replace X' by a scheme of the form $X - Y$, where Y is a closed subset of X that has codimension ≥ 2 in every component of **63**
X that contains it. When this restriction on Y is dropped, there should still be an analogous finiteness result, provided that "tame"-type restrictions are imposed on the ramification at the maximal points of the irreducible components of Y that have codimension 1.

Problem XIII.3.4. *Let A be a complete noetherian local ring of dimension 2, with algebraically closed residue field. Again let $X = \mathrm{Spec}(A)$ and $X' = X - \{\mathfrak{a}\}$. Find special structural properties of $\pi_1(X')$ in the case where A is a complete intersection.*

(*) *Editor's note.* The analogous statement is true for (connected) schemes X of finite type over a separably closed field k, under the hypothesis of strong desingularization for all $\bar{k}$-schemes of finite type; in particular, this holds if k has characteristic zero or if X has dimension ≤ 2. One reduces to the case of quasi-projective surfaces by Lefschetz-type techniques developed by Mme Raynaud; cf. the notes *supra*; see SGA 7.1, Theorem II.2.3.1.

(*)The possibility of "resolving"A has now been proved in full generality by Abhyankar [8].

A satisfactory solution of this problem might make it possible to solve the following old problem:

Conjecture 3.5(*)**.** *Find an irreducible curve in* $\mathbf{P}^3_k$ *(where* k *is an algebraically closed field), preferably nonsingular, which is not set-theoretically the intersection of two hypersurfaces.*

(Kneser [4] shows that it can always be obtained as the intersection of three hypersurfaces).

XIII.4 Problems related to higher π_i: local and global Lefschetz theorems for complex analytic spaces(*)

Let X be a scheme locally of finite type over the field of complex numbers $\mathbf{C}$. One can associate with it a complex analytic space X^h, and hence homotopy and homology invariants $\pi_i(X^h)$, $\mathrm{H}_i(X^h)$, $\mathrm{H}^i(X^h)$, etc. Moreover, X is known to be connected if and only if X^h is connected, so there is a bijection

$$\pi_0(X^h) \to \pi_0(X).$$

Similarly, since every étale cover X' of X defines an étale cover X'^h of X^h , there is a canonical **64**
homomorphism

$$\pi_1(X^h) \to \pi_1(X),$$

which, by a theorem of Grauert–Remmert, is known to identify the second group with the completion of the first for the topology defined by subgroups of finite index. This simply expresses the fact that $X' \longmapsto X'^h$ is an equivalence from the category of étale covers of X to the category of finite étale covers of X^h. It follows that the results of this seminar, obtained by purely algebraic methods, concerning $\pi_0(X)$ and $\pi_1(X)$ imply results for $\pi_0(X^h)$ and $\pi_1(X^h)$, which are transcendental in nature. Moreover, if X is proper, the well-known exact sequence $0 \to \mathbf{Z} \to \mathbf{C} \to \mathbf{C}^* \to 0$ shows that the Néron–Severi group of X (the quotient of its Picard group by the connected component of the identity) is isomorphic to a subgroup of $\mathrm{H}^2(X^h, \mathbf{Z})$; in the nonsingular Kähler case, it is the subgroup denoted $\mathrm{H}^{(1,1)}(X^h, \mathbf{Z})$ (classes of type $(1,1)$):

$$\mathrm{Pic}(X)/\mathrm{Pic}^0(X) \subset \mathrm{H}^2(X, \mathbf{Z}).$$

Consequently, the information that we have obtained about Picard groups implies information, admittedly very partial, about the groups $\mathrm{H}^2(X^h, \mathbf{Z})$. It is tempting to complete all these fragmentary results with conjectures.

Very precise indications pointing in the same direction as those just mentioned are supplied by a classical theorem of Lefschetz [7]. It asserts that if X is a nonsingular irreducible projective analytic space of dimension n, and if Y is a nonsingular hyperplane section, then the inclusion

$$Y^{n-1} \to X^n$$

induces a homomorphism **65**

$$\pi_i(Y^{n-1}) \to \pi_i(X^n)$$

that is an *isomorphism for* $i \leq n-2$ and an *epimorphism for* $i = n-1$. The analogous statement follows for the homomorphisms

$$\mathrm{H}_i(Y^{n-1}) \to \mathrm{H}_i(X^n)$$

(*) *Editor's note.* As of autumn 2004, this problem is still open.

(*) *Editor's note.* The statements are made more precise in the Comments (Section XIII.6). The conjectures appearing there have become theorems; cf. the footnotes to Section XIII.6.

on homology (integral homology, to be definite), while in cohomology

$$\mathrm{H}^i(X^n) \to \mathrm{H}^i(Y^{n-1})$$

is an isomorphism in degree $i \leq n-2$ and a monomorphism in degree $i = n-1$. We have obtained variants of these results in the setting of schemes, for π_0, π_1, and Pic, valid moreover, to a large extent, without nonsingularity hypotheses; cf. Exp. XII. Furthermore, in removing the nonsingularity hypotheses, we made essential use of "local"variants of these global Lefschetz theorems. All this suggests the following problems, which will doubtless have to be attacked simultaneously(*).

Problem XIII.4.1. *Let X be an analytic space and Y a closed analytic subset of X (or merely a closed subset?)(*) such that for every $x \in Y$, the local ring $\mathcal{O}_{X,x}$ is a complete intersection. Let n be the complex codimension of Y in X. Is the canonical homomorphism*

$$\pi_i(X - Y) \to \pi_i(X)$$

an isomorphism for $i \leq n-2$ and an epimorphism for $i = n-1$?

In this problem and the following ones, a base point is of course implicitly understood to have been chosen in order to define the homotopy groups. To state the following problem, one must define, for an analytic space X (more generally, for a locally path-connected space) and a point $x \in X$, local invariants $\pi_i^x(X)$(*). To do this, choose a nonconstant map f from the **66**
interval $[0,1]$ to X such that $f(0) = x$ and $f(t) \neq x$ for $t \neq 0$; such a map exists if x is not an isolated point. Then, for every neighborhood U of x, there is an $\varepsilon > 0$ such that $0 < t < \varepsilon$ implies $f(t) \in U$, and the homotopy groups $\pi_i(U - x, f(t))$ are essentially independent of t: as t varies, they are connected by a transitive system of isomorphisms. They may be denoted by $\pi_i(U - x, f)$. Set

$$\pi_i^x(X) = \varprojlim_U \pi_{i-1}(U - x, f),$$

where the inverse limit is taken over the system of open neighborhoods U of x. Strictly speaking, this limit depends on f and should be denoted $\pi_i^x(X, f)$, but one verifies that as f varies these groups are isomorphic to one another(*); more precisely, they form a local system on the space of paths of the kind under consideration issuing from x. These invariants are the homotopical counterpart of the local cohomology invariants $\mathrm{H}_x^i(F)$ for a sheaf F on X, introduced in I, and should play the role of *relative local homotopy groups* of X modulo $X - x$. Their vanishing for $i \leq n$ and every $x \in Y$, where Y is a closed subset of X of topological dimension $\leq d$, should imply that the homomorphisms

$$\pi_i(X - Y) \to \pi_i(X)$$

are bijective for $i < n - d$ and surjective for $i = n - d$(*). From this point of view, XIII.4.1 would imply, when Y is reduced to a point, a conjecture of purely local nature, expressed by

$$\pi_i^x(X) = 0 \text{ for } \mathrm{i} \leq \mathrm{n} - 1$$

when X is a complete intersection of dimension n at x.

As an example of the local invariants $\pi_i^x(X)$, note that if x is a nonsingular point of X of complex dimension n, then **67**

(*)The formulations XIII.4.1 to XIII.4.3 below are provisional. See Conjectures A to D below, in the "Comments on Exp. XIII", for more satisfactory formulations, and also Exp. XIV.

(*)*Editor's note.* The meaning of this question is not clear; indeed, the statement of the problem itself does not seem to make sense in this case, since the codimension of Y in X is not defined when Y is no longer assumed analytic.

(*)If $i \geq 2$. For the case $i \leq 1$, cf. the *Comments* in no. 6 below, page 128.

(*)At least if x does not locally disconnect X; cf. the *Comments* below, page 128.

(*)For a corrected formulation, cf. the *Comments* below, page 128.

$$\pi_i^x(X) = \pi_{i-1}(S^{2n-1}),$$

where S^{2n-1} denotes the sphere of dimension $2n-1$. In particular, in this case $\pi_i^x(X) = 0$ for $i \leq 2n-1$, which corresponds to the fact that if a closed subset Y of codimension $\geq m$ is removed from a topological *manifold* X, then $\pi_i(X-Y) \to \pi_i(X)$ is an isomorphism for $i \leq m-2$ and an epimorphism for $i = m-1$.

This being said:

Problem XIII.4.2. *Let X be an analytic space, let $x \in X$, let t be a section of $\mathcal{O}_X$ vanishing at x, and let Y be the zero set of t. Suppose that the following conditions are satisfied:*

a) t is regular at x (i.e. it is not a zero divisor at x; this hypothesis may moreover be superfluous).

b) At the points x' of $X-Y$ near x, $\mathcal{O}_{X,x'}$ is a complete intersection. If XIII.4.1 is true, this hypothesis should be replaceable by the following more general one: for x' as above, $\pi_i^{x'}(X) = 0$ for $i \leq n-1$.

c) At the points y of $Y - \{x\}$ near x, one has

$$\operatorname{prof} \mathcal{O}_{X,y} \geq n$$

(it suffices, for example, that $\operatorname{prof} \mathcal{O}_{X,x} \geq n$*).*

Under these conditions, is the canonical homomorphism

$$\pi_i^x(Y) \to \pi_i^x(X)$$

an isomorphism for $i \leq n-2$ and an epimorphism for $i = n-1$?

Here, finally, is a global variant of XIII.4.2, which should follow from it by considering the affine cone at its vertex and which would generalize the classical Lefschetz theorems:

Problem XIII.4.3. *Let X be a projective analytic space, equipped with an ample invertible Module L; let t be a section of L, and let Y be the zero set of t. Suppose that:* **68**

a) t is a regular section (a hypothesis that may be superfluous).

b) For every $x \in X-Y$, $\mathcal{O}_{X,x}$ is a complete intersection. This should be replaceable by $\pi_i^x(X) = 0$ for $i \leq n-1$.

c) For every $x \in Y$, $\operatorname{prof} \mathcal{O}_{X,x} \geq n$.

Under these conditions, is the homomorphism

$$\pi_i(Y) \to \pi_i(X)$$

an isomorphism for $i \leq n-2$ and an epimorphism for $i = n-1$?

We leave it to the reader to state analogous conjectures of a *cohomological* nature(*), with hypotheses and conclusions concerning local cohomological invariants (with coefficients in a given group). In any event, the key result seems to be XIII.4.2, with hypothesis b) taken in the more general form just indicated, whether one adopts the point of view of homology or that of homotopy.

We have stated these conjectures in the transcendental setting in the hope of interesting topologists in them and convincing them that "Lefschetz"-type questions are far from closed. Of course, now that we are about to possess a good theory of the cohomology of schemes (with finite coefficients), thanks to the recent work of M. Artin, the same questions arise in the setting of schemes, and it is difficult to doubt that they will receive an affirmative answer in the near future(*).

(*) Cf. Exp. XIV for the corresponding results in the theory of schemes.

(*) See the preceding note.

XIII.5 Problems related to local Picard groups

A first fundamental problem, pointed out for the first time by Mumford [5] in a particular **69**
case, is the following. Let A be a complete local ring with residue field k, let $X = \mathrm{Spec}(A)$, and let $U = \mathrm{Spec}(A) - \{a\}$, where a is the maximal ideal of A, i.e. the closed point of $\mathrm{Spec}(A)$. The problem is to construct a strict projective system G of locally algebraic groups G_i over k, and a natural isomorphism

$$\mathrm{Pic}(U) \simeq G(k) \qquad (+)$$

where, of course, we set $G(k) = \varprojlim G_i(k)$. Heuristically, the problem is to "put an algebraic-group structure"(or at least a pro-algebraic one, in a suitable sense) on the group $\mathrm{Pic}(U)$.

It is clear that, as stated, the problem is not sufficiently precise, since the datum of an isomorphism $(+)$ is far from characterizing the pro-object G. If A contains a subfield, again denoted by k, that is a coefficient field, the problem can be made more precise by requiring that, for a varying extension k' of k, there be an isomorphism, functorial in k':

$$\mathrm{Pic}(U') \simeq G(k') \qquad (+')$$

where U' is the open subset analogous to U in $\mathrm{Spec}(A')$, with $A' = A\hat{\otimes}_k k'$. One can proceed analogously even if A has no coefficient field, provided that k is perfect, since one can then construct an A' functorially "by residue-field extension k'/k". Moreover, when A admits a coefficient field, the algebraic structure obtained on $\mathrm{Pic}(U)$ will depend essentially on the choice of that coefficient field (as one sees already for the projective cone over an elliptic curve); it therefore seems necessary to start from a "pro-algebraic ring over k"in the sense of Greenberg [3], in order to define the pro-object G. It is moreover conceivable that, when no coefficient field is given, one obtains only a projective system of *quasi*-algebraic groups in **70**
the sense of Serre, or rather quasi-locally algebraic groups (the groups G_i obtained will not in general be of finite type over k, but only locally of finite type over k). It is even possible that in general one obtains only a still weaker structure on $\mathrm{Pic}(U)$, of the kind encountered by Néron [6] in his theory of degeneration of abelian varieties defined over local fields.

One method of attacking the problem, likewise introduced by Mumford, consists in resolving the singularities of X, i.e. in considering a projective birational morphism $Y \to X$ with Y regular. When U is regular (i.e. a is an isolated singular point), one can often find Y such that $Y_{|}U = V \to U$ is an isomorphism. In that case one therefore has

$$\mathrm{Pic}(U) \simeq \mathrm{Pic}(V) \simeq \mathrm{Pic}(Y)/\Im\mathbf{Z}^I,$$

where I is the set of irreducible components of the fiber Y_a (each of which defines an element of $\mathrm{Pic}(Y)$, since it is a locally principal divisor, by the regularity of Y). On the other hand, using the techniques of Formal Geometry in EGA III 4 and 5, in particular the existence theorem, one finds

$$\mathrm{Pic}(Y) \simeq \varprojlim \mathrm{Pic}(Y_n),$$

where $Y_n = Y \otimes_A A_n$ and $A_n = A/\mathfrak{m}^{n+1}$. When A admits a coefficient field k, the theory of Picard schemes of the projective k-schemes Y_n is available, and hence

$$\mathrm{Pic}(Y_n) \simeq \mathbf{Pic}_{Y_n/k}(k).$$

This therefore gives a construction of a projective system of locally algebraic groups $\mathbf{Pic}_{Y_n/k}/\Im\mathbf{Z}^I$, which is the desired system. (*) In the case considered here, one can moreover see (using the fact that a is an isolated singular point) that the connected components of the

universal-image subgroups in this projective system form an *essentially constant* projective system; thus, in this case, one obtains a locally algebraic group G as a solution of the problem. If one further assumes that A is normal of dimension 2, then a remark of Mumford (stating that the intersection matrix of the components of Y_a in X is negative definite(*)) implies that G is even an algebraic group, i.e. of finite type over k (the number of its connected components being equal, moreover, to the determinant of the intersection matrix just considered). **71**

If, on the contrary, a is not an isolated singularity, examples (with A of dimension 2) show that one obtains a projective system of algebraic groups which does not reduce to a single algebraic group.

Once a good notion of "*local Picard scheme*"were available, one should strengthen the notion of parafactoriality by saying that A is "*geometrically parafactorial*"when not only A, and even $\widehat{A}$, is parafactorial, but the local Picard scheme $G(\widehat{A})$ is the trivial group (which, when the residue field is not algebraically closed, is stronger than saying that G has no rational point over k other than the identity). The necessity of a strengthened notion of parafactoriality becomes apparent on recalling that there exist complete normal local rings of dimension 2 which are factorial but admit finite étale algebras which are not(*). A "*geometrically factorial*"local ring would then be a normal ring A such that all the localizations of dimension ≥ 2 are geometrically parafactorial, or better, such that the localizations of $\widehat{A}$ are parafactorial(*). Of course, it would be interesting to find a "good"definition of these notions, independent of the still undeveloped theory of local Picard schemes.(*)

It is in any event plausible that these notions will be needed if one wishes to obtain statements of the following type: Let A be a "good ring"(for example an algebra of finite type over $\mathbf{Z}$, or over a complete local ring, for example over a field). Let U be the set of $x \in X = \mathrm{Spec}(A)$ such that $\mathcal{O}_{X,x}$ is "geometrically factorial"; is U open? Or again: Let $f\colon X \to Y$ be a flat morphism of finite type with Y locally noetherian, and let U be the set of $x \in X$ such that $\mathcal{O}_{X_{f(x)},x}$ is "geometrically factorial"; is U open, at least under suitable additional hypotheses on f? I doubt that true statements of this type exist with the usual notion of factorial ring. **72**

We have raised here, in a particular case, the question of studying the geometric properties of "varying"local rings, for example the $\mathcal{O}_{X,x}$ as x ranges over a prescheme X. When X is, for example, a scheme of finite type over a field, it is known(*) that there exists on X a projective system of finite algebras $P^n_{X/k}$ (obtained by completing $X \times_k X$ along the diagonal),

(*)*Editor's note.* The question was greatly clarified by the results of Boutot (Boutot J.-F., *Schéma de Picard local*, Lect. Notes in Math., vol. 632, Springer, Berlin, 1978). In particular, if A is a complete (noetherian) local k-algebra of depth ≥ 2 such that $H^2_{\mathfrak{m}}(A)$ has finite dimension over k, the local Picard group is a group scheme locally of finite type over k, whose tangent space at the origin is $H^2_{\mathfrak{m}}(A)$. If A is moreover normal of dimension ≥ 3, Serre's normality criterion XI XI.3.12, together with Corollary V 3.6, ensures the required finiteness and hence the existence of the local Picard scheme. See also (Lipman J., "The Picard group of a scheme over an Artin ring", *Publ. Math. Inst. Hautes Études Sci.* **46** (1976), p. 15–86) for an approach closer to Grothendieck's outline above.

(*)*Editor's note.* Mumford D., "The topology of normal singularities of an algebraic surface and a criterion for simplicity", *Publ. Math. Inst. Hautes Études Sci.* **9** (1961), p. 5–22.

(*)*Editor's note.* Factorial rings with nonfactorial henselization arise naturally in the study of moduli spaces of vector bundles. See, for example, (Drézet J.-M., "Groupe de Picard des variétés de modules de faisceaux semi-stables sur $\mathbf{P}_2$", in *Singularities, representation of algebras, and vector bundles (Lambrecht, 1985)*, Lect. Notes in Math., vol. 1273, Springer, Berlin, 1987, p. 337–362). *Strictly speaking*, Drézet shows that the completion is not factorial, but in fact the proof gives the result for the henselization: the point is that Luna's étale slice theorem (Luna D., "Slices étales", in *Sur les groupes algébriques*, Mém. Soc. math. France, vol. 33, Société mathématique de France, Paris, 1973, p. 81–105) describes the local ring of a quotient in the sense of invariant theory near a semistable point, locally for the étale topology.

(*)For a more flexible notion of "geometrically factorial"local ring, cf. *Comments*, page 127.

(*)*Editor's note.* See page 127: a local ring is geometrically factorial (resp. parafactorial) if its strict henselization is factorial (resp. parafactorial).

(*)*Editor's note.* See EGA IV.16.

whose fiber at every point $x \in X$ rational over k is isomorphic to the projective system of the $\mathcal{O}_{X,x}/\mathfrak{m}_X^{n+1}$. It is then natural to relate the study of the completions of the local rings $\mathcal{O}_{X,x}$, for varying x, to that of the "algebraic family of complete local rings"given by the P^n, noting that for every $x \in X$ (rational over k or not), one obtains a complete local ring

$$P^\infty(x) = \varprojlim P^n(x)$$

(where $P^n(x)$ = reduced fiber $P^n \otimes_{\mathcal{O}_{X,x}} k(x)$). Of particular interest, for example, will be the complete ring thus associated with the generic point, and one expects its algebro-geometric properties (expressed, for example, by its local Picard, homotopy, or homology groups) to be essentially those of the completions $\widehat{\mathcal{O}}_{X,x}$ for x in a suitable dense open subset U.

More generally, one may undertake the simultaneous study of the complete local rings obtained in this way from an adic projective system (P_n) of finite algebras over a given scheme X. It is plausible that, under certain regularity conditions (such as flatness of the P_n), the **73** local homotopy groups arise from a projective system of finite group schemes over X, and that analogous results hold for the local Picard groups. As regards the latter, a first interesting case deserving investigation is that in which one starts with an algebraic surface X having singular curves, and seeks to study the local Picard schemes at varying points of these curves in terms of a suitable pro-group scheme defined over the singular locus.

XIII.6 Comments(*)

The viewpoint of the "étale cohomology"of schemes, and recent progress in this theory, leads **74** us both to make some of the problems posed more precise and to broaden them. For the notion of "topology"and of the "étale topology of a scheme", I refer to M. Artin, *Grothendieck Topologies*, Harvard University 1962 (mimeographed notes)(*).

By providing a finer notion of localization than that furnished by the traditional "Zariski topology", this theory gives particular importance to *strictly local rings* , i.e. henselian local rings with separably closed residue field. For every local ring A with residue field k, and every separable closure k' of k, one can find a local homomorphism from A to a strictly local ring A', the *strict local closure* of A, with residue field k', having an evident universal property. The ring A' is henselian and flat over A, and $A' \otimes_A k \simeq k'$; it is noetherian if and only if A is. (Cf. *loc. cit.* Chap. III, section 4)(*). If X is a prescheme, x a point of X, and x' a point above x which is the spectrum of a separable closure k' of $k = k(x)$, one is led to define the *strictly local ring* of X at x', $\mathcal{O}'_{X,x'}$, as the strict local closure of the usual local ring $\mathcal{O}_{X,x}$, relative to the residue-field extension k'/k. It is the *strictly* local rings at the "geometric"points of X which, from the viewpoint of the étale topology, are meant to reflect the local properties of the prescheme X. They also play, in many respects, the role formerly assigned to the *completions* of the local rings of X (say, at points with algebraically closed residue field), while remaining "closer"to X and allowing an easier passage to "neighboring points".

One should therefore reconsider for noetherian henselian local rings (resp. noetherian strictly local rings) many questions generally posed for complete local rings (possibly restricted to those having an algebraically closed residue field). Thus the topological problems raised **75** in n$^{\text{os}}$ XIII.2 and XIII.3 arise more generally for strictly local rings. One can moreover state conjecturally, for "good"strictly local rings, certain simple-connectedness and acyclicity properties for the geometric fibers of the canonical morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$; these would show that, for many properties of a "topological"nature, it amounts to the same thing to

(*)Written in March 1963.
(*)Or preferably to SGA 4.
(*)Or EGA IV 18.8.

prove them for the ring A or for its completion $\widehat{A}$. Certain results already obtained in this direction(*) give reason to hope that complete results in this direction will soon be available.

The notion of étale localization provides a seemingly reasonable definition of a "*geometrically parafactorial*"or "*geometrically factorial*"local ring (the need for which was indicated in no. XIII.5, p. 125): this will mean a local ring whose strict local closure is parafactorial, resp. factorial. Hypotheses of this nature arise naturally in the study of the étale cohomology of preschemes(*). Thus, if X is a locally noetherian prescheme whose strictly local rings are factorial (i.e. whose ordinary local rings are "geometrically factorial"), one proves that the $\mathrm{H}^i(X_{\text{ét}}, \mathbf{G}_m)$ are torsion groups for $i \geq 2$ (which sometimes makes it possible to express these groups in terms of cohomology groups with coefficients in the groups $\boldsymbol{\mu}_n$ of n-th roots of unity), and if X is integral with fraction field K, the natural homomorphism $\mathrm{H}^2(X_{\text{ét}}, \mathbf{G}_m) \to \mathrm{H}^2(K, \mathbf{G}_m) = \mathrm{Br}(K)$ is injective(*); examples show that these conclusions may fail, even for local X, if one assumes only that X is factorial rather than geometrically factorial(*).

As regards the problems of local and global Lefschetz type raised in XIII.3.4, and their analogues in scheme theory, the homological version of these questions has been considerably clarified: everything follows formally from three general theorems. The first concerns the cohomological dimension of certain affine schemes (resp. Stein spaces), such as affine schemes **76**
X of finite type over an algebraically closed field: their cohomological dimension is $\leq \dim X$ ("affine Lefschetz theorem")(*); the second is a duality theorem for the cohomology (with discrete coefficients) of a projective morphism(*); and the last is a *local duality* theorem of an analogous nature(*). In Algebraic Geometry, only the last of these has not been proved at the time of writing (it is, however, proved in characteristic 0, using resolution of singularities by Hironaka). Moreover, in the transcendental setting, global and local duality are already available, having recently been proved by Verdier(*). Let us merely indicate that, in the statements of the homological versions of Problems XIII.4.2 and XIII.4.3 (which henceforth deserve the name of conjectures), the "at infinity"conditions a) and c) are certainly superfluous; only the *local cohomological structure* of $X - Y$ is important, and one may assume, for example, that it is locally a complete intersection of dimension $\geq n$. Moreover, in XIII.4.3, say, the fact that Y is a hyperplane section should play no role and should be replaceable by the sole

(*)Cf. M. Artin in SGA 4 XIX

(*)*Editor's note.* See, for example, (Strano R., "The Brauer group of a scheme", *Ann. Mat. Pura Appl. (4)* **121** (1979), p. 157–169), where the hypothesis that the local rings of a scheme X are geometrically parafactorial sometimes makes it possible to prove that the Brauer groups of X (computed in terms of Azumaya algebras) coincide with the cohomological Brauer group of X.

(*)*Editor's note.* The connection between the Brauer group and the Picard group is a close one. In this connection, let us cite the following results of Saito (Saito S., "Arithmetic on two-dimensional local rings", *Invent. Math.* **85** (1986), no. 2, p. 379–414) in the case of surfaces, the first local and the other global. Let A be an excellent normal henselian local ring of dimension 2 with *finite* residue field, and let X be the complement of the closed point in $\mathrm{Spec}(A)$. Then there is a perfect duality of *torsion* groups $\mathrm{Pic}(X) \times \mathrm{Br}(X) \to \mathbf{Q}/\mathbf{Z}$ — by the Brauer group of X we mean the *cohomological* Brauer group $\mathrm{Br}(X) = H^2_{\text{ét}}(X, G_m)$. In the global case one has the following generalization of a result of Lichtenbaum (Lichtenbaum S., "Duality theorems for curves over p-adic fields", *Invent. Math.* **7** (1969), p. 120–136): let k be the fraction field of a complete discrete valuation ring $\mathcal{O}$ with finite residue field, and let X be a projective, smooth, and geometrically complete curve over k. The group $\mathrm{Pic}^0(X)$ is endowed with the topology induced by the adic topology of k, and $\mathrm{Pic}(X)$ is the topological group for which $\mathrm{Pic}^0(X)$ is an open subgroup. Then there is a perfect duality of topological groups $\mathrm{Pic}(X) \times \mathrm{Br}(X) \to \mathbf{Q}/\mathbf{Z}$. Note that this statement, which concerns curves, is of course proved by considering a regular (proper and flat) model of X over $\mathcal{O}$: it is a result about surfaces.

(*)Cf. A. Grothendieck, le groupe de Brauer II (Séminaire Bourbaki no. 297, Nov. 1965), notably 1.8 and 1.11 b.

(*)Cf. SGA 4 XIV.

(*)Cf. SGA 4 XVIII.

(*)Cf. SGA 5 I.

(*)*Editor's note.* See Verdier J.-L., "Dualité dans la cohomologie des espaces localement compacts", in *Séminaire Bourbaki*, vol. 9, Société mathématique de France, Paris, 1995, Exp. 300, p. 337–349.

hypothesis that X is compact and $X - Y$ is Stein (i.e. in Algebraic Geometry, X is proper over k and $X - Y$ is affine; as we said, the homological version of this conjecture is proved for algebraic spaces over the field $\mathbf{C}$)$^{(*)}$.

In the definition (p. 122) of the $\pi_i^x(X)$, one must assume $i \geq 2$. For $i = 0, 1$, there is no reasonable definition of the $\pi_i^x(X)$; they should be replaced by $\mathrm{H}_0^x(X)$ and $\mathrm{H}_1^x(X)$, defined respectively as the cokernel and kernel in the natural homomorphism

$$\varprojlim \mathrm{H}_0(U - (x), \mathbf{Z}) \to \varprojlim \mathrm{H}_0(U, \mathbf{Z}).$$

At a pinch, and for convenience of formulation, one may set $\pi_i^x(X) = \mathrm{H}_i^x(X)$ for $i \leq 1$; otherwise, subsequent assertions concerning the π_i^x must be supplemented by the corresponding assertions for $\mathrm{H}_0^x, \mathrm{H}_1^x$. If x is an isolated point of X, one should set $\pi_i^x(X) = 0$ for $i \neq 0$ and $\pi_0^x(X) = \mathrm{H}_0^x(X) = \mathbf{Z}$.

The assertion that the $\pi_i^x(U, f)$ are mutually isomorphic is true only when X is not **77** disconnected by x in a neighborhood of x, i.e. if $\pi_i^x(X) = 0$ for $i = 0, 1$. In the general case $\pi_i^x(X)$ can designate only a *family* of groups, not necessarily mutually isomorphic; nevertheless the notation $\pi_i^x(X) = 0$ retains an evident meaning.

Page 122, where I predict that the vanishing of the local homotopy invariants $\pi_i^x(X)$ for $x \in Y$, $i \leq n$, should imply that $\pi_i(X - Y) \to \pi_i(X)$ is bijective for $i < n - d$ and surjective for $i = n - d$, one must be cautious, since in the present context (as in Algebraic Geometry) one does not have at one's disposal "general"points at which the local conditions would also have to apply. For this reason it will doubtless be necessary to use relative local homotopy invariants

$$\pi_i^Y(X, f) = \pi_i^Y(X, x) = \varprojlim_U \pi_{i-1}(U - U \cap Y, f(t)) \qquad \text{for } i \geq 2,$$

(with an *ad hoc* definition as above for $i = 0, 1$), where Y is a closed subset of X; or else to compensate for the absence of general points by expressing the hypotheses on X in terms of properties, topological in nature (for the étale topology), of the spectra of the local rings of X, which makes it possible to recover general points. The same reservation applies to the generalization of Conjectures XIII.4.2 and XIII.4.3 to the case in which $X - Y$ is not assumed to be locally a complete intersection, a generalization suggested in the statements of conditions b) of those conjectures.

To formulate the expurgated versions of Conjectures XIII.4.2 and XIII.4.3 suggested by the results mentioned above, we make the following

Definition XIII.6.1. Let X be a topological space, let Y be a locally closed subset of X, and let n be an integer. One says that X has *homotopical depth* $\geq n$ along Y, and writes $\operatorname{prof} \operatorname{hpt}_Y(X) \geq n$, if for every $x \in Y$ one has $\pi_i^Y(X, x) = 0$ for $i < n$.

It should be equivalent to say that, for every open subset X' of X and every $x \in X' \cap U = U'$ **78** (where $U = X - Y$), the canonical homomorphism

$$\pi_i(U', x) \to \pi_i(X', x)$$

is an isomorphism for $i < n - 1$ and a monomorphism for $i = n - 1$$^{(*)}$.

Definition XIII.6.2. Let X be a complex analytic space and let n be an integer. One says that the *rectified homotopical depth* of X is $n$$^{(*)}$ if, for every locally closed analytic subset Y of X, one has

$$\operatorname{prof} \operatorname{htp}_Y(X) \geq n - \dim Y \qquad (x)$$

$^{(*)}$Cf. Exp. XIV.

$^{(*)}$*Editor's note.* When the pair (X, Y) is moreover polyhedral, this equivalence holds; cf. (Eyral C., "Profondeur homotopique et conjecture de Grothendieck", *Ann. Sci. Éc. Norm. Sup. (4)* **33** (2000), no. 6, p. 823–836).

$^{(*)}$In the first edition of these notes, we used the term "true homotopical depth". In the present version we follow EGA IV 10.8.1.

(where, of course, $\dim Y$ denotes the *complex* dimension of Y).

It should be equivalent to say that, for every irreducible locally closed analytic subset Y of X, there exists a closed analytic subset Z of Y, of dimension $< \dim Y$, such that relation (x) holds with $Y - Z$ in place of Y. This would make it possible, for example, to restrict in Definition XIII.6.2 to the case where Y is nonsingular. (*)

The following conjecture, purely topological in nature, is in the spirit of a "*local Hurewicz theorem*".

Conjecture A ("local Hurewicz theorem"(*)). *Let X be a topological space and Y a locally closed subset, subject if necessary to "smoothness"conditions such as local triangulability of the pair (X, Y), and let n be an integer ≥ 3. In order that* $\operatorname{prof} \operatorname{htp}_Y(X) \geq n$, *it is necessary and sufficient that* **79**

$$\mathrm{H}^i_Y(\mathbf{Z}_X) = 0 \quad \textit{for } i < n$$

(one then says that X has cohomological depth $\geq n$ *along Y), and that the local fundamental groups*

$$\pi_2^Y(X, x) = \varprojlim_{U \ni x} \pi_1(U - U \cap Y)$$

vanish (one then says that X is "pure"along Y).

Note that if X is an analytic space, Y an analytic subspace, and X is pure along Y, then for every $x \in Y$ the local ring $\mathcal{O}_{X,x}$, as well as its localizations with respect to prime ideals containing the ideal defining the germ Y at x (i.e. in the inverse image Y_x of Y under $\operatorname{Spec}(\mathcal{O}_{X,x}) = X_x \to X$), are pure in the sense of Exp. X; it seems plausible that the converse is also true. Analogous remarks apply to cohomological depth, it being understood that one works with the étale topology on the $\operatorname{Spec}(\mathcal{O}_{X,x})$.

Conjecture XIII.4.1 then generalizes to

Conjecture B ("Purity"(*)). *Let E be an analytic space and X an analytic subset of E. Suppose that E is nonsingular of dimension N at $x \in X$, and that X can be described by p analytic equations in a neighborhood of every point. Then the rectified homotopical depth of X is $\geq N - p$.*

In particular, a local complete intersection of dimension n at every point would have rectified homotopical depth $\geq n$, which is precisely Conjecture XIII.4.1.

Conjectures XIII.4.2 and XIII.4.3 generalize respectively to:

Conjecture C ("Local Lefschetz"(*)). *Let X be an analytic space, let Y be a closed analytic subset, and let x be a point of Y. Suppose that $X - Y$ is Stein in a neighborhood of x (for example, that Y is defined by one equation at x), and that $X - Y$ has rectified homotopical depth $\geq n$ in a neighborhood of x (for example, that at every point of $X - Y$ near x it is a complete intersection of dimension $\geq n$; cf. Conjecture B). Then the canonical homomorphism*

$$\pi_i^x(Y) \to \pi_i^x(X)$$

(*) *Editor's note.* All the conjectures that follow, suitably rectified, if I may say so, became theorems through the work of Hamm and Lê Dũng Tráng (Hamm H.A. & Lê Dũng Tráng, "Rectified homotopical depth and Grothendieck conjectures", in *The Grothendieck Festschrift, Vol. II*, Progr. Math., vol. 87, Birkhäuser, Boston, 1990, p. 311–351), cited as [HL] below. As for the two conjecturally equivalent definitions of rectified depth, they are in fact equivalent to a third, expressed in terms of Whitney stratification (cf. *loc. cit.*, Theorem 1.4).

(*) *Editor's note.* As observed in [HL], Example 3.1.3, this conjecture is already false for $X = \{\boldsymbol{z} \in \mathbf{C}^n \mid z_1^2 + z_2^3 + \cdots + z_n^3 = 0\}, n \geq 4$, and Y reduced to the origin. Suitably modified, however, it is true (Theorem 3.1.4 of *loc. cit.*).

(*) *Editor's note.* This conjecture is proved, even when E is singular, in [HL]: it is Theorem 3.2.1.

(*) *Editor's note.* This conjecture is proved in [HL], indeed even in the strong form of the remark that follows; cf. Theorem 3.3.1 of *loc. cit.*.

is an isomorphism for $i < n-1$ and an epimorphism for $i = n-1$.

Conjecture D ("Global Lefschetz"(*)). *Let X be a* compact *analytic space and let* **80**
Y be an analytic subspace of X such that $U = X - Y$ is Stein and has rectified homotopical depth $\geq n$ (for example, is a complete intersection of dimension $\geq n$ at every point). Then the canonical homomorphism

$$\pi_i(Y) \to \pi_i(X)$$

is an isomorphism for $i < n-1$ and an epimorphism for $i = n-1$.

Remark. When, in the statements C and D, the hypothesis that $X - Y$ is Stein is replaced by the hypothesis that $X - Y$ is a union of $c+1$ Stein open subsets (which plays the role of a topological "concavity"hypothesis), the conclusions should simply be modified by replacing n there by $n - c$.(*)

Finally, let us make explicit, in the "global case"D, the conjecture concerning the fundamental group (obtained by taking $n = 3$):

Conjecture D′ (Global Lefschetz for the fundamental group(*)). *Let X be a compact analytic space over the field of complex numbers and let Y be a closed analytic subset such that $U = X - Y$ is Stein. Suppose moreover that the following conditions hold:*

(i) For every $x \in U$, the local fundamental group $\pi_2^x(X, x)$ vanishes (i.e. X is "pure at *x"), or merely the local ring $\mathcal{O}_{X,x}$ is pure.*

(ii) The local rings at the points of U are "connected in dimension ≥ 2".

(iii) The local rings at the points of U have dimension ≥ 3.

Under these conditions, for every $x \in Y$, the homomorphism

$$\pi_1(Y, x) \to \pi_1(X, x)$$

is an isomorphism (and $\pi_2(Y, x) \to \pi_2(X, x)$ is an epimorphism). **81**

Note that the local conditions (i), (ii), and (iii) on U hold if U is locally a complete intersection of dimension ≥ 3. From the viewpoint of Algebraic Geometry (when U arises from a scheme, again denoted by U), conditions (i) through (iii) correspond to hypotheses on the local invariants $\pi_i^x(U)$, namely $\pi_i^x(U) = 0$ for $i < 3 - \deg \operatorname{tr} k(x)/k$, at points x for which one has respectively $\deg \operatorname{tr} k(x)/k = 0, 1, 2$. The global condition on U (U Stein) is satisfied if X is projective and Y is a hyperplane section.

(*) *Editor's note.* This conjecture is again proved in [HL], indeed even in the strong form of the remark that follows; cf. Theorem 3.4.1 of *loc. cit.*.

(*) *Editor's note.* Finally, let us mention the following result of Fulton, to be compared with the Fulton–Hansen result cited in the editor's note ((*)) on page 107: let X and H be closed subschemes of $\mathbf{P}^m_{\mathbf{C}}$, let n be the dimension of X, and let d be the codimension of H. Then the map

$$\pi_i(X, X \cap H) \to \pi_i(\mathbf{P}^n_{\mathbf{C}}, H)$$

is an isomorphism if $i \leq n-d$ and is surjective if $i = n-d-1$; see (Fulton W., "Connectivity and its applications in algebraic geometry", in *Algebraic geometry (Chicago, Ill., 1980)*, Lect. Notes in Math., vol. 862, Springer, Berlin–New York, 1981, p. 26–92).

(*) *Editor's note.* This conjecture is proved in [HL]; cf. Theorem 3.5.1 of *loc. cit.*.

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Exposé XIV

Depth and Lefschetz theorems in étale cohomology

by Mrs M. Raynaud(*)

In no. XIV.1, we define a notion of "*étale depth*"that is the analogue in étale cohomology **83**
of the notion of depth studied in III, in the cohomology of coherent sheaves. After a technical part, we prove "Lefschetz theorems"in no. XIV.4, the central theorem being XIV.4.2. Let X be a scheme, Y a closed subset of X, U the complementary open subset $X - Y$, and F an abelian sheaf on X for the étale topology; in general, the purpose of Lefschetz theorems is to show that, if F satisfies certain local conditions on U, expressible in terms of étale depth at the points of U, then, under certain additional conditions of a global nature on U (for example, U *affine*), the natural map of étale cohomology groups

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, F_{|Y})$$

is an isomorphism for values $i < n$, where n is a certain specified integer. By taking F to be a constant sheaf, one thereby obtains conditions for $\pi_0(X)$ to equal $\pi_0(Y)$ and conditions for the abelianized fundamental groups of X and Y to be the same. In no. XIV.5, the introduction of a notion of "geometric depth"makes it possible to give useful special cases of the Lefschetz theorems (XIV.5.7). Finally, in no. XIV.6, we mention some conjectures, concerning in particular "noncommutative"variants of the theorems obtained.

XIV.1 Cohomological and homotopical depth

XIV.1.0

Fix the following notation. Let X be a scheme(*), Y a closed subset of X, U the complementary **84**
open subset, and $i : Y = X - U \to X$ the canonical immersion. Let $\underline{\Gamma}_Y$ be the functor that associates to an abelian sheaf on X the "sheaf of sections with support in Y", that is, $\underline{\Gamma}_Y = i_* i^!$ (cf. SGA 4 IV 3.8 and VIII 6.6), and let Γ_Y be the functor $\Gamma \cdot \underline{\Gamma}_Y$ (where Γ is the "global sections"functor). Consider the derived category $D^+(X)$ and the derived functor $\mathrm{R}\underline{\Gamma}_Y$ (resp. $\mathrm{R}\Gamma_Y$) of $\underline{\Gamma}_Y$ (resp. of Γ_Y) (cf. [3]). Given a complex of abelian sheaves F on X, bounded below, it may be regarded as an element of $D^+(X)$; we then denote by $\underline{H}^p_Y(F)$ the p-th cohomology sheaf of $\mathrm{R}\underline{\Gamma}_Y(F)$ and by $\mathrm{H}^p_Y(X, F)$ the p-th cohomology group of $\mathrm{R}\Gamma_Y(F)$. The results of (SGA 4 V 4.3 and 4.4) extend trivially to $\underline{H}^p_Y(F)$ and $\mathrm{H}^p_Y(X, F)$.

(*)Based on unpublished notes by A. Grothendieck.

(*)In accordance with the new terminology (cf. the reissue of EGA I), here we shall call "scheme"what was formerly called "prescheme", and "separated scheme"what was called "scheme".

Proposition XIV.1.1. *Let X be a scheme, Y a closed subset of X, U the complementary open subset, and $i : U \to X$ the canonical immersion. Let F denote either a sheaf of sets on X, a sheaf of groups on X, or a complex of abelian sheaves on X, bounded below. Fix the following notation: if $X' \to X$ is a morphism, U' and F' denote the inverse images of U and F on X'; moreover, if $\bar{y}$ is a geometric point of X, $\bar{X}$ denotes the strict localization of X at $\bar{y}$, and $\bar{U}$ and $\bar{F}$ the inverse images of U and F in $\bar{X}$.*

1°) *Let F be a sheaf of sets on X and n an integer ≤ 2; then the following conditions are* **85**
equivalent:

(i) *The canonical morphism*

$$F \to i_* i^* F$$

is injective if $n \geq 1$, and bijective if $n \geq 2$.

(ii) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^0(X', F') \to \mathrm{H}^0(U', F')$$

is injective if $n \geq 1$, and bijective if $n \geq 2$.

Suppose in addition that the open subset U is retrocompact in X; then the preceding conditions are equivalent to the following one:

(iii) *For every geometric point $\bar{y}$ of Y, the canonical morphism*

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{U}, \bar{F})$$

is injective if $n \geq 1$, and bijective if $n \geq 2$.

2°) *Let F be a sheaf of groups on X and n an integer ≤ 3; then the following conditions are equivalent:*

(i) *The canonical morphism*

$$F \to i_* i^* F$$

is injective if $n \geq 1$, bijective if $n \geq 2$, and if $n \geq 3$, in addition to the preceding conditions, the sheaf of pointed sets $\mathrm{R}^1 i_*(i^* F)$ is zero.

(ii) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^0(X', F') \to \mathrm{H}^0(U', F')$$

is injective if $n \geq 1$, bijective if $n \geq 2$, and moreover the canonical morphism **86**

$$\mathrm{H}^1(X', F') \to \mathrm{H}^1(U', F')$$

is injective if $n \geq 2$, and bijective if $n \geq 3$.

(ii bis) *The same as* (ii)*, except in the case $n = 2$, where it is only assumed that $\mathrm{H}^0(X', F') \to \mathrm{H}^0(U', F')$ is bijective.*

Suppose in addition that U is retrocompact in X; then the preceding conditions are also equivalent to the following one:

(iii) *For every geometric point $\bar{y}$ of Y, the canonical morphism*

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{U}, \bar{F})$$

is injective if $n \geq 1$, and bijective if $n \geq 2$; finally, if $n \geq 3$, in addition to the preceding conditions, $\mathrm{H}^1(\bar{U}, \bar{F})$ is zero.

3°) *Let F be a complex of abelian sheaves, bounded below, and let n be an integer; then the following conditions are equivalent:*

(i) *We have $\underline{H}^p_Y(F) = 0$ for $p < n$ (cf. 1.0).*

(ii) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^p(X', F') \to \mathrm{H}^p(U', F')$$

is bijective for $p < n-1$, and injective for $p = n-1$.

Suppose that U is retrocompact in X; then the preceding conditions are also equivalent to the following one:

(iii) *For every geometric point $\bar{y}$ of Y, the canonical morphism*

$$\mathrm{H}^p(\bar{X}, \bar{F}) \to \mathrm{H}^p(\bar{U}, \bar{F})$$

is bijective for $p < n-1$, and injective for $p = n-1$.

In the case where F is an abelian sheaf and $n \geq 2$, conditions (i) *and* (ii) *are also equivalent* **87**
to the following one:

(ii bis) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^p(X', F') \to \mathrm{H}^p(U', F')$$

is bijective for $p < n-1$.

Proof. 1°) It is clear that (i) $\iff$ (ii). Let us show that, if U is quasi-compact in X, (i) $\iff$ (iii). Indeed, (i) is equivalent to saying that, for every geometric point $\bar{y}$ of X, the morphism $F_{\bar{y}} \to (i_* i^* F)_{\bar{y}}$ is injective if $n \leq 1$ and bijective if $n \leq 2$ (SGA 4 VIII 3.6). Since this morphism is in any case bijective when $\bar{y}$ is a geometric point of U, it is enough to consider the geometric points $\bar{y}$ of Y. Now it follows from the fact that i is quasi-compact and from (SGA 4 VIII 5.3) that the morphism

$$F_{\bar{y}} \to (i_* i^* F)_{\bar{y}}$$

is canonically identified with the morphism

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{U}, \bar{F}),$$

which proves the equivalence of (i) and (iii).

2°) (i) $\Longrightarrow$ (ii). The assertions concerning H^0 follow from 1°). Let i' be the canonical immersion of U' into X'; the assertions concerning H^1 follow from the exact sequence (SGA 4 XII 3.2)

$$0 \to \mathrm{H}^1(X', i'_* i'^* F') \to \mathrm{H}^1(U', F') \to \mathrm{H}^0(X', \mathrm{R}^1 i'_* (i'^* F')).$$

(ii bis) $\Longrightarrow$ (i) . By 1°), it is enough to show that, for $n \geq 3$, one has $\mathrm{R}^1 i_*(i^* F) = 0$. Now **88**
$\mathrm{R}^1 i_*(i^* F)$ is the sheaf associated with the presheaf $X' \longmapsto \mathrm{H}^1(U', F')$, that is, by hypothesis, the sheaf associated with the presheaf $X' \longmapsto \mathrm{H}^1(X', F')$, and the latter is zero.

(i) $\iff$ (iii). Taking 1°) into account, the only thing left to verify is that the relation $\mathrm{R}^1 i_*(i^* F) = 0$ is equivalent to the condition that $\mathrm{H}^1(\bar{U}, \bar{F}) = 0$ for every geometric point $\bar{y}$ of Y. Since $\mathrm{R}^1 i_*(i^* F))$ is zero outside Y, saying that $\mathrm{R}^1 i_*(i^* F) = 0$ is equivalent to saying that $(\mathrm{R}^1 i_*(i^* F))_{\bar{y}} = 0$ for every geometric point $\bar{y}$ of Y. It is then enough to note that, since i is quasi-compact, one has $(\mathrm{R}^1 i_*(i^* F))_{\bar{y}} = \mathrm{H}^1(\bar{U}, \bar{F})$ (SGA 4 VIII 5.3).

3°) (i) $\Longrightarrow$ (ii). Let X' be a scheme ètale over X; one has the exact sequence (SGA 4 V 4.5)

$$\to \mathrm{H}^p_{Y'}(X', F') \to H^p(X', F') \to H^p(U', F') \to; \qquad (*)$$

hence (ii) is equivalent to $H^p_{Y'}(X', F') = 0$ for $p < n$ and for every scheme X' ètale over X. Consider now the spectral sequence

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(X', \underline{H}^q_Y(F)) \Longrightarrow \mathrm{H}^*_{Y'}(X', F');$$

by hypothesis, $\underline{H}^q_Y(F) = 0$ for $q < n$, whence $\mathrm{E}_2^{pq} = 0$ for $p + q < n$ and consequently $\mathrm{H}^p_{Y'}(X', F') = 0$ for $p < n$.

(ii) $\Longrightarrow$ (i). The sheaf $\underline{H}^p_Y(F)$ is associated with the presheaf $X' \longmapsto \mathrm{H}^p_{Y'}(X', F')$; since we have already observed that (ii) is equivalent to the relation $\mathrm{H}^p_{Y'}(X', F') = 0$ for $p < n$ and for every scheme X' ètale over X, it follows that $\underline{H}^p_Y(F) = 0$ for $p < n$.

(i) $\Longleftrightarrow$ (iii). The sheaves $\underline{H}^p_Y(F)$ are concentrated on Y; consequently, saying that $\underline{H}^p_Y(F) = 0$ is equivalent to saying that, for every geometric point $\bar{y}$ of Y, the fiber $(\underline{H}^p_Y(F))_{\bar{y}}$ **89**
is zero. Since i is quasi-compact, (SGA 4 VIII 5.2) gives $(\underline{H}^p_Y(F))_{\bar{y}} = \underline{H}^p_{\bar{Y}}(\bar{X}, \bar{F})$. The equivalence of (i) and (iii) follows, in view of the analogue on $\bar{X}$ of the exact sequence $(*)$.

(ii bis) $\Longrightarrow$ (ii) when F is an abelian sheaf. The only thing left to show is that $\underline{H}^{n-1}_Y(F) = 0$. Now, for $n > 2$, the sheaf $\underline{H}^{n-1}_Y(F)$ is associated with the presheaf $X' \longmapsto \mathrm{H}^{n-2}(U', F') = \mathrm{H}^{n-2}(X', F')$ and is therefore zero. The case $n = 2$ follows from the fact that $\underline{H}^1_Y(F)$ is the cokernel of the morphism $F \to i_* i^* F$.

Definition XIV.1.2. The notation is that of XIV.1.1. One says that F has ètale Y-depth $\geq n$ and writes

$$\mathrm{prof}_Y(F) \geq n$$

if F satisfies the equivalent conditions (i) and (ii) of XIV.1.1. If F is a complex of abelian sheaves, the ètale Y-depth of F is the supremum of the n for which $\mathrm{prof}_Y(F) \geq n$; the same notation will be used if F is a sheaf of sets resp. of not necessarily commutative groups (so that one then has $0 \leq \mathrm{prof}_Y(F) \leq 2$, resp. $0 \leq \mathrm{prof}_Y(F) \leq 3$, when the context does not make clear which of the three variants considered here is being used).

If $\mathbf{L}$ is a set of prime numbers, one says that the ètale Y-depth of X for $\mathbf{L}$ is $\geq n$ and writes

$$\mathrm{prof}^{\mathbf{L}}_Y(X) \geq n$$

if, for every constant sheaf of the form $\mathbf{Z}/\ell\mathbf{Z}$ with $\ell \in \mathbf{L}$, one has $\mathrm{prof}_Y(\mathbf{Z}/\ell\mathbf{Z}) \geq n$. The ètale **90**
Y-depth of X for $\mathbf{L}$ is defined in the evident way. If $\mathbf{L} = \mathbf{P}$, the set of all prime numbers, and there is no risk of confusion with the notation of (EGA IV 5.7.1) (which concerns the case $F = \mathcal{O}_X$), $\mathbf{L}$ is omitted from the notation; otherwise one writes $\mathrm{prof}\,\text{ét}(X)$.

Finally, one says that X has homotopical Y-depth ≥ 3 for $\mathbf{L}$ and writes

$$\mathrm{prof\,hop}^{\mathbf{L}}_Y(X) \geq 3$$

if, for every constant finite sheaf F of $\mathbf{L}$-groups on X, one has $\mathrm{prof}_Y(F) \geq 3$. If $\mathbf{L} = \mathbf{P}$, $\mathbf{L}$ is omitted from the notation.

Corollary XIV.1.3. *Under the conditions of XIV.1.1, if* $\mathrm{prof}_Y(F) \geq n$*, then, for every closed subset* Z *of* Y*, one has*

$$\mathrm{prof}_Z(F) \geq n.$$

For example, let us give the argument when F is a complex of abelian sheaves bounded below. We use XIV.1.1 3°) (ii). Let $V = X - Z$ and, for every integer p, consider the sequence of morphisms

$$\mathrm{H}^p(X, F) \xrightarrow{f} \mathrm{H}^p(V, F) \xrightarrow{g} \mathrm{H}^p(U, F).$$

By hypothesis, g and $f \circ g$ are bijective for $p < n - 1$ and injective for $p = n - 1$; hence the same is true of f. Since the argument remains valid when X is replaced by a scheme X' ètale over X, this proves XIV.1.3.

Corollary XIV.1.4. *The notation is that of XIV.1.1* 2°)*. If* X' *is a scheme over* X*, let* Φ' **91**
denote the functor that assigns to an ètale covering of X' *its restriction to* U'*, and let* $\Phi'_{F'}$

denote the functor that assigns to a torsor(*) *under* F' *its restriction to* U'*. Then the following conditions are equivalent:*

(i) *One has* $\mathrm{prof}_Y(F) \geq 1$ *(resp.* $\mathrm{prof}_Y(F) \geq 2$*, resp.* $\mathrm{prof}_Y(F) \geq 3$*).*

(ii) *For every scheme* X' *ètale over* X*, the functor* $\Phi'_{F'}$ *is faithful (resp. fully faithful, resp. an equivalence of categories).*

In particular, one has $\mathrm{prof}_Y(X) \geq 1$ *(resp.* $\mathrm{prof}_Y(X) \geq 2$*, resp.* $\mathrm{prof\,hop}_Y(X) \geq 3$*) if and only if the functor* Φ' *is faithful (resp. fully faithful, resp. an equivalence of categories).*

This follows from XIV.1.1 2°) (ii), in view of the interpretation of $\mathrm{H}^1(X', F')$ as the set of classes (up to isomorphism) of torsors under F' (SGA 4 VII 2), and of ètale coverings Z of degree n of a scheme as being associated with principal Galois coverings whose group is the symmetric group $\mathfrak{S}_n$: to Z one associates the covering $\underline{\mathrm{Isom}}_X(Z_0, Z)$, where Z_0 is the trivial covering of X of degree n.

Corollary XIV.1.5. *Under the conditions of XIV.1.1* 3°)*, suppose that* $\mathrm{prof}_Y(F) \geq n$*; then*

$$\mathrm{H}^n_Y(X, F) \simeq \mathrm{H}^0(X, \underline{H}^n_Y(F)).$$

The corollary follows from the spectral sequence

$$\mathrm{E}_2^{pq} = H^p(X, \underline{H}^q_Y(F)) \Longrightarrow H^*_Y(X, F).$$

Indeed, by hypothesis $\mathrm{E}_2^{pq} = 0$ for $q < n$, and hence

$$\mathrm{H}^n_Y(X, F) = \mathrm{E}_2^{0n} = \mathrm{H}^0(X, \underline{H}^n_Y(F)).$$

Remarks XIV.1.6. a) The notion of Y-depth, in the form of the equivalent conditions (i) and **92**
(ii) of XIV.1.1, makes sense on any site. In the particular case in which X is a locally Noetherian scheme equipped with the Zariski topology and F is a sheaf of coherent $\mathcal{O}_X$-modules, one recovers the usual notion of Y-depth as the infimum of the depths at the points of Y (III).

b) For $n \leq 2$, the notion of ètale Y-depth of X is independent of $\mathbf{L}$. For $n = 1$, it simply means that U is dense in X. Indeed, this condition is necessary for $\mathrm{prof}_Y(F) \geq 1$, and it is also sufficient because one may assume that X is reduced, in which case the condition that U be dense in X is preserved by ètale base change (EGA IV 11.10.5) (ii) b)). If U is quasi-compact in X, the relation $\mathrm{prof}_Y(X) \geq 1$ is also equivalent to saying that Y contains no generic point of X (EGA I 6.6.5). For $n = 2$ and U quasi-compact in X, the condition $\mathrm{prof}_Y(X) \geq 2$ is equivalent to the condition that, for every geometric point $\bar{y}$ of Y, $\bar{U}$ be nonempty and connected, that is, that "Y does not disconnect X locally for the ètale topology".

c) If X has Y-depth $\geq n$ for $\mathbf{L}$ and U is quasi-compact in X, then for every locally constant abelian sheaf of $\mathbf{L}$-torsion F on X, one has $\mathrm{prof}_Y(X) \geq n$. Indeed, since the property $\mathrm{prof}_Y(F) \geq n$ is local for the ètale topology, one may assume that F is constant; then F is a filtered direct limit of sheaves that are finite direct sums of sheaves of the form $\mathbf{Z}/p^m\mathbf{Z}$, where m is an integer > 0 and $p \in \mathbf{L}$. Using XIV.1.1 (iii) and (SGA 4 VII 3.3), one sees that one may reduce to the case $F = \mathbf{Z}/p^m\mathbf{Z}$, and then , by induction on m, to the case $F = \mathbf{Z}/p\mathbf{Z}$, for **93**
which the assertion follows from the definition.

d) By XIV.1.4, if $\mathrm{prof}_Y(X) \geq 3$, the pair (X, Y) is *pure* in the sense of X 3.1. In fact, the pure pairs encountered in practice (cf. X 3.4) satisfy the stronger condition of homotopical depth ≥ 3, and this notion can therefore advantageously replace that of a pure pair.

e) Let F be a complex of abelian sheaves and $T(F)$ the complex obtained by applying the translation functor to F ([3]); then one plainly has:

$$\mathrm{prof}_Y(T(F)) = \mathrm{prof}_Y(F) - 1.$$

(*)That is, a "homogeneous principal bundle"in older terminology.

f) Let us point out that the recent work of Artin–Mazur ([1]) makes it possible to define the notion of homotopical depth $\geq n$ for every integer n (not only for $n \geq 3$).

g) Under the conditions of XIV.1.1 3), one has $\mathrm{prof}_Y(F) = \infty$ if and only if $F \xrightarrow{\sim} Ri_*(i^*F)$ in $D^+(X)$. Indeed, the $\underline{H}^p_Y(F)$ are the cohomology sheaves of the cone (= mapping cylinder) of the canonical morphism $F \to Ri_*i^*(F)$.

Definition XIV.1.7. Let X be a scheme, x a point of X, $\bar{x}$ a geometric point above x, and $\bar{X}$ the strict localization of X at $\bar{x}$. As before, F denotes either a sheaf of sets on X, a sheaf of groups on X, or a bounded-below complex of abelian sheaves on X; $\bar{F}$ denotes its inverse image on $\bar{X}$, and $\mathbf{L}$ a set of prime numbers. We say that F has *étale depth* $\geq n$ *at the point* x (respectively that the *étale depth for* $\mathbf{L}$ *of* X *at* x *is* $\geq n$, respectively that the **94**
homotopical depth for $\mathbf{L}$ *of* X *at* x *is* ≥ 3) and write $\mathrm{prof}_x(F) \geq n$ (respectively $\mathrm{prof}^{\mathbf{L}}_x(X) \geq n$, respectively $\mathrm{prof\,hop}^{\mathbf{L}}_x(X) \geq 3$) if $\mathrm{prof}_{\bar{x}}(\bar{F}) > n$ (respectively $\mathrm{prof}^{\mathbf{L}}_{\bar{x}}(\bar{X}) > n$, respectively $\mathrm{prof\,hop}^{\mathbf{L}}_{\bar{x}}(\bar{X}) \geq 3$). The integer $\mathrm{prof}^{\mathbf{L}}_x(X)$ and, if F is a complex of abelian sheaves, the integer $\mathrm{prof}_x(F)$ are defined in the evident way. If $\mathbf{L}$ is the set of all prime numbers, we omit $\mathbf{L}$ from the notation $\mathrm{prof}^{\mathbf{L}}_x(X)$, unless there is a risk of confusion with the notation of (EGA IV 5.7.1), in which case we write $\mathrm{prof\,ét}_x(X)$.

We then have the following *pointwise characterization* of depth:

Theorem XIV.1.8. *Let X be a scheme and Y a closed subset of X such that the open subset $U = X - Y$ is retrocompact in X. If F is either a sheaf of sets on X, a sheaf of groups on X, or a bounded-below complex of abelian sheaves on X, then*

$$\mathrm{prof}_Y(F) = \inf_{y \in Y} \mathrm{prof}_y(F).$$

Let us first show that, for every point y of Y, one has the inequality $\mathrm{prof}_y(F) \geq \mathrm{prof}_Y(F)$. Indeed, let $\bar{y}$ be a geometric point above y, $\bar{X}$ the strict localization of X at $\bar{y}$, and $\bar{F}$ and $\bar{Y}$ the inverse images of F and Y on $\bar{X}$. By XIV.1.7 and XIV.1.3, one has

$$\mathrm{prof}_y(F) = \mathrm{prof}_{\bar{y}}(\bar{F}) \geq \mathrm{prof}_{\bar{Y}}(\bar{F}) \geq \mathrm{prof}_Y(F),$$

the last inequality using the hypothesis "U retrocompact", through conditions (iii) in XIV.1.1 and transitivity in the formation of strict localizations.

Conversely, suppose that, for every point y of Y, one has $\mathrm{prof}_y(F) \geq n$ (n an integer), and let us show that $\mathrm{prof}_Y(F) \geq n$.

First recall the following well-known results (SGA 4 VIII): **95**

Lemma XIV.1.8.1. *Let X be a scheme and F a sheaf of sets on X (respectively let $G \to F$ be a monomorphism of sheaves of sets on X). Then two sections s and s' of F coincide (respectively a section s of F comes from a section of G) if and only if this is so locally. In particular, if s and s' are two sections of F, there is a largest open subset V of X on which they coincide (respectively, if s is a section of F on X, there is a largest open subset V of X such that $s|V$ comes from a section of G on V). This open subset is also the set of points x of X such that, denoting by $\bar{x}$ a geometric point above x, the sections s and s' have the same image in the fiber $F_{\bar{x}}$ (respectively the image of s in $F_{\bar{x}}$ comes from an element of $G_{\bar{x}}$).*

Let us return to the proof of XIV.1.8.

1°) Case where F is a sheaf of sets. If $n = 1$, it is enough to show that the canonical morphism

$$\mathrm{H}^0(X, F) \to \mathrm{H}^0(U, F)$$

is injective, since the result also applies when X is replaced by a scheme étale over X. Let s and s' be two sections of F over X which have the same image in $\mathrm{H}^0(U, F)$, and let V be the largest open subset over which they are equal; clearly $V \supset U$. Suppose $V \neq X$, and let y be a maximal point of $X - V$, $\bar{y}$ a geometric point above y, $\bar{X}$ the strict localization of X at $\bar{y}$, **96** and $\bar{V}$ and $\bar{F}$ the inverse images of V and F on $\bar{X}$. By the choice of y, one has $\bar{X} - \bar{y} = \bar{V}$, and by hypothesis the morphism

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{X} - \bar{y}, \bar{F}) = \mathrm{H}^0(\bar{V}, \bar{F})$$

is injective. It follows that s and s' coincide at the point y, which is absurd. If $n = 2$, it is enough, in view of the preceding, to show that the morphism

$$\mathrm{H}^0(X, F) \to \mathrm{H}^0(U, F) = \mathrm{H}^0(X, i_* i^* F)$$

is surjective (where i is the canonical immersion of U into X). Let s be a section of $i_* i^* F$ over X, and let V be the largest open subset over which it comes from a section of F. Suppose $V \neq X$, and let y be a maximal point of $X - V$; with the preceding notation, it follows from the hypothesis that the canonical morphism

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{X} - \bar{y}, \bar{F}) = \mathrm{H}^0(\bar{V}, \bar{F})$$

is bijective; consequently $s|\bar{V}$ extends to the point $\bar{y}$, which is absurd and completes the proof in case 1°).

2°) Case where F is a sheaf of groups. In view of 1°), the only thing left to show is that, in the case $n = 3$, the morphism

$$\mathrm{H}^1(X, F) = \mathrm{H}^1(X, i_* i^* F) \to \mathrm{H}^1(U, F)$$

is bijective. We already know that it is injective, by 1°) and XIV.1.1 2°) (ii bis). For surjectivity, we use the exact sequence (SGA 4 XII 3.2)

$$0 \to \mathrm{H}^1(X, i_* i^* F) \to \mathrm{H}^1(U, F) \xrightarrow{d} \mathrm{H}^0(X, \mathrm{R}^1 i_*(i^* F)).$$

Let $s \in \mathrm{H}^1(U, F)$, and let $V \supset U$ be the largest open subset over which $d(s) = 0$; this is also **97** the largest open subset such that s comes from an element of $\mathrm{H}^1(V, F)$. Suppose $V \neq X$, and let y be a maximal point of $X - V$; if $\bar{X}$ is the strict localization of X at a geometric point $\bar{y}$ above y, then, with the evident notation, we have the exact sequence

$$0 \to \mathrm{H}^1(\bar{X}, \bar{i}_*(\bar{i}^* \bar{F})) \to \mathrm{H}^1(\bar{U}, \bar{F}) \xrightarrow{\bar{d}} \mathrm{H}^0(\bar{X}, \mathrm{R}^1 \bar{i}_*(\bar{i}^* \bar{F})).$$

Since $i : U \to X$ is quasi-compact, $\mathrm{R}^1 \bar{i}_*(\bar{i}^* \bar{F})$ is the inverse image of $\mathrm{R}^1 i_*(i^* F)$ under the morphism $\bar{X} \to X$, whence $\mathrm{H}^0(\bar{X}, \mathrm{R}^1 \bar{i}_*(\bar{i}^* \bar{F})) = (\mathrm{R}^1 i_*(i^* F))_{\bar{y}}$. By hypothesis, and since $y \in Y$, the morphism

$$\mathrm{H}^1(\bar{X}, \bar{F}) \to \mathrm{H}^1(\bar{V}, \bar{F})$$

is bijective. The image $\bar{s}$ of s in $\mathrm{H}^1(\bar{U}, \bar{F})$, which extends to $\bar{V}$ by the definition of V, therefore also extends to $\bar{X}$; it follows that $\bar{d}(\bar{s}) = 0$, and hence the image of $d(s)$ in the geometric fiber $(\mathrm{R}^1 i_*(i^* F))_{\bar{y}}$ is zero; but this contradicts the definition of V, proving case 2°).

3°) Case where F is a bounded-below complex of abelian sheaves. We argue by induction on n. The conclusion holds for sufficiently small n, since F is bounded below. Suppose,

therefore, that $\mathrm{prof}_Y(F) \geq n-1$, and let us show that $\mathrm{prof}_Y(F) \geq n$, knowing that for every point y of Y one has $\mathrm{prof}_y(F) \geq n$. It is enough to see that the canonical morphism

$$\mathrm{H}^{n-2}(X,F) \to \mathrm{H}^{n-2}(U,F) \qquad (*)$$

is surjective and that

$$\mathrm{H}^{n-1}(X,F) \to \mathrm{H}^{n-1}(U,F) \qquad (**)$$

is injective (the result also applying when X is replaced by a scheme étale over X). **98**

a) Surjectivity of $(*)$. The proof is analogous to that of 2°). In view of XIV.1.5 and (SGA 4 V 4.5), one has the exact sequence

$$\mathrm{H}^{n-2}(X,F) \to \mathrm{H}^{n-2}(U,F) \xrightarrow{d} \mathrm{H}^{n-1}_Y(X,F) = \mathrm{H}^0(X, \underline{H}^{n-1}_Y(F)).$$

Let $s \in \mathrm{H}^{n-2}(U,F)$, and let $V \supset U$ be the largest open subset over which $d(s)=0$, which is also the largest open subset such that s extends to $\mathrm{H}^{n-2}(V,F)$. Suppose $V \neq X$, and let y be a maximal point of $X-V$ and $\bar{X}$ the strict localization of X at a geometric point $\bar{y}$ above y. Since $i: U \to X$ is quasi-compact, the formation of $\underline{H}^{n-1}_Y(F)$ commutes with the base change $\bar{X} \to X$, and hence one has (with the evident notation) the exact sequence

$$\mathrm{H}^{n-2}(\bar{X},\bar{F}) \to \mathrm{H}^{n-2}(\bar{U},\bar{F}) \xrightarrow{\bar{d}} \mathrm{H}^{n-1}_{\bar{Y}}(\bar{X},\bar{F}) = (\underline{H}^{n-1}_Y(F))_{\bar{y}},$$

the last equality following from the retrocompactness hypothesis on U.

But by hypothesis one has the isomorphism

$$\mathrm{H}^{n-2}(\bar{X},\bar{F}) \xrightarrow{\sim} \mathrm{H}^{n-2}(\bar{X}-\bar{y},\bar{F}) = \mathrm{H}^{n-2}(\bar{V},\bar{F});$$

consequently the image $\bar{s}$ of s in $\mathrm{H}^{n-2}(\bar{U},\bar{F})$, which extends (by definition of V) to $\mathrm{H}^{n-2}(\bar{V},\bar{F})$, also extends to $\mathrm{H}^{n-2}(\bar{X},\bar{F})$; but this shows that $\bar{d}(\bar{s})=0$, that is, that $d(s)$ is zero at $\bar{y}$, which is absurd.

b) Injectivity of $(**)$. Using the surjectivity of $(*)$, one obtains the exact sequence

$$0 \to \mathrm{H}^0(X, \underline{H}^{n-1}_Y(F)) \to \mathrm{H}^{n-1}(X,F) \to \mathrm{H}^{n-1}(U,F)$$

and we must show that every element $s \in \mathrm{H}^0(X, \underline{H}^{n-1}_Y(F))$ is zero. Let V be the largest open **99** subset over which $s=0$. Suppose that $V \neq X$, and let y be a maximal point of $X-V$ and $\bar{X}$ a strict localization of X at a geometric point $\bar{y}$ above y. By the induction hypothesis and 84, one has $\mathrm{prof}_{\bar{Y}}(\bar{F}) \geq \mathrm{prof}_Y(F) \geq n-1$, and hence the map e in the following diagram is injective:

$$\mathrm{H}^0(\bar{X}, \underline{H}^{n-1}_Y(\bar{F})) = (\underline{H}^{n-1}_Y(F))_{\bar{y}} \xrightarrow{e} \mathrm{H}^{n-1}(\bar{X},\bar{F}) \xrightarrow{f} \mathrm{H}^{n-1}(\bar{V},\bar{F}).$$

The same is true of f by the hypothesis; the equality on the left follows from the retrocompactness hypothesis on U. Let $\bar{s}$ be the image of s in $(\underline{H}^{n-1}_Y(F))_{\bar{y}}$; since s vanishes over V, one has $f \cdot e(\bar{s}) = 0$, whence $\bar{s}=0$, which contradicts the choice of y and completes the proof.

Remark XIV.1.9. A result analogous to XIV.1.8 is probably valid when the étale topos of a scheme X is replaced by a "locally finite-type topos", that is, one definable by a locally finite-type site (SGA 4 VI 1.1). To see this, one must use a result of P. Deligne (SGA 4 VI.9) asserting that there are "enough fiber functors"in such a topos.

We shall deduce from XIV.1.8 important cases in which the étale depth can be determined.

Theorem 1.10 (Cohomological semipurity theorem$^{(*)}$). *Let X denote either a scheme smooth over a field k, or an excellent regular scheme* (EGA IV 7.8.2) *of characteristic zero (N.B. if resolution of singularities in the sense of* (SGA 4 XIX) *is assumed, it is enough, more*

generally, to suppose that X is an excellent regular scheme of equal characteristic). Let Y **100**
be a closed subset of X and $\mathbf{L}$ the set of prime numbers distinct from the characteristic of X. Then

$$\operatorname{prof}_Y^{\mathbf{L}}(X) = 2\operatorname{codim}(Y, X).$$

Proof. It follows from XIV.1.8 that

$$\operatorname{prof}_Y^{\mathbf{L}}(X) = \inf_{y\in Y} \operatorname{prof}_y(X).$$

On the other hand, since $\operatorname{codim}(Y, X) = \inf_{y\in Y} \dim \mathcal{O}_{X,y}$, we are reduced to showing that

$$\operatorname{prof}_y^{\mathbf{L}}(X) = 2\dim \mathcal{O}_{X,y},$$

which follows from (SGA 4 XVI 3.7 and XIX 3.2).

Theorem XIV.1.11 (Homotopical purity theorem(*))**.** *If X is a locally noetherian scheme that is regular (resp. whose local rings are complete intersections), and Y is a closed subset of X such that* $\operatorname{codim}(Y, X) \geq 2$ *(resp.* $\operatorname{codim}(Y, X) \geq 3$*), then one has*

$$\operatorname{prof}\operatorname{hop}_Y(X) \geq 3.$$

Indeed, it follows from XIV.1.8 that $\operatorname{prof}\operatorname{hop}_Y(X) = \inf_{y\in Y} \operatorname{prof}\operatorname{hop}_y(X)$. But the strictly local rings of X at the various points of Y are regular rings of dimension ≥ 2 (resp. complete intersections of dimension ≥ 3). It then follows from the purity theorem X 3.4 that $\operatorname{prof}\operatorname{hop}_y(X) \geq 3$, which proves the theorem.

Example XIV.1.12. Let X be a locally noetherian scheme, Y a closed subset of X, and $n = 1$ or 2. Then, if $\operatorname{prof}_Y(\mathcal{O}_X) \geq n$ ($\operatorname{prof}_Y(\mathcal{O}_X)$ denoting the Y-depth in the sense of coherent sheaves (cf. 1.6 a)), one also has $\operatorname{prof}_Y(X) \geq n$; this is obvious for $n = 1$, and for $n = 2$ it is precisely Hartshorne's theorem (III 1). On the other hand, the analogous assertion is false for **101**
$n \geq 3$. For example, take an affine space of dimension ≥ 3 over a field of characteristic $\neq 2$ and let the group $\mathbf{Z}/2\mathbf{Z}$ act by reflection through the origin. Let X be the quotient and $Y = \{x\}$ the image of the origin in X. Then $\mathcal{O}_{X,x}$ is a Cohen–Macaulay ring, so $\operatorname{prof}_x(\mathcal{O}_X) \geq 3$; but the affine space with the origin removed is an étale covering of $X - \{x\}$ that does not extend to an étale covering of X; hence, by XIV.1.4, $\operatorname{prof}_Y(X) = 2$.

The following theorem is the analogue of (EGA IV 6.3.1):

(*) *Editor's note.* Gabber subsequently proved — in 1994 — Grothendieck's absolute cohomological purity conjecture: if Y is a closed subscheme of absolute Noetherian schemes of pure codimension c and n is an integer invertible on X, then $\mathcal{H}^q_Y(\Lambda)$ is zero if $q \neq 2c$ and is $\Lambda_Y(-c)$ (Tate twist) otherwise, where $\Lambda = \mathbf{Z}/n\mathbf{Z}$. See (Fujiwara K., "A Proof of the Absolute Purity Conjecture (after Gabber)", in *Algebraic geometry 2000, Azumino (Hotaka)*, Adv. Stud. in Pure Math., vol. 36, 2002, p. 153-183). For applications to the existence of the dualizing complex, see (SGA 5, Lect. Notes in Math., vol. 589, Springer-Verlag, 1977, p. 1672), Exposé 1 and *loc. cit.*, §8. This conjecture had been proved in the case $n = \ell^\nu$, with ℓ a sufficiently large prime invertible on X, by making crucial use of K-theory (Thomason R.W., "Absolute cohomological purity", *Bull. Soc. Math. France* **112** (1984), no. 3, p. 397–406). K-theory reappears in Gabber's proof through the Atiyah-Hirzebruch-Thomason spectral sequence relating étale cohomology and K-theory, a method already used in Thomason's approach. Besides this result, the other fundamental argument is the generalization of the Lefschetz theorem cited in note ((*)), page 150.

(*) *Editor's note.* recently, de Jong and Oort obtained the following purity statement: let $\tilde{S} \to S$ be a resolution of the singularities of the spectrum S of a normal noetherian local ring of dimension 2, and let U be the complement of the closed point s in S. Suppose in addition that $k(s)$ is algebraically closed. Then, for every prime number p, in particular if S has characteristic p, the restriction morphism $H^1_{\text{ét}}(\tilde{S}, \mathbf{Q}_p) \to H^1_{\text{ét}}(U, \mathbf{Q}_p)$ is bijective (de Jong A.J. & Oort F, "Purity of the stratification by Newton polygons", *J. Amer. Math. Soc.* **13** (2000), no. 1, p. 209–241, Theorem 3.2). If $k = \mathbf{C}$ and A is the completion of a surface singularity, this result is due to Mumford (see page 131, [5]).

Theorem XIV.1.13. *Let $f : X \to S$ be a morphism of schemes, Y a closed subset of X, and Z a closed subset of S, such that $f(Y) \subset Z$. Suppose that the local rings of X at the various points of Y are noetherian and that the open subsets $X - Y$ and $S - Z$ are retrocompact in X and S, respectively. Let p, q, r be integers such that $p \geq -r$, $q \geq 0$, let $\mathbf{L}$ be a set of prime numbers, and let F be a complex of abelian sheaves of $\mathbf{L}$-torsion on S, such that the cohomology sheaves $\mathrm{H}^i(F)$ vanish for $i < -r$. Suppose that*

a) The morphism f is locally $(p+q+r-2)$-acyclic for $\mathbf{L}$ (SGA 4 XV 1.11).

b) One has

$$\mathrm{prof}_Z(F) \geq p.$$

c) For every point s of Z, one has

$$\mathrm{prof}^{\mathbf{L}}_{Y_s}(X_s) \geq q.$$

Then one has **102**

$$\mathrm{prof}_Y(f^*F) \geq p + q.$$

We shall need the following lemma:

Lemma XIV.1.13.1. *Let $\mathbf{L}$ be a set of prime numbers, let n and r be integers, and let $f : X \to S$ be a morphism locally n-acyclic for $\mathbf{L}$. Let F be a complex of abelian sheaves whose cohomology sheaves are of $\mathbf{L}$-torsion, such that $\mathrm{H}^i(F) = 0$ for $i < -r$, let Z be a closed subset of S such that $S - Z$ is retrocompact in S, and let $T = f^{-1}(Z)$. Then the canonical morphism*

$$f^*(\underline{H}^i_Z(F)) \to \underline{H}^i_T(f^*F)$$

is bijective for $i < n - r + 2$ and injective for $i = n - r + 2$.

Set $U = S - Z$ and $V = X - T$, so that one has the cartesian square

$$\begin{array}{ccc} V & \xrightarrow{g} & U \\ {\scriptstyle k}\downarrow & & \downarrow{\scriptstyle j} \\ X & \xrightarrow{f} & S \end{array}$$

Consider the following commutative diagram whose rows are exact

$$\begin{array}{ccccccc} \longrightarrow f^*(\underline{H}^i_Z(F)) & \longrightarrow & f^*(\underline{H}^i(F)) & \longrightarrow & f^*(\underline{H}^i(\mathrm{R}j_*(j^*F))) & \longrightarrow \\ \downarrow & & \downarrow\wr & & \downarrow & \\ \longrightarrow \underline{H}^i_T(f^*F) & \longrightarrow & \underline{H}^i(f^*F) & \longrightarrow & \underline{H}^i(\mathrm{R}k_*(k^*f^*F)) & \longrightarrow ; \end{array}$$

it follows that one is reduced to showing that the morphism

$$f^*(\underline{H}^i(\mathrm{R}j_*(j^*F))) \to \underline{H}^i(\mathrm{R}k_*(k^*f^*F))$$

is bijective for $i < n - r + 1$ and injective for $i = n - r + 1$. But such a morphism arises from **103**
the following morphism between hypercohomology spectral sequences

$$\begin{array}{ccc} f^*E_2^{p,q} = f^*(\mathrm{R}^p j_*(\underline{H}^q(j^*F))) & \Longrightarrow & f^*(\underline{H}^*(\mathrm{R}j_*(j^*F))) \\ \downarrow & & \downarrow \\ E_2'^{p,q} = \mathrm{R}^p k_*(\underline{H}^q(k^*f^*F)) & \Longrightarrow & \underline{H}^*(\mathrm{R}k_*(k^*f^*F)). \end{array}$$

Since j is quasi-compact, it follows from (SGA 4 XV 1.10) that the morphism $f^*(E_2^{p,q}) \to E_2'^{p,q}$ is bijective for $p \leq n$ and injective for $p = n+1$; in particular, it is bijective for $p+q \leq n-r$ and injective for $p+q = n-r+1$. The conclusion follows at once.

Let us return to the proof of XIV.1.13. Let $T = f^{-1}(Z)$. By XIV.1.13.1 and condition a), the canonical morphism $f^*(\underline{H}^i_Z(F)) \to \underline{H}^i_T(f^*F)$ is an isomorphism for $i \leq p+q$. It therefore follows from b) that $\underline{H}^i_T(f^*F) = 0$ for $i < p$ and, for $i < p+q$, the restriction of $\underline{H}^i_T(f^*F)$ to T is the inverse image of a sheaf G^i on Z. Let

$$f_T : T \to Z$$

be the restriction of f to T. It then follows from c) and the corollary below that $\underline{H}^j_Y(f_T^*(G^i)) = 0$ for $j < q$. We conclude that

$$\underline{H}^j_Y(\underline{H}^i_T(f^*F)) = 0 \text{ for } i+j < p+q,$$

because the inequality $i+j < p+q$ implies either $i < p$, in which case $\underline{H}^i_T(f^*F) = 0$, or $j < q$, in which case $\underline{H}^j_Y(\underline{H}^i_T(f^*F)) = 0$. Given that, with the notation of XIV.1.0, $\underline{\Gamma}_Y = \underline{\Gamma}_Y.\underline{\Gamma}_T$, one has the spectral sequence

$$E_2^{i,j} = \underline{H}^j_Y(\underline{H}^i_T(f^*F)) \Longrightarrow \underline{H}^*_Y(f^*F); \tag{1.13.2}$$

since $E_2^{i,j} = 0$ for $i+j < p+q$, one sees that $\underline{H}^k_Y(f^*F) = 0$ for $k < p+q$. The theorem will therefore be proved if we prove the following corollary (which is the special case of 1.13 obtained by taking $Z = S$, $r = p = 0$ and F concentrated in degree 0). **104**

Corollary XIV.1.14. *Let $f : X \to S$ be a morphism, let Y be a closed subset such that the complementary open subset $X - Y$ is quasi-compact in X and the local rings of X at the various points of Y are Noetherian. Let $\mathbf{L}$ be a set of prime numbers, let q be an integer, and let F be an abelian sheaf of $\mathbf{L}$-torsion on S. Suppose that f is locally $(q-2)$-acyclic for $\mathbf{L}$ and that, for every point s of S, one has $\operatorname{prof}^{\mathbf{L}}_{Y_s}(X_s) \geq q$. Then $\operatorname{prof}_Y(f^*F) \geq q$.*

1°) Reduction to the case in which X and S are strictly local schemes, f is a local morphism, and Y consists of a closed point of X.

By XIV.1.8, to prove XIV.1.14, one must show that, for every point y of Y, one has:

$$\operatorname{prof}_y(f^*F) \geq q.$$

Let $s = f(y)$, let $\bar{s}$ be a geometric point above s, let $\bar{y}$ be a geometric point above y and $\bar{s}$, let $\bar{X}$ and $\bar{S}$ be the strict localizations of X and S at $\bar{y}$ and $\bar{s}$, respectively, let $\bar{f} : \bar{X} \to \bar{S}$ be the canonical morphism, and let $\bar{F}$ be the inverse image of F on $\bar{S}$. Since $\operatorname{prof}_y(f^*F) = \operatorname{prof}_{\bar{y}}(\bar{f}^*\bar{F})$, it is enough to show that the hypotheses of XIV.1.14 are preserved when f (resp. Y, resp. F) is replaced by $\bar{f}$ (resp. $\{\bar{y}\}$, resp. $\bar{F}$). The quasi-compactness condition follows from the Noetherian hypothesis on $\mathcal{O}_{X,x}$, which implies that $\bar{X}$ is Noetherian . By (SGA 4 XV 1.10 (i)), $\bar{f}$ is still locally $(q-2)$-acyclic for $\mathbf{L}$. Moreover, the fiber $(\bar{X})_{\bar{s}}$ of $\bar{X}$ above $\bar{s}$ identifies with the strict localization of X_s at $\bar{y}$, and therefore satisfies $\operatorname{prof}_{\bar{y}}((\bar{X})_{\bar{s}}) \geq q$. Since an analogous relation is trivially satisfied for the fibers of the $\bar{S}$-scheme $\bar{X}$ other than the closed fiber, this completes the reduction. **105**

2°) The case in which X and S are strictly local, f is a local homomorphism, and Y consists of the closed point of X. Let

$$g : Y = X - \{y\} \to S$$

be the structural morphism of U. We must show that the canonical morphism

$$u_i : H^i(X, f^*F) \to H^i(U, f^*F)$$

is bijective for $i \leq q-2$ and injective for $i = q-1$. Consider the commutative diagram

$$\begin{array}{ccc} H^i(X, f^*F) & \xrightarrow{\quad u_i \quad} & H^i(U, f^*F) \\ & {\scriptstyle v_i}\nwarrow \quad \nearrow{\scriptstyle w_i} & \\ & H^i(S, F) & \end{array}$$

The morphism v_i is plainly bijective for every i. Moreover, g is locally $(q-2)$-acyclic for $\mathbf{L}$; its fibers are also $(q-2)$-acyclic for $\mathbf{L}$, as follows from the fact that $\mathrm{prof}_y(X_s) \geq q$ and that the fibers of f are $(q-2)$-acyclic for $\mathbf{L}$. Since g is quasi-compact because X is Noetherian, (SGA 4 XV 1.16) shows that g is $(q-2)$-acyclic for $\mathbf{L}$. Consequently w_i, and hence also u_i, is bijective for $i \leq q-2$ and injective for $i = q-1$, which completes the proof of XIV.1.14. **106**

Corollary XIV.1.15. *Let $f : X \to S$ be a morphism of schemes, let $\mathbf{L}$ be a set of prime numbers, let m and r be integers, and let F be a complex of abelian sheaves of $\mathbf{L}$-torsion on S such that $\underline{H}^i(F) = 0$ for $i < -r$. Let x be a point of X, let $s = f(x)$, and suppose that the local ring $\mathcal{O}_{X,x}$ is Noetherian. Then, if f is locally m-acyclic for $\mathbf{L}$, one has*

$$\mathrm{prof}_x(f^*F) \geq \inf(\mathrm{prof}_s(F) + \mathrm{prof}_x^{\mathbf{L}} X_s), n) \text{ where } n = m - r + 2. \qquad (*)$$

In particular, if $n \geq \mathrm{prof}_s(F) + \mathrm{prof}_x^{\mathbf{L}}(X_s)$, for example if f is locally acyclic for $\mathbf{L}$, then

$$\mathrm{prof}_x(f^*F) \geq \mathrm{prof}_s(F) + \mathrm{prof}_x^{\mathbf{L}}(X_s). \qquad (**)$$

If $\mathbf{L}$ consists of a single element ℓ and $n \geq \mathrm{prof}_s(F) + \mathrm{prof}_x^{\mathbf{L}}(X_s)$, the preceding inequality is an equality.

One reduces to the case in which s and x are closed points by taking the strict localizations of S and x at geometric points $\bar{s}$ above s and $\bar{x}$ above x and $\bar{s}$. If $n \geq \mathrm{prof}_s(F) + \mathrm{prof}_x^{\mathbf{L}}(X_s)$, then $(*)$ follows from XIV.1.13 by taking $p = \mathrm{prof}_s(F)$ and $q = \mathrm{prof}_x^{\mathbf{L}}(X_s)$ there (the hypothesis that $S - \{s\}$ is quasi-compact in S follows from the fact that $X - X_s$ is quasi-compact in X and that f is surjective, since it is (-1)-acyclic, except possibly when the conclusion of 1.15 is empty). If $n < \mathrm{prof}_s(F) + \mathrm{prof}_x^{\mathbf{L}}(X_s)$, inequality $(*)$ again follows from XIV.1.13, for example by taking $p = \mathrm{prof}_s(F)$ and $q = n - p$. It remains to prove the last assertion. Let **107** $p = \mathrm{prof}_s(F)$ and $q = \mathrm{prof}_x(X_s)$; it follows from (1.13.2) that

$$\underline{H}_x^{p+q}(f^*(F)) \simeq \underline{H}_x^q(f^*(\underline{H}_s^p(F))).$$

Since $\mathrm{prof}_s(F) = p$, the sheaf $\underline{H}_s^p(F)$ is a nonzero sheaf of ℓ-torsion, constant on s. Consequently the sheaf $G = f^*(\underline{H}_s^p(F))$ is a nonzero sheaf of ℓ-torsion, constant on X_s, and therefore contains a subsheaf isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$; since $\underline{H}_x^q(\mathbf{Z}/\ell\mathbf{Z})$ is nonzero, one indeed has $\underline{H}_x^q(G) \neq 0$.

Corollary XIV.1.16. *Let $f : X \to S$ be a regular morphism of excellent schemes (*EGA IV 7.8.2*) of characteristic zero, let ℓ be a prime number, and let F be a complex of sheaves of ℓ-torsion on S. Let $x \in X$ and $s = f(x)$; then*

$$\mathrm{prof}_x(f^*F) = \mathrm{prof}_s(F) + 2\dim(\mathcal{O}_{X,x}).$$

Indeed, f is locally acyclic (SGA 4 XIX 4.1). It then follows from XIV.1.15 that

$$\mathrm{prof}_x(f^*F) = \mathrm{prof}_s(F) + \mathrm{prof}_x(X_s).$$

But by XIV.1.10,

$$\mathrm{prof}_x(X_s) = 2\dim \mathcal{O}_{X_s,x},$$

which gives the result.

Remark XIV.1.17. It follows from XIV.1.15 that XIV.1.13 remains valid when b) and c) are replaced by the conditions: **108**

b′) For every point $s \in f(Y)$, one has $\mathrm{prof}_s(F) \geq p$.

c′) For every point $x \in Y$, if $s = f(x)$, one has $\mathrm{prof}_x^{\mathbf{L}}(X_s) \geq q$.

For a sheaf of sets or groups, one has the following theorem, analogous to XIV.1.13.

Theorem XIV.1.18. *Let $f : X \to S$ be a morphism of schemes and let Y be a closed subset of X such that $X - Y$ is quasi-compact in X and, for every point x of Y, the local ring $\mathcal{O}_{X,x}$ is Noetherian.*

1°) *Let F be a sheaf of sets on S and let n be an integer equal to* 1 *or* 2. *Suppose that f is locally $(n-2)$-acyclic and that, for every point s of $f(Y)$,*

$$\mathrm{prof}_{Y_s}(X_s) + \mathrm{prof}_s(F) \geq n.$$

Then:

$$\mathrm{prof}_Y(f^*F) \geq n.$$

2°) *Let* **L** *be a set of prime numbers and let F be a sheaf of ind-***L***-groups. Suppose that f is locally* 1*-aspherical for* **L** *(*SGA 4 XV 1.11*) and that, for every point s of $f(Y)$,*

$$\mathrm{prof\,hop}_{Y_s}^{\mathbf{L}}(X_s) + \mathrm{prof}_s(F) \geq 3.$$

Then:

$$\mathrm{prof}_Y(f^*F) \geq 3.$$

Proof. We reduce, as in XIV.1.14 and XIV.1.15, to the case where X and S are strictly local schemes, f is a local morphism, and Y is the closed point x of X. Let $s = f(x)$ be the closed point of S; we have the commutative diagram: **109**

$$\begin{array}{ccccc} X - X_s & \xrightarrow{i} & X - \{x\} & \xrightarrow{j} & X \\ {\scriptstyle h}\downarrow & & & \searrow^{g} & \downarrow{\scriptstyle f} \\ S - \{s\} & & \xrightarrow[k]{} & & S. \end{array}$$

1°) a) *Case $n = 1$.*

If $\mathrm{prof}_s(F) \geq 1$, then the morphism $F \to k_*k^*F$ is injective, and hence the morphism $f^*F \to f^*(k_*k^*F)$ is also injective. On the other hand, it follows from the fact that f is locally (-1)-acyclic that the morphism $f^*(k_*k^*F) \to (j.i) * (f^*F_{|X-X_s})$ is injective. Finally, the composite morphism $f^*F \to (j.i)*(f^*F_{|X-X_s})$ is injective, which shows that $\mathrm{prof}_{X_s}(f^*F) \geq 1$, and consequently also $\mathrm{prof}_x(f^*F) \geq 1$.

If $\mathrm{prof}_x(X_s) \geq 1$, consider the commutative diagram

$$\begin{array}{ccc} H^0(X, f^*F) & \xrightarrow{v} & H^0(X - \{x\}, f^*F) \\ \wr\downarrow & & \downarrow \\ H^0(X_s, f^*F) & \xrightarrow{v'} & H^0(X - \{x\}, f^*F); \end{array} \qquad (*)$$

By hypothesis, v' is injective, and therefore so is v.

b) *Case $n = 2$.* Consider the commutative diagram

$$\begin{array}{ccccc} H^0(S, F) & & \xrightarrow{\qquad u \qquad} & & H^0(S - \{s\}, F) \\ {\scriptstyle m}\downarrow\wr & & & & {\scriptstyle n}\downarrow\wr \\ H^0(X, f^*F) & \xrightarrow{v} & H^0(X - \{x\}, f^*F) & \xrightarrow{w} & H^0(X - X_s, f^*F); \end{array} \qquad (**)$$

we must show that v is bijective. The morphism m is obviously bijective, and, since f is 0-acyclic, n is also bijective. **110**

If $\mathrm{prof}_s(F) \geq 2$, u is bijective. As we saw in a), the sole hypothesis $\mathrm{prof}_s(F) \geq 1$ implies the relation $\mathrm{prof}_{X_s}(f^*F) \geq 1$; consequently v and w are injective, and it then follows from $(**)$ that v is bijective.

If $\mathrm{prof}_x(X_s) \geq 2$, then g is 0-acyclic (because it is locally 0-acyclic and its fibers are 0-acyclic). It follows that $v \cdot m$ is bijective, and therefore v is bijective.

If $\mathrm{prof}_s(F) \geq 1$ and $\mathrm{prof}_x(X_s) \geq 1$, then we already know that v and w are injective. Let z be a maximal point of $X_s - \{x\}$ (such a point exists by the hypothesis $\mathrm{prof}_x(X_s) \geq 1$), let $\bar{Z}$ be the strict localization of X at a geometric point above z, and let $\overline{f^*F}$ be the inverse image of f^*F on $\bar{Z}$. Consider the commutative diagram

$$\begin{array}{ccccc}
 & & H^0(S,F) & \longrightarrow & H^0(S-\{s\},F) \\
 & \overset{m}{\swarrow}\sim & & & \sim\overset{n}{\searrow} \\
H^0(X,f^*F) & \xrightarrow{v} & H^0(X-\{x\},f^*F) & \xrightarrow{w} & H^0(X-X_s,f^*F) \\
 & \underset{m'}{\searrow}\sim & \underset{r}{\swarrow} & & \sim\underset{n'}{\swarrow} \\
 & & H^0(\bar{Z},f^*F) & \longrightarrow & H^0(\bar{Z}-\{\bar{z}\},\overline{f^*F})
\end{array}$$

the morphism $m' \cdot m$ is obviously bijective, and it follows from the fact that f is locally 0-acyclic that $n' \cdot n$ is bijective; consequently m' and n' are also bijective. Since w is injective, r is also injective, and therefore v is bijective.

2°) In view of b), we already know that $\mathrm{prof}_x(f^*F) \geq 2$.

If $\mathrm{prof}_s(F) \geq 3$, then $\mathrm{R}^1k_*(k^*F) = 1$(*). Since f is locally 1-aspherical, one has $\mathrm{R}^1(j \cdot i)_*(f^*F_{|X-X_s}) = f^*(\mathrm{R}^1k_*(k^*F)) = 1$. Thus $\mathrm{prof}_{X_s}(f^*F) \geq 3$ and consequently $\mathrm{prof}_x(f^*F) \geq 3$. **111**

If $\mathrm{prof}_x(X_s) \geq 3$, then g is 1-aspherical (because g is locally 1-aspherical and its fibers are 1-aspherical). Thus $H^1(X - \{x\}, f^*F) = H^1(S, F) = 1$, and consequently $\mathrm{prof}_x(f^*F) \geq 3$.

If $\mathrm{prof}_s(F) \geq 2$ and $\mathrm{prof}_x(X_s) \geq 1$, we use the exact sequence (SGA 4 XII 3.2):

$$1 \to \mathrm{R}^1j_*(i_*(f^*F_{|X-X_s})) \to \mathrm{R}^1(j.i)_*(f^*F_{|X-X_s}) \to j_*(\mathrm{R}^1i_*(f * F_{|X-X_s})).$$

Since f and g are locally 1-aspherical, one has

$$\mathrm{R}^1(j \cdot i)_*(f^*F_{|X-X_s}) \simeq f^*(\mathrm{R}^1k_*(k^*F))$$
$$\mathrm{R}^1i_*(f^*F_{|X-X_s}) \simeq g^*(\mathrm{R}^1k_*(k^*F));$$

the preceding exact sequence can then be written in the form

$$1 \to \mathrm{R}^1j_*(i_*(f^*F_{|X-X_s})) \to f^*(\mathrm{R}^1k_*(k^*F)) \xrightarrow{a} j_*(j^*(f^*(\mathrm{R}^1k_*(k^*F)))). \qquad (***)$$

The hypothesis $\mathrm{prof}_s(F) \geq 2$ shows that the morphism $F \to k_*k^*F$ is bijective; applying g^*, and taking into account the fact that g is locally 0-acyclic, we find $f^*F_{|X-\{x\}} = i_*(f^*F_{|X-X_s})$. The hypothesis $\mathrm{prof}_x(X_s) \geq 1$ shows that the morphism a is injective (note that $f^*(\mathrm{R}^1k_*(k^*F))$ is a sheaf equal to 1 outside X_s and constant on X_s). It then follows from $(***)$ that $\mathrm{R}^1j_*(f^*F_{|X-\{x\}}) = 1$, and hence $\mathrm{prof}_x(f^*F) \geq 3$. **112**

If $\mathrm{prof}_s(F) \geq 1$ and $\mathrm{prof}_x(f^*F) \geq 2$, consider the sheaf of homogeneous spaces G defined by the exact sequence

$$1 \to F \to k_*k * F \to G \to 1.$$

(*) *Editor's note.* the trivial torsor is subsequently denoted successively by 0 or 1; we have retained this double notation, which, on reflection, causes no ambiguity.

Applying the exact functor g^* to this exact sequence and using (SGA 4 XII 3.1), we obtain the following commutative diagram with exact rows:

$$\begin{array}{ccccccc} f^*(k_*k^*F) & \longrightarrow & f^*G & \longrightarrow & 1 & & \\ \downarrow & & \downarrow b & & & & \\ j_*(g^*(k_*k^*F)) & \xrightarrow{u} & j_*(g^*G) & \longrightarrow & \mathrm{R}^1j_*(g^*F) & \longrightarrow & \mathrm{R}^1j_*(g^*(k_*k^*F)). \end{array}$$

Since $\mathrm{prof}_x(X_s) \geq 2$, the morphism b is bijective, so u is surjective, and we thus have a map whose kernel is reduced to the neutral element:

$$1 \to \mathrm{R}^1j_*(g^*F) \to \mathrm{R}^1j_*(g^*(k_*k^*F)) = R.$$

Since $g^*(k_*k^*F) \simeq i_*(f^*F_{|X-X_s})$ (because g is locally 0-acyclic), R is identified with the first term of the exact sequence $(***)$; but in the preceding case we saw that $R = 1$ as soon as $\mathrm{prof}_x(X_s) \geq 1$, which proves that $\mathrm{prof}_x(f^*F) \geq 3$ and completes the proof of XIV.1.18.

The following corollaries are generalizations of (SGA 4 XVI 3.2 and 3.3).

Corollary XIV.1.19. *Let $f : X \to S$ be a flat morphism with separable fibers between locally Noetherian schemes, and let Y be a closed subset of X. Suppose that, for every point $s \in f(Y)$,* **113**
the fiber Y_s is rare[(*)] in X_s and that one of the following two conditions holds:

a) the closure of $f(Y)$ is rare in S.

b) X_s is geometrically unibranch at the points of Y_s.

Then

$$\mathrm{prof}_Y(X) \geq 2.$$

Indeed, it follows from the hypothesis on f that f is locally 0-acyclic (SGA 4 XV 4.1). We then apply XIV.1.13. The hypothesis that Y_s is rare in X_s (respectively that $\overline{f(Y)}$ is rare in S) is equivalent, by XIV.1.6 b), to the relation $\mathrm{prof}_{Y_s}(X_s) \geq 1$ (respectively $\mathrm{prof}_{\overline{f(Y)}}(S) \geq 1$). The hypothesis that X_s is geometrically unibranch at every point of Y_s is equivalent to saying that the strict localization of X_s at a geometric point of Y_s is irreducible; knowing that Y_s is rare in X_s, this clearly implies $\mathrm{prof}_{Y_s}(X_s) \geq 2$, by XIV.1.8. In either case XIV.1.13 indeed gives $\mathrm{prof}_Y(X) \geq 2$.

Corollary XIV.1.20. *Let $f : X \to S$ be a regular morphism (*EGA IV 6.8.1*) between locally Noetherian schemes, and let Y be a closed subset of X. Suppose that, for every point $s \in f(Y)$, one of the following conditions holds:*

a) One has $\mathrm{codim}(Y_s, X_s) \geq 2$.

b) One has $\mathrm{codim}(Y_s, X_s) \geq 1$ and $\mathrm{prof}_s(S) \geq 1$.

c) One has $\mathrm{prof\,hop}_s(S) \geq 3$.

Then **114**

$$\mathrm{prof\,hop}_Y(X) \geq 3.$$

Indeed, this follows from XIV.1.18, since hypothesis a) implies $\mathrm{prof\,hop}_{Y_s}(X_s) \geq 3$ (cf. XIV.1.11), and the condition $\mathrm{codim}(Y_s, X_s) \geq 1$ clearly implies $\mathrm{prof}_Y(X) \geq 2$.

(*) *Editor's note.* "rare"= "having empty interior", cf. Bourbaki TG IX.52.

XIV.2 Technical lemmas

XIV.2.1

Let S be a locally noetherian scheme, $f : X \to S$ a morphism locally of finite type, and t a point of S. If $x \in X$ is such that $s = f(x) \in \operatorname{Spec} \mathcal{O}_{S,t}$, set

$$\delta_t(x) = \deg \operatorname{tr} k(x)/k(s) + \dim(\overline{\{s\}}),$$

where $\overline{\{s\}}$ denotes the closure of s in $\operatorname{Spec} \mathcal{O}_{S,t}$, and $k(x)$ and $k(s)$ the residue fields of x and s, respectively. If S is a local ring with closed point t, we also write $\delta(x)$ instead of $\delta_t(x)$ (cf. SGA 4 XIV 2.2).

Lemma XIV.2.1.1. *Let there be a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array},$$

where S and S' are noetherian local rings, with closed points t and t', respectively, g is a faithfully flat morphism such that $g^{-1}(t) = t'$, and f is a morphism locally of finite type. Let **115**
$x' \in X'$, $x = h(x')$, $s = f(x)$, $s' = f'(x')$; then one has

$$\delta(x') \leq \delta(x).$$

Moreover, the preceding inequality is an inequality if and only if one has:

$$\deg \operatorname{tr} k(x)/k(s) = \deg \operatorname{tr} k(x')/k(s') \text{ and } \dim(\overline{\{s\}}) = \dim(\overline{\{s'\}}).$$

In particular, given $x \in X$, one can find x' such that $\delta(x) = \delta(x')$.

Indeed, one has (EGA IV 6.11)

$$\dim(\overline{\{s\}}) = \dim g^{-1}(\overline{\{s\}}).$$

It follows that, for every point s' of $g^{-1}(s)$, one has the relation $\dim(\overline{\{s'\}}) \leq \dim(\overline{\{s\}})$, and that, given s, one can find $s' \in g^{-1}(s)$ for which equality holds. Let Z then denote the schematic closure of x in the fiber X_s of X at s, and let $Z' = Z \times_{\operatorname{Spec} k(s)} \operatorname{Spec} k(s')$. Then Z' is equidimensional of dimension $\deg \operatorname{tr} k(x)/k(s)$; therefore, for every point $x' \in Z'_x$, one has

$$\deg \operatorname{tr} k(x')/k(s') \leq \deg \operatorname{tr} k(x)/k(s), \text{ with equality}$$

when x' is a maximal point of Z'_x. The asserted conclusion follows at once.

XIV.2.2

Let $f : X \to S$ be a morphism locally of finite type and T a closed subset of S. Let $x \in X, s = f(x)$; we set

$$\delta_T(x) = \deg \operatorname{tr} k(x)/k(s) + \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) = \inf_{t \in T \cap \overline{\{s\}}} \delta_t(x).$$

Lemma XIV.2.2.1. *Let there be a cartesian square* **116**

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array},$$

where the schemes S and S' are locally noetherian and catenary, the morphism f is locally of finite type, and g is faithfully flat. Let T be a closed subset of S and T' a closed subset of S', such that $g(T') \subset T$, let x' be an element of X', let $x = h(x')$, and let

$$h_{x'} : \operatorname{Spec} \mathcal{O}_{X',x'} \to \operatorname{Spec} \mathcal{O}_{X,x}$$

be the morphism induced by h. Then one has:

$$\delta_T(x) - \delta_{T'}(x') \leq \dim h_{x'}^{-1}(x).$$

Let $s' = f'(x'), s = f(x)$. By definition, one has:

$$\delta_T(x) - \delta_{T'}(x') = \deg \operatorname{tr} k(x)/k(s) - \deg \operatorname{tr} k(x')/k(s') + \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) - \operatorname{codim}(\overline{\{s'\}} \cap T', \overline{\{s'\}}).$$

Since g is faithfully flat, it follows from (EGA IV 6.1.4) that one has

$$\begin{aligned} \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) &= \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap g^{-1}(T), g^{-1}(\overline{\{s\}})) \\ &\leq \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap T', g^{-1}(\overline{\{s\}})); \end{aligned} \qquad (*)$$

since S' is catenary, one has, by (EGA 0_{IV} 14.3.2 b)):

$$\begin{aligned} \operatorname{codim}(\overline{\{s'\}} \cap T', g^{-1}(\overline{\{s\}})) &= \operatorname{codim}(\overline{\{s'\}} \cap T', \overline{\{s'\}}) + \operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})) \\ &= \operatorname{codim}(\overline{\{s'\}} \cap T', g^{-1}(\overline{\{s\}}) \cap T') + \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap T', g^{-1}(\overline{\{s\}})). \end{aligned}$$

From this relation and $(*)$, one deduces **117**

$$\delta_T(x) - \delta_{T'}(x') \leq \deg \operatorname{tr} k(x)/k(s) - \deg \operatorname{tr} k(x')/k(s') + \operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})).$$

Let us calculate $\operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})) = \dim \mathcal{O}_{S'_s,s'}$ (where S'_s is the fiber of S' at s). Let Z be the closed image of x in X_s and let $Z' \subset X'_s$ be the scheme defined by the cartesian square

$$\begin{array}{ccc} Z' & \longrightarrow & Z \\ \downarrow & & \downarrow \\ S'_s & \longrightarrow & \operatorname{Spec} k(s) \end{array} .$$

The morphism $Z \to \operatorname{Spec} k(s)$ is flat, locally of finite type, and $\dim Z = \deg \operatorname{tr} k(x)/k(s)$. It then follows from (EGA IV 6.1.2) that

$$\dim(\mathcal{O}_{Z',x'}) = \dim(\mathcal{O}_{S'_s,s'}) + \deg \operatorname{tr} k(x)/k(s) - \deg \operatorname{tr} k(x')/k(s');$$

taking into account the fact that $Z'_{s'} \simeq Z \otimes_{k(s)} k(s')$, one then obtains:

$$\delta_T(x) - \delta_{T'}(x') \leq \dim(\mathcal{O}_{Z',x'}).$$

But $\operatorname{Spec}(\mathcal{O}_{Z',x'})$ identifies with the fiber at x of the morphism

$$(\operatorname{Spec}(\mathcal{O}_{X',x'}))_s \to (\operatorname{Spec}(\mathcal{O}_{X,x}))_s,$$

and hence also with the fiber at x of $h_{x'}$, which proves the theorem.

XIV.2.3

The proofs of the theorems in no. XIV.4 are based on duality theory; they use the following lemmas. Let m be an integer that is a power of a prime number ℓ; if X is a scheme, all the **118** sheaves considered on X are sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules; there is then a notion of a dualizing complex on X (SGA 5 I 1.7). Suppose that such a complex K exists on X; then, for every geometric point $\bar{x}$ over a point x of X, one deduces from K (cf. SGA 5 I 4.5) a dualizing complex $K_{\bar{x}}$ on $\operatorname{Spec} k(\bar{x})$, so that $K_{\bar{x}} \simeq \mathbf{Z}/m\mathbf{Z}[n]$ (the brackets denoting the translation functor) for a certain integer n depending only on x. We set

$$\delta_K(x) = n.$$

If K is normalized at the point x (SGA 5 I 4.5), then $n = 0$.

Lemma XIV.2.3.1. *Let X be a locally noetherian scheme endowed with a dualizing complex K. If x and x' are two points of X, such that x is a specialization of x' and* $\operatorname{codim}(\overline{\{x\}}, \overline{\{x'\}}) = 1$, *then one has*

$$\delta^K(x) = \delta^K(x') - 2.$$

One may first reduce to the case where X is a strictly local scheme. Indeed, let $\bar{X}$ be the strict localization of X at a geometric point $\bar{x}$ over x, let $i : \bar{X} \to X$ be the canonical morphism, and let $\bar{x}'$ be a geometric point of $\bar{X}$ over x'. Then i^*K is a dualizing complex on $\bar{X}$, and one has (SGA 5 I 4.5)

$$(i^*K)_{\bar{x}} \simeq K_{\bar{x}} \text{ and } (i^*K)_{\bar{x}'} \simeq K_{\bar{x}'},$$

which completes the reduction to the strictly local case. **119**

If $j : \overline{\{x'\}} \to X$ denotes the immersion of the reduced closed subscheme of X whose underlying space is $\overline{\{x'\}}$, then $\mathrm{R}^!j(K)$ is a dualizing complex on $\overline{\{x'\}}$, and one sees at once, using (SGA 5 I 4.5), that it suffices to prove the lemma for $\overline{\{x'\}}$. One is thus reduced to the case where X is an integral strictly local scheme of dimension 1.

Let X' then be the normalization of X and let $f : X' \to X$ be the canonical morphism; f is integral, surjective, and radicial, and it follows that f^*K is a dualizing complex on X' and that it suffices to prove the lemma for X' and for the points over x and x'. One is thus reduced to the case where X is a local, integral, regular scheme of dimension 1, but it is known (cf. SGA 5 I 4.6.2 and 5.1) that then $\mu_m[2]$ and $\mathbf{Z}/m\mathbf{Z}$ are dualizing complexes, normalized at the points x and x', respectively; the lemma follows at once.

Lemma XIV.2.3.2. *Let S be a noetherian local scheme and $f : X \to S$ a morphism of finite type. If K is a dualizing complex on S, normalized at the closed point t of S, and if $\mathrm{R}^!f(K) = K'$ is a dualizing complex on X (cf.* SGA *5 I 3.4.3), one has, for every point x of X:*

$$\delta^{K'}(x) = 2\delta(x).$$

Indeed, let $s = f(x)$ and let x' be a closed point of the fiber X_s; then $\delta^{K'}(x') = \delta^K(s)$ and, by XIV.2.3.1,

$$\delta^K(s) = 2\operatorname{codim}(\overline{\{t\}}, \overline{\{s\}}) = 2\dim(\overline{\{s\}}).$$

Since x' may be chosen to be a specialization of x, one has, by XIV.2.3.1, **120**

$$\delta^{K'}(x) = \delta^{K'}(x') + 2\operatorname{codim}(\overline{\{x\}}, \overline{\{x'\}}) = \delta^{K'}(x') + 2\deg \operatorname{tr} k(x)/k(s);$$

the lemma follows at once.

The following lemma will be used only for the converse of the Lefschetz theorem, in no. XIV.4:

Lemma XIV.2.3.3. *Let there be a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array},$$

where S is an excellent strictly local scheme of characteristic zero, S' is the completion of S, and f is a morphism of finite type. Let ℓ be a prime number, $x \in X$, let Z be the schematic closure of X'_x in X', and let $i : X'_x \to Z$, $j : Z \to X$ be the canonical morphisms. Then, if $k : X' \to R$ is a closed immersion of X' into an excellent regular scheme R of characteristic zero, the complex

$$K' = i^*(R^!(k.j)(\mathbf{Z}/\ell\mathbf{Z}))$$

is a constant dualizing complex on X'_x (that is, it has only one nonzero cohomology sheaf, isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$).

In view of (SGA 5 I 3.4.3), the only thing to prove is that K' is constant. But, since Z is excellent, the set of points of Z whose local rings are regular is an open subset U (EGA IV 7.8.3 (iv)), and U plainly contains X'_x, which is regular. Let **121**

$$u : U \to R$$

then be the canonical immersion of U into R; it follows from the purity theorem (SGA 4 XIX 3.2 and 3.4) and the isomorphism

$$(\mu_l)_S \simeq (\mathbf{Z}/\ell\mathbf{Z})_S$$

(S strictly local) that one has

$$\mathrm{R}^!u(\mathbf{Z}/\ell\mathbf{Z}) \simeq \mathbf{Z}/\ell\mathbf{Z}[2c],$$

where c is a locally constant function on U, necessarily constant in a neighborhood of X'_x, because the fibers of g are geometrically integral by (EGA IV 18.9.1), and hence X'_x is integral. The lemma follows at once.

XIV.3 Converse to the affine Lefschetz theorem

The present section will be used in no. XIV.4 to prove a converse to the "Lefschetz theorem"; a reader interested only in the direct part of that theorem may therefore omit the present section.

XIV.3.1

Recall the statement of the affine Lefschetz theorem(*) (SGA 4 XIX 6.1 bis):

Let S be a strictly local excellent scheme of characteristic zero, $f : X \to S$ an affine morphism of finite type, and F a torsion sheaf on X. Then, setting **122**

(*) *Editor's note.* Gabber proved the following generalization. Let Y be a strictly local scheme of arithmetic type over a regular Noetherian scheme S of dimension ≤ 1. Let $f : X \to Y$ be an affine morphism of finite type, $\Lambda = \mathbf{Z}/n\mathbf{Z}$ with n invertible on X, and F a Λ-sheaf. Then $H^q(X, F) = 0$ if $q > \delta(F)$. The following local Lefschetz theorem follows. Let $\mathcal{O}$ be strictly local of arithmetic type over S. For every non-zero-divisor $f \in \mathcal{O}$ and every Λ-sheaf F on $\mathrm{Spec}(\mathcal{O}[f^{-1}])$, one has $H^q(\mathrm{Spec}(\mathcal{O}[f^{-1}]), F) = 0$ for $q > \dim(\mathcal{O})$. Cf. (Fujiwara K., "A Proof of the Absolute Purity Conjecture (after Gabber)", in *Algebraic geometry 2000, Azumino (Hotaka)*, Adv. Stud. in Pure Math., vol. 36, 2002, p. 153-183, §5), and especially Illusie's article (Illusie L., "Perversité et variation", *Manuscripta Math.* **112** (2003), p. 271-295). This result is one of the crucial points used by Gabber in his proof of Grothendieck's purity theorem (cf. note ((*)), page 140).

$$\delta(F) = \sup\{\delta(x) | x \in X \text{ and } F_{\bar{x}} \neq 0\},$$

one has

$$H^q(X, F) = 0 \text{ for } q > \delta(F).$$

Before stating the converse, let us prove a few lemmas.

Lemma XIV.3.2. *Let K be a field, ℓ a prime number distinct from the characteristic of K, and F a nonzero constructible sheaf of ℓ-torsion on K. Suppose that the ℓ-cohomological dimension of K (*SGA 4 X 1*) is equal to n (this holds, for example, if K is the field of fractions of an integral strictly local excellent ring of characteristic zero and dimension n (*SGA 4 XIX 6.3*), or if K is an extension of finite type and transcendence degree n over a separably closed field (*SGA 4 X 2.1*)). Then one can find a finite separable extension L of K such that:*

$$H^n(L, F_{|L}) \neq 0.$$

One can find a finite extension K' of K such that the restrictions of F and μ_ℓ to $\operatorname{Spec} K'$ are constant sheaves. Then $\operatorname{cd}_\ell(K') = \operatorname{cd}_\ell(K) = n$ (SGA 4 X 2.1), and it follows from ([2] II §3 Prop. 4 (iii)) that one can find a finite extension L of K' such that **123**

$$H^n(L, \mu_\ell) \neq 0, \quad \text{i.e. } H^n(L, \mathbf{Z}/\ell\mathbf{Z}) \neq 0.$$

Now the functor $H^n(L, \cdot)$ is right exact on the category of sheaves of ℓ-torsion, since $\operatorname{cd}_\ell(L) = n$; as F has a quotient isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$, one also has $H^n(L, F_{|L}) \neq 0$.

Corollary XIV.3.3. *Let k be a field, K an extension of finite type and transcendence degree n over k, and F a nonzero constructible sheaf of ℓ-torsion on K, with ℓ prime to the characteristic of k. Then one can find a finite separable extension L of K such that, if $u : \operatorname{Spec} L \to \operatorname{Spec} k$ denotes the canonical morphism, one has*

$$R^n u_*(F_{|\operatorname{Spec} L}) \neq 0.$$

When the field k is separably closed, the corollary is a special case of XIV.3.2. In the general case, one can find a finite separable extension k_1 of k such that the irreducible components of $K \otimes_k k_1$ are geometrically irreducible (EGA IV 4.5.11); let K_1 be one of them. If k' is a separable closure of k_1, then $K' = K_1 \otimes_{k_1} k'$ is a field, and by (EGA IV 4.2) one has

$$\deg \operatorname{tr} K'/k' = \deg \operatorname{tr} K/k = n.$$

It then follows from XIV.3.2 that one can find a finite separable extension L' of K' such that $H^n(L', F_{|L'}) \neq 0$. But $k' = \varinjlim_i k_i$, where k_i runs through the finite extensions of k_1 contained in k', and consequently $K' = \varinjlim_i (k_i \otimes_{k_1} K_1)$. It follows that one can find an index i and a **124** finite separable extension L of $k_i \otimes_{k_1} K_1 = K_i$ such that $L' \simeq L \otimes_{K_i} K'$. The extension L of K has the required property; indeed, it follows from the commutative diagram

$$\begin{array}{ccc} & \operatorname{Spec} L & \\ {\scriptstyle v}\swarrow & & \searrow{\scriptstyle u} \\ \operatorname{Spec} k_i & \xrightarrow{\quad w \quad} & \operatorname{Spec} k\,, \end{array}$$

with w finite and hence $R^q w_* = 0$ if $q > 0$, that

$$R^n u_*(F_{|\operatorname{Spec} L}) \simeq w_*(R^n v_*(F_{|\operatorname{Spec} L})).$$

Now $R^n v_*(F_{|\operatorname{Spec} L}) \neq 0$, since $H^n(L', F_{|L'}) \neq 0$; therefore one also has $R^n u_*(F_{|\operatorname{Spec} L}) \neq 0$.

Recall the following familiar lemma (cf. EGA 0_{III} 10.3.1.2 and EGA IV 18.2.3):

Lemma XIV.3.4. *Let X be a scheme, x a point of X, and K a finite separable extension of $k(x)$. Then there exist a scheme X_1 étale over X and affine, and a point $x_1 \in X_1$ above x, such that $k(x_1)$ is $k(x)$-isomorphic to K.*

In no. XIV.4 we shall use the following technical form of the converse to XIV.3.1.

Proposition XIV.3.5. *Let there be a Cartesian square* **125**

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S, \end{array}$$

*where the schemes S and S' are strictly local excellent schemes of characteristic zero, the morphism f is locally of finite type, g is regular (*EGA IV 6.8.1*) and surjective, and the closed fiber of g is reduced to the closed point of S'. Given an S-scheme X_1 (respectively an S-morphism f_1, etc.), we denote by X_1' (respectively f_1', etc.) the scheme $X_1 \times_S S'$ (respectively the morphism $(f_1)_{(S')}$, etc.). Let F be a sheaf of $\mathbf{Z}/m\mathbf{Z}$-modules on X' (m a power of a prime number ℓ), constructible and satisfying the following conditions:*

(i) For every point $x \in X$, one can find a finite separable extension K of $k(x)$ such that the restriction of F to the fiber $(X')_{(\operatorname{Spec} K)}$ comes by inverse image from a constructible sheaf on $\operatorname{Spec} K$.

(ii) For every morphism $f_1 : X_1 \to S$, with X_1 affine and étale over X, for every point $s \in S$, and for every integer $q > 0$, one can find a finite separable extension K of $k(s)$ such that the restriction of $R^q f_{1*}'(F|X_1')$ to the fiber $S'_{(\operatorname{Spec} K)}$ comes by inverse image from a constructible sheaf on $\operatorname{Spec} K$.

Let n be an integer, and suppose that for every affine scheme X_1 étale over X one has **126**

$$H^i(X_1', F) = 0 \text{ for } i > n.$$

Then, if $\bar{x}'$ is a geometric point above the point $x' \in X'$ such that $F_{\bar{x}'} \neq 0$, one has

$$\delta(x') \leq n.$$

Let Z' be the set of points x' of X' such that $F_{\bar{x}'} = 0$. Then, if $Z = h(Z')$, one has, by (i), $Z' = h^{-1}(Z)$; let $x' \in X'$, $x = h(x')$, $s' = f'(x')$, $s = f(x)$. It follows from XIV.2.1.1 and the fact that the function δ decreases under specialization that it suffices to prove the inequality $\delta(x') \leq n$ when x is a maximal point of Z and x' is such that

$$r = \deg \operatorname{tr} k(x)/k(s) = \deg \operatorname{tr} k(x')/k(s') \text{ and } d = \dim \overline{\{s\}} = \dim \overline{\{s'\}}$$

Let x' be such a point; it suffices to show that one can find an affine scheme X_1 étale over X such that

$$H^{d+r}(X_1', F) \neq 0.$$

The set Z' is constructible (SGA 4 IX 2.4), and hence so is Z (EGA IV 1.9.12); one may then suppose, after restricting X to a neighborhood of x, that Z is an irreducible closed subset with generic point x. Let $T = f(Z)$; T is a constructible set contained in $\overline{\{s\}}$; one can therefore find an affine open subset U of S such that $s \in U$ and $T \cap U = T_U$ is an irreducible closed subset of U with generic point s.

Let V then be an affine scheme étale over X, whose image in X contains x and whose **127** image in S is contained in U; let Z_V be the inverse image of Z in V and let $u : Z_V \to T_U$ be

the canonical morphism. Let W be an affine scheme étale over U; denote by T_W the inverse image of T_U in W and let $X_1 = W \times_U V$. Since F is zero outside Z', one has the spectral sequence

$$E_2^{pq} = H^p((T_W)', R^q u'_*(F_{|(Z_V)'})) \Longrightarrow H^*(X'_1, F).$$

We shall show that V and W may be chosen so that

a) $E_2^{pq} = 0$ for $p > d$ and for $q > r$.

b) $E_2^{dr} \neq 0$.

The relation $H^{d+r}(X'_1, F) \neq 0$ will then follow from the spectral sequence.

1°) Set $G_q = R^q u'_*(F_{|(Z_V)'})$; then one has:

$$(G_q)_{\bar{s}'} = H^q((Z_V)'_{\bar{s}'}, F_{|(Z_V)'\,\bar{s}'}),$$

because s' is a maximal point of $(T_U)'$. Since the fiber $(Z_V)'_{\bar{s}'}$ is an affine scheme of finite type of dimension r over a separably closed field, it follows from XIV.3.1 that

$$(G_q)_{\bar{s}'} = 0 \text{ for } q > r.$$

For $q > r$, let Y'_q be the set of points of $(T_U)'$ where the geometric fiber of G_q is $\neq 0$, **128**
and let $Y_q = g(Y'_q)$; then $Y'_q = g^{-1}(Y_q)$ by (ii), so Y_q is a constructible subset of T_U (SGA 4 XIX 5.1 and EGA 1.9.12) that does not contain s; after restricting U to an open neighborhood of s, one may suppose that $G_q = 0$ for $q > r$, and hence that $E_2^{pq} = 0$ for $q > r$.

Moreover, since $(T_W)'$ is an affine scheme of finite type over $g^{-1}(\overline{\{s\}})$, one has, for every q (cf. XIV.3.1):

$$H^p((T_W)', G_q) = 0 \text{ for } p > \dim g^{-1}(\overline{\{s\}}) = d,$$

whence condition a).

2°) Let us show that V may be chosen so that $(G_r)_{\bar{s}'} \neq 0$. By (i), there exists a constructible sheaf I, defined over a finite separable extension K of $k(x)$, whose inverse image on $(X')_{(\operatorname{Spec} K)}$ is isomorphic to $F_{|(X')_{(\operatorname{Spec} K)}}$. By XIV.3.3, one finds a finite separable extension L of K such that, if $v : \operatorname{Spec} L \to \operatorname{Spec} k(s)$ denotes the canonical morphism, then $R^r v_*(I) \neq 0$. Since the morphism $S'_s \to \operatorname{Spec} k(s)$ is regular, one has, by (SGA 4 XIX 4.2):

$$R^r v'_*(F_{|(\operatorname{Spec} L)'}) \simeq (R^r v_*(I))' \neq 0.$$

By Lemma XIV.3.4, one can find an affine scheme X_2 étale over X, and a point x_2 of X_2 over x, such that L is $k(x)$-isomorphic to $k(x_2)$, and one may suppose that X_2 lies over U. Since x is a maximal point of Z, one has **129**

$$\operatorname{Spec} L \simeq \varprojlim_V Z_V,$$

where V ranges over the affine open neighborhoods of x_2. By passage to the limit (SGA 4 VII 5.8), after restriction to the geometric fiber at $\bar{s}'$, one deduces:

$$(R^r v'_*(F_{|\operatorname{Spec} I})')_{\bar{s}'} = \varinjlim_{V'} (R^r u'_*(F_{|(Z_V)'})_{\bar{s}'},$$

which indeed shows that one can find V such that $(G_r)_{\bar{s}'} \neq 0$.

3°) With the scheme V having been chosen in 2°), let us show that the scheme W may be chosen so that

$$E_2^{dr} = H^d((T_W)', G_r) \neq 0.$$

By (ii), there exists a constructible sheaf J, defined over a finite separable extension K of $k(s)$, whose inverse image on $(S')_{(\operatorname{Spec} K)}$ is isomorphic to $G_{r|(S')_{(\operatorname{Spec} K)}}$. By Lemma XIV.3.2, one can find a finite separable extension L of K such that $H^d(\operatorname{Spec} L, J) \neq 0$. Since the morphism $(S')_{(\operatorname{Spec} L)} \to \operatorname{Spec} L$ is acyclic (SGA 4 XIX 4.1 and XV 1.10 and 1.16), one has

$$H^d((\operatorname{Spec} L)', G_{r|(S')_{(\operatorname{Spec} L)'}}) = H^d(\operatorname{Spec} L, J) \neq 0.$$

By XIV.3.4, one can find an affine scheme U_1 étale over U, and a point s_1 over s, such that $k(s_1)$ is $k(s)$-isomorphic to L. Since s is a maximal point of T_U, one has

$$\operatorname{Spec} L \simeq \varprojlim_W T_W,$$

where W ranges over the affine open neighborhoods of s_1. One deduces that $(\operatorname{Spec} L)' \simeq$ **130**
$\varprojlim_W (T_W)'$, and by passage to the limit (SGA 4 VII 5.8):

$$H^d((\operatorname{Spec} L)', G_{r|(\operatorname{Spec} L)'}) \simeq \varinjlim_W H^d((T_W)', G_{r|(T_W)'});$$

consequently, one can find W such that

$$H^d((T_W)', G_{r|(T_W)'}) \neq 0,$$

which completes the proof of the theorem.

Corollary XIV.3.6. *The hypotheses concerning S, S', f, f', m are those of XIV.3.5. Let F now denote a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X', bounded below and with constructible cohomology, and whose cohomology sheaves satisfy conditions* (i) *and* (ii) *of XIV.3.5. Let n be an integer, and suppose that, for every affine scheme X_1 étale over X, one has*

$$H^i(X_1', F) = 0 \text{ for } i > n.$$

Then, if $\bar{x}'$ is a geometric point lying over a point x' of X' such that, for an integer j, $(\underline{H}^j(F))_{\bar{x}'} \neq 0$, one has

$$\delta(x') \leq n - j.$$

Let T' be the set of points of X' where the conclusion of XIV.3.7 fails, and suppose $T' \neq \emptyset$; let $T = f(T')$, let x be a maximal point of T, and let x' be a point of X' lying over x. Let j be the largest integer such that $(\underline{H}^j(F))_{\bar{x}'} \neq 0$; thus $r = \delta(x) > n - j$. Let Z_q' be the set **131**
of points where the geometric fiber of $\underline{H}^q(F)$ is $= 0$, and let $Z_q = h(Z_q')$; as in the proof of XIV.3.5, one sees that Z_q is constructible. Clearly $Z_q' = \emptyset$ for $q > n$ and for q sufficiently small. The other values of q fall into three subsets. Let

$$Q_1 = \{q \mid x \in Z_q \text{ and a generization of } x, \text{ distinct from } x, \notin Z_q\}.$$

One has $j \in Q_1$, and one can find an affine open neighborhood U_1 of x such that, for every $q \in Q_1$, $U_1 \cap Z_q$ is an irreducible closed subset with generic point x. If $q \in Q_1$, one has

$$\delta(\underline{H}^q(F)_{|U_1'}) = \delta(x) \quad \text{(for the definition of } \delta(\underline{H}^q(F)), \text{ cf. XIV.3.1).} \qquad (*)$$

Let

$$Q_2 = \{q \mid \text{no generization of } x \text{ belongs to } Z_q\}.$$

Then, if $j < q \leq n$, one has $q \in Q_2$, and one can find an affine open neighborhood U_2 of x such that, for every $q \in Q_2$, $Z_q \cap U_2 = \emptyset$; thus one has

$$\underline{H}^q(F)_{|U_2'} = 0 \text{ for } q \in Q_2. \qquad (**)$$

Finally, let

$$Q_3 = \{q \mid Z_q \text{ contains strict generizations of } x\}.$$

Then one can find an affine open neighborhood of x, U_3, such that, for every $q \in Q_3$, all the maximal points of $Z_q \cap U_3$ are generizations of x. If $q \in Q_3$, one has

$$\delta(\underline{H}^q(F)_{|U_3'}) \leq n - q. \qquad (***)$$

For every affine scheme X_1 étale over $U_1 \cap U_2 \cap U_3$, consider the hypercohomology spectral **132** sequence

$$E_2^{pq} = H^p(X_1', \underline{H}^q(F)) \Longrightarrow H^*(X_1', F).$$

One has $E_2^{pq} = 0$ for $q \in Q_2$ by $(**)$. One has $E_2^{pq} = 0$ for $p + q \geq r + j$ except possibly for $p = r, q = j$. Indeed, this is clear if $q \in Q_2$; if $q \in Q_1$, then $p > r$ except when $p = r, q = j$, and this follows from XIV.3.1 in view of $(*)$; finally, if $q \in Q_3$, since $r > n - j$, one has $p > n - q$, and the assertion follows from XIV.3.1 in view of $(***)$. Since $H^{r+j}(X_1', F) = 0$, it follows from the spectral sequence that

$$H^r(X_1', \underline{H}^j(F)) = 0;$$

but, by XIV.3.5, this implies $\delta(x) < r$, which is absurd.

Corollary XIV.3.7. *Let S be an excellent strictly local scheme of characteristic zero, $f : X \to S$ a morphism locally of finite type, m a power of a prime number, F a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below, with constructible cohomology, and n an integer. Then the following conditions are equivalent:*

(i) For every affine scheme X_1 étale over X, one has

$$H^i(X_1, F) = 0 \text{ for } i > n.$$

(ii) For every geometric point $\bar{x}$ lying over the point x of X, and for every integer j such **133** *that $(\underline{H}^j(F))_{\bar{x}} \neq 0$, one has*

$$\delta(x) \leq n - j.$$

(i) $\Longrightarrow$ (ii) is the special case of XIV.3.6 obtained by taking $S = S'$.
(ii) $\Longrightarrow$ (i) follows immediately from XIV.3.1, using the hypercohomology spectral sequence

$$H^p(X_1, \underline{H}^q(F)) \Longrightarrow H^*(X_1, F).$$

XIV.4 Main theorem and variants

XIV.4.0

Let $g\colon X \to S$ be a separated morphism of finite type, let T be a closed subset of S, let $Z = g^{-1}(T)$, and let F be a complex of abelian sheaves on X, bounded below. We call the group

$$\mathrm{H}^i_{Z!}(X/S, F) = \mathrm{H}^i_T(S, \mathrm{R}_!g(F))$$

the *i-th cohomology group of F, with proper support and support in Z*, where $\mathrm{R}_!g$ denotes the "direct image with proper support"(SGA 4 XVII). In the particular case in which g is proper, one simply has

$$\mathrm{H}^i_{Z!}(X/S, F) = \mathrm{H}^i_Z(X, F).$$

Proposition XIV.4.1. *Let $f\colon U \to S$ be a morphism of finite type and let F be a complex of abelian sheaves on U, bounded below. Suppose that a factorization of f is given:* **134**

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & X \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array}$$

where i is an open immersion and g is a separated morphism of finite type, and let G be a complex of abelian sheaves on X, bounded below, that extends F. Let Y be a closed subscheme of X whose underlying space is $X - U$, so that one has a commutative diagram:

$$\begin{array}{ccc} Y & \xrightarrow{\ j\ } & X \\ & {\scriptstyle h}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array}$$

Finally, let n be an integer and let T be a closed subset of S. Then the following conditions are equivalent:

(i) One has $\operatorname{prof}_T(\mathrm{R}_!f(F)) \geq n$.

(ii) The canonical morphism

$$\underline{H}^i_T(\mathrm{R}_!g(G)) \to \underline{H}^i_T(\mathrm{R}_!h(j^*G))$$

is bijective for $i < n-1$ and injective for $i = n-1$.

(iii) For every scheme S' ètale over S, if X' (resp. f', resp. etc.) denotes the scheme $X \times_S S'$ (resp. the morphism $f_{(S')}$, resp. etc.), the canonical morphism **135**

$$\mathrm{H}^i_{g'^{-1}(T')!}(X'/S', G') \to \mathrm{H}^i_{h'^{-1}(T')!}(Y'/S', j'^*G')$$

is bijective for $i < n-1$ and injective for $i = n-1$.

In the derived category $\mathrm{D}^+(X)$ (cf. [3]), consider the distinguished triangle

$$\begin{array}{ccc} & j_*j^*G & \\ & \swarrow \quad \nwarrow & \\ i_!F & \longrightarrow & G. \end{array}$$

Applying the functor $\mathrm{R}_!g$ to this triangle gives the triangle

$$\begin{array}{ccc} & \mathrm{R}_!h(j^*G) & \\ & \swarrow \quad \nwarrow & \\ \mathrm{R}_!f(F) & \longrightarrow & \mathrm{R}_!g(G). \end{array} \qquad (*)$$

Let us show that (i) $\Longleftrightarrow$ (ii). Indeed, by Definition XIV.1.2, (i) is equivalent to the relation

$$\underline{H}^i_T(\mathrm{R}_!f(F)) = 0 \quad \text{for } i < n;$$

while $(*)$ gives the exact sequence of sheaves

$$\to \underline{H}^i_T(\mathrm{R}_!f(F)) \to \underline{H}^i_T(\mathrm{R}_!g(G)) \to \underline{H}^i_T(\mathrm{R}_!h(j^*G)) \to,$$

which proves the equivalence of (i) and (ii). **136**

(i) $\Longleftrightarrow$ (iii). Indeed, (i) is equivalent to saying that, for every scheme S' ètale over S, one has

$$\mathrm{H}^i_{T'}(S', \mathrm{R}_! f'(F')) = 0 \quad \text{for } i < n. \tag{$**$}$$

But $(*)$ gives the exact sequence of abelian groups

$$\to \mathrm{H}^i_{T'}(S', \mathrm{R}_! f'(F')) \to \mathrm{H}^i_{T'}(S', \mathrm{R}_! g'(G')) \to \mathrm{H}^i_{T'}(S', \mathrm{R}_! h'(j'^* G')) \to ;$$

in view of XIV.4.0, this exact sequence takes the form

$$\to \mathrm{H}^i_{T'}(S', \mathrm{R}_! f'(F')) \to \mathrm{H}^i_{g'^{-1}(T')!}(X'/S', G') \to \mathrm{H}^i_{h'^{-1}(T')!}(Y'/S', j'^* G') \to .$$

The equivalence of (i) and (iii) follows, taking into account the form $(**)$ of (i).

When $f\colon U \to S$ is affine, we shall give *local* conditions on F under which conditions (i) through (iii) of XIV.4.1 hold. In what follows, all schemes considered are excellent schemes of characteristic zero, and all sheaves are sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules, where m is a power of a prime number. If resolution of singularities in the sense of (SGA 4 XIX) were available, the stated results and their proofs would remain valid for excellent schemes of equal characteristic, with m prime to the characteristic.

Theorem XIV.4.2. *Let S be an excellent scheme of characteristic zero and let $f\colon U \to S$* **137** *be a separated morphism of finite type. Let F be a bounded-below complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on U with constructible cohomology, let n be an integer, and let T be a closed subset of S. Then the following conditions are equivalent:*

(i) For every scheme U_1 ètale over U and affine over S, denoting by f_1 the structural morphism of U_1 and by F_1 the restriction of F to U_1, one has

$$\mathrm{prof}_T(\mathrm{R}_! f_1(F_1)) \geq n$$

(cf. Proposition. XIV.4.1 for the meaning of this relation).

(ii) For every point u of U, one has

$$\mathrm{prof}_u(F) \geq n - \delta_T(u),$$

where one sets (cf. XIV.2.2): $\delta_T(u) = \deg \mathrm{tr}(k(x)/k(s)) + \mathrm{codim}\,(\overline{\{s\}} \cap T, \overline{\{s\}})$.

Proof. 1°) Let t be a point of T, let $\overline{S}$ be the strict localization of S at a geometric point above t, and let S' be the completion of $\overline{S}$, with closed point t'; then $\overline{S}$ is excellent by (EGA IV 7.9.5), so S' is an excellent complete strictly local scheme. Given a scheme U over X (respectively an S-morphism f, respectively etc.), we shall denote by U' (respectively f', respectively etc.) the scheme $U \times_S S'$ (respectively the morphism $f_{(S')}$, respectively etc.). We have the Cartesian square

$$\begin{array}{ccc} U' & \xrightarrow{h} & U \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S, \end{array}$$

in which the morphism g is regular (EGA IV 7.8.2). Let us show that it is enough to prove **138** that (for every point $t \in T$) the following two properties are equivalent:

(i)$_t$ For every scheme U_1 étale over U and affine over S, setting $f_1 : U_1 \to S$, one has

$$\operatorname{prof}_{t'}(\mathrm{R}_! f_1'(F_1')) \geq n.$$

(ii)$_t$ For every point u' of U', one has

$$\operatorname{prof}_{u'}(F') \geq n - \delta_{t'}(u').$$

It is enough to prove the following lemma:

Lemma XIV.4.2.1. *One has* (i) $\iff$ (i)$_t$ *for every* $t \in T$ *and* (ii) $\iff$ (ii)$_t$ *for every* $t \in T$.

(i) $\iff$ (i)$_t$ for every $t \in T$. Indeed, (i) is equivalent to saying that, for every scheme U_1 étale over U and affine over S, one has

$$\operatorname{prof}_T(\mathrm{R}_! f_1(F_1)) \geq n \ ;$$

now, by XIV.1.8,

$$\operatorname{prof}_T(\mathrm{R}_! f_1(F_1)) = \inf_{t \in T} \operatorname{prof}_t(\mathrm{R}_! f_1(F_1)).$$

Since $g^*(\mathrm{R}_! f_1(F_1)) \simeq \mathrm{R}_! f_1'(F_1')$ (SGA 4 XVII), one has, by XIV.1.16,

$$\operatorname{prof}_t(\mathrm{R}_! f_1(F_1)) = \operatorname{prof}_{t'}(\mathrm{R}_! f_1'(F_1')),$$

and hence (i) is equivalent to saying that, for every $t \in T$, one has $\operatorname{prof}_{t'}(\mathrm{R}_! f_1'(F_1')) \geq n$, which **139**
is precisely (i)$_t$.

(ii)$_t$ for every $t \in T \implies$ (ii). Indeed, let $u \in U$; we must show the relation

$$\operatorname{prof}_u(F) \geq n - \delta_T(u),$$

where $\delta_T(u) = \inf_{t \in T \cap \overline{\{s\}}} \delta_t(u)$ (cf. XIV.2.2); we are therefore reduced to showing that, for every $t \in T \cap \overline{\{s\}}$, one has

$$\operatorname{prof}_u(F) \geq n - \delta_t(u).$$

Let u' be a point of U' such that $h(u') = u$ and $\delta_{t'}(u') = \delta_t(u)$ (cf. XIV.2.1.1). Since h is locally acyclic (SGA 4 XIX 4.1), it follows from XIV.1.16 and the fact that u' is a generic point of U'_u that

$$\operatorname{prof}_{u'}(F') = \operatorname{prof}_u(F).$$

But by (ii)$_t$ one has $\operatorname{prof}_u(F) = \operatorname{prof}_{u'}(F') \geq n - \delta_t(u)$, which proves (ii).

(ii) $\implies$ (ii)$_t$ for every t. With the notation of XIV.2.2.1, for every point u' of U' one has, by XIV.1.16,

$$\operatorname{prof}_{u'}(F') \geq \operatorname{prof}_u(F) + 2 \dim h_{u'}^{-1}(u) \geq \operatorname{prof}_u(F) + \dim h_{u'}^{-1}(u).$$

Taking into account XIV.2.2.1 and (ii), we obtain

$$\operatorname{prof}_{u'}(F') \geq n - \delta_T(u) + \dim h_{u'}^{-1}(u) \geq n - \delta_{t'}(u'),$$

which is precisely (ii)$_t$. **140**

2°) (ii)$_t$ $\iff$ (i)$_t$. We reduce immediately to the case where F is bounded, by truncating F at a sufficiently high degree. We can realize S' as a closed subscheme of a complete regular local scheme, hence an excellent one; it then follows from (SGA 5 I 3.4.3) that there exists a dualizing complex K on S' and that $\mathrm{R}^! f'(K) = K'$ is a dualizing complex on U'. We choose K so that $\delta^K(t') = 0$ (for the definition of $\delta^K(t')$, cf. XIV.2.3), and denote by $\mathrm{D}\, F'$ the dual of F' with respect to K'. We can reformulate the hypothesis (ii)$_t$ as follows:

Lemma XIV.4.2.2. *Let u' be a point of U'; then the following conditions are equivalent:*

(i) One has $\operatorname{prof}_{u'}(F') \geq n - \delta_{t'}(u')$.

(ii) One has $(\underline{H}^q(\mathrm{D}\,F'))_{\overline{u}'} = 0$ *for* $q > -n - \delta_{t'}(u')$ *($\overline{u}'$ a geometric point above u').*

Let $\overline{U}'$ be the strict localization of U' at $\overline{u}'$, and let $\overline{F}'$ be the inverse image of F under the morphism $\overline{U}' \to U'$. The relation $\operatorname{prof}_{u'}(F') \geq n - \delta_{t'}(u')$ is, by definition, equivalent to the following:

$$\mathrm{H}^i_{\overline{u}'}(\overline{F}') = 0 \quad \text{for } i > n - \delta_{t'}(u'). \tag{$*$}$$

Let $\mathrm{D}\,(\mathrm{H}^i_{\overline{u}'}(\overline{F}'))$ be the dual of the abelian group $\mathrm{H}^i_{\overline{u}'}(\overline{F}')$ with respect to $\mathbf{Z}/m\mathbf{Z}$. By XIV.2.3.2, $K'[-2\delta_{t'}(u')] = K''$ satisfies $\delta^{K''}(u') = 0$; since F' has constructible cohomology, one has $\overline{\mathrm{D}\,F'} = \mathrm{D}(\overline{F}')$, and the local duality theorem (SGA 5 I 4.5.3) then shows that **141**

$$\mathrm{D}\,(\mathrm{H}^i_{\overline{u}'}(\overline{F}')) \simeq (\mathrm{H}^{-i-2\delta_{t'}(u')}(\mathrm{D}\,F'))_{\overline{u}'}.$$

Thus $(*)$ is equivalent to the relation

$$(\underline{H}^q(\mathrm{D}\,F'))_{\overline{u}'} = 0 \quad \text{for } q > -n - \delta_{t'}(u'). \tag{$**$}$$

We are now in a position to prove the theorem. The relation $(\mathrm{ii})_t$ is equivalent to relation $(**)$. Let $G^q = \underline{H}^q(\mathrm{D}\,F')$; the affine Lefschetz theorem (XIV.3.1) implies in particular that, for every scheme U_1 étale over U and affine over S, one has

$$\mathrm{H}^p(U'_1, G^q) = 0 \quad \text{for } p > \delta(G^q),$$

where $\delta(G^q)$ is the upper bound of the $\delta_{t'}(u')$ for those u' such that $G^q_{\overline{u}'} \neq 0$; by $(**)$, one has $\delta(G^q) \leq -n - q$, so $(\mathrm{ii})_t$ implies the relation

$$\mathrm{H}^p(U'_1, \underline{H}^q(\mathrm{D}\,F')) = 0 \quad \text{for } p > -q - n.$$

Taking into account the hypercohomology spectral sequence for the functor "sections on U'_1"with respect to the complex $\mathrm{D}\,F'$:

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(U'_1, \underline{H}^q(\mathrm{D}\,F')) \Longrightarrow \mathrm{H}^*(U'_1, \mathrm{D}\,F'),$$

we obtain the relation

$$\mathrm{H}^i(U'_1, \mathrm{D}\,F') = 0 \quad \text{for } i > -n. \tag{$***$}$$

Conversely, suppose that the preceding relation holds for every U_1 étale over U and **142** affine over S. Apply Proposition XIV.3.6, replacing S there by $\overline{S}'$; the hypotheses of XIV.3.6 concerning $\overline{S}$ are satisfied, since, for every affine scheme $\overline{U}_1$ étale over $\overline{U}$, one can find a scheme above $\overline{U}_1$ which comes by inverse image from a scheme étale over U and affine over S; as for the hypotheses concerning F, they are satisfied by XIV.2.3.3. Thus, for every point u' of U' such that $(\underline{H}^q(\mathrm{D}\,F'))_{\overline{u}'} \neq 0$, one has:

$$\delta_{t'}(u') \leq -n - q,$$

which is precisely relation $(**)$; we have therefore proved the equivalence

$$(\mathrm{ii})_t \Longleftrightarrow (***).$$

We shall transform relation $(***)$; first, one has

$$\mathrm{H}^i(U'_1, \mathrm{D}\,F') = (\underline{H}^i(\mathrm{R}f'_{1*}(\mathrm{D}\,F'_1)))_{t'}\ ;$$

but by (SGA 5 I 1.12), there is a canonical isomorphism

$$\mathrm{R}f'_{1*}(\mathrm{D}\,F'_1) \simeq \mathrm{D}\,(\mathrm{R}_!f'_1(F'_1)),$$

where $\mathrm{D}\,(\mathrm{R}_!f'_1(F'_1))$ denotes the dual of $\mathrm{R}_!f'_1(F'_1)$ with respect to K. We thus see that $(\mathrm{ii})_t$ is equivalent to

$$(\underline{H}^i(\mathrm{D}(\mathrm{R}_!f'_1(F'_1))))_{t'} = 0 \quad \text{for } i > -n.$$

Applying the local duality theorem (SGA 5 I 4.5.3) once again, but this time at the point t', **143** we find that

$$(\underline{H}^i(\mathrm{D}(\mathrm{R}_!f'_1(F'_1))))_{t'} \simeq \mathrm{D}\,(\mathrm{H}^{-i}_{t'}(\mathrm{R}_!f'_1(F'_1))),$$

and finally $(\mathrm{ii})_t$ is equivalent to the relation

$$\mathrm{H}^i_{t'}(\mathrm{R}_!f'_1(F'_1)) = 0 \quad \text{for } i < n,$$

that is, $\mathrm{prof}_{t'}(\mathrm{R}_!f'_1(F'_1)) \geq n$, which completes the proof of the theorem.

Remark XIV.4.2.3. The argument simplifies quite considerably when one assumes that S admits (at least locally) a dualizing complex (for example, that it is locally immersible in a regular scheme). This avoids recourse to a completion (the passage to the case where S is strictly local being immediate), to XIV.2.3.3, and to the rather unappealing technical statement XIV.3.6, which may then be replaced by the more congenial reference XIV.3.7.

Corollary XIV.4.3. *Let S be an excellent scheme of characteristic zero and $f\colon U \to S$ a separated morphism of finite type such that U is the* union of $c+1$ open subsets affine over S. *Let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules, bounded below and with constructible cohomology, let n be an integer, and let T be a closed subset of S. Suppose that, for every point $u \in U$, one has*

$$\mathrm{prof}_u(F) \geq n - \delta_T(u).$$

Then one has **144**

$$\mathrm{prof}_T(\mathrm{R}_!f(F)) \geq n - c.$$

Indeed, let U_j, $0 \leq j \leq c$, be a covering of U by open subsets U_j affine over S. Adopting again the notation of the proof of XIV.4.2, one has, for every j,

$$\mathrm{H}^i(U'_j, \underline{H}^q(\mathrm{D}\,F')) = 0 \quad \text{for } i > -n.$$

Using the spectral sequence that relates the cohomology of U to that of the covering formed by the U_j (SGA 4 V 2.4), the preceding relation shows that

$$\mathrm{H}^i(U', \underline{H}^q(\mathrm{D}\,F')) = 0 \quad \text{for } i > -n + c.$$

The corollary then follows from the end of the proof of XIV.4.2.

Corollary XIV.4.4. *Let S be an excellent scheme of characteristic zero, $g\colon X \to S$ a morphism, U an open subset of X that is the union of $c+1$ open subsets affine over S, Y a closed subscheme with underlying space $X - U$, and $j\colon Y \to X$ the natural morphism. Let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below and with constructible cohomology, let T be a closed subset of S, and let n be an integer. Suppose that, for every point u of U, one has*

$$\mathrm{prof}_u(F) \geq n - \delta_T(u).$$

Then the canonical morphism **145**

$$\mathrm{H}^i_{g^{-1}(T)!}(X/S, F) \to \mathrm{H}^i_{(g^{-1}(T)\cap Y)!}(Y/S, j^*F)$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

This follows immediately from XIV.4.1 and XIV.4.3.

Corollary XIV.4.5 (Local Lefschetz theorem)**.** *Let S be an excellent henselian local scheme of characteristic zero, t the closed point of S, X a scheme proper over $S' = S - \{t\}$, and U an open subset of X that is the union of $c+1$ affine open subsets. Let Y be a closed subscheme of X with underlying space $X - U$, let $j\colon Y \to X$ be the canonical morphism, let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below and with constructible cohomology, and let n be an integer. Suppose that, for every point u of U, one has*

$$\operatorname{prof}_u(F) \geq n - \delta'_t(u), \quad \text{where } \delta'_t(u) = \delta_t(u) - 1.$$

Then the canonical morphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, j^*F)$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

Let $f\colon U \to S$ be the canonical morphism; it follows from XIV.4.2, applied with n replaced by $n+1$, that

$$\operatorname{prof}_t(\mathrm{R}_!f(F_{|U})) \geq n + 1 - c.$$

The preceding relation shows that the canonical morphism **146**

$$\mathrm{H}^i(S, \mathrm{R}_!f(F_{|U})) \to \mathrm{H}^i(S', \mathrm{R}_!f(F_{|U}))$$

is bijective for $i < n - c$, and injective for $i = n - c$. Since $\mathrm{R}_!f(F_{|U})$ is zero outside S', one has $\mathrm{H}^i(S, \mathrm{R}_!f(F_{|U})) \simeq (\underline{H}^i(\mathrm{R}_!f(F_{|U})))_t = 0$, and consequently

$$\mathrm{H}^i(S', \mathrm{R}_!f(F_{|U})) = 0 \quad \text{for } i < n - c. \tag{$*$}$$

Let $g\colon X \to S'$, $h\colon Y \to S'$, $f'\colon U \to S'$ be the canonical morphisms. It follows from the distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}h_*(j^*F) & \\ \swarrow & & \nwarrow \\ \mathrm{R}_!f'(F_{|U}) & \longrightarrow & \mathrm{R}g_*(F) \end{array}$$

that condition $(*)$ is equivalent to the assertion that the morphism

$$\mathrm{H}^i(S', \mathrm{R}g_*(F)) \to \mathrm{H}^i(S', \mathrm{R}h_*(j^*F))$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$. Since this morphism identifies canonically with the morphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, j^*F),$$

the conclusion follows at once.

Corollary XIV.4.6 (Global Lefschetz theorem)**.** *Let S be the spectrum of a field, X a* **147** *scheme proper over S, and U an open subset of X that is the union of $c+1$ affine open subsets. Let Y be a closed subscheme of X with underlying space $X - U$, let $j\colon Y \to X$ be the canonical morphism, let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below and with constructible cohomology, and let n be an integer. Suppose that, for every point u of U, one has*

$$\operatorname{prof}_u(F) \geq n - \dim(\overline{\{u\}}).$$

Then the canonical morphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, j^*F)$$

is bijective for $i < n-c-1$, and injective for $i = n-c-1$.

More generally, if $g\colon X \to S$ is a separated morphism of finite type, with the hypotheses on S, U, Y, and F being the same as above, then the canonical morphism

$$\mathrm{H}^i_!(X/S, F) \to \mathrm{H}^i_!(Y/S, j^*F)$$

(where $\mathrm{H}^i_!$ denotes cohomology with proper support, that is, $\mathrm{H}^i_!(X/S,F) = \mathrm{H}^i(S, \mathrm{R}_!g(F))$) is bijective for $i < n-c-1$, and injective for $i = n-c-1$.

The corollary is a special case of XIV.4.4, with $T = S$.

Here is a partial converse to XIV.4.3:

Proposition XIV.4.7. *Let S be a Noetherian scheme and let $f\colon U \to S$ be a morphism of finite type. Suppose that there exists a dualizing complex K on S and that $\mathrm{R}^!f(K)$ is a dualizing complex on U. Let T be a closed subset of S and let c be an integer. Then the following conditions are equivalent:* **148**

(i) For every bounded-below complex F of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on U with constructible cohomology, and every integer n such that, for every point u of U,

$$\mathrm{prof}_u(F) \geq n - \delta_T(u),$$

one has

$$\mathrm{prof}_T(\mathrm{R}_!f(F)) \geq n - c.$$

(ii) For every constructible sheaf G of $\mathbf{Z}/m\mathbf{Z}$-modules on U, and every point $t \in T$, one has

$$(\mathrm{R}^p f_*(G))_{\bar{t}} = 0 \quad \text{for } p > \delta(G, f, t) + c$$

*(recall from (*SGA 4 XIX 6.0*) that $\delta(G,f,t) = \sup\{\delta_t(u) | t \in \overline{\{u\}} \text{ and } G_{\bar{u}} \neq 0\}$).*

N.B. Condition (ii) is satisfied by XIV.3.1 if f is separated and, locally on S for the ètale topology, U is a union of $c+1$ open subsets affine over S; thus XIV.4.7 contains XIV.4.3$^{(*)}$.

We may plainly assume that S is local and that T is the closed point t of S. The proof of (ii) $\Longrightarrow$ (i) is essentially identical to part 2°) of the proof of XIV.4.2. Let us briefly prove that (i) $\Longrightarrow$ (ii). The local duality theorem (SGA 5 I 4.3.2), applied to $\mathrm{D}\,G$, shows that **149**

$$\mathrm{D}\left(\mathrm{H}^i_{\bar{u}}(\mathrm{D}\,G)\right) \simeq (\underline{H}^{-i-2\delta_t(u)}(G))_{\bar{u}}.$$

Since G is concentrated in degree 0, one therefore has $\mathrm{H}^i_{\bar{u}}(\mathrm{D}\,G) = 0$ except possibly for $i = -2\delta_t(u)$; more precisely,

$$\mathrm{prof}_u(\mathrm{D}\,G) = \begin{cases} -2\delta_t(u) & \text{if } G_{\bar{u}} \neq 0, \\ \infty & \text{if } G_{\bar{u}} = 0. \end{cases}$$

It follows that, for every $u \in U$,

$$\mathrm{prof}_u(\mathrm{D}\,G) \geq -n - \delta_t(u).$$

Hypothesis (i) then gives $\mathrm{prof}_t(\mathrm{R}_!f(\mathrm{D}\,G)) \geq -n-c$. We transform this relation using the isomorphism $\mathrm{R}_!f(\mathrm{D}\,G) \simeq \mathrm{D}(\mathrm{R}f_*(G))$ (SGA 5 I 1.12) and applying the local duality theorem at t; this gives

$$(\underline{H}^i(\mathrm{R}f_*(G))_{\bar{t}} = 0 \quad \text{for } i > n + c,$$

which is precisely (ii).

XIV.4.8

Under the hypotheses of XIV.4.4 with g proper (resp. XIV.4.5, resp. XIV.4.6 with g proper), if V is an open neighborhood of Y in X, the morphism **150**

$$\mathrm{H}^i(V,F) \to \mathrm{H}^i(Y,j^*F)$$

is bijective for $i < n-c-1$ and injective for $i = n-c-1$. Indeed, if $\iota\colon V \to X$ is the canonical morphism, it is enough to apply XIV.4.4 (resp. XIV.4.5, resp. XIV.4.6) to the complex $\mathrm{R}\iota_*(F_{|V})$. One may ask whether the preceding morphism is bijective for $i = n-c-1$ and injective for $i = n-c$. It plainly suffices that the hypotheses hold with n replaced by $n+1$; the following proposition shows that slightly less is enough.

Proposition XIV.4.9. *Let S be an excellent local scheme of characteristic zero with closed point t (resp. assume in addition to the preceding conditions that S is Henselian), let $f\colon X \to S$ be a scheme proper over S (resp. proper over $S-\{t\}$), and let U be an open subset of X that is the union of $c+1$ affine open subsets. Let Y be a closed subscheme of X whose underlying space is $X-U$, let $j\colon Y \to X$ be the canonical morphism, let F be a bounded-below complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X with constructible cohomology, and let n be an integer. Suppose that, for every point u of U,*

$$\mathrm{prof}_u(F) \geq \inf(n-1, n-\delta_t(u)) \quad (\textit{resp.}\ \ \mathrm{prof}_u(F) \geq \inf(n-1, n+1-\delta_t(u))).$$

Then, for every open neighborhood V of Y in X, the canonical morphism **151**

$$\mathrm{H}^i_{f^{-1}(t)}(V,F) \to \mathrm{H}^i_{f^{-1}(t)\cap Y}(Y,j^*F) \quad (\textit{resp.}\ \ \mathrm{H}^i(V,F) \to \mathrm{H}^i(Y,j^*F))$$

is bijective for $i < n-c-2$ and injective for $i = n-c-2$. Moreover, there exists an open neighborhood V_0 of Y in X such that, for every such V with $V \subset V_0$, the canonical morphism

$$\mathrm{H}^i_{f^{-1}(t)\cap V}(V,F) \to \mathrm{H}^i_{f^{-1}(t)\cap Y}(Y,j^*F) \quad (\textit{resp.}\ \ \mathrm{H}^i(V,F) \to \mathrm{H}^i(Y,j^*F))$$

is bijective for $i < n-c-1$ and injective for $i = n-c-1$.

Proof. For simplicity, set $\delta'_t(u) = \delta_t(u)$ (resp. $\delta'_t(u) = \delta_t(u)-1$). The first assertion of XIV.4.9 follows from XIV.4.8, since the hypotheses of XIV.4.4 (resp. XIV.4.5) hold there with n replaced by $n-1$. They also hold for n itself, except at points u such that $\delta'_t(u) = 0$. Now, for $u \in U$, the condition $\delta'_t(u) = 0$ is equivalent to saying that u is a *closed point of U_t* (resp. a *closed point of X*). Let E be the set of points u of U such that $\delta'_t(u) = 0$; let us show that *for all but finitely many points $u \in E$*, one has $\mathrm{prof}_u(F) \geq n$. Let $\overline{S}$ be the strict localization of S at t, let S' be the completion of $\overline{S}$, with closed point t', and consider the Cartesian square

$$\begin{array}{ccc} U' & \xrightarrow{h} & U \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S. \end{array}$$

The depth hypotheses at the points of U are preserved when U is replaced by U' and F **152** by its inverse image F' on U'. Indeed, let $u' \in U'$ and $u = h(u')$. If $u \notin E$, one has $\mathrm{prof}_u(F) \geq n - \delta'_t(u)$, and XIV.4.2.1 shows that this implies $\mathrm{prof}_{u'}(F') \geq n - \delta'_t(u')$. If $u \in E$, then u' is a closed point of U'_t (resp. a point of $X' = X \times_S S'$ closed in X'), and, since the fiber U'_u of h at u has dimension zero and h is regular, XIV.1.16 gives $\mathrm{prof}_{u'}(F') = \mathrm{prof}_u(F) \geq n-1$.

(*) At least in the case where S locally admits a dualizing complex, for example when S is locally embeddable in a regular scheme.

Let K be a dualizing complex on S' normalized at the closed point t', and let $\mathrm{D}\,F'$ be the dual of F' with respect to $\mathrm{R}^!f(K)$. By XIV.4.2.2, the ètale-depth hypotheses at the points of U' translate into the relations

$$(\underline{H}^q(\mathrm{D}\,F'))_{\overline{u}'} = 0 \quad \text{for } q > -n - \delta'_t(u') \quad (\text{resp. } q > -n-2-\delta'_{t'}(u')),$$

if u' is not a point of $E' = h^{-1}(E)$, and

$$(\underline{H}^q(\mathrm{D}\,F'))_{\overline{u}'} = 0 \quad \text{for } q > -(n-1) \quad (\text{resp. } q > -n-1), \text{ if } u' \in E'.$$

Let $G = \underline{H}^{-(n-1)}(\mathrm{D}\,F')$ (resp. $G = \underline{H}^{-n-1}(\mathrm{D}\,F')$). Since G is a constructible sheaf, the set of points at which its geometric fiber is nonzero is constructible (SGA 4 IX 2.4 (iv)). By hypothesis this set is contained in the set E' of closed points of $U'_{t'}$ (resp. of points of U' closed in X'); it follows from XIV.4.9.1 below that this set consists of finitely many points. Applying XIV.4.2.2, one sees that $\mathrm{prof}_{u'}(F') \geq n$ for all but finitely many points of E'. By XIV.1.16, it follows that, for all but finitely many points of E, **153**

$$\mathrm{prof}_u(F) \geq n.$$

Let V be an open neighborhood of Y in X, contained in the complement in X of the finite set of points u of E for which $\mathrm{prof}_u(F) = n-1$. If $\iota\colon V \to X$ is the canonical immersion, set

$$F_1 = \mathrm{R}\iota_*(F_{|V})\ ;$$

then F_1 is a bounded-below complex of sheaves on X with constructible cohomology (SGA 4 XIX 5.1). We shall see that, for every point u of U, the complex F_1 satisfies

$$\mathrm{prof}_u(F_1) \geq n - \delta'_t(u). \qquad (*)$$

If $u \in U \cap V$, then $\mathrm{prof}_u(F_1) = \mathrm{prof}_u(F)$, and relation $(*)$ holds by hypothesis at the points of U that do not belong to E; at the latter points, it also holds by the choice of V. Finally, if $u \in U$ and $u \notin V$, then by XIV.1.6 g), $\mathrm{prof}_u(F_1) = \infty$. Apply XIV.4.4 (resp. XIV.4.5) with F replaced by F_1; this gives the asserted result, since, for every i,

$$\mathrm{H}^i_{f^{-1}(t)}(X, \mathrm{R}\iota_*(F_{|V})) \simeq \mathrm{H}^i_{f^{-1}(t)\cap V}(V, F) \quad (\text{resp. } \mathrm{H}^i(X, \mathrm{R}\iota_*(F_{|V})) \simeq \mathrm{H}^i(V, F).$$

Lemma XIV.4.9.1. *A constructible set E contained in the set of closed points of a Noetherian scheme X consists of finitely many points .* **154**

Indeed, E is a finite union of sets of the form $U \cap \complement V$, where U and V are open subsets of X; by hypothesis, all the points of $U \cap \complement V$ are generic points of this set, so there are only finitely many of them.

XIV.5 Geometric depth

To apply XIV.4.2 and its corollaries in practice, one needs a convenient criterion for verifying the étale-depth hypotheses at the points of U. We shall give such a criterion, using the local Lefschetz theorem XIV.4.5.

XIV.5.1

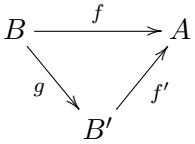
Let A be a Noetherian local ring; when we speak of the étale depth of A, we mean the depth at the closed point. We shall introduce a notion of "geometric depth of A"and use XIV.4.5 to compare it with the étale depth $\operatorname{prof}_{\text{ét}}(A)$.

Proposition XIV.5.2. *Let A be a Noetherian local ring; suppose that A is isomorphic to a quotient of a regular local ring B by an ideal I (this is true, for example, when A is complete, by Cohen's theorem (*EGA 0_{IV} 19.8.8*)). Let q be the minimal number of generators of I; then the number $\dim(B) - q$ is independent of the choice of B.*

The minimal number of generators of I is also equal to the dimension of the k-vector space $I \otimes_B k$, where k denotes the residue field of A. We reduce at once to the case where A is complete, since $\widehat{A} \simeq \widehat{B}/\widehat{I}$ with $\dim \widehat{B} = \dim B$ and $\operatorname{rg}_k(I \otimes_B k) = \operatorname{rg}_k(\widehat{I} \otimes_{\widehat{B}} k)$; for the same reason we may suppose that the ring B is complete. Let B and B' be two complete **155** regular local rings, $f\colon B \to A$ and $f'\colon B' \to A$ two surjective homomorphisms, and $I = \operatorname{Ker}(f)$, $I' = \operatorname{Ker}(f')$. We must show that

$$\dim B - \operatorname{rg}_k(I \otimes_B k) = \dim B' - \operatorname{rg}_k(I' \otimes_{B'} k).$$

First consider the case in which there is a factorization of the form



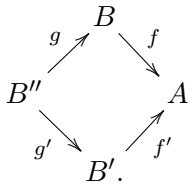
with g surjective. Let $J = \operatorname{Ker}(g)$; then $J \subset I$ and $I/J = I'$. Since B' is regular, $\dim(B') = \dim(B) - \operatorname{rg}_k(J \otimes_B k)$, and J is generated by elements belonging to a regular system of parameters of B. It follows that there is an exact sequence

$$0 \to J \otimes_B k \to I \otimes_B k \to J/I \otimes_{B'} k \to 0,$$

and consequently

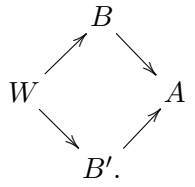
$$\dim B - \operatorname{rg}_k(I \otimes_B k) = \dim B - \operatorname{rg}_k(J \otimes_B k) - \operatorname{rg}_k(J/I \otimes_{B'} k) = \dim B' - \operatorname{rg}_k(I' \otimes_{B'} k).$$

The general case reduces to the preceding one; to see this, it is enough to show that one can find a complete regular local ring B'' and surjective homomorphisms $g\colon B'' \to B$ and $g'\colon B'' \to B'$ making the diagram commutative **156**



(∗)

Now, if W is a Cohen ring with residue field k, there is a local morphism $W \to A$ which lifts to B and B' (EGA IV 19.8.6), so that one has the commutative diagram



One can find integers n and n' and surjective morphisms $h\colon W[\![T_1,\dots,T_n]\!]\to B$ and $h'\colon W[\![T'_1,\dots,T'_{n'}]\!]\to B'$ which are morphisms of W-algebras (EGA 0_{IV} 19.8.8); if we then set $B''=W[\![T_1,\dots,T_n,T'_1,\dots,T'_{n'}]\!]$ and define g and g' as morphisms of W-algebras such that

$$g(T_i)=h(T_i),\quad g(T'_i)=b_i,\quad g'(T_i)=b'_i,\quad g'(T'_i)=h'(T'_i),$$

where b_i (respectively b'_i) is an element of B (respectively of B') lifting $(f'\circ h')(T'_i)$ (respectively $(f\circ h)(T_i)$), the diagram $(*)$ is indeed commutative.

Proposition XIV.5.2 justifies the following definition:

Definition XIV.5.3. Let A be a Noetherian local ring and $\widehat{A}$ its completion, which is thus **157** isomorphic to the quotient of a complete regular local ring B by an ideal I; if q is the minimal number of generators of I, the *geometric depth* of A is the number

$$\operatorname{prof}\text{ géom}(A)=\dim B-q$$

Proposition XIV.5.4. *Let A be a Noetherian local ring. Then*

$$\operatorname{prof}\text{ géom}(A)\le\dim A,$$

with equality if and only if A is a complete intersection.

We may suppose that A is complete. Let $A=B/I$, where B is a complete regular local ring and I an ideal of B. If $(x_1,\dots,x_q)$ is a minimal system of generators of I, then $\dim A\ge\dim B-q$, and saying that $\dim A=\dim B-q$ is equivalent to saying that $(x_1,\dots,x_q)$ belongs to a system of parameters of B (EGA 0_{IV} 16.3.7); the proposition follows immediately.

Proposition XIV.5.5. *Let A and A' be Noetherian local rings, and $f\colon A\to A'$ a local homomorphism. Suppose that f is flat and that, denoting by k the residue field of A, $A'\otimes_A k$ is a field, separable over k. Then*

$$\operatorname{prof}\text{ géom}(A)=\operatorname{prof}\text{ géom}(A').$$

After replacing A and A' by their completions, we may suppose that A and A' are complete **158** (it follows from (EGA 0_{III} 10.2.1) that the flatness hypothesis is preserved, and this is evident for the other hypotheses). Let $A=B/I$, where B is a regular local ring and I an ideal of B. Since A' is formally smooth over A (EGA 0_{IV} 19.8.2), it follows from (EGA 0_{IV} 19.7.2) that one can find a complete Noetherian local ring B' and a local homomorphism $B\to B'$ such that B' is a flat B-module and $B'\otimes_B A\simeq A'$. Thus $A'\simeq B'/IB'$; moreover the ring B' is regular: indeed, if $\mathfrak{m}$ is the maximal ideal of B, $\mathfrak{m}B'$ is the maximal ideal of B', and, since $\mathfrak{m}$ is generated by a regular sequence by the definition of "regular", $\mathfrak{m}B'$ is generated by a B'-regular sequence (EGA 0_{IV} 15.1.14). Since one clearly has $\dim B=\dim B'$, and since I and IB' have the same minimal number of generators, the assertion follows.

Theorem 5.6$^{(*)}$**.** *Let A be an excellent local ring of characteristic zero. Then*

$$\operatorname{prof}\text{ ét}(A)\ge\operatorname{prof}\text{ géom}(A).$$

One may suppose that A is strictly local and complete, since geometric depth and étale depth are preserved under passage to the strict henselization and to the completion by XIV.5.5 and XIV.1.16. Let $A\simeq B/I$, where B is a complete regular local ring, and let $(f_1,\dots,f_q)$ be a minimal system of generators of the ideal I. Thus one has **159**

$^{(*)}$*Editor's note.* Illusie subsequently proved the inequality $\operatorname{prof}_x(\mathbf{Z}/\ell^\nu\mathbf{Z})\ge\operatorname{prof}\text{ géom}_x(X/S)-\delta(x)+1$ for x a point of X, where X is a scheme of finite type over a trait S whose residual characteristic is prime to ℓ, and $\nu\ge1$. If S has characteristic zero, this is a consequence of Theorem XIV.5.6; see (Illusie L., "Perversité et variation", *Manuscripta Math.* **112** (2003), p. 271-295).

$$\pi = \text{prof géom}(A) = \dim B - q.$$

Consider the closed immersion

$$Y = \operatorname{Spec} A \to X = \operatorname{Spec} B,$$

and let $U = X - Y = \bigcup_{1 \le i \le q} X_{f_i}$. If a denotes the closed point of X, one must show that, for every prime number p, one has

$$\mathrm{H}^i_a(Y, \mathbf{Z}/p\mathbf{Z}) = 0 \quad \text{for } i < \pi.$$

Since B is regular and excellent, one has prof ét$(B) = 2 \dim X$ (cf. XIV.1.10) and consequently $\mathrm{H}^i_a(X, \mathbf{Z}/p\mathbf{Z}) = 0$ for $i < 2 \dim X$. Thus, to prove the theorem, it suffices to prove that the morphism

$$\mathrm{H}^i_a(X, \mathbf{Z}/p\mathbf{Z}) \to \mathrm{H}^i_a(Y, \mathbf{Z}/p\mathbf{Z}) \qquad (*)$$

is bijective for $i < \pi$. For this, apply the local Lefschetz theorem XIV.4.5 with $n = \pi + q - 1$, $c = q - 1$, and hence $n - c = \pi$. Note that $U = X - Y$ is the union of the q affine open subsets X_{f_i}. Let us show that, for every point u of U, one has:

$$\text{prof ét}_u(X) \ge \pi + q - 1 - \dim\left(\overline{\{u\}}\right) = \dim \mathcal{O}_{X,u}$$

(where $\overline{\{u\}}$ denotes the closure of u *in* $X - \overline{\{a\}}$). Indeed, it follows from XIV.1.10 that

$$\text{prof ét}_u(X) = 2 \dim \mathcal{O}_{X,u} \ge \dim \mathcal{O}_{X,u}.$$

Using XIV.4.5, one sees that $(*)$ is bijective for $i < \pi$, which completes the proof of the theorem. **160**

Corollary XIV.5.7. *Let S be the spectrum of a field of characteristic zero (resp. an excellent henselian local scheme of characteristic zero), and let $f\colon X \to S$ exhibit X as a scheme proper over S (resp. over $S - \{s\}$). Let U be a union of $c+1$ open subsets of X that are* affine*, let Y be a closed subscheme with underlying space $X - U$, and let n and m be integers > 0. Suppose that, for every point u of U, one has*

$$\text{prof géom}(\mathcal{O}_{X,u}) \ge n - \dim\left(\overline{\{u\}}\right)$$

($\overline{\{u\}}$ being the closure of u in X). Then the canonical morphism

$$\mathrm{H}^i(X, \mathbf{Z}/m\mathbf{Z}) \to \mathrm{H}^i(Y, \mathbf{Z}/m\mathbf{Z})$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

Apply XIV.4.5 and XIV.4.6. The hypotheses on étale depth at the points of U are satisfied because, by XIV.5.6, one has

$$\text{prof ét}_u(X) \ge \text{prof géom}(\mathcal{O}_{X,u}) \ge n - \dim\left(\overline{\{u\}}\right).$$

XIV.6 Open questions

XIV.6.1

One may ask whether the implication (ii) $\Longrightarrow$ (i) of XIV.4.2 holds more generally for torsion sheaves F, not necessarily annihilated by a given integer m and not necessarily constructible . In the case where S is not of characteristic zero, it seems possible that this implication remains valid, even for sheaves of p-torsion (p being the residue characteristic). Finally, it is also unclear whether the hypothesis that S is excellent can be removed. **161**

XIV.6.2

Let X be a scheme proper over a field k, or else the complement of the closed point of a henselian local scheme, and let $j : Y \to X$ be a closed subscheme of X whose complement U is affine. Then, if F is a sheaf of sets on X or a sheaf of not necessarily commutative groups, the statements XIV.4.5 or XIV.4.6 and XIV.4.9 still make sense for such an F, provided that one restricts to small values of n. If u is a point of U, denote by $\bar{u}$ a geometric point lying over u, by $X(\bar{u})$ the strict localization of X at $\bar{u}$, and by $F_{\bar{u}}$ the fiber of F at $\bar{u}$. Then, possibly making certain hypotheses on X and F, for example by assuming X excellent (possibly of characteristic zero, or of equal characteristic using resolution of singularities) and F ind-finite (or, if necessary, even **L**-ind-finite with **L** prime to the characteristic of X), one would like to prove the following statements:

a) Let F be a sheaf of sets (resp. a sheaf of groups), and suppose that, for every point u of U, one has

$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) \leq 1$$

(that is, for such a u, one has $\mathrm{prof}_u(F) \geq 1$). Then, as V ranges over the set of open neighborhoods of Y, the canonical morphism **162**

$$\varinjlim_V H^0(V, F) \to H^0(Y, j^*F)$$

is bijective (resp. one has the preceding conclusion and, moreover, the morphism $\varinjlim_V H^1(V, F) \to H^1(Y, j^*F)$ is injective). If F is constructible, one may replace the $\varinjlim$ by the cohomology of V for V "sufficiently small".

b) Let F be a sheaf of sets (resp. a sheaf of groups), and suppose that, for every point u of U, one has $\mathrm{prof}_u(F) \geq 2 - \dim(\overline{\{u\}})$, which is also expressed by the relations

$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ is bijective if } \dim(\overline{\{u\}}) = 0$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ is injective if } \dim(\overline{\{u\}}) = 1.$$

Then the canonical morphism

$$H^0(X, F) \to H^0(Y, j^*F)$$

is bijective (resp. one has the preceding conclusion and, moreover, the morphism $H^1(X, F) \to H^1(Y, j^*F)$ is injective).

c) Let F be an ind-finite sheaf of groups. Suppose that, for every point u of U, one has

$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective if } \dim(\overline{\{u\}}) = 0 \text{ or } 1,$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) = 2.$$

Then , as V ranges over the set of open neighborhoods of Y, the canonical morphisms **163**

$$\varinjlim_V H^0(V, F) \to H^0(Y, j^*F) \text{ and } \varinjlim_V H^1(V, F) \to H^1(Y, j^*F)$$

are bijective. If F is constructible, one may replace the $\varinjlim$ by the cohomology of V for V sufficiently small.

d) Let F be a sheaf of groups. Suppose that, for every point u of U, one has $\mathrm{prof}_u(F) \geq 3 - \dim(\overline{\{u\}})$, which is also expressed by the conditions

$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective, and } H^1(X(\bar{u}) - \bar{u}, F) = 0 \text{ if } \dim(\overline{\{u\}}) = 0,$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective if } \dim(\overline{\{u\}}) = 1,$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) = 2.$$

Then the canonical morphisms

$$H^0(X, F) \to H^0(Y, j^*F) \text{ and } H^1(X, F) \to H^1(Y, j^*F)$$

are bijective.

As evidence in favor of these statements (*), let us mention XIII XIII.2.1, X 3.4, and XII XII.3.5. Note that, by the argument of XIV.4.8 and XIV.4.9, statement a) (resp. c)) would follow from b) (resp. d)).

XIV.6.3

The following statement, analogous to XIV.5.6, would follow from d): if A is a noetherian **164**
local ring (possibly excellent) and $\mathrm{prof\,géom}(A) \geq 3$, then $\mathrm{prof\,hop}(A) \geq 3$. To see this, realize $Y' = \mathrm{Spec}\, A$ as a closed subscheme of a regular local scheme $X' = \mathrm{Spec}\, B$, whose complement is the union of q affine open subsets, with the relation $\dim B - q = \mathrm{prof\,géom}(A)$. For every point x of X', if $n = \dim B$, one has $\mathrm{prof\,hop}_x(X) \geq \inf(3, n - \dim(\overline{\{x\}}))$ (cf. XIV.1.11), and one deduces from d) that this implies $\mathrm{prof\,hop}_y(Y') \geq \inf(3, n - q - \dim(\overline{\{y\}}))$ for every point y of Y'. The result is then obtained by taking y to be the closed point of Y'.

XIV.6.4

A variant of XIV.4.2, at least of the implication (ii) $\Longrightarrow$ (i), should also hold in the complex analytic case, provided one works with "analytically constructible"sheaves (cf. XIII); the proof would be analogous to that of XIV.4.2, using the duality theory of J.-L. Verdier. Note, however, that for the complex analytic analogue of the noncommutative variants mentioned in XIV.6.2, not even a method of approach is available for the statements concerning the fundamental group suggested by the results of Exposés X, XII, and XIII, recalled at the end of XIV.6.2. Indeed, the methods of the Seminar seem inextricably tied to the case of *finite* coverings (which may be studied in terms of coherent sheaves of algebras).

(*) *Editor's note.* all the statements of XIV.6.2, apart from the constructible variants, were proved by Mrs Raynaud; see (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental", *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52, Corollary III.1.3).

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Index of notation

$\Gamma_Z,\ \underline{\Gamma}_Z$	I 1
$i_!,\ i^!$	I 1
$\mathbf{Z}_{Z,X}$	I 1.6
$H^i_Z(X,F), \underline{H}^i_Z(F)$	I 2
$H^i_J(M), H^i((f),M), H^i(M)$	IV 5.4, V 2 (p. 41)
$\operatorname{Ass} M$, $\operatorname{Supp} M$, $\operatorname{Ann}\ M, \mathfrak{r}(\mathfrak{a})$	III 1.1
$\operatorname{prof}_I(M), \operatorname{prof}_Y(F)$	III 2.3, 2.8
$\mathbf{Ab}$	IV 1 (p. 27)
$\operatorname{Hom}^{\cdot}(F^{\cdot}, G^{\cdot})$	V 1.1
$\operatorname{Ext}^i_Z(X;F,G), \underline{\operatorname{Ext}}^i_Z(F,G)$	VI 1.1
$\mathbf{Et}(X), \mathbf{L}(Z)$	X (p. 76)
$\operatorname{Lef}(X,Y), \operatorname{Leff}(X,Y)$	X 2 (p. 77)
$\underline{P}(Z), \operatorname{Pic}(Z)$	XI (p. 85)
$\Pi^x_i(X)$	XIII (p. 122)
$D^+(X)$	XIV XIV.1.0
$\operatorname{prof}_Y(F) \geq n$	XIV XIV.1.2
$\operatorname{prof}^{\mathbf{L}}_Y(X) \geq n$	XIV XIV.1.2
$\operatorname{prof}\ \text{ét}_Y(X)$	XIV XIV.1.2
$\mathbf{L}$	XIV XIV.1.2
$\operatorname{prof}_x(F) \geq n$	XIV XIV.1.7
$\operatorname{prof}^{\mathbf{L}}_x(X) \geq n$	XIV XIV.1.7
$\operatorname{prof}\ \operatorname{hop}^{\mathbf{L}}_x(X) \geq 3$	XIV XIV.1.7
$\operatorname{prof}\ \text{ét}_x(X)$	XIV XIV.1.7
$\delta_t(x),\ \delta(n)$	XIV XIV.2.1
$H^i_{Z!}(X/S,F)$	XIV XIV.4.0

Terminological index

Auslander-Buchsbaum (théorème de -)		XI XI.3.14
comparaison (théorème de -)	comparison theorem	IX 1.1, XII XII.2.1
dualité locale (théorème de -)	local duality theorem	V 2.1
dualité projective (théorème de -)	projective duality theorem	XII XII.1.1
enveloppe injective	injective envelope	IV (p. 34)
excision (théorème d' -)	excision theorem	I 2.2, VI 1.3
existence (théorème d' -)	existence theorem	IX 2.1, XII XII.3.1
extension essentielle	essential extension	IV (p. 34)
finitude (théorème de -)	finiteness theorem	VIII VIII.2.1, XII XII.1.5
foncteur dualisant	dualizing functor	IV 4.1, 4.2
forme résidu	residue form	IV 5.5
géométriquement factoriel, resp.	geometrically factorial (resp.	
géom. parafactoriel (anneau local -)	geom. parafactorial) local ring	XIII (p. 20, 24)
groupes d'homotopie locale	local homotopy groups	XIII (p. 122)
Gysin (homomorphisme de -)	Gysin homomorphism	I (p. 11)
Hartshorne (théorème de -)		III 3.6
Lefschetz (condition de -)	Lefschetz condition	X 2 (p. 77)
Lefschetz (condition de - effective)	effective Lefschetz condition	X 2 (p. 77)
Lefschetz (théorème de - affine)	affine Lefschetz theorem	XIV XIV.3.1
module dualisant	dualizing module	IV 4.1
orthogonal d'un sous-module	orthogonal of a submodule	IV 5
parafactoriel (couple - de préschémas)	parafactorial pair of preschemes	XI XI.3.1
parafactoriel (anneau local -)	parafactorial local ring	XI XI.3.2
profondeur	depth	III 2.3
profondeur étale	etale depth	XIV XIV.1.2
profondeur géométrique d'un anneau	geometrical depth of a noetherian	
local noethérien	local ring	XIV XIV.5.3
profondeur homotopique	homotopical depth	XIII XIII.6 def. 1 (p. 128)
profondeur homotopique (pour $\mathbf{L}$)	homotopical depth (with respect to $\mathbf{L}$)	XIV XIV.1.2
profondeur homotopique rectifiée	rectified homotopical depth	XIII def. 2 (p. 128)

pur (anneau local -)	pure local ring	X 3.2
pure (couple - de préschémas)	pure pair of preschemes	X 3.1
pureté (théorème de - de Zariski-Nagata)	purity theorem of Zariski-Nagata	X 3.4
pureté cohomologique (théorème de -)	cohomological purity theorem	XIV XIV.1.11
régulier (M-régulier)	M-regular	2.1
Samuel (conjecture de -)		XI XI.3.15
semi-pureté cohomologique (théorème de -)	cohomological semi-purity theorem	XIV XIV.1.10
strictement local (anneau -)	strictly local ring	XIII XIII.6 (p. 126)
Zariski-Nagata (théorème de pureté de -)	purity theorem of Zariski-Nagata	X 3.4

Seminar on Algebraic Geometry 3

Group Schemes

English Translation

M. Demazure and A. Grothendieck, editors

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Editorial Notice

We present here a slightly revised edition of the original Seminar, whose purpose and contents are described in the Introduction. The revision consisted mainly of correcting typographical errors, adding a few supplementary remarks or references in footnotes, dividing the work in its present form into three volumes, each with a detailed table of contents and an index of notation, and adding a terminological index at the end of Volume 3. In addition, Exposé VI_{B}, by J.-E. Bertin, was partially rewritten by the author, especially Sections 5 and 10, so some references to that Exposé differ from the references to the original version. A list of the Exposés in the Seminar appears at the beginning of this volume.

Since the first edition of the present Seminar appeared, all of Chapter IV of the *Éléments de Géométrie Algébrique* has been published, making certain passages of the Seminar unnecessary. We have occasionally indicated in footnotes the relevant references to EGA IV that allow those passages to be bypassed.

For another treatment of algebraic groups that systematically uses the language of schemes, we refer to M. Demazure and P. Gabriel, *Groupes Algébriques* (North-Holland & Masson et Cie). Unlike the present Seminar, that book assumes no knowledge of algebraic geometry, but includes all the required preliminaries from scheme theory; it can therefore serve as an introduction to the study of this Seminar. (It also treats subjects not covered in the Seminar, such as the Dieudonné-style structure theory of commutative affine algebraic groups, in *loc. cit.*, Chapter V.)

Bures-sur-Yvette, March 1970

Introduction

1. The purpose of this Seminar is twofold.

On the one hand, we aim to provide convenient foundations for the theory of group schemes in general. Exposés I–IV will supply the indispensable preliminary exercises in schematic and categorical syntax. To obtain a language that fits geometric intuition without effort, and to avoid circumlocutions that become intolerable in the long run, we always identify a prescheme X over another prescheme S with the functor

$$(\mathbf{Sch})^{\circ}_{/S} \longrightarrow \mathbf{Set}$$

that it represents*. It is therefore necessary to formulate many definitions so that they apply to arbitrary functors, whether representable or not. Moreover, nearly all the functors we shall use will be "sheaves" (for the "faithfully flat quasi-compact topology"). Exposé IV, which treats groups only incidentally, sketches the language of "localization" and sheaves; this language also proves very convenient in questions concerning the representability of functors. Above all, that Exposé will provide the most convenient framework for the questions of passage to a quotient that arise later.

Exposé V gives some general results on the existence of quotients. They are taken up again in Exposé $\mathrm{VI_A}$ for the quotient of an algebraic group over a field (or, more generally, over an Artinian ring) by a subgroup**. That Exposé and the following Exposé $\mathrm{VI_B}$ also contain various elementary results specific to algebraic groups over a field that are used routinely later.

Exposé VII studies certain phenomena related to the characteristic of the base field and, in particular, develops with the appropriate generality the correspondence between radicial group schemes of height 1 and restricted p-Lie algebras.

Finally, Exposé XVIII contains the scheme-theoretic generalization of Weil's theorem on the "birational" definition of algebraic groups.

On the other hand, we intend to generalize the Borel–Chevalley structure theory of affine algebraic groups to groups over an arbitrary base prescheme. In fact, while the Seminar notes were being written, it became apparent that the affine hypothesis was unnecessary for many results of the theory. The most complete results are of course obtained for "semisimple

*Such a viewpoint seems to have been considered for the first time eight or nine years ago, in connection with the theory of formal groups, by P. Cartier, who unfortunately did not take the trouble to state it precisely and systematize it as it deserved.

**For a more thorough study of passage to the quotient, especially by reductive groups, see D. Mumford's important work *Geometric Invariant Theory*, Ergebnisse der Mathematik, vol. 34, Springer, 1965. We note that on one important point the terminology of that book does not agree with ours: over a field of characteristic $p > 0$, the groups Mumford calls "reductive"[(1)] are the smooth groups of multiplicative type in the sense of the Seminar (cf. Exposé IX). One may reasonably suppose that Mumford adopted this meaning of "reductive," which loses its significance over a base that is not a field, provisionally and as a kind of stopgap (and this is approximately what Mumford explains, for other reasons, as early as the second paragraph of his preface). [(1)] *Editor's note:* See the remarks added at the end of this Introduction.

group schemes," or more generally "reductive group schemes," with which we shall be exclusively concerned beginning with Exposé XIX. Chevalley himself had already constructed the "Tôhoku groups" over the ring of integers, and that construction will be taken up in the present Seminar. The principal uniqueness theorem[(2)] gives a simple characterization of the "twisted" variants of these Tôhoku groups over a base prescheme S: they are the affine, smooth group schemes over S whose geometric fibers are connected semisimple groups in the usual sense$^{***(2)}$.

2.

As in the familiar theory over an algebraically closed field, the subtori of the group schemes under consideration play a crucial role. Thus the preliminary study of tori and, more generally, of "group schemes of multiplicative type"—both intrinsically and as multiplicative-type subgroups of a given group—occupies a fairly large part of this Seminar (Exposés VIII–XII). Their remarkable rigidity (in some respects even greater than that of abelian schemes or semisimple group schemes) makes them highly effective tools for studying certain more general groups.

3.

Beginning with Exposé XII (apart from Exposé XVIII, already mentioned), we shall routinely use the theory of affine algebraic groups over an algebraically closed field, which the reader will find in the 1956 Chevalley Seminar, especially in Exposés IV–IX of that Seminar. We shall also use, though to a lesser extent, the later Exposés of the Chevalley Seminar devoted to the structure of semisimple algebraic groups. Indeed, we shall redevelop Chevalley's theory directly within the framework of schemes; it will be seen that this approach makes the exposition simpler and more precise, even over a base field.

4.

The principal purpose of this Seminar is, of course, to develop techniques that apply to the study of group schemes over an arbitrary base, that is, essentially, to the study of families of algebraic groups. The infinitesimal properties of such families, and in particular the case of an Artinian base scheme, therefore play an important role. These properties arise even in the study of algebraic groups over a field k when the field is not perfect, especially when applying descent in the non-Galois case. Among the new results obtained in this setting, let us mention the existence of maximal tori and Cartan subgroups in an arbitrary smooth algebraic group, the rationality of the variety of maximal tori, and various related results (Exposé XIV[(3)]); another example is the correspondence between the "forms" of a semisimple group and homogeneous principal bundles under a suitable semisimple algebraic group (generally not connected) (Exposé XXIV, 1.16–1.20). In general, one may say that the methods required for work over a nonperfect base field are essentially those used over arbitrary base preschemes and thus lie outside the framework of classical algebraic geometry.

[(2)] *Editor's note:* This refers to Corollary XXIII.5.6, which follows easily from the uniqueness theorem for split reductive groups (cf. XXIII, Theorem 4.1 and Corollary 5.2), since every semisimple S-group is a "twisted form" (for the étale topology) of a Tôhoku group (cf. XXII, Corollary 2.3).

***This is the essential result of M. Demazure's thesis, *Schémas en groupes réductifs*, Bull. Soc. Math. France **93** (1965), 369–413.

[(3)] *Editor's note:* In particular, Theorems 1.1 and 6.1.

5.

It did not seem useful to indicate, at the beginning of the written Exposés, the date or dates of the corresponding oral presentations in the Seminar. We shall merely say that the order of the mimeographed Exposés (I–XXVI) does indeed agree with the order of the oral presentations. Moreover, the final written text was sometimes prepared considerably later than the oral presentation and often differs from it quite substantially: the written text is generally more detailed and more complete (as in Exposés IV and $\mathrm{VII_B}$), or even appreciably more general (as in Exposés XII and $\mathrm{VII_B}$), than the oral presentation. Other written Exposés correspond to no oral presentation ($\mathrm{VI_B}$, $\mathrm{VII_A}$, XV, XVI, XVII, and most of XXVI); they were written and inserted into the mimeographed Seminar either to provide convenient references for various other Exposés (notably in the case of $\mathrm{VI_B}$) or because they form a natural continuation of the concepts and techniques already developed. Consequently, Exposés $\mathrm{VII_A}$, $\mathrm{VII_B}$, XV, XVI, and XVII need not be read in order to study the rest of the Seminar, particularly the part devoted to reductive group schemes.

6.

From scheme theory we shall use primarily the general language of schemes set out in EGA I; the notions of flat, étale, and smooth morphisms set out in SGA 1, I–V; and the theory of faithfully flat descent in SGA 1, VIII. As far as possible, we have avoided imposing unnecessary Noetherian hypotheses. In return, this has required us to replace the usual hypothesis "of finite type" with the hypothesis "of finite presentation." For the notion of a morphism of finite presentation, the reader should consult EGA IV, 1.4 and 1.6. The results of SGA 1, I–IV, which are most often stated in the Noetherian setting, will be developed in general in EGA IV****; standard methods for reducing certain kinds of statements involving finite-presentation hypotheses to the Noetherian case will also be developed there in detail (EGA IV, Sections 8, 9, and 11). A reader who does not wish to assume these results from EGA IV may simplify certain statements or their proofs by assuming the base prescheme to be locally Noetherian. This does, however, lead to difficulties when a proof proceeds by descent from $\widehat{A}$ to A, where $\widehat{A}$ is the completion of a Noetherian local ring A, because that method introduces the generally non-Noetherian ring

$$\widehat{A} \otimes_A \widehat{A}.$$

7.

References follow the usual decimal system: the reference 5.7.11 denotes the proposition (or lemma, definition, and so forth) with that number in the same Exposé; in a reference such as XVII 7.8, the Roman numeral denotes the Exposé. We use the following abbreviations for our standard references:

Bible Chevalley Seminar, *Groupes de Lie algébriques*, 1956/58.

EGA X**,** $x.y.z$ J. Dieudonné and A. Grothendieck, *Éléments de Géométrie Algébrique*, Chapter X, statement $x.y.z$ (or subsection $x.y$, and so forth).

SGA n**,** X $y.z$ *Séminaire de Géométrie Algébrique du Bois-Marie*, year n, statement $y.z$ of Exposé X.

****Since this Introduction was written, all four parts (Sections 1–21) of EGA IV have appeared.

TDTE A. Grothendieck, *Techniques de descente et Théorèmes d'existence en Géométrie Algébrique*, Exposés delivered in the Bourbaki Seminar between 1959 and 1962.

(1) *Editor's note.* Concerning quotients by a reductive group, the situation has evolved considerably since A. Grothendieck wrote Note (∗∗). For an affine algebraic group over an arbitrary field k, the appropriate notion, introduced by Mumford, is that of a "geometrically reductive" group. (On the other hand, G is said to be "linearly reductive" if every rational representation of G is completely reducible; as noted in Note (∗∗), however, that condition is too restrictive when $\mathrm{char}(k) > 0$.) By results of M. Nagata and W. J. Haboush ([Na64], [NM64], [Ha75]), the geometrically reductive k-groups are precisely the reductive k-groups in the sense of the present Seminar. For all of this, see the later editions of Mumford's book ([MF82]). The notion of geometric reductivity over an arbitrary base, together with its consequences for passage to a quotient, was extended to this setting by C. S. Seshadri ([Se77]); additions due to M. Raynaud appear in the article [CTS79] by J.-L. Colliot-Thélène and J.-J. Sansuc. Finally, for more recent developments on passage to the quotient by an algebraic group or, more generally, by a groupoid (cf. Exposé V of the present Seminar), we refer to the articles [Ko97] and [KM97] by J. Kollár and by S. Keel and S. Mori.

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Exposé I
Algebraic Structures. Group Cohomology

by M. Demazure

This exposé consists of two parts: the first collects a number of general definitions and establishes notation that will often be used in the sequel; the second treats group cohomology and culminates in Theorem 5.3.3, on the vanishing of the cohomology of diagonalizable groups. [1]

We choose once and for all a Universe.[2] All definitions and constructions will be relative to this Universe. We shall systematically allow ourselves the following abuse of language. To define a functor $f : \mathscr{C} \to \mathscr{C}'$, we shall be content to define the object $f(S)$ of $\mathscr{C}'$ for every object S of $\mathscr{C}$, whenever there is no ambiguity about how to define $f(h)$ for an arrow h of $\mathscr{C}$. In practice, we shall say: let $f : \mathscr{C} \to \mathscr{C}'$ be the functor defined by $f(S) = \cdots$.

1. Generalities

1.1.

Let $\mathscr{C}$ be a category. We denote by $\widehat{\mathscr{C}}$ the category

$$\mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

of contravariant functors from $\mathscr{C}$ to the category **Set** of sets.[3] There is a canonical functor

$$\mathbf{h} : \mathscr{C} \longrightarrow \widehat{\mathscr{C}}$$

which associates with every $X \in \mathrm{Ob}(\mathscr{C})$ the functor $\mathbf{h}_X$ such that

$$\mathbf{h}_X(S) = \mathrm{Hom}(S, X).$$

For every functor $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, one has (see, for example, EGA 0_{III}, 8.1.4) a bijection

$$\mathrm{Hom}(\mathbf{h}_X, \mathbf{F}) \xrightarrow{\sim} \mathbf{F}(X).$$

[4]

[1] Editor's note: version of October 13, 2024.

[2] Editor's note: see SGA 4, Exposé I, §0 and the Appendix; see also the discussion in [DG70], p. xxvi.

[3] Editor's note: this is called the category of presheaves on $\mathscr{C}$; see IV, 4.3.1.

[4] Editor's note: this result is often called the Yoneda lemma; we use that terminology in other editor's notes.

In particular, for every pair X, Y of objects of $\mathscr{C}$, the canonical map

$$\mathrm{Hom}_{\mathscr{C}}(X,Y) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{h}_X, \mathbf{h}_Y)$$

is bijective; that is, the functor $\mathbf{h}$ is fully faithful. It therefore defines an isomorphism of $\mathscr{C}$ onto a full subcategory of $\widehat{\mathscr{C}}$, and an equivalence of $\mathscr{C}$ with the full subcategory of $\widehat{\mathscr{C}}$ formed by the representable functors (that is, the functors isomorphic to one of the form $\mathbf{h}_X$). In the sequel we shall often identify X with $\mathbf{h}_X$. The following paragraphs are intended to show that this identification can be made safely.

Remark 1.1.1.[5] We sometimes need the following variant. Let $\mathscr{D}$ be a full subcategory of $\mathscr{C}$, and let $X, Y \in \mathrm{Ob}(\mathscr{D})$. Denote by $\mathbf{h}'_X$ and $\mathbf{h}'_Y$ the restrictions of $\mathbf{h}_X$ and $\mathbf{h}_Y$ to $\mathscr{D}$. Then

$$\mathrm{Hom}_{\mathscr{C}}(X,Y) = \mathrm{Hom}_{\mathscr{D}}(X,Y) = \mathrm{Hom}_{\widehat{\mathscr{D}}}(\mathbf{h}'_X, \mathbf{h}'_Y).$$

Thus, to give a morphism $X \to Y$ "is the same thing" as to give, for every $T \in \mathrm{Ob}(\mathscr{D})$, a map

$$\phi(T) : \mathrm{Hom}(T,X) \longrightarrow \mathrm{Hom}(T,Y)$$

functorially in T.

1.2.

We shall say that $\mathbf{F}$ is a *subobject* (or a subfunctor) of $\mathbf{G}$ if $\mathbf{F}(S)$ is a subset of $\mathbf{G}(S)$ for every S.[6]

In $\widehat{\mathscr{C}}$, arbitrary inverse limits exist and are computed pointwise:

$$\left(\varprojlim_i \mathbf{F}_i\right)(S) = \varprojlim_i \mathbf{F}_i(S).$$

[7] In particular, fiber products are defined by

$$(\mathbf{F} \times_{\mathbf{G}} \mathbf{F}')(S) = \mathbf{F}(S) \times_{\mathbf{G}(S)} \mathbf{F}'(S).$$

[8]

As a final object of $\widehat{\mathscr{C}}$, we choose the functor $\underline{\mathbf{e}}$ such that $\underline{\mathbf{e}}(S) = \{\varnothing\}$.[9] Every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$ has a unique morphism to $\underline{\mathbf{e}}$, and we set

$$\mathbf{F} \times \mathbf{F}' = \mathbf{F} \times_{\underline{\mathbf{e}}} \mathbf{F}'.$$

The functor $\mathbf{h}$ commutes with inverse limits. In particular, for $X \times X'$ to exist, where $X, X' \in \mathrm{Ob}(\mathscr{C})$, respectively for $\mathscr{C}$ to admit a final object e, it is necessary and sufficient that $\mathbf{h}_X \times \mathbf{h}_{X'}$, respectively $\underline{\mathbf{e}}$, be representable; in that case

$$\mathbf{h}_X \times \mathbf{h}_{X'} \simeq \mathbf{h}_{X \times X'}, \qquad \mathbf{h}_e \simeq \underline{\mathbf{e}}.$$

[5] Editor's note: this remark has been added.

[6] Editor's note: the order has been changed so that fiber products are introduced before monomorphisms; see editor's note (9).

[7] Editor's note: similarly, arbitrary direct limits exist and are computed pointwise: $\left(\varinjlim_i \mathbf{F}_i\right)(S) = \varinjlim_i \mathbf{F}_i(S)$. In general, however, the functor $\mathbf{h}$ does not commute with direct limits.

[8] Editor's note: in particular, the equalizer of a pair of morphisms $u, v : \mathbf{F} \rightrightarrows \mathbf{G}$ is the subfunctor $\mathrm{Ker}(u,v)$ of $\mathbf{F}$ defined by $\mathrm{Ker}(u,v)(S) = \{x \in \mathbf{F}(S) \mid u(x) = v(x)\}$.

[9] Editor's note: $\{\varnothing\}$, the power set of the empty set, is a one-element set.

A monomorphism in $\widehat{\mathscr{C}}$ is precisely a morphism $\mathbf{F} \to \mathbf{G}$ such that, for every $S \in \mathrm{Ob}(\mathscr{C})$, the corresponding map of sets $\mathbf{F}(S) \to \mathbf{G}(S)$ is injective.[10]

The functor Γ. For every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, set

$$\Gamma(\mathbf{F}) = \mathrm{Hom}(\underline{\mathbf{e}}, \mathbf{F}).$$

An element of $\Gamma(\mathbf{F})$ is therefore a family $(\gamma_S)_{S \in \mathrm{Ob}(\mathscr{C})}$, with $\gamma_S \in \mathbf{F}(S)$, such that for every arrow $f : S' \to S''$ of $\mathscr{C}$,

$$\mathbf{F}(f)(\gamma_{S''}) = \gamma_{S'}.$$

For $X \in \mathrm{Ob}(\mathscr{C})$, set $\Gamma(X) = \Gamma(\mathbf{h}_X)$. If $\mathscr{C}$ has a final object e, then

$$\Gamma(X) \simeq \mathrm{Hom}(e, X).$$

1.3.

Let $S \in \mathrm{Ob}(\mathscr{C})$. We denote by $\mathscr{C}_{/S}$ the category of objects of $\mathscr{C}$ over S: its objects are the arrows $f : T \to S$ of $\mathscr{C}$, and $\mathrm{Hom}(f, f')$ is the subset of $\mathrm{Hom}(T, T')$ formed by the arrows u such that $f = f' \circ u$. If $\mathscr{C}$ has a final object e, then $\mathscr{C}_{/e}$ is isomorphic to $\mathscr{C}$. The category $\mathscr{C}_{/S}$ has a final object, namely the identity arrow $S \to S$.

If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, one may form the category $(\mathscr{C}_{/S})_{/f}$, which by abuse of language we denote by $(\mathscr{C}_{/S})_{/T}$, and there is a canonical isomorphism

$$\mathscr{C}_{/T} \simeq (\mathscr{C}_{/S})_{/T}.$$

This construction also applies to $\widehat{\mathscr{C}}$; in particular, one obtains the category $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. On the other hand, one may form the category $\widehat{\mathscr{C}_{/S}}$.

If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, then $\Gamma(f)$ is identified with the set $\Gamma(T/S)$ of sections of T over S, that is, arrows $S \to T$ that are right inverses to f. Moreover, $\mathbf{h}_f : \mathbf{h}_T \to \mathbf{h}_S$ is then an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and

$$\Gamma(\mathbf{h}_f) \simeq \Gamma(\mathbf{h}_T/\mathbf{h}_S) \simeq \Gamma(T/S) \simeq \Gamma(f).$$

1.4.

We now define an equivalence between the categories $\widehat{\mathscr{C}_{/S}}$ and $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. In other words, we shall prove that to give a functor on the category of objects of $\mathscr{C}$ over S is "the same thing" as to give a functor on $\mathscr{C}$ equipped with a morphism to $\mathbf{h}_S$.

(i) Construction of $\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \to \widehat{\mathscr{C}_{/S}}$.

First let $H : \mathbf{F} \to \mathbf{h}_S$ be an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. We must define a functor $\alpha_S(H)$ on $\mathscr{C}_{/S}$. For an object $f : T \to S$ of $\mathscr{C}_{/S}$, define $\alpha_S(H)(f)$ to be the inverse image of $f \in \mathbf{h}_S(T)$ under the map

$$H(T) : \mathbf{F}(T) \longrightarrow \mathbf{h}_S(T).$$

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[10] Editor's note: if $\mathbf{F}(S) \to \mathbf{G}(S)$ is injective for every S, it is clear that $\mathbf{F} \to \mathbf{G}$ is a monomorphism. The converse follows by considering $\mathbf{F} \times_{\mathbf{G}} \mathbf{F} \rightrightarrows \mathbf{F} \to \mathbf{G}$. Thus $\mathbf{F} \to \mathbf{G}$ is a monomorphism if and only if the diagonal morphism $\mathbf{F} \to \mathbf{F} \times_{\mathbf{G}} \mathbf{F}$ is an isomorphism (see EGA I, 5.3.8). Similarly, a pointwise surjective morphism is an epimorphism, and the converse follows by considering the pushout $\mathbf{G} \amalg_{\mathbf{F}} \mathbf{G}$; see the proof of Lemma 4.4.4 in Exposé IV.

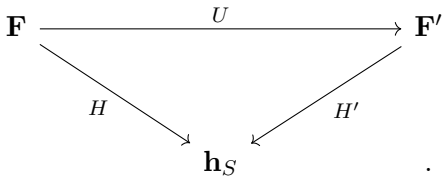
[11] Editor's note: for example, if $\mathbf{F} = \mathbf{h}_X$, then H corresponds to a morphism $h : X \to S$, and $H(T) : \mathrm{Hom}_{\mathscr{C}}(T, X) \to \mathrm{Hom}_{\mathscr{C}}(T, S)$ is the map $g \mapsto h \circ g$. Thus $\alpha_S(\mathbf{h}_X) = \mathrm{Hom}_{\mathscr{C}_{/S}}(-, X)$.

Next, let $u : f \to f'$ be an arrow of $\mathscr{C}_{/S}$. Then $\mathbf{F}(u) : \mathbf{F}(T') \to \mathbf{F}(T)$ induces a map from $\alpha_S(H)(f')$ to $\alpha_S(H)(f)$, denoted $\alpha_S(H)(u)$. It is immediate that the assignments

$$f \longmapsto \alpha_S(H)(f), \qquad u \longmapsto \alpha_S(H)(u)$$

define a functor on $\mathscr{C}_{/S}$, hence an object $\alpha_S(H)$ of $\widehat{\mathscr{C}_{/S}}$.

Finally, let $H : \mathbf{F} \to \mathbf{h}_S$ and $H' : \mathbf{F}' \to \mathbf{h}_S$ be two objects of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and let $U : H \to H'$ be a morphism in that category:



For every $f : T \to S$, the map $U(T) : \mathbf{F}(T) \to \mathbf{F}'(T)$ induces a map

$$\alpha_S(U)(f) : \alpha_S(H)(f) \longrightarrow \alpha_S(H')(f),$$

which defines a morphism of functors

$$\alpha_S(U) : \alpha_S(H) \longrightarrow \alpha_S(H').$$

It is easy to verify that the assignments

$$H \longmapsto \alpha_S(H), \qquad U \longmapsto \alpha_S(U)$$

define a functor

$$\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \longrightarrow \widehat{\mathscr{C}_{/S}}.$$

(ii) Proposition 1.4.1. *The functor $\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \to \widehat{\mathscr{C}_{/S}}$ is an equivalence of categories.*

We indicate only the principle of the construction of a quasi-inverse functor

$$\beta_S : \widehat{\mathscr{C}_{/S}} \longrightarrow \widehat{\mathscr{C}}_{/\mathbf{h}_S}.$$

Let $\mathbf{G}$ be a functor on $\mathscr{C}_{/S}$. For every object T of $\mathscr{C}$, set

$$\beta_S(\mathbf{G})(T) = \coprod_{f \in \mathrm{Hom}(T,S)} \mathbf{G}(f), \qquad \mathrm{Hom}(T,S) = \mathbf{h}_S(T).$$

This defines a functor $\beta_S(\mathbf{G})$ on $\mathscr{C}$, equipped with the evident projection to $\mathbf{h}_S$.

1.5.

The equivalence α_S commutes with the functors Γ. In other words, if $H : \mathbf{F} \to \mathbf{h}_S$ is an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$ and $\alpha_S(H)$ is the corresponding object of $\widehat{\mathscr{C}_{/S}}$, then

$$\Gamma(\alpha_S(H)) \simeq \Gamma(H) \simeq \Gamma(\mathbf{F}/\mathbf{h}_S).$$

The equivalence α_S also commutes with the functors $\mathbf{h}$. That is, if $f : T \to S$ is an object of $\mathscr{C}_{/S}$, then $\mathbf{h}_f : \mathbf{h}_T \to \mathbf{h}_S$ is an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and its image under α_S is precisely $\mathbf{h}_{\mathscr{C}_{/S}}(f)$, where

$$\mathbf{h}_{\mathscr{C}_{/S}} : \mathscr{C}_{/S} \longrightarrow \widehat{\mathscr{C}_{/S}}$$

is the canonical functor.[12] Consequently:

Proposition 1.5.1. *Let $H : \mathbf{F} \to \mathbf{h}_S$ be an object of $\widehat{\mathscr{C}_{/\mathbf{h}_S}}$. The functor*

$$\alpha_S(H) : (\mathscr{C}_{/S})^\circ \longrightarrow \mathbf{Set}$$

is representable if and only if $\mathbf{F} : \mathscr{C}^\circ \to \mathbf{Set}$ is representable. If $\mathbf{F} \simeq \mathbf{h}_T$, then $\alpha_S(H)$ is represented by the object $T \to S$ of $\mathscr{C}_{/S}$.

The equivalence α_S is transitive in S. If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, there is a commutative diagram of equivalences:

$$\begin{array}{ccccc} (\widehat{\mathscr{C}_{/\mathbf{h}_S}})_{/\mathbf{h}_T} & \xrightarrow{\alpha_{S/\mathbf{h}_T}} & (\widehat{\mathscr{C}_{/S}})_{/\mathbf{h}_T} & \xrightarrow{\alpha_f} & \widehat{(\mathscr{C}_{/S})_{/T}} \\ & \searrow^{\simeq} & & \swarrow_{\simeq} & \\ & & \widehat{\mathscr{C}_{/\mathbf{h}_T}} & \xrightarrow{\alpha_T} & \widehat{\mathscr{C}_{/T}} \end{array} ,$$

Here $\alpha_{S/\mathbf{h}_T}$ denotes, provisionally, the restriction (see 1.6) of the functor α_S to the objects over $\mathbf{h}_T$.

1.6. Base change in a functor

For every $S \in \mathrm{Ob}(\mathscr{C})$, there is a canonical functor

$$i_S : \mathscr{C}_{/S} \longrightarrow \mathscr{C}$$

defined by $i_S(f) = T$ when f is the arrow $T \to S$. If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, we denote by $i_{T/S} = i_f$ the functor

$$i_{T/S} : (\mathscr{C}_{/S})_{/T} \longrightarrow \mathscr{C}_{/S},$$

and there is a commutative diagram:

$$\begin{array}{ccccc} (\mathscr{C}_{/S})_{/T} & \xrightarrow{i_{T/S}} & \mathscr{C}_{/S} & \xrightarrow{i_S} & \mathscr{C} \\ & \searrow^{\simeq} & & \nearrow_{i_T} & \\ & & \mathscr{C}_{/T} & & \end{array} ,$$

Thus, identifying $(\mathscr{C}_{/S})_{/T}$ with $\mathscr{C}_{/T}$, as we shall henceforth do,

$$i_S \circ i_{T/S} = i_T.$$

Likewise, if $\mathscr{C}$ is identified with $\mathscr{C}_{/e}$ when $\mathscr{C}$ has a final object e, then $i_{S/e} : \mathscr{C}_{/S} \to \mathscr{C}_{/e}$ is identified with i_S.

For $X \in \mathrm{Ob}(\mathscr{C})$ (respectively, $Y \in \mathrm{Ob}(\mathscr{C}_{/S})$), let $p_S(X)$ (respectively, $p_{T/S}(Y)$) be the object of $\mathscr{C}_{/S}$ (respectively, of $\mathscr{C}_{/T}$), when it exists, defined by $X \times S$ (respectively, $Y \times_S T$) equipped with its second projection:

[12] Editor's note: see editor's note (10).

$$\begin{array}{ccc} X \times S \\ {\scriptstyle p_S(X)}\downarrow \\ S \end{array} \qquad \text{resp.} \qquad \begin{array}{ccc} Y \times_S T \\ \downarrow{\scriptstyle p_{T/S}(Y)} \\ T \end{array} .$$

The partially defined functor p_S (respectively, $p_{T/S}$) is called the *base-change functor*. By the definition of a product (respectively, a fiber product), it is right adjoint to i_S (respectively, to $i_{T/S}$). [13] We also write

$$p_S(X) = X_S, \qquad p_{T/S}(Y) = Y_T.$$

The functor i_S defines a restriction functor

$$i_S^* : \widehat{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}_{/S}}.$$

We write

$$\mathbf{F}_S = i_S^*(\mathbf{F}) = \mathbf{F} \circ i_S.$$

Plainly,

$$i_{T/S}^* \circ i_S^* = i_T^*,$$

that is, for every functor $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$,

$$(\mathbf{F}_S)_T = \mathbf{F}_T.$$

The notation is justified by the following result.

Proposition 1.6.1. *The functor*

$$(\mathbf{h}_X)_S : (\mathscr{C}_{/S})^\circ \longrightarrow \mathbf{Set}$$

is representable if and only if the product $X \times S$ exists. In that case

$$(\mathbf{h}_X)_S \simeq \mathbf{h}_{X_S}.$$

This shows that $\mathbf{F}_S$ has two interpretations: it is the restriction of $\mathbf{F}$ to $\mathscr{C}_{/S}$, and it is the functor obtained by the base change $e \leftarrow S$. This leads to the following notation:

$$\begin{array}{ccccc} \mathbf{F} & \longrightarrow & \mathbf{F}_S & \longrightarrow & \mathbf{F}_T \\ \downarrow & & \downarrow & & \downarrow \\ e & \longleftarrow & S & \longleftarrow & T \end{array}$$

This diagram reflects both of the preceding interpretations. Notice that

$$\Gamma(\mathbf{F}_S) \simeq \mathrm{Hom}(\mathbf{h}_S, \mathbf{F}) \simeq \mathbf{F}(S),$$

and, in particular,

$$\Gamma(X_S) \simeq \mathrm{Hom}(S, X).$$

[13] Editor's note: that is, if $g : U \to S$ (respectively, $h : V \to T$) is an object of $\mathscr{C}_{/S}$ (respectively, of $\mathscr{C}_{/T}$), then $i_S(g) = U$, while $i_{T/S}(h)$ is the object $f \circ h : V \to S$ of $\mathscr{C}_{/S}$, and

$$\mathrm{Hom}_{\mathscr{C}_{/S}}(U, X \times S) \simeq \mathrm{Hom}_{\mathscr{C}}(U, X),$$

respectively,

$$\mathrm{Hom}_{\mathscr{C}_{/T}}(V, X \times_S T) \simeq \mathrm{Hom}_{\mathscr{C}_{/S}}(V, X).$$

1.7.0.

Let $\mathbf{E}$ be an object of $\widehat{\mathscr{C}}$. [14] Consider the category $\mathscr{C}_{/\mathbf{E}}$ of *objects of* $\mathscr{C}$ over $\mathbf{E}$. Its objects are pairs (V, ρ) formed by an object V of $\mathscr{C}$ and a morphism $\rho : \mathbf{h}_V \to \mathbf{E}$ in $\widehat{\mathscr{C}}$, that is, $\rho \in \mathbf{E}(V)$. A morphism from (V, ρ) to (V', ρ') is a morphism $f : V \to V'$ in $\mathscr{C}$ such that

$$\rho' \circ f = \rho, \qquad \text{that is,} \qquad \mathbf{E}(f)(\rho') = \rho.$$

Let $\mathbf{L}$ be the functor

$$\mathbf{L} = \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V.$$

Thus, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\mathbf{L}(S) = \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V(S)$$

is the set of equivalence classes of triples (V, ρ, v), where $v : S \to V$ is a morphism in $\mathscr{C}$, and where (V, ρ, v) is identified with $(V', \rho', f \circ v)$ for every morphism $f : V \to V'$ in $\mathscr{C}$ such that $\rho' \circ f = \rho$.

The map $\phi_{\mathbf{E}}(S)$ which associates with the class of (V, ρ, v) the element $\rho \circ v$ of $\mathbf{E}(S)$ is well defined and gives a morphism of functors

$$\phi_{\mathbf{E}} : \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V \longrightarrow \mathbf{E}.$$

Lemma. *The morphism $\phi_{\mathbf{E}}$ is an isomorphism.*

Indeed, let $S \in \mathrm{Ob}(\mathscr{C})$. Every $x \in \mathbf{E}(S)$ is the image under $\phi_{\mathbf{E}}(S)$ of the triple (S, x, id_S); hence $\phi_{\mathbf{E}}(S)$ is surjective. Conversely, let $\ell_1 = (V_1, \rho_1, v_1)$ and $\ell_2 = (V_2, \rho_2, v_2)$ be two elements of $\mathbf{L}(S)$ having the same image in $\mathbf{E}(S)$, and set

$$\gamma = \rho_1 \circ v_1 = \rho_2 \circ v_2.$$

Then ℓ_1 and ℓ_2 are both equal in $\mathbf{L}(S)$ to the class of $(S, \gamma, \mathrm{id}_S)$. Thus $\phi_{\mathbf{E}}(S)$ is injective.

Corollary. *For every object $\mathbf{F}$ of $\widehat{\mathscr{C}}$,*

$$\mathrm{Hom}(\mathbf{E}, \mathbf{F}) = \varprojlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{F}(V).$$

1.7. The objects Hom, Isom, etc.

Let $\mathbf{F}$ and $\mathbf{G}$ be two objects of $\widehat{\mathscr{C}}$. We define another object of $\widehat{\mathscr{C}}$ by

$$\begin{aligned}\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})(S) &= \mathrm{Hom}_{\widehat{\mathscr{C}_{/S}}}(\mathbf{F}_S, \mathbf{G}_S) \\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}_{/\mathbf{h}_S}}}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G} \times \mathbf{h}_S) \\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G}).\end{aligned}$$

The object $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$ has the following properties:

[14] Editor's note: the lemma below has been added (see SGA 4, I.3.4); it is used in the proof of 1.7.1 and will be useful repeatedly in the sequel.

(i) $\underline{\mathrm{Hom}}(\mathbf{e}, \mathbf{G}) \simeq \mathbf{G}$. [15]

(ii) The formation of $\underline{\mathrm{Hom}}$ commutes with base extension:

$$\underline{\mathrm{Hom}}(\mathbf{F}_S, \mathbf{G}_S) \simeq \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})_S.$$

(iii) $(\mathbf{F}, \mathbf{G}) \mapsto \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$ is a bifunctor, contravariant in $\mathbf{F}$ and covariant in $\mathbf{G}$.

These three properties follow immediately from the definitions.

We shall prove that, for every object $\mathbf{E}$ of $\widehat{\mathscr{C}}$,

$$\mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

Let $\phi : \mathbf{E} \times \mathbf{F} \to \mathbf{G}$. We must associate with it a morphism from $\mathbf{E}$ to $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$. For an arrow $S' \to S$ of $\mathscr{C}$, there are maps

$$\mathbf{E}(S) \times \mathbf{F}(S') \longrightarrow \mathbf{E}(S') \times \mathbf{F}(S') \xrightarrow{\phi(S')} \mathbf{G}(S').$$

Every element e of $\mathbf{E}(S)$ therefore defines, for each $S' \to S$, a map $\mathbf{F}(S') \to \mathbf{G}(S')$ functorial in S'; this is an element $\theta_\phi(e)$ of $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})(S)$. We have thus obtained a map

$$\phi \longmapsto \theta_\phi, \qquad \mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \longrightarrow \mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big),$$

which is functorial in $\mathbf{E}$.

Proposition 1.7.1. [16] *Let $\mathbf{E}, \mathbf{F}, \mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$.*

(a) The map $\phi \mapsto \theta_\phi$ is a bijection:

$$\mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \xrightarrow{\sim} \mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big).$$

(b) Moreover, there is an isomorphism of functors

$$\underline{\mathrm{Hom}}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \underline{\mathrm{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

For (a), regard both sides as functors of $\mathbf{E}$. The asserted result is true when $\mathbf{E} = \mathbf{h}_X$, since it is then precisely the definition of the functor $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$. Moreover, both sides, as functors of $\mathbf{E}$, take direct limits to inverse limits. Finally, by Lemma 1.7.0, every object $\mathbf{E}$ of $\widehat{\mathscr{C}}$ is isomorphic to the direct limit of the $\mathbf{h}_X$, as X ranges over $\mathscr{C}_{/\mathbf{E}}$. This proves (a).

Let us also sketch a direct proof of (a). To every

$$\theta \in \mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big)$$

associate $\phi_\theta \in \mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G})$ as follows. For every $S \in \mathrm{Ob}(\mathscr{C})$, there is a map

$$\theta(S) : \mathbf{E}(S) \longrightarrow \mathrm{Hom}(\mathbf{F} \times S, \mathbf{G}),$$

[15] Editor's note: if $\mathbf{E}$ is a third object of $\widehat{\mathscr{C}}$, then

$$\underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{F} \times \mathbf{G}) \simeq \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{F}) \times \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G}).$$

[16] Editor's note: part (b), which will be useful in II.1 and II.3.11, has been added. A second, more direct proof of part (a) has also been sketched.

functorial in S. If $(e, f) \in \mathbf{E}(S) \times \mathbf{F}(S)$, then f is a morphism $S \to \mathbf{F}$, so $f \times \mathrm{id}_S$ is a morphism $S \to \mathbf{F} \times S$. On the other hand, $\theta(S)(e)$ is a morphism $\mathbf{F} \times S \to \mathbf{G}$. Composition therefore gives a morphism

$$\theta(S)(e) \circ (f \times \mathrm{id}_S) : S \longrightarrow \mathbf{G},$$

that is, an element $\phi_\theta(S)(e, f)$ of $\mathbf{G}(S)$. It is easy to check that $S \mapsto \phi_\theta(S)$ is functorial in S, and therefore defines a morphism ϕ_θ from $\mathbf{E} \times \mathbf{F}$ to $\mathbf{G}$. We leave to the reader the verification that $\theta \mapsto \phi_\theta$ and $\phi \mapsto \theta_\phi$ are mutually inverse bijections.

For (b), if $S \in \mathrm{Ob}(\mathscr{C})$, then by 1.7(ii) and part (a), applied to $\mathscr{C}_{/S}$,

$$\begin{aligned}
&\underline{\mathrm{Hom}}(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G}))(S) \\
&\quad \simeq \mathrm{Hom}_S(\mathbf{E}_S, \underline{\mathrm{Hom}}_S(\mathbf{F}_S, \mathbf{G}_S)) \\
&\quad \simeq \mathrm{Hom}_S(\mathbf{E}_S \times_S \mathbf{F}_S, \mathbf{G}_S) \\
&\quad \simeq \mathrm{Hom}(\mathbf{E} \times \mathbf{F} \times S, \mathbf{G}) \\
&\quad \simeq \underline{\mathrm{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G})(S),
\end{aligned}$$

and these isomorphisms are functorial in S.

Corollary 1.7.2. *There are isomorphisms*

$$\mathrm{Hom}(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \mathrm{Hom}(\mathbf{F}, \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G})),$$

and

$$\underline{\mathrm{Hom}}(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\mathrm{Hom}}(\mathbf{F}, \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G})).$$

In particular, taking $\mathbf{E} = \underline{\mathbf{e}}$ and using $\underline{\mathrm{Hom}}(\underline{\mathbf{e}}, \mathbf{G}) \simeq \mathbf{G}$, we obtain

$$\Gamma\big(\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \mathrm{Hom}(\mathbf{F}, \mathbf{G}).$$

Composition supplies functorial morphisms

$$\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{H}) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{H}).$$

If $\mathbf{F}$ and $\mathbf{G}$ are two objects of $\widehat{\mathscr{C}}$, let $\mathrm{Isom}(\mathbf{F}, \mathbf{G})$ denote the subset of $\mathrm{Hom}(\mathbf{F}, \mathbf{G})$ formed by the isomorphisms from $\mathbf{F}$ onto $\mathbf{G}$. Define a subobject $\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})$ of $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$ by

$$\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})(S) = \mathrm{Isom}(\mathbf{F}_S, \mathbf{G}_S).$$

Then

$$\Gamma\big(\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})\big) \simeq \mathrm{Isom}(\mathbf{F}, \mathbf{G}), \qquad \underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G}) \simeq \underline{\mathrm{Isom}}(\mathbf{G}, \mathbf{F}).$$

In the special case $\mathbf{F} = \mathbf{G}$, set

$$\begin{aligned}
\underline{\mathrm{End}}(\mathbf{F}) &= \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{F}), & \mathrm{End}(\mathbf{F}) &= \mathrm{Hom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma\big(\underline{\mathrm{End}}(\mathbf{F})\big), \\
\underline{\mathrm{Aut}}(\mathbf{F}) &= \underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{F}), & \mathrm{Aut}(\mathbf{F}) &= \mathrm{Isom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma\big(\underline{\mathrm{Aut}}(\mathbf{F})\big).
\end{aligned}$$

Formation of the objects $\underline{\mathrm{Hom}}$, $\underline{\mathrm{Isom}}$, $\underline{\mathrm{Aut}}$, and $\underline{\mathrm{End}}$ commutes with base change.

Remark 1.7.3. [17] An object isomorphic to $\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})$ may also be constructed as follows. There is a morphism

$$\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\mathrm{End}}(\mathbf{F}).$$

[17] Editor's note: the numbering 1.7.3 has been added for later references.

Interchanging $\mathbf{F}$ and $\mathbf{G}$ gives a morphism

$$\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G},\mathbf{F}) \longrightarrow \underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G}).$$

On the other hand, the identity morphism of $\mathbf{F}$ is an element of $\mathrm{End}(\mathbf{F})$, and hence defines a morphism $\underline{\mathbf{e}} \to \underline{\mathrm{End}}(\mathbf{F})$. Doing the same for $\mathbf{G}$ and taking the product gives a morphism

$$\underline{\mathbf{e}} \longrightarrow \underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G}).$$

It is then immediate that the fiber product of $\underline{\mathbf{e}}$ and

$$\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G},\mathbf{F})$$

over $\underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G})$ is isomorphic to $\underline{\mathrm{Isom}}(\mathbf{F},\mathbf{G})$.

All these definitions apply, in particular, when $\mathbf{F} = \mathbf{h}_X$ and $\mathbf{G} = \mathbf{h}_Y$. If $\underline{\mathrm{Hom}}(\mathbf{h}_X, \mathbf{h}_Y)$ is representable by an object of $\mathscr{C}$, we denote that object by $\underline{\mathrm{Hom}}(X,Y)$. It has the following property: if $Z \times X$ exists, then

$$\mathrm{Hom}\big(Z, \underline{\mathrm{Hom}}(X,Y)\big) \simeq \mathrm{Hom}(Z \times X, Y).$$

This property characterizes it when products exist in $\mathscr{C}$.

Similarly, one defines, when they exist, the objects

$$\underline{\mathrm{Isom}}(X,Y), \qquad \underline{\mathrm{End}}(X), \qquad \underline{\mathrm{Aut}}(X).$$

We merely note that the construction above shows that $\underline{\mathrm{Isom}}(X,Y)$ exists whenever fiber products exist in $\mathscr{C}$ and the objects

$$\underline{\mathrm{Hom}}(X,Y), \quad \underline{\mathrm{Hom}}(Y,X), \quad \underline{\mathrm{End}}(X), \quad \text{and} \quad \underline{\mathrm{End}}(Y)$$

exist.

Everything above also applies to categories of the form $\mathscr{C}_{/S}$. The corresponding objects will be denoted as explicitly as possible by appropriate symbols. For example, if T and T' are two objects of $\mathscr{C}$ over S, we denote by $\underline{\mathrm{Hom}}_S(T,T')$ the object

$$\underline{\mathrm{Hom}}_{\mathscr{C}_{/S}}(T/S, T'/S).$$

1.8. Constant objects

Let $\mathscr{C}$ be a category in which direct sums and fiber products exist and in which direct sums commute with base change (for example, the category of schemes).[18] For every set E and every object S of $\mathscr{C}$, set

$$E_S = \Big\{\text{the direct sum of a family } (S_i)_{i\in E} \text{ of objects} \quad \text{of } \mathscr{C}, \text{ all isomorphic to } S. \tag{1.8.1}$$

This object is characterized by the formula

$$\mathrm{Hom}_{\mathscr{C}}(E_S, T) = \mathrm{Hom}_{\mathbf{Set}}\big(E, \mathrm{Hom}_{\mathscr{C}}(S,T)\big), \tag{1.8.2}$$

for every $T \in \mathrm{Ob}(\mathscr{C})$.

[18] Editor's note: the former terminology "preschemes/schemes" has been replaced by the current terminology "schemes/separated schemes."

The object E_S has a canonical projection to S, so that $E \mapsto E_S$ is in fact a functor from **Set** to $\mathscr{C}_{/S}$.[19] If $S' \to S$ is an arrow of $\mathscr{C}$, then, because direct sums commute with base change,

$$E_{S'} = (E_S)_{S'}.$$

In particular, if $\mathscr{C}$ has a final object e, then

$$E_S = (E_e)_S.$$

The functor $E \mapsto E_S/S$, from **Set** to $\mathscr{C}_{/S}$, commutes with finite products. It is enough to prove that

$$E_S \times_S F_S = (E \times F)_S. \qquad (\times)$$

Indeed, by the results of 1.7 applied to $\mathscr{C}_{/S}$, for every $T \in \mathrm{Ob}(\mathscr{C}_{/S})$ there are natural isomorphisms (all unspecified Hom's being taken in $\mathscr{C}_{/S}$)

$$\begin{aligned}\mathrm{Hom}((E \times F)_S, T) &\simeq \mathrm{Hom}_{\mathbf{Set}}(E \times F, \mathrm{Hom}(S, T)) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}_{\mathbf{Set}}(F, \mathrm{Hom}(S, T))) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}(F_S, T)),\end{aligned}$$

and

$$\begin{aligned}\mathrm{Hom}(E_S \times_S F_S, T) &\simeq \mathrm{Hom}(E_S, \underline{\mathrm{Hom}}(F_S, T)) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}(S, \underline{\mathrm{Hom}}(F_S, T))).\end{aligned}$$

But

$$\mathrm{Hom}(S, \underline{\mathrm{Hom}}(F_S, T)) \simeq \mathrm{Hom}(F_S, T),$$

which proves $(\times)$.

Suppose that, with $\varnothing$ denoting an initial object of $\mathscr{C}$, the following diagram is cartesian:

$$(*) \qquad \begin{array}{ccc} \varnothing & \longrightarrow & S \\ \downarrow & & \downarrow \\ S & \longrightarrow & S \amalg S \end{array}$$

This is the case for the category of schemes.[20] Then the functor $E \mapsto E_S$ commutes with finite inverse limits.

Indeed, in view of $(\times)$, it is enough to show that $E \mapsto E_S$ commutes with fiber products. Let $u : E \to G$ and $v : F \to G$ be two maps of sets. Since direct sums in $\mathscr{C}$ commute with base change, one has

$$F_S \times_{G_S} E_S \simeq \coprod_{f \in F} (S_f \times_{G_S} E_S) \simeq \coprod_{\substack{f \in F \\ x \in E}} (S_f \times_{G_S} S_x). \qquad (1)$$

If $v(f) \neq u(x)$, there is a morphism in $\mathscr{C}_{/S}$

$$S_f \times_{G_S} S_x \longrightarrow S_f \times_{S_{v(f)} \coprod S_{u(x)}} S_x.$$

[19] Editor's note: in the next three paragraphs, the order of the sentences has been changed and a few details have been added concerning the role of condition $(*)$ below.

[20] Editor's note: Condition $(*)$ is not satisfied if $\mathscr{C}$ is the category whose arrows are $A \to B$, id_A, and id_B. In that case $B \coprod B = B$ and $B \times_B B = B \not\simeq A$.

By condition $(*)$, the term on the right is $\varnothing$. Hence

$$S_f \times_{G_S} S_x \simeq \varnothing \qquad \text{if } v(f) \neq u(x). \tag{2}$$

On the other hand, if $v(f) = u(x)$, there is a morphism in $\mathscr{C}_{/S}$

$$S \longrightarrow S_f \times_{G_S} S_x.$$

Since $S \xrightarrow{\mathrm{id}} S$ is the final object of $\mathscr{C}_{/S}$, it follows that

$$S_f \times_{G_S} S_x \simeq S \qquad \text{if } v(f) = u(x). \tag{3}$$

Combining (1), (2), and (3), we obtain an isomorphism functorial in S:

$$F_S \times_{G_S} E_S \simeq \coprod_{F \times_G E} S = (F \times_G E)_S.$$

An object of the form E_S is called a *constant object*. There is a morphism, functorial in E,

$$E \longrightarrow \Gamma(E_S/S),$$

which associates with each $i \in E$ the section of E_S over S defined by the isomorphism from S onto S_i. Suppose condition $(*)$ is satisfied for every object S of $\mathscr{C}$. Then $E \to \Gamma(E_S/S)$ is a monomorphism whenever $S \not\simeq \varnothing$.

If $\mathscr{C}$ is the category of schemes, then $\Gamma(E_S/S)$ is identified with the locally constant maps from the topological space S to the set E, the preceding map associating with each element of E the corresponding constant map. It follows that E_S may equivalently be defined as representing the functor that assigns to every S' over S the set of locally constant functions from the topological space S' to E.[21]

2. Algebraic structures

Given a species of algebraic structure in the category of sets, we shall extend it to the category $\mathscr{C}$. We first treat the example of groups.

2.1. Group structures

We retain the notation of the preceding paragraph.

Definition 2.1.1. Let $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$. A *$\widehat{\mathscr{C}}$-group structure* on $\mathbf{G}$ is the datum, for every $S \in \mathrm{Ob}(\mathscr{C})$, of a group structure on the set $\mathbf{G}(S)$, such that for every arrow $f : S' \to S''$ of $\mathscr{C}$, the map

$$\mathbf{G}(f) : \mathbf{G}(S'') \longrightarrow \mathbf{G}(S')$$

is a group homomorphism. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathscr{C}}$-groups, a *morphism of $\widehat{\mathscr{C}}$-groups* from $\mathbf{G}$ to $\mathbf{H}$ is a morphism $u \in \mathrm{Hom}(\mathbf{G}, \mathbf{H})$ such that $u(S) : \mathbf{G}(S) \to \mathbf{H}(S)$ is a group homomorphism for every $S \in \mathrm{Ob}(\mathscr{C})$.

We denote by $\mathrm{Hom}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ the set of these morphisms and by $(\widehat{\mathscr{C}}\text{-Gr})$ their category.

Examples. For $\mathbf{E} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, the object $\underline{\mathrm{Aut}}(\mathbf{E})$ has an evident $\widehat{\mathscr{C}}$-group structure. The final object $\underline{\mathbf{e}}$ has a unique $\widehat{\mathscr{C}}$-group structure, making it a final object of $(\widehat{\mathscr{C}}\text{-Gr})$.

[21] Editor's note: the diagonal morphism $E_S \to E_S \times_S E_S = (E \times E)_S$ is also a closed immersion; that is, E_S is separated over S.

For every $S \in \mathrm{Ob}(\mathscr{C})$, let $e_{\mathbf{G}}(S)$ be the identity element of $\mathbf{G}(S)$. The family of elements $e_{\mathbf{G}}(S)$ defines

$$e_{\mathbf{G}} \in \Gamma(\mathbf{G}) = \mathrm{Hom}(\underline{\mathbf{e}}, \mathbf{G}).$$

It is a morphism of $\widehat{\mathscr{C}}$-groups $\underline{\mathbf{e}} \to \mathbf{G}$, called the *unit section* of $\mathbf{G}$.

To give a $\widehat{\mathscr{C}}$-group structure on $\mathbf{G}$ is equivalently to give a composition law on $\mathbf{G}$, that is, a $\widehat{\mathscr{C}}$-morphism

$$\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}$$

such that $\pi_{\mathbf{G}}(S)$ equips $\mathbf{G}(S)$ with a group structure for every $S \in \mathrm{Ob}(\mathscr{C})$.

Similarly, $f : \mathbf{G} \to \mathbf{H}$ is a morphism of $\widehat{\mathscr{C}}$-groups if and only if the following diagram is commutative:

$$\begin{array}{ccc} \mathbf{G} \times \mathbf{G} & \xrightarrow{\pi_{\mathbf{G}}} & \mathbf{G} \\ {\scriptstyle (f,f)}\downarrow & & \downarrow{\scriptstyle f} \\ \mathbf{H} \times \mathbf{H} & \xrightarrow{\pi_{\mathbf{H}}} & \mathbf{H} \end{array} .$$

A subobject $\mathbf{H}$ of $\mathbf{G}$ such that $\mathbf{H}(S)$ is a subgroup of $\mathbf{G}(S)$ for every S has the evident $\widehat{\mathscr{C}}$-group structure induced by that of $\mathbf{G}$. It is the unique structure for which the monomorphism $\mathbf{H} \to \mathbf{G}$ is a morphism of $\widehat{\mathscr{C}}$-groups. With this structure, $\mathbf{H}$ is called a *sub-$\widehat{\mathscr{C}}$-group* of $\mathbf{G}$.

If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathscr{C}}$-groups, then $\mathbf{G} \times \mathbf{H}$ has the evident product structure: for every S, the set $\mathbf{G}(S) \times \mathbf{H}(S)$ is given the product of the prescribed group structures. The resulting object is called the *product $\widehat{\mathscr{C}}$-group* of $\mathbf{G}$ and $\mathbf{H}$; it is their product in the category of $\widehat{\mathscr{C}}$-groups.

If $\mathbf{G}$ is a $\widehat{\mathscr{C}}$-group, then $\mathbf{G}_S$ is a $\widehat{\mathscr{C}_{/S}}$-group for every $S \in \mathrm{Ob}(\mathscr{C})$. For two $\widehat{\mathscr{C}}$-groups $\mathbf{G}$ and $\mathbf{H}$, define the object $\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ of $\widehat{\mathscr{C}}$ by

$$\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})(S) = \mathrm{Hom}_{\widehat{\mathscr{C}_{/S}}\text{-Gr}}(\mathbf{G}_S, \mathbf{H}_S).$$

Notice that $\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}$ is not in general a $\widehat{\mathscr{C}}$-group, still less the internal $\underline{\mathrm{Hom}}$ object in the category ($\widehat{\mathscr{C}}$-Gr).

One similarly defines

$$\underline{\mathrm{Isom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H}), \qquad \underline{\mathrm{End}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}), \qquad \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}).$$

Definition 2.1.2. Let $G \in \mathrm{Ob}(\mathscr{C})$. A *$\mathscr{C}$-group structure* on G is a $\widehat{\mathscr{C}}$-group structure on $\mathbf{h}_G \in \mathrm{Ob}(\widehat{\mathscr{C}})$. A morphism from the $\mathscr{C}$-group G to the $\mathscr{C}$-group H is an element

$$u \in \mathrm{Hom}(G, H) \simeq \mathrm{Hom}(\mathbf{h}_G, \mathbf{h}_H)$$

that defines a morphism of $\widehat{\mathscr{C}}$-groups from $\mathbf{h}_G$ to $\mathbf{h}_H$.

We denote the category of $\mathscr{C}$-groups by ($\mathscr{C}$-Gr). There is a cartesian square in **Cat**:

$$\begin{array}{ccc} (\mathscr{C}\text{-Gr.}) & \longrightarrow & (\widehat{\mathscr{C}}\text{-Gr.}) \\ \downarrow & & \downarrow \\ \mathscr{C} & \xrightarrow{\mathbf{h}} & \widehat{\mathscr{C}} \end{array} .$$

All the preceding definitions and constructions therefore carry over at once to ($\mathscr{C}$-Gr) whenever the functors they use (products, internal $\underline{\mathrm{Hom}}$ objects, and so on) are representable. They also apply to the categories $\mathscr{C}_{/S}$. In that case we write $\underline{\mathrm{Hom}}_{S\text{-Gr}}$ for $\underline{\mathrm{Hom}}_{\mathscr{C}_{/S}\text{-Gr}}$, and similarly for the other objects.

2.2.

More generally, suppose (T) is a species of structure on n base sets defined by finite inverse limits—for example, by commutative diagrams constructed from cartesian products, as in the structures of a monoid, a group, a set with operators, a module over a ring, or a Lie algebra over a ring. The preceding construction defines the notion of a "structure of species (T)" on n objects $\mathbf{F}_1, \ldots, \mathbf{F}_n$ of $\widehat{\mathscr{C}}$. Such a structure consists, for every $S \in \mathrm{Ob}(\mathscr{C})$, of a structure of species (T) on the sets $\mathbf{F}_1(S), \ldots, \mathbf{F}_n(S)$, in such a way that for every arrow $S' \to S''$ of $\mathscr{C}$, the family of maps

$$\big(\mathbf{F}_i(S'')\big)_i \longrightarrow \big(\mathbf{F}_i(S')\big)_i$$

is a polyhomomorphism for the species (T).

Morphisms of structures of species (T) are defined similarly, giving a category

$$(\widehat{\mathscr{C}} \times \cdots \times \widehat{\mathscr{C}})^{(T)}.$$

The fully faithful functor $\mathbf{h} \times \cdots \times \mathbf{h}$ then defines, by inverse image, the category

$$(\mathscr{C} \times \cdots \times \mathscr{C})^{(T)}.$$

Since it commutes with inverse limits, all the functorial properties, notions, and notation introduced in $\widehat{\mathscr{C}}$ carry over to this category.

Now suppose fiber products exist in $\mathscr{C}$, and let (T) be a species of algebraic structure specified by certain morphisms between cartesian products, subject to axioms consisting of commutative diagrams formed with those morphisms. A structure of species (T) on a family of objects of $\mathscr{C}$ is then defined by corresponding morphisms between cartesian products satisfying the corresponding commutativity conditions. Consequently, if $\mathscr{C}$ and $\mathscr{C}'$ are categories with products and $f : \mathscr{C} \to \mathscr{C}'$ is a functor commuting with products, then a structure of species (T) on a family (F_i) of objects of $\mathscr{C}$ induces one on the family $(f(F_i))$ in $\mathscr{C}'$. Thus every $\mathscr{C}$-group becomes a $\mathscr{C}'$-group, every pair consisting of a $\mathscr{C}$-ring and a module over it becomes the analogous pair in $\mathscr{C}'$, and so on.

In particular, let $\mathscr{C}$ satisfy the conditions of 1.8.[22] The functor $E \mapsto E_S$ defined there commutes with finite inverse limits. It therefore carries groups to S-groups (that is, $\mathscr{C}_{/S}$-groups), rings to S-rings, and so forth.

Remark. It is worth observing that the preceding construction, when applied to $\widehat{\mathscr{C}}$, gives back precisely the notions already defined there. In other words, to give an object of $\widehat{\mathscr{C}}$ a structure of species (T) when it is viewed as a functor on $\mathscr{C}$ is the same as to give the representable functor on $\widehat{\mathscr{C}}$ defined by that object a structure of species (T).[23]

We shall treat two further special cases of this construction: structures with operator groups and modules.

[22] Editor's note: including condition $(*)$.

[23] Editor's note: for group structures, for example, let $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$. If the functor

$$\widehat{\mathscr{C}} \longrightarrow \mathbf{Set}, \qquad \mathbf{F} \longmapsto \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}, \mathbf{G})$$

has a group structure, then so does its restriction to $\mathscr{C}$, namely $X \mapsto \mathbf{G}(X)$. Conversely, if $\mathbf{G}$ is a $\widehat{\mathscr{C}}$-group, its multiplication morphism $\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \to \mathbf{G}$ induces, for every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, a group structure on $\mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}, \mathbf{G})$, functorially in $\mathbf{F}$.

2.3. Structures with operator groups

Definition 2.3.1. Let $\mathbf{E} \in \mathrm{Ob}(\widehat{\mathscr{C}})$ and $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}}\text{-}\mathbf{Gr})$. A structure on $\mathbf{E}$ of an object with $\widehat{\mathscr{C}}$-operator group $\mathbf{G}$ (or of a $\mathbf{G}$-object) consists in giving $\mathbf{E}(S)$, for every $S \in \mathrm{Ob}(\mathscr{C})$, the structure of a set with operator group $\mathbf{G}(S)$, in such a way that, for every arrow $S' \to S''$ in $\mathscr{C}$, the map of sets $\mathbf{E}(S'') \to \mathbf{E}(S')$ is compatible with the homomorphism of operator groups $\mathbf{G}(S'') \to \mathbf{G}(S')$.

As usual, this is equivalent to giving a morphism

$$\mu : \mathbf{G} \times \mathbf{E} \longrightarrow \mathbf{E}$$

which, for every S, equips $\mathbf{E}(S)$ with the structure of a set on which $\mathbf{G}(S)$ acts. But

$$\mathrm{Hom}(\mathbf{G} \times \mathbf{E}, \mathbf{E}) \simeq \mathrm{Hom}(\mathbf{G}, \underline{\mathrm{End}}(\mathbf{E})),$$

so μ defines a morphism $\mathbf{G} \to \underline{\mathrm{End}}(\mathbf{E})$. It is immediate that this morphism maps $\mathbf{G}$ into $\underline{\mathrm{Aut}}(\mathbf{E})$ and is a morphism of $\widehat{\mathscr{C}}$-groups. Consequently, giving $\mathbf{E}$ the structure of an object with $\widehat{\mathscr{C}}$-operator group $\mathbf{G}$ is equivalent to giving a morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}(\mathbf{E}).$$

In particular, every element $g \in \mathbf{G}(S)$ defines an automorphism $\rho(g)$ of the functor $\mathbf{E}_S$, that is, an automorphism of $\mathbf{E} \times \mathbf{h}_S$ commuting with the projection $\mathbf{E} \times \mathbf{h}_S \to \mathbf{h}_S$, and hence, in particular, an automorphism of the set $\mathbf{E}(S')$ for every $S' \to S$.

Definition 2.3.2. We denote by $\mathbf{E}^{\mathbf{G}}$ the subobject of $\mathbf{E}$ defined as follows:

$$\mathbf{E}^{\mathbf{G}}(S) = \{x \in \mathbf{E}(S) \mid x_{S'} \text{ is invariant under } \mathbf{G}(S') \text{ for every } S' \to S\},$$

where $x_{S'}$ denotes the image of x under $\mathbf{E}(S) \to \mathbf{E}(S')$.

Thus $\mathbf{E}^{\mathbf{G}}$, the "subobject of invariants of $\mathbf{G}$," is the largest subobject of $\mathbf{E}$ on which $\mathbf{G}$ acts trivially.

Definition 2.3.3. Let $\mathbf{F}$ be a subobject of $\mathbf{E}$. We denote by $\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F}$ and $\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F}$ the sub-$\widehat{\mathscr{C}}$-groups of $\mathbf{G}$ defined by

$$\begin{aligned}
(\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}_S = \mathbf{F}_S\} \\
&= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}(S') = \mathbf{F}(S') \text{ for every } S' \to S\}, \\
(\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}_S} = \mathrm{id}\} \\
&= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}(S')} = \mathrm{id} \text{ for every } S' \to S\}.
\end{aligned}$$

Here the vertical bar after $\rho(g)$ denotes restriction. [24]

Scholium 2.3.3.1. [25] In particular, let $x \in \Gamma(\mathbf{E})$, that is (cf. 1.2), a collection of elements $x_S \in \mathbf{E}(S)$, $S \in \mathrm{Ob}(\mathscr{C})$, such that for every arrow $f : S' \to S$ one has $\mathbf{E}(f)(x_S) = x_{S'}$. If $\mathscr{C}$ has a final object S_0, then $\Gamma(\mathbf{E}) = \mathbf{E}(S_0)$. The element x then defines a subfunctor of $\mathbf{E}$, denoted by $\mathbf{x}$, and

$$\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{x} = \underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{x}.$$

[24] Editor's note: the original has been corrected by replacing the inclusion $\rho(g)\mathbf{F}_S \subset \mathbf{F}_S$ with an equality, so as to ensure that $\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{F})$ is indeed a group. See VI$_B$, 6.4 for conditions under which the "transporter" coincides with the "strict transporter."

[25] Editor's note: Scholium 2.3.3.1 and Remark 2.3.3.2 have been added.

We denote this functor by $\underline{\mathrm{Stab}}_{\mathbf{G}}(x)$ and call it the *stabilizer* of x; thus, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\underline{\mathrm{Stab}}_{\mathbf{G}}(x)(S) = \{g \in \mathbf{G}(S) \mid \rho(g)x_S = x_S\}.$$

Suppose that fiber products exist in $\mathscr{C}$. If $\mathbf{G} = \mathbf{h}_G$ (respectively, $\mathbf{E} = \mathbf{h}_E$), where G is a $\mathscr{C}$-group (respectively, $E \in \mathrm{Ob}(\mathscr{C})$), and if $\mathscr{C}$ has a final object S_0, so that x is a morphism $S_0 \to E$, then $\underline{\mathrm{Stab}}_{\mathbf{G}}(x)$ is represented by the fiber product $G \times_E S_0$, where the morphism $G \to E$ is the composite of

$$\mathrm{id}_G \times x : G = G \times S_0 \longrightarrow G \times E$$

and

$$\mu : G \times E \longrightarrow E.$$

Remark 2.3.3.2.[25] The formation of $\mathbf{E}^{\mathbf{G}}$, $\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F}$, and $\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F}$ commutes with base change; that is, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$(\mathbf{E}^{\mathbf{G}})_S = (\mathbf{E}_S)^{\mathbf{G}_S}, \qquad (\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\mathrm{Norm}}_{\mathbf{G}_S}\mathbf{F}_S, \qquad (\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\mathrm{Centr}}_{\mathbf{G}_S}\mathbf{F}_S.$$

Definition 2.3.4. If G is a $\mathscr{C}$-group and $\mathbf{E}$ is an object of $\widehat{\mathscr{C}}$ (respectively, E is an object of $\mathscr{C}$), a G-object structure on $\mathbf{E}$ (respectively, on E) is an $\mathbf{h}_G$-object structure on $\mathbf{E}$ (respectively, on $\mathbf{h}_E$).

In view of this definition, all the notions and notation defined above carry over to $\mathscr{C}$ whenever they involve only representable functors. For example, if $\underline{\mathrm{Norm}}_{\mathbf{h}_G}(\mathbf{h}_F)$ is representable, then there is a unique subobject of G representing it; it is then a sub-$\mathscr{C}$-group of G, and is denoted by $\mathrm{Norm}_G(F)$, and so forth.

Definition 2.3.5.

(a) We say that the $\widehat{\mathscr{C}}$-group $\mathbf{G}$ acts on the $\widehat{\mathscr{C}}$-group $\mathbf{H}$ if $\mathbf{H}$ is equipped with a $\mathbf{G}$-object structure such that, for every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{H}(S)$ defined by g is a group automorphism.

Equivalently, for every $g \in \mathbf{G}(S)$, the automorphism $\rho(g)$ of $\mathbf{H}_S$ is an automorphism of $\widehat{\mathscr{C}_{/S}}$-groups; or equivalently, the morphism of $\widehat{\mathscr{C}}$-groups $\mathbf{G} \to \underline{\mathrm{Aut}}(\mathbf{H})$ maps $\mathbf{G}$ into

$$\underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-}\mathbf{Gr}}(\mathbf{H}).$$

(b) In the situation above, there is a unique $\widehat{\mathscr{C}}$-group structure on $\mathbf{H} \times \mathbf{G}$ such that, for every S, $(\mathbf{H} \times \mathbf{G})(S)$ is the semidirect product of the groups $\mathbf{H}(S)$ and $\mathbf{G}(S)$ for the given action of $\mathbf{G}(S)$ on $\mathbf{H}(S)$. We denote this $\widehat{\mathscr{C}}$-group by $\mathbf{H} \cdot \mathbf{G}$ and call it the *semidirect product of* $\mathbf{H}$ *by* $\mathbf{G}$. Thus, by definition,

$$(\mathbf{H} \cdot \mathbf{G})(S) = \mathbf{H}(S) \cdot \mathbf{G}(S).$$

Let $\mathbf{G}$ be a $\widehat{\mathscr{C}}$-group. For every arrow $S' \to S$ in $\mathscr{C}$ and every $g \in \mathbf{G}(S)$, let $\mathrm{Int}(g)$ be the automorphism of $\mathbf{G}(S')$ defined by

$$\mathrm{Int}(g)h = ghg^{-1}.$$

This definition extends to a morphism of $\widehat{\mathscr{C}}$-groups

$$\mathrm{Int} : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-}\mathbf{Gr}}(\mathbf{G}) \subset \underline{\mathrm{Aut}}(\mathbf{G}).$$

Definition 2.3.3 therefore applies, and for every subobject $\mathbf{E}$ of $\mathbf{G}$ one has the sub-$\widehat{\mathscr{C}}$-groups of $\mathbf{G}$

$$\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{E}) \qquad \text{and} \qquad \underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{E}).$$

Definition 2.3.6. The *center* of $\mathbf{G}$, denoted $\underline{\mathrm{Centr}}(\mathbf{G})$, is the sub-$\widehat{\mathscr{C}}$-group $\underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{G})$ of $\mathbf{G}$. We say that $\mathbf{G}$ is commutative if $\underline{\mathrm{Centr}}(\mathbf{G}) = \mathbf{G}$, or, equivalently, if $\mathbf{G}(S)$ is commutative for every S.

We say that the sub-$\widehat{\mathscr{C}}$-group $\mathbf{H}$ of $\mathbf{G}$ is invariant in $\mathbf{G}$ if $\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is invariant in $\mathbf{G}(S)$ for every S. [26]

Definition 2.3.6.1. [27] Let $f : \mathbf{G} \to \mathbf{G}'$ be a morphism of $\widehat{\mathscr{C}}$-groups. The *kernel* of f, denoted $\mathrm{Ker}\, f$, is the sub-$\widehat{\mathscr{C}}$-group of $\mathbf{G}$ defined by

$$(\mathrm{Ker}\, f)(S) = \{x \in \mathbf{G}(S) \mid f(S)(x) = 1\} = \mathrm{Ker}\, f(S)$$

for every $S \in \mathrm{Ob}(\mathscr{C})$; it is an invariant sub-$\widehat{\mathscr{C}}$-group.

Notice that if $\mathbf{G} = \mathbf{h}_G$ and $\mathbf{G}' = \mathbf{h}_{G'}$, if $\mathscr{C}$ has a final object S_0, and if fiber products exist in $\mathscr{C}$, then $\mathrm{Ker}(f)$ is represented by $S_0 \times_{G'} G$.

Definition 2.3.6.2.[27] Let $\mathbf{E} \in \widehat{\mathscr{C}}$, and let $\mathbf{G}$ be a $\widehat{\mathscr{C}}$-group acting on $\mathbf{E}$. We say that the action of $\mathbf{G}$ on $\mathbf{E}$ is *faithful* if the kernel of the morphism $\mathbf{G} \to \underline{\mathrm{Aut}}(\mathbf{E})$ is trivial; that is, if, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $g \in \mathbf{G}(S)$, the condition

$$g_{S'} \cdot x = x \qquad \text{for every } S' \to S \text{ and every } x \in \mathbf{E}(S')$$

implies $g = 1$.

Many definitions and propositions from elementary group theory carry over readily. We record only the following one, which will be useful to us.

Proposition 2.3.7. *Let $f : \mathbf{W} \to \mathbf{G}$ be a morphism of $\widehat{\mathscr{C}}$-groups. Put $\mathbf{H}(S) = \mathrm{Ker}\, f(S)$. Let $u : \mathbf{G} \to \mathbf{W}$ be a morphism of $\widehat{\mathscr{C}}$-groups that is a section of f (and is therefore necessarily a monomorphism). Then $\mathbf{W}$ is identified with the semidirect product of $\mathbf{H}$ by $\mathbf{G}$ for the action of $\mathbf{G}$ on $\mathbf{H}$ defined by*

$$(g, h) \longmapsto \mathrm{Int}(u(g))h$$

for $g \in \mathbf{G}(S)$, $h \in \mathbf{H}(S)$, and $S \in \mathrm{Ob}(\mathscr{C})$.

All these definitions and propositions carry over, as usual, to $\mathscr{C}$. In particular, the semidirect product of two $\mathscr{C}$-groups H and G, with G acting on H, is defined whenever the Cartesian product $H \times G$ exists, and the analog of Proposition 2.3.7 takes the following form.

Proposition 2.3.8. *Let*

$$H \xrightarrow{i} W \xrightarrow{f} G$$

be a sequence of morphisms of $\mathscr{C}$-groups such that, for every $S \in \mathrm{Ob}(\mathscr{C})$, the pair $(H(S), i(S))$ is a kernel of $f(S) : W(S) \to G(S)$. Let $u : G \to W$ be a morphism of $\mathscr{C}$-groups that is a section of f. Then W is identified with the semidirect product of H by G for the action of G on H such that, if $S \in \mathrm{Ob}(\mathscr{C})$, $g \in G(S)$, and $h \in H(S)$, then

$$\mathrm{Int}(u(g))i(h) = i({}^g h).$$

[26] Editor's note: in addition, $\mathbf{H}$ is said to be *central* in $\mathbf{G}$ if $\underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is central in $\mathbf{G}(S)$ for every S.

[27] Editor's note: Definitions 2.3.6.1 and 2.3.6.2 have been added.

3. The category of O-modules, the category of G-O-modules

Definition 3.1. Let $\mathbf{O}$ and $\mathbf{F}$ be two objects of $\widehat{\mathscr{C}}$. We say that $\mathbf{F}$ is a $\widehat{\mathscr{C}}$-module over the $\widehat{\mathscr{C}}$-ring $\mathbf{O}$, or, for short, an $\mathbf{O}$-module, if, for every $S \in \mathrm{Ob}(\mathscr{C})$, the set $\mathbf{O}(S)$ has been endowed with a ring structure and $\mathbf{F}(S)$ with a module structure over that ring, in such a way that for every arrow $S' \to S''$ in $\mathscr{C}$, the map $\mathbf{O}(S'') \to \mathbf{O}(S')$ is a ring homomorphism and $\mathbf{F}(S'') \to \mathbf{F}(S')$ is a homomorphism of abelian groups compatible with that ring homomorphism.

Once $\mathbf{O}$ is fixed, one defines in the usual way a morphism between two $\mathbf{O}$-modules $\mathbf{F}$ and $\mathbf{F}'$, hence the abelian group $\mathrm{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and the category of $\mathbf{O}$-modules, denoted ($\mathbf{O}$-Mod.).

Lemma 3.1.1.[28] *The category* ($\mathbf{O}$-*Mod.*) *has the structure of an abelian category, defined objectwise. Moreover,* ($\mathbf{O}$-*Mod.*) *satisfies axiom* (AB 5) *(see* [Gr57, 1.5]*); that is, arbitrary direct sums exist, and if* $\mathbf{M}$ *is an* $\mathbf{O}$*-module,* $\mathbf{N}$ *a submodule, and* $(\mathbf{F}_i)_{i\in I}$ *an increasing filtered family of submodules of* $\mathbf{M}$*, then*

$$\bigcup_{i\in I}(\mathbf{F}_i \cap \mathbf{N}) = \left(\bigcup_{i\in I}\mathbf{F}_i\right) \cap \mathbf{N}.$$

Indeed, let $f : \mathbf{F} \to \mathbf{F}'$ be a morphism of $\mathbf{O}$-modules. Define the $\mathbf{O}$-modules $\mathrm{Ker}\, f$ (respectively, $\mathrm{Im}\, f$ and $\mathrm{Coker}\, f$) by setting, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$(\mathrm{Ker}\, f)(S) = \mathrm{Ker}\, f(S)$$

(respectively, the analogous formulas). Then $\mathrm{Ker}\, f$ (respectively, $\mathrm{Coker}\, f$) is a kernel (respectively, a cokernel) of f, and there is an isomorphism of $\mathbf{O}$-modules

$$\mathbf{F}/\mathrm{Ker}\, f \xrightarrow{\sim} \mathrm{Im}\, f.$$

This proves that ($\mathbf{O}$-Mod.) is an abelian category.

Arbitrary direct sums exist and are defined objectwise. Finally, if $\mathbf{M}$ is an $\mathbf{O}$-module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i\in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then the inclusion

$$\bigcup_{i\in I}(\mathbf{F}_i \cap \mathbf{N}) \subset \left(\bigcup_{i\in I}\mathbf{F}_i\right) \cap \mathbf{N}$$

is an equality. Indeed, if $S \in \mathrm{Ob}(\mathscr{C})$ and

$$x \in \mathbf{N}(S) \cap \bigcup_i \mathbf{F}_i(S),$$

there is an $i \in I$ such that $x \in \mathbf{N}(S) \cap \mathbf{F}_i(S)$.

Proposition 3.1.2.[28] *Suppose that the category* $\mathscr{C}$ *is small, that is, that* $\mathrm{Ob}(\mathscr{C})$ *is a set. Then* ($\mathbf{O}$-*Mod.*) *has a generator, and consequently has enough injective objects.*

Proof. For every object S of $\mathscr{C}$, define the $\mathbf{O}$-module $\mathbf{O}_S$ as follows. For every object S', $\mathbf{O}_S(S')$ is a direct sum of copies of $\mathbf{O}(S')$ indexed by $\mathrm{Hom}(S', S)$, that is,

$$\mathbf{O}_S(S') = \bigoplus_{g:S'\to S} \mathbf{O}(S')g.$$

[28] Editor's note: Statements 3.1.1 and 3.1.2 have been added to show that the category ($\mathbf{O}$-Mod.) is abelian and satisfies axiom (AB 5), and moreover has enough injective objects when the category $\mathscr{C}$ is small. An error in 3.1.2, pointed out in 2016 by Linyuan Liu, is corrected here.

For every morphism $f : S'' \to S'$, writing f^* for the ring homomorphism $\mathbf{O}(S') \to \mathbf{O}(S'')$, the morphism

$$\mathbf{O}_S(f) : \mathbf{O}_S(S') \longrightarrow \mathbf{O}_S(S'')$$

sends each element ag of $\mathbf{O}_S(S')$, where $a \in \mathbf{O}(S')$ and $g : S' \to S$, to the element $f^*(a)(g \circ f)$ of $\mathbf{O}_S(S'')$. It is readily checked that this defines a functor in $\mathbf{O}$-modules.

Since $\mathrm{Ob}(\mathscr{C})$ is a set by hypothesis, we may form the direct sum

$$\mathbf{G} = \bigoplus_{S \in \mathrm{Ob}(\mathscr{C})} \mathbf{O}_S.$$

Lemma 3.1.2.1. *Let $\mathbf{F}$ be an $\mathbf{O}$-module. For every $S \in \mathrm{Ob}(\mathscr{C})$, every element $x \in \mathbf{F}(S)$ defines a unique morphism of $\mathbf{O}$-modules*

$$\mathbf{x} : \mathbf{O}_S \longrightarrow \mathbf{F}$$

such that $\mathbf{x}(1\,\mathrm{id}_S) = x$.

Proof. Let $\phi : \mathbf{O}_S \to \mathbf{F}$ be a morphism of $\mathbf{O}$-modules such that $\phi(1\,\mathrm{id}_S) = x$. Let $g : S' \to S$ and $a \in \mathbf{O}(S')$. The $\mathbf{O}(S')$-linearity and the commutativity of the diagram

$$\begin{array}{ccc} \mathbf{O}_S(S) & \xrightarrow{\phi} & \mathbf{F}(S) \\ {\scriptstyle \mathbf{O}_S(g)}\downarrow & & \downarrow{\scriptstyle \mathbf{F}(g)} \\ \mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \end{array}$$

imply that

$$\phi(ag) = a\phi(1g) = a\mathbf{F}(g)(\phi(1\,\mathrm{id}_S)) = a\mathbf{F}(g)(x).$$

Thus ϕ, if it exists, is determined by the preceding equality. Conversely, if ϕ is defined in this way, then for every $f : S'' \to S'$ one has

$$\begin{aligned} \phi \circ \mathbf{O}_S(f)(ag) &= \phi(f^*(a)(g \circ f)) \\ &= f^*(a)\mathbf{F}(g \circ f)(x) \\ &= \mathbf{F}(f)(a\mathbf{F}(g)(x)) \\ &= \mathbf{F}(f) \circ \phi(ag), \end{aligned}$$

that is, the diagram

$$\begin{array}{ccc} \mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \\ {\scriptstyle \mathbf{O}_S(f)}\downarrow & & \downarrow{\scriptstyle \mathbf{F}(f)} \\ \mathbf{O}_S(S'') & \xrightarrow{\phi} & \mathbf{F}(S'') \end{array}$$

is commutative. This proves the lemma.

Now let $\mathbf{F}$ be an $\mathbf{O}$-module. For each $S \in \mathrm{Ob}(\mathscr{C})$, let $F_0(S)$ be a set of generators of the $\mathbf{O}(S)$-module $\mathbf{F}(S)$, and let $\mathbf{L}_S$ be the direct sum, indexed by $F_0(S)$, of copies of $\mathbf{O}_S$. By the lemma, we obtain a morphism of $\mathbf{O}$-modules

$$\mathbf{L}_S \longrightarrow \mathbf{F}$$

such that the morphism $\mathbf{L}_S(S) \to \mathbf{F}(S)$ is surjective, and hence is an epimorphism in the category of $\mathbf{O}(S)$-modules.

Set

$$I = \coprod_{S \in \mathrm{Ob}(\mathscr{C})} F_0(S).$$

We then obtain a morphism of $\mathbf{O}$-modules

$$\mathbf{G}^{\oplus I} \longrightarrow \bigoplus_{S \in \mathrm{Ob}(\mathscr{C})} \mathbf{L}_S \longrightarrow \mathbf{F}$$

which is an epimorphism objectwise, and hence an epimorphism of ($\mathbf{O}$-Mod.). This shows that $\mathbf{G}$ is a generator of ($\mathbf{O}$-Mod.) (see [Gr57, 1.9.1]). Since ($\mathbf{O}$-Mod.) satisfies (AB 5), it follows from [Gr57, 1.10.1] that ($\mathbf{O}$-Mod.) has enough injective objects.

Remark 3.1.3. If $\mathbf{O}_0$ is the $\widehat{\mathscr{C}}$-ring defined by

$$\mathbf{O}_0(S) = \mathbb{Z}$$

(which must not be confused with the functor associated with the constant object $\mathbb{Z}$), then the category of $\mathbf{O}_0$-modules is isomorphic to the category of commutative $\widehat{\mathscr{C}}$-groups.

Definition 3.1.4. Notice that if $\mathbf{F}$ is an $\mathbf{O}$-module, then for every $S \in \mathrm{Ob}(\mathscr{C})$, $\mathbf{F}_S$ is an $\mathbf{O}_S$-module. We may therefore define an abelian $\widehat{\mathscr{C}}$-group $\underline{\mathrm{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$ by

$$\underline{\mathrm{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')(S) = \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{F}_S, \mathbf{F}'_S).$$

In the same way one defines the objects

$$\underline{\mathrm{Isom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}'), \qquad \underline{\mathrm{End}}_{\mathbf{O}}(\mathbf{F}), \qquad \text{and} \qquad \underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F}),$$

the last of which is a $\widehat{\mathscr{C}}$-group under the structure induced by composition of automorphisms.

Definition 3.2. Let $\mathbf{O}$ be a $\widehat{\mathscr{C}}$-ring, $\mathbf{F}$ an $\mathbf{O}$-module, and $\mathbf{G}$ a $\widehat{\mathscr{C}}$-group. A $\mathbf{G}$-$\mathbf{O}$-module structure on $\mathbf{F}$ is a $\mathbf{G}$-object structure such that, for every $S \in \mathrm{Ob}(\mathscr{C})$ and every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{F}(S)$ defined by g is an automorphism of its $\mathbf{O}(S)$-module structure.

Equivalently, the morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}(\mathbf{F})$$

defined in 2.3 maps $\mathbf{G}$ into the $\widehat{\mathscr{C}}$-subgroup $\underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F})$ of $\underline{\mathrm{Aut}}(\mathbf{F})$. Thus, giving a $\mathbf{G}$-$\mathbf{O}$-module structure on the $\mathbf{O}$-module $\mathbf{F}$ is the same as giving a morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F}).$$

One defines naturally the abelian group $\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and hence the additive category of $\mathbf{G}$-$\mathbf{O}$-modules, denoted ($\mathbf{G}$-$\mathbf{O}$-Mod.).

Remark 3.2.1. The reader will notice that ($\mathbf{G}$-$\mathbf{O}$-Mod.) may also be defined as the category of $\mathbf{O}[\mathbf{G}]$-modules, where $\mathbf{O}[\mathbf{G}]$ is the algebra of the $\widehat{\mathscr{C}}$-group $\mathbf{G}$ over the $\widehat{\mathscr{C}}$-ring $\mathbf{O}$, whose definition is clear. [29] Consequently, by 3.1.1 and 3.1.2, ($\mathbf{G}$-$\mathbf{O}$-Mod.) is an abelian category satisfying axiom (AB 5); moreover, if $\mathscr{C}$ is small, ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects.

All the constructions of this section specialize immediately to the case where $\mathbf{G}$ (or $\mathbf{O}$, or both) is representable by an object of $\mathscr{C}$, which is thereby endowed with the corresponding algebraic structure.

[29] Editor's note: the sentence which follows has been added; it will be used in Section 5.

We have treated briefly the principal algebraic structures that occur later in this seminar. For the others (the structure of an **O**-Lie algebra, for example), we believe that the reader has now seen enough examples in this section to make the general mechanism sketched in 2.2 work in each particular case.

We shall now apply what we have just done to the category of schemes, denoted (**Sch**), and more generally to the categories $(\mathbf{Sch})_{/S}$, also denoted $(\mathbf{Sch}_{/S})$.

4. Algebraic structures in the category of schemes

Whenever there is no ambiguity, we shall permit ourselves the following abuses of language. We shall denote by T the object $T \xrightarrow{f} S$ of $\mathscr{C}_{/S}$, with the datum of f (the structural morphism of T) understood, and we shall identify $\mathscr{C}$ with a subcategory of $\widehat{\mathscr{C}}$. In view of the compatibilities stated in the preceding sections, these identifications may be made without danger.

We shall also simplify terminology according to the following pattern: a (**Sch**)-group will also be called a *group scheme*; a $(\mathbf{Sch})_{/S}$-group will be called a *group scheme over* S, an S-*group scheme*, or an S-*group*; and it will be called an A-*group* when S is the spectrum of the ring A.

4.1. Constant schemes

The category of schemes satisfies the conditions required in 1.8. Constant objects may therefore be defined in it. For every set E, there is a scheme $E_{\mathbb{Z}}$, and for every scheme S, an S-scheme

$$E_S = (E_{\mathbb{Z}})_S.$$

Recall (cf. *loc. cit.*) that for every S-scheme T,

$$\mathrm{Hom}_S(T, E_S)$$

is the set of locally constant maps from the topological space underlying T to E.

The functor $E \mapsto E_S$ commutes with finite inverse limits. In particular, if G is a group, then G_S is an S-group scheme; if O is a ring, then O_S is an S-ring scheme; and similarly for other algebraic structures.

4.2. Affine S-groups

Let us recall some facts about affine S-schemes (EGA II, Section 1). An S-scheme T is said to be *affine over* S if the inverse image of every affine open subset of S is affine. The $\mathcal{O}_S$-algebra $f_*\mathcal{O}_T$, denoted by $\mathscr{A}(T)$, is then quasi-coherent, where f denotes the structural morphism of T. Conversely, to every quasi-coherent $\mathcal{O}_S$-algebra $\mathscr{A}$, one can associate an S-scheme affine over S, denoted $\mathrm{Spec}(\mathscr{A})$. The functors

$$T \longmapsto \mathscr{A}(T), \qquad \mathscr{A} \longmapsto \mathrm{Spec}(\mathscr{A})$$

are quasi-inverse equivalences between the category of S-schemes affine over S and the category opposite to that of quasi-coherent $\mathcal{O}_S$-algebras.

It follows that giving an algebraic structure on an S-scheme T affine over S is equivalent to giving the corresponding structure on $\mathscr{A}(T)$ in the category opposite to that of quasi-coherent $\mathcal{O}_S$-algebras. In particular, if G is an S-group affine over S, then $\mathscr{A}(G)$ is endowed with the structure of an augmented $\mathcal{O}_S$-bialgebra: there are two morphisms of $\mathcal{O}_S$-algebras

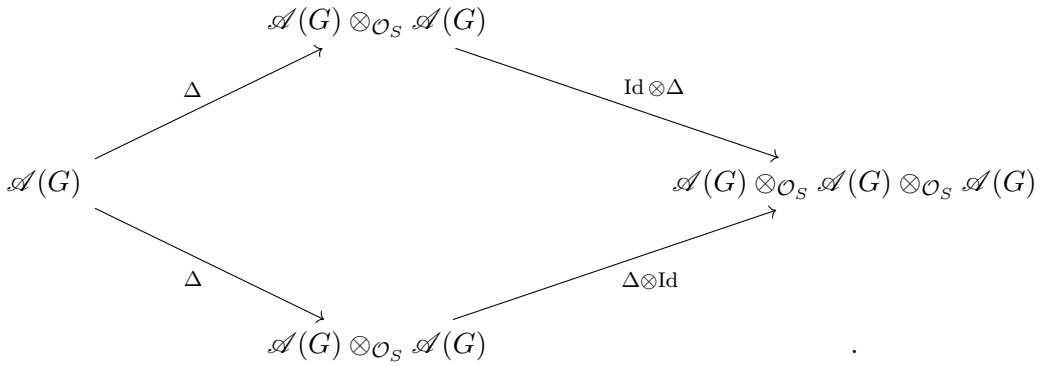
$$\Delta : \mathscr{A}(G) \longrightarrow \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G), \qquad \varepsilon : \mathscr{A}(G) \longrightarrow \mathcal{O}_S,$$

corresponding to the morphisms of S-schemes

$$\pi : G \times G \longrightarrow G, \qquad e_G : S \longrightarrow G.$$

The maps Δ and ε satisfy the following conditions, which express that G is an S-monoid. [30]

(HA 1) Δ is coassociative: the following diagram is commutative.



(HA 2) Δ is compatible with ε: the following two composites are the identity:

$$\mathscr{A}(G) \xrightarrow{\Delta} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\mathrm{Id} \otimes \varepsilon} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathscr{A}(G),$$

$$\mathscr{A}(G) \xrightarrow{\Delta} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\varepsilon \otimes \mathrm{Id}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\sim} \mathscr{A}(G).$$

Let us take this opportunity to note once again that it follows from the definition of an S-group structure that, in order to give such a structure on an S-scheme G affine over S, it is not necessary to verify anything on $\mathscr{A}(G)$; one need only endow every $G(S')$, for S' over S, with a group structure functorial in S'. This observation applies *mutatis mutandis* to morphisms.

4.3. The groups $\mathbb{G}_a$ and $\mathbb{G}_m$. The ring O

4.3.1.

Let $\mathbb{G}_a$ be the functor from $(\mathbf{Sch})^\circ$ to **Set** defined by

$$\mathbb{G}_a(S) = \Gamma(S, \mathcal{O}_S),$$

endowed with the $\widehat{(\mathbf{Sch})}$-group structure defined by the additive-group structure of the ring $\Gamma(S, \mathcal{O}_S)$. It is representable by an affine scheme, which we denote by $\mathbb{G}_a$, and which is therefore a group scheme:

$$\mathbb{G}_a = \operatorname{Spec} \mathbb{Z}[T].$$

Indeed,

$$\mathbb{G}_a(S) = \operatorname{Hom}(S, \mathbb{G}_a) = \operatorname{Hom}_{\text{Alg.}}(\mathbb{Z}[T], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S).$$

For every scheme S, one therefore has an S-group affine over S, denoted $\mathbb{G}_{a,S}$, corresponding to the $\mathcal{O}_S$-bialgebra $\mathcal{O}_S[T]$, whose diagonal map and augmentation are defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \qquad \varepsilon(T) = 0.$$

[30] Editor's note: of course, the inversion morphism $G \to G$ induces a morphism of $\mathcal{O}_S$-algebras $\tau : \mathscr{A}(G) \to \mathscr{A}(G)$ which, together with Δ and ε, makes $\mathscr{A}(G)$ into a Hopf $\mathcal{O}_S$-algebra.

4.3.2.

Let $\mathbb{G}_m$ be the functor from $(\mathbf{Sch})^\circ$ to **Set** defined by

$$\mathbb{G}_m(S) = \Gamma(S, \mathcal{O}_S)^\times,$$

where $\Gamma(S, \mathcal{O}_S)^\times$ denotes the multiplicative group of invertible elements of the ring $\Gamma(S, \mathcal{O}_S)$, endowed with its natural $\widehat{(\mathbf{Sch})}$-group structure. It is representable by an affine scheme denoted by $\mathbb{G}_m$:

$$\mathbb{G}_m = \operatorname{Spec} \mathbb{Z}[T, T^{-1}] = \operatorname{Spec} \mathbb{Z}[\mathbb{Z}],$$

where $\mathbb{Z}[\mathbb{Z}]$ denotes the group algebra of $\mathbb{Z}$ over the ring $\mathbb{Z}$. Indeed,

$$\mathbb{G}_m(S) = \operatorname{Hom}_{\text{Alg.}}(\mathbb{Z}[T, T^{-1}], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S)^\times.$$

For every scheme S, one therefore has an S-group affine over S, denoted $\mathbb{G}_{m,S}$, corresponding to the $\mathcal{O}_S$-algebra $\mathcal{O}_S[\mathbb{Z}]$, whose diagonal map and augmentation are defined by

$$\Delta(x) = x \otimes x, \qquad \varepsilon(x) = 1, \qquad x \in \mathbb{Z}.$$

4.3.3.

Let $\mathbf{O}$ be the functor $\mathbb{G}_a$ endowed with its $\widehat{(\mathbf{Sch})}$-ring structure. It is represented by the scheme $\operatorname{Spec} \mathbb{Z}[T]$, which we denote by $\mathbf{O}$ when it is regarded as endowed with its ring-scheme structure.

For every scheme S,

$$\mathbf{O}_S = S \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Z}[T] = \operatorname{Spec}(\mathcal{O}_S[T])$$

is therefore an S-ring scheme affine over S. (Note: in EGA II, 1.7.13, $\mathbf{O}_S$ is denoted by $S[T]$.)

4.3.3.1. [31] For every object X of $\widehat{(\mathbf{Sch})}$,

$$\mathbf{O}(X) = \operatorname{Hom}(X, \mathbf{O})$$

has a ring structure, functorial in X. In particular, if X' is a scheme and morphisms $x : X' \to X$ and $f : X \to \mathbf{O}$ are given (that is, $f \in \mathbf{O}(X)$), then

$$f(x) = f \circ x \in \mathbf{O}(X') = \Gamma(X', \mathcal{O}_{X'}).$$

Definition. Let $\pi : M \to X$ be a morphism in $\widehat{(\mathbf{Sch})}$, and set

$$\mathbf{O}_X = \mathbf{O} \times X.$$

We say that M is an $\mathbf{O}_X$-module if, for every X-scheme X', the set

$$\operatorname{Hom}_X(X', M)$$

has been endowed with an $\mathbf{O}(X')$-module structure, functorial in X'.

This is equivalent to giving an X-abelian-group law

$$\mu : M \times_X M \longrightarrow M$$

[31] Editor's note: this paragraph has been added; it will be useful later (cf. II, 1.3).

and an *external law*

$$\mathbf{O} \times M = \mathbf{O}_X \times_X M \longrightarrow M, \qquad (f, m) \longmapsto f \cdot m,$$

which is an X-morphism, that is, $\pi(f \cdot m) = \pi(m)$, and which, for every $x \in X(S)$, endows

$$M(x) = \{m \in M(S) \mid \pi(m) = x\}$$

with an $\mathbf{O}(S)$-module structure.

In that case, for every $Y \in \mathrm{Ob}(\widehat{(\mathbf{Sch})}_{/X})$, not necessarily representable,

$$\mathrm{Hom}_X(Y, M) = \Gamma(M_Y/Y)$$

is an $\mathbf{O}(Y)$-module, functorially in Y.

4.4. Diagonalizable groups

4.4.1. The construction of $\mathbb{G}_m$ generalizes as follows. Let M be a commutative group, and let $M_{\mathbb{Z}}$ be the group scheme associated with it by the procedure of 4.1. Consider the functor defined by

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{groups}}(M, \mathbb{G}_m(S)) \simeq \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This is a commutative $\widehat{(\mathbf{Sch})}$-group represented by a group scheme denoted $D(M)$; thus, by definition,

$$D(M) \simeq \underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m).$$

Indeed, set

$$D(M) = \mathrm{Spec}\, \mathbb{Z}[M],$$

where $\mathbb{Z}[M]$ is the group algebra of M over $\mathbb{Z}$. Then

$$D(M)(S) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[M], \Gamma(S, \mathcal{O}_S)) \simeq \mathrm{Hom}_{\mathrm{groups}}(M, \Gamma(S, \mathcal{O}_S)^\times)$$

by the very definition of the algebra $\mathbb{Z}[M]$.

4.4.2. For every scheme S, one therefore has an S-group scheme affine over S,

$$D_S(M) = D(M)_S = \left(\underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m)\right)_S = \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

It is associated with the $\mathcal{O}_S$-bialgebra $\mathcal{O}_S[M]$, endowed with the diagonal map and augmentation defined by

$$\Delta(x) = x \otimes x, \qquad \varepsilon(x) = 1, \qquad x \in M.$$

4.4.3. If $f : M \to N$ is a homomorphism of commutative groups, one defines in the evident way a morphism of S-groups

$$D_S(f) : D_S(N) \longrightarrow D_S(M),$$

and hence a functor

$$D_S : M \longmapsto D_S(M)$$

from the category of abelian groups to the category of S-groups affine over S. It may also be defined as the composite of the functor $M \mapsto M_S$ and the functor

$$M_S \longmapsto \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This functor commutes with base extension.

An S-group isomorphic to a group of the form $D_S(M)$ is said to be *diagonalizable*. Notice that the elements of M are interpreted as certain characters of $D_S(M)$, that is, as certain elements of

$$\mathrm{Hom}_{S\text{-}\mathbf{Gr}}(D_S(M), \mathbb{G}_{m,S}).$$

(Compare VIII, 1.)

4.4.4. Let us give some examples of diagonalizable groups. First,

$$D(\mathbb{Z}) = \mathbb{G}_m, \qquad D(\mathbb{Z}^r) = (\mathbb{G}_m)^r.$$

Set

$$\boldsymbol{\mu}_n = D(\mathbb{Z}/n\mathbb{Z}),$$

and call it the *group of n-th roots of unity*. Indeed,

$$\boldsymbol{\mu}_n(S) = \mathrm{Hom}_{\mathrm{groups}}(\mathbb{Z}/n\mathbb{Z}, \Gamma(S, \mathcal{O}_S)^\times) = \{f \in \Gamma(S, \mathcal{O}_S) \mid f^n = 1\}.$$

The S-group $\boldsymbol{\mu}_{n,S}$ corresponds to the $\mathcal{O}_S$-algebra $\mathcal{O}_S[T]/(T^n - 1)$. Suppose, in particular, that S is the spectrum of a field k of characteristic $p = n$. On setting $T - 1 = s$, one obtains

$$k[T]/(T^p - 1) = k[s]/(s^p),$$

which shows that the underlying topological space of $\boldsymbol{\mu}_{p,k}$ consists of one point, whose local ring is the Artinian k-algebra $k[s]/(s^p)$. In the same vein, note that the S-schemes $\mathbb{G}_{a,S}$, $\mathbb{G}_{m,S}$, and $\mathbf{O}_S$ are smooth over S, and that $D_S(M)$ is

flat over S. Moreover, it is formally smooth (respectively, smooth) over S if and only if no residue characteristic of S divides the torsion of M (respectively, and if in addition M is of finite type); see Exposé VIII, 2.1.

4.5. Other examples of groups

The preceding procedure applies to the "classical groups" (linear groups GL_n, symplectic groups Sp_n, orthogonal groups O_n, and so forth). For example, one defines GL_n as representing the functor $\mathbf{GL}_n$ such that

$$\mathbf{GL}_n(S) = \mathrm{GL}(n, \Gamma(S, \mathcal{O}_S)) = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{O}_S^n).$$

It may be constructed, for example, as the open subscheme of $\mathrm{Spec}\,\mathbb{Z}[T_{ij}]$, $1 \leq i, j \leq n$, defined by the function

$$f = \det((T_{ij})_{i,j=1}^n),$$

that is, as $\mathrm{Spec}\,\mathbb{Z}[T_{ij}, f^{-1}]$.

4.6. Functor-modules in the category of schemes

We shall associate with every $\mathcal{O}_S$-module on a scheme S a $\mathbf{O}_S$-module, where $\mathbf{O}_S$ denotes the functor-ring introduced in 4.3.3. This can be done in two different ways, as follows.

Definition 4.6.1. Let S be a scheme. For every $\mathcal{O}_S$-module $\mathscr{F}$, denote by $\mathbf{V}(\mathscr{F})$ and $\mathbf{W}(\mathscr{F})$ the contravariant functors on $(\mathbf{Sch})_{/S}$ defined by

$$\mathbf{V}(\mathscr{F})(S') = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}, \mathcal{O}_{S'}),$$
$$\mathbf{W}(\mathscr{F})(S') = \Gamma(S', \mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

where $\mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ denotes the inverse image on S' of the $\mathcal{O}_S$-module $\mathscr{F}$.

Then $\mathbf{V}(\mathscr{F})$ and $\mathbf{W}(\mathscr{F})$ are endowed in the evident way with $\mathbf{O}_S$-module structures (recall that $\mathbf{O}_S(S') = \Gamma(S', \mathcal{O}_{S'}) = \mathbf{W}(\mathcal{O}_S)(S')$). Thus one has in fact defined two functors $\mathbf{V}$ and $\mathbf{W}$ from the category of $\mathcal{O}_S$-modules to the category of $\mathbf{O}_S$-modules, with $\mathbf{V}$ contravariant and $\mathbf{W}$ covariant.

For the remainder of this paragraph, we restrict to the case in which the $\mathcal{O}_S$-modules in question are quasi-coherent. Thus we regard $\mathbf{V}$ and $\mathbf{W}$ as functors

$$\mathbf{V} : (\mathcal{O}_S\text{-}\mathbf{Mod.q.c.})^\circ \longrightarrow (\mathbf{O}_S\text{-}\mathbf{Mod.}), \qquad \mathbf{W} : (\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \longrightarrow (\mathbf{O}_S\text{-}\mathbf{Mod.}).$$

Remark 4.6.1.1. [32] The reader will notice that, in what follows, every proposition involving only the functor $\mathbf{W}$ remains valid for arbitrary $\mathcal{O}_S$-modules, not necessarily quasi-coherent.

Proposition 4.6.2.

(i) The functors $\mathbf{V}$ and $\mathbf{W}$ commute with base extension: if S' lies over S and $\mathscr{F}$ is a quasi-coherent $\mathcal{O}_S$-module, then

$$\mathbf{V}(\mathscr{F} \otimes \mathcal{O}_{S'}) \simeq \mathbf{V}(\mathscr{F})_{S'}, \qquad \mathbf{W}(\mathscr{F} \otimes \mathcal{O}_{S'}) \simeq \mathbf{W}(\mathscr{F})_{S'}.$$

(ii) The functors $\mathbf{V}$ and $\mathbf{W}$ are fully faithful: the canonical maps

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F})),$$

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}'))$$

are bijective.

(iii) The functors $\mathbf{V}$ and $\mathbf{W}$ are additive:

$$\mathbf{V}(\mathscr{F} \oplus \mathscr{F}') \simeq \mathbf{V}(\mathscr{F}) \times_S \mathbf{V}(\mathscr{F}'), \qquad \mathbf{W}(\mathscr{F} \oplus \mathscr{F}') \simeq \mathbf{W}(\mathscr{F}) \times_S \mathbf{W}(\mathscr{F}').$$

Parts (i) and (iii) are immediate from the definitions. For (ii), take the S' to be open subschemes of S. We leave the proof to the reader (for $\mathbf{V}$, use EGA II, 1.7.14).

Recall (cf. 3.1.4) that if $\mathbf{F}$ and $\mathbf{F}'$ are $\mathbf{O}_S$-modules, then $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{F}, \mathbf{F}')$ denotes the S-functor in abelian groups that assigns to every $S' \to S$ the group

$$\mathrm{Hom}_{\mathbf{O}_{S'}}(\mathbf{F}_{S'}, \mathbf{F}'_{S'}).$$

Proposition 4.6.3. *There are canonical morphisms in* $(\mathbf{O}_S\text{-}\mathbf{Mod.})$*:* [33]

$$\begin{array}{ccc} \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}')) & = & \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F})) \\ & \nwarrow \quad \nearrow & \\ & \mathbf{W}(\mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}')) & \end{array} .$$

This follows immediately from 4.6.2(i) and (ii).

[32] Editor's note: the numbering 4.6.1.1 has been added for later references.

[33] Editor's note: the isomorphism $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}')) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F}))$ has been added.

Notation 4.6.3.1. Let $\mathscr{F}$ be a quasi-coherent $\mathcal{O}_S$-module. By EGA II, 1.7.8, the S-functor $\mathbf{V}(\mathscr{F})$ is represented by an S-scheme affine over S, denoted $\mathbb{V}(\mathscr{F})$, and called the *vector fibration* [34] defined by $\mathscr{F}$:

$$\mathbb{V}(\mathscr{F}) = \mathrm{Spec}(\mathcal{S}(\mathscr{F})),$$

where $\mathcal{S}(\mathscr{F})$ denotes the symmetric algebra of the $\mathcal{O}_S$-module $\mathscr{F}$. [35]

Proposition 4.6.4. *Let $\mathscr{F}$ and $\mathscr{F}'$ be two quasi-coherent $\mathcal{O}_S$-modules, and let $\mathscr{A}$ be a quasi-coherent $\mathcal{O}_S$-algebra. There is a functorial isomorphism*

$$\mathrm{Hom}_S(\mathrm{Spec}(\mathscr{A}), \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}'), \mathbf{W}(\mathscr{F}))) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}', \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}).$$

Indeed, put $X = \mathrm{Spec}(\mathscr{A})$. The left-hand side is canonically isomorphic to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}'), \mathbf{W}(\mathscr{F}))(X),$$

that is, by definition, to

$$\begin{aligned}&\mathrm{Hom}_{\mathbf{O}_X}(\mathbf{W}(\mathscr{F}')_X, \mathbf{W}(\mathscr{F})_X)\\ &\quad\simeq \mathrm{Hom}_{\mathbf{O}_X}(\mathbf{W}(\mathscr{F}' \otimes \mathcal{O}_X), \mathbf{W}(\mathscr{F} \otimes \mathcal{O}_X))\end{aligned}$$

(cf. 4.6.2(i)). By 4.6.2(ii), this may also be written as

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}' \otimes \mathcal{O}_X, \mathscr{F} \otimes \mathcal{O}_X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}', \pi_*\pi^*(\mathscr{F})),$$

where $\pi : X \to S$ is the structural morphism. But EGA II, 1.4.7 gives

$$\pi_*\pi^*(\mathscr{F}) \simeq \mathscr{F} \otimes \mathscr{A},$$

which completes the proof.

Corollary 4.6.4.1. *There is a canonical isomorphism*

$$\mathbf{W}(\mathscr{F} \otimes \mathscr{A}) \simeq \underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathscr{A}), \mathbf{W}(\mathscr{F})).$$

Indeed, [36] let $f : S' \to S$ be an S-scheme and put $X' = X \times_S S'$. There is a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & X \\ {\scriptstyle \pi'}\downarrow & & \downarrow{\scriptstyle \pi} \\ S' & \xrightarrow{f} & S \end{array}$$

By EGA II, 1.5.2, X' is affine over S' and $\pi'_*\mathcal{O}_{X'} = f^*(\mathscr{A})$. Hence

$$\begin{aligned}&\underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathscr{A}), \mathbf{W}(\mathscr{F}))(S')\\ &\quad= \mathrm{Hom}_{S'}(\mathrm{Spec}(f^*\mathscr{A}), \mathbf{W}(f^*\mathscr{F})).\end{aligned}$$

[34] Editor's note: "vector bundle" has been replaced by "vector fibration." Current usage calls a vector fibration that is locally trivial of rank r a "vector bundle of rank r," that is, one whose sheaf of sections is locally isomorphic to $\mathcal{O}_S^{\oplus r}$.

[35] Editor's note: the articles [Ni04] and [Ni02], respectively, show that if S is Noetherian and $\mathscr{F}$ is a coherent $\mathcal{O}_S$-module, then $\mathbf{W}(\mathscr{F})$, respectively the S-group assigning $\mathrm{Aut}_{\mathcal{O}_T}(\mathscr{F} \otimes \mathcal{O}_T)$ to every $T \to S$, is representable if and only if $\mathscr{F}$ is locally free.

[36] Editor's note: the original has been expanded in what follows.

By 4.6.4, applied to $f^*\mathscr{F}$, $\mathscr{F}' = \mathcal{O}_{S'}$, and $f^*\mathscr{A}$, this is equal to

$$\begin{aligned}\Gamma(S', f^*\mathscr{F} \otimes f^*\mathscr{A}) &= \Gamma(S', f^*(\mathscr{F} \otimes \mathscr{A})) \\ &= \mathbf{W}(\mathscr{F} \otimes \mathscr{A})(S').\end{aligned}$$

Proposition 4.6.5. *If $\mathscr{F}$ and $\mathscr{F}'$ are two $\mathcal{O}_S$-modules locally free of finite type, the morphisms of 4.6.3 are isomorphisms.*

Indeed, for every $S' \to S$,

$$\begin{aligned}&\mathbf{W}(\mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}'))(S') \\ &\quad = \Gamma\big(S', \mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}') \otimes \mathcal{O}_{S'}\big) \\ &\quad = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathscr{F} \otimes \mathcal{O}_{S'}, \mathscr{F}' \otimes \mathcal{O}_{S'}).\end{aligned}$$

The last member is isomorphic both to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}'))(S')$$

and to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F}))(S')$$

by 4.6.2(i) and (ii).

Corollary 4.6.5.1. *Let $\mathscr{F}$ be an $\mathcal{O}_S$-module locally free of finite type. Put $\mathscr{F}^\vee = \mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathcal{O}_S)$. There are canonical isomorphisms*

$$\mathbf{W}(\mathscr{F}^\vee) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{O}_S) \simeq \mathbf{V}(\mathscr{F}),$$

$$\mathbf{V}(\mathscr{F}^\vee) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}), \mathbf{O}_S) \simeq \mathbf{W}(\mathscr{F}).$$

Finally, we have the following proposition.

Proposition 4.6.6. *Let $f : \mathscr{F} \to \mathscr{F}'$ be a morphism of $\mathcal{O}_S$-modules locally free of finite rank. The morphism*

$$\mathbf{W}(f) : \mathbf{W}(\mathscr{F}) \longrightarrow \mathbf{W}(\mathscr{F}')$$

is a monomorphism if and only if f identifies $\mathscr{F}$ locally with a direct summand of $\mathscr{F}'$.

The direct assertion is essentially contained in EGA 0_{I}, 5.5.5. [37] Conversely, if $\mathscr{F}$ is locally a direct summand of $\mathscr{F}'$, then for every $\pi : S' \to S$, the module $\pi^*\mathscr{F}$ is a submodule of $\pi^*\mathscr{F}'$, since this holds locally as a direct summand. Thus

$$\mathbf{W}(\mathscr{F})(S') = \Gamma(S', \pi^*\mathscr{F})$$

is a submodule of

$$\mathbf{W}(\mathscr{F}')(S') = \Gamma(S', \pi^*\mathscr{F}').$$

4.7. The category of G-$\mathcal{O}_S$-modules

Let G be an S-group and $\mathscr{F}$ an $\mathcal{O}_S$-module; $\mathbf{W}(\mathscr{F})$ has a $\mathbf{O}_S$-module structure.

Definition 4.7.1. A *G-$\mathcal{O}_S$-module structure* on $\mathscr{F}$ is an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{W}(\mathscr{F})$ (cf. 3.2). A morphism of G-$\mathcal{O}_S$-modules is, by definition, a morphism of the associated $\mathbf{h}_G$-$\mathbf{O}_S$-modules. This gives the category (G-$\mathcal{O}_S$-**Mod**.); we denote by (G-$\mathcal{O}_S$-**Mod**.**q**.**c**.) its full subcategory consisting of those G-$\mathcal{O}_S$-modules that are quasi-coherent as $\mathcal{O}_S$-modules.

[37] Editor's note: the following sentence has been expanded.

By 3.2, giving a G-$\mathcal{O}_S$-module structure on $\mathscr{F}$ is therefore equivalent to giving a morphism of $\widehat{(\mathbf{Sch})}_{/S}$-groups

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F})).$$

Remark 4.7.1.0. [38] Since 4.6.2 gives an anti-isomorphism of S-functors in groups

$$(\dagger) \qquad \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F})) \simeq \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F})),$$

it is equivalent to give an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{W}(\mathscr{F})$ or on $\mathbf{V}(\mathscr{F})$. Indeed, let

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}))$$

be as above. For every $T \to S$ and $g \in G(T)$, denote by $\rho^*(g)$ the image of $\rho(g)$ under the anti-isomorphism (†). Then

$$\rho^*(gh) = \rho^*(h) \circ \rho^*(g),$$

so ρ^* defines a *right* $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{V}(\mathscr{F})$. Setting

$$\rho^\vee(g) = \rho^*(g^{-1}),$$

one obtains an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{V}(\mathscr{F})$, whose datum is equivalent to that of ρ.

Remark 4.7.1.1. One may say that the categories just defined have been constructed by the following Cartesian squares.

$$\begin{array}{ccccc}
(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod}.\mathbf{q}.\mathbf{c}.) & \hookrightarrow & (G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod}.) & \longrightarrow & (\mathbf{h}_G\text{-}\mathbf{O}_S\text{-}\mathbf{Mod}.) \\
\downarrow & & \downarrow & & \downarrow{\scriptstyle \text{forget}} \\
(\mathcal{O}_S\text{-}\mathbf{Mod}.\mathbf{q}.\mathbf{c}.) & \hookrightarrow & (\mathcal{O}_S\text{-}\mathbf{Mod}.) & \xrightarrow{\mathbf{W}} & (\mathbf{O}_S\text{-}\mathbf{Mod}.)
\end{array} .$$

The categories $(\mathcal{O}_S\text{-}\mathbf{Mod}.)$ and $(\mathbf{O}_S\text{-}\mathbf{Mod}.)$ are abelian, but one should remember that in general the functor $\mathbf{W}$ is neither left nor right exact. [39]

Remark 4.7.1.2. [40] Let $\mathscr{F}$ be a G-$\mathcal{O}_S$-module. The subsheaf of invariants $\mathscr{F}^G$ is defined as follows: for every open subset U of S,

$$\mathscr{F}^G(U) = \mathbf{W}(\mathscr{F})^G(U) = \{x \in \mathscr{F}(U) \mid g \cdot x_{S'} = x_{S'} \text{ for every } S' \xrightarrow{f} U,\ g \in G(S')\},$$

where $x_{S'}$ denotes the image of x in

$$\Gamma(S', f^*\mathscr{F}) = \Gamma(U, f_* f^* \mathscr{F}).$$

One should note that the natural morphism

$$\mathbf{W}(\mathscr{F}^G) \longrightarrow \mathbf{W}(\mathscr{F})^G$$

is not an isomorphism in general. For example, let $S = \mathrm{Spec}(\mathbb{Z})$, and let G be the constant group $\mathbb{Z}/2\mathbb{Z} = \{1, \tau\}$ acting on $\mathscr{F} = \mathcal{O}_S$ by $\tau \cdot 1 = -1$. Then $\mathscr{F}^G = 0$, whereas, if R is an $\mathbb{F}_2$-algebra,

$$\mathbf{W}(\mathscr{F})^G(\mathrm{Spec}(R)) = R.$$

[38] Editor's note: this remark has been added from [DG], II, Section 2, 1.1.

[39] Editor's note: the original has been corrected by deleting the assertion that the category $(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod}.)$ is abelian; see 4.7.2.1 below.

[40] Editor's note: this remark has been added.

4.7.2. From now until the end of 4.7, suppose that G is affine over S. [41] By 4.6.4, giving a morphism of S-functors

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{End}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}))$$

is equivalent to giving a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathscr{F} \longrightarrow \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G).$$

Saying that ρ is a morphism of $\widehat{(\mathbf{Sch})}_{/S}$-groups is then equivalent to saying that μ satisfies the following axioms.

(CM 1) The following diagram is commutative.

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\quad\mu\quad} & \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \\ {\scriptstyle\mu}\downarrow & & \downarrow{\scriptstyle \mathrm{Id}\otimes\Delta} \\ \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G) & \xrightarrow{\mu\otimes\mathrm{Id}} & \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \end{array} .$$

(CM 2) The following composite is the identity.

$$\mathscr{F} \xrightarrow{\ \mu\ } \mathscr{F} \otimes \mathscr{A}(G) \xrightarrow{\ \mathrm{Id}\otimes\varepsilon\ } \mathscr{F} \otimes \mathcal{O}_S \xrightarrow{\ \sim\ } \mathscr{F} \quad .$$

These axioms (CM 1) and (CM 2) are those of a right *comodule* structure over the bialgebra $\mathscr{A}(G)$. [42]

Put $\mathscr{A} = \mathscr{A}(G)$. If $\mathscr{F}$ and $\mathscr{F}'$ are $\mathscr{A}$-comodules, a morphism of comodules $f : \mathscr{F} \to \mathscr{F}'$ is a morphism of $\mathcal{O}_S$-modules such that the following diagram is

commutative.

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\quad f\quad} & \mathscr{F}' \\ {\scriptstyle\mu_{\mathscr{F}}}\downarrow & & \downarrow{\scriptstyle \mu_{\mathscr{F}'}} \\ \mathscr{F} \otimes \mathscr{A} & \xrightarrow{f\otimes\mathrm{Id}} & \mathscr{F}' \otimes \mathscr{A} \end{array} .$$

This gives the category ($\mathscr{A}$-**Comod**.); we denote by ($\mathscr{A}$-**Comod.q.c.**) its full subcategory consisting of those $\mathscr{A}$-comodules that are quasi-coherent as $\mathcal{O}_S$-modules. We have therefore proved the following statement.

Proposition 4.7.2. *Let G be an S-group affine over S. There are equivalences of categories*

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod}.) \simeq (\mathscr{A}(G)\text{-}\mathbf{Comod}.),$$

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \simeq (\mathscr{A}(G)\text{-}\mathbf{Comod.q.c.}).$$

If, in addition, $S = \mathrm{Spec}(\Lambda)$ is affine and one sets $\Lambda[G] = \Gamma(S, \mathscr{A}(G))$, there is an equivalence of categories

$$(\mathscr{A}(G)\text{-}\mathbf{Comod.q.c.}) \simeq (\Lambda[G]\text{-}\mathbf{Comod}.).$$

[41] Editor's note: see VI_B, Sections 11.1–11.6 for the extension of the results of 4.7.2 to the case where G is not necessarily affine but G and $\mathscr{F}$ are assumed flat over S.

[42] Editor's note: left G-$\mathcal{O}_S$-modules correspond naturally to right $\mathscr{A}(G)$-comodules.

[43]

Suppose, moreover, that $\mathscr{A} = \mathscr{A}(G)$ is a flat $\mathcal{O}_S$-module. [44] Let $\mathscr{E}$ be an $\mathscr{A}$-comodule and $\mathscr{F}$ an $\mathcal{O}_S$-submodule of $\mathscr{E}$. Since $\mathscr{A}$ is flat over $\mathcal{O}_S$, one may identify $\mathscr{F}\otimes\mathscr{A}$ (respectively, $\mathscr{F}\otimes\mathscr{A}\otimes\mathscr{A}$) with an $\mathcal{O}_S$-submodule of $\mathscr{E}\otimes\mathscr{A}$ (respectively, of $\mathscr{E}\otimes\mathscr{A}\otimes\mathscr{A}$). Suppose that $\mu_{\mathscr{E}}$ maps $\mathscr{F}$ into $\mathscr{F}\otimes\mathscr{A}$. Then its restriction

$$\mu_{\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{F}\otimes\mathscr{A}$$

endows $\mathscr{F}$ with an $\mathscr{A}$-comodule structure; one says that $\mathscr{F}$ is a *subcomodule* of $\mathscr{E}$. Passing to the quotient, $\mu_{\mathscr{E}}$ defines a morphism of $\mathcal{O}_S$-modules

$$\mathscr{E}/\mathscr{F} \longrightarrow (\mathscr{E}/\mathscr{F})\otimes\mathscr{A},$$

which endows $\mathscr{E}/\mathscr{F}$ with an $\mathscr{A}$-comodule structure. If $f : \mathscr{E} \to \mathscr{E}'$ is a morphism of $\mathscr{A}$-comodules, then $\operatorname{Ker} f$ (respectively, $\operatorname{Im} f$) is an $\mathscr{A}$-subcomodule of $\mathscr{E}$ (respectively, of $\mathscr{E}'$), and f induces an isomorphism of $\mathscr{A}$-comodules

$$\mathscr{E}/\operatorname{Ker} f \xrightarrow{\sim} \operatorname{Im} f.$$

Moreover, if $\mathscr{E}$ and $\mathscr{E}'$ are quasi-coherent $\mathcal{O}_S$-modules, then so are $\operatorname{Ker} f$ and $\operatorname{Im} f$. Consequently, ($\mathscr{A}$-**Comod**.) and ($\mathscr{A}$-**Comod**.**q**.**c**.) are abelian categories.

Corollary 4.7.2.1. *Suppose that G is affine and flat over S. Then the category (G-$\mathcal{O}_S$-**Mod**.**q**.**c**.) (respectively, (G-$\mathcal{O}_S$-**Mod**.)), equivalent to the category of $\mathscr{A}(G)$-comodules that are quasi-coherent over $\mathcal{O}_S$ (respectively, of all $\mathscr{A}(G)$-comodules), is abelian.*

4.7.3. Suppose now that G is a *diagonalizable group*, that is, that $\mathscr{A}(G)$ is the algebra of a commutative group M over the sheaf of rings $\mathcal{O}_S$. If $\mathscr{F}$ is an $\mathcal{O}_S$-module, then

$$\mathscr{F}\otimes\mathscr{A}(G) = \coprod_{m\in M} \mathscr{F}\otimes m\mathcal{O}_S.$$

Giving a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathscr{F} \longrightarrow \mathscr{F}\otimes\mathscr{A}(G)$$

is therefore equivalent to giving $\mathcal{O}_S$-endomorphisms $(\mu_m)_{m\in M}$ of $\mathscr{F}$ such that, for every section x of $\mathscr{F}$ on an open subset of S, the family $(\mu_m(x))$ is a section of the direct sum $\coprod_{m\in M}\mathscr{F}$. This means that on every sufficiently small open subset, only finitely many restrictions of the $\mu_m(x)$ are nonzero.

For μ defined by

$$\mu(x) = \sum_{m\in M} \mu_m(x)\otimes m,$$

condition (CM 1), respectively (CM 2), holds if and only if

$$\mu_m\circ\mu_{m'} = \delta_{mm'}\mu_m, \qquad \left(\text{respectively, } \sum_{m\in M}\mu_m = \mathrm{Id}_{\mathscr{F}}\right).$$

Thus the μ_m are pairwise orthogonal projectors whose sum is the identity. We have proved the following result.

[43] Editor's note: the preceding sentence has been added.

[44] Editor's note: the following paragraph has been added from [Se68, Section 1.3].

Proposition 4.7.3. *If $G = D_S(M)$ is a diagonalizable S-group, the category of quasi-coherent G-$\mathcal{O}_S$-modules (respectively, of all G-$\mathcal{O}_S$-modules) is equivalent to the category of quasi-coherent $\mathcal{O}_S$-modules (respectively, of all $\mathcal{O}_S$-modules) graded by M.*

Remark. If $\mathscr{F}$ has an $\mathcal{O}_S$-algebra structure preserved by the operations of G, then the grading of $\mathscr{F}$ is an algebra grading. More precisely:

Corollary 4.7.3.1. *The functor $\mathscr{A} \mapsto \mathrm{Spec}(\mathscr{A})$ induces an equivalence between the category of quasi-coherent $\mathcal{O}_S$-algebras graded by M and the category opposite to that of S-schemes affine over S with S-operator group $G = D_S(M)$.*

Proposition 4.7.4. *Let G be a diagonalizable S-group. If*

$$0 \longrightarrow \mathscr{F}_1 \longrightarrow \mathscr{F}_2 \longrightarrow \mathscr{F}_3 \longrightarrow 0$$

is an exact sequence of quasi-coherent G-$\mathcal{O}_S$-modules that splits as a sequence of $\mathcal{O}_S$-modules, then it also splits as a sequence of G-$\mathcal{O}_S$-modules.

Indeed, if $G = D_S(M)$, each $\mathscr{F}_i$ is graded by the $(\mathscr{F}_i)_m$, and for every $m \in M$ the sequence

$$0 \longrightarrow (\mathscr{F}_1)_m \longrightarrow (\mathscr{F}_2)_m \longrightarrow (\mathscr{F}_3)_m \longrightarrow 0$$

of $\mathcal{O}_S$-modules is split. The preceding proposition then gives the result.

5. Group cohomology

5.1. The standard complex

Let $\mathscr{C}$ be a category, $\mathbf{G}$ a $\widehat{\mathscr{C}}$-group, $\mathbf{O}$ a $\widehat{\mathscr{C}}$-ring, and $\mathbf{F}$ a $\mathbf{G}$-$\mathbf{O}$-module.[45] For $n \geq 0$, set

$$C^n(\mathbf{G}, \mathbf{F}) = \mathrm{Hom}(\mathbf{G}^n, \mathbf{F}), \qquad \underline{C}^n(\mathbf{G}, \mathbf{F}) = \underline{\mathrm{Hom}}(\mathbf{G}^n, \mathbf{F}),$$

where $\mathbf{G}^0$ is the final object $\underline{\mathbf{e}}$. Then $\underline{C}^n(\mathbf{G}, \mathbf{F})$ (respectively, $C^n(\mathbf{G}, \mathbf{F})$) is endowed in the evident way with an $\mathbf{O}$-module structure (respectively, a $\Gamma(\mathbf{O})$-module structure), and

$$C^n(\mathbf{G}, \mathbf{F}) \simeq \Gamma(\underline{C}^n(\mathbf{G}, \mathbf{F})), \qquad \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S).$$

To give an element of $C^n(\mathbf{G}, \mathbf{F})$ is to give, for every $S \in \mathrm{Ob}(\mathscr{C})$, an n-cochain of $\mathbf{G}(S)$ with values in $\mathbf{F}(S)$, functorially in S. Recall that the boundary operator

$$\partial : C^n(\mathbf{G}(S), \mathbf{F}(S)) \longrightarrow C^{n+1}(\mathbf{G}(S), \mathbf{F}(S))$$

is given by

$$\begin{aligned} \partial f(g_1, \ldots, g_{n+1}) = {} & g_1 f(g_2, \ldots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \ldots, g_n). \end{aligned}$$

It is functorial in S, and hence defines a homomorphism

$$\partial : C^n(\mathbf{G}, \mathbf{F}) \longrightarrow C^{n+1}(\mathbf{G}, \mathbf{F})$$

[45] Editor's note: This complex is often called the Hochschild complex; see, for example, Section II.3 of [DG70].

such that $\partial \circ \partial = 0$. We have thus defined a complex of abelian groups (indeed, of $\Gamma(\mathbf{O})$-modules), denoted $C^*(\mathbf{G}, \mathbf{F})$. In the same way one defines the complex of $\mathbf{O}$-modules $\underline{C}^*(\mathbf{G}, \mathbf{F})$, and

$$C^*(\mathbf{G}, \mathbf{F}) = \Gamma(\underline{C}^*(\mathbf{G}, \mathbf{F})).$$

We denote by $H^n(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{H}^n(\mathbf{G}, \mathbf{F})$) the groups (respectively, the $\widehat{\mathscr{C}}$-groups) of cohomology of $C^*(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{C}^*(\mathbf{G}, \mathbf{F})$). In particular,

$$\underline{H}^0(\mathbf{G}, \mathbf{F}) = \mathbf{F}^{\mathbf{G}}, \qquad H^0(\mathbf{G}, \mathbf{F}) = \Gamma(\mathbf{F}^{\mathbf{G}}).$$

Remark 5.1.1. The "set-theoretic" description of ∂ given above is convenient for checking that $\partial \circ \partial = 0$. One may also define ∂ in terms of the multiplication $m : \mathbf{G} \times \mathbf{G} \to \mathbf{G}$ and the action $\mu : \mathbf{G} \times \mathbf{F} \to \mathbf{F}$ as follows: for every $f \in C^n(\mathbf{G}, \mathbf{F})$,

$$\partial f = \mu \circ (\mathrm{id}_{\mathbf{G}} \times f) + \sum_{i=1}^{n} (-1)^i f \circ (\mathrm{id}_{\mathbf{G}^{i-1}} \times m \times \mathrm{id}_{\mathbf{G}^{n-i}}) + (-1)^{n+1} f \circ \mathrm{pr}_{[1,n]},$$

where $\mathrm{pr}_{[1,n]}$ denotes the projection of $\mathbf{G}^{n+1} = \mathbf{G}^n \times \mathbf{G}$ onto $\mathbf{G}^n$. Similarly, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $f \in \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S)$,

$$\partial f = \mu_S \circ (\mathrm{id}_{\mathbf{G}_S} \times f) + \sum_{i=1}^{n} (-1)^i f \circ (\mathrm{id}_{\mathbf{G}_S^{i-1}} \times m_S \times \mathrm{id}_{\mathbf{G}_S^{n-i}}) + (-1)^{n+1} f \circ \mathrm{pr}_{[1,n]},$$

where m_S and μ_S are obtained from m and μ by base change. [46]

5.2.

Recall (see Section 3) that ($\mathbf{G}$-$\mathbf{O}$-Mod.) is an abelian category, with its structure defined objectwise. Thus

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

is an exact sequence of $\mathbf{G}$-$\mathbf{O}$-modules if and only if the sequence of abelian groups

$$0 \longrightarrow \mathbf{F}'(S) \longrightarrow \mathbf{F}(S) \longrightarrow \mathbf{F}''(S) \longrightarrow 0$$

is exact for every $S \in \mathrm{Ob}(\mathscr{C})$. [47]

Suppose that $\mathscr{C}$ is small. Then, by 3.2.1, ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects, so the derived functors of the left exact functors $\underline{H}^0$ and H^0 are defined. We shall now show that the functors $\underline{H}^n$ (respectively, H^n) are indeed the derived functors of $\underline{H}^0$ (respectively, H^0). [48]

Definition 5.2.0. For every $\mathbf{O}$-module $\mathbf{P}$, let $E(\mathbf{P})$ denote the object $\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})$ of $\widehat{\mathscr{C}}$, endowed with the following $\mathbf{G}$-$\mathbf{O}$-module structure. For every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})(S) = \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S),$$

and $g \in \mathbf{G}(S)$ and $a \in \mathbf{O}(S)$ act on $\phi \in \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ by

$$(g \cdot \phi)(h) = \phi(hg), \qquad (a \cdot \phi)(h) = a\phi(h),$$

[46] Editor's note: This remark has been added.
[47] Editor's note: The preceding reminder has been added.
[48] Editor's note: The preceding sentence has been added.

for every $h \in \mathbf{G}(S')$, $S' \to S$. Moreover, for every $\phi \in \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$, set

$$\varepsilon(\phi) = \phi(1) \in \mathbf{P}(S),$$

where 1 denotes the unit element of $\mathbf{G}(S)$. This defines a functor

$$E : (\mathbf{O}\text{-Mod.}) \longrightarrow (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$$

and a natural transformation $\varepsilon : E \to \mathrm{Id}$, where Id denotes the identity functor of ($\mathbf{O}$-Mod.). [49]

Remark 5.2.0.1. In what follows, let $\mathbf{G}_1$ and $\mathbf{G}_2$ denote two copies of $\mathbf{G}$. The morphism

$$\mathbf{G}_1 \times E(\mathbf{P}) \longrightarrow E(\mathbf{P}), \qquad (g_1, \phi) \longmapsto (g_2 \longmapsto \phi(g_2 g_1))$$

corresponds, via the isomorphisms

$$\begin{aligned}\mathrm{Hom}\big(\mathbf{G}_1 \times E(\mathbf{P}), E(\mathbf{P})\big) &\simeq \mathrm{Hom}\Big(E(\mathbf{P}), \underline{\mathrm{Hom}}\big(\mathbf{G}_1, \underline{\mathrm{Hom}}(\mathbf{G}_2, \mathbf{P})\big)\Big) \\ &\simeq \mathrm{Hom}\Big(E(\mathbf{P}), \underline{\mathrm{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})\Big),\end{aligned}$$

to the morphism

$$\phi \longmapsto \big((g_2, g_1) \longmapsto \phi(g_2 g_1)\big),$$

that is, to the morphism

$$\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})$$

induced by the multiplication

$$\mu_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}, \qquad (g_2, g_1) \longmapsto g_2 g_1.$$

Lemma 5.2.0.2. *(i) The functor E is right adjoint to the forgetful functor*

$$(\mathbf{G}\text{-}\mathbf{O}\text{-}\textit{Mod.}) \longrightarrow (\mathbf{O}\text{-}\textit{Mod.}).$$

More precisely, $\varepsilon : E \to \mathrm{Id}$ induces, for every $\mathbf{M} \in (\mathbf{G}\text{-}\mathbf{O}\text{-}\textit{Mod.})$ and $\mathbf{P} \in \mathrm{Ob}(\mathbf{O}\text{-}\textit{Mod.})$, a bijection

$$\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}\big(\mathbf{M}, E(\mathbf{P})\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{O}\text{-Mod.}}(\mathbf{M}, \mathbf{P})$$

functorial in $\mathbf{M}$ and $\mathbf{P}$.

(ii) Consequently, if $\mathbf{I}$ is an injective object of ($\mathbf{O}$-Mod.), then $E(\mathbf{I})$ is an injective object of ($\mathbf{G}$-$\mathbf{O}$-Mod.).

Proof. To every $\mathbf{O}$-morphism $f : \mathbf{M} \to \mathbf{P}$, associate the element ϕ_f of $\mathrm{Hom}_{\mathbf{O}}(\mathbf{M}, E(\mathbf{P}))$ defined as follows. For every $S \in \mathrm{Ob}(\mathscr{C})$ and $m \in \mathbf{M}(S)$, the element $\phi_f(m)$ of $\mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $g \in \mathbf{G}(S')$, $S' \to S$, by

$$\phi_f(m)(g) = f(gm) \in \mathbf{P}(S').$$

Then, for every $h \in \mathbf{G}(S)$, one has $\phi_f(hm) = h \cdot f(m)$; that is, ϕ_f is an element of

$$\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}\big(\mathbf{M}, E(\mathbf{P})\big).$$

[49] Editor's note: The original has been modified so as to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

If $\phi \in \mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P}))$, and if for every $m \in \mathbf{M}(S)$ we write $f(m) = \phi(m)(1)$, then

$$\phi_f(m)(g) = f(gm) = \phi(gm)(1) = (g \cdot \phi(m))(1) = \phi(m)(g);$$

thus $\phi_f = \phi$. Conversely, it is clear that $\phi_f(m)(1) = f(m)$. This proves (i), and (ii) follows at once.

Definition 5.2.0.3. Let $\mathbf{M}$ be a $\mathbf{G}$-$\mathbf{O}$-module. The identity map of $\mathbf{M}$, considered as an $\mathbf{O}$-module, corresponds under the adjunction to the morphism of $\mathbf{G}$-$\mathbf{O}$-modules

$$\mu_{\mathbf{M}} : \mathbf{M} \longrightarrow E(\mathbf{M})$$

such that, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $m \in \mathbf{M}(S)$, $\mu_{\mathbf{M}}(m)$ is the morphism $\mathbf{G}_S \to \mathbf{M}_S$ defined as follows: for every $S' \to S$ and $g \in \mathbf{G}(S')$,

$$\mu_{\mathbf{M}}(m)(g) = g \cdot m_{S'} \in \mathbf{M}(S').$$

Notice that $\mu_{\mathbf{M}}$ is a monomorphism. Indeed, $\varepsilon_{\mathbf{M}} : E(\mathbf{M}) \to \mathbf{M}$ is a morphism of $\mathbf{O}$-modules such that $\varepsilon_{\mathbf{M}} \circ \mu_{\mathbf{M}} = \mathrm{id}_{\mathbf{M}}$. Consequently, as an $\mathbf{O}$-module, $\mathbf{M}$ is a direct summand of $E(\mathbf{M})$.

Proposition 5.2.1. *Suppose that $\mathscr{C}$ is small, that finite products exist in $\mathscr{C}$, and that $\mathbf{G}$ is representable. Then the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) are the derived functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$) on the category of $\mathbf{G}$-$\mathbf{O}$-modules.*

By the well-known general results, [50] it suffices to verify that the $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0. Let

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

be an exact sequence of $\mathbf{G}$-$\mathbf{O}$-modules, and let $S \in \mathrm{Ob}(\mathscr{C})$. By hypothesis, $\mathbf{G}$ is represented by an object $G \in \mathrm{Ob}(\mathscr{C})$, and finite products exist in $\mathscr{C}$; in particular, $\mathscr{C}$ has a final object e. Hence every $\mathbf{G}^n \times \mathbf{h}_S$ is represented by $G^n \times S$ (with $G^0 = e$), and the sequence

$$0 \longrightarrow \mathbf{F}'(G^n \times S) \longrightarrow \mathbf{F}(G^n \times S) \longrightarrow \mathbf{F}''(G^n \times S) \longrightarrow 0$$

is exact. This shows that the sequence of $\mathbf{O}$-modules

$$0 \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}') \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}) \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}'') \longrightarrow 0$$

is exact. It follows that $\underline{C}^*(\mathbf{G}, -)$, regarded as a functor on ($\mathbf{G}$-$\mathbf{O}$-Mod.) with values in the category of complexes of ($\mathbf{O}$-Mod.), is exact. Thus the functors $\underline{H}^n(\mathbf{G}, -)$ are cohomological.

Since Γ is exact, the same is true of the functors $H^n(\mathbf{G}, -)$. It remains only to prove the following.

Lemma 5.2.2. *For every $\mathbf{P} \in \mathrm{Ob}(\mathbf{O}\text{-Mod.})$, one has*

$$\underline{H}^n(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \qquad H^n(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \qquad n > 0.$$

It is enough to show that $\underline{C}^*(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$ and $C^*(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$ are homotopically trivial in degrees > 0. It is enough, in fact, to do this for the second complex, since the corresponding result for the first follows by base change. [51] For every $n \geq 0$, define a morphism

$$\sigma : C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})) \longrightarrow C^n(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$$

[50] Editor's note: See [Gr57, 2.2.1] and [Gr57, 2.3]. The original has also been expanded in what follows.
[51] Editor's note: The original has been expanded in what follows.

as follows. Let $f \in C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$. For every $S \in \mathrm{Ob}(\mathscr{C})$ and $g_1, \dots, g_n \in \mathbf{G}(S)$, the element $\sigma(f)(g_1, \dots, g_n)$ of $\mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $S' \to S$ and $x \in \mathbf{G}(S')$, by

$$\sigma(f)(g_1, \dots, g_n)(x) = f(x, g_1, \dots, g_n)(e) \in \mathbf{P}(S'),$$

where e denotes the unit element of $\mathbf{G}(S')$. Then σ is a homotopy operator in degrees > 0. Indeed, for every $g_1, \dots, g_{n+1} \in \mathbf{G}(S)$ and $x \in \mathbf{G}(S')$, on the one hand,

$$\begin{aligned}\partial\sigma(f)(g_1, \dots, g_{n+1})(x) = {} & f(xg_1, g_2, \dots, g_{n+1})(e) \\ & + \sum_{i=1}^{n} (-1)^i f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1})(e) \\ & + (-1)^{n+1} f(x, g_1, \dots, g_n)(e),\end{aligned}$$

and, on the other hand,

$$\begin{aligned}\sigma(\partial f)(g_1, \dots, g_{n+1})(x) = {} & \big(xf(g_1, g_2, \dots, g_{n+1})\big)(e) - f(xg_1, g_2, \dots, g_{n+1})(e) \\ & + \sum_{i=1}^{n} (-1)^{i+1} f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+2} f(x, g_1, \dots, g_n)(e).\end{aligned}$$

Consequently,

$$\big(\partial\sigma(f) + \sigma(\partial f)\big)(g_1, \dots, g_{n+1})(x) = f(g_1, \dots, g_{n+1})(x);$$

that is, $\partial\sigma + \sigma\partial$ is the identity map of $C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$ for every $n \geq 0$.

Remark 5.2.3. The hypothesis that $\mathscr{C}$ be small is used only to ensure the existence of the derived functors $R^n\underline{H}^0$ and R^nH^0. In every case, what precedes shows that the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0, and hence are the (right) satellite functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$), in the sense of [Gr57, 2.2]. If (**G**-**O**-Mod.) has enough injective objects (as it does when $\mathscr{C}$ is small), they coincide with the derived functors (*loc. cit.*, 2.3). [52]

5.3. Cohomology of G-$\mathscr{O}_S$-modules

Let S be a scheme, G an S-group, and $\mathscr{F}$ a quasi-coherent G-$\mathscr{O}_S$-module. The cohomology groups of G with values in $\mathscr{F}$ are defined by

$$H^n(G, \mathscr{F}) = H^n(\mathbf{h}_G, \mathbf{W}(\mathscr{F}))$$

(for the notation, see 4.6).

Suppose that G is affine over S. In view of Proposition 4.6.4, this cohomology is computed as follows: $H^n(G, \mathscr{F})$ is the n-th cohomology group of the complex $C^*(G, \mathscr{F})$, whose n-th term is

$$C^n(G, \mathscr{F}) = \Gamma\Big(S, \mathscr{F} \otimes \underbrace{\mathscr{A}(G) \otimes \cdots \otimes \mathscr{A}(G)}_{n \text{ times}}\Big).$$

If f (respectively, a_i) is a section of $\mathscr{F}$ (respectively, of $\mathscr{A}(G)$) over an open subset of S, then

$$\begin{aligned}\partial(f \otimes a_1 \otimes \cdots \otimes a_n) = {} & \mu_{\mathscr{F}}(f) \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \\ & + \sum_{i=1}^{n} (-1)^i f \otimes a_1 \otimes \cdots \otimes \Delta a_i \otimes \cdots \otimes a_n \\ & + (-1)^{n+1} f \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \otimes 1,\end{aligned}$$

[52] Editor's note: This remark has been added.

where

$$\Delta : \mathscr{A}(G) \longrightarrow \mathscr{A}(G) \otimes \mathscr{A}(G), \qquad \mu_{\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A}(G)$$

describe the coalgebra structure of $\mathscr{A}(G)$ and the comodule structure of $\mathscr{F}$. Notice in passing that the cohomology of G with values in $\mathscr{F}$ therefore depends only on the comodule structure of $\mathscr{F}$, and in particular only on the S-monoid structure of G.

In particular,

$$H^0(G, \mathscr{F}) = \Gamma(S, \mathscr{F}^G),$$

where $\mathscr{F}^G$, the *sheaf of invariants of* $\mathscr{F}$, is the sheaf whose sections over an open subset U of S are those sections of $\mathscr{F}$ over U whose inverse image in every S' over U is invariant under $G(S')$ (see 4.7.1.2).

Theorem 5.3.1. *Let S be an affine scheme and G an S-group that is affine and flat over S. The functors $H^n(G, -)$ are the derived functors of $H^0(G, -)$ on the category of quasi-coherent G-$\mathscr{O}_S$-modules.*

Proof. Since G is affine and flat over S, 4.7.2.1 shows that the category (G-$\mathscr{O}_S$-Mod.q.c.) is equivalent to the category ($\mathscr{A}(G)$-Comod.q.c.) of quasi-coherent $\mathscr{A}(G)$-comodules over $\mathscr{O}_S$, and is therefore abelian. Moreover, since $\mathscr{A}(G)$ is a flat $\mathscr{O}_S$-module, each functor

$$\mathscr{F} \longmapsto \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{A}(G)^{\otimes n}$$

is exact; since S is affine, it follows that $C^*(G, -)$ is an exact functor on (G-$\mathscr{O}_S$-Mod.q.c.).[53]

Let Δ (respectively, η) denote the comultiplication (respectively, the augmentation) of $\mathscr{A}(G)$. For every quasi-coherent $\mathscr{O}_S$-module $\mathscr{P}$, let

$$\mathrm{Ind}(\mathscr{P}) = \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G),$$

endowed with the $\mathscr{A}(G)$-comodule structure defined by

$$\mathrm{id}_{\mathscr{P}} \otimes \Delta : \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \otimes_{\mathscr{O}_S} \mathscr{A}(G).$$

This defines a functor

$$\mathrm{Ind} : (\mathscr{O}_S\text{-Mod.q.c.}) \longrightarrow (G\text{-}\mathscr{O}_S\text{-Mod.q.c.}).$$

It follows from 4.6.4.1, 5.2.0, and 5.2.0.1 that there is an isomorphism of $\mathbf{G}$-$\mathbf{O}_S$-modules

$$\mathbf{W}(\mathrm{Ind}(\mathscr{P})) \simeq E(\mathbf{W}(\mathscr{P})) = \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{W}(\mathscr{P})). \qquad (*)$$

Under this identification, the morphism $\varepsilon : E(\mathbf{W}(\mathscr{P})) \to \mathbf{W}(\mathscr{P})$ corresponds to the morphism of $\mathscr{O}_S$-modules

$$\mathrm{id}_{\mathscr{P}} \otimes \eta : \mathrm{Ind}(\mathscr{P}) \longrightarrow \mathscr{P}.$$

We have already used the fact that the functor

$$\mathbf{W} : (\mathscr{O}_S\text{-Mod.}) \longrightarrow (\mathbf{O}_S\text{-Mod.})$$

is fully faithful. By Definition 4.7.1, the same is true of its restriction to (G-$\mathscr{O}_S$-Mod.): if $\mathscr{M}$ and $\mathscr{M}'$ are G-$\mathscr{O}_S$-modules, there is a functorial isomorphism

$$\mathrm{Hom}_{G\text{-}\mathscr{O}_S\text{-Mod.}}(\mathscr{M}, \mathscr{M}') \simeq \mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}_S\text{-Mod.}}(\mathbf{W}(\mathscr{M}), \mathbf{W}(\mathscr{M}')).$$

[53] Editor's note: The original has been modified to show that the category (G-$\mathscr{O}_S$-Mod.q.c.) is abelian and has enough injective objects. Compare [Ja03], Part I, 3.3–3.4, 3.9, 4.2, and 4.14–4.16 (bearing in mind that "k-group scheme" there means "affine k-group scheme"; see *loc. cit.*, 2.1).

Consequently, Lemma 5.2.0.2 gives the following result.

Corollary 5.3.1.1. *(i) The functor* Ind *is right adjoint to the forgetful functor*

$$(G\text{-}\mathscr{O}_S\text{-Mod.q.c.}) \longrightarrow (\mathscr{O}_S\text{-Mod.q.c.}).$$

More precisely, the map $\mathrm{id}_{\mathscr{P}} \otimes \eta : \mathrm{Ind}(\mathscr{P}) \to \mathscr{P}$ *induces, for every object* $\mathscr{M}$ *of* $(G\text{-}\mathscr{O}_S\text{-Mod.q.c.})$*, a bijection*

$$\mathrm{Hom}_{G\text{-}\mathscr{O}_S\text{-Mod.}}(\mathscr{M}, \mathrm{Ind}(\mathscr{P})) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_S}(\mathscr{M}, \mathscr{P}).$$

(ii) Consequently, if $\mathscr{I}$ *is an injective object of* $(\mathscr{O}_S\text{-Mod.q.c.})$*, then* $\mathrm{Ind}(\mathscr{I})$ *is an injective object of* $(G\text{-}\mathscr{O}_S\text{-Mod.q.c.})$*.*

Let $\mathscr{F}$ be a G-$\mathscr{O}_S$-module, and let $\mu_{\mathscr{F}} : \mathscr{F} \to \mathrm{Ind}(\mathscr{F})$ be the map defining its $\mathscr{A}(G)$-comodule structure. It follows from 5.2.0.3 (or directly from axioms (CM 1) and (CM 2) of 4.7.2) that $\mu_{\mathscr{F}}$ is a morphism of G-$\mathscr{O}_S$-modules, and that

$$(\mathrm{id}_{\mathscr{F}} \otimes \eta) \circ \mu_{\mathscr{F}} = \mathrm{id}_{\mathscr{F}}\,.$$

Thus $\mathscr{F}$ is a direct summand of $\mathrm{Ind}(\mathscr{F})$ as an $\mathscr{O}_S$-module; in particular, $\mu_{\mathscr{F}}$ is a monomorphism. By (*) and 5.2.2,

$$H^n(\mathbf{G}, \mathbf{W}(\mathrm{Ind}(\mathscr{F}))) \simeq H^n\Big(\mathbf{G}, \underline{\mathrm{Hom}}_S(\mathbf{G}, \mathbf{W}(\mathscr{F}))\Big) = 0 \qquad (n > 0).$$

Therefore $H^n(G, -)$ is effaceable for $n > 0$.

Finally, since S is affine, $(\mathscr{O}_S$-Mod.q.c.) has enough injective objects. Let $\mathscr{F} \hookrightarrow \mathscr{I}$ be a monomorphism of $\mathscr{O}_S$-modules, where $\mathscr{I}$ is injective in $(\mathscr{O}_S$-Mod.q.c.). Since $\mathscr{A}(G)$ is flat over $\mathscr{O}_S$, $\mathrm{Ind}(\mathscr{F})$ is a G-$\mathscr{O}_S$-submodule of $\mathrm{Ind}(\mathscr{I})$. Hence:

Corollary 5.3.1.2. *The abelian category* $(G\text{-}\mathscr{O}_S\text{-Mod.q.c.})$ *has enough injective objects.*

Taking account of [Gr57, 2.2.1 and 2.3], already used in the proof of 5.2.1, completes the proof of Theorem 5.3.1.

Remark 5.3.1.3. Corollary 5.3.1.1 may also be proved by the following calculation. To every morphism of G-$\mathscr{O}_S$-modules

$$\phi : \mathscr{M} \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G)$$

associate the $\mathscr{O}_S$-morphism

$$(\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \phi : \mathscr{M} \longrightarrow \mathscr{P}.$$

Conversely, to every $\mathscr{O}_S$-morphism $f : \mathscr{M} \to \mathscr{P}$, associate the morphism of G-$\mathscr{O}_S$-modules

$$(f \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} : \mathscr{M} \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G).$$

One sees at once that

$$(\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ (f \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} = (f \otimes \mathrm{id}_{\mathscr{O}_S}) \circ (\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \mu_{\mathscr{M}} = f.$$

On the other hand, for every ϕ, the following diagram is commutative.

$$\begin{array}{ccc} \mathscr{M} & \xrightarrow{\quad\phi\quad} & \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \\ {\scriptstyle \mu_{\mathscr{M}}}\downarrow & & \downarrow{\scriptstyle \mathrm{Id}_{\mathscr{P}} \otimes \Delta} \\ \mathscr{M} \otimes_{\mathscr{O}_S} \mathscr{A}(G) & \xrightarrow{\phi \otimes \mathrm{Id}_{\mathscr{A}(G)}} & \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \otimes_{\mathscr{O}_S} \mathscr{A}(G) \end{array} \quad .$$

It follows that

$$\begin{aligned}\Big(((\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \phi) \otimes \mathrm{id}_{\mathscr{A}(G)}\Big) \circ \mu_{\mathscr{M}} \\ = (\mathrm{id}_{\mathscr{P}} \otimes \eta \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ (\phi \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} \\ = (\mathrm{id}_{\mathscr{P}} \otimes \eta \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ (\mathrm{id}_{\mathscr{P}} \otimes \Delta) \circ \phi = \phi.\end{aligned}$$

This proves Corollary 5.3.1.1(i), and (ii) follows.

Let $\mathscr{F}$ be a G-$\mathscr{O}_S$-module. We saw above that axiom (CM 2) of 4.7.2 shows that, as an $\mathscr{O}_S$-module, $\mathscr{F}$ is a direct summand of $E(\mathscr{F})$. This gives:

Proposition 5.3.2. *Let S be an affine scheme and G an S-group that is affine and flat. Suppose that every exact sequence*

$$0 \longrightarrow \mathscr{F}_1 \longrightarrow \mathscr{F}_2 \longrightarrow \mathscr{F}_3 \longrightarrow 0$$

of quasi-coherent G-$\mathscr{O}_S$-modules which splits as a sequence of $\mathscr{O}_S$-modules also splits as a sequence of G-$\mathscr{O}_S$-modules. Then the functors $H^n(G, -)$ vanish for $n > 0$ (or, equivalently, the functor $H^0(G, -)$ is exact).

Indeed, by hypothesis, the sequence of G-$\mathscr{O}_S$-modules

$$0 \longrightarrow \mathscr{F} \longrightarrow E(\mathscr{F}) \longrightarrow E(\mathscr{F})/\mathscr{F} \longrightarrow 0$$

is split. Thus $\mathscr{F}$ is a direct summand of $E(\mathscr{F})$ as a G-$\mathscr{O}_S$-module, while the cohomology of the latter vanishes.

Combining this with Proposition 4.7.4 immediately gives:

Theorem 5.3.3. *Let S be an affine scheme and G a diagonalizable S-group. For every quasi-coherent G-$\mathscr{O}_S$-module $\mathscr{F}$, one has*

$$H^n(G, \mathscr{F}) = 0 \qquad (n > 0).$$

Remark. Proposition 5.3.2 remains valid when G is not necessarily flat over S; the proof then uses relative cohomology. [54]

6. G-equivariant objects and modules

Let $\mathscr{C}$ be a category having a final object and in which fiber products exist.[55] Let G be a $\widehat{\mathscr{C}}$-group, $\pi : M \to X$ a morphism in $\widehat{\mathscr{C}}$, and $\lambda = \lambda_X : G \times X \to X$ an action of G on X. In what follows, $Y \times_f M$ denotes the fiber product of $\pi : M \to X$ and an X-functor $f : Y \to X$.

For every $U \in \mathrm{Ob}(\mathscr{C})$ and $x \in X(U)$, write $M_x = U \times_x M$; thus, for every $\phi : U' \to U$,

$$M_x(U') = \{m \in M(U') \mid \pi(m) = x_{U'} = \phi^*(x)\}.$$

Finally, if $g \in G(U)$, we also write gx for the element $\lambda(g, x)$ of $X(U)$.

Definition 6.1. *(a) We say that M is a G-equivariant X-object if an action $\Lambda : G \times M \to M$ of G on M, lifting λ, has been given; that is, if the following square commutes.*

[54] Editor's note: The editors have not attempted to develop this remark.

[55] Editor's note: This section has been added.

$$\begin{array}{ccc} G \times M & \xrightarrow{\Lambda} & M \\ {\scriptstyle \mathrm{id}_G \times \pi}\downarrow & & \downarrow{\scriptstyle \pi} \\ G \times X & \xrightarrow{\lambda} & X \end{array}$$

Equivalently, for every morphism $(g, x) : U \to G \times X$, one is given maps

$$\Lambda_x^U(g) : M_x(U) \longrightarrow M_{gx}(U), \qquad m \longmapsto g \cdot m,$$

which satisfy $1 \cdot m = m$ and $g \cdot (h \cdot m) = (gh) \cdot m$, and which are functorial in the $(G \times X)$-object U. Equivalently again, one is given morphisms of U-objects

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}$$

satisfying $\Lambda_x(1) = \mathrm{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$.

(b) Now let $\mathbf{O}$ be a $\widehat{\mathscr{C}}$-ring and put $\mathbf{O}_X = \mathbf{O} \times X$. Under the hypotheses of (a), we say that M is a G-equivariant $\mathbf{O}_X$-module if it is an $\mathbf{O}_X$-module (cf. Definition 4.3.3.1, which is valid for every ring functor on a category $\mathscr{C}$) and if the action Λ is compatible with the $\mathbf{O}_X$-module structure on M. Thus, for every $(g, x) \in G(U) \times X(U)$, $U \in \mathrm{Ob}(\mathscr{C})$, the map

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \qquad m \longmapsto g \cdot m,$$

is a morphism of $\mathbf{O}_U$-modules.

Remark 6.2. (i) In (a), the conditions $\Lambda_x(1) = \mathrm{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ plainly imply that each $\Lambda_x(g)$ is an isomorphism, with inverse $\Lambda_{gx}(g^{-1})$. Conversely, if each $\Lambda_x(g)$ is an isomorphism, applying the second condition with $h = 1$ gives $\Lambda_x(1) = \mathrm{id}$.

(ii) Let M be an $\mathbf{O}_X$-module. Giving a morphism $\Lambda : G \times M \to M$ which makes the diagram of Definition 6.1 commute and for which every morphism

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \qquad m \longmapsto g \cdot m,$$

is an isomorphism of $\mathbf{O}_U$-modules is equivalent to giving an isomorphism of $\mathbf{O}_{G \times X}$-modules

$$\theta : G \times M = (G \times X) \times_{\mathrm{pr}_X} M \xrightarrow{\sim} (G \times X) \times_{\lambda} M, \qquad (g, x, m) \longmapsto (g, x, g \cdot m).$$

Since each $\Lambda_x(g)$ is assumed to be an isomorphism, the equality $\Lambda_x(1) = \mathrm{id}$ follows from $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ with $h = 1$. Consequently, Λ is an action of G on M, that is, $g \cdot (h \cdot m) = (gh) \cdot m$, if and only if the following diagram of $(G \times G \times X)$-isomorphisms commutes. Here μ denotes multiplication in G, and $f^*(\theta)$ is obtained from θ by base change along $f : G \times G \times X \to G \times X$.

$$\begin{array}{ccc} (G \times G \times X) \underset{\mathrm{pr}_X \circ \mathrm{pr}_{2,3}}{\times} M & \xrightarrow[\sim]{\mathrm{pr}_{2,3}^*(\theta)} & (G \times G \times X) \underset{\lambda \circ \mathrm{pr}_{2,3}}{\times} M \\ \| & & \| \\ (G \times G \times X) \underset{\mathrm{pr}_X \circ (\mu \times \mathrm{id}_X)}{\times} M & & (G \times G \times X) \underset{\mathrm{pr}_X \circ (\mathrm{id}_G \times \lambda)}{\times} M \\ {\scriptstyle (\mu \times \mathrm{id}_X)^*(\theta)}\downarrow \sim & & \sim \downarrow{\scriptstyle (\mathrm{id}_G \times \lambda)^*(\theta)} \\ (G \times G \times X) \underset{\lambda \circ (\mu \times \mathrm{id}_X)}{\times} M & = & (G \times G \times X) \underset{\lambda \circ (\mathrm{id}_G \times \lambda)}{\times} M \end{array} .$$

Thus, giving a G-equivariant $\mathbf{O}_X$-module structure on M is equivalent to giving an isomorphism θ of $\mathbf{O}_{G\times X}$-modules as above for which this diagram commutes.

(iii) Everything above extends to the case in which G is merely a $\widehat{\mathscr{C}}$-monoid. In that case, giving an action $\Lambda : G \times M \to M$ lifting λ, such that every $\Lambda_x(g) : M_x \to M_{gx}$ is a morphism of $\mathbf{O}_U$-modules, is equivalent to giving a morphism θ of $\mathbf{O}_{G\times X}$-modules as in (ii), for which the diagram above commutes without the isomorphism signs under the arrows and which satisfies

$$p_M \circ \theta \circ (\varepsilon_G \times \mathrm{id}_M) = \mathrm{id}_M,$$

where ε_G is the unit section of G and p_M is the projection onto M.

6.3. G-equivariant morphisms

Let Y be a second object of $\widehat{\mathscr{C}}$, endowed with an action $\lambda_Y : G \times Y \to Y$, and let N be a second G-equivariant $\mathbf{O}_X$-module. A morphism $f : Y \to X$ in $\widehat{\mathscr{C}}$ (respectively, a morphism of $\mathbf{O}_X$-modules $\phi : M \to N$) is called *G-equivariant* if it commutes with the action of G. In set-theoretic notation, this says

$$f(g \cdot y) = g \cdot f(y) \qquad (\text{respectively, } \phi(g \cdot m) = g \cdot \phi(m)),$$

or, equivalently,

$$f \circ \lambda_Y = \lambda_X \circ (\mathrm{id}_G \times f), \qquad \phi \circ \Lambda_M = \Lambda_N \circ (\mathrm{id}_G \times \phi).$$

The following lemma is immediate.

Lemma 6.3.1. *Let $f : Y \to X$ be a G-equivariant morphism and let M be a G-equivariant $\mathbf{O}_X$-module. Then the inverse image $f^*(M) = Y \times_f M$ is a G-equivariant $\mathbf{O}_Y$-module.*

On the other hand, G acts on $\mathrm{Hom}_{\widehat{\mathscr{C}}}(Y, X)$. Indeed, let $T \in \mathrm{Ob}(\mathscr{C})$, $g \in G(T)$, and let ϕ be a T-morphism $Y_T \to X_T$. Then g defines automorphisms $\lambda_Y(g)$ and $\lambda_X(g)$ of Y_T and X_T, and we write $g \cdot \phi$, or $g\phi g^{-1}$, for the morphism

$$\lambda_X(g) \circ \phi \circ \lambda_Y(g^{-1}).$$

This defines an action of $G(T)$ on $\mathrm{Hom}_T(Y_T, X_T)$, functorially in T.

Definition 6.3.2. *If $\phi : Y \to X$ is an arbitrary morphism in $\widehat{\mathscr{C}}$, we may therefore consider its stabilizer* $\mathrm{Stab}_G(\phi)$ *(cf. 2.3.3.1). For every $T \in \mathrm{Ob}(\mathscr{C})$, $\mathrm{Stab}_G(\phi)(T)$ is the subgroup of $G(T)$ formed by those g for which $g \circ \phi_T = \phi_T \circ g$, that is, for which the following diagram commutes.*

$$\begin{array}{ccc} Y_T & \xrightarrow{\phi_T} & X_T \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g} \\ Y_T & \xrightarrow{\phi_T} & X_T \end{array}$$

Then $\phi : Y \to X$ is equivariant under $\mathrm{Stab}_G(\phi)$.

6.4. Global sections

Let M be a G-equivariant $\mathbf{O}_X$-module. Let S_0 denote the final object of $\mathscr{C}$, and denote by $\prod_{X/S_0} M$ the *functor of sections of M over X* (cf. Exposé II, 1.1): to every $T \in \mathrm{Ob}(\mathscr{C})$ it assigns

$$\mathrm{Hom}_X(X_T, M) = \mathrm{Hom}_{X_T}(X_T, M_T) = \Gamma(M_T/X_T).$$

Recall also that every morphism $g : Z \to Y$ of $\widehat{\mathscr{C}}$-objects over X induces a morphism of abelian groups

$$M(g) : M(Y) \longrightarrow M(Z),$$

compatible with the ring homomorphism $g^* : \mathbf{O}(Y) \to \mathbf{O}(Z)$. In particular, when $Z = Y$, so that g is an X-endomorphism of Y, one obtains a morphism of abelian groups

$$M(g) : M(Y) \longrightarrow M(Y)$$

which is not generally a morphism of $\mathbf{O}(Y)$-modules, but which satisfies

$$M(g)(\alpha \cdot m) = g^*(\alpha) \cdot M(g)(m)$$

for every $m \in M(Y)$ and $\alpha \in \mathbf{O}(Y)$. In what follows, we simply write g^* instead of $M(g)$.

Let $T \in \mathrm{Ob}(\mathscr{C})$, put $X_T = X \times T$, and let $\mathrm{pr}_X : X_T \to X$ be the projection. For $\alpha \in \mathbf{O}(X_T)$ and $g \in G(T)$, set

$$g(\alpha) = (g^{-1})^*(\alpha).$$

Set-theoretically, this element of $\mathbf{O}(X_T)$ is defined, for every $S \in \mathrm{Ob}(\mathscr{C})$, $x \in X(S)$, and $t \in T(S)$, by

$$g(\alpha)(x, t) = \alpha(g^{-1}x, t).$$

This gives a left action of $G(T)$ by ring automorphisms on $\mathbf{O}(X_T)$, functorially in T, and $g(\alpha) = \alpha$ when α is the image in $\mathbf{O}(X_T)$ of an element of $\mathbf{O}(T)$.

Let ϕ now be the identity morphism of X_T (cf. 6.3 and the generalization below to a morphism $\phi : Y \to X$). Denote $\mathrm{Hom}_X(X_T, M)$ by $M(\phi g^{-1})$ or $M(\phi)$ according as X_T is regarded as an X-object through $\mathrm{pr}_X \circ \lambda(g^{-1})_T$ or through pr_X, respectively.

Then $\lambda(g^{-1})_T$ is an X-morphism between these two X-objects. Hence there is a morphism of abelian groups

$$(g^{-1})^* : M(\phi) \longrightarrow M(\phi g^{-1}), \qquad m \longmapsto m \circ \lambda(g^{-1})_T,$$

and

$$(g^{-1})^*(\alpha \cdot m) = (g\alpha) \cdot (g^{-1})^*(m)$$

for $m \in M(\phi)$ and $\alpha \in \mathbf{O}(X_T)$. If m is a section of M_T over X_T, then $(g^{-1})^*(m)$ is the section defined set-theoretically by $(x, t) \mapsto m(g^{-1}x, t)$. In particular, $(g^{-1})^*$ is a morphism of $\mathbf{O}(T)$-modules.

The functoriality of the $\mathbf{O}(X_T)$-module morphisms $\Lambda_x(g)$ gives the following commutative diagram.

$$\begin{array}{ccc} M(\phi) & \xrightarrow{(g^{-1})^*} & M(\phi g^{-1}) \\ \Big\downarrow{\scriptstyle \Lambda_\phi(g)} & & \Big\downarrow{\scriptstyle \Lambda_{\phi g^{-1}}(g)} \\ M(g\phi) & \xrightarrow{(g^{-1})^*} & M(g\phi g^{-1}) \end{array}$$

Moreover, $g\phi g^{-1} = \phi$, because ϕ is the identity. Writing

$$A(g) = (g^{-1})^* \circ \Lambda_\phi(g),$$

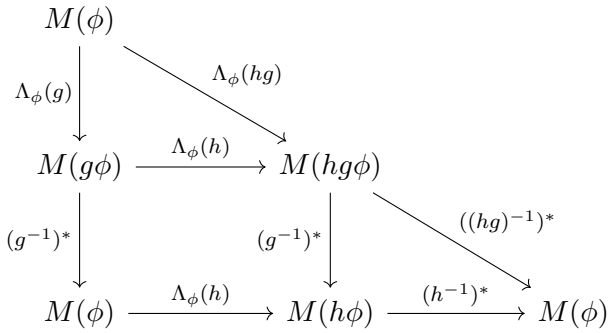
we obtain a morphism of abelian groups

$$A(g) : M(\phi) \longrightarrow M(\phi)$$

which is compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$, namely

$$A(g)(\alpha \cdot m) = (g\alpha) \cdot A(g)(m).$$

Finally, if h is a second element of $G(T)$, functoriality of M and of the morphisms $\Lambda_x(g)$ gives the following commutative diagram.



It follows that $A(h) \circ A(g) = A(hg)$, and hence:

Proposition 6.4.1. *For every T, the $\mathbf{O}(X_T)$-module*

$$\left(\prod_{X/S_0} M\right)(T) = \Gamma(M_T/X_T)$$

is endowed, functorially in T, with an action of $G(T)$ compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$; that is,

$$A(g)(\alpha \cdot m) = g(\alpha) \cdot A(g)(m).$$

Since $g(\alpha) = \alpha$ for $\alpha \in \mathbf{O}(T)$, it follows in particular that $\prod_{X/S_0} M$ is a G-$\mathbf{O}_{S_0}$-module.

More generally, let Y be a second G-equivariant $\widehat{\mathscr{C}}$-object, let $\phi : Y \to X$ be a morphism in $\widehat{\mathscr{C}}$, and put $H = \mathrm{Stab}_G(\phi)$, as in 6.3. Then

$$M_\phi = Y \times_\phi M$$

is an H-equivariant $\mathbf{O}_Y$-module. We obtain:

Corollary 6.4.2. *The functor $\prod_{Y/S_0} M_\phi$ is a $\mathrm{Stab}_G(\phi)$-$\mathbf{O}_{S_0}$-module.*

6.5. G-equivariant $\mathcal{O}_X$-modules

Apply the preceding discussion in the following situation. Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be an $\mathcal{O}_X$-module, not necessarily quasi-coherent.

Definition 6.5.1. *We say that $\mathcal{F}$ is a G-equivariant $\mathcal{O}_X$-module if the $\mathbf{O}_X$-module $M = W(\mathcal{F})$ is G-equivariant. Thus, for every morphism $(g, x) : U \to G \times_S X$, one is given isomorphisms of $\mathcal{O}_U$-modules*

$$\Lambda_x(g) : W\big(x^*(\mathcal{F})\big) \xrightarrow{\sim} W\big((gx)^*(\mathcal{F})\big),$$

functorial in U and satisfying $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$. Because W is fully faithful (cf. 4.6.1.1 and 4.6.2), these yield isomorphisms of $\mathcal{O}_U$-modules

$$\Lambda_x(g) : x^*(\mathcal{F}) \xrightarrow{\sim} (gx)^*(\mathcal{F}),$$

where gx denotes $\lambda \circ (g, x) : U \to X$. Applying this to the identity morphism of $G \times_S X$ gives an isomorphism of $\mathcal{O}_{G\times_S X}$-modules

$$(\star) \qquad \theta : \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

such that the following diagram $(\star\star)$ of morphisms of sheaves on $G \times_S G \times_S X$ commutes.

$$\begin{array}{ccccc}
\mathrm{pr}_{2,3}^* \circ \mathrm{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{\mathrm{pr}_{2,3}^*(\theta)} & \mathrm{pr}_{2,3}^* \circ \lambda^*(\mathcal{F}) & = & (\mathrm{id}_G \times\lambda)^* \circ \mathrm{pr}_X^*(\mathcal{F}) \\
\| & & & & \sim \downarrow (\mathrm{id}_G \times\lambda)^*(\theta) \\
(\mu \times \mathrm{id}_X)^* \circ \mathrm{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{(\mu\times \mathrm{id}_X)^*(\theta)} & (\mu \times \mathrm{id}_X)^* \circ \lambda^*(\mathcal{F}) & = & (\mathrm{id}_G \times\lambda)^* \circ \lambda^*(\mathcal{F})
\end{array} .$$

If $\mathcal{E}$ is a second G-equivariant $\mathcal{O}_X$-module, a morphism $\phi : \mathcal{E} \to \mathcal{F}$ of $\mathcal{O}_X$-modules is called *G-equivariant* if $W(\phi) : W(\mathcal{E}) \to W(\mathcal{F})$ is G-equivariant. With the notation above, this is equivalent to

$$\theta_{\mathcal{F}} \circ \mathrm{pr}_X^*(\phi) = \lambda^*(\phi) \circ \theta_{\mathcal{E}}.$$

Remark 6.5.2. If $f : Y \to X$ is G-equivariant, Lemma 6.3.1 shows that $f^*(\mathcal{F})$ is a G-equivariant $\mathcal{O}_Y$-module.

Remark 6.5.3. Suppose in addition that $\mathcal{F}$ is quasi-coherent. Giving a G-equivariant $\mathcal{O}_X$-module structure on $M = W(\mathcal{F})$ is then equivalent to giving such a structure on $V(\mathcal{F})$, and this in turn is equivalent to giving an action of G on the vector fibration $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

Indeed, let

$$\varphi : \lambda^*(\mathcal{F}) \xrightarrow{\sim} \mathrm{pr}_X^*(\mathcal{F})$$

be the inverse of θ. For every morphism $(g, x) : U \to G \times_S X$, the isomorphism of $\mathcal{O}_U$-modules

$$\varphi_x(g) = \Lambda_x(g)^{-1} = \Lambda_{gx}(g^{-1})$$

from $(gx)^*(\mathcal{F})$ to $x^*(\mathcal{F})$ induces an isomorphism of $\mathcal{O}_U$-modules

$$V\big((gx)^*(\mathcal{F})\big) \xrightarrow{\sim} V\big(x^*(\mathcal{F})\big).$$

Denote the latter by ${}^t\varphi_x(g)$. Then

$${}^t\varphi_{gx}(h) \circ {}^t\varphi_x(g) = {}^t\big(\Lambda_{gx}(h) \circ \varphi_x(g)\big)^{-1} = \Lambda_x(hg)^{-1} = {}^t\varphi_x(hg).$$

For every X-scheme $f : Y \to X$, one has

$$V(\mathcal{F}) \times_f Y \simeq V\big(f^*(\mathcal{F})\big) \qquad \text{(cf. 4.6.2)}.$$

Therefore the isomorphism

$$\begin{aligned} {}^t\varphi : \quad G \times_S V(\mathcal{F}) &= (G \times_S X) \times_{\mathrm{pr}_X} V(\mathcal{F}) = V\big(\mathrm{pr}_X^*(\mathcal{F})\big) \\ &\xrightarrow{\sim} V\big(\lambda^*(\mathcal{F})\big) = (G \times_S X) \times_\lambda V(\mathcal{F}) \end{aligned}$$

endows $V(\mathcal{F})$ with a G-equivariant $\mathcal{O}_X$-module structure. Identifying every functor $V(-)$ with the vector fibration which represents it, and composing ${}^t\varphi$ with the projection onto $V(\mathcal{F})$, gives an action of G on $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

This recovers the definition in [GIT, Chapter 1, Section 3], except that Mumford there considers a locally free $\mathcal{O}_X$-module $\mathcal{E}$ of finite rank and defines an action of G on $V(\mathcal{E}) = W(\mathcal{E}^\vee)$. Indeed, the preceding diagram $(\star\star)$, with θ replaced by φ and the directions of the arrows reversed, is exactly the diagram in *loc. cit.*, Definition 1.6, and the isomorphism ${}^t\varphi$ above coincides with the isomorphism Φ of *loc. cit.*, p. 31.

Remark 6.5.4. Consider in particular the case $X = S$, with the trivial action of G. A G-equivariant $\mathcal{O}_S$-module $\mathcal{F}$ is then the same thing as a G-$\mathcal{O}_S$-module (cf. 4.7.1). If $f : G \to S$ denotes the morphism which here equals both pr_X and λ, the isomorphism

$$(\star) \qquad \theta : f^*(\mathcal{F}) \xrightarrow{\sim} f^*(\mathcal{F})$$

is an element of

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_G}\big(f^*(\mathcal{F}), f^*(\mathcal{F})\big) &= \mathrm{Hom}_{\mathcal{O}_G}\big(W(f^*(\mathcal{F})), W(f^*(\mathcal{F}))\big)\\ &= \mathrm{End}_{\mathcal{O}_S}\big(W(\mathcal{F})\big)(G).\end{aligned}$$

It is precisely the morphism

$$\rho : G \longrightarrow \mathrm{End}_{\mathcal{O}_S}\big(W(\mathcal{F})\big)$$

which defines the action of G on $W(\mathcal{F})$. Moreover, θ corresponds by adjunction to the morphism of $\mathcal{O}_S$-modules

$$f_*(\theta) \circ \tau : \mathcal{F} \longrightarrow f_*f^*(\mathcal{F}),$$

where $\tau : \mathcal{F} \to f_*f^*(\mathcal{F})$ is the unit morphism of the adjunction. This will be used in Exposé $\mathrm{VI_B}$, 11.10 bis.

6.6. The functors $\prod_{X/S} W(\mathcal{F})$ and $W(p_*(\mathcal{F}))$

Let S be a scheme, let G be an S-group scheme acting on S-schemes X and Y through

$$\lambda : G \times_S X \to X, \qquad \mu : G \times_S Y \to Y,$$

and let $p : X \to Y$ be a G-equivariant morphism. Suppose that $p : X \to Y$ and $\nu_Y : Y \to S$ are quasi-compact and quasi-separated and that $\pi : G \to S$ is flat.

The projection $\mathrm{pr}_Y : G \times_S Y \to Y$ is then flat, as is μ, since μ is the composite of pr_Y and the automorphism $(g, y) \mapsto (g, gy)$. For a variable S-scheme $f : T \to S$, denote by $p_T : X_T \to T$ and $f_X : X_T \to X$ the morphisms obtained from p and f by base change.

Let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module, and let

$$\theta : \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

be the isomorphism of 6.5.1 $(\star)$. Since p is quasi-compact and quasi-separated, $p_*(\mathcal{F})$ is quasi-coherent (cf. EGA I, 9.2.1).

Lemma 6.6.1. *The quasi-coherent $\mathcal{O}_Y$-module $p_*(\mathcal{F})$ is G-equivariant.*

Indeed, one has the following two cartesian squares.

$$\begin{array}{ccccc} G \times_S X & \xrightarrow{\mathrm{pr}_X} & X & \xleftarrow{\lambda} & G \times_S X \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ G \times_S Y & \xrightarrow{\mathrm{pr}_Y} & Y & \xleftarrow{\mu} & G \times_S Y \end{array} .$$

Since p is quasi-compact and quasi-separated and pr_Y and μ are flat, EGA III, 1.4.15, completed by EGA IV_1, 1.7.21, gives

$$q_*\mathrm{pr}_X^*(\mathcal{F}) = \mathrm{pr}_Y^* p_*(\mathcal{F}), \qquad q_*\lambda^*(\mathcal{F}) = \mu^* p_*(\mathcal{F}).$$

Consequently, θ induces an isomorphism

$$\theta_Y : \mathrm{pr}_Y^* p_*(\mathcal{F}) \xrightarrow{\sim} \mu^* p_*(\mathcal{F}).$$

The same argument shows that θ_Y satisfies the cocycle condition 6.5.1 $(\star\star)$, because the morphisms $G \times_S G \times_S Y \to G \times_S Y$ involved are flat. This proves the lemma.

In particular, take $Y = S$ with the trivial action of G. By Remark 6.5.4, $p_*(\mathcal{F})$ is then a G-$\mathcal{O}_S$-module. If, in addition, G is affine over S, and if

$$\mathscr{A}(G) = \pi_*(\mathcal{O}_G),$$

then $p_*(\mathcal{F})$ is an $\mathscr{A}(G)$-comodule, by 4.7.2.

On the other hand, by Proposition 6.4.1, the functor $\prod_{X/S} W(\mathcal{F})$, which assigns to every $f : T \to S$

$$W(f_X^*(\mathcal{F}))(X_T) = \Gamma(X_T, f_X^*(\mathcal{F})) = \Gamma(T, p_{T*} f_X^*(\mathcal{F})),$$

is a G-$\mathcal{O}_S$-module. There is moreover a canonical morphism

$$\tau : W(p_*(\mathcal{F})) \longrightarrow \prod_{X/S} W(\mathcal{F}),$$

which, for each $f : T \to S$, is given by the canonical morphism

$$\Gamma(T, f^* p_*(\mathcal{F})) \longrightarrow \Gamma(T, p_{T*} f_X^*(\mathcal{F})).$$

It is an isomorphism when restricted to the full subcategory of schemes T flat over S, and it is immediate that τ is a morphism of G-$\mathcal{O}_S$-modules. We have therefore proved the following proposition; for part (ii), compare [GIT], p. 32.

Proposition 6.6.2. *Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Suppose that $\pi : G \to S$ is flat and $p : X \to S$ is quasi-compact and quasi-separated.*

(i) Then $p_*(\mathcal{F})$ is a quasi-coherent G-$\mathcal{O}_S$-module. The canonical morphism

$$W(p_*(\mathcal{F})) \longrightarrow \prod_{X/S} W(\mathcal{F})$$

is a morphism of G-$\mathcal{O}_S$-modules, and the two functors coincide on the category of flat S-schemes.

(ii) If, in addition, G is affine over S, and if $\mathscr{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ has the structure of an $\mathscr{A}(G)$-comodule.[56]

[56] Editor's note: The same holds if G is flat, quasi-compact, and quasi-separated over S, and if $p_*(\mathcal{F})$ is a flat $\mathcal{O}_S$-module; cf. Exposé $\mathrm{VI_B}$, 11.6.1(ii).

6.7. Stabilizers

Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Let Y be a second S-scheme with an action of G, possibly the trivial action, let $\phi : Y \to X$ be an S-morphism, not necessarily G-equivariant, and let $\mathrm{Stab}_G(\phi)$ be the stabilizer of ϕ in G (cf. 6.3.2).

Let us point out at once (see Exposé $\mathrm{VI_B}$, Section 6) that $\mathrm{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S (cf. *loc. cit.*, Definition 6.2.1). Indeed, consider

$$r : G \times_S Y \longrightarrow X \times_S X, \qquad r(g, y) = (\phi(y), g\phi(g^{-1}y)),$$

put $P = G \times_S Y$, and let P' be the inverse image under r of the diagonal $\Delta_{X/S}$. Then (cf. *loc. cit.*, 6.2.4(a))

$$\mathrm{Stab}_G(\phi) = \prod_{P/G} P'.$$

It follows that $\mathrm{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S. The latter condition is automatic if S is the spectrum of a field or if $Y = S$. Under these hypotheses, $\phi^*(\mathcal{F})$ is a quasi-coherent H-equivariant $\mathcal{O}_Y$-module (cf. 6.5.2).

If, in addition, $\pi : G \to S$ and $p : Y \to S$ are quasi-compact and quasi-separated over S, and π is flat, then $p_*\phi^*(\mathcal{F})$ is an H-$\mathcal{O}_S$-module by Proposition 6.6.2. In particular:

Corollary 6.7.1. *Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Suppose that $\pi : G \to S$ is flat and $X \to S$ is quasi-compact and separated. Let $\tau : S \to X$ be a section of X over S.*

(i) The stabilizer $H = \mathrm{Stab}_G(\tau)$ is a closed subgroup scheme of G, and $\tau^*(\mathcal{F})$ is a quasi-coherent H-$\mathcal{O}_S$-module.

(ii) If H is moreover affine over S, for example if G is, and if $\mathscr{A}(H) = \pi_*(\mathcal{O}_H)$, then $\tau^*(\mathcal{F})$ is an $\mathscr{A}(H)$-comodule.

6.8. G-equivariant sheaves on G

To conclude, we give two results, 6.8.1 and 6.8.6, which will be used in Exposés II and III, in particular in III, 4.25.

Proposition 6.8.1. *Let S be a scheme, let $\pi : G \to S$ be an S-group scheme, and let $\varepsilon : S \to G$ be the unit section. Consider the action of G on itself by left translations. The functors*

$$\mathcal{E} \longmapsto \pi^*(\mathcal{E}), \qquad \mathcal{F} \longmapsto \varepsilon^*(\mathcal{F})$$

induce mutually quasi-inverse equivalences between the category of quasi-coherent $\mathcal{O}_S$-modules and the category of quasi-coherent G-equivariant $\mathcal{O}_G$-modules.

Proof. Let μ denote multiplication in G, and let $\mathrm{pr}_2 : G \times_S G \to G$ be the second projection. Since $\pi \circ \mu = \pi \circ \mathrm{pr}_2$, every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$ has a canonical isomorphism

$$\mu^*\pi^*(\mathcal{E}) = \mathrm{pr}_2^*\pi^*(\mathcal{E}).$$

It is immediate that this isomorphism satisfies the cocycle condition 6.5.1 ($\star\star$), so $\pi^*(\mathcal{E})$ is a G-equivariant $\mathcal{O}_G$-module. Since $\varepsilon^*\pi^*(\mathcal{E}) = \mathcal{E}$, the functor $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ is

fully faithful. It remains to show that, for every G-equivariant $\mathcal{O}_G$-module $\mathcal{F}$,

$$\mathcal{F} \simeq \pi^*\varepsilon^*(\mathcal{F}).$$

By hypothesis, there is an isomorphism

$$\theta : \mathrm{pr}_2^*(\mathcal{F}) \xrightarrow{\sim} \mu^*(\mathcal{F}).$$

Pulling θ back along the morphism

$$\tau : G \longrightarrow G \times_S G, \qquad \tau = (\mathrm{id}_G, \varepsilon \circ \pi),$$

gives an isomorphism $\mathcal{F} \xrightarrow{\sim} \pi^*\varepsilon^*(\mathcal{F})$.

Remarks 6.8.2. (a) Consider the action of $G \times_S G$ on G defined by

$$(g_1, g_2) \cdot g = g_1 g g_2^{-1}.$$

The stabilizer of the unit section $\varepsilon : S \to G$ is the diagonal subgroup H of $G \times_S G$. Consequently, if $\mathcal{F}$ is a quasi-coherent $(G \times_S G)$-equivariant $\mathcal{O}_G$-module, then $\mathcal{E} = \varepsilon^*(\mathcal{F})$ has the structure of an H-$\mathcal{O}_S$-module, by 6.5.2.

(b) One can show that $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ is an equivalence between the category of quasi-coherent $(G \times_S G)$-equivariant $\mathcal{O}_G$-modules and the category of quasi-coherent H-$\mathcal{O}_S$-modules. This is a special case of more general descent results (cf. Exposé IV, Section 2, and SGA 1, VIII); see, for example, [Th87], 1.2–1.3.

Remark 6.8.3. Retain the notation of 6.8.1. For every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, let $\pi_*^G\pi^*(\mathcal{E})$ be the $\mathcal{O}_S$-submodule of $\pi_*\pi^*(\mathcal{E})$ whose sections over an open subset V of S are the elements

$$\gamma \in \Gamma\big(\pi^{-1}(V), \pi^*(\mathcal{E})\big)$$

such that $g \cdot \gamma_{S'} = \gamma_{S'}$ for every $S' \to V$ and $g \in G(S')$. Then the natural morphism

$$\mathcal{E} \longrightarrow \pi_*^G\pi^*(\mathcal{E})$$

is an isomorphism. This is immediate if

$$\pi_*\pi^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_S} \pi_*(\mathcal{O}_G),$$

for example if $G \to S$ is affine, or if $G \to S$ is quasi-compact and quasi-separated and $\mathcal{E}$ is flat; the general case is verified without difficulty.

Remark 6.8.4. Let S be a scheme, let H be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Suppose H is flat over S, and let $W_P(\mathcal{F})$ denote the restriction of $W(\mathcal{F})$ to the full subcategory of flat S-schemes. By 6.5.1, endowing $\mathcal{F}$ with an H-equivariant module structure is equivalent to giving an isomorphism

$$\theta : \mathrm{pr}_X^*(\mathcal{F}) \longrightarrow \lambda^*(\mathcal{F})$$

of sheaves on $H \times_S X$ which satisfies the cocycle condition $(\star\star)$. Hence, to endow $\mathcal{F}$ with an H-equivariant module structure, it suffices to endow $W_P(\mathcal{F})$ with such a structure.

Recollection 6.8.5. Let X be an S-scheme and Y a closed S-subscheme of X. Let $\mathcal{N}_{Y/X}$ denote the conormal sheaf of the immersion $i : Y \hookrightarrow X$ (cf. EGA IV_4, 16.1.2). If $S' \to S$ is flat and $i' : Y' \hookrightarrow X'$ is obtained from i by base change, then *loc. cit.*, 16.2.2(iii), gives

$$\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} = \mathcal{N}_{Y'/X'}.$$

Proposition 6.8.6. *Let S be a scheme, let X be an S-group scheme, and let Y be an S-subgroup scheme of X. Suppose that Y is flat over S. Then the conormal sheaf $\mathcal{N}_{Y/X}$ is a $(Y \times_S Y)$-equivariant $\mathcal{O}_Y$-module.*

Indeed, $H = Y \times_S Y$ is flat over S. By Remark 6.8.4, it is therefore enough to endow $W_P(\mathcal{N}_{Y/X})$ with an H-equivariant module structure. Let S' be a flat S-scheme, let $Y' \hookrightarrow X'$ be the immersion obtained by base change, and put

$$\mathcal{N}' = \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}.$$

By 6.8.5, $\mathcal{N}' = \mathcal{N}_{Y'/X'}$. Every $h \in H(S')$ induces an automorphism of Y', and hence, for every $y \in Y'(S')$, isomorphisms

$$\Gamma(S', y^*(\mathcal{N}')) \xrightarrow{\sim} \Gamma(S', y^*h^*(\mathcal{N}'))$$

which endow $W_P(\mathcal{N}_{Y/X})$ with an H-equivariant module structure (cf. 6.1).

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[57]

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[57] Editor's note: These are additional references cited in this Exposé.

Exposé II
Tangent Bundles – Lie Algebras

by M. Demazure

We propose in this exposé to construct, in the theory of schemes, the analogues of the tangent bundles and Lie algebras of classical theory. It will, however, be useful not to restrict ourselves to schemes proper, but also to consider certain functors on the category of schemes that are not necessarily representable (for example, the functors Hom, Norm, and so forth). As announced in the preceding exposé (cf. I 1.1), we shall identify a scheme with its associated functor.[0]

On the other hand, the constructions presented below go beyond the setting of scheme theory. They are equally valid, for example, in the theory of analytic spaces with nilpotent elements, apart from a few changes of detail.

Before beginning this construction, we must give a few general definitions that supplement those of I 1.7.

1. The functors $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$

We resume the notation of I 1.1. We identify the category $\mathscr{C}$ with a full subcategory of

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

(in particular, we suppress the underlinings that allowed us to distinguish graphically an object of $\widehat{\mathscr{C}}$ from an object of $\mathscr{C}$).[1]

Consider the following situation: four objects of $\widehat{\mathscr{C}}$, denoted by S, X, Y, Z, the first of which is in fact an object of $\mathscr{C}$, with X and Y over Z, and Z over S.

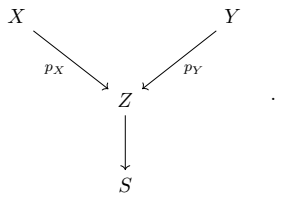

Figure 1: The objects X and Y over Z, and Z over S.

[0] Editor's note: version of October 14, 2024.

[1] Editor's note: in this edition these are rendered by bold letters $\mathbf{F}, \mathbf{G}, \mathbf{V}, \mathbf{W}$; cf. Exposé I. For functors such as Norm and Centr (cf. 5.2), however, the underlinings of the original have been retained, and one has been added to the functor Lie, in order to distinguish the functor $\underline{\mathrm{Lie}}(G/S)$ from the $\Gamma(S, \mathcal{O}_S)$-Lie algebra $\mathrm{Lie}(G/S) = \underline{\mathrm{Lie}}(G/S)(S)$, used, for example, in Exposé VII_A.

Definition 1.1.

We define an object of $\widehat{\mathscr{C}}_{/S}$, denoted $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$, by

$$\underline{\mathrm{Hom}}_{Z/S}(X,Y)(S') = \mathrm{Hom}_{Z_{S'}}(X_{S'},Y_{S'}) = \mathrm{Hom}_Z(X\times_S S',Y), \tag{1}$$

for every object S' of $\mathscr{C}_{/S}$. One sees at once that $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$ is simply the subobject of $\underline{\mathrm{Hom}}_S(X,Y)$ formed by the morphisms compatible with p_X and p_Y; that is, the kernel of the pair of morphisms

$$\underline{\mathrm{Hom}}_S(X,Y) \rightrightarrows \underline{\mathrm{Hom}}_S(X,Z),$$

the first defined by composition with p_Y, and the second being the constant morphism whose "image" is p_X.

As in I 1.7, for every object T of $\widehat{\mathscr{C}}$ over S, there is a natural bijection

$$\mathrm{Hom}_S(T,\underline{\mathrm{Hom}}_{Z/S}(X,Y)) \simeq \mathrm{Hom}_Z(X\times_S T,Y). \tag{2}$$

Moreover, by I 1.7.1, if E,F are objects of $\widehat{\mathscr{C}}$ over Z, then

$$\mathrm{Hom}_Z(E,\underline{\mathrm{Hom}}_Z(F,Y)) \simeq \mathrm{Hom}_Z(E\times_Z F,Y) \simeq \mathrm{Hom}_Z(F,\underline{\mathrm{Hom}}_Z(E,Y)).$$

Applying this with $E=X$ and $F=Z\times_S T$, one obtains natural bijections, for every object T of $\widehat{\mathscr{C}}_{/S}$,

$$\begin{aligned}\mathrm{Hom}_S(T,\underline{\mathrm{Hom}}_{Z/S}(X,Y)) &\simeq \mathrm{Hom}_Z(X\times_S T,Y)\\ &\simeq \begin{cases}\mathrm{Hom}_Z(Z\times_S T,\underline{\mathrm{Hom}}_Z(X,Y)),\\ \mathrm{Hom}_Z(X,\underline{\mathrm{Hom}}_Z(Z\times_S T,Y)).\end{cases}\end{aligned} \tag{3}$$

These bijections are functorial in T, and hence give isomorphisms of S-functors.[2]

(4)
$$\begin{array}{ccc}\underline{\mathrm{Hom}}_S(T,\underline{\mathrm{Hom}}_{Z/S}(X,Y)) & \xrightarrow{\ \sim\ } & \underline{\mathrm{Hom}}_{Z/S}(X,\underline{\mathrm{Hom}}_Z(Z\times_S T,Y))\\ & \searrow^{\sim} \quad \nearrow_{\sim} & \\ & \underline{\mathrm{Hom}}_{Z/S}(X\times_S T,Y) & \end{array}$$

Figure 2: The functorial isomorphisms (4).

Let us point out two special cases of the definition. If $Z=S$, then

$$\underline{\mathrm{Hom}}_{S/S}(X,Y) = \underline{\mathrm{Hom}}_S(X,Y).$$

On the other hand, when $X=Z$, we set

$$\prod_{Z/S} Y = \underline{\mathrm{Hom}}_{Z/S}(Z,Y), \tag{5}$$

so that, by definition,

$$\left(\prod_{Z/S} Y\right)(S') = \mathrm{Hom}_Z(Z\times_S S',Y) \simeq \Gamma(Y_{S'}/Z_{S'}).$$

The functor $\prod_{Z/S}:\widehat{\mathscr{C}}_{/Z}\to\widehat{\mathscr{C}}_{/S}$ is right adjoint to the base-change functor from S to Z: for every S-functor U,

$$\mathrm{Hom}_S\left(U,\prod_{Z/S} Y\right) = \mathrm{Hom}_Z(U\times_S Z,Y).$$

[2]Editor's note: points (2), (3), and (4) have been expanded; in particular, (4) will be used in 3.11.

If $\mathscr{C} = \mathbf{Sch}$ and Z is an S-scheme, the functor $\prod_{Z/S}$ is called the "Weil restriction of scalars." [3] One also has an isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y) \simeq \underline{\mathrm{Hom}}_{X/S}(X, Y \times_Z X) = \prod_{X/S} (Y \times_Z X), \tag{6}$$

which gives in particular, when $Z = S$, an isomorphism

$$\underline{\mathrm{Hom}}_S(X, Y) \simeq \prod_{X/S} Y_X. \tag{7}$$

Remark 1.2.

The functor $Y \mapsto \underline{\mathrm{Hom}}_{Z/S}(X, Y)$ commutes with products in the following sense: there is a functorial isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y \times_Z Y') \simeq \underline{\mathrm{Hom}}_{Z/S}(X, Y) \times_S \underline{\mathrm{Hom}}_{Z/S}(X, Y'). \tag{*}$$

It follows that if Y is a Z-group, respectively a Z-ring, and so on, then $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ is an S-group, respectively an S-ring, and so on.

Remark 1.3.

Let $\pi : M \to Y$ be a Y-functor in $\mathcal{O}_Y$-modules (cf. I 4.3.3.1), and set $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$. Then $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ has a natural $\mathcal{O}_H$-module structure; more precisely, for every H' over H, the set

$$\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$$

has a natural $\mathcal{O}(X \times_S H')$-module structure. [4]

Indeed, let $m : M \times_Y M \to M$ and $\lambda : \mathcal{O}_Y \times_Y M \to M$ denote the morphisms defining the structures of a Y-abelian group and an $\mathcal{O}_Y$-module. Let H' be an S-scheme over $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$; equivalently, one has been given a Z-morphism $f : X \times_S H' \to Y$, which makes $X \times_S H'$ a Y-object. Then

$$\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$$

is the set of Z-morphisms $\phi : X \times_S H' \to M$ such that $\pi \circ \phi = f$, that is, the set of Y-morphisms $X \times_S H' \to M$.

If ϕ, ψ are two such morphisms, define $\phi + \psi$ to be the composite Y-morphism

$$X \times_S H' \xrightarrow{\phi \times \psi} M \times_Y M \xrightarrow{m} M.$$

This makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ an abelian group over $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$. [5]

Similarly, if $a \in \mathcal{O}(X \times_S H')$, that is, if $a : X \times_S H' \to \mathcal{O}_S$ is an S-morphism, define $a\phi$ to be the composite $\lambda \circ (a \times \phi)$, where $a \times \phi$ is the Y-morphism from $X \times_S H'$ to $\mathcal{O}_Y \times_Y M \simeq \mathcal{O}_S \times_S M$ with components a and ϕ. This gives $\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$ an $\mathcal{O}(X \times_S H')$-module structure, functorial in the H-object H'.

[3] Editor's note: the preceding two sentences have been added.

[4] Editor's note: this remark has been added; it is useful for Corollary 3.11.1.

[5] Editor's note: if M is a Y-group, not necessarily abelian, and one sets $\phi \cdot \psi = m \circ (\phi \times \psi)$, this makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ a group over $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$.

2. The schemes $I_S(\mathcal{M})$

Definition 2.1.

Let S be a scheme and $\mathcal{M}$ a quasi-coherent $\mathcal{O}_S$-module. Denote by $\mathscr{D}_{\mathcal{O}_S}(\mathcal{M})$ the quasi-coherent $\mathcal{O}_S$-algebra $\mathcal{O}_S \oplus \mathcal{M}$, where $\mathcal{M}$ is regarded as an ideal of square zero. Denote by

$$I_S(\mathcal{M}) = \operatorname{Spec} \mathscr{D}_{\mathcal{O}_S}(\mathcal{M})$$

the corresponding S-scheme.[6] In particular, put

$$\mathscr{D}_{\mathcal{O}_S} = \mathscr{D}_{\mathcal{O}_S}(\mathcal{O}_S), \qquad I_S = I_S(\mathcal{O}_S),$$

and call these, respectively, the algebra of dual numbers over S and the scheme of dual numbers over S.

The assignment $\mathcal{M} \mapsto I_S(\mathcal{M})$ is a contravariant functor from quasi-coherent $\mathcal{O}_S$-modules to S-schemes. In particular, the morphisms $0 \to \mathcal{M}$ and $\mathcal{M} \to 0$ define, respectively, the structural morphism

$$\rho : I_S(\mathcal{M}) \longrightarrow I_S(0) = S$$

and a section $\varepsilon_{\mathcal{M}}$ of it, called the *zero section*.[7]

2.1.1.

Since $\mathcal{M} \mapsto I_S(\mathcal{M})$ is contravariant, every $a \in \operatorname{End}_{\mathcal{O}_S}(\mathcal{M})$ defines an S-endomorphism a^* of $I_S(\mathcal{M})$, and

$$1^* = \mathrm{id}, \qquad (ab)^* = b^* \circ a^*, \qquad 0^* = \varepsilon_{\mathcal{M}} \circ \rho, \qquad a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}.$$

Consequently, $I_S(\mathcal{M})$ carries a right action of the multiplicative monoid of $\operatorname{End}_{\mathcal{O}_S}(\mathcal{M})$, commuting with the S-morphisms $I_S(\mathcal{M}) \to I_S(\mathcal{M}')$ arising from morphisms $\mathcal{M}' \to \mathcal{M}$; in particular, the operators a^* preserve the zero section.[8]

For every $a \in \operatorname{End}_{\mathcal{O}_S}(\mathcal{M})$, every $f : S' \to S$, and every $m \in I_S(\mathcal{M})(S')$, write $m \cdot a = a^*(m)$. Then

$$m \cdot 1 = m, \qquad (m \cdot a) \cdot b = m \cdot (ab), \qquad m \cdot 0 = \varepsilon_{\mathcal{M}}(\rho(m)),$$

and if $m = \varepsilon_{\mathcal{M}}(f)$, then $m \cdot a = m$. [9]

Remark 2.1.2.

Formation of $I_S(\mathcal{M})$ commutes with base extension: there are canonical isomorphisms

$$I_S(\mathcal{M})_{S'} \simeq I_{S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}).$$

For simplicity write $I_{S'}(\mathcal{M}) = I_S(\mathcal{M})_{S'}$. More generally, if X is an S-functor, not necessarily representable, write $I_X(\mathcal{M}) = I_S(\mathcal{M}) \times_S X$.

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[6] Editor's note: the underlying space of $I_S(\mathcal{M})$ is the same as that of S.

[7] Editor's note: compare 3.1.1.

[8] Editor's note: what follows expands the original; the numbering 2.1.1–2.1.3 has also been added.

[9] Editor's note: later we shall be chiefly interested in $a \in \mathcal{O}(S)$, acting by homotheties on $\mathcal{M}$. For example, if $\mathcal{M}$ is free of rank r, then for every $S' \to S$, $\operatorname{Hom}_S(S', I_S(\mathcal{M}))$ identifies with the set of r-tuples $(\epsilon_1, \ldots, \epsilon_r) \in \mathcal{O}(S')^r$ satisfying $\epsilon_i \epsilon_j = 0$ for all i, j, and $(\epsilon_1, \ldots, \epsilon_r) \cdot a = (a\epsilon_1, \ldots, a\epsilon_r)$ for every $a \in \mathcal{O}(S)$.

2.1.3.

By the preceding discussion, the multiplicative monoid of $\mathcal{O}(S')$ acts on the S'-scheme $I_{S'}(\mathcal{M})$, functorially in $\mathcal{M}$. In other words, the S-scheme $I_S(\mathcal{M})$ is an object with an operator-monoid structure $\mathcal{O}_S$, functorial in $\mathcal{M}$. Thus there is a morphism of S-schemes

$$\lambda : I_S(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_S(\mathcal{M}),$$

satisfying the evident conditions. For every S-functor X, base change gives a morphism of X-functors

$$\lambda_X : I_X(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_X(\mathcal{M}),$$

making $I_X(\mathcal{M})$ an object with operator monoid $\mathcal{O}(X)$. Every $a \in \mathcal{O}(X) = \mathrm{Hom}_S(X, \mathcal{O}_S)$ defines the X-endomorphism

$$a^* = \lambda_X \circ (\mathrm{id}_{I_X(\mathcal{M})} \times (a \circ \mathrm{pr}_X)).$$

Explicitly, if $x \in X(S')$ and $m \in I_S(\mathcal{M})(S') = I_{S'}(\mathcal{M})(S')$, then $a(x) = a \circ x \in \mathcal{O}(S')$ and

$$(m, x) \cdot a = (m \cdot a(x), x).$$

This operation is functorial in $\mathcal{M}$ and preserves the zero section $\varepsilon_{\mathcal{M}} : X \to I_X(\mathcal{M})$; thus $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$. [10]

The operation is also "functorial in X" in the following sense. If $\pi : Y \to X$ is a morphism of S-functors and $u : \mathcal{O}(X) \to \mathcal{O}(Y)$ the corresponding ring morphism, so that $u(a) = a \circ \pi$, then the following diagram is commutative.

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{a^*} & I_X(\mathcal{M}) \\ \uparrow{\scriptstyle \pi} & & \uparrow{\scriptstyle \pi} \\ I_Y(\mathcal{M}) & \xrightarrow{u(a)^*} & I_Y(\mathcal{M}) \end{array} .$$

Figure 3: Functoriality of the operator action under $Y \to X$.

2.2.

Let $\mathcal{M}$ and $\mathcal{N}$ be two quasi-coherent $\mathcal{O}_S$-modules. The commutative diagram below induces the corresponding diagram of S-schemes on the following page.

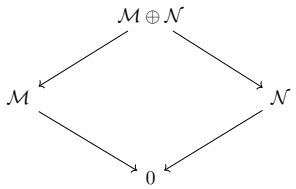

Figure 4: The direct-sum diagram of $\mathcal{O}_S$-modules.

[10] Editor's note: this paragraph has been added; it will be useful in 3.4.2 and 4.6.2.

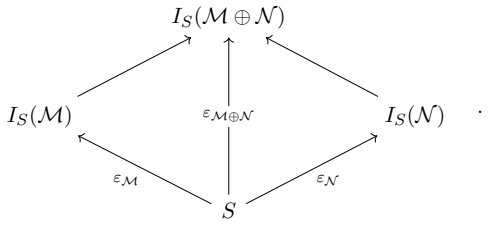

Figure 5: The induced commutative diagram of S-schemes.

Proposition 2.2.

For every S-scheme X, the diagram of functors over S obtained by applying $\underline{\mathrm{Hom}}_S(-, X)$ to the preceding diagram is cartesian.

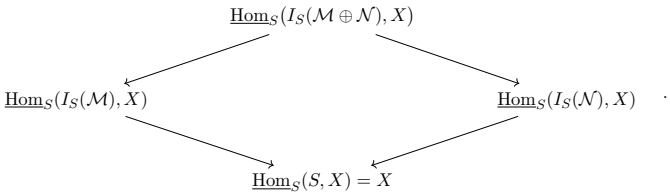

Figure 6: The cartesian diagram of $\underline{\mathrm{Hom}}_S$-functors.

Indeed, for every $S' \to S$, one must verify that the diagram of sets obtained by evaluating the functors on S' is cartesian. Since formation of $I_S(\mathcal{P})$ commutes with base extension, it is enough to take $S' = S$, and hence to verify that the following diagram of sets is cartesian.

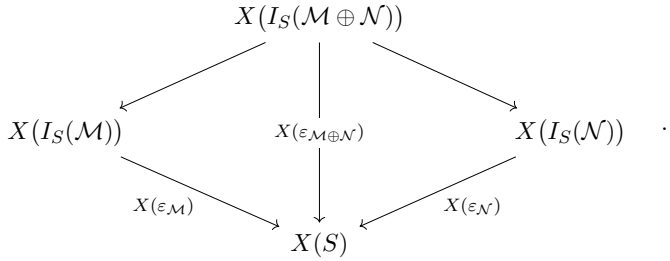

Figure 7: The cartesian diagram after evaluation on S.

If $x \in X(S)$, it follows from SGA 1, III 5.1 that $X(\varepsilon_{\mathcal{M}})^{-1}(x)$ is isomorphic, functorially in $\mathcal{M}$, to

$$\mathrm{Hom}_{\mathcal{O}_S}\big(x^*(\Omega^1_{X/S}), \mathcal{M}\big),$$

where $\Omega^1_{X/S}$ denotes the sheaf of relative differentials of X over S. This latter functor in $\mathcal{M}$ plainly carries a direct sum of $\mathcal{O}_S$-modules to the product of the corresponding sets, which proves the proposition.[11]

Corollary 2.2.1.

Let X be an S-scheme and $\mathcal{M}$ a finite free $\mathcal{O}_S$-module. The functor $\underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)$, as a functor over X, is isomorphic to a finite product over X of copies of $\underline{\mathrm{Hom}}_S(I_S, X)$.

[11] Editor's note: see also the addition 0.1.8 in Exposé III.

Nota 2.2.2.

It follows from the proof of the proposition that $\underline{\mathrm{Hom}}_S(I_S, X)$, as an X-functor, is isomorphic to $\mathbf{V}(\Omega^1_{X/S})$ (I 4.6.1), and is therefore represented by the vector fibration[12]

$$V(\Omega^1_{X/S}).$$

3. The tangent bundle, condition (E)

Throughout this section, unless otherwise stated, the letters $\mathcal{M}, \mathcal{M}', \mathcal{N}, \ldots$ will denote finite free $\mathcal{O}_S$-modules (that is, modules isomorphic to finite direct sums of copies of $\mathcal{O}_S$).

We shall systematically use the identifications established in Exposé I. Thus, by a "functor over S" we shall mean indifferently either a functor equipped with a morphism to $S = h_S$, or a functor on the category of objects over S. We shall use "group functor over S," and similar expressions, in the same way.

Definition 3.1.

Let S be a scheme, let $\mathcal{M}$ be a finite free $\mathcal{O}_S$-module, and let X be a functor over S. The *tangent bundle of X over S, relative to the $\mathcal{O}_S$-module $\mathcal{M}$*, is the S-functor

$$T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X).$$

In particular, the *tangent bundle of X over S* is [13]

$$T_{X/S} = T_{X/S}(\mathcal{O}_S) = \underline{\mathrm{Hom}}_S(I_S, X).$$

The assignment $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$-modules to S-functors. In particular, the morphisms $\mathcal{M} \to 0$ and $0 \to \mathcal{M}$ define, respectively, an S-morphism

$$\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \longrightarrow T_{X/S}(0) \simeq X$$

and a section τ_0 of it, called the *zero section*. [14]

Remark 3.1.1.

[15] The projection $\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \to X$ is induced by the zero section $\varepsilon_{\mathcal{M}} : S \to I_S(\mathcal{M})$, whereas the zero section $\tau_0 : X \to T_{X/S}(\mathcal{M})$ is induced by the structural morphism $\rho : I_S(\mathcal{M}) \to S$. More explicitly, let $t \in T_{X/S}(\mathcal{M})(S')$ correspond to the S-morphism $f : S' \times_S I_S(\mathcal{M}) \to X$, and let $x \in X(S')$ correspond to $g : S' \to X$. Then

$$\pi(t) = f \circ (\mathrm{id}_{S'} \times \varepsilon_{\mathcal{M}}), \qquad \tau_0(x) = g \circ (\mathrm{id}_{S'} \times \rho).$$

It follows from 3.1 that $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$-modules to functors over X. In particular, $\mathcal{O}(S)$ is an operator monoid of the X-functor $T_{X/S}(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$.

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[12] Editor's note: cf. editor's note (33) of Exposé I.

[13] Editor's note: when $S = \mathrm{Spec}(k)$ and X is a k-scheme, one has $T_{X/S}(S') = \mathrm{Hom}_S(S' \otimes_k k[\epsilon], X) = X(S' \otimes_k k[\epsilon])$; this recovers one of the usual definitions of the tangent bundle.

[14] Editor's note: the original order "$0 \to \mathcal{M}$ and $\mathcal{M} \to 0$" has been corrected to "$\mathcal{M} \to 0$ and $0 \to \mathcal{M}$."

[15] Editor's note: paragraphs 3.1.1 and 3.1.2 have been added.

Scholium 3.1.2.

[15] The preceding discussion means, in particular, the following. For every S-morphism $\varphi : X' \to X$, put

$$\Sigma(X', \mathcal{M}) = \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})).$$

The multiplicative monoid $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ acts on $\Sigma(X', \mathcal{M})$; write the action as $(\lambda, x) \mapsto \lambda * x$. Then

$$\lambda * (\mu * x) = (\lambda\mu) * x, \qquad 1 * x = x, \qquad 0 * x = \tau_0 \circ \varphi.$$

There is likewise an action of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M} \oplus \mathcal{M})$ on $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$. [16]

Let $m : \mathcal{M} \oplus \mathcal{M} \to \mathcal{M}$ be addition, and let $\delta : \mathcal{M} \to \mathcal{M} \oplus \mathcal{M}$ be the diagonal. Write

$$m_{X'} : \Sigma(X', \mathcal{M} \oplus \mathcal{M}) o \Sigma(X', \mathcal{M}), \qquad \delta_{X'} : \Sigma(X', \mathcal{M}) o \Sigma(X', \mathcal{M} \oplus \mathcal{M})$$

for the induced maps. For $\lambda, \mu \in \mathcal{O}(S)$, let ℓ_λ denote multiplication by λ on $\mathcal{M}$, and let $\ell_{\lambda,\mu}$ denote multiplication by (λ, μ) on $\mathcal{M} \oplus \mathcal{M}$. Since

$$m \circ \ell_{\lambda,\lambda} = \ell_\lambda \circ m, \qquad m \circ \ell_{\lambda,\mu} \circ \delta = \ell_{\lambda+\mu},$$

we have, for every $z \in \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ and $x \in \Sigma(X', \mathcal{M})$,

$$\lambda * m(z) = m((\lambda, \lambda) * z), \qquad m((\lambda, \mu) * \delta(x)) = (\lambda + \mu) * x. \tag{$\dagger$}$$

Definition 3.2.

Let $u \in X(S) = \mathrm{Hom}_S(S, X) = \Gamma(X/S)$. The *tangent space of X over S at u, relative to $\mathcal{M}$*, denoted $L^u_{X/S}(\mathcal{M})$, is the S-functor obtained from the X-functor $T_{X/S}(\mathcal{M})$ by pullback along $u : S \to X$.

$$\begin{array}{ccc} L^u_{X/S}(\mathcal{M}) & \longrightarrow & T_{X/S}(\mathcal{M}) \\ \downarrow & & \downarrow \pi \\ S & \xrightarrow[u]{} & X \end{array}$$

Figure 8: The cartesian square defining $L^u_{X/S}(\mathcal{M})$.

In particular, $L^u_{X/S}(\mathcal{O}_S)$ is denoted $L^u_{X/S}$ and is called the *tangent space of X over S at u*.

Remark 3.2.1.

[17] It follows from 3.1.1 that, for every $t : S' \to S$, $L^u_{X/S}(\mathcal{M})(S')$ is the set of S-morphisms $f : I_{S'}(\mathcal{M}) \to X$ such that $f \circ \varepsilon_{\mathcal{M}} = u \circ t$, where $\varepsilon_{\mathcal{M}} : S' \to I_{S'}(\mathcal{M})$ is the zero section.

[16] Editor's note: in fact, only the actions of $\mathcal{O}(S)$ (respectively $\mathcal{O}(S) \times \mathcal{O}(S)$) on $\Sigma(X', \mathcal{M})$ (respectively $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$) will be used; see below and the proof of 3.6.

[17] Editor's note: this remark has been added.

Proposition 3.3.

If X is represented by an S-scheme $\widetilde{X}$, then $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ are representable. In particular, $T_{X/S}$ and $L^u_{X/S}$ are represented by the vector fibrations

$$V(\Omega^1_{\widetilde{X}/S}) \qquad \text{and} \qquad V(u^*\Omega^1_{\widetilde{X}/S})$$

over $\widetilde{X}$ and S, respectively.

It plainly suffices to prove the assertion for $T_{X/S}(\mathcal{M})$, since the corresponding assertion for $L^u_{X/S}(\mathcal{M})$ follows by pullback. By Corollary 2.2.1, it even suffices to treat $T_{X/S}$, in which case the result is exactly Nota 2.2.2.

Remark 3.3.1.

The proposition gives a particularly simple description of the vector fibration representing $L^u_{X/S}$. The image of the section u of X over S is locally closed, [18] and hence is defined by a quasi-coherent ideal $\mathfrak{m}$ in the scheme induced on an open subscheme of X. The quotient $\mathfrak{m}/\mathfrak{m}^2$ may be regarded as a quasi-coherent module on S; it is this module that defines the vector fibration in question.

For example, let X be an algebraic scheme over a field k, and let u be a k-rational point of X. Let $\mathfrak{m}$ be the maximal ideal of the local ring $\mathcal{O}_{X,u}$, and let $\mathfrak{t}$ be the k-vector space dual to $\mathfrak{m}/\mathfrak{m}^2$; this is the Zariski tangent space of $\mathcal{O}_{X,u}$ at u. In the notation of I 4.6.5.1,

$$L^u_{X/k} = V(\mathfrak{m}/\mathfrak{m}^2) = W(\mathfrak{t}).$$

Returning to the general situation, observe first that $L^u_{X/S}(\mathcal{M})$ is covariant from finite free $\mathcal{O}_S$-modules to functors over S. In particular, $\mathcal{O}(S)$ is a set of operators of the S-functor $L^u_{X/S}(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$. [19]

Proposition 3.4.

Formation of $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ commutes with base extension. For every S-scheme S', there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X_{S'}/S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} T_{X/S}(\mathcal{M})_{S'},$$

$$L^{u'}_{X_{S'}/S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} L^u_{X/S}(\mathcal{M})_{S'}, \qquad u' = u_{S'}.$$

This follows immediately from the compatibility of $\underline{\mathrm{Hom}}$ with base extension.

Corollary 3.4.1.

The X-functor $T_{X/S}(\mathcal{M})$, respectively the S-functor $L^u_{X/S}(\mathcal{M})$, is naturally an object with operators $\mathcal{O}_X$, respectively $\mathcal{O}_S$, and these structures are functorial in $\mathcal{M}$.

For $L^u_{X/S}(\mathcal{M})$, this is immediate. For each S' over S, the ring $\mathcal{O}(S')$ acts on $\mathcal{M} \otimes \mathcal{O}_{S'}$, hence on

$$L^{u'}_{X_{S'}/S'}(\mathcal{M} \otimes \mathcal{O}_{S'}) = L^u_{X/S}(\mathcal{M})_{S'};$$

this action is functorial in S'.

[18] Editor's note: cf. EGA I, 5.3.11.

[19] Editor's note: cf. 3.1.2.

The case of $T_{X/S}(\mathcal{M})$ requires a little more detail. For an X-object $f : X' \to X$, put $T_{X/S}(\mathcal{M})_{X'} = T_{X/S}(\mathcal{M}) \times_X X'$. We must equip

$$T_{X/S}(\mathcal{M})_{X'}(X') = \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$$

with an action of $\mathcal{O}(X')$, functorially in X'. Let $X_{X'} = X \times_S X'$, and let f' be the section of $X_{X'}$ over X' defined by f. The following diagram provides the required identification.

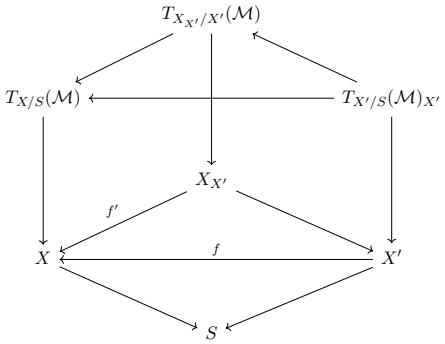

Figure 9: The base-change diagram used to define the $\mathcal{O}_X$-operator structure.

Together with 3.2.1, the diagram identifies $T_{X/S}(\mathcal{M})_{X'}(X')$ with

$$L^{f'}_{X_{X'}/X'}(\mathcal{M})(X') = \left\{ \; X'\text{-morphisms } \psi : I_{X'}(\mathcal{M}) \to X_{X'} \;\middle|\; \psi \circ \varepsilon_{\mathcal{M}} = f' \; \right\}.$$

Every $a \in \mathcal{O}(X')$ acts through its action on $I_{X'}(\mathcal{M})$: in the notation of 2.1.1,

$$a\psi = \psi \circ a^*, \qquad (a\psi)(x) = \psi(x \cdot a)$$

for every $X'' \to X'$ and every $x \in I_{X'}(\mathcal{M})(X'')$. This construction is plainly functorial in X'. [20] The isomorphisms of Proposition 3.4 are therefore, by construction, isomorphisms for the corresponding $\mathcal{O}_{X_{S'}}$- and $\mathcal{O}_{S'}$-operator structures.

Remark 3.4.2.

[21] The action of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$ can be seen more directly as follows. For every $f : X' \to X$,

$$\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})) = \{\varphi \in \mathrm{Hom}_S(I_{X'}(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = f\}.$$

By 2.1.3, the S-functor $I_{X'}(\mathcal{M})$ carries an action of the monoid $\mathcal{O}(X')$ preserving the zero section. Thus, if a^* denotes the X'-endomorphism defined by $a \in \mathcal{O}(X')$, then

$$a\varphi = \varphi \circ a^*, \qquad (a\varphi)(m, x') = \varphi(m \cdot a(x'), x')$$

for every $S' \to S$ and every $(m, x') \in \mathrm{Hom}_S(S', I_S(\mathcal{M}) \times_S X')$. Notice that $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$, so that $(a\varphi) \circ \varepsilon_{\mathcal{M}} = f$.

Likewise, for $t : S' \to S$, the set $L^u_{X/S}(\mathcal{M})(S')$ consists of the S-morphisms $\varphi : I_{S'}(\mathcal{M}) \to X$ satisfying $\varphi \circ \varepsilon_{\mathcal{M}} = u \circ t$, and $a\varphi = \varphi \circ a^*$ for $a \in \mathcal{O}(S')$.

Remark 3.4.3.

[21] The observations of 3.1.2 apply equally to the action of $\mathcal{O}_S$ on $L^u_{X/S}(\mathcal{M})$ and to that of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$.

[20] Editor's note: the original has been expanded in what follows.
[21] Editor's note: Remarks 3.4.2 and 3.4.3 have been added.

Definition 3.5.

Let S be a scheme and X an S-functor. We say that X *satisfies condition (E) relative to* S if, for every S' over S and every pair of finite free $\mathcal{O}_{S'}$-modules $\mathcal{M}, \mathcal{N}$, the diagram of sets obtained by applying X to diagram $(*)$ of 2.1 is cartesian. [22]

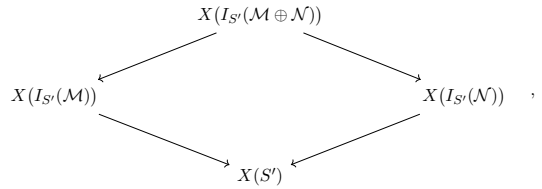

Figure 10: The cartesian diagram in condition (E).

3.5.1.

[23] Equivalently, the functor $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ carries direct sums of finite free $\mathcal{O}_S$-modules to products of X-functors. In that case, the same is true of

$$\mathcal{M} \longmapsto L^u_{X/S}(\mathcal{M}) = S \times_X T_{X/S}(\mathcal{M})$$

for every $u \in \Gamma(X/S)$.

Proposition 2.2 shows that every representable functor satisfies condition (E). We shall sometimes abbreviate "X satisfies condition (E) relative to S" to "X/S satisfies condition (E)."

If X/S satisfies (E), then $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ commutes with products and hence carries groups to groups. In particular, $T_{X/S}(\mathcal{M})$ is a commutative X-group; for the same reason, $L^u_{X/S}(\mathcal{M})$ is a commutative S-group.

Proposition 3.6.

If X/S satisfies (E), then the abelian-group structure on $T_{X/S}(\mathcal{M})$, respectively on $L^u_{X/S}(\mathcal{M})$, together with the action of $\mathcal{O}_X$, respectively of $\mathcal{O}_S$, makes it an $\mathcal{O}_X$-module, respectively an $\mathcal{O}_S$-module.

The operator action is functorial in $\mathcal{M}$, and therefore respects the abelian-group structure obtained by functoriality from that of $\mathcal{M}$. [24] Indeed, retaining the notation of 3.1.2, the abelian X-group law on $T_{X/S}(\mathcal{M})$ is the composite

$$T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M}) \simeq T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \xrightarrow{m} T_{X/S}(\mathcal{M}),$$

whereas

$$T_{X/S}(\mathcal{M}) \xrightarrow{\delta} T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \simeq T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M})$$

is the diagonal morphism.

By Remark 3.4.3 and the identities (†) of 3.1.2,

$$\lambda(x+y) = \lambda x + \lambda y, \qquad (\lambda+\mu)x = \lambda x + \mu x$$

for every $f : X' \to X$, every $x, y \in \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$, and every $\lambda, \mu \in \mathcal{O}(X')$.

[22] Editor's note: for examples of S-functors and S-groups that do not satisfy (E), see 6.2 and 6.3.
[23] Editor's note: what follows has been added.
[24] Editor's note: what follows has been expanded.

Remark 3.6.1.

If X is representable, then it satisfies (E), and $T_{X/S}$ and $L^u_{X/S}$ are represented by vector fibrations. The preceding laws are then exactly those induced by the vector-fibration structures (cf. I 4.6). [25]

Proposition 3.4 bis.

If X/S satisfies (E), then $X_{S'}/S'$ satisfies (E), and the isomorphisms of 3.4 respect the $\mathcal{O}_{X_{S'}}$-module and $\mathcal{O}_{S'}$-module structures, respectively.

No further comment is needed.

Proposition 3.7.

The functors $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ are functorial in X. Thus, if $f : X \to X'$ is an S-morphism, the following diagrams commute.

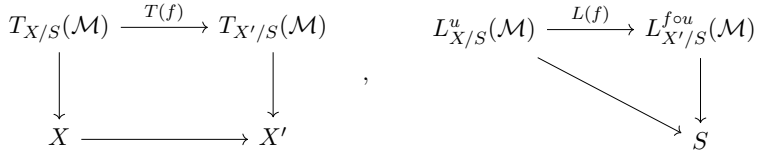

Figure 11: Functoriality of tangent bundles and tangent spaces in X.

Moreover, if f is a monomorphism, then so are $T(f)$ and $L(f)$. [26] The existence of $T(f)$ and $L(f)$, and the last assertion, follow immediately from the definitions. Commutativity follows from functoriality in $\mathcal{M}$ and from $T_{X/S}(0) = X$.

Remark 3.7.1.

[27] Suppose that X and X' are representable, and let r be the rank of $\mathcal{M}$. By 2.2.2, $T_{X/S}(\mathcal{M})$ is isomorphic to the product over X of r copies of $V(\Omega^1_{X/S})$, and similarly for $T_{X'/S}(\mathcal{M})$. Hence the square in 3.7 is cartesian when f is an open immersion, or more generally when $f^*\Omega^1_{X'/S} = \Omega^1_{X/S}$, for example when f is étale. Under these conditions there is an isomorphism of S-functors

$$L^u_{X/S}(\mathcal{M}) \xrightarrow{\sim} L^{f\circ u}_{X'/S}(\mathcal{M}).$$

In general, the square in 3.7 defines the following morphism of X-functors.

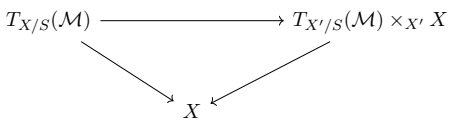

Figure 12: The induced morphism over X in the general case.

[25] Editor's note: explicitly, for every S-morphism $f : X' \to X$, the action of $\mathcal{O}(X')$ on $\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$ corresponds, under the identification

$$\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})) \simeq \mathrm{Hom}_{\mathcal{O}_{X'}}(f^*\Omega^1_{X/S}, \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{X'}),$$

to the natural action on the right. This follows from the proof of SGA 1, III 5.1; see also the addition 0.1.8 in Exposé III.

[26] Editor's note: the preceding sentence has been added.

[27] Editor's note: the hypothesis that X and X' be representable has been added, and what follows has been expanded.

Proposition 3.7 bis.

Let $f : X \to X'$ be an S-morphism. If X and X' satisfy (E) relative to S, then

$$T_{X/S}(\mathcal{M}) \xrightarrow{T(f)} T_{X'/S}(\mathcal{M})_X, \qquad \text{respectively} \qquad L^u_{X/S}(\mathcal{M}) \xrightarrow{L(f)} L^{f\circ u}_{X'/S}(\mathcal{M}),$$

is a morphism of $\mathcal{O}_X$-modules, respectively of $\mathcal{O}_S$-modules.

This follows from Proposition 3.7 by functoriality in $\mathcal{M}$.

Proposition 3.8.

For two functors X, Y over S, there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X/S}(\mathcal{M}) \times_S T_{Y/S}(\mathcal{M}) \xrightarrow{\sim} T_{(X\times_S Y)/S}(\mathcal{M}), \tag{3.8.1}$$

$$L^u_{X/S}(\mathcal{M}) \times_S L^v_{Y/S}(\mathcal{M}) \xrightarrow{\sim} L^{(u,v)}_{(X\times_S Y)/S}(\mathcal{M}). \tag{3.8.2}$$

The first isomorphism follows from 1.2 $(*)$; [28] the second follows from it by base change along $(u, v) : S \to X \times_S Y$.

Remark 3.8.0.

The isomorphism (3.8.1) may also be read as an isomorphism of $X \times_S Y$-functors

$$\big(T_{X/S}(\mathcal{M}) \times_X (X \times_S Y)\big) \underset{X\times_S Y}{\times} \big(T_{Y/S}(\mathcal{M}) \times_Y (X \times_S Y)\big) \xrightarrow{\sim} T_{(X\times_S Y)/S}(\mathcal{M}).$$

Corollary 3.8.1.

If X/S carries an algebraic structure defined by finite cartesian products, then $T_{X/S}(\mathcal{M})$ carries a structure of the same kind, and the projection $T_{X/S}(\mathcal{M}) \to X$ is a morphism for that structure.

Proposition 3.8 bis.

If X/S and Y/S satisfy (E), then $(X \times_S Y)/S$ satisfies (E), and (3.8.1), respectively (3.8.2), is an isomorphism of $\mathcal{O}_{X\times_S Y}$-modules, respectively of $\mathcal{O}_S$-modules.

Proof. [29] By 3.5.1 and (3.8.1), the product $(X \times_S Y)/S$ satisfies (E). Let us prove the assertion about (3.8.1). Given an S-morphism $(x, y) : Z \to X \times_S Y$, Remark 3.4.2 reduces the claim to the $\mathcal{O}(Z)$-linearity of the map

$$\{\varphi \in \mathrm{Hom}_S(I_Z(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = x\} \times \{\psi \in \mathrm{Hom}_S(I_Z(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = y\} \longrightarrow \{\theta \in \mathrm{Hom}_S(I_Z(\mathcal{M}), X \times_S Y) \mid \theta \circ \varepsilon_{\mathcal{M}} = (x, y)\},$$

which sends (φ, ψ) to $\varphi \times \psi$. This is clear, since for $a \in \mathcal{O}(Z)$,

$$(\varphi \circ a^*) \times (\psi \circ a^*) = (\varphi \times \psi) \circ a^* = a \cdot (\varphi \times \psi).$$

The assertion about (3.8.2) follows similarly from 3.2.1.

[28] Editor's note: what follows has been expanded.

[29] Editor's note: this proof has been added.

3.9.0.

[30] If X is an S-group with unit section $e : S \to X$, write

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) = L^e_{X/S}(\mathcal{M}).$$

Thus $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is defined by the following cartesian square.

$$\begin{array}{ccc} \underline{\mathrm{Lie}}(X/S, \mathcal{M}) & \xrightarrow{\ i\ } & T_{X/S}(\mathcal{M}) \\ \downarrow & & \downarrow p \\ S & \xrightarrow{\ e\ } & X \end{array} .$$

Figure 13: The cartesian square defining $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$.

By Corollary 3.8.1, the projection $p : T_{X/S}(\mathcal{M}) \to X$ is a morphism of S-groups. Hence $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is an S-group, identified by i with the kernel of p.

If X/S also satisfies (E), Proposition 3.9 will show that this group structure, induced by that of X, agrees with the abelian-group structure induced by functoriality in $\mathcal{M}$ (cf. 3.5.1). In fact, the result holds under the weaker assumption that X be an S-functor in monoids, or more generally an S-functor in H-sets, as defined next.

Definitions 3.9.0.1.

(a) An *H-set* is a set X equipped with a composition law having a two-sided identity, denoted e_X or simply e. [31] If $f : X \to Y$ is a morphism of H-sets, its kernel is $\ker f = f^{-1}(e_Y)$, a sub-H-set of X.

(b) An *H-object* in a category $\mathscr{C}$ is an object X equipped with a morphism $X \times X \to X$ and a section of X over the final object that is a two-sided identity. Every $\mathscr{C}$-monoid, and in particular every $\mathscr{C}$-group, is a $\mathscr{C}$-H-object. An H-object in the category of functors over S will be called an *S-H-functor*.

(c) If X is an S-H-functor (for example, an S-group), and $e : S \to X$ is its identity section, write

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) = L^e_{X/S}(\mathcal{M}), \qquad \underline{\mathrm{Lie}}(X/S) = \underline{\mathrm{Lie}}(X/S, \mathcal{O}_S).$$

We now spell out the following special case of 3.8.1. [32]

Corollary 3.9.0.2.

If X is an S-H-functor, respectively an S-group, then $T_{X/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ are likewise S-H-functors, respectively S-groups, and there are morphisms of S-H-functors, respectively of S-groups, as follows.

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) \xrightarrow{i} T_{X/S}(\mathcal{M}) \underset{s}{\overset{p}{\leftrightarrows}} X,$$

Figure 14: The inclusion, projection, and zero section for the tangent functor.

Here i identifies $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ with $\ker p$, and s is a section of p.

[30] Editor's note: this paragraph has been added to explain the introduction of the notion of an S-H-functor.
[31] Editor's note: this terminology is inspired by the notion of an H-space in topology.
[32] Editor's note: Corollary 3.9.0.2 has been added; it will be used in 4.1.

Proposition 3.9.

Let X be an S-H-functor satisfying (E) relative to S. The S-H-functor structure on $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ induced by that of X agrees with the S-group structure defined in 3.5.1.

The preceding discussion shows that $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is an H-object in the category of $\mathcal{O}_S$-modules. The proposition follows from the next lemma. [33]

Lemma 3.10.

Let $\mathscr{C}$ be a category. Let G be an H-object in the category of $\mathscr{C}$-H-objects. Thus G is a $\mathscr{C}$-H-object, whose composition law we denote $f : G \times G \to G$, together with a morphism of $\mathscr{C}$-H-objects $h : G \times G \to G$. [34] Then $f = h$, and f is commutative.

Evaluating the functors on a variable argument reduces the proof, as usual, to the case in which $\mathscr{C}$ is the category of sets. We then have a set G and two maps $f, h : G \times G \to G$ satisfying

$$h(f(x,y), f(z,t)) = f(h(x,z), h(y,t)).$$

Let $e, u \in G$ be the respective identities for f and h, so that

$$f(e,x) = f(x,e) = x, \qquad h(u,x) = h(x,u) = x.$$

First,

$$h(f(u,y), f(x,u)) = f(x,y) = h(f(x,u), f(u,y)).$$

Putting $y = e$, respectively $x = e$, gives

$$\begin{aligned} x = f(x,e) = h(f(u,e), f(x,u)) = h(u, f(x,u)) = f(x,u), \\ y = f(e,y) = h(f(e,u), f(u,y)) = h(u, f(u,y)) = f(u,y). \end{aligned}$$

Substituting these identities into the original relation yields

$$h(y,x) = f(x,y) = h(x,y).$$

This proves the lemma and Proposition 3.9. The following corollaries now follow from 3.9.

Corollary 3.9.1.

If X is an S-H-functor satisfying (E) relative to S, every element of $X(I_S(\mathcal{M}))$ that maps to the identity element of $X(S)$ is invertible.

Corollary 3.9.2.

If X is an S-monoid satisfying (E) relative to S, an element of $X(I_S(\mathcal{M}))$ is invertible if and only if its image in $X(S)$ is invertible.

Corollary 3.9.3.

If X is an S-group satisfying (E) relative to S, the two S-group laws on $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ coincide.

[33] Editor's note: this is inspired by the standard proof that the fundamental group of an H-space is abelian.

[34] Editor's note: $G \times G$ is equipped with the law $(G \times G) \times (G \times G) \to G \times G$, $((x,z),(y,t)) \mapsto (f(x,y), f(z,t))$.

Corollary 3.9.4.

[35] Let G be an S-group satisfying (E) relative to S. For $n \in \mathbb{Z}$, let $n_G : G \to G$ be the morphism of S-functors defined by $g \mapsto g^n$. Then the derived morphism

$$L(n_G) : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S)$$

is multiplication by n; that is, it sends $x \in \underline{\mathrm{Lie}}(G/S)(S')$ to nx.

The map n_G is not generally a group homomorphism, but it preserves the identity section $e : S \to G$, so its derivative indeed carries $\underline{\mathrm{Lie}}(G/S) = L^e_{G/S}$ to itself. If i denotes the inclusion $\underline{\mathrm{Lie}}(G/S) \hookrightarrow T_{G/S}$, then $L(n_G)$ is characterized by

$$i(L(n_G)(x)) = i(x)^n.$$

By 3.9, the group laws on $\underline{\mathrm{Lie}}(G/S)$ coming from (E) and from the law on G agree. Hence $i(x)^n = i(nx)$, and therefore $L(n_G)(x) = nx$.

Before drawing further consequences from Proposition 3.9, we prove one more functoriality result.

Proposition 3.11.

In the situation of Section 1, there is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

Indeed, by definition,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X,Y)).$$

Applying the isomorphism (4) of 1.1 with $T = I_S(\mathcal{M})$ gives

$$\underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X,Y)) \simeq \underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(Z \times_S I_S(\mathcal{M}), Y)).$$

Since $Z \times_S I_S(\mathcal{M}) \simeq I_Z(\mathcal{M})$, this is

$$\underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(I_Z(\mathcal{M}), Y)) = \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

Corollary 3.11.1.

If Y/Z satisfies (E), then $\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S$ satisfies (E), and the isomorphism of 3.11 respects the module structures over $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$. [36]

Proof. Let $\mathcal{M}, \mathcal{N}$ be finite free $\mathcal{O}_S$-modules. If Y/Z satisfies (E), then

$$T_{Y/Z}(\mathcal{M} \oplus \mathcal{N}) \simeq T_{Y/Z}(\mathcal{M}) \times_Y T_{Y/Z}(\mathcal{N}).$$

The right-hand side is a subfunctor of $T_{Y/Z}(\mathcal{M}) \times_S T_{Y/Z}(\mathcal{N})$, and 1.2 $(*)$ therefore gives

$$\underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M} \oplus \mathcal{N})) \simeq \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \underset{\underline{\mathrm{Hom}}_{Z/S}(X,Y)}{\times} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{N})).$$

[35] Editor's note: this corollary has been added; it amplifies Remark 4.1.1.2 below.

[36] Editor's note: writing $H = \underline{\mathrm{Hom}}_{Z/S}(X,Y)$, this implies in particular that $T_{H/S}(\mathcal{M})$ is naturally a module over $\prod_{X \times_S H/H} \mathcal{O}_H$. This result appeared somewhat surprising to the editors, so the proof has been given in detail.

Together with 3.11, this says that $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$ satisfies (E), proving the first assertion.

For the second, put $H = \underline{\mathrm{Hom}}_{Z/S}(X,Y)$, and let $\Delta : H' \to H$ be an S-morphism. Equivalently, let $\delta : H' \times_S X \to Y$ be the corresponding Z-morphism; it makes $H' \times_S X$ a Y-object. On the one hand, we have the following commutative diagram.

$$\begin{array}{ccc}
\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))) & \longrightarrow & \mathrm{Hom}_S(H', \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))) \\
\| & & \| \\
\mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M})) & \longrightarrow & \mathrm{Hom}_Z(H' \times_S X, T_{Y/Z}(\mathcal{M})) \\
\| & & \| \\
\{\psi \in \mathrm{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\} & \longrightarrow & \mathrm{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y)
\end{array} .$$

Figure 15: The first comparison diagram in the proof of Corollary 3.11.1.

By 1.3, the action of $\alpha \in \mathcal{O}(H' \times_S X)$ on $\Psi \in \mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M}))$ is given, for every $U \to S$ and every $(h,x) \in \mathrm{Hom}_S(U, H' \times_S X)$, by

$$(\alpha\Psi)(h,x) = \alpha(h,x)\Psi(h,x),$$

where U is viewed over Y through $\delta \circ (h,x)$, and $\alpha(h,x) \in \mathcal{O}(U)$ acts through the $\mathcal{O}_Y$-module structure on $T_{Y/Z}(\mathcal{M})$. By 3.4.2, under the identification

$$\mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M})) = \{\psi \in \mathrm{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\},$$

this action is

$$(\alpha\psi)(m,h,x) = \psi(m \cdot \alpha(h,x), h, x) \tag{1}$$

for every $(m,h,x) \in \mathrm{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$.

On the other hand, consider $T_{H/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), H)$. We have a second commutative diagram.

$$\begin{array}{ccc}
\mathrm{Hom}_H(H', T_{H/S}(\mathcal{M})) & \longrightarrow & \mathrm{Hom}_S(H', T_{H/S}(\mathcal{M})) \\
\| & & \| \\
\{\Phi \in \mathrm{Hom}_S(I_{H'}(\mathcal{M}), H) \mid \Phi \circ \varepsilon_{\mathcal{M}} = \Delta\} & \longrightarrow & \mathrm{Hom}_S(I_{H'}(\mathcal{M}), H) \\
{\scriptstyle (*)}\| & & \| \\
\{\phi \in \mathrm{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y) \mid \phi \circ \varepsilon_{\mathcal{M}} = \delta\} & \longrightarrow & \mathrm{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y)
\end{array} ,$$

Figure 16: The second comparison diagram in the proof of Corollary 3.11.1.

The bijection $(*)$ in the diagram is described as follows (cf. 1.1(2) and I 1.7.1). For every $U \to S$ and $(m,h,x) \in \mathrm{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$, where U lies over Z through $U \xrightarrow{x} X \to Z$, one has $\Phi(m,h) \in \mathrm{Hom}_Z(X \times_S U, Y)$ and

$$\varphi(m,h,x) = \Phi(m,h) \circ (x \times \mathrm{id}_U) \in \mathrm{Hom}_Z(U,Y). \tag{$\dagger$}$$

By 3.4.2, with X replaced by H and X' by H', the action of $a \in \mathcal{O}(H')$ on $\Phi \in \mathrm{Hom}_S(I_{H'}(\mathcal{M}), H)$ is

$$(a\Phi)(m,h) = \Phi(m \cdot a(h), h).$$

Consequently, if φ, respectively $a\varphi$, corresponds to Φ, respectively $a\Phi$, then (†) gives

$$(a\varphi)(m,h,x) = \Phi(m\cdot a(h),h)\circ(x\times \mathrm{id}_U) = \varphi(m\cdot a(h),h,x). \tag{2}$$

Together with (1), this proves that the isomorphism

$$T_{H/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))$$

of 3.11.1 is an isomorphism of $\mathcal{O}(H)$-modules. Moreover, for every $H'\to H$, the $\mathcal{O}(H')$-module structure on $\mathrm{Hom}_H(H', T_{H/S}(\mathcal{M}))$ extends, functorially in H', to an $\mathcal{O}(H'\times_S X)$-module structure.

Taking $Z = S$ yields the next corollary.

Corollary 3.11.2.

There is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_S(X, T_{Y/S}(\mathcal{M})).$$

If Y/S satisfies (E), then $\underline{\mathrm{Hom}}_S(X,Y)/S$ satisfies (E), and this isomorphism respects the module structures over $\underline{\mathrm{Hom}}_S(X,Y)$.

Let $u : X\to Y$ be an S-morphism, identified with the constant morphism $u : S\to \underline{\mathrm{Hom}}_S(X,Y)$ such that $u(f) = u$ for every $f : S'\to S$. The fiber product of u with

$$\underline{\mathrm{Hom}}_S(X, T_{Y/S}(\mathcal{M})) \longrightarrow \underline{\mathrm{Hom}}_S(X,Y)$$

identifies with $\underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M}))$, where X lies over Y through u. Hence the definition of the tangent space and Corollary 3.11.2 give the following result. [37]

Corollary 3.11.3.

For an S-morphism $u : X\to Y$, there is an isomorphism, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})),$$

where on the right X lies over Y through u. If Y/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$-modules. [38]

In particular, for $Y = X$, the functor $\underline{\mathrm{End}}_S(X)$ is an S-functor in monoids, hence an S-H-functor. Recall that $\underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M})$ denotes $L^e_{\underline{\mathrm{End}}_S(X)/S}(\mathcal{M})$, with e the identity section.

Corollary 3.11.4.

There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}).$$

If X/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$-modules. [38]

[37] Editor's note: the preceding sentences have been added.
[38] Editor's note: the preceding sentence has been added.

Remark 3.11.5.

[39] Suppose that X/S satisfies (E). Then

$$\prod_{X/S} T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))$$

is a module over $\prod_{X/S} \mathcal{O}_X$. Explicitly, for every $S' \to S$,

$$\underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))(S') \\ = \{\psi \in \mathrm{Hom}_X(I_{S'}(\mathcal{M}) \times_S X, X) \mid \psi \circ (\varepsilon_{\mathcal{M}} \times \mathrm{id}_X) = \mathrm{pr}_X\}$$

is functorially an $\mathcal{O}(X \times_S S')$-module. This follows either from 3.6 and the properties of $\prod_{X/S}$ (cf. 1.2), or from the proof of 3.11.1.

We now give a geometric interpretation of the tangent bundle. Let U be an S-functor. [40] By I 1.7.2, there are isomorphisms, functorial in $\mathcal{M}$,

$$\begin{aligned} T_{X/S}(\mathcal{M})(U) &= \mathrm{Hom}_S\big(U, \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)\big) \\ &\simeq \mathrm{Hom}_S\big(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_S(U, X)\big) \\ &\simeq \mathrm{Hom}_{I_S(\mathcal{M})}\big(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}\big). \end{aligned}$$

In particular, the morphism $\mathcal{M} \to 0$ gives the following commutative diagram.

$$\begin{array}{ccc} \mathrm{Hom}_S(U, T_{X/S}(\mathcal{M})) & \xrightarrow{\sim} & \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \\ \Big\downarrow{\scriptstyle \pi_{\mathcal{M}}} & & \Big\downarrow \\ \mathrm{Hom}_S(U, X) & \xrightarrow{\text{identity}} & \mathrm{Hom}_S(U, X) \end{array} ,$$

Figure 17: The geometric interpretation of the projection $\pi_{\mathcal{M}}$.

The second vertical arrow is obtained by base change along $\varepsilon_{\mathcal{M}} : S \to I_S(\mathcal{M})$. [41]

Proposition 3.12.

Let $h_0 : U \to X$ be an S-morphism. Then $\mathrm{Hom}_X(U, T_{X/S}(\mathcal{M}))$ identifies with the set of $I_S(\mathcal{M})$-morphisms

$$U_{I_S(\mathcal{M})} \longrightarrow X_{I_S(\mathcal{M})}$$

whose restriction to U is h_0, where U is regarded as a subobject of $U \times_S I_S(\mathcal{M})$ through $\mathrm{id}_U \times_S \varepsilon_{\mathcal{M}}$.

Taking $U = X$ and $h_0 = \mathrm{id}_X$ gives the following. [42]

[39] Editor's note: this remark has been added.

[40] Editor's note: the original has been simplified in what follows.

[41] Editor's note: via

$$\mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \simeq \mathrm{Hom}_S(U \times_S I_S(\mathcal{M}), X),$$

this is equivalent to Remark 3.1.1. Likewise, 3.12 is equivalent to the first part of Remark 3.4.2.

[42] Editor's note: the original has been expanded in what follows.

Corollary 3.12.1.

The set $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$-endomorphisms ϕ of $X_{I_S(\mathcal{M})}$ inducing the identity on X; equivalently, the following diagram commutes. [43]

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{\phi} & I_X(\mathcal{M}) \\ {\scriptstyle \varepsilon_{\mathcal{M}}}\uparrow & & \uparrow{\scriptstyle \varepsilon_{\mathcal{M}}} \\ X & \xrightarrow{\mathrm{id}} & X \end{array} .$$

Figure 18: An infinitesimal endomorphism inducing the identity on X.

On the other hand, by 3.11.2,

$$\Gamma(T_{X/S}(\mathcal{M})/X) \simeq \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M})(S).$$

If X/S satisfies (E), then $\underline{\mathrm{End}}_S(X)/S$ satisfies (E), so the latter Lie functor is an $\mathcal{O}_S$-module by 3.6 (indeed, by 3.11.5, it is a $\prod_{X/S}\mathcal{O}_X$-module). Applying 3.9 gives the next proposition. [44]

Proposition 3.13.

If X/S satisfies (E), the abelian group $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$-endomorphisms of $X_{I_S(\mathcal{M})}$ that induce the identity on X. Consequently, every such endomorphism is an automorphism.

Corollary 3.13.1.

Let $u : X \to Y$ be an S-isomorphism, with Y/S satisfying (E). Every $I_S(\mathcal{M})$-morphism $X_{I_S(\mathcal{M})} \to Y_{I_S(\mathcal{M})}$ extending u is an isomorphism.

Corollary 3.13.2.

If Y/S satisfies (E), the monomorphism

$$\underline{\mathrm{Isom}}_S(X, Y) \longrightarrow \underline{\mathrm{Hom}}_S(X, Y)$$

induces, for every $u \in \underline{\mathrm{Isom}}_S(X, Y)$, an isomorphism

$$L^u_{\underline{\mathrm{Isom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}).$$

Proof. [45] For every $S' \to S$, we must show that this map is a bijection on S'-points. By base change (cf. 3.4), it suffices to take $S' = S$. The right-hand side then parametrizes the $I_S(\mathcal{M})$-morphisms $X_{I_S(\mathcal{M})} \to Y_{I_S(\mathcal{M})}$ extending u, whereas the left-hand side parametrizes those that are isomorphisms. The assertion is therefore Corollary 3.13.1.

[43] Editor's note: when $\mathcal{M} = \mathcal{O}_X$, these are called *infinitesimal endomorphisms* of X; see 6.2 at the end of this exposé.

[44] Editor's note: the original has been expanded in what follows.

[45] Editor's note: the following proof has been added.

Corollary 3.13.3.

If X/S satisfies (E), the monomorphism $\underline{\mathrm{Aut}}_S(X) \to \underline{\mathrm{End}}_S(X)$ induces, for every $u \in \underline{\mathrm{Aut}}_S(X)$, an isomorphism

$$L^u_{\underline{\mathrm{Aut}}_S(X)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{End}}_S(X)/S}(\mathcal{M}).$$

In particular,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}),$$

so that $\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S, \mathcal{M})$ is a $\prod_{X/S} \mathcal{O}_X$-module. [46]

3.14.

Finally, suppose that X is representable. [47] By 2.2.2, the X-functor $T_{X/S}$ is represented by $V(\Omega^1_{X/S})$, whence bijections

$$\Gamma(T_{X/S}/X) \simeq \mathrm{Hom}_X(\Omega^1_{X/S}, \mathcal{O}_X) \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X).$$

This also follows from the preceding results. By 3.13, $\Gamma(T_{X/S}/X)$ identifies with the set of infinitesimal endomorphisms of X, that is, the I_S-endomorphisms of X_{I_S} that induce the identity on X. The schemes X and X_{I_S} have the same underlying topological space, and their structure sheaves are $\mathcal{O}_X$ and

$$\mathscr{D}_{\mathcal{O}_X} = \mathcal{O}_X \oplus \mathcal{M}, \qquad \mathcal{M} = \mathcal{O}_X,$$

where $\mathcal{M}$ is a square-zero ideal. Let $\pi : \mathscr{D}_{\mathcal{O}_X} \to \mathcal{O}_X$ be the morphism of $\mathcal{O}_X$-algebras vanishing on $\mathcal{M}$. Giving an infinitesimal endomorphism of X is equivalent to giving a morphism of $\mathcal{O}_S$-algebras $\phi : \mathcal{O}_X \to \mathscr{D}_{\mathcal{O}_X}$ satisfying $\pi \circ \phi = \mathrm{id}_{\mathcal{O}_X}$, hence to giving an $\mathcal{O}_S$-derivation of $\mathcal{O}_X$.

If $D, D' \in \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$, and ϕ_D denotes the infinitesimal endomorphism corresponding to D, then

$$\phi_{D+D'} = \phi_D \circ \phi_{D'}.$$

Thus the identification

$$\{\text{infinitesimal endomorphisms of } X\} \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$$

is an isomorphism of abelian groups. In view of 3.13 and 3.11.5, we have therefore constructed an isomorphism of abelian groups, indeed of $\mathcal{O}(X)$-modules,

$$\Gamma(T_{X/S}/X) \xrightarrow{\sim} \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X),$$

recovering the classical interpretation of tangent vector fields as derivations of the structure sheaf. [48] Finally,

$$\Gamma(T_{X/S}/X) = H^0(X, \mathfrak{g}_{X/S}),$$

where $\mathfrak{g}_{X/S}$ is the dual of $\Omega^1_{X/S}$.

[46] Editor's note: the preceding assertion has been added.
[47] Editor's note: the reminder of 2.2.2 has been added, and the sequel has been expanded.
[48] Editor's note: this isomorphism is also functorial in S. If $S' \to S$ and $X' = X_{S'}$, then

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S)(S') \simeq \Gamma(T_{X'/S'}/X') \simeq \mathrm{Der}_{\mathcal{O}_{S'}}(\mathcal{O}_{X'}).$$

4. Tangent Space to a Group — Lie Algebras

4.1.

Let G be a group functor over S. By 3.9.0.2, $T_{G/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ are endowed with group structures over S, and there are group morphisms

$$\underline{\mathrm{Lie}}(G/S, \mathcal{M}) \xrightarrow{i} T_{G/S}(\mathcal{M}) \underset{s}{\overset{p}{\leftrightarrows}} G;$$

Figure 19: The split tangent sequence of the group functor G.

By definition, i identifies $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ with the kernel of p, and s is a section of p. It follows from I 2.3.7 that this sequence identifies $T_{G/S}(\mathcal{M})$ with the semidirect product of G by $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Definition 4.1.A.

The corresponding action of G on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is denoted

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}\big(\underline{\mathrm{Lie}}(G/S, \mathcal{M})\big)$$

and is called the *adjoint representation* of G, relative to $\mathcal{M}$. Thus, by definition, for $x \in G(S')$ and $X \in \underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$,

$$\mathrm{Ad}(x)X = i^{-1}\big(s(x)i(X)s(x)^{-1}\big).$$

If G is commutative, then $T_{G/S}(\mathcal{M})$ is commutative as well and $\mathrm{Ad}(x)X = X$. [49]

Definition 4.1.B.

Let G, H be group functors over S, and let $f : G \to H$ be a group morphism. Functoriality gives a morphism of exact sequences compatible with the canonical sections.

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \underline{\mathrm{Lie}}(G/S, \mathcal{M}) & \longrightarrow & T_{G/S}(\mathcal{M}) & \longrightarrow & G & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle L(f)} & & \downarrow{\scriptstyle T(f)} & & \downarrow{\scriptstyle f} & & \\ 1 & \longrightarrow & \underline{\mathrm{Lie}}(H/S, \mathcal{M}) & \longrightarrow & T_{H/S}(\mathcal{M}) & \longrightarrow & H & \longrightarrow & 1 \end{array} ;$$

Figure 20: Functoriality of the split tangent sequence.

The morphism $L(f)$, also denoted $\underline{\mathrm{Lie}}(f)$, is called the *derived morphism* of f. [50]

Remark 4.1.C.

If G/S and H/S satisfy condition E, then $L(f)$ respects the $\mathcal{O}_S$-module structures obtained from "functoriality in $\mathcal{M}$" (cf. 3.6). [51]

[49] Editor's note: the numbering 4.1.A and 4.1.B has been added in order to bring these definitions into relief.

[50] Editor's note: if f is a monomorphism, then so are $L(f)$ and $T(f)$; cf. 3.7.

[51] Editor's note: this remark has been added.

Proposition 4.1.1.

Let $g \in G(S)$. Then

$$\mathrm{Ad}(g) : \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$

is the derived morphism of $\mathrm{Int}(g) : G \to G$.

Indeed, $\mathrm{Ad}(g)X = i^{-1}(\mathrm{Int}(g)i(X))$, which is precisely $L(\mathrm{Int}(g))(X)$ by the definition of the derived morphism.

Suppose that G/S satisfies E. By Proposition 3.9, the group structure just defined on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is exactly the one induced by its $\mathcal{O}_S$-module structure (which is defined by (E)). Proposition 4.1.1 and the functoriality of the $\mathcal{O}_S$-action (cf. 3.6) therefore yield the following corollary.

Corollary 4.1.1.1.

Suppose that G/S satisfies E. Then Ad maps G into the subgroup

$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

of $\underline{\mathrm{Aut}}_{S\text{-gr.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$. In other words, for every $g \in G(S')$, the automorphism $\mathrm{Ad}(g)$ respects the $\mathcal{O}_{S'}$-module structure on $\underline{\mathrm{Lie}}(G'/S', \mathcal{M})$, where $G' = G \times_S S'$. Thus Ad is a *linear representation* (cf. I 3.2) of G on the $\mathcal{O}_S$-module $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Remarks 4.1.1.2.

a) The functor G/S satisfies E if and only if, for every pair $(\mathcal{M}, \mathcal{N})$ of finite free $\mathcal{O}_S$-modules, the following diagram, obtained by applying $\underline{\mathrm{Lie}}(G/S, -)$ to diagram $(*)$ of 2.1, is cartesian.

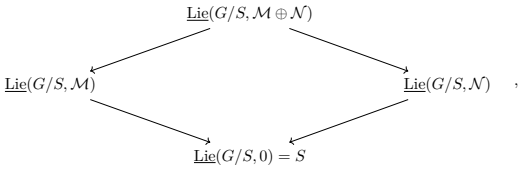

Figure 21: The cartesian criterion for condition E.

b) Suppose that G/S satisfies E. The derived morphism of the group law $\pi : G \times_S G \to G$ is the addition law on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$. Notice that π is not a group morphism, but $\pi(e, e) = e$; hence the derived morphism $L(\pi)$ maps

$$T^{(e,e)}_{(G\times_S G)/S}(\mathcal{M}) = \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$

to $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$; cf. 3.7 and 3.8. Likewise, for $n \in \mathbb{Z}$, let $n_G : G \to G$ denote the S-functor morphism $g \mapsto g^n$. Then $L(n_G)$ is multiplication by n on $\underline{\mathrm{Lie}}(G/S)$; cf. 3.9.4.

4.1.2.0.

Consider the S-functor $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$.[52] For every $S' \to S$, one has $T_{G/S}(\mathcal{M})_{S'} \simeq T_{G_{S'}/S'}(\mathcal{M})$ (cf. 3.4), and therefore

$$\begin{aligned}&\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))(S')\\ &\quad\simeq \mathrm{Hom}_{G_{S'}}(G_{S'}, T_{G_{S'}/S'}(\mathcal{M})) = \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}).\end{aligned}$$

[52] Editor's note: the original has been expanded in what follows to display the action of G on the functors $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$ and $\underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$.

First note the isomorphism, functorial in S',

$$\mathrm{Hom}_{S'}(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})) \xrightarrow{\sim} \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}), \qquad (*)$$

which sends $f : G_{S'} \to \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ to the section $s_f : G_{S'} \to T_{G_{S'}/S'}(\mathcal{M})$ defined, for every $S'' \to S'$ and $g \in G(S'')$, by

$$s_f(g) = i(f(g))s(g).$$

Let h be an automorphism of the functor $G_{S'}$ over S', not necessarily preserving the group structure. To every section τ of $T_{G_{S'}/S'}(\mathcal{M})$ one associates, by transport of structure, a section $h(\tau)$: it is the unique section that makes the following diagram commute.

$$\begin{array}{ccc} G_{S'} & \xrightarrow{\tau} & T_{G_{S'}/S'}(\mathcal{M}) \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle T(h)} \\ G_{S'} & \xrightarrow{h(\tau)} & T_{G_{S'}/S'}(\mathcal{M}) \end{array} .$$

Figure 22: Transport of a tangent section by an automorphism h.

In particular, take $h = t_x$, right translation by an element $x \in G(S')$; thus $h(g) = t_x(g) = g \cdot x$ for all $g \in G(S'')$, $S'' \to S'$. One then has

$$t_x(s_f) = s_{t_x(f)},$$

where $t_x(f) : G_{S'} \to \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ is given by

$$t_x(f)(g) = f(g \cdot x^{-1}).$$

Let G act by right translations on

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \quad \text{and} \quad \underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

as follows. For $S' \to S$ and $x \in G(S')$, let

$$\sigma \in \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}),$$
$$f \in \mathrm{Hom}_{S'}(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})).$$

Set

$$(\sigma \cdot x)(g) = \sigma(g \cdot x^{-1}) \cdot s(x), \qquad (f \cdot x)(g) = f(g \cdot x^{-1})$$

for every $g \in G(S'')$, $S'' \to S'$. The isomorphism $(*)$ respects these G-actions.

Consequently, under this isomorphism, the elements of

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$$

correspond to the constant morphisms from $G_{S'}$ to $\underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$, that is, to the morphisms factoring through $G_{S'} \to S'$; equivalently, they correspond to the elements of $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$.

Terminology. The elements of $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ are called the sections of $T_{G_{S'}/S'}(\mathcal{M})$ invariant under right translation.

Proposition 4.1.2.

The map

$$\underline{\mathrm{Lie}}(G/S,\mathcal{M})(S) \longrightarrow \Gamma(T_{G/S}(\mathcal{M})/G), \qquad X \longmapsto (x \mapsto X\cdot x),$$

is a bijection onto the set of sections invariant under right translation. [53]

Likewise, let G act on the right on $\underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$ by

$$(u\cdot x)(g) = u(g\cdot x^{-1})\cdot x,$$

for $S'\to S$, $x\in G(S')$, $u\in \mathrm{End}_{I_{S'}(\mathcal{M})}(G_{I_{S'}(\mathcal{M})})$, and $g\in G(S'')$, $S''\to I_{S'}(\mathcal{M})$. The morphism of 3.12.1

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \longrightarrow \underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$$

respects the G-actions. For each $S'\to S$, it therefore induces a bijection between $\Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and the set of $I_{S'}(\mathcal{M})$-endomorphisms u of $G_{I_{S'}(\mathcal{M})}$ that induce the identity on G and "commute with right translations," meaning that $u_{S''}\cdot x = u_{S''}$ for every $S''\to S'$ and $x\in G(S'')$.

Proposition 4.1.3.

There is a bijection, functorial in G, between $\underline{\mathrm{Lie}}(G/S,\mathcal{M})(S)$ and the set of $I_S(\mathcal{M})$-endomorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G, in the sense just specified.

Taking 3.13 into account gives the following theorem.

Theorem 4.1.4.

Let G be an S-functor in groups, and suppose that G/S satisfies E. Then $\underline{\mathrm{Lie}}(G/S,\mathcal{M})(S)$ is identified, functorially in G, with the group of $I_S(\mathcal{M})$-automorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G.

For $\mathcal{M}=\mathcal{O}_S$, this recovers one of the classical definitions of the Lie algebra of a group.

4.2.0.

Before proceeding, let us establish further consequences of 3.11.[54] Let X, Y be over Z, with Z over S, as in Section 1. As was seen in 3.11, the isomorphisms 1.1(4)

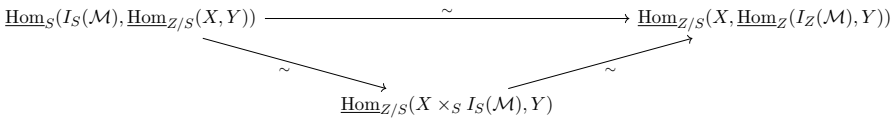

Figure 23: The adjunction isomorphisms of 1.1(4).

induce the following isomorphism θ.

[53] Statements 4.1.2, 4.1.3, and 4.1.4 can be obtained more simply by observing that the automorphisms of G invariant under right translations are the left translations. Editor's note: see, for example, the proofs of Propositions 4.6 and 2.5 of [DG70], §II.4.

[54] Editor's note: paragraph 4.2.0 has been added; its results are used implicitly in 4.2 and 4.7 of the original.

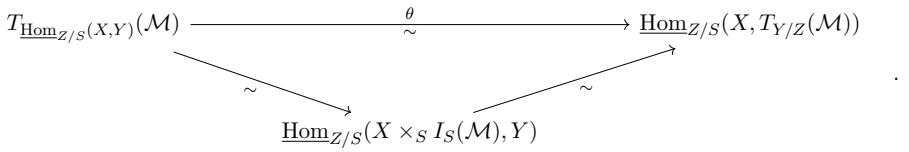

Figure 24: The induced tangent-space isomorphism θ.

By 1.3, if Y is a Z-group, then $\underline{\mathrm{Hom}}_Z(V, Y)$ is a Z-group for every $V \to Z$, in particular for $V = I_Z(\mathcal{M})$. Explicitly, if $Z'' \to Z' \to Z$ and $\varphi, \psi \in \mathrm{Hom}_Z(V_{Z'}, Y)(Z')$, their product is defined by

$$(\varphi \cdot \psi)(v) = \varphi(v)\psi(v) \qquad (v \in V_{Z'}(Z'')).$$

Suppose now that X and Y are Z-groups.

Definition 4.2.0.1.

We denote by $\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)$ the subfunctor of $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ defined, for every $S' \to S$, by

$$\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}\text{-gr.}}(X_{S'}, Y_{S'}). \tag{3}$$

The same definition applies with Y replaced by the Z-group $T_{Y/Z}(\mathcal{M})$.

Under the isomorphisms (2) above, $T_{\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X,Y)/S}(\mathcal{M})(S')$ corresponds to the $Z_{S'}$-morphisms

$$\varphi : X_{S'} \times_{S'} I_{S'}(\mathcal{M}) \longrightarrow Y_{S'}$$

that are multiplicative in X, that is,

$$\varphi(x_1 x_2, m) = \varphi(x_1, m)\varphi(x_2, m).$$

These in turn correspond to the morphisms of $Z_{S'}$-groups $X_{S'} \to T_{Y/Z}(\mathcal{M})_{S'}$. Hence:

Proposition 4.2.0.2.

Let X, Y be Z-groups, where Z is over S. There are isomorphisms of S-functors, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, T_{Y/Z}(\mathcal{M})).$$

Taking $Z = S$ gives the following consequence. If Y is a commutative S-group, then so is $T_{Y/S}(\mathcal{M})$, and hence so are $H = \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, Y)$, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M}))$, and $T_{H/S}(\mathcal{M})$.

Corollary 4.2.0.3.

Let X, Y be S-groups. There are isomorphisms of S-functors, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M})).$$

If Y is commutative, these are moreover isomorphisms of abelian S-groups.

Definition 4.2.0.4.

If Y is an $\mathcal{O}_S$-module, the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$, carries an $\mathcal{O}_S$-module structure induced by that of Y. Equipped with this structure, it will be denoted $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$. [55]

Consequently, if X, Y are $\mathcal{O}_S$-modules, then $T'_{Y/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), Y)$. If $H = \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)$, then $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M}))$ and $T'_{H/S}(\mathcal{M})$ also carry $\mathcal{O}_S$-module structures.

Corollary 4.2.0.5.

If X, Y are $\mathcal{O}_S$-modules, there are isomorphisms of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$T'_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})).$$

Definition 4.2.A.

Let X, L be S-groups, with X acting on L by group automorphisms (cf. I 2.3.5). We define the subfunctor $\underline{Z}^1_S(X,L)$ of $\underline{\mathrm{Hom}}_S(X,L)$ by

$$\underline{Z}^1_S(X,L)(S') = \{\, \varphi \in \mathrm{Hom}_{S'}(X_{S'}, L_{S'}) \mid \varphi(x_1x_2) = \varphi(x_1)\big(x_1 \cdot \varphi(x_2)\big), \\ x_1, x_2 \in X(S''),\ S'' \to S' \,\}.$$

It is called the *functor of crossed homomorphisms* from X to L. [56]

Remark 4.2.B.

a) If L' is another S-group on which X acts by group automorphisms, then

$$\underline{Z}^1_S(X, L \times_S L') \simeq \underline{Z}^1_S(X,L) \times_S \underline{Z}^1_S(X,L').$$

b) If L is an X-$\mathcal{O}_S$-module, then $\underline{Z}^1_S(X,L)$ is the kernel of the differential

$$\partial : \underline{\mathrm{Hom}}_S(X,L) \longrightarrow \underline{\mathrm{Hom}}_S(X^2, L)$$

defined in I 5.1. In particular, $\underline{Z}^1_S(X,L)$ is then an $\mathcal{O}_S$-module.

4.2.

Let $u : X \to Y$ be a morphism of S-groups. We describe $L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M})$. First, 3.11.3 gives an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})). \qquad (\dagger)$$

Since Y is an S-group,

$$T_{Y/S}(\mathcal{M}) = \underline{\mathrm{Lie}}(Y/S,\mathcal{M}) \cdot Y = \underline{\mathrm{Lie}}(Y/S,\mathcal{M})_Y,$$

[55] Editor's note: the prime is used to distinguish the $\mathcal{O}_S$-module structure induced by that of Y from the one that will be obtained from condition E.

[56] Editor's note: the original has been expanded below; in particular, Definition 4.2.A and Remark 4.2.B have been added.

and one obtains an isomorphism, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

For $S' \to S$, let $u' : X' \to Y'$ be the base change of u. The S-functor[57]

$$\underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y)$$

is defined by

$$\begin{aligned}&\underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y)(S') \\ &\quad = \mathrm{Hom}_{Y'\text{-gr.}}(X', (\underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y)_{S'}) \\ &\quad = \mathrm{Hom}_{Y'\text{-gr.}}(X', \underline{\mathrm{Lie}}(Y'/S', \mathcal{M}) \cdot Y').\end{aligned}$$

Thus (†) induces an isomorphism

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y). \qquad (\dagger')$$

On the other hand, the composite

$$X \xrightarrow{\ u\ } Y \xrightarrow{\ \mathrm{Ad}\ } \underline{\mathrm{Aut}}_{S\text{-gr.}}(\underline{\mathrm{Lie}}(Y/S, \mathcal{M}))$$

Figure 25: The action of X on $\underline{\mathrm{Lie}}(Y/S, \mathcal{M})$.

defines an action of X on $L = \underline{\mathrm{Lie}}(Y/S, \mathcal{M})$ by group automorphisms.

Let $\Phi \in \underline{\mathrm{Hom}}_{Y/S}(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y)$. For $S'' \to S' \to S$ and $x \in X(S'')$, there is a unique expression

$$\Phi(S')(x) = \varphi(S')(x) \cdot u'(x), \qquad \varphi(S')(x) \in \underline{\mathrm{Lie}}(Y'/S', \mathcal{M})(S'').$$

This determines an element $\varphi \in \underline{\mathrm{Hom}}_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}))$. The map $\Phi(S')$ is a group morphism if and only if, for all $x_1, x_2 \in X(S'')$,

$$\begin{aligned}\varphi(S')(x_1x_2) &= \varphi(S')(x_1)(u(x_1)\varphi(S')(x_2)u(x_1)^{-1}) \\ &= \varphi(S')(x_1)(x_1 \cdot \varphi(S')(x_2)),\end{aligned}$$

that is, precisely when $\varphi \in \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}))$.

Proposition 4.2.

Let $u : X \to Y$ be a morphism of S-groups. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

Suppose in addition that Y/S satisfies E. It follows from 4.2.0.3, just as in the proof of 3.11.1, that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X, Y)/S$ satisfies E. Thus (cf. 3.5.1)

$$\begin{aligned}L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M} \oplus \mathcal{N}) \simeq\ & L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \\ & \times_S L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{N}).\end{aligned}$$

This also follows directly from Proposition 4.2 and Remark 4.2.B*a*. Consequently, both sides of the isomorphism in Proposition 4.2 carry $\mathcal{O}_S$-module structures obtained from functoriality in $\mathcal{M}$, and that isomorphism respects them.

[57] Editor's note: this is Definition 4.2.0.1 applied with $Z = Y$, to the Y-groups $u : X \to Y$ and $T_{Y/S}(\mathcal{M}) \to Y$.

Proposition 4.2 bis.

Let $u : X \to Y$ be a morphism of S-groups, and suppose that Y/S satisfies E. There is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

[58]

When Y/S satisfies E, 3.13.1 also gives, just as in the proof of 3.13.2, an isomorphism functorial in $\mathcal{M}$, for every $u \in \underline{\mathrm{Isom}}_{S\text{-gr.}}(X, Y)$,

$$L^u_{\underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}). \qquad (*)$$

This yields the following two corollaries.

Corollary 4.2.1.

Let $u : X \to Y$ be a morphism of S-groups. If Y/S satisfies E, there is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

Corollary 4.2.2.

Let X be an S-group. If X/S satisfies E, there is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{S\text{-gr.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(X/S, \mathcal{M})).$$

If Y is commutative, its adjoint action on $L = \underline{\mathrm{Lie}}(Y/S, \mathcal{M})$ is trivial, so $\underline{Z}^1_S(X, L) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, L)$. [59]

Corollary 4.2.3.

Let Y be a commutative S-group. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

4.3.

We now consider the case in which X and Y are $\mathcal{O}_S$-modules. Recall (cf. 4.2.0.4) that $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}'(Y/S, \mathcal{M})$, denotes the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}(Y/S, \mathcal{M})$, equipped with the $\mathcal{O}_S$-module structure induced by that of Y.

When Y/S satisfies E, we shall always write $\underline{\mathrm{Lie}}(Y/S, \mathcal{M})$ for this functor equipped instead with the $\mathcal{O}_S$-module structure defined for every functor satisfying E. By 3.9, the underlying abelian-group structures of $\underline{\mathrm{Lie}}(Y/S, \mathcal{M})$ and $\underline{\mathrm{Lie}}'(Y/S, \mathcal{M})$ coincide, but their module structures need not coincide (see the counterexample in 6.3). For $S' \to S$, $a \in \mathcal{O}(S')$, and $m \in \underline{\mathrm{Lie}}(Y/S, \mathcal{M})(S')$, we write $a \cdot' m$, respectively $a \cdot m$, for the action coming from Y, respectively the action obtained from condition E. We use the same convention for the two actions on $T_{Y/S}(\mathcal{M})$.

[58] Editor's note: the original has been expanded from this point; in particular, Proposition 4.2 bis has been added.

[59] Editor's note: the following sentence has been added.

There is an isomorphism of $\mathcal{O}_S$-modules

$$T'_{Y/S}(\mathcal{M}) \simeq \underline{\mathrm{Lie}}'(Y/S, \mathcal{M}) \oplus Y.$$

Exactly as for Propositions 4.2 and 4.2 bis, one therefore obtains:

Proposition 4.3.

Let $u : X \to Y$ be a morphism of $\mathcal{O}_S$-modules. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, \underline{\mathrm{Lie}}'(Y/S, \mathcal{M})). \qquad (*)$$

If Y/S satisfies E, then $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)/S$ satisfies E, and $(*)$ is an isomorphism of $\mathcal{O}_S$-modules when both sides carry the structures obtained from functoriality in $\mathcal{M}$. [60] [61]

Remark 4.3 bis.

Let $u : X \to Y$ be a morphism of $\mathcal{O}_S$-modules.[62] Denote by τ_u the map that sends each $\mathcal{O}_S$-linear morphism $\varphi : X \to \underline{\mathrm{Lie}}'(Y/S, \mathcal{M})$ to

$$u \oplus \varphi : X \longrightarrow T'_{Y/S}(\mathcal{M}) = Y \oplus \underline{\mathrm{Lie}}'(Y/S, \mathcal{M}).$$

The isomorphism of 4.3 belongs to the following commutative diagram, functorial in $\mathcal{M}$.

$$\begin{array}{ccc} L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, \underline{\mathrm{Lie}}'(Y/S, \mathcal{M})) \\ \downarrow & & \downarrow{\scriptstyle \tau_u} \\ T_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})) \end{array} .$$

Figure 26: Compatibility of the isomorphism of 4.3 with τ_u.

Moreover, when Y/S satisfies E, 3.13.1 gives, just as in the proof of 3.13.2, for every $u \in \underline{\mathrm{Isom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)$,

$$L^u_{\underline{\mathrm{Isom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) = L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}). \qquad (*)$$

Corollary 4.3.1.

Let X be an $\mathcal{O}_S$-module satisfying E over S. There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, \underline{\mathrm{Lie}}'(X/S, \mathcal{M})),$$

and this isomorphism respects the $\mathcal{O}_S$-module structures obtained from functoriality in $\mathcal{M}$. In particular, $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S$ satisfies E. [63]

[60] Editor's note: the following sentence has been added.

[61] Editor's note: on the right-hand side, the scalar action is $(a\varphi)(x) = a\varphi(x)$, where $a\varphi(x)$ denotes the action of a on $\varphi(x) \in \underline{\mathrm{Lie}}(Y/S, \mathcal{M})$. This differs from the action $(a \cdot' \varphi)(x) = a \cdot' \varphi(x) = \varphi(ax)$, but the two agree when $\underline{\mathrm{Lie}}(Y/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(Y/S, \mathcal{M})$.

[62] Editor's note: this remark has been added; it will be useful in 4.5.1.

[63] Editor's note: compare editor's note (61). The final sentence and its proof have also been added.

Proof. The first assertion follows from (∗) and 4.3. For the second, since X/S satisfies E, there are isomorphisms of $\mathcal{O}_S$-modules

$$\underline{\mathrm{Lie}}'(X/S, \mathcal{M} \oplus \mathcal{N}) \simeq \underline{\mathrm{Lie}}'(X/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(X/S, \mathcal{N}).$$

Consequently,

$$\begin{aligned} &\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M} \oplus \mathcal{N}) \\ &\quad \simeq \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{N}). \end{aligned}$$

Remark 4.1.1.2*a* now proves the assertion.

4.3.2.

Before continuing in this direction, let us examine more closely the relations among Y, $\underline{\mathrm{Lie}}(Y/S)$, and $\underline{\mathrm{Lie}}'(Y/S)$. First,

$$\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(\mathcal{O}_S/S, \mathcal{M}) = \mathbf{W}(\mathcal{M}), \tag{1}$$

where $\mathbf{W}(\mathcal{M})$ is defined in I 4.6. There is therefore a canonical isomorphism

$$d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S). \tag{2}$$

Let F be an $\mathcal{O}_S$-module. For every $S_2 \to S_1 \to S$, there is a dihomomorphism

$$\begin{cases} F(S_1) \longrightarrow F(S_2), \\ \mathcal{O}(S_1) \longrightarrow \mathcal{O}(S_2), \end{cases} \tag{3}$$

and hence a morphism of $\mathcal{O}(S_2)$-modules

$$F(S_1) \otimes_{\mathcal{O}(S_1)} \mathcal{O}(S_2) \longrightarrow F(S_2).$$

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In particular, take $S_1 = S'$ and $S_2 = I_{S'}(\mathcal{M})$. One obtains morphisms of $\mathcal{O}(S')$-modules, functorial in $\mathcal{M}$,

$$F(S') \otimes_{\mathcal{O}(S')} T_{\mathcal{O}_S/S}(\mathcal{M})(S') \longrightarrow T'_{F/S}(\mathcal{M})(S').$$

As S' varies, these give morphisms of $\mathcal{O}_S$-modules

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}), \tag{4}$$

functorial in $\mathcal{M}$. Compatibility with the tangent-bundle projections onto their bases then yields morphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}), \tag{5}$$

and the following diagram commutes.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) & \longrightarrow & F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) & \longrightarrow & T'_{F/S}(\mathcal{M}) & \longrightarrow & F & \longrightarrow & 0 \end{array} .$$

Figure 27: The morphisms (4) and (5) in the split tangent sequences.

[64] Editor's note: $S' \to S$ has been replaced by $S_2 \to S_1 \to S$, since both S_2 and S_1 must vary below. The original has also been expanded in what follows.

The morphisms (5) may be regarded as morphisms of abelian S-groups

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}). \tag{6}$$

Tensoring F with the isomorphism $d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S)$ gives, when $\mathcal{M} = \mathcal{O}_S$, a morphism of abelian S-groups

$$F \xrightarrow{\sim} F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S) \longrightarrow \underline{\mathrm{Lie}}(F/S), \tag{7}$$

also denoted $d : F \to \underline{\mathrm{Lie}}(F/S)$.

Remark 4.3.3.

When F/S satisfies E, the morphisms (6) and (7) need not be $\mathcal{O}_S$-linear if both sides are equipped with the module structures obtained from $\mathcal{M}$ by condition E.

For example, let k be a field of characteristic $p > 0$, let $S = \mathrm{Spec}(k)$, and let F be the $\mathcal{O}_S$-module that assigns to every S-scheme T the set $F(T) = \Gamma(T, \mathcal{O}_T)$, with scalars acting through the p-th power:

$$r \cdot f = r^p f \qquad (r \in \mathcal{O}(T),\ f \in F(T)).$$

As an S-functor in groups, $F \simeq \mathbf{G}_{a,S}$. Hence F satisfies E and $\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbf{G}_{a,S}/S) \simeq \mathcal{O}_S$. For every $T \to S$, the canonical morphism $d : F(T) \to \mathcal{O}(T)$ is the identity: it respects the abelian-group structure but not the $\mathcal{O}_S$-module structure. [65]

Remark 4.3.4.

The morphisms (4) and (5) can be made explicit as follows. The morphism

$$\Theta : F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), F)$$

is defined, for $S' \to S$, $\alpha \in \mathcal{O}(I_{S'}(\mathcal{M}))$, and $f : S' \to F$, by

$$\Theta(f \otimes \alpha) = \alpha \cdot (\tau_0 \circ f) = \alpha \cdot (f \circ \rho),$$

where $\tau_0 : F \to T'_{F/S}(\mathcal{M})$ is the zero section and $\rho : I_{S'}(\mathcal{M}) \to S'$ is the structural morphism.

The morphism Θ induces

$$\theta : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}).$$

This follows from the functoriality in $\mathcal{M}$ noted after (4), and can also be seen directly. Indeed,

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M})(S') = \{\, \varphi \in \mathrm{Hom}_S(I_{S'}(\mathcal{M}), F) \mid \varphi \circ \varepsilon_{\mathcal{M}} = e \,\},$$

where $e : S' \to F$ is the unit section. On the other hand, $\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M})(S')$ equals $\Gamma(S', \mathcal{M})$, the kernel of the augmentation

$$\eta : \mathcal{O}(I_{S'}(\mathcal{M})) \longrightarrow \mathcal{O}(S').$$

It remains to see that if $f \in F(S')$ and $\alpha \in \Gamma(S', \mathcal{M})$, then $\alpha \cdot (f \circ \rho) \circ \varepsilon_{\mathcal{M}} = e$.

Apply the dihomomorphism (3) to the S-morphism $\varepsilon_{\mathcal{M}} : S' \to I_{S'}(\mathcal{M})$. The following diagram commutes.

[65] Editor's note: the original asserted that, when F/S satisfies E, (6), and hence (7), are morphisms of $\mathcal{O}_S$-modules. The counterexample in 6.3 contradicts this. That assertion has been removed, the preceding example inserted, and Definition 4.4 modified accordingly. This does not affect what follows.

$$\begin{array}{ccc} F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S') \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\eta(\alpha)} \\ F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S') \end{array} .$$

Figure 28: Compatibility of scalar action with $F(\varepsilon_{\mathcal{M}})$.

Thus, for every $\varphi : I_{S'}(\mathcal{M}) \to F$,

$$(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = \eta(\alpha) \cdot (\varphi \circ \varepsilon_{\mathcal{M}}),$$

and therefore $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = e$ when $\eta(\alpha) = 0$.

In particular, taking $\mathcal{M} = t\mathcal{O}_S$, one has $\underline{\mathrm{Lie}}(\mathcal{O}_S/S) = t\mathcal{O}_S$, and the morphism $F \to \underline{\mathrm{Lie}}'(F/S)$ is $f \mapsto t \cdot (f \circ \rho)$.
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Remark 4.3.5.

Let F be an $\mathcal{O}_S$-module, set

$$E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

and denote by d_F and d_E the $\mathcal{O}_S$-linear morphisms given by 4.3.2(5):

$$d_F : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}),$$
$$d_E : E \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(E/S, \mathcal{M}).$$

Remark 4.3.4 gives the following commutative diagram of $\mathcal{O}_S$-linear morphisms.

$$\begin{array}{ccc} E \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) & \xrightarrow{\quad d_E \quad} & \underline{\mathrm{Lie}}'(E/S, \mathcal{M}) \\ \| & & \sim\uparrow{\scriptstyle(*)} \\ \underline{\mathrm{Hom}}_{\mathcal{O}_S}(F, F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \underline{\mathrm{Hom}}_{\mathcal{O}_S}(F, \underline{\mathrm{Lie}}'(F/S, \mathcal{M})) \end{array}$$

Figure 29: Compatibility of d_E and d_F.

The right vertical arrow is the isomorphism $(*)$ of 4.3. Thus, by that result, if F/S satisfies E, then E/S satisfies E, and $(*)$ is also an isomorphism of $\mathcal{O}_S$-modules when the right-hand terms are equipped with the module structures obtained from E. [67]

Remark 4.4.0.

In 4.3.2, the morphisms (4) are isomorphisms if and only if the morphisms (5) are. If these equivalent conditions hold, then F/S satisfies E. Indeed, it is enough to verify that

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M} \oplus \mathcal{N}) \simeq \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(F/S, \mathcal{N}).$$

Under the hypothesis, the horizontal arrows in the following commutative diagram are isomorphisms.

[66] Editor's note: this remark has been added; it will be useful in 4.6.2.
[67] Editor's note: this remark has been added; it will be useful in 4.5 and 4.7.

$$\begin{array}{ccc} F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M} \oplus \mathcal{N}) & \xrightarrow{\sim} & \underline{\mathrm{Lie}}'(F/S, \mathcal{M} \oplus \mathcal{N}) \\ \downarrow \wr & & \downarrow \\ F \otimes_{\mathcal{O}_S} (\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{N})) & \xrightarrow{\sim} & \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(F/S, \mathcal{N}) \end{array} ;$$

Figure 30: The direct-sum comparison used to verify condition E.

The second vertical arrow is therefore an isomorphism as well, which is exactly condition E for F/S. [68]

Definition 4.4.

An $\mathcal{O}_S$-module F is called *good* if the morphisms

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T_{F/S}(\mathcal{M}),$$

or, equivalently,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}),$$

are isomorphisms of abelian S-groups (so that F/S satisfies (E)), and if moreover they respect the $\mathcal{O}_S$-module structures obtained from condition E.

Corollary 4.4.1.

Let F be an $\mathcal{O}_S$-module, and consider the following conditions:[69]

(i) F is a good $\mathcal{O}_S$-module;

(ii) F/S satisfies E, and $d : F \to \underline{\mathrm{Lie}}(F/S)$ is an isomorphism of $\mathcal{O}_S$-modules;

(iii) $\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$.

Then

$$\text{(i)} \iff \text{(ii)} \implies \text{(iii)}.$$

Proof. The implication (i) $\Rightarrow$ (ii) follows from the definition. To prove (ii) $\Rightarrow$ (i), it remains to show that the morphisms of abelian S-groups, functorial in $\mathcal{M}$,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(F/S, \mathcal{M})$$

are isomorphisms of $\mathcal{O}_S$-modules. Since F/S satisfies E, both sides carry finite direct sums of copies of $\mathcal{O}_S$ to finite products of abelian S-groups. We are therefore reduced to the case $\mathcal{M} = \mathcal{O}_S$, which is precisely the hypothesis.

Finally, (i) $\Rightarrow$ (iii) follows from the definition and from the fact that the isomorphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$$

of 4.3.2(5) are $\mathcal{O}_S$-linear.

[68] Editor's note: in view of the changes made in Definition 4.4 (cf. editor's note (65)), this remark has been inserted here; in the original it occurred in the proof of 4.4.1.

[69] Editor's note: the original proved that (i) implies (ii) and (iii); the implication (ii) implies (i) has been added.

Examples 4.4.2.

For every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, the $\mathcal{O}_S$-modules $\mathbf{V}(\mathcal{E})$ and $\mathbf{W}(\mathcal{E})$ defined in I 4.6 are good.

Indeed, for every $f : S' \to S$, the morphisms

$$\mathbf{V}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) \longrightarrow T_{\mathbf{V}(\mathcal{E})/S}(\mathcal{M})(S'),$$
$$\mathbf{W}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) \longrightarrow T_{\mathbf{W}(\mathcal{E})/S}(\mathcal{M})(S')$$

correspond respectively to

$$\begin{aligned} &\mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), \mathcal{O}_{S'}) \otimes_{\mathcal{O}(S')} \Gamma\big(S', D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \\ &\qquad \longrightarrow \mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}\big(f^*(\mathcal{E}), D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \qquad \text{(V)} \\ &\Gamma(S', f^*(\mathcal{E})) \otimes_{\mathcal{O}(S')} \Gamma\big(S', D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \\ &\qquad \longrightarrow \Gamma\big(S', f^*(\mathcal{E}) \otimes_{\mathcal{O}_{S'}} D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \qquad \text{(W)}. \end{aligned}$$

These are isomorphisms because, as an $\mathcal{O}_{S'}$-module, $D_{\mathcal{O}_{S'}}(\mathcal{M})$ is isomorphic to a finite direct sum of copies of $\mathcal{O}_{S'}$. [70]

Proposition 4.5. *Let F be a good $\mathcal{O}_S$-module. Then:*

(i) The S-group $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ satisfies condition (E), and there is a functorial isomorphism

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}\left(F, \underline{\mathrm{Lie}}(F/S, \mathcal{M})\right),$$

which respects the $\mathcal{O}_S$-module structures obtained from condition (E). In particular, there is an isomorphism of $\mathcal{O}_S$-modules

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \xrightarrow{\sim} \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

(ii)[71] *Moreover, $\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$ is a good $\mathcal{O}_S$-module.*

Indeed, by 4.4.1, F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \simeq F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathbf{O}_S/S, \mathcal{M}). \tag{1}$$

Assertion (i) now follows from 4.3.1. Put $E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$. By (1) and Remark 4.3.5, one has the following commutative diagram of morphisms of abelian S-groups.

$$\begin{array}{ccc} \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F) \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) & \xrightarrow{d_E} & \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \\ \| & & \sim\uparrow(*) \\ \underline{\mathrm{Hom}}_{\mathcal{O}_S}(F, F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \underline{\mathrm{Hom}}_{\mathcal{O}_S}(F, \underline{\mathrm{Lie}}(F/S, \mathcal{M})) \end{array}$$

Figure 31: The comparison diagram for the differential d_E.

Here d_F and the vertical isomorphism $(*)$ are isomorphisms of $\mathcal{O}_S$-modules; consequently, so is d_E. This proves (ii).

Scholium 4.5.1. [72] Put (cf. 2.1) $\mathcal{O}_{I_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$, with $t^2 = 0$, and let F be a good $\mathcal{O}_S$-module. For every $S' \to S$, the morphism

$$F(S') \oplus tF(S') = F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}) \longrightarrow F(I_{S'}) = F(S') \oplus \underline{\mathrm{Lie}}(F/S)(S'),$$

[70] Editor's note: the preceding proof has been added.

[71] Editor's note: part (ii), an immediate consequence of what precedes, has been added.

[72] Editor's note: this scholium, implicit in the original, has been added for use in 4.7.1. Here and below, t, t', and so forth denote square-zero variables; the letter ε is reserved for the unit section of a group.

which is the identity on $F(S')$, induces an isomorphism of $\mathcal{O}(S')$-modules

$$tF(S') \xrightarrow{\sim} \underline{\mathrm{Lie}}(F/S)(S').$$

As S' varies, this gives an isomorphism that we may write

$$\underline{\mathrm{Lie}}(F/S) \simeq tF.$$

Thus, for every $S' \to S$, Proposition 4.5 gives a commutative diagram.

$$\begin{array}{ccccc}
\underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}, tF_{S'}) & \xrightarrow{\sim} & \underline{\mathrm{Lie}}\left(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S\right)(S') \\
 & & \downarrow & & \downarrow \\
 & & \underline{\mathrm{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}}) & = & T_{\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S}(S')
\end{array}$$

Figure 32: Endomorphisms and automorphisms over the dual numbers.

It follows from 4.3 bis that every

$$X \in \underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$$

corresponds to $\mathrm{id} + tX \in \underline{\mathrm{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$.

Definition 4.6. An S-functor in groups G is called *good* if G/S satisfies condition (E) and $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module.

[73] Notice that if F is a good $\mathcal{O}_S$-module, then it is a good S-group: indeed, F/S satisfies (E), and $\underline{\mathrm{Lie}}(F/S) \simeq F$ is a good $\mathcal{O}_S$-module.

Example 4.6.1. *If G is representable, it is good. Indeed, G/S satisfies (E), and $\underline{\mathrm{Lie}}(G/S)$ has the form $V(E)$, hence is good by 4.4.2.*

Lemma 4.6.2. [74] *Let G be an S-functor in groups such that G/S satisfies (E), and put $F = \underline{\mathrm{Lie}}(G/S)$. Then F satisfies (E), and the morphism of abelian groups*

$$d : F \longrightarrow \underline{\mathrm{Lie}}(F/S)$$

respects the $\mathcal{O}_S$-module structures. Consequently, G is good if and only if $F \to \underline{\mathrm{Lie}}(F/S)$ is bijective.

Proof. Suppose that G/S satisfies (E). Let $\mathcal{M}, \mathcal{N}$ be finite free $\mathcal{O}_S$-modules, write

$$F(\mathcal{N}) = \underline{\mathrm{Lie}}(G/S, \mathcal{N}),$$

and let e denote the unit section of G. For every $S' \to S$,

$$F(\mathcal{N})(S') = \left\{ g \in \mathrm{Hom}_S(I_{S'}(\mathcal{N}), G) \mid g \circ \varepsilon_{\mathcal{N}} = e \right\}.$$

Moreover, $\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M})(S')$ identifies with the set of morphisms

$$\left\{ \Phi \in \mathrm{Hom}_S(I_{S'}(\mathcal{N}) \times_{S'} I_{S'}(\mathcal{M}), G) \;\middle|\; \begin{array}{l} \Phi \circ (\varepsilon_{\mathcal{N}} \times \mathrm{id}_{I_{S'}(\mathcal{M})}) = e \in G(I_{S'}(\mathcal{M})), \\ \Phi \circ (\mathrm{id}_{I_{S'}(\mathcal{N})} \times \varepsilon_{\mathcal{M}}) = e \in G(I_{S'}(\mathcal{N})). \end{array} \right\}.$$

[73] Editor's note: the following sentence has been added.
[74] Editor's note: this lemma has been added.

This proves

$$\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}) \simeq \underline{\mathrm{Lie}}(F(\mathcal{M})/S, \mathcal{N}). \tag{1}$$

Since G/S satisfies (E), it follows that

$$\begin{aligned}\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) &\simeq \underline{\mathrm{Lie}}(F(\mathcal{M}_1 \oplus \mathcal{M}_2)/S, \mathcal{N}) \\ &\simeq \underline{\mathrm{Lie}}((F(\mathcal{M}_1) \times_S F(\mathcal{M}_2))/S, \mathcal{N}).\end{aligned}$$

By 3.8, the last term is isomorphic to

$$\begin{aligned}&\underline{\mathrm{Lie}}(F(\mathcal{M}_1)/S, \mathcal{N}) \times_S \underline{\mathrm{Lie}}(F(\mathcal{M}_2)/S, \mathcal{N}) \\ &\quad \simeq \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2).\end{aligned}$$

Hence

$$\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) \simeq \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2), \tag{2}$$

so $F(\mathcal{N})/S$ satisfies (E), by Remark 4.1.1.2(a).

It remains to show that the morphism of abelian groups

$$d : F(\mathcal{N}) \longrightarrow \underline{\mathrm{Lie}}(F(\mathcal{N})/S)$$

respects the $\mathcal{O}_S$-module structures. Consider the free $\mathcal{O}_S$-module $\mathcal{M} = t\mathcal{O}_S$. Then

$$\mathcal{O}(I_{S'}(\mathcal{M})) = \mathcal{O}(S')[t]/(t^2) = \mathcal{O}(S') \oplus t\mathcal{O}(S'), \qquad \underline{\mathrm{Lie}}(\mathbf{O}_S/S) \simeq t\mathcal{O}_S.$$

Let $\rho_t : I_S(\mathcal{M}) \to S$ be the structural morphism, corresponding to the injection

$$u_t : \mathcal{O}_S \hookrightarrow D_{\mathcal{O}_S}(\mathcal{M}) = \mathcal{O}_S \oplus t\mathcal{O}_S.$$

Recall (cf. 3.4.2) that for every S-functor X, $F(\mathcal{N})(X)$ is the set of S-morphisms $\phi : I_X(\mathcal{N}) \to G$ such that $\phi \circ \varepsilon_{\mathcal{N}} = e$, and the action of $a \in \mathcal{O}(X)$ is

$$a \cdot \phi = \phi \circ a^*,$$

where a^* is the endomorphism of $I_X(\mathcal{N})$ associated with a; see 2.1.3.

Consequently, by 4.3.4, the composite

$$d : F(\mathcal{N}) \xrightarrow{\sim} F(\mathcal{N}) \otimes_{\mathcal{O}_S} t\mathcal{O}_S \longrightarrow \underline{\mathrm{Lie}}(F(\mathcal{N})/S)$$

is given, for $S' \to S$ and $f \in \mathrm{Hom}_S(S', F(\mathcal{N}))$, by

$$f \longmapsto f \otimes t \longmapsto t \cdot (f \circ \rho_t) = f \circ \rho_t \circ t^*.$$

Here $f \circ \rho_t \in F(\mathcal{N})(I_{S'}(\mathcal{M}))$ is viewed as an S-morphism

$$\phi : I_{I_{S'}(\mathcal{M})}(\mathcal{N}) \longrightarrow G$$

such that $\phi \circ \varepsilon_{\mathcal{N}} = e$. We must prove that d is $\mathcal{O}_S$-linear, or equivalently that

$$d(a \cdot f) = u_t(a) \cdot d(f) \qquad (a \in \mathcal{O}(S')).$$

Viewing $f \in F(\mathcal{N})(S')$ as a morphism $I_{S'}(\mathcal{N}) \to G$, one likewise has $a \cdot f = f \circ a^*$. Functoriality in X of the action of $\mathcal{O}(X)$ on $I_X(\mathcal{N})$ gives the following commutative diagram.

$$\begin{array}{ccc} I_{S'}(\mathcal{N}) & \xrightarrow{a^*} & I_{S'}(\mathcal{N}) \\ {\scriptstyle \rho_t}\downarrow & & \downarrow{\scriptstyle \rho_t} \\ I_{I_{S'}(\mathcal{M})}(\mathcal{N}) & \xrightarrow{w_t(a)^*} & I_{I_{S'}(\mathcal{M})}(\mathcal{N}) \end{array}$$

Figure 33: Compatibility of scalar action with the structural morphism ρ_t.

Therefore

$$\begin{aligned} d(a \cdot f) &= f \circ a^* \circ \rho_t \circ t^* \\ &= f \circ \rho_t \circ u_t(a)^* \circ t^* \\ &= f \circ \rho_t \circ t^* \circ u_t(a)^* = u_t(a) \cdot d(f). \end{aligned}$$

The penultimate equality follows from the commutativity of $\mathcal{O}$. This completes the proof of Lemma 4.6.2.

Theorem 4.7. *If F is a good $\mathcal{O}_S$-module, then the S-group $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-}mod.}(F)$ is good.*

[75] Indeed, by 4.5, $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S$ satisfies (E), while

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a good $\mathcal{O}_S$-module.

4.7.1.

[76] Let G be an S-group and F a good $\mathcal{O}_S$-module. Suppose given a linear representation of G in F, that is (I 4.7.1), a morphism of S-groups

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F).$$

If G/S satisfies (E), then 4.1.C and 4.5 give a morphism of $\mathcal{O}_S$-modules, denoted ρ' or $d\rho$,

$$\underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

[77] Furthermore, put $\mathcal{O}_{I_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$, with $t^2 = 0$. It follows from 4.5.1 that, if $S' \to S$ and

$$X \in \underline{\mathrm{Lie}}(G/S)(S') \subset G(I_{S'}),$$

then in $\underline{\mathrm{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$ one has

$$\rho(X) = \mathrm{id} + t\rho'(X). \qquad (*)$$

In other words, for every $S'' \to I_{S'}$ and $f \in F(S'')$,

$$\rho(X)(f) = f + t\rho'(X)(f) \quad \text{in } F(S'').$$

Definition 4.7.2. Let G be a good S-group. Then $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module, and there is a morphism of S-groups

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S)).$$

[75] Editor's note: the original has been simplified here by using the additions made in 4.3.1 and 4.5.
[76] Editor's note: the numbering 4.7.1, 4.7.2, and 4.7.3 has been added.
[77] Editor's note: the following sentence has been added.

By 4.7.1, it induces a morphism of $\mathcal{O}_S$-modules

$$\mathrm{ad} : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S)\big),$$

or, equivalently, an $\mathcal{O}_S$-bilinear morphism

$$\underline{\mathrm{Lie}}(G/S) \times_S \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S), \qquad (x, y) \longmapsto [x, y] = \mathrm{ad}(x) \cdot y.$$

Here x, y are arbitrary elements of

$$\underline{\mathrm{Lie}}(G/S)(S') = \underline{\mathrm{Lie}}(G_{S'}/S')(S').$$

If G is commutative, then $[x, y] = 0$.

4.7.3.

The bracket admits the following equivalent description. It is enough to consider sections over S, that is,

$$x, y \in \underline{\mathrm{Lie}}(G/S)(S).$$

There is a canonical isomorphism $I_S \times_S I_S \simeq I_{I_S}$. To avoid confusion, let I and I' denote two copies of I_S, and put

$$\mathcal{O}_I = \mathcal{O}_S[t], \qquad \mathcal{O}_{I'} = \mathcal{O}_S[t'], \qquad t^2 = 0 = (t')^2.$$

There is then a commutative square

$$\begin{array}{ccc} I \times_S I' & \longrightarrow & I' \\ \downarrow & & \downarrow \\ I & \longrightarrow & S \end{array}$$

Figure 34: The two dual-number structures on $I \times_S I'$.

in which the two arrows from $I \times I'$ identify it with the scheme of dual numbers over I or over I', respectively. Writing $L = \underline{\mathrm{Lie}}(G/S)$, one obtains the following commutative diagram of groups.

$$\begin{array}{ccccccccc} & & & & 1 & & 1 & & \\ & & & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & L(I') & \longrightarrow & G(I \times I') & \longrightarrow & G(I') & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & L(S) & \longrightarrow & G(I) & \longrightarrow & G(S) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 1 & & 1 & & 1 & & \end{array}$$

Figure 35: The first exact-sequence diagram used to construct the bracket.

The ninth object in this diagram is $\underline{\mathrm{Lie}}(L/S)(S)$. If G is good, this is $L(S)$, and hence one obtains the next commutative diagram. Its rows and columns are exact sequences of groups; the five groups denoted $L(-)$ are commutative.[78] Under the identifications

$$L(I) = L(S) \oplus tL(S), \qquad L(I') = L(S) \oplus t'L(S),$$

[78] Editor's note: the original has been expanded in what follows.

the injections $L(S) \hookrightarrow L(I)$ and $L(S) \hookrightarrow L(I')$ are $u \mapsto tu$ and $u \mapsto t'u$, respectively.

$$\begin{array}{ccccc}
z \in L(S) & \xrightarrow{\ t\ } & L(I) & \longrightarrow & L(S) \ni x \\
\Big\downarrow{\scriptstyle t'} & & \Big\downarrow & & \Big\downarrow \\
L(I') & \longrightarrow & G(I \times I') & \longrightarrow & G(I') \\
\Big\downarrow & & \Big\downarrow & & \Big\downarrow \\
y \in L(S) & \longrightarrow & G(I) & \longrightarrow & G(S)
\end{array}$$

Figure 36: The exact-sequence diagram defining $[x, y]$.

In such a diagram, take x, y as indicated and choose arbitrary lifts $\widetilde{x} \in L(I)$ of x and $y' \in L(I')$ of y. The commutator

$$\widetilde{x}\, y'\, \widetilde{x}^{-1}(y')^{-1} \quad \text{in } G(I \times I')$$

is independent of the chosen lifts and is the image of a uniquely determined element z as shown. One has $z = [x, y]$. [79]

Indeed, use the same letters x and y for their images under the canonical sections $L(S) \to L(I)$ and $L(S) \to L(I')$. Then

$$\widetilde{x} = xu, \qquad y' = yv, \qquad u, v \in L(S) = L(I) \cap L(I').$$

Since $L(I)$ and $L(I')$ are commutative,

$$\begin{aligned}
\widetilde{x}\, y'\, \widetilde{x}^{-1}(y')^{-1} &= xuyvu^{-1}x^{-1}v^{-1}y^{-1} \\
&= xuyu^{-1}vx^{-1}v^{-1}y^{-1} \\
&= xyx^{-1}y^{-1}.
\end{aligned}$$

This element maps to the unit in both $G(I)$ and $G(I')$, and therefore comes from the unique z indicated above. Finally, viewing y as an element of $L_{I'}(I')$, and x as an element of $L(S) \subset G(I')$, formula 4.7.1 $(*)$ gives

$$xyx^{-1} = \mathrm{Ad}(x)(y) = (\mathrm{id} + t' \,\mathrm{ad}(x))(y) = y + t'[x, y].$$

Thus $xyx^{-1}y^{-1} \in L(I')$ is the image of $z = [x, y] \in L(S)$.

Two properties are now apparent from this construction:

(i) the bracket is functorial in G. More precisely,

$$G \longmapsto \underline{\mathrm{Lie}}(G/S)$$

is a functor from good S-groups to good $\mathcal{O}_S$-modules equipped with an $\mathcal{O}_S$-bilinear composition law;

(ii) $[x, y] + [y, x] = 0$, since the diagram is symmetric about its main diagonal.[80]

[79] Editor's note: the original said in a footnote that the author, at Gabriel's request, acknowledged that the exercise was not immediate, which was precisely why it had not been placed in the text. The details have been supplied here, taking account of the addition in 4.7.1; see also [DG70], §II.4, 4.2.

[80] Editor's note: that is, x and y play symmetric roles. The image of $[y, x]$ in $G(I \times I')$ is the commutator $y'\widetilde{x}(y')^{-1}\widetilde{x}^{-1}$, the inverse of $\widetilde{x}y'\widetilde{x}^{-1}(y')^{-1} = [x, y]$. On the other hand (cf. 4.10 below), it is not known at this point whether $[x, x] = 0$ necessarily holds: if x is represented by $\widetilde{x} \in L(I)$ and by $x' \in L(I')$, it is not a priori clear that $\widetilde{x}$ and x' commute.

Proposition 4.8. *Let F be a good $\mathcal{O}_S$-module. Under the identification*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

one has

$$\mathrm{Ad}(g)\cdot Y = g\circ Y\circ g^{-1}, \qquad [X,Y] = X\circ Y - Y\circ X,$$

for every $S'\to S$, $g\in \underline{\mathrm{Aut}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$, and

$$X,Y\in \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S)\,(S') = \underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}).$$

Proof.[81] After base change, we may assume $S'=S$. Put $I=I_S$ and $\mathcal{O}_I=\mathcal{O}_S[t]$, with $t^2=0$. Recall from 4.5.1 that the inclusion

$$i:\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)\hookrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_I\text{-mod.}}(F_I)$$

sends Y to $\mathrm{id}+tY$. By the definition of $\mathrm{Ad}(g)$ in 4.1.A,

$$\mathrm{id}+t\,\mathrm{Ad}(g)(Y) = g\circ(\mathrm{id}+tY)\circ g^{-1} = \mathrm{id}+t(g\circ Y\circ g^{-1}),$$

and therefore $\mathrm{Ad}(g)(Y)=g\circ Y\circ g^{-1}$.

Let I' be a second copy of I_S, and put $\mathcal{O}_{I'}=\mathcal{O}_S[t']$, with $(t')^2=0$. Apply 4.7.3 to

$$G=\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F), \qquad L=\underline{\mathrm{Lie}}(G/S)=\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

Here $L=\mathrm{End}(F)$ is the differential identification of Proposition 4.5. Identify X with its image under the canonical section $L(S)\hookrightarrow L(I)$. Its image in $G(I\times I')$ is $\mathrm{id}+t'X$, whose inverse is $\mathrm{id}-t'X$. Similarly, Y lifts to $\mathrm{id}+tY$, with inverse $\mathrm{id}-tY$. Their commutator is

$$\begin{aligned}(\mathrm{id}+t'X)\circ(\mathrm{id}+tY)\circ(\mathrm{id}-t'X)\circ(\mathrm{id}-tY)\\ = \mathrm{id}+tt'(X\circ Y - Y\circ X).\end{aligned}$$

This is the image in $G(I\times I')$ of $Z=X\circ Y-Y\circ X\in L(S)$: first Z maps to $tZ\in L(I)$, and then to $\mathrm{id}+t't Z\in G(I\times I')$. By 4.7.3, $[X,Y]=X\circ Y-Y\circ X$.

Corollary 4.8.1. *Let G be a good S-group and let $x,y,z\in \underline{\mathrm{Lie}}(G/S)(S')$. Then*

$$[x,[y,z]]+[y,[z,x]]+[z,[x,y]]=0.$$

[82] Indeed, since G is good, $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module; hence Theorem 4.7 shows that

$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S)\big)$$

is a good S-group. The morphism

$$\mathrm{Ad}: G\longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S)\big)$$

and the functoriality in 4.7.3(i) yield

$$\mathrm{ad}[x,y] = [\mathrm{ad}\,x,\mathrm{ad}\,y].$$

Together with Proposition 4.8, this gives

$$\mathrm{ad}[x,y] = \mathrm{ad}\,x\circ\mathrm{ad}\,y - \mathrm{ad}\,y\circ\mathrm{ad}\,x.$$

[81] Editor's note: the original has been expanded and simplified in what follows.
[82] Editor's note: the original has been expanded in what follows.

Applying both sides to z gives the Jacobi identity.

Corollary 4.8.2. *Let G be a good S-group acting linearly on a good $\mathcal{O}_S$-module F; equivalently, let F be a G-$\mathcal{O}_S$-module, with G and F good. Then the linear map*

$$\rho' : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a representation; that is,

$$\rho'([x,y]) = \rho'(x) \circ \rho'(y) - \rho'(y) \circ \rho'(x).$$

Scholium 4.9. To every good S-group G (for example, every representable one) we have functorially associated a good $\mathcal{O}_S$-module $\underline{\mathrm{Lie}}(G/S)$ equipped with a bilinear map satisfying

$$[x,y] + [y,x] = 0, \qquad [x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0.$$

We shall call $\underline{\mathrm{Lie}}(G/S)$, with this structure, the *Lie algebra* of G over S. The qualification is that it is not yet known whether it is an $\mathcal{O}_S$-Lie algebra in the strict sense.[83] Every linear representation of G in a good $\mathcal{O}_S$-module F induces a representation of its Lie algebra. In particular, the adjoint representation of G induces the adjoint representation of its Lie algebra.

Definition 4.10. An S-functor in groups G is called *very good* if it is good and $\underline{\mathrm{Lie}}(G/S)$ is an $\mathcal{O}_S$-Lie algebra, that is, if $[x,x] = 0$ identically.

Examples 4.10.1. The following S-groups are very good: $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ for every good $\mathcal{O}_S$-module F (by 4.7 and 4.8); every representable group (as shown below); every good S-group admitting a monomorphism into a very good S-group, in particular every good subgroup functor of a representable group; and every good S-group admitting a faithful linear representation in a good $\mathcal{O}_S$-module, for example every good S-group for which Ad is a monomorphism.

4.11.

Suppose now that G is a group scheme over S. By 4.1.4, $\underline{\mathrm{Lie}}(G/S)(S)$ identifies with the group of infinitesimal automorphisms of G/S invariant under right translation; by 3.14, this is the group of derivations of $\mathcal{O}_G$ over $\mathcal{O}_S$ invariant under right translation. This identification respects the module structure and is an anti-isomorphism of Lie algebras.[84]

To see this, argue as in 4.7.3. Put

$$\mathcal{O}_I = \mathcal{O}_S[t], \qquad \mathcal{O}_{I'} = \mathcal{O}_S[t'], \qquad t^2 = 0 = (t')^2,$$

and let $x \in L(I)$, $y \in L(I')$, where $L = \underline{\mathrm{Lie}}(G/S)$. Left translation λ_x, respectively λ_y, is an S-automorphism of $G_{I\times I'}$ inducing the identity on $G_{I'}$, respectively on G_I. It corresponds to the $\mathcal{O}_S$-automorphism

$$u = \mathrm{id} + t\, d_x, \qquad \text{respectively} \qquad v = \mathrm{id} + t'\, d_y$$

of

$$\mathcal{O}_{G_{I\times I'}} = \mathcal{O}_G[t,t']/(t^2,(t')^2),$$

[83] Editor's note: the unresolved point is whether $[x,x] = 0$; see 4.10.

[84] Editor's note: a sign error in the original has been corrected here; cf. [DG70], §II.4, 4.4 and 4.6.

where d_x, d_y are $\mathcal{O}_S$-derivations of $\mathcal{O}_G$ invariant under right translation. The correspondence between S-automorphisms of $G_{I\times I'}$ and $\mathcal{O}_S$-automorphisms of $\mathcal{O}_{G_{I\times I'}}$ is contravariant. Therefore

$$\lambda_x\lambda_y\lambda_x^{-1}\lambda_y^{-1}$$

corresponds to

$$v^{-1}u^{-1}vu = \mathrm{id} + tt'(d_yd_x - d_xd_y).$$

It follows from 4.7.3 that

$$x \longmapsto -d_x$$

is an isomorphism of Lie algebras; for further details, see [DG70], §II.4, 4.4 and 4.6. The same argument applies to

$$\underline{\mathrm{Lie}}(G/S)(S') = \underline{\mathrm{Lie}}(G_{S'}/S')(S')$$

for every $S' \to S$. Thus one recovers the classical definition.[85]

Scholium 4.11.1. *Via the isomorphism $x \mapsto -d_x$, $\underline{\mathrm{Lie}}(G/S)$ identifies with the functor that assigns to each $S' \to S$ the $\mathcal{O}(S')$-Lie algebra of derivations of $G_{S'}$ relative to S' invariant under right translation.*

Since every representable group is already known to be good by 4.6.1, the preceding discussion gives:

Corollary 4.11.2. *Every representable group is very good.*

Let $\varepsilon : S \to G$ be the unit section of G, and put

$$\omega^1_{G/S} = \varepsilon^*(\Omega^1_{G/S}).$$

Recall from 3.3 that $\underline{\mathrm{Lie}}(G/S)$ is represented by the vector fibration $V(\omega^1_{G/S})$.

Scholium 4.11.3. *Thus to every S-group scheme G we have functorially associated a vector fibration*

$$\mathrm{Lie}(G/S) = V(\omega^1_{G/S})$$

over S, representing the functor $\underline{\mathrm{Lie}}(G/S)$, and hence endowed with the structure of an S-scheme in $\mathcal{O}_S$-Lie algebras. Moreover, this construction commutes with base extension and finite products (cf. 3.4 and 3.8).

Remarks 4.11.4. [86] Let $\pi : G \to S$ be the structural morphism.

(a) The $\mathcal{O}_G$-module $\Omega^1_{G/S}$ is naturally $(G\times_S G)$-equivariant (cf. I, §6). Hence, by I 6.8.1,

$$\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S}).$$

For example, $\Omega^1_{G/S}$ is locally free, respectively locally free of finite type, whenever $\omega^1_{G/S}$ is. This holds in particular when S is the spectrum of a field, respectively when S is the spectrum of a field and G is locally of finite type over S.

(b) By I 6.8.2, $\omega^1_{G/S}$ has a canonical G-$\mathcal{O}_S$-module structure, which induces the adjoint action on

$$V(\omega^1_{G/S}) = \mathrm{Lie}(G/S).$$

[85] Editor's note: the original has been expanded in what follows. In particular, the numbering 4.11.1–4.11.8 has been added for later reference.

[86] Editor's note: what follows has been expanded, taking account of the additions made in Expose I, §6.8.

[87]

(c) By EGA I, 5.3.11 and 5.4.6, $\varepsilon : S \to G$ is an immersion; it is a closed immersion if G is separated over S. Thus $\omega^1_{G/S}$ identifies with $\mathcal{I}/\mathcal{I}^2$, where $\mathcal{I}$ is the quasi-coherent ideal defining $\varepsilon(S)$ in an open subscheme $U \subset G$ in which $\varepsilon(S)$ is closed. If $G \to S$ is separated, one may take $U = G$. If $G = \operatorname{Spec} A(G)$ is affine over S, then $\mathcal{I}$ is the augmentation ideal of $A(G)$, namely the kernel of

$$\varepsilon^\natural : A(G) \longrightarrow \mathcal{O}_S$$

(cf. I 4.2).[88]

Remark 4.11.5. The isomorphism $\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S})$ identifies the $\mathcal{O}_S$-module $\omega^1_{G/S}$ with the sheaf

$$\pi_*(\Omega^1_{G/S})^G$$

of differentials of G relative to S invariant under right translation: its sections over an open $U \subset S$ are the sections of $\Omega^1_{G/S}$ over $\pi^{-1}(U)$ invariant under right translation (cf. I 6.8.3; compare also VII$_\mathrm{A}$, 2.4).[89]

Notation 4.11.6. Write $\mathscr{L}ie(G/S)$ for the sheaf of sections of the vector fibration $\mathrm{Lie}(G/S) \to S$. It is the $\mathcal{O}_S$-module

$$\mathscr{L}ie(G/S) = (\omega^1_{G/S})^\vee = \mathcal{H}om_{\mathcal{O}_S}(\omega^1_{G/S}, \mathcal{O}_S),$$

dual to $\omega^1_{G/S}$ (cf. EGA II, 1.7.9), and it carries a structure of $\mathcal{O}_S$-Lie algebra.

In general this construction does not commute with base extension. Consequently, the Lie-algebra structure on this module does not determine the structure of an S-scheme in $\mathcal{O}_S$-Lie algebras on $\mathrm{Lie}(G/S)$.[90]

Nevertheless:

Lemma 4.11.7. *Suppose that $\omega^1_{G/S}$ is locally free of finite type. This is the case, in particular, if G is smooth over S (cf. EGA IV$_4$, 17.2.4), or if S is the spectrum of a field and G is locally of finite type over S. Then*

$$\mathscr{L}ie(G/S)^\vee \simeq (\omega^1_{G/S})^{\vee\vee} \simeq \omega^1_{G/S},$$

and therefore

$$\mathrm{Lie}(G/S) = V(\omega^1_{G/S}) = V\big(\mathscr{L}ie(G/S)^\vee\big) = W\big(\mathscr{L}ie(G/S)\big),$$

where the final equality follows from I 4.6.5.1.

Finally, let $G \to H$ be a monomorphism of functors in groups. Then

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(H/S)$$

[87] Editor's note: for this reason, the linear action of G on $\omega^1_{G/S}$ could be called the pre-adjoint action; by an abuse of terminology it will still be called the adjoint action. A slightly more general construction, used in Expose III (especially III 0.8), is as follows. Suppose that to every $Y \to S$ one assigns an $\mathcal{O}_Y$-module $\mathcal{N}(Y)$, and to every S-morphism $\phi : Z \to Y$ a functorial morphism of $\mathcal{O}_Z$-modules $\mathcal{N}(\phi) : \phi^*\mathcal{N}(Y) \to \mathcal{N}(Z)$. For example, if $\mathcal{J}$ is a quasi-coherent ideal of $\mathcal{O}_S$, one may take $\mathcal{N}(Y) = \mathcal{J}\mathcal{O}_Y$. The action of G on $\omega^1_{G/S}$ then induces a generalized adjoint action of G on the S-functor in abelian groups that assigns to $f : Y \to S$ the group $\mathrm{Hom}_{\mathcal{O}_Y}(f^*\omega^1_{G/S}, \mathcal{N}(Y))$.

[88] Editor's note: the preceding description of $\omega^1_{G/S}$, which is useful later (for example in VII$_\mathrm{A}$, 5.5), has been added.

[89] Editor's note: this description of $\omega^1_{G/S}$ in terms of invariant differentials is not used below.

[90] Editor's note: the reason is that $\mathscr{L}ie(G/S) \simeq (\omega^1_{G/S})^\vee$ need not determine $\omega^1_{G/S}$. For example, let $S = \mathbb{A}^1_k$, where k is a field, and let G be the S-group whose fiber at $s = 0$ is $\mathbb{G}_{a,k}$, while all other fibers are the unit group. Then $\mathscr{L}ie(G/S) = 0$, but $\mathrm{Lie}(G_s/k)(k) = k$.

is also a monomorphism (cf. 3.7). Every vector monomorphism of vector fibrations is a closed immersion.[91] It follows that:

Corollary 4.11.8. *Let $G \hookrightarrow H$ be a monomorphism of S-group schemes.*
(i) The morphism

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(H/S)$$

is a closed immersion; consequently

$$\omega^1_{H/S} \longrightarrow \omega^1_{G/S}$$

is an epimorphism.
(ii) If $\omega^1_{G/S}$ is locally free of finite type, then the corresponding morphism

$$\mathscr{L}ie(G/S) \longrightarrow \mathscr{L}ie(H/S)$$

identifies $\mathscr{L}ie(G/S)$ with an $\mathcal{O}_S$-submodule of $\mathscr{L}ie(H/S)$ that is locally a direct summand.

5. Computation of some Lie algebras

5.1. Examples of Lie algebras: diagonalizable groups

Let $G = D_S(M)$ be a diagonalizable group over S (I 4.4). Since the formation of $\underline{\mathrm{Lie}}(G/S)$ commutes with base extension, it is enough to carry out the construction for $G = D(M)$. We then have

$$G(I_S) = \mathrm{Hom}_{\mathrm{gr}}\big(M, \Gamma(I_S, \mathscr{O}_{I_S})^\times\big) = \mathrm{Hom}_{\mathrm{gr}}\big(M, \Gamma(S, \mathscr{D}_{\mathscr{O}_S})^\times\big).$$

There is a split exact sequence

$$1 \longrightarrow \Gamma(S, \mathscr{O}_S) \longrightarrow \Gamma(S, \mathscr{D}_{\mathscr{O}_S})^\times \longrightarrow \Gamma(S, \mathscr{O}_S)^\times \longrightarrow 1,$$

which shows that $\underline{\mathrm{Lie}}(G)(S)$ identifies with $\mathrm{Hom}_{\mathrm{gr}}(M, \mathscr{O}(S))$, endowed with its evident $\mathscr{O}(S)$-module structure. After base change we therefore obtain:

Proposition 5.1. *There are isomorphisms*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, \mathbf{O}_S) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S)$$

and

$$\mathscr{H}om_{\mathrm{gr}}(\widetilde{M}_S, \mathscr{O}_S) \xrightarrow{\sim} \mathscr{L}ie(D_S(M)/S),$$

where, in the second isomorphism, $\widetilde{M}_S$ denotes the constant sheaf of groups on S defined by M, and $\mathscr{H}om_{\mathrm{gr}}$ denotes the sheaf of homomorphisms of sheaves of groups.

Corollary 5.1.1. *If M is free of finite type—or, as we shall say later, if $D_S(M)$ is a split torus—then, with W as in I 4.6.5,*

$$W\big(\mathscr{L}ie(D_S(M)/S)\big) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S), \qquad M^\vee \otimes_{\mathbb{Z}} \mathscr{O}_S \xrightarrow{\sim} \mathscr{L}ie(D_S(M)/S).$$

Here $M^\vee$ denotes the dual of the abelian group M. In particular,

$$\mathbf{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathbb{G}_{m,S}/S), \qquad \mathscr{O}_S \xrightarrow{\sim} \mathscr{L}ie(\mathbb{G}_{m,S}/S).$$

[91] Editor's note: let $f : M \to N$ be a morphism of $\mathcal{O}_S$-modules and let $P = \mathrm{Coker}(f)$. If $V(N) \to V(M)$ is a monomorphism, the surjective morphism $\mathrm{Sym}(N) \to \mathrm{Sym}(P)$ factors through $\mathcal{O}_S$; hence $P = 0$.

5.2. Normalizers and centralizers

We first prove a few lemmas. Recall (I 3.1.1) that a sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

of $\mathbf{O}_S$-modules is called *exact* if, for every $S' \to S$, the sequence

$$0 \longrightarrow F'(S') \longrightarrow F(S') \longrightarrow F''(S') \longrightarrow 0$$

of $\mathbf{O}(S')$-modules is exact. Likewise, a sequence $1 \to G' \to G \to G'' \to 1$ of S-groups is called exact if the corresponding sequence of groups on S'-points is exact for every $S' \to S$.

Lemma 5.2.1. *Let $1 \to G' \to G \to G'' \to 1$ be an exact sequence of S-groups.* [92]

(i) *The sequences*

$$1 \longrightarrow T_{G'/S}(\mathscr{M}) \longrightarrow T_{G/S}(\mathscr{M}) \longrightarrow T_{G''/S}(\mathscr{M}) \longrightarrow 1$$

and

$$1 \longrightarrow \underline{\mathrm{Lie}}(G'/S, \mathscr{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathscr{M}) \longrightarrow \underline{\mathrm{Lie}}(G''/S, \mathscr{M}) \longrightarrow 1$$

are exact.

(ii) *Let $1 \to H' \to H \to H'' \to 1$ be a second sequence of groups. It is exact if and only if*

$$1 \longrightarrow G' \times_S H' \longrightarrow G \times_S H \longrightarrow G'' \times_S H'' \longrightarrow 1$$

is exact.

(iii) *If two of G', G, G'' satisfy condition (E), then so does the third.*

(iv) *If $0 \to F' \to F \to F'' \to 0$ is exact and two of the $\mathbf{O}_S$-modules F', F, F'' are good, then so is the third.*

(v) *If two of the S-groups G', G, G'' are good, then so is the third.*

The first part of (i) is immediate, and the second follows from it. Likewise, (ii) is immediate. We prove (iii). For brevity, write $L(\mathscr{M}) = \underline{\mathrm{Lie}}(G, \mathscr{M})$, $L'(\mathscr{M}) = \underline{\mathrm{Lie}}(G', \mathscr{M})$, and similarly for L''. We have the following commutative diagram. [93]

$$\begin{array}{ccccccccc}
1 & \longrightarrow & L'(\mathcal{M}\oplus\mathcal{N}) & \longrightarrow & L(\mathcal{M}\oplus\mathcal{N}) & \longrightarrow & L''(\mathcal{M}\oplus\mathcal{N}) & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & L'(\mathcal{M}) \times_S L'(\mathcal{N}) & \longrightarrow & L(\mathcal{M}) \times_S L(\mathcal{N}) & \longrightarrow & L''(\mathcal{M}) \times_S L''(\mathcal{N}) & \longrightarrow & 1
\end{array}$$

Figure 37: The exact-sequence diagram used to prove Lemma 5.2.1(iii).

The upper row is exact by (i), and the lower row is exact by (i) and (ii). Assertion (iii) follows.

We prove (iv). There is a commutative diagram

[92] Editor's note: The statement and proof of this lemma have been expanded; this will be useful in 5.3.1.
[93] Editor's note: The proofs of (iii) and (v) have been added.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F'' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & T_{F'/S}(\mathcal{M}) & \longrightarrow & T_{F/S}(\mathcal{M}) & \longrightarrow & T_{F''/S}(\mathcal{M}) & \longrightarrow & 0 \end{array}$$

Figure 38: The exact-sequence diagram used to prove Lemma 5.2.1(iv).

with exact rows: the first because $T_{\mathbf{O}_S/S}(\mathscr{M})$ is a free, hence flat, $\mathbf{O}_S$-module, and the second by (i). Thus, if two of F', F, F'' are good, so is the third. Finally, (v) follows from (iii) and (iv).

Lemma 5.2.2. *Let G be an S-group, let E, H be two G-objects, and let F be a G-$\mathbf{O}_S$-module.*

(i) *The canonical homomorphism $E^G \times_S H^G \to (E \times_S H)^G$ is an isomorphism.*

(ii) *If F is a good $\mathbf{O}_S$-module, then so is F^G.*

The first assertion is immediate; we prove the second. For every $S' \to S$, there is a commutative diagram

$$\begin{array}{ccc} F^G(I_{S'}(\mathcal{M})) & \longrightarrow & F(I_{S'}(\mathcal{M})) \\ \varphi \downarrow & & \uparrow \varphi_1 \sim \\ F^G(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) & \xrightarrow{\sim} & F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) \end{array}$$

Figure 39: The invariants-and-base-change square in the proof of Lemma 5.2.2.

and we must prove that ϕ is bijective. It is plainly injective. To prove surjectivity, let $(t_0, \ldots, t_n)$ be a basis of the free $\mathbf{O}(S')$-module $\mathbf{O}(I_{S'}(\mathscr{M}))$, and let

$$u = \sum_{i=0}^{n} f_i \otimes t_i$$

be an element of $F(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathscr{M}))$ such that $\phi_1(u)$ belongs to $F^G(I_{S'}(\mathscr{M}))$. We show that each f_i belongs to $F^G(S')$.

Let $S'' \to S'$. Via the zero section $\varepsilon_{\mathscr{M}}$, we may regard S'' as lying over $I_{S'}(\mathscr{M})$. Then, for every $g \in G(S'')$,

$$u_{S''} = g \cdot u_{S''} = \sum_{i=0}^{n} g \cdot (f_i)_{S''} \otimes (t_i)_{S''}.$$

Since the $(t_i)_{S''}$ form a basis of

$$\mathbf{O}(I_{S''}(\mathscr{M})) \simeq \mathbf{O}(I_{S'}(\mathscr{M})) \otimes_{\mathbf{O}(S')} \mathbf{O}(S'')$$

over $\mathbf{O}(S'')$, it follows that $g \cdot (f_i)_{S''} = (f_i)_{S''}$, hence $f_i \in F^G(S')$ for all i.

Notation. If E is an S-group and F a sub-S-group, write E/F for the S-functor sending $S' \to S$ to the set $E(S')/F(S')$ of classes $\bar{x} = xF(S')$. If E is commutative, then E/F is a commutative S-group. [94]

[94] Editor's note: These notations have been expanded.

Now let G be an S-group and K a sub-S-group, and set $E = \underline{\mathrm{Lie}}(G/S, \mathscr{M})$ and $F = \underline{\mathrm{Lie}}(K/S, \mathscr{M})$. The adjoint action of K on E stabilizes F, and therefore induces an action on E/F. For every $S' \to S$,

$$(E/F)^K(S') = \{\bar{x} \in E(S')/F(S') \mid \text{for every } f : S'' \to S' \text{ and } k \in K(S''), \\ f^*(\bar{x}^{-1})\,\mathrm{Ad}(k)(f^*(\bar{x})) \in F(S'')\},$$

where $f^*(\bar{x})$ denotes the image of $\bar{x}$ in $E(S'')$.

Theorem 5.2.3. *Let G be an S-group and K a sub-S-group. Put (I 2.3.3)*

$$N = \underline{\mathrm{Norm}}_G(K), \qquad Z = \underline{\mathrm{Centr}}_G(K),$$

and let K act on $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ through the adjoint representation of G.

(i) *If the group law of $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ is commutative,*[95][96] *then*

$$\frac{\underline{\mathrm{Lie}}(N/S, \mathscr{M})}{\underline{\mathrm{Lie}}(K/S, \mathscr{M})} = \left(\frac{\underline{\mathrm{Lie}}(G/S, \mathscr{M})}{\underline{\mathrm{Lie}}(K/S, \mathscr{M})}\right)^K.$$

(ii) *If the same group law is commutative, then*

$$\underline{\mathrm{Lie}}(Z/S, \mathscr{M}) = \underline{\mathrm{Lie}}(G/S, \mathscr{M})^K.$$

(iii) *If G satisfies (E), then Z satisfies (E); if G and K satisfy (E), then N satisfies (E).*

(iv) *If G is good, then Z is good; if, moreover, G is very good, then Z is very good.*

(v) *If G and K are good, then N is good; if, moreover, G is very good, then N is very good.*

To prove (i) and (ii),[97] we use the following lemma, which follows from the diagram of exact sequences considered in 4.1.B, with G and H interchanged.

Lemma 5.2.3.0. *Let G be an S-group, H a sub-S-group, and $\mathscr{M}$ a free $\mathscr{O}_S$-module of finite type. Then $T_{H/S}(\mathscr{M})$ and $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ are sub-S-groups of $T_{G/S}(\mathscr{M})$, and*

$$T_{H/S}(\mathscr{M}) \cap \underline{\mathrm{Lie}}(G/S, \mathscr{M}) = \underline{\mathrm{Lie}}(H/S, \mathscr{M}),$$

where

$$T_{H/S}(\mathscr{M}) \cap \underline{\mathrm{Lie}}(G/S, \mathscr{M}) := T_{H/S}(\mathscr{M}) \times_{T_{G/S}(\mathscr{M})} \underline{\mathrm{Lie}}(G/S, \mathscr{M}).$$

Since the functors in (i) and (ii) commute with base extension, it is enough to prove the equalities on S-points. Put $H = N$ (respectively, $H = Z$) and let $\alpha = \pm 1$. By the preceding lemma and the definitions of N and Z (I 2.3.3),

$$\underline{\mathrm{Lie}}(H/S, \mathscr{M})(S) = \big\{X \in \underline{\mathrm{Lie}}(G/S, \mathscr{M})(S) \subset G(I_S(\mathscr{M})) \;\big|\; \\ \text{for every } f : S' \to I_S(\mathscr{M}) \text{ and } u \in K(S'), \\ f^*(X^\alpha)\, u\, f^*(X^{-\alpha})\, u^{-1} \in K(S'), \\ \text{respectively } f^*(X)\, u\, f^*(X^{-1})\, u^{-1} = 1\big\}. \tag{*}$$

[95] This condition is automatic when G satisfies (E), for example when G is representable.

[96] Editor's note: For an example in which the S-group $\underline{\mathrm{Lie}}(G/S)$ is noncommutative, see 6.3.

[97] Editor's note: The proof has been expanded, and in particular Lemma 5.2.3.0, implicit in the original, has been added.

Here $f^*(X)$ denotes the image of X in $G(S')$.

For brevity, write

$$\mathfrak{g} = \underline{\mathrm{Lie}}(G/S, \mathscr{M}), \quad \mathfrak{k} = \underline{\mathrm{Lie}}(K/S, \mathscr{M}), \quad \mathfrak{n} = \underline{\mathrm{Lie}}(N/S, \mathscr{M}), \quad \mathfrak{z} = \underline{\mathrm{Lie}}(Z/S, \mathscr{M}).$$

If $X \in \underline{\mathrm{Lie}}(H/S, \mathscr{M})(S)$, the equalities (*) hold for every $f : S' \to S$, because f factors through $I_S(\mathscr{M})$ as

$$f = \rho \circ \varepsilon_{\mathscr{M}} \circ f,$$

where $\rho : I_S(\mathscr{M}) \to S$ is the structural morphism and $\varepsilon_{\mathscr{M}} : S \to I_S(\mathscr{M})$ is the zero section. Consequently,

$$\mathfrak{n}(S)/\mathfrak{k}(S) \subset (\mathfrak{g}/\mathfrak{k})^K(S), \qquad \mathfrak{z}(S) \subset \mathfrak{g}^K(S).$$

For the reverse inclusions, assume henceforth that $\mathfrak{g}$ is commutative. Then $\mathfrak{g}^K$ and $(\mathfrak{g}/\mathfrak{k})^K$ are commutative S-groups. Let $X \in \mathfrak{g}(S)$, and write $\bar{X}$ for its image in $(\mathfrak{g}/\mathfrak{k})(S)$. Suppose that $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$ (respectively, $X \in \mathfrak{g}^K(S)$); we show that $X \in \mathfrak{n}(S)$ (respectively, $X \in \mathfrak{z}(S)$).

Let $f : S' \to I_S(\mathscr{M})$. Consider the cartesian square

$$\begin{array}{ccc} I_{S'}(\mathcal{M}) & \xrightarrow{\ p\ } & I_S(\mathcal{M}) \\ {\scriptstyle \rho'}\downarrow & & \downarrow{\scriptstyle \rho} \\ S' & \xrightarrow[\rho \circ f]{} & S \end{array}$$

Figure 40: The cartesian square used in the reverse-inclusion argument.

and let h be the section of ρ' defined by f. It is enough to show that, for every $v \in K(I_{S'}(\mathscr{M}))$,

$$\left.\begin{array}{rc} & p^*(X^\alpha)\, v\, p^*(X^{-\alpha})\, v^{-1} \in K(I_{S'}(\mathscr{M})), \\ \text{respectively} & p^*(X)\, v\, p^*(X^{-1})\, v^{-1} = 1. \end{array}\right\} \qquad (**)$$

Indeed, take $v = \rho'^*(u)$ and apply h^* to (**). This gives (*), because $\rho' \circ h = \mathrm{id}_{S'}$ and $p \circ h = f$.

We now prove (**), writing X for $p^*(X)$. Every $v \in K(I_{S'}(\mathscr{M}))$ is uniquely of the form Yk, with $Y \in \underline{\mathrm{Lie}}(K/S, \mathscr{M})(S')$ and $k \in K(S')$. The expression $X^\alpha v X^{-\alpha} v^{-1}$ becomes

$$X^\alpha Y (k X^{-\alpha} k^{-1}) Y^{-1} = X^\alpha \,\mathrm{Ad}(k)(X^{-\alpha}),$$

because $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ is commutative. This element lies in $\underline{\mathrm{Lie}}(G/S, \mathscr{M})(S')$. By Lemma 5.2.3.0, condition (**) is therefore equivalent, for every $k \in K(S')$, to

$$\begin{cases} \mathrm{Ad}(k)(X) = X, & H = Z, \\ X^\alpha \,\mathrm{Ad}(k)(X^{-\alpha}) \in \underline{\mathrm{Lie}}(K/S, \mathscr{M})(S'), & H = N. \end{cases}$$

For $H = Z$, this follows from $X \in \mathfrak{g}^K(S)$. For $H = N$, the condition may equally be written

$$\mathrm{Ad}(k)(\bar{X}) = \bar{X}, \qquad \mathrm{Ad}(k)(\bar{X}^{-1}) = \bar{X}^{-1}. \qquad (*')$$

The second equality follows from the first because the action of K on $\mathfrak{g}/\mathfrak{k}$ respects its group structure. Thus (**) also follows from $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$, proving (i) and (ii).

For (iii)–(v), put

$$\mathfrak{g}(\mathscr{M}) = \underline{\mathrm{Lie}}(G/S, \mathscr{M})$$

and define $\mathfrak{k}(\mathscr{M})$, $\mathfrak{z}(\mathscr{M})$, and $\mathfrak{n}(\mathscr{M})$ similarly. If G/S satisfies (E), then

$$\mathfrak{g}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{g}(\mathscr{M}) \times_S \mathfrak{g}(\mathscr{N}),$$

and hence, by Lemma 5.2.2(i),

$$\mathfrak{g}(\mathscr{M} \oplus \mathscr{N})^K \simeq \mathfrak{g}(\mathscr{M})^K \times_S \mathfrak{g}(\mathscr{N})^K.$$

It follows that $\mathfrak{z}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{z}(\mathscr{M}) \times_S \mathfrak{z}(\mathscr{N})$, so Z satisfies (E). If K/S also satisfies (E), then

$$(\mathfrak{g}/\mathfrak{k})(\mathscr{M} \oplus \mathscr{N}) \simeq (\mathfrak{g}/\mathfrak{k})(\mathscr{M}) \times_S (\mathfrak{g}/\mathfrak{k})(\mathscr{N}),$$
$$(\mathfrak{g}/\mathfrak{k})^K(\mathscr{M} \oplus \mathscr{N}) \simeq (\mathfrak{g}/\mathfrak{k})(\mathscr{M})^K \times_S (\mathfrak{g}/\mathfrak{k})(\mathscr{N})^K,$$

and we have the following commutative diagram.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \ell(\mathcal{M}\oplus\mathcal{N}) & \longrightarrow & n(\mathcal{M}\oplus\mathcal{N}) & \longrightarrow & (n/\ell)(\mathcal{M}\oplus\mathcal{N}) & \longrightarrow & 0 \\
 & & \downarrow\sim & & \downarrow & & \downarrow\sim & & \\
0 & \longrightarrow & \ell(\mathcal{M})\times_S\ell(\mathcal{N}) & \longrightarrow & n(\mathcal{M})\times_S n(\mathcal{N}) & \longrightarrow & (n/\ell)(\mathcal{M})\times_S(n/\ell)(\mathcal{N}) & \longrightarrow & 0
\end{array}$$

Figure 41: The exact-sequence comparison proving condition (E) for N.

It follows that

$$\mathfrak{n}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{n}(\mathscr{M}) \times_S \mathfrak{n}(\mathscr{N}),$$

so N satisfies (E).

Now write $\mathfrak{g} = \mathfrak{g}(\mathscr{O}_S) = \underline{\mathrm{Lie}}(G/S)$, and similarly $\mathfrak{k}, \mathfrak{z}, \mathfrak{n}$. If G is good, then $\mathfrak{g}$ is a good $\mathbf{O}_S$-module, and Lemma 5.2.2(ii) shows that $\mathfrak{z} \simeq \mathfrak{g}^K$ is good; since Z satisfies (E), Z is good. If K is also good, then $\mathfrak{k}$ is good, and Lemmas 5.2.1(iv) and 5.2.2(ii) show that $\mathfrak{g}/\mathfrak{k}$ and $(\mathfrak{g}/\mathfrak{k})^K$ are good. From the exact sequence

$$0 \longrightarrow \mathfrak{k} \longrightarrow \mathfrak{n} \longrightarrow (\mathfrak{g}/\mathfrak{k})^K \longrightarrow 0$$

and Lemma 5.2.1(iv), $\mathfrak{n}$ is good. Finally, if G is very good, so that $[x, x] = 0$ identically for $x \in \mathfrak{g}$, then Z and N are plainly very good. This proves (iii)–(v).

Corollary 5.2.3.1. *If the group law of* $\underline{\mathrm{Lie}}(G/S)$ *is commutative, then*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Centr}}(G)/S) = \underline{\mathrm{Lie}}(G/S)^G.$$

Corollary 5.2.3.2. *If the group law of* $\underline{\mathrm{Lie}}(G/S)$ *is commutative and* K *is a normal subgroup of* G*, then*

$$(\underline{\mathrm{Lie}}(G/S)/\underline{\mathrm{Lie}}(K/S))^K = \underline{\mathrm{Lie}}(G/S)/\underline{\mathrm{Lie}}(K/S).$$

5.3. Linear representations

Let G be a good S-group acting linearly on a good $\mathbf{O}_S$-module F through

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S\text{-mod}}(F).$$

In 4.7.1 and 4.8.2 we defined the corresponding linear representation

$$\rho' : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathbf{O}_S\text{-mod}}(F).$$

The sub-S-groups $\underline{\mathrm{Norm}}_G(E)$ and $\underline{\mathrm{Centr}}_G(E)$ are defined for every subset E of F, and in particular for every $\mathbf{O}_S$-submodule E.

Definition 5.3.0. For every $S' \to S$, put

$$\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} \subset E_{S'}\},$$
$$\underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} = 0\}.$$

This construction works for every linear representation of an $\mathbf{O}_S$-Lie algebra in the sense of 4.9, and the two subobjects obtained are $\mathbf{O}_S$-submodules stable under the bracket.

Theorem 5.3.1. *Let G be a good S-group acting linearly on a good $\mathbf{O}_S$-module F, and let E be an $\mathbf{O}_S$-submodule of F.*

(i) *One has*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Centr}}_G(E)/S) = \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E),$$

and $\underline{\mathrm{Centr}}_G(E)$ is a good S-group; it is very good if G is very good.

(ii) *Suppose that E is a good $\mathbf{O}_S$-module. Then*[98]

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Norm}}_G(E)/S) = \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E),$$

and $\underline{\mathrm{Norm}}_G(E)$ is a good S-group; it is very good if G is very good.

The proof is left to the reader.

5.3.2. Let G be a good S-group. The preceding results apply in particular to the adjoint representation. Let E be a good $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G/S)$.[99] It has a centralizer and a normalizer in G, and by 5.3.1 their Lie algebras are the usual centralizer and normalizer of E, computed using the bracket:

$$\underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] = 0\},$$
$$\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] \subset E_{S'}\}.$$

5.3.3. Let K be a sub-S-group of G. Then $\underline{\mathrm{Lie}}(K/S)$ is an $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G/S)$. Suppose that it is a good $\mathbf{O}_S$-module,[100] as happens when K is a good S-group. Plainly,

$$\underline{\mathrm{Centr}}_G(K) \subset \underline{\mathrm{Centr}}_G\big(\underline{\mathrm{Lie}}(K/S)\big),$$

$$\underline{\mathrm{Norm}}_G(K) \subset \underline{\mathrm{Norm}}_G\big(\underline{\mathrm{Lie}}(K/S)\big).$$

Hence, by 5.3.1,

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Centr}}_G(K)/S\big) \subset \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}\big(\underline{\mathrm{Lie}}(K/S)\big),$$

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Norm}}_G(K)/S\big) \subset \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}\big(\underline{\mathrm{Lie}}(K/S)\big).$$

None of these four inclusions is an equality in general; many examples will appear later. In particular, if K is a normal subgroup of G, then $\underline{\mathrm{Lie}}(K/S)$ is an ideal of $\underline{\mathrm{Lie}}(G/S)$.

[98] Editor's note: The original stated the following equality without a hypothesis on E; this has been corrected. See 6.5.1 for counterexamples.

[99] Editor's note: This hypothesis has been added; cf. 6.5.1.

[100] Editor's note: This hypothesis has been added; cf. 6.5.1.

6. Miscellaneous remarks

6.1.

The bracket of two infinitesimal automorphisms can be defined for an S-functor X which is not necessarily a group: apply the results of this exposé to $\underline{\mathrm{Aut}}_S(X)$. To obtain a convenient formalism, one is led to assume that X is good, meaning that the $\mathbf{O}_X$-module $T_{X/S}$ is good. If X is an S-group, this agrees with Definition 4.6.

6.2.

There exist functors with infinitesimal endomorphisms that are not automorphisms and hence, a fortiori, do not satisfy condition (E). [101] For a pointed set (E, x_0), let $M_A(E)$ be the free abelian monoid generated by E, and let $M_{A,P}(E, x_0)$ be the quotient of $M_A(E)$ by the equivalence relation generated by $m \sim x_0 + m$. The assignment

$$(E, x_0) \longmapsto M_{A,P}(E, x_0)$$

is left adjoint to the forgetful functor from abelian monoids to pointed sets; $M_{A,P}(E, x_0)$ is the free abelian monoid on the pointed set (E, x_0).

Now let X be the functor assigning to every scheme S the free abelian monoid on $\mathbf{O}(S)$, pointed by zero. Every morphism

$$f : S \longrightarrow I_{\mathbb{Z}} = \mathrm{Spec}(\mathbb{Z}[\varepsilon])$$

corresponds to a square-zero element $u_f \in \mathbf{O}(S)$, and hence defines an endomorphism of $X(S)$ by $x \mapsto x + u_f$, where the sum is taken in $M_{A,P}(\mathbf{O}(S), 0)$. Thus we obtain an endomorphism ϕ of

$$X_{I_{\mathbb{Z}}} = X \times_{\mathbb{Z}} I_{\mathbb{Z}}$$

defined, for $f \in I_{\mathbb{Z}}(S)$ and $x \in X(S)$, by

$$\phi((x, f)) = (x + u_f, f).$$

If $f_0 : S \to I_{\mathbb{Z}}$ is the composite of the structural morphism $S \to \mathrm{Spec}(\mathbb{Z})$ and the zero section, then $u_{f_0} = 0$, so $\phi((x, f_0)) = (x, f_0)$. Thus ϕ induces the identity on X: it is an infinitesimal endomorphism of X which is plainly not an automorphism.

6.3.

There exist modules which are not good. First, the preceding counterexample can be modified slightly.[102] Let $F(S)$ be the free $\mathbf{O}(S)$-module with basis the elements of $\mathbf{O}(S)$. Then F does not satisfy (E) relative to $\mathrm{Spec}(\mathbb{Z})$. Moreover, let G be the $\mathbb{Z}$-group defined by

$$G(S) = \mathrm{GL}_n(R(S)), \qquad n \geq 2,$$

where $R(S)$ is the $\mathbf{O}(S)$-algebra of the abelian group $\mathbf{O}(S)$. Then the $\mathbb{Z}$-group $\underline{\mathrm{Lie}}(G/\mathbb{Z})$ is not commutative.

There are also the following counterexamples.[103] Let $S = \mathrm{Spec}(k)$, where k is a field of characteristic $p > 0$.

[101] Editor's note: The original has been expanded here. In particular, O. Gabber pointed out the correct definition of the free abelian monoid on a pointed set.

[102] Editor's note: The following observation has been added.

[103] Editor's note: Example (a) has been expanded and example (b) has been added.

(a) Let F be the $\mathbf{O}_S$-module assigning to every S-scheme T the group

$$F(T) = \Gamma(T, \mathscr{O}_T),$$

with its $\mathbf{O}(T)$-module structure defined by $r \cdot f = r^p f$. As an S-functor of groups, F is isomorphic to $\mathbb{G}_{a,S}$. Thus F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbb{G}_{a,S}/S) \simeq \mathbf{O}_S.$$

For each $T \to S$, the canonical morphism $F \to \underline{\mathrm{Lie}}(F/S)$ is the identity map $F(T) \to \mathbf{O}(T)$. It respects the abelian-group structure but not the module structure, so F is not good (4.4.1).

(b) Let $\alpha_{p,k}$ be the k-functor of groups defined by

$$\alpha_{p,k}(T) = \{x \in \mathbf{O}(T) \mid x^p = 0\}.$$

It is represented by $\mathrm{Spec}(k[X]/(X^p))$, hence is a very good S-group. It also carries an $\mathbf{O}_S$-module structure, but is not a good $\mathbf{O}_S$-module, because the canonical morphism

$$\alpha_{p,k} \longrightarrow \underline{\mathrm{Lie}}(\alpha_{p,k}/k) = \mathbb{G}_{a,k}$$

is not bijective.

6.4.

For a group-valued functor G over S, the definitions give

$$(G/S \text{ satisfies (E)}) \Longleftarrow (G \text{ is good}) \Longleftarrow (G \text{ is very good}).$$

It would be interesting to prove or disprove the reverse implications. [104]

6.5.

Let Nil be the $\mathbb{Z}$-functor of groups defined by

$$\mathrm{Nil}(S) = \{x \in \mathbf{O}(S) \mid \text{there exists } n \in \mathbb{N} \text{ with } x^n = 0\}.$$

[105] The functor Nil is very good but not representable. Let Nil^2, $\mathbf{O}_{\mathrm{red}}$, and F be the $\mathbb{Z}$-functors of groups assigning to S, respectively, the ideal $\mathrm{Nil}(S)^2$ and the groups

$$\mathbf{O}_{\mathrm{red}}(S) = \mathbf{O}(S)/\mathrm{Nil}(S), \qquad F(S) = \mathbf{O}(S)/\mathrm{Nil}(S)^2.$$

It is easy to see that

$$\underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{red}}/\mathbb{Z}) = 0,$$

so the $\mathbf{O}_{\mathbb{Z}}$-module $\mathbf{O}_{\mathrm{red}}$ is not good, although it is a good $\mathbb{Z}$-group. If $\mathscr{M}, \mathscr{N}$ are free $\mathbb{Z}$-modules of finite rank, then

$$\begin{aligned}\mathrm{Nil}^2(I_S(\mathscr{M} \oplus \mathscr{N})) \simeq\ & \mathrm{Nil}^2(S) \oplus \mathrm{Nil}(S) \otimes_{\mathbb{Z}} \mathscr{M} \\ & \oplus \mathrm{Nil}(S) \otimes_{\mathbb{Z}} \mathscr{N},\end{aligned}$$

[104] Editor's note: A representable G is very good (4.11). For representability criteria see, for example, [MO67]. For automorphisms of an affine algebraic group over an algebraically closed field of characteristic zero, see [HM69].

[105] Editor's note: This paragraph has been added.

and hence

$$F(I_S(\mathscr{M} \oplus \mathscr{N})) \simeq F(S) \oplus \mathbf{O}_{\mathrm{red}}(S) \otimes_{\mathbb{Z}} \mathscr{M} \\ \oplus \mathbf{O}_{\mathrm{red}}(S) \otimes_{\mathbb{Z}} \mathscr{N}.$$

Thus F satisfies condition (E), while

$$\underline{\mathrm{Lie}}(F/\mathbb{Z}) = \mathbf{O}_{\mathrm{red}}.$$

Since $\mathbf{O}_{\mathrm{red}}$ is not a good $\mathbf{O}_{\mathbb{Z}}$-module, F is a $\mathbb{Z}$-group which satisfies (E) but is not good.

6.5.1. We now give the counterexamples announced in 5.3.1–5.3.3. Let S be a scheme, let $F = \mathbf{O}_S^{\oplus 2}$ with the natural action of the good S-group $G = \mathrm{GL}_{2,S}$, and let E be the $\mathbf{O}_S$-submodule of F formed by the pairs (x_1, x_2) for which x_2 is nilpotent. Put

$$N = \underline{\mathrm{Norm}}_G(E).$$

Then

$$\underline{\mathrm{Lie}}(N/S) = \underline{\mathrm{Lie}}(G/S),$$

whereas, for every $S' \to S$,

$$\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \left\{ \begin{pmatrix} a & b \\ x & c \end{pmatrix} \,\middle|\, a, b, c, x \in \mathbf{O}(S'),\ x \text{ nilpotent} \right\}.$$

Consequently,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Norm}}_G(E)/S) \neq \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E).$$

Taking the semidirect product $G_1 = F \rtimes G$ gives an analogous counterexample in which E is an $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G_1/S)$. Moreover, with the notation above,

$$E = \underline{\mathrm{Lie}}(K/S),$$

where K is the subgroup $\mathbf{O}_S \oplus \mathrm{Nil}^2$ of F; that is, $K(S')$ consists of the pairs (x_1, x_2) with $x_2 \in \mathrm{Nil}(S')^2$.

Bibliography

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[**HM69**] G. Hochschild and G. D. Mostow, *Automorphisms of affine algebraic groups*, *J. Algebra* **13** (1969), 535–543.

[**MO67**] H. Matsumura and F. Oort, *Representability of group functors and automorphisms of algebraic schemes*, *Invent. Math.* **4** (1967), 1–25.

[106] Editor's note: Additional references cited in this exposé.

Exposé III
Infinitesimal Extensions

by M. Demazure

In this exposé we work in the following general situation. We have a scheme S and a coherent nilpotent ideal $\mathscr{I}$ on S. We denote by S_n the closed subscheme of S defined by the ideal $\mathscr{I}^{n+1}$, for $n \geq 0$. In particular, S_0 is defined by $\mathscr{I}$. Since $\mathscr{I}$ is nilpotent, $S_n = S$ for all sufficiently large n, and the S_i have the same underlying topological space. A typical example is the following: S is the spectrum of a local Artinian ring A, and $\mathscr{I}$ is the ideal defined by the radical of A, so that S_0 is the spectrum of the residue field of A.[0]

In this situation one is given certain data over S_0 and seeks data over S that lift them, that is, data whose base change from S to S_0 recovers the original data. This is done step by step, through the schemes S_n. At each step we shall define the obstructions that arise and classify the resulting solutions when they exist.

The passage from S_n to S_{n+1} may be generalized as follows. We have a scheme S and two ideals $\mathscr{I} \supset \mathscr{J}$ such that $\mathscr{I}\mathscr{J} = 0$. In the preceding case S, $\mathscr{I}$, and $\mathscr{J}$ are respectively

$$S_{n+1}, \qquad \mathscr{I}/\mathscr{I}^{n+2}, \qquad \mathscr{I}^{n+1}/\mathscr{I}^{n+2}.$$

We denote by S_0, respectively $S_{\mathscr{J}}$, the closed subscheme of S defined by $\mathscr{I}$, respectively $\mathscr{J}$, and pose a problem of extending data from $S_{\mathscr{J}}$ to S.

SGA 1, Exposé III, treated the problems of extending morphisms of schemes and of extending schemes. Here we consider the problems of extending group morphisms, group structures, and subgroups.

In §0 we have collected those results of SGA 1, Exposé III, that will be useful to us, putting them into the form best suited to our purpose and sparing the reader constant reference to that exposé.[1] Section 1 collects computations in group cohomology that will be used later and have nothing to do with scheme theory. Sections 2 and 3 treat, respectively, the extension of group morphisms and the extension of group structures. In §4 we briefly recall the proof of a result stated in TDTE IV concerning the extension of subschemes, and apply it to the problem of extending subgroups. For the remainder of the Seminar, only the result of §2 concerning the extension of group morphisms is indispensable.[2]

The idea of reducing problems of infinitesimal extension to the usual cohomological computations in group extensions was suggested by J. Giraud during the oral exposé, whose computations were appreciably more complicated and less transparent. Unfortunately, this

[0] Editor's note: version of October 14, 2024.

[1] Editor's note: the reader may prefer to begin with §1, which is easier and may serve as motivation and a guide to the results of §0.

[2] Editor's note: nevertheless, 3.10 is used in XXIV 1.13. Moreover, 4.37 is used in the proof of IX 3.2 bis and 3.6 bis, although those results also follow from results of Exposé X that do not use 4.37.

method seems to work well only for the first two problems studied, and we have been unable to avoid rather laborious computations for extensions of subgroups.

To simplify the language, we shall call a Y-functor, respectively a Y-scheme, and so on, a functor, respectively a scheme, and so on, equipped with a morphism to the functor Y. This extends the definitions of Exposé I, which concerned only representable Y.

0. Reminders from SGA 1, Exposé III, and miscellaneous remarks

We begin with a general definition.

Definition 0.1. *Let $\mathscr{C}$ be a category, X an object of $\widehat{\mathscr{C}}$, and G a $\widehat{\mathscr{C}}$-group acting on X. We say that X is* formally principal homogeneous[3] *under G if the following equivalent conditions hold:*

(i) *for every object T of $\mathscr{C}$, the set $X(T)$ is either empty or principal homogeneous under $G(T)$;*

(ii) *the morphism of functors*

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x),$$

is an isomorphism.

We shall now put the results of SGA 1, III §5[4] into the form most useful here. Throughout this section we use the following notation. On a scheme S we have two quasi-coherent ideals $\mathscr{I}$ and $\mathscr{J}$ such that

$$\mathscr{I} \supset \mathscr{J}, \qquad \mathscr{I}\,\mathscr{J} = 0.$$

In particular, $\mathscr{J}^2 = 0$. We denote by S_0, respectively $S_{\mathscr{J}}$, the closed subscheme of S defined by $\mathscr{I}$, respectively $\mathscr{J}$. For every S-functor X, we systematically denote by X_0 and $X_{\mathscr{J}}$ the functors obtained by base change from S to S_0 and $S_{\mathscr{J}}$. We use the same notation for a morphism.

Definition 0.1.1.[5] *Let X be an S-functor. Define a functor X^+ over S by*

$$\mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}}) = \mathrm{Hom}_S(Y_{\mathscr{J}}, X)$$

for a variable S-scheme Y. In the notation of Exposé II, §1,

$$X^+ \simeq \prod_{S_{\mathscr{J}}/S} X_{\mathscr{J}}.$$

The identity morphism of $X_{\mathscr{J}}$ defines, by construction, an S-morphism

$$p_X : X \longrightarrow X^+.$$

Explicitly,[6] for every S-scheme Y, the map

$$p_X(Y) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_S(Y_{\mathscr{J}}, X)$$

[3] Editor's note: one also calls such an object a *pseudo-torsor*; see EGA IV$_4$, 16.5.15. The more general notion of a formally homogeneous object, not necessarily principal homogeneous, is defined in Exposé IV, 6.7.1.

[4] Editor's note: see also EGA IV$_4$, 16.5.14–18.

[5] Editor's note: the numbering 0.1.1 through 0.1.13 has been added to make the definitions and results contained here more visible.

[6] Editor's note: the following sentence has been added, and the next two remarks have been expanded.

is induced by the morphism $Y_{\mathscr{J}} \to Y$.

Remark 0.1.2. If X is an S-group functor, then $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group functor. The defining formula for X^+ therefore equips X^+ with an S-group-functor structure, and the description above shows that

$$p_X : X \longrightarrow X^+$$

is a morphism of S-group functors.

Remark 0.1.3. For every S-group functor Y, one has

$$\mathrm{Hom}_{S\text{-gr}}(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}\text{-gr}}(Y_{\mathscr{J}}, X_{\mathscr{J}}).$$

Indeed, let $f \in \mathrm{Hom}_S(Y, X^+)$ correspond to $f_{\mathscr{J}} \in \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}})$. The condition that f be an S-group morphism is that, for every $T \to S$ and every $y, y' \in Y(T)$,

$$f(y \cdot y') = f(y) \cdot f(y'),$$

which is equivalent to

$$f_{\mathscr{J}}(y_{\mathscr{J}}) \cdot f_{\mathscr{J}}(y'_{\mathscr{J}}) = f_{\mathscr{J}}((y \cdot y')_{\mathscr{J}}).$$

Since $(y \cdot y')_{\mathscr{J}} = y_{\mathscr{J}} \cdot y'_{\mathscr{J}}$, this is precisely the condition that $f_{\mathscr{J}}$ be a group morphism. Applying this to $Y = X$, we again see that p_X, which corresponds to $\mathrm{id}_{X_{\mathscr{J}}}$, is a morphism of S-group functors.

Return now to the general case and suppose that X is an S-scheme. An S-morphism from a variable S-scheme Y to X^+ is, by definition, an $S_{\mathscr{J}}$-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$. We define an X^+-functor in abelian groups L_X as follows.

Scholium 0.1.4.[7] If $\pi : Y \to S$ is a morphism of schemes and M is an $\mathscr{O}_S$-module, we denote the inverse image $\pi^* M$ by

$$M \otimes_{\mathscr{O}_S} \mathscr{O}_Y.$$

If $\mathscr{J}$ is an ideal of $\mathscr{O}_S$, we denote by $\mathscr{J}\mathscr{O}_Y$ the ideal of $\mathscr{O}_Y$ that is the image of

$$\pi^*(\mathscr{J}) = \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Y \longrightarrow \mathscr{O}_Y.$$

For every morphism of S-schemes $f : Z \to Y$, there is an epimorphism of $\mathscr{O}_Z$-modules

$$f^*(\mathscr{J}\mathscr{O}_Y) \longrightarrow \mathscr{J}\mathscr{O}_Z. \tag{0.1.4}$$

It follows from the commutative diagram below.

$$\begin{array}{ccc} \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Y \otimes_{\mathscr{O}_Y} \mathscr{O}_Z & = & \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Z \\ \downarrow & & \downarrow \\ f^*(\mathscr{J}\mathscr{O}_Y) = (\mathscr{J}\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{O}_Z & \longrightarrow & \mathscr{J}\mathscr{O}_Z. \end{array}$$

Figure 1: The tensor-product diagram yielding the epimorphism (0.1.4).

Definition 0.1.5.[8] Let X be an S-scheme. For every X^+-scheme Y, given by a morphism

$$g_{\mathscr{J}} : Y_{\mathscr{J}} \longrightarrow X_{\mathscr{J}},$$

[7] Editor's note: this scholium has been added.

[8] Editor's note: introducing the functor L_X gives an extension, functorial in Y and in X, of SGA 1, III, Proposition 5.1; see 0.1.8 and 0.1.9 below, as well as 0.1.10 and 0.2. Moreover, when X is an S-group functor, L_X gives rise, under certain hypotheses, to an S-group functor L'_X and to an exact sequence $1 \to L'_X \to X \to X^+$; see 0.4, 0.9, and 0.11. These play an essential role in this exposé; see Theorems 2.1, 3.5, and 4.21.

set

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y),$$

where $\Omega^1_{X_0/S_0}$ is the module of relative differentials of X_0 over S_0, and $\mathscr{J}\mathscr{O}_Y$ is regarded as an $\mathscr{O}_{Y_0}$-module because it is annihilated by $\mathscr{I}$.

Then L_X is an X^+-functor in abelian groups.[9] Indeed, for every X^+-morphism $f : Z \to Y$, the functor f_0^* and the morphism

$$f_0^*(\mathscr{J}\mathscr{O}_Y) \longrightarrow \mathscr{J}\mathscr{O}_Z$$

of (0.1.4) induce a natural homomorphism $L_X(f)$:

$$\begin{aligned}\mathrm{Hom}_{\mathscr{O}_{Y_0}}&(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y)\\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_{Z_0}}(f_0^* g_0^*(\Omega^1_{X_0/S_0}), f_0^*(\mathscr{J}\mathscr{O}_Y))\\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_{Z_0}}(f_0^* g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Z).\end{aligned}$$

Finally, $L_X(f)$ has the following simple local description. The schemes Y and $Y_{\mathscr{J}}$ have the same underlying topological space, as do V and $V_{\mathscr{J}}$ for every open subset V of Y. Let

$$U = \mathrm{Spec}(A), \qquad V = \mathrm{Spec}(B), \qquad W = \mathrm{Spec}(C)$$

be affine opens such that U lies over $\mathrm{Spec}(\Lambda) \subset S$, $g_{\mathscr{J}}(V_{\mathscr{J}}) \subset U$, and $W \subset f^{-1}(V)$. Write I, J for the ideals of Λ corresponding to $\mathscr{I}$, $\mathscr{J}$. Then f, respectively $g_{\mathscr{J}}$, induces a morphism of Λ-algebras

$$\theta : B \longrightarrow C, \qquad \text{respectively} \qquad \varphi : A \longrightarrow B/JB,$$

and clearly $\theta(JB) \subset JC$. On the other hand,

$$m \in \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y)$$

induces an element D of

$$\begin{aligned}\mathrm{Hom}_{\mathscr{O}_{V_0}}(g_0^*(\Omega^1_{U/S}), \mathscr{J}\mathscr{O}_V) &= \mathrm{Hom}_{B/IB}(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB)\\ &= \mathrm{Der}_\Lambda(A, JB),\end{aligned}$$

and the image of $L_X(f)(m)$ in

$$\begin{aligned}\mathrm{Hom}_{\mathscr{O}_{W_0}}(f_0^* g_0^*(\Omega^1_{U/S}), \mathscr{J}\mathscr{O}_Z) &= \mathrm{Hom}_{C/IC}(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC)\\ &= \mathrm{Der}_\Lambda(A, JC)\end{aligned}$$

is simply $\theta \circ D$.

Remark 0.1.6.[10] Let $f : X \to W$ be an S-morphism. It induces an S-morphism

$$f^+ : X^+ \longrightarrow W^+$$

defined as follows. If $g \in \mathrm{Hom}_S(Y, X^+)$ corresponds to an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, then $f^+(g)$ is the element $f \circ g_{\mathscr{J}}$ of $\mathrm{Hom}_S(Y, W^+)$. The following diagram is commutative.

[9] Editor's note: what follows has been added.

[10] Editor's note: this remark has been expanded.

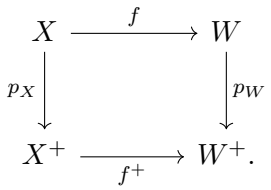

Figure 2: Naturality of the morphisms p_X and p_W.

Reminders 0.1.7.[11] In this paragraph, given an S-scheme X, we "recall" certain functorial properties of the module of differentials $\Omega^1_{X/S}$ and of the first infinitesimal neighborhood of the diagonal, $\Delta^{(1)}_{X/S}$. These properties follow readily from EGA IV_4, §§16.1–16.4, but do not appear there explicitly.

(a) Let us begin by recalling the following facts (cf. EGA II, §§1.2–1.5). Let $g : Y \to X$ be a morphism of schemes, let $\pi : X' \to X$ be an affine X-scheme, and let $\mathscr{B} = \pi_*(\mathscr{O}_{X'})$ be the corresponding quasi-coherent $\mathscr{O}_X$-algebra. Then the Y-scheme $Y \times_X X'$ is affine and corresponds to the quasi-coherent $\mathscr{O}_Y$-algebra $g^*(\mathscr{B})$, and there is a commutative diagram of bijections:

$$\begin{array}{ccc} \mathrm{Hom}_X(Y, X') & \xrightarrow{\sim} & \mathrm{Hom}_Y(Y, Y \times_X X') \\ \wr\downarrow & & \downarrow\wr \\ \mathrm{Hom}_{\mathscr{O}_X\text{-alg.}}(\mathscr{B}, g_*(\mathscr{O}_Y)) & \xrightarrow[\sim]{} & \mathrm{Hom}_{\mathscr{O}_Y\text{-alg.}}(g^*(\mathscr{B}), \mathscr{O}_Y)\,. \end{array}$$

Figure 3: The affine-scheme/algebra bijections after base change by g.

Moreover, these bijections are functorial in the pair (X, X'). More precisely, let W' be an affine W-scheme corresponding to the quasi-coherent $\mathscr{O}_W$-algebra $\mathscr{A}$, and suppose that one has a commutative diagram of morphisms of schemes as follows.

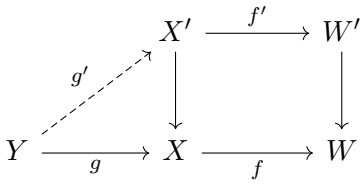

Figure 4: The scheme morphisms used to express functoriality in (X, X').

Denote by

$$\varphi : \mathscr{A} \longrightarrow f_*(\mathscr{B}), \qquad \varphi^\sharp : f^*(\mathscr{A}) \longrightarrow \mathscr{B}$$

the algebra morphisms associated with f', and by

$$\theta : \mathscr{B} \longrightarrow g_*(\mathscr{O}_Y), \qquad \theta^\sharp : g^*(\mathscr{B}) \longrightarrow \mathscr{O}_Y$$

those associated with a variable X-morphism $g' : Y \to X'$. Then the following diagram is commutative, where Y is viewed as a W-scheme through $f \circ g$.

[11] Editor's note: This paragraph has been added (see also [DG70], §I.4, nos. 1–2).

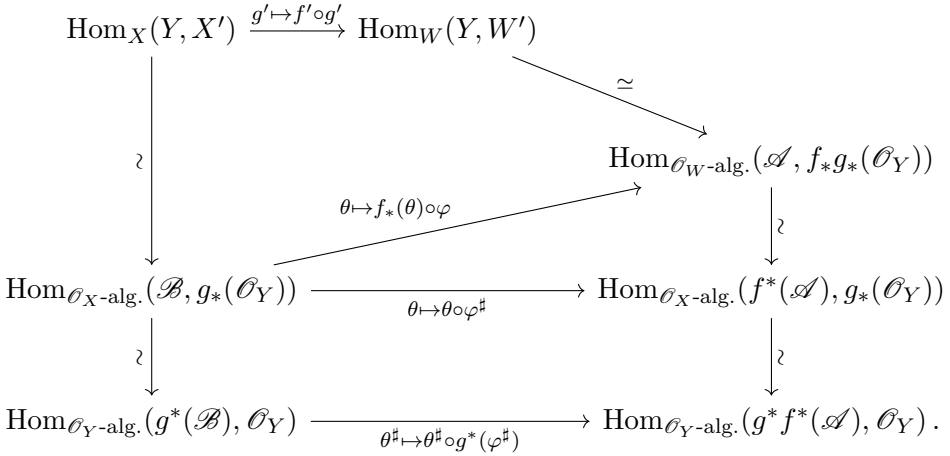

Figure 5: Functoriality of the affine-scheme/algebra correspondences.

(b) Let now X be an S-scheme. Let $\Omega^1_{X/S}$ be the module of differentials of X over S, and let $\Delta^{(1)}_{X/S}$ be the first infinitesimal neighborhood of the diagonal immersion $\delta_X : X \to X \times_S X$. It is a subscheme of $X \times_S X$, in which the diagonal $\Delta_{X/S}$ is a closed subscheme. Write pr^i_X, $i = 1, 2$, for the two projections $X \times_S X \to X$, and write π_X for the restriction of pr^1_X to $\Delta^{(1)}_{X/S}$.

On the one hand, every morphism $f : X \to W$ of S-schemes induces an S-morphism

$$\Delta^{(1)} f : \Delta^{(1)}_{X/S} \longrightarrow \Delta^{(1)}_{W/S}$$

for which the following diagram is commutative.

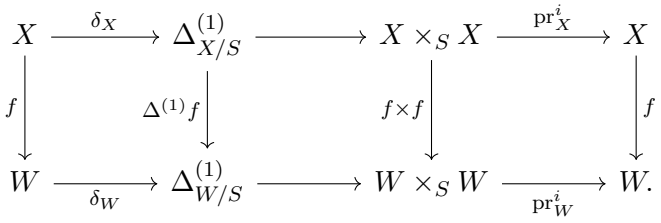

Figure 6: Functoriality of the first infinitesimal neighborhood of the diagonal.

On the other hand, through the projection π_X, the scheme $\Delta^{(1)}_{X/S}$ is an affine X-scheme: it is the spectrum of the augmented quasi-coherent $\mathscr{O}_X$-algebra

$$\mathscr{P}^1_{X/S} = \mathscr{O}_X \oplus \Omega^1_{X/S},$$

where $\Omega^1_{X/S}$ is a square-zero ideal. The augmentation is the $\mathscr{O}_X$-algebra morphism

$$\varepsilon_X : \mathscr{P}^1_{X/S} \longrightarrow \mathscr{O}_X$$

that vanishes on $\Omega^1_{X/S}$ and corresponds to the closed immersion $\delta_X : X \hookrightarrow \Delta^{(1)}_{X/S}$. Hence every morphism of S-schemes $f : X \to W$ induces a morphism of augmented $\mathscr{O}_X$-algebras

$$f^*(\mathscr{P}^1_{W/S}) = \mathscr{O}_X \oplus f^*(\Omega^1_{W/S}) \longrightarrow \mathscr{P}^1_{X/S} = \mathscr{O}_X \oplus \Omega^1_{X/S},$$

or, equivalently, a morphism of $\mathscr{O}_X$-modules

$$f_{X/W/S} : f^*(\Omega^1_{W/S}) \longrightarrow \Omega^1_{X/S};$$

see EGA IV$_4$, (16.4.3.6), and (16.4.18.2) for the notation $f_{X/W/S}$.

Since $\pi_X : \Delta^{(1)}_{X/S} \to X$ is affine, part (a) shows that $\Delta^{(1)}f$ is completely determined by $f_{X/W/S}$. For every X-scheme $g : Y \to X$, the set

$$\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \simeq \mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{O}_Y \oplus g^*(\Omega^1_{X/S}), \mathscr{O}_Y)$$

identifies with a subset of $\mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{O}_Y)$, namely

$$\widetilde{\mathrm{Hom}}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{O}_Y),$$

the subset formed by the $\mathscr{O}_Y$-morphisms $\psi : g^*(\Omega^1_{X/S}) \to \mathscr{O}_Y$ for which $\mathrm{Im}(\psi)$ is a square-zero ideal of $\mathscr{O}_Y$.[12]

Consequently, apply part (a) to the following diagram.

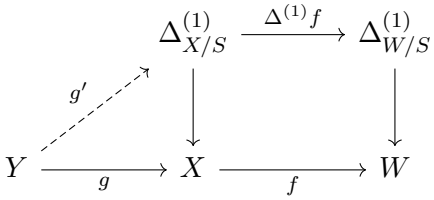

Figure 7: The diagram to which the affine functoriality statement is applied.

Taking into account that $\Delta^{(1)}f$ is the restriction of $f \times f$ to $\Delta^{(1)}_{X/S}$, one obtains the following commutative diagram, functorial in the X-scheme $g : Y \to X$.

Figure 8: The functorial comparison of first-neighborhood morphisms and differential maps, equation 0.1.7 $(*)$.

Remark 0.1.7.1. Let us conclude this paragraph with the following remark, which will be useful later (cf. 0.1.10). Denote by $\widetilde{L}_X$ the X-functor that associates to every X-scheme $g : Y \to X$ the set

$$\mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{O}_Y),$$

and denote by $\widetilde{L}_f : \widetilde{L}_X \to \widetilde{L}_W$ the morphism of functors defined above: it sends $\psi \in \widetilde{L}_X(Y)$ to $\psi \circ g^*(f_{X/W/S})$. The preceding discussion gives a commutative diagram of functors.

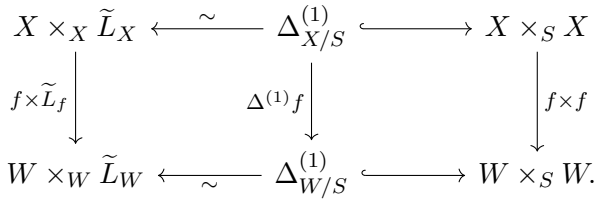

Figure 9: Functoriality of $\widetilde{L}_X$ and the first infinitesimal neighborhood.

[12] Editor's note: In the proof of SGA 1, III 5.1, the sentence "But the algebra homomorphisms $\mathscr{A} \to \mathscr{O}_Y$ correspond ..." must be corrected accordingly; the remainder of the cited passage is unchanged.

Theorem 0.1.8. (SGA 1, III 5.1)[13] *Let Y and X be two S-schemes, let $\mathscr{J}$ be a quasi-coherent square-zero ideal of $\mathscr{O}_Y$, let $Y_{\mathscr{J}}$ be the closed subscheme of Y defined by $\mathscr{J}$, and let $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$ be an S-morphism.*

(a) *The set $P(g_{\mathscr{J}})$ of S-morphisms $g : Y \to X$ extending $g_{\mathscr{J}}$ is either empty or principal homogeneous under the abelian group*

$$\mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}).$$

(b) *Suppose that $i : Y_0 \hookrightarrow Y_{\mathscr{J}}$ is the closed immersion defined by a quasi-coherent ideal $\mathscr{I} \supset \mathscr{J}$ such that $\mathscr{I}\mathscr{J} = 0$, and set $g_0 = g_{\mathscr{J}} \circ i$. Then the preceding abelian group is isomorphic to*

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X/S}), \mathscr{J}).$$

Proof. Part (b) follows at once from part (a). Indeed, since $\mathscr{J}$ is annihilated by $\mathscr{I}$, it may be regarded as an $\mathscr{O}_{Y_0}$-module, and adjunction gives

$$\mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(i^* g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}).$$

To prove part (a), we may suppose that $P(g_{\mathscr{J}}) \neq \emptyset$, that is, that an S-morphism $g : Y \to X$ extending $g_{\mathscr{J}}$ exists. Write $j : Y_{\mathscr{J}} \hookrightarrow Y$ for the immersion. Then $P(g_{\mathscr{J}})$ is the set of S-morphisms $g' : Y \to X$ such that $g' \circ j = g_{\mathscr{J}}$. Giving such a g' is equivalent to giving an S-morphism

$$h : Y \longrightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ h = g$ and $h_{\mathscr{J}} = \delta \circ g_{\mathscr{J}}$, where $h_{\mathscr{J}} = h \circ j$ and $\delta : X \hookrightarrow X \times_S X$ is the diagonal immersion.

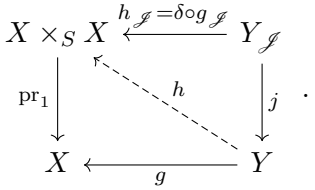

Figure 10: The morphism h determined by two extensions of $g_{\mathscr{J}}$.

Since $h_{\mathscr{J}}$ factors through δ, and since Y lies in the first infinitesimal neighborhood of the immersion $j : Y_{\mathscr{J}} \to Y$ because $\mathscr{J}^2 = 0$, functoriality (EGA IV_4, 16.2.2(i)) shows that the desired morphisms h factor uniquely through $\Delta^{(1)}_{X/S}$; see 0.1.7. Put

$$Y' = \Delta^{(1)}_{X/S} \times_X Y, \qquad Y'_{\mathscr{J}} = Y' \times_Y Y_{\mathscr{J}} = \Delta^{(1)}_{X/S} \times_X Y_{\mathscr{J}}.$$

The desired morphisms h are then in bijection with the sections u of $Y' \to Y$ extending the section $u_{\mathscr{J}} = (\delta \circ g_{\mathscr{J}}, \mathrm{id})$ of $Y'_{\mathscr{J}} \to Y_{\mathscr{J}}$. On the other hand, Y' (respectively $Y'_{\mathscr{J}}$) is affine over Y (respectively $Y_{\mathscr{J}}$) and corresponds to the quasi-coherent algebra

$$\mathscr{A} = \mathscr{O}_Y \oplus g^*(\Omega^1_{X/S}), \qquad \text{respectively} \qquad \mathscr{A}_{\mathscr{J}} = \mathscr{A} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_{Y_{\mathscr{J}}} \oplus g^*_{\mathscr{J}}(\Omega^1_{X/S}).$$

[13] Editor's note: The statement and proof of SGA 1, III 5.1 are reproduced here, since they are needed below to prove Proposition 0.2; see already the following Corollary 0.1.9. Moreover, $\mathscr{J}$ (respectively $\mathscr{I}$) is another calligraphic form of the letter J (respectively I). Returning to the earlier notation, with $\mathscr{J} \subset \mathscr{I}$ ideals of $\mathscr{O}_S$ such that $\mathscr{I}\mathscr{J} = 0$, this result will be applied with $\mathscr{J} = J\mathscr{O}_Y$ (respectively $\mathscr{I} = I\mathscr{O}_Y$).

Let $\varepsilon : \mathscr{A} \to \mathscr{O}_Y$ be the canonical augmentation, that is, the $\mathscr{O}_Y$-algebra morphism that vanishes on $g^*(\Omega^1_{X/S})$, and define $\varepsilon_{\mathscr{J}} : \mathscr{A}_{\mathscr{J}} \to \mathscr{O}_{Y_{\mathscr{J}}}$ similarly. Then

$$\Gamma(Y'/Y) \simeq \mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{A}, \mathscr{O}_Y), \qquad \Gamma(Y'_{\mathscr{J}}/Y_{\mathscr{J}}) \simeq \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}\text{-alg}}(\mathscr{A}_{\mathscr{J}}, \mathscr{O}_{Y_{\mathscr{J}}}).$$

Under these isomorphisms, the section $u = (\delta \circ g, \mathrm{id})$ (respectively $u_{\mathscr{J}}$) corresponds to ε (respectively $\varepsilon_{\mathscr{J}}$). Consequently, $P(g_{\mathscr{J}})$ is in bijection with the algebra morphisms $\mathscr{A} \to \mathscr{O}_Y$ that reduce to $\varepsilon_{\mathscr{J}}$; under this bijection, g corresponds to ε.

Put $M = g^*(\Omega^1_{X/S})$. Then $\mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{A}, \mathscr{O}_Y)$ identifies with the set of $\mathscr{O}_Y$-morphisms $\psi : M \to \mathscr{O}_Y$ whose image is a square-zero ideal. We are interested in those morphisms that induce the zero morphism $M \to \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_Y/\mathscr{J}$, that is, those that carry M into $\mathscr{J}$. Conversely, since $\mathscr{J}^2 = 0$, every $\mathscr{O}_Y$-morphism $\phi : M \to \mathscr{J}$ comes from a unique algebra morphism $\mathscr{A} \to \mathscr{O}_Y$ reducing to $\varepsilon_{\mathscr{J}}$. Finally,

$$\mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{J}) = \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J})$$

because $\mathscr{J}^2 = 0$, as in the proof of part (b). Thus one obtains a bijection

$$P(g_{\mathscr{J}}) \simeq \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J})$$

under which g corresponds to the zero morphism.

For every

$$m \in \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}),$$

write $m \cdot g$ for the element of $P(g_{\mathscr{J}})$ associated with g and m by the preceding bijection. We have already seen that $0 \cdot g = g$; it remains to prove

$$m' \cdot (m \cdot g) = (m + m') \cdot g. \qquad (0.1.8\,(*))$$

This may be checked locally.[14] Indeed, the two preceding morphisms $Y \to X$ induce the same continuous map as g on the underlying topological spaces. It is therefore enough to check that, for every affine open $U = \mathrm{Spec}(A)$ of X lying over an affine open $\mathrm{Spec}(\Lambda)$ of S, and every affine open $V = \mathrm{Spec}(B)$ of $g^{-1}(U)$, they induce the same morphism of Λ-algebras $A \to B$.

Set $J = \Gamma(V, \mathscr{J})$, and let $\varphi, \psi, \eta : A \to B$ be the morphisms induced respectively by g, $m \cdot g$, and $m' \cdot (m \cdot g)$. They agree modulo J. There are unique expressions

$$\psi = \varphi + D, \qquad \eta = \psi + D',$$

where D belongs to

$$\mathrm{Der}_\varphi(A, J) = \{\delta \in \mathrm{Hom}_\Lambda(A, J) \mid \delta(ab) = \varphi(a)\delta(b) + \varphi(b)\delta(a)\},$$

and D' belongs to $\mathrm{Der}_\psi(A, J)$. But $\mathrm{Der}_\varphi(A, J) = \mathrm{Der}_\psi(A, J)$, since $J^2 = 0$, and both identify with

$$\mathrm{Hom}_{B/J}(\Omega^1_{A/\Lambda} \otimes_A B/J, J).$$

Under this identification, D corresponds to m and D' to m'. Hence $\eta = \varphi + D + D'$, while $D + D'$ corresponds to $m + m'$, proving (0.1.8 (*)).

[14] Editor's note: The preceding argument is reproduced almost word for word from SGA 1, III 5.1; what follows has been expanded.

Corollary 0.1.9.[15] *Let X be an S-scheme, and retain the notation of 0.1.5. Then X carries a left action of the abelian X^+-group L_X, making X formally principal homogeneous under L_X over X^+. In other words, there is an isomorphism of X^+-functors*

$$L_X \times_{X^+} X \xrightarrow{\sim} X \times_{X^+} X, \qquad (m, x) \longmapsto (x, m \cdot x).$$

Proof. Let $i_0 : X_0 \hookrightarrow X$ be the immersion. Since $X_0 = X \times_S S_0$, one first has

$$i_0^*(\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0}$$

(EGA IV, 16.4.5).

Let Y be an X^+-scheme, given by an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, and let $g_0 : Y_0 \to X_0$ be obtained by base change. By 0.1.8, if $\mathrm{Hom}_{X^+}(Y, X)$ is nonempty, it is principal homogeneous under

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^* i_0^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y),$$

which identifies with

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y) = L_X(Y).$$

Thus there is a bijection

$$L_X(Y) \times \mathrm{Hom}_{X^+}(Y, X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(Y, X \times_{X^+} X), \qquad (m, g) \longmapsto (g, m \cdot g).$$

We show that this is functorial in Y. Let $f : Z \to Y$ be a morphism of S-schemes. It is enough to prove that the following diagram is commutative.

$$\begin{array}{ccc} L_X(Y) \times \mathrm{Hom}_{X^+}(Y, X) & \xrightarrow{\sim} & \mathrm{Hom}_{X^+}(Y, X \times_{X^+} X) \\ \downarrow{\scriptstyle L_X(f) \times f} & & \downarrow{\scriptstyle f \times f} \\ L_X(Z) \times \mathrm{Hom}_{X^+}(Z, X) & \xrightarrow[\sim]{} & \mathrm{Hom}_{X^+}(Z, X \times_{X^+} X)\,. \end{array}$$

Figure 11: Functoriality in Y of the principal-homogeneous bijection.

If $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$, there is nothing to prove. It is therefore enough to show that, for every S-morphism $g : Y \to X$ extending $g_{\mathscr{J}}$ and every $m \in L_X(Y)$, one has

$$(m \cdot g) \circ f = L_X(f)(m) \cdot (g \circ f). \qquad (0.1.9\,(*))$$

These two S-morphisms $Z \to X$ agree on $Z_{\mathscr{J}}$ with $g_{\mathscr{J}} \circ f_{\mathscr{J}}$; in particular, they induce the same continuous map as $g \circ f$ on the underlying topological spaces. It therefore suffices to check the equality on local rings. Let $z \in Z$, put $y = f(z)$, $x = g(y)$, and let A, B, C denote respectively the local rings $\mathscr{O}_{X,x}, \mathscr{O}_{Y,y}, \mathscr{O}_{Z,z}$. Let s be the image of x in S, put $\Lambda = \mathscr{O}_{S,s}$, let J and I be the ideals of Λ corresponding to $\mathscr{J}$ and $\mathscr{I}$, and let $\varphi, \psi : A \to B$ and $\theta : B \to C$ be the Λ-algebra morphisms induced by g, $m \cdot g$, and f, respectively. Then m induces an element D of

$$\mathrm{Hom}_{B/IB}(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB) = \mathrm{Der}_\Lambda(A, JB),$$

and $\psi = \varphi + D$. Thus $(m \cdot g) \circ f$ corresponds to

$$\theta \circ \psi = \theta \circ \varphi + \theta \circ D.$$

[15] Editor's note: This corollary has been added to SGA 1, III 5.1; it proves part (i) of Proposition 0.2 below.

By 0.1.5, $\theta \circ D$ is the image of $L_X(f)(m)$ in

$$\mathrm{Hom}_{C/IC}(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC) = \mathrm{Der}_\Lambda(A, JC).$$

Consequently, $\theta \circ \varphi + \theta \circ D$ is the image of $L_X(f)(m) \cdot (g \circ f)$. This proves (0.1.9 (*)).

Corollary 0.1.10.[16]

(a) *The functor L_X depends functorially on X. For every S-morphism $f : X \to W$, there is an S-morphism $L_f : L_X \to L_W$ that is a morphism of abelian groups over f^+; that is, the following diagram is commutative,*

$$\begin{array}{ccc} L_X & \xrightarrow{L_f} & L_W \\ \downarrow & & \downarrow \\ X^+ & \xrightarrow[f^+]{} & W^+ \end{array}$$

Figure 12: The morphism L_f over f^+.

and, for every $Y \to X^+$, the map

$$L_f(Y) : \mathrm{Hom}_{X^+}(Y, L_X) \longrightarrow \mathrm{Hom}_{W^+}(Y, L_W),$$

where Y lies over W^+ through f^+, is a homomorphism of abelian groups.

(b) *Moreover, the following diagram is commutative.*

$$\begin{array}{ccc} L_X \times_{X^+} X & \xrightarrow{\sim} & X \times_{X^+} X \\ {\scriptstyle L_f \times f}\downarrow & & \downarrow{\scriptstyle f \times f} \\ L_W \times_{W^+} W & \xrightarrow[\sim]{} & W \times_{W^+} W \end{array} .$$

Figure 13: Compatibility of L_f with the formally principal homogeneous actions.

Proof. For part (a), L_f is induced by the morphism of $\mathscr{O}_{X_0}$-modules

$$f_{X_0/W_0/S_0} : f_0^*(\Omega^1_{W_0/S_0}) \longrightarrow \Omega^1_{X_0/S_0}$$

(cf. 0.1.7(b)). For every X^+-scheme Y, given by an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, one has the following commutative diagram, functorial in Y.

$$\begin{array}{ccc} \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y\right) & \xrightarrow{L_f(Y)} & \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(g_0^* f_0^*(\Omega^1_{W_0/S_0}), \mathscr{J}\mathscr{O}_Y\right) \\ \downarrow & & \downarrow \\ \{g_{\mathscr{J}}\} & \longrightarrow & \{f_{\mathscr{J}} \circ g_{\mathscr{J}}\} \end{array}$$

Figure 14: The homomorphism $L_f(Y)$ on the differential-module descriptions.

[16] Editor's note: The functoriality of L_X in X has been made explicit, in particular in part (b) below.

Here $L_f(Y)$ is the map

$$\psi \longmapsto \psi \circ g_0^*(f_{X_0/W_0/S_0}),$$

which is indeed a homomorphism of abelian groups.[17]

We now prove part (b). Let Y be an X^+-scheme. If $\mathrm{Hom}_{X^+}(Y,X)=\emptyset$, there is nothing to prove. Thus let $g \in \mathrm{Hom}_{X^+}(Y,X)$. We must show that, for every $m \in L_X(Y)$,

$$f \circ (m \cdot g) = L_f(Y)(m) \cdot (f \circ g). \qquad (0.1.10\,(*))$$

With g fixed, $\mathrm{Hom}_X(Y, X \times_{X^+} X)$ is a subset of $\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S})$, and

$$\begin{aligned} L_X(Y) &= \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y) \\ &= \mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y), \end{aligned}$$

which is a subset of $\widetilde{L}_X(Y)$; see 0.1.7. Moreover, $L_f(Y)$ is the restriction of $\widetilde{L}_f(Y)$ to $L_X(Y)$. Finally, the bijection

$$L_X(Y) \xrightarrow{\sim} \mathrm{Hom}_X(Y, X \times_{X^+} X), \qquad m \longmapsto (g, m \cdot g),$$

is the inverse of the restriction to $L_X(Y) \subset \widetilde{L}_X(Y)$ of the bijection

$$\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \xrightarrow{\sim} \{g\} \times \widetilde{L}_X(Y)$$

considered in 0.1.7.1. Consequently, equality (0.1.10 (*)) follows from (0.1.7 (*)). Indeed, let g' be the X-morphism $Y \to \Delta^{(1)}_{X/S}$ defined by $(g, m\cdot g)$. The element of $\widetilde{L}_W(Y)$ corresponding to $(f\circ g, f\circ g')$ is $\widetilde{L}_f(Y)(m) = L_f(Y)(m)$. Thus

$$L_f(m) \cdot (f \circ g) = f \circ (m \cdot g).$$

Lemma 0.1.11. *Let X, X' be two S-schemes. There is a commutative diagram*

$$\begin{array}{ccc} L_X \times_S L_{X'} & \xrightarrow{\sim} & L_{X\times_S X'} \\ \downarrow & & \downarrow \\ X^+ \times_S X'^+ & \xrightarrow[\sim]{} & (X \times_S X')^+ \end{array} .$$

Figure 15: Compatibility of L_X and X^+ with products.

Proof.[18] For every S-scheme Y,

$$\begin{aligned} \mathrm{Hom}_S(Y, X^+ \times_S X'^+) &= \mathrm{Hom}_S(Y, X^+) \times \mathrm{Hom}_S(Y, X'^+) \\ &\simeq \mathrm{Hom}_S(Y_{\mathscr{J}}, X) \times \mathrm{Hom}_S(Y_{\mathscr{J}}, X') \\ &= \mathrm{Hom}_S(Y, (X \times_S X')^+). \end{aligned}$$

Thus

$$X^+ \times_S X'^+ \simeq (X \times_S X')^+.$$

Now let Y be a scheme over $X^+ \times_S X'^+$, represented by a morphism

$$h : Y_{\mathscr{J}} \longrightarrow X \times_S X'.$$

[17] Editor's note: Observe that L_f depends only on $f_0 : X_0 \to W_0$.
[18] Editor's note: the proof has been expanded.

Let p, q be the projections from $X \times_S X'$ to X, X', and put $f = p \circ h$, $g = q \circ h$. Since

$$\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0} \simeq p_0^*(\Omega^1_{X_0/S_0}) \oplus q_0^*(\Omega^1_{X'_0/S_0})$$

(EGA IV$_4$, 16.4.23), there is a natural isomorphism

$$\begin{aligned}\mathrm{Hom}_{\mathscr{O}_{Y_0}}(f_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y) \times \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X'_0/S_0}), \mathscr{J}\mathscr{O}_Y) \\ \simeq \mathrm{Hom}_{\mathscr{O}_{Y_0}}(h_0^*(\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0}), \mathscr{J}\mathscr{O}_Y).\end{aligned}$$

In other words,

$$L_X(Y) \times L_{X'}(Y) \simeq L_{X \times_S X'}(Y).$$

Remark 0.1.12.[19] Let $\mathscr{C}$ be a category stable under fiber products, let S be an object of $\mathscr{C}$, and let T_1, T_2 be two objects over S. For $i = 1, 2$, let L_i and X_i be two objects over T_i, as in the diagram below.

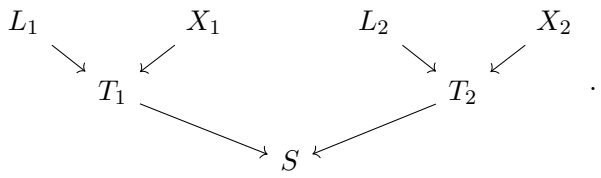

Figure 16: The objects L_i, X_i over T_i, and T_i over S.

There is then a natural isomorphism

$$(L_1 \times_{T_1} X_1) \times_S (L_2 \times_{T_2} X_2) \simeq (L_1 \times_S L_2) \times_{T_1 \times_S T_2} (X_1 \times_S X_2).$$

Consequently, the preceding lemma gives:

Corollary 0.1.13. *Let X_1, X_2 be two S-schemes. There is a commutative diagram of isomorphisms:*

$$\begin{array}{ccc} L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & & \\ \Big\downarrow{\scriptstyle (0.1.11)}\,\simeq & \searrow^{\simeq} & \\ (L_{X_1} \times_S L_{X_2}) \times_{(X_1^+ \times_S X_2^+)} (X_1 \times_S X_2) & \xrightarrow[(0.1.12)]{\simeq} & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2). \end{array}$$

Figure 17: The product isomorphisms of Corollary 0.1.13.

We can now state:

Proposition 0.2. *For every S-scheme X, one can define a left action of the X^+-abelian group L_X on the X^+-object X, with the following properties:*

(i) *this action makes X formally principal homogeneous under L_X over X^+; that is, the morphism*

$$L_X \times_{X^+} X \longrightarrow X \times_{X^+} X$$

is an isomorphism of X^+-functors;

[19] Editor's note: this remark and the corollary that follows have been added.

(ii) *the action is functorial in the S-scheme X: for every S-morphism $f : X \to W$, the following diagram is commutative;*

$$\begin{array}{ccc} L_X \times_{X^+} X & \longrightarrow & X \\ {\scriptstyle L_f \times f}\downarrow & & \downarrow{\scriptstyle f} \\ L_W \times_{W^+} W & \longrightarrow & W \end{array} ;$$

Figure 18: Functoriality of the action in Proposition 0.2(ii).

(iii) *the action commutes with fiber products: for all S-schemes X_1, X_2, the following diagram is commutative.*

$$\begin{array}{ccccc} L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & & \longrightarrow & & X_1 \times_S X_2 \\ \simeq\downarrow & \searrow & & \nearrow & \\ (L_{X_1} \times_S L_{X_2}) \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & \xrightarrow{\simeq} & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2) & & \end{array} .$$

Figure 19: Compatibility of the action with fiber products.

Proof.[20] Parts (i) and (ii) follow respectively from Corollaries 0.1.9 and 0.1.10. To prove (iii), write

$$P(X) = L_X \times_{X^+} X$$

for every S-scheme X. Applying (ii) to the projections

$$p_i : X_1 \times_S X_2 \longrightarrow X_i$$

gives the following commutative squares.

$$\begin{array}{ccc} P(X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ {\scriptstyle L_{p_i} \times p_i}\downarrow & & \downarrow{\scriptstyle p_i} \\ P(X_i) & \longrightarrow & X_i \end{array}$$

Figure 20: The square induced by each projection p_i.

For $i = 1, 2$, these yield the commutative square

$$\begin{array}{ccc} P(X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ \downarrow & & \| \\ P(X_1) \times_S P(X_2) & \longrightarrow & X_1 \times_S X_2 \end{array} .$$

Figure 21: The product square in the proof of Proposition 0.2(iii).

[20] Editor's note: the original has been modified and expanded to account for the preceding additions; in particular, those additions incorporate, with details, the content of original page 89.

Combining this square with Corollary 0.1.13 shows that its vertical arrow is an isomorphism and gives precisely the commutative diagram asserted in (iii).

Remark 0.3. Suppose that the X^+-scheme Y is flat over S (cf. SGA 1, IV). We may then write

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}).$$

Remark 0.4. Let $\pi_0 : X_0 \to S_0$ be the structure morphism, and suppose that there is an $\mathscr{O}_{S_0}$-module $\omega^1_{X_0/S_0}$ such that

$$\Omega^1_{X_0/S_0} \simeq \pi_0^*(\omega^1_{X_0/S_0}).$$

(This occurs, in particular, when X_0 is an S_0-group; cf. II 4.11.) Define a functor L'_X over S by

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_Y). \tag{0.4.1}$$

Then

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X)$$

for every X^+-scheme Y; in other words,

$$L_X = L'_X \times_S X^+.$$

Since $L_X \times_{X^+} X = L'_X \times_S X$, the action of L_X on X induces an action of L'_X on X, and this action respects $p_X : X \to X^+$.[21] Indeed, if Y is an S-scheme, $h : Y \to X$ is an S-morphism, and $m \in L'_X(Y)$, then h and $m \cdot h$ have the same restriction to $Y_{\mathscr{J}}$; that is,

$$p_X(m \cdot h) = p_X(h).$$

Remark 0.5. Keep the hypotheses and notation of 0.4, and suppose in addition that Y is an S-scheme flat over S. Then

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{S_0}(Y_0, L_{0X}),$$

where the S_0-functor in abelian groups L_{0X} is defined by the following identity (with T the variable S_0-scheme):

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{\mathscr{O}_T}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T). \tag{0.5.1}$$

Thus, in the notation of II 1, we have shown that L'_X and $\prod_{S_0/S} L_{0X}$ have the same restriction to the full subcategory of $(\mathbf{Sch})/S$ whose objects are the S-schemes flat over S.

Remark 0.6. Keep the hypotheses and notation of 0.5,[22] and suppose in addition that $\pi_0 : X_0 \to S_0$ has a section ε_0. Then

$$\omega^1_{X_0/S_0} \simeq \varepsilon_0^*(\Omega^1_{X_0/S_0}).$$

First, independently of this last hypothesis, one has

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma\Big(T, \mathscr{H}om_{\mathscr{O}_T}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T)\Big).$$

Suppose now that $\omega^1_{X_0/S_0}$ admits a finite presentation (cf. EGA 0_{I}, 5.2.5), as is in particular the case when X_0 is locally of finite presentation over S_0 (cf. EGA IV_4, 16.4.22). If T is flat over S_0, it then follows from EGA 0_{I}, 6.7.6 that

$$\mathscr{H}om_{\mathscr{O}_T}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T) \simeq \mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T,$$

[21] Editor's note: what follows has been expanded.

[22] Editor's note: the following hypothesis, which was implicit in the original, has been added.

whence

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma\Big(T, \mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\Big).$$

Using the notation $\mathbf{W}(\,\cdot\,)$ of I 4.6.1, we have therefore proved that, for every S_0-scheme T flat over S_0,

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{S_0}\Big(T, \mathbf{W}\Big(\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big)\Big).$$

In summary, if $\omega^1_{X_0/S_0}$ admits a finite presentation and one restricts to the category of S-schemes flat over S, then

$$L'_X = \prod_{S_0/S} \mathbf{W}\Big(\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big), \tag{0.6.1}$$

and $\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})$ is a quasi-coherent $\mathscr{O}_{S_0}$-module, by EGA I, 9.1.1.

Finally, suppose moreover that $\omega^1_{X_0/S_0}$ is locally free of finite rank, for example that X_0 is smooth over S_0 (in which case it is automatically locally of finite presentation over S_0). Then

$$\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \simeq \underline{\mathrm{Lie}}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}, \tag{0.6.2}$$

where, by an abuse of language (since X_0 need not be an S_0-group), $\underline{\mathrm{Lie}}(X_0/S_0)$ denotes the dual of the $\mathscr{O}_{S_0}$-module $\omega^1_{X_0/S_0}$.[23]

Proposition 0.2 (and its proof) has two important corollaries.[24]

Corollary 0.7. *Let X be an S-scheme.*

(a) *Every S-endomorphism of X inducing the identity on $X_{\mathscr{J}}$ is an automorphism.*

(b) *There is an exact sequence of groups*

$$0 \longrightarrow \mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathscr{J}}}(X_{\mathscr{J}}).$$

(c) *Let $\mathrm{Aut}_S(X)$ act on the first group by transport of structure. Then, for every $u \in \mathrm{Aut}_S(X)$ and $m \in \mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X)$,*

$$i(u \cdot m) = u\, i(m)\, u^{-1}.$$

Proof. By 0.2(i), $\mathrm{Hom}_{X^+}(X, X)$ is a principal homogeneous set under $\mathrm{Hom}_{X^+}(X, L_X)$, since it is certainly nonempty: it contains the marked point given by the identity automorphism of X.[25] Consequently, the map $m \mapsto m \cdot \mathrm{id}_X$ induces a bijection

$$\mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X) = L_X(X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, X).$$

Let $m \in L_X(X)$, and let $f = m' \cdot \mathrm{id}_X$ be an element of $\mathrm{Hom}_{X^+}(X, X)$. Applying 0.2(ii) to f gives

$$f \circ (m \cdot \mathrm{id}_X) = L_f(X)(m) \cdot f = L_f(X)(m) \cdot (m' \cdot \mathrm{id}_X).$$

[23] Editor's note: this is justified by II 3.3 and 4.11. Indeed, the functor $L^{\varepsilon_0}_{X_0/S_0}$, the "tangent space to X_0 over S_0 at ε_0," is represented over S_0 by the vector bundle $\mathbf{V}(\varepsilon_0^*(\Omega^1_{X_0/S_0})) = \mathbf{V}(\omega^1_{X_0/S_0})$. Its sheaf of sections is the dual module $\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{O}_{S_0})$; this is $\underline{\mathrm{Lie}}(X_0/S_0)$ when X_0 is an S_0-group and ε_0 is the identity section.

[24] Editor's note: Corollary 0.7 bis, which was used implicitly in the proof of 0.8, has been added to 0.7. We point out that 0.8–0.12, as well as 0.17, which play an important technical role later in this exposé, are consequences of 0.7 and 0.7 bis.

[25] Editor's note: what follows has been expanded.

On the other hand, because f is an X^+-endomorphism of X, one has $f_{\mathscr{J}} = \mathrm{id}_{X_{\mathscr{J}}}$, hence $f_0 = \mathrm{id}_{X_0}$. Since L_f depends only on f_0 (cf. the editor's note (17) in 0.1.10), it follows that $L_f(X)(m) = m$. Thus the preceding equality becomes

$$(m' \cdot \mathrm{id}_X) \circ (m \cdot \mathrm{id}_X) = m \cdot (m' \cdot \mathrm{id}_X) = (m + m') \cdot \mathrm{id}_X \,.$$

This shows that the bijection $m \mapsto m \cdot \mathrm{id}_X$ carries the group law of $\mathrm{Hom}_{X^+}(X, L_X)$ to composition of X^+-endomorphisms of X.

It follows first that every element of $\mathrm{Hom}_{X^+}(X, X)$ is invertible, proving part (a), and then that there is an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathscr{J}}}(X_{\mathscr{J}}),$$

which proves part (b).

The morphism i just defined is functorial in X with respect to isomorphisms, because it is defined structurally from the action of L_X on X over X^+, an action that is itself functorial in X by Proposition 0.2(ii).[26] Hence every automorphism u of X over S induces, by transport of structure, isomorphisms

$$h : \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, L_X) \quad \text{and} \quad f : \mathrm{Aut}_S(X) \xrightarrow{\sim} \mathrm{Aut}_S(X)$$

such that the following diagram commutes.

$$\begin{array}{ccc} \mathrm{Hom}_{X^+}(X, L_X) & \xrightarrow{\quad i \quad} & \mathrm{Aut}_S(X) \\ \Big\downarrow{\scriptstyle h} & & \Big\downarrow{\scriptstyle f} \\ \mathrm{Hom}_{X^+}(X, L_X) & \xrightarrow{\quad i \quad} & \mathrm{Aut}_S(X) \end{array}$$

Figure 22: Functoriality of i under transport of structure.

Thus $f \circ i = i \circ h$. On the other hand, f is described by the following commutative diagram.

$$\begin{array}{ccc} X & \xrightarrow{\quad a \quad} & X \\ \Big\downarrow{\scriptstyle u} & & \Big\downarrow{\scriptstyle u} \\ X & \xrightarrow{\quad f(a) \quad} & X \end{array}$$

Figure 23: The automorphism f acts by conjugation with u.

In other words,

$$f(a) = u \circ a \circ u^{-1} \qquad \text{for every } a \in \mathrm{Aut}_S(X).$$

Writing $i(h(m)) = f(i(m))$ now gives the asserted formula.

Corollary 0.7 bis. *Let X be an S-scheme such that $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-monoid. Then L_X has the structure of an S-monoid, and there is a split exact sequence of S-monoids:*

[26] Editor's note: what follows has been expanded.

$$1 \longrightarrow L'_X \xrightarrow{i} L_X \underset{s}{\overset{p}{\rightleftarrows}} X^+ \longrightarrow 1$$

Figure 24: The split exact sequence of S-monoids in Corollary 0.7 bis.

The monoid law induced on L'_X coincides with its abelian-group structure. In particular, if $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group, then L_X is an S-group and is the semidirect product of X^+ and L'_X.
Proof. Since $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-monoid,

$$X^+ = \prod_{S_{\mathscr{J}}/S} X_{\mathscr{J}}$$

is an S-monoid; indeed, $X^+(Y) = X_{\mathscr{J}}(Y_{\mathscr{J}})$ for every $Y \to S$. For every S-scheme Y, let $\widetilde{Y}_{\mathscr{J}}$ denote the affine $Y_{\mathscr{J}}$-scheme corresponding to the quasi-coherent $\mathscr{O}_{Y_{\mathscr{J}}}$-algebra

$$\mathscr{O}_{Y_{\mathscr{J}}} \oplus \mathscr{J}\mathscr{O}_Y,$$

that is, the graded algebra associated with the filtration $\mathscr{O}_Y \supset \mathscr{J}\mathscr{O}_Y$. Then $L_X(Y)$ identifies with $X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}})$, and $L'_X(Y)$ identifies with the kernel of the morphism

$$p : X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}}) \longrightarrow X_{\mathscr{J}}(Y_{\mathscr{J}})$$

induced by the *zero section* $Y_{\mathscr{J}} \to \widetilde{Y}_{\mathscr{J}}$. Equivalently, that section is induced by the morphism of $\mathscr{O}_{Y_{\mathscr{J}}}$-algebras

$$\mathscr{O}_{\widetilde{Y}_{\mathscr{J}}} \longrightarrow \mathscr{O}_{Y_{\mathscr{J}}}$$

that vanishes on the ideal $\mathscr{J}\mathscr{O}_Y$. Thus, for every $Y \to S$, there is a split exact sequence of monoids, functorial in Y:

$$1 \longrightarrow L'_X(Y) \xrightarrow{i} L_X(Y) \underset{s}{\overset{p}{\rightleftarrows}} X^+(Y) \longrightarrow 1 \ .$$

Figure 25: The pointwise split exact sequence for $L_X(Y)$.

It remains to show that the monoid law induced on L'_X coincides with its abelian-group structure. Let μ be the monoid law of L_X, and let e be its unit section. We must prove that, for every $m, m' \in L'_X(Y)$,

$$\mu(m \cdot e, m' \cdot e) = (m + m') \cdot e.$$

This may be seen in either of two ways. First, repeat the proof of (0.1.10 $(*)$), replacing the morphism $f : X \to W$ there by

$$\mu : L_X \times_S L_X \longrightarrow L_X.$$

Identify $X^+(Y) = X_{\mathscr{J}}(Y_{\mathscr{J}})$ with its image under s in $L_X(Y) = X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}})$. For all $g, g' \in X_{\mathscr{J}}(Y_{\mathscr{J}})$ and $m, m' \in L'_X(Y)$, one then obtains

$$\mu(m \cdot g, m' \cdot g') = L_\mu^{(g,g')}(m, m') \cdot \mu(g, g'). \qquad (\star)$$

Here $L_\mu^{(g,g')}$ denotes the morphism derived from μ at the point (g, g'); in other words, $\widetilde{Y}_{\mathscr{J}}$ lies over $L_X \times_S L_X$ through (g, g'). In particular,

$$\mu(m \cdot e, m' \cdot e) = L_\mu^{(e,e)}(m, m') \cdot e.$$

But

$$L_\mu^{(e,e)}(m,m') = L_{\ell_e}(m') + L_{r_e}(m),$$

where ℓ_e, respectively r_e, denotes left, respectively right, translation by e. Each is the identity map of $X_{\mathscr{J}}$, and therefore $L_\mu^{(e,e)}(m,m') = m + m'$.

Alternatively, one may argue as follows (cf. the proof of [DG70], §II.4, Theorem 3.5). By Lemma 0.1.11, formation of X^+ and of L_X commutes with products, and hence so does formation of L'_X. It follows that the morphism

$$\mu' : L'_X \times L'_X \longrightarrow L'_X$$

induced by μ is a homomorphism for the abelian-group structure. Lemma 3.10 of Exposé II then shows that μ' coincides with the abelian-group law.

0.8.[27] Now let X be an S-scheme such that $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group. Suppose that there is an S-morphism

$$P : X \times_S X \longrightarrow X$$

whose base change

$$P_{\mathscr{J}} : X_{\mathscr{J}} \times_{S_{\mathscr{J}}} X_{\mathscr{J}} \longrightarrow X_{\mathscr{J}}$$

is the group law of $X_{\mathscr{J}}$. An important special case is that in which X is an S-group and P is its group law. We obtain a morphism

$$L_P : L_X \times_S L_X \simeq L_{X\times_S X} \longrightarrow L_X.$$

In fact, this morphism does not depend on P, since it can be computed from the group law $P_{\mathscr{J}}$ of $X_{\mathscr{J}}$, as we now show.[28]

By Proposition 0.2(ii) and (iii), for every $Y \to S$, every $x, x' \in X(Y)$, and every $m, m' \in L'_X(Y)$,

$$\begin{aligned} P(m\cdot x, m'\cdot x') &= P\big((m,m')\cdot(x,x')\big) \\ &= L_P^{(x,x')}(m,m')\cdot\mu(g,g'), \end{aligned}$$

where g, respectively g', is the image of x, respectively x', in $X^+(Y)$. Moreover (cf. the proof of 0.10), $L_P^{(x,x')} = L_\mu^{(g,g')}$, and by 0.7 bis ($\star$) this is the element of $L'_X(Y)$ defined by the equality

$$L_\mu^{(g,g')}(m,m')\cdot\mu(g,g') = \mu(m\cdot g, m'\cdot g')$$

in $L_X(Y)$.

Write $\times$, rather than μ, for the group law of L_X, and write Ad for the *adjoint action* of X^+ on L'_X. This action factors through X_0 and is induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$. Then

$$\begin{aligned} L_\mu^{(g,g')}(m,m')\times g\times g' &= m\times g\times m'\times g' \\ &= \big(m\times \mathrm{Ad}(g)(m')\big)\times g\times g', \end{aligned}$$

and hence

$$L_P^{(x,x')}(m,m') = m\times \mathrm{Ad}(g)(m').$$

We have therefore proved:

[27] Editor's note: the number 0.8 has been inserted here to mark the return to the original text.
[28] Editor's note: the original has been expanded in what follows.

Proposition 0.8. *Let $P : X \times_S X \to X$ be an S-morphism such that $P_{\mathscr{J}}$ equips $X_{\mathscr{J}}$ with an $S_{\mathscr{J}}$-group structure. Let $\times$ denote the group law of L'_X, let $(m, x) \mapsto m \cdot x$ denote the morphism $L'_X \times_S X \to X$ defining the action of L'_X on X, and let*

$$\mathrm{Ad} : X^+ \longrightarrow \mathrm{Aut}_{S\text{-gr}}(L'_X)$$

be the adjoint action of X^+ on L'_X, induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$. Then, for every $S' \to S$, every $x, x' \in X(S')$, and every $m, m' \in L'_X(S')$,

$$P(m \cdot x, m' \cdot x') = (m \times \mathrm{Ad}(p_X(x))(m')) \cdot P(x, x'). \tag{0.8.1}$$

If X is an S-group, we shall write $*$ for its law, e for its unit section, and i for the S-morphism defined by

$$i(m) = m \cdot e$$

for every $S' \to S$ and every $m \in L'_X(S')$.

Corollary 0.9. *Let X be an S-group. Then X^+ is naturally an S-group, and p_X is a morphism of S-groups. Moreover, the S-morphism*

$$i : L'_X \longrightarrow X, \qquad m \longmapsto m \cdot e,$$

is an isomorphism of S-groups from L'_X onto $\mathrm{Ker}(p_X)$, and, for every $S' \to S$, every $x' \in X(S')$, and every $m \in L'_X(S')$,

$$m \cdot x' = (m \cdot e) * x' = i(m) * x'. \tag{0.9.1}$$

The first two assertions were already proved in 0.1.2. Since X is formally principal homogeneous over X^+ under $L_X = L'_X \times_S X^+$, the morphism i is indeed an isomorphism of S-functors from L'_X onto the kernel of p_X. The fact that i is a group morphism and formula (0.9.1) follow from (0.8.1), applied respectively with $x = x' = e$, and with $x = e,\ m' = 1$.

Corollary 0.10. *Let X be an S-group. With the preceding notation, for every $S' \to S$, every $x \in X(S')$, and every $m' \in L'_X(S')$,*

$$x * i(m') * x^{-1} = i(\mathrm{Ad}(p_X(x))(m')). \tag{0.10.1}$$

This follows from the equality $i(m') * x^{-1} = m' \cdot x^{-1}$ and from (0.8.1), applied with $m = 1$ and $x' = x^{-1}$.

Thus, when X is an S-group, we have explicitly determined the kernel of $X \to X^+$ and the action of the inner automorphisms of X on this kernel. We shall now see that the same can be done for certain, not necessarily representable, S-group functors. One case will be useful to us, namely the functors $\underline{\mathrm{Aut}}$ of Exposé I, 1.7. We state it at once in the next proposition.

Proposition 0.11. *Let E be an S-scheme, and put $X = \underline{\mathrm{Aut}}_S(E)$. The kernel of the morphism of S-group functors*

$$p_X : X \longrightarrow X^+$$

is canonically identified with the commutative S-group functor L'_X defined by

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{E_0 \times_{S_0} Y_0}}\left(\Omega^1_{E_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J}\,\mathscr{O}_{E \times_S Y}\right),$$

where Y denotes a variable S-scheme.

Indeed, for a variable S-scheme Y, we have

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$$

and

$$\begin{aligned}\mathrm{Hom}_S(Y, X^+) &= \mathrm{Hom}_S(Y_{\mathscr{J}}, X)\\ &= \mathrm{Aut}_{Y_{\mathscr{J}}}(E \times_S Y_{\mathscr{J}})\\ &= \mathrm{Aut}_{Y_{\mathscr{J}}}((E \times_S Y) \times_Y Y_{\mathscr{J}}).\end{aligned}$$

Applying 0.7 (b) to the Y-scheme $E \times_S Y$, we obtain an isomorphism of groups

$$\mathrm{Hom}_S(Y, L'_X) \simeq \mathrm{Ker}\Big(\mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)\Big).$$

This isomorphism is readily checked to be functorial in the S-scheme Y. Hence it yields an isomorphism of S-groups

$$L'_X \simeq \mathrm{Ker}(X \longrightarrow X^+),$$

which completes the proof of Proposition 0.11.

Corollary 0.12.[29] *Retain the notation of 0.11: E is an S-scheme and $X = \underline{\mathrm{Aut}}_S(E)$. There is a natural action f of X on L'_X, defined as follows. For every S-scheme Y,*

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$$

and

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{E_0 \times_{S_0} Y_0}}\left(\Omega^1_{E_0 \times_{S_0} Y_0/Y_0}, \mathscr{J}\mathscr{O}_{E \times_S Y}\right).$$

Here

$$\Omega^1_{E_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0} \simeq \Omega^1_{E_0 \times_{S_0} Y_0/Y_0}$$

(see EGA IV_4, 16.4.5). The first group acts on the second by transport of structure, and this action is functorial in Y. One then has

$$x\, i(m)\, x^{-1} = i(f(x)m), \tag{0.12.1}$$

for every $Y \to S$, $x \in \mathrm{Hom}_S(Y, X)$, and $m \in \mathrm{Hom}_S(Y, L'_X)$.

Indeed, this follows from 0.7 (c), applied to the Y-scheme $E \times_S Y$.

Reminder 0.13. The direct image of a quasi-coherent module under a morphism of finite presentation is quasi-coherent. Under the same hypotheses, formation of the direct image commutes with flat base change. In the situation

$$\begin{array}{ccc} T & \xleftarrow{\;g'\;} & T' = T \times_S S' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{\;g\;} & S' \end{array}$$

Figure 26: The flat base-change square in Reminder 0.13.

if f (and hence f') is of finite presentation and g (and hence g') is flat, then for every quasi-coherent $\mathscr{O}_T$-module $\mathscr{F}$,

$$f_*(\mathscr{F}) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} = f'_*(\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}),$$

[29] Editor's note: "Remark" has been changed to "Corollary".

or, more elegantly,

$$g^*(f_*\mathscr{F}) = f'_*(g'^*\mathscr{F}).$$

These two facts hold more generally for a quasi-compact and quasi-separated morphism f; see EGA I, 9.2.1, and EGA III_1, 1.4.15 in the quasi-compact separated case (taking EGA III_2, $\mathrm{Err}_{\mathrm{III}}$ 25 into account), as well as EGA IV_1, 1.7.4 and 1.7.21.

Remark 0.14.[30] Return to the notation of 0.11: let E be an S-scheme, put $X = \underline{\mathrm{Aut}}_S(E)$, and let L'_X be the commutative S-group functor defined by

$$\begin{aligned} L'_X(Y) &= \mathrm{Hom}_{\mathscr{O}_{E_0\times_{S_0}Y_0}}\left(\Omega^1_{E_0/S_0}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_{E\times_S Y}\right) \\ &= \mathrm{Hom}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, \mathscr{J}\mathscr{O}_{E\times_S Y}\right) \\ &= \Gamma\Big(E\times_S Y, \widetilde{\mathrm{Hom}}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, \mathscr{J}\mathscr{O}_{E\times_S Y}\right)\Big). \end{aligned}$$

Suppose that Y is flat over S. There are then isomorphisms

$$\mathscr{J}\mathscr{O}_{E\times_S Y} \xleftarrow{\sim} (\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_S}\mathscr{O}_Y \xrightarrow{\sim} (\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}.$$

Suppose in addition that E is of finite presentation over S. Then $\Omega^1_{E/S}$ is an $\mathscr{O}_E$-module of finite presentation (EGA IV_4, 16.4.22), and therefore, by EGA 0_{I}, 6.7.6,

$$\begin{aligned} &\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, (\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_S}\mathscr{O}_Y\right) \\ &\quad\simeq \widetilde{\mathrm{Hom}}_{\mathscr{O}_E}\left(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E\right)\otimes_{\mathscr{O}_S}\mathscr{O}_Y. \end{aligned}$$

Let $\pi : E \to S$ and $g : Y \to S$ be the structural morphisms. Apply 0.13 to the diagram

$$\begin{array}{ccc} E & \xleftarrow{\ g'\ } & E\times_S Y \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ S & \xleftarrow{\ g\ } & Y \end{array}$$

Figure 27: The structural base-change square used in Remark 0.14.

and to the $\mathscr{O}_E$-module

$$\mathscr{F} = \widetilde{\mathrm{Hom}}_{\mathscr{O}_E}\left(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E\right).$$

We obtain

$$\begin{aligned} \Gamma(E\times_S Y, g'^*\mathscr{F}) &= \Gamma(Y, \pi'_* g'^*\mathscr{F}) \\ &= \Gamma(Y, g^*\pi_*\mathscr{F}) \\ &= W(\pi_*\mathscr{F})(Y). \end{aligned}$$

Thus, if E is of finite presentation over S, we have

$$L'_X = W\left(\pi_*\widetilde{\mathrm{Hom}}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\right) \tag{0.14.1}$$

on the category of S-schemes flat over S. Notice also that the module to which W is applied is quasi-coherent, by EGA I, 9.1.1 and 9.2.1.

[30] Editor's note: The first part of this remark has been expanded, and a second part has been added.

Let L_0 be the S_0-functor[31]

$$L_0 = W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0}, \mathscr{J}\mathscr{O}_E)\Big).$$

Returning to the definition of $L'_X(Y)$, and using the isomorphism

$$\mathscr{J}\mathscr{O}_{E\times_S Y} \simeq (\mathscr{J}\mathscr{O}_E) \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0},$$

the same reasoning gives

$$L'_X(Y) = L_0(Y_0) = L_0(Y \times_S S_0) = \left(\prod_{S_0/S} L_0\right)(Y).$$

Consequently, on the category of S-schemes flat over S,

$$L'_X = \prod_{S_0/S} W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0}, \mathscr{J}\mathscr{O}_E)\Big).$$

It is not obvious that the action of X on L'_X defined in 0.12 comes from an action of $X_0 = \underline{\mathrm{Aut}}_{S_0}(E_0)$ on L_0. This is nevertheless the case when E is also flat over S.

Indeed, in that case there are canonical isomorphisms

$$\mathscr{J}\mathscr{O}_E \simeq \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_E \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0},$$

and hence

$$L_0 \simeq W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}\left(\Omega^1_{E_0/S_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0}\right)\Big),$$

and

$$L'_X = \prod_{S_0/S} W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}\left(\Omega^1_{E_0/S_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0}\right)\Big). \tag{0.14.2}$$

For every S_0-scheme T, we then have

$$L_0(T) \simeq \mathrm{Hom}_{\mathscr{O}_{E_0\times_{S_0}T}}\left(\Omega^1_{E_0\times_{S_0}T/T}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0\times_{S_0}T}\right).$$

Moreover,

$$\mathrm{Hom}_{S_0}(T, X_0) = \mathrm{Aut}_T(E_0 \times_{S_0} T)$$

acts on $L_0(T)$ by transport of structure, functorially in T. Finally, for every S-scheme Y flat over S, the action by transport of structure of

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$$

on $L'_X(Y) = L_0(Y_0)$ factors through $\mathrm{Aut}_{Y_0}(E_0 \times_{S_0} Y_0)$.

Finally, let us extract the following two propositions from SGA 1, Exposé III.

Proposition 0.15. *(SGA 1, Exposé III, 6.8)*[32] *For every affine $S_{\mathscr{J}}$-scheme Y smooth over $S_{\mathscr{J}}$, there exists an S-scheme X, smooth over S, such that $X \times_S S_{\mathscr{J}} \simeq Y$; such an X is unique up to a non-unique isomorphism.*

[31] Editor's note: What follows has been added.

[32] Editor's note: This "global" lifting result uses the "local" lifting result of loc. cit., 4.1, which is stated for S locally Noetherian; see EGA IV_4, 18.1.1 for the general case.

Proposition 0.16. *(SGA 1, Exposé III, 5.5)*[33] *Let X be an S-scheme smooth over S. For every affine S-scheme Y, the canonical map*

$$p_X(Y) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_\mathscr{J}}(Y_\mathscr{J}, X_\mathscr{J})$$

is surjective.

Corollary 0.17. *Let E be an affine S-scheme smooth over S, and put $X = \underline{\mathrm{Aut}}_S(E)$. For every affine S-scheme Y, the canonical map*

$$\begin{aligned}\mathrm{Aut}_Y(E \times_S Y) = \mathrm{Hom}_S(Y, X) &\longrightarrow \mathrm{Hom}_S(Y, X^+) \\ &= \mathrm{Aut}_{Y_\mathscr{J}}(E_\mathscr{J} \times_{S_\mathscr{J}} Y_\mathscr{J})\end{aligned}$$

is surjective.

Indeed, $Y \times_S E$ is affine over the affine scheme Y, hence is affine. Applying 0.16, we see that every $S_\mathscr{J}$-morphism

$$Y_\mathscr{J} \times_{S_\mathscr{J}} E_\mathscr{J} \longrightarrow E_\mathscr{J}$$

extends to an S-morphism

$$Y \times_S E \longrightarrow E.$$

In other words,[34] every $Y_\mathscr{J}$-endomorphism of $Y_\mathscr{J} \times_{S_\mathscr{J}} E_\mathscr{J}$ lifts to a Y-endomorphism of $Y \times_S E$. Then 0.7 (a) shows that every $Y_\mathscr{J}$-automorphism of $Y_\mathscr{J} \times_{S_\mathscr{J}} E_\mathscr{J}$ lifts to a Y-automorphism of $Y \times_S E$, which is the asserted property.

1. Extensions and Cohomology

1.1.

Let $\mathscr{C}$ be a category stable under fiber products.[35] Let S be an object of $\mathscr{C}$, let G be a representable S-group, and let F be an S-functor in commutative groups on which G acts. The cohomology groups $H^n(G, F)$ were defined in I 5.1. Recall that they are the cohomology groups of a complex denoted $C^*(G, F)$. Writing

$$(G/S)^n = G \times_S \cdots \times_S G \qquad (n \text{ factors}),$$

one has

$$C^n(G, F) = \mathrm{Hom}_S((G/S)^n, F).$$

Since G, and hence each $(G/S)^n$, is representable, one also has

$$C^n(G, F) = F((G/S)^n).$$

It follows from this and from the definition of the coboundary operator that the complex $C^*(G, F)$ depends only on the restriction of F to the full subcategory of $\mathscr{C}/S$ whose objects are the Cartesian powers of G over S. Consequently:

Lemma 1.1.1. *Let $\mathscr{C}$ be a category stable under fiber products, let S be an object of $\mathscr{C}$, and let G be a representable S-group. Denote by $\mathscr{C}(G)$ the full subcategory of $\mathscr{C}/S$ whose objects*

[33] Editor's note: This is an immediate consequence of the definition of "(formally) smooth" adopted in EGA IV_4, 17.3.1 (and 17.1.1).

[34] Editor's note: The original has been modified slightly so that the hypotheses of 0.7 apply exactly.

[35] Editor's note: the hypothesis that $\mathscr{C}$ is stable under fiber products has been added. It holds in the applications in which $\mathscr{C}$ is the category of schemes (**Sch**), or the category of functors (**Sch**)$^\circ \to$ (**Ens**). If this hypothesis is omitted, one must instead assume below that the fiber products $G \times_S \cdots \times_S G$ are representable.

are the Cartesian powers of G over S. Let F and F' be two S-functors in commutative groups on which G acts. If F and F' have the same restriction to $\mathscr{C}(G)$, there is a canonical isomorphism

$$H^*(G,F) \xrightarrow{\sim} H^*(G,F').$$

1.1.2. Cohomology and restriction of scalars.[36]

We record another comparison result. Let $T \to S$ be a morphism in $\mathscr{C}$. If F is a T-functor in commutative groups, then the functor obtained by restriction of scalars (cf. Exposé II, 1),

$$F_1 = \prod_{T/S} F,$$

is an S-functor in commutative groups, and there is a morphism of S-functors in groups[37]

$$u : \prod_{T/S} \underline{\mathrm{Aut}}_{T\text{-gr.}}(F) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(F_1).$$

Now let G be an S-functor in groups, and suppose an action of G_T on F is given by

$$G_T \longrightarrow \underline{\mathrm{Aut}}_{T\text{-gr.}}(F).$$

By the definition of $\prod_{T/S}$, this gives a morphism of S-functors in groups

$$G \longrightarrow \prod_{T/S} \underline{\mathrm{Aut}}_{T\text{-gr.}}(F).$$

Composing with u therefore gives an action of G on $F_1 = \prod_{T/S} F$.[38]

Lemma 1.1.2. *Under the preceding assumptions, there is a canonical isomorphism*

$$H^*\left(G, \prod_{T/S} F\right) \simeq H^*(G_T, F).$$

Indeed, by the definition of cohomology, the standard complexes are canonically isomorphic.

[36] Editor's note: the heading of this paragraph has been added.

[37] Editor's note: explicitly, let $S' \to S$ and $\alpha \in \underline{\mathrm{Aut}}_{(T\times_S S')\text{-gr.}}(F_{T\times_S S'})$. For every $U \to T \times_S S'$, let $\alpha_U \in \mathrm{Aut}_{\mathrm{gr.}}(F_{T\times_S S'}(U))$ be the element defined by α. Then $u(S')(\alpha)$ is the element β of

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(F_1)(S') = \underline{\mathrm{Aut}}_{S'\text{-gr.}}\left(\prod_{T\times_S S'/S'} F_{T\times_S S'}\right)$$

such that $\beta_{S''} = \alpha_{T\times_S S''}$ for every $S'' \to S'$.

[38] Editor's note: explicitly, for every $S' \to S$, the action of $G(S')$ on $F_1(S') = F(T \times_S S')$ is given by the action of $G_T(T \times_S S') = G(T \times_S S')$ on $F(T \times_S S')$, together with the group morphism $G(S') \to G(T \times_S S')$ corresponding to the projection $T \times_S S' \to S'$. Equivalently, the action of G_T on F is a morphism $G_T \times_T F \to F$. Applying $\prod_{T/S}$, and writing $G_1 = \prod_{T/S} G_T$, gives, by II 1.2, a morphism $G_1 \times_S F_1 \to F_1$, hence an action of G_1 on F_1. The action of G is obtained through the natural morphism $G \to G_1$, which corresponds by adjunction to id_G.

1.2. Lifting group morphisms.[39]

Following the general principles, we make the following definition.

Definition 1.2.1. *Let*

$$1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be a sequence of morphisms of $\widehat{\mathscr{C}}$-groups. It is called exact *if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the following sequence of ordinary groups is exact:*

$$1 \longrightarrow M(S) \xrightarrow{u(S)} E(S) \xrightarrow{v(S)} G(S);$$

(ii) *for every object H of $\widehat{\mathscr{C}}$, the following sequence of ordinary groups is exact:*

$$1 \longrightarrow \mathrm{Hom}(H, M) \xrightarrow{u(H)} \mathrm{Hom}(H, E) \xrightarrow{v(H)} \mathrm{Hom}(H, G).$$

Taking $H = G$ in (ii), one sees that the set of sections of v (which need not preserve the group structures) is either empty or principal homogeneous under $\mathrm{Hom}(G, M)$. Suppose that it is nonempty, and let

$$s : G \longrightarrow E$$

be a section of v. For every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $x \in G(S)$, the element $s(x) \in E(S)$ defines an inner automorphism of E_S that normalizes M_S (more precisely, the image of M_S under u_S), and hence an automorphism of M_S.

Scholium 1.2.1.1. [40] *If M is commutative, one sees set-theoretically that this automorphism is independent of the chosen section and depends only on x, and that it depends multiplicatively on x. In summary, every exact sequence*

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

in which M is commutative and v admits a section determines a morphism of $\widehat{\mathscr{C}}$-groups

$$G \longrightarrow \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M),$$

called the action of G on M defined by the extension (E).

Definition 1.2.1.2. We saw in I 2.3.7 that v admits a section that is a morphism of $\widehat{\mathscr{C}}$-groups if and only if the extension (E) is isomorphic, as an extension, to the semidirect product of M by G for the preceding action. Such a section of v is called a *section of the extension* (E), or simply a *section of* (E).

If s is a section of (E) and

$$m \in \Gamma(M) \simeq \mathrm{Ker}\big(\Gamma(E) \to \Gamma(G)\big)$$

(for the definition of Γ, see I 1.2), then the morphism $G \to E$ defined by

$$x \longmapsto u(m)s(x)u(m)^{-1}$$

[39] Editor's note: this heading has been added.

[40] Editor's note: the numbering 1.2.1.1 and 1.2.1.2 has been added to highlight the notions introduced here.

is also a section of (E).[41] It is called the section deduced from s by the inner automorphism defined by m, or by $u(m)$.

Lemma 1.2.2. *Let*

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups in which M is commutative and v admits a section. Let G act on M in the manner defined by (E).

(i) *The extension (E) canonically defines a class $c(E) \in H^2(G, M)$, whose vanishing is necessary and sufficient for the existence of a section of (E).*

(ii) *If $c(E) = 0$, the set of sections of (E) is principal homogeneous under $Z^1(G, M)$, and the set of sections of (E), modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$, is principal homogeneous under $H^1(G, M)$.*

(iii) [42] *Let s be a section of (E). The set of conjugates of s by the inner automorphisms defined by $\Gamma(M)$ is in bijection with $\Gamma(M)/H^0(G, M)$.*

The proof is exactly the same as for ordinary groups; starting from a section of v ensures the functoriality of the set-theoretic computations. We briefly indicate the principal steps.

(a) To every section s of v, associate the morphism

$$D_s : G \times G \longrightarrow M$$

defined set-theoretically by

$$u(D_s(x, y)) = s(xy)s(y)^{-1}s(x)^{-1}.$$

The following computation shows that D_s is a 2-cocycle.[43] By the definition of the differential in the standard complex (I 5.1),

$$(\partial^2 D_s)(x, y, z) = \big(s(x)D_s(y, z)s(x)^{-1}\big) \cdot D_s(x, y)^{-1} \cdot D_s(xy, z)^{-1} \cdot D_s(x, yz).$$

Substituting the definition of D_s into this formula gives, without using commutativity,

$$D_s \in Z^2(G, M).$$

(b) If s and s' are two sections of v, there is a morphism $f : G \to M$ such that $s(x) = f(x)s'(x)$. Then

$$\begin{aligned} D_{s'}(x, y) &= f^{-1}(xy)D_s(x, y)s(x)f(y)s(x)^{-1}f(x) \\ &= D_s(x, y) \cdot f^{-1}(xy) \cdot \big(s(x)f(y)s(x)^{-1}\big) \cdot f(x), \end{aligned}$$

and therefore

$$D_{s'} = D_s \cdot \partial^1 f.$$

This shows that the class of D_s in $H^2(G, M)$ is independent of the chosen section s of v; it is the class $c(E)$ of the extension (E).[44]

[41] Editor's note: in the right-hand side, x has been corrected to $s(x)$.

[42] Editor's note: this item has been added as the counterpart of 1.2.4(iii). The assertion is valid for every section of v; see the proof.

[43] Editor's note: the original has been corrected to make it compatible with I 5.1.

[44] Editor's note: the preceding sentence has been added.

(c) Let s and s' be two sections of v, and let $m \in \Gamma(M)$. The equality

$$s(x) = m^{-1}s'(x)m \qquad (x \in G(S),\ S \in \mathrm{Ob}\,\mathscr{C})$$

is equivalent to

$$s(x) = m^{-1}s'(x)m\,s'(x)^{-1}s'(x), \qquad \text{that is,} \qquad s = \partial^0 m \cdot s'.$$

In particular, the stabilizer of s in $\Gamma(M)$ is the subgroup of elements m such that $\partial^0 m = e_M$, namely $H^0(G, M)$. This already proves (iii).

(d) The argument is now familiar.[45] Let s_0 be an arbitrary section of v. There exists a section s, necessarily of the form $s = f \cdot s_0$, that is a group morphism, that is, satisfies $D_s = 0$, if and only if

$$(D_{s_0})^{-1} = \partial^1 f,$$

for some $f : G \to M$, or equivalently if and only if $c(E) = 0$. This proves (i).

In that case, the sections of (E) are precisely the sections $s' = h \cdot s$, where $h : G \to M$ satisfies $\partial^1 h = 0$, that is, $h \in Z^1(G, M)$. Moreover, by (c), two such sections $h_1 \cdot s$ and $h_2 \cdot s$ are conjugate under $\Gamma(M)$ if and only if h_1 and h_2 have the same image in $H^1(G, M)$. This proves (ii).

Still let

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups with M commutative, and let

$$f : H \longrightarrow G$$

be a morphism of $\widehat{\mathscr{C}}$-groups.

Consider $E_f = H \times_G E$. It is a $\widehat{\mathscr{C}}$-group, and the projection $v_f : E_f \to H$ is a morphism of $\widehat{\mathscr{C}}$-groups; the same holds for $e_f : E_f \to E$. Sending M into E by u and into H by the unit morphism also defines a morphism of $\widehat{\mathscr{C}}$-groups $u_f : M \to E_f$. We have therefore constructed the following commutative diagram of $\widehat{\mathscr{C}}$-groups.

$$\begin{array}{lccccccc}
(E) & 1 & \longrightarrow & M & \xrightarrow{u} & E & \xrightarrow{v} & G \\
 & & & \uparrow{\scriptstyle \mathrm{id}} & & \uparrow{\scriptstyle e_f} & & \uparrow{\scriptstyle f} \\
(E_f) & 1 & \longrightarrow & M & \xrightarrow{u_f} & E_f & \xrightarrow{v_f} & H
\end{array}$$

Figure 28: The pullback extension (E_f) and its morphism to (E).

It follows immediately that:

Lemma 1.2.3.

(i) *The sequence (E_f) is exact.*

(ii) *The map $s \mapsto e_f \circ s = f'$ gives a bijective correspondence between the morphisms*

$$s : H \longrightarrow E_f$$

such that $v_f \circ s = \mathrm{id}$, that is, the sections of v_f, and the morphisms

$$f' : H \longrightarrow E$$

such that $v \circ f' = f$, that is, the morphisms f' lifting f.

[45] Editor's note: what follows has been expanded.

(iii) *Under this correspondence, the sections of* (E_f) *correspond to the group morphisms* f' *lifting* f.

Applying Lemma 1.2.2 to the extension (E_f), and taking 1.2.3 into account, gives the following proposition, which formally contains Lemma 1.2.2.

Proposition 1.2.4. *Let*

$$(E) \qquad 1 \longrightarrow M \longrightarrow E \stackrel{v}{\to} G$$

be an exact sequence of $\widehat{\mathscr{C}}$*-groups with* M *commutative. Let* $f : H \to G$ *be a morphism of* $\widehat{\mathscr{C}}$*-groups, and suppose that it lifts to a morphism, not necessarily a group morphism,* $f' : H \to E$. *Let* H *act on* M *by the composite morphism, which is multiplicative and independent of the choice of* f',

$$H \stackrel{f'}{\longrightarrow} E \stackrel{\text{int}}{\longrightarrow} \underline{\text{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M).$$

(i) *The morphism* f *canonically defines a class* $c(f) \in H^2(H, M)$, *whose vanishing is necessary and sufficient for the existence of a morphism of* $\widehat{\mathscr{C}}$*-groups*

$$f' : H \longrightarrow E$$

lifting f.

(ii) *If* $c(f) = 0$, *the set of morphisms of* $\widehat{\mathscr{C}}$*-groups* f' *lifting* f, *modulo the action of the inner automorphisms defined by elements of* $\Gamma(M)$ *(that is, by elements* $m \in \Gamma(E)$ *such that* $v(m) = e$*), is principal homogeneous under* $H^1(H, M)$.

(iii) *If* $f' : H \to E$ *is a group morphism lifting* f, *the set of transforms of* f' *by the inner automorphisms defined by the elements of* $\Gamma(M)$ *is isomorphic to*

$$\Gamma(M)/\Gamma(M^H) = \Gamma(M)/H^0(H, M).$$

1.3. Extensions of group laws.

Consider the following situation: one has a morphism in $\widehat{\mathscr{C}}$,

$$(\dagger) \qquad p : X \longrightarrow Y,$$

and a commutative $\widehat{\mathscr{C}}$-group M acting on X, such that X is formally principal homogeneous over Y under M_Y.

If $g : Y \to Z$ is an arbitrary morphism in $\widehat{\mathscr{C}}$, then $g \circ p : X \to Z$ is invariant under M: for every $S \in \text{Ob}\,\mathscr{C}$, the map $(g \circ p)(S)$ is invariant under the action of $M(S)$ on $X(S)$. Conversely, we shall assume that the following condition holds for $n = 1, 2, 3, 4$:

$(+)_n$: *Every morphism from* X^n *to* M *that is invariant under the action of* M^n *on* X^n *factors uniquely through* $p^n : X^n \to Y^n$, *where the superscript* n *denotes a Cartesian power.*

Lemma 1.3.1.

(i) *If* h *is a morphism from* Y *to* M, *the automorphism* u_h *of* X, *defined set-theoretically by* $x \mapsto h(p(x)) \cdot x$, *preserves the fibers of* p *and commutes with the operations of* M *on* X.[46] *Thus, for every* $S \in \text{Ob}\,\mathscr{C}$, $x \in X(S)$, *and* $m \in M(S)$, *one has*

$$p(h(p(x)) \cdot x) = p(x), \qquad m \cdot h(p(x)) \cdot x = h(p(m \cdot x)) \cdot m \cdot x.$$

[46] Editor's note: here and below, "commutes with p and the operations of M" has been replaced by "preserves the fibers of p and commutes with the operations of M." The equalities that follow have also been added.

(ii) *This construction gives a bijective correspondence between morphisms from Y to M and automorphisms of X that preserve the fibers of p and commute with the operations of M.*

The first assertion is clear, since $p(m \cdot x) = p(x)$ and M is commutative. Conversely, an automorphism u of X preserving the fibers of p can be written set-theoretically as $x \mapsto g(x)x$, where g is a certain morphism from X to M. If u commutes with the operations of M, then g is invariant under M,[47] and the conclusion follows from condition $(+)_1$.

Suppose now, in addition, that a group law on Y and an action of Y on M are given, that is, a morphism of $\widehat{\mathscr{C}}$-groups

$$(\ddagger) \qquad f : Y \longrightarrow \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M).$$

Definition 1.3.2. *A composition law on X,*

$$P : X \times X \longrightarrow X,$$

is called admissible *if it satisfies the following two conditions:*

(i) *P lifts the group law of Y; that is, the following diagram is commutative.*

$$\begin{array}{ccc} X \times X & \xrightarrow{\quad P \quad} & X \\ {\scriptstyle (p,p)}\downarrow & & \downarrow{\scriptstyle p} \\ Y \times Y & \longrightarrow & Y \end{array}$$

Figure 29: The compatibility of an admissible composition law with the group law on Y.

(ii) *For every $S \in \mathrm{Ob}\,\mathscr{C}$ and all $x, y \in X(S)$, $m, n \in M(S)$, one has*

$$(++) \qquad P(m \cdot x, n \cdot y) = m \cdot f(p(x))(n) \cdot P(x, y).$$

Proposition 1.3.3. *For a group law $*$ on X to be admissible, it is necessary and sufficient that the following four conditions hold:*

(i) *$p : X \to Y$ is a group morphism.*

(ii) *The morphism $i : M \to X$, defined by $i(m) = m \cdot e_X$, is a group isomorphism from M onto* $\ker(p)$*; in other words, set-theoretically,*

$$(m \cdot e_X) * (n \cdot e_X) = (mn) \cdot e_X.$$

(iii) *One has $m \cdot x = (m \cdot e_X) * x = i(m) * x$ for every $m \in M(S)$ and $x \in X(S)$.*

(iv) *The inner automorphisms of X act on* $\ker(p)$ *according to the set-theoretic formula*

$$x * i(m) * x^{-1} = i\big(f(p(x))m\big).$$

[47] Editor's note: indeed, $mg(x)x = mu(x) = u(mx) = g(mx)mx$, and hence $g(mx) = g(x)$.

The proof is immediate.

Lemma 1.3.4.[48] *Let $h : Y \to M$ be a morphism and let u_h be the automorphism $x \mapsto h(p(x)) \cdot x$ of X (cf. 1.3.1). Let P be an admissible composition law (respectively, an admissible group law) on X, and let P' be the composition law on X obtained from P by means of u_h, that is,*

$$P'(x, y) = u_h^{-1}(P(u_h(x), u_h(y))).$$

Then:

(i) *P' is an admissible composition law (respectively, an admissible group law).*

(ii) *For every $x, y \in X(S)$, with $S \in \mathrm{Ob}\,\mathscr{C}$, set $v = p(x)$ and $w = p(y)$. Then*

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

Proof. One has $u_h^{-1} = u_{h^\vee}$, where $h^\vee : Y \to M$ is defined by $h^\vee(y) = h(y)^{-1}$. By 1.3.2(i) and (ii),

$$P(h(v) \cdot x, h(w) \cdot y) = h(v) \cdot f(v)(h(w)) \cdot P(x, y), \qquad p(P(x, y)) = vw,$$

and hence

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

It follows immediately that P' satisfies conditions (i) and (ii) of 1.3.2.

Definition 1.3.5. *Two admissible composition laws obtained from one another by the procedure of 1.3.4 are called* equivalent.[49]

Proposition 1.3.6. *Suppose that an admissible composition law on X exists. Then:*

(i) *There exists a canonically determined class $c \in H^3(Y, M)$, whose vanishing is necessary and sufficient for the existence of an admissible associative composition law on X.*

(ii) *If $c = 0$, the set of admissible associative composition laws on X (respectively, the set of their equivalence classes) is principal homogeneous under $Z^2(Y, M)$ (respectively, $H^2(Y, M)$).*

The proof proceeds in several steps.

(a) Let P be an admissible composition law on X. Since P lifts the associative composition law of Y, there is a unique morphism $a : X^3 \to M$ such that

$$(*) \qquad P(x, P(y, z)) = a(x, y, z) \cdot P(P(x, y), z).$$

Applying conditions 1.3.2(i) and (ii), one sees at once that a is invariant under the action of M^3 on X^3.[50] Condition $(+)_3$ therefore gives:

[48] Editor's note: the statement and its proof have been expanded with a view to step (e) in the proof of 1.3.6.

[49] Editor's note: this is indeed an equivalence relation, since $u_h^{-1} = u_{h^\vee}$.

[50] Editor's note: for $x, y, z \in X(S)$, set $u = p(x)$, $v = p(y)$, and $w = p(z)$. Then

$$\begin{aligned} P(mx, P(m'y, m''z)) &= P(mx, m'f(v)(m'')P(y, z)) \\ &= mf(u)(m')f(uv)(m'')P(x, P(y, z)). \end{aligned}$$

On the other hand, since $p(P(x, y)) = uv$, one also has

$$\begin{aligned} P(P(mx, m'y), m''z) &= P(mf(u)(m')P(x, y), m''z) \\ &= mf(u)(m')f(uv)(m'')P(P(x, y), z). \end{aligned}$$

Comparison of these equalities with $(*)$ gives $a(mx, m'y, m''z) = a(x, y, z)$.

(1) There exists a unique morphism $DP : Y^3 \to M$ such that

$$P(x, P(y, z)) = DP(p(x), p(y), p(z)) \cdot P(P(x, y), z),$$

and P is associative if and only if $DP = 0$.

(b) Let us compute $P(P(P(x, y), z), t)$ step by step using the preceding formula. Put

$$p(x) = u, \qquad p(y) = v, \qquad p(z) = w, \qquad p(t) = h.$$

One obtains[51] the following pentagonal diagram, in which an arrow $a \xrightarrow{m} b$ means that $b = m \cdot a$.

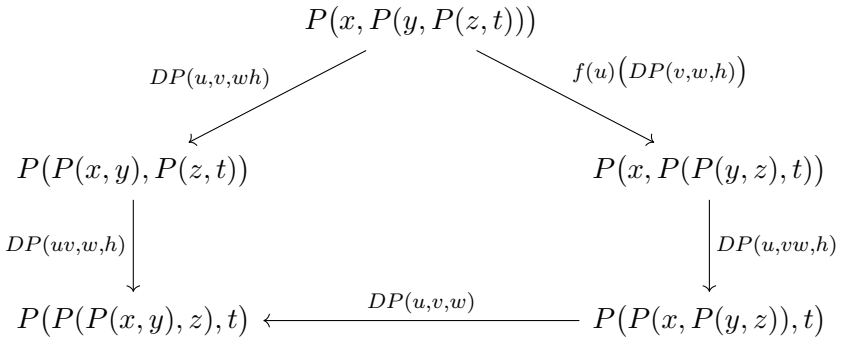

Figure 30: The pentagonal comparison used to compute the associativity obstruction.

Thus one finds

$$DP(u, v, w) \cdot DP(u, vw, h) \cdot f(u)(DP(v, w, h)) \cdot DP(u, v, wh)^{-1} \cdot DP(uv, w, h)^{-1} = e_M,$$

that is, $\partial^3 DP(u, v, w, h) = e_M$. Moreover, the left-hand side of the preceding formula can be expressed, by means of P and a, as the value at (x, y, z, t) of a certain morphism $X^4 \to M$.

It follows from the uniqueness assertion in $(+)_4$ that $\partial^3 DP$ and e_M, which factor the same morphism, are equal. Hence:

(2) DP is a cocycle; that is, $DP \in Z^3(Y, M)$.

(c) If P and P' are two admissible composition laws on X, there is a unique morphism

$$b : X^2 \longrightarrow M$$

such that $P'(x, y) = b(x, y)P(x, y)$. Applying 1.3.2(ii) to P and P', one sees that b is invariant under M^2, and hence, by $(+)_2$:

(3) For every pair (P, P') of admissible composition laws there exists a unique morphism $d(P, P') : Y^2 \to M$ such that

$$P'(x, y) = d(P, P')(p(x), p(y)) \cdot P(x, y),$$

and the set of admissible composition laws is thereby made principal homogeneous under

$$\mathrm{Hom}(Y^2, M) = C^2(Y, M).$$

(d) Under the preceding conditions one has

$$(4) \qquad DP' - DP = \partial^2 d(P, P').$$

[51] Editor's note: the diagram that follows has been added.

(e) The laws P and P' are equivalent if and only if there is a morphism

$$h \in C^1(Y, M) = \mathrm{Hom}(Y, M)$$

such that $d(P, P') = \partial^1 h$. This follows from the definition of equivalence and from 1.3.4(ii).

(f) It remains only to conclude. We seek an associative P', that is, one for which $DP' = e_M$. Now DP is a cocycle whose class in $H^3(Y, M)$ does not depend on the chosen admissible composition law P, by (3) and (4). This class is the required obstruction c. An appropriate P' can be chosen if and only if $c = 0$. Indeed, after choosing an arbitrary P, one must solve, by (1),

$$0 = DP' = DP + \partial^2 d(P, P'),$$

which is possible, by (3) and (4), if and only if $c = 0$. Again by (3) and (4), the set of associative laws P' is principal homogeneous under $Z^2(Y, M)$. By (e), the set of associative laws P' modulo equivalence is principal homogeneous under $H^2(Y, M)$.

2. Infinitesimal extensions of a morphism of group schemes

Retain the notation of §0. Let Y and X be two S-group functors, and let M be the kernel of the group homomorphism $p_X : X \to X^+$. Thus there is an exact sequence of S-group functors

$$1 \longrightarrow M \longrightarrow X \xrightarrow{p_X} X^+.$$

By the definition of X^+, there are isomorphisms

$$\mathrm{Hom}_S(Y, X^+) \xrightarrow{\sim} \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}}),$$
$$\mathrm{Hom}_{S\text{-gr}}(Y, X^+) \xrightarrow{\sim} \mathrm{Hom}_{S_{\mathscr{J}}\text{-gr}}(Y_{\mathscr{J}}, X_{\mathscr{J}}).$$

The map

$$\mathrm{Hom}_S(Y, p_X) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)$$

sends an S-morphism $f : Y \to X$ to the S-morphism $f^+ : Y \to X^+$ corresponding under these isomorphisms to the $S_{\mathscr{J}}$-morphism $f_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$ obtained from f by base change. If M is commutative, Proposition 1.2.4 applies to this situation.

2.0.[52] In what follows we shall consider the case in which Y is flat over S, and X is an S-group functor of one of the following two types:

a X is an S-group scheme;

b $X = \underline{\mathrm{Aut}}_S(E)$, where E is an S-scheme of finite presentation over S.

Write (Flat)/S for the category of S-schemes flat over S. In case a, respectively b, the S-group functor $M = \mathrm{Ker}(X \to X^+)$, its restriction L to (Flat)/S, and the actions on M of the inner automorphisms of X were calculated in 0.9, 0.5, and 0.10, respectively in 0.11, 0.14, and 0.12.

More explicitly, in case a, let L_0 be the S_0-functor in commutative groups defined, for every S_0-scheme T_0, by

$$\mathrm{Hom}_{S_0}(T_0, L_0) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\right).$$

[52] Editor's note: The numbering 2.0 has been added for later references.

The group X_0 acts on this functor through its adjoint representation on $\omega^1_{X_0/S_0}$, and

$$L = \prod_{S_0/S} L_0; \qquad L(T) = L_0(T \times_S S_0)$$

for every S-scheme T.

In case b, let $\pi : E \to S$ be the structural morphism. Then L is the abelian-group functor on (Flat)/S defined by

$$\mathrm{Hom}_S(T, L) = \Gamma\Big(T, \pi_* \mathrm{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E) \otimes_{\mathscr{O}_S} \mathscr{O}_T\Big),$$

on which X, regarded as a functor on (Flat)/S, acts as described in 0.12. In either case there is an exact sequence of group functors on (Flat)/S:

$$1 \longrightarrow L \longrightarrow X \longrightarrow X^+. \tag{E}$$

Since Y is flat over S, Lemma 1.1.1 shows that the groups $H^i(Y, M)$ depend only on the restriction L of M to (Flat)/S. In case a, where $L = \prod_{S_0/S} L_0$, Lemma 1.1.2 gives isomorphisms

$$H^i(Y, L) \simeq H^i(Y_0, L_0).$$

Taking all this into account, Proposition 1.2.4 yields the following result.[53]

Theorem 2.1. *Let S be a scheme, and let $\mathscr{I} \supset \mathscr{J}$ be two quasi-coherent ideals satisfying $\mathscr{I}\mathscr{J} = 0$, defining the closed subschemes S_0 and $S_{\mathscr{J}}$. Let*

X be an S-group functor of type a or b, with L_0 and L as above;

Y be an S-group scheme flat over S, and $f_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$ a morphism of $S_{\mathscr{J}}$-groups.

Then:

i The morphism $f_{\mathscr{J}}$ lifts to a morphism of S-groups $Y \to X$ if and only if the following two conditions hold:

i_1 $f_{\mathscr{J}}$ lifts to a morphism of S-functors $Y \to X$. By Proposition 1.2.4 this defines an action of Y on L, independent of the chosen lift. In case a, the resulting action of Y_0 on L_0 comes from $f_0 : Y_0 \to X_0$ and the adjoint action of X_0 on L_0.

i_2 The canonically defined obstruction $c(f_{\mathscr{J}}) \in H^2(Y, L)$ vanishes. In case a, this group is isomorphic to $H^2(Y_0, L_0)$.

(ii) *If the conditions in i hold, the set $\mathcal{E}$ of morphisms of S-group functors $Y \to X$ extending $f_{\mathscr{J}}$ is principal homogeneous under $Z^1(Y, L)$, isomorphic in case a to $Z^1(Y_0, L_0)$. Moreover, the quotient of $\mathcal{E}$ by the action of inner automorphisms of X defined by sections over S that induce the unit section of $X_{\mathscr{J}}$ over $S_{\mathscr{J}}$ is principal homogeneous under $H^1(Y, L)$, isomorphic in case a to $H^1(Y_0, L_0)$.*

(iii) *If $f : Y \to X$ is a morphism of S-group functors extending $f_{\mathscr{J}}$, the set of morphisms $Y \to X$ obtained from f by those inner automorphisms is isomorphic to*

$$\Gamma(L)/H^0(Y, L),$$

and in case a to $\Gamma(L_0)/H^0(Y_0, L_0)$.

[53] Editor's note: The definitions that appeared in the original statement of Theorem 2.1 and occupied most of original page 113 have been moved above.

Remark 2.1.1.[54] If $f, f' : Y \to X$ are morphisms of S-group functors extending $f_{\mathscr{J}}$, one obtains a cocycle

$$d(f, f') \in Z^1(Y, L) \quad (\simeq Z^1(Y_0, L_0) \text{ in case (a)})$$

such that

$$f' = d(f, f') \cdot f. \tag{*}$$

[55] Write $\bar{d}(f, f')$ for the image of $d(f, f')$ in $H^1(Y, L)$, or equivalently in $H^1(Y_0, L_0)$ in case a.

Remark 2.2. Keep the preceding notation; in particular, Y is flat over S. In case b, formula (0.14.1) identifies L with the restriction to (Flat)/S of

$$\mathbf{W}\Big(\pi_* \operatorname{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\Big).$$

In case a, suppose in addition that X is locally of finite presentation over S. Formula (0.6.1) then identifies L with the restriction to (Flat)/S of

$$\prod_{S_0/S} \mathbf{W}\Big(\operatorname{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big).$$

In both cases the module to which $\mathbf{W}$ is applied is quasi-coherent, by EGA I, 9.1.1. Suppose further that Y is affine over S.[56] By I, 5.3, one obtains

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i\Big(Y_0, \operatorname{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big) \qquad \text{in case (a)},$$
$$H^i(Y, L) = H^i\Big(Y, \pi_* \operatorname{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\Big) \qquad \text{in case (b)}.$$

Remark 2.3.

(1) By 0.16 and 0.17, condition i_1 is automatic when Y is affine and

$$\begin{cases} X \text{ is smooth over } S, & \text{in case (a)}, \\ E \text{ is smooth and affine over } S, & \text{in case (b)}. \end{cases}$$

(2) Under these hypotheses, with Y still flat over S as in 2.0, formulas 2.2(a) and (0.6.2) give in case a

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i\Big(Y_0, \operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),$$

and in case b, by (0.14.2), 1.1.2, and I, 5.3,

$$H^i(Y, L) = H^i\Big(Y_0, \pi_{0*} \operatorname{Hom}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0})\Big).$$

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[54] Editor's note: This remark, analogous to 4.5.1, has been added to introduce the notation $d(f, f')$ and $\bar{d}(f, f')$, used in 4.38. Consequently, the part of 2.1(ii) concerning $\mathcal{E}$ itself was also added.

[55] Editor's note: We follow the sign conventions of the original. One could instead write $f' = d(f', f) \cdot f$, but then, in order to retain the equality $d_1(d(i, i')) = d(Y, i'(Y))$ in 4.27 for two immersions $i, i' : Y \hookrightarrow X$, the sign of the class $d(Y, Y')$ introduced in 4.5.1 would have to be changed. This would in turn alter the signs in 4.8, 4.14, and 4.17. We have preferred to retain all the signs of the original, which are correct.

[56] Editor's note: This hypothesis and the reference to I, 5.3 have been added.

[57] Editor's note: The preceding material has been added; see Editor's Note 31 in 0.14.

We now state several corollaries for a diagonalizable S-group Y (I, 4.4). If S is affine, then, by I, 5.3.3, $H^n(Y, F) = 0$ for every $n > 0$ and every quasi-coherent $\mathscr{O}_S$-module F.

Corollary 2.4. *Let $S_0 \hookrightarrow S$ be a closed subscheme defined by a nilpotent ideal, and let Y be a diagonalizable S-group. Let*

a X be an S-group locally of finite presentation over S, or

b $X = \underline{\mathrm{Aut}}_S(E)$, where E is an S-scheme locally of finite presentation.

If $f : Y \to X$ is a morphism of S-groups whose base change $f_0 : Y_0 \to X_0$ is the unit morphism, then f is the unit morphism.

Indeed, the question is local on S, and in case b also local on E. We may therefore suppose that S is affine and, in case b, that E is of finite presentation over S. Introduce the closed subschemes $S_n \subset S$ defined by the powers of the ideal defining S_0. This reduces the assertion to the case in which S_0 is defined by a square-zero ideal, where it follows from Theorem 2.1 and Remark 2.2.

If f_0 is not required to be the unit morphism, one has the following statement.

Corollary 2.5. *Let S and S_0 be as in 2.4, with S affine. Let Y be a diagonalizable S-group, X an S-group functor, and $f_0 : Y_0 \to X_0$ a morphism of S_0-group functors.*

i [58]

(ii) *Suppose either that X is a smooth S-group, or that $X = \underline{\mathrm{Aut}}_S(E)$ with E smooth and affine over S. Then f_0 extends to a morphism of S-groups $Y \to X$, and any two extensions are conjugate by an inner automorphism of X defined by a section over S inducing the unit section of X_0 over S_0.*

Introduce S_n as above.[59] For (ii), a scheme smooth over S is necessarily locally of finite presentation. Thus, in case b, a smooth affine E/S is of finite presentation, so the hypothesis in 2.0(b) is indeed satisfied.

Condition i_1 of Theorem 2.1 is automatic by 0.16 and 0.17. Moreover, every section of X_{S_n} over S_n lifts to a section of $X_{S_{n+1}}$ over S_{n+1}: this follows from smoothness in case a, and from 0.17 in case b. Consequently, if f and f' are two lifts of f_0, one may arrange step by step that $f_n = f'_n$ by lifting the inner automorphism supplied by Theorem 2.1(ii). This proves the assertion.

The same argument, together with Remark 2.3, gives:

Corollary 2.6. *Let $\mathscr{I}$ be a nilpotent ideal of S defining S_0, let Y be an affine S-group flat over S, and let X be a smooth S-group.*

i If, for every $n \geq 0$,

$$H^2\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then every S_0-group morphism $f_0 : Y_0 \to X_0$ lifts to an S-group morphism $f : Y \to X$.

[58] Editor's note: The original stated: "Suppose one of the following two properties holds: a X is an S-group locally of finite presentation; b $X = \underline{\mathrm{Aut}}_S(E)$, where E is of finite presentation over S. Then f_0 extends to a morphism of S-groups $Y \to X$ if and only if it extends to a morphism of S-functors $Y \to X$." It added that this followed step by step from part (ii) of the theorem. That argument appears insufficient. For example, if $\mathscr{I}^3 = 0$ and $\mathscr{J} = \mathscr{I}^2$, and if $f : Y \to X$ is an S-functor morphism lifting f_0, then f_0 lifts to an $S_{\mathscr{J}}$-group morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$; but must $g_{\mathscr{J}}$ then lift to an S-functor morphism $g : Y \to X$? In any event, assertion i is not used later, where X is always assumed smooth over S.

[59] Editor's note: What follows has been slightly modified and expanded from the original.

(ii) *If, for every* $n \geq 0$,

$$H^1\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then any two such lifts are conjugate by an inner automorphism of X *defined by a section over* S *inducing the unit section of* X_0 *over* S_0.

Lemma 2.7. *Let* S *be affine,* G *an affine* S*-group,* $\mathscr{F}$ *a quasi-coherent* $\mathscr{O}_S$*-module, and* $\mathscr{L}$ *a locally free* $\mathscr{O}_S$*-module. Suppose* G *acts on* $\mathscr{F}$ *in the sense of Exposé I, hence acts on* $\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{L}$. *Let* Λ *be the ring of* S, *and let* L *be the projective* Λ*-module defining* $\mathscr{L}$. *Then there is a canonical isomorphism*

$$H^*(G, \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{L}) \simeq H^*(G, \mathscr{F}) \otimes_\Lambda L.$$

60

Let $\mathscr{A}$ be the $\mathscr{O}_S$-algebra $\mathscr{A}(G)$, and consider the complex $\mathscr{C}$ of quasi-coherent $\mathscr{O}_S$-modules

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A} \longrightarrow \mathscr{F} \otimes \mathscr{A} \otimes \mathscr{A} \longrightarrow \cdots .$$

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By I, 5.3, $H^*(G, \mathscr{F})$, respectively $H^*(G, \mathscr{F} \otimes \mathscr{L})$, is the cohomology of $\Gamma(S, \mathscr{C})$, respectively $\Gamma(S, \mathscr{C} \otimes \mathscr{L})$. Since S is affine, EGA I, 1.3.12 gives

$$\Gamma(S, \mathscr{C} \otimes \mathscr{L}) \simeq \Gamma(S, \mathscr{C}) \otimes_\Lambda L.$$

Because L is projective, hence flat,

$$H^*(\Gamma(S, \mathscr{C}) \otimes_\Lambda L) \simeq H^*(\Gamma(S, \mathscr{C})) \otimes_\Lambda L,$$

which proves the lemma.

Using the lemma, Corollary 2.6 becomes:

Corollary 2.8. *Let* S *be affine and let* $\mathscr{I}$ *be a nilpotent ideal defining* S_0. *Suppose every* $\mathscr{I}^{n+1}/\mathscr{I}^{n+2}$ *is locally free on* S_0. *Let* Y *be an affine* S*-group flat over* S, *let* X *be a smooth* S*-group, and let* $f_0 : Y_0 \to X_0$ *be an* S_0*-group morphism.*

i If $H^2(Y_0, \mathrm{Lie}(X_0/S_0)) = 0$, *then* f_0 *lifts to an* S*-group morphism* $Y \to X$.

(ii) *If* $H^1(Y_0, \mathrm{Lie}(X_0/S_0)) = 0$, *then any two such lifts are conjugate by an inner automorphism of* X *defined by a section over* S *inducing the unit section of* X_0 *over* S_0.

Taking $Y = X$ gives:

Corollary 2.9. *Let* S *and* S_0 *be as above, and let* X *be a smooth affine* S*-group.*

i If $H^1(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, *every endomorphism of* X *over* S *that induces the identity on* X_0 *is the inner automorphism defined by a section over* S *inducing the unit section of* X_0 *over* S_0.

[60] Editor's note: Here $\mathscr{L}$ is given the trivial action of G.

[61] Editor's note: The original proof has been expanded and simplified. It also invoked the isomorphisms $H^n(\mathscr{C} \otimes \mathscr{L}) \simeq H^n(\mathscr{C}) \otimes \mathscr{L}$ and $H_n(\Gamma(-)) \simeq \Gamma(H_n(-))$, where $H^n(\mathscr{C})$ denotes the cohomology sheaf of the complex $\mathscr{C}$.

(ii) *If* $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, *every* S_0*-automorphism of* X_0 *extends to an* S*-automorphism of* X.[62]

Remark 2.10. The assertions involving H^1 have converses by Theorem 2.1. For example, let $S = I_{S_0}$ be the dual-number scheme over S_0 (II, 2.1), and let X be a flat S-group such that every automorphism over S inducing the identity over S_0 is an inner automorphism defined by a section over S inducing the unit section of X_0. Then

$$H^1(X_0, \mathrm{Lie}(X_0/S_0)) = 0.$$

[63]

Corollary 2.11. *Let* $S, \mathscr{I}, \mathscr{J}$ *be as in 2.1. Let* Y *be an* S*-group scheme flat over* S*, let* X *be an* S*-group scheme, and let* $f : Y \to X$ *be an* S*-group morphism. The set of morphisms from* Y *to* X *obtained from* f *by conjugation with elements* $x \in X(S)$ *inducing the unit of* $X(S_{\mathscr{J}})$ *is isomorphic to the quotient*

$$\mathcal{E} = \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})/\,\mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})^{\mathrm{ad}(Y_0)}.$$

Here the second group consists of those $\mathscr{O}_{S_0}$*-morphisms* $\omega^1_{X_0/S_0} \to \mathscr{J}$ *which, after every base change* $S' \to S_0$*, yield morphisms*

$$\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{S'} \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{S'}$$

invariant under the action of $Y_0(S')$ *on the first factor.*

By Theorem 2.1(iii), this set is $\Gamma(L_0)/H^0(Y_0, L_0)$. Now

$$\Gamma(L_0) = \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}),$$

and $H^0(Y_0, L_0)$ is precisely $\Gamma(L_0)^{\mathrm{ad}(Y_0)}$ in the sense just stated.

Corollary 2.12. *Under the hypotheses of 2.11, suppose in addition that* $\omega^1_{X_0/S_0}$ *is locally free of finite rank. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big)\,/H^0\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

Indeed,[64] local freeness of finite rank gives

$$\mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \simeq \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}.$$

Corollary 2.13. *Suppose in addition that* Y_0 *is diagonalizable. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big)\,/\Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),$$

where $\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}$ *is the factor of the decomposition in I, 4.7.3 corresponding to the zero character of* Y_0.

[62] Editor's note: Let f_0 be an S_0-automorphism of X_0, with inverse g_0. By 2.8(i), lift f_0 and g_0 to S-endomorphisms f and g of X. Then $g \circ f$ and $f \circ g$ induce the identity on X_0, so by 0.7 they are S-automorphisms. Hence f and g are S-automorphisms as well.

[63] Editor's note: This is used in Exposé XXIV, 1.13.

[64] Editor's note: The following sentence has been added.

Indeed, if $Y_0 \simeq D_{S_0}(M)$, the cited result gives a direct-sum decomposition

$$\mathrm{Lie}(X_0/S_0) = \mathrm{Lie}(X_0/S_0)_0 \oplus \bigoplus_{\substack{m\in M\\ m\neq 0}} \mathrm{Lie}(X_0/S_0)_m.$$

Tensoring with $\mathscr{J}$ gives the analogous decomposition for $\mathrm{Lie}(X_0/S_0)\otimes\mathscr{J}$, whence

$$H^0(Y_0, \mathrm{Lie}(X_0/S_0)\otimes\mathscr{J}) \simeq \Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0)_0 \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

Corollary 2.14. *Suppose in addition that S_0 is affine. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \Big[\mathrm{Lie}(X_0/S_0)/\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}\Big] \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

3. Infinitesimal extensions of a group scheme

Still using the notation of §0 ($S, \mathscr{I}, \mathscr{J}$, and so on), let X be an S-scheme, and suppose that $X_{\mathscr{J}}$ is endowed with a group structure. We propose to determine the S-group structures on X that induce the given structure on $X_{\mathscr{J}}$.

From now on, assume that X is flat over S. Let $\mathscr{C}$ be the category of S-schemes flat over S; thus $X \in \mathrm{Ob}\,\mathscr{C}$. Denote by Y, respectively M, the functor on $\mathscr{C}$ defined by X^+, respectively L'_X. The canonical morphism $p_X : X \to X^+$ defines a morphism in $\widehat{\mathscr{C}}$

$$p : X \longrightarrow Y,$$

and the action of L'_X on X in $\widehat{(\mathrm{Sch})/S}$ defines an action of M on X in $\widehat{\mathscr{C}}$. One checks at once that X is thereby formally principal homogeneous under M_Y over Y (cf. 0.2(i) and 0.4).

The action of X^+ on L'_X defined in 0.8 (denoted Ad there) defines an action, denoted $\widetilde{f}$, of Y on M. On the other hand, by 0.5,

$$\mathrm{Hom}_{\widehat{\mathscr{C}}}(Z, M) \simeq \mathrm{Hom}_{S_0}(Z_0, L_0), \qquad Z \in \mathrm{Ob}\,\mathscr{C},$$

where L_0 is the functor defined in 0.5.

Lemma 3.1.

(i) *Condition $(+)_n$ of 1.3 holds for every positive integer n.*

(ii) *Let the S_0-group X_0 act on the S_0-functor L_0 through its adjoint representation. Then there is a canonical isomorphism*

$$H^*(X_0, L_0) \simeq H^*(Y, M),$$

where the first cohomology is computed in $(\mathrm{Sch})/S_0$, and the second in $\mathscr{C}$.

Both parts of the lemma follow from the relation

$$\begin{aligned}\mathrm{Hom}_{\widehat{\mathscr{C}}}(Y, M) &\simeq \mathrm{Hom}_{\widehat{(\mathrm{Sch})/S_0}}(X^+ \times_S S_0, L_0)\\ &\simeq \mathrm{Hom}_{S_0}(X_0, L_0)\\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}}}(X, M),\end{aligned}$$

which follows at once from the definition of M as $\prod_{S_0/S} L_0$. More generally, the same relation holds with X, Y replaced by X^n, Y^n. Consequently, every morphism $X^n \to M$ factors

uniquely through Y^n, which gives $(+)_n$. It also gives the relation $C^*(Y, M) = C^*(X_0, L_0)$, and hence (ii).

We may therefore apply the constructions of 1.3. In particular:

Lemma 3.2. *Let $P : X \times_S X \to X$ be a morphism. Then P induces the group law of $X_{\mathscr{J}}$ if and only if P is an admissible composition law on X (cf. 1.3.2).*

Indeed, P induces the group law of $X_{\mathscr{J}}$ if and only if it lifts the group law of X^+, or equivalently that of Y. Thus it remains only to show that every morphism P lifting the group law of $X_{\mathscr{J}}$ satisfies identity ((++)) of 1.3.2(ii), which is exactly what was proved in 0.8.

Proposition 3.3. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let X be an S-scheme that is flat and either quasi-compact or locally of finite presentation over S. Let $P : X \times_S X \to X$ be a composition law on X. Then P is a group law if and only if the following two conditions hold:*

(i) *P is associative;*

(ii) *P induces a group law on $X_0 = X \times_S S_0$.*

These conditions are clearly necessary. We prove that they are sufficient. Suppose first that $X \to S$ has a section. Since $X(S')$ is then nonempty for every $S' \to S$, it suffices [65] to show that, for every $x \in X(S')$, left and right translation by x are isomorphisms of $X_{S'}$. [66]

We may plainly assume $S' = S$. The translation in question, t, induces on X_0 a translation t_0, which is an automorphism since X_0 is a group. The conclusion follows by flatness (SGA 1 III, 4.2). [67]

Now drop the assumption that X has a section over S, and suppose that there exists $S' \to S$ such that $X_{S'}$ has a section over S'. By what has just been proved, $X_{S'}$ is then an S'-group; let e' be its unit section. The inverse image of e' under $\mathrm{pr}_i : S' \times_S S' \to S'$, $i = 1, 2$, is the unit section of $X_{S''}$ for the group law obtained by pulling $P_{S'}$ back along pr_i. But since P is "defined over S," these two group laws coincide, and so do their unit sections. Hence

$$\mathrm{pr}_1^*(e') = \mathrm{pr}_2^*(e').$$

If $S' \to S$ is a descent morphism (cf. Exposé IV, no. 2), there is a section of X whose base extension is e', and the proof is complete. Since X_X has a section over X, namely the diagonal section, it is now enough to prove that $X \to S$ is a descent morphism. It is flat and surjective, and is either quasi-compact or locally of finite presentation; it is therefore an (fpqc) covering, hence a descent morphism (Exposé IV, no. 6).

[65] Editor's note: The original has been corrected by deleting the inappropriate reference to an exercise of Bourbaki on semigroups (cf. [BAlg], §I.2, Exercises 9–13) and by indicating the role of the left and right translations; see the following editor's note.

[66] Editor's note: Let E be a nonempty set endowed with an associative composition law such that every left translation ℓ_x is bijective, and fix $x_0 \in E$. There is a unique $e \in E$ such that $x_0 e = x_0$; then $x_0 ex = x_0 x$ implies $ex = x$ for every $x \in E$. Moreover, for every x there is a unique x' such that $xx' = e$. Suppose in addition that there is a $b \in E$ for which right translation r_b is injective. Then $xeb = xb$ gives $xe = x$, so e is a unit, and $xx'x = x = xe$ gives $x'x = e$; hence x' is both a left and a right inverse of x, and E is a group.

The hypothesis that r_b be injective is necessary: on any set E, define $xy = y$ for all $x, y \in E$. Every left translation is then the identity (and the law is associative), whereas $r_y(E) = \{y\}$ for every y; thus E is not a group when $|E| > 1$.

[67] Editor's note: Since X and X_0 have the same underlying topological space and t_0 is an automorphism, t is a homeomorphism, hence an affine morphism; cf. Exposé $\mathrm{VI_B}$, 2.9.1, or EGA $\mathrm{IV_4}$, 18.12.7.1. It therefore suffices to show the following. Let $\mathscr{J}$ be a nilpotent ideal of a ring Λ, and let $\varphi : A \to B$ be a morphism of Λ-algebras, with B flat over Λ, such that $\varphi \otimes_\Lambda (\Lambda/\mathscr{J})$ is bijective. Then φ is bijective. By the "nilpotent Nakayama lemma," φ is surjective; moreover, since B is flat over Λ, one also has $\mathrm{Ker}(\varphi) \otimes_\Lambda (\Lambda/\mathscr{J}) = 0$, whence $\mathrm{Ker}(\varphi) = 0$. Thus φ is bijective.

Remark. In fact, the hypothesis that $X \to S$ be quasi-compact or locally of finite presentation is superfluous, by the following result, which the reader may prove as an exercise on Exposé IV:

Under the stated hypotheses on S and S_0, if $X \to S$ is flat and $X_0 \to S_0$ is an (fpqc) covering, then $X \to S$ is a descent morphism.

Lemma 3.4. *Two admissible composition laws on X are equivalent (cf. 1.3.5) if and only if one is obtained from the other by an automorphism of X over S that induces the identity on $X_{\mathscr{J}}$.*

Indeed, the morphisms constructed in 1.3.1 are precisely those in the preceding statement, by 0.7. [68]

Taking account of all the preceding results, Proposition 1.3.6 gives:

Theorem 3.5. *Let S be a scheme, and let $\mathscr{I}$ and $\mathscr{J}$ be two ideals on S such that $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let S_0 and $S_{\mathscr{J}}$ be the closed subschemes of S that they define. Let X be an S-scheme flat over S (and either locally of finite presentation or quasi-compact over S), and let X_0 and $X_{\mathscr{J}}$ be the schemes obtained by base change. Suppose $X_{\mathscr{J}}$ is endowed with an $S_{\mathscr{J}}$-group structure, and let L_0 denote the S_0-functor in commutative groups defined by*

$$\mathrm{Hom}_{S_0}(T, L_0) = \mathrm{Hom}_{\mathscr{O}_T}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right),$$

on which X_0 acts through its adjoint representation.

(i) *An S-group structure exists on X that induces the given structure on $X_{\mathscr{J}}$ if and only if the following conditions hold:*

  (i_1) *there is a morphism of S-schemes $X \times_S X \to X$ inducing the group law of $X_{\mathscr{J}}$;*

  (i_2) *a certain obstruction class in $H^3(X_0, L_0)$, canonically defined by X and the group law of $X_{\mathscr{J}}$, is zero.*

(ii) *If the conditions in (i) hold, the set $\mathcal{E}$ of group laws on X inducing the given law on $X_{\mathscr{J}}$ is principal homogeneous under $Z^2(X_0, L_0)$. Moreover, the quotient of $\mathcal{E}$ by the S-automorphisms of X inducing the identity on $X_{\mathscr{J}}$ is principal homogeneous under $H^2(X_0, L_0)$.*

[69] Indeed, by 3.2 every morphism of S-schemes $f : X \times_S X \to X$ inducing the group law of $X_{\mathscr{J}}$ is an admissible composition law on X. By 1.3.6(i), the existence of an associative admissible composition law $P : X \times_S X \to X$ is equivalent to the vanishing of a certain class $c(f) \in H^3(X_0, L_0)$; in that case, 3.3 shows that P is a group law. This proves (i), and (ii) then follows from 3.3 and 1.3.6(ii).

Remark 3.5.1. [70] If μ and μ' are group laws on X inducing the given law on $X_{\mathscr{J}}$, one obtains a cocycle $\delta(\mu, \mu') \in Z^2(X_0, L_0)$. The chosen sign convention is $\mu' = \delta(\mu, \mu') \cdot \mu$; that is, for every $S' \to S$ and $x, y \in X(S')$,

$$\mu'(x, y) = \delta(\mu, \mu')(x_0, y_0) \cdot \mu(x, y).$$

[68] Editor's note: Indeed, by the proof of 0.7, the S-endomorphisms of X inducing the identity on $X_{\mathscr{J}}$ are the automorphisms $m \cdot \mathrm{id}_X$, where m ranges over $M(X) = \mathrm{Hom}_S(X, M)$. For every $S' \to S$ and every $x \in X(S')$, one has $(m \cdot \mathrm{id}_X)(x) = m(x) \cdot x$. By the proof of 3.1, each $m : X \to M$ factors uniquely through a morphism $h : Y = X^+ \to M$; hence $m \cdot \mathrm{id}_X$ is the automorphism u_h introduced in 1.3.1. The lemma now follows from the definition of equivalence; cf. 1.3.4 and 1.3.5.

[69] Editor's note: The following argument has been added.

[70] Editor's note: This remark, analogous to 4.5.1, has been added to introduce the notation $\delta(\mu, \mu')$, or $\delta(X, X')$, used in 4.38. Consequently, the part of 3.5(ii) concerning $\mathcal{E}$ itself has also been added.

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Write $\bar{\delta}(\mu,\mu')$ for the image of $\delta(\mu,\mu')$ in $H^2(X_0,L_0)$. Finally, if X endowed with μ, respectively μ', is denoted simply by X, respectively X', write $\delta(X,X')$ in place of $\delta(\mu,\mu')$, and similarly for $\bar{\delta}(X,X')$.

Remark 3.6. Let $X_{\mathscr{J}}$ be a smooth affine $S_{\mathscr{J}}$-scheme. By 0.15, up to isomorphism there is a unique S-scheme X, smooth over S, whose reduction modulo $\mathscr{J}$ is $X_{\mathscr{J}}$. If $X_{\mathscr{J}}$ is endowed with an $S_{\mathscr{J}}$-group structure, condition (i_1) is automatic by 0.16. Moreover, 0.6 simplifies the definition of L_0 and gives:

Corollary 3.7. *Let $S,\mathscr{I},\mathscr{J}$ be as in 3.1, and let $X_{\mathscr{J}}$ be a smooth affine $S_{\mathscr{J}}$-group.*

(i) *The set of smooth S-groups whose reduction modulo $\mathscr{J}$ is $X_{\mathscr{J}}$, up to isomorphism inducing the identity on $X_{\mathscr{J}}$, is either empty or principal homogeneous under*

$$H^2\Big(X_0,\mathrm{Lie}(X_0/S_0)\otimes_{\mathscr{O}_{S_0}}\mathscr{J}\Big).$$

(ii) *A smooth S-group whose reduction modulo $\mathscr{J}$ is $X_{\mathscr{J}}$ exists if and only if a certain obstruction in*

$$H^3\Big(X_0,\mathrm{Lie}(X_0/S_0)\otimes_{\mathscr{O}_{S_0}}\mathscr{J}\Big)$$

is zero.

As usual, one obtains the following corollaries.

Corollary 3.8. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Let X_0 be a smooth affine S_0-group.*

(i) *If, for every $n\geq 0$,*

$$H^2\Big(X_0,\mathrm{Lie}(X_0/S_0)\otimes_{\mathscr{O}_{S_0}}\mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big)=0,$$

then any two smooth S-groups whose reduction is X_0 are isomorphic by an isomorphism inducing the identity on X_0.

(ii) *If, for every $n\geq 0$,*

$$H^3\Big(X_0,\mathrm{Lie}(X_0/S_0)\otimes_{\mathscr{O}_{S_0}}\mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big)=0,$$

then there exists a smooth S-group whose reduction is X_0.

Corollary 3.9. *Let S be affine and S_0 a closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Suppose that every $\mathscr{I}^{n+1}/\mathscr{I}^{n+2}$ is locally free on S_0. Let X_0 be a smooth affine S_0-group.*

(i) *If $H^2(X_0,\mathrm{Lie}(X_0/S_0))=0$, then any two smooth S-groups whose reduction is X_0 are isomorphic.*

(ii) *If $H^3(X_0,\mathrm{Lie}(X_0/S_0))=0$, then there exists a smooth S-group whose reduction is X_0.*

[71] Editor's note: We follow the sign conventions of the original in order to have, in 4.38(5), the equality $\partial^1\bar{d}(X,X')=\bar{\delta}(X,X')$; see also Editor's Note 54.

Corollary 3.10. *Let S_0 be a scheme and let $S = I_{S_0}$ be the dual-number scheme over S_0. Let X_0 be a smooth S_0-group. Then every smooth S-group Y for which Y_0 is S_0-isomorphic to X_0 is S-isomorphic to $X = X_0 \times_{S_0} S$ if and only if*

$$H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0.$$

[72]

Indeed, by 3.5 the set of classes of such groups Y, up to an S-group isomorphism inducing the identity on X_0, is in bijection with $H^2(X_0, \mathrm{Lie}(X_0/S_0))$. Hence their classes up to an arbitrary S-group isomorphism are in bijection with

$$H^2(X_0, \mathrm{Lie}(X_0/S_0))/\Gamma_0,$$

where

$$\Gamma_0 = \mathrm{Aut}_{S_0\text{-gr.}}(X_0),$$

which acts in the evident way on H^2. The conclusion follows at once. [73]

4. Infinitesimal extensions of closed subgroups

We first state a result valid in an arbitrary abelian category.

Lemma 4.1. *Let*

$$0 \longrightarrow A' \xrightarrow{i} A \xrightarrow{p} A'' \longrightarrow 0$$

be an exact sequence, let $\phi : A' \to Q$ be a morphism, and let $\pi : A'' \to P$ be an epimorphism with kernel C. Let E be the set, up to isomorphism, of quadruples (B, f, g, h) such that the sequence

$$0 \longrightarrow Q \xrightarrow{f} B \xrightarrow{g} P \longrightarrow 0$$

is exact and the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle \pi} & & \\ 0 & \longrightarrow & Q & \xrightarrow{f} & B & \xrightarrow{g} & P & \longrightarrow & 0 \end{array} .$$

(i) *The set E is nonempty if and only if the image in $\mathrm{Ext}^1(C, Q)$ of the element A of $\mathrm{Ext}^1(A'', A')$ is zero.*

(ii) *Under these conditions, E is a principal homogeneous set under the abelian group $\mathrm{Hom}(C, Q)$.*

Introduce the amalgamated sum

$$B' = A \underset{A'}{\sqcup} Q.$$

There is then a commutative diagram with exact rows: [74]

[72] Editor's note: This result is used in XXIV, 1.13.

[73] Editor's note: Indeed, $\mathrm{Aut}_{S_0\text{-gr.}}(X_0)$ acts by group automorphisms on the abelian group $H^2(X_0, \mathrm{Lie}(X_0/S_0))$, so the orbit of 0 is the singleton $\{0\}$. Consequently, the quotient set is a singleton if and only if $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = \{0\}$.

[74] Editor's note: One has $\mathrm{Coker}(j) = B' \underset{Q}{\sqcup} 0 = A \underset{A'}{\sqcup} 0 = A''$, and $\mathrm{Ker}(j) \simeq \mathrm{Ker}(i) = 0$, as one sees by reasoning "as if the category were a category of modules." For a proof entirely in terms of arrows, see for example [Fr64], Theorem 2.5.4 (*).

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \phi} & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & Q & \xrightarrow{j} & B' & \longrightarrow & A'' & \longrightarrow & 0 \end{array} ,$$

It is clear that the category of solutions to the problem is canonically isomorphic to the category of solutions to the corresponding problem for the sequence

$$0 \longrightarrow Q \longrightarrow B' \longrightarrow A'' \longrightarrow 0$$

and the morphisms id_Q and $\pi : A'' \to P$. [75] In this case, E is in bijection with the set of subobjects N of B' for which $B' \to A''$ induces an isomorphism of N with the kernel C of $A'' \to P$; equivalently, it is the set of morphisms $e : C \to B'$ lifting the canonical morphism $C \to A''$. The abelian group $G = \mathrm{Hom}(C, Q)$ acts on E by $g \cdot e = g + e$, with addition in $\mathrm{Hom}(C, B')$; if $E \neq \varnothing$, this makes E a principal homogeneous set under G.

We deduce the following.

Proposition 4.2. [76] *Let S be a scheme, let $S_{\mathscr{J}}$ be the closed subscheme defined by a quasi-coherent square-zero ideal $\mathscr{J}$, let X be an S-scheme, and let $\mathscr{F}$ be an $\mathscr{O}_X$-module. Put*

$$X_{\mathscr{J}} = X \times_S S_{\mathscr{J}}, \qquad \mathscr{F}_{\mathscr{J}} = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S_{\mathscr{J}}},$$

and let $\mathscr{G}_{\mathscr{J}} = \mathscr{F}_{\mathscr{J}}/\mathscr{H}_{\mathscr{J}}$ be a quotient module of $\mathscr{F}_{\mathscr{J}}$. Suppose given a morphism of $\mathscr{O}_{X_{\mathscr{J}}}$-modules

$$f : \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}} \longrightarrow \mathscr{Q}.$$

Let $\mathcal{E}$ be the sheaf of sets on X defined as follows: for each open subset U of X, $\mathcal{E}(U)$ is the set of quotient modules $\mathscr{G}$ of $\mathscr{F}|_U$ such that $\mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$ and for which there exists an isomorphism

$$h : \mathscr{J}\mathscr{G} \xrightarrow{\sim} \mathscr{Q}|_U$$

making the following diagram commute:

$$\begin{array}{ccc} \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} (\mathscr{G}_{\mathscr{J}}|_U) & \xrightarrow{f|_U} & \mathscr{Q}|_U \\ \Big\downarrow{\scriptstyle \text{can.}} & \nearrow_{h}^{\sim} & \\ \mathscr{J}\mathscr{G} & & \end{array} .$$

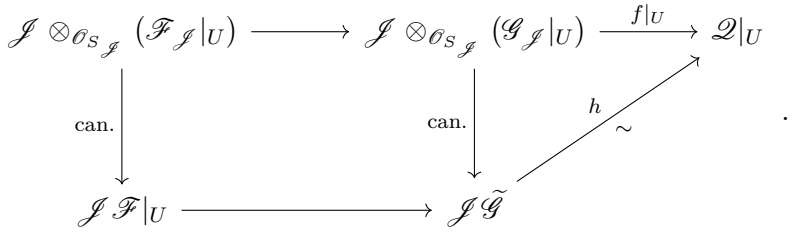
The isomorphism h is then unique, since $\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} (\mathscr{G}_{\mathscr{J}}|_U) \to \mathscr{J}\mathscr{G}$ is an epimorphism. Then $\mathcal{E}$ is a sheaf formally principal homogeneous under the sheaf of commutative groups

$$\mathscr{A} = \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q}) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q}).$$

Proof. If $\mathcal{E}(U) = \varnothing$, there is nothing to prove; we may therefore suppose that $\mathcal{E}(U)$ contains an element $\widetilde{\mathscr{G}}$. In the following diagram, h is an isomorphism and all the arrows are epimorphisms:

[75] Editor's note: In what follows, A has been replaced by B', and the end of the argument has been expanded.

[76] Editor's note: The statement has been rewritten so as to fit exactly the application made in 4.3; the proof has also been expanded, following indications supplied by M. Demazure.



Thus the morphism

$$\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} (\mathscr{F}_{\mathscr{J}}|_U) \longrightarrow \mathscr{Q}|_U$$

induces a necessarily unique epimorphism $\phi : \mathscr{J}\mathscr{F}|_U \to \mathscr{Q}|_U$. If $\mathscr{G}$ is an $\mathscr{O}_U$-module such that $\mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$ and there is a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{J}\mathscr{F}|_U & \longrightarrow & \mathscr{F}|_U & \longrightarrow & \mathscr{F}_{\mathscr{J}}|_U & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \phi} & & \downarrow & & \downarrow{\scriptstyle \pi} & & \\ 0 & \longrightarrow & \mathscr{Q}|_U & \longrightarrow & \mathscr{G} & \xrightarrow{p_{\mathscr{J}}} & \mathscr{G}_{\mathscr{J}}|_U & \longrightarrow & 0 \end{array}$$

(where $p_{\mathscr{J}}$ is the projection $\mathscr{G} \to \mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$, so that $\mathscr{Q}|_U = \mathrm{Ker}(p_{\mathscr{J}}) = \mathscr{J}\mathscr{G}$), then $\mathscr{G}$ may be identified with a quotient module of $\mathscr{F}|_U$. Consequently, by 4.1(ii), the set $\mathcal{E}(U)$ is principal homogeneous under the abelian group

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q})(U) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q})(U).$$

Proposition 4.3. *(TDTE IV 5.1) Let S be a scheme, let $S_{\mathscr{J}}$ be the closed subscheme defined by a quasi-coherent square-zero ideal $\mathscr{J}$, let X be an S-scheme, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. Put*

$$X_{\mathscr{J}} = X \times_S S_{\mathscr{J}}, \qquad \mathscr{F}_{\mathscr{J}} = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S_{\mathscr{J}}}.$$

Let $\mathscr{G}_{\mathscr{J}} = \mathscr{F}_{\mathscr{J}}/\mathscr{H}_{\mathscr{J}}$ be a quasi-coherent quotient module of $\mathscr{F}_{\mathscr{J}}$, flat over $S_{\mathscr{J}}$. For each open subset U of X, let $\mathcal{E}(U)$ be the set of quasi-coherent quotient modules $\mathscr{G}$[77] of $\mathscr{F}|_U$, flat over S, such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}|_U$. Then the $\mathcal{E}(U)$ form a sheaf of sets $\mathcal{E}$ on X, formally principal homogeneous under the sheaf of commutative groups

$$\mathscr{A} = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{H}_{\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}}\right).$$

Proof. Let $\pi : X \to S$ be the structural morphism. Let U be an open subset of X, and let $\mathscr{G}$ be an $\mathscr{O}_U$-module flat over S, such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}|_U$. For every $x \in U$, the stalk $\mathscr{G}_x$ is a flat module over the local ring $\mathscr{O}_{S,s}$, where $s = \pi(x)$. Hence the morphism

$$\mathscr{J}_s \otimes_{\mathscr{O}_{S,s}} (\mathscr{G}/\mathscr{J}\mathscr{G})_x = \mathscr{J}_s \otimes_{\mathscr{O}_{S,s}} \mathscr{G}_x \longrightarrow (\mathscr{J}\mathscr{G})_x$$

is bijective. We therefore have an exact sequence

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_S} (\mathscr{G}_{\mathscr{J}}|_U) \longrightarrow \mathscr{G} \longrightarrow \mathscr{G}_{\mathscr{J}}|_U \longrightarrow 0.$$

[77] Editor's note: To lighten the statement, the hypothesis that $\mathscr{G}$ be quasi-coherent has been added here; the remark that this hypothesis is automatically satisfied has been moved to the proof, which has been expanded accordingly.

Since $\mathscr{J} \otimes_{\mathscr{O}_S} (\mathscr{G}_{\mathscr{J}}|_U)$ and $\mathscr{G}_{\mathscr{J}}|_U$ are quasi-coherent $\mathscr{O}_U$-modules, $\mathscr{G}$ is quasi-coherent as well (EGA III, 1.4.17).

Conversely, since $\mathscr{G}_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$, if $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_U$-module such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}$ and the morphism

$$\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}} \longrightarrow \mathscr{J}\mathscr{G}$$

is bijective, then $\mathscr{G}$ is flat over S, by the "fundamental criterion of flatness" (SGA 1, IV, 5.5). [78]

Consequently, the set $\mathcal{E}(U)$ considered here is the set considered in 4.2 when f is the identity morphism of $\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}}$; the conclusion therefore follows from 4.2. □

Retain the preceding notation. [79] Let $Y_{\mathscr{J}}$ be a closed subscheme of $X_{\mathscr{J}}$, defined by a quasi-coherent ideal $\mathscr{I}_{Y_{\mathscr{J}}}$, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Applying 4.3 to $\mathscr{F} = \mathscr{O}_X$ and

$$\mathscr{G}_{\mathscr{J}} = \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_{X_{\mathscr{J}}}/\mathscr{I}_{Y_{\mathscr{J}}},$$

we obtain the following corollary.

Corollary 4.3.1. *Let $S, S_{\mathscr{J}}, \mathscr{J}, X, X_{\mathscr{J}}, Y_{\mathscr{J}}$, and $\mathscr{I}_{Y_{\mathscr{J}}}$ be as above, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Let $\mathscr{A}_{\mathscr{J}}$ be the sheaf of commutative groups*

$$\mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right)$$

on $X_{\mathscr{J}}$, and put $\mathscr{A} = i_*(\mathscr{A}_{\mathscr{J}})$, where $i : X_{\mathscr{J}} \hookrightarrow X$ is the immersion. For each open subset U of X, let $\mathcal{E}(U)$ be the set of closed subschemes Y of U, flat over S, such that

$$Y \times_S S_{\mathscr{J}} = Y_{\mathscr{J}} \cap U.$$

Then $\mathcal{E}$ is an $\mathscr{A}$-pseudo-torsor.

If, moreover, such a Y exists locally—that is, if every $x \in X$ has an open neighborhood U with $\mathcal{E}(U) \neq \varnothing$—then $\mathcal{E}$ is an $\mathscr{A}$-torsor. The $\mathscr{A}$-torsors on X are parametrized by

$$H^1(X, \mathscr{A}) = H^1(X_{\mathscr{J}}, \mathscr{A}_{\mathscr{J}})$$

(see, for example, EGA IV$_4$, 16.5.15), and $\mathcal{E}$ has a global section—that is, $\mathcal{E}(X) \neq \varnothing$—if and only if its corresponding cohomology class is zero. Thus we obtain:

Corollary 4.4. *Let $S, S_{\mathscr{J}}, \mathscr{J}, X, X_{\mathscr{J}}, Y_{\mathscr{J}}$, and $\mathscr{I}_{Y_{\mathscr{J}}}$ be as above, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Let E be the set of closed subschemes Y of X, flat over S, such that $Y \times_S S_{\mathscr{J}} = Y_{\mathscr{J}}$.*

(i) *The set E is either empty or principal homogeneous under the abelian group*

$$H^0(X, \mathscr{A}) = H^0(X_{\mathscr{J}}, \mathscr{A}_{\mathscr{J}}) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right).$$

(ii) *The set E is nonempty if and only if both of the following conditions hold:*

(a) *The problem has a solution locally on X.*

(b) *A certain obstruction vanishes; it lies in*

$$H^1\Big(X_{\mathscr{J}}, \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right)\Big).$$

[78] Editor's note: See also [BAC], §III.5, Theorem 1.

[79] Editor's note: The discussion that follows has been expanded and Corollary 4.3.1 has been added. Recall also that "pseudo-torsor" is synonymous with "formally principal homogeneous."

Complement 4.4.1. [80] Retain the notation of 4.4, and suppose that E contains an element Y. Let $\mathscr{I}_Y$ be the ideal of $\mathscr{O}_X$ defining Y, and let $\mathscr{I}_{Y_\mathscr{J}}$ be its image in $\mathscr{O}_{X_\mathscr{J}}$. As we saw in the proof of 4.2, there is a commutative diagram

$$\begin{array}{ccc} \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{X_\mathscr{J}} & \twoheadrightarrow & \mathscr{J}\mathscr{O}_X \\ \downarrow & \swarrow & \downarrow \\ \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{Y_\mathscr{J}} & \xrightarrow{\sim} & \mathscr{J}\mathscr{O}_Y \end{array}$$

and hence an epimorphism of $\mathscr{O}_X$-modules

$$\phi : \mathscr{J}\mathscr{O}_X \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{Y_\mathscr{J}}.$$

Let $\mathscr{K}$ be its kernel. For every element Y' of E, the morphism $\mathscr{O}_X \to \mathscr{O}_{Y'}$ then factors through $\mathscr{O}_X/\mathscr{K}$, which is the amalgamated sum B' in the proof of Lemma 4.1. If $\mathscr{I}_{Y'}$ denotes the ideal of Y' in $\mathscr{O}_X$, there is a commutative diagram

$$\begin{array}{ccccccccc} & & & & 0 & & 0 & & \\ & & & & \downarrow & & \downarrow & & \\ & & & & \mathscr{I}_{Y'}/\mathscr{K} & \xrightarrow{\sim} & \mathscr{I}_{Y_\mathscr{J}} & & \\ & & & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & (\mathscr{J}\mathscr{O}_X)/\mathscr{K} & \longrightarrow & \mathscr{O}_X/\mathscr{K} & \longrightarrow & \mathscr{O}_{X_\mathscr{J}} & \longrightarrow & 0 \\ & & \downarrow \wr & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{Y_\mathscr{J}} & \longrightarrow & \mathscr{O}_{Y'} & \longrightarrow & \mathscr{O}_{Y_\mathscr{J}} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}.$$

Replacing X by the closed subscheme defined by $\mathscr{K}$, we are thus reduced to the case $\mathscr{K} = 0$. The datum of Y' is then equivalent to that of the $\mathscr{O}_X$-submodule $\mathscr{I}_{Y'}$ of $\mathscr{O}_X$ which maps bijectively onto $\mathscr{I}_{Y_\mathscr{J}}$ under the projection $p : \mathscr{O}_X \to \mathscr{O}_{X_\mathscr{J}}$. Let

$$f' : \mathscr{I}_{Y_\mathscr{J}} \xrightarrow{\sim} \mathscr{I}_{Y'}, \qquad f : \mathscr{I}_{Y_\mathscr{J}} \xrightarrow{\sim} \mathscr{I}_Y$$

denote the respective inverse isomorphisms. Then $f' - f$ is an element of

$$\mathrm{Hom}_{\mathscr{O}_{X_\mathscr{J}}}\left(\mathscr{I}_{Y_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{Y_\mathscr{J}}\right) = \mathrm{Hom}_{\mathscr{O}_{X_\mathscr{J}}}(\mathscr{I}_{Y_\mathscr{J}}, \mathscr{J}\mathscr{O}_Y),$$

which we denote by $d(Y', Y)$. Notice that $d(Y, Y') = -d(Y', Y)$.

For our fixed Y and variable Y', consider the morphism

$$\mathscr{I}_{Y'} \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_Y = \mathscr{O}_X/\mathscr{I}_Y.$$

[80] Editor's note: This complement has been added; it is useful in proving Proposition 4.8(ii).

Since its composite with $\mathscr{O}_Y \to \mathscr{O}_{Y_\mathscr{J}}$ is zero, it takes values in

$$\mathscr{J}\mathscr{O}_Y = \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}.$$

More precisely, let V be an open subset of X, let x' be a section of $\mathscr{I}_{Y'}$ on V, and let $x_\mathscr{J}$ be its image in $\Gamma(V, \mathscr{I}_{Y_\mathscr{J}})$. Then

$$x' = f'(x_\mathscr{J}) = f(x_\mathscr{J}) + (f' - f)(x_\mathscr{J}) = f(x_\mathscr{J}) + d(Y', Y)(x_\mathscr{J}).$$

Consequently, the morphism $\mathscr{I}_{Y'} \to \mathscr{J}\mathscr{O}_Y$ is given by $d(Y', Y)$.

4.5.0.[81] Retain the notation of 4.3.1 and 4.4, and make the following transformations. The quasi-coherent $\mathscr{O}_{X_\mathscr{J}}$-module

$$\mathscr{I}_{Y_\mathscr{J}}/\mathscr{I}_{Y_\mathscr{J}}^2$$

is annihilated by $\mathscr{I}_{Y_\mathscr{J}}$, and is therefore the direct image of a quasi-coherent $\mathscr{O}_{Y_\mathscr{J}}$-module, denoted by $\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}$, and called the *conormal sheaf* of $Y_\mathscr{J}$ in $X_\mathscr{J}$.[82] Since $\mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}$ is annihilated by $\mathscr{I}_{Y_\mathscr{J}}$, the sheaf of commutative groups $\mathscr{A}_\mathscr{J}$ of 4.3.1 identifies with

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{I}_{Y_\mathscr{J}}/\mathscr{I}_{Y_\mathscr{J}}^2, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right)\\ &\quad = \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right),\end{aligned}$$

and hence, for every $i \geq 0$,

$$H^i(X_\mathscr{J}, \mathscr{A}_\mathscr{J}) = H^i\Big(Y_\mathscr{J}, \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right)\Big).$$

[83] One may now drop the hypothesis that Y be closed, as follows. First observe that every open subset $U_\mathscr{J}$ of $X_\mathscr{J}$ is obtained by base change from the open subscheme U of X having the same underlying topological space as $U_\mathscr{J}$. Let $Y_\mathscr{J}$ now be a closed subscheme of $U_\mathscr{J}$, flat over $S_\mathscr{J}$, and let $\mathscr{I}_{Y_\mathscr{J}}$ be the quasi-coherent ideal of $\mathscr{O}_{U_\mathscr{J}}$ defining $Y_\mathscr{J}$. If $Y_\mathscr{J}$ lifts to a subscheme Y of X, then Y, having the same underlying topological space as $Y_\mathscr{J}$, is a closed subscheme of U. Consequently, the obstruction to lifting $Y_\mathscr{J}$ to a subscheme, flat over S, of X or of U is the same; it lies in

$$H^1\Big(Y_\mathscr{J}, \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right)\Big).$$

Finally, return to the notation of §0. Let $\mathscr{I}$ be a quasi-coherent ideal of $\mathscr{O}_S$ such that $\mathscr{J} \subset \mathscr{I}$ and $\mathscr{I}\mathscr{J} = 0$, and let S_0 be the closed subscheme of $S_\mathscr{J}$ defined by $\mathscr{I}$. For every S-scheme Z, write

$$Z_\mathscr{J} = Z \times_S S_\mathscr{J}, \qquad Z_0 = Z \times_S S_0.$$

Since $\mathscr{J}$ is annihilated by $\mathscr{I}$, the notation of 4.4 gives

$$\mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}} = \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0},$$

and

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right)\\ &\quad = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}} \otimes_{\mathscr{O}_{Y_\mathscr{J}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right),\end{aligned}$$

[81] Editor's note: The numbering 4.5.0 has been added to mark the return to the original.
[82] Editor's note: The following sentence has been corrected.
[83] Editor's note: The discussion that follows has been expanded.

and similarly for the other terms. Thus one obtains:

Proposition 4.5. *Let S be a scheme, and let S_0 and $S_{\mathscr{J}}$ be the closed subschemes defined by quasi-coherent ideals $\mathscr{I}$ and $\mathscr{J}$, where $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let X be an S-scheme and let $Y_{\mathscr{J}}$ be a subscheme of $X_{\mathscr{J}}$, flat over $S_{\mathscr{J}}$. Define the $\mathscr{O}_{Y_0}$-module*

$$\mathscr{A}_0 = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right).$$

Then:

(i) *A subscheme Y of X, reducing to $Y_{\mathscr{J}}$ and flat over S, exists if and only if the following two conditions hold:*

  (a) *Such a Y exists locally on X.*

  (b) *A certain obstruction in $R^1\Gamma(Y_0, \mathscr{A}_0)$ vanishes.*[84]

(ii) *Under these conditions, the set of subschemes Y satisfying the stated requirements is principal homogeneous under the abelian group $\Gamma(Y_0, \mathscr{A}_0)$.*

Remark 4.5.1.[85] It follows from 4.5(ii) that for every pair (Y, Y') of subschemes[86] of X, flat over S and reducing to $Y_{\mathscr{J}}$, there is a "deviation"

$$\begin{aligned} d(Y', Y) &\in \Gamma(Y_0, \mathscr{A}_0) \\ &= \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right). \end{aligned}$$

With the sign convention adopted in 4.4.1, $d(Y', Y)$ corresponds to the morphism of $\mathscr{O}_X$-modules

$$\mathscr{I}_{Y'} \hookrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_Y,$$

which takes values in $\mathscr{J}\mathscr{O}_Y \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}$, and factors through $\mathscr{I}_{Y'_{\mathscr{J}}} = \mathscr{I}_{Y_{\mathscr{J}}}$, then through $\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}$.

Remark 4.6.[87] *If X is flat over S and $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$, then condition (a) is always satisfied, and every Y flat over S lifting $Y_{\mathscr{J}}$ is locally complete intersection in X. If, moreover, Y_0 is affine, condition (b) is also satisfied.*

Definition 4.6.1. *(cf. SGA 6, VII 1.1) Let B be a commutative ring, let $f : E \to B$ be a B-linear morphism, where E is a free B-module of finite rank d, and let $I = f(E)$. If a basis of E is chosen, f is given by a d-tuple $(f_1, \ldots, f_d)$ of elements of B, and I is the ideal generated by the f_i. The* Koszul complex *$K_\bullet(f)$ is the graded B-module $\bigwedge_B^\bullet E$, equipped with the differential of degree -1*

$$x_1 \wedge \cdots \wedge x_i \longmapsto \sum_{j=1}^{i} (-1)^{j-1} f(x_j) x_1 \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge x_i.$$

[84] Editor's note: Here $R^1\Gamma(Y_0, \mathscr{A}_0)$ denotes the "coherent" cohomology group $H^1(Y_0, \mathscr{A}_0)$ of the $\mathscr{O}_{Y_0}$-module $\mathscr{A}_0$, in order to distinguish it from the "Hochschild" cohomology groups $H^i(Y_0, M_0)$, where Y_0 is an S_0-group and M_0 an $\mathscr{O}_{S_0}$-module, which will be considered beginning in 4.16.

[85] Editor's note: This remark has been placed here; it replaces Remark 4.7 of the original.

[86] Editor's note: "closed subschemes" has been corrected to "subschemes."

[87] Editor's note: We have retained, for reference, Remark 4.6 of the original, which did not include the definition of "locally complete intersection." It is followed by the "correct" definition, taken from SGA 6, VII 1.4 (and replacing that of EGA IV$_4$, 16.9.2), and by proofs of the three assertions made in the remark.

Thus there is an augmented chain complex, with B/I in degree -1,

$$\cdots \longrightarrow \bigwedge^2 E \longrightarrow E \xrightarrow{f} B \longrightarrow B/I \longrightarrow 0,$$

which is exact in degree 0 by definition, since $I = f(E)$. The morphism f is said to be *regular* if $K_\bullet(f)$ is acyclic in degrees > 0, that is, if the augmented complex above is a resolution of $C = B/I$.

In that case, the proof of SGA 6, VII 1.2(b) shows that the C-modules I^n/I^{n+1}, for $n \in \mathbb{N}$, are free, with I/I^2 of rank d.

Definition 4.6.2. *(cf. SGA 6, VII 1.4) Let X be a scheme, Y a subscheme, and U an open subset of X such that Y is a closed subscheme of U, defined by the quasi-coherent ideal $\mathscr{I}_Y$.*

The subscheme Y is said to be *locally complete intersection* in X if $Y \hookrightarrow X$ is a regular immersion in the sense of SGA 6, VII 1.4; that is, if for every $y \in Y$ there exist an affine open neighborhood V of y in U, a finite free $\mathscr{O}_V$-module $\mathscr{E}$, and a regular morphism $f : \mathscr{E} \to \mathscr{O}_V$ with image $\mathscr{I}_Y|_V$, such that $K_\bullet(f)$ is a resolution of $\mathscr{O}_{Y\cap V}$.

This implies that the immersion $Y \hookrightarrow X$ is locally of finite presentation and, by 4.6.1, that the conormal sheaf

$$\mathscr{N}_{Y/X} = \mathscr{I}_Y/\mathscr{I}_Y^2$$

is a finite locally free $\mathscr{O}_Y$-module.

Lemma 4.6.3.[88] *Let A be a ring, let J be a square-zero ideal of A, put $\overline{A} = A/J$, let B be a flat A-algebra, let E be a free B-module of finite rank, and let $f : E \to B$ be a morphism of B-modules. Suppose that the morphism*

$$g : \overline{E} = E \otimes_A \overline{A} \longrightarrow \overline{B} = B \otimes_A \overline{A}$$

induced by f is regular and that $\overline{B}/g(\overline{E})$ is flat over $\overline{A}$. Then f is regular and $B/f(E)$ is flat over A.

Proof. Put

$$C = B/f(E), \qquad \overline{C} = C \otimes_A \overline{A} = \overline{B}/g(\overline{E}).$$

First, the $\bigwedge_B^i E$ are free B-modules, and hence flat A-modules, since B is flat over A. Since

$$\bigwedge_B^\bullet E \otimes_A \overline{A} \simeq \bigwedge_{\overline{B}}^\bullet \overline{E},$$

there is an exact sequence of complexes

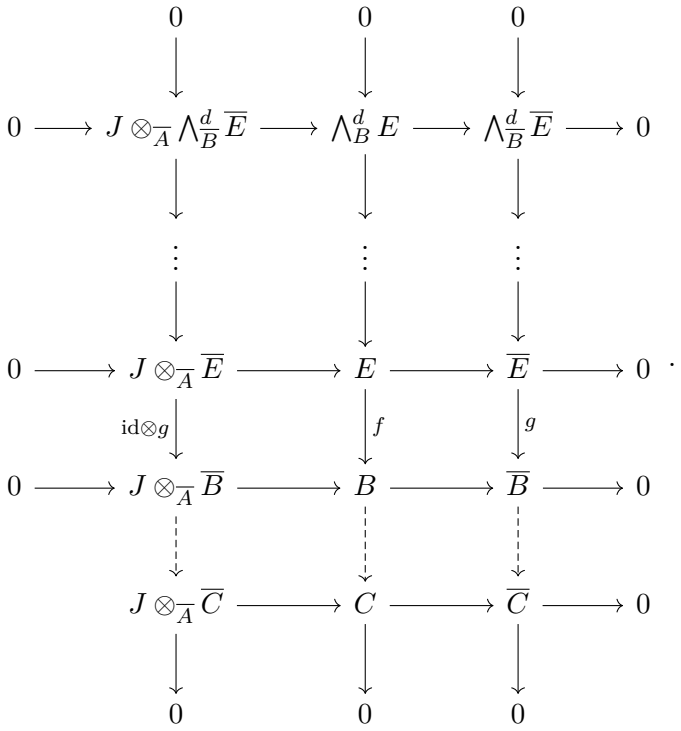
$$0 \longrightarrow J \otimes_A \bigwedge_B^\bullet E \longrightarrow \bigwedge_B^\bullet E \longrightarrow \bigwedge_{\overline{B}}^\bullet \overline{E} \longrightarrow 0.$$

Moreover, since $J^2 = 0$, for every A-module M one has

$$J \otimes_A M \simeq J \otimes_{\overline{A}} (\overline{A} \otimes_A M).$$

Writing the augmentation arrows as dashed arrows, and denoting the rank of E by d, one obtains the following bicomplex, whose rows are exact.

[88] Editor's note: To prove the results stated in Remark 4.6, Lemmas 4.6.3 and 4.6.4, Proposition 4.6.5, and Remark 4.6.6 have been added.



The right and left columns are also exact, since $K_\bullet(g)$ is a resolution of $\overline{C}$, and $\overline{C}$ is flat over $\overline{A}$. The long exact homology sequence associated with the exact sequence of unaugmented complexes therefore shows that $K_\bullet(f)$ is acyclic in degrees > 0, and in degree 0 gives an exact sequence

$$0 \longrightarrow J \otimes_{\overline{A}} \overline{C} \longrightarrow C \longrightarrow \overline{C} \longrightarrow 0.$$

It follows from the "fundamental criterion of flatness" that C is flat over A (cf. [BAC], §III.5, Theorem 1). □

Lemma 4.6.4.[88] *Let A be a commutative ring, let J be a nilpotent ideal, and let $N \subset M$ be A-modules such that M/N is flat over A. If $x_1, \ldots, x_n$ are elements of N whose images generate the image $\overline{N}$ of N in M/JM, then they generate N.*

Indeed, let N' be the submodule of N generated by the x_i, and put $Q = N/N'$. The morphism $N' \otimes_A (A/J) \to \overline{N}$ is surjective. On the other hand, since M/N is flat over A, the morphism $N \otimes_A (A/J) \to \overline{N}$ is bijective. Thus $Q \otimes_A (A/J) = 0$, whence $Q = 0$ by the "nilpotent Nakayama lemma":

$$Q = JQ = J^2Q = \cdots = 0.$$

We can now prove the following proposition.

Proposition 4.6.5.[88] *Let $S, \mathscr{I}, \mathscr{J}$, and $X, Y_{\mathscr{J}}$ be as in 4.5. Suppose in addition that X is flat over S and that $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$.*

(a) *Condition (a) of 4.5(i) is then satisfied; moreover, every Y flat over S lifting $Y_{\mathscr{J}}$ is locally complete intersection in X.*

(b) *If, in addition, Y_0 is affine, condition (b) of the cited result is also satisfied.*

Proof. The first assertion of (a) follows from Lemma 4.6.3, and the second then follows from Lemma 4.6.4. Moreover, the hypothesis implies (cf. 4.6.2) that $\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}$ is a finite locally

free $\mathscr{O}_{Y_{\mathscr{J}}}$-module. Hence the $\mathscr{O}_{Y_0}$-module

$$\mathscr{A}_0 = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right)$$

is quasi-coherent (cf. EGA I, 1.3.12), and therefore $R^1\Gamma(Y_0, \mathscr{A}_0) = 0$ when Y_0 is affine. □

Remark 4.6.6.[88] Let us conclude this paragraph with an example showing that, under the hypotheses of Lemma 4.6.3, a regular sequence $(\overline{g}_1, \overline{g}_2)$ generating the ideal $\overline{I} = g(\overline{E})$ need not lift to a regular sequence in B.

Let k be a field, put $\overline{A} = k[X, Y]$, let $k\varepsilon$ denote the $\overline{A}$-module $\overline{A}/(X, Y)$, so that

$$P\varepsilon = P(0,0)\varepsilon \qquad (P \in \overline{A}),$$

and let $A = \overline{A} \oplus k\varepsilon$, where $J = k\varepsilon$ is a square-zero ideal. Thus $A/J = \overline{A}$.

The algebra $B = A \otimes_k k[Z, T]$ is free, and hence flat, over A, and $\overline{B} = k[X, Y, Z, T]$. Put

$$\overline{g}_1 = XZ - YT, \qquad \overline{g}_2 = XZ - 1.$$

Since $\overline{g}_1$ is irreducible, $\overline{B}/(\overline{g}_1)$ is an integral domain; hence $(\overline{g}_1, \overline{g}_2)$ is a regular sequence in $\overline{B}$, generating the ideal

$$\overline{I} = (XZ - 1, YT - 1).$$

Therefore

$$\overline{C} = \overline{B}/\overline{I} = k[X, Y, X^{-1}, Y^{-1}] = \overline{A}[X^{-1}, Y^{-1}]$$

is a flat $\overline{A}$-algebra, and also a flat A-algebra. But every lift of $\overline{g}_1$ to B has the form

$$XZ - YT + \lambda\varepsilon, \qquad \lambda \in k[Z, T],$$

and therefore annihilates ε.

4.7. Remark 4.7 has been suppressed here and placed in 4.5.1.

Remark 4.8.0.[89] *Let S be a scheme, S' a closed subscheme of S, X an S-scheme, Y an S-subscheme of X, and put $X' = X \times_S S'$ and $Y' = Y \times_S S'$. Then there is a surjective morphism of $\mathscr{O}_{Y'}$-modules*

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \twoheadrightarrow \mathscr{N}_{Y'/X'}.$$

Indeed, after replacing X by a suitable open subscheme, one may suppose that Y is closed and is defined by an ideal $\mathscr{I}_Y$ of $\mathscr{O}_X$. The image of $\mathscr{I}_Y$ in $\mathscr{O}_{X'}$ is then the ideal $\mathscr{I}_{Y'}$ defining Y', and there is a surjective morphism of $\mathscr{O}_{Y'}$-modules

$$\pi : (\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \twoheadrightarrow \mathscr{I}_{Y'}/\mathscr{I}_{Y'}^2.$$

Suppose in addition that $\mathscr{O}_Y = \mathscr{O}_X/\mathscr{I}_Y$ is flat over $\mathscr{O}_S$. The natural morphism

$$\mathscr{I}_Y \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{I}_{Y'}$$

is then bijective (cf. EGA IV_2, 2.1.8).

There is therefore the following commutative diagram with exact rows:

[89] Editor's note: This remark, used in the proposition that follows, has been inserted here; it appeared in 4.10 of the original.

$$\begin{array}{ccccccccc}
\mathscr{I}_Y^2 \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} & \longrightarrow & \mathscr{I}_Y \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} & \longrightarrow & (\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} & \longrightarrow & 0 \\
\downarrow{\scriptstyle \text{surj.}} & & \downarrow{\scriptstyle \wr} & & \downarrow{\scriptstyle \pi}\ {\scriptstyle \text{surj.}} & & \\
0 \longrightarrow \mathscr{I}_{Y'}^2 & \longrightarrow & \mathscr{I}_{Y'} & \longrightarrow & \mathscr{I}_{Y'}/\mathscr{I}_{Y'}^2 & \longrightarrow & 0
\end{array}$$

It follows from the snake lemma that[90]

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \xrightarrow{\sim} \mathscr{N}_{Y'/X'} \qquad \text{if } Y \text{ is flat over } S. \tag{4.8.0}$$

Proposition 4.8. *Let $S, S_0, S_{\mathscr{J}}$, and $\mathscr{I}, \mathscr{J}$ be as in 4.5.*[91] *Let X be an S-scheme, let Y be a subscheme of X, and let $i : Y \hookrightarrow X$ be the immersion.*

(i) *For every S-morphism $f : T \to X$ such that $f_{\mathscr{J}} : T_{\mathscr{J}} \to X_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, one can define an obstruction*

$$c(X, Y, f) \in \operatorname{Hom}_{\mathscr{O}_{T_0}} \left(f_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T \right) \tag{*}$$

whose vanishing is equivalent to the existence of a factorization of f through Y.

(ii) *Let Y' be a second subscheme of X. Suppose that $Y'_{\mathscr{J}} = Y_{\mathscr{J}}$ and that Y and Y' are flat over S. Then there are isomorphisms (cf. 4.8.0)*

$$\mathscr{J}\mathscr{O}_Y \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0} \simeq \mathscr{J}\mathscr{O}_{Y'}, \qquad \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_{\mathscr{J}}} \xrightarrow{\sim} \mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}},$$

and hence an isomorphism

$$\begin{aligned} u : \operatorname{Hom}_{\mathscr{O}_{Y_0}} & \left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_Y \right) \\ & \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_{Y_0}} \left(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_{Y'} \right). \end{aligned}$$

Writing $i' : Y' \hookrightarrow X$ for the canonical immersion and $d(Y, Y')$ for the deviation of 4.5.1, one has[92]

$$c(X, Y, i') = u(d(Y, Y')). \tag{**}$$

(iii) *The canonical morphism*

$$\mathscr{N}_{Y/X} \xrightarrow{D} i^*(\Omega^1_{X/S})$$

(cf. SGA 1, II, formula 4.3)[93] *induces a morphism*

$$D_0 : \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0},$$

[90] Editor's note: In the original, this was stated in Remark 4.10 under the additional hypothesis that Y' be locally complete intersection in X'. Consequently, that hypothesis also appeared in statements 4.12–4.14; it seems in fact superfluous, and has been deleted from those statements.

[91] Editor's note: The hypothesis that $\mathscr{I}$ be nilpotent has been deleted, since it appears to be superfluous (cf. the proof).

[92] Editor's note: See also 4.27 below.

[93] Editor's note: See also EGA IV$_4$, 16.4.21. Recall that if U is an affine open subset of X such that $Y \cap U$ is defined by the ideal I of $A = \mathscr{O}_X(U)$, if d denotes the differential $A \to \Gamma(U, \Omega^1_{X/S})$, and if $x \in I$, then $D(x + I^2)$ is the element $d(x) \otimes 1$ of $\Gamma(U, \Omega^1_{X/S}) \otimes_A (A/I)$.

and hence, for every S-morphism $f : T \to X$ as in (i), a morphism

$$v_{f_0} : \mathrm{Hom}_{\mathscr{O}_{T_0}}\left(f_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_T\right) \longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\left(f_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T\right), \quad a \longmapsto a \circ f_0^*(D_0),$$

where the first group above is $\mathrm{Hom}_{X^+}(T, L_X)$*, cf. 0.1.5. For* $a \in \mathrm{Hom}_{X^+}(T, L_X)$*, one has*

$$c(X, Y, a \cdot f) - c(X, Y, f) = v_{f_0}(a), \qquad (***)$$

where $a \cdot f$ denotes the composite morphism

$$T \xrightarrow{a \times f} L_X \times_{X^+} X \longrightarrow X.$$

We shall prove part (i) of the proposition, leaving it to the reader to (not) verify assertions (ii) and (iii); this verification proceeds by reduction to the affine case and then by comparison of the explicit definitions.[94]

Let us therefore prove (i). The morphism $f : T \to X$ defines a morphism of sheaves of rings[95]

$$\phi : \mathscr{O}_X \longrightarrow f_*(\mathscr{O}_T).$$

Let U be an open subscheme of X in which Y is closed. Since T, respectively $Y_{\mathscr{J}}$, has the same underlying space as $T_{\mathscr{J}}$, respectively Y, the continuous map underlying f sends T into U. Since U is open in X, the morphism ϕ induces a morphism of sheaves of rings

$$\mathscr{O}_U = \mathscr{O}_X|_U \longrightarrow f_*(\mathscr{O}_T),$$

that is, f factors through U.

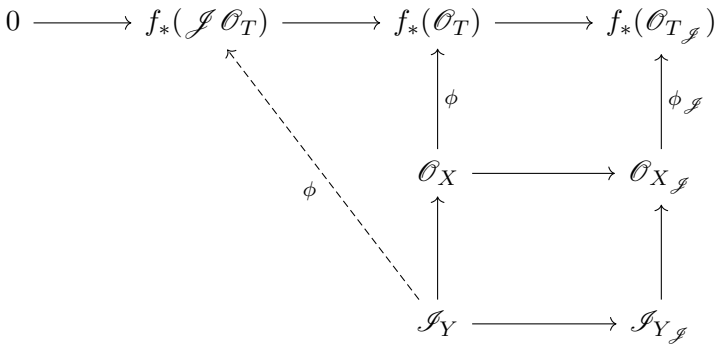
We may therefore restrict to the case in which Y is closed, hence defined by a sheaf of ideals $\mathscr{I}_Y$. The morphism f factors through Y if and only if the composite

$$\mathscr{I}_Y \longrightarrow \mathscr{O}_X \longrightarrow f_*(\mathscr{O}_T)$$

is zero. Since $f_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, the composite

$$\mathscr{I}_{Y_{\mathscr{J}}} \longrightarrow \mathscr{O}_{X_{\mathscr{J}}} \longrightarrow f_*(\mathscr{O}_{T_{\mathscr{J}}})$$

is zero. Consider the following commutative diagram, whose first row is exact:



[94] Editor's note: These verifications have been carried out below.

[95] Editor's note: On the one hand, the hypothesis that $\mathscr{I}$ be nilpotent—that is, that X_0 have the same underlying topological space as X—has been deleted; on the other hand, the sentence that follows has been expanded.

It follows that ϕ maps $\mathscr{I}_Y$ into $f_*(\mathscr{J}\mathscr{O}_T)$.[96] Since $\mathscr{J}^2 = 0$, the $\mathscr{O}_X$-module $f_*(\mathscr{J}\mathscr{O}_T)$, with its $\mathscr{O}_X$-module structure induced by ϕ, is annihilated by $\mathscr{I}_Y$. Consequently, ϕ induces a morphism of $\mathscr{O}_X$-modules

$$h : i_*(\mathscr{N}_{Y/X}) = \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow f_*(\mathscr{J}\mathscr{O}_T).$$

On the other hand, one has the following cartesian squares:

$$\begin{array}{ccccc} T_0 & \xrightarrow{f_0} & X_0 & \xleftarrow{i_0} & Y_0 \\ \downarrow{\scriptstyle \tau_{T_0}} & & \downarrow{\scriptstyle \tau_{X_0}} & & \downarrow{\scriptstyle \tau_{Y_0}} \\ T & \xrightarrow{f} & X & \xleftarrow{i} & Y \end{array} .$$

Here $\tau_{T_0}, \tau_{X_0}, \tau_{Y_0}$ are the closed immersions obtained by base change from $S_0 \hookrightarrow S$. Since $\mathscr{J}\mathscr{O}_T$ is a quasi-coherent $\mathscr{O}_T$-module annihilated by $\mathscr{I}$, there is an isomorphism

$$\mathscr{J}\mathscr{O}_T \simeq (\tau_{T_0})_*\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T),$$

and hence

$$f_*(\mathscr{J}\mathscr{O}_T) \simeq (\tau_{X_0})_*(f_0)_*\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T).$$

Thus, by adjunction, h corresponds to a morphism of $\mathscr{O}_{T_0}$-modules

$$h_0 : f_0^*\tau_{X_0}^* i_*(\mathscr{N}_{Y/X}) \longrightarrow \tau_{T_0}^*(\mathscr{J}\mathscr{O}_T).$$

Now

$$\tau_{X_0}^* i_*(\mathscr{N}_{Y/X}) \simeq (i_0)_*\tau_{Y_0}^*(\mathscr{N}_{Y/X}) = \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}.$$

Returning to the customary abuse of notation $\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T) = \mathscr{J}\mathscr{O}_T$, the morphism h_0 identifies with a morphism of $\mathscr{O}_{T_0}$-modules

$$h_0 : f_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}) \longrightarrow \mathscr{J}\mathscr{O}_T,$$

which is the required obstruction $c(X, Y, f)$. This proves (i).

When f is the immersion $i' : Y' \hookrightarrow X$, one sees that $c(X, Y, i')$ comes from the morphism

$$\mathscr{I}_Y \hookrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_{Y'},$$

and hence corresponds, by 4.4.1 and 4.5.1, to the class $d(Y, Y')$. This proves (ii).

Let us prove (iii). First, $D : \mathscr{N}_{Y/X} \to i^*(\Omega^1_{X/S})$ induces a morphism

$$\begin{aligned} D_0 : \quad \tau_{Y_0}^*(\mathscr{N}_{Y/X}) &\longrightarrow \tau_{Y_0}^* i^*(\Omega^1_{X/S}) \\ &= i_0^*\tau_{X_0}^*(\Omega^1_{X/S}). \end{aligned}$$

Since $X_0 = X \times_S S_0$, one has

$$\tau_{X_0}^*(\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0} \qquad \text{(cf. EGA IV}_4\text{, 16.4.5).}$$

This gives the asserted morphism

$$D_0 : \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0}.$$

[96] Editor's note: The argument that follows has been expanded.

We shall verify equality $(***)$ after the following remark.

Remark 4.9. The obstruction $c(X, Y, f)$ is computed locally on T. Let $U = \mathrm{Spec}(C)$ be an affine open subset of T lying over an affine open subset $\mathrm{Spec}(A)$ of X, itself lying over an affine open subset $\mathrm{Spec}(\Lambda)$ of S. Let $J \subset I \subset \Lambda$, respectively $I_Y \subset A$, be the ideals corresponding to $\mathscr{J} \subset \mathscr{I}$, respectively $\mathscr{I}_Y$, put $B = A/I_Y$, and let $\phi : A \to C$ be the morphism of Λ-algebras corresponding to $f : T \to X$. Since $f(T_{\mathscr{J}}) \subset Y_{\mathscr{J}}$, one has $\phi(I_Y) \subset JC$, and ϕ therefore induces a morphism of Λ-algebras

$$B \longrightarrow C/JC \longrightarrow C_0 = C/IC.$$

The obstruction $c = c(X, Y, f)$ is then computed by the following commutative diagram:

$$\begin{array}{ccccc} I_Y & \longrightarrow & A & \xrightarrow{\phi} & C \\ \downarrow & & & & \uparrow \\ I_Y/I_Y^2 & \longrightarrow & (I_Y/I_Y^2) \otimes_B C_0 & \xrightarrow{c} & JC \end{array} ,$$

That is, over U, it is the unique element of

$$\mathrm{Hom}_{C_0}\left(I_Y/I_Y^2 \otimes_B C_0, JC\right) = \mathrm{Hom}_{B_0}\left(I_Y/I_Y^2 \otimes_B B_0, JC\right)$$

such that, with the evident notation,

$$c(\overline{x} \otimes_B 1) = \phi(x) \qquad \text{for every } x \in I_Y.$$

[97] We can now complete the proof of 4.8(iii). Equality $(***)$ may be checked locally on T, so we are reduced to the affine situation described above. Let $d_{A/\Lambda} : A \to \Omega^1_{A/\Lambda}$ denote the differential. Over U, the element a corresponds to an element a_U of

$$\begin{aligned} &\mathrm{Hom}_{C_0}\left(\Omega^1_{A_0/\Lambda_0} \otimes_{A_0} C_0, JC\right) \\ &\quad \simeq \mathrm{Hom}_{B_0}\left(\Omega^1_{A/\Lambda} \otimes_A B_0, JC\right) \simeq \mathrm{Hom}_A(\Omega^1_{A/\Lambda}, JC). \end{aligned}$$

On the one hand, $v_{f_0}(a)$ corresponds over U to $a_U \circ D_0$, where D_0 is the morphism of B-modules[98]

$$\begin{aligned} I_Y/I_Y^2 &\longrightarrow \Omega^1_{A/\Lambda} \otimes_A B_0, \\ x + I_Y^2 &\longmapsto d_{A/\Lambda}(x) \otimes 1. \end{aligned}$$

On the other hand (cf. the proofs of 0.1.8 and 0.1.9), the morphism of Λ-algebras $\phi' : A \to C$ corresponding to $a \cdot f$ differs from ϕ by the Λ-derivation $A \to JC$ associated with a_U; that is,

$$\phi' = \phi + a_U \circ d_{A/\Lambda} = \phi + a_U \circ (d_{A/\Lambda} \otimes 1).$$

Consequently, writing $c' = c(X, Y, a \cdot f)$, for every $x \in I_Y$, and writing $\overline{x}$ for its image in I_Y/I_Y^2, one has

$$\begin{aligned} (c' - c)(\overline{x} \otimes 1) &= a_U\big(d_{A/\Lambda}(x) \otimes 1\big) \\ &= (a_U \circ D_0)(\overline{x}) = v_{f_0}(a)(\overline{x}). \end{aligned}$$

[97] Editor's note: What follows has been added.
[98] Editor's note: Cf. Editor's Note 93.

This proves that $c' - c = v_{f_0}(a)$.

4.10. Remark 4.10 of the original has been suppressed, since it was made obsolete by the addition of Remark 4.8.0.

4.11. We now propose to study the following situation. Let S, $S_{\mathscr{J}}$, and S_0 be as in 4.8. We have three S-schemes X, X', T, a subscheme Y of X (respectively a subscheme Y' of X'), and morphisms $f : T \to X'$ and $g : X' \to X$.

$$\begin{array}{ccccc} & & Y' & & Y \\ & & \big\downarrow{\scriptstyle i'} & & \big\downarrow{\scriptstyle i} \\ T & \xrightarrow{f} & X' & \xrightarrow{\quad g \quad} & X \end{array}$$

We suppose that, after reduction modulo $\mathscr{J}$, this diagram can be completed to a commutative diagram

$$\begin{array}{ccccc} & & Y'_{\mathscr{J}} & \dashrightarrow & Y_{\mathscr{J}} \\ & \nearrow & \big\downarrow{\scriptstyle i'_{\mathscr{J}}} & & \big\downarrow{\scriptstyle i_{\mathscr{J}}} \\ T_{\mathscr{J}} & \xrightarrow{f_{\mathscr{J}}} & X'_{\mathscr{J}} & \xrightarrow{\quad g_{\mathscr{J}} \quad} & X_{\mathscr{J}} \end{array}$$

By 4.8 we therefore have the obstructions

$$\begin{aligned} c(X, Y, g \circ i') &\in \mathrm{Hom}_{\mathscr{O}_{Y'_0}}(i'^*_0 g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{Y'}), \\ c(X', Y', f) &\in \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_T), \\ c(X, Y, g \circ f) &\in \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T), \end{aligned}$$

whose relations we seek to compute. [99]

Lemma 4.12. *Suppose that Y' is flat over S, so that*

$$\mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y'_0} = \mathscr{J}\mathscr{O}_{Y'}.$$

Then:

(i) *There is a natural morphism*

$$\begin{aligned} b_{f_0} : \mathrm{Hom}_{\mathscr{O}_{Y'_0}}(i'^*_0 g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{Y'}) \\ \longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T). \end{aligned}$$

(ii) *There is also a natural morphism, functorial in T,* [100]

$$\begin{aligned} a_{g_0}(f_0) : \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_T) \\ \longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T). \end{aligned}$$

[99] Editor's note: Starting with 4.17, this will be applied when X is an S-group, $g : X \times_S X \to X$ is multiplication, Y is a subscheme of X such that $Y_{\mathscr{J}}$ is a subgroup of $X_{\mathscr{J}}$, $Y' = Y \times_S Y$, and the two morphisms $Y^3 \to X^2$ send (y_1, y_2, y_3) respectively to $(y_1 y_2, y_3)$ and $(y_1, y_2 y_3)$. In that case, comparison of the obstructions above will show that the obstruction to Y being a subgroup of X lies in a certain Hochschild cohomology group $H^2(Y_0, N_0)$.

[100] Editor's note: We have removed the hypothesis that Y'_0 be locally complete intersection in X'_0, which is superfluous by 4.8.0. We have also added that $a_{g_0}(f_0)$ is "functorial in T", a fact that plays a crucial role in the proof of 4.17.

Proof. [101] First note that, with X, X', Y, Y' fixed, giving a T as above is equivalent to giving a morphism $(f, f_{\mathscr{J}}) : T \to X' \times_{X'^+} Y'^+$. Put $Z = X' \times_{X'^+} Y'^+$, and let M and M' be the Z-functors defined, for every $f : T \to Z$, by

$$M(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}),\, \mathscr{J}\,\mathscr{O}_T\big),$$
$$M'(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}),\, \mathscr{J}\,\mathscr{O}_T\big).$$

[102]

In all cases there is a commutative diagram

$$\begin{array}{ccc} f_0^*\big(\mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y'_0}\big) & = & \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0} \\ \downarrow & & \downarrow \\ f_0^*(\mathscr{J}\,\mathscr{O}_{Y'}) & \dashrightarrow & \mathscr{J}\,\mathscr{O}_T \end{array}$$

Since Y' is flat over S, the left vertical arrow is an isomorphism, and hence we obtain a morphism of $\mathscr{O}_{T_0}$-modules

$$f_0^*(\mathscr{J}\,\mathscr{O}_{Y'}) \longrightarrow \mathscr{J}\,\mathscr{O}_T.$$

It induces a morphism of abelian groups

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), f_0^*(\mathscr{J}\,\mathscr{O}_{Y'})\big) \longrightarrow M(T),$$

and, on composing with the morphism

$$M(Y') \longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), f_0^*(\mathscr{J}\,\mathscr{O}_{Y'})\big)$$

induced by f_0^*, we obtain the morphism $b_{f_0} : M(Y') \to M(T)$.

Likewise, in all cases there is a diagram

$$\begin{array}{ccc} g_0^*\big(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}\big) & \dashrightarrow & \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0} \\ \downarrow & & \downarrow \\ g_0^*(\mathscr{N}_{Y_0/X_0}) & \longrightarrow & \mathscr{N}_{Y'_0/X'_0} \end{array}$$

Since Y' is flat over S, the second vertical arrow is an isomorphism by 4.8.0. We therefore obtain an $\mathscr{O}_{Y'_0}$-morphism

$$i_0'^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}) \longrightarrow \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}.$$

This induces a morphism $a_{g_0}(\mathrm{id}_{Y'}) : M'(Y') \to M(Y')$ and, for every $f : T \to Z$, a morphism $a_{g_0}(f) : M'(T) \to M(T)$, such that the following diagram commutes:

[101] Editor's note: The proof has been expanded to make the "functoriality in T" of a_{g_0} explicit.

[102] Editor's note: The situation simplifies from 4.16 onward: we shall restrict to schemes flat over S, Y will be a flat S-group, and $Y' = Y \times_S Y$; this will give S_0-functors N_0 and N'_0.

$$\begin{array}{ccc} M'(Y') & \xrightarrow{a_{g_0}(\mathrm{id}_{Y'_0})} & M(Y') \\ \Big\downarrow{\scriptstyle b'_{f_0}} & & \Big\downarrow{\scriptstyle b_{f_0}} \\ M'(T) & \xrightarrow[a_{g_0}(f_0)]{} & M(T) \end{array}$$

Here b'_{f_0} is defined in the same way as b_{f_0}. Q.E.D.

Remark 4.12.1. [103] *Let M_0 and M'_0 be the Y'_0-functors defined, for every $f : T_0 \to Y'_0$, by*

$$M_0(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\big),$$
$$M'_0(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\big).$$

Note at once that $Z_0 = Y'_0$, and that on the category of Z-schemes T flat over S, the functors M and M' coincide respectively with $\prod_{S_0/S} M_0$ and $\prod_{S_0/S} M'_0$. In that case, b_{f_0} is simply the morphism

$$f_0^* : M_0(Y'_0) \longrightarrow M(T_0)$$

induced by f_0. Moreover, for every morphism $u : U \to T$, if $h = f \circ u$, the following diagram commutes:

$$\begin{array}{ccc} M'_0(T_0) & \xrightarrow{a_{g_0}(f_0)} & M_0(T_0) \\ \Big\downarrow{\scriptstyle u_0^*} & & \Big\downarrow{\scriptstyle u_0^*} \\ M'_0(U_0) & \xrightarrow[a_{g_0}(h_0)]{} & M_0(U_0) \end{array}$$

Thus a_{g_0} becomes a morphism of functors

$$\prod_{S_0/S} M'_0 \longrightarrow \prod_{S_0/S} M_0.$$

Proposition 4.13. *Suppose that Y' is flat over S. Then*

$$c(X, Y, g \circ f) = a_{g_0}\big(c(X', Y', f)\big) + b_{f_0}\big(c(X, Y, g \circ i')\big).$$

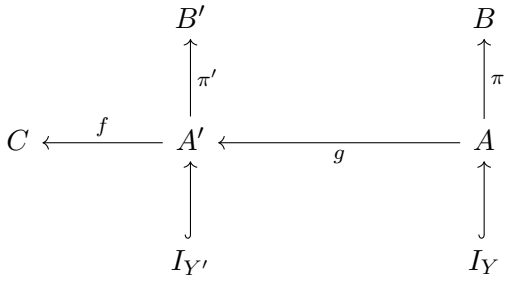
Since the definitions of the various obstructions and of the morphisms a_{g_0} and b_{f_0} are local, it plainly suffices to verify the formula when all the schemes in question are affine. Write

$$S = \mathrm{Spec}(\Lambda), \qquad S_{\mathscr{J}} = \mathrm{Spec}(\Lambda/\mathscr{J}), \qquad S_0 = \mathrm{Spec}(\Lambda/\mathscr{I}),$$
$$T = \mathrm{Spec}(C), \qquad X = \mathrm{Spec}(A), \qquad Y = \mathrm{Spec}(A/I_Y) = \mathrm{Spec}(B),$$
$$X' = \mathrm{Spec}(A'), \qquad Y' = \mathrm{Spec}(A'/I_{Y'}) = \mathrm{Spec}(B').$$

There is therefore a diagram of rings and ideals [104]

[103] Editor's note: This remark has been added for use in the proof of 4.17.

[104] Editor's note: We retain the notation of the original, writing $f : A' \to C$ and $g : A \to A'$ for the ring homomorphisms corresponding to $f : T \to X'$ and $g : X' \to X$. This explains the formula $c(X, Y, g \circ f)(\bar{x} \otimes 1) = f(g(x))$, for $x \in I_Y$.



Let us study the terms of the formula to be proved. In what follows, for $x \in I_Y$ (respectively $u \in I_{Y'}$), write $\bar{x}$ (respectively $\bar{u}$) for its image in I_Y/I_Y^2 (respectively $I_{Y'}/I_{Y'}^2$). If m belongs to a Λ-module M, write $\bar{m}_0$ for its image in $M_0 = M/\mathscr{I}M$.

We saw that $c = c(X, Y, g \circ f)$ is the unique C_0-morphism

$$I_Y/I_Y^2 \otimes_B C_0 \longrightarrow \mathscr{J}C$$

such that $c(\bar{x} \otimes 1) = f(g(x))$ for every $x \in I_Y$.

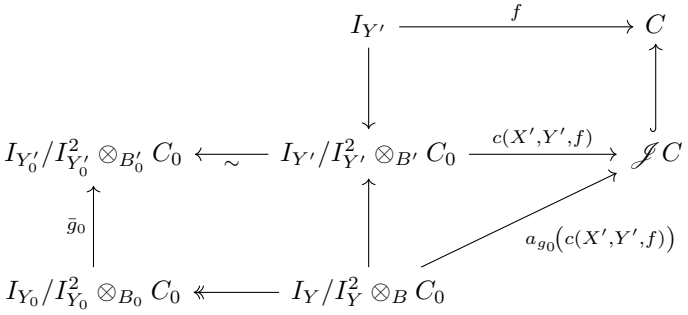
Fix $x \in I_Y$. Since $g_{\mathscr{J}}(Y'_{\mathscr{J}}) \subset Y_{\mathscr{J}}$, one has $g(x) \in I_{Y'} + \mathscr{J}A'$. Write

$$g(x) = x' + \sum_i \lambda_i a'_i, \qquad x' \in I_{Y'}, \quad \lambda_i \in \mathscr{J}, \quad a'_i \in A'.$$

Then

$$c(X, Y, g \circ f)(\bar{x} \otimes 1) = f(g(x)) = f(x') + \sum_i \lambda_i f(a'_i). \tag{1}$$

Consider $a_{g_0}(c(X', Y', f))$. By the definitions just given, it is defined by the following diagram:



Thus $a_{g_0}(c(X', Y', f))(\bar{x} \otimes 1) = f(u)$, where $u \in I_{Y'}$ has image $\bar{u}$ in $I_{Y'_0}/I_{Y'_0}^2$ satisfying

$$\bar{u}_0 \otimes 1 = \bar{g}_0(\bar{x}_0) \otimes 1 = g_0(\bar{x}_0) \otimes 1.$$

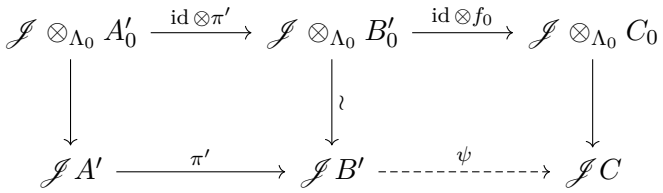
We may therefore take $u = x'$, which gives

$$a_{g_0}(c(X', Y', f))(\bar{x} \otimes 1) = f(x'). \tag{2}$$

Finally consider $b_{f_0}(c(X, Y, g \circ i'))$. By hypothesis, the morphism of Λ_0-algebras $f_0 : A'_0 \to C_0$ factors through B'_0. Since Y' is flat, and hence

$$\mathscr{J} \otimes_{\Lambda_0} B'_0 \xrightarrow{\sim} \mathscr{J}B',$$

we obtain a morphism of B'_0-modules $\psi : \mathscr{J}B' \to \mathscr{J}C$ making the following diagram commute:



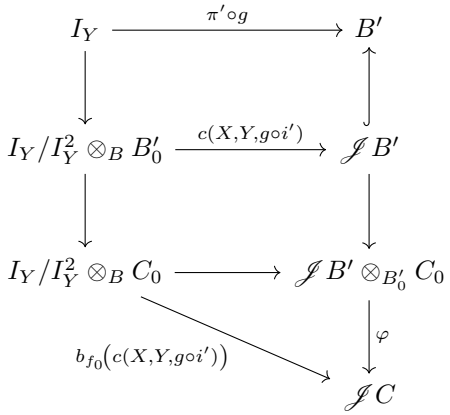
Let

$$\varphi : \mathscr{J}B' \otimes_{B'_0} C_0 \longrightarrow \mathscr{J}C$$

be the morphism of C_0-modules induced by ψ. Then, for every $a' \in A'$ and $\lambda \in \mathscr{J}$,

$$\varphi(\lambda\pi'(a') \otimes 1) = \lambda f(a'). \tag{†}$$

The element $b_{f_0}(c(X, Y, g \circ i'))$ is then defined by the commutative diagram



Consequently,

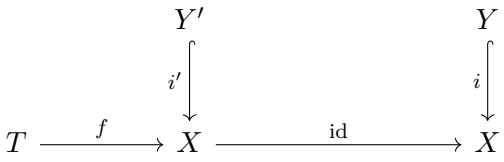
$$b_{f_0}(c(X, Y, g \circ i'))(\bar{x} \otimes 1) = \varphi\left(\sum_i \lambda_i \pi'(a'_i) \otimes 1\right) = \sum_i \lambda_i f(a'_i), \tag{3}$$

the last equality following from (†). Comparison of the three explicit formulas (1), (2), and (3) gives the required formula.

Corollary 4.14. *Let Y, Y' be two flat subschemes of X, both reducing to $Y_{\mathscr{J}}$, and suppose that Y_0 is locally complete intersection in X_0. If $f : T \to X$ is an S-morphism such that $f_{\mathscr{J}}$ factors through $Y_{\mathscr{J}} \to X_{\mathscr{J}}$, then*

$$c(X, Y, f) - c(X, Y', f) = b_{f_0}\big(d(Y, Y')\big).$$

Indeed, applying the preceding formula to the diagram



we find $c(X, Y, f) - c(X, Y', f) = b_{f_0}(c(X, Y, i'))$. Moreover, by 4.8(ii), $c(X, Y, i') = d(Y, Y')$.

Proposition 4.15. *Let X be an S-group smooth over S, and let Y be a sub-S-group flat and locally of finite presentation over S. Then Y is locally complete intersection in X (cf. 4.6.2).*

Proof. [105] We show that the immersion $Y \to X$ is regular in the sense of EGA IV_4, 16.9.2. By EGA IV_4, 19.5.1, this implies that it is regular also in the sense of 4.6.2 (indeed, by loc. cit., the two definitions are equivalent when S is locally noetherian). Thus "regular immersion" will here mean regular in the sense of EGA IV_4, 16.9.2. Since X and Y are flat and locally of finite presentation over S, EGA IV_4, 19.2.4 reduces the assertion to proving that $Y_s \to X_s$ is a regular immersion for every $s \in S$. By EGA IV_4, 19.1.5(ii), it is enough to check this on the geometric fibers of S, and hence when S is the spectrum of an algebraically closed field k.

By Exposé $\mathrm{VI_A}$, 3.2, the quotient X/Y exists and is smooth, the morphism $\pi : X \to X/Y$ is flat, and the following square is cartesian:

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ \bar{e} & \xrightarrow{i} & X/Y \end{array}$$

Here $\bar{e}$ is the image in X/Y of the unit point of X. By flat base change (cf. EGA IV_4, 19.1.5(iii)), it therefore suffices to show that i is a regular immersion. This is immediate: the noetherian local ring $\mathscr{O}_{X/Y,\bar{e}}$ is smooth (hence regular), so its maximal ideal is generated by a regular sequence.

4.16. [106] Let X be an S-group smooth over S, and denote its group law by $\mu : X \times_S X \to X$. Let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. By 4.15, $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$. Hence, by 4.6.5, every flat [107] S-scheme Y lifting $Y_{\mathscr{J}}$ is locally complete intersection in X. For such a Y, 4.8.0 gives

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} = \mathscr{N}_{Y_0/X_0} = \mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}. \tag{4.16.1}$$

On the other hand, let $\varepsilon_0 : S_0 \to Y_0$ be the unit section of Y_0, and let $\mathbf{n}_{Y_0/X_0}$ be the quasi-coherent $\mathscr{O}_{S_0}$-module

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathscr{N}_{Y_0/X_0}).$$

[105] Editor's note: We have added to the statement the hypothesis that Y be locally of finite presentation over S, and supplied the proof that follows, which is more direct than the one sketched in the original. For completeness, we also spell out the latter. As in the proof above, first reduce to the case $S = \mathrm{Spec}(k)$, where k is algebraically closed. By EGA IV_4, 16.9.10 and 19.3.2, it suffices to see that, for every $y \in Y$, the completion of the local ring $\mathscr{O}_{Y,y}$ is a quotient of a complete noetherian local ring by a regular sequence. By loc. cit., 19.3.3, the set of $y \in Y$ with this property is an open subset $U \subset Y$. Since Y is of finite type over k, it suffices to show that U contains every closed point. Because Y is a k-group, a translation argument reduces this to the completion of $\mathscr{O}_{Y,e}$, that is, to the formal group $\widehat{Y}$ corresponding to Y (cf. Exposé $\mathrm{VII_B}$). Since X is smooth, the affine algebra $\mathscr{A}(\widehat{X})$ is a power series algebra $k[[X_1, \ldots, X_n]]$. The Dieudonné structure theorem then shows that $\mathscr{A}(\widehat{Y})$ is isomorphic to a quotient

$$k[[X_1, \ldots, X_{r+s}]]/(X_1^{p^{n_1}}, \ldots, X_r^{p^{n_r}}),$$

where p is the characteristic exponent of k and $r + s \leq n$; cf. Exposé $\mathrm{VII_B}$, Remark 5.5.2(b).

[106] Editor's note: We have reorganized 4.16 by collecting here, on the one hand, the hypotheses stated at the end of 4.15 and, on the other, the definition of the obstruction DY.

[107] Editor's note: The original has been corrected by adding the word "flat".

Since Y_0 and X_0 are S_0-groups, it is immediate that $\mathscr{N}_{Y_0/X_0}$ is invariant under, say, left translations by Y_0. Hence [108] it is the inverse image of $\mathbf{n}_{Y_0/X_0}$ under $Y_0 \to S_0$; that is,

$$\mathscr{N}_{Y_0/X_0} = \mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}. \tag{4.16.2}$$

Taking (4.16.1) and (4.16.2) into account, we deduce first from 4.5 that the set of subschemes Y of X, flat over S and lifting $Y_{\mathscr{J}}$, is either empty or principal homogeneous under

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}). \tag{4.16.3}$$

On the other hand, 4.8(i) shows that, for every such Y and every S-morphism $f : T \to X$ such that $f_{\mathscr{J}} : T_{\mathscr{J}} \to X_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, the obstruction $c(X, Y, f)$ to f factoring through Y belongs to

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J}\mathscr{O}_T).$$

If T is also flat over S, this last group is equal to

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}).$$

This leads us to introduce the following group functor N_0.

Definition 4.16.1. *Let N_0 be the S_0-functor in commutative groups defined, for every $Z \in \mathrm{Ob}(\mathbf{Sch})_{/S_0}$, by*

$$\mathrm{Hom}_{S_0}(Z, N_0) = \mathrm{Hom}_{\mathscr{O}_Z}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_Z, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_Z). \tag{*}$$

Then the set of subschemes Y of X, flat over S and lifting $Y_{\mathscr{J}}$, is either empty or principal homogeneous under

$$\mathrm{Hom}_{S_0}(Y_0, N_0) = C^1(Y_0, N_0).$$

For every such Y, consider the diagram

$$\begin{array}{ccc} Y \times_S Y & & Y \\ {\scriptstyle (i,i)}\downarrow & & \downarrow{\scriptstyle i} \\ X \times_S X & \xrightarrow{\qquad\mu\qquad} & X \end{array}$$

and let

$$DY = c\big(X, Y, \mu \circ (i, i)\big)$$

be the obstruction to $\mu \circ (i, i)$ factoring through Y, that is, to Y being stable under the group law of X. By what precedes, DY is an element of

$$N_0(Y_0 \times_{S_0} Y_0) = C^2(Y_0, N_0).$$

Lemma 4.17.[109] *Let X be an S-group smooth over S, and let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. For every subscheme Y of X, flat over S and lifting $Y_{\mathscr{J}}$, consider the obstruction defined in 4.16.1:*

$$DY \in \mathrm{Hom}_{S_0}(Y_0 \times_{S_0} Y_0, N_0) = C^2(Y_0, N_0).$$

(i) *For Y to be a sub-S-group of X, it is necessary and sufficient that $DY = 0$.*

[108] Editor's note: See 4.25 below.

[109] Editor's note: Sections 4.17 and 4.18 have been modified to take account of the additions made in 4.16.

(ii) *If Y_0 is made to act on N_0, by functoriality, through the inner automorphisms of Y_0, then $DY \in Z^2(Y_0, N_0)$.*

(iii) *If Y and Y' are two subschemes of X, flat over S, lifting $Y_{\mathscr{J}}$ (so that the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ is defined; cf. 4.5.1), then*

$$DY' - DY = \partial^1 d(Y, Y').$$

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We prove these assertions in succession.

4.18. — Proof of 4.17(i). By definition, $DY = 0$ if and only if Y is stable under the group law of X. Thus $DY = 0$ if Y is a subgroup of X. Conversely, if $DY = 0$, then Y is endowed with the induced law μ_Y, which is associative and reduces modulo $\mathscr{J}$ to the group law on $X_{\mathscr{J}}$. Since Y is flat and locally of finite presentation over S, it follows from 3.3 that μ_Y is a group law.

4.19. — Proof of 4.17(ii). We compare the two values of $u = c(X, Y, \mu^2 \circ (i, i, i))$ computed in the following two diagrams (D_j), $j = 1, 2$:

$$(D_j) \qquad \begin{array}{ccccc} Y \times_S Y \times_S Y & & Y \times_S Y & & Y \\ \downarrow{\scriptstyle (i,i,i)} & & \downarrow{\scriptstyle (i,i)} & & \downarrow{\scriptstyle i} \\ X \times_S X \times_S X & \xrightarrow{f_j} & X \times_S X & \xrightarrow{\mu} & X \end{array}$$

Here $f_1 = (1, \pi)$, $f_2 = (\pi, 1)$, and μ^2 denotes the morphism

$$\mu \circ f_1 = \mu \circ f_2 : X \times_S X \times_S X \longrightarrow X.$$

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$$\mu_Y = \mu \circ (i, i), \qquad f_j^Y = f_j \circ (i, i, i), \qquad \mu^{2,Y} = \mu^2 \circ (i, i, i).$$

For $j = 1, 2$, let a_j and b_j be the morphisms

$$a_j = a_{\mu_0}((f_j^Y)_0), \qquad b_j = b_{(f_j^Y)_0},$$

associated by Lemma 4.12 with the pair of morphisms (f_j^Y, μ). Thus

$$\begin{cases} a_j : \mathrm{Hom}_{\mathscr{O}_{Y_0^3}}\left((f_j^Y)_0^* \mathscr{N}_{Y_0 \times Y_0 / X_0 \times X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\right) \longrightarrow \\ \qquad\qquad \mathrm{Hom}_{\mathscr{O}_{Y_0^3}}\left((\mu^{2,Y})_0^* \mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\right), \\ b_j : \mathrm{Hom}_{\mathscr{O}_{Y_0^2}}\left((\mu_Y)_0^* \mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^2}\right) \longrightarrow \\ \qquad\qquad \mathrm{Hom}_{\mathscr{O}_{Y_0^3}}\left((\mu^{2,Y})_0^* \mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\right). \end{cases} \tag{†}$$

Since

$$\mathscr{N}_{Y_0 \times Y_0 / X_0 \times X_0} \simeq \mathrm{pr}_1^* \mathscr{N}_{Y_0/X_0} \oplus \mathrm{pr}_2^* \mathscr{N}_{Y_0/X_0}$$

[110] Editor's note: The original has $DY' - DY = -\partial d(Y, Y')$, but the operator ∂ there is the negative of the differential ∂^1 defined in I, 5.1.

[111] Editor's note: The notation has been slightly modified, and the beginning of the argument has been expanded.

(because X_0 and Y_0 are flat over S_0), and

$$\mathscr{N}_{Y_0/X_0} \simeq \mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0},$$

we have

$$(f_j^Y)_0^* \mathscr{N}_{Y_0\times Y_0/X_0\times X_0} \simeq (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathscr{O}_{Y_0^3},$$

and likewise

$$(\mu^{2,Y})_0^* \mathscr{N}_{Y_0/X_0} \simeq \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^3},$$
$$(\mu_Y)_0^* \mathscr{N}_{Y_0/X_0} \simeq \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^2}.$$

Moreover, since Y_0^2 and Y_0^3 are flat over S_0, (†) may be rewritten as

$$\begin{cases} a_j : \mathrm{Hom}_{S_0}(Y_0^3, N_0 \oplus N_0) \longrightarrow \mathrm{Hom}_{S_0}(Y_0^3, N_0), \\ b_j : \mathrm{Hom}_{S_0}(Y_0^2, N_0) \longrightarrow \mathrm{Hom}_{S_0}(Y_0^3, N_0). \end{cases} \tag{‡}$$

Applying 4.13 twice to $c(X, Y, \mu^{2,Y}) = u$, we obtain

$$a_1\big(c(X^2, Y^2, f_1)\big) + b_1\big(c(X, Y, \mu_Y)\big) = u$$
$$= a_2\big(c(X^2, Y^2, f_2)\big) + b_2\big(c(X, Y, \mu_Y)\big).$$

Now $c(X, Y, \mu_Y) = DY$, and, since $f_1 = (1, \mu)$ and $f_2 = (\mu, 1)$, we have, with the evident notation,

$$c(X^2, Y^2, f_1) = (0, DY), \qquad c(X^2, Y^2, f_2) = (DY, 0).$$

Consequently,

$$u = a_1((0, DY)) + b_1(DY) = a_2((DY, 0)) + b_2(DY).$$

The first point to note is that b_j is simply $\mathrm{Hom}_{S_0}((f_j^Y)_0, N_0)$, that is, the morphism deduced from $(f_j^Y)_0$ by functoriality. The preceding identity therefore becomes

$$a_1((0, DY)) - \mathrm{Hom}((\mu, 1), N_0)(DY) + \mathrm{Hom}((1, \mu), N_0)(DY)$$
$$- a_2((DY, 0)) = 0.$$

The two middle terms are the $DY(xy, z)$ and $DY(x, yz)$ parts of the formula for the 2-coboundary. It remains only to identify the other two terms.

We first compute the map a_j. It is obtained, by pullback along $(f_j^Y)_0$, from the morphism of $\mathscr{O}_{Y_0^2}$-modules

$$P : \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^2} \longrightarrow (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathscr{O}_{Y_0^2}$$

induced by multiplication in Y_0. Consider the vector bundle $V = \mathbf{V}(\mathbf{n}_{Y_0/X_0})$. By duality, P gives

$$\mathbf{V}(P) : V \times_{S_0} V \times_{S_0} Y_0 \times_{S_0} Y_0 \longrightarrow V \times_{S_0} Y_0 \times_{S_0} Y_0,$$

which is expressed set-theoretically by

$$\mathbf{V}(P)(u, v, a, b) = (u + \mathrm{Ad}(a)v, ab, b).$$

[112] This is proved exactly as is the corresponding fact for Lie algebras, that is, for the module $\omega^1_{Y_0/S_0}$. First observe that, by functoriality in Y_0, V is endowed with a group structure in the category of vector bundles over S_0. By the lemma already used for Lie algebras (Exposé II,

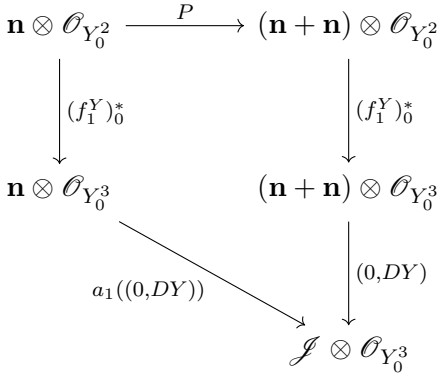
[112] Editor's note: The entries a, b have been replaced by ab, b to show that $\mathbf{V}(P)$ is obtained, by pullback to Y_0^2, from the multiplication morphism $V_{Y_0} \times_{S_0} V_{Y_0} \to V_{Y_0}$.

3.10), this structure coincides with the group structure underlying its $\mathscr{O}_S$-module structure. Next,

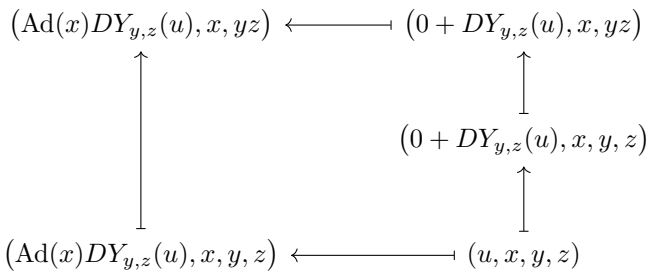
$$\mathbf{V}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}) = \mathbf{V}(\mathscr{N}_{Y_0/X_0})$$

is itself endowed with an S_0-group structure, namely the semidirect product of the structure on V by that on Y_0. It remains only to identify the operations of Y_0 on V to obtain the desired formula.

We now compute the two remaining terms. Consider first $a_1((0, DY))$. It is computed by the following diagram, in which $\mathbf{n}$ denotes $\mathbf{n}_{Y_0/X_0}$:



Regard the vector bundles defined by these modules as schemes over S_0, and take points with values in an arbitrary scheme. If (u, x, y, z) denotes a point of $\mathbf{V}(\mathscr{J}) \times Y_0^3$, we obtain the following diagram:



Thus

$$a_1((0, DY))(x, y, z) = \mathrm{Ad}(x)DY(y, z),$$

which is indeed the first term of the coboundary. Likewise, $a_2((DY, 0))(x, y, z) = DY(x, y)$, and therefore

$$\begin{aligned} 0 = {} & \mathrm{Ad}(x)DY(y, z) - DY(xy, z) + DY(x, yz) - DY(x, y) \\ = {} & (\partial^2 DY)(x, y, z). \end{aligned}$$

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4.20. — Proof of 4.17(iii). [114] We compare the two values of $v = c(X, Y, \mu \circ (i', i'))$ computed in the following two diagrams:

[113] Editor's note: The signs have been changed to make them compatible with I, 5.1.
[114] Editor's note: The argument that follows has been expanded from the original.

$$(*)\qquad \begin{array}{ccccc} & & Y' & & Y \\ & & \downarrow{\scriptstyle i'} & & \downarrow{\scriptstyle i} \\ Y' \times_S Y' & \xrightarrow{\mu\circ(i',i')} & X & = & X \end{array}$$

$$(\dagger)\qquad \begin{array}{ccccc} & & Y \times_S Y & & Y \\ & & \downarrow{\scriptstyle (i,i)} & & \downarrow{\scriptstyle i} \\ Y' \times_S Y' & \xrightarrow{(i',i')} & X \times_S X & \xrightarrow{\mu} & X \end{array}$$

Let $f = \mu \circ (i', i')$. Diagram $(*)$ gives

$$v = DY' + f_0^*(c(X, Y, i')). \tag{1}$$

Now $Y_0' = Y_0$, and f_0 is the multiplication $Y_0^2 \to Y_0$; hence

$$f_0^*(c(X, Y, i'))(x_0, y_0) = c(X, Y, i')(x_0 y_0). \tag{2}$$

Put $c = c(X, Y, i')$. Under the identification $N_0' \simeq N_0 \oplus N_0$, the element

$$c(X \times_S X, Y \times_S Y, (i', i'))$$

identifies with the pair (c, c). Thus, putting $h = (i', i')$, diagram $(\dagger)$ gives

$$v = h_0^*(DY) + a_{\mu_0}(c, c). \tag{3}$$

Since h_0 is the identity map on Y_0^2, we have $h_0^*(DY) = DY$. Finally, by the preceding calculation of a_{μ_0}, for every $S' \to S$ and every $x_0, y_0 \in Y_0(S_0')$,

$$a_{\mu_0}(c, c)(x_0, y_0) = c(x_0) + \mathrm{Ad}(x_0)(c(y_0)). \tag{4}$$

It follows that

$$\begin{aligned} (DY' - DY)(x_0, y_0) = {} & \mathrm{Ad}(x_0)(c(X, Y, i')(y_0)) - c(X, Y, i')(x_0 y_0) + c(X, Y, i')(x_0) \\ = {} & (\partial^1 c(X, Y, i'))(x_0, y_0). \end{aligned} \tag{5}$$

Since $c(X, Y, i') = d(Y, Y')$ (cf. 4.8(iii)), this proves that

$$DY' - DY = \partial^1 d(Y, Y').$$

Theorem 4.21. *Let S be a scheme and let $\mathscr{I}$ and $\mathscr{J}$ be two ideals*[115] *on S, such that $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let X be an S-group smooth over S, and let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. Consider the S_0-functor of commutative groups N_0 defined by*

$$\mathrm{Hom}_{S_0}(T, N_0) = \mathrm{Hom}_{\mathscr{O}_T}\left(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right), \qquad T \in \mathrm{Ob}(\mathbf{Sch}/S_0),$$

on which Y_0 acts through the inner automorphisms of X_0.

[115] Editor's note: The hypothesis that $\mathscr{I}$ be nilpotent, which appears superfluous, has been removed.

(i) *For there to exist a sub-S-group of X, flat over S, which reduces to $Y_{\mathscr{J}}$, it is necessary and sufficient that the following two conditions hold:*

(i_1) *There exists a subscheme Y of X, flat over S, lifting $Y_{\mathscr{J}}$. This condition is automatically satisfied when Y_0 is affine (cf. 4.6.5).*

(i_2) *A certain canonical obstruction, an element of $H^2(Y_0, N_0)$, vanishes.*

(ii) *If the conditions in (i) hold, the set of sub-S-groups Y of X, flat over S and reducing to $Y_{\mathscr{J}}$, is a principal homogeneous set under $Z^1(Y_0, N_0)$.*[116]

Indeed, condition (i_1) is necessary. Suppose that it holds, and let Y, flat over S, lift $Y_{\mathscr{J}}$. We must find a Y', also flat over S and lifting $Y_{\mathscr{J}}$, such that $DY' = 0$,[117] cf. 4.17(i). By 4.17(iii), this amounts to finding an element $d(Y', Y) \in C^1(Y_0, N_0)$ such that

$$DY = \partial^1 d(Y', Y).$$

[118] Let $c \in H^2(Y_0, N_0)$ be the class represented by DY, which is a cocycle by 4.17(ii). By 4.17(iii), this class does not depend on the choice of Y, and its vanishing is necessary and sufficient for the existence of $d(Y', Y)$ satisfying the preceding equation. This proves (i).

If Y has now been chosen so that $DY = 0$, the equation to be solved becomes $\partial^1 d(Y', Y) = 0$, which proves (ii).

Remark 4.22. Retain the notation of 4.21. By 4.15, Y_0 is locally a complete intersection in X_0; hence $\mathscr{N}_{Y_0/X_0}$ is a finite locally free $\mathscr{O}_{Y_0}$-module, and therefore

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathscr{N}_{Y_0/X_0})$$

is a finite locally free $\mathscr{O}_{S_0}$-module. Thus, writing

$$\mathbf{n}^\vee_{Y_0/X_0} = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{O}_{Y_0}),$$

we have

$$\begin{aligned}\mathrm{Hom}_{\mathscr{O}_T}\left(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right)\\ \simeq \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\end{aligned}$$

for every $T \to S_0$.[119] Consequently, the S_0-functor N_0 is isomorphic to

$$\mathbf{W}\left(\mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\right) \simeq \mathbf{W}\left(\mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\right).$$

It follows that there are isomorphisms[120]

$$\begin{aligned}H^2(Y_0, N_0) &\simeq H^2\left(Y_0, \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\right)\\ &\simeq H^2\left(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\right),\\ Z^1(Y_0, N_0) &\simeq Z^1\left(Y_0, \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\right)\\ &\simeq Z^1\left(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\right).\end{aligned}$$

[116] Editor's note: The question whether this set, modulo conjugation by the elements $x \in X(S)$ inducing the identity element of $X(S_{\mathscr{J}})$, is a principal homogeneous set under $H^1(Y_0, N_0)$, occupies 4.23–4.36.

[117] Editor's note: The expression $\partial DY'$ in the original has been corrected to DY'.

[118] Editor's note: See the editor's note to 4.17(iii).

[119] Editor's note: The preceding argument has been expanded. It shows that the following isomorphism is valid without a flatness hypothesis. On the other hand, since 4.16 the discussion has been restricted to S-schemes $f : T \to S$ flat over S, in order to ensure that the group

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\left(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J}\mathscr{O}_T\right),$$

in which the obstruction $c(X, Y, f)$ lies, coincides with $N_0(T_0)$; see the end of 4.16.

[120] Editor's note: With the notation of I, 5.3, and assuming that Y_0 is affine over S_0.

4.23. — Still under the hypotheses of 4.21, we shall now study how the set of subgroups Y lifting $Y_{\mathscr{J}}$ behaves under conjugation by sections of X. If x is a section of X over S inducing the unit section of $X_{\mathscr{J}}$, the inner automorphism $\mathrm{Int}(x)$ defined by x sends flat subgroups of X lifting $Y_{\mathscr{J}}$ to flat subgroups of X lifting $Y_{\mathscr{J}}$. Under the conditions of 4.21(ii), the set of these subgroups is a principal homogeneous space under $Z^1(Y_0, N_0)$. We shall see that there is then a subgroup Γ of $B^1(Y_0, N_0)$[121] such that two subgroups of X, flat over S and lifting $Y_{\mathscr{J}}$, are conjugate by an $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$ if and only if their *difference* in $Z^1(Y_0, N_0)$ belongs to Γ. In the best cases we shall show that $\Gamma = B^1(Y_0, N_0)$. Thus the set of flat subgroups of X lifting $Y_{\mathscr{J}}$, modulo conjugation by the elements $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$, is either empty or a principal homogeneous space under $H^1(Y_0, N_0)$ (cf. 4.29 and 4.36).

4.24. — We retain the notation of 4.21. Let Y be a flat subgroup of X whose reduction is $Y_{\mathscr{J}}$. Recall that in 0.5 we introduced the functor L_0^X (respectively L_0^Y) defined, for a variable S_0-scheme T, by the identity

$$\mathrm{Hom}_{S_0}(T, L_0^X) = \mathrm{Hom}_{\mathscr{O}_T}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right),$$

and similarly with Y in place of X, as well as the functor

$$L'_X = \prod_{S_0/S} L_0^X.$$

We then have:

Lemma 4.25. *There is a canonical exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules*

$$\mathbf{n}_{Y_0/X_0} \xrightarrow{d} \omega^1_{X_0/S_0} \longrightarrow \omega^1_{Y_0/S_0} \longrightarrow 0 \qquad (+)$$

having the following properties:

(i) *After pullback to Y_0, d gives the morphism $\overline{D}_0$ of 4.8(iii).*

(ii) *If X_0 and Y_0 are smooth over S_0, then d is injective. Since the two modules of differentials are then locally free of finite type, the same is true of $\mathbf{n}_{Y_0/X_0}$, and the sequence is locally split.*

Proof.[122] Let $\pi_0 : Y_0 \to S_0$ denote the structure morphism. By SGA 1 II, formula (4.3) (see also EGA IV_4, 16.4.21), there is a canonical exact sequence of $\mathscr{O}_{Y_0}$-modules

$$\mathscr{N}_{Y_0/X_0} \xrightarrow{\overline{D}_0} \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{Y_0/S_0} \longrightarrow 0. \qquad (\dagger)$$

Since this sequence consists of $(Y_0 \times_S Y_0)$-equivariant modules and morphisms, its pullback $(+)$ by ε_0^* is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules, and $(\dagger)$ is the pullback of $(+)$ by π_0^* (cf. Exposé I, Section 6.8). This proves (i).

Suppose in addition that X_0 and Y_0 are smooth over S_0. Then, by SGA 1 II, 4.10 (see also EGA IV_4, 17.2.3(i) and 17.2.5), $\overline{D}_0$ is injective and the sequence $(\dagger)$ consists of $\mathscr{O}_{Y_0}$-modules locally free of finite type (and is therefore locally split). By the equivalence of categories I, 6.8.1, d is also injective, and consequently the sequence $(+)$ has the asserted properties.

[121] Editor's note: $Z^1(Y_0, N_0)$ has been replaced by $B^1(Y_0, N_0)$, since the proof shows that Γ is a subgroup of $B^1(Y_0, N_0)$; cf. 4.27–4.29.

[122] Editor's note: The proof has been expanded to take account of the additions made in Exposé I, Section 6.8.

4.26. —[123] For every S_0-scheme $f : T \to S_0$, the sequence (+) gives an exact sequence of $Y_0(T)$-$\mathscr{O}(T)$-modules

$$0 \longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\omega^1_{Y_0/S_0}, f^*\mathscr{J}\right) \longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\omega^1_{X_0/S_0}, f^*\mathscr{J}\right) \longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\mathbf{n}_{Y_0/X_0}, f^*\mathscr{J}\right).$$

Thus there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow L_0^Y \longrightarrow L_0^X \xrightarrow{d} N_0. \tag{4.26.1}$$

It gives an exact sequence of complexes of abelian groups

$$0 \longrightarrow C^*(Y_0, L_0^Y) \longrightarrow C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0),$$

and, in particular, the following commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^0(Y_0, L_{0Y}) & \longrightarrow & C^0(Y_0, L_{0X}) & \xrightarrow{d^0} & C^0(Y_0, N_0) \\ & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} \\ 0 & \longrightarrow & C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \xrightarrow{d^1} & C^1(Y_0, N_0) \end{array}$$

Observe that $C^0(Y_0, L_0^Y)$ (respectively $C^0(Y_0, L_0^X)$) is just

$$\mathrm{Hom}_{S_0}(S_0, L_0^Y) = \mathrm{Hom}_S(S, L'_Y)$$

(respectively the analogous group with X in place of Y); by 0.9, it is the group of sections of Y (respectively X) over S inducing the unit section of $X_{\mathscr{J}}$. Also, d^1 is precisely the morphism $v_{i_{Y_0}}$ of 4.8(iii), where $i_{Y_0} : Y_0 \to X_0$ is the canonical immersion.[124]

Lemma 4.27. *Under the conditions of 4.21 for S, $\mathscr{I}$, $\mathscr{J}$, and X, let Y be a subgroup of X, flat over S and lifting $Y_{\mathscr{J}}$. Let $i : Y \hookrightarrow X$ denote the canonical immersion.*[125]

(i) *Let $i' : Y \to X$ be a morphism of S-schemes lifting i_0 (so that i' is also an immersion), let $Y' = i'(Y)$, and let $d(i, i')$ be the element of $C^1(Y_0, L_0^X)$ such that $i' = d(i, i') \cdot i$ (cf. 1.2.4). Then the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by*

$$d(Y, Y') = d^1\big(d(i, i')\big).$$

(ii) *Let $x \in C^0(Y_0, L_0^X)$ be a section of X over S inducing the unit section of $X_{\mathscr{J}}$ over $S_{\mathscr{J}}$. Then the deviation $d(Y, \mathrm{Int}(x)Y) \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by*

$$-d\big(Y, \mathrm{Int}(x)Y\big) = d^1\partial x = \partial d^0 x.$$

Indeed, Y' is the image of Y under the composite morphism [126]

$$Y \xrightarrow{(d(i,i'),i)} L'_X \times_S X \longrightarrow X,$$

[123] Editor's note: The original has been expanded to make clear that this is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules.

[124] Editor's note: This follows from the definition of $d : \mathbf{n}_{Y_0/X_0} \to \Omega^1_{X_0/S_0}$ (cf. 4.25) and from the definition of $v_{i_{Y_0}}$ (cf. 4.8).

[125] Editor's note: Part (i) below has been added; it will be useful in 4.35.1 and then in 4.38(4) and (5).

[126] Editor's note: Recall that $L'_X = \prod_{S_0/S} L_0^X$.

which is denoted $d(i,i')\cdot i$ in 4.8(iii). By loc. cit. and the equality $v_{i_0} = d^1$, we therefore have

$$\begin{aligned} &c(X,Y',d(i,i')\cdot i) - c(X,Y',i) \\ &\quad = v_{i_0}(d(i,i')) = d^1(d(i,i')). \end{aligned}$$

But $d(i,i')\cdot i = i'$ factors through Y' by definition, so the first term is zero. Moreover, by 4.8(ii),

$$c(X,Y',i) = d(Y',Y) = -d(Y,Y').$$

Hence $d(Y,Y') = d^1(d(i,i'))$, which proves (i).

Now let x be as in (ii). From the formula

$$xyx^{-1} = xyx^{-1}y^{-1}y = (x - \mathrm{Ad}(y)x)y = (-\partial x)(y)\cdot y,$$

we see that Y' is the image of Y under the immersion $i' = (-\partial x)\cdot i_Y$. Part (i) therefore gives

$$-d(Y, \mathrm{Int}(x)Y) = d^1\partial x = \partial d^0 x.$$

Corollary 4.28. *Two subgroups Y and Y' of X, flat over S and lifting $Y_{\mathscr{J}}$, are conjugate by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ if and only if*

$$d(Y,Y') \in \partial d^0 C^0(Y_0, L_0^X) \subset \partial C^0(Y_0, N_0) = B^1(Y_0, N_0).$$

Corollary 4.29. *If d^0 is surjective, then Y and Y' as above are conjugate by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ if and only if $d(Y,Y') \in B^1(Y_0,N_0)$.*

Corollary 4.30. *Let Y be as in 4.27. The set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$d^1\partial C^0(Y_0, L_0^X) = C^0(Y_0, L_0^X)/\ker(d^1\partial).$$

Observe now that $C^0(Y_0, L_0^X)/\ker(d^1\partial)$ is computed using only the left-hand square of the commutative diagram in 4.26. In particular, it can also be computed in any diagram of the same type having the same left-hand square. Consider the functor L_0^X/L_0^Y over S_0, defined by

$$\mathrm{Hom}_{S_0}(T, L_0^X/L_0^Y) = \mathrm{Hom}_{S_0}(T, L_0^X)/\mathrm{Hom}_{S_0}(T, L_0^Y).$$

There is a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C^0(Y_0, L_{0Y}) & \longrightarrow & C^0(Y_0, L_{0X}) & \longrightarrow & C^0(Y_0, L_{0X}/L_{0Y}) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} & & \\ 0 & \longrightarrow & C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \longrightarrow & C^1(Y_0, L_{0X}/L_{0Y}) & \longrightarrow & 0 \end{array}$$

whence, by the preceding observation:

Corollary 4.31. *Let Y be as in 4.27. The set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$E = \partial C^0(Y_0, L_0^X/L_0^Y) = C^0(Y_0, L_0^X/L_0^Y)/H^0(Y_0, L_0^X/L_0^Y).$$

Corollary 4.32. *Suppose in addition that S_0 is affine and that $\omega^1_{Y_0/S_0}$ is finite locally free.*[127] *If*

$$\mathscr{F}_0 = \left(\mathrm{Lie}(X_0/S_0)/\mathrm{Lie}(Y_0/S_0)\right) \otimes_{\mathscr{O}_{S_0}} \mathscr{J},$$

then $E = \Gamma(S_0, \mathscr{F}_0)/H^0(Y_0, \mathscr{F}_0)$.

Indeed,[128] since $\omega^1_{Y_0/S_0}$ is finite locally free, as is $\omega^1_{X_0/S_0}$ (because X is assumed smooth

[127] Editor's note: $\mathrm{Lie}(Y_0/S_0)$ has been corrected to $\omega^1_{Y_0/S_0}$.

[128] Editor's note: The original has been expanded in what follows.

over S), 0.6 gives

$$L_0^Y = \mathbf{W}(\mathrm{Lie}(Y_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}), \qquad L_0^X = \mathbf{W}(\mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}).$$

On the other hand, by 4.25 there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow \mathscr{K} \longrightarrow \omega^1_{X_0/S_0} \xrightarrow{\varphi} \omega^1_{Y_0/S_0} \longrightarrow 0, \qquad \mathscr{K} = \ker(\varphi).$$

Since $\omega^1_{Y_0/S_0}$ and $\omega^1_{X_0/S_0}$ are finite locally free, there is a locally split exact sequence

$$0 \longrightarrow \mathrm{Lie}(Y_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \longrightarrow \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \longrightarrow \mathscr{F}_0 \longrightarrow 0.$$

It follows that there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow L_0^Y \longrightarrow L_0^X \longrightarrow \mathbf{W}(\mathscr{F}_0).$$

By the argument used to prove 4.31, we may compute E as the image of the composite map

$$C^0(Y_0, L_0^X) \xrightarrow{\pi} C^0(Y_0, \mathbf{W}(\mathscr{F}_0)) \xrightarrow{\partial} C^1(Y_0, \mathbf{W}(\mathscr{F}_0)).$$

Now π is the map

$$\Gamma(S_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}) \longrightarrow \Gamma(S_0, \mathscr{F}_0).$$

Since S_0 is affine, π is surjective, and this proves the assertion.

Corollary 4.33. *Let $S, \mathscr{I}, \mathscr{J}$, and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose that $\omega^1_{Y_0/S_0}$ is finite locally free and that S_0 is affine.*[129] *The set of subgroups of X conjugate to Y by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$E = \Gamma\left(S_0, \frac{\mathrm{Lie}(X_0/S_0)}{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}} \otimes_{\Gamma(S_0, \mathscr{O}_{S_0})} \Gamma(S_0, \mathscr{J})\right),$$

that is, to $L_0^X(Y_0)/H^0(Y_0, L_0^X)$.[130]

Indeed, by I, 4.7.3 (cf. 2.13), write

$$\mathrm{Lie}(X_0/S_0) = \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)} \oplus \mathscr{R}.$$

Since Y_0 is commutative,

$$\mathrm{Lie}(Y_0/S_0) \subset \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)},$$

and hence

$$\mathscr{F}_0 = \left(\frac{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}}{\mathrm{Lie}(Y_0/S_0)}\right) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \ \oplus \ \mathscr{R} \otimes_{\mathscr{O}_{S_0}} \mathscr{J},$$

$$\mathscr{F}_0^{\mathrm{ad}(Y_0)} = \left(\frac{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}}{\mathrm{Lie}(Y_0/S_0)}\right) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}.$$

By 4.32, therefore, $E \simeq \Gamma(S_0, \mathscr{R} \otimes_{\mathscr{O}_{S_0}} \mathscr{J})$. Returning to the definition of $\mathscr{R}$ completes the proof.

Corollary 4.34. *Let $S, \mathscr{I}, \mathscr{J}$, and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose that $\omega^1_{Y_0/S_0}$ is finite locally free and that S_0 is affine.*[131] *If $x \in X(S)$ induces the unit section of $X_{\mathscr{J}}$ and normalizes Y, then it centralizes Y.*

[129] Editor's note: The hypothesis on $\omega^1_{Y_0/S_0}$ has been added, and the hypothesis "S affine" has been replaced by "S_0 affine."

[130] Editor's note: What follows has been added; cf. 4.34.

[131] Editor's note: Cf. Editor's Note 129.

This follows immediately by comparing the preceding corollary with 2.14. Indeed, 4.33 shows that the elements of $C^0(Y_0, L_0^X)$ which globally preserve Y are the elements of $H^0(Y_0, L_0^X)$, and 2.14 shows that these are precisely the elements which act trivially on the canonical immersion $Y \longrightarrow X$.

4.35. — Return to the general situation of 4.21 and suppose that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$. By 4.25(ii), there is then an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow \mathrm{Lie}(Y_0/S_0) \longrightarrow \mathrm{Lie}(X_0/S_0) \longrightarrow \mathbf{n}^{\vee}_{Y_0/X_0} \longrightarrow 0, \qquad (*)$$

and these are finite locally free $\mathscr{O}_{S_0}$-modules.

On the other hand, by SGA 1, II, 4.10, every subscheme Y of X lifting $Y_{\mathscr{J}}$ and flat over S is smooth over S.[132] Suppose in addition that S_0 and $Y_{\mathscr{J}}$ are affine. Since $\mathbf{n}^{\vee}_{Y_0/X_0}$ is a locally free $\mathscr{O}_{S_0}$-module, the sequence $(*)$ remains exact after tensoring with $\mathscr{J}$ over $\mathscr{O}_{S_0}$, and then pulling back to Y_0^n. Since the Y_0^n are affine, this gives an exact sequence of complexes of abelian groups

$$0 \longrightarrow C^*(Y_0, L_0^Y) \longrightarrow C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0) \longrightarrow 0,$$

and, in particular, a commutative diagram with exact rows:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \xrightarrow{d^1} & C^1(Y_0, N_0) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle \partial} & & \\
0 & \longrightarrow & C^2(Y_0, L_{0Y}) & \longrightarrow & C^2(Y_0, L_{0X}) & \xrightarrow{d^2} & C^2(Y_0, N_0) & \longrightarrow & 0
\end{array}$$

Let now Y and Y' be two subgroups of X lifting $Y_{\mathscr{J}}$ and flat, hence smooth, over S. Since $Y_{\mathscr{J}}$ is affine, 0.15 shows that Y and Y' are isomorphic as schemes lifting $Y_{\mathscr{J}}$; that is, there is an isomorphism of S-schemes

$$f : Y \xrightarrow{\sim} Y'$$

inducing the identity on $Y_{\mathscr{J}}$. On the one hand, by 1.2.4, f defines an element $a \in C^1(Y_0, L_0^X)$ such that

$$f(y) = a(y_0)y$$

for every $y \in Y(S')$, $S' \to S$, and by 4.27(i),

$$d^1(a) = d(Y, Y').$$

Moreover, because Y and Y' are subgroups of X, the last element belongs to $Z^1(Y_0, N_0)$ (cf. 4.21). Thus ∂a is an element of $Z^2(Y_0, L_0^Y)$, whose image $\overline{\partial a}$ in $H^2(Y_0, L_0^Y)$ depends only on the class $\overline{d(Y, Y')} \in H^1(Y_0, N_0)$. This is precisely the definition of the connecting map

$$\partial^1 : H^1(Y_0, N_0) \longrightarrow H^2(Y_0, L_0^Y),$$

so that

$$\partial^1\big(\overline{d(Y, Y')}\big) = \overline{\partial a}.$$

On the other hand, transport the group structure of Y' by f, and let Y_1 be the resulting group, whose underlying scheme is therefore Y. Thus its group law μ_1 is defined, for every $S' \to S$ and $x, y \in Y(S')$, by

$$\mu_1(x, y) = f^{-1}\big(f(x)f(y)\big).$$

[132] Editor's note: What follows, together with Proposition 4.35.1, has been added; it is implicit in the original. Cf. 4.38(5).

By 3.5.1, Y_1 defines a cocycle

$$\delta(Y, Y_1) \in Z^2\big(Y_0, \mathrm{Lie}(Y_0/S_0)\big)$$

such that, for every $S' \to S$ and $x, y \in Y(S')$,

$$\delta(Y, Y_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}\big(f(x)f(y)\big).$$

Put $b = \delta(Y, Y_1)$. For every $S' \to S$ and $x, y \in Y(S')$, we have $(b(x_0, y_0)xy)_0 = x_0y_0$. Hence $f(b(x_0, y_0)xy)$ is equal, on the one hand, to

$$a(x_0y_0)b(x_0, y_0)xy,$$

and, on the other hand, to

$$f(x)f(y) = a(x_0)x\,a(y_0)y = a(x_0)\,\mathrm{Ad}(x_0)\big(a(y_0)\big)xy.$$

Comparing these two expressions, we find that $b(x_0, y_0)$ equals

$$\begin{aligned} &a(x_0y_0)^{-1}a(x_0)\,\mathrm{Ad}(x_0)\big(a(y_0)\big) \\ &\quad = \mathrm{Ad}(x_0)\big(a(y_0)\big) - a(x_0y_0) + a(x_0) = (\partial a)(x_0, y_0). \end{aligned}$$

Thus $\delta(Y, Y_1) = \partial a$, and we have proved:

Proposition 4.35.1.[132] *Under the hypotheses of 4.21, suppose in addition that S_0 is affine and that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$ and affine. Let Y, Y' be two subgroups of X lifting $Y_{\mathscr{J}}$ and flat (hence smooth) over S. Let $f : Y \xrightarrow{\sim} Y'$ be an isomorphism of S-schemes inducing the identity on $Y_{\mathscr{J}}$, and let Y_1 denote the group obtained by transporting the group structure of Y' by f. Then*

$$\partial^1\big(\overline{d(Y, Y')}\big) = \overline{\delta(Y, Y_1)}.$$

Proposition 4.36. *Under the hypotheses of 4.21, suppose in addition that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$ and that S_0 is affine. The set of sub-S-groups Y of X, flat (or smooth) over S, reducing to $Y_{\mathscr{J}}$, modulo conjugation by sections of X over S inducing the unit section of $X_{\mathscr{J}}$, is either empty or a principal homogeneous set under the group*

$$H^1\Big(Y_0, \big[\mathrm{Lie}(X_0/S_0)/\,\mathrm{Lie}(Y_0/S_0)\big] \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

It is enough to verify that Corollary 4.29 applies, that is, that

$$d^0 : \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \longrightarrow \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})$$

is surjective. This follows from the fact that the sequence (+) of 4.25(ii) is split, since S_0 is affine. [133]

Let us finally state a corollary common to 4.21 and 4.36. In fact, this will be the only form in which we shall subsequently use the general results of this section. [134]

Corollary 4.37. *Let S be a scheme and let S_0 be the closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Let X be an S-group smooth over S, and let Y_0 be a sub-S_0-group of X_0, flat over S_0.*

[133] Editor's note: This also follows from the proof of 4.32.

[134] Editor's note: For example, 4.37 is used in Exposé IX to prove statements 3.2 bis and 3.6 bis.

(i) *If S_0 is affine, Y_0 is smooth over S_0, and if*

$$H^1\Big(Y_0, [\mathrm{Lie}(X_0/S_0)/\mathrm{Lie}(Y_0/S_0)] \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0$$

for every $n \geq 0$, then any two sub-S-groups of X, flat (or smooth) over S and reducing to Y_0, are conjugate by a section of X over S inducing the unit section of X_0.

(ii) *If Y_0 is affine and of finite presentation, and if*[135]

$$H^2\Big(Y_0, \mathbf{n}^{\vee}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0$$

for every $n \geq 0$, then there exists a sub-S-group of X, flat over S, reducing to Y_0.

4.38. It remains to relate the three constructions made in this Exposé. To avoid inessential complications, we shall place ourselves in the following situation: S_0 is the spectrum of a field k, S is the spectrum of the dual numbers $D(k)$, G is an S-group smooth over S, and K is a sub-S-group, smooth over S and affine. [136]

Put

$$\mathfrak{g}_0 = \mathrm{Lie}\, G_0 = \Gamma(S_0, \mathrm{Lie}\, G_0) = \mathrm{Lie}(G_0/S_0)(S_0), \qquad \mathfrak{k}_0 = \mathrm{Lie}\, K_0.$$

We have an exact sequence of k-vector spaces [137]

$$0 \longrightarrow \mathfrak{k}_0 \xrightarrow{i} \mathfrak{g}_0 \xrightarrow{d} \mathbf{n}^{\vee}_{K_0/G_0} \longrightarrow 0,$$

which gives rise to an exact cohomology sequence

$$\begin{aligned} 0 &\longrightarrow H^0(K_0, \mathfrak{k}_0) \xrightarrow{i^0} H^0(K_0, \mathfrak{g}_0) \xrightarrow{d^0} H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) \\ &\xrightarrow{\partial^0} H^1(K_0, \mathfrak{k}_0) \xrightarrow{i^1} H^1(K_0, \mathfrak{g}_0) \xrightarrow{d^1} H^1(K_0, \mathbf{n}^{\vee}_{K_0/G_0}) \xrightarrow{\partial^1} H^2(K_0, \mathfrak{k}_0). \end{aligned}$$

These various groups all have a geometric meaning.

a) $H^0(K_0, \mathfrak{k}_0) = \mathrm{Lie}\, \underline{\mathrm{Centr}}(K_0)$ [138] by II, 5.2.3.

b) $H^0(K_0, \mathfrak{g}_0) = \mathrm{Lie}\, \underline{\mathrm{Centr}}_{G_0}(K_0)$ [138] (idem).

c) $H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) = \mathrm{Lie}\, \underline{\mathrm{Norm}}_{G_0}(K_0)/\mathfrak{k}_0$ [138] (idem).

d)

$$H^1(K_0, \mathfrak{k}_0) = \mathrm{Lie}\Big(\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)\Big) \,/\, \mathrm{Im}(\mathfrak{k}_0),$$

where $\mathrm{Im}(\mathfrak{k}_0)$ denotes the image of $\mathfrak{k}_0$ under the morphism $\mathrm{Lie}(\mathrm{Int}_0)$ deduced from

$$\mathrm{Int}_0 : K_0 \longrightarrow \underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0).$$

Indeed, it follows from 2.1(ii), applied with $Y = X = K$ and $f_0 = \mathrm{id}_{K_0}$, that $Z^1(K_0, \mathfrak{k}_0)$ is the group of infinitesimal automorphisms of the S_0-group K_0, and that $H^1(K_0, \mathfrak{k}_0)$

[135] Editor's note: We have replaced $\mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{I}^{n+1}/\mathscr{I}^{n+2})$ by $\mathbf{n}^{\vee}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}$, in accordance with Remark 4.22.

[136] Editor's note: We have slightly modified the original in what follows. In particular, we have replaced X by G and Y by K, and denoted their Lie algebras by $\mathfrak{g}_0$ and $\mathfrak{k}_0$. We have also written $H^i(K_0, \cdot)$ explicitly in place of the abbreviation $H^i(\cdot)$ used in the original.

[137] Editor's note: Equipped with the adjoint action of K_0.

[138] Editor's note: Since the formation of centralizers and normalizers commutes with base change (cf. I, 2.3.3.1), we have written $\underline{\mathrm{Centr}}(K_0)$ in place of $\underline{\mathrm{Centr}}(K)_0$ in the original, and likewise $\underline{\mathrm{Centr}}_{G_0}(K_0)$ and $\underline{\mathrm{Norm}}_{G_0}(K_0)$ in place of $\underline{\mathrm{Centr}}_G(K)_0$ and $\underline{\mathrm{Norm}}_G(K)_0$.

is obtained by quotienting by inner infinitesimal automorphisms, that is, by the image of $\mathfrak{k}_0$. Moreover, by II, 4.2.2, we also have

$$Z^1(K_0, \mathfrak{k}_0) = \mathrm{Lie}\Big(\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)\Big).$$

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e) By 2.1(ii), $H^1(K_0, \mathfrak{g}_0)$ is the group of deviations between homomorphisms $K \to G$ extending the canonical immersion $i_0 : K_0 \to G_0$, modulo the deviations obtained through the action of the inner automorphisms of G defined by elements of $G(S)$ which give the unit of $G(S_0)$ (that is, elements of $\mathfrak{g}_0$).

f) By 4.36, $H^1(K_0, \mathbf{n}^{\vee}_{K_0/G_0})$ is the group of deviations between subgroups K' of G extending K_0 and flat over S (hence smooth over S, cf. SGA 1, II, 4.10), modulo the deviations obtained through the action of the inner automorphisms of G constructed as above.

g) By 3.5(ii), $H^2(K_0, \mathfrak{k}_0)$ is the group of deviations between group structures on K extending that of K_0, modulo the S-automorphisms of K inducing the identity on K_0.

We now propose to make explicit the six morphisms of the preceding exact sequence in terms of the geometric interpretation just given.

1) The morphisms i^0 and d^0 are simply those obtained by passing to the Lie algebra (and then, for d^0, to the quotient) in the canonical monomorphisms

$$\underline{\mathrm{Centr}}(K_0) \longrightarrow \underline{\mathrm{Centr}}_{G_0}(K_0) \longrightarrow \underline{\mathrm{Norm}}_{G_0}(K_0).$$

This follows immediately from the definition of the identifications **a)**, **b)**, and **c)**.

2) The morphism ∂^0 is constructed as follows. Let

$$\bar{x} \in \mathrm{Lie}\,\underline{\mathrm{Norm}}_{G_0}(K_0)/\mathfrak{k}_0.$$

Lift it to an element

$$x \in \mathrm{Lie}\,\underline{\mathrm{Norm}}_{G_0}(K_0) \subset \underline{\mathrm{Norm}}_G(K)(S).$$

Then $\mathrm{Int}(x)$ defines an automorphism of K inducing the identity on K_0, hence an element of $\mathrm{Lie}\,\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)$. Denote by $\overline{\mathrm{Int}(x)}$ the image of this element in

$$\mathrm{Lie}\,\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)/\,\mathrm{Im}(\mathfrak{k}_0).$$

Then

$$\partial^0(\bar{x}) = -\overline{\mathrm{Int}(x)} = \overline{\mathrm{Int}(x^{-1})}. \qquad (*)$$

Indeed, let us compute the element of $\mathrm{Lie}\,\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)$ defined by $\mathrm{Int}(x)$. By definition it corresponds to an element $a \in Z^1(K_0, \mathfrak{k}_0)$ such that

$$xyx^{-1} = a(y_0)y, \qquad y \in K(S'), \quad S' \longrightarrow S.$$

[139] Editor's note: This is the Lie algebra $\mathrm{Der}_k(\mathfrak{k}_0)$ of derivations of $\mathfrak{k}_0$; hence $H^1(K_0, \mathfrak{k}_0)$ is the quotient of $\mathrm{Der}_k(\mathfrak{k}_0)$ by the inner derivations (that is, by the image of $\mathrm{ad} : \mathfrak{k}_0 \to \mathrm{Der}_k(\mathfrak{k}_0)$).

But this can also be written

$$a(y_0) = xyx^{-1}y^{-1} = x - \mathrm{Ad}(y)x = -\partial(x)(y_0),$$

whence $a = -\partial(x)$.

On the other hand,[140] the image under ∂ of

$$x \in \mathrm{Lie}\,\underline{\mathrm{Norm}}_{G_0}(K_0) \subset \mathfrak{g}_0$$

is an element of $Z^1(K_0, \mathfrak{k}_0)$, whose image $\overline{\partial(x)}$ in $H^1(K_0, \mathfrak{k}_0)$ depends only on $\bar{x}$. By definition of the connecting map ∂^0,

$$\partial^0(\bar{x}) = \overline{\partial(x)}.$$

Combined with $a = -\partial(x)$, this proves $(*)$.

3) [141] Let $i : K \to G$ denote the canonical immersion. Let $\bar{u} \in H^1(K_0, \mathfrak{k}_0)$ be the image of an element

$$u \in \mathrm{Lie}\,\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(K).$$

Then

$$i^1(\bar{u}) = \bar{d}(i, i \circ u), \qquad (**)$$

where $\bar{d}(i, i \circ u)$ is the class defined in 2.1.1.

Indeed, $\bar{u}$ is the image of an element $v \in Z^1(K_0, \mathfrak{k}_0)$ such that $u(y) = v(y_0)y$, and $i^1(\bar{u})$ is the image in $H^1(K_0, \mathfrak{g}_0)$ of the cocycle $i \circ v \in Z^1(K_0, \mathfrak{g}_0)$. Since i is a group morphism, the equality $u(y) = v(y_0)y$ implies

$$i(u(y)) = i(v(y_0))i(y).$$

It follows that $i \circ v = d(i, i \circ u)$, proving $(**)$.

4) Let $i' : K \to G$ be a group morphism lifting i_0, let $\bar{d}(i, i')$ be the class defined in 2.1.1, and let

$$d(K, i'(K)) \in C^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$$

be the deviation defined in 4.5.1. By 4.21, $d(K, i'(K))$ belongs to $Z^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$. Denote its image in $H^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$ by $\bar{d}(K, i'(K))$. Then, by 4.27(i),

$$d^1(\bar{d}(i, i')) = \bar{d}(K, i'(K)). \qquad (\dagger)$$

5) Finally, let K' be a subgroup of G lifting K_0 and flat, hence smooth, over S. We have assumed that K_0 is affine. We therefore know that K and K' are isomorphic as schemes extending K_0 (cf. 0.15), so there exists an isomorphism of S-schemes

$$f : K \xrightarrow{\sim} K'$$

inducing the identity on K_0. Transport the group structure of K' by f, and let K_1 be the group thereby obtained (whose underlying scheme is thus K); that is, the group law μ_1 of K_1 is defined, for every $S' \to S$ and every $x, y \in K(S')$, by

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

[140] Editor's note: We have added the following sentence.

[141] Editor's note: We have expanded the original in what follows, and in $(**)$ we have corrected $u \circ i$ to $i \circ u$.

By 3.5.1,[142] K_1 defines a cocycle $\delta(K, K_1) \in Z^2(K_0, \mathfrak{k}_0)$ such that, for every $S' \to S$ and every $x, y \in K(S')$,

$$\delta(K, K_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}\big(f(x)f(y)\big).$$

Then, by 4.35.1,

$$\partial^1 \bar{d}(K, K') = \bar{\delta}(K, K_1). \qquad (\ddagger)$$

Bibliography

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[142] Editor's note: We have modified the original in what follows, taking account of the additions made in 3.5.1 and 4.35.1.

[143] Editor's note: Additional references cited in this Exposé.

Exposé IV
Topologies and Sheaves

by M. Demazure[*]

This exposé is intended to acquaint the reader with the essentials of the language of topologies and sheaves (without cohomology), a language particularly convenient for questions concerning the formation of quotients, among others.[0]

The first three sections develop the language of forming quotients. The fourth, which is the central part, presents the theory of sheaves, oriented mainly toward its application to questions about quotients; the fifth applies this theory to the formation of quotients in groups and to principal homogeneous bundles. The final section concerns the category of schemes more specifically and defines several useful topologies on that category.

The reader will profitably consult [AS], [MA], [D], and SGA 4; in particular, [D] for the applications of topologies to descent theory, and SGA 4 for questions concerning universes (which are treated especially poorly in this exposé).

1. Universal Effective Epimorphisms

Throughout the remainder of this exposé, a category $\mathscr{C}$ is fixed.

Definition 1.1. *A morphism $u : T \to S$ is called an* epimorphism *if, for every object X, the corresponding map*

$$X(S) = \mathrm{Hom}(S, X) \longrightarrow X(T) = \mathrm{Hom}(T, X)$$

is injective.[1] *It is called a* universal epimorphism *if, for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists and $u' : T' \to S'$ is an epimorphism.*

Definition 1.2. *A diagram of maps of sets*

$$A \xrightarrow{u} B \underset{v_2}{\overset{v_1}{\rightrightarrows}} C$$

is said to be exact *if u is injective and its image consists of those elements $b \in B$ for which $v_1(b) = v_2(b)$. A diagram of the same form in $\mathscr{C}$ is said to be* exact *if, for every object X of $\mathscr{C}$, the corresponding diagram of sets*

$$A(X) \longrightarrow B(X) \rightrightarrows C(X)$$

[*]This text develops the substance of two lectures given by A. Grothendieck, supplementing them at several important points that had been passed over in silence or only briefly touched upon.

[0]Editor's note: version of October 13, 2024.

[1]Editor's note: that is, if u is right-cancellable.

is exact; one then also says that u makes A a kernel *of the pair of arrows (v_1, v_2). Dually, a diagram in $\mathscr{C}$*

$$C \overset{v_1}{\underset{v_2}{\rightrightarrows}} B \xrightarrow{u} A$$

is said to be exact if it is exact as a diagram in the opposite category $\mathscr{C}^\circ$; that is, if for every object X of $\mathscr{C}$, the corresponding diagram of sets

$$X(A) \longrightarrow X(B) \rightrightarrows X(C)$$

is exact.[2] *One also says that u makes A a* cokernel *of the pair of arrows (v_1, v_2).*

Definition 1.3. *A morphism $u : T \to S$ is called an* effective epimorphism *if the fiber square $T \times_S T$ exists and the diagram*

$$T \times_S T \overset{\mathrm{pr}_1}{\underset{\mathrm{pr}_2}{\rightrightarrows}} T \xrightarrow{u} S$$

is exact; that is, if u makes S a cokernel of $(\mathrm{pr}_1, \mathrm{pr}_2)$. It is called a universal effective epimorphism *if, for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists and the morphism $u' : T' \to S'$ is an effective epimorphism.*

There are evidently implications

$$\begin{array}{ccc} \text{universal effective epimorphism} & \Longrightarrow & \text{effective epimorphism} \\ \Downarrow & & \Downarrow \\ \text{universal epimorphism} & \Longrightarrow & \text{epimorphism,} \end{array}$$

but in general no other implication holds.[3]

Definition 1.4.0.[4] We "recall" that a morphism $u : T \to S$ is said to be *squarable* if, for every morphism $S' \to S$, the fiber product $T \times_S S'$ exists.

Lemma 1.4. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then:*

(a) *if u and v are epimorphisms, then uv is an epimorphism; and if uv is an epimorphism, then u is an epimorphism;*

(b) *if u and v are universal epimorphisms, then uv is a universal epimorphism; and if uv is a universal epimorphism and u is squarable, then u is a universal epimorphism.*

Lemma 1.4 follows immediately from the definitions. We deduce:

Corollary 1.5. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and*

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

[2] Editor's note: this implies, in particular, that u is an epimorphism.

[3] Editor's note: for example, if $\mathscr{C} = \mathbf{Sch}$ is the category of schemes, it is easy to see that every universal epimorphism is surjective. Let

$$T = \coprod_{p \text{ prime}} \mathrm{Spec}(\mathbf{F}_p), \qquad S = \mathrm{Spec}(\mathbf{Z}).$$

Then the morphism $u : T \to S$ is an epimorphism that is not universal. On the other hand, $T \times_S T$ identifies with T, so the two projections $T \times_S T \to T$ coincide. Since id_T does not descend to a morphism $S \to T$, this shows that u is not an effective epimorphism.

[4] Editor's note: the numbering 1.4.0 has been added for later reference.

is a universal epimorphism.

We also record the following.

Definition 1.6.0.[5] An object S of $\mathscr{C}$ is called *squarable* if its product with every object of $\mathscr{C}$ exists. (If $\mathscr{C}$ has a final object e, this is equivalent to saying that the morphism $S \to e$ is squarable; cf. 1.4.0.)

Lemma 1.6. *Let $u : X \to Y$ be a morphism in $\mathscr{C}_{/S}$. For it to be an epimorphism (respectively, a universal epimorphism, an effective epimorphism, or a universal effective epimorphism), it is sufficient that the corresponding morphism in $\mathscr{C}$ have the same property; it is also necessary if S is assumed to be a squarable object of $\mathscr{C}$.*

The proof is immediate and is left to the reader. The hypothesis that S is squarable is used to interpret the $\mathscr{C}$-morphisms from an object Y of $\mathscr{C}_{/S}$ to an object Z of $\mathscr{C}$ as the $\mathscr{C}_{/S}$-morphisms from Y to $Z \times S$.

Lemma 1.7. *With the notation of 1.4, if u and v are effective epimorphisms and v is a universal epimorphism, then uv is an effective epimorphism.*

To see this, consider the diagram

$$\begin{array}{ccccc} S & \xleftarrow{u} & T & \leftleftarrows & T \times_S T \\ & & {\scriptstyle v}\uparrow & & {\scriptstyle v\times_S v}\uparrow \\ & & U & \leftleftarrows & U \times_S U \\ & & \uparrow\uparrow & \nearrow & \\ & & U \times_T U & & \end{array} \tag{1.7.1}$$

By hypothesis, the first row and the first column are exact; moreover, by 1.5 and 1.6, $v \times_S v$ is an epimorphism, since v is a universal epimorphism. The conclusion follows by an evident diagram chase. Indeed, if an element of $X(U)$ has the same two images in $X(U \times_S U)$, then it has, *a fortiori*, the same two images in $X(U \times_T U)$, and hence comes from an element of $X(T)$, since the first column is exact.

Since the first row is exact, it remains only to verify that the element in question has the same two images in $X(T \times_S T)$. Because $v \times_S v$ is an epimorphism, it is enough to verify that their images in $X(U \times_S U)$ are the same, which is indeed the case.

Proposition 1.8. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. If u and v are universal effective epimorphisms, then uv is a universal effective epimorphism. If uv is a universal effective epimorphism and u is squarable, then u is a universal effective epimorphism.*

The first implication follows at once from 1.7. For the second, consider the following diagram (of "bisimplicial" type):

$$\begin{array}{ccccccc} S & \xleftarrow{u} & T & \leftleftarrows & T \times_S T \\ {\scriptstyle uv}\uparrow & & \uparrow & & \uparrow \\ U & \longleftarrow & U \times_S T & \leftleftarrows & U \times_S T \times_S T \\ \uparrow\uparrow & & \uparrow\uparrow & & \uparrow\uparrow \\ U \times_S U & \longleftarrow & U \times_S U \times_S T & \leftleftarrows & U \times_S U \times_S T \times_S T. \end{array} \tag{1.8.1}$$

The first, second, and third columns are exact by the hypothesis that uv is a universal effective epimorphism. The second row is exact because $U \times_S T \to U$ is an effective epimorphism (it has a section over U), and the same is true of the third row, for the same reason. An evident diagram chase then shows that the first row is exact, that is, that u is an effective epimorphism. Since the hypotheses are invariant under an arbitrary base change $S' \to S$, it follows that u is in fact a universal effective epimorphism.

[5] Editor's note: the numbering 1.6.0 has been added for later reference.

Corollary 1.9. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal effective epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and*

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

is a universal effective epimorphism.

The proof is the same as for 1.5, using the diagram

$$\begin{array}{ccccc} & & Y' & \xleftarrow{u'} & X' \\ & & \uparrow & & \uparrow \\ X & \longleftarrow & X \times Y' & \longleftarrow & X \times X' \\ {\scriptstyle u}\downarrow & & \downarrow & \swarrow_{u\times u'} & \\ Y & \longleftarrow & Y \times Y' & & . \end{array} \tag{1.9.1}$$

Corollary 1.10. *Let $u : T \to S$ be a squarable morphism, and let $S' \to S$ be a base-change morphism that is a universal effective epimorphism. Then u is a universal effective epimorphism if and only if*

$$u' : T' = T \times_S S' \longrightarrow S'$$

is one:

$$\begin{array}{ccc} T & \xleftarrow{v'} & T' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u'} \\ S & \xleftarrow{v} & S'. \end{array} \tag{1.10.1}$$

Only sufficiency requires proof. If u' is a universal effective epimorphism, then so is vu', by 1.8. Since $vu' = uv'$, another application of 1.8 shows that u is a universal effective epimorphism.

Remark 1.11. *The same reasoning shows that in 1.10 one may replace "universal effective epimorphism" by "universal epimorphism," by "universal and effective epimorphism," or simply by "epimorphism" (and in the last case the hypothesis that u be squarable is plainly unnecessary).*

In the proof of 1.8 we used the following result, which deserves to be stated explicitly.

Proposition 1.12. *Let $u : T \to S$ be a morphism admitting a section. Then u is an epimorphism; if $T \times_S T$ exists, it is an effective epimorphism; and if, in addition, u is squarable, it is a universal effective epimorphism.*

The first assertion is contained in 1.4(a), and the third follows at once from the second, so it is enough to establish the second. In fact, there is a stronger conclusion: for every functor $F : \mathscr{C}^\circ \to \mathbf{Set}$, not necessarily representable, the diagram of sets

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact. This may be regarded as a special case of the formalism of Čech cohomology (in dimension 0!), which we merely recall here. Assume only that $T \times_S T$ exists, and set

$$\check{H}^0(T/S, F) = \mathrm{Ker}\big(F(T) \rightrightarrows F(T \times_S T)\big).$$

One may plainly regard $\check{H}^0(T/S, F)$ as a contravariant functor of the variable T in $\mathscr{C}_{/S}$: every S-morphism $T' \to T$ defines a map

$$\check{H}^0(T/S, F) \longrightarrow \check{H}^0(T'/S, F). \tag{+}$$

Fix T, T' in $\mathscr{C}_{/S}$. A familiar calculation shows that, if there exists an S-morphism $T' \to T$, the corresponding map (+) is independent of the choice of that morphism.[6]

Thus $\check{H}^0(T/S, F)$ may be regarded as a functor on the category associated with the set $\mathrm{Ob}(\mathscr{C}_{/S})$, preordered by the relation of domination: T' dominates T if there exists an S-morphism $T' \to T$. In particular, if T and T' are isomorphic in this latter category, that is, if each dominates the other, then (+) is an isomorphism of sets. This applies in particular when T' is the final object of $\mathscr{C}_{/S}$, essentially S itself. The object T always dominates $T' = S$, while the converse holds precisely when T/S has a section. This proves 1.12 in the stronger form announced.

Remark 1.13. *For various applications, the notions introduced and the results stated in this exposé must be developed more generally for a family of morphisms $u_i : T_i \to S$ with the same target, rather than for a single morphism $u : T \to S$. Such a family is called* epimorphic *if, for every object X of $\mathscr{C}$, the corresponding map*

$$X(S) \longrightarrow \prod_i X(T_i)$$

is injective. One similarly introduces the notion of an effective epimorphic family and the universal variants of these notions. Whenever needed below, we shall take for granted that the results of this exposé extend to this more general situation.

Remark 1.14. *Every morphism that is both a monomorphism and an effective epimorphism is an isomorphism.*

Indeed, in the notation of 1.3, the fact that $T \to S$ is a monomorphism implies that the two morphisms

$$\mathrm{pr}_1, \mathrm{pr}_2 : T \times_S T \rightrightarrows T$$

are equal (and are isomorphisms). But a diagram

$$C \overset{v}{\underset{v}{\rightrightarrows}} B \xrightarrow{u} A$$

is exact only if u is an isomorphism, as follows immediately from the definition.[7]

2. Descent Morphisms

Recall the following definitions.

Definition 2.1. *Let $f : S' \to S$ be a morphism such that $S'' = S' \times_S S'$ exists, and let $u' : X' \to S'$ be an object over S'. A* gluing datum *on X'/S', relative to f, is an S''-isomorphism*

$$c : X''_1 \xrightarrow{\sim} X''_2,$$

[6] Editor's note: this is the following argument, communicated by M. Demazure. Let $f, g : T' \to T$, and let $\phi : T' \to T \times_S T$ be the morphism with components f and g, so that $p_1 \circ \phi = f$ and $p_2 \circ \phi = g$. Then $F(\phi) : F(T \times_S T) \to F(T')$ satisfies $F(\phi) \circ F(p_1) = F(f)$ and $F(\phi) \circ F(p_2) = F(g)$. For every $x \in \check{H}^0(T/S, F)$, however, one has $F(p_1)(x) = F(p_2)(x)$. Applying $F(\phi)$ to both sides gives $F(f)(x) = F(g)(x)$, which shows that f and g induce the same map.

[7] Editor's note: a monomorphism that is an epimorphism need not be an isomorphism. For example, in $\mathscr{C} = \mathbf{Sch}$, the morphism

$$\mathrm{Spec}(\mathbf{F}_p) \amalg \mathrm{Spec}(\mathbf{Z}[1/p]) \longrightarrow \mathrm{Spec}(\mathbf{Z})$$

is a monomorphism and a surjective epimorphism, but is not an isomorphism.

where X''_i, for $i = 1, 2$, denotes the inverse image, assumed to exist, of X'/S' under the projection $\mathrm{pr}_i : S'' \to S'$. *The gluing datum c is called a* descent datum *if it satisfies the "cocycle condition"*

$$\mathrm{pr}_{3,1}^*(c) = \mathrm{pr}_{3,2}^*(c)\ \mathrm{pr}_{2,1}^*(c),$$

where $\mathrm{pr}_{i,j}$, *for* $1 \leq j < i \leq 3$, *are the canonical projections from*

$$S''' = S' \times_S S' \times_S S'$$

to S'' (and we now assume that S''' also exists), where $\mathrm{pr}_{i,j}^*(c)$ *is the inverse image of c, viewed as an S'''-morphism from X'''_j to X'''_i, and where, for each integer k from 1 to 3, X'''_k denotes the inverse image, assumed to exist, of X'/S' under the k-th projection $q_k : S''' \to S'$.*

In the second part of the definition we have therefore used customary identifications and abuses of notation,[8] which experience shows to be harmless, but which must plainly be avoided in a rigorous exposition of descent theory—a theory that must itself justify, to some extent, these customary abuses of language. Such a formal exposition ([D]) was written by Giraud, with a view to justifying and making precise SGA 1, Exposé VII, which was never written. For a detailed account of the faithfully flat descent results used constantly in this Seminar, one may consult SGA 1, Exposé VIII.

Continue to let $f : S' \to S$ be a morphism for which $S'' = S' \times_S S'$ exists, and let X be an object over S such that $X' = X \times_S S'$ and $X'' = X \times_S S''$ exist. The inverse images of X' under pr_i, for $i = 1, 2$, then exist and are canonically isomorphic; consequently, X'/S' carries a canonical gluing datum relative to f. When S''' and $X''' = X \times_S S'''$ exist, this is in fact a descent datum. If Y is another object over S satisfying the same conditions as X/S, then for every S-morphism $X \to Y$, the corresponding S'-morphism $X' \to Y'$ is compatible with the canonical gluing data on X' and Y'. In particular, if $S' \to S$ is squarable, then

$$X \longmapsto X' = X \times_S S'$$

is a functor from $\mathscr{C}_{/S}$ to the "category of objects over S' equipped with a descent datum relative to f"—a category whose definition is left to the reader and which is a full subcategory of the "category of objects over S' equipped with a gluing datum relative to f."

With this understood, we make the following definition.

Definition 2.2. *A morphism $f : S' \to S$ is called a* descent morphism *(respectively, an* effective descent morphism*) if f is squarable (that is, $X \times_S S'$ exists for every $X \to S$) and the preceding functor*

$$X \longmapsto X' = X \times_S S'$$

from the category $\mathscr{C}_{/S}$ of objects over S to the category of objects over S' equipped with a descent datum relative to f is fully faithful (respectively, an equivalence of categories).

The first of these notions can be expressed using gluing data alone, and therefore without invoking the triple fiber product S''': f is a descent morphism precisely when it is squarable and $X \mapsto X'$ is a fully faithful functor from $\mathscr{C}_{/S}$ to the category of objects over S' equipped with a gluing datum relative to f. Written out, this means that for any two objects X, Y over S, the following diagram of sets is exact:

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S'}(X', Y') \underset{p_2}{\overset{p_1}{\rightrightarrows}} \mathrm{Hom}_{S''}(X'', Y''). \tag{x}$$

[8] Editor's note: for example, we have identified, on the one hand, $\mathrm{pr}_{2,1}^*(X''_1) = \mathrm{pr}_{3,1}^*(X''_1)$, and, on the other hand, $\mathrm{pr}_{2,1}^*(X''_2) = \mathrm{pr}_{3,2}^*(X''_2)$.

Here $p_i(u')$ denotes the inverse image of $u' \in \mathrm{Hom}_{S'}(X', Y')$ under the projection $\mathrm{pr}_i : S'' \to S'$, for $i = 1, 2$. Indeed, by definition, the kernel of the pair (p_1, p_2) is precisely the set of S'-morphisms $X' \to Y'$ compatible with the canonical gluing data.

By the definition of the inverse images Y' and Y'', there are canonical bijections

$$\mathrm{Hom}_{S'}(X', Y') \simeq \mathrm{Hom}_S(X', Y), \qquad \mathrm{Hom}_{S''}(X'', Y'') \simeq \mathrm{Hom}_S(X'', Y).$$

Thus exactness of (x) is equivalent to exactness of

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_S(X', Y) \rightrightarrows \mathrm{Hom}_S(X'', Y), \tag{xx}$$

which is obtained by applying $\mathrm{Hom}_S(-, Y)$ to the diagram in $\mathscr{C}_{/S}$

$$X'' \rightrightarrows X' \longrightarrow X, \tag{xxx}$$

itself obtained from

$$S'' \rightrightarrows S' \longrightarrow S$$

by the base change $X \to S$. In view of 1.6, this proves the first part of the following proposition.

Proposition 2.3. *Let $f : S' \to S$ be a morphism. If f is a universal effective epimorphism, then it is a descent morphism. Conversely, every descent morphism is a universal effective epimorphism if S is a squarable object of $\mathscr{C}$, that is, if its product with every object of $\mathscr{C}$ exists.*

It remains to prove the converse. If f is a descent morphism, we must show that it is a universal effective epimorphism: for every morphism $X \to S$, diagram (xxx) must be exact, or equivalently, its image under $\mathrm{Hom}(-, Z)$ must be an exact diagram of sets for every object Z of $\mathscr{C}$. By hypothesis, $Z \times S$ exists. Let Y be the object of $\mathscr{C}_{/S}$ that it defines. The image of (xxx) under $\mathrm{Hom}(-, Z)$ is then isomorphic to its image under $\mathrm{Hom}_S(-, Y)$, and the latter is exact by the hypothesis on f.

The results concerning descent morphisms may therefore be applied to universal effective epimorphisms. For example:

Proposition 2.4. *Let $f : S' \to S$ be a descent morphism (for example, a universal effective epimorphism). Then:*

(a) *For every S-morphism $u : X \to Y$, the morphism u is an isomorphism (respectively, a monomorphism) if and only if $u' : X' \to Y'$ is one.*

(b) *Let X, Y be two subobjects of S, and let X', Y' be their inverse images in S'. Then X is contained in Y (respectively, is equal to Y) if and only if the same is true of X' and Y'.*

For (a), the definition shows that if u' is an isomorphism in the category of objects with gluing datum, then u is an isomorphism. Every isomorphism of objects over S' that is compatible with gluing data is, however, an isomorphism of objects with gluing datum: its inverse is also compatible with the gluing data. For (b), it remains to show that if X' is contained in Y', that is, if there exists an S'-morphism $X' \to Y'$, then the same is true of X, Y over S. Since $Y' \to S'$, and hence $Y'' \to S''$, is a monomorphism, the morphism $X' \to Y'$ is automatically compatible with the gluing data and therefore comes from an S-morphism $X \to Y$.

The proof applies more generally to two objects X, Y over S, with $Y \to S$ a monomorphism, when one asks whether $X \to S$ factors through Y: it is enough that $X' \to S'$ factor through Y'.

Corollary 2.5. *Let $f : S' \to S$ be a universal effective epimorphism and $g : S \to T$ a morphism such that $S \times_T S$ exists. Suppose that $S'' = S' \times_S S'$ is also a fiber product of S' with itself over T, that is, that the canonical morphism*

$$S' \times_S S' \xrightarrow{\sim} S' \times_T S'$$

is an isomorphism. Then $g : S \to T$ is a monomorphism (and the converse is of course true).

Indeed, consider the cartesian diagram

$$\begin{array}{ccc} S' \times_S S' & \xrightarrow{\sim} & S' \times_T S' \\ \downarrow & & \downarrow f \\ S & \longrightarrow & S \times_T S, \end{array} \tag{2.5.1}$$

where the second horizontal arrow is the diagonal morphism. By 1.9, the second vertical arrow f is a universal effective epimorphism; by hypothesis, the first horizontal arrow is an isomorphism. It follows from 2.4(a), or equally from 2.4(b),[9] that $S \to S \times_T S$ is also an isomorphism, which says precisely that $g : S \to T$ is a monomorphism.

Remark 2.6. *The notions introduced in 2.1 in terms of morphisms between certain inverse limits have an evident expression in terms of the contravariant functors defined by the objects S, S', X' under consideration. Subject to the existence of the fiber products considered in 2.1, there is a one-to-one correspondence between gluing data (respectively, descent data) on X'/S' relative to $S' \to S$, and gluing data (respectively, descent data) for the corresponding objects of*

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set}).$$

This makes it possible, if desired, to extend these notions to the case in which no existence hypothesis is imposed on fiber products in $\mathscr{C}$.

Remark 2.7. *The notions introduced in this section generalize to families of morphisms. They can also be presented more intrinsically by means of the notion of a sieve (4.1). For these matters the reader is referred to [D].*

3. Universal Effective Equivalence Relations

3.1. Equivalence Relations: Definitions

Definition 3.1.1. *A $\mathscr{C}$-equivalence relation in an object $X \in \mathrm{Ob}\,\mathscr{C}$ is a representable subfunctor R of the functor $X \times X$ such that, for every $S \in \mathrm{Ob}\,\mathscr{C}$, $R(S)$ is the graph of an equivalence relation on $X(S)$.*

This definition applies in particular to the category $\widehat{\mathscr{C}}$. If X is regarded as an object of $\widehat{\mathscr{C}}$, then a $\widehat{\mathscr{C}}$-equivalence relation in X is simply a subfunctor R of $X \times X$ such that $R(S)$ is the graph of an equivalence relation on $X(S)$, for every object S of $\mathscr{C}$.[10]

If R is a $\mathscr{C}$-equivalence relation in X, let p_i denote the morphism from R to X induced by the projection $\mathrm{pr}_i : X \times X \to X$. Thus there is a diagram

$$p_1, p_2 : R \rightrightarrows X.$$

[9] Editor's note: applied to $f : S' \times_T S' \to S \times_T S = Y$.

[10] Editor's note: the condition is plainly necessary. Conversely, if $R(S)$ is the graph of an equivalence relation for every $S \in \mathrm{Ob}\,\mathscr{C}$, this equivalence relation extends to $R(F)$ for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$: two morphisms $\varphi, \psi : F \to R$ are declared equivalent if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $x \in F(S)$, the elements $\varphi(x)$ and $\psi(x)$ are equivalent in $X(S)$.

Definition 3.1.2. *A morphism $u : X \to Z$ is said to be* compatible with R *if $up_1 = up_2$. An object of $\mathscr{C}$ that is a cokernel of the pair (p_1, p_2) is also called a* quotient object *of X by R, and is denoted X/R.*

Thus one has an exact diagram

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{p} X/R,$$

and X/R represents the covariant functor

$$\mathrm{Hom}_{\mathscr{C}}(X/R, Z) = \{\text{morphisms from } X \text{ to } Z \text{ compatible with } R\}.$$

Since quotient objects are taken in $\mathscr{C}$, the quotient X/R is unique whenever it exists.

These definitions extend at once to a $\widehat{\mathscr{C}}$-equivalence relation in X. One should note, however, that the identification of the objects of $\mathscr{C}$ with their images in $\widehat{\mathscr{C}}$ does not commute with the formation of quotients: the quotient X/R of X by R in $\mathscr{C}$ is not *a priori* a quotient of X by R in $\widehat{\mathscr{C}}$. One must therefore avoid identifying $\mathscr{C}$ uncritically with its image in $\widehat{\mathscr{C}}$ in questions involving passage to a quotient.[11]

In what follows, we shall simply say *equivalence relation* to mean a $\widehat{\mathscr{C}}$-equivalence relation; whenever necessary, we shall specify that a $\mathscr{C}$-equivalence relation is intended.[12]

Definition 3.1.3. *If X is an object of $\mathscr{C}$ over S, an* equivalence relation in X over S *is an equivalence relation R in X for which the structure morphism $X \to S$ is compatible with R.*

The canonical monomorphism $R \to X \times X$ then factors through the monomorphism

$$X \times_S X \longrightarrow X \times X$$

and defines an equivalence relation in the object $X \to S$ of $\mathscr{C}_{/S}$. Whenever the quotient X/R exists, it is equipped with a canonical morphism to S, and the corresponding object of $\mathscr{C}_{/S}$ is a quotient of $X \in \mathrm{Ob}\,\mathscr{C}_{/S}$ by the preceding equivalence relation. Conversely, if S is a squarable object of $\mathscr{C}$ and $Y \to S$ is a quotient of X by this equivalence relation in $\mathscr{C}_{/S}$, then Y is a quotient of X by R in $\mathscr{C}$. In any event, we shall never have to consider quotients in $\mathscr{C}_{/S}$ that are not already quotients in $\mathscr{C}$.

Definition 3.1.4. *Let X (respectively X') be an object of $\mathscr{C}$ equipped with an equivalence relation R (respectively R'). A morphism*

$$u : X \longrightarrow X'$$

is said to be compatible with R and R' *if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, two points of $X(S)$ congruent modulo $R(S)$ are carried by u to two points of $X'(S)$ congruent modulo $R'(S)$;*

[11] Editor's note: here is an illustration that also previews the sequel of this exposé. Let G be a $\mathscr{C}$-group, let H be a $\mathscr{C}$-subgroup, and let R be the equivalence relation in G defined by $G \times H \to G \times G$, $(g, h) \mapsto (g, gh)$ (cf. 3.2). The functor Q defined by $Q(S) = G(S)/H(S)$ is a quotient in $\widehat{\mathscr{C}}$ (by 4.4.9 applied to the coarsest topology; cf. 4.4.2), but it is not in general the quotient one wants. For example, when $\mathscr{C} = \mathbf{Sch}$, there is an exact sequence of affine group schemes

$$1 \longrightarrow \mu_2 \longrightarrow \mathbf{G}_m \xrightarrow{p} \mathbf{G}_m \longrightarrow 1$$

that identifies $\mathbf{G}_m$ with the quotient $\mathbf{G}_m/\mu_2$. Moreover, because p is finite and locally free, $\mathbf{G}_m$ is the sheaf quotient of $\mathbf{G}_m$ by μ_2 in the larger category of sheaves for the fppf topology; cf. 4.6.6(ii) and 6.3.2. By contrast, the quotient Q in $\widehat{\mathscr{C}}$ is not isomorphic to $\mathbf{G}_m$, since, for example, $Q(\mathbf{Z}) = \{1\}$, whereas $\mathbf{G}_m(\mathbf{Z}) = \{\pm 1\}$. Thus Q is not an fppf sheaf and, *a fortiori*, is not representable.

[12] Editor's note: even for a $\widehat{\mathscr{C}}$-equivalence relation in X, the question of interest is the existence of a quotient in $\mathscr{C}$.

(ii) *there exists a necessarily unique morphism $R \to R'$ making the diagram*

$$\begin{array}{ccc} R & \longrightarrow & R' \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow{u \times u} & X' \times X' \end{array}$$

commute.

By the universal property of X/R, whenever X/R and X'/R' exist there is a unique morphism v making the diagram

$$\begin{array}{ccc} X & \xrightarrow{p} & X/R \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ X' & \xrightarrow{p'} & X'/R' \end{array}$$

commute.

Definition 3.1.5. *A subobject (that is, a representable subfunctor) Y of X is said to be* stable under the equivalence relation R *if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the subset $Y(S)$ of $X(S)$ is stable under $R(S)$;*

(ii) *the two subobjects of R that are the inverse images of Y under p_1 and p_2 are identical.*

An important special case of a stable subobject of X is the following: the quotient X/R exists and Y is the inverse image in X of a subobject of X/R.

Definition 3.1.6. *Let R be an equivalence relation in X, and let $X' \to X$ be a morphism. The equivalence relation R' in X' obtained from the cartesian diagram*

$$\begin{array}{ccc} R' & \longrightarrow & R \\ \downarrow & & \downarrow \\ X' \times X' & \longrightarrow & X \times X \end{array}$$

is called the equivalence relation in X' that is the inverse image *of the equivalence relation R in X. In particular, if X' is a subobject of X, it is called the equivalence relation* induced *on X' by R, and is denoted $R_{X'}$.*

The morphism $X' \to X$ is compatible with R' and R; hence, when the quotients exist, it induces a morphism $X'/R' \to X/R$ (3.1.4). If X' is a subobject of X, we shall see later that in certain cases one can prove that $X'/R' \to X/R$ is a monomorphism and therefore identifies X'/R' with a subobject of X/R. In that situation, the inverse image of this subobject in X is a subobject containing X' and stable under R: the *saturation* of X' with respect to the equivalence relation R.

Proposition 3.1.7. *If the subobject Y of X is stable under R, then for $i = 1, 2$ the following two squares are cartesian:*

$$\begin{array}{ccc} R_Y & \longrightarrow & R \\ {\scriptstyle p_i}\downarrow & & \downarrow{\scriptstyle p_i} \\ Y & \longrightarrow & X. \end{array}$$

The proof is immediate.

3.2. Equivalence Relation Defined by a Group Acting Freely

Definition 3.2.1. *Let X be an object of $\mathscr{C}$, and let H be a $\mathscr{C}$-group acting on X, that is, equipped with a morphism of $\widehat{\mathscr{C}}$-groups*

$$H \longrightarrow \mathrm{Aut}(X).$$

The group H is said to act freely *on X if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the group $H(S)$ acts freely on $X(S)$;*

(ii) *the morphism of functors*

$$H \times X \longrightarrow X \times X$$

defined on sets by $(h, x) \mapsto (hx, x)$ is a monomorphism.

In the preceding definition the group H was assumed to act on X on the left. There is of course an analogous notion when the group acts on the right, that is, when one is given a group morphism from the opposite group H° to $\mathrm{Aut}(X)$.

If H acts freely on X, the image of $H \times X$ under the morphism in (ii) is an equivalence relation in X, called the *equivalence relation defined by the action of H on X*. When the quotient of X by this equivalence relation exists, it is denoted $H\backslash X$ (or X/H when H acts on the right). It represents the following covariant functor: for every object Z of $\mathscr{C}$,

$$\mathrm{Hom}(H\backslash X, Z) = \{\text{morphisms from } X \text{ to } Z \text{ invariant under } H\},$$

where a morphism $f : X \to Z$ is said to be invariant under H if, for every $S \in \mathrm{Ob}\,\mathscr{C}$, the corresponding morphism $X(S) \to Z(S)$ is invariant under the group $H(S)$.

Lemma 3.2.2. *Under the assumptions of 3.2.1, let Y be a subobject of X. The following conditions are equivalent:*

(i) *Y is stable under the equivalence relation defined by H (3.1.5);*

(ii) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the subset $Y(S)$ of $X(S)$ is stable under $H(S)$;*

(iii) *there exists a necessarily unique morphism f making the diagram*

$$\begin{array}{ccc} H \times Y & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ H \times X & \longrightarrow & X \end{array}$$

commute.

Under these conditions, f defines a morphism of $\widehat{\mathscr{C}}$-groups

$$H \longrightarrow \mathrm{Aut}(Y),$$

and the equivalence relation in Y defined by this action of H is precisely the equivalence relation induced on Y by the equivalence relation in X defined by the action of H.

The proof is immediate. There is of course an analogous statement for a right action. The action of H on Y defined above will be called the action induced on Y by the given action of H on X.

Consider now two $\mathscr{C}$-groups H and G, together with a group morphism

$$u : H \longrightarrow G.$$

Then H acts on G by translations (set-theoretically, $hg = u(h)g$), and it acts freely if and only if u is a monomorphism. When it exists, the quotient of G by this action of H is denoted $H\backslash G$. Similarly, one defines a right action of H on G and a quotient G/H. These quotients are functorial with respect to the groups involved. More precisely, one has the following lemma, stated for right quotients.

Lemma 3.2.3. *Let $u : H \to G$ and $u' : H' \to G'$ be monomorphisms of $\mathscr{C}$-groups, and suppose that a morphism of $\mathscr{C}$-groups*

$$f : G \longrightarrow G'$$

is given. The following conditions are equivalent:

(i) *f is compatible with the equivalence relations in G and G' defined by H and H';*

(ii) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, one has $fu(H(S)) \subseteq u'(H'(S))$;*

(iii) *there exists a necessarily unique multiplicative morphism $g : H \to H'$ such that the diagram*

$$\begin{array}{ccc} H & \xrightarrow{g} & H' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u'} \\ G & \xrightarrow{f} & G' \end{array}$$

commutes.

Under these conditions, if the quotients G/H and G'/H' exist, there is a unique morphism $\overline{f}$ making the diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & G' \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p'} \\ G/H & \xrightarrow{\overline{f}} & G'/H' \end{array}$$

commute.

The first assertion is proved by reduction to the set-theoretic case. The second follows immediately from (i).

The notions introduced above for general equivalence relations could all be translated into the present setting. We record only the following lemma, whose proof is immediate by reduction to the set-theoretic case.

Lemma 3.2.4. *Let $u : H \to G$ be a monomorphism of $\mathscr{C}$-groups, and let G' be a $\mathscr{C}$-subgroup of G. The subobject G' of G is stable under the equivalence relation defined by H if and only if u factors through the canonical monomorphism $G' \to G$. Under this condition, the action of H on G' induced by the action on G defined by u is precisely the action arising from the monomorphism $H \to G'$ through which u factors.*

3.3. Universally Effective Equivalence Relations

Definition 3.3.1. *Let $f : X \to Y$ be a morphism. The* equivalence relation defined by f in X, *denoted $\mathscr{R}(f)$, is the $\widehat{\mathscr{C}}$-equivalence relation in X that is the image of the canonical monomorphism*

$$X \times_Y X \longrightarrow X \times X.$$

Definition 3.3.2. *An equivalence relation R in X is said to be* effective *if:*

(i) *R is representable, that is, it is a $\mathscr{C}$-equivalence relation;*

(ii) *the quotient $Y = X/R$ exists in $\mathscr{C}$;*[13]

(iii) *the diagram*

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{p} Y$$

makes R the fiber square of X over Y, that is, R is the equivalence relation defined by p.

Scholium 3.3.2.1.[14] *If R is an effective equivalence relation in X, then p is an effective epimorphism (1.3). If $f : X \to Y$ is an effective epimorphism, then $\mathscr{R}(f)$ is an effective equivalence relation in X, one quotient of which is Y. Thus there is a one-to-one "Galois" correspondence between effective equivalence relations in X and effective quotients of X, that is, equivalence classes of effective epimorphisms with source X.*

Definition 3.3.3. *An equivalence relation R in X is said to be* universally effective *if the quotient $Y = X/R$ exists and, for every $Y' \to Y$, the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist and R' is the fiber square of X' over Y'. Equivalently, R is effective and $p : X \to X/R$ is a universal effective epimorphism.*

Scholium 3.3.3.1.[14] *As above, there is a one-to-one correspondence between universally effective equivalence relations in X and universal effective quotients of X.*

Remark 3.3.3.2.[14] Suppose that $\mathscr{C}$ is the category of S-schemes, and write $\mathbf{A}^1_S$ for affine one-space over S. Let $R \subset X \times_S X$ be a universally effective equivalence relation, and let $p : X \to Y$ be its quotient. Then, for every open subset U of Y,

$$\mathcal{O}(U) = \mathrm{Hom}_S(U, \mathbf{A}^1_S)$$

is the set of elements φ of

$$\mathcal{O}(p^{-1}(U)) = \mathrm{Hom}_S(p^{-1}(U), \mathbf{A}^1_S)$$

such that $\varphi \circ \mathrm{pr}_1 = \varphi \circ \mathrm{pr}_2$. In particular, if R is defined by the action of a group H acting freely on the right on X (cf. 3.2.1), then $\mathcal{O}(U)$ is the set of all $\varphi \in \mathcal{O}(p^{-1}(U))$ such that

$$\varphi(xh) = \varphi(x)$$

for every $S' \to S$, $x \in X(S')$, and $h \in H(S')$.

Proposition 3.3.4. *Let R be a universally effective equivalence relation in X. Let $f : X \to Z$ be a morphism compatible with R, hence factoring through a morphism $g : X/R \to Z$. The following conditions are equivalent:*

(i) *g is a monomorphism;*

(ii) *R is the equivalence relation defined by f.*

Indeed, (i) plainly implies (ii), while the converse is precisely 2.5.

Definition 3.3.5. *Let H be a $\mathscr{C}$-group acting freely on X. One says that H acts* effectively*, or that the action of H on X is* effective *(respectively* universally effective*), if the equivalence relation in X defined by the action of H is effective (respectively universally effective).*

[13] Editor's note: the words "in $\mathscr{C}$" have been added.

[14] Editor's note: the numbering 3.3.2.1 (respectively 3.3.3.1) has been added for later reference; Remark 3.3.3.2 has also been added.

3.4. (M)-Effectivity

In practice, universal effective epimorphisms are most often difficult to characterize. Nevertheless, one frequently has at hand a number of morphisms of this kind–in scheme theory, for example, faithfully flat quasi-compact morphisms. This leads to the developments below.

3.4.1. We first state several conditions on a family (M) of morphisms of $\mathscr{C}$:

(a) (M) is stable under base change: every $u : T \to S$ belonging to (M) is squarable (cf. 1.4.0), and for every $S' \to S$, the base change

$$u' : T \times_S S' \longrightarrow S'$$

belongs to (M);

(b) the composite of two elements of (M) belongs to (M);

(c) every isomorphism belongs to (M);

(d) every element of (M) is an effective epimorphism.

Conditions (a) and (b) imply:

(a′) the cartesian product of two elements of (M) belongs to (M). More precisely, let $u : X \to Y$ and $u' : X' \to Y'$ be two S-morphisms belonging to (M). If $Y \times_S Y'$ exists, then $X \times_S X'$ exists and $u \times_S u'$ belongs to (M).

This follows from the diagram

$$\begin{array}{ccccc} & & Y' & \xleftarrow{u'} & X' \\ & & \uparrow & & \uparrow \\ X & \longleftarrow & X \times_S Y' & \longleftarrow & X \times_S X' \\ {\scriptstyle u}\downarrow & & \downarrow & & \swarrow_{u \times_S u'} \\ Y & \longleftarrow & Y \times_S Y' & & \end{array}$$

Likewise, (a) and (d) imply:

(d′) every element of (M) is a universal effective epimorphism.

3.4.2. The family (M_0) of universal effective epimorphisms satisfies conditions (a)–(d) of 3.4.1. Indeed, (a), (c), and (d) hold by definition, and (b) follows from 1.8. Henceforth we assume given a family (M) of morphisms of $\mathscr{C}$ satisfying these conditions; our results apply in particular to the family (M_0).

Definition 3.4.3. An equivalence relation R in X is said to be *of type* (M) if it is representable and $p_1 \in (M)$; conditions (b) and (c) then imply that $p_2 \in (M)$.

The relation R is said to be (M)-*effective* if it is effective and the canonical morphism $X \to X/R$ belongs to (M).

A quotient Y of X is said to be (M)-*effective* if the canonical morphism $X \to Y$ belongs to (M).

The following consequences are immediate from the definition.[15]

Proposition 3.4.3.1.

[15]Editor's note: the numbering 3.4.3.1 has been added for later reference, and the proof of part (i) has been expanded.

(i) *An (M)-effective equivalence relation is of type (M) and is universally effective.*

(ii) *An (M)-effective quotient is universally effective (cf. 3.3.3).*

(iii) *The maps $R \mapsto X/R$ and $p \mapsto \mathscr{R}(p)$ establish a one-to-one correspondence between (M)-effective equivalence relations in X and (M)-effective quotients of X.*

(iv) *To be (M_0)-effective is equivalent to being universally effective.*

We prove (i). Since R is (M)-effective, there is a cartesian square

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow{p} & X/R, \end{array}$$

and $p \in (M)$. By 3.4.1(a), p_1 and p_2 belong to (M), so R is of type (M).

Set $Y = X/R$, and let $Y' \to Y$ be any morphism. By 3.4.1(a), the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist, and the morphisms $X' \to Y'$ and $p'_i : R' \to X'$ belong to (M). Finally, since $R = X \times_Y X$, associativity of the fiber product gives

$$R' = X \times_Y X \times_Y Y' = X' \times_{Y'} X'.$$

This shows that R' is (M)-effective; in particular, R is universally effective. This proves (i), as well as (iv). Parts (ii) and (iii) follow, taking Definition 3.3.2 into account.

3.4.4. Let H be an S-group whose structure morphism belongs to (M). If H acts freely on the S-object X, then it defines an equivalence relation of type (M) in X. Indeed, by (a), the fiber product $H \times_S X$ exists and $\mathrm{pr}_2 : H \times_S X \to X$ belongs to (M). The action of H is said to be (M)-*effective* if the equivalence relation in X defined by this action is (M)-effective.

Proposition 3.4.5 ((M)-effectivity and base change). *Let R be an (M)-effective equivalence relation in X over S, and set $Y = X/R$. Let $S' \to S$ be a base change for which $Y' = Y \times_S S'$ exists. Then $X' = X \times_S S'$ exists, $R' = R \times_S S'$ is an (M)-effective equivalence relation in X' over S', and*

$$X'/R' \simeq (X/R)'.$$

Indeed, the canonical morphisms $X \to Y$ and $R \to Y$ belong to (M); hence, by (a′), X' and R' are representable. By associativity of the fiber product, R' is the equivalence relation in X' defined by the canonical morphism $X' \to Y'$, which belongs to (M). This proves the assertion.

Proposition 3.4.6 ((M)-effectivity and cartesian products). *Let R (respectively R') be an (M)-effective equivalence relation in X (respectively X') over S. If $(X/R) \times_S (X'/R')$ exists, then $X \times_S X'$ exists, $R \times_S R'$ is an (M)-effective equivalence relation in $X \times_S X'$ over S, and*

$$(X \times_S X')/(R \times_S R') \simeq (X/R) \times_S (X'/R').$$

Set $Y = X/R$ and $Y' = X'/R'$. By (a′), the fiber product $X \times_S X'$ exists and the canonical morphism

$$q : X \times_S X' \longrightarrow Y \times_S Y'$$

belongs to (M). The formula

$$(X \times_S X') \times_{(Y \times_S Y')} (X \times_S X') \simeq (X \times_Y X) \times_S (X' \times_{Y'} X')$$

(all unlabelled fiber products being taken over S) shows that $R \times_S R'$ is the equivalence relation in $X \times_S X'$ defined by q, which completes the proof.

3.4.7.[16] Suppose that $\mathscr{C}$ has a final object e, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups belonging to (M). By 3.4.1(a), the kernel $\mathrm{Ker}(f)$ is represented by $e \times_{G'} G$, and the morphism $\mathrm{Ker}(f) \to e$ belongs to (M).

On the other hand, the equivalence relation defined by f is the same as that defined by the action of $\mathrm{Ker}(f)$, say on the right, on G: it is the image of the morphism

$$G \times \mathrm{Ker}(f) \longrightarrow G \times G$$

defined on sets by $(g, h) \mapsto (g, gh)$. Consequently, Scholium 3.3.2.1 gives the following corollary.

Corollary 3.4.7.1. *Suppose that $\mathscr{C}$ has a final object e, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups belonging to (M). Then the action of $\mathrm{Ker}(f)$ on G is (M)-effective, and G' is the quotient $G/\mathrm{Ker}(f)$.*

3.5. Construction of Quotients by Descent

It frequently happens that one cannot construct a quotient directly, but can do so after a suitable base change. This subsection gives a useful criterion for that situation.

In 2.1 we defined a descent datum on an object X' over S', relative to a morphism $S' \to S$.

Definition 3.5.1. *A descent datum on X', relative to $S' \to S$, is said to be* effective *if X', equipped with this descent datum, is isomorphic to the inverse image over S' of an object X over S, equipped with its canonical descent datum.*

If $S' \to S$ is a descent morphism (2.2), then the object X in the definition is unique up to unique isomorphism. To say that $S' \to S$ is an effective descent morphism (2.2) means that it is a descent morphism and that every descent datum relative to it is effective.

Now consider an equivalence relation R in an object X over S. Let X', X'', and X''' be the inverse images of X over

$$S', \qquad S'' = S' \times_S S', \qquad S''' = S' \times_S S' \times_S S',$$

respectively, and let R', R'', and R''' be the equivalence relations obtained from R by inverse image. Suppose that R' in X' is (M)-effective, and consider the quotient $Y' = X'/R'$, an object over S'. By 3.4.5, its two inverse images over S'' are isomorphic to X''/R''. Thus the S'-object Y' has a canonical gluing datum. Applying in the same way the uniqueness of X'''/R''', one sees that this is in fact a descent datum. (In this argument it has been tacitly assumed that all the displayed fiber products exist. This holds, in particular, if $S' \to S$ is squarable, for example if it is a descent morphism.)

Proposition 3.5.2. *Let R be an equivalence relation in the object X over S, and let $S' \to S$ be a universal effective epimorphism. Assume that every S-morphism whose inverse image over S' belongs to (M) itself belongs to (M). The following conditions are equivalent:*

(i) *R is (M)-effective in X;*

(ii) *R' is (M)-effective in X', and the canonical descent datum on X'/R' is effective.*

[16] Editor's note: this paragraph has been added.

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R.

The implication (i)⇒(ii) is simply 3.4.4 translated into the language of descent. Once the converse is proved, the last assertion of the proposition follows from the fact that a universal effective epimorphism is a descent morphism (2.3), and hence the descended object is unique up to unique isomorphism.

We prove (ii)⇒(i). Let $Y' = X'/R'$, and let Y be the descended object. The canonical morphism $p' : X' \to X'/R' = Y'$ is compatible by construction with the descent data: its two inverse images over S'' coincide with the canonical morphism $X'' \to X''/R''$. It is therefore the inverse image over S' of an S-morphism $p : X \to Y$. Since p' belongs to (M), the assumption on $S' \to S$ implies that p also belongs to (M). Moreover, because p' is compatible with the equivalence relation R', the morphism p is compatible with R, again because a universal effective epimorphism is a descent morphism. Hence there is a morphism

$$R \longrightarrow X \times_Y X.$$

To prove that R is (M)-effective and that Y is isomorphic to X/R, it suffices to prove that this morphism is an isomorphism. After base change from S to S', it becomes an isomorphism because R' is effective; it is therefore an isomorphism for the same reason as in 2.4.

Notice that the hypothesis in the proposition is satisfied if (M) is the family (M_0) of universal effective epimorphisms and if $\mathscr{C}$ has fiber products (1.10). We obtain the following.

Corollary 3.5.3. *Suppose that $\mathscr{C}$ has fiber products (fiber products over S would suffice). Let R be an equivalence relation in X over S, and let $S' \to S$ be a universal effective epimorphism. The following conditions are equivalent:*

(i) *R is universally effective in X;*

(ii) *R' is universally effective in X', and the canonical descent datum on X'/R' is effective.*

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R.

4. Topologies and Sheaves

The notion of a sieve, and the presentation of the notion of a topology adopted here (4.2.1), which is more intrinsic and in many respects more convenient than the presentation by covering families in [MA], are due to J. Giraud [AS].

4.1. Sieves

Definition 4.1.1. *A* sieve *of the category $\mathscr{C}$ is a subfunctor C of the final functor*

$$e : \mathscr{C}^\circ \longrightarrow \mathbf{Set}.$$

To every sieve C of $\mathscr{C}$ one associates the set $E(C)$ of objects X of $\mathscr{C}$ such that $C(X) \neq \varnothing$, that is, such that the structural morphism $X \to e$ factors through C. Thus

$$\begin{cases} X \in E(C) & \Longleftrightarrow \quad C(X) = e(X) = \{\varnothing\}, \\ X \notin E(C) & \Longleftrightarrow \quad C(X) = \varnothing. \end{cases} \tag{+}$$

The set $E = E(C)$ has the following property:

$$X \in E \text{ and } \mathrm{Hom}(Y, X) \neq \varnothing \quad \Longrightarrow \quad Y \in E. \tag{++}$$

(If the set $\mathrm{Ob}\,\mathscr{C}$ is endowed with its natural preorder, with Y dominating X when there is an arrow from Y to X, then the sets E satisfying (++) are precisely the subsets of $\mathrm{Ob}\,\mathscr{C}$ that contain every object dominating one of their elements.)[17]

Conversely, if E is a subset of $\mathrm{Ob}\,\mathscr{C}$ having property (++), then E can be written uniquely in the form $E(C)$, with C defined by the formulas (+). Thus sieves of $\mathscr{C}$ correspond bijectively to subsets of $\mathrm{Ob}\,\mathscr{C}$ satisfying (++). By abuse of language, we shall sometimes call the set $E(C)$ itself a sieve of $\mathscr{C}$.

Let C and C' be two sieves of $\mathscr{C}$.[18] Since both are subfunctors of the final functor e, it is equivalent to say that C dominates C' (for the domination relation in $\mathrm{Ob}\,\widehat{\mathscr{C}}$), that C is a subfunctor of C', or that $E(C) \subset E(C')$. In this case we say that C is *finer* than C'. This is an order relation on the set of sieves of $\mathscr{C}$. Moreover,

$$E(C) \cap E(C') = E(C \times C'),$$

so the set of sieves of $\mathscr{C}$ is filtered for the order relation "is finer than."

Every subset E of $\mathrm{Ob}\,\widehat{\mathscr{C}}$, and in particular every subset of $\mathrm{Ob}\,\mathscr{C}$, defines a sieve $C(E)$: the set of $X \in \mathrm{Ob}\,\mathscr{C}$ such that $F(X) \neq \varnothing$ for at least one $F \in E$ satisfies (++), and hence defines the desired sieve.

This sieve may also be defined as the image of the family of morphisms $\{F \to e \mid F \in E\}$, in the sense of the following definition.

Definition 4.1.2. *Let $\{F_i \to F\}$ be a family of morphisms of $\widehat{\mathscr{C}}$ with common target F. The* image *of this family is the subfunctor of F defined by*

$$S \longmapsto \bigcup_i \mathrm{Im}\, F_i(S) \subset F(S).$$

Proposition 4.1.3. *Formation of the image commutes with base change. In the preceding notation let I be the image of the family $\{F_i \to F\}$. For every morphism $G \to F$ in $\widehat{\mathscr{C}}$, the image of the family*

$$\{F_i \times_F G \longrightarrow G\}$$

is the subfunctor $I \times_F G$ of G.

Definition 4.1.4.0.[19] *Let C be a sieve of $\mathscr{C}$. If E is a subset of $\mathrm{Ob}\,\mathscr{C}$ such that $C(E) = C$, then E is called a* base *of C. Every sieve C has a base, for example $E(C)$.*

We now describe $\mathrm{Hom}(C, F)$, where C is a sieve of $\mathscr{C}$ and $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, by means of a base $\{S_i\}$ of C. For every pair (i, j) there is a diagram in $\widehat{\mathscr{C}}$:

$$\begin{array}{ccc} S_i \times S_j & \longrightarrow & S_i \\ \downarrow & & \downarrow \\ S_j & \longrightarrow & e. \end{array}$$

[17] Editor's note: Here "dominating" is meant in the sense of the preorder just mentioned: Y dominates X if there is an arrow $Y \to X$. On the other hand, if X, Y are two subobjects of an object Z, one says (cf. 2.4) that Y contains X if $X \subset Y$. To avoid any ambiguity between these usages, the sequel uses "dominating" in the first case and "containing" in the second.

[18] Editor's note: The following sentence has been expanded.

[19] Editor's note: The numbering 4.1.4.0 has been added to highlight this definition.

It gives a diagram of sets

$$\Gamma(F) = \mathrm{Hom}(e, F) \xrightarrow{\sigma} \prod_i \mathrm{Hom}(S_i, F) \underset{\tau_2}{\overset{\tau_1}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F),$$

with $\tau_1\sigma = \tau_2\sigma$. Hence there is a morphism

$$\mathrm{Hom}(e, F) \longrightarrow \mathrm{Ker}\left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)\right).$$

One checks immediately:

Proposition 4.1.4. *There is an isomorphism, functorial in F,*

$$\mathrm{Hom}(C, F) \xrightarrow{\sim} \mathrm{Ker}\left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)\right),$$

for which the diagram

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Ker}\left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)\right) \\ \| & & \uparrow \wr \\ \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Hom}(C, F) \end{array}$$

is commutative; the bottom arrow is induced by the canonical morphism $C \to e$.

Corollary 4.1.5. *Suppose that the fiber products $S_i \times S_j$ are representable, for example that the S_i are squarable. Then, for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, there is an isomorphism*

$$\mathrm{Hom}(C, F) \xrightarrow{\sim} \mathrm{Ker}\left(\prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times S_j)\right).$$

Remark 4.1.6. Let R be a sieve of $\mathscr{C}$, let $\overline{R}$ denote the full subcategory of $\mathscr{C}$ whose object set is $E(R)$, and let

$$i_R : \overline{R} \longrightarrow \mathscr{C}$$

be the inclusion functor. There is an isomorphism, functorial in $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$,

$$\mathrm{Hom}(R, F) \xrightarrow{\sim} \Gamma(F \circ i_R),$$

for which the diagram

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Hom}(R, F) \\ \downarrow \wr & & \downarrow \wr \\ \Gamma(F) & \longrightarrow & \Gamma(F \circ i_R) \end{array}$$

is commutative; the bottom arrow is induced by i_R.

Definition 4.1.7. *Let $\mathscr{C}$ be a category. A* sieve of the object S *of $\mathscr{C}$ is a sieve of the category $\mathscr{C}_{/S}$.*

A sieve of S is therefore a subobject of S in $\widehat{\mathscr{C}}$. It corresponds canonically to a subset of $\mathrm{Ob}\,\mathscr{C}_{/S}$ that contains the source of every arrow whose target it contains. By abuse of language, such a subset will also be called a sieve of S.

4.2. Topologies: Definitions

Definition 4.2.1. *A* topology *on a category $\mathscr{C}$ consists, for every object S of $\mathscr{C}$, of a set $J(S)$ of sieves of S, called* covering sieves *or* refinements *of S, satisfying the following axioms:*

(T 1) *For every refinement R of S and every morphism $T \to S$, the sieve $R \times_S T$ of T is covering ("stability under base change").*

(T 2) *If R, C are two sieves of S, if R is covering, and if for every $T \in \mathrm{Ob}\,\mathscr{C}$ and every morphism $T \to R$, the sieve $C \times_S T$ of T is covering, then C is a refinement of S.*[20]

(T 3) *If $C \supset R$ are two sieves of S and R is covering, then C is covering.*

(T 4) *For every S, S is a refinement of S.*

These axioms may be reformulated as follows. Suppose that a topology $S \mapsto J(S)$ on $\mathscr{C}$ is given. For every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, let $J(F)$ be the set of subfunctors R of F such that, for every morphism $T \to F$ in $\widehat{\mathscr{C}}$ with T representable, the sieve $R \times_F T$ of T is covering. By (T 1), this notation agrees with the preceding one. An $R \in J(F)$ is again called a *refinement* of F. The preceding axioms immediately imply:

($T'0$) If $F \supset G$ are objects of $\widehat{\mathscr{C}}$, and if $G \times_F S \in J(S)$ for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every morphism $S \to F$, then $G \in J(F)$.

($T'1$) If $G \in J(F)$ and $H \to F$ is a morphism in $\widehat{\mathscr{C}}$, then $G \times_F H \in J(H)$.

($T'2$) If $F \supset G \supset H$, if $G \in J(F)$, and if $H \in J(G)$, then $H \in J(F)$.

($T'3$) If $F \supset G \supset H$ and $H \in J(F)$, then $G \in J(F)$.

($T'4$) For every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, one has $F \in J(F)$.

Conversely, if for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$ one is given a set $J(F)$ of subobjects of F satisfying $T'0$–$T'4$, then $S \mapsto J(S)$ defines a topology on $\mathscr{C}$, and these two constructions are inverse to each other.

Properties $T'1$, $T'2$, and $T'3$ imply the further property[21]:

($T'5$) If G and H are two subobjects of F and $G, H \in J(F)$, then $G \cap H \in J(F)$.

Thus $J(F)$, ordered by $\supset$, is filtered. This observation will be useful below.

4.2.2. The topology defined by J is said to be *finer* than the topology defined by J' if $J(S) \supset J'(S)$ for every $S \in \mathrm{Ob}\,\mathscr{C}$. Equivalently, $J(F) \supset J'(F)$ for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$.

Every set of topologies on $\mathscr{C}$ has a greatest lower bound. More precisely, let I be an index set and, for each $i \in I$, let $S \mapsto J_i(S)$ be a topology on $\mathscr{C}$. Then

$$J(S) = \bigcap_{i \in I} J_i(S)$$

defines a topology, and it is the greatest lower bound of the given set.

In particular, suppose that for every $S \in \mathrm{Ob}\,\mathscr{C}$ a set $E(S)$ of sieves of S is given. The *topology generated by these sets* is the least fine topology for which every element of $E(S)$ is a refinement of S.

[20] Editor's note: In other words, if C is covering "locally with respect to the covering sieve R," then C is covering.

[21] Editor's note: $T'1$ and $T'2$ suffice: $G \cap H = G \times_F H$ belongs to $J(H)$ by $T'1$, and hence to $J(F)$ by $T'2$.

Definition 4.2.3. *Let $\{F_i \to F\}$ be a family of morphisms in $\widehat{\mathscr{C}}$, and let $G \subset F$ be its image (4.1.2). The family is* covering *if $G \in J(F)$. A morphism is covering if the one-member family consisting of that morphism is covering.*

This definition applies in particular to an inclusion: a sieve C of S is covering if and only if the canonical morphism $C \to S$ is covering.

Axioms $T'0$–$T'5$ give the following properties of covering families:

(C 0) If $\{F_i \to F\}$ is a family in $\widehat{\mathscr{C}}$ such that every representable base change $\{F_i \times_F S \to S\}$ is covering, then the original family is covering.

(C 1) If $\{F_i \to F\}$ is covering and $G \to F$ is any morphism, then $\{F_i \times_F G \to G\}$ is covering ("stability under base change").

(C 2) If $\{F_i \to F\}$ is covering and, for every i, $\{F_{ij} \to F_i\}$ is covering, then the composite family $\{F_{ij} \to F\}$ is covering ("stability under composition").

(C 3) If $\{G_j \to F\}$ is covering and $\{F_i \to F\}$ is a family such that for every j there is an i for which $G_j \to F$ factors through $F_i \to F$, then $\{F_i \to F\}$ is covering ("saturation").

(C 4) Every one-member family consisting of an isomorphism is covering.

Notice that (C 2) and (C 3) also imply:

(C 5) If $\{F_i \to F\}$ is a family for which there exists a covering family $\{G_j \to F\}$ such that $\{F_i \times_F G_j \to G_j\}$ is covering for every j, then $\{F_i \to F\}$ is covering ("a locally covering family is covering").

4.2.4. Conversely, let $\mathscr{C}$ be a category having fiber products, and suppose that for every $S \in \mathrm{Ob}\,\mathscr{C}$ one is given a set of families of morphisms in $\mathscr{C}$ with target S, called covering families, satisfying (C 1)–(C 4), and hence also (C 5). Let $J(S)$ be the set of sieves of S having a covering base (or, equivalently by (C 3), all of whose bases are covering). Then $S \mapsto J(S)$ defines a topology on $\mathscr{C}$, and the two preceding constructions are inverse to one another.

In applications it is inconvenient to consider all covering families, since one often has a fairly simple description of a "sufficient" number of them. This leads to the following definitions.

Definition 4.2.5.0.[22] *Let $\mathscr{C}$ be a category, and suppose given, for every $S \in \mathrm{Ob}\,\mathscr{C}$, a set $P(S)$ of families of morphisms in $\mathscr{C}$ with target S. The* topology generated by P *is the least fine topology for which the given families are covering.*

Definition 4.2.5. *A* pretopology *on a category $\mathscr{C}$ consists, for every $S \in \mathrm{Ob}\,\mathscr{C}$, of a set $R(S)$ of families $\{S_i \to S\}$, called* covering for the pretopology*, satisfying:*

(P 1) *For every $\{S_i \to S\} \in R(S)$ and every morphism $T \to S$, the fiber products $S_i \times_S T$ exist and $\{S_i \times_S T \to T\} \in R(T)$.*

(P 2) *If $\{S_i \to S\} \in R(S)$ and, for each i, $\{T_{ij} \to S_i\} \in R(S_i)$, then the composite family $\{T_{ij} \to S\}$ belongs to $R(S)$.*

(P 3) *Every one-member family consisting of an isomorphism is covering.*

[22] Editor's note: This definition, which appeared after 4.2.5 in the original, has been placed here because it will be used in 6.2.1 in a slightly more general setting than that of 4.2.5.

Proposition 4.2.6. *For each S, let $J(S)$ be the set of sieves of S that are covering for the topology generated by the pretopology R. Let $J_R(S)$ be the subset of $J(S)$ consisting of the sieves defined by the families in $R(S)$. Then $J_R(S)$ is cofinal in $J(S)$: every refinement of S contains a sieve defined by a family in $R(S)$.*

For every S, let $J'(S)$ be the set of sieves of S containing a sieve in $J_R(S)$. Clearly $J'(S) \subset J(S)$. To prove $J(S) = J'(S)$, it suffices to prove that the $J'(S)$ define a topology, that is, satisfy (T 1)–(T 4). Axioms (T 1), (T 3), and (T 4) are immediate; only (T 2) remains.[23]

Let $U \in J'(S)$, let C be a sieve of S, and suppose that $C \times_S T \in J'(T)$ for every $T \to U$. We must prove $C \in J'(S)$. By definition, U contains a refinement U' defined by a family $\{S_i \to S\} \in R(S)$. Since (T 3) has already been verified, it is enough to prove $U' \cap C \in J'(S)$; hence we may suppose $U = U'$. By hypothesis, $C \times_S S_i \in J'(S_i)$ for every i. Thus for each i there is a covering family $\{T_{ij} \to S_i\} \in R(S_i)$ such that $T_{ij} \to S_i$ factors through $C \times_S S_i \to S_i$. Consequently $T_{ij} \to S$ factors through $C \to S$. Hence C contains the sieve defined by the composite family $\{T_{ij} \to S\}$, and (P 2) completes the proof.

Axioms (P 1)–(P 3) are those of [MA]. Because pretopologies are so useful in practice, we shall interpret every important result in terms of a pretopology defining the topology under consideration.

Remark 4.2.7. One may introduce a slightly more general notion: for every S, one specifies a set of covering families satisfying (P 1), (P 3), and the conclusion of Proposition 4.2.6. This occurs in particular when the given families satisfy (P 1), (P 3), and (C 5). See [D].

Definition 4.2.8. *Let $\mathscr{C}$ be equipped with a topology, let S be an object of $\mathscr{C}$, and let $P(S')$ be a condition on an argument $S' \in \mathrm{Ob}\,\mathscr{C}_{/S}$. Suppose that $\mathrm{Hom}(S'', S') \neq \varnothing$ implies $P(S') \Rightarrow P(S'')$. The condition P is said to hold* locally on S *for the topology under consideration if the following equivalent conditions hold:*

(i) *The set of $S' \to S$ for which $P(S')$ holds is a refinement of S.*

(ii) *There is a refinement of S such that $P(S')$ holds for every S' in that refinement.*

(iii) *If the topology is defined by a pretopology, there is a covering family for that pretopology such that $P(S')$ holds for every S' in the family.*

Example 4.2.9. *Let $f : X \to Y$ be an S-morphism. We say that f is* locally an isomorphism *if there is a covering family $\{S_i \to S\}$ such that $f \times_S S_i$ is an isomorphism for every i. Equivalently, there is a refinement R of S such that $X(T) \to Y(T)$ is an isomorphism for every $T \to R$.*

Many other examples of this "local" language will appear below.

4.3. Presheaves, Sheaves, and the Sheaf Associated to a Presheaf

Definition 4.3.1. *Let $\mathscr{C}$ be a category. A* presheaf of sets *on $\mathscr{C}$ is a contravariant functor from $\mathscr{C}$ to the category of sets. The category*

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

is called the category of presheaves *on $\mathscr{C}$. If $\mathscr{C}$ is equipped with a topology, a presheaf P is* separated *(respectively, is a* sheaf*) if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $R \in J(S)$, the canonical map*

$$P(S) = \mathrm{Hom}(S, P) \longrightarrow \mathrm{Hom}(R, P) \qquad (+)$$

[23] Editor's note: Typographical errors in the original have been corrected in what follows.

is injective (respectively, bijective). The full subcategory of $\widehat{\mathscr{C}}$ whose objects are the sheaves is called the category of sheaves *and is denoted by $\widetilde{\mathscr{C}}$.*[24]

Proposition 4.3.2. *Let P be a separated presheaf (respectively, a sheaf). For every $H \in \mathrm{Ob}\,\widehat{\mathscr{C}}$ and every $R \in J(H)$, the canonical map*

$$\mathrm{Hom}(H, P) \longrightarrow \mathrm{Hom}(R, P) \qquad (+)$$

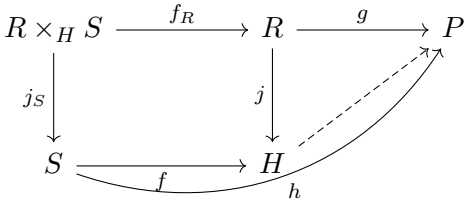
is injective (respectively, bijective).

Let P be separated, let H be a presheaf, let $R \in J(H)$, and let $u, v : H \to P$ satisfy $uj = vj$. For every $f : S \to H$, with $S \in \mathrm{Ob}\,\mathscr{C}$, the sieve $R \times_H S$ is a refinement of S, and $uf j_S = v f j_S$. This is expressed by the diagram

$$\begin{array}{ccccc} R \times_H S & \xrightarrow{j_S} & S & & \\ {\scriptstyle f_R}\downarrow & & {\scriptstyle f}\downarrow & & \\ R & \xrightarrow{j} & H & \overset{u}{\underset{v}{\rightrightarrows}} & P. \end{array}$$

Since P is separated, it follows that $uf = vf$. This holds for every representable S, so $u = v$.

Now suppose that P is a sheaf. Let $g : R \to P$; we show that it factors through H. For every $f : S \to H$, with $S \in \mathrm{Ob}\,\mathscr{C}$, the morphism $g \circ f_R : R \times_H S \to P$ factors uniquely through S, thereby defining a morphism $h : S \to P$. By uniqueness, h is plainly functorial in f:



Thus, for every S, we have defined a map $H(S) \to P(S)$ functorial in S, and hence a morphism $H \to P$ with the required properties.

Corollary 4.3.2.1.[25] *Let R and F be two sheaves. If R is a refinement of F, then $R = F$.*

Indeed, suppose that R is a refinement of F, and write $j : R \hookrightarrow F$ for the inclusion. By 4.3.2, $\mathrm{Hom}(F, R) = \mathrm{End}(R)$; hence there is a $\pi : F \to R$ such that $\pi j = \mathrm{id}_R$. Likewise, $\mathrm{End}(F) = \mathrm{Hom}(R, F)$, and $j\pi j = j$ implies $j\pi = \mathrm{id}_F$. Thus j is an isomorphism.

Proposition 4.3.3 ([AS], 1.3). *Let $\mathscr{C}$ be a category and P a presheaf on $\mathscr{C}$. For every $S \in \mathrm{Ob}\,\mathscr{C}$, let $J(S)$ be the set of sieves R of S such that, for every $T \to S$, the map*

$$\mathrm{Hom}(T, P) \longrightarrow \mathrm{Hom}(R \times_S T, P) \qquad (+)$$

is injective (respectively, bijective). Then the $J(S)$ define a topology on $\mathscr{C}$; that is, they satisfy (T 1)–(T 4).

Corollary 4.3.4. *For every $S \in \mathrm{Ob}\,\mathscr{C}$, let $K(S)$ be a family of sieves satisfying (T 1), and let P be a presheaf on $\mathscr{C}$. Then P is separated (respectively, a sheaf) for the topology generated by the $K(S)$ if and only if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $R \in K(S)$, the canonical map*

$$\mathrm{Hom}(S, P) \longrightarrow \mathrm{Hom}(R, P) \qquad (+)$$

[24] Editor's note: If Q is a subpresheaf of a separated presheaf P, then Q is separated. Indeed, for every sieve R of S, the composite $Q(S) \hookrightarrow P(S) \hookrightarrow P(R)$ is injective and factors through $Q(S) \to Q(R)$.

[25] Editor's note: This corollary has been added.

is injective (respectively, bijective).

Corollary 4.3.5. *For each $S \in \mathrm{Ob}\,\mathscr{C}$, let $R(S)$ be a set of families of morphisms in $\mathscr{C}$ with target S, satisfying (P 1), for example a pretopology. Let P be a presheaf on $\mathscr{C}$. Then P is separated (respectively, a sheaf) for the topology generated by R if and only if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every family $\{S_i \to S\} \in R(S)$, the map*

$$P(S) \longrightarrow \prod_i P(S_i)$$

is injective (respectively, the diagram

$$P(S) \longrightarrow \prod_i P(S_i) \rightrightarrows \prod_{i,j} P(S_i \times_S S_j)$$

is exact).

Definition 4.3.6. *The* canonical topology *on a category $\mathscr{C}$ is the finest topology for which every representable functor is a sheaf.*

Corollary 4.3.7. *A sieve R of S is a refinement for the canonical topology if and only if, for every morphism $T \to S$ in $\mathscr{C}$ and every $X \in \mathrm{Ob}\,\mathscr{C}$, the canonical map*

$$\mathrm{Hom}(T, X) \longrightarrow \mathrm{Hom}(R \times_S T, X)$$

is bijective.

Definition 4.3.8. *A sieve covering for the canonical topology is called a* universal effective epimorphic sieve.

Corollary 4.3.9. *A universal effective epimorphic family defines a universal effective epimorphic sieve. Conversely, every squarable family defining a universal effective epimorphic sieve is universal effective epimorphic.*

Return now to a category $\mathscr{C}$ equipped with an arbitrary topology and to the construction of the sheaf associated to a presheaf P. Let S be an object of $\mathscr{C}$. If $R \supset R'$ are two refinements of S, there is a diagram

$$\begin{array}{ccc} \mathrm{Hom}(S, P) & \longrightarrow & \mathrm{Hom}(R, P) \\ & \searrow & \downarrow \\ & & \mathrm{Hom}(R', P). \end{array}$$

As already observed, the ordered set $J(S)$ is filtered. Since $S \in J(S)$, there is an evident morphism

$$\mathrm{Hom}(S, P) \longrightarrow \varinjlim_{R \in J(S)} \mathrm{Hom}(R, P).$$

Definition 4.3.10.0.[26] Set

$$\check{H}^0(S, P) = \varinjlim_{R \in J(S)} \mathrm{Hom}(R, P).$$

One checks that $\check{H}^0(S, P)$ depends functorially on S, and hence defines a functor LP by

$$\mathrm{Hom}(S, LP) = \check{H}^0(S, P) = \varinjlim_{R \in J(S)} \mathrm{Hom}(R, P). \qquad (++)$$

[26] Editor's note: The numbering 4.3.10.0 has been added for later reference. Moreover, it follows from the definition that if $Q \to P$ is a monomorphism, then the same is true of $LP \to LQ$; thus L "preserves monomorphisms" (see also 4.3.16 for the more general statement that L "commutes with finite inverse limits").

By construction there are morphisms

$$\ell_P : P \longrightarrow LP, \qquad z_R : \mathrm{Hom}(R, P) \longrightarrow \mathrm{Hom}(S, LP).$$

Lemma 4.3.10.

(i) *For every refinement R of S and every $u : R \to P$, the diagram*

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ {\scriptstyle u}\uparrow & & \uparrow{\scriptstyle z_R(u)} \\ R & \xrightarrow{i_R} & S \end{array}$$

is commutative.

(ii) *For every morphism $v : S \to LP$, there exist a refinement R of S and a morphism $u : R \to P$ such that $v = z_R(u)$.*

(iii) *Let Q be a functor and let $u, v : Q \to P$ satisfy $\ell_P u = \ell_P v$. Then the kernel of the pair (u, v) is a refinement of Q.*

(iv) *Let $u : R \to P$ and $u' : R' \to P$. The equality $z_R(u) = z_{R'}(u')$ holds if and only if there is a refinement $R'' \subset R \cap R'$ of S on which u and u' coincide.*

Proof. For (i), we must prove $z_R(u) i_R = \ell_P u$. It is enough to compare the composites of these two morphisms with every morphism $g : T \to R$, where T is representable. Put $f = i_R g$ and $R' = R \times_S T$.[27] We have

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ {\scriptstyle u}\uparrow & & \uparrow{\scriptstyle z_R(u)} \\ R & \xrightarrow{i_R} & S \\ {\scriptstyle p}\uparrow & \overset{g}{\nwarrow} & \uparrow{\scriptstyle f} \\ R' & \xrightarrow[\sim]{i_{R'}} & T. \end{array}$$

By the definition of ℓ_P, $\ell_P u g = z_R(u) f$; this is the special case of the assertion in which i_R is an isomorphism. But $z_R(u) f = z_R(u) i_R g$, proving (i).

Assertions (ii) and (iv) merely express the definition of $\mathrm{Hom}(S, LP)$ as a direct limit.

For (iii), let K be the kernel of (u, v). For every morphism $f : S \to Q$, with S representable, $K \times_Q S$ is a subfunctor of the kernel of $uf, vf : S \to P$. By $T'0$, it is therefore enough to prove the assertion when $Q = S$ is representable. In that case, (ii) and (iv) show that K contains a refinement of S, and hence is itself a refinement of S.

One finally checks that $P \mapsto LP$ defines a functor

$$L : \widehat{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}},$$

and that $P \mapsto \ell_P$ defines a morphism of functors

$$\ell : \mathrm{Id}_{\widehat{\mathscr{C}}} \longrightarrow L.$$

[27] Editor's note: Let $p : R' \to R$ be the projection. Since i_R is a monomorphism, $p = g i_{R'}$. Let g' be the section of $i_{R'}$ defined by g. Then $pg' = g$, hence $pg' i_{R'} = g i_{R'} = p$. This implies $g' i_{R'} = \mathrm{id}_{R'}$, so $i_{R'} : R' \to T$ is an isomorphism with inverse g'.

We now state the essential result.

Proposition 4.3.11.

(i) *For every presheaf P, the presheaf LP is separated and $\ell_P : P \to LP$ is covering (4.2.3).*

(ii) *If P is a sheaf, then $P \to LP$ is an isomorphism.*

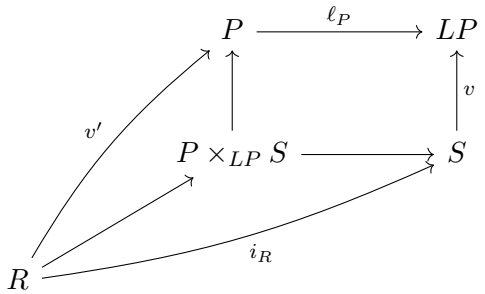
(iii) *For every presheaf P and every separated presheaf (respectively, sheaf) F, the map*

$$\mathrm{Hom}(\ell_P, F) : \mathrm{Hom}(LP, F) \longrightarrow \mathrm{Hom}(P, F)$$

is injective (respectively, bijective).

(iv) *If P is separated, then $\ell_P : P \to LP$ is a covering monomorphism (so P is a refinement of LP), and LP is a sheaf.*

Proof. (i) First, $P \to LP$ is covering. Indeed, for every $v : S \to LP$, Lemma 4.3.10(i),(ii) supplies a refinement R of S and a commutative diagram (the map from R to the fiber product is induced by $v' : R \to P$ and $i_R : R \to S$)

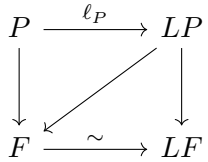


so $P \times_{LP} S \to S$ is covering. By (C 0), $P \to LP$ is covering.

Now suppose that two morphisms $v_1, v_2 : S \to LP$ induce the same morphism on a refinement R of S. There are refinements R_i, $i = 1, 2$, and morphisms $u_i : R_i \to P$ with $z_{R_i}(u_i) = v_i$. Replacing R by a smaller refinement, we may suppose $R_1 = R_2 = R$. The commutative diagram of Lemma 4.3.10(i) then gives $\ell_P u_1 = \ell_P u_2$. By 4.3.10(iii), u_1 and u_2 coincide on a refinement of R, hence on a refinement of S; by 4.3.10(iv), $z_R(u_1) = z_R(u_2)$. Thus LP is separated.

(ii) This is clear: if P is a sheaf, then $\mathrm{Hom}(S, P) \to \mathrm{Hom}(R, P)$ is already an isomorphism for every refinement R of S.

(iii) Let $u, v : LP \to F$ satisfy $u\ell_P = v\ell_P$. To show $u = v$, it is enough to prove $uf = vf$ for every $f : S \to LP$ with S representable. There are a refinement R of S and a morphism $g : R \to P$ with $f = z_R(g)$. The morphisms uf and vf agree on R, where both equal $u\ell_P g = v\ell_P g$. If F is separated, they therefore agree on S, proving injectivity. If F is a sheaf, the commutative diagram



shows that $\mathrm{Hom}(\ell_P, F)$ is also surjective.

(iv) Suppose that P is separated. To show that $P \to LP$ is a monomorphism, it is enough to consider $u, v : S \to P$, with S representable, satisfying $\ell_P u = \ell_P v$. By 4.3.10(iii),

u and v agree on a refinement of S, and therefore agree because P is separated. Thus $P \to LP$ is a monomorphism; since it is covering by (i), P is a refinement of LP.

It remains to show that LP is a sheaf. We already know from (i) that it is separated. Let $S \in \mathrm{Ob}\,\mathscr{C}$, let R be a refinement of S, and let $h : R \to LP$. We must find $u : S \to LP$ with $ui_R = h$. Since P is a refinement of LP, $R' = P \times_{LP} R$ is a refinement of R, and hence of S. Put $u = z_{R'}(h')$; the situation is

$$\begin{array}{ccccc} P & \xrightarrow{\ell_P} & LP & & \\ {\scriptstyle h'}\uparrow & & {\scriptstyle h}\uparrow & \nwarrow^{u} & \\ R' & \xrightarrow{j} & R & \xrightarrow{i_R} & S. \end{array}$$

Here the diagonal arrow is $u : S \to LP$. We have $ui_{R'} = \ell_P h' = hj$, hence $ui_R j = hj$. Since R' is a refinement of R and LP is separated, Proposition 4.3.2 gives $ui_R = h$.

Corollary 4.3.12.[28] *Let F be a sheaf and let R be a subobject of F in $\widehat{\mathscr{C}}$. Then R is a separated presheaf, $\ell_R : R \to LR$ is a covering monomorphism, and there is a commutative diagram*

$$\begin{array}{ccc} R & \xrightarrow{i} & F \\ & {\scriptstyle \ell_R}\searrow & {\scriptstyle j}\uparrow \\ & & LR. \end{array}$$

Consequently, R is a refinement of F if and only if j is an isomorphism.

We already observed that R is separated and that j is a monomorphism; see editor notes (24) and (26). By Proposition 4.3.11(iv), ℓ_R is a covering monomorphism. Thus, if j is an isomorphism, R is a refinement of F. Conversely, if i is covering, then so is j, and Proposition 4.3.2.1 shows that j is an isomorphism.

Remark 4.3.13. If $J'(S)$ is a cofinal subset of $J(S)$, then

$$\mathrm{Hom}(S, LP) = \varinjlim_{R \in J'(S)} \mathrm{Hom}(R, P).$$

In particular, let $S \mapsto R(S)$ be a pretopology generating the given topology. The functor L can be described using the covering families belonging to $R(S)$. Writing out the preceding formula recovers the construction of [MA].

Let $i : \widetilde{\mathscr{C}} \to \widehat{\mathscr{C}}$ be the inclusion functor. Proposition 4.3.11 gives the following theorem.

Theorem 4.3.14. *There is a unique functor $a : \widehat{\mathscr{C}} \to \widetilde{\mathscr{C}}$ for which the diagram*

$$\begin{array}{ccc} \widehat{\mathscr{C}} & \xrightarrow{L} & \widehat{\mathscr{C}} \\ {\scriptstyle a}\downarrow & & {\scriptstyle L}\downarrow \\ \widetilde{\mathscr{C}} & \xrightarrow{i} & \widehat{\mathscr{C}} \end{array}$$

is commutative; that is, $L(L(P))$ is a sheaf for every presheaf P. The functors i and a are adjoint: for every presheaf P and every sheaf F, there is an isomorphism, functorial in P and F,

$$\mathrm{Hom}_{\widehat{\mathscr{C}}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathscr{C}}}(a(P), F),$$

or simply

$$\mathrm{Hom}(P, F) \simeq \mathrm{Hom}(a(P), F).$$

[28] Editor's note: The statement and proof of this corollary have been expanded.

Definition 4.3.15. *The sheaf $a(P)$ is called the sheaf* associated *to the presheaf P.*

Remark 4.3.16. Since L is constructed using inverse limits and filtered direct limits, it commutes with finite inverse limits.[29]

Moreover, under the identification $L(P\times P) = LP\times LP$, the morphism $\ell_{P\times P}$ is identified with $\ell_P \times \ell_P$. Thus, for example, if P is a presheaf of groups, then LP is canonically a presheaf of groups and the canonical morphism $P \to LP$ is a morphism of groups. The same holds for the functor a. Consequently, if P is a presheaf of groups and F a sheaf of groups, there is an isomorphism

$$\mathrm{Hom}_{\widehat{\mathscr{C}}\text{-gr.}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathscr{C}}\text{-gr.}}(a(P), F).$$

See [D] for further details.

4.3.17. Let $\mathscr{V}$ be any category. A *presheaf on $\mathscr{C}$ with values in $\mathscr{V}$* is a contravariant functor from $\mathscr{C}$ to $\mathscr{V}$. Before defining sheaves with values in $\mathscr{V}$, we recall the definition of the inverse limit of a functor. Let $\mathscr{R}$ and $\mathscr{V}$ be categories, and let

$$F : \mathscr{R}^\circ \longrightarrow \mathscr{V}$$

be a contravariant functor. Write $\varprojlim F$ for the object of $\widehat{\mathscr{V}}$ defined by

$$\begin{aligned}(\varprojlim F)(X) &= \mathrm{Hom}_{\widehat{\mathscr{V}}}(X, \varprojlim F)\\ &= \varprojlim_{U\in \mathrm{Ob}\,\mathscr{R}} \mathrm{Hom}_{\mathscr{V}}(F(U), X)\\ &= \mathrm{Hom}(c_X, F),\end{aligned}$$

where X varies over the objects of $\mathscr{V}$, where c_X denotes the contravariant functor from $\mathscr{R}$ to $\mathscr{V}$ that sends every object to X and every arrow to id_X, and where the last Hom is taken in $\mathrm{Hom}(\mathscr{R}^\circ, \mathscr{V})$. If $\mathscr{R}$ has a final object $e_{\mathscr{R}}$, then $\varprojlim F = F(e_{\mathscr{R}})$. If $\mathscr{V} = \mathbf{Set}$, the functor $\varprojlim$ is identified with Γ.

Let S be an object of $\mathscr{C}$ and R a sieve of S. Let $\overline{R}$ be the full subcategory of $\mathscr{C}_{/S}$ whose object set is $E(R)$, and let $i_R : \overline{R} \to \mathscr{C}_{/S}$ be the canonical functor. A presheaf P on $\mathscr{C}$ with values in $\mathscr{V}$ restricts to a functor

$$P_S : (\mathscr{C}_{/S})^\circ \longrightarrow \mathscr{V}.$$

The functor i_R induces a morphism in $\widehat{\mathscr{V}}$:

$$P(S) = P_S(S) = \varprojlim P_S \longrightarrow \varprojlim(P_S \circ i_R).$$

We write

$$\varprojlim_R P = \varprojlim(P_S \circ i_R)$$

for the latter object of $\widehat{\mathscr{V}}$. In view of 4.1.6, Definition 4.3.1 generalizes as follows.

Definition 4.3.18. *A presheaf P on $\mathscr{C}$ with values in $\mathscr{V}$ is* separated *(respectively, a* sheaf*) if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $R \in J(S)$, the canonical morphism in $\widehat{\mathscr{V}}$*

$$P(S) \longrightarrow \varprojlim_R P$$

[29] Editor's note: In particular, if K is the kernel of a pair of morphisms of presheaves $u, v : Q \rightrightarrows P$, then LK is the kernel of $Lu, Lv : LQ \rightrightarrows LP$. This will be used in 4.4.5.

is a monomorphism (respectively, an isomorphism).

When $\mathscr{V}$ is the category **Gr** of groups (or any other category of sets endowed with algebraic structures defined by finite inverse limits), the following notions are equivalent (cf. [D]): a presheaf on $\mathscr{C}$ with values in **Gr** whose underlying presheaf of sets is a sheaf, and a group object in the category of sheaves of sets. With these identifications understood, we shall always regard sheaves with values in a category of sets endowed with algebraic structures defined by finite inverse limits as sheaves of sets equipped, in $\widetilde{\mathscr{C}}$, with the corresponding algebraic structure.

4.4. Exactness Properties of the Category of Sheaves

Theorem 4.4.1.

(i) *Arbitrary inverse limits exist in $\widetilde{\mathscr{C}}$, and "are computed in $\widehat{\mathscr{C}}$"; that is, the inclusion functor*

$$i : \widetilde{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}}$$

commutes with inverse limits. More explicitly, if (X_α) is an inverse system of sheaves, the presheaf

$$\varprojlim_\alpha i(X_\alpha) : S \longmapsto \varprojlim_\alpha X_\alpha(S)$$

is a sheaf, and

$$i\left(\varprojlim_\alpha X_\alpha\right) = \varprojlim_\alpha i(X_\alpha).$$

(ii) *Arbitrary direct limits exist in $\widetilde{\mathscr{C}}$. If (X_α) is a direct system of sheaves, then*

$$\varinjlim_\alpha X_\alpha = a\left(\varinjlim_\alpha i(X_\alpha)\right),$$

where $\varinjlim_\alpha i(X_\alpha)$ is the presheaf direct limit of the $i(X_\alpha)$:

$$\varinjlim_\alpha i(X_\alpha) : S \longmapsto \varinjlim_\alpha X_\alpha(S).$$

(iii) *The functor*

$$a : \widehat{\mathscr{C}} \longrightarrow \widetilde{\mathscr{C}}$$

commutes with arbitrary direct limits and finite inverse limits.

Assertions (i) and (ii) follow formally from the adjunction formula (4.3.14), and[30] the first assertion of (iii) follows from (ii). The second assertion of (iii) was already noted in 4.3.16.

Scholium 4.4.2. *This theorem makes it possible to use the following method for proving in $\widetilde{\mathscr{C}}$ a statement involving both arbitrary direct limits and finite inverse limits (for example, "every epimorphism is universally effective"; cf. below). One begins by proving the corresponding statement in the category of sets, then extends it pointwise to the category of presheaves. One then uses the preceding theorem to pass from the category of presheaves to the category of sheaves. Many examples of this method will occur below (4.4.3, 4.4.6, 4.4.9, and so forth).*

[30] Editor's note: the original has been modified here.

Finally, observe that statements concerning the category of presheaves are formally corollaries of the corresponding statements concerning the category of sheaves. It is enough to take the coarsest ("chaotic") topology, namely the topology defined by

$$J(S) = \{S\} \qquad \text{for every } S \in \mathrm{Ob}\,\mathscr{C};$$

every functor is a sheaf for this topology.

Proposition 4.4.3. *Let $\mathscr{F} = \{F_i \to F\}$ be a family of morphisms of sheaves. The following conditions are equivalent:*

(i) *$\mathscr{F}$ is an epimorphic family;*

(ii) *$\mathscr{F}$ is a universal effective epimorphic family (1.13);*

(iii) *$\mathscr{F}$ is covering (4.2.3);*

(iv) *the sheaf image of $\mathscr{F}$, that is, the sheaf associated with the presheaf image of $\mathscr{F}$ (4.1.2), is F.*

The equivalence of (iii) and (iv) follows from 4.3.12. The remaining equivalences will follow from the next lemmas.

Lemma 4.4.4. *Let $f : X \to Y$ be a monomorphism of sheaves that is also an epimorphism. Then f is an isomorphism.*

The lemma is clear first of all in the category of sets. Let us next prove it in the category of presheaves. Consider the presheaf

$$V : S \longmapsto Y(S) \amalg_{X(S)} Y(S);$$

this is the amalgamated sum of Y and Y over X in the category of presheaves.[31] Let $i_1, i_2 : Y \rightrightarrows V$ be the two coordinate morphisms. If $X \to Y$ is an epimorphism in the category of presheaves, then $i_1 = i_2$. In that case, for every S, the map $X(S) \to Y(S)$ is surjective; since it is also injective, it is bijective, and hence $X \to Y$ is an isomorphism.

Finally, work in the category $\widetilde{\mathscr{C}}$ of sheaves. By 4.4.1(ii), the amalgamated sum in $\widetilde{\mathscr{C}}$ of the sheaves Y and Y over X is the sheaf Z associated with the presheaf V. Consider the diagram

$$X \xrightarrow{f} Y \underset{i_2}{\overset{i_1}{\rightrightarrows}} V \xrightarrow{\tau} Z = a(V).$$

We have $i_1 f = i_2 f$, hence $\tau i_1 f = \tau i_2 f$, and therefore $\tau i_1 = \tau i_2$, since f is an epimorphism in $\widetilde{\mathscr{C}}$. By part (iii) of the following lemma, the presheaf V is separated, so τ is a monomorphism (4.3.11(iv)). Thus $i_1 = i_2$, and we saw above that this implies that $f : X \to Y$ is an isomorphism. This proves Lemma 4.4.4, once it has been verified that V is separated.

Let $R_\varnothing$ denote the presheaf that assigns the empty set to every $S \in \mathrm{Ob}\,\mathscr{C}$; it is an initial object of $\widehat{\mathscr{C}}$.[32]

Lemma 4.4.5.

[31] Editor's note: the remainder of the proof has been slightly modified.

[32] Editor's note: part (i) of Lemma 4.4.5 has been corrected, and the proofs of all three parts have been given in detail.

(i) *Suppose that* $R_\varnothing \notin J(S)$ *for every* $S \in \mathrm{Ob}\,\mathscr{C}$. *If* (X_i) *is a family of separated presheaves, then the coproduct presheaf* $\coprod_i X_i$ *is separated.*[33]

(ii) *Consider an equivalence relation in the category of presheaves*

$$X \overset{u}{\underset{v}{\rightrightarrows}} Y,$$

and let $w : Y \to Z$ *be the quotient. If* X *is a sheaf and* Y *is separated, then* Z *is separated.*

(iii) *Consider an amalgamated sum in the category of presheaves, in which* u *and* u' *are monomorphisms:*

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ {\scriptstyle u'}\downarrow & & \downarrow \\ Y' & \longrightarrow & V. \end{array}$$

If Y *and* Y' *are separated and* X *is a sheaf, then* V *is separated.*

Proof. For (i), set $X = \coprod_i X_i$. Let $S \in \mathrm{Ob}\,\mathscr{C}$, let $\tau : R \hookrightarrow S$ be a refinement of S, and let $x_1, x_2 \in X(S)$ satisfy $x_1 \circ \tau = x_2 \circ \tau$. There are indices i, j such that $x_1 \in X_i(S)$ and $x_2 \in X_j(S)$. Since $R \neq R_\varnothing$, there is a morphism $\phi : T \to R$ with $T \in \mathrm{Ob}\,\mathscr{C}$. Then

$$x_1 \circ \tau \circ \phi = x_2 \circ \tau \circ \phi.$$

Since $X(T)$ is the disjoint union of the $X_k(T)$, it follows that $i = j$. Because X_i is separated and is a subobject of X, the map

$$X_i(S) \longrightarrow X_i(R) \longrightarrow X(R)$$

is injective. Hence $x_1 = x_2$, which proves that X is separated.

We next prove (iii). Let $i : Y \to V$ and $j : Y' \to V$ be the canonical morphisms, and let K be the kernel of the pair

$$Y \times Y' \overset{p}{\underset{q}{\rightrightarrows}} V, \qquad p = i \circ \mathrm{pr}_1, \quad q = j \circ \mathrm{pr}_2\,.$$

For $S \in \mathrm{Ob}\,\mathscr{C}$, the monomorphisms u, u' allow us to identify $X(S)$ with its image in $Y(S)$, respectively in $Y'(S)$. Write $Z(S)$, respectively $Z'(S)$, for the complement. Then $V(S)$ identifies with the disjoint union of $Z(S)$, $Z'(S)$, and $X(S)$. It follows at once that

$$i(S) : Y(S) \longrightarrow V(S), \qquad j(S) : Y'(S) \longrightarrow V(S)$$

are injective and that

$$(u \times u')(S) : X(S) \longrightarrow K(S)$$

is bijective. Consequently i and j are monomorphisms, and $u \times u' : X \to K$ is an isomorphism.

[33] Editor's note: in general, the coproduct of two sheaves F, G is not a sheaf. Indeed, let $S_1, S_2 \in \mathrm{Ob}\,\mathscr{C}$. Suppose that the coproduct $S = S_1 \coprod S_2$ exists in $\mathscr{C}$, and that the fiber product $S_1 \times_S S_2$ is an initial object $\varnothing$ of $\mathscr{C}$ (cf. I, 1.8). Let R be the sieve of S with basis $\{S_1, S_2\}$. Then $(F \coprod G)(R)$ is the disjoint union of $F(S) \coprod G(S)$ and the sets $F(S_i) \times G(S_j)$, for $i \neq j$; thus $F \coprod G$ is not a sheaf in general. On the other hand, let $\mathscr{C}$ be the category with a single object S and id_S as its only morphism, equipped with the topology defined by $J(S) = \{R_\varnothing, S\}$. Then the only separated presheaves are $R_\varnothing$ and $X = \mathbf{h}_S$ (which is a sheaf), and $X \coprod X$ is not separated.

Set $U = Y \coprod Y'$. For a refinement $\tau : R \hookrightarrow S$, there is a commutative diagram

$$\begin{array}{ccc} U(S) & \xrightarrow{U(\tau)} & U(R) \\ \downarrow & & \downarrow \\ V(S) & \xrightarrow{V(\tau)} & V(R). \end{array}$$

Let $v_1, v_2 \in V(S)$ have the same image in $V(R)$. By the definition of V, they lift to elements $y_1, y_2 \in U(S)$; let z_1, z_2 denote their images in $U(R)$. Then z_1, z_2 have the same image in $V(R)$.

Since $i : Y \to V$ and $j : Y' \to V$ are monomorphisms, the maps $Y(R) \to V(R)$ and $Y'(R) \to V(R)$ are injective. Thus, because Y and Y' are separated, if y_1, y_2 both belong to $Y(S)$, or both belong to $Y'(S)$, then $y_1 = y_2$. Otherwise we may suppose that $y_1 \in Y(S)$ and $y_2 \in Y'(S)$, so that $z_1 \in Y(R)$ and $z_2 \in Y'(R)$. Since $i \circ z_1 = j \circ z_2$, the morphism

$$z_1 \boxtimes z_2 : R \longrightarrow Y \times Y'$$

factors through $K = X$. Moreover, since X is a sheaf, the restriction map identifies $X(S)$ with $X(R)$. Hence there is an $x \in X(S)$ such that

$$u(x) \circ \tau = z_1 = y_1 \circ \tau, \qquad u'(x) \circ \tau = z_2 = y_2 \circ \tau.$$

Since Y and Y' are separated, $u(x) = y_1$ and $u'(x) = y_2$, and therefore $v_1 = v_2$. This proves that V is separated.

Finally, we prove (ii). First observe that the morphism of presheaves

$$X \xrightarrow{u \boxtimes v} K,$$

where K is the kernel of the pair

$$w \circ \mathrm{pr}_i : Y \times Y \rightrightarrows Z,$$

is an isomorphism. Indeed, for every $T \in \mathrm{Ob}\,\mathscr{C}$, $X(T)$ is an equivalence relation on $Y(T)$, and hence the diagram

$$X(T) \xrightarrow{u \boxtimes v} Y(T) \times Y(T) \underset{w \circ \mathrm{pr}_2}{\overset{w \circ \mathrm{pr}_1}{\rightrightarrows}} Z(T)$$

is exact in the category of sets.

Now let $S \in \mathrm{Ob}\,\mathscr{C}$, and let $g_1, g_2 : S \to Z$ be two morphisms that agree on a refinement $\tau : R \hookrightarrow S$. By the construction of Z,

$$Z(S) = Y(S)/X(S),$$

so there are morphisms $f_1, f_2 : S \to Y$ such that $wf_i = g_i$. We have $wf_1\tau = wf_2\tau$; by the preceding observation, there is a morphism $\phi : R \to X$ such that

$$u\phi = f_1\tau, \qquad v\phi = f_2\tau.$$

Since X is a sheaf, there is a morphism $\psi : S \to X$ with $\phi = \psi\tau$. Thus, in $Y(R)$,

$$u\psi\tau = f_1\tau, \qquad v\psi\tau = f_2\tau.$$

Because $Y(S) \to Y(R)$ is injective, Y being separated, it follows that $u\psi = f_1$ and $v\psi = f_2$, hence $g_1 = g_2$. Thus Z is separated.

Lemma 4.4.6.[34]

[34]Editor's note: the statement of the lemma and its proof have been given in detail.

(i) *Let $\{F_i \xrightarrow{f_i} F\}$ be a family of morphisms of presheaves, and let $G \subset F$ be its presheaf image. Then the following diagram in $\widehat{\mathscr{C}}$ is exact:*

$$\left(\coprod_i F_i\right) \times_G \left(\coprod_j F_j\right) \overset{\mathrm{pr}_1}{\underset{\mathrm{pr}_2}{\rightrightarrows}} \coprod_i F_i \longrightarrow G.$$

Equivalently, for every presheaf H, the following diagram of sets is exact:

$$\mathrm{Hom}(G,H) \longrightarrow \prod_i \mathrm{Hom}(F_i,H) \overset{p}{\underset{q}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(F_i \times_G F_j, H). \qquad (*)$$

(ii) *Every covering family of morphisms of sheaves is a universal effective epimorphic family.*

Proof. For (i), let H be a presheaf. The map

$$f^* : \mathrm{Hom}(G,H) \longrightarrow \prod_i \mathrm{Hom}(F_i,H), \qquad \phi \longmapsto (\phi \circ f_i)_i,$$

is injective. Indeed, for each $S \in \mathrm{Ob}\,\mathscr{C}$, the map $\phi(S)$ is determined by the maps $(\phi \circ f_i)(S)$, since the family

$$f_i(S) : F_i(S) \longrightarrow G(S)$$

is surjective. It is clear that the image of f^* is contained in $\ker(p,q)$.

Conversely, let $\phi_i : F_i \to H$ be a family of morphisms such that, for every i,j, the diagram

$$\begin{array}{ccc} F_i \times_G F_j & \longrightarrow & F_j \\ \downarrow & & \downarrow \phi_j \\ F_i & \xrightarrow{\phi_i} & H \end{array}$$

commutes. For every S, the map

$$\coprod_i F_i(S) \longrightarrow H(S)$$

then factors uniquely through a map $\phi(S) : G(S) \to H(S)$. These maps define a morphism $\phi : G \to H$ satisfying $f^*(\phi) = (\phi_i)$. This proves the exactness of $(*)$, and hence (i).

We prove (ii). Since covering families are stable under base change, it is enough to show that every covering family is effective epimorphic. Let $\{F_i \to F\}$ be a covering family of morphisms of sheaves, and let G be its presheaf image. Because the family is covering, the monomorphism $G \hookrightarrow F$ is covering as well; therefore, by 4.3.12,

$$a(G) = F.$$

Moreover, since $G \hookrightarrow F$ is a monomorphism, the fiber products $F_i \times_G F_j$ and $F_i \times_F F_j$ coincide. Part (i) therefore shows that, for every presheaf H, the diagram

$$\mathrm{Hom}(G,H) \longrightarrow \prod_i \mathrm{Hom}(F_i,H) \overset{p}{\underset{q}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(F_i \times_F F_j, H) \qquad (**)$$

is exact. If, in addition, H is a sheaf, then

$$\mathrm{Hom}(G,H) = \mathrm{Hom}(a(G),H) = \mathrm{Hom}(F,H).$$

Thus (∗∗) shows that $\{F_i \to F\}$ is an effective epimorphic family in the category $\widetilde{\mathscr{C}}$ of sheaves.

Lemma 4.4.7. *Every family of morphisms of sheaves $\{F_i \to F\}$ factors as a covering family $\{F_i \to G\}$ followed by a monomorphism $G \to F$.*

Indeed, it is enough to take G to be the sheaf image of the given family.

Proof of Proposition 4.4.3. We saw in 4.4.6 that (iii) implies (ii), while (ii) evidently implies (i). Finally, let $\{F_i \to F\}$ be an epimorphic family. By Lemma 4.4.7 it factors as a covering family followed by a monomorphism. The latter monomorphism, being dominated[35] by an epimorphic family, is itself an epimorphism; hence it is an isomorphism by 4.4.4.

Remark 4.4.8.[36] Because the presheaf image of the family $\mathscr{F} = \{F_i \to F\}$ is separated, the construction of the associated sheaf shows that the conditions of Proposition 4.4.3 are also equivalent to the following:

(v) for every $S \in \mathrm{Ob}\,\mathscr{C}$, each $f \in F(S)$ lies locally in the image of $\mathscr{F}$; that is,

(vi) for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $f \in F(S)$, the set of morphisms $S' \to S$ such that the image of f in $F(S')$ belongs to the image of one of the $F_i(S')$ is a refinement of S;

(vii) if the topology is defined by a pretopology $S \mapsto \mathscr{R}(S)$, then, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $f \in F(S)$, there is a family $\{S_j \to S\} \in \mathscr{R}(S)$ such that, for each j, the image f_j of f in $F(S_j)$ belongs to the image of one of the $F_i(S_j)$.

Remark 4.4.8 bis. If the sheaf F is representable, the preceding conditions are also equivalent to:

(viii) the set of morphisms $T \to F$, with $T \in \mathrm{Ob}\,\mathscr{C}$, for which there exist an i and a commutative diagram

$$\begin{array}{ccc} & & F_i \\ & & \downarrow \\ T & \longrightarrow & F \end{array}$$

is a refinement of F.

Indeed, if (viii) holds, the presheaf image of the F_i is dominated by[37] a refinement of F, which implies that the family is covering. Conversely, apply (vi) to $\mathrm{id}_F \in F(F)$.

In pictorial language, this condition says: *locally on F, there is an i for which $F_i \to F$ has a section.* In particular, a morphism

$$G \longrightarrow F,$$

where G is a sheaf and F is a representable sheaf, is covering if and only if it has a section locally on F.

The following lemmas will be useful in VI_B, 8.1 and 8.2.[38]

Lemma 4.4.8.1. *Let $Q \subset P$ be presheaves of groups on $\mathscr{C}$, with Q normal in P. Suppose that P is separated and that Q is a sheaf. Then the presheaf of groups P/Q is separated. Consequently,*

$$a(P/Q) = L(P/Q) = \varinjlim_{R \in J(S)} (P/Q)(R).$$

[35] Editor's note: recall (cf. editor's note 17) that a morphism $g : G \to H$ is said to be *dominated* by a family of morphisms $F_i \to H$ if that family factors through g.

[36] Editor's note: the numbering of the original, which had two paragraphs numbered 4.4.7, has been corrected to agree with later references.

[37] Editor's note: "contains" has been replaced by "is dominated by"; cf. editor's note 17.

[38] Editor's note: Lemmas 4.4.8.1 and 4.4.8.2 have been added.

It is enough to prove the first assertion, since the second follows from it by 4.3.11(iv) and (ii). Let $S \in \mathrm{Ob}\,\mathscr{C}$, let R be a sieve of S, and let $y \in P(S)/Q(S)$ have identity image in $(P/Q)(R)$. We must show that $y = 1$.

By hypothesis, the morphism $P(S) \to P(R)$ is injective, while $Q(S) \to Q(R)$ is an isomorphism, and the upper row in the following commutative diagram is exact:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Q(S) & \longrightarrow & P(S) & \longrightarrow & P(S)/Q(S) & \longrightarrow & 1 \\ & & \downarrow\sim & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & Q(R) & \longrightarrow & P(R) & \longrightarrow & (P/Q)(R). & & \end{array}$$

The result follows if the lower row is exact. Let $f : R \to P$ have identity image in $(P/Q)(R)$, and let $\phi : T \to R$ with $T \in \mathrm{Ob}\,\mathscr{C}$. Then $f \circ \phi$ is the identity in

$$(P/Q)(T) = P(T)/Q(T),$$

that is, $f \circ \phi \in Q(T)$. Hence $\phi \mapsto f \circ \phi$ is a functorial map $R(T) \to Q(T)$, and so defines a morphism of functors $\pi : R \to Q$ such that $\pi(\mathrm{id}_R) = f$. Thus $f \in Q(R)$. This proves the exactness of the lower row and establishes the lemma.

Lemma 4.4.8.2. *Let $H \subset G$ be presheaves of groups on $\mathscr{C}$.*

(i) *If H is normal in G, then $L(H)$ is normal in $L(G)$.*

(ii) *If H is central in G, then $L(H)$ is central in $L(G)$.*

Let $S \in \mathrm{Ob}\,\mathscr{C}$. We must show (cf. I 2.3.6) that $L(H)(S)$ is normal, respectively central, in $L(G)(S)$. Let

$$h \in L(H)(S), \qquad g \in L(G)(S).$$

There are a sieve R of S and elements $h' \in H(R)$, $g' \in G(R)$ such that

$$h = z_R(h'), \qquad g = z_R(g')$$

in the notation of 4.3.10. Since z_R is a group morphism,

$$ghg^{-1}h^{-1} = z_R(g'h'g'^{-1}h'^{-1}).$$

In case (i),

$$g'h'g'^{-1}h'^{-1} \in H(R),$$

and therefore $ghg^{-1}h^{-1} \in L(H)(S)$. In case (ii), $g'h'g'^{-1}h'^{-1} = 1$, and hence $ghg^{-1}h^{-1} = 1$.

Proposition 4.4.9. *Every equivalence relation in $\widetilde{\mathscr{C}}$ is universally effective (3.3.3). More precisely, let R be a $\widetilde{\mathscr{C}}$-equivalence relation on the sheaf X. Then the sheaf associated with the separated presheaf*

$$i(X)/i(R) : S \longmapsto X(S)/R(S)$$

is a universal effective quotient of X by R.

Let X/R denote the sheaf quotient of X by R, which exists by 4.4.1(ii):

$$X/R = a\big(i(X)/i(R)\big).$$

We must show that $X \to X/R$ is a universal effective epimorphism and that the morphism

$$f : R \longrightarrow X \times_{X/R} X$$

is an isomorphism. The first assertion has already been proved in 4.4.3. As for f, it is obtained by applying the functor a to the morphism

$$i(R) \longrightarrow i(X) \times_{i(X/R)} i(X),$$

or, since $i(X)/i(R)$ is separated by 4.4.5(ii), so that

$$i(X)/i(R) \longrightarrow i(X/R)$$

is a monomorphism, to the canonical morphism

$$i(R) \longrightarrow i(X) \times_{i(X)/i(R)} i(X).$$

Thus we are reduced to proving the same statement in the category of presheaves. But $i(X)/i(R)$ is the presheaf

$$S \longmapsto X(S)/R(S),$$

so the assertion reduces to its set-theoretic analogue, where it is immediate.

Proposition 4.4.10. *Under the hypotheses of 4.4.9, let Y be a subsheaf of X, and write R_Y for the equivalence relation induced on Y by R. Then the canonical morphism (3.1.6)*

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. It identifies Y/R_Y with a subsheaf of X/R, namely the sheaf image of the composite

$$Y \longrightarrow X \longrightarrow X/R.$$

The morphism of presheaves

$$i(Y)/i(R_Y) = i(Y)/i(R)_{i(Y)} \longrightarrow i(X)/i(R)$$

is a monomorphism. Since the functor a is left exact (4.3.16), it carries monomorphisms to monomorphisms; hence

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. The final assertion follows from the commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y/R_Y & \longrightarrow & X/R \end{array}$$

and the fact that $Y \to Y/R_Y$ is covering.

In view of this proposition, we shall always identify Y/R_Y with a subsheaf of X/R.

Proposition 4.4.11. *Let R be a $\widetilde{\mathscr{C}}$-equivalence relation on the sheaf X. For every subsheaf Y of X stable under R, let $Y' = Y/R_Y$, regarded as a subsheaf of $X' = X/R$. Then*

$$Y = Y' \times_{X'} X,$$

and the maps

$$Y \longmapsto Y/R_Y, \qquad Y' \longmapsto Y' \times_{X'} X$$

define a bijective correspondence between the subsheaves Y of X stable under R and the subsheaves Y' of X'.

If Y' is a subsheaf of X', then $Y' \times_{X'} X$ is a subsheaf of X stable under R.[39] If Y' is obtained by passage to the quotient from a subsheaf Y of X, then Y is a subobject of $Y' \times_{X'} X$. It is therefore enough to prove the following: if Y and Y_1 are two subsheaves of X, both stable under R, with Y_1 containing Y, and if their quotients Y/R_Y and Y_1/R_{Y_1} are identical, then $Y = Y_1$.

We may plainly reduce to the case $Y_1 = X$. Write

$$P = i(X)/i(R), \qquad Q = i(Y)/i(R_Y)$$

for the corresponding presheaves. The diagram

$$\begin{array}{ccc} X & \longrightarrow & P \\ \uparrow & & \uparrow \\ Y & \longrightarrow & Q \end{array}$$

is Cartesian. On the other hand, there is a commutative diagram

$$\begin{array}{ccc} Q & \longrightarrow & a(Q) \\ \downarrow & & \downarrow\sim \\ P & \longrightarrow & a(P). \end{array}$$

Since $Q \hookrightarrow a(Q)$ is covering by 4.3.11, the monomorphism $Q \hookrightarrow P$ is covering, so Q is a refinement of P. After base change, Y is a refinement of X. Because X and Y are sheaves, 4.3.12 then gives $Y = X$.

4.4.12. In particular, if Y is a subsheaf of X and $Y' = Y/R_Y$, the preceding correspondence defines a subsheaf $\overline{Y}$ of X, stable under R, containing Y, and minimal with these properties. It is called the *saturation* of Y for the equivalence relation R.

4.5. The Case of a Topology Coarser than the Canonical Topology

By 4.3.6 and 4.3.8, the following conditions on a topology $\mathcal{T}$ on $\mathscr{C}$ are equivalent:

(i) $\mathcal{T}$ is coarser than the canonical topology of $\mathscr{C}$;

(ii) every representable presheaf is a sheaf for $\mathcal{T}$;

(iii) every covering sieve for $\mathcal{T}$ is universal effective epimorphic.

If $\mathcal{T}$ is defined by a pretopology $S \mapsto \mathscr{R}(S)$, these conditions are also equivalent to:

(iv) every family belonging to $\mathscr{R}(S)$ is universal effective epimorphic.

When these conditions hold, the canonical functor $\mathscr{C} \to \widehat{\mathscr{C}}$ factors through a functor

$$j_{\mathscr{C}} = j : \mathscr{C} \longrightarrow \widetilde{\mathscr{C}};$$

we also write $j(S) = \widetilde{S}$.[40]

Proposition 4.5.1. *The functor $j : \mathscr{C} \to \widetilde{\mathscr{C}}$ is fully faithful and commutes with arbitrary inverse limits. In particular, it is left exact and therefore preserves algebraic structures defined by finite inverse limits.*

[39] Editor's note: moreover, $(Y' \times_{X'} X)/R = Y'$.

[40] Editor's note: we shall also write $\mathbf{h}_S = \widehat{S}$; cf. the first commutative diagram in 4.5.4.

This follows immediately from the commutative factorization

$$\begin{array}{ccc} & \mathscr{C} & \\ {\scriptstyle j}\swarrow & & \searrow \\ \widetilde{\mathscr{C}} & \xrightarrow{i} & \widehat{\mathscr{C}} \end{array}$$

and from 4.4.1(i).

Before giving further properties of j, we must define the topology induced on a category $\mathscr{C}/S$. We no longer suppose here that the given topology is necessarily coarser than the canonical topology. A sieve of an object T in $\mathscr{C}$, together with a morphism $T \to S$, naturally defines a sieve of T in $\mathscr{C}/S$, because the definition of a sieve of T depends only on the category

$$\mathscr{C}/T = (\mathscr{C}/S)/T.$$

For example, if the sieve is defined by a family $\{T_i \to T\}$, its image in $\mathscr{C}/S$ is defined by the same family, now regarded as a family of morphisms in $\mathscr{C}/S$.

Declaring these images of covering sieves to be covering defines a topology on $\mathscr{C}/S$, called the *topology induced* by the given topology. In the terminology of [AS], 2.3, it is the coarsest topology on $\mathscr{C}/S$ for which the canonical functor

$$i_S : \mathscr{C}/S \longrightarrow \mathscr{C}$$

is a comorphism.[41] By definition, the identifications

$$(\mathscr{C}/S)/T = \mathscr{C}/T$$

respect these topologies.

Proposition 4.5.2. *Let S be a representable sheaf on $\mathscr{C}$, and let $F \to S$ be a morphism in $\widehat{\mathscr{C}}$. The presheaf*

$$S' \longmapsto \mathrm{Hom}_S(S', F)$$

on $\mathscr{C}/S$ is separated (respectively, is a sheaf) if and only if F is a separated presheaf (respectively, a sheaf) on $\mathscr{C}$.

For every functor P, one has (I, 1.4.1)

$$\mathrm{Hom}(P, F) = \coprod_{f \in \mathrm{Hom}(P,S)} \mathrm{Hom}_f(P, F).$$

[42] Let $S' \in \mathrm{Ob}(\mathscr{C})$, and let $\eta : C' \to S'$ be a covering sieve. Since S is a sheaf, the map $f \mapsto f \circ \eta$ is a bijection

$$\mathrm{Hom}(S', S) \xrightarrow{\sim} \mathrm{Hom}(C', S).$$

Consequently, the map

$$\mathrm{Hom}_{\widehat{\mathscr{C}}}(S', F) \longrightarrow \mathrm{Hom}_{\widehat{\mathscr{C}}}(C', F)$$

decomposes as the "disjoint union" of the maps

$$\mathrm{Hom}_f(S', F) \longrightarrow \mathrm{Hom}_{f\circ\eta}(C', F).$$

The proposition follows.

[41] Editor's note: that is, for every object $T \to S$ of $\mathscr{C}/S$, every covering sieve R' of $i_S(T \to S) = T$, when regarded as a sieve of the object $(T \to S) \in \mathrm{Ob}(\mathscr{C}/S)$, is covering.

[42] Editor's note: the argument implicit in the original has been expanded below.

Corollary 4.5.3. *If a topology on $\mathscr{C}$ is coarser than the canonical topology, then the topology it induces on $\mathscr{C}/S$ is coarser than the canonical topology of $\mathscr{C}/S$.*

Corollary 4.5.4. *Suppose that the given topology on $\mathscr{C}$ is coarser than the canonical topology. For every $S \in \mathrm{Ob}(\mathscr{C})$, there is an equivalence of categories*

$$\widetilde{\mathscr{C}_{/S}} \xrightarrow{\sim} \widetilde{\mathscr{C}/S}.$$

The following diagrams commute up to isomorphism; every unlabelled arrow in them is an equivalence:

$$\begin{array}{ccccccc}
\mathscr{C}_{/S} & \xrightarrow{(j_{\mathscr{C}})_{/S}} & \widetilde{\mathscr{C}}_{/\widetilde{S}} & \xrightarrow{(i_{\mathscr{C}})_{/\widetilde{S}}} & \widehat{\mathscr{C}}_{/\widehat{S}} & \xrightarrow{(a_{\mathscr{C}})_{/\widehat{S}}} & \widetilde{\mathscr{C}}_{/\widetilde{S}} \\
\| & & \downarrow\wr & & \downarrow\wr & & \downarrow\wr \\
\mathscr{C}_{/S} & \xrightarrow{j_{\mathscr{C}/S}} & \widetilde{\mathscr{C}/S} & \xrightarrow{i_{\mathscr{C}/S}} & \widehat{\mathscr{C}/S} & \xrightarrow{a_{\mathscr{C}/S}} & \widetilde{\mathscr{C}/S}.
\end{array}$$

If $T \to S$ is an object of $\mathscr{C}/S$, one also has the commutative diagram

$$\begin{array}{ccccc}
(\widetilde{\mathscr{C}_{/\widetilde{S}}})_{/\widetilde{T}} & \longrightarrow & (\widetilde{\mathscr{C}/S})_{/\widetilde{T}} & & \\
\downarrow\wr & & & \searrow & (\widetilde{\mathscr{C}/S})_{/T}\,. \\
\widetilde{\mathscr{C}}_{/\widetilde{T}} & \longrightarrow & \widetilde{\mathscr{C}/T} & \overset{\sim}{\nearrow} &
\end{array}$$

The commutativity of the first two squares follows from the definition of the equivalence $\widetilde{\mathscr{C}_{/\widetilde{S}}} \to \widetilde{\mathscr{C}/S}$. To prove the commutativity of the last square, it is enough to observe that the following square commutes:

$$\begin{array}{ccc}
\widehat{\mathscr{C}}_{/\widehat{S}} & \xrightarrow{L'} & \widehat{\mathscr{C}}_{/\widehat{S}} \\
\downarrow & & \downarrow \\
\widehat{\mathscr{C}/S} & \xrightarrow{L''} & \widehat{\mathscr{C}/S}.
\end{array}$$

Here L' is the restriction of the functor $L_{\mathscr{C}}$ to $\widehat{\mathscr{C}}_{/\widehat{S}}$, and L'' is the functor $L_{\mathscr{C}/S}$. This is seen immediately by returning to the definition of the functors L, given after 4.3.9. The second diagram is simply the restriction to the categories of sheaves of the corresponding diagram for the categories of presheaves (Exposé I, no. 1), which is commutative.

Scholium 4.5.5. The preceding assertions show that, when the given topology is coarser than the canonical topology, one may identify $\mathscr{C}$ with a full subcategory of $\widetilde{\mathscr{C}}$, itself a full subcategory of $\widehat{\mathscr{C}}$. Under these identifications one may make the customary abuses of language concerning $\mathscr{C} \to \widehat{\mathscr{C}}$, justified by the preceding commutative diagrams. In particular, the first diagram of 4.5.4 shows that the functor a may be used without any special precaution.

In the next subsection we shall see that the identification of $\mathscr{C}$ with a full subcategory of $\widetilde{\mathscr{C}}$, unlike the corresponding identification with $\widehat{\mathscr{C}}$, commutes with the formation of certain direct limits; we shall then explain how this fact is to be used.

From now on, unless expressly stated otherwise, we assume that the given topology is coarser than the canonical topology, and we systematically make the identifications described above.

Proposition 4.5.6. *Let F and G be two sheaves over S, and let $f : F \to G$ be an S-morphism. The following conditions on f are equivalent:*

(i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\widetilde{\mathscr{C}}$;*

(ii) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\widetilde{\mathscr{C}}_{/S} = \widetilde{\mathscr{C}/S}$.*

For monomorphisms and isomorphisms this is immediate, since it is a question about presheaves. For epimorphisms it follows from the description of epimorphisms as covering morphisms (4.4.3) and from the fact that, by the definition of the induced topology, these are the same in $\widetilde{\mathscr{C}}$ and in $\widetilde{\mathscr{C}/S}$.

Proposition 4.5.7. *Let $f : F \to G$ be a morphism of sheaves. The following conditions are equivalent:*

(i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism);*

(ii) *for every $S \in \mathrm{Ob}(\mathscr{C})$, the morphism $f_S : F_S \to G_S$ is locally a monomorphism (respectively, an epimorphism, respectively, an isomorphism), that is:*

(iii) *for every $S \in \mathrm{Ob}(\mathscr{C})$, the set of morphisms $T \to S$ for which $F_T \to G_T$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) is a refinement of S.*

If the given topology is defined by a pretopology $\mathscr{R}$, these conditions are also equivalent to:

(iv) *for every $S \in \mathrm{Ob}(\mathscr{C})$, there is a covering family $\{S_i \to S\} \in \mathscr{R}(S)$ such that, for every i, $F_{S_i} \to G_{S_i}$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism).*

If $\mathscr{C}$ has a final object e, it is enough in conditions (ii), (iii), and (iv) to take $S = e$.

Clearly (ii), (iii), and (iv) are equivalent. To prove the equivalence of (i) and (ii), as well as the assertion concerning a final object, it is enough to show that (ii) implies (i) and that the properties in question are stable under base change. We prove the latter point first. For monomorphisms and isomorphisms it is immediate, since it is a question about presheaves. For epimorphisms it follows from the fact that every epimorphism of sheaves is universal (4.4.3).

It remains to show that (ii) implies (i). Suppose first that $f_S : F_S \to G_S$ is locally a monomorphism (respectively, an isomorphism). There is then a covering sieve C of S such that f_T is a monomorphism (respectively, an isomorphism) for every $T \to S$ belonging to C. Since an inverse limit of monomorphisms (respectively, isomorphisms) is again one, $\mathrm{Hom}(C, f)$ is a monomorphism (respectively, an isomorphism); see 4.1.4. The commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(C,F) & \xrightarrow{\mathrm{Hom}(C,f)} & \mathrm{Hom}(C,G) \\ \uparrow\wr & & \uparrow\wr \\ F(S) & \xrightarrow{f(S)} & G(S) \end{array}$$

therefore shows that $f(S)$ is injective (respectively, bijective).

Finally, suppose that $f : F \to G$ is locally an epimorphism, and let $H \subset G$ be its image. For every $S \to G$, the sheaf $H \times_G S$ is the image of

$$f \times_G S : F \times_G S \longrightarrow S.$$

To prove that f is an epimorphism, it is enough to prove that H is a refinement of G, or equivalently that $H \times_G S$ is a refinement of S for every S. But this holds after each base change $T \to S$ belonging to a refinement of S, since $f \times_G S$ is locally covering. Hence $H \times_G S$ is itself a refinement of S, by Axiom (T 2).

Corollary 4.5.8. *Let F and G be two sheaves over S, and let $f : F \to G$ be an S-morphism. Then f is a monomorphism, respectively an epimorphism, respectively an isomorphism, if and only if it is so locally on S.*

Remark 4.5.9. The proof of the proposition shows that its assertion about monomorphisms (respectively, isomorphisms) remains valid if one assumes only that F is a separated presheaf (respectively, a sheaf) and G is an arbitrary presheaf (respectively, a separated presheaf).

We return temporarily to the case of an arbitrary topology and make the following definition.

Definition 4.5.10. *Let $G \to F$ be a morphism in $\widehat{\mathscr{C}}$. We say that G is a* relative sheaf over F *if, for every F-functor H and every refinement R of H, the canonical map*

$$\mathrm{Hom}_F(H, G) \xrightarrow{\sim} \mathrm{Hom}_F(R, G) \qquad (+)$$

is bijective.

Proposition 4.5.2 immediately gives the following generalization.

Proposition 4.5.11. *If F is a sheaf, then G is a relative sheaf over F if and only if G is a sheaf.*

Lemma 4.5.12. *In a sequence*

$$X \longrightarrow T \longrightarrow S,$$

where X, T, S are three objects of $\widehat{\mathscr{C}}$, suppose that X is a relative sheaf over T. Then

$$U = \prod_{T/S} X$$

is a relative sheaf over S.

Indeed, for every S-functor Y, one has

$$\mathrm{Hom}_S(Y, U) = \mathrm{Hom}_T(T \times_S Y, X).$$

If C is a sieve of Y, then $T \times_S C$ is a sieve of $T \times_S Y$, and the conclusion is immediate.

Corollary 4.5.13. *The presheaves $\mathrm{Hom}_{T/S}(X, Y)$, $\mathrm{Isom}_S(X, Y)$, and the analogous presheaves are sheaves whenever all the arguments occurring in them are sheaves.*

Indeed, all these presheaves are constructed using fiber products and presheaves $\prod_{T/S} X$ (I, 1.7 and II, 1). It is therefore enough to verify the assertion for a presheaf $\prod_{T/S} X$; in that case it follows from 4.5.11 and 4.5.12.

4.6. Description of the Quotient of a Sheaf by an Equivalence Relation

Recall that the given topology $\mathcal{T}$ is assumed to be coarser than the canonical topology.

Proposition 4.6.1. *Let $R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X$ be a $\widetilde{\mathscr{C}}$-equivalence relation in the sheaf X. Define $F \in \mathrm{Ob}(\widehat{\mathscr{C}})$ as follows: for each $S \in \mathrm{Ob}(\mathscr{C})$,*[43]

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ whose quotient} \\ \text{by } R_Z \text{ is } S, \text{ that is, for which } R_Z \rightrightarrows Z \to S \text{ is exact} \end{array} \right\}.$$

[43] Editor's note: $R \times S$ denotes the equivalence relation in $X \times S$ defined by $R \times S_{\mathrm{diag}} \subset X \times X \times S \times S$, and R_Z denotes the equivalence relation that it induces on Z (cf. 3.1.6).

For every sheaf Y, $\mathrm{Hom}(Y, F)$ *is naturally identified with the set*

$$\left\{ \begin{array}{l} \text{sub-}Y\text{-sheaves of } X \times Y, \text{ stable under } R \times Y, \\ \text{and having quotient } Y \end{array} \right\}.$$

In particular, the subsheaf $R \subset X \times X$ corresponds to an element $p \in \mathrm{Hom}(X, F)$, and the exact diagram

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \overset{p}{\to} F$$

identifies F with the sheaf quotient X/R.

Indeed, put $Q = X/R$. For every sheaf Y and every morphism $f \in \mathrm{Hom}(Y, Q)$, corresponding to a section $s : Y \hookrightarrow Q \times Y$, consider the diagram

$$\begin{array}{ccccc} R \times Y & \rightrightarrows & X \times Y & \longrightarrow & Q \times Y \\ & & \uparrow & & \uparrow s \\ & & Z & \longrightarrow & Y, \end{array} \qquad (*)$$

in which the right-hand square is cartesian. By 4.4.11, Z is immediately seen to be a sub-Y-sheaf of $X \times Y$, stable under $R \times Y$, with quotient Y. Conversely, every Z of this kind arises from a unique section of $Q \times Y$ over Y. Taking Y first to be representable gives an isomorphism $Q \simeq F$; taking arbitrary Y then gives the asserted description of $\mathrm{Hom}(Y, F)$. Finally, the canonical morphism $X \to Q$ corresponds to the sub-X-sheaf $R \subset X \times X$, which completes the proof.

Corollary 4.6.2. *Let G be any subfunctor of F such that* $\mathrm{Hom}(X, G) \subset \mathrm{Hom}(X, F)$ *contains R. Then the canonical morphism $p : X \to F$ factors through G.*

Since p is covering (4.4.9 and 4.4.3), it follows that G is a refinement of F. In particular, every subsheaf G of F satisfying the preceding condition equals F (4.3.12).

4.6.3. We now consider the case in which X and R are representable. We begin with some terminology. Besides the conditions (a)–(d) introduced in 3.4.1, we shall use the following conditions on a family (M) of morphisms of $\mathscr{C}$; for completeness, we repeat conditions (a)–(c):

(a) (M) is stable under base change.

(b) The composite of two elements of (M) belongs to (M).

(c) Every isomorphism belongs to (M).

($d_{\mathcal{T}}$) Every element of (M) is covering. [44]

($e_{\mathcal{T}}$) Let $f : X \to Y$ be a morphism of $\mathscr{C}$. If there is a refinement R of Y such that, for every $Y' \to R$, the morphism $X \times_Y Y' \to Y'$ belongs to (M), then f belongs to (M).

Conditions (a) and (b) imply

(a') the fiber product of two elements of (M) belongs to (M).

On the other hand, by 4.3.9, conditions (a) and ($d_{\mathcal{T}}$) imply

(d') every element of (M) is a universal effective epimorphism.

[44] Editor's note: $\mathcal{T}$ denotes the given topology on $\mathscr{C}$, which is coarser than the canonical topology.

4.6.4. When $\mathscr{C}$ has fiber products, the family of covering morphisms, denoted $(M_{\mathcal{T}})$, satisfies the preceding conditions. Indeed (cf. 4.2.3), (a) follows from (C 1), (b) from (C 2), (c) from (C 4), $(d_{\mathcal{T}})$ from the definition, and $(e_{\mathcal{T}})$ from (C 5). Thus the results established below for a family satisfying these conditions apply in particular to $(M_{\mathcal{T}})$. In particular, $\mathcal{T}$ may be taken to be the canonical topology and (M) to be the family of universal effective epimorphisms.

Lemma 4.6.5. *Let (M) be a family of morphisms satisfying conditions (a)–$(e_{\mathcal{T}})$ above. Let R be a $\mathscr{C}$-equivalence relation of type (M) in $X \in \mathrm{Ob}(\mathscr{C})$. Let $\widetilde{X}$ be the sheaf defined by X, let $\widetilde{R}$ be the $\widetilde{\mathscr{C}}$-equivalence relation in $\widetilde{X}$ defined by R, and let $\widetilde{X}/\widetilde{R}$ be the sheaf quotient. Then R is (M)-effective if and only if $\widetilde{X}/\widetilde{R}$ is representable. When these conditions hold, $\widetilde{X}/\widetilde{R}$ is represented by the quotient X/R.*

Suppose first that R is (M)-effective, and put $Y = X/R$. The canonical morphism $p : X \to Y$ belongs to (M), hence is covering by $(d_{\mathcal{T}})$. The corresponding morphism

$$\widetilde{p} : \widetilde{X} \longrightarrow \widetilde{X}/\widetilde{R}$$

is therefore a universal effective epimorphism in $\widetilde{\mathscr{C}}$ (4.4.3). Consequently it identifies $\widetilde{X}/\widetilde{R}$ with the quotient of $\widetilde{X}$ by the equivalence relation R' defined by $\widetilde{p}$. Since the canonical functor $\mathscr{C} \to \widetilde{\mathscr{C}}$ commutes with fiber products, R' is precisely $\widetilde{R}$, because R is the equivalence relation defined by p.

Conversely, suppose that $\widetilde{X}/\widetilde{R}$ is represented by an object Y of $\mathscr{C}$. Let $p : X \to Y$ be the morphism deduced from the canonical morphism $\widetilde{X} \to \widetilde{X}/\widetilde{R}$; by 4.4.3 it is covering. As before, R is plainly the equivalence relation defined by p. It remains only to show that $p \in (M)$. The cartesian diagram

$$\begin{array}{ccccc} R & \xrightarrow{\sim} & X \times_Y X & \longrightarrow & X \\ {\scriptstyle p_2}\searrow & & \downarrow{\scriptstyle \mathrm{pr}_2} & & \downarrow{\scriptstyle p} \\ & & X & \xrightarrow{p} & Y \end{array}$$

shows that, after the covering base change p, the morphism p becomes p_2, which belongs to (M). The assertion follows from $(e_{\mathcal{T}})$.

Corollary 4.6.5.1. [45] *Let (M) satisfy conditions (a)–$(e_{\mathcal{T}})$ above, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups such that $f \in (M)$. Suppose that $\mathrm{Ker}(f)$ is representable (as is the case when $\mathscr{C}$ has a final object e). Then the equivalence relation in G defined by $H = \mathrm{Ker}(f)$ is (M)-effective, and G' represents the sheaf quotient $\widetilde{G}/\widetilde{H}$ for the topology $\mathcal{T}$.*

This follows from 4.6.5 and 3.4.7.1.

We can now state the main result of this subsection.

Theorem 4.6.6. *Let (M) be a family of morphisms satisfying Axioms (a)–$(e_{\mathcal{T}})$ of 4.6.3. Let R be a $\mathscr{C}$-equivalence relation of type (M) (cf. 3.4.3) in an object X of $\mathscr{C}$. Consider the functor $F \in \mathrm{Ob}(\widehat{\mathscr{C}})$ defined by*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S, \text{ stable under } R \times S, \\ \text{whose quotient by } R_Z \text{ is } S \end{array} \right\}.$$

Let F_0 be the subfunctor of F defined as follows: $F_0(S)$ consists of the representable $Z \in F(S)$, that is,

$$F_0(S) = \left\{ \begin{array}{l} \text{sub-}\mathscr{C}/S\text{-objects } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ R_Z \text{ is } (M)\text{-effective with quotient } S, \text{ i.e. such that} \\ Z \to S \text{ belongs to } (M) \text{ and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

Then:

[45] Editor's note: This corollary has been added.

(i) *The morphism* $p \in \mathrm{Hom}(X, F) = F(X)$ *defined by the subobject* $R \subset X \times X$ *identifies* F *with the sheaf quotient of* X *by* R.

(ii) *The following conditions are equivalent:*

(a) F *is representable;*

(b) F_0 *is representable;*

(c) R *is* (M)*-effective.*

Under these conditions, $F = F_0 = X/R$.

(iii) *Let* (N) *be a family of morphisms stable under base change, such that for every covering family* $\{S_i \to S\}$ *and every family* $\{T_i \to S_i\}$ *of morphisms in* (N)*, every descent datum on the* T_i *relative to* $\{S_i \to S\}$ *is effective. Suppose that* X *is squarable (cf. 1.6.0) and that* $R \to X \times X$ *belongs to* (N)*. Then* $F_0 = F$.

Proof. Assertion (i) was proved in 4.6.1.

For (ii), we have already proved the equivalence of (a) and (c), as well as the equality $F = X/R$. It remains to prove that either (b) or (c) implies $F_0 = F$. First note, as stated in the theorem, that F_0 is indeed a subfunctor of F: if $S \in \mathrm{Ob}(\mathscr{C})$ and $Z \in F_0(S)$, then $Z \to S$ is squarable, and hence $Z \times_S S' \in F_0(S')$ for every $S' \to S$. Since $R \in F(X)$ belongs to $F_0(X)$, Corollary 4.6.2 shows that (b) implies $F_0 = F$.

Now suppose that (c) holds, and let Q be an object of $\mathscr{C}$ representing X/R. Then $X \to Q$ belongs to (M). For every $S \in \mathrm{Ob}(\mathscr{C})$ and every $Z \in F(S)$, diagram $(*)$ of 4.6.1 gives

$$Z = S \times_{Q \times S} (X \times S).$$

Thus Z is representable and $Z \to S$ belongs to (M), so $Z \in F_0(S)$.

For (iii), let $f \in \mathrm{Hom}(S, F)$ correspond to $Z \in F(S)$. We must show that f factors through F_0, or equivalently that Z is representable. This is immediate when f factors through X, by the following lemma.

Lemma 4.6.7. *Let* $x_0 \in X(S)$*. The image of* x_0 *in* $F(S)$ *corresponds to the subsheaf* $Z \subset X_S$ *defined by the two cartesian squares*

$$\begin{array}{ccccc} X_S & \xrightarrow{\mathrm{id}_{X_S} \times x_0} & X_S \times_S X_S & \longrightarrow & X \times X \\ \uparrow & & \uparrow & & \uparrow \\ Z & \longrightarrow & R_S & \longrightarrow & R. \end{array}$$

This follows at once from the description of the morphism $X \to F$.

Returning to the proof of the theorem, suppose first that f factors through X. Then Z is representable; moreover, since $R \to X \times X$ belongs to (N), so does $Z \to X_S$.

In general, f need not factor through X. Since $X \to F$ is covering (4.4.3), however, 4.4.8(vii) gives a covering family $\{S_i \to S\}$ and, for each i, a morphism $S_i \to X$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & F \\ \uparrow & & \uparrow f \\ S_i & \longrightarrow & S \end{array}$$

commute. By the preceding argument, the morphism $f_i : S_i \to F$ defined by this diagram belongs to $\mathrm{Hom}(S_i, F_0)$ and corresponds to the subsheaf $Z \times_S S_i \subset X_{S_i}$. The morphism $Z \times_S S_i \to X_{S_i}$ belongs to (N), and the family $\{X_{S_i} \to X_S\}$ is covering. It therefore remains only to prove the following proposition.

Proposition 4.6.8. *Let $\{S_i \to S\}$ be a covering family and let Z be a sheaf over S. Suppose that, for every i, the S_i-functor $Z \times_S S_i$ is represented by an object T_i. Then the family (T_i) carries a canonical descent datum relative to $\{S_i \to S\}$. The sheaf Z is representable if and only if this datum is effective; in that case the descended object represents Z.*

First observe that, by 4.4.3, $\{S_i \to S\}$ is a universal effective epimorphic family in $\widetilde{\mathscr{C}}$, and hence a descent family in $\widehat{\mathscr{C}}$ (2.3). If Z is represented by an object T, then $T \times_S S_i$, regarded as a sheaf, is isomorphic to $Z \times_S S_i$. Thus the descent datum on the T_i is effective, and its uniquely determined descended object is isomorphic to T, hence represents Z.

Conversely, suppose that the canonical descent datum on the T_i is effective, and let T be the descended object. Because $\{S_i \to S\}$ is a descent family in $\widetilde{\mathscr{C}}$, there is an S-morphism $T \to Z$ whose base change to each S_i is the canonical morphism $T_i \to Z \times_S S_i$. This morphism is locally an isomorphism; since both T and Z are sheaves, 4.5.8 implies that it is an isomorphism.

Corollary 4.6.9. *Let R be an (M)-effective equivalence relation in X. For every sheaf F, the map*

$$\mathrm{Hom}(X/R, F) \longrightarrow \mathrm{Hom}(X, F)$$

identifies the first set with the subset of the second consisting of morphisms compatible with R.

Corollary 4.6.10. *Let $\mathcal{T}'$ be a topology coarser than $\mathcal{T}$, for which the morphisms in (M) are covering. Under the hypotheses of 4.6.6(iii), X/R is also the sheaf quotient of X by R for every topology lying between $\mathcal{T}'$ and the canonical topology.*

Remark 4.6.11. If, in the statement of 4.6.6(iii), one assumes in addition that the descended morphism $T \to S$ belongs to (N) whenever the hypotheses there are satisfied, then the inclusion morphisms $Z \hookrightarrow X_S$ also belong to (N). This follows immediately from the construction of Z by descent.

Remark 4.6.12. The implications

$$\text{(c)} \Longrightarrow \text{(b)} \Longrightarrow \text{(a)} \qquad \text{and} \qquad \text{(c)} \Longrightarrow [F_0 = F = X/R]$$

were proved without using the sufficiency direction of Lemma 4.6.5, the only place where condition $(e_{\mathcal{T}})$ enters. They therefore remain valid when (M) satisfies only conditions (a)–$(d_{\mathcal{T}})$. The family of squarable covering morphisms is an example of such a family (M) (compare 4.6.4). For the canonical topology these are precisely the universal effective epimorphisms. Hence:

Corollary 4.6.13. *Let R be a universal effective equivalence relation in X. Then the object X/R of $\mathscr{C}$ is the sheaf quotient of X by R for the canonical topology. It represents the following functor: $(X/R)(S)$ is the set of sub-$\mathscr{C}/S$-objects Z of X_S, stable under $R \times S$, for which the induced equivalence relation is universal effective and has quotient S.*

Similarly, for an arbitrary topology:

Corollary 4.6.14. *Let (M) be the family of squarable covering morphisms. If R is an (M)-effective equivalence relation in X, then the object X/R of $\mathscr{C}$ is the sheaf quotient of X by R, and it represents the functor F_0 of 4.6.6.*

Scholium 4.6.15. We can now make the following clarifications to 4.5.5. In questions involving inverse limits alone—fiber products, algebraic structures, and so forth—the results of Exposé I and 4.5.5 allow us to identify $\mathscr{C}$, without distinction, with a full subcategory of either $\widetilde{\mathscr{C}}$ or $\widehat{\mathscr{C}}$. The same is not true for questions that mix inverse and direct limits.

Whenever both inverse and direct limits occur, and especially when passing to quotients (for example, when putting a group structure on the quotient of a group by an invariant subgroup), we shall regard the given category as embedded in the category of sheaves. Thus, if R is a $\mathscr{C}$-equivalence relation in an object X of $\mathscr{C}$, the notation X/R will mean the sheaf quotient of X by R, formerly denoted $j(X)/j(R)$, and, when this sheaf is representable, also the object that represents it. The preceding results show that, in the most important cases, a quotient in $\mathscr{C}$ is also a quotient in the category of sheaves. In any event, we shall not use the notation X/R for a quotient in $\mathscr{C}$ that does not coincide with the quotient in $\widetilde{\mathscr{C}}$—for instance, one that is not universal. This modifies the definitions of no. 3.

To study a problem of this kind, one therefore works first in the category of sheaves, where all the usual results hold (cf. no. 4.4), and then specializes the resulting statements to the original category, using the results of the present subsection and, when available, criteria for the effectivity of descent. Examples of this method will appear in the following subsections.

4.7. Use of Effectivity Criteria: Isomorphism Theorem

In this subsection we give an example of the use of effectivity criteria. The initial data are a topology $\mathcal{T}$ on $\mathscr{C}$, still assumed coarser than the canonical topology; a family (M) of morphisms of $\mathscr{C}$ satisfying Axioms (a)–$(e_{\mathcal{T}})$ of 4.6.3; and a family (N) of morphisms of $\mathscr{C}$ which may satisfy the following axioms:

(a) (N) is stable under base change;

$(f_{\mathcal{T}})$ morphisms in (N) descend for the given topology. More precisely, for every $S \in \mathrm{Ob}(\mathscr{C})$, every covering family $\{S_i \to S\}$, and every family $\{T_i \to S_i\}$ of morphisms in (N), every descent datum on the T_i relative to $\{S_i \to S\}$ is effective; if T denotes the descended object, then $T \to S$ belongs to (N).

Since every element of (M) is covering (condition 4.6.3 $(d_{\mathcal{T}})$), Axiom $(f_{\mathcal{T}})$ implies the following axiom:[46]

(f_M) if $Y \to X$ belongs to (N) and $X \to X_1$ belongs to (M), every descent datum on Y relative to $X \to X_1$ is effective; if Y_1 denotes the descended object, then $Y_1 \to X_1$ belongs to (N).

Here is an example, to be treated later: $\mathscr{C}$ is the category of schemes, $\mathcal{T}$ is the faithfully flat quasi-compact topology, (M) is the family of faithfully flat quasi-compact morphisms, and (N) is the family of closed immersions, or the family of quasi-compact immersions.[47]

Taking 4.6.11 into account, the principal result of 4.6.6 may be recalled as follows.

Proposition 4.7.1. *Let X be a squarable object of $\mathscr{C}$, and let R be an equivalence relation of type (M) in X. Suppose that $R \to X \times X$ belongs to (N), where (N) satisfies (a) and $(f_{\mathcal{T}})$. Then the sheaf quotient X/R is given by*

$$(X/R)(S) = \{Z \subseteq X_S \mid Z \text{ is an } S\text{-subobject stable under } R \times S,\ Z \to X_S \text{ belongs to } (N),\quad Z \to S \text{ is covering or belongs to } (M),\ R_Z \simeq Z \times_S Z\}.$$

We also have the following result.

[46] Editor's note: Axiom (f_M), which originally appeared before Proposition 4.7.2, has been moved here.

[47] Editor's note: cf. §6.4; see also VI_A, 5.3.1.

Proposition 4.7.2. *Let $X \in \mathrm{Ob}(\mathscr{C})$, let R be an (M)-effective equivalence relation in X, and let (N) be a family of morphisms satisfying (a) and (f_M). For every subobject Y of X which is stable under R and for which $Y \to X$ belongs to (N), the equivalence relation R_Y induced in Y by R is (M)-effective. Moreover, the quotient*

$$Y/R_Y = Y'$$

is a subobject of $X' = X/R$, and $Y' \to X'$ belongs to (N).

The map $Y \mapsto Y'$ is a bijection from the set of subobjects Y of X which are stable under R and for which $Y \to X$ belongs to (N), onto the set of subobjects Y' of X' for which $Y' \to X'$ belongs to (N). Its inverse is

$$Y' \longmapsto Y' \times_{X'} X.$$

Indeed, since R is (M)-effective, the morphism $X \to X'$ belongs to (M). Let Y' be a subobject of X' such that the canonical morphism $Y' \to X'$ belongs to (N). Then $Y = Y' \times_{X'} X$ is a subobject of X stable under R, and the morphisms $Y \to X$ and $Y \to Y'$ belong respectively to (N) and (M), since both families are stable under base change. Let R_Y denote the equivalence relation induced in Y by R. By 4.4.11, the sheaf quotient Y/R_Y is represented by Y'; hence 4.6.5 shows that R_Y is (M)-effective.

Conversely, let Y be a subobject of X, stable under R, such that $Y \to X$ belongs to (N). Since Y is stable under R, its two inverse images in

$$R = X \times_{X'} X$$

are identical, and Y is equipped with a descent datum relative to $X \to X'$. The required conclusion follows from Axiom (f_M) for the family (N).

Corollary 4.7.3. *Let $X \in \mathrm{Ob}(\mathscr{C})$, and let R be an (M)-effective equivalence relation in X. Suppose in addition that $R \to X \times X$ belongs to (N), where (N) satisfies (a) and (f_T). Then, for every Y as in 4.7.2, $R_Y \to Y \times Y$ also belongs to (N), and consequently 4.7.1 gives*

$$\begin{aligned}(Y/R_Y)(S) = \{Z \subseteq Y_S \mid\ & Z \text{ is an } S\text{-subobject stable under } R_Y \times S,\\ & Z \to Y_S \text{ belongs to } (N) \text{ (hence so does } Z \to X_S),\\ & Z \to S \text{ is covering}, \quad R_Z \simeq Z \times_S Z\,\}.\end{aligned}$$

5. Passage to the Quotient and Algebraic Structures

5.1. Principal Homogeneous Bundles

Definition 5.1.0.[48] We recall (III 0.1) that an object X equipped with a right action of H is said to be *formally principal homogeneous*[49] under H if the canonical morphism of functors

$$X \times H \longrightarrow X \times X, \qquad (x, h) \longmapsto (x, xh),$$

is an isomorphism. Equivalently (cf. *loc. cit.*), for every $S \in \mathrm{Ob}\,\mathscr{C}$, the set $X(S)$ is formally principal homogeneous under $H(S)$: it is either empty or principal homogeneous under

[48] Editor's note: the numbering 5.1.0 has been introduced for later reference.

[49] Editor's note: one also says "pseudo-torsor"; cf. EGA IV$_4$, 16.5.15. The more general notion of a formally homogeneous object (not necessarily principal homogeneous) is defined in Addendum 6.7.1 at the end of this exposé.

$H(S)$. In particular, when H acts on itself by right translations, it is formally principal homogeneous under itself.

Definition 5.1.1. An object X with an action of H is said to be *trivial* if it is isomorphic, as an H-object, to H with its translation action.

Proposition 5.1.2. *If X is formally principal homogeneous under H, there is an isomorphism*

$$\Gamma(X) \xrightarrow{\sim} \mathrm{Isom}_{H\text{-obj.}}(H, X)$$

of principal homogeneous sets under $\Gamma(H)$.

To a section x of X, associate the morphism $H \to X$ defined on sets by $h \mapsto xh$. The assertion follows immediately by reduction to the set-theoretic case.

Corollary 5.1.3. *There is an isomorphism of H-objects*

$$X \xrightarrow{\sim} \mathrm{Isom}_{H\text{-obj.}}(H, X).$$

Corollary 5.1.4. *An object with a group action is trivial if and only if it is formally principal homogeneous and has a section.*

Definition 5.1.5. Let $\mathscr{C}$ be a category equipped with a topology. An S-object X with an action of the S-group H is called a *principal homogeneous bundle under H* if it is locally trivial, that is, if the following equivalent conditions hold:[50]

(i) the collection of morphisms $T \to S$ for which the functor $X \times_S T$ is trivial under $H \times_S T$ is a refinement of S;

(ii) for a pretopology defining the given topology, there is a covering family $\{S_i \to S\}$ such that, for every i, the S_i-functor $X \times_S S_i$ is trivial under the S_i-group functor $H \times_S S_i$, equivalently, it has a section over S_i.

Proposition 5.1.6. *Let $\mathscr{C}$ be a category equipped with a topology $\mathcal{T}$, and let (M) be a family of morphisms of $\mathscr{C}$ satisfying axioms (a)–($e_{\mathcal{T}}$) of 4.6.3. Let H be an S-group whose structure morphism $H \to S$ belongs to (M), and let P be an S-object with an action of H. The following conditions are equivalent:*

(i) *P is a principal homogeneous bundle under H (Definition 5.1.5);*

(ii) *P is formally principal homogeneous under H, and its structure morphism $P \to S$ belongs to (M);*

(iii) *there exists a morphism $S' \to S$ belonging to (M) such that, after base change from S to S', P becomes trivial; that is, $P \times_S S'$ is trivial under $H \times_S S'$;*

(iv) *H acts freely and (M)-effectively on P, and the quotient P/H is isomorphic to S.*

First, (ii) and (iv) are equivalent. In either case $P \to S$ belongs to (M), hence is squarable, which ensures that the fiber products $H \times_S P$ and $P \times_S P$ are representable. Condition (ii) plainly implies (iii): one may take $S' = P$, since formal principal homogeneity implies that $P \times_S P$ is trivial under $H \times_S P$ by 5.1.4, its diagonal section providing a section. Condition (iii) plainly implies (i), because $\{S' \to S\}$ is a covering family by axiom $(d_{\mathcal{T}})$.

It remains to prove (i)⇒(ii). The morphism of sheaves

$$P \times_S H \longrightarrow P \times_S P$$

[50] Editor's note: in this case one also says that X is an H-torsor.

is locally an isomorphism and hence is an isomorphism (4.5.8); thus P is formally principal homogeneous. The structure morphism $P \to S$ is locally isomorphic to the structure morphism $H \to S$, which belongs to (M). It therefore belongs to (M) by axiom $(e_{\mathcal{T}})$.

The equivalence between (i) and (iv) admits the following generalization.

Proposition 5.1.7. *Under the same assumptions on $\mathscr{C}$ and (M), let H be an S-group and let X be an S-object on which H acts on the right. Suppose that $H \to S$ belongs to (M). The following conditions are equivalent:*

(i) *H acts freely and (M)-effectively on X;*

(ii) *there exists an S-morphism $p : X \to Y$, compatible with the equivalence relation on X defined by the action of H, such that the induced action of $H \times_S Y$ on X over Y makes X a principal homogeneous bundle under H_Y over Y.*

Under these conditions, p identifies Y with the quotient X/H.

If $p : X \to Y$ is compatible with the action of H, then the induced action of $H \times_S Y$ on X over Y defines the same equivalence relation on X as the original action of H, by the isomorphism

$$H_Y \times_Y X \xrightarrow{\sim} H \times_S X.$$

The proposition follows from this observation and the preceding equivalence (iv)⇔(i).

Corollary 5.1.7.1.[51] *Let $\mathscr{C}$ be a category having a final object and fiber products, and let it be equipped with a topology $\mathcal{T}$ coarser than the canonical topology. Let $f : G \to H$ be a morphism of $\mathscr{C}$-groups, put $K = \mathrm{Ker}(f)$, and suppose that f is covering for $\mathcal{T}$. Then H represents the sheaf quotient G/K, and f is a K_H-torsor. Equivalently, one may say that G is a K-torsor over H.*

Indeed, because f is covering, it is a universal effective epimorphism (4.4.3). Hence, by 3.3.3.1, H is the quotient of G by the equivalence relation

$$\mathscr{R}(f) = G \times_H G.$$

Moreover, the morphism

$$G \times K \longrightarrow G \times_H G, \qquad (g, k) \longmapsto (g, gk),$$

is an isomorphism of $K_G = G \times_H K_H$-objects; its inverse is $(g, g') \mapsto (g, g^{-1}g')$. Thus $\mathscr{R}(f)$ is the equivalence relation defined by K. Since $f : G \to H$ is covering, it is a K_H-torsor by 5.1.6(ii), or directly by Definition 5.1.5(ii).

We can now sharpen Theorem 4.6.6 in the case of a quotient by a group action.

Proposition 5.1.8. *Under the assumptions of 5.1.7, let F_0 denote the functor over S defined as follows: for each $S' \to S$, $F_0(S')$ is the set of representable S'-subfunctors Z of $X \times_S S'$ that are stable under $H \times_S S'$ and are principal homogeneous bundles under this S'-group for the induced action (3.2.2).*

(i) *The following conditions are equivalent:*

  (a) *the action of H on X is (M)-effective and free;*[52]

  (b) *F_0 is representable.*

[51] Editor's note: this special case has been added as a corollary; it will be used several times in the following exposés.

[52] Editor's note: the words "and free" have been added.

Under these conditions, $F_0 = X/H$.

(ii) *Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S'_i \to S'\}$ and every family of morphisms $\{T_i \to S'_i\}$ in (N), every descent datum on the T_i relative to $\{S'_i \to S'\}$ is effective. Suppose that $X \times_S H \to X \times_S X$ belongs to (N), and that X is squarable. Then the element $p \in \mathrm{Hom}(X, F_0)$ corresponding to the subobject $X \times_S H$ of $X \times_S X$ identifies F_0 with the sheaf quotient X/H.*

5.2. Group Structures and Passage to the Quotient

In this subsection we are concerned with the algebraic structures that can be placed on the quotient (G/H) of a group by a subgroup. We first work in the category of sheaves on $\mathscr{C}$ for an arbitrary topology. Taking the canonical topology and using 4.5.12 will then give results concerning universal effective passage to the quotient in $\mathscr{C}$.

Proposition 5.2.1. *Let $u : H \to G$ be a monomorphism of sheaves of groups. There is a unique G-object structure on the quotient sheaf G/H for which the canonical morphism*

$$p : G \longrightarrow G/H$$

is a morphism of G-objects. This structure is functorial in the pair (G, H): given a commutative diagram

$$\begin{array}{ccc} H & \longrightarrow & G \\ \downarrow & & \downarrow f \\ H' & \longrightarrow & G', \end{array}$$

the morphism $f : G/H \to G'/H'$ of 3.2.3 is compatible with the morphism f between the acting groups.

Indeed, G/H is the sheaf associated with the presheaf

$$i(G)/i(H) : S \longmapsto G(S)/H(S).$$

Since the functor a is left exact, it carries objects with a group action to objects with a group action. The presheaf $i(G)/i(H)$ is an $i(G)$-object, and therefore

$$G/H = a\big(i(G)/i(H)\big)$$

is an $a(i(G)) = G$-object. This construction plainly has all the stated properties.

Corollary 5.2.2. *Let $u : H \to G$ be a monomorphism of $\mathscr{C}$-groups, and suppose that the action of H on G is universally effective. There is a unique G-object structure on the quotient object $G/H \in \mathrm{Ob}\,\mathscr{C}$ for which $p : G \to G/H$ is a morphism of objects with an action. In the sense of the preceding proposition, this structure is functorial in the pair (H, G), with H acting universally effectively on G.*

Proposition 5.2.3. *Let $u : H \to G$ be a monomorphism of sheaves of groups that identifies H with an invariant subgroup sheaf of G. There is a unique sheaf-of-groups structure on the quotient sheaf G/H for which the canonical morphism $p : G \to G/H$ is a group morphism. This structure is functorial in the pair (H, G), with H invariant.*

The proof is similar to that of 5.2.1.

Corollary 5.2.4. *Let $u : H \to G$ be a monomorphism of $\mathscr{C}$-groups that identifies H with an invariant subgroup of G. Suppose that the action of H on G is universally effective. There*

is a unique group structure on the quotient object $G/H \in \mathrm{Ob}\,\mathscr{C}$ for which the canonical morphism $G \to G/H$ is a group morphism. This structure is functorial in the pair (H, G), with H invariant and acting universally effectively.

The group structure of G/H can be characterized more explicitly.

Proposition 5.2.5. *Under the assumptions of 5.2.4, let K be a $\mathscr{C}$-group and let $f : G \to K$ be a morphism. The following conditions are equivalent:*

(i) *f is a group morphism compatible with the equivalence relation defined by H;*

(ii) *f is a group morphism whose restriction $H \to K$ is the trivial morphism;*

(iii) *f factors through a group morphism $G/H \to K$.*

In particular, there is an isomorphism, functorial in the group K,

$$\mathrm{Hom}_{\mathscr{C}\text{-gr.}}(G/H, K) \xrightarrow{\sim} \{f \in \mathrm{Hom}_{\mathscr{C}\text{-gr.}}(G, K) \mid f \circ u = e\}.$$

The equivalence of (i) and (ii) is proved set-theoretically, and (iii)⇒(ii) is immediate. The equivalence of (iii) and (ii) follows from the formula

$$\mathrm{Hom}(G/H, K) \simeq \mathrm{Hom}\big(i(G)/i(H), K\big)$$

and the definition of the group structure on G/H.

Remark 5.2.6. In the preceding situation, if the kernel of f is exactly H, then the morphism $G/H \to K$ through which f factors is a monomorphism. This follows at once from 3.3.4.

For sheaves of groups, 4.4.11 can be sharpened as follows.

Proposition 5.2.7. *Let G be a sheaf of groups and H an invariant subgroup sheaf. For every subgroup sheaf K of G containing H, let $K' = K/H$, regarded as a subgroup of $G' = G/H$. Then*

$$K = K' \times_{G'} G,$$

and the maps

$$K \longmapsto K/H, \qquad K' \longmapsto K' \times_{G'} G$$

give a bijection between the subgroup sheaves of G containing H and the subgroup sheaves of G'. Under this correspondence, the invariant subgroup sheaves of G and of G' correspond.

The first assertion follows readily from 4.4.11 and 3.2.4. It remains to show that K is invariant in G if and only if K' is invariant in G'. If K is invariant in G, then the presheaf $i(K)/i(H)$ is invariant in $i(G)/i(H)$; by the usual argument, the same is true of the associated sheaves. Conversely, if K' is invariant in G', then the fiber product $K' \times_{G'} G$ is invariant in G, as is immediate.

Now let L be an arbitrary subgroup sheaf of G. Denote by $\overline{L}$ the saturation of L for the equivalence relation defined by H; we also write $\overline{L} = L \cdot H$.

Proposition 5.2.8. *Under the preceding assumptions, $L \cdot H$ is a subgroup sheaf of G containing H, and the image of L in G/H is identified with*

$$(L \cdot H)/H \simeq L/(H \cap L).$$

Indeed, let L' denote the image sheaf of L in G/H. Under the correspondence of the preceding proposition, it is the subgroup sheaf of G/H corresponding to $L \cdot H$. Since $L \to L'$ is covering, and hence a universal effective epimorphism of sheaves, it follows from 4.4.9 that L' is identified with the quotient of L by the kernel of $L \to L'$, which is plainly $H \cap L$.

Finally, consider a sheaf of groups G, a subgroup sheaf K, and an invariant subgroup sheaf H of K. We first define a right action of the sheaf of groups $H\backslash K$ $(= K/H)$ on G/H. The group K acts on G by right translations. Since H is invariant in K, this action is compatible with the equivalence relation defined by the action of H, and therefore induces an action of K on G/H, that is, a morphism

$$K^\circ \longrightarrow \mathrm{Aut}(G/H)$$

from the group opposite to K. Since the target is a sheaf (4.5.13) and this morphism is trivial on H, it factors through K/H and defines the desired action. The right and left actions of G on itself commute; hence the actions of G and K/H on G/H commute.

Proposition 5.2.9. *Under the preceding assumptions, K/H acts freely on the right on G/H, and there is a canonical isomorphism of G-objects*

$$(G/H)/(K/H) \simeq G/K.$$

If K is invariant in G, in which case K/H is invariant in G/H by 5.2.7, this isomorphism respects the group structures on both sides.

There is an isomorphism of presheaves

$$\big(i(G)/i(H)\big)/\big(i(K)/i(H)\big) \xleftarrow{\sim} i(G)/i(K),$$

which respects the $i(G)$-object structures. Applying the functor a to this relation gives the asserted result.

Corollary 5.2.10. *Let G be a $\mathscr{C}$-group, K a $\mathscr{C}$-subgroup of G, and H an invariant $\mathscr{C}$-subgroup of K. Let (M) be a family of morphisms of $\mathscr{C}$ satisfying axioms (a)–($e_{\mathcal{T}}$). Suppose that the right action of H on G (respectively, on K) is (M)-effective. Then K/H acts naturally and freely on the right on G/H, and this action commutes with that of G. The following conditions are equivalent:*

(i) *the action of K on G is (M)-effective;*

(ii) *the action of K/H on G/H is (M)-effective.*

Under these conditions, there is an isomorphism of objects[53] *with a G-action*

$$(G/H)/(K/H) \simeq G/K.$$

5.3. Use of Effectivity Criteria: Noether's Theorem

[54] Let $\mathscr{C}$, $\mathcal{T}$, and (M) be as usual. Let (N) be a family of morphisms satisfying axioms (a) and (f_M) of 4.7. Combining 5.2.7 and 4.7.2 gives the following result.

Proposition 5.3.1. *Let G be a $\mathscr{C}$-group, and let H be an invariant $\mathscr{C}$-subgroup of G that acts (M)-effectively on G. For every $\mathscr{C}$-subgroup K of G containing H such that $K \to G$ belongs to (N), the action of H on K is (M)-effective, and the quotient $K/H = K'$ is a $\mathscr{C}$-subgroup of $G/H = G'$ such that $K' \to G'$ belongs to (N). The map $K \mapsto K'$ is a bijection*

[53] Editor's note: of $\mathscr{C}$.

[54] Editor's note: in fact, the isomorphisms established in 5.2.8–5.2.10 deserve to be called "Noether isomorphism theorems."

between the set of $\mathscr{C}$-subgroups K of G that contain H and for which $K \to G$ belongs to (N), and the set of $\mathscr{C}$-subgroups K' of G' for which $K' \to G'$ belongs to (N). Its inverse is

$$K' \longmapsto K \times_{G'} G.$$

Under this correspondence, the invariant subgroups of G and of G' correspond.

Corollary 5.3.2. *If $H \to G$ belongs to (N), then $\mathscr{C}$ has a final object e, and the unit section $e \to G/H$ belongs to (N).*[55]

6. Topologies in the Category of Schemes

6.1. The Zariski Topology

This is the topology generated by the following pretopology: a family of morphisms $\{S_i \to S\}$ is covering if each morphism is an open immersion and the union of the images of the S_i is all of S. It is denoted by (Zar).

Definition 6.1.1. *A sheaf for the Zariski topology is also called a* functor of local nature*: it is a contravariant functor from* **Sch** *to* **Set** *such that, for every scheme S and every covering of S by open subschemes S_i, the diagram*

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \cap S_j)$$

is exact.

In particular, a functor of local nature carries direct sums to products. Since every representable functor is a sheaf, the Zariski topology is coarser than the canonical topology.

As a matter of terminology, whenever we say "local" or "locally" without further qualification, the reference is to the Zariski topology and hence to the usual meaning of these words.

6.2. A Procedure for Constructing Topologies

Proposition 6.2.1. *Let $\mathscr{C}$ be a category, let $\mathscr{C}'$ be a full subcategory, and let P be a set of families of morphisms in $\mathscr{C}$ with common target, stable under base change and composition (that is, satisfying axioms (P 1) and (P 2) of 4.2.5). Let P' be a set of families of morphisms in $\mathscr{C}'$ that contains every singleton family consisting of an identity isomorphism. Equip $\mathscr{C}$ with the topology generated by P and P' (cf. 4.2.5.0), and suppose that the following three conditions hold:*

(a) *If $\{S_i \to S\} \in P'$, so that $S_i, S \in \mathrm{Ob}\,\mathscr{C}'$, and $T \to S$ is a morphism in $\mathscr{C}'$, then the fiber products $S_i \times_S T$, formed in $\mathscr{C}$, exist and the family $\{S_i \times_S T \to T\}$ belongs to P'; in particular, $S_i \times_S T \in \mathrm{Ob}\,\mathscr{C}'$.*

This condition implies that P' is stable under base change in $\mathscr{C}'$, but is not equivalent to that assertion: it also requires the inclusion functor $\mathscr{C}' \hookrightarrow \mathscr{C}$ to commute with certain fiber products.

(b) *For every $S \in \mathrm{Ob}\,\mathscr{C}$, there exists a family $\{S_i \to S\} \in P$ with $S_i \in \mathrm{Ob}\,\mathscr{C}'$ for every i.*

[55] Editor's note: this follows by applying the proposition with $K = H$.

(c) *In a situation*

$$\begin{array}{ccc} & & S_{ijk} \\ & & {\scriptstyle (P')}\downarrow \\ S_i & \xleftarrow{(P)} & S_{ij} \\ {\scriptstyle (P')}\downarrow & & \\ S & & \end{array} \tag{6.2.1.c}$$

where S, S_i, S_{ij}, S_{ijk} are objects of $\mathscr{C}'$, where $\{S_i \to S\} \in P'$, $\{S_{ij} \to S_i\} \in P$ for each i, and $\{S_{ijk} \to S_{ij}\} \in P'$ for each pair ij, there exists a family $\{T_n \to S\} \in P'$, and for each n a multi-index (ijk), fitting into a commutative diagram

$$\begin{array}{ccc} S_{ijk} & \longleftarrow & T_n \\ & \searrow & \downarrow \\ & & S. \end{array} \tag{6.2.1.d}$$

Under these hypotheses, a sieve R of an object $S \in \mathrm{Ob}\,\mathscr{C}$ is covering if and only if there is a composite family

$$\begin{array}{ccc} R & \dashleftarrow & S_{ij} \\ \downarrow & & {\scriptstyle (P')}\downarrow \\ S & \xleftarrow{(P)} & S_i \end{array} \tag{6.2.1.e}$$

where $S_i, S_{ij} \in \mathrm{Ob}\,\mathscr{C}'$, $\{S_i \to S\} \in P$, and $\{S_{ij} \to S_i\} \in P'$ for every i, such that the resulting morphisms $S_{ij} \to S$ factor through R. Equivalently, the sieve generated by this composite family is contained in R.

Proof. Families belonging to P and P' are covering, and therefore so is a composite of such families, by (C 2). A sieve of the displayed form is thus covering because it contains a covering sieve.

Conversely, it is enough to show that the sieves of this form constitute a topology, that is, to verify axioms (T 1)–(T 4) of 4.2.1.

Axiom (T 4). Let $S \in \mathrm{Ob}\,\mathscr{C}$. By (b), there exists a family $\{S_i \to S\} \in P$ with $S_i \in \mathrm{Ob}\,\mathscr{C}'$. The singleton families $\{S_i \xrightarrow{\mathrm{id}_{S_i}} S_i\}$ belong to P' by hypothesis. Hence the maximal sieve S of S has the required form:

$$\begin{array}{ccc} S_i & \longrightarrow & S \\ {\scriptstyle (P')}\downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \mathrm{id}_S} \\ S_i & \xrightarrow{(P)} & S. \end{array} \tag{6.2.1.T4}$$

Axiom (T 3). This is evident.

Axiom (T 2). Let R be a sieve of S of the required form, and let C be a sieve[56] of S such that, for every $T \in \mathrm{Ob}\,\mathscr{C}$ and every morphism $T \to S$ factoring through R, the sieve $C \times_S T$ of T has the required form. Since $S_{ij} \to S$ factors through R, the sieve

$$C_{ij} = C \times_S S_{ij}$$

[56]Editor's note: the original has been corrected here.

of S_{ij} has the required form; it is represented schematically by

$$\begin{array}{ccccc} & & S_{ij} & \longleftarrow & C_{ij} \\ & \swarrow & {\scriptstyle (P')}\downarrow & & \\ & & S_i & & \downarrow \\ & & {\scriptstyle (P)}\downarrow & & \\ R & \longrightarrow & S & \longleftarrow & C. \end{array} \tag{6.2.1.T2a}$$

Consequently, for every pair ij, there is a diagram of the form

$$\begin{array}{ccc} S_{ijkl} & & \\ {\scriptstyle (P')}\downarrow & \searrow & \\ S_{ijk} & & \\ {\scriptstyle (P)}\downarrow & & \\ S_{ij} & \longleftarrow & C_{ij}. \end{array} \tag{6.2.1.T2b}$$

We have therefore produced a composite family

$$S_{ijkl} \xrightarrow{(P')} S_{ijk} \xrightarrow{(P)} S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S \tag{6.2.1.T2c}$$

belonging to $P \circ P' \circ P \circ P'$, factoring through C, and having every object other than S in $\mathscr{C}'$.

Apply condition (c) to each family $\{S_{ijkl} \to S_i\}$. For every i we obtain a family $\{T_{in} \to S_i\} \in P'$ such that $T_{in} \to S$ factors through one of the S_{ijkl}, and hence through C:

$$\begin{array}{ccc} T_{in} & \longrightarrow & S_{ijkl} \\ {\scriptstyle (P')}\downarrow & & \downarrow \\ S_i & & C \\ {\scriptstyle (P)}\downarrow & \swarrow & \\ S & & \end{array} \tag{6.2.1.T2d}$$

Thus the sieve C of S has the required form, completing the verification of (T 2).

Axiom (T 1). Let R be a sieve of S of the prescribed form and let $T \to S$ be a morphism in $\mathscr{C}$. We show that the sieve $R \times_S T$ of T again has the prescribed form. Consider the commutative configuration

$$\begin{array}{ccccccc} & & S_{ij} & \longleftarrow\!\!\!

shows that the morphisms $U_{ikj} \to T$ factor through the sieve $T \times_S R$ of T. This sieve therefore has the required form, and the proof is complete.

Corollary 6.2.2. *Let $S \in \mathrm{Ob}\,\mathscr{C}'$, and let R be a sieve of S. Then R is covering if and only if there exists a family $\{T_i \to S\} \in P'$ that factors through R.*

Such a sieve is plainly covering. Conversely, apply condition (c) with the family $\{S_i \to S\}$ reduced to the identity isomorphism of S; the proposition then shows that every covering sieve has the stated form.

Corollary 6.2.3. *A presheaf F on $\mathscr{C}$ is separated (respectively, is a sheaf) if and only if the map*

$$F(S) \longrightarrow \prod_i F(S_i)$$

is injective (respectively, the diagram

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times_S S_j)$$

is exact) in each of the following two cases:

(i) $\{S_i \to S\} \in P$;

(ii) $S, S_i \in \mathrm{Ob}\,\mathscr{C}'$ *and* $\{S_i \to S\} \in P'$.

The conditions are necessary because the families in question are covering. Let C be the sieve of S that is the image of a composite family

$$\{S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S\}.$$

A straightforward diagram chase shows that the conditions of the corollary imply that

$$\mathrm{Hom}(S, F) \xrightarrow{g} \mathrm{Hom}(C, F)$$

is injective, respectively bijective. Every covering sieve R of S contains a sieve C of this form, and there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(S,F) & \xrightarrow{f} & \mathrm{Hom}(R,F) \\ & \searrow^{g} & \downarrow{\scriptstyle h} \\ & & \mathrm{Hom}(C,F). \end{array} \tag{6.2.3.1}$$

Since g is injective, so is f, and hence F is separated. Moreover, R refines C, so h is also injective. If g is bijective, then f is bijective as well, and F is a sheaf.

Remark 6.2.4. *The preceding corollary does not follow from 4.3.5, since P' is not stable under base extension.*

Remark 6.2.5. *Condition (c) holds in particular when:*

(i) *P' is stable under composition;*

(ii) *whenever $\{S_j \to S\}$ is a family of morphisms in $\mathscr{C}'$ belonging to P, it contains a subfamily belonging to P'.*

6.3. Application to the Category of Schemes

Take $\mathscr{C}$ to be the category of schemes, $\mathscr{C}'$ the full subcategory of affine schemes, and P the set of surjective families of open immersions. We shall consider the following sets P_i':

P_1' : finite surjective families of flat morphisms;
P_2' : finite surjective families of flat, finitely presented, quasi-finite morphisms;
P_3' : finite surjective families of étale morphisms;
P_4' : finite surjective families of finite étale morphisms.

The description of P_2' is accompanied by the following clarification.[57]

For each P_i', except P_4', the conditions of Proposition 6.2.1 hold. Condition (c) follows from 6.2.5: since an affine scheme is quasi-compact, every family of morphisms in $\mathscr{C}'$ belonging to P contains a finite subfamily that still belongs to P, and hence belongs to P_i' for $i = 1, 2, 3$. The topology T_i generated by P and P_i' is denoted and named as follows:[57]

T_1 = (fpqc) the faithfully flat quasi-compact topology,
T_2 = (fppf) the faithfully flat topology (locally of finite presentation),
T_3 = (ét) the étale topology,
T_4 = (étf) the finite étale topology.

Since $P_1' \supset P_2' \supset P_3' \supset P_4'$, one has

$$(\text{fpqc}) \geq (\text{fppf}) \geq (\text{ét}) \geq (\text{étf}) \geq (\text{Zar}).$$

Proposition 6.3.1.

(i) *A sieve R of S is covering for T_i, $1 \leq i \leq 3$, if and only if there exist a covering (S_p) of S by affine open subschemes and, for every p, a family $\{S_{pq} \to S_p\} \in P_i'$, with all S_{pq} affine, such that every morphism $S_{pq} \to S$ factors through R.* [58]

(ii) *A presheaf F on* **Sch** *is a sheaf for* (fpqc) *(respectively,* (fppf), (ét), *or* (étf)*) if and only if:*

(a) *F is a sheaf for* (Zar)*, that is, a functor of local nature;*

(b) *for every faithfully flat (respectively, faithfully flat, finitely presented and quasi-finite; surjective étale; or surjective finite étale) morphism $T \to S$, with T and S affine, the diagram*

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact.

(iii) *The topologies T_i, for $i = 1, 2, 3, 4$, are coarser than the canonical topology.*

(iv) *Every surjective family of flat open morphisms (respectively, flat morphisms locally of finite presentation; étale morphisms; or finite étale morphisms) is covering for* (fpqc) *(respectively,* (fppf), (ét), *or* (étf)*).*

[57] Editor's note: let P_2'' be the set of finite surjective families of morphisms in $\mathscr{C}'$ that are flat and of finite presentation. By Proposition 6.3.1 below, the topology T_2 generated by P and P_2' coincides with the topology generated by P and P_2''. This follows from the results of EGA IV$_4$, §17.16 on quasi-sections; see the proof of 6.3.1. We recall the following special case of EGA IV$_4$, 17.16.2. Let S be affine and let $f : X \to S$ be a surjective flat morphism locally of finite presentation. Then there exist a faithfully flat, finitely presented, quasi-finite morphism $S' \to S$, with S' affine, and an S-morphism $S' \to X$.

[58] Editor's note: every family $\{S_{pq} \to S_p\} \in P_i'$ is finite. Hence $S_p' = \coprod_q S_{pq}$ is affine, and the family may be replaced by the morphism $S_p' \to S_p$, which still belongs to P_i'.

(v) *Every finite surjective family of flat quasi-compact morphisms is covering for* (fpqc). *In particular, every faithfully flat quasi-compact morphism is covering for* (fpqc).

Proof. Part (i) follows from 6.2.1 and part (ii) from 6.2.3, taking into account that a sheaf for the Zariski topology carries direct sums to products. Every representable functor is a sheaf for (Zar) and satisfies condition (ii)(b), by SGA 1, VIII, 5.3. Thus T_1 is coarser than the canonical topology, which proves (iii).

We prove (iv). Let $\{S_i \to S\}$ be a family as in the statement. By covering S with affine open subschemes, we reduce immediately to the case in which S is affine.[59]

First suppose that the morphisms $S_i \to S$ are flat and open (respectively, étale). Cover each S_i by affine open subschemes S_{ij}. Since the morphisms are open, the images T_{ij} of the S_{ij} form an open covering of S. The affine scheme S is quasi-compact, so finitely many T_{ij}, indexed by a finite set E, cover it. Then

$$S' = \coprod_{(i,j)\in E} S_{ij}$$

is affine, and $S' \to S$ belongs to P_1', respectively P_3', and is therefore covering. Since it factors through the given family, that family is covering.

In the (étf) case, every S_i is finite over S and hence affine. In the preceding argument one may take the singleton covering $\{S_i\}$ of S_i, which gives $S' \to S \in P_4'$.

Now suppose that the morphisms $f_i : S_i \to S$ are flat and locally of finite presentation. For every $s \in S$, the proof of EGA IV_4, 17.16.2 supplies an affine subscheme $X(s)$ of some S_i, with $s \in f_i(X(s))$, such that the restriction

$$g_i : X(s) \longrightarrow S$$

is flat, finitely presented, and quasi-finite. Its image $g_i(X(s))$ is an open neighborhood $U(s)$ of s, by EGA IV_2, 2.4.6. Since S is affine, finitely many such opens $U(s_j)$, $j = 1, \ldots, n$, cover S. Consequently,

$$X' = \coprod_{j=1}^{n} X(s_j)$$

is affine and $X' \to S$ is surjective, flat, finitely presented, and quasi-finite, so it belongs to P_2'.[60] This completes the proof of (iv).

We prove (v). Let $\{S_i \to S\}$ be a finite surjective family of flat quasi-compact morphisms, and cover S by affine open subschemes T_j. The schemes

$$S_{ij} = T_j \times_S S_i$$

are quasi-compact, and therefore admit finite affine open coverings T_{ijk}. Each $T_{ijk} \to T_j$ is flat, and the family $\{T_{ijk} \to T_j\}$ is finite and surjective, hence covering for T_1. By composition, $\{T_{ijk} \to S\}$ is covering as well. It factors through the given family, which is therefore covering:

$$\begin{array}{ccccc} S_i & \longleftarrow & S_{ij} & \longleftarrow & T_{ijk} \\ \downarrow & & \downarrow & \swarrow & \\ S & \longleftarrow & T_j & & \end{array} \qquad (6.3.1.\mathrm{v})$$

[59] Editor's note: the argument that follows has been simplified by using the fact that S is now assumed affine.

[60] Editor's note: this shows that if P_2'' denotes the set of finite surjective families of flat, finitely presented morphisms in $\mathscr{C}'$, then the topology generated by P and P_2'' equals T_2. Moreover, in the notation used at the beginning of the proof of (iv), if each S_i is covered by affine opens of finite presentation over S, then the resulting morphism $S' \to S$ belongs to P_2''.

Corollary 6.3.2. *Let (M_i) be the following families of morphisms:*

(M_1) : faithfully flat quasi-compact morphisms;

(M_2) : faithfully flat morphisms locally of finite presentation;

(M_3) : surjective étale morphisms;

(M_4) : surjective finite étale morphisms.

The family (M_i) satisfies axioms (a), (b), (c), (d_{T_i}), and (e_{T_i}) of 4.6.3. [61]

Conditions (a), (b), and (c) are classical (see EGA and SGA 1, passim).[62] By 6.3.1(iv),(v), (M_i) satisfies (d_{T_i}). It remains to verify (e_{T_i}). It is enough to verify (e_{T_1}), which implies all the others; this follows from SGA 1, VIII, §§4–5.

Corollary 6.3.3. *Let X be a scheme and let R be an equivalence relation in X of type (M_i). Then R is (M_i)-effective if and only if the sheaf quotient of X by R for T_i is representable; in that case it is represented by the quotient X/R.*

This is 4.6.5.

6.4. Effectivity Conditions

We now seek families (N) of morphisms satisfying axiom (f_T) of 4.7. Observe first that (f_{T_1}) implies (f_{T_i}), so it is enough to consider the (fpqc) topology.

Lemma 6.4.1. *The following families of morphisms satisfy axiom (f_{T_1}) of 4.7, that is, they "descend by* (fpqc)*":*

(N) : open immersions;

(N') : closed immersions;

(N'') : quasi-compact immersions.

By 6.3.1(ii), it is enough to verify that these families descend for the Zariski topology and along a faithfully flat quasi-compact morphism. The first assertion is clear. For (N), the second is SGA 1, VIII, 4.4; for (N'), it is loc. cit., 1.9; for (N''), argue as in loc. cit., 5.5, using the two preceding results.

Corollary 6.4.2. *The same result holds for quasi-compact open immersions.*

These results make the general results of 4.7.1, 4.7.2, 5.1.8, 5.3.1, and so on applicable to the present situation. We state the first as an example.

Corollary 6.4.3. (= 4.7.1 + 4.6.10). *Let X be a scheme and R an equivalence relation in X. Suppose that $R \to X$ is faithfully flat and quasi-compact, and that $R \to X \times X$ is a closed immersion (respectively, open, quasi-compact, or quasi-compact open). Then the sheaf quotient X/R is the same for the* (fpqc) *topology as for the canonical topology. For every scheme S,*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{closed (respectively, open, retrocompact, or open retrocompact)} \\ \text{subschemes } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ Z \to S \text{ is faithfully flat and quasi-compact and} \\ R_Z \rightrightarrows Z \to S \text{ is exact} \end{array} \right\}. \tag{6.4.3.1}$$

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[61] Editor's note: the original has been corrected by adding surjectivity to (M_3) and (M_4); it is automatically satisfied in the other two cases.

[62] Editor's note: see EGA I, 6.6.4 for "quasi-compact"; EGA II, 6.1.5 for "finite"; EGA IV_1, 1.6.2 for "locally of finite presentation"; EGA IV_2, 2.2.13 for "faithfully flat"; and EGA IV_4, 17.3.3 for "étale".

[63] Editor's note: recall that a subscheme Z of a scheme T is called *retrocompact* if the immersion $Z \hookrightarrow T$ is quasi-compact; see EGA 0_{III}, 9.1.1.

6.5. Principal Homogeneous Bundles

We merely record the terminology:

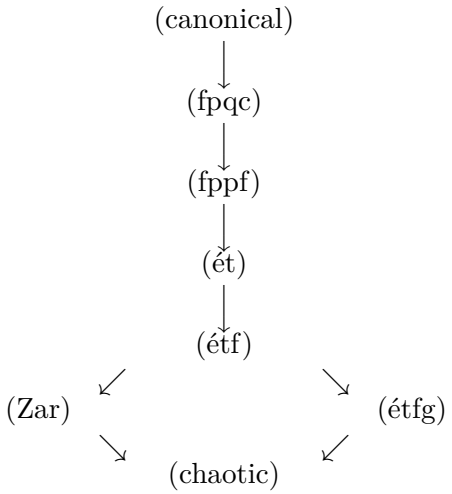
topology	principal homogeneous bundles
(fpqc)	principal homogeneous bundles (without qualifier)
(ét)	quasi-isotrivial principal homogeneous bundles
(étf)	locally isotrivial principal homogeneous bundles
(Zar)	locally trivial principal homogeneous bundles.

(6.5.1)

6.6. Other Topologies

Other topologies on the category of schemes are sometimes useful. One is the *global finite étale topology* (étfg), generated by the pretopology whose covering families are the surjective families of finite étale morphisms. It is not finer than the Zariski topology. The corresponding principal homogeneous bundles are called *isotrivial*.

The position of this topology, and of the chaotic topology, in the lattice is as follows:



(6.6.1)

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6.7. Homogeneous Spaces

Let[65] G be an S-group scheme, let X be an S-scheme equipped with a left action of G, and let

$$\Phi : G \times_S X \longrightarrow X \times_S X$$

be the morphism of S-schemes defined on points by

$$(g, x) \longmapsto (gx, x).$$

Recall (cf. 5.1.0 and III.0.1) that X is called a *formally principal homogeneous space* under G if the following equivalent conditions hold:

(i) for every $T \to S$, the set $X(T)$ is empty or is a principal homogeneous space under $G(T)$;

[64] Editor's note: recall (cf. 4.4.2) that the chaotic topology is the coarsest topology, defined by $J(S) = \{S\}$ for every $S \in \mathrm{Ob}\,\mathscr{C}$.

[65] Editor's note: the numbering that follows has been added.

(ii) Φ is an isomorphism of S-functors;

(iii) Φ is an isomorphism of S-schemes.

The equivalence (i)$\Longleftrightarrow$(ii) is clear, and (ii)$\Longleftrightarrow$(iii) holds because $\mathscr{C} = (\mathbf{Sch}/S)$ is a full subcategory of $\widehat{\mathscr{C}}$.

The notion of a *formally homogeneous space*, not necessarily principal homogeneous, is obtained by requiring Φ to be an epimorphism in the category of sheaves for an appropriate topology $\mathcal{T}$. Indeed, requiring Φ to be an epimorphism of S-functors is equivalent to requiring that, for every $T \to S$, the set $X(T)$ be empty or homogeneous (not necessarily principal homogeneous) under $G(T)$. This is too restrictive, as the following simple example shows.

Let

$$S = \operatorname{Spec} R, \qquad G = \mathbf{G}_{m,R}, \qquad X = \mathbf{G}_{m,R},$$

and let G act on X by $t \cdot x = t^2 x$. Then Φ is surjective, finite, and étale, and hence is an epimorphism in the category of sheaves for the (étf) topology (and, *a fortiori*, an epimorphism of S-schemes). On the other hand, the points 1 and -1 of $X(R)$ are not conjugate under any element of $G(R)$. Thus

$$G(R) \times X(R) \longrightarrow X(R) \times X(R)$$

is not surjective.[66] We are therefore led to the following definition.

Definition 6.7.1. *Let G be an S-group, let X be an S-scheme equipped with an action of G, and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The scheme X is called a* formally homogeneous space under G, *relative to $\mathcal{T}$, if the following equivalent conditions hold:*

(i) *the morphism $\Phi : G \times_S X \to X \times_S X$ is an epimorphism in the category of sheaves for $\mathcal{T}$;*

(ii) *for every $T \to S$ and every $x, y \in X(T)$, there exist a $\mathcal{T}$-covering morphism $T' \to T$ and an element $g \in G(T')$ such that $y_{T'} = g \cdot x_{T'}$.*

Remark 6.7.2. *Condition (i) implies in particular that Φ is a universal effective epimorphism in $(\mathbf{Sch}/S)$, by 4.4.3. As is readily seen, it follows that Φ is surjective; cf. 1.3, Editor's note (3).*

Proposition and Definition 6.7.3. [67] *Let G be an S-group, let X be an S-scheme equipped with an action of G, and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The following conditions are equivalent:*

(i) *X satisfies both of the following conditions:*

(1) *the morphism $\Phi : G \times_S X \to X \times_S X$ is covering, that is, X is a formally homogeneous G-space;*

[66] Editor's note: this difficulty plainly comes from the following fact. Let $\mathscr{C}'$ be a full subcategory of $\widehat{\mathscr{C}}$ containing $\mathscr{C}$, for example the category $\widetilde{\mathscr{C}_{\mathcal{T}}}$ of sheaves on $\mathscr{C}$ for a topology $\mathcal{T}$ coarser than the canonical topology. If $f : X \to Y$ is a morphism in $\mathscr{C}$, then the implications

$$f \text{ epimorphic in } \widehat{\mathscr{C}} \Longrightarrow f \text{ epimorphic in } \mathscr{C}' \Longrightarrow f \text{ epimorphic in } \mathscr{C}$$

are generally strict.

[67] Editor's note: cf. [Ray70], Definition VI.1.1.

(2) *the structural morphism $X \to S$ is also covering, that is, locally for $\mathcal{T}$ it has a section (cf. 4.4.8 bis, (b)).*

(ii) *Locally on S for the topology $\mathcal{T}$, the scheme X, with its G-action, is isomorphic to the quotient sheaf of G by a subgroup scheme H, for $\mathcal{T}$. In other words, there is a covering family $\{S_i \to S\}$ such that every $X \times_S S_i$ represents the sheaf quotient of $G \times_S S_i$ by some subgroup scheme H_i.*

Under these conditions, X is called a G-homogeneous space, *relative to $\mathcal{T}$.*

Proof. Assume (ii). Put

$$G_i = G \times_S S_i, \qquad X_i = X \times_S S_i.$$

The scheme X_i has a section over S_i, namely the composite of the unit section $\varepsilon_i : S_i \to G_i$ with the projection $\pi_i : G_i \to X_i = G_i/H_i$. It follows that $X \to S$ is covering.

Moreover, π_i is covering, so $\pi_i \times \pi_i$ is covering as well, by 4.2.3 (C 1),(C 2). There is a commutative diagram

$$\begin{array}{ccc} G_i \times_{S_i} X_i & \xrightarrow{\Phi_i} & X_i \times_{S_i} X_i \\ {\scriptstyle \mathrm{id} \times \pi_i}\uparrow & & {\scriptstyle \pi_i \times \pi_i}\uparrow \\ G_i \times_{S_i} G_i & \xrightarrow[\sim]{\Psi_i} & G_i \times_{S_i} G_i, \end{array} \tag{6.7.3.1}$$

where Φ_i is obtained from Φ by the base change $S_i \to S$, and Ψ_i is the isomorphism defined on points by

$$(g, g') \longmapsto (gg', g).$$

The composite $(\pi_i \times \pi_i) \circ \Psi_i$ is covering; hence Φ_i is covering by 4.2.3 (C 3). Thus Φ is locally covering and is therefore covering, by 4.2.3 (C 5). This proves (ii)$\Longrightarrow$(i).

Conversely, assume (i), and first suppose in addition that the structural morphism $X \to S$ has a section σ. By EGA I, 5.3.13, σ is an immersion. Define $H = G \times_X S$ by the following diagram, whose two squares are cartesian:

$$\begin{array}{ccccc} H & \longrightarrow & G & \xrightarrow{\mathrm{id}_G \boxtimes \sigma} & G \times_S X \\ \downarrow & & {\scriptstyle \pi}\downarrow & & {\scriptstyle \Phi}\downarrow \\ S & \xrightarrow{\sigma} & X & \xrightarrow{\mathrm{id}_X \boxtimes \sigma} & X \times_S X. \end{array} \tag{6.7.3.2}$$

Here π, $\mathrm{id}_G \boxtimes \sigma$, and $\mathrm{id}_X \boxtimes \sigma$ are the morphisms defined on points, for $T \to S$, $g \in G(T)$, and $x \in X(T)$, by

$$\pi(g) = g \cdot \sigma_T, \qquad (\mathrm{id}_G \boxtimes \sigma)(g) = (g, \sigma_T), \qquad (\mathrm{id}_X \boxtimes \sigma)(x) = (x, \sigma_T).$$

Then π is covering, and H is a subgroup scheme of G representing the stabilizer $\mathrm{Stab}_G(\sigma)$ of σ (cf. I, 2.3.3). Thus, for every $T \to S$,

$$H(T) = \{g \in G(T) \mid g \cdot \sigma_T = \sigma_T\}.$$

Let G/H denote the presheaf

$$T \longmapsto G(T)/H(T),$$

and let $a(G/H)$ be its associated sheaf for $\mathcal{T}$. The preceding construction gives a commutative diagram of presheaves equipped with a G-action:

$$\begin{array}{ccc} G & \xrightarrow{\pi} & X \\ \downarrow & \nearrow_{\bar{\pi}} & \\ G/H & & \end{array} \tag{6.7.3.3}$$

where $\bar{\pi}$ is a monomorphism. Since π is covering, $\bar{\pi}$ is covering as well. By 4.3.12, $\bar{\pi}$ therefore induces an isomorphism

$$a(G/H) \xrightarrow{\sim} X.$$

We have proved: if X is a G-homogeneous space such that $X \to S$ has a section σ, then X represents the sheaf quotient G/H, where $H = G \times_X S$ is the stabilizer of σ.

In general, by hypothesis there is a covering family $\{S_i \to S\}$ such that every morphism

$$X_i = X \times_S S_i \longrightarrow S_i$$

has a section σ_i. Put $G_i = G \times_S S_i$. The morphism

$$\Phi_i : G_i \times_{S_i} X_i \longrightarrow X_i \times_{S_i} X_i$$

obtained from Φ by base change along $S_i \to S$ is still covering.

The preceding argument then gives

$$X_i \simeq G_i/H_i,$$

where H_i is the stabilizer of σ_i in G_i. This completes the proof of (i)$\Longrightarrow$(ii).

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[68] Editor's note: this reference has been added.

Exposé V
Construction of Quotient Schemes

by P. Gabriel

The purpose of this exposé is to prove the theorems stated in TDTE III.[1] If X and T are two objects of a category $\mathscr{C}$, we write $X(T)$ instead of $\mathrm{Hom}_{\mathscr{C}}(T, X)$. Similarly, if $\varphi : Y \to X$ is an arrow (respectively, if T is an object) of $\mathscr{C}$, then $\varphi(T)$ denotes the map $g \mapsto \varphi \circ g$ from $Y(T)$ to $X(T)$:

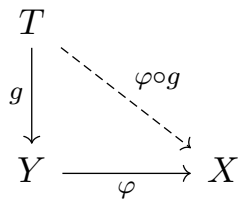

Figure 1: The map $\varphi(T) : Y(T) \to X(T)$.

Likewise, $T(\varphi)$ denotes the map $g \mapsto g \circ \varphi$ from $T(X)$ to $T(Y)$:

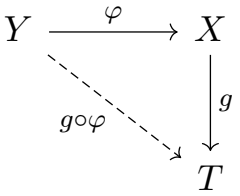

Figure 2: The map $T(\varphi) : T(X) \to T(Y)$.

Finally, if P is a scheme, we denote by $\underline{P}$ the underlying set of P.

As an exception, in the present exposé we do not follow the convention on quotient notation stated in IV 4.6.15 (*loc. cit.*, top of p. 227 of the original), because we wish to give here a construction of quotients that also applies to "pre-equivalence relations"[2] that are not equivalence relations.

[1] Editor's note: namely, Theorems 5.1, 5.3, 6.1, 6.2, and 7.2 of TDTE III. The first two (respectively, the next two) correspond to Theorem 4.1 (respectively, Theorems 7.1 and 8.1) of this exposé. Theorem 7.2 of TDTE III is proved in Exposé $\mathrm{VI_A}$, 3.2 and 3.3.

[2] Editor's note: that is, groupoids with base X; see the terminology at the end of Section 1. When $\mathscr{C}$ is the category of schemes, the quotient $p : X \to Y$ of a groupoid X_* with base X exists under certain hypotheses (see 4.1, 6.1, and 7.1). If, in addition, X_* is an equivalence relation, then under the same hypotheses p is faithfully flat and quasi-compact, hence a universal epimorphism; see *loc. cit.*.

1. $\mathscr{C}$-groupoids

a) Let $\mathscr{C}$ be a category in which products and fiber products exist. Recall first that a diagram

$$X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0 \xrightarrow{\ p\ } Y$$

Figure 3: A parallel pair followed by its proposed cokernel.

in $\mathscr{C}$ is said to be *exact* if $pd_0 = pd_1$ and if, for every $T \in \mathscr{C}$, the map $T(p)$ is a bijection from $T(Y)$ onto the subset of $T(X_0)$ consisting of the arrows $f : X_0 \to T$ such that $fd_0 = fd_1$. One also says that (Y, p) is the *cokernel* of (d_0, d_1), and writes

$$(Y, p) = \mathrm{Coker}(d_0, d_1).$$

b) For example, let $\mathscr{C}$ be the category (**Esp.An**) of ringed spaces. In this case a cokernel (Y, p) always exists, and it may be described as follows. The underlying topological space of Y is obtained from X_0 by identifying the points $d_0(x)$ and $d_1(x)$, and by giving Y the quotient topology. The canonical map $\pi : X_0 \to Y$, together with d_0, d_1, then induces a pair of arrows of sheaves of rings on Y:

$$\pi_*(\mathcal{O}_0) \underset{\delta_1}{\overset{\delta_0}{\rightrightarrows}} \pi_*(d_{0*}\mathcal{O}_1) = \pi_*(d_{1*}\mathcal{O}_1)$$

Figure 4: The pair of induced arrows between direct-image sheaves.

Here $\mathcal{O}_i$ is the structure sheaf of X_i. As the sheaf of rings on Y, one chooses the subsheaf of $\pi_*(\mathcal{O}_0)$ whose sections s satisfy $\delta_0(s) = \delta_1(s)$. The arrow p is defined in the evident way.

Consider[3] the following diagram in (**Esp.An**):

$$X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0$$

Figure 5: The pair $(d_0, d_1) : X_1 \rightrightarrows X_0$.

Let (Y, p) be its cokernel. An open subset U of X_0 is said to be *saturated* if

$$d_0^{-1}(U) = d_1^{-1}(U),$$

which is equivalent to saying $U = p^{-1}(p(U))$. In this case, since Y has the quotient topology, $p(U)$ is an open subset of Y.

Lemma 1.1. *Let U be a saturated open subset of X, and set $V = p(U)$. Denote by $U_1 = d_0^{-1}(U) = d_1^{-1}(U)$ the corresponding open subset of X_1, and by $\widetilde{d}_0, \widetilde{d}_1$, and $\widetilde{p}$ the restrictions of d_0, d_1 to U_1, and of p to U, respectively. Then $(V, \widetilde{p})$ is a cokernel in* (**Esp.An**) *of:*[4]

[3] Editor's note: Lemmas 1.1 and 1.2 have been added; they are used several times in Sections 5–9.

[4] Editor's note: this is not the case in the category of schemes. For example, let $S = \mathrm{Spec}(\mathbf{C})$, let $X_0 = \mathbf{A}^2_S = \mathrm{Spec}(\mathbf{C}[x_1, x_2])$, let $d_1 : \mathbf{G}_{m,S} \times_S \mathbf{A}^2_S \to \mathbf{A}^2_S$ be the action of $\mathbf{G}_{m,S}$ on $\mathbf{A}^2_S$ by homotheties, let d_0 be the projection onto the second factor, and let $U = \mathbf{A}^2_S \setminus \{m\}$, where $m = (0, 0)$. Then the projective line

$$U_1 \underset{\widetilde{d_0}}{\overset{\widetilde{d_1}}{\rightrightarrows}} U \xrightarrow{\widetilde{p}} V$$

Figure 6: The restricted pair and its cokernel in Lemma 1.1.

The verification is easy.

Lemma 1.2.— *Let $d_0, d_1 : X_1 \rightrightarrows X_0$ be a diagram in* (**Sch**)*, and let (Y, p) be its cokernel in* (**Esp.An**)*.*

(i) If Y is a scheme and p is a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in (**Sch**)*.*

(ii) Suppose that every point of X_0 has a saturated open neighborhood U with the following property. Let $\widetilde{d}_0$ and $\widetilde{d}_1$ be the restrictions of d_0 and d_1 to $d_0^{-1}(U) = d_1^{-1}(U)$, and let (Q, q) be the cokernel of $(\widetilde{d}_0, \widetilde{d}_1)$ in (**Esp.An**)*. If Q is a scheme and q is a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in* (**Sch**)*.*

Part (i) is proved in §4(c); since the proof is short, let us repeat it here. Let $f : X_0 \to Z$ be a morphism of schemes such that $fd_0 = fd_1$. By hypothesis there is a unique morphism of ringed spaces $r : Y \to Z$ such that $f = rp$. It remains to show that, for every $y \in Y$, the homomorphism $\mathscr{O}_{r(y)} \to \mathscr{O}_y$ induced by r is local. This follows because p is surjective, so $y = p(x)$ for some x, and because the homomorphism $\mathscr{O}_{f(x)} \to \mathscr{O}_x$ induced by f is local.

Part (ii) follows from part (i) and the preceding lemma.

c) In this exposé we study the existence of $\mathrm{Coker}(d_0, d_1)$ when the double arrow (d_0, d_1) is placed in a richer context. More precisely, let

$$X_2 = X_1 \times_{d_1, d_0} X_1$$

be the fiber product of the diagram

$$\begin{array}{ccc} & & X_1 \\ & & \downarrow{\scriptstyle d_1} \\ X_1 & \xrightarrow{d_0} & X_0 \end{array} \qquad (*)$$

and let $d_0', d_2' : X_2 \to X_1$ be the two canonical projections. Thus, by definition, the square

$$\begin{array}{ccc} X_2 & \xrightarrow{d_0'} & X_1 \\ {\scriptstyle d_2'}\downarrow & & \downarrow{\scriptstyle d_1} \\ X_1 & \xrightarrow{d_0} & X_0 \end{array} \qquad (0)$$

is Cartesian.

In addition, suppose that a third arrow $d_1' : X_2 \to X_1$ is given. We say that

$$(d_0, d_1 : X_1 \rightrightarrows X_0, d_1')$$

$\mathbf{P}^1_S$ is the cokernel of $(\widetilde{d}_0, \widetilde{d}_1)$ in both (**Esp.An**) and (**Sch**), whereas the cokernel Y of (d_0, d_1) in (**Esp.An**) is the union of $\mathbf{P}^1_S$ and the point $y_0 = \{p(m)\}$. The only open subset containing y_0 is Y, and $\Gamma(Y, \mathcal{O}_Y) = \mathbf{C}$. If $f : \mathbf{A}^2_S \to T$ is a morphism of S-schemes such that $fd_0 = fd_1$, and if $V = \mathrm{Spec}(A)$ is an affine open subset of T containing $t_0 = f(y_0)$, then $f^{-1}(V) = \mathbf{A}^2$, and the ring homomorphism $A \to \mathbf{C}[x_1, x_2]$ factors through $\mathbf{C}$. This shows that $S = \mathrm{Spec}(\mathbf{C})$ is the cokernel of (d_0, d_1) in the category (**Sch**/S).

is a *$\mathscr{C}$-groupoid* if, for every object T of $\mathscr{C}$, $X_1(T)$ is the set of arrows of a groupoid $X_*(T)$ whose set of objects is $X_0(T)$, whose source map is $d_1(T)$, whose target map is $d_0(T)$, and whose composition map is $d'_1(T)$. As usual, we identify

$$(X_1 \times_{d_1,d_0} X_1)(T) \simeq X_1(T) \times_{d_1(T),d_0(T)} X_1(T),$$

and recall that a groupoid is a category in which every arrow is invertible.[5]

If φ is an arrow of the groupoid $X_*(T)$, the map $f \mapsto \varphi \circ f$ is a bijection from the set of arrows f whose target equals the source of φ onto the set of arrows having the same target as φ. This is equivalently expressed by saying that the square

$$\begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ {\scriptstyle d'_0}\downarrow & & \downarrow{\scriptstyle d_0} \\ X_1 & \xrightarrow{d_0} & X_0 \end{array} \tag{1}$$

is Cartesian.

Similarly, $g \mapsto g \circ \varphi$ is a bijection from the set of arrows g of $X_*(T)$ whose source equals the target of φ onto the set of arrows having the same source as φ. Equivalently, the square

$$\begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ {\scriptstyle d'_2}\downarrow & & \downarrow{\scriptstyle d_1} \\ X_1 & \xrightarrow{d_1} & X_0 \end{array} \tag{2}$$

is Cartesian.

On the other hand, let $s : X_0 \to X_1$ be the unique arrow of $\mathscr{C}$ such that, for every T, the map $s(T) : X_0(T) \to X_1(T)$ sends each object of $X_*(T)$ to its identity arrow.[6] The arrow s satisfies

$$d_1 s = \mathrm{id}_{X_0}, \tag{3}$$

$$d_0 s = \mathrm{id}_{X_0}. \tag{3 bis}$$

Finally, associativity of the composition maps $d'_1(T)$ is equivalent to commutativity of the diagram

$$\begin{array}{ccc} X_1 \times_{d_1,d_0} X_1 \times_{d_1,d_0} X_1 & \xrightarrow{d'_1 \times X_1} & X_1 \times_{d_1,d_0} X_1 \\ {\scriptstyle X_1 \times d'_1}\downarrow & & \downarrow{\scriptstyle d'_1} \\ X_1 \times_{d_1,d_0} X_1 & \xrightarrow{d'_1} & X_1 \end{array} \tag{4}$$

Conversely, conditions (1), (2), and (4), together with the existence of an arrow s satisfying (3), imply that $(X_1 \rightrightarrows X_0, d'_1)$ is a $\mathscr{C}$-groupoid. Condition (3) is harmless: it merely ensures that $d_1(T) : X_1(T) \to X_0(T)$ is surjective for every $T \in \mathscr{C}$. In the sequel we shall

[5] Editor's note: in this case $X_2(T)$ is the set of pairs (f_2, f_1) of composable arrows, that is, pairs such that $d_0(f_1) = d_1(f_2)$; the maps d'_0, d'_1, d'_2 send (f_2, f_1), respectively, to $f_2, f_2 \circ f_1, f_1$.

[6] Editor's note: $T \mapsto s(T)$ defines an element of $\mathrm{Hom}(h_{X_0}, h_{X_1})$, and the latter is equal to $\mathrm{Hom}(X_0, X_1)$ by the Yoneda lemma.

mainly use the Cartesian squares (0), (1), and (2), summarized in the diagram

$$\begin{array}{ccccc} X_2 & \underset{d'_0}{\overset{d'_1}{\rightrightarrows}} & X_1 & \xrightarrow{d_0} & X_0 \\ {\scriptstyle d'_2}\downarrow & & {\scriptstyle d_1}\downarrow & & \\ X_1 & \underset{d_0}{\overset{d_1}{\rightrightarrows}} & X_0 & & \end{array} \qquad (0,1,2)$$

The two left-hand squares in this diagram (that is, squares (0) and (2)) are Cartesian; the first row is exact; and X_2 is identified with the fiber product $X_1 \times_{d_0,d_0} X_1$.

We use associativity only indirectly, for example to ensure the existence of an arrow s satisfying (3) and (3 bis), or to ensure the existence of an arrow

$$\sigma : X_1 \longrightarrow X_1, \qquad d_0\sigma = d_1, \quad d_1\sigma = d_0. \qquad (\dagger)$$

Here σ is chosen so that $\sigma(T) : X_1(T) \to X_1(T)$ sends every arrow of $X_*(T)$ to its inverse.[7]

By an abuse of language, we shall sometimes call a diagram

$$X_2 \underset{}{\overset{d'_0,d'_1,d'_2}{\rightrightarrows}} X_1 \underset{}{\overset{d_0,d_1}{\rightrightarrows}} X_0$$

a $\mathscr{C}$-groupoid when (0), (1), and (2) are Cartesian, (4) is commutative, and there exists an s satisfying (3). Thus X_2 may be "a" fiber product of (*) without being "the" fiber product of (*).[8]

Terminology. In place of the $\mathscr{C}$-groupoid X_*, we shall also speak of the *groupoid X_* with base X_0*, or of the *pre-equivalence relation X_* in X_0*.

2. Examples of $\mathscr{C}$-groupoids

a) Let X be an object of $\mathscr{C}$, and let G be a $\mathscr{C}$-group acting on the left on X. We denote by $d_0 : G \times X \to X$ the arrow defining the action of G on X, by $d_1 : G \times X \to X$ the projection onto the second factor, and by $\mu : G \times G \to G$ the arrow defining

the $\mathscr{C}$-group structure on G, and finally by $\mathrm{pr}_{2,3}$ the projection of $G\times G\times X = G\times(G\times X)$ onto the second factor. Then

$$G \times G \times X \underset{G\times d_0}{\overset{\mathrm{pr}_{2,3}}{\overset{\mu\times X}{\rightrightarrows}}} G \times X \underset{d_0}{\overset{d_1}{\rightrightarrows}} X$$

Figure 7: The groupoid attached to the action of the $\mathscr{C}$-group G on X.

is a $\mathscr{C}$-groupoid.

b) Let $d_0, d_1 : X_1 \to X_0$ be an *equivalence pair*. In other words, if $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is the arrow with components d_0 and d_1, we suppose that, for every object T of $\mathscr{C}$, the map $(d_0 \boxtimes d_1)(T)$ is a bijection from $X_1(T)$ onto the graph of an equivalence relation on $X_0(T)$. Consequently, $X_1(T)$ is identified with the set of pairs (x, y) of elements of $X_0(T)$ such that $x \sim y$. Likewise,

$$X_2(T) = (X_1 \times_{d_1,d_0} X_1)(T)$$

[7] Editor's note: the Yoneda lemma shows that σ is an involutive automorphism of X_1; this will be used, for example, in 3(e) and in Theorem 4.1.

[8] Editor's note: see Example 2(a) below.

is identified with the set of triples (x, y, z) of elements of $X_0(T)$ such that $x \sim y$ and $y \sim z$. There is therefore a unique arrow $d_1' : X_2 \to X_1$ making squares (1) and (2) commute: $d_1'(T)$ must send $(x, y, z) \in X_2(T)$ to $(x, z) \in X_1(T)$. With this choice of d_1',

$$(d_0, d_1 : X_1 \rightrightarrows X_0, d_1')$$

is a $\mathscr{C}$-groupoid.

Conversely, consider a $\mathscr{C}$-groupoid X_* for which $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is a monomorphism. Then (d_0, d_1) is an equivalence pair, and X_* can be reconstructed from (d_0, d_1) as explained a few lines above.[9]

c) If $p : X \to Y$ is any arrow of $\mathscr{C}$, and if pr_1 and pr_2 are the two projections of $X \times_{p,p} X$ onto X, then

$$(\mathrm{pr}_1, \mathrm{pr}_2) : X \times_{p,p} X \rightrightarrows X$$

is an equivalence pair. The arrow p is called an *effective epimorphism* if the diagram

$$X \times_{p,p} X \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} X \xrightarrow{\ p\ } Y$$

Figure 8: The defining diagram for an effective epimorphism.

is exact, that is, if

$$(Y, p) = \mathrm{Coker}(\mathrm{pr}_1, \mathrm{pr}_2).$$

For example, let S be a Noetherian scheme and let $\mathscr{C}$ be the category of schemes finite over S. We show that an epimorphism in $\mathscr{C}$ need not be effective. Take

$$S = \mathrm{Spec}\, k[T^3, T^5], \qquad Y = S, \qquad X = \mathrm{Spec}\, k[T],$$

where k is a commutative field. If i is the inclusion of $B = k[T^3, T^5]$ into $A = k[T]$, take $p = \mathrm{Spec}\, i$. Then $X \times_{p,p} X$ is identified with $\mathrm{Spec}(A \otimes_B A)$, and $\mathrm{Coker}(\mathrm{pr}_1, \mathrm{pr}_2)$ with $\mathrm{Spec}\, B'$, where B' is the subring of A consisting of the elements a such that $a \otimes_B 1 = 1 \otimes_B a$. But

$$T^7 \otimes_B 1 = (T^2 T^5) \otimes_B 1 = T^2 \otimes_B T^5 = T^2 \otimes_B (T^3 T^2) = T^5 \otimes_B T^2 = 1 \otimes_B T^7.$$

Thus T^7 belongs to B', does not belong to B, and $\mathrm{Spec}\, B'$ is different from $\mathrm{Spec}\, B$, which gives the counterexample.[10]

3. Some sorites concerning $\mathscr{C}$-groupoids

Here are, in no particular order, some remarks used below.

a) Let

$$X_2 \underset{}{\overset{d_0', d_1', d_2'}{\rightrightarrows\!\rightrightarrows}} X_1 \xrightarrow{\ d_0, d_1\ } X_0$$

Figure 9: The $\mathscr{C}$-groupoid X_*.

[9] Editor's note: in particular, if G is a $\mathscr{C}$-group acting on the left on an object X of $\mathscr{C}$, and if X_* is the $\mathscr{C}$-groupoid defined in 2(a), then (d_0, d_1) is an equivalence pair if and only if G acts freely on X; cf. Exposé III, 3.2.1.

[10] Editor's note: the same argument applies to $B = k[T^3, T^4]$ and $T^5 \otimes_B 1$; more generally, it applies to $B = k[T^n, T^{n+r}]$ and the element $T^{n+2r} \otimes_B 1$, provided that n does not divide $2r$.

be a $\mathscr{C}$-groupoid, and let $f_0 : Y_0 \to X_0$ be an arrow of $\mathscr{C}$. We shall define a $\mathscr{C}$-groupoid with base Y_0,

$$Y_2 \overset{e_0',e_1',e_2'}{\rightrightarrows} Y_1 \overset{e_0,e_1}{\rightrightarrows} Y_0$$

Figure 10: The induced $\mathscr{C}$-groupoid Y_*.

which will be said to be *induced by* X_* *and* f_0. We also say that Y_* is the *inverse image of* X_* *under the base change* f_0.

We choose Y_1 to be the fiber product of the diagram

$$\begin{array}{ccc} Y_1 & \xrightarrow{f_1} & X_1 \\ \downarrow & & \downarrow{\scriptstyle d_0 \boxtimes d_1} \\ Y_0 \times Y_0 & \xrightarrow[f_0 \times f_0]{} & X_0 \times X_0 \end{array}$$

Figure 11: The fiber-product definition of Y_1.

and take e_0 and e_1 to be the composites of the canonical arrow $Y_1 \to Y_0 \times Y_0$ with the first and second projections of $Y_0 \times Y_0$, respectively. The morphism $Y_1 \to Y_0 \times Y_0$ is then $e_0 \boxtimes e_1$, and

$$f_0 \circ e_i = d_i \circ f_1 \qquad (i = 0, 1),$$

where f_1 denotes the projection of Y_1 onto X_1.

Set $Y_2 = Y_1 \times_{e_0,e_1} Y_1$, cf. 1(c). One may say that the pair (e_0, e_1) is defined so that, for every $T \in \mathscr{C}$ and every pair (y, x) of elements of $Y_0(T)$, there is a certain bijective correspondence

$$\psi \longmapsto {}_y\psi_x$$

between the arrows ψ of $X_*(T)$ with source $f_0(x)$ and target $f_0(y)$, and the arrows ${}_y\psi_x$ of $Y_*(T)$ with source x and target y. Thus $e_1' : Y_2 \to Y_1$ is determined by defining, for every $T \in \mathscr{C}$, the composition of arrows in $Y_*(T)$ by the formula

$${}_z\psi_y \circ {}_y\varphi_x = {}_z(\psi \circ \varphi)_x.$$

It is clear that this definition makes each $Y_*(T)$ a groupoid.

b) Given the $\mathscr{C}$-groupoid X_* and the base change $f_0 : Y_0 \to X_0$, one can reconstruct the pair $(e_0, e_1) : Y_1 \rightrightarrows Y_0$ in another way.[11] Construct $Y_0 \times_{X_0} X_1$, pr_1, and pr_2 so that the square

$$\begin{array}{ccc} Y_0 \times_{X_0} X_1 & \xrightarrow{\mathrm{pr}_2} & X_1 \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle d_0} \\ Y_0 & \xrightarrow[f_0]{} & X_0 \end{array}$$

Figure 12: The Cartesian square defining $Y_0 \times_{X_0} X_1$.

is Cartesian. A straightforward reduction to the set-theoretic case then shows that the following square is Cartesian:

[11] Editor's note: this second viewpoint is used in 3(f) and in the proof of 6.1.

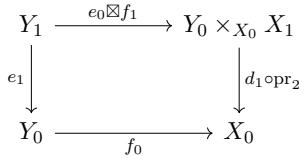

Figure 13: The second Cartesian square in the reconstruction of (e_0, e_1).

Here f_1 denotes the canonical projection of

$$Y_1 = (Y_0 \times Y_0) \times_{(X_0 \times X_0)} X_1$$

onto X_1.

c) We give two examples of inverse images of a $\mathscr{C}$-groupoid. Take $Y_0 = X_1$ and $f_0 = d_0$. For every object T of $\mathscr{C}$, the set $Y_1(T)$ is then identified with the set of diagrams of the form

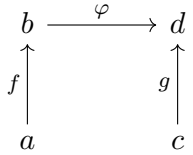

Figure 14: An arrow of the first inverse-image groupoid.

in $X_*(T)$. The source of such a diagram is the arrow f, and its target is the arrow g. These diagrams compose in the evident way.

Now set $Y_0' = X_1$ and $f_0' = d_1$ (we add the primes[12] to avoid any confusion with the preceding example). In this case, for every $T \in \mathscr{C}$, the set $Y_1'(T)$ is identified with the set of diagrams of the form

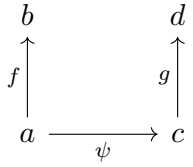

Figure 15: An arrow of the second inverse-image groupoid.

in the groupoid $X_*(T)$. The source of such a diagram is f, its target is g, and these diagrams again compose in the evident way.

It is now clear that the identity map of $Y_0(T)$, together with the map

Figure 16: The map from $Y_1(T)$ to $Y_1'(T)$.

defines an isomorphism from the groupoid $Y_*(T)$ onto $Y_*'(T)$. Moreover, this isomorphism depends functorially on T, so the $\mathscr{C}$-groupoids Y_* and Y_*' are isomorphic.[13]

[12] Editor's note: "accents" in the original.

[13] Editor's note: this plays a crucial role in the proof of Lemma 6.1.

d)

Proposition 3.1.— *Keep the notation of 3(a), and suppose that f_0 is an effective and universal epimorphism. Then* $\mathrm{Coker}(d_0, d_1)$ *exists if and only if* $\mathrm{Coker}(e_0, e_1)$ *exists.*[14] *Moreover, in this case f_0 induces an isomorphism*

$$\mathrm{Coker}(d_0, d_1) \xrightarrow{\sim} \mathrm{Coker}(e_0, e_1).$$

Recall first that an epimorphism $f_0 : Y_0 \to X_0$ is called *universal* if, in every Cartesian square

$$\begin{array}{ccc} Y' & \longrightarrow & Y_0 \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f_0} \\ X' & \longrightarrow & X_0 \end{array}$$

Figure 17: A base change of the universal epimorphism f_0.

the arrow f' is an epimorphism. Now let $C(d_0, d_1)$ denote the covariant functor from $\mathscr{C}$ to sets which assigns to every $T \in \mathscr{C}$ the kernel of the pair

$$T(d_0), T(d_1) : T(X_0) \rightrightarrows T(X_1),$$

and define $C(e_0, e_1)$ in the same way. For every $T \in \mathscr{C}$, there is therefore a commutative diagram

$$\begin{array}{ccccc} C(d_0,d_1)(T) & \longrightarrow & T(X_0) & \underset{T(d_0)}{\overset{T(d_1)}{\rightrightarrows}} & T(X_1) \\ {\scriptstyle T(f)}\downarrow & & \downarrow{\scriptstyle T(f_0)} & & \downarrow{\scriptstyle T(f_1)} \\ C(e_0,e_1)(T) & \longrightarrow & T(Y_0) & \underset{T(e_0)}{\overset{T(e_1)}{\rightrightarrows}} & T(Y_1) \end{array}$$

Figure 18: The comparison of the functors $C(d_0, d_1)$ and $C(e_0, e_1)$.

Here $T(f)$ is the injection induced by the injection $T(f_0)$. If we show that $T(f)$ is surjective for every T, we obtain a functorial isomorphism

$$f : C(d_0, d_1) \xrightarrow{\sim} C(e_0, e_1).$$

Thus one of these functors is representable if and only if the other is, which proves the proposition.

To prove that $T(f)$ is surjective, consider the diagram

$$\begin{array}{ccccc} & & Y_1 & \xrightarrow{f_1} & X_1 \\ & \overset{\Delta}{\nearrow} & {\scriptstyle e_0}\downarrow\downarrow{\scriptstyle e_1} & & {\scriptstyle d_0}\downarrow\downarrow{\scriptstyle d_1} \\ Y_0 \times_{X_0} Y_0 & \underset{\mathrm{pr}_1}{\overset{\mathrm{pr}_2}{\rightrightarrows}} & Y_0 & \xrightarrow[f_0]{} & X_0 \end{array}$$

Figure 19: The diagram used to prove the surjectivity of $T(f)$.

[14] Editor's note: the original has been modified to display explicitly the isomorphism below.

Here Δ is the section of $Y_1 \to Y_0 \times Y_0$ defined by the morphism

$$s \circ f_0 \circ \mathrm{pr}_1 : Y_0 \times Y_0 \longrightarrow X_1,$$

where $s : X_0 \to X_1$ satisfies (3) and (3 bis) of Section 1.

If $g : Y_0 \to T$ satisfies $g \circ e_0 = g \circ e_1$, then

$$g \circ e_0 \circ \Delta = g \circ e_1 \circ \Delta,$$

so $g \circ \mathrm{pr}_1 = g \circ \mathrm{pr}_2$. Since f_0 is an effective epimorphism, g factors as $h \circ f_0$ for an arrow $h : X_0 \to T$; in other words, $g = T(f_0)(h)$. It remains to prove that $h \in C(d_0, d_1)(T)$, or equivalently that $hd_0 = hd_1$. But

$$hd_0f_1 = hf_0e_0 = ge_0 = ge_1 = hf_0e_1 = hd_1f_1,$$

and the desired equality follows because f_1 is an epimorphism (since f_0 is a universal epimorphism).

e) Now let S be a scheme and take $\mathscr{C} = (\mathbf{Sch}/S)$. The data of a $\mathscr{C}$-groupoid

$$X_2 \;\underset{\underset{d_0'}{\overset{d_2'}{\rightrightarrows}}}{\overset{}{\rightrightarrows}}\; X_1 \;\underset{d_0}{\overset{d_1}{\rightrightarrows}}\; X_0$$

Figure 20: A $(\mathbf{Sch}/S)$-groupoid.

defines an equivalence relation on the set $\underline{X_0}$ underlying the scheme X_0: for $x, y \in \underline{X_0}$, write $x \sim y$ if there exists $z \in \underline{X_1}$ such that $x = d_1 z$ and $y = d_0 z$. Reflexivity and symmetry are clear.[15] Let us prove transitivity. If $x \sim y$ and $y \sim z$, there exist $u, v \in \underline{X_1}$ such that

$$x = d_1 u, \qquad y = d_0 u, \qquad y = d_1 v, \qquad z = d_0 v.$$

It follows that (v, u) belongs to the set-theoretic fiber product $\underline{X_1} \times_{\underline{d_1}, \underline{d_0}} \underline{X_1}$. Since the canonical map

$$\underline{X_1 \times_{d_1, d_0} X_1} \longrightarrow \underline{X_1} \times_{\underline{d_1}, \underline{d_0}} \underline{X_1}$$

Figure 21: The canonical map from the underlying set of the fiber product to the fiber product of the underlying sets.

is surjective, (v, u) is the image of some $w \in \underline{X_2}$. Hence

$$x = d_1 d_1' w, \qquad z = d_0 d_1' w,$$

and therefore $x \sim z$.

f) Keep the notation of 3(a) and 3(b), still taking $\mathscr{C} = (\mathbf{Sch}/S)$. If x, y are points of Y_0, then

$$x \sim y \quad \Longleftrightarrow \quad f_0(x) \sim f_0(y).$$

Thus the inverse image of the equivalence relation defined by a groupoid is the equivalence relation defined by the inverse image of that groupoid.

[15] Editor's note: reflexivity follows from the existence of $s : X_0 \to X_1$, which is a section of both d_0 and d_1; symmetry follows from the existence of the involution σ of X_1 which "exchanges d_0 and d_1", that is, which satisfies $d_0\sigma = d_1$ and $d_1\sigma = d_0$; cf. Section 1, (3), (3 bis), and (†).

Indeed, suppose that $x \sim y$. There is a $z \in \underline{Y_1}$ such that $x = e_1(z)$ and $y = e_0(z)$. Since $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, it follows that

$$f_0(x) = d_1 f_1(z), \qquad f_0(y) = d_0 f_1(z),$$

and hence $f_0(x) \sim f_0(y)$.

Conversely, suppose that $f_0(x) \sim f_0(y)$, and let $z \in \underline{X_1}$ satisfy

$$f_0(y) = d_1(z), \qquad f_0(x) = d_0(z).$$

Using the construction and notation of 3(b), there is a point $t \in \underline{Y_0 \times_{X_0} X_1}$ such that $\mathrm{pr}_1(t) = x$ and $\mathrm{pr}_2(t) = z$. Since $f_0(y) = d_1 \,\mathrm{pr}_2(t)$, there is likewise an $s \in \underline{Y_1}$ such that

$$y = e_1(s), \qquad (e_0 \boxtimes f_1)(s) = t.$$

It follows that

$$e_0(s) = \mathrm{pr}_1(e_0 \boxtimes f_1)(s) = \mathrm{pr}_1(t) = x,$$

so $x \sim y$.

4. Passage to the quotient by a finite flat groupoid (proof of a special case)

Theorem 4.1.— *Consider a* $(\mathbf{Sch}/S)$*-groupoid*

$$X_2 \;\underset{\underset{d_0'}{\overset{d_2'}{\rightrightarrows}}}{\overset{d_1'}{\rightrightarrows}}\; X_1 \;\underset{d_0}{\overset{d_1}{\rightrightarrows}}\; X_0$$

and suppose that the following conditions hold:[16]

a) d_1 *is finite locally free;*

b) for every $x \in X_0$*, the set* $d_0 d_1^{-1}(x)$ *is contained in an affine open subset of* X_0*.*[17]

Then:

(i) *there exists a cokernel* (Y, p) *of* (d_0, d_1) *in* $(\mathbf{Sch}/S)$*. Moreover, such a pair* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of all ringed spaces;*

(ii) p *is integral and open, and* Y *is affine when* X_0 *is affine;*[18]

(iii) *the morphism*

$$X_1 \longrightarrow X_0 \times_Y X_0$$

with components d_0 *and* d_1 *is surjective;*

[16] Editor's note: since $d_0 = d_1\sigma$, where σ is an involutive automorphism of X_1, these two conditions are symmetric in d_1 and d_0. Moreover, $d_0 d_1^{-1}(x) = d_1 d_0^{-1}(x)$.

[17] Editor's note: condition b) cannot be omitted. H. Hironaka gave an example of an action of the finite group $G = \mathbb{Z}/2\mathbb{Z}$ on a proper complex variety X_0 for which the quotient X_0/G is an algebraic space that is not a scheme; see [Hi62], and also [Mum65], Chapter 4, §3.

[18] Editor's note: the assertion that p is open has been added by repeating the analogous argument given in 6.1.

(iv) *if (d_0, d_1) is an equivalence pair, then $X_1 \to X_0 \times_Y X_0$ is an isomorphism*[19] *and $p : X_0 \to Y$ is finite locally free.*[20] *Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology and, for every base change $Y' \to Y$, the scheme Y' is the cokernel of the groupoid $X_* \times_Y Y'$ obtained from X_* by the base change $X_0 \times_Y Y' \to X_0$.*

In particular, for every base change $S' \to S$, the scheme $Y' = Y \times_S S'$ is the cokernel of the S'-groupoid $X'_* = X_* \times_S S'$. Thus, in this case, formation of the quotient commutes with base change.

It follows immediately from (i) that the underlying topological space of Y is the quotient of the underlying topological space of X_0 by the equivalence relation defined by the $(\mathbf{Sch}/S)$-groupoid X_*.

We shall first prove the theorem when X_0 is affine and d_1 is locally free of constant rank n. We shall then see how to reduce the general situation to this special case.

In the case now under consideration, X_0, X_1, X_2 are affine. We may therefore suppose that

$$X_i = \operatorname{Spec} A_i, \qquad d_j = \operatorname{Spec} \delta_j, \qquad d'_k = \operatorname{Spec} \delta'_k,$$

where the A_i are commutative rings and the δ_j, δ'_k are ring homomorphisms. The diagram (0,1,2) may then be replaced by the following dual diagram:

$$(0,1,2)^* \qquad \begin{array}{ccccc} & & A_2 & \overset{\delta'_1}{\underset{\delta'_0}{\leftleftarrows}} & A_1 & \xleftarrow{\delta_0} & A_0 \\ & & \uparrow{\scriptstyle \delta'_2} & & \uparrow{\scriptstyle \delta_1} & & \\ & & A_1 & \overset{\delta_1}{\underset{\delta_0}{\leftleftarrows}} & A_0 & & \end{array}$$

The two left-hand squares in this diagram are cocartesian.

Let B be the subring of A_0 consisting of the elements $a \in A_0$ for which $\delta_0(a) = \delta_1(a)$.

a) We first show that A_0 is integral over B. For $a \in A_0$, let

$$P_{\delta_1}(T, \delta_0(a)) = T^n - \sigma_1 T^{n-1} + \cdots + (-1)^n \sigma_n \tag{1}$$

be the characteristic polynomial of $\delta_0(a)$, where A_1 is regarded as an A_0-algebra through δ_1; see Bourbaki, *Algèbre*, VIII, §12, and *Algèbre commutative*, II, §5, Exercise 9. Since the two left-hand squares of the diagram $(0,1,2)^*$ are cocartesian, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_0\delta_0(a)), \tag{2}$$

$$\delta_1(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_1\delta_0(a)). \tag{3}$$

Because $\delta'_0\delta_0 = \delta'_1\delta_0$, it follows that

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = \delta_1(P_{\delta_1}(T, \delta_0(a)))\,;$$

equivalently,

$$\delta_0(\sigma_i) = \delta_1(\sigma_i) \qquad \text{for every } i.$$

[19] Editor's note: in this case $X_1 \to X_0 \times_S X_0$ is therefore an immersion (EGA I, 5.3.10); see also Exposé $\mathrm{VI_B}$, 9.2.1. For existence of the quotient—in the category of schemes or in that of algebraic spaces—under the weaker hypothesis that d_0 and d_1 are finite but not necessarily flat, see [An73], §1.1, [Fe03], [Ko08], and the references there.

[20] Editor's note: the consequences that follow have been made explicit; see [Ray67a], Theorem 1(iii), and the proof given below at the end of Section 5.

On the other hand, the Cayley–Hamilton theorem gives

$$\delta_0(a)^n - \delta_1(\sigma_1)\delta_0(a)^{n-1} + \cdots + (-1)^n\delta_1(\sigma_n) = 0.$$

Since $\delta_1(\sigma_i) = \delta_0(\sigma_i)$, one also has

$$\delta_0(a)^n - \delta_0(\sigma_1)\delta_0(a)^{n-1} + \cdots + (-1)^n\delta_0(\sigma_n) = 0.$$

Hence

$$a^n - \sigma_1 a^{n-1} + \cdots + (-1)^n\sigma_n = 0,$$

because there is a homomorphism $\tau : A_1 \to A_0$ such that $\tau\delta_0 = \mathrm{id}_{A_0}$, so δ_0 is injective. It follows that A_0 is integral over B.

b) Consider two prime ideals x and y of A_0. We shall show that the equality

$$x \cap B = y \cap B$$

implies the existence of a prime ideal z of A_1 such that

$$x = d_0(z), \qquad y = d_1(z).$$

Indeed, if this assertion were false, x would be distinct from $\delta_0^{-1}(t)$ for every prime ideal t of A_1 such that $\delta_1^{-1}(t) = y$. For every such t, one would have

$$\delta_0^{-1}(t) \cap B = \delta_1^{-1}(t) \cap B = y \cap B = x \cap B.$$

It would follow from Cohen–Seidenberg (see Bourbaki, *Algèbre commutative*, V, §2, Corollary 1 to Proposition 1) that x is contained in none of the ideals $\delta_0^{-1}(t)$.[21] There are at most n prime ideals t of A_1 satisfying $\delta_1^{-1}(t) = y$ (see *loc. cit.*, Proposition 3). Hence, by the Prime Avoidance Lemma (*loc. cit.*, II, §1, Proposition 3), there would be an element $a \in x$ belonging to none of the ideals $\delta_0^{-1}(t)$.

Consequently, $\delta_0(a)$ would belong to none of these prime ideals t, and the lemma below would imply that the norm $N_{\delta_1}(\delta_0(a))$ does not belong to y. Here the norm is computed by regarding A_1 as an A_0-algebra through δ_1; with the notation of part a),

$$N_{\delta_1}(\delta_0(a)) = \sigma_n.$$

On the other hand,

$$(-1)^{n-1}\sigma_n = a^n + \sum_{i=1}^{n-1}(-1)^i\sigma_i a^{n-i},$$

so this norm belongs to $B \cap x = B \cap y$, a contradiction.

Lemma 4.1.1.— *Let $A \to A'$ be a homomorphism of commutative rings which makes A' a projective A-module of rank n. Let $p \in \mathrm{Spec}(A)$, let $q_1, \ldots, q_r$ be the points of $\mathrm{Spec}(A')$ lying above p, and let $a \in A'$. Then $a \in q_1 \cup \cdots \cup q_r$ if and only if its norm $N(a)$ belongs to p.*

Indeed, after replacing A and A' by the localizations A_p and A'_p, we reduce to the case in which (A, p) is local and A' is semilocal, with

$$\mathrm{Spec}(A') = \{q_1, \ldots, q_r\}.$$

[21] Editor's note: the original has been expanded in what follows; in particular, Lemma 4.1.1, taken from [DG70], III, §2.4, Lemma 4.3, has been added.

In this case A' is a free A-module of rank n (see Bourbaki, *Algèbre commutative*, II, §3.2, Corollary 2 to Proposition 5), and $N(a)$ is the determinant of the endomorphism

$$\ell_a : A' \longrightarrow A', \qquad a' \longmapsto aa'.$$

Thus one has the equivalences

$$N(a) \notin p \iff N(a) \text{ is invertible} \iff \ell_a \text{ is invertible} \iff a \notin q_1 \cup \cdots \cup q_r.$$

c) Proof of (i). Set $Y = \operatorname{Spec} B$ and $p = \operatorname{Spec} i$, where $i : B \hookrightarrow A_0$ is the inclusion. By part a), the morphism $p : X_0 \to Y$ is surjective. We first show that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. Indeed, part b) shows that the set underlying $\operatorname{Spec} B$ is obtained from the set underlying X_0 by identifying the points x and y for which there exists $z \in X_1$ such that

$$d_1 z = y, \qquad d_0 z = x.$$

Moreover, since i is integral, $p = \operatorname{Spec} i$ is closed, and therefore Y carries the quotient topology induced by X_0.

It follows that p is open. Indeed, let U' be any open subset of X_0. Since d_1 is surjective and finite locally free, it is faithfully flat and of finite presentation, hence open. Thus

$$U = d_1\left(d_0^{-1}(U')\right),$$

the saturation of U' for the equivalence relation defined by X_*, is open. Since Y has the quotient topology, $p(U') = p(U)$ is open.

Finally, the choice of B, together with the fact that p, d_0, and d_1 are affine, shows that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*\mathcal{O}_{X_0} \underset{p_*(\delta_0)}{\overset{p_*(\delta_1)}{\rightrightarrows}} p_*d_{0*}\mathcal{O}_{X_1} = p_*d_{1*}\mathcal{O}_{X_1}$$

is exact.

It remains to show that (Y, p) is also the cokernel of (d_0, d_1) in the category of schemes—more generally, in the category of locally ringed spaces. Let $q : X_0 \to Z$ be a morphism of schemes such that $qd_0 = qd_1$. By what precedes, there is a unique morphism of ringed spaces $r : Y \to Z$ such that $q = rp$. We must show that, for every $y \in Y$, the homomorphism

$$\mathcal{O}_{Z,r(y)} \longrightarrow \mathcal{O}_{Y,y}$$

induced by r is local. Since p is surjective, $y = p(x)$ for some $x \in X_0$, and the assertion follows from the fact that the homomorphism

$$\mathcal{O}_{Z,q(x)} \longrightarrow \mathcal{O}_{X_0,x}$$

induced by q is local.

d) Proof of (ii). This follows from parts a) and c).

e) Proof of (iii). Recall that $\underline{P}$ denotes the set underlying a scheme P, and that $\underline{d} : \underline{P} \to \underline{Q}$ denotes the map induced by a morphism $d : P \to Q$.

Lemma 4.1.2.—[22] *Let $(A, \mathfrak{m})$ be a local ring, let k be its residue field, and let K/k be a field extension. There exists a local, flat A-algebra B such that $B/\mathfrak{m}B$ is k-isomorphic to K. If K/k is finite, B may moreover be chosen finite and free over A.*

[22] Editor's note: this lemma, which is used repeatedly in this Exposé and in later Exposés ($\mathrm{VI_A}$, $\mathrm{VI_B}$), has been inserted here. It appeared as Lemma $\mathrm{VI_B}$, 4.5.1, in the original 1965 edition of SGA 3.

This is proved in EGA 0_{III}, 10.3.1, where it is also shown that B may be chosen Noetherian when A is Noetherian. For the reader's convenience, we recall the proof.

Put $A' = A[T]$, where T is an indeterminate. If $K = k(T)$, let $\mathfrak{p} = \mathfrak{m}A'$ and $B = A'_{\mathfrak{p}}$. Then

$$B/\mathfrak{m}B \simeq k[T]_{(0)} = k(T).$$

The ring B is flat over A', which is a free A-module, so B is flat over A.

Suppose next that $K = k(t) = k[t]$, where t is algebraic over k. Set $B = A'/(F)$, where $F \in A'$ is a monic polynomial whose image in $k[T]$ is the minimal polynomial f of t over k. Then B is a free A-module of finite rank

$$\deg(F) = \deg(f).$$

In particular, B is integral over A, and hence every maximal ideal of B contains $\mathfrak{m}$. Since

$$B/\mathfrak{m}B \simeq k[T]/(f) \simeq K,$$

B is local, with maximal ideal $\mathfrak{m}B$. This already proves that B can be chosen finite and free over A when $[K : k] < \infty$.

In the general case, choose a family $(t_i)_{i\in I}$ of generators of K/k, and well-order I; that is, give I a total order $\leq$ in which every nonempty subset has a least element. For $i \in I$, let k_i (respectively $k_{<i}$) be the subfield of K generated by the t_j with $j \leq i$ (respectively $j < i$). By adding one element if necessary, we may suppose that I has a greatest element ξ, so that $K = k_\xi$.

Let $J \subseteq I$ be the set of indices i for which there exists a direct system $(A_j)_{j\leq i}$ of local, flat A-algebras such that

$$A_j/\mathfrak{m}A_j \simeq k_j$$

and A_j is flat over A_ℓ whenever $\ell < j$. Suppose that $I \setminus J$ is nonempty, let i be its least element, and put

$$A' = \varinjlim_{j<i} A_j.$$

Since tensor products commute with direct limits, A' is flat over A and over every A_j, $j < i$, and

$$A'/\mathfrak{m}A' \simeq A' \otimes_A (A/\mathfrak{m}) \simeq k_{<i}.$$

Moreover, A' is local, with maximal ideal $\mathfrak{m}A'$. Indeed, if $x = f_j(x_j)$ is not invertible, then x_j is not invertible and hence belongs to the maximal ideal $\mathfrak{m}A_j$ of A_j; therefore $x \in \mathfrak{m}A'$.

The monogenic case treated above now gives a local, flat A'-algebra A_i such that

$$A_i/\mathfrak{m}A_i \simeq k_{<i}(t_i) = k_i.$$

The algebra A_i is flat over every A_j, $j < i$, so $i \in J$, contrary to the choice of i. This contradiction proves that $J = I$, and A_ξ has the required properties. This proves Lemma 4.1.2.

Let us now prove Theorem 4.1(iii). Recall once more that $\underline{P}$ denotes the set underlying a scheme P, and that $\underline{d} : \underline{P} \to \underline{Q}$ is the map induced by a morphism $d : P \to Q$. Part b) can then be restated by saying that the map

$$\underline{d_0} \boxtimes \underline{d_1} : \underline{X_1} \longrightarrow \underline{X_0} \times_{\underline{Y}} \underline{X_0}$$

with components $\underline{d_0}$ and $\underline{d_1}$ is surjective. This map factors as

$$\underline{X_1} \xrightarrow{\underline{d_0}\boxtimes\underline{d_1}} \underline{X_0} \times_{\underline{Y}} \underline{X_0} \xrightarrow{q} \underline{X_0 \times_Y X_0},$$

where q is the canonical map. The image of $\underline{d_0} \boxtimes \underline{d_1}$ therefore contains every point $v \in \underline{X_0 \times_Y X_0}$ such that

$$\{v\} = q^{-1}(q(v)).$$

This condition is satisfied in particular when v is *rational over* Y, that is, when the residue field $\kappa(v)$ is identified with the residue field $\kappa(w)$ of the image w of v in Y.[23]

If $v \in \underline{X_0 \times_Y X_0}$ is not rational over Y, again let w be its image in Y. By Lemma 4.1.2, there exist a local ring C with radical $\mathfrak{m}$ and a local, flat homomorphism

$$f : \mathcal{O}_{Y,w} \longrightarrow C$$

such that $C/\mathfrak{m}$ is isomorphic to $\kappa(v)$ as a $\kappa(w)$-algebra. Put $Y' = \operatorname{Spec} C$, and let $\pi : Y' \to Y$ be the morphism induced by f. The canonical projection

$$(X_0 \times_Y X_0) \times_Y Y' \longrightarrow X_0 \times_Y X_0$$

sends some point v', rational over Y', to v. Since

$$(X_0 \times_Y X_0) \times_Y Y' \simeq (X_0 \times_Y Y') \times_{Y'} (X_0 \times_Y Y'),$$

and since the hypotheses of Theorem 4.1 and the preceding results, in particular part b), remain valid after the base change $\pi : Y' \to Y$, the point v' is the image of an element

$$u' \in \underline{X_1 \times_Y Y'}$$

under the base change of $d_0 \boxtimes d_1$. If u is the image of u' in X_1, then

$$v = (d_0 \boxtimes d_1)(u),$$

as required.

f) Proof of (iv).[24]

Lemma 4.1.3.— *If a monomorphism of schemes $f : T \to Z$ is finite, it is a closed immersion.*

Covering Z by affine open subsets Z_i, and replacing f by the induced morphisms $f^{-1}(Z_i) \to Z_i$, reduces us—since a finite morphism is affine—to the case

$$Z = \operatorname{Spec} B, \qquad T = \operatorname{Spec} A.$$

Because f is a monomorphism, the diagonal morphism

$$T \longrightarrow T \times_Z T$$

is an isomorphism (EGA I, 5.3.8); equivalently, multiplication gives an isomorphism

$$A \otimes_B A \xrightarrow{\sim} A.$$

Consequently, for every maximal ideal $\mathfrak{m}$ of B, there is an isomorphism

$$(A/\mathfrak{m}A) \otimes_k (A/\mathfrak{m}A) \xrightarrow{\sim} A/\mathfrak{m}A, \qquad k = B/\mathfrak{m}.$$

[23] Editor's note: the pages of Lecture Notes in Mathematics 151 are permuted here; their actual order is 265–266–268–269–267–270–271.

[24] Editor's note: the following lemma has been added from [DG70], I, §5, Proposition 1.5; see also the proof of EGA IV$_3$, 8.11.5. It was used implicitly in the original and explicitly in [DG70], III, §2, 4.6. It is also useful in Theorem 7.1 below.

Since A is finite over B, the k-vector space $A/\mathfrak{m}A$ has finite dimension d, and this isomorphism implies $d = 0$ or $d = 1$. Hence the map

$$k = B/\mathfrak{m} \longrightarrow A/\mathfrak{m}A$$

is surjective. Nakayama's lemma, applied after localization at $\mathfrak{m}$, shows that $B_\mathfrak{m} \to A_\mathfrak{m}$ is surjective. It follows that $B \to A$ is surjective: its cokernel C satisfies $C_\mathfrak{m} = 0$ for every maximal ideal $\mathfrak{m}$, and therefore vanishes. This proves the lemma.

Let us now prove (iv). By hypothesis,

$$X_0 = \operatorname{Spec} A_0, \qquad X_1 = \operatorname{Spec} A_1,$$

and, for $i = 0, 1$, the homomorphism $\delta_i : A_0 \to A_1$ makes A_1 a finite A_0-module. Therefore, a fortiori, the homomorphism

$$A_0 \otimes_B A_0 \longrightarrow A_1$$

is finite.

We assume in addition that

$$d = d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_Y X_0$$

is a monomorphism. By Lemma 4.1.3, the corresponding homomorphism $A_0 \otimes_B A_0 \to A_1$ is surjective. We shall prove that it is an isomorphism; in the course of doing so, we shall also prove that $p : X_0 \to Y$ is finite locally free.

It is enough to prove that, for every prime ideal $\mathfrak{p}$ of B, the homomorphism

$$(A_0)_\mathfrak{p} \otimes_{B_\mathfrak{p}} (A_0)_\mathfrak{p} \longrightarrow (A_1)_\mathfrak{p},$$

with components $\delta_{0\mathfrak{p}}$ and $\delta_{1\mathfrak{p}}$, is bijective. Thus we may suppose that B is local. Part b) then shows that $(A_0)_\mathfrak{p}$ is semilocal. Indeed, if $\mathfrak{m}$ is one of its maximal ideals, the others have the form $\delta_0^{-1}(\mathfrak{n})$, where $\mathfrak{n}$ ranges over the prime ideals of A_1 satisfying $\delta_1^{-1}(\mathfrak{n}) = \mathfrak{m}$. There are at most

$$n = [A_1 : A_0]$$

such prime ideals $\mathfrak{n}$.

After a faithfully flat base change, we may also suppose that the residue field of B is infinite.[25] We can then use the following lemma.

Lemma 4.2.— *Let B be a local ring with infinite residue field, let A be a semilocal ring, and let $i : B \to A$ be a homomorphism which sends the maximal ideal $\mathfrak{n}$ of B into the radical $\mathfrak{r}$ of A. Let M be a free A-module of rank n, and let N be a B-submodule of M which generates M as an A-module. Then N contains an A-basis of M.*

Recall that a sequence $m_1, \ldots, m_n$ of elements of M is an A-basis if and only if its canonical images in $M/\mathfrak{r}M$ form a basis over $A/\mathfrak{r}$. We may therefore replace

$$M \text{ by } M/\mathfrak{r}M, \quad N \text{ by } N/(N \cap \mathfrak{r}M), \quad A \text{ by } A/\mathfrak{r}, \quad B \text{ by } B/\mathfrak{n}.$$

In this case the lemma is elementary. Namely, if

$$A = K_1 \times \cdots \times K_r, \qquad M = K_1^n \times \cdots \times K_r^n,$$

choose, for each j, an element $x_j \in N$ whose j-th component is nonzero. Because the field B is infinite, a suitable B-linear combination x of the x_j has all its components nonzero. Replace M by M/Ax, and proceed by induction on n.

[25] Editor's note: see Lemma 4.1.2.

Apply the lemma with $A = A_0$, with $i : B \hookrightarrow A_0$, with $M = A_1$ regarded as an A_0-module through δ_1, and with $N = \delta_0(A_0)$. Indeed, since

$$d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_Y X_0$$

is a closed immersion, the homomorphism

$$A_0 \otimes_B A_0 \longrightarrow A_1$$

with components δ_0, δ_1 is surjective. This says exactly that $\delta_0(A_0)$ generates the A_0-module A_1.

Choose $a_1, \ldots, a_n \in A_0$ so that $\delta_0(a_1), \ldots, \delta_0(a_n)$ form an A_0-basis of A_1. If $a_1, \ldots, a_n$ form a B-basis of A_0, then $A_0 \otimes_B A_0 \to A_1$ sends the basis

$$(1 \otimes a_i)_{1 \le i \le n}$$

to the basis

$$(\delta_0(a_i))_{1 \le i \le n},$$

and is therefore bijective. Thus, if

$$\varepsilon : \mathbb{Z}^n \longrightarrow A_0$$

is the homomorphism of abelian groups which sends the standard basis of $\mathbb{Z}^n$ to $a_1, \ldots, a_n$, it remains only to prove that the map

$$B \otimes_{\mathbb{Z}} \mathbb{Z}^n \longrightarrow A_0$$

with components i and ε is bijective.

The dual diagram $(0,1,2)^*$ considered at the beginning of this proof induces the following commutative diagram:

$$\begin{array}{ccccc}
A_2 & \underset{\delta_0'}{\overset{\delta_1'}{\leftleftarrows}} & A_1 & \xleftarrow{\ \delta_0\ } & A_0 \\
\uparrow{\scriptstyle u_2} & & \uparrow{\scriptstyle u_1}\ \cong & & \uparrow{\scriptstyle u_0} \\
A_1 \otimes_{\mathbb{Z}} \mathbb{Z}^n & \underset{\delta_0 \otimes \mathbb{Z}^n}{\overset{\delta_1 \otimes \mathbb{Z}^n}{\leftleftarrows}} & A_0 \otimes_{\mathbb{Z}} \mathbb{Z}^n & \xleftarrow{\ i \otimes \mathbb{Z}^n\ } & B \otimes_{\mathbb{Z}} \mathbb{Z}^n
\end{array}$$

Figure 22: The commutative ring-and-module diagram concluding the affine proof of Theorem 4.1(iv).

Here u_0, u_1, u_2 have, respectively, the component pairs

$$(i, \varepsilon), \qquad (\delta_1, \delta_0\varepsilon), \qquad (\delta_2', \delta_0'\delta_0\varepsilon).$$

We know that u_1 is an isomorphism. Since the two left-hand squares of $(0,1,2)^*$ are cocartesian, u_2 is bijective. The two horizontal rows of the diagram are exact, and therefore u_0 is bijective.[26] Thus A_0 is a locally free B-module of rank n. By the preceding reductions, it follows that

$$\delta_0 \otimes \delta_1 : A_0 \otimes_B A_0 \xrightarrow{\sim} A_1$$

is an isomorphism. This completes the proof of Theorem 4.1 in the special case under consideration: X_0 is affine and d_1 is locally free of constant rank n.

[26] Editor's note: the argument which follows has been added.

5. Passage to the quotient by a finite flat groupoid (the general case)

a) Let $U^{(n)}$ be the largest open subset of X_0 over which d_1 is finite locally free of rank n. The scheme X_0 is the disjoint union of the $U^{(n)}$. The two Cartesian squares

$$\begin{array}{ccc} X_2 & \xrightarrow{d_0'} & X_1 \\ {\scriptstyle d_2'}\downarrow & & \downarrow{\scriptstyle d_1} \\ X_1 & \xrightarrow[d_0]{} & X_0 \end{array} \qquad \textit{and} \qquad \begin{array}{ccc} X_2 & \xrightarrow{d_1'} & X_1 \\ {\scriptstyle d_2'}\downarrow & & \downarrow{\scriptstyle d_1} \\ X_1 & \xrightarrow[d_1]{} & X_0 \end{array}$$

Figure 23: The two Cartesian squares controlling the rank strata $U^{(n)}$.

show that the inverse images of $U^{(n)}$ under d_0 and d_1 both coincide with the largest open subset of X_1 over which d_2' is locally free of rank n.[27] Thus

$$d_0^{-1}(U^{(n)}) = d_1^{-1}(U^{(n)}),$$

and the groupoid X_* is the disjoint union of the groupoids $X_*^{(n)}$ induced by X_* on the open-and-closed subschemes $U^{(n)}$. It is therefore enough to prove Theorem 4.1 for each $X_*^{(n)}$. We are reduced to the case in which d_1 is finite locally free of rank n.

b) *We can now prove the theorem in the general case.*

By part a), we may suppose that d_1 is locally free of rank n. Let (Y, p) be a cokernel of (d_0, d_1) in the category of all ringed spaces. The argument at the end of 4(c) shows that, in order to prove Theorem 4.1(i), it is enough to show that Y is a scheme and that $p : X_0 \to Y$ is a morphism of schemes. By Lemma 1.2, this question is local on Y. Let $y \in Y$ and choose $x \in X_0$ with $p(x) = y$. If x has a saturated affine open neighborhood U, then Section 4 shows that $p(U)$ is an affine open subset of Y, and $p|_U : U \to p(U)$ is a morphism of schemes. Thus it remains to prove that every $x \in X_0$ has a saturated affine open neighborhood. We proceed as follows; the argument is taken from SGA 1, VIII, Corollary 7.6.

$$\begin{array}{lccccccc} d_1\left(d_0^{-1}(x)\right) & \mathcal{U} = (V_f)' & \subset & V_f & \subset & V' & \subset & V & \subset & X_0 \\ & \uparrow & & \uparrow & & \uparrow & & \nwarrow & & \\ & \text{largest saturated} & & \text{special affine} & & \text{largest saturated} & & \text{affine open} & & \\ & \text{open in } V_f & & \text{open in } V & & \text{open in } V & & & & \end{array}$$

Figure 24: The nested open subsets used to construct a saturated affine neighborhood of x.

By condition 4.1(b), there is an affine open subset $V \subseteq X_0$ which contains $d_1(d_0^{-1}(x))$.[28] Put $F = X_0 \setminus V$. Since d_1 is integral, $d_1(d_0^{-1}(F))$ is closed, and

$$V' = X_0 \setminus d_1(d_0^{-1}(F))$$

[27] Editor's note: since d_0 (respectively d_1) is surjective, flat, and finite, hence faithfully flat and affine, d_2' has rank n over a neighborhood of a point $x \in X_1$ if and only if d_1 has rank n over a neighborhood of $d_0(x)$ (respectively $d_1(x)$).

[28] Editor's note: $d_1(d_0^{-1}(x)) = d_0(d_1^{-1}(x))$; see editor's note 16 in Theorem 4.1.

is the largest saturated open subset contained in V. The set V' is a neighborhood of the finite set $d_1(d_0^{-1}(x))$. Hence there is a section f of the structure sheaf of V, vanishing on $V \setminus V'$, such that $d_1(d_0^{-1}(x))$ is contained in the principal open subset $V_f \subseteq V$. We shall prove that the largest saturated open subset $(V_f)'$ contained in V_f is affine; it will then be the required neighborhood.

Put $Z(f) = V' \setminus V_f$. Then $d_0^{-1}(Z(f))$ is the set of points of

$$d_0^{-1}(V') = d_1^{-1}(V')$$

at which $d_0^*(f)$ vanishes. The morphism induced by d_1 from this common inverse image to V' is locally free of rank n.[29] Lemma 4.1.1 therefore shows that $d_1(d_0^{-1}(Z(f)))$ is precisely the zero locus of the norm N of $d_0^*(f)$ relative to d_1. It follows that

$$(V_f)' = V' \setminus d_1(d_0^{-1}(Z(f)))$$

is the set of points of V_f at which N does not vanish. Consequently, $(V_f)'$ is affine.

This proves Theorem 4.1(i); assertions (ii), (iii), and the first part of (iv) are then clear. It remains to prove the consequences stated at the end of (iv); see [Ray67a], Theorem 1(iii).

By hypothesis, the groupoid X_* comes from an equivalence relation

$$i : R \longrightarrow X_0 \times X_0,$$

where i is an immersion (see editor's note 19). We have shown that R is effective (see Exposé IV, 3.3.2), and that

$$p : X_0 \longrightarrow Y = X_0/R$$

is surjective and finite locally free; in particular, it is faithfully flat and of finite presentation.

Consequently, if (M) denotes the family of faithfully flat morphisms locally of finite presentation, then R is (M)-effective. By Exposé IV, 6.3.3, (Y, p) therefore represents the quotient sheaf of X_0 by R for the fppf topology, and the base-change assertions follow from Exposé IV, 3.4.3.1.

Remark 5.1.—[30] Keep the hypotheses and notation of Theorem 4.1, and suppose in addition that S is locally Noetherian and that $\pi_0 : X_0 \to S$ is quasi-projective. We shall prove that $\pi : Y \to S$ is quasi-projective.

The preceding hypotheses imply that $Y \to S$ is of finite type; see the proof of 6.1(ii). Let $\mathscr{A}$ be an invertible $\mathcal{O}_{X_0}$-module which is ample for π_0. By EGA II, 6.1.12, $p_*\mathscr{A}$ is an invertible $p_*\mathcal{O}_{X_0}$-module. Hence Y has an affine open cover $(V_i)_{i\in I}$ such that $\mathscr{A}$ is trivial over each of the saturated affine open subsets

$$U_i = p^{-1}(V_i).$$

For each i, put

$$A_{i,0} = \mathcal{O}_{X_0}(U_i),$$

and let $A_{i,1}$ be the ring of the affine open subset $d_0^{-1}(U_i) = d_1^{-1}(U_i)$ of X_1. Let $\delta_{i,0}$ (respectively $\delta_{i,1}$) be the homomorphism $A_{i,0} \to A_{i,1}$ induced by d_0 (respectively d_1), and put

$$B_i = \mathcal{O}_Y(V_i) = \{b \in A_{i,0} \mid \delta_{i,0}(b) = \delta_{i,1}(b)\}.$$

Following EGA II, §6.5, consider the invertible $\mathcal{O}_{X_0}$-module

$$N_{d_1}(d_0^*\mathscr{A}),$$

[29] Editor's note: the reference to Lemma 4.1.1 has been added; see [DG70], III, §5.2, p. 313.

[30] Editor's note: this paragraph has been added.

the norm, relative to the finite locally free morphism $d_1 : X_1 \to X_0$, of the invertible $\mathcal{O}_{X_1}$-module $d_0^*\mathscr{A}$. Suppose that $\mathscr{A}$, relative to the cover $(U_i)_{i\in I}$, is given by transition functions

$$c_{ij} \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times.$$

Then $N_{d_1}(d_0^*\mathscr{A})$ is given by the transition functions

$$N_{\delta_1}(\delta_0(c_{ij})) \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times.$$

By 4(a), these elements belong to $\mathcal{O}_Y(V_i \cap V_j)^\times$, and hence define an invertible $\mathcal{O}_Y$-module $\mathscr{L}$ such that

$$p^*\mathscr{L} = N_{d_1}(d_0^*\mathscr{A}).$$

Moreover, for every $n \in \mathbb{N}^*$,

$$p^*(\mathscr{L}^n) = N_{d_1}(d_0^*(\mathscr{A}^n))\,;$$

see *loc. cit.*, (6.5.2.1).

We now show that $\mathscr{L}$ is ample for $\pi : Y \to S$. Replacing S by an affine open subset, we may suppose that S is affine. Let $y \in Y$, choose $x \in X_0$ with $p(x) = y$, let V be an affine open neighborhood of y in Y, and put $U = p^{-1}(V)$. Since $\mathscr{A}$ is π_0-ample, there exist $n \in \mathbb{N}^*$ and a section

$$s \in \Gamma(X_0, \mathscr{A}^n)$$

such that

$$x \in (X_0)_s \subseteq U.$$

With the notation above, s is represented by sections

$$a_i \in A_{i,0} = \mathcal{O}_{X_0}(U_i), \qquad a_i = c_{ij}a_j \ \text{ on } \ U_i \cap U_j,$$

and $(X_0)_s$ is the union of the open subsets

$$U_i' = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid a_i \notin \mathfrak{p}\}.$$

For every i, put

$$N(a_i) = N_{\delta_1}(\delta_0(a_i)) \in B_i.$$

Theorem 4.1(i) and Lemma 4.1.1 give

$$p(U_i') = p\,d_1 d_0^{-1}(U_i') = p\,d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\})\,,$$

while

$$d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\}) = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid N_{\delta_1}(\delta_{i,0}(a_i)) \notin \mathfrak{p}\}.$$

It follows that

$$p(U_i') = \{\mathfrak{p} \in \operatorname{Spec}(B_i) \mid N(a_i) \notin \mathfrak{p}\}.$$

Consequently,

$$p((X_0)_s) = Y_{N(s)},$$

where $N(s)$ is the section of $\mathscr{L}^n$ on Y defined by the sections $N(a_i) \in \mathcal{O}_Y(V_i)$. Thus

$$y \in p((X_0)_s) = Y_{N(s)} \subseteq p(U) = V. \qquad (*)$$

This proves that $\mathscr{L}$ is ample for $\pi : Y \to S$, and hence completes the proof that π is quasi-projective.

6. Passage to the quotient when a quasi-section exists

We shall now prove a technical lemma that will be used in the proof of the two theorems we have in view. Let S be a scheme and let

$$X_2 \overset{d'_0,d'_1,d'_2}{\rightrightarrows\!\rightrightarrows} X_1 \overset{d_0,d_1}{\rightrightarrows} X_0$$

Figure 25: The groupoid X_*.

be a $(\mathbf{Sch}/S)$-groupoid. We call a *quasi-section* of the groupoid X_* any subscheme U of X_0 for which the following two conditions hold:

(1) The restriction v of d_1 to $d_0^{-1}(U)$ is a finite, locally free, surjective morphism from $d_0^{-1}(U)$ onto X_0.

(2) Every subset E of U consisting of points that are pairwise equivalent for the equivalence relation defined by X_* (§3.e)) is contained in an affine open subset of U.[31]

If U is a quasi-section of X_*, the $(\mathbf{Sch}/S)$-groupoid

$$U_2 \overset{u'_0,u'_1,u'_2}{\rightrightarrows\!\rightrightarrows} U_1 \overset{u_0,u_1}{\rightrightarrows} U$$

Figure 26: The groupoid U_* induced by X_*.

induced by X_* and the inclusion $U \hookrightarrow X_0$ (see §3.a)) satisfies the hypotheses of Theorem 4.1. Indeed, put

$$V = d_0^{-1}(U),$$

and let u and v be the morphisms with source V induced by d_0 and d_1, respectively:

$$X_0 \xleftarrow[v]{} V \xrightarrow{u} U$$

Figure 27: The morphisms $v : V \to X_0$ and $u : V \to U$.

By §3.b), there is a Cartesian square

$$\begin{array}{ccc} U_1 & \longrightarrow & V \\ {\scriptstyle u_1}\downarrow & & \downarrow{\scriptstyle v} \\ U & \xrightarrow[\text{inclusion}]{} & X_0 \end{array}$$

Figure 28: The Cartesian square defining U_1.

so u_1 is surjective and finite locally free by condition (1). Together with condition (2), condition (1) therefore ensures that the groupoid U_* satisfies the hypotheses of Theorem 4.1.

[31] Editor's note: If $x, y \in E$, there is a $z \in X_1$ such that $d_1(z) = x$ and $d_0(z) = y$; thus z belongs to $v^{-1}(x)$, which is finite by (1). Hence E is contained in the finite set $d_0(v^{-1}(x)) = d_0 d_1^{-1}(x) \cap U$.

In particular, $\mathrm{Coker}(u_0, u_1)$ exists in $(\mathbf{Sch}/S)$. Moreover, d_0 has a section, so u is a universal effective epimorphism (see Exposé III, 1.12). Proposition 3.1 then shows that $\mathrm{Coker}(u_0, u_1)$ coincides with the cokernel $\mathrm{Coker}(v_0, v_1)$ of the groupoid V_*:

$$V_2 \;\underset{v_0'}{\overset{v_2'}{\overset{v_1'}{\rightrightarrows}}}\; V_1 \;\underset{v_0}{\overset{v_1}{\rightrightarrows}}\; V$$

Figure 29: The groupoid V_*.

This is the inverse image of U_* under the base change $u : V \to U$, and hence also the inverse image of X_* under the base change

$$V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_0} X_0$$

Figure 30: The base change defining V_*.

By §3.c), V_* is isomorphic to the groupoid V'_*, the inverse image of X_* under the base change

$$v: \qquad V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_1} X_0$$

Figure 31: The base change defining V'_*.

Thus V'_* admits a cokernel in $(\mathbf{Sch}/S)$. Since $v : V \to X_0$ is flat, surjective, and finite, it is faithfully flat and quasi-compact, and therefore a universal effective epimorphism by Exposé III, 6.3.2. Proposition 3.1 consequently shows that X_* admits a cokernel $\mathrm{Coker}(d_0, d_1)$ in $(\mathbf{Sch}/S)$. We have therefore proved the first assertion of part (i) of the following lemma.[32]

Lemma 6.1.—*Suppose that the* $(\mathbf{Sch}/S)$*-groupoid* X_* *possesses a quasi-section. Then:*

(i) *There exists a cokernel* (Y, p) *of* (d_0, d_1) *in* $(\mathbf{Sch}/S)$*. Moreover, every such* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of all ringed spaces.*

(i') *The morphism* p *is surjective; it is open (respectively, universally closed) if* d_0 *is.*

(ii) *Suppose that* S *is locally Noetherian and that* X_0 *is locally of finite type (respectively, of finite type) over* S*. Then* p *and* $Y \to S$ *are locally of finite presentation (respectively, of finite presentation).*

(iii) *The morphism* $X_1 \to X_0 \times_Y X_0$ *with components* d_0 *and* d_1 *is surjective.*

(iv) *If* (d_0, d_1) *is an equivalence pair, then* $X_1 \to X_0 \times_Y X_0$ *is an isomorphism. Moreover, if* $d_0 : X_1 \to X_0$ *is flat, then* p *is faithfully flat.*

Before proving the second assertion of part (i), we shall prove parts (i'), (ii), and (iii).

a) Proof of (i') and (ii).—We have just seen that (Y, p) identifies both with $\mathrm{Coker}(v_0, v_1)$ and with $\mathrm{Coker}(u_0, u_1)$. Let q and r be the canonical epimorphisms from U and V, respectively, to Y:

[32] Editor's note: The argument that follows has been slightly modified. In particular, the additional hypothesis that d_0 be flat has been moved to part (iv); the former part (ii) has been split into parts (i') and (ii); and the second assertion of part (i') has been added.

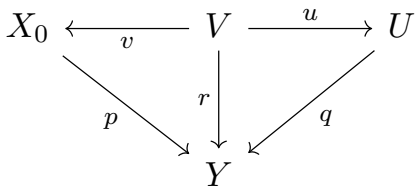

Figure 32: The canonical epimorphisms to Y.

By hypothesis, v is surjective and finite locally free, hence open.

On the other hand, if $d_0 : X_1 \to X_0$ is open (respectively, universally closed), then u, being obtained from d_0 by base change, has the same property. By Theorem 4.1, q is surjective, integral, and open. It follows that r is surjective and is open (respectively, universally closed) if d_0 is. Since v is surjective, the same is true of p. This proves part (i').

Now suppose that S is locally Noetherian and that X_0 is locally of finite type over S, so that X_0 is locally Noetherian.

We show that Y is locally of finite presentation over S. Let

$$S' = \operatorname{Spec} R$$

be an affine open subset of S, let

$$Y' = \operatorname{Spec} B$$

be an affine open subset of Y mapping into S', and let

$$U' = \operatorname{Spec} A$$

be the inverse image of Y' in U. Since R is Noetherian, it is enough to show that B is a finitely generated R-algebra. By §§4 and 5, B is contained in A, which is a finitely generated R-algebra; the assertion therefore follows because R is Noetherian and A is integral over B.

Finally, since $X_0 \to S$ is locally of finite type, so is p (EGA I, 6.6.6). Hence p is locally of finite presentation because Y is locally Noetherian.

It remains to prove the final assertion of part (ii). Suppose in addition that X_0 is of finite type over S. Since p is surjective, Y is then quasi-compact over S, and hence of finite type over S. Because S is locally Noetherian, the morphisms $X_0 \to S$ and $Y \to S$ are of finite presentation, and therefore so is $p : X_0 \to Y$ (EGA IV_1, 1.6.2(v)).

b) Proof of (iii).—The groupoid V_* with base V is isomorphic both to the inverse image of U_* under the base change u and to the inverse image of X_* under the base change v. We therefore have a double Cartesian square:

$$\begin{array}{ccccc} X_1 & \longleftarrow & V_1 & \longrightarrow & U_1 \\ \downarrow{\scriptstyle d_0 \boxtimes d_1} & & \downarrow{\scriptstyle v_0 \boxtimes v_1} & & \downarrow{\scriptstyle u_0 \boxtimes u_1} \\ X_0 \times_Y X_0 & \xleftarrow{v \times v} & V \times_Y V & \xrightarrow{u \times u} & U \times_Y U \end{array}$$

Figure 33: The double Cartesian square used in the proof of Lemma 6.1(iii).

Since $u_0 \boxtimes u_1$ is surjective, so is $v_0 \boxtimes v_1$. Since $v \times v$ is surjective, the composite morphism

$$V_1 \longrightarrow X_0 \times_Y X_0$$

is surjective, and hence so is $d_0 \boxtimes d_1$.

c) Proof of (i).—It remains to prove that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. We first show that Y is obtained from X_0 by identifying those points x and y for which there exists $z \in X_1$ with

$$d_0(z) = x, \qquad d_1(z) = y.$$

Indeed, p is surjective and $pd_0 = pd_1$. Conversely, if $p(x) = p(y)$, there is a point $z' \in X_0 \times_Y X_0$ whose first projection is x and whose second projection is y. If $z \in X_1$ satisfies

$$(d_0 \boxtimes d_1)(z) = z',$$

then $d_0(z) = x$ and $d_1(z) = y$, as required.

On the other hand, if W is a saturated open subset of X_0, then $W \cap U$ is a saturated open subset of U. By Theorem 4.1, $q(W \cap U)$ is open in Y. Since $q(W \cap U) = p(W)$, the topology on Y is the quotient topology induced by X_0.

It remains to show that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*\mathcal{O}_{X_0} \rightrightarrows p_*d_{0*}\mathcal{O}_{X_1} = p_*d_{1*}\mathcal{O}_{X_1} \tag{4}$$

is exact.

Let Y' be an open subset of Y, and put

$$U' = q^{-1}(Y'), \qquad X'_0 = p^{-1}(Y'),$$

and so forth.[33] Then U' is an open subset of U saturated for the equivalence relation defined by the groupoid U_*. It follows from Lemmas 1.1 and 1.2 that Y' is the cokernel, both in $(\mathbf{Sch}/S)$ and in the category of ringed spaces, of the groupoid induced by U_* on U'. Likewise, X'_0 is an open subset of X_0 saturated for the equivalence relation defined by X_*, and we have the following commutative diagram, in which both squares are Cartesian:

$$\begin{array}{ccccc}
X'_0 & \xleftarrow[\widetilde{d}_1]{} & V' = d_0^{-1}(U') & \xrightarrow{\widetilde{d}_0} & U' \\
 & & \downarrow & & \downarrow \\
X_0 & \xleftarrow[d_1]{} & V = d_0^{-1}(U) & \xrightarrow[d_0]{} & U
\end{array}$$

Figure 34: The restricted groupoids over X'_0 and X_0.

Here $\widetilde{d}_1$ is surjective and finite locally free. On the other hand, let $x \in U'$. Since U is a quasi-section, the set

$$E = d_0 d_1^{-1}(x) \cap U$$

is finite and is contained in an affine open subset W of U. Then $E' = E \cap U'$ is a finite set contained in the quasi-affine open subset $W \cap U'$. Consequently there is an affine open subset W' of $W \cap U'$ that contains E'. This proves that U' is a quasi-section of the groupoid X'_* induced by X_* on X'_0. The first assertion of part (i), applied to X'_* and U', now shows that Y' is the cokernel of X'_* in $(\mathbf{Sch}/S)$.

In particular, for every S-scheme T, the sequence

$$T(Y') \xrightarrow{T(p|_{X'_0})} T(X'_0) \underset{T(d_0|_{X'_1})}{\overset{T(d_1|_{X'_1})}{\rightrightarrows}} T(X'_1) \tag{5}$$

[33] Editor's note: The argument that follows has been expanded, in particular the verification that U' is a quasi-section of the groupoid induced on X'_0.

is exact. If T is the affine line $\mathbf{G}_{a,S}$ (Exposé I, 4.3), this sequence identifies with

$$\begin{aligned}\Gamma(Y',\mathcal{O}_Y) \longrightarrow \Gamma(p^{-1}(Y'),\mathcal{O}_{X_0}) \underset{\delta_0}{\overset{\delta_1}{\rightrightarrows}} \Gamma(d_0^{-1}p^{-1}(Y'),\mathcal{O}_{X_1}) \\ = \Gamma(d_1^{-1}p^{-1}(Y'),\mathcal{O}_{X_1}),\end{aligned} \tag{6}$$

which is therefore exact for every open subset Y'. This completes the proof of Lemma 6.1(i).

d) Proof of (iv).—If (d_0, d_1) is an equivalence pair, then so is (u_0, u_1). It follows from Theorem 4.1 that

$$u_0 \boxtimes u_1 : U_1 \longrightarrow U \times_Y U$$

is an isomorphism, and hence so is $v_0 \boxtimes v_1$; see the Cartesian squares in part (b). Since $v \times v$ is faithfully flat and quasi-compact, $d_0 \boxtimes d_1$ is an isomorphism (SGA 1, VIII, 5.4).

Moreover, if d_0 is flat, then u is flat. The morphism q is flat by Theorem 4.1, so r is flat as well. Since v is faithfully flat, p is flat; being also surjective, it is faithfully flat.

7. Quotient by a proper and flat groupoid

Theorem 7.1.—[34]*Let S be a locally Noetherian scheme and let*

$$X_2 \underset{d_0'}{\overset{d_2'}{\overset{d_1'}{\rightrightarrows}}} X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0$$

Figure 35: The groupoid occurring in Theorem 7.1.

be a $(\mathbf{Sch}/S)$-groupoid such that d_1 is proper and flat, X_0 is quasi-projective over S,[35] *and the morphism*

$$d : X_1 \longrightarrow X_0 \times_S X_0$$

with components d_0 and d_1 is quasi-finite. Then:

(i) *There exists a cokernel (Y, p) of (d_0, d_1) in $(\mathbf{Sch}/S)$. Moreover, every such (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*

(ii) *The morphism p is surjective, open, proper, and of finite presentation, and $Y \to S$ is separated and of finite presentation.*[36]

[34]Editor's note: We mention here the article of S. Keel and S. Mori [KM97], which establishes the following theorem. Let X be an algebraic space of finite type over a locally Noetherian base S, and let $j : R \to X \times_S X$ be a flat groupoid whose stabilizer $j^{-1}(\Delta_X)$ is finite over X. There then exists an algebraic space that is a geometric quotient of X by R and a uniform categorical quotient; moreover, if j is separated, this quotient is separated. In particular, if a flat S-group scheme G acts properly on X, with finite stabilizer (that is, if

$$G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (x, g \cdot x)$$

is proper and the stabilizer of the diagonal is finite over X), then there exists a geometric quotient $X \to X/G$. For a reductive S-group scheme G, this is a result of J. Kollár [Ko97].

[35]Editor's note: This hypothesis on X_0 is necessary; see editor's note (17) to Theorem 4.1.

[36]Editor's note: TDTE III, Theorem 6.1, states that $Y \to S$ is quasi-projective if S is Noetherian. The editors did not see how to deduce this from the local existence of quasi-sections.

(iii) *The morphism*

$$X_1 \longrightarrow X_0 \times_Y X_0$$

with components d_0 and d_1 is surjective.

(iv) *If (d_0, d_1) is an equivalence pair, then $X_1 \to X_0 \times_Y X_0$ is an isomorphism and p is faithfully flat.*[37] *Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology, and, for every base change $Y' \to Y$, the scheme Y' is the cokernel of the groupoid $X_* \times_Y Y'$ obtained from X_* by the base change $X_0 \times_Y Y' \to X_0$.*

In particular, for every base change $S' \to S$, the scheme

$$Y' = Y \times_S S'$$

is the cokernel of the S'-groupoid

$$X'_* = X_* \times_S S'.$$

Thus, in this case, "formation of the quotient commutes with base change".

Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Lemma 1.2 shows that, to prove part (i), it is enough to show that every point $z \in X_0$ has a saturated open neighborhood U_z such that, if $\widetilde{d}_0$ and $\widetilde{d}_1$ denote the restrictions of d_0 and d_1 to

$$d_0^{-1}(U_z) = d_1^{-1}(U_z),$$

and if (Q, q) denotes the cokernel of $(\widetilde{d}_0, \widetilde{d}_1)$ in the category of ringed spaces, then Q is a scheme and q is a morphism of schemes.

By Lemma 6.1(i), it is therefore enough to show that every point $z \in X_0$ has a saturated open neighborhood U_z such that the groupoid induced by X_* on U_z has a quasi-section. We may even assume that z is closed in its fiber over S (we shall say that z is *closed relative to* S).[38] The existence of U_z follows from the lemmas below.

Lemma 7.2.—*Let T be an affine Noetherian scheme, let X, Y, Z be T-schemes of finite type, with X quasi-projective over T, and let*

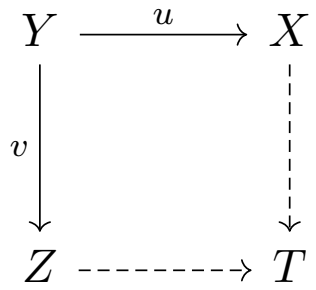

Figure 36: The diagram in Lemma 7.2.

[37] Editor's note: The consequences that follow have been made explicit; see [Ray67a], Theorem 1(iii), and the end of the proof of the theorem. We also mention that another proof of Theorem 7.1 in the case of an equivalence relation, based on the existence of Hilbert schemes, is given in [AK80], Theorem 2.9; see also [BLR90], §8.2, Theorem 12. In that case it is furthermore shown that $Y \to S$ is quasi-projective.

[38] Editor's note: Indeed, if such an open neighborhood U_z has been constructed for every point z that is closed relative to S, then the union of these U_z covers X_0: every fiber over S of the closed complement is a Noetherian scheme with no closed points, and is therefore empty.

be a diagram in (**Sch**/T). *Let z be a point of $v(Y)$ that is closed relative to T, and suppose that v is flat at the points of $v^{-1}(z)$. Then there exists a closed subscheme F of X such that*

$$u\big(u^{-1}(F) \cap v^{-1}(z)\big)$$

is finite and nonempty, and such that the restriction of v to $u^{-1}(F)$ is flat at the points of $v^{-1}(z)$.

Put $T = \operatorname{Spec} A$. We may suppose that

$$X = \operatorname{Proj} \mathscr{S},$$

where $\mathscr{S}$ is the symmetric algebra of a finitely generated A-module E.

If $u(v^{-1}(z))$ is finite, we may take $F = X$. Otherwise, let

$$y_1, \dots, y_n$$

be the points of the fiber $v^{-1}(z)$ associated with its structure sheaf $\mathcal{O}_{v^{-1}(z)}$. Thus, if $\mathcal{O}_i$ denotes the local ring of $v^{-1}(z)$ at y_i, the maximal ideal of $\mathcal{O}_i$ consists of zero divisors. Let t be the image of z in T. The set $u(v^{-1}(z))$ is an infinite constructible subset of the fiber of X over t. Consequently, that fiber contains a closed point $x \in u(v^{-1}(z))$ distinct from $u(y_1), \dots, u(y_n)$. The open subset $X - \{x\}$ is a neighborhood of $u(y_1), \dots, u(y_n)$, and hence contains an open neighborhood of the form $D_+(f)$, where $f \in \mathscr{S}$ is homogeneous of degree d (we use the notation of EGA II, §2.3).

The closed subscheme

$$X_1 = V_+(f)$$

defined by f therefore contains x and avoids the points $u(y_1), \dots, u(y_n)$. Its inverse image

$$Y_1 = u^{-1}(V_+(f))$$

is consequently distinct from Y and meets $v^{-1}(z)$. We shall in fact prove that the restriction v_1 of v to Y_1 is flat at the points of $v^{-1}(z)$. If $u(v_1^{-1}(z))$ is finite, we may then take $F = X_1$. Otherwise, we repeat the argument with Y_1, v_1, and the morphism u_1 induced by u on Y_1, in place of Y, v, u. In this way we obtain a descending sequence

$$X \supset X_1 \supset \cdots$$

of closed subschemes of X. Since such a sequence terminates,

$$u\big(u^{-1}(X_n) \cap v^{-1}(z)\big)$$

is finite and nonempty for some n, and we then take $F = X_n$.

It remains to show that v_1 is flat at the points of $v^{-1}(z)$. Let $y \in Y_1$ lie over z, let $\mathcal{O}_y$ be the local ring of Y at y, let $\overline{\mathcal{O}}_y$ be the local ring of $v^{-1}(z)$ at y, and let $\mathcal{O}_{v(y)}$ be the local ring of Z at $v(y)$. Choose $g \in \mathscr{S}_1$ such that $D_+(g)$ is a neighborhood of $u(y)$ in X, and let φ and $\overline{\varphi}$ be the respective images of f/g^d in $\mathcal{O}_y$ and $\overline{\mathcal{O}}_y$. By the construction of f, the element $\overline{\varphi}$ is not a zero divisor in $\overline{\mathcal{O}}_y$. Since $\mathcal{O}_y$ is flat over $\mathcal{O}_z = \mathcal{O}_{v(y)}$, the element φ is not a zero divisor in $\mathcal{O}_y$, and

$$\mathcal{O}_y/\mathcal{O}_y\varphi$$

is flat over $\mathcal{O}_z$ (SGA 1, IV, 5.7). This quotient is precisely the local ring of Y_1 at y.

Lemma 7.3.—*Retain the notation and hypotheses of Theorem 7.1. Every point $z \in X_0$ that is closed relative to S has a saturated open neighborhood U_z such that the groupoid induced by X_* on U_z has a quasi-section.*

The assertion is local on S, so we may suppose that S is affine and Noetherian and apply the preceding lemma to the diagram

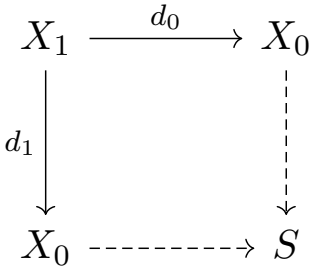

Figure 37: The diagram to which Lemma 7.2 is applied.

in (**Sch**/S). Thus let F be a closed subscheme of X_0 such that

$$d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$$

is finite and nonempty, and such that the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $d_1^{-1}(z)$.

Let F_1 and F_2 be the inverse images of F under d_0 and $d_0 d_0' = d_0 d_1'$, respectively, and let $\widetilde{d}_0, \widetilde{d}_1, \ldots$ denote the morphisms induced by $d_0, d_1, \ldots$. We then have a commutative diagram

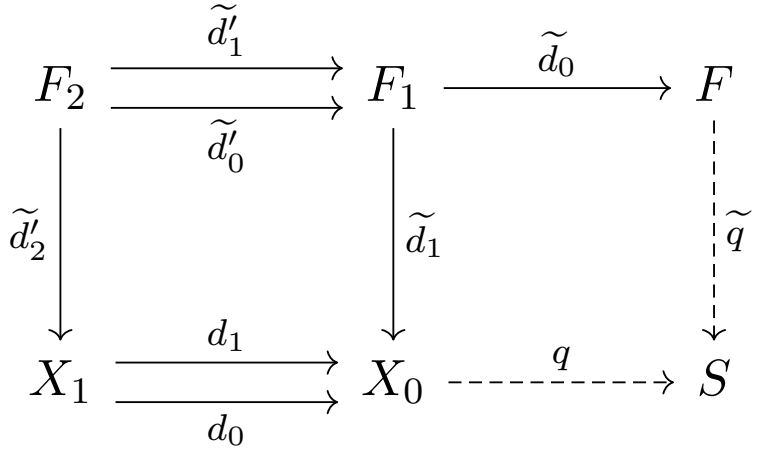

Figure 38: The induced diagram on F_2, F_1, F.

in which the two left-hand squares are Cartesian, the first row is exact (cf. the diagram $(0, 1, 2)$ in §1), and $q, \widetilde{q}$ are the structure morphisms.

We first show that only finitely many points of F_1 lie over z.[39] Let s be the image of z in S. Since F is of finite type over S, the fiber $\widetilde{q}^{-1}(s)$ is a Noetherian scheme. Since $\widetilde{d}_0$ is proper,

$$\widetilde{d}_0(\widetilde{d}_1^{-1}(z))$$

is a closed subscheme of $\widetilde{q}^{-1}(s)$ consisting of finitely many points. Consequently (cf. EGA I, 6.2.2), those points are closed in $\widetilde{q}^{-1}(s)$, and also in the fiber $q^{-1}(s)$ of X_0, because F is closed in X_0.

[39]Editor's note: Details have been added, and the role of the properness hypothesis on d_0 and d_1 in Theorem 7.1 has been made explicit. Compare the statement and proof of Theorem 8.1, where this properness hypothesis is omitted.

Let y be one of these points. Since $q^{-1}(s)$ is of finite type over $\kappa(s)$, it has affine open neighborhoods

$$\operatorname{Spec} B \quad \text{and} \quad \operatorname{Spec} C$$

of y and z, respectively, with B and C finitely generated $\kappa(s)$-algebras. The points y and z correspond to maximal ideals

$$\mathfrak{p} \subset B, \qquad \mathfrak{q} \subset C.$$

The fields $B/\mathfrak{p}$ and $C/\mathfrak{q}$ are finite over $\kappa(s)$, so

$$(B/\mathfrak{p}) \otimes_{\kappa(s)} (C/\mathfrak{q})$$

is a finite-dimensional $\kappa(s)$-algebra. Its maximal ideals correspond precisely to the points of $X_0 \times_S X_0$ whose second projection is z and whose first projection is y. Thus there are only finitely many points $u \in X_0 \times_S X_0$ whose second projection is z and whose first projection belongs to

$$\widetilde{d}_0(\widetilde{d}_1^{-1}(z)).$$

Finally, since $X_1 \to X_0 \times_S X_0$ has finite fibers, each such point u is the image of only finitely many points of X_1. This proves the assertion.

The morphism $\widetilde{d}_1$ is therefore quasi-finite and flat at the points of F_1 lying over z. Since $\widetilde{d}_1$ is of finite type, SGA 1, IV, 6.10 and EGA III, 4.4.10 imply that the set Φ of points of F_1 at which $\widetilde{d}_1$ is not both flat and quasi-finite is closed in F_1, and hence in X_1 because F_1 is closed in X_1.[40] Since d_1 is proper, $\widetilde{d}_1(\Phi)$ is closed and, by what precedes, does not contain z. Put

$$W = \widetilde{d}_1(F_1) - \widetilde{d}_1(\Phi).$$

The restriction of $\widetilde{d}_1$ to $\widetilde{d}_1^{-1}(W)$ is of finite presentation (by the Noetherian hypotheses), flat, proper, and quasi-finite.[41] It is therefore finite, locally free, and open, by EGA III, 4.4.2 and EGA IV$_2$, 2.1.12 and 2.4.6. Consequently, $\widetilde{d}_1(F_1)$ is a neighborhood of z, and W is the largest open subset of X_0 contained in $\widetilde{d}_1(F_1)$ over which $\widetilde{d}_1$ is both flat and quasi-finite.

We shall see in Lemma 7.4 that the inverse images of Φ under $\widetilde{d}'_1$ and $\widetilde{d}'_0$ both identify with the set of points of F_2 at which $\widetilde{d}'_2$ is not both flat and quasi-finite. It follows that

$$d_0^{-1}(W) = \widetilde{d}'_2(F_2) - \widetilde{d}'_2((\widetilde{d}'_0)^{-1}(\Phi)),$$
$$d_1^{-1}(W) = \widetilde{d}'_2(F_2) - \widetilde{d}'_2((\widetilde{d}'_1)^{-1}(\Phi))$$

coincide. Thus W is saturated.

Set

$$W_1 = \widetilde{d}_1^{-1}(W).$$

The equality $d_0^{-1}(W) = d_1^{-1}(W)$ implies

$$(\widetilde{d}'_2)^{-1}(d_0^{-1}(W)) = (\widetilde{d}'_2)^{-1}(d_1^{-1}(W)),$$

that is,

$$(\widetilde{d}'_0)^{-1}(W_1) = (\widetilde{d}'_1)^{-1}(W_1).$$

[40] Editor's note: If $f : X \to Y$ is locally of finite type, the set of points $x \in X$ that are isolated in their fiber $f^{-1}(f(x))$ is open in X. In EGA III, 4.4.10, this is deduced from Zariski's Main Theorem when Y is locally Noetherian. For arbitrary Y, it follows from Chevalley's semicontinuity theorem (EGA IV$_3$, 13.1.3 and 13.1.4). Consequently, f is quasi-finite at x if and only if it is of finite type at x and x is isolated in $f^{-1}(f(x))$. This will be used below; see editor's note (42).

[41] Editor's note: The original has been expanded in what follows.

The morphism $\widetilde{d}_0$ is faithfully flat and quasi-compact, because d_0, like d_1, is surjective, proper, and flat. Moreover, the square

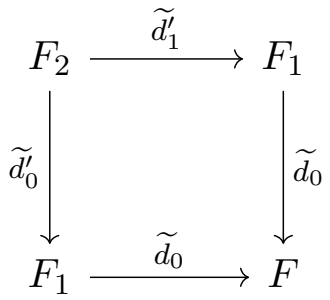

Figure 39: The Cartesian square used to descend W_1.

is Cartesian. It follows that

$$W_1 = \widetilde{d}_0^{-1}(U)$$

for an open subset U of F (SGA 1, VIII, 4.4). This open subset U is a quasi-section of the groupoid with base W induced by X_*. We may therefore take $U_z = W$.

It remains to state the following lemma, whose proof is classical.

Lemma 7.4.—*Consider a Cartesian square of schemes*

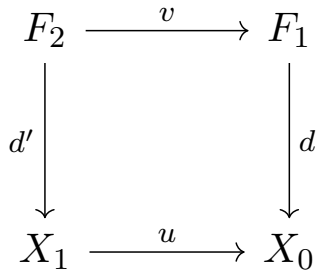

Figure 40: The Cartesian square in Lemma 7.4.

and let x be a point of F_2.

(i) If u is flat, d' is flat at x if and only if d is flat at $v(x)$.

(ii) If d is locally of finite type, d' is quasi-finite at x if and only if d is quasi-finite at $v(x)$.[42]

We have thus proved that X_0 has a covering by saturated open subsets W such that the groupoid W_* induced by X_* on W possesses a quasi-section.[43]

By Lemma 6.1 and the reductions stated after Theorem 7.1, this proves assertions (i) and (iii) of that theorem, as well as the facts that p is surjective and open and that p and

[42]Editor's note: The conditions are sufficient by base change (EGA II, 6.2.4(iii), and EGA IV$_2$, 2.1.4). Conversely, put $y = d'(x)$ and $z = u(y) = d(v(x))$, and suppose that d' is flat at x and that u (hence also v) is flat. Then $\mathcal{O}_{v(x)} \to \mathcal{O}_x$ is faithfully flat, as is $\mathcal{O}_z \to \mathcal{O}_y \to \mathcal{O}_x$. Consequently, $\mathcal{O}_z \to \mathcal{O}_{v(x)}$ is faithfully flat (EGA IV$_2$, 2.2.11(iv)). Finally, suppose that d is locally of finite type and that d' is quasi-finite at x. Then $v(x)$ is isolated in its fiber $d^{-1}(z)$, since x is isolated in its fiber $d'^{-1}(y) = d^{-1}(z) \otimes_{\kappa(z)} \kappa(y)$. By Chevalley's semicontinuity theorem there is therefore an open neighborhood of $v(x)$ every point of which is isolated in its fiber (EGA IV$_3$, 13.1.3 and 13.1.4), and hence d is quasi-finite at $v(x)$.

[43]Editor's note: The sequel has been modified to take advantage of the additions made to Lemma 6.1.

$Y \to S$ are locally of finite presentation. Moreover, because $X_0 \to S$ is quasi-projective, and hence separated and of finite type, p is separated; the proof of Lemma 6.1(ii) shows that p and $Y \to S$ are of finite presentation.

To prove that p is proper, it remains to prove that it is universally closed. Since this assertion is local on Y, we may work over a saturated open subset W such that the groupoid W_* induced by X_* on W possesses a quasi-section U (because X_0 is covered by such open subsets). With the notation of 6(a), there is a commutative diagram

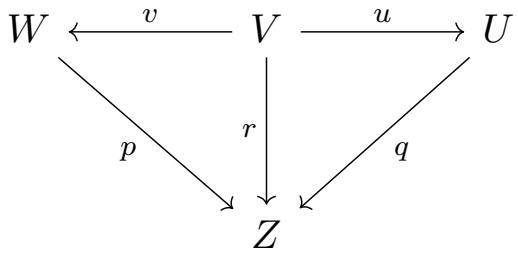

Figure 41: The properness diagram over the open subset $Z \subseteq Y$.

where Z is an open subset of Y, all the arrows are surjective, and q is integral. By hypothesis $d_0 : X_1 \to X_0$ is proper, so u, which is obtained from d_0 by base change, is proper as well. Consequently r is universally closed, and hence so is p, since v is surjective.

Finally, because p is surjective and universally closed and X_0 is quasi-projective, hence separated, the diagonal $\Delta_{Y/S}(Y)$ is closed in $Y \times_S Y$: it is the image under $p \times p$ of the diagonal $\Delta_{X_0/S}(X_0)$. Thus Y is separated over S. This completes the proof of Theorem 7.1(ii).

The assertions in Theorem 7.1(iv) are local on Y. Since the saturated open subsets U_z cover X_0, it is enough to verify them after replacing X and Y by U_z and $V_z = p(U_z)$, respectively. As observed at the beginning of the proof of Theorem 7.1, Lemmas 1.1, 1.2, and 6.1(i) show that $(V_z, p|_{U_z})$ is the cokernel, both in (**Sch**) and in the category of ringed spaces, of the groupoid induced by X_* on U_z. The hypothesis that $d = (d_0, d_1)$ is a monomorphism is preserved under the base change $U_z \to X_0$. The first two assertions of Theorem 7.1(iv) therefore follow from Lemma 6.1(iv).

Let us finally prove the consequences stated at the end of part (iv) (see [Ray67a], Theorem 1(iii)). By hypothesis, the groupoid X_* comes from an equivalence relation $R \to X_0 \times X_0$. We have proved that R is effective (Exposé IV, 3.3.2) and that $p : X_0 \to Y = X_0/R$ is faithfully flat and of finite presentation. Consequently, if (M) denotes the family of faithfully flat morphisms locally of finite presentation, then R is (M)-effective. By Exposé IV, 6.3.3, the pair (Y, p) therefore represents the quotient sheaf of X_0 by R for the fppf topology, and the assertions concerning base change follow from Exposé IV, 3.4.3.1.

8. Passage to the quotient by a flat groupoid that is not necessarily proper

Theorem 8.1.—[44] *Let S be a Noetherian scheme and*

[44] Editor's note: There is a largest open subset W of X_0 satisfying the conclusions of the theorem. Indeed, let W be an open subset as in the theorem and let $W^\sharp$ be a dense saturated open subset of W. Since p is open, $V^\sharp = p(W^\sharp)$ is open in V, and $W^\sharp = p^{-1}(V^\sharp)$, because $W^\sharp$ is saturated. By Lemma 1.1, $V^\sharp$ is a cokernel for the groupoid induced on $W^\sharp$. Thus two open subsets W and W' satisfying the conclusions of the theorem may be glued along their intersection $W^\sharp$, and conditions (i)–(iv), together with the property that p and $V \to S$ are locally of finite presentation, are preserved. Conclusion (ii') follows, as in the proof of Lemma 6.1(ii), from the hypothesis that X_0 is of finite type over the Noetherian scheme S.

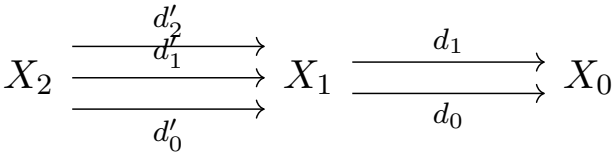

Figure 42: The (**Sch**/S)-groupoid of Theorem 8.1.

a (**Sch**/S)*-groupoid such that* d_1 *is flat and of finite type,* X_0 *is of finite type over* S*, and the morphism* $X_1 \to X_0 \times_S X_0$ *with components* d_0 *and* d_1 *is quasi-finite.*

There exists a dense saturated open subset $W \subseteq X_0$ *having the following properties:*

(i) *if* $W_2 \overset{w_i'}{\rightrightarrows} W_1 \overset{w_j}{\rightrightarrows} W$ *is the groupoid induced by* X_* *on* W*, then* (w_0, w_1) *has a cokernel* (V, p) *in* (**Sch**/S)*; moreover,* (V, p) *is a cokernel of* (w_0, w_1) *in the category of all ringed spaces;*

(ii) p *is surjective and open;*

(ii') p *and* $V \to S$ *are of finite presentation;*

(iii) *the morphism* $W_1 \to W \times_V W$ *with components* w_0 *and* w_1 *is surjective;*

(iv) *if* (d_0, d_1) *is an equivalence pair, then* $W_1 \to W \times_V W$ *is an isomorphism and* p *is faithfully flat.*

We shall prove that W may be chosen so that the (**Sch**/S)-groupoid W_* induced by X_* possesses a quasi-section (see §7). Theorem 8.1 will then follow from Lemma 6.1.

Suppose provisionally that for every point $z \in X_0$ closed relative to S (see §7) there is a saturated open subset W_z that possesses a quasi-section and meets every irreducible component of X_0 passing through z. Then the complement $X_0 - W_z$ is saturated: its saturation $d_1(d_0^{-1}(X_0 - W_z))$ is open and does not meet W_z. If this complement is nonempty, choose in it a point z' closed relative to S, and associate to z' an open subset $W_{z'}$ as above. We may further suppose that $W_{z'} \subseteq X_0 - W_z$. Then W_z and $W_{z'}$ are disjoint, and the groupoid induced by X_* on $W_z \cup W_{z'}$ possesses a quasi-section. The process must stop, because X_0 has only finitely many irreducible components. It therefore remains to construct W_z.

For this purpose we may suppose that S is affine. Let $y \in X_1$ satisfy $d_1(y) = z$, let X be an affine open subset of X_0 containing $d_0(y)$, let Y be the inverse image of X under $d_0 : X_1 \to X_0$, and let $u : Y \to X$ and $v : Y \to X_0$ be the morphisms induced by d_0 and

Lemmas 1.1 and 1.2 also show that the union of all saturated open subsets $U \subseteq X_0$ such that the open subset $p(U) \subseteq Y$ is a scheme and $p|_U : U \to p(U)$ is a morphism of schemes is the largest saturated open subset $\Omega \subseteq X_0$ satisfying condition (i) of Theorem 8.1. The theorem shows that Ω contains a dense open subset W, but it is not immediate that Ω satisfies properties (ii)–(iv). For more precise results, and for the representability of the fppf quotient sheaf $\widetilde{X/R}$ (where R denotes the groupoid X_* with base $X = X_0$), see [Ray67a], [Ray67b], and Appendix I of [An73]. These works use weaker hypotheses: S may be arbitrary, X is locally of finite type over S, and R is an S-groupoid with base X such that d_0 (and hence d_1) is flat and of finite presentation. In particular, if $\widetilde{X/R}$ is represented by an S-scheme Y, then Y is also the cokernel in the category of ringed spaces. The converse is false in general (see [Mum65], Chapter 0, §3, Example 0.4, cited in [Ray67a], Remark 1), but it is true when $d = (d_0, d_1)$ is an immersion. Under this hypothesis, $p : \Omega \to Z := \Omega/R_\Omega$ is faithfully flat and of finite presentation. If, in addition, S is locally Noetherian, then a codimension-one point x of X belongs to Ω if and only if the graph of the groupoid induced on $\mathrm{Spec}(\mathcal{O}_{X,x})$ is closed. See [Ray67a], Proposition 1; [Ray67b], Proposition 1 and Theorems 2, 1, and 4; and [An73], Theorems 5 and 6 on pages 66–67 and Proposition 3.3.1 on page 49. For an action of an algebraic group over an algebraically closed field k, see also [DR81].

d_1. Since X is affine, hence quasi-projective, Lemma 7.2 provides a subscheme $F \subseteq X_0$ such that $d_0^{-1}(F) \cap d_1^{-1}(z)$ is nonempty, $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite, and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $v^{-1}(z)$. We can therefore resume the notation of Lemma 7.3, writing F_1 and F_2 for the inverse images of F in X_1 and X_2, and so on:

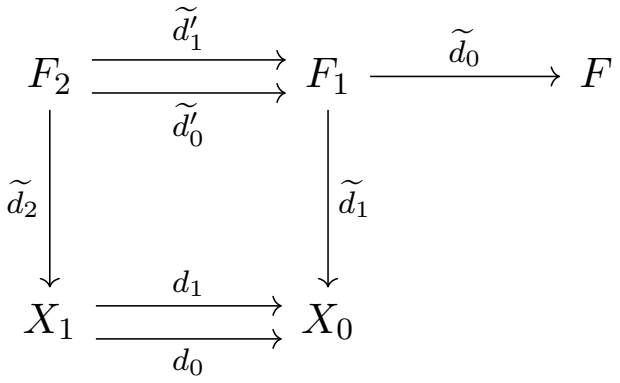

Figure 43: The induced diagram on F_2, F_1, F used in the proof of Theorem 8.1.

As in 7.3, one proves that $\widetilde{d}_1$ is quasi-finite at the points of $\widetilde{d}_1^{-1}(z)$. It is therefore natural to consider the open subset $F_1' \subseteq F_1$ consisting of the points at which $\widetilde{d}_1$ is both flat and quasi-finite. By Lemma 7.4, the inverse images of F_1' under $\widetilde{d}_1'$ and $\widetilde{d}_0'$ both consist of the points of F_2 at which $\widetilde{d}_2'$ is flat and quasi-finite. Thus these inverse images coincide, and F_1' is of the form $\widetilde{d}_0^{-1}(F')$ for an open subset $F' \subseteq F$ (SGA 1, VIII, 4.4). After replacing F by F', we may therefore suppose that $\widetilde{d}_1$ is quasi-finite and flat. Let W_z be the largest open subset of $\widetilde{d}_1(F_1)$ over which $\widetilde{d}_1$ is finite and flat.

The open subset W_z need not contain z, but it contains the generic points of the irreducible components of X_0 passing through z.[45] Since d_0 (and likewise d_1) is faithfully flat and of finite presentation, hence open, SGA 1, VIII, 5.7 shows that $d_0^{-1}(W_z)$ and $d_1^{-1}(W_z)$ both coincide with the largest open subset of $\widetilde{d}_2'(F_2)$ over which $\widetilde{d}_2'$ is finite and flat. As in 7.3, the inverse images of $\widetilde{d}_1^{-1}(W_z)$ under $\widetilde{d}_0'$ and $\widetilde{d}_1'$ therefore coincide. Hence $\widetilde{d}_1^{-1}(W_z) = \widetilde{d}_0^{-1}(U)$ for an open subset $U \subseteq F$, and U is a quasi-section for the groupoid induced by X_* on W_z.

9. Elimination of the Noetherian hypotheses in Theorem 7.1

a) Resume the notation and hypotheses of Lemma 6.1, and let $\pi : S' \to S$ be an arbitrary base change. If $f : X \to Y$ is a morphism of S-schemes, write $f' : X' \to Y'$ for the morphism of S'-schemes obtained from f by extension of the base along π. With this convention, $p' : X_0' \to Y'$ is surjective, as is the morphism

$$X_1' \longrightarrow X_0' \times_{Y'} X_0'$$

with components d_0' and d_1'. Thus the set underlying Y' is the quotient of the set underlying X_0' by the equivalence relation on X_0' defined by the S'-groupoid X_*'. Moreover, $q' : U' \to Y'$ is integral and surjective. The topology of Y' is consequently the quotient topology of that of U', and hence also of that of X_0'; see the proof in §6(c).

On the other hand, U' is plainly a quasi-section of the S'-groupoid X_*', so Lemma 6.1 applies. In particular, X_*' has a cokernel (Y_1, p_1), and the topological space underlying Y_1

[45] Editor's note: Indeed, let η be such a generic point. The hypotheses imply that $\mathcal{O}_{X_0,\eta}$ is an Artinian local ring and that $\mathcal{O}_{F_1,\eta}$ is a finitely generated $\mathcal{O}_{X_0,\eta}$-module. By SGA 1, VIII, 6.5, there is therefore an open neighborhood of η over which $\widetilde{d}_1$ is finite.

is obtained from that underlying X'_0 by identifying the points equivalent under the relation defined by X'_*. It follows that the canonical morphism $Y_1 \to Y'$ is a homeomorphism. In fact, it is a universal homeomorphism. Indeed, if S'' is an S'-scheme, let Y_2 be the cokernel of $(d_0 \times_S S'', d_1 \times_S S'')$. The preceding argument, applied to the base changes $S'' \to S'$ and $S'' \to S$, shows that

$$Y_2 \longrightarrow Y_1 \times_{S'} S'' \qquad \text{and} \qquad Y_2 \longrightarrow Y \times_S S'' \simeq Y' \times_{S'} S''$$

are homeomorphisms. Hence

$$Y_1 \times_{S'} S'' \longrightarrow Y' \times_{S'} S''$$

is a homeomorphism as well.

b) The same remarks plainly apply when the groupoid X_* *locally* possesses quasi-sections (see the proof of Theorem 7.1).[46] For example, we have the following theorem:

Theorem 9.0.—*Let S be an arbitrary scheme and*

$$X_2 \xrightarrow{d'_j} X_1 \xrightarrow{d_i} X_0$$

a $(\mathbf{Sch}/S)$-groupoid such that X_0 is of finite presentation and quasi-projective over S, d_1 is of finite presentation, proper, and flat, and the morphism

$$d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_S X_0$$

is quasi-finite. Then:

(1) *Every point x of X_0 has a saturated open neighborhood W such that the groupoid induced by X_* on W has a quasi-section.*

(2) *Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Then Y is a scheme, p is a morphism of schemes, and (Y, p) is a cokernel of (d_0, d_1) in $(\mathbf{Sch}/S)$.*

(3) *The morphism p is surjective, open, and universally closed.*

(4) *The morphism*
$$d : X_1 \longrightarrow X_0 \times_Y X_0$$
with components d_0 and d_1 is surjective.

(5) *If (d_0, d_1) is an equivalence pair, then:*

   (a) *the morphism*
   $$d : X_1 \longrightarrow X_0 \times_Y X_0$$
   is an isomorphism and p is faithfully flat;

   (b) *the morphisms p and $Y \to S$ are of finite presentation, and (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.*

Proof. For (1), the question is local on S, so we may suppose that $S = \operatorname{Spec} B$ is affine. There then exist a ring A of finite type over $\mathbb{Z}$, a morphism

$$S \longrightarrow T = \operatorname{Spec} A,$$

[46] Editor's note: What follows has been expanded in order to bring out Theorem 9.0 below.

and a $(\mathbf{Sch}/T)$-groupoid Z_* such that

$$X_* \simeq Z_* \times_T S.$$

Indeed, apply EGA IV_3, 8.8.3 to $S_0 = \operatorname{Spec}\mathbb{Z}$ and $S_i = \operatorname{Spec} A_i$, where the A_i range over the finitely generated $\mathbb{Z}$-subalgebras of B. Moreover, by EGA IV_3, 8.10.5, we may suppose that Z_* satisfies the hypotheses of Theorem 7.1. Consequently, Z_* has quasi-sections locally.

By part (a), the same is therefore true of X_*, and assertions (2), (3), (4), and (5)(a) follow from Lemma 6.1, just as in the proof of Theorem 7.1.

c) Proof of (5)(b).—Let us show that $Y \to S$ is of finite presentation.[47] By hypothesis, $(d_0^{X_*}, d_1^{X_*})$ is an equivalence pair; that is,

$$d^{X_*} : X_1 \longrightarrow X_0 \times_S X_0$$

is a monomorphism. By EGA IV_3, 8.10.5, after enlarging A if necessary, we may suppose that

$$d^{Z_*} : Z_1 \longrightarrow Z_0 \times_T Z_0$$

is a monomorphism. Since $T = \operatorname{Spec} A$, with A Noetherian, Theorem 7.1 then shows that the groupoid Z_* has a cokernel (Q, q) in $(\mathbf{Sch}/T)$, that q and $Q \to T$ are of finite presentation, and, moreover, that

$$q : Z_0 \longrightarrow Q$$

is faithfully flat and d^{Z_*} induces an isomorphism

$$Z_1 \xrightarrow{\sim} Z_0 \times_Q Z_0.$$

Put $Q_S = Q \times_T S$. Since $X_i \simeq Z_i \times_T S$, we obtain an isomorphism

$$d^{Z_*} \times_T S : X_1 \xrightarrow{\sim} X_0 \times_{Q_S} X_0.$$

Denote its inverse by ϕ, and let π be the canonical morphism

$$X_0 \times_Y X_0 \longrightarrow X_0 \times_{Q_S} X_0.$$

Then $\phi \circ \pi$ is the inverse of

$$d_0 \boxtimes d_1 : X_1 \xrightarrow{\sim} X_0 \times_Y X_0.$$

It follows that the equivalence relation defined by X_*, that is, the monomorphism shown in Figure 44,

$$X_1 \xrightarrow[\sim]{d_0 \boxtimes d_1} X_0 \times_Y X_0 \hookrightarrow X_0 \times_S X_0$$

Figure 44: The equivalence-relation monomorphism defined by X_*.

identifies with the equivalence relation $\mathscr{R}(q_S)$ defined by $q_S : X_0 \to Q_S$. Since q_S is faithfully flat and of finite presentation, hence a universal effective epimorphism, $\mathscr{R}(q_S)$ has quotient Q_S; see Exposé IV, 3.3.2. Consequently,

$$Y \simeq Q \times_T S,$$

[47] Editor's note: The original states that this follows from Proposition 9.1 below. We were unable to reconstruct that argument. The proof that follows was communicated to us by O. Gabber.

so $Y \to S$ and p are of finite presentation. Moreover, by Exposé IV, 6.3.3, (Y, p) is also a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.

Proposition 9.1.—*Consider morphisms of schemes*

$$X_0 \xrightarrow{p} Y \xrightarrow{q} S$$

such that qp is of finite type (respectively, of finite presentation) and p is faithfully flat and of finite presentation. Then q is of finite type (respectively, of finite presentation).[*]

Since p is surjective and qp is quasi-compact, q is quasi-compact. We may therefore suppose that S, Y, and X_0 are affine, with rings A, B, and C, respectively. We have

$$B = \varinjlim_i B_i,$$

where the B_i range over the finitely generated A-subalgebras of B. Since C is of finite presentation over B, there exist an index i_0, a B_{i_0}-algebra C_{i_0} of finite presentation, and an isomorphism

$$C \simeq C_{i_0} \otimes_{B_{i_0}} B.$$

For $i \geq i_0$, put

$$C_i = C_{i_0} \otimes_{B_{i_0}} B_i;$$

then

$$C \simeq C_i \otimes_{B_i} B.$$

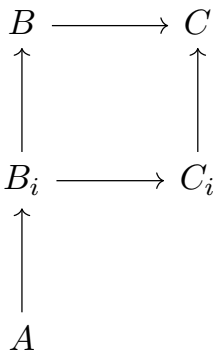

Figure 45: The compatible A-algebras used in the limit argument.

Since C is faithfully flat over B, EGA IV_3, 11.2.6 and 8.10.5(vi) give an index $i_1 \geq i_0$ such that C_{i_1} is faithfully flat over B_{i_1}. Consequently, C_i is faithfully flat over B_i for every $i \geq i_1$. For $i \geq i_1$, the canonical map $C_i \to C$ is then injective, because it is obtained from $B_i \to B$ by faithfully flat base extension.

If C is of finite type over A, it follows that there is an index $j \geq i_1$ such that $C_j = C$, whence $B_j = B$, because C_j is faithfully flat over B_j. Thus B is of finite type over A.

Now suppose that C is of finite presentation over A. By what precedes, B is of finite type over A, and hence has the form

$$B = \overline{B}/I,$$

where $\overline{B}$ is a polynomial algebra over A in finitely many indeterminates and I is an ideal of $\overline{B}$. The ideal I is the union of its finitely generated subideals I_α; consequently,

$$B = \varinjlim_\alpha B_\alpha, \qquad B_\alpha = \overline{B}/I_\alpha.$$

[*]See EGA IV_4, 17.7.5 for a more general result.

Proceeding as above, there exist an index α_0, a B_{α_0}-algebra C_{α_0} of finite presentation, and an isomorphism

$$C \simeq C_{\alpha_0} \otimes_{B_{\alpha_0}} B.$$

For $\alpha \geq \alpha_0$, put

$$C_\alpha = C_{\alpha_0} \otimes_{B_{\alpha_0}} B_\alpha,$$

so that

$$C \simeq C_\alpha \otimes_{B_\alpha} B \qquad (\alpha \geq \alpha_0).$$

Again by EGA IV$_3$, 11.2.6 and 8.10.5(vi), C_α is faithfully flat over B_α for all sufficiently large α. In that case, the kernels of the maps $C_\alpha \to C$ and $C_\alpha \to C_\beta$, respectively, identify with

$$C_\alpha \otimes_{B_\alpha} (I/I_\alpha) \quad \text{and} \quad C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) \qquad (\beta \geq \alpha).$$

Since C_α and C are of finite presentation over A and $C_\alpha \to C$ is surjective,

$$C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$$

is a finitely generated ideal,[48] and it is the union of the ideals

$$C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha).$$

Thus, for all sufficiently large β,

$$C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) = C_\alpha \otimes_{B_\alpha} (I/I_\alpha).$$

Since C_α is faithfully flat over B_α, this also gives $I_\beta = I$. Therefore B is of finite presentation over A.

10. Supplement: quotients by a group scheme

Sections 10.2–10.4 below, written following suggestions of M. Raynaud, apply the preceding theorems to the case of an action of a group scheme. For the reader's convenience, Section 10.1 first reproduces statements 2.1–2.3 of Exposé XVI.

10.1. Representability theorems for quotients.—We first recall the following result.

Theorem 10.1.1.—*Let S be a scheme, let X and Y be two S-schemes, and let $f : X \to Y$ be an S-morphism. Suppose that one of the following two conditions holds:*

α) *the morphism f is locally of finite presentation;*

β) *the scheme S is locally Noetherian and X is locally of finite type over S.*

Then the following conditions are equivalent:

(i) *there exist an S-scheme X' and a factorization*

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S-morphism locally of finite presentation and f'' is a monomorphism;

[48] Editor's note: see EGA IV$_1$, 1.4.4.

(ii) *the first projection*

$$p_1 : X \times_Y X \longrightarrow X$$

is flat.

Moreover, if these conditions hold, then (X', f') is a quotient of X by the equivalence relation defined by f for the fppf topology. Thus the factorization $f = f'' \circ f'$ in (i) is unique up to isomorphism.

The case in which Y is locally Noetherian and X is of finite type over Y is treated in [Mur65], Corollary 2 to Theorem 2. We shall see that the general case reduces to that one.

We first make a few remarks.

a) The implication (i)⇒(ii) is immediate. Indeed, the first projection

$$p'_1 : X \times_{X'} X \longrightarrow X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism because f'' is a monomorphism, and p'_1 is flat because f' is flat; hence p_1 is flat.

b) The assertions of Theorem 10.1.1 are local on Y, and hence local on S. They are also local on X, as follows readily from the fact that a flat morphism locally of finite presentation is open (EGA IV_3, 11.3.1).

c) Under hypothesis α of Theorem 10.1.1, the preceding remarks reduce us to the case where X and Y are affine and f is of finite presentation. Replacing S by Y, if necessary, we may suppose that X and Y are of finite presentation over S. EGA IV_3, 11.2.6 then reduces us to the case where S is Noetherian.

d) Under hypothesis β of Theorem 10.1.1, we may suppose that S, X, and Y are affine, that S is Noetherian, and that X is of finite type over S. Write Y as a filtered inverse limit of affine schemes Y_i of finite type over S. The schemes

$$X \times_{Y_i} X$$

form a filtered decreasing family of closed subschemes of $X \times_S X$, whose inverse limit is $X \times_Y X$. Since $X \times_S X$ is Noetherian, for all sufficiently large i we have

$$X \times_{Y_i} X = X \times_Y X.$$

It follows that

$$f_i : X \xrightarrow{f} Y \longrightarrow Y_i$$

satisfies condition (ii) of Theorem 10.1.1 whenever f does. The equivalence relation on X defined by f is the same as the one defined by f_i. Thus it suffices to prove (ii)⇒(i) for f_i, reducing us to the case where Y is of finite type over S.

Application to group schemes. Let S be a scheme, let G be an S-group scheme locally of finite presentation over S, and suppose that G acts on the left on an S-scheme X. If $X \to S$ has a section ξ, recall that the stabilizer $\mathrm{Stab}_G(\xi)$ is representable by a subgroup scheme of G; see Exposé I, 2.3.3.

Theorem 10.1.2.—*Let S be a scheme and let G be an S-group scheme locally of finite presentation over S, acting on an S-scheme X. Suppose that $X \to S$ has a section ξ whose stabilizer H in G is flat over S. If one of the following hypotheses holds:*

a) X is locally of finite type over S;

b) S is locally Noetherian,

then the fppf quotient sheaf G/H is representable by an S-scheme, locally of finite presentation over S, and the S-morphism

$$f : G \longrightarrow X, \qquad g \longmapsto g \cdot \xi$$

factors as

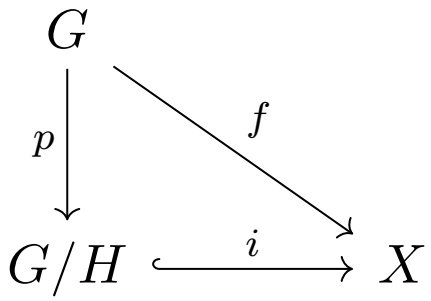

Figure 46: The factorization of the orbit morphism in Theorem 10.1.2.

where p is the canonical projection, which is faithfully flat and locally of finite presentation, and i is a monomorphism.

Proof.— The morphism f makes G into an X-scheme. By definition of the stabilizer of ξ, the morphism

$$G \times_S H \longrightarrow G \times_X G, \qquad (g, h) \longmapsto (g, gh)$$

is an isomorphism. Since H is flat over S, the scheme $G \times_S H$ is flat over G; hence the first projection

$$p_1 : G \times_X G \longrightarrow G$$

is flat. Moreover, if X is locally of finite type over S, then f is locally of finite presentation (EGA IV_1, 1.4.3(v)); otherwise, S is assumed locally Noetherian. It therefore suffices to apply Theorem 10.1.1 to f. It remains only to see that G/H is locally of finite presentation over S, and this follows immediately from Proposition 9.1.

Corollary 10.1.3.— *Let S be a scheme and $u : G \to H$ a morphism of S-group schemes. Suppose that G is locally of finite presentation over S, and that either H is locally of finite type over S or S is locally Noetherian.*

If $K = \ker(u)$ is flat over S, then the quotient group G/K is representable by an S-group scheme locally of finite presentation over S, and u factors as

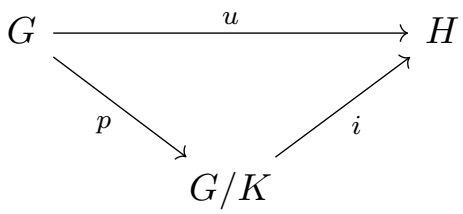

Figure 47: The factorization of u through G/K.

where p is the canonical projection and i is a monomorphism.

Proof.— Apply Theorem 10.1.2 with $X = H$ and with the unit section of H as ξ.

10.2. Stabilizer of the diagonal

Let S be a Noetherian scheme, X an S-scheme of finite type, and G a flat S-group scheme of finite type acting on the left on X. Thus we have an S-action $d_0 : G \times_S X \to X$. Let $d_1 : G \times_S X \to X$ be projection onto the second factor. As in §2(a), we have the groupoid

$$G \times_S G \times_S X \;\underset{G\times d_0}{\overset{\mathrm{pr}_{2,3}}{\underset{\longrightarrow}{\overset{\longrightarrow}{\overset{\mu\times X}{\longrightarrow}}}}}\; G \times_S X \;\underset{d_0}{\overset{d_1}{\rightrightarrows}}\; X$$

Figure 48: The action groupoid attached to (X, G).

whose cokernel, when it exists, is denoted $G\backslash X$.

Definition 10.2.1.— *We denote by $F \subset G \times_S X$ the* stabilizer of the diagonal section*; that is, F is the X-scheme defined by the Cartesian square*

$$\begin{array}{ccc} F & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle \Delta} \\ G \times_S X & \xrightarrow{(d_0,d_1)} & X \times_S X \end{array}$$

Figure 49: The stabilizer F of the diagonal section.

Then F is an X-subgroup scheme of $G \times_S X$. Since $G \times_S X$ is of finite type over the Noetherian scheme S, it is Noetherian; consequently F is of finite type both over S and over X (EGA I, 6.3.5 and 6.3.6). If $X \to S$ is separated, then F is a closed X-subgroup scheme of $G \times_S X$.

Recall that G is said to *act freely* on X when

$$(d_0, d_1) : G \times_S X \longrightarrow X \times_S X$$

is a monomorphism (Exposé III, 3.2.1). Equivalently, F is the trivial group scheme over X.

10.3. The case in which F is quasi-finite over X

Since F is of finite type over X, it is quasi-finite over X if and only if the stabilizers of the geometric points of X are finite.

Theorem 10.3.1.—[49] *Under the hypotheses of 10.2, suppose that F is quasi-finite over X. Then there is a dense, G-saturated open subset $U \subset X$ having the following properties:*

(i) *in $(\mathbf{Sch}/S)$, the cokernel $V = G\backslash U$ exists; moreover, V is a quotient in the category of ringed spaces;*

(ii) *$p : U \to V$ is surjective, open, and of finite presentation;*

[49] Editor's note: here too there is a largest open subset $U \subset X$ satisfying the conclusions of the theorem; see editor's note 44.

(iii) *V is of finite presentation over S;*

(iv) *the morphism*

$$G \times_S U \longrightarrow U \times_V U, \qquad (g, x) \longmapsto (gx, x),$$

is surjective;

(v) *if, in addition, G acts freely on X, then $U \to V$ is a left G-torsor, locally trivial for the fppf topology. In particular, $U \to V$ is faithfully flat.*[50]

Proof.— By hypothesis, the morphism

$$G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (gx, x),$$

is quasi-finite. Theorem 8.1 therefore applies to the groupoid defined by (X, G). Hence there is a dense saturated open subset $U \subset X$ for which the quotient $G\backslash U$ exists, and it has properties (i), (ii), and (iii); the same theorem also gives (iv).

To prove (v), recall that G acts freely on X if and only if (d_0, d_1) is an equivalence pair (Exposé III, 3.2.1). In that case Theorem 8.1(iv) shows that

$$G \times_S U \longrightarrow U \times_V U$$

is an isomorphism and that p is faithfully flat and of finite presentation. Thus U is a G-torsor over V, locally trivial for the fppf topology.

10.4. The case in which F is flat over X

Write

$$d = (d_0, d_1) : G \times_S X \longrightarrow X \times_S X, \qquad d(g, x) = (gx, x).$$

Recall that the sheaf-theoretic graph $\widetilde{\Gamma}$ of the equivalence relation associated with (X, G) is the fppf S-subsheaf of $X \times_S X$ given by the image of (d_0, d_1). It is the fppf sheaf associated with the graph functor

$$T \longmapsto \Gamma(T) = \big\{(x_0, x_1) \in X(T) \times X(T) \mid x_0 \in G(T)x_1\big\}.$$

Put $G_X = G \times_S X$. For every S-scheme T there is a surjective map

$$G_X(T) \longrightarrow \Gamma(T), \qquad (g, x) \longmapsto (gx, x),$$

which induces a bijection

$$\phi(T) : G_X(T)/F(T) \xrightarrow{\sim} \Gamma(T).$$

Indeed, if $(g, x), (g', x') \in G_X(T)$ have the same image, then $x' = x$ and $g^{-1}g'x = x$, so that $(g^{-1}g', x) \in F(T)$, and (g, x) and (g', x) have the same image in $G_X(T)/F(T)$.

By definition (see Exposé IV, 4.4.1(ii), or the proof of 5.2.1), the quotient sheaf G_X/F is the fppf sheaf associated with the presheaf

$$T \longmapsto G_X(T)/F(T) \simeq \Gamma(T).$$

We therefore obtain an isomorphism of sheaves

$$\phi : G_X/F \xrightarrow{\sim} \widetilde{\Gamma}.$$

Theorem 10.4.1.—[51] *Under the hypotheses of 10.2, the following assertions hold:*

[50] Editor's note: if one assumes in addition that G is a reductive S-group scheme and that its free action on X is linearizable, then $G\backslash X$ is known to be representable and $X \to G\backslash X$ is a left G-torsor. This follows from results of Raynaud and Seshadri and appears in [CTS79], Proposition 6.11.

[51] Editor's note: this is part (2) of Theorem 3 of [Ray67b]. That note also sketches another proof of Theorem 10.1.1.

a) the sheaf $\widetilde{\Gamma}$ is representable if and only if F is flat over X;

b) suppose that F is flat over X. Then the morphisms induced by d_1 and d_0

$$G_X/F \underset{\overline{d}_0}{\overset{\overline{d}_1}{\rightrightarrows}} X$$

Figure 50: The two maps induced by d_1 and d_0.

are faithfully flat and of finite presentation.

Proof of a).— Suppose that the fppf sheaf G_X/F is represented by an X-scheme Y. Then, by Exposé IV, 6.3.3, the morphism $p : G_X \to Y$ is faithfully flat and locally of finite presentation, and the second square in the following diagram is Cartesian:

$$\begin{array}{ccccc} F & \longrightarrow & F \times_X G_X & \longrightarrow & G_X \\ \downarrow & & \downarrow & & \downarrow p \\ X & \xrightarrow{e_X} & G_X & \longrightarrow & Y \end{array}$$

Figure 51: The Cartesian square used in the proof of Theorem 10.4.1(a).

The first square is obtained by base change along the unit section $e_X : X \to G_X$. Since p is faithfully flat and locally of finite presentation, the same is true of $F \to X$.

Conversely, suppose F is flat over X, and put $X^2 = X \times_S X$. The morphism $d : G_X \to X^2$ gives the fiber square

$$\begin{array}{ccc} G_X \times_{X^2} G_X & \longrightarrow & G_X \\ \downarrow & & \downarrow \\ G_X & \longrightarrow & X^2 \end{array}$$

Figure 52: The fiber square associated with $d : G_X \to X^2$.

The first projection

$$\mathrm{pr}_1 : G_X \times_{X^2} G_X \longrightarrow G_X$$

is then an $F \times_X G_X$-torsor over G_X, and is therefore flat and of finite type, because F is. By Theorem 10.1.1, d factors uniquely as

$$G_X \xrightarrow{\psi} Y \xrightarrow{\tau} X \times_S X$$

Figure 53: The unique factorization of d.

where ψ is faithfully flat and of finite type, and τ is a monomorphism of schemes.

It follows that the morphism of sheaves $\psi : G_X \to Y$ is F-invariant, and hence induces a morphism

$$\bar{\psi} : G_X/F \longrightarrow Y.$$

Moreover, because ψ is faithfully flat and of finite type, the monomorphism of sheaves τ factors through the sheaf image of d, namely $\widetilde{\Gamma}$. Thus the isomorphism $G_X/F \simeq \widetilde{\Gamma}$ factors through the monomorphism $Y \to \widetilde{\Gamma}$. We conclude that Y represents G_X/F.

Proof of b).— Suppose F is flat over X. By part (a) and its proof, G_X/F is representable, and

$$p : G_X \longrightarrow G_X/F$$

is faithfully flat and of finite presentation. On the other hand, the morphisms $d_i : G_X \to X$, for $i = 0, 1$, are faithfully flat and of finite presentation by hypothesis. Since $d_i = \bar{d}_i \circ p$, it follows from EGA IV$_2$, 2.2.13(iii), and EGA IV$_3$, 11.3.16, that each $\bar{d}_i$ is faithfully flat and of finite presentation.

Theorem 10.4.2.—[52] *Under the hypotheses of 10.2, suppose that F is flat over X. Then there is a dense saturated open subset $U \subset X$ such that the fppf quotient $V = G\backslash U$ is an S-scheme of finite type and $U \to V$ is faithfully flat and of finite presentation.*

Proof.— Theorem 10.4.1 shows that $G_X/F \simeq \widetilde{\Gamma}$ is representable. The fppf sheaf $G\backslash X$ is therefore the quotient sheaf of

$$G_X/F \underset{\bar{d}_0}{\overset{\bar{d}_1}{\rightrightarrows}} X$$

Figure 54: The equivalence pair whose quotient is $G\backslash X$.

By what precedes,

$$\bar{d}_i : G_X/F \longrightarrow X$$

is faithfully flat and of finite presentation for $i = 0, 1$, and the morphism

$$G_X/F \xrightarrow{\sim} \widetilde{\Gamma} \hookrightarrow X \times_S X$$

Figure 55: The monomorphism from the quotient G_X/F to $X \times_S X$.

is a monomorphism. In other words, $(\bar{d}_0, \bar{d}_1)$ is an equivalence pair. Consequently, Theorem 8.1 applies. There is therefore a dense saturated open subset $U \subset X$ such that the fppf quotient $V = G\backslash U$ is an S-scheme of finite type and $U \to V$ is faithfully flat and of finite presentation.

Taking the generic-flatness theorem into account (EGA IV$_2$, 6.9.3), we obtain the following result.

[52] Editor's note: here too there is a largest open subset $U \subset X$ satisfying the conclusions of the theorem. Moreover, a codimension-one point $x \in X$ belongs to U if and only if

$$(G_X/F) \times_X \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{X,x}) \times_S \operatorname{Spec}(\mathcal{O}_{X,x})$$

is a closed immersion; see editor's note 44.

Corollary 10.4.3.— *Under the hypotheses of 10.2, suppose that X is reduced. Then there is a dense saturated open subset $U \subset X$ such that the fppf quotient $G\backslash U$ is an S-scheme of finite type and $U \to G\backslash U$ is faithfully flat and of finite presentation.*

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[53] Editor's note: additional references cited in this exposé.

Exposé VI_A
Generalities on Algebraic Groups

by P. Gabriel

[0]Throughout this chapter, A will denote an Artinian local ring with residue field k. A group scheme G over $\operatorname{Spec} A$ will be called simply an *A-group*. This A-group is said to be *locally of finite type* if its underlying scheme is locally of finite type over A; it is said to be *algebraic* if its underlying scheme is of finite type over A.

0. Preliminary Remarks

0.1. Let us first consider a group scheme G over an arbitrary scheme S. We call the structural morphism

$$\mu : G \times_S G \longrightarrow G$$

multiplication, and we call the morphism $c : G \to G$, defined by $c(T)(x) = x^{-1}$ for every scheme T over S and every $x \in G(T)$, *inversion*. If U and V are subsets of the underlying set of G, we denote by $U \cdot V$ the image under multiplication of the subset of $G \times_S G$ consisting of those points whose first projection belongs to U and whose second projection belongs to V. Likewise, the notations U^{-1} and $c(U)$ are equivalent.

Let pr_1 be the projection of $G \times_S G$ onto the first factor, and let

$$\sigma : G \times_S G \longrightarrow G \times_S G$$

be the morphism with components pr_1 and μ. For every S-scheme T, the map $\sigma(T)$ is

$$(x, y) \longmapsto (x, xy);$$

it follows that σ is an automorphism. The composite of this automorphism with the projection pr_2 of $G \times_S G$ onto the second factor is the multiplication morphism. When G is flat over S, pr_2, and hence μ, are flat morphisms; when G is smooth over S, pr_2, and hence μ, are smooth morphisms, and so on.

0.2. From now on we assume that S is the spectrum of an Artinian local ring A with residue field k. We denote by $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ the category of reduced schemes over k. For every scheme X over A, the reduced scheme X_{red} is an object of $(\mathbf{Sch}_{/k})_{\mathrm{red}}$, and the functor

$$X \longmapsto X_{\mathrm{red}}$$

is right adjoint to the inclusion of $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ into $\mathbf{Sch}_{/A}$. It follows that, for every A-group G, G_{red} is a group in the category $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.[1] In other words, for every reduced k-scheme

[0]Editor's note: version of October 13, 2024.

[1]Editor's note: the sentences that follow have been added.

T, $G_{\mathrm{red}}(T)$ is endowed, functorially in T, with a group structure. One must bear in mind that G_{red} is not necessarily a k-group, because its "multiplication" is only a morphism

$$(G_{\mathrm{red}} \times_k G_{\mathrm{red}})_{\mathrm{red}} \longrightarrow {G_{\mathrm{red}}}^2.$$

[3] However, if k is a perfect field, the inclusion of $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ into $\mathbf{Sch}_{/k}$ commutes with products. Groups in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ may therefore be identified with group schemes over k whose underlying schemes are reduced. In this case, if G is a k-group, G_{red} is a group subscheme of G; in general, this subgroup is not normal in G.

For example, if k is a field of characteristic 3, the constant group $(\mathbb{Z}/2\mathbb{Z})_k$ acts nontrivially on the diagonalizable group $D_k(\mathbb{Z}/3\mathbb{Z})$ (cf. Exposé I, 4.1 and 4.4). If G denotes the semidirect product of $D_k(\mathbb{Z}/3\mathbb{Z})$ by $(\mathbb{Z}/2\mathbb{Z})_k$ defined by this action, then G_{red} is identified with $(\mathbb{Z}/2\mathbb{Z})_k$ and is not normal in G.[4]

Let k be an arbitrary field, let $k^{p^{-\infty}}$ be its perfect closure, and let H be a group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$. Then

$$(H \otimes_k k^{p^{-\infty}})_{\mathrm{red}}$$

is a group scheme over the perfect field $k^{p^{-\infty}}$. Since $H \otimes_k k^{p^{-\infty}}$ and $(H \otimes_k k^{p^{-\infty}})_{\mathrm{red}}$ have the same underlying topological space, groups in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ share with k-groups certain topological properties that are invariant under extension of the base field. For example, it will follow from 0.3 and the preceding remarks that every group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ is separated.

In what follows, we shall encounter groups in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ whenever we deal with a nonempty locally closed subset U of an A-group G such that

$$U \cdot U = U \qquad \text{and} \qquad U^{-1} = U.$$

Indeed, the reduced subscheme of G defined by U is then a group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.

0.3. *An A-group G is always separated, because the unit section $e : \operatorname{Spec} A \to G$ is a closed immersion.* Indeed, let x be the unique point of $\operatorname{Spec} A$, and let $\eta : G \to \operatorname{Spec} A$ be the structural morphism. Since $\eta \circ e = \mathrm{id}_{\operatorname{Spec} A}$, for every affine open $U = \operatorname{Spec} B$ of G containing $e(x)$, the homomorphism $B \to A$ has a section and is therefore surjective. It follows that e is a closed immersion.[5]

The diagonal in $G \times_A G$ is identified with the functor from $(\mathbf{Sch}_{/A})^\circ$ to **Sets** which associates to every A-scheme S the inverse image of the identity element of $G(S)$ under the map

$$\varphi(S) : G(S) \times G(S) \longrightarrow G(S), \qquad (x, y) \longmapsto x \cdot y^{-1}.$$

We therefore have the following cartesian square. Since the diagonal morphism is obtained from a closed immersion by base change, it is itself a closed immersion.

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\ \varphi\ } & G \\ {\scriptstyle \text{diagonal morphism}}\uparrow & & \uparrow{\scriptstyle \text{unit section}} \\ G & \longrightarrow & \operatorname{Spec} A \end{array}$$

[2] Editor's note: for examples of group schemes G over a nonperfect field k for which G_{red} is not a group scheme over k, see 1.3.2 below.

[3] Editor's note: the remainder of 0.2, as well as 0.3, has been slightly modified.

[4] Editor's note: for an analogous example with G connected, one may take, when $\mathrm{char}(k) = p > 0$, the semidirect product of $\alpha_{p,k}$ and $\mathbb{G}_{m,k}$.

[5] Editor's note: this argument is valid for every local ring A of dimension zero, and shows that, if S is a discrete scheme, every S-group is separated; cf. VI$_B$, 5.2. On the other hand (cf. VI$_B$, 5.6.4), if S contains a nonisolated closed point s, the S-scheme G obtained by gluing two copies of S along the open subset $S - \{s\}$ is not separated over S, although it carries an S-group structure.

0.4.[6] Let G be an A-scheme. We shall say that a point g of G is *strictly rational over* A if there exists an A-morphism

$$s : \operatorname{Spec} A \longrightarrow G$$

which sends the unique point of $\operatorname{Spec} A$ to g; that is, if the homomorphism $A \to \mathcal{O}_{G,g}$ admits a retraction. One then has $\kappa(g) = k$, so g is a closed point of G. Indeed, if B is the ring of an affine open neighborhood of g, and if $\mathfrak{p}$ is the prime ideal of B corresponding to g, then $B/\mathfrak{p} \subset k$ is a finite A-algebra, hence an integral Artinian ring, and therefore a field.

Suppose henceforth that G is an A-group. Such a morphism $s : \operatorname{Spec} A \to G$ defines an automorphism r_s of the scheme G over A, called *right translation by* s. For every morphism $\pi : S \to \operatorname{Spec} A$, the automorphism $r_s(\pi)$ of $G(S)$ is defined by

$$x \longmapsto x \cdot G(\pi)(s), \qquad x \in G(S).$$

Similarly, ℓ_s denotes *left translation by* s, namely the automorphism of G defined by

$$\ell_s(\pi)(x) = G(\pi)(s) \cdot x, \qquad x \in G(S).$$

The schemes $G \otimes_A k$ and G have the same underlying topological space $\underline{G}$, $G \otimes_A k$ is a k-group, and $s \otimes_A k$ depends only on g, not on s. Consequently, the automorphisms of $\underline{G}$ induced by r_s and ℓ_s (or by $r_{s\otimes k}$ and $\ell_{s\otimes k}$) depend only on g, not on s. Thus, if P is a subset of $\underline{G}$, we may write

$$r_g(P) \text{ or } P \cdot g \qquad (\text{respectively, } \ell_g(P) \text{ or } g \cdot P)$$

instead of $r_s(P)$ (respectively, $\ell_s(P)$), in agreement with 0.1.

Remark 0.4.1. *If g is a strictly rational point of G, and if $A \to A'$ is a morphism of Artinian local rings, then $G' = G \otimes_A A'$ has a unique point g' above g, and g' is strictly rational over A'. Moreover, if P' denotes the inverse image of P in G', then $P' \cdot g'$ is the inverse image of $P \cdot g$ (cf. EGA I, 3.4.8).*

Proposition 0.5.[7] *Let G be an A-group, and let U, V be two dense open subsets of G. Then $U \cdot V$, that is, the image of $U \times_A V$ under multiplication, is the entire underlying space of G.*

Proof. Since G and $G \otimes_A k$ have the same underlying space, after replacing A by k and G by $G \otimes_A k$, if necessary, we may assume that $A = k$. Let $g \in G$, and set $K = \kappa(g)$. Then left translation ℓ_g is an automorphism of G_K. Since the projection $G_K \to G$ is open, U_K and V_K are dense open subsets of G_K, as is the image of V_K under ℓ_g. Consequently there exists $v \in V_K$ such that $u = \ell_g(v)$ belongs to U_K.

Let L be an extension of K containing $\kappa(v)$, and hence $\kappa(u)$, and let g_L and v_L be the L-points of G_L induced by g and v. Then

$$g_L \cdot v_L = u'$$

is a point of G_L above u, and therefore

$$g_L = u' \cdot v_L^{-1}$$

lies above $U \cdot V$. Thus $g \in U \cdot V$, proving the proposition. □

[6] Editor's note: the original has been expanded in what follows; in particular, Remark 0.4.1 has been added.

[7] Editor's note: the original stated this result under the hypothesis that G be locally of finite type over A. Since the result is useful in the general case, and since the proof is essentially the same, it has been stated and proved here in full generality. It will be used several times below.

Corollary 0.5.1.[8] *If G is an irreducible A-group, then G is quasi-compact.*

Proof. Let U be a nonempty affine open subset of G. Then U is dense in G, so by 0.5 the morphism

$$\mu : U \times_A U \longrightarrow G$$

is surjective. Hence G is quasi-compact, since $U \times_A U$ is. □

Corollary 0.5.2.[9] *Let G be an A-group scheme, and let H be a group subscheme of G over A. Then H is closed.*

Proof. Let k' be the perfect closure of the residue field k of A. Since the underlying topological spaces of G and H are unchanged under the base change $A \to k \to k'$, we may assume that $A = k$ is a perfect field. We may then assume that G and H are reduced, hence geometrically reduced.

Let $\overline{H}$ be the closure of H. Then $\mu^{-1}(\overline{H})$ is a closed subset of $G \times G$ containing $H \times H$. The morphisms

$$H \longrightarrow \operatorname{Spec} k \qquad \text{and} \qquad \overline{H} \longrightarrow \operatorname{Spec} k$$

are universally open. Since H is dense in $\overline{H}$, $H \times H$ is dense in $\overline{H} \times H$, and $\overline{H} \times H$ is dense in $\overline{H} \times \overline{H}$. Thus $H \times H$ is dense in $\overline{H} \times \overline{H}$. Consequently

$$\mu(\overline{H} \times \overline{H}) \subset \overline{H}.$$

Since $\overline{H} \times \overline{H}$ is reduced, μ induces a morphism

$$\mu' : \overline{H} \times \overline{H} \longrightarrow \overline{H}.$$

Now let $g \in \overline{H}$, and put $K = \kappa(g)$. The projection $\overline{H}_K \to \overline{H}$ is open, so H_K and $\ell_g(H_K)$ are two dense open subsets of $\overline{H}_K$. Hence there exist $u, v \in H_K$ such that $\ell_g(v) = u$. As in the proof of 0.5, it follows that g belongs to $H \cdot H = H$. Therefore $\overline{H} = H$. □

1. Local properties of an A-group locally of finite type

We shall first see that, if G is locally of finite type and flat over A, one can "make every closed point of G strictly rational" by means of a finite flat extension of the base.

1.1. Unless expressly stated otherwise, from now on we assume that G is an A-group locally of finite type. When A is a field k, we shall obtain in Exposé VII$_B$ very precise results on the local rings of G.[10] Here we content ourselves with a few elementary results.

Proposition 1.1.1. *Let x be a point of an A-group G locally of finite type and flat over A. Then the local ring $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and there is a system of parameters $a_1, \ldots, a_n$ of $\mathcal{O}_{G,x}$ such that $\mathcal{O}_{G,x}/(a_1, \ldots, a_n)$ is a finite flat A-module (hence finite free).*

We first suppose that A is equal to its residue field k. It then suffices to prove that $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and we may restrict to the case in which x is a closed point (cf. EGA 0$_{\text{IV}}$, 16.5.13). By Lemma 1.1.2 below, G contains a closed point y such that $\mathcal{O}_{G,y}$ is Cohen–Macaulay. By SGA 1, I, § 9, this is equivalent to saying that, for every finite extension K of the base field k and every point $\overline{y}$ of $\overline{G} = G \otimes_k K$ lying above y, the local

[8] Editor's note: this corollary has been inserted here; cf. 2.4 and 2.6.2 below.

[9] Editor's note: this corollary, used implicitly in VIII, 6.7, has been added; see also VI$_B$, 6.2.5.

[10] Editor's note: for example, they are always complete intersections; see Exposé VII$_B$, 5.5.1. Moreover, if $\operatorname{char}(k) = 0$, then G is smooth (Exposé VI$_B$, 1.6.1; see also Exposé VII$_B$, 3.3.1).

ring $\mathcal{O}_{\overline{G},\overline{y}}$ is Cohen–Macaulay. If K has been chosen sufficiently large, that is, if K contains a normal extension of k containing the residue fields $\kappa(y)$ and $\kappa(x)$, then $\overline{y}$ is (strictly) rational over K, as is every point $\overline{x}$ of $\overline{G}$ lying above x.[11] Since the automorphism

$$r_{\overline{x}} \circ (r_{\overline{y}})^{-1}$$

sends $\overline{y}$ to $\overline{x}$, it follows that $\mathcal{O}_{\overline{G},\overline{x}}$, and hence $\mathcal{O}_{G,x}$ (SGA 1, I, § 9), are Cohen–Macaulay.

Now let A again be arbitrary. The preceding argument applies to $k \otimes_A G$, so that $k \otimes_A \mathcal{O}_{G,x}$ is Cohen–Macaulay. If $a_1, \ldots, a_n$ is a sequence of elements of $\mathcal{O}_{G,x}$ whose image in $k \otimes_A \mathcal{O}_{G,x}$ is a system of parameters, it follows from SGA 1, IV, 5.7, or EGA 0_{IV}, 15.1.16, that $a_1, \ldots, a_n$ is an $\mathcal{O}_{G,x}$-regular sequence and that $\mathcal{O}_{G,x}/(a_1, \ldots, a_n)$ is finite and flat (hence finite and free) over A.

Lemma 1.1.2. *Every nonempty scheme X, locally of finite type over an Artinian ring A, contains a closed point x whose local ring is Cohen–Macaulay.*

We may plainly suppose that X is affine, with algebra B, and argue by induction on $\dim X$ (the assertion is clear if X is discrete, since all its local rings are then Artinian). Since B is of finite type over A, if $\dim B > 0$, then B contains an element a that is not invertible and is not a zero divisor.[12] The closed subscheme

$$X' = \operatorname{Spec} B/(a)$$

of X then has dimension strictly less than $\dim X$, and by the induction hypothesis it contains a closed point x such that $\mathcal{O}_{X',x}$ is Cohen–Macaulay. Since $\mathcal{O}_{X',x} = \mathcal{O}_{X,x}/(a)$, and a is neither invertible nor a zero divisor in $\mathcal{O}_{X,x}$, the ring $\mathcal{O}_{X,x}$ is Cohen–Macaulay (see also EGA IV_2, 6.11.3).

Proposition 1.2. *Let A be an Artinian local ring, let G be an A-group locally of finite type and flat over A, and let x be a closed point of G. Then there exists a local A-algebra A', finite and free over A, such that every point of $G \otimes_A A'$ lying above x is strictly rational over A'.*

[13] Indeed, let k_1 be a finite normal extension of k containing the residue field $\kappa(x)$ of x. By Lemma V.4.1.2, there exists a local A-algebra A_1, finite and free over A, with residue field k_1. In this case (cf. Editor's note 11), all the points $g_1, \ldots, g_n$ of $G \otimes_A A_1$ lying above $x \in G$ have k_1 as residue field; that is, $g_1, \ldots, g_n$ are rational over A_1 in the sense of Exposé V, § 4.e).

Let $B_1, \ldots, B_n$ be the local rings of $g_1, \ldots, g_n$. By 1.1.1, the rings $B_1, \ldots, B_n$ have quotients $B'_1, \ldots, B'_n$ that are Artinian and finite and free over A_1. Set

$$A' = B'_1 \otimes_{A_1} \cdots \otimes_{A_1} B'_n.$$

Then A' is local, finite, and free over A_1, and, for each $i = 1, \ldots, n$, there are surjective homomorphisms

$$B_i \otimes_{A_1} A' \twoheadrightarrow B'_i \otimes_{A_1} A' \twoheadrightarrow A',$$

[11] Editor's note: indeed, the hypothesis on K implies that, for every extension L of K, every k-morphism $\kappa(x) \to L$ (respectively, $\kappa(y) \to L$) factors through K. Consequently, all the points of $G \otimes_k K$ lying above x or y have K as residue field; that is, they are (strictly) rational over K.

[12] Editor's note: indeed, B is a Noetherian Jacobson ring (cf. [BAC], V, § 3.4). If every noninvertible element is a zero divisor, then every prime ideal is an associated prime ideal of B; in particular, B has only finitely many maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$. Since B is a Jacobson ring, the intersection of the $\mathfrak{m}_i$ is the nilradical of B. It follows that each $\mathfrak{m}_i$ is a minimal prime ideal of B, and hence that $\dim B = 0$.

[13] Editor's note: the original has been expanded in what follows.

the second induced by the multiplication map

$$B'_i \otimes_{A_1} B'_i \twoheadrightarrow B'_i.$$

Consequently, A' has the required property.

1.3. Let e be the unit element (or origin) of G, that is, the image of the unique point of $\operatorname{Spec} A$ under the unit section $\operatorname{Spec} A \to G$. By definition, e is strictly rational over A.

Proposition 1.3.1.[14] *Let G be a group locally of finite type and flat over an Artinian ring A, and let K (respectively, $\overline{k}$) be the perfect closure (respectively, an algebraic closure) of the residue field k of A.*

(1) *For every closed point x of $\overline{G} = G \otimes_k \overline{k}$, the local rings $\mathcal{O}_{\overline{G},e}$ and $\mathcal{O}_{\overline{G},x}$ are isomorphic. In particular, the tangent spaces $T_e\overline{G}$ and $T_x\overline{G}$ have the same dimension.*

(2) *The following assertions are equivalent:*

(i) *$G \otimes_A K$ is reduced.*

(i bis) *$\mathcal{O}_{G,e} \otimes_A K$ is reduced.*

(ii) *G is smooth over A.*

(ii bis) *G is smooth over A at the origin.*

Proof. (1) Let x be a closed point of $\overline{G}$. There is exactly one $\overline{k}$-morphism

$$s : \operatorname{Spec} \overline{k} \longrightarrow \overline{G}$$

whose image is x. Right translation r_s then induces an isomorphism from

$$\mathcal{O}_{\overline{G},e} = \mathcal{O}_{G,e} \otimes_k \overline{k}$$

onto $\mathcal{O}_{\overline{G},x}$, which proves (1).

We now prove (2). By SGA 1, II.2.1, we reduce at once to the case in which A is a field ($A = k$). The implications

$$\text{(i)} \Rightarrow \text{(i bis)}, \qquad \text{(ii)} \Rightarrow \text{(ii bis)}, \qquad \text{(ii)} \Rightarrow \text{(i)}, \qquad \text{(ii bis)} \Rightarrow \text{(i bis)}$$

are evident, so it remains only to prove that (i bis) implies (ii).

Now (i bis) implies that $\mathcal{O}_{G,e} \otimes_k \overline{k}$ is reduced. Hence, by (1), $\mathcal{O}_{\overline{G},x}$ is reduced for every closed point x of $\overline{G}$, and therefore $\overline{G}$ is reduced. Since $\overline{G}$ is locally of finite type over $\overline{k}$, there is at least one closed point y such that $\mathcal{O}_{\overline{G},y}$ is regular. By (1), the local rings at all closed points of $\overline{G}$ are isomorphic to $\mathcal{O}_{\overline{G},e}$; thus all these local rings are regular. Consequently, $\overline{G}$ is smooth over $\overline{k}$, and hence G is smooth over k.

[15] We may now give the examples below, pointed out by M. Raynaud, of group schemes G over a nonperfect field k for which G_{red} is not a k-group scheme.

Examples 1.3.2. Let k be a nonperfect field of characteristic $p > 0$, let $t \in k - k^p$, let $\overline{k}$ be an algebraic closure of k, and let $\alpha \in \overline{k}$ satisfy $\alpha^p = t$.

[14] Editor's note: part (1), which is useful in Examples 1.3.2 below, has been added.

[15] Editor's note: the following sentence and Examples 1.3.2 have been added.

(1) Consider the additive group $\mathbb{G}_{a,k} = \operatorname{Spec} k[X]$, and let G be the group subscheme, finite over k, defined by the additive polynomial $X^{p^2} - tX^p$. Then

$$G_{\mathrm{red}} = \operatorname{Spec} k[X]/X\,(X^{p(p-1)} - t)$$

is étale at the origin. If it were a group scheme over k, it would be smooth (by 1.3.1); but G_{red} is not geometrically reduced, so it is not a group scheme over k.

(2) Consider

$$\mathbb{G}_{a,k}^4 = \operatorname{Spec} k[X, Y, U, V],$$

and let G be the group subscheme defined by the ideal I generated by the additive polynomials

$$P = X^p - tY^p \qquad \text{and} \qquad Q = U^p - tV^p.$$

Then G is two-dimensional and irreducible, since

$$(G_{\overline{k}})_{\mathrm{red}} \simeq \operatorname{Spec} \overline{k}[Y, V]$$

is irreducible.

Let $A = k[X, Y, U, V]$, and let $\mathfrak{m}$ be its augmentation ideal. Denote by x, y, u, v the images of dX, dY, dU, dV in

$$\Omega^1_{A/k} \otimes_A (A/\mathfrak{m}),$$

viewed as linear forms on the tangent space

$$k^4 = T_0\mathbb{G}_{a,k}^4.$$

We shall show that the subspace $E = T_0 G_{\mathrm{red}}$ equals k^4. Otherwise, there would be a linear form

$$f = ax + by + a'u + b'v, \qquad a, b, a', b' \in k,$$

not all of whose coefficients are zero, that vanishes on E. Recall that the formation of $\Omega^1_{A/k}$, and hence that of tangent spaces, commutes with base change (cf. EGA IV$_4$, 16.4.5), and identify f with its image in $(\overline{k}^4)^*$. Since

$$(G_{\overline{k}})_{\mathrm{red}} \subset (G_{\mathrm{red}})_{\overline{k}},$$

the form f vanishes on the subspace $T_0(G_{\overline{k}})_{\mathrm{red}}$ of $\overline{k}^4$, which is defined by the equations

$$g_1 = x - \alpha y \qquad \text{and} \qquad g_2 = u - \alpha v.$$

Thus

$$f = \lambda g_1 + \mu g_2, \qquad \lambda, \mu \in \overline{k}.$$

But $\lambda g_1 + \mu g_2$ belongs to k^4 only if $\lambda = \mu = 0$. This contradiction shows that $E = k^4$, and hence

$$T_0(G_{\mathrm{red}})_{\overline{k}} = \overline{k}^4.$$

On the other hand, $R = XV - YU$ belongs to $\sqrt{I}$, since

$$R^p = (X^p - tY^p)V^p - Y^p(U^p - tV^p).$$

Consequently, the tangent space F at the point $(\alpha, 1, \alpha, 1)$ of $(G_{\mathrm{red}})_{\overline{k}}$ is contained in the hyperplane H of $\overline{k}^4$ with equation

$$\alpha\, dV + dX - dU - \alpha\, dY = 0,$$

and therefore has dimension at most 3.[16] It follows from part (1) of 1.3.1 that G_{red} is not a group scheme over k.

[16] Editor's note: in fact, it is easy to see that $\sqrt{I}$ is generated by P, Q, R at every point other than 0, so $F = H$.

2. Connected components of an A-group locally of finite type

2.1. Let us first consider an arbitrary A-group G, and let G' be the connected component of the origin e in G. This connected component is plainly closed, so we may identify it with the reduced closed subscheme of G having G' as its underlying space.[17]

Proposition 2.1.1. *For every extension K of the residue field k of A, the underlying space of $G' \otimes_A K$ is the connected component of the origin in the K-group $G \otimes_A K$; in other words, G' is geometrically connected.*

Indeed, let $(G \otimes_A K)'$ be the connected component of the origin in $G \otimes_A K$. Since the image of $(G \otimes_A K)'$ in G is connected and contains the unit element of G, that image is contained in G'. Thus $(G \otimes_A K)'$ is contained in the inverse image $G' \otimes_A K$ of G' in $G \otimes_A K$. The proposition therefore follows from the connectedness of $G' \otimes_A K$, which is proved in the following lemma.

Lemma 2.1.2. *Let X and Y be two connected schemes over a field k. If X contains a rational point, then $X \times_k Y$ is connected.*

We give a direct proof below of this result from EGA IV_2 (4.5.8 and 4.5.14).

Suppose first that Y is nonempty, connected, and affine, with algebra B. Then $X \times_k Y$ is the spectrum of the quasi-coherent $\mathcal{O}_X$-algebra

$$\mathcal{B} = \mathcal{O}_X \otimes_k B.$$

We wish to show that every nonempty subset U of $X \times_k Y$ that is both open and closed equals $X \times_k Y$. Now U is affine over X, and its affine $\mathcal{O}_X$-algebra is a direct factor of $\mathcal{B}$. It follows from Lemma 2.1.3 below that the image of U in X is open and closed, and hence is all of X. In particular, this image contains a rational point x of X, so U meets the inverse image of x in $X \times_k Y$. Since this inverse image is isomorphic to Y, and hence connected, U contains the whole inverse image. The same conclusion would hold for the complement of U in $X \times_k Y$ if U were different from $X \times_k Y$, which is absurd.

If Y is now an arbitrary k-scheme, the preceding argument shows that the fibers of the canonical projection

$$X \times_k Y \longrightarrow Y$$

are connected. If x is a rational point of X, all these fibers meet the subscheme $\{x\} \times_k Y$, which is itself connected. This proves the assertion.

Lemma 2.1.3. *Let X be a scheme, and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_X$-algebra that is locally*[18] *a direct factor of a free $\mathcal{O}_X$-module. Then the image of* $\operatorname{Spec} \mathcal{A}$ *in X is open and closed.*

Let V be this image. It is clear that V is contained in the support of $\mathcal{A}$. Conversely, if x belongs to the support of $\mathcal{A}$, then $\mathcal{A}_x$ is nonzero and is a free $\mathcal{O}_{X,x}$-module, since, by Kaplansky, every projective module over a local ring is free. Consequently, the fiber of $\operatorname{Spec} \mathcal{A}$ at x, which is affine with algebra

$$\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x),$$

is nonempty. Thus the image of $\operatorname{Spec} \mathcal{A}$ coincides with the support of $\mathcal{A}$.

If s is the unit section of $\mathcal{A}$, the equality $s_x = 0$ implies that s, and hence $\mathcal{A}$, vanish in a neighborhood of x. Therefore the support of $\mathcal{A}$ is closed. On the other hand, we may

[17] Editor's note: we shall see below (2.2) that G' is a group in the category $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.

[18] Editor's note: the word "locally" has been added. The reference to Kaplansky is *Projective modules*, Ann. of Math. **68** (1958), 372–377; see also [BAC], § II.3, Ex. 3.

assume that $\mathcal{A}$ is a direct factor of the direct sum $\mathcal{L}$ of a family $(\mathcal{O}_X^\alpha)$ of copies of $\mathcal{O}_X$.[19] For each β, let φ^β denote the restriction to $\mathcal{A}$ of the canonical projection

$$\mathcal{L} = \bigoplus_\alpha \mathcal{O}_X^\alpha \longrightarrow \mathcal{O}_X^\beta.$$

For $x \in X$, let $\mathcal{A}_x^*$ denote the dual module

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}).$$

Since $\mathcal{A}_x$ is a direct factor of $\mathcal{L}_x$, the restriction map

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathcal{O}_{X,x}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}) = \mathcal{A}_x^*$$

is surjective. If $\mathcal{A}_x \neq 0$, then, since $\mathcal{A}_x$ is free and nonzero, there exist $a \in \mathcal{A}_x$ and a linear form $\varphi \in \mathcal{A}_x^*$ such that $\varphi(a) = 1$. There is therefore a family (ξ^α) of elements of $\mathcal{O}_{X,x}$ such that, for every $u \in \mathcal{A}_{X,x}$,

$$\varphi(u) = \sum_\alpha \xi^\alpha \varphi^\alpha(u);$$

this sum is finite because $\varphi^\alpha(u) = 0$ except for finitely many α. Applying this to $u = a$, we find $\alpha_1, \ldots, \alpha_n$ such that

$$1 = \varphi(a) = \xi^{\alpha_1}\varphi^{\alpha_1}(a) + \cdots + \xi^{\alpha_n}\varphi^{\alpha_n}(a).$$

There is then an open neighborhood U of x on which a and the ξ^{α_i} come from sections

$$\widetilde{a} \in \mathcal{A}(U) \qquad \text{and} \qquad \widetilde{\xi}^{\alpha_i} \in \mathcal{O}_X(U).$$

The equality

$$\sum_i \widetilde{\xi}^{\alpha_i}\varphi^{\alpha_i}(\widetilde{a}) = 1 \quad \text{on } U$$

shows that $\widetilde{a}_y \neq 0$ for every $y \in U$. ("The support of a projective module is open.")

2.2. The notation remains that of 2.1. It is clear that G' is a reduced k-scheme. Lemma 2.1.2 shows that $G' \times_k G'$ is connected, so $(G' \times_k G')_{\mathrm{red}}$ is the reduced subscheme of $G \times_A G$ whose underlying space is the connected component of the origin. In particular, the multiplication morphism

$$\mu : G \times_A G \longrightarrow G$$

induces a morphism

$$\mu' : (G' \times_k G')_{\mathrm{red}} \longrightarrow G'$$

that makes G' a group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.

2.2.bis.[20] Recall (cf. Exposé V) that, if P is a scheme, $\underline{P}$ denotes the underlying topological space of P. We define an A-subfunctor G^0 of G by setting, for every A-scheme S,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G'}\}.$$

Let $c : G \to G$ be the inversion morphism. Since $c(\underline{G'}) = \underline{G'}$, we have $c \circ u \in G^0(S)$ for every $u \in G^0(S)$. On the other hand, if $u, v \in G^0(S)$, then $u \boxtimes v$ sends $\underline{S}$ into the subspace of $\underline{G \times_A G}$ consisting of the points whose two projections belong to $\underline{G'}$. This subspace is identified with the underlying space of $G' \times_A G'$, which is connected by Lemma 2.1.2.

[19] Editor's note: the original has been expanded in what follows.

[20] Editor's note: this paragraph has been added in order to make the connection with Exposé VI$_B$, § 3.

Consequently, $\mu \circ (u \boxtimes v)$ sends $\underline{S}$ into $\underline{G'}$. This proves that G^0 is a group subfunctor of G over A.

If the connected component of e is open in G, then the subfunctor G^0 is represented by the subscheme induced by G on this open subset, which is therefore a group subscheme of G; this subscheme will also be denoted by G^0. In this case, with the notation of 2.1,

$$G' = (G^0)_{\mathrm{red}},$$

and the topological spaces $\underline{G'}$ and $\underline{G^0}$ coincide.[21]

2.3. In accordance with our convention in 1.1, we again assume from now on that G is locally of finite type over A. Then G is locally Noetherian, hence locally connected;[22] thus every connected component of G is open.

We shall write G^0 for the subscheme induced by G on the connected component G' of the unit element. By 2.2.bis, G^0 is a group subscheme of G, which we shall call the *neutral component* of G. For every A-scheme S, therefore,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G^0} = \underline{G'}\}.$$

Let G^α be an arbitrary connected component of G, and let

$$\nu^\alpha : G^\alpha \times_A G^0 \longrightarrow G$$

be the morphism defined by

$$\nu^\alpha(S)(g, \gamma) = g\gamma g^{-1}$$

for every $S \in (\mathbf{Sch}_{/A})$, $g \in G^\alpha(S)$, and $\gamma \in G^0(S)$. If e is the origin of G, the restriction of ν^α to $G^\alpha \times_A \{e\}$ is the trivial morphism. Since $G^\alpha \times_A G^0$ is connected by 2.1.2, it follows that ν^α factors through G^0. Thus, for every A-scheme S, $G^0(S)$ is a normal subgroup of $G(S)$. We have proved the following proposition.[23]

Proposition 2.3.1. *Let G be an A-group locally of finite type. Then the neutral component G^0 is an open normal group subscheme of G.*

Proposition 2.4. *Let G be an A-group locally of finite type.*

(i) *G^0 is irreducible, and $G^0 \otimes_A k$ is geometrically irreducible over k.*

(ii) *G^0 is quasi-compact, and hence is of finite type over A.*

[24] *Proof.* (i) Since G^0 and $G^0 \otimes_A k$ have the same underlying topological space, it suffices to prove the second assertion. Let $\overline{k}$ be an algebraic closure of k. By 2.2, $(G \otimes_A \overline{k})_{\mathrm{red}}$ is a reduced $\overline{k}$-group locally of finite type, and hence is smooth over $\overline{k}$ by 1.3.1. A fortiori, the local rings of $(G \otimes_A \overline{k})_{\mathrm{red}}$ are integral domains. Therefore,[25] since $G \otimes_A \overline{k}$ is locally Noetherian, the connected components of $G \otimes_A \overline{k}$ are irreducible (cf. EGA I, 6.1.10). In particular, the connected component $G^0 \otimes_A \overline{k}$ (cf. 2.1.1) is irreducible.

[21] Editor's note: in this exposé, the notation G^0 is reserved for the case in which the connected component of e is open; in Exposé VI_B, § 3, this connected component will be denoted by $\underline{G^0}$ in every case. This is a slight abuse of notation, but it is compatible with the preceding discussion when the connected component is the underlying topological space of an open group subscheme G^0 of G.

[22] Editor's note: indeed, in a Noetherian space there are only finitely many connected components, and each is therefore open; see also EGA I, 6.1.9.

[23] Editor's note: the number 2.3.1 has been added in order to emphasize this statement. Moreover, G^0 is even a characteristic subgroup of G; see 2.6.5(ii).

[24] Editor's note: in the statement, "G^0 is geometrically irreducible" has been replaced by "$G^0 \otimes_A k$ is geometrically irreducible over k," and the proof has been expanded.

[25] Editor's note: the original has been expanded here, and the reference to EGA I, 6.1.10, has been added.

(ii) We now show that G^0 is of finite type over A. Since G^0 is locally of finite type over A, it suffices to prove that G^0 is quasi-compact. Since G^0 is irreducible, this follows from 0.5.1.

Corollary 2.4.1. *Every connected component of G is irreducible,*[26] *of finite type over A,*[27] *and of the same dimension as G^0.*

Indeed, we may assume that A equals its residue field k. Let C be a connected component of G, let x be a closed point of C, let $\kappa(x)$ be the residue field of x, and let k' be a finite normal extension of k containing $\kappa(x)$. The canonical projection

$$\pi : C \otimes_k k' \longrightarrow C$$

is open and closed.[28] Consequently, if C' is a connected component of $C \otimes_k k'$, the projection $C' \to C$ is surjective. Thus C' contains a point $y \in \pi^{-1}(x)$, and such a point is rational over k' (cf. the proof of 1.2); it follows that C' is the image of $G^0 \otimes_k k'$ under right translation r_y. Now $G^0 \otimes_k k'$ is of finite type over k', by 2.4, and $\pi^{-1}(x)$ is finite, of cardinality at most $[k' : k]$. Consequently, $C \otimes_k k'$ is of finite type over k', and hence C is of finite type over k.

Moreover, since $G^0 \otimes_k k'$ is irreducible by 2.4, the same is true of C',[29] and hence also of C, because the projection $C' \to C$ is surjective.

Finally, we saw above that $C \otimes_k k'$ is the disjoint union of finitely many translates of $G^0 \otimes_k k'$. Since dimension is invariant under extension of the base field (cf. EGA IV$_2$, 4.1.4), it follows that C has the same dimension as G^0. (Moreover, by EGA IV$_2$, 5.2.1, one has $\dim_g G = \dim G^0$ for every point $g \in G$.)

2.5. This paragraph has been added. The results that follow appear in Exposé VI$_\mathrm{B}$, but could (or should) have appeared in VI$_\mathrm{A}$; it is useful to have them available already, in order to make Theorem 3.2 below more precise.

Lemma 2.5.1. *Let $(A, \mathfrak{m})$ be an Artinian local ring, and let $k = A/\mathfrak{m}$ be its residue field.*

(i) *If X is an A-scheme such that $X \otimes_A k$ is locally of finite type (respectively, of finite type) over k, then X has the same property over A.*

(ii) *Let $u : X \to Y$ be a morphism of A-schemes. If $u \otimes_A k$ is an immersion (respectively, a closed immersion), then so is u.*

Proof. (i) Suppose that $X \otimes_A k$ is locally of finite type over k. Let $U = \operatorname{Spec} B$ be an affine open subset of X. By hypothesis, there are elements $x_1, \ldots, x_n$ of B whose images generate $B/\mathfrak{m}B$ as a k-algebra, and the "nilpotent Nakayama lemma" then shows that the x_i generate B as an A-algebra. This proves that X is locally of finite type over A. If, in addition, $X \otimes_A k$ is quasi-compact, then so is X, which has the same underlying topological space; hence X is of finite type over A. This proves (i).

We prove (ii). Suppose that $u \otimes_A k$ is an immersion (respectively, a closed immersion). Then u is a homeomorphism of X onto a locally closed (respectively, closed) subset of Y, and, for every $x \in X$, the ring homomorphism

$$\varphi_x : \mathcal{O}_{Y,u(x)} \longrightarrow \mathcal{O}_{X,x}$$

[26] Editor's note: one should bear in mind that a nonneutral connected component is not geometrically connected in general. For example, if $k = \mathbb{R}$, the group $\mu_{3,\mathbb{R}}$, represented by $\mathbb{R}[X]/(X^3 - 1)$, has two connected components: $\{e\} = \operatorname{Spec} \mathbb{R}$ and $C = \operatorname{Spec} \mathbb{R}[X]/(X^2 + X + 1)$; and $C \otimes_{\mathbb{R}} \mathbb{C}$ has two components.

[27] Editor's note: the assertion that follows has been added; cf. Exposé VI$_\mathrm{B}$, 1.5.

[28] Editor's note: the original has been expanded in what follows.

[29] Editor's note: the original has been simplified here.

is such that $\varphi_x \otimes_A k$ is surjective. The nilpotent Nakayama lemma therefore shows that φ_x is surjective, so u is an immersion (respectively, a closed immersion).

Proposition 2.5.2. *Let A be an Artinian local ring with residue field k, and let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type.*

(a) *The set $u(G)$ is closed in H, and its connected components are irreducible and all have the same dimension.*

(b) *One has*
$$\dim G = \dim u(G) + \dim \mathrm{Ker}(u).$$

(c) *If u is a monomorphism, then it is a closed immersion.*

Proof. By the preceding lemma, it suffices to prove the proposition when $A = k$. Moreover, the properties under consideration are stable under fpqc descent, and dimension is invariant under extension of the base field, so we may assume that k is algebraically closed.

We prove (a). Let C be the reduced subscheme of H whose underlying topological space is $u(G)$. Since $u(G)$ is stable under the inversion morphism of H, so is C. Furthermore, $u : G \to C$ is quasi-compact and dominant. Thus, by EGA IV_2, 2.3.7, the same is true of
$$u \times_k \mathrm{id}_G \qquad \text{and} \qquad \mathrm{id}_H \times_k u,$$
and hence of their composite
$$u \times_k u : G \times_k G \longrightarrow C \times_k C.$$
Consequently, multiplication on H sends $C \times_k C$ into C, and therefore C is a group subscheme of H.

Replacing H by C, we are thus reduced to the case in which u is dominant. Then $G(k)$ is dense in H, and hence meets every connected component of H; it therefore acts transitively on the set of those connected components. It is consequently enough to show that $u(G)$ contains H^0. Replacing G by $u^{-1}(H^0)$, we may therefore suppose that $H = H^0$. In that case, 2.4 shows that H is irreducible and of finite type over k, and hence Noetherian. Moreover, u is locally of finite type (cf. EGA I, 6.6.6) and quasi-compact, and is therefore of finite type. By Chevalley's constructibility theorem (cf. EGA IV_1, 1.8.5), $u(G)$ is consequently a constructible (and dense) subset of
$$H = \overline{u(G)},$$
and thus contains a dense open subset U of H (cf. EGA 0_{III}, 9.2.2). By 0.5, we then have
$$H = U \cdot U \subset u(G),$$
whence $u(G) = H$. Together with 2.4.1, this proves (a).

We prove (b). Recall first that the functor $\mathrm{Ker}(u)$ (cf. I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the identity element of H. Since u is of finite type, $\mathrm{Ker}(u)$ is of finite type over k. On the other hand, replacing H by the reduced closed subscheme $u(G)$, we may assume that u is surjective. Let u^0 denote the restriction of u to G^0. Since G and $\mathrm{Ker}(u)$ are equidimensional, and since
$$\mathrm{Ker}(u)^0 \subset \mathrm{Ker}(u^0),$$
we are reduced to the case in which G, and hence also H, is irreducible.

By EGA IV_3, 9.2.6.2 and 10.6.1(ii), the set of $y \in H$ such that
$$\dim u^{-1}(y) = \dim G - \dim H$$

contains a nonempty open subset V. Since u is surjective, $U = u^{-1}(V)$ is then a nonempty open subset of G, and therefore contains a closed point x of G, because G is a Jacobson scheme (cf. EGA IV$_3$, 10.4.8). Right translation r_x is an isomorphism from $\mathrm{Ker}(u)$ onto $u^{-1}(u(x))$, whence

$$\dim \mathrm{Ker}(u) = \dim u^{-1}(u(x)) = \dim G - \dim H.$$

We prove (c), following [DG70], I, § 3.4. (Another proof is given in Exposé VI$_\mathrm{B}$, 1.4.2.) Suppose that u is a monomorphism. If C is a connected component of G, there is a closed point $x \in G$ such that $C = r_x(G^0)$. If u_C (respectively, u^0) denotes the restriction of u to C (respectively, to G^0), then

$$u_C = r_{u(x)} \circ u^0 \circ r_x^{-1}.$$

It therefore suffices to show that u^0 is a closed immersion. We may thus suppose that $G = G^0$, so that G is irreducible and of finite type over k.

Let ξ be the generic point of G. Then $\mathcal{O}_{G,\xi}$ is an Artinian local ring; denote its maximal ideal by $\mathfrak{m}$. Let $h = u(\xi)$, let $\mathfrak{n}$ be the maximal ideal of $\mathcal{O}_{H,h}$, and set

$$A = \mathcal{O}_{G,\xi}/\mathfrak{n}\mathcal{O}_{G,\xi}.$$

Since u is a monomorphism, so is the morphism

$$u_h : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(\kappa(h))$$

obtained by base change. The multiplication morphism

$$A \otimes_{\kappa(h)} A \longrightarrow A$$

is therefore an isomorphism (cf. EGA I, 5.3.8), whence $A = \kappa(h)$. Nakayama's lemma then shows that

$$\mathcal{O}_{H,h} \longrightarrow \mathcal{O}_{G,\xi}$$

is surjective, since $\mathfrak{n}\mathcal{O}_{G,\xi}$ is contained in $\mathfrak{m}$ and is therefore nilpotent.

Let V be an affine open subset of H containing h, let $U \neq \varnothing$ be an affine open subset of G contained in $u^{-1}(V)$, and let

$$\varphi : \mathcal{O}_H(V) \longrightarrow \mathcal{O}_G(U)$$

be the homomorphism of k-algebras induced by u. Let $\mathfrak{p}$ be the prime ideal of $\mathcal{O}_G(U)$ corresponding to ξ,and set $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$. Since G is of finite type over k, the k-algebra $\mathcal{O}_G(U)$ is generated by finitely many elements $a_1, \ldots, a_n$. By the preceding argument, there are elements $b_1, \ldots, b_n, s \in \mathcal{O}_H(V)$, with $s \notin \mathfrak{q}$, such that in $\mathcal{O}_G(U)_\mathfrak{p}$ one has

$$a_i/1 = \varphi(b_i)/\varphi(s).$$

There are therefore elements

$$t_1, \ldots, t_n \in \mathcal{O}_G(U) - \mathfrak{p}$$

such that

$$t_i(a_i\varphi(s) - \varphi(b_i)) = 0.$$

Set

$$t = t_1 \cdots t_n \varphi(s) \in \mathcal{O}_G(U) - \mathfrak{p}.$$

The equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ then already hold in $\mathcal{O}_G(U)_t$. Since

$$t \in \mathcal{O}_G(U) = k[a_1, \ldots, a_n],$$

there is an element $b \in \mathcal{O}_H(V)$ such that, for some $r \in \mathbb{N}$,

$$t/1 = \varphi(b)/\varphi(s)^r.$$

Consequently, φ induces a surjection

$$\mathcal{O}_H(V)_{sb} \longrightarrow \mathcal{O}_G(U)_t,$$

and hence u is a local immersion at ξ.

The open subset W of G consisting of the points at which u is a local immersion is therefore nonempty. Since G is a Jacobson scheme, W contains a closed point y. To prove that $W = G$, it is enough to show that every closed point of G belongs to W. But every closed point x is the image of y under the translation $r_x \circ r_y^{-1}$, and hence belongs to W. Thus $W = G$, which proves that u is a local immersion.

Since G is irreducible, it follows that u is an immersion. Indeed, for every $x \in G$, let U_x and V_x be open subsets of G and H, respectively, such that $x \in U_x$ and u induces a closed immersion of U_x into V_x. Since U_x is dense in G, the subset $u(U_x)$ is dense in $u(G) \cap V_x$. But $u(U_x)$ is closed in V_x, so

$$u(U_x) = V_x.$$

Moreover, since u is injective, $U_x = u^{-1}(V_x)$. Thus u induces a closed immersion of G into the open subscheme of H covered by the V_x. Hence $u : G \to H$ is an immersion. We have already seen that $u(G)$ is closed in H, so u is a closed immersion.

Lemma 2.5.3.[30] *Let A be an Artinian local ring, let k be its residue field, let G be a flat A-group, and let X be an A-scheme equipped with a left action*

$$\mu : G \times_A X \longrightarrow X$$

of G and with a section $s_0 : \operatorname{Spec} A \to X$. (This is the case, for example, if G' is a second A-group and one is given a morphism of A-groups $G \to G'$.)

Let φ be the morphism

$$\mu \circ (\mathrm{id}_G \times s_0) : G = G \times_A A \longrightarrow X.$$

If φ is flat at a point g of G, then φ is flat.

Proof. Since G is flat over A, the fiberwise criterion for flatness (EGA IV$_3$, 11.3.10.2) shows that it is enough to prove that $\varphi \otimes_A k$ is flat. We may therefore suppose that $A = k$. In that case, giving s_0 is equivalent to giving a k-point $x_0 \in X(k)$, and φ is the morphism $h \mapsto hx_0$.

Let $h \in G$; we show that φ is flat at h. Let K be an extension of k containing copies of both $\kappa(g)$ and $\kappa(h)$. We have a cartesian square

$$\begin{array}{ccc} G_K & \xrightarrow{\varphi_K} & X_K \\ \downarrow & & \downarrow \\ G & \xrightarrow[\varphi]{} & X \end{array}$$

in which the two vertical arrows are faithfully flat. Thus, by Exposé V, 7.4(i), φ_K is flat at every point $g' \in G_K$ lying over g; and, in order to show that φ is flat at h, it is enough to show that φ_K is flat at one point h' lying over h. We are therefore reduced to the case in which g and h are rational.

Set $u = hg^{-1}$, and let ℓ_u (respectively, μ_u) be left translation on G (respectively, on X) defined by u. Since $\varphi \circ \ell_u = \mu_u \circ \varphi$, we obtain a commutative square

[30] Editor's note: no finiteness hypotheses are made in this statement; this will be useful later (cf. 6.2 and Exposé VI$_B$, § 12).

$$\begin{array}{ccc} \mathcal{O}_{G,h} & \xrightarrow{\sim} & \mathcal{O}_{G,g} \\ \uparrow & & \uparrow \\ \mathcal{O}_{X,hx_0} & \xrightarrow[\sim]{} & \mathcal{O}_{X,gx_0} \end{array}$$

whose horizontal arrows are isomorphisms. Since the morphism

$$\mathcal{O}_{X,gx_0} \longrightarrow \mathcal{O}_{G,g}$$

is flat, the morphism

$$\mathcal{O}_{X,hx_0} \longrightarrow \mathcal{O}_{G,h}$$

is flat as well.

Proposition 2.5.4. *Let A be an Artinian local ring, let k be its residue field, let G be an A-group locally of finite type, and let $X \neq \varnothing$ be an A-scheme locally of finite type equipped with a left action of G. Suppose that the morphism*

$$\varphi : G \times_A X \longrightarrow X \times_A X, \qquad (g, x) \longmapsto (gx, x),$$

defined on underlying sets by the displayed rule, is surjective. Then:

(i) *The connected components of X are of finite type, irreducible, and all of the same dimension.*

(ii) *More precisely, let $\overline{k}$ be an algebraic closure of k, and let x be a closed point of $X \otimes_A \overline{k}$. Its stabilizer F is a closed group subscheme of $G \otimes_A \overline{k}$, and the dimension of the irreducible components of X is*

$$\dim G - \dim F.$$

In view of Lemma 2.5.1, we may suppose that $A = k$. Suppose first that k is algebraically closed. Then G_{red} is a k-group locally of finite type; hence, replacing G by G_{red} and X by X_{red}, we may suppose that G and X are reduced.

Since $G \times_k X$ is locally of finite type over k, the morphism φ is locally of finite type (cf. EGA I, 6.6.6(v)), and is therefore locally of finite presentation because $X \times_k X$ is locally Noetherian. Let x be a rational point of X. The morphism

$$\varphi_x : G \longrightarrow X$$

obtained from φ by base change is surjective and locally of finite presentation. If η is a maximal point of X, then $\mathcal{O}_{X,\eta}$ is a field, since X is reduced; consequently, φ_x is flat at every point of G lying over η. By Lemma 2.5.3, φ_x is therefore flat. Thus $\varphi_x : G \to X$ is faithfully flat and locally of finite presentation, and hence open (cf. EGA IV$_2$, 2.4.6).

Since G^0 is open in G, irreducible, and quasi-compact (by 2.4), every orbit

$$G^0 x = \varphi_x(G^0),$$

as x ranges over the rational points of X, is an open, irreducible, and quasi-compact subset of X, and is therefore of finite type over k, since X is locally of finite type over k.

Every nonempty open subset of X contains a rational point, so these open subsets cover X. Furthermore, any two of them are either disjoint or equal. Indeed, if

$$\varphi_x(G^0) \cap \varphi_y(G^0)$$

is nonempty, it contains a rational point z. There are then rational points $g, h \in G^0$ such that

$$gx = z = hy,$$

whence $x = g^{-1}z$ and $y = h^{-1}z$, and consequently

$$\varphi_x(G^0) = \varphi_z(G^0) = \varphi_y(G^0).$$

It follows that the orbits $\varphi_x(G^0)$ are also closed, and are therefore at once the connected components and the irreducible components of X.

Finally, let x, y be two rational points of X. Since φ_x is surjective, there is a rational point $g \in G$ such that $y = gx$. Since G^0 is a normal subgroup of G, the orbit G^0y is the image of G^0xunder translation ℓ_g on X; hence G^0y and G^0x have the same dimension.

Moreover, by Exposé I, 2.3.3.1, the stabilizer of x is represented by the closed subscheme F of G defined by the following cartesian square:

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow{\scriptstyle \varphi_x} \\ \operatorname{Spec} k & \longrightarrow & X \end{array}$$

Thus F is a k-group locally of finite type, while $F \cap G^0$ is a k-group of finite type containing F^0. By 2.4.1, the schemes F and $F \cap G^0$ are equidimensional, of the same dimension as F^0. Let

$$C = \varphi_x(G^0)$$

be the irreducible component of X containing x. Arguing as in the proof of part (b) of 2.5.2, we obtain

$$\dim C = \dim G^0 - \dim F^0 = \dim G - \dim F.$$

In the general case, that is, when k is an arbitrary field, let $\overline{k}$ be an algebraic closure of k. Let C be a connected component of X, and let C' be a connected component of $C \otimes_k \overline{k}$. Then C' is a connected component of

$$X' = X \otimes_k \overline{k}.$$

The morphism $\pi : X' \to X$ is open (cf. EGA IV_2, 2.4.10), and, since it is integral, it is also closed; therefore $\pi(C') = C$. Since C' is irreducible and quasi-compact, C is irreducible and quasi-compact, and hence is of finite type over k, because X is locally of finite type over k.

Finally, dimension is invariant under extension of the base field (cf. EGA IV_2, 4.1.4), so $\dim C = \dim C'$. Since all the irreducible components of X' have the same dimension, the same is true of those of X.

2.6. Complements.— This paragraph has been added; it is taken from [Per75], II, §§ 1–2, with complements due to O. Gabber.[31] It shows that the preceding results are valid for every group scheme G over a field k. (This will be used in Sections 5, 6, and 7 of Exposé VI_B.)

Fix a field k. We begin with the following lemma (loc. cit., II, 2.1.1), which does not appear explicitly in EGA IV_2, § 4.4 (although it can probably be read between the lines at the beginning of loc. cit., § 4.4.1).

Lemma 2.6.0. *Let X be an irreducible k-scheme, let K be an extension of k, and let X' be an irreducible component of X_K. The projection $X' \to X$ is surjective.*

[31] Editor's note: these results were communicated to us by O. Gabber, in particular 2.6.6, which plays an important role in Section 5 of Exposé VI_B.

Proof. Let B be a transcendence basis of K over k, and set $L = k(B) \subset K$. By EGA IV_2, 4.3.2 and 4.4.1, X_L is irreducible and X' dominates X_L. Hence $X' \to X_L$ is integral and dominant, and therefore surjective. Since $X_L \to X$ is surjective (loc. cit., 4.4.1), so is $X' \to X$.

For the rest of 2.6, fix a k-group scheme G and an action $\mu : G \times X \to X$ of G on a k-scheme X, satisfying the following condition:

$$\Phi : G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x) \quad \text{is surjective.} \tag{$\star$}$$

(This holds in particular when G acts on itself by left translations.) For short, we shall say: "Let X be a G-scheme satisfying $(\star)$." Finally, write k' for the perfect closure of k.

Proposition 2.6.1. *Let X be a G-scheme satisfying $(\star)$.*

(i) *X is geometrically pointwise irreducible over k: for every extension K of k, each point $x \in X_K$ belongs to a unique irreducible component of X_K.*

(ii) *Every local ring of $(X_{k'})_{\mathrm{red}}$ is normal.*

(iii) *Let η be a maximal point of $(X_{k'})_{\mathrm{red}}$, put $C = \overline{\{\eta\}}$, and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L-scheme and is geometrically irreducible over L; that is, $C \otimes_L K$ is irreducible for every extension K of L.*

(iv) *In particular, if x is a rational point of X, then the irreducible component of X containing x is geometrically irreducible over k.*

Proof. (i) Since hypothesis $(\star)$ is preserved by every base change $k \to K$, it is enough to show that every $x \in X$ belongs to a unique irreducible component of X. Since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is a universal homeomorphism, we may further suppose that k is perfect. We may then suppose that G and X are reduced.

Let η be a maximal point of X, and let z be any point of $Z = \overline{\{\eta\}}$. Since X is reduced, $\mathcal{O}_{X,\eta} = \kappa(\eta)$; and, since k is perfect, $\kappa(\eta) \otimes_k K$ is normal for every extension K of k (cf. EGA IV_2, 6.14.2). Thus every point of X_K lying above η is normal.

Since Φ is surjective, there is a point γ of $G \times X$ such that the two projections of $\Phi(\gamma)$ are z and η. Set $K = \kappa(\gamma)$. There are then rational points $g \in G_K$ and $\eta' \in X_K$, with η' lying above η, such that $z' = g\eta'$ lies above z. By the preceding paragraph, η' is a normal point of X_K, and therefore so is z'. Since $\pi : X_K \to X$ is flat, it follows that $z = \pi(z')$ is a normal point of X (cf. EGA IV_2, 2.1.13). This proves (ii) and (i).

The first assertion of (iii) now follows from (ii). Since L is algebraically closed in $\kappa(\eta)$, C is geometrically irreducible over L, by EGA IV_2, 4.5.9.

Finally, if X has a rational point x, it follows from (iii) that $L = k$, and hence the irreducible component C containing x is geometrically irreducible over k.This can also be seen directly as follows (cf. [Per75], II, 2.1). Let C be the irreducible component of X containing x, and let K be an extension of k. The scheme X_K has a unique point x_K above x, and by (i), x_K belongs to a unique irreducible component C' of X_K. On the other hand, by 2.6.0, every irreducible component of C_K contains x_K, and therefore equals C'.

Notation.— Write provisionally C^0 for the reduced closed subscheme of G whose underlying space is the unique irreducible component of G containing the identity element e.

Corollary 2.6.2. *C^0 is geometrically irreducible over k and is set-theoretically stable under the group law; that is, $(C^0_{k'})_{\mathrm{red}}$ is a subgroup of $(G_{k'})_{\mathrm{red}}$. Consequently, C^0 is quasi-compact.*
Proof. By 2.6.1, C^0 is geometrically irreducible over k, so $C^0 \times C^0$ is irreducible and

$$\nu(C^0 \times C^0) \subset C^0,$$

where ν denotes the morphism $(g,h) \mapsto gh^{-1}$. Since $\operatorname{Spec} k' \to \operatorname{Spec} k$ is a universal homeomorphism, the same conclusion holds for $C^0_{k'}$, and then for

$$H = (C^0_{k'})_{\mathrm{red}}.$$

Since $H \times_{k'} H$ is reduced, ν induces a morphism

$$H \times_{k'} H \longrightarrow H;$$

thus H is a subgroup of $(G_{k'})_{\mathrm{red}}$. Consequently, by 0.5.1, H, and hence C^0, is quasi-compact.

Recall 2.6.3.— Let Y be a scheme. Recall (EGA 0_{III}, 9.1.1) that a subset E of Y is called *retrocompact* if the inclusion $E \hookrightarrow Y$ is quasi-compact. Moreover, by EGA IV_1, 1.9.5(v) and 1.10.1, if U is a retrocompact open subset of Y, its closure $\overline{U}$ is the union of the closures $\overline{\{y\}}$ of the points $y \in U$. Of course, it is enough to let y run through the maximal points of U, that is, the maximal points of Y contained in U.

Consequently, if Y is quasi-separated and η is a maximal point of Y, then the intersection of the closed neighborhoods of η equals $\overline{\{\eta\}}$. Indeed, if $y \in Y - \overline{\{\eta\}}$, then y belongs to an affine open V not containing η. Since Y is quasi-separated, V is retrocompact, and therefore $\overline{V}$ is the union of the $\overline{\{\xi\}}$, for ξ running through the maximal points of Y belonging to V. Hence $\eta \notin \overline{V}$, so $Y - \overline{V}$ is a closed neighborhood of η not containing y.

Proposition 2.6.4. *Let X be a G-scheme satisfying $(\star)$, and let U be an open subset of X.*

(i) *C^0U is an open subset of X, equal to the union of the irreducible components of X whose generic point belongs to U.*

(i′) *Consequently, if X is irreducible, it is quasi-compact.*

(ii) *If, moreover, U is retrocompact in X, then $C^0U = \overline{U}$, and this subset is both open and closed in X.*

Proof. (i) First, C^0U is open, since it is the union, for $g \in C^0$, of the projections of the open subsets

$$g \cdot U_{\kappa(g)} \subset X_{\kappa(g)},$$

and every projection $X_{\kappa(g)} \to X$ is open.

To prove the second assertion of (i), we may replace k by k' (since $\operatorname{Spec} k' \to \operatorname{Spec} k$ is a universal homeomorphism), and therefore suppose that k is perfect. We may then suppose that G and X are reduced, and hence geometrically reduced.

Let η be a maximal point of X contained in U, and put $Z = \overline{\{\eta\}}$. Consider

$$\mu' : C^0 \times Z \longrightarrow X.$$

Since C^0 is geometrically irreducible, $C^0 \times Z$ is irreducible; write γ for its generic point. The morphism μ' sends the point $\varepsilon(\eta)$, where ε denotes the identity section of C^0, to η. Hence γ is sent to a generization of η, and therefore to η. Thus μ' sends the underlying space of $C^0 \times Z$ into Z, and, since $C^0 \times Z$ is reduced, μ' factors through Z.

Now let $z \in Z$, and put $K = \kappa(z)$. The morphism

$$\mu_z : G_K \longrightarrow X_K, \qquad h \longmapsto h \cdot z,$$

is surjective. Let α be a maximal point of X_K. The local ring $\mathcal{O}_{X_K,\alpha}$ is a field, since X_K is reduced, so μ_z is flat at every point of G_K above α. It follows from Lemma 2.5.3 that μ_z is flat.

On the other hand, μ_z sends the generic point ω of C^0_K to a point $t \in Z_K$. Let β be a maximal point of Z_K such that $t \in \overline{\{\beta\}}$. Since μ_z is flat, there is a generization ξ of ω such that $\mu_z(\xi) = \beta$. But ω is a maximal point of G_K, so necessarily $\xi = \omega$. Thus $\mu_z(\omega) = \beta$, and β lies above η, because $Z_K \to Z$ is flat.

Put $L = \kappa(\omega)$, and let ω_L and z_L be the L-points obtained from ω and z. Then

$$\omega_L \cdot z_L = \beta'$$

is a point of Z_L lying above β, and hence

$$z_L = \omega_L^{-1} \cdot \beta' \in C^0_L \cdot U_L.$$

Therefore $z \in C^0 \cdot U$, which proves (i).

Since C^0 is quasi-compact by 2.6.2, assertion (i′) follows: if X is irreducible and U is a nonempty affine open subset, then $X = C^0U$, so X is the image of the morphism $C^0 \times U \to X$, and is therefore quasi-compact. Finally, if U is retrocompact in X, then by 2.6.3, $\overline{U}$ is the union of the $\overline{\{\eta\}}$, where η runs through the maximal points of X contained in U. Hence $\overline{U} = C^0U$. This proves (ii).

One then obtains the following result ([Per75], II, Th. 2.4; see also [Per76], Prop. 4.1.1).

Theorem 2.6.5. *Let k be a field, and let G be a k-group scheme.*

(i) *There is a unique group subscheme G^0 of G, called the* identity component *of G, such that:*

  (a) *the underlying space of G^0 is the irreducible component of the identity element;*
  (b) *$G^0 \to G$ is a flat closed immersion; that is,*

$$\mathcal{O}_{G,g} = \mathcal{O}_{G^0,g} \qquad \text{for every } g \in G^0.$$

(ii) *Moreover, G^0 is quasi-compact, geometrically irreducible, and is a characteristic subgroup of G.*

(iii) *If G is connected, then $G = G^0$.*

Proof. (i) Recall first that G is separated (0.3), and hence, a fortiori, quasi-separated. Let U be an affine open subset of G containing the generic point ω of C^0. By 2.6.3 and 2.6.4,

$$\overline{U} = C^0U$$

is both open and closed, and, set-theoretically,

$$C^0 = \overline{\{\omega\}}$$

is the intersection of these open-and-closed subsets.

As U runs through the affine open subsets containing ω, the G-schemes $\overline{U}$ form a projective system whose transition morphisms are affine, since they are closed immersions. We can therefore form its projective limit G^0 (cf. EGA IV$_3$, 8.2.2). Thus, for every affine open V of G, $G^0 \cap V$ is the spectrum of the algebra

$$\varinjlim_{U} \mathcal{O}_G(V \cap \overline{U}) = \mathcal{O}_G(V) \Big/ \sum_U I_{\overline{U}}(V),$$

where $I_{\overline{U}}(V)$ denotes the kernel of

$$\mathcal{O}_G(V) \longrightarrow \mathcal{O}_G(V \cap \overline{U}).$$

It follows that the underlying space of G^0 is C^0, and that $G^0 \to G$ is a closed immersion.

Moreover, for every $g \in G^0$, $\mathcal{O}_{G^0,g}$ is the inductive limit, as V runs through the affine open subsets of G containing g, of the k-algebras

$$\mathcal{O}_{G^0}(V \cap G^0) = \varinjlim_{U} \mathcal{O}_G(V \cap \overline{U}).$$

This double inductive limit is identified with

$$\varinjlim_{U} \varinjlim_{\substack{V \\ g \in V \subset \overline{U}}} \mathcal{O}_G(V) = \mathcal{O}_{G,g}.$$

Thus

$$\mathcal{O}_{G^0,g} = \mathcal{O}_{G,g}$$

(see also EGA IV_2, 5.13.3(ii)), and consequently $i : G^0 \to G$ is a flat closed immersion. Conversely, this condition implies that $i^*(\mathcal{O}_G) = \mathcal{O}_{G^0}$, so G^0 is uniquely determined by conditions (a) and (b). This proves (i).

The first two assertions of (ii) follow from 2.6.2. Finally, let S be a k-scheme and let ϕ be an automorphism of the S-group G_S. For every $s \in S$, ϕ_s sends $G^0_{\kappa(s)}$ into itself, and therefore $\phi(G^0_S) \subset G^0_S$. Moreover, the closed immersion

$$i_S : G^0_S \hookrightarrow G_S$$

obtained from i by base change is flat, so

$$\mathcal{O}_{G_S,\phi(z)} = \mathcal{O}_{G^0_S,\phi(z)} \qquad \text{for every } z \in G^0_S.$$

Hence $\phi \circ i_S$ factors through G^0_S. This proves that G^0 is a characteristic subgroup of G, and proves (ii). Finally, (iii) is a special case of part (i) of the following proposition.[32]

Proposition 2.6.6. *Let k be a field, and let G be a k-group acting on a k-scheme X in such a way that*

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x)$$

is surjective. Suppose that X is quasi-separated. Then:

(i) *Every connected component C of X is irreducible.*

(ii) *Let η be the generic point of C, and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L-scheme, is geometrically irreducible over L, and the morphism*

$$G^0_L \times_L C \longrightarrow C \times_L C$$

is surjective.

(iii) *In particular, if C contains a rational point x, then C is geometrically irreducible over k, and*

$$\phi : G^0 \longrightarrow C, \qquad g \longmapsto gx,$$

is surjective.

[32] Editor's note: this proposition, as well as the preceding results, was communicated to us by O. Gabber. It will be used to correct the proof of Theorem 5.3 in Exposé VI_B.

Proof. (i) Let C be a connected component of X, let Y be an irreducible component of X contained in C, let η be the generic point of Y, and let U be an affine open subset of X containing η. Since X is quasi-separated, U is retrocompact in X. Hence, by 2.6.4,

$$\overline{U} = G^0 U$$

is an open-and-closed subset of X which meets C, and therefore contains C. By 2.6.3, the intersection of the $\overline{U}$, as U runs through a fundamental system of affine open neighborhoods of η, is $\overline{\{\eta\}}$. It follows that

$$C = \overline{\{\eta\}},$$

which proves (i).

The first assertion of (ii), and also that of (iii), follows from 2.6.1. We first prove the second assertion of (iii). Let ω, respectively η, denote the generic point of G^0, respectively C. Let $z \in C$, and set $K = \kappa(z)$. Since G^0, respectively C, is geometrically irreducible over k, the scheme G^0_K, respectively C_K, is irreducible; denote its generic point by ξ, respectively β. We saw in the proof of 2.6.4 that

$$\mu_z : G_K \longrightarrow C_K, \qquad h \longmapsto h \cdot z,$$

sends ξ to β. Likewise, $\phi(\omega) = \eta$.

Put $L = \kappa(\xi)$, and let ξ_L and z_L be the L-points obtained from ξ and z. Then

$$\xi_L \cdot z_L = \beta'$$

is a point of C_L lying above $\beta \in C_K$, and hence also above $\eta \in C$. Consider the cartesian square

$$\begin{array}{ccc} G^0_L & \xrightarrow{\varphi_L} & C_L \\ {\scriptstyle \pi_{G^0}}\downarrow & & \downarrow{\scriptstyle \pi_C} \\ G^0 & \xrightarrow[\varphi]{} & C \end{array}$$

Since

$$\phi_L(\pi_{G^0}^{-1}(\omega)) = \pi_C^{-1}(\eta)$$

(cf. EGA I, 3.4.8), there is a $g \in G^0_L$ such that $\phi_L(g) = \beta'$. Therefore

$$z_L = \xi_L^{-1} \cdot \phi_L(g) = \phi_L(\xi_L^{-1} g),$$

whence

$$\phi(\pi_{G^0}(\xi_L^{-1} g)) = \pi_C(z_L) = z.$$

This proves that ϕ is surjective.

We now prove the second assertion of (ii). It is enough to show that, for every $z \in C$, the morphism

$$\mu_z : G^0_L \otimes_L \kappa(z) \longrightarrow C \otimes_L \kappa(z)$$

is surjective. But this follows from (iii), since z is a rational point of $C \otimes_L \kappa(z)$.

Remark 2.6.7.— *Under the hypotheses of 2.6.6, if $C(k) = \varnothing$, the morphism*

$$G^0 \times_k C \longrightarrow C \times_k C$$

need not be surjective. For example, let $k = \mathbb{R}$ *and* $G = \{\pm 1\}_{\mathbb{R}}$. *The* G*-torsor*

$$X = \operatorname{Spec} \mathbb{R}[X]/(X^2 + 1)$$

is connected, but

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{R}} X$$

is not surjective. (Here $L = \mathbb{C}$, *and*

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{C}} X$$

is an isomorphism.)

3. Construction of quotients $F\backslash G$ (for G, F of finite type)

3.1.— Let A be an Artinian local ring, and let $u : F \to G$ be a homomorphism of A-groups. If

$$\mu : F \times_A F \longrightarrow F \qquad \text{and} \qquad \nu : G \times_A G \longrightarrow G$$

denote the multiplication morphisms, and if λ denotes the composite morphism

$$F \times_A G \xrightarrow{u \times G} G \times_A G \xrightarrow{\nu} G \ ,$$

recall that the *left quotient* $F\backslash G$ of G by F is the cokernel of the (Sch/A)-groupoid G_* described below:

$$F \times_A F \times_A G \underset{\mathrm{pr}_{2,3}}{\overset{F\times\lambda}{\underset{}{\overset{}{\rightrightarrows}}}} F \times_A G \underset{\mathrm{pr}_2}{\overset{\lambda}{\rightrightarrows}} G$$

Here pr_2 and $\mathrm{pr}_{2,3}$ are the projections of $F \times_A G$ and $F \times_A (F \times_A G)$, respectively, onto the second factors. We shall say that G_* is the *groupoid with base* G *defined by* u (cf. Exposé V, § 2.a; as in Exposé V, we do not follow in this exposé the convention of IV, 4.6.15).

Since the unique A-morphism $F \to \operatorname{Spec} A$ is universally open (EGA IV$_2$, 2.4.9), pr_2 is an open morphism. The same therefore holds for λ, which is the composite of pr_2 with the automorphism σ of $F \times_A G$ defined by

$$\sigma(S)(x, y) = \big(x, u(S)(x) \cdot y\big),$$

where S is a variable A-scheme and $x \in F(S)$, $y \in G(S)$. One sees in the same way that pr_2 and λ are flat when F is flat over A.

Let us also note, to conclude these preliminaries, that every A-morphism $s : \operatorname{Spec} A \to G$ defines an automorphism of the groupoid G_* which induces on G, $F \times_A G$, and $F \times_A F \times_A G$, respectively, the automorphisms

$$r_s, \qquad \mathrm{id}_F \times_A r_s, \qquad \mathrm{id}_F \times_A \mathrm{id}_F \times_A r_s.$$

We shall again write r_s for this automorphism of G_*, and shall say that r_s is the *right translation defined by* s (cf. 0.4).

Theorem 3.2. *Let* F *and* G *be flat groups locally of finite type over an Artinian local ring* A. *Let* $u : F \to G$ *be a quasi-compact homomorphism of* A*-groups whose kernel is finite over* A. *Then:*[33]

[33]Editor's note: we have added (ii′) and expanded point (iv), taking account of the additions made in 2.5.2, 2.5.4, and Exposé V, 6.1.

(i) *The left quotient $F\backslash G$ of G by F exists in* (Sch/A)*, and the sequence*

$$F \times_A G \underset{\mathrm{pr}_2}{\overset{\lambda}{\rightrightarrows}} G \xrightarrow{\ p\ } F\backslash G$$

is exact in the category of all ringed spaces.

(ii) *The canonical morphism $p : G \to F\backslash G$ is surjective and open, and $F\backslash G$ is a direct sum of schemes of finite type over A.*

(ii′) *More precisely, $X = F\backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A, irreducible, and all of dimension* $\dim G - \dim F$.

(iii) *The canonical morphism*

$$F \times_A G \longrightarrow G \times_{(F\backslash G)} G$$

is surjective.

(iv) *If u is a monomorphism,*[34] *then:*

(a) *The morphism*

$$F \times_A G \xrightarrow{(\lambda, \mathrm{pr}_2)} G \times_{(F\backslash G)} G$$

is an isomorphism, and $G \to F\backslash G$ is faithfully flat and locally of finite presentation.

(a′) *$F\backslash G$ represents the* (fppf) *quotient sheaf $\widetilde{F\backslash G}$, and $G \to F\backslash G$ is an F-torsor locally trivial for the* (fppf) *topology.*

(b) *$F\backslash G$ is flat over A, and is smooth over A if G is so.*

(c) *$u : F \to G$ is a closed immersion, and $F\backslash G$ is separated.*

(d) *If, moreover, F is a normal subgroup of G, there is a unique A-group structure on $F\backslash G$ for which $p : G \to F\backslash G$ is a morphism of A-groups.*

In the proof of this theorem, A' will denote an A-algebra which is local, finite, and free over A. If R is a relation involving A', we shall say that "$R(A')$ is true when A' is sufficiently large" if there is an A-algebra A_1, local, finite, and free over A, such that the relation $R(A')$ holds for every A-algebra A' which is local, finite, and free over A_1.

We shall first prove the theorem when F and G are of finite type over A.

3.2.1.— Suppose for a moment that every point of G has an open saturated neighborhood W such that the groupoid induced by G_* on W has a quasi-section (cf. Exposé V, § 6). Then, by Exposé V, 6.1, assertions (i), (ii), (iii), and (iv)(a) hold, and $F\backslash G$ is of finite type over A.Moreover, under the hypothesis of (iv), since

$$G \longrightarrow F\backslash G$$

is faithfully flat and locally of finite presentation, assertion (b) follows from EGA IV, 2.2.14 and 17.7.7. Assertion (iv)(a′) follows from (iv)(a), by Exposé IV, 3.4.3.1, 5.2.2, and 5.1.6. Finally, assertion (iv)(c) will be proved in 3.2.5, while (ii′) and (iv)(d) will be proved in Section 5.

[34]Editor's note: in this case, the hypothesis that G be flat can be omitted; cf. subsection 3.3.

We now prove the following assertion:

every finite subset of $F\backslash G$ is contained in an affine open subset. (†)

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If U is a quasi-section of the groupoid induced by G_* on a saturated open subset W of G, then $U \otimes_A A'$ is a quasi-section of the (Sch/A')-groupoid induced by $G_* \otimes_A A'$ on $W \otimes_A A'$. Moreover, if U_* is the (Sch/A)-groupoid induced by G_* on U, then $U_* \otimes_A A'$ is identified with the (Sch/A')-groupoid induced by $G_* \otimes_A A'$ on $U \otimes_A A'$.[36]

It follows from the proofs in Exposé V that the construction of the quotient

$$X = F\backslash G$$

commutes with a base extension $A \to A'$ of the kind considered here.[37]

Let $x_1, \ldots, x_n$ be points of $X = F\backslash G$, which we may suppose to be closed,[38] and let $g_1, \ldots, g_n$ be closed points of G mapping to $x_1, \ldots, x_n$. Let V be an everywhere dense affine open subset of X,[39] and let U be its inverse image in G. By 1.2, there is an A-algebra A', local, finite, and free over A, such that the points $g'_1, \ldots, g'_p$ of

$$G' = G \otimes_A A'$$

lying above $g_1, \ldots, g_n$ are strictly rational over A'.[40] Since the morphisms $G' \to G$ and $G \to X$ are open,

$$U' = U \otimes_A A'$$

is dense in G'. Hence the open subset

$$\bigcap_{i=1}^{p} (U')^{-1} \cdot g'_i$$

is nonempty and therefore contains a closed point x. By 1.2 (and 0.4.1), after enlarging A' if necessary, we may suppose that x is strictly rational over A'. Since $x \in (U')^{-1} \cdot g'_i$, we have

$$g'_i \in U' \cdot x \qquad (1 \leq i \leq p).$$

[35] Editor's note: let us point out that if $A = k$ is a field, every quasi-compact open subset of $F\backslash G$ is quasi-projective (a result due to Chow for smooth algebraic groups); cf. [Ray70], VI, 2.6. By contrast, over the Artinian local ring $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, there exist abelian A-schemes G which are not projective (loc. cit., XII, 4.2).

[36] Editor's note: the preceding assertion is valid for every base change $A \to A'$.

[37] Editor's note: this is explained in detail in 4.6 below. One must show that formation of direct images by the morphisms p, λ, and pr_2 commutes with flat base changes $A \to A'$. Since F and G are of finite type over the Artinian ring A, the morphisms f in question are all quasi-compact and quasi-separated, and

$$f_*(\mathcal{O}_X) \otimes_A A' = f'_*(\mathcal{O}_{X'})$$

(with the evident notation) follows from EGA IV_1, 1.7.21.

[38] Editor's note: indeed, let $y_1, \ldots, y_n$ be arbitrary points of X. Since X is of finite type over A, the closure of each y_i contains a closed point x_i, and every open subset containing x_i contains y_i.

[39] Editor's note: such an open subset exists because X is of finite type over A: X has only finitely many irreducible components $C_1, \ldots, C_p$, and it is enough, for each i, to choose a nonempty affine open subset contained in

$$C_i - \bigcup_{j \neq i} C_j.$$

(Here, moreover, one knows from (ii′) that the C_i are disjoint.)

[40] Editor's note: the original has been expanded in what follows.

Write V' for the inverse image of V in

$$X' = X \otimes_A A'.$$

It is an affine open subset of X', and is also the image of U' under the projection $G' \to X'$. Since right translation r_x is an automorphism of the groupoid $G_* \otimes_A A'$, it induces an automorphism, again denoted r_x, of the quotient X'. Consequently,

$$V' \cdot x = r_x(V'),$$

the image of $U' \cdot x$ in X', is an affine open subset of X' containing the images $x'_1, \ldots, x'_p$ of $g'_1, \ldots, g'_p$.

Consider the equivalence relation on $X' = X \otimes_A A'$ defined by the projection $X \otimes_A A' \to X$:

$$X \otimes_A A' \otimes_A A' \underset{d_0}{\overset{d_1}{\rightrightarrows}} X \otimes_A A' \longrightarrow X \ ,$$

Here d_0 and d_1 are induced by the two canonical injections of A' into $A' \otimes_A A'$. Since A' is a finite free A-algebra, say of rank n, d_0 and d_1 are finite and locally free of rank n. We may therefore apply the argument of Exposé V, 5.b, taken from the proof of SGA 1, VIII.7.6. It follows that $x'_1, \ldots, x'_p$ are contained in a saturated affine open subset W' contained in the affine open subset $V' \cdot x$. The image of W' in X then contains $x_1, \ldots, x_n$, and is an affine open subset of X, by Exposé V, 4.1(ii).

3.2.2.— For every A-algebra A', local, finite, and free over A, let $U(A')$ denote the set of points of $G \otimes_A A'$ having an open saturated neighborhood W such that the groupoid induced by $G_* \otimes_A A'$ on W has a quasi-section. It is clear that $U(A')$ is saturated under the operations of $G(\operatorname{Spec} A')$ on $G \otimes_A A'$. We shall show that, when A' is sufficiently large,

$$U(A') = G \otimes_A A'.$$

By Theorem V.8.1, $U(A)$ is nonempty, and therefore contains a closed point y. We argue by induction on

$$\dim(G - U(A)).$$

Let $g_1, \ldots, g_n$ be closed points belonging to the various irreducible components of $G - U(A)$. By 1.2, there is an A-algebra A', local, finite, and free over A, such that the points

$$g'_1, \ldots, g'_p \qquad (\text{respectively, } x = x_1, \ldots, x_r)$$

of $G' = G \otimes_A A'$, lying above $g_1, \ldots, g_n$ (respectively, above y), are strictly rational over A'. Then $U(A')$ contains

$$\big(U(A) \otimes_A A'\big) \cdot x^{-1} g'_i \qquad \text{for every } i.$$

Consequently, $U(A')$ contains $g'_1, \ldots, g'_p$, and

$$\dim(G' - U(A')) < \dim(G - U(A)).$$

The induction hypothesis therefore gives an algebra A'', local, finite, and free over A', such that

$$U(A'') = G' \otimes_{A'} A'' = G \otimes_A A''.$$

3.2.3.— We are now in a position to prove the existence of $F\backslash G$ when F and G are of finite type over A. Let A' be sufficiently large over A that $U(A')$ coincides with $G \otimes_A A'$ (cf. 3.2.2). Set

$$A'' = A' \otimes_A A',$$

and, for every A-scheme X, write X' and X'' for the fiber products $X \otimes_A A'$ and $X \otimes_A A''$. By 3.2.1 and 3.2.2, the quotients $F'\backslash G'$ and $F''\backslash G''$ exist, and we have the following commutative diagram, in which the first two rows and columns are exact:

$$(*) \qquad \begin{array}{ccccc}
F'' \times_{A''} G'' & \underset{\lambda''}{\overset{\mathrm{pr}_2''}{\rightrightarrows}} & G'' & \xrightarrow{p''} & F''\backslash G'' \\
{\scriptstyle w_1}\downarrow\downarrow{\scriptstyle w_2} & & {\scriptstyle v_1}\downarrow\downarrow{\scriptstyle v_2} & & {\scriptstyle u_1}\downarrow\downarrow{\scriptstyle u_2} \\
F' \times_{A'} G' & \underset{\lambda'}{\overset{\mathrm{pr}_2'}{\rightrightarrows}} & G' & \xrightarrow{p'} & F'\backslash G' \\
{\scriptstyle h}\downarrow & & {\scriptstyle g}\downarrow & & \\
F \times_A G & \underset{\lambda}{\overset{\mathrm{pr}_2}{\rightrightarrows}} & G. & &
\end{array}$$

In this diagram, pr_2' and λ' (respectively, pr_2'' and λ'') are obtained from pr_2 and λ by the evident base changes; the morphisms g and h are induced by the canonical injection $A \to A'$. We write p' and p'' for the canonical morphisms, while v_1, v_2 and w_1, w_2 are induced by the two canonical injections of A' into A''.

Finally, formation of the quotient $F'\backslash G'$ commutes with the two base changes

$$f_1, f_2 : \operatorname{Spec} A'' \rightrightarrows \operatorname{Spec} A'.$$

Thus, if

$$\pi' : F'\backslash G' \longrightarrow \operatorname{Spec} A'$$

denotes the structure morphism, there are canonical isomorphisms, for $i = 1, 2$,

$$\tau_i : F''\backslash G'' \xrightarrow{\sim} (F'\backslash G') \times_{\pi', f_i} \operatorname{Spec} A'',$$

and u_i is the composite of τ_i with the projection

$$(F'\backslash G') \times_{\pi', f_i} \operatorname{Spec} A'' \longrightarrow F'\backslash G'.$$

Now, for a diagram of type $(*)$ in which the first two rows and columns are exact, one checks readily that $\mathrm{Coker}(\mathrm{pr}_2, \lambda)$ exists if and only if $\mathrm{Coker}(u_1, u_2)$ exists, and that these two cokernels are then identified. The existence of $F\backslash G$ therefore follows from that of $\mathrm{Coker}(u_1, u_2)$.

The compatibility of the formation of $F\backslash G$ with the base extensions considered here (cf. editor's note 37 in 3.2.1 and 4.6 below) shows that the composite

$$(F'\backslash G') \times_{\pi', f_1} \operatorname{Spec} A'' \xrightarrow{\tau_1^{-1}} F''\backslash G'' \xrightarrow{\tau_2} (F'\backslash G') \times_{\pi', f_2} \operatorname{Spec} A''$$

is a descent datum on $F'\backslash G'$ relative to

$$f : \operatorname{Spec} A' \longrightarrow \operatorname{Spec} A.$$

By 3.2.1 (†) and SGA 1, VIII.7.6, this descent datum is effective; that is, $\mathrm{Coker}(u_1, u_2)$ exists. (One could instead apply Exposé V, Theorem 4.1 directly.)

3.2.4.— To complete the proof of assertions (i), (ii), (iii), and (iv)(a) of 3.2 in the case where F and G are of finite type over A, it remains to study the quotient $F\backslash G$. By Exposé V, 6.1, assertions (ii), (iii), and (iv)(a) "become true" after the base change

$$f : \operatorname{Spec} A' \longrightarrow \operatorname{Spec} A.$$

By EGA IV$_2$, 2.6.1, 2.6.2, and 2.7.1, these assertions were therefore true before the base change. Finally, to prove the second assertion of (i), namely that $F\backslash G$ is the cokernel of (pr_2, λ) in the category of all ringed spaces, it suffices to refer to Exposé V, §6(c).

3.2.5.—[41] We now prove assertion (iv)(c) of 3.2, following the proof of Exposé VI$_B$, 9.2.1. Write

$$X = F\backslash G$$

and let d be the morphism

$$F \times_A G \longrightarrow G \times_A G$$

whose components are λ and pr_2.

Since u is a closed immersion by 2.5.2, and since

$$d = \sigma \circ (u \times \mathrm{id}_G), \qquad \sigma(x, y) = (xy, y),$$

where σ is an automorphism of $G \times_A G$, the morphism d is a closed immersion. On the other hand, by (iv)(a), the following square is Cartesian:

$$\begin{array}{ccc} F \times_A G & \xrightarrow{d} & G \times_A G \\ \downarrow & & \downarrow{\scriptstyle p\times p} \\ X & \xrightarrow[\Delta_X]{} & X \times_A X \end{array}$$

Moreover, p, and hence also $p \times p$, is faithfully flat and locally of finite presentation. Therefore, by fppf descent, since d is a closed immersion, so is Δ_X; that is, X is separated.

3.3.—[42] In Theorem 3.2, the hypothesis that G be flat can be omitted when $u : F \to G$ is a monomorphism. This generalization is mentioned in Exposé VI$_B$, Remark 9.3(b), and also in [Ray67a], Example a)(i), p. 82. The proof, given in [An73], Theorem 4, follows from Theorem 3.2 and from the following theorem of Grothendieck (mentioned in [Ray67a], Theorem 1(ii), and proved in [DG70], § III.2, 7.1). If X is a scheme and R an equivalence relation on X, we write $\widetilde{X/R}$ for the (fppf) quotient sheaf of X by R (cf. Exposé IV, 4.4.9).

Theorem 3.3.1 (Grothendieck). *Let A be a ring, let X be an A-scheme, and let*

$$d_0, d_1 : R \rightrightarrows X$$

be an A-equivalence relation on X, with each d_i faithfully flat and of finite presentation. Let X_0 be a saturated closed subscheme of X, defined by a nilpotent ideal, and let R_0 be the equivalence relation induced by R on X_0. If the (fppf) *quotient sheaf $\widetilde{X_0/R_0}$ is representable by an A-scheme, then so is $\widetilde{X/R}$.*

For the proof, see [DG70], § III.2, 7.1. Let us now return to the case where A is an Artinian local ring. Let $u : F \to G$ be a quasi-compact morphism between A-groups locally

[41] Editor's note: we have added this paragraph.

[42] Editor's note: we have added this subsection.

of finite type, and suppose in addition that u is a monomorphism. Then u is a closed immersion by 2.5.2.

We may now state the following variant of Theorem 3.2.

Theorem 3.3.2. *Let A be an Artinian local ring, let G be an A-group locally of finite type, and let F be a closed subgroup of G, flat over A.*[43] *Then:*

(i) *The* (fppf) *quotient sheaf $\widetilde{F\backslash G}$ is representable by a separated A-scheme $F\backslash G$, locally of finite type; moreover, the sequence*

$$F \times_A G \rightrightarrows G \xrightarrow{p} F\backslash G$$

is exact in the category of all ringed spaces.

(ii) *The morphism*

$$F \times_A G \xrightarrow{(\lambda, \mathrm{pr}_2)} G \times_{(F\backslash G)} G$$

is an isomorphism, and $p : G \to F\backslash G$ is faithfully flat and locally of finite presentation; hence p is an F-torsor locally trivial for the (fppf) *topology.*

(iii) *If G is flat (respectively, of finite type; respectively, smooth) over A, then so is $F\backslash G$.*

(iv) *$X = F\backslash G$ is equipped with a right action of G such that $p(e)\cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A, irreducible, and all of dimension* $\dim G - \dim F$.

(v) *If, in addition, F is a normal subgroup of G, there is a unique A-group structure on $F\backslash G$ for which $p : G \to F\backslash G$ is a morphism of A-groups.*

Assertions (i) and (ii) follow from 3.2 and 3.3.1. Since $G \to F\backslash G$ is faithfully flat and locally of finite presentation, assertion (iii) follows from EGA IV, 2.2.14, 2.7.1, and 17.7.7. Assertions (iv) and (v) will be proved in Section 5. Let us record at once the following corollary.

Corollary 3.3.3. *Let A be an Artinian local ring, let G be an A-group locally of finite type, and let H be a closed subgroup of G, flat over A. Write p for the morphism $G \to G/H$, and write λ (respectively,* pr_1*) for the morphism $G\times H \to G$ defined by $\lambda(g,h) = gh$ (respectively, the projection $G \times H \to G$). Then, for every open subset U of G/H,*

$$\mathcal{O}(U) = \{\varphi \in \mathcal{O}(p^{-1}(U)) \mid \varphi \circ \lambda = \varphi \circ \mathrm{pr}_1\}.$$

In other words, $\mathcal{O}(U)$ is the set of $\varphi \in \mathcal{O}(p^{-1}(U))$ such that $\varphi(gh) = \varphi(g)$ for every A-scheme S and every $g \in G(S)$, $h \in H(S)$.

Indeed, $p : G \to G/H$ is faithfully flat and locally of finite presentation, hence a covering for the (fppf) topology; the result follows from Exposé IV, 3.3.3.2.

[43] Editor's note: for an example in which F is not flat and $\widetilde{F\backslash G}$ is not representable, see [DG70], § III.3, no. 3.3.

4. Construction of quotients $F\backslash G$ (general case)

We now assume the hypotheses of Theorem 3.2, without requiring F and G to be of finite type over A.

4.1.— Let G^α be a connected component of G. We first show that the saturation

$$\mathscr{S}(G^\alpha)$$

[44] of G^α under the equivalence relation defined by the groupoid G_* is an open-and-closed subset of G; in other words, it is a union of connected components of G.

This saturation is the image of $F \times_A G^\alpha$ under λ, and is therefore open in G (cf. § 3.1). Let k be the residue field of A, and let $\bar{k}$ be an algebraic closure of k. It remains to show that the image of

$$(F \times_A G^\alpha) \otimes_A k$$

under $\lambda \otimes_A k$ is closed in $G \otimes_A k$, or, equivalently by SGA 1, VIII.4.4, that the image of

$$(F \times_A G^\alpha) \otimes_A \bar{k}$$

under $\lambda \otimes_A \bar{k}$ is closed in $G \otimes_A \bar{k}$. Since $G^\alpha \otimes_A \bar{k}$ is a union of finitely many connected components of $G \otimes_A \bar{k}$, we are reduced to the case where A is an algebraically closed field, which we now assume. In this case, $\mathscr{S}(G^\alpha)$ is the union of the images of G^α under the left translations $\ell_{u(x)}$, where x runs through the closed points of F. The assertion follows because these images are connected components of G.

4.2.— Take G^α in particular to be the connected component G^0 of the identity in G. Then $\mathscr{S}(G^0)$ plainly contains the image of F under u, which is precisely the equivalence class of the identity. On the other hand, if F^β is a connected component of F, then $F^\beta \times_A G^0$ is connected by 2.1.2, so its image under λ is contained in the connected component of $u(F^\beta)$ in G. In other words, $\mathscr{S}(G^0)$ is the union of the connected components that meet the image of F.

Let us also observe that the open subscheme of G whose underlying space is $\mathscr{S}(G^0)$ is a subgroup of G; we continue to denote it by $\mathscr{S}(G^0)$. Indeed, inversion on G preserves the image of F and permutes the connected components of G that meet this image. It therefore suffices to prove that

$$\nu : G \times_A G \longrightarrow G$$

maps $\mathscr{S}(G^0) \times_A \mathscr{S}(G^0)$ into $\mathscr{S}(G^0)$. For this we may suppose that A is an algebraically closed field: with the notation of 4.1, $\mathscr{S}(G^0) \otimes_A \bar{k}$ is the saturation of $(G \otimes_A \bar{k})^0$ for the equivalence relation defined by the homomorphism $u \otimes_A \bar{k}$.

If G^γ and G^δ are connected components of $\mathscr{S}(G^0)$, then $G^\gamma \times_A G^\delta$ is connected and its image under ν meets the image of F. Consequently,

$$\nu(G^\gamma \times_A G^\delta)$$

is contained in a connected component of G that meets $u(F)$.

4.3.— It follows from the preceding discussion that the groupoid G_*, with base G, defined by u, is the direct sum of the groupoids $\mathscr{S}(G^\alpha_*)$ induced by G_* on the various open-and-closed parts of G of the form $\mathscr{S}(G^\alpha)$. The cokernel of G_* is therefore the direct sum of the cokernels of the groupoids $\mathscr{S}(G^\alpha_*)$, which we are thus led to study separately.

[44] Editor's note: we have written $\mathscr{S}(G^\alpha)$, rather than $\overline{G^\alpha}$, for the saturation of G^α.

First consider the groupoid $\mathscr{S}(G^0_*)$ induced by G_* on $\mathscr{S}(G^0)$. It is clear that $\mathscr{S}(G^0_*)$ is the groupoid with base $\mathscr{S}(G^0)$ defined by the homomorphism

$$F \longrightarrow \mathscr{S}(G^0)$$

induced by u (§ 3.1). The cokernel whose existence we wish to prove is thus identified with $F\backslash\mathscr{S}(G^0)$. Consider, on the other hand, the groupoid

$$G_2^0 \underset{\ell'_0}{\overset{\ell'_2}{\overset{\ell'_1}{\rightrightarrows}}} G_1^0 \underset{\ell_0}{\overset{\ell_1}{\rightrightarrows}} G_0^0 = G^0$$

induced by $\mathscr{S}(G^0_*)$ on G^0. Referring to the explicit construction in Exposé V, § 3.b, the object denoted there by $Y_0 \times_{X_0} X_1$ is precisely $F \times_A G^0$. Hence G_1^0 is the inverse image of G^0 under the morphism

$$F \times_A G^0 \longrightarrow \mathscr{S}(G^0)$$

induced by λ.

I claim that this inverse image is $F_0 \times_A G^0$, where F_0 denotes the inverse image of G^0 under u. Indeed, if F^β is a connected component of F_0, then $F^\beta \times_A G^0$ is connected by 2.1.2 and $\lambda(F^\beta \times_A G^0)$ is contained in G^0.Conversely, if F^β is a connected component of F not contained in F_0, then the image of $F^\beta \times_A G^0$ is again connected and contains $u(F^\beta)$; if $u(F^\beta)$ is not contained in G^0, then $\lambda(F^\beta \times_A G^0)$ does not meet G^0.

It follows that the groupoid G^0_* induced by G_* on G^0 is the groupoid with base G^0 defined by the homomorphism

$$F_0 \longrightarrow G^0$$

induced by u. Since G^0, and hence F_0, are of finite type over A, Section 3 shows that G^0_* has a cokernel, namely $F_0\backslash G^0$.

I now claim that $F_0\backslash G^0$ is identified with $F\backslash\mathscr{S}(G^0)$. The proof is analogous to that of the first part of assertion (i) of Exposé V, Lemma 6.1. Consider the diagram

$$\mathscr{S}(G^0) \xleftarrow{\ v\ } F \times_A G^0 \xrightarrow{\ \mathrm{pr}_2\ } G^0\ ,$$

where v is the morphism induced by λ. Since pr_2 has a section, it is a universal effective epimorphism. Consequently, $F_0\backslash G^0$ coincides with $\mathrm{Coker}(v_0, v_1)$, where

$$V_2 \underset{v'_0}{\overset{v'_2}{\overset{v'_1}{\rightrightarrows}}} V_1 \underset{v_0}{\overset{v_1}{\rightrightarrows}} V = F \times_A G^0$$

is the inverse image under pr_2 of the groupoid G^0_* (cf. Exposé V, § 3.a), and hence also the inverse image of $\mathscr{S}(G^0_*)$ under the composite morphism

$$F \times_A G^0 \xrightarrow{\ \text{inclusion}\ } F \times_A \mathscr{S}(G^0) \xrightarrow{\ \mathrm{pr}_2\ } \mathscr{S}(G^0)\ .$$

Similarly, since v is faithfully flat and quasi-compact, $F\backslash\mathscr{S}(G^0)$ coincides with the cokernel of the inverse image of $\mathscr{S}(G^0_*)$ under the base change v. This inverse image is isomorphic to V_* by Exposé V, § 3.c. It follows that the canonical inclusion of G^0_* in $\mathscr{S}(G^0_*)$ induces an isomorphism

$$F_0\backslash G^0 \xrightarrow{\ \sim\ } F\backslash\mathscr{S}(G^0).$$

Finally, formation of $F\backslash\mathscr{S}(G^0)$ commutes with finite locally free base changes, because the same holds for $F_0\backslash G^0$ (cf. editor's note 37 and 4.6 below).

4.4.— It remains to construct the cokernel of the groupoid $\mathscr{S}(G^\alpha_*)$ when G^α is an arbitrary connected component of G. Let A' be a sufficiently large local A-algebra, finite and free over A (cf. 3.2). Then $G^\alpha \otimes_A A'$ is the union of finitely many connected components

$$C^1, \ldots, C^n$$

of $G \otimes_A A'$, each of which has a strictly rational point. For every i, there is therefore a right translation r_i of $G \otimes_A A'$ which maps $G^0 \otimes_A A'$ onto C^i. This translation induces an isomorphism

$$\mathscr{S}(G^0_*) \otimes_A A' \xrightarrow{\sim} \mathscr{S}(C^i_*).$$

Consequently, the groupoid induced by $G_* \otimes_A A'$ on the saturation of C^i has a cokernel.

Since $\mathscr{S}(G^\alpha_*) \otimes_A A'$ is the direct sum of some of the groupoids $\mathscr{S}(C^i_*)$, it too has a cokernel. This cokernel is a direct sum of some number of copies of

$$(F_0 \otimes_A A')\backslash(G^0 \otimes_A A').$$

Thus every finite subset of it is contained in an affine open subset. Moreover, formation of this cokernel commutes with finite locally free extensions of the base (cf. editor's note 37 and 4.6 below). As in 3.2.3, it follows that this cokernel has the form $Y \otimes_A A'$, where Y is a cokernel of $\mathscr{S}(G^\alpha_*)$.

4.5.— We have now constructed $F\backslash G$ and shown that it is a direct sum of schemes of finite type over A. The remaining assertions of Theorem 3.2 reduce directly to assertions about the groupoids $\mathscr{S}(G^\alpha_*)$. As in Exposé V, § 6, the second assertion in (i) follows from the first together with (ii) and (iii); it is therefore enough to prove (ii), (iii), and (iv)(a).

Since A' is a local A-algebra, finite and free over A, the morphism $A \to A'$ is faithfully flat and of finite presentation. By SGA 1, VIII (3.1, 4.6, 5.4), it is therefore enough to verify the corresponding assertions for the groupoid

$$\mathscr{S}(G^\alpha_*) \otimes_A A'.$$

This groupoid is isomorphic to a direct sum of finitely many copies of $\mathscr{S}(G^0_*) \otimes_A A'$ (cf. 4.4), so we are reduced to the groupoid $\mathscr{S}(G^0_*)$. For this last groupoid we continue to follow the proof given in Exposé V, § 6, as we began to do in 4.3.

4.6.— Let us conclude this section with some remarks concerning Lemma 6.1 and § 9(a) of Exposé V. Under the hypotheses and with the notation of Exposé V, § 9(a), we seek a condition under which formation of the cokernel of the (Sch/S)-groupoid X_* commutes with a base extension

$$\pi : S' \longrightarrow S.$$

The cokernels of X_* and X'_* are identified with the cokernels of the groupoids U_* and U'_* induced by X_* and X'_* on the quasi-sections U and U', respectively. We are therefore reduced to the case of a (Sch/S)-groupoid satisfying the hypotheses of Exposé V, Theorem 4.1.

[45]Let Y be the cokernel of U_*, put

$$Y' = Y \times_S S',$$

and let Y_1 be the cokernel of U'_*; we saw in Exposé V, § 9(a), that the canonical morphism

$$Y_1 \longrightarrow Y'$$

[45]Editor's note: we have added the following sentence.

is a homeomorphism (indeed, a universal homeomorphism); we may thus identify Y_1 and Y' as topological spaces. If $p : U \to Y$ is the canonical morphism and $p' : U' \to Y'$ is obtained from it by base change, we want the following sequence of $\mathcal{O}_{Y'}$-modules to be exact:

$$\mathcal{O}_{Y'} \longrightarrow p'_*(\mathcal{O}_{U'}) \rightrightarrows p'_*u'_{1*}(\mathcal{O}_{U'_1}) = p'_*u'_{0*}(\mathcal{O}_{U'_1}). \qquad (*)$$

[46]Since we are under the hypotheses of Exposé V, Theorem 4.1, u_0 and u_1 are finite and locally free; moreover, p is integral by Exposé V, 4.1(ii). Thus p and $p \circ u_i$ are affine, hence separated and quasi-compact.

Consequently, if S' is flat over S, EGA III$_1$, 1.4.15 (taking account of correction Err$_{\text{III}}$ 25 in EGA III$_2$) identifies the sequence (*) with the inverse image of the sequence

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_U) \rightrightarrows p_*u_{1*}(\mathcal{O}_{U_1}) = p_*u_{0*}(\mathcal{O}_{U_1}), \qquad (**)$$

which is exact.[47] An analogous argument applies when the groupoid X_* has quasi-sections "locally" (cf. the proof of Exposé V, Theorem 7.1). We therefore obtain:

Proposition 4.6.1. *Formation of the cokernel of X_* commutes with flat base extensions when X_* has quasi-sections locally.*

4.7.— Consider the groupoid G_* of Theorem 3.2, supposing temporarily that F and G are of finite type over A. By 3.2.2 there is a local A-algebra A', finite and free over A, such that $G_* \otimes_A A'$ has quasi-sections "locally." For every base extension $T \to \operatorname{Spec} A$, the sequence

$$(F''\backslash G'') \times_{\operatorname{Spec} A} T \rightrightarrows (F'\backslash G') \times_{\operatorname{Spec} A} T \longrightarrow (F\backslash G) \times_{\operatorname{Spec} A} T$$

obtained from diagram $(*)$ of 3.2.3 is exact.

If, in addition, T is flat over $\operatorname{Spec} A$, then 4.6 identifies

$$(F''\backslash G'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (F'\backslash G') \times_{\operatorname{Spec} A} T$$

respectively with the cokernels of the groupoids

$$(G_* \otimes_A A'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (G_* \otimes_A A') \times_{\operatorname{Spec} A} T.$$

The diagram obtained from 3.2.3$(*)$ by the base change $T \to \operatorname{Spec} A$ then shows that

$$(F\backslash G) \times_{\operatorname{Spec} A} T$$

is the cokernel of $G_* \times_{\operatorname{Spec} A} T$. The same argument applies in the general case, when F and G are merely locally of finite type over A. Thus:

Proposition 4.7.1. *Under the hypotheses of Theorem 3.2, for every flat A-scheme T, the scheme*

$$(F\backslash G) \times_{\operatorname{Spec} A} T$$

is identified with the left quotient of $G \times_{\operatorname{Spec} A} T$ by $F \times_{\operatorname{Spec} A} T$.[48]

[46]Editor's note: in what follows we have modified the original; the additional hypotheses imposed there on X_* are unnecessary.

[47]Editor's note: we have added both the following sentence and the numbering 4.6.1 below, in order to highlight the stated result.

[48]Editor's note: we have added the numbering 4.7.1 in order to highlight this result.

5. Connections with Exposé IV and consequences

5.1.—[49] We retain the notation of § 3 and the hypotheses of Theorem 3.2. We then have the following commutative diagram:

$$\begin{array}{ccc}
F \times_A G \times_A G & \xrightarrow{F\times\nu} & F \times_A G \\
\Big\downarrow{\scriptstyle \mathrm{pr}_2\times G}\ \Big\downarrow{\scriptstyle \lambda\times G} & & \Big\downarrow{\scriptstyle \mathrm{pr}_2}\ \Big\downarrow{\scriptstyle \lambda} \\
G \times_A G & \xrightarrow{\nu} & G \\
\Big\downarrow{\scriptstyle p\times G} & & \Big\downarrow{\scriptstyle p} \\
(F\backslash G) \times_A G & \xrightarrow{\rho} & F\backslash G
\end{array}$$

It satisfies

$$\mathrm{pr}_2 \circ (F \times \nu) = \nu \circ (\mathrm{pr}_2 \times G), \qquad \lambda \circ (F \times \nu) = \nu \circ (\lambda \times G).$$

Moreover, since G is flat over A, the left-hand vertical sequence is exact by 4.7. Hence ν induces a morphism of A-schemes

$$\rho : (F\backslash G) \times_A G \longrightarrow F\backslash G.$$

The morphism ρ makes G act on the right on $F\backslash G$, as is immediately checked; moreover, the canonical morphism $G \to F\backslash G$ commutes with the right actions of G on G and on $F\backslash G$.[50] This proves the first assertion of Theorem 3.2(ii′). By 2.5.4, the connected components of

$$X = F\backslash G$$

are therefore of finite type, irreducible, and all of the same dimension.

To compute this dimension, we may suppose that $A = k$ and that k is algebraically closed. By Exposé I, 2.3.3.1, the stabilizer of the k-point $p(e)$ is represented by the fiber

$$H = p^{-1}(p(e)).$$

Since $F\backslash G$ is the quotient of G by F in the category of ringed spaces, the underlying space of this fiber is $u(F)$. As $\ker(u)$ is finite, we obtain

$$\dim H = \dim u(F) = \dim F.$$

It follows from 2.5.4(ii) that

$$\dim X = \dim G - \dim F.$$

This proves Theorem 3.2(ii′), and hence also Theorem 3.3.2(iv).

5.2.— When the homomorphism of A-groups $u : F \to G$ is a monomorphism, the result of 5.1 may also be recovered from the results of Exposé IV. Indeed, the canonical morphism

$$p : G \longrightarrow F\backslash G$$

is faithfully flat and open by 3.2; it is therefore a covering for the (fpqc) topology by Exposé IV, 6.3.1, and we may apply Exposé IV, Corollaries 5.2.2 and 5.2.4.

[49] Editor's note: we have changed the title of this section, which was called "Complements" in the original.
[50] Editor's note: we have added what follows.

In particular, suppose, in addition to the hypotheses of 3.2, that u is the inclusion in G of a normal subgroup F. Then there is a unique A-group structure on $F\backslash G$ for which the canonical morphism

$$p : G \longrightarrow F\backslash G$$

is a homomorphism of A-groups.[51] This proves Theorem 3.3.2(v).

5.3.— We now review several statements from Exposé IV.

5.3.1.— The statements of Exposé IV, 5.2.7 and 5.3.1 translate as follows. Let F and G be groups locally of finite type and flat over A, with F a closed normal subgroup of G. The maps

$$H \longmapsto F\backslash H, \qquad H' \longmapsto H' \times_{(F\backslash G)} G$$

define a bijection between the flat A-subgroups of G containing F and the flat A-subgroups of $F\backslash G$. Under this bijection, the closed (respectively, normal) subgroups of G containing F correspond to the closed (respectively, normal) subgroups of $F\backslash G$.[52]

5.3.2.— Exposé IV, Proposition 5.2.9, gives the following result. Let F, H, G be groups locally of finite type and flat over A, and suppose

$$F \subset H \subset G,$$

with F closed in G and normal in H. Then $F\backslash H$ acts freely on the left on $F\backslash G$, the quotient scheme

$$(F\backslash H)\backslash(F\backslash G)$$

exists, and there is a canonical isomorphism of G-schemes

$$(F\backslash H)\backslash(F\backslash G) \xrightarrow{\sim} H\backslash G.$$

5.3.3.— Finally, Exposé IV, 5.2.8, gives the following assertion. Let F, H, G be groups locally of finite type and flat over A. Suppose that F is contained in G, closed and normal in G, that H is contained in G, and that $F \cap H$ is flat over A. Let

$$F \overset{\tau}{\times}_A H$$

be the A-group whose underlying scheme is $F \times_A H$, with multiplication defined by

$$\big((x,h),(y,h')\big) \longmapsto \big(xhyh^{-1}, hh'\big).$$

Likewise, let

$$u : H \cap F \longrightarrow F \overset{\tau}{\times}_A H$$

be the monomorphism $x \mapsto (x^{-1}, x)$, and let $F \cdot H$ be the quotient

$$(F \cap H)\backslash\big(F \overset{\tau}{\times}_A H\big).$$

Then there is a canonical isomorphism

$$F\backslash(F \cdot H) \xrightarrow{\sim} (F \cap H)\backslash H.$$

[51] Editor's note: we have added the following sentence.

[52] Editor's note: in addition to the cited statements of Exposé IV, we use the fact that, since $G \to F\backslash G$ is faithfully flat, an A-subgroup H of G is flat over A if and only if $F\backslash H$ is.

5.4.—[53] Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type, such that the kernel N of u is flat over A. In this case, by 3.3.2 and 5.2, the quotient A-group

$$C = N\backslash G$$

exists, and the morphism $p : G \to C$ is faithfully flat and locally of finite presentation. On the other hand, by Exposé IV, 5.2.6, u induces a monomorphism

$$v : C \longrightarrow H,$$

which is quasi-compact (because u is quasi-compact and $G \to C$ is surjective; cf. EGA IV_1, 1.1.3), and is therefore a closed immersion by 2.5.2. We have thus obtained the following proposition.

Proposition 5.4.1.— *Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type, such that $N = \ker u$ is flat over A. Then u admits the factorization*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ {\scriptstyle p}\downarrow & \nearrow_{i} & \\ N\backslash G & & \end{array}$$

where p is faithfully flat and locally of finite presentation, and i is a closed immersion.

Suppose in addition that G is flat over A. Then, by 3.3.2, $C = N\backslash G$ is flat over A, and hence the quotient

$$X = C\backslash H$$

exists in $(\mathbf{Sch}_{/A})$ and represents the quotient (fppf)-sheaf $\widetilde{C}\backslash H$, while

$$q : H \longrightarrow X$$

is a C-torsor. Consequently, if

$$e : \operatorname{Spec} A \longrightarrow H$$

denotes the unit section of H, then v induces an isomorphism of (fppf)-sheaves between $\widetilde{C}$ and the fiber product of q with

$$q \circ e : \operatorname{Spec} A \longrightarrow X,$$

which is represented by a closed subscheme of H. Consequently, v is an isomorphism of C onto a closed group subscheme K of H, equal to the stabilizer of the A-point $q \circ e$ of X. (This also gives another proof that every quasi-compact monomorphism $v : C \to H$ between A-groups locally of finite type is a closed immersion; cf. 2.5.2 and Exposé VI_B, 1.4.2.)

Suppose further that C is a normal subgroup of H. In this case, the A-group

$$\overline{H} = C\backslash H$$

is the cokernel, in the category of A-groups, of the morphism $u : G \to H$, and K is the kernel of the morphism

$$H \longrightarrow \overline{H}.$$

[53] Editor's note: we have expanded the original in what follows; in particular, we have added Proposition 5.4.1.

When G and H are abelian A-groups, K is the image of u in the category of abelian A-groups, whereas

$$C = (\ker u)\backslash G$$

is the coimage of u. Taking into account the isomorphism

$$C \xrightarrow{\sim} K$$

just established, we obtain the following.[53]

Theorem 5.4.2.— *Let k be a field. The category of commutative algebraic k-groups is abelian.*

Indeed, when k is a field, $\ker(u)$ is flat over k, whatever u may be.

[54] Note that the full subcategory of commutative affine algebraic k-groups is thick. Indeed, consider an exact sequence of commutative algebraic k-groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow G/N \longrightarrow 1.$$

If G is affine, then N is clearly affine, and G/N is also affine by a theorem of Chevalley; cf. Exposé VI_B, 11.17. Conversely, if N and G/N are affine, then G is also affine by Exposé VI_B, 9.2(viii). We therefore obtain the following corollary.

Corollary 5.4.3.— *Let k be a field. The category of commutative affine algebraic k-groups is abelian.*

Let us also point out that the category of all commutative affine k-groups (not necessarily of finite type) is abelian. This follows from Exposé VI_B, 11.17 and 11.18.2 (cf. [DG70], § III.3, 7.4); see also Exposé VII_B, 2.4.2 for a proof using formal groups.

5.5.— Let G be a group locally of finite type and flat over an Artinian local ring A. We know from 2.3 that the connected component G^0 of the identity is a normal open group subscheme of G, hence is also flat over A. It then follows from 3.2 and 5.2 that

$$G^0\backslash G$$

is an A-group scheme, flat over A. Moreover, since every connected component G^α of G is saturated for the equivalence relation defined by G^0, the quotient $G^0\backslash G$ is the disjoint union of the $G^0\backslash G^\alpha$ (cf. 4.3). In particular, the connected component of the identity in $G^0\backslash G$ is precisely

$$G^0\backslash G^0 \simeq \operatorname{Spec} A,$$

and hence the morphism

$$G^0\backslash G \longrightarrow \operatorname{Spec} A$$

is a local isomorphism at the identity. Consequently, $G^0\backslash G$ is étale over $\operatorname{Spec} A$, by Exposé VI_B, 1.3.[55] We therefore obtain the following proposition (for assertion (ii), see [DG70], § II.5, 1.7–1.10).

Proposition 5.5.1.— *Let A be an Artinian local ring and G an A-group locally of finite type and flat.*

(i) *$G^0\backslash G$ is an étale A-group.*

[53] Editor's note: we have added the number 5.4.2 to this theorem.

[54] Editor's note: we have added what follows.

[55] Editor's note: we have expanded the preceding discussion from the original and added Proposition 5.5.1 in order to make this result explicit.

(ii) *Consequently, if $A = k$ is an algebraically closed field, $G^0\backslash G$ is a constant k-group acting simply transitively on the set of connected components of G. Thus, if G is algebraic, $G^0\backslash G$ is finite.*

5.6.— Now let k be a perfect field and G a k-group locally of finite type. We saw in 0.2 that G_{red} is then a group subscheme of G. Moreover, the equivalence class of the identity of G under the left action of G_{red} on G is the whole underlying space of G. Theorem 3.2 therefore gives the following result.

Proposition 5.6.1.— *Let k be a perfect field and G a k-group locally of finite type. Then the k-scheme*

$$G_{\mathrm{red}}\backslash G$$

is the spectrum of a finite local k-algebra with residue field k.

[56] Indeed, by 3.2, the quotient $G_{\mathrm{red}}\backslash G$ has a single point, with residue field k, and is a k-scheme of finite type. It is therefore the spectrum of a finite-dimensional local k-algebra (cf. EGA I, 6.4.4).

Proposition 5.6.2.— *Let $u : F \to G$ be a morphism between groups locally of finite type over a perfect field k. The following assertions are equivalent:*

(i) *u is flat.*

(ii) *The morphism*

$$u^0 : F^0 \longrightarrow G^0$$

is dominant, and the morphism

$$v : F_{\mathrm{red}}\backslash F \longrightarrow G_{\mathrm{red}}\backslash G$$

induced by u is flat.

[57] Indeed, consider the following commutative diagram:

$$\begin{array}{ccc} F & \xrightarrow{\ p\ } & F_{\mathrm{red}}\backslash F \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ G & \xrightarrow[\ q\ ]{} & G_{\mathrm{red}}\backslash G \end{array}$$

where p and q denote the canonical projections. By 3.2(iv), p and q are faithfully flat. Consequently, if u is flat, then

$$q \circ u = v \circ p$$

is flat, and hence v is flat as well.

Conversely, suppose that v is flat and u^0 is dominant. Since u^0 is quasi-compact (the scheme F^0 is of finite type over k by 2.4, and hence Noetherian), it maps the generic point ξ of F^0 to the generic point η of G^0. Let R be the finite local k-algebra such that

$$G_{\mathrm{red}}\backslash G = \operatorname{Spec} R,$$

and let $\mathfrak{m}$ be its maximal ideal. We have local homomorphisms of local rings

$$R \longrightarrow \mathcal{O}_{G,\eta} \longrightarrow \mathcal{O}_{F,\xi}.$$

Observe that the following square is Cartesian:

[56] Editor's note: we have added the number 5.6.1, as well as the proof that follows.

[57] Editor's note: we have expanded the proof of (ii) ⇒ (i) and simplified the diagram below.

$$\begin{array}{ccc} G_{\mathrm{red}} & \longrightarrow & G \\ \downarrow & & \downarrow q \\ \mathrm{Spec}(R/\mathfrak{m}) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

It follows that

$$\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta} \simeq \mathcal{O}_{G_{\mathrm{red}},\eta} = \kappa(\eta),$$

so that

$$\mathcal{O}_{F,\xi}/\mathfrak{m}\mathcal{O}_{F,\xi}$$

is flat over $\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta}$.

On the other hand, since q and $v \circ p$ are flat, G and F are flat over R. The local criterion for flatness (cf. EGA IV$_3$, 11.3.10.2) therefore shows that $\mathcal{O}_{F,\xi}$ is flat over $\mathcal{O}_{G,\eta}$; in other words, u is flat at ξ. It follows from 2.5.3 that u is flat.

6. Complements on k-groups not necessarily of finite type

[58] Let us also record the following results, which will be useful in the addendum, Exposé VI$_B$, § 12. We fix a base field k.

Lemma 6.1.— *Let G be a k-group. For every $x \in G$, there exists a point $u \in G \times G$ such that $\mu(u) = x$ and both projections $p_1(u)$ and $p_2(u)$ are maximal points of G.*

Proof. Set $K = \kappa(x)$. Since the projection $G_K \to G$ sends maximal points to maximal points, we are reduced to the case in which x is rational. Left translation λ_x (respectively, right translation ρ_x) then gives a morphism

$$G \longrightarrow G \times G, \qquad g \longmapsto (\lambda_x(g^{-1}), g)$$

(respectively, $g \longmapsto (g, \rho_x(g^{-1}))$), which induces an isomorphism from G onto $\mu^{-1}(x)$, inverse to p_2 (respectively, p_1). Thus, if u is a maximal point of $\mu^{-1}(x)$, then $p_1(u)$ and $p_2(u)$ are maximal points of G, and u has the required properties.

Corollary 6.2.— *Let $f : G \to H$ be a quasi-compact dominant morphism of k-groups.*

(i) *f is surjective.*

(ii) *If H is reduced, f is faithfully flat.*

Proof. Write μ_H (respectively, μ_G) for the multiplication of H (respectively, G). Let $h \in H$. By 6.1, there exists $u \in H \times H$ such that $\mu_H(u) = h$ and $\alpha = p_1(u)$, $\beta = p_2(u)$ are maximal points of H. Since f is quasi-compact and dominant, $f^{-1}(\alpha)$ and $f^{-1}(\beta)$ are nonempty (cf. EGA IV$_1$, 1.1.5). Hence there exists $v \in G \times G$ such that

$$(f \times f)(v) = u$$

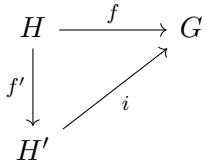
(cf. EGA I, 3.5.2). Then $g = \mu_G(v)$ satisfies $f(g) = h$. This proves that f is surjective.

Suppose in addition that H is reduced. Then $\mathcal{O}_{H,\alpha}$ is a field, and we have seen above that $f^{-1}(\alpha) \neq \varnothing$. Thus f is flat at every point $\xi \in f^{-1}(\alpha)$, and consequently f is flat by Lemma 2.5.3.

[58] Editor's note: we have added the results that follow, taken from [Per75]. Note that Lemma 6.1 can be expressed, in Weil's language, by saying that "every point of G is a product of two generic points."

Reminder 6.3.— Recall (cf. EGA IV$_3$, 11.10.1) that a morphism of schemes $f : X \to Y$ is said to be *schematically dominant* if it satisfies the following condition: for every open subset U of Y, if Z is a closed subscheme of U such that the morphism $f^{-1}(U) \to U$ factors through Z, then $Z = U$. When f is quasi-compact and quasi-separated, this is equivalent to saying that the closed image of X under f is Y (cf. loc. cit., 11.10.3(iv), and EGA I, 9.5.8).

Proposition 6.4.— *Let $f : H \to G$ be a quasi-compact morphism of k-groups. Then the closed image of f is a group subscheme H' of G, and f factors as*



where f' is schematically dominant, quasi-compact, and surjective.

Proof. Since H is separated (0.3), f is quasi-compact and separated. Thus $f_*(\mathcal{O}_H)$ is a quasi-coherent $\mathcal{O}_G$-module, and the closed image H' of f exists and is the closed subscheme of G defined by the quasi-coherent ideal

$$\mathcal{I} = \ker(\mathcal{O}_G \longrightarrow f_*(\mathcal{O}_H))$$

(cf. EGA I, § 9.5).

Write c_G, μ_G (respectively, c_H, μ_H) for the inversion and multiplication morphisms of G (respectively, H). Then $c_G \circ f = f \circ c_H$ factors through the closed subscheme $c_G(H')$, whence

$$H' \subset c_G(H') \qquad \text{and therefore} \qquad H' = c_G(H'),$$

since $c_G^2 = \mathrm{id}_G$. Similarly,

$$f \circ \mu_H = \mu_G \circ (f \times f)$$

factors through H', so $f \times f$ factors through the closed subscheme $\mu_G^{-1}(H')$ of $G \times G$. On the other hand, since formation of the closed image commutes with flat base change (EGA III, 1.4.15, and EGA IV$_1$, 1.7.21), the closed image of $f \times \mathrm{id}_H$ (respectively, $\mathrm{id}_{H'} \times f$) is $H' \times H$ (respectively, $H' \times H'$). Hence, by "transitivity of closed images" (EGA I, 9.5.5), the closed image of $f \times f$ is $H' \times H'$. It is therefore contained in $\mu_G^{-1}(H')$; that is, the restriction of μ_G to $H' \times H'$ factors through H'. This proves that H' is a closed group subscheme of G. Write $i : H' \hookrightarrow G$ for the inclusion.

Now $f = i \circ f'$, where $f' : H \to H'$ is schematically dominant and quasi-compact (because f is quasi-compact and i is separated). By 6.2, f' is therefore surjective. This proves 6.4.

We can now state the following theorem ([Per75], V, 3.1 and 3.2; see also [Per76], 0.0 and 0.1).

Theorem 6.5 (D. Perrin).— *Let G be a quasi-compact k-group.*

(i) *G is the projective limit of a filtered system (G_i) of k-groups of finite type (whose transition morphisms $u_{ij} : G_j \to G_i$ are affine for i sufficiently large), and the morphisms $G \to G_i$ are faithfully flat (and affine for i sufficiently large).*

(ii) *Let H be a closed k-subgroup of G. Then the* fpqc *quotient sheaf $\widetilde{G/H}$ is a k-scheme in either of the following two cases:*

  (1) *the immersion $H \to G$ is of finite presentation; in this case G/H is of finite type over k;*

(2) *H is normal in G.*

For the proof of this theorem, which rests on several intermediate theorems, we refer to [Per75]. For the reader's convenience, however, we prove the two corollaries below; cf. [Per75], V, 3.3–3.4, or [Per76], 4.2.3–4.2.5.

Corollary 6.6.— *Let $f : G \to H$ be a quasi-compact morphism of k-groups.*

(i) *If f is schematically dominant, then it is faithfully flat.*

(ii) *This is so, in particular, if H is affine and the morphism*

$$f^\natural : \mathcal{O}(H) \longrightarrow \mathcal{O}(G)$$

is injective.

Proof. Suppose that f is schematically dominant. By 6.2(i), f is surjective, so by 2.5.3 it is enough to prove that f is flat at the identity element of G. Since

$$\mathcal{O}_{H,e} = \mathcal{O}_{H^0,e}$$

(cf. 2.6.5), we may replace H by H^0, and hence suppose that H is irreducible. Then H is quasi-compact (loc. cit.); since f is quasi-compact, G is also quasi-compact. By 6.5(i),

$$H = \varprojlim_i H_i,$$

where each H_i is an algebraic k-group. Let N_i be the kernel of $G \to H_i$; it is a closed normal k-subgroup of G. Moreover, since the identity section $\{e\} \to H_i$ is of finite presentation, the immersion $N_i \to G$ is of finite presentation. Hence, by 6.5(ii), the fpqc quotient G/N_i is an algebraic k-group. We then have a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{\ f\ } & H \\ {\scriptstyle q_i}\downarrow & & \downarrow{\scriptstyle p_i} \\ G/N_i & \xrightarrow[f_i]{} & H_i \end{array}$$

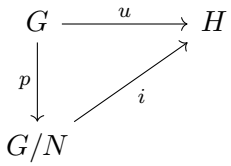
in which f is schematically dominant and p_i, q_i are faithfully flat, hence schematically dominant. Thus

$$f_i \circ q_i = p_i \circ f$$

is schematically dominant, and therefore so is f_i. On the other hand, f_i is a monomorphism, hence a closed immersion because G/N_i and H_i are algebraic (2.5.2). It follows that f_i is an isomorphism, and consequently $G \to H_i$ is faithfully flat. By [BAC], I, § 2.7, Proposition 9, the morphism $G \to H$ is then flat; it is also surjective by 6.2(i), and is therefore faithfully flat. This proves (i). Assertion (ii) follows, because if H is affine and $f^\natural$ is injective, then the closed image of f is H, so f is schematically dominant (cf. 6.4).

Corollary 6.7.— *Let $u : G \to H$ be a morphism of k-groups and set $N = \ker(u)$. Suppose that u is quasi-compact.*

(i) *The* fpqc *quotient sheaf $\widetilde{G/N}$ is represented by a k-group scheme G/N, and u factors as*



where p is faithfully flat and i is a closed immersion.

(ii) *In particular, if u is a monomorphism, then u is a closed immersion; and if u is schematically dominant, then u is faithfully flat.*

Proof. (i) By 6.4 and 6.6, the closed image G' of u is a closed group subscheme of H, and

$$p : G \longrightarrow G'$$

is faithfully flat and quasi-compact. Clearly $\ker(p) = \ker(u) = N$. Thus, by Exposé IV, 3.3.2.1 and 5.1.7, G' represents the fpqc quotient sheaf $\widetilde{G/N}$.

(ii) The second assertion is contained in 6.6; let us prove the first. If u is a monomorphism, then so is p. Thus p is both a monomorphism and an effective epimorphism, and is therefore an isomorphism (cf. Exposé IV, 1.14). This proves 6.7.

Let us finally record the following corollaries (cf. [Per76], 4.2.6–4.2.8).

Corollary 6.8.— *The category of commutative quasi-compact k-group schemes is abelian.*

Taking 6.7 into account, the proof is analogous to that of 5.4.2.

Corollary 6.9.— *If* $\mathrm{char}(k) = 0$*, every k-group scheme G is geometrically reduced.*

Indeed, if $g \in G$ and K is an algebraic closure of $\kappa(g)$, then

$$\mathcal{O}_{G,e} \otimes_k K \simeq \mathcal{O}_{G_K,g_K}.$$

It is therefore enough to prove that $\mathcal{O}_{G,e} = \mathcal{O}_{G^0,e}$ is geometrically reduced. We are thus reduced to the case in which G is connected, hence quasi-compact by 2.6.5. The result then follows from 6.5 and Cartier's theorem for algebraic groups (cf. Exposé VI$_B$, 1.6.1, or [DG70], § II.6, Theorem 1.1).

Corollary 6.10.— *Let G be a quasi-compact k-group, and suppose that k is algebraically closed.*

(i) *If $f : G \to H$ is a faithfully flat morphism of k-groups, then the induced map $G(k) \to H(k)$ is surjective.*

(ii) *The set of rational points is dense in G.*

For the proof, we refer to [Per75], V, 3.7 and 3.9.

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Exposé VI_B
Generalities on Group Schemes

by J.-E. Bertin

[0] This Exposé, which does not correspond to any oral lecture in the seminar, is intended to bring together a number of commonly used technical results concerning group schemes.*

1. Morphisms of groups locally of finite type over a field

1.1.— Let A be an Artinian local ring, let G and H be two A-groups,[2] and let $u : G \to H$ be a morphism of A-groups. Then u induces a group morphism

$$u(A) : G(A) \longrightarrow H(A).$$

[3] Since $H(A)$ acts on H by right translation, restriction along $u(A)$ defines an action of $G(A)$ on H. This action is compatible with u and with the action of $G(A)$ on G by right translation. Since $G(A)$ acts transitively on the strictly rational points of G (see Exposé VI_A, 0.4 for the definition), these points "all behave in the same way with respect to u." The following properties spring from this observation.

Proposition 1.2.— *Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type over A. Then the set $u(G)$ is closed in H, and*

$$\dim G = \dim u(G) + \dim \ker u.$$

[4]

Since u commutes with the inversion morphisms of G and H, the image $u(G)$ is invariant under inversion on H; the same is therefore true of its closure $\overline{u(G)}$ in H. On the other hand, let L be the set of points of $H \times_A H$ whose two projections belong to $u(G)$. It is clear that L is the image of

$$u \times_A u : G \times_A G \longrightarrow H \times_A H.$$

The multiplication morphism of H consequently sends L into $u(G)$; in other words,

$$u(G) \cdot u(G) = u(G).$$

[0] Editor's note: version of October 13, 2024.

*This Exposé was quite substantially revised after its mimeographed edition; in particular, Sections 10 and 11 were completely rewritten.(1)

[1] Editor's note: and also Section 5; cf. Editor's notes 34 and 46.

[2] Editor's note: recall the convention adopted at the beginning of Exposé VI_A: an A-group is a group scheme over A (and every such scheme is separated; cf. Exposé VI_A, 0.3).

[3] Editor's note: we have replaced "homomorphism" by "morphism."

[4] Editor's note: this statement also appears in Exposé VI_A, 2.5.2.

Lemma 1.2.1 below shows that the closure $\overline{L}$ of L in $H \times_A H$ is the set of points whose two projections belong to $\overline{u(G)}$. Hence

$$\overline{u(G)} \cdot \overline{u(G)} = \overline{u(G)}.$$

Thus the reduced subscheme of H whose underlying space is $\overline{u(G)}$ is naturally equipped with a group structure in the category $(\mathbf{Sch}_{/k})_{\mathrm{red}}$, where k is the residue field of A (cf. Exposé VI$_A$, 0.2).

Let us prove the first assertion of 1.2. After replacing A by an algebraic closure of its residue field k, we may suppose that A is an algebraically closed field k (cf. EGA IV$_2$, 2.3.12). Replacing u by

$$u_{\mathrm{red}} : G_{\mathrm{red}} \longrightarrow H_{\mathrm{red}},$$

we may suppose that G and H are reduced. In this case, as we have just seen, $u(G)$ is the underlying space of a reduced group subscheme of H; we may therefore suppose that u is dominant. Then $G(k)$ acts transitively on the set of connected components of H, and it is enough to prove that $u(G) \cap H^0$ is closed. We are thus reduced to the case in which H is connected, hence irreducible and of finite type (Exposé VI$_A$, 2.4). The morphism u is then of finite type, since it is quasi-compact and locally of finite type. As H is Noetherian, $u(G)$ is constructible (EGA IV$_1$, 1.8.5), and therefore contains an open subset V of H (EGA 0$_{\mathrm{III}}$, 9.2.2). By Exposé VI$_A$, 0.5, we then have

$$H = V \cdot V \subset u(G) \cdot u(G) = u(G).$$

Let us prove the second assertion. Recall first that the functor $\ker u$ (cf. Exposé I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the identity element of H. When u is locally of finite type, $\ker u$ is therefore locally of finite type over A. As above, this time using EGA IV$_2$, 4.1.4, we reduce to the case in which A is an algebraically closed field k.

We may further suppose that G and H are irreducible and of finite type and that u is dominant. Indeed, since k is algebraically closed, the connected components of G, on whose set $G(k)$ acts transitively, clearly all have the same dimension; and if u^0 denotes the restriction of u to G^0, then

$$(\ker u)^0 \subset \ker u^0, \qquad \dim \ker u^0 = \dim \ker u.$$

Likewise, u is then of finite type over k. If η is the generic point of H, then

$$\dim u^{-1}(\eta) = \dim G - \dim H$$

(EGA IV$_3$, 10.6.1(ii)). By EGA IV$_3$, 9.2.3 and 9.2.6, the set of $y \in H$ such that

$$\dim u^{-1}(y) = \dim u^{-1}(\eta)$$

contains a nonempty open subset V. Since u is dominant, $U = u^{-1}(V)$ is a nonempty open subset of G; it contains a closed point x, because G is a Jacobson scheme (EGA IV$_3$, 10.4.7). Right translation r_x is an isomorphism from $\ker u$ onto $u^{-1}(u(x))$, and hence

$$\dim \ker u = \dim u^{-1}(u(x)) = \dim u^{-1}(\eta) = \dim G - \dim H.$$

Lemma 1.2.1.— *Let $f : X' \to X$ and $g : Y' \to Y$ be two quasi-compact dominant morphisms of schemes over an Artinian local ring A. Then*

$$f \times_A g : X' \times_A Y' \longrightarrow X \times_A Y$$

is dominant (and quasi-compact).

Indeed,

$$f \times_A g = (\mathrm{id}_X \times_A g) \circ (f \times_A \mathrm{id}_{Y'}).$$

It is therefore enough to prove that $f \times_A \mathrm{id}_{Y'}$ and $\mathrm{id}_X \times_A g$ are dominant. For this purpose we may replace A by its residue field k. Then X and Y' are flat over $A = k$, and $f \times_A \mathrm{id}_{Y'}$ (respectively, $\mathrm{id}_X \times_A g$) is obtained from f (respectively, g) by the flat base change $Y' \to A$ (respectively, $X \to A$). It is therefore dominant (and quasi-compact) by EGA IV_2, 2.3.7.

Counterexample 1.2.2.— *Let k be a field of characteristic zero, let G be the constant k-group $\mathbb{Z}$, and let H be the additive k-group $\mathbb{G}_{a,k}$. Let $u : G \to H$ be a morphism of k-groups. If $u \neq 0$, then $u(G)$ is not closed in H.*

Proposition 1.3.—[5] *Let A be an Artinian local ring with residue field k, let G be an A-group locally of finite type and flat, and let $u : G \to H$ be a morphism of A-groups. The following assertions are equivalent:*

(i) *u is flat (respectively, quasi-finite, unramified, smooth, or étale) at some point of G;*[6]

(ii) *u is flat (respectively, quasi-finite, unramified, smooth, or étale).*

Proof. It is enough to prove that (i) implies (ii). We first establish the following lemma.

Lemma 1.3.0.— *Let $A \to B \to C$ be morphisms of commutative rings, and let $\mathfrak{n}$ be a nilpotent ideal of A. Suppose that $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$-algebra of finite type.*

(i) *Then C is a B-algebra of finite type.*

(ii) *Moreover, if C is flat over A and $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$-algebra of finite presentation, then C is a B-algebra of finite presentation.*

Indeed, let $x_1, \ldots, x_n$ be elements of C whose images generate $C/\mathfrak{n}C$ as a $(B/\mathfrak{n}B)$-algebra. By the nilpotent Nakayama lemma, the x_i generate C as a B-algebra. This proves (i). Let

$$\varphi : B[X_1, \ldots, X_n] \longrightarrow C$$

be the resulting surjective morphism, and put $I = \ker \varphi$.

Suppose now that C is flat over A and that $C/\mathfrak{n}C$ is of finite presentation over $B/\mathfrak{n}B$. On the one hand, $I/\mathfrak{n}I$ identifies with the kernel of

$$\overline{\varphi} = \varphi \otimes_A (A/\mathfrak{n}).$$

On the other hand, by EGA IV_1, 1.4.4, $\ker \overline{\varphi}$ is an ideal of finite type. Choose $P_1, \ldots, P_s \in I$ whose images generate $I/\mathfrak{n}I$ as an ideal. The nilpotent Nakayama lemma shows that the P_i generate I. This proves (ii).

[5] Editor's note: in the original, 1.3, 1.3.1, and 1.3.1.1 were stated with A a field. For completeness, we treat the case of an Artinian local ring, and for this purpose have added Lemma 1.3.0.

[6] Editor's note: recall that a morphism of schemes $f : X \to Y$ is said to be of finite type (respectively, of finite presentation or quasi-finite) at x if there are affine open subsets V, containing $f(x)$, and U, containing x, such that $f(U) \subset V$ and $\mathcal{O}_X(U)$ is an $\mathcal{O}_Y(V)$-algebra of finite type (respectively, of finite presentation, or with the additional condition that $f^{-1}(f(x'))$ is finite for every $x' \in U$); see also Editor's note 40 of Exposé V. The morphism is locally quasi-finite (respectively, locally of finite type or locally of finite presentation) if it has the property at every point x. It is smooth (respectively, unramified or étale) at x if there is an open neighborhood U of x such that $f|_U : U \to Y$ is smooth (respectively, unramified or étale). These definitions show that the loci where f is of finite presentation, of finite type, quasi-finite, smooth, unramified, or étale are open in X. Moreover, because the flatness locus of a morphism locally of finite presentation is open (EGA IV_3, 11.3.1), the locus where f is both of finite presentation and flat is open as well. These facts will be used repeatedly below.

We return to the proof of 1.3. Let x be an arbitrary point of G. Since G is flat over A, the fiberwise flatness criterion, in the form EGA IV_3, 11.3.10.2, shows that u is flat at x if $u \otimes_A k$ is. Likewise, the preceding lemma shows that u is of finite type (respectively, of finite presentation) at x if $u \otimes_A k$ is. The remaining properties can then be checked on the fibers (cf. EGA IV_4, 17.4.1, 17.5.1, and 17.6.1 for the unramified, smooth, and étale cases). We are therefore reduced to the case $A = k$.

Now let x be a point of G at which one of the conditions in 1.3(i) holds. The properties under consideration are preserved by fpqc descent (cf. EGA IV, 2.5.1, 2.7.1, and 17.7.1). Thus, after replacing k by an algebraic closure of $\kappa(x)$, we reduce to the case in which k is algebraically closed and $x \in G(k)$.

Since G is a Jacobson scheme (cf. EGA IV_3, 10.4.7), and the set W of points of G at which u is flat (respectively, quasi-finite, unramified, smooth, or étale) is stable under generization[7] (respectively, open), it is enough to show that every $y \in G(k)$ belongs to W. For such a point y, the translation

$$r_y \circ r_x^{-1}$$

sends x to y, so u has the required property at y. Thus $y \in W$, and the proof is complete.

Corollary 1.3.1.— *Let A be an Artinian local ring with residue field k, and let G be a flat A-group. The following assertions are equivalent:*[8]

(i) *G is locally quasi-finite (respectively, unramified, smooth, or étale) over A at some point;*

(ii) *G is locally quasi-finite (respectively, unramified, smooth, or étale) over A.*

[9] *Proof.* If G satisfies one of the conditions in (i) at a point x, there is an open neighborhood U of x that is of finite type over A. Consequently, it is enough to apply 1.3 in the case where H is the unit A-group, taking the following lemma into account.

Lemma 1.3.1.1.— *Let A be an Artinian local ring and G an A-group. If G contains a nonempty open subset of finite type over A, then G is locally of finite type over A.*

Proof. By Lemma 1.3.0, we may suppose that A equals its residue field k. Moreover, by fpqc descent, we may suppose that k is algebraically closed (cf. EGA IV_2, 2.7.1). Let V be the open subset of G consisting of the points at which G is of finite type over k; by hypothesis, $V \neq \varnothing$. Since G is a Jacobson scheme, V contains a closed point x, and to prove that $V = G$ it is enough to show that every closed point y of G belongs to V. For such a point y, the translation

$$r_y \circ r_x^{-1}$$

sends x to y, and hence $y \in V$.

Corollary 1.3.2.— *Let A be an Artinian local ring, and let $u : G \to H$ be a morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) *u is universally open;*

(ii) *u is open;*

[7] Editor's note: the original stated that the flat locus W is open, which does not seem evident a priori.

[8] Editor's note: it is not enough to suppose that G is flat over A at one point. For example, suppose that $A \neq k$, let H be the constant A-group $(\mathbb{Z}/2\mathbb{Z})_A$, and let G be the closed A-subgroup of H whose nonidentity component is reduced. Then the structural morphism $G \to \operatorname{Spec} A$ is a local isomorphism at the identity point e, but is not flat at the point $g \neq e$.

[9] Editor's note: we have added the sentence that follows and simplified the proof of Lemma 1.3.1.1 by invoking EGA IV, 2.7.1 instead of loc. cit., 17.7.5.

(iii) *u is open at some point of G;*

(iv) *the morphism $u^0 : G^0 \to H^0$ induced by u is dominant;*

(iv′) *u^0 is surjective;*

(v) *there is a connected component C of G such that, if D denotes the connected component of H containing $u(C)$, the induced morphism $u' : C \to D$ is dominant.*

Proof. The implications

$$\text{(i)} \Longrightarrow \text{(ii)} \Longrightarrow \text{(iii)} \qquad \text{and} \qquad \text{(iv}') \Longrightarrow \text{(iv)} \Longrightarrow \text{(v)}$$

are clear. Since G^0 is of finite type over A (Exposé VI_A, 2.4), and hence Noetherian, u^0 is quasi-compact. Thus $u^0(G^0)$ is closed in H^0 by 1.2, and therefore (iv) implies (iv′). On the other hand, G^0 (respectively, C) is open in G (Exposé VI_A, 2.3), while H^0 (respectively, D) is irreducible (Exposé VI_A, 2.4.1). Hence (ii) implies (iv) (respectively, (iii) implies (v)). It remains to prove that (v) implies (i).

We endow the open subset C (respectively, D) of G (respectively, H) with its induced scheme structure, and let u' denote the induced morphism $C \to D$. Let k be the residue field of A.[10] Since the base change

$$\operatorname{Spec} k \longrightarrow \operatorname{Spec} A$$

is a universal homeomorphism, we may suppose that $A = k$. By hypothesis u' is dominant; and since C is of finite type over k (Exposé VI_A, 2.4.1), and hence Noetherian, u' is quasi-compact. By EGA IV_2, 2.3.7,

$$u' \otimes_k \bar{k}$$

is again quasi-compact and dominant, where $\bar{k}$ denotes an algebraic closure of k. Since $C \otimes_k \bar{k}$ is a union of connected components of $G \otimes_k \bar{k}$, the morphism

$$u \otimes_k \bar{k} : G \otimes_k \bar{k} \longrightarrow H \otimes_k \bar{k}$$

satisfies assertion (v). In view of EGA IV_2, 2.6.4, we are thus reduced to the case where $A = k$ is an algebraically closed field.

In this case we may further replace u by u_{red}, and are thereby reduced to the case where H is reduced. Let ξ (respectively, η) be the generic point of C (respectively, D). Since u' is quasi-compact and dominant,

$$u'(\xi) = \eta$$

(cf. EGA IV_1, 1.1.5). Since H is reduced, the local ring $\mathcal{O}_{H,\eta}$ is a field, and hence u' is flat at ξ.[11] It follows from 1.3 that u is flat. Moreover, since u is locally of finite type and H is locally Noetherian, u is locally of finite presentation. Therefore u is universally open by EGA IV_2, 2.4.6.

Proposition 1.4.— *Let A be an Artinian local ring, and let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) *u is proper;*

(ii) *there exists $h \in H$ such that the fiber $u^{-1}(h)$ is nonempty and proper over $\kappa(h)$;*

[10] Editor's note: we have expanded the original in what follows.

[11] Editor's note: the original invoked the generic flatness theorem (EGA IV_2, 6.9.1), which is not necessary here.

(iii) $\ker u$ *is proper over* A.

Proof. It is clear that (i) implies (iii), and that (iii) implies (ii). The hypotheses imply that u is of finite type; and since G is separated (Exposé VI$_A$, 0.3), u is separated as well (EGA I, 5.5.1). It remains to prove that (ii) implies that u is universally closed. We may therefore suppose that A equals its residue field k.[12] Let k' be an algebraic closure of $\kappa(h)$, let

$$u' : G' \longrightarrow H'$$

be obtained from u by base change, and let $h' \in H'$ be a point above h. Then

$$u'^{-1}(h') = u^{-1}(h) \times_{\kappa(h)} k'$$

is nonempty and proper, and it is enough to prove that u' is proper (EGA IV$_2$, 2.6.4). Thus we may suppose that k is algebraically closed and $h \in H(k)$.

By 1.2, $u(G)$ is then the underlying set of a closed reduced group subscheme of H. Since every closed immersion is proper (EGA II, 5.4.2), we may suppose that u is surjective and that H is reduced. The group $G(k)$ then acts transitively on the set of closed points of H. Consequently, for every closed point $y \in H$, the fiber $u^{-1}(y)$ is proper over $\kappa(y)$. By EGA IV$_3$, 9.6.1, the set of $y \in H$ for which $u^{-1}(y)$ is not proper over $\kappa(y)$ is locally constructible. It contains no closed point, and is therefore empty (cf. EGA IV$_3$, 10.3.1 and 10.4.7).

[13] Now consider the generic point η of H^0. By what precedes, the fiber

$$u^{-1}(\eta) = G \times_H \operatorname{Spec} \kappa(\eta)$$

is proper over $\kappa(\eta)$. Since H is reduced, $\kappa(\eta) = \mathcal{O}_{H,\eta}$. The latter ring is the direct limit of the rings $\mathcal{O}_H(V)$, as V ranges over the nonempty open subsets of H^0. It follows from EGA IV$_3$, 8.1.2(a) and 8.10.5(xii), that there exists a nonempty open subset $V \subset H^0$ such that the restriction of u over V is proper. The translates $g \cdot V$, for $g \in G(k)$, form an open covering of H; and for every $g \in G(k)$, the restriction of u over $g \cdot V$ is proper. Hence u is proper (cf. EGA II, 5.4.1).

Corollary 1.4.1.— *Let A be an Artinian local ring, and let $u : G \to H$ be a morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) *u is locally quasi-finite;*

(ii) *u is quasi-finite at some point;*

(iii) $\ker u$ *is discrete;*

(iv) *the restriction of u to each connected component of G is finite.*

If u is quasi-compact, these assertions are also equivalent to

(v) *u is finite.*

It is clear that (iv) implies (iii), that (iii) implies (ii) (EGA I, 6.4.4), and that, when u is quasi-compact, (iv) and (v) are equivalent. We have already seen in 1.3 that (i) and (ii) are equivalent.

It remains to prove that (i) implies (iv). Let C be a connected component of G. Since C is of finite type over A (Exposé VI$_A$, 2.4.1) and G, H are separated (Exposé VI$_A$, 0.3),

[12] Editor's note: we have added the following sentence.

[13] Editor's note: we have expanded the original in what follows.

EGA I, 5.5.1 and 6.3.4 show that the restriction u' of u to C is separated and of finite type.[14] Its fibers are discrete, so u' is quasi-finite (cf. EGA II, 6.2.2). Since every quasi-finite proper morphism is finite (cf. EGA III$_1$, 4.4.2), it is enough to prove that u' is universally closed.

For this, we may suppose that A equals its residue field k. By fpqc descent (cf. EGA IV$_2$, 2.6.4), it is enough to prove that

$$u' \otimes_k \overline{k}$$

is universally closed, where $\overline{k}$ is an algebraic closure of k. Since C is of finite type over k, the scheme $C \otimes_k \overline{k}$ is the disjoint union of finitely many connected components $C'_1, \ldots, C'_n$ of $G \otimes_k \overline{k}$, and it is enough to prove the assertion for each C'_i. We are thus reduced to the case in which k is algebraically closed.

Let g be a closed point of C, and let

$$u^0 : G^0 \longrightarrow H$$

be the restriction of u to G^0. Then

$$u' = r_{u(g)} \circ u^0 \circ r_{g^{-1}},$$

where r_g denotes right translation by g. Thus, to prove that u' is proper, it is enough to prove that u^0 is proper. By hypothesis u is locally quasi-finite, so the nonempty fiber $\ker u$ is discrete. We have seen that u^0 is of finite type, hence its fiber $\ker u^0$ is finite (cf. EGA II, 6.2.2), and is therefore nonempty and proper. Proposition 1.4 now shows that u^0 is proper.

Corollary 1.4.2.— *Let A be an Artinian local ring, and let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type. The following assertions are equivalent:*[15]

(i) *u is a closed immersion;*

(ii) *u is a monomorphism;*

(iii) *$\ker u$ is trivial, that is, isomorphic to the unit A-group.*

In particular, every group subscheme[16] *of H is closed.*

It is clear that (i) implies (ii). Considering the functors represented by G and H, one sees immediately that (ii) and (iii) are equivalent. Finally, if $\ker u$ is the unit A-group, it is a proper nonempty fiber. Proposition 1.4 therefore shows that u is a proper monomorphism; it is of finite presentation because H is locally Noetherian (EGA IV$_1$, 1.6.1), and hence is a closed immersion (EGA IV$_3$, 8.11.5).

The final assertion follows because H is locally Noetherian, so every immersion $G \to H$ is quasi-compact (EGA I, 6.6.4).

Counterexample 1.4.3.— *Let k be a field of characteristic zero, let G be the constant k-group $\mathbb{Z}$, and let H be the additive k-group $\mathbb{G}_{a,k}$. Let $u : G \to H$ be a morphism of k-groups. If $u \neq 0$, then $\ker u = 0$, but u is not a closed immersion (cf. 1.2.2).*

We shall use below the following two results, which should have appeared in Exposé VI$_A$.

Lemma 1.5.— *Let k be a field and let G be a k-group locally of finite type. Then:*

[14] Editor's note: we have expanded the original in what follows.

[15] Editor's note: the hypothesis that G and H be locally of finite type may be removed, because [Per76], 4.2.4, states that every quasi-compact monomorphism $u : G \to H$ between group schemes over a field k is a closed immersion. See also [DG70], III.3.7.2(b), for the case in which G and H are affine; in that case every morphism $G \to H$ is affine (EGA II, 1.6.2(v)), hence quasi-compact.

[16] Editor's note: in this particular case, see also Exposé VI$_A$, 0.5.2, which holds without any finiteness hypothesis.

(i) *all the irreducible components of G have the same dimension;*

(ii) *for every $g \in G$,*

$$\dim_g G = \dim G.$$

[17] Assertion (i) has already been proved in Exposé VI$_A$, 2.4.1, and (ii) follows. Indeed, let $g \in G$, and let C be the connected component of G containing g. By definition (EGA 0_{IV}, 14.1.2), $\dim_g G$ is the infimum of the integers $\dim U$, where U ranges over the open neighborhoods of g. Hence

$$\dim_g G = \dim U_0$$

for some U_0, which we may suppose is contained in C (because $\dim V \leq \dim U$ when $V \subset U$). Since C is irreducible and of finite type over k (Exposé VI$_A$, 2.4.1),

$$\dim U_0 = \dim C = \mathrm{trdeg}_k\, \kappa(\xi),$$

where ξ is the generic point of C (EGA IV$_2$, 5.2.1). Therefore

$$\dim_g G = \dim C = \dim G.$$

Proposition 1.6.—[18] *Let S be a scheme of characteristic zero, and let G be an S-group scheme locally of finite presentation over S at every point of the identity section $\epsilon(S)$. The scheme G is smooth over S at every point of the identity section if and only if the $\mathcal{O}_S$-module*

$$\omega_{G/S} = \epsilon^*(\Omega^1_{G/S}),$$

called the conormal module of the identity section of G, is locally free.

Recall that a scheme S is said to be of characteristic zero if the field $\kappa(x)$ has characteristic zero for every closed point $x \in S$.

Recall also (Exposé II, 4.11) that, if $\pi : G \to S$ is the structural morphism, then

$$\Omega^1_{G/S} = \pi^*(\omega_{G/S}).$$

Thus it is equivalent to say that the $\mathcal{O}_S$-module $\omega_{G/S}$ is locally free or that the $\mathcal{O}_G$-module $\Omega^1_{G/S}$ is locally free.

If there is an open neighborhood U of $\epsilon(S)$ that is smooth over S, then by EGA IV$_4$, 17.2.3, $\Omega^1_{U/S}$ is locally free of finite type, and so is

$$\omega_{G/S} = \epsilon^*(\Omega^1_{U/S}).$$

Conversely, if $\omega_{G/S}$ is locally free, then so is

$$\Omega^1_{G/S} = \pi^*(\omega_{G/S}).$$

Since S has characteristic zero, the Jacobian criterion (EGA IV$_4$, 16.12.2) shows that G is differentially smooth over S. It follows from EGA IV$_4$, 17.12.5, that G is smooth over S at every point of the identity section.

Corollary 1.6.1 (Cartier).— *Let k be a field of characteristic zero. Every k-group locally of finite type over k is smooth over k.*

Indeed, the k-module $\omega_{G/k}$ is then plainly free. By 1.6, G is smooth over k at the identity point e, and hence smooth over k by 1.3.1.

[17] Editor's note: the implication (ii) $\Rightarrow$ (i) is a general fact (cf. EGA 0_{IV}, 14.1.6), while (i) $\Rightarrow$ (ii) follows from the fact that if X is an irreducible scheme of finite type over a field, then $\dim X = \dim U$ for every nonempty open subset U of X. The essential point here is therefore assertion (i), which was already established in an addition to Exposé VI$_A$, 2.4.1. We have modified the statement and proof accordingly.

[18] Editor's note: in the statement, we have replaced "along the identity section" by "at every point of the identity section." At the end of the proof, we have also made explicit the cited results from EGA IV$_4$.

2. "Open properties" of groups and group morphisms locally of finite presentation

2.0.— In all that follows, S will denote an arbitrary scheme; an S-group scheme will be called an S-group. Given an S-group G, we shall denote by ϵ the identity section, by

$$c : G \longrightarrow G$$

the inversion morphism, and by

$$\mu : G \times_S G \longrightarrow G$$

the multiplication morphism. For every S-scheme X, we shall denote by π, or by π_X, the structural morphism

$$X \longrightarrow S.$$

Given a property $\mathscr{P}(u)$ of a morphism of S-schemes

$$u : X \longrightarrow Y,$$

we shall say that $\mathscr{P}(u)$ is *stable under base change* if, whenever u satisfies $\mathscr{P}(u)$, the same is true of the morphism $u_{(Y')}$, for every S-morphism $Y' \to Y$. We say that $\mathscr{P}(u)$ is *local in nature for the topology* $\mathcal{T}$ (cf. Exposé IV, §§ 4 and 6) if $\mathscr{P}(u)$ satisfies the following two conditions:

a) $\mathscr{P}(u)$ is stable under base change;

b) whenever there is a family of S-morphisms $Y_i \to Y$, covering for the topology $\mathcal{T}$, such that each morphism $u_{(Y_i)}$ satisfies $\mathscr{P}(u)$, then u satisfies $\mathscr{P}(u)$.

Proposition 2.1.— *Let $\mathscr{P}(u)$ be a property of a morphism of S-schemes, local in nature for the* fpqc *topology. Let $u : G \to H$ be a morphism of S-groups. Suppose that G is flat and universally open over S.*

Let W be the largest open subset of H over which u satisfies the property $\mathscr{P}(u)$, and let $V = u^{-1}(W)$. Then

$$U = \pi_G(V)$$

is open in S, and V is an open group subscheme of $G|_U$.

The existence of a largest open subset W of H over which u satisfies $\mathscr{P}(u)$ follows from the fact that $\mathscr{P}(u)$ is local in nature for the Zariski topology. Since π_G is universally open, $\pi_G(V)$ is an open subset U of S. It is enough to show that V is a group subscheme of $G|_U$. We may therefore suppose that $U = S$.

Set

$$G' = G \times_S V, \qquad H' = H \times_S V, \qquad V' = V \times_S V, \qquad W' = W \times_S V,$$

and $u' = u_{(V)}$. Let W'_1 be the largest open subset of H' over which u' satisfies $\mathscr{P}(u)$. Since V is flat and universally open over S, the same is true of H' over H, and Lemma 2.1.1 below shows that

$$W'_1 = W'.$$

Now consider the automorphism a (respectively, b) of the V-scheme G' (respectively, H'), namely right translation by the inverse of the diagonal section δ (respectively, by the inverse of $u(\delta)$), defined by

$$a(g, v) = (g \cdot v^{-1}, v), \qquad \big(\text{respectively, } b(h, v) = (h \cdot u(v^{-1}), v)\big),$$

for every morphism $T \to S$, every $g \in G(T)$, $v \in V(T)$, and $h \in H(T)$. It is clear that

$$u' \circ a = b \circ u',$$

which shows that W' is stable under b, hence that V' is stable under a, so that V is a group subscheme of G.

Lemma 2.1.1.— *Let $\mathscr{P}(u)$ be a property of an S-morphism u, local in nature for the* fpqc *topology. Consider a cartesian square of morphisms of S-schemes:*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ \downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & Y \end{array}$$

where g is flat and open. Let W (respectively, W'_1) be the largest open subset of Y (respectively, Y') over which f (respectively, $f' = f_{(Y')}$) satisfies $\mathscr{P}(u)$. Then

$$W'_1 = W \times_Y Y'.$$

Set $W' = W \times_Y Y'$. Since $\mathscr{P}(u)$ is stable under base change, we have

$$W' \subset W'_1.$$

Since g is open,

$$W_1 = g(W'_1)$$

is an open subset of Y. Set

$$V_1 = f^{-1}(W_1), \qquad V'_1 = V_1 \times_{W_1} W'_1;$$

it is clear that $V'_1 = f'^{-1}(W'_1)$. Since g is flat and open, the morphism

$$W'_1 \longrightarrow W_1 = g(W'_1)$$

induced by g is faithfully flat and open, hence is a covering for the fpqc topology (cf. Exposé IV, 6.3.1(iv)). Since the morphism

$$V'_1 \longrightarrow W'_1$$

induced by f' satisfies $\mathscr{P}(u)$, the same is true of the morphism $V_1 \to W_1$ induced by f. Thus

$$W_1 \subset W, \qquad W'_1 \subset g^{-1}(W_1) \subset g^{-1}(W) = W',$$

and therefore $W' = W'_1$.

Remark 2.1.2.— *A great many properties of a morphism are local in nature for the* fpqc *topology; among them are the properties of being flat, (universally) open, (locally) of finite type, of finite presentation, quasi-finite (cf. EGA IV_2, 2.5.1, 2.6.1, and 2.7.1), smooth, étale, and unramified (EGA IV_4, 17.7.3).*

The proof of 2.1 in fact uses only base changes by flat morphisms. The proposition therefore applies to a property satisfying condition b) *of 2.0 relative to the* fpqc *topology and stable under base change by flat morphisms (for example, the property of being quasi-compact and dominant).*

Of course, one can state an analogous proposition concerning properties local in nature for a topology $\mathcal{T}$ finer than the Zariski topology, the condition to be verified by G then being that π_G be universally open and covering for the topology $\mathcal{T}$.

In particular, if G is flat and locally of finite presentation over S, there is an analogous statement for properties stable under base change by flat morphisms locally of finite presentation and satisfying condition b) *of 2.0 relative to the* fpqc *topology (for example, the properties of being regular, reduced, Cohen–Macaulay, and so on; cf. EGA IV$_2$, 6.8).*

Proposition 2.2.— *Let G and H be two S-groups, and let $u : G \to H$ be a morphism of S-groups. Then:*[19]

(i) *Suppose that G or H is flat over S, and that G or H is locally of finite presentation over S. Let V be the largest open subset of G such that the restriction of u to V is flat and locally of finite presentation (respectively, smooth, or étale). Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.*

(ii) *Suppose that G or H is universally open over S, and let V be the largest open subset of G such that the restriction of u to V is universally open. Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.*

We first prove (i). We show that the restriction π_V of π_G to V is flat and locally of finite presentation:

a) if π_G is flat (respectively, locally of finite presentation), then so is π_V;

b) if π_H is flat (respectively, locally of finite presentation), then so is π_V, as the composite of the restriction of u to V with π_H.

Thus, in the four cases contemplated in the statement, π_V is flat and locally of finite presentation, hence universally open (EGA IV$_2$, 2.4.6). Set $U = \pi_V(V)$; then U is open in S. It is therefore enough to show that V is an open group subscheme of $G|_U$, and we may suppose that $U = S$.

Set

$$G' = G \times_S V, \qquad H' = H \times_S V, \qquad V' = V \times_S V, \qquad \text{and} \qquad u' = u_{(V)}.$$

Since V is flat and locally of finite presentation over S, the same is true of H' over H. By EGA IV$_4$, 17.7.4, V' is then the largest open subset of G' such that the restriction of u' to V' is flat and locally of finite presentation (respectively, smooth, or étale). With the automorphisms a and b defined as in the proof of 2.1, it is clear that V' is stable under a, hence that V is a group subscheme of G.

We now prove (ii). The restriction π_V of π_G to V is universally open, either because π_G is universally open, or because it is the composite of the restriction of u to V with π_H, when π_H is universally open. Set $U = \pi_V(V)$; then U is open in S. It is enough to show that V is an open group subscheme of $G|_U$, and we may suppose that $U = S$.

Set, as before,

$$G' = G \times_S V, \qquad H' = H \times_S V, \qquad V' = V \times_S V, \qquad \text{and} \qquad u' = u_{(V)}.$$

[19] Editor's note: in case (i), the open subset V consists of all the points of G at which u is smooth, respectively étale, respectively of finite presentation and flat; cf. editor's note 6. By contrast, in case (ii), V is the largest open subset contained in the set E of points of G at which u is universally open, but E is not necessarily open, as the following example shows (EGA IV$_3$, 14.1.3(i)). Let k be a field, let $H = S = \operatorname{Spec} A$, where $A = k[x]$, and let G be the S-group $\operatorname{Spec} A[y]/(xy)$. Then E is the identity section of G, which is not open.

Then $\pi_V : V \to S$ is surjective and universally open; the same is therefore true of $G' \to G$. Consequently, V' is the largest open subset of G' such that the restriction of u' to V' is universally open, by EGA IV_3, 14.3.4(i),(ii). With the automorphisms a and b defined as before, it is clear that V' is stable under a, hence that V is a group subscheme of G.

Corollary 2.3.— *Let G be an S-group, and let V be the largest open subset of G that is flat and locally of finite presentation (respectively, smooth, étale, or universally open) over S. Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.*

It is enough to apply 2.2 when H is the identity S-group and u is the unique morphism of S-groups $G \to H$, since π_H is then an isomorphism and

$$\pi_G = \pi_H \circ u.$$

Corollary 2.4.— *Let G be an S-group. If there is a neighborhood X of the identity section having one (or more) of the following properties:*

X is flat and locally of finite presentation (respectively, smooth, étale, or universally open) over S,

then there is an open group subscheme of G having the same properties.

It is enough to apply 2.3, observing that here, with the notation of 2.2, we have

$$\epsilon(S) \subset V,$$

and hence $U = S$.

Proposition 2.5.—[20] *Let $u : G \to H$ be a morphism of S-groups.*

(i) *Suppose that G (respectively, H) is of finite presentation and flat (respectively, of finite type) over S at the points of its identity section. Then the sets*

$$U_{\text{plat}} \supset U_{\text{lisse}} \supset U_{\text{ét.}},$$

formed by the points $s \in S$ such that u_s is flat (respectively, smooth, or étale), are open in S.

If, moreover, G (respectively, H) is locally of finite presentation and flat (respectively, locally of finite type) over S, then the set V_{plat} (respectively, V_{lisse} or $V_{\text{ét.}}$) of points of G at which u is flat (respectively, smooth, or étale) is the inverse image under π_G of U_{plat} (respectively, U_{lisse} or $U_{\text{ét.}}$).

(ii) *Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$ (a condition satisfied if G is of finite type over S at the points of the identity section (1.3.1.1)), and that u is locally of finite type (respectively, locally of finite presentation) at the points of the identity section of G. Then the sets*

$$U_{\text{l.q.f.}} \supset U_{\text{n.r.}},$$

formed by the $s \in S$ such that u_s is locally quasi-finite (respectively, unramified), are open in S.

If, moreover, u is locally of finite type (respectively, locally of finite presentation), then the set $V_{\text{l.q.f.}}$ (respectively, $V_{\text{n.r.}}$) of points of G at which u is quasi-finite (respectively, unramified) is the inverse image under π_G of $U_{\text{l.q.f.}}$ (respectively, $U_{\text{n.r.}}$).

[20] Editor's note: we have expanded the statement and proof, and in (i) we have weakened the hypothesis on H, replacing "of finite presentation" by "of finite type."

We prove (i). First note (1.3.1.1) that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Let Y be the open subset of H consisting of the points at which π_H is of finite type, and let X_1 be the open subset of $u^{-1}(Y)$ consisting of the points at which π_G is of finite presentation. By EGA IV$_3$, 11.3.1, the set X of points of X_1 at which π_G is of finite presentation and flat is open in X_1, hence in G.

Let

$$u_X : X \longrightarrow Y$$

be the morphism induced by u. Since Y (respectively, X) contains the identity section of H (respectively, G), the restriction π_X of π_G to X is surjective, and the morphism

$$S \longrightarrow X$$

induced by ϵ_G is an S-section of X.

Consider the sets

$$W_{\text{plat}} \supset W_{\text{lisse}} \supset W_{\text{ét.}},$$

consisting of the points of X at which u_X is flat (respectively, smooth, or étale). It is clear that W_{lisse} and $W_{\text{ét.}}$ are open in X; cf. Editor's note 6. On the other hand, since π_Y is locally of finite type and

$$\pi_X = \pi_Y \circ u_X$$

is locally of finite presentation, EGA IV$_1$, 1.4.3(v), shows that u_X is locally of finite presentation. Consequently, the flatness locus W_X of u_X is open in X (EGA IV$_3$, 11.3.1).

Let $x \in X$ and set $s = \pi_X(x)$. Then, by EGA IV$_2$, 11.3.10, combined with EGA IV$_4$, 17.5.1 (respectively, 17.6.1), x belongs to W_{plat} (respectively, W_{lisse} or $W_{\text{ét.}}$) if and only if u_s is flat (respectively, smooth, or étale) at the point x, or, equivalently by 1.3, if and only if u_s is flat (respectively, smooth, or étale).

Consequently, for

$$\text{“}\flat = \text{flat, smooth, or étale,”}$$

we have

$$U_\flat = \epsilon_G^{-1}(W_\flat), \qquad W_\flat = \pi_X^{-1}(U_\flat).$$

Since $W_\flat$ is open in X, hence in G, the first equality shows that $U_\flat$ is open in S.

The second assertion of (i) follows from what we have just seen, for then

$$Y = H, \qquad X = G, \qquad V_\flat = W_\flat = \pi_G^{-1}(U_\flat).$$

We prove assertion (ii). Let

$$V_{\text{l.t.f.}} \supset V_{\text{l.q.f.}}$$

be the open subsets of G consisting of the points at which u is of finite type (respectively, quasi-finite). Set

$$X = V_{\text{l.t.f.}},$$

and let u_X denote the restriction of u to X. By hypothesis, X contains $\epsilon(S)$. Let $x \in X$ and set $s = \pi_X(x)$.

If u is quasi-finite at x, then, by base change, u_s is quasi-finite at x; and since G_s is assumed to be locally of finite type, u_s is locally quasi-finite by 1.3.1.

Conversely, if u_s is locally quasi-finite, then

$$u_X^{-1}(u_X(x)) \subset u_s^{-1}(u_s(x))$$

is a finite set. Since

$$u_X : X \longrightarrow H$$

is locally of finite type, Chevalley's semicontinuity theorem (EGA IV_3, 13.1.3) shows that there is an open neighborhood W of x in X such that the fiber

$$u_X^{-1}(u_X(x'))$$

is discrete for every $x' \in W$. Hence u_X is quasi-finite at x, by EGA II, 6.2.2 (and EGA III_2, Err_{III} 20).

Consequently,

$$U_{\text{l.q.f.}} = \epsilon_G^{-1}(V_{\text{l.q.f.}}), \qquad V_{\text{l.q.f.}} = \pi_X^{-1}(U_{\text{l.q.f.}}),$$

and the first equality shows that $U_{\text{l.q.f.}}$ is open in S.

If, moreover, u is locally of finite type over S, then $X = G$, and hence $V_{\text{l.q.f.}}$ is the inverse image under π_G of $U_{\text{l.q.f.}}$. This proves the first half of (ii).

Now consider the open subsets

$$V_{\text{l.p.f.}} \supset V_{\text{n.r.}},$$

consisting of the points at which u is of finite presentation (respectively, unramified), and suppose that

$$Z = V_{\text{l.p.f.}}$$

contains the identity section of G. Let $z \in Z$; set

$$s = \pi_Z(z), \qquad h = u_Z(z).$$

If u is unramified at z, then, by base change, u_s is unramified at z; and since G_s is assumed to be locally of finite type, u_s is unramified by 1.3.1.

Conversely, if u_s is unramified at the point z, then the fiber

$$u_s^{-1}(h) = u^{-1}(h)$$

is unramified over $\kappa(h)$ at the point z. Since Z is an open subset of G, the fiber

$$u_Z^{-1}(h) = u_s^{-1}(h) \cap Z$$

is unramified over $\kappa(h)$ at the point z. Since u_Z is locally of finite presentation, EGA IV_4, 17.4.1, implies that u_Z is unramified at the point z.

Thus

$$U_{\text{n.r.}} = \epsilon_G^{-1}(V_{\text{n.r.}}), \qquad V_{\text{n.r.}} = \pi_Z^{-1}(U_{\text{n.r.}}),$$

and the first equality shows that $U_{\text{n.r.}}$ is open in S.

If, moreover, u is locally of finite presentation over S, then $Z = G$, and hence $V_{\text{n.r.}}$ is the inverse image under π_G of $U_{\text{n.r.}}$. This completes the proof of assertion (ii).

Corollary 2.6.— *Let $u : G \to H$ be a morphism of S-groups that is radicial (as is the case if u is a monomorphism; EGA I, 3.5.4). Suppose that G (respectively, H) is locally of finite presentation and flat (respectively, locally of finite type) over S. Then the set U of $s \in S$ such that u_s is an open immersion is open in S, and the restriction of u over U is an open immersion.*

By 2.5(i), the set U' of points $s \in S$ such that u_s is étale is open in S. Since u is radicial, so is u_s, for every $s \in S$; hence EGA IV_4, 17.9.1, gives

$$U = U',$$

which shows that U is open. Finally, by 2.5(i), the restriction of u over U is étale; since u is radicial, this restriction is an open immersion (EGA IV_4, 17.9.1).

Proposition 2.7.— *Let G be an S-group. The following conditions are equivalent:*

(i) *G is unramified over S at the points of the identity section;*

(ii) *the identity section is an open immersion;*

(iii) *G is of finite presentation over S at the points of the identity section, and, for every $s \in S$, G_s is unramified over $\kappa(s)$.*

If, moreover, G is locally of finite presentation over S, then each of the three preceding conditions is equivalent to the following:

(iv) *G is unramified over S.*

The equivalence of assertions (i) and (ii) follows from the more general Lemma 2.7.1 below. Note (1.3.1.1) that either condition (i) or condition (iii) implies that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Thus, by EGA IV_4, 17.4.1, assertion (i) is equivalent to the assertion that, for every $s \in S$, G_s is unramified over $\kappa(s)$ at the point e_s, the identity of G_s, or, equivalently by 1.3.1, to the assertion that G_s is unramified over $\kappa(s)$. Consequently, assertions (i) and (iii) are equivalent. Finally, if G is locally of finite presentation over S, assertions (iii) and (iv) are equivalent (EGA IV_4, 17.4.1).

Lemma 2.7.1.— *Let G be an S-scheme equipped with a section ϵ. In order that G be unramified over S at the points of this section, it is necessary and sufficient that ϵ be an open immersion.*

The condition is necessary by EGA IV_4, 17.4.1, a)⇒b″). Conversely, if ϵ is an open immersion, then the restriction to $\epsilon(S)$ of the structural morphism

$$G \longrightarrow S$$

is an isomorphism; hence G is unramified over S at the points of $\epsilon(S)$.

Corollary 2.8.—[21] *Let $u : G \to H$ be a morphism of S-groups. Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$.*

a) *If u is locally of finite type, the following conditions are equivalent:*

(i) *u is locally quasi-finite;*
(ii) *for every $s \in S$, the morphism $u_s : G_s \to H_s$ is locally quasi-finite;*
(iii) $\operatorname{Ker} u$ *is locally quasi-finite over S;*
(iv) *the fibers of* $\operatorname{Ker} u$ *are discrete.*

b) *If u is locally of finite presentation, the following conditions are equivalent:*

(v) *u is unramified;*
(vi) *for every $s \in S$, the morphism $u_s : G_s \to H_s$ is unramified;*
(vii) $\operatorname{Ker} u$ *is unramified;*
(viii) *the identity section $S \to \operatorname{Ker} u$ is an open immersion.*

The equivalences

$$\text{(i)} \Longleftrightarrow \text{(ii)} \qquad \text{and} \qquad \text{(v)} \Longleftrightarrow \text{(vi)}$$

follow from 2.5(ii), and Reminder 2.8.1 below shows that

$$\text{(i)} \Longrightarrow \text{(iii)} \qquad \text{and} \qquad \text{(v)} \Longrightarrow \text{(vii)}.$$

[21] Editor's note: we have changed the presentation by separating the assertions concerning the "quasi-finite" case from those concerning the "unramified" case.

For every $s \in S$, denote by e_s the identity element of H_s. Then (iii) (respectively, (vii)) implies that, for every $s \in S$,

$$(\operatorname{Ker} u)_s = \operatorname{Ker} u_s = u_s^{-1}(e_s)$$

is locally quasi-finite (respectively, unramified) over $\kappa(s) = \kappa(e_s)$. It follows that u_s is quasi-finite (respectively, unramified) at the identity point of G_s, and hence, by 1.3, that u_s is locally quasi-finite (respectively, unramified). Thus

$$\text{(iii)} \Longrightarrow \text{(ii)} \qquad \text{and} \qquad \text{(vii)} \Longrightarrow \text{(vi)}.$$

Finally,

$$\text{(ii)} \Longleftrightarrow \text{(iv)} \quad \text{by 1.4.1,} \qquad \text{(vii)} \Longleftrightarrow \text{(viii)} \quad \text{by 2.7.}$$

Reminder 2.8.1.— Recall (cf. Exposé I, 2.3.6.1) that, given a morphism $u : G \to H$ of S-groups, the *kernel of* u, denoted $\operatorname{Ker} u$, is the group subfunctor of G defined, for every morphism $f : T \to S$, by

$$(\operatorname{Ker} u)(T) = \{a \in G(T) \mid u \circ a = \epsilon_H \circ f\}.$$

By loc. cit. (or EGA I, 4.4.1), this functor is represented by the S-group

$$G \times_H S = u^{-1}(\epsilon_H(S)),$$

which is denoted simply by $\operatorname{Ker} u$. In particular, the structural morphism

$$\operatorname{Ker} u \longrightarrow S$$

is obtained from u by base change.

Lemma 2.9.— *Let $\pi : X \to S$ be a morphism admitting an S-section ϵ.*

(i) *If π is injective, then it is integral.*[(*)]

(ii) *If π is locally of finite type and, for every $s \in S$, π_s is an isomorphism, then π is an isomorphism.*[(**)]

First observe that, by Lemma 2.9.1 below, π, being a homeomorphism, is an affine morphism.

If π is injective, then ϵ is surjective. Since ϵ is a surjective immersion, $\epsilon(S)$ is isomorphic to the closed subscheme of X defined by a nil ideal $\mathcal{I}$ of $\mathcal{O}_X$. Since ϵ is an S-section of π, there is a direct-sum decomposition

$$\mathcal{O}_X = \mathcal{O}_S \oplus \mathcal{I}$$

of $\mathcal{O}_S$-modules. Since $\mathcal{I}$ is a nil ideal of $\mathcal{O}_X$, it is plainly integral over $\mathcal{O}_S$. Thus $\mathcal{O}_X$ is integral over $\mathcal{O}_S$, and π is integral.

Now suppose that π is locally of finite type. Since $\pi \circ \epsilon = \mathrm{id}_S$, the section ϵ is locally of finite presentation (cf. EGA IV$_1$, 1.4.3(v)); hence $\mathcal{I}$ is an ideal of finite type in $\mathcal{O}_X$ (EGA IV$_1$, 1.4.1). For every $s \in S$, one has

$$\mathcal{O}_{X_s} = \kappa(s) \oplus \mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s).$$

(*) This is also a special case of EGA IV$_4$, 18.12.11, since π is plainly a universal homeomorphism.
(**) This is also an immediate consequence of EGA IV$_4$, 18.12.6.

By hypothesis, ϵ_s is an isomorphism, so

$$\mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s) = 0$$

for every $s \in S$, and therefore, a fortiori,

$$\mathcal{I} \otimes_{\mathcal{O}_X} \kappa(x) = 0$$

for every $x \in X$. Nakayama's lemma now gives $\mathcal{I} = 0$, and consequently π is an isomorphism.

Lemma 2.9.1.— *Let $f : X \to Y$ be a morphism of schemes that is a homeomorphism. Then f is an affine morphism.*(***)

It is enough to show that every point $y \in Y$ has an open neighborhood W such that the restriction of f over W is affine. Let $y \in Y$, and let V be an affine open neighborhood of y in Y. Set

$$V' = f^{-1}(V).$$

Then V' is an open neighborhood of $x = f^{-1}(y)$ in X. There is an affine open neighborhood W' of x in X contained in V'. Set

$$W = f(W').$$

Then W is an open neighborhood of y in Y, contained in the affine scheme V, and therefore W is separated. Since W' is affine, the restriction of f over W is an affine morphism (EGA II, 1.2.3).

Corollary 2.10.— *Let G be an S-group locally of finite type. Suppose that, for every $s \in S$, G_s is the identity $\kappa(s)$-group. Then G is the identity S-group.*

More generally:

Corollary 2.11.— *Let $f : X \to Y$ be an S-morphism locally of finite type. In order that f be a monomorphism, it is necessary and sufficient that f_s be a monomorphism for every $s \in S$.*

The condition is plainly necessary; we prove that it is sufficient. If f_s is a monomorphism for every $s \in S$, then, a fortiori, f_y is a monomorphism for every $y \in Y$. We may therefore assume that $Y = S$.

By EGA I, 5.3.8, to prove that f is a monomorphism it is enough to show that

$$\Delta_f : X \longrightarrow X \times_S X$$

is an isomorphism, or, equivalently, that the first projection

$$p : X \times_S X \longrightarrow X$$

is an isomorphism. If f_s is a monomorphism, EGA I, 5.3.8 likewise shows that the first projection

$$X_s \times_{\kappa(s)} X_s \longrightarrow X_s,$$

which is identified with p_s, is an isomorphism. But p has the S-section Δ_f; hence Lemma 2.9 shows that, if p_s is an isomorphism for every $s \in S$, then p itself is an isomorphism.

Corollary 2.12.— *Let G be an S-scheme admitting an S-section ϵ, and suppose that the structural morphism $\pi : G \to S$ is closed. Let $s \in S$ be such that π is of finite presentation at $\epsilon(s)$, and such that*

$$\epsilon_s : \mathrm{Spec}\big(\kappa(s)\big) \longrightarrow G_s$$

(***)Cf. EGA IV_4, 18.12.7.1 for a slightly more general result, proved by the same argument.

is an isomorphism (or, equivalently, such that

$$\pi_s : G_s \longrightarrow \operatorname{Spec} \kappa(s)$$

is an isomorphism). Then there exists an open neighborhood U of s in S such that

$$\epsilon|_U : U \longrightarrow G \times_S U$$

is an isomorphism.

Let V be the set of points of G at which π is unramified. This set is known to be open (cf. Editor's note 6), and it contains $\epsilon(s)$. Hence

$$W = \epsilon^{-1}(V)$$

is an open subset of S containing s, and π is unramified at $\epsilon(t)$ for every $t \in W$. Since π is closed, the same is true of $\pi|_W$, and we may therefore suppose that $W = S$.

By 2.7.1,

$$X = G - \epsilon(S)$$

is then a closed subset of G that does not meet G_s. Since π is closed,

$$F = \pi(X)$$

is a closed subset of S that does not contain s. Set

$$U = S - F.$$

Then U is an open subset of S, and $\epsilon|_U$ is an isomorphism from U onto $G|_U$.

3. Identity component of a group locally of finite presentation

3.0.— Given a subset A (respectively, B) of an S-scheme X (respectively, Y), by abuse of notation

$$A \times_S B$$

will denote the subset of $X \times_S Y$ consisting of the points whose first projection belongs to A and whose second projection belongs to B.

Given a subset A of an S-group G, we shall say that A is *stable under the group law of* G if

$$c(A) \subset A \qquad \text{and} \qquad \mu(A \times_S A) \subset A.$$

Definition 3.1.— Let G be an S-functor in groups satisfying the following condition:

$$\text{for every } s \in S, \text{ the functor } G_s = G \otimes_S \kappa(s) \text{ is representable.} \qquad (+)$$

We then denote by $\underline{G_s^0}$ the connected component of the identity element of the $\kappa(s)$-group G_s.[22] We define a sub-S-functor in groups of G, called the *identity component of* G and denoted by G^0, by setting, for every morphism $T \to S$,

$$G^0(T) = \{u \in G(T) \mid \text{for every } s \in S,\ u_s(T_s) \subset \underline{G_s^0}\}.$$

[22] Editor's note: this is a slightly abusive notation, but it is compatible with the notation of Exposé VI$_A$, 2.3 when this connected component is the underlying topological space of an open group subscheme G^0 of G; cf. Exposé VI$_A$, 2.2.bis.

Thus we have defined the functor

$$G \longmapsto G^0$$

from

$$(\widehat{\mathbf{Sch}}_{/S})\text{-gr.} \quad \text{to} \quad (\widehat{\mathbf{Sch}}_{/S})\text{-gr.}$$

Remark 3.2.— (i) Let G be an S-functor in groups satisfying condition (+). Then

$$\mathrm{Lie}(G^0/S) = \mathrm{Lie}(G/S)$$

by Exposé II.[23] Indeed, for every S-scheme T, denote by I_T the T-scheme of dual numbers over T, and by

$$\tau : T \longrightarrow I_T$$

the zero section (cf. Exposé II, 2.1). By definition,

$$\mathrm{Lie}(G/S)(T) = \{u \in G(I_T) \mid u \circ \tau = e\},$$

where $e : T \to G$ denotes the composite of $T \to S$ with the identity section $S \to G$. Likewise,

$$\begin{aligned}\mathrm{Lie}(G^0/S)(T) &= \{u \in G^0(I_T) \mid u \circ \tau = e\} \\ &= \{u \in G(I_T) \mid u \circ \tau = e \text{ and } u_s((I_T)_s) \subset \underline{G^0_s} \text{ for every } s \in S\}.\end{aligned}$$

For every $s \in S$, the schemes T_s and

$$(I_T)_s = I_{T_s}$$

have the same underlying set. Hence, if $u \in \mathrm{Lie}(G/S)(T)$, then

$$u_s(I_{T_s}) = e_s,$$

where e_s denotes the identity point of G_s, and therefore $u \in \mathrm{Lie}(G^0/S)(T)$. Thus the inclusion

$$\mathrm{Lie}(G^0/S)(T) \subset \mathrm{Lie}(G/S)(T)$$

is an equality for every T, and consequently

$$\mathrm{Lie}(G^0/S) = \mathrm{Lie}(G/S).$$

(ii) Let G and G' be two S-functors in groups satisfying (+). Then:

a) if $G \subset G'$, then $G^0 \subset G'^0$;

b) if $G \subset G'$ and $G'^0 \subset G$, then $G^0 = G'^0$;

c) if, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$, then G^0 satisfies condition (+), by Exposé VI_A, 2.3, and

$$(G^0)^0 = G^0.$$

Proposition 3.3.— *Let G be an S-functor in groups satisfying condition (+), and let S' be an S-scheme. Then*

$$(G \times_S S')^0 = G^0 \times_S S'.$$

In other words, the functor $G \mapsto G^0$ commutes with base change; that is, the following diagram is commutative:

[23] Editor's note: we have added what follows.

$$\begin{array}{ccc} (\widehat{\mathrm{Sch}}_{/S})\text{-gr.} & \xrightarrow{G\mapsto G^0} & (\widehat{\mathrm{Sch}}_{/S})\text{-gr.} \\ \downarrow & & \downarrow \\ (\widehat{\mathrm{Sch}}_{/S'})\text{-gr.} & \longrightarrow & (\widehat{\mathrm{Sch}}_{/S'})\text{-gr.} \end{array}$$

It is enough to check that, for every $s' \in S'$, whose image in S we denote by s,

$$\begin{aligned} \big((G \times_S S') \otimes_{S'} \kappa(s')\big)^0 &= \big(G_s \otimes_{\kappa(s)} \kappa(s')\big)^0 \\ &= G_s^0 \otimes_{\kappa(s)} \kappa(s'). \end{aligned}$$

This follows from Exposé VI$_A$, 2.1.2. Note, for later use in 4.2, that we have not used the group structure of G_s, but only the fact that G_s^0 has a rational point, namely $\epsilon(s)$, and is therefore geometrically connected (see also EGA IV$_2$, 4.5.14).

Special Case 3.4.— Let G be an S-group scheme, and denote by $\underline{G^0}$ the subset of G equal to the union of the $\underline{G_s^0}$ as s runs through S. Then $\underline{G^0}$ is a subset of G stable under the group law of G (cf. 3.0), and, for every morphism $S' \to S$,

$$G^0(S') = \{u \in G(S') \mid u(S') \subset \underline{G^0}\}.$$

When $\underline{G^0}$ is an open subset of G, it is clear that G^0 is represented by the subscheme of G induced on the open subset $\underline{G^0}$.

Proposition 3.5.— *Let S be a quasi-compact and quasi-separated scheme, and let G be an S-group with fibers locally of finite type. Then there is a quasi-compact open subset U of G containing $\underline{G^0}$.*

Since the identity section ϵ is an immersion, $\epsilon(S)$ is a quasi-compact subspace of G. Thus there is a quasi-compact open subset V of G containing $\epsilon(S)$. Since S is quasi-separated and V is quasi-compact, V is quasi-compact over S (EGA IV$_1$, 1.2.4). Hence $V \times_S V$ is quasi-compact over S, and therefore quasi-compact. It follows that

$$V \cdot V = \mu(V \times_S V)$$

is quasi-compact. Set

$$V_s = V \cap G_s, \qquad V_s^0 = V \cap \underline{G_s^0}.$$

Then V_s^0 is open in $\underline{G_s^0}$, and is dense in $\underline{G_s^0}$ because G_s^0 is irreducible (Exposé VI$_A$, 2.4). Therefore

$$V_s^0 \cdot V_s^0 = \underline{G_s^0} \qquad \text{(Exposé VI}_A\text{, 0.5)},$$

which shows that

$$V_s \cdot V_s \supset \underline{G_s^0}, \qquad \text{and hence} \qquad V \cdot V \supset \underline{G^0}.$$

Finally, since $V \cdot V$ is quasi-compact, there is a quasi-compact open subset U of G containing $V \cdot V$, and therefore containing $\underline{G^0}$.

Corollary 3.6.— *Let G be an S-group whose fibers are locally of finite type and connected. Then G is quasi-compact over S.*

The assertion is local on S (EGA I, 6.6.1), so we are reduced to the case where S is affine. By 3.5, there is then a quasi-compact open subset U of G containing

$$\underline{G^0} = G.$$

Thus G is quasi-compact, and hence quasi-compact over the affine scheme S (EGA I, 6.6.4(v)).

Proposition 3.7.—[24] *Let G be an S-group locally of finite presentation. Then:*

(i) $\underline{G^0}$ *is ind-constructible in G.*

(ii) *If, moreover, G is quasi-separated over S, and S is quasi-compact and quasi-separated, then $\underline{G^0}$ is constructible.*

(iii) *Consequently, if G is quasi-separated over S, then $\underline{G^0}$ is locally constructible.*

We first prove the first assertion. Since

$$\pi : G \longrightarrow S$$

is locally of finite presentation, given $s \in S$, there are an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s such that

$$\pi(U) \subset V$$

and the induced morphism

$$\pi' : U \longrightarrow V$$

is of finite presentation. Then

$$T = \epsilon^{-1}(U)$$

is an open subset of S. Set

$$W = (\pi')^{-1}(T), \qquad \pi'' = \pi'|_W.$$

Thus $\pi'' : W \to T$ is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ.

For every $t \in T$, since $\underline{G_t^0}$ is irreducible (VI_A, 2.4), $W \cap \underline{G_t^0}$ is dense in $\underline{G_t^0}$, hence irreducible and therefore connected. It is thus the connected component of $(\pi'')^{-1}(t)$ containing $\epsilon''(t)$. It follows from EGA IV_3, 9.7.12, that the union W^0 of the $W \cap \underline{G_t^0}$, for $t \in T$, is locally constructible in W. On the other hand, VI_A, 0.5, gives

$$\underline{G^0} \cap \pi^{-1}(T) = W^0 \cdot W^0;$$

that is, $\underline{G^0} \cap \pi^{-1}(T)$ is the image of

$$W^0 \times_T W^0$$

under the morphism

$$\mu'' : W \times_T W \longrightarrow \pi^{-1}(T)$$

[24] Editor's note: we recall the following definitions and results (cf. EGA 0_{III}, §9.1, and EGA IV_1, §§1.8 and 1.9). Let X be a topological space. (i) An open subset U of X is called *retrocompact* if the inclusion $U \hookrightarrow X$ is quasi-compact, i.e. if $U \cap V$ is quasi-compact for every quasi-compact open subset $V \subset X$. (ii) A subset C of X is called *constructible* if it is a finite union of subsets of the form $U - V$, where U and V are retrocompact open subsets of X. (iii) A subset L of X is called *locally constructible* if, for every $x \in X$, there is an open neighborhood U of x in X such that $L \cap U$ is constructible in U. (If X is quasi-compact and quasi-separated, then L is constructible.) (iv) A subset I of X is called *ind-constructible* if, for every $x \in X$, there is an open neighborhood U of x in X such that $I \cap U$ is a union of locally constructible subsets of U. Now let $f : X \to Y$ be a morphism of schemes. By Chevalley's constructibility theorem (cf. EGA IV_1, 1.8.4 and 1.9.5(viii)), if f is of finite presentation (respectively, locally of finite presentation), then the image under f of every locally constructible subset is locally constructible (respectively, ind-constructible).

induced by μ.[25] Since $W \times_T W$ (respectively, $\pi^{-1}(T)$) is of finite presentation (respectively, locally of finite presentation) over T, the morphism μ'' is locally of finite presentation and quasi-separated, by EGA IV_1, 1.4.3 and 1.2.2. If, moreover, $\pi^{-1}(T)$ is quasi-separated over T, then μ'' is quasi-compact (loc. cit., 1.2.4), hence of finite presentation. Since $W^0 \times_T W^0$ is locally constructible in $W \times_T W$ (because W^0 is locally constructible in W), Chevalley's constructibility theorem (loc. cit., 1.8.4 and 1.9.5(viii)) shows that $\underline{G^0} \cap \pi^{-1}(T)$ is ind-constructible in $\pi^{-1}(T)$, and is locally constructible in $\pi^{-1}(T)$ if G, and hence $\pi^{-1}(T)$, is quasi-separated over T. This proves assertions (i) and (iii).

Now suppose that G is quasi-separated over S, and that S is quasi-compact and quasi-separated. By 3.5, there is a quasi-compact open subset U of G containing $\underline{G^0}$. Since G is quasi-separated over S, G is quasi-separated; hence the open subset U is retrocompact (EGA IV_1, 1.2.7), and it suffices to show that $\underline{G^0}$ is constructible in U (EGA 0_{III}, 9.1.8).

Moreover, U is quasi-compact, hence quasi-compact over S (EGA IV_1, 1.2.4), and is quasi-separated over S. Therefore the restriction of π to U is of finite presentation. By EGA IV_3, 9.7.12, $\underline{G^0}$ is locally constructible in U, hence constructible in U, since U is quasi-compact and quasi-separated (EGA IV_1, 1.8.1). This proves (ii), and it follows that, for every quasi-compact and quasi-separated open subset T of S (for example, every affine open subset of S),

$$\underline{G^0} \cap \pi^{-1}(T)$$

is constructible.

Corollary 3.8.— *Let S_0 be a quasi-compact and quasi-separated scheme, I an increasing filtered preordered set, $(\mathcal{A}_i)_{i\in I}$ a direct system of commutative quasi-coherent $\mathcal{O}_{S_0}$-algebras,*

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i, \qquad S_i = \operatorname{Spec} \mathcal{A}_i \quad (i \in I), \qquad S = \operatorname{Spec} \mathcal{A}$$

(cf. EGA II, 1.3.1). *Let G be an S_0-group scheme locally of finite presentation. Then the canonical map*

$$\varinjlim_i G^0(S_i) \longrightarrow G^0(S)$$

is bijective.

Since G is locally of finite presentation over S_0, the canonical map

$$\varinjlim_i G(S_i) \longrightarrow G(S)$$

is bijective, by EGA IV_2, 8.14.2(c). It follows immediately that the canonical map

$$\varinjlim_i G^0(S_i) \longrightarrow G^0(S)$$

is injective. We prove that it is surjective. Let

$$g \in G^0(S) \subset G(S).$$

There is an $i \in I$ such that g factors, by means of an element $g_i \in G(S_i)$, through $S \to S_i$. By hypothesis,

$$g^{-1}(\underline{G^0}) = S.$$

But $\underline{G^0}$ is ind-constructible in G, by 3.7, so

$$g_i^{-1}(\underline{G^0})$$

[25] Editor's note: we have expanded the original in what follows.

is ind-constructible in S_i. It follows from EGA IV_2, 8.3.4, that there is an index $j \geq i$ such that

$$g_j^{-1}(\underline{G^0}) = S_j,$$

where g_j is the map induced by g_i under the base change $S_j \to S_i$. Thus $g_j \in G^0(S_j)$, and hence g comes from an element of

$$\varinjlim_i G^0(S_i).$$

Proposition 3.9.— *Let G be an S-group locally of finite presentation. Suppose that G^0 is representable. Then the canonical morphism*

$$i : G^0 \longrightarrow G$$

is an open immersion; moreover, G^0 is quasi-compact over S.

Since G^0 is a subfunctor of the functor G, the morphism i is a monomorphism, hence is radicial. From the definition of the functor G^0, one checks immediately from the definition (EGA IV_4, 17.1.1) that i is formally étale (noting that G^0 is the image of i in G). Finally, by the characterization (EGA IV_3, 8.14.2(c)) of S-schemes locally of finite presentation in terms of the functors they represent, and by 3.8, G^0 is locally of finite presentation over S, since G is. Hence i is locally of finite presentation (cf. EGA IV_1, 1.4.3). It is therefore a radicial étale morphism, and hence an open immersion (EGA IV_4, 17.9.1).

The last assertion follows from 3.6.

Theorem 3.10.— *Let G be an S-group. The following conditions are equivalent:*

(i) *G is smooth over S at the points of the identity section.*

(ii) *G is flat and locally of finite presentation over S at the points of the identity section, and, for every $s \in S$, G_s is smooth over $\kappa(s)$.*

(iii) *There is an open group subscheme G' of G that is smooth over S.*

(iv) *G^0 is representable by an open subscheme of G that is smooth over S.*

It is clear that

$$\text{(iv)} \Longrightarrow \text{(iii)} \Longrightarrow \text{(i)},$$

and, by 1.3.1 and 2.4, (i) implies both (ii) and (iii). Moreover, (ii) implies (i) by EGA IV_4, 17.5.1.

It remains to prove that (iii) implies (iv). Lemma 3.10.1 below shows that G' contains $\underline{G^0}$ and that

$$G'^0 = G^0.$$

It therefore suffices to show that $\underline{G^0}$ is open in G, for we have already seen in 3.4 that G^0 will then be represented by the smooth group subscheme induced by G' on the open subset

$$\underline{G'^0} = \underline{G^0}.$$

We may thus suppose that $G' = G$.

To show that $\underline{G^0}$ is open, it suffices to show that every $s \in S$ has a neighborhood T in S such that

$$\underline{G^0} \cap \pi^{-1}(T)$$

is open in $\pi^{-1}(T)$. Let $s \in S$. Since $G = G'$, the morphism π is locally of finite presentation. As in the proof of 3.7, we may therefore construct an open subset T of S containing s, and an open subset W of G containing $\epsilon(s)$, such that the induced morphism

$$\pi'' : W \longrightarrow T$$

is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ. For every $t \in T$,

$$W \cap \underline{G_t^0}$$

is then the connected component of $(\pi'')^{-1}(t)$ containing $\epsilon''(t)$. Since π is smooth, so is π'', which is therefore smooth and of finite presentation. By EGA IV_3, 15.6.5, the union W^0 of the $W \cap \underline{G_t^0}$, for $t \in T$, is open in W.

On the other hand, VI_A, 0.5, gives

$$W^0 \cdot W^0 = \underline{G^0} \cap \pi^{-1}(T),$$

and we must show that this is open in $\pi^{-1}(T)$. We may now suppose that $T = S$; it remains to show that $W^0 \cdot W^0$ is open in G. Since π is universally open, so is μ (VI_A, 0.1). Hence, since W^0 is open in G, so is

$$W^0 \cdot W^0 = \mu(W^0 \times_S W^0).$$

This result will be strengthened in 4.4.

Lemma 3.10.1.— *Let G be an S-group whose fibers are locally of finite type. Then every open group subscheme H of G contains $\underline{G^0}$ and satisfies*

$$G^0 = H^0.$$

Let $s \in S$, and set

$$G'_s = H_s \cap \underline{G_s^0}.$$

Then G'_s is an open subset of $\underline{G_s^0}$, and is dense in $\underline{G_s^0}$ since $\underline{G_s^0}$ is irreducible (VI_A, 2.4). Hence

$$G'_s \cdot G'_s = \underline{G_s^0}$$

by VI_A, 0.5, which shows that

$$\underline{G_s^0} = G'_s \cdot G'_s \subset H_s \cdot H_s = H_s.$$

Thus $\underline{G_s^0} = \underline{H_s^0}$ for every s, whence

$$\underline{G^0} \subset H, \qquad G^0 = H^0.$$

Proposition 3.11.—[26] *Let $u : G \to H$ be a morphism between S-groups locally of finite presentation. If u is flat, the induced map*

$$u^0 : G^0 \longrightarrow H^0$$

is surjective. The converse holds if G is flat over S and H has reduced fibers.

[26] Editor's note: compare VI_A, 5.6.

Suppose that u is flat. Then, for every $s \in S$, u_s is flat and locally of finite presentation, hence open (EGA IV_2, 2.4.6). Therefore

$$u_s^0 : \underline{G_s^0} \longrightarrow \underline{H_s^0}$$

is surjective, by 1.3.2.[27] Thus the map

$$u^0 : G^0 \longrightarrow H^0$$

is surjective.

Conversely, suppose that $u^0 : G^0 \to H^0$ is surjective, that G is flat over S, and that H has reduced fibers. Then, for every $s \in S$, the morphism

$$u_s^0 : \underline{G_s^0} \longrightarrow \underline{H_s^0}$$

is surjective, and hence is flat at every point lying above the generic point η of $\underline{H_s^0}$, since

$$\mathcal{O}_{\underline{H_s^0},\eta}$$

is a field. It follows from 1.3 that u_s is flat. Therefore u is flat, by the fiberwise flatness criterion (EGA IV_3, 11.3.11).

4. Dimension of the fibers of groups locally of finite presentation

Proposition 4.1.— *Let G be an S-scheme locally of finite type, equipped with an S-section ϵ, and suppose that, for every $s \in S$,*

$$\dim G_s = \dim_{\epsilon(s)} G_s$$

(as is the case when G is an S-group; 1.5).

(i) *The function*

$$s \longmapsto \dim G_s$$

is upper semicontinuous on S.

(ii) *If, moreover, G is locally of finite presentation over S, this function is locally constructible.*

Let

$$\pi : G \longrightarrow S$$

be the structural morphism. Chevalley's semicontinuity theorem (EGA IV_3, 13.1.3) asserts that the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x))$$

is upper semicontinuous on G. For every $s \in S$, however,

$$\dim G_s = \dim \pi^{-1}(s) = \dim_{\epsilon(s)} \pi^{-1}\big(\pi(\epsilon(s))\big).$$

[27] Editor's note: we have shortened the original by referring here to 1.3.2. We have also simplified the argument below by omitting a reference to the generic flatness theorem (EGA IV_2, 6.9.1), which is not needed here.

Since the map $s \mapsto \epsilon(s)$ is continuous on S, the composite function

$$s \longmapsto \dim G_s$$

is upper semicontinuous on S.

Now suppose that G is locally of finite presentation over S. Arguing as above, to show that $s \mapsto \dim G_s$ is locally constructible, it is enough to show that the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x))$$

is locally constructible on G, which follows from EGA IV_3, 9.9.1.

Proposition 4.2.— *Let $\pi : G \to S$ be an S-scheme locally of finite presentation, equipped with an S-section ϵ, and satisfying the following two conditions* (which hold when G is an S-group, by 1.5 and VI_A, 2.4.1)*:*

(a) *For every $s \in S$ and every $x \in G_s$, one has*

$$\dim G_s = \dim_x G_s$$

(or equivalently (1.5), for every $s \in S$, all the irreducible components of G_s have the same dimension).

(b) *For every $s \in S$, if $\underline{G_s^0}$ denotes the connected component of G_s containing $\epsilon(s)$, then $\underline{G_s^0}$ is geometrically irreducible.*

Let $s \in S$. The following conditions are equivalent:

(i) *G is universally open over S at the points of $\underline{G_s^0}$.*

(i bis) *G is universally open over S at every point of a neighborhood of $\epsilon(s)$ in $\underline{G_s^0}$.*[28]

(ii) *The function*

$$t \longmapsto \dim G_t$$

is constant in a neighborhood of s in S.

(iii) *$\underline{G^0}$ is "universally open over S at the points of $\underline{G_s^0}$," that is, given $S' \to S$, $s' \in S'_s$, $g \in \underline{G_{s'}^0}$, and an open neighborhood V of g in $G' = G_{S'}$, then*

$$\pi(V \cap \underline{G'^0})$$

is an open neighborhood of s' in S'.[29]

Clearly (i) implies (i bis). By EGA IV_3, 14.3.3.1(ii), the set of points of $\underline{G_s^0}$ at which π_G is universally open is closed in $\underline{G_s^0}$. Since $\underline{G_s^0}$ is irreducible, it follows that (i bis) implies (i).

We show that (i) implies (ii). Let T be the set of $t \in S$ such that

$$\dim G_t = \dim G_s.$$

By 4.1, T is locally constructible; hence, by EGA IV_1, 1.10.1, to show that T is a neighborhood of s, it is enough to show that every generalization s' of s belongs to T.

[28] Editor's note: the original reads "at the points of $\epsilon(s)$." It is not enough to suppose that G is universally open over S at $\epsilon(s)$, as the example given in editor's note 19 shows. We have corrected the proof accordingly.

[29] Editor's note: we have added condition (iii), pointed out by O. Gabber; it will be useful later (5.7).

Let x be the generic point of $\underline{G_s^0}$, and let U be an open subset of G containing x. Since π_G is universally open at x, EGA IV$_3$, 14.3.13 gives, for every generalization s' of s,

$$\dim(U \cap G_{s'}) \geq \dim_x(U \cap G_s).$$

In view of hypothesis (a), this implies

$$\dim G_{s'} \geq \dim G_s.$$

On the other hand, by 4.1 the function $s \mapsto \dim G_s$ is upper semicontinuous, so that

$$\dim G_{s'} \leq \dim G_s.$$

Thus $s' \in T$, proving that (i) implies (ii).

It is clear that (iii) implies (i); we prove that (ii) implies (iii). The dimension of the fibers is unchanged by extension of the base field, and the formation of $\underline{G^0}$ commutes with base change (see the proof of 3.3). We may therefore assume that $S' = S$ and $s' = s$. Moreover, every open subset V of G meeting $\underline{G_s^0}$ contains the generic point η of $\underline{G_s^0}$, so we may assume that $g = \eta$.

We may further suppose that S is affine. Let U be an affine open neighborhood of $\epsilon(s)$; it is then of finite presentation over S. Replacing S by $\epsilon^{-1}(U)$, and then U by $U \cap \pi^{-1}(S)$, we reduce to the case in which

$$\pi : U \longrightarrow S$$

is of finite presentation and admits $\epsilon : S \to U$ as a section. By EGA IV$_3$, 9.7.12, $\underline{G^0} \cap U$ is constructible in U. Replacing V by an affine open subset contained in $V \cap U$, we find that $\underline{G^0} \cap V$ is constructible in V. Since $\pi : V \to S$ is of finite presentation, Chevalley's constructibility theorem (EGA IV$_1$, 1.8.4) shows that

$$\pi(\underline{G^0} \cap V)$$

is locally constructible in S.

Consequently, by *loc. cit.*, 1.10.1, to show that $\pi(\underline{G^0} \cap V)$ is an open neighborhood of s, it is enough to show that, for every generalization t of s, there is a generalization ξ of η belonging to $\underline{G^0}$, and hence to $\underline{G^0} \cap V$. The generic point ξ of $\underline{G_t^0}$ is such a generalization of η. Indeed, $\epsilon(s)$ belongs to the closure X of $\{\xi\}$ in G; hence, by Chevalley's semicontinuity theorem (EGA IV$_3$, 13.1.3),

$$\dim_{\epsilon(s)} X_s \geq \dim_\xi X_t.$$

On the other hand, hypothesis (ii) implies

$$\dim_\xi \underline{G_t^0} = \dim_{\epsilon(s)} G_s.$$

It follows that one of the irreducible components of X_s containing $\epsilon(s)$ is equal to $\underline{G_s^0}$, whence

$$\eta \in \overline{\{\xi\}}.$$

This proves that (ii) implies (iii), and hence proves the proposition.

[30]One can also prove that (ii) implies (i) as follows. Since π is locally of finite presentation, there exist an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s, such that $\pi(U) \subset V$ and the morphism

$$\pi' : U \longrightarrow V$$

[30]Editor's note: the preceding argument was communicated to us by O. Gabber. We have also retained the proof of the implication (ii) $\Rightarrow$(i) given in the original.

induced by π is of finite presentation. Set

$$T = \epsilon^{-1}(U), \qquad W = \pi'^{-1}(T) = U \cap \pi^{-1}(T).$$

The morphism

$$\pi'' : W \longrightarrow T$$

induced by π' is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ. Moreover, for every $t \in T$, since $\underline{G_t^0}$ is irreducible, $W \cap \underline{G_t^0}$ is dense in $\underline{G_t^0}$, hence irreducible and therefore connected. It is consequently the connected component of $\pi''^{-1}(t)$ containing $\epsilon''(t)$.

Since $W \cap \underline{G_t^0}$ is a dense open subset of $\underline{G_t^0}$, 1.5 and EGA IV_2, 5.2.1 give

$$\dim(W \cap \underline{G_t^0}) = \dim \underline{G_t^0} = \dim G_t.$$

Thus the function

$$t \longmapsto \dim(W \cap \underline{G_t^0})$$

is constant in a neighborhood of s in T. Finally, we show that $W \cap \underline{G_t^0}$ is geometrically irreducible for every $t \in T$. Let K be an extension of $\kappa(t)$. Then

$$(W \cap \underline{G_t^0}) \otimes_{\kappa(t)} K = (W \otimes_{\kappa(t)} K) \cap (\underline{G_t^0} \otimes_{\kappa(t)} K)$$

is a nonempty open subset of $\underline{G_t^0} \otimes_{\kappa(t)} K$, and is therefore irreducible because $\underline{G_t^0} \otimes_{\kappa(t)} K$ is irreducible.

We are now in the situation of EGA IV_3, 15.6.6(ii), which asserts that π'', and hence π, is universally open at the points of $W \cap \underline{G_s^0}$. But, by EGA IV_3, 14.3.3.1(ii), the subset of G_s consisting of the points at which π is universally open is closed in G_s. Since it contains $W \cap \underline{G_s^0}$, it therefore contains its closure $\underline{G_s^0}$.

Corollary 4.3.— *Let G be a flat S-group locally of finite presentation. Then the function*

$$s \longmapsto \dim G_s$$

is locally constant on S.

This follows immediately from 4.2, since every flat morphism locally of finite presentation is universally open (EGA IV_2, 2.4.6).

Corollary 4.4.— *Let G be an S-group locally of finite presentation over S at the points of the identity section. Consider the following conditions:*

(i) *G is smooth over S at the points of the identity section* (cf. 3.10).

(ii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and the function*

$$s \longmapsto \dim G_s$$

is locally constant on S.

(iii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and there exists a neighborhood V of the identity section such that*

$$\pi_V : V \longrightarrow S$$

is universally open.

(iv) *For every $s \in S$, G^0_s is smooth over $\kappa(s)$, and G^0 is representable by an open group subscheme of G, universally open over S.*

Then the following implications hold:

$$\text{(i)} \Longrightarrow \text{(ii)} \Longleftrightarrow \text{(iii)} \Longleftrightarrow \text{(iv)}.$$

If, moreover, S is reduced, then conditions (i)–(iv) *are equivalent and imply that G^0 is smooth over S.*[31]

Let us show that (i) implies (ii). For every $x \in G$, by 1.5,

$$\dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x)).$$

Consequently, by EGA IV_4, 17.10.2, the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$$

is continuous in a neighborhood of the identity section. Therefore the function $s \mapsto \dim G_s$ is continuous on S, and hence locally constant on S. On the other hand, for every $s \in S$, G_s is smooth over $\kappa(s)$, by 1.3.1.

Let us show that (ii) implies (iv). It is enough to show that $\underline{G^0}$ is open in G, for then, by 3.4, G^0 will be representable by the group subscheme induced on $\underline{G^0}$, and the properties of G^0 stated above will follow from 2.4 and 4.2. Given $s \in S$, construct, as in the proof of 3.10,

$$W, \quad T, \quad \pi'', \quad \epsilon'', \quad \text{and} \quad W^0.$$

Then, by EGA IV_3, 15.6.7, W^0 is open in W. On the other hand, under hypothesis 4.4(ii), it follows from 4.2 that π is universally open at every point of W^0. Therefore, by VI_A, 0.1, μ is universally open at every point of

$$W^0 \times_S W^0,$$

which shows that $W^0 \cdot W^0$ is open; the proof is completed as in the proof of Theorem 3.10.

It is clear that

$$\text{(iv)} \Longrightarrow \text{(iii)},$$

and (iii) $\Longrightarrow$ (ii) follows by applying 4.2 to V.

Finally, suppose that (ii)–(iv) hold, and let us show that G^0 is smooth over S when S is reduced.[32] For this, we may suppose that $G = G^0$. Then G is of finite presentation over S, by 5.5 below. Thus π_G is of finite presentation, has geometrically integral fibers, and has locally constant relative dimension on S. It follows from EGA IV_3, 15.6.7 that the morphism

$$G \times_S S_{\mathrm{red}} \longrightarrow S_{\mathrm{red}}$$

induced by π_G is flat. Hence π_G is flat when S is reduced. In that case, $G = G^0$ is smooth over S, by Theorem 3.10.

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[31] Editor's note: the hypothesis that S be reduced cannot be omitted here. Indeed, let k be a field, let $S = \operatorname{Spec} A$, where $A = k[\delta]$ with $\delta^2 = 0$, and let G be the closed subgroup

$$\operatorname{Spec} A[X]/(\delta X)$$

of $\mathbb{G}_{a,S}$ (which assigns to every A-algebra R the subgroup of those $r \in R$ such that $\delta r = 0$). Then G is an S-group satisfying (ii)–(iv), but it is not flat, and hence not smooth, over S.

[32] Editor's note: we have expanded the original in the argument that follows.

5. Separation of groups and homogeneous spaces

Proposition 5.1.— *For an S-group G to be separated, it is necessary and sufficient that the identity section of G be a closed immersion.*

The condition is necessary (EGA I, 5.4.6); it is sufficient by virtue of the following cartesian diagram (cf. VI_A, 0.3):

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\mu\circ(\mathrm{id}_G \times c)} & G \\ {\scriptstyle \Delta_{G/S}}\uparrow & & \uparrow{\scriptstyle \varepsilon} \\ G & \xrightarrow{\pi} & S \end{array} .$$

Proposition 5.2.— *If S is discrete, every S-group is separated.*

Indeed,

$$S = \coprod_{s\in S} \operatorname{Spec} \mathcal{O}_{S,s},$$

and, by EGA I, 5.5.5, it is enough to show that, for every $s \in S$,

$$G \times_S \operatorname{Spec} \mathcal{O}_{S,s}$$

is separated. This follows from VI_A, 0.3, since $\mathcal{O}_{S,s}$ is a local ring of dimension 0.[33]

Theorem 5.3.— *Let S be a scheme, let G be an S-group scheme locally of finite presentation over S and universally open over S in a neighborhood of the identity section, and let X be an S-scheme on which G acts in such a way that the morphism*

$$\Phi : G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (gx, x),$$

is surjective. Suppose moreover that, for every $s \in S$:

(i) *there exists an open subscheme U of X, separated over S, such that U_s is dense in X_s;*

(ii) *the fiber X_s is locally of finite type over $\kappa(s)$.*[34]

Then X is separated over S.

Corollary 5.4.— *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that X has fibers locally of finite type and connected. Then:*

(i) *X is separated over S.*

(ii) *If there exists an open subset V of X, quasi-compact over S, which meets every nonempty fiber X_s,*[35] *then X is quasi-compact over S.*

[33] Editor's note: we have replaced "Artinian" by "of dimension 0" and mentioned this generalization in VI_A, 0.3.

[34] Editor's note: we have corrected the original by assuming that G is universally open over S in a neighborhood of the identity section and by adding hypothesis (ii); see below for examples, due to O. Gabber, showing that the statements 5.3 and 5.4 of the original are false without additional hypotheses. We also point out that Theorem 5.3 is a reworked version of Theorem 5.3A below, which appears in the 1965 edition of SGAD and is due to M. Raynaud; compare the notes (∗) in Exposé X, 8.5 and 8.8.

Theorem 5.3A (Raynaud).— *Let G be an S-group locally of finite presentation, universally open over S, and with connected fibers. Then G is separated over S. More generally, every S-scheme X locally of finite presentation over S, equipped with an action of G such that the morphism $\Phi : G \times_S X \to X \times_S X$, $(g, x) \mapsto (gx, x)$, is surjective, is separated over S.*

[35] Editor's note: without this hypothesis there is the following counterexample, pointed out by O. Gabber. Let $G = S$ be a local scheme with closed point s, such that $S^* = S - \{s\}$ is not quasi-compact, and let X be the disjoint union of $\{s\}$ and S^*.

Proof. (i) Let $s \in S$ be such that $X_s \neq \varnothing$. The morphism

$$\Phi_s : G_s \times_{\kappa(s)} X_s \longrightarrow X_s \times_{\kappa(s)} X_s$$

obtained from Φ by base change is surjective. Since X_s is connected, it is irreducible by VI_A, 2.5.4. Thus, if U is an affine open subset of X such that U_s is nonempty, then U_s is dense in X_s, and the theorem applies.

To prove (ii), we may suppose that S is affine. Then V is quasi-compact and, by 3.5, there exists a quasi-compact open subset U of G containing $\underline{G^0}$. Let $s \in S$ be such that $X_s \neq \varnothing$. Then X_s is irreducible (VI_A, 2.6.6), and, since U_s contains $\underline{G^0_s}$, the morphism

$$U_s \times_{\kappa(s)} V_s \longrightarrow X_s, \qquad (g, x) \longmapsto gx,$$

is surjective (VI_A, 2.6.4). Consequently the morphism

$$U \times_S V \longrightarrow X$$

is surjective, and X is therefore quasi-compact, since U and V are. As S was assumed affine, hence separated, it follows that X is quasi-compact over S (cf. EGA I, 6.6.4(v)).

Remark 5.4.1.—[36] We shall see in the course of the proof that the conclusion of Theorem 5.3 remains true if one assumes only:

> (i') there exists an open subscheme U of X, separated over S, such that U_s is dense in every irreducible component of X_s.

This follows from (i) if X_s locally has a finite number of irreducible components, as is the case under hypothesis (ii). We shall also see later that 5.4 remains true under the hypothesis that every fiber X_s is quasi-separated and connected.

Corollary 5.5.— *Let S be a scheme, and let G be an S-group scheme, locally of finite presentation, with connected fibers, and universally open over S. Then G is separated and of finite presentation over S.*[37]

Indeed, by 3.6 and 5.4, G is quasi-compact and separated over S. Since G is locally of finite presentation over S, it is therefore of finite presentation over S.

5.6.— *Proof of Theorem 5.3.* Before proving 5.3, let us establish a few lemmas.

Lemma 5.6.0.—[38]

(i) *Let $A \subset B$ be integral domains, with B integral over A, and let $\mathfrak{q} \in \mathrm{Spec}(B)$ be such that A is unibranch at $\mathfrak{p} = \mathfrak{q} \cap A$. Then the morphism*

$$\pi : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

is open at $\mathfrak{q}$.

[36] Editor's note: we have added this remark; compare editor's note 34.

[37] Editor's note: we point out here the following result ([Ray70a], VI, 2.5). If S is normal, G is smooth with connected fibers, and X is an (fppf) homogeneous G-space—that is, the morphisms $X \to S$ and $G \times_S X \to X \times_S X$ are coverings for the (fppf) topology—locally of finite type over S, then X is locally quasi-projective over S. In particular, G is quasi-projective over S. See also editor's note 35 in VI_A.

[38] Editor's note: we have added this lemma, communicated by O. Gabber. It improves EGA IV_3, 14.4.1.2 and corrects the proof of *loc. cit.*, 14.4.1.3 without changing its hypotheses (compare the erratum (ErrIV, 38) in EGA IV_4).

(ii) *Let X, Y be irreducible preschemes, let $f : X \to Y$ be a dominant morphism, and let $x \in X$ be a point at which f is quasi-finite, such that $y = f(x)$ is a unibranch point of Y. Then f is open at x. In particular, if*

$$Y = \mathrm{Spec}(\mathcal{O}_{Y,y})$$

is a local prescheme with closed point y, then $f(U) = Y$ for every neighborhood U of x.

(i) Let K (respectively, L) be the fraction field of A (respectively, B), let A' be the normalization of A, and let B' be the subring of L generated by A' and B. Then B' is integral over A'. Set

$$Y = \mathrm{Spec}(A), \quad X = \mathrm{Spec}(B), \quad Y' = \mathrm{Spec}(A'), \quad X' = \mathrm{Spec}(B').$$

We then have the commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\pi'} & Y' \\ \downarrow & & \downarrow{\scriptstyle \phi} \\ X & \xrightarrow{\pi} & Y \end{array}$$

in which all morphisms are integral and surjective.

Since A is unibranch at $\mathfrak{p}$, Y' has a unique point $\mathfrak{p}'$ above $\mathfrak{p}$. Consequently, if U is an open neighborhood of $\mathfrak{p}'$ in Y', then

$$\phi(Y' - U)$$

is a closed subset not containing $\mathfrak{p}$, so that its open complement is contained in $\phi(U)$. This shows that $\phi : Y' \to Y$ is open at $\mathfrak{p}'$. It is therefore enough to show that π' is open, and we are reduced to the case where $A = A'$ is normal.

Let N be a quasi-Galois extension of K containing L, let

$$\widetilde{X} = \mathrm{Spec}(\widetilde{B}),$$

where $\widetilde{B}$ is the integral closure of B in N, and let $G = \mathrm{Aut}(N/K)$. Since $\widetilde{X} \to X$ is surjective, it is enough to show that

$$\widetilde{\pi} : \widetilde{X} \longrightarrow Y$$

is open. Let U be an open subset of $\widetilde{X}$, and set

$$U' = \bigcup_{g \in G} gU.$$

Because G acts transitively on the fibers of $\widetilde{X} \to Y$ (cf. [BAC], Chap. V, §2.3, Prop. 6), one has

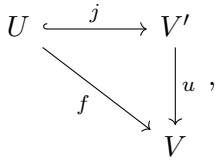
$$\widetilde{\pi}(U) = \widetilde{\pi}(U'),$$

and the latter is the open complement of the closed subset

$$\widetilde{\pi}(\widetilde{X} - U').$$

This proves (i).

(ii) We may suppose that Y and X are reduced. By Zariski's Main Theorem (cf. EGA IV$_3$, 8.12.9), there exist affine open neighborhoods U of x and $V = \mathrm{Spec}(A)$ of y, such that $f(U) \subset V$, and a factorization



where j is an open immersion and u is finite. Replacing V' by the closure of $j(U)$, we may suppose that V' is irreducible, hence that $V' = \mathrm{Spec}(B)$, where B is an integral A-algebra, finite over A. Moreover, since f is dominant, so is u, and the morphism $A \to B$ is therefore injective.

Since A is unibranch at y, it follows from (i) that u is open at $j(x)$, and hence that

$$f = u \circ j$$

is open at x. This proves the first assertion of (ii). The second follows because, if Y is a local prescheme with closed point y, every open subset containing y equals Y. In the case $Y = \mathrm{Spec}(\mathcal{O}_{Y,y})$, one may also use, instead of EGA IV$_3$, 8.12.9, the local form of Zariski's Main Theorem found, for example, in [Pes66] or [Ray70b], Chap. IV, Th. 1.

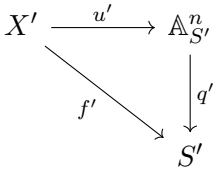
Lemma 5.6.1.0.—[39] *Let $f : X \to S$ be a morphism locally of finite presentation, let $s \in S$, and let $x \in X_s$. Set $n = \dim_x(X_s)$, and let $q : \mathbb{A}^n_S \to S$ be the structure morphism.*

Suppose that a quasi-finite morphism $u : X \to \mathbb{A}^n_S$ is given such that $f = q \circ u$, and suppose that f is universally open at the generic point z of an irreducible component $\mathfrak{z}$ of X_s, containing x and of dimension n. Then u is universally open at the point $x \in X$.

First observe that

(†) *the hypotheses are preserved under every base change $\pi : S' \to S$ covering s*

(that is, such that $\pi^{-1}(s) \neq \varnothing$). Indeed, let $S' \to S$ be such a morphism, let



be the diagram obtained by base change, and let s' be a point of S' above s and x' a point of $X'_{s'}$ above x. Since

$$X'_{s'} = X_s \otimes_{\kappa(s)} \kappa(s'),$$

the lifting of generizations and the invariance of dimension under field extension (EGA IV$_2$, 2.3.4(i) and 4.1.4) show that x' is contained in an irreducible component $\mathfrak{z}'$ of $X'_{s'}$ whose generic point z' lies above z, and

$$n \leq \dim \mathfrak{z}' \leq \dim_{x'} X'_{s'} \leq n.$$

Thus

$$\dim \mathfrak{z}' = n = \dim_{x'} X'_{s'}.$$

Since f is universally open at z, f' is universally open at z'.

Set $Y = \mathbb{A}^n_S$. By EGA IV$_3$, 14.3.3.1(i), to prove that u is universally open at x, it is enough to prove that, for every integer $r \geq 0$ and every point x' of

$$X' = X[T_1, \ldots, T_r]$$

[39] Editor's note: we have made explicit this lemma, which is used in the proof of Lemma 5.6.1.

above x, the morphism

$$u' : X' \longrightarrow Y' = \mathbb{A}^n_S[T_1, \ldots, T_r]$$

is open at x'. Now set

$$S' = S[T_1, \ldots, T_r],$$

and let q' be the projection

$$Y' \simeq \mathbb{A}^n_{S'} \longrightarrow S'.$$

This is the situation obtained by the base change $S' \to S$. Thus, by the preceding observation, we are reduced to showing that u is open at x.

Set $y = u(x)$. Since u is locally of finite presentation (because f and q are; cf. EGA IV_1, 1.4.3), EGA IV_1, 1.10.3 shows that it is enough to prove

$$u(\operatorname{Spec} \mathcal{O}_{X,x}) = \operatorname{Spec} \mathcal{O}_{Y,y}.$$

For this purpose we may assume that S is affine and integral. Let $\pi : S' \to S$ be its normalization, and denote by $u' : X' \to Y'$ the morphism obtained from u by base change. Since

$$\pi_Y : Y' \longrightarrow Y$$

is integral and surjective, one has

$$\operatorname{Spec} \mathcal{O}_{Y,y} = \bigcup_{y'} \pi_Y(\operatorname{Spec} \mathcal{O}_{Y',y'}),$$

where the union ranges over all points y' of Y' above y. It is therefore enough to show that, for every such y' and every $x' \in X'_{y'}$ above x,

$$u'(\operatorname{Spec} \mathcal{O}_{X',x'}) = \operatorname{Spec} \mathcal{O}_{Y',y'}.$$

Since the hypotheses are preserved under the base change $\pi : S' \to S$, we are thereby reduced to the case of S'; that is, we may assume that S is integral and normal.

The hypotheses on f now imply, by EGA IV_3, 14.3.13, that there is an irreducible component Z of X, containing $\mathfrak{z}$ (and hence x), dominating S, and such that

$$\dim_z(Z_s) = n = \dim(Z_\eta),$$

where η is the generic point of S. Let ξ be the generic point of Z. Since u, and hence also

$$u_\eta : Z_\eta \longrightarrow \mathbb{A}^n_\eta,$$

is quasi-finite, the closure in $\mathbb{A}^n_\eta$ of the point

$$u_\eta(\xi) = u(\xi)$$

has dimension n. Thus $u(\xi)$ is the generic point of $\mathbb{A}^n_\eta$, which is also the generic point of $Y = \mathbb{A}^n_S$. Consequently, if g denotes the restriction of u to Z, then

$$g : Z \longrightarrow Y$$

is quasi-finite and dominant. Since Y is normal, Lemma 5.6.0 shows that g is open. Hence u is open at every point of Z, and in particular at x. Consequently,

$$u(\operatorname{Spec} \mathcal{O}_{X,x}) = \operatorname{Spec} \mathcal{O}_{Y,y},$$

which completes the proof of Lemma 5.6.1.0.

The following lemma advantageously replaces EGA IV_3, 14.5.10,[40] in that it is independent of noetherian hypotheses.

Lemma 5.6.1.— *Let S be a scheme, let $f : X \to S$ be an S-scheme locally of finite presentation, let s be a point of S, and let x be a closed point of X_s. Suppose that f is universally open at the generic point z of an irreducible component Z of X_s containing x, and that*

$$\dim_x(X_s) = \dim Z.$$

Then there exists a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & S \\ {\scriptstyle h}\uparrow & \nearrow_{\phi} & \uparrow{\scriptstyle w} \\ S'' & \xrightarrow{\pi} & S' \end{array},$$

where S' is affine, w is étale, π is finite, surjective, and of finite presentation, and $\phi^{-1}(s)$ consists of a single point s'', such that $h(s'') = x$ and $\kappa(s'') = \kappa(x)$.[41]

Proof. First, we may suppose that $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$, where B is an A-algebra of finite presentation. Let $\mathfrak{p}$ and $\mathfrak{q}$ be the prime ideals of A and B corresponding respectively to s and x, so that

$$\mathfrak{p} = A \cap \mathfrak{q}.$$

Set $n = \dim_x(X_s)$, and choose $t_1, \ldots, t_n \in B$ whose images in

$$\mathcal{O}_{X_s,x} = B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$$

form a system of parameters. Then

$$\mathcal{O}_{X_s,x}/(t_1, \ldots, t_n)$$

is finite-dimensional over $\kappa(x)$, and hence also over $\kappa(s)$, because x is a closed point of the algebraic $\kappa(s)$-scheme X_s. Consequently, the S-morphism

$$u : X \longrightarrow \mathbb{A}^n_S = \mathrm{Spec}(A[T_1, \ldots, T_n]), \qquad T_i \longmapsto t_i,$$

is of finite presentation, x is isolated in the fiber $u^{-1}(u(x))$, and

$$u(x) = \tau_0(s),$$

where $\tau_0 : S \to \mathbb{A}^n_S$ denotes the zero section of $\mathbb{A}^n_S \to S$, corresponding to the A-algebra homomorphism

$$A[T_1, \ldots, T_n] \longrightarrow A, \qquad T_i \longmapsto 0.$$

The set of points of X that are isolated in their fibers over $\mathbb{A}^n_S$ is open (EGA IV_3, 13.1.4). Thus, after shrinking X, we may suppose that u is quasi-finite and that $u^{-1}(u(x)) = \{x\}$. By Lemma 5.6.1.0, u is universally open at x.

Let

$$B_0 = B/(t_1, \ldots, t_n), \qquad X_0 = X \times_{\mathbb{A}^n_S} \tau_0(S) = \mathrm{Spec}(B_0),$$

[40] Editor's note: we have corrected the original, which indicated 19.5.10.

[41] Editor's note: consequently, h induces a section σ of $X \times_S S'' \to S''$ such that $\sigma(s'')$ lies above x; compare EGA IV_3, 14.5.10.

and let $u_0 : X_0 \to S$ be obtained from u by the base change $\tau_0 : S \hookrightarrow \mathbb{A}^n_S$. Then u_0 is quasi-finite and of finite presentation, is universally open at x, and x is the unique point of X_0 above s.

Let A' be the henselization of the local ring

$$A_{\mathfrak{p}} = \mathcal{O}_{S,s},$$

and set

$$S' = \mathrm{Spec}(A'), \qquad B'_0 = B_0 \otimes_A A', \qquad X'_0 = X_0 \times_S S' = \mathrm{Spec}(B'_0).$$

The closed point s' of S' is the unique point of S' above s, and $\kappa(s') = \kappa(s)$. The scheme X'_0 has a unique point x' above s'; moreover $\kappa(x') = \kappa(x)$, and x' is also the unique point of X'_0 above both s and x. Since A' is henselian, EGA IV$_4$, 18.5.11 gives a disjoint decomposition into open and closed parts

$$X'_0 = V \amalg W, \qquad V = \mathrm{Spec}(\mathcal{O}_{X'_0,x'}). \tag{*}$$

The local ring $\mathcal{O}_{X'_0,x'}$ is finite and of finite presentation over A'. Hence the restriction π of u'_0 to V is finite and of finite presentation. Moreover, because $u'_0 : X'_0 \to S'$ is open at x',

$$u'_0(V) = \mathrm{Spec}(\mathcal{O}_{S',s'}) = S',$$

so π is surjective.

This proves the assertion when $S = S'$. In general, A' is the filtered direct limit of A-subalgebras A_i that are étale over A and for which

$$S_i = \mathrm{Spec}(A_i)$$

has a unique point s_i above s, with $\kappa(s_i) = \kappa(s)$. We have

$$B'_0 = \varinjlim_i B_i, \qquad B_i = B_0 \otimes_A A_i.$$

Set

$$X_i = \mathrm{Spec}(B_i) = X_0 \times_S S_i, \qquad C = \mathcal{O}_{X'_0,x'}.$$

By $(*)$, one has

$$C \simeq B'_0/fB'_0$$

for some idempotent $f \in B'_0$. There are an index i and an element $f_i \in B_i$ whose image in B'_0 is f. Put

$$C_i = B_i/f_iB_i, \qquad V_i = \mathrm{Spec}(C_i).$$

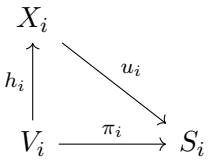
Then

$$C = C_i \otimes_{A_i} A', \qquad V = V_i \times_{S_i} S',$$

and C_i is an A_i-algebra of finite presentation, since the same is true of B_i. Consequently, the morphisms

$$\begin{array}{ccc} X'_0 & & \\ {\scriptstyle h'}\uparrow & \searrow^{u'_0} & \\ V & \xrightarrow{\pi} & S' \end{array}$$

arise, by the base change $S' \to S_i$, from morphisms of finite presentation



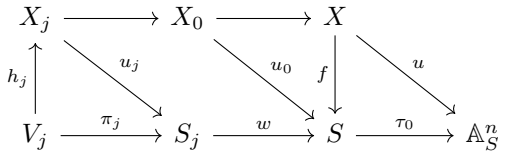
For every $j \geq i$, let

$$V_j = V_i \times_{S_i} S_j$$

and let $\pi_j : V_j \to S_j$ be the morphism of finite presentation obtained from π_i by base change. Since $\pi : V \to S'$ is finite and surjective, EGA IV$_2$, 8.10.5 shows that, for some j, the morphism $\pi_j : V_j \to S_j$ is finite and surjective. Then $w : S_j \to S$ is affine and étale, S_j has a unique point s_j above s, and V_j has a unique point x_j above s_j, since x' is the unique point of

$$V = V_j \times_{S_j} S'$$

above s':



Thus x_j is the unique point of V_j above s, its image under $V_j \to X_j \to X$ is x, and

$$\kappa(x_j) = \kappa(x).$$

Lemma 5.6.2.0.—[42] *Let k be a field and let G be a nonempty k-scheme of finite type.*

(i) *Let K be an extension of k, and let W be a dense open subset of G_K. Denote by π the projection $G_K \to G$. Then*

$$U = \{g \in G \mid W \text{ contains every maximal point of } \pi^{-1}(g)\}$$

is a dense open subset of G. (N.B. If g is a closed point of G belonging to U, then $\pi^{-1}(g) \subset W$.

(ii) *Suppose in addition that G is geometrically irreducible. Let*

$$\mu : G \times X \longrightarrow Y$$

be a morphism of k-schemes, let x be a point of X, and let Ω be an open subset of Y such that

$$\mu_x^{-1}(\Omega_{\kappa(x)}) \neq \varnothing,$$

where μ_x denotes the morphism

$$G_{\kappa(x)} \longrightarrow Y_{\kappa(x)}, \qquad g \longmapsto \mu(g, x).$$

Then

$$U = \{g \in G \mid \mu \text{ sends every maximal point of } g \times x \text{ into } \Omega\}$$

is a dense open subset of G. For every closed point g of G belonging to U, one has

$$\mu(g \times x) \subset \Omega,$$

[42] Editor's note: we have added this lemma, communicated by O. Gabber. It makes it possible to simplify the proof of 5.6.2 and to prove Theorem 5.3, as well as 5.4, in a more general form; see 5.7 and 5.8 below.

and hence

$$\mu(g', x) \in \Omega_{\kappa(x)} \quad \text{(respectively, } \mu(g, x') \in \Omega_{\kappa(g)}\text{)},$$

for every point $g' \in G_{\kappa(x)}$ *above* g (respectively, $x' \in X_{\kappa(g)}$ above x).

(iii) *Suppose that* $G = H^0$, *where* H *is a* k*-group scheme locally of finite type, acting on a nonempty* k*-scheme* X *in such a way that the morphism*

$$H \times_S X \longrightarrow X \times_S X, \qquad (h, x) \longmapsto (hx, x)$$

is surjective. Let U *be an open subset of* X. *Then:*

(a) $G \cdot U$ *is an open subset of* X, *equal to the union of the irreducible components of* X *whose generic point belongs to* U.

(b) *Suppose that* U *is dense in every irreducible component of* X. *Then, for every finite subset* A *of* X, *there exists a closed point* $g \in G$ *such that*

$$g \cdot A' \subset U_{\kappa(g)},$$

where A' *is the inverse image of* A *in* $X_{\kappa(g)}$. *In particular, the morphism*

$$G \times U \longrightarrow X$$

is surjective.

Proof. (i) Let $\bar{k}$ be a separable closure of k, and let L be an extension of k containing a copy of $\bar{k}$ and of K. Denote by π_L (respectively, ϕ) the projection

$$G_L \longrightarrow G \qquad \text{(respectively, } G_L \longrightarrow G_K\text{)}.$$

For every $g \in G$, the map ϕ sends the maximal points of $\pi_L^{-1}(g)$ surjectively onto those of $\pi^{-1}(g)$. Consequently,

$$U = \{g \in G \mid W_L \text{ contains every maximal point of } \pi_L^{-1}(g)\}.$$

Thus, replacing K by L, we are reduced to the case in which K contains $\bar{k}$.

Since the projection

$$p : G_K \longrightarrow G_{\bar{k}}$$

is surjective and open, $V = p(W)$ is a dense open subset of $G_{\bar{k}}$. Moreover,

$$p^{-1}(g) = \operatorname{Spec}(\kappa(g) \otimes_{\bar{k}} K)$$

is irreducible for every $g \in G_{\bar{k}}$ (cf. EGA IV$_2$, 4.3.3). Hence, for every $g \in V$, the generic point of $p^{-1}(g)$ belongs to W.

On the other hand, set

$$\mathscr{G} = \operatorname{Gal}(\bar{k}/k).$$

The projection $q : G_{\bar{k}} \to G$ is surjective, and $\mathscr{G}$ acts transitively on the fibers of q. Therefore

$$U = q(V'),$$

where V' is the intersection of the $\mathscr{G}$-conjugates of V.

Let Z be the closed subset $G_{\bar{k}} - V$, endowed with its reduced closed subscheme structure. Since $G_{\bar{k}}$, and hence Z, is of finite presentation over $\bar{k}$, the scheme Z is obtained by base change from a reduced closed subscheme

$$Z_1 \subset G \otimes_k k_1$$

for some finite Galois extension k_1 of k. Thus the $\mathscr{G}$-conjugates of Z are finite in number, and their union is again a nowhere-dense closed subset F of $G_{\bar{k}}$. It follows that

$$V' = G_{\bar{k}} - F$$

is a dense open subset of $G_{\bar{k}}$. Since $q : G_{\bar{k}} \to G$ is surjective and open, $U = q(V')$ is therefore a dense open subset of G.

Moreover, for every closed point g of G, the fiber $\pi^{-1}(g)$ consists of finitely many closed points of G_K. Thus, if $g \in U$, then

$$\pi^{-1}(g) \subset W.$$

This proves (i), and (ii) follows from it.

Assertion (iii)(a) was proved in VI$_A$, 2.6.4. Finally, if U is dense in every irreducible component of X, then

$$X = G \cdot U.$$

Hence, for every $x \in X$,

$$\mu_x^{-1}(U_{\kappa(x)})$$

is a nonempty open subset of $G_{\kappa(x)}$, and (iii)(b) follows from (ii).

Corollary 5.6.2.— *Let S, G, and X be as in the preliminary hypotheses of 5.3, and let U be an open subset of X, let $s \in S$, and let A be a finite subset of X_s. Suppose that U_s is dense in X_s and that X_s is locally of finite type over $\kappa(s)$.*[43]

Then there are morphisms $f : S'' \to S$ and $h : S'' \to G$, where f factors as a finite surjective morphism $S'' \to S'$ followed by an étale morphism $S' \to S$, such that the inverse image A'' of A in $X \times_S S''$ (i.e., in $X_s \times_{\operatorname{Spec}\kappa(s)} S''_s$) *is contained in*

$$\ell_h^{-1}(U_{S''}),$$

where ℓ_h denotes the translation of $X_{S''} = X \times_S S''$ defined by the element $h \in G(S'')$.

Proof. Since X_s is locally of finite type over $\kappa(s)$, the connected components of X_s are open and irreducible (cf. VI$_A$, 2.5.4). Hence U_s is dense in every irreducible component of X_s.

By Lemma 5.6.2.0, there therefore exists a closed point

$$g \in G_s^0$$

such that

$$g \cdot a' \in U_s \otimes_{\kappa(s)} \kappa(g)$$

for every $a' \in X_{\kappa(g)}$ lying above a point of A.

By Lemma 5.6.1, there are morphisms $f : S'' \to S$ and $h : S'' \to G$, where f factors as a finite surjective morphism $S'' \to S'$ followed by an étale morphism $S' \to S$, such that $f^{-1}(s)$ consists of a single point s_0, and such that

$$h(s_0) = g, \qquad \kappa(s_0) = \kappa(g).$$

Denote by A'' the inverse image of A in

$$X_s \times_{\operatorname{Spec}\kappa(s)} S''_s = X_s \otimes_{\kappa(s)} \kappa(s_0) = X_s \otimes_{\kappa(s)} \kappa(g).$$

[43] Editor's note: we have added the hypothesis on X_s and simplified (and corrected) the proof, taking account of the addition of 5.6.2.0. Moreover, the proof shows that the conclusion remains valid if one assumes only that U_s is dense in every irreducible component of X_s.

Then the translation ℓ_h of $X_{S''}$ sends A'' into

$$U_s \otimes_{\kappa(s)} \kappa(s_0).$$

Lemma 5.6.3.— *Let X be an S-scheme. The following conditions are equivalent:*

(i) *X is separated over S.*

(ii) *For every S-scheme T, every section $\sigma : T \to X_T$ is a closed immersion.*

(iii) *For every reduced S-scheme T, two S-morphisms $f_1, f_2 : T \to X$ that agree on a dense open subset U of T are equal.*

(iv) *For every $s \in S$ and every pair of points x_1, x_2 of X_s, there exist a morphism*

$$\phi : S'' \longrightarrow S' \longrightarrow S$$

and an open subscheme V of $X_{S''}$, separated over S'', such that:

(a) *$S' \to S$ is open, $S'' \to S'$ is closed and surjective, and $\phi^{-1}(s) \neq \varnothing$;*

(b) *the inverse image of $\{x_1, x_2\}$ in $X_{S''}$ is contained in V.*

(iv′) *For every $s \in S$, every pair of points $x_1, x_2 \in X_s$ is contained in an open subset V of X that is separated over S.*[44]

Proof. The implication (ii)⇒(i) is clear (take $T = X$ and let σ be the diagonal section), as are

$$\text{(i)} \Rightarrow \text{(iv}') \Rightarrow \text{(iv)}.$$

On the other hand, (i) ⇒ (ii) by EGA I, 5.4.6.

(iii)⇒(ii). Let $\sigma : T \to X_T$ be a section of $p : X_T \to T$. By EGA I, 5.3.13, σ is an immersion, that is, an isomorphism of T onto a locally closed subscheme E of X_T. To show that E is closed, we may suppose that T and E are reduced. Let $\overline{E}$ be the reduced closed subscheme of X_T whose underlying space is the closure of E, so that E is a dense open subscheme of $\overline{E}$. The immersion

$$i : \overline{E} \hookrightarrow X_T$$

and $\sigma \circ p \circ i$ agree on E, hence on $\overline{E}$ by hypothesis (iii). Thus every point of $\overline{E}$ belongs to $\sigma(T) = E$, and consequently $\overline{E} = E$.

(i)⇒(iii). Suppose that X is separated over S, and let T be a reduced S-scheme and $f_1, f_2 : T \to X$ two S-morphisms that agree on a dense open subset U of T. Let

$$g : T \longrightarrow X \times_S X$$

[44]Editor's note: we have simplified the formulation of condition (iv) and added condition (iv′). We have also added the proof of the implication (i)⇒(iii), which is used in the proof of (iv)⇒(iii). Observe moreover that, if

$$T = \operatorname{Spec} B, \qquad B = k[\varepsilon, x]/(\varepsilon^2, \varepsilon x)$$

(k a field) and

$$X = T_{\mathrm{red}} = \operatorname{Spec} k[x],$$

then the morphisms

$$\phi_\lambda : T \longrightarrow X, \qquad x \longmapsto x + \lambda\varepsilon \quad (\lambda \in k),$$

agree on the dense open subset $\operatorname{Spec} B_x$, but are not equal.

be the morphism with components f_1 and f_2. Since

$$D = \Delta_{X/S}(X)$$

is closed, its inverse image under g is a closed subset of T containing U, and is therefore equal to T. Since T is reduced, g factors through D (cf. EGA I, 5.2.2); consequently

$$f_1 = p_1 \circ g = p_2 \circ g = f_2.$$

(iv)⇒(iii). Let T be a reduced S-scheme, and let $f_1, f_2 : T \to X$ be two S-morphisms that agree on a dense open subset U. Since T is reduced, in order to prove that $f_1 = f_2$, it is enough to prove that $f_1 = f_2$ set-theoretically. Indeed, suppose this has been established, let $t \in T$, let V be an affine open subset of X containing $f_1(t) = f_2(t)$, and let W be the open neighborhood of t that is the inverse image of V under the continuous map underlying both f_1 and f_2. Then the morphisms

$$f_i|_W : W \longrightarrow V$$

agree on the dense open subset $U \cap W$ of W. Since V is separated and W is reduced, this implies that $f_1|_W = f_2|_W$, and hence that $f_1 = f_2$.

Now let $t \in T$, let s be its image in S, and let us show that the points

$$x_1 = f_1(t), \qquad x_2 = f_2(t)$$

of X_s are equal. Let

$$\phi : S'' \longrightarrow S' \longrightarrow S$$

and an open subset V of $X \times_S S''$ be as in (iv). Set

$$T' = T \times_S S', \qquad T'' = T \times_S S'',$$

and denote by

$$g : T'' \longrightarrow T, \qquad f_i'' : T'' \longrightarrow X_{S''} \quad (i = 1, 2)$$

the morphisms obtained by base change.

Since U is dense in T and $T' \to T$ is open, the inverse image U' of U in T' is dense in T'. Let U'' be the inverse image of U' in T'', and let F be the reduced subscheme of T'' whose underlying space is $\overline{U''}$. Since $T'' \to T'$ is surjective and closed, the image of F contains U' and is closed, and is therefore equal to T'. Consequently,

$$F \cap g^{-1}(t)$$

contains a point u.

For $i = 1, 2$, let h_i be the restriction of f_i'' to F. Then $h_i(u)$ is a point of $X_{S''}$ above $f_i(t) = x_i$, and hence belongs to V, since V contains the inverse image of $\{x_1, x_2\}$ in $X_{S''}$.

Thus

$$W = h_1^{-1}(V) \cap h_2^{-1}(V)$$

is an open subset of F containing u, and the S''-morphisms

$$h_i|_W : W \longrightarrow V$$

agree on the dense open subset $U'' \cap W$ of W. Since V is separated over S'', we have

$$h_1|_W = h_2|_W.$$

It follows that $h_1(u) = h_2(u)$, and hence that $f_1(t) = f_2(t)$. This proves (iv) ⇒ (iii).

Theorem 5.3 now follows from 5.6.2 and the implication (iv) ⇒ (i) of 5.6.3.

Counterexample 5.6.4.— *Not every S-group G is separated. Let S be a scheme having a non-isolated closed point s, and let G be the scheme obtained by gluing two copies of S along the open subset $S - \{s\}$. It is easily seen that G is not separated over S, and that it carries a natural S-group structure whose fibers are all trivial, except for the fiber G_s, which is isomorphic to the two-element group $\mathbb{Z}/2\mathbb{Z}$.*[45]

Theorem 5.7.—[46] *Let S be a scheme, let G be an S-group locally of finite presentation over S, such that the function*

$$s \longmapsto \dim G_s$$

is locally constant on S, let X be an S-scheme on which G acts, and let U be an open subset of X, separated over S. Then $\underline{G^0} \cdot U$ is an open subset of X, separated over S.

Proof. Let

$$\mu : G \times_S X \longrightarrow X$$

denote the action of G on X. It is the composite of the automorphism

$$(g, x) \longmapsto (g, gx)$$

of $G \times_S X$ and the projection onto X. Since $G \times_S U$ is an open subset of $G \times_S X$, it follows from 4.2(iii) that

$$V = \underline{G^0} \cdot U$$

is open in X.

Then, arguing as in the proof of 5.6.2, we deduce from the implication (iv)⇒(i) of 5.6.3 that V is separated over S.

Corollary 5.7.1.— *Let S and G be as in 5.7, and let σ, τ be two S-sections of G^0, that is, $\sigma, \tau \in G^0(S)$. Then the coincidence subscheme $S(\sigma, \tau) \subset S$ of σ and τ* (that is, the inverse image of the diagonal of $G \times_S G$ under the morphism (σ, τ)) *is closed.*

Indeed, for every $s \in S$, let U be an affine open subset of G containing $\epsilon(s)$. Then

$$V = \epsilon^{-1}(U)$$

is an open neighborhood of s in S, and, since G^0U is separated, $S(\sigma, \tau) \cap V$ is closed in V.

Remark 5.7.2.— Gabber points out that one can show the following: if S is henselian local, with closed point s, then the intersection of the open subsets G^0U, where U ranges over the affine open neighborhoods of $\epsilon(s)$, is an open group subscheme of G, separated over S.

5.8. Complements

[47]We begin by recalling Proposition VI_A, 2.6.6:

[45]Editor's note: we reproduced this example in VI_A, 0.3, editor's note 5.

[46]Editor's note: on the one hand, we omitted Corollary 5.6.5, which repeated 5.5. On the other hand, the original stated Corollary 5.7.1 below as Remark 5.7, referring for its proof to 4.7, a number that does not exist in *Lecture Notes in Mathematics* 151 (but did appear in the 1965 edition of SGAD, whose numbers 4.5 and 4.6 became 5.6.1 and 5.6.2); we have added Theorem 5.7, communicated by O. Gabber, which makes the aforementioned statement of SGAD more precise.

[47]Editor's note: we have added this paragraph of complements and counterexamples, all communicated by O. Gabber.

Proposition 5.8.1.— *Let k be a field, let G be a k-group locally of finite type acting on a k-scheme X in such a way that the morphism*

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x),$$

is surjective. Suppose that X is quasi-separated. Then the connected components of X are irreducible.

Corollary 5.8.2.— *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that every fiber X_s is quasi-separated and connected. Then X is separated and quasi-compact over S.*

Indeed, by the preceding proposition, every fiber X_s is irreducible, and the rest of the proof is identical to that of 5.4.

Example 5.8.3.— Fix an algebraically closed field k. Recall first that every "locally Boolean" topological space X, that is, every space having a basis of compact open subsets, becomes a k-scheme when equipped with the sheaf of locally constant k-valued functions (cf. [DG70], I, §1, 2.12). Such an X will be called a *locally Boolean k-scheme*.

We also note that every topological space E having a basis of separated open subsets—and likewise every scheme X—has a dense separated open subset. Indeed, every increasing union of separated open subsets is separated (for a scheme X, this follows from 5.6.3), so there is a maximal such open subset U. It must be dense: otherwise there would be a nonempty open subset V with $U \cap V = \varnothing$; the open set V would contain a nonempty separated open subset W, and

$$U \cup W$$

would still be separated, contradicting the maximality of U.

Now let C be the Cantor space, viewed as the underlying space of the group

$$(\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$$

with its product topology. For every $p \in C$, let $C(p)$ be another copy of C, and let X be obtained by gluing each $C(p)$ to C along $C - \{p\}$. Then X is a nonseparated locally Boolean k-scheme, and C is a dense open subset of X.

Let G be the group of automorphisms of the k-scheme X, that is, of homeomorphisms of X. Then G acts transitively on X. Indeed, X is the union of C and, for each point $p \in C$, a second point p'. This second point is characterized as the unique $x \in X - \{p\}$ for which

$$(C - \{p\}) \cup \{x\}$$

is separated. Consequently, every automorphism ϕ of C extends to an automorphism ϕ_X of X satisfying

$$\phi_X(p') = \phi_X(p)'$$

for every p. Moreover, the map

$$\theta_p : X \longrightarrow X$$

that exchanges p and p' and fixes every other point is an automorphism of X. Finally, the automorphism group of C acts transitively on C: using the group structure of C, it is enough to consider translations.

Thus the discrete k-group G, which is locally of finite type, acts on the k-scheme X in such a way that

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x),$$

is surjective, but X is not separated, even though C is a dense separated open subset.

Example 5.8.4.— Retain the notation of the preceding example. Using the description of C as the set of sequences

$$(u_n)_{n\in\mathbb{N}}, \qquad u_n \in \mathbb{Z}/2\mathbb{Z},$$

one sees that C with one point p removed is homeomorphic to a countable disjoint union of copies of C. Indeed, using the group structure of C, for example, translation reduces us to the case in which $p = 0$, the zero sequence. Then

$$C - \{0\} = \coprod_{i\in\mathbb{N}} C_i,$$

where

$$C_i = \{(u_n)_{n\in\mathbb{N}} \mid u_i = 1 \text{ and } u_j = 0 \text{ for } j < i\},$$

and every C_i is homeomorphic to C. It follows by induction on $|F|$ that, for every nonempty finite subset F of C, the space $C - F$ is homeomorphic to a countable disjoint union of copies of C, and hence to C with one point removed.

For each $F \subset C$ of cardinality 2, let $C(F)$ be another copy of C, denote by q_F the point 0 of $C(F)$, and choose a homeomorphism

$$\phi_F : C(F) - \{q_F\} \xrightarrow{\sim} C - F.$$

Let X' be obtained by gluing every $C(F)$ to C by means of ϕ_F. Then X' is a nonseparated locally Boolean k-scheme. Moreover, the construction shows that every locally constant function

$$f : X' \longrightarrow k$$

is constant. Indeed, if $x, y \in C$ and $F = \{x, y\}$, then every neighborhood of q_F meets every neighborhood of x and of y. Thus

$$f(x) = f(q_F) = f(y).$$

If $F' = \{z, t\}$, with $z, t \in C$, then likewise

$$f(q_{F'}) = f(z) = f(q_{\{z,x\}}) = f(x).$$

Consequently, X' is connected.

Every point $x \in X'$ also has a pointed neighborhood homeomorphic to $(C, 0)$. More precisely, for every F, fix a homeomorphism of pointed topological spaces

$$h_F : (C, 0) \xrightarrow{\sim} (C(F), q_F),$$

and let T be the group of translations of C. If $x = q_F$, use h_F; if $x \in C$, the translation $t_x \in T$ is a homeomorphism

$$t_x : (C, 0) \xrightarrow{\sim} (C, x),$$

and it is the unique element of T with this property.

Let L be the free group generated by the h_F, and set

$$G = T * L,$$

the free product (coproduct) of T and L. For

$$h \in \{h_F\}_{|F|=2} \cup T,$$

let $\sigma(h)$ denote the corresponding generator of G, and let $S(h)$ and $B(h)$ be respectively the source and the target of h. It is convenient also to set

$$\sigma(h_F^{-1}) = \sigma(h_F)^{-1}, \qquad S(h_F^{-1}) = B(h_F), \qquad B(h_F^{-1}) = S(h_F),$$

and to let E be the set consisting of the elements of T and all h_F, h_F^{-1}.

On the product space

$$P = G \times X',$$

where G carries the discrete topology, consider the equivalence relation generated by

$$(g\sigma(h), x) \sim (g, h(x)) \qquad \text{when } x \in S(h),$$

for every $h \in E$, and let Z be the quotient space. Thus Z is obtained by gluing open subsets of the disjoint union

$$\coprod_{g \in G} \{g\} \times X'.$$

Consequently, the saturation $\widetilde{\Omega}$ of every open subset Ω of P is open (cf. [BTop], I, §5.1, Example 2). Explicitly, since every open subset of P is the union of its intersections with the slices $\{g\} \times X'$, it is enough to consider $\{1\} \times W$, with W open in X'. Its saturation is the union of $\{1\} \times W$ and the sets

$$\{\sigma(h_1)\} \times h_1^{-1}(W \cap B(h_1)),$$

$$\{\sigma(h_1)\sigma(h_2)\} \times h_2^{-1}\Big(h_1^{-1}(W \cap B(h_1)) \cap B(h_2)\Big),$$

and so on, for all finite sequences $h_1, \ldots, h_n$ of elements of E. It is therefore open, and the projection

$$\pi : P \longrightarrow Z$$

is open.

Moreover, the word

$$\sigma(h_1) \cdots \sigma(h_n)$$

is reduced in G, unless one of the h_i is the identity of T, two consecutive h_i's belong to T, or one of the pairs (h_i, h_{i+1}) is

$$(h_F, h_F^{-1}) \quad \text{or} \quad (h_F^{-1}, h_F).$$

Suppose, therefore, that $x \in X'$ and that

$$\beta = (\sigma(h_1) \cdots \sigma(h_n), h_n^{-1} \cdots h_1^{-1}(x))$$

belongs to $\{1\} \times X'$. We may then suppose that every h_i is a translation t_i. In that case,

$$\sigma(t_1 \cdots t_n) = 1$$

implies $t_1 \cdots t_n = 1$, and hence $\beta = (1, x)$. The equivalence relation is compatible with the action of G on $P = G \times X'$ by left translation on the first factor. It follows that, for every $g \in G$, the restriction of π to $\{g\} \times X'$ is injective.

Now let $z \in Z$ and choose a representative $(g, x) \in P$. By the preceding discussion,

$$U = \pi(\{g\} \times X')$$

is an open neighborhood of z, and the continuous map

$$\{g\} \times X' \longrightarrow U$$

induced by π is open and bijective, hence a homeomorphism. Thus Z is locally isomorphic to X', and hence also to C; it is therefore again a locally Boolean k-scheme.

Finally, G acts transitively on Z. Indeed, every $z \in Z$ is G-conjugate to an element of the form $\pi(1, x)$, so it is enough to show that every $(1, x) \in P$ is equivalent to some $(\sigma(h), 0)$. If $x \in C$, take $h = t_x$; if $x = q_F$, take $h = h_F$. Thus Z is a locally Boolean k-scheme endowed with a transitive action of the discrete group G, but Z is not separated.

6. Subfunctors and group subschemes*

Definition 6.1.— (i) Let X be an S-functor (that is, a functor from $(\mathbf{Sch}/S)^\circ$ to $(\mathbf{Ens})$), let G be an S-functor in groups, and let u and v be two S-morphisms from X to G. The *transporter of u into v*, denoted $\underline{\mathrm{Transp}}(u, v)$, is the sub-$S$-functor of G defined by

$$\begin{aligned}\underline{\mathrm{Transp}}(u,v)(S') &= \{g \in G(S') \mid (\mathrm{int}\, g) \circ u_{S'} = v_{S'}\} \\ &= \left\{ g \in G(S') \;\middle|\; \begin{array}{l} g_{S''} u_{S''}(x) g_{S''}^{-1} = v_{S''}(x), \\ \text{for every } x \in X(S'') \text{ and every } S'' \to S' \end{array} \right\}.\end{aligned}$$

In particular, $\underline{\mathrm{Transp}}(u, u)$ is a group sub-S-functor of G; it is called the *centralizer of u* and is denoted $\underline{\mathrm{Centr}}(u)$.

(ii) Let G be an S-functor in groups, and let X and Y be two sub-S-functors of G. The *transporter of X into Y* (respectively, the *strict transporter of X into Y*), denoted $\underline{\mathrm{Transp}}_G(X, Y)$ (respectively, $\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X, Y)$), is the sub-$S$-functor of G defined by

$$\begin{aligned}\underline{\mathrm{Transp}}_G(X,Y)(S') &= \{g \in G(S') \mid (\mathrm{int}\, g)(X_{S'}) \subset Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} \subset Y(S'') \text{ for every } S'' \to S'\},\end{aligned}$$

and, respectively,

$$\begin{aligned}\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X,Y)(S') &= \{g \in G(S') \mid (\mathrm{int}\, g)(X_{S'}) = Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} = Y(S'') \text{ for every } S'' \to S'\}.\end{aligned}$$

Note that

$$\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X,Y) = \underline{\mathrm{Transp}}_G(X,Y) \cap c\big(\underline{\mathrm{Transp}}_G(Y,X)\big),$$

where c denotes the inversion morphism of G.[49]

*On the same theme, see also the results of Exposé XI, §6, whose natural place would be in the present Exposé VI$_B$. There one finds, in particular, representability criteria for certain group subfunctors of a given group scheme. (48) Editor's note: we have inserted these results below (6.5.2–6.5.5).

[49]Editor's note: if $u : X \to G$ and $v : Y \to G$ are two arbitrary morphisms of S-functors, let $u(X)$ be the image functor of u, defined by

$$u(X)(S') = u(S')\big(X(S')\big) \subset G(S');$$

it is a subfunctor of G, as is the image functor $v(Y)$. One may then consider the transporter of the image of u into the image of v, $\underline{\mathrm{Transp}}_G(u(X), v(Y))$. Thus, in part (ii) of the definition, it is not necessary to restrict to subfunctors, that is, to the case in which u and v are monomorphisms. In the original, this restriction sometimes imposed repetitions in the hypotheses, such as: "Let $u, v : X \to G$ be morphisms of S-functors and let $w : H \to G$ and $w' : K \to G$ be monomorphisms; then $\underline{\mathrm{Transp}}(u, v)$, $\underline{\mathrm{Centr}}_G(w) = \underline{\mathrm{Centr}}_G H$, and $\underline{\mathrm{Transp}}_G(H, K)$ satisfy ...." Such repetitions can be avoided by considering $\underline{\mathrm{Transp}}_G(u(X), v(Y))$. We have made such modifications in 6.2 and, later, in 10.11.

(iii) Let G be an S-functor in groups, let H be a sub-S-functor of G, and let $i : H \to G$ be the canonical S-morphism. The *centralizer* and *normalizer of H in G* are the following group sub-S-functors of G:

$$\underline{\mathrm{Centr}}_G H = \underline{\mathrm{Centr}}(i) = \underline{\mathrm{Transp}}(i,i), \qquad \underline{\mathrm{Norm}}_G H = \underline{\mathrm{Transp}}^{\mathrm{str}}_G(H,H).$$

Finally, the *center of G* is the S-functor in groups

$$\underline{\mathrm{Centr}}(\mathrm{id}_G) = \underline{\mathrm{Centr}}_G G;$$

it is denoted $\underline{\mathrm{Centr}}G$.

Remark 6.1.1.—[50] It follows from the definitions that the functors $\underline{\mathrm{Transp}}(u,v)$ and $\underline{\mathrm{Transp}}_G(X,Y)$ (and hence also $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(X,Y)$) commute with base change: for every $S' \to S$, if G', X', Y', u', v' are obtained from G, X, Y, u, v by base change, then

$$\underline{\mathrm{Transp}}(u,v)_{S'} = \underline{\mathrm{Transp}}(u',v'), \qquad \underline{\mathrm{Transp}}(X,Y)_{S'} = \underline{\mathrm{Transp}}(X',Y').$$

Proposition 6.2.—[51] *Let G be an S-group. For a subfunctor T of the functor G, consider the following property:*

$(+_f)$ for every $s \in S$, T_s is representable by a closed subscheme of G_s.

Let $u, w : X \to G$ and $v : Y \to G$ be morphisms of S-schemes. Then:

(i) $\underline{\mathrm{Transp}}(u,w)$ and $\underline{\mathrm{Centr}}(u) = \underline{\mathrm{Transp}}(u,u)$ satisfy condition $(+_f)$.

(ii) $\underline{\mathrm{Transp}}_G(u(X), v(Y))$ and $\underline{\mathrm{Norm}}_G v(Y)$ satisfy condition $(+_f)$ if, for every $s \in S$, v_s is a closed immersion.

(iii) $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(X,Y)$ satisfies condition $(+_f)$ if, for every $s \in S$, u_s and v_s are closed immersions.

This follows from Remark 6.1.1 and Corollary 6.2.5 below.[52]

Definition 6.2.1.— Let $f : X \to S$ be a morphism of schemes. One says that f is *essentially free*, or that X is *essentially free over S*, if one can find a covering of S by affine open subsets S_i, for every i an affine S_i-scheme S'_i faithfully flat over S_i, and a covering $(X'_{ij})_j$ of

$$X'_i = X \times_S S'_i$$

by affine open subsets X'_{ij}, such that, for every (i,j), the ring of X'_{ij} is a projective[53] module over the ring of S'_i.

Proposition 6.2.2.—

a) *If X is essentially free over S, it is flat over S; the converse holds if S is Artinian.*

[50] Editor's note: we have added this remark, which is used implicitly in Proposition 6.2.

[51] Editor's note: we have modified the statement as indicated in editor's note 49.

[52] Editor's note: the original referred to the results of Exposé VIII, §6. For the reader's convenience, we have reproduced these results (with the exception of VIII, 6.3 and 6.8) in 6.2.1–6.2.5 below. Moreover, this had been suggested by A. Grothendieck in a note at the beginning of VIII.6: "The present section is independent of the theory of diagonalizable groups; its natural place would be in VI_B."

[53] Editor's note: on the one hand, we have replaced the word "free" by "projective", as indicated in VIII, Remark 6.8. On the other hand, the notion of an essentially free S-scheme is closely related to the geometric notion of an S-scheme that is flat and pure, introduced and studied by M. Raynaud and L. Gruson ([RG71]); see the addition 6.9 below.

b) *If S is the spectrum of a field, every S-scheme is essentially free over S.*

c) *If X is essentially free over S, then $X' = X \times_S S'$ is essentially free over S', for every $S' \to S$. The converse holds if $S' \to S$ is faithfully flat and quasi-compact.*

The proof is immediate, using for the converse in a) the fact that a flat module over an Artinian local ring is free.[54]

The introduction of Definition 6.2.1 is justified by the following theorem.

Theorem 6.2.3.— *Let S be a scheme, let Z be an S-scheme that is essentially free, and let Y be a closed subscheme of Z. Consider the functor*

$$F = \prod_{Z/S} Y/Z : (\mathbf{Sch}_{/S})^\circ \longrightarrow \mathbf{Sets}$$

defined by the following condition: $F(S') = \varnothing$ when $Y_{S'} \neq Z_{S'}$, and $F(S')$ is a singleton otherwise.[55]

(i) *This functor is representable by a closed subscheme T of S.*

(ii) *If, moreover, $Y \to Z$ is of finite presentation, then so is $T \to S$.*

Proof. First observe that the functor in question is a sheaf for the fpqc topology. Since $F(S')$ is either $\varnothing$ or $\{\mathrm{pt}\}$ for every S', this amounts to checking the following assertions. If (S_i) is an open covering of S and every

$$Y_{S_i} \longrightarrow Z_{S_i}$$

is an isomorphism, then $Y \to Z$ is an isomorphism; and if $S' \to S$ is faithfully flat and quasi-compact and

$$Y_{S'} \longrightarrow Z_{S'}$$

is an isomorphism, then again $Y \to Z$ is an isomorphism. The first assertion is clear, and the second follows from SGA 1, VIII, 5.4, or EGA IV$_2$, 2.7.1.

Moreover, by SGA 1, VIII, 1.9, faithfully flat and quasi-compact morphisms are effective descent morphisms for the fibered category of closed-immersion arrows. With the notation of 6.2.1, this allows us to restrict to the case $S = S'_i$.

Let (Z_j) be a covering of Z by affine open subsets such that

$$\mathcal{O}(Z_j)$$

is a projective module over $A = \mathcal{O}(S)$. Put

$$Y_j = Y \cap Z_j,$$

[54]Editor's note: indeed, let $(A, \mathfrak{m})$ be an Artinian local ring, k its residue field, M an arbitrary A-module, and let $(x_i)_{i\in I}$ be elements of M whose images form a basis of $M/\mathfrak{m}M$ over k. Let F be the free A-module with basis $(e_i)_{i\in I}$, and let $\varphi : F \to M$ be the A-morphism defined by $\varphi(e_i) = x_i$. Then $Q = \operatorname{Coker}\varphi$ satisfies $Q = \mathfrak{m}Q$, whence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose moreover that M is flat over A; then $K = \operatorname{Ker}\varphi$ satisfies $K \otimes_A k = 0$, that is, $K = \mathfrak{m}K$, whence $K = 0$.

[55]Editor's note: cf. Exposé II, §1, where this functor is denoted $\prod_{Z/S} Y$. For every $S' \to S$,

$$F(S') = \Gamma(Y_{S'}/Z_{S'}),$$

which here is equal to $\{\mathrm{id}_{Z_{S'}}\}$ if $Y_{S'} = Z_{S'}$, and is empty otherwise. We have also added assertion (ii), which is used in the proof of 6.5.3.

and let

$$F_j : (\mathbf{Sch}_{/S})^\circ \longrightarrow \mathbf{Sets}$$

be the functor defined in terms of (Z_j, Y_j) as F was defined in terms of (Z, Y). It is a subfunctor of the final functor, and clearly

$$F = \bigcap_j F_j.$$

It is therefore enough to prove that every F_j is representable by a closed subscheme T_j of S, for then F is represented by the closed subscheme

$$T = \bigcap_j T_j$$

of S.

We may consequently suppose that Z is itself affine, say

$$Z = \operatorname{Spec} B,$$

where B is a projective A-module. Let L be a free A-module with basis $(e_\lambda)_{\lambda\in\Lambda}$ of which B is a direct summand as an A-module, and let

$$\phi_\lambda : L \longrightarrow A$$

be the coordinate forms with respect to this basis. Let E be a set of generators for the ideal J of B defining the closed subscheme Y of Z, and let I be the ideal of A generated by the coordinates

$$\phi_\lambda(x), \qquad x \in E, \quad \lambda \in \Lambda.$$

For every A-algebra C, the morphism

$$B \otimes_A C \longrightarrow (B/J) \otimes_A C$$

is an isomorphism if and only if the image of $x \otimes 1$ in $L \otimes_A C$ is zero for every $x \in E$. This, in turn, is equivalent to saying that the kernel of $A \to C$ contains the ideal I. Hence

$$T = V(I) = \operatorname{Spec}(A/I)$$

has the required property, proving (i).

Finally, if $Y \to Z$ is of finite presentation, the set E may be chosen finite. Then I is a finitely generated ideal of A, that is, the closed immersion $T \to S$ is of finite presentation. This proves (ii).

Examples 6.2.4.— Let us give important examples of functors which reduce to functors

$$\prod_{Z/S} Y/Z$$

of the type considered in 6.2.3, and for which it will be useful to have representability criteria later on. We denote by S a scheme, and by $X, Y, Z, \ldots$ schemes over S.

a) Suppose given an S-morphism

$$q : X \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z), \tag{x}$$

("X acts on Y, with values in Z"), that is, a morphism

$$r : X \times_S Y \longrightarrow Z. \tag{xx}$$

Consider a subscheme Z' of Z, and hence a monomorphism

$$\underline{\mathrm{Hom}}_S(Y, Z') \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z),$$

which makes the first functor a subfunctor of the second. Let X' be the inverse image of this subfunctor under q. Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Z_T$$

factors through Z'_T.

This functor X' may be described as follows. Put

$$P = X \times_S Y,$$

and let P' be the inverse image of Z' under $r : P \to Z$. Then there is an evident isomorphism

$$X' \simeq \prod_{P/X} P'/P. \tag{xxx}$$

It follows that, if Y is essentially free over S and Z' is closed in Z, the subfunctor X' of X is representable by a closed subscheme of X.[56] If, moreover, $Z' \to Z$ is of finite presentation, then so is $X' \to X$.

b) Suppose given two ways of letting X act on Y, with values in Z, that is, two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\mathrm{Hom}}_S(Y, Z),$$

and put

$$X' = \mathrm{Ker}(q_1, q_2).$$

Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms

$$q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$$

are equal. The data of q_1, q_2 are equivalent to the data of a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z \times_S Z),$$

or, equivalently, of a morphism

$$r : X \times_S Y \longrightarrow Z \times_S Z.$$

Put $U = Z \times_S Z$, and let U' be the diagonal subscheme of $Z \times_S Z$. Then X' is precisely the inverse image under q of the subfunctor

$$\underline{\mathrm{Hom}}_S(Y, U') \subset \underline{\mathrm{Hom}}_S(Y, U).$$

It can therefore be put in the form (xxx), with $P = X \times_S Y$ and P' the inverse image of the diagonal under r, that is, the kernel of

$$r_1, r_2 : X \times_S Y \rightrightarrows Z.$$

[56] Editor's note: we have added the sentence that follows.

We are thus under the hypotheses of (a).

Consequently, if Y is essentially free over S and Z is separated over S, the subfunctor X' of X is representable by a closed subscheme of X.[56] If, moreover, $Z \to S$ is locally of finite type, then $X' \to X$ is of finite presentation.

c) Suppose given a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y),$$

that is, "X acts on Y." Let X' be the "kernel" of this morphism: X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Y_T$$

is the identity. This functor falls under (b), as one sees by introducing a second morphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

by letting X act trivially on Y. Hence, if Y is essentially free and separated over S, the subfunctor X', the kernel of q, is representable by a closed subscheme of X.[56] If, moreover, $Y \to S$ is locally of finite type, then $X' \to X$ is of finite presentation.

d) Under the hypotheses of (c), consider the subfunctor Y' of Y "of the invariants under X." Thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism

$$\bar{q}(y) : X_T \longrightarrow Y_T$$

is the constant T-morphism with value y. Introducing $\bar{q}'$ as in (c), and the morphisms corresponding to $\bar{q}$ and $\bar{q}'$,

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that

$$Y' = \mathrm{Ker}(\bar{q}, \bar{q}').$$

It therefore falls again under (b), with the roles of X and Y reversed and with $Z = Y$.

Consequently, if X is essentially free over S and Y is separated over S, the subfunctor Y' of Y of the invariants under X is representable by a closed subscheme of Y.[56] If, moreover, $Y \to S$ is locally of finite type, then $Y' \to Y$ is of finite presentation.

e) Constructions of the kind described in the preceding examples occur especially often in group theory. Thus, when G is an S-group scheme acting on the S-scheme X,

$$q : G \longrightarrow \underline{\mathrm{Aut}}_S(X),$$

the kernel of q, the subgroup of G acting trivially, is a closed subscheme of G, provided X is essentially free and separated over S (example (c)). The subobject X^G of invariants is a closed subscheme of X, provided G is essentially free over S and X is separated over S (example (d)).

Let Y, Z be subschemes of X, and consider the subfunctor $\underline{\mathrm{Transp}}_G(Y, Z)$ of G, the transporter of Y into Z. Its T-valued points, for T over S, are the elements $g \in G(T)$ such that the corresponding automorphism of X_T satisfies

$$g(Y_T) \subset Z_T,$$

that is, induces a morphism $Y_T \to X_T$ which factors through $Y_T \to Z_T$. Hence, if Y is essentially free over S and Z is closed in X, then $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subscheme of G (example (a)).

One may also consider the strict transporter of Y into Z,[57] whose T-valued points, for T over S, are the elements $g \in G(T)$ such that $g(Y_T) = Z_T$. This functor is precisely

$$\underline{\mathrm{Transp}}_G(Y,Z) \cap \sigma(\underline{\mathrm{Transp}}_G(Z,Y)),$$

where σ is the inversion morphism of G. Consequently, if Y and Z are essentially free over S and closed in X, the strict transporter of Y into Z is a closed subscheme of G.

An important case is $X = G$, with G acting on itself by inner automorphisms. If H is a subscheme of G, the strict transporter of H into H is also called the *normalizer of H in G*, and is denoted $\underline{\mathrm{Norm}}_G H$. Hence, if H is a closed group subscheme of G, essentially free over S, then $\underline{\mathrm{Norm}}_G H$ is representable by a closed group subscheme of G.

Finally, let Z be a subscheme of G. Its centralizer $\underline{\mathrm{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure of (d), when one regards Z as acting on G by the operations induced by those of G. Thus, if Z is essentially free over S and G is separated over S, then $\underline{\mathrm{Centr}}_G(Z)$ is a closed group subscheme of G. In particular, if G is essentially free and separated over S, then the center C of G, which is none other than $\underline{\mathrm{Centr}}_G(G)$, is a closed group subscheme of G.

When S is the spectrum of a field, 6.2.2(b) shows that the "essentially free" conditions in examples (a)–(e) above are automatically satisfied, leaving only the separation conditions. Recalling that a group scheme over a field is necessarily separated (VI_A, 0.3), we obtain, for example:

Corollary 6.2.5.— *Let G be a group scheme over a field k, and let Y, Y' be two subschemes of G. Then:*

(i) *The centralizer of Y in G is a closed group subscheme of G; in particular, this holds for the center $\underline{\mathrm{Centr}}_G(G)$ of G.*

(i′) *More generally, if $u, v : X \to G$ are morphisms of schemes, then $\underline{\mathrm{Transp}}_G(u,v)$ is representable by a closed subscheme of G.*

(ii) *If Y is closed, the transporter $\underline{\mathrm{Transp}}_G(Y',Y)$ is a closed subscheme of G. If Y' is also closed, the same conclusion holds for $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(Y',Y)$.*

(iii) *For every group subscheme[58] H of G, $\underline{\mathrm{Norm}}_G(H)$ is a closed group subscheme of G.*

Corollary 6.2.6.—[59] *Let k be a field and let G be a connected algebraic k-group. Then $\underline{\mathrm{Centr}}_G(G)$ is representable by a closed group subscheme Z of G, and G/Z is an affine algebraic k-group.*

Proof. The first assertion is, of course, contained in 6.2.5(i), but we shall see that it also follows from the proof of the second assertion. Indeed, G acts by the adjoint representation on the finite-dimensional k-vector spaces

$$V_n = \mathcal{O}_{G,e}/\mathfrak{m}_e^{n+1},$$

where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$. Let K_n be the kernel of the morphism

$$\rho_n : G \longrightarrow \mathrm{GL}(V_n).$$

[57] Editor's note: denoted $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(Y,Z)$.

[58] Editor's note: indeed, over a field k, every group subscheme of G is closed; see VI_A, 0.5.2. Moreover, 6.2.5 completes the insertion of the results taken from VIII, §6.

[59] Editor's note: we have inserted this corollary here (cf. Exposé XII, 6.1), as it will be useful later. We return in 6.3 to the original text of Exposé VI_B.

By VI$_A$, 5.4.1, ρ_n induces a closed immersion

$$G/K_n \hookrightarrow \mathrm{GL}(V_n),$$

so every G/K_n is affine. Since G is Noetherian, the intersection K of the K_n is equal to one of the K_n; hence G/K is affine.

On the other hand, if Z denotes the center of G, it is clear that

$$Z \subset K.$$

We shall prove the reverse inclusion. Let $\widehat{\mathcal{O}}_{G,e}$ be the completion of $\mathcal{O}_{G,e}$ for the $\mathfrak{m}_e$-adic topology, and put

$$\widehat{S} = \operatorname{Spec} \widehat{\mathcal{O}}_{G,e}, \qquad S = \operatorname{Spec} \mathcal{O}_{G,e}.$$

Since $\widehat{S} \to S$ is faithfully flat, and since the two morphisms

$$K \times_k S \longrightarrow S, \qquad (g,x) \longmapsto gxg^{-1} \quad \text{and} \quad (g,x) \longmapsto x$$

coincide after the base change $\widehat{S} \to S$, they coincide. Thus K acts trivially on $\mathcal{O}_{G,e}$.

By 6.2.4(e), the subobject G^K of points of G fixed by K, which is precisely $\underline{\mathrm{Centr}}_G(K)$, is a closed subscheme of G. It is therefore defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_G$. Since G^K contains

$$S = \operatorname{Spec} \mathcal{O}_{G,e}$$

and $\mathcal{I}$ is of finite type (because G is Noetherian), there is an open neighborhood U of e such that

$$\mathcal{I}|_U = 0.$$

The subgroup

$$G^K = \underline{\mathrm{Centr}}_G(K)$$

therefore contains U, and hence also $U \cdot U$, which is equal to G because G is irreducible (VI$_A$, 0.5). Consequently,

$$\underline{\mathrm{Centr}}_G(K) = G.$$

Thus $K \subset Z$, and therefore $Z = K$, as required.

Remark 6.3.— *Let k be an algebraically closed field, let G be a k-group, and let H be a group subscheme of G. Suppose that G and H are of finite type over k and are reduced. Then $\underline{\mathrm{Norm}}_G H$ (respectively, $\underline{\mathrm{Centr}}_G H$) is representable by a group subscheme of G, whose associated reduced subscheme is precisely the normalizer (respectively, centralizer) of H in G in the sense of the* Bible.

Proposition 6.4.— *Let G be an S-group, and let $u : X \to G$ be a monomorphism of S-schemes. Set*

$$T = \underline{\mathrm{Transp}}_G(X, X).$$

The following conditions are equivalent:

(i) *T is a group sub-S-functor of G.*

(ii)
$$T = \underline{\mathrm{Transp}}^{\mathrm{str}}_G(X, X) = \underline{\mathrm{Norm}}_G X.$$

These conditions hold in each of the following two cases:

a) *X is of finite presentation over S.*

b) *T is representable by a scheme of finite presentation over S.*

The equivalence of (i) and (ii) follows from the fact that, for every morphism $S' \to S$ and all $t, t' \in T(S')$, one has

$$tt' \in T(S'),$$

and from the identity

$$\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X, X) = T \cap c(T)$$

(cf. 6.1(ii)).

Suppose first that a) holds. If $t \in T(S)$, then $\mathrm{int}(t)$ is a monomorphism from X to itself, and therefore an S-automorphism of X by EGA IV$_4$, 17.9.6. Thus

$$t \in \underline{\mathrm{Transp}}_G^{\mathrm{str}}(X, X),$$

which proves the assertion in case a).

In case b), it is clear that

$$\mu(T \times_S T) \subset T,$$

and the assertion follows from the next lemma.

Lemma 6.4.2.—[60] *Let G be an S-scheme of finite presentation, equipped with an associative law (in the sense of EGA 0_{III}, 8.2.5). Suppose that, for every S-scheme S' and every $g \in G(S')$, right and left translation by g in the set $G(S')$ are injective, and suppose that $G(S) \neq \varnothing$. Then G is an S-group.*

Proof. It is enough to prove that $G(S')$ is a group for every S-scheme S'. The hypothesis immediately implies that right and left translation by every $g \in G(S')$ on $G_{S'}$ are S'-monomorphisms

$$G_{S'} \longrightarrow G_{S'}.$$

They are therefore S'-automorphisms, since G is of finite presentation over S (EGA IV$_4$, 17.9.6). Consequently, right and left translation by g in the set $G(S')$ are bijective, and the assertion reduces to the following lemma.

Lemma 6.4.3.— *Let G be a nonempty set with an associative law such that right and left translations are bijective. Then G is a group.*

The proof is left to the reader.

Definition 6.5.— *Let G be an S-group, let H be an S-functor, and let $u : H \to G$ be a monomorphism.*

(i) The *connected centralizer of H in G*, denoted

$$\underline{\mathrm{Centr}}_G^0 H,$$

is the identity component of the functor $\underline{\mathrm{Centr}}_G H$ (cf. 3.1 and 6.1(iii)).

(i′) For every morphism $u : X \to G$, one similarly defines the functor

$$\underline{\mathrm{Centr}}^0(u)$$

(cf. 6.2(iv)).

(ii) When u_s is a closed immersion for every $s \in S$, the *connected normalizer of H in G*, denoted

$$\underline{\mathrm{Norm}}_G^0 H,$$

is the identity component of the functor $\underline{\mathrm{Norm}}_G H$.

[60] Editor's note: we have retained the numbering of the original; there is no number 6.4.1.

N.B. By 1.4.2, the hypothesis in (ii) is satisfied when H is an S-group scheme, G and H are locally of finite type over S, and u is a quasi-compact morphism of S-groups.

Proposition 6.5.1.— *Let G be an S-group locally of finite presentation and quasi-separated over S, let H be a smooth S-group with connected fibers, and let $u : H \to G$ be a monomorphism. Let N be the normalizer of H in G (cf. 6.1). By 6.5.5 below, N is representable by a closed group subscheme of G, of finite presentation over G. Under these circumstances, the following conditions are equivalent:*

(i) *The canonical morphism $H \to N$ is an open immersion.*

(ii) *$N^0 = H$ (cf. 3.10).*

(iii) *For every $s \in S$, one has $H_s = (N_s)^0$.*

Condition (i) implies (ii) by Lemma 3.10.1, since $H^0 = H$. It is also clear that (ii) implies (iii), for

$$H_s = (N^0)_s = (N_s)^0.$$

Finally, let us prove that (iii) implies (i). Since $H_s = (N^0)_s$ for every $s \in S$, the morphism

$$H_s \longrightarrow N_s$$

is an open immersion. Moreover, H and N are locally of finite presentation over S, and H is flat over S. Consequently, $H \to N$ is an open immersion (EGA IV_4, 17.9.5).

For the reader's convenience, the results 6.8–6.11 of Exposé XI have been included below.[61]

Theorem 6.5.2.— *Let X be a smooth scheme over S, with geometrically irreducible fibers.*[62]

(i) *For every closed subscheme Y of X, the functor*

$$\prod_{X/S} Y/X$$

is representable by a closed subscheme T of S.

(ii) *Moreover, if $Y \to X$ is of finite presentation, then so is $T \to S$.*

Proof. Let $f : X \to S$ denote the structure morphism. Since f is faithfully flat and locally of finite presentation, it is a covering for the fpqc topology. Moreover,

$$T = \prod_{X/S} Y/X$$

is evidently a subsheaf of S for the fpqc topology. It follows that the representability of T by a closed subscheme of S is a question local on S for the fpqc topology,[63] and the same is true of the question whether T is of finite presentation over S (EGA IV_2, 2.7.1).

[61] Editor's note: Sections 6.5.2–6.5.5, taken from Exposé XI, 6.8–6.11, have been inserted here. This was in fact suggested by A. Grothendieck in a note at the beginning of XI 6: "The present section does not use the results of Sections 3, 4, and 5 [of XI]; its natural place would be in VI_B."

[62] Editor's note: on the one hand, we have corrected the original by replacing "connected" with "irreducible" (see the proof). On the other hand, according to [RG71], I, 3.3.4(iii) and 4.1.1, it is enough to assume that X is flat over S, with geometrically irreducible fibers and without embedded components.

[63] Editor's note: see, for example, the addition 1.7 in Exposé VIII.

After the base change $S' \to S$ with $S' = X$, we are reduced to the case in which X admits a section

$$\epsilon : S \longrightarrow X.$$

We may also suppose that S is affine, and hence quasi-compact. We then have:

Corollary 6.5.3.— *Under the hypotheses of 6.5.2, suppose that S is quasi-compact, that $X \to S$ admits a section ϵ, and that $Y \to X$ is of finite presentation. Then there exists an integer $n \geq 0$ such that*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n is the n-th infinitesimal neighborhood of the immersion $\epsilon : S \to X$, and $Y_n = Y \cap X_n$.

When Y is of finite presentation over X, this corollary does indeed imply 6.5.2. Since X is smooth over S, X_n is finite and locally free over S, and is therefore, a fortiori, essentially free over S (see 6.2.1). Hence

$$\prod_{X_n/S} Y_n/X_n$$

is representable by a closed subscheme T of S, of finite presentation over S, by 6.2.3.

Let us first prove 6.5.3, and hence 6.5.2, when S is Noetherian. Put

$$T_n = \prod_{X_n/S} Y_n/X_n.$$

The T_n form a decreasing sequence of closed subschemes of S, and since S is Noetherian this sequence is stationary. Let

$$R = \bigcap_{n\geq 0} \prod_{X_n/S} Y_n/X_n$$

be their common value for all sufficiently large n. Clearly $T \subset R$, and it is enough to prove $R \subset T$.

After the base change $R \to S$, we are reduced to the case

$$R = S,$$

that is,

$$Y_n = X_n \quad \text{for every } n, \qquad \text{or equivalently} \qquad Y \supset X_n \quad \text{for every } n.$$

We must then prove $T = S$, that is, $Y = X$.

Now $Y \supset X_n$ for every n, together with the fact that X is locally Noetherian, implies that in a neighborhood of each point of $\epsilon(S)$, Y is an induced open subscheme of X.[64] Consequently, there exists an induced open subscheme U of X, containing $\epsilon(S)$, such that $U \subset Y$. By EGA IV_3, 11.10.10, since the fibers of X/S are integral, U is schematically dense in X. Since Y is a closed subscheme containing U, it follows that $Y = X$. This proves 6.5.3, and hence 6.5.2, when S is Noetherian.

The general case is reduced to the preceding one. For every $s \in S$, there exist an affine open neighborhood U of s and an affine open neighborhood V of $\epsilon(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U. Let S' be an affine open neighborhood of s contained in

$$\epsilon^{-1}(V) \cap f(V),$$

[64] Editor's note: see, for example, the proof of 6.2.6.

and put

$$X' = V \cap f^{-1}(S').$$

Then X' and S' are affine open subsets of X and S, respectively, and X'/S' admits a section. By the local nature of 6.5.2 and 6.5.3, we may suppose that $S = S'$.

Thus X' is an affine open subset of X containing $\epsilon(S)$. Since every fiber X_s is irreducible, X'_s is schematically dense in X_s. Consequently, by EGA IV_3, 11.10.10, X' is schematically dense in X; likewise, after every base change $S_1 \to S$,

$$X' \times_S S_1$$

is schematically dense in $X \times_S S_1$.

It follows that

$$\prod_{X/S} Y/X = \prod_{X'/S} Y'/X', \qquad Y' = Y \cap X'.$$

This reduces us to the case $X = X'$, so we may suppose that both S and X are affine. Moreover, if

$$X = \operatorname{Spec} B$$

and J is the ideal of B defining Y, then J is the filtered colimit of its finitely generated subideals. Thus Y is the intersection of closed subschemes Y_i of X that are of finite presentation over X, and consequently

$$\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X.$$

To prove 6.5.2, we are therefore reduced to the case in which Y is of finite presentation over X, with S and X affine.

In that case X and Y over S are obtained by base change $S \to S_0$ from an analogous situation X_0, Y_0 over a Noetherian scheme S_0. This reduces us to the Noetherian case already treated, and completes the proof of 6.5.2 and 6.5.3.

Corollary 6.5.4.— *Let X be a smooth S-group scheme of finite presentation with connected fibers, let Y be a group scheme of finite presentation over S, and let $i : Y \to X$ be a monomorphism of S-group schemes.*

(i) *The functor*

$$\prod_{X/S} Y/X$$

is representable by a closed subscheme of S of finite presentation.

(ii) *If, moreover, S is quasi-compact, then for all sufficiently large n,*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n denotes the n-th infinitesimal neighborhood of the unit section $\epsilon : S \to X$, and $Y_n = X_n \cap Y$.

The proof is essentially that of 6.5.3.[65] On the one hand, i is locally of finite presentation (EGA IV_1, 1.4.3). On the other hand, the unit sections of Y and X induce bijective immersions

$$S \longrightarrow Y_n \qquad \text{and} \qquad S \longrightarrow X_n,$$

[65] Editor's note: the original has been expanded in what follows, and references to EGA IV have been added.

and hence isomorphisms of S_{red} with $(Y_n)_{\mathrm{red}}$ and $(X_n)_{\mathrm{red}}$. Consequently, i_n is quasi-compact, hence of finite type, and one has the commutative diagram

$$\begin{array}{ccc} (Y_n)_{\mathrm{red}} & \xrightarrow{\tau} & Y_n \\ & \searrow^{\sigma} & \downarrow{\scriptstyle i_n} \\ & & X_n \end{array}$$

where σ and τ are closed immersions and τ is surjective. Since i_n is separated (being a monomorphism), it follows that i_n is proper (EGA II, 5.4.3). Thus i_n is a proper monomorphism of finite presentation, and therefore a closed immersion (EGA IV_3, 8.11.5). It follows from 6.2.3 that

$$\prod_{X_n/S} Y_n/X_n$$

is representable by a closed subscheme T_n of S, of finite presentation over S. It remains to prove the last assertion of 6.5.4 when S is assumed affine as well.

We again reduce immediately to the case where S is Noetherian. With the notation of the proof of 6.5.3, it is enough to prove $R = T$, or equivalently that

$$Y \supset X_n \quad \text{for every } n \qquad \Longrightarrow \qquad Y = X.$$

The hypothesis implies that $i : Y \to X$ is étale at the points of the unit section of Y over S. Thus Y is smooth over S at those points, and the open subset U of points of Y at which Y is smooth over S is an induced open subgroup of Y (cf. 2.3). The morphism

$$\tau : U \longrightarrow X$$

is then an étale monomorphism by 2.5, and hence an open immersion. Since the fibers of X are connected and every open subgroup of an algebraic group is also closed, τ is a surjective open immersion, that is, an isomorphism. Therefore $U = X$, and a fortiori $Y = X$. This completes the proof of 6.5.4.

Proceeding as in 6.2.4(e), one deduces Corollary 6.5.5 from 6.5.4.

Corollary 6.5.5.— *Let G be an S-group scheme locally of finite type and quasi-separated over S, let H be a smooth group scheme of finite presentation over S with connected fibers, and let $i : H \to G$ be a monomorphism of S-groups. Then:*

a) $\underline{\mathrm{Centr}}_G(H)$ *and* $\underline{\mathrm{Norm}}_G(H)$ *are representable by closed subschemes of G, of finite presentation over G.*

a′) *Likewise, for every monomorphism $j : K \to G$ of S-group schemes that is of finite presentation, $\underline{\mathrm{Transp}}_G(H, K)$ is representable by a closed subscheme of G, of finite presentation over G.*

b) *If G is quasi-compact, there exists an integer $n \geq 0$ such that, if H_n denotes the n-th infinitesimal neighborhood of the unit section of H, then*

$$\underline{\mathrm{Centr}}_G(H) = \underline{\mathrm{Centr}}_G(H_n), \qquad \underline{\mathrm{Norm}}_G(H) = \underline{\mathrm{Norm}}_G(H_n),$$

$$\underline{\mathrm{Transp}}_G(H, K) = \underline{\mathrm{Transp}}_G(H_n, K) = \underline{\mathrm{Transp}}_G(H_n, K_n).$$

Proof.[66] First observe that the hypothesis on G implies that the monomorphism $H \to G$ is of finite presentation (EGA IV$_1$, 1.2.4 and 1.4.3), as is the diagonal immersion

$$\Delta_{G/S} : G \longrightarrow G \times_S G$$

(loc. cit., 1.4.3.1). The case of $\underline{\mathrm{Norm}}_G(H)$ therefore reduces to that of the transporter, by taking $H = K$.

In view of 6.2.4(e), we apply 6.5.4 to the group scheme

$$X = H_G = H \times_S G$$

over the base scheme G, and to the following group subscheme Y.

In the case of $\underline{\mathrm{Transp}}_G(H, K)$, take Y to be the inverse image of K_G under the morphism of G-groups

$$H \times_S G \longrightarrow G \times_S G, \qquad (h, g) \longmapsto (ghg^{-1}, g).$$

In the case of $\underline{\mathrm{Centr}}_G(H)$, take Y to be the inverse image of the diagonal subgroup

$$\Delta_{G/S}(G)_G$$

of $(G \times_S G)_G$ under the morphism of G-groups

$$H \times_S G \longrightarrow G \times_S G \times_S G, \qquad (h, g) \longmapsto (h, ghg^{-1}, g).$$

Definition 6.6.— Let G be an S-functor in groups and let H be a group sub-S-functor of G. One says that H is *invariant* (respectively, *central*, respectively, *characteristic*) in G if

$$\underline{\mathrm{Norm}}_G H = G$$

(respectively, if $\underline{\mathrm{Centr}}_G H = G$, respectively, if for every S-scheme T and every automorphism

$$a \in \mathrm{Aut}_{T\text{-gr}}(G_T)$$

one has $a(H_T) \subset H_T$). In other words, for every S-scheme T, the subgroup $H(T)$ of $G(T)$ is normal in $G(T)$ (respectively, central in $G(T)$, respectively, invariant under every automorphism of G_T).

N.B. If H is central (respectively, characteristic), then it is invariant.

Remark 6.7.— Let G and H be two S-groups and let $u : H \to G$ be a monomorphism. The group H is invariant (respectively, central) in G if and only if the morphism

$$\mu \circ \big(c \circ \mathrm{pr}_2,\, \mu \circ (u \times \mathrm{id}_G)\big) : H \times_S G \longrightarrow G$$

(defined by

$$(h, g) \longmapsto g^{-1} h g$$

for $g \in G(S')$ and $h \in H(S')$) factors through u (respectively, is equal to $u \circ \mathrm{pr}_1$). The group H is characteristic in G if and only if, for every S-scheme T and every T-group automorphism a of G_T, the morphism $a \circ u(T)$ factors through $u(T)$.

Example 6.8.— Let G be an S-functor in groups. Then $\underline{\mathrm{Centr}}G$ is characteristic and central. If G_s is representable for every $s \in S$, then G^0 is characteristic. This follows from the definitions and 3.3.

6.9.—[67] In [RG71], I, 3.3.3, the authors introduce the geometric notion of a *pure S-scheme*, which is local on S for the étale topology; we refer to *loc. cit.* for the precise definition. We merely record the following points:

[66] Editor's note: the argument that follows has been expanded. This also completes the insertion of XI, 6.8–6.11; at 6.6 below we return to the original text of VI$_B$.

[67] Editor's note: this subsection has been added.

a) (*loc. cit.*, Theorem 3.3.5) If B is a flat, finitely presented A-algebra, then B is a projective A-module if and only if $\mathrm{Spec}(B)$ is pure over $\mathrm{Spec}(A)$.

b) Consequently, if $X \to S$ is locally of finite presentation, flat, and pure, then X is essentially free over S.

c) (*loc. cit.*, 3.3.4(iii)) If $X \to S$ is locally of finite type, flat, has geometrically irreducible fibers, and has no embedded components, then X is pure over S.

Every group scheme locally of finite type over a field is Cohen–Macaulay (cf. VI_A, 1.1.1), and therefore has no embedded components. We obtain in particular:

d) Every S-group scheme G that is locally of finite presentation, flat, and has connected fibers is pure over S, and hence essentially free over S.

All the statements of 6.2.4(e) may therefore be used with the results of (b) and (d) above taken into account.

7. Generated subgroups; commutator group

Throughout this section, k denotes a fixed field.

Proposition 7.1.— *Let G be a k-group, let $(X_i)_{i\in I}$ be a family of geometrically reduced k-schemes,*[68] *and, for every $i \in I$, let $f_i : X_i \to G$ be a k-morphism.*

(i) *There exists a smallest closed k-group subscheme of G through which every f_i factors. It is denoted*

$$\Gamma_G((f_i)_{i\in I}).$$

It is a geometrically reduced k-scheme, and hence smooth when G is locally of finite type over k (1.3.1).

(ii) *Put*

$$X = \coprod_{i\in I} X_i,$$

and let $f : X \to G$ be the morphism whose restriction to X_i is f_i, for every $i \in I$. Put

$$X^1 = X \amalg X,$$

and let $f^1 : X^1 \to G$ be the morphism whose restrictions to the two copies of X are f and $c \circ f$, respectively. For every $n > 1$, put

$$X^n = X^1 \times_k X^{n-1}, \qquad f^n = \mu \circ (f^1 \times_k f^{n-1}) : X^n \longrightarrow G.$$

Then $\Gamma_G((f_i)_{i\in I})$ is the reduced subscheme of G whose underlying space is the closure of

$$\bigcup_{n\geq 1} f^n(X^n).$$

(iii) *For every k-scheme S, $\Gamma_G((f_i)_{i\in I})_S$ is the smallest closed group subscheme of G_S through which every*

$$f_{i,S} : (X_i)_S \longrightarrow G_S$$

factors.

[68] Editor's note: here and below, the little-used term "separable" has been replaced by the customary term "geometrically reduced"; see EGA IV_2, 4.6.2.

(iii′) *Moreover, $\Gamma_G((f_i)_{i\in I})_S$ is the smallest group subscheme of G_S, not necessarily closed, through which every $f_{i,S}$ factors.*[69]

First note that, to prove (i), (iii), and (iii′), after defining X and f as in (ii), we are reduced to the case in which I consists of a single element.

Let H be the reduced subscheme of G whose underlying set is

$$\overline{\bigcup_{n\geq 1} f^n(X^n)}.$$

The family of morphisms

$$f^n : X^n \longrightarrow H$$

is schematically dominant (cf. EGA IV$_3$, 11.10.4), and therefore every closed subscheme of G through which all the f^n factor also contains H. Moreover, by *loc. cit.*, 11.10.7, H is geometrically reduced. To prove (i) and (ii), it is therefore enough to show that H is a group subscheme of G, that is, that the restriction of c to H and the restriction of μ to $H \times_k H$ factor through the inclusion $H \to G$.

Since H is geometrically reduced, $H \times_k H$ is reduced, and it is enough to verify, set-theoretically, that

$$c(H) \subset H, \qquad \mu(H \times_k H) \subset H.$$

By EGA IV$_3$, 11.10.6, the union of the

$$f^n_{(H)}(X^n \times_k H)$$

is schematically dense in $H \times_k H$. Similarly, for every $n \geq 1$, the union of the

$$f^m_{(X^n)}(X^n \times_k X^m), \qquad m \geq 1,$$

is schematically dense in $X^n \times_k H$. It is therefore enough to prove

$$\mu\Big(f^n_{(H)}\big(f^m_{(X^n)}(X^n \times_k X^m)\big)\Big) \subset H$$

and $c(f^n(X^n)) \subset H$. But

$$\begin{aligned}\mu\Big(f^n_{(H)}\big(f^m_{(X^n)}(X^n \times_k X^m)\big)\Big) &= \mu((f^n \times_k f^m)(X^n \times_k X^m)) \\ &= f^{n+m}(X^{n+m}) \subset H.\end{aligned}$$

Also, since $c(f^1(X^1)) \subset f^1(X^1)$, for every n one has

$$c(f^n(X^n)) \subset f^n(X^n) \subset H.$$

This proves (i) and (ii).

We now prove (iii). Let G' be a closed group subscheme of G_S through which f_S factors. We must show that G' contains H_S, or equivalently that

$$H_S = H_S \times_{G_S} G'.$$

Put

$$H'_S = H_S \times_{G_S} G' = G' \cap H_S.$$

[69] Editor's note: the original stated (iii′) under the additional hypothesis that G be locally of finite type over k; this hypothesis may be omitted in view of VI$_A$, 0.5.2.

Every f_S^n factors through both G' and H_S, and hence through H'_S. Since the family

$$f^n : X^n \longrightarrow H$$

is schematically dominant, EGA IV$_3$, 11.10.6 implies that the base-changed family

$$f_S^n : X_S^n \longrightarrow H_S$$

is schematically dominant as well. Because every f_S^n factors through H'_S, EGA IV$_3$, 11.10.1(c) gives

$$H'_S = H_S.$$

This proves (iii).

Finally, we show that H_S is the smallest group subscheme, not necessarily closed, of G_S through which f_S factors. Let G' be such a group subscheme and again put

$$H'_S = H_S \times_{G_S} G'.$$

We must show $H'_S = H_S$. By (iii), it is enough to show that H'_S is closed in H_S. It is enough in turn to show that H_S and H'_S have the same underlying set, and for this it suffices to show, for every $s \in S$, that

$$H'_s := H'_S \times_S \kappa(s) = H_s \times_{G_s} G'_s = H_s.$$

By VI$_A$, 0.5.2, the $\kappa(s)$-group subscheme G'_s is closed in G_s. Hence H'_s is closed in H_s, and the preceding argument, applied to H'_s, H_s, and the f_s^n, shows that

$$H'_s = H_s.$$

This proves (iii′) and completes the proof.

Corollary 7.1.1.—[70] *Let G be a k-group locally of finite type, let A and B be two k-subgroups of G that are smooth and of finite type, and let i_A (respectively, i_B) be the inclusion of A (respectively, B) into G. Suppose that B normalizes A. Then*

$$A \cdot B = \Gamma_G(i_A, i_B).$$

Proof. Let

$$H = A \rtimes B$$

be the semidirect product of A and B (cf. I, 2.3.5); it is a smooth k-group of finite type. The group morphism

$$u : H \longrightarrow G, \qquad (a, b) \longmapsto ab,$$

is therefore quasi-compact. Hence, by 1.2,

$$u(H) = A \cdot B$$

is a closed reduced subscheme of G, and is a group in the category $(\mathrm{Sch}/k)_{\mathrm{red}}$. By EGA IV$_3$, 11.10.7, $A \cdot B = u(H)$ is geometrically reduced, since H is; it is therefore a closed subgroup of G. The evident inclusion

$$A \cdot B \subset \Gamma_G(i_A, i_B)$$

now proves the corollary.

[70] Editor's note: this corollary has been added to point out this special case of 7.1.

Definitions and Remarks 7.2.—[71]

(i) Let G be a k-group, let $(X_i)_{i\in I}$ be a family of geometrically reduced k-schemes, and, for every $i \in I$, let $f_i : X_i \to G$ be a k-morphism. The *closed group subscheme of G generated by the family $(f_i)_{i\in I}$* is the smallest closed group subscheme of G through which every f_i factors; in this subsection it is denoted

$$\Gamma_G((f_i)_{i\in I}).$$

If X is a subscheme of G, geometrically reduced over k, and $f : X \hookrightarrow G$ is the immersion, we write $\Gamma_G(X)$ instead of $\Gamma_G(f)$.

(i′) With the notation of 7.1(ii), we shall sometimes put

$$\Gamma'_G(f) = \bigcup_{n\geq 1} f^n(X^n).$$

Notice that $\Gamma'_G(f)$ is a subset of G stable under the group law, in the sense of 3.0.

(ii) If X_1 and X_2 are geometrically reduced k-schemes and $f_1 : X_1 \to G$ and $f_2 : X_2 \to G$ are k-morphisms such that

$$f_1(X_1) = f_2(X_2)$$

as sets, then it is clear that

$$\Gamma_G(f_1) = \Gamma_G(f_2).$$

(iii) Let E be a subset of G such that the reduced subscheme $\overline{E}$ of G is geometrically reduced. The *closed group subscheme of G generated by E* is the group subscheme

$$\Gamma_G(E) := \Gamma_G(i),$$

where $i : \overline{E} \hookrightarrow G$ is the inclusion of that reduced subscheme.

(iv) Since every group subscheme of G is closed, by VI_A, 0.5.2,[72] we shall speak of the "generated group subscheme" rather than the "generated closed group subscheme."

(v) Let X be a geometrically reduced k-scheme and let $f : X \to G$ be a k-morphism. Suppose that $f(X)$ contains the unit element e of G. Put

$$X'^1 = X \times_k X, \qquad f'^1 = \mu \circ (f \times_k (c \circ f)),$$

and, for $n > 1$, put

$$X'^n = X'^1 \times_k X'^{n-1}, \qquad f'^n = \mu \circ (f'^1 \times_k f'^{n-1}).$$

Then $\Gamma_G(f)$ is the reduced subscheme of G whose underlying space is the closure of

$$\bigcup_{n\geq 1} f'^n(X'^n).$$

Indeed, recall the notation of 7.1(ii):

$$X^1 = X \amalg X, \qquad X^n = X^1 \times_k X^{n-1} \quad (n \geq 2).$$

[71] Editor's note: item (viii) of the original has been placed in (i′), and items (vi) and (vii) have been singled out as Corollary 7.2.1 and Definition 7.2.2 below.

[72] Editor's note: the hypothesis that G be locally of finite type over k has been removed here.

The hypothesis $e \in f(X)$ gives, for every $n \geq 1$,

$$f^n(X^n) \subset f'^n(X'^n) \subset f^{2n}(X^{2n}).$$

Consequently, there exists an integer n such that $f^n(X^n) = \Gamma_G(f)$ if and only if there exists an integer m such that $f'^m(X'^m) = \Gamma_G(f)$.

From Remark 7.2(v) we deduce the following result.

Corollary 7.2.1.— *Let X be a geometrically reduced and geometrically connected k-scheme, and let $f : X \to G$ be a k-morphism such that $f(X)$ contains the unit element of G. Then the k-group $\Gamma_G(f)$ is connected, and hence irreducible.*

Proof. Each X'^n is connected, so the union of the

$$f'^n(X'^n),$$

all of which contain the unit element, is connected. Its closure $\Gamma_G(f)$ is therefore connected as well. It follows that $\Gamma_G(f)$ is irreducible, by VI_A, 2.4 when G, and hence also $\Gamma_G(f)$, is locally of finite type over k, and by VI_A, 2.6.5(iii) in the general case.

Definition 7.2.2.— Let G be a k-group.

a) Let A and B be two geometrically reduced k-group subschemes of G. The *commutator group subscheme of A and B in G*, denoted

$$(A, B)_G \qquad \text{or simply} \qquad (A, B),$$

is the closed group subscheme of G generated by the morphism

$$\nu : A \times_k B \longrightarrow G$$

defined, for every k-scheme S, by

$$(a, b) \longmapsto aba^{-1}b^{-1}, \qquad a \in A(S), \quad b \in B(S).$$

b) Suppose that G is geometrically reduced over k. The *derived group of G*, denoted $\mathfrak{D}(G)$,[73] is the group

$$\mathfrak{D}(G) = (G, G).$$

N.B. The group G is commutative if and only if $\mathfrak{D}(G)$ is the unit k-group.

Reminders 7.3.0.—[74] Let $\varphi : Y \to Z$ be a morphism of S-schemes. Recall that the image presheaf $\varphi(Y)$ is the S-functor which assigns to every S' over S the subset

$$\varphi\big(Y(S')\big) \subset Z(S').$$

Suppose that T is a subscheme of Z. If the inclusion of presheaves

$$\varphi(Y) \hookrightarrow Z$$

factors through T, that is, if

$$\varphi \circ h \in T(S')$$

[73] Editor's note: the notation $\mathrm{Der}(G)$ of the original has been changed to avoid any risk of confusion with a space of derivations.

[74] Editor's note: these reminders have been added, and the statement of 7.3 has consequently been modified and expanded.

for every S-morphism $h : S' \to Y$, then, set-theoretically, $\varphi(Y) \subset T$: take $h = \mathrm{id}_Y$. Conversely, if Y is reduced and $\varphi(Y) \subset T$ set-theoretically, then φ factors through T.

Recall also that, by 6.2, if H is a closed k-group subscheme of G, then

$$\underline{\mathrm{Centr}}_G H \qquad \text{and} \qquad \underline{\mathrm{Norm}}_G H$$

are representable by closed k-group subschemes of G.

Corollary 7.3.— *Let G be a k-group, let X be a geometrically reduced k-scheme, and let $f : X \to G$ be a k-morphism.*

(i) *Let S be a k-scheme and let u be an endomorphism of the S-group G_S.*

(a) *If*
$$u(f_S(X_S)) \subset \Gamma_G(f)_S$$
set-theoretically, then the morphism
$$u : \Gamma_G(f)_S \longrightarrow G_S$$
factors through $\Gamma_G(f)_S$.

(b) *If u is an automorphism of the S-group G_S and*
$$u(f_S(X_S)) = f_S(X_S)$$
set-theoretically, then u induces an automorphism of $\Gamma_G(f)_S$. In particular, if $g \in G(S)$ satisfies
$$\mathrm{int}(g)(f_S(X_S)) = f_S(X_S)$$
set-theoretically, then
$$g \in \underline{\mathrm{Norm}}_{G_S}(\Gamma_G(f)_S)(S).$$

(c) *If $u \circ f_S = f_S$, then the restriction of u to the subgroup $\Gamma_G(f)_S$ of G_S is the identity. In particular, if $g \in G(S)$ satisfies*
$$\mathrm{int}(g) \circ f_S = f_S,$$
then
$$g \in \underline{\mathrm{Centr}}_{G_S}(\Gamma_G(f)_S)(S).$$

(ii) *Let H be a group subscheme of G. Then H centralizes (respectively, normalizes) $\Gamma_G(f)$ if and only if, for every k-scheme S and every $h \in H(S)$,*
$$\mathrm{int}(h) \circ f_S = f_S$$
(respectively,
$$\mathrm{int}(h)(f_S(X_S)) \subset \Gamma_G(f)_S$$
set-theoretically). Equivalently, for every $S' \to S$ and every $x \in X(S')$, the elements $h_{S'}$ and $f(x)$ of $G(S')$ commute (respectively,
$$h_{S'} f(x) h_{S'}^{-1} \in \Gamma_G(f)(S').$$
).

(iii) *In particular, let Y be a second geometrically reduced k-scheme and let $\varphi : Y \to G$ be a k-morphism. Suppose that, for every k-scheme S', every $x \in X(S')$, and every $y \in Y(S')$, the elements $\varphi(y)$ and $f(x)$ of $G(S')$ commute (respectively,*

$$\mathrm{int}\big(\varphi(y)\big)\big(f(x)\big) \in \Gamma_G(f)(S').$$

). Then $\Gamma_G(\varphi)$ is a k-subgroup of

$$\underline{\mathrm{Centr}}_G(\Gamma_G(f)) \qquad \textit{(respectively, of)} \qquad \underline{\mathrm{Norm}}_G(\Gamma_G(f)).$$

(iv) *If g is a k-point of G, then the k-group $\Gamma_G(\{g\})$ is commutative.*

(v) *Let A and B be two group subschemes of G, geometrically reduced over k. If A and B are invariant (respectively, characteristic), then so is (A, B).*

Proof. We first prove (i)(a). Put

$$\Gamma_S = \Gamma_G(f)_S.$$

Then $u^{-1}(\Gamma_S)$ is a closed S-group subscheme of G_S. By 7.1(iii), it is therefore enough to show that f_S factors through $u^{-1}(\Gamma_S)$. Since

$$u\big(f_S(X_S)\big) \subset \Gamma_S$$

and X_S is reduced, $u \circ f_S$ factors through Γ_S; hence f_S factors through $u^{-1}(\Gamma_S)$. This proves (a).

The first assertion of (b) follows by applying (a) to u and u^{-1}. In fact, it is enough to assume

$$u\big(f_S(X_S)\big) \subset \Gamma_S \qquad \text{and} \qquad u^{-1}\big(f_S(X_S)\big) \subset \Gamma_S.$$

The second assertion is the special case $u = \mathrm{int}(g)$.

We now prove (c). Consider the morphism of S-groups

$$G_S \longrightarrow G_S \times_S G_S, \qquad g \longmapsto (u(g), g),$$

and let H be the inverse image of the diagonal. It is a group subscheme of G_S. Since G is separated over k, by VI_A, 0.3, G_S is separated over S, and hence H is closed in G_S. By hypothesis, f_S factors through H; therefore H contains $\Gamma_G(f)_S$, by 7.1(iii). This proves the first assertion of (c), and the second is the special case $u = \mathrm{int}(g)$.

We next prove (ii). Put

$$\Gamma = \Gamma_G(f)$$

and let $i : \Gamma \hookrightarrow G$ denote the inclusion. The group H is contained in

$$N = \underline{\mathrm{Norm}}_G(\Gamma)$$

if and only if, for every k-scheme S and every $h \in H(S)$,

$$\mathrm{int}(h)(\Gamma_S) \subset \Gamma_S.$$

Applying this condition to h and h^{-1} shows that the inclusion is then an equality. By (i)(a), this condition holds if and only if

$$\mathrm{int}(h)\big(f_S(X_S)\big) \subset \Gamma_S.$$

Similarly, H is contained in

$$C = \underline{\mathrm{Centr}}_G(\Gamma)$$

if and only if, for every k-scheme S and every $h \in H(S)$,

$$\mathrm{int}(h) \circ i_S = i_S.$$

By (i)(c), this holds if and only if

$$\mathrm{int}(h) \circ f_S = f_S.$$

This proves (ii).

We now prove (iii). In view of (ii), the hypothesis implies that φ factors through C (respectively, N). The latter is a closed subgroup of G, by 6.2, and therefore contains $\Gamma_G(\varphi)$, by 7.1(i).

Assertion (iv) follows from (iii) by taking both f and φ to be the closed immersion

$$\mathrm{Spec}\, k \hookrightarrow G$$

defined by g. Indeed, for every k-scheme S, the sets $X(S)$ and $Y(S)$ then each have a single element, which is sent by f, respectively by φ, to g.

Finally, we prove (v). Let

$$\nu : G \times_k G \longrightarrow G, \qquad (g, h) \longmapsto ghg^{-1}h^{-1},$$

and let ν' be its restriction to $A \times_k B$. By 7.2.2,

$$(A, B) = \Gamma_G(\nu').$$

Let S be a k-scheme, and let u be an inner automorphism, respectively an arbitrary S-group automorphism, of G_S. Then

$$u \circ \nu_S = \nu_S \circ (u \times_S u).$$

The hypothesis implies that u induces automorphisms u_1 of A_S and u_2 of B_S. Consequently,

$$u(\nu'_S(A_S \times_S B_S)) = \nu'_S(u_1(A_S) \times_S u_2(B_S)) = \nu'_S(A_S \times_S B_S).$$

By (i)(b), u therefore induces an automorphism of $(A, B)_S$. This proves (v).

Proposition 7.4.— *Let G be a k-group locally of finite type, let X be a finite-type k-scheme that is geometrically reduced and geometrically connected, and let $f : X \to G$ be a k-morphism such that $f(X)$ contains the unit element e of G. Then, with the notation of 7.1*(ii)*, there exists an integer N such that, set-theoretically,*

$$f^N(X^N) = \Gamma_G(f).$$

Proof. By 7.1(iii) and EGA IV_2, 2.6.1, we may assume that k is algebraically closed. By Corollary 7.2.1, we may assume that $G = G^0$. Finally, with the notation of 7.2(v), it is enough to show that there exists an integer N such that

$$f'^N(X'^N) = \Gamma_G(f).$$

First case. Suppose that X is irreducible. The closed irreducible subsets

$$\overline{f'^n(X'^n)}$$

then form an increasing sequence in the space G, which is Noetherian because $G = G^0$ is of finite type over k (VI_A, 2.4). Hence this sequence is stationary, and there exists an integer m such that

$$\overline{f'^m(X'^m)} = \Gamma_G(f).$$

Moreover, because X and G are of finite type over k, the morphisms f'^n are of finite type over k. Consequently, $f'^m(X'^m)$ is constructible in G (EGA IV_1, 1.8.5), and therefore contains an open subset U of its closure $\Gamma_G(f)$ (EGA 0_{IV}, 9.2.3). By VI_A, 0.5, one then has

$$\Gamma_G(f) \subset U \cdot U \subset f'^{2m}(X'^{2m}) \subset \Gamma_G(f),$$

and hence

$$f'^{2m}(X'^{2m}) = \Gamma_G(f).$$

Second case. Suppose that X has exactly two irreducible components A_1 and A_2. Since X is connected and k is algebraically closed, there exists

$$a \in (A_1 \cap A_2)(k).$$

The four irreducible subsets

$$A_i \times_k A_j \qquad (i, j = 1, 2)$$

therefore cover $X \times_k X$, and the image of each under the morphism

$$f'^1 = \mu \circ (f \times_k (c \circ f))$$

contains e. Let f'^1_{ij} denote the restriction of f'^1 to the reduced subscheme $A_i \times_k A_j$, and put

$$Y = (A_1 \times_k A_1) \times_k (A_1 \times_k A_2) \times_k (A_2 \times_k A_2) \times_k (A_2 \times_k A_1),$$
$$g = \mu \circ \Big(\mu \circ (f'^1_{11} \times_k f'^1_{12}) \times_k \mu \circ (f'^1_{22} \times_k f'^1_{21})\Big).$$

The scheme Y is irreducible, reduced, and of finite type. By the first case, there is therefore an integer m such that

$$g'^m(Y'^m) = \Gamma_G(g).$$

For every $n \geq 1$, however,

$$f'^n(X'^n) \subset g'^n(Y'^n) \subset f'^{4n}(X'^{4n}).$$

It follows that

$$\Gamma_G(f) = \Gamma_G(g) \qquad \text{and} \qquad f'^{4m}(X'^{4m}) = \Gamma_G(f).$$

General case. We argue by induction on the number of irreducible components of X; this number is finite because X, being of finite type over k, is Noetherian. Suppose the proposition has been proved when X has at most $r-1$ irreducible components, and suppose that X has r such components, say $A_1, \ldots, A_r$. The unit e belongs to the image of one of the A_i; suppose, for example, that

$$e \in f(A_1).$$

Put

$$X_r = A_1 \cup \cdots \cup A_{r-1}, \qquad Y = \Gamma_G(f_r),$$

where f_r denotes the restriction of f to the reduced subscheme X_r. We choose the numbering of the A_i so that X_r is connected, which is always possible. Then Y is a closed, reduced, irreducible subgroup of G, by Corollary 7.2.1.

Put

$$Z = \overline{f(A_r)}$$

and let $T = Y \cup Z$, endowed with its closed reduced subscheme structure; let $i : T \hookrightarrow G$ be the inclusion. It follows immediately from 7.2(ii) that

$$\Gamma_G(i) = \Gamma_G(f).$$

Moreover, T is connected: since X is connected, X_r and A_r have a common point a, and Y and Z have the common point $f(a)$. Also, $e \in T$, and T has at most two irreducible components, because Y and Z are irreducible.

By the induction hypothesis, there exists an integer m such that

$$f_r'^m(X_r'^m) = \Gamma_G(f_r) = Y.$$

Since $X = X_r \cup A_r$, and since

$$Z = \overline{f(A_r)} \subset \overline{f'^m(X'^m)}$$

because $e \in f(A_1)$, we obtain

$$f(X) \subset Y \cup Z \subset \overline{f'^m(X'^m)}.$$

On the other hand, because T has at most two irreducible components, the second case gives an integer p such that

$$i'^p(T'^p) = \Gamma_G(i) = \Gamma_G(f).$$

Now

$$T \subset \overline{f'^m(X'^m)},$$

and hence

$$\overline{f'^{mp}(X'^{mp})} = \Gamma_G(f).$$

As in the first case, this last equality implies

$$f'^{2mp}(X'^{2mp}) = \Gamma_G(f).$$

This completes the proof.

Lemma 7.5.— *Let S be a scheme, let G be an S-group scheme, let X be an S-prescheme, and let $f : X \to G$ be an S-morphism. Suppose that X and G are locally of finite presentation over S, and that, for every $s \in S$ and every maximal point g of G_s, there exists a point x of X such that $f(x) = g$ and f is flat at x. Then the morphism*

$$\mu \circ (f \times_S f) : X \times_S X \longrightarrow G$$

is a covering for the (fppf) *topology.*

[75] By EGA IV_3, 11.3.1, the set V of points of X at which f is flat is open, and $f|_V$ is an open morphism. Thus

$$U = f(V)$$

is an open subset of G. Moreover, by hypothesis,

$$U \cap G_s$$

[75] Editor's note: the argument in the original has been expanded in what follows.

is dense in G_s for every $s \in S$. Let ϕ denote the restriction of $\mu \circ (f \times_S f)$ to $V \times_S V$.

It is enough to show that ϕ is a covering for the (fppf) topology, and for this it is enough to show that ϕ is faithfully flat and of finite presentation (Exposé IV, 6.3.1(iv)). Now ϕ is the composite

$$V \times_S V \xrightarrow{f|_V \times f|_V} U \times_S U \xrightarrow{\mu} G,$$

where the first morphism is faithfully flat and locally of finite presentation, since $f|_V$ is. It therefore remains only to prove the same assertions for the restriction of μ to $U \times_S U$.

Since $G \to S$ is locally of finite presentation and flat, so is

$$\mu : G \times_S G \longrightarrow G.$$

Indeed, this morphism is isomorphic to the morphism obtained from $G \to S$ by the base change $G \to S$ (VI$_A$, 0.1). The induced morphism

$$U \times_S U \longrightarrow G$$

therefore has the same properties. To prove that this last morphism is surjective, it is enough to check fiber by fiber. There the result follows from VI$_A$, 0.5, since $U \cap G_s$ is a dense open subset of G_s for every $s \in S$.

Remark 7.6.0.—[76] *Let S be a scheme, let H be an S-group, and let $f : X \to H$ be a morphism of S-schemes. The group subpresheaf of H generated by the image of f, denoted*

$$\langle \operatorname{Im} f \rangle,$$

is the group subfunctor of H which assigns to every S-scheme S' the subgroup of $H(S')$ generated by $f(X(S'))$. Since every element of this subgroup can be written as a finite product

$$f(x_1)^{\varepsilon_1} \cdots f(x_n)^{\varepsilon_n}, \qquad n \in \mathbb{N}^*, \quad x_i \in X(S'), \quad \varepsilon_i = \pm 1,$$

it follows that, if one puts

$$X^1 = X \amalg X$$

and defines the morphisms $f^n : X^n \to H$ as in 7.1, then $\langle \operatorname{Im} f \rangle$ is precisely the image presheaf of the morphism

$$f^\infty : X^\infty \longrightarrow H,$$

where

$$X^\infty = \coprod_{n \geq 1} X^n$$

and $f^\infty : X^\infty \to H$ is the S-morphism whose restriction to each X^n is f^n.

Proposition 7.6.—[77] *Let S be a scheme, let X be an S-scheme flat and locally of finite presentation over S, let H be an S-group locally of finite presentation over S and with reduced fibers, let $f : X \to H$ be a morphism of S-schemes, and let $f^\infty : X^\infty \to H$ be the S-morphism introduced above. The following conditions are equivalent:*

(i) *H represents the S-sheaf for the* (fppf) *topology associated with the presheaf $\langle \operatorname{Im} f \rangle$.*

(ii) *f^∞ is a covering for the* (fppf) *topology.*

[76] Editor's note: this definition has been added; in the original it was contained in the statement of Proposition 7.6.

[77] Editor's note: the original stated this result under the hypotheses of the special case, but the more general form was used implicitly in the proof of 10.12; the statement has been rewritten accordingly.

(iii) *f^∞ is surjective, that is,*

$$H = \bigcup_{n\geq 1} f^n(X^n).$$

If, in addition, H is quasi-compact, these conditions are also equivalent to the following one:

(iv) *There exists an integer n such that $f^n : X^n \to H$ is a covering for the* (fppf) *topology, and hence is surjective.*

This applies in particular when X is a geometrically reduced k-scheme locally of finite type, G is a k-group locally of finite type, $\varphi : X \to G$ is a morphism of k-schemes, $H = \Gamma_G(\varphi)$, and $f : X \to H$ is the morphism induced by φ.

Proof. The sheaf considered in (i) is the image sheaf of X^∞ under f^∞. Hence, by IV, 4.4.3, saying that H represents this sheaf is equivalent to saying that f^∞ is a covering; this in turn implies that f^∞ is surjective. Conversely, suppose that f^∞ is surjective. We shall show that it is then a covering for the (fppf) topology.

Let $s \in S$, let η be a maximal point of the fiber H_s, and let $x \in X_s^\infty$ be such that

$$f^\infty(x) = \eta;$$

such an x exists because f^∞ is surjective. Since the fiber H_s is reduced, $\mathcal{O}_{H_s,\eta}$ is a field, and therefore f_s^∞ is flat at x. Since X^∞ and H are locally of finite presentation over S, and X^∞ is flat over S, the fiberwise criterion for flatness (EGA IV_3, 11.3.10) shows that f^∞ is flat at x. Thus, by Lemma 7.5, the morphism

$$\mu \circ (f^\infty \times_S f^\infty) : X^\infty \times_S X^\infty \longrightarrow H$$

is a covering for the (fppf) topology.

Now $X^n \times_S X^m$ is canonically isomorphic to X^{n+m}, and, under this isomorphism,

$$\mu \circ (f^n \times_S f^m)$$

corresponds to f^{n+m}. It is therefore clear that

$$\mu \circ (f^\infty \times_S f^\infty)$$

factors through f^∞, so f^∞ is a covering for the (fppf) topology. This proves (iii) $\Rightarrow$ (ii), and hence the equivalence of conditions (i), (ii), and (iii).

Furthermore, since the morphisms $X \to S$ and $H \to S$ are locally of finite presentation, so is $f : X \to H$ (EGA IV_1, 1.4.3(v)). The multiplication

$$\mu : H \times_S H \longrightarrow H$$

is also locally of finite presentation (VI_A, 0.1), and it follows that each $f^n : X^n \to H$ is locally of finite presentation. Consequently, by EGA IV_1, 1.9.5(viii), the sets

$$f^n(X^n)$$

are ind-constructible subsets of H.

Suppose, in addition, that H is quasi-compact; then S is also quasi-compact, since $H \to S$ is surjective. By EGA IV_1, 1.9.9, there exists an integer p such that

$$f^p(X^p) = H.$$

As above, the fact that the fibers of H are reduced, together with Lemma 7.5, shows that the morphism

$$\mu \circ (f^p \times_S f^p) : X^p \times_S X^p \longrightarrow H$$

is a covering for the (fppf) topology. Since this morphism is

$$f^{2p} : X^{2p} \longrightarrow H,$$

this completes the proof of Proposition 7.6.

Remark 7.6.1.— The equivalent conditions of 7.6 plainly imply that the sheaf F under consideration is representable. The converse is false in general:[78] for example, if k has characteristic 0, let $G = \mathbb{G}_{a,k}$, and let

$$f : \operatorname{Spec} k \longrightarrow G$$

be the morphism defined by the point 1 of $G(k)$. Then F is represented by the constant k-group $\mathbb{Z}_k$, whereas $\Gamma_G(f) = G$; hence the monomorphism

$$\mathbb{Z}_k \hookrightarrow G$$

is not surjective.

For simplicity, suppose now that the hypotheses of the special case in 7.6 hold, and suppose that F is representable. By EGA IV_3, 8.14.2, the sheaf F is locally of finite presentation over k. The question is therefore whether the dominant monomorphism

$$F \longrightarrow \Gamma_G(f)$$

is an isomorphism, or, equivalently, a closed immersion. By 1.4.2, this is so if F is quasi-compact; and by VI_A, 0.5.1, it is so if F is connected. In particular, by 7.2.1, it is so when X is connected and $f(X)$ contains the unit element of G.

Lemma 7.7.— *Let k be an algebraically closed field, let G be a k-group locally of finite type, let X be a geometrically reduced k-scheme locally of finite type, let $f : X \to G$ be a k-morphism, and let H be a group subscheme of G such that $H \subset f(X)$. Put*

$$\Gamma' = \bigcup_{n \geq 1} f^n(X^n), \qquad \Gamma'_0 = \Gamma' \cap G(k), \qquad H_0 = H(k),$$

and suppose that H_0 has finite index in Γ'_0. Then there exists an integer m such that

$$f^m(X^m) = \Gamma_G(f)$$

(cf. 7.6), and $\Gamma_G(f)$ is the union of finitely many translates of H.

For every $n \geq 1$, the subset $f^n(X^n)$ of G is ind-constructible (EGA IV_1, 1.9.5(viii)). The same is therefore true of Γ'. Since G is a Jacobson scheme, Γ'_0 is dense in Γ'. By hypothesis, there are finitely many points

$$a_1, \ldots, a_r \in \Gamma'_0$$

such that

$$\Gamma'_0 = a_1 H_0 \cup \cdots \cup a_r H_0.$$

Consequently,

$$\begin{aligned} \Gamma_G(f) = \overline{\Gamma'} = \overline{\Gamma'_0} &= \overline{a_1 H_0 \cup \cdots \cup a_r H_0} \\ &= \overline{a_1 H_0} \cup \cdots \cup \overline{a_r H_0} \\ &= a_1 \overline{H_0} \cup \cdots \cup a_r \overline{H_0}, \end{aligned}$$

[78] Editor's note: the argument in the original has been expanded in what follows.

where the last equality follows because translation by each a_i is a homeomorphism of G onto itself. Thus

$$\Gamma_G(f) = a_1H \cup \cdots \cup a_rH.$$

It is also clear that there exists an integer p such that every a_i, for $1 \leq i \leq r$, belongs to $f^p(X^p)$. Finally, since $H \subset f(X)$, one has

$$a_iH \subset f^{p+1}(X^{p+1})$$

for every i. Hence

$$f^{p+1}(X^{p+1}) = \Gamma_G(f).$$

Proposition 7.8.— *Let G be a k-group locally of finite type, and let A and B be two geometrically reduced k-group subschemes of G (hence smooth at the generic points of their irreducible components, and therefore smooth by* 1.3*). Suppose that one of the following conditions holds:*

a) *A and B are invariant and of finite type over k;*

b) *A is connected and B is of finite type over k.*

Then (A, B) is of finite type over k, and represents the sheaf associated, for the (fppf) *topology (or the* (fpqc) *topology), with the group presheaf of commutators of A and B in G. Moreover,*[79] *the k-groups (A, B^0) and (A^0, B) are connected, and*

$$(A, B)^0 = (A, B^0) \cdot (A^0, B).$$

Proof. By 7.6, to show that (A, B) is the desired associated sheaf, it is enough to show that there exists an integer n such that

$$\nu^n((A \times_k B)^n) = (A, B),$$

with the notation of 7.2.2. To prove this and the other two assertions, we may suppose that k is algebraically closed.

Indeed, let $\overline{k}$ be an algebraic closure of k. By 7.1(iii) and VI_A, 2.4, one has, with the evident notation,

$$(A, B)_{\overline{k}} = (A_{\overline{k}}, B_{\overline{k}}),$$
$$\big((A, B)^0\big)_{\overline{k}} = (A_{\overline{k}}, B_{\overline{k}})^0,$$
$$(A, B^0)_{\overline{k}} = (A_{\overline{k}}, B^0_{\overline{k}}), \qquad \text{and so on.}$$

Consequently, suppose that $(A_{\overline{k}}, B_{\overline{k}})$ is of finite type over $\overline{k}$; respectively, suppose that $(A_{\overline{k}}, B^0_{\overline{k}})$ and $(A^0_{\overline{k}}, B_{\overline{k}})$ are connected and that the morphism

$$(A_{\overline{k}}, B^0_{\overline{k}}) \times_{\overline{k}} (A^0_{\overline{k}}, B_{\overline{k}}) \longrightarrow (A_{\overline{k}}, B_{\overline{k}})^0$$

is surjective. The analogous assertions then hold over k, by EGA IV_2, 2.7.1 and 2.6.1.

Let

$$B^1, \ldots, B^p$$

be the connected components of B other than the identity component B^0. There are only finitely many, since B is of finite type over k, hence Noetherian. In case a), likewise let

$$A^1, \ldots, A^q$$

[79] Editor's note: the assertions that follow have been made explicit, and the reduction to the algebraically closed case has been expanded in the proof.

be the connected components of A other than A^0. In case b), only $A^0 = A$ is considered. Let ν_{ij} be the restriction of ν to

$$A^i \times_k B^j.$$

Each A^i and B^j is irreducible, by VI_A, 2.4.1; hence so are $A^0 \times_k B^j$ and $A^i \times_k B^0$. Since the unit element of G belongs to A^0 and to B^0, it belongs to

$$\nu_{0j}(A^0 \times_k B^j) \qquad \text{and} \qquad \nu_{i0}(A^i \times_k B^0).$$

Thus every $\Gamma_G(\nu_{0j})$ and every $\Gamma_G(\nu_{i0})$ is connected, by 7.2.1.

Let u_{0j}, respectively u_{i0}, denote the inclusion of $\Gamma_G(\nu_{0j})$, respectively $\Gamma_G(\nu_{i0})$, in G. Then

$$(A^0, B) = \Gamma_G\big((u_{0j})_{j=0}^{p}\big), \qquad (A, B^0) = \Gamma_G\big((u_{i0})_{i=0}^{q}\big),$$

and these groups are connected. Moreover, it follows from 7.4 and the preceding constructions that there is an integer r such that both (A^0, B) and (A, B^0) are contained in

$$\nu^r\big((A \times_k B)^r\big).$$

In case b), one has $(A, B) = (A^0, B)$, and the proof is complete.

We now consider case a).[80] We already know that (A^0, B) and (A, B^0) are smooth connected k-subgroups of G, hence are of finite type over k (VI_A, 2.4). Since A^0 is a characteristic subgroup of A (VI_A, 2.6.5), it is an invariant subgroup of G; hence 7.3(v) shows that (A^0, B) is an invariant subgroup of G, and the same holds for (A, B^0).

By 7.1.1, the subgroup H of G generated by (A, B^0) and (A^0, B) is therefore

$$H = (A, B^0) \cdot (A^0, B).$$

In particular,

$$H \subset \nu^{2r}\big((A \times_k B)^{2r}\big).$$

If T is a subset of G stable under the group law (cf. 3.0), write T_0 for the group of k-points of G belonging to T. Put

$$\Gamma' = \bigcup_{q \geq 1} \nu^q\big((A \times_k B)^q\big).$$

By Proposition 7.9 below,

$$(A^0, B)_0 = (A_0^0, B_0), \qquad (A, B^0)_0 = (A_0, B_0^0), \qquad \Gamma'_0 = (A_0, B_0).$$

Therefore

$$H_0 = (A_0^0, B_0) \cdot (A_0, B_0^0)$$

has finite index in Γ'_0 (Bible, Exposé III, Appendix), since A_0 and B_0 are invariant and A_0^0, respectively B_0^0, has finite index in A_0, respectively B_0.

The hypotheses of Lemma 7.7 now hold. Since

$$H \subset \nu^{2r}\big((A \times_k B)^{2r}\big),$$

there exists an integer m such that

$$\nu^{2rm}\big((A \times_k B)^{2rm}\big) = (A, B),$$

[80] Editor's note: the argument that follows has been expanded to take account of the added Corollary 7.1.1.

and there are finitely many k-points $a_1, \ldots, a_n$ of G such that

$$(A, B) = a_1 H \cup \cdots \cup a_n H.$$

Since H is of finite type over k, every $a_i H$ is quasi-compact. Their union (A, B) is therefore quasi-compact, hence of finite type over k. Finally, H is irreducible, so each $a_i H$ is irreducible; and because the unit e belongs to H, it follows that

$$H = (A, B)^0.$$

Proposition 7.9.—*Let k be an algebraically closed field, and let G be a k-group locally of finite type.*

(i) *Let A and B be two ind-constructible subsets of G. Denote by A_0 the set of rational points of G that belong to A. Then*

$$(A \cdot B)_0 = A_0 \cdot B_0,$$

where the product on the right is taken in the group $G(k)$.

(ii) *Let X be a geometrically reduced k-scheme locally of finite type, and let $f : X \to G$ be a k-morphism. Put*

$$\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n).$$

Then $\Gamma'_G(f)_0$ is the subgroup of $G(k)$ generated by $f(X)_0$.

(iii) *In particular, let A and B be two smooth group subschemes of G, and put*

$$\Gamma' = \bigcup_{n \geq 1} \nu^n((A \times_k B)^n)$$

with the notation of 7.2.2. Then Γ'_0 is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.

Proof. We first prove (i). It is clear that

$$A_0 \cdot B_0 \subset (A \cdot B)_0.$$

Conversely, let $z \in (A \cdot B)_0$. Then $\mu^{-1}(z)$ is a closed subset of $G \times_k G$, while $A \times_k B$ is an ind-constructible subset of $G \times_k G$ (cf. 3.0). Consequently,

$$\mu^{-1}(z) \cap (A \times_k B)$$

is a nonempty ind-constructible subset of $G \times_k G$. By EGA IV_3, 10.4.8, it therefore contains a rational point of $G \times_k G$. Its projections x and y are rational points of G such that $x \in A$, $y \in B$, and $x \cdot y = z$. Hence

$$(A \cdot B)_0 = A_0 \cdot B_0.$$

To prove (ii), observe that, since f^n is locally of finite type, $f^n(X^n)$ is an ind-constructible subset of G (EGA IV_1, 1.9.5(viii)). Assertion (i) then shows by induction that, if

$$A = f(X)_0,$$

one has

$$f^n(X^n)_0 = (A \cup A^{-1})^n.$$

Consequently,

$$\Gamma'_G(f)_0 = \bigcup_{n\geq 1} f^n(X^n)_0 = \bigcup_{n\geq 1} (A \cup A^{-1})^n,$$

which is the subgroup of $G(k)$ generated by $A = f(X)_0$.

Finally, (iii) follows from (ii) and the definitions.

Corollary 7.10.—*Under the hypotheses of 7.8, if k is algebraically closed, then $(A, B)(k)$ is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.*

Indeed, it is enough to apply 7.9(iii), 7.8, and 7.6.

8. Solvable or nilpotent group schemes

8.1.— Let $\mathcal{C}$ be a category equipped with a topology T (cf. Exposé IV, §4). For every presheaf P on $\mathcal{C}$, write $P^\flat$ for the associated sheaf.

Let G be a presheaf of groups on $\mathcal{C}$, let A and B be two group subpresheaves of G, and let $\mathrm{Comm}(A, B)$ be the group presheaf of commutators of A and B in G. Thus, for every $S \in \mathrm{Ob}\,\mathcal{C}$, $\mathrm{Comm}(A, B)(S)$ is the subgroup of $G(S)$ generated by the commutators

$$aba^{-1}b^{-1}, \qquad a \in A(S), \quad b \in B(S).$$

We write

$$\mathrm{Comm}_T(A, B) = \mathrm{Comm}(A, B)^\flat.$$

In the proof of 8.2, we shall need the following results.[81]

Lemma 8.1.1.— *Let $A \subset G$ be sheaves of groups, with A invariant in G.*

(i) $\mathrm{Comm}_T(G, A)$ *is the smallest invariant group subsheaf C of G such that the sheaf $(A/C)^\flat$, associated with the quotient presheaf A/C, is central in $(G/C)^\flat$.*

(ii) *In particular,* $\mathrm{Comm}_T(G, G)$ *is the smallest invariant group subsheaf C of G such that the quotient sheaf $(G/C)^\flat$ is commutative.*

Clearly, (ii) is the special case $A = G$ of (i), so it is enough to prove (i). Let C be a group subsheaf of A, invariant in G, such that the quotient sheaf $(A/C)^\flat$ is central in $(G/C)^\flat$. By Lemma IV, 4.4.8.1, the presheaves A/C and G/C are separated. Hence, by IV, 4.3.11, all the morphisms in the following diagram are monomorphisms:

$$\begin{array}{ccc} A/C & \hookrightarrow & G/C \\ \downarrow & & \downarrow \\ (A/C)^\flat & \hookrightarrow & (G/C)^\flat \end{array}$$

Since $(A/C)^\flat$ is central in $(G/C)^\flat$, the presheaf A/C is central in G/C. It follows that

$$\mathrm{Comm}(G, A) \subset C,$$

and hence, by IV, 4.3.12, C contains $\mathrm{Comm}_T(G, A)$.

[81] Editor's note: these results were stated without proof in the original. We have highlighted them as Lemmas 8.1.1 and 8.1.2 and supplied the details of the proofs.

Conversely, $\mathrm{Comm}(G, A)$ is an invariant, separated group subpresheaf of A (cf. IV, 4.3.1, editor's note 24). Therefore, by IV, 4.4.8.2(i) and IV, 4.3.11,

$$C = \mathrm{Comm}_T(G, A)$$

is an invariant group subsheaf of A containing $\mathrm{Comm}(G, A)$. Consequently, the presheaf A/C is central in G/C, and hence, by IV, 4.4.8.2(ii), $(A/C)^\flat$ is central in $(G/C)^\flat$. This proves Lemma 8.1.1.

Lemma 8.1.2.— *Let G be a sheaf of groups, and let A, B be two group subpresheaves of G.*

(i) *The morphism*

$$\tau : \mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)$$

is a covering monomorphism.

(ii) *Consequently, there is an isomorphism*

$$\mathrm{Comm}_T(A, B) \xrightarrow{\sim} \mathrm{Comm}_T(A^\flat, B^\flat).$$

Proof. (i) Since A (respectively, B) is a subpresheaf of G, the associated sheaf $A^\flat$ (respectively, $B^\flat$) is a subpresheaf of G containing A (respectively, B). It follows that τ is a monomorphism.

We now show that τ is a covering. Let $S \in \mathrm{Ob}\,\mathcal{C}$ and let

$$g \in \mathrm{Comm}(A^\flat, B^\flat)(S).$$

There exist an integer $n \geq 1$ and, for $1 \leq i \leq n$, elements

$$a'_i \in A^\flat(S), \qquad b'_i \in B^\flat(S), \qquad \varepsilon_i \in \{\pm 1\},$$

such that

$$g = (a'_1, b'_1)^{\varepsilon_1} \cdots (a'_n, b'_n)^{\varepsilon_n},$$

where (a, b) denotes the commutator $aba^{-1}b^{-1}$. There is a refinement R of S such that

$$a'_i \in A(R) \qquad \text{and} \qquad b'_i \in B(R)$$

for every i. Then g is the composite morphism

$$R \xrightarrow{(a'_1, \ldots, b'_n)} (A \times B)^n \xrightarrow{\Phi^{\varepsilon_1, \ldots, \varepsilon_n}} \mathrm{Comm}(A, B)$$

where $\Phi = \Phi^{\varepsilon_1, \ldots, \varepsilon_n}$ is the morphism defined set-theoretically by

$$\Phi(a_1, b_1, \ldots, a_n, b_n) = (a_1, b_1)^{\varepsilon_1} \cdots (a_n, b_n)^{\varepsilon_n}$$

for every $T \in \mathrm{Ob}\,\mathcal{C}$, with $a_i \in A(T)$ and $b_i \in B(T)$. This shows that

$$g \in \mathrm{Comm}(A, B)(R).$$

It follows, as in the proof of IV, 4.3.11(i), that

$$\mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)$$

is a covering.

The morphism

$$\mathrm{Comm}(A^\flat, B^\flat) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)^\flat$$

is likewise a covering monomorphism (IV, 4.3.11(iv)). Hence the same is true of

$$\mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)^\flat.$$

By IV, 4.3.12, we therefore obtain an isomorphism

$$\mathrm{Comm}(A, B)^\flat \xrightarrow{\sim} \mathrm{Comm}(A^\flat, B^\flat)^\flat.$$

This proves Lemma 8.1.2.

Proposition 8.2.— *Let $\mathcal{C}$ be a category, let T be a topology on $\mathcal{C}$, let G be a sheaf of groups on $\mathcal{C}$, and let n be an integer with $n \geq 0$. The following conditions are equivalent:*

(i) *Put $K_0 = G$, and, for $p \geq 1$, put*

$$K_p = \mathrm{Comm}(K_{p-1}, K_{p-1}) \qquad (\text{respectively, } K_p = \mathrm{Comm}(G, K_{p-1})).$$

Then K_n is the unit presheaf of groups.

(ii) *Put $K'_0 = G$, and, for $p \geq 1$, put*

$$K'_p = \mathrm{Comm}_T(K'_{p-1}, K'_{p-1}) \qquad (\text{respectively, } K'_p = \mathrm{Comm}_T(G, K'_{p-1})).$$

Then K'_n is the unit sheaf of groups.

(iii) *There is a sequence*

$$G = H_0 \supset H_1 \supset \cdots \supset H_n$$

of invariant group subsheaves of G such that, for every i, the quotient sheaf H_i/H_{i+1} is commutative (respectively, central in G/H_{i+1}), *and H_n is the unit sheaf.*

It is clear that $K_n \subset K'_n$; consequently, (ii) implies (i). We show that (i) implies (ii). One has

$$K'_1 = \mathrm{Comm}_T(G, G) = K_1^\flat,$$

and induction using Lemma 8.1.2 gives

$$K'_p = K_p^\flat$$

for every p. Consequently, if K_n is the unit presheaf, then $K'_n = K_n^\flat$ is the unit sheaf.

Finally, conditions (ii) and (iii) are equivalent by Lemma 8.1.1.

Definition 8.2.1.— *When these conditions hold, the sheaf G is said to be* solvable of class n (respectively, nilpotent of class n). *If there is an integer n for which these conditions hold, G is said to be* solvable (respectively, nilpotent).

Observe that, by condition (i), this does not depend on the topology T.

Proposition 8.3.— *Let k be a field, let S be a nonempty k-scheme, let Ω be an algebraically closed extension of k, and let G be a smooth k-group of finite type. The following conditions are equivalent:*

(i) *G is solvable of class n* (respectively, nilpotent of class n).

(ii) *$G \times_k S$ is solvable of class n* (respectively, nilpotent of class n).

(iii) *The group $G(\Omega)$ is solvable of class n* (respectively, nilpotent of class n).

(iv) *Put $K_0 = G$, and, for $p \geq 1$, consider the k-groups*

$$K_p = (K_{p-1}, K_{p-1}) \qquad (\text{respectively, } \ K_p = (G, K_{p-1}))$$

(cf. 7.2.2). *Then K_n is the unit k-group.*

The equivalence of conditions (i) and (ii) follows from Proposition 8.2, since the formation of the group presheaf of commutators commutes with base change (IV, 4.1.3).

The equivalence of (i) and (iv) follows from the fact that, by 7.8, the k-group

$$K_p = (K_{p-1}, K_{p-1}) \qquad (\text{respectively, } \ K_p = (G, K_{p-1}))$$

represents the sheaf

$$\mathrm{Comm}_T(K_{p-1}, K_{p-1}) \qquad (\text{respectively, } \ \mathrm{Comm}_T(G, K_{p-1})),$$

where T is the (fppf) or (fpqc) topology.

To prove the equivalence of conditions (iii) and (iv), we may assume that $k = \Omega$; the result then follows from 7.10.

Proposition 8.4.— *Let G be an S-group of finite presentation such that, for every $s \in S$, the fiber G_s is smooth over $\kappa(s)$. Let T be the set of points $s \in S$ for which G_s is solvable* (respectively, nilpotent).

(i) *Then T is locally constructible in S.*

(ii) *If, in addition, G is assumed flat and separated over S* (that is, when G is smooth, quasi-compact, and separated over S), *then T is closed in S.*

Proof. We may clearly suppose that S is affine, with ring A. By 10.1 and 10.10b),* there exist a Noetherian subring A' of A and an A'-group G' of finite type such that

$$G' \otimes_{A'} A \simeq G.$$

By EGA IV_3, 11.2.6 and 8.10.5,[82] if G is flat and separated over S, we may suppose that G' is flat and separated over $S' = \mathrm{Spec}\, A'$.[83] Since G is of finite presentation over S, the set of points $s \in S$ for which G_s is geometrically reduced —or, equivalently, smooth over $\kappa(s)$—is locally constructible (EGA IV_3, 9.7.7). Hence, by EGA IV_3, 9.3.3, we may suppose that $G'_{s'}$ is smooth over $\kappa(s')$ for every $s' \in S'$. On the other hand, if s' denotes the image of s in S', then

$$G'_{s'} \otimes_{\kappa(s')} \kappa(s) \simeq G_s.$$

It follows from 8.3 that G_s is solvable (respectively, nilpotent) if and only if the same is true of $G'_{s'}$. We are therefore reduced to the case where S is a Noetherian affine scheme.

We first show that T is constructible. Applying the criterion of EGA 0_{III}, 9.2.3, and arguing as above, we are reduced to showing that, when S is Noetherian and integral, either T or $S - T$ contains a nonempty open subset of S.

* In the course of this proof we shall use results established in Section 10; like Section 9, they do not depend on the present Section 8.

[82] Editor's note: the reference 10.8.5 has been corrected to 8.10.5.

[83] Editor's note: what follows has been expanded.

Suppose, then, that S is integral and Noetherian, with generic point η. For every $s \in S$, put $K_s^0 = G_s$, and put, for $p \geq 1$,

$$K_s^p = (K_s^{p-1}, K_s^{p-1}) \qquad (\text{respectively, } \ K_s^p = (G_s, K_s^{p-1})). \qquad \dagger$$

We first show that the sequence of closed subschemes K_η^p is stationary. By 7.3(v), each K_η^p is invariant, and hence the sequence $(K_\eta^p)_p$ is decreasing. It is therefore stationary, since G_η is Noetherian. Thus there is an integer n such that

$$K_\eta^p = K_\eta^n \qquad \text{for every } p \geq n.$$

On the other hand, by 10.12.1 and 10.13, there exist a nonempty open subset S' of S and an S'-group D of finite presentation such that, for every $s \in S'$,

$$D_s = K_s^n, \qquad (D_s, D_s) = D_s \quad (\text{respectively, } (G_s, D_s) = D_s).$$

We may suppose that $S' = S$. Then, for every $s \in S$ and every $p \geq n$, one has $D_s = K_s^p$. Consequently, G_s is solvable (respectively, nilpotent) if and only if D_s is isomorphic to the unit $\kappa(s)$-group.

By EGA IV_3, 9.6.1(xi), the set of points $s \in S$ for which the structural morphism

$$D_s \longrightarrow \operatorname{Spec} \kappa(s)$$

is an isomorphism is constructible.[84] It is therefore either nowhere dense or contains a nonempty open subset of S. We have thus proved that T is locally constructible.

We now show that, if G is also flat and separated over S, then T is closed. Since T is locally constructible, it is closed if and only if it is stable under specialization (EGA IV_1, 1.10.1).

Let $s \in S$, and let s' be a specialization of s in S. Since we have reduced to the case where S is Noetherian, EGA II, 7.1.9 provides a discrete valuation ring A and a morphism

$$\operatorname{Spec} A \longrightarrow S$$

such that s, respectively s', is the image of the generic point α, respectively the closed point a, of $\operatorname{Spec} A$. It is enough to show that, if

$$G' = G \times_S \operatorname{Spec} A$$

and G'_α is solvable (respectively, nilpotent), then the same is true of G'_a. Since G is flat and separated over S, the A-group G' is flat and separated. We are thus reduced to the case where S is the spectrum of a discrete valuation ring A.

Since G_α is assumed solvable (respectively, nilpotent), there is an integer q such that K_α^q, with the notation introduced above, is isomorphic to the unit $\kappa(\alpha)$-group. For every n, let $\overline{K_\alpha^n}$ denote the schematic closure of K_α^n in G, in the sense of EGA IV_2, 2.8.5. We show by induction on p that

$$(\overline{K_\alpha^p})_a \supset K_a^p.$$

First, since G is flat over A, one has $\overline{G_\alpha} = G$ by EGA IV_2, 2.8.5, and hence

$$(\overline{K_\alpha^0})_a = K_a^0.$$

[84] Editor's note: here one uses the fact that S is assumed affine, and is therefore quasi-compact and quasi-separated (cf. EGA 0_{III}, 9.1.12).

Let $p \geq 0$, and suppose that

$$K_a^p \subset \left(\overline{K_\alpha^p}\right)_a.$$

Following 7.2.2, denote by $\nu_a, \nu_\alpha, \overline{\nu}, \overline{\nu}_a$ the following morphisms. In the solvable case they are

$$\begin{aligned}
\nu_a &: K_a^p \times_{\kappa(a)} K_a^p \longrightarrow G_a, \\
\nu_\alpha &: K_\alpha^p \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha, \\
\overline{\nu} &: \overline{K_\alpha^p} \times_A \overline{K_\alpha^p} \longrightarrow G, \\
\overline{\nu}_a &: \left(\overline{K_\alpha^p}\right)_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a \longrightarrow G_a;
\end{aligned}$$

respectively, in the nilpotent case they are

$$\begin{aligned}
\nu_a &: G_a \times_{\kappa(a)} K_a^p \longrightarrow G_a, \\
\nu_\alpha &: G_\alpha \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha, \\
\overline{\nu} &: G \times_A \overline{K_\alpha^p} \longrightarrow G, \\
\overline{\nu}_a &: G_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a \longrightarrow G_a.
\end{aligned}$$

Since ν_α factors through K_α^{p+1}, $\overline{\nu}$ factors through $\overline{K_\alpha^{p+1}}$. The latter is clearly a group subscheme of G, and therefore contains $\Gamma_G(\overline{\nu})$. By 7.1(iii),

$$\Gamma_G(\overline{\nu})_a = \Gamma_{G_a}(\overline{\nu}_a).$$

Moreover, the induction hypothesis gives

$$\begin{aligned}
K_a^p \times_{\kappa(a)} K_a^p &\subset \left(\overline{K_\alpha^p}\right)_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a && \text{in the solvable case,} \\
G_a \times_{\kappa(a)} K_a^p &\subset G_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a && \text{in the nilpotent case.}
\end{aligned}$$

Consequently,

$$K_a^{p+1} = \Gamma_{G_a}(\nu_a) \subset \Gamma_{G_a}(\overline{\nu}_a) = \Gamma_G(\overline{\nu})_a \subset \left(\overline{K_\alpha^{p+1}}\right)_a.$$

Finally, K_α^q is isomorphic to the unit $\kappa(\alpha)$-group. The unit A-group is flat over A and is isomorphic to a closed subscheme of G, since G is separated over A (cf. 5.1). It follows from EGA IV$_2$, 2.8.5 that $\overline{K_\alpha^q}$ is isomorphic to the unit A-group. Since we have proved that

$$K_a^q \subset \left(\overline{K_\alpha^q}\right)_a,$$

the group K_a^q is isomorphic to the unit $\kappa(a)$-group. Thus G_a is solvable (respectively, nilpotent).

9. Quotient sheaves

The present section is essentially limited to recalling, in the particular case of homogeneous spaces of groups, well-known general facts concerning passage to the quotient by flat equivalence relations (cf. Exposé IV).

Definition 9.1.— *Given a monomorphism $u : G' \to G$ of S-groups, we write G/G'* (respectively, *$G'\backslash G$*) *and call it the* right (respectively, left) quotient sheaf of G by G', *meaning the quotient sheaf, for the* (fpqc) *topology, of G by the equivalence relation defined by the monomorphism*

$$G \times_S G' \xrightarrow{\delta \circ (\mathrm{id}_G \times u)} G \times_S G,$$

(respectively,

$$G' \times_S G \xrightarrow{\gamma \circ (u \times \mathrm{id}_G)} G \times_S G),$$

where δ (respectively, γ) *denotes the automorphism of* $G \times_S G$ *defined, for* $g, h \in G(T)$*, by*

$$(g,h) \longmapsto (g,gh) \qquad (\text{respectively, } (h,g) \longmapsto (hg,g)).$$

Proposition 9.2.— *Let* $u : G' \to G$ *be a monomorphism of* S*-groups. Suppose that* G/G' *is representable by an* S*-scheme* G''*. Then:*

(i) *The canonical morphism* $p : G \to G''$ *is a covering for the* (fpqc) *topology.*

(ii) *If* $\varepsilon'' = p \circ \varepsilon$ (this morphism is called the unit section of G'')*, the following diagrams are cartesian:*

$$\begin{array}{ccc} G \times_S G' & \xrightarrow{\mu\circ(\mathrm{id}_G \times u)} & G \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle p} \\ G & \xrightarrow[p]{} & G'' \end{array} , \qquad \begin{array}{ccc} G' & \xrightarrow{u} & G \\ {\scriptstyle \pi'}\downarrow & & \downarrow{\scriptstyle p} \\ S & \xrightarrow[\varepsilon'']{} & G'' \end{array} .$$

In particular, u *is an immersion.*

(iii) *There exists on* G'' *a unique structure of an* S*-scheme with left operator group* G *such that* p *is a morphism of* S*-schemes with operator group* G*.*

(iv) *If, in addition,* G' *is assumed invariant in* G*, there exists on* G'' *a unique* S*-group structure such that* p *is a morphism of* S*-groups.*

(v) *Let* S_0 *be an* S*-scheme, and put* $G_0 = G \times_S S_0$ *and* $G'_0 = G' \times_S S_0$*. Then* G_0/G'_0 *is representable by* $G''_0 = G'' \times_S S_0$*.*[85]

(vi) *Let* $\mathscr{P}$ *be a property of an* S*-morphism. Suppose that* $\mathscr{P}$ *is stable under base change. If* $p : G \to G''$ *has property* $\mathscr{P}$*, then the structural morphism* $\pi' : G' \to S$ *has property* $\mathscr{P}$ *as well.*

(vii) *Let* $\mathscr{P}$ *be a property of an* S*-morphism. Suppose that* $\mathscr{P}$ *is local for the* (fpqc) *topology* (cf. 2.0 and 2.1.2)*. Then* $p : G \to G''$ *has property* $\mathscr{P}$ *if and only if* $\pi' : G' \to S$ *has property* $\mathscr{P}$*.*

(viii) *Let* $\mathscr{P}$ *be a property of an* S*-morphism. Suppose that* $\mathscr{P}$ *is local for the* (fpqc) *topology and stable under composition. If the structural morphisms* $G'' \to S$ *and* $G' \to S$ *have property* $\mathscr{P}$*, then the structural morphism* $G \to S$ *has property* $\mathscr{P}$ *as well.*

(ix) *If* G *is reduced, then* G'' *is reduced.*

(x) *For* G'' *to be separated over* S*, it is necessary and sufficient that* u (or, equivalently, ε'') *be a closed immersion.*

(xi) *For* G' *to be flat over* S*, it is necessary and sufficient that* p *be flat* (or, equivalently, faithfully flat)*.*

In this case, for G'' *to be flat over* S*, it is necessary and sufficient that* G *be flat over* S*.*

[85] Editor's note: that is, the quotient is universal; cf. Exposé IV, §3.

(xii) *For G' to be flat and locally of finite presentation over S, it is necessary and sufficient that $p : G \to G''$ be faithfully flat and locally of finite presentation.*

In this case, for G'' to be locally of finite presentation (respectively, locally of finite type, of finite type, smooth, étale, unramified, locally quasi-finite, quasi-finite) *over S, it is enough that G have the same property over S* (and the condition is also necessary in the first two cases; cf. (viii)).

(xiii) *Suppose that G' is flat and of finite presentation over S.*

a) *Then p is of finite presentation and faithfully flat.*

b) *Moreover, for G to be of finite presentation over S, it is necessary and sufficient that G'' be so.*

Proof. Recall that the equivalence relation under consideration is universally effective (IV, 4.4.9). Assertions (i), (iii), (iv), (v), and the first assertion of (ii) then follow, respectively, from IV, 4.4.3, 5.2.2, 5.2.4, 3.4.5, and 3.3.2(iii). The second assertion of (ii) follows from the first, as shown by the following cartesian diagram, since

$$(G \times_S G') \times_G S$$

is isomorphic to G':

$$\begin{array}{ccccc} G' & \xrightarrow{((\varepsilon\circ\pi'),\mathrm{id}_{G'})} & G \times_S G' & \xrightarrow{\mu\circ(\mathrm{id}_G \times u)} & G \\ \downarrow{\scriptstyle \pi'} & & \downarrow{\scriptstyle \mathrm{pr}_1} & & \downarrow{\scriptstyle p} \\ S & \xrightarrow[\varepsilon]{} & G & \xrightarrow[p]{} & G'' \end{array}$$

Indeed, ε'' is an S-section of G'', hence an immersion (EGA I, 5.3.13); by the preceding cartesian diagram the same is true of u, which completes the proof of (ii). Moreover, (vi) is an immediate consequence of the second cartesian diagram in (ii).

We prove (vii). By (i), p is a covering for the (fpqc) topology. Hence, by (ii), in order to show that p has property $\mathscr{P}$, it is enough to show that the first projection

$$\mathrm{pr}_1 : G \times_S G' \longrightarrow G$$

has property $\mathscr{P}$. This follows from the stability of $\mathscr{P}$ under base change, since pr_1 is obtained from π' by base change.

It is clear that (viii) follows from (vii), since

$$\pi = \pi'' \circ p,$$

where $\pi'' : G'' \to S$ denotes the structural morphism.

We prove (ix). By (i), p is an epimorphism. Since G is reduced, p factors through the immersion

$$G''_{\mathrm{red}} \longrightarrow G'',$$

which is therefore also an epimorphism, and hence an isomorphism (IV, 4.4.4).

We prove (x). If G'' is separated over S, then ε'' is a closed immersion by EGA I, 5.4.6. On the other hand, we saw in (ii) that ε'' is a closed immersion if and only if u is. Finally, if u is a closed immersion, then so is

$$\delta \circ (\mathrm{id}_G \times u) : G \times_S G' \longrightarrow G \times_S G.$$

It follows from Lemma 9.2.1 below that G'' is separated over S.

Assertion (xi) follows from (vii) and EGA IV$_2$, 2.2.13.

Assertion (xii) follows from (vii), EGA IV$_4$, 17.7.5 and 17.7.7, and the fact that p is universally open: for every $s \in S$, if the underlying space of G_s is discrete, then the same is true of the underlying space of G''_s.

Finally, assertion (xiii) follows from (vii), (viii), and EGA IV$_4$, 17.7.5.

Lemma 9.2.1.— *Let X be an S-scheme, and let R be an equivalence relation on X defined by the monomorphism $v : R \to X \times_S X$. Suppose that R is effective. Then:*

(i) *The morphism v is an immersion; it is a closed immersion if $Y = X/R$ is separated.*

(ii) *Suppose, in addition, that Y represents the* (fpqc) *quotient sheaf of X by R,*[86] *and that v is a closed immersion. Then $Y = X/R$ is separated over S.*

Recall (IV, Definition 3.3.2) that the hypothesis "R is effective" means that there is a morphism of S-schemes

$$p : X \longrightarrow Y$$

such that the natural morphism

$$R \longrightarrow X \times_Y X$$

is an isomorphism. It follows (EGA I, 5.3.5) that the following diagram is cartesian:

$$\begin{array}{ccc} R & \xrightarrow{v} & X \times_S X \\ \downarrow & & \downarrow{\scriptstyle p\times p} \\ Y & \xrightarrow[\Delta_{Y/S}]{} & Y \times_S Y \end{array}$$

Since $\Delta_{Y/S}$ is an immersion (EGA I, 5.3.9), the same is true of v. Likewise, if Y is separated over S, then $\Delta_{Y/S}$ is a closed immersion, and hence so is v.

Conversely, suppose that v is a closed immersion and that Y represents the (fpqc) quotient sheaf of X by R. Then p is a covering for the (fpqc) topology (IV, 4.4.3), and therefore $p \times p$ is also a covering: by base change, $p \times \mathrm{id}_X$ and $\mathrm{id}_Y \times p$ are coverings, and so is their composite $p \times p$. Thus, by (fpqc) descent (cf. EGA IV$_2$, 2.7.1), $\Delta_{Y/S}$ is a closed immersion; in other words, Y is separated over S.

Remark 9.2.2.— *Under the general hypotheses of 9.2, if G' is flat and locally of finite presentation over S, then p is a covering for the* (fppf) *topology,*[87] *by 9.2*(vii)*. Consequently, assertions 9.2*(vii) *and 9.2*(viii) *extend to properties $\mathscr{P}$ that are local for the* (fppf) *topology.*

Remark 9.3.— a) The question whether a quotient G/G' is representable is often delicate; in this seminar we prove the representability of certain particular quotients.

In general, in order to conclude that G/G' is representable, it is not enough to suppose that G and G' are of finite presentation over S and that G' is flat over S. Indeed, suppose in addition that G is smooth with connected fibers. In that case, if G/G' is a scheme, it is separated by Corollary 5.4; therefore $G' \hookrightarrow G$ is a closed immersion by 9.2(x). Consequently, if G' is not closed in G, then G/G' is not representable.

To obtain such a counterexample, take S to be the spectrum of a discrete valuation ring and put $G = \mathbb{G}_{m,S}$. Let $n > 1$ be an integer invertible on S. Then

$$\mu_n = \ker(G \xrightarrow{n} G)$$

[86] Editor's note: as O. Gabber pointed out, this condition is used in the proof and must be included among the hypotheses.

[87] Editor's note: (fpqc) has been corrected to (fppf).

is a closed subgroup of G, étale over S (cf. Exposé VII$_A$[88]). Let G' be the open subgroup of μ_n obtained by removing from μ_n the closed subset of its closed fiber that is complementary to the origin. Then G' is not closed in G, and hence G/G' is not representable. One can also construct such examples in which G' is smooth with connected fibers.

b) It remains possible, however, that G/G' is representable when G and G' are of finite presentation over S, and G' is flat over S and closed in G[*] [89]. Under these hypotheses, G/G' is known to be representable in the following particular cases:

1°— S is the spectrum of an Artinian ring (cf. VI$_A$, 3.2 and 3.3.2).

2°— G' is proper over S, and G is quasi-projective over S (cf. V, 7.1).

3°— S is locally Noetherian of dimension 1 (cf. [An73], Theorem 4.C).

10. Passage to the projective limit in group schemes and schemes with an operator group

10.0.— Recall the essential result of EGA IV$_3$, §8.8. Suppose that the following data are given: a quasi-compact and quasi-separated scheme S_0, an upward filtered preordered set I, a direct system $(\mathcal{A}_i)_{i\in I}$ of quasi-coherent commutative $\mathcal{O}_{S_0}$-algebras,

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i, \qquad S_i = \operatorname{Spec} \mathcal{A}_i \quad (i \in I), \qquad S = \operatorname{Spec} \mathcal{A}^{90}.$$

Then the category of S-schemes of finite presentation is determined, up to equivalence, by the categories of S_i-schemes of finite presentation, the functors between these categories

$$\rho_{ji} : X_i \longmapsto X_i \times_{S_i} S_j \qquad (i \leq j),$$

and the transitivity isomorphisms

$$\rho_{kj} \circ \rho_{ji} \xrightarrow{\sim} \rho_{ki}.$$

More precisely, given $j \in I$ and an S_j-scheme X_j of finite presentation, put, for every $i \in I$ with $i \geq j$,

$$X_i = X_j \times_{S_j} S_i, \qquad X = X_j \times_{S_j} S.$$

Then EGA IV$_3$, 8.8.2 gives:

(i) Given $j \in I$ and two S_j-schemes X_j and Y_j of finite presentation, the canonical map

$$\varinjlim_{i\geq j} \operatorname{Hom}_{S_i}(X_i, Y_i) \longrightarrow \operatorname{Hom}_S(X, Y)$$

is bijective.

(ii) For every S-scheme X of finite presentation, there are an index $j \in I$, an S_j-scheme X_j of finite presentation, and an S-isomorphism

$$X \xrightarrow{\sim} X_j \times_{S_j} S.$$

[88] Editor's note: see Exposé VII$_A$, 8.4, or Exposé VIII, 2.1.

[*] This is too optimistic, as M. Raynaud shows in his thesis (loc. cit., X, 14).

[89] Editor's note: the starred remark refers to counterexample X, 14 in [Ray70a]. The base there is a regular local scheme of dimension 2.

[90] Editor's note: since S is affine over S_0, it is quasi-compact and quasi-separated; cf. Editor's note 92 below.

It follows from EGA IV$_3$, 8.8.3 that whenever a diagram $\mathcal{D}$ involves finitely many objects and arrows in the category of S-schemes of finite presentation, one can find an index $i \in I$ and a diagram $\mathcal{D}_i$ in the category of S_i-schemes of finite presentation such that $\mathcal{D}$, up to isomorphism, is obtained from $\mathcal{D}_i$ by the base change $S \to S_i$. One can even find i and $\mathcal{D}_i$ such that every cartesian square in $\mathcal{D}$ is obtained from a cartesian square in $\mathcal{D}_i$.

10.1.— Moreover, many familiar properties of a morphism that are stable under base change have the following property:

> *Let $u : X \to Y$ be an S-morphism between S-schemes of finite presentation. By 10.0, it is obtained by base change from an S_j-morphism $u_j : X_j \to Y_j$ between S_j-schemes of finite presentation. Then u has the property $\mathcal{P}$ if and only if there is an $i \geq j$ such that*
>
> $$u_i = u_j \times_{S_j} S_i$$
>
> *has the property $\mathcal{P}$.*

This applies when $\mathcal{P}$ is any of the following properties of a morphism: separated, surjective, radicial, affine, quasi-affine, finite, quasi-finite, proper, projective, quasi-projective, an isomorphism, a monomorphism, an immersion, an open immersion, a closed immersion (EGA IV$_3$, 8.10.5), flat (EGA IV$_3$, 11.2.6), smooth, unramified, or étale (EGA IV$_4$, 17.7.8).[91]

This also applies when $\mathcal{P}$ is the property of being a covering for the (fppf) topology. Indeed, given two S-schemes X and Y of finite presentation and an S-morphism $u : X \to Y$, it follows from Exposé IV, 6.3.1(i)[92] that u is a covering for the (fppf) topology if and only if there are an S-scheme Z and a faithfully flat S-morphism $v : Z \to Y$ of finite presentation that factors through u.

The purpose of this Section 10 is to give variants of results of this kind for the category of S-groups of finite presentation, for the category of S-schemes with an operator group, and for certain properties of group monomorphisms (being invariant, being central with representable quotient sheaf, and so on).

The first two preliminary results of this kind follow. In 10.2 through 10.9, we retain the notation introduced in 10.0.

Lemma 10.2.— *Let G_j and H_j be two S_j-groups of finite presentation. For every $i \geq j$, put*

$$G_i = G_j \times_{S_j} S_i, \qquad G = G_j \times_{S_j} S,$$

and define H_i and H similarly. Then the canonical map

$$\varinjlim_{i \geq j} \mathrm{Hom}_{S_i\text{-gr}}(G_i, H_i) \longrightarrow \mathrm{Hom}_{S\text{-gr}}(G, H)$$

is bijective.

Lemma 10.3.— *Let G be an S-group of finite presentation. Then there are an index $j \in I$, an S_j-group G_j of finite presentation, and an isomorphism of S-groups*

$$G \xrightarrow{\sim} G_j \times_{S_j} S.$$

Assertions 10.2 and 10.3 are easy consequences of 10.0 and 10.1, taking into account the interpretation[93] of the structure of an S-group given in EGA 0$_{\mathrm{III}}$, 8.2.5 and 8.2.6.

[91] Editor's note: the word "flat" has been added, and the reference 17.7.6 has been corrected to 17.7.8.

[92] Editor's note: one also uses the fact that Y, being of finite presentation over S, is quasi-compact.

[93] Editor's note: in terms of commutative diagrams of S-morphisms.

Lemma 10.4.— *Let $u : G \to H$ be a morphism of S-groups between S-groups of finite presentation. By 10.3 and 10.2, u is obtained by base change from a morphism $u_j : G_j \to H_j$ between S_j-groups of finite presentation. Then u is a central monomorphism* (respectively, an invariant monomorphism) *if and only if there is an $i \geq j$ such that*

$$u_i = u_j \times_{S_j} S_i$$

has the same property.

This follows immediately from 10.0 and 10.1, in view of the characterization in 6.7 of central or invariant group monomorphisms.

Corollary 10.5.— *Let G_j be an S_j-group of finite presentation. Then $G_j \times_{S_j} S$ is commutative if and only if there is an $i \geq j$ such that $G_j \times_{S_j} S_i$ is commutative.*

Indeed, to say that an S-group is commutative is the same as saying that it is central when regarded as a group subscheme of itself.

Proposition 10.6.— *Let G_j be an S_j-group of finite presentation, and let G'_j be a group subscheme of G_j, flat and of finite presentation over S_j. Then the quotient*

$$(G_j \times_{S_j} S)/(G'_j \times_{S_j} S)$$

is representable for the (fpqc) *topology if and only if there is an $i \geq j$ such that*

$$(G_j \times_{S_j} S_i)/(G'_j \times_{S_j} S_i)$$

is representable.

This is a consequence of the following more general lemma.

Lemma 10.7.— *Let X_j be an S_j-scheme of finite presentation, and let R_j be an equivalence relation on X_j, flat and of finite presentation.*[94] *Then the quotient sheaf*

$$(X_j \times_{S_j} S)/(R_j \times_{S_j} S)$$

for the topology $\mathcal{T}$ = (fppf) *or* (fpqc) *is representable if and only if there is an $i \geq j$ such that the quotient sheaf*

$$(X_j \times_{S_j} S_i)/(R_j \times_{S_j} S_i)$$

for the topology $\mathcal{T}$ is representable.

In view of the statements EGA IV_2, 8.8.2, 8.8.3, 8.10.5, and 11.2.6 recalled in 10.0, this lemma follows from the next result.

Lemma 10.8.— *Let $\mathcal{T}$ be the* (fppf) *or* (fpqc) *topology. Let X be an S-scheme of finite presentation* (respectively, locally of finite presentation)*, and let R be an equivalence relation on X defined by a monomorphism*

$$v : R \longrightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ v$ is flat and of finite presentation (respectively, flat and locally of finite presentation)*. Then the following conditions are equivalent:*

(i) *The quotient sheaf X/R for the topology $\mathcal{T}$ is representable.*

[94] Editor's note: that is, such that the composite

$$R_j \hookrightarrow X_j \times_{S_j} X_j \xrightarrow{\mathrm{pr}_1} X_j$$

is flat and of finite presentation.

(ii) *There are an S-scheme Y of finite presentation* (respectively, locally of finite presentation) *and a faithfully flat morphism $p : X \to Y$ such that the diagram*

$$\text{(D)} \qquad \begin{array}{ccc} R & \xrightarrow{\mathrm{pr}_1 \circ v} & X \\ {\scriptstyle \mathrm{pr}_2 \circ v}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow[p]{} & Y \end{array}$$

is cartesian.

First note that, by IV, 3.3.2 and 4.4.3, the sheaf X/R for the topology $\mathcal{T}$ is representable by Y if and only if diagram (D) is cartesian and p is a covering for the topology $\mathcal{T}$.

Let us prove that (i) implies (ii). Hypothesis (i) implies that diagram (D) is cartesian. Thus $\mathrm{pr}_1 \circ v$ is obtained from p by base change along p, and p is a covering for the (fpqc) topology. Consequently, by (fpqc) descent (EGA IV_2, 2.7.1), since $\mathrm{pr}_1 \circ v$ is faithfully flat and (locally) of finite presentation, the same is true of p. Then EGA IV_4, 17.7.5 shows that, since X is (locally) of finite presentation over S, so is Y.

Let us prove that (ii) implies (i). It is enough to show that p is a covering for the (fppf) topology. But p is faithfully flat by hypothesis, and it is locally of finite presentation because X and Y are locally of finite presentation over S (EGA IV_1, 1.4.3(v)).

Lemma 10.9.— *Let G_j be an S_j-group of finite presentation, and put*

$$G = G_j \times_{S_j} S.$$

Then G^0 is representable if and only if there is an $i \geq j$ such that

$$(G_i)^0 = (G_j \times_{S_j} S_i)^0$$

is representable.

The condition is sufficient, since the functor $G \longmapsto G^0$ commutes with base change, by 3.3.

Conversely, suppose that G^0 is representable. Then, by 3.9, G^0 is open in G and quasi-compact over S, and hence is of finite presentation over S, since G is. By 10.3 and 10.1, there are an $i \geq j$ and an open group subscheme H_i of G_i such that

$$H_i \times_{S_i} S = G^0.$$

The structural morphism $G^0 \to S$ has geometrically connected fibers (VI_A, 2.1.1). Consequently, by EGA IV_3, 9.3.3 and 9.7.7, after increasing i we may suppose that the structural morphism $H_i \to S_i$ has geometrically connected fibers. It follows from 3.10.1 that the underlying space of H_i is precisely $(G_i)^0$; hence H_i represents $(G_i)^0$.

10.10.— Recall two very useful special cases of the situation stated in 10.0 (cf. EGA IV_3, 8.1.2(a) and (c)):

a) Given a point x of a scheme X, put $S_0 = \operatorname{Spec} \mathbb{Z}$, and consider the decreasing filtered inverse system $(S_i)_{i \in I}$ of affine open neighborhoods of x. Then

$$S = \operatorname{Spec} \mathcal{O}_{X,x}.$$

In particular, if x is the generic point of an integral scheme X, then

$$S = \operatorname{Spec} \kappa(x).$$

b) Put $S_0 = \operatorname{Spec}\mathbb{Z}$, and consider the family $(\mathcal{A}_i)_{i\in I}$, preordered by inclusion, of the finitely generated $\mathbb{Z}$-subalgebras of the ring of an affine scheme S. Since the $\mathcal{A}_i$ are Noetherian rings, this makes it possible in many cases to pass from the Noetherian case to the general case.

We shall now give two results concerning the special case considered in (a).[95]

Proposition 10.11.—[96] *Let S be an integral scheme with generic point η. Let G be an S-group and let Y, Z be S-schemes, all of finite presentation. Let*

$$u, v, i : Y \longrightarrow G, \qquad j : Z \longrightarrow G$$

be morphisms of S-schemes. Suppose that

$$i_\eta : Y_\eta \longrightarrow G_\eta, \qquad j_\eta : Z_\eta \longrightarrow G_\eta$$

are closed immersions.

Then there is a nonempty open subset S' of S such that the morphisms

$$i' : Y' \longrightarrow G', \qquad j' : Z' \longrightarrow G',$$

obtained by base change, are closed immersions, and such that the functors

$$\underline{\mathrm{Transp}}(u', v'), \qquad \underline{\mathrm{Transp}}_{G'}(i'(Y'), j'(Z')),$$
$$\underline{\mathrm{Transp}}^{\mathrm{str}}_{G'}(i'(Y'), j'(Z'))$$

respectively, the functors

$$\underline{\mathrm{Centr}}(u'), \qquad \underline{\mathrm{Norm}}_{G'} i'(Y'),$$

are representable by closed S'-subschemes (respectively, closed S'-subgroups) *of G', of finite presentation over S'.*

Proof. We shall apply the results of 10.1, first in the situation of 10.10(a), and then in that of 10.10(b). The schemes G_η, Y_η, Z_η are flat over the field $\kappa(\eta)$; the group G_η is separated over $\kappa(\eta)$ (VI_A, 0.3); and i_η, j_η are closed immersions. Therefore, by 10.1, there are an affine open subset $S' = \operatorname{Spec}(A')$ of S, a Noetherian subring A of A', A-schemes G_A, Y_A, Z_A, flat and of finite presentation over A, and morphisms

$$u_A, v_A, i_A : Y_A \longrightarrow G_A, \qquad j_A : Z_A \longrightarrow G_A,$$

such that G_A is an A-group separated over A, i_A and j_A are closed immersions, and

$$G \times_S S' = G_A \otimes_A A',$$

with analogous equalities for the remaining schemes and morphisms. Since the functors under consideration over S' are obtained by base change from the analogous functors over $\operatorname{Spec}(A)$, it is enough to prove the result for the latter.

By EGA IV_2, 6.9.2, after replacing A by a localization A_f (and hence S' by the affine open subset S'_f), we may suppose that G_A, Y_A, Z_A are essentially free over A, in the sense of 6.2.1.[97] It then follows from 6.2.4(b) and 6.2.4(e) that, under the hypotheses of the

[95] Editor's note: observe that the proof also uses case (b).

[96] Editor's note: the statement has been simplified, and the case of group subschemes has been treated separately in Corollary 10.11.1.

[97] Editor's note: indeed, since G_A is flat and of finite presentation over A, it is covered by affine open subsets $G_1, \ldots, G_n$ such that each $\mathcal{O}(G_i)$ is a flat A-algebra of finite presentation. By EGA IV_2, 6.9.2, there is an $f_i \in A$ such that $\mathcal{O}(G_i)_{f_i}$ is a free A_{f_i}-module. One may therefore replace $\operatorname{Spec}(A)$ by the affine open subset $D(f)$, where $f = f_1 \cdots f_n$, and proceed in the same way for Y_A and Z_A.

proposition, the functors under consideration are representable by closed A-subschemes of G_A. These subschemes are of finite presentation over A, since A is Noetherian and G_A is of finite presentation over A; and in the cases of $\underline{\mathrm{Centr}}(u_A)$ and $\underline{\mathrm{Norm}}_{G_A} i_A(Y_A)$, they are closed A-subgroups of G_A.

Corollary 10.11.1.— *Let S be an integral scheme with generic point η. Let G, H, K be S-groups of finite presentation, and let*

$$i : H \longrightarrow G, \qquad j : K \longrightarrow G$$

be two quasi-compact monomorphisms of S-groups. Then there is a nonempty open subset S' of S such that the morphisms

$$i' : H' \longrightarrow G', \qquad j' : K' \longrightarrow G',$$

obtained by base change, are closed immersions, and such that the functors

$$\underline{\mathrm{Transp}}_{G'}(H', K'), \qquad \underline{\mathrm{Transp}}^{\mathrm{str}}_{G'}(H', K'),$$

respectively, the functors

$$\underline{\mathrm{Centr}}_{G'} H', \qquad \underline{\mathrm{Norm}}_{G'} H',$$

are representable by closed S'-subschemes (respectively, closed S'-subgroups) *of G', of finite presentation over S'.*

This follows from the preceding proposition, since the hypotheses imply, by 1.4.2, that i_η and j_η are closed immersions.

Proposition 10.12.— *Let S be an integral scheme, let G be an S-group of finite presentation, and let A and B be two group subschemes of G, of finite presentation over S and with smooth generic fibers. Suppose moreover that one of the following conditions holds:*

a) *A has connected generic fiber;*

b) *A and B are invariant in G.*

Then there are a nonempty open subset S' of S and a closed group subscheme D' of

$$G' = G|_{S'},$$

of finite presentation over S' and with smooth fibers, that represents the (fppf) *sheaf associated with the presheaf of groups of commutators of*

$$A' = A|_{S'} \qquad \textit{and} \qquad B' = B|_{S'}$$

in G'. Moreover, D' has connected fibers in case (a)*, and is invariant in G' in case* (b)*.*

In particular, for every $s \in S'$,

$$D'_s = (A_s, B_s),$$

with the notation of 7.2.2.

Proof. Let η be the generic point of S, and put

$$H_\eta = (A_\eta, B_\eta).$$

Since A_η and B_η are smooth, 7.8 in case (a) and 7.3(v) in case (b) show that H_η is connected, respectively invariant in G_η.

We are in the situation of 10.0 corresponding to 10.10(a). Consequently, by 10.3 and 10.1, there are a nonempty open subset S' of S and an S'-group subscheme D', of finite presentation and closed in G', such that

$$D'_\eta = D' \otimes_{S'} \kappa(\eta) = H_\eta.$$

Moreover, by EGA IV$_3$, 9.7.7 and 9.3.3, we may suppose that D' has geometrically reduced fibers. In case (a), another application of EGA IV$_3$, 9.7.7 and 9.3.3 allows us to suppose that D' has connected fibers, hence geometrically connected fibers (cf. VI$_A$, 2.1.1). In case (b), we may suppose, by 10.4, that D' is invariant in G'.

We also saw, in the course of the proof of 7.8, that there is an integer n such that

$$\nu_\eta^n \left((A_\eta \times_{\kappa(\eta)} B_\eta)^n \right) = D'_\eta,$$

where ν_η and ν_η^n are defined as in 7.2.2(a) and 7.1(ii). The same formulas define morphisms

$$\nu' : A' \times_{S'} B' \longrightarrow G', \qquad \nu'^n : (A' \times_{S'} B')^n \longrightarrow G',$$

and

$$\nu'^n \otimes_{S'} \kappa(\eta) = \nu_\eta^n.$$

It follows from 10.1 that we may choose S' so that the morphism ν'^n is flat and factors through D', and so that the resulting morphism

$$\nu'^n : (A' \times_{S'} B')^n \longrightarrow D'$$

is surjective. By 7.5, the morphism[98]

$$\nu'^{2n} = \mu \circ (\nu'^n \times_{S'} \nu'^n) : (A' \times_{S'} B')^{2n} \longrightarrow D'$$

is a covering for the (fppf) topology. Therefore, by 7.6, D' represents the (fppf) sheaf associated with the presheaf of commutators of A' and B' in G'.

Furthermore, for every $s \in S'$, ν'^n induces a surjective morphism

$$\nu_s^n : (A_s \times_{\kappa(s)} B_s)^n \longrightarrow D'_s.$$

[99] Thus D'_s is a closed subgroup of G_s containing

$$\nu_s(A_s \times_{\kappa(s)} B_s),$$

and hence containing (A_s, B_s). Since ν_s^n is surjective, $D'_s = (A_s, B_s)$; and by 7.6, it represents the (fppf) sheaf of commutators of A_s and B_s in G_s.

Corollary 10.12.1.—[100] *Let S be an integral scheme with generic point η, and let G be an S-group of finite presentation with smooth fibers. Put*

$$K_\eta^0 = G_\eta, \qquad K_\eta^p = (K_\eta^{p-1}, K_\eta^{p-1}) \quad \text{(respectively, } K_\eta^p = (G_\eta, K_\eta^{p-1})\text{)}$$

for every $p \in \mathbb{N}^*$.Fix $n \in \mathbb{N}^*$. Then there are a nonempty open subset S' of S and a group subscheme D, invariant in $G|_{S'}$, of finite presentation and with smooth fibers, such that

$$D_s = K_s^n$$

[98] Editor's note: $n+1$ has been corrected below to $2n$.
[99] Editor's note: the argument that follows has been expanded from the original.
[100] Editor's note: this corollary, which is used in the proof of 8.4, has been added.

for every $s \in S'$.

This follows from 10.12, by induction on n.

Corollary 10.13.— *Let S be an integral scheme with generic point η. Let G be an S-group and let H be an invariant group subscheme of G. Suppose that G and H are of finite presentation over S and have smooth generic fibers.*[101]

If

$$(H_\eta, H_\eta) = H_\eta \qquad \text{(respectively, } (G_\eta, H_\eta) = H_\eta),$$

then there is a nonempty open subset S' of S such that, for every $s \in S'$,

$$(H_s, H_s) = H_s \qquad \text{(respectively, } (G_s, H_s) = H_s).$$

Indeed, the proof of 10.12 gives a nonempty open subset S' of S and an S'-group subscheme D of $G|_{S'}$, of finite presentation and with smooth fibers, such that, for every $s \in S'$,

$$D_s = (H_s, H_s) \qquad \text{(respectively, } D_s = (G_s, H_s)).$$

On the other hand, since $D_\eta = H_\eta$ and D and H are of finite presentation over S', EGA IV_3, 8.8.2.5 gives a nonempty open subset S'' of S' such that

$$D|_{S''} = H|_{S''}.$$

Consequently, for every $s \in S''$,

$$H_s = D_s = (H_s, H_s) \qquad \text{(respectively, } H_s = D_s = (G_s, H_s)).$$

10.14.— Statements 10.2 and 10.3, concerning the category of S-groups of finite presentation, extend to the category of pairs consisting of an S-group of finite presentation and an S-scheme of finite presentation with operator group G. More precisely, in the situation recalled at the beginning of 10.0:

(i) let $j \in I$, and let G_j and G'_j be two S_j-groups of finite presentation; let H_j (respectively, H'_j) be an S_j-scheme of finite presentation with operator group G_j (respectively, G'_j). For $i \in I$, $i \geq j$, put

$$G_i = G_j \times_{S_j} S_i, \qquad G = G_j \times_{S_j} S,$$

and define G'_i, G', H_i, H, H'_i, and H' similarly. Denote by

$$\mathrm{Dihom}_{S\text{-gr.}}\big((G, H), (G', H')\big)$$

the set of di-morphisms of S-groups and S-schemes with operator groups from the pair (G, H) to the pair (G', H'). Then the canonical map

$$\varinjlim_{i \geq j} \mathrm{Dihom}_{S_i\text{-gr.}}\big((G_i, H_i), (G'_i, H'_i)\big) \longrightarrow \mathrm{Dihom}_{S\text{-gr.}}\big((G, H), (G', H')\big)$$

is bijective.

(ii) let G be an S-group of finite presentation and H an S-scheme of finite presentation with operator group G. Then there are an index $j \in I$, an S_j-group G_j of finite presentation, an S_j-scheme H_j of finite presentation with operator group G_j, and a di-isomorphism of S-groups and S-schemes with operator groups from

$$\big(G_j \times_{S_j} S,\ H_j \times_{S_j} S\big)$$

onto (G, H).

[101] Editor's note: the original assumed that G and H were of finite presentation and that H had smooth fibers; the hypothesis has been modified in order to apply 10.12. The proof has also been expanded.

Definition 10.15.—[102] *Let $\mathcal{T}$ be a topology on* (**Sch**/S) *coarser than the canonical topology. Given an S-group scheme G and an S-scheme X with operator group G, one says that X is a* formally homogeneous space under G (relative to the topology $\mathcal{T}$) *if the morphism*

$$\Phi : G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (gx, x),$$

defined for every $S' \to S$, $g \in G(S')$, and $x \in X(S')$, is an epimorphism in the category of sheaves for the topology $\mathcal{T}$; equivalently, Φ is a covering for the topology $\mathcal{T}$ (cf. IV, 4.4.3).

One says that X is a homogeneous space *if it is formally homogeneous and, in addition, the morphism $X \to S$ is also a covering for the topology $\mathcal{T}$.*

In particular, one says that X is a formally principal homogeneous space under G *if Φ is an isomorphism, and that X is a* principal homogeneous bundle (or a G-torsor) *if Φ is an isomorphism and, in addition, the morphism $X \to S$ is a covering for the topology $\mathcal{T}$* (cf. IV, 5.1.5 and 5.1.6(ii)).

Proposition 10.16.— *Retain the situation considered at the beginning of 10.0. Let $j \in I$, let G_j be an S_j-group, and let X_j be an S_j-scheme with operator group G_j. Suppose that G_j and X_j are of finite presentation over S_j.*

For

$$X = X_j \times_{S_j} S$$

to be a homogeneous space (respectively, a principal homogeneous bundle) *under*

$$G = G_j \times_{S_j} S$$

for the (fppf) *topology, it is necessary and sufficient that there be an index $i \geq j$ such that*

$$X_i = X_j \times_{S_j} S_i$$

is a homogeneous space (respectively, a principal homogeneous bundle) *under*

$$G_i = G_j \times_{S_j} S_i.$$

Taking into account 10.14 and EGA IV$_3$, 8.8.2, 8.8.3, and 8.10.5, the assertion follows from the property of covering morphisms for the (fppf) topology recalled in 10.1.[103]

11. Affine group schemes

11.0. Reminders.—[104] Let $q : X \to S$ be a quasi-compact and quasi-separated morphism of schemes (cf. EGA IV$_1$, 1.1 and 1.2), and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Recall that $q_*(\mathcal{F})$ is a quasi-coherent $\mathcal{O}_S$-module (EGA I, 9.2.1). Moreover, by EGA III, 1.4.15 (completed by EGA IV$_1$, 1.7.21), one has (c) below, and the proof in *loc. cit.* also gives (a) and (b):

(a) If $\mathcal{F}$ is a filtered direct limit of quasi-coherent submodules $\mathcal{F}_\alpha$, then

$$q_*(\mathcal{F}) = \varinjlim_\alpha q_*(\mathcal{F}_\alpha).$$

[102] Editor's note: the original has been corrected by distinguishing the notions of a "formally homogeneous" object and a "homogeneous" object; see IV, §§5.1 and 6.7.

[103] Editor's note: for limit-passage properties of torsors expressed in terms of Čech cohomology, see also SGA 4, VII, 5.7 and its corollaries. This is developed in detail in [Ma07].

[104] Editor's note: this paragraph of reminders has been added.

(b) If $\mathcal{E}$ is a flat $\mathcal{O}_S$-module, the canonical morphism

$$\mathcal{E} \otimes_{\mathcal{O}_S} q_*(\mathcal{O}_X) \longrightarrow q_* q^*(\mathcal{E})$$

is an isomorphism.

(c) Let $p : S' \to S$ be a flat morphism, let $q' : X' \to S'$ be obtained from q by base change, and let $\mathcal{F}'$ be the inverse image of $\mathcal{F}$ on X'. Then the canonical morphism

$$p^* q_*(\mathcal{F}) \longrightarrow q'_*(\mathcal{F}')$$

is an isomorphism.

Indeed, let $U = \mathrm{Spec}(A)$ be an arbitrary affine open subset of S. By hypothesis, $q^{-1}(U)$ is the union of affine open subsets

$$V_i = \mathrm{Spec}(B_i), \qquad i = 1, \ldots, n,$$

and every intersection $V_i \cap V_j$ is the union of finitely many affine open subsets

$$W_{ijk} = \mathrm{Spec}(C_{ijk}).$$

Then $\Gamma(U, q_*(\mathcal{F})) = \Gamma(q^{-1}(U), \mathcal{F})$ is the kernel of the morphism

$$\bigoplus_{i=1}^{n} \Gamma(V_i, \mathcal{F}) \longrightarrow \bigoplus_{i,j,k} \Gamma(W_{ijk}, \mathcal{F}).$$

Assertion (a) follows, since every term above commutes with filtered direct limits (the V_i and W_{ijk} being affine, hence quasi-compact).

Let us prove (b). The module $E = \Gamma(U, \mathcal{E})$ is flat over A, and $\Gamma(U, q_* q^*(\mathcal{E}))$ is the kernel $K(E)$ of the morphism

$$\bigoplus_{i=1}^{n} B_i \otimes_A E \longrightarrow \bigoplus_{i,j,k} C_{ijk} \otimes_A E.$$

Since E is flat over A, this kernel is identified with

$$K(A) \otimes_A E = \mathcal{O}_X(q^{-1}(U)) \otimes_A E.$$

Finally, if U' is an arbitrary affine open subset of S' lying over U, then $A' = \mathcal{O}(U')$ is a flat A-algebra, and the same argument gives

$$\mathcal{F}(q^{-1}(U)) \otimes_A A' \xrightarrow{\sim} \mathcal{F}'(q'^{-1}(U')).$$

Notation.— Let S be a scheme, let X be an S-scheme, and let $f : X \to S$ be the structural morphism. We shall put

$$\mathcal{A}(X) = f_*(\mathcal{O}_X).$$

Lemma 11.1.— *Let X and Y be two quasi-compact and quasi-separated S-schemes, and let $f : X \to S$ and $g : Y \to S$ be their structural morphisms. Then the canonical homomorphism*

$$\varphi : \mathcal{A}(X) \otimes_{\mathcal{O}_S} \mathcal{A}(Y) \longrightarrow \mathcal{A}(X \times_S Y)$$

is an isomorphism in each of the following cases:[105]

[105] Editor's note: if k is a field and X is an infinite disjoint union of copies of $S = \mathrm{Spec}\, k$ (so that X is not quasi-compact), then $\mathcal{A}(X) = k^X$, and the canonical morphism

$$k^X \otimes_k k^X \longrightarrow k^{X \times X}$$

is not surjective.

(a) *f and g are affine;*

(b) *g* (or *f*) *is flat and affine;*

(c) *g is flat and $f_*(\mathcal{O}_X)$ is a flat $\mathcal{O}_S$-module.*

In case (b), we shall suppose that it is g that is flat and affine. Put

$$S' = \operatorname{Spec} \mathcal{A}(X), \qquad Y' = Y \times_S S', \qquad g' = g \times_S S',$$

and write $v : S' \to S$ for the structural morphism:

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ \operatorname{Spec} \mathcal{A}(X) = S' & \xrightarrow[v]{} & S \end{array}$$

In cases (a) and (b), the morphism g is affine, and hence EGA II, 1.5.2 gives

$$g'_*(\mathcal{O}_{Y'}) = v^* g_*(\mathcal{O}_Y) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}. \tag{1}$$

The same equality holds in case (c), by 11.0(c), since S' is flat over S and g is quasi-compact and quasi-separated.

On the other hand, by EGA II, 1.2.7, the morphism $f : X \to S$ factors through v by means of a morphism $p : X \to S'$, and

$$p_*(\mathcal{O}_X) = \mathcal{O}_{S'}, \qquad X \times_S Y = X \times_{S'} Y'.$$

Since f is quasi-separated, so is p (EGA IV_1, 1.2.2); and since f is quasi-compact and v is quasi-separated, p is also quasi-compact (EGA IV_1, 1.2.4). Consider the cartesian square

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{p'} & Y' \\ \downarrow & & \downarrow{\scriptstyle g'} \\ X & \xrightarrow[p]{} & S' \end{array}$$

In cases (b) and (c), the scheme Y is flat over S, and hence Y' is flat over S'. Applying 11.0(c) once again, we obtain

$$p'_*(\mathcal{O}_{X \times_S Y}) = g'^* p_*(\mathcal{O}_X) = g'^*(\mathcal{O}_{S'}) = \mathcal{O}_{Y'}. \tag{2}$$

The same equality holds in case (a), because p and p' are then isomorphisms.

Finally, since v is affine, EGA II, 1.4.7 gives

$$v_*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(X)$$

for every quasi-coherent $\mathcal{O}_S$-module $\mathcal{F}$. Combined with (2) and (1), this gives

$$\begin{aligned} \mathcal{A}(X \times_S Y) &= v_* g'_* p'_*(\mathcal{O}_{X \times_S Y}) \\ &\overset{(2)}{=} v_* g'_*(\mathcal{O}_{Y'}) \overset{(1)}{=} v_*(\mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \\ &= \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{A}(X). \end{aligned}$$

Corollary 11.2.— *The functor*

$$X \longmapsto \operatorname{Spec} \mathcal{A}(X),$$

from the full subcategory of $(\mathbf{Sch}/S)$ *formed by the* S*-schemes* X *that are flat, quasi-compact, and quasi-separated over* S*, and for which* $\mathcal{A}(X)$ *is a flat* $\mathcal{O}_S$*-module, to the category of flat affine* S*-schemes, commutes with finite products and therefore takes* S*-groups to* S*-groups.*

Definition 11.3.— *Given a flat* S*-group* G*, quasi-compact and quasi-separated over* S*, such that* $\mathcal{A}(G)$ *is flat over* $\mathcal{O}_S$*,*[106] *we write* G_{af}*, and call the* S*-group*

$$G_{\mathrm{af}} = \operatorname{Spec} \mathcal{A}(G)$$

the affine envelope *of* G*.*

Proposition 11.3.1.—[107] *The canonical morphism*

$$\tau_G : G \longrightarrow G_{\mathrm{af}}$$

is a morphism of S*-groups. Moreover, it has the following universal property:*

(i) *For every morphism of* S*-schemes* $\varphi : G \to H$*, where* H *is affine over* S*, there is a unique morphism of* S*-schemes* $\varphi' : G_{\mathrm{af}} \to H$ *such that*

$$\varphi = \varphi' \circ \tau_G.$$

(ii) *If, moreover,* H *is an* S*-group and* φ *is a morphism of* S*-groups, then the same is true of* φ'*.*

11.4.—[108] Let $\mathcal{E}$ and $\mathcal{F}$ be two quasi-coherent $\mathcal{O}_S$-modules. Consider the S-functor

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))$$

(cf. I, 3.1.4), that is, for every S-scheme $f : X \to S$,

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))(X) = \mathrm{Hom}_{\mathcal{O}_X}(\mathrm{W}(\mathcal{E})_X, \mathrm{W}(\mathcal{F})_X).$$

Moreover, by I, 4.6.2, one has $\mathrm{W}(\mathcal{E})_X = \mathrm{W}(f^*(\mathcal{E}))$ (and similarly for $\mathcal{F}$), and

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathrm{W}(f^*(\mathcal{E})), \mathrm{W}(f^*(\mathcal{F}))) = \mathrm{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})).$$

We therefore obtain (using the adjunction formula for the last equality)

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))(X) = \mathrm{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{F})). \qquad (\dagger)$$

Proposition 11.5.— *Let* X *be an* S*-scheme that is quasi-compact and quasi-separated over* S*, let* $f : X \to S$ *be the structural morphism, and let* $\mathcal{E}$ *and* $\mathcal{E}'$ *be two quasi-coherent* $\mathcal{O}_S$*-modules. Suppose that one of the following two conditions holds:*

a) *f is affine;*

[106] Editor's note: if S is a regular locally Noetherian scheme of dimension at most 2, and X is an S-scheme that is flat, quasi-compact, and quasi-separated, then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$-module; cf. [Ray70a], VII, 3.2.

[107] Editor's note: this proposition has been added; see also the additional Section 12 below for a study of the morphism $G \to G_{\mathrm{af}}$ and of its kernel.

[108] Editor's note: in 11.4–11.6, we have considered $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))$ instead of $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{V}(\mathcal{F}), \mathrm{V}(\mathcal{E}))$ (cf. I, 4.6.3), and simplified the original by taking account of the addition 11.0(b).

b) *$\mathcal{E}$ is flat over $\mathcal{O}_S$.*

Then the canonical morphism

$$\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X) \longrightarrow f_* f^*(\mathcal{E})$$

is an isomorphism, and hence

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}'), \mathrm{W}(\mathcal{E}))(X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X)).$$

Indeed, the second assertion follows from 11.4 and the first; the latter follows from EGA II, 1.4.7 in case (a), and from 11.0(b) in case (b).[109]

11.6.—[110] Let G be an S-group and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$-module. Giving $\mathcal{E}$ a G-$\mathcal{O}_S$-module structure (that is, an $\mathcal{O}_S$-linear action of G on $\mathrm{W}(\mathcal{E})$, cf. I, 4.7.1) is equivalent to giving a morphism of S-functors in monoids

$$\rho : G \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}));$$

indeed, such a ρ necessarily takes G into $\underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$.

By 11.4, giving a morphism of S-functors

$$\rho : G \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$$

is equivalent to giving an element θ of

$$\mathrm{Hom}_{\mathcal{O}_G}(f^*(\mathcal{E}), f^*(\mathcal{E})),$$

which corresponds by adjunction to an element δ of

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{E})),$$

where f denotes the projection $G \to S$.

Let $m : G \times_S G \to G$ be multiplication, let δ_G be the morphism

$$\mathcal{O}_G \longrightarrow m_*(\mathcal{O}_{G \times_S G}),$$

and let $\varphi : G \times_S G \to S$ be the projection (which equals $f \circ m$). It is convenient to write $\boxtimes$ for the "external" tensor product. We thus obtain a morphism

$$\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \longrightarrow \mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G}).$$

By abuse of notation, we again write $\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G$ for the composite of this morphism with the canonical morphism

$$\mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G}) \longrightarrow m_* m^* f^*(\mathcal{E}) = m_* \varphi^*(\mathcal{E}).$$

On the other hand, let $h : G \to S$ denote a second copy of $f : G \to S$, and consider the following commutative diagram, in which p, q denote the two projections:

[109] Editor's note: we have simplified the original, which used the isomorphism

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}\big(\mathrm{W}(\mathcal{E}'), \mathrm{W}(\mathcal{E})\big) \simeq \underline{\mathrm{Hom}}_{\mathcal{O}_S}\big(\mathrm{V}(\mathcal{E}), \mathrm{V}(\mathcal{E}')\big),$$

then the inclusion of the right-hand term in

$$\underline{\mathrm{Hom}}_S\big(\mathrm{V}(\mathcal{E}), \mathrm{V}(\mathcal{E}')\big) = \mathrm{Hom}_{\mathcal{O}_S}\big(\mathcal{E}', f_* f^*(\mathrm{Sym}(\mathcal{E}))\big),$$

and applied EGA III, 4.1.15 to $\mathrm{V}(\mathcal{E}) = \mathrm{Spec}(\mathrm{Sym}(\mathcal{E}))$ to deduce 11.0(b).

[110] Editor's note: we have expanded 11.6 and displayed the results obtained in the form of Proposition 11.6.1.

$$\begin{array}{ccc} G \times_S G & \xrightarrow{q} & G \\ {\scriptstyle p}\downarrow & \swarrow^{\varphi} & \downarrow{\scriptstyle h} \\ G & \xrightarrow[f]{} & S \end{array}$$

Writing δ again for the morphism $\mathcal{E} \to h_*h^*(\mathcal{E})$, we obtain the morphism

$$\delta \boxtimes \mathrm{id}_{\mathcal{O}_G} : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \longrightarrow h_*h^*(\mathcal{E}) \boxtimes_{\mathcal{O}_S} \mathcal{O}_G = f^*h_*h^*(\mathcal{E}).$$

By abuse of notation, we again write $\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}$ for the composite of this morphism with the canonical morphism

$$f^*h_*h^*(\mathcal{E}) \longrightarrow p_*\varphi^*(\mathcal{E}).$$

The condition that ρ be compatible with multiplication is then equivalent to saying that, for every open subset U of S, the following diagram is commutative:

$$(1) \qquad \begin{array}{ccc} \Gamma(U,\mathcal{E}) & \xrightarrow{\delta} & \Gamma\big(U \times_S G,\, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G\big) \\ {\scriptstyle \delta}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_{\mathcal{E}} \boxtimes \delta_G} \\ \Gamma\big(U \times_S G,\, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G\big) & \xrightarrow[\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}]{} & \Gamma\big(U \times_S G \times_S G,\, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \boxtimes_{\mathcal{O}_S} \mathcal{O}_G\big) \end{array}$$

Furthermore, the unit section $\varepsilon : S \to G$ induces a morphism u from $\mathcal{O}_G$ to

$$\varepsilon_*\varepsilon^*(\mathcal{O}_G) = \varepsilon_*(\mathcal{O}_S),$$

and the condition that ρ preserve unit elements is equivalent to the commutativity of the diagram

$$(2) \qquad \begin{array}{ccc} \Gamma(U,\mathcal{E}) & \xrightarrow{\qquad\delta\qquad} & \Gamma(U \times_S G,\, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \\ & {\scriptstyle \sim}\searrow \quad \swarrow{\scriptstyle \mathrm{id}_{\mathcal{E}} \boxtimes u} & \\ & \Gamma(U \times_S S,\, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_S)\,. & \end{array}$$

Thus, giving $\mathcal{E}$ a G-$\mathcal{O}_S$-module structure is equivalent to giving a morphism of $\mathcal{O}_S$-modules

$$\delta : \mathcal{E} \longrightarrow f_*f^*(\mathcal{E})$$

satisfying conditions (1) and (2) above. In that case, the morphism

$$\theta : f^*(\mathcal{E}) \longrightarrow f^*(\mathcal{E})$$

deduced from δ by adjunction is an isomorphism, because it corresponds to the isomorphism

$$G \times_S \mathrm{W}(\mathcal{E}) \xrightarrow{\sim} G \times_S \mathrm{W}(\mathcal{E})$$

defined set-theoretically by $(g, x) \longmapsto (g, gx)$; see also I, 6.5.4.

Now suppose that G is flat, quasi-compact, and quasi-separated over S, and that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module. Then, by 11.1(c), the canonical morphism

$$\mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow \mathcal{A}(G \times_S G)$$

is an isomorphism, and we write Δ for the morphism

$$\delta_G : \mathcal{A}(G) \longrightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

If, moreover,

$$\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G) = f_* f^*(\mathcal{E}),$$

as is the case by 11.5 if $G \to S$ is affine or if $\mathcal{E}$ is flat over $\mathcal{O}_S$, then conditions (1) and (2) are equivalent to the following conditions, which express that

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$$

makes $\mathcal{E}$ a right $\mathcal{A}(G)$-comodule (cf. I, 4.7.2):
(CM 1) Put $\mathcal{A} = \mathcal{A}(G)$. The following diagram is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\delta} & \mathcal{E} \otimes \mathcal{A} \\ {\scriptstyle\delta}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_{\mathcal{E}} \otimes \Delta} \\ \mathcal{E} \otimes \mathcal{A} & \xrightarrow[\delta \otimes \mathrm{id}_{\mathcal{A}}]{} & \mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A} \end{array}$$

(CM 2) Write $\eta : \mathcal{A} \to \mathcal{O}_S$ for the morphism $\mathcal{A}(\varepsilon)$. The following diagram is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\quad\delta\quad} & \mathcal{E} \otimes \mathcal{A} \\ & {\scriptstyle\sim}\searrow \quad \swarrow{\scriptstyle \mathrm{id}_{\mathcal{E}} \otimes \eta} & \\ & \mathcal{E} \otimes \mathcal{O}_S. & \end{array}$$

Remark 11.6.A.— Recall that $\mathrm{V}(\mathcal{E})$ denotes the vector fibration over S representing the functor $\mathrm{V}(\mathcal{E})$; that is, for every $S' \to S$,

$$\mathrm{V}(\mathcal{E})(S') = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'}).$$

Since I, 4.6.2 gives an anti-isomorphism of S-functors in monoids

$$\underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E})) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{V}(\mathcal{E})),$$

if $\mathcal{E}$ is a left G-$\mathcal{O}_S$-module, then there is a right action

$$\mu : \mathrm{V}(\mathcal{E}) \times_S G \longrightarrow \mathrm{V}(\mathcal{E})$$

of G on $\mathrm{V}(\mathcal{E})$, defined set-theoretically by

$$(\phi g)(x) = \phi(gx)$$

for every $g \in G(S')$, $x \in \Gamma(S', \mathcal{E}_{S'})$, and $\phi \in \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'})$. We therefore have the commutative diagrams

$$\begin{array}{ccc} \mathrm{V}(\mathcal{E}) \times_S G \times_S G & \xrightarrow{\mu \times \mathrm{id}_G} & \mathrm{V}(\mathcal{E}) \times_S G \\ {\scriptstyle \mathrm{id}_{\mathrm{V}(\mathcal{E})} \times m}\downarrow & & \downarrow{\scriptstyle \mu} \\ \mathrm{V}(\mathcal{E}) \times_S G & \xrightarrow[\mu]{} & \mathrm{V}(\mathcal{E}) \end{array} \qquad \begin{array}{ccc} \mathrm{V}(\mathcal{E}) & \xleftarrow{\ \mu\ } & \mathrm{V}(\mathcal{E}) \times_S G \\ & {\scriptstyle \mathrm{id}_{\mathrm{V}(\mathcal{E})} \times \varepsilon}\nwarrow & \uparrow \\ & & \mathrm{V}(\mathcal{E}) \times_S S. \end{array}$$

When G is flat, quasi-compact, and quasi-separated over S, when $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module, and when one of the conditions of 11.5 is satisfied, one similarly recovers conditions (CM 1) and (CM 2).

Consequently, we have proved:

Proposition 11.6.1.— *Let G be an S-group that is flat, quasi-compact, and quasi-separated over S, such that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module, and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$-module.*

(i) *Giving an $\mathcal{A}(G)$-comodule structure*

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$$

is equivalent to giving $\mathcal{E}$ a G_{af}-$\mathcal{O}_S$-module structure (that is, an $\mathcal{O}_S$-linear action of G_{af} on $\mathcal{E}$). *By composition with the morphism of S-groups $G \to G_{\mathrm{af}}$, this defines a G-$\mathcal{O}_S$-module structure on $\mathcal{E}$.*

(ii) *If, moreover, $\mathcal{E}$ is flat, every $\mathcal{O}_S$-linear action of G on $\mathcal{E}$ factors through G_{af} and corresponds to a unique $\mathcal{A}(G)$-comodule structure on $\mathcal{E}$.*

Lemma 11.7.— *Let G be an S-group that is flat, quasi-compact, and quasi-separated over S, such that $\mathcal{A} = \mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module. Let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$-module, let*

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{A}$$

be an $\mathcal{A}$-comodule structure, and let

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$$

be the associated action of G on $\mathcal{E}$. Let $\mathcal{E}_0$ be a quasi-coherent $\mathcal{O}_S$-submodule of $\mathcal{E}$ such that the restriction δ_0 of δ to $\mathcal{E}_0$ factors through

$$\mathcal{E}_0 \longrightarrow \mathcal{E}_0 \otimes \mathcal{A};$$

that is, suppose that the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{E}_0 & \hookrightarrow & \mathcal{E} \\ \downarrow{\scriptstyle \delta_0} & & \downarrow{\scriptstyle \delta} \\ \mathcal{E}_0 \otimes \mathcal{A} & \hookrightarrow & \mathcal{E} \otimes \mathcal{A}. \end{array}$$

(N.B. The morphism $\mathcal{E}_0 \otimes \mathcal{A} \to \mathcal{E} \otimes \mathcal{A}$ is injective because $\mathcal{A}$ is flat over $\mathcal{O}_S$.) *Then δ_0 makes $\mathcal{E}_0$ an $\mathcal{A}(G)$-comodule, and hence defines an action ρ_0 of G on $\mathcal{E}_0$. This is called the action on $\mathcal{E}_0$ induced by ρ, and we say that $\mathcal{E}_0$ is stable under ρ.*

This follows immediately from the definitions and from 11.6. Notice, however, that in general the canonical map

$$\mathrm{W}(\mathcal{E}_0) \longrightarrow \mathrm{W}(\mathcal{E})$$

is not a monomorphism.

Remark 11.7.bis.—[111] Let G be a flat S-group and let $\mathcal{E}$ be a quasi-coherent G-$\mathcal{O}_S$-module. Write f for the morphism $G \to S$ and δ for the morphism

$$\mathcal{E} \longrightarrow f_* f^*(\mathcal{E})$$

defined in 11.6. Let $\mathcal{E}_0$ be a quasi-coherent $\mathcal{O}_S$-submodule of $\mathcal{E}$. Since f is flat, $f_* f^*(\mathcal{E}_0)$ is an $\mathcal{O}_S$-submodule of $f_* f^*(\mathcal{E})$, and the same holds for $\varphi = f \times f$. Consequently, if the restriction δ_0 of δ to $\mathcal{E}_0$ factors through $f_* f^*(\mathcal{E}_0)$, then it makes $\mathcal{E}_0$ a G-$\mathcal{O}_S$-module. In this case, we say that $\mathcal{E}_0$ is a G-stable submodule of $\mathcal{E}$.

Definition 11.8.0.—[112] Let S be a scheme. An *$\mathcal{O}_S$-coalgebra* is an $\mathcal{O}_S$-module $\mathcal{C}$, together with two morphisms of $\mathcal{O}_S$-modules

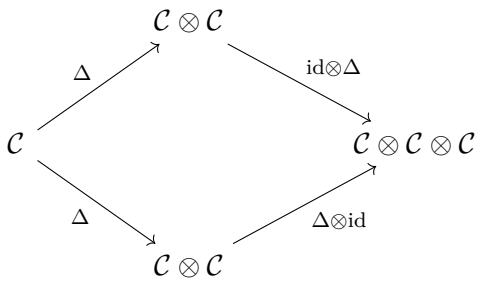
$$\Delta : \mathcal{C} \longrightarrow \mathcal{C} \otimes \mathcal{C}, \qquad \varepsilon : \mathcal{C} \longrightarrow \mathcal{O}_S,$$

satisfying the following two axioms (cf. I, 4.2):

[111] Editor's note: we have added this remark, which generalizes 11.7 and will be useful in 11.10.bis.

[112] Editor's note: the statements 11.8 and 11.9 concern only the notion of a comodule over a coalgebra. We have therefore introduced Definition 11.8.0 and reformulated 11.8 and 11.9 accordingly.

(CO 1) Δ is coassociative: the following diagram is commutative:



(CO 2) ε is a counit; that is, the following two composites are the identity:

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C}\otimes\mathcal{C} \xrightarrow{\mathrm{id}\otimes\varepsilon} \mathcal{C}\otimes\mathcal{O}_S \xrightarrow{\sim} \mathcal{C},$$

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C}\otimes\mathcal{C} \xrightarrow{\varepsilon\otimes\mathrm{id}} \mathcal{O}_S\otimes\mathcal{C} \xrightarrow{\sim} \mathcal{C}.$$

A $\mathcal{C}$-*comodule* (on the right) is an $\mathcal{O}_S$-module $\mathcal{E}$, together with a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathcal{E} \longrightarrow \mathcal{E}\otimes\mathcal{C}$$

satisfying axioms (CM 1) and (CM 2) of 11.6.

We say that $\mathcal{C}$ (respectively, $\mathcal{E}$) is a *quasi-coherent coalgebra* (respectively, a *quasi-coherent comodule*) if it is a quasi-coherent $\mathcal{O}_S$-module.

Let A be a commutative ring and let C be an A-coalgebra. Then

$$C^\vee = \mathrm{Hom}_A(C, A)$$

is an A-algebra. We denote by

$$\mathrm{ev} : C\otimes_A C^\vee \longrightarrow A$$

the natural evaluation map.

Lemma 11.8.— *Let C be an A-coalgebra, V a C-comodule, and M an A-submodule of V. Suppose that C is a projective A-module.*[113] *Let $c(M)$ be the image of the morphism of A-modules*

$$\theta : M\otimes_A C^\vee \xrightarrow{\mu\otimes\mathrm{id}} V\otimes_A C\otimes_A C^\vee \xrightarrow{\mathrm{id}\otimes\mathrm{ev}} V.$$

Then $c(M)$ is the smallest subcomodule of V containing M, and it is a finitely generated A-module if M is. We call $c(M)$ the subcomodule generated by M.

Moreover, for every homomorphism of rings $A \to A'$, if M' denotes the image of $M\otimes_A A'$ in $V' = V\otimes_A A'$, then $c(M')$ is the image of $c(M)\otimes_A A'$ in V'. Thus the formation of $c(M)$ commutes with base change.

First, $M \subset c(M)$ by (CM 2). If N is a subcomodule of V containing M, then $\mu(M) \subset N\otimes C$, and hence $c(M) \subset N$.

By hypothesis, C is a direct summand of a free A-module L with basis $(e_i)_{i\in I}$. Let φ_i denote the restriction to C of the linear form e_i^*, defined by

$$e_i^*(e_j) = \delta_{ij}.$$

[113]Editor's note: in the original, C is assumed to be a free A-module. The generalization to the case where C is a projective A-module, pointed out by J.-P. Serre, is mentioned in Remark 11.10.1. We have incorporated this generalization here and in 11.9, and have expanded the proofs accordingly.

Let $x \in M$. We may write

$$\mu(x) = \sum_{i \in J} x_i \otimes e_i, \tag{1}$$

where $x_i \in V$ and J is a finite subset of I. Then

$$x_i = \theta(x \otimes \varphi_i) \in c(Ax),$$

and consequently

$$c(Ax) = \sum_{i \in J} Ax_i.$$

Since C is a direct summand of L, say $L = C \oplus R$, whence

$$V \otimes L = (V \otimes C) \oplus (V \otimes R),$$

we obtain

$$(c(Ax) \otimes L) \cap (V \otimes C) = c(Ax) \otimes C.$$

Consequently, $\mu(x)$ may also be written in the form

$$\mu(x) = \sum_{j \in J} x_j \otimes b_j, \tag{2}$$

with $b_j \in C$. Write

$$\Delta(b_j) = \sum_{i \in I} b_{ij} \otimes e_i, \qquad b_{ij} \in C.$$

Applying $\mu \otimes \mathrm{id}$ to (1) and $\mathrm{id} \otimes \Delta$ to (2), and using axiom (CM 1), we obtain, for every $i \in J$,

$$\mu(x_i) = \sum_{j \in J} x_j \otimes b_{ij} \in c(Ax) \otimes C.$$

This proves that $c(M)$ is a subcomodule of V, and therefore that it is the smallest subcomodule of V containing M.

It is clear that $c(M)$ is a finitely generated A-module whenever M is: if

$$M = Ax_1 + \cdots + Ax_n, \qquad \mu(x_k) = \sum_i x_{ik} \otimes e_i,$$

then the x_{ik}, for $k = 1, \ldots, n$ and i ranging over a finite subset of I, generate $c(M)$.

Finally, let $A \to A'$ be a homomorphism of rings, and let M' be the image of $M \otimes_A A'$ in $V' = V \otimes_A A'$. Then $c(M')$, respectively the image of $c(M) \otimes_A A'$ in V', is the image of the morphism θ' below, respectively of the composite $\theta' \circ \tau$:

$$M \otimes_A A' \otimes_A C^\vee \xrightarrow{\tau} M \otimes_A \mathrm{Hom}_A(C, A') \xrightarrow{\theta'} V'.$$

These two morphisms have the same image. Indeed, let $\psi \in \mathrm{Hom}_A(C, A')$ and $x \in M$, and write

$$\mu(x) = \sum_{i \in J} x_i \otimes e_i.$$

Then

$$\theta'(x \otimes \psi) = \sum_{i \in J} \psi(e_i) x_i$$

is the image under $\theta' \circ \tau$ of the element

$$\sum_{i \in J} x \otimes \psi(e_i) \otimes \varphi_i \in M \otimes_A A' \otimes_A C^\vee.$$

This proves the lemma.

We also have the following proposition.

Proposition 11.8.bis.—[114] *Let A be a Noetherian ring, let C be an A-coalgebra flat over A, let V be a C-comodule, and let M be a finitely generated A-submodule of V. Then there is a subcomodule W of V, finitely generated over A, that contains M.*

Indeed, since M is finitely generated, so is $\Delta_V(M)$. Hence there is a finitely generated A-submodule M' of V such that

$$\Delta_V(M) \subset M' \otimes_A C.$$

Let $\pi : V \to V/M'$ be the projection, put

$$\overline{\Delta}_V = (\pi \otimes \mathrm{id}_C)\Delta_V,$$

and set

$$W = \{x \in V \mid \Delta_V(x) \in M' \otimes_A C\} = \ker \overline{\Delta}_V.$$

This is an A-submodule of V that contains M and is contained in M', because

$$x = (\mathrm{id}_V \otimes \varepsilon)\Delta_V(x).$$

It is therefore finitely generated over A. Moreover,

$$(\overline{\Delta}_V \otimes \mathrm{id}_C)\Delta_V = (\pi \otimes \Delta_C)\Delta_V$$

vanishes on W; in other words, $\Delta_V(W)$ is contained in the kernel K of $\overline{\Delta}_V \otimes \mathrm{id}_C$. Since C is flat over A, one has

$$K = W \otimes_A C.$$

Thus W is a subcomodule of V.

Lemma 11.8.1.—[115] *Let C be an A-coalgebra, let V be a C-comodule, let M be an A-submodule of V, and let $f : A \to A'$ be a faithfully flat homomorphism of rings. Suppose that $C' = C \otimes_A A'$ is a projective A'-module.*

(i) *There is a smallest subcomodule $t(M)$ of V containing M, and $t(M)$ is a finitely generated A-module if M is. Moreover, the formation of $t(M)$ commutes with base change.*

(ii) *More precisely, C is a projective A-module, and $t(M) = c(M)$.*

Proof. For (ii), [RG71] (see Proposition 11.8.2 below) shows that C is a projective A-module. We may therefore apply Lemma 11.8: $c(M)$ is the smallest subcomodule of V containing M, it is a finitely generated A-module if M is, and its formation commutes with base change.

To avoid an anachronism ([*RG*71] postdates SGA 3), let us sketch a direct proof of (i). Since $A \to A'$ is flat,

$$M' = M \otimes_A A'$$

is an A'-submodule of $V' = V \otimes_A A'$. Since C' is a projective A'-module, $c(M')$ is the smallest subcomodule of V' containing M'. Denote by $V' \otimes_A A'$ and $A' \otimes_A V'$ the two $A' \otimes_A A'$-comodule structures on

$$V'' = V' \otimes_{A'} (A' \otimes_A A')$$

[114] Editor's note: this proposition, taken from [Se68], Section 1.5, Proposition 2, has been added.

[115] Editor's note: this lemma, which is used in the proof of 11.9, has been added.

obtained by the two base changes

$$A' \rightrightarrows A' \otimes_A A', \qquad a' \longmapsto a' \otimes 1, \quad a' \longmapsto 1 \otimes a'.$$

The A'-comodule V' carries an isomorphism of $A' \otimes_A A'$-comodules

$$\varphi : V' \otimes_A A' \xrightarrow{\sim} A' \otimes_A V', \qquad (x \otimes a') \otimes b' \longmapsto b' \otimes (x \otimes a'),$$

which is a descent datum, that is,

$$\varphi_{31} = \varphi_{32} \circ \varphi_{21}.$$

Since $M' = M \otimes_A A'$, the isomorphism φ takes $M' \otimes_A A'$ onto $A' \otimes_A M'$, and hence takes $c(M' \otimes_A A')$ onto $c(A' \otimes_A M')$. Because the formation of $c(M')$ commutes with base change,

$$c(M' \otimes_A A') = c(M') \otimes_A A', \qquad c(A' \otimes_A M') = A' \otimes_A c(M').$$

Thus

$$\varphi(c(M') \otimes_A A') = A' \otimes_A c(M'),$$

and φ equips $c(M')$ with a descent datum. By fpqc descent, there is a unique subcomodule $t(M)$ of V such that

$$c(M') = t(M) \otimes_A A'.$$

It contains M, since $t(M) \otimes_A A'$ contains M'. Moreover, if N is a subcomodule of V containing M, then N contains $t(M)$, because $N \otimes_A A'$ contains

$$c(M') = t(M) \otimes_A A'.$$

Hence $t(M)$ is the smallest subcomodule of V containing M.

Finally, let $A \to B$ be a homomorphism of rings. Put

$$B' = B \otimes_A A',$$

and let M_B, respectively $M'_{B'}$, be the image of $M \otimes_A B$ in $V_B = V \otimes_A B$, respectively of $M' \otimes_{A'} B'$ in $V' \otimes_{A'} B' = V \otimes_A B'$. Then

$$M_B \otimes_B B' = M'_{B'}.$$

On the one hand, the preceding construction, applied to C_B and $B \to B'$, gives

$$c(M_B \otimes_B B') = t(M_B) \otimes_B B' = t(M_B) \otimes_A A'.$$

On the other hand, because the formation of $c(M')$ commutes with base change, $c(M'_{B'})$ is the image in

$$V' \otimes_{A'} B' = V \otimes_A B \otimes_A A'$$

of

$$c(M') \otimes_{A'} B' = t(M) \otimes_A B \otimes_A A'.$$

It follows that $t(M_B)$ is the image of $t(M) \otimes_A B$ in V_B.

Proposition 11.8.2 (Gruson–Raynaud).— *Let $f : A \to A'$ be a faithfully flat homomorphism. Then f "descends projectivity": if M is an A-module and $M \otimes_A A'$ is a projective A'-module, then M is a projective A-module.*

Indeed, by [RG71], II, 2.5.1, f descends the Mittag–Leffler condition, and therefore, by *loc. cit.*, II, 3.1.3, f descends projectivity.[116]

Proposition 11.9.—[117] *Let $\mathcal{C}$ be an $\mathcal{O}_S$-coalgebra, let $\mathcal{E}$ be a $\mathcal{C}$-comodule, and let $\mathcal{F}$ be an $\mathcal{O}_S$-submodule of $\mathcal{E}$, all of them quasi-coherent. Suppose that S is covered by affine open subsets $U_\alpha = \operatorname{Spec} A_\alpha$, and that for each α there is a faithfully flat homomorphism of rings $A_\alpha \to A'_\alpha$ such that*

$$\Gamma(U_\alpha, \mathcal{C}) \otimes_{A_\alpha} A'_\alpha$$

is a projective A'_α-module.[(*)]

Then there is a smallest quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$, and $t(\mathcal{F})$ is a finitely generated $\mathcal{O}_S$-module if $\mathcal{F}$ is. Moreover, for every base change $S' \to S$, if $\mathcal{F}'$ denotes the image of $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ in

$$\mathcal{E}' = \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'},$$

then $t(\mathcal{F}')$ is the image of $t(\mathcal{F}) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ in $\mathcal{E}'$. In other words, the formation of $t(\mathcal{F})$ commutes with base change.

Proof. For each α, let T_α be the A_α-module

$$T_\alpha = t(\Gamma(U_\alpha, \mathcal{F})).$$

By 11.8.1(i), the associated $\mathcal{O}_{U_\alpha}$-module $\widetilde{T}_\alpha$ is the smallest quasi-coherent subcomodule of $\mathcal{E}|_{U_\alpha}$ containing $\mathcal{F}|_{U_\alpha}$; and it is a finitely generated $\mathcal{O}_{U_\alpha}$-module if $\mathcal{F}|_{U_\alpha}$ is.

For every α, β, put

$$U_{\alpha\beta} = U_\alpha \cap U_\beta.$$

Since the construction of $t(M)$ commutes with base change, there are canonical isomorphisms of $\mathcal{O}_{U_{\alpha\beta}}$-modules

$$\varphi_{\alpha\beta} : \widetilde{T}_\beta \otimes_{A_\beta} \mathcal{O}_{U_{\alpha\beta}} \xrightarrow{\sim} \widetilde{T}_\alpha \otimes_{A_\alpha} \mathcal{O}_{U_{\alpha\beta}}.$$

They satisfy the cocycle condition

$$\varphi'_{\alpha\gamma} = \varphi'_{\alpha\beta} \circ \varphi'_{\beta\gamma},$$

where $\varphi'_{\alpha\gamma}$, respectively the other maps, denotes the restriction of $\varphi_{\alpha\gamma}$, respectively the other maps, to $U_\alpha \cap U_\beta \cap U_\gamma$.

Consequently, the $\widetilde{T}_\alpha$ glue to a quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$. We leave it to the reader to verify that $t(\mathcal{F})$ is the smallest quasi-coherent subcomodule of $\mathcal{E}$ containing $\mathcal{F}$, and that its formation commutes with base change.

Definition 11.9.1.—[118] *Let S be a scheme and let $\mathcal{P}$ be a quasi-coherent $\mathcal{O}_S$-module. The following conditions are equivalent:*

(i) *For every affine open subset U of S, the module $\Gamma(U, \mathcal{P})$ is a projective $\mathcal{O}_S(U)$-module.*

[116] Editor's note: let us also point out that assertion II, 2.5.2 of *loc. cit.*, which is more general than II, 2.5.1, is corrected in [Gr73]; this does not affect the case of faithfully flat homomorphisms.

[117] Editor's note: as indicated in editor's note 112, we have rewritten the statement for an $\mathcal{O}_S$-coalgebra $\mathcal{C}$ (rather than for an S-group G satisfying the hypotheses in Corollary 11.10). We have also expanded the proof (the original merely said: "$\cdots$ the proposition follows from Lemma 11.8").

[*] This holds, for example, when $\mathcal{C} = \mathcal{A}(G)$, where G is a reductive S-group, as we shall see in Exposé XXII, 5.7.8.

[118] Editor's note: this definition, taken from the bottom of page 82 of [RG71], has been added. Thus the hypothesis in Proposition 11.9 is that the coalgebra $\mathcal{C}$ be a locally projective $\mathcal{O}_S$-module, and we have used this terminology in Corollary 11.10.

(ii) *There is a covering* (U_α) *of* S *by affine open subsets such that every* $\Gamma(U_\alpha, \mathcal{P})$ *is a projective* $\mathcal{O}_S(U_\alpha)$*-module.*

(iii) *There is a covering* (U_α) *of* S *by affine open subsets, and faithfully flat homomorphisms of rings*

$$A_\alpha = \mathcal{O}_S(U_\alpha) \longrightarrow A'_\alpha,$$

such that, for every α*,*

$$\Gamma(U_\alpha, \mathcal{P}) \otimes_{A_\alpha} A'_\alpha$$

is a projective A'_α*-module.*

It is clear that

$$\text{(i)} \Longrightarrow \text{(ii)} \Longrightarrow \text{(iii)}.$$

Conversely, if (iii) holds, 11.8.2 implies that every $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$-module, and hence (ii) holds.

Finally, suppose that (ii) holds, and let

$$V = \operatorname{Spec} A$$

be an arbitrary affine open subset. It is covered by finitely many affine open subsets

$$V_1, \ldots, V_n, \qquad V_i = \operatorname{Spec} A_i,$$

each contained in at least one $V \cap U_\alpha$, so that $\Gamma(V_i, \mathcal{P})$ is a projective A_i-module. Put

$$A' = A_1 \times \cdots \times A_n.$$

Then $A \to A'$ is faithfully flat and

$$\Gamma(V, \mathcal{P}) \otimes_A A'$$

is a projective A'-module. Therefore 11.8.2 implies that $\Gamma(V, \mathcal{P})$ is a projective A-module.

When these equivalent conditions hold, we say that $\mathcal{P}$ is a *locally projective* $\mathcal{O}_S$-module.

Corollary 11.10.— *Let* S *be a quasi-compact and quasi-separated scheme, let* G *be an* S*-group, and let* ρ *be a linear action of* G *on a quasi-coherent* $\mathcal{O}_S$*-module* $\mathcal{E}$*. Suppose that:*

(i) G *satisfies one of the following conditions:*

(a) G *is affine and flat over* S*;*

(b) G *is flat, quasi-compact, and quasi-separated over* S*, and* $\mathcal{E}$ *is flat;*

(ii) $\mathcal{A}(G)$ *is a locally projective* $\mathcal{O}_S$*-module.*

Then $\mathcal{E}$ *is the direct limit of an increasing filtered family of finitely generated quasi-coherent* $\mathcal{O}_S$*-submodules of* $\mathcal{E}$ *that are stable under* G*.*

By hypothesis (i) and 11.6.1, $\mathcal{E}$ is endowed with an $\mathcal{A}(G)$-comodule structure. On the other hand, since S is quasi-compact and quasi-separated, $\mathcal{E}$ is the direct limit of its finitely generated quasi-coherent submodules (EGA I, 9.4.9 and EGA IV_1, 1.7.7). Consequently, the corollary follows from Proposition 11.9, applied to the coalgebra $\mathcal{A}(G)$.

We also have the following proposition.

Proposition 11.10.bis.—[119] *Let S be a Noetherian scheme, let G be an S-group that is flat, quasi-compact, and quasi-separated over S, let $\mathcal{E}$ be a quasi-coherent G-$\mathcal{O}_S$-module, and let $\mathcal{M}$ be a coherent $\mathcal{O}_S$-submodule of $\mathcal{E}$. Then $\mathcal{M}$ is contained in a coherent $\mathcal{O}_S$-submodule stable under G.*

Proof. Write f for the morphism $G \to S$, and τ for the adjunction morphism

$$\mathcal{E} \longrightarrow f_* f^*(\mathcal{E}).$$

By 11.6, the G-$\mathcal{O}_S$-module structure on $\mathcal{E}$ is given by an automorphism θ of the $\mathcal{O}_G$-module $f^*(\mathcal{E})$, such that the morphism

$$\delta = \theta \circ \tau$$

satisfies conditions (1) and (2) of 11.6. (The situation considered in [Th87] is more general: the author considers a G-scheme X and a G-equivariant $\mathcal{O}_X$-module $\mathcal{E}$ (cf. Exposé I, Section 6); here $X = S$, endowed with the trivial action of G.)

Since S is Noetherian, $\mathcal{E}$ is the filtered direct limit of its coherent submodules $\mathcal{F}_\alpha$ (cf. EGA I, 9.4.9). Then $f^*(\mathcal{E})$ is the filtered direct limit of the $f^*(\mathcal{F}_\alpha)$, which are submodules of $f^*(\mathcal{E})$ because f is flat. Moreover, since f is quasi-compact and quasi-separated, by 11.0 the $\mathcal{O}_S$-module $f_* f^*(\mathcal{E})$ is quasi-coherent and is the filtered direct limit of the quasi-coherent submodules $f_* f^*(\mathcal{F}_\alpha)$. Consequently, $\mathcal{E}$ is the filtered direct limit of the quasi-coherent $\mathcal{O}_S$-submodules $\mathcal{E}_\alpha$, where $\mathcal{E}_\alpha$ denotes the inverse image under δ of $f_* f^*(\mathcal{F}_\alpha)$; equivalently, it is the kernel of the composite

$$\overline{\delta}_\alpha : \quad \mathcal{E} \xrightarrow{\delta} f_* f^*(\mathcal{E}) \longrightarrow f_* f^*(\mathcal{E}/\mathcal{F}_\alpha).$$

Since $\mathcal{M}$ is coherent and S is Noetherian, every increasing sequence of submodules of $\mathcal{M}$ is stationary; hence $\mathcal{M}$ is contained in some $\mathcal{E}_\alpha$. We shall show that every $\mathcal{E}_\alpha$ is coherent and G-stable.

Let

$$u : f^*(\mathcal{E}) \longrightarrow \varepsilon_*(\mathcal{E})$$

be the morphism corresponding by adjunction to the identity morphism from $\mathcal{E}$ to $f_* \varepsilon_*(\mathcal{E}) = \mathcal{E}$. Then the composite

$$\mathcal{E} \xrightarrow{\tau} f_* f^*(\mathcal{E}) \xrightarrow{f_*(u)} f_* \varepsilon_*(\mathcal{E}) = \mathcal{E}$$

is the identity (cf. 11.6(2)). We have the commutative diagram

$$\begin{array}{ccccc}
\mathcal{E}_\alpha & \xrightarrow{\delta} & f_* f^*(\mathcal{F}_\alpha) & \xrightarrow{f_*(u)} & \mathcal{F}_\alpha \\
\downarrow{\scriptstyle i_\alpha} & & \downarrow & & \downarrow{\scriptstyle j_\alpha} \\
\mathcal{E} & \xrightarrow[\tau]{} & f_* f^*(\mathcal{E}) & \xrightarrow[f_*(u)]{} & \mathcal{E},
\end{array}$$

where i_α (respectively j_α) denotes the inclusion of $\mathcal{E}_\alpha$ (respectively $\mathcal{F}_\alpha$) in $\mathcal{E}$. It follows that i_α factors through j_α; thus $\mathcal{E}_\alpha$ is a submodule of $\mathcal{F}_\alpha$, and is therefore coherent.

Finally, let us show that $\mathcal{E}_\alpha$ is G-stable (cf. 11.7.bis). Let $h : G \to S$ denote a second copy of $f : G \to S$, and consider the following commutative diagram:

[119] Editor's note: this proposition has been added; it is a special case of [Th87], 1.4–1.5. The author there refers to an argument of Deligne (cf. [Kn71], III, Th. 1.1); one may also note its similarity with Serre's argument ([Se68], Prop. 2) recalled in 11.8.bis.

$$\begin{array}{ccc} G \times_S G & \xrightarrow{q} & G \\ {\scriptstyle p}\downarrow & \swarrow^{\varphi} & \downarrow{\scriptstyle h} \\ G & \xrightarrow[f]{} & S. \end{array}$$

The exact sequence

$$0 \longrightarrow \mathcal{E}_\alpha \longrightarrow \mathcal{E} \xrightarrow{\bar{\delta}_\alpha} h_* h^*(\mathcal{E}/\mathcal{F}_\alpha)$$

therefore gives, since f is flat, the exact sequence

$$0 \longrightarrow f_* f^*(\mathcal{E}_\alpha) \longrightarrow f_* f^*(\mathcal{E}) \xrightarrow{f_* f^*(\bar{\delta}_\alpha)} f_* f^* h_* h^*(\mathcal{E}/\mathcal{F}_\alpha). \qquad (\dagger)$$

Moreover, since h is quasi-compact and quasi-separated and f is flat, the canonical morphism

$$f^* h_* h^*(\mathcal{E}/\mathcal{F}_\alpha) \longrightarrow p_* q^* h^*(\mathcal{E}/\mathcal{F}_\alpha) = p_* \varphi^*(\mathcal{E}/\mathcal{F}_\alpha)$$

is an isomorphism. Thus the right-hand term in (†) is $\varphi_* \varphi^*(\mathcal{E}/\mathcal{F}_\alpha)$. Using the notation of 11.6 and writing π_α for the projection $\mathcal{E} \to \mathcal{E}/\mathcal{F}_\alpha$, we obtain the following commutative diagram, whose bottom row is exact:

$$\begin{array}{ccccc} & & \mathcal{E}_\alpha & \xrightarrow{\delta} & f_* f^*(\mathcal{E}) \\ & & {\scriptstyle \delta}\downarrow & & \downarrow{\scriptstyle f_*(f^*(\pi_\alpha) \boxtimes \delta_G)} \\ f_* f^*(\mathcal{E}_\alpha) & \longrightarrow & f_* f^*(\mathcal{E}) & \xrightarrow[f_* f^*(\bar{\delta}_\alpha)]{} & \varphi_* \varphi^*(\mathcal{E}/\mathcal{F}_\alpha) \end{array}$$

Since

$$f_*\big(f^*(\pi_\alpha) \boxtimes \delta_G\big) \circ \delta$$

vanishes on $\mathcal{E}_\alpha$, it follows that δ carries $\mathcal{E}_\alpha$ into $f_* f^*(\mathcal{E}_\alpha)$; in other words, $\mathcal{E}_\alpha$ is G-stable. This proves the proposition.

Remark 11.10.1.—[120] In 11.8, it is not enough to assume that C is a flat A-module, even when A is a principal ideal domain. Indeed, we have the following counterexamples, pointed out to us independently by O. Gabber and J.-P. Serre.

(a) Let $(A, \mathfrak{m})$ be a discrete valuation ring, let K be its field of fractions, and let G be the A-group obtained by "extension by zero" from the K-group $(\mathbb{Z}/2\mathbb{Z})_K$. The constant group $(\mathbb{Z}/2\mathbb{Z})_A$, and hence also its subgroup G, acts on the free A-module V with basis v_1, v_2 by interchanging v_1 and v_2. Then Av_1 is not a G-submodule of V, but it is the intersection of the G-submodules

$$Av_1 + \mathfrak{m}^n v_2, \qquad n \in \mathbb{N}^*.$$

Thus there is no smallest G-submodule of V containing v_1.

(b) Let A be an integral domain distinct from its field of fractions K, and let G be the flat affine A-group corresponding to the Hopf algebra

$$\mathcal{A}(G) = \{P \in K[T] \mid P(0) \in A\}.$$

Its comultiplication, counit, and antipode are respectively defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \qquad \varepsilon(T) = 0, \qquad \tau(T) = -T.$$

[120] Editor's note: the first part of the original Remark 11.10.1 has been incorporated into 11.8 and 11.9 (with "free" replaced by "projective"); the counterexamples below correct its second part.

(Thus $G \otimes_A K = \mathbb{G}_{a,K}$.)

Let $V = Av_1 \oplus Av_2$ be free over A, and let u be the endomorphism of V defined by

$$u(v_1) = v_2, \qquad u(v_2) = 0,$$

so that $u^2 = 0$. Then V has the $\mathcal{A}(G)$-comodule structure

$$\mu(m) = 1 \otimes m + T \otimes u(m).$$

The G-submodules of V containing v_1 are precisely the A-submodules $Av_1 \oplus Iv_2$, where I is a nonzero ideal of A. Their intersection is Av_1, which is not a G-submodule. Hence there is no smallest G-submodule of V containing v_1. Moreover,

$$C = A \oplus K \cdot T$$

is a subcoalgebra of $\mathcal{A}(G)$, flat over A, and the coaction

$$\mu : V \longrightarrow V \otimes \mathcal{A}(G)$$

factors through $V \otimes C$. Thus the "very simple" coalgebra C also gives a counterexample.

Finally, the preceding two examples are special cases of the following construction. Let A be an integral domain distinct from its field of fractions K, and let B be a Hopf A-algebra, free over A. Write

$$\varepsilon : B \longrightarrow A, \qquad I = \ker(\varepsilon)$$

for the augmentation and the augmentation ideal. Since $B = A \cdot 1 \oplus I$, one readily sees that

$$B' = \{b \in B \otimes_A K \mid \varepsilon(b) \in A\}$$

is a Hopf subalgebra of B_K. Suppose that V is a B-comodule, free with basis $(v_1, \ldots, v_n)$ as an A-module, and that $\mu_V(v_1) \neq v_1 \otimes 1$. Then Av_1 is not a subcomodule of V, but it is the intersection of the subcomodules

$$Av_1 + J \cdot V$$

as J ranges over the nonzero ideals of A. Thus there is no smallest subcomodule of V containing v_1.

Proposition 11.11.— *Let G be an algebraic group over a field k. The following conditions are equivalent:*

(i) *G is affine.*

(ii) *G is quasi-affine.*

(iii) *G acts faithfully on a quasi-affine k-scheme X.*

(iv) *G acts linearly and faithfully on a k-vector space* (not necessarily finite-dimensional).

(v) *G is isomorphic to a closed subgroup of* $\mathrm{GL}(n)_k$.

Proof. The implication (i) ⇒ (ii) is immediate, and (ii) ⇒ (iii) because G acts faithfully on itself by translations.

Suppose that G acts faithfully on the right on a quasi-affine k-scheme X.[121] Since X is quasi-affine, it is separated and quasi-compact.[122] Similarly, G is separated (VI$_A$, 0.3)

[121] Editor's note: the original has been expanded in what follows. We have also chosen a right action of G on X, so as to obtain a left linear action of G on $\mathcal{O}(X)$.

[122] Editor's note: recall that a k-scheme X is called quasi-affine if it is isomorphic to a quasi-compact open subscheme of an affine k-scheme (EGA II, 5.1.1).

and quasi-compact (since it is of finite type over k). Hence, by 11.1(c), there are canonical isomorphisms

$$\mathcal{O}(X \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G), \qquad \mathcal{O}(X \times G \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G).$$

It follows that the morphism

$$\mu : \mathcal{O}(X) \longrightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$$

induced by $X \times G \to X$ gives $\mathcal{O}(X)$ the structure of a right $\mathcal{O}(G)$-comodule; that is, G acts linearly on the left on the k-algebra $\mathcal{O}(X)$. Consequently, G also acts on the right on the affine envelope

$$X_{\mathrm{af}} = \operatorname{Spec} \mathcal{O}(X)$$

of X, and the canonical morphism $X \to X_{\mathrm{af}}$ is G-equivariant.

Since X is quasi-affine, the morphism $X \to X_{\mathrm{af}}$ is an open immersion (EGA II, 5.1.2). Hence, a fortiori, G acts faithfully on X_{af}, and therefore the left linear action of G on the k-algebra $\mathcal{O}(X) = \mathcal{O}(X_{\mathrm{af}})$ is faithful. This proves (iii) ⇒ (iv).

Now suppose that G acts faithfully on a k-vector space V. By 11.10, V is the direct limit of finite-dimensional vector subspaces V_i stable under G. Let K_i be the kernel of the induced action of G on V_i; that is, the kernel of $G \to \mathrm{Aut}(V_i)$. Then K_i is a closed subscheme of G, and the faithfulness of the action of G means that the intersection of the K_i is the identity subgroup of G. Since G is Noetherian, one of the K_i is already the identity subgroup. Thus $G \to \mathrm{Aut}(V_i)$ is a monomorphism, and hence, by 1.4.2, a closed immersion. This proves (iv) ⇒ (v). Finally, (v) ⇒ (i) is immediate, and the proposition follows.

Remark 11.11.1.— Proposition 11.11 admits the following generalization. Let S be a regular, locally Noetherian scheme of dimension at most 1, and let G be a group scheme that is flat, quasi-compact, and quasi-separated over S. (In this case, $\mathcal{A}(G)$ is a torsion-free $\mathcal{O}_S$-module, and hence is flat.)

(a) The following conditions are equivalent:[123]

  (i) G is affine over S.

  (ii) G is quasi-affine over S.

  (iii) G acts faithfully on a quasi-affine flat S-scheme.

  (iv) G acts linearly and faithfully on a flat quasi-coherent $\mathcal{O}_S$-module.

(b) If, moreover, G is of finite type over S and S is Noetherian, these conditions imply that G is isomorphic to a closed subgroup of $\mathrm{Aut}(\mathcal{E})$, where $\mathcal{E}$ is a finite-rank locally free $\mathcal{O}_S$-module.[124]

Lemma 11.12.— *Let k be a field and let G be an affine k-group. Put $A = \mathcal{A}(G)$. Given $x \in A$, there is a finitely generated k-subalgebra B of A such that*

$$x \in B, \qquad \Delta(B) \subset B \otimes_k B, \qquad u(B) \subset B,$$

where u denotes the involution of A corresponding to the inversion morphism of G.[125]

[123] Editor's note: the equivalence of these conditions is proved in the added Section 12.

[124] Editor's note: this is proved, with various generalizations, in the added Section 13.

[125] Editor's note: on the one hand, this is generalized in Section 13 to the case where G is affine and flat over a regular base of dimension at most 2. On the other hand, the original has been expanded and corrected in what follows.

We may suppose that $x \neq 0$. Then $\Delta(x) \neq 0$, since

$$(\varepsilon \otimes \mathrm{id})\Delta(x) = x,$$

where ε denotes the augmentation (counit) of A. Write

$$\Delta(x) = \sum_{j=1}^{n} e_j \otimes a_j, \qquad n \text{ minimal.} \tag{1}$$

The e_j (respectively the a_j) are then linearly independent. Complete $(e_1, \ldots, e_n)$ to a basis $(e_i)_{i\in I}$ of A, and, for

$$j \in J = \{1, \ldots, n\},$$

write

$$\Delta(e_j) = \sum_{i\in I} e_i \otimes b_{ij}.$$

Applying $\Delta \otimes \mathrm{id}$ and $\mathrm{id} \otimes \Delta$ to (1), the axiom (CO 1) of 11.8.0 (see also (HA 1) in I, 4.2) gives

$$\sum_{j\in J} e_j \otimes \Delta(a_j) = \sum_{\ell\in J} \Delta(e_\ell) \otimes a_\ell = \sum_{i\in I} e_i \otimes \left(\sum_{\ell\in J} b_{i\ell} \otimes a_\ell \right).$$

Since the e_i are linearly independent, it follows that

$$\Delta(a_j) = \sum_{\ell\in J} b_{j\ell} \otimes a_\ell \qquad (j \in J). \tag{2}$$

Let B be the finitely generated k-subalgebra of A generated by the b_{ij} and the $u(b_{ij})$, for $i, j \in J$. It is clear that $u(B) = B$.

Applying $\Delta \otimes \mathrm{id}$ and $\mathrm{id} \otimes \Delta$ to (2), we obtain once again from (CO 1) the equalities

$$\sum_{\ell\in J} \Delta(b_{j\ell}) \otimes a_\ell = \sum_{i\in J} b_{ji} \otimes \Delta(a_i) = \sum_{i,\ell\in J} b_{ji} \otimes b_{i\ell} \otimes a_\ell.$$

Since the a_ℓ are linearly independent, this yields

$$\Delta(b_{j\ell}) = \sum_{i\in J} b_{ji} \otimes b_{i\ell} \qquad (j, \ell \in J). \tag{3}$$

Moreover,

$$\Delta \circ u = (u \otimes u) \circ v \circ \Delta, \qquad v(a \otimes b) = b \otimes a,$$

and hence

$$\Delta(u(b_{j\ell})) = \sum_{i\in J} u(b_{i\ell}) \otimes u(b_{ji}) \qquad (j, \ell \in J). \tag{4}$$

Since Δ is an algebra homomorphism, (3) and (4) imply

$$\Delta(B) \subset B \otimes_k B.$$

Finally, axiom (CO 2) of 11.8.0 (see also (HA 2) in I, 4.2), applied to (2) and (1), gives

$$a_j = \sum_{i\in J} b_{ji}\varepsilon(a_i), \qquad x = \sum_{j\in J} \varepsilon(e_j)a_j.$$

Thus $x \in B$, proving the lemma.

Proposition 11.13.— *Let k be a field and let G be an affine k-group with algebra A. Then G is the projective limit of an increasing filtered system of affine k-groups of finite type whose transition morphisms are faithfully flat.*

If B and B' are two finitely generated subalgebras of A stable under Δ and u, the subalgebra generated by B and B' has the same properties. Hence, by Lemma 11.12, A is the direct limit of an increasing filtered family $(B_i)_{i\in I}$ of finitely generated subalgebras stable under Δ and u. Each B_i, equipped with the restriction of u and with the morphism

$$B_i \longrightarrow B_i \otimes_k B_i$$

induced by Δ, is a Hopf algebra. Thus, by I, 4.2, it is the algebra of an affine k-group G_i of finite type over k. Finally, since

$$A = \varinjlim_i B_i,$$

we have

$$G = \varprojlim_i G_i$$

(cf. EGA IV_3, 8.2.3). The transition morphisms are faithfully flat by the following lemma.

Lemma 11.14.— *Let k be a field, let $u : G \to H$ be a morphism between affine k-groups of finite type, and let*

$$u^\natural : B \longrightarrow A$$

be the corresponding morphism of k-algebras.[126] *Then u is faithfully flat if and only if $u^\natural$ is injective.*

The condition is plainly necessary (cf. EGA 0_{I}, 6.6.1). Conversely, suppose that $u^\natural$ is injective, and put

$$N = \mathrm{Ker}(u).$$

By VI_A, 3.3.2 and VI_A, 5.4.1, G/N is a k-group of finite type and u factors as

$$G \xrightarrow{p} G/N \xrightarrow{v} H,$$

where p is faithfully flat and v is a closed immersion. Since H is affine, G/N is affine, and

$$v^\natural : B \longrightarrow \mathcal{O}(G/N)$$

is surjective (cf. EGA I, 4.2.3). But

$$u^\natural = p^\natural \circ v^\natural$$

is injective, so $v^\natural$ is also injective. Hence $v^\natural$, and therefore v, is an isomorphism. Since p is faithfully flat, so is u.

Definition 11.15.—[127] *Let k be a field, let G be a quasi-compact k-group, and let V be a k-vector space endowed with a k-linear action of G, hence, by 11.6.1, with an $\mathcal{A}(G)$-comodule structure*

$$\delta : V \longrightarrow V \otimes \mathcal{A}(G).$$

Let $v \in V$ be nonzero. The following conditions are equivalent:

[126] Editor's note: we write $u^\natural$ (instead of u°) for the morphism $B \to A$ corresponding to $u : \mathrm{Spec}(A) \to \mathrm{Spec}(B)$.

[127] Editor's note: we have added the hypothesis that G be quasi-compact and have spelled out the equivalence of conditions (i) and (ii).

(i) *There is a $\lambda \in \mathcal{A}(G)$, necessarily unique, such that*

$$\delta(v) = v \otimes \lambda.$$

(ii) *For every k-algebra R and every $g \in G(R)$, one has $g \cdot v \in Rv$; that is, there is a necessarily unique $f \in R$ such that, in $V \otimes R$,*

$$g \cdot v = v \otimes f.$$

It is clear that (i) implies (ii). Conversely, if (ii) holds, apply it with

$$R = \mathcal{A}(G), \qquad g = \mathrm{id}_{\mathcal{A}(G)}\,.$$

There is then a unique $\lambda \in \mathcal{A}(G)$ such that $\delta(v) = v \otimes \lambda$.

When v satisfies these conditions, we call v a *semi-invariant vector under G*, and λ the *weight* of v. We also say that v is a *semi-invariant of weight λ*.

Let Δ denote the comultiplication of $\mathcal{A}(G)$. The equality

$$v \otimes \lambda \otimes \lambda = (\delta \otimes \mathrm{id})(\delta(v)) = (\mathrm{id} \otimes \Delta)(\delta(v)) = v \otimes \Delta(\lambda)$$

implies

$$\Delta(\lambda) = \lambda \otimes \lambda.$$

Consequently, λ defines a morphism of Hopf algebras

$$\mathcal{A}(\mathbb{G}_{m,k}) = k[T, T^{-1}] \longrightarrow \mathcal{A}(G), \qquad T \longmapsto \lambda,$$

and hence a morphism of k-groups

$$\lambda : G \longrightarrow \mathbb{G}_{m,k}.$$

Thus λ is a *character* of G, called the *character associated with the semi-invariant vector v*.

Lemma 11.16.0.—[128] *Let k be a field, let H be an affine k-group, let V be an H-module of dimension n, and let U be a d-dimensional vector subspace of V. Consider the line*

$$D = \bigwedge^d U \subset \bigwedge^d V.$$

Then U is stable under H if and only if D is stable under H.

Necessity is clear. To prove sufficiency, we may assume $d < n$. Let $(e_1, \ldots, e_d)$ be a basis of U, and complete it to a basis $(e_1, \ldots, e_n)$ of V. For every k-algebra R, the module

$$V_R = V \otimes R$$

is free over R, and

$$U_R = \{\, v \in V_R \mid v \wedge (e_1 \wedge \cdots \wedge e_d) = 0 \,\}.$$

Indeed, for $i > d$, the elements

$$e_i \wedge e_1 \wedge \cdots \wedge e_d$$

are linearly independent in $\bigwedge_R^{d+1} V_R$. Since $H(R)$ acts on $\bigwedge_R^\bullet V_R$ by

$$h(x_1 \wedge \cdots \wedge x_s) = h(x_1) \wedge \cdots \wedge h(x_s),$$

[128] Editor's note: this lemma, taken from [DG70], Section II.2, 3.5, has been added.

it follows that, if $H(R)$ stabilizes

$$D_R = R(e_1 \wedge \cdots \wedge e_d),$$

then it also stabilizes U_R.

Theorem 11.16 (Chevalley).— *Let k be a field, let G be an affine algebraic k-group, and let H be a closed group subscheme of G.*[129] *Then there are a finite-dimensional G-module V and a line D in V such that*

$$H = \mathrm{Norm}_G(D);$$

that is, for every k-algebra R,

$$H(R) = \{\, g \in G(R) \mid g(D_R) = D_R \,\}.$$

Equivalently, there are finitely many elements $a_1, \ldots, a_n \in \mathcal{A}(G)$, all semi-invariant of the same weight λ for the "right" action of H; that is,

$$a_i(gh) = \lambda(h) a_i(g)$$

for every k-algebra R, $g \in G(R)$, and $h \in H(R)$, such that H is the largest closed group subscheme of G under which the a_i are semi-invariant.

Proof. Let Δ and ε denote, respectively, the comultiplication and the augmentation of

$$A = \mathcal{A}(G).$$

There is an ideal I of A, contained in $\operatorname{Ker}\varepsilon$, such that

$$H = \mathrm{Spec}(A/I), \qquad \Delta(I) \subset I \otimes A + A \otimes I.$$

Put $B = A/I$, and let $\pi : A \to B$ be the projection. Consider the left action of H on A given by

$$(h\phi)(g) = \phi(gh).$$

The corresponding B-comodule structure is

$$\overline{\Delta}: \quad A \xrightarrow{\Delta} A \otimes A \xrightarrow{\mathrm{id}_A \otimes \pi} A \otimes B.$$

Then I is an H-submodule of A, since

$$\overline{\Delta}(I) \subset I \otimes B.$$

The k-algebra A is finitely generated, and hence Noetherian, so I has a finite system of generators $(x_1, \ldots, x_r)$. By 11.8, the x_i are contained in a finite-dimensional G-submodule V of A. Then

$$W = V \cap I$$

is a finite-dimensional H-module; write

$$d = \dim_k W.$$

Since V contains every x_i, W generates the ideal I.

[129] Editor's note: first, recall that every group subscheme of G is closed (1.4.2). Second, we have stated the result in its usual form, namely that "H is the stabilizer of a line in a representation of G," while retaining the original formulation in terms of a sequence $a_1, \ldots, a_n$ of semi-invariants in $\mathcal{A}(G)$.

Put

$$E = \bigwedge^d V,$$

choose a basis $(w_1, \ldots, w_d)$ of W, and choose a basis $(e_0, \ldots, e_n)$ of E containing

$$e_0 = w_1 \wedge \cdots \wedge w_d.$$

The action of G on V canonically induces an action of G on E, hence an A-comodule structure

$$\rho : E \longrightarrow E \otimes A.$$

For $j = 0, \ldots, n$, write

$$\rho(e_j) = \sum_{i=0}^{n} e_i \otimes b_{ij}.$$

As in the proof of 11.12, one has

$$\Delta(b_{ij}) = \sum_{\ell=0}^{n} b_{i\ell} \otimes b_{\ell j}. \qquad (*)$$

Put $a_i = b_{i0}$. Equivalently, if $(e_0^*, \ldots, e_n^*)$ is the dual basis, then the a_i are the matrix coefficients

$$g \longmapsto e_i^*(ge_0).$$

The action of H on E corresponds to

$$\overline{\rho} : \quad E \xrightarrow{\rho} E \otimes A \xrightarrow{\mathrm{id}_E \otimes \pi} E \otimes B.$$

Since W is stable under H, the vector e_0 is semi-invariant under H. Hence

$$\overline{\rho}(e_0) = e_0 \otimes \pi(a_0), \qquad a_i = b_{i0} \in I \quad (1 \leq i \leq n).$$

Substitution in $(*)$ gives

$$\overline{\Delta}(a_i) = a_i \otimes \pi(a_0) \qquad (0 \leq i \leq n).$$

Thus the a_i are semi-invariant under H, all of weight $\pi(a_0)$. Moreover, after replacing E, if necessary, by the G-submodule generated by e_0 (cf. 11.8), we may suppose that the a_i are linearly independent.

Conversely, let

$$H' = \operatorname{Spec} B', \qquad B' = A/I',$$

be a closed group subscheme of G under which every a_i is semi-invariant, of weight $\lambda_i \in B'$. This includes, in particular, the case in which e_0 is semi-invariant under H' of weight λ'. We shall prove that $H' \subset H$. Let $\pi' : A \to B'$ be the projection. The hypothesis gives

$$a_i \otimes \lambda_i = (\mathrm{id}_A \otimes \pi')\Delta(b_{i0}) = \sum_{\ell=0}^{n} b_{i\ell} \otimes \pi'(a_\ell).$$

It follows that

$$\lambda_i = \pi'(a_0), \qquad a_\ell \in I' \quad (1 \leq \ell \leq n),$$

and hence e_0 is semi-invariant under H'. By Lemma 11.16.0, W is stable under H'. Since W generates I, the ideal I is also stable under H', and therefore

$$\Delta(I) \subset I \otimes A + A \otimes I'.$$

Finally,

$$I \subset \operatorname{Ker} \varepsilon, \qquad (\varepsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A,$$

so $I \subset I'$, whence $H' \subset H$. This completes the proof.

Lemma 11.17.0.—[130] *Let k be an algebraically closed field, let G be a reduced affine algebraic k-group, let V be a finite-dimensional G-module over k, and let $v \in V$. Let E be the vector subspace of V spanned by the vectors gv, for $g \in G(k)$. Then E is the smallest G-submodule of V containing v. Consequently, the morphism*

$$G \times E \longrightarrow V, \qquad (g, x) \longmapsto gx,$$

factors through E.

Proof. By 11.8, there is a smallest G-submodule U of V containing v. Indeed, if

$$\mu : V \longrightarrow V \otimes \mathcal{A}(G)$$

denotes the comodule structure and

$$\mu(v) = \sum_{i=1}^{n} v_i \otimes a_i$$

with the a_i linearly independent, then

$$U = \operatorname{Span}_k(v_1, \ldots, v_n).$$

It is clear that U contains E, and that the morphism $G \times U \to V$ factors through U.

Conversely, the inverse image of E under the morphism

$$\mu_v : G \longrightarrow V, \qquad g \longmapsto gv,$$

is a closed subset of G containing all the rational points. Those points are dense in G, since G is of finite type over k (cf. EGA IV_3, 10.4.8). Hence

$$\mu_v^{-1}(E) = G.$$

Since G is reduced, μ_v therefore factors through E, whence $E = U$.

Theorem 11.17 (Chevalley).— *Let k be a field, let G be an affine k-group* (not necessarily of finite type)*, and let N be a closed group subscheme of G, invariant in G. Then the* (fpqc) *sheaf quotient G/N is representable by an affine k-group.*[131]

Proof. Suppose first that G is of finite type. By VI_A, 3.2 and 5.2, the (fpqc) sheaf quotient G/N is representable by a k-group Q; it remains to prove that Q is affine. We proceed in several steps, first supposing that k is algebraically closed.[132]

(a) Suppose, in addition, that G is reduced and connected, and that N is reduced. By 11.16, there are a finite-dimensional G-module V over k and a line $D = ke_0$ such that

$$\operatorname{Norm}_G(D) = N.$$

[130] Editor's note: we have added this lemma, drawn from the proof of Theorem 5.6 of [DG70], Section III.3. By an abuse of notation, the same letter E denotes a k-vector space and the k-module scheme $W(E) = \operatorname{Spec} S(E^*)$.

[131] Editor's note: for another proof of this theorem that does not use the results of Exposé VI_A, see [Ta72], Theorem 5.2 (see also Remark 11.18.5).

[132] Editor's note: in what follows, we have expanded the original and corrected the erroneous assertion $(G/N)_{\mathrm{red}} = G_{\mathrm{red}}/N_{\mathrm{red}}$, drawing on [DG70], Section III.3, 5.6.

In particular, N acts on D through a character

$$\chi : N \longrightarrow \mathbb{G}_{m,k}.$$

Fix $h \in N(k)$. For every $g \in G(k)$, one has

$$hge_0 = g(g^{-1}hg)e_0 = \chi(g^{-1}hg)\, ge_0.$$

Thus $\chi(g^{-1}hg)$ is an eigenvalue of h. The continuous map

$$\varphi : G(k) \longrightarrow k, \qquad g \longmapsto \chi(g^{-1}hg),$$

therefore takes only finitely many values. Since $G(k)$ is irreducible (it is dense in G), it follows that

$$\varphi(g) = \varphi(e) = \chi(h) \qquad \text{for every } g \in G(k).$$

Consequently,

$$\chi(g^{-1}hg) = \chi(h) \qquad \text{for all } g \in G(k),\ h \in N(k).$$

Let E be the vector subspace of V spanned by the vectors ge_0, for $g \in G(k)$. By Lemma 11.17.0, this is the G-submodule of V generated by e_0.

By the preceding identity, the two morphisms

$$N \times E \longrightarrow E, \qquad (h, x) \longmapsto hx \quad \text{and} \quad (h, x) \longmapsto \chi(h)x,$$

agree on

$$(N \times E)(k) = N(k) \times E(k),$$

which is dense in $N \times E$. Since $N \times E$ is reduced (and E is separated), the two morphisms are equal. Thus N acts on E by scalar multiplications.

It follows that N is contained in the kernel K of the morphism

$$\rho : G \longrightarrow \mathrm{GL}\big(\mathrm{End}_k(E)\big)$$

defined, for every k-algebra R, by

$$\rho(g)(u) = gug^{-1} \qquad \big(g \in G(R),\ u \in \mathrm{End}_R(E \otimes_k R)\big).$$

Conversely, if $g \in K(R)$, then

$$g(Re_0) = Re_0,$$

and hence $g \in N(R)$. Thus $N = K$. By VI_A, 5.4.1, the morphism

$$G/N \longrightarrow \mathrm{GL}\big(\mathrm{End}_k(E)\big)$$

is a closed immersion. Therefore G/N is affine.

(b) Suppose now that G and N are reduced, without assuming that G is connected. Put

$$N' = N \cap G^0.$$

Then G^0/N' is affine by (a). Moreover, NG^0 is an invariant subgroup of G, and G/NG^0, being a quotient of the finite constant group G/G^0 (cf. VI_A, 5.5.1), is itself a finite constant group. The scheme G/N is therefore the disjoint union of the fibers of

$$G/N \longrightarrow G/NG^0.$$

By VI$_A$, 5.3.3, all these fibers are isomorphic to NG^0/N, hence to G^0/N'. Thus G/N is affine.

(c) Suppose that G is reduced, with N arbitrary. The morphism

$$G \times N \longrightarrow N, \qquad (g,h) \longmapsto ghg^{-1},$$

induces a morphism

$$(G \times N)_{\mathrm{red}} \longrightarrow N_{\mathrm{red}}.$$

Since G is reduced and k is algebraically closed,

$$(G \times N)_{\mathrm{red}} = G \times N_{\mathrm{red}},$$

so N_{red} is an invariant subgroup of G. (This fails when G is not reduced; see VI$_A$, 0.2.)

Hence, by (b),

$$G' = G/N_{\mathrm{red}}$$

is affine. By VI$_A$, 5.6.1,

$$N' = N/N_{\mathrm{red}}$$

is a finite k-group. Hence, by Theorem 4.1 of Exposé V,

$$G/N = G'/N'$$

is affine.

(d) For arbitrary G and N, base change of the equivalence relation

$$G \times N \rightrightarrows G$$

along $G_{\mathrm{red}} \to G$ gives

$$G_{\mathrm{red}} \times N' \rightrightarrows G_{\mathrm{red}}, \qquad N' = N \cap G_{\mathrm{red}}.$$

Since the underlying spaces are the same (and the quotient is the quotient ringed space), the morphism

$$G_{\mathrm{red}}/N' \longrightarrow G/N$$

is a homeomorphism. The quotient G_{red}/N' is reduced, because

$$p : G_{\mathrm{red}} \longrightarrow G_{\mathrm{red}}/N'$$

is faithfully flat. It follows that

$$(G/N)_{\mathrm{red}} \simeq G_{\mathrm{red}}/N'.$$

The scheme on the right is affine by (c). Since G/N is of finite type over k (cf. VI$_A$, 3.3.2), EGA I, 5.1.10 implies that G/N is affine.

Finally, let k be arbitrary and let $\overline{k}$ be an algebraic closure. By 9.2(v), one has

$$(G \otimes_k \overline{k})/(N \otimes_k \overline{k}) \simeq (G/N) \otimes_k \overline{k}.$$

The first scheme is affine, hence so is the second. Therefore G/N is affine by (fpqc) descent (cf. EGA IV$_2$, 2.7.1). This proves Theorem 11.17 when G is of finite type. To pass to the general case, we need the following lemma.[133]

[133] Editor's note: we have added this lemma, taken from [DG70], Section III.3, 7.1.

Lemma 11.17.1.— *Let $(C_i \to A_i)_{i\in I}$ be a filtered direct system of ring homomorphisms, all faithfully flat. Then*

$$A = \varinjlim_i A_i$$

is faithfully flat over

$$C = \varinjlim_i C_i.$$

Proof. By [BAC], Section I.3, Proposition 9, a ring homomorphism $B \to B'$ is faithfully flat if and only if it is injective and B'/B is a flat B-module. Since each $C_i \to A_i$ is faithfully flat, there are exact sequences

$$0 \longrightarrow C_i \longrightarrow A_i \longrightarrow A_i/C_i \longrightarrow 0,$$

and A_i/C_i is a flat C_i-module. Thus

$$(A_i/C_i) \otimes_{C_i} C$$

is a flat C-module. Since direct limits are exact and commute with tensor products, we obtain an exact sequence

$$0 \longrightarrow C \longrightarrow A \longrightarrow A/C \longrightarrow 0$$

and an isomorphism

$$\varinjlim_i ((A_i/C_i) \otimes_{C_i} C) = (A/C) \otimes_C C = A/C.$$

It follows that A/C is a flat C-module. Hence A is faithfully flat over C.

We now return to the proof of Theorem 11.17 in the general case, where G is not assumed to be of finite type. Put

$$\mathcal{A}(G) = A, \qquad \mathcal{A}(N) = B = A/J.$$

By 11.13, A is the direct limit of an increasing filtered family $(A_i)_{i\in I}$ of finitely generated sub-Hopf algebras. Thus

$$G = \varprojlim_i G_i, \qquad G_i = \mathrm{Spec}(A_i),$$

where the G_i are affine algebraic k-groups. Let Δ and τ denote, respectively, the comultiplication and antipode of A, and put

$$\Delta^2 = (\Delta \otimes \mathrm{id}_A) \circ \Delta.$$

For each i,

$$B_i = A_i/(J \cap A_i)$$

is a quotient Hopf algebra of A_i, so

$$N_i = \mathrm{Spec}(B_i)$$

is a closed subgroup of G_i. Since N is invariant in G, the morphism

$$G \times N \longrightarrow G, \qquad (g, n) \longmapsto gng^{-1},$$

factors through N. Equivalently, the pair (A, J) satisfies

$$(m_{13} \circ (\Delta \otimes \tau) \circ \Delta)(J) \subset A \otimes_k J,$$

where

$$m_{13}(a_1 \otimes a_2 \otimes a_3) = a_1 a_3 \otimes a_2.$$

It follows that $(A_i, A_i \cap J)$ satisfies the analogous property, so N_i is invariant in G_i. Moreover,

$$\varinjlim_i B_i = B, \qquad \varprojlim_i N_i = N.$$

By the finite-type case already proved, each (fpqc) sheaf quotient G_i/N_i is represented by an affine k-group

$$Q_i = \mathrm{Spec}(C_i).$$

Put

$$C = \varinjlim_i C_i.$$

The Q_i form a filtered projective system of affine k-groups, and its projective limit

$$Q = \varprojlim_i Q_i$$

is the k-group $\mathrm{Spec}(C)$ (cf. EGA IV$_3$, 8.2.3). We have an exact sequence of k-groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q.$$

We claim that Q represents the (fpqc) sheaf quotient of G by N. It is enough to verify that

$$G \longrightarrow Q$$

is a covering for the (fpqc) topology (cf. IV, 3.3.2.1 and 5.1.7.1). Every morphism

$$G_i \longrightarrow Q_i$$

is faithfully flat (cf. 9.2(xi)); equivalently, A_i is faithfully flat over C_i. Since

$$A = \varinjlim_i A_i, \qquad C = \varinjlim_i C_i,$$

Lemma 11.17.1 shows that A is faithfully flat over C. Thus $G \to Q$ is faithfully flat. It is also affine, hence quasi-compact, and is therefore an (fpqc) covering. This completes the proof of Theorem 11.17.

11.18. Complements.—[134] Theorem 11.17 and its proof also give the following results, taken from [DG70], Section III.3.7. Let k be a field. We begin with a lemma (cf. [An73], 2.3.3.2) that will be used later (see 12.10).

Lemma 11.18.1.— *Let $u : G \to G'$ be a morphism of k-groups and put $N = \mathrm{Ker}(u)$. Suppose that G' is affine and that G is of finite type.*

(i) *The morphism $\tau : G/N \to G'$ is a closed immersion. In particular, G/N is affine.*

(ii) *If, moreover, the morphism $u^\natural : \mathcal{O}(G') \to \mathcal{O}(G)$ is injective, then τ is an isomorphism. Thus G' is of finite type and u is faithfully flat.*

[134] Editor's note: we have added this subsection.

Indeed, by 11.13, G' is the projective limit of a filtered system of affine algebraic k-groups G'_i. Let

$$u_i : G \longrightarrow G' \longrightarrow G'_i$$

be the composite, and let N_i be its kernel. The N_i form a filtered decreasing system of closed subgroups of G, whose intersection is N. Since G is Noetherian, there is an index i such that

$$N = N_i.$$

The groups G and G'_i are of finite type. Therefore, by VI_A, 3.2 and 5.4.1, the quotient G/N is a k-group of finite type, and u_i is the composite of the faithfully flat projection

$$p : G \longrightarrow G/N$$

and a closed immersion

$$\tau_i : G/N \hookrightarrow G'_i.$$

Consider the commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{u} & G' \\ {\scriptstyle p}\downarrow & \nearrow{\scriptstyle \tau} & \downarrow{\scriptstyle q_i} \\ G/N & \xrightarrow[\tau_i]{} & G'_i. \end{array}$$

Since

$$q_i \circ \tau = \tau_i$$

is a closed immersion and q_i is separated (G' is separated; cf. VI_A, 0.3), τ is a closed immersion (cf. EGA I, 5.4.4). Hence G/N is affine and

$$\tau^\natural : \mathcal{O}(G') \longrightarrow \mathcal{O}(G/N)$$

is surjective, proving (i).

If $u^\natural : \mathcal{O}(G') \to \mathcal{O}(G)$ is also injective, then so is $\tau^\natural$. It follows that $\tau^\natural$ is an isomorphism, and hence so is τ, since G' and G/N are affine. This proves (ii).

Theorem 11.18.2.— *Let G and G' be affine k-groups, with algebras A and A', and let $u : G \to G'$ be a morphism of k-groups. Put $N = \mathrm{Ker}(u)$, and let $\varphi : A' \to A$ be the morphism induced by u.*

(i) *If φ is injective, then u is faithfully flat and identifies G' with G/N.*

(ii) *One has $G/N = \mathrm{Spec}(B)$, where $B = \varphi(A')$. Thus u is the composite of the faithfully flat morphism $G \to G/N$, corresponding to the inclusion $B \hookrightarrow A$, and the closed immersion $G/N \hookrightarrow G'$, corresponding to the surjection $A' \to B$. Moreover, N is defined in G by the ideal AB_+, where B_+ denotes the augmentation ideal of B.*

(iii) *In particular, if u is a monomorphism, it is a closed immersion.*

Proof. (i) Suppose that φ is injective and identify A' with a sub-Hopf algebra of A. By 11.13, A is the filtered union of finitely generated sub-Hopf algebras

$$A_i = \mathcal{O}(G_i).$$

Put

$$A'_i = A' \cap A_i, \qquad G'_i = \mathrm{Spec}(A'_i),$$

and let N_i be the kernel of the morphism

$$G_i \longrightarrow G'_i$$

induced by $A'_i \hookrightarrow A_i$. By Lemma 11.18.1,

$$G_i/N_i \simeq G'_i.$$

Thus, for each i, there is an exact sequence

$$1 \longrightarrow N_i \longrightarrow G_i \xrightarrow{p_i} G'_i \longrightarrow 1,$$

where p_i is faithfully flat. Since

$$G = \varprojlim_i G_i, \qquad G' = \varprojlim_i G'_i,$$

we obtain an exact sequence

$$1 \longrightarrow N \longrightarrow G \xrightarrow{p} G',$$

where $N = \varprojlim_i N_i$. By Lemma 11.17.1, p is faithfully flat (and affine). Therefore G' represents the (fpqc) sheaf quotient of G by N, proving (i).

In general, $B = \varphi(A')$ is a sub-Hopf algebra of A. Let

$$H = \mathrm{Spec}(B),$$

and let N' be the kernel of the morphism $G \to H$ induced by $B \hookrightarrow A$. By (i), H is identified with G/N', and u is the composite of the projection $G \to G/N'$ and the closed immersion

$$G/N' \hookrightarrow G'$$

induced by the surjection $A' \to B$. Hence $N' = N$. Moreover, by 9.2(ii), the square

$$\begin{array}{ccc} N & \longrightarrow & G \\ \downarrow & & \downarrow p \\ \mathrm{Spec}(k) & \xrightarrow[\varepsilon]{} & G' \end{array}$$

is cartesian. Here ε is the unit section of G', corresponding to the augmentation $B \to k$. It follows that N is defined in G by the ideal AB_+. This proves (ii), and (iii) follows.

Remark 11.18.3.— Let G be an affine k-group and N an invariant k-subgroup. Since

$$p : G \longrightarrow G/N$$

is faithfully flat and quasi-compact, $\mathcal{O}(G/N)$ is, by IV, 3.3.3.2, the subalgebra of $\mathcal{O}(G)$ formed by the functions ϕ that are right N-invariant; that is,

$$\phi(gh) = \phi(g)$$

for every k-scheme S, every $g \in G(S)$, and every $h \in N(S)$. Let J be the ideal of

$$A = \mathcal{O}(G)$$

defining N. The preceding condition is equivalent to

$$\Delta(\phi) - \phi \otimes 1 \in \mathcal{O}(G) \otimes J,$$

where Δ is the comultiplication of A.

The preceding theorem may therefore be reformulated in terms of Hopf algebras as follows.

Corollary 11.18.4.— *Let k be a field, let A be a commutative k-Hopf algebra, and put $G = \mathrm{Spec}(A)$.*

(i) *If B is a sub-Hopf algebra of A, then A is faithfully flat over B.*

(ii) *The map*

$$N \longmapsto \mathcal{O}(G/N)$$

is a bijection from the set of invariant subgroups of G to the set of sub-Hopf algebras of A. Its inverse is

$$B \longmapsto \mathrm{Spec}(A/AB_+).$$

Moreover, if J is the ideal of A defining N, then

$$\mathcal{O}(G/N) = \{\, x \in A \mid \Delta(x) - x \otimes 1 \in A \otimes J \,\}.$$

Remarks 11.18.5.—

(a) One consequence of the preceding theorem is that the category of commutative affine k-groups is abelian. For this, as well as for other results on affine k-groups, we refer to [DG70], § III.3, 7.4–7.8.

(b) Finally, let us point out that M. Takeuchi gave another proof of the results of 11.17–11.18.4; see [Ta72], § 5. He moreover strengthened 11.18.4(i) above by showing that A is even a projective B-module; see [Ta79], Theorem 5 (see also [MW94], Theorem 3.6).

12. Complements on G_{af} and the "anti-affine" groups

[135] We begin with the following lemma, which extends 11.18.1 to the case where G is not assumed to be of finite type.[136]

Lemma 12.1.— *Let k be a field and let $u : G \to H$ be a monomorphism of k-groups, with H affine. Suppose that u is quasi-compact. Then u is a closed immersion.*

By (fpqc) descent, we may suppose that k is algebraically closed. By VI_A, 6.4, the closed image I of u is a closed group subscheme of H, hence is still affine. Replacing H by I, we may therefore suppose that u is schematically dominant. Since k is algebraically closed, $H' = H_{\mathrm{red}}$ is a group subscheme of H. Put $G' = G \times_H H'$. The morphism

$$u' : G' \longrightarrow H'$$

deduced from u by base change is a quasi-compact monomorphism and is dominant (the underlying continuous map being the same for u and u'). Hence, by VI_A, 6.2, u' is faithfully

[135] Editor's note: this section has been added.

[136] Editor's note: this lemma was communicated to us by M. Raynaud; it will be used in the proof of Proposition 12.9.

flat. It is therefore a quasi-compact faithfully flat monomorphism, and thus an isomorphism (cf. IV, 1.14).

Consequently $u : G \to H$ is a homeomorphism, and hence is affine by 2.9.1. Thus G is affine, and u is a closed immersion by 11.18.2(iii).

Theorem 12.2.— *Let G be an algebraic k-group. Denote by $\rho : G \to G_{\mathrm{af}}$ the canonical morphism and by N its kernel.*

(i) *The canonical morphism $G/N \to G_{\mathrm{af}}$ is an isomorphism. Consequently G_{af} is an affine algebraic group and ρ is faithfully flat.*

(ii) *There is a canonical isomorphism $(G/N)_{\mathrm{af}} = G_{\mathrm{af}}$.*

(iii) *N is a characteristic subgroup of G.*

(iv) *$\mathcal{O}(N) = k$.*

(v) *N is smooth, connected, and commutative.*

Assertion (i) is a special case of 11.18.1, and (ii) follows from the universal properties of G_{af} and $(G/N)_{\mathrm{af}}$.

Let us prove (iii). For an arbitrary k-scheme $\pi : S \to \operatorname{Spec} k$, consider the cartesian square

$$\begin{array}{ccc} G_S & \longrightarrow & G \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ S & \xrightarrow{\pi} & \operatorname{Spec} k, \end{array}$$

Since p is quasi-compact and separated and π is flat, EGA III, 1.4.15 and EGA IV_1, 1.7.21 give

$$q_*(\mathcal{O}_{G_S}) = \pi^*(\mathcal{O}(G)) = \pi^*(\mathcal{O}(G_{\mathrm{af}})).$$

Hence EGA II, 1.5.2 gives

$$(G_S)_{\mathrm{af}} = (G_{\mathrm{af}})_S.$$

Thus N_S, being the kernel of the canonical morphism $G_S \to (G_S)_{\mathrm{af}}$, is invariant under every automorphism of G_S; in other words, N is a characteristic subgroup of G.

To prove (iv), put

$$N' = \operatorname{Ker}(N \longrightarrow N_{\mathrm{af}}).$$

By (ii), this is an invariant subgroup of G. Since N is algebraic (being a closed subgroup of G), (i) gives $N/N' \simeq N_{\mathrm{af}}$. Moreover, by VI_A, 3.2 and 5.3.2, there is an isomorphism of k-groups

$$(G/N')/N_{\mathrm{af}} \simeq (G/N')/(N/N') \simeq G/N.$$

Since N_{af} is affine, the projection $G/N' \to G/N$ is affine by 9.2(vii); and since $G/N = G_{\mathrm{af}}$ is affine, G/N' is affine as well. The universal property of G_{af} therefore makes the projection

$$p' : G \longrightarrow G/N'$$

factor through $G_{\mathrm{af}} = G/N$. Hence $N \subset N'$, and consequently $N = N'$. Thus N_{af} is the trivial group, and $\mathcal{O}(N) = k$.

Finally, (v) follows from the next lemma.

Lemma 12.3.— *Let k be a field and let N be an algebraic k-group such that $\mathcal{O}(N) = k$. Then N is smooth, connected, and commutative.*

Indeed, we may suppose that k is algebraically closed. Then

$$H = N^0_{\mathrm{red}}$$

is a k-subgroup of N, and the quotient k-scheme $X = N/H$ is finite (hence affine) over k, by VI_A, 5.5.1 and 5.6.1. Since $p : N \to X$ is faithfully flat,

$$\mathcal{O}(X) \subset \mathcal{O}(N) = k.$$

It follows that $N = N^0_{\mathrm{red}}$, and hence that N is smooth (VI_A, 1.3.1) and connected.

Let Z be the center of N. By 6.2.6, N/Z is affine, and the preceding argument gives $\mathcal{O}(N/Z) = k$. Hence $N = Z$. This proves the lemma and completes the proof of 12.2.

We also record, without proof, the following theorem. Recall that an *abelian variety* over a field k is a proper, smooth, connected k-group scheme.

Theorem 12.4 (Chevalley).— *Let k be a perfect field and let G be a smooth, connected algebraic k-group. Then there exists a smooth, connected affine k-subgroup L, invariant in G, such that G/L is an abelian variety. Moreover, L is unique and its formation commutes with extension of the base field.*

Remarks 12.5.—

(1) This theorem was announced in 1953 by C. Chevalley, who published his proof in 1960 ([Ch60]). Meanwhile, other proofs were obtained independently by I. Barsotti and M. Rosenlicht ([Ba55, Ro56]); see [Se99] for historical comments.

(2) A modern version (that is, in the language of schemes) of Chevalley's proof was given by B. Conrad ([Co02]). (In *loc. cit.*, "algebraic group" means a smooth connected k-group scheme.)

(3) A modern version of Rosenlicht's proof, on the other hand, was given by Ngô B.-C. in a course at Orsay in 2005–2006.

(4) If the hypothesis that k be perfect is omitted, there still exists a smallest invariant connected affine subgroup L, not necessarily smooth, such that G/L is an abelian variety ([BLR], § 9.2, Theorem 1).

(5) One may also omit the hypothesis that G be smooth over k. Indeed, by VII_A, 8.3, there is an integer $n \geq 1$ such that the quotient

$$G' = G/(\mathrm{Fr}^n\, G)$$

is smooth. Then G' contains a subgroup L' as in (4) above, and the inverse image of L' in G again has the same properties. Thus, for every connected algebraic group G over a field k, there is an exact sequence

$$1 \longrightarrow H \longrightarrow G \longrightarrow A \longrightarrow 1,$$

where H is an affine k-group and A is an abelian k-variety. Moreover, by [Per76], Corollary 4.2.9, such an exact sequence exists for every connected k-group G, not necessarily algebraic.

(6) Let k be an algebraically closed field and let G be the semidirect product of an elliptic curve E by the constant k-group $\{\pm 1\}_k$, for the action defined by $(-1)\cdot x = -x$. In this case, if L is a closed invariant subgroup of G such that G/L is connected, then $L = G$.

Remark 12.6.— Following [Br09], a k-group N will be called *anti-affine* if $\mathcal{O}(N) = k$. By 12.3 and 12.4, if k is perfect, every anti-affine algebraic k-group is an extension of an abelian variety by a smooth, connected, commutative affine algebraic k-group. For the precise structure of anti-affine algebraic groups over a perfect field, and various consequences, see the recent articles of M. Brion and C. & F. Sancho de Salas ([Br09, SS09]).

To conclude this section, we shall prove two results due to M. Raynaud, the first being Remark 11.11.1, and the second Proposition 2.1 of Exposé XVII, Appendix III. We shall need the following lemma,[137] which improves, for a complete discrete valuation ring R, the flatness criteria given in [BAC], § III.5.

Lemma 12.7.— *Let R be a discrete valuation ring, K its fraction field, and π a uniformizer. Let A be a flat R-algebra and let M be an A-module flat over R. Suppose that*

(i) *$M/\pi M$ is flat over $\overline{A} = A/\pi A$,*

(ii) *$M \otimes_R K$ is flat over $A \otimes_R K$.*

Then M is flat over A.

By the flatness criterion in the nilpotent case (cf. [BAC], III § 5.2, Theorem 1),

$$M/\pi^n M$$

is flat over $A/\pi^n A$ for every $n \in \mathbb{N}^*$. It then follows from [RG71], II, Lemma 1.4.2.1, that M is flat over A. For the reader's convenience, let us briefly indicate the proof. Put

$$s = \pi^n, \qquad P = M/sM.$$

Since P is flat over A/sA, and A/sA has projective dimension 1 over A, the spectral sequence of composite functors shows that P has Tor-dimension at most 1 over A. Since M is flat over R, and thus has no π-torsion, M_K/M is the direct limit of the A-modules

$$\pi^{-n}M/M \simeq M/\pi^n M.$$

Consequently M_K/M also has Tor-dimension at most 1. From the exact sequence

$$0 \longrightarrow M \longrightarrow M_K \longrightarrow M_K/M \longrightarrow 0$$

and the hypothesis that M_K is flat over A_K, hence over A, it follows that M is flat.

For completeness, let us also give the following simpler proof, pointed out by O. Gabber. Let I be a finitely generated ideal of A. We must show that

$$u : M \otimes_A I \longrightarrow M$$

is injective. By (ii), $u \otimes_R K$ is injective, so $\mathrm{Ker}(u)$ is an R-module of π-torsion. It therefore suffices to show that the π-torsion submodule of $M \otimes_A I$ is zero. This submodule is a quotient of $\mathrm{Tor}_1^A(M, I/\pi I)$, as is seen by tensoring with M the exact sequence

$$0 \longrightarrow I \xrightarrow{\pi} I \longrightarrow I/\pi I \longrightarrow 0.$$

[137] Editor's note: this is a version improved by O. Gabber of a statement communicated by M. Raynaud.

On the other hand, since M has no π-torsion (being flat over R), one has

$$\mathrm{Tor}_i^A(M, \overline{A}) = 0 \qquad (i \geq 1).$$

Consequently, if $P_\bullet$ is a projective resolution of the A-module M, then

$$P_\bullet \otimes_A \overline{A}$$

is a projective resolution of the $\overline{A}$-module $\overline{M} = M/\pi M$. Thus, for every $\overline{A}$-module Q,

$$\mathrm{Tor}_i^A(M, Q) = \mathrm{Tor}_i^{\overline{A}}(\overline{M}, Q),$$

which vanishes for $i \geq 1$, since $\overline{M}$ is flat over $\overline{A}$. Hence $\mathrm{Tor}_1^A(M, I/\pi I) = 0$, proving the lemma.

Remark 12.8.— Let S be a regular locally noetherian scheme of dimension 1, and let X be an S-scheme that is flat, quasi-separated, and quasi-compact over S. Then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$-module. Indeed, we may suppose that S is local. Let s be its closed point, let $i : X_s \hookrightarrow X$ be the inclusion, and let π be a uniformizer of $R = \mathcal{O}(S)$. Since X is flat over S, there is an exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{\pi} \mathcal{O}_X \longrightarrow i_*(\mathcal{O}_{X_s}) \longrightarrow 0.$$

Taking global sections shows that $\mathcal{A}(X)$ has no π-torsion as an R-module, and hence is flat.[138] Moreover, the morphism

$$\mathcal{A}((X_{\mathrm{af}})_s) = \mathcal{A}(X)/\pi\mathcal{A}(X) \longrightarrow \mathcal{A}(X_s),$$

induced by $X \to X_{\mathrm{af}}$, is injective.

We can now prove the following proposition (cf. Remark 11.11.1).

Proposition 12.9.— *Let S be a regular locally noetherian scheme of dimension at most 1, and let G be an S-group scheme that is flat, quasi-separated, and quasi-compact over S. Then the following conditions are equivalent:*

(i) *G is affine over S.*

(ii) *G is quasi-affine over S.*

(iii) *G acts faithfully on a quasi-affine and flat S-scheme X.*

(iv) *G acts linearly and faithfully on a quasi-coherent module $\mathcal{E}$ that is flat over S.*

(v) *The morphism $\rho : G \to G_{\mathrm{af}}$ is a monomorphism.*

The implication (i)⇒ (ii) is clear, as is (ii)⇒ (iii) (take $X = G$).

Suppose (iii) holds. Since $\mathcal{A}(G)$ and $\mathcal{A}(X)$ are flat $\mathcal{O}_S$-modules, the argument of 11.11 shows that G acts faithfully on the right on X_{af}, and hence acts linearly and faithfully on the left on the flat quasi-coherent $\mathcal{O}_S$-module $\mathcal{A}(X)$.

If (iv) holds, 11.6.1(ii) shows that the monomorphism

$$G \longrightarrow \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{E})$$

factors through G_{af}. Hence $G \to G_{\mathrm{af}}$ is a monomorphism.

[138] Editor's note: this is true more generally when S is regular, locally noetherian, and of dimension at most 2; see [Ray70a], VII 3.2.

Finally, suppose (v) holds, and let us prove that $\rho : G \to G_{\mathrm{af}}$ is an isomorphism. Replacing S by one of its connected components, we may suppose that S is irreducible, with generic point η. Since the formation of G_{af} commutes with flat base change,

$$(G_{\mathrm{af}})_\eta = (G_\eta)_{\mathrm{af}}.$$

Therefore $G_\eta \to (G_\eta)_{\mathrm{af}}$ is a monomorphism, hence a closed immersion by 12.1, and consequently an isomorphism because

$$\mathcal{O}((G_\eta)_{\mathrm{af}}) = \mathcal{O}(G_\eta).$$

If $S = \mathrm{Spec}(\kappa(\eta))$, the proof is complete; we may therefore suppose that $\dim S = 1$.

Let s be a closed point of S. We shall show that

$$G_s \longrightarrow (G_{\mathrm{af}})_s$$

is an isomorphism and that ρ is flat at every point of G_s. We may suppose that S is local, with closed point s. The base-change morphism

$$\rho_s : G_s \longrightarrow (G_{\mathrm{af}})_s$$

is a monomorphism, hence a closed immersion by 12.1. Thus the morphism

$$\mathcal{O}((G_{\mathrm{af}})_s) \longrightarrow \mathcal{O}(G_s)$$

induced by ρ_s is surjective. By the preceding remark it is also injective, and hence is an isomorphism. In particular, ρ is surjective.

It follows from Lemma 12.7 that $\rho : G \to G_{\mathrm{af}}$ is faithfully flat. Since G is quasi-compact over S and G_{af} is separated over S, ρ is also quasi-compact (cf. EGA I, 6.6.4). Therefore ρ is a quasi-compact faithfully flat monomorphism, and is thus an isomorphism. This proves the proposition.

Finally, we prove Proposition 2.1 of Exposé XVII, Appendix III. Taking [Per76], Corollary 4.2.5 into account, we have replaced "of finite type" in the hypotheses by "quasi-compact and quasi-separated". (If G is assumed to be of finite type, 12.1 may be used in place of *loc. cit.*)

Proposition 12.10.— *Let S be a regular locally noetherian scheme of dimension at most 1, and let G be a flat, quasi-compact, and quasi-separated S-group scheme.*

(i) *The canonical morphism $\rho : G \to G_{\mathrm{af}}$ is faithfully flat and quasi-compact. Consequently, G_{af} represents the* (fpqc) *sheaf quotient of G by $N = \mathrm{Ker}(\rho)$.*

(ii) *If moreover G is of finite type over S, then ρ is of finite presentation, G_{af} represents the* (fppf) *sheaf quotient of G by N, and G_{af} is of finite type over S.*

(iii) *Suppose moreover that G_η is affine for every maximal point η of S. Then N is an étale S-group and is the unit group if G is separated over S.*

Since G_{af} is affine, and therefore separated over S, the morphism ρ is quasi-compact (cf. EGA I, 6.6.4), and its kernel

$$N = \mathrm{Ker}(\rho)$$

is a closed subgroup of G. Replacing S by one of its connected components, we may suppose that S is irreducible, with generic point η.

To prove (i) and (ii), it is enough to show that ρ is faithfully flat. Indeed, Exposé IV, 5.1.7.1 then shows that G_{af} represents the (fpqc) sheaf quotient of G by N. If G is moreover of finite type over S, it is of finite presentation because S is locally noetherian; by 9.2(xiii), ρ is then of finite presentation, as is $G_{\mathrm{af}} \to S$, and ρ is therefore a covering for the (fppf) topology.

We may thus suppose that

$$S = \mathrm{Spec}(R),$$

where R is a discrete valuation ring if $\dim S = 1$, and is the field $\kappa(\eta)$ if $\dim S = 0$. Let s be the closed point of S. Since formation of G_{af} commutes with flat base change, the canonical morphism $G_\eta \to (G_\eta)_{\mathrm{af}}$ is identified with

$$\rho_\eta : G_\eta \longrightarrow (G_{\mathrm{af}})_\eta.$$

Since

$$\mathcal{O}((G_{\mathrm{af}})_\eta) = \mathcal{O}((G_\eta)_{\mathrm{af}}) = \mathcal{O}(G_\eta),$$

ρ_η is faithfully flat by [Per76], 4.2.5 (see also the addition VI$_A$, 6.6). If $\dim S = 1$, the same argument shows that ρ_s is faithfully flat, because the morphism

$$\mathcal{O}((G_{\mathrm{af}})_s) \longrightarrow \mathcal{O}(G_s)$$

is injective by Remark 12.8. Lemma 12.7 now shows that ρ is faithfully flat. This proves (i) and (ii). In particular, N is flat over S.

Suppose now that G_η is affine. Since ρ_η coincides with the canonical morphism

$$G_\eta \longrightarrow (G_\eta)_{\mathrm{af}},$$

its kernel N_η is the unit group. Let us prove that N is étale over S. Since N is flat over S, it remains to show that N_s is étale over $\kappa(s)$ for every point $s \neq \eta$ of S. There is nothing to prove when $S = \mathrm{Spec}(\kappa(\eta))$, so we may suppose that $S = \mathrm{Spec}(R)$, where R is a discrete valuation ring. Let s be the closed point of S, let K be the fraction field of R, and let π be a uniformizer.

Let $x \in N_s$, and let U_x be an affine open neighborhood of x in N. Since N is flat over S, the intersection $U_x \cap N_\eta$ is nonempty, and hence equals $\{\epsilon(\eta)\}$, where ϵ denotes the unit section. Thus

$$A = \mathcal{O}(U_x)$$

is a flat R-algebra such that $A_K = K$ and $\pi^{-1} \notin A$, since π belongs to the maximal ideal of $\mathcal{O}_{U_x,x}$. It follows that $A = R$, and hence the projection $U_x \to S$ is an isomorphism. Thus N is étale over S. If G is also separated over S, the inverse isomorphism $S \to U_x$ is the unit section, since the two coincide on the dense open subset $\{\eta\}$ of $S = \mathrm{Spec}(R)$. Hence N is the unit group. This proves the proposition.

In particular, we obtain the following corollary, two other proofs of which may be found in [An73], Proposition 2.3.1, and [PY06], Proposition 3.1.

Corollary 12.10.1.— *Let R be a discrete valuation ring, K its fraction field, and G a separated, flat R-group scheme of finite type over R. If G_K is affine, then G is affine.*

Remarks 12.10.2.—

(a) O. Gabber has pointed out to us examples in which G is a flat group of finite type over a discrete valuation ring, its generic fiber is an abelian variety, and the kernel N of $G \to G_{\mathrm{af}}$ is not smooth.

(b) M. Raynaud, on the other hand, gave an example, with S the affine plane of dimension 2 over a field k, of a smooth quasi-affine S-group scheme with affine, connected fibers that is not affine over S; see [Ray70a], § VII.3, p. 116.

13. Flat affine groups over a regular base of dimension at most 2

Let us begin by observing[139] that the familiar argument showing that every affine algebraic group over a field k is linear, as well as Lemma 11.12, extends to a group scheme G that is affine, flat, and of finite type over a Noetherian regular base scheme S of dimension at most 2. For S of dimension at most 1, this proves Remark 11.11.1(b). The extension to $\dim S = 2$ rests on the following lemma, communicated to us by O. Gabber.

Lemma 13.1.— *Let S be a normal Noetherian scheme, let $\mathcal{A}$ be a flat quasi-coherent $\mathcal{O}_S$-module, let $\mathcal{F}$ be a finite-type $\mathcal{O}_S$-submodule of $\mathcal{A}$, let $\mathcal{F}^{**}$ be its bidual, and let U be the flatness locus of $\mathcal{F}$, that is, the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$-module.*

(i) *U is open in S, and*
$$\mathcal{F}^{**} = j_* j^*(\mathcal{F}),$$
where $j : U \hookrightarrow S$ denotes the inclusion.

(ii) *For every quasi-coherent $\mathcal{O}_U$-module $\mathcal{E}$, the canonical morphism*
$$j_*(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow j_*(\mathcal{E} \otimes_{\mathcal{O}_U} j^*(\mathcal{A}))$$
is an isomorphism.

(iii) *In particular,*
$$\mathcal{V} = j_* j^*(\mathcal{F}) \subset \mathcal{A} = j_* j^*(\mathcal{A}),$$
and the canonical morphism
$$\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow j_* j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A})$$
is an isomorphism.

Proof. Replacing S by one of its connected components, we may suppose that S is integral. By EGA IV_2, 2.1.12, the flatness locus of $\mathcal{F}$, namely the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$-module, is an open subset U of S; write $j : U \hookrightarrow S$ for its inclusion. Since $\mathcal{A}_s$ is flat, hence torsion-free, the same is true of $\mathcal{F}_s$. Thus U contains every point of codimension at most 1. Consequently, by [BAC], VII, § 4.2, Corollary to Theorem 1,
$$\mathcal{F}^{**} = j_* j^*(\mathcal{F}),$$
and hence there is a monomorphism
$$\mathcal{F}^{**} \longrightarrow j_* j^*(\mathcal{A}).$$
The proof of (ii) is analogous to that of EGA III, 1.4.15, recalled in 11.0. Since S is normal, the morphism
$$\mathcal{O}_S \longrightarrow j_* j^*(\mathcal{O}_S)$$
is an isomorphism (cf. EGA IV_2, 5.8.6 and 5.10.5). Applying (ii) to $\mathcal{E} = \mathcal{O}_U$ therefore gives
$$\mathcal{A} = j_* j^*(\mathcal{A}).$$
Finally, since
$$j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}) = j^*(\mathcal{F}) \otimes_{\mathcal{O}_U} j^*(\mathcal{A}),$$

[139] Editor's note: this section has been added.

the last assertion of (iii) follows from (ii) applied to $\mathcal{E} = j^*(\mathcal{F})$. This proves the lemma.

Recall also that a finitely generated R-module M is called *reflexive* if the canonical morphism from M to its bidual M^{**} is an isomorphism. If R is a Noetherian regular ring of dimension at most 2, this implies that M is projective. Indeed, for every finitely generated R-module N, choose a resolution

$$L_1 \longrightarrow L_0 \longrightarrow N \longrightarrow 0,$$

where L_0 and L_1 are finite free R-modules. There is then an exact sequence

$$0 \longrightarrow N^* \longrightarrow L_0^* \longrightarrow L_1^* \longrightarrow Q \longrightarrow 0,$$

where Q is the cokernel of $L_0^* \to L_1^*$. Since R has homological dimension at most 2 ([BAC], X, § 4.2, Corollary 1 to Theorem 1), it follows that N^* is projective.

Proposition 13.2.— *Let S be a Noetherian regular scheme of dimension at most 2, let G be an affine flat S-group, and let $\mathcal{A}(G)$ be its affine algebra.*

(i) *If G is of finite type over S, it is isomorphic to a closed subgroup of*

$$H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

for some locally free $\mathcal{O}_S$-module $\mathcal{V}$ of finite rank. If, moreover, S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d, so that $H = \mathrm{GL}_{d,S}$.

(ii) *$\mathcal{A}(G)$ is the filtered direct limit of flat finite-type $\mathcal{O}_S$-sub-Hopf algebras.*

Proof. Let $\mathcal{B}$ be a finite-type $\mathcal{O}_S$-subalgebra of $\mathcal{A}(G)$. Since every coherent module on an open subset of S extends to a coherent module on S (EGA I, 9.4.5), there is a coherent $\mathcal{O}_S$-submodule $\mathcal{M}$ of $\mathcal{B}$ that generates $\mathcal{B}$ as an $\mathcal{O}_S$-algebra (*loc. cit.*, 9.6.5). By 11.10.bis, $\mathcal{M}$ is contained in a coherent G-stable $\mathcal{O}_S$-submodule $\mathcal{F}$. (N.B. Since G is affine over S, the proof of *loc. cit.* is simpler here: one may replace $f_*f^*(\mathcal{E})$ by $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$, and so on.)

Let $j : U \hookrightarrow S$ be the inclusion, where U is the flatness locus of $\mathcal{F}$. By Lemma 13.1 and the preceding observation,

$$\mathcal{V} = j_*j^*(\mathcal{F})$$

is a locally free $\mathcal{O}_S$-submodule of $\mathcal{A}(G)$, and the canonical morphism

$$\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow j_*j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G))$$

is an isomorphism. Thus $\mathcal{V}$ is a G-submodule of $\mathcal{A}(G)$. The action of G on $\mathcal{V}$ induces a morphism of affine S-groups

$$\rho_{\mathcal{V}} : G \longrightarrow H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

and therefore a morphism of $\mathcal{O}_S$-Hopf algebras

$$\varphi_{\mathcal{V}} : \mathcal{A}(H) \longrightarrow \mathcal{A}(G).$$

Write $\mathcal{A}_{\mathcal{V}}$ for the image of $\varphi_{\mathcal{V}}$. It is the affine algebra of a closed subgroup $G_{\mathcal{V}}$ of H, the closed image of $\rho_{\mathcal{V}}$. We shall show that $\mathcal{A}_{\mathcal{V}}$ contains $\mathcal{B}$.

The question is local on S, so we may suppose that $S = \mathrm{Spec}(R)$ and that

$$V = \Gamma(S, \mathcal{V})$$

is a free R-module with basis $v_1, \ldots, v_n$. Then $H \simeq \mathrm{GL}_{n,R}$, and $\mathcal{A}(H)$ is generated as an R-algebra by the "matrix coefficients" c_{ij} and by d^{-1}, where d denotes the determinant. Let Δ (respectively, ε) denote the comultiplication (respectively, augmentation) of $A = \Gamma(S, \mathcal{A}(G))$. For $j = 1, \ldots, n$, write

$$\Delta(v_j) = \sum_{i=1}^{n} v_i \otimes a_{ij}.$$

Then $a_{ij} = \varphi(c_{ij})$ belongs to $\mathrm{Im}(\varphi)$. Since V is an R-submodule of A, the identity

$$(\varepsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A$$

gives

$$v_j = \sum_i \varepsilon(v_i) a_{ij} \in \mathrm{Im}(\varphi).$$

As V contains a set of generators of $B = \Gamma(S, \mathcal{B})$, it follows that

$$B \subset \mathrm{Im}(\varphi).$$

If G is of finite type over S, take $\mathcal{B} = \mathcal{A}(G)$. Then φ is surjective, so the morphism

$$G \longrightarrow H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

is a closed immersion.

If S is also affine, there is a locally free $\mathcal{O}_S$-module $\mathcal{V}'$ of finite rank such that

$$\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^{\oplus d}.$$

Regard $\mathcal{V}'$ as a trivial G-module and replace $\mathcal{V}$ by $\mathcal{O}_S^{\oplus d}$. This realizes G as a closed subgroup of $\mathrm{GL}_{d,S}$.

Finally, return to an arbitrary affine flat S-group G. By EGA I, 9.4.9, $\mathcal{A}(G)$ is the union of its coherent $\mathcal{O}_S$-submodules $\mathcal{M}$, hence also of the $\mathcal{O}_S$-sub-Hopf algebras $\mathcal{A}_{\mathcal{V}}$ constructed above. This proves (ii).

Example 13.3.— *Let R be a discrete valuation ring with uniformizer π and fraction field K. Consider the filtered projective system of R-groups*

$$\cdots \longrightarrow \mathbb{G}_{a,R} \xrightarrow{\times\pi} \mathbb{G}_{a,R} \xrightarrow{\times\pi} \mathbb{G}_{a,R}$$

corresponding to the direct system

$$R[X_0] \longrightarrow R[X_1] \longrightarrow R[X_2] \longrightarrow \cdots,$$

whose transition morphisms are given by $X_n = \pi X_{n+1}$. Its projective limit G is a flat affine R-group scheme, not of finite type, with trivial special fiber and generic fiber $\mathbb{G}_{a,K}$. The affine R-algebra $\mathcal{A}(G)$ is the subring of $K[X]$ consisting of the polynomials whose constant coefficient lies in R. (N.B. We already encountered this example in Remark 11.10.1.) *Furthermore, G represents the functor assigning to every R-algebra B the set of sequences $(x_n)_{n\in\mathbb{N}}$ of elements of B such that*

$$x_n = \pi x_{n+1} \qquad \text{for every } n.$$

(In particular, every x_n is indefinitely π-divisible.)

Now let S be a Noetherian scheme such that every coherent $\mathcal{O}_S$-module is a quotient of a locally free $\mathcal{O}_S$-module of finite type. This holds, for example, when S is a separated

Noetherian regular scheme; see SGA 6, Exposé II, 2.1.1 and 2.2.7. (One can show that every Noetherian regular scheme of dimension at most 1 also has this property. It does not hold, however, when S is the affine plane over a field k with the origin doubled; see *loc. cit.*, 2.2.7.2.)

Definition 13.4.— *Let G be an affine flat S-group. Following R. W. Thomason* ([Th87], 2.1)*, we say that the pair (G, S) has the* equivariant resolution property*, or satisfies* (RE)*, if, for every coherent G-$\mathcal{O}_S$-module $\mathcal{F}$, there are a locally free G-$\mathcal{O}_S$-module $\mathcal{E}$ of finite rank and a G-equivariant epimorphism*

$$\mathcal{E} \longrightarrow \mathcal{F}.$$

In *loc. cit.*, Theorem 3.1, Thomason proves the result below under the hypothesis that G be essentially free over S (see the remark below). Gabber pointed out to us the simpler proof given here, which does not use that hypothesis.

Proposition 13.5.— *Let S be a Noetherian scheme, and let G be an affine S-group, flat and of finite type, with affine algebra $\mathcal{A}(G)$. Suppose that (G, S) satisfies* (RE)*. Then:*

(i) *G is isomorphic to a closed subgroup of*

$$H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

for some locally free $\mathcal{O}_S$-module $\mathcal{V}$ of finite rank.

(ii) *If, moreover, S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d, so that $H = \mathrm{GL}_{d,S}$.*

The proof is analogous to that of 13.2. As there, there is a coherent G-stable $\mathcal{O}_S$-submodule $\mathcal{F}$ that generates $\mathcal{A} = \mathcal{A}(G)$ as an $\mathcal{O}_S$-algebra. Replacing S by one of its connected components, we may suppose that S is connected. By (RE), there are a locally free G-$\mathcal{O}_S$-module $\mathcal{E}$ of rank n and an epimorphism of $\mathcal{A}$-comodules

$$\pi : \mathcal{E} \longrightarrow \mathcal{F}.$$

Put $H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{E})$. This is an S-group scheme locally isomorphic to $\mathrm{GL}_{n,S}$. The action of G on $\mathcal{E}$ induces a morphism of affine S-groups

$$\rho : G \longrightarrow H,$$

corresponding to a morphism of $\mathcal{O}_S$-Hopf algebras

$$\varphi : \mathcal{A}(H) \longrightarrow \mathcal{A}(G).$$

We shall show that ρ is a closed immersion.

The question is local on S, so suppose that $S = \mathrm{Spec}(R)$ and that

$$V = \Gamma(S, \mathcal{E})$$

is a free R-module with basis $v_1, \ldots, v_n$. Then $H \simeq \mathrm{GL}_{n,R}$, and

$$B = \Gamma(S, \mathcal{A}(H))$$

is generated as an R-algebra by the matrix coefficients c_{ij} and by d^{-1}, where d denotes the determinant. Let Δ (respectively, ε) be the comultiplication (respectively, augmentation) of

$$A = \Gamma(S, \mathcal{A}(G)),$$

let

$$\mu : V \longrightarrow V \otimes_R A$$

be the A-comodule structure on V, and put

$$F = \Gamma(S, \mathcal{F}).$$

For $j = 1, \ldots, n$, write

$$\mu(v_j) = \sum_{i=1}^{n} v_i \otimes a_{ij}.$$

Then $\varphi(c_{ij}) = a_{ij}$. Since $\pi : V \to F$ is a morphism of A-comodules, one has

$$\sum_{i=1}^{n} \pi(v_i) \otimes a_{ij} = (\pi \otimes_R \mathrm{id}_A)(\mu(v_j)) = \Delta(\pi(v_j)),$$

and therefore

$$\begin{aligned} \pi(v_j) &= (\varepsilon \otimes_R \mathrm{id}_A)\Delta(\pi(v_j)) \\ &= \sum_{i=1}^{n} \varepsilon(\pi(v_i)) a_{ij} \\ &= \varphi\left(\sum_{i=1}^{n} \varepsilon(\pi(v_i)) c_{ij}\right). \end{aligned}$$

Thus $\varphi(B)$ contains $\pi(V) = F$, which generates A as an R-algebra. Hence φ is surjective, and ρ is a closed immersion. This proves (i).

If S is affine, there is a locally free $\mathcal{O}_S$-module $\mathcal{V}'$ of finite rank such that

$$\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^{\oplus d}.$$

Regard $\mathcal{V}'$ as a trivial G-module and replace $\mathcal{V}$ by $\mathcal{O}_S^{\oplus d}$. This realizes G as a closed subgroup of $\mathrm{GL}_{d,S}$ and completes the proof.

Remarks 13.6.—

(a) For completeness, let us briefly sketch Thomason's argument ([Th87], Theorem 3.1), retaining the preceding notation. The G-equivariant epimorphism

$$\pi : \mathcal{E} \longrightarrow \mathcal{F}$$

induces a closed immersion

$$\tau : G \longrightarrow V(\mathcal{E})$$

such that

$$\tau(g'g) = \tau(g') \cdot g.$$

(N.B. G acts on the right on $V(\mathcal{E})$.) There is an isomorphism

$$H \simeq \mathrm{Aut}_{\mathcal{O}_S}(V(\mathcal{E}))^{\mathrm{op}}$$

compatible with the left action of G on $\mathcal{E}$ and its right action on $V(\mathcal{E})$. Let

$$N = \mathrm{Transp}_H^{\mathrm{str}}(\tau(G), \tau(G))$$

be the strict transporter. If G is essentially free over S, it follows from 6.2.4(e) that N is a closed subgroup scheme of H, hence affine over S. Moreover, ρ factors through a morphism of S-groups

$$\rho' : G \longrightarrow N.$$

For every $S' \to S$ and $h \in N(S')$, put

$$\pi(h) = \tau(1) \cdot h,$$

where 1 is the unit element of $G(S')$. This defines a morphism of S-schemes

$$\pi : N \longrightarrow \tau(G)$$

that is a retraction of ρ', after identifying G with $\tau(G)$. Since N is separated over S, it follows that ρ' is a closed immersion.

(b) The proof of [Th87], Theorem 3.1, appears to require the hypothesis that G be essentially free over S, although that hypothesis is not stated there (the author invokes instead the fact that H is essentially free). It is, however, satisfied when G is reductive (cf. Exposé XXII, 5.7.8), and hence in all cases considered in *loc. cit.*, Corollary 3.2. In particular, Thomason proves in *loc. cit.*, 2.5, that if S is separated, Noetherian, and regular of dimension at most 2, if G is affine and of finite presentation, and if $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$-module, then (G, S) satisfies (RE). By 13.5, this gives 13.2 under a slightly more restrictive hypothesis.

To conclude, let us point out that the proof of [Th87], 2.5, can be simplified slightly as follows. (For brevity, we work in the affine situation $S = \mathrm{Spec}(R)$.)

Proposition 13.7.— *Let R be a Noetherian regular ring of dimension at most 2, let C be an R-coalgebra that is projective as an R-module, and let F be a C-comodule of finite type over R. Then F is a quotient of a C-comodule V that is projective and of finite type over R.*

Proof. Replacing $\mathrm{Spec}(R)$ by one of its connected components, we may suppose that R is integral, with fraction field K. Let Δ (respectively, ε) denote the comultiplication (respectively, augmentation) of C, and let

$$\rho : F \longrightarrow F \otimes C$$

be the comodule structure on F. Let

$$\pi : W \longrightarrow F$$

be a surjective morphism, with W a finite free R-module. Give $W \otimes C$ the comodule structure defined by $\mathrm{id}_W \otimes \Delta$, and do the same for $F \otimes C$. Then $\rho : F \to F \otimes C$ is a morphism of C-comodules with retraction $\mathrm{id}_F \otimes \varepsilon$.

Let W' be the C-comodule defined by the following cartesian square:

$$\begin{array}{ccc} W' & \longrightarrow & F \\ \downarrow & & \downarrow \rho \\ W \otimes C & \xrightarrow{\pi \otimes \mathrm{id}_C} & F \otimes C \end{array}$$

Thus W' identifies with the kernel of the morphism

$$W \otimes C \longrightarrow (F \otimes C)/\rho(F),$$

and the projection

$$\pi' : W' \longrightarrow F, \qquad x \longmapsto (\pi \otimes \varepsilon)(x),$$

is surjective. Since F is a finitely generated R-module, there is a C-subcomodule V' of W', finitely generated over R, such that

$$\pi'(V') = F.$$

Since $W \otimes C$ is R-torsion-free, so is V'. Replacing F by V', we may therefore suppose from the outset that F is torsion-free.

Apply the preceding construction to this new F, obtaining V' as above. Consider the subcomodule V that is the kernel of the morphism

$$W \otimes C \longrightarrow E = \frac{W \otimes C \otimes K}{V' \otimes K}.$$

Then V contains V', and

$$Q = (W \otimes C)/V$$

is an R-submodule of the K-vector space E. Put

$$Q' = E/Q.$$

Since $W \otimes C$ and E are flat R-modules, for every R-module N one obtains

$$\mathrm{Tor}_1^R(V, N) \simeq \mathrm{Tor}_2^R(Q, N) \simeq \mathrm{Tor}_3^R(Q', N).$$

Because R is regular of dimension at most 2, the right-hand term vanishes. Hence V is a flat R-module.

It remains to prove that V is finitely generated. Put

$$M = W \otimes C.$$

By construction,

$$V/V' \simeq (M/V')_{\mathrm{tors}},$$

the R-torsion submodule of M/V'. Since M is a projective R-module, there is a projective R-module P such that

$$M \oplus P = L$$

is free. Then

$$(M/V')_{\mathrm{tors}} \simeq (L/V')_{\mathrm{tors}}.$$

Since V' is finitely generated, there is a direct summand

$$L' \simeq R^n$$

of L containing V'. Hence

$$(L/V')_{\mathrm{tors}} \simeq (L'/V')_{\mathrm{tors}},$$

and the latter is finitely generated because L'/V' is. Thus V is a finitely generated flat R-module, hence a finitely generated projective R-module, since R is Noetherian. This proves Proposition 13.7.

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Exposé VII_A
Infinitesimal Study of Group Schemes

by P. Gabriel

In Exposé II we restricted ourselves to the study of first-order differential invariants, and we did not address certain phenomena special to characteristic $p > 0$ or to characteristic zero. Our purpose in Part A of this exposé is to fill that gap.

Moreover, the infinitesimal study of a group scheme to arbitrary order is related to that of the associated formal group; the purpose of the second part of this exposé is to present the first definitions and properties concerning formal groups.

A) Differential Operators and Restricted p-Lie Algebras*

1. Differential Operators

In this section, as well as in Sections 2 and 3, S denotes a fixed scheme, and all products under consideration are cartesian products in the category of S-schemes.[1] If X is an S-scheme, we write $p_{X/S}$, p_X, or simply p for the structural morphism from X to S.

1.1. Let $u : Y \to X$ be a morphism of S-schemes, and endow the direct image $u_*(\mathcal{O}_Y)$ of the structure sheaf of Y with the $\mathcal{O}_X$-module structure induced by u. The sheaf

$$\mathcal{H} = \mathcal{H}om_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y))$$

of homomorphisms of $p_X^{-1}(\mathcal{O}_S)$-modules from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$ is therefore naturally an $\mathcal{O}_X$-bimodule. If U is an open subset of X, and f and d are sections over U of $\mathcal{O}_X$ and $\mathcal{H}$, respectively, then fd and df are the morphisms

$$g \longmapsto f\, d(g) \qquad \text{and} \qquad g \longmapsto d(fg)$$

from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$. From now on we write $(\operatorname{ad} f)(d)$ in place of $fd - df$.

Definition 1.1.1. *An S-deviation of order $\leq n$ is a pair $D = (u, d)$, consisting of a morphism of S-schemes $u : Y \to X$ and a morphism of $p_X^{-1}(\mathcal{O}_S)$-modules*

$$d : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y),$$

*Part A of the present exposé was not treated seriously in the oral seminars.

[1] In particular, if X and Y are two S-schemes, $X \times_S Y$ is denoted simply by $X \times Y$. For the material in Sections 1 and 2, one may also consult *Groupes algébriques* by Demazure and Gabriel, II.4, nos. 5–6; see also Jantzen, *Representations of Algebraic Groups*, I.7.

such that, for every open subset U of X and every sequence of $n+1$ sections $f_0, \dots, f_n \in \mathcal{O}_X(U)$, one has

$$(\operatorname{ad} f_0)(\operatorname{ad} f_1)\cdots(\operatorname{ad} f_n)(d) = 0. \qquad (*_n)$$

[2]

In this case we also say that d is an S-deviation of u of order $\leq n$. In particular, an S-deviation of u of order ≤ 0 is a morphism of $\mathcal{O}_X$-modules from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$, that is, an element of $\Gamma(Y, \mathcal{O}_Y)$.

Definition 1.1.2. *A morphism of $p_X^{-1}(\mathcal{O}_S)$-modules*

$$d : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y)$$

is an S-deviation of u if, for every point $y \in Y$, there exist an open neighborhood U of $u(y)$ in X and an open neighborhood V of y in Y satisfying:

a) *$u(V) \subset U$;*

b) *if $v : V \to U$ is induced by u, there is an integer n such that the morphism $\mathcal{O}_U \to v_*(\mathcal{O}_V)$ induced by d is an S-deviation of v of order $\leq n$.*

[3]

If d is an S-deviation of u, we also call the pair $D = (u, d)$ an S-deviation, and we may write

$$Y \xrightarrow{D} X \qquad \text{or} \qquad Y \xrightarrow[u]{d} X.$$

When d is the homomorphism of algebras

$$u^\natural : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y)$$

corresponding to the morphism $u : Y \to X$, we also write u in place of D.

Remarks 1.1.3.[4] Let $\operatorname{Dev}(u)$ (respectively, $\operatorname{Dev}_{\leq n}(u)$) be the set of S-deviations of u (respectively, of order $\leq n$). It has a natural $\mathcal{O}_Y(Y)$-module structure: if $\lambda \in \mathcal{O}_Y(Y)$, then λd is the deviation that sends f to $\lambda d(f)$ for every local section f of $\mathcal{O}_X$.

For every open subset V of Y, set $\mathcal{D}\rceil\sqsubseteq(u)(V) = \operatorname{Dev}(u|_V)$. Thus $\mathcal{D}\rceil\sqsubseteq(u)(V)$ is the set of

$$\begin{aligned} d_V &\in \operatorname{Hom}_{p_X^{-1}(\mathcal{O}_S)}\big(\mathcal{O}_X, (u|_V)_*(\mathcal{O}_V)\big) \\ &\simeq \operatorname{Hom}_{p_Y^{-1}(\mathcal{O}_S)}\big((u|_V)^{-1}\mathcal{O}_X, \mathcal{O}_V\big) \\ &\simeq \operatorname{Hom}_{p_Y^{-1}(\mathcal{O}_S)}\big(u^{-1}\mathcal{O}_X, \mathcal{O}_Y\big)(V) \end{aligned}$$

[2] Equivalently, for every $x \in X$ and $f_0, \dots, f_n, g \in \mathcal{O}_{X,x}$,

$$(\operatorname{ad} f_0)\cdots(\operatorname{ad} f_n)(d_x)(g) = 0.$$

Recall also that the adjunction isomorphism

$$\theta : \operatorname{Hom}_{p_X^{-1}(\mathcal{O}_S)}\big(\mathcal{O}_X, u_*(\mathcal{O}_Y)\big) \xrightarrow{\sim} \operatorname{Hom}_{p_Y^{-1}(\mathcal{O}_S)}\big(u^{-1}\mathcal{O}_X, \mathcal{O}_Y\big)$$

sends d to $d' = \varepsilon \circ u^{-1}(d)$, where $\varepsilon : u^{-1}u_*(\mathcal{O}_Y) \to \mathcal{O}_Y$ is canonical. Conversely, $\theta^{-1}(d') = u_*(d') \circ \eta$, with $\eta : \mathcal{O}_X \to u_*u^{-1}\mathcal{O}_X$ canonical. Hence d satisfies $(*_n)$ if and only if d' satisfies the analogous identity for all local sections of $u^{-1}\mathcal{O}_X$.

[3] If X and u are quasi-compact, every S-deviation of u is therefore of order $\leq n$ for some integer n.

[4] These remarks were added by the editors because they are useful in 1.3, 1.4, and 2.1.

such that, for every open $U \subset X$, the map

$$d_V(U) : \mathcal{O}_X(U) \longrightarrow \mathcal{O}_Y(u^{-1}(U) \cap V)$$

satisfies $(*_n)$. This defines a presheaf of $\mathcal{O}_Y$-modules on Y, and it is easily seen to be a sheaf (more precisely, a subsheaf of $\mathcal{H}om_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)$).

1.2. Consider two S-deviations $D = (u, d)$ and $E = (v, e)$:

$$Z \xrightarrow[v]{e} Y \xrightarrow[u]{d} X.$$

As U ranges over the open subsets of X, the composites

$$\Gamma(U, \mathcal{O}_X) \xrightarrow{d(U)} \Gamma(u^{-1}U, \mathcal{O}_Y) \xrightarrow{e(u^{-1}U)} \Gamma(v^{-1}u^{-1}U, \mathcal{O}_Z)$$

define an S-deviation of uv, denoted by de. If d has order $\leq m$ and e has order $\leq n$, then de has order $\leq m + n$. We also write

$$D \circ E = (uv, de) \qquad (\dagger)$$

[5] and call $D \circ E$, or DE, the *composite S-deviation*. When $d = u^\natural$, so that $D = u$ under the convention of 1.1, we also say that DE is the *image of E under u*.

The map $(D, E) \mapsto D \circ E$ allows us to speak from now on of the *category of S-deviations*, whose objects are S-schemes and whose morphisms are S-deviations.[6]

Definition 1.2.0.[7] *Let $w : Z \to X$ be an S-morphism. An S-derivation of w, or an S-derivation of $\mathcal{O}_X$ into $w_*(\mathcal{O}_Z)$, is a morphism of $p^{-1}(\mathcal{O}_S)$-modules*

$$d : \mathcal{O}_X \longrightarrow w_*(\mathcal{O}_Z)$$

such that, for every open $U \subset X$ and $f, g \in \mathcal{O}_X(U)$,

$$d(fg) = w^\natural(f)d(g) + w^\natural(g)d(f).$$

Then d is a deviation of w of order ≤ 1, vanishing on the unit section of $\mathcal{O}_X$. We denote the set of S-derivations of w by $\mathrm{Der}_S(w)$*; it is an $\mathcal{O}(Z)$-module.*

With the notation of 1.2, take

$$Y = I_Z = \operatorname{Spec} \mathcal{O}_Z[t], \qquad t^2 = 0,$$

let v be the zero section $\tau : Z \to I_Z$, defined by the homomorphism $\mathcal{O}_Z[t] \to \mathcal{O}_Z$ sending t to zero, and let e be the morphism of $\mathcal{O}_Z$-modules $\sigma : \mathcal{O}_Z[t] \to \mathcal{O}_Z$ defined by $\sigma(1) = 0$ and $\sigma(t) = 1$, which it is convenient to denote by ∂_t.[8]

If $u : I_Z \to X$ is a morphism satisfying $w = u \circ \tau$, then $\sigma \circ u^\natural$ is an S-derivation from $\mathcal{O}_X$ to $w_*(\mathcal{O}_Z)$. Conversely, an S-derivation d determines the morphism $u : I_Z \to X$ which equals w on the underlying spaces and satisfies

$$u^\natural(f) = w^\natural(f) + d(f)t$$

[5] With this notation, *de* means the composite "d followed by e."

[6] Often one considers only the S-deviations of id_X; they form the algebra of S-differential operators on X, cf. 1.4 below. The more general framework provides a convenient functorial language for statements such as the following: if G is an S-group, the algebra of left-translation-invariant S-differential operators on G is isomorphic to the algebra of S-deviations of the unit section $\varepsilon : S \to G$; cf. 2.1 and 2.4.

[7] The editors expanded this paragraph and assigned the number 1.2.0 to this definition and 1.2.1 to the lemma that follows.

[8] The editors added this notation ∂_t.

for every local section f of $\mathcal{O}_X$.

Lemma 1.2.1. *Let $E = (\tau, \partial_t)$ be the deviation of $\tau : Z \to I_Z$ defined above. For every S-morphism $w : Z \to X$, the map $u \mapsto u \circ E$ is a bijection between the S-morphisms $u : I_Z \to X$ such that $u \circ \tau = w$, and the S-derivations of w.*

1.2.2. Let d be an S-deviation of $u : Y \to X$. On the one hand, d is plainly an S'-deviation of u for every morphism $s : S \to S'$.

On the other hand, let $t : T \to S$ be a morphism with target S, let $u_T : Y_T \to X_T$ be obtained from u by base change, and let $t_Y : Y_T \to Y$ and $t_X : X_T \to X$ be the canonical projections. There exists a unique T-deviation of u_T, denoted d_T or $d \times T$, which satisfies

$$t_X d_T = d t_Y$$

in the sense of (†). Thus, for every open $U \subset X$, the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{t_X^\natural} & \mathcal{O}(U \times T) \\ {\scriptstyle d(U)}\downarrow & & \downarrow{\scriptstyle d_T(U\times T)} \\ \mathcal{O}(u^{-1}U) & \xrightarrow{t_Y^\natural} & \mathcal{O}(u^{-1}U \times T)\,. \end{array}$$

[9] If $D = (u, d)$, we also write $D_T = (u_T, d_T)$, and say that d_T and D_T are obtained from d and D by base change.

1.2.3. For example, let $u : Y \to X$ and $v : Z \to T$ be two S-morphisms, and let d and e be S-deviations of u and v. There is a commutative diagram

$$\begin{array}{ccc} X \times T & \xleftarrow{u_T} & Y \times T \\ {\scriptstyle v_X}\uparrow & \nwarrow^{u\times v} & \uparrow{\scriptstyle v_Y} \\ X \times Z & \xleftarrow{u_Z} & Y \times Z \end{array}$$

and we denote by $d \times e$, the *product of d and e*, the S-deviation of $u \times v$ equal to

$$d_T e_Y = e_X d_Z$$

under the convention of (†). More explicitly, for every open $U \subset X \times T$, set

$$W = v_Y^{-1} u_T^{-1} U = u_Z^{-1} v_X^{-1} U.$$

Then the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{d_T(U)} & \mathcal{O}(u_T^{-1}U) \\ {\scriptstyle e_X(U)}\downarrow & \overset{(d\times e)(U)}{\searrow} & \downarrow{\scriptstyle e_Y(u_T^{-1}U)} \\ \mathcal{O}(v_X^{-1}U) & \xrightarrow{d_Z(v_X^{-1}U)} & \mathcal{O}(W)\,. \end{array}$$

[9] Explicitly, if V is an affine open subset of S, and U (respectively, U') is an affine open subset of X (respectively, T) over V, then

$$\mathcal{O}_{X\times T}(U \times U') = \mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U'),$$

and $d_T(U \times U')$ is the composite

$$\mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') \xrightarrow{d(U)\otimes \mathrm{id}} \mathcal{O}_Y(u^{-1}U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') \longrightarrow \mathcal{O}_{Y\times T}(u^{-1}U \times U').$$

The author left to the reader the verification that d_T is well-defined, and the editors do the same.

If $D = (u, d)$ and $E = (v, e)$, we also write

$$D \times E = (u \times v, d \times e).$$

1.3.[10] Let $u : Y \to X$ be a morphism of S-schemes. Recall that the adjunction isomorphism

$$\mathrm{Hom}_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y)) \xrightarrow{\sim} \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)$$

sends a morphism of $p^{-1}(\mathcal{O}_S)$-modules

$$d : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y)$$

to $d' = \varepsilon \circ u^{-1}(d)$, where $\varepsilon : u^{-1}u_*(\mathcal{O}_Y) \to \mathcal{O}_Y$ is the canonical morphism.

Write $\mathcal{J}_u$ (respectively, $\mathcal{I}_u$) for the kernel of the homomorphism of algebras

$$u^\natural : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y) \quad \Big(\text{respectively, } u^{\natural\prime} : u^{-1}\mathcal{O}_X \longrightarrow \mathcal{O}_Y\Big),$$

and let $d : \mathcal{O}_X \to u_*(\mathcal{O}_Y)$ be a morphism of $p^{-1}(\mathcal{O}_S)$-modules. If $U \subset X$ is open and $f_0, \ldots, f_n, g \in \mathcal{O}_X(U)$, induction on n shows that condition $(*_n)$ is equivalent to

$$0 = \sum_{I \subset \{0,\ldots,n\}} (-1)^{|I|} u^\natural(f_{\{0,\ldots,n\}\setminus I}) d(f_I g), \qquad (**_n)$$

where f_I denotes the product of the f_i for $i \in I$. Consequently, if d satisfies $(*_n)$, then d vanishes on the ideal $\mathcal{J}_u^{n+1}$.

Now suppose that $Y = S$. Then $u : S \to X$ is a section of $p : X \to S$, hence an immersion. On the one hand,

$$\varepsilon : u^{-1}u_*\mathcal{O}_S \longrightarrow \mathcal{O}_S$$

is an isomorphism, and therefore $u^{-1}\mathcal{J}_u = \mathcal{I}_u$. On the other hand, there is an isomorphism

$$u^{-1}\mathcal{O}_X \simeq \mathcal{O}_S \oplus \mathcal{I}_u. \qquad (\star)$$

Suppose that d vanishes on $\mathcal{J}_u^{n+1}$. Then $d' = \varepsilon \circ u^{-1}(d)$ vanishes on $\mathcal{I}_u^{n+1}$, and therefore satisfies the analogues $(**'_n)$ and $(*'_n)$ of $(**_n)$ and $(*_n)$ when

$$f_0, \ldots, f_n \in \mathcal{I}_u(u^{-1}(U)).$$

Moreover,

$$(\mathrm{ad}\, a)(\varphi) = 0$$

for every $a \in \mathcal{O}_S(u^{-1}U)$ and every morphism of $\mathcal{O}_{u^{-1}U}$-modules $\varphi : u^{-1}\mathcal{O}_U \to \mathcal{O}_{u^{-1}U}$. The splitting $(\star)$ therefore implies that d' satisfies $(*'_n)$, and hence that d satisfies $(*_n)$.

Lemma. *If $u : S \to X$ is a section of $p : X \to S$, then d is an S-deviation of u of order $\leq n$ if and only if d' vanishes on $\mathcal{I}_u^{n+1}$.*

This interpretation generalizes as follows. Let $u : Y \to X$ be any S-morphism, and let $\Gamma_u : Y \to Y \times X$ be its graph, with components id_Y and u. Every S-deviation d of u of order $\leq n$ gives, by composition,

$$Y \xrightarrow{\mathrm{diag}} Y \times Y \xrightarrow[u_Y]{d_Y} Y \times X,$$

a Y-deviation of Γ_u of order $\leq n$, denoted Γ_d, the graph of d.

[10] The editors expanded the original in this paragraph; see also their note 2 to Definition 1.1.1.

Conversely, a Y-deviation e of Γ_u gives the composite S-deviation

$$e_X = \mathrm{pr}_2 \circ e : \qquad Y \xrightarrow[\Gamma_u]{e} Y \times X \xrightarrow{\mathrm{pr}_2} X.$$

One has immediately $(\Gamma_d)_X = d$. The identity $\Gamma_{e_X} = e$ follows from the $\mathcal{O}_Y$-linearity of e.[11] Thus there is an isomorphism of $\mathcal{O}_Y(Y)$-modules

$$\left\{ \begin{array}{c} S\text{-deviations of } u \\ \text{of order } \leq n \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{c} Y\text{-deviations of } \Gamma_u \\ \text{of order } \leq n \end{array} \right\}, \qquad d \longmapsto \Gamma_d.$$

Moreover, d is an S-derivation of u if and only if Γ_d is a Y-derivation of Γ_u.

Let $\mathcal{I}_{\Gamma_u}$ denote the kernel of the homomorphism of algebras

$$\Gamma_u^{-1}\mathcal{O}_{Y\times X} \longrightarrow \mathcal{O}_Y$$

corresponding to Γ_u. The preceding lemma yields:

Proposition. *Let $u : Y \to X$ be an S-morphism and $\Gamma_u : Y \to Y \times X$ its graph. The S-deviations of u of order $\leq n$ identify with the Y-deviations of Γ_u of order $\leq n$, and these are in bijection with*

$$\mathrm{Hom}_{\mathcal{O}_Y}\left(\frac{\Gamma_u^{-1}\mathcal{O}_{Y\times X}}{\mathcal{I}_{\Gamma_u}^{n+1}}, \mathcal{O}_Y\right).$$

1.3.1.[12] Return to the case in which $u : S \to X$ is a section of $p : X \to S$. The homomorphism

$$\varphi : u^{-1}\mathcal{O}_X \longrightarrow \mathcal{O}_S$$

has a section, denoted simply by $g \mapsto g \cdot 1$. Thus, with the notation of 1.3,

$$u^{-1}\mathcal{O}_X \simeq \mathcal{O}_S \oplus \mathcal{I}_u,$$

and, for every section f of $u^{-1}\mathcal{O}_X$,

$$f - \varphi(f) \cdot 1 \in \mathcal{I}_u.$$

Let d be an S-deviation of u of order ≤ 1, and let $d' : u^{-1}\mathcal{O}_X \to \mathcal{O}_S$ be the $\mathcal{O}_S$-morphism corresponding to d. If a, b are sections of $u^{-1}\mathcal{O}_X$, then

$$\begin{aligned} 0 &= d'((a - \varphi(a)\cdot 1)(b - \varphi(b)\cdot 1)) \\ &= d'(ab) - \varphi(a)d'(b) - \varphi(b)d'(a) + \varphi(ab)d'(1). \end{aligned}$$

Consequently, d is an S-derivation of u if and only if $d'(1) = 0$.

Lemma. *The S-derivations of u are precisely the S-deviations of u of order 1 which vanish on the unit section of $\mathcal{O}_X$. They correspond to the $\mathcal{O}_S(S)$-module*

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}_u/\mathcal{I}_u^2, \mathcal{O}_S),$$

and there is an isomorphism of $\mathcal{O}_S(S)$-modules

$$\mathrm{Dev}_{\leq 1}(u) \simeq \mathcal{O}_S(S) \oplus \mathrm{Der}_S(u).$$

[11] If λ and f are local sections of $\mathcal{O}_Y$ and $\mathcal{O}_X$, then

$$(\Gamma_{e_X})(\lambda \otimes f) = \lambda\, e(1 \otimes f) = e(\lambda \otimes f).$$

The French note prints g in the two right-hand tensor expressions; the variables just introduced require f.

[12] This paragraph was added by the editors.

Returning to the general case gives:

Corollary. *Let $u : Y \to X$ be an S-morphism and $\Gamma_u : Y \to Y \times X$ its graph. There is a canonical isomorphism of $\mathcal{O}_Y(Y)$-modules*

$$\mathrm{Der}_S(u) \simeq \mathrm{Der}_Y(\Gamma_u) \simeq \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_{\Gamma_u}/\mathcal{I}_{\Gamma_u}^2, \mathcal{O}_Y).$$

Definition 1.4. *Let X be an S-scheme. An S-differential operator (respectively, an S-differential operator of order $\leq n$) on X is an S-deviation (respectively, an S-deviation of order $\leq n$) of the identity morphism of X.*

By 1.1, an S-differential operator of order $\leq n$ is therefore an endomorphism of the $p^{-1}(\mathcal{O}_S)$-module $\mathcal{O}_X$ satisfying the identities $(*_n)$ of 1.1. We write $\mathrm{Dif}^n_{X/S}$ for the $\Gamma(\mathcal{O}_S)$-module[13] of S-differential operators of order $\leq n$, and $\mathrm{Dif}_{X/S}$ for the module of all S-differential operators.

As observed in 1.2, S-deviations of id_X can be composed. This makes $\mathrm{Dif}_{X/S}$ a $\Gamma(\mathcal{O}_S)$-algebra, called the *algebra of differential operators of X/S*.

Similarly, for every open subset $V \subset X$, set

$$\mathcal{D}\rangle\{_{X/S}(V) = \mathrm{Dif}_{V/S} = \mathrm{Dev}(\mathrm{id}_V).$$

By 1.1.3 this defines a sheaf of $\mathcal{O}_X$-modules, called the *sheaf of S-differential operators on X*.[14]

1.4.1. As in 1.3, differential operators on X/S may be interpreted using the graph of the identity morphism of X, namely the diagonal

$$\Delta = \Delta_{X/S} : X \longrightarrow X \times X.$$

Endow $\mathcal{O}_{X\times X}$ with the $\mathrm{pr}_1^{-1}(\mathcal{O}_X)$-algebra structure defined by pr_1. Then $\Delta^{-1}\mathcal{O}_{X\times X}$ is an algebra over

$$\mathcal{O}_X = \Delta^{-1}\,\mathrm{pr}_1^{-1}\,\mathcal{O}_X.$$

Let $\mathcal{I}_{X/S}$ be the kernel of the homomorphism

$$\Delta^{-1}\mathcal{O}_{X\times X} \xrightarrow{m} \mathcal{O}_X$$

adjoint to $\mathcal{O}_{X\times X} \to \Delta_*(\mathcal{O}_X)$, and set

$$\mathcal{P}^m_{X/S} = \Delta^{-1}\mathcal{O}_{X\times X}/\mathcal{I}^{m+1}_{X/S}.$$

If $V \subset S$ and $U \subset X$ are affine open subsets with U above V, and if

$$k = \Gamma(V, \mathcal{O}_S), \qquad A = \Gamma(U, \mathcal{O}_X),$$

then

$$\Gamma(U, \mathcal{P}^m_{X/S}) = (A \otimes_k A)/I^{m+1},$$

where I is generated by the elements $a \otimes 1 - 1 \otimes a$, for $a \in A$.

By 1.3 there is an isomorphism of $\mathcal{O}_X(X)$-modules

$$j_X : \mathrm{Dif}^m_{X/S} \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}^m_{X/S}, \mathcal{O}_X).$$

[13] In this exposé the ring $\Gamma(S, \mathcal{O}_S) = \mathcal{O}_S(S)$ is denoted by $\Gamma(\mathcal{O}_S)$.

[14] The editors modified the original here. It mentioned the sheaf $U \mapsto \mathrm{Dif}_{X_U/U}$, where U ranges over open subsets of S; that sheaf is the direct image of $\mathcal{D}\rangle\{_{X/S}$ under $p_X : X \to S$.

It is given explicitly as follows: if $d \in \mathrm{Dif}^m_{X/S}$ and a section c of $\mathcal{P}^m_{X/S}$ over U has the form

$$c = a \otimes b + I^{m+1},$$

then $j_X(d)(c) = a\, d(b)$.[15]

1.4.2. Let d be a differential operator and u a section of X over S. The *value of d at u* is the composite S-deviation

$$S \xrightarrow{u} X \xrightarrow[\mathrm{id}_X]{d} X.$$

By 1.3 and 1.4.1, if d has order $\leq m$, then du (respectively, d) is canonically associated with a morphism of $\mathcal{O}_S$-modules

$$d' : u^{-1}\mathcal{O}_X/\mathcal{I}_u^{m+1} \longrightarrow \mathcal{O}_S$$

(respectively, a morphism of $\mathcal{O}_X$-modules $d'' : \mathcal{P}^m_{X/S} \to \mathcal{O}_X$).

The morphism d' may be recovered from d'' as follows. The square

$$\begin{array}{ccc} X \simeq S \times X & \xrightarrow{u\times X} & X \times X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_1} \\ S & \xrightarrow{\quad u \quad} & X \end{array}$$

is cartesian. Hence X identifies with $S \times_X (X \times X)$, u with $S \times_X \Delta$, and $u^*(\mathcal{P}^m_{X/S})$ with $u^{-1}\mathcal{O}_X/\mathcal{I}_u^{m+1}$. Under these identifications, $u^*(d'')$ is the morphism

$$u^{-1}\mathcal{O}_X/\mathcal{I}_u^{m+1} \longrightarrow \mathcal{O}_S$$

which is precisely d'.

1.5. Set, as usual,

$$I_S = \mathrm{Spec}\, \mathcal{O}_S[T]/(T^2).$$

Let $\tau : S \to I_S$ be the zero section, and let σ be the canonical deviation of τ defined in 1.2.0: it is the $\mathcal{O}_S$-module homomorphism which vanishes on the unit section of $\mathcal{O}_S[T]/(T^2)$ and sends the class t of T modulo T^2 to the unit section of $\mathcal{O}_S$.

Let X be an S-scheme. Every I_S-automorphism u of $I_S \times X$ inducing the identity on X determines, by composition, a differential operator D_u on X:

$$X \simeq S \times X \xrightarrow{\sigma\times X} I_S \times X \xrightarrow{\ u\ } I_S \times X \xrightarrow{\mathrm{pr}_2} X\,.$$

By Exposé II, 3.14, the map $u \mapsto D_u$ is an isomorphism from the $\Gamma(\mathcal{O}_S)$-Lie algebra

$$\mathrm{Lie}(\mathrm{Aut}\, X) := \mathrm{Lie}(\mathrm{Aut}\, X)(S)$$

onto the $\Gamma(\mathcal{O}_S)$-Lie algebra of $p^{-1}(\mathcal{O}_S)$-derivations of $\mathcal{O}_X$. Its inverse sends a derivation D to the automorphism of $I_S \times X$ corresponding to

$$a + bt \longmapsto a + (Da + b)t$$

on $\mathcal{O}_X[T]/(T^2)$.

[15] Under this isomorphism, the X-derivations of $\Delta_{X/S}$ correspond, by 1.3.1, to the S-derivations of id_X, that is, to the $p^{-1}(\mathcal{O}_S)$-derivations of $\mathcal{O}_X$.

2. Invariant Differential Operators on Group Schemes

2.1. Let G be an S-group scheme; we denote its unit section by ε, or by $\varepsilon_G : S \to G$.

Definition. *Let $U(G)$ be the $\Gamma(\mathcal{O}_S)$-module of S-deviations of ε_G (or S-deviations of the origin), as in 1.1.*

If $d, e \in U(G)$, then $d \times e$ is an S-deviation of

$$\varepsilon \times \varepsilon : S \simeq S \times S \longrightarrow G \times G.$$

The image of $d \times e$ under the multiplication morphism $m : G \times G \to G$, as in 1.2, is called the product *of d and e and is denoted $d \cdot e$.*

The $\Gamma(\mathcal{O}_S)$-module $U(G)$ is thereby made into an associative $\Gamma(\mathcal{O}_S)$-algebra with unit ε_G. We call $U(G)$ the infinitesimal algebra *of G.*[16]

As T ranges over schemes above S, the infinitesimal algebra $U(G_T)$ of the T-group $G \times T$ plainly varies contravariantly with T; thus one may speak of the *infinitesimal-algebra functor*.

Restricting T to the open subsets of S gives a presheaf $T \mapsto U(G_T)$ of $\mathcal{O}_S$-algebras. By 1.1.3 it is a sheaf. We denote it by $\mathcal{U}(G)$ and call it the *sheaf of infinitesimal algebras of G*.

The algebra $U(G)$ is also covariantly functorial in G. Indeed, if $u : G \to H$ is a homomorphism of S-groups and d is an S-deviation of ε_G, its image under u is the element

$$U(u)(d) = ud$$

of $U(H)$. The resulting map $U(u) : U(G) \to U(H)$ is a homomorphism of $\Gamma(\mathcal{O}_S)$-algebras. Likewise one obtains a homomorphism $\mathcal{U}(u) : \mathcal{U}(G) \to \mathcal{U}(H)$.

2.2. Let $d \in U(G)$, that is, let d be an S-deviation of the origin of G. The base change of d is the S-deviation

$$d \times G$$

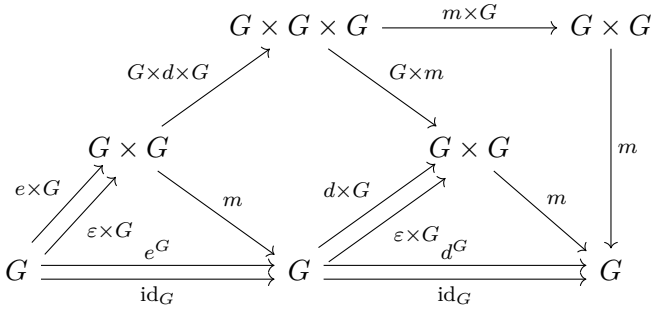
of $\varepsilon \times G : G \simeq S \times G \to G \times G$, as in 1.2.2. Its image under multiplication $m : G \times G \to G$ is an S-deviation of

$$m \circ (\varepsilon \times \mathrm{id}_G) = \mathrm{id}_G,$$

hence an element of $\mathrm{Dif}_{G/S}$, which we denote by d^G.

The map $d \mapsto d^G$ is plainly $\Gamma(\mathcal{O}_S)$-linear. The following commutative diagram shows that

$$(e \cdot d)^G = d^G \cdot e^G.$$



[16] This is now usually called the algebra of distributions (at the origin) of G; see Demazure–Gabriel, II.4, 6.1, and Jantzen, I.7.7.

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The two lower triangles commute by the definitions of d^G and e^G. The composite S-deviation of $e \times G$ and $G \times d \times G$ is $(e \times d) \times G$, by 1.2.2. Its image under $m \times G$ is $(e \cdot d) \times G$, and the image of the latter under m is therefore $(e \cdot d)^G$.

Thus one obtains an antihomomorphism

$$U(G) \longrightarrow \mathrm{Dif}_{G/S}$$

of $\Gamma(\mathcal{O}_S)$-algebras, called *right translation*.[18]

If $\mathcal{D}\rangle\{_{G/S}$ denotes the sheaf of S-differential operators on G, as in 1.4, and $p : G \to S$ is the structural morphism, one likewise defines a right translation

$$\mathcal{U}(G) \longrightarrow p_*(\mathcal{D}\rangle\{_{G/S}).$$

2.3. We now characterize the differential operators on G/S of the form d^G. Let $g : S \to G$ be a section of the structural morphism and let g_G denote right translation by g, namely

$$g_G : G \simeq G \times S \xrightarrow{G \times g} G \times G \xrightarrow{m} G.$$

For every differential operator D on G/S, the composite

$$g_G^{-1} D g_G$$

is again an S-deviation of id_G, hence an element of $\mathrm{Dif}_{G/S}$. We write

$$D^g = g_G^{-1} D g_G.$$

We call D *right-invariant* if for every base change $t : T \to S$ and every section $g : T \to G \times T$, one has

$$(D_T)^g = D_T.$$

Lemma. *For every differential operator D on G/S, the following assertions are equivalent, where m is the multiplication morphism of G:*

(i) D is right-invariant.

(ii) The following two deviations of m are equal:

$$Dm = m(D \times G).$$

$(ii) \Rightarrow (i)$. Since condition (ii) is stable under base change, it suffices to show that it implies $D^g = D$ for every section $g : S \to G$. Let

$$h = G \times g : G \simeq G \times S \longrightarrow G \times G,$$

so that $m \circ h = g_G$. The equality $D^g = D$ is equivalent to $g_G \circ D = D \circ g_G$, and this follows from the commutative diagram

[17] The editors corrected the original diagram by replacing $d \times G \times G$ with $G \times d \times G$. The composite on the left side of the upper triangle is then $(e \times d) \times G$, and $d \mapsto d^G$ becomes an anti-isomorphism from $U(G)$ to the right-invariant differential operators (2.3–2.4). If instead ${}^G d$ is defined as the image of $G \times d$ under m, one similarly obtains an isomorphism onto the left-invariant differential operators. Sections 2.4 and 2.5 have been corrected accordingly.

[18] It would be preferable to call this a *left action*. For example, if d is an S-derivation of the origin, then by 1.2.1 it is the composite of $(\tau, \partial_t) : S \to I_S$ with a morphism $x : I_S \to G$ satisfying $x \circ \tau = \varepsilon$. The derivation d^G of $\mathcal{O}_G$ sends a local section φ to the section $g \mapsto \partial_t \varphi(xg)$. With this terminology one could say that the left action commutes with right translations.

$$\begin{array}{ccccc} G & \xleftarrow{m} & G\times G & \xleftarrow{h} & G. \\ {\scriptstyle D}\Big\downarrow\Big\downarrow{\scriptstyle \mathrm{id}_G} & & {\scriptstyle D\times G}\Big\downarrow\Big\downarrow{\scriptstyle \mathrm{id}_{(G\times G)}} & & {\scriptstyle D}\Big\downarrow\Big\downarrow{\scriptstyle \mathrm{id}_G} \\ G & \xleftarrow{m} & G\times G & \xleftarrow{h} & G \end{array}$$

$(i) \Rightarrow (ii)$. Take $t : T \to S$ to be the structural morphism $p : G \to S$, and take the section $g : T \to G \times T$ to be the diagonal $\Delta : G \to G \times G$. Right translation

$$\Delta_{G\times G} : G \times G \longrightarrow G \times G$$

then has components m and pr_2. The equality $(D_G)^\Delta = D_G$ is equivalent to commutativity of the first square in

$$\begin{array}{ccccc} G\times G & \xrightarrow{\Delta_{G\times G}} & G\times G & \xrightarrow{\mathrm{pr}_1} & G. \\ {\scriptstyle D_G}\Big\downarrow\Big\downarrow{\scriptstyle \mathrm{id}_{G\times G}} & & {\scriptstyle D_G}\Big\downarrow\Big\downarrow{\scriptstyle \mathrm{id}_{G\times G}} & & {\scriptstyle D}\Big\downarrow\Big\downarrow{\scriptstyle \mathrm{id}_G} \\ G\times G & \xrightarrow{\Delta_{G\times G}} & G\times G & \xrightarrow{\mathrm{pr}_1} & G \end{array}$$

because $m = \mathrm{pr}_1 \circ \Delta_{G\times G}$. This proves (ii).

Now consider $d \in U(G)$. Every square in the following diagram is commutative:

$$\begin{array}{ccccccc} G\times G & = & S\times G\times G & \overset{d\times G\times G}{\underset{\varepsilon\times G\times G}{\rightrightarrows}} & G\times G\times G & \xrightarrow{m\times G} & G\times G \\ {\scriptstyle m}\downarrow & & {\scriptstyle S\times m}\downarrow & & \downarrow{\scriptstyle G\times m} & & \downarrow{\scriptstyle m} \\ G & = & S\times G & \underset{\varepsilon\times G}{\overset{d\times G}{\rightrightarrows}} & G\times G & \xrightarrow{m} & G \end{array}$$

Since

$$m \circ (d \times G) = d^G \qquad \text{and} \qquad (m \times G) \circ (d \times G \times G) = d^G \times G,$$

one also has

$$d^G \circ m = m \circ (d^G \times G).$$

Thus, for every S-deviation d of the origin, d^G is a right-invariant differential operator.

Theorem 2.4.

(i) The map $d \mapsto d^G$ is an anti-isomorphism of the infinitesimal algebra $U(G)$ onto the subalgebra $\mathrm{Dif}^G_{G/S}$ of $\mathrm{Dif}_{G/S}$ formed by the right-invariant differential operators.

(ii) Likewise, the map $d \mapsto {}^G d$ is an isomorphism from $U(G)$ onto the subalgebra of $\mathrm{Dif}_{G/S}$ formed by the left-invariant differential operators.

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Let D be an arbitrary differential operator on G/S, and let D_0 denote its *value at the origin*, that is, the composite deviation

$$S \overset{\varepsilon}{\to} G \underset{\mathrm{id}_G}{\overset{D}{\to}} G.$$

The right-invariant differential operator $(D_0)^G$ is obtained by the composite

$$G \simeq S \times G \xrightarrow{\varepsilon\times G} G \times G \underset{\mathrm{id}_{G\times G}}{\overset{D\times G}{\rightrightarrows}} G \times G \xrightarrow{m} G.$$

[19] The editors corrected "isomorphism" to "anti-isomorphism" and added assertion (ii); see the preceding editorial note.

If D is right-invariant, then $Dm = m(D \times G)$, and hence

$$D = Dm(\varepsilon \times G) = m(D \times G)(\varepsilon \times G) = (D_0)^G.$$

Thus $d \mapsto d^G$ is surjective.

Conversely, for every S-deviation d of the origin one has the commutative square

$$\begin{array}{ccc} G \times G & \xleftarrow{d \times G} & G \\ {\scriptstyle G \times \varepsilon}\uparrow & & \uparrow{\scriptstyle \varepsilon} \\ G \times S \simeq G & \xleftarrow{d} & S \end{array}$$

from which

$$d = m(G \times \varepsilon)d = m(d \times G)\varepsilon = (d^G)_0.$$

A fortiori, $d \mapsto d^G$ is injective. This proves the theorem.

As S varies, Theorem 2.4 also shows that right translation

$$\mathcal{U}(G) \longrightarrow p_*(\mathcal{D}\rangle\{_{G/S})$$

is an anti-isomorphism from $\mathcal{U}(G)$ onto the sheaf of $\mathcal{O}_S$-algebras

$$p_*(\mathcal{D}\rangle\{_{G/S})^G,$$

which assigns to every open $V \subset S$ the algebra $\mathrm{Dif}^{G_V}_{G_V/V}$.

Remark 2.4.1. Consider the commutative diagram

$$\begin{array}{ccc} G & \xleftarrow{\eta} & G \times G \\ {\scriptstyle p}\downarrow\uparrow{\scriptstyle \varepsilon} & & {\scriptstyle \mathrm{pr}_1}\downarrow\uparrow{\scriptstyle \Delta} \\ S & \xleftarrow{p} & G, \end{array}$$

where η denotes the morphism $(x, y) \mapsto yx^{-1}$.[20] It induces morphisms

$$\eta' : \eta^{-1}(\mathcal{O}_G) \longrightarrow \mathcal{O}_{G \times G}$$

and

$$\Delta^{-1}(\eta') : p^{-1}\varepsilon^{-1}(\mathcal{O}_G) \longrightarrow \Delta^{-1}(\mathcal{O}_{G \times G}).$$

For every integer $n \geq 1$, set

$$\mathfrak{p}^n_{G/S} = \varepsilon^{-1}(\mathcal{O}_G)/\mathcal{I}_\varepsilon^{n+1}.$$

[21] Since the square formed by $\varepsilon, \eta, \Delta, p$ is cartesian, $\Delta^{-1}(\eta')$ induces an isomorphism of $\mathcal{O}_G$-modules

$$p^*(\mathfrak{p}^n_{G/S}) \xrightarrow{\sim} \mathcal{P}^n_{G/S}.$$

The differential operators on G/S of order $\leq n$ therefore correspond bijectively to the homomorphisms of $\mathcal{O}_G$-modules

$$p^*(\mathfrak{p}^n_{G/S}) \longrightarrow \mathcal{O}_G,$$

or equivalently to the homomorphisms of $\mathcal{O}_S$-modules

$$\mathfrak{p}^n_{G/S} \longrightarrow p_*(\mathcal{O}_G).$$

Under this bijection the right-invariant differential operators correspond to the composites

[20] Thus G acts on itself on the left by right translations.

[21] The editors corrected the original here: the relevant cartesian square is formed by $\varepsilon, \eta, \Delta, p$, not by $p, p, \eta, \mathrm{pr}_1$.

$$\mathfrak{p}^n_{G/S} \longrightarrow \mathcal{O}_S \xrightarrow{\text{can.}} p_*(\mathcal{O}_G)\,.$$

This recovers the isomorphism of Theorem 2.4.

2.5. [22] Let $\mathrm{Lie}(G)$ be the Lie algebra of G.[23] We shall define a homomorphism of $\Gamma(\mathcal{O}_S)$-Lie algebras

$$\alpha : \mathrm{Lie}(G) \longrightarrow U(G).$$

Let $s : S \to I_S$ be the zero section and let σ be the deviation of s defined in 1.2.0. Recall from II, 4.1 that $\mathrm{Lie}(G)$ is the set of morphisms $x : I_S \to G$ such that $x \circ s = \varepsilon_G$. The composite

$$S \underset{s}{\overset{\sigma}{\rightrightarrows}} I_S \xrightarrow{\ x\ } G$$

is an S-deviation of ε_G, hence an element of $U(G)$; in the notation of 1.2 it is denoted σx. Moreover, by 1.2.1 the map

$$\alpha : x \longmapsto \sigma x$$

is an isomorphism of $\mathcal{O}_S(S)$-modules from $\mathrm{Lie}(G)$ onto the submodule $\mathrm{Der}(\varepsilon_G) \subset U(G)$ formed by the S-derivations of ε_G. We show that α is a homomorphism of Lie algebras.[24]

Let

$$\rho' : U(G) \longrightarrow \mathrm{Dif}_{G/S}$$

be the algebra homomorphism which sends an S-deviation d of ε_G to the left-invariant differential operator ${}^G d$, as in 2.2.

Let

$$\rho : G \longrightarrow \mathrm{Aut}_S(G)$$

be the homomorphism of group functors which sends an S-morphism $g : T \to G$ to right translation by g on G_T, namely

$$G_T \simeq T \times_T G_T \xrightarrow{G_T \times g} G_T \times_T G_T \xrightarrow{m_T} G_T\,.$$

Recall also from 1.5 and II, 3.14 that

$$\mathrm{Lie}(\mathrm{Aut}\, G) = \mathrm{Lie}(\mathrm{Aut}_S(G)/S)(S)$$

identifies with the infinitesimal automorphisms of G, that is, the automorphisms of $I_S \times G$ which induce the identity on G.

Since ρ is a monomorphism, so is

$$\mathrm{Lie}(\rho) : \mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(\mathrm{Aut}_S(G)/S)$$

(see, for example, Exposé II, editorial note 50). Consequently

$$\mathrm{Lie}(\rho) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(\mathrm{Aut}\, G)$$

is injective.

On the other hand, by 1.5 the map β, which assigns to every infinitesimal automorphism u of G the differential operator D_u,

[22] The editors reordered this paragraph, beginning with the definition of $\alpha : \mathrm{Lie}(G) \to U(G)$, and corrected the original as indicated in the note to 2.2.

[23] In this exposé, for an S-group scheme G (respectively, an S-scheme X), $\mathrm{Lie}(G)$ (respectively, $\mathrm{Lie}(\mathrm{Aut}\, X)$) denotes, in the notation of Exposé II, $\mathrm{Lie}(G/S)(S)$ (respectively, $\mathrm{Lie}(\mathrm{Aut}_S(X)/S)(S)$). It is a $\Gamma(\mathcal{O}_S)$-Lie algebra by II, 4.11 and 3.14.

[24] See also Exposé II, 4.11.

$$G \simeq S \times G \xrightarrow{\sigma \times G} I_S \times G \xrightarrow{u} I_S \times G \xrightarrow{\mathrm{pr}_2} G$$

is an isomorphism from $\mathrm{Lie}(\mathrm{Aut}\, G)$ onto the Lie subalgebra of $\mathrm{Dif}_{G/S}$ formed by the $p^{-1}(\mathcal{O}_S)$-derivations of $\mathcal{O}_G$.

For every $x \in \mathrm{Lie}(G)$, the following commutative square determines the image of x under $\mathrm{Lie}(\rho)$:

$$\begin{array}{ccc} I_S \times G & \xrightarrow{\mathrm{Lie}(\rho)(x)} & I_S \times G \\ {\scriptstyle x\times G}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ G \times G & \xrightarrow{\quad m \quad} & G\,. \end{array}$$

It follows from this diagram that the image of $\mathrm{Lie}(\rho)(x)$ under β is the composite deviation

$$G \simeq S \times G \underset{s\times G}{\overset{\sigma\times G}{\rightrightarrows}} I_S \times G \xrightarrow{G\times x} G \times G \xrightarrow{m} G$$

which, by the editorial note to 2.2, is precisely

$$^G(\sigma x) = \rho'(\alpha(x)).$$

Thus one obtains the commutative diagram

$$\begin{array}{ccc} \mathrm{Lie}(G) & \xhookrightarrow{\mathrm{Lie}(\rho)} & \mathrm{Lie}(\mathrm{Aut}\, G) \\ {\scriptstyle \alpha}\downarrow & & \downarrow{\scriptstyle \beta} \\ U(G) & \xhookrightarrow{\quad \rho' \quad} & \mathrm{Dif}_{G/S} \end{array}$$

in which $\mathrm{Lie}(\rho)$, β, and ρ' are homomorphisms of Lie algebras. Since ρ' is injective, α is also a homomorphism of Lie algebras. Hence:

Proposition. *The map α is an isomorphism of $\mathcal{O}_S(S)$-Lie algebras from* $\mathrm{Lie}(G)$ *onto the Lie algebra of S-derivations of ε_G; the latter is itself isomorphic, via* $\mathrm{Lie}(\rho)$*, to the Lie algebra of left-invariant S-derivations of G.* [25]

3. Coalgebras and Cartier Duality

3.1. Let S be a scheme (or, more generally, a ringed space). An *$\mathcal{O}_S$-coalgebra*[26] is a pair $(\mathcal{U}, \Delta_{\mathcal{U}})$ consisting of an $\mathcal{O}_S$-module $\mathcal{U}$ and a morphism of $\mathcal{O}_S$-modules

$$\Delta_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U},$$

called the *diagonal morphism* or *comultiplication*, such that:

(i) $\sigma \circ \Delta_{\mathcal{U}} = \Delta_{\mathcal{U}}$, where $\sigma(a \otimes b) = b \otimes a$.

(ii) The square

[25] There exist Lie algebras $\mathfrak{g}$ over a ring A for which the canonical map $\mathfrak{g} \to U(\mathfrak{g})$ is not injective; see Bourbaki, *Lie Groups and Lie Algebras*, I.2, Exercise 9. The result above shows, since α factors through

$$\mathrm{Lie}(G) \longrightarrow U(\mathrm{Lie}(G)) \longrightarrow U(G),$$

that this cannot happen for "algebraic" Lie algebras, namely those of the form $\mathrm{Lie}(G)$ for an A-group scheme G.

[26] The term *cogebra* is also used; see Bourbaki, *Algebra*, III, §11.1. Notice also that in this exposé, as well as in Exposé VII$_B$, we work in the category of cocommutative coalgebras (that is, coalgebras satisfying condition (i)). This is crucial for defining the product and the notion of a coalgebra in groups; see 3.1.0 and 3.2.

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}}} & \mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{U} \\ {\scriptstyle\Delta_{\mathcal{U}}}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_{\mathcal{U}}\otimes\Delta_{\mathcal{U}}} \\ \mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}} & \mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{U} \end{array}$$

is commutative.

(iii) There exists a morphism of $\mathcal{O}_S$-modules

$$\varepsilon_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{O}_S,$$

called the *augmentation*, such that the two composites

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{U} \xrightarrow{\mathrm{id}_{\mathcal{U}}\otimes\varepsilon_{\mathcal{U}}} \mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{O}_S \simeq \mathcal{U}$$

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{U} \xrightarrow{\varepsilon_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}} \mathcal{O}_S\otimes_{\mathcal{O}_S}\mathcal{U} \simeq \mathcal{U}$$

are the identity morphism of $\mathcal{U}$.

If $\varepsilon_{\mathcal{U}}$ and $\varepsilon'_{\mathcal{U}}$ are two augmentations, then

$$\varepsilon_{\mathcal{U}} = (\varepsilon_{\mathcal{U}}\otimes\varepsilon'_{\mathcal{U}})\circ\Delta_{\mathcal{U}} = \varepsilon'_{\mathcal{U}};$$

the augmentation is therefore uniquely determined by (iii).

If $(\mathcal{U},\Delta_{\mathcal{U}})$ and $(\mathcal{V},\Delta_{\mathcal{V}})$ are two $\mathcal{O}_S$-coalgebras, a *morphism* from the first to the second is a morphism of $\mathcal{O}_S$-modules $f : \mathcal{U}\to\mathcal{V}$ for which the diagrams

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{f} & \mathcal{V} \\ {\scriptstyle\Delta_{\mathcal{U}}}\downarrow & & \downarrow{\scriptstyle\Delta_{\mathcal{V}}} \\ \mathcal{U}\otimes\mathcal{U} & \xrightarrow{f\otimes f} & \mathcal{V}\otimes\mathcal{V} \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathcal{U} & & \xrightarrow{f} & & \mathcal{V} \\ & {\scriptstyle\varepsilon_{\mathcal{U}}}\searrow & & \swarrow{\scriptstyle\varepsilon_{\mathcal{V}}} & \\ & & \mathcal{O}_S & & \end{array}$$

are commutative. Morphisms of coalgebras compose like morphisms of $\mathcal{O}_S$-modules, so we may speak of the *category of $\mathcal{O}_S$-coalgebras*.

3.1.0. [27] This category has finite products: its final object is the $\mathcal{O}_S$-module $\mathcal{O}_S$, with the identity as comultiplication. The product of two coalgebras $(\mathcal{U},\Delta_{\mathcal{U}})$ and $(\mathcal{V},\Delta_{\mathcal{V}})$ is the tensor product $\mathcal{U}\otimes_{\mathcal{O}_S}\mathcal{V}$, whose comultiplication is the composite

$$\mathcal{U}\otimes\mathcal{V} \xrightarrow{\Delta_{\mathcal{U}}\otimes\Delta_{\mathcal{V}}} \mathcal{U}\otimes\mathcal{U}\otimes\mathcal{V}\otimes\mathcal{V} \xrightarrow{\mathrm{id}_{\mathcal{U}}\otimes\sigma\otimes\mathrm{id}_{\mathcal{V}}} \mathcal{U}\otimes\mathcal{V}\otimes\mathcal{U}\otimes\mathcal{V}$$

where $\sigma(a\otimes b) = b\otimes a$. The canonical projections of $\mathcal{U}\otimes\mathcal{V}$ onto the factors $\mathcal{U}$ and $\mathcal{V}$ are $\mathrm{id}_{\mathcal{U}}\otimes\varepsilon_{\mathcal{V}}$ and $\varepsilon_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{V}}$, respectively.[28] The "diagonal morphism" $\mathcal{U}\to\mathcal{U}\otimes\mathcal{U}$, corresponding to the pair $(\mathrm{id}_{\mathcal{U}},\mathrm{id}_{\mathcal{U}})$, is precisely the comultiplication $\Delta_{\mathcal{U}}$.

3.1.1. Let $\mathcal{A}$ be a commutative $\mathcal{O}_S$-algebra which, as an $\mathcal{O}_S$-module, is locally free and of finite type. Put

$$\mathcal{A}^* = \mathcal{H}om_{\mathcal{O}_S\text{-Mod.}}(\mathcal{A},\mathcal{O}_S).$$

[27] The editors added the numbering 3.1.0 for later references.

[28] The preceding assertion was added by the editors. Recall also that, to show that $\mathcal{U}\otimes\mathcal{V}$ is indeed the product of $\mathcal{U}$ and $\mathcal{V}$ in the category of cocommutative $\mathcal{O}_S$-cogebras, one argues as follows. Given an arbitrary $\mathcal{O}_S$-cogebra $\mathcal{E}$ and cogebra morphisms $f : \mathcal{E}\to\mathcal{U}$ and $g : \mathcal{E}\to\mathcal{V}$, any cogebra morphism $\varphi : \mathcal{E}\to\mathcal{U}\otimes\mathcal{V}$ satisfying $\mathrm{pr}_{\mathcal{U}}\circ\varphi = f$ and $\mathrm{pr}_{\mathcal{V}}\circ\varphi = g$ must equal $(f\otimes g)\circ\Delta_{\mathcal{E}}$; and this is a cogebra morphism if and only if it also equals $(g\otimes f)\circ\Delta_{\mathcal{E}}$.

The canonical morphism

$$\varphi : \mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}^* \longrightarrow (\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{A})^*$$

is invertible. If $m : \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ is the morphism defining the multiplication of $\mathcal{A}$, composition gives a diagonal morphism

$$\Delta_{\mathcal{A}^*} : \quad \mathcal{A}^* \xrightarrow{m^*} (\mathcal{A} \otimes \mathcal{A})^* \xrightarrow{\varphi^{-1}} \mathcal{A}^* \otimes \mathcal{A}^* .$$

This diagonal morphism plainly makes $\mathcal{A}^*$ into an $\mathcal{O}_S$-coalgebra whose augmentation is the transpose of the morphism $\mathcal{O}_S \to \mathcal{A}$ defined by the unit section of $\mathcal{A}$. Moreover, it is clear that:

The functor $\mathcal{A} \mapsto \mathcal{A}^*$ is an anti-equivalence from the category of $\mathcal{O}_S$-algebras which are locally free and of finite type as $\mathcal{O}_S$-modules onto the category of $\mathcal{O}_S$-coalgebras which are locally free and of finite type as $\mathcal{O}_S$-modules.

3.1.2. To every $\mathcal{O}_S$-coalgebra $\mathcal{U}$ there is canonically associated an S-functor

$$\mathrm{Spec}^* \mathcal{U} : (\mathbf{Sch}/S)^\circ \longrightarrow (\mathbf{Ens}) .$$

Indeed, for every S-scheme $q : T \to S$,

$$q^*(\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U}) \simeq q^*(\mathcal{U}) \otimes_{\mathcal{O}_T} q^*(\mathcal{U}),$$

so $q^*(\Delta_{\mathcal{U}})$ makes $\mathcal{U}_T = q^*(\mathcal{U})$ into an $\mathcal{O}_T$-coalgebra. Thus, by definition and with an obvious abuse of notation,[29]

$$(\mathrm{Spec}^* \mathcal{U})(T) = \{x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1, \quad \Delta_{\mathcal{U}_T}(x) = x \otimes x\} .$$

Sections x of $\mathcal{U}_T$ plainly correspond to morphisms of $\mathcal{O}_T$-modules $\xi : \mathcal{O}_T \to \mathcal{U}_T$. The conditions $\varepsilon(x) = 1$ and $\Delta(x) = x \otimes x$ say exactly that ξ is a morphism of coalgebras. Hence also

$$(\mathrm{Spec}^* \mathcal{U})(T) = \mathrm{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{U}_T).$$

In particular:

Proposition 3.1.2.1. [30] *Let $\mathcal{A}$ be a commutative $\mathcal{O}_S$-algebra which is locally free of finite type as an $\mathcal{O}_S$-module. Then the S-functor* $\mathrm{Spec}^* \mathcal{A}^*$ *is represented by* $\mathrm{Spec}\, \mathcal{A}$.

Indeed, for every S-scheme T, there are canonical isomorphisms

$$\begin{aligned}(\mathrm{Spec}^* \mathcal{A}^*)(T) &= \mathrm{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{A}_T^*) \\ &\simeq \mathrm{Hom}_{\mathcal{O}_T\text{-alg.}}(\mathcal{A}_T, \mathcal{O}_T) \simeq (\mathrm{Spec}\, \mathcal{A})(T).\end{aligned}$$

3.2. An *$\mathcal{O}_S$-coalgebra in groups*, that is, a group in the category of $\mathcal{O}_S$-coalgebras, consists of an $\mathcal{O}_S$-coalgebra $(\mathcal{U}, \Delta_{\mathcal{U}})$ together with three morphisms of $\mathcal{O}_S$-coalgebras

$$m_{\mathcal{U}} : \mathcal{U} \otimes \mathcal{U} \longrightarrow \mathcal{U}, \qquad \eta_{\mathcal{U}} : \mathcal{O}_S \longrightarrow \mathcal{U}, \qquad c_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{U},$$

satisfying conditions (ii)*, (iii)*, and (vi) below. The requirement that $m_{\mathcal{U}}$ be a morphism of coalgebras is expressed by the commutativity of diagrams (iv) and (v):

[29] For $x \otimes y \in \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \Gamma(T, \mathcal{U}_T)$, its image in $\Gamma(T, \mathcal{U}_T \otimes_{\mathcal{O}_T} \mathcal{U}_T)$ is again denoted $x \otimes y$.

[30] The editors added the numbering 3.1.2.1 for later references. Notice also that the S-functor $\mathrm{Spec}^* \mathcal{U}$ is a sheaf for the Zariski topology; it is even an fpqc sheaf if $\mathcal{U}$ is a quasi-coherent $\mathcal{O}_S$-module.

(iv)
$$\begin{array}{ccc} \mathcal{U}\otimes\mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \\ {\scriptstyle \Delta_{\mathcal{U}}\otimes\Delta_{\mathcal{U}}}\downarrow & & \\ \mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U} & & \downarrow{\scriptstyle \Delta_{\mathcal{U}}} \\ {\scriptstyle \mathrm{id}_{\mathcal{U}}\otimes\sigma\otimes\mathrm{id}_{\mathcal{U}}}\downarrow & & \\ \mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U} & \xrightarrow{m_{\mathcal{U}}\otimes m_{\mathcal{U}}} & \mathcal{U}\otimes\mathcal{U} \end{array}$$

(v)
$$\begin{array}{ccc} \mathcal{U}\otimes\mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \\ & {\scriptstyle \varepsilon_{\mathcal{U}}\otimes\varepsilon_{\mathcal{U}}}\searrow \quad \swarrow{\scriptstyle \varepsilon_{\mathcal{U}}} & \\ & \mathcal{O}_S & \end{array}$$

(ii)* The square

$$\begin{array}{ccc} \mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U} & \xrightarrow{\mathrm{id}_{\mathcal{U}}\otimes m_{\mathcal{U}}} & \mathcal{U}\otimes\mathcal{U} \\ {\scriptstyle m_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}}\downarrow & & \downarrow{\scriptstyle m_{\mathcal{U}}} \\ \mathcal{U}\otimes\mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \end{array}$$

is commutative.

(iii)* The two composites

$$\mathcal{U}\simeq\mathcal{U}\otimes\mathcal{O}_S \xrightarrow{\mathrm{id}_{\mathcal{U}}\otimes\eta_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}$$

$$\mathcal{U}\simeq\mathcal{O}_S\otimes\mathcal{U} \xrightarrow{\eta_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}.$$

are the identity morphism of $\mathcal{U}$.

(vi) The composite

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{c_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}\,.$$

is equal to $\eta_{\mathcal{U}}\circ\varepsilon_{\mathcal{U}}$.

3.2.1. The morphisms $\eta_{\mathcal{U}}$ and $c_{\mathcal{U}}$ are uniquely determined by $m_{\mathcal{U}}$. Conditions (ii)* and (iii)* simply say that $m_{\mathcal{U}}$ makes $\mathcal{U}$ into an $\mathcal{O}_S$-algebra whose unit section is the image under $\eta_{\mathcal{U}}$ of the unit section of $\mathcal{O}_S$. Condition (iv) also says that the diagonal morphism $\Delta_{\mathcal{U}}$ is compatible with multiplication. Indeed,

$$\Delta_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{U}\otimes\mathcal{U}$$

must be a homomorphism of coalgebras in groups, which also implies the commutativity of the triangle

(v)*
$$\begin{array}{ccc} & \mathcal{O}_S & \\ & {\scriptstyle \eta_{\mathcal{U}}}\swarrow \quad \searrow{\scriptstyle \eta_{\mathcal{U}}\otimes\eta_{\mathcal{U}}} & \\ \mathcal{U} & \xrightarrow[\Delta_{\mathcal{U}}]{} & \mathcal{U}\otimes\mathcal{U}\,. \end{array}$$

On the other hand, as in every category, the antipode $c_{\mathcal{U}}$ is an isomorphism from $\mathcal{U}$ onto the "opposite" group object;[31] in particular, $c_{\mathcal{U}}$ induces an algebra isomorphism from $\mathcal{U}$ onto the opposite algebra $\mathcal{U}^\circ$.

[31] That is, the object endowed with the multiplication $m'_{\mathcal{U}} = m_{\mathcal{U}}\circ\sigma$.

3.2.2. Since the functor $\mathcal{U} \mapsto \mathrm{Spec}^*\mathcal{U}$ commutes with finite products, it carries a coalgebra in groups to a group-valued S-functor. Indeed, for every S-scheme T, the elements

$$x \in \Gamma(T, \mathcal{U}_T)$$

belonging to $(\mathrm{Spec}^*\mathcal{U})(T)$ form a group under the multiplication of the algebra $\Gamma(T, \mathcal{U}_T)$, and the inverse of x is $c_{\mathcal{U}}(x)$. By 3.1.2.1:

Scholium 3.2.2.1. [32] *Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups, finite and locally free as an $\mathcal{O}_S$-module. Then the group-valued S-functor* $\mathrm{Spec}^*\mathcal{U}$ *is represented by the finite locally free S-group* $\mathrm{Spec}\mathcal{U}^*$.

Remark 3.2.2.2. Let $\mathcal{L}$ be an $\mathcal{O}_S$-Lie algebra and $\mathcal{U}(\mathcal{L})$ its enveloping algebra, that is, the sheaf on S associated with the presheaf assigning to every open subset V the enveloping algebra

$$U\big(\Gamma(V, \mathcal{L})\big)$$

of the Lie algebra $\Gamma(V, \mathcal{L})$.

Every homomorphism from $\mathcal{L}$ to the Lie algebra underlying an associative $\mathcal{O}_S$-algebra factors uniquely through the canonical morphism $\mathcal{L} \to \mathcal{U}(\mathcal{L})$. Besides the functoriality of $\mathcal{U}(\mathcal{L})$ in $\mathcal{L}$, this universal property also shows that the enveloping algebra of a product of Lie algebras is identified with the tensor product of their enveloping algebras.

In particular, the diagonal morphism $\delta : \mathcal{L} \to \mathcal{L} \times \mathcal{L}$ induces an algebra homomorphism

$$\Delta : \mathcal{U}(\mathcal{L}) \longrightarrow \mathcal{U}(\mathcal{L} \times \mathcal{L}) \simeq \mathcal{U}(\mathcal{L}) \otimes \mathcal{U}(\mathcal{L}).$$

The zero morphism $\mathcal{L} \to 0$ induces a homomorphism

$$\varepsilon : \mathcal{U}(\mathcal{L}) \longrightarrow \mathcal{U}(0) \simeq \mathcal{O}_S.$$

The isomorphism $x \mapsto -x$ from $\mathcal{L}$ onto the opposite Lie algebra $\mathcal{L}^\circ$ induces an anti-isomorphism c of the algebra $\mathcal{U}(\mathcal{L})$. One then checks readily that the multiplication m of $\mathcal{U}(\mathcal{L})$ makes $(\mathcal{U}(\mathcal{L}), \Delta)$ into an $\mathcal{O}_S$-coalgebra in groups with augmentation ε and antipode c.[33]

3.2.3. [34] Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups. We shall see that the group-valued S-functor

$$G = \mathrm{Spec}^*\mathcal{U}$$

is *very good*, in the sense of Exposé II, 4.6 and 4.10.

Let $\mathcal{M}$ be a free $\mathcal{O}_S$-module of rank r, and let $T \to S$ be an S-scheme. Since

$$I_T(\mathcal{M}) = \mathrm{Spec}(\mathcal{O}_T \oplus \mathcal{M}_T),$$

the morphism $\pi : I_T(\mathcal{M}) \to T$ is affine, and

$$\pi_*\big(\mathcal{U}_{I_T(\mathcal{M})}\big) = \mathcal{U}_T \otimes_{\mathcal{O}_T} \pi_*\big(\mathcal{O}_{I_T(\mathcal{M})}\big) = \mathcal{U}_T \otimes_{\mathcal{O}_T} (\mathcal{O}_T \oplus \mathcal{M}_T).$$

Consequently,

$$\Gamma\big(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}\big) \simeq \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \big(\mathcal{O}(T) \oplus \Gamma(T, \mathcal{M}_T)\big). \tag{1}$$

[32] The editors added this scholium, which is implicit in the original.

[33] The group-valued S-functor $\mathrm{Spec}^*\mathcal{U}(\mathcal{L})$ is not representable in general. Later, in 5.5, it will be shown that if S is a scheme of characteristic p, if $\mathcal{L}$ is finite locally free over $\mathcal{O}_S$, and if $\mathcal{U}_p(\mathcal{L})$ is its restricted enveloping algebra (see 5.3), then $\mathrm{Spec}^*\mathcal{U}_p(\mathcal{L})$ is represented by a finite locally free S-group.

[34] The editors expanded this paragraph.

Let $(d_1, \dots, d_r)$ be a basis of $\mathcal{M}$. An element

$$u_0 + \sum_i u_i d_i \in \Gamma\big(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}\big)$$

belongs to $G(I_T(\mathcal{M}))$ if and only if

$$1 = \varepsilon\left(u_0 + \sum_i u_i d_i\right) = \varepsilon(u_0) + \sum_i \varepsilon(u_i) d_i$$

and

$$\left(u_0 + \sum_i u_i d_i\right) \otimes \left(u_0 + \sum_i u_i d_i\right) = \Delta\left(u_0 + \sum_i u_i d_i\right) = \Delta(u_0) + \sum_i \Delta(u_i) d_i.$$

Equivalently,

$$\begin{cases} \varepsilon(u_0) = 1, & \Delta u_0 = u_0 \otimes u_0 \qquad\quad \big(\text{that is, } u_0 \in G(T)\big), \\ \varepsilon(u_i) = 0, & \Delta(u_i) = u_i \otimes u_0 + u_0 \otimes u_i \quad (i = 1, \dots, r). \end{cases} \tag{2}$$

Moreover, the morphism $G(I_T(\mathcal{M})) \to G(T)$ corresponding to the zero section of $I_T(\mathcal{M}) \to T$ is given by

$$u_0 + \sum_i u_i d_i \longmapsto u_0.$$

Combining this with (1) and (2), one sees that if $\mathcal{N}$ is a second free $\mathcal{O}_S$-module of finite rank, then the diagram of sets

$$\begin{array}{ccc} G(I_T(\mathcal{M} \oplus \mathcal{N})) & \longrightarrow & G(I_T(\mathcal{N})) \\ \downarrow & & \downarrow \\ G(I_T(\mathcal{M})) & \longrightarrow & G(T) \end{array}$$

is cartesian; that is, G satisfies condition (E) of Exposé II, 3.5.

Write

$$\operatorname{Prim} \Gamma(T, \mathcal{U}_T)$$

for the $\mathcal{O}(T)$-submodule of $\Gamma(T, \mathcal{U}_T)$ formed by the *primitive elements*, that is, the elements u satisfying, with the abuse of notation noted in 3.1.2,

$$\Delta u = u \otimes 1 + 1 \otimes u, \qquad \varepsilon(u) = 0.$$

[35]

Since $(\operatorname{Lie} G)(T)$ is the set of elements $u_0 + ud \in G(I_T)$ above the unit element $u_0 = 1$ of $G(T)$, there is an isomorphism of $\mathcal{O}(T)$-modules, functorial in T,[36]

$$(\operatorname{Lie} G)(T) \simeq \operatorname{Prim} \Gamma(T, \mathcal{U}_T).$$

On the other hand, (1) yields

$$\operatorname{Prim} \Gamma\big(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}\big) \simeq \operatorname{Prim} \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})).$$

[35] The second condition follows from the first, since the first gives $u = (\mathrm{id} \otimes \varepsilon)\Delta(u) = u + \varepsilon(u)$, and hence $\varepsilon(u) = 0$.

[36] The $\mathcal{O}_S$-module structure on $\operatorname{Lie} G$ is defined in Exposé II, Proposition 3.6.

It follows that the natural morphism of $\mathcal{O}(I_T(\mathcal{M}))$-modules

$$(\mathrm{Lie}\, G)(T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})) \longrightarrow (\mathrm{Lie}\, G)(I_T(\mathcal{M}))$$

is an isomorphism; thus $\mathrm{Lie}\, G$ is a good $\mathcal{O}_S$-module (see Exposé II, Definition 4.4).

Therefore G is a good group-valued S-functor (see Exposé II, Definition 4.6), and by Exposé II, 4.7.2, $\mathrm{Lie}\, G$ carries an $\mathcal{O}_S$-bilinear Lie bracket satisfying the Jacobi identity. It remains to show that G is very good, that is, that the bracket on $(\mathrm{Lie}\, G)(T)$ satisfies $[u,u]=0$ for every $u \in (\mathrm{Lie}\, G)(T)$; see Exposé II, 4.10.

Let u, v be two elements of $(\mathrm{Lie}\, G)(T)$, that is, two primitive elements of $\Gamma(T,\mathcal{U}_T)$. Put

$$I = \mathrm{Spec}\, \mathcal{O}_S[d]/(d^2), \qquad I' = \mathrm{Spec}\, \mathcal{O}_S[d']/(d'^2).$$

Since the composition law on $G(I \times I')$ is induced by the multiplication in $\mathcal{U}_{I\times I'}$, one has in $G(I \times I')$

$$\begin{aligned}(1+ud)(1+vd')(1+ud)^{-1}(1+vd')^{-1} &\\ = (1+ud)(1+vd')(1-ud)(1-vd') &\\ = 1+(uv-vu)dd'.&\end{aligned}$$

By the description of the bracket $[u,v]$ given before Proposition 4.8 of Exposé II, it follows that

$$[u,v] = uv - vu,$$

where the right-hand side is the commutator of u and v in the algebra $\Gamma(T,\mathcal{U}_T)$. Hence $[u,u]=0$, and we have proved:

Proposition. *Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups. The group-valued S-functor $G = \mathrm{Spec}^*\,\mathcal{U}$ is very good, and there is an isomorphism*

$$\mathrm{Lie}\, G \simeq \mathrm{Prim}\, \mathcal{W}(\mathcal{U})$$

of $\mathcal{O}_S$-Lie algebras, where $\mathrm{Prim}\, \mathcal{W}(\mathcal{U})$ denotes the functor which assigns to every $T \to S$ the $\mathcal{O}(T)$-Lie algebra of primitive elements of $\mathcal{W}(\mathcal{U})(T) = \Gamma(T,\mathcal{U}_T)$. [37]

3.3. Suppose finally that $\mathcal{U}$ is a *coalgebra in commutative groups*, that is, that the triangle

(i)*
$$\begin{array}{ccc} \mathcal{U}\otimes\mathcal{U} & \xrightarrow{\quad\sigma\quad} & \mathcal{U}\otimes\mathcal{U} \\ & {\scriptstyle m_{\mathcal{U}}}\searrow \quad \swarrow{\scriptstyle m_{\mathcal{U}}} & \\ & \mathcal{U} & \end{array}$$

is commutative, or equivalently that $m_{\mathcal{U}}$ makes $\mathcal{U}$ into a commutative $\mathcal{O}_S$-algebra. Conditions (i), (ii), (iii), (iv), (v), (vi), (i)*, (ii)*, (iii)*, and (v)* then also say that $\mathcal{U}$ is a cogroup in the category of commutative $\mathcal{O}_S$-algebras. Thus, if in addition $\mathcal{U}$ is a quasi-coherent $\mathcal{O}_S$-module, the affine S-scheme $\mathrm{Spec}\,\mathcal{U}$ is a commutative S-group scheme.

In this case the diagonal morphism Δ' of $\mathcal{O}_S[T,T^{-1}]$ sends T to $T\otimes T$. Consequently, morphisms of S-groups from $\mathrm{Spec}\,\mathcal{U}$ to $\mathbb{G}_{m,S}$ (Exposé I, 4.3.2) correspond bijectively to morphisms of unital $\mathcal{O}_S$-algebras

$$\varphi : \mathcal{O}_S[T,T^{-1}] \longrightarrow \mathcal{U}$$

such that

$$(\varphi\otimes\varphi)\circ\Delta' = \Delta_{\mathcal{U}}\circ\varphi.$$

[37] The editors added this proposition to summarize the preceding discussion.

In this situation, $\varepsilon_{\mathcal{U}} \circ \varphi$ is the neutral element of $\mathbb{G}_{m,S}(S)$, that is, the augmentation ε'. Such a morphism φ is determined by the image $\varphi(T)$, which must be an invertible element $x \in \mathcal{U}$ satisfying

$$\Delta_{\mathcal{U}} x = x \otimes x, \qquad \varepsilon_{\mathcal{U}}(x) = \varepsilon'(T) = 1.$$

Therefore

$$\mathrm{Hom}_{S\text{-gr.}}(\mathrm{Spec}\mathcal{U}, \mathbb{G}_{m,S}) \simeq (\mathrm{Spec}^*\mathcal{U})(S).$$

Since this formula remains valid after arbitrary base change, it gives

$$\mathrm{Spec}^*\mathcal{U} \simeq \mathrm{Hom}_{S\text{-gr.}}(\mathrm{Spec}\mathcal{U}, \mathbb{G}_{m,S}).$$

We have therefore proved:

Proposition 3.3.0. *If $\mathcal{U}$ is an $\mathcal{O}_S$-coalgebra in commutative groups, quasi-coherent as an $\mathcal{O}_S$-module, then the affine S-scheme $G = \mathrm{Spec}\mathcal{U}$ is a commutative S-group scheme, and there is an isomorphism of group-valued S-functors*

$$\mathrm{Spec}^*\mathcal{U} \simeq \mathrm{Hom}_{S\text{-}gr.}(G, \mathbb{G}_{m,S}).$$

If, moreover, $\mathcal{U}$ is locally free of finite type as an $\mathcal{O}_S$-module, then by 3.1.2.1 the group-valued S-functor $\mathrm{Spec}^*\mathcal{U}$ is represented by $\mathrm{Spec}\mathcal{U}^*$. Hence:

Proposition 3.3.1 (Cartier duality). *The functor*

$$\mathscr{A}(G) \longmapsto \mathscr{A}(G)^* = \mathcal{H}om_{\mathcal{O}_S\text{-}Mod.}(\mathscr{A}(G), \mathcal{O}_S)$$

induces a duality on the category of commutative finite locally free S-group schemes; it associates to G the S-group* $\mathrm{Hom}_{S\text{-}gr.}(G, \mathbb{G}_{m,S})$.

4. "Frobeniuseries"

Fix a prime number p, and let $(\mathbf{Sch}/\mathbb{F}_p)$ be the category of schemes of characteristic p, that is, schemes over the prime field $\mathbb{F}_p$. Following the general conventions of this seminar, we identify $(\mathbf{Sch}/\mathbb{F}_p)$ with a subcategory of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$ by means of the functor h of Exposé I, 1.1. We likewise use the isomorphism

$$\mathrm{Hom}(h_X, F) \xrightarrow{\sim} F(X)$$

defined there to identify these sets whenever X is an $\mathbb{F}_p$-scheme and F is an object of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$.

Notation 4.0. [38] *If T is an $\mathbb{F}_p$-scheme, a T-functor is a morphism $q : F \to T$ in $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$ with target T. For every T-scheme $r : X \to T$, the set of T-morphisms $X \to F$, that is, of $\mathbb{F}_p$-morphisms $s : X \to F$ satisfying $q \circ s = r$, is denoted*

$$q(r), \qquad q(X/T), \qquad F(r), \qquad \text{or} \qquad F(X/T).$$

It may also be denoted $F(X)$ when no confusion with $\mathrm{Hom}(h_X, F)$ *can result.*

---

*A duality of a category $\mathscr{C}$ is a pair (D, φ) consisting of a contravariant functor $D : \mathscr{C} \to \mathscr{C}$ and a natural isomorphism $\varphi : \mathrm{Id}_{\mathscr{C}} \to DD$, such that the isomorphisms $\varphi D : D \to DDD$ and $D\varphi^{-1} : DDD \to D$ are inverse to one another. The editors corrected $D\varphi$ to $D\varphi^{-1}$.

[38] The editors added the number 4.0 for later references.

4.1. For every scheme S of characteristic p, let $\mathrm{fr}(S)$, or simply fr, denote the endomorphism of S which is the identity on the underlying topological space and which sends a section x of $\mathcal{O}_S$ over an open subset to x^p.

The assignment

$$\mathrm{fr} : S \longmapsto \mathrm{fr}(S)$$

is an endomorphism of the identity functor of $(\mathbf{Sch}/\mathbb{F}_p)$.[39] This has the following consequences. If E is an $\mathbb{F}_p$-functor, the assignment which sends every $\mathbb{F}_p$-scheme S to the endomorphism $E(\mathrm{fr}(S))$ of $E(S)$ is a functorial endomorphism of E, denoted $\mathrm{fr}(E)$ or simply fr. This notation is compatible with the identification of $(\mathbf{Sch}/\mathbb{F}_p)$ with a subcategory of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$. Moreover, $E \mapsto \mathrm{fr}(E)$ is itself an endomorphism of the identity functor of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$, again denoted fr.[40]

For every $\mathbb{F}_p$-scheme S and every S-functor $q : X \to S$, let $X^{(p/S)}$, or $X^{(p)}$, denote the pullback of X along $\mathrm{fr}(S)$:

$$\begin{array}{ccc} X^{(p/S)} & \xrightarrow{\mathrm{pr}_X} & X \\ \downarrow & & \downarrow{\scriptstyle q} \\ S & \xrightarrow{\mathrm{fr}(S)} & S\,. \end{array}$$

The commutative square

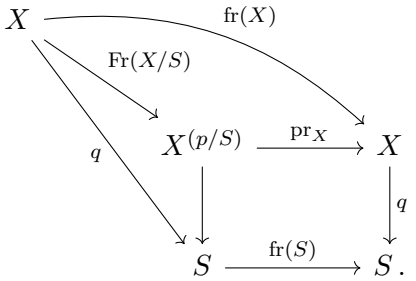
$$\begin{array}{ccc} X & \xrightarrow{\mathrm{fr}(X)} & X \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle q} \\ S & \xrightarrow{\mathrm{fr}(S)} & S \end{array}$$

induces an S-morphism

$$\mathrm{Fr}(X/S) : X \longrightarrow X^{(p/S)}$$

such that

$$\mathrm{fr}(X) = \mathrm{pr}_X \circ \mathrm{Fr}(X/S) :$$



We call $\mathrm{Fr}(X/S)$ the *Frobenius morphism of X relative to S*. Plainly the assignment

$$\mathrm{Fr} : X \longmapsto \mathrm{Fr}(X/S)$$

[39] Thus, for every morphism $f : Y \to X$ of $\mathbb{F}_p$-schemes, the following square commutes:

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ {\scriptstyle \mathrm{fr}(Y)}\downarrow & & \downarrow{\scriptstyle \mathrm{fr}(X)} \\ Y & \xrightarrow{f} & X\,. \end{array}$$

[40] The morphism $\mathrm{fr}(X)$ is called the *absolute* Frobenius morphism of X, to distinguish it from the relative Frobenius morphism $\mathrm{Fr}(X/S)$ introduced below.

is a functorial homomorphism.

[41] Let $r : T \to S$ be an S-scheme. For every

$$\varphi \in X(r) = \mathrm{Hom}_S(T, X),$$

one has the commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{\mathrm{Fr}(X/S)} & X^{(p/S)} & \xrightarrow{\mathrm{pr}_X} & X \\ {\scriptstyle\varphi}\uparrow & \searrow^{q} & \downarrow{\scriptstyle q^{(p/S)}} & & \downarrow{\scriptstyle q} \\ T & \xrightarrow{r} & S & \xrightarrow{\mathrm{fr}(S)} & S\,. \end{array}$$

By the definition of $X^{(p/S)}$ as a fiber product, pr_X induces a bijection

$$X^{(p/S)}(r) = \mathrm{Hom}_S(T, X^{(p/S)}) \xrightarrow{\sim} \mathrm{Hom}_S(T, X) = X(\mathrm{fr}(S) \circ r).$$

On the other hand,

$$r \circ \mathrm{fr}(T) = \mathrm{fr}(S) \circ r,$$

because fr is an endomorphism of the identity functor. Hence

$$\mathrm{Fr}(X/S)(r) : X(r) \longrightarrow X^{(p/S)}(r)$$

is characterized by commutativity of

$$\begin{array}{ccc} X(r) & \xrightarrow{\mathrm{Fr}(X/S)(r)} & X^{(p/S)}(r) \\ {\scriptstyle X(\mathrm{fr}(T))}\downarrow & & \downarrow{\scriptstyle \wr} \\ X(r \circ \mathrm{fr}(T)) & = & X(\mathrm{fr}(S) \circ r)\,. \end{array}$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, then $X^{(p)}$ is the subscheme defined by the ideal $\mathcal{I}^{\{p\}}$ generated by the p-th powers of local sections of $\mathcal{I}$. Moreover, $\mathrm{Fr}(X/S)$ is the canonical immersion

$$\mathrm{Spec}(\mathcal{O}_S/\mathcal{I}) \longrightarrow \mathrm{Spec}(\mathcal{O}_S/\mathcal{I}^{\{p\}}).$$

4.1.1. [42] Let $t : T \to S$ be a base change and put

$$X_T = X \times_{q,t} T.$$

Consider the pullback of X_T along $\mathrm{fr}(T)$:

$$\begin{array}{ccccc} (X_T)^{(p/T)} & \longrightarrow & X_T & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle q} \\ T & \xrightarrow{\mathrm{fr}(T)} & T & \xrightarrow{t} & S\,. \end{array}$$

[41] The editors expanded the original in the remainder of this paragraph.

[42] The editors expanded the original in this paragraph.

Since

$$t \circ \mathrm{fr}(T) = \mathrm{fr}(S) \circ t,$$

$(X_T)^{(p/T)}$ identifies with the pullback of $X^{(p/S)}$ along t. Equivalently, there is a canonical isomorphism

$$X_T^{(p/T)} \xrightarrow{\sim} (X^{(p/S)})_T.$$

Under this identification, $\mathrm{Fr}(X_T/T)$ identifies with the pullback $\mathrm{Fr}(X/S)_T$.

4.1.1.1. In particular, if $S = \operatorname{Spec} \mathbb{F}_p$, then $X^{(p/S)} = X$ and $\mathrm{Fr}(X/S) = \mathrm{fr}(X)$. Consequently, $X_T^{(p/T)}$ identifies with X_T, and $\mathrm{Fr}(X_T/T)$ with $\mathrm{fr}(X)_T$.

For example, if E is a set and E_T the constant T-scheme of type E, then

$$E_T^{(p/T)} \simeq E_T, \qquad \mathrm{Fr}(E_T/T) \simeq \mathrm{id}_{E_T}.$$

4.1.2. The functor $X \mapsto X^{(p/S)}$ plainly commutes with products. It therefore sends an S-group G to an S-group $G^{(p/S)}$. Since Fr is functorial,

$$\mathrm{Fr}(G/S) : G \longrightarrow G^{(p/S)}$$

is a homomorphism of S-groups. We denote its kernel by $\operatorname{Fr} G$.

If $r : T \to S$ is an S-scheme, the pointwise Frobenius diagram shows that the value of $\operatorname{Fr} G$ at r is the kernel of

$$G(\mathrm{fr}(T)) : G(r) \longrightarrow G(r \circ \mathrm{fr}(T)).$$

When $T = I_R$ is the scheme of dual numbers over an S-scheme R, the absolute Frobenius of I_R factors as

$$I_R \xrightarrow{\text{can.}} R \xrightarrow{\mathrm{fr}(R)} R \xrightarrow{\quad s \quad} I_R,$$

where s is the zero section. It follows that $(\operatorname{Fr} G)(I_R)$ contains the kernel $\mathrm{Lie}(G/S)(R)$ of

$$G(s) : G(I_R) \longrightarrow G(R),$$

and hence

$$\mathrm{Lie}(G/S) = \mathrm{Lie}(\operatorname{Fr} G/S).$$

4.1.3.

More generally, for every S-functor X, define the S-functor $X^{(p^n)}$ recursively in n by

$$X^{(p)} = X^{(p/S)} \qquad \text{and} \qquad X^{(p^n)} = \big(X^{(p^{n-1})}\big)^{(p)}.$$

Likewise, $\mathrm{Fr}^n(X/S)$, or simply Fr^n, denotes the composite functorial homomorphism

$$X \xrightarrow{\mathrm{Fr}(X/S)} X^{(p)} \xrightarrow{\mathrm{Fr}(X^{(p)}/S)} X^{(p^2)} \longrightarrow \cdots \longrightarrow X^{(p^{n-1})} \xrightarrow{\mathrm{Fr}(X^{(p^{n-1})}/S)} X^{(p^n)}.$$

By 4.1.1, $\mathrm{Fr}(X^{(p)}/S)$ coincides with $\mathrm{Fr}(X/S)^{(p)}$; in other words, the following diagram commutes:

$$\begin{array}{ccc} X^{(p)} & \longrightarrow & X \\ \downarrow{\scriptstyle \mathrm{Fr}(X^{(p)}/S)} & & \downarrow{\scriptstyle \mathrm{Fr}(X/S)} \\ X^{(p^2)} & \longrightarrow & X^{(p)}. \end{array}$$

If G is a group S-functor, then $G^{(p^n)}$ is again a group S-functor and $\mathrm{Fr}^n(G/S)$ is a homomorphism of group S-functors.

Definition. *We denote the kernel of* $\mathrm{Fr}^n(G/S)$ *by* $\mathrm{Fr}^n G$*, and say that* G *has height* $\leq n$ *if* $\mathrm{Fr}^n(G/S)$ *is zero, that is, if* $\mathrm{Fr}^n G = G$.

Lemma. *The group subfunctor* $\mathrm{Fr}^n G$ *of* G *is characteristic: for every* S*-scheme* T*, every endomorphism* φ *of the group* T*-functor* G_T *induces an endomorphism of* $(\mathrm{Fr}^n G)_T$.

Proof. By 4.1.1, the constructions of $G^{(p^n)}$ and $\mathrm{Fr}^n(G/S)$ commute with base change. We may therefore suppose that $T = S$. The assertion then follows from the fact that $\mathrm{Fr}^n(G/S)$ is a functorial homomorphism. □

4.1.4. Examples.

a) Let M first be an "abstract" abelian group, and let

$$G = D_S(M)$$

be the diagonalizable group of type M (I, 4.4). For every S-scheme T,

$$G(T) = \mathrm{Hom}_{(\mathbf{Ab})}(M, \Gamma(T, \mathcal{O}_T)^\times).$$

Since G is the pullback of the diagonalizable group $D(M)$ over $\mathbb{F}_p$, $G^{(p)}$ identifies with G, and $\mathrm{Fr}(G/S)(T)$ identifies with the endomorphism $x \mapsto x^p$ of $G(T)$ (4.1.1). In particular, when $M = \mathbb{Z}$, one has $D_S(M) = \mathbb{G}_{m,S}$. Hence:

> $\mathrm{Fr}\,\mathbb{G}_{m,S}$ is the S-group $\boldsymbol{\mu}_{p,S}$ which associates to every S-scheme T the group of p-th roots of unity in $\Gamma(T, \mathcal{O}_T)^\times$.

b) Now let S be a scheme of characteristic p, and let $\mathcal{E}$ be a sheaf of modules on S. By I, 4.6.2 there is a canonical isomorphism

$$\mathcal{W}(\mathcal{E})^{(p)} \xrightarrow{\sim} \mathcal{W}(\mathcal{E}^{(p)}),$$

where $\mathcal{E}^{(p)}$ is the pullback of $\mathcal{E}$ along $\mathrm{fr}(S)$. For every S-scheme $\pi : T \to S$, the map $\mathrm{Fr}(\mathcal{W}(\mathcal{E})/S)(\pi)$ is determined, by 4.1 (†), by the commutative triangle

$$\begin{array}{ccc} \Gamma(T, \pi^* \mathrm{fr}(S)^* \mathcal{E}) & \xrightarrow[\text{can.}]{\sim} & \Gamma(T, \mathrm{fr}(T)^* \pi^* \mathcal{E}) \\ & {\scriptstyle \mathrm{Fr}(\mathcal{W}(\mathcal{E})/S)(\pi)} \nwarrow \quad \nearrow {\scriptstyle f'} & \\ & \Gamma(T, \pi^* \mathcal{E}) & \end{array} ,$$

where f' is the map induced by $\mathrm{fr}(T)$.

In particular, if $\mathcal{E} = \mathcal{O}_S$, then $\mathcal{W}(\mathcal{E})$ identifies with the additive group $\mathbb{G}_{a,S}$. In this case $\mathcal{E}^{(p)} = \mathcal{E} = \mathcal{O}_S$, and the Frobenius morphism $\mathrm{Fr}(\mathbb{G}_{a,S}/S)$ sends $x \in \Gamma(T, \mathcal{O}_T)$ to x^p. Thus:

> $\mathrm{Fr}\,\mathbb{G}_{a,S}$ is the S-group $\boldsymbol{\alpha}_{p,S}$ which associates to every S-scheme T the group
>
> $$\{x \in \Gamma(T, \mathcal{O}_T) \mid x^p = 0\}.$$

c) Similarly, for every quasi-coherent $\mathcal{O}_S$-algebra $\mathcal{A}$, the scheme $(\operatorname{Spec}\mathcal{A})^{(p)}$ identifies with $\operatorname{Spec}\mathcal{A}^{(p)}$, the spectrum of the pullback of $\mathcal{A}$ along $\mathrm{fr}(S)$. If π denotes the endomorphism $x \mapsto x^p$ of the sheaf of rings $\mathcal{O}_S$, then

$$\mathcal{A}^{(p)} = \mathcal{A} \otimes_\pi \mathcal{O}_S, \tag{44}$$

and $\mathrm{Fr}((\operatorname{Spec}\mathcal{A})/S)$ is induced by the morphism of $\mathcal{O}_S$-algebras

$$\mathcal{A} \otimes_\pi \mathcal{O}_S \longrightarrow \mathcal{A}, \qquad a \otimes_\pi x \longmapsto a^p x.$$

[43]

Finally, for every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, there are canonical isomorphisms

$$\mathcal{V}(\mathcal{E})^{(p)} \xrightarrow{\sim} \mathcal{V}(\mathcal{E}^{(p)}) \qquad \text{and} \qquad \mathcal{S}(\mathcal{E})^{(p)} \xrightarrow{\sim} \mathcal{S}(\mathcal{E}^{(p)}),$$

where $\mathcal{S}(\mathcal{E})$ denotes the symmetric algebra of the $\mathcal{O}_S$-module $\mathcal{E}$.

d) Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra (3.1), and let T be a scheme of characteristic p. If $\mathcal{U}^{(p/S)}$, or $\mathcal{U}^{(p)}$, denotes the pullback of the coalgebra $\mathcal{U}$ along $\mathrm{fr}(S)$, then, as in *b)*, there is a canonical isomorphism

$$(\operatorname{Spec}^*\mathcal{U})^{(p)} \xrightarrow{\sim} \operatorname{Spec}^*\mathcal{U}^{(p)}.$$

If $\mathcal{U}$ is a group coalgebra, then for an S-scheme T, the value of $\mathrm{Fr}(\operatorname{Spec}^*\mathcal{U})$, that is, of the kernel of the Frobenius morphism

$$\operatorname{Spec}^*\mathcal{U} \longrightarrow (\operatorname{Spec}^*\mathcal{U})^{(p)},$$

is the set of elements γ of

$$(\operatorname{Spec}^*\mathcal{U})(T) = \{x \in \Gamma(T,\mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1,\ \Delta_{\mathcal{U}_T}x = x \otimes x\}$$

such that the image in $\Gamma(T,\mathcal{U}_T \otimes_{\mathrm{fr}(T)} \mathcal{O}_T)$ of the element

$$\gamma \otimes_{\mathrm{fr}(T)} 1 \in \Gamma(T,\mathcal{U}_T) \otimes_{\mathrm{fr}(T)} \mathcal{O}(T)$$

is equal to 1.

4.2. [44]

We now consider a construction closely related to the preceding one. Let S be a scheme of characteristic p, let X be an S-scheme, and let X_S^p be the product, in $(\mathbf{Sch}/S)$, of p copies of X.

Denote by $U^p(X)$ the open subscheme of X_S^p which is the union of the products U_S^p as U ranges over the affine open subsets of X. Thus a point x of X_S^p belongs to $U^p(X)$ if and only if its projections $\mathrm{pr}_i\, x$ on the factors of X_S^p belong to a common affine open subset of X. For example, if every finite subset of X is contained in an affine open subset, then

$$U^p(X) = X_S^p.$$

The symmetric group $\mathfrak{S}_p$ on p letters acts on X_S^p by permuting the factors and leaves $U^p(X)$ stable. We call the quotient of X_S^p by $\mathfrak{S}_p$ in the category of ringed spaces the *p-fold symmetric product of X*, and denote it by $\Sigma^p X$. Let

$$q(X), \text{ or simply } q : X_S^p \longrightarrow \Sigma^p X$$

[43] $\mathcal{A} \otimes_\pi \mathcal{O}_S$ denotes the $\mathcal{O}_S$-algebra obtained by extension of scalars along $\pi : \mathcal{O}_S \to \mathcal{O}_S$; thus $ax \otimes_\pi 1 = a \otimes_\pi x^p$ and $x \cdot (a \otimes_\pi 1) = a \otimes_\pi x$.

[44] For the material in 4.2 and 4.3, see also [DG70], IV, §3, nos. 4–6.

be the canonical projection.

The morphism q maps $U^p(X)$ onto an open subset $V^p(X)$ of the symmetric product, which may be described as follows (cf. V, 4.1). The structure sheaf of $\Sigma^p X$ induces a scheme structure on $V^p(X)$, and the morphism

$$q'(X) : U^p(X) \longrightarrow V^p(X)$$

induced by $q(X)$ is affine, indeed integral. As U ranges over the affine open subsets of X which map into a varying affine open V of S, the schemes $\Sigma^p U$ form an affine open covering of $V^p(X)$. If R is the affine algebra of V, and A that of U, the affine algebra of $\Sigma^p U$ is the subalgebra $\Sigma^p A$ of $\bigotimes_R^p A$ consisting of the symmetric tensors.

Now consider the diagonal morphism

$$\delta : X \longrightarrow U^p(X).$$

If $V = \operatorname{Spec} R$ is an affine open subset of S and $U = \operatorname{Spec} A$ is an affine open subset of X lying over V, the restriction of δ to U is defined by the algebra homomorphism

$$\eta : \bigotimes_R^p A \longrightarrow A, \qquad a_1 \otimes \cdots \otimes a_p \longmapsto a_1 a_2 \cdots a_p.$$

Consequently, if N denotes the symmetrization operator,

$$\begin{aligned} \eta\big(N(a_1 \otimes \cdots \otimes a_p)\big) &= \eta\left(\sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}\right) \\ &= p!\, a_1 \cdots a_p = 0. \end{aligned}$$

In other words, η vanishes on the subspace

$$N\left(\bigotimes_R^p A\right) \subseteq \Sigma^p A$$

consisting of symmetrized tensors. Moreover, if f is a symmetric tensor, then plainly $N(fa) = fN(a)$. Hence $N(\bigotimes_R^p A)$ is an ideal of $\Sigma^p A$. We put

$$U^{[p/S]} = \operatorname{Spec}\left(\Sigma^p A / N\left(\bigotimes_R^p A\right)\right).$$

This is a closed subscheme of $\Sigma^p(U) = V^p(U)$. As U ranges over the affine open subsets of X which map into a varying affine open V of S, the schemes $U^{[p/S]}$ glue to a closed subscheme of $V^p(X)$, denoted $X^{[p/S]}$.

If $i(X)$ denotes the inclusion of $X^{[p/S]}$ in $V^p(X)$, the preceding calculation shows that $q'(X) \circ \delta$ factors through $X^{[p/S]}$. We therefore obtain a morphism

$$F^{[p]}(X/S) : X \longrightarrow X^{[p/S]}.$$

[45] The factorization is displayed by the commutative diagram

[45] In the original, this morphism, respectively the relative Frobenius morphism, was denoted F', respectively F.

$$\begin{array}{ccccc} X_S^p & \supset & \mathrm{U}^p(X) & \xleftarrow{\delta(X)} & X \\ \Big\downarrow{\scriptstyle q(X)} & & \Big\downarrow{\scriptstyle q'(X)} & & \Big\downarrow{\scriptstyle F^{[p]}(X/S)} \\ \Sigma^p(X) & \supset & \mathrm{V}^p(X) & \xleftarrow{i(X)} & X^{[p/S]}. \end{array}$$

It is clear that $X^{[p/S]}$ is functorial in X, and that

$$F^{[p]} : X \longmapsto F^{[p]}(X/S)$$

is a functorial homomorphism.

4.2.1.

The schemes $X^{[p/S]}$ and $X^{(p/S)}$ are naturally related. Let V be an affine open subset of S, with affine algebra R, and let U be an affine open subset of X lying over V, with affine algebra A. If π denotes the endomorphism $x \mapsto x^p$ of R, the affine algebra of $U^{(p/S)}$ is $A \otimes_\pi R$. The map

$$a \otimes_\pi \lambda \longmapsto \left(\lambda a \otimes \cdots \otimes a \bmod N\left(\bigotimes_R^p A\right)\right)$$

is a homomorphism of R-algebras

$$A \otimes_\pi R \longrightarrow \Sigma^p A / N\left(\bigotimes_R^p A\right).$$

It induces a morphism

$$\varphi(U) : U^{[p/S]} \longrightarrow U^{(p/S)}$$

satisfying

$$\varphi(U) \circ F^{[p]}(U/S) = \mathrm{Fr}(U/S).$$

Gluing these morphisms gives the commutative triangle

$$\begin{array}{ccc} & X & \\ {\scriptstyle F^{[p]}(X/S)}\swarrow & & \searrow{\scriptstyle \mathrm{Fr}(X/S)} \\ X^{[p/S]} & \xrightarrow{\varphi(X)} & X^{(p/S)}. \end{array}$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, then $F^{[p]}(X/S)$ identifies with the identity morphism of X. Thus $\varphi(X)$ is the canonical immersion

$$\mathrm{Spec}(\mathcal{O}_S/\mathcal{I}) \longrightarrow \mathrm{Spec}(\mathcal{O}_S/\mathcal{I}^{\{p\}}).$$

This shows that $\varphi(X)$ need not be an isomorphism.

On the other hand, when M is a free R-module, the map

$$M \otimes_\pi R \longrightarrow \Sigma^p M / N\left(\bigotimes_R^p M\right),$$

$$m \otimes_\pi \lambda \longmapsto \left(\lambda m \otimes \cdots \otimes m \bmod N\left(\bigotimes_R^p M\right)\right)$$

is plainly bijective. It remains bijective when M is R-flat, because every flat module is a filtered direct limit of free modules.[46] It follows that

$$\boxed{\varphi(X) : X^{[p/S]} \longrightarrow X^{(p/S)} \text{ is an isomorphism if } X \text{ is flat over } S.}$$

4.3.

Finally, let G be an S-group scheme in abelian groups. The composite morphism

$$\mu(G) : U^p(G) \hookrightarrow G_S^p \longrightarrow G,$$

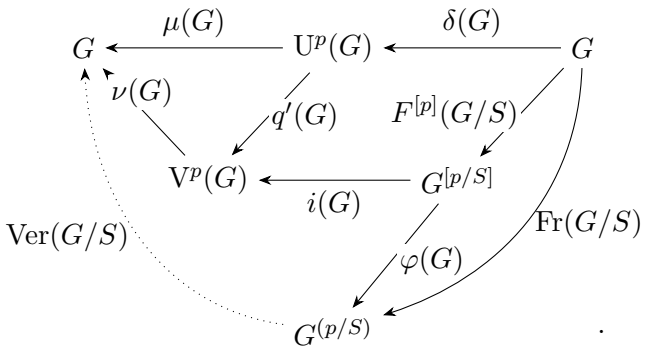
defined by multiplication, factors through $V^p(G)$. Thus there is a morphism

$$\nu(G) : V^p(G) \longrightarrow G$$

such that

$$\nu(G) \circ q'(G) = \mu(G).$$

The resulting commutative diagram is



When G is flat over S, $\varphi(G)$ is an isomorphism, and we may define the morphism called the *Verschiebung*

$$\mathrm{Ver}(G/S) : G^{(p/S)} \longrightarrow G$$

by

$$\mathrm{Ver}(G/S) = \nu(G) \circ i(G) \circ \varphi(G)^{-1}.$$

As G ranges over the flat commutative S-group schemes, the assignment

$$\mathrm{Ver} : G \longmapsto \mathrm{Ver}(G/S)$$

is plainly a functorial homomorphism. Consequently, $\mathrm{Ver}(G/S)$ is a group homomorphism.

For every S-scheme T, the composite

$$G(T) \xrightarrow{\delta(G)(T)} U^p(G)(T) \xrightarrow{\mu(G)(T)} G(T)$$

sends $x \in G(T)$ to $p \cdot x$. We may therefore write $p \cdot \mathrm{id}_G$ in place of $\mu(G) \circ \delta(G)$, obtaining the classical formula

$$\mathrm{Ver}(G/S) \circ \mathrm{Fr}(G/S) = p \cdot \mathrm{id}_G\,. \qquad (*)$$

Examples 4.3.1.

[46] D. Lazard, *Sur les modules plats*, C. R. Acad. Sci. Paris **258** (1964), 6313–6316. See also D. Lazard, Bull. Soc. Math. France **97** (1969), 81–128, or [BAlg], X, §1.6, Th. 1.

a) If G is a constant S-scheme in abelian groups, then $\mathrm{Fr}(G/S)$ identifies with the identity morphism of G (cf. 4.1.1.1). Hence

$$\mathrm{Ver}(G/S) = p \cdot \mathrm{id}_G\,.$$

b) If G is the diagonalizable S-group of type M, then $\mathrm{Fr}(G/S) = p \cdot \mathrm{id}_G$ by 4.1.4 *a)*. It follows at once that $\mathrm{Ver}(G/S)$ is the identity morphism of G.

c) If $\mathcal{E}$ is a flat $\mathcal{O}_S$-module and $G = \mathcal{V}(\mathcal{E})$, then $\mathrm{Ver}(G/S)$ is zero, as is $p \cdot \mathrm{id}_G$. In Exposé VII$_\mathrm{B}$ we shall see that a commutative algebraic group G over a field k is "unipotent" if and only if, for some n, the composite homomorphism

$$G^{(p^n)} \xrightarrow{\mathrm{Ver}(G^{(p^{n-1})}/S)} G^{(p^{n-1})} \longrightarrow \cdots \longrightarrow G^{(p)} \xrightarrow{\mathrm{Ver}(G/S)} G$$

is zero. Here

$$G^{(p^n)} = \left(G^{(p^{n-1})}\right)^{(p)}$$

as in 4.1.3.[47]

4.3.2.

Since $\mathrm{Ver} : G \mapsto \mathrm{Ver}(G/S)$ is a functorial homomorphism on the category of flat commutative S-group schemes, the square

$$\begin{array}{ccc} G^{(p)} & \xrightarrow{\mathrm{Ver}(G/S)} & G \\ {\scriptstyle \mathrm{Fr}(G/S)^{(p)}}\downarrow & & \downarrow{\scriptstyle \mathrm{Fr}(G/S)} \\ G^{(p^2)} & \xrightarrow{\mathrm{Ver}(G^{(p)}/S)} & G^{(p)} \end{array}$$

commutes. Here $\mathrm{Fr}(G/S)^{(p)}$ denotes the pullback of $\mathrm{Fr}(G/S)$ along the base change $\mathrm{fr}(S)$. By 4.1.1,

$$\mathrm{Fr}(G/S)^{(p)} = \mathrm{Fr}(G^{(p)}/S).$$

Applying formula $(*)$ of 4.3 to $G^{(p)}$, we obtain

$$\mathrm{Fr}(G/S) \circ \mathrm{Ver}(G/S) = \mathrm{Ver}(G^{(p)}/S) \circ \mathrm{Fr}(G^{(p)}/S) = p \cdot \mathrm{id}_{G^{(p)}}\,. \qquad (**)$$

4.3.3.

Suppose finally that G is a commutative S-group which is finite and locally free. Let $\mathcal{A}$ be its affine $\mathcal{O}_S$-algebra, and let π be the endomorphism of the sheaf of rings $\mathcal{O}_S$ which sends a section x to x^p.[48] Denote by $\Sigma^p\mathcal{A}$ the subalgebra of $\bigotimes^p_{\mathcal{O}_S} \mathcal{A}$ consisting of the sections invariant under the symmetric-group action, and by

$$i(\mathcal{A}) : \Sigma^p\mathcal{A} \longrightarrow \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A}$$

its inclusion.

Let

$$\Delta^p(\mathcal{A}) : \mathcal{A} \longrightarrow \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A}$$

[47] This criterion is not stated explicitly in Exposé VII$_\mathrm{B}$; see [DG70], IV, §3, Prop. 4.11.

[48] The editors changed the order of presentation, introducing first the objects which occur in the diagram below.

be the morphism obtained by iterating the diagonal morphism of the coalgebra $\mathcal{A}$. It corresponds to the multiplication morphism $U^p(G) = G_S^p \to G$. By the beginning of 4.3, $\Delta^p(\mathcal{A})$ factors through $\Sigma^p\mathcal{A}$; that is, it induces a morphism

$$a(\mathcal{A}) : \mathcal{A} \longrightarrow \Sigma^p\mathcal{A}$$

such that

$$i(\mathcal{A}) \circ a(\mathcal{A}) = \Delta^p(\mathcal{A}).$$

On the other hand, let $S^p(\mathcal{A})$ be the homogeneous component of degree p in the symmetric algebra of $\mathcal{A}$, and let

$$q(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A} \longrightarrow S^p(\mathcal{A})$$

be the canonical projection. The multiplication

$$m^p(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A} \longrightarrow \mathcal{A}$$

factors through $S^p(\mathcal{A})$; equivalently, it induces a map

$$b(\mathcal{A}) : S^p(\mathcal{A}) \longrightarrow \mathcal{A}$$

such that

$$b(\mathcal{A}) \circ q(\mathcal{A}) = m^p(\mathcal{A}).$$

Since $\Sigma^p\mathcal{A}$ is the affine algebra of $V^p(G)$, the composite $i(G) \circ \varphi(G)^{-1}$ induces, again by the beginning of 4.3, an algebra homomorphism

$$r(\mathcal{A}) : \Sigma^p\mathcal{A} \longrightarrow \mathcal{A} \otimes_\pi \mathcal{O}_S.$$

It vanishes on sections of the form

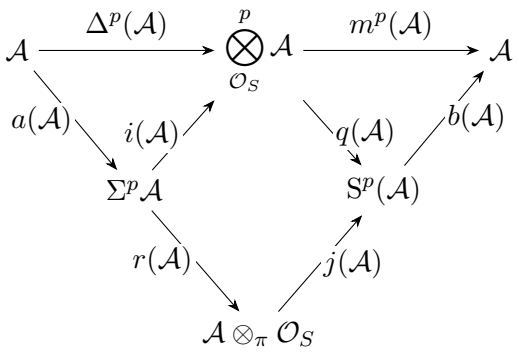
$$\sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}$$

and sends $a \otimes \cdots \otimes a$ to $a \otimes_\pi 1$. Similarly, $j(\mathcal{A})$ is the morphism of $\mathcal{O}_S$-modules

$$a \otimes_\pi 1 \longmapsto q(a \otimes \cdots \otimes a).$$

These maps form the commutative diagram

(A)



.

The composite $r(\mathcal{A}) \circ a(\mathcal{A})$ corresponds to the Verschiebung morphism $\mathrm{Ver}(G/S)$, while $b(\mathcal{A}) \circ j(\mathcal{A})$ corresponds to the Frobenius morphism $\mathrm{Fr}(G/S)$.

The preceding commutative diagram $(\mathcal{A})$ is self-dual. Indeed, let D be the functor which associates to each $\mathcal{O}_S$-module $\mathcal{M}$ its dual

$$D\mathcal{M} = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{M}, \mathcal{O}_S).$$

The image of diagram $(\mathcal{A})$ under D is precisely diagram $(D\mathcal{A})$. The morphisms

$$Dr(\mathcal{A}), \quad Da(\mathcal{A}), \quad Dj(\mathcal{A}), \quad Db(\mathcal{A})$$

identify respectively with

$$j(D\mathcal{A}), \quad b(D\mathcal{A}), \quad r(D\mathcal{A}), \quad a(D\mathcal{A}).$$

By 3.3.1, it follows that:

> *In the category of commutative S-groups which are finite and locally free, Cartier duality interchanges the Frobenius morphism and the Verschiebung.*

[49]

5. Restricted p-Lie Algebras

We first recall some results from the *Séminaire Sophus Lie.*[50]

5.1.

Let p be a prime number, let R be a commutative ring of characteristic p, and let A be an associative R-algebra, not necessarily commutative. For $a, b \in A$, put

$$[a, b] = ab - ba, \qquad a \cdot b = L_a(b) = R_b(a).$$

Then

$$\begin{aligned}(\mathrm{ad}\, x^p)(y) &= [x^p, y] \\ &= (L_x^p - R_x^p)(y) \\ &= (L_x - R_x)^p(y) \\ &= (\mathrm{ad}\, x)^p(y),\end{aligned}$$

which gives Jacobson's first formula:

$$\mathrm{ad}(x^p) = (\mathrm{ad}\, x)^p. \tag{i}$$

If $a_1, \ldots, a_p$ are arbitrary elements of A, and N is the symmetrization operator (cf. 4.2), then

$$\begin{aligned}N(a_1 \otimes \cdots \otimes a_p) &= \sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \cdots a_{\sigma(p)} \\ &= \sum_{\tau \in \mathfrak{S}_{p-1}} [a_{\tau(1)}, [a_{\tau(2)}, [\cdots, [a_{\tau(p-1)}, a_p] \cdots]]].\end{aligned} \tag{*}$$

Here σ ranges over the permutations of p letters and τ over those of $p-1$ letters.

[49] See also [DG70], IV, §3, 4.9.

[50] See P. Cartier, *Exemples d'hyperalgèbres*, Séminaire Sophus Lie 1955/56, Exp. 3.

Indeed, the final expression is equal to

$$\sum_{\tau}\sum_{r=0}^{p-1}\sum_{i_1<\cdots<i_r}(-1)^s a_{\tau(i_1)}\cdots a_{\tau(i_r)}a_p a_{\tau(j_s)}\cdots a_{\tau(j_1)},$$

where τ ranges over the permutations of $p-1$ letters; $i_1,\dots,i_r$ ranges over the strictly increasing sequences in $[1,p-1]$; and $j_1,\dots,j_s$ is the strictly increasing sequence of those integers of $[1,p-1]$ which are different from $i_1,\dots,i_r$. For fixed r, the sum of the terms

$$(-1)^s a_{\tau(i_1)}\cdots a_{\tau(i_r)}a_p a_{\tau(j_s)}\cdots a_{\tau(j_1)}$$

is plainly

$$(-1)^s\binom{p-1}{s}\sum_{\rho}a_{\rho(1)}\cdots a_{\rho(r)}a_p a_{\rho(r+1)}\cdots a_{\rho(p-1)},$$

where ρ ranges over the permutations of $p-1$ letters. But

$$(-1)^s\binom{p-1}{s}=1 \quad \text{in } \mathbb{F}_p,$$

because in $\mathbb{F}_p[x]$,

$$(x-1)^p = x^p-1 = (x-1)(x^{p-1}+\cdots+1),$$

and hence

$$(x-1)^{p-1} = x^{p-1}+\cdots+1.$$

This proves $(*)$.

On the other hand, if $x_0, x_1 \in A$, then

$$(x_0+x_1)^p = x_0^p + x_1^p + \sum_z x_{z(1)}x_{z(2)}\cdots x_{z(p)},$$

where z ranges over the nonconstant maps $[1,p]\to\{0,1\}$. It follows that

$$(x_0+x_1)^p = x_0^p + x_1^p + \sum_{0<r<p}\frac{1}{r!(p-r)!}N(\underbrace{x_0,\dots,x_0}_{r},\underbrace{x_1,\dots,x_1}_{p-r}).$$

[51] By $(*)$,

$$N(\underbrace{x_0,\dots,x_0}_{r},\underbrace{x_1,\dots,x_1}_{p-r}) = r!(p-1-r)!\sum_t [x_{t(1)},[x_{t(2)},[\cdots,[x_{t(p-1)},x_1]\cdots]]],$$

where t ranges over the maps $[1,p-1]\to\{0,1\}$ which take the value 0 exactly r times. We obtain Jacobson's second formula:

$$(x_0+x_1)^p = x_0^p + x_1^p - \sum_{0<r<p}\sum_t \frac{1}{r}[x_{t(1)},[x_{t(2)},[\cdots,[x_{t(p-1)},x_1]\cdots]]], \tag{ii}$$

where t has the same meaning.

[51] The editors inserted the following explanation from [DG70], III.7, 3.2.

5.2.

Let now $\mathfrak{g}$ be an R-Lie algebra. A map

$$x \longmapsto x^{(p)}$$

from $\mathfrak{g}$ to itself is said to make $\mathfrak{g}$ a *restricted p-Lie algebra over R* if the following conditions hold:

$$(0) \quad (\lambda x)^{(p)} = \lambda^p x^{(p)} \quad (\lambda \in R,\ x \in \mathfrak{g}),$$
$$(i) \quad \operatorname{ad} x^{(p)} = (\operatorname{ad} x)^p \quad (x \in \mathfrak{g}),$$

and

$$(ii) \quad (x_0 + x_1)^{(p)} = x_0^{(p)} + x_1^{(p)} - \sum_{0<r<p} \sum_t \frac{1}{r}[x_{t(1)}, [x_{t(2)}, [\cdots, [x_{t(p-1)}, x_1]\cdots]]],$$

where t ranges over the maps $[1, p-1] \to \{0, 1\}$ which take the value 0 exactly r times, and $x_0, x_1 \in \mathfrak{g}$. The map $x \mapsto x^{(p)}$ is called the *symbolic p-th power*.

For example, if A is an associative R-algebra, 5.1 shows that the R-module underlying A, endowed with

$$[x, y] = xy - yx, \qquad x^{(p)} = x^p,$$

is a p-Lie algebra, denoted A_{Lie}. We call it the p-Lie algebra underlying A.

We shall chiefly consider restricted p-Lie subalgebras of algebras of the form A_{Lie}. Here is one example. Let S be a scheme of characteristic $p > 0$, and let X be an S-scheme. Recall that a derivation of X over S is an endomorphism D of the sheaf of abelian groups $\mathcal{O}_X$ such that

$$D(\lambda s) = \lambda D(s), \qquad D(st) = (Ds)t + s(Dt),$$

whenever λ, respectively s, t, are sections of $\mathcal{O}_S$, respectively $\mathcal{O}_X$, on open subsets for which the formulas make sense. Leibniz's formula

$$D^n(st) = \sum_{i=0}^{n} \binom{n}{i} (D^i s)(D^{n-i} t)$$

shows that D^p is again a derivation of X over S, since

$$\binom{p}{i} \equiv 0 \pmod{p} \qquad (i \neq 0, p).$$

It follows that:

> *The algebra* $\mathrm{Der}_{X/S}$ *of derivations of X over S is a restricted p-Lie subalgebra of the $\Gamma(S, \mathcal{O}_S)$-algebra of differential operators of X over S.*

5.2.1. If $\mathfrak{g}$ and $\mathfrak{h}$ are two restricted p-Lie algebras, a *homomorphism*

$$h : \mathfrak{g} \longrightarrow \mathfrak{h}$$

is an R-linear map such that, for $x, y \in \mathfrak{g}$,

$$h([x, y]) = [h(x), h(y)], \qquad h(x^{(p)}) = h(x)^{(p)}.$$

The composite of two homomorphisms is again a homomorphism, so we may speak of the *category of restricted p-Lie algebras over R*.

If $(X, \mathcal{R})$ is a ringed space, we say that an $\mathcal{R}$-module $\mathfrak{g}$ is endowed with the structure of a restricted p-Lie algebra over $\mathcal{R}$ if, for every open subset U, the module $\Gamma(U, \mathfrak{g})$ is a restricted p-Lie algebra over $\Gamma(U, \mathcal{R})$, and if all restriction maps are homomorphisms.

5.3.

We are now interested in the left adjoint of the functor

$$A \longmapsto A_{\mathrm{Lie}}$$

of 5.2. Let $\mathfrak{g}$ be a restricted p-Lie algebra over a ring R of characteristic p, let $U(\mathfrak{g})$ be the enveloping algebra of the Lie algebra underlying $\mathfrak{g}$ (cf. [BLie], I, §2.1), and let

$$i_{\mathfrak{g}} : \mathfrak{g} \longrightarrow U(\mathfrak{g})$$

(or simply i) be the canonical map.

Let A be a unital associative R-algebra. For every Lie algebra homomorphism

$$\varphi : \mathfrak{g} \longrightarrow A_{\mathrm{Lie}},$$

there is a unique homomorphism of unital R-algebras

$$\psi : U(\mathfrak{g}) \longrightarrow A$$

such that $\psi \circ i = \varphi$. Moreover, φ is a homomorphism of restricted p-Lie algebras if and only if ψ vanishes on all the elements

$$i(x)^p - i\big(x^{(p)}\big), \qquad x \in \mathfrak{g}.$$

Definition. The quotient of $U(\mathfrak{g})$ by the two-sided ideal generated by the elements

$$i(x)^p - i\big(x^{(p)}\big)$$

is denoted $U_p^R(\mathfrak{g})$, or simply $U_p(\mathfrak{g})$. We denote by $j_{\mathfrak{g}}$, or simply j, the composite

$$\mathfrak{g} \xrightarrow{i} U(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g}).$$

The algebra $U_p(\mathfrak{g})$ is called the *restricted enveloping algebra* of $\mathfrak{g}$.

The preceding discussion gives the following result.

Proposition. *For every unital associative R-algebra A and every homomorphism of restricted p-Lie algebras*

$$\varphi : \mathfrak{g} \longrightarrow A_{\mathrm{Lie}},$$

there is a unique homomorphism of unital algebras

$$\psi : U_p(\mathfrak{g}) \longrightarrow A$$

such that $\psi \circ j = \varphi$. In other words, the functor $\mathfrak{g} \mapsto U_p(\mathfrak{g})$ is left adjoint to the forgetful functor $A \mapsto A_{\mathrm{Lie}}$.

5.3.1.

With the notation of 5.3, put

$$\beta(x) = i(x)^p - i\big(x^{(p)}\big).$$

For every $y \in \mathfrak{g}$, formulas 5.1 (i) and 5.2 (i) give

$$\begin{aligned}\beta(x)i(y) &= i(y)\beta(x) + [\beta(x), i(y)] \\ &= i(y)\beta(x) + (\operatorname{ad} i(x))^p i(y) - i\big((\operatorname{ad} x)^p y\big) \\ &= i(y)\beta(x).\end{aligned}$$

Thus $\beta(x)$ belongs to the center of $U(\mathfrak{g})$; in particular, the left ideal generated by the elements $\beta(x)$ is already two-sided.

On the other hand,

$$\beta(\lambda x) = \lambda^p \beta(x) \qquad (\lambda \in R),$$

and formulas 5.1 (ii) and 5.2 (ii) give

$$\beta(x + y) = \beta(x) + \beta(y) \qquad (x, y \in \mathfrak{g}).$$

Consequently, if (x_α) is a family of generators of the R-module $\mathfrak{g}$, the left ideal generated by all $\beta(x)$ is already generated by the elements $\beta(x_\alpha)$.

5.3.2. Proposition.[52] *Let $\mathfrak{g}$ be an R-Lie algebra whose underlying R-module is free with basis (x_α). Then the restricted p-Lie algebra structures on $\mathfrak{g}$ correspond bijectively to the families (y_α) of elements of $\mathfrak{g}$ such that*

$$\operatorname{ad} y_\alpha = (\operatorname{ad} x_\alpha)^p.$$

Indeed, suppose that $\mathfrak{g}$ is endowed with a restricted p-Lie algebra structure $x \mapsto x^{(p)}$. By 5.2 (i), (0), and (ii), the elements

$$y_\alpha = x_\alpha^{(p)}$$

satisfy

$$\operatorname{ad} y_\alpha = (\operatorname{ad} x_\alpha)^p,$$

and they determine the restricted p-Lie algebra structure.

We prove the converse. Since $\mathfrak{g}$ is a free R-module, the canonical map

$$i : \mathfrak{g} \longrightarrow U(\mathfrak{g})$$

is injective by the Poincaré–Birkhoff–Witt theorem (cf. [BLie], I, §2.7). We may therefore identify $\mathfrak{g}$ with an R-submodule of $U(\mathfrak{g})$. Suppose that (y_α) is a family of elements of $\mathfrak{g}$ such that

$$\operatorname{ad} y_\alpha = (\operatorname{ad} x_\alpha)^p.$$

Let

$$\pi : R \longrightarrow R, \qquad r \longmapsto r^p,$$

and let $\mathfrak{g} \otimes_\pi R$ be the R-Lie algebra obtained by extension of scalars along π.[53]

There is an R-linear map

$$\gamma : \mathfrak{g} \otimes_\pi R \longrightarrow U(\mathfrak{g})$$

which sends $x_\alpha \otimes_\pi 1$ to $x_\alpha^p - y_\alpha$. Moreover, for every $x \in \mathfrak{g}$,

$$(\operatorname{ad} x_\alpha^p)(x) = (\operatorname{ad} x_\alpha)^p(x) = (\operatorname{ad} y_\alpha)(x).$$

[52] In this paragraph, the editors changed the order, stating the result first and then giving the detailed proof.

[53] Thus $xr \otimes_\pi 1 = x \otimes_\pi r^p$ and $r \cdot (x \otimes_\pi 1) = x \otimes_\pi r$, for $x \in \mathfrak{g}$ and $r \in R$.

It follows that γ maps $\mathfrak{g} \otimes_\pi R$ into the center of $U(\mathfrak{g})$. For $x \in \mathfrak{g}$, put

$$x^{(p)} = x^p - \gamma(x \otimes_\pi 1).$$

Then $x_\alpha^{(p)} = y_\alpha$ for every α. If

$$x = \sum_\alpha \lambda_\alpha x_\alpha,$$

formula 5.1 (ii), applied inductively to the number of indices α for which $\lambda_\alpha \neq 0$, gives

$$x^p - \sum_\alpha \lambda_\alpha^p x_\alpha^p \in \mathfrak{g}.$$

Denote this element by z. Then

$$x^{(p)} = \sum_\alpha \lambda_\alpha^p y_\alpha + z,$$

and hence $x^{(p)} \in \mathfrak{g}$.

It is clear that

$$(\lambda x)^{(p)} = \lambda^p x^{(p)}.$$

Since $\gamma(x \otimes_\pi 1)$ is central, we have $\operatorname{ad} x^{(p)} = \operatorname{ad} x^p$. Jacobson's first formula, 5.1 (i), therefore yields

$$\operatorname{ad} x^{(p)} = (\operatorname{ad} x)^p.$$

Finally, Jacobson's second formula, 5.1 (ii), shows that the map $x \mapsto x^{(p)}$ satisfies condition (ii) of 5.2. It thus makes $\mathfrak{g}$ into a restricted p-Lie algebra. This proves the proposition.

5.3.3. Proposition. *Let $\mathfrak{g}$ be a restricted p-Lie algebra over R whose underlying module is free with basis (x_α). Then*

$$j : \mathfrak{g} \longrightarrow U_p(\mathfrak{g})$$

is injective. If $z_\alpha = j(x_\alpha)$, then $U_p(\mathfrak{g})$ has as a basis the monomials

$$\prod_\alpha z_\alpha^{n_\alpha}, \qquad 0 \leq n_\alpha < p.$$

Here the n_α are zero except for finitely many α; the basis is assumed to be totally ordered, and the products are taken in increasing order.

Identify $\mathfrak{g}$ with a submodule of the enveloping algebra $U(\mathfrak{g})$ by means of the canonical map i. For every family

$$n = (n_\alpha)$$

of nonnegative integers, all but finitely many zero, put

$$|n| = \sum_\alpha n_\alpha, \qquad x^n = \prod_\alpha x_\alpha^{n_\alpha}.$$

Write

$$n_\alpha = m_\alpha + p\ell_\alpha, \qquad 0 \leq m_\alpha < p,$$

and put

$$T_n = \prod_\alpha x_\alpha^{m_\alpha} \beta(x_\alpha)^{\ell_\alpha},$$

where

$$\beta(x) = x^p - x^{(p)}$$

is the map $\mathfrak{g} \to U(\mathfrak{g})$ defined in 5.3.1.

For $r \in \mathbb{N}$, let U^r be the R-submodule of $U(\mathfrak{g})$ generated by the x^n for which $|n| \leq r$. Since the graded ring

$$\bigoplus_r U^r/U^{r-1}$$

is commutative (cf. [BLie], I, §2.6), for every n we have

$$T_n - \prod_\alpha x_\alpha^{n_\alpha} \in U^{|n|-1}.$$

For every $s \in \mathbb{N}$, the x^n with $|n| = s$ form a basis of U^s/U^{s-1} by the Poincaré–Birkhoff–Witt theorem (*loc. cit.*, §2.7); the same is therefore true of the T_n with $|n| = s$.

As $s = |n|$ varies, the T_n consequently form a basis of $U(\mathfrak{g})$. The kernel J of the canonical map

$$U(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g})$$

is the left ideal of $U(\mathfrak{g})$ generated by the central elements $\beta(x_\alpha)$, by 5.3.1. The T_n for which $\ell = (\ell_\alpha) \neq 0$ therefore form a basis of J, while the T_n for which $n_\alpha < p$ for every α form a basis of

$$U_p(\mathfrak{g}) = U(\mathfrak{g})/J.$$

5.3.3 bis.

Let $\mathfrak{g}$ be a restricted p-Lie algebra over R, and let $f : R \to R'$ be an extension of the base ring. We claim that the R'-module

$$R' \otimes_R \mathfrak{g}$$

admits one and only one restricted p-Lie algebra structure such that

$$[\lambda \otimes x, \mu \otimes y] = \lambda\mu \otimes [x, y], \qquad (\lambda \otimes x)^{(p)} = \lambda^p \otimes x^{(p)}. \tag{$*$}$$

In particular, it will follow that the functor

$$\mathfrak{g} \longmapsto R' \otimes_R \mathfrak{g}$$

is left adjoint to restriction of scalars from R' to R.

The uniqueness of the structure defined by $(*)$ is clear; we prove its existence. If $\mathfrak{g}$ is free with basis (x_α), Proposition 5.3.2 gives a unique restricted p-Lie algebra structure on the Lie algebra $R' \otimes_R \mathfrak{g}$ such that

$$(1 \otimes x_\alpha)^{(p)} = 1 \otimes x_\alpha^{(p)}.$$

This is the structure sought.

Suppose now that $\mathfrak{g}$ is arbitrary. There is a restricted p-Lie algebra L_0, free as an R-module, together with a surjective homomorphism

$$q_0 : L_0 \longrightarrow \mathfrak{g}.$$

For example, we may take

$$L_0 = R \otimes_{\mathbb{F}_p} \mathfrak{g}$$

and let q_0 be the homomorphism

$$\lambda \otimes x \longmapsto \lambda x;$$

here $\mathfrak{g}$ is free over the prime field $\mathbb{F}_p$. The kernel of q_0 is a p-ideal of L_0, that is, an ideal of the Lie algebra L_0 stable under $x \mapsto x^{(p)}$. Hence there is also a restricted p-Lie algebra L_1, free as an R-module, and a homomorphism $q_1 : L_1 \to L_0$ whose image is $\operatorname{Ker} q_0$. This gives the exact sequence

$$L_1 \xrightarrow{q_1} L_0 \xrightarrow{q_0} \mathfrak{g} \longrightarrow 0.$$

and therefore an exact sequence of R'-Lie algebras

$$R' \otimes_R L_1 \xrightarrow{R' \otimes_R q_1} R' \otimes_R L_0 \xrightarrow{R' \otimes_R q_0} R' \otimes_R \mathfrak{g} \longrightarrow 0.$$

Since $R' \otimes_R q_1$ is manifestly a homomorphism of restricted p-Lie algebras, the kernel of $R' \otimes_R q_0$ is a p-ideal. Thus the symbolic p-th power operation on $R' \otimes_R L_0$ induces, by passage to the quotient, a map from $R' \otimes_R \mathfrak{g}$ to itself (use formula (ii) of 5.2). This map equips $R' \otimes_R \mathfrak{g}$ with the required restricted p-Lie algebra structure.

5.3.4.

The canonical map

$$j_{\mathfrak{g}} : \mathfrak{g} \longrightarrow U_p(\mathfrak{g})$$

induces, for every extension $R \to R'$ of the base ring, a homomorphism

$$R' \otimes_R j_{\mathfrak{g}} : R' \otimes_R \mathfrak{g} \longrightarrow R' \otimes_R U_p(\mathfrak{g}).$$

It therefore induces a homomorphism

$$h : U_p(R' \otimes_R \mathfrak{g}) \longrightarrow R' \otimes_R U_p(\mathfrak{g})$$

such that

$$h \circ j_{R' \otimes_R \mathfrak{g}} = R' \otimes_R j_{\mathfrak{g}}.$$

The universal properties of $R' \otimes_R \mathfrak{g}$ and of the restricted enveloping algebra show immediately that h is an isomorphism. We may therefore identify

$$U_p(R' \otimes_R \mathfrak{g}) \simeq R' \otimes_R U_p(\mathfrak{g}).$$

In particular, let $r \in R$, and let $R' = R_r$ be the localized ring. Then

$$\mathfrak{g}_r = R_r \otimes_R \mathfrak{g}$$

has a canonical restricted p-Lie algebra structure over R_r. Consequently, the sheaf $\widetilde{\mathfrak{g}}$ on $\operatorname{Spec} R$ is a quasi-coherent restricted p-Lie algebra on $\operatorname{Spec} R$. Moreover,

$$U_p^{R_r}(\mathfrak{g}_r) \simeq U_p^R(\mathfrak{g})_r,$$

so the sheaf associated with the presheaf

$$V \longmapsto U_p(\Gamma(V, \widetilde{\mathfrak{g}}))$$

is quasi-coherent.

Definition. More generally, let S be a scheme of characteristic p, and let $\mathcal{G}$ be a quasi-coherent restricted p-Lie algebra over $\mathcal{O}_S$. The sheaf associated with the presheaf

$$V \longmapsto U_p(\Gamma(V, \mathcal{G}))$$

is quasi-coherent. It is denoted $\mathcal{U}_p(\mathcal{G})$ and is called the *restricted enveloping algebra* of $\mathcal{G}$. If V is affine, then

$$U_p(\Gamma(V, \mathcal{G})) \simeq \Gamma(V, \mathcal{U}_p(\mathcal{G})).$$

5.4.

The universal property of $U_p(\mathfrak{g})$ implies that $U_p(\mathfrak{g})$ is functorial in $\mathfrak{g}$. Every homomorphism of restricted p-Lie algebras

$$\varphi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

induces a unique homomorphism of unital algebras $U_p(\varphi)$ such that

$$j_{\mathfrak{h}} \circ \varphi = U_p(\varphi) \circ j_{\mathfrak{g}}.$$

Here are some examples.

a) If $\mathfrak{h} = 0$, then $U_p(\mathfrak{h})$ identifies with the base ring, and $U_p(\varphi)$ is an algebra homomorphism

$$\varepsilon_{\mathfrak{g}} : U_p(\mathfrak{g}) \longrightarrow R$$

called the *augmentation*.

b) Let now $\mathfrak{h} = \mathfrak{g}^\circ$, the Lie algebra opposite to $\mathfrak{g}$. Thus $\mathfrak{g}^\circ$ has the same underlying module and the same symbolic p-th power as $\mathfrak{g}$, while its bracket is the negative of the bracket of $\mathfrak{g}$. We may identify $U_p(\mathfrak{g}^\circ)$ with the algebra opposite to $U_p(\mathfrak{g})$. Moreover, the isomorphism

$$x \longmapsto -x$$

from $\mathfrak{g}$ onto $\mathfrak{g}^\circ$ induces an isomorphism

$$c_{\mathfrak{g}} : U_p(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g}^\circ) \simeq U_p(\mathfrak{g})^\circ.$$

We call $c_{\mathfrak{g}}$ the *antipode* of $U_p(\mathfrak{g})$.

c) Finally, let $\mathfrak{f}$ and $\mathfrak{g}$ be two restricted p-Lie algebras, and let $\mathfrak{h} = \mathfrak{f} \times \mathfrak{g}$ be their product. Its underlying R-module is the direct product, and

$$[(x, y), (x', y')] = ([x, x'], [y, y']), \qquad (x, y)^{(p)} = (x^{(p)}, y^{(p)}).$$

If

$$h_1 : \mathfrak{f} \to \mathfrak{k}, \qquad h_2 : \mathfrak{g} \to \mathfrak{k}$$

are restricted p-Lie algebra homomorphisms satisfying

$$[h_1(x), h_2(y)] = 0 \quad (x \in \mathfrak{f},\ y \in \mathfrak{g}),$$

then

$$h_1 + h_2 : (x, y) \longmapsto h_1(x) + h_2(y)$$

is a restricted p-Lie algebra homomorphism. Conversely, every homomorphism $\mathfrak{f} \times \mathfrak{g} \to \mathfrak{k}$ is of this form. This characterizes $\mathfrak{f} \times \mathfrak{g}$ by a universal property.

For example, the maps

$$h_1 : x \longmapsto j_{\mathfrak{f}}(x) \otimes 1, \qquad h_2 : y \longmapsto 1 \otimes j_{\mathfrak{g}}(y)$$

induce a homomorphism $h_1 + h_2$ from $\mathfrak{f} \times \mathfrak{g}$ to the restricted p-Lie algebra underlying

$$U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g}).$$

The universal properties of $\mathfrak{f} \times \mathfrak{g}$ and of the restricted enveloping algebras show that this homomorphism extends to an isomorphism

$$\varphi : U_p(\mathfrak{f} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g}).$$

Definition. If $\mathfrak{f} = \mathfrak{g}$, the diagonal map

$$\delta : x \longmapsto (x, x)$$

from $\mathfrak{g}$ to $\mathfrak{g} \times \mathfrak{g}$ induces a homomorphism from $U_p(\mathfrak{g})$ to $U_p(\mathfrak{g} \times \mathfrak{g})$. We denote by

$$\Delta_{\mathfrak{g}} : U_p(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g}) \otimes U_p(\mathfrak{g})$$

the composite of this homomorphism with φ. [54]

It is then easy to see that $\Delta_{\mathfrak{g}}$, together with the multiplication of $U_p(\mathfrak{g})$, makes $U_p(\mathfrak{g})$ an R-coalgebra in groups (cf. 3.2), with augmentation $\varepsilon_{\mathfrak{g}}$ and antipode $c_{\mathfrak{g}}$.

5.5. [55]

Let S be a scheme of characteristic p. First let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups and put

$$G = \mathrm{Spec}^*\, \mathcal{U}.$$

We saw in 3.2.3 that, for every $T \to S$, $(\mathrm{Lie}\, G)(T)$ is the Lie subalgebra of $\Gamma(T, \mathcal{U}_T)$ consisting of primitive elements. If x is primitive, then

$$\Delta(x^p) = x^p \otimes 1 + 1 \otimes x^p,$$

since $\binom{p}{i} \equiv 0 \pmod{p}$ for $0 < i < p$. Thus x^p is again primitive. By 5.1 and 5.2, the map $x \mapsto x^p$ makes $(\mathrm{Lie}\, G)(T)$ into an $\mathcal{O}(T)$-restricted p-Lie algebra.

Now let $\mathcal{L}$ be a quasi-coherent restricted p-Lie algebra over $\mathcal{O}_S$. As V ranges over the open subsets of S, the group-coalgebra structures defined above on

$$U_p(\Gamma(V, \mathcal{L}))$$

induce on the associated sheaf $\mathcal{U}_p(\mathcal{L})$ a structure of $\mathcal{O}_S$-coalgebra in groups. Moreover, for every S-scheme T, there is an isomorphism

$$\mathcal{U}_p(\mathcal{L}_T) \xrightarrow{\sim} \mathcal{U}_p(\mathcal{L})_T.$$

Let $\mathrm{Prim}\, \mathcal{U}_p(\mathcal{L})$ denote the subpresheaf which assigns to every open subset V the primitive elements of $\mathcal{U}_p(\mathcal{L})(V)$; this is readily seen to be a sheaf. As V ranges over the open subsets of S, the composites

$$\Gamma(V, \mathcal{L}) \xrightarrow{j} \mathrm{Prim}\, \mathrm{U}_p(\Gamma(V, \mathcal{L})) \longrightarrow \mathrm{Prim}\, \mathcal{U}_p(\mathcal{L})(V)$$

define a morphism

$$\mathcal{L} \longrightarrow \mathrm{Prim}\, \mathcal{U}_p(\mathcal{L}),$$

again denoted j or $j_{\mathcal{L}}$. It in turn defines a morphism of $\mathcal{O}_S$-restricted p-Lie algebras

$$\mathcal{W}(\mathcal{L}) \longrightarrow \mathrm{Prim}\, \mathcal{W}(\mathcal{U}_p(\mathcal{L}))$$

(cf. 3.2.3).

Proposition 5.5.1. *Let $\mathcal{L}$ be an $\mathcal{O}_S$-restricted p-Lie algebra, locally free as an $\mathcal{O}_S$-module. Then $j_{\mathcal{L}}$ induces an isomorphism of $\mathcal{O}_S$-restricted p-Lie algebras*

$$\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} \mathrm{Prim}\, \mathcal{W}(\mathcal{U}_p(\mathcal{L})).$$

[54] Thus $\Delta_{\mathfrak{g}}(x) = x \otimes 1 + 1 \otimes x$ for every $x \in \mathfrak{g}$; in particular, $\Delta_{\mathfrak{g}}$ is cocommutative.

[55] The editors reorganized §5.4.1 of the original into this §5.5. Proposition 5.5.1 combines results from Section 5 and Proposition 3.2.3 and incorporates the original Lemma 7.3; the proof of Proposition 5.5.3(ii) expands the proof of $(i) \Rightarrow (ii)$ in the original Theorem 7.4.

Proof. Let T be an S-scheme. In view of the identification

$$\mathcal{U}_p(\mathcal{L}_T) = \mathcal{U}_p(\mathcal{L})_T,$$

we must show that

$$\Gamma(T, \mathcal{L}_T) \longrightarrow \operatorname{Prim}\Gamma(T, \mathcal{U}_p(\mathcal{L}_T))$$

is bijective. Replacing S by T, we reduce to $T = S$; it then suffices to prove that the morphism of sheaves

$$j_{\mathcal{L}} : \mathcal{L} \longrightarrow \operatorname{Prim}\mathcal{U}_p(\mathcal{L})$$

is an isomorphism. The question is local on S, so suppose that $S = \operatorname{Spec} R$ and that $\mathcal{L}$ is the sheaf associated with an R-restricted p-Lie algebra L with a totally ordered basis (x_α).

As in 5.3.3, let z_α be the image of x_α in $U = U_p(L)$. For a family $n = (n_\alpha)$, with $0 \leq n_\alpha \leq p-1$ and only finitely many nonzero terms, put

$$z^{(n)} = \prod_\alpha \frac{z_\alpha^{n_\alpha}}{n_\alpha!},$$

the product being taken in increasing order. Since

$$\Delta(z_\alpha) = z_\alpha \otimes 1 + 1 \otimes z_\alpha,$$

one obtains

$$\Delta(z^{(n)}) = \sum_r z^{(n-r)} \otimes z^{(r)},$$

where the sum is over the finite set of r satisfying

$$0 \leq r_\alpha \leq n_\alpha \quad \text{for every } \alpha.$$

The $z^{(n)}$, respectively the $z^{(n)} \otimes z^{(m)}$, form a basis of U, respectively of $U \otimes U$. It follows that

$$\Delta(u) = u \otimes 1 + 1 \otimes u$$

holds for $u \in U$ if and only if u is a linear combination of the z_α. This proves the proposition. □

Remark 5.5.2. Recall from 3.2.2 and 3.2.3 that the group S-functor

$$G = \operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$$

is very good and that

$$\operatorname{Lie} G = \operatorname{Prim} \mathcal{W}(\mathcal{U}_p(\mathcal{L})).$$

Thus Proposition 5.5.1 says that $j_{\mathcal{L}}$ induces an isomorphism

$$\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} \operatorname{Lie} G.$$

If, moreover, $\mathcal{L}$ is locally free of finite rank, then $\mathcal{U}_p(\mathcal{L})$ is finite locally free over $\mathcal{O}_S$ by 5.3.3. Hence $\operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by the S-group

$$\mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec} \mathcal{U}_p(\mathcal{L})^*$$

(cf. 3.2.2.1). We obtain the following more precise statement.

Proposition 5.5.3. *Let $\mathcal{L}$ be an $\mathcal{O}_S$-restricted p-Lie algebra, locally free of finite rank as an $\mathcal{O}_S$-module. Put*

$$\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*, \qquad G = \mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec}\mathcal{A}.$$

Then:

(i) $j_{\mathcal{L}}$ induces an isomorphism

$$\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} \operatorname{Lie} G$$

of $\mathcal{O}_S$-restricted p-Lie algebras.

(ii) Let $\mathcal{I}$ be the augmentation ideal of $\mathcal{A}$ and put $\omega_G = \mathcal{I}/\mathcal{I}^2$ (cf. II, 4.11.4). Then

$$\omega_G \simeq \mathcal{L}^* = \operatorname{Hom}_{\mathcal{O}_S}(\mathcal{L}, \mathcal{O}_S).$$

In particular, ω_G is locally free of finite rank, and $\omega_{G/S}^* \simeq \mathcal{L}$.

Proof. Part (i) follows from Remark 5.5.2. We prove (ii). Let $\eta_{\mathcal{U}}$ and $\varepsilon_{\mathcal{U}}$ denote the unit and augmentation of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, and let $\eta_{\mathcal{A}}$ and $\varepsilon_{\mathcal{A}}$ denote those of $\mathcal{A}$. Put

$$\mathcal{J} = \ker \varepsilon_{\mathcal{U}}.$$

Then

$$\mathcal{U} = \eta_{\mathcal{U}}(\mathcal{O}_S) \oplus \mathcal{J}. \tag{1}$$

Let δ be the morphism defined by the diagram

$$\begin{array}{ccc} \mathcal{J} & \xrightarrow{\delta} & \mathcal{J}\otimes\mathcal{J} \\ {\scriptstyle\tau}\downarrow & & \uparrow{\scriptstyle\pi} \\ \mathcal{U} & \xrightarrow{\Delta} & \mathcal{U}\otimes\mathcal{U} \end{array}$$

where τ and π are the inclusion and projection deduced from decomposition (1). There is an exact sequence

$$(*) \qquad 0 \longrightarrow \mathcal{L}^* \xrightarrow{j_{\mathcal{L}}} \mathcal{J} \xrightarrow{\delta} \mathcal{J}\otimes\mathcal{J}.$$

By 5.3.3, the $\mathcal{O}_S$-module $\mathcal{J}/\mathcal{L}$ is locally free, and by 5.5.1 the preceding sequence remains exact after every base change. Consequently, δ induces an isomorphism from $\mathcal{J}/\mathcal{L}$ onto a locally direct summand Q of $\mathcal{J}\otimes\mathcal{J}$. By duality, the exact sequence therefore becomes

$$(**) \qquad 0 \longleftarrow \mathcal{L} \xleftarrow{{}^t j_{\mathcal{L}}} \mathcal{I} \xleftarrow{{}^t\delta} \mathcal{I}\otimes\mathcal{I}.$$

Decomposition (1) corresponds by duality to

$$\mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{I}. \tag{2}$$

The transpose of Δ is the multiplication

$$m_{\mathcal{A}} : \mathcal{A}\otimes\mathcal{A} \longrightarrow \mathcal{A}.$$

Since $\mathcal{I}$ is an ideal of $\mathcal{A}$, $m_{\mathcal{A}}$ sends $\mathcal{I}\otimes\mathcal{I}$ into $\mathcal{I}$. More precisely, decomposition (2) gives the commutative square

$$\begin{array}{ccc} \mathcal{I} & \xleftarrow{m'} & \mathcal{I}\otimes\mathcal{I} \\ {\scriptstyle {}^t\tau}\uparrow & & \downarrow{\scriptstyle {}^t\pi} \\ \mathcal{A} & \xleftarrow{m_{\mathcal{A}}} & \mathcal{A}\otimes\mathcal{A}. \end{array}$$

This shows that the restriction m' of $m_{\mathcal{A}}$ to $\mathcal{I} \otimes \mathcal{I}$ is the transpose of δ. The dual exact sequence therefore yields

$$\mathcal{I}/\mathcal{I}^2 \simeq \mathcal{L}^*,$$

which proves the proposition. □

6. The p-Lie Algebra of an S-Group Scheme

Let S be a scheme of characteristic $p > 0$. In 5.5 we associated with every quasi-coherent $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$ the group S-functor

$$\mathfrak{G}_p(\mathcal{L}) = \mathrm{Spec}^*\, \mathcal{U}_p(\mathcal{L}).$$

We shall now see that, for every S-group scheme G, the $\mathcal{O}_S$-Lie algebra $\mathrm{Lie}(G/S)$ defined in II, 4.11 is naturally endowed with the structure of an $\mathcal{O}_S$-p-Lie algebra.

6.1.

First identify $\mathrm{Lie}(G/S)(S)$ and $\mathrm{Lie}(\mathrm{Aut}\, G/S)(S)$, respectively, with Lie subalgebras of $\mathrm{U}(G)$ and $\mathrm{Dif}_{G/S}$ by means of the injections α and β of 2.5. Thus $\mathrm{Lie}(\mathrm{Aut}\, G/S)(S)$ is identified with the $\Gamma(S, \mathcal{O}_S)$-Lie algebra of S-derivations of $\mathcal{O}_G$. By 5.2, the latter is a restricted p-Lie subalgebra of $\mathrm{Dif}_{G/S}$.

On the other hand, the image of

$$L = \mathrm{Lie}(G/S)(S)$$

under the injective algebra homomorphism

$$\ell : \mathrm{U}(G) \longrightarrow \mathrm{Dif}_{G/S}, \qquad d \longmapsto {}_G d,$$

consists of the left-invariant derivations (cf. 2.2, editor's note (17), 2.4, and 2.5). If $x \in L$, then $\ell(x)^p = \ell(x^p)$ by the cited passages. Since $\ell(x)^p$ is again a derivation, x^p belongs to $\mathrm{Lie}(G/S)(S)$. Hence

$\mathrm{Lie}(G/S)(S)$ *is a restricted p-Lie subalgebra of the infinitesimal algebra* $\mathrm{U}(G)$.

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6.1.1.

Let $\varphi : G \to H$ be a homomorphism of S-group schemes. The homomorphisms $\mathrm{Lie}(\varphi/S)(S)$ and $\mathrm{U}(\varphi)$ are compatible with the identifications of $\mathrm{Lie}(G/S)(S)$ and $\mathrm{Lie}(H/S)(S)$ with restricted p-Lie subalgebras of $\mathrm{U}(G)$ and $\mathrm{U}(H)$. Since $\mathrm{U}(\varphi)$ is an algebra homomorphism, $\mathrm{Lie}(\varphi/S)(S)$ is a homomorphism of restricted p-Lie algebras.

Likewise, if $s : T \to S$ is a base change, the map from $\mathrm{Lie}(G/S)(S)$ to $\mathrm{Lie}(G/S)(T)$ induced by s is a homomorphism of restricted p-Lie algebras. Equivalently, the functor $\mathrm{Lie}(G/S)$ is endowed with a structure of $\mathcal{O}_S$-p-Lie algebra. In particular, as T ranges over the open subsets of S, the sheaf $\mathrm{Lie}(G/S)$ is endowed with a structure of $\mathcal{O}_S$-p-Lie algebra.

6.2.

Following an idea of Demazure, we shall generalize the preceding construction to certain group S-functors which need not be representable. We first give another definition of the symbolic p-th power in the Lie algebra of an S-group scheme G.

[56] One may also prove directly, without passing through $\mathrm{Dif}_{G/S}$, that the Lie algebra of derivations of G at the origin (isomorphic to $\mathrm{Lie}(G/S)(S)$ by 2.5) is stable under taking p-th powers in $\mathrm{U}(G)$. This is done in 6.2 below.

Let D be a derivation of G at the origin.[57] By 1.2.1, D is the composite of the S-derivation

$$\delta = (\tau, \partial_t)$$

of the zero section $\tau : S \to I_S$, and a morphism $x : I_S \to G$ such that $x \circ \tau = \varepsilon$, that is, $x \in \mathrm{Lie}(G/S)(S)$. By the definition in 2.1, D^p is the following composite deviation:

$$S \simeq \underbrace{S \times S \times \cdots \times S}_{p} \xrightarrow{\delta\times\cdots\times\delta} I_S \times \cdots \times I_S \xrightarrow{x\times\cdots\times x} G \times \cdots \times G \xrightarrow{m^{(p)}} G$$

Here $m^{(p)}$ is induced by the multiplication $m : G \times G \to G$. Since $I_S \times \cdots \times I_S$ is affine over S, with affine algebra

$$\mathcal{B} = \mathcal{O}_S[d_1, \ldots, d_p]/(d_1^2, \ldots, d_p^2),$$

the deviation $\delta \times \cdots \times \delta$ is defined by the homomorphism of $\mathcal{O}_S$-modules

$$\phi : \mathcal{B} \longrightarrow \mathcal{O}_S$$

which sends the monomial $d_1 d_2 \cdots d_p$ to 1, and all the other monomials $d_{i_1} \cdots d_{i_r}$, for $0 \leq r < p$, to 0.

Let pr_i denote the projection of I_S^p onto its i-th factor, and let x_i be the image of x in $G(I_S^p)$ under $G(\mathrm{pr}_i)$. Then

$$m^{(p)} \circ (x \times \cdots \times x)$$

is precisely the product $x_1 x_2 \cdots x_p$. Consequently, D^p is also the composite deviation

$$S \xrightarrow{\delta\times\cdots\times\delta} I_S \times \cdots \times I_S \xrightarrow{x_1 x_2 \cdots x_p} G.$$

This description gives another proof that D^p is a derivation of G at the origin. Since G is a very good group (II, 4.11), the images $G(\mathrm{pr}_1)(x)$ and $G(\mathrm{pr}_2)(x)$ of x in $G(I_S \times I_S)$ commute. It follows that the elements $x_i \in G(I_S^p)$ commute pairwise. Hence, for every permutation γ of the factors of I_S^p,

$$(x_1 \cdots x_p) \circ \gamma = x_1 \cdots x_p.$$

Thus $x_1 \cdots x_p$ factors through the canonical projection from I_S^p to the symmetric product $\Sigma^p I_S$ (cf. 4.2).

The affine algebra of $\Sigma^p I_S$ is the subalgebra $\mathcal{A}$ of $\mathcal{B}$ having as an $\mathcal{O}_S$-basis the elementary symmetric functions

$$1 = \sigma_0, \sigma_1, \ldots, \sigma_p$$

in $d_1, \ldots, d_p$. Let $\kappa : \mathcal{A} \hookrightarrow \mathcal{B}$ be the inclusion, and let

$$\pi : \mathcal{A} \longrightarrow \mathcal{O}_S[t]/(t^2)$$

be the homomorphism of $\mathcal{O}_S$-algebras which kills $\sigma_1, \ldots, \sigma_{p-1}$ and sends σ_p to t. Then

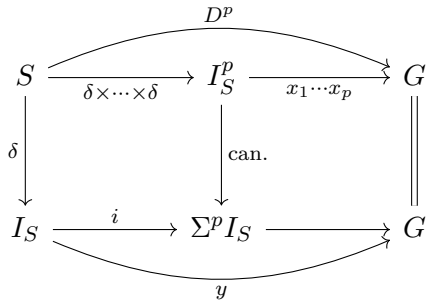
$$\phi \circ \kappa = \partial_t \circ \pi,$$

where $\partial_t : \mathcal{O}_S[t]/(t^2) \to \mathcal{O}_S$ kills 1 and sends t to 1. If

$$i : I_S \hookrightarrow \Sigma^p I_S$$

denotes the closed immersion defined by π, we obtain the commutative diagram

[57] The editors expanded the argument in the original text in what follows.



which shows that D^p has the form $y \circ \delta$, and is therefore indeed a derivation of G at the origin.

6.3.

Let $\mathfrak{S}_p$ be the symmetric group of degree p, and let $I_S^p \times \mathfrak{S}_p$ denote the disjoint union of a family of copies of I_S^p indexed by $\mathfrak{S}_p$. Let

$$\pi : I_S^p \times \mathfrak{S}_p \longrightarrow I_S^p$$

be the canonical projection, and let

$$\mu : I_S^p \times \mathfrak{S}_p \longrightarrow I_S^p$$

be the morphism defining the action of $\mathfrak{S}_p$ on I_S^p: if $\rho \in \mathfrak{S}_p$, the restriction of μ to $I_S^p \times \rho$ has $\mathrm{pr}_{\rho(j)}$ as its j-th component.

Definition. A functor

$$X : (\mathrm{Sch}\,/S)^\circ \longrightarrow (\mathrm{Ens})$$

satisfies *condition (F)* if:

a) X takes finite disjoint unions to direct products;

b) for every S-scheme T, the following sequence is exact:

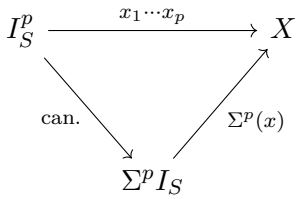
$$X(T \times \Sigma^p I_S) \longrightarrow X(T \times I_S^p) \underset{X(\mathrm{id}_T \times \mu)}{\overset{X(\mathrm{id}_T \times \pi)}{\rightrightarrows}} X(T \times I_S^p \times \mathfrak{S}_p)\,.$$

Every S-scheme satisfies (F). If $\mathcal{F}$ is an $\mathcal{O}_S$-module, then $\mathcal{W}(\mathcal{F})$ satisfies (F). Every projective limit of functors satisfying (F) again satisfies (F). Finally, if Y satisfies (F) and X is an arbitrary S-functor, then $\mathrm{Hom}_S(X, Y)$ satisfies (F).

Let X be a very good group (II, 4.10) satisfying condition (F). Let $x : I_S \to X$ be a morphism extending the unit section of X, and use the notation of 6.2. As above,

$$x_1 \cdots x_p : I_S^p \longrightarrow X$$

factors through $\Sigma^p I_S$:



and therefore defines, by composition, a morphism

$$x^{(p)} : \quad I_S \xrightarrow{i} \Sigma^p I_S \xrightarrow{\Sigma^p(x)} X$$

which we call the *symbolic p-th power* of x.

The endomorphism

$$x \longmapsto x^{(p)}$$

of $\mathrm{Lie}(G/S)(S)$ is plainly compatible with base change and functorial in G. It would be interesting to know for which G this endomorphism makes $\mathrm{Lie}(G/S)(S)$ into a restricted p-Lie algebra.

6.4.

The preceding definition of the symbolic p-th power is particularly well suited to computation. Here are some examples.

6.4.1.

Let M be an "abstract" abelian group, and let $D_S(M)$ be the diagonalizable S-group of type M (I, 4.4.2). For every S-scheme T,

$$D_S(M)(T) = \mathrm{Hom}_{(\mathrm{Ab})}\big(M, \mathcal{O}(T)^\times\big).$$

Let $x \in \mathrm{Lie}(D_S(M)/S)(S)$, that is, an abelian-group homomorphism

$$x : M \longrightarrow \Gamma(S, \mathcal{O}_S + d\mathcal{O}_S)^\times$$

of the form

$$m \longmapsto 1 + d\xi(m), \qquad \xi \in \mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S)).$$

With the notation of 6.2 and 6.3, the product $x_1 \cdots x_p$ associates with $m \in M$ the expression

$$(1 + d_1\xi(m)) \cdots (1 + d_p\xi(m)),$$

that is,

$$1 + \sigma_1\xi(m) + \sigma_2\xi(m)^2 + \cdots + \sigma_p\xi(m)^p.$$

This expression belongs to $\mathcal{O}(\Sigma^p I_S)$. Projecting it to $\mathcal{O}(S)[d]/(d^2)$, killing $\sigma_1, \ldots, \sigma_{p-1}$, and sending σ_p to d, we see that $x^{(p)}$ is the homomorphism

$$m \longmapsto 1 + d\xi(m)^p.$$

Thus, under the identification

$$\mathrm{Lie}(D_S(M)/S)(S) \simeq \mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S))$$

of II, 5.1, the symbolic p-th power sends ξ to

$$\xi^{(p)} : m \longmapsto \xi(m)^p.$$

6.4.2.

Let $\mathcal{F}$ be an $\mathcal{O}_S$-module, and let $G = \mathcal{W}(\mathcal{F})$ be the group S-functor in abelian groups (cf. I, 4.6). Let

$$y \in \mathcal{W}(\mathcal{F})(S) = \Gamma(S, \mathcal{F}),$$

and let y' be its image in $\mathcal{W}(\mathcal{F})(I_S)$ under $\mathcal{W}(\mathcal{F})(I_S \to S)$.

By II, 4.4.2 and 4.5.1, the map

$$y \longmapsto dy'$$

is an isomorphism of $\mathcal{O}(S)$-modules from $\mathcal{W}(\mathcal{F})(S)$ onto $\mathrm{Lie}(\mathcal{W}(\mathcal{F})/S)(S)$. Put $x = dy'$. Then the element x_i of 6.2 is $d_i y''$, where y'' denotes the canonical image of y'[58] in $\mathcal{W}(\mathcal{F})(I_S^p)$. Hence

$$x_1 + \cdots + x_p = (d_1 + \cdots + d_p)y'' = \sigma_1 y'',$$

and this belongs to $\mathcal{W}(\mathcal{F})(\Sigma^p I_S)$. Since the homomorphism

$$\mathcal{O}(\Sigma^p I_S) \longrightarrow \mathcal{O}(I_S)$$

defining the morphism i kills σ_1, we have $x^{(p)} = 0$. Therefore

For every $\mathcal{O}_S$-module $\mathcal{F}$, the operation $x \mapsto x^{(p)}$ on $\mathrm{Lie}\,\mathcal{W}(\mathcal{F})$ is zero.

6.4.3.

Let X be an S-scheme, let $G = \mathrm{Aut}_S X$, and let D be an S-derivation of the structure sheaf $\mathcal{O}_X$. By 1.5, D may be identified with an I_S-automorphism x of X_{I_S} inducing the identity on X. It can be described as follows. If f is a section of $\mathcal{O}_S[d]/(d^2)$ of the form $a + bd$, put

$$D_{I_S} f = Da + (Db)d.$$

Thus D_{I_S} is obtained from D by the base change $I_S \to S$, and the automorphism of X_{I_S} is associated with the endomorphism

$$f \longmapsto f + (D_{I_S} f)d = a + (D(a) + b)d$$

of $\mathcal{O}_S[d]/(d^2)$.

Similarly, let $D_{I_S^p}$ be the differential operator on $X_{I_S^p}$ obtained from D by the base change $I_S^p \to S$. With the notation of 6.2, the automorphism x_i of $X_{I_S^p}$ is associated with the endomorphism

$$f \longmapsto f + (D_{I_S^p} f)d_i$$

of

$$\mathcal{O}_S[d_1, \ldots, d_p]/(d_1^2, \ldots, d_p^2).$$

The product $x_1 \cdots x_p$ is therefore associated with

$$(1 + d_1 D_{I_S^p})(1 + d_2 D_{I_S^p}) \cdots (1 + d_p D_{I_S^p}),$$

that is, with

$$1 + \sigma_1 D_{I_S^p} + \sigma_2 D_{I_S^p}^2 + \cdots + \sigma_p D_{I_S^p}^p.$$

The coefficient of σ_p is $D_{I_S^p}^p$. Consequently, the Lie-algebra isomorphism

$$\mathrm{Der}_S(\mathcal{O}_X) \xrightarrow{\sim} \mathrm{Lie}(\mathrm{Aut}_S X), \qquad D \longmapsto x$$

(cf. 1.5) is also an isomorphism of restricted p-Lie algebras.

6.4.4.

The same method shows that, for every $\mathcal{O}_S$-module $\mathcal{F}$, the Lie-algebra isomorphism

$$\mathrm{Lie}\big(\mathrm{Aut}_{\mathcal{O}_S\text{-mod.}}\,\mathcal{W}(\mathcal{F})/S\big)(S) \xrightarrow{\sim} \big(\mathrm{End}_{\mathcal{O}_S\text{-mod.}}\,\mathcal{W}(\mathcal{F})\big)(S)$$

(cf. II, 4.8) is also an isomorphism of restricted p-Lie algebras.

[58] The editors corrected x to y'.

6.4.5.[59]

Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups, and let

$$G = \mathrm{Spec}^*\,\mathcal{U}$$

be the group S-functor; suppose that G is representable. For every $T \to S$, 5.5 and 6.1.1 define two restricted p-Lie algebra structures on

$$L(T) = \mathrm{Lie}(G)(T).$$

There is a commutative diagram

$$\begin{array}{ccc} L(T) & \xrightarrow{\ \tau\ } & \Gamma(T,\mathcal{U}_T) \\ \Big\downarrow{\scriptstyle \alpha} & \searrow^{i} & \Big\uparrow{\scriptstyle \psi} \\ \mathrm{U}(G_T) & \xleftarrow{\ \varphi\ } & \mathrm{U}(L(T)) \end{array}$$

in which $\mathrm{U}(L(T))$ is the enveloping algebra of $L(T)$, and φ, ψ are the algebra homomorphisms induced by α, τ. It follows that the two restricted p-Lie algebra structures coincide: if $x \in L(T)$ is identified with its image in $\mathrm{U}(G_T)$, respectively in $\Gamma(T,\mathcal{U}_T)$, then $x^{(p)}$ is the image under φ, respectively under ψ, of the element $x^p \in \mathrm{U}(L(T))$.

7. Radicial Groups of Height 1

Let S be a scheme of characteristic $p > 0$.[60] We shall say that an $\mathcal{O}_S$-algebra $\mathcal{A}$ (respectively, an $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$) is *finite locally free* if its underlying $\mathcal{O}_S$-module is locally free and of finite type. If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, we know (cf. 5.5.2) that the group S-functor

$$\mathrm{Spec}^*\,\mathcal{U}_p(\mathcal{L})$$

is represented by a finite locally free S-group scheme $\mathfrak{G}_p(\mathcal{L})$. We shall see that this S-group scheme solves a universal problem (7.2), and we shall characterize the S-group schemes of the form $\mathfrak{G}_p(\mathcal{L})$ (7.4).

Definition 7.0.[61] *Let $H = \mathrm{Spec}\,\mathcal{A}$ be a finite locally free S-group scheme. We say that H is* infinitesimal *if its unit section $\varepsilon_H : S \to H$ is a homeomorphism; equivalently, the augmentation ideal of $\mathcal{A}$ is locally nilpotent.*

7.1.[62]

Let $\mathcal{L}$ be a finite locally free $\mathcal{O}_S$-p-Lie algebra, and let $\mathfrak{G}_p(\mathcal{L})$ be the affine S-group $\mathrm{Spec}\,\mathcal{U}_p(\mathcal{L})$. By 5.5, $\mathcal{L}$ is identified with $\mathrm{Lie}\,\mathfrak{G}_p(\mathcal{L})$.

Now let G be a very good group S-functor satisfying condition (F) of 6.3, and let

$$\varphi : \mathfrak{G}_p(\mathcal{L}) \longrightarrow G$$

be a morphism of group S-functors. By 6.3, the morphism of $\mathcal{O}_S$-Lie algebras

$$\mathrm{Lie}\,\varphi : \mathrm{Lie}\,\mathfrak{G}_p(\mathcal{L}) \longrightarrow \mathrm{Lie}\,G$$

[59] The editors supplied the number 6.4.5 and expanded the original text.

[60] For the results of this section, one may also consult [DG70], § II.7, nos. 3–4.

[61] The editors added this definition (cf. [DG70], § II.4, 7.1), which will be used in 7.2.1.

[62] The editors simplified the original, taking into account the additions made in 5.5.

is compatible with the symbolic p-th power. If $\mathrm{Hom}_p(\mathcal{L}, \mathrm{Lie}\, G)$ denotes the set of morphisms of $\mathcal{O}_S$-Lie algebras compatible with the symbolic p-th power, we therefore have a map

$$\mathrm{Lie} : \mathrm{Hom}_{S\text{-Gr.}}(\mathfrak{G}_p(\mathcal{L}), G) \longrightarrow \mathrm{Hom}_p(\mathcal{L}, \mathrm{Lie}\, G), \qquad \varphi \longmapsto \mathrm{Lie}\, \varphi.$$

7.2. Theorem.
If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, the map

$$\mathrm{Hom}_{S\text{-gr.}}(\mathfrak{G}_p(\mathcal{L}), G) \longrightarrow \mathrm{Hom}_p(\mathcal{L}, \mathrm{Lie}\, G)$$

is bijective in each of the following cases:

(i) G is an S-group scheme;

(ii) G is of the form $\mathrm{Aut}_S X$, where X is an S-scheme;

(iii) G is of one of the forms $\mathcal{W}(\mathcal{F})$ or $\mathrm{Aut}_{\mathcal{O}_S\text{-mod.}}\mathcal{W}(\mathcal{F})$, where $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_S$-module.

The proof of the theorem rests on the following lemma.

Lemma.
If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, the S-group

$$G = \mathfrak{G}_p(\mathcal{L})$$

is annihilated by the Frobenius morphism $\mathrm{Fr} : G \to G^{(p)}$. In particular, G is infinitesimal.

Indeed, let $\mathcal{U}$ be the restricted enveloping algebra of $\mathcal{L}$, let $\mathcal{A} = \mathcal{U}^*$ be the affine algebra of G, and let $\mathcal{I} = \ker \varepsilon_{\mathcal{A}}$ be the augmentation ideal of $\mathcal{A}$.[63] We have

$$\mathcal{A} = \mathcal{I} \oplus \eta_{\mathcal{A}}(\mathcal{O}_S), \tag{1}$$

where $\eta_{\mathcal{A}}$ denotes the unit section of $\mathcal{A}$. Since $\varepsilon_{\mathcal{A}}$ (respectively, $\eta_{\mathcal{A}}$) is the transpose of $\eta_{\mathcal{U}}$ (respectively, $\varepsilon_{\mathcal{U}}$), this decomposition corresponds by duality to

$$\mathcal{U} = \mathcal{J} \oplus \eta_{\mathcal{U}}(\mathcal{O}_S), \tag{2}$$

where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$; thus $\mathcal{J} = \mathcal{I}^*$.

Let π denote the endomorphism $x \mapsto x^p$ of $\mathcal{O}_S$. We must show that $\mathrm{Fr} : G \to G^{(p)}$ factors through the unit section of $G^{(p)}$. By 4.1.4 (c), this is equivalent to saying that the morphism

$$\Phi : \mathcal{I} \otimes_\pi \mathcal{O}_S \longrightarrow \mathcal{A}, \qquad a \otimes_\pi x \longmapsto a^p x,$$

is zero. Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, it suffices to show that the transpose ${}^t\Phi$ is zero.

Now Φ is precisely the following composite:

$$\mathcal{J} \otimes_\pi \mathcal{O}_S \xrightarrow{\tau} \mathcal{A} \otimes_\pi \mathcal{O}_S \xrightarrow{j(\mathcal{A})} \mathrm{S}^p\mathcal{A} \xrightarrow{b(\mathcal{A})} \mathcal{A},$$

Here τ is induced by the inclusion $\mathcal{I} \hookrightarrow \mathcal{A}$, while $b(\mathcal{A})$ and $j(\mathcal{A})$ are defined as in 4.3.3: $b(\mathcal{A})$ is induced by multiplication in $\mathcal{A}$, and $j(\mathcal{A})$ sends $a \otimes_\pi 1$ to the image of $a \otimes \cdots \otimes a$ in $\mathrm{S}^p\mathcal{A}$. The $\mathcal{O}_S$-dual of $\mathrm{S}^p\mathcal{A}$ is the submodule $\Sigma^p\mathcal{U}$ of $\bigotimes^p \mathcal{U}$ consisting of the sections invariant under the symmetric group of degree p. Hence ${}^t\Phi$ is the following composite:

[63] The editors supplied details in what follows. For another proof, see [DG70], § II.7, 3.9.

$$\mathcal{U} \xrightarrow{a(\mathcal{U})} \Sigma^p\mathcal{U} \xrightarrow{r(\mathcal{U})} \mathcal{U} \otimes_\pi \mathcal{O}_S \xrightarrow{q} \mathcal{J} \otimes_\pi \mathcal{O}_S,$$

Here q is induced by the projection $\mathcal{U} \to \mathcal{J}$ with kernel $\eta_{\mathcal{U}}(\mathcal{O}_S)$, $a(\mathcal{U})$ is induced by the comultiplication of $\mathcal{U}$, and $r(\mathcal{U})$ vanishes on symmetrized tensors and sends a section $x \otimes \cdots \otimes x$ to $x \otimes_\pi 1$ (cf. 4.3.3).

It is clear that ${}^t\Phi \circ \eta_{\mathcal{U}} = 0$; by (2), it remains to show that ${}^t\Phi$ annihilates the augmentation ideal $\mathcal{J}$. Since ${}^t\Phi$ is an algebra morphism and $\mathcal{J}$ is generated by $\mathcal{L}$, identified with its image in $\mathcal{U}$, it suffices to show that ${}^t\Phi(x) = 0$ for every section x of $\mathcal{L}$. But

$$-a(\mathcal{U})(x) = (p-1)!\, a(\mathcal{U})(x)$$

is the symmetrization of $x \otimes 1 \otimes \cdots \otimes 1$, so its image under $r(\mathcal{U})$ is zero. This proves the first assertion of the lemma.

The second follows. Every local section of $\mathcal{I}$ has zero p-th power, and $\mathcal{I}$ is an $\mathcal{O}_S$-module of finite type; hence $\mathcal{I}$ is locally nilpotent. Explicitly, if V is an affine open subset of S such that

$$I = \Gamma(V, \mathcal{I})$$

is generated by r elements, then

$$I^{r(p-1)+1} = 0.$$

Thus $G_{\mathrm{red}} = S_{\mathrm{red}}$, and the unit section $\varepsilon_G : S \to G$ is a homeomorphism.

7.2.1.[64]

We first prove assertion (ii) of Theorem 7.2. Let $\pi : X \to S$ be an S-scheme, and first consider an arbitrary infinitesimal S-group H.

Morphisms $\varphi : H \to \operatorname{Aut} X$ correspond bijectively to left actions $\mu : H \times X \to X$ of H on X. For such an action, if ε is the unit section of H, the composite

$$X \simeq S \times X \xrightarrow{\varepsilon \times X} H \times X \xrightarrow{\mu} X$$

must be the identity. Since $(H \times X)_{\mathrm{red}}$ is identified with X_{red}, the morphism μ must induce the identity on the associated reduced schemes. In particular, μ induces an action of H on every open subset of X. Thus, for every open $U \subset X$ affine over S, we obtain a morphism of unital associative algebras

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(H) \otimes \mathcal{A}(U)$$

which makes $\mathcal{A}(U)$ a left $\mathcal{A}(H)$-comodule, in such a way that the restriction maps

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(U'), \qquad U' \subset U,$$

are comodule morphisms. Conversely, every datum of this kind comes from a unique left action $\mu : H \times X \to X$.

We also have the following lemma.

Lemma.

Let $X = \operatorname{Spec} \mathcal{C}$ be an affine S-scheme, let $H = \operatorname{Spec} \mathcal{A}$ be an infinitesimal S-group, and put

$$\mathcal{U} = \mathcal{A}^* = \operatorname{Hom}_{\mathcal{O}_S}(\mathcal{A}, \mathcal{O}_S).$$

[64]The editors supplied details in what follows.

Left actions of H on X correspond bijectively to representations of the algebra $\mathcal{U}$ in the $\mathcal{O}_S$-module $\mathcal{C}$ such that

$$\begin{aligned} &(a) \quad u(1_{\mathcal{C}}) = \varepsilon(u)\, 1_{\mathcal{C}}, \\ &(b) \quad u(xy) = \sum_i v_i(x) w_i(y) \quad \text{if} \quad \Delta u = \sum_i v_i \otimes w_i. \end{aligned}$$

Here u is an arbitrary section of $\mathcal{U}$ over an affine open subset V of S, while x and y are sections of $\mathcal{C}$ over V. The symbol $1_{\mathcal{C}}$ denotes the unit section of $\mathcal{C}$, and ε and Δ denote the augmentation and the diagonal morphism of $\mathcal{U}$, respectively.

Indeed, a left action μ of H on X is defined by a morphism of unital associative algebras

$$\lambda : \mathcal{C} \longrightarrow \mathcal{A} \otimes \mathcal{C}$$

which makes $\mathcal{C}$ a left $\mathcal{A}$-comodule. Let α be the composite

$$\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\mathcal{U}\otimes\lambda} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\gamma\otimes\mathcal{C}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{C} \simeq \mathcal{C}$$

where γ is the "contraction" from $\mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}$ to $\mathcal{O}_S$. Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, the map

$$\lambda \longmapsto (\gamma \otimes \mathcal{C})(\mathcal{U} \otimes \lambda)$$

is a bijection from

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{C}, \mathcal{A} \otimes \mathcal{C}) \quad \text{onto} \quad \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{U} \otimes \mathcal{C}, \mathcal{C}).$$

Moreover, by duality, the condition that λ define a left $\mathcal{A}$-comodule structure (respectively, be a morphism of unital associative algebras) is equivalent to the condition that α be a representation of $\mathcal{U}$ in $\mathcal{C}$ (respectively, satisfy conditions (a) and (b)). This proves the lemma.

For any representation of $\mathcal{U}$ in the $\mathcal{O}_S$-module $\mathcal{C}$, it is also clear that the sections u of $\mathcal{U}$ satisfying conditions (a) and (b) form a subalgebra of $\mathcal{U}$.

In the particular case

$$H = \mathfrak{G}_p(\mathcal{L})$$

which concerns us, these conditions will therefore hold for every section u of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$ if they hold for the sections u of $\mathcal{L}$, identified with its image in $\mathcal{U}_p(\mathcal{L})$.

If u is a section of $\mathcal{L}$, conditions (a) and (b) simply mean

$$u(1_{\mathcal{C}}) = 0, \qquad u(xy) = u(x)y + xu(y);$$

that is, $\alpha(u)$ is an $\mathcal{O}_S$-derivation of $\mathcal{C} = \mathcal{A}(X)$. Assertion (ii) of 7.2 follows. Indeed, every morphism

$$\varphi : \mathfrak{G}_p = \mathfrak{G}_p(\mathcal{L}) \longrightarrow \mathrm{Aut}\, X$$

defines a morphism of p-Lie algebras $\mathrm{Lie}\,\varphi$ from $\mathcal{L} = \mathrm{Lie}\,\mathfrak{G}_p$ to

$$\pi_*(\mathrm{Der}_{\mathcal{O}_S} \mathcal{O}_X),$$

and conversely every datum of this kind comes, by what precedes, from a unique action $\mu : G \times X \to X$.

7.2.2.

We now show how assertion (i) of Theorem 7.2 follows from (ii). Let G be an S-group scheme. If T is an S-scheme and $x \in G(T)$, let ℓ_x^T (respectively, r_x^T) denote the left (respectively, right) translation of G_T defined by x. The maps

$$\ell^T : x \longmapsto \ell_x^T$$

therefore determine a homomorphism $\ell : G \to \operatorname{Aut} G$.

On the other hand, let f be a T-automorphism of G_T, and define

$${}^x f = (r_x^T)^{-1} f r_x^T;$$

thus, for every $T' \to T$ and $g \in G(T')$,

$$({}^x f)(g) = f(gx)x^{-1}.$$

In this way G acts on the left on the group S-functor $\operatorname{Aut} G$, and hence also on the functors

$$T \longmapsto \operatorname{Hom}_{T\text{-Gr.}}\big(\mathfrak{G}_p(\mathcal{L}_T), \operatorname{Aut} G_T\big)$$

and

$$T \longmapsto \operatorname{Hom}_p\big(\mathcal{L}_T, \operatorname{Lie}(\operatorname{Aut} G_T/T)\big).$$

The morphism $\ell^T : G_T \to \operatorname{Aut} G_T$ identifies G_T with the group of automorphisms of the T-scheme G_T which commute with right translations. Its derived morphism $\operatorname{Lie}(\ell^T)$ identifies $\operatorname{Lie} G_T$ with the p-Lie algebra of $\mathcal{O}_T$-derivations of $\mathcal{O}_{G_T}$ which commute with right translations (cf. II, 4.11.1). They therefore induce the commutative squares

$$\begin{array}{ccc}
\operatorname{Hom}_{T\text{-Gr.}}\big(\mathfrak{G}_p(\mathcal{L}_T), G_T\big) & \xrightarrow{\ \operatorname{Lie}\ } & \operatorname{Hom}_p\big(\mathcal{L}_T, \operatorname{Lie}(G_T/T)\big) \\
\Big\downarrow{\scriptstyle \ell_T} & & \Big\downarrow{\scriptstyle \operatorname{Lie}\ell_T} \\
\operatorname{Hom}_{T\text{-Gr.}}\big(\mathfrak{G}_p(\mathcal{L}_T), \underline{\operatorname{Aut}}\, G_T\big) & \xrightarrow{\ \operatorname{Lie}\ } & \operatorname{Hom}_p\big(\mathcal{L}_T, \operatorname{Lie}(\underline{\operatorname{Aut}}\, G_T/T)\big).
\end{array}$$

The images of the two vertical arrows are the subfunctors formed by the invariants under the action of the S-group G. Since the lower horizontal arrow is invertible by 7.2.1 and is compatible with the action of G, the upper horizontal arrow is invertible as well. This proves 7.2 (i).

7.2.3.

Consider the case in which $G = \operatorname{Aut}_{\mathcal{O}_S\text{-mod.}} \mathcal{W}(\mathcal{F})$.[65] Put

$$\mathcal{U} = \mathcal{U}_p(\mathcal{L}), \qquad \mathcal{A} = \mathcal{U}^*, \qquad H = \mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec} \mathcal{A}.$$

Since H is affine over S, VIB, 11.6.1 shows that a morphism of S-groups from H to $\operatorname{Aut}_{\mathcal{O}_S\text{-mod.}} \mathcal{W}(\mathcal{F})$ is the same as a right $\mathcal{A}$-comodule structure

$$\mu : \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}.$$

Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, this is in turn equivalent to giving a representation

$$\alpha : \mathcal{U} \longrightarrow \operatorname{End}_{\mathcal{O}_S}(\mathcal{F})$$

[65] In what follows, the editors expanded and simplified the original, taking VIB, 11.6.1 into account.

of $\mathcal{U}$ in $\mathcal{F}$. Finally, by the universal property of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, giving such a morphism α is equivalent to giving its restriction ρ to $\mathcal{L}$, identified with its image in $\mathcal{U}$, and ρ is a morphism of p-Lie algebras from $\mathcal{L}$ to $\mathrm{End}_{\mathcal{O}_S}(\mathcal{F})$.

Finally, consider the case $G = \mathcal{W}(\mathcal{F})$, with the preceding notation.[66] To give a morphism of S-functors $\varphi : H \to \mathcal{W}(\mathcal{F})$ is first of all equivalent to giving an element θ of $\Gamma(H, \mathcal{F}\otimes\mathcal{O}_H)$. Since H is finite locally free over S, we have

$$\Gamma(H, \mathcal{F} \otimes \mathcal{O}_H) = \Gamma(S, \mathcal{F} \otimes \mathcal{A}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{U}, \mathcal{F}).$$

The condition that φ be a group morphism means that θ, regarded as a morphism of $\mathcal{O}_S$-modules $\mathcal{U} \to \mathcal{F}$, vanishes on $1_{\mathcal{U}}$ and on $\mathcal{J}^2$, where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$. Consequently,

$$\mathrm{Hom}_{S\text{-gr.}}(H, \mathcal{W}(\mathcal{F})) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{J}/\mathcal{J}^2, \mathcal{F}). \tag{1}$$

Next consider the quasi-coherent sheaf $[\mathcal{L}, \mathcal{L}]$, the image of the morphism

$$\mathcal{L} \otimes \mathcal{L} \longrightarrow \mathcal{L}, \qquad x \otimes y \longmapsto [x, y].$$

For every affine open subset V of S, we have

$$[\mathcal{L}, \mathcal{L}](V) = [\mathcal{L}(V), \mathcal{L}(V)].$$

There is an exact sequence

$$(\dagger) \qquad 0 \longrightarrow [\mathcal{L}, \mathcal{L}] \longrightarrow \mathcal{L} \xrightarrow{\pi} \mathcal{J}/\mathcal{J}^2 \longrightarrow 0,$$

in which π is the composite of the inclusion $\mathcal{L} \hookrightarrow \mathcal{J}$ and the projection $\mathcal{J} \to \mathcal{J}/\mathcal{J}^2$.

Indeed, the question is local on S, so we may suppose that S is affine with ring R, and that

$$L = \mathcal{L}(S)$$

is free with basis $(x_1, \ldots, x_r)$. Identify L with its image in $U = U_p(L)$, and let K be the sub-R-module of U which is the direct sum of $[L, L]$ and the submodule with basis the monomials

$$x_1^{n_1} \cdots x_r^{n_r} \quad \text{such that} \quad n_1 + \cdots + n_r \geq 2.$$

One checks that K is a two-sided ideal of U. Since K is contained in J^2, where J is the augmentation ideal of U, and contains all the products $x_i x_j$, which generate J^2, it follows that $K = J^2$. Hence

$$J^2 \cap L = [L, L],$$

and the displayed sequence is exact.

By 6.4.2, $\mathrm{Lie}\,\mathcal{W}(\mathcal{F})$ is simply $\mathcal{W}(\mathcal{F})$, with zero Lie bracket and zero symbolic p-th power. From this and the preceding discussion, we obtain

$$\mathrm{Hom}_p(\mathcal{L}, \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{L}/[\mathcal{L}, \mathcal{L}], \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{J}/\mathcal{J}^2, \mathcal{F}). \tag{2}$$

Together with (1), this completes the proof of Theorem 7.2.

7.3. Lemma.

If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, the morphism

$$j_{\mathcal{L}} : \mathcal{L} \longrightarrow \mathrm{Lie}\,\mathfrak{G}_p(\mathcal{L})$$

[66] The editors added what follows. The original said that "the case of $\mathcal{W}(\mathcal{F})$ is analogous."

of 5.5 is invertible.[67]

For the proof, see 5.5.1.

7.4.

To conclude this section, we give a characterization of the S-group schemes of the form $\mathfrak{G}_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra.

Let G be an S-group scheme, let ε_G be its unit section, and let $\mathcal{I}'$ be the kernel of the morphism

$$\varepsilon_G^{-1}(\mathcal{O}_G) \longrightarrow \mathcal{O}_S$$

corresponding to ε_G. By 2.5 and 1.3.1, the image of $\mathrm{Lie}(G/S)(S)$ in $\mathrm{U}(G)$ is identified with the morphisms of $\mathcal{O}_S$-modules from $\varepsilon_G^{-1}(\mathcal{O}_G)$ to $\mathcal{O}_S$ which vanish on the unit section of $\varepsilon_G^{-1}(\mathcal{O}_G)$ and on $\mathcal{I}'^2$.

We thereby recover the canonical isomorphism

$$\mathrm{Lie}(G/S)(S) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}'/\mathcal{I}'^2, \mathcal{O}_S)$$

of II, 3.3 and 4.11.4.[68] As in the cited passage, put

$$\omega_{G/S} = \mathcal{I}'/\mathcal{I}'^2.$$

Thus the sheaf $\mathrm{Lie}(G/S)$ is identified with

$$\omega_{G/S}^* = \mathrm{Hom}_{\mathcal{O}_S}(\omega_{G/S}, \mathcal{O}_S).$$

Moreover,[69] if $G = \mathfrak{G}_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, then 5.5.3 gives

$$\omega_{G/S} = \mathcal{L}^*.$$

Theorem.

If G is a group scheme over a scheme S of characteristic $p > 0$, the following assertions are equivalent:

(i) There exists a finite locally free $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$ such that $G \simeq \mathfrak{G}_p(\mathcal{L})$.

(i′) The $\mathcal{O}_S$-p-Lie algebra $\mathrm{Lie}(G)$ is finite locally free, and $G \simeq \mathfrak{G}_p(\mathrm{Lie}(G))$.

(ii) G is affine over S, the $\mathcal{O}_S$-module $\omega_{G/S}$ is locally free of finite type, and the affine algebra of G is locally isomorphic to the quotient of the symmetric algebra $\mathrm{S}_{\mathcal{O}_S}(\omega_{G/S})$ by the ideal generated by the p-th powers of the sections of $\omega_{G/S}$.

(iii) G is locally of finite presentation over S, is of height ≤ 1, and $\omega_{G/S}$ is locally free.

(iii′) G is locally of finite type over S, is of height ≤ 1, and $\omega_{G/S}$ is locally free.

(iv) G is locally of finite presentation and flat over S, and is of height ≤ 1.[70]

[67] To avoid changing the numbering, the editors retained statement 7.3, although they included it, together with its proof, in 5.5.1.

[68] If G is affine over S and $\mathcal{I}$ denotes the augmentation ideal of $\mathcal{A}(G)$, then $\mathcal{I}'/\mathcal{I}'^2$ is identified with $\varepsilon_G^*(\mathcal{I}/\mathcal{I}^2)$; cf. loc. cit.

[69] The editors added the following sentence.

[70] The editors added assertion (i′), implicit in the original, and assertions (iii′) and (iv), pointed out by O. Gabber. Assertion (iv) takes up a footnote in the original, which stated: "The condition on $\omega_{G/S}$ is in fact unnecessary, as is easily seen by reducing to the case where S is local with residue field k, and applying the theorem to G_k." As Gabber pointed out, this is incorrect without a flatness hypothesis. Indeed, let A

7.4.1.

The equivalence (i) $\Leftrightarrow$ (i′) follows from 5.5.3 (i), and the implications (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iii′) are clear. We also have (i) $\Rightarrow$ (iv), since $\mathfrak{G}_p(\mathcal{L})$ is finite locally free and of height ≤ 1, by 5.5.2 and the lemma in 7.2. We show that (i) implies (ii).

Let $\mathcal{I}$ be the augmentation ideal of

$$\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*.$$

We saw in 5.5.3 (ii) that $\omega_{G/S} = \mathcal{I}/\mathcal{I}^2$ is identified with $\mathcal{L}^*$, and hence is finite locally free.

Now suppose that S is affine. There is a section

$$\sigma : \omega_{G/S} \longrightarrow \mathcal{I}$$

of the projection $\mathcal{I} \to \mathcal{I}/\mathcal{I}^2$. It induces an algebra morphism

$$\sigma' : \mathrm{S}_{\mathcal{O}_S}(\omega_{G/S}) \longrightarrow \mathcal{A},$$

and, by the lemma in 7.2, σ' factors through a morphism

$$\varphi : \mathrm{S}_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K} \longrightarrow \mathcal{A},$$

where $\mathcal{K}$ denotes the ideal generated by the p-th powers of sections of $\omega_{G/S}$. Filter $\mathcal{A}$ by powers of $\mathcal{I}$, and filter $\mathrm{S}_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K}$ by powers of the ideal generated by $\omega_{G/S}$. The morphism φ plainly induces an epimorphism on the associated graded modules. Hence φ is an epimorphism between locally free $\mathcal{O}_S$-modules of the same rank (cf. 5.3.3), and is therefore an isomorphism. This proves that (i) implies (ii).

7.4.2.

Suppose now that G is of height ≤ 1 and locally of finite presentation over S.[71] The Frobenius morphism $\mathrm{Fr} : G \to G^{(p)}$ is integral and factors through the unit section of $G^{(p)}$; consequently, G is integral, and hence affine, over S. Put $\mathcal{A} = \mathcal{A}(G)$. Since G is locally of finite presentation over S, it follows that G is finite and of finite presentation over S, and hence that $\mathcal{A}(G)$ is an $\mathcal{O}_S$-module of finite presentation (cf. EGA IV$_1$, 1.4.7).

Let $\mathcal{I}$ be the augmentation ideal of $\mathcal{A}$. Since

$$\mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{I},$$

where $\eta_{\mathcal{A}}$ is the unit section of $\mathcal{A}$, the module $\mathcal{I}$ is of finite presentation over $\mathcal{O}_S$, as is $\omega_G = \mathcal{I}/\mathcal{I}^2$. If instead G is assumed to be of height ≤ 1 and locally of finite type over S, the same argument shows that $\mathcal{A}(G)$, $\mathcal{I}$, and $\omega_G = \mathcal{I}/\mathcal{I}^2$ are $\mathcal{O}_S$-modules of finite type.

Thus, under hypothesis (iii′), $\omega_{G/S}$ is finite locally free over $\mathcal{O}_S$, as is

$$\mathcal{L} = \mathrm{Lie}(G/S) = \omega^*_{G/S}.$$

Put

$$\mathcal{B} = \mathcal{U}_p(\mathcal{L})^*, \qquad H = \mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec} \mathcal{B}.$$

---

be an Artinian local ring of characteristic $p > 0$, let J be a proper nonzero ideal of A, and let H be the subgroup

$$\operatorname{Spec} A[x]/(x^p, Jx)$$

of $\alpha_{p,A}$; that is, for every A-algebra R,

$$H(R) = \{x \in R \mid x^p = 0 \text{ and } Jx = 0\}.$$

Then H is not flat over A, and hence is not of the form $\mathfrak{G}_p(\mathcal{L})$ with $\mathcal{L}$ a finite-rank free p-Lie algebra over A.

[71] The editors supplied details and simplified the original in what follows.

By Theorem 7.2, the identity map of $\mathcal{L}$ corresponds to a group morphism from $H = \mathfrak{G}_p(\mathcal{L})$ to G, and hence to a morphism of $\mathcal{O}_S$-algebras

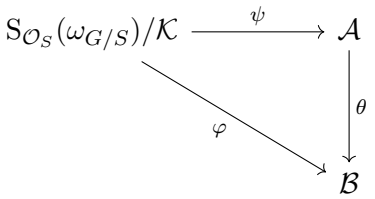
$$\theta : \mathcal{A} \longrightarrow \mathcal{B}.$$

By definition, θ induces an isomorphism from $\omega_{G/S}$ to $\omega_{H/S}$; we must show that θ itself is an isomorphism.

We may restrict to the case in which S is affine. There is then a section τ of the projection $\mathcal{I} \to \omega_{G/S}$, inducing an algebra morphism

$$\tau' : \mathrm{S}_{\mathcal{O}_S}(\omega_{G/S}) \longrightarrow \mathcal{A}.$$

Every local section of $\mathcal{I}$ has zero p-th power, because $\mathrm{Fr} : G \to G^{(p)}$ factors through the unit section of $G^{(p)}$. Thus τ' induces a morphism of $\mathcal{O}_S$-algebras ψ fitting into the commutative diagram



where $\mathcal{K}$ is the ideal generated by the p-th powers of sections of $\omega_{G/S}$. As in 7.4.1, ψ is an epimorphism of $\mathcal{O}_S$-modules. On the other hand, 7.4.1 shows that $\varphi = \theta \circ \psi$ is an isomorphism. Hence θ is an isomorphism as well. This proves (iii′) ⇒ (i).

7.4.3.[72]

We finally prove that (iv) implies (iii). It suffices to show that $\omega_{G/S}$ is locally free, so we may suppose that S is affine with ring R. As observed at the beginning of 7.4.2, hypothesis (iv) then implies that $G = \operatorname{Spec} A$, where A is an R-algebra which is an R-module of finite presentation, as is

$$\omega_{G/A} = I/I^2,$$

where I is the augmentation ideal of A. Since G is also assumed flat over S, the algebra A is a finite locally free R-module (cf. [BAC] II, § 5.2, Theorem 1 and Corollary 2). By the cited result, to show that $\omega_{G/A}$ is locally free of finite rank, it suffices to show that

$$(\omega_{G/A})_{\mathfrak{m}}$$

is flat for every maximal ideal $\mathfrak{m}$ of R. Thus we may suppose that R is local and that A is free of rank $n+1$; then I is free of rank n. Let $\mathfrak{m}$ be the maximal ideal of R, and put

$$k = R/\mathfrak{m}.$$

Let I_k be the augmentation ideal of A_k, and let r be the dimension of

$$\omega_{G_k/k} = I_k/I_k^2.$$

Choose a basis $(e_1, \ldots, e_n)$ of I_k such that $(e_{r+1}, \ldots, e_n)$ is a basis of I_k^2, and choose elements $x_1, \ldots, x_n$ of I lifting the e_i. Nakayama's lemma shows that $(x_1, \ldots, x_n)$ is a basis of I over R.

[72] The editors added this paragraph to prove (iv) ⇒ (iii); cf. editor's note (71).

Let N be the sub-R-module of I with basis $(x_1, \ldots, x_r)$, and let B be the quotient of the symmetric algebra of N by the ideal generated by the elements x^p, for $x \in N$. Since every element of I has zero p-th power, we obtain a morphism of R-algebras

$$\psi : B \longrightarrow A.$$

By 7.4.2, $\psi \otimes k$ is an isomorphism. It follows that $\operatorname{Coker} \psi = 0$, and, if $K = \ker \psi$, the morphism

$$\tau : K \otimes k \longrightarrow B \otimes k$$

is zero. But ψ is surjective and A is flat over R, so τ is also injective. Hence $K \otimes k = 0$.

Since A is an R-module of finite presentation, K is an R-module of finite type (cf. [BAC] I, § 2.8, Lemma 9). Nakayama's lemma now gives $K = 0$. Thus ψ is an isomorphism of R-algebras. Since $\psi^{-1}(I)$ contains the augmentation ideal J of B, it follows that $\psi^{-1}(I) = J$. Therefore ψ^{-1} induces an isomorphism of R-modules

$$I/I^2 \xrightarrow{\sim} J/J^2 = N.$$

This proves that $\omega_{G/S}$ is finite locally free, and hence proves (iv) $\Rightarrow$ (iii). The proof of Theorem 7.4 is complete.

Remark 7.5.[73]

It follows immediately from Theorems 7.2 and 7.4 that the functors

$$G \longmapsto \operatorname{Lie}(G), \qquad \mathcal{L} \longmapsto \mathfrak{G}_p(\mathcal{L})$$

induce quasi-inverse equivalences between the category of S-groups which are locally of finite presentation, flat, and of height ≤ 1, and the full subcategory of the category of $\mathcal{O}_S$-p-Lie algebras consisting of the finite locally free $\mathcal{O}_S$-p-Lie algebras.

8. The Case of a Base Field

8.1.

We now summarize the results obtained when S is the spectrum of a field k of characteristic $p > 0$. We call a k-group scheme *algebraic* if its underlying scheme is of finite type over k. Theorem 7.2 then gives the following result.[74]

Theorem 8.1.1. *The functor $\mathfrak{G}_p$, which associates with every finite-dimensional restricted p-Lie algebra $\mathcal{L}$ over k the k-group $\mathfrak{G}_p(\mathcal{L})$, is left adjoint to the functor which associates* $\operatorname{Lie}(G)$ *with every algebraic k-group G.*

Combining this with Theorem 7.4 gives:

Theorem 8.1.2. *The functors*

$$\mathfrak{G}_p : \mathcal{L} \longmapsto \mathfrak{G}_p(\mathcal{L}), \qquad G \longmapsto \operatorname{Lie}(G),$$

induce quasi-inverse equivalences between the category of finite-dimensional restricted p-Lie algebras over k and the category of algebraic k-groups of height at most 1.

Since $\mathfrak{G}_p$ is a left adjoint, it commutes with inductive limits,[75] and in particular with cokernels. Let

$$\varphi : G \longrightarrow H, \qquad \varphi' : G' \longrightarrow H$$

[73] The editors added this remark.

[74] The editors supplied the numbering 8.1.1 through 8.1.3 in order to highlight the results stated here.

[75] The editors expanded the argument which follows, as well as the proof of Corollary 8.1.3.

be two morphisms between algebraic k-groups of height at most 1. The fiber product $G \times_H G'$ is again an algebraic k-group of height at most 1, because the Frobenius morphism

$$\mathrm{Fr} : G \longrightarrow G^{(p)}$$

commutes with fiber products. Thus the inclusion of the category of algebraic k-groups of height at most 1 into the category of all algebraic k-groups commutes with fiber products, and hence in particular with kernels. We obtain:

Corollary 8.1.3. *The functor $\mathfrak{G}_p$ is exact in the following sense. Let*

$$\pi : \mathcal{L}_1 \longrightarrow \mathcal{L}_2$$

be a surjective homomorphism between finite-dimensional restricted p-Lie algebras over k, and let i be the inclusion of $\mathcal{L}_0 = \ker \pi$ into $\mathcal{L}_1$. Then the following sequence of algebraic k-groups is exact:

$$1 \longrightarrow \mathfrak{G}_p(\mathcal{L}_0) \xrightarrow{\mathfrak{G}_p(i)} \mathfrak{G}_p(\mathcal{L}_1) \xrightarrow{\mathfrak{G}_p(\pi)} \mathfrak{G}_p(\mathcal{L}_2) \longrightarrow 1.$$

[76]

Indeed, by the preceding discussion, $\mathfrak{G}_p(i)$ induces an isomorphism from $\mathfrak{G}_p(\mathcal{L}_0)$ onto $\ker(\mathfrak{G}_p(\pi))$. This kernel is the same whether it is taken in the category of all algebraic k-groups or in the subcategory of those of height at most 1. Furthermore,

$$\mathfrak{G}_p(\pi) : \mathfrak{G}_p(\mathcal{L}_1) \longrightarrow \mathfrak{G}_p(\mathcal{L}_2)$$

identifies $\mathfrak{G}_p(\mathcal{L}_2)$ with the quotient of $\mathfrak{G}_p(\mathcal{L}_1)$ by $\mathfrak{G}_p(\mathcal{L}_0)$ in the category of algebraic k-groups.

Remark 8.1.4.[77]

Let $\varphi : G \to H$ be a morphism of k-groups and put $K = \ker(\varphi)$. Suppose that φ is covering for the (fpqc) topology. This is in particular the case if it is covering for a coarser topology, such as the (fppf) topology. Then φ is a K-torsor over H (cf. IV, 5.1.7.1). Moreover, by IV, 6.3.1, there exists a covering of H by affine open subsets S_i, and for each i an affine faithfully flat morphism $T_i \to S_i$ factoring through φ. Now $G \times_H T_i$ is T_i-isomorphic to $K \times T_i$, hence faithfully flat over T_i. By (fpqc) descent, $G \times_H S_i \to S_i$ is faithfully flat, and therefore φ is faithfully flat.

Conversely, if φ is faithfully flat and quasi-compact, respectively faithfully flat and locally of finite presentation, then it is covering for the (fpqc), respectively (fppf), topology (cf. IV, 6.3.1). Finally, a morphism of sheaves is covering if and only if it is an epimorphism (cf. IV, 4.4.3). In particular, a quasi-compact morphism of k-groups is faithfully flat if and only if it is an epimorphism of (fpqc) sheaves.

8.2. Proposition. *Consider an exact sequence*[78] *of algebraic groups over a field k of characteristic $p > 0$:*

$$1 \longrightarrow G' \xrightarrow{\tau} G \xrightarrow{\pi} G'' \longrightarrow 1.$$

Consider the following assertions:

(i) *π is smooth;*

[76] Moreover, by VIA, 3.2, $\mathfrak{G}_p(\mathcal{L}_2)$ represents the (fppf) quotient sheaf of $\mathfrak{G}_p(\mathcal{L}_1)$ by $\mathfrak{G}_p(\mathcal{L}_0)$.

[77] The editors added this remark, communicated by O. Gabber, for use in 8.3.1.

[78] Thus π is faithfully flat and τ identifies G' with $\ker \pi$, so G'' represents the (fppf) quotient sheaf of G by G'; cf. VIA, 3.2 and 5.2.

(ii) *G' is smooth;*

(iii) *for every integer $n > 0$, the sequence induced by τ and π,*

$$1 \longrightarrow \mathrm{Fr}^n G' \longrightarrow \mathrm{Fr}^n G \longrightarrow \mathrm{Fr}^n G'' \longrightarrow 1.$$

is exact;

(iv) *the morphism*

$$\mathrm{Fr}\,\pi : \mathrm{Fr}\,G \longrightarrow \mathrm{Fr}\,G''$$

is an epimorphism of (fppf) sheaves;

(v) *the morphism*

$$\mathrm{Lie}(\pi) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(G'')$$

is surjective.

Then

$$\text{(i)} \iff \text{(ii)} \implies \text{(iii)} \implies \text{(iv)} \iff \text{(v)},$$

and all five assertions are equivalent when G is smooth over k.

Indeed, (i) is equivalent to (ii) by VIB, 9.2 (vii), and it is clear that (iii) implies (iv). The equivalence of (iv) and (v) follows from 8.1.3.

The implication (ii) ⇒ (iii) follows from the diagram

$$\begin{array}{ccccccccc}
1 & \longrightarrow & G' & \xrightarrow{\tau} & G & \xrightarrow{\pi} & G'' & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle \mathrm{Fr}^n(G'/k)} & & \downarrow{\scriptstyle \mathrm{Fr}^n(G/k)} & & \downarrow{\scriptstyle \mathrm{Fr}^n(G''/k)} & & \\
1 & \longrightarrow & G'^{(p^n)} & \xrightarrow[\tau^{(p^n)}]{} & G^{(p^n)} & \xrightarrow[\pi^{(p^n)}]{} & G''^{(p^n)} & \longrightarrow & 1
\end{array}$$

whose two rows are exact. Since $\mathrm{Fr}^n(G'/k)$ is an epimorphism of (fppf) sheaves by Corollary 8.3.1 below, π induces an epimorphism from $\mathrm{Fr}^n G$ onto $\mathrm{Fr}^n G''$; here one uses the snake lemma for sheaves of groups, not necessarily commutative.

Likewise, suppose that G is smooth over k. Then $\mathrm{Fr}(G/k)$ is an epimorphism. If $\mathrm{Fr}\,\pi$ is also an epimorphism, the same snake lemma, applied to the preceding diagram with $n = 1$, shows that $\mathrm{Fr}(G'/k)$ is an epimorphism. Hence G' is smooth over k by 8.3.1 below.

8.3. Proposition.[79] *Let G be a group locally of finite type over a field k of characteristic $p > 0$. There exists an integer n_0 such that*

$$G/(\mathrm{Fr}^n G)$$

is smooth over k for every $n \geq n_0$.

The construction of $G/(\mathrm{Fr}^n G)$ commutes with extension of the base field (4.1.1 and VIA, 3.3.2), so we may suppose that k is perfect. Then G_{red} is a k-group locally of finite type (cf. VIA, 0.2), and, putting

$$H = G_{\mathrm{red}} \backslash G,$$

we have the following commutative exact diagram:

[79] The editors replaced "algebraic" by "locally of finite type."

$$\begin{array}{ccccccc} 1 & \longrightarrow & G_{\mathrm{red}} & \longrightarrow & G & \longrightarrow & H \\ & & \downarrow{\scriptstyle \mathrm{Fr}^n(G_{\mathrm{red}}/k)} & & \downarrow{\scriptstyle \mathrm{Fr}^n(G/k)} & & \downarrow{\scriptstyle \mathrm{Fr}^n(H/k)} \\ 1 & \longrightarrow & G_{\mathrm{red}}^{(p^n)} & \longrightarrow & G^{(p^n)} & \longrightarrow & H^{(p^n)}\,. \end{array}$$

The scheme H is the spectrum of a finite local k-algebra with residue field k (cf. VIA, 5.6.1). There is therefore an integer n_0 such that, for every $n \geq n_0$, $\mathrm{Fr}^n(H/k)$ factors through the unit section of $H^{(p^n)}$. It follows that, for $n \geq n_0$, $\mathrm{Fr}^n(G/k)$ factors through $G_{\mathrm{red}}^{(p^n)}$, and VIA, 5.4.1 gives the commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{\mathrm{Fr}^n(G/k)} & G_{\mathrm{red}}^{(p^n)} \\ {\scriptstyle \pi}\downarrow & \nearrow{\scriptstyle i} & \\ G/(\mathrm{Fr}^n G) & & \end{array}$$

where i is a closed immersion and π is the canonical projection. Moreover, i induces a homeomorphism of the underlying topological spaces, so it is an isomorphism. Since k is perfect, $G_{\mathrm{red}}^{(p^n)}$ is smooth over k (VIA, 1.3.1). Thus $G/(\mathrm{Fr}^n G)$ is smooth over k for every $n \geq n_0$.

8.3.1. Corollary. *Let G be a group locally of finite type over a field k of characteristic $p > 0$, and let $n \geq 1$.*[80] *The following conditions are equivalent:*

(i) *G is smooth over k;*

(ii) *$\mathrm{Fr}^n(G/k) : G \to G^{(p^n)}$ is an epimorphism of (fppf) sheaves;*

(iii) *$\mathrm{Fr}^n(G/k) : G \to G^{(p^n)}$ is faithfully flat.*

Since G is locally of finite type over k, $\mathrm{Fr}^n(G/k)$ is of finite presentation, so the equivalence of (ii) and (iii) follows from 8.1.4. Suppose that G is smooth over k, and hence reduced. Since $\mathrm{Fr}^n(G/k)$ is surjective, it is faithfully flat (cf. VIA, 6.2 or VIB, 1.3).

Conversely, suppose that $\mathrm{Fr}^n(G/k)$ is faithfully flat. The morphism

$$\mathrm{Fr}^n(G^{(p^n)}/k)$$

is obtained from $\mathrm{Fr}^n(G/k)$ by base change (cf. 4.1.3), so it too is faithfully flat, as is the composite

$$\mathrm{Fr}^{2n}(G/k) : G \longrightarrow G^{(p^n)} \longrightarrow G^{(p^{2n})}.$$

It follows that, for every $m \in \mathbb{N}$,

$$\mathrm{Fr}^{mn}(G/k) : G \longrightarrow G^{(p^{mn})}$$

is faithfully flat. Hence it induces an isomorphism

$$G/(\mathrm{Fr}^{mn} G) \simeq G^{(p^{mn})}$$

[80] The editors made explicit the equivalence of (ii) and (iii) and expanded the proof.

(cf. VIA, 5.4.1). By Proposition 8.3, $G^{(p^{mn})}$ is smooth over k for m sufficiently large, and therefore G is smooth as well by (fpqc) descent (cf. EGA IV$_4$, 17.7.1).

8.4.

In the two statements which conclude this Exposé, we return to the case of a field k of arbitrary characteristic.

If k has characteristic 0, let $n \geq 1$. If k has characteristic $p > 0$, let $n \geq 1$ be prime to p. In both cases we simply say that n is *prime to the characteristic of* k. If G is a group scheme over k, we denote by

$$n_G : G \longrightarrow G$$

the morphism of k-schemes which sends $x \in G(T)$ to $x^n \in G(T)$, for every k-scheme T.

Proposition. *Let G be an algebraic group over a field k, and let n be prime to the characteristic of k. Then*

$$n_G : G \longrightarrow G$$

is an étale morphism.

By VIB, 1.3, it suffices to show that n_G is étale at the origin.[81] Let A be the local ring of G at the origin and I its maximal ideal. By II, 3.9.4, the map

$$\mathrm{Lie}(n_G) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(G)$$

induced by n_G is multiplication by n. It is therefore an isomorphism, as is the endomorphism induced by n_G on

$$I/I^2 = \mathrm{Lie}(G)^*.$$

Suppose first that $\mathrm{char}\, k = 0$. The group G is smooth over k (VIB, 1.6.1; see also VIIB, 3.3.1), and the canonical homomorphism

$$\mathrm{Sym}(I/I^2) \longrightarrow \mathrm{gr}_I(A)$$

is an isomorphism, where $\mathrm{gr}_I(A)$ is the graded ring associated with the I-adic filtration. Hence n_G induces an automorphism of $\mathrm{gr}_I(A)$, and therefore also of the completion $\widehat{A}$. Thus n_G is étale at the origin (cf. EGA IV$_4$, 17.6.3).

Suppose next that $\mathrm{char}\, k = p > 0$ and G has height at most 1. Then A is isomorphic to the quotient of the symmetric algebra on

$$\omega_{G/k} = I/I^2$$

by the ideal generated by the p-th powers of the elements of $\omega_{G/k}$ (cf. 7.4). The same argument as in characteristic 0 shows that n_G induces an automorphism of A.

Suppose that G has height at most r, and that the result has been proved for groups of height at most $r-1$. Let B, A, A' be the affine algebras of $\mathrm{Fr}\, G, G$, and

$$G' = \mathrm{Fr}\, G \backslash G,$$

and let $n_B, n_A, n_{A'}$ be the endomorphisms induced by $n_{\mathrm{Fr}\, G}, n_G, n_{G'}$, respectively. Put

$$I' = I \cap A',$$

the maximal ideal of A'. The cartesian square

[81] The editors changed "étale at the origin" to "étale" in the statement and added the preceding sentence.

$$\begin{array}{ccc} \mathrm{Fr}\, G & \longrightarrow & G \\ \downarrow & & \downarrow \\ e & \longrightarrow & G' \end{array}$$

gives

$$B = A/I'A.$$

[82] The endomorphism $n_{A'}$, respectively n_B, is the endomorphism induced by n_A on A', respectively on B. By VIA, 3.2, A is a faithfully flat A'-module. Since A' is an Artinian local ring, because G' is an algebraic k-group of height at most $r-1$, A is a free A'-module. The restriction of n_A to A' is $n_{A'}$, which is an isomorphism by the induction hypothesis. Nakayama's lemma therefore shows that n_A is an automorphism as soon as the endomorphism it induces on $A/I'A$ is an automorphism.

That endomorphism is n_B, which is an automorphism because $\mathrm{Fr}\, G$ has height at most 1. Hence n_A is an automorphism.

Finally, let G be any algebraic group over a field of characteristic $p > 0$. The preceding argument shows that n_G induces automorphisms of the k-schemes $\mathrm{Fr}^r\, G$. These schemes are affine over k, and their algebras are the quotients of the local algebra A by the ideal

$$I^{\{p^r\}}$$

generated by the p^r-th powers of the elements of I. Since n_G induces automorphisms of all the algebras $A/I^{\{p^r\}}$, passage to the projective limit shows that it induces an automorphism of $\widehat{A}$. Thus n_G is étale at the origin (EGA IV$_4$, 17.6.3).

8.5. Proposition. *Let G be a finite algebraic group of rank n over the field k. Then*

$$n_G : G \longrightarrow G$$

is the zero morphism of G.

Combining 8.4 and 8.5 immediately gives the following corollary.

Corollary 8.5.1. *Let G be a finite algebraic group of rank n over the field k. If n is prime to the characteristic of k, then G is étale over k.* [83]

We now prove 8.5. Let H be a normal subgroup of G, of rank m over k, and let

$$\lambda : H \times G \longrightarrow G$$

be induced by the multiplication of G. With the notation of VIA, 3.2, there is a cartesian square

$$\begin{array}{ccc} H \times G & \xrightarrow{\lambda} & G \\ {\scriptstyle \mathrm{pr}_2}\downarrow & & \downarrow{\scriptstyle \pi} \\ G & \xrightarrow{\pi} & H\backslash G \end{array}$$

[82] The editors expanded the original argument in what follows.

[83] The editors added this corollary, which the original indicated implicitly by "cf. 8.4." For another proof which does not use 8.5, see [TO70], Lemma 5.

Since $\pi : G \to H\backslash G$ is faithfully flat and quasi-compact (VIA, 3.2), while pr_2 is locally free of rank m, EGA IV$_2$, 2.5.2 shows that $G \to H\backslash G$ is locally free of rank m. If

$$r = \mathrm{rk}_k(H\backslash G),$$

then

$$n = \mathrm{rk}_k(G) = rm.$$

On the other hand, for every k-scheme T there is an exact sequence of abstract groups

$$1 \longrightarrow H(T) \longrightarrow G(T) \longrightarrow (H\backslash G)(T).$$

It is therefore clear that n_G is zero whenever m_H and $r_{H\backslash G}$ are zero. Taking $H = G^0$, the identity component, the quotient $G^0\backslash G$ is étale (cf. VIA, 5.5.1). We may consequently assume that G is étale over k, or that it is infinitesimal (cf. 7.0).

If G is étale, extension of the base field reduces us to the case where k is algebraically closed. Then G is a constant group (cf. I, 4.1), and the assertion is classical.

If G is nonzero and infinitesimal, then k necessarily has characteristic $p > 0$ (cf. VIB, 1.6.1 or VIIB, 3.3.1). The subgroups $\mathrm{Fr}^r G$ form a composition series of G whose successive quotients have height at most 1.

We are thus reduced to the case where G has height at most 1. Let A, respectively L, be the affine algebra, respectively the Lie algebra, of G, and put

$$U = U_p(L).$$

By 7.4, $G = \mathfrak{G}_p(L)$, and hence $A = U^*$. Therefore, if $\dim_k L = r$, the rank of G over k is p^r (cf. 5.3.3). We shall study the morphism

$$p_G : G \longrightarrow G$$

defined by taking the p-th power. It induces an endomorphism p_A of A, and, by duality, an endomorphism p_U of U.

Let I be the augmentation ideal of A. We shall prove

$$p_A(I) \subset I^p. \qquad (*)$$

Assuming this, one has

$$p_A^r(I) \subset I^{p^r}.$$

Moreover,

$$I^{r(p-1)+1} = 0,$$

because I is generated by r elements whose p-th powers vanish. Since $p^r > r(p-1)$, we obtain $p_A^r(I) = 0$, and therefore p_G^r is the zero morphism.

For every integer $s \geq 1$, let

$$m_A^{s-1} : A^{\otimes s} \longrightarrow A, \qquad \Delta_U^{s-1} : U \longrightarrow U^{\otimes s},$$

denote the maps induced by the multiplication m_A of A and the comultiplication Δ_U of U, respectively. Then p_A is the composite

$$A \xrightarrow{\Delta_A^{p-1}} A^{\otimes p} \xrightarrow{m_A^{p-1}} A.$$

Since the transpose of m_A, respectively of Δ_A, is Δ_U, respectively m_U, the endomorphism

$$p_U = {}^t p_A$$

is the composite

$$U \xrightarrow{\Delta_U^{p-1}} U^{\otimes p} \xrightarrow{m_U^{p-1}} U.$$

Let J be the augmentation ideal of U.[84] Then

$$U = k1_U \oplus J,$$

and we write $\pi : U \to J$ for the projection with kernel $k1_U$. For $s \geq 1$, let $(I^s)^\perp$ be the orthogonal of I^s in $A^* = U$. Thus $(I^s)^\perp$ consists of the elements $u \in U$ for which the composite

$$I^{\otimes s} \xrightarrow{m_A^{s-1}} I \xrightarrow{u} k$$

is zero. Since the transpose of m_A is Δ_U, $(I^s)^\perp$ is the vector subspace P_{s-1} consisting of those $u \in U$ for which $\Delta_U^{s-1}(u)$ vanishes on $I^{\otimes s}$. If

$$\overline{\Delta}_U^{s-1} = \pi^{\otimes s} \circ \Delta_U^{s-1},$$

then

$$(I^s)^\perp = P_{s-1} = \ker \overline{\Delta}_U^{s-1}$$

(see also VIIB, 1.3.6).

To prove $(*)$, it is enough to show that the transpose $p_U = {}^t p_A$ maps P_{p-1} into

$$I^\perp = k1_U.$$

Since $p_U(1_U) = 1_U$, it suffices to prove

$$p_U(P_{p-1}^+) = 0, \qquad P_{p-1}^+ = J \cap P_{p-1}. \tag{**}$$

An easy induction on s shows that P_{s-1}^+ is spanned by the products

$$x_1 \cdots x_t, \qquad 1 \leq t \leq s-1, \quad x_i \in L$$

(cf. VIIB, 4.3).

For $x_1, \ldots, x_t \in L$, one has

$$p_U(x_1 x_2 \cdots x_t) = m_U^{p-1}\left(\prod_{j=1}^{t}\sum_{i=1}^{p} 1 \otimes \cdots \otimes \underset{\text{position } i}{x_j} \otimes \cdots \otimes 1\right).$$

The expression inside parentheses is a sum of p^t terms x_h, indexed by the maps

$$h : \{1, \ldots, t\} \longrightarrow \{1, \ldots, p\}.$$

Such a map determines an ordered partition $\mathcal{P}_h$ of $\{1, \ldots, t\}$ into at most p parts.[85] Indeed, let

$$i_1 < \cdots < i_q$$

be the elements in the image of h, put

$$I_a = h^{-1}(i_a) \quad (1 \leq a \leq q), \qquad x_{I_a} = \prod_{j \in I_a} x_j,$$

[84] The editors expanded the original argument in this paragraph.

[85] The editors expanded the original argument and replaced the notion of a preorder by the equivalent notion of an ordered partition.

where the product is taken in increasing order. The map h corresponds to the p-tensor

$$1 \otimes \cdots \otimes x_{I_1} \otimes \cdots \otimes x_{I_q} \otimes \cdots \otimes 1,$$

where x_{I_a} occupies position i_a. Its image under the iterated multiplication is the product

$$x_{I_1} \cdots x_{I_q}.$$

This depends only on the ordered partition

$$\mathcal{P} = (I_1, \ldots, I_q);$$

denote it by $x_{\mathcal{P}}$. For a fixed $\mathcal{P}$, the same term is obtained for every choice of $i_1 < \cdots < i_q$ in $\{1, \ldots, p\}$. Consequently,

$$p_U(x_1 x_2 \cdots x_t) = \sum_{\mathcal{P}} \binom{p}{n(\mathcal{P})} x_{\mathcal{P}},$$

where $\mathcal{P}$ ranges over the ordered partitions of $\{1, \ldots, t\}$ into at most p parts and $n(\mathcal{P})$ is the number of parts. Here

$$1 \leq n(\mathcal{P}) \leq \min(t, p).$$

If $t < p$, all the coefficients $\binom{p}{n(\mathcal{P})}$ vanish in k, and hence

$$p_U(x_1 \cdots x_t) = 0.$$

Thus p_U vanishes on P^+_{p-1}. This proves $(**)$, then $(*)$, and completes the proof of 8.5.

Corollary 8.5.2.[86] *Let S be a reduced scheme and G a finite locally free S-group of rank n. Then*

$$n_G : G \longrightarrow G$$

is the zero morphism of G.

Let S' be the disjoint union of the $\operatorname{Spec} \mathcal{O}_{S,\eta}$, where η ranges over the maximal points of S. Since S is reduced, the morphism $S' \to S$ is schematically dominant. The same is therefore true of

$$f : G_{S'} \longrightarrow G,$$

because G is finite locally free over S (cf. EGA IV$_3$, 11.10.5). Since $G \to S$ is affine, and hence separated, the coincidence locus of n_G and the zero morphism is a closed subscheme of G. By 8.5 it contains the image of f, and therefore equals G. Thus n_G is the zero morphism.

Remark 8.5.3.

A theorem of P. Deligne (see [TO70], p. 4) also says that if G is a commutative finite locally free S-group of rank n over an arbitrary base S, then $n_G = 0$.

[86] The editors added this corollary, which is noted in Exposé VIII, Remark 7.3.1.

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Exposé VII_B
Infinitesimal Study of Group Schemes

by P. Gabriel

B) Formal Groups

[0] The study of formal groups is usually extremely simple. If this does not appear clearly in the pages that follow, the responsibility lies with an arithmetician who claims to know formal groups over "something other than fields."[1] We have therefore developed, for formal groups "locally free over projective limits of Artinian rings," the generalities ordinarily stated for formal groups defined over a field. For a more detailed study of the latter, we refer to the 1964/65 Heidelberg–Strasbourg algebraic geometry seminar.[2]

0. Review of Pseudocompact Rings and Modules

This section contains a few technical preliminaries. We recall and complete here several results from [CA] (*Des catégories abéliennes*, Bull. Soc. Math. France **90** (1962)).

0.1. A *left pseudocompact ring* is a separated, complete topological ring with identity which has a basis of neighborhoods of 0 consisting of left ideals $\mathfrak{l}$ of finite colength, that is,

$$\mathrm{length}_A(A/\mathfrak{l}) < +\infty.$$

We shall assume here that A is commutative, so that no distinction need be made "between left and right."

In particular, the quotients $A/\mathfrak{l}$ are Artinian rings, and A identifies with the topological projective limit of these rings, each endowed with the discrete topology.

A complete Noetherian local ring $(A, \mathfrak{m})$ is plainly pseudocompact.[3]

[0] Editors' note: Version of 13 October 2024.

[1] The interest of formal groups over a complete Noetherian local ring appears, for example, in the work of Lubin and Tate (cf. [LT65]). The study of formal groups over an arbitrary base, and of lifting and deformation questions—in particular for Barsotti–Tate groups ("p-divisible groups")—has given rise to an abundant literature; see, for example, [LT66, Ta67, Gr74, Me72, La75, Fo77, Il85, Br00]. In particular, the results of the present exposé are largely taken up again in Chapter I of [Fo77].

[2] The editors found only such a seminar dated 1965/66 and entitled "Linear Algebraic Groups," in which the notion of a formal group does not appear; see instead [De72].

[3] When it is endowed with the $\mathfrak{m}$-adic topology.

0.1.1. Every closed ideal I of A is the intersection of the open ideals that contain it.[4] Every closed maximal ideal is therefore open. Moreover, if $\mathfrak{l}$ is an open ideal of A, the maximal ideals of $A/\mathfrak{l}$ correspond bijectively to the maximal ideals $\mathfrak{m}$ of A containing $\mathfrak{l}$; the latter are thus open and closed. Consequently, every closed maximal ideal is an open maximal ideal (and hence closed), the converse being clear. We denote the set of these ideals by $\Upsilon(A)$.

If $\mathfrak{l}$ is an open ideal of A and $\mathfrak{m} \in \Upsilon(A)$, the localization $(A/\mathfrak{l})_\mathfrak{m}$ is a local ring when $\mathfrak{m}$ contains $\mathfrak{l}$, and is zero otherwise. Since $A/\mathfrak{l}$ is Artinian, it is a direct product of finitely many local rings; thus

$$A/\mathfrak{l} \simeq \prod_{\mathfrak{m}\in\Upsilon(A)} (A/\mathfrak{l})_\mathfrak{m}.$$

It follows that there are "canonical" isomorphisms

$$A \simeq \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} \prod_{\mathfrak{m}} (A/\mathfrak{l})_\mathfrak{m} \simeq \prod_{\mathfrak{m}} \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_\mathfrak{m} \simeq \prod_{\mathfrak{m}} A_\mathfrak{m},$$

where

$$A_\mathfrak{m} = \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_\mathfrak{m}.$$

This *local component* $A_\mathfrak{m}$ is a filtered projective limit of Artinian local rings endowed with the discrete topology. It is therefore a local ring which is pseudocompact for the projective-limit topology.[5]

0.1.2. Let $r(A)$ be the intersection of the open maximal ideals of A, that is, the cartesian product of the ideals $\mathfrak{m}A_\mathfrak{m}$ under the identification

$$A = \prod_{\mathfrak{m}} A_\mathfrak{m}.$$

For every open ideal $\mathfrak{l}$ of A, the image of $r(A)$ in $A/\mathfrak{l}$ is contained in the radical of $A/\mathfrak{l}$. Some power of this image is therefore zero, so $r(A)^n \subset \mathfrak{l}$ for all sufficiently large n. Thus the sequence $r(A)^n$ tends to 0.

The same is true of x^n when $x \in r(A)$. In other words, every element of $r(A)$ is topologically nilpotent, and the converse is clear. It follows that the sequence with general term

$$1 + x + \cdots + x^n$$

converges, and converges to $1/(1-x)$, whenever $x \in r(A)$. This shows that $r(A)$ is the Jacobson radical of A, that is, the intersection of all maximal ideals of A (cf. Bourbaki, *Algèbre*, Chap. 8, §6, Th. 1).[6]

Remarks.[7]

a) If $\mathfrak{p}$ is an open prime ideal of A, then, since $A/\mathfrak{p}$ is Artinian, $\mathfrak{p}$ is maximal. Consequently, $\Upsilon(A)$ is the set of open prime ideals of A.

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[4] Indeed, if $x \notin I$, there is an open ideal $\mathfrak{l}$ such that $(x+\mathfrak{l}) \cap I = \varnothing$; then $I + \mathfrak{l}$ is an open ideal that does not contain x. In what follows, the editors have also made explicit the fact that every "closed maximal" ideal is maximal and open.

[5] Notice that $A_\mathfrak{m}$ is the localization of A at the maximal ideal $\mathfrak{m}$. Indeed, the identity element $e_\mathfrak{m}$ of $A_\mathfrak{m}$ is an idempotent of A not belonging to $\mathfrak{m}$. Since $A = A_\mathfrak{m} \times B$, where $B = A(1-e_\mathfrak{m})$, the ring $A_\mathfrak{m}$ identifies with the localization $A_{e_\mathfrak{m}}$, and hence also with $S^{-1}A$, where $S = A \setminus \mathfrak{m}$. Furthermore, $e_\mathfrak{m}(1-e_\mathfrak{m}) = 0$, so $\mathfrak{m}$ contains $1 - e_\mathfrak{m}$, hence also B; thus $\mathfrak{m}$ identifies with $\mathfrak{m}A_\mathfrak{m} \times B$.

[6] Indeed, let $x \in r(A)$. If $\mathfrak{m}$ were a maximal ideal not containing x, there would be $y \in A$ such that $1 - xy \in \mathfrak{m}$. But $xy \in r(A)$, so $1 - xy$ is invertible, a contradiction. Notice the following consequence: if $\Upsilon(A) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_r\}$ is finite, then the $\mathfrak{m}_i$ are all the maximal ideals of A.

[7] The editors added these remarks in order to compare the definition of the formal spectrum $\mathrm{Spf}(A)$ given in 1.1 with those in EGA I, 10.1.2 and 10.4.2.

b) Each $\mathfrak{m}A_\mathfrak{m}$ is an ideal of definition of $A_\mathfrak{m}$, that is, an open ideal I for which I^n tends to 0 (cf. EGA 0_I, 7.1.2). Consequently,

$$\mathrm{Spec}(A_\mathfrak{m}/\mathfrak{m}A_\mathfrak{m}),$$

endowed with the topological ring $A_\mathfrak{m}$, is an affine formal scheme in the sense of EGA I, 10.1.2.

c) The topological ring A is admissible in the sense of EGA 0_I, 7.1.2 if and only if $r(A)$ is open (and hence an ideal of definition), and this holds if and only if $\Upsilon(A)$ is finite. In that case the affine formal scheme

$$\mathrm{Spf}(A) = \mathrm{Spec}(A/r(A))$$

(cf. EGA I, 10.1.2) has $\Upsilon(A)$, endowed with the discrete topology, as its underlying space; the sections of its structure sheaf over a subset $E \subset \Upsilon(A)$ form the ring

$$\prod_{\mathfrak{m}\in E} A_\mathfrak{m}.$$

d) Let A be an arbitrary pseudocompact ring. In 1.1 below, the space $\Upsilon(A)$ is endowed with the discrete topology and with the sheaf of rings whose ring of sections over any subset E is

$$\prod_{\mathfrak{m}\in E} A_\mathfrak{m}.$$

By *b)*, every point then has an open neighborhood which is an affine formal scheme; this therefore defines a formal scheme (EGA I, 10.4.2), denoted by $\mathrm{Spf}(A)$. For this formal scheme to be affine, it must be quasi-compact, so $\Upsilon(A)$ must be finite; in that case $\mathrm{Spf}(A)$ agrees with the definition of EGA I, 10.1.2.

0.1.3. If A and B are pseudocompact rings, a *homomorphism* from A to B is, by definition, a continuous map compatible with addition, multiplication, and the identity elements. Such a homomorphism sends a topologically nilpotent element of A to a topologically nilpotent element of B; it therefore maps the radical $r(A)$ into the radical $r(B)$.

0.2. Let A be a commutative pseudocompact ring. A *pseudocompact A-module* M is a separated, complete topological A-module having a basis of neighborhoods of 0 consisting of submodules M' for which M/M' has finite length over A.

If M and N are pseudocompact A-modules, a *morphism* from M to N is, by definition, a continuous A-linear map. We write

$$\mathrm{Hom}_c(M, N)$$

for the group of these morphisms.

Proposition 0.2.B.[8]

(i) *The pseudocompact A-modules form an abelian category, denoted* $\mathrm{PC}(A)$. *In particular, for every morphism $f : M \to N$, the submodule* $\mathrm{Im}(f)$ *is complete and hence closed in N.*

(ii) *The pseudocompact A-modules of finite length, whose topology is therefore discrete, form a system of cogenerators of* $\mathrm{PC}(A)$.

[8]The editors have stated the results of this paragraph separately in the following proposition, and have then indicated the steps of the proof; cf. [CA], § IV.3, or [DG70], § V.2.

(iii) *Infinite products and filtered projective limits are exact; in other words,* PC(A) *satisfies axiom* (AB5*).[9]

For the reader's convenience, we briefly indicate the steps of the proof. First we use the following lemma ([CA], § IV.3, Lemma 1; for the proof, see [BEns], III, § 7.4, Theorem 1 and Example 2).

Lemma 0.2.C. *Let B be a ring, let I be a filtered ordered set, and let (M_i) and (N_i) be two projective systems of left B-modules indexed by I. Let*

$$(s_i) : (M_i) \longrightarrow (N_i)$$

be a morphism of projective systems such that s_i is surjective and has Artinian kernel for every i. Then the map

$$\varprojlim_i s_i : \varprojlim_i M_i \longrightarrow \varprojlim_i N_i$$

is surjective.

Corollary 0.2.D *(*[CA]*, § IV.3, Propositions 10 and 11). Let M be a pseudocompact A-module.*

(i) *If K is a closed submodule of M, then M/K, endowed with the quotient topology, is a pseudocompact A-module.*

(ii) *Let (M_i) be a filtered decreasing family of closed submodules of M.*

(a) *The canonical map*

$$M \longrightarrow \varprojlim_i M/M_i$$

is surjective and has kernel $\bigcap_i M_i$.

(b) *For every closed submodule N of M, one has*

$$N + \bigcap_i M_i = \bigcap_i (N + M_i).$$

Proof. Let (L_j) be the filtered decreasing family of open submodules of M. We endow M/K with the quotient topology; that is, a basis of neighborhoods of 0 is formed by the open submodules $(K+L_j)/K$. Since K is closed, it is equal to the intersection of the $K+L_j$, and hence the map

$$\tau : M/K \longrightarrow \varprojlim_j M/(K + L_j)$$

is injective. It is also open, since the right-hand side is the inverse limit of the discrete modules $M/(K + L_j)$. Moreover, for each j, the map

$$t_j : M/L_j \longrightarrow M/(K + L_j)$$

is surjective and has Artinian kernel. Thus, by the preceding lemma, the map t in the following commutative diagram is surjective:

[9] Cf. [Gr57], I, § 1.5 and Proposition 1.8. One may also consult, for example, [Po73], § 2.8, or [We95], Appendix A.4.

$$\begin{array}{ccc} M & \xrightarrow[\sim]{p} & \varprojlim_j M/L_j \\ \downarrow & & \downarrow t \\ M/K & \xrightarrow{\tau} & \varprojlim_j M/(K+L_j). \end{array}$$

Since p is an isomorphism because M is complete, it follows that τ is surjective, and hence is an isomorphism. This proves (i).

Let us prove (ii)(a). By the foregoing, for every i there is an isomorphism

$$M/M_i \xrightarrow{\sim} \varprojlim_j M/(M_i+L_j),$$

so the two horizontal arrows in the following commutative diagram are isomorphisms:

$$\begin{array}{ccc} M & \xrightarrow[\sim]{p} & \varprojlim_j M/L_j \\ \downarrow g & & \downarrow s \\ \varprojlim_i M/M_i & \xrightarrow[\sim]{} & \varprojlim_{i,j} M/(M_i+L_j). \end{array}$$

Moreover, for each j, the family of submodules $(M_i+L_j)/L_j$ has a least element, since M/L_j is Artinian. Consequently, the morphism

$$s_j : M/L_j \longrightarrow \varprojlim_i M/(L_j+M_i)$$

is surjective; therefore, by the preceding lemma, s is surjective. It follows that g is surjective. Finally, the kernel of g is the inverse limit of the M_i, that is, their intersection. This proves (a).

Let us deduce (b) from this. Since N is a closed submodule, and hence separated and complete, it is a pseudocompact module for the topology induced by that of M. Therefore, by (a), the morphisms f and g in the following commutative exact diagram are surjective:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N & \longrightarrow & M & \longrightarrow & M/N & \longrightarrow & 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h & & \\ 0 & \longrightarrow & \varprojlim_i N/(N\cap M_i) & \longrightarrow & \varprojlim_i M/M_i & \longrightarrow & \varprojlim_i M/(N+M_i)\,. & & \end{array}$$

The "snake lemma" then shows that the sequence

$$0 \longrightarrow \ker f \longrightarrow \ker g \longrightarrow \ker h \longrightarrow 0$$

is exact, and the equality

$$N + \bigcap_i M_i = \bigcap_i (N+M_i)$$

follows.

We can now prove Proposition 0.2.B. Let $f : M \to N$ be a morphism of pseudocompact A-modules. Then $K = \ker(f)$ is a closed submodule of M, and hence is separated and complete; thus K, with the topology induced by that of M, is a pseudocompact module. By 0.2.D (i), the module M/K, endowed with the quotient topology, is pseudocompact.

Let us show that the continuous bijective morphism

$$M/K \longrightarrow \mathrm{Im}(f)$$

is bicontinuous. After identifying M/K with $\mathrm{Im}(f)$, it is enough to show that the topology T induced by that of N is finer than the quotient topology Q. Let (L_j), respectively (N_i), be the filtered decreasing families of open submodules of M, respectively N, and put

$$N'_i = N_i \cap \mathrm{Im}(f).$$

Let $P = (K + L_j)/K$ be a submodule of M/K open for Q. Since $M/(K + L_j)$ is Artinian, the family $N'_i + P$ has a least element $N'_{i_0} + P$. The N'_i are open, and hence closed, for T, and therefore also closed for Q. It follows from 0.2.D (ii)(b) that

$$N'_{i_0} + P = \bigcap_i (N'_i + P) = P + \bigcap_i N'_i = P,$$

whence $N'_{i_0} \subset P$. Thus P is open for T, and $M/K \to \mathrm{Im}(f)$ is consequently an isomorphism of pseudocompact modules.

In particular, $\mathrm{Im}(f)$ is complete for T, and hence closed in N. By 0.2.D (i) once more, $\mathrm{Coker}(f)$, endowed with the quotient topology, is pseudocompact. This proves that $\mathrm{PC}(A)$ is an abelian category.

Arbitrary inverse limits also exist in $\mathrm{PC}(A)$: if (M_i) is an inverse system of pseudocompact modules, its inverse limit has as underlying module the inverse limit of the underlying modules, endowed with the inverse-limit topology. Moreover, if one has a family of exact sequences in $\mathrm{PC}(A)$,

$$0 \longrightarrow K_i \longrightarrow M_i \longrightarrow Q_i \longrightarrow 0,$$

then the sequence

$$0 \longrightarrow \prod_i K_i \longrightarrow \prod_i M_i \longrightarrow \prod_i Q_i \longrightarrow 0$$

is exact. Assertion (iii) of 0.2.B follows, since in every abelian category in which arbitrary products exist, conditions (a) and (b) of 0.2.D are equivalent, and are equivalent to the exactness of filtered inverse limits (cf. [Mi65], III, 1.2–1.9, or [Po73], Chap. 2, Th. 8.6).

Finally, every pseudocompact module M is a submodule of the product

$$\prod_L M/L,$$

where L ranges over the open submodules of M. Thus the objects of finite length form a system of cogenerators of $\mathrm{PC}(A)$. (Moreover, every object of length n is isomorphic to a quotient A^n/L, where L is an open submodule of A^n of colength n; these quotients therefore form a *set* of cogenerators.) This completes the proof of 0.2.B.

Let (Ab)[10] be the category of abelian groups and let $\mathrm{LF}(A)$ be the full subcategory of $\mathrm{PC}(A)$ formed by the objects of finite length. For every object M of $\mathrm{PC}(A)$, let h_c^M denote the functor

$$\mathrm{LF}(A) \longrightarrow (\mathrm{Ab}), \qquad N \longmapsto \mathrm{Hom}_c(M, N).$$

By [CA], § II.4, Theorem 1, Lemma 4, and Corollary 1, we have the following results.[11]

[10] The editors inserted Proposition 0.2.E and Corollary 0.2.F into this paragraph; they will be useful in 0.2.2. In the original, these results appeared in 0.3.

[11] Indeed, $\mathrm{PC}(A)$ has a set of Artinian cogenerators, filtered projective limits are exact in it, and $\mathrm{LF}(A)$ is the subcategory of Artinian objects. The dual category $\mathrm{PC}(A)^\circ$ therefore has a set of Noetherian generators, and filtered inductive limits are exact in it. By the proof of [CA], § II.4, Theorem 1, the functor $M \mapsto \mathrm{Hom}_c(M, -)$ is an anti-equivalence from $\mathrm{PC}(A)$ to $\mathrm{Lex}(\mathrm{LF}(A), (\mathrm{Ab}))$. Likewise, Lemma 4 and Corollary 1 of *loc. cit.* state results "dual" to those of Corollary 0.2.F. For a "direct" proof, see [DG70], § V.2, Theorem 3.1, Lemma 3.5, and Corollaries 3.3 and 3.4.

Proposition 0.2.E. *The functor $M \mapsto h_c^M$ is an anti-equivalence from* $\mathrm{PC}(A)$ *to the category*

$$\mathrm{Lex}(\mathrm{LF}(A), (\mathrm{Ab}))$$

of left-exact functors $\mathrm{LF}(A) \longrightarrow (\mathrm{Ab})$.

Corollary 0.2.F.

(i) *An object P of* $\mathrm{PC}(A)$ *is projective if and only if the functor h_c^P is exact; that is, if and only if the functor* $\mathrm{Hom}_c(P, -)$ *is exact on* $\mathrm{LF}(A)$.

(ii) *Let (M_i) be a filtered projective system*[12] *of objects of* $\mathrm{PC}(A)$. *For every object N of* $\mathrm{LF}(A)$, *there is an isomorphism, functorial in N,*

$$\mathrm{Hom}_c\left(\varprojlim_i M_i, N\right) \simeq \varinjlim_i \mathrm{Hom}_c(M_i, N).$$

(iii) *Every filtered projective limit and every product*[12] *of projective objects of* $\mathrm{PC}(A)$ *is a projective object of* $\mathrm{PC}(A)$.

Finally, Corollary 0.2.F gives the following result.

Corollary 0.2.G. *Let $(M_i)_{i\in I}$ be a family of objects of* $\mathrm{PC}(A)$. *Then $\prod_{i\in I} M_i$ is projective if and only if each M_i is projective.*

Indeed, for every $N \in \mathrm{Ob}\,\mathrm{LF}(A)$, one has

$$\mathrm{Hom}_c\left(\prod_i M_i, N\right) \simeq \bigoplus_i \mathrm{Hom}_c(M_i, N).$$

0.2.1. Each local component $A_\mathfrak{m}$ of A is a direct factor of A, hence a projective object of $\mathrm{PC}(A)$ (A is manifestly projective). Moreover,

$$S_\mathfrak{m} = A_\mathfrak{m}/\mathfrak{m}A_\mathfrak{m}$$

is the unique simple quotient of $A_\mathfrak{m}$, so $A_\mathfrak{m}$ is indecomposable. On the other hand, every simple object of $\mathrm{PC}(A)$ is isomorphic to a unique $S_\mathfrak{m}$. By [CA], IV, § 3, Corollary 1 of Theorem 3,[13] we therefore have:

Proposition.

(i) *Every projective object of* $\mathrm{PC}(A)$ *is a direct product of indecomposable projective objects, uniquely determined up to isomorphism.*

(ii) *Every indecomposable projective object is isomorphic to a unique $A_\mathfrak{m}$, where* $\mathfrak{m} \in \Upsilon(A)$.

Definition. A pseudocompact A-module M is said to be *topologically free* if it is isomorphic to the product of a family (A_i) of copies of A.

In this case, a family (m_i) of elements of M is called a *pseudobasis* of M if the A-linear maps $A_i \longrightarrow M$ sending the unit element of A_i to m_i extend to an isomorphism

$$\prod_i A_i \xrightarrow{\sim} M.$$

[12] Since every infinite product is a filtered projective limit of finite products, every additive functor that commutes with filtered projective limits also commutes with infinite products.

[13] This refers to the "dual" statements established in *loc. cit.*, § 2, Theorem 2, and § 1, Proposition 2. For a "direct" proof, see [DG70], V, § 2, Theorem 4.5 and Example 4.6(b).

0.2.2. [14] If M is a pseudocompact A-module, we write $M^\dagger$ for the A-module, without topology,

$$\mathrm{Hom}_c(M, A).$$

Conversely, if N is an A-module, we write

$$N^* = \mathrm{Hom}_A(N, A)$$

for its dual, endowed with the topology of pointwise convergence. Thus a basis of neighborhoods of 0 in N^* is formed by the submodules

$$\mathcal{V}(x, \mathfrak{l}) = \{f \in N^* \mid f(x) \in \mathfrak{l}\},$$

where $x \in N$ and $\mathfrak{l}$ is an open ideal of A.

This makes N^* a pseudocompact A-module. Indeed, first suppose that $N = A$. Then $N^* = A$, endowed with its topology as a pseudocompact ring. If N is a free module $A^{(I)}$, then N^* is the product A^I, endowed with the product topology. Furthermore, for every morphism $\varphi : N_1 \longrightarrow N_2$, the transpose

$${}^t\varphi : N_2^* \longrightarrow N_1^*$$

is continuous, because the inverse image under ${}^t\varphi$ of a submodule $\mathcal{V}(x_1, \mathfrak{l})$ of N_1^* is precisely the submodule $\mathcal{V}(\varphi(x_1), \mathfrak{l})$ of N_2^*. For arbitrary N, choose a presentation

$$A^{(J)} \xrightarrow{\varphi} A^{(I)} \xrightarrow{\pi} N \longrightarrow 0.$$

Then N^* is the kernel of the continuous morphism

$${}^t\varphi : A^I \longrightarrow A^J,$$

and hence N^* is pseudocompact.

When A is Artinian (in which case one may take $\mathfrak{l} = 0$ above), Corollary 0.2.F gives:

Proposition. *When A is Artinian, the functors*

$$P \longmapsto P^\dagger \qquad \textit{and} \qquad Q \longmapsto Q^*,$$

where P, respectively Q, is a projective object of $\mathrm{PC}(A)$*, respectively a projective A-module, establish an anti-equivalence between the category of projective pseudocompact A-modules and the category of projective A-modules.*[15]

In particular, when A is a field k, the functor

$$P \longmapsto P^\dagger$$

is an anti-equivalence between the category of all pseudocompact k-modules (also called *linearly compact k-vector spaces*) and the category of k-vector spaces.[16]

0.3.[17] Let L and M be two pseudocompact A-modules. The functor

$$\mathrm{LF}(A) \longrightarrow (\mathrm{Ab}), \qquad N \longmapsto \mathrm{Bil}_c(L \times M, N),$$

[14] The editors have spelled out the results of this paragraph, which play an important role below; cf. 1.2.3, 1.3.5, 2.2.1, and so on.

[15] Recall also that, over an Artinian ring, a module is projective if and only if it is flat; see, for example, Exposé VI$_B$, editors' note (54), or 0.3.7 below.

[16] The direct sum in $\mathrm{PC}(k)$ of a family $(V_i)_{i\in I}$ of linearly compact k-vector spaces is $\left(\prod_{i\in I} V_i^\dagger\right)^*$.

[17] Editors' note: This paragraph has been modified to take account of the additions made in 0.2.

where $\mathrm{Bil}_c(L \times M, N)$ denotes the abelian group of continuous A-bilinear maps from $L \times M$ to N, is left exact.

By 0.2.E, there therefore exists a pseudocompact A-module $L\widehat{\otimes}_A M$, unique up to unique isomorphism, representing this functor; that is, for every object N of $\mathrm{LF}(A)$, there is a functorial isomorphism

$$\mathrm{Hom}_c(L\widehat{\otimes}_A M, N) \simeq \mathrm{Bil}_c(L \times M, N).$$

Moreover, $L\widehat{\otimes}_A M$ identifies with the projective limit

$$P = \widehat{L \otimes_A M}$$

of the discrete A-modules

$$(L/L') \otimes_A (M/M'),$$

where L' and M' range respectively over the open submodules of L and M.

Indeed, let $\varphi : L \times M \to N$ be a continuous bilinear map into a discrete A-module N of finite length. By Lemma 0.3.1 below, there exist open submodules L' and M' of L and M such that

$$\varphi(L' \times M) = \varphi(L \times M') = \{0\}.$$

This means that the map

$$\overline{\varphi} : L \otimes_A M \longrightarrow N$$

induced by φ has the form $\varphi' \circ p$, where p is the canonical projection from $L \otimes_A M$ onto $(L/L') \otimes_A (M/M')$. If $\widehat{\varphi}$ denotes the composite

$$P \longrightarrow (L/L') \otimes_A (M/M') \xrightarrow{\varphi'} N,$$

then the map $\varphi \mapsto \widehat{\varphi}$ is a functorial bijection from $\mathrm{Bil}_c(L \times M, N)$ onto $\mathrm{Hom}_c(P, N)$. Hence

$$P \simeq L\widehat{\otimes}_A M.$$

Thus the pseudocompact module $L\widehat{\otimes}_A M$ is the separated completion of $L \otimes_A M$ for the linear topology defined by the kernels of the canonical projections

$$L \otimes_A M \longrightarrow (L/L') \otimes_A (M/M'),$$

and it will be called the *completed tensor product* of L and M.

If $x \in L$ and $y \in M$, the image of $x \otimes_A y$ in $L\widehat{\otimes}_A M$ will be denoted by $x\widehat{\otimes}_A y$.

Lemma 0.3.1. *Let L, M, and N be pseudocompact A-modules, with N of finite length. If $\varphi : L \times M \to N$ is a continuous A-bilinear map, there exist open submodules L' and M' of L and M such that*

$$\varphi(L' \times M) = \varphi(L \times M') = \{0\}.$$

Proof. The inverse image $\varphi^{-1}(0)$ is an open neighborhood of $(0, 0)$, and therefore contains an open set of the form $L_1 \times M_1$, where L_1 and M_1 are open submodules of L and M. Since L/L_1 has finite length, there are elements $x_1, \ldots, x_r$ of L such that

$$L_1 + Ax_1 + \cdots + Ax_r = L.$$

If $M' \subset M_1$ is sufficiently small, then $\varphi(x_i, M') = 0$ for every i, since the map $y \mapsto \varphi(x_i, y)$ is continuous. It follows that $\varphi(L, M') = \{0\}$. Likewise, $\varphi(L', M) = \{0\}$ when L' is sufficiently small. □

Corollary 0.3.1.1.[18] *Let M be a pseudocompact A-module.*

[18]Editors' note: This corollary has been added.

(i) *For every open submodule M', there exists an open ideal $\mathfrak{l}$ of A such that*

$$\overline{\mathfrak{l}M} \subset M'.$$

(ii) *Consequently,*

$$M \simeq \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M},$$

where $\mathfrak{l}$ ranges over the filtered projective system of open ideals of A, and every $M/\overline{\mathfrak{l}M}$ is endowed with the quotient topology, which makes it a pseudocompact module; cf. 0.2.D.

Proof. Consider the map

$$\varphi : A \times M \longrightarrow M/M', \qquad (a, m) \longmapsto am + M'.$$

By 0.3.1, there is an open ideal $\mathfrak{l}$ of A such that $\mathfrak{l}M \subset M'$. Since M' is also closed, it contains $\overline{\mathfrak{l}M}$. Since the intersection of the open submodules of M is zero, it follows that

$$\bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M} = (0).$$

On the other hand, by 0.2.D the map

$$\phi : M \longrightarrow \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$$

is surjective. By the proof of 0.2.B, ϕ therefore induces an isomorphism

$$M/\operatorname{Ker}(\phi) \xrightarrow{\sim} \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}.$$

But we have just seen that

$$\operatorname{Ker}(\phi) = \bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M} = (0).$$

□

Remark 0.3.1.2.[19] The completed tensor product satisfies the usual associativity condition: if L, M, and P are pseudocompact A-modules, there is a canonical isomorphism

$$(L\widehat{\otimes}_A M)\widehat{\otimes}_A P \simeq L\widehat{\otimes}_A(M\widehat{\otimes}_A P).$$

Indeed, both objects represent the functor which associates to every object N of $\mathrm{LF}(A)$ the abelian group of continuous A-trilinear maps from $L \times M \times P$ to N.

0.3.2. Let

$$L' \xrightarrow{f} L \xrightarrow{g} L'' \longrightarrow 0$$

be an exact sequence, and let M be an object of $\mathrm{PC}(A)$. It is clear that for every object N of $\mathrm{LF}(A)$, the induced sequences

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Bil}_c(L'' \times M, N) & \longrightarrow & \mathrm{Bil}_c(L \times M, N) & \longrightarrow & \mathrm{Bil}_c(L' \times M, N) \\
 & & \| & & \| & & \| \\
0 & \longrightarrow & \mathrm{Hom}_c(L''\widehat{\otimes}_A M, N) & \longrightarrow & \mathrm{Hom}_c(L\widehat{\otimes}_A M, N) & \longrightarrow & \mathrm{Hom}_c(L'\widehat{\otimes}_A M, N)
\end{array}$$

[19] Editors' note: This remark has been added.

are exact. By 0.2.E, this is equivalent to saying that the sequence

$$L'\widehat{\otimes}_A M \xrightarrow{f\widehat{\otimes}M} L\widehat{\otimes}_A M \xrightarrow{g\widehat{\otimes}M} L''\widehat{\otimes}_A M \longrightarrow 0 \qquad (*)$$

is exact. Hence:

Corollary. *For every pseudocompact A-module M, the functor $-\widehat{\otimes}_A M$ is right exact.*

In particular, take $L = A$, let f be the inclusion of a closed ideal $\mathfrak{a}$ into A, and let g be the canonical projection of A onto $A/\mathfrak{a}$. We may then identify $A\widehat{\otimes}_A M$ with M by means of the map

$$x\widehat{\otimes}_A m \longmapsto xm.$$

Since the image of $\mathfrak{a}\widehat{\otimes}_A M$ is closed in M (cf. 0.2.B), and the image of $\mathfrak{a} \otimes_A M$ is dense in $\mathfrak{a}\widehat{\otimes}_A M$, the image of $f\widehat{\otimes}M$ is precisely the closure $\overline{\mathfrak{a}M}$ of $\mathfrak{a}M$ in M. The exact sequence $(*)$ therefore gives the isomorphism

$$(A/\mathfrak{a})\widehat{\otimes}_A M \xrightarrow{\sim} M/\overline{\mathfrak{a}M}.$$

0.3.3. Nakayama's Lemma. *Let A be a pseudocompact ring, M a pseudocompact A-module, and $\mathfrak{a}$ an ideal of A contained in the radical $r(A)$. Then*

$$\overline{\mathfrak{a}M} = M \qquad \Longrightarrow \qquad M = 0.$$

Proof. Suppose that $\overline{\mathfrak{a}M} = M$.[20] Let M' be an open submodule of M, and put $M'' = M/M'$. Since M'' is discrete, $\mathfrak{a}M''$ is closed in M'', and hence equals $\overline{\mathfrak{a}M''}$. By 0.3.2, the canonical map

$$M/\overline{\mathfrak{a}M} \longrightarrow M''/\overline{\mathfrak{a}M''}$$

is surjective, so

$$\mathfrak{a}M'' = \overline{\mathfrak{a}M''} = M''.$$

Since M'' is a finitely generated A-module and $\mathfrak{a} \subset r(A)$, the usual Nakayama lemma implies that $M'' = 0$. Consequently, every open submodule M' of M is equal to M, and therefore $M = 0$.[21] □

0.3.4. The usual consequences of Nakayama's lemma follow.

Corollary. *Let $\mathfrak{a}$ be a closed ideal contained in $r(A)$, and let $f : M \to N$ be a morphism of pseudocompact A-modules.*

(i) *The map f is surjective if the induced map*

$$f' : M/\overline{\mathfrak{a}M} \longrightarrow N/\overline{\mathfrak{a}N}$$

is surjective.[22]

(ii) *If N is projective, then f is an isomorphism whenever f' is an isomorphism.*

[20] Editors' note: Replacing $\mathfrak{a}$ by its closure if necessary, we may assume that $\mathfrak{a}$ is closed.

[21] Editors' note: Indeed, M is the projective limit of the quotients $M/M' = 0$.

[22] Editors' note: Consequently, every pseudocompact A-module is a quotient of a topologically free A-module. One first reduces to the case in which A is local; cf. the proof of 0.3.7.

Proof. Part (i) follows by applying Lemma 0.3.3 to $\mathrm{Coker}(f)$. For (ii), suppose that f' is an isomorphism. By (i), f is then surjective, and hence has a section. Apply Lemma 0.3.3 to $\mathrm{Ker}(f)$. □

When A is local with maximal ideal $\mathfrak{m}$, one may also deduce from 0.3.3 the following exchange theorem.

Theorem. *Let A be a local pseudocompact ring, let $\mathfrak{m}$ be its maximal ideal, let M be a topologically free A-module with pseudobasis $(m_i)_{i\in I}$ (0.2.1), and let N be a closed direct summand of M. There is a pseudobasis of M consisting of elements of N together with some of the m_i.*

Proof. This is clear when A is a field: use the duality of 0.2.2 and apply the usual exchange theorem. Consequently,

$$N/\overline{\mathfrak{m}N}$$

has a complementary topologically free $A/\mathfrak{m}$-module with pseudobasis $(\overline{m}_i)_{i\in J}$, where $\overline{m}_i$ is the image of m_i in

$$M/\overline{\mathfrak{m}M},$$

and J is a subset of I.[23] If

$$P = \prod_{i\in J} Am_i,$$

then the canonical map

$$N \oplus P \longrightarrow M$$

is "bijective modulo $\mathfrak{m}$." It is therefore bijective by what precedes. For another proof, see [CA], § IV.2, Proposition 8. □

0.3.5. Consider three pseudocompact A-modules L, M, N, where N has finite length. Give the A-module $\mathrm{Hom}_c(M,N)$ the discrete topology. Every element

$$\psi \in \mathrm{Hom}_c(L, \mathrm{Hom}_c(M,N))$$

defines a continuous bilinear map

$$\psi' : L \times M \longrightarrow N, \qquad (\ell, m) \longmapsto \psi(\ell)(m).$$

This gives a natural isomorphism

$$\mathrm{Hom}_c(L, \mathrm{Hom}_c(M,N)) \xrightarrow{\sim} \mathrm{Hom}_c(L\widehat{\otimes}_A M, N). \tag{1}$$

This is another characterization of $\mathrm{Hom}_c(L\widehat{\otimes}_A M, N)$, which we now use when M is the projective limit of a filtered projective system (M_i) of pseudocompact A-modules. By (1) and 0.2.F (ii), there are natural isomorphisms

$$\begin{aligned}\mathrm{Hom}_c\left(L\widehat{\otimes}_A \varprojlim_i M_i, N\right) &\simeq \mathrm{Hom}_c\left(L, \mathrm{Hom}_c(\varprojlim_i M_i, N)\right)\\ &\simeq \mathrm{Hom}_c\left(L, \varinjlim_i \mathrm{Hom}_c(M_i, N)\right).\end{aligned} \tag{2}$$

Moreover, because $\varinjlim_i \mathrm{Hom}_c(M_i,N)$ is discrete, every continuous map from L into this module factors through a finite-length quotient of L. Hence the natural map

$$\varinjlim_i \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i,N)) \xrightarrow{\sim} \mathrm{Hom}_c\left(L, \varinjlim_i \mathrm{Hom}_c(M_i,N)\right) \tag{3}$$

[23] Editors' note: We have corrected $N/\mathfrak{m}N$ to $N/\overline{\mathfrak{m}N}$, and likewise for $M/\overline{\mathfrak{m}M}$ below.

is an isomorphism. Finally, (1) and 0.2.F (ii) again give natural isomorphisms

$$\varinjlim_i \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) \simeq \varinjlim_i \mathrm{Hom}_c(L\widehat{\otimes}_A M_i, N)$$
$$\simeq \mathrm{Hom}_c\left(\varprojlim_i (L\widehat{\otimes}_A M_i), N\right). \qquad (4)$$

Combining (2), (3), and (4) yields the following result.

Proposition. *Let (M_i) be a filtered projective system of objects of* $\mathrm{PC}(A)$*, let $L \in \mathrm{PC}(A)$, and let $N \in \mathrm{LF}(A)$. There is an isomorphism, functorial in N,*

$$\mathrm{Hom}_c\left(L\widehat{\otimes}_A \varprojlim_i M_i, N\right) \simeq \mathrm{Hom}_c\left(\varprojlim_i (L\widehat{\otimes}_A M_i), N\right),$$

and consequently an isomorphism

$$L\widehat{\otimes}_A \varprojlim_i M_i \simeq \varprojlim_i (L\widehat{\otimes}_A M_i).$$

Thus the completed tensor product commutes with filtered projective limits.[24]

0.3.6. In particular,[25] the completed tensor product commutes with arbitrary products. For example, since A is the product of its local components $A_{\mathfrak{m}}$ (0.1.1), every pseudocompact A-module

$$M \simeq A\widehat{\otimes}_A M$$

identifies with the product of the modules

$$M_{\mathfrak{m}} = A_{\mathfrak{m}}\widehat{\otimes}_A M,$$

which are called the local components of M.

Likewise, let M and N be pseudocompact A-modules. Recall from 0.2.2 that

$$M^\dagger = \mathrm{Hom}_c(M, A).$$

Consider the map

$$\varphi : M^\dagger \otimes_A N^\dagger \longrightarrow (M\widehat{\otimes}_A N)^\dagger$$

which sends $f \otimes g$ to the map

$$m\widehat{\otimes}n \longmapsto f(m)g(n)$$

from $M\widehat{\otimes}_A N$ to A. The map φ is bijective when $M \simeq A$.

Corollary. *If A is Artinian, then φ is an isomorphism whenever M is topologically free, or more generally projective.*

Indeed, with N fixed, each of the functors

$$M \longmapsto (M\widehat{\otimes}_A N)^\dagger \qquad \text{and} \qquad M \longmapsto M^\dagger \otimes_A N^\dagger$$

takes arbitrary products to direct sums, by the preceding result and 0.2.F.

[24] This also recovers 0.3.1.1:

$$L = L\widehat{\otimes}_A A = L\widehat{\otimes}_A \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} L/\mathfrak{l}L,$$

where $\mathfrak{l}$ ranges over the open ideals of A.

[25] Cf. editor's note (12).

Remark 0.3.6.A.[26] *The same use of 0.2.F gives the following result. Let A be an Artinian ring, let $M, Q \in \mathrm{PC}(A)$, and let $N \in \mathrm{LF}(A)$. If Q is projective, then there are natural isomorphisms*

$$\mathrm{Hom}_c(M, Q) \xrightarrow{\sim} \mathrm{Hom}_A(Q^\dagger, M^\dagger), \qquad Q^\dagger \otimes_A N \xrightarrow{\sim} \mathrm{Hom}_c(Q, N).$$

0.3.7. For every $\mathfrak{m} \in \Upsilon(A)$, the functor

$$M \longmapsto A_\mathfrak{m} \widehat{\otimes}_A M$$

is plainly exact. Since every projective pseudocompact A-module P is a product of modules of the form $A_\mathfrak{m}$, it follows that

$$M \longmapsto P \widehat{\otimes}_A M$$

is exact whenever P is projective. The converse is also true.

Proposition. *Let A be a pseudocompact ring and P a pseudocompact A-module. The following conditions are equivalent:*

(i) *P is a projective object of* $\mathrm{PC}(A)$*;*

(ii) *every local component $P_\mathfrak{m}$ is a topologically free $A_\mathfrak{m}$-module;*

(iii) *the functor $M \mapsto P \widehat{\otimes}_A M$ is exact.*

The equivalence of (i) and (ii) follows from 0.2.F (iii) and 0.2.1, and we have just proved (ii) $\Rightarrow$ (iii). Suppose that the functor $M \mapsto P \widehat{\otimes}_A M$ is exact. Since $P \widehat{\otimes}_A M$ is the product of its local components,

$$(P \widehat{\otimes}_A M)_\mathfrak{m} \simeq P_\mathfrak{m} \widehat{\otimes}_{A_\mathfrak{m}} M_\mathfrak{m},$$

we reduce to the case in which A is local. We shall prove that P is then topologically free.

Let $\mathfrak{m}$ be the maximal ideal of A. The quotient $P/\overline{\mathfrak{m}}P$ is a linearly compact vector space over $A/\mathfrak{m}$, and hence a product of copies of $A/\mathfrak{m}$ (0.2.2). Therefore there is a family $(A_i)_{i \in I}$ of copies of A and an isomorphism

$$\phi : \prod_{i \in I} (A_i/\mathfrak{m}) \xrightarrow{\sim} P/\overline{\mathfrak{m}}P.$$

Since $\prod_{i \in I} A_i$ is projective, there is a commutative square

$$\begin{array}{ccc} \prod\limits_{i \in I} A_i & \xrightarrow{\quad\psi\quad} & P \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ \prod\limits_{i \in I} (A_i/\mathfrak{m}) & \xrightarrow[\sim]{\varphi} & P/\mathfrak{m}P \end{array}$$

where p and q are the canonical projections. Apply Nakayama's lemma to $\mathrm{Coker}\,\psi$. Since

$$(A/\mathfrak{m}) \widehat{\otimes}_A \psi$$

is precisely ϕ, the map ψ is surjective.[27]

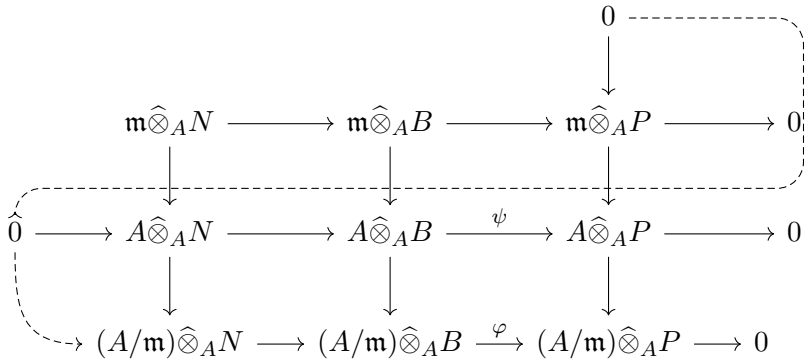
[26] The editors added this remark, which will be useful in 2.3.1.

[27] This proves the result announced in editor's note (22): every pseudocompact A-module M is a quotient of a topologically free A-module. Since products are exact in $\mathrm{PC}(A)$, one reduces to the local case treated here.

Put

$$B = \prod_{i \in I} A_i, \qquad N = \ker \psi.$$

There is then the following commutative exact diagram:



The snake lemma, applied to the first two rows, shows that in the bottom row the morphism

$$(A/\mathfrak{m})\widehat{\otimes}_A N \longrightarrow (A/\mathfrak{m})\widehat{\otimes}_A B$$

is a monomorphism. But ϕ is an isomorphism, so $(A/\mathfrak{m})\widehat{\otimes}_A N = 0$. Hence $N = 0$ by 0.3.3, and ψ is an isomorphism.[28]

0.3.8. Corollary. *Let A be a complete Noetherian local ring and P a pseudocompact A-module. Then P is topologically free if and only if P is flat over A.*

Indeed, the canonical map

$$M \otimes_A P \longrightarrow M\widehat{\otimes}_A P$$

is bijective when $M = A$, and hence also when M is Noetherian: take a finite presentation of M and use the right exactness of both the tensor product and the completed tensor product.

Now P is flat if and only if

$$M \longmapsto M \otimes_A P$$

is exact on Noetherian modules. Likewise, the proof of Proposition 0.3.7 shows that P is topologically free provided the sequence

$$0 \longrightarrow \mathfrak{m}\widehat{\otimes}_A P \longrightarrow A\widehat{\otimes}_A P \longrightarrow (A/\mathfrak{m})\widehat{\otimes}_A P \longrightarrow 0$$

is exact. Thus P is topologically free if and only if the functor

$$M \longmapsto M\widehat{\otimes}_A P$$

is exact on Noetherian modules. The corollary follows from the equality

$$M \otimes_A P = M\widehat{\otimes}_A P$$

established above.

[28] A completely analogous proof, using the "nilpotent Nakayama lemma," shows that if A is Artinian and P is a flat A-module, then every local component of P is a free A-module (cf. [BAC], II, § 3.2, Corollary 2 to Proposition 5).

0.4. Profinite Algebras

Let k be a pseudocompact ring. A *topological k-algebra* is a commutative topological ring A, together with a morphism of topological rings $k \to A$. The algebra A is called a *profinite k-algebra* if its underlying topological k-module is pseudocompact.

Let $\mathfrak{l}$ be an open k-submodule of such an A. The composite map

$$\varphi : A \times A \xrightarrow{\text{mult}} A \longrightarrow A/\mathfrak{l}$$

is continuous. Lemma 0.3.1 therefore gives an open k-submodule $\mathfrak{n}$ of A such that $\varphi(A \times \mathfrak{n}) = 0$. Thus $\mathfrak{l}$ contains the open ideal $A\mathfrak{n}$, and A is a pseudocompact ring.

Likewise, let M be a topological A-module whose underlying k-module is pseudocompact. If M' is an open k-submodule of M, applying Lemma 0.3.1 to

$$A \times M \xrightarrow{\text{mult}} M \longrightarrow M/M'$$

shows that M' contains an open A-submodule. Hence M is also a pseudocompact A-module.[29] Conversely:

Lemma. *Let A be a profinite k-algebra and M a pseudocompact A-module. The k-module $M|_k$, obtained by restriction of scalars, is pseudocompact.*

Indeed, every finite-length pseudocompact A-module is isomorphic to a quotient

$$A^n/L,$$

where L is an open submodule of A^n, and is therefore a pseudocompact k-module. Since $M|_k$ is a projective limit of modules of this kind, it is a pseudocompact k-module.

0.4.1. If A and B are profinite k-algebras, a *morphism* from A to B is, by definition, a continuous homomorphism of k-algebras. We denote the category of profinite k-algebras by

$$\text{Alp}\,/k.$$

It plainly has projective limits: the algebra underlying a projective limit is the projective limit of the underlying algebras, endowed with the projective-limit topology.

It also has finite direct limits.[30] For example, if

$$f : A \longrightarrow B, \qquad g : A \longrightarrow C$$

are morphisms of profinite k-algebras, the amalgamated sum of B and C over A has

$$B \widehat{\otimes}_A C$$

as its underlying topological A-module. Here 0.4 makes B and C pseudocompact A-modules via f and g, and the multiplication is characterized by

$$(b\widehat{\otimes}c)(b'\widehat{\otimes}c') = bb'\widehat{\otimes}cc'$$

for $b, b' \in B$ and $c, c' \in C$.

Definition 0.4.2. *A profinite k-algebra C is said to have* finite length *if its underlying k-module has finite length, and is therefore discrete. We denote by* Alf $/k$ *the full subcategory of* Alp $/k$ *formed by the k-algebras of finite length.*[31]

[29] The editors added the following lemma; cf. editor's note (36) in Proposition 0.5.

[30] In 0.4.2 we shall see that it has arbitrary direct limits.

[31] Caution: k is an object of Alf $/k$ only when k is Artinian; cf. 1.2.2 below.

For every profinite k-algebra A, let h_A be the functor

$$h_A : \mathrm{Alf}\,/k \longrightarrow (\mathbf{Sets}), \qquad C \longmapsto \mathrm{Hom}_{\mathrm{Alp}\,/k}(A, C).$$

The functor h_A is left exact,[32] and the canonical projections $A \to A/\mathfrak{l}$, where $\mathfrak{l}$ ranges over the open ideals of A, induce for every $C \in \mathrm{Alf}\,/k$ an isomorphism

$$\varinjlim_{\mathfrak{l}} \mathrm{Hom}_{\mathrm{Alf}\,/k}(A/\mathfrak{l}, C) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Alp}\,/k}(A, C),$$

functorial in C. In other words, h_A is the direct limit of the representable functors $h_{A/\mathfrak{l}}$:

$$h_A \simeq \varinjlim_{\mathfrak{l}} h_{A/\mathfrak{l}}. \qquad (*)$$

If B is another profinite k-algebra, the general properties of the bifunctor Hom, together with

$$\mathrm{Hom}(h_C, h_B) = h_B(C)$$

for C of finite length, give isomorphisms

$$\begin{aligned} \mathrm{Hom}_{\mathrm{Alp}\,/k}(B, A) &\simeq \varprojlim_{\mathfrak{l}} \mathrm{Hom}_{\mathrm{Alp}\,/k}(B, A/\mathfrak{l}) \\ &\simeq \varprojlim_{\mathfrak{l}} \mathrm{Hom}(h_{A/\mathfrak{l}}, h_B) \\ &\simeq \mathrm{Hom}\left(\varinjlim_{\mathfrak{l}} h_{A/\mathfrak{l}}, h_B\right). \end{aligned}$$

Together with $(*)$, this proves that the contravariant functor $A \mapsto h_A$ is fully faithful. In fact:

Proposition. *The functor $A \mapsto h_A$ is an anti-equivalence from* $\mathrm{Alp}\,/k$ *onto the category of left-exact functors from* $\mathrm{Alf}\,/k$ *to* (**Sets**).

By what precedes, it remains only to show that every left-exact functor

$$F : \mathrm{Alf}\,/k \longrightarrow (\mathbf{Sets})$$

is isomorphic to a functor h_A. The algebra A may be constructed as follows (cf. TDTE II, § 3).

Because F is left exact, for every finite-length k-algebra C and every $\xi \in F(C)$, there is a smallest subalgebra C' of C such that ξ belongs to the image of $F(C')$ in $F(C)$. If $C' = C$, the pair (C, ξ) is called *minimal*.

The minimal pairs form a category when a morphism from (C, ξ) to (D, η) is defined to be a homomorphism $\phi : C \to D$ such that

$$(F\phi)(\xi) = \eta.$$

Such a ϕ is necessarily surjective, and the category of minimal pairs is left filtered. Moreover, one may restrict to pairs (C, ξ) for which C belongs to a set containing one finite-length k-algebra of every isomorphism type.[33] The functor $(C, \xi) \mapsto C$, from the category of minimal

[32] That is, it commutes with finite projective limits.

[33] Indeed, every discrete k-module of length n is isomorphic to k^n/L, where L is an open submodule of k^n of colength n. One may then consider the set of isomorphism classes of profinite k-algebra structures on each such module.

pairs to the category of profinite k-algebras, therefore has a projective limit; take A to be this limit.

Corollary. *The category* Alp $/k$ *has arbitrary direct limits.*

Indeed, the category of left-exact functors from Alf $/k$ to (**Sets**) has projective limits, computed pointwise: if (F_i) is a projective system of such functors, then for every $C \in$ Alf $/k$,

$$(\varprojlim_i F_i)(C) = \varprojlim_i F_i(C).$$

34

0.5. Base Change

Let

$$\phi : k \longrightarrow \ell$$

be a homomorphism of pseudocompact rings (0.1.3). The construction of 0.3 admits the following generalization.[35]

Definition 0.5.A. *For every $M \in \mathrm{PC}(k)$ and $N \in \mathrm{PC}(\ell)$, let*

$$M \widehat{\otimes}_k N$$

be the separated completion of $M \otimes_k N$ for the linear topology defined by the kernels of the projections

$$M \otimes_k N \longrightarrow (M/M') \otimes_k (N/N'),$$

where M', respectively N', ranges over the open submodules of M, respectively N. Then $M \widehat{\otimes}_k N$ is a pseudocompact ℓ-module. If $m \in M$ and $x \in N$, the canonical image of $m \otimes_k x$ in $M \widehat{\otimes}_k N$ is denoted by

$$m \widehat{\otimes}_k x.$$

This applies in particular to $N = \ell$. In that case

$$M \widehat{\otimes}_k \ell$$

is called the pseudocompact ℓ-module obtained from M by the base change $k \to \ell$.

Remarks 0.5.B.

(i) In such a base change, ℓ need not be a profinite k-algebra. The typical example is that in which k is a field and $\ell = K$ is an arbitrary field extension of k.

(ii) If, however, the k-module underlying N is pseudocompact—for example, if ℓ is a profinite k-algebra—then 0.4 shows that every open k-submodule of N contains an open ℓ-submodule. In that case $M \widehat{\otimes}_k N$ agrees with the completed tensor product of 0.3 for the pseudocompact k-modules M and N, so the notation is unambiguous.

In every case, the k-module $N|_k$ obtained by restriction of scalars is a topological k-module: the map

$$k \times N \longrightarrow N, \qquad (t, n) \longmapsto \phi(t)n$$

[34] If $(A_i)_{i \in I}$ is a direct system in Alp $/k$, its direct limit in Alp $/k$ is the separated completion of the k-algebra B, the direct limit of the underlying k-algebras, for the topology defined by the ideals I such that $I \cap A_i$ is open in A_i for every i and $\mathrm{length}_k(B/I) < \infty$. Notice, for example, that if K/k is an algebraic extension of fields of infinite degree and (k_i) is the filtered direct system of finite-degree subextensions, then the direct limit of the system (k_i) in Alp $/k$ is the zero ring.

[35] The editors expanded the details of this paragraph.

is continuous. We write $\mathrm{Hom}_c(M, N|_k)$ for the abelian group of continuous k-linear maps from M to $N|_k$.

Proposition 0.5.C. *For every $M \in \mathrm{PC}(k)$ and $N \in \mathrm{PC}(\ell)$, there is a functorial isomorphism*

$$\mathrm{Hom}_{\mathrm{PC}(\ell)}(M\widehat{\otimes}_k\ell, N) \simeq \mathrm{Hom}_c(M, N|_k).$$

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Indeed, let $u : M \to N|_k$ be a continuous homomorphism. The map

$$u' : M \times \ell \longrightarrow N, \qquad (m, \lambda) \longmapsto \lambda u(m)$$

is continuous and "bilinear," meaning k-linear in the first factor and ℓ-linear in the second. If N' is an open ℓ-submodule of N, the argument of Lemma 0.3.1 gives an open submodule $M' \subset M$ and an open ideal $\ell' \subset \ell$ such that

$$u'(M' \times \ell) \subset N', \qquad u'(M \times \ell') \subset N'.$$

Thus u induces a continuous ℓ-linear map

$$U : M\widehat{\otimes}_k\ell \longrightarrow N$$

satisfying

$$U(m\widehat{\otimes}_k\lambda) = \lambda u(m).$$

Conversely, every morphism

$$f : M\widehat{\otimes}_k\ell \longrightarrow N$$

gives the morphism

$$f' : M \longrightarrow N|_k, \qquad m \longmapsto f(m\widehat{\otimes}_k 1).$$

As in 0.3.2, 0.3.5, and 0.3.1.2, this yields:

Corollary 0.5.D. *The functor*

$$\mathrm{PC}(k) \longrightarrow \mathrm{PC}(\ell), \qquad M \longmapsto M\widehat{\otimes}_k\ell$$

is right exact and commutes with filtered projective limits. Thus, if (M_i) is a filtered projective system in $\mathrm{PC}(k)$, there is a canonical isomorphism

$$(\varprojlim_i M_i)\widehat{\otimes}_k\ell \simeq \varprojlim_i (M_i\widehat{\otimes}_k\ell).$$

Moreover, if M and N are pseudocompact k-modules, there is a canonical isomorphism

$$(M\widehat{\otimes}_k N)\widehat{\otimes}_k\ell \simeq (M\widehat{\otimes}_k\ell)\widehat{\otimes}_\ell(N\widehat{\otimes}_k\ell).$$

Definition 0.5.E. *Finally, if A is a profinite k-algebra, there is a unique profinite ℓ-algebra structure on $A\widehat{\otimes}_k\ell$ such that*

$$(a\widehat{\otimes}_k\lambda)(b\widehat{\otimes}_k\mu) = ab\widehat{\otimes}_k(\lambda\mu)$$

for $a, b \in A$ and $\lambda, \mu \in \ell$. The profinite ℓ-algebra $A\widehat{\otimes}_k\ell$ is said to be obtained from A by extension of scalars, or base change, along $k \to \ell$.

[36] Consequently, when ℓ is a profinite k-algebra, the functor $M \mapsto M\widehat{\otimes}_k\ell$ is left adjoint to the restriction-of-scalars functor.

1. Formal Varieties over a Pseudocompact Ring

1.1. To every pseudocompact ring A one may associate a formal scheme (EGA I, 10.4.2) as follows. Its underlying topological space is the set $\Upsilon(A)$ of open prime ideals of A, which are therefore maximal, endowed with the discrete topology. For every subset $E \subset \Upsilon(A)$, the space of sections of the structure sheaf over E is the Cartesian product

$$\prod_{\mathfrak{m}\in E} A_{\mathfrak{m}}.$$

The formal scheme so obtained is denoted by $\mathrm{Spf}(A)$, the *formal spectrum* of A.[37]

If A and B are two pseudocompact rings, a *morphism* from $\mathrm{Spf}(B)$ to $\mathrm{Spf}(A)$ consists of a map

$$f : \Upsilon(B) \longrightarrow \Upsilon(A)$$

and a family of continuous homomorphisms

$$f_y^\natural : A_{f(y)} \longrightarrow B_y, \qquad y \in \Upsilon(B).$$

Such a morphism induces a continuous homomorphism

$$f^\natural : A = \prod_{x\in\Upsilon(A)} A_x \longrightarrow B = \prod_{y\in\Upsilon(B)} B_y.$$

The converse is true:

Proposition. *The contravariant functor $A \mapsto \mathrm{Spf}(A)$ is fully faithful.*

Indeed, let $\phi : A \to B$ be a continuous homomorphism. The inverse image $\phi^{-1}(\mathfrak{n})$ of an open maximal ideal of B is an open prime ideal of A, and hence is maximal in A. We therefore obtain a map

$$\mathfrak{n} \longmapsto \phi^{-1}(\mathfrak{n})$$

from $\Upsilon(B)$ to $\Upsilon(A)$, and ϕ induces a continuous homomorphism

$$A_{\phi^{-1}(\mathfrak{n})} \longrightarrow B_{\mathfrak{n}}.$$

Thus ϕ induces a morphism of formal schemes

$$\mathrm{Spf}(\phi) : \mathrm{Spf}(B) \longrightarrow \mathrm{Spf}(A).$$

It is readily verified that

$$\mathrm{Spf}(\phi)^\natural = \phi, \qquad \mathrm{Spf}(f^\natural) = f$$

for every morphism $f : \mathrm{Spf}(B) \to \mathrm{Spf}(A)$, which proves the proposition.

Although we shall speak here only of formal schemes of the form $\mathrm{Spf}(A)$, we shall use the language of formal schemes rather than that of pseudocompact rings, so as to ground our assertions in geometric intuition.

1.2. Let k be a pseudocompact ring.

Definition 1.2.A.[38] A *formal variety over k* is a formal scheme X over $\mathrm{Spf}(k)$ which is isomorphic to a formal k-scheme $\mathrm{Spf}(A)$ for some profinite k-algebra A. The algebra A is then isomorphic to the *affine algebra* of X, that is, the algebra of sections of the structure sheaf $\mathcal{O}_X$ of X.

[37] Cf. Remarks 0.1.2.

[38] The editors added the numbering 1.2.A–1.2.E for later reference.

We denote by $\mathrm{Vaf}\,/k$ the full subcategory of the category of formal schemes over $\mathrm{Spf}(k)$ whose objects are the formal k-varieties.[39]

By 1.1, the functor $A \mapsto \mathrm{Spf}(A)$ is an anti-equivalence from $\mathrm{Alp}\,/k$ (0.4.1) onto $\mathrm{Vaf}\,/k$. Hence, by the corollary of 0.4.2:

Proposition 1.2.B. *The category* $\mathrm{Vaf}\,/k$ *has projective and direct limits.*[40]

For example, let $X \in \mathrm{Ob}(\mathrm{Vaf}\,/k)$, let

$$f : Y \longrightarrow X, \qquad g : Z \longrightarrow X$$

be two formal k-varieties over X, and let A, B, C be the affine algebras of X, Y, Z, respectively. By 0.4.1, the affine algebra of the fiber product $Y \times_X Z$ identifies with $B\widehat{\otimes}_A C$,[41] Consequently, the inclusion of $\mathrm{Vaf}\,/k$ into the category of all formal k-schemes commutes with finite projective limits (cf. EGA I, 10.7).

Direct limits of formal k-varieties correspond to projective limits of their affine algebras.

Example 1.2.C (Cokernels). Let $d, e : X \rightrightarrows Y$ be a double arrow in $\mathrm{Vaf}\,/k$. The affine algebra of $\mathrm{Coker}(d, e)$ is isomorphic to the kernel of the homomorphisms induced on the affine algebras of X and Y. There is also the following construction of $\mathrm{Coker}(d, e)$. Its underlying topological space is the cokernel of the underlying spaces.[42] Let p be the canonical projection from the set underlying Y to this cokernel, and let z be a point of the cokernel. The local algebra of $\mathrm{Coker}(d, e)$ at z is the kernel of the double arrow

$$d^\natural, e^\natural : \quad \textstyle\prod_{p(y)=z} \mathcal{O}_{Y,y} \rightrightarrows \prod_{q(x)=z} \mathcal{O}_{X,x}$$

where

$$q = p \circ d = p \circ e,$$

and $d^\natural$ and $e^\natural$ are induced by the homomorphisms $d_x^\natural$ and $e_x^\natural$, in the notation of 1.1.

Definition 1.2.D. Let $\phi : k \to \ell$ be a homomorphism of pseudocompact rings and let X be a formal k-variety with affine algebra A. The formal scheme

$$X \times_{\mathrm{Spf}(k)} \mathrm{Spf}(\ell),$$

obtained by base change, is a formal ℓ-variety. We also denote it by

$$X\widehat{\otimes}_k \ell.$$

Its affine algebra is the completed tensor product $A\widehat{\otimes}_k \ell$ (cf. 0.5 and EGA I, § 10).

Remark 1.2.E. Since every formal variety over k decomposes into formal varieties over the local components of k, *in certain proofs we shall assume that k is a local pseudocompact ring.*

We now give some examples while fixing our terminology.

1.2.1. A *k-functor* is, by definition, a covariant functor from $\mathrm{Alf}\,/k$ to (**Sets**). By 1.1 and 0.4.2, we may identify

$$\mathrm{Vaf}\,/k \simeq (\mathrm{Alp}\,/k)^\circ$$

[39] By the definition of morphisms in 1.1, every $X \in \mathrm{Ob}(\mathrm{Vaf}\,/k)$ is the direct sum in $\mathrm{Vaf}\,/k$ of the one-point formal varieties $\mathrm{Spf}(\mathcal{O}_{X,x})$, for $x \in X$.

[40] In particular, cokernels exist in $\mathrm{Vaf}\,/k$; see below. Moreover, a filtered projective limit in $\mathrm{Vaf}\,/k$, all of whose transition morphisms are surjective, can be empty; cf. editor's note (34).

[41] The formal variety $Y \times_X Z$ is the direct sum, over $x \in X$, of the formal varieties $B_x \widehat{\otimes}_{\mathcal{O}_{X,x}} C_x$, where B_x is the product of the $\mathcal{O}_{Y,y}$ for $y \in Y$ such that $f(y) = x$, and C_x is defined analogously. This will be used in 2.5.1.

[42] That is, it is the quotient set of Y by the identifications $d(x) = e(x)$, for every $x \in X$, endowed with the quotient topology, which here is discrete.

with a full subcategory of the category of k-functors by associating to every formal k-variety X the functor

$$\mathrm{Alf}\,/k \longrightarrow (\mathbf{Sets}), \qquad C \longmapsto X(C) = \mathrm{Hom}_{\mathrm{Vaf}\,/k}\big(\mathrm{Spf}(C), X\big).$$

We shall later encounter k-functors $\mathbf{F}$ which associate to every object C of $\mathrm{Alf}\,/k$ a C-module $\mathbf{F}(C)$, and to every morphism $\phi : C \to D$ in $\mathrm{Alf}\,/k$ a k-linear map

$$\mathbf{F}(\phi) : \mathbf{F}(C) \longrightarrow \mathbf{F}(D)$$

such that, for $x \in \mathbf{F}(C)$ and $\lambda \in C$,

$$\mathbf{F}(\phi)(\lambda x) = \phi(\lambda)\mathbf{F}(\phi)(x).$$

By Exposé I, 3.1, such an $\mathbf{F}$ is endowed with the structure of an $\mathbf{O}_k$-module, where $\mathbf{O}_k$ denotes the k-functor in rings which assigns to every object C of $\mathrm{Alf}\,/k$ its underlying ring.

Definitions.
(i) An $\mathbf{O}_k$-module $\mathbf{F}$ is called *admissible* if every morphism $\phi : C \to D$ in $\mathrm{Alf}\,/k$ induces a bijection from $D \otimes_C \mathbf{F}(C)$ onto $\mathbf{F}(D)$.[43]

(ii) The module $\mathbf{F}$ is called *flat* if it is admissible and if, for every object C of $\mathrm{Alf}\,/k$, the C-module $\mathbf{F}(C)$ is flat.[44]

For example, if M is a k-module, not necessarily pseudocompact, we denote by $\mathbf{W}(M)$, as in Exposé I, 4.6.1, the $\mathbf{O}_k$-module

$$C \longmapsto C \otimes_k M.$$

Then $\mathbf{W}(M)$ is flat whenever M is flat over k.

Moreover, the functor $M \mapsto \mathbf{W}(M)$ has as right adjoint the functor which assigns to every $\mathbf{O}_k$-module $\mathbf{F}$ the k-module

$$\varprojlim_{\mathfrak{l}} \mathbf{F}(k/\mathfrak{l}),$$

where $\mathfrak{l}$ ranges over the open ideals of k.

1.2.2. In what follows, an $\mathbf{O}_k$-module will always be denoted by a boldface letter such as $\mathbf{F}$. When k is Artinian, we shall simply write F instead of $\mathbf{F}(k)$. In that case, it is clear that the functor

$$\mathbf{F} \longmapsto F$$

induces an equivalence from the category of flat $\mathbf{O}_k$-modules onto the category of flat k-modules! The purpose of this terminology is solely to allow us to reason "as if k were always Artinian."

In accordance with Exposé I, § 3.1, we shall use analogous conventions for other algebraic structures. Thus an $\mathbf{O}_k$-algebra, respectively an $\mathbf{O}_k$-coalgebra, an $\mathbf{O}_k$-Lie algebra, or an $\mathbf{O}_k$-p-Lie algebra, consists of an $\mathbf{O}_k$-module $\mathbf{M}$ together with, for every object C of $\mathrm{Alf}\,/k$, a structure of C-algebra, respectively C-coalgebra, C-Lie algebra, or C-p-Lie algebra, on $\mathbf{M}(C)$. We further require that for every morphism $\phi : C \to D$ in $\mathrm{Alf}\,/k$, the map

$$D \otimes_C \mathbf{M}(C) \longrightarrow \mathbf{M}(D)$$

induced by $\mathbf{M}(\phi)$ be a homomorphism of D-algebras, respectively of D-coalgebras, and so on.

[43] The editors introduced this terminology; cf. editor's note (47) in 1.2.3.
[44] It is therefore projective, since C is Artinian.

Finally, if $\mathbf{F}$ and $\mathbf{G}$ are two $\mathbf{O}_k$-modules, then $\mathbf{F} \otimes_k \mathbf{G}$ denotes the $\mathbf{O}_k$-module

$$C \longmapsto \mathbf{F}(C) \otimes_C \mathbf{G}(C).$$

1.2.3.[45] We begin with the following lemma.

Lemma 1.2.3.A. *Let $k \longrightarrow K$ be a morphism of pseudocompact rings, let B be a pseudocompact K-module, and let M be a projective k-module, without topology. There is a canonical isomorphism of pseudocompact K-modules*

$$\theta : \mathrm{Hom}_k(M,k)\widehat{\otimes}_k B \xrightarrow{\sim} \mathrm{Hom}_k(M,B). \tag{2}$$

Here $M^* = \mathrm{Hom}_k(M,k)$ is endowed with the topology defined in 0.2.2, which makes it a pseudocompact k-module. The topology on $\mathrm{Hom}_k(M,B)$ is defined analogously: a basis of neighborhoods of 0 is formed by the following K-submodules, where $x \in M$ and B' is an open K-submodule of B:

$$\mathscr{H}(x,B') = \{f \in \mathrm{Hom}_k(M,B) \mid f(x) \in B'\}.$$

Finally, $M^*\widehat{\otimes}_k B$ is the pseudocompact K-module obtained from M^* by base change; cf. 0.5.A.

With these conventions, θ is plainly an isomorphism when $M = k$. Moreover, both sides of (2), regarded as functors in M, carry direct sums to products; in particular, they commute with finite direct sums. It follows that (2) is an isomorphism first when M is a free k-module, and then when M is projective.

Definition 1.2.3.B. Let N be a pseudocompact k-module. We write $\mathbf{V}_k^f(N)$, or $\mathbf{N}^\dagger$,[46] for the $\mathbf{O}_k$-module which assigns to every $C \in \mathrm{Ob}(\mathrm{Alf}\,/k)$ the C-module

$$(C\widehat{\otimes}_k N)^\dagger,$$

the dual of $C\widehat{\otimes}_k N$ in the sense of 0.2.2; that is, the C-module

$$\mathrm{Hom}_c(C\widehat{\otimes}_k N, C) \simeq \mathrm{Hom}_c(N, C|_k),$$

where $C|_k$ denotes the k-module obtained from C by restriction of scalars. This $\mathbf{O}_k$-module $\mathbf{V}_k^f(N)$ is called the *dual* of N.

If $k_\mathfrak{m}$ is a local component of k, then for every object C of Alf $/k$,

$$\mathrm{Hom}_c(k_\mathfrak{m}, C|_k) = C_\mathfrak{m} = C \otimes_k k_\mathfrak{m}$$

is a projective C-module. Moreover,

$$D \otimes_C C_\mathfrak{m} = D_\mathfrak{m}$$

for every morphism $C \longrightarrow D$ in Alf $/k$. Thus $\mathbf{V}_k^f(k_\mathfrak{m})$ is a flat $\mathbf{O}_k$-module; cf. 1.2.1. Since every projective pseudocompact k-module is a product of modules $k_\mathfrak{m}$ (Proposition 0.2.1), Corollary 0.2.F gives:

Corollary 1.2.3.C. *The $\mathbf{O}_k$-module $\mathbf{V}_k^f(N)$ is flat whenever N is a projective pseudocompact k-module.*

[45] The editors added the following lemma and introduced the numbering 1.2.3.A through 1.2.3.F for later references.

[46] In fact, the second notation will not be used; see editors' note (47).

Definition 1.2.3.D. Conversely, let $\mathbf{M}$ be an admissible $\mathbf{O}_k$-module (cf. 1.2.1).[47] We call the following pseudocompact k-module the *dual* of $\mathbf{M}$, and denote it by $\Gamma^*(\mathbf{M})$. As $\mathfrak{l}$ ranges over the open ideals of k, endow each $k/\mathfrak{l}$-module

$$\mathbf{M}(k/\mathfrak{l})^* = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l})$$

with the topology described in 0.2.2; it is then a pseudocompact k-module. Since

$$\mathbf{M}(k/\mathfrak{l}) \simeq \mathbf{M}(k/\mathfrak{l}') \otimes_{k/\mathfrak{l}'} (k/\mathfrak{l}) \qquad (\mathfrak{l} \supset \mathfrak{l}'),$$

there are transition morphisms

$$\begin{aligned}\mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}') &\longrightarrow \mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}) \\ &= \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l}).\end{aligned}$$

By definition, $\Gamma^*(\mathbf{M})$ is the projective limit of this filtered projective system of pseudocompact k-modules.

From now on, suppose in addition that $\mathbf{M}$ is a flat $\mathbf{O}_k$-module (cf. 1.2.1). Each $\mathbf{M}(k/\mathfrak{l}')$ is then a projective $k/\mathfrak{l}'$-module, so the transition morphisms above are surjective. Consequently the projections

$$\Gamma^*(\mathbf{M}) \longrightarrow \mathbf{M}(k/\mathfrak{l})^*$$

are also surjective, since filtered projective limits are exact in $\mathrm{PC}(k)$.

Let $\tau : k \longrightarrow K$ be a morphism of pseudocompact rings. We write $\mathbf{M} \otimes_k K$, or simply $\mathbf{M}_K$, for the $\mathbf{O}_K$-functor defined as follows. If C is a finite-length K-algebra, then the kernel of $k \longrightarrow C$ is an open ideal $\mathfrak{l}$, and we set

$$\mathbf{M}_K(C) = \mathbf{M}(k/\mathfrak{l}) \otimes_k C.$$

The same value is obtained from $\mathbf{M}(k/\mathfrak{l}') \otimes_k C$ for every open ideal $\mathfrak{l}' \subset \mathfrak{l}$. We then define $\Gamma^*_K(\mathbf{M}_K)$ as the projective limit, with I ranging over the open ideals of K, of the pseudocompact K-modules

$$\mathrm{Hom}_{K/I}(\mathbf{M}_K(K/I), K/I) = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), K/I),$$

where on the right $\mathfrak{l}$ is any open ideal of k such that $\tau(\mathfrak{l}) \subset I$.

By Lemma 1.2.3.A, the right-hand term identifies with

$$\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k (K/I).$$

Since the projections $\Gamma^*(\mathbf{M}) \longrightarrow \mathbf{M}(k/\mathfrak{l})^*$ are surjective, the projective limit of these latter modules is precisely the pseudocompact K-module

$$\Gamma^*(\mathbf{M}) \widehat{\otimes}_k K$$

(cf. 0.5.A). We have therefore proved that, for every flat $\mathbf{O}_k$-module $\mathbf{M}$, formation of $\Gamma^*(\mathbf{M})$ commutes with extension of the base; that is,

$$\Gamma^*_K(\mathbf{M} \otimes_k K) \simeq \Gamma^*(\mathbf{M}) \widehat{\otimes}_k K. \qquad (\star)$$

Proposition 1.2.3.E.

[47] The editors do not understand the definition that follows unless $\mathbf{M}$ is assumed to be admissible, which is why this hypothesis has been added. They also write $\Gamma^*(\mathbf{M})$, rather than $\mathbf{M}^*$, to avoid confusion between $\mathbf{M}^*$, the module dual to the functor $\mathbf{M}$, and $\mathbf{N}^\dagger$, the functor dual to the module N; cf. editors' note (46). Finally, the editors have spelled out the construction of $\Gamma^*(\mathbf{M})$.

(i) *The functors*

$$N \longmapsto \mathbf{V}_k^f(N), \qquad \mathbf{M} \longmapsto \Gamma^*(\mathbf{M})$$

define an anti-equivalence between the category of projective pseudocompact k-modules and the category of flat $\mathbf{O}_k$-modules.[48]

(ii) *Moreover, if $k \longrightarrow K$ is a morphism of pseudocompact rings, the preceding anti-equivalence "commutes with base change" in the following sense: if $N \simeq \Gamma^*(\mathbf{M})$, then*

$$N\widehat{\otimes}_k K \simeq \Gamma_K^*(\mathbf{M} \otimes_k K).$$

Proof. First, there is a natural transformation

$$\phi_N : N \longrightarrow \Gamma^*(\mathbf{V}_k^f(N)).$$

The functor $\Gamma^* \circ \mathbf{V}_k^f$ commutes with products by 0.3.5 and 0.2.F, so it is enough to verify that ϕ_N is an isomorphism when N is a local component $k_{\mathfrak{m}}$ of k. In this case, for every open ideal $\mathfrak{l}$ of k contained in $\mathfrak{m}$, the natural morphism

$$(k/\mathfrak{l})_{\mathfrak{m}} \longrightarrow \mathrm{Hom}_{k/\mathfrak{l}}\left(\mathrm{Hom}_c(k_{\mathfrak{m}}\widehat{\otimes}_k k/\mathfrak{l}, k/\mathfrak{l}), k/\mathfrak{l}\right)$$

is an isomorphism, which proves the assertion.

Conversely, let $\mathbf{M}$ be a flat $\mathbf{O}_k$-module. We show that

$$\Gamma^*(\mathbf{M}) = \varprojlim_{\mathfrak{l}} \mathbf{M}(k/\mathfrak{l})^*$$

is a projective object of $\mathrm{PC}(k)$. Let $N \longrightarrow N'$ be a surjective morphism between objects of $\mathrm{LF}(k)$. By 0.2.F (i) and (ii), it is enough to prove that the natural map

$$\varinjlim_{\mathfrak{l}} \mathrm{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N) \longrightarrow \varinjlim_{\mathfrak{l}} \mathrm{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N')$$

is surjective. This is clear: N and N' are $k/\mathfrak{l}_0$-modules for some open ideal $\mathfrak{l}_0$, and if $\mathfrak{l} \subset \mathfrak{l}_0$, every morphism $\mathbf{M}(k/\mathfrak{l})^* \longrightarrow N'$ lifts to a morphism $\mathbf{M}(k/\mathfrak{l})^* \longrightarrow N$, since $\mathbf{M}(k/\mathfrak{l})^*$ is a projective object of $\mathrm{PC}(k/\mathfrak{l})$.

There is a morphism

$$\psi : \mathbf{M} \longrightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))$$

of functors from $\mathrm{Alf}\,/k$ to (**Sets**). Let B be an object of $\mathrm{Alf}\,/k$. We claim that

$$\begin{aligned} \psi(B) : \mathbf{M}(B) &\longrightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))(B) \\ &= \mathrm{Hom}_c\left(\varprojlim_{\mathfrak{l}} (\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k B), B\right) \end{aligned} \tag{1}$$

is a bijection. In the equality above we used the fact that $\widehat{\otimes}$ commutes with filtered projective limits. Let $\mathfrak{l}_0$ be an open ideal of k contained in the kernel of $k \longrightarrow B$. By Lemma 1.2.3.A, for every $\mathfrak{l} \subset \mathfrak{l}_0$, there is a canonical isomorphism of pseudocompact B-modules

$$\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k B \xrightarrow{\sim} \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), B).$$

[48] The editors added assertion (ii) below and the proof that follows. The original said: "If $\mathbf{M}$ is a flat $\mathbf{O}_k$-module, it is clear that $\mathbf{M}$ is the dual of $\mathbf{M}^*$; hence the functors $N \mapsto N^*$ and $\mathbf{M} \mapsto \mathbf{M}^*$ define an anti-equivalence . . . "

Since

$$\mathbf{M}(B) = \mathbf{M}(k/\mathfrak{l}) \otimes_{k/\mathfrak{l}} B,$$

the right-hand side is $\mathrm{Hom}_B(\mathbf{M}(B), B)$. Thus the projective system in (1) is constant for $\mathfrak{l} \subset \mathfrak{l}_0$, and (1) reduces to the canonical morphism

$$\mathbf{M}(B) \longrightarrow \mathrm{Hom}_c\big(\mathrm{Hom}_B(\mathbf{M}(B), B), B\big).$$

This is an isomorphism by 0.2.2, since B is Artinian and $\mathbf{M}(B)$ is a projective B-module. This proves assertion (i) of the proposition; assertion (ii) follows from the isomorphism $(\star)$ established above. □

Remark 1.2.3.F. Let us return to the statement of the proposition and suppose, in addition, that N is a topologically free pseudocompact k-module. In this case one can choose, "coherently," bases for the C-modules $\mathbf{V}_k^f(N)(C)$.

Indeed, let (n_i) be a pseudobasis of N (0.2.1), and let n_i^C be the canonical image of n_i in $C\widehat{\otimes}_k N$, for $C \in \mathrm{Alf}\,/k$. Define $\delta_i^C \in (C\widehat{\otimes}_k N)^\dagger$ by

$$\delta_i^C(n_i^C) = 1, \qquad \delta_i^C(n_j^C) = 0 \quad (i \neq j).$$

Then (δ_i^C) is a basis of $\mathbf{V}_k^f(N)(C)$. Moreover, for every morphism $\varphi : C \longrightarrow D$ in $\mathrm{Alf}\,/k$, the map $\mathbf{V}_k^f(N)(\varphi)$ sends δ_i^C to δ_i^D.

1.2.4.[49] For every pseudocompact k-module E, let $\widehat{S}_k(E)$ denote the *completed symmetric algebra* of E, defined as follows. Let $T_k(E)$ be the direct sum of the pseudocompact k-modules

$$\widehat{\bigotimes}_k^n E = E\widehat{\otimes}_k \cdots \widehat{\otimes}_k E \qquad \left(n \geq 0; \quad \widehat{\bigotimes}_k^0 E = k\right).$$

Give $T_k(E)$ its usual graded k-algebra multiplication. Let $S_k(E)$ be the quotient of $T_k(E)$ by the homogeneous ideal whose n-th component is the closed k-submodule of $\widehat{\bigotimes}_k^n E$ generated by the elements

$$\begin{aligned} &x_1\widehat{\otimes}\cdots\widehat{\otimes}x_i\widehat{\otimes}x_{i+1}\widehat{\otimes}\cdots\widehat{\otimes}x_n \\ &\quad - x_1\widehat{\otimes}\cdots\widehat{\otimes}x_{i+1}\widehat{\otimes}x_i\widehat{\otimes}\cdots\widehat{\otimes}x_n. \end{aligned}$$

If this quotient is denoted by $S_k^n(E)$, then $S_k(E)$ is a graded k-algebra whose n-th component is $S_k^n(E)$.

Endow $S_k(E)$ with the linear topology defined by the set Υ of ideals $\mathfrak{l}$ such that $S_k(E)/\mathfrak{l}$ is a k-module of finite length and

$$S_k^n(E) \cap \mathfrak{l}$$

is an open submodule of $S_k^n(E)$ for every n. The profinite algebra $\widehat{S}_k(E)$ is the separated completion of $S_k(E)$ for this topology.[50]

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[49] Editors' note: In this paragraph the order has been changed: $\widehat{S}_k(E)$ is defined first, and it is then stated that $\mathbf{V}_k^f(E)$ represents $\mathbf{V}_k^f(E)$.

[50] Editors' note: For example, let k be a field, let E be a pseudocompact k-vector space, and put $V = \mathrm{Hom}_c(E, k)$; then $E \simeq V^*$ (cf. 0.2.2). For every finite-dimensional subspace W of V, let $\mathcal{F}(W)$ be the set of closed points of the k-scheme $\mathbf{V}(W) = \mathrm{Spec}\, S_k(W^*)$, and let $\mathcal{F}(V) = \varinjlim_W \mathcal{F}(W)$. Then $\widehat{S}_k(E)$ is the product, over $x \in \mathcal{F}(V)$, of the pseudocompact k-algebras

$$\widehat{S}_k(E)_x = \varprojlim_W \widehat{\mathcal{O}}_{W,x},$$

where W ranges over the finite-dimensional subspaces of V such that $x \in \mathcal{F}(W)$, and $\widehat{\mathcal{O}}_{W,x}$ is the separated completion of the local ring $\mathcal{O}_{\mathbf{V}(W),x}$ for the topology defined by the ideals of finite codimension (which here coincides with the $\mathfrak{m}$-adic topology). If k is algebraically closed and $(v_i)_{i\in I}$ is a basis of V, so that E has a pseudobasis $(e_i)_{i\in I}$, then every local component $\widehat{S}_k(E)_x$ is isomorphic to the formal power-series ring $k[[e_i,\ i \in I]]$, endowed with the topology defined by the ideals of finite codimension.

We write

$$\mathbf{V}_k^f(E) = \mathrm{Spf}(\widehat{S}_k(E))$$

for the corresponding formal k-variety. It represents the k-functor $\mathbf{V}_k^f(E)$; that is, for every object C of Alf $/k$, there is a canonical isomorphism

$$\begin{aligned}\mathbf{V}_k^f(E)(C) &= \mathrm{Hom}_c(E, C|_k) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Alp}\,/k}(\widehat{S}_k(E), C)\\ &= \mathrm{Hom}_{\mathrm{Vaf}\,/k}(\mathrm{Spf}(C), \mathbf{V}_k^f(E)).\end{aligned}$$

In what follows, we identify the formal variety $\mathbf{V}_k^f(E)$ with the k-functor $\mathbf{V}_k^f(E)$.

1.2.5. By 1.2.4, the zero morphism from E to k corresponds to a morphism of profinite algebras

$$\pi : \widehat{S}_k(E) \longrightarrow k.$$

The morphism π induces the zero map on $S_k^n(E)$ for $n \geq 1$, and defines a section of the structure morphism

$$\mathbf{V}_k^f(E) \longrightarrow \mathrm{Spf}(k).$$

We denote by $\mathbf{V}_k^{f,0}(E)$ the formal variety whose points are the images of the points of $\mathrm{Spf}(k)$ under the section $\mathrm{Spf}(\pi)$, and whose local algebras at those points are the same as those of $\mathbf{V}_k^f(E)$. [51]

The affine algebra of $\mathbf{V}_k^{f,0}(E)$ is thus the separated completion of $S_k(E)$ for the topology defined by the ideals $\mathfrak{l} \in \Upsilon$ (cf. 1.2.4) which contain $S_k^n(E)$ for all sufficiently large n. Hence it is the infinite product

$$k[[E]] = k \times E \times S_k^2(E) \times S_k^3(E) \times \cdots .$$

Let C be an object of Alf $/k$, let

$$u \in \mathbf{V}_k^f(E)(C) = \mathrm{Hom}_c(E, C),$$

and let

$$\phi : \widehat{S}_k(E) \longrightarrow C$$

be the corresponding morphism of profinite k-algebras. Then $\mathrm{Ker}(\phi)$ is an open ideal—that is, ϕ belongs to $\mathbf{V}_k^{f,0}(E)(C)$—if and only if $\mathrm{Ker}(\phi)$ contains $S_k^n(E)$ for all sufficiently large n, or equivalently if and only if $u(E)$ is contained in the radical $r(C)$ of C. Consequently, for every object C of Alf $/k$, there are canonical isomorphisms

$$\mathbf{V}_k^{f,0}(E)(C) \simeq \mathrm{Hom}_{\mathrm{Alp}\,/k}(k[[E]], C) \simeq \mathrm{Hom}_c(E, r(C)).$$

1.2.6. A formal k-variety V is said to be *of finite length* if its affine algebra has finite length. Similarly, if S is a scheme, an S-scheme X is said to be *of finite length* if X is finite over S and the direct image of $\mathcal{O}_X$ on S is an $\mathcal{O}_S$-module of finite length.[52] Thus giving an S-scheme X of finite length is the same as giving a finite set

$$\{s_1, \ldots, s_n\}$$

of closed points of S, together with an $\mathcal{O}_{S,s_i}$-algebra of finite length at each such point.

[51] Editors' note: The remainder of this paragraph has been spelled out in detail.

[52] Editors' note: The following sentence has been added.

It follows that the S-schemes of finite length identify with the formal varieties of finite length over the following formal scheme $\widehat{S}$. The underlying topological space of $\widehat{S}$ is the set of closed points of S, endowed with the discrete topology. If s is such a closed point, the stalk $\mathcal{O}_{\widehat{S},s}$ of the structure sheaf is the separated completion

$$\widehat{\mathcal{O}}_{S,s}$$

of $\mathcal{O}_{S,s}$ for the linear topology defined by the ideals of finite colength. Thus

$$\widehat{S} = \operatorname{Spf} \mathcal{A}(\widehat{S}),$$

where $\mathcal{A}(\widehat{S})$ is the product of the $\widehat{\mathcal{O}}_{S,s}$, as s ranges over the closed points of S, endowed with the product topology.

Definition. *If X is an S-scheme, let $\widehat{X}/\widehat{S}$ denote the formal variety over*

$$k = \mathcal{A}(\widehat{S})$$

defined as follows. Its underlying topological space consists of the points $x \in X$ such that

$$[\kappa(x) : \kappa(s)] < \infty,$$

where s is the image of x in S. The local ring $\mathcal{O}_{\widehat{X}/\widehat{S},x}$ is the separated completion of $\mathcal{O}_{X,x}$ for the linear topology defined by the ideals I of $\mathcal{O}_{X,x}$ for which $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{S,s}$-module. Since $[\kappa(x) : \kappa(s)] < \infty$, this is equivalent to requiring that $\mathcal{O}_{X,x}/I$ have finite length as an $\mathcal{O}_{X,x}$-module.

Let $\mathrm{Vaf}^{\ell f}/\widehat{S}$ be the category of formal varieties of finite length over $\widehat{S}$, identified with the category of S-schemes of finite length. By 1.1 and 1.2.1, the category $\mathrm{Vaf}/\widehat{S}$ of formal varieties over $\widehat{S}$ is equivalent to the category of left-exact contravariant functors

$$\mathrm{Vaf}^{\ell f}/\widehat{S} \longrightarrow (\mathbf{Sets}).$$

In particular, for every S-scheme X, the functor

$$T \longmapsto \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, X)$$

from $\mathrm{Vaf}^{\ell f}/\widehat{S}$ to (**Sets**) is left exact and therefore corresponds to a formal variety over $\widehat{S}$. This variety is precisely $\widehat{X}/\widehat{S}$.[53]

Proposition. *For every S-scheme X, the functors*

$$\mathrm{Hom}_{\mathrm{Vaf}/\widehat{S}}(-, \widehat{X}/\widehat{S}) \qquad \textit{and} \qquad \mathrm{Hom}_{(\mathrm{Sch}/S)}(-, X)$$

have the same restriction to $\mathrm{Vaf}^{\ell f}/\widehat{S}$. This defines a functor

$$X \longmapsto \widehat{X}/\widehat{S}$$

from (Sch/S) to $\mathrm{Vaf}/\widehat{S}$, and this functor commutes with finite projective limits.

Proof. It is immediate that the formal variety $\widehat{X}/\widehat{S}$ has the required property and that the correspondence $X \mapsto \widehat{X}/\widehat{S}$ is functorial. It remains to prove the second assertion.

[53] Editors' note: The following proposition has been expanded by adding the fact that the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with finite projective limits. In the original this appeared in the proof of 1.3.4; the proof given here is more direct than the original one. This result will also be used in Section 2 and in 3.3.1.

Put

$$\mathscr{S} = (\mathrm{Sch}\,/S), \qquad \mathscr{V} = \mathrm{Vaf}\,/\widehat{S},$$

and write $\widehat{X}$ for $\widehat{X}/\widehat{S}$. By 1.2.B, the category $\mathscr{V}$ has arbitrary projective limits. Let $(X_i)_{i\in I}$ be a projective system in $\mathscr{S}$, and suppose that

$$X = \varprojlim_i X_i$$

exists in $\mathscr{S}$, as it does when I is finite. The functor which associates with every object Y of $\mathscr{S}$ the functor

$$h_Y = \mathrm{Hom}_{\mathscr{S}}(-, Y)$$

and, respectively, with every object V of $\mathscr{V}$ the functor

$$h_V = \mathrm{Hom}_{\mathscr{V}}(-, V)$$

commutes with projective limits. Hence, for every S-scheme T of finite length, there are functorial isomorphisms

$$\begin{aligned}\mathrm{Hom}_{\mathscr{S}}(T, X) &\simeq \varprojlim_i \mathrm{Hom}_{\mathscr{S}}(T, X_i)\\ &\simeq \varprojlim_i \mathrm{Hom}_{\mathscr{V}}(T, \widehat{X}_i) \simeq \mathrm{Hom}_{\mathscr{V}}\left(T, \varprojlim_i \widehat{X}_i\right).\end{aligned}$$

Consequently, the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with projective limits whenever they exist in $(\mathrm{Sch}\,/S)$, and in particular with finite projective limits. □

1.3

Proposition 1.3. *Let $f : X \to Y$ be a morphism of formal varieties over k, let A and B be the affine algebras of X and Y, and let $g : B \to A$ be the morphism induced by f. Then f is a monomorphism in* Vaf $/k$ *if and only if g is surjective.*

By 1.1,[54] the functor $A \mapsto \mathrm{Spf}(A)$ is an anti-equivalence from Alp $/k$ to Vaf $/k$. Consequently, f is a monomorphism if and only if g is an epimorphism, which it certainly is when g is surjective.

Conversely, suppose that g is an epimorphism. For every $\mathfrak{n} \in \Upsilon(B)$, put

$$A_{\mathfrak{n}} = A \widehat{\otimes}_B B_{\mathfrak{n}}.$$

By 0.4, A is a pseudocompact B-module, and hence is the product of the $A_{\mathfrak{n}}$ (0.3.6). The morphism g is therefore the product of the morphisms

$$g_{\mathfrak{n}} : B_{\mathfrak{n}} \longrightarrow A_{\mathfrak{n}}$$

obtained from g by base change. These remain epimorphisms. We are thus reduced to the case in which B is local, with maximal ideal $\mathfrak{n}$; set $K = B/\mathfrak{n}$.

By Nakayama's lemma 0.3.3, it is enough to prove that

$$g \widehat{\otimes}_B K$$

is surjective. This map is obtained from g by base change and is therefore an epimorphism in Alp $/K$. We may thus suppose that $B = K$ is a field. Now f is a monomorphism if and only if the diagonal

$$X \longrightarrow X \times_Y X$$

[54] The editors expanded the beginning of the proof.

is an isomorphism; equivalently, the homomorphism

$$A\widehat{\otimes}_K A \longrightarrow A, \qquad x\widehat{\otimes}_K x' \longmapsto xx',$$

is an isomorphism. Since K is a field, this implies $A = K$.

Remark. *It follows from the proposition that every monomorphism $f : X \to Y$ of formal varieties is an isomorphism from X onto a necessarily closed formal subvariety of Y.*[55]

1.3.1

The preceding proposition implies, in particular, that every monomorphism $f : X \to Y$ in Vaf $/k$ is effective (cf. IV 1.3).[56] The same is not true of epimorphisms, as one sees by slightly modifying the counterexample of Exposé V, § 2.c).[57] This is why we now consider a useful class of effective epimorphisms.

Lemma. [58] *Let $f : X \to Y$ be a morphism of formal k-varieties, let A and B be the affine algebras of X and Y, and let $f^\natural : B \to A$ be the morphism induced by f. The following conditions are equivalent:*

(i) *For every $x \in X$, the homomorphism*

$$f_x^\natural : \mathcal{O}_{Y,f(x)} \longrightarrow \mathcal{O}_{X,x}$$

makes $\mathcal{O}_{X,x}$ a topologically free pseudocompact $\mathcal{O}_{Y,f(x)}$-module.

(ii) *For every $y \in Y$, the local component*

$$A_y = \prod_{f(x)=y} \mathcal{O}_{X,x}$$

is a topologically free pseudocompact B_y-module.

(iii) *The map $f^\natural : B \to A$ makes A a projective pseudocompact B-module.*

(iv) *The functor*

$$\mathrm{PC}(B) \longrightarrow \mathrm{PC}(A), \qquad M \longmapsto M\widehat{\otimes}_B A,$$

is exact.

When these conditions hold, f is said to be topologically flat.

The implications

$$\text{(i)} \Longrightarrow \text{(ii)} \Longleftrightarrow \text{(iii)} \Longleftrightarrow \text{(iv)}$$

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[55] The editors added this remark.

[56] In the opposite category $(\mathrm{Vaf}\,/k)^\circ = \mathrm{Alp}\,/k$, the corresponding map $g : B \to A$ is an effective epimorphism: since it is surjective, it induces an isomorphism of profinite k-algebras $B/I \xrightarrow{\sim} A$, where $I = \ker g$. Hence every morphism $\phi : B \to C$ in Alp $/k$ that vanishes on I descends to a morphism $\bar{\phi} : A \to C$ such that $\phi = \bar{\phi} \circ g$.

[57] Let k be a field, $X = \mathrm{Spf}(k[[T]])$, and $Y = \mathrm{Spf}(B)$, where B is the k-subalgebra of $A = k[[T]]$ topologically generated by T^3 and T^4; thus B consists of the series $\sum a_n T^n$ with $a_n = 0$ for $n = 1, 2, 5$. Then $X \to Y$ is an epimorphism that is not effective: the cokernel of $X \times_Y X \rightrightarrows X$ is $\mathrm{Spf}(B')$, where

$$B' = \{a \in A \mid a\widehat{\otimes}1 = 1\widehat{\otimes}a\},$$

and B' contains T^5.

[58] The editors added this lemma, which explains the term "topologically flat."

follow from 0.2.F (iii) and 0.3.7. Conversely, suppose that (ii) holds, and let $x \in X$, $y = f(x)$. Since $\mathcal{O}_{X,x}$ is a direct factor of A_y, it is a projective pseudocompact B_y-module, and therefore is topologically free by 0.2.1, because B_y is local.

A morphism $f : X \to Y$ of formal k-varieties is called *surjective* if it induces a surjection of the underlying sets.

Proposition. *Let $f : X \to Y$ be a surjective, topologically flat morphism of formal k-varieties. Then f is an effective epimorphism* (cf. IV 1.3).

Let A and B be the affine algebras of X and Y, and let $g : B \to A$ be the morphism induced by f. We must prove that Y identifies with the cokernel of

$$X \times_Y X \rightrightarrows X;$$

that is, for every $\mathfrak{n} \in \Upsilon(B)$, $B_\mathfrak{n}$ must identify with the subring of

$$A_\mathfrak{n} = A \widehat{\otimes}_B B_\mathfrak{n}$$

formed by the elements a such that

$$a \widehat{\otimes} 1 = 1 \widehat{\otimes} a.$$

We may suppose that B is local, with maximal ideal $\mathfrak{n}$. The hypothesis then says that g makes A a nonzero topologically free B-module. By Nakayama's lemma 0.3.3, $A/\overline{\mathfrak{n}A} \neq 0$, and hence the induced morphism

$$g' : B/\mathfrak{n} \longrightarrow A/\overline{\mathfrak{n}A}$$

is injective. By Lemma 1.3.2 below, B is a direct factor of A as a B-module; write $A = B \oplus A'$. It follows that B identifies with the set of $a \in A$ such that

$$a \widehat{\otimes}_B 1 = 1 \widehat{\otimes}_B a.$$

1.3.2

Lemma 1.3.2. *Let B be a local pseudocompact ring, let $\mathfrak{n}$ be its maximal ideal, let M, N be projective pseudocompact B-modules, and let $g : M \to N$ be a morphism. If*

$$(B/\mathfrak{n}) \widehat{\otimes}_B g$$

is injective, then g is an isomorphism from M onto a direct factor of N.

Indeed, suppose

$$g' = (B/\mathfrak{n}) \widehat{\otimes}_B g$$

is injective. Since $B/\mathfrak{n}$ is a field, g' has a retraction r'. Let p and q be the canonical projections from M and N onto

$$M/\overline{\mathfrak{n}M} \qquad \text{and} \qquad N/\overline{\mathfrak{n}N}.$$

Since N is projective, there exists $r : N \to M$ such that

$$p \circ r = r' \circ q;$$

thus r' is obtained from r by passage to the quotient. Since $r' \circ g'$ is an isomorphism, 0.3.4 shows that the same is true of $r \circ g$, because M is projective. If s denotes the inverse of $r \circ g$, then $s \circ r$ is a retraction of g.

1.3.3

Proposition 1.3.3. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of formal k-varieties.*

(i) *If f and g are topologically flat, so is $g \circ f$.*

(ii) *If f and $g \circ f$ are topologically flat and f is surjective, then g is topologically flat.*

(iii) *If f is topologically flat, then so is $f \times_Y Y'$ after every base change $Y' \to Y$.*

Assertions (i) and (iii) are clear. To prove (ii), let A, B, C be the affine algebras of X, Y, Z, and let $f' : B \to A$, $g' : C \to B$ be the homomorphisms induced by f, g. Since $g \circ f$ is topologically flat, $f' \circ g'$ makes A a projective pseudocompact C-module. Likewise, f' makes A a faithful projective pseudocompact B-module. Thus, as P ranges over pseudocompact C-modules and N over pseudocompact B-modules, the functors

$$P \longmapsto P\widehat{\otimes}_C A, \qquad N \longmapsto N\widehat{\otimes}_B A$$

are exact; the second is also faithful. Hence the functor

$$P \longmapsto P\widehat{\otimes}_C B$$

is exact, and 0.3.7 implies that B is a projective pseudocompact C-module.

1.3.4

Proposition 1.3.4. *Let S be a scheme, let Y be a locally Noetherian S-scheme, and let $f : X \to Y$ be a faithfully flat S-morphism locally of finite type. Thus f is an effective epimorphism; equivalently, the following sequence is exact* (cf. IV 6.3.1 (iv) and IV 1.3)*:*

$$(*) \qquad X \times_Y X \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} X \xrightarrow{f} Y.$$

Then the morphism of formal $\widehat{S}$-varieties

$$\widehat{f} : \widehat{X}/\widehat{S} \longrightarrow \widehat{Y}/\widehat{S}$$

(cf. 1.2.6) *is surjective and topologically flat, and the following sequence, obtained from the preceding one, is exact:*

$$\widehat{(*)} \qquad \widehat{X \times_Y X}/\widehat{S} \underset{\widehat{\mathrm{pr}_2}}{\overset{\widehat{\mathrm{pr}_1}}{\rightrightarrows}} \widehat{X}/\widehat{S} \xrightarrow{\widehat{f}} \widehat{Y}/\widehat{S}.$$

Let y be a point of Y, with image s in S, such that $\kappa(y)$ is a finite extension of the residue field $\kappa(s)$. Since f is surjective and locally of finite type, $f^{-1}(y)$ is nonempty and locally of finite type over $\kappa(y)$. Its closed points are precisely the points of $\widehat{X}/\widehat{S}$ mapping to y. This proves that $\widehat{f}$ is surjective.

Let x be a closed point of $f^{-1}(y)$.[59] Since Y is locally Noetherian and f is locally of finite type, the local rings $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{X,x}$ are Noetherian. The powers of their maximal ideals therefore have finite colength, and the local rings of $\widehat{Y}/\widehat{S}$ at y and of $\widehat{X}/\widehat{S}$ at x are the completions

$$\widehat{\mathcal{O}}_{Y,y} \qquad \text{and} \qquad \widehat{\mathcal{O}}_{X,x}$$

for their adic topologies. Since f is flat, $\widehat{\mathcal{O}}_{X,x}$ is flat over $\widehat{\mathcal{O}}_{Y,y}$ by SGA 1, IV 5.8. By 0.3.8 it is therefore a topologically free $\widehat{\mathcal{O}}_{Y,y}$-module. Thus $\widehat{f}$ is topologically flat.

Proposition 1.3.1 now shows that $\widehat{f}$ is an effective epimorphism. In other words, the following sequence is exact, where $\widehat{X}$ is written for $\widehat{X}/\widehat{S}$:

[59] The editors expanded the argument that follows and then used the material added in 1.2.6.

$$\widehat{X} \times_{\widehat{Y}} \widehat{X} \underset{\widehat{\mathrm{pr}}_2}{\overset{\widehat{\mathrm{pr}}_1}{\rightrightarrows}} \widehat{X} \xrightarrow{\widehat{f}} \widehat{Y}.$$

Moreover, 1.2.6 gives a natural isomorphism, compatible in particular with the projections onto $\widehat{X}$:

$$\widehat{X \times_Y X} \simeq \widehat{X} \times_{\widehat{Y}} \widehat{X}.$$

Consequently, the completed sequence displayed in the proposition is exact.

1.3.5

Let k be a pseudocompact ring. A formal variety X over k is said to be *topologically flat* if its affine algebra A is a *projective* pseudocompact k-module, that is, if the structure morphism

$$X \longrightarrow \mathrm{Spf}(k)$$

is topologically flat.

Let us first note[60] that 0.2.2 and 0.3.6 imply the following result, analogous to VII$_\mathrm{A}$, 3.1.1.

Lemma 1.3.5.A. *Suppose that k is Artinian. The functors*

$$A \longmapsto A^\dagger = \mathrm{Hom}_c(A, k), \qquad C \longmapsto C^* = \mathrm{Hom}_k(C, k)$$

define an anti-equivalence between the category of topologically flat profinite k-algebras and the category of flat k-coalgebras.

Indeed, if A is a topologically flat profinite k-algebra, then, by 0.3.6, there is an isomorphism of k-modules

$$A^\dagger \otimes_k A^\dagger \xrightarrow{\sim} (A \widehat{\otimes} A)^\dagger.$$

Thus the multiplication

$$A \widehat{\otimes} A \longrightarrow A$$

induces by duality a k-coalgebra structure on $A^\dagger$. The rest then follows from Proposition 0.2.2.

Let us return to the case of an arbitrary pseudocompact ring k.

Definition 1.3.5.B. To every formal k-variety X whose affine algebra A is a projective pseudocompact k-module, one associates a flat $\mathbf{O}_k$-coalgebra $\mathbf{H}(X)$, defined as follows.

Denote by $\mathbf{H}(X)$ the $\mathbf{O}_k$-module $\mathbf{V}_k^f(A)$, the "dual of A". It is a flat $\mathbf{O}_k$-module, since the pseudocompact k-module underlying A is projective (cf. 1.2.3.C). Moreover, by 0.3.6, there is an isomorphism

$$\mathbf{V}_k^f(A \widehat{\otimes} A) \simeq \mathbf{V}_k^f(A) \otimes \mathbf{V}_k^f(A).$$

Consequently, the multiplication of A induces by transposition a diagonal morphism

$$\mathbf{H}(X) = \mathbf{V}_k^f(A) \longrightarrow \mathbf{V}_k^f(A \widehat{\otimes} A) = \mathbf{H}(X) \otimes \mathbf{H}(X),$$

which makes $\mathbf{H}(X)$ a flat $\mathbf{O}_k$-coalgebra. We shall say that $\mathbf{H}(X)$ is the *coalgebra of X over* $\mathbf{O}_k$.

[60] The editors added the following lemma, used implicitly in the original; they also introduced the numbering 1.3.5.A–1.3.5.D for later references.

Definition 1.3.5.C. Conversely, to every $\mathbf{O}_k$-coalgebra $\mathbf{C}$ one can associate a k-functor (cf. 1.2.1) $\mathrm{Spf}^*(\mathbf{C})$, defined as follows. For every object B of $\mathrm{Alf}\,/k$, set, with the notation of VII$_\mathrm{A}$ 3.1,

$$\begin{aligned}\mathrm{Spf}^*(\mathbf{C})(B) &= \mathrm{Hom}_{B\text{-coalg.}}\big(B, \mathbf{C}(B)\big)\\ &= \{x \in \mathbf{C}(B) \mid \varepsilon_{\mathbf{C}(B)}(x) = 1 \text{ and } \Delta_{\mathbf{C}(B)}(x) = x \otimes x\}.\end{aligned} \tag{1}$$

Suppose in addition[61] that the $\mathbf{O}_k$-module underlying $\mathbf{C}$ is admissible (cf. 1.2.1), and put

$$A = \Gamma^*(\mathbf{C}) = \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^*. \tag{2}$$

The algebra structures on the modules $\mathbf{C}(k/\mathfrak{l})^*$ endow A with the structure of a profinite k-algebra. For every object B of $\mathrm{Alf}\,/k$, one has

$$\begin{aligned}\mathrm{Hom}_{\mathrm{Vaf}\,/k}\big(\mathrm{Spf}(B), \mathrm{Spf}(A)\big) &= \mathrm{Hom}_{\mathrm{Alp}\,/k}(A, B)\\ &= \mathrm{Hom}_{\mathrm{Alp}\,/B}\big(A\widehat{\otimes}B, B\big).\end{aligned} \tag{3}$$

Suppose finally that $\mathbf{C}$ is a flat $\mathbf{O}_k$-module. Then, by 1.2.3.E, $A = \Gamma^*(\mathbf{C})$ is a projective pseudocompact k-module. Moreover, we saw in the proof of *loc. cit.* that if $\mathfrak{l}_0$ is an open ideal of k contained in the kernel of $k \to B$, there are isomorphisms

$$\begin{aligned}A\widehat{\otimes}B &= \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^*\widehat{\otimes}_k B\\ &\simeq \mathrm{Hom}_{k/\mathfrak{l}_0}\big(\mathbf{C}(k/\mathfrak{l}_0), B\big)\\ &\simeq \mathrm{Hom}_B\big(\mathbf{C}(B), B\big).\end{aligned} \tag{4}$$

We shall denote the term on the right by $\mathbf{C}(B)^*$. Finally, Lemma 1.3.5.A, applied to the Artinian ring B, gives a natural isomorphism

$$\mathrm{Hom}_{B\text{-coalg.}}\big(B, \mathbf{C}(B)\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Alp}\,/B}\big(\mathbf{C}(B)^*, B\big). \tag{5}$$

Combining (1), (5), (4), (3), and (2), we obtain, when $\mathbf{C}$ is a flat $\mathbf{O}_k$-coalgebra, an isomorphism of functors

$$\mathrm{Spf}^*(\mathbf{C}) \simeq \mathrm{Spf}(A) = \mathrm{Spf}\big(\Gamma^*(\mathbf{C})\big). \tag{$\star$}$$

Consequently, if $\mathcal{A}(X)$ denotes the affine algebra of a formal k-variety X, then, taking 1.2.3.E into account, we obtain the following.

Proposition 1.3.5.D.

(i) *The functors*

$$X \longmapsto \mathbf{H}(X) = \mathbf{V}_k^f\big(\mathcal{A}(X)\big), \qquad \mathbf{C} \longmapsto \mathrm{Spf}^*(\mathbf{C}) = \mathrm{Spf}\big(\Gamma^*(\mathbf{C})\big)$$

define an equivalence between the category of topologically flat formal k-varieties and the category of flat $\mathbf{O}_k$-coalgebras.

(ii) *Moreover, this equivalence "commutes with base change": if $k \to K$ is a morphism of pseudocompact rings, then $X\widehat{\otimes}_k K$ corresponds to $\mathbf{H}(X) \otimes_k K$.*

[61] The editors supplied the details that follow.

1.3.6

In the remainder of this Exposé, we shall several times define a topologically flat formal k-variety X by exhibiting the coalgebra $\mathbf{H}(X)$. We shall then have to interpret, by means of $\mathbf{H}(X)$, certain geometric properties of X. Here is one example of this situation: suppose that a section σ of the structure morphism $X \to \mathrm{Spf}(k)$ is given, and ask under what condition σ induces an isomorphism of the underlying topological spaces.[62]

To begin, suppose that k is Artinian. Let (H, Δ, ε) be a flat k-coalgebra, put $H^+ = \ker(\varepsilon)$, and let $A = H^*$ be the profinite k-algebra dual to H. Suppose that a morphism of k-coalgebras $k \to H$ is given; that is, an element $\phi \in H$ such that

$$\varepsilon(\phi) = 1, \qquad \Delta(\phi) = \phi \otimes \phi.$$

On the one hand, ϕ defines a continuous algebra homomorphism $\Phi : A \to k$, and hence a section

$$\sigma : \mathrm{Spf}(k) \longrightarrow \mathrm{Spf}(A)$$

of the projection $\mathrm{Spf}(A) \to \mathrm{Spf}(k)$.

On the other hand, define k-submodules of H by setting $H_0 = k\phi$ and, for $n \geq 1$,

$$H_n = \{x \in H \mid \Delta(x) - x \otimes \phi \in H_{n-1} \otimes H^+\}.$$

This formula also applies when $n = 0$ if $H_{-1} = 0$. Induction on n shows that $H_{n-1} \subset H_n$. We call

$$H_0 \subset H_1 \subset \cdots$$

the *filtration of H defined by ϕ.*

Remark. Since

$$\Delta(H_n) \subset H_n \otimes H_0 \oplus H_{n-1} \otimes H^+,$$

we have $\Delta(H_n) \subset H_n \otimes H$. Since Δ is cocommutative—that is, $\tau \circ \Delta = \Delta$, where $\tau(a \otimes b) = b \otimes a$—we likewise have $\Delta(H_n) \subset H \otimes H_n$. When H/H_n is flat over k, it follows that H_n is a subcoalgebra of H (see also 1.3.6.A (iii) below). In general, however, the map

$$\Delta : H_n \longrightarrow H_n \otimes H$$

does not factor through $H_n \otimes H_n$.[63]

Lemma 1.3.6.A. *Let k be an Artinian ring, let H be a flat k-coalgebra, and let $A = H^*$ be its dual profinite k-algebra. Let $\phi \in H$ satisfy $\varepsilon(\phi) = 1$ and $\Delta(\phi) = \phi \otimes \phi$. Let $\Phi : A \to k$ be the continuous algebra homomorphism, let $\sigma : \mathrm{Spf}(k) \to \mathrm{Spf}(A)$ be the section of $\mathrm{Spf}(A) \to \mathrm{Spf}(k)$, and let (H_n) be the filtration of H defined by ϕ. Put $I = \ker \Phi$.*

(i) *For every $n \geq 1$, H_{n-1} is the orthogonal in H of the closure $\overline{I^n}$ of I^n.*

(ii) *Consequently, σ induces a bijection of the underlying sets if and only if*

$$H = \bigcup_n H_n.$$

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[62] In Lemma 1.3.6.A below, the editors supplied detailed proofs of parts (i) and (ii) and added part (iii).

[63] For example, let k_0 be a field, let $k = k_0[T]/(T^n)$, where $n \geq 4$, and let $H = k\phi \oplus kx$, with $\varepsilon(x) = 0$ and

$$\Delta(x) = x \otimes \phi + \phi \otimes x + t\, x \otimes x,$$

where t is the image of T in k. Then $H_i = k\phi \oplus t^{\,n-i}x$ for $i = 0, \ldots, n$, but for $2 \leq i \leq n-2$, the element $\Delta(t^{\,n-i}x)$ does not belong to the image of $H_i \otimes H_i$ in $H \otimes H$.

[64] In this case the coalgebra H is said to be *connected*; cf. Addition 2.9.

(iii) *If, in addition, every H/H_n is flat over k, then for every $n \geq 0$,*

$$\Delta(H_n) \subset \sum_{i=0}^{n} H_i \otimes H_{n-i}. \qquad (*)$$

In particular, every H_n is then a subcoalgebra of H.

Proof. For every $x \in H$, the map

$$A \longrightarrow k, \qquad f \longmapsto f(x),$$

is continuous. Thus, if I^n is annihilated by x, so is its closure $\overline{I^n}$. For $n \geq 1$, put

$$(\overline{I^n})^{\perp} = \{x \in H \mid f(x) = 0 \text{ for every } f \in \overline{I^n}\}.$$

Suppose first that

$$\sigma : \mathfrak{m} \longmapsto \Phi^{-1}(\mathfrak{m})$$

is a bijection from $\mathrm{Spf}(k)$ onto $\mathrm{Spf}(A)$. Since I is contained in the intersection of the ideals $\Phi^{-1}(\mathfrak{m})$, it follows from 0.1.2 that the sequence of ideals $(\overline{I^n})$ tends to zero. Let $x \in H$. The submodule

$$J(x) = \{f \in A \mid f(x) = 0\}$$

is open and closed in A; hence it contains $\overline{I^n}$ for all sufficiently large n. In other words, $x \in (\overline{I^n})^{\perp}$ for all sufficiently large n.

Conversely, suppose that

$$H = \bigcup_n (\overline{I^n})^{\perp}.$$

Let $\mathfrak{p}$ be an open prime ideal of A. By the definition of the topology on A (0.2.2), $\mathfrak{p}$ contains an open k-submodule of the form

$$\mathscr{V}(x_1, \ldots, x_s) = \{f \in A \mid f(x_1) = \cdots = f(x_s) = 0\}.$$

By hypothesis, there is an integer n such that

$$x_1, \ldots, x_s \in (\overline{I^n})^{\perp};$$

therefore $\overline{I^n} \subset \mathfrak{p}$. Moreover, since k is Artinian,

$$\mathrm{Spf}(k) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_r\}$$

is finite, and there is an integer $t \geq 1$ such that

$$\left(\prod_i \mathfrak{m}_i\right)^t = 0.$$

It follows that

$$\prod_i \Phi^{-1}(\mathfrak{m}_i)^t \subset I.$$

Thus $\mathfrak{p}$ contains the product of the ideals $\Phi^{-1}(\mathfrak{m}_i)^{tn}$. Since $\mathfrak{p}$ is prime, it contains, and hence equals, one of the $\Phi^{-1}(\mathfrak{m}_i)$. We have therefore proved that [65]

$$\sigma \text{ is a bijection} \quad \Longleftrightarrow \quad H = \bigcup_n (\overline{I^n})^{\perp}. \qquad (65)$$

We also have $H = k\phi \oplus H^+$; let $\pi : H \to H^+$ be the projection with kernel $k\phi$. For $n \geq 0$, let

$$\Delta^n : H \longrightarrow H^{\otimes(n+1)}$$

be the iterated comultiplication, let $\overline{\Delta}^n$ be the composite of Δ^n with the projection

$$\pi^{\otimes(n+1)} : H^{\otimes(n+1)} \longrightarrow (H^+)^{\otimes(n+1)},$$

and put

$$H'_n = \ker(\overline{\Delta}^n) = \left\{ x \in H \;\middle|\; \Delta^n(x) \in \sum_{i=0}^{n} H^{\otimes(n-i)} \otimes H_0 \otimes H^{\otimes i} \right\}.$$

We set $\overline{\Delta}^0 = \mathrm{id}_H$, so that $H'_0 = H_0$. A straightforward induction on n gives

$$H_n \subset H'_n \subset (\overline{I^{n+1}})^{\perp}. \tag{*}$$

Up to this point we have not used the hypothesis that H is flat over k. Assume now that H is flat, and hence projective, over k. Then $A^\dagger = H$ by 0.2.2. We shall prove that

$$H_n = (\overline{I^{n+1}})^{\perp}.$$

This is clear for $n = 0$. Suppose it has been proved for $n < r$. Then H_{r-1} is the kernel of the homomorphism

$$H \longrightarrow (\overline{I^r})^\dagger.$$

Since H^+ is flat, the homomorphism

$$(H/H_{r-1}) \otimes H^+ \longrightarrow (\overline{I^r})^\dagger \otimes H^+$$

is injective. On the other hand, the hypothesis implies that I is a projective pseudocompact k-module, since it is a direct factor of $A = H^*$. Thus, by 0.3.6,

$$\left(\overline{I^r}\widehat{\otimes} I\right)^\dagger \simeq (\overline{I^r})^\dagger \otimes I^\dagger = (\overline{I^r})^\dagger \otimes H^+.$$

The exact sequence

$$\overline{I^r}\widehat{\otimes} I \longrightarrow A \longrightarrow A/\overline{I^{r+1}} \longrightarrow 0$$

therefore gives, by duality, the exact sequence

$$(\overline{I^r})^\dagger \otimes H^+ \xleftarrow{\delta} H \longleftarrow (A/\overline{I^{r+1}})^\dagger \longleftarrow 0, \tag{1}$$

where δ is obtained by composing Δ_H with the projection

$$H \otimes H \longrightarrow H \otimes H^+ \longrightarrow (H/H_{r-1}) \otimes H^+ \hookrightarrow (\overline{I^r})^\dagger \otimes H^+. \tag{2}$$

For every $u \in H$, the projection of $\Delta(u)$ onto $H \otimes H^+$ is $\Delta(u) - u \otimes \phi$. Equations (1) and (2) show that if

$$u \in (A/\overline{I^{r+1}})^\dagger = (\overline{I^{r+1}})^{\perp},$$

[65] This remains valid without assuming that H is cocommutative, with k still a commutative Artinian ring. In that case a basis of neighborhoods of zero in $A = H^*$ is given by the two-sided ideals J such that A/J has finite length over k, and the preceding proof shows that $H = \bigcup_n (\overline{I^n})^{\perp}$ if and only if the $\Phi^{-1}(\mathfrak{m}_i)$ are the only open prime ideals of A.

then $\Delta(u) - u \otimes \phi$ belongs to the kernel of

$$H \otimes H^+ \longrightarrow (H/H_{r-1}) \otimes H^+,$$

namely $H_{r-1} \otimes H^+$. Thus $u \in H_r$. This completes the proof of (i) and (ii), and also proves that

$$H_n = \ker(\overline{\Delta}^n).$$

We now prove (iii). For $i \geq 0$, put

$$H_i^+ = H_i \cap H^+.$$

Let $n \geq 1$. For every $x \in H_n$, the element $\overline{x} = x - \varepsilon(x)\phi$ belongs to H_n^+, and

$$\Delta(x) = \varepsilon(x)\phi \otimes \phi + \overline{x} \otimes \phi + \phi \otimes \overline{x} + \overline{\Delta}(\overline{x}).$$

It is therefore enough to prove that

$$\overline{\Delta}(H_n^+) \subset \sum_{i=1}^{n-1} H_i^+ \otimes H_{n-i}^+.$$

For $i = 0, \ldots, n-1$, the map $\overline{\Delta}^n$ factors as follows:

$$\begin{array}{ccccc} H^+ & \xrightarrow{\overline{\Delta}} & H^+ \otimes H^+ & \xrightarrow{\overline{\Delta}^i \otimes \overline{\Delta}^{n-i-1}} & (H^+)^{\otimes(i+1)} \otimes (H^+)^{\otimes(n-i)} \\ & & \downarrow & & \uparrow g \\ & & \dfrac{H^+}{H_i^+} \otimes \dfrac{H^+}{H_{n-i-1}^+} & \xrightarrow{f} & \dfrac{H^+}{H_i^+} \otimes (H^+)^{\otimes(n-i)}. \end{array}$$

Moreover, since H^+/H_i^+ and $(H^+)^{\otimes(n-i)}$ are flat, the maps f and g in this diagram are injective. It follows that

$$\overline{\Delta}(H_n^+) \subset H_i^+ \otimes H^+ + H^+ \otimes H_{n-i-1}^+$$

for every $i = 0, \ldots, n-1$. Part (iii) now follows from the next lemma, applied with $M = H^+$, $E_0 = 0$, and $E_i = H_{i-1}^+$ for $i \geq 1$.

Lemma 1.3.6.B. *Let k be a ring, and let*

$$0 = E_0 \subset E_1 \subset \cdots \subset E_n \subset M$$

be k-modules. Suppose that M/E_i is flat for every i. Then

$$\bigcap_{i=0}^{n} (E_i \otimes M + M \otimes E_{n-i}) = \sum_{i=1}^{n} E_i \otimes E_{n-i+1}.$$

Proof. Denote the left-hand and right-hand sides by K and S, respectively. Clearly $S \subset K$; we prove the reverse inclusion. For $i = 0, \ldots, n$, put

$$K_i = K \cap (E_i \otimes M).$$

For $i = 1, \ldots, n$, both M/E_{n-i+1} and E_i/E_{i-1} are flat. Consequently, the homomorphism τ_i below is injective, and the composite

$$(E_i/E_{i-1}) \otimes M \longrightarrow (E_i/E_{i-1}) \otimes (M/E_{n-i+1}) \xhookrightarrow{\tau_i} (M/E_{i-1}) \otimes (M/E_{n-i+1})$$

has kernel

$$(E_i/E_{i-1}) \otimes E_{n-i+1}.$$

The image of K_i in $(E_i \otimes M)/(E_{i-1} \otimes M)$ is both contained in and contains this kernel. Hence

$$K_i = K_{i-1} + E_i \otimes E_{n-i+1},$$

which proves the lemma.

To conclude this paragraph, return to an arbitrary pseudocompact ring k.[66] Let $(\mathbf{H}, \Delta, \varepsilon)$ be a flat $\mathbf{O}_k$-coalgebra, put $\mathbf{H}^+ = \ker(\varepsilon)$, and let

$$A = \Gamma^*(\mathbf{H})$$

be the dual profinite k-algebra. Put $X = \mathrm{Spf}(A)$, so that $\mathbf{H} = \mathbf{H}(X)$ (cf. 1.3.5). Suppose that a morphism of $\mathbf{O}_k$-coalgebras

$$\phi : \mathbf{O}_k \longrightarrow \mathbf{H}$$

is given. It defines a continuous homomorphism of k-algebras $A \to k$, and hence a section σ of the structure morphism $X \to \mathrm{Spf}(k)$.

For every object B of Alf $/k$, put

$$\mathbf{H}_0(B) = \phi(B) = B\phi_B, \qquad \phi_B = \phi(1_B) \in \mathbf{H}(B).$$

For $n \geq 1$, define $\mathbf{O}_k$-submodules $\mathbf{H}_n$ of $\mathbf{H}$ by

$$\mathbf{H}_n(B) = \left\{ u \in \mathbf{H}(B) \;\middle|\; \Delta(u) - u \otimes \phi_B \in \mathbf{H}_{n-1}(B) \otimes \mathbf{H}^+(B) \right\}.$$

This gives a filtration

$$\mathbf{H}_0 \subset \mathbf{H}_1 \subset \cdots$$

of $\mathbf{H}(X)$. The preceding result gives:

Proposition. *The section σ induces an isomorphism of the underlying topological spaces if and only if*

$$\mathbf{H}(X) = \bigcup_n \mathbf{H}_n.$$

1.4

Theorem 1.4. *Let k be a pseudocompact ring and let*

$$d_0, d_1 : X_1 \rightrightarrows X$$

be an equivalence pair in Vaf $/k$ (cf. Exposé V, § 2.b) *such that d_1 is topologically flat.*

(i) *The canonical projection from X onto $X/X_1 = \mathrm{Coker}(d_0, d_1)$ is surjective and topologically flat, and the morphism*

$$X_1 \longrightarrow X \times_{X/X_1} X$$

with components d_0 and d_1 is an isomorphism.

(ii) *If X is topologically flat over k, then so is X/X_1.*

First note that (ii) follows from (i) by 1.3.3 (ii). The proof of (i) occupies 1.4.1, 1.4.2, and 1.4.3.

[66] The editors added what follows; the original treated only the case in which k is Artinian.

1.4.1

We first show that we may reduce to the case in which X has only one point. Since we are dealing with an equivalence pair, one sees, as in Exposé V, § 3.e), that an equivalence relation is defined on the underlying set of X by declaring two points x, y to be equivalent if there exists a point $z \in X_1$ such that

$$d_0(z) = y \qquad \text{and} \qquad d_1(z) = x.$$

We may plainly suppose, without loss of generality, that X contains a single equivalence class for this relation; in other words, that X/X_1 has only one point (see the construction of X/X_1 given in 1.2).

In this case, let x be any point of X, and let U be the formal variety having x as its only point and the same local ring as X at x. As in Exposé V, § 6, one then sees that the equivalence relation induced by (d_0, d_1) on U still satisfies the hypotheses of the theorem and that it is enough to prove the theorem for this latter equivalence relation (U is a "quasi-section").

Let us briefly recall the principle of the proof given in Exposé V, § 6. Put

$$V = d_0^{-1}(U) = U \times_{i,d_0} X_1,$$

where i is the inclusion of U into X, and let u and v be the morphisms with source V induced respectively by d_0 and d_1:

$$X \xleftarrow{v} V \xrightarrow{u} U.$$

It is clear that u and v are topologically flat and that u is surjective; since X contains a single equivalence class, v is surjective. If (v_0, v_1) is the inverse image of the equivalence pair (d_0, d_1) under v (cf. V, 3.a)), it follows from V, 3.c) and 3.d) that X/X_1 and the quotient of U by the equivalence relation induced by (d_0, d_1) both identify with $\mathrm{Coker}(v_0, v_1)$. As in the proof of V, 6.1, one then sees that if the conclusion of Theorem 1.4 holds for U, it also holds for X.

1.4.2

We are thus reduced to the case in which X has only one point.[67] Consider the following commutative diagram (cf. V, § 1, (0,1,2)):

$$\begin{array}{ccccc}
X_2 & \underset{d_0'}{\overset{d_1'}{\rightrightarrows}} & X_1 & \xrightarrow{d_0} & X \\
\downarrow{\scriptstyle d_2'} & & \downarrow{\scriptstyle d_1} & & \downarrow \\
X_1 & \underset{d_0}{\overset{d_1}{\rightrightarrows}} & X & \longrightarrow & X/X_1
\end{array}$$

Here

$$X_2 = X_1 \times_{d_1,d_0} X_1,$$

and d_0', d_1', d_2' are respectively the morphisms[68]

$$(x, y, z) \longmapsto (x, y), \qquad (x, y, z) \longmapsto (x, z), \qquad (x, y, z) \longmapsto (y, z).$$

If B, A, A_1, A_2 denote respectively the affine algebras of $X/X_1, X, X_1, X_2$, the preceding diagram induces the commutative diagram

[67] Editors' note: Thus one may suppose that k is local.

[68] Editors' note: Cf. Exposé V, § 2.b).

$$\begin{array}{ccccc} A_2 & \underset{j_0}{\overset{j_1}{\leftleftarrows}} & A_1 & \xleftarrow{i_0} & A \\ \uparrow{\scriptstyle j_2} & & \uparrow{\scriptstyle i_1} & & \uparrow{\scriptstyle i} \\ A_1 & \underset{i_0}{\overset{i_1}{\leftleftarrows}} & A & \xleftarrow{i} & B \end{array}$$

in which both rows are exact and the squares determined by (i_0, j_0) and (i_1, j_1) are cocartesian. Since the morphism

$$X_1 \longrightarrow X \times X$$

with components d_0 and d_1 is a monomorphism by hypothesis, the morphism

$$A \widehat{\otimes}_k A \longrightarrow A_1$$

with components i_0 and i_1 is surjective by Proposition 1.3.

This means that i_1 makes A_1 a pseudocompact A-module, topologically free by hypothesis, generated by $i_0(A)$. Since A is local, Lemma 1.4.3 below gives a topologically free k-module V and a morphism of pseudocompact k-modules $f : V \to A$ such that the morphism

$$\alpha_1 : A \widehat{\otimes}_k V \longrightarrow A_1, \qquad a \widehat{\otimes} v \longmapsto i_1(a) \cdot i_0\big(f(v)\big)$$

is an isomorphism. Let

$$\alpha : B \widehat{\otimes}_k V \longrightarrow A \qquad \text{and} \qquad \alpha_2 : A_1 \widehat{\otimes}_k V \longrightarrow A_2$$

be the morphisms

$$b \widehat{\otimes} v \longmapsto i(b) \cdot f(v) \qquad \text{and} \qquad a_1 \widehat{\otimes} v \longmapsto j_2(a_1) \cdot j_0 i_0\big(f(v)\big).$$

In the following commutative diagram, both rows are exact and the two left-hand squares are cocartesian. Since α_1 is an isomorphism, the same is true of α_2, and hence of α.

$$\begin{array}{ccccc} A_2 & \underset{j_0}{\overset{j_1}{\leftleftarrows}} & A_1 & \xleftarrow{i_0} & A \\ \uparrow{\scriptstyle \alpha_2} & & \uparrow{\scriptstyle \alpha_1} & & \uparrow{\scriptstyle \alpha} \\ A_1 \widehat{\otimes}_k V & \underset{i_0 \widehat{\otimes} V}{\overset{i_1 \widehat{\otimes} V}{\leftleftarrows}} & A \widehat{\otimes}_k V & \xleftarrow{i \widehat{\otimes} V} & B \widehat{\otimes}_k V\,. \end{array}$$

This shows, on the one hand, that A is topologically free over B, with pseudobasis $f(V)$ (cf. 0.2.1), and that a pseudobasis of A_1 over A, where A_1 is regarded as an A-module through i_1, is obtained by taking the image under i_0 of $f(V)$. It follows that the morphism

$$A \widehat{\otimes}_B A \longrightarrow A_1$$

with components i_1 and i_0 is an isomorphism:

$$A \widehat{\otimes}_B A \simeq A \widehat{\otimes}_B B \widehat{\otimes}_k V \simeq A \widehat{\otimes}_k V \simeq A_1.$$

This proves Theorem 1.4, modulo the following lemma.

1.4.3

Lemma 1.4.3. *Let k be a pseudocompact ring, let A be a local profinite k-algebra, let A_1 be a topologically free A-module, and let*

$$i_0 : M \longrightarrow A_1$$

be a morphism of pseudocompact k-modules. Suppose that the map

$$A\widehat{\otimes}_k M \longrightarrow A_1, \qquad a\widehat{\otimes}m \longmapsto a \cdot i_0(m),$$

is surjective. Then there exist a topologically free k-module V and a morphism of pseudocompact k-modules

$$f : V \longrightarrow M$$

such that the map

$$A\widehat{\otimes}_k V \longrightarrow A_1, \qquad a\widehat{\otimes}v \longmapsto a \cdot i_0\big(f(v)\big),$$

is bijective.

Since every pseudocompact k-module is a quotient of a topologically free k-module (cf. Editors' note (27)), we may suppose, without loss of generality, that M is topologically free. Thus write

$$M = \prod_{i\in I} M_i,$$

where each M_i is a copy of k. In this case,

$$A\widehat{\otimes}_k M = \prod_{i\in I} A\widehat{\otimes}_k M_i.$$

Since the map

$$a\widehat{\otimes}m \longmapsto a \cdot i_0(m)$$

is surjective and A_1 is projective, its kernel is a direct factor of $A\widehat{\otimes}_k M$. As A is local, the exchange theorem (0.3.4) shows that this kernel has as a complement a partial product

$$\prod_{i\in J} A\widehat{\otimes}_k M_i,$$

where J is a subset of I. One may therefore take

$$V = \prod_{i\in J} M_i.$$

1.5

Let k be a pseudocompact ring.

Definition. We shall say that a family of morphisms

$$f_i : X_i \longrightarrow X$$

in Vaf $/k$ is a *surjective topologically flat family* if the morphism

$$\coprod_i X_i \longrightarrow X$$

induced by the f_i is surjective and topologically flat. This means that every f_i is topologically flat and that every point of X lies in the image of at least one X_i.

It follows from 1.3.3 that surjective topologically flat families define a pretopology on Vaf $/k$ (IV, 4.2.5). The corresponding topology will be called the *flat topology* on Vaf $/k$.

By IV, 4.3.5, a functor

$$F : (\mathrm{Vaf}\,/k)^{\mathrm{op}} \longrightarrow (\mathbf{Sets})$$

is a sheaf for the flat topology if and only if F carries every direct sum to a direct product and the sequence

$$F(Y) \xrightarrow{F(f)} F(X) \underset{F(\mathrm{pr}_2)}{\overset{F(\mathrm{pr}_1)}{\rightrightarrows}} F(X \times_Y X)$$

is exact for every surjective topologically flat morphism $f : X \to Y$.

By IV, 4.5,[69] Proposition 1.3.1 implies that the flat topology is coarser than the canonical topology. In other words, for every object X of Vaf $/k$, the functor

$$h_X : T \longmapsto \mathrm{Hom}_{\mathrm{Vaf}\,/k}(T, X)$$

is a sheaf for the flat topology. In what follows, as usual (cf. Exposé I), we identify X with h_X.

By IV, 4.6.5, Theorem 1.4 may be reformulated as follows.

Theorem. *Let k be a pseudocompact ring, let $d_0, d_1 : X_1 \rightrightarrows X$ be an equivalence pair in* Vaf $/k$*, and let X/X_1 be the quotient formal variety* (that is, $\mathrm{Coker}(d_0, d_1)$, cf. 1.2)*. If d_1 is topologically flat, then X/X_1 represents the quotient sheaf for the flat topology.*

1.6

To complete these generalities on formal varieties, it remains to define briefly the formal varieties that are étale over k. [70]

Recollections 1.6.A. *(i)* Recall first (cf. EGA 0_{IV}, 19.10.2) that a topological k-algebra A is said to be *formally étale over k*—for the given topologies on k and A, respectively for the discrete topologies—if, for every discrete topological k-algebra C (not necessarily Artinian) and every nilpotent ideal J of C, every continuous morphism of k-algebras

$$A \longrightarrow C/J$$

lifts uniquely to a continuous morphism $A \to C$, where A is endowed with the given topology, respectively with the discrete topology.

This property is plainly preserved under base change. Thus, for every morphism $k \to k'$ of pseudocompact rings,

$$A\widehat{\otimes}_k k'$$

is formally étale over k'. It is also enough to verify the lifting condition for square-zero ideals J; cf. EGA IV_4, 17.1.2 (ii).

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[69] The editors modified the original in what follows. In particular, they replaced the statement: "if $d_0, d_1 : X_1 \rightrightarrows X$ is an equivalence relation such that d_1 is topologically flat, formation of the quotient commutes with h," by the theorem below.

[70] The original continued as follows: "A formal variety X over k is said to be étale if the diagonal morphism $\Delta_X : X \to X \times X$ is a local isomorphism; that is, if Δ_X induces an isomorphism from $\mathcal{O}_{X\times X, \Delta_X(x)}$ onto $\mathcal{O}_{X,x}$ for every point x of X. It follows readily from SGA I that this formulation is equivalent to each of the following two: the formal variety X is topologically flat and, for every point $x \in X$ projecting to $s \in \mathrm{Spf}(k)$, $\mathcal{O}_{X,x}\widehat{\otimes}_k \kappa(s)$ is a finite separable extension of the residue field $\kappa(s)$; or, if A denotes the affine algebra of X, the local components (0.1) of $A\widehat{\otimes}_k(k/\mathfrak{l})$ are finite étale algebras over $k/\mathfrak{l}$, for every open ideal $\mathfrak{l}$ of k." In what follows, the editors corrected the omission of the flatness hypothesis from the first condition above and supplied details for the equivalence of the conditions.

The algebra A is said to be *étale over* k if it is formally étale over k for the discrete topologies and is, in addition, a k-algebra of finite presentation; cf. EGA IV$_4$, 17.3.2 (ii). In what follows, when k is a pseudocompact ring and A a profinite k-algebra, "formally étale" refers to the given topologies unless the contrary is stated.

(ii) Recall also that if $F \in k[T]$ is a separable monic polynomial of degree $d \geq 1$, meaning that the ideal generated by F and its derivative F' is all of $k[T]$, then

$$B = k[T]/(F),$$

which is free of rank d over k and is endowed with the product topology, is formally étale over k.

Indeed, let C be a discrete k-algebra, so that the kernel $\mathfrak{l}$ of $k \to C$ is an open ideal of k; let J be a square-zero ideal of C; and let $\phi : B \to C/J$ be a continuous morphism of k-algebras. Since B is a free k-module of finite rank, $\mathfrak{l}B$ is an open ideal of B, and hence every lift $\Phi : B \to C$ of ϕ is automatically continuous.

Let t be the image of T in B, and choose any lift $u_0 \in C$ of $\phi(t)$. Then $F(u_0) \in J$, since $\phi(t)$ is a root of F. There exist $G, H \in k[T]$ such that

$$GF + HF' = 1.$$

Consequently,

$$H(u_0)F'(u_0) = 1 - G(u_0)F(u_0).$$

The right-hand side is invertible because $F(u_0)$ is square-zero, and therefore $F'(u_0)$ is invertible. We seek $h \in J$ such that $u = u_0 + h$ is a root of F. This is equivalent to

$$0 = F(u) = F(u_0) + F'(u_0)h,$$

whose unique solution is

$$h = -F'(u_0)^{-1}F(u_0) \in J.$$

The same proof, with no continuity assumptions on the morphisms $k \to C$, ϕ, and Φ, also shows that B is an étale k-algebra.

(iii) Finally, if

$$A = A_1 \times \cdots \times A_n$$

is a finite product, then A is formally étale over k if and only if every A_i is.[71] It is enough to prove this for $n = 2$. Let $e = 1_{A_1}$ be the idempotent such that $A_1 = Ae$ and $A_2 = A(1-e)$, and suppose that a continuous morphism

$$\phi : A \longrightarrow C/J$$

is given, where J is square-zero. The polynomial $F = X^2 - X$ is separable, since

$$F' = 2X - 1 \qquad \text{and} \qquad (F')^2 - 4F = 1.$$

Therefore the idempotent $\phi(e)$ of C/J lifts uniquely to an idempotent f of C. Hence

$$C = Cf \oplus C(1-f),$$

and giving a lift Φ is equivalent to giving two morphisms

$$\Phi_1 : A_1 \longrightarrow Cf, \qquad \Phi_2 : A_2 \longrightarrow C(1-f),$$

[71] The editors point out that the proof given in EGA 0_{IV}, 19.3.5 (v), is erroneous.

lifting the respective restrictions of ϕ. The same argument shows that if e is an idempotent of k and $A = Ae$, then A is formally étale over k if and only if it is formally étale over the local component k_e, identified with ke.

Now let X be a formal k-variety with affine algebra A. If x is a point of X, that is, an open maximal ideal $\mathfrak{m}$ of A, and s is its image in $\mathrm{Spf}(k)$, we denote by $k_\mathfrak{m}$, or k_s, the local component of k corresponding to s.

Definition 1.6.B. *The following conditions are equivalent:*

(a) A is formally étale over k.

(b) For every $\mathfrak{m} \in \Upsilon(A)$, $A_\mathfrak{m}$ is formally étale over k, or equivalently over $k_\mathfrak{m}$.

(c) For every open ideal $\mathfrak{l}$ of k,
$$A\widehat{\otimes}_k(k/\mathfrak{l}) = A/A\mathfrak{l}$$
is formally étale over $k/\mathfrak{l}$.

(d) For every $\mathfrak{m} \in \Upsilon(A)$ and every open ideal $\mathfrak{l}$ of $k_\mathfrak{m}$, $A_\mathfrak{m}/A_\mathfrak{m}\mathfrak{l}$ is formally étale over $k_\mathfrak{m}/\mathfrak{l}$.

We say that X is *étale over* k if these conditions hold, and write

$$\mathrm{Vaf}^{\text{ét}}\,/k$$

for the full subcategory of Vaf $/k$ consisting of formal varieties that are étale over k. [72]

Indeed, suppose that
$$\phi : A \longrightarrow C/J$$
is a continuous morphism of k-algebras, where C is a discrete k-algebra and J is square-zero. Then $I = \mathrm{Ker}(\phi)$ is an open ideal of A, so A/I is Artinian. Thus I is contained in only finitely many open maximal ideals
$$\mathfrak{m}_1, \ldots, \mathfrak{m}_r.$$
It therefore contains the product of the components $A_\mathfrak{m}$ for $\mathfrak{m} \notin \{\mathfrak{m}_1, \ldots, \mathfrak{m}_r\}$. This product is $A(1-e)$, where e is the idempotent of A for which
$$Ae = \prod_{i=1}^{r} A_{\mathfrak{m}_i}.$$
Hence $\phi(e) = 1_{C/J}$, and giving a continuous lift of ϕ is equivalent to giving a continuous lift of the induced morphism
$$Ae \simeq A/A(1-e) \longrightarrow C/J.$$

On the other hand, by editors' note (24),
$$A \simeq \varprojlim_{\mathfrak{l}} A/A\mathfrak{l}.$$

The equivalence of the stated conditions follows readily from these observations and the preceding recollections.

Definitions 1.6.C. Put
$$\kappa(k) = \prod_{s \in \mathrm{Spf}(k)} \kappa(s),$$

[72] The category Vaf$^{\text{ét}}$ $/k$ is stable under finite projective limits.

endowed with the product topology. Thus the formal variety $\mathrm{Spf}(\kappa(k))$ is the direct sum of the schemes $\mathrm{Spec}\,\kappa(s)$, for $s \in \mathrm{Spf}(k)$. We write $S_{\kappa(k)}$ for the scheme that is the direct sum of the same $\mathrm{Spec}\,\kappa(s)$.

For every formal variety X over k, put

$$X_\kappa = X\widehat{\otimes}_k\kappa(k).$$

This is the formal variety over $\kappa(k)$ obtained by base change. It has the same points as X, and if $x \in X$ projects to $s \in \mathrm{Spf}(k)$, then

$$\mathcal{O}_{X_\kappa,x} = \mathcal{O}_{X,x}\widehat{\otimes}_k\kappa(s).$$

The base-change functor

$$\mathrm{Vaf}\,/k \longrightarrow \mathrm{Vaf}\,/\kappa(k)$$

sends $\mathrm{Vaf}^{\text{ét}}\,/k$ into $\mathrm{Vaf}^{\text{ét}}\,/\kappa(k)$, by 1.6.A (i).

One then has the following result (cf. SGA 1, I 6.2).

Lemma 1.6.D. *For every*

$$Y \in \mathrm{Ob}(\mathrm{Vaf}^{\text{ét}}\,/k) \qquad \textit{and} \qquad X \in \mathrm{Ob}(\mathrm{Vaf}\,/k),$$

the canonical map

$$\mathrm{Hom}_{\mathrm{Vaf}\,/k}(X,Y) \longrightarrow \mathrm{Hom}_{\mathrm{Vaf}\,/\kappa(k)}(X_\kappa,Y_\kappa)$$

is bijective. In particular, the functor

$$\mathrm{Vaf}^{\text{ét}}\,/k \longrightarrow \mathrm{Vaf}^{\text{ét}}\,/\kappa(k)$$

is fully faithful; below it will be shown to be an equivalence.

Indeed, let A, respectively B, be the affine algebra of X, respectively Y, and let r be the radical of k. Suppose that a morphism

$$B\widehat{\otimes}_k\kappa(k) \longrightarrow A\widehat{\otimes}_k\kappa(k)$$

is given, or equivalently a morphism

$$\phi : B \longrightarrow A/rA.$$

For every open ideal $\mathfrak{l}$ of k, there is an integer $n \geq 1$ such that

$$r^n \subset \mathfrak{l}.$$

Because the multiplication map

$$m^{n-1} : A^n \longrightarrow A$$

is continuous, it follows that

$$(rA)^n \subset \mathfrak{l}A.$$

Thus $rA/\mathfrak{l}A$ is a nilpotent ideal of $A/\mathfrak{l}A$. Consequently, ϕ lifts uniquely to a morphism

$$\phi_\mathfrak{l} : B \longrightarrow A/\mathfrak{l}A.$$

By uniqueness, these morphisms form a projective system and therefore give a continuous morphism

$$\Phi : B \longrightarrow \varprojlim_{\mathfrak{l}} A/\mathfrak{l}A = A.$$

Moreover, Φ is unique: if Φ' is another lift of ϕ, then Φ' and Φ agree modulo $\mathfrak{l}A$ for every $\mathfrak{l}$, and hence are equal.

Proposition 1.6.E. *(a) Let X be a formal variety over k, with affine algebra A. The following conditions are equivalent:*

(i) *X is étale over k.*

(ii) *X is topologically flat over k, and the diagonal morphism*

$$\Delta : X \longrightarrow X \times X$$

is a local isomorphism; that is, for every point x of X, Δ_X induces an isomorphism

$$\mathcal{O}_{X\times X,\Delta(x)} \xrightarrow{\sim} \mathcal{O}_{X,x}.$$

(iii) *For every $x \in X$ projecting to $s \in \mathrm{Spf}(k)$, there is an isomorphism*

$$\mathcal{O}_{X,x} \simeq k_s[T]/(F),$$

where $F \in k_s[T]$ is a separable monic polynomial (cf. 1.6.A (ii)).

(iv) *X is topologically flat over k, and, for every $x \in X$ projecting to $s \in \mathrm{Spf}(k)$,*

$$\mathcal{O}_{X,x}\widehat{\otimes}_k\kappa(s)$$

is a finite separable extension of $\kappa(s)$.

(v) *For every open ideal $\mathfrak{l}$ of k, each local component of*

$$A\widehat{\otimes}_k(k/\mathfrak{l})$$

is a finite étale algebra over the Artinian ring $k/\mathfrak{l}$.

(b) The category $\mathrm{Vaf}^{\text{ét}}/\kappa(k)$ *identifies with the category of étale schemes over $S_{\kappa(k)}$* (cf. 1.6.C)*, and the functor $X \mapsto X_\kappa$ induces an equivalence*

$$\mathrm{Vaf}^{\text{ét}}/k \xrightarrow{\sim} \mathrm{Vaf}^{\text{ét}}/\kappa(k)$$

(cf. SGA 1, I 6.1).

Proof. Let I be the kernel of the multiplication morphism

$$m : A\widehat{\otimes}_k A \longrightarrow A.$$

Suppose first that X is étale over k, so that A is formally étale over k. By EGA 0_{IV}, 20.7.4, the separated completion of I/I^2, for the quotient topology induced from I, is zero; hence

$$I = I^2.$$

For every $x \in X$, the ideal I is contained in the maximal ideal $\mathfrak{m}_{\Delta(x)}$ of $A\widehat{\otimes}_k A$. Thus its localization $I_{\mathfrak{m}_{\Delta(x)}}$ lies in the maximal ideal of $\mathcal{O}_{X\times X,\Delta(x)}$. Nakayama's lemma 0.3 therefore gives

$$I_{\mathfrak{m}_{\Delta(x)}} = 0.$$

Consequently, $\Delta : X \to X \times X$ is a local isomorphism.

Suppose next that Δ is a local isomorphism and that k is a field κ. We show that every $\mathcal{O}_{X,x}$ is a finite-dimensional étale κ-algebra. Replacing X by $\mathrm{Spf}(\mathcal{O}_{X,x})$, we may assume that

$$A = \mathcal{O}_{X,x}$$

is local. We proceed as in the proof of EGA IV$_4$, 17.4.1, (b)⇒(d"). Let K be a finite normal extension of κ containing the residue field $\kappa(x)$, and put

$$B = A \otimes_\kappa K = A\widehat{\otimes}_\kappa K, \qquad X_K = \mathrm{Spf}(B) = X\widehat{\otimes}_\kappa K.$$

The two tensor products agree because $[K : \kappa] < \infty$.

The morphism

$$\Delta_K : X_K \longrightarrow X_K \times_K X_K$$

is again a local isomorphism. Hence, for every $y \in X_K$, multiplication induces an isomorphism

$$\left(B_y\widehat{\otimes}_K B_y\right)_{\mathfrak{m}_{\Delta_K(y)}} \xrightarrow{\sim} B_y.$$

The residue field of B_y is K; cf., for example, VI$_A$, 1.1.1, editors' note (11). Moreover,

$$C = B_y\widehat{\otimes}_K B_y$$

is already local. Indeed,

$$\mathfrak{n} = \mathfrak{m}_y\widehat{\otimes}_K B_y + B_y\widehat{\otimes}_K\mathfrak{m}_y$$

consists of topologically nilpotent elements and therefore lies in the radical of C, while

$$C/\mathfrak{n} = K\widehat{\otimes}_K K = K$$

is a field. It follows that multiplication itself is an isomorphism

$$B_y\widehat{\otimes}_K B_y \xrightarrow{\sim} B_y.$$

Choosing a pseudobasis of B_y over K that contains the unit element 1, we deduce that $B_y = K$. Since B is finite over A, the set X_K is finite. Thus

$$B = A \otimes_\kappa K$$

is a product of finitely many copies of K, and A is a finite étale κ-algebra.

We have therefore shown that, over a field κ, every profinite κ-algebra A that is étale over κ is a product of finite separable extensions K_i/κ, endowed with the product topology. Consequently, $\mathrm{Spf}(A)$ is the direct sum of the schemes

$$\mathrm{Spf}(K_i) = \mathrm{Spec}(K_i),$$

and $\mathrm{Vaf}^{\text{ét}}/\kappa$ identifies with the category of étale κ-schemes.

The preceding argument proves (ii)⇒(iv), because $\mathcal{O}_{X,x}\widehat{\otimes}_k\kappa(s)$ is formally étale over $\kappa(s)$, and also proves part (b). Indeed, return to an arbitrary pseudocompact ring k. The category $\mathrm{Vaf}^{\text{ét}}/\kappa(k)$ has just been identified with the category of étale schemes over $S_{\kappa(k)}$. By 1.6.D, to show that

$$\mathrm{Vaf}^{\text{ét}}/k \longrightarrow \mathrm{Vaf}^{\text{ét}}/\kappa(k)$$

is an equivalence, it remains only to prove essential surjectivity.

Let $s \in \mathrm{Spf}(k)$, and let $K/\kappa(s)$ be a finite separable extension. Choose a primitive element $\xi \in K$, let $n = [K : \kappa(s)]$, and choose a monic polynomial

$$F \in k_s[T]$$

of degree n whose image $\overline{F} \in \kappa(s)[T]$ is the minimal polynomial of ξ. Since $\overline{F}'$ is invertible in $\kappa(s)[T]/(\overline{F})$, Nakayama's lemma shows that F' is invertible in $k_s[T]/(F)$. Thus F is separable, and 1.6.A (ii) shows that

$$A = k_s[T]/(F)$$

is an étale k_s-algebra satisfying

$$A\widehat{\otimes}_k\kappa(s) \simeq K.$$

It follows that every local profinite étale k_s-algebra is free of finite rank over k_s, and therefore is topologically free over k. Hence every profinite étale k-algebra is topologically free over k.

Condition (v) implies condition (d) of 1.6.B, and therefore implies (i). We have thus proved that (i), (iii), and (v) are equivalent and imply (ii), which in turn implies (iv).

Finally, let A be a profinite k-algebra satisfying (iv). We show that A is formally étale over k. We may suppose that A and k are local, and write κ for the residue field of k. By hypothesis,

$$K = A\widehat{\otimes}_k\kappa$$

is a finite separable extension of κ, say of degree n. By the preceding argument and Lemma 1.6.D, there is a k-algebra B, free of rank n and formally étale over k, together with a continuous morphism

$$\phi : B \longrightarrow A$$

such that

$$\phi\widehat{\otimes}_k\kappa$$

is an isomorphism. Since A is topologically flat over k, this implies

$$\mathrm{Coker}(\phi)\widehat{\otimes}_k\kappa = 0 \qquad \text{and} \qquad \mathrm{Ker}(\phi)\widehat{\otimes}_k\kappa = 0.$$

Nakayama's lemma 0.3 gives

$$\mathrm{Coker}(\phi) = 0 = \mathrm{Ker}(\phi).$$

Thus ϕ is an isomorphism (cf. the proof of 0.2.B), which completes the proof of 1.6.E.

Let X be a formal variety over k. If $x \in X$ projects to $s \in \mathrm{Spf}(k)$, then the residue field $\kappa(x)$ is a finite extension of $\kappa(s)$. We denote by $\kappa_e(x)$ the separable closure of $\kappa(s)$ in $\kappa(x)$.[73]

Proposition 1.6.F. *(i) The inclusion*

$$\mathrm{Vaf}^{\text{ét}}/k \longrightarrow \mathrm{Vaf}/k$$

has a left adjoint $X \mapsto X_e$. The formal variety X_e has the same points as X. For each $x \in X$ projecting to $s \in \mathrm{Spf}(k)$, choose a primitive element ξ of $\kappa_e(x)/\kappa(s)$, let n be its degree, choose any lift $u \in \mathcal{O}_{X,x}$ of ξ, and let $F \in k[T]$ be a monic polynomial of degree n that annihilates u. Then put

$$\mathcal{O}_{X_e,x} = k[T]/(F).$$

For every $Y \in \mathrm{Ob}(\mathrm{Vaf}^{\text{ét}}/k)$, the canonical maps in the following diagram are bijective:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathrm{Vaf}/k}(X, Y) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathrm{Vaf}/\kappa(k)}(X_\kappa, Y_\kappa) \\ & & \downarrow \wr \\ \mathrm{Hom}_{\mathrm{Vaf}/k}(X_e, Y) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathrm{Vaf}/\kappa(k)}((X_e)_\kappa, Y_\kappa)\,, \end{array}$$

[73] In what follows, the editors used the preceding additions to give details of the construction of the functor $X \mapsto X_e$ and to prove that it commutes with finite products; this is used in 2.5.1.

The vertical arrow is induced by the inclusions

$$\kappa_e(x) \hookrightarrow \kappa(x), \qquad x \in X.$$

In particular, this defines a morphism

$$p : X \longrightarrow X_e,$$

and every morphism of formal k-varieties $\phi : X \to Y$, with Y étale over k, factors uniquely through p.

(ii) The functor $X \mapsto X_e$ commutes with finite products.

Proof. Assertion (i) follows from 1.6.E (b) and 1.6.D. We prove (ii). By the equivalence of categories in 1.6.E (b), we may suppose that k is a field. In that case, if X is a semilocal formal k-variety, so that its affine algebra A is semilocal, the affine algebra of X_e is the largest subalgebra of A that is étale over k; denote it by A_e.

It is therefore enough to show that, if K and L are two finite extensions of k, then

$$K_e \otimes_k L_e = (K \otimes_k L)_e.$$

Let p be the characteristic exponent of k, so that $p = 1$ if $\mathrm{char}(k) = 0$, and $p = \mathrm{char}(k)$ otherwise. For every $x \in K \otimes_k L$, there is an integer $n \geq 1$ such that

$$x^{p^n} \in K_e \otimes_k L_e.$$

Thus $K \otimes_k L$ is radicial over $K_e \otimes_k L_e$, and its largest étale k-subalgebra is precisely $K_e \otimes_k L_e$. Hence

$$K_e \otimes_k L_e = (K \otimes_k L)_e.$$

Keeping this notation, observe that $\mathcal{O}_{X_e,x}$ is not necessarily a subalgebra of $\mathcal{O}_{X,x}$. It is a subalgebra when X is topologically flat over k, by the proposition that follows.

1.6.1

Let Y be an étale formal k-variety and let $f : X \to Y$ be a morphism in Vaf/k. Then the following square is Cartesian, where Γ_f is the graph morphism $X \to X \times Y$, with components id_X and f:

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times Y \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f \times \mathrm{id}_Y} \\ Y & \xrightarrow{\Delta_Y} & Y \times Y\,. \end{array}$$

It follows that Γ_f is a local isomorphism, and hence that

$$f = \mathrm{pr}_Y \circ \Gamma_f$$

is topologically flat if pr_Y is; this holds, for example, if X is topologically flat over k.

Conversely, since $Y \to k$ is topologically flat, $X \to k$ will also be topologically flat if f is (cf. 1.3.3). Taking in particular for f the canonical morphism

$$p : X \longrightarrow X_e$$

of 1.6, we obtain the following.

Proposition. *Let X be a formal variety over k. The morphism $X \to X_e$ is topologically flat if and only if X is topologically flat over k.*

Remark 1.6.2.[74] When k is a *perfect* field, the functor

$$X \longmapsto X_e$$

is also right adjoint to the inclusion

$$\mathrm{Vaf}^{\text{ét}}\,/k \hookrightarrow \mathrm{Vaf}\,/k.$$

Indeed, in this case

$$\mathcal{O}_{X_e,x} = \kappa(x)$$

for every $x \in X$, and the canonical projections

$$\mathcal{O}_{X,x} \longrightarrow \kappa(x)$$

define a morphism $s : X_e \to X$, which is a section of $p : X \to X_e$. For every morphism $f : X \to Y$, the following diagrams are commutative:

$$\begin{array}{ccc} \mathcal{O}_{Y,f(x)} & \longrightarrow & \mathcal{O}_{X,x} \\ \downarrow & & \downarrow \\ \kappa(f(x)) & \longrightarrow & \kappa(x) \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle s_X}\uparrow & & \uparrow{\scriptstyle s_Y} \\ X_e & \xrightarrow{f_e} & Y_e \end{array}$$

Thus s is functorial in X, and it follows that $X \mapsto X_e$ is right adjoint to the inclusion

$$\mathrm{Vaf}^{\text{ét}}\,/k \hookrightarrow \mathrm{Vaf}\,/k.$$

Consequently, being a right adjoint, the functor $X \mapsto X_e$ commutes with projective limits whenever they exist in $\mathrm{Vaf}^{\text{ét}}\,/k$,[75] and hence, in particular, with finite products.
(This can also be verified directly: for every formal k-variety X, with affine algebra A, the affine algebra of X_e is the quotient of A by its radical $r(A)$; and, since k is perfect, the quotient of $\mathcal{O}_{X,x}\widehat{\otimes}\mathcal{O}_{Y,y}$ by its radical is the algebra $\kappa(x) \otimes_k \kappa(y)$, because the latter is semisimple.)

2. Generalities on Formal Groups

2.1

Let k be a pseudocompact ring and let G be a formal k-group, that is, a group object in the category $\mathrm{Vaf}\,/k$ of formal varieties over k. Let A be the affine algebra of G. The composition law of G plainly defines a diagonal morphism, namely a homomorphism of profinite k-algebras

$$\Delta_A : A \longrightarrow A\widehat{\otimes}_k A.$$

This homomorphism has the following properties.

(i) The diagram

[74] The editors inserted this remark here; in the original it appeared in 2.5.2.
[75] This is the case for finite projective limits.

$$\begin{array}{ccc} A & \xrightarrow{\Delta_A} & A\widehat{\otimes}_k A \\ {\scriptstyle \Delta_A}\downarrow & & \downarrow{\scriptstyle \Delta_A\widehat{\otimes}\,\mathrm{id}_A} \\ A\widehat{\otimes}_k A & \xrightarrow{\mathrm{id}_A\,\widehat{\otimes}\Delta_A} & A\widehat{\otimes}_k A\widehat{\otimes}_k A \end{array}$$

is commutative.

(ii) There is a necessarily unique *augmentation*, that is, a homomorphism of profinite k-algebras

$$\varepsilon_A : A \longrightarrow k,$$

such that the two composites displayed below are the identity maps of A:

$$A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{\varepsilon_A\widehat{\otimes}\,\mathrm{id}_A} k\widehat{\otimes}_k A \simeq A$$

$$\text{and} \qquad A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{\mathrm{id}_A\,\widehat{\otimes}\varepsilon_A} A\widehat{\otimes}_k k \simeq A$$

(iii) There is a necessarily unique *antipode*, that is, a homomorphism of profinite k-algebras

$$c_A : A \longrightarrow A,$$

such that the composite

$$A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{c_A\widehat{\otimes}\,\mathrm{id}_A} A\widehat{\otimes}_k A \xrightarrow{m_A} A$$

equals $\eta_A \circ \varepsilon_A$. Here m_A is the linear map that sends $a\widehat{\otimes}b$ to ab, and $\eta_A : k \to A$ sends λ to $\lambda 1_A$.

Conversely, data

$$(\Delta_A, \varepsilon_A, c_A)$$

satisfying (i)–(iii) endow G with the structure of a formal k-group.[76] Explicitly, for every profinite k-algebra B, the set

$$\mathrm{Hom}_c(A, B)$$

of continuous morphisms of k-modules $\phi : A \to B$ carries a group structure, functorial in B, defined by

$$\phi \cdot \phi' = m_B \circ (\phi\widehat{\otimes}\phi') \circ \Delta_A.$$

The identity element is $\eta_B \circ \varepsilon_A$, where m_B is the multiplication of B and $\eta_B : k \to B$ sends λ to $\lambda 1_B$; the inverse of ϕ is $\phi \circ c_A$. The set

$$\mathrm{Hom}_{\mathrm{Alp}\,/k}(A, B)$$

of continuous morphisms of k-algebras $A \to B$ is a subgroup, because B is commutative.

Definition. A *morphism of formal k-groups* $\theta : K \to G$ is, by definition, a morphism of formal k-varieties that respects the group structures. Let B, respectively A, be the affine

[76] One also says that A is a *cogroup* in the category of profinite k-algebras. The editors expanded the material that follows, in particular by making explicit what a morphism of formal groups $K \to G$ means; cf. Proposition 2.3.1.

algebra of K, respectively G, and let $f : A \to B$ be the morphism corresponding to θ. The condition is equivalently that f be compatible with comultiplication:

$$(f \widehat{\otimes} f) \circ \Delta_A = \Delta_B \circ f.$$

The identities

$$\varepsilon_B \circ f = \varepsilon_A, \qquad c_B \circ f = f \circ c_A$$

then follow automatically. We write Grf $/k$ for the category of formal k-groups.

Notation. In what follows, the ideal

$$I_A = \mathrm{Ker}(\varepsilon_A)$$

will be called the *augmentation ideal* of A, and we write

$$\omega_{G/k} = I_A/\overline{I_A^2}$$

for the pseudocompact k-module obtained by quotienting I_A by the closed ideal generated by the products xy, with $x, y \in I_A$.

2.2

Let $\mathbf{H}$ be a group object in the category of coalgebras over $\mathbf{O}_k$. Thus, for every object C of Alf $/k$, the object $\mathbf{H}(C)$ carries the structure of a C-coalgebra in groups (cf. VII$_\mathrm{A}$ 3.2). Following Manin, we shall say *bialgebra*[77] instead of "coalgebra in groups." Moreover, for every morphism

$$\phi : C \longrightarrow D$$

in Alf $/k$, the map

$$D \otimes_C \mathbf{H}(C) \longrightarrow \mathbf{H}(D)$$

is a homomorphism of D-bialgebras.

Definition. These properties will be summarized by saying that $\mathbf{H}$ is a *bialgebra over* $\mathbf{O}_k$.

The functor

$$\mathbf{H} \longmapsto \mathrm{Spf}^*(\mathbf{H})$$

of 1.3.5 plainly commutes with finite products. It therefore sends a bialgebra over $\mathbf{O}_k$ to a k-group functor, that is, a covariant functor from Alf $/k$ to the category of groups.

Indeed, for every k-algebra C of finite length, the elements of

$$\begin{aligned}\mathrm{Spf}^*(\mathbf{H})(C) &= \mathrm{Spf}^*(\mathbf{H}(C)) \\ &= \{x \in \mathbf{H}(C) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}\end{aligned}$$

form a group under multiplication in the algebra $\mathbf{H}(C)$; cf. VII$_\mathrm{A}$ 3.2.2. Notice also that $\Delta(x) = x \otimes x$ implies

$$\varepsilon(x) = \varepsilon(x)^2.$$

Thus $\varepsilon(x) = 1$ whenever C is local and $x \neq 0$.[78]

[77] The term "bigebra" is also used (cf. [BAlg], III § 11.4). Recall (cf. VII$_\mathrm{A}$, 3.1, editors' note (26)) that all bialgebras considered here are assumed to be cocommutative and equipped with an antipode; they are therefore cocommutative Hopf algebras.

[78] Since $\Delta(x) = x \otimes x$, one has $x = \varepsilon(x)x$; hence $\varepsilon(x) = 0$ can occur only when $x = 0$.

2.2.1

A bialgebra $\mathbf{H}$ over $\mathbf{O}_k$ is said to be *flat* if its underlying module is flat (cf. 1.2.1).[79] If $\mathbf{H}$ is flat, then, by 1.3.5,

$$A = \Gamma^*(\mathbf{H})$$

is a topologically flat profinite k-algebra, and $\mathrm{Spf}^*(\mathbf{H})$, as a functor from

$$(\mathrm{Alf}\,/k)^{\mathrm{op}} = \mathrm{Vaf}\,/k$$

to the category of sets, is isomorphic to the functor

$$\mathrm{Spf}(A) : C \longmapsto \mathrm{Hom}_{\mathrm{Vaf}\,/k}(\mathrm{Spf}(C), \mathrm{Spf}(A)).$$

The group structure on $\mathrm{Spf}^*(\mathbf{H})$ therefore endows

$$\mathscr{G}(\mathbf{H}) = \mathrm{Spf}(A)$$

with the structure of a formal group, described explicitly as follows.

For every object C of $\mathrm{Alf}\,/k$, the C-module underlying $\mathbf{H}(C)$ is projective. Lemma 1.2.3.A therefore gives, by induction on n, natural isomorphisms

$$\begin{aligned}(\mathbf{H}(C)^*)^{\widehat{\otimes}(n+1)} &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^*)^{\widehat{\otimes}n})\\ &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^{\otimes n})^*)\\ &\simeq (\mathbf{H}(C)^{\otimes(n+1)})^*.\end{aligned}$$

For $n = 1, 2$, it follows that the C-algebra structure on $\mathbf{H}(C)$ endows $\mathbf{H}(C)^*$ with a diagonal map satisfying conditions 2.1(i)–(iii), functorially in C.

Consequently,

$$A = \Gamma^*(\mathbf{H}) = \varprojlim_{\mathfrak{l}} \mathbf{H}(k/\mathfrak{l})^*$$

is endowed with the structure of a cogroup in $\mathrm{Alp}\,/k$, which defines the announced formal-group structure on $\mathscr{G}(\mathbf{H})$.

Conversely, let G be a topologically flat formal k-group with affine algebra A, and write

$$\mathbf{H}(G) = \mathbf{V}_k^f(A)$$

for its $\mathbf{O}_k$-coalgebra (cf. 1.2.3). The diagonal morphism

$$\Delta_A : A \longrightarrow A\widehat{\otimes}_k A$$

then induces, for every finite-length k-algebra C, a C-linear map

$$\mathbf{V}_k^f(A)(C) \otimes_C \mathbf{V}_k^f(A)(C) \longrightarrow \mathbf{V}_k^f(A)(C),$$

which makes the coalgebra $\mathbf{V}_k^f(A)(C)$ into a C-bialgebra. We shall call $\mathbf{H}(G)$ the *covariant bialgebra* of the formal group G.[80] Thus, by Proposition 1.3.5.D, we obtain the following.

Proposition.

[79] The editors have expanded the original in what follows.

[80] The editors added the adjective "covariant" to make clear that the functor $G \mapsto \mathbf{H}(G)$ is covariant; this terminology is used in [Di73], I § 2.14.

(i) *The functors*

$$G \longmapsto \mathbf{H}(G), \qquad \mathbf{H} \longmapsto \mathscr{G}(\mathbf{H})$$

define an equivalence between the category of topologically flat formal k-groups and the category of flat $\mathbf{O}_k$-bialgebras.[81]

(ii) *This equivalence "commutes with base change": if $k \to K$ is a morphism of pseudo-compact rings, then*

$$\mathbf{H}\big(G\widehat{\otimes}_k K\big) = \mathbf{H}(G) \otimes_k K$$

and

$$\mathscr{G}\big(\mathbf{H} \otimes_k K\big) = \mathscr{G}(\mathbf{H})\widehat{\otimes}_k K.$$

When k is an Artinian ring and G is a topologically flat formal k-group, the functor $\mathbf{H}(G)$ is plainly determined by its value

$$\mathbf{H}(G) = \mathbf{H}(G)(k)$$

at k. We shall also call $\mathbf{H}(G)$ the *(covariant) bialgebra* of G.[82] Let $\mathcal{H}_k$ denote the category of k-Hopf algebras, and let

$$\mathcal{H}^{\mathrm{cocom}}_{\mathrm{flat}/k}$$

be its full subcategory formed by the k-Hopf algebras that are flat over k and cocommutative. We then obtain the following.

Corollary. *Let k be an Artinian ring.*

(i) *The functors*

$$G \longmapsto \mathbf{H}(G), \qquad \mathbf{H} \longmapsto \mathscr{G}(\mathbf{H}) = \mathrm{Spf}^*(\mathbf{H})$$

define an equivalence between the category of topologically flat formal k-groups and $\mathcal{H}^{\mathrm{cocom}}_{\mathrm{flat}/k}$.

(ii) *This equivalence "commutes with base change": if $k \to K$ is a morphism of Artinian rings, then*

$$\mathbf{H}\big(G\widehat{\otimes}_k K\big) = \mathbf{H}(G) \otimes_k K$$

and

$$\mathscr{G}\big(\mathbf{H} \otimes_k K\big) = \mathscr{G}(\mathbf{H})\widehat{\otimes}_k K.$$

On the other hand, let

$$\mathcal{H}^{\mathrm{com}}_{\mathrm{flat}/k}$$

denote the full subcategory of $\mathcal{H}_k$ formed by the k-Hopf algebras that are flat over k and commutative. Recall that the functor

$$T \longmapsto \mathcal{O}(T)$$

is an anti-equivalence from the category of affine k-group schemes onto $\mathcal{H}^{\mathrm{com}}_{\mathrm{flat}/k}$.

[81] Recall that in this Exposé, "bialgebra" means "cocommutative Hopf algebra."

[82] The editors have introduced here the notation $\mathcal{H}^{\mathrm{cocom}}_{\mathrm{flat}/k}$, which will be useful below.

2.2.2

Suppose, for simplicity, that k is Artinian, and let G be a topologically flat formal k-group.[83] Then G is commutative if and only if its affine algebra $\mathcal{A}(G)$ has a cocommutative comultiplication, which is equivalent to saying that the bialgebra $\mathbf{H}(G)$ has a commutative multiplication. In that case, $\mathbf{H}(G)$ is a commutative and cocommutative Hopf algebra, flat over k. Thus, if

$$D'(G) = \operatorname{Spec} \mathbf{H}(G),$$

then $D'(G)$ is a commutative k-group scheme, affine and flat over k.

Conversely, if T is such a k-group scheme, its affine algebra $\mathcal{O}(T)$ is a commutative group object in the category of cocommutative k-coalgebras flat over k. Hence 1.3.5.C gives a topologically flat formal k-group $D(T)$, defined by

$$D(T) = \operatorname{Spf}^* \mathcal{O}(T) = \operatorname{Spf}(\mathcal{A}), \qquad \mathcal{A} = \mathcal{O}(T)^*.$$

By 1.3.5.D, there are canonical isomorphisms

$$G = \operatorname{Spf}^* \mathbf{H}(G), \qquad \mathbf{H}(D(T)) = \mathcal{O}(T).$$

We therefore obtain canonical isomorphisms

$$D(D'(G)) = \operatorname{Spf}^* \mathcal{O}(D'(G)) = \operatorname{Spf}^* \mathbf{H}(G) = G,$$
$$D'(D(T)) = \operatorname{Spec} \mathbf{H}(D(T)) = \operatorname{Spec} \mathcal{O}(T) = T.$$

Moreover, if k- Gr. denotes the category of k-group schemes, Corollary 2.2.1 gives functorial isomorphisms

$$\operatorname{Hom}_{\mathrm{Grf}\,/k}\big(G, D(T)\big) \simeq \operatorname{Hom}_{\mathcal{H}_k}\big(\mathbf{H}(G), \mathcal{O}(T)\big) \simeq \operatorname{Hom}_{k\text{- Gr.}}\big(T, D'(G)\big).$$

This proves the following.

Proposition (Cartier duality). *Let k be an Artinian ring.*

(i) *The functors*

$$G \longmapsto D'(G) = \operatorname{Spec} \mathbf{H}(G), \qquad T \longmapsto D(T) = \operatorname{Spf}^* \mathcal{O}(T)$$

induce an anti-equivalence between the category $\mathcal{FC}_k$ of commutative topologically flat formal k-groups and the category $\mathcal{AC}_k$ of commutative affine flat k-group schemes. In other words, G and $D'(G)$, respectively T and $D(T)$, are related by the equalities [84]

$$\mathbf{H}(G) = \mathcal{O}(D'(G)), \qquad \mathcal{O}(T) = \mathbf{H}(D(T)).$$

(ii) *This anti-equivalence "commutes with base change": if $k \to K$ is a morphism of Artinian rings, then*

$$D'\big(G \widehat{\otimes}_k K\big) = D'(G) \otimes_k K$$

and

$$D\big(T \otimes_k K\big) = D(T) \widehat{\otimes}_k K.$$

(iii) *In particular, if k is a field, one obtains an anti-equivalence between the category of commutative formal k-groups and the category of commutative affine k-group schemes, and this anti-equivalence commutes with extension of the base field.*

[83] The editors have expanded the original in what follows in order to introduce the notation $D'(G)$ and $D(K)$; cf. [Ca62], § 14.

[84] Let $\mathcal{FC}_k^f$, respectively $\mathcal{AC}_k^f$, denote the full subcategory of $\mathcal{FC}_k$, respectively $\mathcal{AC}_k$, consisting of the objects G, respectively T, for which $\mathbf{H}(G)$, respectively $\mathcal{O}(T)$, is a finite k-module (and hence finite locally free). Then $\mathcal{FC}_k^f$ and $\mathcal{AC}_k^f$ both have the same objects as the category $\mathcal{C}$ of commutative and cocommutative Hopf k-algebras that are finite and flat over k. The correspondence $G \mapsto \mathbf{H}(G)$, respectively $T \mapsto \mathcal{O}(T)$, is covariant, respectively contravariant. Thus one recovers the "Cartier duality" of the category $\mathcal{AC}_k$, already seen in VII$_A$ 3.3.1.

2.3

Let G now be an arbitrary formal k-group,[85] with affine algebra A. We continue to write

$$\mathbf{H}(G) = \mathbf{V}_k^f(A)$$

for the $\mathbf{O}_k$-module dual to A, and denote by

$$\phi_G : \mathbf{H}(G) \otimes_k \mathbf{H}(G) \longrightarrow \mathbf{H}(G \times G)$$

the functorial homomorphism induced, for every object C of Alf $/k$, by the natural map of 0.3.6:

$$(A\widehat{\otimes}_k C)^\dagger \otimes_C (A\widehat{\otimes}_k C)^\dagger \longrightarrow \left((A\widehat{\otimes}_k C)\widehat{\otimes}_C(A\widehat{\otimes}_k C)\right)^\dagger .$$

If

$$m : G \times G \longrightarrow G$$

is the multiplication of G, the composite

$$\mathbf{H}(G) \otimes_k \mathbf{H}(G) \xrightarrow{\phi_G} \mathbf{H}(G \times G) \xrightarrow{\mathbf{H}(m)} \mathbf{H}(G)$$

endows $\mathbf{H}(G)$ with an algebra structure over $\mathbf{O}_k$. For every $C \in$ Alf $/k$, the unit element of

$$\mathbf{H}(G)(C) = (A\widehat{\otimes}_k C)^\dagger$$

is the augmentation of $A\widehat{\otimes}_k C$; cf. 2.1. [86]

If G is not topologically flat over k, the morphism ϕ_G need not be an isomorphism. Consequently, the morphism

$$\delta_G : \mathbf{H}(G) \longrightarrow \mathbf{H}(G \times G)$$

induced by the diagonal $x \mapsto (x, x)$ of G need not factor through

$$\mathbf{H}(G) \otimes_k \mathbf{H}(G).$$

Thus $\mathbf{H}(G)$ need not be a bialgebra over $\mathbf{O}_k$. For this reason, in the general case we simply call $\mathbf{H}(G)$ the *covariant algebra* of the formal group G.

When G is topologically flat over k, of course, ϕ_G is an isomorphism, and one recovers the $\mathbf{O}_k$-bialgebra structure on $\mathbf{H}(G)$ defined in 2.2.1.

Proposition 2.3.1. *Let K and G be two formal k-groups, with affine algebras B and A, respectively. Suppose that K is topologically flat over k. Then there is a canonical bijection from*

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(K, G)$$

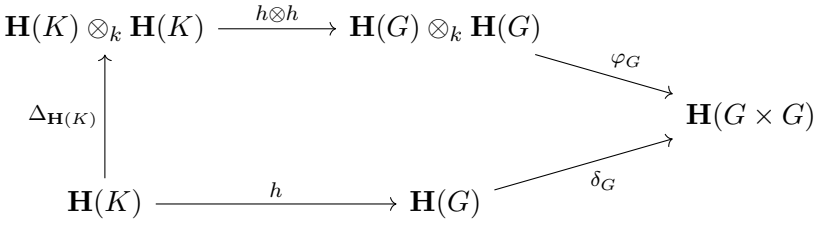
onto the set of homomorphisms of unital $\mathbf{O}_k$-algebras

$$h : \mathbf{H}(K) \longrightarrow \mathbf{H}(G)$$

for which the diagram

[85] That is, G is not necessarily topologically flat over k.

[86] The editors changed the order of the sentences that follow.



is commutative.

Since K is topologically flat, $\mathbf{H}(K)$ is endowed with a bialgebra structure (cf. 2.2), and $\Delta_{\mathbf{H}(K)}$ is its diagonal morphism. In other words, with the notation of 2.3,

$$\Delta_{\mathbf{H}(K)} = \phi_K^{-1} \circ \delta_K.$$

When G too is topologically flat over k, the proposition follows from the equivalence of categories established in 2.2.1.

In the general case, we may suppose that k is Artinian and argue with the algebras

$$\mathbf{H}(K) = B^\dagger, \qquad \mathbf{H}(G) = A^\dagger.$$

Let $\mathrm{Hom}_c(A, B)$ denote the set of continuous k-linear maps from A to B, and let $\mathrm{Hom}_k(B^\dagger, A^\dagger)$ denote the set of k-linear maps from $B^\dagger$ to $A^\dagger$. [87]

By 0.3.6.A, if M and P are pseudocompact k-modules and P is projective, the canonical map

$$\mathrm{Hom}_c(M, P) \longrightarrow \mathrm{Hom}_k(P^\dagger, M^\dagger), \qquad f \longmapsto {}^t f,$$

where ${}^t f$ denotes the transpose of f, is bijective. We shall apply this either with

$$M = A\widehat{\otimes}A, \quad P = B,$$

or with

$$M = A, \quad P = B\widehat{\otimes}B.$$

Let $f \in \mathrm{Hom}_c(A, B)$. Consider the following diagrams. The squares marked (0) are commutative, and the two unnamed vertical arrows in the second diagram are ${}^t(f\widehat{\otimes}f)$.

$$\begin{array}{ccccc}
A\widehat{\otimes}A & \xrightarrow{m_A} & A & \xrightarrow{\Delta_A} & A\widehat{\otimes}A \\
\downarrow{\scriptstyle f\widehat{\otimes}f} & (1) & \downarrow{\scriptstyle f} & (2) & \downarrow{\scriptstyle f\widehat{\otimes}f} \\
B\widehat{\otimes}B & \xrightarrow{m_B} & B & \xrightarrow{\Delta_B} & B\widehat{\otimes}B
\end{array}$$

$$\begin{array}{ccccccccc}
A^\dagger \otimes A^\dagger & \xrightarrow{\varphi_G} & (A\widehat{\otimes}A)^\dagger & \xleftarrow{\delta_G = m_A^t} & A^\dagger & \xleftarrow{\Delta_A^t} & (A\widehat{\otimes}A)^\dagger & \xleftarrow{\varphi_G} & A^\dagger \otimes A^\dagger \\
\uparrow{\scriptstyle {}^t f \otimes {}^t f} & (0) & \uparrow & (1') & \uparrow{\scriptstyle {}^t f} & (2') & \uparrow & (0) & \uparrow{\scriptstyle {}^t f \otimes {}^t f} \\
B^\dagger \otimes B^\dagger & = & (B\widehat{\otimes}B)^\dagger & \xleftarrow[\delta_K = m_B^t]{} & B^\dagger & \xleftarrow[\Delta_B^t]{} & (B\widehat{\otimes}B)^\dagger & = & B^\dagger \otimes B^\dagger
\end{array}$$

If $f : A \to B$ corresponds to a morphism of formal groups $K \to G$, then the squares (1) and (2) are commutative and

$$\varepsilon_B \circ f = \varepsilon_A.$$

[87] The editors supplied the details of the remainder of the proof.

Consequently, the squares (1′) and (2′) are commutative, and ${}^t f$ sends the unit of $B^\dagger = \mathbf{H}(K)$ to that of $A^\dagger = \mathbf{H}(G)$. Thus ${}^t f$ is a homomorphism of unital k-algebras

$$\mathbf{H}(K) \longrightarrow \mathbf{H}(G)$$

for which the diagram in the proposition is commutative.

Conversely, suppose that ${}^t f$ satisfies these conditions. Then

$$\varepsilon_B \circ f = \varepsilon_A$$

and the squares (1′) and (2′) are commutative. For

$$M = A\widehat{\otimes}A, \quad P = B \qquad \text{(respectively } M = A,\ P = B\widehat{\otimes}B),$$

the map $g \mapsto {}^t g$ is injective. Hence the squares (1) and (2) are commutative, so f is compatible with the multiplications and diagonal morphisms of A and B.

It remains to prove that $f(1_A) = 1_B$. What precedes gives

$$\varepsilon_B f(1) = 1, \qquad \Delta_B f(1) = f(1)\widehat{\otimes}f(1), \qquad f(1)\cdot f(1) = f(1).$$

By 2.1(iii), the first two conditions imply that $f(1)$ has inverse $c_B f(1)$ in B. The equality $f(1)\cdot f(1) = f(1)$ therefore implies $f(1) = 1$. Thus

$$f : A \longrightarrow B$$

is a morphism in $\mathrm{Alp}\,/k$, compatible with the comultiplications of A and B.

2.3.2

Let us now suppose, for simplicity, that the ring k is Artinian. When G is topologically flat over k, the algebra

$$\mathbf{H}(G) = \mathbf{H}(G)(k)$$

may be characterized by a universal property due to Cartier. Recall (cf. 1.2.1) that, if U is a k-module, we write $\mathbf{W}(U)$ for the functor that assigns to every finite-length k-algebra C the C-module

$$U \otimes_k C.$$

[88] If U is a k-algebra (associative and with a unit element), then so is $U \otimes_k C$. We shall write $\mathbf{W}(U)^\times$ for the k-group functor that assigns to every $C \in \mathrm{Ob}(\mathrm{Alf}\,/k)$ the multiplicative group of invertible elements of the algebra $U \otimes_k C$:

$$\mathbf{W}(U)^\times(C) = (U \otimes_k C)^\times.$$

Moreover, identify G with the k-group functor

$$C \longmapsto \mathrm{Hom}_{\mathrm{Vaf}\,/k}\big(\mathrm{Spf}(C), G\big),$$

and write

$$\mathrm{Hom}_{k\text{- Gr.}}\big(G, \mathbf{W}(U)^\times\big)$$

for the set of homomorphisms of k-group functors from G to $\mathbf{W}(U)^\times$. We have the following.

[88] The editors added the preceding sentence and, in what follows, wrote $\mathbf{W}(U)^\times$ in place of U_m. Recall also (cf. 1.2.1) that a "k-functor" is a covariant functor from $\mathrm{Alf}\,/k$ to (**Sets**).

Proposition. *Let k be an Artinian ring. For every formal group G topologically flat over k, and every k-algebra U, there is a canonical isomorphism*

$$\mathrm{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times) \xrightarrow{\sim} \mathrm{Hom}_{k\text{-Alg.}}(\mathbf{H}(G), U).$$

Let A be the affine algebra of G. By hypothesis, A is a projective object of $\mathrm{PC}(k)$, and

$$\mathbf{H}(G) = A^\dagger.$$

For every object P of $\mathrm{PC}(k)$, write

$$U \widehat{\otimes}_k P$$

for the inverse limit of the k-modules

$$U \otimes_k (P/N),$$

where N runs through the open submodules of P. There are linear maps

$$U \otimes_k (P/N) \longrightarrow \mathrm{Hom}_k((P/N)^*, U)$$

that send $u \otimes x$ to the k-linear map

$$f \longmapsto f(x)u$$

and form a filtered inverse system. Passing to the inverse limit therefore gives a morphism

$$U \widehat{\otimes}_k P \xrightarrow{\psi_P} \mathrm{Hom}_k(P^\dagger, U). \tag{1}$$

When $P = k$, the morphism ψ_k is plainly an isomorphism. Moreover, the two sides of (1), considered as functors of P, commute with infinite products (every product being a filtered inverse limit of finite products). Thus (1) is an isomorphism when P is a product of copies of k, and then when P is a projective object of $\mathrm{PC}(k)$ (the two sides of (1) commute with finite direct sums).

Now let

$$\mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$$

denote the set of morphisms of k-functors from G to $\mathbf{W}(U)$. Since

$$G = \mathrm{Spf}(A) = \varinjlim_{\mathfrak{l}} \mathrm{Spf}(A/\mathfrak{l}),$$

where $\mathfrak{l}$ runs through the open ideals of A, there are canonical isomorphisms

$$\begin{aligned}\mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) &= \varprojlim_{\mathfrak{l}} \mathrm{Hom}_{\mathcal{F}}(\mathrm{Spf}(A/\mathfrak{l}), \mathbf{W}(U)) \\ &= \varprojlim_{\mathfrak{l}} U \otimes_k (A/\mathfrak{l}) = U \widehat{\otimes}_k A.\end{aligned}$$

The preceding argument therefore gives a canonical isomorphism

$$\mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) = U \widehat{\otimes}_k A \xrightarrow{\psi_A} \mathrm{Hom}_k(\mathbf{H}(G), U). \tag{2}$$

For every finite-length k-algebra C, multiplication makes $U \otimes_k C$ a monoid with a unit element, and every homomorphism of monoids with unit

$$G(C) \longrightarrow U \otimes_k C$$

is necessarily a group homomorphism

$$G(C) \longrightarrow (U \otimes_k C)^\times.$$

It follows that

$$\mathrm{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times)$$

is the part of $\mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$ consisting of morphisms of k-functors into monoids with a unit element.

It remains to show that these morphisms correspond to the k-linear maps from $\mathbf{H}(G)$ to U that preserve multiplication and unit elements. [89] To simplify the notation, write

$$H = \mathbf{H}(G) = A^\dagger$$

and write $\widehat{\otimes}$ instead of $\widehat{\otimes}_k$. Denote by

$$\Delta_A,\ m_A,\ \varepsilon_A \qquad \text{and} \qquad \Delta_H,\ m_H,\ \varepsilon_H$$

the comultiplication, multiplication, and augmentation of A and H, respectively. Let

$$\theta \in \mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)),$$

let γ be its image in $U\widehat{\otimes}A$, and let

$$\phi : H \longrightarrow U$$

be the associated k-linear map. The morphism θ sends the unit section $s \in G(k)$ to an element u of U. Since s corresponds to the augmentation

$$\varepsilon : A \longrightarrow k,$$

which is the unit element 1_H of H, we have

$$\theta(s) = 1_U \quad \Longleftrightarrow \quad \phi(1_H) = 1_U.$$

The morphism

$$\theta \circ m_G : G \times G \longrightarrow \mathbf{W}(U)$$

corresponds to the element

$$(\mathrm{id}_U \,\widehat{\otimes}\Delta_A)(\gamma),$$

which, by duality, corresponds to the map

$$\phi \circ m_H : H \otimes H \longrightarrow U.$$

On the other hand, the morphism

$$\theta \circ \mathrm{pr}_1 : G \times G \longrightarrow \mathbf{W}(U)$$

corresponds to the element

$$\gamma\widehat{\otimes}1_A \in U\widehat{\otimes}A\widehat{\otimes}A,$$

which, by duality, corresponds to

$$\phi \circ (\mathrm{id}_H \otimes\varepsilon_H) : H \otimes H \longrightarrow U.$$

[89] The original stated that this "follows from the functoriality of ψ_A." The editors gave the details that follow.

Similarly, $\theta \circ \mathrm{pr}_2$ corresponds to the element

$$\tau(\gamma \widehat{\otimes} 1_A) \in U \widehat{\otimes} A \widehat{\otimes} A,$$

where

$$\tau(u \widehat{\otimes} a \widehat{\otimes} b) = u \widehat{\otimes} b \widehat{\otimes} a,$$

and this element corresponds by duality to

$$\phi \circ (\varepsilon_H \otimes \mathrm{id}_H) : H \otimes H \longrightarrow U.$$

Finally, the multiplication map

$$\mu = m_{U \widehat{\otimes} A \widehat{\otimes} A}$$

given by

$$\begin{aligned} &(U \widehat{\otimes} A \widehat{\otimes} A) \times (U \widehat{\otimes} A \widehat{\otimes} A) \longrightarrow U \widehat{\otimes} A \widehat{\otimes} A, \\ &(u \widehat{\otimes} a_1 \widehat{\otimes} a_2,\, u' \widehat{\otimes} a_3 \widehat{\otimes} a_4) \longmapsto u u' \widehat{\otimes} a_1 a_3 \widehat{\otimes} a_2 a_4 \end{aligned}$$

may be viewed as the composite of the endomorphism σ_{23} of

$$(U \otimes U) \widehat{\otimes} A^{\widehat{\otimes} 4},$$

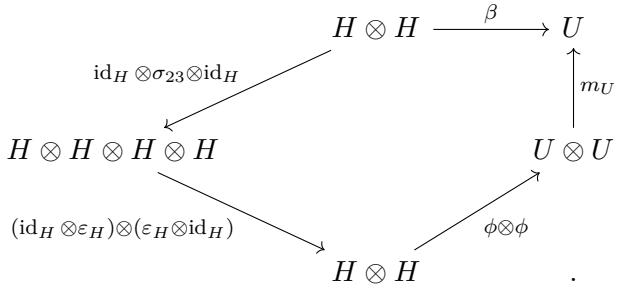
which "exchanges the factors a_2 and a_3," and the map

$$(U \otimes U) \widehat{\otimes} A^{\widehat{\otimes} 2} \widehat{\otimes} A^{\widehat{\otimes} 2} \xrightarrow{m_U \widehat{\otimes} m_A \widehat{\otimes} m_A} U \widehat{\otimes} A \widehat{\otimes} A.$$

It follows that the map

$$\mu \circ (\phi \circ \mathrm{pr}_1,\, \phi \circ \mathrm{pr}_2) : G \times G \longrightarrow \mathbf{W}(U)$$

corresponds to the composite map β below:



Finally, θ is compatible with the laws of G and $\mathbf{W}(U)^{\times}$ if and only if

$$\theta \circ m_G = \mu \circ (\phi \circ \mathrm{pr}_1,\, \phi \circ \mathrm{pr}_2),$$

which, by the preceding argument, is equivalent to

$$\phi \circ m_H = \beta.$$

But plainly

$$(\phi \circ m_H)(x \otimes y) = \phi(xy),$$

whereas

$$\beta(x \otimes y) = \phi(x)\phi(y).$$

2.4

We now return to an arbitrary pseudocompact ring k, in order to apply to formal groups the results of 1.4–1.5 on passage to the quotient by a topologically flat equivalence relation. [90]

Let $u : H \to G$ be a monomorphism of formal k-groups, let $\mu : G \times G \to G$ be the multiplication morphism of G, and let λ be the composite

$$\lambda : \quad G \times H \xrightarrow{\mathrm{id}_G \times u} G \times G \xrightarrow{\mu} G.$$

Since u is a monomorphism, the pair

$$G \times H \underset{\lambda}{\overset{\mathrm{pr}_1}{\rightrightarrows}} G$$

is an equivalence pair in Vaf $/k$ (cf. V, 2.b). Recall (cf. 1.2.C) that the cokernel G/H of this pair is defined as follows.

Let $\mathcal{O}(G)$ and $\mathcal{O}(H)$ be the affine algebras of G and H, let

$$\Delta : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \widehat{\otimes}_k \mathcal{O}(G)$$

be the diagonal morphism of $\mathcal{O}(G)$, and let I be the kernel of

$$u^\natural : \mathcal{O}(G) \longrightarrow \mathcal{O}(H).$$

By Proposition 1.3, the morphism $u^\natural$ induces an isomorphism

$$\mathcal{O}(G)/I \xrightarrow{\sim} \mathcal{O}(H).$$

The affine algebra $\mathcal{O}(G/H)$ of G/H is then the kernel of the pair

$$\mathcal{O}(G) \underset{(\mathrm{id} \widehat{\otimes} u^\natural)\Delta}{\overset{\tau_1}{\rightrightarrows}} \mathcal{O}(G) \widehat{\otimes}_k \mathcal{O}(H),$$

where $\tau_1(x) = x \widehat{\otimes} 1$. Thus

$$\mathcal{O}(G/H) = \{x \in \mathcal{O}(G) \mid \Delta(x) - x \widehat{\otimes} 1 \in \mathcal{O}(G) \widehat{\otimes} I\}.$$

If, moreover, H is topologically flat over k, then pr_1 is topologically flat, and Theorem 1.4 gives the following result.

Theorem. *Let $u : H \to G$ be a monomorphism of formal k-groups. Suppose that H is topologically flat over k. Then the projection*

$$p : G \longrightarrow G/H$$

is surjective and topologically flat, there is an isomorphism

$$G \times H \xrightarrow{\sim} G \times_{G/H} G, \tag{*}$$

and G/H represents the quotient sheaf for the flat topology.

Consequently, G/H has a canonical structure of an object on which G acts, and $p : G \to G/H$ is an equivariant morphism. If, in addition, u identifies H with a normal subgroup

[90] The editors supplied the details of the original argument that follow.

of G, then G/H has a canonical structure of formal k-group for which $p : G \to G/H$ is a morphism of formal k-groups, and H is the kernel of p.

Indeed, the first assertion follows from 1.4, and the other two from Exposé IV, Corollaries 5.2.2 and 5.2.4.

Corollary. [91] *Let G be a formal k-group and H a formal subgroup of G. Let A, A/J, and B be the affine algebras of G, H, and G/H, respectively; let I_A be the augmentation ideal of A, and put $I_B = B \cap I_A$. Suppose that H is topologically flat over k. Then*

$$J = AI_B,$$

the closed ideal generated by I_B.

Indeed, the projection $B \to B/I_B$ corresponds to the unit section

$$e : \mathrm{Spf}(k) \longrightarrow G/H$$

of G/H. By (*), H is identified with the fiber product

$$\mathrm{Spf}(k) \times_{G/H} G.$$

Its affine algebra A/J is therefore identified with

$$(B/I_B)\widehat{\otimes}_B A \simeq A/AI_B.$$

2.4.A

Let G and Q be formal k-groups that are topologically flat. Suppose that there are homomorphisms

$$\sigma : Q \longrightarrow G \qquad \text{and} \qquad \pi : G \longrightarrow Q$$

such that

$$\pi \circ \sigma = \mathrm{id}_Q\,.$$

[92] In particular, σ is a monomorphism, so Q is a formal subgroup of G (cf. Remark 1.3). Let

$$N = \mathrm{Ker}(\pi)$$

and let σ' be the inclusion $N \hookrightarrow G$. Then G is the semidirect product of N by Q (cf. I, 2.3.8). In other words, for every

$$B \in \mathrm{Ob}(\mathrm{Alf}\,/k),$$

identify $N(B)$ and $Q(B)$ with their images in $G(B)$ under σ' and σ. Then $N(B)$ is a normal subgroup of $G(B)$, and the map

$$\mu : N(B) \times Q(B) \longrightarrow G(B), \qquad (x, q) \longmapsto xq \tag{1}$$

is bijective.

The morphism of formal k-varieties

$$\theta : G(B) \longrightarrow N(B), \qquad g \longmapsto g \cdot \sigma\pi(g^{-1}) \tag{2}$$

is then a retraction of σ', and the inverse of μ is

$$g \longmapsto (\theta(g), \pi(g)). \tag{3}$$

[91] The editors made this corollary explicit; it is used, for example, in 5.2.1 and 5.2.3.

[92] The editors added paragraphs 2.4.A and 2.4.B, which will be useful in 5.1.3 and 2.9, respectively.

If $p : G \to G/Q$ denotes the quotient projection, then

$$p \circ \sigma' : N \longrightarrow G/Q$$

is an isomorphism of formal k-varieties.

In particular, N is topologically flat over k, by 1.4 (ii). Let

$$\alpha : Q \times N \longrightarrow N$$

and

$$\beta : G \times N = N \times Q \times N \longrightarrow N$$

be the maps defined on points by

$$\alpha(q, y) = qyq^{-1}, \qquad \beta(x, q, y) = x\alpha(q, y).$$

Then the following diagram is commutative:

(4)
$$\begin{array}{ccccc} G \times G & \xrightarrow{\mathrm{id} \times \theta} & G \times N & \xrightarrow{\theta \times \mathrm{id}} & N \times N \\ \downarrow{\scriptstyle m_G} & & \downarrow{\scriptstyle \beta} & \swarrow{\scriptstyle m_N} & \\ G & \xrightarrow{\theta} & N & & \end{array} .$$

This can be expressed as follows in terms of the affine algebras A, A_0, and A' of G, Q, and N (cf. 5.1.3 below). Let

$$\rho' : A \longrightarrow A', \qquad \rho : A \longrightarrow A_0, \qquad \tau : A_0 \longrightarrow A$$

be the homomorphisms of k-bialgebras corresponding to σ', σ, and π, and put

$$I = \mathrm{Ker}(\rho).$$

By the preceding corollary, A' is identified with

$$A/\overline{A\,\tau(J_0)},$$

where J_0 is the augmentation ideal of A_0.

On the other hand, let B be the affine algebra of G/Q. It is the kernel of the pair of morphisms

$$A \underset{(\mathrm{id}_A \,\widehat{\otimes}\rho)\circ\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A\widehat{\otimes}_k A_0,$$

where $\tau_1(x) = x\widehat{\otimes}1$. Thus

$$B = \{x \in A \mid \Delta_A(x) - x\widehat{\otimes}1 \in A\widehat{\otimes}I\}.$$

Let

$$\gamma = \theta^\natural : A' \longrightarrow A$$

be the corresponding continuous homomorphism of k-algebras. It is a section of ρ', and it is an isomorphism of profinite k-algebras from A' onto B. Moreover,

$$\tau = \pi^\natural$$

identifies A_0 with a sub-bialgebra of A, namely the affine algebra of the quotient $N\backslash G$.

It follows from equations (1) and (3) that there is an isomorphism of profinite k-algebras

$$A' \widehat{\otimes}_k A_0 \xrightarrow{\sim} A, \qquad a' \widehat{\otimes} a_0 \longmapsto \gamma(a')\tau(a_0), \tag{*}$$

whose inverse is the map

$$a \longmapsto (\rho' \widehat{\otimes} \rho)\Delta_A(a).$$

Finally, identify A' with its image in A under γ, so that the projection $A \to A'$ is then $\gamma\rho'$. Writing $\Delta_{A'}$ for the comultiplication of A', one deduces from (4) that

$$\Delta_A(A') \subset A \widehat{\otimes} A'$$

and that the following diagram is commutative:

$$\begin{array}{ccc} A' & \xrightarrow{\Delta_A} & A \widehat{\otimes} A' \\ \| & & \downarrow{\scriptstyle \gamma\rho' \widehat{\otimes} \mathrm{id}} \\ A' & \xrightarrow{\Delta_{A'}} & A' \widehat{\otimes} A' \end{array}$$

Consequently, one also has

$$\gamma\rho' \circ c_A = c_{A'},$$

where c_A and $c_{A'}$ are the antipodes of A and A', respectively. On the other hand, writing m_A for multiplication in A, equation (2) gives, for every $a \in A$,

$$\gamma\rho'(a) = (m_A \circ (\mathrm{id}_A \widehat{\otimes} \tau\rho c_A) \circ \Delta_A)(a).$$

2.4.B

Suppose, for simplicity, that k is Artinian. The preceding discussion can then be expressed more simply in terms of the covariant bialgebras of G, Q, and N. Indeed, since

$$G = N \times Q$$

as formal k-varieties, one has

$$\mathbf{H}(G) = \mathbf{H}(N) \otimes_k \mathbf{H}(Q)$$

as k-coalgebras. Moreover, since multiplication in G is given by

$$(x, q) \cdot (x', q') = (x\alpha(q, x'), qq'), \qquad \alpha(q, x') = qx'q^{-1},$$

multiplication in $\mathbf{H}(G)$ is given, for $x \in \mathbf{H}(N)$ and $q \in \mathbf{H}(Q)$, by

$$(x \otimes q) \cdot (x' \otimes q') = x\phi(q, x') \otimes qq',$$

where

$$\phi : \mathbf{H}(Q) \otimes_k \mathbf{H}(N) \longrightarrow \mathbf{H}(N)$$

is the morphism of k-coalgebras induced by α.

The morphism α is the following composite. Here δ_G and m_G denote, respectively, the diagonal and multiplication morphisms of G, c_Q is inversion in Q, and

$$v(q \otimes q' \otimes x) = q \otimes x \otimes q'.$$

$$Q \times N \xrightarrow{v\circ(\delta_G\times\mathrm{id})} Q \times N \times Q \xrightarrow{\mathrm{id}\times\mathrm{id}\times c_Q} Q \times N \times Q \xrightarrow{m_G} G,$$

Writing c_Q also for the antipode of $\mathbf{H}(Q)$, we obtain

$$\phi(q \otimes x') = \sum_i q_i x' c_Q(q_i') \quad \text{if} \quad \Delta_{\mathbf{H}(Q)}(q) = \sum_i q_i \otimes q_i'. \tag{$\star$}$$

In particular, let M be an abstract group and suppose that $\mathbf{H}(Q)$ is the group algebra kM, so that

$$Q = \mathrm{Spf}^*\, kM$$

is the constant formal k-group M_k. Then for every $\gamma \in M$ and $x' \in \mathbf{H}(N)$,

$$\phi(\gamma \otimes x') = \gamma x' \gamma^{-1}.$$

This defines an action of M on $\mathbf{H}(N)$ by Hopf-algebra automorphisms.

Proposition 2.4.1. *Let $f : G \to K$ be a morphism of formal k-groups. If*

$$H = \ker(f)$$

is topologically flat over k, the homomorphism

$$f' : G/H \longrightarrow K$$

induced by f is a monomorphism.

This is a consequence of the results of Exposé IV,[93] but we give a direct proof. Let T be a formal variety of finite length over k, and let $t \in (G/H)(T)$ be such that $f' \circ t$ is the identity element of $K(T)$. We must show that t is the identity element of $(G/H)(T)$. Let

$$p : G \longrightarrow G/H$$

be the projection, and let

$$X = T \times_{G/H} G.$$

By 2.4, p is surjective and topologically flat. The same is therefore true of

$$\mathrm{pr}_1 : X \longrightarrow T.$$

By Proposition 1.3.1, pr_1 is consequently an epimorphism, so it is enough to show that $t\circ\mathrm{pr}_1$ is the identity element of $(G/H)(X)$.

Let $\mathrm{pr}_2 : X \to G$ be the second projection. We have

$$t \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2,$$

and hence

$$1 = f' \circ t \circ \mathrm{pr}_1 = f' \circ p \circ \mathrm{pr}_2 = f \circ \mathrm{pr}_2\,.$$

The exact sequence

$$1 \longrightarrow H \longrightarrow G \xrightarrow{f} K$$

[93] See Exposé IV, 5.2.6.

therefore shows that pr_2 factors through H. Thus $p \circ \mathrm{pr}_2$ is the zero morphism. Since

$$p \circ \mathrm{pr}_2 = t \circ \mathrm{pr}_1,$$

this proves the proposition.

The following corollary follows. Write $\mathcal{O}(G)$, $\mathcal{O}(K)$, and $\mathcal{O}(G/H)$ for the affine algebras of G, K, and G/H, respectively. We saw in 2.4 that p induces an injection

$$\mathcal{O}(G/H) \hookrightarrow \mathcal{O}(G).$$

Moreover, since $f' : G/H \to K$ is a monomorphism, Proposition 1.3 shows that

$$\mathcal{O}(K) \longrightarrow \mathcal{O}(G/H)$$

is surjective. Hence:

Corollary. *Let $f : G \to K$ be a morphism of formal k-groups and let $H = \ker(f)$. If H is topologically flat over k, then*

$$\mathcal{O}(G/H)$$

is the image of $\mathcal{O}(K)$ in $\mathcal{O}(G)$.

2.4.2

Retain the preceding notation and suppose that H and G are topologically flat over k. Then G is topologically flat both over k and over G/H; hence, by 1.3.3, G/H is topologically flat over k. Consequently, by 2.4, the canonical projection

$$q : K \longrightarrow \mathrm{Coker}(f')$$

is topologically flat, and f' is an isomorphism from G/H onto $\ker(q)$. It is also clear that

$$\mathrm{Coker}(f) = \mathrm{Coker}(f').$$

Thus, under the hypothesis that G and $H = \ker(f)$ are topologically flat over k, we have obtained an isomorphism between $\ker(q)$, the image of f, and

$$G/\ker(f),$$

the coimage of f. This proves the following theorem.

Theorem. *Let k be a field. The commutative formal k-groups form an abelian category.*

Corollary. *Let k be a field. The commutative affine k-group schemes form an abelian category.* [94]

This follows from the theorem and the equivalence of categories in 2.2.2.

2.5

A formal k-group is said to be *étale* if its underlying formal variety is étale (cf. 1.6). Such formal groups have a very simple structure. Suppose that k is local, let κ be its residue field, let κ_s be a separable closure of κ, and let

$$\Gamma = \mathrm{Gal}(\kappa_s/\kappa)$$

[94] See also the remarks following Exposé VI$_A$, 5.4.3.

be the topological Galois group. A Γ-*set* means a set E with a continuous action of Γ; equivalently, the stabilizer of every element $x \in E$ is an open subgroup of Γ.

For every formal k-variety X, put

$$X(\kappa_s) = \varinjlim_{\ell} X(\ell),$$

where ℓ ranges over the finite extensions of κ contained in κ_s.[95] The group Γ acts continuously on every $X(\ell)$, and hence on $X(\kappa_s)$. Moreover, if

$$X_\kappa = X \widehat{\otimes}_k \kappa \qquad \text{(cf. 1.6.C)},$$

then $X(\ell) = X_\kappa(\ell)$ for every ℓ, and therefore

$$X(\kappa_s) = X_\kappa(\kappa_s).$$

Now suppose that X is étale over k. Then X_κ is the formal κ-variety that is the disjoint union of the $\operatorname{Spec}\kappa(x)$, for $x \in X$. If X'_κ denotes the κ-scheme given by the same disjoint union, then

$$\begin{aligned} X_\kappa(\kappa_s) &= X'_\kappa(\kappa_s) \\ &= \operatorname{Hom}_{\mathrm{Sch}/\kappa}(\operatorname{Spec}\kappa_s, X'_\kappa). \end{aligned}$$

Let $\mathcal{C}$ be the full subcategory

$$\mathrm{Sch}^{\text{ét}}/\kappa \subset \mathrm{Sch}/\kappa$$

of étale κ-schemes. It is known that

$$X' \longmapsto X'(\kappa_s)$$

is an equivalence from $\mathcal{C}$ onto the category $\mathcal{C}'$ of Γ-sets (cf. SGA 1, V, §§7–8, or [*DG*70], §I.4, 6.4). It therefore induces an equivalence between the category of group objects in $\mathcal{C}$ and the category of group objects in $\mathcal{C}'$. A group object in $\mathcal{C}'$ is simply an abstract group endowed with a continuous action of Γ by group automorphisms; we call this a Γ-*group*.

Taking into account the equivalences of 1.6.E,

$$\mathrm{Vaf}^{\text{ét}}/k \xrightarrow{\sim} \mathrm{Vaf}^{\text{ét}}/\kappa \xrightarrow{\sim} \mathrm{Sch}^{\text{ét}}/\kappa,$$

we obtain the following.

Proposition. *Let k be a local pseudocompact ring, κ its residue field, κ_s a separable closure of κ, and $\Gamma = \mathrm{Gal}(\kappa_s/\kappa)$.*

(i) *The functor $X \mapsto X(\kappa_s)$ is an equivalence from the category of étale formal k-varieties onto the category of Γ-sets.*

(ii) *It induces an equivalence from the category of étale formal k-groups onto the category of Γ-groups.*

Remark 2.5.A. [96] Let k be a field, let k_s be a separable closure of k, let G be an étale formal k-group, and put

$$M = G(k_s).$$

[95] If $[\kappa_s : \kappa] = \infty$, then κ_s is not a profinite κ-algebra, so the preceding notation is an abuse. The editors also supplied the details that follow.

[96] The editors added this remark.

Choose a set X of representatives for the orbits of $\Gamma = \mathrm{Gal}(k_s/k)$ in M. For each $x \in X$, let Γ_x be its stabilizer, an open subgroup of finite index in Γ, and put

$$L_x = k_s^{\Gamma_x}.$$

Then L_x/k has degree $[\Gamma : \Gamma_x]$ (see, for example, [*BAlg*], V, §10.10).

By the preceding equivalence of categories, the affine algebra $\mathcal{A}(G)$ of G is the product of the L_x, endowed with the product topology. The L_x are therefore exactly the simple quotients

$$\mathcal{A}(G)/\mathfrak{m},$$

where $\mathfrak{m}$ ranges over the open maximal ideals of $\mathcal{A}(G)$. Since these quotients correspond by duality to the simple subcoalgebras of $\mathbf{H}(G)$, the coalgebra $\mathbf{H}(G)$ is pointed—that is,

$$\dim_k C = 1$$

for every simple subcoalgebra C—if and only if $L_x = k$ for every x. In that case, $\mathcal{A}(G)$ is the topological algebra k^M, so $\mathbf{H}(G)$ is the group algebra kM, and

$$M = \{x \in \mathbf{H}(G) \mid \varepsilon(x) = 1,\ \Delta(x) = x \otimes x\} = G(k).$$

2.5.1

Suppose again that k is an arbitrary pseudocompact ring. Since the functor

$$X \longmapsto X_e$$

of 1.6.F commutes with finite products, it sends every formal group G to an étale formal group G_e. The morphism $p : X \to X_e$ of that result is functorial in X, so in this case

$$p : G \longrightarrow G_e$$

is a morphism of formal groups.

Consider its kernel $\ker(p)$.[97] It is the fiber product of the diagram

$$\begin{array}{ccc} & & G \\ & & \big\downarrow p \\ \mathrm{Spf}(k) & \xrightarrow{\ \varepsilon\ } & G_e \end{array}$$

where ε is the identity section of G_e. Since p induces a bijection on the underlying sets, it follows (cf. 1.2, editorial note 41) that the underlying set of $\ker(p)$ is the image of $\mathrm{Spf}(k)$ under ε, and that, for every point s of $\mathrm{Spf}(k)$, the local algebra of $\ker(p)$ at $\varepsilon(s)$ is

$$\mathcal{O}_{G,\varepsilon(s)} \widehat{\otimes}_{\mathcal{O}_{G_e,\varepsilon(s)}} \kappa(s).$$

At each point $\varepsilon(s)$, the residue field of $\mathcal{O}_{G,\varepsilon(s)}$ is $\kappa(s)$, and hence

$$\mathcal{O}_{G_e,\varepsilon(s)} = \kappa(s), \qquad \mathcal{O}_{\ker(p),\varepsilon(s)} = \mathcal{O}_{G,\varepsilon(s)}.$$

For these reasons, we call $\ker(p)$ the *infinitesimal neighborhood of the identity* of G, and write

$$\ker(p) = G_0.$$

We have thus obtained the exact sequence

[97] The editors supplied the details of the original argument that follow.

$$1 \longrightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e.$$

In what follows, we say that G is *infinitesimal* if

$$G = G_0.$$

[98] This is equivalent to saying that, for every $s \in \mathrm{Spf}(k)$, or equivalently for every continuous morphism $k \to \kappa$, where κ is a field with the discrete topology, the identity is the unique element of $G(\kappa(s))$, respectively of $G(\kappa)$.

Suppose in addition that G is topologically flat over k.[99] By 1.6.1, the morphism

$$p : G \longrightarrow G_e$$

is topologically flat. Since it is bijective, Proposition 1.3.1 shows that it is an effective epimorphism. Consequently, G_e is identified with the quotient G/G_0. We have proved:

Corollary. *Let G be a formal group topologically flat over k. Then G_e is identified with the quotient G/G_0; in other words, there is an exact sequence of formal groups*

$$1 \longrightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e \longrightarrow 1.$$

2.5.2

Suppose that k is a perfect field. In this case, we saw in 1.6.2 that the morphism

$$p : X \longmapsto X_e$$

has a section

$$s : X_e \longrightarrow X$$

that depends functorially on X. Consequently, when X is a formal group, this section is a morphism of formal groups. We thus obtain:

Proposition. *When k is a perfect field, every formal k-group G is canonically isomorphic to the semidirect product of an infinitesimal group G_0 and an étale group G_e acting on G_0.*[100]

If, moreover, G is commutative, then G is the product of G_0 and G_e. By 2.2.2, this canonical decomposition of commutative formal k-groups corresponds to an analogous decomposition of affine commutative k-group schemes; see 2.5.3 below.

Remark 2.5.2.A. [101] When k is not perfect, the exact sequence

$$1 \longrightarrow G_0 \longrightarrow G \longrightarrow G_e \longrightarrow 1$$

need not split. Consider the following example, taken from [*DG*70], §III.6, 8.6. Let k be a nonperfect field of characteristic $p > 0$, choose

$$\lambda \in k \setminus k^p,$$

[98] The editors added the sentence that follows.

[99] The editors supplied the details that follow.

[100] The same argument also shows that, for an arbitrary field k, if all the residue fields of G are equal to k, then G_e is the constant k-group $(M)_k$, where

$$M = G(k) = \mathrm{Spf}^*\, \mathbf{H}(G),$$

and $\mathbf{H}(G)$ is the semidirect product of $\mathbf{H}(G_0)$ by the group algebra kM; see the addition in 2.9 below.

[101] The editors added this remark.

and let L_λ be the abelian restricted p-Lie algebra with basis (x, y), whose symbolic p-th power (cf. VII$_A$, 5.2) is given by

$$x^{(p)} = x, \qquad y^{(p)} = \lambda x.$$

Let

$$U_p(L_\lambda) = k[x, y]/(x^p - x,\ y^p - \lambda x)$$

be the restricted enveloping algebra of L_λ (cf. VII$_A$, 5.3), and let G_λ be the commutative formal k-group whose affine algebra is $U_p(L_\lambda)$.

Then G_λ is a nonsplit extension of the constant étale k-group $(\mathbb{Z}/p\mathbb{Z})_k$ by the infinitesimal k-group

$$\alpha_{p,k} = \operatorname{Spf} k[x]/(x^p).$$

Thus there is a nonsplit exact sequence of commutative formal k-groups:[102]

$$0 \longrightarrow \boldsymbol{\alpha}_{p,k} \longrightarrow G_\lambda \longrightarrow (\mathbb{Z}/p\mathbb{Z})_k \longrightarrow 0.$$

By duality, this corresponds to a nonsplit exact sequence of commutative algebraic k-groups:

$$0 \longrightarrow \boldsymbol{\mu}_{p,k} \longrightarrow D'(G_\lambda) \longrightarrow \boldsymbol{\alpha}_{p,k} \longrightarrow 0$$

where

$$\mu_{p,k} = \operatorname{Spec} k[t]/(t^p - 1).$$

In this way one obtains all extensions of $\alpha_{p,k}$ by $\mu_{p,k}$; see [*DG*70], §III.6, 8.6.

2.5.3

Let k be a field.[103]

Definition 2.5.3.A. A commutative k-group scheme is said to be *unipotent* if it is isomorphic to

$$\operatorname{Spec} \mathbf{H}(G),$$

where G is an infinitesimal commutative formal k-group.[104]

On the other hand, following Exposé IX, Definition 1.1,[105] a k-group scheme H is said to be *of multiplicative type* if there exists a scheme S, faithfully flat and quasi-compact over $\operatorname{Spec} k$, such that H_S is a diagonalizable S-group; that is,

$$H_S \simeq D_S(M)$$

for some "abstract" abelian group M. By fpqc descent, this implies that H is affine and commutative. Moreover, since S may be replaced by the residue field of one of its points, H is of multiplicative type if and only if there exists an extension K of k such that H_K is a diagonalizable K-group.

Proposition 2.5.3.B. *Let T be an affine commutative k-group scheme, and let k_s be a separable closure of k.*

[102]The groups $\alpha_{p,k}$, $(\mathbb{Z}/p\mathbb{Z})_k$, and G_λ are also finite flat algebraic k-groups; see editorial note 84 in 2.2.2.

[103]The editors introduced the numbering 2.5.3 and 2.5.3.A–2.5.3.C.

[104]This agrees with the "usual" notion of a unipotent k-group scheme; cf. the addition in 2.8 at the end of Section 2.

[105]The original stated: "Let us say that a commutative k-group scheme is of multiplicative type if it is isomorphic to $\operatorname{Spec} \mathbf{H}(G)$, where G is an étale commutative formal k-group." The editors have given here the "usual" definition, taken from Exposé IX, 1.1, and proved its equivalence with the preceding condition; see also [*DG*70], §IV.1, Theorem 2.2.

(i) *The scheme T is of multiplicative type if and only if its dual $D(T)$ is an étale commutative formal k-group.*

(ii) *Consequently, T is of multiplicative type if and only if*

$$T \otimes_k k_s$$

is diagonalizable.

Proof. Let $\mathcal{A}$ denote the affine algebra of $D(T)$, and let $\mathcal{A}_0$ be its local component at the identity. Then $D(T)$ is étale over k if and only if

$$\mathcal{A}_0 = k.$$

If K is an extension of k, then $\mathcal{A}_0 \widehat{\otimes}_k K$ is local, being an inverse limit of Artinian local rings, and hence coincides with the local component

$$(\mathcal{A}\widehat{\otimes}_k K)_0.$$

Moreover, since the formation of $D(T)$ commutes with base change (cf. 2.2.2), one also has

$$\mathcal{A}\widehat{\otimes}_k K \simeq \mathcal{A}(D(T_K)).$$

Suppose that T is of multiplicative type. Then there exists an extension K of k such that

$$\mathcal{O}(T) \otimes_k K$$

is isomorphic, as a Hopf K-algebra, to the group algebra KM for some abelian group M. It follows that $\mathcal{A}\widehat{\otimes}_k K$ is isomorphic to the algebra K^M, endowed with the product topology. Thus

$$K = (\mathcal{A}\widehat{\otimes}_k K)_0 = \mathcal{A}_0 \widehat{\otimes}_k K.$$

This implies that $\mathcal{A}_0 = k$, and hence that $D(T)$ is étale.

Conversely, suppose that $D(T)$ is étale. Then

$$D(T)\widehat{\otimes}_k k_s = D(T_{k_s})$$

is the constant k_s-group associated with a group M. Therefore its covariant bialgebra is the group algebra $k_s M$ (cf. Remark 2.5.A), and consequently

$$\mathcal{O}(T) \otimes_k k_s = k_s M.$$

This proves that T is of multiplicative type and is split by the extension $k \to k_s$. The proposition follows.

By 2.2.2, 2.5.1, and 2.5, we obtain:

Corollary 2.5.3.C. *Let k be a field, and let G be an affine commutative k-group scheme.*

(i) *The group G contains a subgroup G_m of multiplicative type such that G/G_m is unipotent.*

(ii) *When k is perfect, there is in addition a canonical retraction of G onto G_m, so that G is the product of a unipotent group and a group of multiplicative type.*

(iii) [106] *Let k_s be a separable closure of k, and put*

$$\Gamma = \mathrm{Gal}(k_s/k).$$

The category of k-group schemes of multiplicative type is anti-equivalent to the category of continuous Γ-modules.

[106] The editors added item (iii), a consequence of Proposition 2.5.

2.6

We shall now study infinitesimal formal groups, to which the sections that follow are devoted. In this study, Lie algebras play a fundamental role.

Suppose first that the base ring k is Artinian, and let G be a formal group over k. The Lie algebra of G admits three different interpretations, all of which we shall use:

a) Let

$$D = k[d]/(d^2)$$

be the algebra of dual numbers over k, and let

$$\delta : D \longrightarrow k$$

be the homomorphism that annihilates d. For every formal group G over k, $\mathrm{Lie}(G)$ is the kernel of $G(\delta)$. Thus, by definition, there is an exact sequence of groups

$$1 \longrightarrow \mathrm{Lie}(G) \longrightarrow G(D) \xrightarrow{G(\delta)} G(k) \longrightarrow 1.$$

b) Let A be the affine algebra of G, and let

$$I_A = \ker \varepsilon_A$$

be its augmentation ideal. The elements of $G(D)$ are the morphisms of profinite k-algebras

$$f : A \longrightarrow D.$$

The condition $G(\delta)(f) = 1$ is equivalent to

$$\delta \circ f = \varepsilon_A.$$

Since

$$x - \varepsilon_A(x) \cdot 1_A \in I_A \qquad \text{for every } x \in A,$$

this is equivalent to $f(I_A) \subset k \cdot d$. Hence $\ker(f)$ contains I_A^2, and therefore also its closure $\overline{I_A^2}$. Thus f induces a continuous linear map

$$f' : I_A/\overline{I_A^2} = \omega_{G/k} \longrightarrow k$$

such that, for every $x \in A$,

$$f(x) = \varepsilon_A(x) \cdot 1_D + f'(\bar{x}) \cdot d,$$

where $\bar{x}$ denotes the image of $x - \varepsilon_A(x) \cdot 1_A$ in $I_A/\overline{I_A^2}$. It follows that the map

$$f \longmapsto f'$$

defines a bijection from $\mathrm{Lie}(G)$ onto the topological dual

$$\omega_{G/k}^{\dagger}$$

of $\omega_{G/k}$ (cf. 0.2.2).

This bijection respects the group structures. Indeed, let f and g be two elements of $\mathrm{Lie}(G)$. Their product $f \cdot g$ is the composite $h \circ \Delta_A$, where

$$h : A \widehat{\otimes} A \longrightarrow D$$

is defined by

$$h(b\widehat{\otimes}b') = f(b)g(b').$$

If $a \in I_A$, then, by 2.1(ii),

$$\Delta_A(a) - a\widehat{\otimes}1 - 1\widehat{\otimes}a \in I_A\widehat{\otimes}I_A.$$

Consequently,

$$(f \cdot g)(a) = f(a) + g(a)$$

(cf. also II, 3.10).

Henceforth we identify

$$\mathrm{Lie}(G) = \omega^{\dagger}_{G/k}$$

by means of the bijection $f \mapsto f'$ just described. The group $\mathrm{Lie}(G)$ is thereby endowed with a k-module structure.

c) Let $A^{\dagger}$ and $D^{\dagger}$ be the k-modules dual to A and D, and let

$$\{1^{\dagger}_D, d^{\dagger}\}$$

be the basis dual to the basis $\{1_D, d\}$ of D over k; thus $1^{\dagger}_D = \delta$. Since D is free of finite rank over k, the canonical map

$$\mathrm{Hom}_c(A, D) \longrightarrow \mathrm{Hom}_k(D^{\dagger}, A^{\dagger}), \qquad f \longmapsto {}^t f,$$

is bijective. On the other hand, ${}^t f$ is determined by the values

$${}^t f(1^{\dagger}_D) \qquad \text{and} \qquad {}^t f(d^{\dagger}) = x.$$

The condition $G(\delta)(f) = 1$ is equivalent to

$${}^t f(1^{\dagger}_D) = \varepsilon_A.$$

Moreover, one sees readily that f is compatible with multiplication if and only if (cf. 2.3)

$$\delta_G(x) = \phi_G(x \otimes 1 + 1 \otimes x). \qquad (*)$$

Finally, it is clear that a continuous linear map

$$f : A \longrightarrow D$$

that is compatible with multiplication and satisfies $\delta \circ f = \varepsilon_A$ sends the unit element of A to that of D.[107] The map $f \mapsto x$ therefore allows us to identify $\mathrm{Lie}(G)$ with the set of "primitive elements" of $\mathbf{H}(G)$, that is, the elements $x \in \mathbf{H}(G)$ satisfying relation (*). If x and y are two such elements, then

$$\begin{aligned}\delta_G(xy) &= \delta_G(x)\delta_G(y)\\ &= \phi_G\big((x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y)\big)\\ &= \phi_G\big(xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy\big),\end{aligned}$$

whence

$$\delta_G(xy - yx) = \phi_G\big((xy - yx) \otimes 1 + 1 \otimes (xy - yx)\big).$$

This shows that the k-module $\mathrm{Lie}(G)$ is identified with a Lie subalgebra of $\mathbf{H}(G)$; we call $\mathrm{Lie}(G)$ the *Lie algebra* of G.[108]

[107] Indeed, $\delta \circ f = \varepsilon_A$ implies $f(1) = 1 + \lambda d$, with $\lambda \in k$, and then

$$f(1) = f(1)^2 = 1 + 2\lambda d$$

gives $\lambda = 0$.

[108] Comparing with VII$_A$, 2.5, we see that if K is a group scheme of finite type over k and G is the formal

2.6.1

Let k now be an arbitrary pseudocompact ring, and let G be a formal group over k. We call the functor $\mathrm{Lie}(G)$ that assigns to every object C of $\mathrm{Alf}\,/k$ the C-Lie algebra of the C-formal group

$$G' = G\widehat{\otimes}_k C$$

the $\mathbf{O}_k$-*Lie algebra of* G.[109] Put

$$A' = A\widehat{\otimes}_k C.$$

Since I_A is a direct factor of A, we have

$$I_{A'} = I_A\widehat{\otimes}_k C = I_A\widehat{\otimes}_A A'.$$

Moreover,

$$\omega_{G/k} = I_A\widehat{\otimes}_A k, \qquad \omega_{G'/C} = I_{A'}\widehat{\otimes}_{A'} C.$$

It follows that

$$\omega_{G'/C} = \omega_{G/k}\widehat{\otimes}_A C,$$

and hence

$$\mathrm{Lie}(G)(C) = \mathrm{Hom}_c(\omega_{G/k}\widehat{\otimes}_A C, C),$$

$$\text{i.e.} \qquad \mathrm{Lie}(G) = \mathbf{V}_k^f(\omega_{G/k}).$$

With the notation of 1.2.3.B, Proposition 1.2.3.E therefore shows that $\mathrm{Lie}(G)$ is flat over $\mathbf{O}_k$ if and only if $\omega_{G/k}$ is a projective pseudocompact k-module.

2.6.2

Conversely, every Lie algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a k-group functor. Indeed, let $U(\mathbf{L})$ denote the functor assigning to every object C of $\mathrm{Alf}\,/k$ the enveloping algebra

$$U\big(\mathbf{L}(C)\big)$$

of the C-Lie algebra $\mathbf{L}(C)$. By VII$_\mathrm{A}$, 3.2.2, $U(\mathbf{L})$ is a bialgebra over $\mathbf{O}_k$, and hence, by 2.2, determines the k-group functor

$$\mathrm{Spf}^*\, U(\mathbf{L}),$$

which from now on we denote by $\mathscr{G}(\mathbf{L})$. Thus $\mathscr{G}(\mathbf{L})(C)$ is the group of elements $z \in U(\mathbf{L}(C))$ having augmentation 1 and satisfying

$$\Delta_{U(\mathbf{L}(C))}(z) = z \otimes z.$$

When $\mathbf{L}$ is flat over $\mathbf{O}_k$, we have the following proposition.

Proposition.[110] *Let* $\mathbf{L}$ *be a flat Lie algebra over* $\mathbf{O}_k$.

---

completion of K at the identity—that is, if $\mathcal{A}(G)$ is the completion of the local ring $\mathcal{O}_{K,e}$ for the $\mathfrak{m}$-adic topology, where

$$\mathfrak{m} = \ker\big(\varepsilon : \mathcal{O}_{K,e} \longrightarrow k\big)$$

—then $\mathbf{H}(G)$ is identified with the algebra of distributions $U(K)$, and $\mathrm{Lie}(G)$ with $\mathrm{Lie}(K)$. The hypothesis that K is of finite type over k is used to ensure that $\mathfrak{m}/\mathfrak{m}^2$ is a k-module of finite length, and hence discrete, so that its topological dual coincides with its ordinary dual. In particular, when G is a finite formal k-group—that is, when $\mathcal{A}(G)$ is a finite k-module—in which case G may also be regarded as the k-group scheme $\mathrm{Spec}\,\mathcal{A}(G)$, the two definitions of $\mathrm{Lie}(G)$ coincide.

[109] The editors expanded the original in what follows.

[110] The editors set off parts (i) and (ii) and added part (iii), which will be used in 2.6.3 and 3.3.2.

(i) *$\mathscr{G}(\mathbf{L})$ is a formal group topologically flat over k, whose $\mathbf{O}_k$-bialgebra is $U(\mathbf{L})$.*

(ii) *$\mathscr{G}(\mathbf{L})$ is infinitesimal.*

(iii) *For every object C of* Alf $/k$, $\mathrm{Lie}(\mathscr{G}(\mathbf{L}))(C)$ *is identified with the set*

$$\operatorname{Prim} U(\mathbf{L}(C)) = \{x \in U(\mathbf{L}(C)) \,|\, \varepsilon(x) = 0, \quad \Delta(x) = x \otimes 1 + 1 \otimes x\}$$

of primitive elements of $U(\mathbf{L}(C))$. In particular, there is a natural morphism of Lie algebras over $\mathbf{O}_k$,

$$\tau_{\mathbf{L}} : \mathbf{L} \longrightarrow \mathrm{Lie}(\mathscr{G}(\mathbf{L})).$$

Proof. The hypothesis that $\mathbf{L}$ is flat over $\mathbf{O}_k$ means that, for every morphism $C \to D$ in Alf $/k$,

$$\mathbf{L}(D) = \mathbf{L}(C) \otimes_C D,$$

and that, for every local component C' of C, the C'-module $\mathbf{L}(C')$ is free. The first condition, together with the universal properties of tensor products and enveloping algebras, gives

$$U(\mathbf{L}(D)) = U(\mathbf{L}(C)) \otimes_C D.$$

The second condition and the Poincaré–Birkhoff–Witt theorem (cf. Bourbaki, *Groupes et algèbres de Lie*, I, 2.7) show that $U(\mathbf{L}(C'))$ is a free C'-module. Consequently, the bialgebra $U(\mathbf{L})$ is flat over $\mathbf{O}_k$, proving (i).

To prove (ii) and (iii), we may suppose that k is Artinian. Put

$$L = \mathbf{L}(k), \qquad U = U(L), \qquad U_0 = k \cdot 1_U,$$

and let U^+ be the two-sided ideal of U generated by the image of L. For every $n \geq 1$, put

$$U_n = \left\{ u \in U \,\middle|\, \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+ \right\}.$$

By 1.3.6, it is enough to show that U is the union of the U_n. Identify L with its canonical image in U. Then $L \subset U_1$. If $x_1, \ldots, x_n \in L$, then

$$\Delta_U(x_1 \cdots x_n) = (x_1 \otimes 1 + 1 \otimes x_1) \cdots (x_n \otimes 1 + 1 \otimes x_n).$$

Induction on n therefore shows that $x_1 \cdots x_n \in U_n$, and hence

$$U = \bigcup_n U_n.$$

This proves (ii).

Finally, let

$$D = k[d]/(d^2)$$

be the algebra of dual numbers over k. By hypothesis,

$$\mathbf{L}(D) \simeq L \otimes_k D, \qquad U(\mathbf{L}(D)) \simeq U \otimes_k D.$$

It follows that $\mathrm{Lie}(\mathscr{G}(\mathbf{L}))(k)$ is identified with the set of elements

$$z = 1 + xd \in U \oplus Ud \qquad (x \in U)$$

such that $\varepsilon(z) = 1$ and $\Delta(z) = z \otimes z$. These conditions are equivalent to

$$\varepsilon(x) = 0, \qquad \Delta(x) = x \otimes 1 + 1 \otimes x,$$

that is, to $x \in \operatorname{Prim} U$. In particular,

$$\tau_{\mathbf{L}} : x \longmapsto 1 + dx$$

is a morphism of Lie algebras over $\mathbf{O}_k$ from $\mathbf{L}$ to $\mathrm{Lie}(\mathscr{G}(\mathbf{L}))$.

2.6.3

If $\mathbf{L}$ is a flat Lie algebra over $\mathbf{O}_k$, the formal group $\mathscr{G}(\mathbf{L})$ can be characterized by a universal property.[111] Indeed, every morphism

$$\phi : \mathscr{G}(\mathbf{L}) \longrightarrow G$$

to a formal group G induces a morphism

$$\mathrm{Lie}(\phi) : \mathrm{Lie}(\mathscr{G}(\mathbf{L})) \longrightarrow \mathrm{Lie}(G).$$

Composing it with

$$\tau_{\mathbf{L}} : \mathbf{L} \longrightarrow \mathrm{Lie}(\mathscr{G}(\mathbf{L}))$$

(cf. 2.6.2) gives a morphism

$$\phi' : \mathbf{L} \longrightarrow \mathrm{Lie}(G).$$

We then have:

Proposition. *Let G be a formal k-group and let $\mathbf{L}$ be a flat Lie algebra over $\mathbf{O}_k$. The map $\phi \mapsto \phi'$ defined above is a bijection*

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(\mathscr{G}(\mathbf{L}), G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Lie}}(\mathbf{L}, \mathrm{Lie}(G)),$$

where the right-hand side is the set of morphisms of Lie algebras over $\mathbf{O}_k$ from $\mathbf{L}$ to $\mathrm{Lie}(G)$.

Proof. We may immediately reduce to the case in which k is Artinian. Put $L = \mathbf{L}(k)$. By 2.3.1,

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(\mathscr{G}(\mathbf{L}), G)$$

is in bijection with the set of unital algebra morphisms

$$h : U(L) \longrightarrow \mathbf{H}(G)$$

for which the following diagram is commutative:

$$\begin{array}{ccccc} U(L) & \xrightarrow{\quad h \quad} & \mathbf{H}(G) & & \\ & & & \searrow^{\delta_G} & \\ \downarrow{\scriptstyle \Delta_{U(L)}} & & & & \mathbf{H}(G \times G) \\ & & & \nearrow_{\varphi_G} & \\ U(L) \otimes U(L) & \xrightarrow{\quad h \otimes h \quad} & \mathbf{H}(G) \otimes \mathbf{H}(G) & & \end{array} \quad .$$

Now h is determined by its restriction to L, which is a morphism from L to the Lie algebra underlying $\mathbf{H}(G)$. Commutativity of the diagram says precisely that h sends L into the subset of $\mathbf{H}(G)$ formed by the elements x satisfying

$$\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x).$$

This subset is exactly $\mathrm{Lie}(G)$; cf. 2.6(c). The proposition follows.

[111] The editors modified what follows to take advantage of the addition made in 2.6.2.

2.7

We conclude these generalities with a statement that goes back to S. Lie and that will be used in 5.1. A *formal monoid* over k is, by definition, a pair (M, m) consisting of a formal variety M and a morphism

$$m : M \times M \longrightarrow M$$

such that $m(C)$ makes $M(C)$ a monoid for every object C of Alf $/k$.[112] In particular, the "unit section," which assigns to every object C the identity element of $M(C)$, defines a section

$$\varepsilon_M : \mathrm{Spf}(k) \longrightarrow M$$

of the canonical projection $M \to \mathrm{Spf}(k)$. We shall say that the formal monoid M is *infinitesimal* if ε_M induces a bijection of the underlying sets.

Proposition. *Every topologically flat infinitesimal formal k-monoid M is a formal k-group.*[113]

Proof. We must show that $M(C)$ is a group for every object C of Alf $/k$. We therefore reduce immediately to the case in which k is Artinian. Let

$$U = \mathbf{H}(M)$$

be the coalgebra of M (1.3.5). The multiplication

$$m : M \times M \longrightarrow M$$

induces a morphism of coalgebras

$$m_U : U \otimes U \longrightarrow U$$

that makes U an associative algebra over k. The identity element of this algebra is the image of the identity element of k under the map $k \to U$ induced by the unit section ε_M of M. Likewise, the projection

$$M \longrightarrow \mathrm{Spf}(k)$$

induces a homomorphism

$$\varepsilon_U : U \longrightarrow k;$$

we write U^+ for the kernel of ε_U.

We must show that there is an antipode, that is, a morphism of coalgebras

$$c_U : U \longrightarrow U$$

such that, for every $u \in U$,

$$(m_U \circ (c_U \otimes \mathrm{id}_U) \circ \Delta_U)(u) = \varepsilon_U(u)\, 1_U. \qquad (*)$$

[112] Here and in what follows, the editors have written "monoid" instead of "monoid with identity element" (recall that a monoid is, by definition, a set equipped with an associative composition law and possessing an identity element).

[113] By the equivalence of categories in 1.3.5.D, a monoid in the category of topologically flat formal k-varieties "is the same thing" as a monoid in the category of flat cocommutative $\mathbf{O}_k$-coalgebras, that is, a cocommutative $\mathbf{O}_k$-bialgebra (in the usual sense, hence not necessarily equipped with an antipode). Moreover, by 1.3.6, the hypothesis that M be infinitesimal is equivalent to saying that the corresponding bialgebra is connected. Thus, if k is an Artinian ring, the proposition is equivalent to the following statement: every connected cocommutative k-bialgebra that is flat over k is a k-Hopf algebra, that is, it possesses an antipode. In fact, the cocommutativity hypothesis is superfluous; cf. the proof.

Let (U_n) be the filtration of U defined in 1.3.6, and put

$$U_n^+ = U^+ \cap U_n.$$

Since M is infinitesimal, U^+ is the union of the submodules U_n^+.[114] Set

$$c_0(1) = 1$$

and

$$c_1(x) = -x \qquad \text{if } x \in U_1^+,$$

that is, if x is primitive. Suppose that

$$c_{n-1} : U_{n-1} \longrightarrow U$$

has been constructed so as to satisfy (*), with c_{n-1} in place of c_U, for every $x \in U_{n-1}$, and let $x \in U_n^+$. By the proof of Lemma 1.3.6.A,

$$\Delta_U(x) - x \otimes 1 \in U_{n-1} \otimes U^+;$$

this is where the hypothesis that U is flat over k enters. We may therefore write

$$\Delta_U(x) = x \otimes 1 + \sum_i y_i \otimes z_i, \qquad y_i \in U_{n-1},$$

and then set

$$c_n(x) = -\sum_i c_{n-1}(y_i) z_i.$$

Together with $c_n(1) = 1$, this prescription extends c_{n-1} coherently to U_n. In this way we obtain a k-linear map

$$c : U \longrightarrow U$$

that is the left inverse of id_U for the monoid law on $\mathrm{End}_k(U)$ defined by

$$f \cdot g = m_U \circ (f \otimes g) \circ \Delta_U;$$

the identity element is the map

$$\eta : u \longmapsto \varepsilon_U(u)\, 1_U.$$

It follows that c is uniquely determined and is also the right inverse of id_U; that is,

$$m_U \circ (\mathrm{id}_U \otimes c) \circ \Delta_U = \eta,$$

without any assumption that U be cocommutative. Thus $c_U := c$ is the required antipode, and M is a formal k-group.

[114] The original continued as follows: "one then shows easily, by induction on n, that there exists one and only one linear map $c_n : U_n \to U_n$ such that the composite

$$m_U \circ (c_n \otimes \mathrm{id}_U) \circ \Delta_U : U_n^+ \longrightarrow U$$

is zero." The editors have supplied the details of the proof, which rests on that of Lemma 1.3.6.A.

2.8. Unipotent group schemes over a field

115

Let k be a field. Recall that a k-group scheme G is called *unipotent* if it satisfies the following two conditions (cf. [DG70], § IV.2, Proposition 2.5):

(a) G is affine.

(b) Every simple $\mathcal{O}(G)$-comodule is trivial. Explicitly, if

$$\rho : V \longrightarrow V \otimes_k \mathcal{O}(G)$$

is an $\mathcal{O}(G)$-comodule structure on a nonzero k-vector space V, and there is no nonzero proper subspace $W \subset V$ such that

$$\rho(W) \subset W \otimes_k \mathcal{O}(G),$$

then V is one-dimensional and

$$\rho(v) = v \otimes 1 \qquad \text{for every } v \in V.$$

According to the cited result, when G is of finite type over k, this is equivalent to the definition given in Exposé XVII, § 1: G has a finite composition series whose successive quotients are isomorphic to k-subgroups of $\mathbb{G}_{a,k}$.

For every affine k-group scheme G, the comultiplication of $\mathcal{O}(G)$ endows the pseudo-compact k-module

$$A = \mathcal{O}(G)^*$$

with the structure of a profinite k-algebra, not necessarily commutative; its unit 1_A is the augmentation

$$\varepsilon : \mathcal{O}(G) \longrightarrow k.$$

Set

$$I = \{f \in A \mid f(1_{\mathcal{O}(G)}) = 0\}.$$

Then I is a two-sided ideal of A, and

$$A = k \cdot 1_A \oplus I;$$

cf. 1.3.6.

Let V be a finite-codimensional subspace of A, and consider the continuous k-bilinear map

$$\varphi : A \times A \longrightarrow A/V, \qquad (a, b) \longmapsto ab + V.$$

By Lemma 0.3.1, there are finite-codimensional subspaces $L_1, L_2 \subset A$ such that V contains AL_2 and L_1A. Thus

$$L = L_1 \cap L_2$$

is finite-codimensional, and V contains the two-sided ideal ALA, which is itself finite-codimensional. Hence the finite-codimensional two-sided ideals form a neighborhood basis of zero in A.

It follows that an $\mathcal{O}(G)$-comodule is the same thing as a continuous A-module: an A-module V, equipped with the discrete topology, for which the map

$$A \times V \longrightarrow V$$

[115] The editors added this subsection.

is continuous. Every such module is the union of finite-dimensional A-submodules V_i, each of which is a module over a finite-dimensional quotient k-algebra A_i of A.

Consequently, a simple continuous A-module M is finite-dimensional over k and is a faithful simple module over the finite-dimensional algebra $A/\operatorname{Ann}(M)$. The latter is therefore a finite-dimensional simple k-algebra, so $\operatorname{Ann}(M)$ is an open maximal ideal of A.

Conversely, let P be an open prime ideal of A.[116] Then A/P is a finite-dimensional k-algebra in which the zero ideal is prime, and hence is a finite-dimensional simple k-algebra. Up to isomorphism, there is therefore a unique simple continuous A-module with annihilator P. Thus

$$M \longmapsto \operatorname{Ann}(M)$$

is a bijection from the isomorphism classes of simple continuous A-modules to the open prime ideals of A.

In particular, the A-module A/I, which is one-dimensional over k, is called the *trivial module*. It corresponds to the trivial one-dimensional $\mathcal{O}(G)$-comodule V, for which

$$\rho(v) = v \otimes 1_{\mathcal{O}(G)} \qquad \text{for every } v \in V.$$

We have proved:

Proposition 2.8.1. *Let k be a field, let G be an affine k-group scheme, and put $A = \mathcal{O}(G)^*$.*

(i) *G is unipotent if and only if I is the unique open prime ideal of A.*[117]

(ii) *In particular, suppose that G is commutative, so that*

$$\mathcal{O}(G) = \mathbf{H}(D(G)),$$

where $D(G) = \operatorname{Spf}(A)$ is the Cartier dual of G. Then G is unipotent if and only if $D(G)$ is infinitesimal.

2.9. Pointed cocommutative Hopf algebras over a field

[118]

Let k be a field, let C be a k-coalgebra, and endow

$$A = C^*$$

with the structure of a profinite k-algebra, not necessarily commutative, described in 2.8. By 0.2.2,

$$C = A^\dagger = \operatorname{Hom}_c(A, k).$$

The map

$$D \longmapsto D^\perp = \{f \in A \mid f(D) = 0\}$$

is therefore a bijection from the subcoalgebras of C to the closed two-sided ideals of A (which we shall simply call ideals below). Its inverse is

$$I \longmapsto I^\perp = \{x \in C = A^\dagger \mid x(I) = 0\}.$$

[116] Recall that a two-sided ideal P of a ring A is called prime if, in A/P, the product of any two nonzero two-sided ideals is nonzero.

[117] Equivalently, $k \cdot 1_{\mathcal{O}(G)}$ is the unique simple k-subcoalgebra of $\mathcal{O}(G)$; see, for example, [Ab80], 3.1.4. The editors also note a misprint in the cited work, p. 130, line 4: $M \simeq C^*/\operatorname{ann} M$ should be replaced by $C^*/\operatorname{ann} M \simeq \operatorname{End}_k(M)$.

[118] The editors added this subsection to express Proposition 2.5.2, together with the variant noted in editorial note 100, in the language of cocommutative Hopf algebras.

Since every maximal closed ideal is a maximal open ideal (cf. 0.2.1), every subcoalgebra contains a simple subcoalgebra, necessarily finite-dimensional.

Recall that a subcoalgebra $D \subset C$ is called *irreducible* if it contains a unique simple subcoalgebra S_0. Equivalently,

$$\mathfrak{m}_0 = S_0^{\perp}$$

is the unique open maximal ideal containing $D^{\perp}$, or, equivalently,

$$D^{\perp} + \mathfrak{m} = A \qquad \text{for every open maximal ideal } \mathfrak{m} \neq \mathfrak{m}_0.$$

This applies in particular when $D = S_0$.

Let Σ_0 be the sum of all irreducible subcoalgebras C_i containing S_0. It is plainly a subcoalgebra, and it is irreducible because, for every $\mathfrak{m} \neq \mathfrak{m}_0$, 0.2.D gives

$$\mathfrak{m} + \bigcap_i C_i^{\perp} = \bigcap_i (\mathfrak{m} + C_i^{\perp}) = A.$$

The coalgebra Σ_0 is called the *irreducible component of C corresponding to S_0*.

A coalgebra C is called *pointed* if every simple k-subcoalgebra of C is one-dimensional. Equivalently,

$$A/\mathfrak{m} = k$$

for every open maximal ideal $\mathfrak{m}$ of A. Recall also that C is called *connected* if it is irreducible and pointed. If C is a bialgebra, then it is connected if and only if it is irreducible, since $k \cdot 1_C$ is a simple subcoalgebra.

Suppose from now on that C is cocommutative. Then A is commutative and hence is the product of its local components

$$A_{\mathfrak{m}}, \qquad \mathfrak{m} \in \Upsilon(A);$$

cf. 0.1.1. Put

$$S_{\mathfrak{m}} = \mathfrak{m}^{\perp} \simeq (A/\mathfrak{m})^*,$$

and let $\Sigma_{\mathfrak{m}}$ be its irreducible component. This component can be described explicitly. Let

$$J_{\mathfrak{m}} = \ker(A \longrightarrow A_{\mathfrak{m}}).$$

Then $J_{\mathfrak{m}} \subset \mathfrak{m}$, and it is the smallest closed ideal $I \subset \mathfrak{m}$ satisfying

$$I + \mathfrak{m}' = A \qquad \text{for every open maximal ideal } \mathfrak{m}' \neq \mathfrak{m}.$$

Indeed, any such I contains $A_{\mathfrak{m}'}$ for every $\mathfrak{m}' \neq \mathfrak{m}$, and hence contains $J_{\mathfrak{m}}$. Since

$$A = J_{\mathfrak{m}} \oplus A_{\mathfrak{m}},$$

we obtain

$$\Sigma_{\mathfrak{m}} = J_{\mathfrak{m}}^{\perp} \simeq A_{\mathfrak{m}}^{\dagger}.$$

We can now prove:

Proposition 2.9.1. *Let k be a field.*

(i) *Let G be a formal k-group all of whose residue fields are equal to k. Then G_e is the constant k-group M_k, where*

$$M = G(k) = \{x \in \mathbf{H}(G) \,|\, \varepsilon(x) = 1, \quad \Delta(x) = x \otimes x\},$$

and $\mathbf{H}(G)$ is the semidirect product of $\mathbf{H}(G_0)$ by the group algebra kM (cf. 2.4.B).

(ii) *Equivalently, let H be a pointed cocommutative Hopf k-algebra. Then H is the semidirect product of the irreducible component H_0 of the unit 1_H by kM, where*

$$M = \{x \in H \mid \varepsilon(x) = 1,\ \Delta(x) = x \otimes x\}.$$

Proof. We first prove (i). Since every residue field of G is equal to k, the projection

$$\pi : G \longrightarrow G_e$$

has a section $s : G_e \to G$, defined at every point $g \in G$ by

$$\mathcal{O}_{G,g} \longrightarrow \kappa(g) = k.$$

Moreover, for all $g, h \in G$, the algebra

$$\mathcal{O}_{G,g} \widehat{\otimes}_k \mathcal{O}_{G,h}$$

is local with residue field k. Thus $s \times s$ is a section of

$$\pi \times \pi : G \times G \longrightarrow (G \times G)_e = G_e \times G_e.$$

Since π is a group morphism,

$$\pi \circ m_G \circ (s \times s) = m_{G_e} = \pi \circ s \circ m_{G_e}.$$

As π is an epimorphism, it follows that s is a group morphism. Hence

$$G = G_0 \rtimes G_e,$$

and $\mathbf{H}(G)$ is the semidirect product of $\mathbf{H}(G_0)$ by $\mathbf{H}(G_e)$.

Since every residue field of G is k, $\mathbf{H}(G_e)$ is the group algebra kM, where

$$M = G_e(k);$$

cf. 2.5.A. Finally, because G_0 is infinitesimal, the map

$$G(k) \longrightarrow G_e(k)$$

is injective. It is also surjective, since it has the section induced by s. Thus $M = G(k)$, proving (i).

For (ii), it remains only to show that

$$H_0 = \mathbf{H}(G_0).$$

The unit 1_H of H is the augmentation

$$\varepsilon_{\mathcal{A}} : \mathcal{A} \longrightarrow k,$$

which is the identity section of $G(k)$. Therefore the local component of the affine algebra $\mathcal{A}(G)$ corresponding to H_0 is precisely $\mathcal{A}(G_0)$. By the description above,

$$H_0 = \mathcal{A}(G_0)^\dagger = \mathbf{H}(G_0).$$

This proves the proposition.

Remarks 2.9.2.

(a) The preceding proposition, which is implicit in 2.5.2, was obtained independently by B. Kostant (cf. [Sw69], Preface). Combined with Cartier's Theorem 3.3 below (cf. [Ca62], § 12, Theorem 3), which was also obtained by Kostant (cf. [Sw69], loc. cit.), this result is often called the *Cartier–Gabriel–Kostant theorem*.

(b) In the form (ii), Proposition 2.9.1 was extended by R. G. Heyneman and M. E. Sweedler to the case in which H is assumed to be pointed and the direct sum of its irreducible components, but is not necessarily cocommutative; see [HS69], Theorem 3.5.8.

3. Phenomena specific to characteristic 0

Throughout Section 3, we assume that the pseudocompact ring k contains the field of rational numbers $\mathbb{Q}$.

3.1. Lemma

Lemma 3.1. *Let C be a commutative unital $\mathbb{Q}$-algebra, and let L be a Lie algebra over C whose underlying C-module is free. Then the canonical map*

$$L \longrightarrow U(L)$$

identifies L with the set of primitive elements of $U(L)$.

Proof. Identify L with its canonical image in $U(L)$. Let I be a totally ordered set, and let

$$(x_i)_{i\in I}$$

be a basis of L indexed by I. Denote by $\mathbb{N}^{(I)}$ the set of families

$$n = (n_i)_{i\in I}$$

of nonnegative integers for which $n_i = 0$ except possibly at finitely many indices

$$i_1 < i_2 < \cdots < i_s;$$

these indices depend on n. Put

$$x^n = x_{i_1}^{n_{i_1}} x_{i_2}^{n_{i_2}} \cdots x_{i_s}^{n_{i_s}}, \qquad n! = (n_{i_1}!)(n_{i_2}!)\cdots(n_{i_s}!).$$

By the Poincaré–Birkhoff–Witt theorem, the elements x^n form a basis of $U(L)$. Moreover,

$$\Delta_{U(L)}\left(\frac{x^n}{n!}\right) = \sum_{0\le m\le n} \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!}. \qquad (*)$$

Here the sum ranges over all $m \in \mathbb{N}^{(I)}$ such that $0 \le m \le n$, that is,

$$0 \le m_i \le n_i \qquad \text{for every } i.$$

It follows immediately that

$$\Delta_{U(L)}(u) = u \otimes 1 + 1 \otimes u$$

if and only if u is a linear combination of the x_i. This proves the lemma.

3.2

Suppose now that C is Artinian, with radical $\mathfrak{r}$. For every C-algebra U (associative and unital), the ideal $\mathfrak{r}U$ therefore consists of nilpotent elements. If $x \in \mathfrak{r}U$, put

$$\exp_U(x) = 1 + x + \frac{x^2}{2!} + \cdots .$$

[119] This gives a bijection from $\mathfrak{r}U$ onto $1 + \mathfrak{r}U$; its inverse sends $1 + y \in 1 + \mathfrak{r}U$ to

$$\log_U(1 + y) = y - \frac{y^2}{2} + \frac{y^3}{3} - \cdots .$$

[119] If $x, x' \in \mathfrak{r}U$ commute, then

$$\exp_U(x + x') = \exp_U(x)\exp_U(x').$$

Moreover, the map $\exp_U$ is plainly functorial in U.[120]

Still assuming C Artinian, suppose that U is endowed with a bialgebra structure over C (cf. 2.2). For every primitive element $x \in \mathfrak{r}U$ (cf. VII$_A$, 3.2.3), we then have

$$\begin{aligned}\Delta_U\big(\exp_U(x)\big) &= \exp_{U\otimes_C U}\big(\Delta_U(x)\big)\\ &= \exp_{U\otimes_C U}(x\otimes 1 + 1\otimes x)\\ &= \exp_{U\otimes_C U}(x\otimes 1)\ \exp_{U\otimes_C U}(1\otimes x)\\ &= \big(\exp_U(x)\otimes 1\big)\big(1\otimes \exp_U(x)\big)\\ &= \exp_U(x)\otimes \exp_U(x).\end{aligned}$$

Thus the bijection $\exp_U$ sends a primitive element of $\mathfrak{r}U$ to an element $z \in 1+\mathfrak{r}U$ satisfying

$$\Delta_U(z) = z\otimes z.$$

Conversely, if z satisfies these conditions, then, on putting $x = \log_U(z)$, the preceding calculation shows that

$$\exp_{U\otimes_C U}(x\otimes 1 + 1\otimes x) = z\otimes z = \Delta_U\big(\exp_U(x)\big) = \exp_{U\otimes_C U}\big(\Delta_U(x)\big);$$

hence

$$x\otimes 1 + 1\otimes x = \Delta_U(x).$$

[121] Furthermore, if $z \in 1+\mathfrak{r}U$ satisfies $\Delta_U(z) = z\otimes z$, then

$$\varepsilon_U(z)^2 = \varepsilon_U(z).$$

Since $\varepsilon_U(z)$ is invertible (because z is invertible, and $\mathfrak{r}$ is nilpotent), it follows that $\varepsilon_U(z) = 1$.

In particular, let L be a Lie algebra flat over C, take for U the enveloping algebra $U(L)$ of L over C, and identify L with its canonical image in U. By Lemma 3.1, L is the set of primitive elements of U: indeed, on every local component of C, L is a free module. Consider the C-formal group

$$\mathscr{G}(L) = \mathrm{Spf}^*\, U(L),$$

whose covariant bialgebra is $U = U(L)$ (cf. 2.6.2). Let $\mathfrak{m}$ be a maximal ideal of C, and put

$$\overline{C} = C/\mathfrak{m}, \qquad \overline{U} = U\otimes_C \overline{C}.$$

Since $\mathscr{G}(L)$ is infinitesimal (loc. cit.), the unit element of $\overline{U}$ is the only element $\overline{z} \in \overline{U}$ such that[122]

$$\varepsilon_{\overline{U}}(\overline{z}) = 1, \qquad \Delta_{\overline{U}}(\overline{z}) = \overline{z}\otimes\overline{z}.$$

It follows that every $z \in U$ satisfying

$$\varepsilon_U(z) = 1, \qquad \Delta_U(z) = z\otimes z$$

necessarily belongs to $1+\mathfrak{r}U$.

Finally, since

$$L \cap \mathfrak{r}U = \mathfrak{r}L = L\otimes_C \mathfrak{r},$$

[120] That is, for every homomorphism $\phi : U \to V$ of C-algebras,

$$\phi\big(\exp_U(x)\big) = \exp_V\big(\phi(x)\big).$$

[121] The editors supplied the details of the preceding argument and added the sentence that follows.

[122] The editors added the condition $\varepsilon_{\overline{U}}(\overline{z}) = 1$, which was omitted in the original.

we conclude that $\exp_U$ defines a bijection from $L \otimes_C \mathfrak{r}$ onto the group $\mathscr{G}(L)(C)$. We summarize this as follows.

Proposition 3.2. *Let k be a pseudocompact ring containing $\mathbb{Q}$, and let $\mathbf{L}$ be a flat Lie algebra over $\mathbf{O}_k$.*

(i) *For every object C of* Alf $/k$, *let $\mathfrak{r}(C)$ denote its radical. Then the map*

$$\exp_{U(\mathbf{L}(C))} : \mathbf{L}(C) \otimes_C \mathfrak{r}(C) \longrightarrow \mathscr{G}(\mathbf{L})(C)$$

is bijective and functorial in C and $\mathbf{L}$.

(ii) *The natural morphism*

$$\mathbf{L} \longrightarrow \mathrm{Lie}(\mathscr{G}(\mathbf{L}))$$

(cf. 2.6.2) *is an isomorphism of Lie algebras over $\mathbf{O}_k$.*

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3.2.1

The bijection $\exp_{U(L(C))}$ allows us, by transport of structure, to define a group law on the set

$$L(C) \otimes_C \mathfrak{r}(C),$$

which, as in 3.2, we identify with a subset of $U(L(C))$. If x and y are two elements of $L(C) \otimes_C \mathfrak{r}(C)$, this law is given by

$$\begin{aligned} x \cdot y = \log((\exp x)(\exp y)) &= \log\left(1 + \sum_{p+q>0} \frac{x^p y^q}{p!\,q!}\right) \\ &= \sum_{m\geq 1} \sum_{p_i+q_i>0} \frac{(-1)^{m-1}}{m} \frac{x^{p_1} y^{q_1}}{p_1!\,q_1!} \cdots \frac{x^{p_m} y^{q_m}}{p_m!\,q_m!} \\ &= \sum_{\ell \geq 1} P_\ell(x,y). \end{aligned}$$

Here $P_\ell(x,y)$ denotes the sum of the monomials of total degree ℓ in x and y. For example,

$$P_1(x,y) = x + y,$$

and

$$\begin{aligned} P_2(x,y) &= \underbrace{\frac{x^2}{2} + xy + \frac{y^2}{2}}_{m=1} - \underbrace{\frac{1}{2}(x^2 + xy + yx + y^2)}_{m=2} \\ &= \frac{1}{2}(xy - yx) = \frac{1}{2}[x,y]. \end{aligned}$$

Moreover, $P_3(x,y)$ is the sum of the following three terms:

$$\begin{aligned} P_3(x,y) = &\underbrace{\frac{x^3}{6} + \frac{x^2 y}{2} + \frac{xy^2}{2} + \frac{y^3}{6}}_{m=1} \\ &- \underbrace{\frac{1}{2}\left(x^3 + \frac{3}{2}x^2 y + \frac{1}{2}yx^2 + xyx + yxy + \frac{1}{2}y^2 x + \frac{3}{2}xy^2 + y^3\right)}_{m=2} \\ &+ \underbrace{\frac{1}{3}\left(x^3 + x^2 y + yx^2 + xyx + yxy + y^2 x + xy^2 + y^3\right)}_{m=3}. \end{aligned}$$

[123] The editors added part (ii), which follows immediately from 3.1.

Consequently,

$$P_3(x,y) = \frac{1}{12}(x^2y + yx^2 - 2xyx - 2yxy + y^2x + xy^2) = \frac{1}{12}([[y,x],x] + [y,[y,x]]).$$

More generally, one can prove the Campbell–Hausdorff formula:[124]

$$P_\ell(x,y) = \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m\cdot\ell} \sum_{\substack{p_1,\dots,p_m \\ q_1,\dots,q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!} \frac{(\operatorname{ad} y)^{q_i}}{q_i!} \right) \frac{(\operatorname{ad} x)^{p_m}}{p_m!}(y)$$
$$+ \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m\cdot\ell} \sum_{\substack{p_1,\dots,p_{m-1} \\ q_1,\dots,q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!} \frac{(\operatorname{ad} y)^{q_i}}{q_i!} \right)(x).$$

Here the $p_j, q_i \in \mathbb{N}$ satisfy

$$p_i + q_i \geq 1 \quad (1 \leq i \leq m-1), \qquad p_m + \sum_{i=1}^{m-1}(p_i + q_i) = \ell - 1.$$

In other words, in the sums above every nonzero "Lie monomial" has total degree ℓ. For a proof, see N. Jacobson, *Lie Algebras* (Interscience, 1962), §V.5, or [BLie], II, §6.4, Theorem 2.

3.3

If G is a formal k-group with affine algebra A, recall that I_A denotes the augmentation ideal of A, and that

$$\omega_{G/k} = I_A/I_A^2 \simeq I_A \widehat{\otimes}_A k$$

is a pseudocompact k-module.

Theorem 3.3 (Cartier). *Let k be a pseudocompact ring containing $\mathbb{Q}$,*[125] *and let G be a formal k-group. The following assertions are equivalent:*

(i) *There exists an $\mathcal{O}_k$-Lie algebra $\mathbf{L}$, flat over $\mathcal{O}_k$, such that G is isomorphic to $\mathscr{G}(\mathbf{L})$. In this case $\mathbf{L} = \mathrm{Lie}(G)$ by 3.2.*

(ii) *There exists a projective pseudocompact k-module ω such that the formal variety underlying G is isomorphic to*

$$\mathbf{V}_k^{f,0}(\omega)$$

(cf. 1.2.5)*, whose affine algebra is $k[[\omega]]$. In this case $\omega \simeq \omega_{G/k}$.*

(iii) *The group G is infinitesimal, and $\omega_{G/k}$ is a projective pseudocompact k-module.*

(iv) *The group G is infinitesimal and topologically flat over k.*

Proof of (i) $\Rightarrow$ (ii). Let

$$\omega = \Gamma^*(\mathbf{L})$$

be the pseudocompact k-module dual to $\mathbf{L}$ (cf. 1.2.3.D). For every $C \in \mathrm{Alf}/k$, we must construct, functorially in C, an isomorphism

$$\mathbf{V}_k^{f,0}(\omega)(C) \xrightarrow{\sim} \mathscr{G}(\mathbf{L})(C).$$

[124] The editors corrected the formula given in the original, which was erroneous, and added the reference [BLie].

[125] The editors removed the hypothesis that k be local; the proof reduces to that case.

By 1.2.5, the left-hand side identifies with

$$\mathrm{Hom}_c(\omega, \mathfrak{r}(C)),$$

where $\mathfrak{r}(C)$ is the radical of C. This set is naturally isomorphic to the set

$$\mathrm{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$$

of continuous C-linear maps. Since $\omega \widehat{\otimes}_k C$ is a projective pseudocompact C-module, the canonical map

$$(\omega \widehat{\otimes}_k C)^\dagger \otimes_C \mathfrak{r}(C) \longrightarrow \mathrm{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$$

is bijective (cf. 0.3.6.A). By 1.2.3.E,

$$\mathbf{L}(C) \simeq \mathbf{V}_k^f(\Gamma^*(\mathbf{L}))(C) = (\omega \widehat{\otimes}_k C)^\dagger.$$

Thus $\mathbf{V}_k^{f,0}(\omega)(C)$ is canonically isomorphic to

$$\mathbf{L}(C) \otimes_C \mathfrak{r}(C),$$

which Proposition 3.2 identifies canonically with $\mathscr{G}(\mathbf{L})(C)$. This proves (i) $\Rightarrow$ (ii).

Proof of (ii) $\Rightarrow$ (iii). Let ω be a projective object of PC(k), and let

$$h : k[[\omega]] \xrightarrow{\sim} A$$

be an isomorphism onto the affine algebra of G. The composite

$$\varepsilon_A \circ h : k[[\omega]] \longrightarrow k$$

is determined by its restriction $\lambda : \omega \to \mathfrak{r}(k)$. The map

$$x \longmapsto x - \lambda(x)$$

from ω to the radical of $k[[\omega]]$ extends, by the universal property of $k[[\omega]]$, to an endomorphism ℓ_λ of $k[[\omega]]$. Since

$$\ell_\lambda \circ \ell_{-\lambda} = \ell_{-\lambda} \circ \ell_\lambda = \mathrm{id},$$

the map ℓ_λ is an automorphism. Replacing h by $h \circ \ell_\lambda$, we may assume that $\varepsilon_A \circ h$ vanishes on the closed ideal I of $k[[\omega]]$ generated by ω. Then h induces an isomorphism

$$I/I^2 \xrightarrow{\sim} I_A/I_A^2.$$

Since $I/I^2 \simeq \omega$, we obtain

$$\omega_{G/k} \simeq \omega,$$

so $\omega_{G/k}$ is projective. The formal variety $\mathbf{V}_k^{f,0}(\omega)$, and hence G, is infinitesimal.

Proof of (iii) $\Rightarrow$ (i). Assume that G is infinitesimal and $\omega_{G/k}$ is projective. Let $\mathbf{L}$ be the $\mathcal{O}_k$-Lie algebra of G. By 2.6.1, its underlying $\mathcal{O}_k$-module is

$$\mathbf{V}_k^f(\omega_{G/k}).$$

Consequently, $\mathbf{L}$ is flat over $\mathcal{O}_k$, and Proposition 1.2.3.E gives

$$\Gamma^*(\mathbf{L}) \simeq \omega_{G/k}.$$

The proof of (i) $\Rightarrow$ (ii) therefore identifies the affine algebra of $\mathscr{G}(\mathbf{L})$ with $k[[\omega_{G/k}]]$. By 2.6.3, the identity morphism of $\mathbf{L}$ corresponds canonically to a morphism

$$\mathscr{G}(\mathbf{L}) \longrightarrow G,$$

and hence to a continuous k-algebra morphism

$$\varphi : A \longrightarrow k[[\omega_{G/k}]].$$

Let I be the closed ideal generated by $\omega_{G/k}$. Filter $k[[\omega_{G/k}]]$ by the closures of the powers I^n, and A by the closures of the powers I_A^n. By definition, φ induces the identity on

$$\omega_{G/k} = I_A/I_A^2 = I/I^2.$$

Because $\omega_{G/k}$ is projective, the canonical projection

$$I_A \longrightarrow \omega_{G/k}$$

has a section τ. By the universal property of $k[[\omega_{G/k}]]$, this section defines a continuous algebra morphism

$$\psi : k[[\omega_{G/k}]] \longrightarrow A.$$

The composite $\varphi \circ \psi$ induces the identity on $\omega_{G/k}$, and hence on the associated graded algebra of $k[[\omega_{G/k}]]$. It follows from [CA], §V.7, Lemma 1 that $\varphi \circ \psi$ is an isomorphism.[126]

Moreover, ψ induces an isomorphism

$$I/I^2 \xrightarrow{\sim} I_A/I_A^2,$$

and hence a surjection on associated graded algebras. Since G is radicial, I_A is contained in the radical of A, so

$$\bigcap_n I_A^n = 0.$$

The cited lemma therefore shows that ψ is surjective. Since $\varphi \circ \psi$ is an isomorphism and ψ is surjective, both ψ and φ are isomorphisms. This proves (iii) $\Rightarrow$ (i).

Finally, (i) or (ii) plainly implies (iv), so it remains to prove (iv) $\Rightarrow$ (ii). We may assume that k is local, with residue field k_0. Put

$$G_0 = G\widehat{\otimes}_k k_0, \qquad \omega = \omega_{G/k}, \qquad \omega_0 = \omega\widehat{\otimes}_k k_0.$$

[127] The pseudocompact k_0-module ω_0 has a pseudobasis $(y_\lambda)_{\lambda\in\Lambda}$. Let

$$\omega' = \prod_{\lambda\in\Lambda} k_\lambda$$

be the topologically free k-module with that index set, and lift each y_λ to an element $x_\lambda \in \omega$. This gives a continuous k-linear map

$$f : \omega' \longrightarrow \omega$$

such that

$$f_0 = f\widehat{\otimes}_k k_0$$

[126] Compare 5.1.5 below, and also [BAC], III, §2.8, Theorem 1 and its corollaries. The editors supplied the details of the original argument that follow.

[127] The editors supplied the details of the original argument that follow.

is invertible.[128] Since ω' is projective, f lifts to a continuous k-linear map

$$g : \omega' \longrightarrow I_A$$

with $\pi \circ g = f$, where $\pi : I_A \to \omega$ is the projection. The universal property of $k[[\omega']]$ gives a morphism of topological algebras

$$\phi : k[[\omega']] \longrightarrow A.$$

Now

$$\omega' \widehat{\otimes}_k k_0 \simeq \omega \widehat{\otimes}_k k_0 = \omega_{G_0/k_0},$$

and consequently

$$k[[\omega']] \widehat{\otimes}_k k_0 \simeq k_0[[\omega_{G_0/k_0}]].$$

The pair k_0, G_0 satisfies (iii), and the proof of (iii) $\Rightarrow$ (i) shows that

$$\phi_0 = \phi \widehat{\otimes}_k k_0$$

is invertible. Both $k[[\omega']]$ and A are projective pseudocompact k-modules, so 0.3.4 implies that ϕ itself is invertible. In particular, if I is the augmentation ideal of $k[[\omega']]$, then ϕ induces an isomorphism

$$\omega' = I/I^2 \xrightarrow{\sim} I_A/I_A^2 = \omega.$$

This completes the proof.

3.3.1. Corollary

Let S be a locally Noetherian scheme over $\mathbb{Q}$, and let G be an S-group scheme that is flat and locally of finite type.[129] *If $\omega_{G/S}$ is a locally free $\mathcal{O}_S$-module, then G is smooth over S at every point of the unit section.*

Proof. Let x be a point of the unit section, let s be its image in S, and denote the local rings at x and s by $\mathcal{O}_x$ and $\mathcal{O}_s$, respectively.[130] By hypothesis, $\mathcal{O}_s$ and $\mathcal{O}_x$ are Noetherian local rings. The maximal-ideal-adic topology on each of them therefore agrees with the "pseudocompact" topology defined by the ideals of finite colength. Write

$$\widehat{\mathcal{O}}_x \qquad \text{and} \qquad \widehat{\mathcal{O}}_s$$

for their completions in this topology. By EGA IV$_4$, 17.5.3, G is smooth over S at x if and only if $\widehat{\mathcal{O}}_x$ is formally smooth over $\widehat{\mathcal{O}}_s$, both algebras being endowed with their maximal-ideal-adic topologies. It is therefore enough to prove this last assertion.

The rings $\widehat{\mathcal{O}}_x$ and $\widehat{\mathcal{O}}_s$ are the local rings at x and s of the formal varieties $\widehat{G}/\widehat{S}$ and $\widehat{S}$ defined in 1.2.6. Put

$$k = \widehat{\mathcal{O}}_s, \qquad H = \mathrm{Spf}(\widehat{\mathcal{O}}_x).$$

Then H is an infinitesimal formal k-variety. Since the formation of $\widehat{G}/\widehat{S}$ commutes with products (loc. cit.), H is in fact an infinitesimal formal k-group.

Let I be the augmentation ideal of $\mathcal{O}_x$. By hypothesis,

$$\omega_{G/S,x} = I/I^2$$

[128] One does not know a priori that ω is a projective pseudocompact k-module; this will follow from the argument. Compare the proof of (iv) $\Rightarrow$ (iii) in VIIA 7.4.

[129] The editors added the flatness hypothesis, which was omitted from the original.

[130] The editors supplied the details of the argument that follows.

is a free $\mathcal{O}_s$-module of finite rank n. Since $\mathcal{O}_s$ is Noetherian, the module $\omega_{H/k}$, which is the completion of $\omega_{G/S,x}$, is identified with

$$\omega_{G/S,x} \otimes_{\mathcal{O}_s} \widehat{\mathcal{O}}_s.$$

It is therefore a topologically free k-module of rank n. By the implication (iv)⇒(ii) of Theorem 3.3,

$$\widehat{\mathcal{O}}_x \simeq k[[\omega_{H/k}]] \simeq k[[t_1, \dots, t_n]].$$

Finally, the latter formal power-series algebra is formally smooth over k, by EGA 0_{IV}, 19.3.3. This proves the corollary.

Thus we recover a result obtained independently for group schemes locally of finite presentation over an arbitrary scheme S; cf. VI$_{\mathrm{B}}$, 1.6.

3.3.2 Corollary

Corollary 3.3.2. *Let k be a field of characteristic 0. The functor*

$$\mathbf{L} \longmapsto \mathscr{G}(\mathbf{L})$$

is an equivalence from the category of k-Lie algebras to the category of infinitesimal formal k-groups. [131]

Indeed, as G ranges over the infinitesimal formal k-groups, Theorem 3.3 identifies the functor

$$G \longmapsto \mathscr{G}\big(\operatorname{Lie} G\big)$$

with the identity functor. Likewise, 3.2(ii) identifies

$$\mathbf{L} \longmapsto \operatorname{Lie}\big(\mathscr{G}(\mathbf{L})\big) = \operatorname{Prim} U(\mathbf{L})$$

with the identity functor on k-Lie algebras.

3.3.3

Continue to suppose that k is a field of characteristic 0. Let $\overline{k}$ be an algebraic closure of k, and let Γ be the topological Galois group of $\overline{k}$ over k.

[131] By 2.7, 2.2.1, and 1.3.6, an infinitesimal formal k-group G is "the same thing" as a connected cocommutative k-bialgebra H (cf. 2.9). The corollary is therefore equivalent to the theorem below, obtained independently by Kostant (cf. 2.9.2). J. W. Milnor and J. C. Moore had earlier proved it in [MM65] under the additional hypothesis that H be generated as an algebra by its primitive elements. Although published in 1965, their text had circulated before 1960 (see the review [Ja65]); the result is consequently often called the Cartier–Kostant–Milnor–Moore theorem.

Cartier–Kostant–Milnor–Moore Theorem. Let k be a field of characteristic 0. The functors

$$\mathbf{L} \longmapsto U(\mathbf{L}), \qquad H \longmapsto \operatorname{Prim}(H),$$

define an equivalence between the category of k-Lie algebras and the category of connected cocommutative k-bialgebras.

If k is an Artinian ring containing $\mathbb{Q}$, then 3.2(ii) and 3.3, combined with 2.7, 2.2.1, and 1.3.6, likewise show that

$$\mathbf{L} \longmapsto \mathscr{G}(\mathbf{L})$$

(respectively, $\mathbf{L} \mapsto U(\mathbf{L})$) is an equivalence from the category of k-Lie algebras flat over k to the category of topologically flat infinitesimal formal k-groups (respectively, to the category of connected cocommutative k-bialgebras flat over k).

For every formal k-group G, denote by $\mathbf{Aut}_k(G)$ the k-group functor assigning to every finite-dimensional commutative k-algebra C the group of automorphisms of the formal C-group

$$G\widehat{\otimes}_k C.$$

[132] Since k is a field, G is topologically flat over k. Equivalently, its affine algebra

$$A = \mathcal{A}(G)$$

is topologically flat over k, or its covariant bialgebra

$$H = \mathbf{H}(G)$$

is flat over k. The functor $\mathbf{Aut}_k(G)$ is therefore a formal k-group.

Indeed, $\mathbf{Aut}_k(G)$ is the fiber product of the following diagram (cf. Exposé I, 1.7.3):

$$\begin{array}{ccc} & & \underline{\mathrm{End}}_k(G) \times \underline{\mathrm{End}}_k(G) \\ & & \downarrow \\ \mathrm{Spf}(k) & \longrightarrow & \underline{\mathrm{End}}_k(G) \times \underline{\mathrm{End}}_k(G) \end{array}$$

The vertical arrow sends

$$(\phi, \psi) \longmapsto (\phi \circ \psi, \psi \circ \phi),$$

whereas the horizontal arrow is the unit section. It is therefore enough to show that the k-functor $\mathbf{End}_k(G)$ is represented by an object of $\mathrm{Vaf}\,/k$; by 1.1 and 0.4.2, this amounts to showing that it is left exact, or, equivalently, that it commutes with fiber products of k-algebras.

To give an element of $\mathbf{End}_k(G)(C)$ is equivalent to giving a k-algebra morphism

$$\phi : H \longrightarrow H \otimes_k C$$

that also respects the coalgebra structure, that is, for which the usual compatibility diagrams commute. Since H is flat over k, the functor

$$H \otimes_k -$$

commutes with fiber products of k-algebras. Thus $\mathbf{End}_k(G)$ is left exact and can be treated as a formal k-group.

If L is a k-Lie algebra and $G = \mathscr{G}(L)$, Theorem 3.3 shows that $\mathbf{Aut}_k(G)$ is isomorphic to the k-group functor $\mathbf{Aut}_k(L)$ which assigns to each finite-dimensional k-algebra C the group of automorphisms of the C-Lie algebra

$$C \otimes_k L.$$

Now let G be an arbitrary formal k-group. We saw in 2.5.2 that G decomposes canonically as a semidirect product of an étale formal group G_e and an infinitesimal formal group G_0. This semidirect product is determined by the data of

$$L = \mathrm{Lie}(G) = \mathrm{Lie}(G_0),$$

[132] The original stated: "This k-functor is manifestly left exact (H is topologically flat over k!)." The editors supplied the details that follow.

the Γ-group

$$M = G_e(\overline{k}) = G(\overline{k}),$$

and a morphism of formal k-groups

$$\Phi : G_e \longrightarrow \mathbf{Aut}_k(G_0) \simeq \mathbf{Aut}_k(L).$$

Such a morphism sends G_e into the étale part $\mathbf{Aut}_k(L)_e$ of $\mathbf{Aut}_k(L)$ (cf. 2.5.1). It is consequently determined by a morphism of Γ-groups

$$\phi = \Phi(\overline{k}) : M \longrightarrow (\mathbf{Aut}_k L)(\overline{k}) = \mathrm{Aut}_{\overline{k}}(L \otimes_k \overline{k}).$$

Let Γ act on $L \otimes_k \overline{k}$ by

$$\gamma(\ell \otimes t) = \ell \otimes \gamma(t).$$

Then ϕ makes M act on $L \otimes_k \overline{k}$ by automorphisms of $\overline{k}$-Lie algebras in such a way that

$$\phi\big(\gamma(m)\big) = \gamma \circ \phi(m) \circ \gamma^{-1}.$$

Equivalently,

$$\gamma(m)(\ell \otimes t) = \gamma\Big(m\big(\ell \otimes \gamma^{-1}(t)\big)\Big)$$

for every $m \in M$, $\gamma \in \Gamma$, and $\ell \otimes t \in L \otimes_k \overline{k}$. We express this last condition by saying that M acts on $L \otimes_k \overline{k}$ *compatibly with* Γ.

The situation can be summarized in a "highbrow" statement. A Γ-*Lie algebra over* k is a triple

$$(L, M, \phi)$$

consisting of a k-Lie algebra L, a Γ-group M, and an action ϕ of M on $L \otimes_k \overline{k}$ that is compatible with the actions of Γ on M and on $L \otimes_k \overline{k}$.

If (L, M, ϕ) and (L', M', ϕ') are two such Γ-Lie algebras, a morphism from the first to the second is a pair (f, θ), where

$$f : L \longrightarrow L'$$

is a morphism of k-Lie algebras and

$$\theta : M \longrightarrow M'$$

is a morphism of Γ-groups, such that

$$(f \otimes 1)(m \cdot x) = \theta(m) \cdot (f \otimes 1)(x)$$

for every $x \in L \otimes_k \overline{k}$ and $m \in M$. We have therefore proved:

Theorem. *Let k be a field of characteristic 0. The functor that assigns to a formal k-group G the triple*

$$\big(\mathrm{Lie}(G), G(\overline{k}), \Phi(\overline{k})\big)$$

is an equivalence from the category of formal k-groups to the category of Γ-Lie algebras.[133]

4. Phenomena specific to characteristic $p > 0$

Throughout Section 4, p denotes a prime number, k a pseudocompact ring of characteristic p, and

$$\pi : k \longrightarrow k, \qquad x \longmapsto x^p,$$

the continuous Frobenius endomorphism of k.

[133] In particular, when $k = \overline{k}$, this recovers the Cartier–Gabriel–Kostant theorem mentioned in 2.9.2.

4.1

Let X be a formal k-variety with affine algebra A. We write $X^{(p/k)}$, or simply $X^{(p)}$, for the formal k-variety

$$X\widehat{\otimes}_\pi k$$

obtained by the base change $\pi : k \to k$ (cf. 1.2.D). Its affine algebra is the completed tensor product

$$A\widehat{\otimes}_\pi k.$$

The continuous morphism

$$A\widehat{\otimes}_\pi k \longrightarrow A, \qquad a\widehat{\otimes}_\pi \lambda \longmapsto a^p\lambda,$$

therefore induces a morphism of formal k-varieties

$$\mathrm{Fr}(X/k) : X \longrightarrow X^{(p/k)}.$$

We call $\mathrm{Fr}(X/k)$ the *Frobenius morphism of X relative to k*, and below we shall often write Fr in place of $\mathrm{Fr}(X/k)$.

4.1.1

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Let S be a scheme over the prime field $\mathbb{F}_p$, and let $\eta : X \to S$ be an S-scheme. Write $\mathrm{fr}(S)$ for the "absolute" Frobenius morphism of S: it induces the identity on the underlying topological space and sends every section f of the structure sheaf to f^p; cf. VII$_\mathrm{A}$, 4.1. Let

$$X^{(p)} = X \times_{\mathrm{fr}(S)} S$$

be the corresponding S-scheme (VII$_\mathrm{A}$, loc. cit.); thus its structural morphism to S is $\mathrm{fr}(S)\circ\eta$.

Let $\widehat{S}$ be the formal scheme whose underlying topological space is the set of closed points of S, endowed with the discrete topology, and whose local ring at a closed point s is the separated completion

$$\mathcal{O}_{\widehat{S},s} = \widehat{\mathcal{O}}_{S,s}$$

of $\mathcal{O}_{S,s}$ for the linear topology defined by ideals of finite colength (cf. 1.2.6). Its affine algebra is therefore

$$k = \mathcal{A}(\widehat{S}) = \prod_{s\in S_{\mathrm{cl}}} \widehat{\mathcal{O}}_{S,s}.$$

Recall (loc. cit.) that $\widehat{X}/\widehat{S}$ denotes the formal k-variety consisting of those points $x \in X$ for which

$$[\kappa(x) : \kappa(s)] < \infty,$$

where s is the image of x in S. Its local ring at x is the separated completion

$$\mathcal{O}_{\widehat{X}/\widehat{S},x} = \widehat{\mathcal{O}}_{X,x}$$

[134] The editors corrected the original, which gave the inclusion

$$(\widehat{X}/\widehat{S})^{(p)} \subset \widehat{X^{(p)}}/\widehat{S}$$

instead of the reverse inclusion. They also note that this paragraph is not used below.

of $\mathcal{O}_{X,x}$ for the linear topology defined by those ideals I such that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{X,x}$-module, and hence also as an $\mathcal{O}_{S,s}$-module. By base change we can therefore form the formal k-variety

$$(\widehat{X}/\widehat{S})^{(p)} = (\widehat{X}/\widehat{S})\widehat{\otimes}_\pi k.$$

We can also consider the formal k-variety $\widehat{X^{(p)}}/\widehat{S}$. Its underlying set consists of the points $x \in X^{(p)} = X$ such that, if s denotes the image of x in S, the composite

$$\kappa(s) \xrightarrow{\text{fr}} \kappa(s) \xrightarrow{\eta^\natural} \kappa(x)$$

makes $\kappa(x)$ a finite extension of $\kappa(s)$. In that case the same is true of

$$\eta^\natural : \kappa(s) \longrightarrow \kappa(x),$$

so x is a point of $\widehat{X}/\widehat{S}$, and

$$\mathcal{O}_{\widehat{X^{(p)}}/\widehat{S},x} = \widehat{\mathcal{O}}_{X,x}\widehat{\otimes}_\pi\widehat{\mathcal{O}}_{S,s} = \widehat{\mathcal{O}}_{X^{(p)},x}.$$

The second equality follows because the construction $X \mapsto \widehat{X}/\widehat{S}$ commutes with products (cf. 1.2.6).

It follows that

$$\widehat{X^{(p)}}/\widehat{S} \ \subset \ (\widehat{X}/\widehat{S})^{(p)}$$

as a formal subvariety. Equality holds if and only if, for every closed point s of S, the field $\kappa(s)$ has finite degree over $\kappa(s)^p$. This is the case, for example, if S is locally of finite type over a field κ such that

$$[\kappa : \kappa^p] < \infty.$$

4.1.2

Let G be a formal k-group. Since the functor

$$X \longmapsto X^{(p)} = X\widehat{\otimes}_\pi k$$

commutes with finite products, it carries formal k-groups to formal k-groups. Moreover, $\mathrm{Fr}(X/k)$ is functorial in X, so

$$\mathrm{Fr} : G \longrightarrow G^{(p)}$$

is a homomorphism of formal k-groups.

Define recursively

$$G^{(p^{n+1})} = \big(G^{(p^n)}\big)^{(p)}.$$

Then the same is true of the composite

$$\mathrm{Fr}^n = \mathrm{Fr}^n(G/k) : G \xrightarrow{\mathrm{Fr}} G^{(p)} \xrightarrow{\mathrm{Fr}} G^{(p^2)} \xrightarrow{\mathrm{Fr}} \cdots \xrightarrow{\mathrm{Fr}} G^{(p^n)}.$$

Let A be the affine algebra of G, and let I_A be its augmentation ideal.

Definitions.

(a) The kernel of Fr^n is denoted

$$\mathrm{Fr}_n\, G.$$

It follows directly from the definitions that $\mathrm{Fr}_n\, G$ is infinitesimal and has affine algebra

$$A/I_A^{\{p^n\}},$$

where $I_A^{\{p^n\}}$ is the closed ideal of A generated by the p^n-th powers of the elements of I_A.

(b) The group G is said to have *height at most* n if

$$I_A^{\{p^n\}} = 0;$$

equivalently,

$$\mathrm{Fr}_n\, G = G.$$

4.2

It is clear that the Lie algebra $\mathrm{Lie}(G)$ of a formal k-group G is a p-Lie subalgebra of $\mathbf{H}(G)$ (cf. 2.3). Indeed, we may immediately reduce to the case in which k is Artinian. Then $\mathrm{Lie}(G)$ is the subset of $\mathbf{H}(G)$ consisting of the elements x such that

$$\varphi_G(x \otimes 1 + 1 \otimes x) = \Delta_G(x),$$

with the notation of 2.3 and 2.6(c). Hence

$$\begin{aligned}\varphi_G(x^p \otimes 1 + 1 \otimes x^p) &= \varphi_G\big((x \otimes 1 + 1 \otimes x)^p\big)\\ &= \varphi_G(x \otimes 1 + 1 \otimes x)^p\\ &= \Delta_G(x)^p = \Delta_G(x^p).\end{aligned}$$

4.2.1

Conversely, every p-Lie algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a k-group functor. Let $U_p(\mathbf{L})$ denote the functor assigning to every object C of $\mathrm{Alf}\,/k$ the restricted enveloping algebra

$$U_p(\mathbf{L}(C))$$

of the p-Lie algebra $\mathbf{L}(C)$ over C (VII$_\mathrm{A}$, 5.3). By VII$_\mathrm{A}$, 5.4, $U_p(\mathbf{L})$ is a bialgebra over $\mathbf{O}_k$, and by 2.2 it therefore determines the k-group functor

$$\mathrm{Spf}^*\, U_p(\mathbf{L}),$$

which we henceforth denote by $\mathscr{G}_p(\mathbf{L})$.

Thus $\mathscr{G}_p(\mathbf{L})(C)$ is the group of elements

$$z \in U_p(\mathbf{L}(C))$$

having augmentation 1 and satisfying

$$\Delta_{U_p(\mathbf{L}(C))}(z) = z \otimes z.$$

4.2.2

Suppose that $\mathbf{L}$ is a flat p-Lie algebra over $\mathbf{O}_k$. Taking VII$_\mathrm{A}$, 5.3.3 into account, the argument of part (i) of Proposition 2.6.2 shows that $U_p(\mathbf{L})$ is flat over $\mathbf{O}_k$. Consequently, $\mathscr{G}_p(\mathbf{L})$ is a formal k-group topologically flat over k, with $U_p(\mathbf{L})$ as its covariant $\mathbf{O}_k$-bialgebra.

[135] By the proof of 2.6.2(iii), for every finite-length k-algebra C, the Lie algebra $\mathrm{Lie}(\mathscr{G}_p(\mathbf{L}))(C)$ is the set of primitive elements of

$$U_p(\mathbf{L})(C) = U_p(\mathbf{L}(C));$$

[135] The editors supplied the details that follow.

see also VII$_A$, 3.2.3. Moreover, by VII$_A$, 5.5.3, the canonical morphism $\mathbf{L} \to U_p(\mathbf{L})$ induces an isomorphism of p-Lie algebras

$$\tau_{p,\mathbf{L}} : \mathbf{L} \xrightarrow{\sim} \mathrm{Lie}\big(\mathscr{G}_p(\mathbf{L})\big).$$

Compare this with 3.1 in characteristic 0.

The formal group $\mathscr{G}_p(\mathbf{L})$ can be characterized by a universal property. Every morphism

$$h : \mathscr{G}_p(\mathbf{L}) \longrightarrow G$$

to a formal group G induces a morphism

$$\mathrm{Lie}(h) : \mathrm{Lie}\big(\mathscr{G}_p(\mathbf{L})\big) \longrightarrow \mathrm{Lie}(G).$$

Composing this morphism with $\tau_{p,\mathbf{L}}$ gives

$$h' : \mathbf{L} \longrightarrow \mathrm{Lie}(G).$$

Proposition. *Let k be a pseudocompact ring of characteristic $p > 0$, let G be a formal k-group, and let $\mathbf{L}$ be a flat p-Lie algebra over $\mathbf{O}_k$. The map $h \mapsto h'$ defined above is a bijection*

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}\big(\mathscr{G}_p(\mathbf{L}), G\big) \xrightarrow{\sim} \mathrm{Hom}_{p\text{-}\mathrm{Lie}}\big(\mathbf{L}, \mathrm{Lie}(G)\big),$$

where the right-hand side is the set of morphisms of p-Lie algebras from $\mathbf{L}$ to $\mathrm{Lie}(G)$.

Proof.[136] We may immediately reduce to the case in which k is Artinian. Put

$$L = \mathbf{L}(k).$$

By 2.3.1,

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}\big(\mathscr{G}_p(\mathbf{L}), G\big)$$

is in bijection with the set of unital algebra morphisms

$$h : U_p(L) \longrightarrow \mathbf{H}(G)$$

for which the following diagram commutes:

$$\begin{array}{ccccc}
U_p(L) & & \xrightarrow{\quad h \quad} & & \mathbf{H}(G) \\
 & & & \searrow^{\delta_G} & \\
\Big\downarrow{\scriptstyle \Delta_{U_p(L)}} & & & & \mathbf{H}(G \times G) \\
 & & & \nearrow_{\varphi_G} & \\
U_p(L) \otimes U_p(L) & & \xrightarrow{\quad h \otimes h \quad} & & \mathbf{H}(G) \otimes \mathbf{H}(G)
\end{array} \qquad .$$

The morphism h is determined by its restriction to L, which is a morphism of p-Lie algebras from L to the underlying p-Lie algebra of $\mathbf{H}(G)$. The commutativity of the diagram says precisely that h sends L into the subset of $\mathbf{H}(G)$ consisting of those x such that

$$\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x).$$

This subset is $\mathrm{Lie}(G)$, by 2.6(c). The proposition follows.

[136] The proof is identical to that of 2.6.3.

4.3

We now study the formal k-group $\mathscr{G}_p(\mathbf{L})$ in greater detail when $\mathbf{L}$ is a flat p-Lie algebra over $\mathbf{O}_k$.

For this purpose, first let C be a ring of characteristic p, and let L be a p-Lie algebra over C whose underlying module is free with basis

$$(x_i)_{i\in I}.$$

Let $\mathcal{P}$ be the set of families

$$n = (n_i)_{i\in I}$$

of nonnegative integers such that

$$0 \leq n_i < p$$

and $n_i = 0$ except possibly for finitely many indices. Endow I with a total order, and let

$$i_1 < i_2 < \cdots < i_r$$

be the indices for which $n_i \neq 0$. Put

$$|n| = n_{i_1} + \cdots + n_{i_r},$$

and

$$x^n = x_{i_1}^{n_{i_1}} x_{i_2}^{n_{i_2}} \cdots x_{i_r}^{n_{i_r}}, \qquad n! = (n_{i_1}!)\cdots(n_{i_r}!).$$

By VII$_\mathrm{A}$, 5.3.3, the monomials $x^n/n!$ form a basis of $U_p(L)$, and

$$\Delta\left(\frac{x^n}{n!}\right) = \sum_{m\leq n} \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!}. \qquad (*)$$

The sum ranges over all $m \in \mathcal{P}$ such that $m \leq n$, that is,

$$m_i \leq n_i \qquad \text{for every } i \in I.$$

Formula (*) gives an immediate description of the natural filtration of $U_p(L)$ (cf. 1.3.6). Put

$$U = U_p(L),$$

let U^+ be the two-sided ideal generated by L, and put

$$U_0 = C \cdot 1_U.$$

As in 1.3.6, define a filtration of U by C-subcoalgebras by setting, for $n \geq 1$,

$$U_n = \left\{ u \in U \;\middle|\; \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+ \right\}.$$

Formula (*) then shows that U_n is the free C-module with basis the monomials x^m satisfying

$$|m| \leq n.$$

4.3.1

With the notation of 4.3, let us determine the elements $\xi \in U = U_p(L)$ such that

$$\varepsilon_U(\xi) = 1, \qquad \Delta_U(\xi) = \xi \otimes \xi.$$

Every element $\xi \in U$ has a unique expression

$$\xi = \sum_{n \in \mathcal{P}} \xi_n \frac{x^n}{n!}, \qquad \xi_n \in C.$$

The condition $\varepsilon_U(\xi) = 1$ gives

$$\xi_0 = 1,$$

where 0 denotes the element of $\mathcal{P}$ all of whose components vanish. The condition $\Delta_{U_p(L)}(\xi) = \xi \otimes \xi$ gives

$$\sum_{\substack{n \in \mathcal{P} \\ m \leq n}} \xi_n \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!} = \sum_{q,r \in \mathcal{P}} \xi_q \xi_r \frac{x^q}{q!} \otimes \frac{x^r}{r!}.$$

Equivalently,

$$\xi_{q+r} = \xi_q \xi_r \quad \text{if } q_i + r_i < p \text{ for every } i,$$

and

$$\xi_q \xi_r = 0 \quad \text{otherwise.}$$

Let ξ_i denote the coefficient ξ_n for which $n_i = 1$ and $n_j = 0$ whenever $j \neq i$. The preceding conditions say precisely that

$$\xi_n = \prod_i \xi_i^{n_i} \quad (n \in \mathcal{P}), \qquad \xi_i^p = 0 \quad (i \in I).$$

We obtain:

Proposition. *Let k be a local pseudocompact ring*[137] *of characteristic $p > 0$, let $\mathbf{L}$ be a flat p-Lie algebra over $\mathbf{O}_k$, and let C be an object of* Alf $/k$. *Write*

$$\sqrt[p]{0_C} = \{x \in C \mid x^p = 0\}.$$

There is a bijection, "functorial in C,"

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathscr{G}_p(\mathbf{L})(C).$$

Proof. By Remark 1.2.3.F, one may choose a basis

$$({}^C x_i)_{i \in I}$$

of $\mathbf{L}(C)$ over C in such a way that, for every morphism $\phi : C \to D$ in Alf $/k$, the map $\mathbf{L}(\phi)$ sends ${}^C x_i$ to ${}^D x_i$. The preceding calculation shows that

$$\sum_i ({}^C x_i) \otimes \xi_i \longmapsto \sum_{n \in \mathcal{P}} \left(\prod_i \xi_i^{n_i} \right) \frac{({}^C x)^n}{n!}$$

is a functorial bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathscr{G}_p(\mathbf{L})(C)$.

[137] The editors added the hypothesis that k be local, so that every topologically flat pseudocompact k-module is topologically free; see the proof.

4.3.2

"There is no Campbell–Hausdorff formula in characteristic $p > 0$." Let us explain. The functorial isomorphism of 4.3.1 depends on the choice of the bases $({}^C x_i)_{i\in I}$. Unlike the situation in 3.2, in characteristic 0, there is in general no bijection

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \longrightarrow \mathscr{G}_p(\mathbf{L})(C)$$

that is "functorial both in C and in $\mathbf{L}$." The following argument shows, for example, that no such isomorphism exists when k is a field containing all (p^2-1)-st roots of unity.

Choose $\mathbf{L}$ so that, for every $C \in \mathrm{Alf}\,/k$, the algebra $\mathbf{L}(C)$ is the abelian p-Lie algebra over C generated by an element x subject to

$$x^{(p^2)} = 0.$$

Thus

$$\mathbf{L}(C) = Cx \oplus Cx^{(p)}, \qquad U_p(\mathbf{L}(C)) \simeq C[x]/(x^{p^2}).$$

We shall show that the only functorial morphism

$$\chi_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \longrightarrow U_p(\mathbf{L}(C))$$

which is compatible with the endomorphisms of $\mathbf{L}$ and sends $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ into $\mathscr{G}_p(\mathbf{L})(C)$ is the constant morphism with value 1.

Let

$$\psi_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathscr{G}_p(\mathbf{L})(C)$$

be the bijection of 4.3.1, namely

$$x \otimes a + x^{(p)} \otimes b \longmapsto \sum_{0\le i,j<p} a^i b^j x^{\,i+pj}.$$

Composing χ_C with ψ_C^{-1} gives a morphism, functorial in C,

$$\theta_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \longrightarrow \mathbf{L}(C) \otimes_C \sqrt[p]{0_C},$$
$$x \otimes a + x^{(p)} \otimes b \longmapsto x \otimes P(a,b) + x^{(p)} \otimes Q(a,b).$$

Functoriality in C implies that $P(a,b)$ and $Q(a,b)$ are the values at (a,b) of two polynomials

$$P, Q \in k[X,Y]$$

which may be assumed to have degree less than p in each of X and Y.[138]

Let $\alpha \in k$, and let ℓ_α be the endomorphism of the p-Lie algebra $\mathbf{L}$ sending

$$x \longmapsto \alpha x, \qquad x^{(p)} \longmapsto \alpha^p x^{(p)}.$$

Then $U_p(\ell_\alpha)$ sends x to αx, and hence sends x^i to $\alpha^i x^i$ for $i < p^2$. The following square commutes:

$$\begin{array}{ccc} \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} & \xrightarrow{\psi_C} & U_p(\mathbf{L}(C)) \\ \downarrow{\scriptstyle \ell_\alpha(C)\otimes_C \mathrm{id}} & & \downarrow{\scriptstyle U_p(\ell_\alpha)(C)} \\ \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} & \xrightarrow{\psi_C} & U_p(\mathbf{L}(C)) \end{array}$$

[138] The editors expanded the original in the argument that follows.

The hypothesis

$$\chi_C \circ (\ell_\alpha(C) \otimes \mathrm{id}) = U_p(\ell_\alpha)(C) \circ \chi_C$$

therefore gives

$$P(\alpha a, \alpha^p b) = \alpha P(a,b), \qquad Q(\alpha a, \alpha^p b) = \alpha^p Q(a,b).$$

Write

$$P = \sum_{i,j<p} \lambda_{ij} X^i Y^j, \qquad Q = \sum_{i,j<p} \mu_{ij} X^i Y^j,$$

and take

$$C = k[X,Y]/(X^p, Y^p).$$

If $\lambda_{ij} \neq 0$, then

$$\alpha^{\,i-1+pj} = 1;$$

if $\mu_{ij} \neq 0$, then

$$\alpha^{\,i+p(j-1)} = 1.$$

Taking α to be a primitive root of unity of order p^2-1, we conclude that

$$P = \lambda X, \qquad Q = \mu Y$$

for some $\lambda, \mu \in k$. Consequently,

$$\chi_C(x \otimes a + x^{(p)} \otimes b) = \sum_{0 \leq i,j<p} (\lambda a)^i (\mu b)^j x^{\,i+pj}.$$

Finally, consider the endomorphism f of $\mathbf{L}$ sending

$$x \longmapsto x + x^{(p)}.$$

Take $C = k[a]/(a^3)$ and compare the coefficients of x^p and x^{p+1} in

$$(\chi_C \circ f(C))(x \otimes a) \qquad \text{and} \qquad \big(U_p(f)(C) \circ \chi_C\big)(x \otimes a).$$

This gives

$$\lambda = \mu, \qquad \lambda^2 a^2 = 0,$$

and hence $\lambda = \mu = 0$, as claimed.

4.4 Theorem

Theorem 4.4. *Let k be a pseudocompact ring of characteristic $p > 0$, and let G be a formal k-group. The following assertions are equivalent:*

(i) *There exists a p-Lie algebra $\mathbf{L}$, flat over $\mathcal{O}_k$, such that G is isomorphic to $\mathscr{G}_p(\mathbf{L})$. In this case, $\mathbf{L} = \mathrm{Lie}(G)$ by 4.2.2.*

(ii) *There exists a projective pseudocompact k-module ω such that the affine algebra of G is isomorphic to the quotient of $k[[\omega]]$ by the closed ideal generated by the elements x^p, $x \in \omega$. In this case, $\omega \simeq \omega_{G/k}$.*

(iii) *The group G has height at most 1, and $\omega_{G/k}$ is a projective pseudocompact k-module.*

(iv) *The group G has height at most 1, and is topologically flat over k.*

Proof of (i) $\Rightarrow$ (ii). Set

$$\omega = \Gamma^*(\mathbf{L})$$

(cf. 1.2.3.D), and let A be the quotient of $k[[\omega]]$ by the closed ideal generated by the elements x^p, $x \in \omega$. Every continuous morphism

$$h : A \longrightarrow C, \qquad C \in \mathrm{Alf}\,/k,$$

is determined by its restriction h' to ω, and this restriction maps ω into $\sqrt[p]{0_C}$. Thus there is a canonical bijection from

$$\mathrm{Hom}_{\mathrm{Alp}\,/k}(A, C)$$

onto the set

$$\mathrm{Hom}_c(\omega, \sqrt[p]{0_C})$$

of continuous k-linear maps from ω into $\sqrt[p]{0_C}$. The latter set is canonically isomorphic to

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$$

(see the proof of 3.3). The implication (i) $\Rightarrow$ (ii) therefore follows from the functorial bijection

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathscr{G}_p(\mathbf{L})(C)$$

established in 4.3.1.

For the remaining implications, consult the analogous proofs of Theorems 3.3 and VIIA 7.4.1.

Remark 4.4.A. Let G be an infinitesimal formal k-group, with affine algebra A, such that

$$\omega_{G/k} = I_A/I_A^2$$

is a projective pseudocompact k-module. Then the projection

$$I_A \longrightarrow \omega_{G/k}$$

has a section

$$\tau : \omega_{G/k} \longrightarrow I_A,$$

and τ induces a surjective continuous morphism of algebras

$$\psi : k[[\omega_{G/k}]] \longrightarrow A;$$

compare the proof of the implication (iii) $\Rightarrow$ (i) in 3.3.[139]

4.4.1 Corollary

Corollary 4.4.1. *If k is a field of characteristic $p > 0$, the functor*

$$\mathbf{L} \longmapsto \mathscr{G}_p(\mathbf{L})$$

is an equivalence from the category of p-Lie algebras over k to the category of formal k-groups of height at most 1.

Indeed, as G ranges over the formal groups of height at most 1, Theorem 4.4 identifies the functor

$$G \longmapsto \mathscr{G}_p(\mathrm{Lie}\, G)$$

[139] The editors added this remark, which is used in 4.4.2.

with the identity functor. Likewise, 4.2.2 identifies the functor

$$\mathbf{L} \longmapsto \mathrm{Lie}\,\mathscr{G}_p(\mathbf{L})$$

with the identity functor (see also VIIA 5.5.3).[140]

4.4.2

Continue to assume that k is a field of characteristic p. Let G be an infinitesimal formal k-group, with affine algebra A. Since G is infinitesimal, every open ideal of A contains

$$I_A^{\{p^n\}}$$

for all sufficiently large n. Thus A is the projective limit of the affine algebras

$$A/I_A^{\{p^n\}}$$

of the groups $\mathrm{Fr}_n\, G$ (cf. 4.1.2). Equivalently, every infinitesimal formal k-group is an inductive limit of formal k-groups of finite height.

Suppose that G has height at most n, and write

$$H = G/\,\mathrm{Fr}\, G.$$

[141] By 2.4 and 2.4.1, the morphism

$$\mathrm{Fr} : G \longrightarrow G^{(p)}$$

factors as an epimorphism

$$\pi : G \longrightarrow H$$

followed by a monomorphism

$$i : H \longrightarrow G^{(p)}.$$

We therefore have the following commutative diagram:

$$\begin{array}{ccccc} G & \xrightarrow{\pi} & H & \xrightarrow{\mathrm{Fr}^{n-1}} & H^{(p^{n-1})} \\ & \searrow^{\mathrm{Fr}} & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i^{(p^{n-1})}} \\ & & G^{(p)} & \xrightarrow{\mathrm{Fr}^{n-1}} & G^{(p^n)} \end{array}$$

Since the functor $X \mapsto X^{(p)}$ commutes with fiber products, the morphism

$$i^{(p^{n-1})}$$

is again a monomorphism. Moreover, $\mathrm{Fr}^n(G/k)$ is zero and π is an epimorphism. Hence

$$\mathrm{Fr}^{n-1}\big(G^{(p)}/k\big) \circ i = i^{(p^{n-1})} \circ \mathrm{Fr}^{n-1}(H/k)$$

is zero. Because $i^{(p^{n-1})}$ is a monomorphism,

$$\mathrm{Fr}^{n-1}(H/k) = 0,$$

[140] If k is an Artinian ring of characteristic $p > 0$, the same proof gives an equivalence between the category of p-Lie algebras flat over k and the category of formal k-groups of height at most 1 that are topologically flat over k.

[141] The editors supplied the details of the original argument in what follows.

so H has height at most $n-1$.

It follows that every formal k-group of finite height has a composition series whose quotients have height at most 1, and therefore can be described by p-Lie algebras over k.

Finally, the affine algebra A of G is a quotient of

$$k[[\omega_{G/k}]]$$

by Remark 4.4.A. If $\omega_{G/k}$ is finite-dimensional over k, all the algebras

$$A/I_A^{\{p^n\}}$$

are finite-dimensional k-vector spaces. Consequently, every infinitesimal formal group G over a field of characteristic $p>0$ for which $\omega_{G/k}$ is finite-dimensional over k is an inductive limit of finite formal groups (that is, groups of finite length; cf. 1.2.6).

5. Homogeneous spaces of infinitesimal formal groups over a field

5.0

For simplicity, suppose that k is a field.[142] Let G be a formal k-group with affine algebra

$$A = \mathscr{A}(G).$$

Let A^+ be the augmentation ideal of A. For $a \in A$, write

$$\overline{a} = a - \varepsilon_A(a)1_A$$

for its projection onto A^+. Let F be a formal subgroup of G, defined by a closed ideal J of A, and let

$$\pi : A \longrightarrow A/J$$

be the projection. Write

$$\overline{\pi} : A \longrightarrow A^+/J$$

for the composite of the projections

$$A \longrightarrow A^+ \longrightarrow A^+/J.$$

For every $a \in A$, the projection of $\Delta_A(a)$ onto $A\widehat{\otimes}A^+$ is

$$\Delta_A(a) - a\widehat{\otimes}1.$$

By Theorem 2.4, the quotient formal k-variety $X = G/F$ exists. Its affine algebra is

$$\begin{aligned} B &= \mathrm{Eq}\left(A \underset{(\mathrm{id}_A \widehat{\otimes}\pi)\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A\widehat{\otimes}_k(A/J) \right) \\ &= \left\{ a \in A \;\middle|\; (\mathrm{id}_A \widehat{\otimes}\pi)\Delta_A(a) = a\widehat{\otimes}\pi(1) \right\} \\ &= \ker\left((\mathrm{id}_A \widehat{\otimes}\overline{\pi})\Delta_A\right), \end{aligned}$$

[142] The editors added subsection 5.0 in order to express Proposition 5.1 below in the language of cocommutative Hopf algebras and to cite results subsequently obtained in this direction.

where

$$\tau_1(a) = a\widehat{\otimes}\pi(1).$$

Equivalently, B is the subalgebra of A consisting of the elements φ such that

$$\varphi(gh) = \varphi(g)$$

for every $C \in \mathrm{Alf}\,/k$, $g \in G(C)$, and $h \in F(C)$. For $g, g' \in G(C)$ and $h \in F(C)$, one has

$$\varphi(gg'h) = \varphi(gg').$$

Thus $\Delta_A(\varphi)$ belongs to the kernel of

$$\mathrm{id}_A\,\widehat{\otimes}_k((\mathrm{id}_A\,\widehat{\otimes}\overline{\pi})\Delta_A),$$

which equals $A\widehat{\otimes}_k B$, since A is topologically flat over k. Consequently,

$$\Delta_A(B) \subset A\widehat{\otimes}_k B;$$

that is, the closed subalgebra B is also a left coideal.

Conversely, B determines F, because Corollary 2.4.1 gives

$$J = AB^+, \qquad B^+ = B \cap A^+;$$

in other words, J is the closed ideal generated by B^+. We therefore obtain an injective map

$$\mathcal{Q} : \mathcal{F} \longrightarrow \mathcal{B}$$

from the set $\mathcal{F}$ of formal subgroups of G to the set $\mathcal{B}$ of closed subalgebras of A that are left coideals. Proposition 5.1 below shows that $\mathcal{Q}$ is bijective when G is infinitesimal. In fact, it is bijective for every formal k-group G.

Recall from 2.2.1 that the functor

$$G \longmapsto \mathbf{H}(G)$$

is an equivalence from formal k-groups to cocommutative k-Hopf algebras. If F is defined by the closed ideal $J \subset A$, then the Hopf subalgebra

$$\mathbf{H}(F) \subset H = \mathbf{H}(G)$$

is the orthogonal complement of J for the duality

$$A = H^*, \qquad H = A^\dagger$$

(cf. 0.2.2). Conversely, if B is a closed subalgebra of A that is also a left coideal, then

$$I = B^\perp$$

is a coideal and a left ideal of H; explicitly,

$$\Delta_H(I) \subset I \otimes H + H \otimes I, \qquad \varepsilon_H(I) = 0.$$

Let $\mathcal{H}$ be the set of Hopf subalgebras of H, and let $\mathcal{I}$ be the set of left ideals of H that are also coideals. For $I \in \mathcal{I}$, denote by

$$\pi_I : H \longrightarrow H/I$$

the projection, and by

$$\overline{\pi}_I : H \longrightarrow H^+/I$$

the composite

$$H \longrightarrow H^+ \longrightarrow H^+/I,$$

where H^+ is the augmentation ideal of H.

Let K be a Hopf subalgebra of H, and put

$$K^+ = K \cap H^+.$$

If F is the formal subgroup corresponding to K, then

$$J = K^\perp$$

and A^+/J identifies with the dual of K^+. Since $B = \mathcal{Q}(F)$ is the kernel of

$$A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{\mathrm{id}\,\widehat{\otimes}\overline{\pi}} A\widehat{\otimes}_k(A^+/J),$$

duality identifies $\mathcal{Q}$ with the map

$$\Phi : \mathcal{H} \longrightarrow \mathcal{I}, \qquad K \longmapsto HK^+.$$

Indeed, HK^+ is the image of

$$H \otimes_k K^+ \xrightarrow{\mathrm{id}\otimes\mathrm{can}} H \otimes_k H \xrightarrow{m_H} H,$$

and is both a left ideal and a coideal.

Similarly, the map that assigns AB^+ to $B \in \mathcal{B}$ corresponds by duality to

$$\Psi : \mathcal{I} \longrightarrow \mathcal{H},$$

where

$$\Psi(I) = \{x \in H \mid (\mathrm{id}_H \otimes \pi_I)\Delta_H(x) = x \otimes \pi_I(1)\}.$$

Theorem 5.0.1. *Let k be a field and H a cocommutative Hopf algebra. The maps Φ and Ψ above are mutually inverse bijections between the Hopf subalgebras of H and the left ideals of H that are coideals.*

This theorem was first proved by K. Newman ([Ne75], Theorem 4.1, where the word "cocommutative" was omitted). Newman's proof uses the Cartier–Gabriel–Kostant theorem (cf. 2.9) to reduce to the case in which H is connected, and then uses the existence of a "Sweedler basis" in that case ([Sw67], Theorem 3), a result dual to the Dieudonné–Cartier Theorem 5.2.2 below. A second, shorter proof was given by H. J. Schneider ([Sch90], Theorem 4.15). A. Masuoka subsequently generalized the result to the case in which only the coradical H_0 of H, the sum of its simple subcoalgebras, is assumed commutative ([Ma91], Theorem 1.3 (3)).

Finally, for a commutative k-Hopf algebra corresponding to an affine k-group scheme G, one cannot expect an analogue of Theorem 5.0.1 without additional hypotheses: for a k-subgroup $F \subset G$, the quotient G/F need not be affine. Nevertheless, M. Takeuchi established analogous bijections. In [Tak72], Theorem 4.3, invariant k-subgroups F correspond to subalgebras $B \subset \mathcal{O}(G)$ satisfying

$$\Delta(B) \subset A \otimes B$$

and stable under the antipode. In [Tak79], Theorem 3, the k-subgroups F for which G/F is affine correspond to such subalgebras B for which $B \to A$ is faithfully flat.

5.1

Let k be a pseudocompact ring.[143] Let G be a topologically flat infinitesimal formal k-group, let A be its affine algebra, and let $B \subset A$ be a closed subalgebra. Put

$$X = \mathrm{Spf}(B),$$

and let

$$r : G \longrightarrow X$$

be the epimorphism induced by the inclusion $B \subset A$. We shall determine when r makes X the quotient of G on the right by a subgroup H (cf. 2.4).[144]

Proposition 5.1. *Let G be a topologically flat infinitesimal formal k-group, let A be its affine algebra, let I_A be the augmentation ideal of A, and let B be a closed subalgebra of A. Put*

$$I_B = B \cap I_A, \qquad J_B = AI_B.$$

Assume that A, and also

$$J_B^n/J_B^{n+1}$$

for every $n \geq 0$, are topologically flat over k. Then the following two assertions are equivalent:

(i) *For every $x \in B$,*

$$\Delta_A(x) - x\widehat{\otimes}1 \in A\widehat{\otimes}_k J_B.$$

(ii) *Let*

$$\pi : A \longrightarrow A/J_B$$

be the projection, and put

$$\tau_1(a) = a\widehat{\otimes}1.$$

The sequence

$$B \longrightarrow A \underset{(\mathrm{id}_A\,\widehat{\otimes}\pi)\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A\widehat{\otimes}_k(A/J_B) \qquad (*)$$

is exact. Equivalently, B is the set of all $x \in A$ for which

$$\Delta_A(x) - x\widehat{\otimes}1 \in A\widehat{\otimes}_k J_B.$$

In this case,

$$H = \mathrm{Spf}(A/J_B)$$

is a formal subgroup of G, and the sequence

$$G \times H \underset{\lambda}{\overset{\mathrm{pr}_1}{\rightrightarrows}} G \longrightarrow \mathrm{Spf}(B) \qquad (**)$$

is exact, where λ is the restriction of the multiplication of G to $G \times H$. Thus

$$\mathrm{Spf}(B) \simeq G/H.$$

[143] The original assumes in 5.1 that k is a field. In fact, that hypothesis can be replaced by flatness hypotheses; the editors modified 5.1–5.1.5 accordingly.

[144] The editors replaced "left" by "right" and modified the statement of Proposition 5.1 so as to display both the equivalent conditions (i), (ii) and the conclusion $\mathrm{Spf}(B) \simeq G/H$ more clearly.

Proof. Put

$$\overline{A} = A/J_B, \qquad H = \mathrm{Spf}(\overline{A}).$$

Then H is a formal subvariety of G. Since $J_B \subset I_A$, the augmentation ε_A induces a continuous k-algebra morphism

$$\overline{\varepsilon} : \overline{A} \longrightarrow k.$$

If (i) holds, then

$$\Delta_A(I_B) \subset I_B \widehat{\otimes} 1 + A \widehat{\otimes} J_B.$$

Therefore Δ_A induces, after passage to the quotient, a diagonal morphism

$$\overline{\Delta} : \overline{A} \longrightarrow \overline{A} \widehat{\otimes}_k \overline{A}.$$

The maps $\overline{\Delta}$ and $\overline{\varepsilon}$ make H a formal submonoid of G. Since G is infinitesimal, so is H; Proposition 2.7 then shows that H is a formal subgroup of G. By the definition of G/H (cf. 2.4), assertion (ii) implies the final assertion of the proposition.

It is clear that (ii) implies (i). The proof of the converse occupies 5.1.1–5.1.5.

5.1.1

Consider the following category $\mathcal{C}$. An object of $\mathcal{C}$ is a pair (A, J) consisting of a profinite k-algebra A and a closed ideal $J \subset A$. A morphism

$$\psi : (A, J) \longrightarrow (A', J')$$

in $\mathcal{C}$ is a continuous k-algebra homomorphism

$$A \longrightarrow A'$$

that maps J into J'.

The assignment

$$(A, J) \longmapsto (\mathrm{Spf}(A/J), \mathrm{Spf}(A))$$

is an antiequivalence from $\mathcal{C}$ to the category of pairs (Z, Y) consisting of a formal k-variety Y and a formal subvariety $Z \subset Y$. A morphism

$$\varphi : (Z, Y) \longrightarrow (Z', Y')$$

is a morphism of formal k-varieties

$$Y \longrightarrow Y'$$

that maps Z into Z'.

A cogroup structure on an object (A, J) of $\mathcal{C}$ consists of a formal-group structure on $\mathrm{Spf}(A)$ satisfying the following conditions (with the notation of 2.1):

(1) $\Delta_A(J) \subset J \widehat{\otimes}_k A + A \widehat{\otimes}_k J$,
(2) $\varepsilon_A(J) = 0$,
(3) $c_A(J) \subset J$.

These conditions also say that

$$H = \mathrm{Spf}(A/J)$$

is a formal subgroup of

$$G = \mathrm{Spf}(A).$$

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Suppose in addition that A is local; equivalently, $\mathrm{Spf}(A)$ is an infinitesimal formal group. If $J \neq A$, then conditions (2) and (3) follow from (1). Indeed, a proper closed ideal J of A is contained in the augmentation ideal I_A, so (2) holds. Moreover,

$$M = \mathrm{Spf}(A/J)$$

is a formal submonoid of G. Since G is infinitesimal, Proposition 2.7 shows that M is a formal subgroup of G; hence (3) also holds.

5.1.2

Denote by $\mathbf{Alpg}/k$ the category of *graded profinite k-algebras*. An object of this category consists of a sequence

$$A_0, A_1, \ldots, A_n, \ldots$$

of pseudocompact k-modules, together with a profinite algebra structure on

$$\prod_{n\geq 0} A_n$$

such that

$$A_n A_m \subset A_{m+n}.$$

Here A_n is identified with a subset of $\prod_{i\geq 0} A_i$ by means of the canonical injection. A morphism

$$\psi : (A_n) \longrightarrow (B_n)$$

is a sequence of continuous linear maps

$$\psi_n : A_n \longrightarrow B_n$$

such that

$$\psi_{m+n}(aa') = \psi_m(a)\psi_n(a')$$

for $a \in A_m$ and $a' \in A_n$.

Definitions. Two graded profinite k-algebras (A_n) and (B_n) clearly have a coproduct[146] in $\mathbf{Alpg}/k$. Its n-th component is

$$\prod_{i=0}^{n} A_i \widehat{\otimes}_k B_{n-i},$$

the product of the pseudocompact k-modules

$$A_i \widehat{\otimes}_k B_{n-i}.$$

This coproduct will be denoted by

$$(A_n)\widehat{\otimes}_k(B_n).$$

[145] If (A', J') is a second cogroup of $\mathcal{C}$, corresponding to a pair $H' \subset G'$ of formal k-groups, then giving a cogroup morphism

$$(A', J') \longrightarrow (A, J)$$

is equivalent to giving a morphism of formal k-groups

$$G \longrightarrow G'$$

that maps H into H'.

[146] The editors replaced "direct sum" by "coproduct".

A cogroup structure on an object (A_n) of $\mathbf{Alpg}/k$ is therefore the datum of continuous k-linear maps

$$\Delta_n : A_n \longrightarrow \prod_{i=0}^{n} A_i \widehat{\otimes}_k A_{n-i}, \qquad \varepsilon : A_0 \longrightarrow k,$$

which induce on $\prod_{n\geq 0} A_n$, upon setting $\varepsilon(A_i) = 0$ for $i \geq 1$, a cogroup structure in $\mathbf{Alp}/k$.

Finally, for every object (A, J) of $\mathcal{C}$, denote by

$$\mathrm{Gr}_J(A)$$

the graded profinite algebra associated with the filtration of A by the closures $\overline{J^n}$ of the powers of J. Thus

$$\mathrm{Gr}_J(A)_n = \overline{J^n}/\overline{J^{n+1}},$$

and the multiplication of $\mathrm{Gr}_J(A)$ is induced by that of A.

Lemma. [147] *Let U, V be two pseudocompact k-modules, with U topologically flat, and let*

$$U = U_0 \supset U_1 \supset \cdots, \qquad V = V_0 \supset V_1 \supset \cdots$$

be two decreasing sequences of closed k-submodules. Filter the completed tensor product

$$W = U \widehat{\otimes}_k V$$

by means of the closed submodules

$$W_n = U_n \widehat{\otimes}_k V_0 + U_{n-1} \widehat{\otimes}_k V_1 + \cdots + U_0 \widehat{\otimes}_k V_n.$$

Suppose that every U_i/U_{i+1} is topologically flat over k, so that U/U_n, and hence also U_n, is topologically flat for every n. Then, for every n, there is an isomorphism

$$W_n/W_{n+1} \simeq \bigoplus_{i+j=n} (U_i/U_{i+1}) \widehat{\otimes}_k (V_j/V_{j+1}).$$

Proof. For every $i, j \geq 0$, put

$$W_{i,j} = U_i \widehat{\otimes} V_j, \qquad \overline{W}_{i,j} = (U_i/U_{i+1}) \widehat{\otimes}_k (V_j/V_{j+1}).$$

We prove by induction on n that the natural map

$$\pi_n : W_n \longrightarrow \bigoplus_{i+j=n} \overline{W}_{i,j}$$

is surjective and that the inclusion

$$W_{n+1} \subset \ker(\pi_n)$$

is an equality. For $n = 0$, the projection

$$\pi_0 : U_0 \widehat{\otimes}_k V_0 \longrightarrow (U_0/U_1) \widehat{\otimes}_k (V_0/V_1)$$

is surjective. Since U_0, U_0/U_1, and hence U_1, are topologically flat over k, we have

$$\ker(\pi_0) = U_0 \widehat{\otimes}_k V_1 + U_1 \widehat{\otimes}_k V_0,$$

[147] In the original, the lemma is stated when k is a field, and in that case the proof is left to the reader.

and moreover

$$(U_0\widehat{\otimes}_k V_1)\cap(U_1\widehat{\otimes}_k V_0)=U_1\widehat{\otimes}_k V_1.$$

Suppose now that $n>0$ and that the result has been proved for $n-1$. Put

$$M_0=U_0\widehat{\otimes}_k V_n,\qquad S_0=\sum_{i=1}^{n}U_i\widehat{\otimes}_k V_{n-i}.$$

We have $S_0\subset U_1\widehat{\otimes}_k V_0$. Applying the preceding argument to $V_0\supset V_n$ in place of $V_0\supset V_1$, we obtain

$$M_0\cap S_0\subset(U_0\widehat{\otimes}_k V_n)\cap(U_1\widehat{\otimes}_k V)=U_1\widehat{\otimes}_k V_n.$$

It follows that

$$M_0\cap S_0=U_1\widehat{\otimes}_k V_n.$$

Since $W_n=M_0+S_0$, we obtain a commutative diagram with exact rows, where

$$U_i'=U_{i+1},\qquad W'_{i,n-1-i}=W_{i+1,n-1-i}\quad(0\le i\le n-1):$$

$$\begin{array}{ccccccccc}
0 & \longrightarrow & S_0 & \longrightarrow & W_n & \longrightarrow & (U_0/U_1)\widehat{\otimes}_k V_n & \longrightarrow & 0\\
 & & \downarrow{\scriptstyle \pi'_{n-1}} & & \downarrow{\scriptstyle \pi_n} & & \downarrow{\scriptstyle p} & & \\
0 & \longrightarrow & \bigoplus_{i=0}^{n-1}\overline{W}'_{i,n-1-i} & \longrightarrow & \bigoplus_{i=0}^{n}\overline{W}_{i,n-1} & \longrightarrow & \overline{W}_{0,n} & \longrightarrow & 0
\end{array}$$

Then p is surjective, with kernel

$$(U_0/U_1)\widehat{\otimes}_k V_{n+1}.$$

Moreover, by the induction hypothesis applied to the sequence (U_i'), the map π'_{n-1} is surjective, with kernel

$$W_n'=\sum_{i=1}^{n}W_{i,n+1-i}.$$

It follows that π_n is surjective and that

$$W_{n+1}\subset\ker(\pi_n)$$

is an equality. This proves the lemma.

Let us return to an object (A,J) of $\mathcal{C}$. By 0.2.G, the hypothesis that every

$$\overline{J^n}/\overline{J^{n+1}}$$

is topologically flat over k is equivalent to saying that $\mathrm{Gr}_J(A)$ is topologically flat over k.

Corollary. *Let $\mathcal{P}$ be the full subcategory of $\mathcal{C}$ consisting of the objects (A,J) such that A and $\mathrm{Gr}_J(A)$ are topologically flat over k. Then the functor*

$$\mathcal{P}\longrightarrow\mathbf{Alpg}/k,\qquad(A,J)\longmapsto\mathrm{Gr}_J(A),$$

commutes with finite coproducts and hence carries a cogroup of $\mathcal{P}$ to a cogroup of $\mathbf{Alpg}/k$.

In particular, if k is a field, then for every cogroup (A,J) of $\mathcal{C}$, the algebra $\mathrm{Gr}_J(A)$ is a cogroup of $\mathbf{Alpg}/k$. [148]

[148]Thus, to a pair $H\subset G$ of formal k-groups one associates the "formal completion $\widehat{G}_H$ of G along H", which is a formal k-group. Moreover, we shall see in 5.1.3–5.1.4 that the inclusion $\sigma: H\hookrightarrow\widehat{G}_H$ has a retraction $\pi:\widehat{G}_H\to H$, and that the formal k-group $N=\ker(\pi)$ identifies, as a formal variety, with the completion of the homogeneous space G/H along the unit section. This will be useful in 5.2.2.

5.1.3

Identify every profinite k-algebra Γ with the graded profinite k-algebra $(\Gamma_n)_{n\geq 0}$ such that

$$\Gamma_0 = \Gamma, \qquad \Gamma_n = 0 \quad \text{if } n > 0.$$

In particular, if $(A_n)_{n\geq 0}$ is a graded profinite k-algebra, we shall regard A_0, without distinction, as either a profinite k-algebra or a graded profinite k-algebra. Denote by

$$\rho : (A_n) \longrightarrow A_0$$

the morphism of $\mathbf{Alpg}/k$ such that

$$\rho_0 = \mathrm{id}_{A_0}, \qquad \rho_n = 0 \quad \text{if } n > 0.$$

Similarly,

$$\tau : A_0 \longrightarrow (A_n)$$

will denote the section of ρ such that

$$\tau_0 = \mathrm{id}_{A_0}, \qquad \tau_n = 0 \quad \text{if } n > 0.$$

Every cogroup structure on $(A_n)_{n\in\mathbb{N}}$ induces a cogroup structure on A_0 for which ρ and τ are cogroup homomorphisms. In this case, let I_0 be the augmentation ideal of A_0, and put

$$A'_n = A_n/I_0A_n \qquad (n \geq 0),$$

so that $A'_0 = k$.

Then $(A'_n)_{n\in\mathbb{N}}$ is a cogroup in $\mathbf{Alpg}/k$. Indeed, since

$$A_0 = I_0 \oplus k \cdot 1,$$

the direct summand

$$I_0\widehat{\otimes}_k A_n \simeq I_0A_n$$

of A_n is defined for every n. Since τ is a section of ρ, it follows from 2.4.A that the cogroup $(A_n)_{n\in\mathbb{N}}$ is the "semidirect coproduct" of A_0 and the cogroup $(A'_n)_{n\in\mathbb{N}}$.

More precisely, $(A'_n)_{n\in\mathbb{N}}$ is isomorphic, as an object of $\mathbf{Alpg}/k$, to the kernel of the following pair:

$$(A_n) \underset{(\mathrm{id}\,\widehat{\otimes}\rho)\Delta}{\overset{\tau_1}{\rightrightarrows}} (A_n)\widehat{\otimes}_k A_0$$

Here

$$\Delta : (A_n) \longrightarrow (A_n)\widehat{\otimes}(A_n)$$

is the comultiplication of (A_n), and

$$\tau_1(x) = x\widehat{\otimes}1.$$

Identifying A'_n with its image in A_n, the map

$$A'_n\widehat{\otimes}_k A_0 \longrightarrow A_n, \qquad a'_n\widehat{\otimes}a_0 \longmapsto a'_na_0,$$

is an isomorphism in $\mathbf{Alpg}/k$.

This is not an isomorphism of cogroups. Rather,

$$\Delta(A') \subset A\widehat{\otimes}A',$$

and

$$(\gamma'_\rho\,\widehat{\otimes}\,\mathrm{id}) \circ \Delta\big|_{A'} = \Delta',$$

where Δ' is the comultiplication of A' and $\gamma'_\rho : A \to A'$ is the projection (cf. 2.4.A).

5.1.4

Let (A, J) be an object of $\mathcal{C}$, and let

$$(A_n) = \mathrm{Gr}_J(A)$$

be the associated object of $\mathbf{Alpg}/k$; thus

$$A_n = \overline{J^n}/\overline{J^{n+1}} \qquad (n \geq 0).$$

It is clear that the algebra

$$\mathscr{A} = \prod_{n\geq 0} A_n$$

is generated by A_0 and A_1. In other words, for $n \geq 1$, the multiplication map

$$A_1 \mathbin{\widehat{\otimes}}_{A_0} A_1 \mathbin{\widehat{\otimes}}_{A_0} \cdots \mathbin{\widehat{\otimes}}_{A_0} A_1 \longrightarrow A_n \tag{1}$$

is surjective.

Suppose further that (A, J) is a cogroup of $\mathcal{C}$ and that A and the quotients

$$\overline{J^n}/\overline{J^{n+1}}$$

are topologically flat over k. By Corollary 5.1.2, $\mathrm{Gr}_J(A)$ is then a cogroup of $\mathbf{Alpg}/k$. Hence, by 5.1.3, if one puts

$$A'_n = \left\{ x \in \overline{J^n}/\overline{J^{n+1}} \;\middle|\; \Delta(x) - x\widehat{\otimes}1 \in \bigoplus_{i=1}^{n} (\overline{J^{n-i}}/\overline{J^{n-i+1}})\widehat{\otimes}_k (\overline{J^i}/\overline{J^{i+1}}) \right\}, \tag{2}$$

then the map

$$A'_n \mathbin{\widehat{\otimes}}_k A_0 \longrightarrow A_n, \qquad a'_n \mathbin{\widehat{\otimes}} a_0 \longmapsto a'_n a_0,$$

is an isomorphism in $\mathbf{Alpg}/k$.[149]

Let I_0 denote the augmentation ideal of A_0. It follows from (1) and the following commutative diagram, in which

$$A_1'^{\,\widehat{\otimes} n} = \underbrace{A'_1 \mathbin{\widehat{\otimes}}_k \cdots \mathbin{\widehat{\otimes}}_k A'_1}_{n \text{ factors}},$$

that:

$$\begin{array}{ccccc}
A_1 \mathbin{\widehat{\otimes}}_{A_0} \cdots \mathbin{\widehat{\otimes}}_{A_0} A_1 & \xleftarrow{\sim} & A_1'^{\,\widehat{\otimes} n} \mathbin{\widehat{\otimes}}_k A_0 & \xleftarrow{\sim} & (A_1'^{\,\widehat{\otimes} n} \mathbin{\widehat{\otimes}}_k I_0) \oplus A_1'^{\,\widehat{\otimes} n} \\
\downarrow{\scriptstyle m} & & \downarrow{\scriptstyle m' \widehat{\otimes} \mathrm{id}} & & \downarrow{\scriptstyle (m' \widehat{\otimes} \mathrm{id}) \oplus m'} \\
A_n & \xleftarrow{\sim} & A'_n \mathbin{\widehat{\otimes}}_k A_0 & \xleftarrow{\sim} & (A'_n \mathbin{\widehat{\otimes}}_k I_0) \oplus A'_n
\end{array}$$

[149] Let G, H, and $\widehat{G}_H$ be the formal k-groups corresponding to A, $A_0 = A/J$, and $\mathrm{Gr}_J(A)$, respectively. Then $\tau : A_0 \hookrightarrow \mathrm{Gr}_J(A)$ corresponds to a retraction $\pi : \widehat{G}_H \to H$ of the inclusion $H \hookrightarrow \widehat{G}_H$, and the preceding assertion means that $\widehat{G}_H$ is the semidirect product of $N = \ker(\pi)$ by H.

the multiplication map

$$m' : A'_1 \widehat{\otimes}_k \cdots \widehat{\otimes}_k A'_1 \longrightarrow A'_n \tag{3}$$

is surjective. In other words, the profinite k-algebra

$$\mathscr{A}' = \prod_{n\geq 0} A'_n$$

is generated by its degree-one terms.

Let us now return to hypothesis (i) of Proposition 5.1.[150] Let G be a topologically flat infinitesimal formal k-group, A its affine algebra, I_A the augmentation ideal of A, B a closed subalgebra of A, and

$$J = AI_B, \qquad I_B = B \cap I_A.$$

Let H be the formal subgroup $\mathrm{Spf}(A/J)$, and let

$$\pi : A \longrightarrow A/J$$

be the projection. Suppose that

$$A/\overline{J^n}$$

is topologically flat over k for every $n \geq 1$, and that B is contained in the kernel $\widetilde{B}$ of the following pair:

$$A \underset{(\mathrm{id}_A \,\widehat{\otimes}_k \pi)\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A\widehat{\otimes}_k(A/J).$$

Put

$$(A_n) = \mathrm{Gr}_J(A),$$

let

$$I_0 = I_A/J$$

be the augmentation ideal of $A_0 = A/J$, and define $(A'_n)_{n\in\mathbb{N}}$ as in (2) above. Let $(B_n)_{n\in\mathbb{N}}$, respectively $(\widetilde{B}_n)_{n\in\mathbb{N}}$, be the object of $\mathbf{Alpg}/k$ associated with the filtration of B, respectively of $\widetilde{B}$, induced by that of A; these filtrations are defined by the ideals

$$B \cap \overline{J^n}, \qquad \text{respectively} \qquad \widetilde{B} \cap \overline{J^n}.$$

It is clear that

$$B_n \subset \widetilde{B}_n \subset A'_n \qquad \text{for every } n,$$

and that

$$\mathscr{B} = \prod_{n\geq 0} B_n \subset \widetilde{\mathscr{B}} = \prod_{n\geq 0} \widetilde{B}_n$$

are subalgebras of

$$\mathscr{A}' = \prod_{n\geq 0} A'_n.$$

On the other hand, J, respectively $\overline{J^2}$, is the image in A of

$$I_B\widehat{\otimes}_k A, \qquad \text{respectively} \qquad \overline{I_B^2}\widehat{\otimes}_k A.$$

[150] The editors expanded the original in what follows and moved to the end the "supplement" $\overline{I_B^n} = B \cap \overline{J^n}$, which is not needed to prove Proposition 5.1.

Consequently, the map

$$(I_B/\overline{I_B^2})\widehat{\otimes}_k A \longrightarrow J/\overline{J^2} = A_1 \simeq A_1'\widehat{\otimes}_k A_0$$

is surjective, and it factors through

$$(I_B/\overline{I_B^2})\widehat{\otimes}_k A_0.$$

Since

$$A_0 = k\cdot 1 \oplus I_0,$$

the following commutative exact diagram shows that A_1' is the image of $I_B/\overline{I_B^2}$:

$$\begin{array}{ccc}
(I_B/\overline{I_B^2})\widehat{\otimes}_k A_0 & \xleftarrow{\sim} & \big((I_B/\overline{I_B^2})\widehat{\otimes}_k I_0\big) \oplus (I_B/\overline{I_B^2}) \\
\downarrow & & \downarrow \\
J/\overline{J^2} & \xleftarrow{\sim} & (A_1'\widehat{\otimes}_k I_0) \oplus A_1' \\
\downarrow & & \downarrow \\
0 & & 0
\end{array}$$

Thus

$$B_1 = A_1'.$$

Since $\mathscr{A}'$ is generated by A_1', it follows that, for every $n \geq 1$,

$$A_n' \subset B_n \subset \widetilde{B}_n \subset A_n',$$

and hence

$$B_n = \widetilde{B}_n = A_n'.$$

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Finally, because G is infinitesimal,

$$\mathfrak{r}(A) = I_A,$$

and hence the ideals $\overline{I_A^n}$ tend to zero (cf. 0.1.2). A fortiori, the ideals $\overline{J^n}$ tend to zero, so the induced filtrations on $\widetilde{B}$ and B are separated. Moreover, since B is a closed subalgebra of A, it is complete for the topology defined by the ideals $B \cap \overline{J^n}$. It therefore follows from [CA], § V.7, Lemma 1 (see also 5.1.5 below) that

$$B = \widetilde{B}.$$

This completes the proof of Proposition 5.1.

There is also the following supplement. For every n,

$$\overline{J^n} = A\overline{I_B^n}$$

is the image in A both of

$$A\widehat{\otimes}_B \overline{I_B^n}$$

[151] With the notation of editorial note 149, this implies that the formal completion of G/H along the unit section, whose affine algebra is $\prod_n B_n$, is isomorphic as a formal variety to the formal k-group N.

and of

$$A\widehat{\otimes}_B(B\cap\overline{J^n}).$$

By hypothesis, the affine algebra A/J of the formal subgroup H is topologically flat over k. Thus, by Theorem 2.4, the morphism

$$G=\mathrm{Spf}(A)\longrightarrow G/H=\mathrm{Spf}(B)$$

is surjective and topologically flat. Consequently,

$$A\widehat{\otimes}_B\overline{I_B^n}=\overline{J^n}=A\widehat{\otimes}_B(B\cap\overline{J^n}),$$

which implies

$$\overline{I_B^n}=B\cap\overline{J^n}\qquad\text{for every } n.$$

This also follows from the surjectivity of the maps

$$\overline{I_B^n}/\overline{I_B^{n+1}}\longrightarrow A'_n=(B\cap\overline{J^n})/(B\cap\overline{J^{n+1}})$$

and from 5.1.5(ii) below, applied to

$$B'_n=\overline{I_B^n},\qquad B_n=B\cap\overline{J^n}.$$

5.1.5

Lemma 5.1.5. [152]

(i) *Let M and N be two abelian groups filtered by decreasing sequences of subgroups*

$$(M_n)_{n\in\mathbb{Z}},\qquad (N_n)_{n\in\mathbb{Z}}.$$

Suppose that

$$\bigcup_{n\in\mathbb{Z}}M_n=M,\qquad \bigcup_{n\in\mathbb{Z}}N_n=N,$$

that

$$\bigcap_{n\in\mathbb{Z}}M_n=0,\qquad \bigcap_{n\in\mathbb{Z}}N_n=0,$$

and that M is complete for the topology defined by the M_n. Let

$$f:M\longrightarrow N$$

be a homomorphism of filtered groups.

a) *If f induces a surjection on the associated graded groups, then f is surjective and N is complete for the topology defined by the N_n.*

b) *If f induces an injection on the associated graded groups, then f is injective.*

[152] The original stated only part (ii). For the reader's convenience, the editors stated in part (i) Lemma 1 of [CA], § V.7.

(ii) *Let B be an abelian group, and let*

$$B = B'_0 \supset B'_1 \supset \cdots, \qquad B = B_0 \supset B_1 \supset \cdots$$

be two separated filtrations of B by subgroups such that

$$B'_n \subset B_n \qquad \textit{for every } n.$$

Suppose that B is complete for the topology defined by the filtration (B'_n). If the map

$$B'_i/B'_{i+1} \longrightarrow B_i/B_{i+1}$$

is surjective for every i, then

$$B'_n = B_n \qquad \textit{for every } n.$$

Proof. Part (i) is Lemma 1 of [CA], § V.7 (see also [BAC], III, § 2.8). Part (ii) follows by taking

$$M = B'_n \supset B'_{n+1} \supset \cdots, \qquad N = B_n \supset B_{n+1} \supset \cdots.$$

5.2

Throughout the rest of Section 5, k denotes a perfect field of characteristic $p > 0$.

Put

$$\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}.$$

If B is a profinite k-algebra and $r \in \overline{\mathbb{N}}$, denote by

$$\big((x^{p^r})\big)_{x \in \mathfrak{r}(B)}$$

the closed ideal of B generated by the elements x^{p^r}, where x ranges over the radical $\mathfrak{r}(B)$ of B. If $r = \infty$, use the same notation with the convention that

$$\big((x^{p^\infty})\big)_{x \in \mathfrak{r}(B)}$$

is the zero ideal. In both cases, B_r denotes the quotient

$$B_r = B/\big((x^{p^r})\big)_{x \in \mathfrak{r}(B)}.$$

We say that B is *of height at most r* if

$$\big((x^{p^r})\big)_{x \in \mathfrak{r}(B)}$$

is the zero ideal. If this holds and r is finite, we say that B is *of finite height*.

Consider in particular the case

$$B = k[[\omega]],$$

where ω is a pseudocompact k-vector space (cf. 1.2.5).[153] We then say that B is a *formal power series algebra* and that B_r is a *truncated formal power series algebra* $\big(r \in \overline{\mathbb{N}}\big)$; thus, by convention,

$$B = B_\infty$$

[153] In the original, the author uses "linearly compact vector space," which is equivalent to "pseudocompact vector space" (cf. [BAC], §III.2, Exercises 15 a), 19 a), and 20 d)). The editors preferred to retain the terminology "pseudocompact," used up to this point.

is also called "truncated." If $B = k[[\omega]]$, we also write

$$\big((x^{p^r})\big)_{x\in\omega}$$

instead of

$$\big((x^{p^r})\big)_{x\in\mathfrak{r}(B)}.$$

Notation. Let ω be a pseudocompact k-vector space filtered by an increasing sequence of closed vector subspaces

$$0 = \omega_0 \subset \omega_1 \subset \omega_2 \subset \omega_3 \subset \cdots .$$

(a) The closed ideal of $k[[\omega]]$ generated by the elements x^{p^r}, where r ranges over $\mathbb{N}$ and x ranges over ω_r, will be denoted by

$$\big((x^{p^r})\big)_{x\in\omega_r}.$$

(b) For $r \in \mathbb{N}$, denote by ${}^r\omega$ the filtered pseudocompact vector space defined by

$${}^r\omega_i = \omega_i \quad \text{if } i < r, \qquad {}^r\omega_i = \omega \quad \text{if } i \geq r.$$

Theorem (Dieudonné–Cartier). *Let*

$$H \longrightarrow G$$

be a monomorphism of infinitesimal formal groups over a perfect field k of characteristic $p > 0$. Let B be the affine algebra of the homogeneous space G/H, and suppose that one of the following three conditions holds:[154]

(i) *B has finite height* (in particular, this holds if G has finite height).

(ii) *B is a complete Noetherian local ring.*

(iii) *B is a reduced ring.*

Then B is isomorphic to the completed tensor product of a finite family of truncated formal power series algebras.

The proof of this theorem occupies 5.2.1–5.2.5.

5.2.1

Let A be the affine algebra of G, let I_A be its augmentation ideal, and put

$$I = B \cap I_A.$$

By 2.4,

$$H = \mathrm{Spf}(A/\overline{AI})$$

and

$$B = \left\{ x \in A \;\middle|\; \Delta(x) - x\widehat{\otimes}1 \in A\widehat{\otimes}\overline{AI} \right\}.$$

Put

$$\omega = I/\overline{I^2}.$$

[154] The editors replaced $H\backslash G$ by G/H, and did the same in the proof; they also added condition (iii).

For each r, denote by I_r the closed subideal of I consisting of the elements x such that

$$x^{p^r} = 0,$$

and denote by ω_r the canonical image of I_r in ω. We shall prove the following proposition.

Proposition. *If there exists a continuous section*

$$\sigma : \omega \longrightarrow I$$

of the projection $I \to I/\overline{I^2}$ such that

$$\sigma(\omega_r) \subset I_r$$

for every r, then B is isomorphic to

$$k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r}.$$

Indeed, such a section extends to a continuous morphism

$$k[[\omega]] \longrightarrow B,$$

which factors through

$$B' = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r}.$$

In 5.2.2–5.2.5 we prove that the resulting morphism

$$\varphi : B' = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r} \longrightarrow B$$

is an isomorphism.

5.2.1.A. [155] For every $r \in \mathbb{N}$, put

$$\mathcal{I}_r = \overline{I^2} + I_r,$$

so that

$$\mathcal{I}_r/\overline{I^2} \simeq \omega_r.$$

There is then a commutative diagram with exact rows:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{I}_{r-1} & \longrightarrow & \mathcal{I}_r & \longrightarrow & \mathcal{I}_r/\mathcal{I}_{r-1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow\wr & & \\
0 & \longrightarrow & \omega_{r-1} & \longrightarrow & \omega_r & \longrightarrow & \omega_r/\omega_{r-1} & \longrightarrow & 0
\end{array}$$

Since k is a field, the rows split. Thus a pseudobasis $\mathscr{B}_{r-1}$ of $\mathcal{I}_{r-1}$ can be completed to a pseudobasis

$$\mathscr{B}_{r-1} \cup \mathscr{B}'_r$$

of $\mathcal{I}_r$. The closed subspace S_r with pseudobasis $\mathscr{B}'_r$ is then a complement of $\mathcal{I}_{r-1}$ in $\mathcal{I}_r$, and the projection

$$\pi : I \longrightarrow \omega$$

induces an isomorphism from S_r onto a complement ω'_r of ω_{r-1} in ω_r.

[155] The editors added 5.2.1.A and 5.2.1.B.

Let $\mathcal{I}_\infty$ be the closed ideal

$$\mathcal{I}_\infty = \overline{\bigcup_r \mathcal{I}_r}.$$

It likewise has a complement S_∞ in I, and π induces an isomorphism from

$$\mathcal{I}_\infty/\overline{I^2}$$

onto the closure

$$\omega_\infty = \overline{\bigcup_r \omega_r}$$

of the union of the ω_r, and an isomorphism from S_∞ onto a complement ω'_∞ of ω_∞ in ω. Denote the latter isomorphism by

$$\eta : S_\infty \xrightarrow{\sim} \omega'_\infty.$$

We thus obtain continuous linear maps:

$$\begin{array}{ccc} \overline{I^2} \times S_\infty \times \widehat{\bigoplus}_r S_r & \xrightarrow{\phi} & I \\ \Big\downarrow{\scriptstyle \eta} \times\theta & & \\ \omega = \omega'_\infty \times \omega_\infty & & \end{array}$$

Here

$$\widehat{\bigoplus}_r S_r$$

is the direct sum of the S_r in $\mathrm{PC}(k)$, that is,

$$\left(\prod_r S_r^\dagger\right)^*$$

(cf. editorial note (16) to 0.2.2), and

$$\theta : \widehat{\bigoplus}_r S_r \longrightarrow \omega_\infty$$

is induced by the maps

$$S_r \xrightarrow{\sim} \omega'_r \hookrightarrow \omega_\infty.$$

It follows that a sufficient condition (but not a necessary one, as seen below) for obtaining a section

$$\sigma : \omega \longrightarrow I$$

of the desired kind is that θ be an isomorphism. By duality (cf. 0.2.2), this is equivalent to saying that the linear map

$$\omega_\infty^\dagger \longrightarrow \prod_r S_r^\dagger$$

is bijective.

5.2.1.B. As above, put

$$\omega_\infty = \overline{\bigcup_{n\in\mathbb{N}} \omega_n}.$$

A second case in which a section

$$\sigma : \omega \longrightarrow I$$

of the desired kind exists is the case in which ω_∞ has a pseudobasis $\mathscr{B}_\infty$ that is the union, for $n \in \mathbb{N}^*$, of pseudobases of the spaces

$$\omega_n/\omega_{n-1}.$$

One can then complete it by a pseudobasis $\mathscr{B}'_\infty$ of ω/ω_∞, obtaining a pseudobasis of ω compatible with the filtration.

Put

$$V = \omega_\infty^\dagger,$$

and let V_n be the orthogonal complement of ω_n in V. The preceding condition is equivalent to saying that, in the category of ordinary k-vector spaces, the separated decreasing filtration

$$V = V_0 \supset V_1 \supset V_2 \supset \cdots$$

splits: namely, V is the direct sum, over $n \in \mathbb{N}$, of subspaces F_n such that

$$F_n \simeq V_n/V_{n+1}.$$

This need not hold. For example, let

$$V = \mathscr{S} = k^{\mathbb{N}}$$

be the space of sequences of elements of k, and let $\mathscr{S}_n$ be the subspace of sequences (u_i) such that $u_i = 0$ for $i < n$. Then

$$\dim \mathscr{S}_n/\mathscr{S}_{n+1} = 1,$$

but $\mathscr{S}$ is not isomorphic to the direct sum of the $\mathscr{S}_n/\mathscr{S}_{n+1}$, because $\mathscr{S}$ is not countable-dimensional. In this example, however, $\mathscr{S}$ is the product of the $\mathscr{S}_n/\mathscr{S}_{n+1}$ (cf. 5.2.1.A).

The splitting does hold when V is countable-dimensional.[156] Indeed, let

$$(e_n)_{n\in\mathbb{N}}$$

be a basis of V. We shall construct inductively an increasing function

$$g : \mathbb{N} \longrightarrow \mathbb{N}$$

and subspaces F_i, for $i = 0, \ldots, g(n)$, such that

$$F_i \simeq V_i/V_{i+1}$$

and

$$F_{\leq g(n)} = \bigoplus_{i=0}^{g(n)} F_i$$

[156]This paragraph arose from discussions with J.-M. Fontaine and E. Bouscaren; in particular, Bouscaren indicated the proof that follows.

is a complement of $V_{g(n)+1}$ containing

$$e_0, \ldots, e_n.$$

It will then follow that

$$V = \bigoplus_{i \geq 0} F_i.$$

Let $n+1 \in \mathbb{N}$, and assume the assertion already proved for n (it is vacuous for $n = -1$). If

$$e_{n+1} \in F_{\leq g(n)},$$

put $g(n+1) = g(n)$. Otherwise write

$$e_{n+1} = f + x, \qquad f \in F_{\leq g(n)}, \qquad 0 \neq x \in V_{g(n)+1}.$$

Let j be the least integer such that

$$x \in V_j \setminus V_{j+1}.$$

For

$$i = g(n)+1, \ldots, j,$$

choose a complement F_i of V_{i+1} in V_i, choosing F_j so that $x \in F_j$. Then

$$e_{n+1} \in F_{\leq j};$$

put $g(n+1) = j$.

5.2.1.C. [157] In particular, the two preceding conditions (5.2.1.A and 5.2.1.B) hold when the filtration of ω is stationary, that is, when there exists an integer n_0 such that

$$\omega_n = \omega_{n_0} \qquad \text{for } n_0 \leq n < +\infty.$$

In this case,

$$k[[\omega]]/\big((x^{p^r})\big)_{x \in \omega_r}$$

is isomorphic to the completed tensor product

$$\frac{k[[\omega_1]]}{\big((x^p)\big)_{x \in \omega_1}} \mathbin{\widehat{\otimes}_k} \frac{k[[\omega'_2]]}{\big((x^{p^2})\big)_{x \in \omega'_2}} \mathbin{\widehat{\otimes}_k} \cdots \mathbin{\widehat{\otimes}_k} \frac{k[[\omega'_{n_0}]]}{\big((x^{p^{n_0}})\big)_{x \in \omega'_{n_0}}} \mathbin{\widehat{\otimes}_k} k[[\omega'_\infty]],$$

where ω'_n is a complement of ω_{n-1} in ω_n, and ω'_∞ is a complement of

$$\omega_\infty = \omega_{n_0}$$

in ω.

The filtration of ω is plainly stationary in case (i), where $\omega_r = \omega$ for all sufficiently large r; in case (ii), where ω is finite-dimensional; and also in case (iii), where

$$I_r = 0$$

for every r. In the last case, B is isomorphic to the formal power series algebra $k[[\omega]]$. The preceding remarks therefore imply the theorem, subject to 5.2.2–5.2.5 below.

[157] Here the editors return to the original, shortened to take account of the preceding additions.

5.2.2

Suppose first that B has height at most 1; that is,

$$x^p = 0 \qquad \text{for every } x \in I.$$

By 5.1.4, the graded algebra

$$\mathrm{Gr}_I(B)$$

associated with the filtration

$$I \supset I^2 \supset I^3 \supset \cdots$$

of B is endowed with a cogroup structure in the category **Alpg**/k. In other words, the profinite algebra

$$C = \prod_{n \geq 0} \mathrm{Gr}_I(B)_n$$

is the affine algebra of a formal k-group N. Clearly,

$$\omega_{N/k} = I/I^2,$$

and N is infinitesimal of height at most 1.

By 4.4, the identity map of $\omega_{N/k}$ therefore induces an isomorphism

$$B' = k[[\omega_{N/k}]]/\big((x^p)\big)_{x \in \omega_{N/k}} \xrightarrow{\sim} C.$$

In particular, the map ϕ of 5.2.1 induces an isomorphism between the associated graded algebras of B' and B, when both are filtered by the powers of their augmentation ideals. It follows from [CA], §V.7, Lemma 1 (see also 5.1.5) that ϕ is an isomorphism.

5.2.3

Suppose now that B has finite height at most r. Let

$$\pi : k \xrightarrow{\sim} k, \qquad x \longmapsto x^p,$$

be the Frobenius automorphism. The linear map

$$B \widehat{\otimes}_\pi k \longrightarrow B, \qquad b \widehat{\otimes}_\pi x \longmapsto b^p x = (b x^{1/p})^p,$$

has closed image equal to the closed subalgebra

$$B^p = \{\, b^p \mid b \in B \,\}$$

of B. Put

$$J = B^p \cap I = B^p \cap I_A.$$

[158] Let G_1 be the kernel of the Frobenius morphism

$$\mathrm{Fr} : G \longrightarrow G^{(p)},$$

and let HG_1 be the formal subgroup of G that is the inverse image of the formal subgroup $H^{(p)} \subset G^{(p)}$. Then HG_1 is defined by the closed ideal generated by the p-th powers of the

[158] The editors added what follows.

elements of AI; this ideal is AJ. Moreover, the formation of G/H commutes with base change, because G/H represents the quotient sheaf for the flat topology (cf. 2.4). Hence

$$(G/H)^{(p)} = G^{(p)}/H^{(p)},$$

and we have the following commutative diagrams:

$$\begin{array}{ccc} G & \xrightarrow{\mathrm{Fr}} & G^{(p)} \\ \downarrow & & \downarrow \\ G/H & \xrightarrow[\mathrm{Fr}]{} & G^{(p)}/H^{(p)} \end{array} \qquad \begin{array}{ccc} A & \xleftarrow{a\widehat{\otimes}1\mapsto a^p} & A\widehat{\otimes}_\pi k \\ \uparrow & & \uparrow \\ B & \xleftarrow{b\widehat{\otimes}1\mapsto b^p} & B\widehat{\otimes}_\pi k \end{array}$$

It follows that B^p is the affine algebra of the quotient G/HG_1.[159] Temporarily denote by C the affine algebra of the quotient HG_1/H. Since the formation of G/H commutes with base change, the square

$$\begin{array}{ccc} HG_1 & \longrightarrow & G \\ \downarrow & & \downarrow \\ HG_1/H & \longrightarrow & G/H \end{array}$$

is Cartesian. Consequently,

$$A\widehat{\otimes}_B C \simeq A/AJ = A\widehat{\otimes}_B(B/BJ).$$

Since A is topologically free over B (by 2.4, because A and B are local), the natural morphism

$$B/BJ \longrightarrow C$$

is an isomorphism. Thus B/BJ is the affine algebra of HG_1/H, and of course

$$B/BJ = B_1$$

has height at most 1, since

$$J = \big((x^p)\big)_{x\in\mathfrak{r}(B)}.$$

Let

$$B' = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r},$$

let $\phi : B' \to B$ be the morphism introduced in 5.2.1, let

$$B'^p = \{\, x^p \mid x \in B' \,\},$$

and let J' be the augmentation ideal of B'^p. We then have the commutative diagram

$$\begin{array}{ccc} B' & \xrightarrow{\psi} & B \\ \downarrow & & \downarrow \\ B'_1 = B'/\overline{B'J'} & \xrightarrow{\phi_1} & B_1 = B/\overline{BJ} \end{array}$$

[159] As indicated in the original, this also follows from Proposition 5.1. The editors preferred to give the preceding argument, which does not use the implication (i) ⇒ (ii) of that proposition.

and, by 5.2.2, ϕ_1 is an isomorphism.

On the other hand, by 2.4, A is topologically flat over

$$B = \mathscr{A}(G/H) \qquad \text{and over} \qquad B^p = \mathscr{A}(G/HG_1).$$

It follows from 1.3.3 that B is topologically flat over B^p. Moreover, 5.2.4 below shows that the morphism

$$B'^p \longrightarrow B^p$$

induced by ϕ is an isomorphism. We may therefore apply 0.3.4 to the pseudocompact ring $B'^p = B^p$ and the pseudocompact B^p-modules $M = B'$, $N = B$. By what precedes,

$$\phi_1 = \phi \widehat{\otimes}_{B^p} k$$

is an isomorphism, and it follows that ϕ itself is an isomorphism. This proves 5.2 when B has finite height, subject only to 5.2.4 below.

5.2.4

For every pseudocompact vector space V over k, denote by

$${}_\pi V = V \widehat{\otimes}_\pi k$$

the space obtained from V by extension of scalars along $x \mapsto x^p$.[160] We then have a commutative diagram with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & {}_\pi\overline{I^2} & \xrightarrow{\alpha} & {}_\pi I & \xrightarrow{\beta} & {}_\pi\omega & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle v} & & \downarrow{\scriptstyle w} & & \\ 0 & \longrightarrow & \overline{J^2} & \xrightarrow{\gamma} & J & \xrightarrow{\delta} & \overline{\omega} & \longrightarrow & 0\,, \end{array}$$

Here

$$\overline{\omega} = J/J^2,$$

and the maps u, v, w are induced by the linear map

$${}_\pi B \longrightarrow B^p, \qquad x \widehat{\otimes} a \longmapsto x^p a.$$

Since u and v are surjective, so is w, and its kernel is the image ${}_\pi\omega_1$ of

$${}_\pi I_1 = \ker(v).$$

Put

$$J_n = \{\, x \in J \mid x^{p^n} = 0 \,\}, \qquad \overline{\omega}_n = \delta(J_n).$$

Then, for every $n \geq 0$,

$$J_n = v({}_\pi I_{n+1}), \qquad \overline{\omega}_n = w({}_\pi\omega_{n+1}).$$

The section

$${}_\pi\sigma : {}_\pi\omega \longrightarrow {}_\pi I$$

induced by the section σ of 5.2.1 therefore descends to a section

$$\tau : \overline{\omega} \longrightarrow J$$

[160] Since k is perfect, one may identify ${}_\pi V$ with the abelian group V on which k acts by $\lambda \cdot v = \lambda^{1/p} v$.

compatible with the filtrations of J and $\overline{\omega}$. Since B^p has height at most $r-1$, the induction hypothesis shows that this section induces an isomorphism

$$\psi : B'' = k[[\overline{\omega}]]/\big((x^{p^n})\big)_{x\in\overline{\omega}_n} \xrightarrow{\sim} B^p.$$

But B'' identifies with B'^p, and ψ identifies with the morphism $B'^p \to B^p$ induced by ϕ. Thus the theorem is proved when B has finite height.

Remark 5.2.4.A. [161] Suppose that B has height at most $r+1$, with $r \in \mathbb{N}^*$. Then $I_{r+1} = I$, and there is an isomorphism

$$B \simeq \frac{k[[S_1]]}{\big((x_1^p)\big)_{x_1\in S_1}} \widehat{\otimes}_k \cdots \widehat{\otimes}_k \frac{k[[S_r]]}{\big((x_r^{p^r})\big)_{x_r\in S_r}} \widehat{\otimes}_k \frac{k[[S_{r+1}]]}{\big((x_{r+1}^{p^{r+1}})\big)_{x_{r+1}\in S_{r+1}}}, \tag{1}$$

where each S_n is a complement of

$$I^2 + I_{n-1}$$

in $I^2 + I_n$. Thus $\omega = I/I^2$ identifies with

$$\prod_{i=1}^{r+1} S_i,$$

and it is immediate that, for $n = 1, \ldots, r+1$, the image ω_n of I_n in ω identifies with

$$\prod_{i=1}^{n} S_i.$$

This has the following consequence. Let

$$B_r = B/\check{J}_r, \qquad \check{J}_r = \big((x^{p^r})\big)_{x\in\mathfrak{r}(B)},$$

and let

$$\mathfrak{m} = I/\check{J}_r$$

be the augmentation ideal of B_r. Since $\check{J}_r \subset I^2$, the space

$$\omega(r) = \mathfrak{m}/\mathfrak{m}^2$$

identifies with ω. For $n = 1, \ldots, r$, denote by $\omega(r)_n$ the image in $\omega(r)$ of

$$\mathfrak{m}_n = \{\, y \in \mathfrak{m} \mid y^{p^n} = 0 \,\}.$$

It is also the image in ω of

$$\{\, x \in I \mid x^{p^n} \in \check{J}_r \,\};$$

hence $\omega(r)_n$ contains ω_n.

On the other hand, isomorphism (1) gives

$$\check{J}_r = \big((x_{r+1}^{p^r})\big)_{x_{r+1}\in S_{r+1}},$$

and therefore

$$B_r \simeq \frac{k[[S_1]]}{\big((x_1^p)\big)_{x_1\in S_1}} \widehat{\otimes}_k \cdots \widehat{\otimes}_k \frac{k[[S_r]]}{\big((x_r^{p^r})\big)_{x_r\in S_r}} \widehat{\otimes}_k \frac{k[[S_{r+1}]]}{\big((x_{r+1}^{p^r})\big)_{x_{r+1}\in S_{r+1}}}. \tag{2}$$

It follows that $\omega(r)_n$ identifies with $\prod_{i=1}^n S_i$ for $n = 1, \ldots, r-1$. Thus the inclusion

$$\omega_n \subset \omega(r)_n$$

is an equality for $n = 1, \ldots, r-1$.

[161] The editors added this remark, which is used in 5.2.5.

5.2.5

It remains to consider the case in which B has infinite height and the projection $I \to \omega$ admits a section σ compatible with the filtrations of ω and I. Consider the morphism induced by σ:

$$\phi : B' = k[[\omega]]/\left((x^{p^n})\right)_{x\in\omega_n} \longrightarrow B.$$

It is enough to prove that, for every $r \in \overline{\mathbb{N}}$, the map

$$\phi_r : B'_r \longrightarrow B_r$$

induced by ϕ is an isomorphism.[162]

For every $r \in \mathbb{N}^*$, let G_r be the kernel of the iterated Frobenius morphism

$$G \longrightarrow G^{(p^r)},$$

and let HG_r be the formal subgroup of G that is the inverse image of $H^{(p^r)} \subset G^{(p^r)}$. Thus HG_r is defined by the closed ideal generated by the p^r-th powers of the elements of AI, namely

$$A\check{J}_r, \qquad \check{J}_r = \{\, x^{p^r} \mid x \in I \,\}.$$

Exactly as in 5.2.3, it follows that

$$B_r = B/B\check{J}_r$$

is the affine algebra of HG_r/H; in particular, it has height at most r.

Let

$$\mathfrak{m}(r) = I/B\check{J}_r$$

be the augmentation ideal of B_r. The canonical projection $B \to B_r$ induces an isomorphism

$$\omega = I/I^2 \xrightarrow{\sim} \omega(r) = \mathfrak{m}(r)/\mathfrak{m}(r)^2,$$

which we use to identify the two spaces. Let $\omega(r)_n$ be the image in $\omega(r)$ of the closed ideal

$$\mathfrak{m}(r)_n = \{\, y \in \mathfrak{m}(r) \mid y^{p^n} = 0 \,\}.$$

It is also the image in ω of the closed ideal

$$I(r)_n = \{\, x \in I \mid x^{p^n} \in B\check{J}_r \,\}.$$

[163]

Clearly,

$$\omega(r)_n = \omega \qquad \text{if } n \geq r.$$

We shall prove that

$$\omega(r)_n = \omega_n \qquad \text{if } n < r.$$

For every r, n, the sequence

$$0 \longrightarrow I(r)_n \cap I^2 \longrightarrow I(r)_n \longrightarrow \omega(r)_n \longrightarrow 0$$

is exact. For fixed n,

$$\bigcap_r I(r)_n = I_n,$$

[162]Indeed, B', respectively B, is the inverse limit of the B'_r, respectively the B_r. Moreover, the editors modified the original in what follows in order to take account of the addition made in 5.2.3.

[163]The editors expanded the original in what follows.

because $\bigcap_r B\check{J}_r = 0$. Filtered inverse limits are exact in $\mathrm{PC}(k)$ (cf. 0.2), so for every n,

$$\omega_n = \bigcap_r \omega(r)_n. \tag{*}$$

On the other hand, Remark 5.2.4.A gives

$$\omega(r)_n = \omega(r+1)_n \qquad \text{if } n < r.$$

Together with (*), this implies

$$\omega(r)_n = \omega_n \qquad \text{if } n < r.$$

Consequently, the vector space ω, filtered by the subspaces $\big(\omega(r)_n\big)_{n\geq 0}$, is precisely ${}^r\omega$ in the notation of 5.2. The composite

$$\sigma(r) : \omega \xrightarrow{\sigma} I \longrightarrow \mathfrak{m}(r)$$

is therefore compatible with the filtrations $\big(\omega(r)_n\big)$ and $\big(\mathfrak{m}(r)_n\big)$. Since

$$k[[{}^r\omega]]/\big((x^{p^n})\big)_{x\in {}^r\omega_n}$$

is exactly B'_r, and $\phi_r : B'_r \to B_r$ is induced by $\sigma(r)$, the result already proved for finite-height algebras shows that ϕ_r is an isomorphism.

5.2.6

Definition 5.2.6. [164] Let $(A_\lambda)_{\lambda\in\Lambda}$ be a family of profinite k-algebras, each endowed with an augmentation

$$\varepsilon_\lambda : A_\lambda \longrightarrow k.$$

(This holds in particular if every A_λ is local with residue field k.) Define the infinite completed tensor product

$$\mathcal{A} = \widehat{\bigotimes}^c_{\lambda\in\Lambda} A_\lambda$$

to be the inverse limit in $\mathrm{Alp}\,/k$ of

$$\mathcal{A}_F = \widehat{\bigotimes}^c_{\lambda\in F} A_\lambda,$$

where F ranges through the finite subsets of Λ. For $F' = F \cup \{\lambda\}$, the transition morphism

$$\mathcal{A}_{F'} \longrightarrow \mathcal{A}_F$$

is

$$\mathrm{id}\,\hat{\otimes}\varepsilon_\lambda.$$

In particular, suppose $\Lambda = \mathbb{N}^*$, and write

$$X_n = \mathrm{Spf}(A_n).$$

Then $\mathrm{Spf}(\mathcal{A})$ represents the functor that assigns to every $C \in \mathrm{Alf}\,/k$ the set of *finite* sequences in $\prod_{n\geq 1} X_n(C)$; that is, the sequences

$$(x_1, x_2, \ldots) \in \prod_{n\geq 1} X_n(C)$$

[164] The editors added this number in order to define the infinite tensor products used in 5.3(a).

such that

$$x_n = \varepsilon_n$$

for all sufficiently large n. Here, by abuse of notation, ε_n denotes the composite of

$$\varepsilon_n : A_n \longrightarrow k$$

with the structural morphism $k \to C$.

If, in addition, every A_n is a quotient of an algebra $k[[\omega'_n]]$, one may write 0 for the unique morphism

$$A_n \longrightarrow C$$

that vanishes on ω'_n. The preceding description then becomes the set of sequences for which $x_n = 0$ for all sufficiently large n.

5.3. Remarks

(a) Call a profinite k-algebra *stationary* if it is the completed tensor product of a family of truncated formal power series algebras.[165]

Let G be an infinitesimal formal k-group and let B be the affine algebra of a homogeneous space of G. It follows from Theorem 5.2 that, for every $r \in \mathbb{N}$, the algebra

$$B/\big((x^{p^r})\big)_{x\in\mathfrak{r}(B)}$$

is stationary. In particular, B is an inverse limit of stationary algebras.[166]

(b) I do not know whether, with the notation of 5.2.1, A and B can be chosen so that there exists no section

$$\sigma : \omega \longrightarrow I$$

compatible with the filtrations.[167] Nevertheless, ω may be any pseudocompact vector space filtered by an increasing sequence of closed subspaces. Indeed, given

$$\omega_1 \subset \omega_2 \subset \cdots \subset \omega,$$

one can define on

$$B = A = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r}$$

[165]The author doubtless had in mind a completed tensor product

$$\mathcal{A} = \widehat{\bigotimes}^{c}_{n\in\mathbb{N}^*} \frac{k[[\omega'_n]]}{\big((x^{p^n})\big)_{x\in\omega'_n}},$$

where the ω'_n are arbitrary pseudocompact vector spaces. Then $\omega = \omega_{\mathcal{A}/k}$ identifies with $\prod_n \omega'_n$, and the filtration is $\omega_n = \prod_{i=1}^n \omega'_i$; this is the case of 5.2.1.B. For this reason these algebras would be better called *stable*, rather than *stationary*; cf. [Di73], II, §2.9, p. 75. If, for example, $\dim \omega'_n = 1$ for every $n \in \mathbb{N}^*$, then $\mathcal{A}$ represents the functor that assigns to $C \in \mathrm{Alf}\,/k$ the sequences $(x_n)_{n\in\mathbb{N}^*}$ of elements of C such that $x_n^{p^n} = 0$ and $x_n = 0$ for all sufficiently large n. This case, where $\omega = \prod_{n\in\mathbb{N}^*} \omega'_n$, corresponds in the dual setting of connected cocommutative Hopf algebras to the case studied by M. E. Sweedler; cf. [Sw67], Theorem 3.

[166]Such an inverse limit need not itself be a stable profinite k-algebra in the sense of the preceding editor's note. For example, let S be the ordinary k-vector space of sequences $(u_1, u_2, \ldots)$ of elements of k, and put $\omega = S^*$. Then ω is the direct sum in $\mathrm{PC}(k)$ of copies kx_n of k, for $n \in \mathbb{N}^*$, where $x_n(u) = u_n$. Thus this is the case of 5.2.1.A. The algebra

$$\mathcal{A} = k[[\omega]]\big/\big((x_n^{p^n})\big)_{n\in\mathbb{N}^*}$$

has $\omega_{\mathcal{A}/k} = \omega$ and $\omega_n = \prod_{i=1}^n kx_i$, but it is not stable: $\mathrm{Spf}(\mathcal{A})$ represents the functor that assigns to every $C \in \mathrm{Alf}\,/k$ the infinite sequences $(x_n)_{n\in\mathbb{N}^*}$ of elements of C such that $x_n^{p^n} = 0$ for every n.

[167]The editors do not know either, outside the cases considered in 5.2.1.A and 5.2.1.B.

a diagonal morphism

$$\Delta_A : A \longrightarrow A\widehat{\otimes}_k A$$

satisfying conditions (i), (ii), and (iii) of 2.1. It suffices to put

$$\Delta_A(y) = y\widehat{\otimes}1 + 1\widehat{\otimes}y$$

when y is the image in A of an element of ω.

5.4. Corollary

Corollary 5.4. *Let G be an algebraic group over a perfect field k of characteristic $p > 0$, let H be an algebraic subgroup of G, let e be the image of the identity element of G in G/H, and let A be the local ring of G/H at e. Then $\widehat{A}$ is isomorphic to an algebra of the form*

$$k[[X_1, \ldots, X_r, \ldots, X_s]]/(X_1^{p^{n_1}}, \ldots, X_r^{p^{n_r}}).$$

Proof. Consider the infinitesimal formal groups

$$\widehat{G} = \mathrm{Spf}(\widehat{\mathcal{O}}_{G,e}), \qquad \widehat{H} = \mathrm{Spf}(\widehat{\mathcal{O}}_{H,e}).$$

By 1.3.4, the completion $\widehat{A}$ of

$$A = \mathcal{O}_{G/H,e}$$

is isomorphic to

$$\mathscr{A}(\widehat{G}/\widehat{H}),$$

and the corollary follows from Theorem 5.2(ii).[168]

5.5. Complements

[169]

Recall the following definitions. A Noetherian local ring A is called a *complete intersection* if its completion $\widehat{A}$ is the quotient of a complete regular Noetherian local ring B by an ideal I generated by a regular sequence of elements of B (cf. EGA IV$_4$, 19.3.1).

On the other hand, let

$$\tau : Y \hookrightarrow X$$

be a closed immersion of schemes. If $y \in Y$, then τ is called a *regular immersion at the point y* if the kernel of

$$\mathcal{O}_{X,y} \longrightarrow \mathcal{O}_{Y,y}$$

is generated by a regular sequence. If, moreover, X is locally Noetherian and τ is a regular immersion at every point, then τ is called a *regular immersion* (cf. *loc. cit.*, Proposition 16.9.10 and Definition 16.9.2).

Corollary 5.5.1. *If G is an algebraic group over a field k, then the local ring $\mathcal{O}_{G,e}$ is a complete intersection.*

Proof. By EGA IV$_4$, 19.3.4, we may assume that k is algebraically closed. If $\mathrm{char}(k) = 0$, then G is already known to be smooth (cf. 3.3.1 or VIB, 1.6.1); hence $\widehat{\mathcal{O}}_{G,e}$ is a k-algebra of formal power series, by EGA IV$_4$, 17.5.3 (d$''$). If $\mathrm{char}(k) = p > 0$, then 5.4, applied with $H = \{e\}$, shows that $\widehat{\mathcal{O}}_{G,e}$ is a complete intersection.

[168] See also [DG70], §III.3, Theorem 6.1.

[169] The editors added this subsection in order to give some consequences of 5.4 mentioned in Exposés III and VIA.

Remarks 5.5.2. *Let k be a field, let G be a smooth algebraic k-group, and let H be a closed subgroup of G.*

(a) We saw in Exposé III, 4.15, that the immersion

$$H \hookrightarrow G$$

is regular. This can also be deduced from 5.4 as follows. As in *loc. cit.*, we may assume that k is algebraically closed, and it is enough to show that the kernel I of

$$\mathcal{O}_{G,e} \longrightarrow \mathcal{O}_{H,e}$$

is generated by a regular sequence. Put

$$A = \mathcal{O}_{G,e}, \qquad \widehat{G} = \mathrm{Spf}(\widehat{A}), \qquad \widehat{H} = \mathrm{Spf}(\widehat{\mathcal{O}}_{H,e}).$$

Since A is Noetherian, there is an exact sequence

$$0 \longrightarrow I \otimes_A \widehat{A} \longrightarrow \widehat{A} \xrightarrow{\pi} \mathscr{A}(\widehat{H}) \longrightarrow 0,$$

and $\widehat{A}$ is faithfully flat over A. Therefore, by EGA IV$_4$, 16.9.10(ii) and 19.1.5(ii), it is enough to show that the kernel

$$\widehat{I} = I \otimes_A \widehat{A}$$

of π is generated by a regular sequence of elements of $\widehat{A}$.

Since G is smooth, $\widehat{A}$ is reduced. By 5.4, the subalgebra

$$B = \mathscr{A}(\widehat{G}/\widehat{H})$$

is therefore isomorphic to a formal power series algebra

$$k[[x_1, \ldots, x_n]].$$

Consequently, the identity section of $\widehat{G}/\widehat{H}$ is defined in B by the regular sequence $(x_1, \ldots, x_n)$. Since $\widehat{A}$ is Noetherian, the ideal J of $\widehat{A}$ generated by $x_1, \ldots, x_n$ is closed, and hence equals $\widehat{I}$, by Corollary 1.4. Moreover, $\widehat{A}$ is topologically flat, and hence flat over B (cf. 0.3.8). Thus $(x_1, \ldots, x_n)$ is a regular sequence in $\widehat{A}$, by EGA IV$_4$, 19.1.5(ii).

(b) The following more precise result can also be deduced from 5.2(ii). Suppose that k is perfect. Applying 5.2(ii) to the algebra

$$C = \mathscr{A}(H),$$

there are a basis

$$(y_1, \ldots, y_{r+s})$$

of ω_H and integers

$$1 \leq n_1 \leq \cdots \leq n_r$$

such that $\mathscr{A}(H)$ is isomorphic to the quotient of

$$k[[y_1, \ldots, y_{r+s}]]$$

by the ideal generated by

$$y_i^{p^{n_i}}, \qquad i = 1, \ldots, r.$$

Lift the y_i to elements x_i of ω_G, and extend $(x_1, \ldots, x_{r+s})$ to a basis $(x_1, \ldots, x_n)$ of ω_G. Since $\mathscr{A}(G)$ is reduced, the morphism

$$k[[x_1, \ldots, x_n]] \longrightarrow \mathscr{A}(G)$$

is an isomorphism, by 5.2(iii).

Thus there exists a "coordinate system" $(x_1, \ldots, x_n)$ on G, that is, an isomorphism

$$\mathscr{A}(G) \simeq k[[x_1, \ldots, x_n]],$$

such that H is defined by the equations

$$x_i^{p^{n_i}} = 0 \quad (i = 1, \ldots, r), \qquad x_i = 0 \quad (i > r + s).$$

This is "Dieudonné's theorem"; compare [Di55], §19, Theorem 6, and [Di73], II, §3.2, Proposition 3 and the material preceding it.

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Exposé VIII
Diagonalizable Groups

by A. Grothendieck[0]

1. Biduality

Let $\mathscr{C}$ be a category, which as usual we identify with a full subcategory of

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

(cf. Exposé I). Let I be a commutative group functor, that is, an object of $\widehat{\mathscr{C}}$ endowed with a commutative group structure (cf. I, 2.1).[1] For every $X \in \mathrm{Ob}(\widehat{\mathscr{C}})$, the object $\mathrm{Hom}(X, I)$ carries the commutative group structure induced by that of I. For every group G in $\widehat{\mathscr{C}}$, put

$$D(G) = \mathrm{Hom}_{\mathrm{gr}}(G, I),$$

the subobject of $\mathrm{Hom}(G, I)$ defined, for every $S \in \mathrm{Ob}(\mathscr{C})$, by

$$\text{(x)} \qquad D(G)(S) = \mathrm{Hom}_{S\text{-gr}}(G_S, I_S),$$

where $G_S = G \times S$ and $I_S = I \times S$ are regarded as S-groups, that is, as groups in $\widehat{\mathscr{C}}_{/S}$. Thus $D(G)$ is a $\widehat{\mathscr{C}}$-subgroup of $\mathrm{Hom}(G, I)$. In this way we obtain a contravariant functor D from the category of $\widehat{\mathscr{C}}$-groups to the category of commutative $\widehat{\mathscr{C}}$-groups.

The right-hand side of (x) may also be interpreted as the subset of $\mathrm{Hom}(G \times S, I)$ consisting of morphisms $G \times S \to I$ that are "multiplicative in the first argument G." Moreover, the preceding formulas remain valid when S is an arbitrary object of $\widehat{\mathscr{C}}$, not necessarily one arising from $\mathscr{C}$.

Now take for S a group in $\widehat{\mathscr{C}}$, denoted G'. In the left-hand side $\mathrm{Hom}(G', D(G))$ of (x), consider the subset $\mathrm{Hom}_{\mathrm{gr}}(G', D(G))$ consisting of morphisms that respect the group structures of G' and $D(G)$.

It corresponds to the subset of $\mathrm{Hom}(G \times G', I)$ consisting of morphisms that are multiplicative in each of the two arguments—which one may call *bilinear morphisms from* $G \times G'$ *to* I, or *pairings of* G *and* G' *with values in* I. We therefore have

$$\text{(xx)} \qquad \mathrm{Hom}_{\mathrm{gr}}(G', D(G)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{bil}}(G \times G', I),$$

[0] Editor's note: version 1.1 of November 8, 2009. Additions were made in 1.2, 1.4, 1.7, 3.1, 3.4, 4.5.1, 6.4, and 6.8; 1.5.1 and Section 7 remain to be revised.

[1] Editor's note: the original has been expanded in what follows.

functorially in the pair (G, G'). Since the right-hand side is symmetric in G and G', this gives a functorial bijection

$$\text{(xxx)} \qquad \mathrm{Hom}_{\mathrm{gr}}(G', D(G)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(G, D(G')).$$

In other words, giving a group homomorphism $G' \to D(G)$ "amounts to the same thing" as giving a group homomorphism $G \to D(G')$: both amount to giving a pairing $G \times G' \to I$.

Apply this with $G' = D(G)$ and with the identity homomorphism $G' \to D(G)$. We obtain a canonical homomorphism

$$\text{(xxxx)} \qquad G \longrightarrow D(D(G)).$$

Definition 1.0. [2] *The group G is called* reflexive *(relative to I) if the preceding homomorphism is an isomorphism. This implies, in particular, that G is commutative.*

We therefore have:

Proposition 1.0.1. *The functor D induces an anti-equivalence from the category of reflexive $\widehat{\mathscr{C}}$-groups to itself.*

In particular, if G and H are reflexive groups, D induces an isomorphism

$$\mathrm{Hom}_{\mathrm{gr}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(D(H), D(G)).$$

It is enough, in fact, that H be reflexive, as follows from (xxx).

Definition 1.0.2. *As usual, a $\mathscr{C}$-group G is called* reflexive *if it is reflexive as a $\widehat{\mathscr{C}}$-group, regardless of whether $D(G)$ is representable.*

Thus D gives an anti-equivalence from the category of reflexive $\mathscr{C}$-groups G for which $D(G)$ is representable to itself.

Remark 1.0.3. To conclude these generalities, note that formation of the dual $D(G)$ commutes with base extension, which therefore carries reflexive groups to reflexive groups.

We shall henceforth work in the following setting. Let $\mathscr{C} = (\mathbf{Sch})_{/S}$ be the category of preschemes over S, and let $I = \mathbf{G}_{m,S}$ be the "multiplicative group over S" (cf. Exposé I). For every ordinary group M, consider the constant S-group M_S. For every group prescheme J over S, there is a canonical isomorphism, functorial in M and J and compatible with base extension,

$$\mathrm{Hom}_{S\text{-gr}}(M_S, J) = \mathrm{Hom}_{\mathrm{gr}}(M, J(S)).$$

Applying this with $J = I = \mathbf{G}_{m,S}$ and with a variable S' over S, we obtain a functorial isomorphism

$$\text{(1.0.4)} \qquad D(M_S)(S') \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(M, \mathbf{G}_m(S')).$$

We thus recover the functor already considered in I, 4.4, also denoted $D_S(M)$, which is representable when M is commutative, since

$$D_S(M) = D(M_S) = \mathrm{Spec}\, \mathscr{O}_S(M),$$

where $\mathscr{O}_S(M)$ denotes the group algebra of M over $\mathscr{O}_S$. In general, moreover, $D(M)$ is unchanged when M is replaced by its abelianization, so no information is lost by assuming that M is commutative.

[2] Editor's note: the numbering 1.0, 1.0.1, and so on has been added to make the definitions and statements contained here explicit.

Definition 1.1. *A group prescheme G over S is called* diagonalizable *if it is isomorphic to a scheme of the form*

$$D_S(M) = D(M_S) = \mathrm{Hom}_{S\text{-gr}}(M_S, \mathbf{G}_m)$$

for a suitable commutative group M. It is called locally diagonalizable *if every point of S has an open neighborhood U such that $G|_U$ is diagonalizable.*

Theorem 1.2. *Let Γ be a constant commutative group scheme over S; that is, suppose that it is isomorphic to a group scheme of the form M_S, where M is an ordinary commutative group. Then Γ is reflexive: the canonical homomorphism*

$$\Gamma \longrightarrow D(D(\Gamma))$$

is an isomorphism.[3] *Hence the diagonalizable group $D(M_S)$ is also reflexive.*

In view of the definitions, this follows from the next statement, applied to a prescheme S' over S.

Corollary 1.3. *Let $G = D(M_S)$. Every homomorphism of S-groups*

$$u : G \longrightarrow \mathbf{G}_{m,S}$$

is defined by a unique section of M_S over S, equivalently by a unique locally constant map from S to M.

Proof. By definition,

$$\mathbf{G}_{m,S} = \mathrm{GL}(1)_S = \mathrm{Aut}_{\mathscr{O}_S\text{-mod}}(\mathscr{O}_S).$$

Thus giving a group homomorphism $G \to \mathbf{G}_{m,S}$ is equivalent to giving $\mathscr{O}_S$ the structure of a G-$\mathscr{O}_S$-module compatible with its natural $\mathscr{O}_S$-module structure (cf. I, 4.7). By I, 4.7.3, this is also equivalent to giving an M-grading on $\mathscr{O}_S$, that is, a direct sum decomposition

$$\mathscr{O}_S = \bigoplus_{m \in M} \mathscr{L}_m.$$

It is well known that a direct summand of a locally free module of finite type is locally free of finite type. Hence, near each point of S, every $\mathscr{L}_m$ is either zero or free of rank one; in the latter case it identifies with $\mathscr{O}_S$ on that neighborhood. Let S_m be the open subset of S on which the latter alternative occurs. The direct-sum decomposition shows that the S_m cover S and are pairwise disjoint.

Consequently, giving a group homomorphism $G \to \mathbf{G}_{m,S}$ is equivalent to giving a decomposition of S as a union of pairwise disjoint open subsets S_m, $m \in M$, hence to giving a locally constant map from S to M. This proves Corollary 1.3 and therefore Theorem 1.2. □

Corollary 1.4. *A diagonalizable group is reflexive, and therefore so is a locally diagonalizable group. If M and N are ordinary commutative groups, the natural homomorphism*

$$\mathrm{Hom}_{S\text{-gr}}(M_S, N_S) \longrightarrow \mathrm{Hom}_{S\text{-gr}}(D(N_S), D(M_S))$$

is bijective.

[3] Editor's note: the sentence that follows has been added.

[4] Since the preceding isomorphism is compatible with base extension, it yields an isomorphism of S-functors in groups

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S)). \tag{1}$$

For every S-prescheme T,

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S)(T) = \mathrm{Hom}_{\mathrm{gr}}\big(M, \Gamma(N_T/T)\big),$$

and by I, 1.8, $\Gamma(N_T/T)$ is the abelian group of locally constant maps $T \to N$. On the other hand, let $\mathrm{Hom}_{\mathrm{gr}}(M, N)_S$ be the constant S-group associated with the ordinary abelian group $\mathrm{Hom}_{\mathrm{gr}}(M, N)$. There is an evident homomorphism of S-functors in commutative groups

$$\mathrm{Hom}_{\mathrm{gr}}(M, N)_S \xrightarrow{\theta} \underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S), \tag{2}$$

which is always a monomorphism and is an isomorphism if M is of finite type.[5]

Part (a) of the following corollary follows from the preceding discussion; part (b) follows from it by the descent results recalled in 1.7. [6]

Corollary 1.5.

(a) *Let M and N be ordinary commutative groups, with M of finite type. Then there is an isomorphism*

$$\mathrm{Hom}_{\mathrm{gr}}(M, N)_S \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S));$$

consequently, $\underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S))$ is representable.

(b) *More generally, if G and H are locally diagonalizable and H is of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr}}(G, H)$ is representable.*

[7] Corollary 1.5 gives:

Corollary 1.6. *Under the hypotheses of 1.5, if S is connected, then*

$$\mathrm{Hom}_{S\text{-gr}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(M, N)$$

and

$$\mathrm{Isom}_{S\text{-gr}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Isom}_{\mathrm{gr}}(M, N).$$

[4] Editor's note: the discussion that follows has been expanded.

[5] Editor's note: write $F(M)$ and $G(M)$ for the left- and right-hand sides of (2), respectively. If $M = M_1 \oplus M_2$, there are canonical isomorphisms $F(M) = F(M_1) \oplus F(M_2)$ and $G(M) = G(M_1) \oplus G(M_2)$. It is therefore enough to check that θ is an isomorphism for $M = \mathbf{Z}/r\mathbf{Z}$, $r \geq 0$. In that case $F(M) = ({}_rN)_S$, where ${}_rN$ is the kernel of $r\,\mathrm{id}_N$, and for every $T \to S$ the homomorphism

$$F(M)(T) = \Gamma({}_rN_T/T) \longrightarrow G(M)(T) = {}_r\Gamma(N_T/T)$$

is bijective.

[6] Editor's note: the original stated after 1.5 that, more generally, if G and H are locally diagonalizable and H is of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr}}(G, H)$ is representable. That assertion has been included in 1.5 and its proof made explicit in 1.7.

[7] Editor's note: a statement 1.5.1 treating the functor $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, H)$, considered in X, 5.10 and 5.11, should be added.

1.7. Descent of representability. [8] We recall here several descent results that will be used frequently below.

Scholium 1.7.1. *Let S be a prescheme and let X, Y, T be S-preschemes. If (T_i) is an open covering of T, and if $T_{ij} = T_i \cap T_j = T_i \times_T T_j$, then, because giving a morphism of T-preschemes $X_T \to Y_T$ is local on T, there is an exact sequence of sets*

$$(1) \qquad \mathrm{Hom}_T(X_T, Y_T) \longrightarrow \prod_i \mathrm{Hom}_{T_i}(X_{T_i}, Y_{T_i}) \rightrightarrows \prod_{i,j} \mathrm{Hom}_{T_{ij}}(X_{T_{ij}}, Y_{T_{ij}}).$$

Thus $\underline{\mathrm{Hom}}_S(X, Y)$ is a local S-functor, that is, a sheaf on $(\mathbf{Sch})_{/S}$ for the Zariski topology.

More generally, by IV, 4.5.13, $\underline{\mathrm{Hom}}_S(X, Y)$ is a sheaf on $(\mathbf{Sch})_{/S}$ for every topology coarser than the canonical topology, for example the fpqc topology. If G and H are group preschemes over S, it follows that the subfunctor $\underline{\mathrm{Hom}}_{S\text{-gr}}(G, H)$ is an fpqc sheaf, hence a fortiori a local functor.

Lemma 1.7.2. [9] *Let F be a local S-functor.*

(i) *Suppose that there is an open covering (S_i) of S such that the restriction $F_i = F \times_S S_i$ of F to each S_i is representable by an S_i-prescheme X_i. Then F is representable by an S-prescheme X.*

(ii) *Suppose that F is an fpqc sheaf and that there is a faithfully flat quasi-compact morphism $S' \to S$ such that $F' = F \times_S S'$ is representable by an S'-prescheme X'. Then X' carries a descent datum relative to $S' \to S$ (cf. IV, 2.1). If this descent datum is effective—as it is when X' is affine over S'—then F is representable by an S-prescheme X.*

Proof. (i) The hypothesis implies that $X_i \times_S S_j$ and $X_j \times_S S_i$ both represent the restriction of F to $S_{ij} = S_i \times_S S_j$. The Yoneda lemma therefore gives a unique isomorphism of S_{ij}-preschemes

$$c_{ji} : X_i \times_S S_j \xrightarrow{\sim} X_j \times_S S_i.$$

Over $S_{ijk} = S_i \times_S S_j \times_S S_k$, we obtain the commutative diagram of isomorphisms

$$\begin{array}{ccccc} X_i \times_S S_j \times_S S_k & \xrightarrow{c_{ji}\times \mathrm{id}_{S_k}} & X_j \times_S S_i \times_S S_k & = & X_j \times_S S_k \times_S S_i \\ \| & & & & \downarrow{\scriptstyle c_{kj}\times \mathrm{id}_{S_i}} \\ X_i \times_S S_k \times_S S_j & \xrightarrow{c_{ki}\times \mathrm{id}_{S_j}} & X_k \times_S S_i \times_S S_j & = & X_k \times_S S_j \times_S S_i. \end{array}$$

Indeed, all of its objects represent the restriction of F to S_{ijk}. Thus the c_{ji} satisfy the usual cocycle relation $c_{kj} \circ c_{ji} = c_{ki}$.

It follows that the X_i glue to an S-prescheme X with $X \times_S S_i = X_i$ for every i. For every Y over S_i,

$$(*) \qquad F(Y) = F_i(Y) = \mathrm{Hom}_{S_i}(Y, X \times_S S_i) = \mathrm{Hom}_S(Y, X) = h_X(Y).$$

For an arbitrary $Y \to S$, the $Y_i = Y \times_S S_i$ form an open covering of Y; put $Y_{ij} = Y_i \times_Y Y_j = Y \times_S S_{ij}$. Since F is local by hypothesis and h_X is local because the Zariski topology is

[8] Editor's note: this paragraph has been added to make explicit the "local on S" character of the representability of certain sheaves on S, which is used repeatedly below and was used implicitly in the original.

[9] Editor's note: see also Remark XI, 3.4.

coarser than the canonical topology, both $F(Y)$ and $h_X(Y)$, in view of $(*)$, identify with the kernels of the respective double arrows

$$\prod_i F(Y_i) \rightrightarrows \prod_{i,j} F(Y_{ij}), \qquad \prod_i h_X(Y_i) \rightrightarrows \prod_{i,j} h_X(Y_{ij}).$$

This proves (i).

(ii) Put $S'' = S' \times_S S'$, and let S''_1 and S''_2 denote S'' regarded as an S'-prescheme through the first and second projections. The functors $F''_1 = F' \times_{S'} S''_1$ and $F''_2 = F' \times_{S'} S''_2$ are represented respectively by $X''_1 = X' \times_{S'} S''_1$ and $X''_2 = X' \times_{S'} S''_2$. Since

$$F''_1 = F \times_S S'' = F''_2,$$

there is a unique S''-isomorphism

$$c : X''_1 \xrightarrow{\sim} X''_2.$$

Let q_i, respectively p_{ji}, be the projection from $S''' = S' \times_S S' \times_S S'$ onto its i-th factor, respectively onto its i-th and j-th factors. Let $X'''_i = X' \times_{S'} S'''_i$, where S'''_i means S''' viewed as an S'-prescheme through q_i, and let $p^*_{ji}(c) : X'''_i \xrightarrow{\sim} X'''_j$ be obtained from c by base change. We obtain the diagram

$$\begin{array}{ccc} X'''_1 & \xrightarrow{p^*_{21}(c)} & X'''_2 \\ & \searrow^{p^*_{31}(c)} & \downarrow{\scriptstyle p^*_{32}(c)} \\ & & X'''_3. \end{array}$$

All three objects represent the restriction of F to S''', so the diagram commutes:

$$p^*_{32}(c) \circ p^*_{21}(c) = p^*_{31}(c).$$

Thus c is a descent datum on X' relative to $S' \to S$ (cf. IV, 2.1).

Suppose that this descent datum is effective, so there is an S-prescheme X such that $X' \simeq X \times_S S'$. This is the case when X' is affine over S', by SGA 1, VIII, 2.1.[10] For every $Y \to S'$, one then has

$$(**) \qquad F(Y) = F'(Y) = \mathrm{Hom}_{S'}(Y, X \times_S S') = \mathrm{Hom}_S(Y, X) = h_X(Y).$$

For arbitrary $Y \to S$, put $Y' = Y \times_S S'$ and $Y'' = Y' \times_Y Y' \simeq Y \times_S S''$. Like $S' \to S$, the morphism $Y' \to Y$ is faithfully flat and quasi-compact, hence an $\mathcal{M}$-effective epimorphism, where $\mathcal{M}$ is the family of faithfully flat quasi-compact morphisms. Thus the equivalence relation

$$Y' \times_Y Y' \rightrightarrows Y'$$

is $\mathcal{M}$-effective and has quotient Y. Since F is an fpqc sheaf by hypothesis, while h_X is one because the fpqc topology is coarser than the canonical topology, both $F(Y)$ and $h_X(Y)$, in view of $(**)$, identify with the respective kernels of

$$F(Y') \rightrightarrows F(Y' \times_Y Y'), \qquad h_X(Y') \rightrightarrows h_X(Y' \times_Y Y').$$

This proves (ii). □

[10] Editor's note: for another effectivity criterion, see X, 5.4–5.6.

Corollary 1.7.3. *Let F be an fpqc sheaf on $(\mathbf{Sch})_{/S}$. Suppose there is an open covering (S_i) of S, and for each i a faithfully flat quasi-compact morphism $S'_i \to S_i$, such that $F'_i = F \times_S S'_i$ is representable by an S'_i-prescheme X'_i affine over S'_i. Then F is representable by an S-prescheme X affine over S, with $X \times_S S'_i = X'_i$ for every i. If, moreover, every $X'_i \to S'_i$ is a closed immersion (respectively, a finite étale morphism), then so is $X \to S$.*

The first assertion follows from 1.7.2. For the second, it is enough to check that every morphism $X \times_S S_i \to S_i$ is a closed immersion (respectively, finite and étale); this follows from EGA IV_2, 2.7.1 (respectively, together with IV_4, 17.7.3).

Remark 1.7.4. Assertion 1.5(b) follows, as announced, from 1.7.1 and 1.7.2(i).

2. Scheme-theoretic Properties of Diagonalizable Groups

These properties are summarized in the following proposition.

Proposition 2.1. *Let S be a nonempty prescheme, let M be an ordinary commutative group, and let $G = D(M_S)$ be the diagonalizable S-group defined by M. Then:*

(a) *G is faithfully flat over S and affine over S (a fortiori, quasi-compact over S);*

(b) *M is of finite type if and only if G is of finite type over S, if and only if G is of finite presentation over S;*

(c) *M is finite if and only if G is finite over S, if and only if G is of finite type over S and is annihilated by an integer $n > 0$. In that case,*

$$\deg(G/S) = \mathrm{Card}(M);$$

(c') *M is a torsion group if and only if G is integral over S;*

(d) *$M = 0$ if and only if G is the unit S-group;*

(e) *M is of finite type and the order of its torsion subgroup is prime to the residue characteristics of S if and only if G is smooth over S.*

The verification of (a) through (d) is immediate and is left to the reader. Let us prove (e). If G is smooth over S, it is locally of finite presentation over S, hence of finite presentation over S, since it is affine over S; thus M is of finite type. We may therefore assume from the outset that M is of finite type, and hence that G is of finite presentation over S. Then[11] G is smooth over S if and only if its geometric fibers are smooth, reducing us to the case in which S is the spectrum of an algebraically closed field k.

Write $M = T \times L$, where T is the torsion subgroup and L is free, with $L \simeq \mathbb{Z}^r$. Then

$$D(M) = D(T) \times D(L), \qquad D(L) \simeq \mathbb{G}_m^r,$$

and $D(L)$ is smooth over k. Thus $G = D(M)$ is smooth over k if and only if $D(T)$ is smooth. Since $D(T)$ is finite over k of degree equal to the order n of T, this means that $D(T)(k)$ has n elements. Now T is isomorphic to a sum of groups $\mathbb{Z}/n_i\mathbb{Z}$, where $n = \prod_i n_i$; consequently

$$D(T) = \prod_i D(\mathbb{Z}/n_i\mathbb{Z}) = \prod_i \mu_{n_i},$$

[11] Editor's note: this uses the fact that G is flat over S, by (a).

where μ_{n_i} is the group scheme of n_i-th roots of unity. Hence

$$\operatorname{card}(D(T)(k)) = \prod_i \operatorname{card}(\mu_{n_i}(k)).$$

Here $\operatorname{card}(\mu_{n_i}(k))$, the number of n_i-th roots of unity in k, is at most n_i, and equality holds if and only if n_i is prime to the characteristic p of k. It follows that

$$\operatorname{card}(D(T)(k)) = n, \qquad n = \prod_i n_i,$$

if and only if every n_i is prime to p, that is, if and only if n is prime to p. □

3. Exactness Properties of the Functor D_S

Theorem 3.1. *Let S be a prescheme, and let*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be an exact sequence of ordinary commutative groups. Consider the sequence of transposed homomorphisms

$$0 \longrightarrow D_S(M'') \xrightarrow{v^t} D_S(M) \xrightarrow{u^t} D_S(M') \longrightarrow 0.$$

(i) *The homomorphism v^t induces an isomorphism from $D_S(M'')$ onto the kernel of u^t, and u^t is faithfully flat and quasi-compact.*

(ii) [12] *$D_S(M')$ represents the quotient sheaf $D_S(M)/D_S(M'')$ for the* (fpqc) *topology.*

Let $\mathcal{M}$ denote the family of faithfully flat quasi-compact morphisms. First, (ii) follows from (i) (cf. IV, 4.6.5.1). Indeed, the equivalence relation on $D_S(M)$ defined by u^t is the same as that defined by the subgroup

$$\operatorname{Ker}(u^t) = D_S(M'').$$

Since $u^t \in \mathcal{M}$, this equivalence relation is $\mathcal{M}$-effective (cf. IV, 3.3.2.1); therefore $D_S(M')$ represents the quotient sheaf for the (fpqc) topology (cf. IV, 4.6.5).

The first assertion of (i) is an immediate consequence of the definition of the functors $D_S(-)$. More generally, every exact sequence

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

with no zero on the left gives rise to an exact transposed sequence

$$0 \longrightarrow D_S(M'') \longrightarrow D_S(M) \longrightarrow D_S(M').$$

(This remains valid in the more general setting considered at the beginning of §1.) On the other hand, because $D_S(M)$ and $D_S(M')$ are affine over S, the morphism u^t is necessarily affine, and a fortiori quasi-compact, whatever the homomorphism $u : M' \to M$. The second assertion of (i) will therefore follow from part (a) of the following corollary.

Corollary 3.2. *Let S be a nonempty prescheme, let $u : M' \to M$ be a homomorphism of ordinary commutative groups, and let $u^t : G \to G'$ be the transposed homomorphism. Then:*

[12] Editor's note: what follows has been added, and the proof has been expanded accordingly.

(a) *u is a monomorphism if and only if u^t is faithfully flat;*

(b) *u is an epimorphism if and only if u^t is a monomorphism; in that case u^t is in fact a closed immersion.*

To prove (a), observe that if u is a monomorphism, then $\mathcal{O}_S(M)$, as a module over $\mathcal{O}_S(M')$, has a nonempty basis: namely, the system of sections defined by any system of representatives of M modulo M'. It is therefore faithfully flat. Conversely, if $\mathcal{O}_S(M)$ is faithfully flat over $\mathcal{O}_S(M')$, then

$$u^t : \mathcal{O}_S(M') \longrightarrow \mathcal{O}_S(M)$$

is injective, which, when $S \neq \varnothing$, implies that $u : M' \to M$ is injective.

To prove (b), observe that if u is an epimorphism, then $\mathcal{O}_S(M') \to \mathcal{O}_S(M)$ is surjective, so u^t is a closed immersion and, a fortiori, a monomorphism. Conversely, if u^t is a monomorphism, then $\operatorname{Ker} u^t$ is the unit group. Putting $M'' = \operatorname{Coker} u$, we have seen that

$$\operatorname{Ker} u^t \simeq D_S(M'').$$

Thus 2.1(d) gives $M'' = 0$, and u is an epimorphism.

The usual argument now gives the following consequence of 3.1.

Corollary 3.3. *Let*

$$M' \xrightarrow{u} M \xrightarrow{v} M''$$

be an exact sequence of ordinary commutative groups, and consider the transposed sequence

$$G'' \xrightarrow{v^t} G \xrightarrow{u^t} G'.$$

Then v^t induces a faithfully flat quasi-compact morphism from G'' to $\operatorname{Ker} u^t$, and this kernel is a diagonalizable group isomorphic to

$$D_S(v(M)) = D_S(\operatorname{Coker} u).$$

Corollary 3.4. *Let S be a prescheme, let $u : G \to H$ be a homomorphism of S-group preschemes that are locally diagonalizable, with H of finite type over S, and put $G' = \operatorname{Ker} u$. Then:*

(a) *G' is locally diagonalizable, and is of finite type over S if G is;*

(b) *the quotient G/G' "exists": more precisely, the equivalence relation on G defined by G' is $\mathcal{M}$-effective, where $\mathcal{M}$ is the family of faithfully flat quasi-compact morphisms (cf. IV, 3.4). Moreover, G/G' is locally diagonalizable and of finite type over S;*

(c) *the homomorphism $u : G \to H$ factors uniquely as*

$$G \xrightarrow{v} G/G' \xrightarrow{w} H,$$

where v is the canonical homomorphism, and is therefore faithfully flat and quasi-compact. Moreover, w is a closed immersion, and a fortiori a monomorphism.

Finally, the quotient

$$H' = H/\operatorname{Im} w = \operatorname{Coker} w = \operatorname{Coker} u$$

exists: more precisely, the equivalence relation on H defined by G/G' is $\mathcal{M}$-effective, and H' is of finite type over S.

The first assertion of (c) follows from (b), by the definition of the quotient G/G' (cf. IV, 3.2.3). [13] Let us show that the (fpqc) quotient sheaf $\widetilde{G}/\widetilde{G'}$ is representable. This can be checked locally on S, as can all the other assertions; we may therefore suppose that G and H are diagonalizable, of the form $D_S(M)$ and $D_S(N)$.

Since H is of finite type over S, the group N is of finite type by 2.1(b). Therefore, by 1.5, u is defined by a homomorphism $u' : N \to M$. It follows from 3.1 and 3.2 that

$$G' \simeq D_S(\operatorname{Coker} u')$$

and that $\widetilde{G}/\widetilde{G'}$ is represented by $D_S(\operatorname{Im} u')$. Moreover, on considering the exact sequence

$$0 \longrightarrow \operatorname{Ker} u' \longrightarrow N \xrightarrow{w'} \operatorname{Im} u' \longrightarrow 0,$$

we find that w is a closed immersion and that

$$H' = H/\operatorname{Im} w = D_S(\operatorname{Ker} u').$$

The latter is of finite type over S, since N, and hence $\operatorname{Ker} u'$, is of finite type.

Remarks. The existence result for quotients in 3.4 will be substantially generalized in §5.

On the other hand, note that in the present section and the preceding one, the hypothesis $S \neq \varnothing$ was used only to ensure the validity of certain converses, allowing properties of the corresponding ordinary groups to be deduced from certain hypotheses on diagonalizable S-groups. The results in the "forward" direction hold without any restriction on S, and the proofs given here apply in general.

Corollary 3.5. *Let G be a diagonalizable group prescheme over S, and let $n \neq 0$ be an integer. Then the subgroup ${}_nG$ of G, the kernel of the homomorphism*

$$n \cdot \mathrm{id}_G : G \longrightarrow G,$$

is integral over S, and is finite over S if G is of finite type over S.

Indeed, if $G = D_S(M)$, then

$${}_nG = D_S(M/nM)$$

by 3.1, and the result follows from 2.1(b), (c), and (c').

4. Torsors under a Diagonalizable Group

Let S be a prescheme and let $G = D_S(M)$ be a diagonalizable group over S. We shall determine the G-torsors (or principal homogeneous G-bundles) over S for the faithfully flat quasi-compact topology; cf. Exposé IV, 5.1. Recall that a prescheme P over S, with S-operator group G, is called a *torsor* or a *principal homogeneous bundle* if every point of S has an open neighborhood U and a faithfully flat quasi-compact morphism $S' \to U$ such that

$$P' = P \times_S S'$$

is isomorphic, as an object with operators, to

$$G' = G \times_S S',$$

[13] Editor's note: the original has been expanded in what follows.

where G' acts on itself by right translations. Since G is affine over S, SGA 1, VIII 5.6 implies that P is necessarily affine over S. Also, because G itself is faithfully flat and quasi-compact over S, P is principal homogeneous under G if and only if it is formally principal homogeneous and is, in addition, faithfully flat and quasi-compact over S (cf. IV, 5.1.6).

Recall also (Exposé I, 4.7.3) that giving an S-prescheme P, affine over S, with S-operator group $G = D_S(M)$, amounts to giving a commutative quasi-coherent algebra graded by M over S. In other words, it amounts to giving a quasi-coherent algebra $\mathscr{A}$ over S, together with a direct-sum decomposition as a module

$$\mathscr{A} = \coprod_{m \in M} \mathscr{A}_m,$$

such that

$$\mathscr{A}_m \cdot \mathscr{A}_{m'} \subseteq \mathscr{A}_{m+m'} \qquad \text{for } m, m' \in M.$$

With this understood, the answer to the preceding problem is as follows.

Proposition 4.1. *Let P be the prescheme with S-operator group $G = D_S(M)$ defined by the M-graded algebra $\mathscr{A}$. Then P is a principal homogeneous bundle under G if and only if $\mathscr{A}$ satisfies the following conditions:*

(a) *for every $m \in M$, $\mathscr{A}_m$ is an invertible module over S;*

(b) *for $m, m' \in M$, the homomorphism*

$$\mathscr{A}_m \otimes_{\mathcal{O}_S} \mathscr{A}_{m'} \longrightarrow \mathscr{A}_{m+m'}$$

induced by multiplication in $\mathscr{A}$ is an isomorphism.

The necessity of these conditions follows immediately by descent, since they hold when P is the trivial principal homogeneous bundle, that is, when $\mathscr{A} = \mathcal{O}_S(M)$. For sufficiency, note that (a) already implies that P is faithfully flat over S; it is in any event quasi-compact over S, since it is affine over S. It therefore remains to check that P is formally principal homogeneous under G, that is, that the familiar morphism

$$P \times_S G \longrightarrow P \times_S P$$

is an isomorphism. On affine algebras this morphism is given explicitly by the homomorphism

$$\mathscr{A} \otimes \mathscr{A} \longrightarrow \mathscr{A}(M) = \mathscr{A} \otimes \mathcal{O}_S(M).$$

In bidegree (m, n), where $m, n \in M$, it is given by

$$x_m \otimes y_n \longmapsto x_m y_n \otimes e_n.$$

At the level of degrees, this homomorphism is compatible with the homomorphism $M \times M \to M \times M$ given by

$$(m, n) \longmapsto (m + n, n),$$

which is an isomorphism. This shows that (b) expresses precisely, and independently of (a), that P is formally principal homogeneous, and proves 4.1.

Faithfully flat descent also gives:

Corollary 4.2. *The conditions of 4.1 imply that the homomorphism*

$$\mathcal{O}_S \longrightarrow \mathscr{A}_0$$

is an isomorphism.

For example, if $M = \mathbb{Z}$, then, under the conditions of 4.1, the algebra $\mathscr{A}$ is essentially determined by $\mathscr{A}_1 = \mathscr{L}$, namely

$$\mathscr{A} \simeq \coprod_{n\in\mathbb{Z}} \mathscr{L}^{\otimes n}$$

as graded algebras. Thus we recover the familiar result:

Corollary 4.3. *The category of principal homogeneous bundles P over S under $\mathbb{G}_{m,S}$ is equivalent to the category of invertible modules $\mathscr{L}$ over S, where the morphisms in both categories are the isomorphisms of the structures in question. Quasi-inverse functors are obtained by assigning to P the degree-1 component of its $\mathbb{Z}$-graded affine algebra, and by assigning to $\mathscr{L}$ the spectrum of the $\mathbb{Z}$-graded algebra*

$$\coprod_{n\in\mathbb{Z}} \mathscr{L}^{\otimes n}.$$

In particular:

Corollary 4.4. *The group of isomorphism classes of principal homogeneous bundles over S under $\mathbb{G}_{m,S}$ is isomorphic to the group* $\mathrm{Pic}(S)$ *of isomorphism classes of invertible modules over S, that is, to*

$$H^1(S, \mathcal{O}_S^\times).$$

Since $\mathbb{G}_{m,S}$ is the scheme of automorphisms of the module $\mathcal{O}_S$, Corollary 4.4 is equivalent to the following statement, one of the forms of Hilbert's "Theorem 90."

Corollary 4.5. *Every principal homogeneous bundle over S under $\mathbb{G}_m$ is locally trivial for the Zariski topology.*

Remark 4.5.1. The preceding statement is no longer true in general for a group such as μ_n, or for a "twisted form" of $\mathbb{G}_m$. For example, the unique twisted form of $\mathbb{G}_m$ over the field $\mathbb{R}$ of real numbers has a first cohomology group equal to $\mathbb{Z}/2\mathbb{Z}$.

[14] Indeed, let $\mathbb{S}^1$ be the kernel of the norm morphism

$$N : \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \longrightarrow \mathbb{G}_{m,\mathbb{R}};$$

it is a $\mathbb{C}/\mathbb{R}$-twisted form of $\mathbb{G}_{m,\mathbb{R}}$. The equation $N(z) = -1$ in $\prod_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})$ defines an $\mathbb{S}^1$-torsor X over $\mathrm{Spec}(\mathbb{R})$, locally trivial for the étale topology, but not trivial, since $X(\mathbb{R}) = \varnothing$. Let us show that

$$H^1_{\text{ét}}(\mathbb{R}, \mathbb{S}^1) \simeq \mathbb{Z}/2\mathbb{Z}.$$

There is an exact sequence of smooth commutative algebraic $\mathbb{R}$-groups

$$1 \longrightarrow \mathbb{S}^1 \longrightarrow \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \longrightarrow \mathbb{G}_{m,\mathbb{R}} \longrightarrow 1,$$

which gives rise to a long exact sequence in étale cohomology (or in Galois cohomology):

$$\begin{aligned} 0 \longrightarrow \mathbb{S}^1(\mathbb{R}) \longrightarrow \mathbb{C}^\times \xrightarrow{N} \mathbb{R}^\times \longrightarrow H^1_{\text{ét}}(\mathbb{R}, \mathbb{S}^1) \\ \longrightarrow H^1_{\text{ét}}\left(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}\right) \longrightarrow \cdots . \end{aligned}$$

[14] Editor's note: what follows has been added.

Now (see, for example, XXIV, 8.4),

$$H^1_{\text{ét}}\left(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}\right) \simeq H^1_{\text{ét}}(\mathbb{C}, \mathbb{G}_{m,\mathbb{C}}),$$

and the latter group is zero by 4.5 (or, in this case, because $\mathbb{C}$ is algebraically closed). We therefore obtain an isomorphism

$$H^1_{\text{ét}}(\mathbb{R}, \mathbb{S}^1) \simeq \mathbb{R}^\times / N(\mathbb{C}^\times) \simeq \{\pm 1\}.$$

We shall need the following result in the next section.

Proposition 4.6. *Under the conditions of 4.1, conditions (a) and (b) are equivalent to the following conditions:*

(a') *the homomorphism $\mathcal{O}_S \to \mathscr{A}_0$ is an isomorphism;*

(b') *for every $m \in M$ (it is enough to require this for the members of a system of generators of M), one has*

$$\mathscr{A}_m \cdot \mathscr{A}_{-m} = \mathscr{A}_0.$$

The necessity is clear in view of 4.2.[15] We are thus reduced to proving the following corollary.

Corollary 4.7. *Let*

$$A = \coprod_{n \in \mathbb{Z}} A_n$$

be a $\mathbb{Z}$-graded ring such that

$$A_1 \cdot A_{-1} = A_0.$$

Then the A_n are invertible A_0-modules, and, for $n, n' \in \mathbb{Z}$, the homomorphism

$$A_n \otimes_{A_0} A_{n'} \longrightarrow A_{n+n'}$$

induced by multiplication in A is an isomorphism.

By hypothesis there exist $f_i \in A_1$ and $g_i \in A_{-1}$ such that

$$(*) \qquad \sum_i f_i g_i = 1.$$

Since the assertion to be proved is local on $\operatorname{Spec}(A_0)$,[16] and since $(*)$ shows that $\operatorname{Spec}(A_0)$ is covered by the affine open subsets $D(f_i g_i)$, we are reduced to the case in which there is an element $f \in A_1$ that is invertible in A. Then, for every $n \in \mathbb{Z}$, f^n is an invertible element of A_n, and therefore defines an isomorphism

$$A_0 \xrightarrow{\sim} A_n, \qquad h \longmapsto f^n h.$$

Moreover, this shows that one obtains an isomorphism

$$A_0[t, t^{-1}] \longrightarrow A$$

of graded A_0-algebras by sending t to f, which completes the proof of 4.7.

[15] Editor's note: "4.1" has been corrected to "4.2."

[16] Editor's note: the following passage has been expanded.

Under the hypotheses of 4.6, 4.7 already shows that the $\mathscr{A}_m$, for $m \in M$, are invertible. To prove condition 4.1(b), we may therefore suppose that $\mathscr{A}_m$ and $\mathscr{A}_{m'}$ have bases f_m and $f_{m'}$, with inverses

$$f_m^{-1} \in \Gamma(\mathscr{A}_{-m}) \qquad \text{and} \qquad f_{m'}^{-1} \in \Gamma(\mathscr{A}_{-m'}).$$

Multiplication by

$$f_m^{-1} f_{m'}^{-1} \in \Gamma(\mathscr{A}_{-m-m'})$$

then defines a homomorphism

$$\mathscr{A}_{m+m'} \longrightarrow \mathscr{A}_0 \simeq \mathcal{O}_S$$

that sends the image of $f_m \otimes f_{m'}$ to the unit section of $\mathcal{O}_S$. In the diagram

$$\mathscr{A}_m \otimes \mathscr{A}_{m'} \xrightarrow{u} \mathscr{A}_{m+m'} \xrightarrow{v} \mathscr{A}_0 \simeq \mathcal{O}_S, \qquad (w : \mathscr{A}_m \otimes \mathscr{A}_{m'} \to \mathscr{A}_0 \simeq \mathcal{O}_S)$$

the maps w and v are therefore epimorphisms of invertible sheaves, hence isomorphisms; consequently u is an isomorphism. □

5. Quotient of an Affine Scheme by a Diagonalizable Group Acting Freely

[17] We denote by $\mathcal{M}$ the set of faithfully flat quasi-compact morphisms, and recall that torsors are understood with respect to the (fpqc) topology.

Theorem 5.1. *Let S be a prescheme, M an ordinary commutative group, $G = D_S(M)$ the diagonalizable group over S that it defines, and P an S-prescheme affine over S, on which G acts freely on the right.*

Then the equivalence relation defined by G on P is $\mathcal{M}$-effective (cf. IV, 3.4); that is, the quotient $X = P/G$ exists and P is a torsor over X under $G_X = D_X(M)$. Moreover, P/G is affine over S. More precisely, if P is defined by the M-graded algebra $\mathcal{A}$, then P/G is isomorphic to $\mathrm{Spec}(\mathcal{A}_0)$*, where $\mathcal{A}_0 = \mathcal{A}^G$ is the degree-zero component of $\mathcal{A}$.*

Proof. Put $X = \mathrm{Spec}(\mathcal{A}_0)$. The inclusion $\mathcal{A}_0 \to \mathcal{A}$ gives a natural morphism $P \to X$, which is invariant under the action of G. Thus P becomes an X-prescheme, affine over X, with operator group $G_X = D_X(M)$; the hypothesis that G acts freely on P/S implies that G_X acts freely on P/X. It remains only to show that P is a principal homogeneous bundle under G_X, using the fact that $\mathcal{B}_0 = \mathcal{O}_X$, where $\mathcal{B}$ is the M-graded algebra over X defining P/X.

We may therefore suppose that $X = S$ and that S is affine. Then P is affine, defined by an M-graded ring A; write A_m for its homogeneous part of degree m, so that $S = \mathrm{Spec}(A_0)$. In view of 4.6, it remains to verify

(x) $$A_m \cdot A_{-m} = A_0 \qquad \text{for every } m \in M.$$

A direct calculation also shows that (x) is equivalent to saying that

$$P \times_S G \longrightarrow P \times_S P$$

[17] Editor's note: We have added the following sentence.

is a closed immersion, rather than merely a monomorphism (as follows from the hypothesis that G acts freely); equivalently, the homomorphism of affine rings

$$\theta : A \otimes_{A_0} A \longrightarrow A(M)$$

is surjective.[18] This proves 5.1 if one assumes that the equivalence relation defined by G on P is closed. We shall show, however, that this hypothesis already follows from the fact that G acts freely. This is also implicit in Theorem 5.1: the scheme $G \times_S P$ must be isomorphic to $P \times_X P$, which is closed in $P \times_S P$ because X is affine, hence separated, over S.

Let $R = P \times_S G$. The hypothesis that G acts freely, that is, that $R \to P \times_S P$ is a monomorphism, says that the diagonal morphism

$$R \longrightarrow R \times_{(P \times_S P)} R = R'$$

is an isomorphism. We have $R = \mathrm{Spec}(A(M))$ and

$$R' = \mathrm{Spec}\big(A(M \times M)/K\big),$$

[19] where K is the ideal generated by the elements

$$x_m(e_{m,0} - e_{0,m}), \qquad m \in M, \quad x_m \in A_m.$$

Let

$$\phi : A(M \times M) \longrightarrow A(M)$$

be the surjective ring homomorphism defined by

$$xe_{m,n} \longmapsto xe_{m+n} \qquad (m, n \in M,\ x \in A),$$

where the e_m, respectively $e_{m,n} = e_m \otimes e_n$, are the elements of the canonical basis of $A(M)$, respectively $A(M \times M)$. The diagonal morphism $R \to R'$ then corresponds to the homomorphism

$$\bar{\phi} : A(M \times M)/K \longrightarrow A(M)$$

obtained by passing to the quotient by K. The kernel of ϕ is the ideal K' generated by

$$d_m = e_{m,0} - e_{0,m}.$$

We have $K \subseteq K'$, and the hypothesis that G acts freely on P, that is, that $\bar{\phi}$ is an isomorphism, is equivalent to $K' = K$, expressed by the relations

(xx) $$d_m \in K = \sum_p A(M \times M)\, A_p\, d_p \qquad \text{for every } m \in M.$$

Using the natural trigrading of $A(M \times M)$, and the fact that the first degree of d_m is zero, this says that d_m can be written as a sum of elements of the form

$$f e_{r,s}(e_{p,0} - e_{0,p}), \qquad f \in A_{-p} \cdot A_p.$$

[18] Editor's note: Indeed, for every $b \in A$ and $a_m \in A_m$, one has $\theta(b \otimes a_m) = ba_m \otimes e_m$; hence the surjectivity of θ is equivalent to (x).

[19] Editor's note: We have denoted by K the ideal denoted by J in the original, in order to distinguish it from the ideals J_p of A_0 occurring in (xxx). We have also made explicit below that the relations (xx) are equivalent to the equality $K = K'$; see what follows.

Since the total degree of d_m is m, it is enough to use terms satisfying

$$r + s + p = m.$$

Thus, for every $m \in M$, there must be an expression

$$\text{(xxx)} \qquad \begin{cases} d_m = e_{m,0} - e_{0,m} = \sum_{r,s} \lambda_{r,s}(e_{m-s,s} - e_{r,m-r}), \\ \qquad\qquad \lambda_{r,s} \in J_p = A_p \cdot A_{-p} \subseteq A_0, \qquad p = m - (r+s). \end{cases}$$

We must deduce (x), that is, the relations

$$\text{(xxxx)} \qquad J_n = A_0 \qquad \text{for every } n \in M.$$

For this it is enough to establish the same relation modulo every maximal ideal of A_0. Since all the hypotheses are preserved under such a reduction, we may assume that A_0 is a field.

Lemma 5.2. *Under the preceding conditions, with A_0 a field, if $M \neq 0$, there exists $p \in M - \{0\}$ such that $J_p = A_0$.*

Indeed, otherwise the sum on the right-hand side of (xxx) would vanish for every $m \in M$, which is absurd.

Corollary 5.3. *Under the preceding conditions, but without assuming that A_0 is a field, there are finitely many elements $p_i \in M - \{0\}$ such that*

$$\sum_i J_{p_i} = A_0.$$

Indeed, apply 5.2 to the situations obtained from A/A_0 by reduction modulo the maximal ideals of A_0.[20]

Corollary 5.4. *Suppose again that A_0 is a field. For every subgroup N of M with $N \neq M$, there exists $p \in M - N$ such that $J_p = A_0$.*

Indeed, let $M' = M/N$, and consider the M'-graded ring A', whose underlying ring is A and whose grading is given by

$$A'_{m'} = \coprod_{m \in h^{-1}(m')} A_m,$$

where $h : M \to M' = M/N$ is the canonical homomorphism. Geometrically, this construction amounts to considering the subgroup $G' = D_S(M')$ of G, and the structure on P of a scheme with operator group G' induced by the action of G. It is then clear that G' acts freely on P ; that is, the pair (M', A') satisfies the hypotheses of 5.3. Thus

$$1 = \sum_i f_i g_i, \qquad f_i \in A'_{m'_i}, \quad g_i \in A'_{-m'_i}, \quad m'_i \in M' - \{0\}.$$

Taking the components of the right-hand side along A_0, and using the fact that A_0 is a field, immediately gives 5.4.

Now observe that

$$J_p J_q \subseteq J_{p+q} \qquad \text{and} \qquad J_p = J_{-p}.$$

[20] Editor's note: It follows from 5.2 that $\sum_{p \neq 0} J_p = A_0$; hence 1 can be written as a finite sum $\sum_i x_i$, with $x_i \in J_{p_i}$.

Thus, if N denotes the set of $m \in M$ such that $J_m = A_0$, then N is a subgroup of M. By 5.4, it equals M. This completes the proof of Theorem 5.1.

As noted in the course of the proof, Theorem 5.1 implies:

Corollary 5.5. *Under the hypotheses of 5.1, the graph morphism*

$$P \times_S G \longrightarrow P \times_S P$$

is a closed immersion.

It follows immediately that:

Corollary 5.6. *Let σ be a section of P over S. Then the morphism $g \mapsto \sigma \cdot g$ from G to P defined by σ is a closed immersion.*

Corollary 5.7. *Let G and H be two S-groups, with G diagonalizable and H affine over S, and let $u : G \to H$ be a homomorphism of S-groups that is a monomorphism. Then u is a closed immersion, $H/G = X$ exists, H is a principal homogeneous bundle over X under G_X, and X is affine over S.*

Corollary 5.8. *Under the hypotheses of 5.1, if P is of finite type (respectively, of finite presentation) over S, then the same is true of $X = P/G$.*

Indeed, the hypothesis implies that the fibers of G_X are of finite type. Hence G_X is of finite presentation over X, by 2.1(b); since P is a torsor under G_X, it is therefore of finite presentation over X.[21] It is also faithfully flat over X, so the conclusion follows from Exposé V, Proposition 9.1.

6. Essentially Free Morphisms and Representability of Certain Functors of the Form $\prod_{Y/S} Z/Y$ (*)

Definition 6.1. Let $f : X \to S$ be a morphism of preschemes. We say that f is *essentially free*, or that X is *essentially free over S*, if there are an affine open covering (S_i) of S, for each i an S_i-prescheme S'_i that is affine and faithfully flat over S_i, and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine opens X'_{ij}, such that, for every (i, j), the ring of X'_{ij} is a free module over the ring of S'_i.[22]

Proposition 6.2.

(a) *If X is essentially free over S, then it is flat over S. The converse holds if S is Artinian.*

(b) *If S is the spectrum of a field, every S-prescheme is essentially free over S.*

(c) *If X is essentially free over S, and if $S' \to S$ is a base-change morphism, then $X' = X \times_S S'$ is essentially free over S'. The converse holds if $S' \to S$ is faithfully flat and quasi-compact.*

[21] Editor's note: Cf. EGA II, 2.7.1(vi).

(*) This section is independent of the theory of diagonalizable groups; its natural place would be in Exposé VI_B.

[22] Editor's note: "Free" may be replaced by "projective"; cf. 6.8 below. This notion should also be compared with that of an S-prescheme that is flat and pure, introduced and developed in [RG71]; see in particular *loc. cit.*, Part I, 3.3.12, and Part II, 3.1.4.1.

The proof is immediate, using for the converse in (a) the fact that a flat module over an Artinian local ring is free.[23]

Proposition 6.3. *Let H be a diagonalizable S-group prescheme (more generally, one that becomes diagonalizable after a suitable faithfully flat quasi-compact extension of every affine open of S; that is, H is "of multiplicative type," cf. IX, 1.1). Then H is essentially free over S.*

Indeed, if H is diagonalizable, it is affine over S and is defined by an algebra that is a free $\mathcal{O}_S$-module.

The introduction of Definition 6.1 is justified by the following theorem.

Theorem 6.4. *Let S be a prescheme, Z an S-prescheme essentially free over S, and Y a closed subprescheme of Z. Consider the functor*[24]

$$F = \prod_{Z/S} Y/Z : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Set},$$

$$F(S') = \Gamma(Y_{S'}/Z_{S'}) = \begin{cases} \varnothing, & Z_{S'} \neq Y_{S'}, \\ \{\mathrm{id}_{Z_{S'}}\}, & Z_{S'} = Y_{S'}. \end{cases}$$

This functor is representable by a closed subprescheme of S.[25]

[26] First note that F is a sheaf for the (fpqc) topology. Since $F(S')$ is either $\varnothing$ or $\{\mathrm{pt}\}$ for every S', it is enough to check the following assertion: if (S_i) is an open covering of S (respectively, if $S' \to S$ is faithfully flat and quasi-compact), and every $Y_{S_i} \to Z_{S_i}$ is an isomorphism (respectively, $Y_{S'} \to Z_{S'}$ is an isomorphism), then $Y \to Z$ is an isomorphism. This is clear in the first case and follows from SGA 1, VIII, 5.4, or EGA IV_2, 2.7.1, in the second.

Moreover, by SGA 1, VIII, 2.1 and 5.5, faithfully flat quasi-compact morphisms are morphisms of effective descent for the fibered category of arrows that are closed immersions. Thus, with the notation of 6.1, we may reduce to the case $S = S'_i$.

Let (Z_j) be a covering of Z by affine opens such that $\mathcal{O}(Z_j)$ is a free module over $A = \mathcal{O}(S)$, put $Y_j = Y \cap Z_j$, and let

$$F_j : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Set}$$

be the functor defined from (Z_j, Y_j) as F is defined from (Z, Y). It is a subfunctor of the terminal functor, and clearly

$$F = \bigcap_j F_j.$$

It therefore suffices to prove that every F_j is representable by a closed subscheme T_j of S, for then F is represented by the closed subscheme $T = \bigcap_j T_j$. We may consequently suppose that Z is affine as well, say $Z = \mathrm{Spec}(B)$, where B is a free A-module. Let J be a subset of B defining the closed subprescheme Y of Z, and let K be the ideal of A

[23] Editor's note: Indeed, let $(A, \mathfrak{m})$ be an Artinian local ring, k its residue field, M an arbitrary A-module, and $(x_i)_{i\in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k. Let F be the free A-module with basis $(e_i)_{i\in I}$, and let $\phi : F \to M$ be the A-morphism defined by $\phi(e_i) = x_i$. Then $Q = \mathrm{Coker}\,\phi$ satisfies $Q = \mathfrak{m}Q$, whence $Q = 0$, since $\mathfrak{m}$ is nilpotent. Suppose moreover that M is flat over A. Then $K = \mathrm{Ker}\,\phi$ satisfies $K \otimes_A k = 0$, that is, $K = \mathfrak{m}K$, whence $K = 0$.

[24] Editor's note: Cf. II.1, where this functor is denoted $\prod_{Z/S} Y$.

[25] Editor's note: We have corrected Z to S.

[26] Editor's note: We have expanded the original in what follows; see also 1.7.3.

generated by the $u_i(J) \subseteq A$, where the $u_i : B \to A$ are the coordinate forms for the chosen basis. It is immediate that

$$T = \mathcal{V}(K) = \mathrm{Spec}(A/K)$$

has the required property. This completes the proof.

Examples 6.5. We give some important examples of functors that reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.4, and for which criteria of representability will be useful below. Here S denotes a prescheme, and $X, Y, Z, \ldots$ denote preschemes over S.

(a) Suppose given an S-morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z), \tag{x}$$

("X acts on Y with values in Z"), that is, a morphism

$$r : X \times_S Y \longrightarrow Z. \tag{xx}$$

Let Z' be a subprescheme of Z. It gives a monomorphism

$$\underline{\mathrm{Hom}}_S(Y, Z') \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z),$$

making the first functor a subfunctor of the second. Let X' be the inverse image of this subfunctor under (x). Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Z_T$$

factors through Z'_T. This functor has the following description. Put $P = X \times_S Y$, and let P' be the inverse image of Z' under $r : P \to Z$. Then there is an evident isomorphism

$$X' \simeq \prod_{P/X} P'/P. \tag{xxx}$$

It follows that *if Y is essentially free over S and Z' is closed in Z, then the subfunctor X' of X is representable by a closed subprescheme of X.*

(b) Suppose given two actions of X on Y with values in Z, that is, two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\mathrm{Hom}}_S(Y, Z),$$

and put $X' = \mathrm{Ker}(q_1, q_2)$. This is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms

$$q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$$

are equal. Giving q_1, q_2 is equivalent to giving a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z \times_S Z),$$

or, equivalently, a morphism

$$r : X \times_S Y \longrightarrow Z \times_S Z.$$

Put $U = Z \times_S Z$, and let U' be the diagonal subprescheme of $Z \times_S Z$. Then X' is the inverse image under q of the subfunctor

$$\underline{\mathrm{Hom}}_S(Y, U') \longrightarrow \underline{\mathrm{Hom}}_S(Y, U).$$

It can therefore be written in the form (xxx), with $P = X \times_S Y$ and P' the inverse image of the diagonal under r, that is, the kernel of

$$X \times_S Y \overset{r_1}{\underset{r_2}{\rightrightarrows}} Z.$$

Thus we are in the situation of (a).

Consequently, *if Y is essentially free over S and Z is separated over S, then the subfunctor X' of X is representable by a closed subprescheme of X.*

(c) Suppose given a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y);$$

that is, "X acts on Y." Let X' be the "kernel" of this morphism, namely the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \to Y_T$ is the identity. This functor falls under (b), as one sees by introducing a second morphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

that "makes X act trivially on Y." Hence, *if Y is essentially free and separated over S, then the kernel subfunctor of q is representable by a closed subprescheme of X.*

(d) Under the hypotheses of (c), consider the subfunctor Y' of Y of "invariants under X." Thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $\bar{q}(y) : X_T \to Y_T$ is the "T-constant morphism with value y." Introducing q' as in (c), and writing $\bar{q}$ and $\bar{q}'$ for the corresponding morphisms,

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that $Y' = \mathrm{Ker}(\bar{q}, \bar{q}')$. It therefore again falls under (b), with the roles of X and Y reversed and $Z = Y$. Consequently, *if X is essentially free over S and Y is separated over S, then the subfunctor Y' of Y of invariants under X is representable by a closed subprescheme of Y.*

(e) Constructions of the type described in the preceding examples occur especially often in group theory. Thus, when a group prescheme G over S acts on the S-prescheme X,

$$q : G \longrightarrow \underline{\mathrm{Aut}}_S(X),$$

the kernel of q, the "subgroup of G acting trivially," is a closed subprescheme of G provided that X is essentially free and separated over S, by (c). The subobject X^G of invariants is a closed subprescheme of X, provided that G is essentially free over S and X is separated over S,[27] by (d).

Let Y, Z be subpreschemes of X, and consider the subfunctor $\underline{\mathrm{Transp}}_G(Y, Z)$ of G, the "transporter of Y into Z." Its points with values in an S-prescheme T are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subseteq Z_T$, that is, induces a morphism $Y_T \to X_T$ factoring through Z_T. Hence, *if Y is essentially free over S, and Z is closed in X, then $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subprescheme of G*, by (a).

One may also consider the *strict transporter* of Y into Z,[28] whose T-valued points are the $g \in G(T)$ such that $g(Y_T) = Z_T$. It is precisely

$$\underline{\mathrm{Transp}}_G(Y, Z) \cap \sigma\big(\underline{\mathrm{Transp}}_G(Z, Y)\big),$$

[27] Editor's note: We have corrected S_X to S.

[28] Editor's note: It is denoted $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(Y, Z)$.

where σ is the symmetry of G. Consequently, if Y and Z are essentially free over S and closed in X, then the strict transporter of Y into Z is a closed subprescheme of G.

An important case is $X = G$, with G acting on itself by inner automorphisms. If H is a subprescheme of G, the strict transporter of H into itself is also called the *normalizer of H in G*, and is denoted $\underline{\mathrm{Norm}}_G H$. Hence, *if H is a closed subgroup prescheme of G, essentially free over S, then $\underline{\mathrm{Norm}}_G H$ is representable by a closed subgroup prescheme of G.*

Finally, let Z be a subprescheme of G. Its *centralizer* $\underline{\mathrm{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure in (d), regarding "Z acts on G" through the action induced by that of G. Thus, *if Z is essentially free over S and G is separated over S, then $\underline{\mathrm{Centr}}_G(Z)$ is a closed subgroup prescheme of G.* In particular, *if G is essentially free and separated over S, then the center C of G, which is none other than $\underline{\mathrm{Centr}}_G(G)$, is a closed subgroup prescheme of G.*

When S is the spectrum of a field, 6.2(b) shows that in examples (a)–(e) above the "essentially free" hypotheses are automatically satisfied; only separation hypotheses remain. Recalling that a group prescheme over a field is necessarily separated, one obtains, for example:

Corollary 6.7. [29] *Let G be a group prescheme over a field k. Then:*

- *For every subprescheme Z of G, the centralizer of Z in G is a closed subgroup prescheme of G. In particular, this holds for the center $\underline{\mathrm{Centr}}_G(G)$ of G.*
- *More generally, if $u, v : X \to G$ are morphisms of preschemes, then $\underline{\mathrm{Transp}}_G(u, v)$ is representable by a closed subprescheme of G.*
- *For two subpreschemes Y, Z of G, with Z closed, $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subprescheme of G. If Y is also closed, the same conclusion holds for $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(Y, Z)$.*
- *For every subgroup prescheme*[30] *H of G, the normalizer $\underline{\mathrm{Norm}}_G(H)$ is a closed subgroup prescheme of G.*

Remark 6.8. [31] Let A be a commutative ring, let M be an A-module, and put

$$M^\vee = \mathrm{Hom}_A(M, A).$$

Endow the A-module $\mathrm{End}_A(M)$ with the topology of discrete pointwise convergence. Thus a basis of neighborhoods of zero is formed by the following A-submodules, where $n \in \mathbb{N}$ and $m_1, \ldots, m_n \in M$:

$$K(m_1, \ldots, m_n) = \{u \in \mathrm{End}_A(M) \mid u(m_i) = 0 \text{ for } i = 1, \ldots, n\}.$$

We say that M is a *quasi-free A-module* if the image of the canonical morphism

$$\Theta : M \otimes_A M^\vee \longrightarrow \mathrm{End}_A(M)$$

contains id_M in its closure; that is, if the following condition holds:

[29] Editor's note: We have retained the numbering of the original; there is no 6.6.

[30] Editor's note: Indeed, over a field k, every subgroup prescheme of G is closed; cf. Exposé VI_A, 0.6.1.

[31] Editor's note: We have expanded the original in what follows; in particular, we have added Lemma 6.8.1.

$$(*)\qquad \begin{cases} \text{for all } m_1,\dots,m_n \in M, \text{ there exist } x_1,\dots,x_r \in M \text{ and } f_1,\dots,f_r \in M^\vee \\ \text{such that} \quad m_i = \Theta\left(\sum_{s=1}^{r} x_s \otimes f_s\right)(m_i) = \sum_{s=1}^{r} f_s(m_i)x_s \quad \text{for } i = 1,\dots,n. \end{cases}$$

(In this case $\operatorname{Im}\Theta$ is dense in $\operatorname{End}_A(M)$, since for every $u \in \operatorname{End}_A(M)$ one has $u(m_i) = \sum_{s=1}^{r} f_s(m_i)u(x_s) = \Theta(\sum_{s=1}^{r} u(x_s) \otimes f_s)\,(m_i)$.)

First note that this property is stable under base change. Indeed, let $\phi : A \to A'$ be a ring homomorphism, put $M' = M \otimes_A A'$, and let $m'_1,\dots,m'_n \in M'$. Write

$$m'_i = \sum_j m_{ij} \otimes b_{ij}, \qquad m_{ij} \in M, \quad b_{ij} \in A'.$$

By hypothesis there exist $x_1,\dots,x_r \in M$ and $f_1,\dots,f_r \in \operatorname{Hom}_A(M,A)$ such that

$$m_{ij} = \sum_s x_s f_s(m_{ij}) \qquad \text{for all } i,j.$$

Write $\phi \circ f_s$ for the image of f_s in

$$\operatorname{Hom}_A(M,A') = \operatorname{Hom}_{A'}(M',A').$$

Then, for $i = 1,\dots,n$,

$$\begin{aligned} \left(\sum_s x_s \otimes \phi \circ f_s\right)(m'_i) &= \sum_{s,j} x_s \otimes \phi(f_s(m_{ij}))\, b_{ij} \\ &= \sum_j \left(\sum_s x_s f_s(m_{ij})\right) \otimes b_{ij} \\ &= m'_i. \end{aligned}$$

This proves that M' is quasi-free over A'.

Also, every projective A-module P is quasi-free. Indeed, there are A-morphisms

$$A^{(I)} \xrightarrow{\pi} P \xrightarrow{\tau} A^{(I)} \qquad \text{such that} \qquad \pi \circ \tau = \mathrm{id}_P\,.$$

Let (e_i) be the canonical basis of $A^{(I)}$, and put $f_i = e_i^* \circ \tau$, a linear form on P. Given $m_1,\dots,m_n \in P$, there is a finite subset J of I such that

$$m_k = \sum_{i \in J} f_i(m_k)\pi(e_i) \qquad \text{for } k = 1,\dots,n.$$

Theorem 6.4 remains valid if, in the statement of Definition 6.1, "free" is replaced by "projective" or, more generally, by "quasi-free." Proceeding as in the proof of 6.4, one reduces to the following lemma.

Lemma 6.8.1. *Let M be a quasi-free A-module, N a submodule, and F the covariant functor*

$$F : (A\text{-algebras}) \longrightarrow \mathbf{Set}$$

such that $F(B) = \{\mathrm{pt}\}$ if $M_B = (M/N)_B$, and $F(B) = \varnothing$ otherwise. Then there is an ideal K of A such that $F(B) = \{\mathrm{pt}\}$ if and only if the morphism $A \to B$ factors through A/K; that is, there is an isomorphism, functorial in B,

$$F(B) \simeq \operatorname{Hom}_{A\text{-alg.}}(A/K, B).$$

Proof. Let (n_j) be a system of generators of N, and let F_j be the subfunctor of the terminal functor e, where $e(B) = \{\mathrm{pt}\}$ for every B, corresponding to the submodule An_j. We have $F(B) = \{\mathrm{pt}\}$ if and only if the image of every n_j in M_B is zero. Hence F is the intersection of the functors F_j, reducing us to the case in which N is generated by one element n.

Let B be an A-algebra, with structure map $\phi : A \to B$. If the image $n \otimes 1$ of n in M_B is zero, then for every $f \in M^\vee = \mathrm{Hom}_A(M, A)$,

$$0 = f(n) \otimes 1 = \phi(f(n)).$$

Conversely, because M is quasi-free, there exist $x_1, \ldots, x_r \in M$ and $f_1, \ldots, f_r \in M^\vee$ such that

$$n = \sum_s f_s(n) x_s,$$

whence

$$n \otimes 1 = \sum_s x_s \otimes \phi(f_s(n)).$$

It follows that $n \otimes 1 = 0$ if and only if ϕ factors through A/K, where K is the ideal generated by the $f_s(n)$, for $s = 1, \ldots, r$. This proves the lemma.

7. Appendix: On Monomorphisms of Group Preschemes

The result proved in this section is not needed in the remainder of the seminar, except in X, 8.8 and Exposés XV and XVI. It depends essentially on the existence theorem for quotient groups in Exposé VI_A. [32]

Corollary 5.7 raises the question of when a monomorphism $u : G \to H$ of S-groups can be asserted to be an immersion, or even a closed immersion.

We saw in VI_B, 1.4.2 that this is so when S is the spectrum of a field k, provided that G is of finite type over k and H is locally of finite type over k. It follows readily that the same result remains valid when S is merely assumed Artinian.[33]

On the other hand, examples of bijective monomorphisms that are not immersions are easy to give, for instance over the affine line over a field or over the spectrum of a discrete valuation ring. Take

$$H = (\mathbf{Z}/2\mathbf{Z})_S, \qquad G = G_1 \times_S G_2,$$

where G_1 is the open subgroup of H complementary to the closed point $x \neq 0$ in the fiber $H_s \simeq \mathbf{Z}/2\mathbf{Z}$, with s a fixed closed point of S, and G_2 is the closed subscheme of H which is the union of the subscheme given by the identity section and the reduced closed subscheme defined by x. One checks immediately that G_2 is stable under multiplication in H, and hence is a group scheme. The immersions $G_i \to H$, $i = 1, 2$, then define a homomorphism of S-groups

$$G = G_1 \times_S G_2 \longrightarrow H$$

which is visibly a bijective monomorphism, and even a local immersion, but is not an immersion. Indeed, G and H are reduced; G has three disjoint irreducible components, whereas H has only two.

Notice also that $G_1 \to H$ is an example of an open subgroup of H that is not closed, in contrast to the situation for algebraic groups over a field. The theory of degenerations

[32] Editor's note: cf. Exposé VI_A, Theorems 3.2 and 3.3.2.

[33] Editor's note: the additions made in Exposé VI_B should be taken into account.

of elliptic curves gives further examples of this phenomenon in which H is smooth over S of relative dimension one and G has connected fibers.

One might hope(*)that once G is assumed flat over S, and, say, G and H are of finite presentation over S, every monomorphism $u : G \to H$ of S-groups is automatically an immersion. We shall prove a result of this kind under additional hypotheses.

First, we may assume that S is affine and, by the finite-presentation hypothesis on G and H, that S is Noetherian: one reduces to the spectrum of a ring of finite type over $\mathbf{Z}$.[34] Then G and H are Noetherian. Saying that $u : G \to H$ is a closed immersion (respectively, an immersion) is then equivalent to saying that u is a monomorphism—which holds by hypothesis—and is proper (respectively, proper at every point of $u(G)$).[35]

The valuative criterion for properness shows that it is enough to check, after every base change $S' \to S$ with S' the spectrum of a discrete valuation ring, complete if desired, that the morphism $u' : G' \to H'$ has the corresponding properness property. The case of local properness was omitted from EGA II, 7.3 and was to appear as an erratum in EGA IV.(*)We are therefore reduced to the case in which S itself is the spectrum of a complete discrete valuation ring, subject to the proviso that any additional hypotheses imposed on G, H, u be stable under base change.

Let s, respectively s', be the closed, respectively generic, point of S. The induced homomorphisms on the fibers,

$$u_s : G_s \longrightarrow H_s, \qquad u_{s'} : G_{s'} \longrightarrow H_{s'},$$

are closed immersions because they are monomorphisms of group schemes of finite type over fields (VI_B, 1.4.2). We may therefore identify $G_{s'}$ with a closed subscheme of $H_{s'}$. We shall use the following result.

Lemma 7.1. *Let S be the spectrum of a discrete valuation ring, with generic point s'. Let H be an S-prescheme and let L' be a closed subprescheme of the generic fiber $H_{s'}$, hence also a subprescheme of H. Its schematic closure L in H—the smallest closed subprescheme of H containing L', cf. EGA I, 9.5—exists. It is also the unique closed subprescheme of H, flat over S, whose generic fiber is L'. Formation of L is functorial in the pair (H, L') and commutes with fiber products over S. In particular, if H is an S-group and L' is a subgroup of $H_{s'}$, then L is a subgroup of H.*

The proof is immediate and is left to the reader.(**)Apply this to the pair $(H, G_{s'})$. The monomorphism $u : G \to H$ factors as

$$G \xrightarrow{u'} L \longrightarrow H,$$

where L is a subgroup of H, a closed subprescheme flat over S, and $u' : G \to L$ induces an isomorphism on generic fibers. Thus $u : G \to H$ is an immersion (respectively, a closed immersion) if and only if $u' : G \to L$ is one. We are reduced to the case in which H is flat over S and $u_{s'} : G_{s'} \to H_{s'}$ is an isomorphism, again subject to the requirement that any additional hypotheses remain valid when H is replaced by a closed subgroup prescheme.

Since H is then the schematic closure of $H_{s'}$, if u is an immersion its image $u(G)$ is a subprescheme[36] of H containing $H_{s'}$. Its schematic closure is therefore H, so $u(G)$ is an

(*)In fact, M. Raynaud constructed a counterexample with G smooth and with connected fibers; cf. XVI, 1.1(c). Without the connected fiber hypothesis, one may take $S = \mathrm{Spec}(\mathbf{Z}_2)$, let $G = (\mathbf{Z}/2\mathbf{Z})_S$ with its nonidentity closed point removed, and let $H = (\mu_2)_S$.

[34]Editor's note: cf. EGA IV_3, §8 and Exposé VI_B, §10.

[35]Editor's note: cf. EGA IV_3, 8.11.5 and 15.7.1.

(*)Cf. EGA IV_3, 15.7.

(**)Cf. EGA IV_2, 2.8.

[36]Editor's note: the word "closed" has been removed.

open subprescheme of H. We must consequently prove that u is an open immersion, or an isomorphism if the desired conclusion is that it is a closed immersion.

Because G and H are flat over S, this is equivalent to the induced maps on the fibers being open immersions, respectively isomorphisms (cf. SGA 1, I, 5.7). It already holds for $u_{s'}$, so it remains to prove that u_s is an open immersion, respectively an isomorphism.

Since G and H are flat over S, their fiber dimensions are constant (VI_B, 4.3). Their generic fibers agree, hence these dimensions are the same for G and H. Thus $u_s : G_s \to H_s$ is a monomorphism of algebraic groups of the same dimension. It follows that G_s is open in H_s, and set-theoretically is a union of connected components of H_s. Indeed, after passing to an algebraically closed base field and reducing G and H, both are smooth over the field and the assertion is immediate.

The complement $H_s - G_s$ is closed in H_s, hence in H. Its complement H' in H is open and is stable under the group law. Upon replacing H by H', we may therefore reduce, with the usual proviso, to the case in which u is bijective, that is, $u_s : G_s \to H_s$ is bijective. In every case we are reduced to proving that u, equivalently u_s, is an isomorphism, possibly under this bijectivity hypothesis.

Assume first that u is bijective. If H_s is reduced, then u_s is an isomorphism, because G_s identifies with a closed subscheme of H_s having the same underlying set. In particular, if $k = \kappa(s)$ has characteristic zero, every algebraic group over k is reduced by Cartier's theorem (cf. VI_B, 1.6.1; VII_B, 3.3.1; or EGA IV_4, 16.12.2 and 17.12.5). We have proved:

Proposition 7.2. *Let $u : G \to H$ be a homomorphism of group preschemes of finite presentation over S. If u is a monomorphism, G is flat over S, and every residue field of S has characteristic zero, then u is an immersion.*

Suppose now that $\kappa(s)$ has characteristic $p > 0$, and restrict to the case in which G is commutative. Under the preceding reductions, H is also commutative, because it is the schematic closure of $H_{s'} \simeq G_{s'}$. For $n \geq 0$, set

$$S_n = \mathrm{Spec}(V/\mathfrak{m}^{n+1}), \qquad G_n = G \times_S S_n, \qquad H_n = H \times_S S_n,$$

where V is the valuation ring defining S and $\mathfrak{m}$ its maximal ideal. For $m > 0$, denote by ${}_mG$ and ${}_mH$ the kernels of multiplication by m in G and H. Define similarly

$${}_m(G_n) = ({}_mG)_n,$$

which we abbreviate to ${}_mG_n$, and likewise for H.

By VI_A, 3.2, the quotients

$$Q_n = H_n/G_n$$

exist. Each Q_n is a commutative group scheme over S_n, flat over S_n, and $H_n \to Q_n$ is faithfully flat with kernel G_n. Formation of the quotient commutes with base extension,[37] so

$$Q_n \simeq Q_m \times_{S_m} S_n \qquad (m \geq n).$$

In particular, $Q_n \times_{S_n} S_0 = Q_0 = H_0/G_0$. Since G_0 and H_0 have the same underlying set, Q_0 is set-theoretically a single point: it is a purely infinitesimal group.

It follows that every Q_n is finite and flat over S_n. Let it be defined by an algebra C_n over $V_n = V/\mathfrak{m}^{n+1}$, finite free as a V_n-module. For $m \geq n$,

$$C_n = C_m \otimes_{V_m} V_n,$$

[37] Editor's note: this should be made explicit in Exposés V and VI_A.

compatibly also with the comultiplication maps. Hence

$$C = \varprojlim C_n \quad \text{is a finite free module over} \quad V = \varprojlim V_n,$$

and the comultiplications on the C_n induce one on C. Thus $Q = \mathrm{Spec}(C)$ is a finite flat group scheme over S satisfying

$$Q \times_S S_n = Q_n$$

for every n.

Lemma 7.3. [38] *Let K be a field and let Q be a finite group scheme over K, of degree n. Then the n-th power morphism on Q is zero.*

See VII_A, 8.5.

Remark 7.3.1. The statement of Lemma 7.3 still makes sense for a finite locally free group scheme Q/S over an arbitrary base prescheme S. It would be interesting to find a proof in that generality.

Lemma 7.3 (that is, VII_A, 8.5) does show that the proposed statement holds whenever S is reduced: apply it to the fibers of Q at the maximal points of S, equivalently the generic points of its irreducible components.[39]

In the preceding situation, where S is the spectrum of a discrete valuation ring and Q is commutative, it follows that $n\,\mathrm{id}_Q = 0$. Here n is a power p^{ν_0} of the residue characteristic. Thus:

Corollary 7.4. *With Q and the Q_n as above, suppose that their common degree is p^{ν_0}. Then*

$$p^{\nu_0}\,\mathrm{id}_Q = 0, \qquad p^{\nu_0}\,\mathrm{id}_{Q_n} = 0 \quad \textit{for every } n.$$

Corollary 7.5. *Suppose in addition that G_0 is smooth over k and that $p\,\mathrm{id}_{G_0}$ is flat. Equivalently, by the structure theory of algebraic groups over an algebraically closed field $\overline{k}$, $G_{\overline{k}}$ contains no subgroup isomorphic to the additive group. Then, for every $\nu \geq \nu_0$ and every $n > 0$, the group ${}_{p^\nu}H_n$ is flat over S_n.*

Indeed, $p^{\nu_0}\,\mathrm{id}_{Q_n} = 0$ implies that $p^\nu\,\mathrm{id}_{H_n}$ factors through G_n, giving the commutative diagram

$$\begin{array}{ccc} H_n & \xrightarrow{p^\nu\,\mathrm{id}_{H_n}} & H_n \\ {\scriptstyle u_n}\uparrow & \overset{v_n}{\searrow} & \uparrow{\scriptstyle u_n} \\ G_n & \xrightarrow{p^\nu\,\mathrm{id}_{G_n}} & G_n. \end{array}$$

We claim that v_n is flat. Since H_n and G_n are flat over S_n, it is enough to check that v_0 is flat (SGA 1, IV, 5.9). Thus we may take $n = 0$, so that $S_n = \mathrm{Spec}(k)$. Since $p\,\mathrm{id}_{G_0}$, and hence $p^\nu\,\mathrm{id}_{G_0}$, is flat, its image is an induced open subgroup G_0' of G_0. The map $u_0 : G_0 \to H_0$ is surjective, so v_0 takes its values in G_0' and may be regarded as a homomorphism to G_0'. Its composite with u_0 is an epimorphism; hence v_0 itself is an epimorphism, and therefore a flat homomorphism into G_0', and thus into G_0. Consequently v_n is flat and $\ker v_n = {}_{p^\nu}H_n$ is flat over S_n. □

Remark. We did not explicitly use the assumption that G_0 is smooth over k. It is, however, an easy consequence of the flatness of $p\,\mathrm{id}_{G_0}$; this is why the condition was stated explicitly in Corollary 7.5.

[38] Editor's note: the hypothesis that Q be commutative has been removed here, and 7.3.1 modified accordingly. If Q is not commutative, the n-th power morphism need not be a group homomorphism.

[39] Editor's note: this should be added to Exposé VII_A. On the other hand, the statement is true for every S when Q is commutative; cf. Deligne's theorem in [TO70], p. 4.

Lemma 7.6. *Let $u : G \to H$ be a surjective monomorphism of commutative algebraic groups over a field k of characteristic $p > 0$, and put $Q = H/G$, a purely infinitesimal group. There is an integer ν_1 such that, for every $\nu \geq \nu_1$, the sequence*

$$0 \longrightarrow {}_{p^\nu}G \longrightarrow {}_{p^\nu}H \longrightarrow Q \longrightarrow 0$$

is exact.

It is enough to establish exactness at Q, which in turn amounts to the injectivity of the homomorphism

$$\mathscr{O}_{Q,e} \longrightarrow \mathscr{O}_{{}_{p^\nu}H,e} \tag{x}$$

of local rings at the identity elements. Recall that Q is set-theoretically supported at the single point e. There is a natural homomorphism

$$\mathscr{O}_{{}_{p^\nu}H,e} \longrightarrow \mathscr{O}_{H,e}/\mathfrak{m}^{p^\nu}, \tag{xx}$$

where $\mathfrak{m}$ is the augmentation ideal, equivalently the maximal ideal, of $\mathscr{O}_{H,e}$. Indeed, $p^\nu \operatorname{id}_H$ vanishes on the kernel of the iterated Frobenius homomorphism F^ν. The composite of (x) and (xx) is the natural composite

$$\mathscr{O}_{Q,e} \longrightarrow \mathscr{O}_{H,e} \longrightarrow \mathscr{O}_{H,e}/\mathfrak{m}^{p^\nu}. \tag{xxx}$$

Now $\mathscr{O}_{Q,e} \to \mathscr{O}_{H,e}$ is injective, because $H \to Q$ is an epimorphism and therefore flat. Moreover, $\mathscr{O}_{Q,e}$ is Artinian, while $\bigcap_\nu \mathfrak{m}^{p^\nu} = 0$ in $\mathscr{O}_{H,e}$, so the intersection of the corresponding ideals in $\mathscr{O}_{Q,e}$ is also zero. One of them is therefore zero. Thus (xxx) is injective, and a fortiori so is (x).

Lemma 7.7. *Under the hypotheses of 7.5, there is an integer ν_1 such that, for every $\nu \geq \nu_1$ and every n, the sequence of S_n-groups*

$$0 \longrightarrow {}_{p^\nu}G_n \longrightarrow {}_{p^\nu}H_n \xrightarrow{w_n} Q_n \longrightarrow 0$$

is exact. More precisely, w_n is faithfully flat and its kernel is ${}_{p^\nu}G_n$.

Take ν_1 as in Lemma 7.6, applied to $u_0 : G_0 \to H_0$, and require $\nu_1 \geq \nu_0$, where $p^{\nu_0} = \operatorname{rank} Q_0$. It remains only to check that w_n is faithfully flat. By Corollary 7.5, ${}_{p^\nu}H_n$ is flat over S_n, as is Q_n; it is therefore enough to check that

$$w_0 : {}_{p^\nu}H_0 \longrightarrow Q_0$$

is faithfully flat, equivalently an epimorphism. This follows from the choice of ν_1.

Corollary 7.8. *Suppose in addition that H is separated over S, or more generally that ${}_{p^\nu}H$ is separated over S for every ν, and that ${}_{p^\nu}G$ is finite over S for every ν. Then, for $\nu \geq \nu_1$, the group schemes ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are finite and flat over S, and there is an exact sequence*

$$0 \longrightarrow {}_{p^\nu}G \longrightarrow {}_{p^\nu}H \xrightarrow{w} Q \longrightarrow 0.$$

Because $u : G \to H$ is surjective, the induced morphism ${}_{p^\nu}G \to {}_{p^\nu}H$ is also surjective. The source is finite over S and the target is separated over S, hence the target is finite over S. To prove that ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are flat over S, it is enough to prove this after every base change to S_n, where it is contained in Corollary 7.5. Finally, the exact sequence above is obtained from the exact sequences in Lemma 7.7 as n varies.

Taking generic fibers in the sequence of Corollary 7.8, and recalling that $u_{s'} : G_{s'} \to H_{s'}$ is an isomorphism, gives that $Q_{s'}$ is the unit group. Since Q is flat over S, Q itself is the unit group; therefore so is Q_s, and $u_s : G_s \to H_s$ is an isomorphism. We have proved:

Proposition 7.9. *Let $u : G \to H$ be a homomorphism of group preschemes of finite presentation over a prescheme S. Assume:*

(a) *u is a monomorphism;*

(b) *G is flat over S;*

(c) *for every $s \in S$ whose residue field has characteristic $p > 0$, let $S' = \mathrm{Spec}(\mathscr{O}_{S,s})$, and let $u' : G' \to H'$ be obtained from u by base change to S'. Then G' is commutative, the special fiber $G'_0 = G_s$ is smooth over $\kappa(s)$, and, for every $\nu > 0$, the group ${}_{p^\nu}G'$ is finite over S' while ${}_{p^\nu}H'$ is separated over S'.*

Under these assumptions, u is an immersion.

Indeed, the finiteness of ${}_pG'$ over S' implies the finiteness of its special fiber over $\kappa(s)$. Hence $G'_0 \otimes_{\kappa(s)} \overline{\kappa(s)}$ contains no subgroup isomorphic to the additive group, and the hypotheses of Corollary 7.5 apply.

Remark 7.10. Examples of M. Raynaud (XVI, 1.1(a),(b)) show that in condition (c) one cannot omit either the finiteness of the ${}_{p^\nu}G'$ over S' or the separatedness of the ${}_{p^\nu}H'$ over S'.

We now seek conditions under which u is a closed immersion. Retain the hypotheses preceding Proposition 7.9, but no longer suppose that u is surjective; assume only that it is an isomorphism on generic fibers. We already know that $u : G \to H$ is an open immersion. The same is therefore true of $u_0 : G_0 \to H_0$, whose image contains the identity component $(H_0)^0$. Since G_0 is smooth over k and has no additive component after extending to $\overline{k}$, the same is true of H_0. We use:

Lemma 7.11. *Let H be a commutative algebraic group over a field k, and suppose that $H \otimes_k \overline{k}$ contains no subgroup isomorphic to $\mathbf{G}_a$. If $n = \deg(H/H^0)$, then*

$$ {}_nH \longrightarrow H/H^0 $$

is surjective.

We may assume that k is algebraically closed, in which case this follows from the familiar fact that $H^0(k)$ is a divisible group.

Return to $u : G \to H$, and suppose that the ${}_nG$, $n > 0$, are proper over S, while the ${}_nH$ are separated over S. The ${}_nH$ are also flat over S. Indeed, it is enough to check flatness at the points above s, and hence to check that $n\,\mathrm{id}_{H_0}$ is flat. As noted above, this is equivalent to $(H_0)^0$ being smooth[40] over k and having no $\mathbf{G}_a$-component over $\overline{k}$. Since $(H_0)^0 = (G_0)^0$, this follows from the corresponding hypothesis on G_0.

Because H is separated over S, so are the ${}_nH$. The morphism ${}_nG \to {}_nH$ is proper and therefore has closed image. Its image contains the generic fiber of ${}_nH$, because $u_{s'} : G_{s'} \to H_{s'}$ is an isomorphism. Since ${}_nH$ is flat over S, it is the schematic closure of its generic fiber. Hence ${}_nG \to {}_nH$, and therefore ${}_nG_0 \to {}_nH_0$, is surjective. Together with $G_0 \supset (H_0)^0$, Lemma 7.11 now implies that $G_0 \to H_0$ is surjective. Hence u is surjective. We obtain:

Proposition 7.12. *With the notation of 7.9, suppose that conditions (a) and (b) hold, together with the stronger condition (c′) obtained from (c) by requiring, for every integer $n > 0$, not only those of the form p^ν, that ${}_nG'$ be finite over S' and ${}_nH'$ separated over S'. Then u is a closed immersion.*

Remark 7.13.

[40] Editor's note: the terminology has been updated, replacing "simple" by "smooth"; see, for example, A. Grothendieck's note (1) in SGA 1, Exposé II.

(a) The separatedness hypothesis on the ${}_nH$ easily implies that H itself is separated over S. This is also formally contained in Proposition 7.12, by taking G to be the unit S-group. If S is locally Noetherian, in Proposition 7.9 it is enough to assume that H is locally of finite type over S, rather than of finite type; the proof applies verbatim. For Proposition 7.12, one also assumes that the fibers of H are of finite type.

(b) Using Proposition 7.12, it is not difficult to prove that if $u : G \to H$ is a monomorphism of group preschemes of finite presentation over S, G is diagonalizable—or, more generally, of multiplicative type—and H is separated over S, then u is a closed immersion. This is a satisfactory generalization of the first conclusion in 5.7. If G is smooth over S, apply Proposition 7.12 directly; the general case reduces readily to this one. If H is not assumed separated over S, one can still show that u is an immersion; for a torus G, this also follows from the material that comes next.

(c) If G is assumed to have connected fibers in Proposition 7.9, the separatedness of ${}_{p^\nu}H'$ over S' may be omitted from condition (c). Indeed, after the reductions in the proof, H may be assumed flat over S and u bijective. Thus H has connected fibers, and Raynaud's theorem (VI_B, 5.5) implies that H is separated over S.[41]

Bibliography

[42]

[RG71] M. Raynaud and L. Gruson, *Critères de platitude et de projectivité*, Invent. Math. **13** (1971), 1–89.

[TO70] J. Tate and F. Oort, *Group schemes of prime order*, Ann. Sci. Éc. Norm. Supér. (4) **3** (1970), 1–21.

[41] Editor's note: the attribution, which appeared in SGAD 1965, should be made explicit in Exposé VI_B, §5, together with the changes between SGAD and Lecture Notes 151.

[42] Editor's note: additional references cited in this exposé.

Exposé IX
Groups of Multiplicative Type: Homomorphisms into a Group Scheme

by A. Grothendieck[0]

1. Definitions

Definition 1.1. — *Let S be a prescheme and let G be a group prescheme over S. The group G is said to be* of multiplicative type *if it is locally diagonalizable for the faithfully flat quasi-compact topology (cf. IV 6.3); that is, if for every $s \in S$, there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that*

$$G' = G \times_S S'$$

is a diagonalizable S'-group (I 4.4 and VIII 1.1).

The group G is said to be of quasi-isotrivial multiplicative type *if it is locally diagonalizable even for the étale topology (IV 6.3); that is, if in the preceding definition one can take $S' \to U$ to be étale and surjective. Equivalently—as one sees by taking the disjoint union of the schemes S' attached to the various $s \in S$—there is an étale surjective morphism $S' \to S$ such that $G' = G \times_S S'$ is a locally diagonalizable S'-group.*

If one can moreover choose $S' \to S$ to be finite, étale, and surjective, then G is said to be of isotrivial multiplicative type.

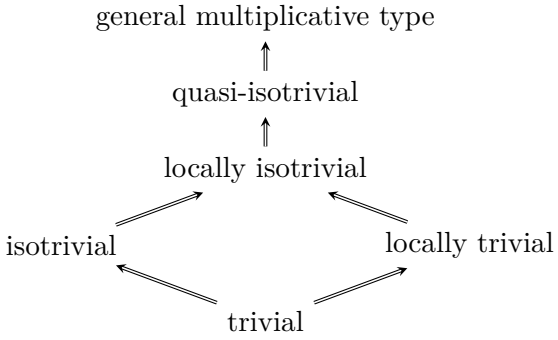
Finally, G is said to be a group of locally trivial multiplicative type *(respectively,* of locally isotrivial multiplicative type*) if every $s \in S$ has an open neighborhood U such that $G \times_S U$ is a diagonalizable U-group (respectively, is of isotrivial multiplicative type; in other words, there is a finite étale surjective morphism $S' \to U$ such that $G \times_S S'$ is a diagonalizable S'-group).*

Comments 1.2. — The five preceding notions are all obtained from the notion of a diagonalizable group by localization, with respect to five different "topologies" associated with (**Sch**).

It is understood in general that, when the word "locally" is not qualified by naming the topology under consideration, the Zariski topology is meant, in accordance with established terminology. In the terminology introduced here, "of locally trivial multiplicative

[0]Editor's note: version 1.0 of November 8, 2009. Additions were made in the proof of 3.6 bis, in 4.4–7, 5.0–6, 6.1, 7.1, and 8.2. Sections 8.1 and 4.5 remain to be revised.

type" is equivalent to "locally diagonalizable" in VIII 1.1; likewise, "trivial" translates as "diagonalizable." The five notions introduced above fit into the following implication diagram, which follows from the corresponding relations among the topologies involved:



For practical purposes, let us point out at once that every group of multiplicative type that we shall encounter will be quasi-isotrivial. Thus we shall see in the next exposé (X 4.5) that, when G is of finite type over S, it is automatically quasi-isotrivial; we shall nevertheless give examples in which it is not locally isotrivial. We shall likewise see that G may be locally trivial without being isotrivial, and hence *a fortiori* without being trivial. This readily implies that the inclusions in the preceding diagram are strict.

On the other hand, we shall also see there (X 5.16) that, when S is locally noetherian and normal—or, more generally, geometrically unibranch—every group of multiplicative type and finite type over S is necessarily isotrivial; moreover, it is trivial as soon as it is locally trivial (or even merely trivial over a dense open subset). This explains why most groups of multiplicative type encountered in practice will doubtless be isotrivial, all the more so because we shall later see that the maximal tori of semisimple group schemes are automatically isotrivial.

Definition 1.3. — *Let S and G be as in 1.1. The group G is called a* torus *if, for the faithfully flat quasi-compact topology, it is locally isomorphic to a group of the form $\mathbf{G}_m^r$, where r is an integer with $r \geq 0$.*

With the notation of 1.1, this says that S' may be chosen so that G' is isomorphic to a group of the form $(\mathbf{G}_{m,S'})^r$. The integer r depends on $s \in S$: it is the dimension of the fiber

$$G_s = G \otimes_S \kappa(s).$$

It is a locally constant function of s, as is seen at once. This observation should be generalized as follows.

Definition 1.4. — *Let G be a diagonalizable group scheme over a field k. Thus G is isomorphic to a group $D_k(M)$, where M is an ordinary commutative group determined up to isomorphism by this condition—more precisely,*

$$M \simeq \mathrm{Hom}_{k\text{-gr}}(G, \mathbf{G}_{m,k})$$

(VIII 1.3). The isomorphism class of M is called the type *of the diagonalizable group G; it is plainly invariant under extension of the base field.*

Now let G be a group scheme of multiplicative type over an arbitrary prescheme S. For every $s \in S$, there is an extension k of $\kappa(s)$ such that $G \times_S \mathrm{Spec}(k)$ is a diagonalizable k-group.[1] *Its type is independent of the chosen extension by the preceding observation and*

[1] Editor's note: indeed, by hypothesis there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that $G_{S'}$ is diagonalizable. For every $s' \in S'$ above s, the field $\kappa(s')$ is an extension of $\kappa(s)$, and $G_{s'} = G \times_S \mathrm{Spec}(\kappa(s'))$ is diagonalizable.

is called the type of G at s, *or the* type of G_s. *In particular, if G itself is diagonalizable, and hence isomorphic to a group of the form $D_S(M)$, then for every $s \in S$ the type of G at s is the isomorphism class of M.*

More generally, if G is a group of multiplicative type over S and M is an ordinary commutative group, then G is said to be of type M *if it is of type M at every point $s \in S$; equivalently, if G is locally isomorphic to $D_S(M)$ for the faithfully flat quasi-compact topology.*

Remark 1.4.1. [2] *a) For a group G of multiplicative type over S, the function*

$$s \longmapsto \text{the type of } G_s$$

is locally constant on S. Indeed, with the notation of 1.1, if G' is of type M, then G is of type M over the neighborhood U of s. Consequently S has a canonical decomposition as a disjoint union of preschemes S_i such that, for every i, G_{S_i} is of type M_i, where the commutative groups M_i are pairwise nonisomorphic.

b) In particular, if S is connected, the type of the fibers of G is constant; that is, there is a commutative group M such that G is of type M. Finally, if G is a torus, the type of G at s is determined by the integer $\dim G_s$*: indeed, G_s is of type $\mathbb{Z}^n$, where $n = \dim G_s$.*

Remark 1.5. — *a) It follows immediately from Definitions 1.1 and 1.3 that these notions are "compatible with base extension." Thus, if G is a group prescheme over S and $S' \to S$ is a base-change morphism, then whenever G is of multiplicative type (respectively, of isotrivial multiplicative type, and so on), the same is true of G'.*

b) If, moreover, $S' \to S$ is faithfully flat and quasi-compact, then G' being of multiplicative type, respectively a torus, implies the same for G. If $S' \to S$ is also étale (respectively, finite étale), then G' being quasi-isotrivial (respectively, isotrivial) implies the same for G.

c) Finally, return to an arbitrary base-change morphism $f : S' \to S$. If $s' \in S'$ and $s = f(s')$, then the type of G' at s' is the same as that of G at s, since

$$G'_{s'} = G_s \otimes_{\kappa(s)} \kappa(s').$$

2. Extension of Certain Properties of Diagonalizable Groups to Groups of Multiplicative Type

The extensions in question are essentially trivial consequences of the results of the preceding Exposé, in view of the definitions in 1.1 and the "local" nature (for the faithfully flat and quasi-compact topology) of the results under consideration.

Definition 2.0. [3] *We denote by $\mathcal{M}$ the set of faithfully flat and quasi-compact morphisms.*

Proposition 2.1. *Let G be a group of multiplicative type over a prescheme S. Then:*

(a) *G is faithfully flat over S, and affine over S (a fortiori, quasi-compact over S);*

(b) *G is of finite type over S if and only if G is of finite presentation over S, if and only if, for every $s \in S$, the type of G at s is given by a commutative group of finite type;*

[2] Editor's note: the number 1.4.1 has been added to the remarks that follow, for use in later references.
[3] Editor's note: This definition has been added; it occurs in Propositions 2.3 and 2.7.

(c) *G is finite over S if and only if, for every $s \in S$, the type of G at s is given by a finite commutative group, if and only if (when S is quasi-compact) G is of finite type over S and is annihilated by an integer $n > 0$;*

(c′) *G is integral over S if and only if, for every $s \in S$, the type of G at s is given by a torsion commutative group;*

(d) *G is the unit group if and only if, for every $s \in S$, the type of G at s is given by the zero commutative group;*

(e) *G is smooth over S if and only if, for every $s \in S$, the type of G at s is given by a commutative group of finite type whose torsion subgroup has order prime to the characteristic of $\kappa(s)$.*

These statements follow from VIII 2.1, taking into account that the properties in question descend under faithfully flat and quasi-compact morphisms (cf. SGA 1, VIII, or EGA IV_2, §2).

Using VIII 3.5, one likewise obtains:

Proposition 2.2. *Let G be a group of multiplicative type and of finite type over S. For every integer $n \neq 0$, the kernel ${}_nG = \ker(n \cdot \mathrm{id}_G)$ is a group of multiplicative type, finite over S.*

Proposition 2.3. *Let G be a group of multiplicative type over the prescheme S, acting freely on the right on an S-prescheme X that is affine over S. Then:*

(a) *the equivalence relation defined by G in X is $\mathcal{M}$-effective (IV 3.4), and $Y = X/G$ is affine over S;*

(b) *if, moreover, X is of finite presentation (respectively, of finite type) over S, then the same is true of Y.*

The first assertion follows from VIII 5.1, which treats the case in which G is diagonalizable, and from IV 3.5.2, which allows one to reduce to that case. Indeed, faithfully flat and quasi-compact morphisms $S' \to S$ are effective descent morphisms for the fibered category of affine schemes over schemes: if Y' is affine over S' and is equipped with a descent datum relative to $S' \to S$, then this datum is effective, that is, Y' comes from a scheme Y affine over S (cf. SGA 1, VIII 2.1).

For the second assertion, one is likewise reduced to the diagonalizable case of VIII 5.8, since the finiteness conditions under consideration descend under faithfully flat and quasi-compact morphisms (SGA 1, VIII 3.3 and 3.6[4]). Proceeding as in VIII, Corollaries 5.5–5.7, one deduces from 2.3:

Corollary 2.4. *Under the hypotheses of 2.3, the graph morphism*

$$X \times_S G \longrightarrow X \times_S X$$

is a closed immersion. For every section σ of X over S, the corresponding morphism

$$G \longrightarrow X, \qquad g \longmapsto \sigma \cdot g,$$

is a closed immersion.

In particular:

[4]Editor's note: See also V, 9.1, or EGA IV_2, 2.7.1.

Corollary 2.5. *Let $u : G \to H$ be a monomorphism of S-preschemes in groups, with G of multiplicative type and H affine over S. Then u is a closed immersion, $H/G = Y$ exists and is affine over S, and H is a principal homogeneous bundle over Y under G_Y.*

Remark 2.6. Let $u : G \to H$ be a monomorphism of S-preschemes in groups, where G is of multiplicative type and of finite type over S, and H is of finite presentation and separated over S. One can then show that u is a closed immersion by using VIII 7.12 (see Remark VIII 7.13(b)).

Proposition 2.7. *Let $u : G \to H$ be a homomorphism of S-groups of multiplicative type, with H of finite type over S. Then:*

(i) *$G_0 = \ker u$ is an S-group of multiplicative type; the equivalence relation defined by G_0 in G is $\mathcal{M}$-effective, and hence (IV 3.4) u factors as*

$$G \longrightarrow G/G_0 = I \longrightarrow H,$$

where $I \to H$ is a monomorphism[5] of S-groups. The group I is of multiplicative type, the equivalence relation in H defined by I is $\mathcal{M}$-effective, and consequently $H_0 = H/I$ exists and is also of multiplicative type;

(ii) *H_0 and I are of finite type, and so is G_0 when G is.*

Proof. Proceeding as for 2.3, one is reduced to the case in which G and H are diagonalizable; then 2.7 reduces to VIII 3.4.

We record the following corollaries.

Corollary 2.8.

(a) *Let S be a prescheme. The category of S-groups of multiplicative type and of finite type is an abelian category.*

(b) *Let $u : G \to H$ be a homomorphism in this category. It is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in this category if and only if u is a monomorphism of preschemes (respectively, u is faithfully flat, respectively, u is an isomorphism of preschemes).*

It is enough to note that the product of two S-groups of multiplicative type is again of multiplicative type (which is immediate), and is plainly of finite type over S if both factors are. The rest of 2.8 follows immediately from 2.7; the details are left to the reader.

Corollary 2.9. *Let $u : G \to H$ be a homomorphism of S-groups of multiplicative type and of finite type. Let U be the set of points $s \in S$ for which $u_s : G_s \to H_s$ is a monomorphism (respectively, faithfully flat, respectively, an isomorphism). Then U is both open and closed, and the induced homomorphism $u_U : G_U \to H_U$ is a monomorphism (respectively, faithfully flat, respectively, an isomorphism).*

Let P (respectively, Q) be the kernel (respectively, cokernel) of u. By 2.7, Q exists and P and Q are of multiplicative type; moreover, formation of P and Q commutes with every base change $S' \to S$, in particular with the formation of fibers. On the other hand, u is a monomorphism (respectively, faithfully flat, respectively, an isomorphism) if and only if $P = 0$ (respectively, $Q = 0$, respectively, $P = Q = 0$). These remarks reduce us to checking the following assertion: if R is an S-group of multiplicative type, then the set U

[5] Editor's note: Indeed, even a closed immersion.

of points $s \in S$ for which $R_s = 0$ is open and closed, and $R_U = 0$. This is contained in Remark 1.4.1(a).

Corollary 2.10. *Let G be an S-group of multiplicative type and of finite type, and let H and H' be two subgroups of multiplicative type.*

(a) *$H'' = H \cap H'$ is a subgroup of G of multiplicative type.*

(b) *Let U be the set of $s \in S$ for which $H_s \subseteq H'_s$ (respectively, $H_s = H'_s$). Then U is open and closed, and $H_U \subseteq H'_U$ (respectively, $H_U = H'_U$).*

Here, of course, the intersection sign in $H \cap H'$ denotes the intersection in the functorial sense, namely $H \times_G H'$, which is plainly an S-subgroup of G. Likewise, the inclusion sign (respectively, the equality sign) denotes the order relation (respectively, equality) between subfunctors of H, not inclusion (respectively, equality) of the underlying sets.

Applying 2.7 first to the inclusion morphism $H \to G$, one finds that H is of finite type; the same holds for H'. It then follows from 2.8 that $H \cap H'$ is of multiplicative type and of finite type. (Note that the canonical functor from the category considered in 2.8 to the category $(\mathbf{Sch})/S$ commutes with finite inverse limits.)

Formation of $H \cap H'$ commutes with base extension, in particular with formation of the fibers. Moreover, $H \subseteq H'$ (respectively, $H = H'$) is equivalent to $H'' = H$ (respectively, to $H'' = H$ and $H'' = H'$). Consider the canonical homomorphisms $H'' \to H$ and $H'' \to H'$. Then U is the set of points $s \in S$ for which the induced homomorphism $H''_s \to H_s$ is an isomorphism (respectively, for which $H''_s \to H_s$ and $H''_s \to H'_s$ are isomorphisms).

By 2.9, it follows that U is open and closed, and that $H''_U \to H_U$ (respectively, $H''_U \to H_U$ and $H''_U \to H'_U$) are isomorphisms. This gives the desired conclusion.

Proposition 2.11. *Let S be a prescheme, let G be an S-group of multiplicative type and of finite type over S, let H be a subgroup of G of multiplicative type, and put $K = G/H$, a quotient group of G of multiplicative type.*

(i) *Suppose G is trivial, that is, diagonalizable. There is a partition of S into open and closed subsets S_i such that, for every i, H_{S_i} and K_{S_i} are diagonalizable. In particular, if S is connected, H and K are diagonalizable.*

(ii) *The same assertion holds with "diagonalizable" replaced by "isotrivial," provided that S is connected, locally connected, or quasi-compact.*

(iii) *If G is locally trivial (respectively, locally isotrivial, respectively, quasi-isotrivial), then the same is true of K and H.*

Proof. (i) By hypothesis, $G = D_S(M)$, where M is a commutative group of finite type. For each quotient group M_i of M, let $H_i = D_S(M_i)$ be the corresponding diagonalizable subgroup of G (VIII 3.1). Let S_i be the set of points $s \in S$ for which $H_s = (H_i)_s$. By 2.10, S_i is open and closed, and $H_{S_i} = (H_i)_{S_i}$; thus H_{S_i} is diagonalizable, and hence so is K_{S_i} (VIII 3.1). The S_i are plainly pairwise disjoint, and we claim that they cover S. Indeed, for every s, the group H_s, being a subgroup of the diagonalizable group G_s, is diagonalizable (cf. 8.1 below[6]). Hence H_s has the form $D_{\kappa(s)}(M_i)$, by VIII 1.5 and 3.2(b). Restricting to the family of nonempty S_i, assertion (i) follows.

(ii) By hypothesis, there exists a finite surjective étale morphism $S' \to S$ such that $G_{S'}$ is diagonalizable. Every point of S' therefore has an open and closed neighborhood U'

[6]Editor's note: The erroneous reference to IV 4.7.5 has been corrected. Note that Section 8 of the present Exposé is independent of Sections 3–7.

such that $H_{U'}$ and $K_{U'}$ are diagonalizable. The image U of U' in S is open and closed, and $S' \to S$ still induces a finite surjective étale morphism $U' \to U$. Thus every point $s \in S$ has an open and closed neighborhood U such that H_U and K_U are isotrivial. Assertion (ii) follows at once.

(iii) This follows at once from (i), (ii), and the definitions. Moreover, the quasi-isotrivial case will also follow from the more general fact "finite type $\Longrightarrow$ quasi-isotrivial," announced in 1.2 (cf. X 4.5).

3. Infinitesimal Properties: Lifting and Conjugation Theorems

The fundamental infinitesimal properties of groups of multiplicative type follow from the following theorem.

Theorem 3.1. *Let S be an affine scheme, let H be an S-group of multiplicative type, and let $\mathscr{F}$ be a quasi-coherent sheaf on S on which H acts (I 4.7). Then*

$$\mathrm{H}^i(H, \mathscr{F}) = 0 \qquad \text{for } i > 0,$$

where H^i denotes the Hochschild cohomology studied in Exposé I, §5.

Indeed, by loc. cit., 5.3, Hochschild cohomology is expressed, in terms of the affine rings A and B of S and G, and the A-module M defining $\mathscr{F}$, as the cohomology of a complex of A-modules $C^\bullet(H, M)$, whose formation commutes with every base change $A \to A'$. Consequently, for a base change $A \to A'$ with A' flat over A, one has

$$\mathrm{H}^i(H', \mathscr{F}') = \mathrm{H}^i(H, \mathscr{F}) \otimes_A A'.$$

If A' is moreover faithfully flat over A, it is therefore enough, in order to prove that $\mathrm{H}^i(H, \mathscr{F})$ vanishes, to prove that $\mathrm{H}^i(H', \mathscr{F}')$ vanishes. This observation reduces us to checking 3.1 in the case where G is diagonalizable; in that case it was proved in I 5.3.3.

Using the results of Exposé III, we shall derive from 3.1 several consequences of a geometric nature.

Theorem 3.2. *Let S be an affine scheme, let S_0 be an affine subscheme defined by an ideal J, let H be a group of multiplicative type over S, and let G be an arbitrary group prescheme over S. Let $u, v : H \to G$ be two group homomorphisms, and let $u_0, v_0 : H_0 \to G_0$ be the homomorphisms obtained from u, v by the base change $S_0 \to S$. If $J^2 = 0$ and $u_0 = v_0$, then there exists $g \in G(S)$ such that*

$$v = \mathrm{int}(g) \circ u \qquad \textit{and} \qquad g_0 = e.$$

This follows from III 2.1(ii) and 3.1 (the vanishing of H^1).

Corollary 3.3. *Under the preliminary hypotheses of 3.2, suppose in addition that G is smooth over S, but assume only that J is nilpotent (not necessarily square-zero). If there exists $g_0 \in G_0(S_0)$ such that $v_0 = \mathrm{int}(g_0) \circ u_0$, then g_0 lifts to an element $g \in G(S)$ such that $v = \mathrm{int}(g) \circ u$.*

By induction on an integer n such that $J^n = 0$, one is reduced to the case in which J is square-zero. Since G is smooth over S, one may lift g_0 to an element g' of $G(S)$. Replacing v by $v' = \mathrm{int}(g') \circ u$, one is then reduced to the situation of 3.2.

Corollary 3.4. *Under the preliminary hypotheses of 3.2, with J nilpotent, suppose in addition that u is central (for example, that G is commutative). Then $u_0 = v_0$ implies $u = v$.*

Indeed, the reduction to the case in which J is square-zero is again immediate, and then it is enough to apply 3.2. In particular:

Corollary 3.5. *Let S, S_0, H, G be as in 3.2, with J nilpotent. Let $u : H \to G$ be a homomorphism of S-groups such that $u_0 : H_0 \to G_0$ is the unit homomorphism. Then u is the unit homomorphism.*

Theorem 3.6. *Let S be an affine scheme, let S_0 be a subscheme defined by a nilpotent ideal J, let H be a group of multiplicative type over S, let G be a group prescheme smooth over S, and let $u_0 : H_0 \to G_0$ be a homomorphism of the S_0-groups obtained by base change $S_0 \to S$. Then there exists a homomorphism $u : H \to G$ of S-groups that lifts u_0. Moreover, by 3.3, any two such liftings u, u' are conjugate by an element of $G(S)$ whose reduction in $G(S_0)$ is the identity element.*

This follows from III 2.1 and 2.3 and from 3.1: the vanishing of H^2 gives the existence of the lifting of u_0, and the vanishing of H^1 gives uniqueness up to conjugation.

One may prove in the same way the following variants of 3.2–3.6. In fact, these variants imply the preceding statements when applied to the graph subgroups of the homomorphisms considered there.

Theorem 3.2 bis. *Let S be an affine scheme, let S_0 be a subscheme defined by an ideal J, let G be an S-group prescheme, and let H, H' be subgroup preschemes of multiplicative type. Let G_0, H_0, H'_0 be the groups over S_0 obtained by base change $S_0 \to S$. If $J^2 = 0$ and $H_0 = H'_0$, then there exists $g \in G(S)$ such that*

$$H' = \mathrm{int}(g)(H) \qquad \text{and} \qquad g_0 = e.$$

Corollary 3.3 bis. *Under the preliminary hypotheses of 3.2 bis, suppose in addition that G is smooth over S, and that J is nilpotent (not necessarily square-zero). If $g_0 \in G_0(S_0)$ satisfies $H'_0 = \mathrm{int}(g_0)(H_0)$, then g_0 lifts to an element $g \in G(S)$ such that $H' = \mathrm{int}(g)(H)$.*

Corollary 3.4 bis. *Under the preliminary hypotheses of 3.2 bis, suppose that J is nilpotent and that H is central (for example, that G is commutative). Then $H_0 = H'_0$ implies $H = H'$.*

Theorem 3.6 bis. *Let S be an affine scheme, let S_0 be a subscheme defined by a nilpotent ideal J, let G be an S-group prescheme smooth over S, and let H_0 be a subgroup prescheme of $G_0 = G \times_S S_0$, of multiplicative type. Then:*

(a) *there exists a subgroup prescheme H of G, flat over S, such that $H \times_S S_0 = H_0$;*

(b) *such an H is necessarily of multiplicative type;*

(c) *finally, by 3.2 bis, any two such liftings H, H' of H_0 are conjugate by an element $g \in G(S)$ whose reduction in $G(S_0)$ is the identity element.*

Let us indicate how one can prove 3.2 bis (of which 3.3 bis and 3.4 bis are immediate consequences) and assertion (a) of 3.6 bis. The latter follows from 3.1 (the vanishing of H^2) and III 4.37(ii). Similarly, 3.2 bis follows from 3.1 (the vanishing of H^1) and III 4.37(i), at least when G and H are smooth over S.

Using the results of the following Exposé, however, one can very simply deduce 3.2 bis and 3.6 bis, in the general form stated here, from 3.2 and 3.6, respectively. For 3.2 bis, it is enough to note that, by X 2.1, H and H' are isomorphic, which reduces us to 3.2.

For 3.6 bis, note that H_0 is necessarily of finite type over S_0:[7] indeed, as a subgroup prescheme of G_0, which is smooth over S_0, the group H_0 is locally of finite type over S_0. But H_0 is affine over S_0 by 2.1(a), and is therefore of finite type over S_0. By X 4.5, H_0 is consequently quasi-isotrivial, and hence comes, by X 2.1, from a group H of multiplicative type over S.[8] By 3.6, there exists a homomorphism of S-groups $u : H \to G$ lifting the immersion $H_0 \hookrightarrow G_0$. Since H and H_0 (respectively, G and G_0) have the same underlying topological space, and since, for every $h \in H$, the morphism

$$\mathscr{O}_{G,u(h)} \longrightarrow \mathscr{O}_{H,h}$$

is surjective (because it is so after reduction modulo the nilpotent ideal J), the morphism u is also an immersion (cf. EGA I, 4.2.2).

Finally, every lifting H of H_0 to a flat subgroup of G is necessarily of multiplicative type by X 2.3,[9] which also proves assertion (b) of 3.6 bis. The reader may verify that the results 3.2 bis–3.6 bis are not used in Exposé X, so there is no vicious circle.

Proposition 3.8. [10] *Let S be a prescheme, let S_0 be a closed subscheme defined by a locally nilpotent ideal J, let G be an S-group of multiplicative type, and let X be an S-prescheme locally of finite presentation. Suppose that G acts on X in such a way that $G_0 = G \times_S S_0$ acts trivially on $X_0 = X \times_S S_0$. Then G acts trivially on X.*

The proof is the proof of III 2.4(b). One may also reduce to loc. cit. by observing that 3.8 immediately reduces to the case in which G is diagonalizable, which is the case treated there.

Corollary 3.9. *Let S, S_0 be as above, and let $u : G \to H$ be a homomorphism of S-groups, with G of multiplicative type and H locally of finite presentation over S. Suppose that the homomorphism $u_0 : G_0 \to H_0$ obtained by base change $S_0 \to S$ is central. Then u is central.*

Indeed, it is enough to apply 3.8 with $X = H$, letting G act on H by

$$(g, h) \longmapsto \mathrm{int}(u(g)) \cdot h = u(g)h\,u(g)^{-1}.$$

4. The Density Theorem

This theorem (cf. 4.7 below)*, together with the theorem[11] of Section 7, will be the essential tool in the present exposé and the two following ones for passing from the infinitesimal properties of groups of multiplicative type developed above to their "finite" properties.

Definition 4.1. *Let X be a prescheme. A family $(Z_i)_{i\in I}$ of subpreschemes of X is said to be* schematically dense *if, for every open subset U of X and every closed subprescheme U_0 of U containing all the $Z_i \cap U$, one has $U_0 = U$.*

A subprescheme Z of X is said to be schematically dense in X *if the one-member family consisting of Z is schematically dense.*

[7] Editor's note: The original said, "by 2.1(b), since its fibers are." The editors did not understand this argument and have replaced it by the argument that follows.

[8] Editor's note: The sentence that follows has been expanded.

[9] Editor's note: X 2.3 depends essentially on Theorem 3.6 of the present Exposé.

[10] Editor's note: The numbering of the original has been retained; there is no 3.7.

* All the results from 4.1 through 4.6 are contained in EGA IV_3, 11.10, to which we refer the reader for a systematic account of the notion of schematic density.

[11] Editor's note: the "algebraicity" theorem 7.1.

It is immediate (cf. EGA IV$_3$, 11.10.1) that the definition is equivalent to saying that, for every open subset U of X, every section f of $\mathscr{O}_U$ that vanishes on each $Z_i \cap U$ is zero. Equivalently, the intersection of the kernels of the canonical homomorphisms

$$u_i : \mathscr{O}_X \longrightarrow (v_i)_*(\mathscr{O}_{Z_i})$$

is zero, where $v_i : Z_i \to X$ is the canonical immersion.[12] When X lies over a prescheme S, this is also equivalent to the following assertion: for every open subset U of X, every S-prescheme Y separated over S, and every pair of morphisms $u, v : U \to Y$ that coincide on the $Z_i \cap U$, one has $u = v$. Indeed, $u = v$ is equivalent to $U_0 = U$, where U_0 is the inverse image of the diagonal in $Y \times_S Y$ under the S-morphism $U \to Y \times_S Y$ defined by (u, v). The diagonal is a closed subprescheme of $Y \times_S Y$, so U_0 is a closed subprescheme of U containing all the $Z_i \cap U$. Thus schematic density gives $U_0 = U$, hence $u = v$. Conversely, take $Y = \operatorname{Spec} \mathscr{O}_S[T]$.

In the terminology introduced at the end of EGA I, 9.5, saying that the subprescheme Z is schematically dense in X also means that X is the *closure subprescheme* of Z in X.

Lemma 4.2. *Let X be a flat S-prescheme, and let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, flat over S. Let S_0 be a subprescheme of S defined by a nilpotent ideal $\mathscr{J}$, and suppose that the modules $\mathscr{J}^n/\mathscr{J}^{n+1}$ are locally free over S_0. Let X_0, Z_{i0} be obtained from X, Z_i by the base change $S_0 \to S$. If $(Z_{i0})_{i\in I}$ is schematically dense in X_0, then $(Z_i)_{i\in I}$ is schematically dense in X.*

Proof. Suppose $\mathscr{J}^{n+1} = 0$, where $n \geq 0$, and argue by induction on n; the assertion is trivial for $n = 0$. Define S_m, X_m, Z_{im} by reduction modulo $\mathscr{J}^{m+1}$, as usual. The induction hypothesis already implies that $(Z_{i,n-1})_{i\in I}$ is schematically dense in X_{n-1}. After replacing X by an induced open subset, it is enough to prove that every section f of $\mathscr{O}_X$ that vanishes on all the Z_i is zero.

The section f_{n-1} of

$$\mathscr{O}_{X_{n-1}} = \mathscr{O}_X/\mathscr{J}^n\mathscr{O}_X$$

defined by f vanishes on the $Z_{i,n-1}$, hence is zero. Consequently, f is a section of $\mathscr{J}^n\mathscr{O}_X$. Since X is flat over S, there is an isomorphism

$$\mathscr{J}^n\mathscr{O}_X \xleftarrow{\sim} \mathscr{E} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{X_0}, \qquad \mathscr{E} = \mathscr{J}^n = \mathscr{J}^n/\mathscr{J}^{n+1}.$$

Similarly, since each Z_i is flat over S, the restriction f_i of f to Z_i may be regarded as a section of

$$\mathscr{J}^n\mathscr{O}_{Z_i} \xleftarrow{\sim} \mathscr{E} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Z_{i0}}.$$

By hypothesis, all the f_i are zero. The module $\mathscr{E}$ is locally free by assumption, and therefore so is

$$\mathscr{F} = \mathscr{E} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{X_0}.$$

Thus f is a section of the locally free module $\mathscr{F}$ on X_0 whose restriction to every Z_{i0} is zero. Since the family $(Z_{i0})_{i\in I}$ is schematically dense in X_0, it follows at once that $f = 0$. □

Corollary 4.3. *Let X be a locally Noetherian S-prescheme, flat over S, and let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, flat over S. Suppose that, for every $s \in S$, the family of fibers $(Z_{is})_{i\in I}$ is schematically dense in X_s. Then $(Z_i)_{i\in I}$ is schematically dense in X.*

Proof. After replacing X by an induced open subset, it is enough to prove that every section f of $\mathscr{O}_X$ whose restriction to all the Z_i is zero is itself zero.[13] For this, it suffices to

[12] Editor's note: all the results on schematic density stated in EGA IV$_3$, §11.10 follow from the results on "separating families of homomorphisms" proved in *loc. cit.*, §11.9, to which some of the editor's notes below will refer.

[13] Editor's note: the original has been expanded in what follows.

show that the image of f in $\mathscr{O}_{X,x}$ is zero for every $x \in X$. Let $\mathfrak{m}_x$ be the maximal ideal of $\mathscr{O}_{X,x}$, and let $\mathfrak{m}_s$ be that of $\mathscr{O}_{S,s}$, where s is the image of x in S. Since $\mathscr{O}_{X,x}$ is Noetherian,

$$\bigcap_{n\in\mathbb{N}} \mathfrak{m}_x^n = 0, \qquad \text{and hence, a fortiori,} \qquad \bigcap_{n\in\mathbb{N}} \mathscr{O}_{X,x}\mathfrak{m}_s^n = 0.$$

It is therefore enough to show, for every integer n, that the section induced by f on

$$X \times_S \operatorname{Spec}(\mathscr{O}_{S,s}/\mathfrak{m}_s^{n+1})$$

is zero. By hypothesis,

$$Z_{is} = Z_i \otimes_{\mathscr{O}_{S,s}} \kappa(s)$$

is a schematically dense family in

$$X_s = X \otimes_{\mathscr{O}_{S,s}} \kappa(s).$$

We are thus reduced to the case in which S is the spectrum of a local ring whose maximal ideal $\mathscr{J}$ is nilpotent, $S_0 = \operatorname{Spec}\kappa(s)$, and the hypotheses of Lemma 4.2 hold. The conclusion follows. □

Lemma 4.4. *Let S be a locally Noetherian prescheme, let X be a locally Noetherian S-prescheme, flat over S, and let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, flat over S. Suppose that, for every $s \in S$, the family of fibers $(Z_{is})_{i\in I}$ is schematically dense in X_s. Then, for every base change $S' \to S$, with S' not necessarily locally Noetherian, the family $(Z'_i)_{i\in I}$ is schematically dense in X'. In other words, $(Z_i)_{i\in I}$ is universally schematically dense in X relative to S (cf. EGA IV$_3$, Definition 11.10.8).*

Proof.[14] Let f' be a section of $\mathscr{O}_{X'}$ over an open subset U' of X', vanishing on the Z'_i; we must show that $f' = 0$. Fix $s' \in S'$. It is enough to prove that f' vanishes at every point of U' above s'. If s is the image of s' in S, then, after replacing S, S' by the spectra of their local rings at s, s', we may suppose that S, S' are local and that s, s' are their closed points. We may also suppose that X is affine, so that

$$S = \operatorname{Spec}(A), \qquad S' = \operatorname{Spec}(A'),$$
$$X = \operatorname{Spec}(B), \qquad X' = \operatorname{Spec}(B'), \qquad B' = B \otimes_A A',$$

where A, A' are local, $A \to A'$ is a local homomorphism, and A, B are Noetherian. We may further suppose that U' is affine, of the form $X'_{g'}$, with $g' \in B'$.

For every A-subalgebra A'' of A', put

$$S'' = \operatorname{Spec}(A'')$$

and let X'', Z''_i be obtained from X, Z_i by the base change $S'' \to S$. We obtain the diagram

$$\begin{array}{ccccc}
Z^i & \longleftarrow & Z_n^{i''} & \longleftarrow & Z_n^{i'} \\
\downarrow & & \downarrow & & \downarrow \\
X & \longleftarrow & X'' & \longleftarrow & X' \\
\downarrow & & \downarrow & & \downarrow \\
S & \longleftarrow & S'' & \longleftarrow & S'.
\end{array}$$

[14] Editor's note: for another proof, reducing to the case in which S' is locally Noetherian and using 4.3 and 4.5 as here, see EGA IV$_3$, 11.9.16 and 11.9.12. In the last line of the proof of 11.9.16, replace 11.9.5 by 11.9.12.

In what follows, we restrict to those A'' that are localizations of A-subalgebras A''_1 of A', of finite type over A, at $\mathfrak{m}' \cap A''_1$, where $\mathfrak{m}'$ is the maximal ideal of A'. Thus the homomorphisms $A \to A'' \to A'$ are local. These A'' form an increasing filtered family of subalgebras whose union, and hence whose direct limit, is A'. Likewise,

$$B' = \varinjlim_{A''} (B \otimes_A A'').$$

There therefore exist such an A'' and an element $g'' \in B'' = B \otimes_A A''$ whose image in B' is g'.

By Lemma 4.5 below, the property that $(Z_{is})_{i\in I}$ be schematically dense in X_s is preserved by every base change $S'' \to S$.[15] Consequently, because each S'' is locally Noetherian, replacing S by a suitable S'' and X, Z_i by X'', Z''_i preserves the hypotheses of Lemma 4.4.

We may therefore suppose that g' comes from an element $g \in B$. After replacing X by the open subset $X_g = U$, we may suppose that $U' = X'$. In the same way, we may suppose that f' comes from a section f of $\mathscr{O}_X$ over X.

Let $Y = V(f)$ be the subscheme of X defined by f, and put

$$Y_i = Z_i \times_X Y.$$

Thus Y_i is the closed subscheme $V(f_i)$ of Z_i, where f_i is the section of $\mathscr{O}_{Z_i}$ induced by f. Denoting by Y', Y'_i the S'-schemes obtained by base change, we have

$$Y' = V(f'), \qquad Y'_i = Z'_i \times_{X'} Y' = V(f'_i),$$

and analogous relations hold for Y'', Y''_i. The assertion that f' vanishes on the Z'_i is equivalent to

$$Y'_i = Z'_i \qquad \text{for every } i.$$

[16]

Let $\mathfrak{m}$ be the maximal ideal of A. For every $n \geq 0$, introduce the closed subscheme

$$S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1})$$

of S, and let X_n, Y_n, Z^i_n, Y^i_n be obtained from X, Y, Z_i, Y_i by the base change $S_n \to S$. More generally, for every S-prescheme P, write $P_n = P \times_S S_n$.

For every n and $i \in I$, consider the functor

$$F^i_n = \prod_{Z^i_n/S_n} Y^i_n/Z^i_n : (\mathbf{Sch})^\circ_{/S_n} \longrightarrow \mathbf{Set},$$

defined, for every P over S_n, by

$$F^i_n(P) = \Gamma\big((Y^i_n)_P/(Z^i_n)_P\big) = \begin{cases} \varnothing, & (Y^i_n)_P \neq (Z^i_n)_P, \\ \{\mathrm{id}\}, & (Y^i_n)_P = (Z^i_n)_P. \end{cases}$$

Since S_n is local Artinian and Z^i_n is flat, hence essentially free, over S_n, VIII, 6.4 shows that each F^i_n is represented by a closed subprescheme T^i_n of S_n.

[15] Editor's note: the preceding sentence has been added, and the original has been expanded in what follows.

[16] Editor's note: the original added "at least at every point x' of X' above s'," but this restriction seems unnecessary.

The completion $\widehat{A}$ of the local ring A is Noetherian because A is, and it is the projective limit of the rings A_n.[17] Let K_n^i be the ideal of

$$A_n = A/\mathfrak{m}^{n+1}$$

defining T_n^i, and let $K^i = \varprojlim_n K_n^i$; this is an ideal of $\widehat{A}$.

For fixed i, integers $m \geq n$, and any S_n-prescheme P,

$$(Y_n^i)_P = Y_i \times_S S_n \times_{S_n} P = Y_i \times_S P = (Y_m^i)_P,$$

and similarly $(Z_n^i)_P = (Z_m^i)_P$. It follows that F_n^i is the restriction $F_m^i \times_{S_m} S_n$ of F_m^i to $(\mathbf{Sch})_{/S_n}$, and therefore

$$T_n^i = T_m^i \times_{S_m} S_n.$$

We thus have a commutative diagram with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K_m^i & \longrightarrow & A_m & \longrightarrow & \mathscr{O}(T_m^i) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & K_n^i & \longrightarrow & A_n & \longrightarrow & \mathscr{O}(T_n^i) & \longrightarrow & 0 \end{array}$$

All the vertical arrows are surjective. It follows, first, that the projective system $(K_n^i)_{n\in\mathbb{N}}$ satisfies the Mittag–Leffler condition, so that

$$\varprojlim_n \mathscr{O}(T_n^i) \simeq \widehat{A}/K^i$$

as topological rings (cf. EGA 0_{III}, §13.2). Second, the map $K^i \to K_n^i$ is surjective (cf. [BEns], III, §7.4, Proposition 5). Hence

$$K_n^i \simeq \frac{K^i + \mathfrak{m}^{n+1}\widehat{A}}{\mathfrak{m}^{n+1}\widehat{A}}, \qquad \mathscr{O}(T_n^i) \simeq (\widehat{A}/K^i) \otimes_A (A/\mathfrak{m}^{n+1}).$$

In other words, $(T_n^i)_n$ is an inductive system of affine Artinian schemes, and its inductive limit

$$T^i = \varinjlim_n T_n^i$$

is a closed formal subscheme of

$$\widehat{S} = \mathrm{Spf}(\widehat{A})$$

(cf. EGA I, §10), whose reduction modulo $\mathfrak{m}^{n+1}$ is T_n^i.

Let T be the closed formal subscheme of $\widehat{S}$ given by the intersection of the T^i, that is, defined by

$$K = \sum_i K^i.$$

Since $\widehat{A}$ is Noetherian, there is a finite subset J of I such that

$$K = \sum_{i\in J} K^i, \qquad \text{equivalently} \qquad T = \bigcap_{i\in J} T^i.$$

[17] Editor's note: the original has been expanded and corrected to show that the projective limit of the rings $\mathscr{O}(T_n^i)$ defines a closed formal subscheme of $\mathrm{Spf}(\widehat{A})$.

For every n, therefore,

$$K_n = \sum_{i\in J} K_n^i,$$

where K_n is the image of K in A_n.

Recall that f_i is the image of $f \in B = \mathscr{O}(X)$ in $\mathscr{O}(Z_i)$, and that f_i' is its image in

$$\mathscr{O}(Z_i') = \mathscr{O}(Z_i) \otimes_A A'.$$

The relation $Y_i' = Z_i'$ is equivalent to $f_i' = 0$. Now $\mathscr{O}(Z_i) \otimes_A A'$ is the direct limit of the subalgebras $\mathscr{O}(Z_i) \otimes_A A''$, for A'' as above. Consequently, there is an A'' for which $Y_i'' = Z_i''$. A priori A'' depends on i, but, because J is finite, one A'' can be chosen that works for every $i \in J$. Put

$$S'' = \mathrm{Spec}(A''), \qquad S''_{(n)} = S'' \times_S S_n.$$

[18]

For every $i \in J$, the equality

$$Y_i'' \times_{S''} S''_{(n)} = (Y_n^i)_{S''_{(n)}} = Z_i'' \times_{S''} S''_{(n)} = (Z_n^i)_{S''_{(n)}}$$

shows, by the definition of T_n^i, that $S''_{(n)} \to S_n$ factors through T_n^i. It therefore factors through

$$T_n = \bigcap_{i\in J} T_n^i,$$

and hence through T_n^i for every $i \in I$. Let f'' be the image of f in $B'' = B \otimes_A A''$, and let $f''_{(n)}$ be its image in

$$B''_{(n)} = B''/\mathfrak{m}^{n+1}B''.$$

The preceding factorization means that, for every n, the section $f''_{(n)}$ vanishes on all the $Z_i \times_S S''_{(n)}$.

Because $A \to A'' \to A'$ are local homomorphisms, the image s'' of s' in S'' is the closed point of S'', corresponding to the maximal ideal $\mathfrak{m}''$ of A'', and its image in S is s. Fix n, and now let S_n'' denote the n-th infinitesimal neighborhood of s'' in S'':

$$S_n'' = \mathrm{Spec}(A''/\mathfrak{m}''^{n+1}).$$

Put

$$B_n'' = B''/\mathfrak{m}''^{n+1}B'', \qquad Z_n^{i''} = Z_i \times_S S_n''.$$

Then the image f_n'' of $f''_{(n)}$ in B_n'' vanishes on every $Z_n^{i''}$. By Lemma 4.5 below, the family of fibers

$$Z_0^{i'''} = Z_n^{i''} \times_{S_n''} \kappa(s'') = Z_{is} \times_{\kappa(s)} \kappa(s'')$$

is schematically dense in

$$X_0''' = X_n'' \times_{S_n''} \kappa(s'') = X_s \times_{\kappa(s)} \kappa(s'').$$

Thus $S_n'', X_n'', (Z_n^{i''})_{i\in I}$, and

$$S_0''' = \mathrm{Spec}\,\kappa(s'')$$

[18] Editor's note: the original has been expanded and corrected, distinguishing $S''_{(n)}$ from the subscheme $S_n'' = S_n''(s'')$ introduced below.

satisfy the hypotheses of Lemma 4.2. It follows that $f''_n = 0$, that is,

$$f'' \in \mathfrak{m}''^{n+1} B'' \qquad \text{for every } n.$$

Since B is Noetherian and $B'' = B \otimes_A A''$ is a localization of a finite-type B-algebra, B'' is Noetherian. Its local rings are therefore separated for their usual topology. Hence f'' is zero at every point x'' of X'' for which $\mathfrak{m}''B''$ is contained in the maximal ideal of $\mathscr{O}_{X'',x''}$, that is, at every point of X'' above s''. A fortiori, f' is zero at every point of X' above s'. □

Lemma 4.5.0.[19] *Let X be an S-prescheme, let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, and let $S' \to S$ be faithfully flat. If $(Z'_i)_{i\in I}$ is schematically dense in X', then $(Z_i)_{i\in I}$ is schematically dense in X.*

This follows from EGA IV$_3$, 11.10.5(i) and 11.9.10(i). It remains to give the proof of the following lemma.

Lemma 4.5. *Let X be a locally Noetherian prescheme*[20] *over a field k, let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, let k'/k be a field extension, and let X', Z'_i be obtained from X, Z_i by base change. Then $(Z_i)_{i\in I}$ is schematically dense in X if and only if $(Z'_i)_{i\in I}$ is schematically dense in X'.*

Proof. The "if" direction follows from Lemma 4.5.0; let us prove the converse.[21] First, we may suppose $X = \mathrm{Spec}(B)$ affine. For every i, let $(Z_{ij})_{j\in J_i}$ be an affine open covering of Z_i. Replacing $(Z_i)_{i\in I}$ by $(Z_{ij})_{(i,j)\in J}$, where

$$J = \coprod_{i\in I} J_i,$$

we may also suppose that every Z_i is affine.

Let

$$g \in B' = B \otimes_k k', \qquad t' \in B'_g$$

be a section of $\mathscr{O}_{X'}$ on the affine open $U' = X'_g$ whose restriction to each $Z'_i \cap U'$ is zero. Thus its image in every

$$\left(\mathscr{O}(Z_i) \otimes_k k'\right)_g$$

is zero. There exists a finite-type k-subalgebra A of k' such that

$$g \in B_A = B \otimes_k A, \qquad t' \in (B_A)_g.$$

The map

$$\mathscr{O}(Z_i) \otimes_k A \longrightarrow \mathscr{O}(Z_i) \otimes_k k'$$

is injective, and therefore so is

$$\left(\mathscr{O}(Z_i) \otimes_k A\right)_g \longrightarrow \left(\mathscr{O}(Z_i) \otimes_k k'\right)_g.$$

The hypothesis consequently implies that the image of t' in every $\left(\mathscr{O}(Z_i) \otimes_k A\right)_g$ is zero. It remains to show that $(Z_{iA})_{i\in I}$ is schematically dense in X_A, which follows from EGA IV$_3$, 11.9.10(b).[22] □

[19] Editor's note: this lemma has been inserted here because it is used in the proofs of 4.5 and 4.7.

[20] Editor's note: this hypothesis is in fact superfluous; cf. EGA IV$_3$, 11.10.6 and 11.9.13.

[21] Editor's note: the original has been expanded in what follows.

[22] Editor's note: the original continued: "Using 4.3 for the $X_{A'}$, where A' is a quotient of A by a power of a maximal ideal, and Hilbert's Nullstellensatz, one is reduced to the case where k' is a finite extension of k." This reduction could be expanded and that argument pursued, since the case of a finite extension k'/k is somewhat simpler than the more general case treated in EGA IV$_3$, 11.9.10(b).

Let us record in passing the following result, which will be used in a later exposé.[23]

Corollary 4.6. *Let X be a flat S-prescheme and U an open subset of X. Suppose that U_s is schematically dense in X_s for every $s \in S$, and that either X is locally Noetherian or X is locally of finite presentation over S. Then U is schematically dense in X.*

Proof. For simplicity, suppose U is retrocompact in X (cf. EGA IV_3, 11.10.10 and 11.9.17 for the general case). The locally Noetherian case is included only for reference; it is contained in 4.3.

In the second case, we may plainly suppose S and X affine. Then X and U are of finite presentation over $S = \mathrm{Spec}(A)$. The ring A is the direct limit of its finite-type $\mathbb{Z}$-subalgebras A_i. The "patented procedure" already used (cf. EGA IV_3, 8.8.2 and 8.10.5(iii)) shows that there are an index i, an affine scheme X_i over

$$S_i = \mathrm{Spec}(A_i),$$

and an open subset U_i of X_i, from which X, U are obtained by the base change $S \to S_i$. Let E_i be the subset of S_i consisting of those $s \in S_i$ for which $(U_i)_s$ is schematically dense in $(X_i)_s$.

By EGA IV_2, 5.9.9 and 5.10.2, E_i is the set of points $s \in S_i$ for which $(U_i)_s$ contains the set

$$\mathrm{Ass}\, \mathscr{O}_{(X_i)_s}$$

of points associated with the structure sheaf of $(X_i)_s$. By EGA IV_3, 11.9.17.1, this is a constructible condition; hence E_i is constructible.[24]

By 4.5, the inverse image of E_i under $S_j \to S_i$ is E_j, and its inverse image under $S \to S_i$ is the set E of those $s \in S$ for which U_s is schematically dense in X_s.[24] By hypothesis, $E = S$, which is also the inverse image of S_i under every $S \to S_i$. EGA IV_3, 8.3.11 therefore implies that there is a $j \geq i$ such that $E_j = S_j$; that is, $(U_j)_s$ is schematically dense in $(X_j)_s$ for every $s \in S_j$.

Apply 4.4 to (S_j, X_j, U_j) and the base change $S \to S_j$. It follows that U is schematically dense in X. Notice that here 4.4 is used only for a family with a finite index set (in fact, a singleton); in this case its proof simplifies considerably. □

Theorem 4.7. *Let S be a prescheme, and let G be a group of multiplicative type and of finite type over S. For every integer $n > 0$, let ${}_nG$ be the kernel of $n \cdot \mathrm{id}_G$. Then the family of subpreschemes $({}_nG)_{n>0}$ is schematically dense in G in the sense of Definition 4.1.*

Proof. We distinguish two cases.

(a) S locally Noetherian. By 4.3, we are reduced to the case in which S is the spectrum of a field k. By 4.5.0, we may suppose that k is algebraically closed and that G is diagonalizable, say

$$G = D_k(M),$$

where M is a commutative group of finite type. Then

$$M \simeq \Gamma \times \mathbb{Z}^r$$

with Γ finite, and consequently

$$G \simeq G_1 \times G_2, \qquad G_1 = D(\Gamma), \qquad G_2 = \mathbf{G}_m^r.$$

[23] Editor's note: this reference remains to be specified.

[24] Editor's note: the original has been expanded in this paragraph and the next.

For n sufficiently divisible—namely, for n a multiple of the order of Γ—one has

$$ {}_nG = G_1 \times {}_nG_2, \qquad \text{since} \qquad {}_nG_1 = G_1. $$

Applying 4.3 again to the projection $G \to G_1$, we are reduced to G_2, hence to the case $G = \mathbf{G}_m^r$. Since G is then reduced, schematic density of $({}_nG)_{n>0}$ is equivalent to density of $\bigcup_n {}_nG$ in the ordinary topology. As

$$ {}_nG = ({}_n\mathbf{G}_m)^r, $$

we are reduced to $G = \mathbf{G}_m$, an irreducible group of dimension one. The assertion now follows because $\bigcup_n {}_nG$, the set of roots of unity in k, is infinite.

(b) The general case. For every $s \in S$, there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that $G' = G_{S'}$ is diagonalizable, say $G' = D_{S'}(M)$. By 4.5.0 we may reduce to G', and therefore suppose that G itself is diagonalizable. It is obtained by base change

$$ S \longrightarrow \mathrm{Spec}(\mathbb{Z}) $$

from the absolute group $H = D_{\mathbb{Z}}(M)$. Part (a) shows that, for every $s \in \mathrm{Spec}(\mathbb{Z})$, the family $({}_nH_s)_{n>0}$ is schematically dense in H_s. It remains only to apply 4.4.[25] □

Corollary 4.8.

(a) *Let $u, v : H \to G$ be two homomorphisms of S-preschemes in groups, with H of multiplicative type*[26] *and of finite type. Suppose that, for every integer $n > 0$, the restrictions of u and v to ${}_nH$ are equal. Then $u = v$.*

(b) *Let H_1, H_2 be two subgroups of multiplicative type and of finite type of a prescheme in groups G. If ${}_nH_1 = {}_nH_2$ for every integer $n > 0$, then $H_1 = H_2$.*

Proof. Part (a) follows from part (b) by considering the subgroups H_1, H_2 of $H \times G$ that are the graphs of u, v. For part (b), put

$$ H = H_1 \cap H_2. $$

This is a subgroup prescheme of H_i, for $i = 1, 2$, and we must show that it equals H_i. The hypothesis says precisely that H contains all the ${}_nH_i$. We are therefore reduced to the next corollary. □

Corollary 4.9. *Let G be an S-group of multiplicative type and of finite type, and let H be a subgroup prescheme containing ${}_nG$ for every $n > 0$. Then $H = G$.*

Proof. By 4.7, it is enough to prove that H is closed, or equivalently that it equals G set-theoretically. This reduces us to the case in which S is the spectrum of a field. Every subgroup prescheme of G is then closed, by VI_B, 1.4.2. □

Remark 4.10. Under the hypotheses of 4.7, let $m > 0$ be an integer with the following properties: for every $s \in S$, m is not a power of the characteristic of $\kappa(s)$; and if G_s has type M, every prime divisor of the torsion subgroup of M divides m. The second condition is automatically satisfied if G is a torus. The proof shows that in Theorem 4.7 and Corollary 4.8 and Corollary 4.9 one may restrict to the subgroups

$$ {}_{m^r}G, \qquad r > 0. $$

[25] Editor's note: this result will be needed in X, 4.3 for a non-locally-Noetherian base, namely $S = \mathrm{Spec}(\widehat{A} \otimes_A \widehat{A})$, where $\widehat{A}$ is the completion of the Noetherian local ring A.

[26] Editor's note: "diagonalizable" has been replaced by "of multiplicative type."

Moreover, if G is smooth over S, then for every $s \in S$ there are an open neighborhood U of s and an integer $m > 0$, prime to every residue characteristic on U, that satisfy the preceding conditions for G_U. For instance, take m to be the order of the torsion subgroup of the type of G at s (cf. 1.4.1(a) and 2.1(e)). After restriction to U, the groups ${}_{m^r}G$ are then finite and étale over S, and their family is schematically dense. In certain applications—in particular those involving Theorem 3.2 and Theorem 3.2 bis (cf. XI, 6), though not those involving Theorem 3.6 and Theorem 3.6 bis—this observation allows one to avoid Theorem 3.1 and its use of Hochschild cohomology.

5. Central Homomorphisms of Groups of Multiplicative Type

Lemma 5.0.[27] *Let $(A, \mathfrak{m})$ be a Noetherian local ring, put $S = \mathrm{Spec}(A)$, and let H be a finite scheme over S, say $H = \mathrm{Spec}(B)$, where B is finite over A. Let K be a subscheme of H such that reduction modulo $\mathfrak{m}^{n+1}$ gives $K_n = H_n$ for every n. Then $K = H$.*

Proof. Let s be the closed point of S. First, K is a closed subscheme of H. Indeed, a priori it is a closed subscheme of an open subset U of H. But K, and hence U, contains the fiber

$$K_s = H_s.$$

Since $H \to S$ is finite, and therefore closed, the complement of U is empty; thus $U = H$. Hence K is defined by an ideal I of B. The hypothesis implies

$$I \subset \mathfrak{m}^n B \qquad \text{for every } n.$$

Since B is a finite A-module, it is separated for the $\mathfrak{m}$-adic topology. Therefore $I = 0$. □

Theorem 5.1. *Let $u, v : H \to G$ be two homomorphisms of S-preschemes in groups, where H is of multiplicative type and of finite type. Suppose either that S is locally Noetherian or that G is of finite presentation over S. Let $s \in S$ satisfy $u_s = v_s$, and suppose that u_s is central. Then there is an open neighborhood U of s such that $u_U = v_U$.*

Proof. We distinguish the two cases.

(a) S locally Noetherian.[28] Let

$$K = \mathrm{Ker}(u, v)$$

be the inverse image of the diagonal of $G \times_S G$ under $(u, v) : H \to G \times_S G$. It is a subgroup prescheme of H. We seek an open U for which $K_U = H_U$. Because S is locally Noetherian and H is of finite type over S, the prescheme H is locally Noetherian. Hence the immersion $K \hookrightarrow H$ is of finite type (cf. EGA I, 6.3.5), so K is of finite type, and therefore of finite presentation, over S. By EGA IV_3, 8.8.2.4, it is enough to prove that

$$K_{S_0} = H_{S_0}, \qquad S_0 = \mathrm{Spec}(\mathscr{O}_{S,s}).$$

We may thus suppose that S is local with closed point s. Replacing the Noetherian local ring A by its completion $\widehat{A}$, if desired, gives a faithfully flat quasi-compact base change $\widehat{S} \to S$, and allows us to suppose that A is complete.[29]

[27] Editor's note: this lemma has been made explicit because it is used several times below; it was implicit in the original.

[28] Editor's note: the original has been expanded in what follows.

[29] Editor's note: this is because $K = H$ if $K_{\widehat{S}} = H_{\widehat{S}}$; cf. SGA 1, VIII, 5.4. The argument below does not use completeness of A.

By 3.4, one has $u_n = v_n$ for every n, where the subscript denotes reduction modulo $\mathfrak{m}^{n+1}$, with $\mathfrak{m}$ the maximal ideal of A. For every $m > 0$, let

$$ {}_mu, {}_mv : {}_mH \longrightarrow G$$

be the homomorphisms induced by u, v. Then

$$ ({}_mu)_n = ({}_mv)_n \qquad \text{for every } n.$$

The group ${}_mH$ is finite over S, by 2.2, so 5.0 gives ${}_mu = {}_mv$. This holds for every m, and therefore 4.7 gives $u = v$.

(b) G of finite presentation.[30] The group H is also of finite presentation over S. By EGA IV_3, 8.8.2, we may again suppose that S is local with closed point s, and it suffices to prove $u = v$. Let $f : S' \to S$ be a faithfully flat quasi-compact morphism. Write

$$ f_H : H' \to H, \qquad f_G : G' \to G, \qquad u', v' : H' \to G'$$

for the resulting morphisms. If $u' = v'$, then

$$ u \circ f_H = f_G \circ u' = f_G \circ v' = v \circ f_H,$$

and $u = v$ because f_H is an epimorphism. After a faithfully flat quasi-compact extension of the base, we may therefore suppose that H is diagonalizable:

$$ H = D_S(M)$$

for a commutative group M of finite type.

As in the proof of 4.6, introduce the increasing filtered family of $\mathbb{Z}$-subalgebras A_i of $A = \mathscr{O}_{S,s}$ that are of finite type, and put

$$ S_i = \mathrm{Spec}(A_i).$$

[30] For every i, the group $H = D_S(M)$ comes from

$$ H_i = D_{S_i}(M).$$

Since G is of finite presentation over S, EGA IV_3, 8.8.2 (see also VI_B, 10.2 and 10.3) gives an index i, an S_i-group prescheme G_i of finite presentation, and homomorphisms

$$ u_i, v_i : H_i \longrightarrow G_i$$

whose base changes are u, v. Let s_i be the image of s in S_i, and define

$$ \rho_i : H_i \times_{S_i} G_i \longrightarrow G_i, \qquad \rho_i(h, g) = u_i(h) g\, u_i(h)^{-1}.$$

Because u_s is central,

$$ \rho_s = \rho_i \times_{\kappa(s_i)} \kappa(s)$$

is the second projection. The extension $\kappa(s_i) \to \kappa(s)$ is faithfully flat, so ρ_i is also the second projection over s_i; hence $(u_i)_{s_i}$ is central. Likewise $u_s = v_s$ implies

$$ (u_i)_{s_i} = (v_i)_{s_i}.$$

[30] Editor's note: the original has been expanded in what follows.

Part (a), applied over S_i, now gives the conclusion. □

Corollary 5.2. *Let $u : H \to G$ be a homomorphism of S-preschemes in groups, where H is of multiplicative type and of finite type, and suppose either that S is locally Noetherian or that G is of finite presentation over S. If $s \in S$ and u_s is the unit homomorphism, then there is an open neighborhood U of s on which u_U is the unit homomorphism.*

Corollary 5.3. *Let u, H, G be as in 5.2, and suppose in addition that G is separated over S. Let U be the set of points $s \in S$ for which u_s is the unit homomorphism. Then U is open and closed in S, and $u_U : H_U \to G_U$ is the unit homomorphism.*

Proof. Only closedness remains to be proved. Put $K = \mathrm{Ker}(u)$. Since G is separated over S, K is a closed subprescheme of H, and U is the set of points $s \in S$ for which $K_s = H_s$. It follows readily that U is closed, for example by VIII, 6.4, since H is essentially free over S by VIII, 6.3. Alternatively, we may suppose that S is reduced and that U is dense in S, hence schematically dense. Then H_U is schematically dense in H because H is flat over S,[31] and u and the unit homomorphism, being equal on H_U, are equal on H. □

Corollary 5.4. *Let S be a prescheme; let H, G be S-groups, with H of multiplicative type and of finite type and G separated and of finite presentation over S; and let $\pi : S' \to S$ be a*[32] *universal effective epimorphism—for example, a faithfully flat quasi-compact morphism or a morphism admitting a section (cf. IV, 1.12)—whose fibers are geometrically connected.*[33]

Let $u' : H' \to G'$ be a central homomorphism of the S'-groups obtained from H, G by base change. Then there is a unique homomorphism $u : H \to G$ of S-groups such that $u \times_S S' = u'$. If S'/S admits a section g, then u is the morphism obtained from u' by the base change $g : S \to S'$.

Proof. Because π is a universal effective epimorphism, so is $H' \to H$; uniqueness follows as at the beginning of the proof of 5.1(b). If π has a section g, then

$$u' = \pi^* u \quad \Longrightarrow \quad u = g^* \pi^* u = g^* u'.$$

For existence, IV, 2.3 reduces us to showing that the two homomorphisms

$$u_1'', u_2'' : H'' \longrightarrow G''$$

of S''-groups obtained from u' by the two base changes

$$\mathrm{pr}_1, \mathrm{pr}_2 : S'' = S' \times_S S' \longrightarrow S'$$

are equal. Their pullbacks along the diagonal $S' \to S''$ are equal, since both are u'.[34] Because u_1'', u_2'' are central, we may apply 5.3 to

$$u_1''(u_2'')^{-1}.$$

There is therefore an open-and-closed subset U of S'', containing the diagonal, over which u_1'' and u_2'' coincide. The fibers of S'/S are geometrically connected, hence so are those

[31] Editor's note: and of finite presentation over S; cf. EGA IV$_3$, 11.10.5(ii) and 11.9.10(ii).

[32] Editor's note: "covering morphism for the faithfully flat quasi-compact topology" has been replaced by "universal effective epimorphism," so that the result applies to the morphism $K \to S$ in 5.5; the beginning of the proof has been modified accordingly.

[33] Editor's note: this holds, for example, when $S = \mathrm{Spec}(k)$ and $S' = \mathrm{Spec}(k')$, where k is a field and k'/k is radicial; cf. X, Proposition 1.4.

[34] Editor's note: the preceding sentence has been added.

of S''/S, and in particular they are connected. Since U contains the diagonals of these fibers, it contains every fiber and therefore equals S''. This proves existence. □

Corollary 5.5. *Let S be a prescheme, let K be an S-group prescheme locally of finite type*[35] *and with connected fibers, and let H be an invariant subgroup of K, of multiplicative type and of finite type. Then H is a central subgroup of K.*

Proof.[36] First, $\pi : K \to S$ has geometrically connected fibers, because for a group scheme locally of finite type over a field, connectedness implies geometric connectedness (cf. VI$_A$, 2.4). Apply 5.4 with $G = H$ and $S' = K$ to the homomorphism of K-groups

$$u' : H_K \longrightarrow H_K, \qquad (h, k) \longmapsto (khk^{-1}, k).$$

It is central because H is commutative. Its pullback along the unit section $\epsilon : S \to K$ is the identity homomorphism of H. Corollary 5.4 therefore implies that u' itself is the identity on H_K. Thus H is central in K. □

Let us state the variants of the preceding results for central subgroups of multiplicative type. Arguing as above, and using 3.2 bis, gives:

Theorem 5.1 bis. *Let G be an S-group prescheme, and let H_1, H_2 be two subgroup preschemes of multiplicative type and of finite type, with $(H_1)_s$ central. Suppose either that S is locally Noetherian or that G is of finite presentation over S. For every $s \in S$ such that $(H_1)_s = (H_2)_s$, there is an open neighborhood U of s for which $H_1|_U = H_2|_U$.*

Corollary 5.3 bis. *Under the preceding hypotheses, suppose moreover that G is separated over S. Let U be the set of points $s \in S$ for which $(H_1)_s = (H_2)_s$. Then U is open and closed in S, and $H_1|_U = H_2|_U$.*

Corollary 5.4 bis. *Let S be a prescheme, let G be a group prescheme separated and of finite presentation over S, let $S' \to S$ be a covering morphism for the faithfully flat quasi-compact topology with geometrically connected fibers, and let H' be a subgroup of multiplicative type and of finite type of $G' = G \times_S S'$. Then there is a unique subgroup H of G such that $H' = H \times_S S'$. The subgroup H is of multiplicative type and of finite type over S, by 1.1 and 2.1(b). If $S' \to S$ admits a section g, then $H = g^*H'$.*

Theorem 5.6. *Let S be a prescheme and let $u : H \to G$ be a homomorphism of S-preschemes in groups, where H is of multiplicative type and of finite type and G is of finite presentation over S.*

(a) *Suppose that G has connected fibers. If $s \in S$ and $u_s : H_s \to G_s$ is central, then there is an open neighborhood U of s such that the homomorphism $H|_U \to G|_U$ induced by u is central.*

(b) *If $u_s : H_s \to G_s$ is central for every $s \in S$, then u is central.*

Proof. We first prove (a).[37] Arguing exactly as in the proof of 5.1(b), we may suppose first that S is local with closed point s, then that $H = D_S(M)$ is diagonalizable, and finally that there exist a finite-type $\mathbb{Z}$-subalgebra A_i of $A = \mathscr{O}_{S,s}$ and a homomorphism

$$u_i : D_{S_i}(M) \longrightarrow G_i$$

[35] Editor's note: the hypothesis "locally of finite type" has been added to agree with VI$_A$, 2.4. In fact, over a field every connected group scheme is geometrically connected.

[36] Editor's note: the original has been expanded in what follows.

[37] Editor's note: the original has been expanded in the proof of (a).

whose base change is u and for which $(u_i)_{s_i}$ is central, where s_i is the image of s in S_i. We are thereby reduced to the case in which S is Noetherian local with closed point s, and we must prove that u is central.

Let $K = \mathrm{Ker}(v, w)$, where

$$v, w : H \times_S G \rightrightarrows G$$

are defined by

$$v(h, g) = g, \qquad w(h, g) = \mathrm{int}(u(h)) \cdot g = u(h) g\, u(h)^{-1}.$$

Then K is a subgroup over G of

$$H_G = H \times_S G,$$

and we must prove that $K = H_G$. By 4.9, it is enough to show that K contains

$$_m(H_G) = (_m H) \times_S G$$

for every $m > 0$. We are thus reduced to the case $H = {}_m H$, so that H is finite over S.

Let e be the unit element of the fiber G_s, and put

$$S^0 = \mathrm{Spec}(\mathscr{O}_{G,e}), \qquad S^0_n = S^0 \times_S S_n, \qquad S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1}).$$

The scheme S^0 is Noetherian local because G is of finite presentation over the Noetherian scheme S. The base change

$$K_{S^0} = K \times_G S^0$$

is a subscheme of

$$H_{S^0} = H \times_S S^0,$$

and 3.9 gives

$$K_{S^0_n} = H_{S^0_n} \qquad \text{for every } n.$$

[38] Since H_{S^0} is finite over S^0, 5.0, applied to the Noetherian local ring $\mathscr{O}_{G,e}$, gives

$$K_{S^0} = H_{S^0}.$$

On the other hand, $H \times_S G$ is Noetherian, being of finite presentation over the Noetherian scheme S. Thus the immersion

$$K \hookrightarrow H \times_S G$$

is of finite type (cf. EGA I, 6.3.5), and K is of finite presentation over G. The equality

$$K_{S^0} = H_G \times_G S^0$$

therefore implies, by EGA IV_3, 8.8.2.4, that there is an open neighborhood W of e in G such that

$$K \times_G W = H_G \times_G W = H \times_S W.$$

Hence K contains the open neighborhood

$$V = H \times_S W$$

of the unit section of $G_H = G \times_S H$ over H.

[38] Editor's note: the original has been expanded and corrected, replacing "the unit section of $(G_H)_n$ over H_n" by $H_{S^0_n}$.

For every $t \in H$, the fiber G_t is an algebraic group over $\kappa(t)$, hence Cohen–Macaulay by VI_A, 1.1.1 and therefore has no embedded components. It is also connected, hence irreducible by VI_A, 2.4, so its generic point is its unique associated point. EGA IV_2, 3.1.8 therefore shows that V_t is schematically dense in G_t. By 4.3, V is schematically dense in G_H.

Moreover, since K is an H-subgroup of G_H, it induces on each fiber G_h a subgroup K_h. This subgroup contains an open neighborhood of the unit element; because G_h is connected, it follows that $K_h = G_h$. Thus $K = G_H$ set-theoretically. Finally, K is a closed subprescheme of G_H containing the schematically dense subprescheme V, and hence $K = G_H$. This proves (a).

Part (b) follows directly from 3.9: to verify that a subprescheme K of $P = G \times_S H$ equals P, it is enough to verify, after every base change $S' \to S$ in which S' is the spectrum of an Artinian quotient of a local ring of S, that $K' = P'$. □

6. Monomorphisms of Groups of Multiplicative Type, and Canonical Factorization of a Homomorphism from Such a Group

Lemma 6.1. *Let S be a quasi-compact prescheme, and let G be a commutative S-group prescheme, of finite presentation and quasi-finite over S. Then there is an integer $n > 0$ such that*

$$n \cdot \mathrm{id}_G = 0, \qquad \textit{that is,} \qquad G = {}_nG.$$

Proof. If S is the spectrum of a field k, then G is finite over k, and by VII_A 8.5 it is enough to take

$$n = \deg(G/k).$$

Suppose now that $S = \mathrm{Spec}(A)$, where A is local Artinian. Let k be the residue field of A, put

$$G_0 = G \otimes_A k, \qquad n_0 = \deg(G_0/k),$$

and distinguish two cases.

(a) k has characteristic zero. Then G_0 is separable over k, and hence G is unramified over S. The unit section of G is therefore an open immersion, so

$${}_nG = \mathrm{Ker}(n \cdot \mathrm{id}_G)$$

is an open subscheme of G. For it to equal G, it is consequently enough that it do so set-theoretically; thus one may take $n = n_0$.

(b) k has characteristic $p > 0$. Let $\mathfrak{m}$ be the maximal ideal of A, and put

$$S_r = \mathrm{Spec}(A/\mathfrak{m}^{r+1}) \qquad (r \geq 0).$$

Choose an integer m such that $\mathfrak{m}^{m+1} = 0$. We claim that one may take $n = p^m n_0$. Indeed, arguing by induction on m and putting $n' = p^{m-1} n_0$, we may suppose that $n' \cdot \mathrm{id}_G$ induces the zero endomorphism on

$$G_{m-1} = G \times_S S_{m-1}.$$

[39] Let E be the set of endomorphisms u of G having this property, and let K be the kernel of the morphism of group functors

$$G \longrightarrow \prod_{S_{m-1}/S} G$$

(cf. III 0.1.2). Then

$$E = \mathrm{Hom}_{S\text{-gr}}(G, K);$$

in particular, the abelian group law on E is induced by that on K. By III 0.9, K is the S-group functor which, to every $g : T \to S$, assigns the k-vector space

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}(g_0^*(\Omega^1_{G_0/S_0}), \mathfrak{m}^m \mathscr{O}_T).$$

Thus $pu = 0$ for every $u \in E$. It follows that

$$pn' \cdot \mathrm{id}_G = 0, \qquad \text{that is,} \qquad p^m n_0 \cdot \mathrm{id}_G = 0.$$

Suppose now that S is Noetherian; this is the case to which the general assertion reduces by the usual reduction to the Noetherian case.[40] It is enough to combine the foregoing with the following lemma.

Lemma 6.2. *Let X be a quasi-finite prescheme over a Noetherian prescheme S, and let $(X_i)_{i\in I}$ be an increasing filtered family of subpreschemes having the following property:*

$$\begin{cases} \text{for every } s \in S \text{ and every } n \geq 0, \text{ putting} \\ S_{s,n} = \mathrm{Spec}(\mathscr{O}_{S,s}/\mathfrak{m}^{n+1}), \text{ there is an } i \in I \text{ such that} \\ X_i \times_S S_{s,n} = X \times_S S_{s,n}. \end{cases}$$

Then there is an $i \in I$ such that $X_i = X$.

Proof. Since S is Noetherian, there is a maximal open subset U on which

$$X|_U = X_i|_U$$

for all sufficiently large i. We shall prove that $U = S$. In other words, if $U \neq S$, we shall find an open subset U' strictly larger than U, and an i, such that

$$X|_{U'} = X_i|_{U'}.$$

Localizing at a maximal point s of $S - U$, we reduce to the case in which S is local with closed point s, and

$$U = S - \{s\}.$$

Indeed, put $S' = \mathrm{Spec}(\mathscr{O}_{S,s})$. If

$$X \times_S S' = X_i \times_S S'$$

for some i,[41] there is an open neighborhood V of s such that

$$X|_V = X_i|_V.$$

[39] Editor's note: the original has been expanded in what follows, with the results of Exposé III that are used made explicit.

[40] Editor's note: since S is quasi-compact, it is covered by finitely many affine open subsets, so one reduces to $S = \mathrm{Spec}(A)$. Since G is of finite presentation over S, EGA IV$_3$, 8.8.2 may then be applied.

[41] Editor's note: $X - \{s\}$ has been corrected above to $S - \{s\}$. Recall also that a quasi-finite morphism is assumed to be of finite type; cf. EGA II, 6.2.3, and EGA III$_2$, Err$_{\mathrm{III}}$ 20 for the definition of "locally quasi-finite." Thus here, since S is Noetherian, X is of finite presentation over S; every immersion $X_i \hookrightarrow X$ is of finite type by EGA I, 6.3.5, so X_i is of finite presentation over S, and EGA IV$_3$, 8.8.2 applies.

Taking i large enough that $X_i|_U = X|_U$, we then have

$$X|_W = X_i|_W, \qquad W = U \cup V.$$

For all sufficiently large i, the equality $X|_U = X_i|_U$ shows that X_i is a closed subprescheme of X, defined by an ideal $\mathscr{I}^{(i)}$ whose support is contained in

$$X_s = X_0 = X \times_S S_0,$$

where

$$S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1}), \qquad S = \mathrm{Spec}(A).$$

Since X_0 is quasi-finite over S_0, it is a finite closed subset of the Noetherian prescheme X. Hence $\mathscr{I}^{(i)}$ is a module of finite length. There is therefore an integer $n \geq 0$ such that

$$\mathscr{I}^{(i)} \cap \mathfrak{m}^{n+1}\mathscr{O}_X = 0.$$

On the other hand, by the hypothesis of the lemma, after increasing i if necessary we may suppose that the image of $\mathscr{I}^{(i)}$ in

$$\mathscr{O}_{X_n} = \mathscr{O}_X/\mathfrak{m}^{n+1}\mathscr{O}_X$$

is zero. It follows that $\mathscr{I}^{(i)} = 0$, and therefore $X_i = X$. □

Lemma 6.3. *Let S be a prescheme, let K be a group prescheme over S, locally of finite presentation, and let $s \in S$. Suppose that K_s is quasi-finite (respectively, unramified) over $\kappa(s)$ at the unit element. Then there is an open neighborhood U of s such that $K|_U$ is locally quasi-finite (respectively, unramified) over U.**

Proof. Let V be the set of points $x \in K$ having the following property: if t is the image of x in S, then K_t is quasi-finite (respectively, unramified) over $\kappa(t)$ at x; that is, x is isolated in K_t (respectively, it is isolated and its local ring in K_t is a separable extension of $\kappa(t)$). The set V is open because K is locally of finite presentation over S.[42] If ϵ denotes the unit section of K, then $\epsilon^{-1}(V)$ is open. By hypothesis it contains s, and is therefore an open neighborhood U of s. This U has the required property. Indeed, if $t \in U$, then K_t, being a group locally of finite type over $\kappa(t)$, is locally quasi-finite (respectively, unramified) over $\kappa(t)$ because it has that property at $\epsilon(t)$; cf. $\mathrm{VI_B}$ 1.3. □

Combining 6.1 and 6.3, one obtains:

Theorem 6.4. *Let S be a prescheme, let H be an S-group of multiplicative type and of finite type, and let K be a closed group subprescheme of H, of finite presentation over S. Let $s \in S$ be such that K_s is finite. Then there are an open neighborhood U of s and an integer $n > 0$ such that*

$$K|_U \subset {}_nH|_U.$$

In particular, since ${}_nH$ is finite over S, the group $K|_U$ is finite over U.

Nakayama's lemma now gives:

Corollary 6.5. *With the preceding notation, suppose that K_s is the unit group. Then there is an open neighborhood U of s such that $K|_U$ is the unit group.*

Corollary 6.6. *Let $u : H \to G$ be a homomorphism of S-group preschemes, where H is of multiplicative type and of finite type and G is separated over S. Suppose in addition*

*Cf. $\mathrm{VI_B}$ 2.5(ii) for a more general statement.

[42]Editor's note: cf. EGA $\mathrm{IV_3}$, 13.1.4 and EGA $\mathrm{IV_4}$, 17.4.1. In addition, t, rather than s, has been used for the image of x.

that either S is locally Noetherian or G is locally of finite presentation over S.[43] *If $s \in S$ and*

$$u_s : H_s \longrightarrow G_s$$

is a monomorphism, then there is an open neighborhood U of s such that u induces a monomorphism

$$H|_U \longrightarrow G|_U.$$

Proof. Put $K = \mathrm{Ker}(u)$. The hypothesis on u_s says that K_s is the unit group, and the desired conclusion says that $K|_U$ is the unit group for a suitable U. Since G is separated over S, the group K is a closed subgroup of H. If S is not assumed locally Noetherian, but G is assumed locally of finite presentation over S, then K is locally of finite presentation over S.[43] It is therefore of finite presentation over S, since it is separated over S (as H is) and quasi-compact over S (being closed in H, which is quasi-compact over S). Corollary 6.5 now applies. □

Remark 6.6.1.[44] *Under the hypotheses of 6.6, note that if, in addition, G is affine over S (respectively, of finite presentation over S), then*

$$H|_U \longrightarrow G|_U$$

is even a closed immersion, as was observed in 2.5 (respectively, 2.6).

Corollary 6.7. *Let $u : H \to G$ be a homomorphism of S-group preschemes, where H is of multiplicative type and of finite type. Suppose that either S is locally Noetherian or G is locally of finite presentation over S. If every homomorphism on fibers*

$$u_s : H_s \longrightarrow G_s$$

is a monomorphism, then u is a monomorphism.

Proof. The argument is the same as in 6.6. Here the hypothesis that G be separated over S is unnecessary in order to ensure that $K = \mathrm{Ker}(u)$ is closed in H, since the assumption that all the u_s are monomorphisms implies that K is set-theoretically reduced to the unit section. □

Theorem 6.8. *Let $u : H \to G$ be a homomorphism of S-group preschemes, where H is of multiplicative type and of finite type and G is separated over S. Suppose in addition that either S is locally Noetherian or G is of finite presentation over S. Then*

$$K = \mathrm{Ker}(u)$$

is a subgroup of H of multiplicative type and of finite type, and u factors as

$$H \xrightarrow{u'} H/K \xrightarrow{u''} G,$$

where H/K is of multiplicative type and of finite type, u' is the canonical homomorphism and is faithfully flat (and affine), and u'' is a monomorphism.

N.B. As noted in 6.6.1, the homomorphism u'' is a closed immersion if G is affine over S, or if G is of finite presentation over S.

[43] Editor's note: it is enough to suppose that G is locally of finite type over S. By EGA IV_4, 1.4.3(v), this implies, since $H \to S$ is of finite presentation, that $H \to G$ is locally of finite presentation; the same is then true of $K \to S$, which is obtained from it by base change.

[44] Editor's note: the number 6.6.1 has been added for use in later references.

Proof. It is enough to prove that K is of multiplicative type; the rest then follows from 2.7 and IV 5.2.6.

Suppose first that G is of finite presentation over S. Since this hypothesis is stable under base change, in proving that K is of multiplicative type we may reduce to the case in which H is diagonalizable:

$$H = D_S(M),$$

where M is a commutative group of finite type. Let $s \in S$. The group K_s is a closed subgroup of

$$H_s = D_{\kappa(s)}(M).$$

By 8.1,[45] it is therefore of the form $D_{\kappa(s)}(N)$, where N is a quotient of M. Put

$$K' = D_S(N).$$

Then K' is a subgroup of H of multiplicative type. Let

$$v' : K' \longrightarrow G$$

be the homomorphism induced by u.[46] By construction, v'_s is the unit homomorphism. Corollary 5.2 therefore gives an open neighborhood U of s such that

$$v'|_U : K'|_U \longrightarrow G|_U$$

is the unit homomorphism. Replacing S by U, we may suppose that v' itself is the unit homomorphism, and hence that u factors as

$$H \xrightarrow{u'} H/K' \xrightarrow{u''} G.$$

The homomorphism u''_s is obtained from u_s by factoring through

$$H_s \longrightarrow H_s/\operatorname{Ker}(u_s),$$

and is therefore a monomorphism by IV 5.2.6. By 6.6, after restricting to a suitable open neighborhood U of s, the homomorphism

$$u''|_U : (H/K')|_U \longrightarrow G|_U$$

is a monomorphism. Thus, after this restriction,

$$\operatorname{Ker}(u) = \operatorname{Ker}(u') = K',$$

which proves that $\operatorname{Ker}(u)$ is of multiplicative type.

The same proof applies if, instead of assuming G of finite presentation over S, one assumes S locally Noetherian, at least when H is diagonalizable. Without this additional hypothesis on H, one must show that there is a covering base change

$$S' \longrightarrow S,$$

with S' locally Noetherian, which trivializes H. This will be proved in the next exposé, X 4.6. □

[45] Editor's note: the erroneous reference to IV 4.7.5 has been corrected. Note that Section 8 of the present exposé is independent of Sections 2 through 7.

[46] Editor's note: "induced by G" has been corrected to "induced by u."

7. Algebraicity of Formal Homomorphisms into an Affine Group

Theorem 7.1. *Let A be a Noetherian ring equipped with an ideal I for which A is separated and complete in the I-adic topology. Put*

$$S = \mathrm{Spec}(A), \qquad S_n = \mathrm{Spec}(A/I^{n+1}),$$

and let H, G be S-group preschemes, with H of isotrivial multiplicative type and G affine. Then the canonical map

$$\theta : \mathrm{Hom}_{S\text{-gr}}(H, G) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}(H_n, G_n) \tag{x}$$

is bijective, where H_n, G_n are the S_n-groups obtained from H, G by the base change $S_n \to S$.

Proof. Suppose first that H is diagonalizable, hence of the form

$$H = \mathrm{Spec}\, A(M),$$

where $B = A(M)$ is the algebra of the commutative group M with coefficients in A. We may also write

$$G = \mathrm{Spec}(C),$$

where C is an A-algebra equipped with a diagonal map satisfying the usual axioms. Homomorphisms of S-groups $H \to G$ correspond bijectively to homomorphisms of A-algebras

$$\varphi : C \longrightarrow B$$

compatible with the diagonal maps; that is, such that, for every $f \in C$,

$$\Delta_H(\varphi(f)) = (\varphi \otimes \varphi)(\Delta_G(f)).$$

There is an analogous description of the homomorphisms of S_n-groups $H_n \to G_n$, given by homomorphisms of A_n-algebras

$$\varphi_n : C_n \longrightarrow B_n,$$

where

$$A_n = A/I^{n+1}, \qquad B_n = B \otimes_A A_n, \qquad C_n = C \otimes_A A_n.$$

Put

$$\widehat{B} = \varprojlim_n B_n, \qquad \widehat{C} = \varprojlim_n C_n.$$

Then

$$\widehat{B} = \varprojlim_n A_n(M)$$

identifies with a submodule of the product A^M, namely the submodule of families $(a_m)_{m\in M}$ of elements of A that tend to zero for the I-adic topology along the filter of complements of finite subsets of M. In particular, the canonical homomorphism

$$B \longrightarrow \widehat{B}$$

is injective.[47] The projective system

$$\theta(\varphi) = (\varphi_n)_{n\geq 0}$$

[47] Editor's note: the argument that follows has been expanded.

defines a homomorphism

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B}$$

such that the following diagram is commutative:

$$\begin{array}{ccc} C & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ \widehat{C} & \xrightarrow{\widehat{\varphi}} & \widehat{B}. \end{array}$$

We prove that θ is surjective. Let $(\varphi_n)_{n\geq 0}$ be a projective system; we must show that it is obtained by reduction from some φ in the left-hand side of (x). A priori, the system (φ_n) defines a homomorphism on the completed algebras,

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B}.$$

It is enough to prove that the composite

$$\Phi : C \longrightarrow \widehat{C} \xrightarrow{\widehat{\varphi}} \widehat{B}$$

sends C into B. Indeed, in that case it gives a homomorphism of A-algebras

$$\varphi : C \longrightarrow B$$

whose reductions are the φ_n. This homomorphism is immediately seen to be compatible with the diagonal maps, since the φ_n are so, and since

$$B \otimes_A B \longrightarrow \widehat{B \otimes_A B}$$

is injective, as follows from the same argument after replacing M by $M \times M$.

Passing to the completions, the diagonal homomorphism Δ_H defines

$$\widehat{\Delta}_H : \widehat{B} \longrightarrow \widehat{B \otimes_A B}.$$

Taking the projective limit of the analogous diagrams defined by the φ_n, one obtains the commutative diagram

$$\begin{array}{ccc} C & \xrightarrow{\Phi} & \widehat{B} \\ {\scriptstyle \Delta_G}\downarrow & & \downarrow{\scriptstyle \widehat{\Delta}_H} \\ C \otimes_A C & \xrightarrow{\Psi} & \widehat{B \otimes_A B}, \end{array}$$

where Ψ is the composite

$$C \otimes_A C \xrightarrow{\Phi\otimes\Phi} \widehat{B} \otimes_A \widehat{B} \longrightarrow \widehat{B \otimes_A B};$$

the last arrow is the evident canonical homomorphism. Consequently, for every $f \in C$, the element $\Phi(f) \in \widehat{B}$ has the property that its image under $\widehat{\Delta}_H$ is a "decomposable" element of $\widehat{B \otimes_A B}$, namely an element in the image of $\widehat{B} \otimes_A \widehat{B}$.

Let $(e_m)_{m\in M}$ be the canonical basis of $A(M)$, and let $(e_{m,m'})$ be that of

$$A(M \times M) = A(M) \otimes_A A(M).$$

Since

$$\Delta_H(e_m) = e_m \otimes e_m = e_{m,m} \qquad (m \in M),$$

it remains only to apply the following lemma.

Lemma 7.2. *Let A be a Noetherian ring, let M be a set, and let $(a_{m,m'})$ be a family of elements of A, indexed by $M \times M$, satisfying:*

(i) $a_{m,m'} = 0$ *if* $m \neq m'$*; that is, the support of the family is contained in the diagonal of* $M \times M$*;*

(ii) one has

$$a_{m,m'} = \sum_i b_m^i c_{m'}^i,$$

where the b^i, c^i *are finitely many elements of* A^M*; in other words,* $(a_{m,m'})$ *belongs to the image of the canonical homomorphism*

$$A^M \otimes_A A^M \longrightarrow A^{M \times M}.$$

Then the support of $(a_{m,m'})$ *is finite.*

Proof. By (i), the family $(a_{m,m'})$ is determined by the elements

$$a_m = a_{m,m}.$$

For $x = (x_n)_{n \in M} \in A(M)$, put

$$(u \cdot x)_m = \sum_{m'} a_{m,m'} x_{m'}.$$

Thus $(a_{m,m'})$ is viewed as the matrix of a homomorphism

$$u : A(M) \longrightarrow A^M.$$

Condition (i) gives simply

$$(u \cdot x)_m = a_m x_m.$$

On the other hand, condition (ii) gives

$$u \cdot x \in \sum_i A \cdot b^i.$$

Hence $u(A(M))$ lies in a finitely generated A-module. If $(e_m)_{m \in M}$ denotes the canonical basis of

$$A(M) \subset A^M,$$

then all the elements $a_m e_m$ lie in a finitely generated A-module. Since A is Noetherian, the module generated by these elements is itself finitely generated. Because the e_m are linearly independent, all but finitely many of the a_m are zero. This proves the lemma, and hence Theorem 7.1 when H is diagonalizable. □

We now prove the general case of 7.1, in which H is only assumed isotrivial. There is then a finite étale surjective morphism

$$S' \longrightarrow S$$

such that

$$H' = H \times_S S'$$

is diagonalizable. We shall use only the fact that $S' \to S$ is finite and a covering for the faithfully flat quasi-compact topology, or simply for the canonical topology on (**Sch**). Thus "étale" could be replaced by "flat."

Put

$$S'' = S' \times_S S',$$

and similarly introduce

$$S'_n \quad \text{and} \quad S''_n = S'' \times_S S_n = S'_n \times_{S_n} S'_n,$$

together with $H', G', H'', G'', H'_n, G'_n, H''_n, G''_n$, obtained from H, G by the evident base changes. The groups H' and H'' are diagonalizable. Moreover, S', and hence S'', is affine. If

$$S' = \mathrm{Spec}(A'), \qquad S'' = \mathrm{Spec}(A''),$$

then A' and A'' are separated and complete for the topologies defined by IA' and IA'', respectively, because both are finite over A.

Since $S' \to S$ and $S'_n \to S_n$ are covering morphisms, we obtain a commutative diagram of maps of sets whose two rows are exact:

$$\begin{array}{ccccc}
\mathrm{Hom}_{S\text{-gr}}(H, G) & \longrightarrow & \mathrm{Hom}_{S'\text{-gr}}(H', G') & \rightrightarrows & \mathrm{Hom}_{S''\text{-gr}}(H'', G'') \\
\downarrow u & & \downarrow u' & & \downarrow u'' \\
\varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}(H_n, G_n) & \longrightarrow & \varprojlim_n \mathrm{Hom}_{S'_n\text{-gr}}(H'_n, G'_n) & \rightrightarrows & \varprojlim_n \mathrm{Hom}_{S''_n\text{-gr}}(H''_n, G''_n).
\end{array}$$

The second row is exact because it is the projective limit of the exact diagrams associated with the various $S'_n \to S_n$. By the diagonalizable case already proved, u' and u'' are bijective. Thus u is bijective as well, completing the proof of 7.1. □

Corollary 7.3. *Under the hypotheses of 7.1, suppose in addition that G is smooth over S, and let*

$$u_0 : H_0 \longrightarrow G_0$$

be a homomorphism of S_0-groups. Then there is a homomorphism of S-groups

$$u : H \longrightarrow G$$

lifting u_0. Any two such liftings u, u' are conjugate by an element $g \in G(S)$ whose reduction is the unit element of $G_0(S_0)$.

Proof. This follows by combining 3.6 and 7.1. To construct u, construct the u_n successively, as is possible by 3.6; Theorem 7.1 then shows that the system (u_n) comes from an S-homomorphism u. Given two liftings u, u', construct successively elements g_n such that

$$g_0 = 1, \qquad g_n \text{ is the reduction of } g_{n+1}, \qquad u'_n = \mathrm{int}(g_n)u_n.$$

Again, this is possible by 3.6. Since A is separated and complete, the g_n come from an element $g \in G(S)$. Finally, the injectivity in 7.1 gives

$$u' = \mathrm{int}(g)u.$$

□

Remark 7.4. *Compare 7.1 with EGA III, 5.4.1, which implies that the statement of 7.1 remains true if, instead of assuming H of multiplicative type and G affine, one assumes that H is proper over S, and that G is separated and locally of finite type over S. A statement such as 7.1 without a properness hypothesis is rather exceptional, and should be understood here as one aspect of the great "rigidity" of the structure of a group of multiplicative type.*

The analogous statement with

$$G = H = \mathbf{G}_a$$

is false in general. Indeed, take A of characteristic $p > 0$, and define the u_n, by reduction modulo I^{n+1}, from an additive formal power series

$$u(T) = \sum_{n \geq 0} a_n T^{p^n},$$

where the $a_n \in A$ tend to zero for the I-adic topology, but not for the discrete topology.

The statement of 7.1 also becomes false if the hypothesis that G be affine is omitted, even when $H = \mathbf{G}_m$. An example, with A a complete discrete valuation ring, is obtained from an elliptic curve over the fraction field K of A whose reduction, in the Néron–Kodaira theory, say, is the group $\mathbf{G}_m$ over the residue field k.[48] *One then has a smooth commutative group scheme G over S, with special fiber $\mathbf{G}_{m,k}$. By 3.6 this permits the construction of a projective system of isomorphisms*

$$u_n : H_n \xrightarrow{\sim} G_n, \qquad H = \mathbf{G}_{m,A},$$

while the generic fiber of G is an abelian variety. Consequently, there is no homomorphism of S-groups $H \to G$ other than the zero homomorphism.

8. Subgroups, Quotient Groups, and Extensions of Groups of Multiplicative Type over a Field*

Scholium 8.0. [49] — *Let k be a field and let H be an affine k-group scheme. Write $k[H]$ for its affine algebra and*

$$X(H) = \mathrm{Hom}_{k\text{-gr}}(H, \mathbf{G}_{m,k})$$

for the group of characters of H. By the independence lemma for characters, $X(H)$ is a linearly independent subset of $k[H]$. Consequently, H is diagonalizable if and only if $X(H)$ spans $k[H]$ as a k-vector space.

Proposition 8.1. — *Let G be a diagonalizable group scheme (respectively, a group scheme of multiplicative type) over a field k.*[50] *Then, for every subgroup scheme H of G, the schemes H and G/H are diagonalizable (respectively, of multiplicative type).*

Definition 1.1 immediately reduces the assertion to the case without the parenthetical qualification. An easy passage to the limit, using VI_B 11.13 and VIII 3.1, reduces us to the case in which G is of finite type over k.[51] By VI_B 11.16,[52] one can find a finite family of nonzero elements

$$f_i = \sum_m a_{im} m, \qquad a_{im} \in k, \tag{8.1.1}$$

[48] Editor's note: such an example could be made explicit.

* Added in July 1969. This afterthought section is independent of Sections 3 through 7.

[49] Editor's note: this scholium has been added.

[50] Editor's note: the words "over a field k," implicit in the original, have been added. In addition, "algebraic group" has been replaced by "group scheme," since G is not assumed to be of finite type.

[51] Editor's note: this "passage to the limit" should be spelled out. It uses 8.0 and the equality

$$G/H = \varprojlim_i G_i/H_i$$

(cf. VI_B, proof of 11.17). To see that $H = \varprojlim_i (H \cap G_i)$, does one use the fact that H is closed in G? Using this fact, one can give a direct proof.

[52] Editor's note: VI_B 11.16 should doubtless be rewritten in the usual, more agreeable form. In particular, a single f_i suffices below.

of the affine ring $k(M)$ of $G = D_k(M)$, such that the points of H, with values in an arbitrary k-algebra k', are precisely the points $g \in G(k')$ for which

$$\tau_g f_i = \lambda_i(g) f_i, \qquad \lambda_i(g) \in k', \tag{8.1.2}$$

where τ_g denotes translation by g. But

$$\tau_g f_i = \sum_m a_{im} \chi_m(g) m, \tag{8.1.3}$$

where $\chi_m : G \to \mathbf{G}_{m,k}$ is the character associated with m. Thus (8.1.2) is equivalent to

$$\chi_m(g) = \chi_{m'}(g) \qquad \text{if } m, m' \in Z_i, \tag{8.1.4}$$

where Z_i denotes the set of $m \in M$ for which $a_{im} \neq 0$. The same relation may be written

$$\chi_{m'-m}(g) = 1 \qquad \text{if } m, m' \in Z_i.$$

Let N be the subgroup of M generated by all the elements $m' - m$, where i varies and $m, m' \in Z_i$. It follows from the definition of $D_k(M) \simeq G$ that H identifies with $D_k(M/N)$. Hence VIII 3.1 identifies G/H with $D_k(N)$. □

Proposition 8.2. [53] — *Let k be a field; let H and K be k-group schemes of multiplicative type and finite type; and let G be a k-group scheme fitting into an exact sequence*

$$1 \longrightarrow H \longrightarrow G \longrightarrow K \longrightarrow 1$$

(which implies that G is of finite type over k).

(i) If G is commutative or K is connected, then G is of multiplicative type.

(ii) If K and H are diagonalizable and K is a torus, then G is diagonalizable.

For the proof of (i), the reader is referred to XVII 7.1.1, of which 8.2(i) is a special case. The case of a field is treated in part (i) of the proof of XVII 7.1.1, without using the results of the later exposés.

For (ii), now suppose that K and H are diagonalizable:

$$K \simeq D_k(M) \qquad \text{and} \qquad H \simeq D_k(N),$$

and suppose that G is of multiplicative type, as follows from (i) when K is a torus. By X 1.4, G is isotrivial over k; that is, there is a finite separable extension k'/k such that

$$G' = G \times_k k'$$

is diagonalizable. Hence $G' = D_{k'}(E)$ for a commutative group E, and there is an exact sequence

$$0 \longrightarrow D_{k'}(N) \longrightarrow D_{k'}(E) \longrightarrow D_{k'}(M) \longrightarrow 0. \tag{1}$$

By VIII 3.1 and 3.2, M is therefore a subgroup of E, and one has an exact sequence

$$0 \longrightarrow M \longrightarrow E \longrightarrow N \longrightarrow 0. \tag{2}$$

For a given extension E_0 of N by M, consider the diagonalizable k-group $G_0 = D_k(E_0)$. The k-group functor A of automorphisms of the extension

$$1 \longrightarrow H \longrightarrow G_0 \longrightarrow K \longrightarrow 1, \tag{3}$$

[53] Editor's note: both the statement and its proof have been expanded.

that is, the group subfunctor of $\mathrm{Aut}_{k\text{-gr}}(G_0)$ whose points on a k-prescheme T are the

$$\phi \in \mathrm{Aut}_{T\text{-gr}}(G_T)$$

inducing the identity on H_T and K_T, identifies with $\mathrm{Hom}_{k\text{-gr}}(K, H)$. By VIII 1.5, this is the constant k-group with value

$$L = \mathrm{Hom}_{\mathrm{gr}}(N, M).$$

It follows that the extensions G of K by H which become isomorphic to (3) over a separable closure k_s of k are classified in the same way as the k-torsors, for the étale topology, under the constant group L_k. Put $\Gamma = \mathrm{Gal}(k_s/k)$. Since L is a trivial Γ-module, these torsors are classified by

$$H^1_{\text{ét}}(k, L_k) = H^1(\Gamma, L) = \mathrm{Hom}_{\text{gr. top.}}(\Gamma, L).$$

If K is moreover a torus, then M is torsion-free, and hence so is L. Therefore

$$\mathrm{Hom}_{\text{gr. top.}}(\Gamma, L) = 0,$$

and every extension of K by H is already diagonalizable.

Remark 8.3. — *As already noted in the proof of 8.2, the first assertion is stated and proved over an arbitrary base in XVII 7.1.1. The second assertion generalizes, with essentially the same proof, to the case of an integral normal base scheme—or, more generally, a geometrically unibranch one—by using X 5.13 below.*

Finally, 8.1 may also be regarded as a corollary of the distinctly less trivial result 6.8, which is likewise valid over an arbitrary base scheme.[54]

Bibliography

[55]

[BEns] N. Bourbaki, *Théorie des ensembles*, Chapters I–IV, Hermann, 1970.

[54] Editor's note: the proof of 6.8 nevertheless uses 8.1.

[55] Editor's note: additional references cited in this exposé.

Exposé X
Characterization and Classification of Groups of Multiplicative Type

by A. Grothendieck

1. Classification of Isotrivial Groups. Case of a Base Field

Recall (IX 1.1) that a group of multiplicative type H over the prescheme S is said to be *isotrivial* if there exists a finite étale surjective morphism $S' \to S$ such that $H' = H \times_S S'$ is diagonalizable. When S is connected, S' decomposes as a finite disjoint sum of connected components S'_i, and hence, after replacing S' by one of the S'_i, we may suppose that S' is connected. Finally, S' can be dominated by a connected finite étale S'_1 that is Galois, that is, a principal homogeneous bundle over S with group Γ_S, where Γ is an ordinary finite group (cf. SGA 1, V, nos. 7 and 3 when S is locally noetherian, and EGA IV_4, 18.2.9 in general).[1] We suppose S' chosen in this way and propose to determine the groups of multiplicative type H over S that are "trivialized" by S', that is, those for which $H' = H \times_S S'$ is diagonalizable.[2] By descent theory (cf. SGA 1, VIII, 5.4), the category of such H is equivalent to the category of diagonalizable groups H' over S', equipped with actions of Γ on H' compatible with the actions of Γ on S'. (N.B. Since the groups under consideration are affine over the base, every descent datum here is effective; cf. SGA 1, VIII, 2.1.) Since S' is connected, the contravariant functor

$$M \longmapsto D_{S'}(M)$$

is an anti-equivalence from the category of ordinary commutative groups to the category of diagonalizable groups over S' (cf. VIII 1.6); a quasi-inverse is[3]

$$H \longmapsto \mathrm{Hom}_{S'\text{-gr}}(H, \mathbb{G}_{m,S'}).$$

Proposition 1.1. — *Let S be a connected prescheme and let S' be a connected principal cover of S with finite group Γ. Then the category of groups of multiplicative type over S*

[0] xy version of November 6, 2009: Addenda placed in Section 9; reviewed through 8.8.

[1] Editor's note: More precisely, this follows from the "axiomatic conditions of a Galois theory," cf. SGA 1, V, no. 4 g); when S is locally noetherian, the axioms are verified in *loc. cit.*, nos. 7 and 3, especially 3.7, which rests on SGA 1, I, 10.9. The latter result is proved without noetherian hypotheses in EGA IV_4, 18.2.9.

[2] Editor's note: In what follows, an S-group of multiplicative type will be called "split" if it is "trivial" in the sense of IX 1.2, that is, if it is a diagonalizable S-group.

[3] Editor's note: S has been corrected to S'.

split by S' is anti-equivalent to the category of Γ-modules, that is, of ordinary commutative groups M equipped with a homomorphism

$$\Gamma \longrightarrow \mathrm{Aut}_{\mathrm{gr}}(M).$$

The following standard consequence results.

Corollary 1.2. — *Let S be a connected prescheme, and let $\xi : \mathrm{Spec}(\Omega) \to S$ be a "geometric point" of S, that is, a morphism into S from the spectrum of an algebraically closed field Ω. Consider the fundamental group of S at ξ (cf. SGA 1, V, no. 7):*

$$\pi = \pi_1(S, \xi).$$

Then the category of isotrivial groups of multiplicative type H over S is anti-equivalent to the category of "Galois modules" under π, that is, of π-modules M such that the stabilizer in π of every element of M is an open subgroup.

Under this correspondence, the group associated with an isotrivial group of multiplicative type H is

$$M = \mathrm{Hom}_{\Omega\text{-gr}}(H_\xi, \mathbb{G}_{m,\Omega}),$$

where H_ξ is the fiber of H at ξ; this group is naturally endowed with an action of $\pi_1(S, \xi)$.

Remark 1.3. — We shall see below (cf. 5.16) that if S is normal, or more generally geometrically unibranch, then every group of multiplicative type and of finite type over S is necessarily isotrivial. Consequently, the classification principle of 1.2 applies to groups of multiplicative type and of finite type over S; they correspond to Galois π-modules that are of finite type over $\mathbb{Z}$. For the moment, let us confine ourselves to the following result.

Proposition 1.4. — *Let k be a field and H a group of multiplicative type and of finite type over k. Then H is isotrivial; that is, there exists a finite separable extension k' of k that splits H.*

Consequently, by 1.2, if π is the topological Galois group of an algebraic closure Ω of k, the category of groups H of multiplicative type and of finite type over k is anti-equivalent to the category of Galois modules M under π that are of finite type as $\mathbb{Z}$-modules.

Since H is of finite type over k, the "principle of finite extension" (cf. EGA IV_3, 9.1.4) first gives a finite extension k' of k that splits H. Recall the argument. By hypothesis, there exist a diagonalizable group G of finite type over k, a faithfully flat S' over $S = \mathrm{Spec}(k)$, and an isomorphism of S'-groups

$$H_{S'} \simeq G_{S'}.$$

After replacing S' by the residue field of one of its points, we may suppose that S' is the spectrum of an extension K of k. The field K is the filtered direct limit of its finitely generated k-subalgebras A_i; it follows readily (cf. EGA IV_3, 8.8.2.4) that the given isomorphism u comes, for sufficiently large i, from an A_i-isomorphism

$$u_i : H_{A_i} \simeq G_{A_i}.$$

By the Nullstellensatz, A_i has a quotient ring k' that is a finite extension of k. Thus k' splits H.

Now k' is a radicial extension of a separable extension k'_s of k. By IX 5.4, the isomorphism

$$u' : H_{k'} \simeq G_{k'}$$

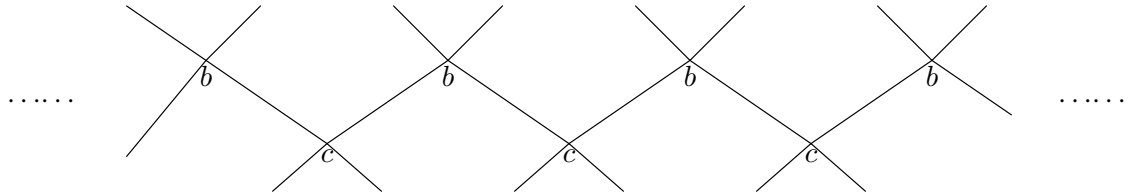
comes from an isomorphism $H_{k'_s} \simeq G_{k'_s}$. Thus k'_s splits H, proving 1.4.

Remark 1.5. — Statement 1.2 gives, in particular, a characterization of the isotrivial tori over S of relative dimension n: setting $\pi = \pi_1(S, \xi)$, they correspond to classes, up to "equivalence," of representations

$$\pi \longrightarrow \mathrm{GL}(n, \mathbb{Z})$$

whose kernel is an open subgroup of π.

1.6. Even when S is an algebraic curve, there may exist tori over S (of relative dimension 2) that are not locally isotrivial (and *a fortiori* not isotrivial); there may also exist locally trivial tori that are not isotrivial. (Observe, however, that such phenomena can occur only when S is not normal, as already noted in 1.3.) For example, let S be an irreducible algebraic curve (over an algebraically closed field, to fix ideas) with an ordinary double point a, let S' be its normalization, and let b and c be the two points of S' above a. A connected principal homogeneous bundle P over S, with structural group $\mathbb{Z}$, is obtained by attaching infinitely many copies of S' according to the diagram



(N.B. This is a principal bundle for the étale topology.) There is a homomorphism

$$\varphi : \mathbb{Z} \longrightarrow \mathrm{GL}(2, \mathbb{Z}), \qquad \varphi(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix},$$

which makes it possible to construct a torus T over S, of relative dimension 2, from the trivial torus $\mathbb{G}_m^2$ over P and the descent datum on it deduced from φ. (N.B. The projection $P \to S$ is a covering for the étale topology and *a fortiori* for the canonical topology of (**Sch**); the descent datum in question is necessarily effective because $\mathbb{G}_{m,P}^2$ is affine over P.) It is not difficult to prove that T is not isotrivial in a neighborhood of a[4] (although it is trivial on $S - \{a\}$).

A variant of this construction is obtained by taking for S a curve with two irreducible components S_1 and S_2 meeting in two points a' and a''. This makes it possible to construct a connected, locally trivial principal homogeneous bundle P over S with group $\mathbb{Z}_S$, and hence an associated torus T that is locally trivial but not isotrivial.

2. Infinitesimal Variations of Structure

[5] Let us begin by recalling the following result (cf. SGA 1, I, 8.3 when S is locally noetherian, and EGA IV_4, 18.1.2 in general).

Recall 2.0. — *Let S be a prescheme and S_0 a subprescheme with the same underlying set. Then the functor*

$$X \longmapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale preschemes over S and the analogous category over S_0.

[4] Editor's note: See 7.3 below.

[5] Editor's note: This "recall" has been added for later references.

Proposition 2.1. — *Let S be a prescheme and S_0 a subprescheme with the same underlying set (that is, defined by a nilideal $\mathscr{I}$). Then the functor*

$$H \longmapsto H_0 = H \times_S S_0$$

from the category of group preschemes of multiplicative type over S to the analogous category over S_0 is fully faithful.

Moreover, it induces an equivalence between the category of quasi-isotrivial groups of multiplicative type over S and the analogous category over S_0.

We first prove full faithfulness, that is, if H and G over S are of multiplicative type, then

$$\mathrm{Hom}_{S\text{-gr}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(H_0, G_0)$$

is bijective. The question is local on S, so we may suppose that S is affine. There then exists a faithfully flat quasi-compact morphism $S' \to S$ that splits H and G. Let $S'' = S' \times_S S'$; write H', G', respectively H'', G'', for the groups obtained from H, G by the base changes $S' \to S$, respectively $S'' \to S$. Define S'_0 and S''_0 similarly; the latter is also isomorphic to $S'_0 \times_{S_0} S'_0$. We obtain a commutative diagram with exact rows:

$$\begin{array}{ccccc}
\mathrm{Hom}_{S\text{-gr.}}(H, G) & \longrightarrow & \mathrm{Hom}_{S'\text{-gr.}}(H', G') & \rightrightarrows & \mathrm{Hom}_{S''\text{-gr.}}(H'', G'') \\
\downarrow{\scriptstyle u} & & \downarrow{\scriptstyle u'} & & \downarrow{\scriptstyle u''} \\
\mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0) & \longrightarrow & \mathrm{Hom}_{S'_0\text{-gr.}}(H'_0, G'_0) & \rightrightarrows & \mathrm{Hom}_{S''_0\text{-gr.}}(H''_0, G''_0),
\end{array}$$

Thus, to prove that u is bijective, it is enough to prove that u' and u'' are bijective. This reduces us to the case where H and G are diagonalizable, hence of the form $D_S(M)$ and $D_S(N)$, where M and N are ordinary commutative groups. Likewise,

$$H_0 = D_{S_0}(M), \qquad G_0 = D_{S_0}(N).$$

We then have a commutative diagram[6]

$$\begin{array}{ccc}
\mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) & \xrightarrow{\sim} & \mathrm{Hom}_{S\text{-gr.}}\big(D_S(M), D_S(N)\big) \\
\downarrow & & \downarrow \\
\mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0}) & \xrightarrow{\sim} & \mathrm{Hom}_{S_0\text{-gr.}}\big(D_{S_0}(M), D_{S_0}(N)\big),
\end{array}$$

whose horizontal arrows are isomorphisms by VIII 1.4. We are therefore reduced to proving that the homomorphism

$$\mathrm{Hom}_{S\text{-gr}}(N_S, M_S) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(N_{S_0}, M_{S_0}) \tag{x}$$

is bijective; in other words, the functor $M_S \mapsto M_{S_0}$, from constant commutative group preschemes over S to constant commutative group preschemes over S_0, is fully faithful. Now (x) is also identified with the natural map

$$\mathrm{Hom}_{\mathrm{gr}}(N, \Gamma(M_S)) \longrightarrow \mathrm{Hom}_{\mathrm{gr}}(N, \Gamma(M_{S_0}))$$

deduced from $\Gamma(M_S) \to \Gamma(M_{S_0})$. The latter map is plainly bijective, since $\Gamma(M_S)$, the set of locally constant maps from S to M, depends only on the underlying topological space of S. This proves the desired assertion.

[6]Editor's note: The original has been corrected by interchanging M and N in the right-hand terms.

To prove the second assertion of 2.1, it remains to show that every quasi-isotrivial group H_0 of multiplicative type over S_0 comes from a quasi-isotrivial group H of multiplicative type over S. Let $S'_0 \to S_0$ be an étale surjective morphism that splits H_0.

By 2.0, there exist an étale morphism $S' \to S$ and an S_0-isomorphism

$$S' \times_S S_0 \simeq S'_0.$$

Thus we may suppose that S'_0 is obtained from S' by reduction. Since H'_0 is diagonalizable, it is immediately seen to be isomorphic to the group obtained by the base change $S'_0 \to S'$ from a diagonalizable group H' over S'. (N.B. If $H'_0 = D_{S'_0}(M)$, take $H' = D_{S'}(M)$.) As usual, set

$$S'' = S' \times_S S', \qquad S''' = S' \times_S S' \times_S S',$$

and similarly define S''_0 and S'''_0 by base change from S to S_0. They are also isomorphic to the fibered square and fibered cube, respectively, of S'_0 over S_0. Applying the full faithfulness already proved to (S'', S''_0) and (S''', S'''_0), we see that the natural descent datum on H'_0 relative to $S'_0 \to S_0$ (cf. IV 2.1) comes from a uniquely determined descent datum on H' relative to $S' \to S$. This descent datum is effective because H' is affine over S' (SGA 1, VIII, 2.1). Hence there exists an S-group H such that

$$H \times_S S' = H' = D_{S'}(M),$$

and H is therefore a quasi-isotrivial group of multiplicative type.

Using the full-faithfulness result for (S', S'_0), one now checks readily that the given isomorphism between H'_0 and $H' \times_{S'} S'_0$ comes from an isomorphism between H_0 and $H \times_S S_0$. (For a more formal presentation of results of this kind, see Giraud's article in preparation on descent theory.*)

Corollary 2.2. — *Let H be an S-group of multiplicative type, and set $H_0 = H \times_S S_0$. Then H is quasi-isotrivial (respectively, locally isotrivial, isotrivial, locally trivial, or trivial) if and only if H_0 has the corresponding property.*

The "only if" direction is immediate. The "if" direction has already been proved in the quasi-isotrivial case: by full faithfulness, it is enough to know that every quasi-isotrivial group over S_0 lifts to a quasi-isotrivial group over S. The same argument applies to "trivial." For "isotrivial," repeat the argument establishing the second assertion of 2.1, but take $S'_0 \to S_0$ to be finite étale and surjective. The locally isotrivial and locally trivial cases follow at once from the isotrivial and trivial cases.

When $\mathscr{I}$ is nilpotent, 2.2 can be generalized slightly without assuming *a priori* that H is of multiplicative type.[7]

Corollary 2.3. — *Suppose that the ideal $\mathscr{I}$ defining S_0 is locally nilpotent. Let H be a group prescheme over S, flat over S, and set $H_0 = H \times_S S_0$. Then H is of quasi-isotrivial multiplicative type if and only if H_0 is so.*

Suppose that H_0 is of quasi-isotrivial multiplicative type. We prove that the same is true of H. The question is local for the étale topology, and by 2.0 the category of étale preschemes over S is equivalent to the category of étale preschemes over S_0 by the functor $S' \mapsto S' \times_S S_0$. We are therefore immediately reduced to the case where H_0 is diagonalizable, hence isomorphic to $D_{S_0}(M)$. Let $G = D_S(M)$. We then have an isomorphism

$$u_0 : H_0 \xrightarrow{\sim} G_0.$$

*Cf. J. Giraud, "Méthode de la descente," *Bull. Soc. Math. France*, Mémoire 2 (1964).

[7]Editor's note: Recall that 2.3 is used in proving Theorem IX 3.6 bis.

I claim that u_0 comes from a unique homomorphism $u : H \to G$, which is necessarily an isomorphism because u_0 is one, H and G are flat over S, and $\mathscr{I}$ is locally nilpotent.[8] This will prove 2.3. We have (cf. VIII 1 (xxx))

$$\mathrm{Hom}_{S\text{-gr}}(H, G) \simeq \mathrm{Hom}_{S\text{-gr}}\big(M_S, \underline{\mathrm{Hom}}_{S\text{-gr}}(H, \mathbb{G}_{m,S})\big),$$

and the right-hand side is also identified with

$$\mathrm{Hom}_{\mathrm{gr}}\big(M, \mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_{m,S})\big).$$

Thus the homomorphism

$$\mathrm{Hom}_{S\text{-gr}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(H_0, G_0) \tag{x}$$

is isomorphic to the homomorphism

$$\mathrm{Hom}_{\mathrm{gr}}\big(M, \mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_{m,S})\big) \longrightarrow \mathrm{Hom}_{\mathrm{gr}}\big(M, \mathrm{Hom}_{S_0\text{-gr}}(H_0, \mathbb{G}_{m,S_0})\big)$$

deduced from the restriction homomorphism

$$\mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_{m,S}) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(H_0, \mathbb{G}_{m,S_0}). \tag{xx}$$

It is therefore enough to prove that (xx) is bijective. The question is local on S, so we may suppose that S is affine. Since $\mathbb{G}_{m,S}$ is commutative and smooth over S, IX 3.6 applies and completes the proof.

Corollary 2.4. — *Let A be an artinian local ring with residue field k, $S = \mathrm{Spec}(A)$, and $S_0 = \mathrm{Spec}(k)$.*

(i) [9] *Let H be an S-group prescheme, flat and locally of finite type, such that $H_0 = H \times_S S_0$ is of multiplicative type. Then H is of multiplicative type, of finite type, and isotrivial. In particular, every S-group prescheme H of multiplicative type and of finite type is isotrivial.*

(ii) *The functor $H \mapsto H_0$ is an equivalence between the category of groups of multiplicative type and of finite type over A and the analogous category over k.*

Indeed, let H be as in (i). Then H_0 is of multiplicative type and locally of finite type, hence of finite type, and therefore isotrivial by 1.4. By 2.3 and 2.2, H is consequently of multiplicative type (and hence of finite type) and isotrivial. Assertion (ii) now follows from 2.1.

Corollary 2.5. — *Let $S' \to S$ be a faithfully flat radicial morphism.*

(i) *The functor*

$$H \longmapsto H' = H \times_S S'$$

from the category of groups of multiplicative type over S to the analogous category over S' is fully faithful. Moreover, it induces an equivalence between the subcategories of quasi-isotrivial groups of multiplicative type.

[8] Editor's note: Since u_0 is an isomorphism, it is enough to see that for every $h \in H$, the morphism $\varphi : \mathscr{O}_{G,u(h)} \to \mathscr{O}_{H,h}$ is bijective. It is bijective after reduction modulo $\mathscr{I} = \mathscr{I}_s$, where s is the image of h in S; hence its cokernel C satisfies $C = \mathscr{I}C$, so $C = 0$ because $\mathscr{I}$ is nilpotent. Since $\mathscr{O}_{H,h}$ is flat over $\mathscr{O}_{S,s}$, the kernel K of φ likewise satisfies $K = \mathscr{I}K$, and hence $K = 0$.

[9] Editor's note: The statement has been rewritten, retaining only the nontrivial implication in order to highlight it.

(ii) *If H is of multiplicative type, then H is quasi-isotrivial (respectively, locally isotrivial, isotrivial, locally trivial, or trivial) if and only if H' has the corresponding property.*

Indeed, let

$$S'' = S' \times_S S', \qquad S''' = S' \times_S S' \times_S S'.$$

The assumption that $S' \to S$ is radicial implies that the diagonal immersions $S' \to S''$ and $S' \to S'''$ are surjective, so base change by either immersion is governed by 2.1 and 2.2. Taking into account that $S' \to S$ is a morphism of effective descent for the fibered category of groups of multiplicative type over preschemes (because it is faithfully flat and quasi-compact[10]), the assertion follows formally from 2.1 and 2.2 (cf. the already cited article of Giraud for a formal argument).

3. Finite Variations of Structure: Complete Base Ring

Lemma 3.1. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology; set $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/I)$. Let G and H be two S-group preschemes, with G of multiplicative type and isotrivial, and with H affine over S, flat over S at the points of $H_0 = H \times_S S_0$, where H_0 is of quasi-isotrivial multiplicative type. Then the natural map*

$$\mathrm{Hom}_{S\text{-gr}}(G, H) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(G_0, H_0)$$

is bijective.

For every integer $n \geq 0$, let

$$S_n = \mathrm{Spec}(A/I^{n+1}),$$

and let G_n, H_n be obtained from G, H by the base change $S_n \to S$. Since G/S is of isotrivial multiplicative type and H is affine over S, IX 7.1 shows that the natural homomorphism

$$\mathrm{Hom}_{S\text{-gr}}(G, H) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}(G_n, H_n)$$

is bijective. On the other hand, 2.3 shows that the H_n are of quasi-isotrivial multiplicative type, and 2.1 shows that all transition homomorphisms in the inverse system

$$\big(\mathrm{Hom}_{S_n\text{-gr}}(G_n, H_n)\big)_n$$

are isomorphisms. Lemma 3.1 follows at once.

Theorem 3.2. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, and set $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/I)$. Then:*

(i) *The functor*

$$H \longmapsto H_0 = H \times_S S_0$$

is an equivalence between the category of isotrivial groups of multiplicative type over S and the analogous category over S_0.

(ii) *Let H be an S-group of multiplicative type and of finite type. Then H is isotrivial if and only if H_0 is isotrivial.*

[10] Editor's note: Spell out this point ...

Proof. For (i), one may either repeat the proof of 2.1 or use 1.2. In either case one uses the fact that the functor

$$S' \longmapsto S'_0 = S' \times_S S_0$$

from the category of finite étale schemes over S to the category of finite étale schemes over S_0 is an equivalence (SGA 1, I, 8.4). Equivalently, after reducing to the case where S is connected (that is, S_0 is connected) and choosing a geometric point ξ of S_0, the canonical homomorphism

$$\pi_1(S_0, \xi) \longrightarrow \pi_1(S, \xi)$$

is an isomorphism.

We prove (ii), namely that if H_0 is isotrivial, then H is isotrivial. By (i), there exist an isotrivial group G of multiplicative type over S and an S_0-isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

[11] Since H is of finite type, so are H_0 and G_0. Therefore, by IX 2.1 b), the type of G at every point of S is a finitely generated abelian group, and hence G is of finite type over S. Moreover, by 3.1, u_0 comes from a homomorphism of S-groups

$$u : G \longrightarrow H.$$

Finally, G and H are groups of multiplicative type and of finite type over S, and u_0 is an isomorphism. It follows from IX 2.9 that u is an isomorphism, since every neighborhood of S_0 in S is equal to S.

Corollary 3.3. — *Let A be a complete noetherian local ring with residue field k.*

(i) *Every group of multiplicative type and of finite type over A is isotrivial.*

(ii) *The functor*

$$H \longmapsto H \otimes_A k = H_0$$

is an equivalence between the categories of groups of multiplicative type and of finite type over A and over k.

[12] First, (i) follows from 3.2 (ii) and 1.4. Then (ii) follows from 3.2 (i), taking into account that H is of finite type if and only if H_0 is of finite type (cf. the proof of 3.2 (ii)).

Remark 3.3.1. — By 1.4, statement 3.3 gives a complete classification of groups of multiplicative type and of finite type over A in terms of the topological Galois group of an algebraic closure Ω of k.

Remarks 3.4. — Under the hypotheses of 3.2 (that is, A noetherian, separated and complete for the I-adic topology, with no further hypothesis on A/I), it will follow from no. 5 that the functor

$$H \longmapsto H_0$$

from the category of groups of multiplicative type and of finite type over S to the analogous category over S_0 is still fully faithful, without any isotriviality hypothesis.[12]

It is not, however, essentially surjective in general. Indeed, there may exist a group H_0 of multiplicative type and of finite type over S_0, even locally trivial if desired (but not isotrivial), that does not arise by reduction from a group H of multiplicative type over S.

[11] Editor's note: The original has been expanded in what follows.

[12] Editor's note: Specify this reference, possibly adding a corollary in no. 5 ...

To see this, take either example from 1.6 of a non-isotrivial group of multiplicative type over a nonnormal curve. The curve may plainly be taken affine, say S_0, and embedded in affine 2-space; it is therefore defined by an ideal J in $k[X,Y]$. Take A to be the completion of $k[X,Y]$ for the J-adic topology. Then A is a regular ring of dimension 2, and *a fortiori* normal. We shall see in 5.16 that every group of multiplicative type and of finite type over $S = \mathrm{Spec}(A)$ is consequently isotrivial. Thus H_0, being of finite type and non-isotrivial, cannot come from a group H of multiplicative type over S, since such an H would necessarily be of finite type and hence isotrivial.

Lemma 3.5.* — *Let S be a prescheme and $u : G \to H$ a homomorphism of S-group preschemes that are locally of finite presentation and flat over S. Let U be the set of $s \in S$ such that the homomorphism induced on the fibers,*

$$u_s : G_s \longrightarrow H_s,$$

is flat (respectively, smooth, unramified, étale, or quasi-finite). Then U is open, and the restriction

$$u|_U : G|_U \longrightarrow H|_U$$

is flat (respectively, smooth, unramified, étale, or quasi-finite).

Let V be the set of points at which u is flat (respectively, ...). It is known that V is open (cf. SGA 1, I through IV in the locally noetherian case, and EGA IV in general[13]), and for $x \in G$ above $s \in S$, one has $x \in V$ if and only if $u_s : G_s \to H_s$ is flat (respectively, ...) at x (same reference; in the flat, smooth, and étale cases, the flatness of G and H over S is used here). Since u_s is a homomorphism of locally algebraic groups, this is also equivalent to u_s being flat (respectively, ...) everywhere ($\mathrm{VI_B}$, 1.3), that is, to $s \in U$. Therefore U is the inverse image of the open set V under the unit section of G, and hence is open. Moreover, $V = G|_U$, so $G|_U \to H|_U$ is flat (respectively, ...), completing the proof. (N.B. In the unramified and quasi-finite cases, the flatness hypothesis on G and H is unnecessary.)

Lemma 3.6. — *Let H be a commutative algebraic group over a field k that admits an open subgroup G of multiplicative type. Then the family of subschemes ${}_nH$, $n > 0$, is schematically dense in H. In particular, if ${}_nH = {}_nG$ for every $n > 0$, then $H = G$.*

Here ${}_nH$ denotes the kernel of $n \cdot \mathrm{id}_H$.[14] Let $\bar{k}$ be an algebraic closure of k. It is enough to show that the family $({}_nH_{\bar{k}})_{n>0}$ is schematically dense in $H_{\bar{k}}$, for then $({}_nH)_{n>0}$ is schematically dense in H (cf. IX 4.5). We may therefore suppose that k is algebraically closed, so that $G = D_k(M)$, where M is an ordinary finitely generated commutative group.

Let

$$M_0 = M/\,\mathrm{Torsion}(M).$$

Then $T = D_k(M_0)$ is the largest torus contained in G, and H/T is finite. Hence $H(k)/T(k)$ is annihilated by an integer $\nu > 0$. There exist finitely many $g_i \in H(k)$ such that

$$H = \coprod_i g_i \cdot G,$$

and $g_i^\nu \in T(k)$. Since k is algebraically closed, the map $\nu \cdot \mathrm{id}_T$ is surjective on

$$T(k) \simeq (k^\times)^d.$$

*Cf. also $\mathrm{VI_B}$, 2.5, for more systematic developments of this kind.

[13]Editor's note: Specify these references. Since G, H are locally of finite presentation, u is locally of finite presentation ($\mathrm{IV_1}$, 1.4.3 (v)); one then applies $\mathrm{IV_3}$, 11.3.10 and 13.1.4 for flat and quasi-finite, and $\mathrm{IV_4}$, 17.4.1, 17.5.1, and 17.6.1 for unramified, smooth, and étale; compare $\mathrm{VI_B}$, 2.5.

[14]Editor's note: The preceding sentence has been added.

After replacing g_i by $g_i t_i^{-1}$, where $t_i \in T(k)$ satisfies $t_i^\nu = g_i^\nu$, we may therefore suppose that $g_i^\nu = 1$. If n is now a multiple of ν, then

$$ {}_nH \supset g_i \cdot {}_nG.$$

As $n > 0$ varies, the family ${}_nG$ is schematically dense in G by IX 4.7, and the conclusion of 3.6 follows.

Theorem 3.7. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology; set $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/I)$. Let H be an affine S-group scheme of finite type, flat over S at the points of H_0, such that $H_0 = H \times_S S_0$ is of multiplicative type and isotrivial. Then there exists an open and closed subgroup G of H, of isotrivial multiplicative type (and of finite type), such that $G_0 = H_0$.*

By 3.2 (i), there exist an isotrivial group G of multiplicative type over S and an isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

By 3.1, u_0 comes from a unique homomorphism of S-groups

$$u : G \longrightarrow H.$$

Using IX 6.6, we see that u is a monomorphism: if U is the set of $s \in S$ such that $u_s : G_s \to H_s$ is a monomorphism, then U is an open neighborhood of S_0, hence equals S, and $G|_U \to H|_U$ is a monomorphism. By IX 2.5, u is even a closed immersion.

Thus G is of finite type, and therefore of finite presentation, over S.[15] Applying Lemma 3.5 in the étale case, there is an open neighborhood U of S_0, hence $U = S$, such that $G|_U \to H|_U$ is étale. Thus u is étale and therefore an open immersion, since it is an étale monomorphism.[16] This completes the proof.

Corollary 3.8. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology; set $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/I)$. Let H be an S-group prescheme of finite type, affine and flat over S.*

Then H is of multiplicative type and isotrivial if and only if H_0 is so and H satisfies either one of the following conditions (a) and (b), which are therefore equivalent under the preceding hypotheses:

(a) *The fibers of H are of multiplicative type and have constant type on each connected component of S.*

(b) *H is commutative and the ${}_nH$, $n > 0$, are finite over S.*

Both of these conditions are implied by the following one:

(c) *The fibers of H are connected.*

Of course, if H is of multiplicative type (and isotrivial), then (a) and (b) hold by IX 1.4.1 a) and 2.1 c). Thus only the "if" direction requires proof. We use the subgroup G provided by 3.7. When (c) holds, plainly $G = H$, and there is nothing to prove.

In case (b), the immersion $u : G \to H$ induces an open immersion

$$ {}_nu : {}_nG \longrightarrow {}_nH$$

[15] Editor's note: The preceding sentence has been added.
[16] Editor's note: Cf. EGA IV_4, 17.9.1.

and an isomorphism

$$({}_nG)_0 \xrightarrow{\sim} ({}_nH)_0.$$

Since ${}_nH$ is finite over S, it follows at once that ${}_nu$ is an isomorphism: the complement of its image is finite over S and reduces to the empty set. By 3.6, the induced morphisms on fibers

$$u_s : G_s \longrightarrow H_s$$

are isomorphisms. Hence u is surjective and therefore an isomorphism.

Finally, in case (a), we may suppose that S is connected.[17] It follows that, for every $s \in S$,

$$u_s : G_s \longrightarrow H_s$$

is a monomorphism of algebraic groups of multiplicative type and of the same type over $\kappa(s)$.[18] Such a homomorphism is necessarily an isomorphism, which again completes the proof.[19] After extending the base field, we may suppose that both groups over $\kappa(s)$ are diagonalizable. The assertion then follows from VIII 3.2 b) and the fact that a surjective homomorphism between isomorphic finitely generated $\mathbb{Z}$-modules

$$M \longrightarrow N$$

is necessarily bijective.

Corollary 3.9. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, set $S = \mathrm{Spec}(A)$, and let H be an S-group prescheme of finite type, affine and flat over S, with connected fibers. Then H is an isotrivial torus if and only if $H_0 = H \times_S \mathrm{Spec}(A/I)$ is an isotrivial torus.*

4. Case of an arbitrary base. Quasi-isotriviality theorem

Let A be a local ring. Recall, following Nagata, that A is said to be *henselian* if every algebra B finite over A is a product of local algebras B_i finite over A.

Recall 4.0.[20] *Let A be a henselian local ring, k its residue field, $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(k)$, and ξ a geometric point of S_0. Then the functor*

$$X \longmapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale coverings of S and the analogous category over S_0 (cf. EGA IV_4, § 18.5). Consequently (cf. SGA 1, V),

$$\pi_1(S_0, \xi) = \pi_1(S, \xi).$$

Suppose in addition that A is noetherian, and write A' for its completion and $S' = \mathrm{Spec}(A')$. Then A' is a complete noetherian local ring, hence henselian (*loc. cit.*, 18.5.14), and the functor

$$X \longmapsto X' = X \times_S S'$$

[17] Editor's note: The original said "that is, S_0 connected." Since A is separated for the I-adic topology, S_0 meets every connected component of S; and since A is complete, the connected components of S_0 and S are in bijection.

[18] Editor's note: This is because G_s and H_s have the same type for every $s \in S_0$, and hence for every s.

[19] Editor's note: Indeed, u will again be a surjective open immersion.

[20] *Editors' note.* This recall has been spelled out, and Remark 4.0.1 has been added.

from the category of étale coverings of S to the analogous category over S' fits into a commutative diagram

$$\begin{array}{ccc} \text{ÉtCov}(S) & \longrightarrow & \text{ÉtCov}(S') \\ & \searrow^{\sim} \quad \swarrow_{\sim} & \\ & \text{ÉtCov}(S_0) & \end{array}$$

and is therefore also an equivalence of categories, whence $\pi_1(S_0, \xi) = \pi_1(S', \xi)$.

Remark 4.0.1. Since S is connected (A being local), it follows from 1.2 that the category of isotrivial groups of multiplicative type over S is equivalent to the analogous category over S_0 (and also over S' if A is noetherian).

Lemma 4.1. *Let A be a henselian local ring with residue field k, $S = \mathrm{Spec}(A)$, and $S_0 = \mathrm{Spec}(k)$.*

(i) *The functor*

$$H \longmapsto H_0 = H \times_S S_0$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S_0.

(ii) *If, in addition, A is noetherian, writing A' for its completion and $S' = \mathrm{Spec}(A')$, the functor*

$$H \longmapsto H' = H \times_S S'$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S'.

As in 4.0, the second assertion follows from the first; let us prove the first. We already know that the functor under consideration is essentially surjective: every group of multiplicative type H_0 over $S_0 = \mathrm{Spec}(k)$, being finite and hence of finite type over k, is isotrivial by 1.4, and therefore comes from an isotrivial group of multiplicative type over S, by 4.0.1.

It remains to prove full faithfulness, that is, that for two finite groups G, H of multiplicative type over S, the following map is bijective:

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(G_0, H_0).$$

In other words,[21] if F denotes the S-functor $\mathrm{Hom}_{S\text{-gr.}}(G, H)$, the natural map

$$\mathrm{Hom}_S(S, F) \longrightarrow \mathrm{Hom}_{S_0}(S_0, F_0)$$

induced by the base change $S_0 \to S$ is bijective. In view of Recall 4.0, it is enough to prove the following lemma.

Lemma 4.2. *Let G, H be two finite groups of multiplicative type over a prescheme S. Then $F = \mathrm{Hom}_{S\text{-}gr.}(G, H)$ is representable by a finite étale scheme over S.*

[22] Let $f : S' \to S$ be a faithfully flat quasi-compact morphism such that G' and H' are diagonalizable. It is enough to show that $F_{S'}$ is represented by a prescheme X' étale and finite (hence affine) over S'. Indeed, the descent datum on X' relative to f (cf. VIII 1.7.2) is then effective (by SGA 1, VIII 2.1), so there is an S-prescheme X such that

[21] *Editors' note.* What follows has been added.

[22] *Editors' note.* The original has been expanded in what follows.

$X \times_S S' = X'$, representing F, and X is étale and finite over S (cf. SGA 1, V 5.7 and EGA IV_4, 17.7.3 (ii)).

We may therefore suppose that $G = D_S(M)$ and $H = D_S(N)$, where M and N are finite commutative groups (cf. VIII 2.1 c)). Then $K = \mathrm{Hom}_{\mathrm{gr.}}(N, M)$ is a finite abelian group and, by VIII 1.5, there is an isomorphism

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \simeq K_S,$$

which completes the proof of 4.2 and hence of 4.1.

Proposition 4.3.0.[23] *Let S be a henselian local scheme, s its closed point, $g : X \to S$ a morphism locally of finite type, and x an isolated point of the fiber X_s.*

(i) *Then $\mathcal{O}_{X,x}$ is finite over $\mathcal{O}_{S,s}$. (In particular, if the residue-field extension $\kappa(x)/\kappa(s)$ is trivial, then $\mathcal{O}_{S,s} \to \mathcal{O}_{X,x}$ is surjective, by Nakayama's lemma.)*

(ii) *If g is also separated, then $X' = \mathrm{Spec}(\mathcal{O}_{X,x})$ is an open and closed subscheme of X; that is, there is a decomposition into a disjoint sum $X = X' \amalg X''$.*

Proof. By the local form of Zariski's Main Theorem found in [Pes66], or in [Ray70, Ch. IV, Th. 1], x has an affine open neighborhood $U = \mathrm{Spec}(B)$, of finite type and quasi-finite over $A = \mathcal{O}_{S,s}$, and there is an open immersion $U \hookrightarrow Y = \mathrm{Spec}(C)$, where C is a finite A-algebra. (N.B. To obtain this conclusion one may also use Chevalley's semicontinuity theorem (EGA IV_3, 13.1.4), followed by the form of Zariski's Main Theorem given in *loc. cit.*, 8.12.8.)

Since A is henselian, Y is the disjoint sum of local schemes $Y_1, \ldots, Y_n$, each finite over S, and the points of Y above s are their closed points $y_1, \ldots, y_n$. Thus $x = y_i$ for some i, and $\mathcal{O}_{X,x} = \mathcal{O}_{U,x} = C_i$ is finite over A. Moreover, $X' = \mathrm{Spec}(C_i)$ is an open subscheme of U, hence of X.

Suppose in addition that g is separated. Since $X' \to S$ is finite (C_i being finite over A), so is the immersion $X' \to X$ (cf. EGA II, 6.1.5 (v)); therefore X' is also closed in X. □

Lemma 4.3. *Let A be a noetherian henselian local ring, A' its completion, $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, and s the closed point of S. Let G, H be two S-preschemes in groups, with G of multiplicative type and of finite type over S, and with H locally of finite type and separated over S, H_s of multiplicative type, and H flat over S at the points of H_s.*

Let G', H' be obtained from G, H by the base change $S' \to S$. Then the following natural map is bijective:

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{S'\text{-gr.}}(G', H').$$

Since S' is faithfully flat and quasi-compact over S, SGA 1, VIII 5.2 shows that this map identifies the first member with the subset of the second formed by the $u' : G' \to H'$ for which the two inverse images $u'_1, u'_2 : G'' \to H''$ of u' on $S'' = S' \times_S S'$, under the projections $\mathrm{pr}_1, \mathrm{pr}_2 : S'' \to S'$, are equal.

It therefore remains to prove that $u'_1 = u'_2$ for every homomorphism of S'-groups $u' : G' \to H'$. By the density theorem IX 4.8, it is enough to prove that u'_1 and u'_2 agree on ${}_nG''$ for every integer $n > 0$. (N.B. Here the density theorem is needed essentially in a case where the base prescheme, namely S'', is not locally noetherian.)

Replacing G by ${}_nG$, this reduces us to the case in which $n \cdot \mathrm{id}_G = 0$ for some integer $n > 0$, and hence G is finite over S. Likewise let ${}_nH$ be the kernel of the n-th power

[23] *Editors' note.* This recall, which is used in the proof of 4.3, has been added (the reference to EGA IV_4, 18.5.11 was not entirely satisfactory).

morphism φ_n on H. (N.B. We have not assumed H commutative, so φ_n, respectively ${}_nH$, need not be a group homomorphism, respectively a subgroup of H.) Since ${}_nH$ is defined as the fiber product of $\varphi_n : H \to H$ and the unit section $\varepsilon : S \to H$, its formation commutes with every base change $T \to S$:

$$({}_nH)_T = {}_n(H_T).$$

[24] We use a right subscript m for reduction modulo $\mathfrak{m}^{m+1}$, where $\mathfrak{m}$ is the maximal ideal of A. Then H_m is flat over S_m (because H is flat over S at the points of H_0); hence, by 2.4, H_m is of multiplicative type and of finite type, since H_0 is. Thus every $({}_nH)_m = {}_n(H_m)$ is a subgroup of H_m, of multiplicative type, finite and flat over S_m.

In particular, $({}_nH)_0 = {}_n(H_0)$ is finite over S_0. Since S is henselian local and ${}_nH$, like H, is locally of finite type and separated over S, Proposition 4.3.0 shows that the local rings of ${}_nH$ at the points of $({}_nH)_0$ are finite over S, and that there is a decomposition as a disjoint sum of open subsets

$$(*) \qquad {}_nH = {}_nH^+ \amalg {}_nH^-,$$

where $Z = {}_nH^+$ is finite over S, while ${}_nH^-$ lies above $S - \{s\}$.

Observe that for every finite S-scheme Y (such as G), every S-morphism $Y \to {}_nH$ factors through Z, and that the formation of the decomposition $(*)$ commutes with the base change $S' \to S$, where $S' = \mathrm{Spec}(A')$ and A' is the completion of A.

Then $Z' = Z \times_S S'$ is finite over S', as is $P' = Z' \times_{S'} Z'$. Let ν denote the restriction to P' of the multiplication on H', and let σ be the automorphism of P' interchanging the two factors. Since P' is finite over S', while H' is separated and locally of finite type over S', EGA II 5.4.3 and IV_1 1.1.3 show that $Y = \nu(P')$ is a closed subscheme of H', universally closed and quasi-compact, hence of finite type and therefore proper over S'. Moreover, $Y \to S'$ has finite fibers (because $P' \to S'$ does). Since S' is noetherian, $Y \to S'$ is therefore finite (cf. EGA III, 4.4.2). Since $Z' \subseteq Y$ and $Z'_m = Y_m$ for every m, Lemma IX 5.0 gives $Z' = Y$; thus Z' is a subgroup of H'. Similarly, the kernel $K = \mathrm{Ker}(\nu, \nu \circ \sigma)$ is a closed subscheme of P' such that $K_m = P'_m$ for every m (because $Z'_m = {}_nH_m$ is commutative). Hence $K = P'$, that is, Z' is a commutative subgroup of H'.

On the other hand, every reduction $Z'_m = {}_nH_m$ is flat over S_m. The "local criterion for flatness" (cf. EGA 0_{III}, 10.2.2, or [BAC], III § 5, Example 1 and Th. 1) therefore shows that Z' is flat over S'. Moreover, $Z'_0 = {}_nH_0$ is a finite group of multiplicative type over S_0, and hence isotrivial by 1.4. It follows from 3.8 b) that Z' is of multiplicative type (and isotrivial) over S'. Since $S' \to S$ is faithfully flat and quasi-compact, the multiplication $Z \times_S Z \to H$ factors through Z and makes Z a subgroup of H, finite over S and of multiplicative type (because Z' is).

Finally, by the observations made above concerning $(*)$, Z' is the "finite part" of ${}_nH'$, and therefore $u' : G' \to {}_nH'$ takes its values in Z'. Since Z, as well as $G = {}_nG$, is finite over S and of multiplicative type, Lemma 4.1 shows that u' comes from a $u : G \to Z$, and hence $u''_1 = u''_2$. This completes the proof of 4.3.

Lemma 4.4.0.[25] *Let A be a ring, $S = \mathrm{Spec}(A)$, and let H be an S-group scheme of multiplicative type and of finite type, quasi-isotrivial (respectively isotrivial). Let (A_i) be a filtered family of subrings of A whose inductive limit is A, and put $S_i = \mathrm{Spec}(A_i)$.*

Then there are an index i and an S_i-group scheme H_i, of multiplicative type and of finite type, quasi-isotrivial (respectively isotrivial), such that $H = H_i \times_{S_i} S$.

[24] *Editors' note.* The original has been expanded in what follows, on the basis of indications by M. Raynaud.

[25] *Editors' note.* This lemma, used several times below, has been added.

Theorem 4.4. *Let S be a prescheme, let H be an affine S-prescheme in groups of finite presentation over S, and let $s \in S$. Suppose:*

α) H is flat over S at the points of H_s;

β) H_s is of multiplicative type.

Then there exist an étale morphism $S' \to S$, a point $s' \in S'$ above s such that $\kappa(s')/\kappa(s)$ is trivial, and an open and closed subgroup G' of $H' = H \times_S S'$, of multiplicative type and finite type, and isotrivial, such that $G'_{s'} = H'_{s'}$.

[26] **a)** Write provisionally $T = \mathrm{Spec}(\mathcal{O}_{S,s})$, and let us show, by "descent of the conclusions", that one may reduce to the case $S = T$. Suppose indeed that we have found an étale morphism $g : T' \to T$, a point $s' \in T'$, and a subgroup G' of $H' = H_{T'}$ satisfying the conditions of the statement. Replacing T' by an affine open neighborhood of s', we may assume that T' is of finite presentation over T.

Since G' is isotrivial, there is a finite étale surjective morphism $f : \widetilde{T} \to T'$ such that $\widetilde{G} = G' \times_{T'} \widetilde{T}$ is isomorphic to $D_{\widetilde{T}}(M)$, where M is a finitely generated abelian group. Since T', H, $\widetilde{T}$, and G' are of finite presentation over $T = \mathrm{Spec}(\mathcal{O}_{S,s})$, and $\mathcal{O}_{S,s}$ is the inductive limit of the subalgebras $A_i = \mathcal{O}_S(S_i)$, where S_i runs through the affine open neighborhoods of s in S, EGA IV$_3$, 8.8.2 (and Exposé VI$_\mathrm{B}$, 10.2 and 10.3) gives an index i, S_i-preschemes $S'_i, \widetilde{T}_i$ of finite presentation (respectively an S_i-prescheme in groups G'_i of finite presentation), and morphisms

$$g_i : S'_i \to S_i, \qquad f_i : \widetilde{T}_i \to S'_i, \qquad u_i : G'_i \to H \times_S S'_i,$$

from which g, f, respectively $u : G' \to H'$, are obtained by base change. Taking i large enough, g_i is étale, f_i is finite étale surjective, and u_i is an open and closed immersion (cf. EGA IV, 8.10.5 and 17.7.8).

Now $\widetilde{G}$ comes from both $\widetilde{G}_i = G'_i \times_{S'_i} \widetilde{T}_i$ and $D_{\widetilde{T}_i}(M)$. By EGA IV$_3$, 8.8.2 (i) (and VI$_\mathrm{B}$, 10.2), there is an index $j \geq i$ such that $\widetilde{G}_j \simeq D_{\widetilde{T}_j}(M)$, so G'_j is an isotrivial group of multiplicative type. Let s'_j be the image of s' in S'_j. Then the étale morphism $S'_j \to S_j \hookrightarrow S$, the point s'_j, and the open and closed subgroup G'_j of $H \times_S S'_j$ satisfy the conditions of the statement. Thus we may assume that $S = \mathrm{Spec}(A)$ is local, with closed point s.

b) The ring A is the inductive limit of local rings A_i which are localizations of finitely generated $\mathbb{Z}$-algebras; put $S_i = \mathrm{Spec}(A_i)$. Let us show that hypotheses α and β "descend" to some A_i. Since $H \to S$ is of finite presentation, the set of $x \in H$ at which H is flat over S is an open set W. It contains H_s by hypothesis, and hence contains a quasi-compact open set V containing H_s (since the affine scheme H_s can be covered by finitely many affine open subsets contained in W). The open immersion $\tau : V \hookrightarrow H$ is of finite presentation, so $V \to S$ is also of finite presentation.

By EGA IV$_3$, 8.8.2 (and Exposé VI$_\mathrm{B}$, 10.2 and 10.3), there are an index i, an S_i-prescheme V_i of finite presentation (respectively an S_i-prescheme in groups H_i of finite presentation), and an S_i-morphism $\tau_i : V_i \to H_i$, from which V, H, τ are obtained by base change $S \to S_i$. Taking i large enough, τ_i is an open immersion and V_i is flat over S_i, by EGA IV$_3$, 8.10.5 and 11.2.6. Let s_i be the image of s in S_i. Since the open immersion $(V_i)_{s_i} \hookrightarrow (H_i)_{s_i}$, after base change $\kappa(s_i) \to \kappa(s)$, gives the equality $V_s = H_s$, one already has $(V_i)_{s_i} = (H_i)_{s_i}$. Thus H_i is flat over S_i at the points of $(H_i)_{s_i}$. Finally, $(H_i)_{s_i}$ is of multiplicative type, because

$$H_s = (H_i)_{s_i} \otimes_{\kappa(s_i)} \kappa(s)$$

[26] *Editors' note.* The reductions that follow have been spelled out (the original said: "One reduces at once to the case where S is the spectrum of a noetherian local ring A, and where s is the closed point of S.")

is. Hence (S_i, H_i, s_i) satisfies the hypotheses of 4.4, and if the desired assertion holds for this triple, it also holds for (S, H, s) by base change. We are reduced to the case in which A is local and noetherian. We now distinguish two cases.

1° *A is local noetherian and henselian.* Let $\widehat{A}$ be its completion, $\widehat{S} = \mathrm{Spec}(\widehat{A})$, and $\widehat{H} = H \times_S \widehat{S}$. Applying Theorem 3.7, we find an isotrivial $\widehat{S}$-group $\widehat{G}$ of multiplicative type and of finite type, together with a homomorphism

$$\widehat{u} : \widehat{G} \longrightarrow \widehat{H}$$

which is both an open and a closed immersion, and which induces an isomorphism

$$\widehat{u}_0 : \widehat{G}_0 \xrightarrow{\sim} \widehat{H}_0.$$

By Remark 4.0.1, base change along $\widehat{S} \to S$ induces an equivalence between the categories of isotrivial groups of multiplicative type over S and over $\widehat{S}$. In particular, $\widehat{G}$ comes from an isotrivial S-group G of multiplicative type and of finite type. By 4.3, $\widehat{u}$ comes from a homomorphism

$$u : G \longrightarrow H.$$

Moreover, u is an open and closed immersion and induces an isomorphism $u_0 : G_0 \to H_0$, since this is true after the faithfully flat quasi-compact base change $\widehat{S} \to S$. This proves 4.4 in this case (taking $S' = S$ and $s' = s$ in its conclusion).

2° *A is local noetherian.* The reduction to case 1° is immediate, by applying that case to the henselization A^h of A. More precisely, one sees easily (using SGA 1, I § 5 *) that the local rings $\mathcal{O}_{S',s'}$ of étale S-preschemes S', equipped with a point s' above s for which $\kappa(s')/\kappa(s)$ is trivial, form a filtered inductive system whose inductive limit is a noetherian henselian local ring A^h (called the henselization of A); for details of this construction, due to Nagata in the normal case, see SGA 4, VIII § 4. The standard arguments of EGA IV$_3$, § 8 then allow one, as in part a), to deduce from the known result over the inductive limit A^h of the $\mathcal{O}_{S',s'}$ an analogous result over one of the pairs (S', s'). This proves 4.4.

Corollary 4.5. *Let S be a prescheme and H an S-prescheme in groups of multiplicative type and of finite type. Then H is quasi-isotrivial; that is, H is split by an étale surjective morphism $S' \to S$.*

Indeed, let $s \in S$. By 4.4, there are an étale morphism $S' \to S$, a point $s' \in S'$ above s with $\kappa(s) = \kappa(s')$, and an isotrivial subgroup G' of H', of multiplicative type and of finite type, such that $G'_{s'} = H'_{s'}$. Since G' and H' are of multiplicative type and finite type, IX 2.9 gives an open neighborhood U' of s' such that $G'|_{U'} = H'|_{U'}$.

[27] Suppose in addition that S is henselian local, with closed point s. By EGA IV$_4$, 18.5.11 b), there is a section σ of $S' \to S$ such that $\sigma(s) = s'$. (N.B. One can see directly that $B = \mathcal{O}_{S',s'}$ equals $A = \mathcal{O}_{S,s}$ as follows. By 4.3.0 (i), $B \simeq A/I$. Since B is a flat A-algebra of finite presentation, I is a finitely generated ideal (cf. EGA IV$_1$, 1.4.7), and $I = I\mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal of A; hence $I = 0$.) Thus H is already isotrivial. We obtain:

Corollary 4.6. *Let A be a henselian local ring, k its residue field, and π the topological Galois group of an algebraic closure of k.*

(i) *Every group of multiplicative type and of finite type over $S = \mathrm{Spec}(A)$ is isotrivial.*

*Or EGA IV$_4$, § 18.6.

[27] *Editors' note.* The original has been expanded in what precedes and what follows.

(ii) *The category of these groups over S is equivalent, through the functor $H \mapsto H \otimes_A k$, to the analogous category over $S_0 = \mathrm{Spec}(k)$. It is therefore anti-equivalent, by 1.4, to the category of Galois modules under π which are finitely generated as $\mathbb{Z}$-modules.*

Corollary 4.7. *Under the hypotheses of 4.4, suppose in addition that one of the following conditions holds:*

a) for every generization t of s, H_t is of multiplicative type and has the same type as H_s;

b) H is commutative and the ${}_nH$, $n > 0$, are finite over S in a neighborhood of s;

c) for every generization t of s, the fiber H_t is connected.

Then there is an open neighborhood U of s such that $H|_U$ is of multiplicative type.

Since an étale morphism is open, it is enough to show, with the notation of the conclusion of 4.4, that there is an open neighborhood U' of s' such that $G'|_{U'} = H'|_{U'}$. Put $S'' = \mathrm{Spec}(\mathcal{O}_{S',s'})$. Since G' and H' are of finite presentation over S', EGA IV$_3$, 8.8.2 reduces the matter to showing that $G'' = H''$. We may therefore assume that $S = S''$; then conditions a), b), c) become those of the following lemma, whose proof is the same as that of 3.8.

Lemma 4.7.1. *Let S be a local scheme with closed point s, let H be an S-group scheme of finite type, and let G be an open and closed subgroup of H, of multiplicative type, such that $G_s = H_s$. Suppose in addition that one of the following conditions holds:*

a) the fibers of H are of multiplicative type and have the same type as H_s;

b) H is commutative and the ${}_nH$, $n > 0$, are finite over S;

c) the fibers of H are connected.

Then $G = H$.

Corollary 4.8. *Let S be a prescheme and let H be an affine S-prescheme in groups, flat and of finite presentation over S, with fibers of multiplicative type.*

For H to be of multiplicative type, it is necessary and sufficient that it satisfy one of the following two conditions (which are equivalent under the preceding hypotheses):

a) the type of H_s (cf. IX 1.4) is a locally constant function of $s \in S$;

b) H is commutative and the ${}_nH$, $n > 0$, are finite over S.

Moreover, a) and b) are implied by:

c) the fibers of H are connected.

In particular:

Corollary 4.9. *Let S be a prescheme and H an S-prescheme in groups, flat and of finite presentation over S. Suppose in addition that H is affine over S and has connected fibers.*[28] *If $s \in S$ is such that H_s is a torus, there is an open neighborhood U of s such that $H|_U$ is a torus.*

In particular, if all the fibers of H are tori, H is a torus.

5. Scheme of homomorphisms from one group of multiplicative type to another. Twisted constant groups and groups of multiplicative type

Definition 5.1. *a) Let S be a prescheme and R a prescheme in groups over S. We say that R is a* twisted constant group *over S if it is locally isomorphic, for the faithfully flat*

[28] *Editors' note.* This result is generalized in 8.1: it is enough to suppose that the fibers of H are affine and connected.

quasi-compact topology, to a constant group scheme, that is, one of the form M_S for an ordinary group M.

b) A twisted constant group R over S is said to be quasi-isotrivial*, respectively* isotrivial*, respectively* locally isotrivial*, respectively* locally trivial*, respectively* trivial*, if in the preceding definition one can replace the faithfully flat quasi-compact topology by the étale topology, respectively the global finite étale topology, respectively the finite étale topology, respectively the Zariski topology, respectively the coarsest (or "chaotic") topology; cf. IX 1.1 and 1.2, and IV 6.6.*

Thus, to say that R is quasi-isotrivial (respectively isotrivial) means that there is an étale surjective (respectively also finite) morphism $S' \to S$ such that $R' = R \times_S S'$ is a constant group over S'; to say that R is trivial means that R is a constant group.

c) As in VIII 1.4, one defines the type *of a twisted constant group R over S at a point $s \in S$. It is an isomorphism class of ordinary groups and, as s varies, is a locally constant function of s, hence a constant function if S is connected. We also say that R is "of type M" if all the fibers of R are of type M.*

One should beware that R is quasi-compact over S only if it is finite over S, that is, only if its type at every $s \in S$ is a finite group.[29]

d) The case of greatest interest to us is that in which R is commutative. We then say that R is finitely generated *if its type at every point $s \in S$ is given by a finitely generated $\mathbb{Z}$-module. This notion must not be confused with the scheme- theoretic notion "R of finite type over S" (cf. above).*

Remark 5.2. We shall also have to consider S-preschemes X which are locally isomorphic, for the faithfully flat quasi-compact topology, to constant preschemes, independently of any group structure. We then call X a *twisted constant bundle* over S, and extend to these preschemes the terminology introduced in 5.1. Of course, when X is endowed with an S-group structure, the meaning of the expressions "twisted constant", "isotrivial", and so on changes according as one does or does not take the group structure over S into account. The same applies when X carries any other kind of algebraic structure (for example, that of a Galois principal bundle, to be considered in the next section).

Proposition 5.3. *Let R be a commutative twisted constant group over S.*

(i) *The functor $H = D_S(R)$ (cf. VIII 1) is representable and is a group of multiplicative type over S.*

(ii) *For every $s \in S$, the type of R at s equals the type of H at s.*

(iii) *The group R is quasi-isotrivial (respectively isotrivial, respectively trivial, respectively locally isotrivial, respectively locally trivial) if and only if H is so.*

Remark 5.3.1.[30] One may be concerned that in 5.3 the word "type" is being used in two different senses, according as it refers to R or to H. Fortunately, when an S-prescheme in groups G is both a twisted constant group and a group of multiplicative type, its type in either sense is the same, thanks to the fact that an ordinary finite commutative group is isomorphic to its dual.

Proof. Covering families for the faithfully flat quasi-compact topology are families of effective descent for the fibered category of affine preschemes in groups over variable base preschemes (SGA 1, VIII 2.1). Consequently H is representable (and affine over S), since this is true "locally"[31] (when R is constant, H is a diagonalizable group).

[29] *Editors' note.* See Lemma 5.12 below.

[30] *Editors' note.* This remark, which appeared after the proof in the original, has been moved here.

[31] *Editors' note.* Cf. VIII § 1.7.

That H is of multiplicative type is then immediate from the definition, as is the equality of the types of R and H at $s \in S$. Finally, since

$$H_{S'} = D_{S'}(R_{S'}),$$

the last assertion reduces to the "trivial" case, namely to checking that R is trivial if and only if H is trivial; this follows at once from the biduality theorem VIII 1.2. □

To complete the description of the correspondence between twisted constant groups and groups of multiplicative type, one must start with a group of multiplicative type H and study $R = D_S(H)$. If R is representable, it is plainly a twisted constant group and $H \simeq D_S(R)$. In other words:

Scholium 5.4.0. *The functor $R \mapsto D_S(R)$ is an anti-equivalence*[32] *between the category of twisted constant groups over S and the category of groups of multiplicative type H over S for which $D_S(H)$ is representable.*

I do not know whether this condition on H is always satisfied. We shall see, however, that it is satisfied when H is quasi-isotrivial, and in particular when H is of finite type.

Lemma 5.4. *Let $S' \to S$ be a faithfully flat morphism locally of finite presentation, and let X' be an S'-prescheme separated, locally of finite presentation, and locally quasi-finite over S'.*

Then every descent datum on X' relative to $S' \to S$ is effective: there are an S-prescheme X and an S'-isomorphism $X \times_S S' \simeq X'$ compatible with the descent datum.

We have already observed that the hypothesis on $S' \to S$ implies that it is a covering morphism for the faithfully flat quasi-compact topology (IV 6.3.1 (iv)), and hence *a fortiori* a morphism of effective descent.

When $X' \to S'$ is quasi-compact, and hence of finite presentation and quasi-finite, it is a quasi-affine morphism (cf. SGA 1, VIII 6.2[33]), and effectivity follows in this case from SGA 1, VIII 7.9. In the general case, one reduces at once to the case where S and S' are affine. Cover X' by affine open subsets U'_i, and let V'_i be the saturation of U'_i under the equivalence relation on X' defined by the descent datum:

$$V'_i = q_2(q_1^{-1}(U'_i)).$$

Here q_1, q_2 are the two projections to X' of

$$X''_1 = X' \times_{S'} (S'', p_1) \quad \text{and} \quad X''_2 = X' \times_{S'} (S'', p_2),$$

respectively: $q_1 = \mathrm{pr}_1$, and q_2 is obtained from the first projection of X''_2 by means of the given descent isomorphism $X''_1 \simeq X''_2$. Since $S' \to S$ is faithfully flat, quasi-compact, and locally of finite presentation, the same is true of

$$p_1 : S'' = S' \times_S S' \longrightarrow S',$$

and hence of q_1, q_2. These are consequently open morphisms (SGA 1, IV 6.6[34]). Therefore V'_i is an open quasi-compact part of X'. By the case already proved, the induced descent data on the V'_i are effective; hence so is the descent datum on X' (SGA 1, VIII 7.2).

Corollary 5.5. *A faithfully flat morphism of finite presentation $S' \to S$ is a morphism of effective descent for the fibered category of twisted constant groups over variable base preschemes.*

[32] *Editors' note.* "Equivalence" has been corrected to "anti-equivalence".
[33] *Editors' note.* When S' is locally noetherian; EGA IV$_3$, 8.11.2 in general.
[34] *Editors' note.* When S' is locally noetherian; EGA IV$_2$, 2.4.6 in general.

Indeed, this amounts to the effectivity assertion of 5.4 when X' is a constant S'-prescheme.

Theorem 5.6. *Let S be a prescheme, and let G, H be two quasi-isotrivial S-preschemes in groups of multiplicative type, with G*[35] *of finite type.*

Then $\mathrm{Hom}_{S\text{-}gr.}(H, G)$ *is representable and is a quasi-isotrivial twisted constant group over S;*[36] *for every $s \in S$, if the type of G at s is M and the type of H at s is N, then the type of* $\mathrm{Hom}_{S\text{-}gr.}(H, G)$ *is* $\mathrm{Hom}_{\mathrm{gr.}}(M, N)$.

The proof proceeds as in 4.2, using the fact that the assertion is known (VIII 1.5) when G and H are trivial. The required effectivity criterion is provided by 5.5, applied to an étale surjective morphism $S' \to S$.

Taking $G = \mathbb{G}_{m,S}$, in particular, gives:

Corollary 5.7. (i) *Let H be a quasi-isotrivial S-group of multiplicative type. Then the S-group $D_S(H)$ is representable and is a quasi-isotrivial twisted constant group over S.*

(ii) *The functors $R \mapsto D_S(R)$ and $H \mapsto D_S(H)$ are mutually quasi-inverse anti-equivalences between the category of quasi-isotrivial twisted constant S-groups and the category of quasi-isotrivial S-groups of multiplicative type.*

(iii) *These functors induce anti-equivalences between the subcategories of isotrivial, respectively locally isotrivial, respectively locally trivial, respectively trivial groups.*

The last assertion is included only for reference, since it is already contained in 5.3.

Moreover, by 4.5 every S-group of multiplicative type and of finite type is quasi-isotrivial. Theorem 5.6 therefore gives:

Corollary 5.8. *Let S be a prescheme, and let G, H be two S-preschemes in groups of multiplicative type and of finite type. Then* $\mathrm{Hom}_{S\text{-}gr.}(H, G)$ *is representable and is a finitely generated quasi-isotrivial twisted constant S-group.*

Also, in 5.3, R is finitely generated if and only if $H = D_S(R)$ is of finite type (IX 2.1 b)). By 4.5, H is then quasi-isotrivial, and hence so is R. Thus:

Corollary 5.9. *The functors of 5.7 induce mutually quasi-inverse anti-equivalences between the category of S-groups H of multiplicative type and of finite type and the category of finitely generated twisted constant S-groups R. Moreover, every such group R is quasi-isotrivial.*

Corollary 5.10. *Let H, G be two S-preschemes in groups of multiplicative type and of finite type.*

(i) $\mathrm{Isom}_{S\text{-}gr.}(H, G)$ *is representable by an open and closed subprescheme of* $\mathrm{Hom}_{S\text{-}gr.}(H, G)$, *and is a twisted constant S-prescheme.*

(ii) *In particular,* $\mathrm{Aut}_{S\text{-}gr.}(H)$ *is representable and is a twisted constant S-group, in general noncommutative.*

This follows from 5.8 and VIII 1.6.[37]

Recall 5.11.0.[38] Recall that if X is a locally noetherian prescheme, its connected components (which are always closed) are open. Indeed, let C be a connected component of X, let $x \in C$, and let U be a noetherian open neighborhood of x. The prescheme U has

[35] *Editors' note.* H has been corrected to G.

[36] *Editors' note.* What follows has been added.

[37] *Editors' note.* This reference should be corrected by treating the functor $\mathrm{Isom}_{S\text{-gr.}}(H, G)$ in an addition 1.5.1 to VIII ...

[38] *Editors' note.* This recall, used several times below, has been added. See also EGA I, 6.1.9; note, however, that the proof in *loc. cit.* seems unnecessarily complicated.

only finitely many connected components, so the connected component U_x of x in U is open in U, hence in X. Since $U_x \subseteq C$, this shows that C is open in X.

Proposition 5.11. *Let S be a prescheme, let R be a commutative twisted constant group over S, and let $H = D_S(R)$ be the group of multiplicative type that it defines. Consider the following conditions:*

(i) *H is isotrivial (that is, R is isotrivial);*

(ii) *R is a union of open and closed subpreschemes R_i which are quasi-compact over S (and then necessarily finite over S);*

(iii) *the connected components of R are finite over S.*

a) Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii).[39]

b) If S is locally noetherian, then (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii).

c) Finally, if R is finitely generated (that is, if H is of finite type over S), then (i) $\Leftrightarrow$ (ii), *at least when S is quasi-compact or when its connected components are open.*

First decompose S as a prescheme sum of preschemes S_i on each of which H has constant type (cf. IX 1.4.1). We are reduced to the case where H, and therefore R, has constant type M. We shall use:

Lemma 5.12. *Let S be a prescheme and R a twisted constant prescheme over S. Every closed subprescheme Z of R which is quasi-compact over S is finite over S.*

Indeed, we reduce to the case where R is constant, hence of the form I_S for a set I. It is the filtered increasing union of the J_S, where J runs through the finite subsets of I. We may also assume that S is affine. Then Z is quasi-compact, and is therefore contained in some J_S. Since J_S is finite over S, so is Z.

Lemma 5.12 already proves the parenthetical assertion in 5.11 (ii).[40] The implication (ii) $\Rightarrow$ (iii) is then clear, since the connected components of R are closed. If S is locally noetherian, they are open and closed by 5.11.0 (R being étale and hence locally of finite type over S); this gives (iii) $\Rightarrow$ (ii) in that case.

Let us prove (i) $\Rightarrow$ (ii). Choose a finite étale surjective morphism $S' \to S$ which splits H, and hence R (cf. 5.3), so that

$$R' \simeq M_{S'} = \coprod_{m \in M} R'_m,$$

where the R'_m are disjoint open subsets of R', each S'-isomorphic to S'. Let $g : R' \to R$ be the projection. It is a finite étale surjective morphism, and hence both open and closed. The subsets $R_m = g(R'_m)$ are therefore open and closed parts of R, and are plainly quasi-compact over S, since the R'_m are.

Finally, suppose that H is of finite type over S, and let us prove (ii) $\Rightarrow$ (i). If the connected components of S are open, the matter reduces at once to the case where S is connected; thus we may assume that S is quasi-compact or connected. Since M is finitely generated, write

$$M = \mathbb{Z}^r \times N,$$

where $r \geq 0$ and N is a finite abelian group. Let $G = D_S(M)$, and consider the preschemes

$$P = \mathrm{Isom}_{S\text{-gr.}}(H, G) \subset \mathrm{Hom}_{S\text{-gr.}}(H, G) = Q$$

[39] *Editors' note.* Compare the examples of 1.6 ...

[40] *Editors' note.* The original has been expanded in what follows.

(cf. 5.8 and 5.10). There are isomorphisms

$$\begin{aligned} Q &\simeq \mathrm{Hom}_{S\text{-gr.}}(M_S, R) \\ &\simeq \mathrm{Hom}_{S\text{-gr.}}(\mathbb{Z}_S^r, R) \times \mathrm{Hom}_{S\text{-gr.}}(N_S, R) \\ &\simeq R^r \times E, \end{aligned}$$

where

$$E = \mathrm{Hom}_{S\text{-gr.}}(N_S, R)$$

is finite over S.[41] It follows that Q is a union of open and closed subpreschemes Q_i finite over S. Hence P is a union of the open and closed subpreschemes $P_i = P \cap Q_i$, finite over S. Since they are étale over S, their images in S are open and closed parts S_i, and these cover S. If S is connected or quasi-compact, there is a finite set of indices i for which the S_i cover S. Let S' be the union of the corresponding P_i. Then $S' \to S$ is finite étale surjective. Putting

$$P' = P \times_S S' = \mathrm{Isom}_{S'\text{-gr.}}(H', G'),$$

we see that P' has a section over S'; hence there is an isomorphism of S'-groups

$$H' = H \times_S S' \xrightarrow{\sim} G' = G \times_S S' = D_{S'}(M).$$

Thus S' splits H, completing the proof of 5.11.[42]

Remark 5.11 bis. If H is of finite type over S and of constant type, one can also give the following isotriviality criterion, with no restriction on S: $H = D_S(R)$ is isotrivial if and only if R is the union of a sequence of open and closed parts finite over S.[43]

Lemma 5.13. *Let S be a locally noetherian connected prescheme, let P be a quasi-isotrivial twisted constant S-prescheme, and let Z be an open and closed part of P. If Z_s is finite for some $s \in S$, then Z is finite over S.*

First, do not assume S connected. Let U be the set of $s \in S$ for which Z_s is finite. We shall prove that U is both open and closed, and that $Z|_U$ is finite over U. This statement is essentially equivalent to 5.13, but has the advantage of being local in S for the étale topology, say. We are therefore reduced to the case where P is constant, of the form I_S for a set I. (N.B. The locally noetherian hypothesis is preserved by étale base change; this is where the quasi-isotriviality of P over S is used.)

We may also suppose that S is connected, since its connected components are open (S being locally noetherian). Then necessarily $Z = J_S$ for a subset $J \subseteq I$, and consequently $U = \varnothing$ or $U = S$ according as J is infinite or finite. This proves the assertion.

Recalls 5.14.0.[44] Let S be a locally noetherian prescheme and let $\widetilde{S}$ be the normalization of S_{red} (cf. EGA II, 6.3.8). Recall that S is said to be *geometrically unibranch* (cf. EGA 0_{IV}, § 23.2 and IV_2, § 6.15) if the canonical morphism $\widetilde{S} \to S$ is radicial, and hence a universal homeomorphism. In particular, the connected components of S are irreducible.

[41] *Editors' note.* Indeed, it is a twisted constant group of type $\mathrm{End}_{\mathrm{gr.}}(N)$; cf. 5.6 and 5.8.

[42] *Editors' note.* This is modulo verifying (ii) ⇒ (i) when S is locally noetherian and R is not finitely generated ...

[43] *Editors' note.* To spell this out: since M is a finitely generated $\mathbb{Z}$-module, it is countable, and the proof of 5.11 (i) ⇒ (ii) shows that R is the union of a countable family of open and closed parts finite over S. The converse would have to be checked ...

[44] *Editors' note.* The original treated in 5.14 the case where S is locally noetherian and normal, and observed in Remark 5.15 that the argument applies more generally when S is merely geometrically unibranch rather than normal. The statement of 5.14 (and also 5.16) has accordingly been modified, and these "recalls", drawn from EGA IV_4, 18.10.6 and 18.10.7, have been added; they show that the proof of 5.14 applies verbatim to the geometrically unibranch case.

Suppose now that S is connected, and hence irreducible. Let η be its generic point and let $f : P \to S$ be a flat, locally quasi-finite morphism. Let P_i be the irreducible components of P, with generic points ξ_i. Since P is flat over S, every ξ_i lies above η (cf. EGA IV_2, 2.3.4), and

$$(P_i)_\eta = P_i \cap P_\eta$$

is the closure of ξ_i in P_η. Since the fiber P_η is discrete, one has $(P_i)_\eta = \{\xi_i\}$. This applies in particular when f is étale. In that case P is also locally noetherian and geometrically unibranch (cf. EGA IV_4, 17.5.7), so its irreducible components are its connected components and are open (and closed).

Corollary 5.14. *Let S be a locally noetherian geometrically unibranch prescheme, and let P be a quasi-isotrivial twisted constant S-prescheme. Then the connected components of P are finite over S.*

We may plainly suppose that S is connected and hence irreducible, and write η for its generic point. By the preceding recalls, every connected component P_i of P is open and closed and meets the fiber P_η in a single point. Lemma 5.13 therefore applies and shows that every P_i is finite over S.

Theorem 5.16.[45] *Let S be a locally noetherian geometrically unibranch prescheme. Then every S-group of multiplicative type and of finite type is isotrivial.*

Indeed, we may suppose S connected, and hence H of constant type M. It is enough to apply 5.14 to

$$P = \mathrm{Isom}_{S\text{-gr.}}(H, G), \qquad G = D_S(M),$$

and then argue as in the proof of 5.11 (ii) $\Rightarrow$ (i). Alternatively, one may apply 5.14 to $P = R = D_S(H)$ (cf. 5.9), and then use 5.11.

6. Infinite Galois principal bundles and the enlarged fundamental group

(The results of the present section and of the next will not be used again in the remainder of this Seminar.)

Let S be a prescheme. We propose to determine the principal homogeneous bundles P over S whose structural group is of the form G_S, the constant S-group defined by an ordinary group G, not necessarily finite. We shall also call them *Galois principal bundles over S with group G*. We use "principal bundle" in the sense of the faithfully flat quasi-compact topology (cf. Exposé IV, Def. 5.1.5). For such a P, however, the structural morphism $P \to S$ is necessarily étale and surjective, hence a covering for the étale topology; consequently P is also locally trivial for the étale topology (cf. IV, Prop. 5.1.6).[46]

We assume that S is a sum of connected preschemes, that is, that its connected components are open; this reduces us at once to the case in which S is connected. Choose a "geometric point" ξ of S, that is, an S-scheme ξ which is the spectrum of an algebraically closed field $\Omega = \kappa(\xi)$. For every Galois principal bundle P over S with group G, P_ξ is a Galois principal bundle over the algebraically closed field $\kappa(\xi)$, and is therefore trivial. We refine the initial problem by asking for the category of Galois principal bundles over S *pointed* above ξ, that is, endowed with an S-morphism $\xi \to P$, or equivalently with a

[45] *Editors' note.* Remark 5.15, rendered obsolete by the addition of 5.14.0 (cf. the preceding editors' note), has been deleted, and in 5.16 "normal" has been replaced by "geometrically unibranch".

[46] *Editors' note.* Notice that P is assumed to be a prescheme. In the *a priori* more general case of an fpqc sheaf P which is a G_S-torsor, is P necessarily representable?

trivialization of P_ξ. For fixed G, let $\bar{\pi}^1(S,\xi;G)$ denote the set of classes of such bundles, up to an isomorphism respecting the base point. Then the set $\bar{\pi}^1(S;G)$ of isomorphism classes of Galois principal bundles over S with group G, without a specified base point, is the set of orbits of G in $\bar{\pi}^1(S,\xi;G)$. Here G acts naturally by translating the marked point of a pointed Galois principal bundle:

$$\bar{\pi}^1(S;G) = \bar{\pi}^1(S,\xi;G)/G.$$

Let $S' \to S$ be any morphism which is a morphism of universal effective descent for the fibered category of twisted constant preschemes over a variable base (for example, $S' \to S$ faithfully flat and locally of finite presentation, cf. 5.4; for other examples, cf. SGA 1 IX). We propose to determine the subsets

$$\bar{\pi}^1(S'/S,\xi;G) \quad \text{and} \quad \bar{\pi}^1(S'/S;G)$$

of the preceding sets, formed by the Galois principal bundles over S which become trivial over S', or, as one says, are "split" by S'. In fact, we shall determine the category of Galois principal bundles P over S split by S'. Of course,

$$\bar{\pi}^1(S,\xi;G) = \varinjlim_{S'} \bar{\pi}^1(S'/S,\xi;G),$$

where S' runs through a cofinal system among the S'/S which cover for the étale topology (for example, if S is quasi-compact, among the quasi-compact S' over S whose structural morphism is étale and surjective). Likewise, the category of Galois principal bundles over S is the inductive limit of the subcategories defined by the S', formed by the bundles split over S'.

By the assumption on S'/S, the category of Galois principal bundles over S split by S' is equivalent to the category of trivial Galois principal bundles over S', hence of the form $G_{S'}$, with G acting by right translations, equipped with descent data relative to $S' \to S$. In terms of the corresponding trivial bundle P' over S' and its descent datum, a base point on a Galois principal bundle P over S split by S' amounts to a trivialization of $P' \times_{S'} S'_\xi$ compatible with the induced descent datum relative to $S'_\xi \to \xi$, where $S'_\xi = S' \times_S \xi$; equivalently, it is a section σ of P'_ξ over S'_ξ compatible with the descent datum.

It is therefore useful, for an arbitrary S-prescheme S', without now assuming that $S' \to S$ is a morphism of universal effective descent for the fibered category of twisted constant bundles, to define $\bar{\pi}^1(S'/S;G)$ and $\bar{\pi}^1(S'/S,\xi;G)$ as the sets of isomorphism classes of the structures with descent data just specified. These functors in G then have a very simple simplicial description in terms of the fibered square and cube S'' and S''' of S' over S; we shall sketch it below (cf. 6.3).

The important conclusion is the following.

Proposition 6.1. *Suppose that the connected components of S' and S'' are open, and, for example, that the quotient of $\pi_0(S')$ by the equivalence relation induced by the two projections $\pi_0(S'') \to \pi_0(S')$ is a single point.*[47]

(i) *The functor*

$$G \longmapsto \bar{\pi}^1(S'/S,\xi;G)$$

[47] *Editors' note.* The hypothesis that the connected components of S' be open, as well as the second hypothesis, has been added. The latter says that the simplicial set $K_\bullet = \pi_0(S_\bullet)$ defined in 6.3 is connected. In fact a weaker hypothesis suffices: the cone $\widetilde{K}_\bullet$ of a certain morphism of simplicial sets $k_\bullet \to K_\bullet$ need only be connected (cf. *loc. cit.*).

from the category of groups to the category of sets is representable by a group, denoted $\pi_1(S'/S,\xi)$ *and called the* fundamental group of S at ξ relative to $S' \to S$. *Thus there is a functorial bijection*

$$\bar{\pi}^1(S'/S,\xi;G) \simeq \mathrm{Hom}_{\mathrm{gr.}}(\pi_1(S'/S,\xi),G).$$

(ii) *This group has a set of generators in bijection with* $\pi_0(S'')$, *and is described in terms of those generators by relations in bijection with the elements of* $\pi_0(S''')$.[48] *In particular,* $\pi_1(S'/S,\xi)$ *is finitely generated (respectively finitely presented) if* $\pi_0(S'')$ *is finite (respectively if both* $\pi_0(S'')$ *and* $\pi_0(S''')$ *are finite).*

(iii) *The category of Galois principal bundles over* S *split by* S', *with base point above* ξ, *is equivalent to the category of ordinary groups* G *equipped with a homomorphism* $\pi_1(S'/S,\xi) \to G$.

The proof is given below; cf. ...

6.2. When S is a connected locally noetherian prescheme, every étale prescheme S' over S is locally noetherian and therefore has open connected components. It follows from the preceding discussion[49] that the functor

$$G \longmapsto \bar{\pi}^1(S,\xi;G)$$

from ordinary groups to sets is *strictly pro-representable* (cf. *Séminaire Bourbaki*, February 1960, No. 195, §§ A.2 and A.3). Thus there is a projective system

$$\Pi = \Pi_1(S;\xi) = (\pi_i)_{i\in I}$$

of ordinary groups over an increasing filtered index set I, which is "strict", meaning that its transition morphisms $\pi_j \to \pi_i$ are surjective, together with an isomorphism of functors in G

$$\bar{\pi}^1(S,\xi;G) \simeq \varinjlim_i \mathrm{Hom}_{\mathrm{gr.}}(\pi_i,G).$$

The second member is also denoted $\mathrm{Hom}_{\text{pro-gr.}}(\Pi,G)$ (cf. *loc. cit.*).

If the projective limit

$$\pi = \varprojlim_i \pi_i$$

is "large enough", more precisely if the canonical homomorphisms $\pi \to \pi_i$ are surjective, it is natural to equip π with the projective-limit topology of the discrete topologies on the π_i. The preceding isomorphism can then also be written

$$\bar{\pi}^1(S,\xi;G) \simeq \mathrm{Hom}_{\mathrm{gr.\,top.}}(\pi,G),$$

where the right-hand side is the set of homomorphisms of topological groups, G being given the discrete topology.

The hypothesis just formulated on the projective system Π holds, as is well known, when the π_i are finite groups (cf. [BEns], III § 7.4, Th. 1). This last condition plainly also means that every Galois principal bundle over S is isotrivial, that is, split by a finite étale

[48] *Editors' note.* The superfluous hypothesis that the connected components of S''' be open has been deleted. On the other hand, the later description (cf. ...) gives the natural set of generators $\pi_0(S'') \amalg \pi_0(S'_\xi)$; the relations between these generators arising from the 2-cells then reduce it to $\pi_0(S'')$. See, for example, [Kan58], § 19 for a finer description.

[49] *Editors' note.* This deduction could be spelled out: I is in bijection with a cofinal set of morphisms $S' \to S$ covering for the étale topology ...

surjective morphism. This is the case when S is geometrically unibranch (for example normal), as follows at once from 5.14.[50] When the π_i are finite, π also coincides with the fundamental group $\pi_1(S,\xi)$ introduced in SGA 1, V.

Thus, in the favorable case in which the maps $\pi \to \pi_i$ are surjective, one might call π the *enlarged fundamental group of S at ξ*. Outside this favorable case, π itself is of little interest, and the role of the usual fundamental group is played by the projective system Π itself, which we call the *enlarged fundamental pro-group of S at ξ*. (Any better terminological suggestion than "enlarged" is welcome![51]) Knowledge of this pro-group is more precise than knowledge of the usual fundamental group $\pi_1(S,\xi)$ of SGA 1, V; more precisely, the latter is the projective limit of the projective system formed by the finite quotients of the π_i.

6.3. Let us indicate briefly the "calculation" of $\pi_1(S'/S,\xi)$. Let S_i be the $(i+1)$-st fibered power of S' over S, so that $S_0 = S'$, $S_1 = S''$, and so on. The evident simplicial operations between the S_i make $(S_i)_{i\in\mathbb{N}}$ a simplicial object of $(\mathrm{Sch})/S$.

Applying to this simplicial object the functor "set of connected components"

$$\pi_0 : (\mathrm{Sch})/S \longrightarrow (\mathrm{Ens}),$$

we obtain a simplicial set $K_\bullet = (K_i)_{i\in\mathbb{N}}$, with

$$K_i = \pi_0(S_i).$$

Likewise, the $S_{i,\xi}$, the $(i+1)$-st fibered powers of S'_ξ over ξ, form a simplicial object of $(\mathrm{Sch})/\xi$, hence of $(\mathrm{Sch})/S$, together with a natural morphism of simplicial objects into $(S_i)_{i\in\mathbb{N}}$. Thus we obtain a simplicial set $k_\bullet$, with

$$k_i = \pi_0(S_{i,\xi}),$$

and a canonical morphism

$$k_\bullet \longrightarrow K_\bullet.$$

Taking the cone of this morphism (cf. 9.5.1), we form a new simplicial set

$$\widetilde{K}_\bullet = \mathrm{Cone}(k_\bullet \longrightarrow K_\bullet).$$

[52] In this way we obtain a "pointed simplicial set" $\widetilde{K}_\bullet$, that is, a simplicial set equipped with a morphism

$$\widetilde{\xi} : e_\bullet \longrightarrow \widetilde{K}_\bullet,$$

where $e_\bullet$ is the final simplicial set. We may form its familiar combinatorial invariants

$$\pi_0(\widetilde{K}_\bullet, \widetilde{\xi}) \quad \text{and} \quad \pi_1(\widetilde{K}_\bullet, \widetilde{\xi}),$$

whose constructions involve only the components of degree at most 1 and at most 2, respectively. These invariants are defined without any restriction on S or S'. When the connected components of S_0 and S_1 are open and $\widetilde{K}_\bullet$ is connected,[53] one checks without difficulty that $\pi_1(\widetilde{K}_\bullet, \widetilde{\xi})$ represents the functor $G \mapsto \bar{\pi}^1(S'/S,\xi;G)$; that is,

$$\pi_1(S'/S;\xi) \simeq \pi_1(\widetilde{K}_\bullet, \widetilde{\xi}).$$

[50] *Editors' note.* 5.13 has been corrected to 5.14.

[51] *Editors' note.* Some authors speak of the "true" fundamental group.

[52] *Editors' note.* The original, which takes $\widetilde{K}_\bullet = K_\bullet/k_\bullet$, has been corrected: on the one hand $k_\bullet$ is already contractible (cf. 9.7.1), and on the other hand $k_\bullet \to K_\bullet$ is not in general injective. It is an epimorphism if S'/S is finite étale (cf. 9.7.3).

[53] *Editors' note.* The hypothesis that $\widetilde{K}_\bullet$ be connected has been added; for the proof, see the addendum below (Section 9).

We also note that if $S' \to S$ is "universally submersive" (cf. SGA 1, IV 2.1) and the connected components of S' are open, then the simplicial set $K_\bullet$, and hence also $\widetilde{K}_\bullet$, is connected.[54]

Examples 6.4. It remains to give examples of enlarged fundamental groups. Return to the examples of 1.6. Let k be an algebraically closed field; let C_1 be a complete rational curve over k with exactly one singular point, that point being an ordinary double point; and let C_2 be a curve which is the union of two irreducible components, each isomorphic to $\mathbb{P}^1_k$, meeting in exactly two points, which are ordinary double points of C_2. In either case, the enlarged fundamental group of the curve is isomorphic to $\mathbb{Z}$.[55]

More generally, one should take up, simplifying and correcting them, the results of SGA 1, IX 5 in the setting of the enlarged fundamental group. The examples of *loc. cit.*, 5.5 would give as many examples of enlarged fundamental pro-groups which are not profinite. Thus, if in the first example the ordinary double point were replaced by a double point with n distinct branches,[56] the enlarged fundamental group would be the free (discrete!) group on $n-1$ generators.

7. Classification of twisted constant preschemes and of groups of multiplicative type of finite type in terms of the enlarged fundamental group

7.0.

Let S be a prescheme, which we continue to assume locally noetherian, in order to ensure that S, and certain preschemes over S that we shall consider (notably those étale over S, and more generally those locally of finite type over S), are locally connected.

Proposition 7.0.1.— *Every twisted constant prescheme X over S that is locally trivial for the* (fppf) *topology (that is, that is split by a faithfully flat morphism $S' \to S$ locally of finite presentation) is quasi-isotrivial (that is, one may even choose $S' \to S$ étale and surjective).*

Indeed, we may suppose that S is connected, and hence that X is of type I, where I is a fixed set. Thus

$$X' = X \times_S S'$$

is isomorphic to $I_{S'}$, so $I_{S'}$ is endowed with descent data relative to $S' \to S$; in other words, there is an isomorphism

$$I_{S''} \xrightarrow{\sim} I_{S''}$$

[54] *Editors' note.* The sequel has been modified to take account of the correction made above; cf. Editors' note (53). (The original read: "$\pi_0(\widetilde{K}_\bullet, \widetilde{\xi})$ is also canonically isomorphic to the set $\pi_0(S, \xi)$ of connected components of S, pointed by the connected component of ξ in S.")

[55] *Editors' note.* This could be spelled out. First, by 5.14, the enlarged fundamental group of the projective line $\mathbb{P}^1_k$ is zero; that is, $\mathbb{P}^1_k$ is "truly" simply connected. One would then have to extend Corollary 5.4 of SGA 1, IX, and the discussion following it, to the enlarged fundamental group and to the category of principal bundles not necessarily finite. Let Γ_1, Γ_2 be two copies of $\mathbb{P}^1_k$, and let a_i, b_i be two distinct points of Γ_i. Let C_2'' be the disjoint union of Γ_1 and Γ_2, and let C_2' be the curve obtained by identifying a_1 with a_2; then C_2 is obtained from C_2' by also identifying b_1 with b_2. The discussion following *loc. cit.*, extended to the enlarged case, shows that the enlarged fundamental group of C_2', respectively C_2, is zero, respectively $\mathbb{Z}$.

[56] *Editors' note.* This means an n-fold point with n distinct tangents, for example the curve $X^n - Y^n = X^{n+1}$.

satisfying the usual transitivity condition. Now $S'' = S' \times_S S'$ is locally noetherian and hence locally connected. It follows that the automorphisms of $I_{S''}$ correspond to the sections of $G_{S''}$, where $G = \mathrm{Aut}(I)$ is the group of permutations of I.

In this way one obtains descent data on $G_{S'}$ (regarded as a trivial Galois principal bundle) relative to $S' \to S$. By 5.4 these descent data are effective, and therefore give a Galois principal bundle P over S with group G. By construction, P represents the functor $\mathrm{Isom}_S(I_S, X)$ in the category of preschemes over S that are locally noetherian. Consequently, the étale surjective base change $P \to S$ splits X, so X is indeed quasi-isotrivial.

Remark 7.0.2.— One should bear in mind that, even when S is the spectrum of a field, it is not true in general that every twisted constant bundle over S is quasi-isotrivial. For example, one may take X to be the disjoint union of a sequence of schemes of the form $\mathrm{Spec}(k_i)$, where the k_i are separable extensions of k of strictly increasing degrees.

The proof just given shows at the same time that the classification of quasi-isotrivial twisted constant bundles X over S of type I is equivalent to that of Galois principal bundles over S with group $G = \mathrm{Aut}(I)$. Indeed, this is an equivalence of categories.

It can be put into a more convenient form, as in SGA 1, V. To this end, suppose that S is connected and equipped with a geometric point ξ. The enlarged fundamental pro-group $\Pi = \Pi_1(S, \xi)$ is then defined. For every quasi-isotrivial twisted constant bundle X over S, let $I = X(\xi)$ be its set-theoretic fiber at ξ. Thus X is of type I, and is consequently associated, as above, with a Galois principal bundle

$$P = \mathrm{Isom}_S(I_S, X)$$

over S, with group $G = \mathrm{Aut}(I)$.

By the definition of Π, one therefore obtains a canonical homomorphism from Π to G, that is, from one of the Π_i to G. Since G is the group of permutations of $I = X(\xi)$, this says that Π "acts continuously on $I = X(\xi)$", meaning that the π_i, for i sufficiently large, act on I compatibly with the transition morphisms.

We leave it to the reader to verify that every S-morphism $X \to Y$ between quasi-isotrivial twisted constant bundles over S induces a map $X(\xi) \to Y(\xi)$ compatible with the actions of Π, and that the resulting functor is an equivalence of categories:

Proposition 7.0.3.— *Let S be a locally noetherian connected prescheme, let ξ be a geometric point of S, and let $\Pi = \Pi_1(S, \xi)$ be the enlarged fundamental pro-group of S at ξ. Then the functor*

$$X \longmapsto X(\xi)$$

is an equivalence between the category of quasi-isotrivial twisted constant bundles over S and the category of sets on which Π acts continuously.

This functor is compatible with finite sums and finite inverse limits. It follows, for example, that quasi-isotrivial twisted constant groups (or rings, and so forth) over S correspond to ordinary groups (respectively rings, and so forth) on which the pro-group Π acts continuously. In particular:

Corollary 7.0.4.— *The category of quasi-isotrivial twisted constant commutative groups over S is equivalent to the category of "Π-modules", that is, commutative groups M on which Π acts continuously.*

Using 5.7, we now obtain:

Theorem 7.1.— *Let S be a locally noetherian connected prescheme, let ξ be a geometric point of S, and let $\Pi = \Pi_1(S, \xi)$ be the enlarged fundamental pro-group of S at ξ. Then*

the functor

$$G \longmapsto \mathrm{Hom}_{\kappa(\xi)\text{-gr}}(G_\xi, \mathbf{G}_{m,\xi})$$

induces an anti-equivalence between the category of quasi-isotrivial groups of multiplicative type over S and the category of Π-modules.

Using 4.5, we obtain:

Corollary 7.2.— *The preceding functor induces an anti-equivalence between the category of groups of multiplicative type and of finite type over S, and the category of Π-modules that are of finite type over $\mathbb{Z}$.*

Example 7.3.— Take for S, for example, a complete rational curve over an algebraically closed field, having exactly one multiple point with $n+1$ distinct branches. By 6.4, the enlarged fundamental group $\Pi(S,\xi)$ is a free group on n generators. Thus, by 7.2, the classification of tori of relative dimension m over S is equivalent to the classification of systems of n endomorphisms $A_1, \ldots, A_n$ of the $\mathbb{Z}$-module $\mathbb{Z}^m$, up to automorphism of $\mathbb{Z}^m$. Except when $n \leq 1$ or $m \leq 1$, an explicit classification of such systems seems hopeless. One can at least define a multitude of nontrivial invariants for such a system, such as the characteristic polynomials of the A_i.[58]

Remark 7.4.— Without any hypothesis on S, it remains true that, for a given ordinary commutative group M of finite type over $\mathbb{Z}$, the category of groups of multiplicative type of type M over S is anti-equivalent to the category of Galois principal bundles over S with group $G = \mathrm{Aut}_{\mathrm{gr}}(M)$. This follows easily from 5.9 and 5.10.

Remark 7.5.— The theory of the fundamental pro-group sketched in the present two sections is more advantageously formulated in the setting of general sites. In that form it applies equally well, for example, to ordinary topological spaces, and gives a satisfactory theory at least for a locally connected topological space (not necessarily locally simply connected). Here too it seems that one cannot make do with a fundamental group, and that a pro-group is needed. Finally, once one has at one's disposal the language of topologies and descent (which really lies at the heart of these questions), the account sketched here is also technically simpler than that of SGA 1, V, and should therefore in principle replace it.

8. Appendix: Elimination of certain affineness hypotheses

Our aim is to prove the following generalization of 4.9.

Theorem 8.1.— *Let S be a prescheme, and let H be a flat S-prescheme in groups of finite presentation, with connected affine fibers.*[59] *Let $s \in S$, and suppose that H_s is a torus. Then there exists an open neighborhood U of s such that $H|_U$ is a torus.*

We immediately obtain:

Corollary 8.2.— *Let H be an S-prescheme in groups, flat and of finite presentation over S. Then H is a torus if and only if its fibers are tori.*

Remark 8.3.— Even when S is the spectrum of a discrete valuation ring, the hypothesis in 8.1 that the fibers of H (here, the generic fiber) be affine cannot be dropped: there

[58] *Editors' note.* In particular, this shows that the tori of relative dimension 2 considered in 1.6 are non-isotrivial.

[59] *Editors' note.* Observe that the hypotheses imply that H is separated over S, by $\mathrm{VI_B}$, Theorem 5.3 or Corollary 5.5.

are examples of smooth groups over S whose generic fiber is an elliptic curve and whose special fiber is $\mathbf{G}_m$.

Proof of 8.1. We may clearly suppose that $S = \mathrm{Spec}(A)$ is affine. By the standard procedure (cf. EGA IV_2, 8.8.2), this reduces us to the case in which S is moreover noetherian. We begin by proving 8.2 in this case.

By 4.9, we are reduced to proving that H is affine over S. We may therefore suppose[60] that A is noetherian and local. Since its completion $\widehat{A}$ is faithfully flat over A, descent reduces us to the case in which A is a complete noetherian local ring. Let S' be the normalization of S_{red}. Nagata's theorem says that it is finite over S (EGA 0_{IV}, 23.1.5). Moreover, $S' \to S$ is surjective, so

$$H' = H \times_S S' \longrightarrow H$$

is finite and surjective. Hence, to prove that H is affine, it suffices to show that H' is affine (EGA II, 6.7.1).

Replacing S by a connected component of S', we are reduced to the case in which A is a noetherian (semilocal), normal, integral ring.[61] Moreover, after possibly replacing A by its normalization in a finite separable extension of its field of fractions, we may suppose that the generic fiber H_η of H is diagonalizable, that is, that there is an isomorphism

$$u_\eta : H_\eta \xrightarrow{\sim} T_\eta, \qquad T = \mathbf{G}_{m,S}^r.$$

We now have:

Lemma 8.4.— *Let S be a locally noetherian normal irreducible prescheme with generic point η. Let H be a smooth prescheme in groups over S, with connected fibers, let T be a prescheme in groups of multiplicative type and of finite type over S, and let $u_\eta : H_\eta \to T_\eta$ be a homomorphism of algebraic groups over $\kappa(\eta)$. Then u_η extends to a homomorphism of groups $u : H \to T$.*

After possibly replacing S by its normalization in a finite separable extension of its function field, we may suppose that T is diagonalizable (which is already the case in the application we have in mind). Then T is a closed subgroup of a group of the form $\mathbf{G}_{m,S}^r$, reducing us to the case $T = \mathbf{G}_{m,S}$.

It remains only to prove that u_η, regarded as a rational map from H to $T = \mathbf{G}_{m,S}$, is everywhere defined (for then the morphism $u : H \to T$ extending it is necessarily a group homomorphism). We may regard u_η as an invertible section f of the structure sheaf of H_η, and must show that it extends to an invertible section of the structure sheaf of H. Since H is smooth over the normal prescheme S, it is normal (SGA 1, II 3.1). It therefore suffices to find a closed subset Z of H, of codimension at least 2, such that f extends to an invertible section of the structure sheaf of $H - Z$. By localizing at the codimension-one points of S, this at once reduces us to the case in which S is the spectrum of a discrete valuation ring A.

Let t be a uniformizer of A, and let t' be the section of $\mathcal{O}_H$ that it defines, so that the special fiber H_0 equals $V(t')$. By hypothesis, H_0 is smooth and connected over the residue field k. The function f is then a rational function on H with neither zeros nor poles in $H - H_0$. Since $H_0 = V(t')$ is an irreducible divisor, there is an integer $n \in \mathbb{Z}$ such that

$$t'^n f = t^n f$$

[60] *Editors' note.* By EGA IV_2, 8.8.2 once again.

[61] *Editors' note.* The original has been expanded in what follows.

has neither zeros nor poles, and hence is an invertible section of $\mathcal{O}_H$. It therefore defines a morphism $v : H \to \mathbf{G}_{m,S}$. Since $v_\eta = {t'}^n u_\eta$, and u_η is a group homomorphism, v sends the identity section of H to a section of $\mathbf{G}_{m,S}$ whose value at the generic point of S is t^n. Because this is a section of $\mathbf{G}_{m,S}$, t^n must be a unit; thus $n = 0$. Hence v extends u_η, completing the proof of 8.4.

Applying this lemma in the present situation, we obtain a group homomorphism

$$u : H \longrightarrow T = \mathbf{G}_{m,S}^r$$

that induces an isomorphism on the generic fibers. We shall prove that u is an isomorphism.

Lemma 8.5.— *For every integer $n > 0$ prime to the residual characteristic of A, the group*

$$_nH = \ker(n \cdot \mathrm{id}_H)$$

is finite over S.

If n is prime to the residual characteristic of A, then it is prime to all the residual characteristics of the points of S. Consequently, $n \cdot \mathrm{id}_H$ induces an étale morphism on every fiber of H, and hence is itself étale (SGA 1, I 5.9); its kernel $_nH$ is therefore étale over S. On the other hand, $_nH$ is separated over S, because H is. * [62] Moreover, all its fibers have the same rank n^r, since the fibers of H are tori, all of the same dimension r (as H is smooth over S). We conclude that $_nH$ is finite over S (SGA 1, I 10.9, or EGA IV_4, 18.2.9).

By the preceding lemma, $u(_nH)$ is therefore a closed subset of T. On the generic fiber T_η, this subset is identical to $_n(T_\eta)$, so it contains the closure of that subset, namely $_nT$. For every $s \in S$, the groups $_n(T_s)$, as n varies, are dense in T_s; since $u_s(H_s)$ is a closed subset ($\mathrm{VI_B}$, 1.4.2) containing them, we have $u_s(H_s) = T_s$. Thus u is surjective. A surjective homomorphism between tori of the same dimension over a field is flat,[63] and hence u induces a flat morphism on every fiber. Therefore u is flat (SGA 1, I 5.9). Consequently $K = \ker(u)$ is flat over S,[64] and hence equals the closure of its generic fiber K_η. By construction, K_η is the identity group. Since K is separated over S (because H is), its identity section is closed, and it follows that K is the identity group. Thus u is a monomorphism; since we have seen that it is faithfully flat, it is an isomorphism (cf. SGA 1, I 5.1, or EGA IV_4, 17.9.1). This proves that H is a torus, and completes the proof of 8.2.[65]

Remark 8.5.1.— Instead of invoking 8.5, one can also invoke Zariski's "Main Theorem", which directly implies that u is an open immersion and hence an isomorphism.[66]

To prove 8.1, the quasi-isotriviality theorem reduces us to the case in which S is local and s is its closed point,[67] and it remains to prove that, under the other hypotheses, H is then a torus. By 8.2, already proved, we are reduced to showing that the fibers of H are tori. We may suppose that S is the spectrum of a complete noetherian local ring A. By

*By Raynaud's theorem, $\mathrm{VI_B}$, 5.3.

[62] *Editors' note.* This point should be expanded in connection with the modifications in $\mathrm{VI_B}$, §5.

[63] *Editors' note.* This follows, for example, from VIII 3.2(a). More generally, if $f : G \to H$ is a surjective morphism of algebraic groups over a field k, and H is reduced, then f is flat (cf. $\mathrm{VI_B}$, 1.3).

[64] *Editors' note.* The original has been expanded in what follows.

[65] *Editors' note.* At least in the case considered up to this point, namely S locally noetherian.

[66] *Editors' note.* The editors did not understand this remark, since they did not see why u would *a priori* have finite and surjective fibers.

[67] *Editors' note.* Expand this reduction.

3.3, for every $n > 0$ there exists a group of multiplicative type T_n, finite over S, and an isomorphism

$$(T_n)_s \simeq {}_n(H_s),$$

where s is the closed point of S. Arguing as in 3.1 and using the fact that T_n is finite over S,[68] we see that the preceding isomorphism comes from a homomorphism $u_n : T_n \to H$, which is moreover uniquely determined. (Pass to the limit over the artinian quotients of A.)

Moreover, by the uniqueness properties, when m is a multiple of n, the homomorphism $u_n : T_n \to H$ is obtained from $u_m : T_m \to H$ by restriction to ${}_n(T_m) \simeq T_n$. It follows from IX 6.6 that the u_n are monomorphisms, so the T_n are subgroups of H, and for m a multiple of n one has

$$T_n = {}_n(T_m).$$

Thus, for every $t \in S$, the groups $(T_n)_t$ are subgroups of H_t, of type $(\mathbb{Z}/n\mathbb{Z})^r$, where $r = \dim H_s = \dim H_t$, and when m is a multiple of n one has

$$(T_n)_t = {}_n(T_m)_t.$$

The fact that H_t is a torus now follows from:

Lemma 8.6.— *Let H be a smooth affine algebraic group over an algebraically closed field k, and let $(T_n)_{n>0}$ be a family of subgroups of multiplicative type of H, such that for every integer $n > 0$, T_n is of type $(\mathbb{Z}/n\mathbb{Z})^r$, and for every multiple m of n, one has $T_n = {}_n(T_m)$. Under these conditions, H contains a torus of dimension at least r containing the T_n. Consequently, if H is connected and has dimension at most r, then H is a torus of dimension r.*

This is an exercise on affine algebraic groups, which we shall handle by repeated reference to *Bible*. We restrict attention to the T_n for n prime to the characteristic. Let K be the closure of the union of the T_n in H, endowed with the induced reduced structure. Standard arguments then show that K is a commutative algebraic subgroup of H. By *Bible*, §4.5, Theorem 4, K is isomorphic to a product $K_u \times K_s$, with K_u "unipotent" and K_s diagonalizable. Every diagonalizable subgroup of K is contained in K_s, so the T_n are subgroups of K_s, and therefore $K = K_s$. Write $K = D(M)$, where M is an ordinary commutative group of finite type over $\mathbb{Z}$. The inclusion $T_n \subseteq K$ means that M has a quotient group isomorphic to $(\mathbb{Z}/n\mathbb{Z})^r$. Since this holds for every integer n prime to the characteristic of k (powers of a single fixed prime would suffice), M has rank at least r. Thus K contains a torus T of dimension r. When H is connected of dimension r, it follows that $T = H$, completing the proof of 8.6. Theorem 8.1 is now proved.

Remarks 8.7.— Using 8.1, it should not be difficult to give analogous generalizations of 4.7 and 4.8. A more interesting problem would be to study the situation of 8.1 after dropping the hypothesis that the fibers of H be affine. One can show that there is then an open neighborhood U of s such that $H|_U$ is commutative and, for every $t \in U$, the geometric fiber H_t is an extension of an abelian variety by a torus.[69] Of course, in questions of this kind one may restrict to the case in which S is the spectrum of a discrete valuation ring, s its closed point, and t its generic point.

This result can be generalized as follows. For every smooth connected algebraic group G over an algebraically closed field k, a well-known theorem of Chevalley says that G is, in a unique way, an extension of an abelian variety A by a smooth connected affine group

[68] *Editors' note.* Expand this point.

[69] *Editors' note.* Give a reference here?

V. Define the *abelian rank* (respectively, *reductive rank*, *nilpotent rank*, and *semisimple dimension*) of G, denoted $\rho_{\mathrm{ab}}(G)$ (respectively, $\rho_r(G)$, $\rho_n(G)$, and $d_s(G)$), to be the dimension of A (respectively, the dimension of the maximal tori of G, the dimension of the Cartan subgroups[70] of G, and the dimension of the quotient of G, or equivalently of V, by its radical); see *Bible* for all these notions. Also define the *unipotent rank* by

$$\rho_u(G) = \rho_u(V) = \rho_n(G) - \rho_r(G) - \rho_{\mathrm{ab}}(G).$$

When the ground field is not algebraically closed, the same names and notations designate the corresponding invariants of $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k.

Now let G be a smooth group scheme over the spectrum S of a discrete valuation ring. Let ρ_{ab}, and so on (respectively, ρ'_{ab}, and so on), denote the invariants associated with the special fiber (respectively, the generic fiber). Then one has the inequalities

$$\begin{cases} \rho_{\mathrm{ab}} \leq \rho'_{\mathrm{ab}}, \\ \rho_r + \rho_{\mathrm{ab}} \leq \rho'_r + \rho'_{\mathrm{ab}}, \\ d_s \leq d'_s, \end{cases} \qquad \begin{cases} \rho_n \geq \rho'_n, \\ \rho_u \geq \rho'_u. \end{cases}$$

Equivalently, if G is smooth of finite type over an arbitrary base S, the functions

$$s \longmapsto \rho_{\mathrm{ab}}(s), \qquad s \longmapsto \rho_{\mathrm{ab}}(s) + \rho_r(s), \qquad s \longmapsto d_s(s)$$

are lower semicontinuous, while the functions

$$s \longmapsto \rho_n(s), \qquad s \longmapsto \rho_u(s)$$

are upper semicontinuous.[71]

The same results probably remain valid without assuming G smooth over S, but assuming only that it is flat and of finite presentation over S, provided that, for an algebraic group G over an algebraically closed field k, $\rho_{\mathrm{ab}}(G)$, and so on, are understood to mean the corresponding invariants of G^0_{red}.

In this Seminar we present some results of this kind when G is affine over S, or more generally has affine fibers. In this case we shall verify the semicontinuity properties for ρ_r and ρ_n, hence for $\rho_u = \rho_n - \rho_r$, and the continuity of ρ_s in a neighborhood of a point s whose fiber is a reductive group.[71]

When G is already assumed commutative, 8.2 can be generalized as follows.

Theorem 8.8.— [72]*Let G be a commutative S-prescheme in groups, flat and of finite presentation over S, with affine connected fibers. Let $s \in S$, and suppose that:*

a) if $\bar{k}$ denotes an algebraic closure of $\kappa(s)$, then $(G_{\bar{k}})_{\mathrm{red}}$ is a torus;

b) there is a generization t of s such that G_t is smooth over $\kappa(t)$.

Under these conditions, there exists an open neighborhood U of s such that $G|_U$ is a torus.

(Observe moreover that, if one assumes only that G_t is affine, respectively connected, for every generization t of s, it follows easily that the same conclusion holds for t in an open neighborhood of s.)

[70] *Editors' note.* Recall the definition of "Cartan subgroup".

[71] *Editors' note.* Give a reference for these results?

[72] *Editors' note.* G. Prasad and J.-K. Yu generalized this result (subject to some additional hypotheses) without assuming G commutative, and with "torus" replaced by "reductive group" in hypothesis (a) and in the conclusion; cf. [PY06], Theorem 6.2.

Proof of 8.8. It suffices to prove that G_s is smooth over $\kappa(s)$. Indeed, since G is flat and of finite presentation over S, it then follows that G is smooth over S above a neighborhood of s (cf. 3.5), and we are then in the situation of 8.1.

To prove that G_s is smooth over $\kappa(s)$, the usual procedure reduces us to the case in which S is affine and noetherian. Choose a morphism $S' \to S$, where S' is the spectrum of a discrete valuation ring, sending the closed point to s and the generic point η to t. We are thereby reduced to the case in which S is itself the spectrum of a discrete valuation ring; we may moreover suppose that it is complete and has algebraically closed residue field, and that s and $t = \eta$ are respectively the closed point and the generic point of S.

Thus G is flat, separated, and of finite type over S; the generic fiber G_η is smooth and connected, and the special fiber G_0 is such that

$$T_0 = (G_0)_{\mathrm{red}}$$

is a torus. Let $m > 0$ be an integer prime to the residual characteristic at s, and hence also to that at η. Then $m \cdot \mathrm{id}_G$ is étale on every fiber (VII_A, 8.4), and is therefore étale because G is flat over S (SGA 1, I 5.9). Its kernel ${}_mG$ is thus étale over S; since G is separated * [73] and of finite type over S, the same holds for ${}_mG$. Since

$$({}_mG)_0 = {}_m(T_0),$$

its degree is m^r, where

$$r = \dim T_0 = \dim G_0 = \dim G.$$

It follows that the rank of $({}_mG)_\eta = {}_m(G_\eta)$ is at least m^r, which already proves, using 8.6, that G_η is a torus of dimension r, since the two fibers of ${}_mG$ have the same rank, and hence, as in 8.5, that ${}_mG$ is finite over S.[74]

Since S is complete with algebraically closed residue field, the finite étale cover ${}_mG$ splits completely. Thus, through every point of ${}_mG_0$ there passes a section of G over S; in particular, the set of points of G_0 through which there passes a section of G over S is everywhere dense. We now use the following lemma.

Lemma 8.9.— *Let S be a locally noetherian regular prescheme of dimension 1, and let G be an S-prescheme in groups, flat and locally of finite type, such that G_η is smooth over $\kappa(\eta)$ for every maximal point η of S (which implies that G is reduced). Suppose that the normalization X of G is finite over G (this is the case, by Nagata, when S is the spectrum of a complete discrete valuation ring; cf. EGA IV₂, 7.7.4), and let G' be the open subset of X formed by the points at which X is smooth over S. With these notations:*

a) If the projection $G' \to S$ is surjective, then G' carries a unique structure of an S-prescheme in groups for which the canonical morphism $G' \to G$ is a group homomorphism.

b) Suppose that, for every closed point s of S, the set of points of G'^0_s through which an étale quasi-section passes is dense in G'^0_s for the Zariski topology. Then X is regular, the hypotheses of (a) *hold, and the map*

$$\Gamma(G'/S) \longrightarrow \Gamma(G/S)$$

is bijective.

*By Raynaud's theorem, VI_B, 5.3.

[73] *Editors' note.* See Editors' note (62) in 8.5.

[74] *Editors' note.* Revise the preceding sentence.

In the case that concerns us, this lemma applies and gives a group homomorphism $u : G' \to G$, where G' is smooth over S and

$$u_\eta : G'_\eta \xrightarrow{\sim} G_\eta$$

is an isomorphism. After possibly replacing G' by an open subgroup, we may suppose that G'_0 is connected. Since $u_0 : G'_0 \to G_0$ induces a surjective morphism $G'_0 \to T_0$, where T_0 is a torus of the same dimension as G'_0, it follows easily that G'_0 is a torus (for example, by using 8.6, or by referring to *Bible*, §7.3, Theorem 3(a)). Consequently, by 8.2, G' is a torus. By IX 6.8, $\ker u$ is then a subgroup of multiplicative type of G'; since its generic fiber is the identity group, it is itself the identity group, and therefore u is a monomorphism. It follows from VIII 7.9 that u is an immersion. Since it is surjective and G is reduced, u is an isomorphism, completing the proof of 8.8.

It remains to prove 8.9. For part (a), the uniqueness of the group law on G' for which $u : G' \to G$ is a group homomorphism is clear, since the group law on the generic fiber of G' is known (we may suppose S irreducible). For existence, we reduce easily to the case in which S is local, the spectrum of a discrete valuation ring A. By uniqueness, and because integral closure commutes with an étale extension of the base, we may make étale base extensions on S. This reduces us to the case in which A is "strictly local", that is, henselian with separably closed residue field. The same reduction applies to part (b); under the hypothesis made there, "étale quasi-section" can now be replaced by "section".

Since G' is smooth over S and X is normal, the prescheme $G' \times_S X$ is normal (being smooth over the normal prescheme X). Thus the composite morphism

$$G' \times_S X \longrightarrow G \times_S G \longrightarrow G$$

factors as

$$p : G' \times_S X \longrightarrow X.$$

We shall prove that the restriction of this morphism to the open subset $G' \times_S G'$ of $G' \times_S X$ induces a morphism

$$G' \times_S G' \longrightarrow G'.$$

It is therefore enough to show that $G'_0 \times_k G'_0$ maps into the open subset G'_0 of X_0. It suffices to check that, for every k-valued point g'_0 of G'_0, the morphism

$$(\times) \qquad h'_0 \longmapsto p_0(g'_0, h'_0)$$

from G'_0 to X_0 takes its values in G'_0. Since G' is smooth over S and S is henselian, every such g'_0 is induced by a section g' of G' over S. The morphism $(\times)$ is then induced by the morphism

$$h' \longmapsto p(g', h')$$

from G' to X, which itself is obtained by transport of structure from the automorphism $h \mapsto g \cdot h$ of G, namely left translation by the section g of G that is the image of g'. Thus $h' \mapsto p(g', h')$ is itself an automorphism of X, and hence maps G' into G'. This proves the assertion.

It remains to prove that the composition law thereby obtained on G' is a group law. Associativity follows at once from associativity on the generic fiber (which is isomorphic to that of G). Moreover, the inversion automorphism of the S-prescheme G induces an automorphism of X, which therefore preserves G' and induces an automorphism σ of G'. One verifies that σ has the properties of an inverse for the composition law on G'. Indeed, this again amounts to verifying the commutativity of certain diagrams involving fiber

powers of G' over S; since these are smooth over S, it suffices to check commutativity on the generic fiber, where it is clear. This proves part (a) of 8.9.

We prove part (b). Let Z' be the set of $x \in X$ for which $\mathcal{O}_{X,x}$ is not regular. It is closed by a theorem of Nagata (EGA IV_2, 6.12.6), and is clearly contained in X_0. Let Z be its image in G, which is therefore a closed subset of G_0. Then Z is nowhere dense in G_0, that is, it contains no maximal point y of G_0. Indeed, G_0 is defined in G by an equation $t = 0$, where t is a uniformizer of the discrete valuation ring A. By the Hauptidealsatz, $\mathcal{O}_{G,y}$ has dimension 1. Hence, for every $x \in X$ above y, $\mathcal{O}_{X,x}$ has dimension 1, and is therefore a discrete valuation ring, since X is normal and hence regular in codimension 1.

On the other hand, for every section g of G over S, the set Z' is plainly stable under the automorphism of X obtained by transport of structure from left translation by g on G. Thus Z is stable under the left translation on G_0 defined by g_0. By hypothesis, the set of these g_0 is dense in G_0^0. Since Z is stable under translation by these g_0 and is a nowhere-dense closed subset, it follows at once that $Z = \varnothing$, whence $Z' = \varnothing$, so X is regular. Consequently X is smooth over S at every point through which a section passes. Every section of G over S lifts uniquely to a section of X over S, and hence to a section of G' over S. This applies in particular to the identity section, proving that the image of G' in S is all of S, that is, that the hypothesis of (a) holds. This completes the proof of 8.9, and hence of 8.8.

9. Addenda

[75]

0.9.1 Simplicial sets, topoi, groupoids, and topological spaces

Notations 9.1.1.— *(1) Let E be a simplicial set. The following objects can be associated with it:*

(2) a topos $T = E^{\sim}$, obtained from the topoi naturally associated with the sets E_i, by the procedure described in [Del74]*, 6.3.1 (see also* [Ill72]*, VI.5.2 and SGA 4, VI.7);*

(3) a groupoid $G = \Pi(E)$, whose objects are the elements of E_0 (the "vertices") and whose arrows are defined in [GZ67]*, II.7;*

(4) a topological space $X = |E|$, a cellular complex called the "geometric realization" (or "topological realization") (cf. loc. cit.*, III.1).*

Observe that a sheaf on T is nothing other than a simplicial set over E.

0.9.2 Locally constant sheaves; descent data

Definitions 9.2.1.— A *locally constant sheaf* on

(1) a simplicial set E is a morphism of simplicial sets $E' \to E$ such that, for every $i \in \mathbb{N}$ and every $e \in E_i$, the face operators d induce isomorphisms

$$E'_e \xrightarrow{\sim} E'_{d(e)}$$

between the fibers (cf. [AM69], §10);

[75] *Editors' note.* This additional section was written by Fabrice Orgogozo (following indications by Ofer Gabber).

(2) a topos T is an object F of T such that there exist an epimorphism $U \to 1$ and an isomorphism

$$F \times U \simeq f^*I \times U,$$

where I is a set and $f : T \to \mathbf{Ens}$ is the final morphism (cf. SGA 4, IX.2);

(3) a groupoid G is a presheaf on G, that is, a contravariant functor from G to the category of sets (cf. [GZ67], Appendix I.1.2);

(4) a topological space X is a sheaf of sets on X that is locally constant in the usual sense.

Finally:

(5) a *descent datum* on a simplicial set E consists of a sheaf F on $E_0^\sim$ (that is, a set-valued function on E_0) and an isomorphism

$$\alpha : p_1^*F \xrightarrow{\sim} p_2^*F,$$

where $p_1, p_2 : E_1^\sim \to E_0^\sim$ are the morphisms induced by the faces, satisfying the usual cocycle relation (cf. Exposé IV, 2.1 [1] and below).

The morphisms between these five kinds of objects, as well as the associated inverse-image functors, are defined in the evident way.

0.9.3 Some equivalences of categories

Let E be a simplicial set, and let T, G, and X be, respectively, the associated topos, groupoid, and topological space.

Proposition 9.3.1.— *The categories of locally constant sheaves on E, T, G, and X, as well as the category of descent data on E, are equivalent.*

Sketch of proof. Denote by (1) through (5) the categories of objects defined in the preceding paragraph.

(1) $\iff$ (5). This is a special case of [AM69], 10.6 (see also [GZ67], Appendix I.2.3, [Fri82], ?.5.6, or [Ill72], VI.8.1.6). An equivalence of categories is given by the functor that associates with the object $E' \to E$ the pair (E'_0, α), where E'_0 is regarded as a sheaf on E_0 and α is the unique isomorphism

$$p_1^*E'_0 \xrightarrow{\sim} p_2^*E'_0$$

whose fiber at each $y \in E_1$, with images denoted x_1 and x_2 under the two projections, is given by the isomorphisms

$$(E'_0)_{x_1} \xleftarrow{\sim} (E'_1)_y \xrightarrow{\sim} (E'_0)_{x_2}.$$

(5) $\iff$ (3). This is evident: one of the two relations defining the morphisms in the groupoid G associated with E is a cocycle relation.

(1) $\iff$ (4). Cf. [GZ67], Appendix I.3.2.1.

[1] PP: I have replaced "FGA, *Technique de descente* I, 1.6" by "Exposé IV, 2.1".

(1) $\Longleftrightarrow$ (2). One must show that an object over E is a locally constant sheaf in the simplicial sense if and only if it is one as a sheaf on $T = E^{\sim}$. This follows from the fact that the locally constant sheaves $F_\bullet$ on T are precisely the cartesian sheaves, that is, those for which the arrows

$$([n] \to [m])^* F_m \longrightarrow F_n$$

are isomorphisms. This last assertion is a special case of a general fact about simplicial topoi, together with the fact that every sheaf on $E_0^{\sim}$ is locally constant. (See also [Ill72], VI.8.1.6.)

Q.E.D.

For every group H, the preceding equivalences of categories induce equivalences between the categories of H-torsors. Here these are locally constant sheaves equipped with an action of H, whose fibers are isomorphic to H acting on itself by translation.

0.9.4 Fundamental groups and groupoids

It follows from the preceding equivalences of categories that, for every group H and every simplicial set E, the sets of isomorphism classes of H-torsors

$$H^1(E, H), \qquad H^1(E^{\sim}, H), \qquad \pi_0\big(\mathrm{Hom}(\Pi(E), BH)\big), \qquad H^1(|E|, H)$$

are naturally in bijection. Recall that BH denotes the one-object groupoid associated with H, and that π_0 is the functor that associates with a category the set of isomorphism classes of its objects.

Likewise, if e is a point of E, the preceding equivalences induce bijections between the sets of isomorphism classes of H-torsors trivialized over e, denoted respectively by

$$H^1(E \operatorname{rel} e, H), \qquad H^1(E^{\sim} \operatorname{rel} e^{\sim}, H), \qquad \mathrm{Hom}\big(\pi_1(\Pi(E), e), H\big), \qquad H^1(|E| \operatorname{rel} |e|, H).$$

Recall that

$$\pi_1(\Pi(E), e) = \mathrm{Isom}_{\Pi(E)}(e, e).$$

For variable H, these functors are represented, in the connected case, by a group denoted $\pi_1(E, e)$. The group $\pi_1(E, e)$ is isomorphic to $\pi_1(\Pi(E), e)$ and to $\pi_1(|E|, |e|)$, and hence also to the fundamental group of a simplicial set as defined by Kan (cf., for example, [May67], 16.1, or [Ill72], I.2.1.1). Recall also that $H^1(E, H)$ is in turn isomorphic to the set $H^1(\pi_1(E, e), H)$ of homomorphisms to H modulo conjugation, also denoted

$$\mathrm{Hom\,ext}(\pi_1(E, e), H).$$

0.9.5 Cones

Definitions 9.5.1.— *(1) Let $f : E' \to E$ be a morphism of simplicial sets. Recall (cf., for example,* [Del74]*, 6.3.1) that $C(f)$ denotes the pointed simplicial set whose underlying set in degree $n \geq 0$ is*

$$E_n \amalg \left(\coprod_{i<n} E'_i \right) \amalg \star,$$

where $\star$ is a singleton. We leave it to the reader to define the simplicial maps; the faces in degrees at most two are recalled below. (See also [GZ67]*, VI.2, for a pointed variant.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant*

sheaves on E equipped with a trivialization of the inverse image on E'. The simplicial set $C(f)$ is naturally pointed by $\star \to C(f)$.

(2) Let $f : T' \to T$ be a morphism of topoi. Denote by $C(f)$ the topos whose objects are quintuples

$$(F, F', A, \alpha : f^*F \to F', \beta : A \to \Gamma(T', F')),$$

where F (respectively F') is an object of T (respectively T'), A is a set, and α (respectively β) is a morphism in T' (respectively in **Ens**). *(See also* [Del80], *4.3.4 and* [Ill72], *III.4 for a variant of this construction.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on T equipped with a trivialization of the inverse image on T'. The topos $C(f)$ is naturally pointed by the fiber functor that sends the quintuple above to the set A.*

(3) Let $f : G' \to G$ be a morphism of groupoids. Denote by $C(f)$ the colimit of the diagram

$$G \longleftarrow G' \longrightarrow B1,$$

where $B1$ is the one-object category (one object, one arrow). The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on G equipped with a trivialization of the inverse image on G'.

(4) Let $f : X' \to X$ be a morphism of topological spaces. Denote by $C(f)$ the colimit of the diagram

$$\star \longleftarrow X' \times \{0\} \longrightarrow X' \times [0,1] \longleftarrow X' \times \{1\} \longrightarrow X.$$

The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on X equipped with a trivialization of the inverse image on X'. The topological space $C(f)$ is naturally pointed by $\star \to C(f)$.

9.5.2. Descent data on the cone of a simplicial map

Let $f : E' \to E$ be a morphism of simplicial sets, and let $C(f)$ be its cone (cf. above, (1)). Use p (respectively q, r) to denote the face maps of E' (respectively $E, C(f)$). Before stating the proposition below, we describe the faces r in degrees at most two in terms of p and q. By convention, p_{ij} (respectively p_i) is the face map

$$E'_2 = E'_{\{1,2,3\}} \longrightarrow E'_1 = E'_{\{1,2\}} \quad \Big(\text{respectively } E'_1 = E'_{\{1,2\}} \longrightarrow E'_0 = E'_{\{1\}}\Big)$$

corresponding to the increasing map

$$\{1,2\} \longrightarrow \{1,2,3\} \quad (\text{respectively } \{1\} \longrightarrow \{1,2\})$$

whose image is $\{i,j\}$ (respectively $\{i\}$). [2] The same convention is adopted for the faces of E and $C(f)$.

The morphism

$$r_1 : C(f)_1 = E_1 \amalg E'_0 \amalg \star \longrightarrow C(f)_0 = E_0 \amalg \star$$

(respectively r_2) is:

- $q_1 : E_1 \to E_0$ (respectively $q_2 : E_1 \to E_0$) on E_1;

[2] This convention departs from the present simplicial convention but is closer to the notation used by Grothendieck in his theory of descent.

- $E'_0 \to \star$ (respectively $f_0 : E'_0 \to E_0$) on E'_0;
- $\star \to \star$ on $\star$.

Likewise, the morphism

$$r_{21} : C(f)_2 = E_2 \amalg E'_1 \amalg E'_0 \amalg \star \longrightarrow C(f)_1 = E_1 \amalg E'_0 \amalg \star$$

(respectively r_{32}, r_{31}) is:

- $q_{21} : E_2 \to E_1$ (respectively q_{32}, q_{31}) on E_2;
- $p_1 : E'_1 \to E'_0$ (respectively $f_1 : E'_1 \to E_1$, $p_2 : E'_1 \to E'_0$) on E'_1;
- $E'_0 \to \star$ (respectively $\mathrm{Id} : E'_0 \to E'_0$, $\mathrm{Id} : E'_0 \to E'_0$) on E'_0.

Let H be a descent datum on $C(f)$, that is, a sheaf H_0 on

$$C(f)_0 = E_0 \amalg \star$$

equipped with an isomorphism

$$\gamma : r_1^* H_0 \xrightarrow{\sim} r_2^* H_0$$

between its inverse images on

$$C(f)_1 = E_1 \amalg E'_0 \amalg \star,$$

satisfying the cocycle relation

$$r_{31}^* \gamma = r_{32}^* \gamma \circ r_{21}^* \gamma.$$

The restriction of γ to E_1 (respectively E'_0) is a descent datum

$$\beta : q_1^* G_0 \xrightarrow{\sim} q_2^* G_0,$$

where G_0 is the restriction of H_0 to E_0 (respectively, a trivialization

$$t : H_0(\star) \xrightarrow{\sim} f^* G_0 =: F_0,$$

where $H_0(\star)$ is the constant sheaf with stalk $H_0(\star)$). The restriction of the cocycle relation satisfied by γ to E_2 (respectively E'_1) is the cocycle relation for β (respectively,

$$p_2^*(t) = f_1^*(\beta) \circ p_1^*(t)).$$

This latter relation says that the trivialization t is compatible with the descent datum $f_1^*(\beta)$ induced by β on F_0. It follows that:

Proposition 9.5.3.— *Let $f : E' \to E$ be a morphism of simplicial sets, and let $C(f)$ be its cone, pointed by $\star$. The category of descent data on $C(f)$ trivialized over $\star$ is equivalent to the category of descent data on E equipped with a trivialization of the descent datum induced on E'.*

0.9.6 Representability of the functor $\overline{\pi}^1(S'/S, \xi; -)$

The notation is that of page 90, and from now on we suppose that the connected components of

$$S_0 = S' \qquad \text{and} \qquad S_1 = S' \times_S S'$$

are open. Under this hypothesis we have the following explicit combinatorial description, which follows immediately from the preceding recall.

Proposition 9.6.1.— *For every group G, the set $\overline{\pi}^1(S'/S, \xi; G)$ of isomorphism classes of descent data equipped with a trivialization over ξ is naturally in bijection with the relative nonabelian cohomology set*

$$H^1(K \operatorname{rel} k, G) := H^1(\widetilde{K} \operatorname{rel} \star, G).$$

It follows that this functor is representable by a group, denoted $\pi_1(K, k)$, the *relative fundamental group*, as soon as the simplicial set

$$\widetilde{K} = C(k \to K)$$

is connected. One verifies immediately that this holds if and only if the map $\pi_0(k) \to \pi_0(K)$ is surjective. It is enough, in particular, that the simplicial set K be connected. As indicated in the text, this is the case when the scheme S is connected and the morphism $S' \to S$ is universally submersive.

0.9.7 Contractibility of k; counterexample to the injectivity of $k_\bullet \to K_\bullet$

The following two propositions make Editors' note (53) more precise.

Proposition 9.7.1.— *Let κ be an algebraically closed field, and let X be a connected κ-scheme. For every integer $i \geq 0$, let X_i denote the $(i+1)$-st fiber power of X over κ. For every integer $i \geq 0$, the canonical map*

$$\pi_0(X_i) \longrightarrow \pi_0(X)^{i+1}$$

is a bijection. In particular, the simplicial set $\pi_0(X_\bullet)$ is contractible.

It suffices to prove the following lemma, which follows by passage to the limit from the Künneth formula. [Expand?]

Lemma 9.7.2.— *Let κ be an algebraically closed field, and let X and Y be two connected κ-schemes. Then the fiber product $X \times_\kappa Y$ is connected.*

Proposition 9.7.3.— *Let X be a connected scheme, let $X' \to X$ be a finite étale morphism, and let $\bar{x}$ be a geometric point of X localized at a point x. The canonical morphism*

$$\pi_0(X'_{\bar{x}}) \longrightarrow \pi_0(X')$$

is surjective.

Indeed, the image of every connected component of X' is open and closed in X, and therefore meets the fiber X'_x. Thus

$$\pi_0(X'_x) \longrightarrow \pi_0(X')$$

is surjective, as is

$$\pi_0(X'_{\bar{x}}) \longrightarrow \pi_0(X'_x).$$

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[76] *Editors' note.* Additional references cited in this Exposé; the references [AM69] and following concern the Addenda.

Exposé XI
Representability Criteria Applications to Multiplicative-Type Subgroups of Affine Group Schemes

by A. Grothendieck[0]

0. Introduction

As we have already seen in Exposé X, nos. 4 and 5, the representability of certain functors—for example, functors of the form $\mathrm{Hom}_S(X, Y)$ and various related constructions—plays an important role in many questions concerning group preschemes.

Among the results that are particularly useful in this theory, besides the representability of quotients studied in Exposés V and VI_B and in Exposé VIII, no. 5, let us mention the representability of functors of the form

$$\prod_{X/S} Y/X,$$

where Y is a subobject of X. This was studied in a very elementary case in Exposé VIII, no. 6; variants will be given in no. 6 of the present exposé. These results provide the representability of various centralizers, normalizers, and transporters.

Less elementary representability criteria, using results that will appear in EGA VI, are indicated in 6.12 and in Exposés XV and XVI. There we shall give a criterion for the representability of quotients G/H in cases not covered by the preceding exposés; this criterion was not developed in the oral lectures.

Our principal purpose in the present exposé is to prove Theorems 4.1 and 4.2, which provide a typical example of a nonprojective construction technique, close to the technique that will be developed in EGA VI. Since the oral lecture and the drafting of the present text, it has moreover become apparent that the affine hypotheses in 4.1 and 4.2 can be removed to a large extent (cf. Exposé XV), and that the main part of the theory developed in the following exposé can dispense with 4.1 and 4.2. Finally, it might be interesting to prove analogues of these results for a general reductive group prescheme (for example, a semisimple one) in place of a group of multiplicative type; in that case 4.1 and 4.2 would doubtless be the key result in the proof.

[0]Editor's note: xy version of December 5, 2008.

1. Review of Smooth, Étale, and Unramified Morphisms

The reader is referred to EGA IV, §§17 and 18, and, pending its publication, to SGA 1, Exposés I, II, and III. Certain noetherian hypotheses used there, which are inconvenient in applications, should however be replaced by finite-presentation hypotheses.

Definition 1.1. — *Let S be a prescheme and let*

$$F : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

be a functor. The functor F is said to be formally smooth *(respectively,* formally unramified[1]*, respectively,* formally étale*) if, for every S-prescheme S' that is affine in the absolute sense and every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal $\mathscr{J}$, the map*

$$F(S') \longrightarrow F(S'_0)$$

is surjective (respectively, injective, respectively, bijective).

An S-prescheme X is said to be formally smooth over S (respectively, formally unramified over S, respectively, formally étale over S) if the corresponding functor has that property. It is said to be smooth over S (respectively, unramified over S, respectively, étale over S) if it satisfies the preceding condition and is, in addition, locally of finite presentation over S.

The usefulness of these definitions for X lies, on the one hand, in their remarkably simple expression in terms of the functor represented by X (and in practice X is often given as the object over S representing an explicitly described functor), and, on the other hand, in the remarkable local structural properties that they express. We recall those properties in the following statements; for their proofs, see the references just cited.

Proposition 1.2. — *Let X be a prescheme locally of finite presentation over S. Then:*

(i) *X is smooth over S if and only if it is flat over S and its geometric fibers*

$$X \otimes_S \operatorname{Spec} \kappa(s)$$

are regular schemes. More generally, X is smooth over S in a neighborhood of $x \in X$ (one then says that X is smooth over S at x) if and only if X is flat over S at x and X_s is smooth over $\kappa(s)$ at x; that is,

$$X_s \otimes_{\kappa(s)} \overline{\kappa(s)}$$

is regular at the points (or merely at one point) lying over x.

(ii) *Suppose that S, and hence X, is locally noetherian. Let $x \in X$, and let $s \in S$ be its image. The smoothness of X over S at x can then be detected from the local homomorphism*

$$A = \mathscr{O}_{S,s} \longrightarrow B = \mathscr{O}_{X,x}$$

of noetherian local rings (or even from the local homomorphism $\widehat{A} \to \widehat{B}$ of their completions), by the following characteristic property: B is flat over A, and $B \otimes_A k$ is geometrically regular over k, the residue field of A. Equivalently, for every finite extension k' of k,

$$(B \otimes_A k) \otimes_k k' = B \otimes_A k'$$

[1] One would now rather say "net" in place of "unramified."

is a regular semilocal ring. When the residue-field extension $k(B)/k(A)$ is trivial, these conditions are also equivalent to the following one: $\widehat{B}$ is isomorphic, as an $\widehat{A}$-algebra, to an algebra of formal power series

$$\widehat{A}[[t_1, \ldots, t_n]].$$

Thus, from the "formal" point of view, the structure of X over S is that of the standard affine space $S[t_1, \ldots, t_n]$ over S.

Proposition 1.3. — *Let X be a prescheme locally of finite presentation over S. Then:*

(i) *The following conditions are equivalent:*

(a) *X is unramified over S.*

(b) *The diagonal morphism $X \to X \times_S X$ is an open immersion.*

(c) $\Omega^1_{X/S} = 0$.

(d) *The geometric fibers of X/S are discrete and reduced; that is, they are isomorphic to disjoint sums of copies of the base field.*

(e) *For every $s \in S$, the fiber*

$$X \otimes_S \operatorname{Spec} \kappa(s) = X_s$$

is unramified over $\kappa(s)$, or equivalently is isomorphic to a disjoint sum of spectra of finite separable extensions of $\kappa(s)$.

(ii) *There are analogous pointwise conditions for X to be unramified over S at a given point x, that is, in a neighborhood of that point. For example, it is necessary and sufficient that $\mathscr{O}_{X_s,x}$ be a finite separable extension of $\kappa(s)$. In terms of the local homomorphism*

$$A = \mathscr{O}_{S,s} \longrightarrow B = \mathscr{O}_{X,x},$$

this says that $B \otimes_A k$ is a finite separable extension of the residue field k of A; equivalently,

$$\mathfrak{r}(A)B = \mathfrak{r}(B) \quad \textit{and} \quad k(B)/k(A) \textit{ is finite separable.}$$

When A, and hence B, is noetherian and $k(A) \xrightarrow{\sim} k(B)$, this also means that $\widehat{A} \to \widehat{B}$ is surjective.

Thus, from the "formal" point of view, saying that X is unramified over S means that X is essentially a subprescheme of S. Observe also that conditions (d) and (e) are expressed solely in terms of the fibers of X/S.

Proposition 1.4. — *Let X be a prescheme locally of finite presentation over S. Then X is étale over S if and only if it is smooth and unramified over S (which is immediate from the definition), so the criteria in 1.2 and 1.3 apply. In particular:*

(i) *X is étale over S if and only if it is flat and unramified over S. There is an analogous local criterion for X to be étale over S at a given point x.*

(ii) *Suppose in addition that S, and hence X, is locally noetherian. Whether X is étale over S at x can then be detected from the local homomorphism*

$$A = \mathscr{O}_{S,s} \longrightarrow B = \mathscr{O}_{X,x}$$

(and even from $\widehat{A} \to \widehat{B}$) by the following characteristic property: B is flat over A, and $B \otimes_A k$ is a finite separable extension of k, the residue field[2] of A. When $k(A) \xrightarrow{\sim} k(B)$, this condition simply says that $\widehat{A} \to \widehat{B}$ is an isomorphism.

Thus, from the "formal" point of view, saying that X is étale over S simply means that X is locally isomorphic to S.

Remark 1.5. — When X is locally of finite presentation over S and a point $x \in X$ is given, the fact that X is smooth (respectively, unramified, respectively, étale) over S at x, that is, in a neighborhood of x, can also be detected from the functor

$$X : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

by the following property. For every S-scheme S' that is the spectrum of a local ring, every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal, and every S-morphism $u_0 : S'_0 \to X$ sending the closed point s' of S' to x, there exists at least one (respectively, at most one, respectively, exactly one) S-morphism $u : S' \to X$ extending u_0.

This statement shows in particular that, in Definition 1.1, one may restrict to local schemes S', provided the functor under consideration is representable by a prescheme locally of finite presentation over S. Moreover, when S is locally noetherian, one may even restrict the preceding pointwise criterion to local artinian schemes S', and the same restriction may therefore be made in Definition 1.1, provided that S is locally noetherian and that the functor in question is represented by a prescheme locally of finite type over S.

Remark 1.6. — Of course, given a morphism $f : X \to Y$ of preschemes, one says that f is smooth (respectively, unramified, respectively, étale) if f makes X a smooth (respectively, unramified, respectively, étale) Y-prescheme. When X and Y are S-preschemes and f is an S-morphism of finite presentation, these properties of f can be expressed immediately in terms of the morphism of functors

$$F, G : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

defined by f. Namely, f is smooth (respectively, unramified, respectively, étale) if, for every S-prescheme S' that is affine in the absolute sense and every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal $\mathscr{J}$, the map

$$F(S') \longrightarrow F(S'_0) \times_{G(S'_0)} G(S')$$

deduced from the commutative square

$$\begin{array}{ccc} F(S') & \longrightarrow & G(S') \\ \downarrow & & \downarrow \\ F(S'_0) & \longrightarrow & G(S'_0) \end{array}$$

[2]Editor's note: "residue extension" was replaced by "residue field."

is surjective (respectively, injective, respectively, bijective). Equivalently, for every commutative square of morphisms of functors over S, where S' and S'_0 are as above and $i : S'_0 \to S'$ is the canonical immersion,

$$\text{(Q)} \qquad \begin{array}{ccc} S'_0 & \xrightarrow{\ i\ } & S' \\ \downarrow{\scriptstyle u_0} & & \downarrow{\scriptstyle v} \\ F & \xrightarrow{\ f\ } & G, \end{array}$$

there exists at least one (respectively, at most one, respectively, exactly one) morphism

$$\begin{array}{ccc} & & S' \\ & \overset{u}{\swarrow} & \\ F & & \end{array}$$

that makes the two corresponding triangles commute:

$$u_0 = u \circ i \qquad \text{and} \qquad v = f \circ u.$$

This property for a homomorphism of functors $f : F \to G$ over S retains its meaning even when the functors under consideration are not representable. If it holds, the homomorphism $f : F \to G$ is said to be *formally smooth* (respectively, *formally unramified*, respectively, *formally étale*). Notice that this property depends only on the homomorphism of the underlying functors

$$(\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens})$$

defined by F and G (cf. Exposé I, 1.4.1), and not on the structure morphisms $F \to S$ and $G \to S$. An equivalent formulation of the preceding definition, somewhat easier to use in applications, is the following: a morphism of functors $F \to G$ over S is formally smooth (respectively, formally unramified, respectively, formally étale) if, for every S' over S and every morphism $S' \to G$, the functor

$$F' = F \times_G S'$$

over S' is formally smooth (respectively, formally unramified, respectively, formally étale).

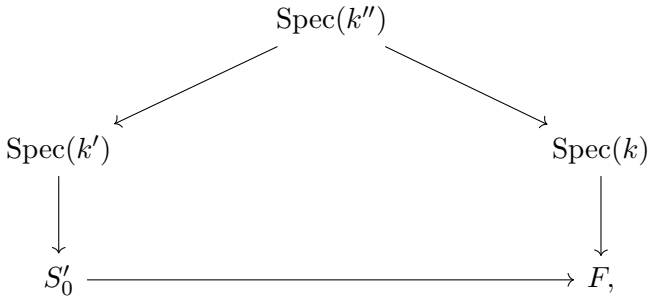
Remark 1.7. — Under the hypotheses of 1.6, suppose that we are given a "point of F with values in a field k," that is, an element $x \in F(\operatorname{Spec} k)$, or equivalently a morphism

$$x : \operatorname{Spec} k \longrightarrow F.$$

We likewise say that f is *formally smooth* (respectively, *formally unramified*, respectively, *formally étale*) *at* x if the condition in the preceding definition is satisfied whenever S' is a local scheme and the morphism $S'_0 \to F$ in diagram (Q) above is "compatible with the marked points" in the following sense. Let k' denote the residue field of S' and S'_0 at their closed point. Then the diagram

$$\begin{array}{ccc} \operatorname{Spec}(k') & & \operatorname{Spec}(k) \\ \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle x} \\ S'_0 & \xrightarrow{\ u_0\ } & F \end{array}$$

in which $j : \mathrm{Spec}(k') \to S'_0$ is the canonical immersion, must admit a completion to a commutative diagram



where k'' is a field.

When F and G are represented by S-preschemes X and Y, and the S-morphism $f : X \to Y$ is locally of finite presentation, this condition says precisely, by 1.5, that $f : X \to Y$ is smooth at the point of X that is the image of $\mathrm{Spec}(k)$ under $x : \mathrm{Spec}(k) \to X$.

Remark 1.8. — When the condition in the preceding definition is satisfied after restricting to local artinian schemes S', we say that $F \to G$ is *infinitesimally smooth* (respectively, *infinitesimally unramified*, respectively, *infinitesimally étale*) *at* x. We say that $F \to G$ is infinitesimally smooth (respectively, infinitesimally unramified, respectively, infinitesimally étale) if it has the corresponding property at every point x; equivalently, if the condition considered in 1.6 is satisfied whenever S' is local artinian.

This variant of the preceding notions is technically useful. Since it is weaker, it is often easier to verify, while it is nevertheless frequently sufficient to imply the stronger condition; this is the case, for example, when $F \to G$ is a morphism locally of finite presentation and G is represented by a locally noetherian prescheme.

Proposition 1.9. — *Let $f : X \to S$ be a smooth morphism of preschemes. Then $\Omega^1_{X/S}$ is a locally free module of finite type on X, and its rank at a point $x \in X$ is equal to the dimension of the fiber X_s, where $s = f(x)$, in a neighborhood of x.*

This dimension is called the *relative dimension of X over S at x*. Notice that it is zero (when f is smooth at x) if and only if f is étale at x. It can also be computed directly in terms of the functor F represented by X, as follows. Let ξ be a point of F with values in a field k, "localized at x," that is, a morphism

$$\mathrm{Spec}(k) \longrightarrow F = X$$

whose image is x. Consider the algebra

$$D(k) = k[t]/(t^2)$$

of dual numbers over k, regarded as an S-prescheme, and the map

$$F(D(k)) \longrightarrow F(k)$$

induced by the augmentation $D(k) \to k$. Finally, let $F(D(k), \xi)$ be the inverse image of $\xi \in F(k)$ under this map. This set is naturally endowed with the structure of a vector space over k. In fact, it is the vector-space dual of

$$\Omega^1_{X/S} \otimes_{\mathcal{O}_{X,x}} k,$$

and its dimension is the relative dimension of X over S at x.

To describe the vector-space law on $F(D(k), \xi)$ explicitly, it is useful to introduce, more generally and as in Exposé II, for every vector space V over k, the algebra

$$D_k(V) = k + V,$$

where V is an ideal of square zero, and to regard $F(D_k(V), \xi)$, the inverse image of ξ under

$$F(D_k(V)) \longrightarrow F(k),$$

as a covariant functor of V with values in (**Ens**). It is then enough that this functor commute with products of two factors. This means that F carries certain amalgamated sums of a very special kind to fiber products; compare Exposé II. This condition always holds when F is representable. It follows that the sets $F(D_k(V), \xi)$, and in particular $F(D(k), \xi)$, are endowed with vector-space structures over k. Thus one can define the relative dimension of F over X at the "point" ξ under conditions appreciably more general than the representability of F.

In the present Exposé, the fact that certain functors to be described explicitly are representable by preschemes smooth over S will be used chiefly through the following result. It will be our technical means of passing from constructions over the completion of the local ring A of a point s of a noetherian scheme S to a local ring A' over A that is étale over A, and hence, in particular, of passing from s to neighboring points.

Proposition 1.10. — *Let $f : X \to S$ be a smooth morphism of preschemes, let $s \in S$, and let $x \in X$ lie over s, with $\kappa(x)$ a finite separable extension of $\kappa(s)$. Then there exists a subscheme $S' \subset X$, étale over S, passing through x. Thus one can find an étale morphism $S' \to S$, a point $s' \in S'$ above s whose residue-field extension is $\kappa(x)/\kappa(s)$, and an S-morphism $S' \to X$ carrying s' to x.*

To construct S', simply choose sections $g_1, \ldots, g_n$ of $\mathscr{O}_X$ on a neighborhood of x whose images at x form a regular system of parameters for the local ring of the fiber X_s at x. The subscheme S' defined by the g_i is then étale over S at x, and, after shrinking S', it is étale over S.

We shall use 1.10 when $\kappa(x) = \kappa(s)$, that is, when x is rational over $\kappa(s)$, or equivalently when x may be regarded as a section of X_s over $\operatorname{Spec} \kappa(s)$. In that case, 1.10 has the character of an extension theorem for sections, after an étale extension of the base. It takes a particularly simple form in the following special case.

Corollary 1.11. — *Under the hypotheses of 1.10, suppose that S is the spectrum of a henselian local ring and that $\kappa(x) = \kappa(s)$. Then there exists a section of X over S passing through x. It is uniquely determined if X is in fact étale over S at x.*

Indeed, since S is henselian, under the hypotheses of 1.10 the prescheme S' contains an open subscheme that is finite over S and whose fiber at s is reduced to x. Since this open subscheme is étale over S, it is isomorphic to S, which proves the assertion. Notice that when S is the spectrum of a complete local ring, 1.10 or 1.11 is more or less equivalent to the classical "Hensel's lemma," and one sometimes refers to the result by that name.

2. Examples of Formally Smooth Functors from the Theory of Groups of Multiplicative Type

We shall interpret, in the language introduced in the preceding number, the results stated in Exposé IX, no. 3, concerning infinitesimal extensions of a homomorphism from a group

of multiplicative type. Those results are consequences of the vanishing of the Hochschild cohomology of such a group, established in Exposé I.

Proposition 2.1. — *Let S be a prescheme, let H be a group of multiplicative type over S, and let G be an S-group prescheme smooth over S. Consider the functor over S*

$$M_H = \mathrm{Hom}_{S\text{-gr}}(H, G).$$

Its value at an S' over S is the set of homomorphisms of S'-groups from $H_{S'}$ to $G_{S'}$. This functor is formally smooth over S.

Cf. Exposé IX, 3.6. More generally:

Corollary 2.2. — *Let S and G be as above, and consider a homomorphism $u : H_1 \to H_2$ of group schemes of multiplicative type over S. With the preceding notation this induces a morphism of functors over S*

$$M_{H_2} \longrightarrow M_{H_1}, \qquad M_{H_i} = \mathrm{Hom}_{S\text{-gr}}(H_i, G),$$

given by $w \mapsto w \circ u$. This morphism is formally smooth.

Indeed, by the definitions in 1.6, this is equivalent to the following assertion. Suppose that S is affine and that $S_0 \subset S$ is a subscheme defined by a nilpotent ideal. Let

$$v : H_1 \longrightarrow G$$

be a homomorphism of S-groups, and let

$$w_0 : (H_2)_{S_0} \longrightarrow G_{S_0}$$

be a homomorphism of S_0-groups such that

$$w_0 u_{S_0} = v_{S_0}.$$

Then there exists a homomorphism of S-groups

$$w : H_2 \longrightarrow G$$

extending w_0 and satisfying $wu = v$.

To see this, first extend w_0 to a homomorphism of S-groups

$$w' : H_2 \longrightarrow G,$$

which is possible by 2.1. Set

$$v' = w'u : H_1 \longrightarrow G.$$

By the hypothesis on w_0, one has $v'_{S_0} = v_{S_0}$. Consequently, by Exposé VIII, 3.6, there exists an element $g \in G(S)$ whose image in $G(S_0)$ is the identity element and such that

$$v = \mathrm{int}(g)v'.$$

It follows that

$$v = \big(\mathrm{int}(g)w'\big)u,$$

and it is therefore enough to take

$$w = \mathrm{int}(g)w'.$$

Corollary 2.3. — *With the notation of 2.1, regard $M_H = M$ as a functor with operator group G, where G acts by*

$$(v, g) \longmapsto \mathrm{int}(g) \circ v.$$

Then the corresponding morphism

$$R : G \times_S M \longrightarrow M \times_S M,$$

defined by

$$(g, v) \longmapsto (\mathrm{int}(g) \circ v, v),$$

is formally smooth.

After a base change $S' \to S$, this is equivalent to the following assertion.

Corollary 2.4. — *With the notation of 2.1, let $v_1, v_2 : H \to G$ be two morphisms of S-groups, and let* $\mathrm{Transp}(v_1, v_2)$ *be the subfunctor of G consisting of the g such that*

$$\mathrm{int}(g) \circ v_1 = v_2.$$

Then this functor is formally smooth over S. In particular, if $v_1 = v_2 = v$, the functor $\mathrm{Centr}(v)$*, the subgroup of G consisting of the h such that* $\mathrm{int}(h)v = v$*, is formally smooth over S.*

N.B. The pair (v_1, v_2) may be regarded as a section over S of the second factor in the codomain of the morphism R in 2.3, and $\mathrm{Transp}(v_1, v_2)$ as the inverse-image functor of that section under R.

The assertion in 2.4 is itself equivalent to the following. Suppose that S is affine and that $S_0 \subset S$ is a subscheme defined by a nilpotent ideal. If $g_0 \in G(S_0)$ satisfies

$$\mathrm{int}(g_0)(v_1)_{S_0} = (v_2)_{S_0},$$

then g_0 extends to an element $g \in G(S)$ such that

$$\mathrm{int}(g)v_1 = v_2.$$

To prove this, first extend g_0 to a section g' of G over S, which is possible because G is smooth over S, and put

$$v_2' = \mathrm{int}(g')v_1.$$

The morphisms v_2 and v_2' have the same restriction over S_0. Hence, by Exposé IX, 3.6, there exists $g'' \in G(S)$ inducing the identity section over S_0 and such that

$$v_2 = \mathrm{int}(g'')v_2'.$$

Therefore

$$v_2 = \mathrm{int}(g'')\,\mathrm{int}(g')v_1 = \mathrm{int}(g''g')v_1,$$

and it is enough to take $g = g''g'$.

Proposition 2.1 bis. — *Let S be a prescheme and let G be an S-group prescheme smooth over S. Consider the functor*

$$M : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens}),$$

defined by

$$M(S') = \left\{ \begin{array}{c} \text{multiplicative-type subgroups} \\ \text{of } G_{S'} \end{array} \right\}.$$

Then M is formally smooth over S.

Cf. Exposé IX, 3.6 bis.

Corollary 2.2 bis. — *Let n be an integer, and consider the morphism of functors*

$$\varphi_n : M \longrightarrow M$$

defined by

$$\varphi_n(H) = {}_nH = \ker(n \cdot \mathrm{id}_H).$$

Then φ_n is formally smooth. For every integer $p > 0$, let M_p denote the subfunctor of M for which $M_p(S')$ is the set of multiplicative-type subgroups H of $G_{S'}$ satisfying ${}_pH = H$. Then the morphism induced by φ_n

$$M_{np} \longrightarrow M_n$$

is formally smooth.

The second assertion is trivially contained in the first and is included only for the convenience of a later reference. The proof of the first assertion is analogous to that of 2.2, now invoking Exposé IX, 3.6 bis.

Corollary 2.3 bis. — *With the notation of 2.1 bis, regard M as a functor with operator group G, where G acts by*

$$(g, H) \longmapsto \mathrm{int}(g)(H).$$

Then the corresponding morphism

$$R : G \times_S M \longrightarrow M \times_S M,$$

defined by

$$(g, v) \longmapsto (\mathrm{int}(g)v, v),$$

is formally smooth.

This is equivalent to the following assertion.

Corollary 2.4 bis. — *With the notation of 2.1 bis, let H_1 and H_2 be two multiplicative-type subgroups of G, and let $\mathrm{Transp}_G(H_1, H_2)$ be the subfunctor of G consisting of the g such that*

$$\mathrm{int}(g)H_1 = H_2.$$

Then this functor is formally smooth over S. In particular, if $H_1 = H_2 = H$, the subfunctor $\mathrm{Norm}_G(H)$ of G, the normalizer of H, is formally smooth over S.

The proof is analogous to that of 2.4, again invoking Exposé IX, 3.6 bis.

Proposition 2.5. — *Let S be a prescheme, let G be an S-group prescheme smooth over S, let K be an S-group subprescheme that is smooth over S or of multiplicative type, let H be an S-group prescheme of multiplicative type, and let $u : H \to G$ be a homomorphism of S-groups. Denote by $\mathrm{Transp}_G(u, K)$ the subfunctor of G whose value at an S' over S consists of the elements $g \in G(S')$ for which*

$$\mathrm{int}(g)u_{S'} : H_{S'} \longrightarrow G_{S'}$$

factors through $K_{S'}$. Then this functor is formally smooth over S.

The proof is analogous to that of 2.2, invoking Exposé IX, 3.6 and Exposé X, 2.1, the latter in the case where K is of multiplicative type.

3. Auxiliary Representability Results

Proposition 3.1. — *Let $F \to S$ be a functor over the prescheme S. The following conditions are equivalent:*

(i) *F is representable and $F \to S$ is an open immersion (one also says simply that $F \to S$ is an open immersion).*

(ii) *F is a sheaf for the faithfully flat quasi-compact topology, "commutes with filtered direct limits of rings," and $F \to S$ is a monomorphism. Moreover, the following condition holds: for every local prescheme S' over S, with residue field k, and every S-morphism*

$$\mathrm{Spec}(k) = S'_0 \longrightarrow F,$$

there exists an S-morphism $S' \to F$ extending it.

(iii) *Suppose that S is locally noetherian. Then F is a sheaf for the faithfully flat quasi-compact topology, commutes with filtered direct limits of rings, commutes with adic inverse limits of local artinian rings, and $F \to S$ is a monomorphism and is infinitesimally étale (cf. 1.8).*

Let us first make two points of terminology precise.

Remark 3.2. — A functor[3] F over S is said to *commute with direct limits of rings* (understood to mean filtered direct limits) if the following holds. Let $(S'_i)_{i\in I}$ be any filtered inverse system of affine S-schemes lying over an affine open subscheme of S, and let A'_i be their rings. Then the natural homomorphism

$$(*) \qquad \varinjlim_i F(S'_i) \longrightarrow F(S'), \qquad S' = \mathrm{Spec}(A'), \quad A' = \varinjlim_i A'_i$$

is bijective.

Notice that S' is precisely the inverse limit of the S'_i in the category of preschemes, and even in the category of all ringed spaces. Thus the condition under consideration has the character of a right-exactness condition—commutation with certain direct limits in $(\mathbf{Sch})^\circ/S$—just as does the condition of being a sheaf for a topology.

Care must be taken that this condition is essentially relative: it involves the morphism $F \to S$, not merely the functor

$$F : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens}).$$

More precisely, in $(*)$, $F(S'_i)$ and $F(S')$ mean $\mathrm{Hom}_S(S'_i, F)$ and $\mathrm{Hom}_S(S', F)$. Consequently, when F is representable, the condition says that F is locally of finite presentation over S. We used several times in the preceding two exposés the fact that a functor represented by an S-prescheme locally of finite presentation commutes with direct limits of rings.

Remark 3.3. — A functor F over S is said to *commute with adic inverse limits of local artinian rings* if the following condition holds. For every S' over S that is the spectrum of a complete noetherian local ring A', set

$$S'_n = \mathrm{Spec}\big(A'/\operatorname{rad}(A')^{n+1}\big).$$

[3]Editor's note: here and in Remark 3.3, the functor is contravariant.

Then the natural map

$$(**) \qquad F(S') \longrightarrow \varprojlim_n F(S'_n)$$

is required to be bijective.

This condition has the character of a left-exactness condition and is satisfied whenever F is representable. In contrast to commutation with direct limits of rings, it is intrinsic to F as an object of $\widehat{\mathbf{Sch}}$; that is, it does not involve the morphism $F \to S$.

Remark 3.4. — Let F be a functor over S that is a sheaf for the Zariski topology, or, as one also says, is "of local nature." It is enough for this that F be a sheaf for a finer topology, such as the faithfully flat quasi-compact topology. Let (S_i) be an open covering of S. A straightforward gluing argument shows that F is representable if and only if each

$$F_i = F \times_S S_i$$

is representable; this allows one, for example, to reduce to the case where S is affine.

Suppose that the functor F, of local nature, commutes with direct limits of rings. Then F is representable if and only if its restriction to the category of preschemes locally of finite presentation over S is representable. Necessity was noted in 3.2. Sufficiency amounts to the following assertion: if X is a prescheme locally of finite presentation over S, and $X \to F$ is a morphism such that, for every S' locally of finite presentation over S, the induced map

$$\mathrm{Hom}_S(S', X) \longrightarrow \mathrm{Hom}_S(S', F)$$

is bijective, then $X \to F$ is an isomorphism. This follows readily from the fact that X and F are both functors of local nature that commute with direct limits of rings.

Proof of Proposition 3.1. The implications (i) ⇒ (ii) and (i) ⇒ (iii) are immediate. We prove the converse implications.

Assume (ii). Let U be the set of points $s \in S$ for which the canonical monomorphism

$$\mathrm{Spec}\,\kappa(s) \longrightarrow S$$

factors through F. By the last condition in (ii), for every $s \in U$, the canonical map $\mathrm{Spec}\,\mathscr{O}_{S,s} \to S$, equivalently its factorization through U, factors through F. The local ring $\mathscr{O}_{S,s}$ is the direct limit of the rings of affine neighborhoods of s. Since F commutes with direct limits of rings, for every $s \in U$ there is an open neighborhood U^s such that the canonical immersion $U^s \to S$ factors through F. It follows that $U^s \subset U$, and hence that U is open.

Because $F \to S$ is a monomorphism and F is of local nature, the S-morphisms $U^s \to F$ agree on the intersections $U^s \cap U^{s'}$, for $s, s' \in U$, and therefore glue to an S-morphism $U \to F$. It remains to prove that this morphism is an isomorphism, or equivalently that every S-morphism $S' \to F$ factors uniquely through U, where S' is an S-prescheme. Since both $F \to S$ and $U \to S$ are monomorphisms, it is equivalent to say that the structure morphism $S' \to S$ factors through U.

By locality, we are reduced to the case in which S' is the spectrum of a field and hence has a single point s'. Let s be the point of S below s'. We claim that the given S-morphism $S' \to F$ factors through

$$\mathrm{Spec}\,\kappa(s) = S_0 \longrightarrow F.$$

This implies $s \in U$ and proves the assertion. Since $S' \to S_0$ is an fpqc covering and F is a sheaf for the fpqc topology, it is enough that the two composites

$$S'' = S' \times_{S_0} S' \rightrightarrows S' \longrightarrow F$$

coincide. They do coincide because $F \to S$ is a monomorphism. This proves (ii) $\Rightarrow$ (i).

Now assume (iii), with S locally noetherian. It is enough to prove the last condition of (ii); by the preceding proof, it is enough to do so when

$$S' = \operatorname{Spec} \mathscr{O}_{S,s}$$

for some $s \in S$. Put

$$A = \mathscr{O}_{S,s}, \qquad A_n = A/\mathfrak{m}^{n+1}, \qquad S_n = \operatorname{Spec}(A_n).$$

Because $F \to S$ is infinitesimally smooth, the given morphism $S'_0 \to F$ extends to morphisms $S_n \to F$. Since $F \to S$ is a monomorphism, these morphisms form an element of $\varprojlim_n F(S_n)$. Since F commutes with adic inverse limits of local artinian rings, they arise from a morphism

$$\operatorname{Spec}(\widehat{A}) = \widehat{S}' \longrightarrow F.$$

Finally, $F \to S$ is a monomorphism, F is an fpqc sheaf, and $\widehat{S}' \to S'$ is an fpqc covering. The last morphism therefore descends to $S' \to F$, completing the proof of (iii) $\Rightarrow$ (ii). □

Proposition 3.5. — *Let S be a locally noetherian prescheme, let $F \to S$ be a functor over S, and let $(X_i, u_i)_{i\in I}$ be a family of S-morphisms $u_i : X_i \to F$, where the X_i are preschemes locally of finite type over S. Suppose that the following conditions hold:*

(a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with direct limits of rings, and commutes with adic inverse limits of local artinian rings.*

(b) *The morphisms $u_i : X_i \to F$ are monomorphisms and are infinitesimally étale (cf. 1.8).*

(c) *The family (u_i) is "set-theoretically surjective."*

Under these conditions, F is representable by a prescheme locally of finite type over S. Moreover, the u_i are open immersions and the family (X_i) is an open covering of F.

Remark 3.6. — Proceeding as at the end of Remark 1.7, one defines, for every functor

$$F : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens}),$$

an "underlying set" $\operatorname{ens}(F)$: it is the quotient of the set of field-valued points of F by the equivalence relation described in 1.7. When F is represented by X, this recovers the underlying set of X.

Plainly, $\operatorname{ens}(F)$ depends functorially on F. Thus, if $G \to F$ is a morphism of functors, we say that it is *set-theoretically surjective* if the induced map

$$\operatorname{ens}(G) \longrightarrow \operatorname{ens}(F)$$

is surjective. Equivalently, every point of F with values in a field k "comes from" a point of G with values in a suitable extension of k. This definition extends immediately to a family of morphisms $G_i \to F$, and thereby specifies the meaning of condition (c).

Proof of Proposition 3.5. For $(i,j) \in I \times I$, put

$$X_{i,j} = X_i \times_F X_j$$

and consider the projections

$$v_{i,j} : X_{i,j} \longrightarrow X_i, \qquad w_{i,j} : X_{i,j} \longrightarrow X_j.$$

We claim that these morphisms are representable by open immersions. Apply criterion 3.1(iii). The functor $X_{i,j}$ satisfies the three exactness conditions: it is a sheaf, it commutes with direct limits of rings, and it commutes with adic inverse limits of local artinian rings. Indeed, F, X_i, and X_j satisfy these conditions, and the conditions are stable under finite inverse limits, in particular under fiber products. Since every $X_i \to F$ is a monomorphism, both projections are monomorphisms, each being obtained from one of these maps by base change. Finally, being infinitesimally étale is also preserved by base change. Thus the hypotheses of 3.1(iii) are satisfied.

We may now use the X_i, $X_{i,j}$, $v_{i,j}$, and $w_{i,j}$ to construct in the usual way an S-prescheme X, in which the X_i are identified with open subschemes, the $X_{i,j}$ with the intersections $X_i \cap X_j$, and $v_{i,j}$, $w_{i,j}$ with the canonical immersions.

Put

$$Y = \coprod_i X_i.$$

Then X is also the quotient of Y by the equivalence relation

$$R = \coprod_{i,j} X_{i,j} \rightrightarrows Y,$$

whose two projections v, w are defined by the $v_{i,j}$ and $w_{i,j}$, respectively.

More precisely, because F is an fpqc sheaf, the morphisms $u_i : X_i \to F$ induce a morphism $u : Y \to F$, and

$$R = Y \times_F Y$$

is exactly the equivalence relation defined by u. The quotient $X = Y/R$ is also a quotient in the category of fpqc sheaves, and even in the category of Zariski sheaves; this follows directly from the definitions of quotient and sheaf. Consequently, u factors uniquely through a morphism

$$\overline{u} : X \longrightarrow F,$$

and this morphism is a monomorphism.

It remains to show that $\overline{u}$ is an isomorphism. Since F is of local nature, we may assume that S is affine. Since F also commutes with direct limits of rings, it is enough, by 3.4, to verify that, for every affine T of finite type over S, every morphism $T \to F$ factors through X. Consider

$$G = X \times_F T \longrightarrow T.$$

It is enough to prove that this morphism is an isomorphism. Since T is noetherian, the argument above shows that it is an open immersion. Here one uses that $X \to F$ is infinitesimally étale, which follows immediately because the induced morphisms $u_i : X_i \to F$ are infinitesimally étale. By hypothesis, $X \to F$ is set-theoretically surjective, and this

condition is stable under base change. Hence $G \to T$ is set-theoretically surjective. Being also an open immersion, it is an isomorphism. □

Proposition 3.7. — *Let S be a locally noetherian prescheme, let I be a directed partially ordered index set, let $(T_i)_{i\in I}$ be a projective system of S-preschemes locally of finite type, let*

$$T = \varprojlim_{i\in I} T_i$$

be the projective-limit functor, let F be a functor over S, and let $u : F \to T$ be an S-morphism. Suppose that the following conditions hold:

(a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, and commutes with adic projective limits of local artinian rings.*

(b) *The morphism $u : F \to T$ is a monomorphism.*

(b') *The morphism $u : F \to T$ is infinitesimally étale.*

(c) *For every point ξ of F with values in the spectrum of a field k, let $\xi_i \in T_i(\operatorname{Spec} k)$ be its image and let t_i be the corresponding point of T_i. There exists $i \in I$ such that, for every $j \geq i$, the transition morphism*

$$p_{i,j} : T_j \longrightarrow T_i$$

is étale at t_j.

(d) *For every prescheme X locally of finite type over S and every S-morphism $X \to F$, the set of points $x \in X$ at which this morphism is infinitesimally étale is open.*

Under these conditions, F is representable by a prescheme locally of finite type over S.

Let us note at once that, in the case to be considered in the next section, conditions (c) and (d) will be verified by means of the following corollary.

Corollary 3.8. — *Assuming (a), (b), and (b'), conditions (c) and (d) are implied by the following conditions:*

(c') *The T_i are smooth over S, and the transition morphisms $p_{i,j} : T_j \to T_i$ are smooth.*

(d') *For every point ξ of F with values in the spectrum of a field k, let $t_i(\xi)$ be the point of T_i defined by ξ, let $d_i(\xi)$ be the relative dimension of T_i over S at $t_i(\xi)$, and put*

$$d(\xi) = \sup_i d_i(\xi).$$

Then:

1° *for every ξ as above, $d(\xi) < +\infty$; and*

2° *for every prescheme X locally of finite type over S and every S-monomorphism $v : X \to F$, the function*

$$x \longmapsto d(\xi_x)$$

on X is locally constant, where, for $x \in X$, ξ_x denotes the point of F with values in $\kappa(x)$ induced by v.

Proof of Proposition 3.7. Work under condition (c). Let t be the element of $\mathrm{ens}(F)$ defined by ξ (cf. 3.6), and put

$$\mathscr{O}_t = \varinjlim_i \mathscr{O}_{T_i,t_i}.$$

Using condition (c), one sees readily that $\mathscr{O}_t$ is a noetherian local ring (EGA 0_{IV}, 10.3.1.3). Its residue field $\kappa(t)$ is the inductive limit of the residue fields $\kappa(t_i)$, and k is an extension of it. If η denotes $\operatorname{Spec}\kappa(t)$, and ξ denotes $\operatorname{Spec}k$, there is a commutative diagram

$$\begin{array}{ccc} \xi & \longrightarrow & F \\ \downarrow & & \downarrow \\ \eta & \longrightarrow & T, \end{array}$$

whose definition is evident and is left to the reader. Since $\xi \to \eta$ is a covering for the fpqc topology, F is a sheaf for that topology by (a), and $F \to T$ is a monomorphism by (b), the morphism $\eta \to T$ factors uniquely through a morphism $\eta \to F$. We now record the following lemma.

Lemma 3.9. — *Under the hypotheses of 3.7, let*

$$T' = \operatorname{Spec}(A) \quad \textit{and} \quad T'_0 = \operatorname{Spec}(k(A))$$

with T' a noetherian local scheme, and let $i : T'_0 \to T'$ be the canonical immersion. Suppose that a commutative square of morphisms is given:

$$\begin{array}{ccc} T'_0 & \longrightarrow & F \\ {\scriptstyle i}\downarrow & & \downarrow \\ T' & \longrightarrow & T. \end{array}$$

Then there exists a unique T-morphism $T' \to F$.

The proof is the proof of 3.1, (iii) $\Rightarrow$ (ii), in which the representability of the S occurring in that statement was not used. One uses the facts that F is a sheaf for the fpqc topology, that it commutes with adic projective limits of local artinian rings, here the rings

$$A/\operatorname{rad}(A)^{n+1},$$

and that $F \to T$ is both a monomorphism and infinitesimally étale.

We apply Lemma 3.9 with

$$T' = \operatorname{Spec}(\mathscr{O}_t), \qquad T'_0 = \operatorname{Spec}(\kappa(t)).$$

We have just constructed $T'_0 \to F$, and the canonical homomorphisms

$$\mathscr{O}_{T_i,t_i} \longrightarrow \mathscr{O}_t$$

define a projective system of morphisms $\operatorname{Spec}(\mathscr{O}_t) \to T_i$, hence a canonical morphism $\operatorname{Spec}(\mathscr{O}_t) \to T$. The corresponding square (x) plainly commutes: by the definition of T, it is enough to check this after replacing T by T_i. We therefore obtain a unique T-morphism

$$v' : T' = \operatorname{Spec}(\mathscr{O}_t) \longrightarrow F.$$

Since F commutes with inductive limits of rings, this morphism factors, for sufficiently large i, through

$$v_i' : T_i' = \mathrm{Spec}(\mathscr{O}_{T_i,t_i}) \longrightarrow F.$$

For such an i, one has

$$T' \xrightarrow{\sim} T_i', \qquad \text{that is,} \qquad \mathscr{O}_{T_i,t_i} \xrightarrow{\sim} \mathscr{O}_t.$$

Indeed, $T' \to T_i'$ is faithfully flat and quasi-compact, hence an effective epimorphism, so it is enough to show that it is a monomorphism. Suppose that two morphisms $R \rightrightarrows T'$, with R representable, have the same composite with $T' \to T_i'$. Their composites with $T' \to T$, which factors through $T' \to T_i'$, are then equal. Hence their composites with $T' \to T_j$, and therefore with $T' \to T_j'$, are equal for every j. The two original morphisms are equal because T' is the projective limit of the T_j' in the category (**Sch**).

Consider the diagram

$$\begin{array}{ccc} T' & \xrightarrow{v'} & F \\ \downarrow & \nearrow_{v_i'} & \downarrow \\ T_i' & \longrightarrow & T_i, \end{array}$$

where the unlabeled arrows are the evident ones. The square and the upper triangle commute. Since $T' \to T_i'$ is an epimorphism, the lower triangle also commutes.

The ring $\mathscr{O}_{T_i,t_i}$ is the inductive limit of the rings of affine open neighborhoods of t_i in T_i. Since F commutes with inductive limits of rings, v_i' comes from a morphism

$$v_t : U_t \longrightarrow F$$

defined on an open neighborhood U_t of t_i in T_i. After shrinking this neighborhood if necessary, v_t is a T_i-morphism. Indeed, T_i, being locally of finite type over S, also commutes with inductive limits of rings. The composite of v_t with $F \to T_i$ is a monomorphism, and therefore v_t is a monomorphism.

We claim that v_t is infinitesimally étale at t_i. Since $F \to T$ is infinitesimally étale, this is equivalent to the composite

$$T_i' \xrightarrow{v_i'} F \longrightarrow T$$

being infinitesimally étale at t_i, or to the induced morphism $T_i' \to T$ being infinitesimally étale. Since $T' \xrightarrow{\sim} T_i'$, it is in turn equivalent to $T' \to T$ being infinitesimally étale. This is immediate, because that morphism is the projective limit of the infinitesimally étale morphisms $T_j' \to T_j$.

Apply condition (d), which has not yet been used. It follows that v_t is infinitesimally étale in a neighborhood of t_i. After replacing U_t by a smaller open, we may therefore suppose that v_t is a monomorphism that is infinitesimally étale.

As t varies in $\mathrm{ens}(F)$, the family of morphisms v_t satisfies Proposition 3.5, which gives the conclusion of Proposition 3.7.

Proof of Corollary 3.8. Assume conditions (c') and (d'). With the notation of (d'), $d_i(\xi)$ is constant for all sufficiently large i. Consequently, the relative dimension of T_j over T_i at t_j, which is

$$d_j(\xi) - d_i(\xi),$$

is zero. Thus $T_j \to T_i$ is étale at t_j, which proves condition (c).

The preceding proof now gives, for each $t \in \mathrm{ens}(F)$, an index i, an open neighborhood U_t of t_i, and a morphism $U_t \to F$ that is a monomorphism and is infinitesimally étale at t_i. It remains to show that this morphism is infinitesimally étale throughout a neighborhood of t_i. In condition (d'), 2°, take $X = U_t$. Replacing U_t, if necessary, by the connected component of t_i in U_t, we may suppose that

$$d(\xi_x) = d(\xi_{t_i}) \qquad \text{for every } x \in U_t.$$

Now $d(\xi_{t_i})$ is the relative dimension of T_i over S at t_i. Since that relative dimension remains constant on U_t, we also have, for every $x \in U_t$,

$$(*) \qquad d(\xi_x) = \text{the relative dimension of } U_t \text{ over } S \text{ at } x.$$

On the other hand, using the notation of the proof of 3.7, consider any local noetherian prescheme

$$R' = \mathrm{Spec}(A), \qquad R'_0 = \mathrm{Spec}(k(A)).$$

Every morphism $R' \to F$ whose induced point $R'_0 \to F$ represents $t \in \mathrm{ens}(F)$, that is, sends the closed point of R' to t, factors uniquely through T'. The proof is the proof of 3.1 or of Lemma 3.9: one uses that $T' \to F$ is a monomorphism infinitesimally étale at t, and that T' is an fpqc sheaf commuting with adic projective limits.

Apply this result to the morphism

$$\mathrm{Spec}(\mathscr{O}_{U_t,x}) = R' \longrightarrow F$$

induced by v_t, and to the point $t_x \in \mathrm{ens}(F)$ that is the image of the closed point of R', or equivalently the image of x under v_t. We obtain a factorization

$$\mathrm{Spec}(\mathscr{O}_{U_t,x}) \longrightarrow \mathrm{Spec}(\mathscr{O}_{t_x}) \longrightarrow F.$$

The second arrow is infinitesimally étale at t_x. To prove that the composite is infinitesimally étale at x, it is therefore enough to prove the same for the first arrow at x.

By (x) above and formula $(*)$, the first arrow is induced by a local S-homomorphism between the localizations, at x and t_x, of the smooth S-preschemes U_t and U_{t_x}, which have the same relative dimension d over S at those points. By EGA I, 6.5.1, this local homomorphism is induced by an S-morphism

$$w : U \longrightarrow V,$$

where U is an open neighborhood of x in U_t and $V = U_{t_x}$. It remains to prove that w is étale at x. Moreover, U and V carry monomorphisms to F, and, after shrinking U once more, w is an F-morphism. Hence w is a monomorphism. It is now enough to prove the following lemma.

Lemma 3.10. — *Let U and V be two smooth S-preschemes of the same relative dimension d over S, and let $w : U \to V$ be an S-morphism that is a monomorphism. Then w is an open immersion, and hence is étale.*

By SGA 1, I, 5.7, we reduce to the case in which S is the spectrum of a field, which may be assumed algebraically closed. By SGA 1, I, 5.1, it is enough to prove that w is étale, and it is enough to do so at the closed points of U. Let x be such a point and put $y = w(x)$. Choose a regular system of parameters $f_1, \ldots, f_d$ of $\mathscr{O}_{V,y}$. Then

$$A = \mathscr{O}_{U,x} / \sum_i f_i \mathscr{O}_{U,x}$$

is the trivial extension of

$$k = \mathscr{O}_{V,y}/\sum_i f_i \mathscr{O}_{V,y};$$

that is, $A \simeq k$. Indeed, since w is a monomorphism, so is the structural morphism

$$\mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(k)$$

obtained from it by base change. Since $\mathscr{O}_{U,x}$ is a regular local ring of dimension d, the f_i form a regular system of parameters of that ring. It follows immediately that w is étale at x. This proves Lemma 3.10, and hence Corollary 3.8.

Corollary 3.11. — *Under the hypotheses of 3.8, for every quasi-compact open subfunctor U of F that is separated over S, there exists $i \in I$ such that, for every $j \geq i$, the morphism*

$$u_j|_U : U \longrightarrow T_j$$

is an open immersion. In particular, if the T_i are quasi-affine over S, then every open subfunctor U of F that is quasi-compact over S, that is, of finite type over S, is quasi-affine over S.

The proof of Proposition 3.7 shows that for every $t \in \mathrm{ens}(F)$ there exists $i \in I$ such that $u_i : F \to T_i$ is a local isomorphism at t, and then u_j is a local isomorphism at that point for every $j \geq i$. By quasi-compactness, i may be chosen independently of $x \in U$. It remains to prove that, for sufficiently large i,

$$u_i|_U : U \longrightarrow T_i$$

is a monomorphism.

Since $U \to T$ is a monomorphism, the intersection of the equivalence relations

$$U \times_{T_i} U \ \subset \ U \times_S U$$

is the diagonal. The prescheme $U \times_S U$ is noetherian, and the $U \times_{T_i} U$ are closed subpreschemes. It follows that one of them is already the diagonal; equivalently, $u_i|_U$ is a monomorphism. This proves the first assertion of Corollary 3.11, and the second follows immediately.

Proposition 3.12. — *Let S be a prescheme and G an affine S-group prescheme.*

(a) Let

$$F : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

be the functor such that, for every T over S,

$$F(T) = \left\{ \begin{array}{c} \text{multiplicative-type subgroups of } G_T \\ \text{that are finite over } T \end{array} \right\}.$$

Suppose that S is locally noetherian or that G is of finite presentation over S. Then F is representable and affine over S. If G is of finite presentation over S, then F is locally of finite presentation over S.

(b) Let H be a finite S-group prescheme of multiplicative type. Then $\mathrm{Hom}_{S\text{-gr}}(H, G)$ *is representable and affine over S. If G is of finite type (respectively, of finite presentation) over S, then the same is true of* $\mathrm{Hom}_{S\text{-gr}}(H, G)$.

Remark 3.13. — Apart from the assertion that $\mathrm{Hom}_{S\text{-gr}}$ is affine, and from the case in which G is of finite presentation over S, which is the case that will suffice for us, 3.12 is an immediate consequence of the theory of Hilbert schemes (A. Grothendieck, *Techniques de construction et théorèmes d'existence en Géométrie Algébrique: IV. Les Schémas de Hilbert*, Séminaire Bourbaki, May 1961, no. 221). It is enough, in fact, that G be quasi-projective over S. In case *(a)* one can also represent the larger functor

$$F_0(T) = \left\{ \begin{array}{c} \text{group subpreschemes of } G_T \text{ that are flat, proper,} \\ \text{and of finite presentation over } T \end{array} \right\}.$$

The canonical monomorphism $F \to F_0$ is an "open immersion," as follows from the criterion in 3.1 and Exposé X, 4.7 *(b)*. Thus the representability of F_0 implies that F is representable by an open subprescheme of F_0. In case *(b)* it is enough to assume that H is projective and of finite presentation over S. In both cases one obtains a functor locally of finite presentation over S.

In the present Exposé, 3.12 is only a technical lemma used to prove a key result in the next section. We shall therefore sketch an easy direct proof that does not use Hilbert schemes.

We first prove *(b)*. It is enough to prove that $\mathrm{Hom}_S(H, G)$ is representable, independently of any group structure on H and G, and that it has the additional properties asserted for $\mathrm{Hom}_{S\text{-gr}}(H, G)$. Indeed, applying the same result to $\mathrm{Hom}_S(H \times_S H, G)$, the subfunctor $\mathrm{Hom}_{S\text{-gr}}(H, G)$ of $\mathrm{Hom}_S(H, G)$ can be expressed by a finite inverse limit—in fact, by fiber products—involving G, $\mathrm{Hom}_S(H, G)$, and $\mathrm{Hom}_S(H \times_S H, G)$. We leave this expression to the reader.

Write

$$H = \mathrm{Spec}(B),$$

where B is a sheaf of algebras on S that is locally free as a sheaf of modules; this is the only hypothesis on H that the argument uses. Since the representability question is local on S, we may suppose that S is affine, with ring A. We may also write $G = \mathrm{Spec}(C)$, where C is an A-algebra.

When $G = S[t] = G_0$, with t an indeterminate, the Hom functor is simply

$$T \longmapsto \Gamma(T, B_T).$$

Since B is locally free, this functor is represented by the vector bundle $V(B^\vee)$, where $B^\vee$ is the dual module sheaf. When $G = S[(t_i)_{i\in I}] = G_0^I$, for a not necessarily finite family of indeterminates, it follows that

$$\mathrm{Hom}_S(H, G) = \mathrm{Hom}_S(H, G_0)^I = V(B^\vee)^I = V(B^{\vee(I)}).$$

In general G is isomorphic to a closed subscheme of a scheme of the form $S[(t_i)_{i\in I}]$; equivalently, C is a quotient of an A-algebra $A[(t_i)_{i\in I}]$. Let (F_j) be a family of generators of the ideal by which one takes the quotient. After covering S by smaller affine open subsets, we may assume that B is free of rank n. Choose a basis $(e_k)_{1\le k\le n}$ of B. Writing

$$x_i = \sum_{k=1}^{n} x_{ik} e_k$$

with indeterminate coefficients x_{ik} in an unspecified A-algebra A', expand $F_j((x_i))$ in this basis. For every F_j this gives n polynomials

$$F_{j,k} \qquad (1 \le k \le n)$$

in the variables $(x_{i,k})_{i\in I,\,1\le k\le n}$, with coefficients in A. It follows at once that $\mathrm{Hom}_S(H,G)$ is represented by the spectrum of the quotient of the polynomial ring $A[(x_{i,k})]$ by the ideal generated by the $F_{j,k}$. This proves *(b)*.

We now prove *(a)*. For every ordinary finite commutative group M, let F_M be the subfunctor of F obtained by restricting to those subgroups of G_T that are of multiplicative type and of type M. A gluing argument like the one used in 3.5 (which we should have stated after decomposing the problem a little further) shows that it is enough to prove that each F_M is representable. Then F is represented by the disjoint sum of the F_M, with M ranging over the isomorphism classes of finite commutative groups. Indeed, F is the corresponding sum in the category of sheaves.

Fix M and write F in place of F_M. Put $H = D_S(M)$, and let

$$F' = \mathrm{Imm}_{S\text{-gr}}(H,G) \subset \mathrm{Hom}_{S\text{-gr}}(H,G)$$

be the subfunctor whose value at T is the set of homomorphisms of T-groups $H_T \to G_T$ that are closed immersions. By *(b)*, $\mathrm{Hom}_{S\text{-gr}}(H,G)$ is represented by an affine S-prescheme. Using Exposé IX, 6.8, one sees immediately that F' is represented by an open-and-closed subprescheme of it; in particular, F' is affine over S.

Consider the canonical morphism $F' \to F$, which assigns to every monomorphism $H_T \to G_T$ its image group. Directly from the definitions, F' is thereby a principal homogeneous bundle over F, in the category of fpqc sheaves, under

$$\Gamma_F = \Gamma_S \times_S F, \qquad \Gamma = \mathrm{Aut}_{\mathrm{gr}}(M).$$

It follows by descent that $F' \to F$ is representable: for every morphism $T \to F$ with T representable, the prescheme $T \times_F F'$ is representable over T, in fact by a principal Galois bundle under Γ.

By Exposé IV, 4.6.6, F is therefore representable if and only if the quotient F'/F'', where $F'' = F' \times_F F'$, exists in (**Sch**) and is universally effective for faithfully flat quasi-compact morphisms. Equivalently, the quotient F'/Γ must exist and be universally effective for such morphisms. Since F' is affine over S, Exposé V, 4.1 shows that this condition is satisfied. This proves the representability of F in *(a)*.

Finally, suppose that G is of finite presentation over S. The additional assertion follows immediately from the preceding proof: by *(b)*, F' is then locally of finite presentation over S, so that Exposé V may be applied.

4. The Scheme of Multiplicative-Type Subgroups of a Smooth Affine Group

The principal result of the present Exposé is the following.

Theorem 4.1. — *Let S be a prescheme, let G be an S-group prescheme smooth and affine over S, and let*

$$F\colon (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

be the functor defined by

$$F(T) = \{\text{multiplicative-type subgroups of } G_T\}.$$

Then F is representable, smooth, and separated over S.

Let us record at once the following variant.

Corollary 4.2. — *Let G and H be two S-group preschemes, with G smooth and affine over S, and H of multiplicative type and of finite type. Then* $\mathrm{Hom}_{S\text{-gr}}(H, G)$ *is representable, smooth, and separated over S.*

Indeed, when H is smooth over S, so is $H \times_S G$, and one may apply 4.1 to the latter. By considering graph subgroups, $\mathrm{Hom}_{S\text{-gr}}(H, G)$ becomes a subfunctor F' of the functor F of multiplicative-type subgroups of $H \times_S G$: namely, $F'(T)$ is the set of multiplicative-type subgroups K of

$$(H \times_S G)_T = H_T \times_T G_T$$

such that the homomorphism induced by the first projection,

$$K \longrightarrow H_T,$$

is an isomorphism. By Exposé IX, 2.9, the canonical morphism $F' \to F$ is an open immersion. Since F is representable, the same is true of F', which is represented by an open subprescheme of F; and since F is separated and smooth over S, so is F'.

When H is finite over S, it is enough to apply 3.12(b) to obtain the representability of $\mathrm{Hom}_{S\text{-gr}}(H, G)$. When $H = H_1 \times_S H_2$, with H_1 smooth over S and H_2 finite over S, then $\mathrm{Hom}_{T\text{-gr}}(H_T, G_T)$ is identified with the subset of

$$\mathrm{Hom}_{T\text{-gr}}\big((H_1)_T, G_T\big) \times \mathrm{Hom}_{T\text{-gr}}\big((H_2)_T, G_T\big)$$

formed by the pairs (u_1, u_2) such that $u_1 \cdot u_2 = u_2 \cdot u_1$. It follows that $\mathrm{Hom}_{S\text{-gr}}(H, G)$ is represented by a closed subprescheme of

$$\mathrm{Hom}_{S\text{-gr}}(H_1, G) \times_S \mathrm{Hom}_{S\text{-gr}}(H_2, G) = X.$$

Indeed, apply Exposé VIII, 6.5(b), with $Y = H$, $Z = G$, and with q_1 and q_2 defined respectively by

$$(u_1, u_2) \longmapsto u_1 \cdot u_2 \qquad \text{and} \qquad (u_1, u_2) \longmapsto u_2 \cdot u_1.$$

In the general case, the question is local on S for the Zariski topology, so we may suppose that S is affine and that H has constant type over S. Since H is quasi-isotrivial (Exposé X, 4.5), there is a surjective étale morphism $S' \to S$, with S' affine, that splits H, that is, such that $H' = H_{S'}$ is diagonalizable. The preceding result then applies, because a diagonalizable group is the product of a diagonalizable torus and a finite diagonalizable group. It remains to see that the descent datum obtained on the S'-prescheme

$$\mathrm{Hom}_{S'\text{-gr}}(H', G') = X'$$

is effective. This follows from the argument of Exposé X, 5.4, noting that, there, the hypothesis that $X' \to S'$ be separated and locally quasi-finite was used only to ensure that every open part of X' quasi-compact over S' was quasi-affine over S'. That property still holds here, as follows readily from the fact that it holds for the functor F of 4.1 (as will be seen during the proof of 4.1). This argument—or any similar variant of this small dévissage—establishes the representability of $\mathrm{Hom}_{S\text{-gr}}(H, G)$, and at the same time shows that it is separated over S. It is locally of finite presentation over S, as follows, for example, from the fact already noted in 3.2 that this functor "commutes with filtered direct limits of rings." Finally, the functor is formally smooth over S by 2.1, and is therefore smooth over S.

Remark 4.3. — We have deduced 4.2 from 4.1; this is genuinely immediate only when H is also smooth over S. To make the deduction without contortions in the general case, one

would need the representability assertion in 4.1 without assuming that G is smooth over S, assuming only that it is affine and of finite presentation over S. Of course, F would then no longer be smooth over S in general. There is little doubt that 4.1 remains true under these more general hypotheses, but the proof appears likely to be more delicate, because one can no longer invoke 3.8.[45]

Let us nevertheless point out that when G is a closed subgroup of an affine smooth group G_0 over S, the functor F of multiplicative-type subgroups of G is represented by a closed subprescheme of the prescheme representing the analogous functor F_0 for G_0, which is covered by 4.1; this follows readily by applying Exposé VIII, 6.4. This also raises the question whether an affine S-group scheme G, affine and of finite presentation over S, is isomorphic to a subgroup scheme of $\mathrm{GL}(n)_S$ for some suitable n. This is true when S is the spectrum of a field (cf. Exposé $\mathrm{VI_B}$, 11.11), but is unfortunately false in general, even for tori; see 4.6. Finally, one could also prove 4.2 directly by exactly the same method as 4.1.

We now prove 4.1. Since the functor F is evidently local in nature, we may suppose that S is affine, say $S = \mathrm{Spec}(A)$, where A is a ring. Regarding A as the filtered direct limit of its subrings of finite type over $\mathbb{Z}$, and noting that G descends from a smooth affine group over one such subring (EGA IV, 8), we are reduced to the case in which S is noetherian.

For every integer $n > 0$, let T_n be the functor defined like F, but restricted to multiplicative-type subgroups H of G_T such that $n \cdot \mathrm{id}_H = 0$, that is, such that ${}_nH = H$. Order the set I of positive integers by divisibility. When m is a multiple of n, define

$$p_{n,m} \colon T_m \longrightarrow T_n$$

by

$$p_{n,m}(H) = {}_nH = \mathrm{Ker}(n \cdot \mathrm{id}_H).$$

In this way the T_n form an inverse system of functors over S. By 3.12(a), the functors T_n are representable, affine, and of finite type over S. Similarly define morphisms

$$u_n \colon F \longrightarrow T_n$$

by $u_n(H) = {}_nH$. Thus one obtains a morphism

$$u \colon F \longrightarrow T = \varprojlim_n T_n,$$

where the inverse limit is taken in the category of functors over S. Since the T_n are representable and affine over S, so is T: it is the spectrum of the filtered direct limit of the quasi-coherent algebras on S defining the T_n. Of course, T is not in general of finite type over S.

We shall apply 3.7. It is enough to verify conditions *(a)–(d)* there; these will imply that F is represented by a prescheme locally of finite type over S. It then follows from 2.1 bis that F is even smooth over S. Moreover, since F is a subfunctor of T, which is affine over S, F is separated over S: it is separated over T, and T is separated over S. We

[4] This is indeed proved for G flat and quasi-affine over S, with connected fibres, provided one restricts to the central subtori of G (Exposé XV, 8.8).

[5] Editor's note: when H is smooth (not necessarily affine) over a normal locally noetherian base S, M. Raynaud showed that the largest representable open subfunctor of the functor of subtori of H is a disjoint sum of opens smooth and affine over S. This is Theorem IX.9.26 in *Faisceaux amples sur les schémas en groupes et sur les espaces homogènes*, Lecture Notes in Mathematics 119 (1970).

can also prove at once the supplementary assertion used above: every open $U \subset F$ that is quasi-compact over S is quasi-affine over S. This follows from 3.11 and the fact that the T_n are affine over S.

Let us now verify the conditions of 3.7.

(a) F is a sheaf for the faithfully flat quasi-compact topology, by the descent theory of SGA 1, Exposé VIII, which applies here because groups of multiplicative type over T are affine over T. It commutes with filtered direct limits of rings by the standard machinery of EGA IV, 8. Let us show that it commutes with adic inverse limits of local artinian rings. For the analogous functor considered in 4.2, this is precisely Exposé IX, 7.1, in the special case where A is a complete noetherian local ring endowed with an ideal of definition for its usual topology. Notice that this is exactly where the hypothesis that G be affine enters essentially. In the present case we are reduced to the following lemma.

Lemma 4.4. — *Let A be a complete noetherian local ring with maximal ideal $\mathfrak{m}$, let G be a group scheme affine over $S = \mathrm{Spec}(A)$, and, for every integer $n \geq 0$, put*

$$S_n = \mathrm{Spec}\big(A/\mathfrak{m}^{n+1}\big), \qquad G_n = G \times_S S_n.$$

For every n, let H_n be a multiplicative-type subgroup of finite type of G_n, and suppose that, whenever $m \geq n$, H_n is obtained from H_m by reduction. Then there exists a unique multiplicative-type subgroup H of G whose reductions are the H_n.

By Exposé X, 3.2, there exists a multiplicative-type group H over S, necessarily of finite type and isotrivial and determined up to unique isomorphism, together with an isomorphism

$$H \times_S S_0 \simeq H_0.$$

Here one uses the fact that H_0 is isotrivial, since it is of finite type over a field; see Exposé X, 1.4. By Exposé X, 2.1, for every n the isomorphism $H \times_S S_0 \simeq H_0$ lifts to a unique isomorphism $H \times_S S_n \simeq H_n$. Exposé IX, 7.1, already cited, now shows that the homomorphisms $H_n \to G_n$ arise from a unique homomorphism of S-groups

$$u\colon H \longrightarrow G.$$

By Exposé IX, 6.6, this homomorphism is a monomorphism because $u_0\colon H_0 \to G_0$ is one. This proves 4.4.

(b) The morphism $u\colon F \to T$ is a monomorphism. This follows from the density theorem, Exposé IX, 4.7, in the form of Corollary 4.8(b). It is essential for the present application that this result is available over a base that need not be noetherian. Since the functor T does not commute with filtered direct limits of rings, one cannot reduce to the noetherian case a priori.

(b′) The morphism $u\colon F \to T$ is infinitesimally étale; in other words, we have the following lemma.

Lemma 4.5. — *Let A be an artinian local ring with residue field k, let $S = \mathrm{Spec}(A)$, let $I \subset \mathrm{rad}(A)$ be an ideal, and put $S_0 = \mathrm{Spec}(A/I)$. Let G be an S-group prescheme smooth over S, and put $G_0 = G \times_S S_0$. Suppose given a multiplicative-type subgroup H_0 of G_0 and, for every integer $n > 0$, a multiplicative-type subgroup $H(n)$ of G, in such a way that*

1° *for every multiple m of n, $H(n) = {}_nH(m)$; and*

2° *$H(n)_0 = {}_nH_0$.*

Then there exists a unique multiplicative-type subgroup H of G such that ${}_nH = H(n)$ for every n.

Uniqueness is already contained in *(b)*. For existence, an immediate induction reduces us to the case in which $I\mathfrak{m} = 0$, where $\mathfrak{m}$ is the maximal ideal of A. Put

$$k = A/\mathfrak{m}, \qquad S_0 = \mathrm{Spec}(k), \qquad G_0 = G \times_S S_0,$$

and let $\mathfrak{g}_0$ and $\mathfrak{h}_0$ be the Lie algebras of G_0 and H_0, respectively. There is a canonical isomorphism of groups

$$\mathfrak{g}_0 \otimes_k I \simeq \mathrm{Ker}\big(G(S) \longrightarrow G(S_0)\big);$$

see Exposé III. By Exposé IX, 3.6 bis and 3.7, there exists a multiplicative-type subgroup H of G reducing to H_0, and such an H is determined, up to an inner automorphism by an element of

$$N = \mathrm{Ker}\big(G(S) \longrightarrow G(S_0)\big).$$

Thus the set P of liftings H of H_0 is a principal homogeneous set under N/M, where

$$M = \{g \in N \mid \mathrm{int}(g) \cdot H = H\}.$$

One sees readily (Exposé III) that this subgroup is precisely the vector subspace of $\mathfrak{g}_0 \otimes_k I$ consisting of the invariants under H_0, where H_0 acts on $\mathfrak{g}_0 \otimes_k I$ through the representation induced by the adjoint representation of G_0. Likewise, the set $P(n)$ of liftings of $H(n)_0 = {}_nH_0$ is a principal homogeneous set under $N/M(n)$, where $M(n)$ is the vector subspace of $\mathfrak{g}_0 \otimes_k I = N$ consisting of the invariants under ${}_nH_0$. The density theorem, Exposé IX, 4.7, shows readily that $M = M(n)$ for sufficiently large n, where "large" is understood with respect to the divisibility order on the positive integers. Consequently, the natural map

$$H \longmapsto {}_nH$$

from P to $P(n)$, which is compatible with the actions of N, and hence with the homomorphism

$$N/M \longrightarrow N/M(n)$$

between the acting groups, is bijective for sufficiently large n. The conclusion of 4.5 follows immediately.

To verify conditions *(c)* and *(d)* of 3.7, we use 3.8, which reduces us to checking *(c′)* and *(d′)* below.

(c′) The T_n are smooth over S, and the transition morphisms $p_{n,m}\colon T_m \to T_n$ are smooth.

This is precisely 2.2 bis.

(d′) With the notation of 3.8, a point ξ of F with values in a field k over S is just a multiplicative-type subgroup H_0 of $G_0 = G_k$. Repeating the argument in *(b′)* above, we see that the integer $d(\xi)$ considered in 3.8 is

$$d(\xi) = \dim_k\big(\mathfrak{g}_0/\mathfrak{g}_0^{H_0}\big),$$

where $\mathfrak{g}_0$ is the Lie algebra of G_0 and $\mathfrak{g}_0^{H_0}$ is the vector subspace of invariants under H_0. This is a finite integer, so condition *(d')1°)* of 3.8 holds.

With the notation of *(d')2°)*, giving a morphism $v\colon X \to F$ amounts to giving a multiplicative-type subgroup H of G_X. For $x \in X$, the integer $d(\xi_x)$ is then the dimension of

$$\big(\mathfrak{g} \otimes \kappa(x)\big)/\big(\mathfrak{g} \otimes \kappa(x)\big)^{H_x}.$$

where $\mathfrak{g}$ is the sheaf of Lie algebras of G_X. It is locally free of finite type over X, because G_X is smooth over X, and $\mathfrak{g}\otimes\kappa(x)$ is the Lie algebra of the fibre G_x of G_X at x; here H_x is the fibre of H at x. Now $\mathfrak{g}$ is a locally free module on X on which the multiplicative-type group H acts, and the invariant subfunctor $\mathfrak{g}^H$ is therefore given by a locally direct-summand, hence locally free, subsheaf of $\mathfrak{g}$. By descent, one reduces to the case in which H is diagonalizable, and then applies Exposé I, 4.7.3, noting that the invariant subsheaf corresponds to the degree-zero component. Consequently,

$$d(\xi_x) = \operatorname{rank}_x(\mathfrak{g}/\mathfrak{g}^H),$$

so this is a locally constant function of x. This proves condition *(d′)*.

We have thus verified the conditions of 3.7, completing the proof of 4.1.

Remark 4.6. — When $G = \mathrm{GL}(n)_S$, one can give a much simpler and more explicit direct proof of 4.1, using Exposé I, 4.7.3. The proof moreover shows that, in this case, the moduli scheme is a disjoint sum of affine S-schemes. Proceeding as indicated in 4.3, one obtains the same result whenever G is a closed subgroup of a group of the form $\mathrm{GL}(n)_S$.

One must not, however, suppose that the preschemes representing the functors in 4.1 or 4.2 are always disjoint sums of affine S-schemes. For example, let H be a multiplicative-type group of finite type over a locally noetherian prescheme S. By Exposé X, 5.11, H is isotrivial if and only if

$$\mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_m)$$

is a disjoint sum of preschemes affine over S. But, as noted in Exposé X, 1.6, there may exist nonisotrivial tori H of relative dimension 2. For such a torus, $\mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_m)$ is not a disjoint sum of S-preschemes affine over S, and likewise the "dual" twisted constant group

$$\mathrm{Hom}_{S\text{-gr}}(\mathbb{G}_m, H)$$

is not such a sum. Indeed, if two commutative twisted constant groups R, R' of finite presentation are dual to one another,

$$R' = \mathrm{Hom}_{S\text{-gr}}(R, \mathbb{Z}_S),$$

then one readily sees that one is isotrivial if and only if the other is.

This last point also shows that such an H is not isomorphic to a subgroup of a group of the form $\mathrm{GL}(n)_S$. More precisely, one can show that a multiplicative-type group of finite type over a connected locally noetherian S is isomorphic to a subgroup of a group of the form $\mathrm{Aut}_{\mathrm{Mod}}(E)$, where E is a locally free module of finite type over S, if and only if it is isotrivial. Finally, taking $G = H \times \mathbb{G}_m$ in the two preceding examples gives an example in which the prescheme representing the functor F of 4.1 is not a disjoint sum of affine S-preschemes; if desired, G may be taken to be a torus of relative dimension 3.

5. First Corollaries of the Representability Theorem

Let H and G be as in 4.2. Put

$$M = \mathrm{Hom}_{S\text{-gr}}(H, G),$$

which, by 4.2, is a smooth S-prescheme separated over S. Notice that G acts on the functor $M = \mathrm{Hom}_{S\text{-gr}}(H, G)$ by

$$(g, u) \longmapsto \mathrm{int}(g) \circ u.$$

Hence there is a canonical morphism

$$G \times_S M \longrightarrow M \times_S M, \tag{x}$$

whose components are the preceding morphism $G \times_S M \to M$ and the second projection $G \times_S M \to M$.

Corollary 5.1. — *The preceding morphism* (x) *is smooth.*

This follows from 4.2 and 2.3. This statement is equivalent to the following one.

Corollary 5.2. — *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S-groups. Then the subfunctor* $\mathrm{Transp}(u_1, u_2)$ *of G (cf. 2.4) is representable by a closed subprescheme of G, smooth over S.*

It remains only to prove that $\mathrm{Transp}(u_1, u_2) \to G$ is indeed a closed immersion. This follows because it is the equalizer of a pair of morphisms $G \rightrightarrows M$, namely

$$g \longmapsto \mathrm{int}(g)u_1 \qquad \text{and} \qquad g \longmapsto u_2,$$

the latter being the "constant" morphism, and because M is separated over S. In particular:

Corollary 5.3. — *Let $u : H \to G$ be a morphism of S-groups. Then*

$$\mathrm{Centr}_G(u) = \mathrm{Transp}(u, u)$$

is representable by a closed group subprescheme of G, smooth over S. Moreover, the quotient of G by $\mathrm{Centr}_G(u)$ *is representable by an open subprescheme of M.*

It remains to prove the last assertion. The morphism

$$g \longmapsto \mathrm{int}(g) \circ u$$

from G to M is smooth and of finite type by 5.2, hence is an open morphism (EGA IV, 6.6). Let U denote its image, equipped with the structure induced from M. The induced morphism $G \to U$ is smooth, surjective, and of finite type, and is therefore a covering for the faithfully flat quasi-compact topology. Moreover, it is clear that the preceding morphism $G \to M$ makes G a formally principal homogeneous sheaf under $\mathrm{Centr}_G(u)_M$. It follows that the sheaf $G/\mathrm{Centr}_G(u)$ is represented by U.

Corollary 5.4. — *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S-groups. The following conditions are equivalent:*

(i) *u_1 and u_2 are conjugate locally for the étale topology.*

(i bis) *u_1 and u_2 are conjugate locally for the faithfully flat quasi-compact topology.*

(ii) *For every $s \in S$, let $\overline{s}$ denote the spectrum of an algebraic closure of $\kappa(s)$. The morphisms*

$$u_{1,\overline{s}}, u_{2,\overline{s}} : H_{\overline{s}} \rightrightarrows G_{\overline{s}}$$

are conjugate by an element of $G(\kappa(\overline{s}))$.

(ii bis) *The structural morphism* $\mathrm{Transp}(u_1, u_2) \to S$ *is surjective.*

(iii) $\mathrm{Transp}(u_1, u_2)$ *is a torsor under the action of the smooth S-group prescheme of finite type* $\mathrm{Centr}_G(u_1)$.

The implications (i) ⇒ (i bis) and (ii) ⇒ (ii bis) are immediate, the latter by the Nullstellensatz. The implication (i bis) ⇒ (ii) follows from the "principle of finite extension" (EGA IV, 9). Moreover, (ii bis) ⇒ (iii): since $\mathrm{Transp}(u_1, u_2)$ is smooth over S, it is flat over S, and since it is of finite type, it is quasi-compact over S. Thus it is faithfully flat and quasi-compact over S if and only if its structural morphism is surjective. On the other hand, it is formally principal homogeneous under $\mathrm{Centr}_G(u_1)$, which is faithfully flat and quasi-compact over S. The last condition is therefore also equivalent to saying that $\mathrm{Transp}(u_1, u_2)$ is a torsor under $\mathrm{Centr}_G(u_1)$, in the faithfully flat quasi-compact topology. Finally, (iii) ⇒ (i) follows from the smoothness of $\mathrm{Transp}(u_1, u_2)$ over S and from Hensel's lemma in the form 1.10.

Remark 5.5. — Fix $u_1 = u : H \to G$. Consider the functor

$$(\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

that assigns to every T over S the set of homomorphisms of T-groups $u_2 : H_T \to G_T$ that are locally conjugate, for the étale topology, to $u_T : H_T \to G_T$. This functor is represented precisely by the open subprescheme of M, isomorphic to $G/\mathrm{Centr}_G(u)$, considered in 5.3.

Let us sketch the variants of the preceding results obtained by applying 4.1 instead of 4.2. Thus let G be a group prescheme smooth and affine over S, and now let M denote the smooth S-prescheme separated over S that represents the functor considered in 4.1. Again G acts on M:

$$(g, H) \longmapsto \mathrm{int}(g)(H).$$

Hence, as above, there is a morphism

$$G \times_S M \longrightarrow M \times_S M. \tag{x bis}$$

Corollary 5.1 bis. — *The preceding morphism is smooth.*

This follows from 4.1 and 2.3 bis. We again conclude:

Corollary 5.2 bis. — *Let H_1, H_2 be two multiplicative-type subgroups of G. Then the subfunctor $\mathrm{Transp}_G(H_1, H_2)$ of G is representable by a closed subprescheme of G, smooth over S.*

In particular:

Corollary 5.3 bis. — *Let H be a multiplicative-type subgroup of G. Then the group subfunctor $\mathrm{Norm}_G(H)$ of G is representable by a closed subprescheme of G, smooth over S. Moreover, the quotient $G/\mathrm{Norm}_G(H)$ is representable by an open subprescheme of M.*

Corollary 5.4 bis. — *Let H_1, H_2 be two multiplicative-type subgroups of G. The following conditions are equivalent:*

(i) *H_1 and H_2 are conjugate locally for the étale topology.*

(i bis) *H_1 and H_2 are conjugate locally for the faithfully flat quasi-compact topology.*

(ii) *For every $s \in S$, let $\overline{s}$ denote the spectrum of an algebraic closure of $\kappa(s)$. The subgroups $H_{1,\overline{s}}$ and $H_{2,\overline{s}}$ of $G_{\overline{s}}$ are conjugate by an element of $G\big(\kappa(\overline{s})\big)$.*

(ii bis) *The structural morphism $\mathrm{Transp}(H_1, H_2) \to S$ is surjective.*

(iii) *$\mathrm{Transp}(H_1, H_2)$ is a principal homogeneous bundle under the action of the smooth S-group prescheme of finite type $\mathrm{Norm}_G(H_1)$.*

Remark 5.5 bis. — Remark 5.5 carries over to the present case as well.

Remark 5.6. — To prove 5.2, and hence also the first assertion of 5.3 and the assertions of 5.4, the reference to 4.2 may be replaced by a reference to VIII, 6.4, whose proof is much easier. This also shows that the hypothesis that G be affine over S is unnecessary there. Moreover, in Section 6 we shall show how a variant of this method extends these results to certain groups H more general than groups of multiplicative type. The same observations apply to the variants 5.2 bis and following. By contrast, the result 5.8 below makes essential use of all the hypotheses imposed—in particular, that G be affine and smooth over S, and that H be of multiplicative type—and the lecturer knows no proof other than one using the representability theorems 4.1 or 4.2.[6]

5.7. Since the morphism (x), respectively (x bis), is smooth and hence open, its image is open. Let R be this image, equipped with the structure induced from $M \times_S M$. One checks readily that R is an equivalence relation on M, namely the relation made explicit in 5.4, respectively 5.4 bis. It would be interesting to know whether the quotient sheaf M/R, which is formally étale over S, is representable; if so, it is represented by a prescheme étale over S. This is the case, for example, when S is the spectrum of a field. Notice also that R need not be closed in $M \times_S M$, even when G is quasi-finite over S. Thus, when M/R is representable, it may be nonseparated over S.

Theorem 5.8. — *Let S be a prescheme, let G be a smooth affine S-prescheme, and let $s \in S$. Then:*

(a) *For every multiplicative-type subgroup H_0 of G_s, there exist an étale morphism $S' \to S$, a point $s' \in S'$ lying over s such that the residue-field extension $\kappa(s')/\kappa(s)$ is trivial, and a multiplicative-type subgroup H' of $G' = G \times_S S'$, such that*

$$H'_{s'} = H_0 \otimes_{\kappa(s)} \kappa(s').$$

(b) *For every group homomorphism $u_0 : H_s \to G_s$, where H is a multiplicative-type S-group prescheme of finite type, there exist an étale morphism $S' \to S$, a point $s' \in S'$ lying over s such that $\kappa(s')/\kappa(s)$ is trivial, and a group homomorphism*

$$u' : H' \longrightarrow G', \qquad H' = H \times_S S', \quad G' = G \times_S S',$$

such that

$$u'_{s'} : H'_{s'} \longrightarrow G'_{s'} \quad \text{equals} \quad u_0 \otimes_{\kappa(s)} \kappa(s').$$

This follows from 4.1, respectively 4.2, and from Hensel's lemma in the form 1.10.

Proposition 5.9. — *Let S be a prescheme, let G be a smooth affine S-prescheme, and let H be a multiplicative-type subgroup of G. Consider*

$$\mathrm{Centr}_G(H) \hookrightarrow \mathrm{Norm}_G(H),$$

which, by 5.3 and 5.3 bis, are closed group subpreschemes of G, smooth over S. The first is then an open and closed subprescheme of the second, and the quotient sheaf

$$W_G(H) = \mathrm{Norm}_G(H)/\,\mathrm{Centr}_G(H)$$

is represented by an open group subprescheme of $\mathrm{Aut}_{S\text{-gr}}(H)$. It is therefore a quasi-finite, étale, and separated S-group prescheme.

[6] The situation has changed since this text was written; cf. Exposés XV and XIX, no. 6.

Indeed, consider the evident homomorphism

$$\theta : \mathrm{Norm}_G(H) \longrightarrow \mathrm{Aut}_{S\text{-gr}}(H),$$

whose kernel is, by definition, $\mathrm{Centr}_G(H)$. Since $\mathrm{Aut}_{S\text{-gr}}(H)$ is represented by an étale S-group prescheme separated over S (X, 5.10), its unit section is an open and closed immersion. Its inverse image under θ is therefore an open and closed subgroup of $\mathrm{Norm}_G(H)$.

Moreover, θ is smooth. This follows formally from the definitions and from the facts that $\mathrm{Norm}_G(H)$ is smooth over S and that $\mathrm{Aut}_{S\text{-gr}}(H)$ is étale over S. As in 5.3, the image of θ is consequently an open subprescheme U of $\mathrm{Aut}_{S\text{-gr}}(H)$, and, with the induced structure, U represents the quotient sheaf $\mathrm{Norm}_G(H)/\mathrm{Centr}_G(H)$. The latter is therefore étale and separated over S. It is also quasi-finite over S, since it is quasi-compact over S, being the image of $\mathrm{Norm}_G(H)$, which is quasi-compact. This proves 5.9.

Corollary 5.10. — *For every $s \in S$, put*

$$w(s) = \mathrm{rank}\big(\mathrm{Norm}_{G_s}(H_s)/\mathrm{Centr}_{G_s}(H_s)\big).$$

This is also the index of $\mathrm{Centr}_{G(\overline{k})}(H(\overline{k}))$ in $\mathrm{Norm}_{G(\overline{k})}(H(\overline{k}))$, where $\overline{k}$ is an algebraic closure of $\kappa(s)$. Then the function $s \mapsto w(s)$ is lower semicontinuous. It is constant in a neighborhood of s if and only if $W_G(H)$ is finite over S in a neighborhood of s.

Indeed, for every S-prescheme W that is étale, of finite type, and separated over S, the function

$$s \longmapsto \mathrm{rank}[W_s : \kappa(s)]$$

is lower semicontinuous, and it is constant in a neighborhood of s if and only if W is finite over S in a neighborhood of s. This fact was noted in SGA 1, I, 10.9; its proof, which presents no difficulty, is given in EGA IV.[7]

Remark 5.11.[8] — Let G be a group prescheme affine and smooth over S, and let H be a multiplicative-type subgroup. Then 5.3 and 5.3 bis imply that the quotients

$$G/\mathrm{Centr}_G(H) \qquad \text{and} \qquad G/\mathrm{Norm}_G(H)$$

are preschemes smooth and quasi-affine over S. Indeed, in both cases the moduli prescheme M in which the quotient is embedded has the property that every open $U \subset M$ quasi-compact over S is quasi-affine over S, by 3.11.

We shall moreover see in the next Exposé[9] that for every such U, the schematic closure $\overline{U}$ of U in M is even affine over S. Thus the quotients under consideration are affine over S, provided that the open subprescheme U of the corresponding moduli prescheme—of group homomorphisms from H to G, respectively of multiplicative-type subgroups of G—is closed in M. This is the case, for example, when H is a "maximal torus" in the second case, or when S is the spectrum of a field; cf. XII.5.4.[10] I do not know whether the quotients under consideration are affine over S in general.

[7] EGA IV, 15.5.1 and 18.10.7.

[8] Editor's note: For a generalization to the nonaffine case, see M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes in Mathematics 119 (1970), IX.2.8.

[9] Editor's note: Cf. Remark XII.5.7. We mention here the following generalization. Without assuming G affine, if H is a torus and S is locally noetherian, U is affine and smooth by a theorem of Raynaud (*loc. cit.*, IX.2.6 and IX.2.8). Indeed, $G/\mathrm{Norm}_G(H)$ is an open subprescheme of M, and the largest representable open subfunctor of M is a disjoint sum of open subpreschemes smooth and affine over S.

[10] Editor's note: This assertion assumes that G has locally constant reductive rank.

6. A Rigidity Property for Homomorphisms of Certain Group Schemes, and the Representability of Certain Transporters[11]

Proposition 6.1. — *Let S be a locally noetherian prescheme, let H be a commutative S-group prescheme of finite type over S, and let E be a set of positive integers stable under multiplication. Suppose that the following conditions hold:*

(a) *for every $n \in E$, the subgroup*

$$ {}_nH = \mathrm{Ker}(n \cdot \mathrm{id}_H) $$

is finite and flat over S;

(b) *for every $s \in S$, the family of the ${}_nH_s$, $n \in E$, is schematically dense in the fibre H_s.*

Let Y be a closed subprescheme of H. Then the subfunctor $\prod_{H/S}(Y/H)$ of S (compare Exposé VIII, 6) is representable by a closed subprescheme T of S. If S is noetherian, there exists $n \in E$ such that[12]

$$ \prod_{H/S}(Y/H) = \prod_{{}_nH/S} (({Y \cap {}_nH})/{}_nH). \tag{x} $$

Indeed, by Exposé VIII, 6.4, condition (a) implies that ${}_nH$ is finite and flat, and hence a fortiori "essentially free" over S. Thus the right-hand side of (x) is representable by a closed subprescheme T_n of S. Order E by divisibility. The T_n then form a decreasing family of closed subpreschemes of S; consequently, when S is noetherian—as we may now suppose—this family is stationary for sufficiently large n. Let T be the stable value of T_n. We claim that T represents the left-hand side of (x), which will complete the proof.

It is enough to prove the following. Let S' be a prescheme over S such that

$$ (Y \cap {}_nH)_{S'} = ({}_nH)_{S'} \qquad \text{for every } n \in E, $$

or, equivalently, such that $Y_{S'} \supset ({}_nH)_{S'}$ for every $n \in E$. Then $Y_{S'} = H_{S'}$. This follows at once from Exposé IX, 4.4: in view of conditions (a) and (b), the family of the $({}_nH)_{S'}$ is schematically dense in $H_{S'}$.

Theorem 6.2. — *Let S be a prescheme, let H, E be as in 6.1 and satisfy conditions (a) and (b), let $u : H \to G$ be a homomorphism of S-groups from H to an S-group prescheme locally of finite type over S, and let K be a closed subgroup prescheme of G. Consider the subfunctor $\mathrm{Transp}_G(u, K)$ of G (cf. 2.5). It is representable by a closed subprescheme of G. If G is noetherian—for example, if S is noetherian and G is of finite type over S—there exists $n \in E$ such that*

$$ \mathrm{Transp}_G(u, K) = \mathrm{Transp}_G(u|_{{}_nH}, K). $$

Suppose, finally, that G is smooth over S, that K is smooth over S or is of multiplicative type, and that H, E satisfy the following condition, stronger than (a):

[11] The present section does not use the results of Sections 3, 4, and 5; its natural place would be in Exposé $\mathrm{VI_B}$.

[12] The printed formula writes the right-hand quotient as $(Y \cap {}_nH)/S$. Since the product functor is taken over ${}_nH/S$, and as the proof requires, the quotient is $(Y \cap {}_nH)/{}_nH$.

a' for every $n \in E$, the subgroup ${}_nH$ of H is of multiplicative type.

Then $\mathrm{Transp}_G(u, K)$ *is smooth over* S.

Consider the G-group

$$H_G = H \times_S G,$$

which plainly satisfies the conditions of 6.1, with S replaced by G and H by H_G. Let Y be the closed subprescheme of H_G that is the inverse image of $K \subset G$ under the map

$$H_G = H \times_S G \longrightarrow G, \qquad (h, g) \longmapsto \mathrm{int}(g) \cdot u(h).$$

Then

$$\mathrm{Transp}_G(u, K) = \prod_{H_G/G} (Y/H_G)$$

(compare Exposé VIII, Example 6.5(e)). The first assertions therefore follow from 6.1. Moreover, for every quasi-compact open $U \subset G$, there exists $n \in E$ for which the restrictions to U of the two functors

$$\mathrm{Transp}_G(u, K) \qquad \text{and} \qquad \mathrm{Transp}_G(u|_{{}_nH}, K)$$

coincide. To verify the last assertion of 6.2, we may therefore replace H by some ${}_nH$, after which it is enough to apply 2.5, since ${}_nH$ is assumed to be of multiplicative type over S.

Remark 6.3. — The preceding proof uses only the very elementary result of Exposé VIII, 6.4, together, for its last part, with 2.5; when K is smooth over S, the latter is the infinitesimal result of Exposé IX, 3.6, and hence the vanishing of the cohomology of multiplicative-type groups. In the most important cases (cf. 6.7), the integers $n \in E$ may even be taken prime to the residue characteristics of S, that is, invertible in $\mathcal{O}_S$. The ${}_nH$ are then finite étale over S, and the cohomological result invoked is practically trivial. Thus, in that situation, 6.2 is independent of the theory of multiplicative-type groups.

6.4. — As usual, one sees that 6.1 extends to the following situation. Let S' be an S-prescheme (not necessarily locally noetherian), put $H' = H_{S'}$, and let Y' be a closed subprescheme of H', provided that $Y' \to H'$ is of finite presentation, that is, that the ideal defining Y' is of finite type. Then $\prod_{H'/S'}(Y'/H')$ is representable by a closed subprescheme T' of S', the morphism $T' \to S'$ is of finite presentation, and, if S' is quasi-compact, there exists $n \in E$ for which the analogue of (x) holds. It follows as well that the first assertion of 6.2 remains valid without assuming that G is locally of finite type over S, provided that the immersion $K \to G$ is of finite presentation.

6.5. — As announced in 5.6, Theorem 6.2 makes it possible to extend to group preschemes satisfying (a') and (b) above certain results proved by another method, under more restrictive hypotheses, for groups of multiplicative type. This applies to 5.2, the beginning of 5.3, and 5.4 and their corresponding "bis" variants, and also to 5.9 and 5.10. It also applies to the results of Exposé IX, Sections 5 and 6, with the exception of Exposé IX, 6.8. The latter assertion is already false for a homomorphism of abelian schemes $u : H \to G$ over the spectrum S of a discrete valuation ring of residue characteristic $p > 0$: it can happen that $\mathrm{Ker}\, u$ has the trivial group as its generic fibre and, as its special fibre, a radicial group that is not reduced to the trivial group.

6.6. — We have just given an example of a rigidity property of multiplicative-type groups that is not shared by abelian schemes. Another, extremely important, example is that the existence theorem for infinitesimal extensions of homomorphisms in Exposé IX, 3.6

no longer holds when H is an abelian scheme. Thus a nonzero abelian scheme over a field admits nontrivial infinitesimal variations, in contrast to a multiplicative-type group. This is the infinitesimal aspect of the fact that there is a theory of moduli—one that, moreover, is far from complete—for abelian varieties, whereas the theory of moduli of multiplicative-type groups is empty.

Another, "global," aspect of this infinitesimal difference is the following. Let H be an abelian scheme over a locally noetherian prescheme S, and let G be a commutative S-group prescheme locally of finite type over S. One can show that $\mathrm{Hom}_{S\text{-gr}}(H, G)$ is representable by a prescheme locally of finite type over S. Unlike the corresponding prescheme when H is of multiplicative type, however, it is not étale over S, but only unramified. Thus, for example, if S is the spectrum of a complete discrete valuation ring and H, G are abelian schemes over S, there may exist homomorphisms $H_0 \to G_0$ on the special fibres that do not arise "by specialization" from a homomorphism on the generic fibres.

6.7. — Theorem 6.2 applies whenever H is an abelian scheme over S, or, more generally, an extension of such a scheme by a torus. Indeed, the question is local on S, so one may suppose that there is a prime number ℓ prime to the residue characteristics of S. Taking E to be the set of powers of ℓ then suffices to satisfy conditions (a') and (b).

An argument analogous to that of 6.1 gives the following result.

Theorem 6.8. — *Let X be a prescheme smooth over S, with nonempty connected geometric fibres. For every closed subprescheme Y of X, the functor $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme T of S. If Y is of finite presentation over X, then T is of finite presentation over S.*

Since $f : X \to S$ is faithfully flat and locally of finite presentation, it is a covering for the fpqc topology. On the other hand, $T = \prod_{X/S}(Y/X)$ is plainly a subsheaf of S for the fpqc topology. Hence both the representability of T by a closed subprescheme of S and the question whether T is of finite presentation over S are local on S for the fpqc topology. After the base change $S' \to S$ with $S' = X$, we are reduced to the case in which X admits a section e over S. We may also suppose that S is affine, and hence in particular quasi-compact. We then have:

Corollary 6.9. — *Under the hypotheses of 6.8, suppose that S is quasi-compact and that X admits a section e over S. For every integer $n \geq 0$, let X_n be the order-n infinitesimal neighbourhood of the section e in X. Suppose that Y is of finite presentation over X. Then there exists an integer $n \geq 0$ such that*

$$\prod_{X/S}(Y/X) = \prod_{X_n/S}(Y_n/X_n), \qquad Y_n = Y \cap X_n.$$

This corollary implies 6.8 when Y is of finite presentation over X, by Exposé VIII, 6.4. Indeed, since X is smooth over S, X_n is finite and locally free over S, and a fortiori "essentially free" over S in the sense of Exposé VIII, 6.1. Thus $\prod_{X_n/S}(Y_n/X_n)$ is representable by a closed subprescheme of S. Furthermore, either the proof of the cited result or reduction to the noetherian case shows at once that this closed subprescheme is of finite presentation over S.

Let us first prove 6.9, and hence 6.8, when S is noetherian. Put

$$T_n = \prod_{X_n/S}(Y_n/X_n).$$

The T_n form a decreasing sequence of closed subpreschemes of S, and this sequence is

stationary because S is noetherian. Let

$$R = \bigcap_{n\geq 0} \prod_{X_n/S} (Y_n/X_n)$$

be their common value for large n. Plainly $T \subset R$, and it remains to show that $R \subset T$. After the base change $R \to S$, we are reduced to the case $R = S$, that is, $Y_n = X_n$ for every n, or equivalently $Y \supset X_n$ for every n. We must then prove $T = S$, that is, $Y = X$.

The inclusions $Y \supset X_n$ for all n, together with the fact that X is locally noetherian, imply that in a neighbourhood of each point of $e(S)$, Y is an induced open subprescheme of X. Hence there is an induced open subprescheme U of X such that $e(S) \subset U \subset Y$. By Exposé IX, 4.3, since the fibres of X/S are integral, U is schematically dense in X. As Y is a closed subprescheme containing U, it follows that $Y = X$. This proves 6.9, and hence 6.8, in the noetherian case.

The general case is reduced to the preceding one. For every $s \in S$, there is an affine open neighbourhood U of s and an affine open neighbourhood V of $e(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighbourhood of s contained in U. Choose an affine open neighbourhood S' of s contained in $e^{-1}(V) \cap f(V)$, and put $X' = V \cap f^{-1}(S')$. Then X' and S' are affine open subpreschemes of X and S, respectively, and X'/S' admits a section. By the local nature of 6.8 and 6.9, we may suppose that $S = S'$.

We claim that

$$\prod_{X/S} (Y/X) = \prod_{X'/S'} (Y'/X'), \qquad Y' = Y \cap X'.$$

Indeed, by Exposé IX, 4.6, X' is schematically dense in X, at least when X is quasi-separated over S, so that X' is retrocompact in X. In fact, one can readily prove that Exposé IX, 4.6 remains valid without the retrocompactness hypothesis. Likewise, after every base change $S_1 \to S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$, and the claimed equality follows immediately. We are thus reduced to the case $X = X'$, so we may suppose that both S and X are affine.

Now let $X = \mathrm{Spec}(B)$, and let J be the ideal of B defining Y. The ideal J is the filtered colimit of its finite-type subideals. Consequently, Y is the intersection of closed subpreschemes Y_i of X that are of finite presentation over S, and

$$\prod_{X/S} (Y/X) = \bigcap_i \prod_{X/S} (Y_i/X).$$

To prove 6.8, it is therefore enough to treat the case in which Y is of finite presentation over X. It remains to prove 6.9 with S and X affine. But X and Y over S then arise by base change $S \to S_0$ from an analogous situation X_0, Y_0 over a noetherian prescheme S_0. This reduces us to the already treated case in which S is noetherian, and completes the proof of 6.8 and 6.9.

Corollary 6.10. — *Let X be an S-group prescheme, smooth and of finite presentation, with connected fibres. Let Y be a group prescheme of finite presentation over S, and let $i : Y \to X$ be a monomorphism of S-group preschemes, so that Y is a subgroup of X. Then $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation. If S is quasi-compact, let X_n denote, for every $n \geq 0$, the order-n infinitesimal neighbourhood of the unit section of X, and put $Y_n = X_n \cap Y$. Then, for all sufficiently large n,*

$$\prod_{X/S} (Y/X) = \prod_{X_n/S} (Y_n/X_n).$$

The proof is essentially that of 6.9. The unit sections of X and Y induce bijective immersions $S \to X_n$ and $S \to Y_n$, and hence isomorphisms

$$S_{\mathrm{red}} \xrightarrow{\sim} (X_n)_{\mathrm{red}}, \qquad S_{\mathrm{red}} \xrightarrow{\sim} (Y_n)_{\mathrm{red}}.$$

It follows that the injection $Y_n \to X_n$ is proper. Since it is a monomorphism of finite presentation, it is therefore a closed immersion. Exposé VIII, 6.4, already used above, now shows that $\prod_{X_n/S}(Y_n/X_n)$ is representable by a closed subprescheme of S of finite presentation over S. It remains to prove the last assertion of 6.10 under the additional assumption that S is affine.

We reduce once more to the case in which S is noetherian. With the notation of the proof of 6.9, it is enough to prove $R = T$, or, equivalently, that $Y \supset X_n$ for every n implies $Y = X$. This hypothesis implies that $i : Y \to X$ is étale at the points of the unit section of Y over S, and therefore that Y is smooth over S at those points. Hence the open subprescheme Y' consisting of the points where Y is smooth over S is an induced open subgroup of Y. By Exposé X, 3.5, the morphism $Y' \to X$ is an étale monomorphism and hence an open immersion. Since the fibres of X are connected and every open subgroup of an algebraic group is also closed, this open immersion is surjective, and therefore an isomorphism. Thus $Y' = X$, and a fortiori $Y = X$, completing the proof of 6.10.

Arguing as in Exposé VIII, 6.5, we deduce:

Corollary 6.11. — *Let G be an S-group prescheme locally of finite type and quasi-separated over S, let H be a group prescheme smooth and of finite presentation over S, with connected fibres, and let $i : H \to G$ be a monomorphism of S-groups, so that H is a subgroup of G. Then:*

(a) $\mathrm{Centr}_G(H)$ *and* $\mathrm{Norm}_G(H)$ *are representable by closed subpreschemes of G of finite presentation over G. Likewise, for every monomorphism $j : K \to G$ of finite presentation between S-group preschemes,* $\mathrm{Transp}_G(H, K)$ *is representable by a closed subprescheme of G of finite presentation over G.*

(b) *If G is quasi-compact, then in each of the cases in (a) there exists an integer $n \geq 0$ such that, if H_n denotes the order-n infinitesimal neighbourhood of the unit section of H (and K_n is defined similarly in the transporter case),*

$$\begin{aligned} \mathrm{Centr}_G(H) &= \mathrm{Centr}_G(H_n), \\ \mathrm{Norm}_G(H) &= \mathrm{Norm}_G(H_n), \\ \mathrm{Transp}_G(H, K) &= \mathrm{Transp}_G(H_n, K) = \mathrm{Transp}_G(H_n, K_n). \end{aligned}$$

Apply 6.10 over the base prescheme G to the group prescheme

$$X = H_G = H \times_S G.$$

For the centralizer, take as Y the inverse image of the diagonal subgroup of $(G \times_S G)_G$ under a suitable homomorphism of G-groups from X to $(G \times_S G)_G$. For the transporter, take the inverse image of K_G under a suitable homomorphism of G-groups from X to G_G. The normalizer case reduces to the transporter case by taking $K = H$. The hypothesis on G ensures that $H \to G$ is of finite presentation, and hence that $Y \to X$ is of finite presentation. In the centralizer case, the same hypothesis ensures that the diagonal subgroup of $G \times_S G$ is of finite presentation over $G \times_S G$; again it follows that Y is of finite presentation over X.

Remark 6.12. — One can prove, using rather delicate results of EGA VI,[13] that if X is a prescheme flat and of finite presentation over S, and is either proper over S or has nonempty connected geometric fibres, then for every closed subprescheme Y of X of finite presentation over S, the functor $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation over S. Likewise, if X is an S-group prescheme flat and of finite presentation with connected fibres, and $i : Y \to X$ is a monomorphism of S-groups, with Y an S-group prescheme of finite presentation, then $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation over S. In particular, 6.11(a) remains valid if the hypothesis "H smooth over S" is replaced by "H flat over S."

[13] Cf. also J. P. Murre, "Representation of unramified functors. Applications," *Séminaire Bourbaki*, no. 294 (May 1965), Theorem 3, p. 13.

Exposé XII
Maximal Tori, the Weyl Group, Cartan Subgroups, and the Reductive Center

by A. Grothendieck

From the present Exposé onwards, in contrast to the preceding Exposés, we shall use the known results on the structure of smooth algebraic groups over an algebraically closed field k, and above all Borel's theory of affine algebraic groups, expounded in the 1956/57 Chevalley Seminar, *Classification of Algebraic Lie Groups.*[0] Following the usage in the theory of algebraic groups, we shall refer to that seminar by the siglum *BIBLE*. In the following Exposés we shall need, in particular, the results of *BIBLE* 4, 5, 6, and 7 (the number following *BIBLE* is the number of the Exposé). Moreover, it appears that scheme theory brings no notable simplification to Borel's theory as it is presented in *BIBLE*. For this reason it did not seem useful to reproduce that theory in the present Seminar: our purpose here is to deduce from results known over algebraically closed fields analogous results valid over an arbitrary base prescheme. (The situation will be different for Chevalley's structure theory of semisimple groups, which, it appears, is more advantageously treated *ab ovo* over an arbitrary base prescheme.)

In the oral Exposés (which we followed in Nos. 1 to 4), we restricted ourselves to affine group preschemes over S, relying essentially on the representability theorems of Exposé XI, No. 4. In the present notes (cf. Nos. 6 to 8), we show quickly how the affineness hypotheses can be eliminated by a simpler method that does not use the most delicate results of Exposé XI. For further generalizations of the results contained in the present Exposé, see also Exposés XV and XVI. (It is clear that the contents of Exposés XI, XII, XV, and XVI ought to be completely rewritten.)

0.1 Maximal tori

1.0. Let first G be an algebraic group over an algebraically closed field k. A *maximal torus of* G is an algebraic subgroup T of G which is a torus (here, since k is algebraically closed, this means that it is isomorphic to a group of the form $\mathbf{G}_m^r$) and which is maximal for this property. Notice that, since k is perfect, G_{red} is a subgroup of G, smooth over k, and hence is essentially an algebraic group in the sense of *BIBLE*. Moreover, every reduced

[-1] Version xy of 2 December 2008.

[0] Editor's note: this seminar has since been published, in a form revised by P. Cartier, as volume 3 of Chevalley's collected works (Springer, 2005).

subgroup of G (such as a torus) is automatically a subgroup of G_{red}. Consequently the maximal tori of G are precisely the maximal tori of G_{red}. When G is affine, and therefore G_{red} is affine, a fundamental theorem of Borel says that any two maximal tori of G are conjugate by an element of

$$G(k) = G_{\mathrm{red}}(k)$$

(*BIBLE* 6, Theorem 4(c)); in particular, they have the same dimension. We call the common dimension of the maximal tori of G the *reductive rank of* G. The assumption that G be affine is in fact unnecessary. This follows from a theorem of Chevalley asserting that every connected smooth algebraic group over k is an extension of an abelian variety by an affine group; cf. No. 6. In Nos. 1 to 4 we shall nevertheless most often restrict ourselves to affine group preschemes over the base.

Let now G be a smooth algebraic group over k, and let T be a maximal torus of G. The centralizer $C(T)$ of T in G will be called the *Cartan subgroup of* G *associated with* T. For us it is a group subprescheme defined by VIII 6.7. Since G is smooth over k, however, its Cartan subgroup C is also smooth over k, by XI 5.3. Thus C is in this case the unique group subprescheme of G, smooth over k (that is, an algebraic subgroup of G in the sense of *BIBLE*), for which $C(k)$ is the subgroup of $G(k)$ centralizing $T(k)$; in other words, it is essentially the centralizer in the sense of *BIBLE*.

By the conjugacy theorem just cited, the Cartan subgroups associated with the various maximal tori are conjugate to one another and therefore have the same dimension. We call their common dimension the *nilpotent rank of* G; it is equal to that of G_{red}. If $\rho_r(G)$ and $\rho_n(G)$ denote the reductive and nilpotent ranks of G, then

$$\rho_r(G) \leq \rho_n(G),$$

and the difference

$$\rho_u(G) = \rho_n(G) - \rho_r(G) = \dim(C/T)$$

might be called, when G is affine, the *unipotent rank of* G. If G is smooth, affine, and connected, then C is a connected nilpotent algebraic group (*BIBLE* 6, Theorem 6(a),(c)). Hence, by *BIBLE* 6, Theorem 2, it is isomorphic to the product

$$C_s \times C_u,$$

where $C_s = T$ is the original maximal torus (and *a fortiori* a maximal torus of C), while C_u is a smooth unipotent subgroup, that is, a successive extension of groups isomorphic to $\mathbf{G}_a$ (*BIBLE* 6, Theorem 1, Corollary 1, and *BIBLE* 7, Theorem 4). In this case one also has

$$\rho_u(G) = \dim C_u.$$

Remark 1.1. Besides the three notions of rank just specified for an affine algebraic group, two others are useful. The first is the *semisimple rank* $\rho_s(G)$, defined as the reductive rank of the quotient G/R, where R is the radical of G. The second is the *infinitesimal rank* $\rho_i(G)$, defined as the nilpotent rank of the Lie algebra of G, which will be defined and studied in the following Exposé. We shall not use them in the present Exposé. We record only the inequalities

$$\rho_s \leq \rho_r \leq \rho_n \leq \rho_i.$$

Lemma 1.2. *Let G be an algebraic group over the algebraically closed field k, let T be an algebraic subgroup of G, and let k' be an algebraically closed extension of k. Let G' and T' be the groups obtained from G and T by base extension. Then T is a maximal torus of G if and only if T' is a maximal torus of G'.*

This is an immediate consequence of the "principle of finite extension" (EGA IV, 9.1.1).

Definition 1.3. Let S be a prescheme, let G be an S-group prescheme of finite type, and let T be a group subprescheme of G. One says that T is a *maximal torus of G* if:

a) T is a torus (IX 1.3); and

b) for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure of $\kappa(s)$, then $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Remarks 1.4. It follows from 1.2 that, when S is the spectrum of an algebraically closed field, this recovers the usual definition, and that property 1.3 is stable under arbitrary base change. Notice that, by X 8.8, condition (a) in Definition 1.3 may be replaced by:

a′) T is of finite presentation and flat over S.

One should take care: a maximal torus in the sense of Definition 1.3 is indeed maximal among the subtori of G, as follows immediately from IX 2.9, but the converse cannot hold—say when the base S is connected—unless G actually admits a maximal torus in the sense of 1.3. This is not true in general, even when $S = \mathrm{Spec}(\mathbb{Z})$ and G is "semisimple"; see also 1.6. We shall see in Exposé XIV, however, that the converse is true when S is artinian, or when S is a local scheme and G is "reductive": in that case every torus of G is contained in a maximal torus.

Definition 1.5. Let G be an algebraic group over a field k. The *reductive rank* (respectively the *nilpotent rank*, the *unipotent rank*, and so on) of G is the reductive rank (respectively, and so on) of $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k.

By 1.2 and the compatibility of the formation of $\mathrm{Centr}_G(T)$ with base extension, the notions of rank introduced in 1.5 are invariant under extension of the base field. For algebraically closed k, they coincide with the notions introduced at the beginning of the present number.

Remark 1.6. It is not difficult to construct a smooth affine group scheme G over the spectrum S of a discrete valuation ring whose generic fiber is isomorphic to $\mathbf{G}_m$ and whose special fiber is isomorphic to $\mathbf{G}_a$. Such a G contains no torus other than the trivial torus T, reduced to the identity subgroup, and T is plainly not a maximal torus. More precisely, in the special fiber $G_0 = \mathbf{G}_{a,k}$, the torus T_0 is indeed maximal, whereas in the generic fiber $G_1 = \mathbf{G}_{m,K}$, the torus T_1 is no longer maximal; here k is the residue field and K the fraction field of the valuation ring. This example also shows that the reductive rank of G_s, for $s \in S$, is not a continuous function of s. Nevertheless, one has the following results.

Theorem 1.7. *Let S be a prescheme and let G be a smooth affine S-group prescheme. For every $s \in S$, set*

$$\rho_r(s) = \rho_r(G_s), \qquad \rho_n(s) = \rho_n(G_s),$$

the reductive and nilpotent ranks of G_s as in 1.5. Then:

a) *The function ρ_r on S is lower semicontinuous, while ρ_n is upper semicontinuous. Hence $\rho_u = \rho_n - \rho_r$ is upper semicontinuous.*

b) *The following conditions, each stable under arbitrary base change, are equivalent:*

(i) *the function ρ_r on S is locally constant;*

(ii) *G admits a maximal torus locally for the étale topology;*

(ii bis) *G admits a maximal torus locally for the faithfully flat quasi-compact topology.*

c) *Let T_1, T_2 be two maximal tori of G, which in particular places us under the conditions of (b). Then T_1 and T_2 are locally conjugate for the étale topology: there exists a surjective étale morphism $S' \to S$ such that the subgroups $(T_1)_{S'}$ and $(T_2)_{S'}$ of $G_{S'}$ are conjugate by a section of $G_{S'}$ over S'.*

d) *Under the conditions of (b), that is, when ρ_r is locally constant, ρ_n is locally constant as well, and hence so is $\rho_u = \rho_n - \rho_r$.*

Proof. a) Observe that for every morphism $S' \to S$, if $G' = G \times_S S'$, then the functions ρ'_r, and so on, on S', defined in terms of G' as ρ_r, and so on, are defined in terms of G, are obtained simply by composing the latter functions with $S' \to S$. When $S' \to S$ is faithfully flat and quasi-compact, it follows that ρ' is continuous, respectively upper semicontinuous, respectively lower semicontinuous, if and only if ρ is, since the Zariski topology on S is the quotient of that on S' (SGA 1, VIII 4.3). Consequently, the assertions in (a) are local for the faithfully flat quasi-compact topology.

Let $s \in S$. We wish to show that the set U of $t \in S$ such that

$$\rho_r(t) \geq \rho_r(s) \qquad (\text{respectively } \rho_n(t) \leq \rho_n(s))$$

is a neighborhood of s. By the principle of finite extension, there exists a finite extension k of $\kappa(s)$ such that G_k admits a maximal torus. There then exist an open neighborhood U of s and a finite, surjective, locally free morphism $S' \to U$ such that the fiber S'_s is $\kappa(s)$-isomorphic to $\mathrm{Spec}(k)$ (cf. EGA III, 10.3.2, where the noetherian hypothesis is plainly unnecessary). Since $S' \to S$ is an open morphism (SGA 1, IV 6.6), we are reduced to the case $S' = S$, that is, to the case in which G_s admits a maximal torus T_s. Moreover, by XI 5.8(a), after possibly replacing S once more by an étale S' over S, equipped with a point s' above s, we may suppose that T_s is the fiber at s of a subtorus T of G. Then, for every $t \in S$,

$$\rho_r(t) = \rho_r(G_t) \geq \dim T_t = \dim T_s = \rho_r(G_s) = \rho_r(s),$$

which proves that ρ_r is lower semicontinuous.

On the other hand, by XI 5.3, the functor

$$\mathrm{Centr}_G(T)$$

is represented by

$$C = \mathrm{Centr}_G(T),^{1}$$

a closed group subprescheme of G, smooth over S. There is therefore a neighborhood U of s such that $t \in U$ implies

$$\dim C_t = \dim C_s = \rho_n(s).$$

[1] Editor's note: the text has been modified here to introduce the prescheme $C = \mathrm{Centr}_G(T)$ representing the functor $\mathrm{Centr}_G(T)$.

The upper semicontinuity of ρ_n now follows from the relation

$$\rho_n(t) \leq \dim C_t \qquad \text{for every } t \in S,$$

which is itself contained in the following purely geometric lemma.

Lemma 1.8. *Let k be a field, let G be a smooth affine algebraic group over k, let T be a torus in G, and let C be its centralizer. Then*

$$\rho_n(G) \leq \dim C.$$

Indeed, we may suppose k algebraically closed, so that T is contained in a maximal torus T'. If C' is the centralizer of T', then $C' \subset C$, and hence $\dim C' \leq \dim C$. This proves the lemma.

b) If ρ_r is locally constant, then for every torus T in G and every $s \in S$, if T_s is a maximal torus in G_s, there exists an open neighborhood U of s such that $T|_U$ is a maximal torus in $G|_U$. Applying the argument of (a), we see that (i) implies (ii bis). Conversely, (ii bis) implies (i), because when G admits a maximal torus T, it is clear that

$$\rho_r(s) = \dim T_s$$

is a locally constant function of s; and we observed at the beginning of the proof of (a) that continuity of ρ_r is local for the faithfully flat topology. Thus (i) is equivalent to (ii bis), while (ii) plainly implies (ii bis); it remains to prove the converse implication (i) $\Rightarrow$ (ii).

For this purpose, introduce the functor F of XI 4.1,[2] which is a smooth separated prescheme over S, and consider the subfunctor $\mathcal{T}$ of F whose value on an S' over S is the set of maximal tori in $G_{S'}$. By the preceding observation that, under (i), a torus in $G_{S'}$ which is maximal in the fiber over a point $s \in S'$ is maximal over an open neighborhood of s, the functor $\mathcal{T}$ is represented by an open subprescheme of F, and is therefore smooth and separated over S. Since the structural morphism $\mathcal{T} \to S$ is plainly surjective, it admits a section locally for the étale topology by XI 1.10. This proves (i) $\Rightarrow$ (ii).

c) This is an immediate consequence of XI 5.4 bis, taking account of Borel's conjugacy theorem recalled at the beginning of this number.

d) By the observation at the beginning of the proof of (a), and in view of (b), we may suppose that G admits a maximal torus T. If C is its centralizer, then C is representable and smooth over S by XI 5.3, so the function

$$s \longmapsto \rho_n(s) = \dim C_s$$

is indeed locally constant.

This completes the proof of 1.7. We shall refer to the conditions in 1.7(b) by saying that, in this case, G is *of locally constant reductive rank*. We record the following consequence.

Corollary 1.9. *Let G be as in 1.7, and let $s \in S$ be such that $\rho_u(s) = 0$, that is, $\rho_r(s) = \rho_n(s)$ (equivalently, the Cartan subgroups of $G_{\bar{k}}$ are tori, where $\bar{k}$ is an algebraic closure of $\kappa(s)$). Then there exists an open neighborhood U of s on which ρ_r and ρ_n are constant. In particular, for every $t \in U$, the unipotent rank $\rho_u(t)$ of G_t is zero.*

Indeed, this follows immediately from 1.7(a) and the inequality $\rho_r(t) \leq \rho_n(t)$ for every $t \in S$.

Notice also that, in the course of proving (b), we proved the following result.

[2] Editor's note: this is the functor of multiplicative-type subgroups of G.

Corollary 1.10. *Let G be as in 1.7, and suppose that G has locally constant reductive rank. Consider the functor*

$$\mathcal{T} : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens})$$

such that, for every S' over S,

$$\mathcal{T}(S') = \{\textit{maximal tori of } G_{S'}\}.$$

Then $\mathcal{T}$ is represented by a smooth, separated prescheme of finite type over S.

It remains only to verify that $\mathcal{T}$ is of finite type over S. Since the question is local for the faithfully flat quasi-compact topology, we may suppose that G admits a maximal torus T. By XI 5.3 bis and the bis version of XI 5.5, $\mathrm{Norm}_G(T)$ and $G/\mathrm{Norm}_G(T)$ are represented by preschemes $\mathrm{Norm}_G(T)$ and $G/\mathrm{Norm}_G(T)$, and $\mathcal{T}$ is isomorphic to

$$G/\mathrm{Norm}_G(T).^3$$

The morphism $G \to \mathcal{T}$ defined by $g \mapsto \mathrm{int}(g)(T)$ is surjective. Since G is quasi-compact over S, so is $\mathcal{T}$, which completes the proof.

Remarks 1.11. a) The prescheme $\mathcal{T}$ of 1.10 will properly be called the *prescheme of maximal tori of G*. We shall see in No. 5 that it is in fact affine over S. This can be checked directly when G is isomorphic to a closed group subprescheme of a prescheme of the form $\mathrm{GL}(n)$, using XI 4.6 and observing that, by the argument in the proof of 1.7(b), $\mathcal{T}$ identifies with a subprescheme both open and closed in the prescheme F representing the multiplicative-type subgroups of G.

b) It is possible to prove 1.7, and hence 1.9, without using the delicate results of Exposé XI, using only the elementary results XI 3.12 and XI 6.2 and working solely with multiplicative-type groups finite over the base (morally, the groups ${}_nT$, where T is a maximal torus); compare No. 7. The same observation applies to the proof of 1.10.

We end this number with examples in which there is a unique maximal torus.

Proposition 1.12. *Let G be an S-group prescheme of multiplicative type and finite type. Then G admits a unique maximal torus, and every torus in G is contained in this maximal torus.*

Uniqueness plainly follows from the latter assertion, which characterizes the maximal torus as the largest subtorus of G. Uniqueness also makes existence a question local for the faithfully flat quasi-compact topology. We may therefore suppose that G is diagonalizable, say $G = D_S(M)$, where M is a finitely generated commutative group. Let M_0 be the quotient of M by its torsion subgroup. We claim that the torus

$$T = D_S(M_0)$$

in G is a maximal torus and the largest subtorus. Indeed, a subtorus T' of G is locally diagonalizable for the fpqc topology. Thus, in order to prove $T' \subset T$, we may suppose that T' is diagonalizable, and hence of the form $D_S(N)$, where N is a free quotient of M (VIII 1.4 and 3.2(b)). It follows that N is a quotient of M_0, and therefore that $T' \subset T$. Since the construction of $T = D_S(M_0)$ is compatible with arbitrary base extension, this proves at the same time that T is a maximal torus of G, and completes the proof. When G is smooth over S, Proposition 1.12 has the following generalization.

[3] Editor's note: the text has been modified here to introduce the preschemes $\mathrm{Norm}_G(T)$ and $G/\mathrm{Norm}_G(T)$.

Proposition 1.13. *Let G be an S-group prescheme of finite presentation over S. Suppose that, locally for the faithfully flat quasi-compact topology, G admits a central maximal torus. Then G admits globally a unique maximal torus,*[4] *and it is the largest subtorus of G.*

Here again, uniqueness follows trivially from the last assertion and makes existence a local question for the fpqc topology. We may therefore suppose that G admits a central maximal torus T. We shall show that every torus R of G is contained in T. This follows from the next lemma.

Lemma 1.14. *Let G be an S-group prescheme of finite presentation over S, let T be a maximal torus of G, let R be a subtorus of G, and suppose that T and R commute. Then $R \subset T$.*

Indeed, since R and T commute, the morphism

$$R \times_S T \longrightarrow G, \qquad (r,t) \longmapsto r \cdot t,$$

is a group homomorphism. By IX 6.8 it therefore has an image subgroup T' in G, which is a multiplicative-type group quotient of $R \times_S T$, hence a torus, and plainly contains T. Since T is a maximal torus, $T' = T$ by 1.4, and therefore $R \subset T$. This proves 1.14 and hence 1.13. In particular, using 1.7(b), we obtain:

Corollary 1.15. *Let G be a commutative, smooth, affine S-group prescheme of locally constant reductive rank. Then G admits a unique maximal torus, and this torus contains every subtorus of G.*

Corollary 1.16. *Let G be a smooth affine S-group prescheme. Suppose that, for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure $\bar{k}$ of $\kappa(s)$, the geometric fiber $G_{\bar{s}}$ is a connected nilpotent algebraic group (in the sense of* BIBLE*, that is, the group $G(\bar{k})$ of its $\bar{k}$-valued points is nilpotent). Suppose moreover that G has locally constant reductive rank. Then G admits a unique maximal torus T; moreover, T is central and is the largest subtorus of G.*

By 1.7(b), G admits a maximal torus locally for the fpqc topology, and by 1.13 we are reduced to proving that a maximal torus of G is central. By IX 5.6(b), we are reduced to the case in which S is the spectrum of a field, which we may suppose algebraically closed. Then $T(k)$ lies in the center of $G(k)$ by *BIBLE* 6, Theorem 2. This implies that T lies in the center of G, because $\mathrm{Centr}_G(T)$ is a closed subscheme of G (VIII 6.6) containing all the points of $G(k)$, and is therefore equal to G by the Nullstellensatz, since G is reduced.

Finally, for later reference, we record the following elementary proposition, which we have already used implicitly.

Proposition 1.17. *Let $G \supset H \supset T$ be group preschemes of finite type over S. The following conditions are equivalent:*

(i) *T is a maximal torus of G;*

(ii) *T is a maximal torus of H, and for every $s \in S$ one has equality of reductive ranks*

$$\rho_r(G_s) = \rho_r(H_s).$$

[4] Which is central!

0.2 The Weyl group

Let first k be an algebraically closed field, and let G be a smooth affine algebraic group over k. If T is a maximal torus, C its centralizer, and N its normalizer, then by XI 5.9 these are smooth closed subgroups of G, and C is an open subgroup of N. Thus

$$W = N/C$$

is a finite étale group over k, and is consequently determined by the group $W(k)$ of its k-valued points:

$$W = W(k)_k$$

(the constant group defined by the ordinary finite group $W(k)$). The finite group $W(k)$ will be called the *geometric Weyl group*, or simply the *Weyl group* when no confusion is likely, of G relative to T. By Borel's conjugacy theorem, the Weyl groups relative to the various maximal tori are isomorphic to one another; this is why one sometimes speaks of "the" Weyl group of G, without specifying a maximal torus. Since the formation of C, N, and N/C commutes with arbitrary base extension, if k' is an algebraically closed extension of k, the geometric Weyl group of $G_{k'}$ relative to $T_{k'}$ is canonically isomorphic to that of G relative to T. Consequently, "the" geometric Weyl group of G, properly speaking an isomorphism class of ordinary finite groups, coincides with that of $G_{k'}$.

This permits us, when G is a smooth affine algebraic group over an arbitrary field k, to speak of the *geometric Weyl group of* G as the isomorphism class of the geometric Weyl group of $G_{k'}$, where k' is any algebraically closed extension of k. When G admits a maximal torus T, we can form

$$C = \mathrm{Centr}_G(T), \qquad N = \mathrm{Norm}_G(T), \qquad W = N/C$$

as above. The latter is a finite étale group over k, called the *Weyl group of G relative to T*. The geometric Weyl group is then simply the class of the group of points of W with values in any algebraically closed extension k' of k. In general, knowledge of the geometric Weyl group $W(k')$ plainly no longer suffices to reconstruct the algebraic group W: one must also know the action on $W(k')$ of the Galois group of the separable algebraic closure of k in k'.

Finally, let G be a group prescheme over an arbitrary base S, smooth and affine over S, and let T be a maximal torus of G. By XI 5.9 we may again form the group

$$W(T) = N(T)/C(T),$$

which is an étale, separated, quasi-finite S-group prescheme. Its geometric fibers, relative to algebraic closures of the residue fields $\kappa(s)$, $s \in S$, are the geometric Weyl groups of the fibers G_s. Thus XI 5.10 gives information about the variation of these groups with $s \in S$. We can make this information more precise and generalize it as follows.

Theorem 2.1. *Let S be a prescheme and let G be a smooth affine S-group prescheme. For every $s \in S$, let $w(s)$ be the geometric Weyl group of G_s, hence an isomorphism class of finite groups. On the set E of such classes, introduce the following preorder: $w \leq w'$ if and only if w and w' can be represented by finite groups W and W' such that W is isomorphic to a quotient of a subgroup of W'. With this convention:*

a) *The function $s \mapsto w(s)$ from S to E is lower semicontinuous.*

b) *Suppose that G has locally constant reductive rank. Then the following conditions are equivalent:*

(i) *the function $s \mapsto w(s)$ is locally constant;*

(i bis) *the function $s \mapsto \operatorname{card} w(s)$ is locally constant;*

(ii) *locally for the étale topology (or merely for the fpqc topology), there exists a maximal torus T such that $W(T)$ is finite over its base;*

(ii bis) *for every S' over S and every maximal torus T of $G_{S'}$, the associated Weyl group $W(T)$ is finite over S'.*

Proof. a) Arguing as in 1.7(a), in order to prove that for every $s \in S$ there exists an open neighborhood U of s such that $t \in U$ implies $w(t) \geq w(s)$, we are reduced to the case in which there is a torus R in G such that R_s is a maximal torus in G_s. Put

$$W(R) = \operatorname{Norm}_G(R)/\operatorname{Centr}_G(R).$$

As in XI 5.9, this is an étale, separated, quasi-finite group prescheme over S. For $t \in S$, let $w'(t) \in E$ be its geometric fiber at t. Since R_s is a maximal torus in G_s, and the formation of Norm, Centr, and Norm / Centr commutes with arbitrary base extension, in particular with passage to fibers, one has

$$w(s) = w'(s).$$

We claim moreover that for t near s,

$$w(t) \geq w'(t) \qquad \text{and} \qquad w'(t) \geq w'(s),$$

which will suffice to prove (a). These two inequalities are contained in the following two lemmas.

Lemma 2.2. *Let S be a prescheme, and let W be an étale, separated, quasi-finite S-group prescheme. For every $s \in S$, let $f(s)$ be the class of the geometric fiber of W at s, an element of the ordered set E of isomorphism classes of finite groups introduced in 2.1. Then the function $f : S \to E$ is lower semicontinuous. It is constant in a neighborhood of $s \in S$ if and only if the function $s \mapsto \operatorname{card} f(s)$ is constant there, and this holds if and only if W is finite over S above some open neighborhood U of s.*

This result refines, in group-theoretic terms, the result used in the proof of XI 5.10. We give only a sketch of the proof, which is entirely standard. As usual, we are reduced to the case in which S is affine and noetherian. The function f is immediately seen to be constructible (EGA 0_{III}, 9.3.1 and 9.3.2, together with the sorites of EGA IV, 9), and EGA 0_{III}, 9.3.4 reduces semicontinuity to proving that if t is a generalization of s, then

$$f(t) \geq f(s).$$

By EGA II, 7.1.7, this reduces us to the case in which S is the spectrum of a discrete valuation ring, which may be supposed complete with algebraically closed residue field. Let s be the closed point of S. Since W is étale and separated over S, it contains a subscheme W', both open and closed and finite over S, such that $W'_s = W_s$ (EGA II, 6.2.6); it is immediate here that W' is a subgroup of W. Moreover, since W' is finite étale over $S = \operatorname{Spec}(V)$, with V complete and with algebraically closed residue field, it is a constant group, hence of the form A_S, where

$$A = W'(\kappa(s)) = W(\kappa(s))$$

has class $f(s)$. If B is the geometric fiber of W at the generic point t of S, there is therefore a canonical monomorphism $A \to B$, proving $f(s) \leq f(t)$. Notice that this argument actually proves semicontinuity for an order on E finer than the one specified in 2.1.

The equivalence between constancy of f at s and constancy of $s \mapsto \operatorname{card} f(s)$ follows from the fact that, for $w, w' \in E$,

$$w \leq w' \quad \text{and} \quad \operatorname{card} w = \operatorname{card} w' \quad \Longrightarrow \quad w = w'.$$

The equivalence of this condition with finiteness of W over a neighborhood of s is independent of the group structure on W; it was noted after XI 5.10, and also follows readily from the preceding argument using the valuative criterion of properness (EGA II, 7.3.8).

Lemma 2.3. *Let G be a smooth affine algebraic group over an algebraically closed field k, and let $R \subset T$ be two subtori. Let $W(R)$ and $W(T)$ be the two associated finite groups of XI 5.9, obtained as the quotient of the normalizer by the centralizer. Then $W(R)$ is isomorphic to a quotient of a subgroup of $W(T)$.*

Indeed, consider the diagram

$$\begin{array}{ccccc} T & \hookrightarrow & C(T) & \hookrightarrow & C(R) \\ & & \downarrow & & \downarrow \\ & & N(T) \cap N(R) & \hookrightarrow & N(R) \\ & & \downarrow & & \\ & & N(T) & & \end{array}$$

Then $(N(T) \cap N(R))/C(T)$ is a subgroup of

$$W(T) = N(T)/C(T),$$

and there is an evident homomorphism

$$(N(T) \cap N(R))/C(T) \longrightarrow W(R) = N(R)/C(R).$$

Everything reduces to proving that this last homomorphism is surjective, that is, that for every k-valued point g of $N(R)$, there exists a k-valued point c of $C(R)$ such that cg normalizes T, or equivalently such that

$$\operatorname{int}(c)\big(\operatorname{int}(g)T\big) = T.$$

For this, it suffices to observe that $\operatorname{int}(g)T$ is a torus of $N(R)$, hence of $C(R)$, which is an open subgroup of $N(R)$. The tori T and $\operatorname{int}(g)T$ are maximal tori of $C(R)$, since they are maximal in G, and the assertion follows from Borel's conjugacy theorem.

This proves 2.3, and hence 2.1(a).

b) We have already observed that (i) and (i bis) are plainly equivalent. They imply (ii bis), by the converse part of 2.2 or, equivalently, XI 5.10. Condition (ii bis) implies (ii) by 1.7(b). Finally, (ii) implies (i bis): as in 1.7(a), condition (i) is local for the fpqc topology, so we may suppose that G admits a maximal torus T such that $W(T)$ is finite over S, and XI 5.10 again gives the conclusion. This completes the proof of 2.1.

0.3 Cartan subgroups

Definition 3.1. Let G be a smooth group prescheme of finite type over a prescheme S. A *Cartan subgroup of* G is a group subprescheme C of G, smooth over S, such that for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure of $\kappa(s)$, then $C_{\bar{s}}$ is a Cartan subgroup, in the sense of No. 1, of $G_{\bar{s}}$.

It is immediate that if C is a Cartan subgroup of G, then for every S' over S, $C_{S'}$ is a Cartan subgroup of $G_{S'}$. If S is the spectrum of an algebraically closed field, this recovers the notion recalled in No. 1. Finally, the property that a group subprescheme C of G be a Cartan subgroup is plainly local for the faithfully flat quasi-compact topology.

Theorem 3.2. *Let G over S be as in 3.1, and suppose that G is affine over S and has locally constant reductive rank. Then the map*

$$T \longmapsto \mathrm{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G onto the set of Cartan subgroups[5] *of G. If C corresponds to T, then T is the unique maximal torus of C.*

If T is a maximal torus of G, the functor $\mathrm{Centr}_G(T)$ is represented, by XI 5.3 or 6.2, by a closed smooth subprescheme

$$C = \mathrm{Centr}_G(T)$$

of G, and the definitions show that C is a Cartan subgroup of G. Moreover, T is plainly a maximal torus of C, and, being central, it is the unique maximal torus of C by 1.13. Thus the map $T \mapsto \mathrm{Centr}_G(T)$ from the set of maximal tori to the set of Cartan subgroups is injective; it remains to prove surjectivity.

Let C be a Cartan subgroup of G. We shall prove that $C = \mathrm{Centr}_G(T)$ for a maximal torus T of G. It is enough to find a maximal torus T of C, for then T is a maximal torus of G: for every $s \in S$, the groups G_s and C_s have the same reductive rank. By IX 5.6(b), T lies in the center of C, so

$$C \subset C' = \mathrm{Centr}_G(T).$$

The smooth subgroup C of the smooth group C' over S coincides with C' fiber by fiber; hence $C = C'$.

Since G has locally constant reductive rank, so does C. Therefore C admits a maximal torus locally for the étale topology by 1.7(b); that torus is central by the preceding argument, and 1.13 then shows that C itself admits a maximal torus. This completes the proof.

Corollary 3.3. *Let G be a smooth affine group prescheme over S, of locally constant reductive rank, and let*

$$\mathcal{C} : (\mathrm{Sch})^{\circ}_{/S} \longrightarrow (\mathrm{Ens})$$

be the functor defined by

$$\mathcal{C}(S') = \{\textit{Cartan subgroups of } G_{S'}\}.$$

Then $\mathcal{C}$ is isomorphic to the functor $\mathcal{T}$ of 1.10, and is therefore represented by the same smooth separated prescheme of finite type over S.

[5] The proof given here in fact proves Theorem 3.2 only for closed Cartan subgroups of G. However, 7.1(a) establishes Theorem 3.2 as stated, and implies that the Cartan subgroups of G are closed.

Corollary 3.4. *Under the hypotheses of 3.2, if* $C = \mathrm{Centr}_G(T)$*, then*

$$\mathrm{Norm}_G(C) = \mathrm{Norm}_G(T).$$

Remark 3.5. When G does not have locally constant reductive rank, it may nevertheless admit Cartan subgroups. For example, if G has connected nilpotent fibers, then G is a Cartan subgroup of itself, although it need not have locally constant reductive rank; see 1.6. In Exposé XV we develop the theory of Cartan subgroups without assuming either that G is affine over S or that its reductive rank is locally constant, using the theory of regular elements from the next Exposé. See also Nos. 6 and 7 below for the removal of the affineness hypothesis.

0.4 The reductive center

Definition 4.1. Let S be a prescheme, let G be an S-group of finite presentation over S with affine fibers, and let Z be a group subprescheme of G. We say that Z is a *reductive center* of G if:

(i) Z is central and of multiplicative type;

(ii) for every base change $S' \to S$ and every central homomorphism

$$u : H \longrightarrow G_{S'},$$

where H is a group of multiplicative type and of finite type over S', the homomorphism u factors through $Z_{S'}$.

(For a variant of this notion when the fibers of G are not assumed affine, see 8.6.)

Such a Z is necessarily of finite type over S, since its fibers are, and is therefore uniquely determined as the largest central subgroup of multiplicative type in G. It is easy to give examples in which G, smooth and affine over S, admits a largest central subgroup Z of multiplicative type but Z is not a reductive center—that is, G admits no reductive center; see, for example, 1.6. It follows readily, however, from IX 6.8 that a subgroup Z of G is a reductive center if and only if it is a largest central subgroup of multiplicative type and retains this property after every base change. See also 4.3 below.

It is clear from 4.1 that if Z is the reductive center of G, then $Z_{S'}$ is the reductive center of $G_{S'}$ after every base change $S' \to S$. This fact and the uniqueness of the reductive center, together with faithfully flat quasi-compact descent (SGA 1, VIII), give:

Proposition 4.2. *Let G be an S-group prescheme of finite presentation over S, with affine fibers. If G admits a reductive center locally for the faithfully flat quasi-compact topology, then it admits a reductive center. For Z to be a reductive center of G, it is necessary and sufficient that it be so locally for the fpqc topology.*

We also record:

Proposition 4.3. *Let G be an S-group prescheme of finite presentation with affine fibers, and let Z be a group subprescheme. For Z to be a reductive center of G, it is necessary and sufficient that it be of multiplicative type and that, for every $s \in S$, Z_s be a reductive center of G_s.*

Indeed, IX 5.6(b) first shows that Z is central. Since the conditions under consideration are stable under base change, it remains to prove that every central homomorphism

$$u : H \longrightarrow G,$$

with H of multiplicative type and of finite type, factors through Z. Since Z is central, u and the canonical immersion $v : Z \to G$ define a group homomorphism

$$w : H \times_S Z \longrightarrow G.$$

By IX 6.8, this homomorphism has a group image K, which is a subgroup of multiplicative type in G. Everything reduces to showing that $K = Z$. This holds fiber by fiber by the hypothesis on Z, and IX 5.1 bis now applies, completing the proof of 4.3.

In criterion 4.3, the hypothesis that Z_s be the reductive center of G_s for every $s \in S$ is in fact purely geometric: it is enough to verify it after passing to an algebraic closure of $\kappa(s)$, as follows from the second assertion of 4.2.

Theorem 4.4. *Let G be an affine algebraic group over a field k. Then G admits a reductive center.*

Since the center of G is represented by a closed subgroup of G by VIII 6.7, we are immediately reduced to the case in which G is commutative. Moreover, by 4.2 we may suppose that k is algebraically closed. We shall see in Exposé XVII that in this case G can be written as a product $Z \times U$, where Z is of multiplicative type and U is "unipotent", from which it follows at once that Z is a reductive center of G. We give here a proof independent of the general structure theorem for commutative algebraic groups just indicated.

By IX 6.8, the set of subgroups H of multiplicative type in G is filtered upward. We show that it has a largest element.

When G is smooth over k, the structure theorem BIBLE 4, Th. 4 gives

$$G = G_s \times G_u,$$

where G_s is of multiplicative type and G_u is "unipotent": here this means that G_u has a composition series whose factors are subgroups of $\mathbb{G}_a$, smooth if desired. Indeed, G_u/G_u^0 is a unipotent group by BIBLE 4, Corollary to Th. 3, hence a p-group, where p is the characteristic exponent, by BIBLE 4, Prop. 4. On the other hand, G_u^0 has a composition series with smooth connected one-dimensional quotients by BIBLE 6, Th. 1, Cor. 1, and these quotients are isomorphic to $\mathbb{G}_a$ by BIBLE 7, Th. 4. Every homomorphism from a group of multiplicative type to $\mathbb{G}_a$, and therefore also to G_u, is trivial, as follows from the lemma below. Thus every subgroup of multiplicative type in G is contained in G_s.

In the general case, consider the subgroup

$$G_0 = G_{\mathrm{red}}$$

of G. It is smooth over k, and hence by the preceding case has a largest subgroup Z_0 of multiplicative type. The subgroups of multiplicative type in G containing Z_0 correspond to the subgroups of multiplicative type in $G' = G/Z_0$. We have an exact sequence

$$0 \longrightarrow G_0/Z_0 \longrightarrow G/Z_0 \longrightarrow G/G_0 \longrightarrow 0,$$

where G_0/Z_0 has no subgroup of multiplicative type other than the unit group. Since every subgroup of a group of multiplicative type is again of multiplicative type by IX 8, every subgroup H of multiplicative type in G/Z_0 satisfies

$$H \cap (G_0/Z_0) = 0;$$

thus $H \to G/G_0$ is injective. The algebraic group G/G_0 is radicial, hence finite over k, so H is itself radicial and its rank is bounded above by that of G/G_0. Consequently the family of subgroups of multiplicative type in G has a largest element, say Z.

We claim that G/Z has no subgroup of multiplicative type other than the unit subgroup. Indeed, a commutative extension of two algebraic groups of multiplicative type is of multiplicative type (7.1.1). If Z' is the inverse image in G of a subgroup of multiplicative type in G/Z, then Z' is of multiplicative type. The maximality of Z gives $Z' = Z$, and hence $Z'/Z = 0$.

The "principle of finite extension" now readily shows that for every extension K of k,

$$(G/Z)_K = G_K/Z_K$$

likewise has no subgroup of multiplicative type other than the unit group.

We can now prove that Z is a reductive center of G. Let

$$u : H \longrightarrow G_S$$

be a homomorphism of S-groups, where S is a prescheme over k and H is of multiplicative type and of finite type over S. We must show that u factors through Z_S, or equivalently that the composite

$$H \longrightarrow G_S \longrightarrow (G/Z)_S = G_S/Z_S$$

is trivial. Put $U = G/Z$. Every homomorphism $H \to U_S$ is trivial: by IX 5.2 one reduces to the case where S is the spectrum of a field K, and IX 6.8 then reduces the claim to the fact that U_K has no nontrivial subgroup of multiplicative type. This proves 4.4, modulo the following lemma.

Lemma 4.4.1. *Let H be an S-group prescheme of multiplicative type. Then every homomorphism of S-groups*

$$u : H \longrightarrow \mathbb{G}_a$$

is trivial.

Regard $E = \mathcal{O}_S^2$ as an extension with submodule $E' \simeq \mathcal{O}_S$ and quotient $E'' \simeq \mathcal{O}_S$. Then $\mathbb{G}_a$ identifies with the prescheme of automorphisms of this extension. A homomorphism $u : H \to \mathbb{G}_a$ therefore amounts to an H-module structure on E preserving the extension, with E' stable under H and with the induced actions on E' and E'' trivial. By I 4.7.3, H acts trivially on E; hence u is trivial.

Remark 4.4.2. If G is a nonzero abelian variety over k, then G admits no reductive center in the sense of 4.1 with the restriction "G affine" omitted. Indeed, if n is prime to the characteristic, then ${}_nG$ is étale over k, of order prime to the characteristic, and hence of multiplicative type. The family of the ${}_nG$ is schematically dense in G. A reductive center, if it existed, would therefore equal G, which is impossible because G is not of multiplicative type. This explains the requirement in 4.1 that G have affine fibers.

Lemma 4.5. *Let S be a prescheme, let G be a smooth affine S-group prescheme with connected fibers, let T be a maximal torus of G, and let $u : H \to G$ be a central homomorphism, where H is an S-group prescheme of multiplicative type and of finite type. Then u factors through T.*

Let $C = \mathrm{Centr}_G(T)$. By XI 5.3 it is a closed smooth group subprescheme of G, hence affine over S. Since T is central in C, it is invariant, and the quotient group

$$U = C/T$$

is represented by VIII 5.1. Because u is central it factors through C, and it remains only to show that the composite

$$H \longrightarrow C \longrightarrow U = C/T$$

is trivial. By IX 5.2 we reduce to the case where S is the spectrum of a field, which we may suppose algebraically closed. Then BIBLE 6, Th. 2 shows that U is a connected algebraic group, smooth over k, which is "unipotent"; as recalled above, it has a composition series with quotients isomorphic to $\mathbb{G}_a$. Every homomorphism from the finite-type group H of multiplicative type to U is therefore trivial. This proves 4.5.

Corollary 4.6. *Let G be as in 4.5. If G admits a reductive center, then that center is contained in every maximal torus of G.*

Theorem 4.7. *Let S be a prescheme and let G be an S-group prescheme, smooth and affine over S, with connected fibers.*

(a) *For every $s \in S$, let $z(s)$ denote the type of the reductive center of G_s, which is defined by 4.4. Order the set E of isomorphism classes of finite-type $\mathbb{Z}$-modules by declaring the class of M to be greater than that of M' when M' is isomorphic to a quotient of M. Then the map*

$$S \longrightarrow E, \qquad s \longmapsto z(s),$$

is lower semicontinuous.

(b) *The group G admits a reductive center Z if and only if the preceding function $z : S \to E$ is locally constant. If this holds, then G/Z is representable* (cf. VIII 5.1)*, and its reductive center is the unit subgroup.*

(c) *Suppose that the reductive rank of G is locally constant* (cf. 1.7(b))*. Then G admits a reductive center Z. If G/Z is representable—for example, if G is affine over S—then the maximal tori T of G* (respectively, the Cartan subgroups C of G) *and the maximal tori T' of $G' = G/Z$* (respectively, the Cartan subgroups C' of G') *are in bijective correspondence:*

$$T' = T/Z, \qquad C' = C/Z,$$

and conversely

$$T = \varphi^{-1}(T'), \qquad C = \varphi^{-1}(C'),$$

where $\varphi : G \to G'$ is the canonical homomorphism.

(d) *Let T be a maximal torus of G, let $\mathfrak{g}$ denote the Lie algebra of G, and consider the homomorphism*

$$\theta : T \longrightarrow \mathrm{GL}(\mathfrak{g})$$

induced by the adjoint representation of G (II 4)*. Then $\ker\theta$ is a reductive center of G.*

Proof. (a) and (b). The proof of (a) and of the first assertion of (b) is essentially identical to that of 1.7(a) and (b), and we shall not repeat the argument. We point out only that in (a) one must appeal to IX 5.6, using the fact that G has connected fibers.

We prove the second assertion of (b). Suppose that Z is a reductive center of G, and put $G' = G/Z$. By VIII 5.1 the quotient is representable. It is affine over S, of finite

presentation over S by VIII 5.8, and even smooth over S: this may be checked on the fibers because G' is flat and of finite presentation over S, and $\mathrm{VI_B}$ 9.2(xii) states that a quotient of a smooth algebraic group over a field is smooth. Since G has connected fibers, so does G'. Thus G' satisfies the same hypotheses as G.

By 4.3 we may reduce to the case in which S is the spectrum of a field. Let Z' be a central subgroup of multiplicative type in G'; it remains to prove that Z' is the unit subgroup. Let Z_1 be the inverse image of Z' in G, and consider the action of Z_1 on G by inner automorphisms. Since Z is central, it acts trivially, and hence Z_1 acts through an action of Z' on G. Moreover Z' acts trivially both on Z and on $G/Z = G'$. Since Z is central, the action of Z' on G therefore comes from a group homomorphism

$$u : Z' \longrightarrow \mathrm{Hom}_{S\text{-gr}}(G', Z),$$

characterized by

$$r(z') \cdot g = g \cdot u(z')(\varphi(g)),$$

where $r : Z' \to \mathrm{Aut}_{S\text{-gr}}(G)$ is the representation just considered and $\varphi : G \to G'$ is the canonical homomorphism. Giving u is equivalent to giving a group homomorphism

$$v : G' \longrightarrow \mathrm{Hom}_{S\text{-gr}}(Z', Z).$$

By X 5.8 the group on the right is representable and étale over S. Its unit section is therefore an open immersion, so $\ker v$ is an open subgroup of G'. Since G' has connected fibers, $\ker v = G'$. Thus $v = 0$, hence $u = 0$, and Z', and therefore Z_1, acts trivially on G. Consequently Z_1 is central in G. It is a commutative extension of the group Z' of multiplicative type by the group Z of multiplicative type, and hence, over a field, is itself of multiplicative type by 7.1.1. Being central in G, it is contained in the reductive center Z. Thus $Z_1 = Z$, whence Z' is the unit subgroup. This proves (b).

(d) Put $Z = \ker \theta$, a subgroup of multiplicative type in T, for example by IX 6.8. By 4.5, every central homomorphism $u : H \to G$, with H of multiplicative type and of finite type, factors through T, and therefore through Z. Since formation of Z commutes with every base change, it remains to show that Z is central, or equivalently that its centralizer C is equal to G. By XI 5.3, C is a closed smooth subgroup of G. Since Z acts trivially on $\mathfrak{g}$, we have

$$\mathrm{Lie}(C) = \mathfrak{g}.$$

It follows that $C = G$. Indeed, the immersion $C \to G$ is étale, as may be checked fiberwise: it is an unramified homomorphism of smooth algebraic groups of the same dimension ($\mathrm{VI_B}$ 1.3), so X 3.5 applies. Hence $C \to G$ is an étale closed immersion, and therefore an open immersion by SGA 1, I 5.1. Since it is both open and closed and G has connected fibers, it is an isomorphism.

(c) By 1.7(b), G admits a maximal torus locally for the fpqc topology. Proposition 4.2 and part (d) therefore show that G admits a reductive center Z. Part (d) also shows that every maximal torus T of G contains Z. Clearly $T' = T/Z$ is a torus. To prove that it is a maximal torus of G', Definition 1.3 reduces us to the case in which S is the spectrum of an algebraically closed field; the assertion then follows from *BIBLE* 7, Th. 3(a).

Conversely, let T' be a maximal torus of G', and set $T = \varphi^{-1}(T')$. We prove that T is a maximal torus of G. The question is local for the fpqc topology, so we may suppose that G already admits a maximal torus T_0. By what precedes, $T'_0 = T_0/Z$ is a maximal torus of G'. By the conjugacy theorem 1.7(c), T' and T'_0 are locally conjugate for the fpqc topology. Since $\varphi : G \to G'$ is a covering for this topology, sections of G' lift locally to

sections of G; thus T and T_0 are likewise locally conjugate. Hence T is a maximal torus. The argument for Cartan subgroups is identical. This completes the proof.

Here is a useful reformulation of (d) when T is diagonalizable, say $T = D_S(M)$, where M is a finite-type free $\mathbb{Z}$-module. By I 4.7.3, the T-module $\mathfrak{g}$ decomposes into the direct sum of its T-submodules $\mathfrak{g}_m$, $m \in M$:

$$\mathfrak{g} = \bigoplus_{m \in M} \mathfrak{g}_m.$$

These submodules are necessarily locally free. Suppose that for every $m \in M$ the rank of $\mathfrak{g}_m$ is constant, as is the case in particular when S is connected. Then the set

$$R = \{m \in M \mid \mathfrak{g}_m \neq 0\}$$

of "roots" is finite. We obtain:

Corollary 4.8. *Under the preceding hypotheses and with the preceding notation, the reductive center of G is the intersection of the kernels in T of the root characters $m \in R$. Thus there is an isomorphism*

$$Z \simeq D_S(N),$$

where N is the quotient of M by the subgroup generated by R.

Corollary 4.9. *Let G be an algebraic group over an algebraically closed field k. Suppose that G is smooth, connected, and affine over k, that its reductive center is the unit subgroup, and that its Lie algebra $\mathfrak{g}$ is nilpotent. Then G is "unipotent," that is, it admits a composition series whose quotients are isomorphic to $\mathbb{G}_a$.*

By *BIBLE* 6, Th. 4, Cor. 3, it is enough to prove that a maximal torus T of G is the unit subgroup, or equivalently that its Lie algebra $\mathfrak{t}$ is zero. The T-module $\mathfrak{g}$ decomposes according to the characters of T by I 4.7.3, so for every $t \in \mathfrak{t}$, the endomorphism $\mathrm{ad}_{\mathfrak{g}}(t)$ of $\mathfrak{g}$ is semisimple. If $\mathfrak{g}$ is nilpotent, this endomorphism is also nilpotent, hence zero. By 4.7(d), the hypothesis on the reductive center makes the homomorphism

$$T \longrightarrow \mathrm{GL}(\mathfrak{g})$$

a monomorphism, and it therefore induces an injective map on Lie algebras. By II 4.5, this says that $\mathrm{ad}_{\mathfrak{g}}(t) = 0$ implies $t = 0$. Consequently $\mathfrak{t} = 0$, as required.

Proposition 4.10. *Let G be a smooth connected affine algebraic group over an algebraically closed field k. Then the reductive center of G is the intersection of the maximal tori of G.*

The intersection is meant schematically, or equivalently in the sense of subfunctors of G: it is the largest closed subprescheme of G contained in every maximal torus. Since G is noetherian, it is already the intersection of a suitable finite family of maximal tori.

Let Z denote this intersection. It is a closed subgroup of a torus, hence is of multiplicative type, and by 4.5 it contains the reductive center of G. To prove equality it remains to show that Z is central. Since G is connected, IX 5.5 reduces this to showing that Z is invariant. By construction Z is invariant under every inner automorphism $\mathrm{int}(g)$, with $g \in G(k)$. Its normalizer N, represented by VIII 6.7, is therefore a closed subgroup of G containing all rational points of G. Since G is reduced, $N = G$, which proves the result.

Proposition 4.11. *Let S be a prescheme and let G be a smooth affine S-group prescheme with connected fibers and unipotent rank zero. Then G has locally constant reductive rank by 1.9, and hence the "prescheme of maximal tori of G," denoted by $\mathcal{T}$, is defined by 1.10 and is smooth, separated, and of finite type over S. Let G act on $\mathcal{T}$ through inner automorphisms, giving a homomorphism of group functors*

$$u : G \longrightarrow \mathrm{Aut}_S(\mathcal{T}).$$

Under these hypotheses the following three subfunctors of G are identical:

(i) *the reductive center Z of G;*

(ii) *the center Z' of G;*

(iii) *the kernel $Z'' = \ker u$ of the preceding homomorphism.*

In particular, the center of G is represented by a subgroup of multiplicative type in G.

Proof. Clearly $Z \subset Z' \subset Z''$. It remains to prove $Z'' \subset Z$, or, since the hypotheses are stable under base change, that every section g of G over S which acts trivially on $\mathcal{T}$ is a section of Z. Put $G' = G/Z$. By 4.7(b) and (c), the prescheme $\mathcal{T}'$ of maximal tori of G' is canonically isomorphic to $\mathcal{T}$. We are therefore reduced to the case $G = G'$, in which the reductive center Z is the unit subgroup. Notice that 4.7(c) also shows that the unipotent rank of G' equals that of G, and hence is zero.

We must prove that g is the unit section of G. The usual reduction process reduces us first to the case in which S is noetherian, and then to the case in which S is local artinian: when S is locally noetherian, it is enough to check the assertion after every base change $S' \to S$ with S' local artinian. In this case $Z'' = \ker u$ is representable by VIII 6.2(a) and 6.5(c); alternatively, representability follows from XI 6.8. By Nakayama's lemma it is enough to show that the fiber Z''_0 is the unit subgroup. We may therefore suppose that S is the spectrum of a field k, which we may take to be algebraically closed.

The subgroup Z'' lies in the stabilizer of every point of $\mathcal{T}(k)$, that is, in the normalizer $N(T)$ of every maximal torus T of G. Since the unipotent rank of G is zero, XI 5.9 says that T is an open subgroup of $N(T)$. Thus

$$\mathrm{Lie}\, N(T) = \mathrm{Lie}\, T,$$

and $\mathrm{Lie}\, Z''$ is contained in $\mathrm{Lie}\, T$ for every maximal torus T. Proposition 4.10 shows that the intersection of the Lie algebras of all maximal tori is the Lie algebra of the reductive center Z, which is zero in the present reduction. Hence $\mathrm{Lie}\, Z'' = 0$, so Z'' is étale over k.

Moreover, Z'' is invariant in G. Since G is connected, it follows that Z'' is central in G. Thus it is contained in

$$T = \mathrm{Centr}_G(T)$$

for every maximal torus T, and therefore in their intersection. By 4.10 this intersection is $Z = 0$, completing the proof.

0.5 Application to the prescheme of subgroups of multiplicative type[6]

Theorem 5.1. *Let S be a prescheme, let G be a smooth affine S-group prescheme, and let M be the "prescheme of subgroups of multiplicative type in G," representing the functor*

[6] For a generalization of the results of this section to the case in which G is not assumed affine over S, see M. Raynaud, "Ample sheaves on group schemes and homogeneous spaces," Chapter IX.

described in XI 4.1. For every integer $n > 0$, let T_n be the subfunctor of M for which $T_n(S')$ is the set of subgroup preschemes H of multiplicative type in $G_{S'}$ satisfying

$$n \operatorname{id}_H = 0.$$

Thus T_n is representable and affine over S by XI 3.12(a). Let

$$u_n : M \longrightarrow T_n$$

be the morphism defined by $u_n(H) = {}_nH$, where ${}_nH = \ker(n \operatorname{id}_H)$. Then:

(a) *Every subprescheme U of M of finite type over S is contained in a closed subprescheme of finite type over S. Every closed subprescheme X of M of finite type over S is affine over S.*

(b) *Suppose that S is quasi-compact, and let X be a closed subprescheme of M of finite type over S. Then there is an integer $n > 0$ such that, for every multiple m of n, the induced morphism*

$$u_m|_X : X \longrightarrow T_m$$

is a closed immersion.

Proof. (a) For the first assertion, take for X the schematic closure of U (EGA I, 9.5.1 and 9.5.3). This closure is defined because the immersion $i : U \to M$ is quasi-compact: U is of finite type over S, and M is separated over S by XI 4.1. It remains to prove that such an X is affine over S, which will prove the second assertion at the same time. The question is local on S, so we may suppose that S is affine. Since U is of finite type over S, it is contained in a quasi-compact open of M. We are therefore reduced to the case in which U itself is an open subprescheme of finite type over S, hence quasi-compact. Notice that a closed subscheme of a scheme affine over S is affine over S.

It is enough to show that such a U is contained in a closed affine subprescheme of M. In this form the usual reduction process reduces us immediately to the case in which S is noetherian. The same argument applies to (b), reducing it to the case in which S is affine noetherian.

Recall the inverse limit T of the T_n used in the proof of XI 4.1. It is an affine prescheme over S, though not of finite type, and its local rings are noetherian, as was observed at the beginning of the proof of IX 3.7. More explicitly, if $t = (t_n) \in T$, then the transition morphisms $T_m \to T_n$ are smooth and the dimensions of the T_m are bounded. Hence there is an n_0 such that $T_n \to T_{n_0}$ is étale at t_n for every multiple n of n_0. The proof just cited, or XI 3.11, shows that the canonical morphism

$$u : M \longrightarrow T$$

is an immersion and induces isomorphisms on local rings. It need not be an open immersion or a quasi-compact morphism.

Let U be a quasi-compact open of M. We shall show that its closure in T is contained in M. If X is the schematic closure of U in M, this will make

$$u|_X : X \longrightarrow T$$

a closed immersion. Then X is affine, proving (a). Since U is noetherian it has only finitely many irreducible components. Every $t \in T$ in the closure of U is a specialization

of a point $x \in U$. The local ring of t in T is noetherian, and so is the local ring A of t in $\overline{\{x\}}$, endowed with the induced reduced structure.

The canonical morphism from the noetherian integral local scheme

$$S' = \mathrm{Spec}(A)$$

to T sends its closed point to t and its generic point to $x \in M$. We must show that $t \in M$. Replacing A, if necessary, by the quotient by a minimal prime of a suitable complete local A-algebra flat over A (EGA 0_{III}, 10.3.1), we may suppose that A is complete and has algebraically closed residue field. One could even reduce further to a discrete valuation ring by EGA II, 7.1.7. Renaming S' as S, we are reduced to the following lemma.

Lemma 5.2. *Let S be the spectrum of a complete noetherian integral local ring with algebraically closed residue field. Let G be a smooth affine S-group prescheme, let $\big(H(n)\big)_{n>0}$ be a system of subgroups of multiplicative type in G, indexed by the positive integers, and let η be the generic point of S. Suppose that:*

(a) *if m is a multiple of n, then*
$$H(n) = {}_nH(m);$$

(b) *there is a subgroup H_η of multiplicative type in G_η such that*
$$H(n)_\eta = {}_n(H_\eta)$$
for every $n > 0$.

Then there is a subgroup H of multiplicative type in G such that $H(n) = {}_nH$ for every $n > 0$.

Proof. For every n, let
$$C_n = \mathrm{Centr}_G\big(H(n)\big).$$
This is a closed subgroup prescheme of G, smooth over S, by XI 5.3. The positive integers being ordered by divisibility, the C_n form a decreasing family of closed subpreschemes of G. Since G is noetherian, this family is stationary. Let C be the common value of the C_n for sufficiently divisible n. Then C satisfies the same conditions as G, and all the $H(n)$ are in fact subgroups of C. Replacing G by C, we may therefore suppose that all the $H(n)$ are central subgroups of G.

Let s be the closed point of S, and let Z_s be the reductive center of G_s, defined in 4.4. By the definition in 4.1, Z_s contains every $H(n)_s$. Since A is complete, Z_s lifts to a subgroup prescheme Z of multiplicative type in G, by XI 5.8. We claim that Z contains every $H(n)$. Indeed, $H(n)$ is central and hence commutes with Z in G. The resulting homomorphism
$$Z \times H(n) \longrightarrow G$$
has, by IX 6.8, an image K which is a subgroup of multiplicative type in G containing Z. It remains only to prove $K = Z$; this follows from $K_s = Z_s$ and IX 5.1 bis.

We are thus reduced to proving the analogue of the lemma with G replaced by a group Z of multiplicative type and of finite type over S, not necessarily smooth over S. Since A is complete with algebraically closed residue field, Z is diagonalizable by X 3.3 and 1.4, hence is of the form $D_S(M)$, where M is a finitely generated commutative group. Every subgroup of multiplicative type H of Z is therefore diagonalizable by IX 2.11(i), and so is of the form $D_S(N)$, where N is a quotient group of M, by VIII 3.2. Thus the $H(n)$

correspond to quotient groups $M(n)$ of M, and the existence of a subgroup H such that $H(n) = {}_nH$ for every n is equivalent to the existence of a quotient group N of M such that

$$M(n) = N/nN$$

for every n. This follows at once from the existence of the subgroup H_η of G_η such that $H(n)_\eta = {}_n(H_\eta)$ for every n. This proves Lemma 5.2 and hence Theorem 5.1(a).

(b) We already know that

$$u|_X : X \longrightarrow T$$

is a closed immersion. It follows readily that, for all sufficiently large m, the composite

$$u_m|_X : X \longrightarrow T_m$$

is also a closed immersion. Since we shall not need this fact below, we omit the details of the proof.

Corollary 5.3. *With the notation of 5.1, let U be a subprescheme of M, both open and closed, and of finite type over S. Then U, with the structure induced by M, is affine over S. If S is quasi-compact, there is an integer $n > 0$ such that, for every multiple m of n, the induced morphism*

$$u_m|_U : U \longrightarrow T_m$$

is both an open and a closed immersion; equivalently, it is an isomorphism onto an open-and-closed subprescheme of T_m, endowed with the induced structure.

Proof. The first assertion follows from 5.1(a), and the second from 5.1(b), since $u_m : M \to T_m$ is smooth by XI 2.2 bis and a smooth immersion, that is, an étale immersion, is an open immersion.

Corollary 5.4. *Let S be a prescheme, let G be a smooth affine S-group prescheme whose reductive rank is locally constant* (1.7(b))*, and let $\mathcal{T}$ be the "prescheme of maximal tori of G"* (1.10)*. Then $\mathcal{T}$ is smooth and affine over S. If T is a maximal torus of G and $N(T)$ its normalizer, then $G/N(T)$* (XI 5.3 bis) *is affine over S. The same is true of $G/C(T)$, where $C(T)$ is the centralizer of T, provided that*

$$W(T) = N(T)/C(T)$$

is finite over S; cf. 2.1(b).

Proof. The second assertion is contained in the first, since the conjugacy theorem identifies $G/N(T)$ with $\mathcal{T}$, by XI 5.5 bis. For the first assertion, observe that by construction $\mathcal{T}$ is isomorphic to an open-and-closed subprescheme of M. Indeed, one may suppose that the reductive rank of G is constant and equal to r; then $\mathcal{T}$ is the subprescheme of M corresponding to the subtori of relative dimension r, that is, the largest subprescheme of M over which the universal subgroup of multiplicative type

$$H \subset G_M$$

is a torus of relative dimension r. Corollary 5.3 therefore applies. Finally, $G/C(T)$ is finite over

$$G/N(T) \simeq (G/C(T))/W(T),$$

and hence is affine because $G/N(T)$ is affine.

Corollary 5.5. *Let G be a smooth affine algebraic group over a field k. Then the scheme M of subgroups of multiplicative type in G is a disjoint union of affine k-schemes. For every subgroup H of multiplicative type in G, if C and N denote respectively its centralizer and its normalizer, then the quotients G/C and G/N are affine.*

Proof. By XI 5.1 bis, the saturation under the action of G on M of every finite closed subset of M is open. Indeed, one reduces to the case in which k is algebraically closed, and hence to the orbit of a k-rational point x; then, by the cited result, the morphism

$$G \longrightarrow M, \qquad g \longmapsto g \cdot x$$

is smooth and therefore open, so its image is open.

Let U be the union of the equivalence classes of the closed points of M under the action of G. Then U is open and contains every closed point of M; by the Nullstellensatz, $U = M$. Thus M is a disjoint union of open subpreschemes, which are necessarily also closed. Hence M is the sum prescheme of subpreschemes M_i, each of which is the G-orbit of a closed point. Each M_i is quasi-compact, hence of finite type, and is therefore affine by 5.3.

If H is a subgroup of multiplicative type in G, it corresponds to a k-rational point x of M, and G/N identifies with the orbit of x under G, by XI 5.5 bis. Thus G/N is affine. Since C is an open subgroup of N, by XI 5.9, the quotient G/C is finite over G/N; indeed,

$$G/N \simeq (G/C)/(N/C).$$

Consequently G/C is affine as well.

This proof also gives the following result.

Corollary 5.6. *Under the hypotheses of 5.5 on G and H, the subprescheme U of M consisting of the subgroups of multiplicative type in G which are locally conjugate to H is an open-and-closed subprescheme of M.*

In picturesque language, every subgroup H' of multiplicative type in G which is a limit of subgroups conjugate to H is itself conjugate to H.

Remarks 5.7. Let S be a prescheme, let G be a smooth affine S-group prescheme, let H be an S-prescheme of multiplicative type and of finite type, and put

$$M = \mathrm{Hom}_{S\text{-gr}}(H, G).$$

By XI 4.2, M is representable, smooth over S, and separated over S. One can prove for M a result entirely analogous to 5.1, either by reducing to 5.1 through an argument analogous to that of XI 4.2, or directly by an argument modeled on the proof of 5.1. The corresponding variants of 5.3, 5.5, and 5.6 follow; we leave their formulation to the reader.

0.6 Maximal tori and Cartan subgroups of not necessarily affine algebraic groups (algebraically closed base field)

Lemma 6.1. *Let G be a connected algebraic group over a field k, and let Z be an algebraic subgroup of finite index in the center of G. Then G/Z is affine.*

Proof. We may suppose that Z is the center of G, since a scheme finite over an affine scheme is affine. Consider the vector spaces

$$P^n = P^n(G) = \mathcal{O}_{G,e}/\mathfrak{m}_{G,e}^{n+1} \qquad (n \geq 0),$$

where $\mathfrak{m}_{G,e}$ is the maximal ideal at the identity. The group G acts on the $P^n(G)$ through the adjoint representation. Let Z_n be the kernel of the corresponding homomorphism

$$\mathrm{ad}_n : G \longrightarrow \mathrm{GL}(P^n).$$

Using the connectedness of G, one readily verifies that Z is the intersection of the Z_n. Since G is noetherian, this intersection is equal to one of the Z_n. Passing to the quotient, ad_n defines a monomorphism

$$G/Z_n \longrightarrow \mathrm{GL}(P^n),$$

which is a closed immersion. Hence G/Z_n is affine, and so is G/Z.

Lemma 6.2. *Let G be a smooth algebraic group over a field k, let Z be a central algebraic subgroup, put $G' = G/Z$, and let*

$$u : G \longrightarrow G'$$

be the canonical homomorphism. Let T be a subgroup of multiplicative type in G, let $T' = u(T)$, let C be the centralizer of T in G, and let C' be the centralizer of T' in G'. Then

$$C'^0 \subset u(C).$$

Proof. We may suppose that k is algebraically closed. Put

$$C_1 = \left(u^{-1}(C'^0)\right)_{\mathrm{red}}.$$

It is enough to prove that $C_1 \subset C$. Indeed, $u(C_1)$ is connected and of finite index in C'^0, hence is equal to it set-theoretically and therefore scheme-theoretically, since C', and consequently C'^0, is smooth.

The commutator morphism

$$(g,t) \longmapsto gtg^{-1}t^{-1}$$

from $G \times G$ to G induces a morphism

$$\varphi : C_1 \times T \longrightarrow Z_1, \qquad Z_1 = Z_{\mathrm{red}}.$$

Indeed, the source is reduced, and for $g \in C_1(k)$ and $t \in T(k)$, the element $gtg^{-1}t^{-1}$ belongs to $Z(k)$ because g centralizes T modulo Z.

A calculation on k-rational points shows that $\varphi(g,t)$ is additive in g and in t, hence "bilinear." We claim that this homomorphism is identically zero; this proves $C_1 \subset C$. By the density theorem for T, we reduce to the case in which T is finite, so that $n\,\mathrm{id}_T = 0$ for some integer $n > 0$. The morphism φ is then defined by a group homomorphism

$$C_1 \longrightarrow \mathrm{Hom}_{k\text{-gr}}(T, Z_1).$$

Since Z_1 is commutative and smooth, the group on the right is represented by an étale algebraic group over k. Since C_1 is connected, every group homomorphism from C_1 to that group is zero.

Corollary 6.3. *Under the preceding hypotheses, suppose that C' is connected. Then*

$$C' = u(C), \qquad C = u^{-1}(C').$$

Proof. In this case $C' = C'^0$. Moreover, C plainly contains Z, and hence

$$C = u^{-1}(u(C)).$$

Lemma 6.4. *Let G be an algebraic group over a field k, and let Z be a central algebraic subgroup such that*

$$G/Z = G'$$

is a torus. Then G is commutative. If k is algebraically closed, there is a torus T in G such that $u(T) = G'$, where

$$u : G \longrightarrow G/Z = G'$$

is the canonical homomorphism.

Proof. We may suppose that k is algebraically closed. Consider again the commutator morphism

$$G \times G \longrightarrow G, \qquad (g, h) \longmapsto ghg^{-1}h^{-1}.$$

Since Z is central, this morphism factors through a morphism $G' \times G' \to G$. Since G' is commutative, the latter takes its values in Z, and even in Z_{red}, because $G' \times G'$ is reduced. As above, the resulting morphism

$$\varphi : G' \times G' \longrightarrow Z_{\mathrm{red}}$$

is bilinear and hence zero. Thus G is commutative.

To find a torus T in G with $u(T) = G'$, we may, after replacing G by G^0_{red}, suppose that G is smooth and connected. Replacing Z by Z^0_{red}, we may also suppose that Z is smooth and connected. By a well-known theorem of Chevalley,[7] Z is an extension of an abelian variety by an affine group. We are therefore reduced to the two cases in which Z is affine or Z is an abelian variety. In the first case G is affine and the result is classical (BIBLE 7, Theorem 3(a)).

Suppose then that Z is an abelian variety. Every homomorphism from the additive group $\mathbb{G}_a$ to the torus $G' = R$, or to the abelian variety Z, is trivial. The same is therefore true of every homomorphism from $\mathbb{G}_a$ to G; thus G contains no subgroup isomorphic to $\mathbb{G}_a$. Chevalley's structure theorem gives an exact sequence

$$(*) \qquad 0 \longrightarrow T \longrightarrow G \longrightarrow A \longrightarrow 0,$$

where A is an abelian variety and T is a smooth connected affine group. Since T is commutative and contains no additive subgroup, T is a torus; it is plainly the unique maximal torus of G. It remains to prove that every epimorphism

$$u : G \longrightarrow R$$

to a torus R satisfies $u(T) = R$.

Put

$$\mathrm{Hom}_{\mathrm{gr}}(T, \mathbb{G}_m) = M, \qquad \mathrm{Hom}_{\mathrm{gr}}(R, \mathbb{G}_m) = P.$$

These are finitely generated free $\mathbb{Z}$-modules, and

$$T = D_k(M), \qquad R = D_k(P).$$

Let

$$B = \mathrm{Ext}^1_{k\text{-gr}}(A, \mathbb{G}_m);$$

this is also the set of k-rational points of the abelian variety dual to A. We have

$$(**) \qquad \mathrm{Ext}^1(A, T) = \mathrm{Hom}_{\mathrm{gr}}(M, B), \qquad \mathrm{Ext}^1(A, R) = \mathrm{Hom}_{\mathrm{gr}}(P, B).$$

[7] Editor's note: references should be supplied here.

In particular, the extension G of A by T is determined by a homomorphism

$$\theta : M \longrightarrow B.$$

The exact sequence (*) also gives

$$0 \longrightarrow \mathrm{Hom}(A, R) \longrightarrow \mathrm{Hom}(G, R) \longrightarrow \mathrm{Hom}(T, R) \longrightarrow \mathrm{Ext}^1(A, R).$$

Moreover, $\mathrm{Hom}(A, R) = 0$, and the last map identifies with the homomorphism

$$\mathrm{Hom}(P, M) \longrightarrow \mathrm{Hom}(P, B)$$

induced by $\theta : M \to B$. Setting

$$N = \ker \theta,$$

we obtain canonical bijections

$$\mathrm{Hom}(G, R) \simeq \mathrm{Hom}(P, N) \simeq \mathrm{Hom}(S, R), \qquad S = D_k(N).$$

This has the following immediate geometric description.

Lemma 6.5. *Let G be an extension of an abelian variety A by a torus $T = D_k(M)$, defined by a homomorphism*

$$\theta : M \longrightarrow B = \mathrm{Ext}^1(A, \mathbb{G}_m),$$

over an algebraically closed field k. Put $N = \ker \theta$, and let $S = D_k(N)$ be the corresponding torus, isomorphic to T/U, where $U = D_k(M/N)$. Consider the extension $H = G/U$ of A by S. This extension splits,[8] *so there is a unique projection from H to S, and hence a unique homomorphism*

$$v : G \longrightarrow S$$

extending the canonical homomorphism $T \to S$. For every torus R and every homomorphism $u : G \to R$, there is a unique homomorphism $u' : S \to R$ such that $u = u'v$. In particular,

$$\mathrm{Im}(u) = \mathrm{Im}(u') = u(T).$$

It follows in particular that if u is an epimorphism, then its restriction to T is also an epimorphism. This completes the proof of 6.4.

Theorem 6.6. *Let G be a smooth connected algebraic group over an algebraically closed field k.*

(a) *The maximal tori of G are conjugate.*

(b) *If T is a torus in G, then its centralizer is smooth and connected.*

(c) *The assignment*

$$T \longmapsto \mathrm{Centr}_G(T)$$

gives a bijection between the maximal tori T of G and the Cartan subgroups of G defined in No. 1. An algebraic subgroup C of G is a Cartan subgroup if and only if it

[8] Editor's note: the extension is split.

is smooth, nilpotent, and set-theoretically equal to its connected normalizer; in that case it is even scheme-theoretically equal to its connected normalizer. Moreover,

$$C = \mathrm{Centr}_G(T),$$

where T is the unique maximal torus of C, and

$$\mathrm{Norm}_G(C) = \mathrm{Norm}_G(T).$$

(d) *Let $v : G \to H$ be an epimorphism of smooth connected algebraic groups. The maximal tori, respectively the Cartan subgroups, of H are the images of the maximal tori, respectively the Cartan subgroups, of G. If T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$.*

(e) *Under the hypotheses of* (d)*, suppose that* $\ker v$ *is a central subgroup of G. Then*

$$T \longmapsto v(T), \qquad C \longmapsto v(C)$$

give bijections between the maximal tori, respectively the Cartan subgroups, of G and those of H. The Cartan subgroups of G contain the center of G, and hence $\ker v$*; they are precisely the groups $v^{-1}(C')$, where C' is a Cartan subgroup of H.*

We also record the following immediate consequence of Theorem 6.6(c).

Corollary 6.7. *The group G is nilpotent, that is, $G(k)$ is nilpotent, if and only if every torus in G is central, or equivalently if G has a unique maximal torus. In that case this torus is the largest subtorus of G.*

Proof of Theorem 6.6. Let Z be a central algebraic subgroup of G, put $G' = G/Z$, and let

$$u : G \longrightarrow G'$$

be the canonical homomorphism. Then G' is smooth and connected. If T' is a subtorus of G', it follows from 6.4 that $u^{-1}(T')$ is commutative and that T' is the image of a subtorus of $u^{-1}(T')$, hence of a subtorus of G. Since $u^{-1}(T')$ is commutative, it plainly has a largest subtorus T: the sum of two subtori is a third subtorus containing both. Thus

$$u(T) = T'.$$

It follows immediately that, for every maximal torus T of G, its image $T' = u(T)$ is a maximal torus of G', and that

$$T \longmapsto u(T)$$

is a bijection between the maximal tori of G and those of G'.

Now take

$$Z = \mathrm{Centr}(G)^0_{\mathrm{red}}.$$

Then G' is affine by 6.1. Its maximal tori are conjugate, and therefore so are those of G, which proves (a). Moreover, G is nilpotent, respectively has a unique maximal torus, if and only if G' has the same property. Since G' is affine, these two conditions on G' are equivalent (BIBLE 6, Th. 4, Cor. 2); hence they are also equivalent for G. If G has a unique maximal torus T, then T is invariant and therefore central; and because every torus of G is contained in a maximal torus, every torus is central. Conversely, if every

torus is central, then so are the maximal tori, and the conjugacy theorem (a) shows that there is only one. This proves 6.7.

Let T be any torus of G, let $T' = u(T)$, and put

$$C' = \mathrm{Centr}_{G'}(T').$$

The group C' is connected (BIBLE 6, Th. 6(a)). By 6.3 the centralizer C of T is $u^{-1}(C')$, and is therefore connected because Z is connected. This proves (b). If T is maximal, then T' is maximal; hence C' is nilpotent, and so is C, which is a central extension of C'. Moreover, T is a maximal torus of C, and by 6.7 it is the unique maximal torus of C. Consequently

$$T \longmapsto \mathrm{Centr}_G(T)$$

is a bijection from the maximal tori of G to the Cartan subgroups of G. We also have

$$\mathrm{Centr}_G(T) = C \subset \mathrm{Norm}_G(C) \subset \mathrm{Norm}_G(T).$$

The centralizer of T has finite index in its normalizer (cf. XI 5.9); the argument there remains valid without the affineness hypothesis, because it uses only the representability of the two functors in question, as noted in XI 6.5. Since C is smooth and connected, it follows that

$$C = \mathrm{Norm}_G(C)^0.$$

Furthermore, the bijection between maximal tori and Cartan subgroups shows that the normalizers of C and T have the same k-valued points. Since the latter normalizer is smooth,

$$\mathrm{Norm}_G(T) = \mathrm{Norm}_G(C).$$

To finish the proof of (c), it remains to show that a smooth, connected, nilpotent subgroup C of G which has finite index in its normalizer is a Cartan subgroup. Since Z is central, the normalizer of C contains Z; and since Z is smooth and connected, it follows that $Z \subset C$. Hence

$$C = u^{-1}(C'), \qquad C' = u(C),$$

and

$$\mathrm{Norm}_G(C) = u^{-1}\big(\mathrm{Norm}_{G'}(C')\big).$$

Thus C' is connected and nilpotent and has finite index in its normalizer. By Theorem 1 of BIBLE 7, it is a Cartan subgroup of G', and it follows at once that C is a Cartan subgroup of G.

We next prove (e). We are in the situation considered at the beginning of the proof, with $\ker u = Z$ and $G' = H$, and we have already seen that $T \mapsto u(T)$ is a bijection between the maximal tori of G and those of G'. Using the bijection just proved between maximal tori and Cartan subgroups, we obtain a bijection between Cartan subgroups C of G and Cartan subgroups C' of G': to

$$C = \mathrm{Centr}_G(T)$$

we associate

$$C' = \mathrm{Centr}_{G'}(T'), \qquad T' = u(T).$$

Since C' is connected by (b), 6.3 gives

$$C' = u(C), \qquad C = u^{-1}(C'),$$

which proves (e).

It remains to prove (d). Let T be a maximal torus of G. We first prove that $v(T)$ is a maximal torus of H; by the conjugacy theorem (a), this will imply that all maximal tori of H arise in this way. Let R be a torus of H containing $v(T)$. Replacing H by R and G by $v^{-1}(R)^0_{\mathrm{red}}$, we may suppose that $R = H$, so that H is a torus; it remains to prove $v(T) = H$. Put again

$$Z = \mathrm{Centr}(G)^0_{\mathrm{red}}, \qquad G' = G/Z, \qquad H' = H/v(Z).$$

We have a commutative diagram with exact rows

$$\text{(D)} \qquad \begin{array}{ccccccccc} e & \longrightarrow & Z & \longrightarrow & G & \xrightarrow{u} & G' & \longrightarrow & e \\ & & \downarrow{\scriptstyle v''} & & \downarrow{\scriptstyle v} & & \downarrow{\scriptstyle v'} & & \\ e & \longrightarrow & v(Z) & \longrightarrow & H & \xrightarrow[u']{} & H' & \longrightarrow & e. \end{array}$$

We already know that $u(T)$ is a maximal torus of G'. Because G' is affine, the epimorphism

$$v' : G' \longrightarrow H'$$

is an epimorphism of smooth connected affine groups. Therefore $v'(u(T))$ is a maximal torus of H' (BIBLE 7, Th. 3(a)), and hence equals H'. Thus $u'(v(T)) = H'$, so in order to prove $v(T) = H$ it is enough to show that $v(T) \supset v(Z)$. Now $v(Z)$ is a smooth connected subgroup of H, hence a torus, and

$$v'' : Z \longrightarrow v(Z)$$

is an epimorphism. By 6.4,

$$v(Z) = v''(S)$$

for a torus S of Z. Since S is central in G, it is contained in the maximal torus T, and therefore $v(Z) \subset v(T)$. This proves (d) for maximal tori.

In view of the bijection between maximal tori and Cartan subgroups, it remains to prove that, if T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$. Return to diagram (D), now without assuming that H is a torus, and put

$$T' = u(T), \qquad C' = u(C).$$

By (e), C' is the centralizer of the maximal torus T'. Since G', and hence H', is affine, $v'(C')$ is the centralizer of the maximal torus $v'(T')$ of H' (BIBLE 7, Th. 3(a)). Equivalently, $u'(v(C))$ is the centralizer of $u'(v(T))$. Because C contains Z, the group $v(C)$ contains $v(Z)$. It is therefore the inverse image under u' of $u'(v(C))$, that is, of the centralizer of $u'(v(T))$. By (e), applied to $u' : H \to H'$ and the maximal torus $v(T)$ of H, this inverse image is precisely the centralizer of $v(T)$. This completes the proof of 6.6. □

0.7 Application to smooth group preschemes not necessarily affine

Theorem 7.1. *Let S be a prescheme and let G be an S-group prescheme that is smooth, separated, and of finite type over S. Suppose that G admits a maximal torus locally for the faithfully flat quasi-compact topology. Then:*

(a) The map

$$T \longmapsto \mathrm{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G onto the set of Cartan subgroups of G. If C corresponds to T, then T is the unique maximal torus of C.

(b) Let T, T' be two maximal tori of G, and let C, C' be the corresponding Cartan subgroups. Then

$$\mathrm{Transp}_G(T, T') = \mathrm{Transp}_G(C, C') = \mathrm{Transp}_G(T, C').$$

The first two terms are also identical to the strict transporters. Moreover, the functor in question is representable by a closed subprescheme of G, smooth over S. The tori T, T' and the Cartan subgroups C, C' are locally conjugate for the étale topology.

(c) A maximal torus of G, and a Cartan subgroup of G, exist locally for the étale topology.

(d) Suppose that every finite subset of a fiber G_s is contained in an affine open of G—for example, that G is quasi-projective over S, or that S is artinian. Then the functor

$$\mathcal{T} : (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets})$$

defined in 1.10, the functor of maximal tori of G, is isomorphic to the functor

$$\mathcal{C} : (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets})$$

of Cartan subgroups of G, and is representable by a smooth, separated prescheme of finite type over S. This prescheme is quasi-projective over S when G is, and is affine over S when G is affine over S, or when S is artinian.

(e) Let $u : G \to G'$ be a morphism of S-group preschemes, where G' is smooth, separated, and of finite type over S, and suppose that

$$u_s(G_s) = G'_s$$

for every $s \in S$, that is, that u_s is faithfully flat.[9] *For every maximal torus T of G, $u(T)$ is a maximal torus of G'. If G' has connected fibers, then for every Cartan subgroup C of G, $u(C)$ is a Cartan subgroup of G'; and if C is the centralizer of T, then $u(C)$ is the centralizer of $u(T)$. In either case the induced morphism $T \to u(T)$, respectively $C \to u(C)$, is faithfully flat.*

(f) Under the hypotheses of (e)*, suppose in addition that* $\ker u$ *is a central subgroup of G. Then*

$$T \longmapsto u(T)$$

is a bijection from the set of maximal tori of G onto the set of maximal tori of G'. If G' has connected fibers, then

$$C \longmapsto u(C)$$

is likewise a bijection from the set of Cartan subgroups of G onto the set of Cartan subgroups of G'; if $C' = u(C)$, then $C = u^{-1}(C')$.

[9] The editors did not locate the printed reference in Exposé VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), Proposition II.5.1(c).

Remarks 7.2. We shall see in Exposé XV that the conclusion of Theorem 7.1(d) remains valid without the restrictive condition on the finite subsets of the G_s. We shall also prove there the conclusions concerning Cartan subgroups alone that occur in (b), (c), (d), and (e), under the weaker hypothesis that G admit a Cartan subgroup locally for the faithfully flat quasi-compact topology, but not necessarily a maximal torus. For that purpose we shall moreover need to use 7.1 when S is artinian. The method of Exposé XV also considerably simplifies the proof of (c) and (d) when S is not artinian.

Proof of Theorem 7.1. (a) Proceeding as in 3.2, one sees that for every maximal torus T of G,

$$C = \mathrm{Centr}_G(T)$$

is indeed a Cartan subgroup of G, and that T is determined by C as the unique maximal torus of C. It remains only to show that every Cartan subgroup C of G is defined by a maximal torus of G, or equivalently that C admits a central maximal torus.

The question is still local for the fpqc topology, so we may suppose that G admits a maximal torus T. By XI 6.2,

$$\mathrm{Transp}_G(T, C)$$

is representable by a closed subprescheme S' of G, smooth over S; moreover $S' \to S$ is surjective by 6.6(c). After the base change $S' \to S$, we may therefore suppose that there is a section g of G such that

$$\mathrm{ad}(g) \cdot T \subset C.$$

Then $\mathrm{ad}(g) \cdot T$ is a maximal torus of C. It is central fiber by fiber by 6.6(c), and hence central by IX 5.6(b). This proves (a).

(b) Let T, T' be two maximal tori of G, and C, C' their corresponding Cartan subgroups. The following conditions are equivalent:

$$(1)\ T \subset T', \quad (2)\ T = T', \quad (3)\ T \subset C', \quad (4)\ C \subset C', \quad (5)\ C = C'.$$

This follows immediately from (a). Applying the same result after arbitrary base change gives the asserted identities among the transporters and strict transporters.

We have already observed in (a) that $\mathrm{Transp}_G(T, C')$ is representable by a closed subprescheme of G, smooth over S, whose structural morphism is surjective. By Hensel's lemma this prescheme has a section locally for the étale topology. Thus T, T', on the one hand, and C, C', on the other, are locally conjugate for the étale topology. This proves (b).

(c) Suppose first that S is artinian and local. If G admits a maximal torus T, the conjugacy theorem just proved shows that the functor $\mathcal{T}$ of maximal tori of G is represented by the homogeneous space G/N, where N is the normalizer of T in G; it is smooth by (b). Moreover, as observed in VI_A 3.2.1, every finite subset of $\mathcal{T}$ is contained in an affine open, because $\mathcal{T}$ is a homogeneous space under G.

In general there is a finite extension k'/k of the residue field of S such that $G_{k'}$ has a maximal torus. The extension k'/k is obtained by a finite flat base change $S' \to S$, and the maximal torus of $G_{k'}$ lifts to a maximal torus of $G_{S'}$ by XI 2.1 bis. Thus $\mathcal{T}_{S'}$ is represented by a smooth, separated prescheme of finite type over S', every finite subset of which is contained in an affine open. Its natural descent datum is effective. Hence $\mathcal{T}$ is representable, and descent shows that it is smooth, separated, and of finite type over S. By smoothness and Hensel's lemma, $\mathcal{T}$ has a section locally for the étale topology. This proves (c) and (d) when S is artinian. We shall prove below that in this case $\mathcal{T}$ is in fact affine over S.

Now let S be arbitrary. To prove (c) and (d), which are local assertions on S for the Zariski topology, we may suppose that S is affine, that there is a prime ℓ prime to all residue characteristics of S, and that the reductive rank of the fibers of G is constant, say equal to r. The latter rank is visibly locally constant because of the assumed fpqc-local existence of a maximal torus.

Choose a faithfully flat quasi-compact morphism $S' \to S$, with S' affine, such that $G_{S'}$ admits a maximal torus T_0, and let C_0 be its centralizer. I claim that there is a suitable power n of ℓ such that

$$C_0 = \mathrm{Centr}_{G_{S'}}({}_nT_0).$$

Indeed, this reduces immediately to the case where S' is noetherian, which was treated in XI 6.2. Fix such an n, and put

$$M = (\mathbb{Z}/n\mathbb{Z})^r, \qquad P = \mathrm{Hom}_{S\text{-gr}}(M_S, G).$$

Then P is represented by a closed subprescheme of finite presentation of $({}_nG)^r$, where

$${}_nG = \mathrm{Hom}_{S\text{-gr}}((\mathbb{Z}/n\mathbb{Z})_S, G).$$

This is also the kernel of the n-th-power morphism on G, and is therefore represented by a closed subprescheme of finite presentation of G. Moreover, P is smooth over S by XI 2.1.

Let $s \in S$. From what precedes, there is a finite separable extension $k''/\kappa(s)$ for which $G_{k''}$ has a maximal torus T''. Since ${}_nT''$ is étale over k'', we may, after replacing k'' by a finite separable extension, suppose that

$${}_nT'' \simeq M_{k''}.$$

Choose an étale morphism $S'' \to S$ and a point $s'' \in S''$ above s, inducing the residue-field extension $\kappa(s'') \simeq k''$. We obtain a section of $P_{S''} \otimes \kappa(s'')$ over $\mathrm{Spec}\,\kappa(s'')$. By Hensel's lemma, after replacing the pointed prescheme (S'', s'') by an étale neighborhood if necessary, this section extends to a section of $P_{S''}$. Equivalently, there is a homomorphism

$$M_{S''} \longrightarrow G_{S''}$$

inducing the given isomorphism

$$M_{\kappa(s'')} \simeq {}_nT''.$$

By IX 6.4—which here reduces to Nakayama's lemma—this homomorphism is a closed immersion above an open neighborhood of s'', which we may take to be all of S''. Let H'' be its image, and let C'' be its centralizer in $G_{S''}$; by XI 6.2, C'' is a closed smooth subprescheme of $G_{S''}$.

Lemma 7.3. *Under the preceding hypotheses on ℓ, n, r, for every prescheme S'' over S and every maximal torus T'' of $G_{S''}$, one has*

$$\mathrm{Centr}_{G_{S''}}(T'') = \mathrm{Centr}_{G_{S''}}({}_nT'').$$

Indeed, after faithfully flat descent from $S' \times_S S''$, we reduce to the case in which $G_{S''}$ admits a maximal torus T_1'' for which the displayed equality holds. By (b), T'' and T_1'' are locally conjugate for the étale topology, so the same equality holds for T''.

Apply this result over $\mathrm{Spec}\,\kappa(s'')$. It shows that the fiber $C''_{s''}$ is a Cartan subgroup of $G_{s''}$. Now, since G is smooth over S, the union of the identity components of the fibers G_s is an open subprescheme of G; this is the general EGA IV result for smooth morphisms

recorded by the editors without a precise reference. This open is visibly stable under the group law and hence is a subgroup, denoted G^0. It satisfies the same preliminary hypotheses as G, and maximal tori of G correspond bijectively to maximal tori of G^0. Thus, for the proof of (c) and (d), we may and do suppose that G has connected fibers. Its Cartan subgroups then also have connected fibers by 6.6(a).

I claim that C''^0 is a Cartan subgroup of $G_{S''}$ above an open neighborhood of s''. This will complete the proof of (c). The claim follows from the next lemma.

Lemma 7.4. *Under the hypotheses of 7.1, suppose that G has connected fibers. Let C be a closed group subprescheme of G, smooth over S, and let $s \in S$ be such that C_s is a Cartan subgroup of G_s. Then C^0 is a Cartan subgroup of G over an open neighborhood of s.*

We reduce easily to the case where S is local and s is its closed point, and to proving there that C^0 is a Cartan subgroup of G. By flat descent we may further suppose that G admits a maximal torus T. By XI 6.2,

$$\mathrm{Transp}_G(T, C)$$

is represented by a closed subprescheme of G, smooth over S. The fiber of this transporter at s is nonempty by the hypothesis on C_s and the conjugacy theorem 6.6(a). Faithfully flat descent therefore reduces us to the case where the transporter has a section over S, hence to the case where C contains a maximal torus T of G. Then T is central in C^0 by IX 5.6(a), so

$$C^0 \subset \mathrm{Centr}_G(T).$$

This is an inclusion of smooth group preschemes over S, with connected fibers and the same relative dimension—namely, the dimension of their common fiber at s. It is therefore an equality, proving the lemma and completing (c).

(d) Retain the preceding notation and hypotheses on ℓ, n, r, and the assumption that G has connected fibers. Let

$$Q : (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets})$$

be the functor defined by

$$Q(S') = \left\{ \begin{array}{c} \text{subgroups of multiplicative type in } G_{S'} \\ \text{of type } M = (\mathbb{Z}/n\mathbb{Z})^r \end{array} \right\}$$

in the sense of IX 1.4. The rule

$$T \longmapsto {}_nT$$

defines a morphism

$$\varphi : \mathcal{T} \longrightarrow Q,$$

which is a monomorphism by Lemma 7.3.

I claim that Q is representable by a separated prescheme of finite presentation over S. As noted in the proof of XI 3.12(a), there is an isomorphism

$$Q \simeq P_0/\Gamma,$$

where P_0 is the open and closed subprescheme of

$$P = \mathrm{Hom}_{S\text{-gr}}(M_S, G)$$

corresponding to monomorphisms $M_{S'} \to G_{S'}$, as in IX 6.8, and

$$\Gamma = (\mathrm{Aut}_{\mathrm{gr}}(M))_S.$$

The hypothesis that every finite subset of a fiber G_s lies in an affine open is stable under passage to a closed subprescheme and under fiber products. It is therefore inherited by $({}_nG)^r$, then by P, and then by P_0. Hence Q is represented by a prescheme of finite presentation over S.[10] The same argument shows that Q is quasi-projective, respectively affine, over S when G is. In every case Q is separated over S.

Lemma 7.5. *The preceding morphism $\varphi : \mathcal{T} \to Q$ is representable by an open immersion.*

In other words, let H be a subgroup of multiplicative type in G, of type M. We must prove that there is an open subset $U \subset S$ with the following property: for every $S' \to S$, the group $H_{S'}$ is of the form ${}_nT_{S'}$, for a suitable maximal torus $T_{S'}$ of $G_{S'}$, if and only if $S' \to S$ factors through U.

We may suppose that the nilpotent rank of the fibers of G is constant, say equal to r'; it is locally constant by (b). Put

$$C = \mathrm{Centr}_G(H),$$

a closed group subprescheme smooth over S by XI 6.2. Replacing S by the open and closed subset where C has relative dimension r', we may suppose that this is its relative dimension everywhere. Then $H = {}_nT$ for a maximal torus T of G if and only if C^0 admits a central maximal torus T, everywhere of relative dimension r, and $H = {}_nT$. This gives another description of the subfunctor U.

By flat descent, we may suppose that G admits a maximal torus T_1. Consider the subfunctor

$$R = \mathrm{Transp}_G(T_1, \mathrm{Centr}_{C^0}(C^0)) \subset \mathrm{Transp}_G(T_1, C).$$

The transporter on the right is represented by a smooth prescheme over S by XI 6.2. The one on the left is represented by the induced open subprescheme, as follows immediately from IX 5.6(a); in particular it too is smooth over S. Its structural morphism is therefore open. Replacing S by its image, equipped with the induced structure, we may suppose that this morphism is surjective. Applying 1.13 to C^0, we find that C^0 admits a central maximal torus T, because it admits one fpqc-locally. Its relative dimension is that of T_1, namely r. The remaining condition on H is $H = {}_nT$, and by IX 2.10 this again cuts out a suitable open and closed subset of S. This proves the lemma.

Lemma 7.5 shows that $\mathcal{T}$ is represented by a separated prescheme locally of finite presentation over S, and in fact of finite presentation. For the latter assertion, one may either inspect the proof of Lemma 7.5 to see that φ is a quasi-compact open immersion, or descend fpqc-locally to the case where G has a maximal torus T, when

$$\mathcal{T} \simeq G/\mathrm{Norm}_G(T).$$

This expression, or XI 2.1, also shows that $\mathcal{T}$ is smooth over S. If G is quasi-projective over S, then so are Q and $\mathcal{T}$. If G is affine over S, the affineness of $\mathcal{T}$ over S was proved in 5.4. It is unknown here whether, without the hypothesis that G/S is affine, n can be chosen so that the open immersion of Lemma 7.5 is also closed.

[10] The editors left the printed reference to Exposé V unspecified.

It remains to show that $\mathcal{T}$ is affine over S when S is artinian. By EGA I 6.1.7, we reduce to the case where S is the spectrum of a field, which we may suppose algebraically closed. In view of (f), proved below, it suffices to prove the assertion for

$$G/\operatorname{Centr}(G).$$

This quotient is affine by 6.1, so the preceding affine case applies. This proves (d).

(e) By IX 6.8, there is a subtorus T' of G' such that u induces a faithfully flat morphism

$$T \longrightarrow T'.$$

This characterizes T' as the subsheaf $u(T)$ of G'. Let C and C' be the centralizers of T and T', respectively. We prove that $C \to C'$ is flat, and faithfully flat when G' has connected fibers. Since C and C' are flat and of finite presentation over S, SGA 1, I 5.9 reduces us to a base field, which we may suppose algebraically closed. Replacing G, G' by their identity components does not change C^0, C'^0, so we may suppose G, G' connected. The assertion then follows from 6.6(d), together with (a).

(f) By (a) and (e), it is enough to prove the assertion about Cartan subgroups. A Cartan subgroup C of G, being the centralizer of a maximal torus, contains the center of G, and therefore contains $\ker u$. Thus

$$C = u^{-1}(C'), \qquad C' = u(C),$$

where C' is the Cartan subgroup of G' supplied by (e). Hence $C \mapsto u(C)$ is injective. To prove surjectivity, it suffices to show that $u^{-1}(C')$ is a Cartan subgroup of G for every Cartan subgroup C' of G'.

The question is fpqc-local. We may suppose that G admits a Cartan subgroup C_1. Then $C_1' = u(C_1)$ is a Cartan subgroup of G', and is fpqc-locally conjugate to C' by (b). Since

$$u^{-1}(C_1') = C_1$$

is a Cartan subgroup of G, the same is true of $u^{-1}(C')$. This proves (f) and completes the proof of the theorem. □

Alternatively, to prove that $u^{-1}(C')$ is a Cartan subgroup, observe that it is flat over S, since u is; SGA 1, I 5.9 then reduces the assertion, by definition, to the case of a base field, where 6.6(e) applies.

Corollary 7.6. *Under the hypotheses of 7.1*(e)*, G' admits a maximal torus locally for the étale topology, and therefore satisfies the preliminary hypotheses imposed on G. If moreover* $\ker u$ *is central, then the functors $\mathcal{T}_G, \mathcal{C}_G$ of maximal tori and Cartan subgroups of G are isomorphic to the analogous functors $\mathcal{T}_{G'}, \mathcal{C}_{G'}$ for G'. Consequently, whenever these functors are representable, they are represented by isomorphic S-preschemes.*

Remark 7.7. *(a)* In contrast with the case where G is affine over S—in fact it is enough that G have affine fibers, as will be seen in Exposé XVI—local constancy of the reductive rank of G does not imply that G admits a maximal torus fpqc-locally. For example, let S be the spectrum of a discrete valuation ring, and let G_1, G_2 be smooth separated group schemes of finite type over S, such that the generic fiber of G_1 is an elliptic curve and its special fiber is $\mathbb{G}_m$, while the generic fiber of G_2 is $\mathbb{G}_m$ and its special fiber is $\mathbb{G}_a$. Put

$$G = G_1 \times_S G_2.$$

Both fibers of G have reductive rank 1, but G plainly does not admit a maximal torus fpqc-locally.

It is, on the other hand, very plausible that the following condition on a smooth separated group of finite type over a prescheme S is sufficient for the étale-local existence of a maximal torus: the reductive rank and the abelian rank of the fibers of G are locally constant functions.

(b) In the proof of 7.1, particularly in (a), we used XI 6.2 in cases where S was not assumed locally noetherian. In the instances at hand, however, the reduction to affine noetherian S is immediate.

Here is a variant of 7.1(b).

Proposition 7.8. *Let G be a smooth S-group prescheme of finite presentation with connected fibers, let H be a smooth S-group prescheme, and let*

$$i : H \longrightarrow G$$

be a monomorphism of S-groups, making H an S-subgroup of G. For every Cartan subgroup C of G,[11] *the transporter* $\mathrm{Transp}_G(C, H)$ *is represented by a closed subprescheme of G, smooth over S.*

The representability of the transporter by a closed subprescheme of G, of finite presentation, follows from XI 6.11, because C has connected fibers when G does, by 6.6(b). To prove smoothness, the standard procedure reduces first to affine noetherian S, then to artinian local S, and finally, by descent, to an algebraically closed residue field. In that case C has a maximal torus T, which is also a maximal torus of G. Unless the reductive and nilpotent ranks of the fiber H_0 equal those of G_0, the transporter is empty. Under this equality one sees, using the connectedness of C and the smoothness of the centralizer in H of a maximal torus of H, that

$$\mathrm{Transp}_G(C, H) = \mathrm{Transp}_G(T, H).$$

The right-hand side is smooth by XI 2.5, and therefore so is the left-hand side.

The same argument proves the more general assertion 7.9(b) below.

Proposition 7.9. *Let G and $i : H \to G$ be as in 7.8. Suppose moreover that, for every $s \in S$, H_s is connected and has the same reductive rank and the same nilpotent rank as G_s; equivalently, H_s contains a Cartan subgroup of G_s. Then:*

(a) $\mathrm{Norm}_G(H)$ *is represented by a closed subprescheme of G, of finite presentation over S, and the canonical monomorphism*

$$H \longrightarrow \mathrm{Norm}_G(H)$$

is an open immersion. Consequently i is an immersion and

$$H = \mathrm{Norm}_G(H)^0.$$

(b) For every Cartan subgroup C of G, $\mathrm{Transp}_G(C, H)$ *is a closed subprescheme of G, smooth over S, with surjective structural morphism. If C is the centralizer of a maximal torus T of G, then moreover*

$$\mathrm{Transp}_G(T, H) = \mathrm{Transp}_G(C, H).$$

[11] The proof only requires C to be a smooth subgroup of G whose geometric fibers are connected centralizers of subgroups of maximal tori.

(c) Let C be a group subprescheme of H. Then C is a Cartan subgroup of H if and only if it is a Cartan subgroup of G.

(d) If G admits a Cartan subgroup, respectively a maximal torus, locally for the étale topology or for the fpqc topology, then so does H.

Proof. (a) The representability of $\mathrm{Norm}_G(H)$ by a closed subprescheme of G, of finite presentation over S, is contained in XI 6.11. Since H is smooth, hence flat and locally of finite presentation over S, VI_B 2.6 reduces the assertion that

$$H \longrightarrow \mathrm{Norm}_G(H)$$

is an open immersion to the fibers. We therefore reduce to the case where $S = \operatorname{Spec} k$, with k algebraically closed.

The assertion then reduces to showing that the corresponding map on Lie algebras is an isomorphism,[12] or equivalently that

$$(\mathfrak{g}/\mathfrak{h})^H = 0,$$

where $\mathfrak{g}, \mathfrak{h}$ are the Lie algebras of G, H and the superscript denotes invariants; compare II 5.2.3(i). By hypothesis H contains a Cartan subgroup C of G, the centralizer of a maximal torus T of G. It therefore suffices to prove

$$(\mathfrak{g}/\mathfrak{h})^T = 0.$$

By complete reducibility of representations of T (I 4.7.3), this follows from

$$(\mathfrak{g}/\mathfrak{c})^T = 0, \qquad \mathfrak{c} = \mathrm{Lie}(C).$$

The latter equality is equivalent to

$$\mathfrak{g}^T = \mathfrak{c}.$$

It says that the centralizer C and the normalizer of T have the same Lie algebra, and follows from the fact that C is an open subgroup of that normalizer by XI 5.9. This proves (a).

(b) was proved in 7.8.

(c) Since H is a subprescheme of G, the assertion reduces immediately to an algebraically closed base field, where it follows directly from the hypothesis on H.

(d) follows from (b), (c), and the Hensel lemma XI 1.10. □

Corollary 7.10. *Let G, H be as in 7.9, and suppose that for every algebraically closed field k over S, every element of $G_k(k)$ normalizing H_k belongs to $H_k(k)$. Then H is a closed subprescheme of G and is its own normalizer.*

This follows immediately from 7.9(a). We shall apply it in particular to Borel subgroups, and more generally to parabolic subgroups, of G.

[12] The editors did not locate the printed reference in Exposé VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), Corollary II.5.6.

Corollary 7.11. *Let G, H be as in 7.9. Then H contains every group subprescheme Z of G which is central in G, flat, and of finite presentation over S.*

As usual, we reduce first to affine noetherian S, then to artinian S; the hypotheses of 7.1 then hold. By 7.9(c), we may suppose that H contains a Cartan subgroup C of G, and it remains to show $C \supset Z$. Since S is artinian, the quotient

$$G' = G/Z$$

is represented by a group prescheme of finite type over S; the canonical morphism $u : G \to G'$ is faithfully flat and has kernel Z by VI_A 3.2. The group G' is smooth over S. Theorem 7.1(f) therefore shows that $C = u^{-1}(C')$ for a Cartan subgroup C' of G', and hence that C contains Z.

Corollary 7.12. *Let G, G' be smooth S-group preschemes of finite presentation, and let $u : G \to G'$ be a faithfully flat group homomorphism, meaning that u_s is faithfully flat for every $s \in S$. Suppose that* $\ker u$ *is central and that G has connected fibers. Then*

$$H' \longmapsto H = u^{-1}(H')$$

defines a bijection between:

- *the group subpreschemes H' of G', smooth and of finite presentation over S, having at every $s \in S$ the same reductive rank and the same nilpotent rank as G'; and*
- *the group subpreschemes H of G, smooth and of finite presentation over S, having at every $s \in S$ the same reductive rank and the same nilpotent rank as G.*

Moreover, H has connected fibers if and only if H' does.

Let H be a group subprescheme of G with the stated properties. By 7.11 it contains $Z = \ker u$, so faithfully flat descent gives

$$H = u^{-1}(H')$$

for a uniquely determined group subprescheme H' of G'. Using 6.6(e), one checks immediately that H' has the asserted properties. If H has connected fibers, so does $H' = H/Z$.

Conversely, start with $H' \subset G'$ having the stated properties, and put $H = u^{-1}(H')$. In view of 6.6(e), it remains to show that H is smooth over S, and that it has connected fibers when H' does. Since u is flat and H' is flat over S, H is already flat over S. It therefore suffices to check smoothness, respectively connectedness, on geometric fibers, reducing to an algebraically closed base field.

Then H' contains a Cartan subgroup C' of G', and H contains its inverse image C. By 6.6(e), C is a Cartan subgroup of G, hence is smooth and connected. The conclusion now follows from the next lemma.

Lemma 7.13. *Let G, G' be group preschemes flat and of finite presentation over S, let $u : G \to G'$ be a faithfully flat group homomorphism, and let C' be a group subprescheme of G', of finite presentation over S, such that $C = u^{-1}(C')$ is smooth over S, respectively has connected fibers. For every group subprescheme H' of G', of finite presentation over S, containing C', and smooth over S, respectively with connected fibers, its inverse image $H = u^{-1}(H')$ is smooth over S, respectively has connected fibers.*

As noted above, this reduces at once to the case where S is the spectrum of a field. The group H is then a principal C-bundle over H/C, by VI_B 9, while

$$H/C \simeq H'/C'$$

by Exposé IV. Since H' is smooth, respectively connected, so is H'/C', hence so is H/C. The same property holds for C by hypothesis and therefore for the bundle H. This proves the lemma and Corollary 7.12.

0.8 Semisimple elements; unions and intersections of maximal tori in group schemes not necessarily affine

Throughout this section, G denotes an S-group prescheme that is smooth and of finite presentation over S, with connected fibers.

Suppose first that S is the spectrum of an algebraically closed field k. When G is affine, the notion of a semisimple element of $G(k)$ was defined in *BIBLE* 4, No. 4; one checks at once that this notion is invariant under an algebraically closed extension k'/k of the base field. Moreover, by *BIBLE* 6, Theorem 5(c), an element $g \in G(k)$ is semisimple if and only if it is contained in a maximal torus of G.

When G is no longer assumed affine, Chevalley's theorem gives a canonical expression of G as an extension of an abelian variety by a smooth connected affine algebraic group:

$$(*) \qquad e \longrightarrow V \longrightarrow G \longrightarrow A \longrightarrow e.$$

We shall say that $g \in G(k)$ is *semisimple* if it is a semisimple element of $V(k)$. Since the maximal tori of V are precisely the maximal tori of G, this is equivalent to saying that g belongs to a maximal torus of G. This notion is again invariant under algebraically closed extensions of the base field.

Suppose next that $S = \operatorname{Spec} k$ for an arbitrary field k, and let $g \in G$. After choosing an algebraically closed extension K of $\kappa(g)$, the point g is the image of a geometric point g' of G with values in an algebraically closed extension K of k. We call g semisimple when g' is semisimple. What precedes shows that this is independent of the choice of g'. If k'/k is any extension, the set of semisimple elements of $G_{k'}$ is the inverse image of the set G^{ss} of semisimple elements of G.

Finally, for arbitrary S, a point $g \in G$ is called semisimple when it is semisimple in its fiber G_s. Thus, for every morphism $S' \to S$, the subset $G^{\mathrm{ss}}_{S'}$ is the inverse image of G^{ss}. Suppose that the functor $\mathcal{T}$ of maximal tori defined in 1.10 is representable by a prescheme of finite presentation over S. This is so, for example, if G admits a maximal torus locally for the fpqc topology, by 7.1(d), at least when G is quasi-projective over S. Let T be the canonical maximal torus of $G_{\mathcal{T}}$, the "universal maximal torus of G", and consider the morphism

$$u : T \longrightarrow G$$

induced by the projection $G_{\mathcal{T}} \to G$. It follows immediately from the definition that G^{ss} is the image of this morphism. We obtain:

Proposition 8.1. *The set G^{ss} of semisimple elements of G is locally constructible. It is therefore constructible when S, and hence G, is quasi-compact and quasi-separated.*

As usual, we reduce to affine noetherian S. By the standard noetherian criterion for constructibility (EGA 0_{III}, 9.2.3), we may further suppose that S is integral and need

only show that there is a nonempty open $U \subset S$ such that $G^{\mathrm{ss}}|_U$ is constructible. After shrinking U, and replacing it if necessary by a finite covering, we may suppose that G is separated over S and contains a maximal torus. By 7.1(d), $\mathcal{T}$ is then represented by a prescheme of finite presentation over S; the same is true of its universal torus, whose image in G is therefore constructible.

Suppose again that k is a field, and let H be the subgroup of G generated by the preceding morphism (VI_B 1.2). It is a smooth group subscheme of G, connected because the universal torus is connected, and its formation commutes with every extension of the base field, as does the formation of $\mathcal{T}$.

When k is algebraically closed, H is also the algebraic subgroup of G generated by the maximal tori of G, or, equivalently, by all the tori of G. It is likewise the smallest algebraic subgroup of G containing all semisimple elements of $G(k)$. These characterizations remain valid whenever k is infinite, because, as will be proved in Exposé XIV, the set of k-rational points of $\mathcal{T}$ is dense in $\mathcal{T}$.

The subgroup H is invariant in G. Indeed, it is enough to check this after passing to an algebraically closed field; since H and G are then smooth, it suffices to check stability under the inner automorphisms $\mathrm{int}(g)$, $g \in G(k)$, which is immediate. One can even show that H is characteristic in G, that is, stable under $\mathrm{Aut}_{k\text{-gr}}(G)$.

It follows from 6.6(d) that G/H has reductive rank zero. More precisely, if K is an invariant algebraic subgroup of G, then, over an algebraically closed field, the maximal tori of G/K are the direct images of the maximal tori of G. Hence G/K has reductive rank zero if and only if K contains all maximal tori of G, or, over an arbitrary field, if and only if $K \supset H$. Thus H is the smallest invariant algebraic subgroup of G for which G/H has reductive rank zero. It is also the smallest algebraic subgroup of G having the same reductive rank as G.

Finally, H is affine. This may again be checked after passing to an algebraic closure. In the exact sequence $(*)$, every maximal torus of G is contained in V; consequently $H \subset V$, and V is affine. We have proved:

Proposition 8.2. *Let G be a smooth connected algebraic group over a field k, and let $\bar{k}$ be an algebraic closure of k. There exists an algebraic subgroup H of G such that $H_{\bar{k}}$ is the algebraic subgroup of $G_{\bar{k}}$ generated by the maximal tori, or equivalently by all the tori, of $G_{\bar{k}}$. The group H is also characterized as the smallest algebraic subgroup of G having the same reductive rank as G, or as the smallest invariant algebraic subgroup of G such that G/H has reductive rank zero. It is a smooth, connected, invariant, affine subgroup of G, and its formation commutes with every extension of the base field.*

The characterization in terms of G/H gives the following consequences.

Corollary 8.3. *Let G be a smooth connected algebraic group over an algebraically closed field k. Then G has reductive rank zero —equivalently, in the notation of 8.2, $H = (e)$— if and only if G is an extension of an abelian variety by a smooth connected unipotent algebraic group, that is, a successive extension of groups isomorphic to the additive group $\mathbb{G}_a$.*

Indeed, Chevalley's exact sequence $(*)$ reduces the assertion to the fact that a smooth connected affine group V has reductive rank zero if and only if it is unipotent; this is contained in *BIBLE* 6.4, Theorem 4, Corollary 3. Thus, in the notation of 8.2, H is the smallest invariant algebraic subgroup of G such that $(G/H)_{\bar{k}}$ is an extension of an abelian variety by a smooth connected affine unipotent algebraic group.

Corollary 8.4. *In the notation of 8.2, $G = H$, or equivalently $G_{\bar{k}}$ is generated by its maximal tori, if and only if G is affine and every homomorphism $G_{\bar{k}} \to \mathbb{G}_a$ is trivial.*

Remarks 8.5.

(a) Let V be the largest smooth connected affine algebraic subgroup of $G_{\bar{k}}$, so that $G_{\bar{k}}$ is an extension of an abelian variety by V. It is well known, by Rosenlicht,[13] that when k is not perfect, V is not in general defined over k: there need not exist an algebraic subgroup V_0 of G such that $(V_0)_{\bar{k}} = V$. If, however, V is generated by its maximal tori—equivalently, if V has no quotient isomorphic to $\mathbb{G}_{a,\bar{k}}$—then such a group does exist, namely the group H of 8.2. Thus, in this rationality question as in many others (compare Exposé XIV, No. 6), all the difficulties arise from unipotent groups, hence from the additive group, whereas the presence of sufficiently many groups of multiplicative type makes descent work.

(b) In the notation of 8.2, let H' be the inverse image in G of the commutator subgroup of G/H. Then H' is the smallest invariant algebraic subgroup of G such that $(G/H')_{\bar{k}}$ is a commutative extension of an abelian variety by a smooth connected affine unipotent algebraic group. It is again a smooth connected affine subgroup of G.

Assume that the characteristic p is positive, and let H'' be the inverse image in G of $p^n(G/H')$ for n sufficiently large. Then H''/H' is an abelian variety, and H'' is the smallest invariant algebraic subgroup of G such that G/H'' is a commutative smooth connected affine unipotent algebraic group. It follows that $V = H_{\bar{k}}$, in the notation of (a), if and only if $H'' = G$, or equivalently if and only if every homomorphism $G_{\bar{k}} \to \mathbb{G}_a$ is trivial.

We now generalize to smooth groups with connected fibers the notion of reductive center developed in Section 4, following the idea of 4.10. Let $\mathcal{Z}$ be the subfunctor of G defined by

$$\mathcal{Z}(S') = \{\, g' \in G_{S'}(S') : \text{ for every } S'' \to S' \text{ and every maximal torus } T'' \subset G_{S''},$$
$$\text{the pullback } g'' \text{ of } g' \text{ is a section of } T'' \text{ over } S'' \,\}.$$

Let $\mathcal{T}(S')$ be the set of maximal tori of $G_{S'}$, and let $\widetilde{\mathcal{T}}(S')$ be the set of pairs (T', g') where T' is a maximal torus of $G_{S'}$ and g' is a section of T' over S'. Then $\widetilde{\mathcal{T}}$ is a subfunctor of

$$\mathcal{T} \times_S G = \mathcal{T}_G,$$

and in this notation one may also write

$$\mathcal{Z} = \prod_{\mathcal{T}_G/G} \widetilde{\mathcal{T}}/\mathcal{T}_G.$$

Using XI 6.8 and 7.1(d), which under suitable hypotheses represents $\mathcal{T}$ by a smooth S-prescheme, one could deduce representability of $\mathcal{Z}$ under corresponding hypotheses. We shall instead obtain it more directly below.

[13] The editors ask that references be supplied here.

Definition 8.6. *Let G be an S-group prescheme, smooth and of finite presentation with connected fibers. We say that G admits a* reductive center *if the preceding functor $\mathcal{Z}$, which is evidently a subgroup of G, is representable by a group of multiplicative type. In that case $\mathcal{Z}$ is called the reductive center of G.*

If Z is a reductive center of G, then $Z_{S'}$ is a reductive center of $G_{S'}$ for every base change $S' \to S$. The existence of a reductive center is consequently local for the fpqc topology. The terminology is justified because $\mathcal{Z}$ is invariant under

$$\mathrm{Aut}_S(G),$$

and hence is an invariant subgroup of G; by IX 5.5, it is central.

Lemma 4.5 is replaced here by the following statement.

Lemma 8.7. *Let $u : H \to G$ be a central homomorphism, where H is of multiplicative type and of finite type over S. Suppose that for every algebraically closed field k over S, the subgroup $u_k(H_k)$ is contained in the largest smooth connected affine subgroup of G_k. Then u factors through every maximal torus T of G. In particular, u factors through the subfunctor $\mathcal{Z}$ defined above.*

An easy variant of IX 5.1 bis, with equality replaced by inclusion, reduces the assertion to the case in which S is the spectrum of an algebraically closed field: first reduce to affine noetherian S, then to artinian S, and use IX 3.6. Since T is contained in the largest smooth connected affine subgroup V of G, we are reduced to $G = V$, hence to the affine case proved in 4.5.

Proposition 8.8. *Let G be an S-group prescheme, smooth and of finite presentation with connected fibers.*

(a) If $S = \mathrm{Spec}\, k$, then G admits a reductive center Z. When G is an extension of an abelian variety by a smooth connected affine algebraic group V —for example, when k is algebraically closed—then Z is also the reductive center of V, and it is the largest central subgroup of multiplicative type of V.

(b) Let Z be a group subprescheme of multiplicative type in G. Then Z is a reductive center of G if and only if Z_s is a reductive center of G_s for every $s \in S$. It is then the largest subgroup of multiplicative type K in G such that K_s is contained in the reductive center of G_s for every $s \in S$. More generally, let $u : H \to G$ be a homomorphism with H of multiplicative type and of finite type over S, and suppose that for every algebraically closed field k over S, u_k factors through the largest smooth connected affine subgroup of G_k. Then u factors through Z, and in particular is central.

(c) If G admits a maximal torus locally for the fpqc topology, then G admits a reductive center.

(d) If T is a maximal torus of G, then

$$T \cap \mathrm{Centr}(G) = \ker(T \longrightarrow \mathrm{GL}(\mathfrak{g})),$$

and this group is a reductive center of G.

(e) Let Z be a reductive center of G, and suppose that $G' = G/Z$ is representable, for example when S is artinian. Then the unit subgroup is a reductive center of G', and

$$T' \longmapsto T = u^{-1}(T')$$

is a bijection from the set of maximal tori of G' to the set of maximal tori of G.

Proof. (a) Let $S = \operatorname{Spec} k$. To prove existence, we may pass to an algebraic closure of k, and thus suppose that G is an extension of an abelian variety by a smooth connected affine algebraic group V. For every S' over k, the maximal tori of $G_{S'}$ are those of $V_{S'}$: this follows from IX 5.2 and from the fact that a homomorphism from a torus to an abelian variety over an algebraically closed field is trivial. Consequently the functor $\mathcal{Z}$ defined using G is the same as the one defined using V, and we reduce to affine G.

The functor $\mathcal{T}$ is representable and is necessarily "essentially free" over k (VIII 6.1); hence $\mathcal{T}_G$ is essentially free over G. Since the universal torus is a closed subscheme of $\mathcal{T}_G = G_{\mathcal{T}}$ (for example by VIII 5.7), VIII 6.4 shows that

$$\mathcal{Z} = \prod_{\mathcal{T}_G/G} \widetilde{\mathcal{T}}/\mathcal{T}_G$$

is represented by a closed subscheme of G. It is therefore a group subscheme. It is of multiplicative type: after passing to an algebraic closure, G admits a maximal torus T, and by definition $Z \subset T$; a subgroup of a group of multiplicative type is of multiplicative type (IX 8). Thus Z is a reductive center. Lemma 8.7 shows that it is the largest central subgroup of multiplicative type of V.

(b) Only sufficiency requires proof. Suppose that Z is of multiplicative type and that Z_s is the reductive center of G_s for every $s \in S$. Lemma 8.7 first shows that Z is contained in every maximal torus of G, and this remains true after every base change. It remains to show that if g is a section of G over S whose pullback to every S' over S is a section of every maximal torus of $G_{S'}$, then g is a section of Z.

As usual, since the fpqc sheaf

$$\mathcal{Z}_0 = \prod_{\mathcal{T}_G/G} \widetilde{\mathcal{T}}/\mathcal{T}_G$$

commutes with filtered direct limits of rings, we reduce successively to affine, noetherian, and artinian S. By 7.1(d), $\mathcal{T}$ is then represented by a smooth prescheme over S. Arguing as in (a), $\mathcal{Z}_0$ is represented by a closed subscheme Z_0 of G. We already know $Z \subset Z_0$; by hypothesis $Z_k = (Z_0)_k$ over the residue field k, and Z is flat over S. It follows that $Z = Z_0$, proving that Z is the reductive center. The remaining assertions of (b) follow from 8.7.

(c) follows immediately from (d).

(d) Here $\mathfrak{g}$ denotes the Lie algebra of G, and $T \to \mathrm{GL}(\mathfrak{g})$ is induced by the adjoint representation. Trivially,

$$T \cap \operatorname{Centr}(G) \subset \ker\bigl(T \longrightarrow \mathrm{GL}(\mathfrak{g})\bigr).$$

The proof of the reverse inclusion is the same as in 4.7(d). Put

$$Z = \ker\bigl(T \longrightarrow \mathrm{GL}(\mathfrak{g})\bigr).$$

Then Z is of multiplicative type (IX 6.8). By (b), we may check that it is a reductive center over a field. Let Z_0 be the reductive center supplied by (a). Plainly

$$Z_0 \subset T \cap \operatorname{Centr}(G) = Z.$$

Conversely, Z is a central subgroup of multiplicative type contained in the smooth connected affine group T, so (a) gives $Z \subset Z_0$. Thus $Z = Z_0$.

(e) Under the hypotheses of 7.1(f), put $Z = \ker u$, and suppose that for every algebraically closed field k over S, Z_k is contained in a maximal torus of G_k. One checks readily that

$$T' \longmapsto u^{-1}(T')$$

is a bijection from the maximal tori of G' to those of G. Applying this observation to the situation of (e) proves the assertion. □

Remarks 8.9.

(a) The proof of 8.8 is independent of the results of Section 4; in particular, the proof of 8.8(a) does not use 4.4, whose proof is somewhat laborious.

(b) The subfunctor $\mathcal{Z}$ considered in 8.6 is always central, whether or not it is representable. It may therefore be tempting to call it the reductive center of G in all cases.

(c) One may likewise generalize 4.9. If G is a smooth connected algebraic group over an algebraically closed field, then G is an extension of an abelian variety by a smooth connected unipotent algebraic group—equivalently, G has reductive rank zero, by 8.3—if and only if the reductive center of G is the unit group and the Lie algebra of G is nilpotent.

0.9 Complement: actions of group schemes and fixed points

The purpose of this section, added in January 2008, is to study fixed points under the action of a group scheme.

0.9.1 Representability of the fixed-point functor

Let S be a scheme, let G be an S-group scheme, and suppose that G acts on an S-scheme X. Define the subfunctor X^G of fixed points of X under G by requiring, for every S-scheme T, that $X^G(T)$ be the subset of $X(T)$ consisting of the points fixed under G (VIII 6(e)).

Recall the notion of an S-pure scheme introduced in Raynaud–Gruson, §3.3, and in Raynaud. Let G be a flat S-group scheme of finite presentation. Since the fibers of G have no embedded components, purity takes the following form: G is *S-pure* if, for every strictly henselian local S-scheme T with closed point t, every generic point $\widetilde{x}$ of a fiber of $G \times_S T$ specializes to a point of G_t; equivalently, the closure of $\widetilde{x}$ in $G \times_S T$ meets G_t.

Examples.

(1) If G is quasi-finite and separated over S, then G is S-pure if and only if it is finite over S.

(2) If $S = \operatorname{Spec} R$ and $G = \operatorname{Spec} A$, then G is S-pure if and only if A is a projective R-module (Raynaud–Gruson, 3.3.5). In particular, a diagonalizable G is S-pure.

(3) Every group scheme of multiplicative type is S-pure, since purity is local for the étale topology on S.

(4) The scheme G is S-pure if it is proper over S, if its fibers are irreducible, or if S is semilocal artinian.

(5) In particular, every reductive S-group scheme is S-pure.

Proposition 9.2. *Let G be an S-group scheme of finite presentation, flat and S-pure over S, acting on an S-scheme X of finite presentation. Then the fixed-point functor X^G is represented by a closed subscheme of X, of finite presentation over S.*

Indeed, let $a \in X(S)$, and let H be the group subscheme of G fixing a. The point a is fixed under G if and only if $H = G$. The coincidence functor of H and G—whose value on an S-scheme T is a singleton when

$$H \times_S T \longrightarrow G \times_S T$$

is bijective and is empty otherwise—is represented by a closed subscheme of S, defined by a finite-type sheaf of ideals (Raynaud–Gruson, 4.1.1). Apply this result with $S = X$ and with a the universal point of X, that is, the identity element of

$$X(X) = \mathrm{Hom}_S(X, X).$$

0.9.3 The infinitesimal obstruction

Suppose that G is a flat S-group scheme of finite presentation acting on the left on a smooth S-prescheme X. We shall study the formal smoothness of X^G.

Let S_0 be a closed subscheme of S, defined by a quasi-coherent sheaf of ideals $\mathcal{I}$ with $\mathcal{I}^2 = 0$, and write

$$j : S_0 \longrightarrow S, \qquad G_0 = G \times_S S_0, \qquad X_0 = X \times_S S_0.$$

Let $\epsilon_0 \in X^G(S_0)$ be a point that lifts to $X(S)$. We shall study the obstruction to lifting ϵ_0 to $X^G(S)$.

Since X is smooth, the set of lifts of ϵ_0 to $X(S)$ is a trivial principal homogeneous space under the abelian group

$$\mathrm{Hom}_{\mathcal{O}_{S_0}}(\epsilon_0^*\Omega^1_{X_0/S_0}, \mathcal{I})$$

(III 0.2, 0.3). Put

$$L_0 = \mathrm{Hom}_{\mathcal{O}_{S_0}}(\epsilon_0^*\Omega^1_{X_0/S_0}, \mathcal{I}).$$

This is an $\mathcal{O}_{S_0}$-module. Since ϵ_0 is fixed under G, it is naturally a G_0-$\mathcal{O}_{S_0}$-module. Let

$$\rho_0 : G_0 \longrightarrow \mathrm{Aut}_{S_0}(L_0), \qquad \rho : G \longrightarrow \mathrm{Aut}_S(j_*L_0)$$

be the associated representations.

Lemma 9.4. *There is a canonically defined class*

$$c(\epsilon_0) \in H^1(G, j_*L_0) \simeq H^1(G_0, L_0)$$

such that ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

The displayed adjunction is the one in III 1.1.2. We work on the small fppf site of S. For every S-scheme T, flat and of finite presentation, write

$$G_T = G \times_S T, \qquad X_T = X \times_S T.$$

Consider the sheaf $\mathcal{A}$ on this site defined by

$$\mathcal{A}(T) = \{ \text{ lifts of } \epsilon_0 \times_S T \text{ to } X(T) \}.$$

Since the formation of L_0 commutes with flat base change, $\mathcal{A}(T)$ is a trivial principal homogeneous space under

$$H^0(T, j_* L_0).$$

Moreover, because ϵ_0 is fixed, every $g \in G(T)$ acts by affine automorphisms on $\mathcal{A}(T)$, compatibly with the action of G on $j_* L_0$:

$$g(a_T + v) = g(a_T) + \rho(g)(v), \qquad v \in H^0(T, j_* L_0).$$

Since G is flat and of finite type, we may take $T = G$ and g equal to the universal point id_G. We thereby obtain an action of the fppf sheaf G on $\mathcal{A}$.

Let a be a lift of ϵ_0 to $X(S)$, and let $g^\sharp$ denote the universal point of G. Define

$$v^\sharp \in H^0(G, j_* L_0)$$

by the relation

$$\rho(g^\sharp)(v^\sharp) = g^\sharp \cdot a_G - a_G.$$

For every S-prescheme Y, set

$$z(Y) : G(Y) \longrightarrow H^0(Y, j_* L_0), \qquad g \longmapsto v^\sharp(g).$$

This defines a 1-cocycle $z \in Z^1(G, j_* L_0)$ (Exposé I, §5). Its class $c(\epsilon_0)$ does not depend on the choice of a. In particular, if ϵ_0 lifts to $X^G(S)$, then $c(\epsilon_0) = 0$.

Conversely, if $c(\epsilon_0) = 0$, there is a section

$$w \in H^0(S, j_* L_0)$$

such that, for every S-prescheme Y and every $g \in G(Y)$,

$$z(Y)(g) = g \cdot w_Y - w_Y.$$

Apply this with $Y = G$ and $g = g^\sharp$. It follows that $a - w \in X^G(S)$. Thus ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

Remark 9.5. When S is affine and G is affine, flat, and of finite type over S, this obstruction can be refined to a class

$$\widetilde{c}(\epsilon_0) \in H^1_{G_0}(X_0, L_0),$$

where the right-hand side is the G_0-equivariant cohomology group defined by Wewers (Appendix C). In this case it is unnecessary to assume that ϵ_0 lifts to $X(S)$.

0.9.6 Smoothness of fixed points

Theorem 9.7. *Let G be a flat S-group scheme of finite type over a noetherian base S, acting on a smooth S-scheme X. Suppose that G is S-pure and that, for every geometric point s of S with algebraically closed residue field,*

$$H^1\big(G_s, (\Omega^1_{X_s})^*\big) = 0.$$

Then the fixed-point functor X^G is represented by a closed subscheme of X, smooth over S.

By 9.2, X^G is represented by a closed subscheme of X, of finite presentation over S. We may suppose $S = \operatorname{Spec} A$ with A local. By the smoothness criterion of SGA 1, III 3.1(iii bis), it is enough to verify formal smoothness for

$$S' = \operatorname{Spec} A'$$

and its closed subscheme

$$S'' = \operatorname{Spec}(A'/I'),$$

where A' is an artinian local ring and I' is an ideal of square zero. By dévissage, we may assume that I' is annihilated by the maximal ideal of A'. If k is the residue field of A', then I' is therefore a k-vector space.

Let $\epsilon'' \in X^G(S'')$ be a point that we wish to lift to $X^G(S')$, and put

$$\begin{aligned} X' &= X \times_S S', \qquad & G' &= G \times_S S', \\ X'' &= X \times_S S'', \qquad & G'' &= G \times_S S''. \end{aligned}$$

Since S' is affine and X' is smooth over S', ϵ'' lifts to $X(S')$. By Lemma 9.4, the obstruction to lifting ϵ'' to $X^G(S')$ is a class

$$c(\epsilon'') \in H^1(G'', L''),$$

where

$$L'' = \operatorname{Hom}_{\mathcal{O}_{S''}}\big((\epsilon'')^* \Omega^1_{X''/S''}, I'\big).$$

Because L'' is a k-vector space, there is a canonical isomorphism

$$H^1(G'', L'') \simeq H^1(G'' \times_{A'/I'} k, L'').$$

Let $\bar{k}$ be an algebraic closure of k. By IX 3.1 in the affine case, and by Lemma 9.11 in general, there is an isomorphism

$$H^1(G'' \times_{A'/I'} k, L'') \otimes_k \bar{k} \xrightarrow{\sim} H^1(G'' \times_{A'/I'} \bar{k}, L'' \otimes_k \bar{k}).$$

The representation $L'' \otimes_k \bar{k}$ is a direct sum of copies of

$$(\Omega^1_{X_{\bar{k}}})^*.$$

The group on the right therefore vanishes by hypothesis. It follows that $H^1(G'', L'') = 0$, so the obstruction vanishes and X^G is formally smooth.

Corollary 9.8. *Let G be a flat S-group scheme of finite presentation acting on a smooth S-scheme X of finite presentation. Suppose that G admits a composition series whose factors are of one of the following types:*

(1) an abelian S-scheme, that is, a smooth S-group scheme whose fibers are abelian varieties;

(2) an S-group scheme of multiplicative type;

(3) a finite étale S-group scheme of degree invertible on S;

(4) a reductive S-group scheme, when S is a $\mathbb{Q}$-scheme.

Then the fixed-point functor X^G is represented by a closed subscheme of X, of finite presentation and smooth over S.

Indeed, suppose that

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1$$

is an exact sequence of S-groups satisfying the hypotheses. The group G'' acts on $X^{G'}$, and X^G is the fixed-point functor of $X^{G'}$ under G''. We are therefore reduced to the case of a group of one of the four types listed above. By passage to the limit, we may suppose S noetherian. We verify the cohomological criterion of Theorem 9.7 for a geometric fiber G_s and a finite-dimensional linear representation V of G_s.

For an abelian scheme, every morphism from G_s^n to V is constant; consequently

$$H^i(G_s, V) = 0 \qquad (i > 0).$$

For a group scheme of multiplicative type, G_s is diagonalizable, and I 5.3.3 again gives

$$H^i(G_s, V) = 0 \qquad (i > 0).$$

In the remaining two cases the assertion follows from semisimplicity of the linear representations of G_s. If G_s is finite étale of degree invertible in $\kappa(s)$, it is a constant finite group of invertible order, and every linear representation of G_s is semisimple. In the reductive case, $\kappa(s)$ is algebraically closed of characteristic zero, and every linear representation of G_s is semisimple (Tauvel–Yu, Theorem 27.3.3).

Remark 9.9. This applies in particular to an action of G/S on a smooth S-group scheme H. If G is of multiplicative type and acts by inner automorphisms on H, with H smooth and affine over S, one recovers XI 5.3: the centralizer of G in H is smooth.

Remark 9.10. When $S = \operatorname{Spec} k$ and G is a linearly reductive algebraic group over an algebraically closed field k, the result is due independently to Fogarty (Theorem 5.4) and Iversen (Proposition 1.3).

The proof of Theorem 9.7 used the following lemma, which is well known for affine group schemes (IX 3.1).

Lemma 9.11. *Let G be a group scheme over an affine scheme $S = \operatorname{Spec} A$, and let $f : S' = \operatorname{Spec} A' \to S$ be flat. Put $G' = G \times_S S'$. Then, for every quasi-coherent G-$\mathcal{O}_S$-module M, there are canonical isomorphisms*

$$H^i(G, M) \otimes_A A' \xrightarrow{\sim} H^i(G', M \otimes_A A') \qquad (i \geq 0).$$

For $n \geq 1$, the degree-n Hochschild cochains are determined by their value at

$$(\mathrm{id}, \ldots, \mathrm{id}) \in G(G) \times \cdots \times G(G).$$

Thus there is an isomorphism of A-modules

$$C^n(G, M) \simeq \Gamma\big(G^n, M \otimes_A \mathcal{O}_{G^n}\big).$$

The group $H^n(G, M)$ is the n-th cohomology group of the complex

$$M \longrightarrow \Gamma(G, M \otimes_A \mathcal{O}_G) \longrightarrow \Gamma(G^2, M \otimes_A \mathcal{O}_{G^2}) \longrightarrow \Gamma(G^3, M \otimes_A \mathcal{O}_{G^3}) \longrightarrow \cdots .$$

Since A' is flat over A, EGA IV_1, 1.7.21 gives

$$\Gamma\big(G^n, M \otimes_A \mathcal{O}_{G^n}\big) \otimes_A A' \xrightarrow{\sim} \Gamma\big((G')^n, (M \otimes_A A') \otimes_{A'} \mathcal{O}_{(G')^n}\big).$$

Taking cohomology proves the assertion.

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Exposé XIII
Regular Elements of Algebraic Groups and Lie Algebras

by A. Grothendieck

0.1 An auxiliary lemma on varieties with operators

Let S be a prescheme, let G be an S-group prescheme acting on the left on an S-prescheme V, let W be a closed S-subprescheme of V, and let N be its stabilizer in G. Thus N is the subgroup of G whose points with values in an S' over S are the elements $g \in G(S')$ such that

$$g \cdot W_{S'} = W_{S'}.$$

We endow (Sch/S) with the faithfully flat quasi-compact topology and identify G, V, and W with the corresponding sheaves. We shall therefore reason in the category of sheaves on (Sch/S); throughout this section, "locally" refers to the topology just specified on (Sch/S).

Since N is a sheaf, consider the quotient sheaf G/N. It is immediately seen to be isomorphic to the following functor: to every S' over S, it assigns the set of subsheaves W' of $V_{S'}$ that are locally conjugate to $W_{S'}$ under G. Let X be the subsheaf of $G/N \times_S V$ whose value on an S' over S is the set of pairs (W', v), where W' is as above and v is a section of W' over S' (hence also a section of $V_{S'}$ over S'). Let Z be the inverse image of X in $G \times_S V$. We then have the cartesian diagram

(x)
$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle i} \\ G \times_S V & \longrightarrow & G/N \times_S V, \end{array}$$

where i is the canonical immersion, while the lower horizontal arrow is induced by the canonical morphism $G \to G/N$.

This morphism sends $g \in G(S')$ to the subsheaf $g \cdot W_{S'}$ of $V_{S'}$. Consequently, $Z(S')$ is the set of pairs $(g, v) \in G(S') \times V(S')$ such that $v \in (g \cdot W_{S'})(S')$. It follows that Z is isomorphic to the sheaf $G \times_S W$, via the isomorphism

$$G \times_S W \xrightarrow{\sim} Z, \qquad (g, w) \longmapsto (g, g \cdot w).$$

Thus the preceding cartesian diagram yields the cartesian diagram

(xx)
$$\begin{array}{ccc} G \times_S W & \xrightarrow{q} & X \\ {\scriptstyle\lambda}\downarrow & & \downarrow{\scriptstyle i} \\ G \times_S V & \longrightarrow & G/N \times_S V \end{array}$$

where

$$\lambda(g, w) = (g, g \cdot w), \qquad q(g, w) = (\bar{g}, g \cdot w),$$

where $\bar{g}$ denotes the image of g under the canonical map $G(S') \to (G/N)(S')$.

Diagram (x) also shows that $Z \to X$ makes Z a principal bundle over X, with N acting on the right by

$$(g, v) \cdot n = (gn, v).$$

Accordingly, in diagram (xx), the morphism $q : G \times_S W \to X$ makes $G \times_S W$ a principal bundle over X, with right N-action

$$(g, w) \cdot n = (gn, n^{-1} \cdot w).$$

The principal morphisms just introduced are summarized by

(D)
$$\begin{array}{ccccc} & & G \times_S W & & \\ & & \downarrow{\scriptstyle q} & \searrow^{\varphi} & \\ G/N & \xleftarrow[p]{} & X & \xrightarrow{\psi} & V \\ & {\scriptstyle \mathrm{pr}_1}\swarrow & \downarrow{\scriptstyle i} & \searrow{\scriptstyle \mathrm{pr}_2} & \\ & & G/N \times_S V & & \end{array}$$

where

$$\psi = \mathrm{pr}_2 \circ i, \qquad \varphi = \psi \circ q, \qquad \varphi(g, w) = g \cdot w.$$

If v is a section of V over S, let X_v be its inverse image in X under ψ. For every S' over S, $X_v(S')$ is the set of subsheaves W' of $V_{S'}$ that are locally conjugate to $W_{S'}$ under G and contain the section $v_{S'}$ of $V_{S'}$. The inverse image of v in $G \times_S W$ under φ is isomorphic to the subsheaf M_v of G defined by

$$M_v(S') = \{g \in G(S') \mid v_{S'} \in (g \cdot W_{S'})(S')\} = \{g \in G(S') \mid g^{-1} v_{S'} \in W(S')\}.$$

If v is a section of W, and not merely of V, then M_v evidently contains N.

None of these explicit descriptions used the representability of G, V, or W (nor even the fact that the site in question is defined in terms of preschemes). Suppose now, however, that N is representable, faithfully flat, and quasi-compact over S, and that G/N is representable. When S is the spectrum of a field and G is of finite type over that field, this hypothesis is necessarily satisfied (VIII 6 and $\mathrm{VI_B}$.11.18). Using the cartesian diagram (xx), faithfully flat quasi-compact descent, and the fact that $Z \to G \times_S V$ is a closed immersion, one sees that X is representable: it is obtained by descending the closed subprescheme Z of $G \times_S V$ along the faithfully flat quasi-compact morphism

$$G \times_S V \longrightarrow G/N \times_S V.$$

Thus diagram (D) is a diagram of morphisms of preschemes over S.

We henceforth assume that S is the spectrum of a field k, and that G, V, and W are of finite type over k. Let $\mathfrak{n}$ be the Lie algebra of N. Then $\dim N \leq \operatorname{rank}_k \mathfrak{n}$, with equality if and only if N is smooth over k (Exposé VI[1]). Let $a \in W(k)$, and consider the subscheme M_a of G defined above; it contains N and is isomorphic to $\varphi^{-1}(a)$. We denote by $\mathfrak{m}_a$ its Zariski tangent space at the identity element e of N. Thus

$$\mathfrak{n} \subset \mathfrak{m}_a, \qquad \dim N \leq \operatorname{rank}_k \mathfrak{n} \leq \operatorname{rank}_k \mathfrak{m}_a. \tag{1}$$

Lemma 1.1. — *With the preceding notation:*

a) Consider the following conditions:

(i) $\mathfrak{n} = \mathfrak{m}_a$, and N is smooth over k.

(i bis) $\dim N = \operatorname{rank}_k \mathfrak{m}_a$.

(ii) The morphism $\psi : X \to V$ is unramified at $(\bar{e}, a)$.

(iii) M_a and N coincide in a neighborhood of e.

Then

$$\text{(i)} \iff \text{(i bis)} \implies \text{(ii)} \iff \text{(iii)}.$$

b) Suppose that $\varphi : G \times_S W \to V$ is smooth at (e, a). Then M_a is smooth over k at e, and ψ is smooth at $(\bar{e}, a)$.

Proof. *a)* The equivalence of (i) and (i bis) follows immediately from (1) and from the fact noted above that N is smooth over k if and only if $\dim N = \operatorname{rank}_k \mathfrak{n}$. Consider the inclusion morphism $N \to M_a$. It is well known[2] that if N is smooth over k at e and the tangent map at e is surjective, then $N \to M_a$ is smooth at e; since it is an immersion, it is therefore an isomorphism at e. This proves that (i) implies (iii).

To prove the equivalence of (ii) and (iii), let $X_a = \psi^{-1}(a)$, as above. From the isomorphisms

$$M_a \simeq \varphi^{-1}(a) \simeq q^{-1}(X_a)$$

we obtain a morphism $p_a : M_a \to X_a$ making M_a a principal homogeneous bundle with group N_{X_a}. Consider the diagram

$$\begin{array}{ccc} M_a & \xleftarrow{\;j_a\;} & N \\ {\scriptstyle p_a}\downarrow & & \downarrow \\ X_a & \xrightarrow{\;j'_a\;} & S = \operatorname{Spec}(k). \end{array}$$

Here $\operatorname{Spec}(k) \to X_a$ is defined by the point $\bar{a} = (\bar{e}, a)$ of X, and $j_a : N \to M_a$ is the canonical immersion. To say that ψ is unramified at $(\bar{e}, a)$ means that j'_a is an open immersion, or equivalently that it induces an isomorphism

$$S \xrightarrow{\sim} \operatorname{Spec}(\mathcal{O}_{X_a, \bar{a}}).$$

[1] Editor's note in the French source: since this reference could not be identified, see Theorem II.5.2.1 of M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson (1970).

[2] Editor's note in the French source: see, for example, EGA IV_4, Theorem 17.11.1(d).

Since p_a is flat, this is equivalent to saying that the morphism obtained from the preceding one by the base change

$$\mathrm{Spec}(\mathcal{O}_{M_a,e}) \longrightarrow \mathrm{Spec}(\mathcal{O}_{X_a,\bar{a}})$$

is an isomorphism. The resulting morphism is precisely

$$\mathrm{Spec}(\mathcal{O}_{N,e}) \longrightarrow \mathrm{Spec}(\mathcal{O}_{M_a,e}),$$

which proves the equivalence of (ii) and (iii).

Notice moreover that the proof shows that (ii) and (iii) imply the following condition, apparently stronger than (iii):

(iii bis) N is an open and closed subscheme of M_a.

b) The first assertion follows from the isomorphism $M_a \simeq \varphi^{-1}(a)$, and the second from the fact that q is flat and $q(e,a) = (\bar{e},a)$.

0.2 Density theorem and theory of the regular points of G

We apply the constructions and notation of the preceding section in the following situation. Let G be a smooth connected algebraic group over k; take $V = G$, with G acting on V by inner automorphisms; and let $W = H$, where H is a smooth connected algebraic subgroup of G. We denote the Lie algebras of G and H by $\mathfrak{g}$ and $\mathfrak{h}$, the normalizer of H in G by N, and its Lie algebra by $\mathfrak{n}$. For $a \in G(k)$, we again denote by M_a, as in Section 1, the inverse of the transporter of a into H. Thus, if $a \in H(k)$, then $N \subset M_a$; in that case $\mathfrak{m}_a$ denotes the Zariski tangent space of M_a at the identity element e of G. Note that

$$\mathfrak{h} \subset \mathfrak{n} \subset \mathfrak{m}_a \subset \mathfrak{g} \qquad \text{for } a \in H(k).$$

We shall use the following lemma.

Lemma 2.0. — *The equality $H = N^0$ holds if and only if*

$$(\mathfrak{g}/\mathfrak{h})^H = 0,$$

where the left-hand side denotes the subspace of invariants for the action of H induced by the adjoint representation. When this condition holds, N is smooth and $\dim X = \dim G$. In all cases, $\dim X \leq \dim G$, and equality holds if and only if H has finite index in N.

Indeed, by II 5.2.3(i), $\mathfrak{n}$ is the inverse image of $(\mathfrak{g}/\mathfrak{h})^H$ under $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$. Hence $(\mathfrak{g}/\mathfrak{h})^H = 0$ is equivalent to $\mathfrak{h} = \mathfrak{n}$, which in turn is equivalent to $H = N^0$, since H is a smooth connected algebraic subgroup of the algebraic group N (cf. VI.2). This also shows that N is smooth. Furthermore,

$$\dim X = (\dim G - \dim N) + \dim H = \dim G - (\dim N - \dim H),$$

and therefore

$$\dim X \leq \dim G,$$

with equality if and only if $\dim H = \dim N$, that is, if and only if H has finite index in N. This holds in particular when $H = N^0$, and proves the lemma.

Theorem 2.1. — *Let G be a smooth connected algebraic group over an algebraically closed field k, let H be a smooth connected algebraic subgroup, and let N be its normalizer. Write $\mathfrak{g}, \mathfrak{h}, \mathfrak{n}$ for their Lie algebras, let $X = G \times^N H$ be the scheme introduced in Section*

1 (fibered over G/N with typical fiber H), and let $\psi : X \to G$ be the canonical morphism. Its image is also the image of $\varphi : G \times H \to G$, where

$$\varphi(g, h) = \mathrm{int}(g)h = ghg^{-1}.$$

The following conditions are equivalent:

(i) H contains a Cartan subgroup C of G (XII 1).

(i bis) H has the same reductive rank and the same nilpotent rank (XII 1) as G.

(ii) H contains a maximal torus T of G, and $(\mathfrak{g}/\mathfrak{h})^T = 0$.

(iii) The set of conjugates of H containing a given maximal torus is finite and nonempty, and H has finite index in its normalizer.

(iv) There exists $a \in H(k)$ that belongs to only finitely many conjugates of H (or merely such that $\psi^{-1}(a)$ has an isolated point), and H has finite index in its normalizer.

(iv bis) The morphism $\psi : X \to G$ is generically quasi-finite (that is, there is a dense open subset of X on which ψ is quasi-finite), and H has finite index in its normalizer.

(v) There is a dense open subset U of G such that, for every $x \in U(k)$, the set of conjugates of H containing x is finite and nonempty; equivalently, $\psi : X \to G$ is dominant and generically quasi-finite.

(vi) There is a dense open subset U of G such that every $x \in U(k)$ is contained in a conjugate of H; equivalently, $\psi : X \to G$ is dominant.

(vii) There exists $a \in H(k)$ such that the fixed subspace of $\mathfrak{g}/\mathfrak{h}$ under $\mathrm{ad}_{\mathfrak{g}/\mathfrak{h}}(a)$ is zero.

Moreover, these conditions imply that H is its own connected normalizer: N is smooth and $\dim H = \dim N$. They also imply that $\psi : X \to G$ is generically étale.

Proof. By 2.0, $\dim X \leq \dim G$, with equality if and only if $\dim H = \dim N$, or equivalently if and only if H has finite index in N. The inequality $\dim X \leq \dim G$ shows that ψ is dominant if and only if it is dominant and generically quasi-finite, and also if and only if it is generically quasi-finite and $\dim X = \dim G$. Since the last equality is equivalent, by 2.0, to H having finite index in N, conditions (vi), (v), and (iv bis) are equivalent. The equivalence of (iv) and (iv bis) is immediate.

The equivalence of (i) and (i bis) follows directly from the definitions and is left to the reader. If H contains a Cartan subgroup C of G, it contains the maximal torus T of C, which is a maximal torus of G. Since C is the centralizer of T, its Lie algebra $\mathfrak{c}$ is

$$\mathfrak{c} = \mathfrak{g}^T$$

(II 5.2.3(ii)). Hence $\mathfrak{g}^T = \mathfrak{c} \subset \mathfrak{h}$, and I 4.7.3 gives the relation

(x) $$(\mathfrak{g}/\mathfrak{h})^T = 0.$$

Conversely, suppose that H contains the maximal torus T and that (x) holds, so that $\mathfrak{h} \supset \mathfrak{c}$. We claim that H contains the centralizer C of T, which will prove the equivalence of (i) and (ii). This follows from the next lemma.

Lemma 2.1.1. — *Let G be a smooth algebraic group over a field k, let T be a subgroup of multiplicative type of G, let C be its connected centralizer (equal to the centralizer of*

T when G is connected and T is a torus, by XII 6.6(b)), and let H be a smooth subgroup of G containing T. Then H contains C if and only if its Lie algebra $\mathfrak{h}$ contains the Lie algebra $\mathfrak{c}$ of C.

Indeed, $\mathrm{Centr}_G(T)$ is smooth over k by XI 2.4, and hence C is smooth over k. Likewise, $\mathrm{Centr}_H(T)$ is smooth, and

$$\mathrm{Centr}_H(T) = \mathrm{Centr}_G(T) \cap H$$

has Lie algebra $\mathfrak{h} \cap \mathfrak{c}$. The hypothesis therefore implies that this smooth subgroup has the same Lie algebra as the smooth group $\mathrm{Centr}_G(T)$, so it contains the connected component C of the latter. Thus H contains C. □

We now prove the equivalence of (i) and (iii). If H contains the maximal torus T of G, we must show that $H \supset C$ (equivalently, that (x) holds) if and only if H has finite index in its normalizer and the set of conjugates of H containing T is finite. If $H \supset C$, then (x) implies, a fortiori,

$$(\mathfrak{g}/\mathfrak{h})^H = 0. \tag{xx}$$

The Lie algebra $\mathfrak{n}$ is the inverse image of the left-hand side under the canonical homomorphism $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ (II 5.2.3(i)). Consequently, by 2.0, (xx) says that $H = N^0$; in particular, H has finite index in its normalizer. Consider the diagram of subgroups

$$\begin{array}{ccccc} T & \longrightarrow & N(T) \cap H & \longrightarrow & H \\ & & \downarrow & & \downarrow \\ & & N(T) \cap N(H) & \longrightarrow & N(H) \\ & & \downarrow & & \\ & & N(T). & & \end{array}$$

By the conjugacy theorem for maximal tori in H (XII 6.6(a)), every conjugate of H containing T is obtained from H by conjugation by an element of $N(T)(k)$. Thus the set of such conjugates is in bijection with

$$\big(N(T)/(N(T) \cap N(H))\big)(k).$$

Since $H \supset C$, we have $N(T) \cap H \supset C$, so the preceding set is a quotient of $(N(T)/C)(k)$, which is finite. This proves (i) ⇒ (iii).

Conversely, suppose (iii), so that $N(T)/(N(T) \cap N(H))$ and $N(H)/H$ are finite. The conjugacy theorem in H shows that the homomorphism

$$(N(T) \cap N(H))/(N(T) \cap H) \longrightarrow N(H)/H$$

induced by the preceding diagram is bijective on k-valued points (indeed, it is an isomorphism). Since the latter quotient is finite, so is the former. Hence $N(T) \cap H$ has finite index in $N(T)$ and therefore contains $C = N(T)^0$. Thus $H \supset C$, and (i), (i bis), (ii), and (iii) are equivalent.

We next prove (ii) ⇒ (vii). Both conditions are invariant under an extension $k \to k'$ with k' algebraically closed. We may therefore assume that k has infinite transcendence

degree over its prime subfield. There is then an element $a \in T(k)$ such that the subgroup of $T(k)$ generated by a is Zariski dense in T. It follows that

$$(\mathfrak{g}/\mathfrak{h})^T = (\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(a)}.$$

The left-hand side is zero by hypothesis, which proves (vii).

The implication (vii) $\Rightarrow$ (vi) is contained in the following result, which sharpens 2.1.

Corollary 2.2. — *Let G be a smooth algebraic group over a field k, let H be a smooth algebraic subgroup, let N be its normalizer in G, and let*

$$\varphi : G \times H \longrightarrow G, \qquad \varphi(g,h) = \mathrm{ad}(g)h = ghg^{-1}.$$

Let $\psi : X = G \times^N H \to G$ be the morphism obtained from φ by passage to the quotient (cf. Section 1), and let $a \in H(k)$. The following conditions are equivalent:

(i) φ is smooth at (e,a).

(ii) ψ is étale at $(\bar{e},a)$, and N is smooth over k.

(iii) $(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(a)} = 0$, where $\mathfrak{g}$ and $\mathfrak{h}$ are the Lie algebras of G and H.

These conditions imply $H^0 = N^0$.

The smoothness at a k-rational point of φ, a morphism between smooth k-preschemes, is equivalent to the surjectivity of the tangent map at that point. Under the usual identifications of the tangent spaces of G and H with their Lie algebras, a direct calculation gives

$$d\varphi(\xi,\eta) = (\mathrm{id} - \mathrm{ad}(a))\xi + \eta,$$

as a map from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Its surjectivity is therefore equivalent to the surjectivity of $\mathrm{id} - \mathrm{ad}(a)$ on $\mathfrak{g}/\mathfrak{h}$, which is precisely (iii).

Condition (iii) implies, a fortiori, $(\mathfrak{g}/\mathfrak{h})^H = 0$, and hence $H^0 = N^0$ by 2.0 (the connectedness hypothesis made at the beginning of the section is not needed there). It follows that N is smooth and that $\dim H = \dim N$, whence $\dim X = \dim G$. Since $q : G \times H \to X$ is flat and $\psi \circ q = \varphi$ is smooth at (e,a), the morphism ψ is smooth at $q(e,a) = (\bar{e},a)$, and hence is étale there by dimension. We have proved (i) $\Longleftrightarrow$ (iii) $\Longrightarrow$ (ii). Conversely, (ii) implies (i), because the smoothness of N implies that of q. □

In particular, under condition (vii) of 2.1 the morphism φ is smooth at (e,a), so its image contains a nonempty open subset of G. Thus ψ is dominant, proving (vii) $\Rightarrow$ (vi).

It remains to prove (vi) $\Rightarrow$ (i). Suppose first that G is affine. Let U be a nonempty open subset of G such that every $x \in U(k)$ is contained in a conjugate of H, and let C be a Cartan subgroup of G. Applying the already proved implication (i) $\Rightarrow$ (iv) with C in place of H (Borel's "density theorem"), we find a conjugate of C meeting U. We may therefore assume that $U \cap C \neq \varnothing$; equivalently, there is a nonempty open subset V of C such that every $x \in V(k)$ belongs to a conjugate of H.

Write

$$C = T \cdot C_u,$$

where T is the maximal torus of C, hence a maximal torus of G, and C_u is the unipotent part of C; T lies in the center of C (*BIBLE* 6, Theorem 2). We may again assume that k has infinite transcendence degree over its prime subfield. Choose $t \in T(k)$ in the projection of V onto T, a nonempty open subset of T, so that $tC_u \cap V \neq \varnothing$, and choose it so that it generates a Zariski-dense subgroup of T. Every algebraic subgroup of G containing a

product tu, with $t \in T(k)$ and $u \in C_u(k)$, contains both factors (*BIBLE* 4, Theorem 3). Taking $tu \in V(k)$, we conclude that a conjugate of H contains t, and therefore contains T. After replacing H by that conjugate, we may assume

$$T \subset H.$$

Let W be the inverse image in C_u of V under $u \mapsto tu$. For every $x \in (T \cdot W)(k)$, there is then a conjugate of H containing both T and x. As observed above, every such conjugate is of the form $\mathrm{int}(g)H$ for some $g \in N(T)(k)$. Consider

$$f : N(T) \times H \longrightarrow G, \qquad f(g,h) = \mathrm{int}(g)h = ghg^{-1}.$$

The image of f contains TW. Since C has finite index in $N(T)$, the latter is a finite union of translates Cg_i, with $g_i \in N(T)(k)$. It follows that some dense open subset V' of $C = TC_u$ is contained in the image under f of $(Cg_i) \times H$. Replacing H by $\mathrm{int}(g_i)H$, we may assume $g_i = e$, so that

$$f(C \times H) \supset V'.$$

Thus for every $u \in V'(k)$ there are $v \in C(k)$ and $h \in H(k)$ such that

$$v^{-1}hv = u, \qquad \text{hence} \qquad vuv^{-1} \in H(k).$$

Set

$$C' = C \cap H = \mathrm{Centr}_H(T).$$

Since vuv^{-1} also belongs to C, it lies in C', and hence $u \in \mathrm{int}(v^{-1})C'$. Therefore the union of the conjugates of C' in C, by elements of $C(k)$, is dense. As in the preceding argument for the pair (G,H), now applied to (C,C'), this implies that C' has finite index in its normalizer in C. By the lemma of *BIBLE* 7, Section 1, we obtain $C' = C$, and consequently $C \subset H$. This proves (vi) $\Rightarrow$ (i) when G is affine.

In the general case we argue by induction on $n = \dim G$, the assertion being trivial for $n = 0$. Let Z be the center of G. There are two cases.

1° If $\dim(Z \cap H) > 0$, set $G' = G/(Z \cap H)$. Then $\dim G' < n$, and condition (vi) for H implies the same condition for its image H' in G'. By induction, H' contains a Cartan subgroup C' of G'. Hence H contains the inverse image C of C', which is a Cartan subgroup by XII 6.6(e).

2° If $\dim(Z \cap H) = 0$, the canonical morphism $H \to G/Z$ is finite. Since G/Z is affine by XII 6.1, H is affine. Every homomorphism from H to an abelian variety is therefore zero (indeed, every morphism of preschemes from H to an abelian variety is constant). This follows from the fact that a smooth connected affine algebraic group over an algebraically closed field is a rational variety, or simply from the fact that it is the union of its Borel subgroups (*BIBLE* 6, Theorem 5(b)); the structure theorems *BIBLE* 6.2 and 6.3 show readily that a smooth connected solvable affine group is a rational variety.

Apply Chevalley's structure theorem to G: the group G is an extension of an abelian variety A by a smooth affine group. Since H is affine, its image in A is zero. On the other hand, that image must be all of A, because the union of its conjugates in A is dense. Hence $A = 0$, so G is affine, reducing us to the case already proved.

This completes the proof of 2.1.

Corollary 2.3. — *Assume that the equivalent conditions of 2.1 hold.*

a) Let $k(X)$ and $k(G)$ be the fields of rational functions of X and G, respectively. Then $k(X)$ is a finite separable extension of $k(G)$. Denote its degree by d.

b) Let T be a maximal torus of G contained in H, which exists by 2.1(ii), and let C be the corresponding Cartan subgroup of G. Then $C \subset H$. Moreover, $\mathrm{Norm}_G(T)$ is a smooth subgroup of G, and

$$\mathrm{Norm}_G(T) \cap \mathrm{Norm}_G(H) = \mathrm{Norm}_{\mathrm{Norm}_G(H)}(T)$$

is a smooth subgroup of it, of finite index equal to the degree d defined in (a). The number of conjugates of H containing a given maximal torus, or a given Cartan subgroup, is d.

c) Let U be the largest open subset of G such that $\psi : X \to G$ induces a finite étale morphism

$$\psi^{-1}(U) \longrightarrow U.$$

Then U is dense. For $g \in G(k)$, the following are equivalent:

(a) $g \in U(k)$;

(b) exactly d conjugates of H contain g;

(c) there are at least d distinct conjugates H_i of H containing g, and for every i,

$$(\mathfrak{g}/\mathfrak{h}_i)^{\mathrm{ad}(g)} = 0, \qquad \mathfrak{h}_i = \mathrm{Lie}(H_i).$$

Assertion (a) follows from the generic étaleness of ψ, stated at the end of 2.1. This also shows that the open set U in (c) is nonempty, hence dense, and gives the two stated characterizations of its k-points: here one uses that ψ is separated, that X is integral, that G is integral and normal (SGA 1, I 10.11), and that $(\mathfrak{g}/\mathfrak{h}_i)^{\mathrm{ad}(g)} = 0$ means that ψ is étale at the point $x_i \in \psi^{-1}(g)$ corresponding to H_i.

If H contains the maximal torus T of G, the centralizers of T in H and G have the same dimension and are smooth and connected (XII 6.6(b)); they are therefore equal, proving that $C \subset H$. The normalizer of T in any smooth group containing it is smooth (XI 2.4 bis), so $\mathrm{Norm}_G(T)$ and $\mathrm{Norm}_{\mathrm{Norm}_G(H)}(T)$ are smooth. Recall also that $N = \mathrm{Norm}_G(H)$ is smooth by the last assertion of 2.1. Furthermore, $\mathrm{Norm}_N(T)$ contains C, which has finite index in $\mathrm{Norm}_G(T)$; hence $\mathrm{Norm}_N(T)$ itself has finite index in $\mathrm{Norm}_G(T)$.

The conjugacy theorem for maximal tori in H shows that this index is the number of conjugates of H containing T, equivalently the number containing C. The union of the conjugates of C in G is dense, by 2.1(i) $\Rightarrow$ (vi) applied to C in place of H, and U is stable under inner automorphisms. Hence $C \cap U \neq \varnothing$. Arguing as in the proof of 2.1(vi) $\Rightarrow$ (ii), and making a harmless extension of the base field if necessary, one finds $g \in (C \cap U)(k)$ such that every conjugate of H containing g contains T, and hence C. Thus the conjugates of H containing C are precisely those containing g; since $g \in U(k)$, their number is d. This proves (b).

In fact, the argument shows that the set of conjugates of H containing T is a homogeneous set under the group of rational points of

$$W_G(T) = \mathrm{Norm}_G(T)/\,\mathrm{Centr}_G(T).$$

In particular,

$$d \leq \text{the order of the Weyl group of } G.$$

Corollary 2.4. — *With the notation of 2.1, the following conditions are equivalent:*

(i) $\psi : X \to G$ is birational.

(ii) Exactly one conjugate of H contains a given Cartan subgroup of G.

(iii) H contains a Cartan subgroup C of G, and $\mathrm{Norm}_G(H) \supset \mathrm{Norm}_G(C)$.

(iv) There is a nonempty open subset V of G such that every $g \in V(k)$ belongs to exactly one conjugate of H.

This follows immediately from 2.1 and 2.3.

Corollary 2.5. — *Assume that the conditions of 2.4 hold, and let $g \in G(k)$. The following conditions are equivalent:*

(i) $g \in U(k)$, where U is defined in 2.3(c); that is, g is contained in exactly one conjugate of H.

(ii) The set of conjugates of H containing g is finite and nonempty.

(iii) The scheme $\psi^{-1}(g)$, the "scheme of conjugates of H containing g," has an isolated point.

(iv) There is a conjugate H' of H containing g such that

$$(\mathfrak{g}/\mathfrak{h}')^{\mathrm{ad}(g)} = 0, \qquad \mathfrak{h}' = \mathrm{Lie}(H').$$

Finally, U is also the largest open subset of G on which ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

The equivalence of (i), (ii), and (iii), together with the last assertion, follows from the "Main Theorem"[3] applied to the birational morphism $\psi : X \to G$, since G is normal. Their equivalence with (iv) follows immediately from Corollary 2.2, which characterizes the points of X at which ψ is étale.

Theorem 2.6. — *Let G be a smooth connected algebraic group over an algebraically closed field k, let C be a Cartan subgroup associated with a maximal torus T, and put*

$$N = \mathrm{Norm}_G(C) = \mathrm{Norm}_G(T) \qquad \text{(cf. XII 8.4)}.$$

Let $X = G \times^N C$, where N acts on the left factor G by right translations and on the right factor C by inner automorphisms, and let $\psi : X \to G$ be the canonical morphism.

a) ψ is birational.

b) Let U be the largest open subset of G such that ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U$$

(cf. 2.5), and let

$$\rho = \rho_\nu(G) = \dim C$$

be the nilpotent rank of G. For every $g \in G(k)$, the multiplicity of the eigenvalue 1 in the action of $\mathrm{ad}(g)$ on $\mathfrak{g}$ is at least ρ; it equals ρ if and only if $g \in U(k)$.

[3] Editor's note in the French source: Zariski's Main Theorem.

Proof. Condition 2.1(i) holds, so 2.4(iii) ⇒ (i) proves (a). In *BIBLE* 7, when G is affine, the points of $U(k)$ are called the *regular points* of $G(k)$. We shall use the same terminology and call U the *open subset of regular points* of G. Notice that the proof in *BIBLE* that this set is itself open is incorrect; the present Exposé establishes its openness under more general hypotheses.

To prove (b), for each $g \in G(k)$ introduce the characteristic polynomial

$$P(\mathrm{ad}(g), t) = t^n + c_1(g)t^{n-1} + \cdots + c_n(g).$$

Replacing k by an arbitrary k-algebra shows immediately that the functions $c_i(g)$ arise from uniquely determined sections

$$c_i \in \Gamma(G, \mathcal{O}_G).$$

Suppose that $g \in G(k)$ belongs to a Cartan subgroup, which we may take to be C; this includes the case of a regular element. By 2.5(iv),

$$(\mathfrak{g}/\mathfrak{c})^{\mathrm{ad}(g)} = 0$$

if and only if g is regular, where $\mathfrak{c} = \mathrm{Lie}(C)$. Since C is nilpotent, $\mathrm{ad}_{\mathfrak{c}}(g)$ has only the eigenvalue 1. It follows that the multiplicity of the eigenvalue 1 in $\mathrm{ad}_{\mathfrak{g}}(g)$ is at least ρ, and is exactly $\dim C = \rho$ if and only if g is regular.

In particular, the characteristic polynomial above is divisible by $(t-1)^\rho$. Divisibility by $(t-1)^\rho$ is expressed by linear relations, with integer coefficients, among the coefficients of the polynomial. These relations hold for $g \in U(k)$, where U is dense; since G is reduced, they therefore hold everywhere. Thus

$$(\dagger) \qquad t^n + c_1 t^{n-1} + \cdots + c_n = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho})$$

in the polynomial ring over $\Gamma(G, \mathcal{O}_G)$. In particular, for every field extension K/k and every $g \in G(K)$, the eigenvalue 1 of $\mathrm{ad}(g)$ has multiplicity at least ρ.

We have already seen that equality holds when g is regular; let us prove the converse. Suppose first that G is affine, and write

$$g = g_s g_u$$

as the product of its semisimple and unipotent parts (*BIBLE* 4, Section 4). Then

$$\mathrm{ad}(g) = \mathrm{ad}(g_s)\,\mathrm{ad}(g_u)$$

is the corresponding decomposition of $\mathrm{ad}(g)$ (loc. cit., corollary to Theorem 3). Hence $\mathrm{ad}(g)$ and $\mathrm{ad}(g_s)$ have the same eigenvalues, counted with multiplicity; in particular, the eigenvalue 1 has the same multiplicity in both.

By *BIBLE* 7, Theorem 2, Corollary 1, the element g is regular if and only if g_s is regular. We may therefore assume that $g = g_s$ is semisimple. It is then contained in a maximal torus by *BIBLE* 6, Theorem 5(c), and hence in a Cartan subgroup, a case already treated. This proves (b) when G is affine.

In general, let $Z = \mathrm{Centr}(G)_{\mathrm{red}}$. By XII 6.6(e), the Cartan subgroups of G are the inverse images of those of $G' = G/Z$. Thus g is regular in G if and only if its image g' is regular in G'. Since Z is smooth, the Lie algebra of G' is $\mathfrak{g}' = \mathfrak{g}/\mathfrak{z}$, where $\mathfrak{z} = \mathrm{Lie}(Z)$, and $\mathrm{ad}(g')$ is the endomorphism induced by $\mathrm{ad}(g)$ on $\mathfrak{g}/\mathfrak{z}$. Consequently, the multiplicity of the eigenvalue 1 in $\mathrm{ad}(g)$ is

$$d = \dim Z$$

plus its multiplicity in $\mathrm{ad}(g')$. The former therefore equals the nilpotent rank of G if and only if the latter equals the nilpotent rank of G'. We are reduced to G', which is affine by XII 6.1, and that case has already been proved. This completes the proof of 2.6.

Corollary 2.7. — *With the notation of the preceding proof,*[4] *let*

$$b = 1 + b_1 + \cdots + b_{n-\rho} \in \Gamma(G, \mathcal{O}_G).$$

Then the open subset of regular points of G is

$$U = G_b,$$

the set of points of G at which b is invertible. In particular, U is an affine open subset when G is affine.

Corollary 2.8. — *Let H be a smooth connected algebraic subgroup of G containing a Cartan subgroup of G.*

a) Let C be an algebraic subgroup of H. Then C is a Cartan subgroup of H if and only if it is a Cartan subgroup of G.

b) Let $g \in G(k)$, and let d be the integer introduced in 2.3. Then g is a regular point of G if and only if there are exactly d conjugates H_i of H containing g, and g is a regular element of every H_i. Equivalently, there are at most d conjugates of H containing g, and g is regular in one of them. When these conditions hold, if C is the unique Cartan subgroup of G containing g, then the conjugates of H containing g are precisely the conjugates of H containing C.

c) Let $g \in H(k)$. Then g is regular in G if and only if it is regular in H and

$$(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(g)} = 0.$$

Proof. *a)* Under either hypothesis on C, its unique maximal torus T is a maximal torus of both H and G, since H has the same reductive rank as G. The smooth connected groups $\mathrm{Centr}_H(T) \subset \mathrm{Centr}_G(T)$ have the same dimension, and therefore are equal. Hence C is equal to one of these two centralizers if and only if it is equal to the other, proving (a).

b) Suppose first that g is regular in G, and let C be the unique Cartan subgroup of G containing g. By 2.3(b), exactly d conjugates H_i of H contain C. Since

$$(\mathfrak{g}/\mathfrak{c})^{\mathrm{ad}(g)} = 0,$$

the endomorphism $\mathrm{ad}(g)$ has no eigenvalue 1 on $\mathfrak{g}/\mathfrak{c}$, and a fortiori

$$(\mathfrak{g}/\mathfrak{h}_i)^{\mathrm{ad}(g)} = 0.$$

By 2.3(c), exactly d conjugates of H contain g, namely the H_i. A Cartan subgroup of such an H_i containing g is a Cartan subgroup of G by (a), and hence is C. Thus g is regular in every H_i.

Conversely, suppose that at most d conjugates H_i of H contain g, and that g is regular in one of them, which we may take to be H. Since g is regular in H, it belongs to a unique Cartan subgroup C of H; by (a), C is a Cartan subgroup of G. Let C' be any Cartan subgroup of G containing g. By 2.3(b), exactly d conjugates of H contain C'. They all

[4] Editor's note in the French source: cf. (†) above.

contain g, so they are precisely the H_i; in particular, H contains C'. Thus C and C' are two Cartan subgroups of H containing the same regular element g of H, and therefore $C = C'$. Hence g is regular in G. □

c) For an endomorphism u of a finite-dimensional vector space, let $\nu(u)$ denote the nullity[5] of $\mathrm{id} - u$. Then

$$\nu(\mathrm{ad}(g)_{\mathfrak{g}}) = \nu(\mathrm{ad}(g)_{\mathfrak{h}}) + \nu(\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}).$$

The two terms on the right are respectively at least the nilpotent rank of H, which equals the nilpotent rank ρ of G by (a), and at least 0. Hence $\nu(\mathrm{ad}(g)_{\mathfrak{g}}) = \rho$ if and only if

$$\nu(\mathrm{ad}(g)_{\mathfrak{h}}) = \rho, \qquad \nu(\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}) = 0.$$

This says precisely that g is regular in G if and only if it is regular in H and $\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}$ has no nonzero invariants. □

Remarks 2.9. — In the statement of 2.1, condition (iii) cannot be weakened by assuming only that H contains a maximal torus and has finite index in its normalizer, even if one also requires that normalizer to be smooth, that is, that $H = N^0$, and even when G is affine and solvable. An example is the group G of matrices of the form

$$g = \begin{pmatrix} t & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

and the subgroup H formed by the matrices above for which $b = c = 0$. The Cartan subgroup of G here consists of the matrices g for which $a = c = 0$.

Remarks 2.10. — Let G be a smooth algebraic group over k, and let H be a smooth algebraic subgroup; we no longer assume that G and H are connected. Suppose that H^0 contains a Cartan subgroup of G^0. Then

$$(\mathfrak{g}/\mathfrak{h})^H \subset (\mathfrak{g}/\mathfrak{h})^{H^0} = 0,$$

so $H^0 = N^0$, where N is the normalizer of H; in particular, N is smooth.

Nevertheless, even with G connected, examples are easily constructed in which H has a connected component H_i such that

$$(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(h)} \neq 0 \qquad \text{for every } h \in H_i(k).$$

Equivalently, the morphism

$$G \times H_i \longrightarrow G, \qquad (g, h) \longmapsto \mathrm{ad}(g) \cdot h$$

is nowhere étale, or even quasi-finite. One may take, for example, H to be the normalizer of a maximal torus in $\mathrm{SL}(2)_k$.

Similarly, even if the image H'' of H in the finite group $G'' = G/G^0$ is all of G'', the union of the conjugates of H in G need not be dense. For example, take G to be the semidirect product of $\mathbb{Z}/2\mathbb{Z}$ with $G^0 = \mathrm{SL}(2)_k$, on which it acts by "symmetry," and take H to be the semidirect product $\mathbb{Z}/2\mathbb{Z} \cdot T$, where T is a maximal torus of G^0.

On the other hand, if one does not assume in advance that H^0 contains a Cartan subgroup of G^0, but assumes that the union of the conjugates of H in G is dense, then H^0 necessarily contains a Cartan subgroup of G^0. To see this, one may plainly assume that G is connected and repeat the proof of 2.1(vi) ⇒ (i), which does not require H to be connected.

[5] Editor's note in the French source: that is, the dimension of the nullspace of $\mathrm{id} - u$.

0.3 The case of an arbitrary base prescheme

Suppose first that the base is a field k, not necessarily algebraically closed. Since conditions 2.1 (i bis), (iv bis), (v), and (vi) are invariant under extension of the base field, passage to the algebraic closure $\overline{k}$ of k shows that they are equivalent to one another, and equivalent to the assertion that $H_{\overline{k}}$ contains a Cartan subgroup of $G_{\overline{k}}$. When this condition is satisfied, it is still true (with the notation of 2.3) that $k(X)$ is a finite separable extension of $k(G)$, of degree d independent of every extension of the base. If U is the largest open subset of G such that ψ induces a finite étale morphism

$$\psi^{-1}(U) \longrightarrow U,$$

then the formation of U commutes with extension of the base field. If ψ is birational, then U is also the largest open subset of G such that ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U;$$

and if $g \in U(k)$, there is one and only one subgroup H' of G, conjugate to H over $\overline{k}$, such that $H' \ni g$.

A point $g \in G(k)$ is called *regular* if it is regular as an element of

$$G(\overline{k}) = G_{\overline{k}}(\overline{k}).$$

More generally, the construction of 2.7 gives an open subset of G, whose formation commutes with every extension of the base field, called the *open subset of regular points* of G. It is also characterized by the following property: for every algebraically closed extension K of k and every $g \in G(K)$, the point g is regular in G_K if and only if $g \in U(K)$. If $g \in U(k)$[6], then g is contained in one and only one Cartan subgroup of G, as we shall prove below under more general hypotheses.

Let G be a smooth, separated group prescheme of finite type, with connected fibers over the prescheme S, and consider the functor

$$\mathcal{C} : (\mathrm{Sch})^{\circ}_{/S} \longrightarrow (\mathrm{Ens})$$

defined by

$$\mathcal{C}(S') = \left\{ \text{ Cartan subgroups (XII 3.1) of } G_{S'} \right\}.$$

Suppose that this functor is representable by a prescheme smooth over S. In Exposé XV we give a necessary and sufficient condition for this to hold, but we already know that the hypothesis is satisfied if G is affine over S with locally constant reductive rank (XII 3.3), or, more generally, if G locally admits a maximal torus for the fpqc topology (XII 7.1 a)); this is the case, for example, if S is the spectrum of a field.

Let X be the Cartan subgroup of the $\mathcal{C}$-group prescheme $G_{\mathcal{C}}$, the "universal Cartan subgroup" of G. Thus, as a prescheme over S, X represents the functor

$$X(S') = \left\{ \begin{array}{c|l} (C, g) & C \text{ is a Cartan subgroup of } G_{S'}, \\ & g \text{ is a section of } C \text{ over } S' \end{array} \right\}.$$

Consider the canonical projection morphism $(C, g) \mapsto g$,

$$\psi : X \longrightarrow G.$$

We then have the following theorem.

[6]Editor's note: $G(k)$ has been corrected to $U(k)$.

Theorem 0.3.1. *Under the preceding hypotheses on G, and with the preceding notation, let U be the set of $g \in G$ such that g is a regular element of its fiber G_s. Then U is open, and it is also the largest open subset U of G such that ψ induces an isomorphism*

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Let us first prove that U is open. By the representability of $\mathcal{C}$ as a prescheme smooth over S, and since its structure morphism is plainly surjective, it follows at once that G locally admits a Cartan subgroup for the étale topology and that the nilpotent rank of the fibers of G is locally constant. The same holds for the dimension of the fibers of G. After localizing on S, we may therefore suppose that these two integers are constant, say ρ and n. Consider the Killing polynomial

$$P_G(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t], \qquad A = \Gamma(G, \mathcal{O}_G).$$

The restriction of this polynomial to the fibers G_s of G, and in particular to the fibers at the maximal points of S, is divisible by $(t-1)^\rho$. This says that certain linear combinations of the c_i, with integer coefficients, vanish on the fibers G_s. When S is reduced, as we may assume in order to prove that U is open, it follows that these combinations themselves vanish. Hence the Killing polynomial itself is divisible by $(t-1)^\rho$, say

$$P_G(t) = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho}).$$

Let b be the sum of the coefficients $b_0 = 1, b_1, \ldots, b_{n-\rho}$ of the second factor. By 2.7, applied to the fibers of G, we have

$$U = G_b,$$

which proves that U is open.

To prove that $\psi^{-1}(U) \to U$ is an isomorphism, SGA 1, I 5.7 reduces us to checking this fiber by fiber. We are thereby reduced to the case of a base field, which may be assumed algebraically closed. There is then a Cartan subgroup C of G. If N is its normalizer, the conjugacy theorem XII 7.1 a) and b) identifies $\mathcal{C}$ with G/N, and the morphism $\psi : X \to G$ considered here is precisely the one defined in Section 2. The conclusion follows from 2.6 b). The same argument also shows that U is the largest open subset of G for which ψ induces an isomorphism $\psi^{-1}(U) \to U$.

Corollary 0.3.2. *Under the hypotheses of 3.1, let g be a regular section of G, that is, suppose that $g(s)$ is a regular point of G_s for every $s \in S$. Then there is one and only one Cartan subgroup C of G such that g is a section of C.*

Indeed, the hypothesis on g says that g is a section of U, while the conclusion says that it admits a unique lift to a section of X. This is simply another way of saying that $\psi^{-1}(U) \to U$ is an isomorphism.

Observe now that the open subset $\psi^{-1}(U)$ of the Cartan subgroup X of $G_{\mathcal{C}}$ is precisely the open subset of X consisting of the points regular in $G_{\mathcal{C}}$, by which we mean regular in their fibers. We thus obtain a natural "fibration" of the dense open subset U of regular points of G over the prescheme $\mathcal{C}$; its fibers are dense open subsets of Cartan subgroups of the fibers of $G_{\mathcal{C}}$, namely their open subsets of points regular in G. For example, we obtain the following result, which will be considerably sharpened in the next Exposé.

Corollary 0.3.3. *Let G be a smooth connected algebraic group over the field k, and let $\mathcal{T}$ be the scheme of maximal tori of G, which is isomorphic to the scheme of Cartan subgroups*

of G. Then the function field $k(G)$ of G is isomorphic to the function field of a smooth connected affine nilpotent algebraic group C over the function field $k(\mathcal{T})$ of $\mathcal{T}$, namely

$$C = \text{“the generic Cartan subgroup of } G\text{”}.$$

If G is affine and has unipotent rank zero, that is, if the Cartan subgroups of $G_{\overline{k}}$ are tori, then $k(G)$ is a unirational extension of $k(\mathcal{T})$.

Of course, by the generic Cartan subgroup of G we mean, by an abuse of language, the Cartan subgroup of $G_{k(\mathcal{T})}$ that is the generic fiber of X over $\mathcal{T}$. It remains only to prove the last assertion of 3.3, which is contained in the following well-known result, due to Chevalley.

Lemma 0.3.4. *Let k be a field, T a torus over k, and $k(T)$ the field of rational functions on T. Then $k(T)$ is a unirational extension of k, that is, it is contained in a purely transcendental extension of k.*

Indeed, let k' be a finite separable extension of k that splits[7] T (X 1.4). Then $T \otimes_k k'$ is a rational variety: it contains a dense open subset isomorphic to a dense open subset of affine n-space $\mathbb{A}^n_{k'}$. Hence

$$T' = \prod_{\mathrm{Spec}(k')/\mathrm{Spec}(k)} T_{k'}/\mathrm{Spec}(k')$$

is a rational variety, since it has a dense open subset isomorphic to a dense open subset of

$$\prod_{\mathrm{Spec}(k')/\mathrm{Spec}(k)} \mathbb{A}^n_{k'},$$

which is isomorphic to affine mn-space over k, where $m = [k' : k]$. Consider the norm homomorphism from T' to T, which is defined whenever T is a commutative group scheme over k. The composite

$$T \longrightarrow T' \longrightarrow T$$

is the m-th power map on T, and is therefore dominant. Consequently $T' \to T$ is dominant, proving that T is unirational.

Return to the hypotheses of 3.1, but suppose in addition that G locally admits a maximal torus for the fpqc topology (XII 7.1). Let Y be the maximal torus of the Cartan subgroup X of $G_{\mathcal{C}}$. The morphism $\psi : X \to G$ then induces a morphism $Y \to G$ whose set-theoretic image consists of the semisimple elements of the fibers of G (XII 8). Finally, 3.1 shows that the restriction of ψ to the open subset Y^{reg} of regular points of Y induces a closed immersion

$$Y^{\mathrm{reg}} \longrightarrow U = G^{\mathrm{reg}}.$$

Making explicit the functorial meaning of $Z = Y^{\mathrm{reg}}$, we obtain the following.

Corollary 0.3.5. *Let G be a smooth, separated S-group prescheme of finite type, with connected fibers over the prescheme S, and suppose that G locally admits a maximal torus for the fpqc topology. Let*

$$Z : (\mathrm{Sch})^{\circ}_{/S} \longrightarrow (\mathrm{Ens})$$

be the functor defined by

$$Z(S') = \left\{ \begin{array}{c} \textit{regular sections of } G_{S'} \textit{ over } S' \textit{ that are} \\ \textit{contained in a maximal torus of } G_{S'} \end{array} \right\}.$$

[7] Editor's note: “splits” (?), or: “over which T is split”.

Then Z is representable by a closed subprescheme, smooth over S, of the open subset $U = G^{\mathrm{reg}}$ of G introduced in 3.1.

To conclude, let us record the following result, which sharpens the density theorem 2.1 (i) $\Rightarrow$ (vi).

Corollary 0.3.6. *Under the hypotheses of 3.5, let C be a Cartan subgroup of G, and consider the morphism*

$$\varphi : Z \times C \longrightarrow G$$

defined by

$$\varphi(g, h) = \mathrm{ad}(g)h = ghg^{-1}.$$

Then φ is dominant.

It plainly suffices to prove the assertion fiber by fiber, which reduces us to the case where S is the spectrum of an algebraically closed field k. Let T be the maximal torus of C, let $t_0 \in T(k)$ be regular in G, and let $c_0 \in C(k)$ be regular in G. Consider $\varphi^{-1}(\varphi(t_0, c_0))$. Its k-rational points are the pairs (t, c), with $t \in Z(k)$ and $c \in C(k)$, such that

$$\mathrm{ad}(t)c = \mathrm{ad}(t_0)c_0, \qquad \text{that is,} \qquad c = \mathrm{ad}(t^{-1}t_0)c_0.$$

They are therefore in bijection with the $t \in Z(k)$ such that $\mathrm{ad}(t^{-1}t_0)c_0 \in C$, or equivalently, since c_0 is regular, such that $t^{-1}t_0 \in N$, where N is the normalizer of C; in other words, $t \in N$. Restricting to those $t \in Z(k)$ that belong to $C(k)$ gives an open and closed part of this fiber. We have therefore found a connected component of $\varphi^{-1}(\varphi(t_0, c_0))$ isomorphic to T. The preceding set-theoretic argument indeed gives an isomorphism of schemes: one simply replaces k-valued points by points with values in an arbitrary k-prescheme.

It follows that the generic fiber of φ has dimension at most $\dim T$, and therefore

$$\dim \mathrm{Im}(\varphi) \geq \dim(Z \times C) - \dim T = \dim Z + \dim C - \dim T.$$

But

$$\dim Z = \dim Y = \dim \mathcal{C} + \dim T = \dim G - \dim C + \dim T.$$

Thus $\dim \mathrm{Im}(\varphi) \geq \dim G$, and φ is dominant.

Q.E.D.

Remarks 0.3.7. The argument shows in addition that the connected component at (t_0, c_0) of the fiber $\varphi^{-1}(\varphi(t_0, c_0))$ is isomorphic to T. In particular, it is smooth over k and has the same dimension as the generic fiber. It follows that φ is in fact smooth at (t_0, c_0), something that one should also be able to verify by calculating the tangent map. Consequently, under the hypotheses of 3.6, the induced morphism

$$Z \times_S C^{\mathrm{reg}} \longrightarrow G^{\mathrm{reg}}, \qquad C^{\mathrm{reg}} = C \cap G^{\mathrm{reg}},$$

is smooth. Likewise, the analogous morphism

$$Z \times T^{\mathrm{reg}} \longrightarrow Z,$$

where T is a maximal torus of G, is smooth. More generally, for every smooth connected normal algebraic subgroup H of C containing an element $c_0 \in G(k)$ that is regular in G, the image of $Z \times H \to G$ is dense in the image of $G \times H \to G$.

0.4 Lie algebras over a field: rank, regular elements, Cartan subalgebras

In the remainder of this Exposé we take up the theory developed by Chevalley in his book *Théorie des Groupes de Lie III* (Act. Sc. Ind. 1226, Paris 1955); the methods of schemes allow us to dispense with the hypothesis of characteristic zero. We begin by recalling in this section several well-known notions and results.

Let $\mathfrak{g}$ be a Lie algebra over a ring k. For every $a \in \mathfrak{g}$, denote by $\mathrm{ad}(a)$ the endomorphism

$$\mathrm{ad}(a) \cdot x = [a, x]$$

of $\mathfrak{g}$; it is a derivation of the Lie algebra $\mathfrak{g}$. For every derivation D of $\mathfrak{g}$, the nil-space of D, that is, the union of the kernels of the iterates of D, is a Lie subalgebra of $\mathfrak{g}$, as follows from the Leibniz formula

$$D^m([x, y]) = \sum_{0 \le p \le m} \binom{m}{p} [D^p x, D^{m-p} y].$$

We set

$$\mathrm{Nil}(a, \mathfrak{g}) = \text{nil-space of } \mathrm{ad}(a) = \bigcup_{m \ge 0} \mathrm{Ker}\, \mathrm{ad}(a)^m.$$

When no confusion can arise, we write simply $\mathrm{Nil}(a)$, and call it the *nil-space of a in $\mathfrak{g}$*.

Proposition 0.4.1. *For every $a \in \mathfrak{g}$, its nil-space $\mathrm{Nil}(a, \mathfrak{g})$ is a Lie subalgebra of $\mathfrak{g}$, equal to its own normalizer.*

It remains only to prove that it is its own normalizer. In other words, every element of $\mathfrak{g}/\mathrm{Nil}(a, \mathfrak{g})$ annihilated by the adjoint representation of $\mathrm{Nil}(a, \mathfrak{g})$ on this quotient must be zero. This is immediate, since every element of the quotient annihilated by $\mathrm{ad}(a)$ is zero.

For the remainder of this section, suppose that k is a field and that $\mathfrak{g}$ is finite-dimensional over k. We denote by $W(\mathfrak{g})$ the scheme over k defined by $\mathfrak{g}$; its points in a k-algebra A are the elements of $\mathfrak{g} \otimes_k A$. If $a \in \mathfrak{g}$, the characteristic polynomial of $\mathrm{ad}(a)$ is also called the *characteristic polynomial*, or the *Killing polynomial*, of a in $\mathfrak{g}$:

$$P_{\mathfrak{g}}(a, t) = t^n + c_1(a)t^{n-1} + \cdots + c_n(a),$$

where $n = \mathrm{rank}_k\, \mathfrak{g}$ and $c_i(a) \in k$. Taking the same polynomial for $a \in \mathfrak{g} \otimes_k A$[8], where A is any k-algebra, we see that the $c_i(a)$ arise from uniquely determined sections c_i of the structure sheaf of $W(\mathfrak{g})$, that is, from elements of the symmetric algebra

$$A = \mathrm{Sym}_k(\mathfrak{g}^\vee),$$

where $\mathfrak{g}^\vee$ is the dual of the k-module $\mathfrak{g}$. When k is infinite, the c_i are determined by the corresponding polynomial functions $\mathfrak{g} \to k$, but this is no longer true when k is finite.

Let r be the largest integer such that the Killing polynomial

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t]$$

is divisible by t^r; thus

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_{n-r} t^r, \qquad c_{n-r} \neq 0.$$

The integer r is called the *nilpotent rank* of the Lie algebra $\mathfrak{g}$. It is invariant under extension of the base field.

[8]Editor's note: $W \otimes_k A$ has been corrected to $\mathfrak{g} \otimes_k A$.

Proposition 0.4.2. *Let r be the nilpotent rank of $\mathfrak{g}$, and let $a \in \mathfrak{g}$. Then*

$$\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g}) \geq r,$$

with equality if and only if

$$c_{n-r}(a) \neq 0.$$

In this case $\operatorname{Nil}(a, \mathfrak{g})$ *is a nilpotent Lie algebra. We shall see the converse in 5.7 b) when* $\mathfrak{g}$ *is the Lie algebra of an algebraic group G smooth over k.*

The first assertion is immediate, since by definition

$$\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g}) = \text{the multiplicity of the zero root in } P_{\mathfrak{g}}(a, t).$$

Suppose now that $c_{n-r}(a) \neq 0$. We prove that $\operatorname{Nil}(a)$ is nilpotent, which is equivalent to saying that $\operatorname{ad}_{\operatorname{Nil}(a)}(x)$ is a nilpotent endomorphism for every $x \in \operatorname{Nil}(a)$. We may suppose that k is algebraically closed. Since

$$\operatorname{ad}(a)_{\mathfrak{g}/\operatorname{Nil}(a)}$$

is injective, there is a nonempty open subset $U \subset W(\operatorname{Nil}(a))$ such that, for every $x \in U(k)$, the endomorphism $\operatorname{ad}(x)_{\mathfrak{g}/\operatorname{Nil}(a)}$ is injective and hence $\operatorname{Nil}(x) \subset \operatorname{Nil}(a)$. We may further suppose that U is contained in the open subset on which c_{n-r} does not vanish; that open subset is nonempty because $c_{n-r}(a) \neq 0$. Then $\operatorname{Nil}(x)$ and $\operatorname{Nil}(a)$ have the same dimension, so

$$\operatorname{Nil}(x) = \operatorname{Nil}(a).$$

It follows that $\operatorname{ad}(x)_{\operatorname{Nil}(a)}$ is nilpotent for every $x \in U(k)$. By the principle of extension of algebraic identities, this remains true for every $x \in \operatorname{Nil}(a)$, and hence $\operatorname{Nil}(a)$ is nilpotent.

An element $a \in \mathfrak{g}$ is called *regular* if $c_{n-r}(a) \neq 0$, that is, if

$$\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g}) = r.$$

When k is infinite, this also says that $\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g})$ is as small as possible as a varies in $\mathfrak{g}$. In every case, the notion of a regular element of $\mathfrak{g}$ is invariant under extension of the base field. The set of regular points of $W(\mathfrak{g})$, meaning those that arise from regular points with values in a suitable extension of k, is open, since it is

$$W(\mathfrak{g})_{c_{n-r}},$$

the set of points at which c_{n-r} is invertible.

Corollary 0.4.3. *Let a be a regular element of $\mathfrak{g}$, and let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ containing a. Then $\mathfrak{h}$ is nilpotent if and only if*

$$\mathfrak{h} \subset \operatorname{Nil}(a, \mathfrak{g}).$$

In particular, $\operatorname{Nil}(a, \mathfrak{g})$ *is a maximal nilpotent subalgebra of* $\mathfrak{g}$.

Since $\operatorname{Nil}(a)$ is nilpotent, the inclusion $\mathfrak{h} \subset \operatorname{Nil}(a)$ implies that $\mathfrak{h}$ is nilpotent. Conversely, if $\mathfrak{h}$ is nilpotent, it is contained in the nil-space of its element a; that is, $\mathfrak{h} \subset \operatorname{Nil}(a)$.

Proposition 0.4.4. *Suppose that k is infinite. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. Consider the following conditions:*

(i) $\mathfrak{d}$ is maximal nilpotent and contains a regular element of $\mathfrak{g}$.

(i bis) $\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$ *for some regular element* a *of* $\mathfrak{g}$.

(ii) $\mathfrak{d}$ *is nilpotent and* $\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$ *for some* $a \in \mathfrak{g}$.

(ii bis) $\mathfrak{d}$ *is nilpotent, and there exists* $a \in \mathfrak{d}$ *such that* $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ *is injective.*

(iii) $\mathfrak{d}$ *is nilpotent and equal to its own normalizer.*

Then

$$\text{(i)} \iff \text{(i bis)} \implies \text{(ii)} \iff \text{(ii bis)} \iff \text{(iii)}.$$

We shall see in 5.7 a) that if $\mathfrak{g}$ *is the Lie algebra of a smooth algebraic group, then all the preceding conditions are equivalent.*

The equivalence of (i) and (i bis) follows immediately from 4.3, and these conditions plainly imply (ii). The equivalence of (ii) and (ii bis) is also immediate, as is (ii bis) $\Rightarrow$ (iii) (cf. 4.1). It remains to prove (iii) $\Rightarrow$ (ii bis). This is the only implication that uses the infinitude of k, and it follows at once from the next lemma.

Lemma 0.4.5. *Let* $\mathfrak{d}$ *be a nilpotent Lie algebra over an infinite field* k*, acting on a finite-dimensional vector space* V*. Suppose that the endomorphism* $u(x)$ *is not injective for every* $x \in \mathfrak{d}$*. Then there is a nonzero element* $v \in V$ *annihilated by* $\mathfrak{d}$*.*

We may suppose that k is algebraically closed and that $\mathfrak{d}$ is finite-dimensional. Then V is known to be a direct sum of finitely many nonzero stable vector subspaces V_i, $1 \leq i \leq n$, such that, for every i and every $x \in \mathfrak{d}$, the restriction $u(x)|_{V_i}$ has a single eigenvalue $\lambda_i(x)$ (cf. Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Exercise 22). Let $c_i(x)$ be the constant term of the characteristic polynomial of $u(x)|_{V_i}$, so that

$$\lambda_i(x) = 0 \quad \iff \quad c_i(x) = 0.$$

Each c_i is a polynomial function on $\mathfrak{d}$, and the hypothesis says that $\mathfrak{d}$ is the union of their zero sets. Thus one of the c_i is identically zero. Replacing V by V_i, we are reduced to the case in which all $u(x)$, $x \in \mathfrak{d}$, are nilpotent. Engel's theorem (Bourbaki, loc. cit., Th. 1) then implies that some nonzero $v \in V$ is annihilated by $\mathfrak{d}$.

Q.E.D.

It is easy to see, still assuming that k is infinite, that conditions (i) and (i bis) of 4.4 are invariant under every extension of the base field. When they hold, $\mathfrak{d}$ is called a *Cartan subalgebra* of $\mathfrak{g}$. In general, with k not necessarily infinite, $\mathfrak{d}$ is called a Cartan subalgebra of $\mathfrak{g}$ if it becomes one after one, and hence every, extension $k \to k'$ with k' infinite. It follows in particular that $\mathfrak{d}$ is nilpotent and equal to its own normalizer.

Proposition 0.4.6. a) *Let* $a \in \mathfrak{g}$*. If* a *is regular, then it is contained in one and only one Cartan subalgebra of* $\mathfrak{g}$*. We shall see the converse in 6.1 d), when* k *is algebraically closed and* $\mathfrak{g}$ *is the Lie algebra of a smooth algebraic group.*

b) *Let* $\mathfrak{d}$ *be a Cartan subalgebra of* $\mathfrak{g}$*, and let* $a \in \mathfrak{d}$*. Then* a *is regular in* $\mathfrak{g}$ *if and only if* $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ *is injective.*

For a), if a is regular, then $\mathrm{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$, since this is true after an infinite extension k' of k. It follows at once from 4.3 that every Cartan subalgebra of $\mathfrak{g}$ containing a is equal to this one. For b), observe that the nullity[9] of $\mathrm{ad}(a)_{\mathfrak{g}}$ is the sum of the nullities of $\mathrm{ad}(a)_{\mathfrak{d}}$ and $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$. Since the first is r, their sum is r if and only if $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.

Q.E.D.

[9] Editor's note: that is, the dimension of the nil-space.

Corollary 0.4.7. *Let a be a regular element of $\mathfrak{g}$, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$ containing a, let A be a k-algebra, and let $\mathfrak{g}_A$ and $\mathfrak{d}_A \subset \mathfrak{g}_A$ be the A-Lie algebras obtained from $\mathfrak{g}$ and $\mathfrak{d}$ by base change; write a_A for the image of a in $\mathfrak{g}_A$. Let u be an automorphism of $\mathfrak{g}_A$. Then*

$$u(\mathfrak{d}_A) = \mathfrak{d}_A$$

if and only if

$$u(a_A) \in \mathfrak{d}_A.$$

The condition is plainly necessary; let us prove that it is sufficient. Suppose it holds. Then

$$\mathfrak{d}' = u^{-1}(\mathfrak{d}_A)$$

is a Lie subalgebra containing a_A, and every $b \in \mathfrak{d}'$ is such that $\mathrm{ad}(b)_{\mathfrak{d}'}$ is nilpotent. Indeed, $\mathfrak{d}'$ is isomorphic to $\mathfrak{d}_A$, which has this property, as follows immediately from the definition of "nilpotent" in Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Def. 1. Taking $b = a_A$, we see that $\mathrm{Nil}(b, \mathfrak{g}_A)$ contains $\mathfrak{d}'$. On the other hand,

$$\mathrm{Nil}(b, \mathfrak{g}_A) = \mathfrak{d}_A.$$

Moreover, $\mathfrak{d}'$ is locally a direct summand of $\mathfrak{g}_A$, since $\mathfrak{d}$ is, and hence is locally a direct summand of $\mathfrak{d}_A$. It is a projective module of the same rank r as $\mathfrak{d}_A$, and therefore equals $\mathfrak{d}_A$.

Q.E.D.

Proposition 0.4.8. *Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$.*

a) *If k is infinite, the following conditions are equivalent:*

(i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.

(ii) $\mathfrak{h}$ contains a regular element a of $\mathfrak{g}$, and an element b such that $\mathrm{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ is injective.

(iii) $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$, and contains a regular element of $\mathfrak{g}$.

These conditions are invariant under extension of the base field k.

b) *Suppose that these conditions hold after a suitable infinite extension k' of k. For $a \in \mathfrak{h}$, the element a is regular in $\mathfrak{g}$ if and only if it is regular in $\mathfrak{h}$ and $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective; equivalently, if and only if*

$$\mathrm{Nil}(a, \mathfrak{h}) = \mathrm{Nil}(a, \mathfrak{g})$$

and this common Lie algebra is a Cartan subalgebra of $\mathfrak{h}$.

c) *Under the hypotheses of b), let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$. In order that $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{h}$, it is sufficient that it be a Cartan subalgebra of $\mathfrak{g}$. We shall see in 5.8 that this condition is also necessary when $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G, and $\mathfrak{h}$ is the Lie algebra of a smooth algebraic subgroup H of G.*

Conditions (ii) and (iii) of a) are visibly invariant under extension of the base field k, assumed infinite. In b) and c), we may suppose that k is infinite, and shall do so. If $\mathfrak{h}$ contains the Cartan subalgebra

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}),$$

then a is a regular element of $\mathfrak{g}$ such that $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective. Thus (i) $\Rightarrow$ (ii). Conversely, if (ii) holds, then a "sufficiently general" element a of $\mathfrak{h}$ simultaneously satisfies both conditions in (ii). Hence $\mathrm{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ contained in $\mathfrak{h}$, proving (i). Conditions (i) and (ii) are therefore equivalent.

Suppose they hold, and let a vary in $\mathfrak{h}$. Then

$$\text{(x)} \qquad \operatorname{rank}_k \operatorname{Nil}(a,\mathfrak{g}) = \operatorname{rank}_k \operatorname{Nil}(a,\mathfrak{h}) + \operatorname{rank}_k \operatorname{Nil}(\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{h}}).$$

The two terms on the right are respectively at least r', the nilpotent rank of $\mathfrak{h}$, and at least 0; moreover, both equalities are attained[10] for a sufficiently general element of $\mathfrak{h}$. We also have

$$\operatorname{rank}_k \operatorname{Nil}(a,\mathfrak{g}) \geq r,$$

where r is the nilpotent rank of $\mathfrak{g}$. Equality is attained for a sufficiently general element of $\mathfrak{h}$, and holds if and only if a is regular in $\mathfrak{g}$.

It follows that $r = r'$, and that a is regular if and only if the two terms on the right-hand side of (x) are respectively r' and 0; that is, if and only if a is regular in $\mathfrak{h}$ and $\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective. This proves b), and c) follows immediately by taking an element $a \in \mathfrak{d}$ regular in $\mathfrak{g}$, so that $\operatorname{Nil}(a,\mathfrak{g}) = \mathfrak{d}$.

The preceding result also proves (i) $\Rightarrow$ (iii). Finally, (iii) $\Rightarrow$ (i): under (iii), a sufficiently general $a \in \mathfrak{h}$ is regular both in $\mathfrak{h}$ and in $\mathfrak{g}$. Thus

$$\operatorname{Nil}(a,\mathfrak{h}) \subset \operatorname{Nil}(a,\mathfrak{g})$$

are respectively Cartan subalgebras of $\mathfrak{h}$ and $\mathfrak{g}$. Since they have the same rank over k, they are equal, proving (i). This completes the proof of 4.8.

0.5 The case of the Lie algebra of a smooth algebraic group: the density theorem

Let G be a smooth algebraic group over the field k, and let $\mathfrak{g}$ be its Lie algebra. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Let G act on $W(\mathfrak{g})$ through the adjoint representation, and consider the subscheme $W(\mathfrak{h})$. The construction of Section 1 leads us to introduce

$$N = \operatorname{Norm}_G(\mathfrak{h}) = \operatorname{Norm}_G(W(\mathfrak{h})),$$

which is an algebraic subgroup of G (not necessarily smooth), and

$$\mathfrak{n} = \operatorname{Lie}(N),$$

together with the scheme

$$X = G \times^N W(\mathfrak{h}),$$

the quotient of $\widetilde{X} = G \times W(\mathfrak{h})$ by the right action of N given by

$$(g,x) \cdot n = (gn, \operatorname{Ad}(n^{-1})x).$$

We consider the canonical morphisms

$$\begin{array}{ccc} G \times W(\mathfrak{h}) & & \\ \downarrow{\scriptstyle q} & \searrow{\scriptstyle \varphi} & \\ X & \xrightarrow{\psi} & W(\mathfrak{g}). \end{array}$$

[10] Editor's note: "the equality" has been replaced by "the equalities".

Theorem 0.5.1. *With the preceding notation, suppose that k is infinite. Consider the following conditions:*

(i) $\mathfrak{h}$ *contains a Cartan subalgebra* $\mathfrak{d}$ *of* $\mathfrak{g}$.

(ii) There exists $a \in \mathfrak{h}$ *such that* $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ *is injective.*

(iii) The morphism $\varphi : G \times W(\mathfrak{h}) \to W(\mathfrak{g})$ *is generically smooth.*

(iv) The preceding morphism φ *(or, equivalently,* $\psi : X \to W(\mathfrak{g})$*) is dominant, and* $\mathfrak{h}$ *has the same nilpotent rank as* $\mathfrak{g}$.

(v) The morphism $\psi : X \to W(\mathfrak{g})$ *is generically smooth and* $\mathfrak{h} = \mathfrak{n}$.

(vi) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant, and* $\mathfrak{h} = \mathfrak{n}$.

(vii) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant.*

(viii) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant, and* N *is smooth.*

(ix) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant,* $\mathfrak{h} = \mathfrak{n}$*, and* $\mathfrak{h}$ *is the Lie algebra of a smooth algebraic subgroup* H *of* G.

(x) The morphism $\psi : X \to W(\mathfrak{g})$ *is generically quasi-finite, and* $\mathfrak{n} = \mathfrak{h}$.

(xi) The morphism $\psi : X \to W(\mathfrak{g})$ *is generically étale, and* $\mathfrak{n} = \mathfrak{h}$.

(xii) There exists a smooth algebraic subgroup H *of* G*, with Lie algebra* $\mathfrak{h}$*, such that* $\mathfrak{h}$ *contains a Cartan subalgebra* $\mathfrak{d}$ *of* $\mathfrak{g}$.

(xiii) There exists $a \in \mathfrak{h}$ *such that* N *and the transporter* M_a *of* a *into* $\mathfrak{h}$ *(cf. Section 1) coincide in a neighborhood of* e*, and* $\mathfrak{n} = \mathfrak{h}$.

One then has the following diagram of implications:

$$\begin{array}{ccccccccccc} \text{(xi)} & \Longleftrightarrow & \text{(xii)} & \Longleftrightarrow & \text{(xiii)} & \Longrightarrow & \text{(viii)} & \Longleftrightarrow & \text{(ix)} & \Longleftrightarrow & \text{(x)} \\ & & \Downarrow & & & & & & \Downarrow & & \\ \text{(i)} & \Longleftrightarrow & \text{(ii)} & \Longleftrightarrow & \text{(iii)} & \Longleftrightarrow & \text{(iv)} & \Longrightarrow & \text{(v)} & \Longrightarrow & \text{(vi)} \Longrightarrow \text{(vii)}. \end{array}$$

When k has characteristic zero, all the conditions under consideration are equivalent. Finally,

$$\text{(xi)} \Longleftrightarrow [\text{(i) } \textit{and } \text{(viii)}] \Longleftrightarrow [\text{(v) } \textit{and } \text{(viii)}].$$

Let us first record the trivial implications

$$\text{(v)} \Rightarrow \text{(vi)} \Rightarrow \text{(vii)}, \qquad \text{(ix)} \Rightarrow \text{(vi)}, \qquad \text{(xi)} \Longleftrightarrow [\text{(x) and (v)}].$$

We prove the equivalence of conditions (i) through (iv), and that they imply (v). The implication (i)⇒(ii) is immediate. On the other hand, when k is algebraically closed, condition (iii) says that there is a k-rational point of $G \times W(\mathfrak{h})$ at which the tangent map of φ is surjective. It is clear that, after translating by a suitable element of $G(k)$, this point may be taken to have the form (e, a), with $a \in \mathfrak{h}$. It follows that when k is infinite, though not necessarily algebraically closed, this condition—which is plainly sufficient for generic smoothness—is also necessary.

The tangent map is readily computed. Upon identifying the tangent space of $W(\mathfrak{h})$ at a with $\mathfrak{h}$, it is the map

$$(\xi, x) \longmapsto [\xi, a] + x$$

from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Its surjectivity is equivalent to the surjectivity of $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$, or, what amounts to the same thing, to its injectivity. This proves the equivalence of (ii) and (iii). Moreover, (ii) plainly implies $\mathfrak{h} = \mathfrak{n}$, and (iii) implies that ψ is generically smooth: if φ is smooth at a point u, then, since q is flat, ψ is smooth at $q(u)$. Thus (ii) and (iii) imply (v).

We now show that they imply (i). Since ψ is dominant and the set of regular points of $\mathfrak{g}$ is dense and open, $\mathfrak{h}$ contains regular elements of $\mathfrak{g}$. Hence a sufficiently general element b of $\mathfrak{h}$ is regular in $\mathfrak{g}$ and makes $\mathrm{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ injective. Consequently $\mathrm{Nil}(b, \mathfrak{g}) \subset \mathfrak{h}$, so $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = \mathrm{Nil}(b, \mathfrak{g})$. Conditions (i), (ii), and (iii) are therefore equivalent. Finally, (i)$\Longleftrightarrow$(iv): we have already observed that if $\mathfrak{h}$ contains a Cartan subalgebra, then it has the same nilpotent rank as $\mathfrak{g}$ (4.6), so (i) implies (iv). Conversely, if (iv) holds, then $\mathfrak{h}$ contains a regular element of $\mathfrak{g}$, and since it has the same rank as $\mathfrak{g}$, it contains a Cartan subalgebra by 4.6.

We next prove the equivalence of conditions (viii) through (x). We first note the following facts.

Lemma 0.5.2. *a) If N is smooth, then*

$$\dim X \leq \dim G,$$

with equality if and only if $\mathfrak{h} = \mathfrak{n}$.

b) If $\mathfrak{h} = \mathfrak{n}$, then

$$\dim X \geq \dim G,$$

with equality if and only if N is smooth.

These assertions follow at once from the formula

$$\dim X = \dim G - \dim N + \dim_k \mathfrak{h}$$

and from the fact that $\dim N = \dim_k \mathfrak{n}$ is equivalent to the smoothness of N.

Now (viii) implies (x): by dominance, (viii) gives $\dim X \geq \dim G$, so Lemma 5.2(a) gives equality and $\mathfrak{h} = \mathfrak{n}$, hence (x). The same argument in reverse, using Lemma 5.2(b), shows that (x) implies (viii). Moreover, (viii) implies (ix), since it implies $\mathfrak{h} = \mathfrak{n}$, so $\mathfrak{h}$ is the Lie algebra of the smooth algebraic subgroup N of G. Conversely, (ix) implies (viii): since H normalizes its Lie algebra $\mathfrak{h}$, it is contained in N; and since H is smooth and has the same Lie algebra as N, it follows that N is smooth.

Finally, we prove the equivalence of (xi), (xii), and (xiii), and show that they imply (iii); this will complete the implication diagram. We have (xi)$\Longleftrightarrow$(xiii). Indeed, if $\mathfrak{n} = \mathfrak{h}$, then Lemma 5.2(b) gives $\dim X \geq \dim G$. Since $W(\mathfrak{g})$ is normal, condition (xi) is therefore equivalent to ψ being generically unramified, and this in turn is equivalent to (xiii) by 1.1, (ii)$\Longleftrightarrow$(iii), arguing as above in the proof of (ii)$\Longleftrightarrow$(iii).

Since (xi)$\Rightarrow$(x)$\Rightarrow$(viii), what we have already proved shows that (xi) implies that N is smooth. Thus $q : G \times W(\mathfrak{h}) \to X$ is smooth, and the composite $\varphi = \psi \circ q$ is generically smooth; this is (iii). Since (iii) implies (i), it also follows that (xi)$\Rightarrow$(xii).[11] Finally, (xii) implies (xi). Indeed, (xii) plainly implies (i), and we have seen that (i)$\Rightarrow$(iii)$\Rightarrow$(v); hence

[11] Editor's note: because $H = N$ will do.

(xii) implies (v). It also implies (ix), and (ix) implies (x). Thus (xii) implies both (v) and (x), and therefore (xi), since

$$\text{generically étale} = \text{generically smooth} + \text{generically quasi-finite}.$$

If k has characteristic zero, then (vii) implies (viii), because Cartier's theorem makes N automatically smooth ($\mathrm{VI_B}$ 1.6.1). Moreover, [(viii) and (x)] implies (xi), because in characteristic zero, for a morphism of integral preschemes,

$$\text{generically étale} = \text{dominant and generically quasi-finite}.$$

Thus in this case all conditions (i) through (xiii) are equivalent.

Corollary 0.5.3. *Under the equivalent conditions (viii) through (x), there exists a unique smooth connected algebraic subgroup H of G whose Lie algebra is $\mathfrak{h}$, and*

$$\mathrm{Norm}_G(H) = \mathrm{Norm}_G(\mathfrak{h}) = N, \qquad H = N^0.$$

Indeed, $H = N^0$ has the required properties. Conversely, if H has these properties, then $H \subset N$, since H normalizes its Lie algebra $\mathfrak{h}$. This is an inclusion of smooth groups with the same Lie algebra, and H is connected, so $H = N^0$. To prove $\mathrm{Norm}_G(H) = \mathrm{Norm}_G(\mathfrak{h})$, we may assume that k is algebraically closed. What we have just proved then shows immediately that the two groups have the same k-points. On the other hand, the inclusions

$$H \subset \mathrm{Norm}_G(H) \subset N$$

show that $\mathrm{Norm}_G(H)$ and N have the same Lie algebra; hence they are equal.

Corollary 0.5.4. *Under the equivalent conditions (i) through (iv), let $a \in \mathfrak{h}$. Then the following conditions are equivalent, and they hold whenever a is regular in $\mathfrak{g}$:*

(i) *φ is smooth at (e, a).*

(ii) *$M_a = \mathrm{Transp}_G(a, \mathfrak{h})$ is smooth at e, and $\dim_e(M_a) = \mathrm{rank}_k \mathfrak{h}$.*

(iii) *$\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (equivalently, bijective).*

Under the equivalent conditions (xi) through (xiii), let H be the algebraic subgroup of G considered in Corollary 5.3. The preceding conditions are also equivalent to:

(iv) *ψ is étale at $(\overline{e}, a)$.*

(v) *If M_a^0 denotes the connected component of e in the transporter M_a of a into $\mathfrak{h}$, endowed with the structure induced by M_a, then*

$$H = M_a^0.$$

Plainly (i) implies (ii), since M_a is isomorphic to the fiber $\varphi^{-1}(a)$, with e corresponding to (e, a). Conversely, (ii) implies (i): condition (ii) says that φ is "equidimensional" at (e, a), in the sense that the dimension of the fiber through this point is the dimension of the generic fiber. Since $G \times W(\mathfrak{h})$ and $W(\mathfrak{g})$ are regular, this implies that φ is flat at (e, a), hence smooth there because its fiber is smooth at that point. The equivalence of (i) and (iii) was obtained in the proof of Theorem 5.1 by the elementary calculation of the tangent map. We also saw in 4.8(b) that regularity of a in $\mathfrak{g}$ implies (iii).

Under conditions (xi) through (xiii), the morphism $q : G \times W(\mathfrak{h}) \to X$ is smooth, since N is smooth. Thus (i) is equivalent to the smoothness of ψ at $(\overline{e}, a)$; and since ψ is generically étale, this is equivalent to (iv). Finally, as was noted at the end of the proof of 1.1, condition (iv) implies that N is the prescheme induced on M_a by an open and closed subset of M_a, which gives (v). Conversely, (v) trivially implies (ii). Alternatively, (v) implies (iv) by 1.1, since ψ is dominant and $W(\mathfrak{g})$ is normal, and "unramified" is equivalent here to "étale". This completes the proof of Corollary 5.4.

Corollary 0.5.5. *Let G be a smooth algebraic group over a field k, and let H be a smooth algebraic subgroup such that its Lie algebra $\mathfrak{h}$ contains, at least after a suitable extension of k, a Cartan subalgebra of the Lie algebra $\mathfrak{g}$ of G. Let K be a connected algebraic subgroup of G, not necessarily smooth, with Lie algebra $\mathfrak{k}$, and suppose that $\mathfrak{k}$ contains a regular element a of $\mathfrak{g}$, again at least after a suitable extension of k. Then H contains K if and only if $\mathfrak{h}$ contains $\mathfrak{k}$.*

Indeed, Corollary 5.4, (iii)⇒(v), applied to the connected subgroup supplied by Corollary 5.3, gives $H^0 = M_a^0$. This identity is invariant under extension of the base field, so it remains valid without assuming that the base field is infinite. Moreover, $\mathfrak{k} \subset \mathfrak{h}$ plainly implies $K \subset M_a$. Since K is connected, $K \subset M_a^0 \subset H$. □

Corollary 0.5.6. *Let G and H be as in Corollary 5.5, and let K be an algebraic subgroup of G. Suppose that H is connected and K is smooth. Then K contains H if and only if $\mathfrak{k}$ contains $\mathfrak{h}$.*

Indeed, if $\mathfrak{k} \supset \mathfrak{h}$, then K satisfies, with respect to H, the hypothesis imposed on H in Corollary 5.5. Meanwhile H plainly satisfies, with respect to K, the condition imposed on K there. The conclusion follows from Corollary 5.5.

Corollary 0.5.7. *Let $\mathfrak{g}$ be the Lie algebra of a smooth algebraic group G over a field k. Then:*

a) A Lie subalgebra $\mathfrak{d}$ of $\mathfrak{g}$ is a Cartan subalgebra if and only if it is nilpotent and equal to its own normalizer.

b) An element $a \in \mathfrak{g}$ is regular if and only if $\mathrm{Nil}(a, \mathfrak{g})$ *is nilpotent.*

After extending the base field, we may assume that k is infinite. In view of 4.4, to prove (a) it remains only to show that if $\mathfrak{d}$ is nilpotent and contains an element a for which $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective, then $\mathfrak{d}$ is a Cartan subalgebra. By Theorem 5.1, (ii)⇒(i), $\mathfrak{d}$ contains a Cartan subalgebra

$$\mathfrak{d}' = \mathrm{Nil}(a, \mathfrak{g}).$$

Since $\mathfrak{d}$ is nilpotent, 4.3 now gives $\mathfrak{d} = \mathfrak{d}'$. For (b), one notes that $\mathrm{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ by (a), and therefore a is regular.

Corollary 0.5.8. *Let G be a smooth algebraic group over a field k, and let H be a smooth algebraic subgroup, with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Suppose that after a suitable extension of the base field, $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$. Then $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$ if and only if it is a Cartan subalgebra of $\mathfrak{g}$.*

In view of 4.8(c), it remains to show that if $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$, then it is one of $\mathfrak{g}$. It is enough to show that $\mathfrak{d}$ contains an element a regular in $\mathfrak{g}$, and we may assume that k is algebraically closed. But there is a dense open subset of $\mathfrak{h}$ consisting of points regular in $\mathfrak{g}$, so the assertion follows from Theorem 5.1, (i)⇒(vii), applied to $(\mathfrak{h}, \mathfrak{d})$.

0.6 Cartan subalgebras and subgroups of type (C) associated with a smooth algebraic group

For simplicity, in the following theorem we restrict ourselves to an algebraically closed base field; the case of an arbitrary base prescheme will be treated in the next Exposé.

Theorem 0.6.1. *Let G be a smooth algebraic group over an algebraically closed field k, and let $\mathfrak{g}$ be its Lie algebra. Then:*

a) The Cartan subalgebras of $\mathfrak{g}$ are conjugate.

b) Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Its normalizer N in G is smooth, and $D = N^0$ is the unique smooth connected subgroup of G whose Lie algebra is $\mathfrak{d}$. One has

$$\mathrm{Norm}_G(\mathfrak{d}) = \mathrm{Norm}_G(D) = N, \qquad \textit{and hence} \qquad D = \mathrm{Norm}_G(D)^0.$$

c) With $\mathfrak{d}$ as in (b), set, as in Section 5,

$$X = G \times^N W(\mathfrak{d}),$$

and consider the canonical morphism

$$\psi : X \longrightarrow W(\mathfrak{g}).$$

Its fiber over $a \in \mathfrak{g}$ has as its k-valued points the set of Cartan subalgebras of $\mathfrak{g}$ containing a. Then ψ is birational.

d) With the notation of (c), let U be the largest open subset of $W(\mathfrak{g})$ for which ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Then, for $a \in \mathfrak{g}$, the following conditions are equivalent:

(i) $a \in U(k)$.

(ii) The element a belongs to one and only one Cartan subalgebra of $\mathfrak{g}$.

(iii) The set of Cartan subalgebras of $\mathfrak{g}$ containing a is finite and nonempty.

(iv) The fiber $\psi^{-1}(a)$ has an isolated point.

(v) If $a \in \mathfrak{d}$, the morphism ψ is étale—or merely quasi-finite—at the point $(\overline{e}, a)$.

(vi) The element a is regular in $\mathfrak{g}$.

(vii) With the notation of Corollary 5.4(iii), one has $N = M_a$.

Proof. Apply Theorem 5.1 with the Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$ in place of $\mathfrak{h}$. We show that the strongest conditions, (xi) through (xiii), hold. This is clear either in the form (xiii), by 4.7—which says that if a regular element $a \in \mathfrak{g}$ lies in $\mathfrak{d}$, then $N = M_a$—or in the form

$$(\mathrm{xi}) = (\mathrm{x}) + (\mathrm{i}).$$

Indeed, (i) is automatic, while (x) follows from the fact that a regular point of $\mathfrak{g}$ belongs to a unique Cartan subalgebra of $\mathfrak{g}$ (4.6(a)), and therefore, a fortiori, to a unique conjugate of $\mathfrak{d}$.

Part (b) now follows from Corollary 5.3. Part (c) follows because ψ is generically étale and because a sufficiently general point of $\mathfrak{g}$—more precisely, a regular point—belongs to

a unique Cartan subalgebra of $\mathfrak{g}$. Under these conditions, the equivalence of conditions (i) through (v) on a follows immediately from Zariski's Main Theorem, applied to the separated birational morphism ψ, since $W(\mathfrak{g})$ is normal and X is integral. The equivalence of (v) and (vi) is the special case of Corollary 5.4, (iii)$\Longleftrightarrow$(iv), obtained after carrying a into $\mathfrak{d}$ by a suitable element $g \in G(k)$, in view of 4.6(b). Moreover, Corollary 5.4 shows that (v) and (vi) are also equivalent to

$$N \supset M_a^0.$$

By 4.7 this even implies $N = M_a$. This proves (d).

Of course, (b), (c), and the equivalence of (i), (iv), (v), (vi), and (vii) remain valid over an arbitrary base field. We now prove (a), using the assumption that k is algebraically closed. By Theorem 5.1, (i)$\Rightarrow$(vii), the morphism $\psi : X \to W(\mathfrak{g})$ is dominant. Hence there is a dense open subset V of $W(\mathfrak{g})$ such that every $a \in V(k)$ is the image of a point of $X(k)$, or equivalently, belongs to a conjugate of $\mathfrak{d}$. Apply the same result to a second Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}$. We may choose V so that every element of $V(k)$ is conjugate both to an element of $\mathfrak{d}$ and to an element of $\mathfrak{d}'$. Choose a regular element of $V(k)$. Some conjugate $\mathfrak{d}''$ of $\mathfrak{d}'$ then contains an element of $\mathfrak{d}$ regular in $\mathfrak{g}$, and 4.6(a) gives $\mathfrak{d}'' = \mathfrak{d}$. This proves (a) and completes the proof. □

Definition 0.6.2. Let G be a smooth algebraic group over a field k. A *subgroup of type* (C) of G is a smooth connected algebraic subgroup whose Lie algebra is a Cartan subalgebra of $\mathfrak{g}$. The *infinitesimal rank* of G is the nilpotent rank of its Lie algebra $\mathfrak{g}$; it is equal to the dimension of every subgroup of type (C) of G.

Theorem 6.1(b) immediately gives:

Corollary 0.6.3. *Under the hypotheses of Definition 6.2, the map*

$$D \longmapsto \mathfrak{d} = \mathrm{Lie}(D)$$

establishes a bijection between the subgroups of type (C) *of* G *and the Cartan subalgebras* $\mathfrak{d}$ *of* $\mathfrak{g}$. *If* D *is a subgroup of type* (C) *of* G, *then* D *is its own connected normalizer:*

$$D = \mathrm{Norm}_G(D)^0.$$

Combining Theorem 6.1(a) and Corollary 6.3 gives:

Corollary 0.6.4. *If* k *is algebraically closed, the subgroups of type* (C) *of* G *are conjugate to one another.*

Corollary 0.6.5. *Let* G *be a smooth algebraic group over an algebraically closed field* k, *and let* H *be a smooth algebraic subgroup of* G, *with Lie algebras* $\mathfrak{g}$ *and* $\mathfrak{h}$, *respectively. Then* $\mathfrak{h}$ *contains a Cartan subalgebra of* $\mathfrak{g}$ *if and only if* H *contains a subgroup* D *of type* (C) *of* G.

The condition is plainly sufficient. It is also necessary, since by Corollary 5.5,

$$H \supset D \quad \Longleftrightarrow \quad \mathfrak{h} \supset \mathfrak{d}.$$

Remarks 0.6.6.

a) The connected subgroups H of G described in Corollary 6.5 correspond bijectively to the Lie subalgebras $\mathfrak{h}$ of $\mathfrak{g}$ satisfying the strongest condition, (xii), of Theorem 5.1. By Corollary 5.5, inclusion relations among such subgroups can already be read from their Lie algebras.

b) Continue to suppose that k is algebraically closed, and let D be a subgroup of type (C) of G. It is then easy to show that D contains a Cartan subgroup C of G. Indeed, put $V = \mathfrak{g}/\mathfrak{d}$. For a general element $a \in \mathfrak{d}$, the endomorphism $\mathrm{ad}(a)$ of V is injective; it suffices to take a regular in $\mathfrak{g}$. From this one readily deduces that, for a general $g \in D(k)$, the action of $\mathrm{Ad}(g)$ on V has no nonzero invariant vector. Thus 2.1, (vii)⇒(i), applies. In fact, in the next Exposé we shall prove the more precise result that every Cartan subgroup C of G is contained in one and only one subgroup of type (C) of G.

c) In general one must not confuse Cartan subgroups with subgroups of type (C). The subgroups of type (C) of G are Cartan subgroups if and only if they are nilpotent, since Cartan subgroups are maximal nilpotent subgroups. It may happen that $\mathfrak{g}$ is nilpotent although G is not; an example is $\mathrm{SL}(2)_k$ when k has characteristic 2. Then the Cartan subalgebras of $\mathfrak{g}$ are all equal to $\mathfrak{g}$. Thus, if G is connected, G is a subgroup of type (C) of itself, but it is not a Cartan subgroup of itself.

On the other hand, we shall see in Exposé XIV that if G is an adjoint semisimple group, its subgroups of type (C) are precisely its Cartan subgroups. The same conclusion holds in characteristic zero for every smooth algebraic group G over k. This follows at once from the fact that, in characteristic zero, a connected algebraic group is nilpotent if and only if its Lie algebra is nilpotent. In turn, this follows from the fact that in characteristic zero the center Z of a connected algebraic group G has Lie algebra equal to the center of $\mathfrak{g}$. Indeed, one obtains a priori the space of invariants in $\mathfrak{g}$ under the adjoint representation of G. Since G is connected and k has characteristic zero, these are also the infinitesimal invariants of the adjoint representation of $\mathfrak{g}$, namely the center of $\mathfrak{g}$. Finally, Cartier's theorem (cf. VI_B 1.6.1) shows that Z is smooth.

Exposé XIV
Regular Elements: Continuation Applications to Algebraic Groups

by A. Grothendieck, with an appendix by J.-P. Serre

0.1 Construction of Cartan subgroups and maximal tori for a smooth algebraic group

Theorem 1.1. *Let G be a smooth algebraic group over a field k. Then G admits a maximal torus T, and hence a Cartan subgroup $C = \mathrm{Centr}_G(T)$.*

By XII 3.2, finding a maximal torus T of G is equivalent to finding a Cartan subgroup C of G. Moreover, since the maximal tori of G are those of G^0, we may suppose that G is connected. We distinguish two cases.

1°) *The field k is finite.* Let $\mathcal{T}$ be the scheme of maximal tori of G (XII 1.10); it is smooth over k. The group G acts on $\mathcal{T}$ through inner automorphisms, and the conjugacy theorem (XII 6.6(a)) says that any two $\bar{k}$-rational points of $\mathcal{T}_{\bar{k}}$ are conjugate under $G_{\bar{k}}(\bar{k})$. Since $\mathcal{T}$ is smooth over k, $\mathcal{T}_{\bar{k}}$ is smooth over $\bar{k}$; it follows that $\mathcal{T}_{\bar{k}}$ is isomorphic to

$$G_{\bar{k}}/N,$$

where N is the stabilizer of an element of $\mathcal{T}_{\bar{k}}(\bar{k})$, that is, the normalizer of a maximal torus $\bar{T}$ of $G_{\bar{k}}$. Consequently $\mathcal{T}$ is a homogeneous space under the action of G. A well-known theorem of Lang (*Amer. J. Math.* **78** (1956), 555–563) says that every homogeneous space under a smooth connected algebraic group over a finite field k has a rational point. In particular, $\mathcal{T}$ has a rational point; equivalently, G has a maximal torus T.

2°) *The field k is infinite.* We shall use the following lemma.

Lemma 1.2. *Let G be a smooth algebraic group over a field k. Then G admits a subgroup of type* (C) (XIII 6.2).

By XIII 6.3, this is equivalent to saying that $\mathfrak{g}$ contains a Cartan subalgebra $\mathfrak{d}$. When k is infinite this is immediate: $\mathfrak{g}$ contains a regular element a, and one takes

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}).$$

The case of finite k is treated exactly as in the proof of 1°) above, but it requires first constructing the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ and proving that this scheme is smooth over k; this will be done below in 2.16.

[0]Version xy of 1 December 2008.

To prove 1.1 in case 2°), which is the case now under consideration, it is therefore enough to know 1.2 when k is infinite. We also record the following result for later use.

Lemma 1.3. *Let G be a smooth connected affine algebraic group whose reductive center* (XII 4.1 and 4.4) *is reduced to the identity group. Then G is nilpotent if and only if its Lie algebra $\mathfrak{g}$ is nilpotent.*

This is contained in XII 4.9.

We can now give a procedure for constructing Cartan subgroups of G. The same procedure applies when k is finite, provided Lemma 1.2 is granted in that case. Suppose first that G is affine. We argue by induction on $n = \dim G$, the assertion being trivial when $n = 0$. Suppose therefore that $n > 0$, and that the assertion has been proved in dimensions $n' < n$. Let Z be the reductive center of G, and let

$$u\colon G \longrightarrow G' = G/Z$$

be the canonical homomorphism. By XII 4.7(c), the rule

$$C' \longmapsto C = u^{-1}(C')$$

is a bijection from the Cartan subgroups of G' to those of G. Replacing G by G', we may thus suppose that the reductive center Z of G is reduced to the identity element (the same is true for the reductive center of G', by XII 4.7(b)).

By Lemma 1.2, G admits a subgroup D of type (C). Over the algebraic closure $\bar{k}$ of k, the group $D_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$ (XIII 6.6(b)); hence every Cartan subgroup of D is a Cartan subgroup of G (XIII 2.8(a)). We are therefore reduced to finding a Cartan subgroup of D.

If $\dim D = \dim G$, that is, if $D = G$, then the Lie algebra of G is a Cartan subalgebra of itself and is therefore nilpotent. Lemma 1.3 then shows that G is nilpotent, so G is a Cartan subgroup of itself. If $\dim D < \dim G$, the induction hypothesis gives a Cartan subgroup of D, which is consequently a Cartan subgroup of G. This proves the assertion when G is affine.

In the general case, let Z be the center of G. Then $G/Z = G'$ is affine (XII 6.1), and the inverse image in G of every Cartan subgroup C' of G' is a Cartan subgroup of G (XII 6.6(e)). We are thus reduced to finding a Cartan subgroup of the smooth affine algebraic group G', which is the case already treated.

Corollary 1.4. *Let G be a smooth group scheme of finite type over an Artinian scheme S. Then G admits a maximal torus T, and hence a Cartan subgroup $C = \mathrm{Centr}_G(T)$. Every torus in G is contained in a maximal torus.*

Indeed, we may suppose that S is local, with residue field k. By 1.1, the group

$$G_0 = G \times_S \mathrm{Spec}(k)$$

admits a maximal torus T_0. By IX 3.6 bis and X 2.3, T_0 comes from a torus T of G, which is plainly maximal. The last assertion follows by applying the preceding result to the centralizer of the given torus, which is smooth over k (XI 2.4).

Remarks 1.5. (a) We shall give below, in 3.20, 3.21, and Exposé XV, variants of 1.4 in which S is not assumed Artinian.

(b) I do not know whether every algebraic group G, not necessarily smooth, over a field k admits a maximal torus. The question arises only in characteristic $p > 0$. Applying 1.1 to

a smooth quotient group of the form $G' = G/I$, where I is a suitable radicial subgroup of G (for example, the kernel of a sufficiently high power of the Frobenius homomorphism), and then taking in G the inverse image of a maximal torus of G', reduces the question to the case in which $(G_{\bar{k}})_{\mathrm{red}}$ is a torus; here, as always, $\bar{k}$ denotes the algebraic closure of k. The answer is easily seen to be affirmative when G is commutative, or more generally nilpotent: in that case G has a unique maximal torus, which may for example be constructed by descent from the maximal torus of $G_{\bar{k}}$.*

(c) When G is affine and either k is perfect or G is solvable, 1.1 was already known, by a theorem of Rosenlicht; his proof is quite different from the one given here.

(d) When k is infinite, 1.1 follows from the much more precise result, proved below in 6.1, that the scheme $\mathcal{T}$ of maximal tori of G is a rational variety. The method essentially combines the proof of 1.1 with an explicit description of the structure of the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$. To obtain the desired result, we must first extend to an arbitrary base prescheme certain results of XIII 4–6 (this is the purpose of the next two sections), and then refine the preceding construction proving 1.1, using the fact that every Cartan subgroup of G is contained in one and only one subgroup of type (C) in G (Nos. 4 and 5).

0.2 Lie algebras over an arbitrary prescheme: regular sections and Cartan subalgebras

Lemma 2.1. *Let A be a ring, let $\mathfrak{d}$ be a Lie algebra over A, and for each $s \in \mathrm{Spec}(A)$ let*

$$\mathfrak{d}(s) = \mathfrak{d} \otimes_A k(s)$$

be the corresponding Lie algebra over the residue field $k(s)$. Suppose that the A-module $\mathfrak{d}$ is of finite presentation. The following conditions are equivalent:

(i) *for every $s \in \mathrm{Spec}(A)$, the Lie algebra $\mathfrak{d}(s)$ is nilpotent;*

(ii) *for every $x \in \mathfrak{d}$, the endomorphism $\mathrm{ad}(x)$ is nilpotent;*

(iii) *there is an integer $N \geq 0$ such that, for every sequence $x_1, \ldots, x_N$ of elements of $\mathfrak{d}$, one has*

$$\mathrm{ad}(x_1)\,\mathrm{ad}(x_2)\cdots\mathrm{ad}(x_N) = 0.$$

When A is a field, the equivalence of (i) and (iii) is the definition of nilpotence, while the equivalence of (ii) and (iii) is the familiar consequence of Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §4, No. 2). In the general case, (iii) plainly implies (ii), and (ii) implies (i) by the preceding result and the fact that (ii) is preserved under passage to quotients and under localization. It remains to prove that (i) implies (iii).

Suppose first that A is local Artinian, with maximal ideal $\mathfrak{m}$. Choose $n > 0$ such that $\mathfrak{m}^n = 0$, and choose N such that condition (iii) holds for

$$\mathfrak{d}(s) = \mathfrak{d} \otimes_A A/\mathfrak{m}.$$

Taking $N' = nN$, one sees at once that N' satisfies (iii) for $\mathfrak{d}$.

If A is Noetherian, there are finitely many points $s_i \in \mathrm{Spec}(A)$ and ideals of definition $\mathfrak{q}_i \subset A_{s_i}$ such that the natural map

$$\mathfrak{d} \longrightarrow \prod_i (\mathfrak{d} \otimes_A A_i), \qquad A_i = A_{s_i}/\mathfrak{q}_i,$$

*M. Raynaud gave a negative answer to the question raised here; see XVII, Example 5.9(c).

is injective. By the preceding case, for each i there is an integer N_i satisfying (iii) for the Lie algebra $\mathfrak{d} \otimes_A A_i$. Taking N to be the largest of the N_i, condition (iii) follows for $\mathfrak{d}$. Finally, the general case reduces to the Noetherian case by the usual procedure described in EGA IV 8.

Definition 2.2. Let S be a prescheme and let $\mathfrak{d}$ be a quasi-coherent Lie algebra over S whose underlying module is of finite presentation. We say that $\mathfrak{d}$ is *locally nilpotent* if, for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ over $k(s)$ is nilpotent. We say that $\mathfrak{d}$ is *strictly locally nilpotent* if it is locally free and, on every open subset U of S over which it has constant rank r, its Killing polynomial reduces to

$$P_{\mathfrak{d}}(t) = t^r.$$

Of course, when $\mathfrak{d}$ is a Lie algebra over S that is locally free as a module, its Killing polynomial is defined as a polynomial

$$P_{\mathfrak{d}} \in A[t], \qquad A = \Gamma\Big(\mathrm{Sym}_{\mathcal{O}_S}(\mathfrak{d})\Big) \simeq \Gamma(W(\mathfrak{d}))\,,$$

where A is the ring of sections of the structure sheaf of the vector bundle $W(\mathfrak{d})$ defined by $\mathfrak{d}$.

The two notions introduced in 2.2 are plainly stable under base change and local for the fpqc topology. We record:

Proposition 2.3. *If $\mathfrak{d}$ is strictly locally nilpotent, then it is locally nilpotent. The converse holds when $\mathfrak{d}$ is locally free and S is reduced.*

The proof is immediate.

Definition 2.4. Let S be a prescheme, let $\mathfrak{g}$ be a Lie algebra over S whose underlying module is locally free of finite type, and let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. We call $\mathfrak{d}$ a *Cartan subalgebra* if it satisfies the following conditions:

(i) $\mathfrak{d}$ is locally a direct-factor submodule (and hence is itself locally free of finite type);

(ii) for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ is a Cartan subalgebra of $\mathfrak{g}(s)$.

Definition 2.5. Let S and $\mathfrak{g}$ be as in 2.4. A section a of $\mathfrak{g}$ is called *quasi-regular* if, for every $s \in S$, the element $a(s) \in \mathfrak{g}(s)$ is regular in the Lie algebra $\mathfrak{g}(s)$ over $k(s)$. We call a a *regular section* if it is quasi-regular and is contained in a Cartan subalgebra of $\mathfrak{g}$.

The notions in 2.4 and 2.5 are again stable under base change and local for the fpqc topology. Only the last assertion, and only for the notion of a regular section, requires proof. It follows from the fact that the Cartan subalgebra containing a given regular section is uniquely determined. More precisely:

Proposition 2.6. *Let S and $\mathfrak{g}$ be as in 2.4, and let a be a quasi-regular section of $\mathfrak{g}$. There is at most one Cartan subalgebra of $\mathfrak{g}$ containing a. Such a Cartan subalgebra exists—that is, a is a regular section—if and only if a satisfies the following condition:*

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$$

is locally a direct-factor submodule of $\mathfrak{g}$, and $\mathrm{ad}(a)$ induces an automorphism of $\mathfrak{g}/\mathfrak{d}$. In that case $\mathfrak{d}$ is the unique Cartan subalgebra of $\mathfrak{g}$ containing a.

Suppose first that a is contained in a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$. Then $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is bijective on every fiber. Since $\mathfrak{g}/\mathfrak{d}$ is locally free of finite type, it follows that this endomorphism is an automorphism of $\mathfrak{g}/\mathfrak{d}$. On the other hand, by 2.1 the endomorphism $\mathrm{ad}(a)_{\mathfrak{d}}$ is locally nilpotent. Hence

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}),$$

which proves both the uniqueness of $\mathfrak{d}$ and the necessity of the stated regularity criterion.

For sufficiency, observe that the hypothesis on a implies that the formation of $\mathrm{Nil}(a, \mathfrak{g})$ commutes with base extension, and in particular with passage to fibers. Thus the fibers $\mathfrak{d}(s)$ of $\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$ are Cartan subalgebras of the $\mathfrak{g}(s)$. Moreover, $\mathfrak{d}$ is a Lie subalgebra of $\mathfrak{g}$ by XIII 4.1. It is therefore a Cartan subalgebra.

Corollary 2.7. *Under the hypotheses of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, let a be a section of $\mathfrak{d}$ that is regular in $\mathfrak{g}$, and let u be an automorphism of $\mathfrak{g}$. Then u preserves $\mathfrak{d}$ if and only if $u(a)$ is a section of $\mathfrak{d}$.*

Indeed, by transport of structure $u(a)$ is then a regular section of $\mathfrak{g}$, contained in both Cartan subalgebras $\mathfrak{d}$ and $u(\mathfrak{d})$; Proposition 2.6 therefore shows that these two subalgebras are equal.

Corollary 2.8. *Under the hypotheses of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. For every $s \in S$ such that $k(s)$ is infinite, there exist an open neighborhood V of s and a regular section a of $\mathfrak{g}$ over V such that*

$$\mathfrak{d}|_V = \mathrm{Nil}(a, \mathfrak{g}|_V)\,;$$

equivalently, a is a section of $\mathfrak{d}|_V$.

Indeed, since $k(s)$ is infinite, there is a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}(s)$. After replacing S by an open neighborhood U of s, we may extend a_0 to a section a of $\mathfrak{d}$ over U. The endomorphism induced by $\mathrm{ad}(a)$ on the fiber

$$\mathfrak{g}(s)/\mathfrak{d}(s)$$

is an automorphism. Shrinking U to a smaller open neighborhood V, we may therefore suppose that $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is an automorphism. It follows that a is a quasi-regular section of $\mathfrak{g}|_V$ satisfying the required condition.

Let $\mathfrak{g}$ be as in 2.4. Inspection of its Killing polynomial immediately shows that the function

$$s \longmapsto \text{the nilpotent rank of } \mathfrak{g}(s)$$

on S is upper semicontinuous. We shall be especially interested in the case in which this function is continuous, that is, locally constant. Here are several variants of this property.

Proposition 2.9. *Let $S, \mathfrak{g}$ be as in 2.4. Consider the following conditions:*

(C_0) *The nilpotent rank of $\mathfrak{g}(s)$, for $s \in S$, is a locally constant function of s.*

(C_1) *Locally for the fpqc topology, there exists a Cartan subalgebra of $\mathfrak{g}$.*

(C'_1) *The same as (C_1), with "fpqc topology" replaced by "étale topology".*

(C_2) *Condition (C_0) holds, and for every S' over S, every quasi-regular section of*

$$\mathfrak{g}_{S'} = \mathfrak{g} \otimes_S S'$$

is regular.

(C_3) *Every $s \in S$ has an open neighborhood V on which the Killing polynomial of $\mathfrak{g}$ has the form*

$$t^r\left(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}\right),$$

where, for every $s \in V$,

$$c_{n-r}(s) \in \mathrm{Sym}\big(\mathfrak{g}(s)^\vee\big)$$

is nonzero.

With this notation, one has the implications

$$(C_3) \Longrightarrow (C_2) \Longrightarrow (C_1') \Longrightarrow (C_1) \Longrightarrow (C_0),$$

and, when S is reduced, the five conditions above are equivalent.

Notice also that these conditions are manifestly stable under base change and are local for the fpqc topology.

The implications

$$(C_1') \Longrightarrow (C_1) \Longrightarrow (C_0)$$

are trivial, and the implication $(C_0) \Longrightarrow (C_3)$ when S is reduced is immediate. We note moreover:

Corollary 2.10. *Suppose that condition (C_0) holds. Let U be the set of elements of $W(\mathfrak{g})$ that are regular in their fiber. Then U is open. In particular, for every section a of $\mathfrak{g}$ over S, the set V of $s \in S$ for which $a(s) \in \mathfrak{g}(s)$ is regular is open.*

Indeed, the first assertion follows from the second, applied to $\mathfrak{g}_{S'}$ for every base change S' over S. For the second, we may suppose that S is reduced; then (C_3) holds, and it is enough to inspect the Killing polynomial of a in $\mathfrak{g}$.

The implication $(C_2) \Longrightarrow (C_1')$ now follows easily from part b) of the following more precise statement.

Corollary 2.11. *Suppose that condition (C_2) holds. Then:*

a) *For every $s \in S$ and every Cartan subalgebra $\mathfrak{d}_0$ of $\mathfrak{g}(s)$ such that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$* (a condition automatically satisfied when $k(s)$ is infinite)*, there exist an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ whose fiber at s is $\mathfrak{d}_0$. If S_1 is a subscheme of S containing s, and if $\mathfrak{d}_0$ has already been extended to a Cartan subalgebra $\mathfrak{d}_1$ of $\mathfrak{g} \otimes_S S_1$, then one can find an open neighborhood V of s in S and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ such that*

$$\mathfrak{d} \otimes_V (S_1 \cap V) = \mathfrak{d}_1|_{S_1 \cap V}.$$

b) *For every $s \in S$ such that $\mathfrak{g}(s)$ contains a regular element* (a condition automatically satisfied when $k(s)$ is infinite)*, there exist an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$.*

Part b) follows from part a) by taking

$$\mathfrak{d}_0 = \mathrm{Nil}(a_0, \mathfrak{g}(s))\,,$$

where a_0 is a regular element of $\mathfrak{g}(s)$. To prove part a), in its second formulation, choose a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$. Extend it near s in S_1 to a section of $\mathfrak{d}_1$, and extend that section in turn near s to a section of $\mathfrak{g}$. By 2.10 this section is quasi-regular on an open neighborhood V of s, hence regular by (C_2). Therefore, by 2.6,

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}|_V)$$

has the required properties.

Remark 2.12. I do not know whether the assertions of 2.11 a) and b) remain valid without the hypothesis that regular points exist, when $k(s)$ is finite.

It remains to prove

$$(C_3) \Longrightarrow (C_2).$$

We also record the following equivalent form of (C_3):

(C_3') Condition (C_0) holds, that is, the nilpotent rank of $\mathfrak{g}(s)$, for $s \in S$, is locally constant; and on every open subset V of S on which this rank has the value r, the Killing polynomial of $\mathfrak{g}|_V$ is divisible by t^r.

We must show that this condition implies that every quasi-regular section a of $\mathfrak{g}$ is regular. In view of 2.6, this follows from the implication (iv) $\Longrightarrow$ (iii) in the following lemma, applied to the endomorphism $\mathrm{ad}(a)$ of $\mathfrak{g}$.

Lemma 2.13. *Let A be a ring, let M be a projective A-module of finite type, and let u be an endomorphism of M. The following conditions are equivalent:*

(i) *M is the direct sum of two u-stable submodules M' and M'', such that $u|_{M'}$ is nilpotent and $u|_{M''}$ is an automorphism of M''.*

(ii) *There exists an integer $n > 0$ such that*

$$\operatorname{Im} u^n + \operatorname{Ker} u^n = M.$$

(iii) *The nilspace*

$$\mathrm{Nil}(u) = \bigcup_{n>0} \operatorname{Ker} u^n$$

is a direct factor of M, and

$$M = \mathrm{Nil}(u) + u(M).$$

These conditions are implied by the following one, and are equivalent to it when A is reduced:

(iv) *Locally on* $\mathrm{Spec}(A)$*, for the Zariski topology, the characteristic polynomial $P_u(t)$ of u can be written in the form*

$$t^r\big(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}\big),$$

where c_{n-r} is invertible.

The equivalence of (i), (ii), and (iii) is immediate and is recalled for reference. When A is reduced, (i) implies (iv) because a nilpotent endomorphism of a projective module of rank r then has characteristic polynomial t^r. On the other hand, in all cases the constant term of the characteristic polynomial of an automorphism of a projective module of finite type is, up to sign, the determinant of u, locally on $\mathrm{Spec}(A)$; it is therefore an invertible element of A.

Finally, to prove (iv) $\Longrightarrow$ (i), regard M as a module over the polynomial ring $A[t]$, with t acting as u. The familiar identity

$$P(u) = 0$$

shows that M is annihilated by $PA[t]$, and hence may be regarded as a module over $A[t]/PA[t]$. Write

$$P = t^r Q,$$

where the constant term of Q is invertible. Then

$$PA[t] = t^r A[t] \cap QA[t],$$

so that $A[t]/PA[t]$ decomposes as the product of the rings

$$A[t]/t^r A[t] \qquad \text{and} \qquad A[t]/QA[t].$$

This gives a corresponding decomposition of M into the direct sum of two $A[t]$-modules, or equivalently two u-stable A-submodules M' and M''; this is precisely the decomposition of (i).

This completes the proof of 2.13, and hence that of 2.9.

Corollary 2.14. *When condition* (C_3) *of 2.9 holds, the Cartan subalgebras of* $\mathfrak{g}$ *are strictly nilpotent* (2.2).

The proof is immediate.

Remark 2.15. a) One can prove a converse to 2.14: condition (C_3) is equivalent to the assertion that, for every S' over S, every quasi-regular section of $\mathfrak{g}_{S'}$ is regular and every Cartan subalgebra of $\mathfrak{g}_{S'}$ is strictly nilpotent; equivalently, every quasi-regular section of $\mathfrak{g}_{S'}$ is contained in a strictly nilpotent Cartan subalgebra of $\mathfrak{g}_{S'}$.

b) Unlike the other conditions (C_0) through (C_2), condition (C_3) is of "infinitesimal nature." More precisely, when S is locally noetherian, $\mathfrak{g}$ satisfies (C_3) if and only if, for every base change $S' \to S$ with S' local Artinian, $\mathfrak{g}_{S'}$ satisfies the same condition. If desired, one may take S' to be the spectrum of an Artinian quotient of a local ring of S. Similarly, when (C_0) holds, the condition for a section of $\mathfrak{g}$ to be regular is of infinitesimal nature.

c) When $\mathfrak{g}$ is the Lie algebra of a smooth group prescheme of finite presentation over S, we shall see that the conditions

$$(C_0), \quad (C_1), \quad (C_1'), \quad (C_2)$$

on $\mathfrak{g}$ are equivalent (5.2 a). I do not know what happens in general, except that even when S is local Artinian, (C_0) does not imply (C_1). However, even when $\mathfrak{g}$ comes from a group G and S is local Artinian, it is not true in general that (C_2) implies (C_3), since $\mathfrak{g}$ may be nilpotent without being strictly nilpotent.

An example is obtained from a smooth affine group scheme G over the spectrum S of a discrete valuation ring, such that the Lie algebra of the generic fiber is not nilpotent—for example, the generic fiber may be an adjoint semisimple group—whereas that of the special fiber is nilpotent—for example, the special fiber may be a vector group. Then, for sufficiently large n, the Lie algebra of

$$G_n = G \times_S S_n$$

is not strictly nilpotent, although it is nilpotent.

We continue under the hypotheses of 2.4. Let

$$\mathcal{D}\colon \left((\mathrm{Sch})_{/S}\right)^{\circ} \longrightarrow \mathrm{Sets}$$

be the functor defined by

$$\mathcal{D}(S') = \{\text{Cartan subalgebras of } \mathfrak{g}_{S'}\}.$$

Introduce also the functor

$$X(S') = \left\{ (\mathfrak{d}, a) \;\middle|\; \begin{array}{l} \mathfrak{d} \text{ is a Cartan subalgebra of } \mathfrak{g}_{S'}, \\ a \text{ is a section of } \mathfrak{d} \end{array} \right\}.$$

The two projections

$$(\mathfrak{d}, a) \longmapsto \mathfrak{d} \qquad \text{and} \qquad (\mathfrak{d}, a) \longmapsto a$$

give morphisms

$$p\colon X \longrightarrow \mathcal{D} \qquad \text{and} \qquad \psi\colon X \longrightarrow W(\mathfrak{g}).$$

When (C_0) holds, let U also denote the open subset of regular points of $W(\mathfrak{g})$ (2.10). Condition (C_2) is then equivalent to saying that the morphism

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism; a priori it is a monomorphism by 2.6.

It is immediate that

$$p\colon X \longrightarrow \mathcal{D}$$

is representable by a projection of vector bundles. More explicitly, for every S-morphism $S' \to \mathcal{D}$ corresponding to a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}_{S'}$, the fiber product

$$X \times_{\mathcal{D}} S'$$

is represented by the vector bundle $W(\mathfrak{d})$ over S'. Thus, if $\mathcal{D}$ is representable, then so is X. We have:

Theorem 2.16. *Let S be a prescheme and let $\mathfrak{g}$ be a Lie algebra over S whose underlying $\mathcal{O}_S$-module is locally free of finite type. Suppose that condition (C_0) of 2.9 holds.*

a) *The functor $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$, defined above, is represented by a quasi-projective prescheme of finite presentation over S. The same holds for the functor X.*

b) *If condition (C_2) of 2.9 holds, then $\mathcal{D}$ and X are smooth over S, and the morphism*

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism.

c) *Continue to suppose that (C_2) holds. Let $s \in S$, and let $\mathfrak{d}_0$ be a Cartan subalgebra of $\mathfrak{g}(s)$, corresponding to a $k(s)$-rational point d of $\mathcal{D}$. Suppose that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$, a condition automatically satisfied when $k(s)$ is infinite. If r is the infinitesimal rank of $\mathfrak{g}(s)$ and n its rank over $k(s)$, then there exists an open neighborhood V of d in $\mathcal{D}$ that is S-isomorphic to an open subset V' of*

$$S[t_1, \ldots, t_{n-r}].$$

Proof. We may suppose that $\mathfrak{g}$ has constant rank n and constant nilpotent rank r. The assertions concerning X in a) and b) follow immediately from those concerning $\mathcal{D}$ and from the fact that X is a vector bundle over $\mathcal{D}$, defined by a locally free module; they were included only for completeness.

a) The functor $\mathcal{D}$ is a subfunctor of $\mathrm{Grass}_{n-r}(\mathfrak{g})$, whose value at S' is the set of locally free quotient modules of rank $n-r$ of $\mathfrak{g}_{S'}$. It is well known that the latter functor is represented by a projective prescheme smooth over S; see, for example, Cartan Seminar 1960/61, Exposé 12, Nos. 2 and 3, whose constructions carry over unchanged to preschemes.* [3]

We are therefore reduced to the following relative problem. Given a locally free quotient module of rank $n-r$ of $\mathfrak{g}$, or, equivalently, a locally free submodule $\mathfrak{d}$ of rank r that is locally a direct factor, represent the functor

$$F(S') = \begin{cases} \varnothing, & \text{if } \mathfrak{d}_{S'} \text{ is not a Cartan subalgebra of } \mathfrak{g}_{S'}, \\ \{\varnothing\}, & \text{otherwise.} \end{cases}$$

We shall show that F is represented by a subprescheme of finite presentation over S. This will show that

$$\mathcal{D} \longrightarrow \mathrm{Grass}_{n-r}(\mathfrak{g})$$

is represented by an immersion of finite presentation and will complete the proof of a).

First express the condition that $\mathfrak{d}_{S'}$ be a Lie subalgebra of $\mathfrak{g}_{S'}$. This condition says precisely that $S' \to S$ factors through a certain closed subprescheme S_1 of finite presentation over S; locally on S, its equations are obtained at once from a basis of $\mathfrak{g}$ adapted to $\mathfrak{d}$. We may therefore suppose that $S = S_1$.

Next we must express that $\mathfrak{d}_{S'}$ contains, fpqc-locally, a quasi-regular section of $\mathfrak{g}_{S'}$. For this purpose consider

$$V = W(\mathfrak{d}) \cap U,$$

where U is the open subset of regular points of $W(\mathfrak{g})$ (2.10). Since the structure morphism $V \to S$ is smooth and quasi-compact, its image S_1 is an open subset of S, and the immersion $S_1 \to S$ is quasi-compact, hence of finite presentation. The condition on S' is exactly that $S' \to S$ factor through S_1. We are thus reduced to the case $S = S_1$, and, by descent, to the case in which $\mathfrak{d}$ has a section a that is quasi-regular in $\mathfrak{g}$.

It remains to express that, for the induced section $a_{S'}$, the endomorphism

$$\mathrm{ad}(a_{S'})_{\mathfrak{g}_{S'}/\mathfrak{d}_{S'}}$$

is bijective. This again says that $S' \to S$ factors through an open subprescheme of finite presentation, namely S_D, where

$$D = \det\Big(\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}\Big).$$

But $\mathfrak{d}|_{S_D}$ is then a Cartan subalgebra of $\mathfrak{g}|_{S_D}$. Consequently S_D represents F, which proves a).

b) This follows immediately from 2.11(a) and XI 1.5. It is also, of course, a consequence of the more precise assertion c).

c) Let a_0 be a regular point of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$, and extend it to a section a of $\mathfrak{g}$ over an open neighborhood of s. We may plainly suppose that this neighborhood is all of S.

*See also EGA I, second edition (announced as forthcoming from North-Holland Publishing Company).

[3] Editors' note: See §I.1.3 of M. Demazure and P. Gabriel, *Groupes algébriques*, Masson (1970).

Choose a complement M_0 of $\mathfrak{d}_0$ in the vector space $\mathfrak{g}(s)$. After again replacing S by an open neighborhood of s, there is a submodule M of $\mathfrak{g}$, locally a direct factor, such that $M(s) = M_0$; we again suppose that the neighborhood is S.

Let $\mathcal{V}$ be the subfunctor of $\mathcal{D}$ for which $\mathcal{V}(S')$ consists of the Cartan subalgebras $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ satisfying:

1°) $\mathfrak{d}'$ is a complement of $M_{S'}$;

2°) the unique section in $(a_{S'} + M_{S'}) \cap \mathfrak{d}'$ is a regular section of $\mathfrak{g}_{S'}$.

Condition 1°) defines an open V_1 of $\mathcal{D}$, induced by the open of $\mathrm{Grass}_{n-r}(\mathfrak{g})$ given by the same condition. By 2.10 and (C_2), the conjunction of 1°) and 2°) defines an open of V_1. Thus $\mathcal{V}$ is represented by an open subprescheme V of $\mathcal{D}$, plainly containing the point d.

Now let $\mathcal{V}'$ be the subfunctor of $W(M)$ defined by

$$\mathcal{V}'(S') = \left\{ u' \in \Gamma(S', M_{S'}) \;\middle|\; \begin{array}{l} \text{(i) } a_{S'} + u' \text{ is a regular section of } \mathfrak{g}_{S'}, \\ \text{(ii) the unique Cartan subalgebra } \mathfrak{d}' \text{ of } \mathfrak{g}_{S'} \\ \quad \text{containing } a_{S'} + u' \text{ is a complement of } M_{S'} \end{array} \right\}.$$

Condition (i) defines an open subprescheme V_1' of $W(M)$: it is the inverse image of the open U of regular points of $W(\mathfrak{g})$ (2.10) under the translation morphism $m \mapsto a + m$. The conjunction of (i) and (ii) defines an open V' of V_1', namely the inverse image of V under the evident morphism

$$V_1' \longrightarrow \mathcal{D}$$

that associates with u' the unique Cartan subalgebra $\mathfrak{d}'$ containing $a_{S'} + u'$. Its restriction gives a morphism

$$V' \longrightarrow V,$$

which is plainly an isomorphism. This proves c).

Corollary 2.17. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k. Then the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$* (2.16(a)) *is quasi-projective, smooth, and irreducible. If $\mathfrak{g}$ contains a regular element* (for example, if k is infinite)*, then $\mathcal{D}$ is a rational variety; that is, its function field is a purely transcendental extension of k.*

Indeed, $\mathcal{D}$ is irreducible because there is a surjective morphism

$$\psi^{-1}(U) \longrightarrow \mathcal{D},$$

and $\psi^{-1}(U)$ is irreducible, being isomorphic to the open subset U of $W(\mathfrak{g})$. The assertion concerning the function field follows immediately from 2.16(c).

Remarks 2.18. I do not know whether this conclusion remains true when k is finite without assuming that $\mathfrak{g}$ contains a regular point; compare 2.12. It can be proved when $\mathfrak{g}$ is the Lie algebra of an algebraic group G smooth over k, at least when $G/\,\mathrm{radical}(G)$ is an "adjoint" semisimple group, by using a result of Chevalley cited below (see the Appendix). It is plausible that the result remains valid without restriction on G; for this it would suffice that the cited result of Chevalley be proved for every semisimple algebraic group, not necessarily an adjoint group.

0.3 Subgroups of type (C) of group preschemes over an arbitrary prescheme

Theorem 3.1. *Let S be a prescheme, let G be a smooth S-group prescheme, and let $\mathfrak{g}$ be its Lie algebra, which is a locally free module of finite type over S. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ that, as a module, is locally a direct factor in $\mathfrak{g}$, and suppose that, for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let a be a quasi-regular section of $\mathfrak{g}$ (2.5). Then*

$$M_a = \mathrm{Transp}_G(a, \mathfrak{h}),$$

the subfunctor of G whose S'-valued points are the $g \in G(S')$ such that

$$\mathrm{ad}(g) \cdot a_{S'} \in \Gamma(S', \mathfrak{h}_{S'}),$$

is represented by a closed subprescheme of G, smooth over S, whose structure morphism to S is surjective. [4]

Proof. Consider the canonical morphism

$$\varphi\colon G \times_S W(\mathfrak{h}) \longrightarrow W(\mathfrak{g}), \qquad (g, x) \longmapsto \mathrm{ad}(g) \cdot x.$$

Then M_a is S-isomorphic to $\varphi^{-1}(a)$, the inverse image under φ of a, viewed as a section of $W(\mathfrak{g})$ over S. To prove that M_a is smooth, it is enough to show that φ is smooth at the points of $G \times_S W(\mathfrak{h})$ lying over $\mathrm{Im}(a)$. More generally, φ is smooth at every point lying over a point of $W(\mathfrak{g})$ that is regular in its fiber $W(\mathfrak{g}(s))$ over S.

Indeed, the source and target of φ are smooth over S, and hence are flat and locally of finite presentation over S. We may therefore check the assertion fiber by fiber. This reduces us to the case in which S is the spectrum of an algebraically closed field k, G is a locally algebraic group over k, $\mathfrak{h}$ is a Lie subalgebra of its Lie algebra $\mathfrak{g}$ containing a Cartan subalgebra of $\mathfrak{g}$, and a is a regular point of $\mathfrak{g}$.

Since φ is a G-morphism, we may suppose that the point of $G \times W(\mathfrak{h})$ under consideration is of the form (e, a). We may also suppose that G is connected, and hence of finite type over k. The assertion is then precisely XIII 5.4.

It is also immediate that M_a is a closed subprescheme of G of finite presentation over S: it is the inverse image of $W(\mathfrak{h})$ under the morphism

$$G \longrightarrow W(\mathfrak{g}), \qquad g \longmapsto \mathrm{ad}(g) \cdot a.$$

Finally, the surjectivity of $M_a \to S$ likewise reduces to the case of an algebraically closed base field. By hypothesis, $\mathfrak{h}$ then contains a Cartan subalgebra $\mathfrak{d}'$. By the conjugacy theorem XIII 6.1(a), $\mathfrak{d}'$ is conjugate to

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}).$$

It follows that $\mathfrak{h}$ contains a conjugate of a. This completes the proof.

Corollary 3.2. *Let G and $\mathfrak{g}$ be as in 3.1, with G of finite type over S, and let $\mathfrak{k}$ and $\mathfrak{h}$ be two Lie subalgebras of $\mathfrak{g}$ that are locally direct factors as modules. Suppose that one of the following two hypotheses holds:*

[4] Editors' note: This transporter is denoted $\mathrm{Transp}_G(a, \mathfrak{h})$.

a) *For every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent and contains a regular element of $\mathfrak{g}_{\bar{s}}$, while the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

b) *$\mathfrak{k}$ is a Cartan subalgebra of $\mathfrak{g}$.*

Then $\mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h})$ *is a closed subprescheme of G, smooth over S. Moreover, in case* a), *its structure morphism to S is surjective.*

The representability of the transporter by a closed subprescheme of G of finite presentation is immediate and is left to the reader. We first prove smoothness in case a). Suppose first that $\mathfrak{k}$ has a section a that is quasi-regular in $\mathfrak{g}$. It is then enough to apply 3.1 and the following lemma.

Lemma 3.3. *Under the hypotheses of 3.2*(a), *if a is a section of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$, then*

$$\mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}) = \mathrm{Transp}_G(a, \mathfrak{h}).$$

Indeed, by the definitions it is enough to show that, if a is also a section of $\mathfrak{h}$, then $\mathfrak{k} \subset \mathfrak{h}$. Since $\mathfrak{k}$ is locally nilpotent by hypothesis, 2.1 gives

$$\mathfrak{k} \subset \mathrm{Nil}(a, \mathfrak{g}).$$

On the other hand,

$$\mathrm{Nil}(a, \mathfrak{g}) \subset \mathfrak{h},$$

because

$$\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$$

is injective; this can be checked fiber by fiber by XIII 4.8(b). The lemma follows.

In the general case, the usual standard argument first reduces us to the case where S is affine noetherian, and then to the case where S is local artinian, smoothness being an infinitesimal property. By flat descent we may further suppose that the residue field is infinite. The fiber of $\mathfrak{k}$ then contains an element regular in the corresponding fiber of $\mathfrak{g}$. Lifting this element to a section of $\mathfrak{k}$ reduces us to the preceding case.

Thus the transporter is smooth in case a). Its structure morphism is surjective because this may be checked after reducing to the case where S is the spectrum of an algebraically closed field; $\mathfrak{k}$ then contains a regular point of $\mathfrak{g}$, and 3.3 and 3.1 apply.

For case b), the definition of smoothness (XI 1.1) reduces the assertion to the following lifting statement. Let S be affine, let S_0 be a closed subscheme defined by a quasi-coherent nilpotent ideal J, and let $g_0 \in G(S_0)$ transport $\mathfrak{k}_0$ into $\mathfrak{h}_0$. Then g_0 lifts to an element $g \in G(S)$ that transports $\mathfrak{k}$ into $\mathfrak{h}$. The hypothesis on g_0 places this lifting problem under case a), already proved. This completes the proof.

Of course, if in 3.2(b) the subalgebra $\mathfrak{h}$ satisfies the stronger hypothesis of 3.1, then, and only then, the structure morphism

$$\mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}) \longrightarrow S$$

is surjective. Using Hensel's lemma XI 1.10, one obtains from 3.1 and 3.2 the following corollary.

Corollary 3.4. *Under the hypotheses of 3.1 for G and $\mathfrak{h}$, and assuming that G is of finite type over S, the following assertions hold:*

a) *For every quasi-regular section a of $\mathfrak{g}$, étale locally there is a conjugate of a that is a section of $\mathfrak{h}$.*

b) *For every Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$, étale locally $\mathfrak{d}$ is conjugate to a subalgebra of $\mathfrak{h}$.*

In particular, when $\mathfrak{h}$ is itself a Cartan subalgebra of $\mathfrak{g}$, one obtains the following.

Corollary 3.5. *Let G be a smooth S-prescheme in groups of finite type, let $\mathfrak{g}$ be its Lie algebra, and let $\mathfrak{d}$ and $\mathfrak{d}'$ be two Cartan subalgebras of $\mathfrak{g}$. Then* $\mathrm{Transp}_G(\mathfrak{d},\mathfrak{d}')$ *is identical to the strict transporter of $\mathfrak{d}$ into $\mathfrak{d}'$, and is a closed subprescheme of G, smooth over S, whose structure morphism is surjective. Étale locally, $\mathfrak{d}$ and $\mathfrak{d}'$ are conjugate.*

Here the transporter is identical to the strict transporter simply because $\mathfrak{d}$ and $\mathfrak{d}'$ are locally direct factors in $\mathfrak{g}$ and have the same rank at every point. Thus 3.5 is a special case of 3.4. Taking $\mathfrak{d} = \mathfrak{d}'$ gives the following.

Corollary 3.6. *Let G be as in 3.5, and let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then* $\mathrm{Norm}_G(\mathfrak{d})$ *is a closed subprescheme in groups of G, smooth over S, and its Lie algebra is $\mathfrak{d}$.*

Indeed, the last assertion says that $\mathfrak{d}$ is its own normalizer in $\mathfrak{g}$, and this follows at once because it is true fiber by fiber.

Corollary 3.7. *Let G and $\mathfrak{g}$ be as in 3.5. Then the conditions* (C_2), (C_1'), *and* (C_1) *of 2.9 are equivalent. In other words, if $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally, then every quasi-regular section of $\mathfrak{g}$ is regular.*

Indeed, let a be a quasi-regular section; we prove that it is regular. Since the question is fpqc local, we may suppose that $\mathfrak{g}$ admits a Cartan subalgebra $\mathfrak{d}$. By 3.4(a), étale locally a is conjugate to a section of $\mathfrak{d}$. We are thereby reduced to the case in which a is a section of $\mathfrak{d}$, where the conclusion follows immediately from the definition.

Definition 3.8. *Let G be a smooth prescheme in groups over a prescheme S. A* subgroup of type (C) *of G is a subprescheme in groups $D \subset G$, smooth over S and with connected fibers, such that*

$$\mathfrak{d} = \mathrm{Lie}(D)$$

is a Cartan subalgebra (2.4) *of $\mathfrak{g} = \mathrm{Lie}(G)$. Equivalently, for every $s \in S$, the fiber D_s is a subgroup of type* (C) *of the algebraic group G_s* (XIII 6.2).

Theorem 3.9. *Let G be a smooth S-prescheme in groups of finite presentation, and let $\mathfrak{g}$ be its Lie algebra. Then:*

a) *The map*

$$D \longmapsto \mathfrak{d} = \mathrm{Lie}(D)$$

establishes a bijective correspondence between the subgroups of type (C) *of G and the Cartan subalgebras of $\mathfrak{g}$.*

b) *If D and $\mathfrak{d}$ correspond, then*

$$\mathrm{Norm}_G(D) = \mathrm{Norm}_G(\mathfrak{d}).$$

This is a closed subprescheme of G, smooth over S, and

$$D = \mathrm{Norm}_G(D)^0 = \mathrm{Norm}_G(\mathfrak{d})^0.$$

c) *Any two subgroups D and D' of type* (C) *in G are conjugate étale locally.*

Proof. Let D be a subgroup of type (C) of G, and let $\mathfrak{d}$ be its Lie algebra. Then

$$D \subset \mathrm{Norm}_G(\mathfrak{d}).$$

By Definition 3.8 and Corollary 3.6, this is an inclusion of smooth preschemes in groups over S that induces an isomorphism on Lie algebras. Since D has connected fibers, it follows that

$$D = \mathrm{Norm}_G(\mathfrak{d})^0.$$

Thus the map in a) is injective. We prove that it is surjective. Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. By 3.6,

$$N = \mathrm{Norm}_G(\mathfrak{d})$$

is a closed subprescheme in groups of G, smooth over S, and has Lie algebra $\mathfrak{d}$. Since G is of finite presentation over S, so is N. Consequently, as noted in XII after 7.3, the union of the connected components of the fibers of N is the underlying set of an open subprescheme in groups $N^\circ \subset N$. Evidently N° is a subgroup of type (C) of G with Lie algebra $\mathfrak{d}$. This proves a). The first assertion of b) follows at once, and the formula

$$D = \mathrm{Norm}_G(\mathfrak{d})^0$$

has already been proved. Finally, c) follows from a) and 3.5.

Corollary 3.10. *Suppose that $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally, or equivalently by 3.9*(a)*, that G admits a subgroup of type* (C) *fpqc locally. Consider the functor*

$$\mathcal{D}\colon (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets}),$$

where $\mathcal{D}(S')$ is the set of subgroups of type (C) *of $G_{S'}$. This functor is representable by a quasi-projective prescheme, smooth over S, with connected geometric fibers. If $S = \mathrm{Spec}(k)$, so that G is a smooth algebraic group over k, and if $\mathfrak{g}$ admits a regular point—a condition automatically satisfied when k is infinite—then $\mathcal{D}$ is a rational variety over k.*

Indeed, by 3.9(a) the functor $\mathcal{D}$ is canonically isomorphic to the functor considered in 2.16. On the other hand, by 3.7 condition (C_2) is satisfied. Therefore 3.10 follows from 2.16 and 2.17.

Corollary 3.11. *Let D be a subgroup of type* (C) *of G, and let N be its normalizer in G. Then the quotient sheaf G/N is canonically isomorphic to the functor $\mathcal{D}$ of 3.10. It is therefore representable by a quasi-projective prescheme, smooth over S, with connected geometric fibers.*

Proposition 3.12. *Let G be a smooth S-prescheme in groups of finite presentation over S. Let H and K be two subpreschemes in groups, smooth and of finite presentation over S, with K having connected fibers, and let $\mathfrak{g}, \mathfrak{h}, \mathfrak{k}$ be the corresponding Lie algebras. Suppose that one of the following conditions holds on their geometric fibers:*

a) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$, and $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

b) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

Under these conditions, $H \supset K$ if and only if $\mathfrak{h} \supset \mathfrak{k}$.

It remains only to prove that $\mathfrak{h} \supset \mathfrak{k}$ implies $H \supset K$. In case b), the inclusion $\mathfrak{h} \supset \mathfrak{k}$ places us under the hypotheses of case a), so it is enough to prove a). Proceeding as in 3.2(a), by reduction to the case where S is local artinian, we reduce to the case in which there exists a section a of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$. Arguing then as in XIII 5.5 reduces us to the following statement.

Corollary 3.13. *Let G be a smooth S-prescheme in groups of finite presentation over S, let $H \subset G$ be a subprescheme in groups smooth and of finite presentation over S, and let $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras. Let a be a section of $\mathfrak{h}$ that is quasi-regular in $\mathfrak{g}$. Suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Put*

$$M_a = \mathrm{Transp}_G(a, \mathfrak{h}).$$

This is a closed subprescheme of G, smooth over S (cf. 3.1)*. Hence M_a^0, the union in the fibers of M_a of the connected components containing the identity, is an open subset of M_a, endowed with the structure induced from M_a. Then*

$$H^0 = M_a^0.$$

Clearly $H \subset M_a$, and therefore $H^0 \subset M_a^0$. Since this is an inclusion of preschemes smooth over S, it is enough to check equality on the fibers. This reduces us to the case where S is the spectrum of an algebraically closed field, which was treated in XIII 5.4.

Corollary 3.14. *Let G be a smooth S-prescheme in groups of finite presentation, let D be a subgroup of type* (C) *of G, and let $H \subset G$ be a subprescheme in groups smooth and of finite presentation over S. If $\mathfrak{d}$ and $\mathfrak{h}$ are their Lie algebras, then*

$$\mathrm{Transp}_G(D, H) = \mathrm{Transp}_G(\mathfrak{d}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G, smooth over S.

Indeed, the identity between the two transporters follows from 3.12(b), after which 3.2 applies.

Corollary 3.15. *Let G and H be as in 3.14, and suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Suppose also that $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally. Then, étale locally, H contains a subgroup of type* (C) *of G.*

By 3.7 and 3.9(a), G admits a subgroup of type (C) étale locally. We may therefore suppose that G admits such a subgroup D. The hypothesis on $\mathfrak{h}$ then also says that the structure morphism of the transporter in 3.14 is surjective, by the conjugacy theorem XIII 6.1(a). The conclusion follows from Hensel's lemma XI 1.10.

Corollary 3.16. *Let G, H, K be as in 3.12*(a)*, and suppose in addition that every geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent, that is, that $\mathfrak{k}$ is locally nilpotent. Then*

$$\mathrm{Transp}_G(K, H) = \mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G, smooth over S, whose structure morphism to S is surjective. Étale locally, H contains a subgroup conjugate to K.

The identity between the two transporters is again contained in 3.12(a). The assertion about its structure is precisely 3.2(a), and the last assertion of 3.16 then follows from Hensel's lemma.

Remarks 3.17. a) In 3.12 and 3.16, the hypothesis that K be smooth over S may be replaced by the weaker hypothesis that the sheaf

$$e_K^*(\Omega^1_{K/S}) = \omega^1_K$$

of relative 1-differentials along the identity section be locally free. In this form, 3.12 includes XIII 5.5.

b) Let $G, \mathfrak{g}, \mathfrak{h}$ be as in 3.1, with G of finite presentation over S. Then

$$N = \mathrm{Norm}_G(\mathfrak{h})$$

need not be smooth over S along the identity section. Equivalently, there need not exist a subprescheme in groups $H \subset G$, smooth over S, whose Lie algebra is $\mathfrak{h}$, even when S is the spectrum of a field. When such an H exists, taking H with connected fibers gives

$$N = \mathrm{Norm}_G(H);$$

I do not know whether N is then smooth over S. In this question one may plainly reduce to the case where S is local artinian.

To conclude this section, we examine the case in which G is "semisimple" over S.

Theorem 3.18. *Let S be a prescheme and let G be a smooth S-prescheme in groups whose geometric fibers are semisimple "adjoint" groups, that is, semisimple groups whose reductive center* (XII 4.1 and 4.4) *is the identity group. Then the subgroups of type* (C) *of G are precisely its maximal tori, and hence also precisely its Cartan subgroups* (XII 3.1).

In view of the definitions, it is enough to treat the case where S is the spectrum of an algebraically closed field. We must then show that, for a maximal torus T of G, its Lie algebra $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$. Put $r = \dim T = \mathrm{rank}\,\mathfrak{t}$. Taking into account the inequality

$$\mathrm{nil\,rank}(\mathfrak{g}) \geq \mathrm{nil\,rank}(G) = \dim T = \mathrm{rank}\,\mathfrak{t} = r,$$

it is enough to find $x \in \mathfrak{t}$ such that

$$\dim \mathrm{Nil}(x, \mathfrak{g}) = r.$$

Since $\mathfrak{t}$ is abelian, and therefore nilpotent, this is equivalent to finding $a \in \mathfrak{t}$ for which

$$\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{t}}$$

is injective (XIII 5.7(a)).

Consider the characters α of T occurring in the representation of T induced by the adjoint representation of G. The structure theory of the semisimple group *G (Bible)*, 13, Theorem 1(a), and Theorem 3, Corollary 2, or more precisely the "big cell" of G, shows that the eigenspace of $\mathfrak{g}$ for the trivial character is exactly $\mathfrak{t}$. The other eigenspaces are one-dimensional, and their characters are precisely the roots of G relative to T. Here the big cell is built from T and the subgroups P_α, each isomorphic to the additive group $\mathbf{G}_a$, stable under T, and corresponding to the root characters.

The reductive center of G is the intersection of the kernels of the characters of T occurring in the adjoint representation (XII 4.8). Thus the assertion that G is adjoint says that its roots generate the lattice

$$M = \mathrm{Hom}(T, \mathbf{G}_m).$$

A familiar lemma from root theory says that every root belongs to a system of simple roots, hence to a basis of the group generated by the roots, and therefore to a basis of the dual lattice of T. We obtain the following.

Corollary 3.19. *Let G be an adjoint semisimple algebraic group over an algebraically closed field k, and let T be a maximal torus of G. For every root*

$$\alpha\colon T \longrightarrow \mathbf{G}_m$$

of G relative to T, the corresponding homomorphism

$$\alpha'\colon \mathfrak{t} \longrightarrow k$$

is nonzero.

This assertion is essentially equivalent to Theorem 3.18. Indeed, for $t \in \mathfrak{t}$, the endomorphism $\mathrm{ad}(t)$ is semisimple, and its eigenvalues on $\mathfrak{g}/\mathfrak{t}$ are precisely the numbers $\alpha'(t)$. Hence $\mathrm{ad}(t)_{\mathfrak{g}/\mathfrak{t}}$ is injective if and only if all the $\alpha'(t)$ are nonzero; and such a $t \in \mathfrak{t}$ exists if and only if every α' is nonzero.

Corollary 3.20. *Let S be a prescheme and let G be a smooth S-prescheme in groups of finite presentation over S, whose geometric fibers are connected reductive algebraic groups, that is, extensions of a semisimple group by a torus. For every $s \in S$ there is an open neighborhood U of s such that $G|_U$ admits a maximal torus.*[5]

We shall see in Exposé XVI that the hypothesis on G implies that G is affine over S and has locally constant reductive rank. Hence, by XII 4.7(c), G admits a reductive center Z, the quotient

$$G' = G/Z$$

is a smooth affine group over S whose reductive center is the identity group, and the maximal tori of G correspond bijectively to those of G'. Moreover, the geometric fibers of G' are plainly connected semisimple groups, and are adjoint by definition because their reductive center is trivial.

It is therefore enough to treat the case where G is semisimple and adjoint. By 3.18, the problem reduces to finding an open neighborhood U of s and a subgroup of type (C) of $G|_U$, or equivalently by 3.9(a), a Cartan subalgebra of $\mathfrak{g}|_U$. This is possible when $k(s)$ is infinite: by 3.7, $\mathfrak{g}$ satisfies condition (C_2) of 2.9, and 2.11(b) applies.

In fact, 3.20 remains true without assuming that $k(s)$ is infinite. By the preceding argument, it is enough to know that, for every adjoint semisimple group G over a finite field k, the Lie algebra $\mathfrak{g}$ contains a regular element. Chevalley proved this assertion using the properties of a Coxeter element of the Weyl group; see the appendix below by J.-P. Serre.

Remarks 3.21. a) Assertion 3.20 remains valid, with essentially the same proof, if G is replaced by a closed subprescheme in groups H, smooth over S, whose rank is everywhere the same as that of G (for example, a "parabolic subgroup" of G), provided that $k(s)$ is infinite. I do not know whether the infinitude hypothesis is again superfluous. It can at least be removed when H is parabolic: constructing the radical U of H and the quotient H/U reduces the question to the semisimple case. Unfortunately, the method of regular elements seems powerless here. One readily constructs examples—for instance, the projective group with its "standard" Borel subgroup—in which the Lie algebra $\mathfrak{h}$ of H contains no regular element.

b) When $k(s)$ is infinite, the proof of 3.20 in fact gives the following sharper statement, using 3.9(b): every maximal torus T_0 of G_s is induced by a maximal torus T over an

[5] This result, as well as 2.11 on which it rests, immediately generalizes to the case in which s is replaced by a finite subset of S contained in an affine open.

open neighborhood of s. I do not know whether this remains true without the infinitude hypothesis. The evident difficulty is that the Lie algebra $\mathfrak{t}_0$ of T_0 need not contain a regular element of the Lie algebra $\mathfrak{g}_0$ of the fiber G_s.

An affirmative answer would imply the following assertion, which is known only when the residue field is infinite or when A is separated and complete. Let A be a local ring with residue field k, and let M be an "Azumaya algebra" over A, meaning an algebra that is a finite free A-module and whose reduction

$$M_0 = M \otimes_A k$$

is a central simple algebra over k. Let $D_0 \subset M_0$ be a commutative subalgebra separable over k, with

$$[M_0 : k] = [D_0 : k]^2.$$

Then there should exist a commutative subalgebra $D \subset M$ that is a direct summand as an A-module and satisfies

$$D \otimes_A k = D_0.$$

The datum of M is equivalent to that of a principal homogeneous bundle under the projective group $\mathrm{PGL}(n)_A$, and hence to an inner twisted form G of $\mathrm{PGL}(n)$. The maximal tori of G correspond bijectively to the commutative subalgebras $D \subset M$ that are étale of rank n over A.

c) Applying 3.20 to the centralizer of a subtorus Q of a reductive S-prescheme in groups G, one sees that every such Q is contained, Zariski locally, in a maximal torus of G.

0.4 A Digression on Borel Subgroups

Definition 4.1. *Let G be a smooth algebraic group over an algebraically closed field. A* Borel subgroup *of G is a smooth, solvable, connected algebraic subgroup that is maximal with these properties.*

When G is affine, this agrees with the terminology of *(Bible)*, 6, Definition 1. Observe at once that if Z is a connected smooth subgroup of G contained in its center (or, more generally, if Z is solvable and normal), then, for every Borel subgroup B of G, the image BZ of $B \times Z$ under the morphism

$$(b, z) \longmapsto bz$$

from $B \times Z$ to G is a smooth, solvable, connected subgroup of G containing B. It is therefore equal to B. Thus B contains Z, and consequently B is the inverse image of an algebraic subgroup B' of $G' = G/Z$; it is immediate that B' is a Borel subgroup of G'.

Taking

$$Z = \mathrm{Centr}(G^0)^0_{\mathrm{red}},$$

the quotient $G' = G/Z$ is affine by XII 6.1. Hence the Borel subgroups of G' are conjugate, and G'/B' is a projective variety for every such B' *(Bible)*, 6, Theorem 4. Consequently:

Proposition 4.2. *Let G be as in 4.1. Then the Borel subgroups of G are conjugate. If B is a Borel subgroup, then G/B is a projective variety. The maximal tori of B are maximal tori of G; and, when G is connected, the Cartan subgroups of B are Cartan subgroups of G.*

It remains to prove the last assertion, and we may plainly suppose that $G = G^0$. For Cartan subgroups, it follows from the analogous assertion in G' *(Bible)*, 6, Theorem 4,

Corollary 4 and XII 6.6(e). The assertion for maximal tori follows from this one, since by XII 6.6(c) the maximal tori of a smooth algebraic group are precisely the maximal tori of its Cartan subgroups.

Corollary 4.3. *Suppose that G is connected. Then every element of G is contained in a Borel subgroup of G.*

This reduces to the same assertion in G', where it is well known *(Bible)*, 6, Theorem 5(d).

Corollary 4.4. *Let B be a Borel subgroup of G, let C be a Cartan subgroup of B, and let N be its normalizer in G. Then*

$$N \cap B = C.$$

Indeed,

$$N \cap B = \mathrm{Norm}_B(C),$$

so it remains to show that, when G is connected and solvable, a Cartan subgroup C is its own connected normalizer. With the preceding notation, C is the inverse image of a Cartan subgroup C' of G', and we are therefore reduced to the case in which G is affine. The normalizer of a Cartan subgroup is smooth (C being its own connected normalizer; see, for example, XII 6.6(c)). It is thus enough to check that C and N have the same k-valued points, which is exactly *(Bible)*, Theorem 6(d).

Definition 4.5. *Let G be a smooth prescheme in groups of finite presentation over a prescheme S. A* Borel subgroup *of G is a smooth subprescheme in groups B of G, of finite presentation, such that for every $s \in S$ the geometric fiber $B_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$.*

As is immediately verified, this notion is stable under base change and is local for the fpqc topology. Indeed, if k' is an algebraically closed extension of an algebraically closed field k, then an algebraic subgroup B of a smooth algebraic group G over k is a Borel subgroup of G if and only if $B_{k'}$ is a Borel subgroup of $G_{k'}$.

It follows from the definition that, if G is a smooth algebraic group over an arbitrary field k and B is a Borel subgroup of G, then G/B is a projective variety; every maximal torus T of B is a maximal torus of G; and the normalizer of T in B equals its centralizer C, which is a Cartan subgroup of G when G is connected.

Remarks 4.6. Unfortunately, it is no longer true in general that two Borel subgroups of G are locally conjugate for the fpqc topology. This can fail even when G is affine over S and S is the spectrum of the algebra of dual numbers over an algebraically closed field k.

As a consequence of this regrettable fact, let us point out that if G is a smooth, affine, connected algebraic group over a nonperfect field k, it is in general impossible to define naturally a homogeneous space D under G that plays the role of a flag variety, that is, of the variety of Borel subgroups of G. Over the algebraic closure $\bar{k}$ of k, such a space would have to be isomorphic to $G_{\bar{k}}/B$, where B is a Borel subgroup of $G_{\bar{k}}$.

Indeed, suppose that the quotient of $G_{\bar{k}}$ by its radical $\bar{R}$ is an adjoint semisimple group. The kernel of

$$G_{\bar{k}} \longrightarrow \mathrm{Aut}_{\bar{k}}(\bar{D})$$

is then the radical of $G_{\bar{k}}$. Consequently, if $\bar{D}$ came from a homogeneous space D under G, the radical $\bar{R}$ would come from a subgroup R of G. But examples are easily constructed for which $G_{\bar{k}}/\bar{R}$ is adjoint while $\bar{R}$ is not defined over k.

Under these conditions, the functor

$$\mathfrak{B}\colon (\mathrm{Sch}/S)^\circ \longrightarrow \mathrm{Ens}, \qquad \mathfrak{B}(S') = \{\text{Borel subgroups of } G_{S'}\},$$

is not representable by a smooth S-prescheme. From the infinitesimal point of view (III, §3), the failure of the conjugacy theorem is expressed by the fact that, if B is a Borel subgroup of the smooth algebraic group G, the cohomology group

$$H^1(B, \mathfrak{g}/\mathfrak{b})$$

may be nonzero.[6]

We shall see, on the other hand, in a later exposé that when G is semisimple, or more generally reductive, such unpleasant phenomena do not occur. These phenomena, together no doubt with the absence of good existence theorems, explain why Borel subgroups play only a relatively unobtrusive role in the scheme-theoretic study of general group schemes, whereas they dominate the theory of semisimple group schemes in the later exposés.

Proposition 4.8. *Let G be a smooth S-prescheme in groups of finite presentation with connected fibers, and let B be a Borel subgroup of G. Then B is its own normalizer and is a closed subprescheme of G.*

By XII 7.10, it is enough to prove that, over an algebraically closed field k, every element of $G(k)$ that normalizes B belongs to $B(k)$. When G is affine this is a fundamental result of Chevalley *(Bible)*, 9, Theorem 1; the general case reduces immediately to it by the reduction already used in 4.2.

Remarks 4.8.1. Definition 4.5 may be generalized by also introducing the notion of a *parabolic subgroup* of G. This means a subprescheme in groups P of G, smooth and of finite presentation over S, such that for every $s \in S$ the geometric fiber $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$, that is, contains a Borel subgroup of $G_{\bar{s}}$.

Proposition 4.8 extends to a parabolic subgroup P of G, by the same reduction to the known set-theoretic assertion. We note the following consequence (compare Exposé XVI). If P is a parabolic subgroup of G, then G/P is representable by a quasi-projective prescheme of finite presentation over S; here G is assumed to have connected fibers. Moreover, G/P is plainly smooth over S, and its geometric fibers are connected and proper. It follows readily from EGA III, 5.5.1 that

$$D = G/P$$

is proper, hence projective, over S. If its relative dimension is n, it is known that the inverse of the invertible sheaf $\Omega^n_{D/S}$ induces ample sheaves on the geometric fibers of D/S; hence, by EGA III, 4.7.1,

$$\left(\Omega^n_{D/S}\right)^{-1}$$

is ample on D relative to S.

It is readily seen, by reduction to the affine case and to an algebraically closed base field, that if

$$u\colon G \longrightarrow G'$$

[6] Here $\mathfrak{b}$ denotes the Lie algebra of B.

is an epimorphism of smooth algebraic groups, then $u(B) = B'$ is a Borel subgroup of G' for every Borel subgroup B of G. We now consider the case in which this gives a bijective correspondence between the Borel subgroups of G and those of G'.

Proposition 4.9. *Let G and G' be smooth S-preschemes in groups of finite presentation with connected fibers, and let*

$$u\colon G \longrightarrow G'$$

be a faithfully flat, that is, surjective, homomorphism of groups.[7] *Put $N = \ker u$, and suppose that one of the following two conditions holds:*

a) *N is central in G.*

b) *$S = \mathrm{Spec}(k)$ for a field k, and, if $\bar{k}$ is an algebraic closure of k, then*

$$N_{\bar{k}} = \ker(u_{\bar{k}})$$

is contained in the radical of $G_{\bar{k}}$, that is, in its largest smooth, connected, solvable, normal subgroup.

Then the map

$$B' \longmapsto u^{-1}(B')$$

induces a bijection from the set of Borel subgroups of G' onto the set of Borel subgroups of G.

Case b) follows immediately from the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N, together with the fact that, over an algebraically closed field, the Borel subgroups of G contain its radical. The latter fact follows at once from the argument preceding 4.2.

We prove case a). By XII 7.12, the map

$$H' \longmapsto H = u^{-1}(H')$$

gives a bijection between the subpreschemes in groups H' of G' that are smooth and of finite presentation over S, have connected fibers, and at every $s \in S$ have the same reductive rank and the same nilpotent rank as G', and the subpreschemes in groups H of G having the analogous properties. Borel subgroups of G' and of G have these properties.

It remains to show that, when H' and H correspond, H' is a Borel subgroup of G' if and only if H is a Borel subgroup of G. By definition, this question reduces to the case where S is the spectrum of an algebraically closed field. Since N is central in G, and hence in H, the group H is solvable if and only if H' is solvable. Finally, the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N shows immediately that H' has the maximality property of Definition 4.1 if and only if H does. This completes the proof.

Corollary 4.10. *With the notation of 4.9, if B' and B are corresponding Borel subgroups of G' and G, then*

$$\mathfrak{b} = \mathrm{Lie}(u)^{-1}(\mathfrak{b}'),$$

where $\mathfrak{g}, \mathfrak{g}', \mathfrak{b}, \mathfrak{b}'$ are the Lie algebras of G, G', B, B', respectively, and

$$\mathrm{Lie}(u)\colon \mathfrak{g} \longrightarrow \mathfrak{g}'$$

[7] The editors ask whether a reference should be added here.

is the homomorphism induced by u.

This follows immediately from the definitions and the identity

$$u^{-1}(B') = B.$$

We can now prove the principal result of this section.

Theorem 4.11. *Let G be a smooth algebraic group over an algebraically closed field k, and let $\mathfrak{g}$ be its Lie algebra. Then $\mathfrak{g}$ is the union of the Lie algebras $\mathfrak{b}$ of the Borel subgroups B of G.*

We may plainly suppose that G is connected. Let R be the radical of G, and put $G' = G/R$. By 4.9(b) and 4.10, it is enough to prove the theorem for G'; thus we may suppose that G is semisimple.

Let Z be the center of G, which is its reductive center, and put $G' = G/Z$. The same argument, now using 4.9(a), reduces the theorem to G'. We may therefore suppose that G is semisimple and adjoint.

Let B be a Borel subgroup of G, let T be a maximal torus of B, hence of G, and let $\mathfrak{b}$ and $\mathfrak{t}$ be their Lie algebras. By 3.18, T is a subgroup of type (C) of G, that is, $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$. The union of the conjugates of $\mathfrak{t}$ is consequently dense in $\mathfrak{g}$ (XIII 5.1, (i)⇒(vii)). *A fortiori*, the union of the conjugates of $\mathfrak{b}$ is dense in $\mathfrak{g}$.

Let X be the closed subscheme of

$$G/B \times W(\mathfrak{g})$$

whose k-valued points are the pairs (gB, x) such that

$$x \in \mathrm{Ad}(g)\mathfrak{b}$$

(compare XIII 1). The morphism

$$\psi\colon X \longrightarrow W(\mathfrak{g})$$

induced by the second projection is proper because G/B is proper over k. We have just seen that it is dominant; it is therefore surjective. This proves 4.11.

The only result of this section that we shall use in the remainder of this exposé is the following corollary.

Corollary 4.12. *Let k be an infinite field, let G be a smooth algebraic group over k, and let T be a maximal torus of G. Write $\mathfrak{g}$ and $\mathfrak{t}$ for their Lie algebras. Let*

$$u\colon G \longrightarrow \mathrm{GL}(V)$$

be a linear representation, where V is finite-dimensional over k, and let

$$u'\colon \mathfrak{g} \longrightarrow \mathfrak{gl}(V)$$

be the induced representation of Lie algebras. Then the minimum of the nullity of $u'(x)$, for $x \in \mathfrak{g}$, is attained at an element $x \in \mathfrak{t}$.

We reduce immediately to the case where k is algebraically closed. We may plainly suppose that G is connected, and, after dividing G by $(\ker u)_{\mathrm{red}}$, that G is affine. Using 4.11 and the conjugacy theorem for the maximal tori of G, we reduce to the further case in which G is solvable.

Then G is a semidirect product $T \cdot U$, where U is the unipotent part of G, a smooth connected unipotent group *(Bible)*, 6, Theorem 3. Thus, as a vector space,

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{u}, \qquad \mathfrak{u} = \mathrm{Lie}(U).$$

By the Lie–Kolchin theorem *(Bible)*, 6, Theorem 1, the representation space V has a composition series by stable subspaces V_i, with one-dimensional quotients

$$W_i = V_i/V_{i+1}.$$

For every i there is therefore an induced representation

$$u_i \colon G \longrightarrow \mathrm{GL}(W_i) = \mathbb{G}_m$$

and a corresponding homomorphism of Lie algebras

$$u_i' \colon \mathfrak{g} \longrightarrow k.$$

For every $x \in \mathfrak{g}$, the nullity of $u'(x)$ is the number of indices i for which $u_i'(x) = 0$.

Since U is unipotent, every u_i is trivial on U, and hence every u_i' vanishes on $\mathfrak{u}$. If

$$x = t + v, \qquad t \in \mathfrak{t}, \quad v \in \mathfrak{u},$$

then $u_i'(x) = u_i'(t)$ for every i. Thus $u'(x)$ and $u'(t)$ have the same nullity, and 4.12 follows.

0.5 Relations Between Cartan Subgroups and Cartan Subalgebras

Applying 4.12 to the adjoint representation of G, we obtain:

Theorem 5.1. *Let G be a smooth algebraic group over an infinite field, let T be a maximal torus of G, and let $\mathfrak{g} \supset \mathfrak{t}$ be their Lie algebras. Then $\mathfrak{t}$ contains a regular element of $\mathfrak{g}$.*

Corollary 5.2. *Let G be a smooth S-prescheme in groups of finite presentation, and let $\mathfrak{g}$ be its Lie algebra.*

a) *Conditions* (C_0) *through* (C_2) *of 2.9 for $\mathfrak{g}$ are equivalent. In particular, if the infinitesimal rank of the fibers of $\mathfrak{g}$ at the points of S is locally constant, then, étale locally, $\mathfrak{g}$ admits a Cartan subalgebra and hence, by 3.9*(a)*, G admits a subgroup of type* (C).

b) *Let H be a subprescheme in groups of G, smooth and of finite presentation over S, whose fibers are connected and at every $s \in S$ have the same reductive rank as G; for example, H may be a maximal torus or a Cartan subgroup of G. Let D be a subgroup of type* (C) *of G. Then*

$$H \subset D \quad \Longleftrightarrow \quad \mathfrak{h} \subset \mathfrak{d}.$$

c) *Suppose that condition* (C_0) *holds, that is, the infinitesimal rank of G is locally constant. Let H be a subprescheme in groups of G, smooth and of finite presentation over S, with connected nilpotent fibers, and having at every point the same reductive rank as G; for example, H may be a maximal torus or a Cartan subgroup of G. Then H is contained in a unique subgroup D of type* (C) *of G.*

d) *If G admits a Cartan subgroup fpqc locally, then so does every subgroup D of type* (C) *of G.*

Proof. a) Assume (C_0), and let us prove (C_2), that is, that every quasi-regular section a of $\mathfrak{g}$ is regular. As usual, we reduce first to S affine noetherian and then, since the question is infinitesimal by 2.15(b), to S local Artinian. In this last case (C_0) is automatic. We may moreover suppose that the residue field k of S is infinite.

By 3.7, it is enough to show that $\mathfrak{g}$ admits a Cartan subalgebra. Let T be a maximal torus of G. By 5.1 there is a quasi-regular element t in the Lie algebra $\mathfrak{t}$ of T. We shall show that t is regular.

Consider the linear representation of T on $\mathfrak{g}$ induced by the adjoint representation of G. There is a finite set of characters $(u_i)_{i\in I}$ of T and a direct-sum decomposition

$$\mathfrak{g} = \bigoplus_{i\in I} \mathfrak{g}_i$$

into T-stable submodules such that T acts on $\mathfrak{g}_i$ through u_i (I, §4.7.3). Let

$$u_i' \colon \mathfrak{t} \longrightarrow A$$

be the homomorphism of Lie algebras induced by $u_i \colon T \to \mathbb{G}_m$, where A is the ring of S. Let u_{i0} and u_{i0}' be their reductions under $A \to k$. Put

$$I' = \{i \in I \mid u_{i0}' \neq 0\}, \qquad I'' = I \setminus I'.$$

The regularity of t_0 is expressed by

$$u_{i0}'(t_0) \neq 0 \qquad (i \in I'),$$

and therefore $u_i'(t)$ is invertible for $i \in I'$.

The Cartan subalgebra of $\mathfrak{g}_0$ defined by t_0, that is, the kernel of the semisimple endomorphism $\mathrm{ad}(t_0)$, is

$$\bigoplus_{i\in I''} (\mathfrak{g}_i)_0.$$

Set

$$\mathfrak{d} = \bigoplus_{i\in I''} \mathfrak{g}_i.$$

Then $\mathrm{ad}(t)$ is nilpotent on $\mathfrak{d}$, and is an automorphism of

$$\mathfrak{g}/\mathfrak{d} \simeq \bigoplus_{i\in I'} \mathfrak{g}_i.$$

By 2.6, t is regular.

b) This follows from 3.12(a). By 5.1, H satisfies the required hypothesis that every geometric fiber of $\mathfrak{h}$ contains a regular element of the corresponding fiber of $\mathfrak{g}$.

c) By b), it is enough to show that $\mathfrak{h}$ is contained in a unique Cartan subalgebra of $\mathfrak{g}$. By a), $\mathfrak{g}$ satisfies (C_2). We are therefore reduced to the following lemma.

Lemma 5.3. *Let $\mathfrak{g}$ be a Lie algebra over S, locally free of finite type as a module, and satisfying condition* (C_2) *of 2.9. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ such that $\mathfrak{h}$ is locally a direct summand, is locally nilpotent, and, for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$. Then $\mathfrak{h}$ is contained in a unique Cartan subalgebra of $\mathfrak{g}$.*

In the situation at hand, $\mathfrak{h}$ has these properties: it is locally nilpotent because H has nilpotent fibers, and the condition on regular elements follows from 5.1.

Since 5.3 is fpqc local, it is enough, at a point $s \in S$ with $\kappa(s)$ infinite, to find an open neighborhood U of s on which existence and uniqueness hold after every base change $S' \to S$ factoring through U. Choose a regular element of $\mathfrak{g} \otimes \kappa(s)$ lying in $\mathfrak{h} \otimes \kappa(s)$, and extend it to a section a of $\mathfrak{h}$ over an open neighborhood U of s. Condition (C_2) and 2.10 allow us to shrink U so that a is regular. We may thus suppose $U = S$.

Every Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$ contains a, and hence equals

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$$

by 2.6; this proves uniqueness. Conversely, $\mathfrak{h} \subset \mathfrak{d}$ because $\mathfrak{h}$ is locally nilpotent, which proves existence.

d) This is a special case of XII 7.9(d).

Corollary 5.4. *Let G be a smooth connected algebraic group over an algebraically closed field k. Let H be a connected algebraic subgroup such that $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$, or equivalently such that H contains a subgroup of type* (C) *of G. Then the number of conjugates of H containing a regular element of $G(k)$ equals the number of conjugates of $\mathfrak{h}$ containing a regular element of $\mathfrak{g}$.*

Indeed, let g be a regular element of $G(k)$, and let C be the unique Cartan subgroup of G containing g (XIII 2). The conjugates of H that contain g are exactly those that contain C (XIII 2.8(b)).

Similarly, let x be a regular element of $\mathfrak{g}$. If a conjugate $\mathfrak{h}'$ of $\mathfrak{h}$ contains x, then

$$\mathrm{ad}(x)_{\mathfrak{g}/\mathfrak{h}'}$$

is injective by XIII 5.4, and therefore

$$\mathfrak{h}' \supset \mathrm{Nil}(x, \mathfrak{g}) = \mathfrak{d}.$$

Thus the number of conjugates of $\mathfrak{h}$ containing x equals the number containing the Cartan subalgebra $\mathfrak{d}$.

The subalgebra $\mathfrak{d}$ is the Lie algebra of a subgroup D of type (C) of G. We may suppose that $C \subset D$. It remains to show that

$$H \supset C \quad \Longleftrightarrow \quad \mathfrak{h} \supset \mathfrak{d}.$$

By XIII 5.5, the inclusion $\mathfrak{h} \supset \mathfrak{d}$ implies $H \supset D$, and *a fortiori* $H \supset C$. Conversely, by 5.1 the Lie algebra $\mathfrak{t}$ of C contains a regular element x of $\mathfrak{g}$. Hence $H \supset C$ implies $x \in \mathfrak{h}$, and, as observed above, this implies $\mathfrak{h} \supset \mathfrak{d}$. This completes the proof.

Let G be as in 5.2, and suppose that the infinitesimal rank of its fibers is locally constant, that is, condition (C_0) holds. By 5.2(c), there is a morphism of functors on $(\mathrm{Sch}/S)^\circ$,

$$\mathcal{C} \longrightarrow \mathcal{D},$$

where

$$\mathcal{C}(S') = \{\text{Cartan subgroups of } G_{S'}\},$$
$$\mathcal{D}(S') = \{\text{subgroups of type (C) of } G_{S'}\}.$$

By 3.10 and 5.2(a), the functor $\mathcal{D}$ is represented by a smooth quasi-projective prescheme over S. Let D be the universal subgroup of type (C) of $G_{\mathcal{D}}$. One may then define the functor

$$\mathcal{C}_D \colon (\mathrm{Sch}\,/\mathcal{D})^\circ \longrightarrow \mathrm{Ens}$$

for the $\mathcal{D}$-group D, exactly as $\mathcal{C}$ was defined for the S-group G.

Proposition 5.5. *Under the preceding hypotheses, regard $\mathcal{C}$ as a functor over $\mathcal{D}$. Then $\mathcal{C}$ is $\mathcal{D}$-isomorphic to the functor $\mathcal{C}_D$ of Cartan subgroups of D.*

This follows immediately from the next corollary.

Corollary 5.6. *Let G be a smooth S-prescheme in groups of finite presentation, and let D be a subgroup of type* (C) *of G. There is a bijective correspondence between the Cartan subgroups of G contained in D and the Cartan subgroups of D. More precisely, for a subgroup $H \subset D$, H is a Cartan subgroup of G if and only if it is a Cartan subgroup of D.*

This is a special case of XII 7.9(c), taking into account that, over an algebraically closed field, every subgroup of type (C) of G contains a Cartan subgroup of G.

For the next section, the principal result obtained here is 5.2(c) with H a Cartan subgroup of G. It permits the formulation of 5.5 and thus gives a useful dévissage of $\mathcal{C}$.

0.6 Applications to the Structure of Algebraic Groups

Theorem 6.1. *Let G be a smooth algebraic group over a field k. Consider the scheme $\mathcal{T}$ of maximal tori of G, isomorphic to the scheme $\mathcal{C}$ of Cartan subgroups of G. This is a homogeneous space under G^0 and a smooth affine connected algebraic scheme* (XII 7.1(d)). *Then $\mathcal{C}$ is a rational variety: its field of rational functions is a purely transcendental extension of k.*

We first prove the theorem when k is infinite. We may plainly suppose that G is connected, since $\mathcal{T}$, and hence $\mathcal{C}$, is unchanged when G is replaced by G^0. Moreover, by XII 7.6, $\mathcal{C}$ is unchanged when G is divided by a central subgroup. We may therefore first divide by the center of G and suppose that G is affine (XII 6.1), and then divide by its reductive center (XII 4.1 and 4.4) and suppose that the reductive center of G is trivial (XII 4.7(b)).

We proceed by induction on $n = \dim G$, assuming the result for dimensions $n' < n$. If G is nilpotent, then $\mathcal{C}$ consists of a single k-rational point and the assertion is trivial. Otherwise, the Lie algebra of G is not nilpotent (1.3). Its Cartan subalgebras, and therefore the subgroups of type (C) of G, have some dimension $n' < n$. Consider the morphism

$$\mathcal{C} \longrightarrow \mathcal{D}$$

of 5.5. By 3.10, since the infinite-field Lie algebra $\mathfrak{g}$ contains a regular element, $\mathcal{D}$ is a rational variety. Thus its function field K is purely transcendental over k.

Let x be the generic point of $\mathcal{D}$. By 5.5, the fiber of $\mathcal{C}$ at x is the scheme of Cartan subgroups of a smooth connected algebraic group D_x over $K = \kappa(x)$, namely the generic subgroup of type (C) of G. The function field L of $\mathcal{C}$ is therefore the function field of $\mathcal{C}_{D_x}$. Since $\dim D_x = n' < n$, the induction hypothesis makes L purely transcendental over K, and hence, by transitivity, purely transcendental over k.

Suppose now that k is finite. A different argument is required. We may again suppose that G is affine and connected. Since k is perfect, the radical of $G_{\bar{k}}$ descends to a subgroup R of G. Suppose first that $R \neq G$, so that G is not solvable, and let

$$u\colon G \longrightarrow G' = G/R$$

be the canonical morphism. By XII 7.1(e), it induces

$$v\colon \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}, \qquad C \longmapsto u(C).$$

Let x be the generic point of $\mathcal{C}_{G'}$, and put $K = \kappa(x)$. The fiber $v^{-1}(x)$ is the scheme of Cartan subgroups of G_K whose image in G'_K is the generic Cartan subgroup C'_x of G'. By XII 7.9(c), it is also the scheme of Cartan subgroups of

$$H = u_K^{-1}(C'_x).$$

The field K is infinite. The infinite-field case just proved therefore shows that the function field L of $\mathcal{C}_G$, which is also that of $\mathcal{C}_H$, is purely transcendental over K.

It remains to prove that K is purely transcendental over k, and we are thus reduced to the case in which G is semisimple. Dividing by its reductive center, we may further suppose that G is adjoint. Then 3.18 gives $\mathcal{C}_G \simeq \mathcal{D}_G$, and by 3.10 it is enough to show that $\mathfrak{g}$ has a regular point. As noted in 3.20, this is an unpublished result of Chevalley.[8]

It remains only to treat the case in which k is finite and G is connected, affine, and solvable. In fact, the next result holds over an arbitrary field.

Corollary 6.2. *Let G be a smooth solvable algebraic group over a field k. Then the variety $\mathcal{C}$ of Cartan subgroups of G is isomorphic to an affine space* $\operatorname{Spec} k[t_1, \ldots, t_N]$.

We may suppose that G is connected and affine. Let G_∞ be the terminal group in the descending central series

$$G_0 = G, \qquad G_{i+1} = [G, G_i].$$

Thus G_∞ is the smallest invariant algebraic subgroup of G for which G/G_∞ is nilpotent. Let C be a Cartan subgroup of G, which exists by 1.1. Its image in G/G_∞ is a Cartan subgroup and hence is all of G/G_∞. Consequently the morphism

$$G_\infty \longrightarrow \mathcal{C}, \qquad g \longmapsto \operatorname{Ad}(g)C$$

is an epimorphism and identifies

$$\mathcal{C} \simeq G_\infty/(N \cap G_\infty),$$

where N is the normalizer of C in G. In fact $N = C$, as recalled in 4.4, but that is immaterial here.

The group $U = G_\infty$ is smooth, connected, and unipotent: over $\bar{k}$ it lies in the unipotent part of G, by the known structure of smooth affine solvable groups *Bible* 6, Theorem 3. If k is perfect, as in the finite-field case needed for 6.1, it follows easily that U is k-unipotent: it has a composition series by algebraic subgroups U_i with $U_i/U_{i+1} \simeq \mathbb{G}_a$.

Rosenlicht proved that a group of the form G_∞ above is k-unipotent without any restriction on k (M. Rosenlicht, *Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Matematica, 1963, pp. 97–120, Theorem 4, Corollary 2). We admit this substantially more delicate result and apply the following lemma.

[8] See the Appendix by J.-P. Serre.

Lemma 6.3. *Let U be a smooth connected algebraic group over a field k, and let $X = U/V$ be a homogeneous space under U having a k-rational point. If U is k-unipotent, then, as a k-scheme, X is isomorphic to an affine space* $\operatorname{Spec} k[t_1, \ldots, t_N]$.

Choose a composition series

$$U = U_0 \supset U_1 \supset \cdots \supset U_n = (e)$$

by smooth connected normal subgroups of U, with $U_i/U_{i+1} \simeq \mathbb{G}_a$. The groups

$$K_i = U_i V$$

are algebraic subgroups of U. They need not be smooth or connected when V is not, but this does not matter, and K_{i+1} is normal in K_i. The canonical epimorphism

$$U_i/U_{i+1} \longrightarrow K_i/K_{i+1}$$

shows that K_i/K_{i+1} is either the unit group or isomorphic to $\mathbb{G}_a/H_i$, where H_i is a finite subgroup of $\mathbb{G}_a$. Rosenlicht's Theorem 2 in the cited paper gives

$$K_i/K_{i+1} \simeq \mathbb{G}_a;$$

this last assertion is immediate when k is perfect.

Put $X_i = U/K_i$. We prove inductively that every X_i is an affine space. If K_i/K_{i+1} is trivial, then $X_i = X_{i+1}$. Otherwise X_{i+1} is a principal $\mathbb{G}_a$-bundle over X_i. Since X_i is affine, the bundle is trivial, so

$$X_{i+1} \simeq X_i \times \mathbb{G}_a.$$

The induction proves the lemma. It proves 6.2 and thereby completes the proof of 6.1.

Corollary 6.4. *Let G be a smooth algebraic group over an infinite field k. Then the set of k-rational points of $\mathcal{C}$, in the notation of 6.1, is Zariski dense. The union of the Cartan subgroups of G is dense in G.*

The first assertion holds for every rational variety over an infinite field and is, for us, the most important arithmetic consequence of the rationality result. The second follows from the first and the density theorem XIII 2.1.

Corollary 6.5. *Let G be a smooth connected algebraic group over k. Then the variety Z of regular semisimple points of G* (XIII 3.5) *is unirational. In particular, if k is infinite, its k-rational points are dense in Z.*

Indeed, Z is an open subscheme of a torus over $\mathcal{C}$. Its function field L is therefore the function field of a torus over the function field K of $\mathcal{C}$, namely the generic maximal torus of G. Thus L is unirational over K by XIII 3.4. Since 6.1 makes K purely transcendental over k, L is unirational over k.

Corollary 6.6. *Let G be a smooth connected algebraic group over k, and let H be the subgroup generated by the subvariety Z of regular semisimple points. Equivalently* (XII 8.2), *H is the smallest invariant algebraic subgroup of G for which G/H has reductive rank zero; over $\bar{k}$, G/H is an extension of an abelian variety by a smooth connected unipotent group. Then H is unirational.*

In particular, suppose $G = H$; equivalently (XII 8.4), *suppose G is affine and there is no nontrivial homomorphism $G_{\bar{k}} \to \mathbb{G}_a$. Then G is unirational. If, in addition, k is infinite, $G(k)$ is dense in G.*

This follows immediately from 6.5. Indeed, if Z_i are smooth connected unirational k-preschemes and $u_i\colon Z_i \to G$ are morphisms, then the algebraic subgroup of G generated by the u_i is unirational ($\mathrm{VI_B}$ 7.1).

An interesting special case of either 6.5 or 6.6 is the following.

Corollary 6.7. *Let G be a smooth connected affine algebraic group of unipotent rank zero. Then G is unirational.*

The conclusion of 6.4 may be sharpened as follows.

Corollary 6.8. *Let G be a smooth connected algebraic group over an infinite field k, and let C be a Cartan subgroup of G. Then the union of the conjugates of C by regular semisimple elements of $G(k)$ is dense in G.*

This follows at once from 6.4 and XIII 3.6, which says that the morphism

$$\varphi\colon Z \times C \longrightarrow G, \qquad \varphi(t,c) = \mathrm{Ad}(t)c,$$

is dominant. The same result also implies the next corollary, without any assumption that k is infinite.

Corollary 6.9. *Let G be a smooth connected algebraic group over k, and let H be a smooth connected algebraic subgroup having the same reductive rank and the same nilpotent rank as G. Equivalently, over $\bar{k}$, the group $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$. If H is unirational, so is G. If $H(k)$ is dense in H, then $G(k)$ is dense in G.*

The morphism

$$\varphi\colon Z \times H \longrightarrow G, \qquad \varphi(t,h) = \mathrm{Ad}(t)h,$$

is dominant. By 6.5, Z is unirational, and by hypothesis so is H, hence so are $Z \times H$ and G. The density assertion is proved in the same way.

We recover the following well-known result, due to Chevalley in characteristic zero and to Rosenlicht in characteristic $p > 0$.

Corollary 6.10. *Let G be a smooth connected affine algebraic group over a perfect field k. Then G is unirational. If k is also infinite, $G(k)$ is dense in G.*

By 1.1, G has a Cartan subgroup C. By 6.9, it is enough to prove that C is unirational. Since k is perfect, Galois descent from the algebraically closed case *Bible* 6, Theorem 2 gives

$$C = T \times C_u,$$

where T is the maximal torus of C and C_u is smooth, connected, and unipotent. The torus T is unirational by XIII 3.4. Moreover, C_u is k-unipotent because k is perfect, so 6.3 applies to C_u.

Remarks 6.11. (a) Rosenlicht gave examples of twisted forms of $\mathbb{G}_a$ over a nonperfect field having only finitely many rational points; *a fortiori*, they are not unirational. Chevalley, on the other hand, gave an example of a torus over a field of characteristic zero that is not rational. By contrast, Chevalley's theory of semisimple groups shows that, over an algebraically closed field, every smooth connected affine algebraic group is rational. In any case, the question of unirationality arises only for affine algebraic groups: by Chevalley's structure theorem, a unirational algebraic group is necessarily affine.

(b) With the notation of 6.6, it is tempting to seek a criterion for the unirationality of G in terms of G/H, which is unipotent when G is affine. Clearly G/H must be unirational;

is that condition also sufficient? An example of Rosenlicht shows that a smooth connected unipotent algebraic group U may be rational without being k-unipotent.

(c) Over a finite field k, it would be interesting to study questions of "density" such as the following one, raised by Rosenlicht: if G is a smooth connected algebraic group over k, is G generated by its Cartan subgroups?[9]

Editors' note. The editors did not identify this result in Steinberg's *Collected Papers*; here is a proof based on the Bruhat decomposition.

Let G/k be semisimple over the finite field k, and let $G^\sharp$ be the subgroup generated by its k-subtori. The question is stable under central isogeny and restriction of scalars, so, as in Lemma 1 of the Appendix below, it is enough to treat the geometrically simple case. The group G is quasi-split by Lang's theorem and admits a Killing pair (T, B). By definition $T \subset G^\sharp$, hence T normalizes $G^\sharp$. The big cell

$$R_u(B^-)\, T\, R_u(B)$$

shows that it suffices to prove $R_u(B) \subset G^\sharp$. Let S be the maximal split k-torus of T. For the relative root system $\Phi(G, S)$, choose a basis Δ_k, and denote by $U(\alpha)$ the unipotent subgroup associated with α (A. Borel, *Linear Algebraic Groups*, second edition, 1991, Proposition 21.9). Since $R_u(B)$ is generated by the $U(\alpha)$, $\alpha \in \Delta_k$, it is enough to show $U(\alpha) \subset G^\sharp$.

The classification supplies a quasi-split semisimple subgroup G_α of type A_1, ${}^2A_1$, ${}^3A_1$, or ${}^2A_2$, with

$$U(\alpha) \subset G_\alpha \subset G.$$

Thus it is enough to consider $G = \mathrm{PGL}_2$ or $G = \mathrm{SU}_3(K)$, where K/k is the unique quadratic extension.

For PGL_2, since $G^\sharp$ contains the standard split torus T, the possibilities up to $G(k)$-conjugacy are

$$G^\sharp = T, \qquad G^\sharp = B, \qquad G^\sharp = \mathrm{PGL}_2\,.$$

The first is excluded because $G^\sharp$ contains the k-torus $R_{K/k}(\mathbb{G}_m)/\mathbb{G}_m$. The preceding discussion also shows that $G^\sharp = B$ forces $G^\sharp = \mathrm{PGL}_2$. Hence $G^\sharp = \mathrm{PGL}_2$.

For $G = \mathrm{SU}_3(K)$, one has $\Delta_k = \{\alpha\}$, and the possibilities up to conjugacy are

$$\begin{aligned} G^\sharp = T, \quad B, \quad U(2\alpha) \rtimes T, \\ \langle U(2\alpha), U(-2\alpha), T\rangle = \mathrm{SL}_2 \cdot T, \quad \text{or } G. \end{aligned}$$

The case B is excluded as for PGL_2. The case $\mathrm{SL}_2 \cdot T$ is excluded because $G^\sharp$ contains the k-torus

$$R^1_{K/k}\Big(R^1_{K_3/K}(\mathbb{G}_m)\Big)\,,$$

where K_3/K is the degree-three extension, and the isogeny class of this torus is irreducible. The same torus excludes T and $U(2\alpha) \rtimes T$. Hence $G^\sharp = G$.

For Rosenlicht's question one may reduce to the affine case by dividing by the center. In the semisimple case, an affirmative answer would also follow from a strengthening of Chevalley's existence result for regular points mentioned in 3.20: one would need a regular element of $\mathfrak{g}$ outside $\mathfrak{h} = \mathrm{Lie}(H)$, where $H \neq G$ is a prescribed smooth algebraic subgroup of an adjoint semisimple group G over the finite field k.

[9] This question was subsequently answered in the affirmative by Steinberg.

0.7 Appendix: Existence of Regular Elements over Finite Fields

by J.-P. Serre

Throughout the Appendix, k denotes a finite field, $\bar{k}$ its algebraic closure, and $\mathcal{G} = \mathrm{Gal}(\bar{k}/k)$. If $q = \mathrm{Card}(k)$, then $\mathcal{G}$ is topologically generated by the Frobenius element

$$F\colon x \longmapsto x^q.$$

We shall prove the following theorem.[10]

Theorem. *Let G be an adjoint semisimple group defined over k, and let $\mathfrak{g}$ be its Lie algebra. Then the k-Lie algebra $\mathfrak{g}(k)$ contains a regular element.*

Remarks. (1) Recall that "adjoint" means that the center of G, as a subgroup scheme, is trivial. In view of the Chevalley seminar, this also means that, for a maximal torus T of G, the character group $X(T)$ of $T_{\bar{k}}$ is generated by the set R of roots. In particular, the rank of $\mathfrak{g}$ equals $\dim T$, the rank of G.

(2) The editor does not know whether the adjoint hypothesis is indispensable.

Lemma 1. *It is enough to prove the theorem when G is geometrically simple.*

Here geometrically simple means that $G_{\bar{k}}$ has no smooth normal subgroup scheme other than $G_{\bar{k}}$ and (e); equivalently, the associated root system is irreducible.

We may first suppose that G is indecomposable over k. Then $G_{\bar{k}}$ is a product of geometrically simple components S, permuted transitively by $\mathcal{G}$. If $\mathcal{H} \subset \mathcal{G}$ fixes one component S, and K is its fixed field, then S is defined over K, and the standard restriction-of-scalars argument gives

$$G = R_{K/k}(S).$$

Similarly,

$$\mathfrak{g} = R_{K/k}(\mathfrak{s}),$$

where $\mathfrak{s}$ is the Lie algebra of S. If the theorem holds for S, there is an element

$$x \in \mathfrak{s}(K) = \mathfrak{g}(k)$$

regular in $\mathfrak{s}$, and one checks immediately that it is regular in $\mathfrak{g}$. This proves the lemma.

Henceforth assume that G is geometrically simple, and choose a maximal torus T of G. Let X be the character group of $T_{\bar{k}}$, let R be its root system, W its Weyl group, and E the group of automorphisms of X preserving R. Then W is normal in E.

For another maximal torus T', write X', R', W', E' for the corresponding objects. If $y \in G(\bar{k})$ satisfies

$$yT_{\bar{k}}y^{-1} = T'_{\bar{k}},$$

then $\mathrm{int}(y)$ identifies (X', R', W', E') with (X, R, W, E). Replacing y changes this identification by an element of W.

Frobenius acts on X' while preserving R', and thus defines an element $f_{T'} \in E'$. Put $f = f_T$. After the above identification $E' \simeq E$, the element $f_{T'}$ becomes an element $f' \in E$, well defined up to W-conjugacy. We now compare f' with f.

[10] The theorem is due to Chevalley. The editor wishes to express his gratitude to American Express which, by misplacing a trunk of Chevalley's manuscripts, obliged him to reconstruct the proof.

Lemma 2. *One has $f' \equiv f \pmod{W}$. Conversely, every $f' \in E$ satisfying this condition arises from a maximal torus T' of G.*

Put $\bar{T} = T_{\bar{k}}$ and $\bar{T}' = T'_{\bar{k}}$, and choose $y \in G(\bar{k})$ such that

$$y\bar{T}y^{-1} = \bar{T}'.$$

Since $y^q\bar{T}y^{-q} = \bar{T}'$, the element

$$n = y^{-1}y^q$$

belongs to the normalizer $N(T)(\bar{k})$. The action of f' on the points of $\bar{T}$ is

$$\begin{aligned} f'(t) &= y^{-1}(yty^{-1})^q y \\ &= y^{-1}y^q\, t^q\, y^{-q}y = nt^q n^{-1}. \end{aligned}$$

If $w \in W$ is represented by n, then f' and $f \circ w$ have the same action on $\bar{T}(\bar{k})$, hence on its characters. Thus $f' = f \circ w$, and $f' \equiv f \pmod{W}$.

Conversely, represent a given $w \in W$ by $n \in N(T)(\bar{k})$. Lang's theorem gives

$$n = y^{-1}y^q$$

for some $y \in G(\bar{k})$. Then $y\bar{T}y^{-1}$ is defined over k, and the preceding calculation shows that its corresponding element is $f \circ w$.

Lemma 3. *Let X, R, W, E be as above, with R irreducible, and let $\varphi \in E/W$. There exist a representative $f \in E$ of φ and roots $\theta_1, \ldots, \theta_n$ such that:*

1. *$(\theta_1, \ldots, \theta_n)$ is a basis of X;*
2. *R is the union of the orbits of the θ_i under the powers of f: every $a \in R$ has the form $a = f^m\theta_i$ for suitable m and i.*

The proof will be given below. We first need a lemma of linear algebra.

Lemma 4. *Let V be a vector space over k, and let $(\theta_1, \ldots, \theta_n)$ be a basis of the dual of $V \otimes_k \bar{k}$. Then there is an $x \in V$ such that $\theta_i(x) \neq 0$ for every i.*

Let V^* be the dual of V, and let W_i be the $\bar{k}$-subspace of $V^* \otimes_k \bar{k}$ generated by the Galois conjugates of θ_i. It descends to a k-subspace, again denoted W_i. The natural map

$$W_1 \otimes \cdots \otimes W_n \longrightarrow \bigwedge^n V^*$$

is nonzero: after extension to $\bar{k}$, it contains $\theta_1 \wedge \cdots \wedge \theta_n \neq 0$. Hence V^* has a basis $u_1, \ldots, u_n$ with $u_i \in W_i$. Choose $x \in V$ with $u_i(x) \neq 0$ for all i, for example with $u_i(x) = 1$. If $\theta_i(x) = 0$, every Galois conjugate of θ_i vanishes at x, and so does u_i, a contradiction.

End of the proof of the theorem. By Lemmas 2 and 3, choose a maximal torus T whose element f has the properties of Lemma 3. Let Y be the dual of X. The Lie algebra of T satisfies

$$\mathfrak{t}(\bar{k}) \simeq Y \otimes_{\mathbb{Z}} \bar{k},$$

compatibly with the Galois action. Put $V = \mathfrak{t}(k)$. An element $x \in V$ is regular precisely when it is annihilated by no root $\alpha \in R$, viewed as a linear form in $V^* \otimes_k \bar{k}$.

Lemma 4 supplies $x \in V$ annihilated by none of the θ_i. Every root is a Galois conjugate of one of the θ_i, by property (2) of Lemma 3. Hence no root annihilates x, and x is regular. This proves the theorem.

Proof of Lemma 3. The proof uses Coxeter transformations. Let $a_1, \ldots, a_n$ be a simple system of roots of R, and let r_i be the reflection corresponding to a_i. Put

$$c = r_1 \cdots r_n.$$

Then $c \in W$. Although c depends on the simple system and on its ordering, its conjugacy class does not. It is called the Coxeter element. We shall use, without proof, the fact that 1 is not an eigenvalue of c.

Lemma 5. Put

$$\theta_i = r_n \cdots r_{i+1}(a_i).$$

Then:

(a) the θ_i form a basis of the group X generated by the roots;

(b) $\theta_i > 0$ and $c(\theta_i) < 0$ for every i;

(c) every root $a \in R$ such that $a > 0$ and $c(a) < 0$ is one of the θ_i;

(d) R is the union of the orbits of the θ_i under the powers of c.

Indeed,

$$\theta_i = a_i + \sum_{j>i} m_{ij} a_j, \qquad m_{ij} \in \mathbb{Z},$$

which proves (a). Assertions (b) and (c) follow from the observation that r_i preserves the sign of every root other than $\pm a_i$, and reverses the sign of $\pm a_i$.

For (d), an orbit of c cannot consist entirely of positive roots, nor entirely of negative roots. Otherwise the sum of the roots in that orbit would give a nonzero element of X fixed by c, contrary to the fact that 1 is not an eigenvalue. Hence every orbit contains a root $a > 0$ for which $c(a) < 0$, and (c) applies.

Remark. The proof was recalled only for the reader's convenience; one could instead refer to the standard texts on Coxeter transformations, for example Koszul, Séminaire Bourbaki 1959/1960, Exposé 191. They also show that all orbits of c have the same number h of elements and each contains exactly one θ_i. In particular,

$$h = \frac{\mathrm{Card}(R)}{n}.$$

We return to Lemma 3 and distinguish three cases.

(1) Suppose that $\varphi \in E/W$ is the identity. We must choose $f \in W$; take $f = c$. Lemma 5 shows that this works.

(2) Suppose that R has type A_n, where $n \geq 2$ is even, and that φ is the unique nontrivial element of E/W. Then $-1 \notin W$, and we take

$$f = -c.$$

The Coxeter element c has order $h = n + 1$, which is odd. If $a \in R$, Lemma 5 gives $a = c^m \theta_i$ for suitable m, i. Adding h to m if necessary, we may suppose that m is even. Then

$$a = (-c)^m \theta_i = f^m \theta_i.$$

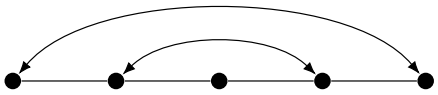
Thus the orbits of the θ_i under f fill R.

(3) Suppose that $\varphi \in E/W$ is nontrivial and that R has one of the following types:

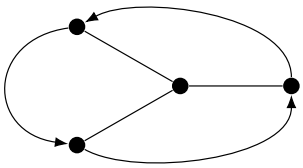
$$A_n \ (n \geq 3 \text{ odd}), \qquad D_4, \qquad D_n \ (n \geq 5), \qquad E_6.$$

These, together with the even A_n case treated separately above, are exactly the cases in which E/W is nontrivial. The following diagrams record the corresponding Dynkin-diagram automorphisms.

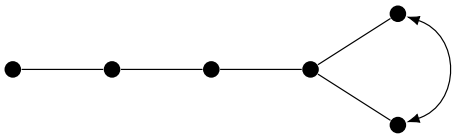
$A_n,\ n$ odd ≥ 3



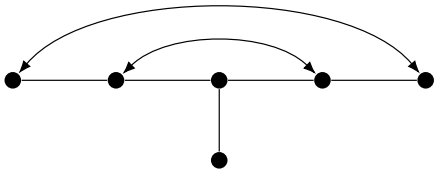
D_4



$D_n,\ n \geq 5$



E_6.



Let S be a simple system of roots, not yet ordered. The group E is the semidirect product of W with the group Ψ of permutations of S preserving the Cartan matrix, or equivalently the Dynkin diagram. Thus $E/W \simeq \Psi$, and φ corresponds to some $\psi \in \Psi$.

Inspection of the diagrams shows that each ψ-orbit in S consists of pairwise orthogonal roots. This fails for A_n with n even, which is why that case was separated. If σ is such an orbit, the reflections r_a, $a \in \sigma$, commute; write

$$\rho_\sigma = \prod_{a \in \sigma} r_a.$$

Plainly ρ_σ commutes with ψ.

Choose a total order on S for which every ψ-orbit is a segment, and write $S = \{a_1, \ldots, a_n\}$. Let $\theta_1, \ldots, \theta_n$ and $c = r_1 \cdots r_n$ be defined as above. The element c is a product of the ρ_σ, with the orbits in some order, so it commutes with ψ. Put

$$f = c\psi.$$

The automorphism ψ permutes the θ_i. Indeed, $\psi(\theta_i) > 0$, and, since c commutes with ψ,

$$c(\psi(\theta_i)) = \psi(c(\theta_i)) < 0.$$

Lemma 5(c) therefore gives $\psi(\theta_i) = \theta_j$ for some j.

Finally, if $a \in R$, Lemma 5 gives $a = c^m\theta_i$ for some m, i. Hence

$$a = f^m(\psi^{-m}\theta_i) = f^m\theta_j$$

for a suitable j. Thus f and the θ_i satisfy both conditions of Lemma 3, completing its proof.

Remark. Except in case (2), one can prove that every orbit of f has exactly

$$h = \frac{\mathrm{Card}(R)}{n}$$

elements and contains exactly one θ_i. In case (2), some of the θ_i are superfluous.

Exposé XV
Complements on Sub-tori of a Group Prescheme
Application to Smooth Groups

by M. Raynaud[1]

0. Introduction

This Exposé supplements and partly recasts Exposés XI and XII; knowledge of Exposés XIII and XIV is not indispensable. Continuing the effort begun in XII, we shall work with S-group preschemes that are not necessarily affine and not necessarily separated over S.

Sections 1, 2, 3, and 4 are devoted to the study of sub-tori of a group prescheme. We obtain infinitesimal lifting theorems (§2) and a global lifting theorem (§4), in which points of finite order (§1) play an essential role.

Sections 5, 6, and 7 are independent of the preceding sections. The consideration of infinitesimal neighbourhoods leads to the representability of the functor of smooth subgroups equal to their connected normalizer (§5). In §§6 and 7 we turn more particularly to Cartan subgroups.

Finally, in §8 we give a necessary and sufficient condition for the functor of sub-tori of a smooth group, or the functor of maximal tori, to be representable.

1. Lifting finite subgroups

1.1. Finite, smooth, central subgroups of multiplicative type

Proposition 1.1.— *Let S be an affine scheme, let S_0 be a closed subscheme of S defined by a square-zero ideal, let G be an S-group prescheme, and let H_0 be a subgroup scheme of $G_0 = G \times_S S_0$ that is smooth and of finite type over S_0, and of multiplicative type. A subgroup scheme of G, of multiplicative type, lifting H_0 exists if and only if there exists a subscheme H' of G, flat over S, lifting H_0.*

The necessity of the condition is clear; let us prove its sufficiency. The group H_0 of multiplicative type is quasi-isotrivial (X 4.5). By Exposé X 2.1, there exist an S-group H

[1]This Exposé and the two following ones (Exposés XVI and XVII) do not correspond to oral lectures delivered at the seminar. With some additional material, they develop the substance of concise manuscript notes written by A. Grothendieck in 1964 for the present seminar.

of multiplicative type and an S_0-isomorphism of groups

$$u_0 : H \times_S S_0 \xrightarrow{\sim} H_0.$$

Since H' is flat over S and $H' \times_S S_0$ is of finite presentation over S_0 (Exposé IX 2.1(b)), H' is of finite presentation over S. Its fibres are smooth, so H' is smooth over S (EGA IV 17.5.1). Since S is affine, u_0 therefore lifts to an S-morphism of preschemes

$$u : H \longrightarrow H'.$$

It then follows from Exposé III 2.1 and Exposé IX 3.1 that the composite

$$v_0 : H \times_S S_0 \xrightarrow{\sim} H_0 \longrightarrow G_0$$

also lifts to an S-homomorphism

$$v : H \longrightarrow G.$$

Since v_0 is an immersion, so is v. The image of H under v is therefore a subgroup scheme of G, of multiplicative type, lifting H_0.

Proposition 1.2.— *Let S be a prescheme, let S_0 be a subprescheme of S defined by a locally nilpotent sheaf of ideals, let G be an S-group prescheme flat and of finite presentation over S, and let H_0 be a subgroup scheme of $G_0 = G \times_S S_0$ that is smooth, finite over S_0, of multiplicative type, and central. Then there exists a unique subgroup scheme H of G, of multiplicative type, lifting H_0. Moreover, H is central. (See XVII, Appendix III, 1.)*

Proposition 1.2 bis.— *Let S, G, S_0, G_0 be as above, let H be an S-group scheme of multiplicative type, smooth and finite over S, and let $u_0 : H_0 = H \times_S S_0 \to G_0$ be a central homomorphism. Then u_0 lifts uniquely to a homomorphism $u : H \to G$. Moreover, u is central.*

The existence of the lifting u in Proposition 1.2 bis follows readily from Proposition 1.2 by considering the graph of u_0. The lifting u is unique and central by Exposé IX 3.4 and Exposé IX 5.1.

Proof of Proposition 1.2. The uniqueness of H, and the fact that H is central, follow from Exposé IX 5.6(b) and Exposé IX 3.4 bis. In view of uniqueness, to prove existence we may assume that S is affine and that S_0 is defined by a square-zero ideal; by Proposition 1.1 it is enough to find a subscheme of G, flat over S, lifting H_0.

Since H_0 is smooth and finite over S_0, after possibly restricting S we may assume that there is an integer $n > 0$, invertible on S, such that $H_0 = {}_nH_0$. Consider the n-th power morphism on G:

$$u : G \longrightarrow G, \qquad x \longmapsto x^n.$$

We continue to denote by ${}_nG$ the "kernel of u", that is, the inverse image under u of the identity section of G. Notice that u is not, in general, a group homomorphism. We first admit the following lemma.

Lemma 1.3.— *Let k be a field, let G be a group scheme locally of finite type over k, let n be an integer prime to the characteristic of k, and let u be the n-th power morphism on G. Then u is étale at every point x of G that belongs to the centre of G.*

Since G is flat and of finite presentation over S, the preceding lemma and EGA IV 17.8.2 show that, if $x \in G$ lies above $s \in S$ and belongs to the centre of G_s, then u is étale at x. If in addition $x \in {}_nG$, then ${}_nG$ is étale over S at x. By hypothesis, H_0

is central and is contained in ${}_nG_0$; it is therefore contained in the largest open subset V of ${}_nG$ that is étale over S. Since H_0 and $V \times_S S_0 = V_0$ are étale over S_0, H_0 is an open subscheme of V_0 (EGA IV 17.8.7 and 17.9.1). The open subprescheme of V having the same underlying space as H_0 is consequently a subscheme of G, flat over S, lifting H_0.

It remains to prove Lemma 1.3. The largest open subset of G on which u is étale is invariant under extension of the base field (EGA IV 17.8.2), so we may reduce to the case in which x is k-rational. Let t_x denote translation by x, a k-automorphism of G. Since x belongs to the centre of G, the diagram

$$\begin{array}{ccc} G & \xrightarrow{u} & G \\ {\scriptstyle t_x}\downarrow & & \downarrow{\scriptstyle t_x^n} \\ G & \xrightarrow{u} & G. \end{array}$$

is commutative. It is therefore enough to prove that u is étale at the identity; this was established in VII_A 8.4.

1.2. Global lifting of finite groups

Lemma 1.4.— *Let A be a local ring, separated and complete for the topology defined by its maximal ideal $\mathfrak{m}$, let*

$$S = \mathrm{Spec}(A), \qquad S_n = \mathrm{Spec}(A/\mathfrak{m}^n).$$

Then, for every prescheme X (respectively, for every S-prescheme X), the canonical map

$$(*) \qquad \mathrm{Hom}(S, X) \longrightarrow \varprojlim_n \mathrm{Hom}(S_n, X)$$

(respectively,

$$(**) \qquad \Gamma(X/S) \longrightarrow \varprojlim_n \Gamma(X_n/S_n), \qquad X_n = X \times_S S_n)$$

is bijective.

Statement $(**)$ is an easy consequence of $(*)$. Let us prove $(*)$. Let

$$u_n : S_n \longrightarrow X \qquad (n \in \mathbb{N})$$

be a coherent system of morphisms. The image y of the closed point of S_n is independent of n, and u_n factors through $\mathrm{Spec}\,\mathcal{O}_y$. On passing to the inverse limit, the morphisms u_n define a ring homomorphism

$$\widetilde{u} : \mathcal{O}_y \longrightarrow \varprojlim_n (A/\mathfrak{m}^n) = A.$$

This proves that $(*)$ is surjective; it is injective as soon as A is separated for the $\mathfrak{m}$-adic topology.

Corollary 1.5.— *Let A be a complete noetherian local ring, let $\mathfrak{m}$ be its maximal ideal, let*

$$S = \mathrm{Spec}(A), \qquad S_n = \mathrm{Spec}(A/\mathfrak{m}^n),$$

let X be a finite scheme over S, and let Y be an S-prescheme. Then the canonical map

$$\mathrm{Hom}_S(X,Y) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n}(X_n,Y_n)$$

(where $X_n = X \times_S S_n$, and similarly for Y_n) is bijective.

Indeed, it follows from EGA II 6.2.5 that X is a finite disjoint sum of local S-schemes finite over S. We are thus reduced to the case in which X itself is the spectrum of a complete noetherian local ring. But

$$\mathrm{Hom}_S(X,Y) = \Gamma(Z/X), \qquad Z = Y \times_S X,$$

and the preceding lemma applies.

Proposition 1.6.— *Let A, S, S_n be as above, and let G and M be two S-group preschemes, with M finite over S. Then:*

a) *The canonical map*

$$\mathrm{Hom}_{S\text{-gr}}(M,G) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}(M_n,G_n)$$

is bijective.

b) *If M is of multiplicative type and G is smooth over S, the canonical map*

$$\varphi : \mathrm{Hom}_{S\text{-gr}}(M,G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(M_0,G_0)$$

is surjective. Moreover, if $\varphi(u) = \varphi(u') = u_0$, then u and u' are conjugate by an element of $G(S)$ that reduces to the identity element of $G(S_0)$.

c) *If M is of multiplicative type and smooth over S, if G is flat and of finite type over S, and if $u_0 : M_0 \to G_0$ is a central homomorphism, then u_0 lifts uniquely to a homomorphism*

$$u : M \longrightarrow G.$$

Moreover, u is central if G has connected fibres.

Proof. *a)* This follows from 1.5, from the fact that $M \times_S M$ is finite over S, and from the characterization of group morphisms as the morphisms for which the familiar diagram

$$\begin{array}{ccc} M \times_S M & \xrightarrow{u\times u} & G \times_S G \\ \downarrow & & \downarrow \\ M & \xrightarrow[u]{} & G \end{array}$$

is commutative.

b) By Exposé IX 3.6, one can construct a coherent system of homomorphisms $u_n : M_n \to G_n$ lifting a homomorphism $u_0 : M_0 \to G_0$. In view of part a), this proves the first assertion of b).

If u and u' are two liftings of u_0, then u_n and u'_n are conjugate by an element $g_n \in G(S_n)$ lifting the identity element of $G(S_0)$ (Exposé IX 3.6). The same reference also implies that the g_n may be chosen coherently; by 1.4 they therefore arise from a section $g \in G(S)$. The morphisms u and $\mathrm{Int}(g)u'$ then agree modulo $\mathfrak{m}^n$ for every n, and hence agree by 1.5.

c) The existence and uniqueness of u follow from part a) and 1.2 bis. If G has connected fibres, u is central by Exposé IX 5.6(a).

2. Infinitesimal lifting of sub-tori

2.1. Statement of the theorem

We shall give an infinitesimal lifting theorem for sub-tori of a group prescheme that does not require smoothness hypotheses (in contrast with Exposé IX 3.6 bis) and that also answers a very natural question:[2] is it enough to be able to lift "sufficiently many" points of finite order of a sub-torus T_0 in order to ensure that T_0 itself can be lifted (infinitesimally, of course)?

Theorem 2.1.— *Let S be a noetherian affine scheme, let S_0 be a closed subscheme of S defined by a square-zero ideal J, let G be an S-group prescheme of finite type, let $G_0 = G \times_S S_0$, let T_0 be a sub-torus of G_0, and let $q > 0$ be an integer invertible on S. Suppose that, for every integer n equal to a power of q, there exists a subscheme M_n of G, flat over S, such that*

$$M_n \times_S S_0 = {}_nT_0.$$

Then there exists a sub-torus T of G such that $T \times_S S_0 = T_0$.

Theorem 2.1 will be useful to us through the following two corollaries.

Corollary 2.2.— *Let S be a locally noetherian prescheme, let S_0 be a closed subprescheme of S defined by a locally nilpotent sheaf of ideals, let G be an S-group prescheme of finite type, let T_0 be a sub-torus of $G_0 = G \times_S S_0$, and let $q > 0$ be an integer invertible on S. As n ranges over the powers of q, let (M_n) be a coherent system of S-subgroup schemes of G, of multiplicative type, lifting the ${}_nT_0$. (The system (M_n) is said to be coherent if*

$$M_m = {}_m(M_n)$$

whenever m divides n.) Then there exists a unique sub-torus T of G such that

$$T \times_S S_0 = T_0 \qquad \text{and} \qquad {}_nT = M_n$$

for every n.

Corollary 2.3.— *Let G be an S-group prescheme, flat and of finite presentation over S, let S_0 be a closed subprescheme of S defined by a locally nilpotent sheaf of ideals of finite type, and let T_0 be a central torus of $G_0 = G \times_S S_0$. Then there exists a unique sub-torus T of G lifting T_0. Moreover, T is central.*

Remark 2.4.— *We leave it to the reader to formulate the analogues of statements 2.1, 2.2, and 2.3 in which, instead of lifting a sub-torus of G_0, one is given a torus T over S and asks to lift a morphism*

$$u_0 : T_0 \longrightarrow G_0.$$

One reduces to the preceding cases by considering the graph of u_0.

Let us show how Corollaries 2.2 and 2.3 follow from Theorem 2.1.

Proof of Corollary 2.2. The uniqueness of T follows from Exposé IX 4.8(b) and Exposé IX 4.10. To prove existence, we may therefore reduce to the case in which S is affine, hence noetherian, and S_0 is defined by a square-zero ideal.

[2] Editor's note: The idea of approximating a torus by its finite subgroups occurs in Grothendieck's proof of the connectedness of centralizers of tori; see §4.6 of *BIBLE*.

Lemma 2.5.— *Let G be an S-group prescheme of finite presentation, and let H be a subgroup scheme of G, of multiplicative type and smooth over S. Then* $\mathrm{Centr}_G(H)$ *is representable by an S-group subprescheme of G, of finite presentation.*

When G is separated over S, the lemma follows from Exposé VIII 6.5(e), without any smoothness hypothesis on H. In the present case, observe that the assertion is local for the fpqc topology. We may therefore assume that H is diagonalizable, hence of the form $D_S(M)$. We may also assume that S is affine and then, by EGA IV 8, noetherian. Since H is smooth and of finite type over S, the order q of the torsion subgroup of M is invertible on S (Exposé VIII 2.1(e)). It is then immediate (cf. Exposé IX 4.10) that the subgroups ${}_nH$, with n ranging over the powers of q, are schematically dense in H (Exposé IX 4.1). But ${}_nH$ is a completely split covering of S, that is, it is isomorphic to a finite disjoint sum of copies of S. Consequently

$$\mathrm{Centr}_G({}_nH) = Z_n$$

is representable as the intersection of the centralizers in G of the sections of ${}_nH$ over S. It remains to apply the following lemma.

Lemma 2.5 bis.— *Let S be a noetherian prescheme, let G be an S-group prescheme of finite type, let H be a subgroup of G of multiplicative type, and let (H_i) be an increasing filtered family of subgroups of G of multiplicative type. Suppose that*

$$Z_i = \mathrm{Centr}_G(H_i)$$

is representable by a group subprescheme of G. Then the family (Z_i) is stationary. If, in addition, the H_i are schematically dense in H, then

$$Z_i = \mathrm{Centr}_G(H)$$

for all sufficiently large i.

To see that the family (Z_i) is stationary, it is enough to prove that the family of underlying sets $\mathrm{ens}(Z_i)$ is stationary. Indeed, the stationary value will then be a closed subset of an open subset U of G. After replacing G by U, we are reduced to studying a decreasing filtered family of closed subpreschemes of a noetherian prescheme. An elementary constructibility argument reduces us to the case in which S is integral. We then have to show that $\mathrm{ens}(Z_i)$ is stationary over some nonempty open subset of S. The generic fibre of G is separated (Exposé $\mathrm{VI_A}$ 0.3); after restricting S, we may therefore assume that G is separated over S (EGA IV 8). Then Z_i is closed in G (Exposé VIII 6.5(e)).

To prove the last assertion of the lemma, let Z denote the stationary value of the family (Z_i). It is clear that $\mathrm{Centr}_G(H)$ is a subfunctor of Z; we show that Z centralizes H. Let E be the subprescheme of $H \times_S Z$ that is the kernel of the pair of morphisms

$$\begin{aligned} H \times_S Z &\rightrightarrows G, \\ (h, c) &\longmapsto c, \\ (h, c) &\longmapsto hch^{-1}. \end{aligned}$$

The prescheme E contains $H_i \times_S Z$ for every i. The H_i are flat over S, so EGA IV 11.10.9 shows that, for every point s of S, the $(H_i)_s$ are schematically dense in H_s, and $H_i \times_S Z$ is schematically dense in $H \times_S Z$. Since G_s is separated, E_s is closed in $(H \times_S Z)_s$, and is therefore equal to it. Thus E is closed in $H \times_S Z$, and hence equals $H \times_S Z$. This says that Z centralizes H.

Returning to the proof of 2.2, Lemma 2.5 shows that

$$Z_n = \mathrm{Centr}_G(M_n)$$

is representable, and Lemma 2.5 bis shows that the decreasing family of subpreschemes (Z_n) is stationary; let Z be its stationary value. The group Z contains T_0 and all the M_n. After replacing G by Z, we may therefore suppose that the M_n are central.

The hypotheses of Theorem 2.1 are now satisfied, so there exists a sub-torus T of G lifting T_0. The groups ${}_nT$ and M_n are two liftings of ${}_nT_0$; they are therefore conjugate (Exposé IX 3.2 bis), and consequently coincide because M_n is central. This torus T has the required properties.

Proof of Corollary 2.3. The uniqueness of T follows from Exposé IX 5.1 bis, and its centrality follows from Exposé IX 5.6(b). By the usual procedure, this observation reduces us first to the case in which S is affine (and hence S_0 is defined by a nilpotent ideal of finite type), and then to the noetherian case. After restricting S if necessary, we may suppose that there exists an integer q invertible on S. Corollary 2.3 then follows from 2.2 and 1.2.

Remark 2.6.— *Corollary 2.3 remains true if the torus T_0 is replaced by a smooth central subgroup of multiplicative type of G_0.*

2.2. Proof of Theorem 2.1

a) Reduction to the case $T_0 = \mathbb{G}_{m,S_0}$.

By 1.1, we may assume that M_n is a subgroup of multiplicative type. Using Exposé IX 3.2 bis, we may suppose that the family (M_n) is coherent (2.2). The centralizer Z_n of M_n in G is representable (2.5), and the filtered family (Z_n) is stationary (2.5 bis). After replacing G by Z_n for sufficiently large n, we may therefore suppose that T_0 and the M_n are central. The uniqueness of a lifting of T_0 is then guaranteed by Exposé IX 5.1 bis.

Proceeding as in the proof of 1.1, we may suppose that there exist an S-torus T and an S_0-isomorphism

$$u_0 : T \times_S S_0 \xrightarrow{\sim} T_0.$$

Lifting u_0 is equivalent to lifting T_0. By uniqueness, it is enough to prove the existence of a lifting of u_0 after an affine faithfully flat extension $S' \to S$ of finite type (fpqc descent). This allows us to suppose that

$$T = \mathbb{G}_{m,S}^r$$

(Exposé X 4.5). If the restriction of u_0 to each factor $\mathbb{G}_m$ lifts to an S-morphism—necessarily central—then these liftings immediately give a lifting of u_0. In short, we may suppose that $T_0 = \mathbb{G}_{m,S_0}$.

b) Definition of the obstruction to the existence of a lifting of T_0.

By 1.1, to prove Theorem 2.1 it is enough to find a subscheme of G, flat over S, lifting T_0. We shall see that the obstruction to the existence of such a lifting can be defined as an element of a certain Ext^1 group of sheaves of modules.

Let U be an open subset of G in which T_0 is closed, and again denote by U (respectively, U_0) the open subscheme of G (respectively, G_0) having this open subset as its underlying space. The sheaf $\mathcal{O}_{T_0}$, regarded as a sheaf on U, is therefore a quotient of $\mathcal{O}_{U_0}$. Let

$$h : \mathcal{O}_{U_0} \longrightarrow \mathcal{O}_{T_0}$$

be the canonical epimorphism.

Lemma 2.7.— *The canonical map*

$$h' = \mathrm{id}_J \otimes h : J \otimes_{S_0} \mathcal{O}_{U_0} \longrightarrow J \otimes_{S_0} \mathcal{O}_{T_0}$$

factors, necessarily uniquely, as $i \circ j_U$*, where* j_U *is the canonical epimorphism*

$$J \otimes_{S_0} \mathcal{O}_{U_0} \longrightarrow J\mathcal{O}_U \simeq J\mathcal{O}_{U_0}.$$

We must show that $h'(K) = 0$, where K is the kernel of j_U. For every integer n equal to a power of q, there is an epimorphism

$$h_n : \mathcal{O}_{T_0} \longrightarrow \mathcal{O}_{{}_nT_0}.$$

Since ${}_nT_0$ lifts to a scheme M_n flat over S, the canonical morphism

$$j_n : J \otimes_{S_0} \mathcal{O}_{{}_nT_0} \xrightarrow{\sim} J\mathcal{O}_{M_n}$$

is an isomorphism. The commutative diagram

$$\begin{array}{ccccccc}
K & \longrightarrow & J \otimes_{S_0} \mathcal{O}_{U_0} & \xrightarrow{j_U} & J\mathcal{O}_U & \hookrightarrow & \mathcal{O}_U \\
 & & \downarrow{\scriptstyle h'} & \swarrow{\scriptstyle i} & & & \\
 & & J \otimes_{S_0} \mathcal{O}_{T_0} & & & & \\
 & & \downarrow{\scriptstyle h'_n} & & & & \\
 & & J \otimes_{S_0} \mathcal{O}_{{}_nT_0} & \xrightarrow[j_n]{\sim} & J\mathcal{O}_{M_n} & \hookrightarrow & \mathcal{O}_{M_n}
\end{array}$$

shows that $h'(K)$ is contained in $\ker h'_n$ for every n. It is therefore contained in $\bigcap_n \ker h'_n$, and it remains to prove that this intersection is zero. Now

$$\mathcal{O}_{T_0} = \mathcal{O}_{S_0}[\mathbb{Z}], \qquad \mathcal{O}_{{}_nT_0} = \mathcal{O}_{S_0}[\mathbb{Z}/n\mathbb{Z}].$$

Let

$$a = \sum_{m \in \mathbb{Z}} a_m \otimes m \in J \otimes_{S_0} \mathcal{O}_{T_0}.$$

The a_m are sections of J, almost all zero. Choose n large enough that the indices m for which $a_m \neq 0$ have distinct images in $\mathbb{Z}/n\mathbb{Z}$. If $a \in \ker h'_n$, then necessarily $a = 0$. Thus $\bigcap_n \ker h'_n = 0$, proving 2.7.

Let K_0 be the kernel of h. Consider the diagram

$$\begin{array}{ccccccccc}
 & & & & & & 0 & & \\
 & & & & & & \downarrow & & \\
 & & J\mathcal{O}_U & & & & K_0 & & \\
 & & \downarrow{\scriptstyle \sim} & & & & \downarrow & & \\
0 & \longrightarrow & J\mathcal{O}_{U_0} & \longrightarrow & \mathcal{O}_U & \longrightarrow & \mathcal{O}_{U_0} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle i} & & & & \downarrow{\scriptstyle h} & & \\
 & & J \otimes_{S_0} \mathcal{O}_{T_0} & & & & \mathcal{O}_{T_0} & & \\
 & & & & & & \downarrow & & \\
 & & & & & & 0. & &
\end{array}$$

The sheaf $\mathcal{O}_U$ defines an element

$$\Phi \in \mathrm{Ext}^1_{\mathcal{O}_U}(\mathcal{O}_{U_0}, J\mathcal{O}_{U_0}).$$

Let Ψ be the element of

$$\mathrm{Ext}^1_{\mathcal{O}_U}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$$

obtained from Φ, by the bifunctoriality of Ext^1, through the morphisms

$$K_0 \longrightarrow \mathcal{O}_{U_0}, \qquad i : J\mathcal{O}_{U_0} \longrightarrow J \otimes_{S_0} \mathcal{O}_{T_0}.$$

It follows from Exposé III 4.1 and the infinitesimal criterion of flatness (cf. Exposé III 4.3) that a subscheme of U, flat over S and lifting T_0, exists if and only if $\Psi = 0$.

Since T_0 is affine, Exposé III 4.5 and 4.6 show that it is enough to construct, locally on U, a subscheme flat over S lifting T_0. It therefore suffices to show that the image of Ψ in the sheaf

$$\mathscr{E} = \mathcal{E}xt^1_{\mathcal{O}_U}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$$

is zero. We again denote this element of $\Gamma(U, \mathscr{E})$ by Ψ.

c) Reduction to the case in which S is local artinian with algebraically closed residue field.

Since K_0 and $J \otimes_{S_0} \mathcal{O}_{T_0}$ are coherent sheaves, so is $\mathscr{E}$; moreover, $\mathscr{E}$ is supported on T_0, because the same is true of $J \otimes_{S_0} \mathcal{O}_{T_0}$. To show that the section Ψ of $\mathscr{E}$ is zero, it is enough to show that, for every point u of T_0, its image in the fibre of $\mathscr{E}$ at u is zero. Formation of the $\mathcal{E}xt^i$ of coherent sheaves commutes with flat extensions of the base,[3] so we are reduced to proving the existence of a lifting of

$$T_0 \cap \mathrm{Spec}\, \mathcal{O}_u$$

flat over S. Let s be the image of u in S. We may replace S by $\mathrm{Spec}\, \mathcal{O}_s$ and G by $G \times_S \mathrm{Spec}\, \mathcal{O}_s$.

After another faithfully flat extension if necessary, we may suppose that $\mathcal{O}_s$ has an algebraically closed residue field (EGA 0_{III}, 10.3.1). Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_s$, and set

$$S_n = \mathrm{Spec}(\mathcal{O}_s/\mathfrak{m}^n).$$

Suppose that, for every $n > 0$, the obstruction to lifting

$$T_0 \times_S S_n = (T_0)_n$$

to a subscheme of $G_n = G \times_S S_n$, flat over S_n, has been shown to vanish, and let T_n be the unique flat lifting of $(T_0)_n$ over S_n that is a sub-torus of G_n. If $n > n'$, it is clear that

$$T_n \times_{S_n} S_{n'} = T_{n'}.$$

If u is a point of T_0 lying over s, the lemma below, applied to the coherent system of liftings

$$T_n \cap \mathrm{Spec}\, \mathcal{O}_u$$

of $(T_0)_n \cap \mathrm{Spec}\, \mathcal{O}_u$, shows that $T_0 \cap \mathrm{Spec}\, \mathcal{O}_u$ indeed has a lifting flat over $\mathcal{O}_s$. We are therefore reduced to proving that $\Psi = 0$ when $S = S_n$ is the spectrum of an artinian local ring with algebraically closed residue field, and to proving the following lemma.

[3] Editor's note: cf. EGA 0_{III}, 12.3.5.

Lemma 2.8.— *Let $A \to B$ be a local homomorphism of noetherian local rings, let $\mathfrak{m}$ be the maximal ideal of A, let J be a square-zero ideal of A, let M be a finite-type B-module, and set*

$$A_0 = A/J, \qquad B_0 = B/JB, \qquad M_0 = M/JM.$$

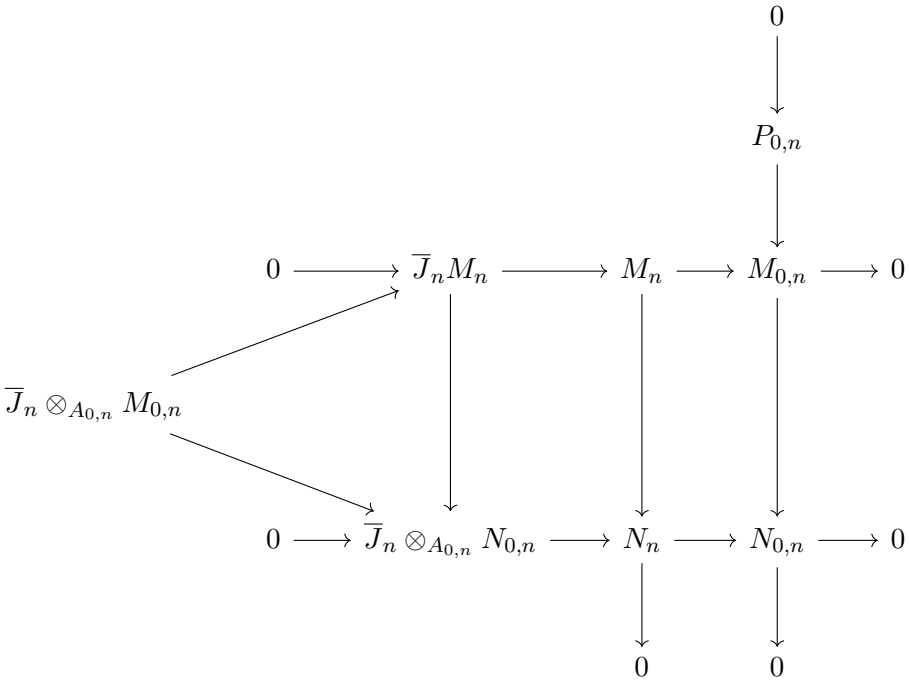
Let N_0 be a quotient B_0-module of M_0, flat over A_0. For every $n > 0$, let $A_n, A_{0,n}$, and so forth denote the objects obtained by the base extension

$$A \longrightarrow A/\mathfrak{m}^n = A_n,$$

and let $\overline{J}_n$ be the image of J in A_n. For every $n > 0$, let N_n be a quotient B_n-module of M_n, flat over A_n, lifting $N_{0,n}$, and suppose that, for $n \geq n'$, $N_{n'}$ is obtained from N_n by the base extension $A_n \to A_{n'}$. Then there exists a quotient B-module N of M, flat over A, lifting N_0.

Proof of 2.8. Let P_0 be the kernel of the epimorphism $M_0 \to N_0$. For every $n > 0$, there is a commutative diagram with exact horizontal rows:

$(*_n)$



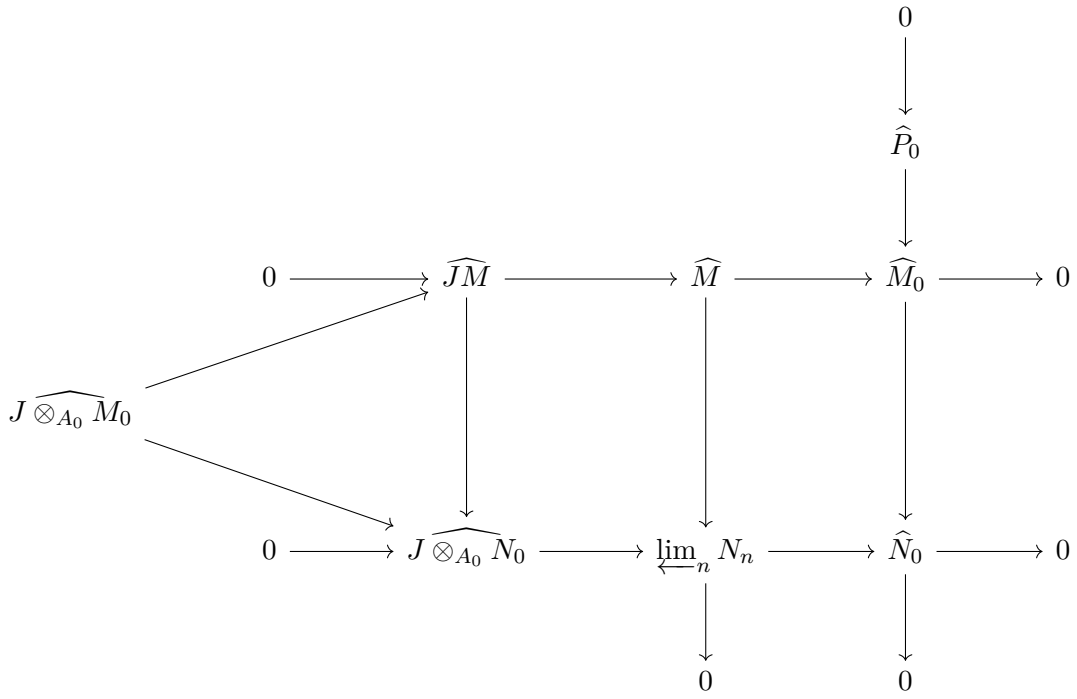
Moreover, by hypothesis, the diagram $(*)_{n+1}$ reduces modulo $\mathfrak{m}^n$ to $(*)_n$.

The Artin–Rees lemma (Bourbaki, *Algèbre commutative*, Chapter III, §3, Corollary 1) shows that the filtrations on JM, $J \otimes_{A_0} M_0$, and $J \otimes_{A_0} N_0$, defined by the kernels of the respective morphisms

$$JM \longrightarrow \overline{J}_n M_n,$$
$$J \otimes_{A_0} M_0 \longrightarrow \overline{J}_n \otimes_{A_{0,n}} M_{0,n},$$
$$J \otimes_{A_0} N_0 \longrightarrow \overline{J}_n \otimes_{A_{0,n}} N_{0,n},$$

are $\mathfrak{m}B$-good. Passing to the inverse limit in the diagrams $(*)_n$, and writing $\widehat{Q}$ for the separated completion of a B-module Q for the $\mathfrak{m}B$-adic topology, we obtain a commutative

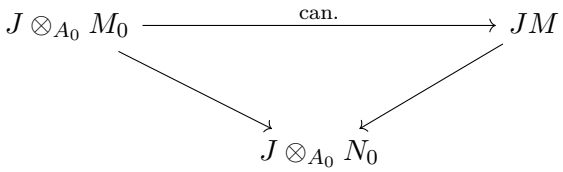
diagram in which the two horizontal rows remain exact:
$(\widehat{*})$



Now $(B, \mathfrak{m}B)$ is a Zariski ring and $J \otimes_{A_0} N_0$ is of finite type over B, hence separated for the $\mathfrak{m}B$-adic topology. The diagram $(\widehat{*})$ then shows that the morphism

$$J \otimes_{A_0} M_0 \longrightarrow J \otimes_{A_0} N_0$$

induced by the epimorphism $M_0 \to N_0$ factors through JM:



Under these conditions, Exposé III 4.1 and 4.3 show that a quotient B-module N of M, flat over A and lifting N_0, exists if and only if a certain element

$$g \in E = \mathrm{Ext}^1_B(P_0, J \otimes_{A_0} N_0)$$

vanishes. More precisely, the exact sequence

$$0 \longrightarrow JM \longrightarrow M \longrightarrow M_0 \longrightarrow 0$$

defines an element $f \in F$, where

$$F = \mathrm{Ext}^1_B(M_0, JM),$$

and g is the image of f under the natural morphism $F \to E$ arising by bifunctoriality from

$$P_0 \longrightarrow M_0, \qquad JM \longrightarrow J \otimes_{A_0} N_0.$$

The diagram $(\hat{*})$ and Exposé III 4.1 show that the image of g in

$$\widehat{E} \simeq \mathrm{Ext}^1_{\widehat{B}}(\widehat{P}_0, J\widehat{\otimes}_{A_0} N_0)$$

is zero. But E is of finite type over B, and hence separated for the $\mathfrak{m}B$-adic topology. Consequently $g = 0$, which completes the proof of 2.8.

d) Study of $\mathscr{E}$.

We now suppose that S is the spectrum of an artinian local ring A with algebraically closed residue field k. Let A_0 be the ring of S_0, and let $\mathfrak{m}_0$ be the maximal ideal of A_0. Since S_0 is artinian, T_0 is closed in G (Exposé $\mathrm{VI_B}$ 1.4.2). We may therefore take $U = G$, so that

$$\mathscr{E} = \mathcal{E}xt^1_{\mathcal{O}_G}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0}).$$

Let B_0 be the ring of the affine S_0-scheme $T_0 = \mathbb{G}_{m,S_0}$, and let $\overline{B}_0$ be the ring of its special fibre

$$T_0 \times_{S_0} \mathrm{Spec}(k) = \mathbb{G}_{m,k}.$$

The sheaf $\mathscr{E}$ is a coherent $\mathcal{O}_{T_0}$-module, hence is defined by a finite-type B_0-module, which we denote by E.

Consider the graded module associated with E and the $\mathfrak{m}_0 B_0$-adic filtration:

$$E_r = \mathfrak{m}_0^r E/\mathfrak{m}_0^{r+1}E.$$

Each E_r is a finite-type $\overline{B}_0$-module, and $E_r = 0$ for all sufficiently large r, since S_0 is artinian.

Let g be a section of G over S whose restriction to S_0 is a section g_0 of T_0. Left translation by g defines an "automorphism of the situation" from the point of view of the obstruction problem. In particular, g induces an S_0-automorphism of the sheaf $\mathscr{E}$ that fixes the obstruction Ψ. More precisely, g defines a semilinear automorphism of the B_0-module E, relative to the A_0-automorphism of B_0 defined by translation by g_0 in T_0. Reducing modulo $\mathfrak{m}_0^{r+1}$, g therefore defines a semilinear automorphism of E_r, relative to the k-automorphism of $\overline{B}_0$ defined by translation by

$$g_0 \times_{S_0} \mathrm{Spec}(k)$$

in $T_0 \times_{S_0} \mathrm{Spec}(k)$.

Lemma 2.9.— *For every integer $r \geq 0$, E_r is a locally free $\overline{B}_0$-module.*

Let x be a point of T_0, let $\kappa(x)$ be its residue field, and let

$$(E_r)_x = E_r \otimes_{\overline{B}_0} \kappa(x)$$

be the "fibre" of E_r at x. Denote by $\ell(x)$ the rank of $(E_r)_x$ over $\kappa(x)$, and by ℓ the maximum of $\ell(x)$ as x ranges over the points of T_0. Let L be the largest closed subscheme of

$$\mathrm{Spec}\,\overline{B}_0 = \mathbb{G}_{m,k}$$

over which E_r is locally free of rank ℓ (TDTE IV, Lemma 3.6). Let $\beta \in L(k)$, which exists because L is of finite type over the algebraically closed field k, and let $\alpha \in \mathbb{G}_{m,k}(k)$ be a point whose order is a power n of q. The point α is k-rational. Since, by hypothesis, ${}_nT_0$ lifts to a subscheme M_n étale over S, there is a section a of G over S lifting α and whose restriction to S_0 is a section of T_0.

The preceding observations show that the fibres of E_r at the points β and $\alpha\beta$ of $\mathbb{G}_{m,k}(k)$ are k-isomorphic. The points whose order is a power of q are dense in $\mathbb{G}_{m,k}$, as are their translates by β. Since L is closed in $\mathbb{G}_{m,k}$ and $\mathbb{G}_{m,k}$ is reduced, $L = \mathbb{G}_{m,k}$, and E_r is locally free of rank ℓ over $\overline{B}_0$.

e) Completion of the proof of Theorem 2.1.

Let $K_0(n)$ be the sheaf of ideals of $\mathcal{O}_{G_0}$ defining the closed subscheme ${}_nT_0$. The sheaf ${K_0}^4$ is therefore a subsheaf of $K_0(n)$. For brevity, set

$$R = J \otimes_{S_0} \mathcal{O}_{T_0}, \qquad R(n) = J \otimes_{S_0} \mathcal{O}_{{}_nT_0}.$$

The sheaf $R(n)$ is a quotient of R, and we have the diagram

$$\begin{array}{ccccccccc} & & & & K_0 & & & & \\ & & & & \downarrow & & & & \\ & & & & K_0(n) & & & & \\ & & & & \downarrow & & & & \\ 0 & \longrightarrow & J\mathcal{O}_G & \longrightarrow & \mathcal{O}_G & \longrightarrow & \mathcal{O}_{G_0} & \longrightarrow & 0 \\ & & \downarrow & & & & & & \\ & & R & & & & & & \\ & & \downarrow & & & & & & \\ & & R(n) & & & & & & \end{array}$$

Using the bifunctoriality of $\mathrm{Ext}^1_{\mathcal{O}_G}$, we obtain the commutative diagram

$$\begin{array}{ccccc} \mathcal{E}xt^1_{\mathcal{O}_G}(\mathcal{O}_{G_0}, J\mathcal{O}_G) & & & & \\ & \searrow & & & \\ & & \mathcal{E}xt^1_{\mathcal{O}_G}(K_0(n), R) & \longrightarrow & \mathcal{E}xt^1_{\mathcal{O}_G}(K_0, R) = \mathscr{E} \\ & & \downarrow & & \downarrow \\ & & \mathcal{E}xt^1_{\mathcal{O}_G}(K_0(n), R(n)) & \longrightarrow & \mathcal{E}xt^1_{\mathcal{O}_G}(K_0, R(n)). \end{array}$$

Again let Φ denote the element of $\mathrm{Ext}^1_{\mathcal{O}_G}(\mathcal{O}_{G_0}, J\mathcal{O}_G)$ defined by the exact sequence

$$0 \longrightarrow J\mathcal{O}_G \longrightarrow \mathcal{O}_G \longrightarrow \mathcal{O}_{G_0} \longrightarrow 0,$$

so that Ψ is the image of Φ in $\mathscr{E}$. Since ${}_nT_0$ lifts to a subscheme M_n of G, flat over S, the image of Φ in

$$\mathrm{Ext}^1_{\mathcal{O}_G}(K_0(n), R(n))$$

is zero (Exposé III 4.1). A fortiori, the image of Φ in $\mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R(n))$, which is also the image of Ψ, is zero.

[4] Editor's note: Introduced after Lemma 2.7.

Lemma 2.10.— *For every integer n, the canonical morphism*

$$\mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R) \otimes_{B_0} \mathcal{O}_{{}_nT_0} \longrightarrow \mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R(n))$$

is injective.

Indeed, the affine scheme $T_0 = \mathbb{G}_{m,S_0}$ has ring

$$B_0 = A_0[T, T^{-1}].$$

The subscheme ${}_nT_0$ is defined by the vanishing of

$$h(n) = T^n - 1.$$

This section is regular (EGA 0_{IV}, 15.2.2) and remains regular after every base change $S'_0 \to S_0$. Hence there is an exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_{T_0} \xrightarrow{h(n)} \mathcal{O}_{T_0} \longrightarrow \mathcal{O}_{{}_nT_0} \longrightarrow 0.$$

Since ${}_nT_0$ is flat over S, tensoring over $\mathcal{O}_{S_0}$ with J gives the exact sequence

$$0 \longrightarrow R \xrightarrow{h(n)} R \longrightarrow R(n) \longrightarrow 0,$$

and then the exact sequence of Ext groups

$$\cdots \longrightarrow \mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R) \xrightarrow{h(n)} \mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R) \longrightarrow \mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R(n)),$$

which proves the lemma.

The foregoing shows that, for every integer n equal to a power of q, the image of Ψ in $\mathscr{E}/h(n)\mathscr{E}$ is zero. To prove that $\Psi = 0$, it is enough to show that

$$\Psi \in \mathfrak{m}_0^r \mathscr{E} \quad \Longrightarrow \quad \Psi \in \mathfrak{m}_0^{r+1} \mathscr{E}.$$

Let $\overline{\Psi}$ be the image of Ψ in E_r. There is an element $\Psi(n) \in \mathscr{E}$ such that

$$\Psi = \Psi(n)h(n) \qquad (n \text{ a power of } q).$$

We observed that the image $\overline{h(n)}$ of $h(n)$ in $\overline{B}_0$ is again $\overline{B}_0$-regular. Since E_s is locally free over $\overline{B}_0$ for every s (2.9), multiplication by $\overline{h(n)}$ on E_s is injective. It follows immediately that $\Psi(n) \in \mathfrak{m}_0^r \mathscr{E}$. Let $\overline{\Psi(n)}$ be its image in E_r; then

$$\overline{\Psi} = \overline{h(n)}\,\overline{\Psi(n)} \qquad (n \text{ a power of } q).$$

Thus the set F of points of $\mathbb{G}_{m,k}$ at which $\overline{\Psi}$ vanishes contains the dense set of points whose order is a power of q. Moreover, F is closed because E_r is locally free over $\overline{B}_0$, and $\mathbb{G}_{m,k}$ is reduced. Therefore $\overline{\Psi} = 0$.

This completes the proof of Theorem 2.1.

3. Characterization of a sub-torus by its underlying set

3.1. Statement of the theorem

Notation.—If X is a prescheme, $\mathrm{ens}(X)$ denotes the underlying set of X. If X and S' are two S-preschemes,

$$X_{S'} = X \times_S S', \qquad u : X_{S'} \longrightarrow X$$

is the canonical morphism, and if E is a subset of $\mathrm{ens}(X)$, we denote by $E_{S'}$ (or $E \times_S S'$) the subset $u^{-1}(E)$ of $\mathrm{ens}(X_{S'})$.

Theorem 3.1.— *Let S be a locally noetherian prescheme, let G be an S-group prescheme of finite type, let $q > 0$ be an integer invertible on S, and let E be a subset of* $\mathrm{ens}(G)$. *Consider the following assertions about E:*

(i) *The set E is the underlying set of a sub-torus T of G.*

(ii) a) *For every point s of S, there is a sub-torus T_s of G_s such that*

$$E \cap \mathrm{ens}(G_s) = E_s$$

is the underlying set of T_s.

b) *As n ranges over the powers of q, there is a coherent family (cf. 2.2) M_n of subgroup schemes of G, of multiplicative type, such that for every point s of S,*

$$(M_n)_s = {}_nT_s.$$

(iii) a) *As in* (ii) a).

b) *The set E is locally closed in* $\mathrm{ens}(G)$, *and the dimension of the fibers of E over S is locally constant.*

(iv) a) *As in* (ii) a).

b) *For every S-scheme S' that is the spectrum of a complete discrete valuation ring with algebraically closed residue field, $E_{S'}$ is the underlying set of a sub-torus of $G_{S'}$.*

Then the following implications hold:

A) (i) $\Longleftrightarrow$ (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (iv).

B) *If G is separated over S, then* (iii) $\Longleftrightarrow$ (iv), *and moreover E is closed.*

C) *Conditions* (i), (ii), *and* (iii) *(and also* (iv) *if G is separated over S) are equivalent in the following two cases:*

First case: a) *The prescheme S is reduced, or G is flat over S; and*
b) *for every point s of S, T_s is a central torus of G_s.*

Second case: *S is normal.*

Moreover, in the two cases above, the torus T with underlying set E is unique.

Remarks 3.2.— a) *When S is reduced, there is no need in* (ii) *to assume that the family M_n is coherent, since this condition follows from the other hypotheses. Indeed, if m divides n, the subgroups ${}_mM_n$ and M_m are étale over S, hence reduced, and have the same underlying space; they therefore coincide.*

b) *If S is no longer assumed normal, it is in general no longer true that* (iii) $\Rightarrow$ (i), *even when S is reduced and geometrically unibranch and G is a smooth group scheme over S. Indeed, consider the Borel subgroup of* $\mathrm{SL}_{2,S}$ *consisting of matrices of the form*

$$\begin{pmatrix} t & u \\ 0 & t^{-1} \end{pmatrix},$$

where S is the affine curve over a field k with ring

$$k[x,y]/(y^2 - x^3).$$

Define E as follows. Above the cusp of S, where $x = y = 0$, take the diagonal torus $u = 0$. Above the complementary open subset $x \neq 0$, take the torus obtained from the diagonal torus by conjugation by

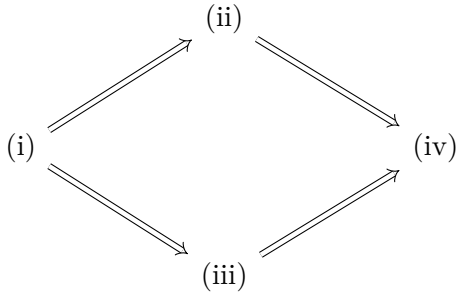
$$\begin{pmatrix} 1 & y/x \\ 0 & 1 \end{pmatrix}.$$

The resulting set E satisfies (iii) a)*; it is also closed in G, and the reduced subscheme having E as its underlying set has equations*

$$\begin{cases} xu + y(t - t^{-1}) = 0, \\ u^2 - x(t - t^{-1})^2 = 0. \end{cases}$$

It is therefore not a sub-torus of G, since its fiber above the cusp is not reduced.

Plan of the proof of 3.1. In a first part we shall establish the following "easy" implications:



and

$$\text{(iv)} \Longrightarrow [\,\text{(iii) and } E \text{ is closed}\,] \quad \text{if } G \text{ is separated over } S.$$

The more delicate implications will be proved in three stages:

I) Reduction of (iii) $\Rightarrow$ (i), under the hypotheses of C), first case, to the case in which S is normal.

II) (iii) $\Rightarrow$ (ii) when S is normal.

III) (ii) $\Rightarrow$ (i).

3.2. Proof of the "easy" assertions in 3.1

The implications (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iii) are trivial.

(iii) $\Rightarrow$ (iv). After replacing S by S', we may assume that S is the spectrum of a discrete valuation ring. Let t be the generic point of S, and let s be its closed point.

Since E is locally closed, there is a reduced subprescheme of G with underlying space E; we also denote it by E. Its generic fiber E_t is therefore the sub-torus T_t of G_t, of rank r, by (iii) a). Let E' be the schematic closure of E_t in E. The prescheme E' is irreducible, and its closed fiber E'_s is nonempty (it contains the unit section of G_s); hence E'_s has dimension r (EGA IV 14.3.10). But E'_s is a closed subscheme of E_s, while E_s has dimension r by (iii) b) and is irreducible because it has the same underlying space as T_s. Thus E'_s and E_s have the same underlying space, and consequently $E' = E$. This proves that E is flat over S.

Let E'' now be the schematic closure of E_t in G. Then E'' is a subgroup prescheme of G, flat over S (Exposé VIII 7.1), which contains E', and therefore E. Its closed fiber E''_s is an algebraic subgroup of G_s of dimension r (*loc. cit.*). Since T_s is a closed irreducible subset of G_s of dimension r, E_s has the same underlying set as the identity component $(E''_s)^0$ of E''_s. Let $(E'')^0$ denote the "identity component" of E'', that is, the open subgroup of E'' complementary to the union of the irreducible components of E''_s that do not contain the origin. Then

$$E = \mathrm{ens}((E'')^0).$$

Since E and E'' are reduced, one even has $E = (E'')^0$. Finally, E is a subgroup prescheme of G, flat and of finite type over S, with connected fibers, hence separated (Exposé $\mathrm{VI_B}$ 5.2); its generic fiber is the torus T_t, and its reduced closed fiber is the torus T_s. It follows that E is a torus (Exposé X 8.8).

(ii) $\Rightarrow$ (iv). Again we may assume that S is the spectrum of a discrete valuation ring, and we retain the notation introduced above. The schematic closure T of T_t in G is a subgroup prescheme of G, flat over S, and it contains M_n for every n that is a power of q. Consequently, its closed fiber is a closed subscheme of G_s containing $(M_n)_s$ for every n, hence containing T_s. For dimension reasons, the "identity component" T^0 of T has E as its underlying set, and the preceding argument shows that T^0 is a sub-torus of G.

(iv) $\Rightarrow$ [(iii) and E is closed], if G is separated over S. Let us show that E is closed.[5] We first prove the following lemma.

Lemma 3.3.— *If the conditions stated in 3.1* (iv) *are satisfied, E is a locally constructible subset of* $\mathrm{ens}(G)$.

By the usual method, we are reduced to the case in which S is noetherian and integral, with generic point η. After shrinking S, we may assume (Exposé $\mathrm{VI_B}$ 10.10) that there is a subgroup scheme H of G, flat over S and with connected fibers, whose generic fiber H_η is T_η. To prove Lemma 3.3 it is enough to show that $\mathrm{ens}(H) = E$.

Let s be a point of S. By EGA II 7.1.9 there is an S-scheme S', the spectrum of a discrete valuation ring which we may assume complete with algebraically closed residue field, whose generic point t' maps to η and whose closed point s' maps to s. By (iv) b), there is a sub-torus $T_{S'}$ of $G_{S'}$ having $E_{S'}$ as its underlying space. The two subgroup preschemes $T_{S'}$ and $H_{S'} = H \times_S S'$ of $G_{S'}$ are flat over S', have connected fibers, and have the same generic fiber $T_{t'}$. Therefore both coincide with the identity component of the schematic closure of $T_{t'}$ in $G_{S'}$. Hence $\mathrm{ens}(H_s) = E_s$, and therefore $\mathrm{ens}(H) = E$, proving the lemma.

Now that E is known to be locally constructible, to see that it is closed it is enough to prove that every specialization in G of a point of E again belongs to E. By the usual technique, we are reduced to the case in which S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. In that case the sub-torus of G with

[5] Editor's note: The implication (iv) $\Rightarrow$ (iii) will appear in the course of the proof.

underlying space E, supplied by (iv) b), is closed in G, because G is separated (Exposé VIII 7.12).

3.3. Continuation of the proof of 3.1

I) Reduction of (iii) ⇒ (i), case C, first case, to the case in which S is normal.

a) Reduction to the case in which S is affine and reduced. We therefore assume that, for every point s of S, E_s is the underlying space of a central sub-torus of G_s. The uniqueness of T then follows from Exposé IX 5.1 bis, and moreover T will necessarily be central (*loc. cit.*). In view of this uniqueness, in order to prove the existence of T we may suppose that S is noetherian and affine, with ring A. If S is not reduced, G is flat over S by hypothesis. By 2.3 it is then enough to solve the problem for S_{red} and $G \times_S S_{\mathrm{red}}$. We may therefore suppose in addition that S is reduced.

b) Reduction to the case in which A is an algebra of finite type over $\mathbb{Z}$. We shall use the following lemma.

Lemma 3.4.— *Let k be a field, G an algebraic k-group, and E a subset of G. Let k'/k be a field extension, and suppose that $E_{k'}$ is the underlying set of a sub-torus $T_{k'}$ of $G_{k'}$. If $T_{k'}$ is central in $G_{k'}$, or if k is perfect, then E is the underlying set of a sub-torus T of G.*

Proof. By fpqc descent it is enough to prove that the two inverse images of $T_{k'}$ in $G_{k''}$, where

$$k'' = k' \otimes_k k',$$

coincide. They have the same underlying set. If $T_{k'}$ is central, they therefore coincide by Exposé IX 5.1 bis. If k is perfect, k'' is reduced; the two inverse images, being smooth over k'', are reduced and consequently coincide.

Remarks 3.5.— (1) *Lemma 3.4 applies in particular when the residue fields of S are perfect.*

(2) *Under the hypotheses of the first case in Theorem 3.1 C, for every S-scheme S' as in 3.1* (iv) b)*, the sub-torus of $G_{S'}$ having $E_{S'}$ as its underlying space is central.*

Lemma 3.6.— *Let S be the projective limit of a filtered inverse system of affine schemes S_i, and let H be an S-group of multiplicative type and of finite type. There exist an index i, a group H_i of multiplicative type over S_i, and an S-isomorphism*

$$H_i \times_{S_i} S \xrightarrow{\sim} H.$$

If H is isotrivial, H_i may be chosen isotrivial.

Proof. Since H is of finite presentation over S, there are an index i, an S_i-group prescheme H_i, of finite presentation over S_i, and an S-isomorphism

$$H_i \times_{S_i} S \xrightarrow{\sim} H$$

(Exposé $\mathrm{VI_B}$ 10.16). After increasing i, we may suppose that H_i is of multiplicative type (Exposé IX 2.11). It remains to prove that H_i can be chosen quasi-isotrivial, and isotrivial when H is isotrivial.

We may thus suppose that H is trivialized by an étale, surjective morphism $S' \to S$ of finite presentation. For i sufficiently large, there are an S_i-prescheme S'_i, an étale, surjective morphism $S'_i \to S_i$ of finite presentation, and an S-isomorphism

$$S'_i \times_{S_i} S \xrightarrow{\sim} S'$$

(EGA IV 17.16). For j sufficiently large set

$$S'_j = S'_i \times_{S_i} S_j, \qquad H'_j = H_j \times_{S_j} S'_j, \qquad H' = H \times_S S'.$$

By the choice of S', there are a finitely generated abelian group M and an S'-isomorphism of group schemes

$$D_{S'}(M) \xrightarrow{\sim} H'.$$

Since the S'_i are quasi-compact and $S' = \varprojlim S'_i$, Exposé $\mathrm{VI_B}$ 10 shows that there are an index j and an S'_j-isomorphism of group schemes

$$D_{S'_j}(M) \xrightarrow{\sim} H'_j.$$

This says precisely that H_j is a quasi-isotrivial group of multiplicative type. When H is isotrivial, one argues similarly using a trivialization of H by a finite étale morphism $S' \to S$.

We can now carry out reduction b). The ring A of S is the filtered direct limit of its subrings A_i of finite type over $\mathbb{Z}$. Put $S_i = \operatorname{Spec} A_i$, and let

$$u_j : S \longrightarrow S_j, \qquad u_{jk} : S_k \longrightarrow S_j \quad (k \geq j)$$

be the transition morphisms. Since S is noetherian and G is of finite presentation over S, there are an index i and an S_i-group prescheme G_i, of finite type over S_i, such that G is S-isomorphic to $G_i \times_{S_i} S$. Likewise, since E is locally closed in G, we may suppose that there is a locally closed subset E_i of G_i such that

$$E = E_i \times_{S_i} S$$

(EGA IV 8.3.11). For $j > i$, put

$$G_j = G_i \times_{S_i} S_j, \qquad E_j = E_i \times_{S_i} S_j,$$

and let Q_j be the set of points s of S_j for which $(E_j)_s$ is the underlying set of a central sub-torus of $(G_j)_s$.

It follows from 3.4 that

$$Q_k = u_{jk}^{-1}(Q_j) \quad (k \geq j),$$

and by hypothesis

$$\operatorname{ens}(S) = u_j^{-1}(Q_j) \quad (j \geq i).$$

I further claim that Q_j is ind-constructible (EGA IV 1.9.4). Indeed, since S_j is noetherian, it is enough (EGA IV 1.9.10) to see the following. Let S be a noetherian integral scheme with generic point η, and suppose that E_η is the underlying set of a central sub-torus T_η of G_η. Then there is a neighborhood U of η such that E_s has the same property for every $s \in U$. After shrinking S, we may suppose that the center Z of G is representable and that there is a subgroup scheme T of G whose generic fiber is T_η (Exposé $\mathrm{VI_B}$, §10). The two locally closed subsets $\operatorname{ens}(T)$ and E of $\operatorname{ens}(G)$ coincide on the generic fiber. We can therefore find a neighborhood U of η such that, above U,

$$\operatorname{ens}(T) = E$$

(EGA IV 8.3.11). For the same reasons, we may suppose that T is central above U; it is a torus by 3.6.

Now that Q_j is known to be ind-constructible, EGA IV 8.3.4 implies that $Q_j = \mathrm{ens}(S_j)$ for j sufficiently large. We may then replace S, G, E by S_j, G_j, E_j, which reduces us to the case in which S is an affine reduced scheme of finite type over $\mathbb{Z}$.

c) Reduction to the case in which S is the spectrum of a complete reduced noetherian local ring. Because the torus with underlying space E is unique, the usual technique (EGA IV 8) and Lemma 3.6 allow us to replace S by the spectrum of the local ring A at a point of S. Let $\widehat{S}$ be the spectrum of the completion $\widehat{A}$ of A for the topology defined by its maximal ideal. Since A is the localization of an algebra of finite type over $\mathbb{Z}$, $\widehat{S}$ is still reduced (EGA IV 7.6.5).

I claim that it is enough to solve the problem after the base extension $\widehat{S} \to S$. Indeed, let $\widehat{T}$ be the sub-torus of $G_{\widehat{S}}$ with underlying space $E_{\widehat{S}}$. Its two inverse images in $G_{\widehat{S}\times_S\widehat{S}}$ are central sub-tori with the same underlying space; they therefore coincide (Exposé IX 5.1 bis). By fpqc descent, $\widehat{T}$ comes from a torus T of G that solves the problem (cf. 3.4).

d) A descent lemma. Recall the following properties of finite morphisms, noted in TDTE I. Let S, S' be two preschemes and $u : S' \to S$ a finite morphism. Then:

(1) The morphism u is an epimorphism if and only if the canonical morphism of sheaves

$$\mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'})$$

is injective.

(2) The morphism u is an effective epimorphism (Exposé IV 1.3) if and only if the canonical diagram

$$0 \longrightarrow \mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'}) \rightrightarrows u_*(\mathcal{O}_{S'}) \otimes_{\mathcal{O}_S} u_*(\mathcal{O}_{S'})$$

is exact.

(3) If in addition S is noetherian and u is an epimorphism, u is the composite of a finite sequence of finite effective epimorphisms.

We can now prove the following lemma, whose proof uses a non-flat descent technique.

Lemma 3.7.— *Let S be a locally noetherian prescheme, G an S-group prescheme of finite type, S' a prescheme, and $u : S' \to S$ a finite morphism. For every S-prescheme T, let* $\mathrm{M}(T)$ *be the set of subgroups of G_T of multiplicative type* (respectively, the set of central subgroups of G_T of multiplicative type)*, so that* M *is naturally a contravariant functor on* Sch/S*. If u is an effective epimorphism* (respectively, an epimorphism)*, then the diagram of sets*

$$(*) \qquad \mathrm{M}(S) \longrightarrow \mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_S S')$$

is exact.

Proof. *i) Reduction to the case in which u is an effective epimorphism.* Here we consider the functor of central subgroups of G of multiplicative type. The injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$ is local on S; once this injectivity is granted, the exactness of $(*)$ is also local on S. We may therefore suppose that S is affine and noetherian.

Consider the case in which $u : S' \to S$ is the composite of two finite epimorphisms

$$S' \xrightarrow{v} S'' \xrightarrow{w} S.$$

I claim that if both diagrams

$$(*') \qquad \mathrm{M}(S'') \longrightarrow \mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_{S''} S')$$

and

$$(*'') \qquad \mathrm{M}(S) \longrightarrow \mathrm{M}(S'') \rightrightarrows \mathrm{M}(S'' \times_S S'')$$

are exact, then so is $(*)$.

Indeed, the injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$ is clear. If T' belongs to

$$\mathrm{Ker}\big(\mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_S S')\big),$$

then, a fortiori, T' belongs to

$$\mathrm{Ker}\big(\mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_{S''} S')\big).$$

By the exactness of $(*')$, it therefore comes from a unique element T'' of $\mathrm{M}(S'')$. It is enough to show that T'' belongs to

$$\mathrm{Ker}\big(\mathrm{M}(S'') \rightrightarrows \mathrm{M}(S'' \times_S S'')\big),$$

for then the exactness of $(*'')$ descends T'' to S. Let T_1'', T_2'' be the two inverse images of T'' in $\mathrm{M}(S'' \times_S S'')$. These are central subgroups of $G_{S'' \times_S S''}$ of multiplicative type, so to prove that $T_1'' = T_2''$ it is enough to see that they have the same fibers (Exposé IX 5.1 bis). Consider the commutative diagram

$$\begin{array}{ccc} S' & \leftleftarrows & S' \times_S S' \\ {\scriptstyle v}\downarrow & & \downarrow{\scriptstyle v \times v} \\ S'' & \leftleftarrows & S'' \times_S S''. \end{array}$$

The morphism v is a finite epimorphism, hence dominant by property (1) above, and closed; it is therefore surjective, and so is $v \times v$. Let x'' be a point of $S'' \times_S S''$, and let x' be a point of $S' \times_S S'$ lying above x''. The commutativity of the diagram shows that the two inverse images of $(T_1'')_{x''}$ and $(T_2'')_{x''}$ in $G_{x'}$ coincide. Thus

$$(T_1'')_{x''} = (T_2'')_{x''}$$

(EGA IV 2.2.15).

The preceding argument and an immediate induction on the number of factors in a factorization of $u : S' \to S$ into effective epimorphisms, supplied by property (3) above, show that in order to prove the exactness of $(*)$ for central subgroups of multiplicative type it is enough to treat the case in which u is an effective epimorphism. Finally, another application of Exposé IX 5.1 bis shows that it is enough to prove 3.7 when M is the functor of all subgroups of G of multiplicative type and u is an effective epimorphism.

ii) Injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$. Since $u : S' \to S$ is an epimorphism, the canonical morphism

$$\mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'})$$

is injective. Moreover, an S-group of multiplicative type is flat over S. The injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$ is therefore a consequence of the following more general lemma.

Lemma 3.8.— *Let $f : X \to S$ and $g : S' \to S$ be morphisms of preschemes, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and put*

$$X' = X \times_S S', \qquad \mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}.$$

Let $\mathrm{Q}(\mathcal{F})$ be the set of quotient $\mathcal{O}_X$-modules $\mathcal{G}$ of $\mathcal{F}$ that are quasi-coherent and flat over S, and define $\mathrm{Q}(\mathcal{F}')$ analogously relative to $\mathcal{F}', X'$, and S'. Suppose that g is quasi-compact and that

$$\mathcal{O}_S \longrightarrow g_*(\mathcal{O}_{S'})$$

is injective. Then the canonical map

$$\begin{aligned} \mathrm{Q}(\mathcal{F}) &\longrightarrow \mathrm{Q}(\mathcal{F}'), \\ \mathcal{G} &\longmapsto \mathcal{G} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \end{aligned}$$

is injective.

Proof. Indeed, we may suppose that S is affine, and then that X is affine. Since S' is quasi-compact, it has a finite covering by affine open subsets. Replacing S' by the disjoint union of these open subsets, we may therefore suppose that S' is affine. We are then reduced to the following lemma, whose proof is immediate.

Lemma 3.9.— *Let $A \to A'$ be an injective homomorphism of rings, let M be an A-module, let $N = M/P$ be a quotient A-module flat over A, and put*

$$M' = M \otimes_A A', \qquad N' = N \otimes_A A' = M'/P'.$$

Then P is the inverse image of P' under the canonical homomorphism $M \to M'$. Thus N and P are determined by N' and P'.

iii) Exactness of 3.7 $(*)$ at $\mathrm{M}(S')$. Let

$$T' \in \mathrm{Ker}(\mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_S S')).$$

Suppose that iii) has been proved when S is the spectrum of a noetherian local ring. For every point s of S, there then exists a subgroup T_s of multiplicative type in $G \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$, obtained by descent from

$$T' \times_{S'} u^{-1}(\mathrm{Spec}(\mathcal{O}_{S,s})).$$

By Exposé $\mathrm{VI_B}$, §10, and 3.6, there is an open neighborhood U of s in S and a subgroup scheme T of G over U, of multiplicative type, that extends T_s. Let U' be the inverse image of U in S'. We then have two subschemes of $G'|_{U'}$, namely $T'|_{U'}$ and $T \times_U U'$, which coincide over $u^{-1}(\mathrm{Spec}(\mathcal{O}_{S,s}))$.

Regarding the latter scheme as the projective limit of the $u^{-1}(V)$, where V runs through the open neighborhoods of s in U, EGA IV 8 shows that there is an open subset $V \subset U$, containing s, over whose inverse image $T \times_U U'$ and T' coincide. Thus, under the stated hypotheses, T' descends locally on S; by the uniqueness proved in ii), it consequently descends globally. In short, it is enough to prove iii) when S is the spectrum of a noetherian local ring.

Let $\widehat{S}$ be the spectrum of the completion of the ring of S, and put

$$S'' = S' \times_S S', \qquad \widehat{S}' = \widehat{S} \times_S S', \qquad \widehat{S}'' = \widehat{S} \times_S S'' = \widehat{S}' \times_S S'.$$

I claim that it is enough to show that

$$(\widehat{*}) \qquad \mathrm{M}(\widehat{S}) \longrightarrow \mathrm{M}(\widehat{S}') \rightrightarrows \mathrm{M}(\widehat{S}'')$$

is exact at $\mathrm{M}(\widehat{S}')$. This follows from the commutative diagram

$$\begin{array}{ccccc}
\mathrm{M}(S) & \longrightarrow & \mathrm{M}(S') & \rightrightarrows & \mathrm{M}(S'') \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{M}(\widehat{S}) & \longrightarrow & \mathrm{M}(\widehat{S}') & \rightrightarrows & \mathrm{M}(\widehat{S}'') \\
\downarrow\downarrow & & \downarrow\downarrow & & \\
\mathrm{M}(\widehat{S} \times_S \widehat{S}) & \xrightarrow{f} & \mathrm{M}(\widehat{S}' \times_{S'} \widehat{S}'). & &
\end{array}$$

Here the second row is exact at $\mathrm{M}(\widehat{S}')$ by hypothesis, the first two columns are exact by fpqc descent, and f is injective by ii), applied to the finite epimorphism

$$\widehat{S}' \times_{S'} \widehat{S}' \longrightarrow \widehat{S} \times_S \widehat{S}$$

deduced from $S' \to S$ by the flat base change $\widehat{S} \times_S \widehat{S} \to S$. The diagram chase is left to the reader. The characterization of finite effective epimorphisms recalled in d)(2) shows that $\widehat{S}' \to \widehat{S}$, obtained from $S' \to S$ by the flat base change $\widehat{S} \to S$, is again a finite effective epimorphism. We are therefore reduced to the case in which S is, in addition, the spectrum of a complete noetherian local ring.

Let S_0 be the reduced closed subscheme of S whose underlying space is the closed point. Let T' be as above and write T'' for its common image in $\mathrm{M}(S'')$. Since

$$S'_0 = S' \times_S S_0$$

is faithfully flat over S_0, there is a subgroup T_0 of G_{S_0}, of multiplicative type, whose inverse image in $G_{S'_0}$ is $T'_{S'_0}$, by fpqc descent. Since S is complete noetherian local, there exist an S-group T of multiplicative type and an S_0-morphism

$$u_0 : T \times_S S_0 \longrightarrow T_0$$

(Exposé X 3.3). The inverse image of u_0 over S'_0 extends uniquely to an S'-isomorphism

$$u' : T_{S'} \xrightarrow{\sim} T',$$

again by Exposé X 3.3. Notice here that, because S' is finite over the complete local scheme S, it is a finite disjoint sum of complete local schemes.

The two inverse images of u' over S'' are morphisms from $T_{S''}$ to T'' that coincide over $S_0 \times_S S''$; hence they coincide everywhere by the same result. Since T is flat over S and $S' \to S$ is a finite effective epimorphism, TDTE I, page 8, gives an exact diagram

$$\mathrm{Hom}_S(T, G) \longrightarrow \mathrm{Hom}_{S'}(T_{S'}, G_{S'}) \rightrightarrows \mathrm{Hom}_{S''}(T_{S''}, G_{S''}).$$

Thus u' comes from a morphism of preschemes $u : T \to G$. Likewise, $T \times_S T$ is flat over S, and therefore

$$\mathrm{Hom}_S(T \times_S T, G) \longrightarrow \mathrm{Hom}_{S'}(T_{S'} \times_{S'} T_{S'}, G_{S'})$$

is injective. It follows at once that u is a homomorphism of groups, since u' is one.

Moreover, u is a monomorphism. First observe that $\mathrm{Ker}(u)$ is flat over S. To prove this we may suppose that S is artinian local (EGA 0_{III}, 10.2.2), hence that G is separated

(Exposé $\mathrm{VI_A}$, 0.3); then $\mathrm{Ker}(u)$ is of multiplicative type (Exposé IX 6.8), and is therefore flat over S. Since

$$\mathrm{Ker}(u) \times_S S' = \mathrm{Ker}(u') = 0,$$

Lemma 3.8 gives $\mathrm{Ker}(u) = 0$. Finally, a monomorphism u is an immersion (Exposé VIII, Remarks 7.13), and its image $u(T)$ is an element of $\mathrm{M}(S)$ whose image in $\mathrm{M}(S')$ is T'. This proves Lemma 3.7.

e) End of the proof of I). By reduction c), we are reduced to the case in which S is the spectrum of a complete reduced noetherian local ring A. Let S' be the spectrum of the normalization A' of A. Nagata's theorem says that A' is finite over A (EGA 0_{IV}, 23.1.5), and $S' \to S$ is an epimorphism because $A \to A'$ is injective. Suppose that $G_{S'}$ has a torus T' whose underlying space is $E_{S'}$. The two inverse images of T' in $G_{S' \times_S S'}$ are central sub-tori with the same underlying space, so they coincide (Exposé IX 5 bis). By Lemma 3.7, T' therefore comes from a central sub-torus T of G_S, which plainly has E as its underlying space. It is thus enough to prove the existence of T'. This reduces us to the normal case and completes the proof of I).

II) Proof that (iii) *implies* (ii) *when S is normal.* We may restrict ourselves to the case in which S is integral; let t be its generic point. For every integer n that is a power of q, let ${}_nG$ be the "kernel" in G of raising to the n-th power. Since E is locally closed in G, by (iii)(b), the intersection of $\mathrm{ens}({}_nG)$ with E is locally closed in $\mathrm{ens}(G)$. Let $E(n)$ denote the reduced subscheme of G whose underlying space is $\mathrm{ens}({}_nG) \cap E$.

We show that the structural morphism

$$E(n) \longrightarrow S$$

is separated and universally open. Both properties have a valuative criterion (EGA II 7.2.3 and EGA IV 14.5.8). Let S' therefore be an S-scheme that is the spectrum of a complete discrete valuation ring with algebraically closed residue field. We have proved that 3.1(iii) implies 3.1(iv), so there is a sub-torus T' of $G_{S'}$ whose underlying space is $E_{S'}$. Now ${}_nT'$ is finite étale over S', hence is separated and universally open over S'; the same is therefore true of $E(n) \times_S S'$, which has the same underlying space.

The fibers of $E(n)$, moreover, all have the same number of geometric points, namely n^r if r is the rank of the torus T_t. Finally, the generic fiber $E(n)_t$, being reduced, is equal to ${}_nT_t$, and hence is étale over $\mathrm{Spec}(t)$. Since S is normal, SGA I 10.11 now shows that $E(n)$ is an étale covering of S.

If s is a point of S, then $E(n)_s$ is étale over $\mathrm{Spec}\,\kappa(s)$, hence reduced, and consequently coincides with the group ${}_nT_s$ of multiplicative type. We show that $E(n)$ is a group subscheme of G. Let

$$m : E(n) \times_S E(n) \longrightarrow G$$

be the morphism induced by multiplication in G. The underlying map $\mathrm{ens}(m)$ factors through $E(n)$, so m itself factors through the scheme $E(n)$, because its source $E(n) \times_S E(n)$ is étale over S, and hence reduced. It then follows from Exposé X 4.8(a) that $E(n)$ is a subgroup of multiplicative type. As observed already in 3.2(a), the family of subgroups $E(n)$ is necessarily coherent. This proves that (iii) implies (ii) when S is normal.

III) Proof that (ii) *implies* (i). In fact, we shall show that there is a unique sub-torus T of G, with underlying space E, such that

$${}_nT = M_n$$

for every n that is a power of q. Uniqueness follows immediately from Exposé IX 4.8(b). In proving existence, in view of this uniqueness, we may successively suppose that:

a) S is noetherian.

b) The M_n are central subgroups. Indeed, it is enough to replace G by

$$Z_n = \mathrm{Centr}_G(M_n)$$

for sufficiently large n, by 2.5 and 2.5 bis.

c) S is the spectrum of a local ring. Suppose that the problem has been solved after every base change $\mathrm{Spec}(\mathcal{O}_{S,s}) \to S$, for $s \in S$, and let T_s be the resulting sub-torus of $G \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$. There is then an open neighborhood U of s and a sub-torus T of $G|_U$ extending T_s, by Exposé $\mathrm{VI_B}$, §10, and 3.6. We have proved that 3.1(ii) implies 3.1(iv); since S is noetherian, E is therefore constructible by 3.3. After shrinking U, we may consequently suppose that

$$\mathrm{ens}(T) = E \times_S U$$

(EGA IV 9.5.2). For every n that is a power of q, the groups ${}_nT$ and $M_n \times_S U$ are then two subgroups of multiplicative type in $G|_U$, with the same fibers; because M_n is central, they coincide by Exposé IX 5.3 bis. Thus the problem has a local solution on S, and uniqueness makes these local solutions into a global one.

d) S is the spectrum of a complete noetherian local ring and, if s is the closed point, T_s is trivial. This follows from EGA 0_{III}, 10.3.1, and fpqc descent.

e) S is reduced. Apply 2.2.

f) S is normal. Apply Nagata's finiteness theorem (EGA 0_{IV}, 23.1.5) and Lemma 3.7 for central sub-tori.

After these reductions, T_s is trivial, and hence $(M_n)_s$ is trivial. It follows from Exposé X 3.3 that M_n itself is trivial. If t is any point of S, all the subgroups ${}_nT_t$ of T_t, for n a power of q, are therefore trivial; Exposé X 1.4 then shows at once that T_t itself is trivial. It remains only to prove the following lemma.

Lemma 3.10.— *Under the hypotheses of 3.1(ii), suppose in addition that S is noetherian and normal and that, for every point s of S, T_s is a trivial torus. Then there exists a unique sub-torus T of G, with underlying space E, such that*

$${}_nT = M_n$$

for every n that is a power of q. Moreover, T is trivial.

Proof. Uniqueness follows from the fact that T, being smooth over S, is reduced. To prove existence, we may suppose that S is irreducible, with generic point η. Let r be the rank of T_η, put

$$T' = \mathbb{G}_{m,S}^r,$$

and choose an isomorphism

$$u_\eta : T'_\eta \xrightarrow{\sim} T_\eta.$$

Write ${}_nu_\eta$ for its restriction to ${}_nT'_\eta$. Since ${}_nT'$ and M_n are trivial, ${}_nu_\eta$ has a unique extension to an S-isomorphism

$${}_nu : {}_nT' \xrightarrow{\sim} M_n.$$

I claim that, for every point s of S, there is a necessarily unique group isomorphism

$$u_s : T'_s \xrightarrow{\sim} T_s$$

that extends ${}_nu)_s$ for every n that is a power of q.

Indeed, let S^1 be an S-scheme that is the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point t_1 maps to η and whose closed point s_1 maps to s (EGA II 7.1.9). The proof that 3.1(ii) implies 3.1(iv) gives a sub-torus T^1 of G_{S^1} such that

$$(M_n)_{S^1} = {}_nT^1$$

for every n that is a power of q. Since S^1 is normal, T^1 is isotrivial by Exposé X 5.16. The classification of isotrivial tori (Exposé X 1.2) and SGA I, Exposé V 8.2, together with the fact that the generic fiber of T^1 is trivial, show that T^1 is trivial. We may therefore extend

$$u_\eta \times_\eta t_1$$

to an S^1-isomorphism of group schemes

$$u^1 : T' \times_S S^1 \xrightarrow{\sim} T^1.$$

On ${}_nT' \times_S S^1$, the morphisms u^1 and ${}_nu \times_S S^1$ agree on the generic fiber, and hence agree everywhere. Restricting u^1 to the closed fiber gives the required u_s after the field extension $\kappa(s) \to \kappa(s_1)$; it descends to $\kappa(s)$ by Exposé IX 4.8(a) and fpqc descent.

The scheme T' is smooth over the normal scheme S, hence is normal. The extension criterion for rational maps in EGA IV_4, 20.4.6, now shows that there is an S-morphism

$$u : T' \longrightarrow G$$

whose restriction to each fiber T'_s is the composite

$$T'_s \xrightarrow{u_s} T_s \longrightarrow G_s.$$

Indeed, for every base change $S^1 \to S$ of the kind used above, the preceding construction gives precisely the required extension over S^1.

It remains to show that u is a homomorphism. Let $m_{T'}$ and m_G denote multiplication in T' and G, respectively. The coincidence subscheme of

$$m_G \circ (u \times_S u) \quad \text{and} \quad u \circ m_{T'}$$

contains every fiber, since every u_s is a homomorphism. It therefore has the same underlying space as $T' \times_S T'$, and so equals $T' \times_S T'$, because the latter is smooth over S and hence reduced. Finally, u is a monomorphism because it is one on every fiber; it is therefore an immersion (Exposé VIII 7.9). The image of T' under u is a sub-torus of G whose underlying set is E.

This completes the proof of Theorem 3.1.

4. Characterization of a sub-torus T by the subgroups ${}_nT$

4.1. Statement of the main theorem

Theorem 4.1.— *Let S be a connected locally noetherian prescheme, let G be an S-group prescheme of finite type over S, let $q > 1$ be an integer invertible on S, and let r be a positive integer. For every integer n that is a power of q, let $M(n)$ be a subgroup scheme of G, of multiplicative type and of type $(\mathbb{Z}/n\mathbb{Z})^r$. Suppose that:*

a) *the family $M(n)$ is coherent: if $m \mid n$, then*

$$ {}_mM(n) = M(m); $$

b) *there exist a point $s \in S$ and a sub-torus T_s of G_s such that*

$$ M(n)_s = {}_nT_s \qquad \text{for every } n; $$

c) *for every point $t \in S$, there exists a closed affine subscheme F_t of G_t containing $M(n)_t$ for every n.*

Then there exists a unique sub-torus T of G such that ${}_nT = M(n)$ for every n that is a power of q.

There is an analogous theorem for lifting morphisms.

Theorem 4.1 bis.— *Let S, G, q be as above, let T be an S-torus, and, for every n that is a power of q, let $u(n) : {}_nT \to G$ be an S-homomorphism. Suppose that:*

a) *the family $u(n)$ is coherent: if $m \mid n$, then*

$$ u(m) = u(n)|_{{}_mT}; $$

b) *there exist a point $s \in S$ and a homomorphism $u_s : T_s \to G_s$ such that*

$$ u_s|_{{}_nT_s} = u(n)_s \qquad \text{for every } n \text{ that is a power of } q; $$

c) *for every point $t \in S$, there exists a closed affine subscheme F_t of G_t containing $u(n)_t({}_nT_t)$ for every n.*

Then there exists a unique homomorphism $u : T \to G$ whose restriction to ${}_nT$ equals $u(n)$ for every n that is a power of q.

Remark 4.2.— *Using the lower semicontinuity of the abelian rank of a flat group prescheme of finite type over the spectrum of a discrete valuation ring (cf. Exposé X 8.7), condition (c) in Theorems 4.1 and 4.1 bis may be weakened by requiring the closed affine subscheme F_t only for every maximal point t of S.*

Let us deduce Theorem 4.1 bis from Theorem 4.1. Put $G' = G \times_S T$. For every n that is a power of q, consider the homomorphism

$$ v(n) : {}_nT \longrightarrow G' $$

whose projections to G and T are respectively $u(n)$ and the canonical immersion ${}_nT \to T$. Thus $v(n)$ is an immersion; let $M(n)$ be its image subgroup. The family $M(n)$ is coherent in the sense of Theorem 4.1. The group $M(n)_s$ equals ${}_nT'_s$, where T'_s is the sub-torus of G'_s that is the graph of u_s, and, for every $t \in S$, the closed affine subscheme $F_t \times_t T_t$ of G'_t contains all the groups $M(n)_t$. By Theorem 4.1 there is therefore a sub-torus T' of G' such that ${}_nT' = M(n)$ for every n.

Let $f : T' \to T$ be the restriction of the projection $G' \to T$, and let $f(n)$ be its restriction to $M(n)$. Then

$$ f(n) \circ v(n) = \mathrm{id}_{{}_nT}\,. $$

The fibre T'_s is the graph of u_s (Exposé IX 4.8(b)), so f_s is an isomorphism. The groups $\ker f$ and $\operatorname{coker} f$ are of multiplicative type (Exposé IX 2.7) and of constant type, since S is connected. They are therefore trivial, and f is an isomorphism. If $v = f^{-1}$, then

$$ v|_{{}_nT} = v(n). $$

Composing v with the projection $G' \to G$ gives the required homomorphism $u : T \to G$. This proves existence; uniqueness follows from Exposé IX 4.8(a).

4.2. Application

We shall generalize Theorem IX 7.1. Let A be a complete noetherian local ring, let I be its maximal ideal, and put

$$S = \operatorname{Spec} A, \qquad S_m = \operatorname{Spec}(A/I^m).$$

For every prescheme X, put $X_m = X \times_S S_m$.

Let G be an S-group prescheme of finite type, let T be an S-torus, let q be an integer invertible on S, and let

$$u_m : T_m \longrightarrow G_m \qquad (m \in \mathbb{N})$$

be a coherent family of homomorphisms. For n ranging over the powers of q, denote by $u_m(n)$ the restriction of u_m to ${}_nT_m$, and by

$$u(n) : {}_nT \longrightarrow G$$

the unique homomorphism extending the morphisms $u_m(n)$ for all m (Proposition 1.6(a)). We call the family $(u_m)_{m\in\mathbb{N}}$ *admissible* if, for every point $t \in S$, there exists a closed affine subprescheme F_t of G_t containing $u(n)_t({}_nT_t)$ for every n that is a power of q. As will be seen, this property does not depend on q.

Proposition 4.3.— *With the preceding notation, the canonical map*

$$\operatorname{Hom}_{S\text{-gr}}(T, G) \longrightarrow \varprojlim_m \operatorname{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

identifies its source with the subset of the target consisting of admissible coherent families.

Indeed, apply Theorem 4.1 bis with s the closed point of S.

Corollary 4.4.— *With the preceding notation, suppose in addition that G has affine fibres. Then the canonical map*

$$\operatorname{Hom}_{S\text{-gr}}(T, G) \longrightarrow \varprojlim_m \operatorname{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

is an isomorphism.

Indeed, when G has affine fibres every coherent family is admissible.

Remark 4.5.— *When G is separated, the ring A in Proposition 4.3 may be replaced by a noetherian I-adic ring: one uses EGA III 5.4.1 in place of Proposition 1.6(a).*

4.3. Proof of Theorem 4.1

Lemma 4.6.— *Let k be a field, let G be a k-algebraic group, let q be an integer prime to the characteristic of k, let $r > 0$, and let $M(n)$, for n ranging over the powers of q, be a coherent family of subgroups of G, of multiplicative type and of type $(\mathbb{Z}/n\mathbb{Z})^r$. There exists a smallest closed subscheme H of G containing all the $M(n)$. Moreover, H is a smooth, connected, commutative algebraic subgroup of G, and its formation commutes with extension of the base field.*

Let $M \subseteq \operatorname{ens}(G)$ be the union of the underlying sets of all the $M(n)$, and let H be the reduced closed subscheme of G whose underlying space is the closure of M. Each $M(n)$ is étale over k, hence reduced, and is therefore contained in H. Thus H is the smallest closed subscheme containing all the $M(n)$.

Let $\bar{k}$ be an algebraically closed extension of k. By construction the $M(n)$ are schematically dense in H (Exposé IX 4.1), hence the $M(n)_{\bar{k}}$ are schematically dense in $H_{\bar{k}}$ (Exposé IX 4.5). Since every $M(n)_{\bar{k}}$ is reduced, $H_{\bar{k}}$ is the reduced closed subscheme of $G_{\bar{k}}$ whose underlying space is the closure of $M_{\bar{k}}$. Thus H is geometrically reduced (EGA IV 4.6.1).

Coherence of the family implies that M is stable under the group law. The subset $M \times_k M$ is dense in $\mathrm{ens}(H \times_k H)$, and $H \times_k H$ is reduced (EGA IV 4.6.5); it follows that H is an algebraic subgroup of G. It is smooth over k, since it is geometrically reduced (Exposé $\mathrm{VI_A}$ 1.3.1), and it is commutative because all the $M(n)$ are commutative.

It remains to prove connectedness. Let H^0 be the identity component of H, let m be the number of geometric points of H/H^0, and let q^s be the q-primary part of m. For every n that is a power of q, the subgroup ${}_{q^s}M(n)$ is contained in H^0. Coherence and the prescribed type of $M(n)$ give

$$ {}_{q^s}M(nq^s) = M(n). $$

Hence H^0 contains every $M(n)$, and therefore $H^0 = H$.

We can now sharpen condition (c) of Theorem 4.1. By Lemma 4.6 there is, for each $t \in S$, a smallest closed subscheme H_t of G_t containing all the $M(n)_t$. Its formation commutes with extension of the residue field. Thus $H_t \otimes_t \bar{t}$ is contained in the affine closed subset F_t, and is affine; hence H_t itself is affine. Lemma 4.6 also says that H_t is a smooth, connected, commutative algebraic group. The structure theorem for such affine groups over an algebraically closed field (*Bible*, 4, Theorem 4) expresses H_t as a direct product of a torus T_t and a unipotent group U_t. Since every $M(n)_t$ is contained in T_t, minimality gives $H_t = T_t$. Consequently H_t is a torus.

We begin the proof of Theorem 4.1.

a) Uniqueness. This is Exposé IX 4.8(b).

b) The case in which S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field k. Let K be the fraction field of A, let s and t be the closed and generic points of S, let J be the maximal ideal of A, and put

$$ S_m = \mathrm{Spec}(A/J^m), \qquad X_m = X \times_S S_m. $$

First case: the point in Theorem 4.1(b) is the generic point t. Let T' be the schematic closure of T_t in G. Its closed fibre T'_s is an algebraic subgroup of G_s, of dimension r, containing every $M(n)_s$, hence containing H_s. The torus H_s contains every $M(n)_s$, so it has rank at least r. Consequently H_s has the same underlying space as the identity component of T'_s.

The identity component $(T')^0$ of T' is then a subgroup prescheme of G, flat and separated (Exposé $\mathrm{VI_B}$ 5.2), whose generic fibre T_t is a torus and whose reduced closed fibre H_s is a torus. Hence $(T')^0$ is a torus (Exposé X 8.8); denote it by T. The groups ${}_nT$ and $M(n)$ are smooth over S, hence reduced, and have the same underlying space. They therefore coincide, and T is the required solution.

Second case: the point in Theorem 4.1(b) is the closed point s. After replacing G by the schematic closure in G of the smallest algebraic subgroup H_t containing the family $M(n)_t$ (Lemma 4.6), we may suppose that G_t is affine.

For every $m \geq 0$, Proposition 2.2 gives a unique sub-torus T_m of G_m lifting T_s and satisfying

$$ {}_nT_m = M(n)_m \qquad \text{for every } n \text{ that is a power of } q. $$

Put $T' = \mathbb{G}^r_{m,S}$. Since k is algebraically closed, T_s is split, and there is a k-isomorphism

$$ u_s : T'_s \xrightarrow{\sim} T_s. $$

It lifts uniquely to an S_m-isomorphism $u_m : T'_m \to T_m$ (Exposé IX 3.3). The coherent family $(u_m)_{m\in\mathbb{N}}$ defines a morphism of formal completions $\widehat{u} : \widehat{T'} \to \widehat{G}$, fitting into

$$\begin{array}{ccc} \widehat{T'} & \xrightarrow{\widehat{u}} & \widehat{G} \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ T' & & G, \end{array}$$

We shall prove that $\widehat{u}$ is algebraizable, reducing first to the case in which G is affine.

Lemma 4.7.— *Let $S = \operatorname{Spec} A$, where A is a discrete valuation ring, and let X, Y be S-preschemes that are quasi-compact, quasi-separated, and flat over S. Then the canonical map*

$$\Gamma(X, \mathcal{O}_X) \otimes_A \Gamma(Y, \mathcal{O}_Y) \longrightarrow \Gamma(X \times_S Y, \mathcal{O}_{X\times_S Y})$$

is an isomorphism.

Let $f : X \to S$ and $g : Y \to S$ be the structural morphisms. Because X and Y are quasi-compact and quasi-separated, EGA I 9.2.2 and EGA IV 1.7.4 show that $f_*\mathcal{O}_X$ and $g_*\mathcal{O}_Y$ are quasi-coherent $\mathcal{O}_S$-algebras. Let $\widetilde{X}$ and $\widetilde{Y}$ be the corresponding affine S-schemes. Since Y is flat over S, EGA III 1.4.15 and EGA IV 1.7.21 give an isomorphism

$$\Gamma(\widetilde{X} \times_S Y, \mathcal{O}_{\widetilde{X}\times_S Y}) \xrightarrow{\sim} \Gamma(X \times_S Y, \mathcal{O}_{X\times_S Y}).$$

The scheme $\widetilde{X}$ is flat over S, because over A flatness is equivalent to torsion-freeness. Applying EGA III 1.4.15 again, with X and Y interchanged, yields

$$\begin{aligned} \Gamma(\widetilde{X}, \mathcal{O}_{\widetilde{X}}) \otimes_A \Gamma(\widetilde{Y}, \mathcal{O}_{\widetilde{Y}}) &\xrightarrow{\sim} \Gamma(\widetilde{X} \times_S \widetilde{Y}, \mathcal{O}_{\widetilde{X}\times_S \widetilde{Y}}) \\ &\xrightarrow{\sim} \Gamma(\widetilde{X} \times_S Y, \mathcal{O}_{\widetilde{X}\times_S Y}), \end{aligned}$$

which proves the lemma.

The S-group G is flat and of finite type over S, hence quasi-compact and quasi-separated. Applying the lemma to $G \times_S G$ gives

$$\Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G) \xrightarrow{\sim} \Gamma(G \times_S G, \mathcal{O}_{G\times_S G}).$$

Thus the multiplication $m_G : G \times_S G \to G$ determines a homomorphism

$$\Gamma(G, \mathcal{O}_G) \longrightarrow \Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G),$$

and hence an S-morphism

$$m_{\widetilde{G}} : \widetilde{G} \times_S \widetilde{G} \longrightarrow \widetilde{G},$$

where $\widetilde{G}$ is the affine S-scheme corresponding to $\Gamma(G, \mathcal{O}_G)$. The usual identities show that $m_{\widetilde{G}}$ makes $\widetilde{G}$ an S-group scheme and that the canonical morphism $v : G \to \widetilde{G}$ is a homomorphism.

Remark 4.8.— *The group $\widetilde{G}$ plays the role of a largest affine quotient of G. In fact it is an fpqc quotient of G, and is therefore of finite type over S (XVII, Appendix III, 2).*

In our situation the generic fibre G_t is affine, so $v_t : G_t \to \widetilde{G}_t$ is an isomorphism (EGA I 9.3.3). Since $\widetilde{G}$ is affine, the coherent family

$$w_m = v_m \circ u_m : T'_m \longrightarrow \widetilde{G}_m \qquad (m \in \mathbb{N})$$

comes from a unique S-homomorphism (Exposé IX 7.1)

$$w : T' \longrightarrow \widetilde{G}.$$

Let T_t be the sub-torus

$$T_t = v_t^{-1}(w_t(T'_t))$$

of G_t. Its rank is at most r. We show that it contains every $M(n)_t$. Let

$$u(n)_m : ({}_nT')_m \xrightarrow{\sim} M(n)_m$$

be the restriction of u_m. This coherent family comes from a unique S-isomorphism

$$u(n) : {}_nT' \xrightarrow{\sim} M(n),$$

because $M(n)$ is finite over S. For every $m \geq 0$,

$$w_m|_{({}_nT')_m} = (v_m \circ u_m)|_{({}_nT')_m} = (v \circ u(n))_m.$$

It follows from Proposition 1.6(a) that

$$w|_{{}_nT'} = v \circ u(n).$$

On the generic fibre,

$$v_t^{-1}(w_t(({}_nT')_t)) = u(n)_t(({}_nT')_t) = M(n)_t.$$

Thus T_t contains every $M(n)_t$, and consequently has rank r.

We conclude as in the first case. Let T'' be the schematic closure of T_t in G. Since T_t contains every $M(n)_t$, the closed fibre T''_s contains every $M(n)_s$, and hence contains T_s by the density theorem. On the other hand T'' is flat over S and T_t has dimension r, so T''_s has dimension r (Exposé $\mathrm{VI_B}$ 4). Therefore T_s has the same underlying space as the identity component of T''_s, and the identity component of T'' is the required sub-torus of G.

c) End of the proof. To prove existence of T, we may assume that S is reduced (Proposition 2.2). By Theorem 3.1, (ii)⇒(i), it is then enough to show that the set U of points $x \in S$ for which there exists a sub-torus T_x of G_x satisfying

$${}_nT_x = M(n)_x \qquad \text{for every } n \text{ that is a power of } q$$

equals $\mathrm{ens}(S)$. Such a torus is necessarily unique, and fpqc descent reduces its existence to existence after an extension of the residue field of x (Exposé IX 4.8(b)).

Since S is locally noetherian and connected and U is nonempty (it contains the point s of Theorem 4.1(b)), it is enough to prove the following: if $x, x' \in S$, if x is a specialization of x', and if one of these two points lies in U, then both lie in U. The usual method (EGA II 7.1.9) reduces this to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, which was treated in part (b). This completes the proof of Theorem 4.1.

5. Representability of the functor of smooth subgroups equal to their connected normalizer

Proposition 5.1.— *Let S be a prescheme, let G be an S-group prescheme of finite presentation, and let H be a subgroup prescheme of G, smooth with connected fibres. Then:*

a) *the normalizer N of H in G is represented by a closed subprescheme of G, of finite presentation over S;*

b) *the following conditions are equivalent:*

   i) *the canonical immersion $H \to N$ is an open immersion;*

   ii) *N is smooth along the identity section, and its identity component, which is then representable (Exposé VI_B 3.10), equals H;*

   iii) *for every $s \in S$, one has $H_s = (N_s)^0$.*

Proof. The group prescheme H is locally of finite presentation over S, since it is smooth, and has connected fibres; it is therefore of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3). Part (a) follows from Exposé XI 6.11. The equivalence in (b), recalled here for reference, was proved in Exposé $\mathrm{VI_B}$ 6.5.1.

We can now state the main theorem of this section.

Theorem 5.2.— *Let S be a prescheme and G an S-group prescheme of finite presentation over S. Let $\mathcal{L}_G$, or simply $\mathcal{L}$, be the S-functor defined, for every S-prescheme S', by*

$$\mathcal{L}_G(S') = \left\{ \begin{array}{c} \text{subgroup preschemes of } G_{S'}, \text{ smooth over } S', \\ \text{with connected fibres, equal to their connected normalizer} \end{array} \right\}.$$

Then $\mathcal{L}$ is represented by an S-prescheme that is the union of an increasing family $(U_i)_{i\in\mathbb{N}}$ of open subpreschemes, each quasi-projective and of finite presentation over S, and hence in particular separated over S.

Initial reductions. For every $r \geq 0$, let $\mathcal{L}_r$ be the subfunctor of $\mathcal{L}$ defined by

$$\mathcal{L}_r(S') = \left\{ \begin{array}{c} H \subseteq G_{S'} : \ H \text{ is smooth with connected fibres,} \\ H \text{ equals its connected normalizer, and } \dim(H/S') = r \end{array} \right\}.$$

The fibre dimension of a smooth group is locally constant. Hence $\mathcal{L}_r \to \mathcal{L}$ is represented by an immersion that is both open and closed. It is therefore enough to prove the asserted representability and finiteness properties for every $\mathcal{L}_r$, for then

$$\mathcal{L} = \coprod_{r\in\mathbb{N}} \mathcal{L}_r.$$

For $n \geq 0$, an S-prescheme S', and a subgroup prescheme H of $G_{S'}$, denote by $H^{(n)}$ the n-th normal invariant of the identity section $S' \to H$ (EGA IV 16.1.2). Thus $H^{(n)}$ is a finite-type $\mathcal{O}_{S'}$-module corresponding to the n-th infinitesimal neighbourhood of the identity section of H. If H is smooth of relative dimension r, then $H^{(n)}$ is locally free of a rank $\varphi(n,r)$ depending only on n, r. Since H is a subprescheme of $G_{S'}$, there is a canonical epimorphism, compatible with base change,

$$G^{(n)} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \xrightarrow{\sim} G^{(n)}_{S'} \longrightarrow H^{(n)}.$$

Introduce the projective S-scheme

$$P_{\varphi(n,r)} = \mathrm{Grass}_{\varphi(n,r)}(G^{(n)})$$

(EGA I, 2nd ed., 9.7; see also Cartan Seminar 1960/61, Exposé 14 by A. Grothendieck). The assignment

$$H \longmapsto H^{(n)}$$

therefore defines a canonical morphism

$$u_{n,r} : \mathcal{L}_r \longrightarrow P_{\varphi(n,r)}.$$

The group G acts naturally on $G^{(n)}$, and hence on $P_{\varphi(n,r)}$, through

$$\mathrm{int} : G \longrightarrow \mathrm{Aut}_{S\text{-gr}}(G), \qquad g \longmapsto \mathrm{int}(g).$$

If S' is quasi-compact over S and $H \in \mathcal{L}_r(S')$, then for sufficiently large n one has (Exposé XI 6.11)

$$\mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}(H^{(n)}).$$

For each $n \geq 0$, define the subfunctor $\mathcal{L}_r^n$ of $\mathcal{L}_r$ by

$$\mathcal{L}_r^n(S') = \left\{ H \in \mathcal{L}_r(S') \;\middle|\; \mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}(H^{(n)}) \right\}.$$

Representability of $\mathcal{L}_r^n$. The integer r remains fixed through the proof, so we write $\mathcal{L}, \mathcal{L}^n, P_n, u_n$ in place of $\mathcal{L}_r, \mathcal{L}_r^n, P_{\varphi(n,r)}, u_{n,r}$. Let

$$v_n : \mathcal{L}^n \longrightarrow P_n$$

be the restriction of u_n. The definition of $\mathcal{L}^n$ and Proposition 5.1(b)(ii) show that v_n is a monomorphism. More precisely:

Lemma 5.3.— *The morphism v_n is an immersion of finite presentation. Consequently $\mathcal{L}^n$ is represented by an S-prescheme, quasi-projective and of finite presentation over S.*

After replacing S by P_n, the usual method reduces the assertion to the following. Let $Q \in P_n(S)$, and define a subfunctor F of the final functor h_S by

$$F(S') = \begin{cases} h_S(S'), & \text{if there exists } H \in \mathcal{L}^n(S') \text{ with } H^{(n)} = Q_{S'}, \\ \varnothing, & \text{otherwise.} \end{cases}$$

We must prove that $F \to S$ is an immersion of finite presentation.

Let N be the normalizer of Q in G. It is a subgroup prescheme of finite presentation over S, namely the inverse image of Q under

$$G \longrightarrow P_n, \qquad g \longmapsto g \cdot Q.$$

We claim that F is equivalently described by

$$F(S') = \begin{cases} h_S(S'), & N_{S'} \text{ is smooth along the identity section, of} \\ & \text{relative dimension } r, \text{ and } (N_{S'})^{(n)} = Q_{S'}, \\ \varnothing, & \text{otherwise.} \end{cases}$$

Call the first and second descriptions F_1 and F_2. Suppose $F_1(S') = h_S(S')$, and let $H \in \mathcal{L}^n(S')$ with $H^{(n)} = Q_{S'}$. Then

$$\begin{aligned}\mathrm{Norm}_{G_{S'}}(H) &= \mathrm{Norm}_{G_{S'}}(H^{(n)}) \\ &= \mathrm{Norm}_{G_{S'}}(Q_{S'}) = N_{S'}.\end{aligned}$$

By Proposition 5.1(b)(ii), $N_{S'}$ is smooth along the identity section and its identity component is H. Thus it has relative dimension r; since H is open in $N_{S'}$,

$$(N_{S'})^{(n)} = H^{(n)} = Q_{S'}.$$

Hence $F_2(S') = h_S(S')$.

Conversely, suppose $F_2(S') = h_S(S')$. The identity component of $N_{S'}$ is then represented (Exposé $\mathrm{VI_B}$ 3.10) by a subgroup prescheme H, smooth over S', with connected fibres of dimension r. Since H is invariant and open in $N_{S'}$,

$$\begin{aligned}N_{S'} &\subseteq \mathrm{Norm}_{G_{S'}}(H) \subseteq \mathrm{Norm}_{G_{S'}}(H^{(n)}) \\ &= \mathrm{Norm}_{G_{S'}}((N_{S'})^{(n)}) = \mathrm{Norm}_{G_{S'}}(Q_{S'}) = N_{S'}.\end{aligned}$$

All inclusions are equalities. The first says that H equals its connected normalizer, and the second says $H \in \mathcal{L}^n(S')$. Hence $F_1 = F_2$.

We retain the second description. Condition (a) is represented by an immersion of finite presentation, by the following lemma.

Lemma 5.4.— *Let S be a prescheme, let X be locally of finite presentation over S, let $\sigma : S \to X$ be a section, let $r \geq 0$, and define the subfunctor L of S by*

$$L(S') = \begin{cases} h_S(S'), & X_{S'} \text{ is smooth along } \sigma_{S'}, \text{ of relative} \\ & \text{dimension } r \text{ at the points of } \sigma_{S'}(S'), \\ \varnothing, & \text{otherwise.} \end{cases}$$

Then:

a) *$L \to S$ is an immersion of finite presentation;*

b) *let $\mathcal{J}$ be the conormal sheaf of $S \to X$. If for every $s \in S$ the vector space $\mathcal{J} \otimes_{\mathcal{O}_{S,s}} \kappa(s)$ has rank at most r, then $L \to S$ is a closed immersion.*

Proof. The functor L is local on S, so we may suppose S affine. After replacing X by a neighbourhood of $\sigma(S)$, we may suppose X of finite presentation, and then reduce to noetherian S by EGA IV 8.9. In part (b), use also compatibility of the conormal sheaf with base change (EGA IV 16.6.4) and constructibility of the rank of the fibres of $\mathcal{J}$.

For an S-prescheme S', let $\mathcal{J}' = \mathcal{J}_{S'}$, let $S^\bullet(\mathcal{J}')$, canonically isomorphic to $S^\bullet(\mathcal{J})_{S'}$, be the symmetric algebra of $\mathcal{J}'$ over $\mathcal{O}_{S'}$, let $\mathrm{Gr}^\bullet(\sigma_{S'})$ be the graded algebra associated with $\sigma_{S'}$ (EGA IV 16), and let

$$\sigma_{n,S'} : S^n(\mathcal{J}') \longrightarrow \mathrm{Gr}^n(\sigma_{S'})$$

be the canonical epimorphism. By EGA IV 17.12.3 and EGA 0_{IV} 19.5.4, $X_{S'}$ is smooth along $\sigma_{S'}$ of relative dimension r if and only if:

i) $\mathcal{J}'$ is locally free of rank r;

ii) $\sigma_{n,S'}$ is an isomorphism for every $n \geq 1$.

TDTE IV, Lemma 3.6, shows that the functor making $\mathcal{J}$ locally free of rank r is represented by a subprescheme S_1, closed in S in case (b). Replacing S by S_1, assume that $\mathcal{J}$ is locally free of rank r. Proceed by induction on n. Suppose a closed subprescheme S_{n-1} represents the functor making $\sigma_{q,S'}$ injective for $q \leq n-1$. The subfunctor[6] making $\sigma_{n,S'}$ injective is represented by a closed subprescheme S_n of S_{n-1}.

Indeed, after replacing S by S_{n-1}, all $\sigma_{q,S}$ for $q \leq n-1$ are isomorphisms. The $(n-1)$-st normal invariant $X^{(n-1)}$ of σ_S has a filtration with successive quotients $\mathrm{Gr}^0(\sigma_S), \ldots, \mathrm{Gr}^{n-1}(\sigma_S)$, all locally free; hence $X^{(n-1)}$ is flat over S. Compatibility of the normal invariants with base change gives

$$\mathrm{Gr}^n(\sigma_{S'}) = \ker\Big(X^{(n)}_{S'} \to X^{(n-1)}_{S'}\Big) = \mathrm{Gr}^n(\sigma)_{S'},$$
$$\sigma_{n,S'} = (\sigma_{n,S})_{S'}.$$

Thus we must represent the functor making $\sigma_{n,S}$ injective. Locally we may suppose S affine and $S^n(\mathcal{J})$ free. The required functor is represented by the closed subscheme defined by the ideal generated by the coordinates of $\ker \sigma_{n,S}$ in a basis of $S^n(\mathcal{J})$.

Since S is noetherian, the descending chain (S_n) stabilizes. Its stable value represents L, proving the lemma.

Return to $\mathcal{L}^n$. After replacing S by the subprescheme provided by Lemma 5.4, assume N smooth along the identity section of relative dimension r. Then $\mathcal{L}^n$ is the coincidence functor of two sections $h, g : S \to P_n$, corresponding to Q and $N^{(n)}$. It is therefore represented by the inverse image of the diagonal of $P_n \times_S P_n$ under the finite-presentation morphism $h \times_S g$. This proves Lemma 5.3.

The transition morphisms $\mathcal{L}^n \to \mathcal{L}^m$. If $H \in \mathcal{L}^n(S')$, then $H \in \mathcal{L}^m(S')$ for every $m \geq n$. Thus there are natural monomorphisms

$$u^m_n : \mathcal{L}^n \longrightarrow \mathcal{L}^m \qquad (m \geq n).$$

Lemma 5.5.— *The morphism u^m_n is an open immersion.*

After a base change, it is enough to consider $H \in \mathcal{L}^m(S)$ and

$$N = \mathrm{Norm}_G(H) = \mathrm{Norm}_G(H^{(m)}), \qquad N' = \mathrm{Norm}_G(H^{(n)}).$$

Let D be the coincidence functor of N, N': $D(S') = h_S(S')$ if $N_{S'} = N'_{S'}$, and $D(S') = \varnothing$ otherwise. We must show that $D \to S$ is an open immersion.

The same functor is obtained by requiring that the immersion $H \to N'$ become open. One implication follows from Proposition 5.1. Conversely, if $H_{S'} \to N'_{S'}$ is open, then, because H has connected fibres, $H_{S'}$ is the identity component of $N'_{S'}$ (Exposé $\mathrm{VI_B}$ 3.10), hence is invariant in $N'_{S'}$. Thus $N'_{S'} \subseteq N_{S'}$. The opposite inclusion always holds, so equality follows. Since H, N' are of finite presentation and H is flat over S, Exposé $\mathrm{VI_B}$ 2.6 shows that $D \to S$ is open.

End of the proof of Theorem 5.2. The functors $\mathcal{L}^n$ are represented, and the transition morphisms u^m_n are compatible open immersions. There is therefore an S-prescheme X, the union of an increasing sequence of open subpreschemes $(X_i)_{i\in\mathbb{N}}$, such that X_i represents

[6] The editors replaced "functor" by "subfunctor" here, and ask whether the intended wording is instead "which makes $\sigma_{q,S'}$ injective for $q \leq n$".

$\mathcal{L}^i$ and $X_i \to X_j$ corresponds to u_i^j. In the category of sheaves for the Zariski topology on Sch/S,

$$X = \varinjlim_i X_i, \qquad \mathcal{L} = \varinjlim_i \mathcal{L}^i.$$

Thus X represents $\mathcal{L}$, completing the proof.

Remark 5.6.— *Suppose all residue characteristics of S are zero. Then, for every $r \geq 0$, $\mathcal{L}_r = \mathcal{L}_r^1$, so $\mathcal{L}_r$ is represented by an S-prescheme quasi-projective and of finite presentation over S.*

Indeed, for $H \in \mathcal{L}_r(S')$, it suffices to prove that

$$H \longrightarrow N = \mathrm{Norm}_{G_{S'}}(H^{(1)})$$

is an open immersion. Because H is flat and both H, N are of finite presentation, this may be checked on the fibres (Exposé $\mathrm{VI_B}$ 2.6). Cartier's theorem (Exposé $\mathrm{VI_B}$ 1.6) gives, for a connected algebraic subgroup H of an algebraic group in characteristic zero,

$$\mathrm{Norm}_G(H) = \mathrm{Norm}_G(\mathrm{Lie}\, H) = \mathrm{Norm}_G(H^{(1)}).$$

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In nonzero residue characteristic the subfunctors $\mathcal{L}_r^n$ can form a strictly increasing sequence, even when S is quasi-compact, and then $\mathcal{L}_r$ is not represented by an S-prescheme quasi-compact over S. For example, over a field k of characteristic $p > 0$, let

$$T = \mathbb{G}_m \times \mathbb{G}_m, \qquad U = \mathbb{G}_a \times \mathbb{G}_a,$$

and let $G = T \ltimes U$, where $(t, t') \cdot (u, u') = (tu, t'u')$. For $n > 0$, let $U_n \subset U$ be defined by $u' = u^{p^n}$, and let $T_n \subset T$ be defined by $t' = t^{p^n}$. Then T_n acts on U_n, and $G_n = T_n U_n$ is smooth, connected, and equal to its normalizer in G. For every m, the distinct groups G_n, $n \geq m$, have the same infinitesimal neighbourhood of order p^m.

Remark 5.7.— *There is a canonical invertible sheaf L on $\mathcal{L}$ whose restriction to every open subset of $\mathcal{L}$ quasi-compact over S is S-ample. One may take*

$$L = (\det(\mathrm{Lie}\, H))^{-1},$$

where $H \subseteq G_{\mathcal{L}}$ is the universal subgroup.

Recall that for a finite-rank locally free sheaf $\mathcal{F}$, its determinant restricts on the open-and-closed locus where $\mathrm{rank}\, \mathcal{F} = r$ to $\bigwedge^r \mathcal{F}$. It is enough to restrict to $\mathcal{L}_r^n$. Let $v_n : \mathcal{L}_r^n \to P_n$ be the canonical immersion, and let Q be the universal locally free quotient on the Grassmannian P_n. By construction,

$$v_n^* Q = H^{(n)}.$$

Since $\det Q$ is the canonical ample sheaf on P_n (EGA I, 2nd ed., 9.7), $\det H^{(n)}$ is ample relative to S (EGA II 4.6.13(i) bis).

Let $\mathcal{J} = H^{(1)}$ be the conormal sheaf, and let $S^q(\mathcal{J})$ be the homogeneous degree-q part of its symmetric algebra. Smoothness of H gives

$$\det H^{(n)} \xrightarrow{\sim} \bigotimes_{1 \leq q \leq n} \det S^q(\mathcal{J}).$$

[7] For example, one may apply Proposition II.6.1 of M. Demazure and P. Gabriel, *Groupes algébriques* I, Masson–North Holland, 1970.

For every locally free $\mathcal{J}$ of rank r and every $q > 0$ there is a canonical isomorphism

$$\det S^q(\mathcal{J}) \xrightarrow{\sim} (\det \mathcal{J})^{\otimes s},$$

where $s > 0$ depends only on r, q. Thus, for some $s > 0$,

$$\det H^{(n)} \simeq (\det \mathcal{J})^{\otimes s}.$$

Since $\det \mathcal{J} = (\det \operatorname{Lie} H)^{-1}$, EGA II 4.5.6 proves that L is S-ample.

Remark 5.8.— *Let G, H be S-group preschemes of finite presentation and let $i : H \to G$ be a monomorphism. Suppose H is smooth over S with connected fibres. The normalizer $N = \operatorname{Norm}_G(H)$ is then represented by a closed subgroup prescheme of G, of finite presentation over S (Exposé XI 6.11). If, in addition, N is smooth along the identity section and has the same relative dimension as H, then $H = N^0$; in particular H is a subgroup prescheme of G.*

Indeed, N^0 is represented by an open subgroup prescheme of N, smooth over S (Exposé $\mathrm{VI_B}$ 3.10), and i factors through N^0. For every $s \in S$, the connected smooth algebraic groups $H_s, (N^0)_s$ have the same dimension and hence coincide. Since H is flat, $H \to N^0$ is an isomorphism (EGA IV 17.9.5). Thus the functor $\mathcal{L}$ of this section may equivalently be described as the functor of subgroup preschemes H of G that are smooth over S, have connected fibres, and equal their connected normalizer.

6. Functor of Cartan subgroups and functor of parabolic subgroups

When G is a smooth connected algebraic group defined over an algebraically closed field k, sub-tori of G, maximal sub-tori, Cartan subgroups (Exposé XII 1), Borel subgroups (Exposé XIV 4.1), and parabolic subgroups (Exposé XIV 4.8 bis) have already been defined. We extend these notions to a group prescheme over an arbitrary base as follows.

Definition 6.1. *Let S be a prescheme, let G be an S-group prescheme of finite presentation, let H be an S-group prescheme, and let $i : H \to G$ be an S-monomorphism making H a subgroup of G. We say that H is a sub-torus of G (respectively, a maximal sub-torus, a Cartan subgroup, a Borel subgroup, or a parabolic subgroup) if:*

i) *i) H is smooth over S.*

ii) *ii) For every geometric point $\bar{s}$ above S, $H_{\bar{s}}$ is a sub-torus of $(G_{\bar{s}})^0_{\mathrm{red}}$ (respectively, a maximal sub-torus, a Cartan subgroup, a Borel subgroup, or a parabolic subgroup).*

Remarks 6.1 bis.

a) If the S-group H is a sub-torus of G (respectively, ...), its fibres are connected, and consequently H is of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3).

b) If H is a sub-torus of G, then H is a torus in the sense of Exposé IX, as follows immediately from Exposé X 8.1. Moreover, the monomorphism $i : H \to G$ is an immersion (see 8.3 below).

c) If G is smooth over S, with connected fibres, and if H is a Cartan subgroup of G (respectively, a Borel subgroup or a parabolic subgroup), the monomorphism $i : H \to G$ is an immersion. Thus our definitions coincide with those introduced in Exposés XII and XIV. Indeed, H is then equal to its connected normalizer (by XII 6.6(c), XIV 4.8 and 4.8 bis), and it suffices to apply 5.8.

Definition 6.1 ter. *Let S be a prescheme, let G be an S-group prescheme locally of finite type, and let s be a point of S. The* ***nilpotent rank*** *of G at s, denoted by $\rho_n(s)$, is the dimension of the Cartan subgroups of $(G_s)^0_{\mathrm{red}}$. The reductive rank $\rho_r(s)$, unipotent rank $\rho_u(s)$, and abelian rank $\rho_{\mathrm{ab}}(s)$ are defined similarly (see Exposé X 8.7).*

If G is now a smooth connected algebraic group defined over an algebraically closed field k, recall that the **radical** of G, denoted by $\mathrm{rad}(G)$, is the largest invariant smooth connected solvable algebraic subgroup of G; $G/\mathrm{rad}(G)$ is then semisimple (use Exposé XII 6.1 to reduce to the affine case). If G is moreover affine, the **unipotent radical** $\mathrm{rad}^u(G)$ is defined as the largest invariant smooth connected unipotent algebraic subgroup of G; $G/\mathrm{rad}^u(G)$ is then reductive.

Proposition 6.2. *Let S be a prescheme, let G and H be two S-group preschemes of finite presentation, and let $i : H \to G$ be an S-monomorphism making H a subgroup of G. Let $P(s)$ be one of the following properties concerning a point s of S:*

i) *$(G_{\bar{s}})^0_{\mathrm{red}}$ is an abelian variety (respectively, is affine, is a torus, or is unipotent).*

ii) *$(H_{\bar{s}})^0_{\mathrm{red}}$ is a maximal torus of $G_{\bar{s}}$.*

iii) *$(H_{\bar{s}})^0_{\mathrm{red}}$ is the centralizer in $(G_{\bar{s}})^0_{\mathrm{red}}$ of a torus of $G_{\bar{s}}$ (respectively, is a Cartan subgroup of $G_{\bar{s}}$).*

iv) *$(H_{\bar{s}})^0_{\mathrm{red}}$ is a Borel subgroup (respectively, a parabolic subgroup) of $G_{\bar{s}}$.*

v) *$(H_{\bar{s}})^0_{\mathrm{red}}$ is the radical of $G_{\bar{s}}$ (respectively, $(G_{\bar{s}})^0_{\mathrm{red}}$ is semisimple).*

vi) *$G_{\bar{s}}$ is affine and $(H_{\bar{s}})^0_{\mathrm{red}}$ is the unipotent radical of $G_{\bar{s}}$ (respectively, $(G_{\bar{s}})^0_{\mathrm{red}}$ is reductive).*

Then the set E of points s of S for which $P(s)$ holds is locally constructible (EGA 0_{III}, 9.1.11).

Remark 6.2.1. This proposition complements Exposé $\mathrm{VI_B}$, §10. The structure of E can be described more precisely by using semicontinuity theorems (see Exposé X 8.7); we shall see an example below.

Proof of 6.2. Note that if S is the spectrum of a field, E is invariant under extension of that field. A standard reduction (EGA IV 9) then allows us to reduce to the case where S is noetherian and integral, with generic point η. One must show that E or $\mathrm{ens}(S) \setminus E$ contains a neighbourhood of η (EGA IV 9.2.1). One may assume S affine of ring A and field of fractions K. If L is a finite extension of K, it is immediate that there exists an A-subalgebra B of L, finite over A, having L as field of fractions. The canonical morphism $S' \to S$, where $S' = \mathrm{Spec}\, B$, is dominant, of finite presentation, so the image of a non-empty open subset of S' contains a non-empty open subset of S (EGA IV 1.8.4). From the viewpoint that interests us, we may therefore replace S by S', hence replace K by a finite extension L. Thus we may choose L so that $(G_L)_{red}$ and $(H_L)_{red}$ are smooth over L (EGA IV 4.6.6). Possibly restricting S', we may assume that G_{red} and H_{red} are group preschemes smooth over S (Exposé $\mathrm{VI_B}$, §10, and EGA IV 17). In view of the properties to be proved, we may replace G and H by their reduced identity components (Exposé $\mathrm{VI_B}$ 10.9), and hence assume that G and H are smooth over S with connected fibres. Finally, we may assume that H is a closed subgroup prescheme of G (Exposé $\mathrm{VI_B}$ 10.4).

Proof of i). Possibly after a finite extension of K, we may assume that G_η admits a "Chevalley decomposition", i.e. is an extension of an abelian variety D_η by a smooth connected linear algebraic group F_η (Séminaire Bourbaki 1956/57, no. 145). By Exposé

$\mathrm{VI_B}$ 10.16, there is an open neighbourhood U of η over which this generic extension comes from an extension

$$1 \longrightarrow F \longrightarrow G|_U \longrightarrow D \longrightarrow 1.$$

One may moreover assume that F and D are smooth over U with connected fibres, that F is affine over U, and that D is proper over U (EGA IV 8, 9 and 17). For every point s of U, (D_s, F_s) is then the Chevalley decomposition of G_s. Consequently, G_s is an abelian variety (respectively, is affine) if and only if F_s (respectively, D_s) is the unit group; this is a constructible condition (EGA IV 9.2.6.1).

To establish the last two assertions of (i), we may, by the preceding, assume that G is affine over S. Let q be a prime number invertible on S, and let ${}_qG$ be the "kernel" of the q-th-power morphism in G. It follows easily from the structure of affine algebraic groups that G_s is a torus (respectively, is unipotent) if and only if

a) ${}_qG_s$ is quasi-finite, which is a constructible property (EGA IV 9.3.2), and

b) ${}_qG_s$ has q^r geometric points, where r is the relative dimension of G over S (respectively, ${}_qG_s$ has a single point).

The function

$$s \longmapsto (\text{number of geometric points of } {}_qG_s)$$

is constructible (EGA IV 9.7.9). This completes the proof of (i).

Proof of iii). **a) Case of the centralizer of a torus.** Suppose

$$H_\eta = \mathrm{Centr}_{G_\eta}(T_\eta),$$

where T_η is a torus of G_η. We shall show that H_s is the centralizer of a sub-torus of G_s for s in a neighbourhood of η. By (i), after possibly restricting S, we may assume that T_η comes from a sub-torus T of G. Then $Z = \mathrm{Centr}_G(T)$ is represented (Exposé XI 6.11) by a subgroup prescheme of G. Since H and Z coincide generically, they coincide over a neighbourhood of η. Thus the set E of points s of S for which H_s is the centralizer in G_s of a sub-torus of G_s is ind-constructible (EGA IV 1). This will suffice in Lemma 6.6 below to prove that E is open in S, and hence, *a fortiori*, locally constructible.

b) Case of a Cartan subgroup. Suppose that H_η is a Cartan subgroup of G_η. We shall show that H_s is a Cartan subgroup of G_s for every s in a neighbourhood of η. The group H_η is the centralizer in G_η of a torus of G_η, and is nilpotent (Exposé XII 6.6). By (a) and Exposé $\mathrm{VI_B}$ 8.4, H_s has the same properties for every s in some neighbourhood U of η. For every $s \in U$, the group H_s therefore has the same reductive rank as G_s, and its unique maximal torus is central (Exposé XII 6.7). Hence H_s is the centralizer of a maximal torus of G_s, that is, a Cartan subgroup of G_s.

Suppose now that H is not a Cartan subgroup of G, and let us show that H_s is not a Cartan subgroup of G_s for s in a neighborhood U of η. In view of the assertion provisionally admitted in (a) above, we may restrict to the case where H is the centralizer in G of a sub-torus T. But then H_η contains a Cartan subgroup C_η of H_η. We have just seen that, possibly restricting S, C_η extends to a Cartan subgroup C of G, which one may assume contained in H. By hypothesis H_η strictly contains C_η, so H_s strictly contains C_s for s in a neighborhood U of η (EGA IV 9.5.2); *a fortiori*, H_s is not

a Cartan subgroup of G_s for s in U.

Proof of ii). Suppose H_η is a maximal torus of G_η and let C_η be its centralizer in G_η. By i) and iii), H is a torus over a neighborhood U of η and $C = \mathrm{Centr}_{G|U}(H|U)$ is a Cartan subgroup of $G|U$. To prove that H is a maximal torus of G over a neighborhood of η, we

may then replace G by C, then by the linear component F of a Chevalley decomposition of C (cf. i)). Let q be an integer invertible on S, ${}_qF$ the kernel of the q-th-power morphism in F. Since F_s is affine, nilpotent, smooth and connected, F_s is the direct product of its maximal torus T_s by a unipotent group (*Bible* 6-04), so ${}_qF_s = {}_qT_s$. Since H_η is a maximal torus, ${}_qH_\eta = {}_qF_\eta$, and consequently ${}_qH = {}_qF$ over a neighborhood V of η. For every point s of V, ${}_qH_s = {}_qT_s$, so $H_s = T_s$ is a maximal torus.

Suppose now that H_η is not a maximal torus of G_η. By i), we may restrict to the case where H_η is a torus, then assume that it is contained in a strictly larger torus T_η. The latter extends to a torus T strictly containing H over a neighborhood U of η. *A fortiori*, H_s is not a maximal torus for $s \in U$.

Proof of iv). Possibly restricting S, we may assume that the center Z of G is representable (Exp. $\mathrm{VI_B}$ 10.11) and flat over S, as well as the quotient G/Z (*loc. cit.*). The property "H_s contains Z_s" is constructible (EGA IV 9.5.2) and every parabolic subgroup of G_s contains Z_s (Exp. XIV 4.9 a)); this allows us to replace G by G/Z, hence assume G affine over S (Exp. XII 6.1 and i)). We may further

assume that G/H is representable, but then H_s is a parabolic subgroup of G_s if and only if $(G/H)_s$ is proper (*Bible* 6 Th. 4 b)), which is an ind-constructible property (EGA IV 9.3.5). So E is ind-constructible, and this will suffice for us to prove that E is open (Lemma 6.6), hence locally constructible.

Let us now examine the case of Borel subgroups. If H_η is a Borel subgroup of G_η, i.e. a solvable parabolic subgroup of G_η, what precedes and Exposé $\mathrm{VI_B}$, 8.4, entail that these properties remain true at every point s of a neighborhood of η. If now H_η is not a Borel subgroup of G_η, to prove that the same holds at points s of a neighborhood of η, we may restrict (in view of what precedes) to the case where H_η is a parabolic subgroup, then assume that H_η contains a Borel subgroup B_η. We have just shown that the latter extends to a Borel subgroup B of G over a neighborhood U of η. Since H_η strictly contains B_η, H_s strictly contains B_s at every point of an open subset V, and H_s is not a Borel subgroup of G_s for $s \in V$.

Proof of v). Suppose H_η is the radical of G_η. The group H_η is then invariant in G_η, solvable (smooth and connected); the same is therefore true for H_s for s belonging to a neighborhood U of η (Exp. $\mathrm{VI_B}$ 8 and 10), so for $s \in U$, H_s is contained in the radical of G_s. Replacing G by G/H (Exp. $\mathrm{VI_B}$ 10), we must

prove that if G_η is semisimple, then G_s is semisimple at every point of a neighborhood V of η. Using i) and ii), one may assume that G is affine over S and possesses a maximal torus T. Let W be the Weyl group of T (Exp. XII 2), which is quasi-finite and étale over S, hence finite and étale over an open subset V. It then follows from the elementary properties of roots (Exp. XIX 1.12) that G is semisimple over V.

Suppose now that H_η is not the radical of G_η. Possibly replacing K by a finite extension L, we may assume that G_η admits a radical R_η. By what precedes, R_η extends to a subgroup prescheme R of $G|_U$ such that for every $s \in U$, R_s is the radical of G_s. By hypothesis, $R_\eta \neq H_\eta$. So $R_s \neq H_s$ for $s \in V$. It remains to prove that if G_η is not semisimple, neither is G_s at neighboring points, but this is a particular case of what precedes (take $H =$ unit group).

Proof of vi). The proof is entirely analogous to that of v), in view of i), and is left to the care of the reader.

Corollary 6.3. *Let S_0 be a quasi-compact prescheme, let $(S_i)_{i\in L}$ be a projective system of S_0-preschemes affine over S_0, and let $S = \varprojlim S_i$ (EGA IV 8.2). Let G_0 be a group prescheme of finite presentation over S_0, set*

$$G_i = G_0 \times_{S_0} S_i, \qquad G = G_0 \times_{S_0} S,$$

and let H be a subgroup of G. If H is a sub-torus of G (respectively, a maximal sub-torus, a Cartan subgroup, a Borel subgroup, or a parabolic subgroup), then there exist an index $i \in L$ and a subgroup H_i of G_i such that

$$H = H_i \times_{S_i} S$$

and H_i is a sub-torus of G_i (respectively, ...).

Indeed, H is smooth with connected fibres, hence of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3). By Exposé $\mathrm{VI_B}$, §10, there exist $i \in L$ and a subgroup H_i of G_i, smooth over S_i, such that $H = H_i \times_{S_i} S$. Corollary 6.3 then follows from Definition 6.1, Proposition 6.2 and EGA IV 9.3.3.

Corollary 6.3 bis. *Let S be a prescheme and let G be an S-group prescheme of finite presentation. Then the functions $\rho_n, \rho_r, \rho_u, \rho_{\mathrm{ab}}$ (cf. 6.1 ter) are locally constructible on S.*

It suffices to show (EGA IV 9.) that if S is a noetherian integral scheme with generic point η, the functions in question are constant on a neighborhood of η. Possibly replacing S by a scheme S' finite over S and dominating S, we may assume that G_η admits a Cartan subgroup C_η with a Chevalley decomposition $1 \to L_\eta \to C_\eta \to A_\eta \to 1$. The argument made in 6.2 i) proves that this decomposition extends to a Chevalley decomposition over a neighborhood of η:

$$1 \longrightarrow L \longrightarrow C \longrightarrow A \longrightarrow 1.$$

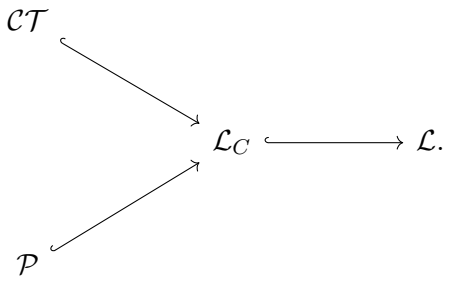
Moreover, one may assume that C is a Cartan subgroup of G (6.3) and that the maximal torus T_η of L_η extends to a maximal torus T of L (6.3). The corollary follows immediately from this and from the definitions.

6.4.0. Let S be a prescheme, G an S-prescheme in groups of finite presentation, and $\mathcal{L}$ the functor of subgroups of G that are smooth, have connected fibres, and are equal to their connected normalizer (cf. §5). The functor $\mathcal{L}$ is representable (5.2 and 5.8). The rest of this section is devoted to certain subfunctors of $\mathcal{L}$. More precisely, define $\mathcal{L}_C$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$) as follows: for every S-prescheme S', the set $\mathcal{L}_C(S')$ (resp. $\mathcal{CT}(S')$, resp. $\mathcal{P}(S')$) consists of the subgroups H of $G_{S'}$ that are smooth over S', have connected fibres, are equal to their connected normalizer, and are such that, for every point s' of S', $H_{s'}$ contains a Cartan subgroup (6.1) of $G_{s'}$ (resp. is the centralizer in $(G_{s'})^0_{\mathrm{red}}$ of a subtorus of $G_{s'}$, resp. is a parabolic subgroup (6.1) of $G_{s'}$).

Theorem 6.4. *Let S be a prescheme and G an S-prescheme in groups of finite presentation. The S-functors $\mathcal{L}_C$, $\mathcal{CT}$, and $\mathcal{P}$ defined in 6.4.0 are representable by S-preschemes that are quasi-projective and of finite presentation over S.*

Remark 6.5. If G has smooth fibres—for example, if G is smooth over S, or if the residue characteristics of S are zero (Exposé $\mathrm{VI_B}$, 1.6.1)—then every subgroup H of G, smooth over S with connected fibres, such that H_s contains a Cartan subgroup of $(G_s)^0$ for every point s of S, is necessarily equal to its connected normalizer (and hence is an element of $\mathcal{L}_C(S)$). Indeed, by 5.1 iii) we may assume that S is the spectrum of a field, in which case the property was noted at the end of the statement of Exposé XIII, 2.1.

There are natural monomorphisms



Let us show that these monomorphisms are open immersions of finite presentation (which will already prove, in view of 5.2, that $\mathcal{L}_C$, $\mathcal{CT}$, $\mathcal{P}$ are representable by S-preschemes, union of an increasing sequence of open subpreschemes, quasi-projective and of finite presentation over S). Now this will follow, by the usual technique, from the following lemma:

Lemma 6.6. *Let S be a prescheme, G an S-prescheme in groups of finite presentation, and H a subgroup of G, smooth with connected fibres. The set of points s of S such that H_s contains a Cartan subgroup of G_s (resp. is equal to the centralizer in $(G_s)^0_{\mathrm{red}}$ of a torus of G_s) is open in S. If, moreover, H is equal to its connected normalizer,*[8] *then the set of points s of S such that H_s is a parabolic subgroup of G_s is also open in S.*

The assertion to be proved is local on S, which allows us to assume S affine, then, by EGA IV 8.1 and 6.3, S noetherian. Denote by E the set of points of S having the property in question. It then follows from the assertions effectively proved in 6.2 that E is ind-constructible. But S is noetherian, so to prove that E is open, it suffices to show that E is stable under generizations (EGA IV 1.10.1). Using EGA II 7.1.9, we are finally led to prove that if S is the spectrum of a discrete valuation ring, and if the closed point s belongs to E, then so does the generic point t.

We shall need the following lemma:

Lemma 6.7. *Let A be a complete noetherian local ring, $S = \operatorname{Spec} A$, s the closed point of S, H an S-prescheme in groups, smooth with connected fibres, and T_s a subtorus of H_s. Then:*

i) there exists a closed subgroup prescheme C of H, smooth with connected fibres, such that $C_s = \operatorname{Centr}_{H_s}(T_s)$;

ii) for every point t of S, C_t is the centralizer in H_t of a subtorus T_t of H_t.

Proof of 6.7. Let T' be an S-torus for which there is an isomorphism $u_0 : T'_s \xrightarrow{\sim} T_s$ (Exposé X, 4.6). Let $\mathfrak{m}$ be the maximal ideal of A, and put $A_n = A/\mathfrak{m}^n$, $S_n = \operatorname{Spec} A_n$, $H_n = H \times_S S_n$, and so on. Since H is smooth over S, for every integer $n > 0$ there is an S_n-group morphism

$$u_n : T'_n \longrightarrow H_n$$

lifting u_0 (Exposé IX, 3.6), and by induction on n we may assume that u_n lifts u_{n-1}. Let q be a prime invertible on S. For every integer ℓ that is a power of q, write ${}_\ell u_n$ for the restriction of u_n to ${}_\ell T'_n$. For fixed ℓ and varying n, the morphisms ${}_\ell u_n$ form a projective system and therefore arise from a unique S-group morphism

$${}_\ell u : {}_\ell T' \longrightarrow H$$

[8] This hypothesis is in fact superfluous; see Exposé XVII, Appendix III, 3.

(1.6 a)). Since H is separated (Exposé $\mathrm{VI_B}$, 5.2) and u_0 is a monomorphism, ${}_\ell u$ is a monomorphism (Exposé IX, 6.8), and even a closed immersion because it is finite, ${}_\ell T'$ being finite over S. Denote its image by $M(\ell)$. The family of subgroups of multiplicative type $M(\ell)$ is clearly coherent in the sense of 4.1. Put

$$C_\ell = \mathrm{Centr}_H(M(\ell)).$$

This is represented by a closed subgroup prescheme (2.5), since H is separated, and is smooth over S (Exposé XI, 2.4). The C_ℓ form a filtered decreasing family of closed subschemes and are therefore stationary, since H is noetherian. Their stationary value is a smooth closed subgroup C such that $C_s = \mathrm{Centr}_{H_s}(T_s)$, by the density theorem (Exposé IX, 4.7). It remains to show that for every point t of S, C_t is the centralizer in H_t of a subtorus T_t; this will also imply that C_t is connected. This follows from the more precise lemma below, applied to the family $M(\ell)_t$ of subgroups of H_t.

Lemma 6.8. *Let G be a connected algebraic group over a field k, let $r > 0$ be an integer, and let q be an integer prime to the characteristic*[9] *of k. Let $M(\ell)$, where ℓ ranges over the powers of q, be a coherent family of subgroups of G, of multiplicative type and of type $(\mathbb{Z}/\ell\mathbb{Z})^r$ (cf. 4.6). Let M be the algebraic subgroup of G generated by the $M(\ell)$ (*loc. cit.*), and let T be the unique maximal torus of M (cf. 3.4). Then, for all sufficiently large ℓ,*

$$\mathrm{Centr}_G(T) = \mathrm{Centr}_G(M) = \mathrm{Centr}_G(M(\ell)).$$

The last equality is clear. To prove the first, introduce the centre Z of G, put $G' = G/Z$, let M' (resp. T') be the image of M (resp. T) in G', and let K be the inverse image of T' in G, that is, the algebraic subgroup of G generated by T and Z. It plainly suffices to prove $K \supset M$, or equivalently $T' = M'$. Now M is smooth and connected (4.6), while G' is affine (Exposé XII, 6.1); hence M' is the direct product of its maximal torus T' (Exposé XII, 6.6 d)) and a unipotent group (*Bible* 4, Th. 4), after assuming k algebraically closed. The image of $M(\ell)$ in M' is therefore contained in T'. Thus the inverse image of T' in M contains $M(\ell)$ for every ℓ, and consequently is equal to M. Hence $M' = T'$. This proves 6.8, and therefore 6.7.

We now prove 6.6. We have reduced to the case where S is the spectrum of a discrete valuation ring A, which we may moreover assume complete with algebraically closed residue field. After replacing A, if necessary, by its normalization in a finite extension of its fraction field, we may assume that $(G_t)_{\mathrm{red}}$ is smooth.[10] To prove 6.6 we may clearly replace G by the identity component of the schematic closure in G of $(G_t)^0_{\mathrm{red}}$, and therefore assume that G is flat over S, has connected fibres, and has smooth generic fibre G_t.

a) Suppose that H_s is the centralizer in $(G_s)_{\mathrm{red}}$ of a torus T_s. We show that H_t is then the centralizer in G_t of a subtorus of G_t. By Lemma 6.7 there is a subgroup scheme C of H, smooth over S, whose closed fibre is H_s, such that $C_t = \mathrm{Centr}_{H_t}(T_t)$ for a subtorus T_t of H_t. Since H is smooth over S with connected fibres, dimension considerations give $C = H$. With the notation of 6.7, $H = C = C_\ell$ for all sufficiently large ℓ. Similarly, put $C'_\ell = \mathrm{Centr}_G(M(\ell))$ (2.5), and let C' be the stationary value of the C'_ℓ for large ℓ (2.5 bis). The group scheme C' contains H, and

$$C'_t = \mathrm{Centr}_{G_t}(T_t) \quad (6.8), \qquad C'_s = \mathrm{Centr}_{G_s}(T_s).$$

The hypothesis on H_s gives $\dim H_s = \dim C'_s$. Moreover, $\dim H_s = \dim H_t$, since H is smooth over S, and $\dim C'_t \leq \dim C'_s$ (Exposé $\mathrm{VI_B}$, 4.1). Hence $\dim H_t = \dim C'_t$.

[9] Editor's note: the word "residual" has been removed.

[10] Editor's note: details or references should be supplied here.

Since G_t is smooth and connected, $C'_t = \mathrm{Centr}_{G_t}(T_t)$ is smooth and connected; finally,

$$H_t = C'_t = \mathrm{Centr}_{G_t}(T_t).$$

b) Suppose that H_s contains a Cartan subgroup C_s of G_s, that is, the centralizer in $(G_s)_{\mathrm{red}}$ of a maximal torus T_s of G_s. By 6.7 there is a subgroup scheme C of H, smooth over S with connected fibres, lifting C_s. It suffices to prove that C_t contains a Cartan subgroup of G_t. By part a), applied with $H = C$, C_t is the centralizer in G_t of a subtorus of G_t, and hence contains a Cartan subgroup of G_t.

c) Suppose that H_s is a parabolic subgroup of G_s. Let $N = \mathrm{Norm}_G(H)$. By hypothesis H is equal to its connected normalizer, so N_s is smooth and consequently

$$N_s = \mathrm{Norm}_{(G_s)_{\mathrm{red}}}(H_s) = H_s \qquad \text{(Exposé XII, 8 bis)}.$$

Thus N is flat over S. We shall see in Exposé XVI that under these conditions G/N is representable. Since $H_s = N_s$ is a parabolic subgroup, $(G/N)_s$ is proper. Since G/N is flat over S with connected fibres, EGA III, 5.5.1 implies that G/N is proper over S. Thus $(G/N)_t = G_t/N_t$ is proper over t, and so is G_t/H_t, because N_t/H_t is finite. It remains only to apply the following lemma.

Lemma 6.9. *Let k be a field, G a k-algebraic group, H a smooth connected algebraic subgroup of G, and $N = \mathrm{Norm}_G(H)$. If $\dim H = \dim N$ and G/H is proper, then H is a parabolic subgroup of G.*

Indeed, we may assume that k is algebraically closed and that G is smooth and connected. The centre Z of G is contained in N, and $\dim H = \dim N$ implies that $Z' = (Z)^0_{\mathrm{red}}$ is contained in H. Hence $G' = G/Z'$ is affine (Exposé XII, 6.1). Replacing G by G' and H by its image in G', we reduce to the case in which G is affine (Exposé XIV, 4.9), and Lemma 6.9 then follows from *Bible* 6, Th. 4. This proves Lemma 6.6.

To complete the proof of 6.4, we must show that the S-preschemes representing $\mathcal{L}_C$ (respectively $\mathcal{CT}$ and $\mathcal{P}$) are of finite presentation over S. This assertion is local on S, so we may assume first that S is affine and then, since G is of finite presentation, that S is noetherian (EGA IV, 8.9). We have just seen that the natural inclusions $\mathcal{CT} \to \mathcal{L}$ and $\mathcal{P} \to \mathcal{L}$ are immersions. The same is therefore true of $\mathcal{CT} \to \mathcal{L}_C$ and $\mathcal{P} \to \mathcal{L}_C$. Consequently, it is enough to prove that $\mathcal{L}_C$ is represented by an S-prescheme of finite presentation.

Resume the notation introduced in 5.2. For every integer $n \geq 0$, let $\mathcal{L}^n$ be the subfunctor of $\mathcal{L}$ defined by

$$\mathcal{L}^n(S') = \left\{ H \in \mathcal{L}(S') \;\middle|\; \mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}\left(H^{(n)}\right) \right\}.$$

The S-functor $\mathcal{L}^n$ is thus represented by an open subprescheme of $\mathcal{L}$, the disjoint union of the $\mathcal{L}^n_r$. Each $\mathcal{L}^n_r$ is of finite presentation over S (5.3), and is empty for

$$r > \sup_{s \in S} \dim G_s,$$

which is a finite number because S is quasi-compact. Hence $\mathcal{L}^n$ is of finite presentation over S. It remains to prove that $\mathcal{L}_C$ is contained in $\mathcal{L}^n$ for all sufficiently large n.

For each point s of S, let $d(s)$ be the least integer n, finite or infinite, such that

$$\mathcal{L}_{C,G_s} \subset \mathcal{L}^n_{G_s}.$$

It is enough to show that d is bounded on S. Indeed, if M is an upper bound, then $\mathcal{L}_C$ is set-theoretically contained in $\mathcal{L}^M$, and hence is contained in $\mathcal{L}^M$, since the latter is open in $\mathcal{L}$. A direct constructibility argument reduces us to showing that, when S is noetherian and integral with generic point t, the function d is bounded on a neighbourhood of t.

a) **Reduction to the case in which G is smooth over S.** Proceeding as in 6.2, we see that, after possibly changing S, we may assume that G_{red} is a group prescheme smooth over S; denote it by G'. Put

$$X = \mathcal{L}_{C,G}, \qquad X' = \mathcal{L}_{C,G'},$$

and let H (respectively H') be the subgroup prescheme of G_X (respectively $G'_{X'}$) universal for the functor $\mathcal{L}_{C,G}$ (respectively $\mathcal{L}_{C,G'}$). Since H is smooth over X, the prescheme $H \times_X X_{\mathrm{red}}$ is reduced, and is therefore contained in $G'_{X_{\mathrm{red}}}$. It is an element of $\mathcal{L}_{C,G'}(X_{\mathrm{red}})$, and hence defines a canonical morphism

$$p : X_{\mathrm{red}} \longrightarrow X'.$$

It is clear that p is a monomorphism; we show that it is in fact an immersion. Let N' be the normalizer of H' in $G_{X'}$. The set of points $s \in X'$ for which $H'_s \to N'_s$ is an open immersion is an open subset U, and

$$H'|_U \longrightarrow N'|_U$$

is an open immersion (Exposé $\mathrm{VI_B}$, 2.5, and EGA IV, 17.9.5). It follows from 5.1(iii) that U is the largest open subset of X' over which H' equals its connected normalizer in $G_{X'}$. Thus $H'|_U \in \mathcal{L}_{C,G}(U)$, and it follows at once that p is an isomorphism of X_{red} onto U_{red}. If $\mathcal{L}_{C,G}$ is known to be of finite type when G is smooth, then X' is of finite type over S; consequently X_{red} is of finite type (since S is noetherian), and is therefore contained in $\mathcal{L}_G^M$ for all sufficiently large M. The same then holds for X.

b) **The case in which S is the spectrum of an algebraically closed field k of characteristic $p > 0$, and G is a smooth algebraic group over k.** Instead of the infinitesimal neighbourhoods $H^{(n)}$ of a subgroup prescheme H of G, we use the radicial subgroups $F_n(H)$, the kernels of the iterates of the Frobenius morphism of H (Exposé $\mathrm{VII_A}$, 4). This is legitimate here because $F_n(H)$ is contained in the infinitesimal neighbourhood of order p^n of the unit section of H. If T is a subtorus of G, then

$$F_n(T) = {}_{p^n}T,$$

and duality immediately shows that $\mathrm{Centr}_G({}_{p^n}T) = \mathrm{Centr}_G(T)$ for all sufficiently large n. We have the following more precise proposition.

Proposition 6.10. *Let k be a field of characteristic $p > 0$, let G be a smooth k-algebraic group, let T be a maximal torus of $G_{\bar{k}}$, and let m be the least integer such that*

$$\mathrm{Centr}_G({}_{p^m}T)^0 = \mathrm{Centr}_G(T)^0.$$

Then, for every k-prescheme S and every $H \in \mathcal{L}_C(S)$, one has

$$\mathrm{Norm}_{G_S}(H) = \mathrm{Norm}_{G_S}(F_m(H)).$$

In particular, $\mathcal{L}_C$ is contained in $\mathcal{L}^{p^m}$.

Since H is smooth and of finite presentation over S, and has constant nilpotent rank, namely that of G, we shall see in the next section (7.3) that the functor $\mathcal{C}_H$ of Cartan subgroups of H is represented by an S-prescheme smooth over S. The reader will verify that the proof given there does not use the finite-type property of $\mathcal{L}_{C,H}$. It follows from Exposé XIII, 3.1, that we may consider the open subset U of regular points of H.

Let S' be an S-prescheme and let $g \in G(S')$ normalize $F_m(H)_{S'}$. To prove that g normalizes $H_{S'}$, it is enough to prove that

$$\mathrm{int}(g)U_{S'} \subset H_{S'}.$$

Indeed, $H_{S'} \cap \mathrm{int}(g)H_{S'}$ would then contain an open subgroup of $\mathrm{int}(g)H_{S'}$, and hence would equal it, since the latter has connected fibres. After replacing S' by a suitable covering S'', and then replacing S'' by S, we are reduced to proving the following: if $g \in G(S)$ normalizes $F_m(H)$ and $u \in U(S)$, then $\mathrm{int}(g)u \in H(S)$.

Let C be the unique Cartan subgroup of H "containing" u (Exposé XIII, 3.2). It is enough to show that

$$C' = \mathrm{int}(g)C \subset H.$$

Since $H \in \mathcal{L}_{C,G}(S)$, the group C is also a Cartan subgroup of G. The latter admits maximal tori, so C is the centralizer in G_S^0 of its unique maximal torus T (Exposé XII, 7.1(a)). By the definition of m, and because any two maximal tori of G are locally conjugate for the fpqc topology (Exposé XII, 7.1),

$$C = \mathrm{Centr}_G(T)^0 = \mathrm{Centr}_G({}_{p^m}T)^0.$$

Conjugating by g, we obtain

$$C' = \mathrm{Centr}_G\big(\mathrm{int}(g)({}_{p^m}T)\big)^0.$$

Now $\mathrm{int}(g)({}_{p^m}T)$ is a subgroup of multiplicative type of

$$\mathrm{int}(g)\big(F_m(H)\big) = F_m(H),$$

because g normalizes $F_m(H)$; it is therefore contained in H. Exposé XIII, 2.1 (proved there when the base is a field, but extending at once to an arbitrary base) then shows that, in order to prove $C' \subset H$, it is enough to prove

$$\mathrm{Lie}\, C' \subset \mathrm{Lie}\, H.$$

But

$$\mathrm{Lie}\, C' = \mathrm{int}(g)(\mathrm{Lie}\, C) \subset \mathrm{int}(g)(\mathrm{Lie}\, H).$$

On the other hand, we may assume $m \geq 1$, and Exposé $\mathrm{VII_A}$, 4.1.2, gives

$$\mathrm{Lie}\, H = \mathrm{Lie}\big(F_m(H)\big) = \mathrm{Lie}\big(\mathrm{int}(g)(F_m(H))\big) = \mathrm{int}(g)(\mathrm{Lie}\, H).$$

Thus $\mathrm{Lie}\, C' \subset \mathrm{Lie}\, H$, which completes the proof of 6.10.

We shall need another description of the integer m introduced in 6.10.

Lemma 6.11. *Let G be a smooth algebraic group over an algebraically closed field k of characteristic $p > 0$, let T be a maximal torus of G, put $\mathfrak{g} = \mathrm{Lie}\, G$, and let R be the set of nonzero characters of T occurring in the representation of T on $\mathfrak{g}$ induced by the*

adjoint representation of G. For $r \in R$, let e_r be the greatest integer n such that p^n divides r in the character group of T. Then

$$m = \sup_{r \in R}(e_r + 1)$$

if $R \neq \varnothing$, and $m = 0$ otherwise.

Indeed, $\mathrm{Centr}_G(T)$ is smooth and is contained in $\mathrm{Centr}_G({}_{p^m}T)$. Hence

$$\begin{aligned} \mathrm{Centr}_G(T)^0 &= \mathrm{Centr}_G({}_{p^m}T)^0 \\ &\iff \mathrm{Lie}\big(\mathrm{Centr}_G(T)\big) = \mathrm{Lie}\big(\mathrm{Centr}_G(F_m(T))\big) \\ &\iff \mathfrak{g}^T = \mathfrak{g}^{{}_{p^m}T} \qquad \text{(Exposé II, 5.2.3).} \end{aligned}$$

With the usual notation,

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{r \in R} \mathfrak{g}_r.$$

Thus $\mathfrak{g}^T = \mathfrak{g}_0$, whereas

$$\mathfrak{g}^{{}_{p^m}T} = \mathfrak{g}_0 \oplus \bigoplus_{r \in R'} \mathfrak{g}_r,$$

where R' is the subset of R formed by the characters whose restriction to ${}_{p^m}T$ is zero. A nonzero character r of T has trivial restriction to ${}_{p^m}T$ if and only if $m \leq e_r$, which proves the lemma.

c) **Return to the proof of 6.4.** By part a) and the paragraph preceding it, we have reduced to the case in which S is a noetherian integral scheme and G is smooth over S. We must show that d is bounded on a neighbourhood of the generic point t of S. After possibly changing S, we may assume that G has a split maximal torus (6.2)

$$T = D_S(M).$$

Let

$$\mathfrak{g} = \bigoplus_{\lambda \in M} \mathfrak{g}^\lambda$$

be the decomposition of the Lie algebra of G according to the characters of T, and let R be the finite set of nonzero $\lambda \in M$ for which $\mathfrak{g}^\lambda \neq 0$. We distinguish two cases.

First case: t, and hence every point of S, has positive residue characteristic p. It is clear from the foregoing that d is then bounded by p^m, where m is defined as in 6.11.

Second case: t has residue characteristic zero. For each $\lambda \in R$, let n_λ be the greatest integer dividing λ in the group M, and put

$$n = \prod_{\lambda \in R} n_\lambda.$$

For every prime number q dividing n, let S_q be the closed subset of S formed by the points of residue characteristic q, and let U be the nonempty open subset, containing t, complementary to the union of the S_q. If $s \in U$, then either s has residue characteristic zero, in which case $d(s) = 1$ by 5.6, or s has residue characteristic $p > 0$ not dividing n. In the latter case the integer m attached to G_s in 6.11 is at most one. Moreover, Exposé VII[11] gives

$$\mathrm{Norm}_{G_{S'}}(F_1(H)) = \mathrm{Norm}_{G_{S'}}(\mathrm{Lie}\, H) = \mathrm{Norm}_{G_{S'}}(H^{(1)}).$$

[11] Editor's note: the argument should be made explicit.

Finally, 6.10 shows that, for $H \in \mathcal{L}_{C,G_s}(S')$,

$$\mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}(H^{(1)}).$$

Consequently $d(s) \leq 1$, so d is bounded by 1 on U. This completes the proof of 6.4.

Corollary 6.12. *Let S be a prescheme and G an S-prescheme in groups of finite presentation. Suppose that the nilpotent rank of the fibres of G (respectively the dimension of their Borel subgroups) is locally constant on S. Then the functor $\mathcal{C}$ of Cartan subgroups of G (respectively the functor $\mathcal{B}$ of Borel subgroups of G) is represented by an S-prescheme of finite presentation over S.*

Indeed, after restricting S, we may assume that the nilpotent rank ν of the fibres is constant. Then $\mathcal{C}$ is represented by the open-and-closed subprescheme of the prescheme X representing $\mathcal{L}_C$ (6.4) over which the universal subgroup of G_X for the functor $\mathcal{L}_C$ has relative dimension ν. The proof is analogous for $\mathcal{B}$, using the representability of $\mathcal{P}$.

7. Cartan subgroups of a smooth group

Proposition 7.1.— *Let S be a prescheme and let G be an S-group prescheme of finite presentation, smooth over S, with connected fibres.*

i) *Let $\mathcal{CT}$ be the S-functor defined by*

$$\mathcal{CT}(S') = \left\{ \begin{array}{c} \text{subgroup preschemes } H \text{ of } G_{S'}, \text{ smooth over } S', \\ \text{such that every } H_{s'} \text{ is the centralizer in } G_{s'} \\ \text{of a sub-torus of } G_{s'} \end{array} \right\}.$$

Then $\mathcal{CT}$ is represented by an S-prescheme of finite presentation, smooth and quasi-projective over S.

ii) *If S is artinian and $H \in \mathcal{CT}(S)$, then H is the centralizer in G of a sub-torus of G. If S is the spectrum of a henselian local ring and $H \in \mathcal{CT}(S)$, then H is the centralizer in G of a subgroup of multiplicative type that is étale over S.*

iii) *Let L be an S-group prescheme smooth and of finite presentation, let $i : L \to G$ be an S-group monomorphism, and let $H \in \mathcal{CT}(S)$. Then $\mathrm{Transp}_G(H, L)$ and $\mathrm{Transpstr}_G(H, L)$ (Exposé VIII 6.5(e)) are represented by closed subpreschemes of G, smooth over S.*

iv) *If $H \in \mathcal{CT}(S)$, then H is closed in G; $N = \mathrm{Norm}_G(H)$ is represented by a closed subgroup prescheme of G, smooth over S; N/H is represented by an S-group prescheme, separated, étale, and of finite type over S; and G/N is represented by an S-prescheme smooth and quasi-projective over S.*

v) *Let G' be an S-group prescheme of finite presentation and let $u : G \to G'$ be faithfully flat. Then G' satisfies the same hypotheses as G (Exposé VI$_B$ 9). If $H \in \mathcal{CT}_G(S)$, its image is represented by a subgroup prescheme $H' \in \mathcal{CT}_{G'}(S)$, and $H \to H'$ is faithfully flat. If $H = \mathrm{Centr}_G(T)$ for a torus T, then*

$$H' = \mathrm{Centr}_{G'}(T'), \qquad T' = u(T).$$

vi) *Under the hypotheses of (v), the S-morphism*

$$\widetilde{u} : \mathcal{CT}_G \longrightarrow \mathcal{CT}_{G'}, \qquad H \longmapsto H' = u(H),$$

is quasi-compact and faithfully flat. If, moreover, $\ker u$ *is central, then* $\widetilde{u}$ *is an isomorphism, with inverse* $H' \mapsto u^{-1}(H')$.

Proof of (ii). For the first assertion we may suppose that S is local artinian, with closed point s. Let $H \in \mathcal{CT}(S)$, and let T_s be the maximal central torus of H_s, which is defined over $\kappa(s)$ (cf. Lemma 3.4). Since

$$H_s = \mathrm{Centr}_{G_s}(T_s),$$

and H is smooth, T_s lifts uniquely to a sub-torus T of H, central in H (Exposé IX 3.6 bis and 5.6). Then $H' = \mathrm{Centr}_G(T)$ contains H and has the same fibre as H. Because H is flat, $H = H'$ (Exposé $\mathrm{VI_B}$ 2.5).

Suppose now that S is the spectrum of a henselian local ring, which the usual reductions allow us to take noetherian. Let s be its closed point, let T_s be the maximal central torus of H_s, let q be invertible on S, let ℓ be a power of q, and let T be an S-torus whose closed fibre is isomorphic to T_s (Exposé X 4.6). Let ${}_\ell H$ denote the "kernel" of the ℓ-th power morphism in H, and let U_ℓ be the largest open subset of ${}_\ell H$ étale over S. Lemma 1.3 and flatness of H show that U_ℓ contains ${}_\ell T_s$. Hensel's lemma gives a unique S-morphism

$$u_\ell : {}_\ell T \longrightarrow U_\ell$$

inducing the canonical immersion ${}_\ell T_s \to (U_\ell)_s$ on the closed fibre. Uniqueness shows that u_ℓ is a central homomorphism (Exposé IX 5.6(a)). Proceeding as in 6.6 and 6.7, one proves that u_ℓ is an immersion and, if $M_\ell = u_\ell({}_\ell T)$, that

$$\mathrm{Centr}_G(M_\ell) = H$$

for all sufficiently large ℓ. This is where noetherianity of S is used.

Proof of (i). If $H \in \mathcal{CT}(S)$, then H has connected fibres (Exposé XII 6.6(b)) and equals its connected normalizer (6.5). Hence the present functor $\mathcal{CT}$ is the functor of 6.4.0. Theorem 6.4 therefore represents it by an S-prescheme of finite presentation, quasi-projective over S. It remains to prove smoothness.

EGA IV 8 reduces us to affine noetherian S. By Exposé XI 1.5 it then suffices to prove the following infinitesimal lifting property: if S is local artinian, S_0 is defined by a nilpotent ideal, and $H_0 \in \mathcal{CT}(S_0)$, then H_0 lifts to a smooth subgroup prescheme H of G. By part (ii),

$$H_0 = \mathrm{Centr}_{G_0}(T_0)$$

for a sub-torus T_0 of G_0. Since G is smooth, T_0 lifts to a sub-torus T of G (Exposé IX 3.6 bis), and $H = \mathrm{Centr}_G(T)$ works; it is smooth by Exposé XI 2.4 and Lemma 2.5.

Proof of (iii). Because H is smooth with connected fibres, Exposé XI 6.11 represents $\mathrm{Transp}_G(H, L)$ by a closed subprescheme of G, of finite presentation over S. Smoothness reduces to the following lifting problem. Let S be local artinian, S_0 a closed subprescheme, and $g_0 \in G(S_0)$ such that

$$\mathrm{int}(g_0)H_0 \subseteq L_0.$$

We must lift g_0 to $g \in G(S)$ with $\mathrm{int}(g)H \subseteq L$.

Smoothness of G gives $g_1 \in G(S)$ lifting g_0. Put $H' = \mathrm{int}(g_1)H$. Then $H' \in \mathcal{CT}(S)$ and $H'_0 \subseteq L_0$. By (ii), $H' = \mathrm{Centr}_G(T')$ for a torus $T' \subseteq G$. Since L is smooth, $T'_0 \subseteq L_0$ lifts to a torus $T'' \subseteq L$ (Exposé IX 3.6 bis). The group $\mathrm{Centr}_L(T'')$ lies in $\mathrm{Centr}_G(T'')$, has the same closed fibre as the latter, namely H'_0, and is smooth. Hence

$$\mathrm{Centr}_L(T'') = \mathrm{Centr}_G(T'').$$

The sub-tori T' and T'' of G lift the same T'_0, so they are conjugate by an element $h \in G(S)$ reducing to the identity (Exposé IX 3.3 bis). Their centralizers are therefore conjugate. Then $g = hg_1$ lifts g_0 and satisfies $\mathrm{int}(g)H \subseteq L$.

For $g \in \mathrm{Transp}_G(H, L)(S)$, the equality $\mathrm{int}(g)H = L$ holds if and only if $\dim H_s = \dim L_s$ for every $s \in S$. If U is the open-and-closed subprescheme over which these dimensions agree, then

$$\mathrm{Transpstr}_G(H, L) = U \times_S \mathrm{Transp}_G(H, L).$$

Proof of (iv). To prove H closed in G, reduce first to affine noetherian S, then, by EGA IV 8, to the spectrum of a complete local ring. Part (ii) writes H as the centralizer in G of a subgroup of multiplicative type, so H is closed because G is separated (Exposé $\mathrm{VI_B}$ 5.2).

By (iii),

$$N = \mathrm{Norm}_G(H) = \mathrm{Transpstr}_G(H, H)$$

is represented by a subgroup prescheme smooth over S and closed in G. The morphism

$$G \longrightarrow \mathcal{CT}, \qquad g \longmapsto \mathrm{int}(g)H$$

is smooth by (iii), so its image is an open subset U of $\mathcal{CT}$. As in Exposé XI 5.3, G/N is represented by U; it is therefore quasi-projective.

We now treat N/H. By EGA IV 8, assume S affine noetherian and then $S = \mathrm{Spec}\, A$, with A local of finite dimension[12]. We use induction on $\dim S$. When $\dim S = 0$, the assertion is Exposé $\mathrm{VI_A}$, §4. If N/H is representable, it is separated because H is closed in N, of finite type and étale because N is smooth of finite type and H is open. It is therefore quasi-affine (SGA 1, VIII 6.2). Effective descent for quasi-affine schemes (*loc. cit.*, 7.9) permits replacement of A by its completion.

Thus assume S complete noetherian local, with closed point s, and put $U = S \setminus \{s\}$. By induction, $(N|_U)/(H|_U)$ is represented by a U-group K. Let T_s be the maximal central torus of H_s, choose q invertible on S, let ℓ be a power of q, and let M_ℓ be the unique central subgroup of multiplicative type of H lifting ${}_\ell T_s$, as in (ii). Choose ℓ so large that

$$\mathrm{Centr}_G(M_\ell) = H,$$

and put $N' = \mathrm{Norm}_G(M_\ell)$.

The torus T_s, and hence $(M_\ell)_s$, is characteristic in H_s. Therefore N_s normalizes $(M_\ell)_s$, and $N_s = N'_s$. The proof of Exposé XI 5.9 represents

$$N'/H = \mathrm{Norm}_G(M_\ell)/\,\mathrm{Centr}_G(M_\ell)$$

by an S-group K'. Since N' is smooth and $N'_s = N_s$, it is an open subgroup of N containing H (Exposé $\mathrm{VI_B}$ 2.5). Its image over U is an open subgroup of K, isomorphic to $K'|_U$. Glue K and K' along this open subgroup to obtain L, and glue the projections $N|_U \to K$ and $N' \to K'$ to obtain $p : N \to L$. Then (L, p) represents N/H.

[12] Krull dimension.

Proof of (v). First suppose $S = \operatorname{Spec} k$. The image H' of H is a smooth subgroup of G'. The required assertion follows from:

Lemma 7.2.— *Let $u : G \to G'$ be an epimorphism of smooth connected algebraic groups over a field k, let T be a torus of G, and let T' be its image in G'. Then*

$$u(\operatorname{Centr}_G(T)) = \operatorname{Centr}_{G'}(T').$$

Put $H = \operatorname{Centr}_G(T)$, $H' = \operatorname{Centr}_{G'}(T')$, and $H'' = u(H)$. We have $H'' \subseteq H'$. After extending the base field, assume k algebraically closed. It suffices to prove that every Cartan subgroup of H' lies in H'': then H'' contains the open subset of regular points of H', so is open, and equals the connected group H'.

Let C' be a Cartan subgroup of H'. It is also a Cartan subgroup of G', since H', being the centralizer of the torus T', has the same reductive and nilpotent ranks as G'. Put

$$K = (u^{-1}(C'))^0_{\mathrm{red}}.$$

Because T' is central in H', C' contains T', and K contains T. Let C be a Cartan subgroup of K containing T. The torus T lies in the unique maximal torus of C, central in C (Exposé XII 6.6(c)); hence $C \subseteq H$. Any two Cartan subgroups of G are conjugate, and the image of a Cartan subgroup of G is a Cartan subgroup of G' (Exposé XII 6.6). It follows that C is a Cartan subgroup of G, and its image is a Cartan subgroup of G' contained in C'. Thus $u(C) = C'$, proving $C' \subseteq H''$.

This proves (v) over a field. In general, finite presentation and the usual reductions permit us to assume S affine noetherian, then local, and finally the spectrum of a complete noetherian local ring A, by fpqc descent of subpreschemes of G'.

Use the notation of (ii). Let T_s be the maximal central torus of H_s, and let $M_\ell \subset H$ lift ${}_\ell T_s$, with $H = \operatorname{Centr}_G(M_\ell)$. Let $T'_s = u_s(T_s)$. Since G' is separated, $u(M_\ell)$ is represented by a subgroup M'_ℓ of multiplicative type (Exposé IX 6.8). Put

$$H' = \operatorname{Centr}_{G'}(M'_\ell),$$

which is smooth. For every power ℓ' of q, a sufficiently large ℓ makes $(M'_\ell)_s$ contain ${}_{\ell'}T'_s$. We may therefore choose ℓ such that

$$H'_s = \operatorname{Centr}_{G'_s}(T'_s) = u_s(H_s),$$

the last equality being Lemma 7.2.

The restriction $v : H \to H'$ of u is flat. Indeed, H, H' are flat over S, $H_s \to H'_s$ is flat, and hence v is flat near H_s (EGA IV 11.3.10 and 11.3.1). It is therefore flat on an open subgroup of H (Exposé $\mathrm{VI_B}$ 2.2), and thus everywhere because the fibres of H are connected. The set-theoretic image of H is consequently an open subset of H', necessarily $(H')^0$, and with its induced structure represents the fpqc image sheaf $u(H)$. Lemma 7.2 shows that it belongs to $\mathcal{CT}_{G'}(S)$.

If $H = \operatorname{Centr}_G(T)$, then $T' = u(T)$ is a sub-torus of G' (Exposé IX 6.8). The image $u(H)$ is contained in $\operatorname{Centr}_{G'}(T')$, agrees with it fibrewise by Lemma 7.2, and is smooth. Hence

$$u(H) = \operatorname{Centr}_{G'}(T').$$

Proof of (vi). Since $\mathcal{CT}_G, \mathcal{CT}_{G'}$ are smooth over S, it is enough to prove faithful flatness on geometric fibres. Thus take $S = \operatorname{Spec} k$, with k algebraically closed. Let $H' \in \mathcal{CT}_{G'}(k)$, let T' be its maximal central torus, choose a sub-torus $T \subset G$ mapping onto T', and put

$$H = \operatorname{Centr}_G(T), \qquad N = \operatorname{Norm}_G(H), \qquad N' = \operatorname{Norm}_{G'}(H').$$

By Lemma 7.2, $H' = u(H)$. Part (iv) identifies G/N and G'/N' with open neighbourhoods of H and H' in the two $\mathcal{CT}$-spaces. On these neighbourhoods, $\widetilde{u}$ is

$$w : G/N \longrightarrow G'/N',$$

obtained from u by passage to quotients. This is an epimorphism of homogeneous spaces under G, hence faithfully flat. Therefore $\widetilde{u}$ is flat and its image is an open subset of $\mathcal{CT}_{G'}$ containing all k-points. Since $\mathcal{CT}_{G'}$ is of finite type, $\widetilde{u}$ is surjective and hence faithfully flat. If $\ker u$ is central, the remaining assertions are Exposé XII 7.12.

The next theorem generalizes Exposé XII 7.1.

Theorem 7.3.— *Let S be a prescheme and let G be an S-group prescheme of finite presentation, smooth over S, with connected fibres. Define the functor*

$$\mathcal{C}(S') = \{\textit{Cartan subgroups of } G_{S'}\}.$$

i) *The following are equivalent:*

   a) *$\mathcal{C}$ is representable;*

   b) *$\mathcal{C}$ is represented by an S-prescheme smooth, quasi-projective, and of finite presentation over S, with affine fibres;*

   c) *G admits a Cartan subgroup locally for the étale topology;*

   d) *G admits a Cartan subgroup locally for the faithfully flat topology;*

   e) *the nilpotent rank of the fibres of G is locally constant on S.*

ii) *If these conditions hold, any two Cartan subgroups of G are locally conjugate for the étale topology. The regular locus of the fibres of G (Exposé XIII 2.7) is an open subprescheme G_{reg}, of finite presentation over S, and every section of G_{reg} over S is contained in a unique Cartan subgroup of G.*

iii) *Let G' be an S-group prescheme of finite presentation and $u : G \to G'$ faithfully flat. Then, for every Cartan subgroup C of G, its image $C' = u(C)$ is represented by a Cartan subgroup of G', and $C \to C'$ is faithfully flat.*

iv) *Under the conditions of (i) and (iii), the morphism*

$$\widetilde{u} : \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}, \qquad C \longmapsto u(C),$$

   is faithfully flat. If $\ker u$ is central, it is an isomorphism.

v) *Further information on transporters and relations with maximal tori is given in Proposition 7.1 and Exposé XII 7.1.*

Proof. For (i) we prove

$$\text{(b)} \Rightarrow \text{(c)} \Rightarrow \text{(d)} \Rightarrow \text{(e)} \Rightarrow \text{(b)} \Rightarrow \text{(a)} \Rightarrow \text{(d)}.$$

Assume (b), and let $s \in S$. Since $\mathcal{C}_s$ is smooth over $\kappa(s)$, it has points with residue field a finite separable extension of $\kappa(s)$. Exposé XI 1.10 gives an open neighbourhood U of s and an étale surjective morphism $U' \to U$ such that $G_{U'}$ admits a Cartan subgroup. Thus (b)⇒(c), while (c)⇒(d) is clear.

Assume (d), and let $s \in S$. There is a flat S-prescheme S', whose image contains s, such that $G_{S'}$ admits a Cartan subgroup. For $s' \in S'$ above s, the nilpotent rank of the

fibres of $G_{S'}$ is constant on $\operatorname{Spec}\mathcal{O}_{S',s'}$. Hence the nilpotent rank of the fibres of G is constant on $\operatorname{Spec}\mathcal{O}_{S,s}$, which is the image of that local spectrum (EGA IV 2.3.4(ii)). Let the value be r. By 6.3 bis, the set E_r where the nilpotent rank equals r is ind-constructible; it contains a neighbourhood of s by EGA IV 1.10.1. Thus (d)⇒(e).

Assume (e). Locally on S, the nilpotent rank is a constant r. For every S', the Cartan subgroups of $G_{S'}$ are exactly the smooth subgroup preschemes of relative dimension r whose geometric fibres are centralizers of tori. By Proposition 7.1, $\mathcal{C}$ is the open-and-closed subprescheme of $\mathcal{CT}$ representing the relative-dimension-r subfunctor. It has all the properties in (b); affineness of its fibres is Exposé XII 7.1(d). This proves (e)⇒(b), and (b)⇒(a) is immediate.

Finally assume (a). By 6.2(iii), $\mathcal{C}$ commutes with filtered inductive limits of rings, so its representing prescheme is locally of finite presentation (EGA IV 8.14.2). To prove it smooth, it is enough to show that, when S is affine and $S_0 \subseteq S$ is defined by a nilpotent ideal, every Cartan subgroup C_0 of G_0 lifts to a Cartan subgroup of G. Existence of C_0 implies (e) on S_0, hence on S, which has the same underlying space. Since (e)⇒(b), $\mathcal{C}$ is smooth. It is also surjective over S, so (a)⇒(d).

For (ii), if C, C' are Cartan subgroups, then $\operatorname{Transp}_G(C, C')$ is represented by a smooth S-prescheme (Proposition 7.1(iii)) with nonempty fibres (Exposé XII 6.6(a),(c)). Hensel's lemma (Exposé XI 1.10) gives local conjugacy for the étale topology. The assertions about regular points follow from Exposé XIII 3.1 and 3.2.

For (iii), Proposition 7.1(v) represents $u(C)$ by a smooth subgroup C' of G'. Its fibres are Cartan subgroups of the fibres of G' (Exposé XII 6.6(d)), so C' is a Cartan subgroup.

For (iv), faithful flatness is proved as in Proposition 7.1(vi). If $\ker u$ is central and C' is a Cartan subgroup of G', then $C = u^{-1}(C')$ is smooth with connected fibres (Proposition 7.1(vi)), and its fibres are Cartan subgroups (Exposé XII 6.6(f)). Thus C is a Cartan subgroup, and Proposition 7.1(vi) completes the proof.

8. Representability criterion for the functor of sub-tori of a smooth group

8.0. In this section, if S is a prescheme and G an S-group prescheme, $\mathcal{T}_G$ (or simply $\mathcal{T}$, if there is no ambiguity) denotes the S-functor

$$(\mathrm{Sch}\,/S)^\circ \longrightarrow \mathrm{Ens}$$

such that, for every S-prescheme S',

$$\mathcal{T}(S') = \{\text{sub-tori of } G_{S'}\}.$$

The functor $\mathcal{T}_C$ of central sub-tori of G is defined similarly.

Proposition 8.1.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type. Then the following conditions are equivalent:*

i) *The functor $\mathcal{T}_G$ "commutes with adic limits of local artinian rings": for every S-prescheme $S' = \operatorname{Spec}(A)$, where A is a complete noetherian local ring for the topology defined by its maximal ideal $\mathfrak{m}$, the canonical map*

$$\mathcal{T}(S') \longrightarrow \varprojlim_n \mathcal{T}(S'_n), \qquad S'_n = \operatorname{Spec}(A/\mathfrak{m}^n),$$

is bijective.

ii) *The assertion in* i), *restricted to the case in which A is a complete discrete valuation ring with algebraically closed residue field.*

iii) *The assertion in* ii), *restricted to the subfunctor $\mathcal{T}^{(1)}$ of $\mathcal{T}$ formed by sub-tori of G of relative dimension* 1.

i bis) *For every S-prescheme S' as in* i) *and every S'-torus T, the canonical map*

$$\mathrm{Hom}_{S'\text{-}gr}(T, G_{S'}) \longrightarrow \varprojlim_n \mathrm{Hom}_{S'_n\text{-}gr}(T_{S'_n}, G_{S'_n})$$

is bijective.

ii bis) *The assertion in* i bis), *restricted to the case in which A is a complete discrete valuation ring with algebraically closed residue field.*

iii bis) *The assertion in* ii bis), *restricted to $T = \mathbb{G}_{m,S'}$.*

Remark 8.2. There is an analogous proposition in which one restricts to central sub-tori of G and to central homomorphisms from a torus into G.

Proof. We shall use the following lemma.

Lemma 8.3.— *Let S be a prescheme, G an S-group prescheme, T an S-torus, and $u : T \to G$ an S-group morphism. Suppose, moreover, either that G is of finite presentation over S, or that S is locally noetherian and G is locally of finite type. Then:*

a) $\ker u$ *is a subgroup of multiplicative type of T;*

b) *the quotient $T' = T/\ker u$ is a torus;*

c) *the canonical monomorphism $T' \to G$ induced by u after passage to the quotient is an immersion.*

When G is separated over S, this lemma follows from Exposé IX 6.8. In the general case, the usual reduction allows us to suppose that S is noetherian. We first show that $K = \ker u$ is flat. We may suppose that S is the spectrum of a local artinian ring (EGA 0_{III}, 10.2.6); then G is separated (Exposé $\mathrm{VI_B}$ 5.2), so K is of multiplicative type (Exposé IX 6.8), and hence is flat over S. We next show that $\ker u$ is closed in T. This reduces us to the case in which S is the spectrum of a discrete valuation ring (EGA II 7.2.1). Since T is flat with connected fibres, u factors through the identity component of the schematic closure in G of the generic fibre of G. We may thus suppose that G is flat with connected fibres; it is then separated (Exposé $\mathrm{VI_B}$ 5.2), and K is closed. It follows from Exposé X 4.8(b) that K is a subgroup of multiplicative type of T. The quotient $T' = T/K$ is therefore representable and is a group of multiplicative type (Exposé IX 2.7) whose fibres are tori; hence T' is a torus. Finally, the monomorphism $T' \to G$ is an immersion by Exposé VIII 7.9.

Proof of Proposition 8.1. i)⇒i bis). Put $T_n = T_{S'_n}$, $G_n = G_{S'_n}$, and let

$$(u_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathrm{Hom}_{S'_n\text{-gr}}(T_n, G_n).$$

For every n, $u_n(T_n)$ is a sub-torus T'_n of G_n (Lemma 8.3). By hypothesis, there is a unique sub-torus T' of G lifting T'_n for every n. Since T' has affine fibres, the assertion follows from 4.4.

i bis)⇒i). Let $(T_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathcal{T}(S'_n)$. By Exposé X 4.6 there are an S'-torus T^0 and an S'_0-isomorphism

$$u_0 : T^0_0 \xrightarrow{\sim} T_0.$$

Since T_n is smooth over S'_n, u_0 lifts to an S'_n-morphism $u_n : T^0_n \to T_n$ (Exposé IX 3.6), and the family $(u_n)_{n\in\mathbb{N}}$ may be chosen coherent. It therefore comes from a morphism $u : T^0 \to G$. By Lemma 8.3, the image of T^0 under u is a sub-torus of G, and it lifts T_n for every n.

The implications

$$\text{i)} \Rightarrow \text{ii)} \Rightarrow \text{iii)} \quad \text{and} \quad \text{i bis)} \Rightarrow \text{ii bis)} \Rightarrow \text{iii bis)}$$

are immediate. The implication iii) ⇒ iii bis) is proved in the same way as i) ⇒ i bis). It remains to prove

$$\text{iii bis)} \Rightarrow \text{ii bis)} \Rightarrow \text{i bis)}.$$

ii bis)⇒i bis). In the terminology of 4.3, assertion i bis) holds if and only if every element

$$(u_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathrm{Hom}_{S'_n\text{-gr}}(T_n, G_n)$$

is admissible. For every point t of S' distinct from its closed point s, there is an S-scheme S'', the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point maps to t and whose closed point maps to s (EGA II 7.1.9). This gives at once a valuative criterion for a family of morphisms to be admissible, and proves the implication.

iii bis)⇒ii bis). Let T be an S-torus, where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. The torus T is split (Exposé X 4.6), say

$$T \simeq \mathbb{G}^r_{m,S}.$$

Let

$$(u_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}\big((\mathbb{G}_m)^r_{S_n}, G_n\big).$$

By hypothesis, the restriction of u_n to each factor of $\mathbb{G}^r_{m,S}$ comes from a group morphism $\mathbb{G}_{m,S} \to G$. Their product, followed by the composition law $G^r \to G$, gives a morphism

$$v : \mathbb{G}^r_{m,S} \longrightarrow G.$$

Clearly $u_n = v_n$ for every n. It remains to prove that v is a group morphism. Equivalently, two evident morphisms

$$f, g : X = \mathbb{G}^r_{m,S} \times_S \mathbb{G}^r_{m,S} \longrightarrow G$$

must coincide. Let Z be their coincidence subscheme. Since $\mathbb{G}^r_{m,S}$ has connected fibres and its image under v contains the unit section, the argument of Lemma 8.3 shows that v factors through the identity component of G. We may therefore suppose that G is separated (Exposé $\mathrm{VI_B}$ 5.2), so Z is closed. On the other hand, v_n is a group morphism, hence $f_n = g_n$ for every n. Thus Z contains a neighbourhood of the closed fibre of X (EGA I 10.9.4), which is schematically dense in X, since X has smooth irreducible fibres (Exposé IX 4.6). Consequently $Z = X$, and $f = g$.

Definition 8.4.— *Let S be a locally noetherian prescheme and G an S-group prescheme of finite type. Let $\mathcal{S}$ be the set of S-schemes S' that are spectra of complete discrete valuation rings with algebraically closed residue field. Write s and t for the closed and generic*

points of S', and require that $(G_t)^0_{\mathrm{red}}$ be smooth and admit a Chevalley decomposition: an extension of an abelian variety by a smooth connected linear algebraic subgroup L_t. This decomposition is then unique. For $S' \in \mathcal{S}$, let G', respectively L, denote the schematic closure in $G_{S'}$ of G_t, respectively of L_t.

We say that "the abelian part of G does not degenerate into a toric part", or briefly that G has property AT*, if, for every $S' \in \mathcal{S}$, L_s has the same reductive rank as G'_s. Intuitively, suppose that the quotient $A = G'/L$ is representable. Then A is a flat group prescheme and $(A_t)^0_{\mathrm{red}}$ is an abelian variety; property* AT *says that A_s has reductive rank zero, so that $(A_s)^0_{\mathrm{red}}$ is an extension of an abelian variety by a unipotent group.*

Similarly, if G has connected fibres, we say that G has property ATC *if, for every $S' \in \mathcal{S}$, the schematic closure Z of the centre Z_t of G_t has property* AT*.*

These technical definitions are justified by the following proposition.

Proposition 8.5.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type.*

a) *The functor $\mathcal{T}$ of sub-tori of G commutes with adic limits of local artinian rings (cf. Proposition 8.1) if and only if G has property* AT*.*

b) *If G has connected fibres, the functor $\mathcal{T}_C$ of central sub-tori of G commutes with adic limits of local artinian rings if and only if G has property* ATC*.*

We need the following technical lemma.

Lemma 8.6.— *Let S be the spectrum of a discrete valuation ring, with closed point s, let G be an S-group prescheme flat and of finite type, and let T_s be a sub-torus of G_s. There are an S-scheme S', the spectrum of a discrete valuation ring faithfully flat over S, with closed point s', and a subgroup scheme C of $G_{S'}$, flat over S', commutative, and with connected fibres, such that $C_{s'}$ contains $T_s \times_s s'$.*

After replacing S by the spectrum S' of a discrete valuation ring faithfully flat over it, we may suppose that $T_s = \mathbb{G}^r_{m,s}$ and that the transcendence degree of $\kappa(s)$ over its prime field is at least r (EGA 0_{III}, 10.3.1, and EGA II 7.1.9). There is then an element $x \in T_s(s)$ such that every algebraic subgroup of G_s containing x contains T_s (cf. Exposé XIII, proof of 2.1, ii) $\Rightarrow$ vii)). Since G is flat over S, a quasi-section passes through x (EGA IV 14.5.8). After another such replacement of S, we may therefore suppose that x lifts to a section $\widetilde{x}$ of G over S. Let C_t be the commutative algebraic subgroup of G_t generated by $\widetilde{x}_t$ (Exposé $\mathrm{VI_B}$ 7), and let C be its schematic closure in G. Clearly C_s contains x, hence contains T_s. The identity component of C is therefore a flat commutative group scheme with connected fibres having the required property.

Proof of Proposition 8.5(a). Suppose first that $\mathcal{T}$ commutes with adic limits of artinian rings. Let $S' \in \mathcal{S}$, and let T_s be a maximal torus of G'_s. We must show that $T_s \subset L_s$. The formation of L and G' plainly commutes with faithfully flat extensions $S'' \to S'$ of discrete valuation rings. By Lemma 8.6, we may therefore replace S' so that there is a flat commutative subgroup scheme C of $G'_{S'}$ for which C_s contains T_s. The torus T_s is central in C_s, and hence lifts infinitesimally to a central sub-torus (Corollary 2.3). By the hypothesis on G, T_s lifts to a sub-torus T of G. Clearly T_t lies in the linear part L_t of G_t, and therefore $T \subset L$.

Conversely, suppose that G has property AT. We prove condition 8.1 iii bis). Let $S = \mathrm{Spec}(A)$, where A is a complete discrete valuation ring with algebraically closed residue field, let $\mathfrak{m}$ be its maximal ideal, put $S_n = \mathrm{Spec}(A/\mathfrak{m}^n)$, and let

$$u_n : (\mathbb{G}_{m,S})_n \longrightarrow G_n = G \times_S S_n, \qquad n \in \mathbb{N},$$

be a coherent system of group morphisms. Let q be a prime invertible on S. As ℓ ranges over the powers of q, there is a unique S-group morphism

$$ {}_\ell u : {}_\ell \mathbb{G}_{m,S} \longrightarrow G $$

lifting the restriction of u_n to ${}_\ell(\mathbb{G}_{m,S})_n$ for every n (Proposition 1.6(a)). Consequently, if an S-group morphism $u : \mathbb{G}_{m,S} \to G$ lifting every u_n exists, its restriction to ${}_\ell\mathbb{G}_{m,S}$ is uniquely determined. The density theorem (Exposé IX 4.8(a)) proves both the uniqueness of u and that, for its existence, we may make a faithfully flat extension of the base.

Now $(G_t)^0_{\mathrm{red}}$ admits a Chevalley decomposition, already defined over a finite extension L of the fraction field K of A. After replacing S by its normalization in L, we may suppose that $S \in \mathcal{S}$. The morphism ${}_\ell u_t$ factors through $(G_t)_{\mathrm{red}}$, and ${}_\ell u$ therefore factors through the schematic closure G'' of $(G_t)_{\mathrm{red}}$ in G'. The density theorem then shows that every u_n factors through G''_n. Since G has property AT, every sub-torus of G'_s, and hence of G''_s, is contained in L_s; thus u_s factors through L_s. In fact every u_n factors through L_n. Indeed, L_t is invariant in G''_t, so L is invariant in G''; moreover L is flat over S, and the quotient

$$ H_n = G''_n / L_n $$

is representable for every n (Exposé $\mathrm{VI_A}$, §4). The image of $(\mathbb{G}_{m,S})_n$ in H_n is a sub-torus (Exposé IX 6.8) whose closed fibre is zero. It is therefore zero, and u_n factors through L_n. Since L_t is affine, Proposition 4.3 now shows that the family (u_n) is admissible.

The proof of Proposition 8.5(b) is entirely analogous, in view of Remark 8.2 and Exposé IX 5.6(a).

Proposition 8.7.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type.*

i) *If G has connected fibres and has property* AT*, then it has property* ATC*.*

ii) *If G has property* AT*, every subgroup prescheme of G has property* AT*.*

iii) *If G is flat over S and the abelian rank of its fibres is a locally constant function on S, then G has property* AT*.*

Assertion (i) follows at once from Proposition 8.5 and Exposé IX 5.6(a).

For (ii), let H be a subgroup prescheme of G. To prove that H has property AT, we may suppose that S is the spectrum of a complete discrete valuation ring. We must show that every coherent family of S_n-group morphisms

$$ u_n : (\mathbb{G}_{m,S})_n \longrightarrow H_n $$

is admissible (Proposition 8.5 and 8.1 iii bis)). By hypothesis, G has property AT, so there is an S-group morphism

$$ u : \mathbb{G}_{m,S} \longrightarrow G $$

lifting every u_n. As in the proof of Proposition 8.5, the restriction of u to ${}_\ell\mathbb{G}_{m,S}$, with ℓ ranging over powers of a prime q invertible on S, factors through H. By density, u_t factors through H_t on the generic fibre. Since $\mathbb{G}_{m,S}$ is reduced, u itself factors through H.

For (iii), let $S' \in \mathcal{S}$, with the notation of Definition 8.4. The group L is flat over S, and its generic fibre L_t is affine. Under these conditions L_s is necessarily affine (Exposé XVII, Appendix III 2). Since G and L are flat, $G = G'$, and the dimensions of the fibres of G and L are constant on S (Exposé $\mathrm{VI_B}$ 4). Consequently

$$ \mathrm{rk_{ab}}\, G_s \leq \dim G_s - \dim L_s = \dim G_t - \dim L_t = \mathrm{rk_{ab}}\, G_t. $$

By hypothesis these inequalities are equalities. Thus $(G_s/L_s)^0_{\mathrm{red}}$ is an abelian variety and has reductive rank zero. It follows that L_s has the same reductive rank as G_s.

We can now prove the principal theorems of this section.

Theorem 8.8.— *Let G be a group prescheme of finite type over a locally noetherian prescheme S. Suppose that G is flat over S and has connected fibres.*

a) *The functor $\mathcal{T}_C$ of central sub-tori of G is representable if and only if G has property* ATC. *In that case, $\mathcal{T}_C$ is represented by an S-prescheme étale and separated over S.*

b) *Under the hypotheses of* (a), *for every S-torus T, the functor* $\mathrm{Hom}^{\mathrm{cent}}_{S\text{-}gr}(T,G)$ *of central homomorphisms from T into G is represented by an S-prescheme étale and separated over S.*

Theorem 8.9.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type, smooth over S.*

a) *The functor $\mathcal{T}$ of sub-tori of G is representable if and only if G has property* AT. *In that case, $\mathcal{T}$ is represented by an S-prescheme smooth and separated over S. More precisely, its structural morphism admits a canonical factorization*

$$\mathcal{T} \xrightarrow{u} Y \xrightarrow{v} S,$$

where v is smooth and quasi-projective (hence of finite type) and u is étale and separated.

b) *Under the hypotheses of* (a), *for every S-torus T, the functor* $\mathrm{Hom}_{S\text{-}gr}(T,G)$ *of homomorphisms from T into G is represented by an S-prescheme smooth and separated over S.*

Proof of Theorem 8.8(a). If the functor $\mathcal{T}_C$ is representable, it commutes with adic limits of artinian rings. Consequently, by 8.2 and Proposition 8.5(b), G has property ATC. To prove the converse, we shall use the following result, which will be proved in EGA VI and may also be found in Murre's exposé, *Séminaire Bourbaki*, May 1965, no. 294, Theorem 1, Corollary 2.

Lemma 8.10.— *Let S be a locally noetherian prescheme and let F be a contravariant functor from* Sch $/S$ *to the category of sets. In order that F be representable by an S-prescheme étale and separated over S, it is necessary and sufficient that F satisfy the following five properties:*

i) *F is a sheaf for the fpqc topology (Exposé IV).*

ii) *F commutes with filtered inductive limits of rings (Exposé XI 3.2).*

iii) *F commutes with adic limits of local artinian rings (8.1*(i)*).*

iv) *F satisfies the valuative criterion for separatedness: for every S-prescheme S' that is the spectrum of a discrete valuation ring, with generic point t, the canonical map*

$$F(S') \longrightarrow F(t)$$

is injective.

v) *F is infinitesimally étale (Exposé XI 1.8).*

We show that the functor $\mathcal{T}_C$ of Theorem 8.8 satisfies the five conditions of Lemma 8.10.

i) The functor $\mathcal{T}$, and likewise $\mathcal{T}_C$, is a sheaf for the fpqc topology whenever G is of finite presentation over S. Indeed, every monomorphism from a torus into G is then an immersion (Exposé VIII 7.9), and the assertion follows from fpqc descent for subpreschemes.

ii) Corollary 6.3 proves that $\mathcal{T}$ commutes with filtered inductive limits of rings when G is of finite presentation over S; the same property for $\mathcal{T}_C$ follows immediately.

iii) By Proposition 8.5, condition (iii) is precisely equivalent to property ATC.

iv) If S is the spectrum of a discrete valuation ring, two sub-tori of G with the same generic fibre coincide. More precisely, each is the identity component of the schematic closure in G of that common generic fibre.

v) This follows from Proposition 2.3 because G is flat over S.

Proof of Theorem 8.8(b). Proceeding as in Exposé XI 4.2, it is enough to prove that the product group $T \times_S G$ again has property ATC, which is immediate from the definition.

Proof of Theorem 8.9(a). After replacing G, if necessary, by its identity component (Exposé $\mathrm{VI_B}$ 3.10), we may suppose that G has connected fibres. If T is a sub-torus of G, its centralizer $\mathrm{Centr}_G(T)$ is represented (Exposé XI 6.11) by a subgroup prescheme of G, smooth over S (Exposé XI 2.4), with connected fibres (Exposé XII 6.6(b)). It is therefore an element of $\mathcal{CT}(S)$, where $\mathcal{CT}$ is the functor defined in 7.1(i). The rule

$$T \longmapsto \mathrm{Centr}_G(T)$$

clearly defines an S-morphism

$$u : \mathcal{T} \longrightarrow \mathcal{CT}.$$

Since $\mathcal{CT}$ is represented by an S-prescheme smooth and quasi-projective over S (7.1(i)), it remains only to prove that u is representable by a separated étale morphism.

After a suitable base change $S' \to S$, with $S' = \mathcal{CT}$ and hence S' locally noetherian, we are reduced to the following problem. Let S be a locally noetherian prescheme, let G be an S-group prescheme smooth and of finite type over S, with connected fibres, and let H be a subgroup of G, smooth over S, with connected fibres. Consider the subfunctor X of $\mathcal{T}_{C,H}$ such that, for every S-prescheme S', $X(S')$ is the set of central sub-tori T of $H_{S'}$ for which

$$\mathrm{Centr}_{G_{S'}}(T) = H_{S'}.$$

We shall show that X is represented by an S-prescheme étale and separated over S.

Indeed, G has property AT by hypothesis, and so does H by Proposition 8.7(ii). Since H has connected fibres, it also has property ATC by Proposition 8.7(i). Moreover, H is smooth over S, hence flat. Theorem 8.8(a) therefore represents $\mathcal{T}_{C,H}$ by an S-prescheme étale and separated over S. It is enough to show that the canonical monomorphism

$$X \longrightarrow \mathcal{T}_{C,H}$$

is represented by an open immersion.

Put $S' = \mathcal{T}_{C,H}$, and let K be the centralizer in $G_{S'}$ of the universal central torus of $H_{S'}$. The group K is a smooth group scheme over S', with connected fibres, and contains $H_{S'}$. By definition,

$$X = \prod_{K/S'} H_{S'}/K,$$

which is represented by the open-and-closed subprescheme of S' over which $H_{S'}$ and K have the same relative dimension.

The proof of Theorem 8.9(b) is analogous to that of Theorem 8.8(b).

Corollary 8.11.— *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S. If the abelian rank of the fibres of G is a locally constant function on S—in particular, if the fibres of G are affine—then the functor of sub-tori of G is represented by an S-prescheme smooth and separated over S. The same holds for the functor* $\mathrm{Hom}_{S\text{-}gr}(T, G)$*, for every S-torus T.*

The assertion is local on S, so we may suppose that S is affine and that the abelian rank of the fibres of G is constant. There are then a noetherian affine scheme S_0 and an S_0-group prescheme G_0, smooth and of finite type over S_0, such that

$$G \simeq G_0 \times_{S_0} S$$

as S-group preschemes (Exposé $\mathrm{VI_B}$, §10). The abelian rank of the fibres of an S-group prescheme of finite presentation is an ind-constructible function (6.3 bis). A standard argument (EGA IV 8) therefore allows us, in the present situation, to choose G_0 so that the abelian rank of its fibres is constant on S_0. Proposition 8.7(iii) gives property AT for G_0, and Theorem 8.9 then represents its functor of sub-tori by an S_0-prescheme smooth and separated over S_0. Base change gives the assertion for G. The argument for $\mathrm{Hom}_{S\text{-gr}}(T, G)$ is analogous.

Generalization of Theorem 8.9. The functor $\mathcal{T}$ of sub-tori of a smooth group G need not be representable. We now give sufficient conditions for a subfunctor of $\mathcal{T}$ to be representable.

Proposition 8.12.— *Let S be a locally noetherian prescheme, let G be an S-group prescheme smooth over S with connected fibres, and let F be an S-subfunctor of the functor $\mathcal{T}$ of sub-tori of G. Suppose that:*

i) *F is a sheaf for the fpqc topology (Exposé IV);*

ii) *F commutes with filtered inductive limits of rings (Exposé XI 3.2);*

iii) *F commutes with adic limits of local artinian rings (Exposé XI 3.3);*

iv) *F is infinitesimally smooth over S (Exposé XI 1.8);*

v) *F is stable under inner automorphisms of G: for every S-prescheme S',*

$$T \in F(S'), \quad g \in G(S') \quad \Longrightarrow \quad \mathrm{int}(g)T \in F(S').$$

Then F is represented by an S-prescheme smooth and separated over S.

Proposition 8.12 bis.— *Let S and G be as in Proposition 8.12, let T be an S-torus, and let F be a subfunctor of* $\mathrm{Hom}_{S\text{-}gr}(T, G)$*. Suppose that F satisfies conditions* (i)–(iv)

of Proposition 8.12 and is stable under inner automorphisms of G: for every S-prescheme S',

$$u \in F(S'), \quad g \in G(S') \quad \Longrightarrow \quad \mathrm{int}(g)u \in F(S').$$

Then F is represented by an S-prescheme smooth and separated over S.

Proof of Proposition 8.12. The proof of Proposition 8.12 bis is entirely analogous and is left to the reader. Let

$$u : F \longrightarrow \mathcal{CT}$$

be the S-morphism which assigns to every $T \in F(S')$ the element $\mathrm{Centr}_{G_{S'}}(T)$ of $\mathcal{CT}(S')$ (7.1(i)). Since $\mathcal{CT}$ is represented by an S-prescheme smooth and quasi-projective over S, we are reduced to proving that u is representable.

After the base change $\mathcal{CT} \to S$, the problem is the following. Given S and G as above and a smooth subgroup H of G with connected fibres, represent the functor F_H defined by

$$F_H(S') = \left\{ T \in F(S') \;\middle|\; H_{S'} = \mathrm{Centr}_{G_{S'}}(T) \right\}.$$

We show that F_H is étale and separated over S by verifying the five conditions of Lemma 8.10.

The first three conditions follow directly from Proposition 8.12(i)–(iii). We have already observed that $\mathcal{T}_{C,H}$ is separated and infinitesimally étale. Thus F_H, being a subfunctor of $\mathcal{T}_{C,H}$, is separated and infinitesimally unramified (Exposé XI 1.8). It remains to prove that F_H is infinitesimally smooth.

Let S be the spectrum of a local artinian ring, let $S_0 \subseteq S$ be defined by a nilpotent ideal, and let $T_0 \in F_H(S_0)$. By Proposition 8.12(iv), T_0 lifts to an element $T' \in F(S)$. Since H is smooth, T_0 also lifts to a sub-torus T'' of H_S (Exposé IX 3.6 bis). This sub-torus is conjugate to T' by an element $g \in G(S)$ (*loc. cit.*), and hence $T'' \in F(S)$ by Proposition 8.12(v). The centralizer $\mathrm{Centr}_{G_S}(T'')$ is smooth over S and agrees with H_{S_0} over S_0. Therefore T'' lies in the centre of H_S (Exposé IX 5.6(a)), and its centralizer in G equals H_S. Thus $T'' \in F_H(S)$, and we take $T = T''$.

Corollary 8.13.— *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S, with connected fibres. Let $\mathcal{T}_D$ be the subfunctor of $\mathcal{T}$ for which $\mathcal{T}_D(S')$ is the set of sub-tori T of $G_{S'}$ such that, for every $s' \in S'$, $T_{s'}$ is contained in the derived subgroup of $G_{s'}$ (Exposé VI$_B$ 7). Then $\mathcal{T}_D$ is represented by an S-prescheme smooth and separated over S.*

Corollary 8.13 bis.— *Let S and G be as in Corollary 8.13, and let T be an S-torus. Let*

$$\mathrm{Hom}^{\mathrm{der}}_{S\text{-}gr}(T, G)$$

be the subfunctor of $\mathrm{Hom}_{S\text{-}gr}(T, G)$ *whose S'-points are the S'-morphisms $u : T_{S'} \to G_{S'}$ such that, for every $s' \in S'$, the morphism $u_{s'}$ factors through the derived group of $G_{s'}$. Then this functor is represented by an S-prescheme smooth and separated over S.*

The two corollaries have analogous proofs; we prove Corollary 8.13 bis. By the usual procedure, we reduce to the case in which S is noetherian. We verify the five conditions of Proposition 8.12 bis.

Conditions (i) and (iv) follow immediately from the corresponding properties of the functor

$$\mathrm{Hom}_{S\text{-gr}}(T, G).$$

Condition (v) holds because the derived group of an algebraic group is invariant (Exposé $\mathrm{VI_B}$, §7).

For condition (ii), a standard reduction (EGA IV 8) reduces us to the following assertion. Let S be an integral noetherian scheme with generic point η, and let $u : T \to G$ be an S-group morphism whose generic fibre factors through the derived group of G_η. Then there is a neighbourhood U of η such that, for every $s \in U$, u_s factors through the derived group of G_s. This is an immediate consequence of Exposé $\mathrm{VI_B}$ 10.12.

For condition (iii), retain the notation of Proposition 4.3. We must show that an element

$$(u_m)_{m\in\mathbb{N}} \in \varprojlim_m \mathrm{Hom}^{\mathrm{der}}_{S_m\text{-gr}}(T_m, G_m)$$

is admissible in the sense of Proposition 4.3 and comes from an element of $\mathrm{Hom}^{\mathrm{der}}_{S\text{-gr}}(T, G)$. As in 8.1, we reduce immediately to the case in which S is the spectrum of a complete discrete valuation ring. Let t be its generic point, s its closed point, and let D be the schematic closure in G of the derived group D_t of G_t.

a) The group D_s contains the derived group of G_s. Indeed, D_t is invariant in G_t, and G is flat over S, so D is invariant in G. Moreover, the morphism

$$G \times_S G \longrightarrow G, \qquad (x, y) \longmapsto xyx^{-1}y^{-1},$$

factors through D_t on the generic fibre and therefore factors through D. Consequently G_s/D_s is commutative.

b) If $u_m \in \mathrm{Hom}^{\mathrm{der}}_{S_m\text{-gr}}(T_m, G_m)$, then u_m factors through D_m. Its closed fibre u_s factors through the derived group of G_s, hence through D_s by (a). Since D_m is flat over S_m and invariant in G_m, the quotient

$$H_m = G_m/D_m$$

is representable (Exposé $\mathrm{VI_A}$, §4). The image of T_m in H_m is a torus (Exposé IX 6.8) with trivial closed fibre, hence is trivial. Thus u_m factors through D_m.

c) The family $(u_m)_{m\in\mathbb{N}}$ is admissible and lifts to a morphism $T \to G$ belonging to $\mathrm{Hom}^{\mathrm{der}}_{S\text{-gr}}(T, G)$. By what precedes and Proposition 4.1 bis, it is enough to prove that D_t is affine. This follows from the next lemma.

Lemma 8.14.— *Let G be a smooth connected algebraic group over a field k. Then the derived group D of G is affine.*

Formation of D commutes with extension of the base field (Exposé $\mathrm{VI_B}$ 7), so we may suppose that k is algebraically closed. The group G is then an extension of an abelian variety A by a linear group L. Since A is commutative, D is necessarily contained in L, and is therefore affine.

Maximal tori.

Theorem 8.15.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme smooth and of finite type over S. The following conditions are equivalent:*

i) *The S-functor $\mathcal{T}_M$, whose S'-points are the maximal tori of $G_{S'}$ (Exposé XII 1.3), is representable.*

ii) *The functor $\mathcal{T}_M$ is represented by an S-prescheme smooth and quasi-projective over S, with affine fibres.*

iii) *The group G admits a maximal torus locally for the étale topology.*

iv) *The group G admits a maximal torus locally for the faithfully flat topology.*

v) *The group G has property* AT*, and the reductive rank of its fibres is a locally constant function on S.*

Proof. (ii)⇒(i) is clear.

(i)⇒(iii). Since G is of finite presentation over S, Corollary 6.3 shows that $\mathcal{T}_M$ commutes with filtered inductive limits of rings. If representable, it is therefore locally of finite presentation (EGA IV 8.14). Moreover, $\mathcal{T}_M$ is formally smooth (Exposé XI 2.1 bis). Hence its representing prescheme is smooth over S, and (iii) follows from Hensel's lemma (Exposé XI 1.10).

(iii)⇒(iv) is clear.

(iv)⇒(v). Let $s \in S$. There is an S-prescheme S', flat over S and with image containing s, such that $G_{S'}$ has a maximal torus T'. If $s' \in S'$ lies over s, the reductive rank of the fibres of $G_{S'}$ is constant on $\operatorname{Spec} \mathcal{O}_{S',s'}$. Thus the reductive rank of the fibres of G is constant on its image $\operatorname{Spec} \mathcal{O}_{S,s}$ (EGA IV 2.3.4(ii)). Since this rank is a locally constructible function on S (6.3 bis), it is constant on a neighbourhood of s (EGA IV 1.10.1).

To prove property AT, let S_1 be an S-scheme which is the spectrum of a discrete valuation ring, whose closed point s_1 maps to s. The prescheme

$$S'_1 = S_1 \times_S S'$$

is faithfully flat over S_1, and $G_{S'_1}$ has a maximal torus. Let A and A' be the rings of S_1 and S'_1, respectively. Writing A' as the filtered inductive limit of its finitely generated A-subalgebras and applying Corollary 6.3, we obtain an S_1-scheme S''_1 such that $G_{S''_1}$ has a maximal torus and $S''_1 \to S_1$ is of finite type and surjective. By EGA II 7.1.9, we may reduce to the case in which S'_1 is the spectrum of a discrete valuation ring. Then $G_{S'_1}$, and hence G_{S_1}, has property AT. Since this holds for every such S_1, G has property AT.

(v)⇒(i). By Theorem 8.9 the functor $\mathcal{T}$ of sub-tori of G is representable. If the reductive rank of G is the constant function r, then $\mathcal{T}_M$ is represented by the open-and-closed subprescheme of $\mathcal{T}$ representing the subfunctor of tori of rank r.

(iii)⇒(ii). Under condition (iii), the results of Exposé XII 7.1 identify $\mathcal{T}_M$ canonically with the functor of Cartan subgroups of G. The assertion follows from Theorem 7.3(i).

Remark 8.16. The prescheme $\mathcal{T}_M$ of maximal tori of G can in fact be shown to be affine over S,[13] generalizing Exposé XII 5.4.

Corollary 8.17.— *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S. If the abelian and reductive ranks of the fibres of G are locally constant functions on S, then G satisfies the equivalent conditions* (i)–(iv) *of Theorem 8.15.*

We may suppose that the abelian and reductive ranks of the fibres of G are constant. Proceeding as in Corollary 8.11 and using 6.3 bis, we reduce to the case in which S is noetherian. Then G has property AT by Proposition 8.7, and the conclusion follows from Theorem 8.15(v).

[13] Cf. M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes* (thesis, forthcoming). Editor's note: see *Lecture Notes in Mathematics* 119 (1970), Springer, especially IX 2.9.

Corollary 8.18.— *Let S be a prescheme and let G be an S-group prescheme smooth over S. The following conditions are equivalent:*

a) *The unipotent rank ρ_u and the abelian rank ρ_{ab} of the fibres of G are locally constant functions on S.*

b) *The unipotent rank ρ_u is locally constant, and G admits maximal tori locally for the fpqc topology.*

c) *The reductive rank ρ_r and the abelian rank ρ_{ab} of the fibres of G are locally constant functions on S.*

Remark 8.19. Under the hypotheses of Corollary 8.18, a more precise argument using lower semicontinuity of the abelian rank, announced in Exposé X 8.7, shows that if two of the three ranks $\rho_u, \rho_r, \rho_{\mathrm{ab}}$ are locally constant, then so is the third.

Proof of Corollary 8.18. After replacing G, if necessary, by its identity component, we may suppose that G is of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3).

(a)⇒(c). Let $s \in S$. Since ρ_{ab} is locally constant, Corollary 8.11 allows us, after an étale extension covering S, to suppose that G has a sub-torus T whose fibre T_s is a maximal torus of G_s. Put

$$C = \mathrm{Centr}_G(T).$$

This is a subgroup prescheme of G, smooth over S, with connected fibres. For every $t \in S$, the group C_t contains a Cartan subgroup C'_t of G_t. After shrinking S, we may suppose that ρ_u, ρ_{ab}, and the relative dimension of C over S are constant. We then have

$$\begin{aligned}\dim C_s = \dim C_t &\geq \dim C'_t \\ &\geq \rho_u(t) + \rho_{\mathrm{ab}}(t) + \dim T_t \\ &= \rho_u(s) + \rho_{\mathrm{ab}}(s) + \dim T_s \\ &= \rho_\nu(s) = \dim C_s.\end{aligned}$$

It follows that $C_t = C'_t$. Thus T is a maximal torus of G, and in particular $\rho_r(t) = \rho_r(s)$.

(c)⇒(b) follows from Corollary 8.17.

(b)⇒(a). Since G admits maximal tori locally for the fpqc topology, it also admits Cartan subgroups locally for that topology. Hence, by Exposé XII 7.3, the nilpotent rank

$$\rho_\nu = \rho_u + \rho_r + \rho_{\mathrm{ab}}$$

is locally constant. The existence of maximal tori makes ρ_r locally constant, while ρ_u is locally constant by hypothesis; therefore ρ_{ab} is locally constant.

Exposé XVI
Groups of Unipotent Rank Zero

by M. Raynaud[(*)]

1. An Immersion Criterion

1.1. Examples of Monomorphisms of Group Preschemes That Are Not Immersions

We shall construct a prescheme S, two S-group preschemes G and H, and an S-group monomorphism

$$u : G \longrightarrow H$$

that is not an immersion. The groups G and H will be of finite presentation over S, flat over S, and G will even be smooth over S (cf. Exposé VIII, the paragraph and note $(*)$ preceding 7.1; see also XVII, Appendix III, 4).

First take S to be the spectrum of a discrete valuation ring A of unequal characteristic and residue characteristic 2. Let t be the generic point of S and s the closed point.

a) Take for G the open subgroup of the constant group

$$G' = (\mathbb{Z}/2\mathbb{Z})_S$$

obtained by removing the point of the closed fiber $(\mathbb{Z}/2\mathbb{Z})_s$ distinct from the identity element, and take for H the group of multiplicative type $(\mu_2)_S$ of square roots of unity (Exposé I, 4.4.4). By the choice of S, H_t is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_t$, whereas H_s is a radicial group. There is an evident morphism $G' \to H$, defined by the section (-1) of H, hence a morphism $u : G \to H$ which is a monomorphism but not an immersion (otherwise it would necessarily be an isomorphism; cf. Exposé VIII, 7). Here Exposé VIII, 7.9 does not apply, since ${}_2G = G$ is not finite over S.

b) Let K denote the étale, nonseparated S-group prescheme obtained from the identity group by "doubling" the unique point of the closed fiber. Take for H the product group

$$(\mu_2)_S \times_S K.$$

Let a (respectively b) be the unique section of μ_2 (respectively K) over S distinct from the identity section, and let $h = (a, b)$ be the corresponding section of H. The datum of h defines an S-group morphism

$$u : G = (\mathbb{Z}/2\mathbb{Z})_S \longrightarrow H.$$

[0] xy version of December 1, 2008.

[*] Cf. the note on page 1 of Exposé XV.

It is clear that u is a monomorphism and is not an immersion. Here Exposé VIII, 7.9 does not apply, since ${}_2H = H$ is not separated over S.

c) More interesting is the following example, in which G is a smooth group with connected fibers (and even $G = (\mathbb{G}_a)_S$); cf. Exposé VIII, 7.10.

Take for S the spectrum of a discrete valuation ring A of equal characteristic 2, and let π be a uniformizer of A and $\mathfrak{m}$ the maximal ideal of A.

Consider the additive group $(\mathbb{G}_a)_S$, the product group $(\mathbb{G}_a \times \mathbb{G}_a)_S$ with ring $A[X, Y]$, and let G_1 be the closed subgroup defined by

$$X^2 + X - \pi Y = 0. \tag{1}$$

The group G_1 is smooth over S; its generic fiber is isomorphic to $(\mathbb{G}_a)_t$, and its closed fiber is isomorphic to the product of $(\mathbb{G}_a)_s$ with $(\mathbb{Z}/2\mathbb{Z})_s$. For the $(\mathbb{Z}/2\mathbb{Z})_s$-factor one may take the subgroup N_s of $(G_1)_s$ defined by

$$Y = X^2 + X = 0. \tag{2}$$

Let N' and N'' be two group subschemes of G_1, isomorphic to $(\mathbb{Z}/2\mathbb{Z})_S$, whose closed fibers coincide with N_s and whose generic fibers are distinct. Thus N' and N'' are defined by two sections of G_1 over S, with coordinates (a', b') and (a'', b''), respectively, such that

$$\begin{cases} a'^2 + a' - \pi b' = a''^2 + a'' - \pi b'' = 0, \\ a' \text{ and } a'' \equiv 1 \pmod{\mathfrak{m}}, \qquad a' \neq a'', \\ b' \text{ and } b'' \equiv 0 \pmod{\mathfrak{m}}. \end{cases} \tag{3}$$

(For example, one may take $(1, 0)$ and $(1 + \pi^2, \pi + \pi^3)$.)

The quotient groups

$$G' = G_1/N' \qquad \text{and} \qquad G'' = G_1/N''$$

are representable (Exposé V, 4.1). Let u be the canonical morphism

$$u : G_1 \longrightarrow G' \times_S G'',$$

and let H be the group subscheme of $G' \times_S G''$ equal to the schematic closure in $G' \times_S G''$ of $u_t((G_1)_t)$, so that u factors through H. The kernel of u is

$$K = N' \cap N'',$$

whose generic fiber is the identity group and whose closed fiber is equal to N_s; in particular, K is not flat over S. On the generic fiber, u_t is therefore an isomorphism from $(G_1)_t$ onto H_t. On the other hand, H_s is not smooth. Indeed, if H_s were smooth, then, since u_s is a morphism with finite kernel and $(G_1)_s$ and H_s have the same dimension, namely 1, u_s would be flat. Since G_1 and H are flat, u would then be flat, and consequently $\operatorname{Ker} u$ would be flat over S, which it is not. It is clear that the restriction of u to the identity component G of G_1 is a monomorphism: the fibers of $K \cap G$ are identity groups, so $K \cap G$ is the identity group (Exposé $\mathrm{VI_B}$, 2.9). But it is not an immersion, for otherwise it would be an open immersion and H_s would be smooth. Here Exposé VIII, 7.9 does not apply, since ${}_2G = G$ is not finite over S.

For lovers of equations, in the preceding example one may take for H the closed subgroup of $(\mathbb{G}_a \times_S \mathbb{G}_a)_S$ with ring $A[V, W]/(F)$, where

$$F = (a''b' - b''a')(a''^2 V - a'^2 W) - (a''V - a'W)^2$$

(where a', a'', b', b'' satisfy (3)). For G one takes the group $(\mathbb{G}_a)_S$ with ring $A[T]$, and for the morphism $u : G \to H$ the S-morphism defined by

$$V \longmapsto a'(a'T + \pi T^2), \qquad W \longmapsto a''(a''T + \pi T^2).$$

Remark 1.2. — The preceding construction is inspired by the method of Koizumi–Shimura.[1] It does not apply when S is the spectrum of a ring of unequal characteristic, because the points of finite order of G are then "too rigid."

1.3. Statement of the Immersion Criterion

Theorem 1.3. — *Let S be a locally noetherian prescheme, let G be an S-group prescheme, smooth and of finite type over S, possessing Cartan subgroups locally for the fpqc topology (Exposé XV, 6.1 and 7.3 (i)), let H be an S-group prescheme locally of finite type, and let $u : G \to H$ be an S-group monomorphism. Then:*

a) If G has connected fibers, then u is an immersion if and only if, for every S-scheme S' that is the spectrum of a complete discrete valuation ring with algebraically closed residue field, and every Cartan subgroup C of $G_{S'}$, the restriction of $u_{S'}$ to C is an immersion.

b) The morphism u is an immersion (respectively a closed immersion) if and only if, for every S' as above and every Cartan subgroup C of the identity component $(G_{S'})^0$ of $G_{S'}$, the restriction of $u_{S'}$ to the normalizer N of C in $G_{S'}$ (Exposé XI, 6.11) is an immersion (respectively a closed immersion).

Before proving 1.3, let us state some applications.

Corollary 1.4. — *Let S be a prescheme, let G be an S-group prescheme, smooth and of finite presentation over S, of unipotent rank zero (Exposé XV, 6.1 ter), possessing maximal tori locally for the fpqc topology, let H be an S-group prescheme, and let $u : G \to H$ be an S-group monomorphism. Suppose in addition either that H is of finite presentation over S, or that S is locally noetherian and H is locally of finite type. Then:*

a) If G has connected fibers, u is an immersion.

b) If H is separated over S, and if for every S' over S and every maximal torus T of $G_{S'}$, the Weyl group

$$W = \mathrm{Norm}_{G_{S'}}(T)/\,\mathrm{Centr}_{G^0_{S'}}(T)$$

is representable (a condition always satisfied when G has connected fibers; Exposé XV, 7.1 (iv)) and finite over S, then u is a closed immersion.

Corollary 1.5. — *Let S be a prescheme, let G be an S-group prescheme, smooth and of finite presentation over S, with connected fibers, let H be an S-group prescheme, and let $u : G \to H$ be an S-group monomorphism. Suppose either that H is of finite presentation over S, or that S is locally noetherian and H is locally of finite type. Then:*

a) If G is reductive, that is, affine over S with reductive fibers (cf. Exposé XIX), u is an immersion, and a closed immersion if H is separated.

[1] Editor's note: S. Koizumi and G. Shimura, "Specialization of abelian varieties," *Sci. Papers Coll. Gen. Ed. Univ. Tokyo* 9 (1959), 187–211.

b) If G has affine, solvable, connected fibers of unipotent rank zero, u is an immersion, and a closed immersion if H is separated.

Proof of 1.4 from 1.3.

Reduction to the noetherian case. If S is locally noetherian, the reduction is immediate, since the properties to be proved are local on S. In the second case G and H are of finite presentation over S. After shrinking S, we may suppose that S is affine. By Exposé $\mathrm{VI_B}$, §10, there exist a noetherian scheme S_0, an S_0-group prescheme G_0, smooth over S_0 (with connected fibers in case a)) and of finite type over S_0, an S_0-group prescheme H_0 of finite type (separated in case b)), and an S_0-group morphism

$$u_0 : G_0 \longrightarrow H_0$$

such that G, H, u are obtained from G_0, H_0, u_0 by the base extension $S \to S_0$. The assertion that u is a monomorphism means that $\operatorname{Ker} u$ is the identity group; we may therefore suppose that u_0 is a monomorphism.

Since G has unipotent rank $\rho_u = 0$ and possesses maximal tori locally for the fpqc topology, the abelian rank ρ_{ab} of the fibers of G is locally constant (Exposé XV, 8.18). But ρ_u and ρ_{ab} are locally constructible functions (Exposé XV, 6.3 bis). A standard argument (cf. EGA IV, 8.3.4) shows that one may choose S_0 and G_0 so that the unipotent rank (respectively the abelian rank) of the fibers of G_0 is zero (respectively locally constant). Then G_0 possesses maximal tori locally for the fpqc topology (Exposé XV, 8.18), and the functor $\mathcal{TM}_{G_0}$ of maximal tori of G_0 is representable by an S_0-scheme X_0 of finite type over S_0 (Exposé XV, 8.15). Let T_0 be the "universal" maximal torus of $(G_0)_{X_0}$, let $X = X_0 \times_{S_0} S$, and let $T = T_0 \times_{S_0} S$, the universal maximal torus for G. By hypothesis, in case b), the Weyl group relative to T is representable and finite over X. These two properties are compatible with projective limits of preschemes (Exposé $\mathrm{VI_B}$, 10.1 iii, and EGA IV, 8.10.5). We may therefore choose S_0 so that the Weyl group of T_0 is finite over X_0. Under these conditions, to prove 1.4 we may replace S, G, u, H by S_0, G_0, u_0, H_0, and hence suppose that S is noetherian.

Use the valuative criterion furnished by 1.3. We are reduced to the case where S is the spectrum of a discrete valuation ring and where G^0 possesses a Cartan subgroup C.

Proof of 1.4 a). We must show that the restriction of u to C is an immersion (1.3 a)), which reduces us to the case where $G = C$ has connected, nilpotent fibers. Standard reductions (cf. the proof of Exposé VIII, 7.1) reduce us to the case where H is flat over S, u is an isomorphism on the generic fiber, and u is an isomorphism of the underlying spaces on the closed fiber. The group H then has connected fibers and is therefore separated over S (Exposé $\mathrm{VI_B}$, 5.2). Grant for a moment the following lemma.

Lemma 1.6. — *Let S be a prescheme and let G be an S-group prescheme, smooth over S, with connected, nilpotent fibers and unipotent rank zero. Then:*

i) G is commutative.

ii) For every integer $n > 0$, ${}_nG$ is an S-group prescheme, flat and quasi-finite over S, and it is finite over S if and only if the abelian rank, or the reductive rank, of G is locally constant on S.

iii) For every integer $q > 0$ invertible on S, the family of subgroups ${}_{q^n}G$ is universally schematically dense in G relative to S (EGA IV, 11.10.8).

Lemma 1.6 applies to G, and since G possesses maximal tori locally, the reductive rank of the fibers of G is locally constant. Thus, by 1.6 ii), ${}_nG$ is finite over S for every integer $n > 0$. It follows from Exposé VIII, 7.9 that u is an immersion.

Proof of 1.6. Let us first examine the case where S is the spectrum of a field.

Lemma 1.7. — *Let k be a field of characteristic p, and let G be a smooth, connected, nilpotent algebraic k-group of unipotent rank zero. Then G is a commutative group, an extension of an abelian variety A by a torus T. For every integer $n > 0$, ${}_nG$ is a finite group, étale if $(n,p) = 1$, defined by a k-algebra of rank $n^{\rho_r+2\rho_{ab}}$. If $(q,p) = 1$, the family of subgroups ${}_{q^n}G$, $n \in \mathbb{N}$, is schematically dense in G.*

Proof of 1.7. Let Z be the center of G. The group G/Z is affine (Exposé XII, 6.1), smooth, connected, and of unipotent rank zero; hence it is a torus (BIBLE 4, th. 4). It follows that G is commutative (Exposé XII, 6.4). If T is the unique maximal torus of G (Exposé XV, 3.4), Chevalley's theorem (Séminaire Bourbaki 1956/57, no. 145) immediately shows that G is an extension of an abelian variety A by T. For every integer $n > 0$, raising to the n-th power is an epimorphism on T, whence the exact sequence

$$0 \longrightarrow {}_nT \longrightarrow {}_nG \longrightarrow {}_nA \longrightarrow 0.$$

A classical theorem of Weil (A. Weil, *Variétés abéliennes et courbes algébriques*, §IX, th. 33, cor. 1) says that ${}_nA$ is a finite group defined by a k-algebra of rank $n^{2\rho_{ab}}$. Since ${}_nT$ is a finite group of rank n^{ρ_r}, the asserted structure of ${}_nG$ follows.

Now let H be the smallest closed subscheme of G containing ${}_{q^n}G$ for every n. It follows from the preceding discussion and Exposé XV, 4.6 that H is a smooth, connected subgroup, hence of the same type as G. Raising to the q-th power on H is an epimorphism (since ${}_qH$ is finite), so there is an exact sequence

$$0 \longrightarrow {}_qH \longrightarrow {}_qG \longrightarrow {}_q(G/H) \longrightarrow 0.$$

It follows that ${}_q(G/H) = 0$, and therefore $G/H = 0$. Thus the subgroups ${}_{q^n}G$ are schematically dense in G.

Continuation of the proof of 1.6 i). To show that G is commutative, the usual procedure reduces us first to the case where S is noetherian, and then to the case where S is the spectrum of a local ring with closed point s. The center of G is represented by a closed group subscheme Z of G (Exposé XI, 6.11). To prove that $Z = G$, it suffices to prove this after reduction modulo every power of the maximal ideal of the ring of S: Z will then be an open subgroup of G, and hence equal to G, since G has connected fibers. This reduces us to the case where S is local Artinian.

Let q be an integer invertible on S. For every integer n, ${}_{q^n}G_s$ is an étale, central subgroup of multiplicative type of G_s (1.7). Since G is smooth, ${}_{q^n}(G_s)$ lifts to an étale, central S-group subscheme M_n of G (Exposé XV, 1.2). The family of flat subgroups M_n, $n \in \mathbb{N}$, is then schematically dense in G (1.7 and EGA IV, 11.10.9). Since Z is closed in G and contains every M_n, we have $Z = G$.

Proof of 1.6 ii). To see that ${}_nG$ is flat and quasi-finite over S, it suffices to show that the n-th power map on G is flat and quasi-finite. Since G is flat over S, we are reduced to the case where S is the spectrum of a field (EGA IV, 11.3.10), and we observed in the proof of 1.7 that the n-th power map is an epimorphism. Moreover, ${}_nG$ is separated over S, since G is separated over S (Exposé $\mathrm{VI_B}$, 5.2). For ${}_nG$ to be finite over S, it is necessary and sufficient that the fibers of ${}_nG$ be spectra of finite algebras of locally constant rank. This is seen immediately by reducing to the case where S is the spectrum of a henselian local

ring. By 1.7, this condition is equivalent to saying that the abelian rank, or the reductive rank, of the fibers of G is locally constant.

Proof of 1.6 iii). To see that the family of subgroups ${}_{q^n}G$, $n \in \mathbb{N}$, is universally schematically dense in G, we again reduce to the noetherian case. In view of 1.7 and 1.6 ii), it suffices to apply EGA IV, 11.10.9.

Proof of 1.4 b). We have reduced to the case where S is the spectrum of a discrete valuation ring and where G^0 possesses a Cartan subgroup C. Let

$$N = \mathrm{Norm}_G(C) = \mathrm{Norm}_G(T),$$

where T is the unique maximal torus of C (Exposé XII, 7.1 a) and b)). Since H is separated over S, it follows from 1.6 ii) and Exposé VIII, 7.12 that the restriction of u to C is a closed immersion. On the other hand, to prove that u is a closed immersion, it suffices to prove this for $u|_N$ (1.3 b)). Since, by hypothesis, $W = N/C$ is represented by a finite group scheme, the assertion follows from the next lemma, applied to the exact sequence

$$0 \longrightarrow C \longrightarrow N \longrightarrow W \longrightarrow 1.$$

(Recall that a proper immersion is a closed immersion.)

Lemma 1.8. — *Let S be a prescheme and let G be an S-group prescheme of finite presentation over S, an extension of an S-group prescheme G'', proper and of finite presentation over S, by an S-group prescheme G', flat and of finite presentation over S:*

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1.$$

Let H be an S-group prescheme of finite presentation over S (or locally of finite presentation if S is locally noetherian), and let $u : G \to H$ be an S-group morphism. If the restriction of u to G' is proper, then u is proper.

Proof of 1.8. As usual, we reduce to the case where S is noetherian (Exposé $\mathrm{VI_B}$, §10, and Exposé XV, 6.2), and we may then apply the valuative criterion of properness (EGA II, 7.3.8). We therefore suppose that S is the spectrum of a discrete valuation ring A, with closed point s and generic point t. Let $\bar{x} \in G(t)$ and $h \in H(S)$ satisfy

$$u_t(\bar{x}) = h(t).$$

We must show that $\bar{x}$ comes from a unique element $x \in G(S)$. It is enough to prove the existence and uniqueness of x after a faithfully flat extension of the discrete valuation ring A. Let $\bar{y}$ be the image of $\bar{x}$ in $G''(t)$. Since G'' is proper over S, $\bar{y}$ comes from a unique element $y \in G''(S)$. Let X be the inverse image of y in G. The prescheme X is faithfully flat over S, since G' is faithfully flat over S, as is the morphism $G \to G''$ (Exposé $\mathrm{VI_B}$, §9). After a faithfully flat change of discrete valuation ring, we may suppose that X has a section ℓ over S (EGA IV, 14.5.8). Replacing $\bar{x}$ by $\bar{x}\ell_t^{-1}$, we may suppose that $\bar{x} \in G'(t)$. The existence and uniqueness of x now follow from the properness of $u|_{G'}$.

Proof of 1.5.

a) We shall see in Exposé XIX that, if G is reductive, then G possesses maximal tori locally for the faithfully flat topology, has a finite Weyl group, and has unipotent rank zero. Assertion a) therefore follows from 1.4.

b) If G has affine fibers and unipotent rank zero, then G possesses maximal tori locally for the fpqc topology (Exposé XV, 8.18). It remains only to apply 1.4, observing that,

since G has connected, smooth, affine, solvable fibers, its Weyl group is the identity group (BIBLE 6, th. 1, cor. 3).

Proof of 1.3 a). Since u is already a monomorphism, to prove that u is an immersion it suffices to prove that u is proper at the points of $u(G)$ (EGA IV, 15.7.1); for this we may use the valuative criterion of local properness (EGA IV, 15.7.5). We are therefore reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, closed point s, and generic point t. Since G is flat over S, standard reductions (cf. the proof of Exposé VIII, 7.1) reduce us to the case where H is flat over S, u_t is an isomorphism, and u_s is an isomorphism of underlying spaces. To say that u is an immersion is then equivalent to saying that u is an isomorphism.

By hypothesis, G possesses Cartan subgroups locally for the fpqc topology, so we may speak of the open subset U of regular points of G (Exposé XV, 7.3 i) and ii)).

Lemma 1.9. — *Under the preceding hypotheses, for u to be an immersion it suffices that $u|_U$ be an immersion.*

Indeed, to say that $u|_U$ is an immersion means that there exist an open subscheme V of H and a closed subscheme F of V such that $u|_U$ factors through F and induces an isomorphism from U onto F. Since H_t, and therefore V_t, is irreducible, one necessarily has $V_t = F_t$. Then F contains the schematic closure of V_t in V, which equals V because H is flat over S. Thus $V = F$. It follows that $u|_U$ is an open immersion, and hence that u is an open immersion (Exposé $\mathrm{VI_B}$, 2.6).

It remains to prove that $u|_U$ is an immersion. For this we apply the valuative criterion of local properness. After replacing S by the spectrum of a discrete valuation ring faithfully flat over S, we must show the following: if h is a section of H over S whose image $h(S)$ is contained in $u(U)$, then h is the image of a section g of U over S. It suffices to show that h is contained in the image of a Cartan subgroup C of G. Indeed, by hypothesis $u|_C$ is an immersion, so h is the image of a section g of C, necessarily a section of U.

Let $a \in U(s)$ and $b \in U(t)$ satisfy, respectively,

$$u_s(a) = h(s), \qquad u_t(b) = h(t).$$

Since u_t is an isomorphism, $h(t)$ is a regular point of H_t, and is therefore contained in a unique Cartan subgroup D_t of H_t (Exposé XIII, 3.2). Let D be the schematic closure of D_t in H.

Lemma 1.10. —

i) *D_s is a nilpotent algebraic group (Exposé VI_B, §8).*

ii) $\dim D_s = \nu$, *where ν is the nilpotent rank of G, and also the nilpotent rank of $(H_s)_{\mathrm{red}}$.*

iii) $h(s) \in D_s(\kappa(s))$.

i) Since H is separated (Exposé $\mathrm{VI_B}$, 5.2), so is D. Moreover, D is flat over S and its generic fiber is nilpotent. It follows from Exposé $\mathrm{VI_B}$, 8.4 that D_s is nilpotent.

ii) Since $(H_s)_{\mathrm{red}} \simeq G_s$, these two groups have the same nilpotent rank. Since G possesses Cartan subgroups locally for the fpqc topology, G_s and G_t have the same nilpotent rank ν. Finally, since D is flat over S,

$$\dim D_s = \dim D_t = \nu$$

(Exposé $\mathrm{VI_B}$, §4).

iii) Since h_t factors through D_t, the section h plainly factors through D, which proves iii).

The next lemma now shows that $(D_s)_{\mathrm{red}}$ is a Cartan subgroup of

$$(H_s)_{\mathrm{red}} \simeq G_s.$$

Lemma 1.11. — *Let G be a smooth, connected algebraic group over an algebraically closed field k, and let D be a smooth, nilpotent algebraic subgroup containing a regular element $a \in G(k)$. Then D^0 is contained in a Cartan subgroup of G. If, in addition, $\dim D$ equals the nilpotent rank of G, then D is a Cartan subgroup of G, and in particular is connected.*

Let Z be the center of G, let $G' = G/Z$, which is affine (Exposé XII, 6.1), and let a' and D' be the images of a and D in G'. The correspondence between Cartan subgroups of G and Cartan subgroups of G' (Exposé XII, 6.6 e)) shows at once that a' is a regular element of G' and that it suffices to prove the lemma for the pair (D', G'). Thus we may suppose that G is affine.

Let s be the semisimple component of a. It belongs to $D(k)$ (BIBLE 4, th. 3) and is regular (BIBLE 7, th. 2, cor. 1). By BIBLE 6, th. 2, s centralizes D^0; hence D^0 is contained in the connected centralizer of s, which is a Cartan subgroup of G (BIBLE 7, th. 2). If now $\dim D$ equals the nilpotent rank of G, then D^0 is a Cartan subgroup of G, equal to $\mathrm{Centr}_G(T)$, where T is the unique maximal torus of D^0 (Exposé XII, 6.6). But D is nilpotent, and therefore centralizes T (BIBLE 6, th. 2), so $D = D^0$.

Return now to the group G. The group

$$C_t = u_t^{-1}(D_t)$$

is the unique Cartan subgroup of G_t containing b. Let C be the schematic closure of C_t in G.

By functoriality of schematic closure, $u|_C$ factors through D, and therefore $u_s(C_s) \subset D_s$. Since u_s is an isomorphism from G_s onto $(H_s)_{\mathrm{red}}$, since

$$\dim C_s = \dim C_t = \nu = \dim D_s$$

by 1.10, and since D_s is connected by 1.11, u_s induces an isomorphism from $(C_s)_{\mathrm{red}}$ onto $(D_s)_{\mathrm{red}}$. Thus $(C_s)_{\mathrm{red}}$ is a Cartan subgroup of G_s. The next lemma shows that C itself is a Cartan subgroup of G.

Lemma 1.12. — *Let S be the spectrum of a discrete valuation ring, let G be a smooth S-group prescheme of finite type, and let C be a group subscheme of G, flat over S, such that the generic fiber C_t is a Cartan subgroup of G_t and the geometric reduced closed fiber $(C_{\bar{s}})_{\mathrm{red}}$ is a Cartan subgroup of $G_{\bar{s}}$. Then C is a Cartan subgroup of G.*

We must prove that C is smooth over S, and it suffices to establish this after a faithfully flat extension of the base. The hypotheses on C imply that the nilpotent rank of the fibers of G is constant; hence we may speak of the open subset U of regular points of G (Exposé XV, 7.3). Since $(C_{\bar{s}})_{\mathrm{red}}$ is a Cartan subgroup of $G_{\bar{s}}$, $C_{\bar{s}}$ contains a regular point $a_{\bar{s}}$ of $G_{\bar{s}}$. Since C is flat over S, after a faithfully flat extension of the base if necessary (EGA IV, 14.5.8), we may suppose that $a_{\bar{s}}$ is an element of $G(s)$ which lifts to a section $a \in C(S)$. Since U is open and contains $a(s)$, it contains a. Let C' be the unique Cartan subgroup of G containing a (Exposé XV, 7.3). Necessarily $C_t = C'_t$, and consequently C' and C coincide with the identity component of the schematic closure of C_t in G (observe that both C' and C are flat over S).

End of the proof of 1.3 a). By hypothesis, the restriction of u to the Cartan subgroup C of G is an immersion. It is clear that $h(S)$ is contained in $u(C) = D$, and hence h is the image of a section $g \in C(S)$. □

Proof of 1.3 b). We shall again use the valuative criterion of local properness (EGA IV, 15.5) in the case of an immersion (respectively, the valuative criterion of properness (EGA II, 7.3.6) in the case of a closed immersion). This reduces us to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. Let s be the closed point and t the generic point of S. Replacing H by the schematic closure in H of $u_t(G_t)$, we may suppose that H is flat over S and that u_t is an isomorphism.

Let G^0 be the connected component of G. By 1.2 a), the restriction $u|_{G^0}$ is an immersion; denote by H^0 the group subprescheme of H that is the image of G^0. Thus

$$H^0_s = (H_s)^0_{\mathrm{red}}.$$

To verify the valuative criterion, we must show that every section h of H over S such that $h(S)$ is contained in $u(G)$, in the case of an immersion (respectively, every section h of H over S, in the case of a closed immersion), is the image of a section g of G over S.

Let C be a Cartan subgroup of G, and let $D = u(C)$ be the Cartan subgroup of H^0 that is the image of C. The S-group

$$D' = \mathrm{int}(h)D$$

has connected fibers, so its underlying space is contained in that of H^0; on the other hand, since it is smooth over S, it is reduced, and hence D' is a smooth group subprescheme of H^0. The fibers of D' are Cartan subgroups of the fibers of H^0; therefore D' is a Cartan subgroup of H^0, and

$$C' = u^{-1}(D')$$

is a Cartan subgroup of G. But $\mathrm{Transpstr}_G(C, C')$ is smooth and surjective over S (Exposé XII, 7.1 b)). Since S is henselian with algebraically closed residue field, there exists $g \in G(S)$ such that $\mathrm{int}(g)C = C'$. Replacing h by $u(g)^{-1}h$, we may suppose that $h(t)$ normalizes D_t (and that $h(s)$ is the image of an element of the normalizer of C_s in G_s, in the case of an immersion). Let

$$N = \mathrm{Norm}_G(C).$$

By hypothesis, $u|_N$ is an immersion (respectively, a closed immersion); hence h is indeed the image of a section of N over S, which completes the proof. □

2. A Representability Theorem for Quotients

Let us "recall" the following result.

Theorem 2.1. — *Let S be a prescheme, let X and Y be two S-preschemes, and let $f : X \to Y$ be an S-morphism. Suppose that one is in one of the following two cases:*

a) The morphism f is locally of finite presentation.

b) The prescheme S is locally noetherian and X is locally of finite type over S.

Then the following conditions are equivalent:

i) There exist an S-prescheme X' and a factorization of f

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S-morphism, locally of finite presentation, and f'' is a monomorphism.

ii) The first projection

$$p_1 : X \times_Y X \longrightarrow X$$

is a flat morphism.

Moreover, if the preceding conditions hold, then (X', f') is a quotient of X, for the fpqc topology, by the equivalence relation defined by f. Consequently the factorization $f = f'' \circ f'$ in i) is unique up to isomorphism.

The proof of this delicate theorem will be given in EGA V. One may also consult the lecture by J.-P. Murre, *Séminaire Bourbaki*, May 1965, no. 294, theorem 2, corollary 2, which treats the case where Y is locally noetherian and X is of finite type over Y. We shall see that one can reduce to that case.

Let us first make a few remarks.

a) The implication i) $\Rightarrow$ ii) is trivial. Indeed, the first projection

$$p_1' : X \times_{X'} X \longrightarrow X$$

factors through $X \times_Y X$:

$$p_1' : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism because f'' is a monomorphism, and p_1' is flat because f' is flat; therefore p_1 is flat.

b) The assertions of 2.1 are local on Y (hence local on S); they are also local on X, as follows readily from the fact that a flat morphism locally of finite presentation is open (EGA IV, 11.3.1).

c) Under the hypotheses of 2.1 a), the foregoing reduces us to the case where X and Y are affine and f is of finite presentation. Replacing S by Y, we may suppose that X and Y are of finite presentation over S. We then reduce to the noetherian case for S by EGA IV, 11.2.6.

d) Under the hypotheses of 2.1 b), we may suppose that S, X, and Y are affine, that S is noetherian, and that X is of finite type over S. Regard Y as a filtered projective limit of affine schemes Y_i of finite type over S. The schemes $X \times_{Y_i} X$ form a decreasing filtered family of closed subschemes of $X \times_S X$, whose projective limit is $X \times_Y X$. Since $X \times_S X$ is noetherian, one has

$$X \times_{Y_i} X = X \times_Y X$$

for all sufficiently large i. Thus

$$f_i : X \xrightarrow{f} Y \longrightarrow Y_i$$

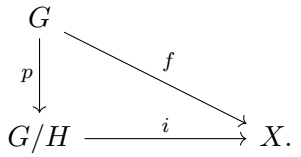
satisfies the hypotheses of 2.1 ii) if f does. Since the equivalence relation on X defined by f coincides with the one defined by f_i, it is clear that it suffices to prove ii) $\Rightarrow$ i) for f_i. This reduces us to the case where Y is of finite type over S.

Application to Group Preschemes

Theorem 2.2. — *Let S be a prescheme and let G be an S-group prescheme locally of finite presentation over S, acting on an S-prescheme X. Let $\xi \in X(S)$ be a section of X over S, such that the stabilizer H of ξ in G is an S-group subprescheme of G, flat over S. If X is locally of finite type over S, or if S is locally noetherian, then the homogeneous quotient space G/H is representable by an S-prescheme locally of finite presentation over S, and the S-morphism*

$$f : G \longrightarrow X, \qquad g \longmapsto g \cdot \xi$$

factors as



where p is the canonical projection, a faithfully flat morphism locally of finite presentation, and i is a monomorphism. (For definiteness, we have supposed that G acts on the left on X.)

Proof. The morphism f makes G into an X-prescheme. By the definition of the stabilizer of ξ, the morphism

$$G \times_S H \longrightarrow G \times_X G, \qquad (g, h) \longmapsto (g, gh)$$

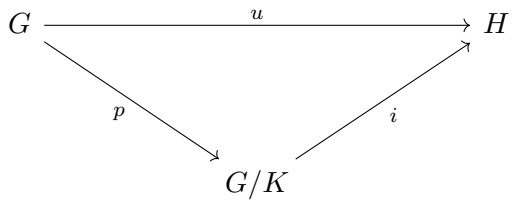
is an isomorphism. Since H is flat over S, $G \times_S H$ is flat over G; hence the first projection

$$p_1 : G \times_X G \longrightarrow G$$

is flat. Moreover, if X is locally of finite type over S, then f is locally of finite presentation (EGA IV, 1.4.3 v)); otherwise S is locally noetherian by hypothesis. It therefore suffices to apply 2.1 to f.

It remains only to show that G/H is locally of finite presentation over S, and this follows immediately from Exposé V, 9.1. □

Corollary 2.3. — *Let S be a prescheme, let G and H be two S-group preschemes, and let $u : G \to H$ be an S-homomorphism. Suppose that G is locally of finite presentation over S, and that either H is locally of finite type over S, or S is locally noetherian. If $K = \mathrm{Ker}(u)$ is flat over S, then the quotient group G/K is representable by an S-group prescheme locally of finite presentation over S, and u factors as*



where p is the canonical projection and i is a monomorphism.

Proof. Apply 2.2 with $X = H$ and with ξ the unit section of H. □

Corollary 2.4. — *Let S be a prescheme, let G be an S-group prescheme of finite presentation over S, let H be an S-group prescheme smooth over S with connected fibers*

(and hence of finite presentation over S, by Exposé VI_B, 5.5), and let $i : H \to G$ be a monomorphism of S-groups, so that H is a subgroup of G. Suppose that

$$N = \mathrm{Norm}_G(H),$$

which is representable by a closed group subprescheme of G, of finite presentation over S (Exposé XI, 6.11), is flat over S. Then G/N is representable by an S-prescheme of finite presentation over S and quasi-projective over S.

All the assertions to be proved, except the last one, are local on S. To establish them, we may therefore suppose that S is quasi-compact and that the relative dimension of H over S is constant, equal to r. Proceed as in Exposé XV, §5. For every integer $n > 0$, let $G^{(n)}$ (respectively $H^{(n)}$) be the n-th normal invariant of the unit section of G (respectively of H) (EGA IV, 16). The sheaf of $\mathcal{O}_S$-modules $H^{(n)}$ is a quotient of $G^{(n)}$. Since H is smooth over S of dimension r, $H^{(n)}$ canonically defines an element ξ_n of

$$X_n = \mathrm{Grass}_{\varphi(n,r)} G^{(n)}$$

(EGA I, second edition, §9), for a suitable integer $\varphi(n, r)$. On the other hand, G acts naturally on $G^{(n)}$, and hence also on X_n, through the representation

$$G \longrightarrow \mathrm{Aut}_{S\text{-gr}}(G), \qquad g \longmapsto \mathrm{int}(g).$$

Since S is quasi-compact, there exists an integer m such that, for $n \geq m$,

$$N = \mathrm{Norm}_G(H) = \mathrm{Norm}_G H^{(n)} \qquad \text{(Exposé XI, 6.11 b))}.$$

In other words, N is the stabilizer of ξ_n for sufficiently large n. The representability of G/N therefore follows from 2.2. The fact that G/N is of finite presentation over S is a consequence of Exposé V, 9.1. It remains to prove that G/N is quasi-projective over S. For this, we shall exhibit a canonical invertible sheaf on G/N that is ample relative to S. Consider the functor

$$\mathscr{F} : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Ens}$$

such that, for every S-prescheme T,

$$\mathscr{F}(T) = \left\{ \begin{array}{c} T\text{-subgroups } H \text{ of } G_T, \text{ representable and smooth over } T, \\ \text{with connected fibers} \end{array} \right\}.$$

Such a group H is of finite presentation over S (Exposé $\mathrm{VI_B}$, 5.5), and $H \to G_T$ is a quasi-affine morphism (EGA IV, 8.11.2). By effective descent for quasi-affine morphisms (SGA 1, VIII, 7.9), it follows that $\mathscr{F}$ is a sheaf for the fpqc topology. Since G/N is the sheaf associated with a subfunctor of $\mathscr{F}$, there is a canonical monomorphism

$$G/N \longrightarrow \mathscr{F}.$$

Consequently there exists a subgroup H' of $G \times_S (G/N)$, representable and smooth over G/N, with connected fibers, that is "universal" for the functor G/N. I claim that the invertible sheaf

$$\mathscr{L} = \left(\det(\mathrm{Lie}\, H')\right)^{-1}$$

is ample relative to S. In this form the assertion is local on S, and its proof is analogous to the one given in Exposé XV, 5.8.

The following corollary was announced in Exposé XIV, 4.8 bis.

Corollary 2.5. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let P be a parabolic subgroup of G (Exposé XIV, 4.8 bis). Then G/P is representable by an S-prescheme smooth and projective over S.*

It remains only to show (*loc. cit.*) that G/P is representable by an S-prescheme that is quasi-projective and of finite presentation. This follows from 2.4, since

$$P = \mathrm{Norm}_G(P)$$

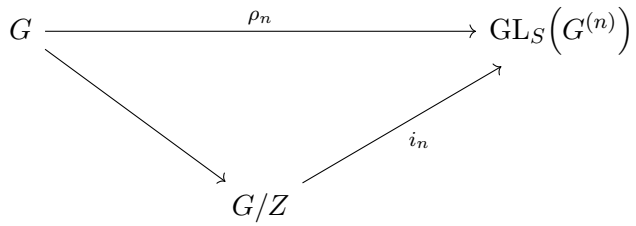
(*loc. cit.*) and is therefore flat over S. □

3. Groups with Flat Center

Proposition 3.1. — *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers. Suppose that the center Z of G, which is representable by Exposé XI, 6.11, is flat over S. For every integer $n > 0$, let $G^{(n)}$ denote the locally free $\mathcal{O}_S$-module equal to the n-th normal invariant of G along the unit section, and let ρ_n be the natural "adjoint representation" of G in $\mathrm{GL}_S(G^{(n)})$. Then:*

a) The quotient $G' = G/Z$ is representable by an S-group prescheme that is smooth, of finite presentation over S, quasi-affine, and has affine connected fibers.

b) The representation ρ_n factors through G':



If S is quasi-compact, then i_n is a monomorphism for all sufficiently large n.

c) The functor $\mathscr{T}_{G'}$ of subtori of G' is representable by an S-prescheme smooth over S, and, if S is quasi-compact, this prescheme is a sum of a family of preschemes affine over S.

Proof. The assertions in a) are local on S, which allows us to suppose that S is quasi-compact. By Exposé XI, 6.11 b), one has

$$Z = \mathrm{Ker}\, \rho_n$$

for all sufficiently large n; this kernel is therefore flat over S. The representability of G/Z and the fact that i_n is a monomorphism now follow from 2.3. Since G is smooth and of finite presentation over S, the same is true of G' (Exposé $\mathrm{VI_B}$, §9). The group $\mathrm{GL}_S(G^{(n)})$ is affine over S, and a monomorphism of finite presentation is quasi-affine (EGA IV, 8.11.2). Hence G' is quasi-affine over S and has affine fibers. Since G' is smooth over S, the functor $\mathscr{T}_{G'}$ is formally smooth (Exposé XI, 2.1 bis). Taking Exposé XI, 4.6 and 4.3 into account, the assertions of 3.1 c) will therefore follow from the next lemma.

Lemma 3.2. — *Let S be a prescheme, let G and H be two S-group preschemes of finite presentation over S, let $u : G \to H$ be an S-group monomorphism, and let $\mathscr{T}_G$ (respectively*

$\mathscr{T}_H$) be the functor of subtori of G (respectively of H) (cf. Exposé XV, §8). Then the map $T \mapsto u(T)$ defines a monomorphism (Exposé XV, 8.3 c))

$$\widetilde{u} : \mathscr{T}_G \longrightarrow \mathscr{T}_H$$

that is representable by a closed immersion of finite presentation.

By the usual procedure, we reduce to the following problem. Let T be a subtorus of H, let

$$T' = G \times_H T$$

be its inverse image in G, and let $u_T : T' \to T$ be the S-group monomorphism induced by u. One must show that the S-functor

$$\prod_{T/S} (T'/T)$$

is representable by a closed subprescheme of S, of finite presentation. But T, being a torus, is smooth over S with connected fibers, and it suffices to apply Exposé XI, 6.10. □

Theorem 3.3. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let Z be the center of G. Suppose that Z is flat over S and that the unipotent rank of G (Exposé XV, 6.1 ter) equals that of Z. Then:*

a) The group $G' = G/Z$ is representable; if S is quasi-compact, the canonical morphism

$$i_n : G' \longrightarrow \mathrm{GL}_S(G^{(n)})$$

(cf. 3.1 b)) is an immersion for all sufficiently large n.

b) The group G' is quasi-affine over S, with affine fibers; its center is the unit group, and G' has unipotent rank zero.

c) Locally for the étale topology, G' possesses maximal tori, and these are also Cartan subgroups of G'. The functor $\mathscr{TM}_{G'}$ of maximal tori of G' (Exposé XV, §8) is representable by an S-prescheme smooth and affine over S.

d) Locally for the étale topology, G possesses Cartan subgroups, and the functor $\mathscr{C}_G$ of Cartan subgroups of G is representable by an S-prescheme smooth and affine over S.

Proof. By 3.1, the group G' is represented by an S-prescheme that is smooth and quasi-affine over S, with affine fibers. In view of the correspondence between Cartan subgroups of the fibers of G and those of the fibers of G' (Exposé XII, 6.6 e)), the hypothesis on the unipotent rank of Z implies that G' has unipotent rank zero. Exposé XV, 8.18 now shows that, locally for the étale topology, G' possesses maximal tori. The fact that i_n is an immersion for all sufficiently large n then follows from 3.1 b) and 1.4 a). This proves a).

We now prove c). Since G' has unipotent rank zero, every maximal torus of G' is plainly also a Cartan subgroup of G'. The functor of maximal tori of G' is represented by a subprescheme that is both open and closed in the functor of subtori of G'; moreover it is of finite type over S (Exposé XV, 8.15). It follows from 3.1 c) that this functor is represented by an S-prescheme smooth and affine over S.

Since, locally for the étale topology, G' possesses Cartan subgroups, the same is true of G; and the functor of Cartan subgroups of G is canonically isomorphic to that of G' (Exposé XV, 7.3 iv)). Thus c) implies d).

It remains to prove b), and more precisely to prove that the center Z' of G' is the unit group. Since Z' is representable (Exposé XI, 6.11) and of finite presentation over S, it suffices to show that the fibers of Z' are reduced to the unit group (Exposé $\mathrm{VI_B}$, 2.9). We are thereby reduced to the case where S is the spectrum of an algebraically closed field. The center Z' is evidently contained in every Cartan subgroup of G', hence, by c), in every maximal torus of G'; it is therefore of multiplicative type. On the other hand, we shall see in Exposé XVII that, if G is a connected algebraic group with center Z, the center Z' of G/Z is unipotent. In the present case, Z', being at once of multiplicative type and unipotent, is reduced to the unit group (cf. Exposé XVII). □

Examples of Groups Whose Center Is Flat

Proposition 3.4. — *Let S be a prescheme, let G be an S-group prescheme of finite presentation over S, smooth, with connected fibers, and let Z be the center of G. Then Z is flat over S in either of the following two cases:*

a) The unipotent rank ρ_u of the fibers of G (Exposé XV, 6.1 ter) is zero.

b) i) S is reduced; ii) the dimension of the fibers of Z is a locally constant function on S; iii) the unipotent rank of G equals that of Z.

Proof of 3.4 a). We shall prove at the same time the following proposition.

Proposition 3.5. — *Under the hypotheses of 3.4 a), suppose moreover that G possesses a maximal torus T, and let*

$$C = \mathrm{Centr}_G(T)$$

be the Cartan subgroup of G associated with T (Exposé XII, 7.1). Then:

(i) $Z \cap T$ is a subgroup of multiplicative type of G.

(ii) C is commutative and equals $T \cdot Z$.

(iii) If the quotients $Z/(Z \cap T)$ and C/T are representable, then they are represented by abelian preschemes (that is, S-group preschemes smooth over S, whose fibers are abelian varieties), and the canonical monomorphism

$$Z/(Z \cap T) \longrightarrow C/T$$

is an isomorphism.

Using the general properties of passage to the limit established in Exposé $\mathrm{VI_B}$, §10 and Exposé XV, 6.2, 6.3, and 6.3 bis, we reduce as usual to the case where S is noetherian (observe that the assertions of 3.4 and 3.5 are local on S). We noted in 3.1 that the hypotheses on G imply that Z is representable. To prove that Z is flat over S, EGA 0_{III}, 10.2.6 reduces us to the case where S is local artinian. Since G is then smooth, it possesses maximal tori locally for the étale topology (Exposé XV, 8.17). After a finite flat extension of S, if necessary (which is permissible when proving that Z is flat), we may therefore suppose that G possesses a maximal torus T.

Let us show, in the same way, that to establish 3.5 we may reduce to the case where S is artinian.

i) Since Z is closed in G (Exposé XI, 6.11), $Z \cap T$ is a closed group subscheme of T. It follows from Exposé X, 4.8 b), that $Z \cap T$ is of multiplicative type if and only if it is flat over S. As above, it suffices to establish that $Z \cap T$ is flat when S is artinian.

ii) Since G has unipotent rank zero, every Cartan subgroup C of G satisfies the hypotheses of Lemma 1.6, and is therefore commutative. The equality $C = T \cdot Z$ will follow from iii).

iii) If C/T is representable, then C/T is smooth over S (Exposé $\mathrm{VI_B}$, §9), and its fibers are abelian varieties (1.7); thus C/T is an abelian prescheme. To show that the canonical monomorphism

$$i : Z/(Z \cap T) \longrightarrow C/T$$

is an isomorphism, it suffices to verify this when S is local artinian. Indeed, one then deduces successively that $Z/(Z \cap T)$ is flat over S (EGA 0_{III}, 10.2.6), that i is flat (EGA IV, 11.3.10), that i is an open immersion (EGA IV, 17.9.1), and finally that i is an isomorphism (because C/T has connected fibers).

We henceforth suppose that S is local artinian, with closed point s, and that G possesses a maximal torus T. Put

$$C = \mathrm{Centr}_G(T).$$

The group C is a Cartan subgroup of G (Exposé XII, 7.1) and contains Z.

The algebraic group $M_s = Z_s \cap T_s$ is a central subgroup of multiplicative type of G_s. Since T is smooth over S, M_s lifts to a group subscheme M of T, of multiplicative type (Exposé IX, 3.6 bis), and contained in the center of G (Exposé IX, 3.9), hence contained in $Z \cap T$. Since S is artinian and M and T are flat over S, the quotient groups

$$(Z \cap T)/M, \qquad Z' = Z/M, \qquad A = C/T$$

are representable (Exposé $\mathrm{VI_A}$, §§4 and 5). By construction, $\big((Z \cap T)/M\big)_s$ is the unit group, so $Z \cap T = M$ (Exposé $\mathrm{VI_B}$, 2.9); in particular it is flat over S, which proves 3.5 i).

Passing to the quotient gives a canonical monomorphism

$$i : Z' \longrightarrow A,$$

which is therefore a closed immersion (Exposé $\mathrm{VI_B}$, 1.4.2). We have already observed that A is an abelian prescheme. It remains to show that i is an isomorphism. This will prove 3.5 iii) and will imply that Z' is flat over S, and hence that Z is flat over S (as an extension of Z' by the flat group M; Exposé $\mathrm{VI_B}$, §9). Since Z_s has the same abelian rank as G_s (Exposé XII, 6.1), i_s is an epimorphism, hence an isomorphism. Let $q > 0$ be an integer invertible on S. By the density theorem of 1.6 iii), to see that $i(Z') = A$, it suffices to show that, for every integer n equal to a power of q, $i(Z')$ contains ${}_nA$. Now let M_s^0 be the identity component of M_s. It follows immediately by duality that raising to the q-th power in M_s^0 is an epimorphism. Consequently, if m_0 is the exponent of q in the prime factorization of $\mathrm{card}(M_s/M_s^0)$, the image of ${}_{nm_0}Z_s$ in

$$A_s \simeq Z_s/M_s$$

contains ${}_nA_s$. There is therefore a subgroup $M_s(n)$ of multiplicative type of Z_s whose image in A_s is ${}_nA_s$. As above, $M_s(n)$ lifts to a central subgroup $M(n)$ of multiplicative type of G. The image of $M(n)$ in A is a subgroup of multiplicative type (Exposé IX, 6.8),

and is necessarily equal to ${}_nA$, since it is so on the reduced fiber (Exposé IX, 5.1 bis). This completes the proof of 3.4 a) and of 3.5.

Proof of 3.4 b). The assertion to be proved is local on S, so we may suppose that S is affine, with ring A. Regarding A as the inductive limit of its finitely generated $\mathbb{Z}$-subalgebras, we reduce as above to the case where S is noetherian and reduced.

To prove that Z is flat, we may use the valuative criterion for flatness (EGA IV, 11.8.1), which reduces us to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with generic point t and closed point s. Let Z' be the schematic closure of Z_t in Z. We must show that $Z' = Z$, and it even suffices to show that $Z_s = Z'_s$ (Exposé $\mathrm{VI_B}$, 2.6). By Exposé $\mathrm{VI_B}$, §4, one has the inequalities

$$\dim Z_t = \dim Z'_t = \dim Z'_s \leq \dim Z_s.$$

By hypothesis, however, $\dim Z_t = \dim Z_s$; hence $\dim Z'_s = \dim Z_s$, and consequently Z'_s contains $(Z_s)^0_{\mathrm{red}}$. It follows that

$$G'_s = G_s/Z'_s$$

is affine (Exposé XII, 6.1). In view of the correspondence between Cartan subgroups of G and G' (Exposé XII, 6.6 e)), hypothesis 3.4 b) iii) implies that the unipotent rank of G'_s is zero, and consequently its Cartan subgroups are also its maximal tori. The image Z''_s of Z_s in G'_s is a central algebraic subgroup of G'_s, hence is contained in every Cartan subgroup of G'_s, that is, in every maximal torus; Z''_s is therefore a subgroup of multiplicative type of G'_s. Under these conditions, we shall show in Exposé XVII, §7, that there exists a finite subgroup M_s of multiplicative type of Z_s whose image in Z_s/Z'_s is Z''_s (we used this fact in the proof of 3.4 a) when Z''_s is étale). Since G is smooth over S and S is the spectrum of a complete local ring, M_s lifts to an S-subgroup M of multiplicative type of G (Exposé IX, 3.6 bis and Exposé XV, 1.6 b)), which is central (Exposé IX, 5.6 a)). Thus M_t is contained in $Z'_t = Z_t$, and, since M is flat, M is contained in Z'. The group M_s is therefore contained in Z'_s, but this implies that Z''_s is the unit group, that is, $Z_s = Z'_s$. This completes the proof of 3.4 b).

Example of a Smooth Group Prescheme with Connected Fibers Whose Center Is Not Flat

Let S be the spectrum of a discrete valuation ring A, let π be a uniformizer of A, and let t and s be the generic and closed points of S, respectively. Let G be the smooth affine S-group whose ring is

$$B = A[T, T^{-1}, U]/F, \qquad F = 1 - T + \pi U,$$

the composition law being defined by

$$(t, u), (t', u') \longmapsto (tt', \pi uu' + u + u').$$

The generic fiber G_t is therefore isomorphic to the multiplicative group $(\mathbb{G}_m)_t$, while the closed fiber G_s is isomorphic to the additive group $(\mathbb{G}_a)_s$. The function T is invertible in $\Gamma(G, \mathcal{O}_G)$ and defines an S-group morphism

$$\varphi : G \longrightarrow (\mathbb{G}_m)_S$$

which is an isomorphism on the generic fiber and the trivial morphism on the closed fiber. The datum of φ permits one to construct the semidirect-product group

$$H = G \cdot (\mathbb{G}_a)_S$$

(note that $(\mathbb{G}_m)_S$ acts on $(\mathbb{G}_a)_S$). The center Z of H is the unit group on the generic fiber and is equal to H_s on the closed fiber, and is therefore not flat over S.

4. Groups with Affine Fibers, of Unipotent Rank Zero

Theorem 4.1. — *Let S be a prescheme and let G be an S-group prescheme smooth over S, of finite presentation, with affine, connected fibers, of unipotent rank zero (Exposé XV, 6.1 ter). Then:*

a) The center Z of G is a group of multiplicative type (and is therefore a reductive center of G; Exposé XII, 4.1).

b) G satisfies conditions a) through d) of 3.3.

c) Locally for the étale topology, G possesses maximal tori, and these are also Cartan subgroups of G. The functor $\mathscr{TM}_G$ of maximal tori of G is representable by an S-prescheme smooth and affine over S.

d) G is quasi-affine over S. Moreover, if T is a maximal torus of G, if

$$N = \mathrm{Norm}_G(T)$$

is its normalizer in G, and if $W = N/T$ is the Weyl group relative to T, the following conditions are equivalent:

(i) G is affine over S;

(ii) $G' = G/Z$ is affine over S;

(iii) N is affine over S;

(iv) W is affine over S.

These conditions are always satisfied if S is locally noetherian of dimension at most 1.

Proof. Since G has affine fibers (and hence abelian rank zero) and unipotent rank zero, G possesses maximal tori locally for the étale topology (Exposé XV, 8.18), and every maximal torus of G is plainly a Cartan subgroup of G. This proves the first part of c). To see that Z is of multiplicative type, we may suppose, by what precedes, that G possesses a maximal torus T. Since T is also a Cartan subgroup, it contains Z (because $T = \mathrm{Centr}_G(T)$ by Exposé XII, 7.1), and

$$Z = Z \cap T$$

is of multiplicative type by 3.5 i). This proves a). Assertion b) is clear in view of a). On the other hand, the functor $\mathscr{TM}_G$ is isomorphic to the functor of Cartan subgroups of G (Exposé XII, 7.1), and is therefore smooth and affine over S by 3.3 c). This finishes the proof of c). It remains to prove d).

Proof of d). Since G' is quasi-affine (3.3 b)) and Z is of multiplicative type, hence affine over S, G is quasi-affine (Exposé $\mathrm{VI_B}$, §9).

i) $\Longleftrightarrow$ ii). If G is affine, then G' is affine by Exposé IX, 2.3. If G' is affine, then G is affine as an extension of an affine group by an affine group (Exposé $\mathrm{VI_B}$, §9).

i) $\Longleftrightarrow$ iii). If G is affine, then N is affine, since it is closed in G (Exposé XI, 6.11 a)). Moreover, G/N is isomorphic to the functor $\mathscr{T}\mathscr{M}_G$ (conjugacy of maximal tori; cf. Exposé XII, 7.1 b)), and is therefore affine over S by c). Thus if N is affine over S, then G is affine, as an "extension" of an affine homogeneous space by an affine group (Exposé $\mathrm{VI_B}$, §9).

iii) $\Longleftrightarrow$ iv). If N is affine, then $W = N/T$ is affine (Exposé IX, 2.3), and the converse follows from Exposé $\mathrm{VI_B}$, §9.

Moreover, N/T is representable by an S-group prescheme that is étale, separated, and of finite type over S (Exposé XV, 7.1 iv)), hence quasi-finite over S. The last assertion of d) therefore follows from the next lemma, applied with $X = W$ and $Y = S$.

Lemma 4.2. — *Let S be a locally noetherian prescheme of dimension 1, let X and Y be two S-preschemes locally of finite type over S, and let $u : X \to Y$ be a quasi-finite and separated S-morphism. Suppose that, for every point s of S, the morphism*

$$u_s : X_s \longrightarrow Y_s$$

induced from u by the base change $\operatorname{Spec}\kappa(s) \to S$ is finite. Then u is an affine morphism.

The assertion to be proved is local on Y, which allows us to suppose that Y (and hence also X) is of finite type over S. By EGA IV, §8, it suffices to prove 4.2 when S is the spectrum of a local ring A. By fpqc descent we may suppose that A is complete, then that A is reduced (EGA II, 1.6.4), and then that A is normal (Nagata's theorem, EGA 0_{IV}, §22, and Chevalley's theorem, EGA II, 6.7.1). If A is a field, then u is finite by hypothesis, hence affine. Otherwise A is a discrete valuation ring; let s and t be the closed and generic points of S, and let π be a uniformizer of A. Let y be a point of Y_s. Applying EGA IV, §8, once more, we may replace Y by the spectrum Y' of $\mathcal{O}_{Y,y}$, and X by

$$X' = X \times_Y Y';$$

we may even suppose that $\mathcal{O}_{Y,y}$ is complete. Since u is quasi-finite and separated, EGA II, 6.2.6, shows that X' is the disjoint union of two schemes X_1 and X_2, where X_1 is finite over Y' (hence affine), while $u(X_2)$ does not contain the closed point y of Y'. I claim that, under these conditions, X_2 does not meet the closed fiber X'_s. Indeed, by hypothesis,

$$X'_s \longrightarrow Y'_s$$

is finite, so the restriction of this morphism to the closed subset $(X_2)_s$ is finite, and its image in $(Y')_s$ is closed. Since this image does not contain the closed point of the local scheme $(Y')_s$, $(X_2)_s$ is empty. Since u_t is finite, the restriction to $(X_2)_t = X_2$ of the morphism

$$X'_t \longrightarrow Y'_t$$

is finite. Moreover $Y'_t = Y'_\pi$, so the open immersion $Y'_t \to Y'$ is affine. In short, the composite morphism

$$X_2 \longrightarrow Y'_t \longrightarrow Y'$$

is affine, and it follows that

$$X' = X_1 \amalg X_2 \longrightarrow Y'$$

is affine.

Corollary 4.3. — *Let S be a prescheme, and let G be an S-group prescheme smooth and of finite presentation over S, with affine, solvable, connected fibers, of unipotent rank*

zero. Then G is affine over S. If, moreover, the center of G is the unit group and S is quasi-compact, the canonical morphism

$$i_n : G \longrightarrow \mathrm{GL}_S(G^{(n)}) \qquad \textit{(cf. 3.1 b))}$$

is a closed immersion for all sufficiently large n.

To prove that G is affine over S, we may suppose that G possesses a maximal torus T (4.1 c)). Since G has solvable fibers, the Weyl group relative to T is the unit group (BIBLE, 6, Th. 1, Cor. 3), and condition 4.1 d) iv) is satisfied. If the center of G is the unit group, i_n is a monomorphism for all sufficiently large n (3.1), and is therefore a closed immersion (1.5 b)).

5. Application to Reductive and Semisimple Groups

Definition 5.1. — *An S-group prescheme G is said to be* reductive *(respectively* semisimple*) if G is smooth and affine over S, with reductive (respectively semisimple) fibers.*

5.1.1. Reductive groups will be studied systematically beginning with Exposé XIX. In this section we shall need the following properties, which will be proved in Exposé XIX (without using the developments of the present Exposé).

a) Let S be a prescheme, let G be an S-group prescheme smooth and affine over S, with connected fibers, and let s be a point of S such that G_s is reductive (respectively semisimple). Then there exists a neighborhood U of s such that $G|_U$ is reductive (respectively semisimple).

b) If G is reductive, then locally for the étale topology G possesses maximal tori; and if T is a maximal torus of G, the Weyl group

$$W = \mathrm{Norm}_G(T)/T$$

is finite over S.

c) A reductive group has unipotent rank zero.

Admitting these facts, we propose to strengthen assertion a) above.

Theorem 5.2. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let s be a point of S. Then:*

(i) If the fibers of G are affine and G_s is reductive, there exists an open neighborhood U of s such that $G|_U$ is reductive.

(ii) If G_s is semisimple, there exists an open neighborhood U of s such that $G|_U$ is semisimple.

Using Exposé XV, 6.2 i), and Exposé $\mathrm{VI_B}$, §10, we reduce to the case where S is noetherian.

In case ii), consider the center Z of G, which is representable (Exposé XI, 6.11 a)). Since G_s is semisimple, it is well known that Z_s is finite. Consequently (Exposé $\mathrm{VI_B}$, §4), there exists a neighborhood U of s such that Z is quasi-finite over U. For every point t of U, G_t is then affine (Exposé XII, 6.1). Shrinking S if necessary, we may therefore suppose that the fibers of G are affine, in case ii) as well as in case i).

In view of a), to prove 5.2 it suffices to prove that G is affine over S. Since G has affine fibers, we know that the functor of subtori of G is representable by a smooth S-prescheme (Exposé XV, 8.11 and 8.9). After making an étale extension covering s, which is permissible, we may therefore suppose that there is a subtorus T of G such that T_s is a maximal torus of G_s. But then

$$C = \mathrm{Centr}_G(T)$$

is a smooth subgroup of G, with connected fibers, containing T, such that $C_s = T_s$ (because C_s is a Cartan subgroup of G_s, and G_s has unipotent rank zero by c)). It follows that $C = T$, so T is a Cartan subgroup and a maximal torus of G; in particular, the unipotent rank of G is zero, and 4.1 applies.

By 4.1 d), it suffices to show that the Weyl group W relative to T is affine over a neighborhood U of s. In fact, we shall show that W is even finite over a neighborhood of s. Since W is quasi-finite over S (Exposé XV, 7.1 iv)), saying that W is finite over a neighborhood U of s is equivalent to saying that the morphism

$$W \longrightarrow S$$

is proper at s (EGA IV, 15.7, and EGA III, 4.4.2). To establish this, the valuative criterion of local properness (EGA IV, 15.7) reduces us to the case where S is the spectrum of a discrete valuation ring, with closed point s. But then, by the last assertion of 4.1 d), G is affine over S, and we may apply property a) recalled above to conclude that G is reductive (respectively semisimple). Applying property b), we conclude that W is indeed finite over S, which completes the proof of 5.2.

6. Applications: Extension of Certain Rigidity Properties of Tori to Groups of Unipotent Rank Zero

Proposition 6.1. — *Let S be a prescheme and let G be an S-group prescheme smooth over S, of finite presentation, with connected affine fibers, and whose center is of multiplicative type (for example, G of unipotent rank zero; cf. 4.1 a)). Then every closed normal subgroup K of G, of finite presentation over S and quasi-finite over S, is finite over S.*

Indeed, for every geometric point $\bar{s}$ above S, $(K_{\bar{s}})_{\mathrm{red}}$ is a finite étale subgroup of $G_{\bar{s}}$, normal and therefore central (since $G_{\bar{s}}$ is connected). Consequently

$$K' = Z \cap K,$$

where Z denotes the center of G, has the same underlying space as K, and it suffices to show that K' is finite over S. Now K' is a closed quasi-finite subgroup of the group of multiplicative type Z, and is therefore finite, as is immediate (locally on S, K' is contained in ${}_nZ$ for a suitable integer n, and ${}_nZ$ is finite over S).

Corollary 6.2. — *Let S and G be as above, let K be an S-group subprescheme of G, of finite presentation over S, normal and closed in G, and let s be a point of S. If K_s is finite (respectively is the unit group), there exists an open neighborhood U of s such that $K|_U$ is finite over S (respectively is the unit group).*

In view of the upper semicontinuity of the dimension of the fibers of K (Exposé VI_B, §4), after shrinking S we may already suppose that K is quasi-finite over S, and hence finite by 6.1. If, moreover, K_s is the unit group, then Nakayama's lemma easily implies that K is the unit group over $\mathrm{Spec}\,\mathcal{O}_s$, and hence over a neighborhood of s.

Proposition 6.3. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with affine, connected fibers, of unipotent rank zero, let H be an S-group prescheme, let s be a point of S, and let*

$$u : G \longrightarrow H$$

be an S-group morphism such that $\operatorname{Ker} u_s$ is central. Suppose, moreover, that H is of finite presentation over S, or that S is locally noetherian and G is locally of finite type over S. Then there exists an open neighborhood U of s such that, if $K = \operatorname{Ker} u$, $K|_U$ is central and of multiplicative type. Moreover,

$$G' = (G|_U)/(K|_U)$$

is representable, and the morphism

$$u' : G' \longrightarrow H|_U$$

induced from u by passage to the quotient is an immersion.

Proof. Let Z be the center of G, and put $K' = K \cap Z$.

a) K' is a subgroup of multiplicative type over a neighborhood U of s. Indeed, after making an étale extension over a neighborhood U of s, which is legitimate, we may suppose that G possesses a maximal torus T (4.1 c)). But then $T \cap K$ is a group of multiplicative type (Exposé XV, 8.3) whose fiber at s is central by hypothesis; therefore $T \cap K$ is central over a neighborhood of s (Exposé IX, 5.6 a)), and consequently coincides with K'.

b) Let us show that $K' = K$ over a neighborhood U of s. By a), we may already suppose that K' is of multiplicative type, and hence flat over S. Since $K'_s = K_s$, the natural immersion

$$K' \longrightarrow K$$

is then open over a neighborhood U of s (Exposé $\mathrm{VI_B}$, 2.6). If $t \in U$, the image of K_t in the group

$$G'_t = G_t/Z_t$$

is a finite étale group, normal in G'_t (since K_t is normal in G_t), hence central, and therefore reduced to the unit group (3.3 b)). This says that K_t is contained in Z_t, and hence is equal to K'_t, whence $K = K'$ over U.

c) The representability of G' now follows from 2.3; the fact that u' is an immersion is contained in 1.3 a), in view of 4.1 c) applied to G'.

Proposition 6.4. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let s be a point of S such that G_s is generated by its subtori (Exposé XII, 8.2) (for example, G_s affine of unipotent rank zero). Let H be an S-group prescheme, and let*

$$u, v : G \longrightarrow H$$

be two S-homomorphisms such that $u_s = v_s$ and u_s is central. Suppose, moreover, that H is of finite presentation over S, or that S is locally noetherian. Then there exists a neighborhood U of s such that

$$u|_U = v|_U.$$

We reduce as usual to the case where S is noetherian (to study the condition "G_s is generated by its subtori," one uses Exposé $\mathrm{VI_B}$, 7.4). Since G has connected fibers, to prove that $u = v$ over a neighborhood U of s, it suffices to prove that $u = v$ after reduction modulo every power of the maximal ideal of the local ring $\mathcal{O}_{S,s}$, which reduces us to the case where S is local artinian.

But then the functor of maximal subtori of G is representable by an S-scheme X, smooth and of finite type over S (Exposé XV, 8.17). Let T be the maximal subtorus of G_X universal for the functor X. The hypothesis on G_s means that the algebraic subgroup of G_s generated (Exposé $\mathrm{VI_B}$, §7) by the $\kappa(s)$-morphism

$$f : T_s \longrightarrow G_s$$

(the composite of the immersion $T_s \to G_{X_s}$ and the canonical projection $G_{X_s} \to G_s$) is all of G_s. But X_s is geometrically connected (for if $\overline{T}$ is a maximal torus of $G_{\bar{s}}$, and N its normalizer in $G_{\bar{s}}$, then $X_{\bar{s}}$ is isomorphic to $G_{\bar{s}}/N$), and the image of T_s under f contains the unit section of G_s. It follows from Exposé $\mathrm{VI_B}$, 7.4, that there exist an integer $N > 0$ and an S-morphism

$$f^N : T^N \longrightarrow G, \qquad T^N = T \times_S \cdots \times_S T \quad (N \text{ factors}),$$

where f^N depends only on f and the multiplication law of G, such that $(f^N)_s$ is surjective.

Consider the base change $X \to S$ and the restrictions of u_X and v_X to the subtorus T of G_X. By hypothesis,

$$u_{X_s} = v_{X_s},$$

and

$$u_{X_s} : T_{X_s} \longrightarrow H_{X_s}$$

is a central homomorphism. It then follows from Exposé IX, 5.1, that

$$u_X|_T = v_X|_T.$$

Making the definition of f^N explicit, and taking account of the fact that u and v are homomorphisms, one immediately deduces that

$$uf^N = vf^N.$$

But the $\kappa(s)$-morphism f_s^N is surjective and G_s is smooth, hence reduced; consequently f_s^N is generically flat. Since T is smooth over S, hence flat, there exists a nonempty open subset V of T^N such that

$$f^N|_V$$

is flat (EGA IV, 11.3.10). The image of V is an open subset W of G (EGA IV, 11.3.1), and

$$f^N : V \longrightarrow W$$

is faithfully flat and of finite presentation, hence covering for the fpqc topology. The equality $uf^N = vf^N$ then implies

$$u|_W = v|_W.$$

Since W is schematically dense in G (EGA IV, 11.10.10), and H is separated over S (Exposé $\mathrm{VI_A}$, 0.3), it follows immediately that $u = v$.

EXPOSÉ XVII

UNIPOTENT ALGEBRAIC GROUPS. EXTENSIONS BETWEEN UNIPOTENT GROUPS AND GROUPS OF MULTIPLICATIVE TYPE

by M. Raynaud[*]

0. Some notation[0]

In this chapter we shall be concerned chiefly with algebraic groups defined over a field k. The number $p \geq 0$ will always denote the characteristic of k; $\mathbf{F}_p$ will denote the prime field with p elements when $p > 0$; $\bar{k}$ will denote an algebraically closed extension of k; and q will denote a prime number different from p.

For every S-prescheme, $(\mathbb{G}_a)_S$ (respectively $(\mathbb{G}_m)_S$) denotes the additive group (respectively the multiplicative group) over S (cf. Exposé I, 4.3). For every integer $n > 0$, $(\boldsymbol{\mu}_n)_S$ (respectively $(\mathbb{Z}/n\mathbb{Z})_S$) denotes the group of n-th roots of unity (Exposé I, 4.4.4) (respectively the constant group over S associated with the abstract group $\mathbb{Z}/n\mathbb{Z}$ (Exposé I, 4.1)). The group $(\mathbb{Z}/n\mathbb{Z})_S$ is finite and étale over S; the group $(\boldsymbol{\mu}_n)_S$ is flat and finite over S, and is étale over S if and only if n is invertible on S (Exposé VIII, 2.1).

Suppose that S is a prescheme of characteristic $p > 0$. For every integer $n > 0$ and every S-group prescheme G, we denote by $F^n(G)$ the radicial S-subgroup prescheme of G equal to the kernel of the n-th iterate of the relative Frobenius morphism of G (Exposé VII_A). In particular, if $G = (\mathbb{G}_a)_S$, we set

$$F^n(G) = (\boldsymbol{\alpha}_{p^n})_S.$$

This is a radicial S-group, flat and finite over S, representing the following functor: for every S-prescheme S', $(\boldsymbol{\alpha}_{p^n})_S(S')$ is the set of $x' \in \Gamma(S', \mathcal{O}_{S'})$ such that $x'^{p^n} = 0$.

The already defined group $(\boldsymbol{\mu}_{p^n})_S$ is canonically isomorphic to $F^n((\mathbb{G}_m)_S)$.

When there is no ambiguity about the base scheme S, we shall simply write $\mathbb{G}_a$, $\mathbb{G}_m$, $\boldsymbol{\alpha}_{p^n}$, and so on, in place of $(\mathbb{G}_a)_S$, $(\mathbb{G}_m)_S$, $(\boldsymbol{\alpha}_{p^n})_S$, and so on.

[*] Cf. the note on page 1 of Exposé XV.

[0] Version x.y of 18 November 2008.

If G is a commutative S-group prescheme, then for every integer $n > 0$, ${}_nG$ denotes the subgroup prescheme of G equal to the kernel of raising to the n-th power in G.

For the reader's convenience, we have collected in an appendix some properties of algebraic groups proved in Exposés VI and VII, together with elementary properties of Hochschild cohomology that will be useful in this chapter.

1. Definition of unipotent algebraic groups

Definition 1.1.— *An algebraic group G defined over an algebraically closed field k is called* unipotent *if G admits a composition series whose successive quotients are isomorphic to algebraic subgroups of $(\mathbb{G}_a)_k$.*

Proposition 1.2.— *Let k be an algebraically closed field, let K be an algebraically closed extension of k, and let G be an algebraic group defined over k. Then G is unipotent if and only if G_K is unipotent.*

The necessity is clear, since a composition series remains a composition series after extension of the base.

The proof of sufficiency is standard. The field K is the filtered direct limit of its finitely generated k-subalgebras. By Exposé $\mathrm{VI_B}$, §10, one can find a finitely generated k-subalgebra A of K, a composition series G_i of G_S, where $S = \operatorname{Spec} A$, and immersions

$$u_i : H_i = G_i/G_{i+1} \longrightarrow (\mathbb{G}_a)_S.$$

To prove that G is unipotent, it therefore suffices to make a k-valued base change $A \to k$. This is possible because $\mathrm{Hom}_{k\text{-alg}}(A, k)$ is nonempty: A is a nonzero finitely generated algebra over the algebraically closed field k.

Definition 1.3.— *Let G be an algebraic group defined over a field k. We say that G is* unipotent *if there exists an algebraically closed extension $\bar{k}$ of k such that $G_{\bar{k}}$ is unipotent in the sense of Definition 1.1.*

By Proposition 1.2, this property is independent of the chosen algebraically closed extension $\bar{k}$.

Definition 1.4.— *Let G be an algebraic group defined over a field k, and let H be an algebraic group defined over an extension k' of k. We say that H is a* form *of G over k' if the algebraic groups $G_{\bar{k}'}$ and $H_{\bar{k}'}$ become isomorphic over an algebraic closure $\bar{k}'$ of k'. As above, this property is independent of the choice of $\bar{k}'$. We shall also say that H is a* twisted form *of G.*

We can now describe the algebraic subgroups of $\mathbb{G}_a$.

Proposition 1.5.— *Let k be a field of characteristic $p \geq 0$. An algebraic subgroup H of $(\mathbb{G}_a)_k$ is of one of the following types:*

(i) *$H = 0$;*

(ii) *$H = \mathbb{G}_a$;*

(iii) *if $p > 0$, H is an extension of a twisted constant group $(\mathbb{Z}/p\mathbb{Z})^r$ by a radicial group $\boldsymbol{\alpha}_{p^n}$, where $r, n \geq 0$ and $r + n > 0$. If, moreover, k is perfect, this extension is necessarily trivial.*

Proof. If $\dim H = 1$, then clearly $H = \mathbb{G}_a$. Otherwise H has dimension zero and is therefore an extension[3] of an étale group H'' by its identity component H', which is radicial. To describe $H''_{\bar{k}}$, it is enough to know the abstract group $H''(\bar{k})$, which is isomorphic to $H(\bar{k})$.

The latter is a finite subgroup of $\mathbb{G}_a(\bar{k})$. It is therefore zero when $p = 0$, and otherwise is of the form $(\mathbb{Z}/p\mathbb{Z})^r$, since it is annihilated by p. The group H' is closed in $\mathbb{G}_a$ and is defined by a single equation (the coordinate ring $\bar{k}[T]$ of $\mathbb{G}_a$ is principal). Since this equation has 0 as its only root over $\bar{k}$, it is of the form $T^n = 0$. Compatibility with the group law implies that $(T+T')^n$ belongs to the ideal generated by T^n and T'^n in $\bar{k}[T, T']$. Thus, if $p = 0$, then $n = 1$ and H' is the trivial group; while if $p > 0$, then $n = p^m$ and $H' = \boldsymbol{\alpha}_{p^m}$.

The last assertion of Proposition 1.5 follows more generally from the following lemma.

Lemma 1.6.— *If k is a perfect field, every extension H of an étale algebraic group H'' by a radicial group H' is trivial. Moreover, H'' has a unique lifting in H, namely H_{red}.*

Indeed, since k is perfect, the reduced k-scheme H_{red} is an algebraic subgroup of H (Exposé $\mathrm{VI_A}$, 0.2), is geometrically reduced and hence smooth over k (Exposé $\mathrm{VI_A}$, 1.3.1), and is therefore étale because H has dimension zero. To see that the canonical projection $H_{\mathrm{red}} \to H''$ is an isomorphism, it is enough to check this after the base-field extension $k \to \bar{k}$; there it is enough to check that it induces an isomorphism on $\bar{k}$-valued points, which is clear. The final assertion follows because every lifting of H'' in H, being étale over k, is reduced and hence necessarily contained in H_{red}.

Notice that $\boldsymbol{\alpha}_{p^n}$ is an iterated extension of groups isomorphic to $\boldsymbol{\alpha}_p$. Proposition 1.5 therefore gives the following corollary.

Corollary 1.7.— *Let G be an algebraic group defined over an algebraically closed field k. Then G is unipotent if and only if it has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$ when $p = 0$, and to one of the groups $\mathbb{G}_a, \mathbb{Z}/p\mathbb{Z}, \boldsymbol{\alpha}_p$ when $p > 0$. We shall call these the* elementary unipotent groups*.*

2. First properties of unipotent groups

Proposition 2.1.— *A unipotent algebraic group defined over a field k is affine over k.*

By fpqc descent for affine morphisms, it suffices to prove the assertion when k is algebraically closed. In that case, G is by definition an iterated extension of affine algebraic groups, and is therefore affine by Exposé $\mathrm{VI_B}$, 9.2 (viii), applied to affine morphisms.[4]

Proposition 2.2.—

(i) *The property that an algebraic group is unipotent is invariant under extension of the base field.*

(ii) *Every algebraic subgroup of a unipotent group is unipotent.*

(iii) *Every algebraic quotient group of a unipotent group is unipotent.*

(iv) *Every extension of a unipotent algebraic group by a unipotent algebraic group is likewise unipotent.*

[3] Editor's note: reference to Exposé $\mathrm{VII_A}$?

[4] Editor's note: verify this reference.

Proof. Assertion (i) follows immediately from Definitions 1.3 and Proposition 1.2. To prove the remaining assertions, we may suppose that k is algebraically closed. Assertion (iv) is then immediate from the definition.

Let G be a unipotent algebraic group, G' an algebraic subgroup of G, G'' an algebraic quotient group of G, and let G_i, $1 \leq i \leq n$, be a composition series of G such that

$$H_i = G_i/G_{i+1}$$

is an elementary unipotent group (Corollary 1.7).

For (ii), consider the composition series of G' induced by that of G: $G'_i = G_i \cap G'$. The group G'_i/G'_{i+1} identifies with an algebraic subgroup of H_i, hence is isomorphic to a subgroup of $\mathbb{G}_a$. It follows that G' is unipotent.

For (iii), consider the composition series of G'' obtained as the image of that of G: let G''_i be the image of G_i in G''. Then G''_i/G''_{i+1} is a quotient of H_i, so it remains only to prove the following lemma.

Lemma 2.3.— *Let H be an elementary unipotent group (Corollary 1.7) defined over a field k. Every algebraic quotient group H'' of H is either zero or isomorphic to H over k.*

Proof. If $H = \mathbb{G}_a$ in characteristic zero, or if $H = \boldsymbol{\alpha}_p$ or $\mathbb{Z}/p\mathbb{Z}$ in characteristic $p > 0$, it is enough to observe from Proposition 1.5 that H has no algebraic subgroups other than 0 and H. Indeed, a nonzero algebraic subgroup of $\boldsymbol{\alpha}_p$ is defined by a k-algebra of rank at least p over k, by Proposition 1.5(iii), and is therefore $\boldsymbol{\alpha}_p$.

Now let $H = \mathbb{G}_a$ in characteristic $p > 0$, so that there is an exact sequence

$$0 \longrightarrow N \longrightarrow \mathbb{G}_a \longrightarrow H'' \longrightarrow 0.$$

If $N = \mathbb{G}_a$, then $H'' = 0$. Otherwise, arguing by induction on the length of a composition series of N, we may suppose that $N = \boldsymbol{\alpha}_p$, or that N is a form of $(\mathbb{Z}/p\mathbb{Z})^r$.

(a) If $N \simeq \boldsymbol{\alpha}_p$, the proof of Proposition 1.5(iii) shows that N is necessarily the kernel of the Frobenius morphism F of $\mathbb{G}_a$, and the assertion follows from the exact sequence

$$0 \longrightarrow \boldsymbol{\alpha}_p \longrightarrow \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 0.$$

(b) If $N \simeq \mathbb{Z}/p\mathbb{Z}$, there is plainly an $a \in k^\times$ such that N is the closed subscheme of $\mathbb{G}_a = \operatorname{Spec} k[X]$ defined by $X^p - aX = 0$. The assertion then follows from the exact sequence

$$0 \longrightarrow N \longrightarrow \mathbb{G}_a \xrightarrow{P} \mathbb{G}_a \longrightarrow 0,$$

where P is the Artin–Schreier morphism $x \mapsto x^p - ax$.

(c) If N is a form of $(\mathbb{Z}/p\mathbb{Z})^r$, there is a finite Galois extension k'/k that trivializes N. By (b) and an evident induction on r, the quotient $\mathbb{G}_a/N$ is a form of $\mathbb{G}_a$ trivialized by k'. It remains to apply the next lemma.

Lemma 2.3 bis.— *Let k be a field, and let G be a k-algebraic group that is a form of $\mathbb{G}_a$, trivialized by a finite separable extension k'/k. Then G is isomorphic to $\mathbb{G}_a$.*

Proof of Lemma 2.3 bis. Indeed, the group of k'-automorphisms of the algebraic group $(\mathbb{G}_a)_{k'}$ is the group of nonzero homotheties $(k')^\times$ (*Bible*, Exposé 9, Lemma 1), and the Galois cohomology group $H^1(k'/k, \mathbb{G}_m)$ is zero (we may suppose that k' is a Galois extension of k), by Hilbert's Theorem 90. Lemma 2.3 bis then follows from the classification of

the k-forms of an algebraic group (cf. J.-P. Serre, *Cohomologie galoisienne*, Chapter III, 1.3).

This completes the proof of 2.2.

Proposition 2.4.— *Let k be a field, let M be a k-group of multiplicative type (Exposé VIII) and of finite type, and let U be a unipotent k-algebraic group. Then:*

(i) $\underline{\mathrm{Hom}}_{k\text{-}gr}(M,U) = \underline{e}$ *(a fortiori* $\mathrm{Hom}_{k\text{-}gr}(M,U) = e$*);*

(ii) $\mathrm{Hom}_{k\text{-}gr}(U,M) = e$.

To prove (i), we must establish that, for every prescheme S over k,

$$\mathrm{Hom}_{S\text{-gr}}(M_S, U_S) = e.$$

This follows from the next lemma.

Lemma 2.5.— *Let S be a prescheme, let M be an S-group of multiplicative type and of finite type over S, and let U be an S-group prescheme, of finite presentation over S, whose fibers are unipotent. Then* $\mathrm{Hom}_{S\text{-}gr}(M,U) = e$.

Proof. Under these hypotheses, for an S-group morphism $u : M \to U$ to be the unit homomorphism, it suffices that the restriction of u to the fibers of M over the points of S be the zero morphism (Exposé IX, 5.2). We are therefore reduced to the case in which S is the spectrum of a field, which we may moreover suppose to be algebraically closed. In view of Definition 1.1, it is enough to take $U = \mathbb{G}_a$, in which case the assertion has already been proved (Exposé XII, 4.4.1).

We now prove Proposition 2.4(ii). Let $u : U \to M$ be a k-group morphism. The image $u(U)$ is represented by an algebraic subgroup U'' of M (Exposé $\mathrm{VI_B}$, 5.4).[5] The group U'' is unipotent as a quotient of a unipotent group (2.2(iii)), and is of multiplicative type as a subgroup of a group of multiplicative type (cf. *Bible*, Exposé 4, Theorem 2, Corollary 1, or Exposé IX, 6.8). Thus U'' is the unit group by Proposition 2.4(i).

Remark 2.6.— *Keeping the notation of Proposition 2.4, it is no longer true in general that the functor* $\underline{\mathrm{Hom}}_{k\text{-}gr}(U,M)$ *is equal to* $\underline{e}$*. Indeed, let S be a prescheme such that* $\Gamma(S,\mathcal{O}_S)$ *contains a nonzero element ε with $\varepsilon^2 = 0$ (for example, the spectrum of the algebra of dual numbers over a ring A). For every S' over S, the map*

$$u \longmapsto 1 + \varepsilon_{S'} u$$

defines, functorially in S', a homomorphism from the additive group $\Gamma(S',\mathcal{O}_{S'})$ *to the multiplicative group* $\Gamma(S',\mathcal{O}_{S'}^{\times})$*. It therefore defines an S-group morphism*

$$(\mathbb{G}_a)_S \longrightarrow (\mathbb{G}_m)_S,$$

and, since $\varepsilon \neq 0$, this morphism is not zero.

Recall (Exposé $\mathrm{VII_A}$, §3) that when G is a commutative S-group prescheme, finite and flat over S, Cartier duality is available and G is reflexive in the sense of Exposé VIII, §1. More precisely, the functor

$$G \longmapsto \underline{\mathrm{Hom}}_{S\text{-gr}}(G, \mathbb{G}_m)$$

is represented

[5] Editor's note: verify this reference.

by a commutative S-group prescheme $D(G)$, finite and flat over S, and the canonical morphism

$$G \longrightarrow D(D(G))$$

is an isomorphism. In particular:

(i) $D(\boldsymbol{\mu}_n) \simeq \mathbb{Z}/n\mathbb{Z}$, and consequently $D(\mathbb{Z}/n\mathbb{Z}) \simeq \boldsymbol{\mu}_n$;

(ii) if S has characteristic $p > 0$, then $D(\boldsymbol{\alpha}_p) \simeq \boldsymbol{\alpha}_p$.

3. Unipotent groups acting on a vector space

Recall (Exposé I, 4.6.1) that if S is a prescheme and M is a sheaf of $\mathcal{O}_S$-modules, $\mathbf{W}(M)$ denotes the S-functor

$$(\mathrm{Sch}\,/S)^\circ \longrightarrow (\mathrm{Ens})$$

defined, for every S-prescheme S', by

$$\mathbf{W}(M)(S') = \Gamma\big(S', M \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}\big)\,.$$

Recall also that if an S-group G acts on an S-functor V, the S-functor V^G of invariants of V under G is the subfunctor of V whose S'-valued points, for S' over S, are those $x \in V(S')$ such that $x_{S''}$ is fixed under $G(S'')$ for every S'' over S'.

We then have the following lemma.

Lemma 3.1.— *Let S be a prescheme, and let G be an S-group prescheme, affine over S, defined by the quasi-coherent $\mathcal{O}_S$-algebra A. Suppose that G acts on a quasi-coherent sheaf M of $\mathcal{O}_S$-modules. Let*

$$\mu : M \longrightarrow A \otimes_{\mathcal{O}_S} M$$

be the comorphism defining the action of G on M (Exposé I, 4.7.2), and let

$$\nu : M \longrightarrow A \otimes_{\mathcal{O}_S} M$$

be the morphism $x \mapsto \mu(x) - 1 \otimes x$. Then:

(i) $\underline{M}^G(S) = \Gamma\,\mathrm{Ker}(\nu)$;

(ii) if S is the spectrum of a field k, then $\underline{M}^G$ is of the form $\mathbf{W}(N)$, where N is a vector subspace of M;

(iii) if S is the spectrum of a field k, every element x of M is contained in a finite-dimensional k-vector subspace of M stable under the action of G.

Proof. Assertion (iii) is recalled here for reference and was already proved in Exposé $\mathrm{VI_B}$, 11.2.

For (i), it is clear that $\underline{M}^G(S)$ contains $\Gamma\,\mathrm{Ker}(\nu)$. To prove the reverse inclusion, we may suppose that S is affine, with ring B. Let $m \in \underline{M}^G(S)$. For every B-algebra B' and every

$$u \in \mathrm{Hom}_{B\text{-alg}}(A, B')$$

(where u corresponds to an element of $G(\mathrm{Spec}\,B')$), we have

$$(u \otimes 1_M)\nu(m) = 0 \qquad \text{in } B' \otimes_B M.$$

Taking $B' = A$ and u to be the identity of A, we obtain $\nu(m) = 0$.

For (ii), let N be the kernel of ν, equal to $\underline{M}^G(k)$ by (i). Every k-prescheme S is flat over k; hence

$$\underline{M}^G(S) = \Gamma \operatorname{Ker}(\nu \otimes_k S) = \Gamma(N \otimes_k S).$$

Thus $\underline{M}^G$ is isomorphic to the functor $\mathbf{W}(N)$.

Proposition 3.2.— *Let G be a unipotent algebraic group, defined over a field k, acting on a k-vector space V. If $V \neq 0$, then $V^G \neq 0$.*

In view of 3.1(ii) and (iii), we may suppose that k is algebraically closed and that V is finite-dimensional over k.

Let

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$

be an exact sequence of algebraic groups, and suppose that G acts on a sheaf V (for the fpqc topology). Of course $V^G \subset V^{G'}$, and the quotient presheaf G/G' acts naturally on $V^{G'}$. But $V^{G'}$ is a sheaf, being the kernel of the familiar pair of morphisms

$$V \rightrightarrows \underline{\operatorname{Hom}}(G', V).$$

Consequently the sheaf associated with G/G', namely G'', acts on $V^{G'}$, and it is immediate that

$$\left(V^{G'}\right)^{G''} = V^G = \left(V^{G'}\right)^{G/G'}.$$

This remark reduces the proof of 3.2 to the case in which G is an elementary unipotent group (1.7).

(a) $G = \mathbb{G}_a$, $p = 0$. It follows from (BIBLE 4, Proposition 4) that a morphism from $\mathbb{G}_a$ to the linear group $\mathrm{GL}(V)$ is given by an exponential map

$$T \longmapsto \sum_{q=0}^{\infty} T^q \frac{n^q}{q!},$$

where n is a nilpotent endomorphism of V. But then $V \neq 0$ implies $\operatorname{Ker} n \neq 0$, and it is clear that every vector of V annihilated by n is fixed by G.

Suppose now that $p > 0$.

(b) $G = \boldsymbol{\alpha}_p$. Since $\boldsymbol{\alpha}_p$ is a radicial group of height 1 (Exposé VII_A, §7), giving a representation of $\boldsymbol{\alpha}_p$ in V amounts to giving a representation of the p-Lie algebra $\boldsymbol{\alpha}_p$ in $\mathfrak{gl}(V)$ (Appendix II, 2.2), that is, in this case, an element $X \in \mathrm{End}(V)$ such that $X^p = 0$ (Appendix II, 2.1). But then

$$V \neq 0 \quad \Longrightarrow \quad W = \operatorname{Ker}(X) \neq 0,$$

and, again by Appendix II, 2.2, one has $W = V^{\boldsymbol{\alpha}_p}$.

(c) $G = \mathbb{Z}/p\mathbb{Z}$. A representation of G in V is equivalent to the choice of an element $x \in \mathrm{Aut}(V)$ such that $x^p = 1$, that is, $(1 - x)^p = 0$. Thus x is of the form $1 + n$, with n nilpotent, and $W = \operatorname{Ker} n$ is fixed by x.

(d) $G = \mathbb{G}_a$. Let G_i, $i \in I$, be the increasing filtered family of étale algebraic subgroups of $\mathbb{G}_a$, hence groups isomorphic to $(\mathbb{Z}/p\mathbb{Z})^{r_i}$ (Proposition 1.5), and put $V_i = V^{G_i}$. Since V has finite nonzero dimension and each V_i is nonzero by (c), the decreasing filtered family of the V_i is stationary, and

$$W = \bigcap_{i \in I} V_i \neq 0.$$

We shall use the following lemma.

Lemma 3.3.— *The family of étale subgroups of $(\mathbb{G}_a)_S$, where S is a prescheme of characteristic $p > 0$, is schematically dense in G (Exposé IX, 4.1).*

By Exposé IX, 4.4, it suffices to prove the lemma when S is the spectrum of the prime field $\mathbb{F}_p$. In that case it is enough to consider the family of étale subgroups G_n, $n \geq 1$, defined by

$$X^{p^n} - X = 0.$$

This family is schematically dense in $(\mathbb{G}_a)_{\mathbb{F}_p}$, since it contains every closed point.

With this established, let us return to the proof of 3.2(d). If $w \in W$, its stabilizer in G is an algebraic subgroup of $\mathbb{G}_a$ containing G_i for every $i \in I$; it is therefore equal to $\mathbb{G}_a$ by 3.3. Consequently $W = V^G$.

Proposition 3.2 immediately gives the following.

Corollary 3.4.— *Let k be a field, and let G be a unipotent k-algebraic group acting on a finite-dimensional k-vector space V. Then V has a sequence of vector subspaces V_i, defined over k and stable under G,*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V,$$

such that G acts trivially on V_{i+1}/V_i. Moreover, the V_i may be chosen so that V_{i+1}/V_i has dimension 1.

We shall now summarize and supplement the properties of unipotent groups already proved in the following theorem.

Theorem 3.5.— *Let G be an algebraic group defined over a field k. The following properties are equivalent:*

(i) G is unipotent.

(ii) G has a composition series defined over k, whose successive quotients are isomorphic to $\mathbb{G}_a$ if $p = 0$, and, respectively, to $\boldsymbol{\alpha}_p$, $\mathbb{G}_a$, or a twisted form of $(\mathbb{Z}/p\mathbb{Z})^r$ (1.4) if $p > 0$.

(iii) As in (ii), with the additional requirement that the composition series be central.

(iv) G has a characteristic composition series (Exposé VI$_B$) defined over k, whose successive quotients are isomorphic to $(\mathbb{G}_a)^r$ if $p = 0$, and, respectively, to $(\boldsymbol{\alpha}_p)^r$, a twisted form of $(\mathbb{G}_a)^s$, or a twisted form of $(\mathbb{Z}/p\mathbb{Z})^t$, taken in this order, if $p > 0$.

(v) G is isomorphic to an algebraic subgroup of the group $\mathrm{Trigstr}(n)_k$ of strictly upper-triangular matrices in the linear group $\mathrm{GL}(n)_k$, for a suitable integer $n \geq 0$.

(vi) G is affine, and for every linear representation of G on a nonzero finite-dimensional k-vector space V, one has $V^G \neq 0$.

Proof. The implication (i)⇒(vi) follows from 2.1 and 3.2.

For (vi)⇒(v), since the algebraic group G is affine, it is an algebraic subgroup of a suitable linear group $\mathrm{GL}(V)$ (Exposé VI$_\mathrm{B}$, 11.3). Applying 3.4 to the representation of G on V defined by this embedding gives (v).

For (v)⇒(iii), the algebraic group $\mathrm{Trigstr}(n)$ is known to have a central composition series whose successive quotients are isomorphic to $\mathbb{G}_a$. The composition series induced on G gives property (iii), in view of 1.5.

The implications

$$\text{(iii)} \Rightarrow \text{(ii)} \Rightarrow \text{(i)} \qquad \text{and} \qquad \text{(iv)} \Rightarrow \text{(i)}$$

are clear.

We shall prove (i)⇒(iv) shortly, but first note some consequences of what has already been proved.

Definition 3.6.— *A p-Lie algebra $\mathfrak{g}$, $p > 0$ (cf. Exposé VII$_A$, §5), is called* unipotent *if the map*

$$x \longmapsto x^{(p)}$$

is nilpotent; that is, if for every $x \in \mathfrak{g}$ there is an integer $n > 0$ such that $x^{(p^n)} = 0$.

Corollary 3.7.— *A unipotent algebraic group G is nilpotent (Exposé VI$_B$, §8); its Lie algebra $\mathfrak{g}$ is nilpotent (Bourbaki,* Groupes et algèbres de Lie*, Chap. I, §4) and is isomorphic to a Lie algebra of nilpotent endomorphisms of a finite-dimensional vector space. In characteristic $p > 0$, $\mathfrak{g}$ is a unipotent p-Lie algebra (3.6).*

Since (i) implies (v), it is enough to prove 3.7 when $G = \mathrm{Trigstr}(n)$. We have already used the fact that $\mathrm{Trigstr}(n)$ is a nilpotent algebraic group. Moreover, the Lie algebra $\mathfrak{h}$ of $\mathrm{Trigstr}(n)$ consists of the upper-triangular endomorphisms of V that have zeros on the main diagonal. These endomorphisms are therefore nilpotent, and consequently $\mathfrak{h}$ is nilpotent (Bourbaki, *loc. cit.*, Chap. I, §4, Cor. 3). If $p > 0$, the p-th power in the p-Lie algebra $\mathfrak{gl}(V) = \mathrm{End}(V)$ coincides with the p-th power of endomorphisms of V (Exposé VII$_\mathrm{A}$, 6.4.4); hence $\mathfrak{h}$ is unipotent.

Corollary 3.8.— *Let k be an algebraically closed field, and let G be a smooth affine algebraic group over k. Then the following properties are equivalent:*

(i) G is unipotent.

(ii) $G(k)$ consists of unipotent elements (BIBLE 4, Prop. 4, Cor. 1); that is, G is unipotent in the sense of BIBLE.

The implication (i)⇒(ii) follows because, by 3.2, (i)⇒(v), G is isomorphic to an algebraic subgroup of a group $\mathrm{Trigstr}(n)$.

For (ii)⇒(i), suppose that G is a unipotent algebraic group in the sense of BIBLE, and write G^0 for its identity component. The maximal tori of G^0, being made up of unipotent elements, are trivial; thus G^0 is equal to each of its Cartan subgroups. It follows that G^0 is solvable (BIBLE 6, Th. 6), and hence triangularizable (BIBLE 6, Th. 1). In short, G^0 is an algebraic subgroup of a group $\mathrm{Trigstr}(n)$, and is therefore unipotent in the sense of this Exposé.

The group $(G/G^0)(k)$ is a finite group consisting of unipotent elements. It is therefore trivial if $p = 0$, and is a finite p-group if $p > 0$ (BIBLE 4, Prop. 4). But then G/G^0 is unipotent in the sense of this Exposé, since it is an iterated extension of groups isomorphic to $\mathbb{Z}/p\mathbb{Z}$. This proves that G is unipotent.

Completion of the proof of 3.5. It remains to prove (i)⇒(iv).

(a) $p > 0$. Consider the increasing sequence of algebraic subgroups of G

$$\{e\} \subset F(G) \subset F^2(G) \subset \cdots \subset F^n(G) \subset G^0 \subset G.$$

This gives a characteristic composition series of G (Appendix II, 1), and for n sufficiently large, $G/F^n(G)$ is smooth (Appendix II, 3.1). Thus its successive quotients are, in order:

(1) radicial groups of height 1;

(2) a smooth connected group;

(3) an étale group.

To prove (i)⇒(iv), it is therefore enough to prove the following lemma.

Lemma 3.9.— *Let G be a unipotent algebraic group defined over a field k of characteristic $p > 0$. Then G has a characteristic composition series, defined over k, whose successive quotients are isomorphic to:*

(i) $(\boldsymbol{\alpha}_p)^r$*, if G is radicial;*

(ii) a twisted form of $(\mathbb{G}_a)^r$*, if G is smooth and connected;*

(iii) a twisted form of $(\mathbb{Z}/p\mathbb{Z})^r$*, if G is étale.*

Proof of 3.9(i). The group G is radicial. Filtering G by the $F^n(G)$, we reduce to the case in which G is radicial of height 1. Since G is nilpotent (3.5, (i)⇒(iii)), and since the center of an algebraic group is representable (Exposé VIII, 6.5(e)), we may consider the ascending central series of G. This series is plainly characteristic in G, and reduces us to the case in which G is also commutative.

Put $\mathfrak{g} = \mathrm{Lie}(G)$. The p-th-power morphism π is then additive on $\mathfrak{g}$ (Exposé VII_A); we shall reduce to the case in which it is zero. For every prescheme S over k, set

$$\mathfrak{g}_S = \mathfrak{g} \otimes_k \mathcal{O}_S,$$

and let $\mathfrak{h}_S$ be the subsheaf of abelian groups of $\mathfrak{g}_S$ that is the image of $\mathfrak{g}_S$ under π_S. Finally, let $\overline{\mathfrak{h}}_S$ be the subsheaf of $\mathcal{O}_S$-modules of $\mathfrak{g}_S$ generated by $\mathfrak{h}_S$. It is clear that

$$\overline{\mathfrak{h}}_S = \overline{\mathfrak{h}}_k \otimes_k \mathcal{O}_S$$

and that $\overline{\mathfrak{h}}_S$[4] is a characteristic p-Lie subalgebra of $\mathfrak{g}_S$; that is, it is stable under the S-functor $\mathrm{Aut}_{p\text{-Lie}}(\mathfrak{g})$. It follows from Appendix II, 2.2 that $\overline{\mathfrak{h}}_k$ is the Lie algebra of an algebraic subgroup H of G that is characteristic in G.

Moreover, in view of 3.7 and Lemma 3.9 bis below, if $G \neq \{e\}$, then $H \neq G$, since $\overline{\mathfrak{h}}_k \neq \mathfrak{g}$. On the other hand, if $G/H = G''$, then

$$\mathrm{Lie}\, G'' = \mathfrak{g}/\overline{\mathfrak{h}}_k \qquad \text{(Appendix II, 2.2)},$$

and consequently the p-th power is zero in $\mathrm{Lie}\, G''$. Induction on $\dim \mathrm{Lie}\, G$ therefore reduces us to the case in which $\mathrm{Lie}\, G$ is a p-Lie algebra whose p-th power is zero. But then $\mathrm{Lie}\, G''$ is isomorphic to $\mathrm{Lie}((\boldsymbol{\alpha}_p)^r)$ for some suitable integer $r \geq 0$ (Appendix II, 2.1), and hence G'' is isomorphic to $(\boldsymbol{\alpha}_p)^r$ (Appendix II, 2.2). It remains to prove the following lemma.

Lemma 3.9 bis.— *Let k be a field of characteristic $p > 0$, let $\mathfrak{g}$ be a commutative unipotent p-Lie algebra (3.6) of finite dimension over k, and let $\mathfrak{h}$ be the p-Lie subalgebra of $\mathfrak{g}$ generated by the image of the p-th power in $\mathfrak{g}$. If $\mathfrak{g} \neq 0$, then $\mathfrak{h} \neq \mathfrak{g}$.*

Indeed, since $\mathfrak{g}$ is commutative, $\mathfrak{h}$ is simply the k-vector subspace of $\mathfrak{g}$ generated by the $X^{(p)}$, $X \in \mathfrak{g}$. If $\mathfrak{g} \neq 0$ and is unipotent, there exists an $X \in \mathfrak{g}$, $X \neq 0$, such that $X^{(p)} = 0$.

[4]Editor's note: $\mathfrak{h}$ has been changed to $\overline{\mathfrak{h}}$.

Let $X_1, \ldots, X_n$ be a basis of a complement in $\mathfrak{g}$ to the line kX. The Lie algebra $\mathfrak{h}$ is then the k-vector subspace of $\mathfrak{g}$ generated by

$$X_1^{(p)}, \ldots, X_n^{(p)};$$

it therefore has dimension at most $n = \dim \mathfrak{g} - 1$.

Proof of 3.9(ii). The group G is smooth and connected. In this case the descending central series of G is represented by smooth, connected, characteristic algebraic subgroups G_i (Exposé $\mathrm{VI_B}$, 8.3 and 7.4), and $G_i = 0$ for sufficiently large i, since G is nilpotent (3.5, (i)⇒(iii)). It is enough to prove 3.9 for the groups G_i/G_{i+1}, which reduces us to the case in which G is also commutative.

For every integer $n > 0$, let G_n be the algebraic subgroup of G that is the image of G under the p^n-th-power morphism. The group G_n is therefore smooth, connected, and characteristic; and it follows from Definition 1.1 of unipotent groups that $G_n = 0$ for sufficiently large n. Replacing G by G_n/G_{n+1}, we may further suppose that G is annihilated by raising to the p-th power. But then, by J.-P. Serre, *Groupes algébriques et corps de classes*, Chap. VII, Prop. 11, G is a form of $(\mathbb{G}_a)^r$ for a suitable integer r.

Proof of 3.9(iii). The group G is étale. Proceeding as in (ii), we reduce first to the case in which G is commutative, and then to the case in which G is annihilated by p. But then

$$G_{\overline{k}} \simeq (\mathbb{Z}/p\mathbb{Z})^r.$$

Proof of 3.5, (i)⇒(iv), in case (b), $p = 0$. The group G is then smooth and connected; proceeding as in 3.9(ii), we reduce to the case in which G is also commutative. We then have the following more precise result.

Lemma 3.9 ter.— *Let k be a field of characteristic 0, let G be a commutative unipotent k-algebraic group, and put $\mathfrak{g} = \operatorname{Lie} G$. There is a canonical isomorphism*

$$\exp : \mathbf{W}(\mathfrak{g}) \xrightarrow{\sim} G.$$

The morphism exp *is the unique homomorphism $\mathbf{W}(\mathfrak{g}) \to G$ that induces the identity on Lie algebras.*

Since G is unipotent, it can be realized as an algebraic subgroup of $\mathrm{Trigstr}(n)$, for a suitable integer n (3.5, (i)⇒(v)). Choosing such an embedding identifies $\mathfrak{g}$ with a Lie subalgebra of $\mathfrak{gl}(n)$ consisting of nilpotent endomorphisms. We thereby obtain a k-morphism

$$\begin{aligned} \exp : \mathbf{W}(\mathfrak{g}) &\longrightarrow \mathrm{GL}(n), \\ T &\longmapsto \sum_{i \geq 0} \frac{T^i}{i!}. \end{aligned}$$

Since G is commutative, the morphism exp is a homomorphism. Let G' be the algebraic group that is the image of $\mathbf{W}(\mathfrak{g})$ under exp. Under the canonical identification $\operatorname{Lie} \mathbf{W}(\mathfrak{g}) = \mathfrak{g}$, the tangent linear map of exp is simply the inclusion $\mathfrak{g} \hookrightarrow \mathfrak{gl}(n)$. Hence

$$\operatorname{Lie}(G \cap G') = \mathfrak{g} \cap \mathfrak{g} = \mathfrak{g}.$$

Since $G \cap G'$ is smooth (Exposé $\mathrm{VI_B}$, 1.6.1) and G is connected—being an iterated extension of groups $\mathbb{G}_a$ (3.5, (i)⇒(ii))—one necessarily has $G = G'$. The kernel $\ker(\exp)$ is an étale unipotent group, and hence is the trivial group. Consequently exp is an isomorphism from $\mathbf{W}(\mathfrak{g})$ onto G.

If $h : \mathbf{W}(\mathfrak{g}) \to G$ is another homomorphism such that $\mathrm{Lie}(h)$ is the identity map of $\mathfrak{g}$, then $h - \exp$ is a homomorphism, since G is commutative, and its tangent linear map is zero. Because k has characteristic 0 and $\mathbf{W}(\mathfrak{g})$ is connected, Cartier's theorem again implies that $h = \exp$.

Proposition 3.10.— *Let G be an algebraic group defined over an algebraically closed field k. The following properties are equivalent:*

(i) Every homomorphism from a group M of multiplicative type into G is trivial.

(i bis) G has no nontrivial subgroup of multiplicative type.

(ii) (a) If $p = 0$, then $G(k)$ has no point of finite order other than e.

(b) If $p \neq 0$, then:

- *for every prime number $q \neq p$, if $x \in G(k)$ and $x^q = e$, then $x = e$;*
- *for every $X \in \mathfrak{g} = \mathrm{Lie}\, G$ such that $X^{(p)} = X$, one has $X = 0$.*

(ii bis) (a) If $p = 0$, the condition is the same as in (ii)(a).

(b) If $p \neq 0$, then:

- *every finite subgroup of $G(k)$ is a p-group;*
- *for every $X \in \mathfrak{g} = \mathrm{Lie}\, G$ such that $X^{(p)} = X$, one has $X = 0$.*

Proof. The equivalence (i)⇔(i bis) follows because the image of a group of multiplicative type is of multiplicative type (Exposé IX, 2.7).

The equivalence (ii)⇔(ii bis) follows from the following observation. If an ordinary finite group H has order that is not a power of p, then there exist a prime number $q \neq p$ and an element $x \in H$, distinct from e, such that $x^q = e$ (Sylow's theorem; cf. J.-P. Serre, *Corps locaux*, Chapter IX, §2).

The implication (i)⇒(ii) follows from the next lemma.

Lemma 3.11.— *Let G be an algebraic group defined over a field k.*

(a) If k contains the n-th roots of unity, where n is prime to p, then

$$\mathrm{Hom}_{k\text{-gr}}(\mu_n, G) \simeq \mathrm{Hom}_{k\text{-gr}}(\mathbb{Z}/n\mathbb{Z}, G) \simeq {}_nG(k),$$

where ${}_nG(k)$ denotes the points of order n in $G(k)$.

(b) If $p > 0$, then

$$\mathrm{Hom}_{k\text{-gr}}(\mu_p, G) \simeq \{X \in \mathfrak{g} = \mathrm{Lie}\, G \mid X^{(p)} = X\}.$$

Indeed, for (a), note that μ_n is then isomorphic over k to $\mathbb{Z}/n\mathbb{Z}$; assertion (b) follows from Appendix II, 2.1.

For (ii)⇒(i bis), Lemma 3.11 shows that (ii) is equivalent to the assertion that, for every prime number r, the group G contains no subgroup μ_r. The implication now follows from:

Lemma 3.12.— *Let G be a diagonalizable S-group, of finite type over S, and distinct from the trivial group. There exist a prime number r and a subgroup of G isomorphic to $(\mu_r)_S$.*

Write $G = D_S(M)$, where M is a finitely generated abelian group, hence an extension of a free group M'' by a finite group M'. If $M'' \neq 0$, it is clear that M has quotients

isomorphic to $\mathbb{Z}/r\mathbb{Z}$ for every integer r. If $M'' = 0$, then M' has a quotient isomorphic to $\mathbb{Z}/r\mathbb{Z}$ for every prime number r dividing the order of M. The lemma follows by duality.

We saw in Proposition 2.4 that a unipotent group satisfies the equivalent conditions of 3.10. The purpose of the next section is to prove the converse.

4. A Characterization of Unipotent Groups

As announced, we shall prove that an algebraic group G defined over an algebraically closed field k, and containing no nontrivial subgroup of multiplicative type, is unipotent. In fact, it is enough that G contain none of certain particular "elementary" subgroups of multiplicative type, which depend on the hypotheses imposed on G. Before stating the general theorem, let us study some special cases in detail.

4.1. Smooth, Connected, Affine Algebraic Groups

Proposition 4.1.1.— *Let k be a field, let G be a smooth, connected, affine k-algebraic group, and put $\mathfrak{g} = \operatorname{Lie} G$. The following properties are equivalent:*

(i) G is unipotent.

(ii) G has a central composition series whose successive quotients are forms of $\mathbb{G}_a$.

(iii) G has a central, characteristic composition series whose successive quotients are forms of $(\mathbb{G}_a)^r$.

(iv) There is an integer $n > 1$ such that $G_{\overline{k}}$ contains no subgroup isomorphic to μ_n.

(v) Every maximal torus of G is the trivial group.

Suppose, moreover, that G is an algebraic subgroup of a linear group $\mathrm{GL}(n)$. The preceding conditions are then also equivalent to:

(vi) $\mathfrak{g} \subset \mathfrak{gl}(n)$ consists of nilpotent endomorphisms.

(vii) $\mathfrak{g}$ is nilpotent, and its center contains no nonzero semisimple endomorphism.

Proof. The implication (ii)⇒(i) is clear, and (i)⇒(iii) was proved in 3.9. The implication (iii)⇒(ii) will follow from the next lemma.

Lemma 4.1.2.— *Let k be a field, and let G be a k-algebraic group that is a form of $(\mathbb{G}_a)^r$. Then:*

(a) G can be realized as an algebraic subgroup of $(\mathbb{G}_a)^n$, for a suitable integer n.

(b) G has a composition series whose successive quotients are forms of $\mathbb{G}_a$.

Indeed, by hypothesis there is an extension k' of k such that $G_{k'}$ is isomorphic to $(\mathbb{G}_{a,k'})^r$. By the principle of finite extension (EGA IV, 9.1.1), we may assume that k'/k is finite. For (a), it then suffices to consider the canonical closed immersion (EGA V[5])

$$G \longrightarrow \prod_{k'/k} (G_{k'})_{/k'} \xrightarrow{\sim} (\mathbb{G}_{a,k})^n, \qquad n = r \deg(k'/k).$$

[5]Editor's note: provide another reference here...

To prove (b), in view of (a), we may assume that G is a closed subgroup of $G' = (\mathbb{G}_a)^n$. If $G \neq 0$, there is a hyperplane $\mathfrak{h}$ in $\mathfrak{g}' = \operatorname{Lie} G'$ that does not contain $\mathfrak{g}$. Let H be the vector subgroup $\mathbf{W}(\mathfrak{h})$ of $\mathbf{W}(\mathfrak{g}') = G'$. Since H is defined by one equation in G', the intersection $G \cap H$ is defined by one equation in G, and

$$\begin{aligned} \dim G - 1 &\leq \dim(G \cap H) \\ &\leq \dim \operatorname{Lie}(G \cap H) = \dim_k(\mathfrak{g} \cap \mathfrak{h}) \\ &= \dim_k \mathfrak{g} - 1 = \dim G - 1. \end{aligned}$$

Thus $\dim(G \cap H) = \dim \operatorname{Lie}(G \cap H)$, and consequently $G \cap H$ is smooth. The group

$$G_1 = (G \cap H)^0$$

is a smooth, connected algebraic subgroup of G such that G/G_1 is smooth, connected, and one-dimensional, hence is a form of $\mathbb{G}_a$ (4.1, (i)⇒(iii)). Induction on the dimension of G completes the proof.

Before continuing the proof of 4.1, note that the equivalence (i)⇔(ii), together with 2.3 bis, gives the following corollary.

Corollary 4.1.3.— *If k is a perfect field, a smooth, connected k-algebraic group is unipotent if and only if it has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.*

Continuation of the proof of 4.1. The implication (i)⇒(iv) follows from 2.4(i).

For (iv)⇒(v), by Exposé XIV, 4.1, the group G has a maximal torus T defined over k. If $r = \dim T$, then

$$({}_nT)_{\overline{k}} \simeq (\mu_n)^r.$$

It follows that $r = 0$.

The implication (v)⇒(i) follows from the observation made in the proof of 3.8.

For (i)⇒(vi), by 3.4 the group G is in fact contained in an algebraic subgroup of $\mathrm{GL}(n)$ isomorphic to $\mathrm{Trigstr}(n)$. Hence $\mathfrak{g}$ consists of nilpotent endomorphisms.

For (vi)⇒(vii), the Lie algebra $\mathfrak{g}$ is nilpotent by Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §4, Corollary 3).

For (vii)⇒(v), let T be a maximal subtorus of G (Exposé XIV, 1.1), and let $\mathfrak{t}$ be its Lie algebra. The embedding of G in $\mathrm{GL}(n)$ defines a representation of T, which is necessarily semisimple; this can be seen after passing to an algebraic closure $\overline{k}$ of k and applying Exposé I, 4.7.3. Thus, if $X \in \mathfrak{t}$, then X is a semisimple endomorphism in $\mathfrak{gl}(n)$. It follows at once that

$$\operatorname{ad} X : \ Y \longmapsto [X, Y]$$

is a semisimple endomorphism of $\mathfrak{gl}(n)$, and hence of $\mathfrak{g}$. This endomorphism is also nilpotent, since $\mathfrak{g}$ is nilpotent. Therefore $\operatorname{ad} X = 0$, so X is central. Consequently $\mathfrak{t}$ is central and consists of semisimple endomorphisms; by hypothesis it is zero. A fortiori, T is the trivial group.

Remark 4.1.4.—

(a) *We shall give below (4.3.1) an infinitesimal characterization of unipotent groups in characteristic $p > 0$ that is independent of an embedding in* $\mathrm{GL}(n)$.

(b) *When k is perfect, conditions (ii) and (iii) of 4.1.1 simplify by virtue of the following lemma.*

Lemma 4.1.5.— *If k is a perfect field, every k-algebraic group G that is a form of $(\mathbb{G}_a)^r$ is isomorphic to $(\mathbb{G}_a)^r$.*

The lemma follows from 3.9 ter when the characteristic p of k is zero, and from 2.3 bis when $r = 1$. In the general case ($p > 0$), realize G as an algebraic subgroup of $(\mathbb{G}_a)^n$, for a suitable integer n (4.1.2), and argue by induction on $n - r$. If $r = n$, then indeed $G = (\mathbb{G}_a)^r$. Otherwise, the quotient $(\mathbb{G}_a)^n/G$ is a nonzero smooth, connected, unipotent group. By 4.1.1, (i)⇒(ii), and 2.3 bis, it has a composition series whose quotients are isomorphic to $\mathbb{G}_a$. It follows that there is a smooth, connected algebraic subgroup G_1 of $(\mathbb{G}_a)^n$, containing G, such that

$$(\mathbb{G}_a)^n/G_1 = \mathbb{G}_a.$$

By induction, it suffices to show that G_1 is isomorphic to $(\mathbb{G}_a)^{n-1}$.

It is immediate that a homomorphism

$$\operatorname{Spec} k[X_1, \dots, X_n] = (\mathbb{G}_a)^n \longrightarrow \mathbb{G}_a = \operatorname{Spec} k[T]$$

is defined by an additive polynomial of the form

$$\sum_{i,j} a_{i,j} X_i^{p^j}.$$

Since G_1 is smooth, the linear part of this polynomial is nonzero. After a linear change of the coordinates X_i, we may assume that G_1 is the algebraic subgroup of $(\mathbb{G}_a)^n$ defined by

$$(*) \qquad P(X) = -X_1 + \sum_{j=1}^{q} a_j X_1^{p^j} + Q(X_2, \dots, X_n) = 0,$$

where

$$Q(X_2, \dots, X_n) = \sum_{\substack{i>1 \\ j>0}} b_{i,j} X_i^{p^j}.$$

We now argue by induction on the degree of P. If $\deg P = 1$, it is clear that

$$G_1 \xrightarrow{\sim} (\mathbb{G}_a)^{n-1}.$$

Otherwise, since k is perfect, one has $Q(X) = Q_1(X)^p$, and we may define an endomorphism v of $(\mathbb{G}_a)^n$ by

$$X_i \longmapsto X_i \qquad \text{for } i > 1,$$
$$X_1 \longmapsto \sum_{j=1}^{q} a_j^{1/p} X_1^{p^{j-1}} + Q_1(X).$$

It is clear that v induces an isomorphism on G_1, and that $v(G_1)$ is defined in $(\mathbb{G}_a)^n$ by the equation

$$(*_1) \qquad P_1(X) = -X_1 + \sum_{j=1}^{q} a_j^{1/p} X_1^{p^j} + Q_1(X_2, \dots, X_n) = 0.$$

We distinguish two cases.

First case. $\deg Q = \deg P > p^q$. Then $\deg P_1 < \deg P$, and the induction hypothesis applies.

Second case. $\deg P = p^q$, with $a_q \neq 0$. Then

$$\deg P_1 = \deg P = p^q \qquad \text{and} \qquad \deg Q_1 < p^q.$$

We cannot have $Q = 0$, for otherwise G_1 would not be connected. If the polynomial Q_1 has no linear part, the preceding transformation may be repeated. Continuing in this fashion, we eventually obtain an equation of the form

$$(*_s) \qquad P_s(X) = -X_1 + \sum_{j=1}^{q} a_j^{1/p^s} X_1^{p^j} + Q_s(X),$$

where

$$Q_s(X) = Q(X_2, \ldots, X_n)^{1/p^s}$$

is an additive polynomial with a nonzero linear part, and moreover $\deg Q_s < p^q$. Suppose, for example, that the coefficient of X_2 in Q_s is nonzero, and let $-L$ be the linear part of $P_s(X)$. After a linear change of coordinates, the equation of G_1 becomes

$$P'(X) = -L + \sum_{j=1}^{q} a_j^{1/p^s} X_1^{p^j} + Q'(L, X_3, \ldots, X_n),$$

where Q' is an additive polynomial without a linear part and $\deg Q' < p^q$. We are thereby reduced to the first case.[6]

4.2. Radicial Groups

Proposition 4.2.1.— *Let G be a radicial algebraic group defined over a field k of characteristic $p > 0$. The following conditions are equivalent:*

(i) G is unipotent.

(ii) G has a central composition series whose successive quotients are isomorphic to $\boldsymbol{\alpha}_p$.

(iii) G has a central, characteristic composition series whose successive quotients are isomorphic to $(\boldsymbol{\alpha}_p)^r$.

(iv) $G_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$.

(v) $\mathfrak{g} = \operatorname{Lie} G$ is a unipotent p-Lie algebra (3.6).

Proof. The implications (iii)⇒(ii)⇒(i) are clear; (i)⇒(iii) is 3.9(i), and (i)⇒(iv) follows from 2.4(i).

We shall need the following lemma on abelian p-Lie algebras.

Lemma 4.2.2.— *Let $\mathfrak{g}$ be an abelian, finite-dimensional p-Lie algebra over a perfect field k. Then $\mathfrak{g}$ can be written uniquely as a direct sum*

$$\mathfrak{g} = \mathfrak{r} \oplus \mathfrak{u}$$

of a p-Lie subalgebra $\mathfrak{r}$ on which the p-th power is bijective and a unipotent p-Lie subalgebra $\mathfrak{u}$ (3.6). The algebra $\mathfrak{r}$ is called the reductive part *of $\mathfrak{g}$, and $\mathfrak{u}$ its* unipotent part.

[6] Editor's note: replace X_1 by L.

The formation of $\mathfrak{r}$ *and* $\mathfrak{u}$ *is compatible with extension of the field* k*. If, moreover,* k *is algebraically closed, then* $\mathfrak{r}$ *has a basis* e_i *such that*

$$e_i^{(p)} = e_i.$$

The proof of this lemma is easy and is left to the reader; see Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §1, Exercise 23. We merely note that $\mathfrak{u}$ is the kernel of a suitable iterate of the map $X \mapsto X^{(p)}$, while $\mathfrak{r}$ is the image of that same iterate.

We now prove (iv)⇒(v). After extending the base field if necessary, we may suppose that k is algebraically closed. Let $X \in \mathfrak{g}$, and let $\mathfrak{h}$ be the p-Lie subalgebra of $\mathfrak{g}$ generated by X. The algebra $\mathfrak{h}$ is plainly commutative, and its reductive part (4.2.2) is zero. Otherwise, by the cited result, $\mathfrak{h}$ would contain an element $Y \neq 0$ such that $Y^{(p)} = Y$, and consequently, by Appendix II, 2.1 and 2.2, G would contain a subgroup isomorphic to $\boldsymbol{\mu}_p$, contrary to the hypothesis. Thus $\mathfrak{h}$, and hence $\mathfrak{g}$, is a unipotent p-Lie algebra.

It remains to prove (v)⇒(i), the least immediate implication in 4.2.1.

(a) The case in which G *has height* 1 (Exposé $\mathrm{VII_A}$, 4.1.3). Since G is radicial, it is affine and hence isomorphic to an algebraic subgroup of a linear group $\mathrm{GL}(V)$ (Exposé $\mathrm{VI_B}$, §11). This embedding identifies $\mathfrak{g}$ with a p-Lie subalgebra of $\mathrm{End}(V)$, and the p-th power of X in $\mathfrak{g}$ agrees with the p-th power of the endomorphism X (Exposé $\mathrm{VII_A}$, 6.4.4). Since $\mathfrak{g}$ is unipotent by hypothesis, it is therefore an algebra of nilpotent endomorphisms of V. By Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §4, Theorem 1), it is a Lie subalgebra of the Lie algebra $\mathfrak{h}$ of the group of strictly upper-triangular matrices $\mathrm{Trigstr}(n)$, relative to a suitable basis of V. Since G has height 1, Appendix II, 2.2 then shows that G itself is an algebraic subgroup of $\mathrm{Trigstr}(n)$, and hence is unipotent by 3.5(v)⇒(i).

(b) The general case. We argue by induction on the height h of G.* The case $h = 1$ has just been treated. Suppose that $h > 1$, and put

$$G' = {}_F G, \qquad G'' = G/G'.$$

The group G' has height 1, and $\mathrm{Lie}\, G' = \mathrm{Lie}\, G$ is unipotent. Consequently G' is unipotent by (a). To prove that G is unipotent, it is therefore enough to show that G'' is unipotent (2.2). But G'' has height $h - 1$, so the induction hypothesis reduces us to proving that $\mathrm{Lie}\, G''$ is unipotent.

In view of (iv)⇒(v), it suffices to prove that $G''_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$. Suppose, then, that such a subgroup has been given, and let H be its inverse image in $G_{\overline{k}}$. Since G' is unipotent, we shall prove in §5 that the extension

$$e \longrightarrow G'_{\overline{k}} \longrightarrow H \longrightarrow \boldsymbol{\mu}_p \longrightarrow e$$

is necessarily trivial; the proof of that fact is independent of the results of the present section. In short, $\boldsymbol{\mu}_p$ lifts to H. Being of height 1, it is then necessarily contained in

$$G'_{\overline{k}} = {}_F(G_{\overline{k}}),$$

which is impossible because G' is unipotent.

*That is, the least integer h such that $F^h = \mathrm{id}_G$; see Appendix II, 1.

4.3. Connected Affine Groups in Characteristic $p > 0$

Proposition 4.3.1.— *Let G be a connected affine algebraic group over a field k of characteristic $p > 0$. The following conditions are equivalent:*

(i) *G is unipotent.*

(ii) *G has a composition series whose successive quotients are isomorphic to $\boldsymbol{\alpha}_p$ and $\mathbb{G}_a$, taken in that order.*

(iii) *G has a characteristic composition series whose successive quotients are isomorphic to $(\boldsymbol{\alpha}_p)^r$ and $(\mathbb{G}_a)^s$, taken in that order.*

(iv) *$G_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$.*

(v) *$\mathfrak{g} = \mathrm{Lie}\, G$ is unipotent (3.6).*

(vi) *$\mathfrak{g}$ is nilpotent, and the reductive part of the center of $\mathfrak{g}$ (4.2.2) is trivial.*

(vi bis) *$\mathfrak{g}$ is nilpotent, and every subgroup of multiplicative type of the identity component of the center of G is trivial.*

(vi ter) *G is nilpotent, and every subgroup of multiplicative type in the identity component of the center of G is trivial.*

Proof. The implications (iii)⇒(ii)⇒(i) are clear. To prove (i)⇒(iii), we shall need the following lemma.

Lemma 4.3.2.— *Let k be a field of characteristic $p > 0$, let $n \in \mathbb{N}$, and let k' be a radicial extension of k such that $(k')^{p^n}$ is contained in k. For every k-prescheme X (respectively, every k'-prescheme X'), denote by $X^{(p^n)}$ (respectively, $(X')^{\phi}$) the k-prescheme obtained from X (respectively, from X') by the base change*

$$F^n : k \longrightarrow k, \qquad x \longmapsto x^{p^n}, \qquad (\textit{respectively, } \phi : k' \longrightarrow k, \qquad x' \longmapsto (x')^{p^n}).$$

Then, for every k-prescheme X, there is a functorial isomorphism

$$(X_{k'})^{\phi} \xrightarrow{\sim} X^{(p^n)}.$$

Consequently, if X and Y are two k-preschemes for which there exists a k'-isomorphism

$$u' : X_{k'} \xrightarrow{\sim} Y_{k'},$$

then there exists a k-isomorphism

$$v : X^{(p^n)} \xrightarrow{\sim} Y^{(p^n)}.$$

If, moreover, X and Y are endowed with structures of k-group preschemes and u' is a k'-homomorphism, then v is a k-homomorphism.

The lemma follows immediately from the transitivity of base changes and from the fact that the composite

$$k \longrightarrow k' \xrightarrow{\phi} k$$

is F^n.

Continuation of the proof of 4.3.1.

(i)⇒(iii). We argue by induction on $\dim G$. If $\dim G = 0$, then, because G is connected, it is radicial, and 3.9(i) applies. If $\dim G > 0$, there exists an integer $m \geq 0$ such that the quotient $G/F_m G$ is a smooth group (Appendix II, 3.1), plainly connected and nonzero. Applying 4.1.1(i)⇒(iii) to this quotient, we see that G has an algebraic subgroup G' which is characteristic and connected, and for which the quotient

$$G'' = G/G'$$

is a form of $(\mathbb{G}_a)^r$, where $r > 0$. By 4.1.5, if K is a perfect closure of k, then

$$G''_K \xrightarrow{\sim} (\mathbb{G}_{a,K})^r.$$

Since G'' is of finite type over k, there exists a finite radicial extension k' of k such that

$$G''_{k'} \xrightarrow{\sim} (\mathbb{G}_{a,k'})^r$$

(Exposé $\mathrm{VI_B}$, §10). Choose $n > 0$ such that $(k')^{p^n} \subset k$. Retaining the notation of 4.3.2, we deduce a k-isomorphism of algebraic groups

$$(\mathbb{G}_{a,k})^r = \left((\mathbb{G}_{a,k'})^r\right)^{\phi} \xrightarrow{\sim} (G'')^{(p^n)}.$$

Now consider the Frobenius homomorphism relative to G'' (Exposé $\mathrm{VII_A}$, §4):

$$F^n : G'' \longrightarrow (G'')^{(p^n)}.$$

Since G'', and hence also $(G'')^{(p^n)}$, is smooth over k, while F^n is radicial, F^n is an epimorphism for the fpqc topology. Thus $(G'')^{(p^n)}$ is identified with $G''/F_n(G'')$, and we have shown that

$$G''/F_n(G'') \xrightarrow{\sim} (\mathbb{G}_a)^r$$

as algebraic groups. The inverse image G'_n of $F_n(G'')$ in G is a connected, characteristic subgroup of G, of dimension strictly less than that of G; the induction hypothesis therefore applies to G'_n.

(i)⇒(iv). This follows from 2.4(i).

(iv)⇒(i). Regard G as an extension of a smooth connected group G'' by a radicial group G' (Appendix II, 3.1). The group G' is unipotent by 4.2.1(iv)⇔(i). It is therefore enough to prove that G'' is unipotent, and for this it is enough to show that $G''_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$, by 4.1.1(i)⇔(iv). If $G''_{\overline{k}}$ contained such a subgroup, it would lift to $G_{\overline{k}}$ by the result 5.1, already used above, whose proof is given in §5. This would contradict (iv).

Continuation of the proof of 4.3.1.

(i)⇒(v). This follows from 3.7.

(v)⇒(vi). Indeed, since

$$(\operatorname{ad} X)^{p^r} = \operatorname{ad}(X^{(p^r)}) \qquad (\text{Exposé } \mathrm{VII_A}, 5.2),$$

$\operatorname{ad} X$ is nilpotent when $\mathfrak{g}$ is unipotent; hence $\mathfrak{g}$ is nilpotent by Engel's theorem. Moreover, if $\mathfrak{g}$ is unipotent, then its center plainly is as well, and the reductive part of that center is therefore trivial (4.2.2).

(vi)⇒(iv). Indeed, if $G_{\overline{k}}$ contained a subgroup isomorphic to $\boldsymbol{\mu}_p$, its Lie algebra would contain a nonzero element X such that

$$X^{(p)} = X \qquad \text{(Appendix II, 2.1)},$$

and hence $X^{(p^r)} = X$ for every $r > 0$. Since $\operatorname{ad} X$ is nilpotent because $\mathfrak{g}$ is nilpotent, and

$$(\operatorname{ad} X)^{p^r} = \operatorname{ad}(X^{(p^r)}),$$

we necessarily have $\operatorname{ad} X = 0$. Thus X belongs to the reductive part of the center of $\mathfrak{g}_{\overline{k}}$, contradicting (vi).

(i)⇒(vi ter). This follows from 2.4(i) and 3.5(i)⇒(iii).

(vi ter)⇒(vi bis). Indeed, if G is nilpotent, then so is its subgroup ${}_FG$. Appendix II, 2.2 shows that ${}_FG$ is nilpotent if and only if

$$\operatorname{Lie}({}_FG) = \operatorname{Lie} G \qquad \text{(Exposé VII}_\text{A}\text{)}$$

is nilpotent.

(vi bis)⇒(vi). Let Z be the identity component of the center of G, and let $\mathfrak{r}$ be the reductive part of the center of $\mathfrak{g}$. We must prove that $\mathfrak{r} = 0$. It is immediate that $\mathfrak{r}$ is a characteristic p-Lie subalgebra of

$$\mathfrak{g} = \operatorname{Lie}({}_FG);$$

that is, it is stable under the functor $\operatorname{Aut}_{p\text{-Lie}}(\mathfrak{g})$. Consequently $\mathfrak{r}$ is the Lie algebra of a characteristic radicial subgroup R of ${}_FG$ (Appendix II, 2.2). The last assertion of 4.2.2 and Appendix II, 2.1 show moreover that R is a form of $(\boldsymbol{\mu}_p)^r$. Since R is characteristic in ${}_FG$, which is itself characteristic in G (Appendix II, 1), it is invariant in G, and hence central because G is connected (Exposé IX, 5.5). By (vi bis), $R = 0$, and therefore $\mathfrak{r} = 0$.

4.4. Étale Groups

The following proposition is an easy consequence of the Sylow theorems and of the structure of finite q-groups (cf. J.-P. Serre, *Corps locaux*, Chapter IX, §1).

Proposition 4.4.1.— *Let G be a finite étale algebraic group over an algebraically closed field. Then G is unipotent if and only if, for every prime number q distinct from the characteristic p of the field, G contains no subgroup isomorphic to $\boldsymbol{\mu}_q$.*

4.5. Abelian Varieties

Let G be an abelian variety over an algebraically closed field k. The following conditions are equivalent:

(i) G is unipotent.

(ii) $G = 0$.

(iii) There exists an integer n, relatively prime to the characteristic p of k, such that G contains no subgroup isomorphic to $\boldsymbol{\mu}_n$.

Indeed, if G is an abelian variety of dimension d, it is known (cf. S. Lang, *Abelian Varieties*, Chapter IV, §3, Theorem 6) that the group ${}_nG(k)$, for n prime to p, is isomorphic to

$$(\mathbb{Z}/n\mathbb{Z})^{2d},$$

and hence to $(\boldsymbol{\mu}_n)^{2d}$. Thus (iii)⇒(ii), while (ii)⇒(i)⇒(iii) are clear.

4.6. The General Case

If G and H are two algebraic groups defined over an algebraically closed field k, we shall denote by $P(G, H)$ the property

"there is no algebraic subgroup of G isomorphic to H."

We then obtain the following characterizations of unipotent groups.

Theorem 4.6.1.— *Let G be an algebraic group defined over an algebraically closed field k of characteristic p. Then:*

(i) If G is smooth, affine, and connected, then

$$G \text{ is unipotent} \iff \text{there exists } n > 1 \text{ such that } P(G, \boldsymbol{\mu}_n) \text{ holds} \iff P(G, \mathbb{G}_m) \text{ holds.}$$

(ii) If G is smooth and connected, then

$$G \text{ is unipotent} \iff \text{there exists } n \text{ prime to } p \text{ such that } P(G, \boldsymbol{\mu}_n) \text{ holds.}$$

(iii) If G is smooth, then

$$G \text{ is unipotent} \iff P(G, \boldsymbol{\mu}_n) \text{ holds for every prime } n \neq p.$$

(iv) If G is affine and connected and $p > 0$, then

$$G \text{ is unipotent} \iff P(G, \boldsymbol{\mu}_p) \text{ holds.}$$

(v) If G is connected and $p > 0$, then

$$G \text{ is unipotent} \iff \text{there exists } n \text{ prime to } p \text{ such that } P(G, \boldsymbol{\mu}_n) \text{ and } P(G, \boldsymbol{\mu}_p) \text{ both hold.}$$

(vi) For an arbitrary algebraic group G,

$$G \text{ is unipotent} \iff P(G, \boldsymbol{\mu}_n) \text{ holds for every prime } n.$$

Proof. Assertion (i) follows from 4.1.1, and (iv) from 4.3.1. We shall prove (vi); assertions (ii), (iii), and (v) are proved in the same way and are left to the reader.

Let G be an algebraic group. If G is unipotent, then $P(G, \boldsymbol{\mu}_n)$ holds for every $n > 1$ by 2.4(i). Conversely, suppose that $P(G, \boldsymbol{\mu}_n)$ holds for every prime n, and let us prove that G is unipotent. Write G^0 for the identity component of G. If G^0 is smooth, a classical theorem of Chevalley[*] shows that G^0 is an extension of an abelian variety A by a smooth connected affine group L. If G is not smooth—which entails $p > 0$ (Exposé $\mathrm{VI_B}$, 1.6.1)—there is an integer $n > 0$ such that

$$G'' = G^0/_{F^n}(G)$$

[*] *Séminaire Bourbaki*, no. 145, 1956/57.

is smooth (Appendix II, 3.1). Thus G'' is an extension of an abelian variety A by a smooth connected linear group L''. Let L be the inverse image of L'' in G^0. It is still affine and connected, since ${}_{F^n}(G)$ is radicial. In every case, therefore, G has a composition series

$$0 \subset L \subset G^0 \subset G$$

such that L is affine and connected, $G^0/L = A$ is an abelian variety, and G/G^0 is an étale group.

If $P(G, \boldsymbol{\mu}_n)$ holds, then a fortiori $P(L, \boldsymbol{\mu}_n)$ holds, and hence L is unipotent (4.1.1 and 4.3.1). If $A \neq 0$, there are a prime number n and a subgroup of A isomorphic to $\boldsymbol{\mu}_n$ (4.5). By 5.1 below, this subgroup lifts to G, contradicting $P(G, \boldsymbol{\mu}_n)$. Hence $A = 0$. Finally, if G/G^0 is not unipotent, there are an integer q and a subgroup of G/G^0 isomorphic to $\boldsymbol{\mu}_q$ (4.4.1). As above, it follows that $P(G, \boldsymbol{\mu}_q)$ does not hold. Thus G/G^0 is unipotent, and consequently so is G.

5. Extensions of Groups of Multiplicative Type by Unipotent Groups

5.1. Statement of the Theorem

Definition 5.1.0.— *Let k be a field and G an algebraic k-group. Following the terminology introduced by Rosenlicht (*Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Matematica 61 (1963)), we shall say that G is* k-solvable *if it satisfies the following equivalent conditions:*

(i) G has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.

(ii) G has a characteristic composition series (G_i) such that the successive quotients G_i/G_{i+1} are commutative and themselves have composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.

The implication (i)⇒(ii) is proved in *loc. cit.* In fact, for the composition series (G_i) one may take the series introduced in the proof of 3.9(ii).

Theorem 5.1.1.— *Let k be a field, U a unipotent algebraic k-group, H a k-group of multiplicative type, and E an algebraic k-group which is an extension of H by U, so that there is an exact sequence*

$$1 \longrightarrow U \longrightarrow E \longrightarrow H \longrightarrow 1.$$

Then:

(i) The extension E is trivial in each of the following cases:

  (a) k is algebraically closed;

  (b) k is perfect and one of the groups U or H is connected;

  (c) U is k-solvable;

  (d) U is smooth and H is connected.

(ii) If H' and H'' are two lifts of H in E, then they are conjugate by an element of $U(k)$ in each of the following cases:

(a) k is algebraically closed and H is smooth;

(b) k is algebraically closed and U is smooth.

(During the proof we shall indicate other cases in which the conclusion of (ii) is true without assuming that k is algebraically closed.)

If U is an algebraic group (respectively, a commutative algebraic group) defined over a field k, we denote by $H^1(k,U)$ (respectively, by $H^i(k,U)$, $i \geq 0$) the pointed set (respectively, the i-th group) of Galois cohomology of k with values in U; see J.-P. Serre, *Cohomologie galoisienne*, Lecture Notes in Mathematics no. 5, Springer.

If S is a prescheme, H a commutative S-group prescheme, and G an S-group prescheme acting on H, we denote by $H^i(G,H)$[7] the i-th Hochschild cohomology group of G with values in H (Appendix I, 1).

To prove 5.1.1, we shall proceed in several steps.

5.2. Proof of 5.1.1(i) and (ii) when U is smooth and H is étale

Lemma 5.2.1.— *With the notation of 5.1.1, suppose that H is étale. The canonical morphism $E \to H$ has a section $s : H \to E$, defined over k, in each of the following cases:*

(a) k is algebraically closed;

(b) k is perfect and U is smooth and connected;

(c) U is k-solvable.

Assertion (a) follows simply from the surjectivity of $E(k) \to H(k)$. We have (b)⇒(c) by 4.1.2(b) and 2.3 bis, so it is enough to treat case (c). Let x be a point of H. It is an open and closed subset of H, and the induced subscheme is isomorphic to $\operatorname{Spec} K$, where K is a finite separable extension of k. Let X be the inverse image in E of the scheme x. The K-scheme X is a principal homogeneous K-space under U_K. But U_K has a composition series whose successive quotients are isomorphic to $(\mathbb{G}_a)_K$; hence

$$H^1(K, U_K) = 0$$

(J.-P. Serre, *Cohomologie galoisienne*, Chapter III, Proposition 6). Consequently X is trivial and has a K-rational point. Thus, over every point x of H, we obtain a k-section of $E \to H$, and these give a section $H \to E$.

Lemma 5.2.3.— [8] *With the notation of 5.1.1, suppose that H is étale. Then:*

(i) The extension E is trivial in each of the following cases:

(a) U is commutative and $E \to H$ has a section;

(b) k is algebraically closed;

(c) k is perfect and U is connected;

(d) U is k-solvable.

Moreover, in each of the preceding cases, two lifts H' and H'' of H in E are conjugate by an element of $U(k)$.

[7] Editor's note: H_0^i has been replaced by H^i, in agreement with the notation of Appendix I and Exposé I.

[8] Editor's note: there is no item numbered 5.2.2.

(ii) Let R be the k-functor

$$(\mathrm{Sch}\,/k)^\circ \longrightarrow \mathrm{Ens}$$

for which, for every k-prescheme S, $R(S)$ is the set of lifts of H_S in E_S. If U is commutative, then R is representable by a nonempty affine k-scheme. The group U acts on R by inner automorphisms, and this action makes R a homogeneous space under U for the fpqc topology (cf. Exposé IV).

Proof of (i). We shall reduce cases (b), (c), and (d) to case (a).

Case (b). Since U has a characteristic composition series with commutative successive quotients (3.5, (i)⇔(iv)), we immediately reduce to the case in which U is commutative. Moreover, $E \to H$ has a section by 5.2.1(a), so we are reduced to case (a).

Case (d). We argue in the same way, using 5.1.0(ii) and 5.2.1(c).

Case (c). Let E_{red} be the reduced subgroup associated with E. Since H is smooth, E_{red} is an extension of H by

$$U_0 = U \cap E_{\mathrm{red}},$$

and we may replace E by E_{red}. The group U, and therefore U_0, is connected, and U_0 is plainly the identity component of E_{red}; in particular it is smooth. Thus U_0 is smooth and connected and k is perfect, so U_0 is k-solvable by 4.1.4(b), reducing us to case (d).

The preceding reductions show that in cases (b), (c), and (d) we may suppose that U has a characteristic composition series (U_i) such that U_i/U_{i+1} is commutative and the maps

$$U_i(k) \longrightarrow (U_i/U_{i+1})(k)$$

are surjective. (In cases (c) and (d), the latter assertion follows from $H^1(k, \mathbb{G}_a) = 0$.) An immediate dévissage then shows that it is enough to prove the conjugacy of two lifts H' and H'' in E when U is commutative. In short, it is enough to prove (i)(a). In that case the extension E is trivial provided that $H^2(H, U) = 0$ (Appendix I, 3.1), and H' and H'' are conjugate provided that $H^1(H, U) = 0$. We now use the following lemma.

Lemma 5.2.4.— *Let S be a prescheme, U a commutative S-group prescheme, and H an étale finite S-group prescheme of rank n acting on U. Then the groups $H^i(H, U)$, for $i > 0$, are annihilated by n in each of the following cases:*

(a) H is a constant S-group (Exposé I, 4.1);

(b) S is the spectrum of a field.

Proof of (a). By hypothesis, H is the constant group associated with an ordinary group $\{H\}$ of order n. It is clear[9] that $H^i(H, U)$ is isomorphic to the i-th cohomology group

$$H^i(\{H\}, U(S))$$

of the group $\{H\}$ with values in the ordinary group $U(S)$. It is classical that these groups are annihilated by n (J.-P. Serre, *Corps locaux*, Chapter VIII, Proposition 4, Corollary 1).

Proof of (b). Let $x \in H^i(H, U)$, where $i > 0$, and represent it by an i-cocycle

$$f : H^{(i)} \longrightarrow U,$$

[9]Editor's note: this is Proposition III.6.4.2 of M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

where $H^{(i)}$ denotes the product over k of i copies of H. If K is a finite Galois extension of k which splits H, assertion (a) shows that nf_K is a coboundary. More precisely, a direct computation shows that, if one defines the K-morphism

$$F_K : H_K^{(i-1)} \longrightarrow U_K$$

by

$$F_K(h_1, \ldots, h_{i-1}) = \sum_{h \in H(K)} f_K(h_1, \ldots, h_{i-1}, h),$$

then, up to sign,

$$d(F_K) = nf_K, \qquad d \text{ denoting the coboundary operator.}$$

An immediate Galois argument shows that F_K descends to a k-morphism $F : H^{(i-1)} \to U$, and consequently

$$d(F) = nf.$$

Corollary 5.2.4 bis.— *With the notation of 5.2.4, suppose in addition that U is flat and of finite presentation over S, with unipotent fibres, and that n is prime to the residue characteristics of S. Then, in cases (a) and (b) above,*

$$H^i(H, U) = 0 \qquad (i > 0).$$

Proof. It is enough to show that the n-th power map on U is an isomorphism, for it will then follow that multiplication by n on $H^i(H, U)$ is both an isomorphism and the zero morphism, and hence that $H^i(H, U) = 0$. Under the hypotheses made on U, it is enough to check that the n-th power map is an isomorphism on the fibres of U (EGA IV, 17.9.5). We are thus reduced to the case in which S is the spectrum of a field k of characteristic p. Since $(n, p) = 1$, the n-th power map on U is étale (Exposé VII); it is also a monomorphism (2.4(i)), and hence an open immersion (EGA IV, 17.9.1). This already shows that its restriction to the identity component U^0 is an isomorphism. Since U/U^0 is a finite p-group, the proof is complete.

This completes the proof of 5.2.3(i)(a), since an étale group of multiplicative type H has order prime to p.

Proof of 5.2.3(ii). It is clear that R is a sheaf for the fpqc topology. Moreover, in view of descent for affine schemes, the assertions of 5.2.3(ii) are local for the fpqc topology. We may therefore suppose that k is algebraically closed. The group H is then split and the extension E is trivial by (i)(b); let H' be a lift of H in E. For every k-prescheme S and every lift H'' of H_S in E_S, the groups H'_S and H'' are conjugate by an element of $U(S)$, since

$$H^1(H_S, U_S) = 0$$

by 5.2.4 bis.

Let $U^{H'}$ be the sheaf of invariants of U under H'. It is representable by an algebraic subgroup of U (Exposé VIII, 6.5(d)). The preceding remarks show that the k-morphism

$$U \longrightarrow R, \qquad u \longmapsto \mathrm{int}(u)H' \quad (u \in U(S))$$

induces a k-isomorphism

$$U/U^{H'} \xrightarrow{\sim} R.$$

This proves that R is representable and affine (2.1).

Remark 5.2.5.— *The assertions of 5.2.3(ii) can also be proved when U is not commutative, but we shall not need this in order to prove 5.1.1.*

5.3. The Case Where H Is Smooth

Proposition 5.3.1.— *The assertions of 5.2.3(i) remain true when the hypothesis "H is étale" is replaced by "H is smooth."*

Proceeding as in the proof of 5.2.3(i), we reduce to the case in which U is also commutative.

Let N' be the set of positive integers prime to p, ordered by divisibility. For every $n \in N'$, the group ${}_nH$ is étale, and the family of groups ${}_nH$, $n \in N'$, is schematically dense in H, since H is smooth (Exposé IX, 4.10). Let E_n be the inverse image of ${}_nH$ in E, so that E_n is an extension of ${}_nH$ by U, and let R_n be the k-functor of lifts of ${}_nH$ in E_n (cf. 5.2.3(ii)). If n divides m, there is plainly a natural k-morphism

$$R_m \longrightarrow R_n,$$

so the R_n form an inverse system of k-functors.

By 5.2.3(ii), R_n is representable by a nonempty affine k-scheme. Since a filtered direct limit of nonzero rings is nonzero, the functor

$$R = \varprojlim_n R_n$$

is representable by a nonempty affine k-scheme (EGA IV, 8 and 1.9.1). There are therefore an extension K of k and a point $u \in R(K)$. The image u_n of u in $R_n(K)$ corresponds to a lift H'_n of $({}_nH)_K$ in $(E_n)_K$. By construction,

$$H'_n = {}_n(H'_m) \qquad \text{if } n \text{ divides } m.$$

Put

$$U_n = (U_K)^{H'_n}.$$

The choice of H'_n identifies $(R_n)_K$ with U_K/U_n. The family of groups H'_n is filtered increasing, and hence the family U_n is filtered decreasing. It is therefore stationary for sufficiently large n, since U_K is noetherian. It follows that the family $(R_n)_K$ is stationary, and consequently so is the family R_n. In short,

$$R_n = R$$

for sufficiently large n.

Under the hypotheses in force, 5.2.3(i) shows that $R_n(k)$ is nonempty. We may therefore find a coherent system of lifts H'_n of ${}_nH$, for $n \in N'$. Let H' be the smallest closed subscheme of E containing H'_n for every n (Exposé $\mathrm{VI_B}$, §7). The argument of Exposé XV, 4.6 shows that H' is a smooth commutative algebraic group whose formation commutes with extension of the base field. To prove that H' is a lift of H in E, we may therefore suppose that k is algebraically closed. According to BIBLE 4, Theorem 4, H' is then the direct product of a smooth group M of multiplicative type and a unipotent group V. The groups H'_n are necessarily contained in M by 2.4; the definition of H' then gives $H' = M$. Thus H' is of multiplicative type, and hence

$$H' \cap U = 0.$$

The morphism $H' \to H$ is consequently a monomorphism, and the density theorem of Exposé IX, 4.10 shows that it is an epimorphism. It is therefore an isomorphism.

Now let H' and H'' be two lifts of H in E. For each $n \in N'$, let

$$T_n = \mathrm{Transp}({}_nH', {}_nH'')$$

be the transporter of ${}_nH'$ into ${}_nH''$. It is representable by a closed subscheme of U (Exposé VIII, 6.5(e)). The T_n form a filtered decreasing family of closed subschemes of U, and they are nonempty by 5.2.3(i)(a). Let T be their stationary value. Under the hypotheses of 5.2.3(i), $T_n(k)$ is nonempty. Thus there is an element $u \in U(k)$ such that

$${}_nH'' = \mathrm{int}(u)({}_nH') \qquad \text{for every } n \in N'.$$

It follows that

$$H'' = \mathrm{int}(u)H'$$

by Exposé IX, 4.8(b).

5.4. The Case Where U Is Radicial

Proposition 5.4.1.— *If k is a perfect field of characteristic $p > 0$, and if U is radicial, then the extension E of 5.1.1 is trivial.*

Using a characteristic composition series of U, we may restrict ourselves to the case

$$U = (\boldsymbol{\alpha}_p)^r$$

by 3.9.

It follows from Appendix II, 2.2 and 2.1 that there are isomorphisms of k-functors

$$\mathrm{Aut}_{k\text{-gr}}\big((\boldsymbol{\alpha}_p)^r\big) \xrightarrow{\sim} \mathrm{Aut}_{p\text{-Lie}}\big(\mathrm{Lie}((\boldsymbol{\alpha}_p)^r)\big) \xrightarrow{\sim} \mathrm{GL}\big(\mathrm{Lie}((\boldsymbol{\alpha}_p)^r)\big).$$

Let V be a k-vector space of rank r, let $\mathbf{W}(V)$ be the corresponding vector k-group (Exposé I, 4.6), and identify $(\boldsymbol{\alpha}_p)^r$ with ${}_FV$. The preceding remarks show that the action of H on ${}_FV$ defined by the extension E comes from a linear representation v of H on V. Consider the exact sequence

$$(*) \qquad 0 \longrightarrow {}_FV \longrightarrow V \xrightarrow{F} V^{(p)} \longrightarrow 0,$$

where F is the Frobenius morphism.

The sequence $(*)$ is an exact sequence of H-groups, provided that H acts on $V^{(p)}$ through the linear representation

$$H \xrightarrow{v} \mathrm{GL}(V) \xrightarrow{F} \mathrm{GL}(V^{(p)}).$$

Since k is perfect, the morphism $F : V \to V^{(p)}$ induces a surjective map

$$V(k) \longrightarrow V^{(p)}(k).$$

It follows from Appendix I, 2.1 that the exact sequence $(*)$ defines an exact sequence

$$(**) \qquad H^1(H, V^{(p)}) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(H, {}_FV) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(H, V).$$

We shall show that

$$\mathrm{Ext}_{\mathrm{alg}}(H, V) = 0.$$

Let E' be an algebraic group which is an extension of H by V:

$$1 \longrightarrow V \longrightarrow E' \xrightarrow{h} H \longrightarrow 1.$$

The scheme E' is a torsor over H under $\mathbb{G}_a^r$, and hence defines an element of

$$H^1(H, \mathcal{O}_H^r)$$

in coherent-sheaf cohomology. Since H is affine,

$$H^1(H, \mathcal{O}_H^r) = 0$$

by EGA III, §1. Thus $E' \to H$ has a section. Consequently, $\mathrm{Ext}_{\mathrm{alg}}(H, V)$ is isomorphic to $H^2(H, V)$ by Appendix I, 3.1. But

$$H^i(H, V) = H^i(H, \mathbf{W}(V)) = 0 \qquad (i > 0)$$

by Exposé IX, 3.1. The exact sequence $(**)$ now gives

$$\mathrm{Ext}_{\mathrm{alg}}(H, {}_F V) = 0,$$

and hence E is a trivial extension.

5.5. Proof of 5.1.1(i)

If k has characteristic 0, then U is k-solvable by 4.1.3 and H is smooth. It follows from 5.3.1 and 5.2.3(d) that E is a trivial extension. The same argument shows that two lifts of H in E are conjugate by an element of $U(k)$.

Henceforth we suppose that k is a field of characteristic $p > 0$.

Proof of (i)(b): the case where k is perfect and U is connected. We shall reduce to the case in which U is radicial. Since k is perfect, H_{red} is smooth; let E' be its inverse image in E. By 5.3.1 and 5.2.3(i)(c), the extension

$$1 \longrightarrow U \longrightarrow E' \longrightarrow H_{\mathrm{red}} \longrightarrow 1$$

is trivial. Let H_1 be a lift of H_{red} in E. By Appendix II, 3.1, there is an integer $n > 0$ such that

$$E^{(n)} = E/{}_{F^n}(E)$$

is smooth. Let E'' be the algebraic subgroup of E generated by H_1 and ${}_{F^n}(E)$; equivalently, E'' is the inverse image in E of the image of H_1 in $E^{(n)}$. Let H'' be the image of E'' in H. We claim that $H'' = H$.

Indeed, let R be the image of ${}_{F^n}(E)$ in H, so that H'' is generated by R and H_{red}. The group H/R is a quotient of $E^{(n)}$, and is therefore smooth. Consequently the canonical morphism

$$H_{\mathrm{red}} \longrightarrow H/R$$

is an epimorphism. Thus H is generated by R and H_{red}, and hence equals H''. We have obtained an exact sequence

$$(\dagger) \qquad 1 \longrightarrow U'' = U \cap E'' \longrightarrow E'' \longrightarrow H \longrightarrow 1.$$

But E'' has the same underlying space as H_1, so U'' is radicial. It is plainly enough to prove that the extension $(\dagger)$ is trivial, and this follows from 5.4.1.

Proof of (i)(b): the case where k is perfect and H is connected. The group U is an extension of an étale group by its identity component U^0. The case where U is connected having just been treated, it is enough to consider the case where U is étale. We shall use the following more precise lemma.

Lemma 5.5.1.— *With the notation of 5.1.1, suppose in addition that U is étale. Then:*

(i) If H is connected, it has a unique lift in E, namely the identity component E^0 of E.

(ii) If k is algebraically closed, then E is trivial, and any two lifts of H in E are conjugate by an element of $U(k)$.

Proof of (i). Since formation of the identity component commutes with extension of the base field, we may restrict ourselves to the case where k is algebraically closed. If H is a torus, then E is trivial by 5.3.1 and 5.2.3(i)(b), and E^0 is plainly the unique lift of H. This already proves that in the general case $E^0 \cap U$ is radicial. It is also étale, since U is étale, and is therefore zero. The morphism

$$E^0 \longrightarrow H$$

is consequently a monomorphism; it is flat because E^0 is open in E, and it is surjective because H is connected. It is therefore an isomorphism. If H' is any other lift of H, then H' is connected, hence contained in E^0, and thus equal to E^0.

Proof of (ii). Let H^0 be the identity component of H. By (i), E^0 is the unique lift of H^0 in E. Put

$$E' = E/E^0, \qquad H' = H/H^0.$$

There is then an extension

$$1 \longrightarrow U \longrightarrow E' \longrightarrow H' \longrightarrow 1.$$

Since H' is étale, this extension is trivial by 5.2.3(i)(b). If H'_1 is a lift of H' in E', and H_1 is its inverse image in E, then H_1 plainly lifts H in E. If H_2 is a second lift of H in E, then it contains E^0. By 5.2.3(i)(b), the image of H_2 in E' is conjugate to H'_1 by an element $u \in U(k)$, and hence

$$H_2 = \mathrm{int}(u)H_1.$$

Remark 5.5.2.— *Under the hypotheses of 5.5.1(i), the group E^0 centralizes U.*

Proof of 5.1.1(i)(a). Using the composition series

$$1 \longrightarrow U^0 \longrightarrow U \longrightarrow U/U^0 \longrightarrow 1,$$

assertion (i)(a) follows by combining (i)(b) with 5.5.1(ii).

Before proving 5.1.1(i)(c) and (d), we shall first establish 5.1.1(ii).

Proof of 5.1.1(ii)(a). For want of a satisfactory general statement, we shall describe a number of cases in which, when H is smooth, two lifts of H in E are conjugate.

Proposition 5.6.1.— *With the notation of 5.1.1, suppose in addition that H is smooth, and let H' and H'' be two lifts of H in E. Then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:*

(a) k is algebraically closed.

(b) U is commutative.

(c) k is perfect and U is connected.

(d) U is k-solvable.

(e) The centralizer of H' in U is k-solvable.

(f) The group of multiplicative type H is split by a finite Galois extension K/k of degree prime to p.

Proof. Parts (a), (b), (c), and (d) follow from 5.3.1.

(e) Let Z be the centralizer of H' in U, and let T be the transporter of H' into H''. By (a), T is nonempty; hence T is a principal homogeneous space under Z. The hypothesis on Z therefore implies that T is trivial.

(f) Proceeding as in 5.3.1, we see that it is enough to consider the case where H is étale. Suppose first that H is diagonalizable, defined by an abstract group M of order m prime to p. The two lifts H' and H'' determine a one-cocycle of M with values in $U(k)$, that is, a map

$$h : M \longrightarrow U(k)$$

such that

$$h(mn) = h(m)\,{}^m h(n)$$

for every pair m, n of elements of M. The groups H' and H'' are conjugate by an element of $U(k)$ if and only if there exists $a \in U(k)$ such that

$$h(m) = a^{-1}({}^m a).$$

Now the abstract group $U(k)$ has a composition series whose successive quotients are commutative and annihilated by a power of p (in view of 5.6.1(c), we may suppose that $p > 0$). It follows at once in this case that h is a coboundary.

Consider now the general case. Again let Z denote the centralizer of H' in U, and T the transporter of H' into H''; thus T is a torsor under Z. By hypothesis there is a finite Galois extension K/k, with Galois group G of order prime to p, which splits H. By the preceding case, H'_K and H''_K are conjugate by an element of $U(K)$, and therefore T_K is trivial. Consequently T is defined by an element of

$$H^1(G, Z).$$

For the same reasons as above, the hypothesis on G implies that

$$H^1(G, Z) = e;$$

hence T is trivial.

5.7. Proof of 5.1.1(ii)(b)

Lemma 5.7.1.— *Let S be a prescheme, let G be an S-group prescheme, separated and smooth over S, and let H be an S-group of multiplicative type acting on G. Then the S-functor*

$$Z = G^H$$

of invariants of G under H is representable by a closed smooth S-group subprescheme of G.

The representability of G^H by a closed group subprescheme of G follows from Exposé VIII, 6.5(d). Consider the semidirect product

$$K = G \rtimes_S H.$$

The centralizer of H in K is then $Z \times_S H$. To prove that Z is smooth, we must verify the following. Suppose that S is affine, that S_0 is a closed subscheme of S defined by a square-zero ideal, and that $u_0 \in Z(S_0)$. We must lift u_0 to an element of $Z(S)$.

Since G is smooth over S, there is an element $u \in G(S)$ lifting u_0. Let i be the canonical immersion of H into K, and consider the two S-group morphisms

$$H \rightrightarrows K, \qquad i \quad \text{and} \quad j = \mathrm{int}(u)i.$$

Since $u_0 \in Z(S_0)$, we have $i_{S_0} = j_{S_0}$. By Exposé IX, 3.2, there exists $v \in K(S)$, lifting the identity section in $K(S_0)$, such that

$$i = \mathrm{int}(v)j = \mathrm{int}(vu)i.$$

Thus

$$vu = (u', v') \in (Z \times_S H)(S).$$

It follows that $u' \in Z(S)$, and u' lifts u_0.

Lemma 5.7.2.— *Let k be a field and H a k-algebraic group, and let $P(H)$ denote the following property:*

> *For every smooth unipotent k-group U, and every extension E of H by U, any two lifts of H in E are conjugate by an element of $U(k)$.*

Suppose that H is an algebraic group which is an extension of a group H'' of multiplicative type by a group H' of multiplicative type. If $P(H')$ and $P(H'')$ hold, then $P(H)$ holds.

Indeed, let U be a smooth unipotent k-group, let E be an extension of H by U, and let H_1, H_2 be two lifts of H in E. Let H_1', H_2' be the corresponding lifts of H'. Since $P(H')$ holds, there exists $u \in U(k)$ such that

$$H_2' = \mathrm{int}(u)H_1'.$$

After replacing H_1 by $\mathrm{int}(u)H_1$, we may suppose that $H_1' = H_2'$; denote this common group simply by H'. Put

$$E' = \mathrm{Centr}_E(H') = U^{H'}H_1 = U^{H'}H_2.$$

By 5.7.1, $U^{H'} = U'$ is smooth. We then have an extension

$$1 \longrightarrow U' \longrightarrow E' \longrightarrow H \longrightarrow 1.$$

By construction H' is central in E', and hence invariant. Passing to the quotient gives an exact sequence

$$1 \longrightarrow U' \longrightarrow E'' \longrightarrow H'' \longrightarrow 1.$$

Since U' is smooth and $P(H'')$ holds, the images of H_1 and H_2 in E'' are conjugate by an element $u \in U'(k)$. It follows that

$$H_1 = \mathrm{int}(u)H_2.$$

To prove 5.1.1(ii)(b), observe that, since k is algebraically closed, H has a composition series whose successive quotients are smooth or, when $p > 0$, isomorphic to $\boldsymbol{\mu}_p$. Repeated application of 5.7.2 reduces us to the case where H is smooth or $H = \boldsymbol{\mu}_p$. In the first case it is enough to apply 5.1.1(ii)(a). It remains to consider $H = \boldsymbol{\mu}_p$. Since U is smooth, U has a characteristic composition series whose successive quotients are étale or isomorphic to $\mathbb{G}_a^r$, by 3.9. If U is étale, apply 5.5.1. We are finally reduced to

$$H = \boldsymbol{\mu}_p, \qquad U = \mathbb{G}_a^r.$$

We must prove that

$$H^1(\boldsymbol{\mu}_p, U) = 0.$$

The method used in 5.4.1 no longer applies, because $\boldsymbol{\mu}_p$ does not in general act linearly on $\mathbb{G}_a^r$. Fix the following notation: H' denotes a lift of H in E, and

$$\mathfrak{e} = \operatorname{Lie} E, \qquad \mathfrak{u} = \operatorname{Lie} U, \qquad \mathfrak{h} = \operatorname{Lie} H, \qquad \mathfrak{h}' = \operatorname{Lie} H'.$$

Let X be a nonzero element of $\mathfrak{h}$ such that

$$X^{(p)} = X$$

(Appendix II, 3.1), and let X' be its lift in $\mathfrak{h}'$.

Since $\boldsymbol{\mu}_p$ is a radicial group of height 1, the set of lifts of H in E is in bijective correspondence with the set A of elements $Y \in \mathfrak{e}$ such that

$$Y^{(p)} = Y$$

and whose image is X (Appendix II, 2.2). Likewise, if $Y \in A$ corresponds to a lift H'' of H in E, and if $u \in U(k)$, then

$$\operatorname{int}(u)H' = H'' \quad \Longleftrightarrow \quad \operatorname{Ad}(u)X' = Y.$$

Let B be the subset of $\mathfrak{e}$ consisting of the elements $\operatorname{Ad}(u)X'$, with $u \in U(k)$. Plainly $B \subset A$, and it remains only to prove $A = B$ when k is algebraically closed.

(a) *Study of A.* Since $\mathfrak{u}$ is commutative and normalized by $\mathfrak{h}'$, Jacobson's formula (Exposé $\mathrm{VII_A}$, 5.2) gives, for $u \in \mathfrak{u}$,

$$(X' + u)^{(p)} = X'^{(p)} + u^{(p)} + (\operatorname{ad} X')^{p-1}(u) = X' + (\operatorname{ad} X')^{p-1}(u).$$

Consequently,

$$X' + u \in A \quad \Longleftrightarrow \quad u = (\operatorname{ad} X')^{p-1}(u).$$

Let

$$\mathfrak{u} = \bigoplus_{n \in \mathbb{Z}/p\mathbb{Z}} \mathfrak{u}_n = \mathfrak{u}_0 \oplus \bigoplus_{n \in (\mathbb{Z}/p\mathbb{Z})^\times} \mathfrak{u}_n$$

be the canonical decomposition of $\mathfrak{u}$ under the action of H'. If $u \in \mathfrak{u}_0$, then $\operatorname{ad} X'(u) = 0$. If $u \in \mathfrak{u}_n$, with $n \in (\mathbb{Z}/p\mathbb{Z})^\times$, then

$$(\operatorname{ad} X')^{p-1}(u) = u.$$

Thus $Y = X' + u$ belongs to A if and only if

$$u \in \bigoplus_{n \in (\mathbb{Z}/p\mathbb{Z})^\times} \mathfrak{u}_n.$$

In particular, A is the set of k-rational points of an irreducible subscheme of $\mathbf{W}(\mathfrak{e})$, of dimension

$$\operatorname{rk}\mathfrak{u} - \operatorname{rk}\mathfrak{u}_0 = \operatorname{rk}\mathfrak{u} - \operatorname{rk}\operatorname{Centr}_{\mathfrak{u}}(X').$$

(b) *Study of B.* We shall need the following lemma.

Lemma 5.7.3 (Rosenlicht).— *Let U be a unipotent algebraic group over a field k, acting on a quasi-affine k-scheme X. Then the orbit of every point $x \in X(k)$ is closed in X. Here, by the orbit of x, we mean the subset of* $\operatorname{ens}(X)$ *which is the image of U under the morphism $g \mapsto g \cdot x$.*

By functoriality, U acts on the affine envelope of X, that is,

$$\operatorname{Spec}\Gamma(X, \mathcal{O}_X),$$

so we may suppose that X is affine. We may then suppose that k is algebraically closed, X is reduced, and U is smooth (observe that U_{red} acts on X_{red} when k is perfect). Let Y be the schematic image of U (EGA I, 9.5.1) under the morphism $g \mapsto g \cdot x$. It is a closed reduced subscheme of X, on which U acts. It follows readily from EGA IV, 1.8.6 that the orbit of x is an open subset Z of Y, dense in Y. We must prove that

$$Z = \operatorname{ens}(Y).$$

Let F be the closed reduced subscheme of Y whose underlying space is $Y \setminus Z$. Thus $F = V(J)$, where J is a nonzero ideal of $\Gamma(Y, \mathcal{O}_Y)$. Since U is smooth, U acts on F, hence on J, and therefore

$$J^U \neq 0$$

by 3.2. If a is a nonzero element of J^U, then a is constant on the orbit Z, and hence on Y, because Z is dense in Y. The ideal J consequently contains k, and $F = \varnothing$.

Apply the preceding lemma to the group U acting on the affine space $\mathbf{W}(\mathfrak{e})$ through the adjoint representation. The orbit of X' is the underlying set of a closed subscheme of $\mathbf{W}(\mathfrak{e})$. Moreover, the stabilizer Z of X' is its centralizer in U, and there is a closed immersion

$$U/Z \longrightarrow \mathbf{W}(\mathfrak{e}).$$

Thus the orbit of X' is the underlying space of a closed subscheme of $\mathbf{W}(\mathfrak{e})$ of dimension

$$\dim U - \dim Z.$$

(c) *End of the proof of 5.1.1(ii)(b).* When k is algebraically closed, the canonical map

$$U(k) \longrightarrow (U/Z)(k)$$

is surjective. By (b), B is therefore the set of k-rational points of a closed subscheme of $\mathbf{W}(\mathfrak{e})$ of dimension $\dim U - \dim Z$. In view of (a), to prove $A = B$ it is enough to show that

$$\operatorname{rk}\mathfrak{u} - \operatorname{rk}\operatorname{Centr}_{\mathfrak{u}}(X') \leq \dim U - \dim \operatorname{Centr}_U(X').$$

Since U is smooth,

$$\dim U = \operatorname{rk}\mathfrak{u}.$$

On the other hand, Exposé II, 5.3.3 gives

$$\dim \operatorname{Centr}_U(X') \leq \operatorname{rk}\operatorname{Centr}_{\mathfrak{u}}(X'),$$

which proves the assertion. Notice in fact that

$$\dim \operatorname{Centr}_U(X') = \operatorname{rk}\operatorname{Centr}_{\mathfrak{u}}(X');$$

this also proves once more that Z is smooth, as in 5.7.1.

5.8. End of the proof of 5.1.1(i)

It remains to prove (i)(c) and (i)(d).

Proof of 5.1.1(i)(d): U smooth and H connected. We shall need the following lemma.

Lemma 5.8.1.— *With the notation of 5.1.1, suppose that U is smooth and H is radicial, and let H_1 be an algebraic subgroup of H which has a lift H_1' in E. Then H has a lift H' in E which contains H_1'.*

Proof. By induction on the height of H/H_1, we may suppose that H/H_1 has height 1. Put

$$C' = \mathrm{Centr}_E(H_1').$$

We claim that the canonical morphism $C' \longrightarrow H$ is an epimorphism. To prove this, we may suppose that k is algebraically closed. The extension E is then trivial, by 5.1.1(i)(a). Let H'' be a lift of H in E, and let H_1'' be the inverse image of H_1 in H''. The groups H_1' and H_1'' are two lifts of H_1 in E; hence they are conjugate by an element of $U(k)$, since k is algebraically closed and U is smooth, by 5.1.1(ii)(b). Consequently, to prove the assertion about C', it is enough to prove it for

$$C'' = \mathrm{Centr}_E(H_1'').$$

But C'' contains H'', so the assertion is clear.

It also follows from 5.7.1 that $C' \cap U$ is smooth. We may therefore replace E by C', and thus suppose that H_1' is central. Passing then to the quotient by H_1', we may suppose that $H_1 = 0$ and that H has height 1.

Since U is smooth, we have the following exact sequence of p-Lie algebras (Appendix II, 3.2):

$$(*) \qquad 0 \longrightarrow \mathrm{Lie}\, U \longrightarrow \mathrm{Lie}\, E \longrightarrow \mathrm{Lie}\, H \longrightarrow 0.$$

In view of Appendix II, 2.2, to say that E is trivial is equivalent to saying that $(*)$ is a trivial extension of p-Lie algebras. Suppose that $H \neq 0$, and hence that $\mathrm{Lie}\, H \neq 0$. Suppose also that we have found a nonzero p-Lie subalgebra

$$\mathfrak{h}_1 \subset \mathfrak{h} = \mathrm{Lie}\, H$$

which lifts to a p-Lie subalgebra $\mathfrak{h}_1' \subset \mathrm{Lie}\, E$. By the cited result, there is a subgroup H_1 of H, with $\mathrm{Lie}\, H_1 = \mathfrak{h}_1$, and a lift H_1' of H_1 in E, with $\mathrm{Lie}\, H_1' = \mathfrak{h}_1'$. Applying once more the reduction just described, we are brought back to the same problem with H replaced by H/H_1. Since H has height 1,

$$\mathrm{Lie}(H/H_1) = \mathrm{Lie}\, H/\mathrm{Lie}\, H_1$$

(loc. cit.), and consequently

$$\mathrm{rk}\,\mathrm{Lie}(H/H_1) \leq \mathrm{rk}\,\mathrm{Lie}\, H - 1.$$

Thus, by induction on the rank of $\mathrm{Lie}\, H$, it is enough, when $\mathfrak{h} \neq 0$, to find a nonzero Lie subalgebra of $\mathrm{Lie}\, H$ which lifts to $\mathrm{Lie}\, E$. This follows from the next lemma.

Lemma 5.8.2.— *Let k be a field of characteristic $p > 0$, and let*

$$\varphi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

be a surjective morphism of finite-rank p-Lie algebras over k. Then:

(i) If $\mathfrak{h}_{\overline{k}}$ is reductive (4.2.2) and nonzero, there exists a reductive Lie subalgebra $\mathfrak{h}_1$ of $\mathfrak{g}$ whose image in $\mathfrak{h}$ is nonzero.

(ii) If k is perfect and u is a unipotent element of $\mathfrak{h}$, that is, if $u^{(p^n)} = 0$ for some n, then u lifts to a unipotent element of $\mathfrak{g}$.

Proof. Let X be a nonzero element of $\mathfrak{h}$ in case (i), and let $X = u$ in case (ii); choose a lift X' of X in $\mathfrak{g}$. The p-Lie subalgebra $\mathfrak{h}'$ of $\mathfrak{g}$ generated by X' is abelian (Exposé VII).

Case (i). The description given in 4.2.2 shows that the reductive part (loc. cit.) of $\mathfrak{h}'_{\overline{k}}$ is already defined over k; denote it by $\mathfrak{r}'$. We claim that the image of $\mathfrak{r}'$ in $\mathfrak{h}$ is nonzero. We may suppose that k is algebraically closed, so that

$$\mathfrak{h}' = \mathfrak{r}' \oplus \mathfrak{u}',$$

where $\mathfrak{u}'$ is the unipotent part of $\mathfrak{h}'$. If the image of $\mathfrak{r}'$ in $\mathfrak{h}$ were zero, then the image of $\mathfrak{h}'$ in $\mathfrak{h}$ would be unipotent, and hence zero because $\mathfrak{h}$ is reductive (2.4(ii)); but by construction that image contains X, a contradiction.

Case (ii). The same argument applies, with the roles of $\mathfrak{r}'$ and $\mathfrak{u}'$ interchanged.

End of the proof of 5.8.1. Suppose that $H \neq 0$. By 5.8.2 there exists a nonzero reductive p-Lie subalgebra $\mathfrak{h}'_1 \subset \operatorname{Lie} E$. Since $\operatorname{Lie} U$ is unipotent, by 4.3(i), we necessarily have

$$(\operatorname{Lie} U) \cap \mathfrak{h}'_1 = 0.$$

Thus $\mathfrak{h}'_1$ is a lift of a p-Lie subalgebra of $\operatorname{Lie} H$.

End of the proof of 5.1.1(i)(d). By 5.8.1 there is a family of algebraic subgroups H'_n ($n \in \mathbb{N}$) of E such that H'_{n+1} contains H'_n, and such that H'_n lifts ${}_{F^n}H$. The descending sequence of subgroups

$$\operatorname{Centr}_E(H'_n)$$

is stationary; let C be its stationary value. The center Z of C contains H'_n for every n. Hence the image of Z in H contains ${}_{F^n}(H)$ for every n, and is therefore an open subgroup of H (Exposé VII$_\mathrm{A}$, §4). It is consequently equal to H, since H is connected. To prove that E is trivial, we may thus replace E by Z, and hence suppose that E is commutative. We shall see in 7.2.1 that E then contains a maximal subgroup M of multiplicative type, whose formation commutes with extension of the base field. Since $E_{\overline{k}}$ is a trivial extension, by 5.1.1(i)(a), and U is unipotent, it is clear that M is the unique lift of H in E.

Proof of 5.1.1(i)(c): U k-solvable. Since $(H/H^0)(\overline{k})$ has order prime to p, duality immediately shows that there exists an integer n such that ${}_nH$ is an étale subgroup and the canonical morphism

$${}_nH \longrightarrow H/H^0$$

is an epimorphism. Thus $H/{}_nH$ is connected. By 5.2.3(d), there exists a lift H' of ${}_nH$ in E. As at the beginning of the proof of 5.8.1, one sees that

$$C = \operatorname{Centr}_E(H')$$

is a subgroup of E such that $C \longrightarrow H$ is an epimorphism and $C \cap U$ is smooth. Replacing E by C and passing to the quotient by H', we are reduced to the case where U is smooth and H is connected, namely case (i)(d).

This completes the proof of 5.1.1.

5.9. Counterexamples

Let us first indicate a procedure for obtaining nontrivial extensions of a k-group H of multiplicative type by a unipotent k-group U. Suppose that an action of H on U is given, and let

$$E = U \rtimes H$$

be the semidirect product. Let, moreover, a class in $\mathrm{H}^1(k, U)$ be represented by a 1-cocycle a. The group U acts on E by inner automorphisms, and a therefore defines a k-form of E, denoted by E_a. Suppose in addition that U is commutative. Then U acts trivially on U and on the quotient $E/U = H$, so E_a is again an extension of H by U.

For simplicity, suppose that H is an étale diagonalizable group, so that the Galois group $\mathscr{G} = \mathrm{Gal}(\overline{k}/k)$ acts trivially on $H(\overline{k})$. Its action on $E_a(\overline{k})$ is then given by

$$^g(u, h) = (^g u + a(g) - {}^h a(g), h) \qquad (g \in \mathscr{G},\ u \in U(\overline{k}),\ h \in H(\overline{k})).$$

If $h \in H(k)$, and X is its inverse image in E_a, then X is a torsor under U defined by the class of the 1-cocycle

$$g \longmapsto a(g) - {}^h a(g)$$

of $\mathscr{G}$ with values in U. It follows that if there is a point $h \in H(k)$ for which this cocycle is nontrivial, then the extension E_a is nontrivial. We apply this construction in the following two particular cases.

(a) A nontrivial extension of an étale diagonalizable group H by $\mathbb{Z}/p\mathbb{Z}$. Take

$$H = (\mathbb{Z}/p\mathbb{Z})^\times, \qquad U = \mathbb{Z}/p\mathbb{Z},$$

with H acting on U by multiplication. Let k be a field of characteristic p, let K/k be a Galois extension whose Galois group $\mathscr{G}$ is isomorphic to $\mathbb{Z}/p\mathbb{Z}$, and take a nonzero element

$$a \in \mathrm{H}^1(\mathscr{G}, U) = \mathrm{Hom}(\mathscr{G}, U).$$

Then E_a is the required example.

(b) A nontrivial extension of an étale diagonalizable group H by a smooth connected unipotent group U. Choose a nonperfect field k admitting a k-form U of $\mathbb{G}_a$ such that $\mathrm{H}^1(k, U) \neq 0$. For example (see J.-P. Serre, *Cohomologie Galoisienne*), one may take for k the fraction field of a discrete valuation ring of equal characteristic $p > 0$, and for U the algebraic subgroup of $\mathbb{G}_a \times_k \mathbb{G}_a$ defined by

$$X^p + X + tY^p = 0,$$

where t is a uniformizer of k. Indeed, suppose for simplicity that k contains the $(p-1)$-st roots of unity. There is then an exact sequence of algebraic groups

$$0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow U \longrightarrow \mathbb{G}_a \longrightarrow 0, \qquad (x, y) \longmapsto y,$$

and hence an exact cohomology sequence

$$\mathbb{G}_a(k) \xrightarrow{d} \mathrm{H}^1(k, \mathbb{Z}/p\mathbb{Z}) \longrightarrow \mathrm{H}^1(k, U) \longrightarrow 0,$$

where d associates to x the homogeneous space under $\mathbb{Z}/p\mathbb{Z}$ defined by

$$X^p + X + tx^p = 0.$$

Moreover,

$$\mathrm{H}^1(k, \mathbb{Z}/p\mathbb{Z}) \simeq k/P(k), \qquad P(x) = x^p + x,$$

and therefore

$$\mathrm{H}^1(k, U) \simeq k/(P(k) + tk^p).$$

If, in addition, $p \neq 2$, then t^2 does not belong to $P(k) + tk^p$; consequently $\mathrm{H}^1(k, U) \neq 0$.

The group $\boldsymbol{\mu}_{p-1}$ acts on U by

$$(h, x, y) \longmapsto (hx, hy).$$

Let $\mathscr{G}$ be the Galois group of the extension K/k defined by

$$X^p + X + t^2 = 0,$$

and let $a \in \mathrm{H}^1(\mathscr{G}, U)$ be the nonzero class described above. It follows immediately that E_a is a nontrivial extension of $\boldsymbol{\mu}_{p-1}$ by U.

(c) A nontrivial extension of $\mathbb{G}_m$ by $\boldsymbol{\alpha}_p$. By 5.1.1(i)(b), such an extension can exist only over a nonperfect field. Let k therefore be nonperfect, and let G be the semidirect product of $U = \mathbb{G}_a$ by $H = \mathbb{G}_m$, where $\mathbb{G}_m$ acts on $\mathbb{G}_a$ by homotheties. Since U is invariant in G, so is ${}_FU$. Put

$$G' = G/{}_FU.$$

The group G' is isomorphic to $\mathbb{G}_a \rtimes \mathbb{G}_m$, where $\mathbb{G}_m$ acts on $\mathbb{G}_a$ by

$$(h, u) \longmapsto h^p u.$$

The functor $\mathscr{T}_G$ (respectively $\mathscr{T}_{G'}$) of subtori of G (respectively G') (see Exposé XV) is isomorphic to $\mathbb{G}_a$, and the morphism

$$\mathscr{T}_G \longrightarrow \mathscr{T}_{G'}$$

induced by $G \longrightarrow G'$ identifies with

$$u \longmapsto u^p.$$

It follows that if T' is a subtorus of G' corresponding to a point $x \in k \simeq \mathbb{G}_a(k)$ such that $x^{1/p} \notin k$, then its inverse image E in G is an extension of the torus T' by ${}_FU = \boldsymbol{\alpha}_p$. It has no maximal torus defined over k, and hence is nontrivial. As a subgroup of

$$G = \operatorname{Spec} k[U, T, T^{-1}],$$

the group E is defined by

$$U^p = x - xT^p.$$

Remark 5.9.1.— *This last example shows that a nonsmooth algebraic group defined over a nonperfect field need not possess a maximal torus, and thus answers the question posed in Exposé XIV, 1.5(b).*

(d) A trivial extension with nonconjugate lifts. We now give an example of a trivial extension E of a group H of multiplicative type by a unipotent group U, and of two lifts H' and H'' of H which are not conjugate by an element of $U(k)$.

Take for E the semidirect product of $U = \mathbb{G}_a$ by

$$\boldsymbol{\mu}_p = \operatorname{Spec} k[T]/(T^p - 1),$$

where the action of $\boldsymbol{\mu}_p$ on $\mathbb{G}_a = \operatorname{Spec} k[U]$ is defined by the comorphism

$$U \longmapsto (U + U^p)T - U^p.$$

The centralizer of $\boldsymbol{\mu}_p$ in U is then the étale group Z defined by

$$U + U^p = 0.$$

It follows that if k is not algebraically closed, the canonical map

$$U(k) \longrightarrow (U/Z)(k)$$

is not in general surjective. In view of 5.1.1(ii)(b), two lifts of $\boldsymbol{\mu}_p$ in E therefore need not be conjugate by an element of $U(k)$.

Here is another example, with k algebraically closed of characteristic $p > 0$. Let G be the radicial semidirect product in which $\boldsymbol{\mu}_p$ acts on $\boldsymbol{\alpha}_p$ by homotheties. There is an exact sequence of algebraic groups with $\boldsymbol{\mu}_p$-operators

$$0 \longrightarrow \boldsymbol{\alpha}_p \longrightarrow \mathbb{G}_a \xrightarrow{x \longmapsto x^p} \mathbb{G}_a \longrightarrow 0,$$

where $\boldsymbol{\mu}_p$ acts by homotheties on the first $\mathbb{G}_a$ and trivially on the second. The exact cohomology sequence (Appendix I, Proposition 11) gives

$$0 \longrightarrow \mathbb{G}_a(k) \longrightarrow \mathrm{H}^1(\boldsymbol{\mu}_p, \boldsymbol{\alpha}_p) \longrightarrow \mathrm{H}^1(\boldsymbol{\mu}_p, \mathbb{G}_a).$$

Since the last term is zero by I, 5.3.3, the group $\mathrm{H}^1(\boldsymbol{\mu}_p, \boldsymbol{\alpha}_p)$ is nonzero. Thus two lifts of $\boldsymbol{\mu}_p$ in G need not be conjugate.

6. Extensions of a Unipotent Group by a Group of Multiplicative Type

6.1. Statement of the Theorem

Theorem 6.1.1.— *Let k be a field, U a unipotent algebraic k-group, H a k-group of multiplicative type, and E an algebraic k-group which is an extension of U by H, so that there is an exact sequence*

$$1 \longrightarrow H \longrightarrow E \longrightarrow U \longrightarrow 1.$$

Then the extension E is trivial and there exists a unique lift of U in E in each of the following cases:

(A) The group U is smooth and one of the following conditions holds:

- *(i) U is connected and the canonical morphism $E \longrightarrow U$ admits a section.*
- *(ii) U has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.*
- *(iii) H is étale.*
- *(iv) k is perfect.*

(B) $U = \boldsymbol{\alpha}_p$ and k is perfect.

(C) E is commutative and k is perfect.

6.2. Proof of 6.1.1(A)

We first establish three lemmas.

Lemma 6.2.1.— *Let S be a prescheme, E an S-group prescheme which is an extension of an S-group prescheme U, with connected fibres, by an S-group H of multiplicative type and finite type (that is, U is the quotient of E by H for the fpqc topology). Then E is a central extension.*

Proof. Since H is commutative, U acts on H by inner automorphisms, through an S-morphism of groups

$$u : U \longrightarrow \mathrm{Aut}_{S\text{-gr}}(H).$$

The functor $\mathrm{Aut}_{S\text{-gr}}(H)$ is representable by an étale S-scheme (Exposé X, 5.10), and consequently its identity section is both an open and a closed immersion. Since U has connected fibres, it follows that u is the unit morphism.

Lemma 6.2.2.— *With the notation of 6.1.1, if E is trivial, there exists a unique lift of U in E.*

Proof. Let U' and U'' be two lifts of U in E. To prove that $U' = U''$, we may suppose that k is algebraically closed, and it is enough to prove that $\mathrm{H}^1(U, H) = 0$. If U is connected, then U centralizes H by 6.2.1, and hence

$$\mathrm{H}^1(U, H) = \mathrm{Hom}_{k\text{-gr}}(U, H) = 0$$

by 2.4(ii).

In the general case, let U'_1 denote the unique lift in E of the identity component U^0 of U, and put

$$N = \mathrm{Norm}_E(U'_1).$$

If $g \in E(k)$, then $\mathrm{int}(g)U'_1$ is a lift of U^0, and is therefore equal to U'_1; thus $N(k) \supset E(k)$. Moreover, N contains H, by 6.2.1, as well as U'_1, and it follows at once that $N = E$. On passing to the quotient by U'_1, we are reduced to the case in which U is étale. Since k is algebraically closed, in this case $\mathrm{H}^1(U, H)$ identifies with the ordinary cohomology group[10]

$$\mathrm{H}^1\big(U(k), H(k)\big),$$

and is therefore zero, since $U(k)$ is a finite p-group and $H(k)$ is uniquely p-divisible.

Corollary 6.2.3.— *To prove that E is trivial, it is enough to prove that it becomes trivial after a finite separable extension of the base field. In particular, we may suppose that H is diagonalizable.*

Lemma 6.2.4.— *In order that every central extension E of U by a diagonalizable k-group H be trivial, it is enough that every central extension of U by $\mathbb{G}_m$ be trivial.*

Proof. Induction on r first shows that the hypothesis implies that E is trivial when $H = \mathbb{G}_m^r$. In the general case, H embeds in $\mathbb{G}_m^r$ for a suitable integer r (this is immediate by duality); put

$$H'' = \mathbb{G}_m^r / H.$$

[10] Editor's note: see, for example, Proposition III.6.4.2 of M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

We obtain the exact sequence (Appendix I, 2.1)

$$\mathrm{Z}^1(U, H'') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(U, H) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(U, \mathbb{G}_m^r) = 0,$$

where U acts trivially on H, H'', and $\mathbb{G}_m^r$. But

$$\mathrm{Z}^1(U, H'') = \mathrm{Hom}_{k\text{-gr}}(U, H'') = 0$$

by 2.4(ii), and hence $\mathrm{Ext}_{\mathrm{alg}}(U, H) = 0$.

Proof of 6.1.1(A)(i). Since $E \longrightarrow U$ admits a section, E defines an element f of $\mathrm{H}^2(U, H)$ (Appendix I, 3.1). We must prove that f is zero. For this it is enough to prove that a 2-cocycle

$$f : U \times_k U \longrightarrow H$$

is a constant morphism, which follows from the next lemma.

Lemma 6.2.5.— *Let U be a smooth connected unipotent algebraic k-group, and H a k-group of multiplicative type. Then every k-morphism of preschemes*

$$f : U \longrightarrow H$$

is constant.

Proof. We may suppose that k is algebraically closed. We argue by induction on $\dim U$. If $\dim U > 0$, then U has a composition series (see 3.9)

$$1 \longrightarrow U' \longrightarrow U \xrightarrow{\pi} U'' \longrightarrow 1, \qquad U' \simeq \mathbb{G}_a, \quad \dim U'' < \dim U.$$

It is enough to show that f factors through U''. Since the graph of the equivalence relation defined by π is smooth over k, and hence reduced, it is enough to show that if $x, y \in U(k)$ have the same image z in $U''(k)$, then $f(x) = f(y)$. Now $\pi^{-1}(z)$ is isomorphic to $\mathbb{G}_a$, so the restriction of f to $\pi^{-1}(z)$ factors through a reduced irreducible component of H, and hence through a k-scheme isomorphic to $\mathbb{G}_m^r$. It remains only to note that every morphism from $\mathbb{G}_a$ to $\mathbb{G}_m^r$ is constant, since every invertible regular function on $\mathbb{G}_a$ is constant.

Proof of 6.1.1(A)(ii). By 6.2.1, 6.2.3, and 6.2.4, we may suppose that $H = \mathbb{G}_m$. By hypothesis, U has a composition series

$$U \supset U_1 \supset \cdots$$

such that U_i/U_{i+1} is isomorphic to $\mathbb{G}_a$. Let E_1 be the inverse image of U_1 in E. By induction on $\dim U$, we may suppose that the extension E_1 is trivial; let U_1' be the unique lift of U_1. Arguing as in the normalizer argument of 6.2.2, we find that U_1' is invariant in E. After taking the quotient by U_1', we are reduced to the case $U = \mathbb{G}_a$. The S-scheme E, where $S = U$, is then a torsor under the S-group $\mathbb{G}_m \times_k S$, and therefore admits a section, since $\mathrm{Pic}(S) = 0$ (Exposé VIII, 4.3). The extension E is then trivial by (A)(i).

Proof of 6.1.1(A)(iii) (U smooth, H étale). Suppose first that U is connected. The group $U_{\overline{k}}$ then has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$, so $E_{\overline{k}}$ is trivial by (A)(ii). Since H is étale, the unique lift of $U_{\overline{k}}$ in $E_{\overline{k}}$ is plainly the identity component of $E_{\overline{k}}$; this lift is therefore already defined over k. In the general case, after quotienting by the identity component of E, we are reduced to the case where E is étale, and then to the case where E is completely split (6.2.3). Since $U(k)$ is a p-group and $H(k)$ has order prime to p, we may take the Sylow p-subgroup of $E(k)$ as a lift of $U(k)$.

Proof of 6.1.1(A)(iv) (U smooth, k perfect). If U is connected, it has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$ (4.1.2(b)), and we apply (A)(ii). In the general case, the preceding argument reduces us to the case in which U is étale, and then to the case in which U is completely split and H is diagonalizable (6.2.3). Using a characteristic composition series of H,

$$H \supset H^0 \supset {}_{F^n}(H) \supset 0,$$

we are reduced to the case in which H is of one of the following three types:

(a) H is étale;

(b) $H = \mathbb{G}_m^r$;

(c) H is radicial.

In case (a) we apply (A)(iii), and in case (c) we apply 1.6. Finally, in case (b), Hilbert's Theorem 90 shows that $E \longrightarrow U$ admits a section, so it is enough to show that

$$\mathrm{H}^2(U, \mathbb{G}_m^r) = \mathrm{H}^2\big(U(k), \mathbb{G}_m^r(k)\big) = 0.$$

But $U(k)$ is a finite p-group, whereas $\mathbb{G}_m^r(k)$ is uniquely p-divisible, since k is perfect.

6.3. Proof of 6.1.1(B) and (C)

By 6.2.1, 6.2.3, and 6.2.4, it is enough to prove (B) when $H = \mathbb{G}_m$. We therefore have an exact sequence

$$1 \longrightarrow \mathbb{G}_m \longrightarrow E \longrightarrow \boldsymbol{\alpha}_p \longrightarrow 1.$$

Since $\mathbb{G}_m$ is smooth, it gives an exact sequence of restricted Lie algebras (Appendix II, 3.2)

$$(*) \qquad 0 \longrightarrow \operatorname{Lie}\mathbb{G}_m \longrightarrow \operatorname{Lie} E \longrightarrow \operatorname{Lie}\boldsymbol{\alpha}_p \longrightarrow 0.$$

The group $\boldsymbol{\alpha}_p$ has height 1, so the extension E is trivial if and only if the exact sequence $(*)$ of restricted Lie algebras splits (Appendix II, 2.2). The algebra $\operatorname{Lie}\boldsymbol{\alpha}_p$ is generated by an element $X \neq 0$ such that $X^{(p)} = 0$ (Appendix II, 2.1). It is therefore enough to show that X lifts to an element Z of $\operatorname{Lie} E$ for which $Z^{(p)} = 0$. By 5.8.2(ii), there is a unipotent element Z of $\operatorname{Lie} E$ which lifts X. Since the unipotent part of $\operatorname{Lie} E$ plainly has dimension at most 1, it follows necessarily that $Z^{(p)} = 0$.

Proof of 6.1.1(C). If U_1' is a lift in E of an algebraic subgroup U_1 of U, then U_1' is invariant in E, since E is assumed commutative, and we may pass to the quotient by U_1'. Using a composition series of U (3.5(ii)), the preceding remark reduces us to the case in which U is smooth or equal to $\boldsymbol{\alpha}_p$. But then E is trivial by (A)(iv) and (B).

6.4. Examples of nontrivial extensions of a unipotent group U by a group H of multiplicative type

In view of 6.1.1(A)(iv), the question arises only in characteristic $p > 0$.

(a) $H = \mathbb{G}_m$, and U is a nontrivial form of $\mathbb{G}_a$ (cf. Appendix III, §5).

Let k be a nonperfect field of characteristic 2, and let $u \in k$ be such that $u^{1/2} \notin k$. Consider the affine group E with coordinate ring

$$k\big[X, Y, (X^2 + uY^2)^{-1}\big],$$

whose multiplication is given by the comorphism

$$(X, Y) \longmapsto (XX' + uYY',\ XY' + YX').$$

The group E is smooth, connected, commutative, and of dimension 2; the subscheme $Y = 0$ defines a subgroup $H \simeq \mathbb{G}_m$. The kernel K of the squaring map on E has equation

$$X^2 + uY^2 = 1,$$

and hence has dimension 1. The group K contains the unipotent radical of E, defined over $\bar{k}$, as well as the contribution of H, which is isomorphic to $\boldsymbol{\mu}_2$. Since K is reduced over $\bar{k}$, the unipotent radical of E is not defined over k, and $U = E/H$ does not lift in E.

One checks immediately that U is the form of $\mathbb{G}_a$ with coordinate ring

$$k[V, W]/(V + uV^2 + W^2),$$

the morphism $E \longrightarrow U$ corresponding to the comorphism

$$V \longmapsto \frac{Y^2}{X^2 + uY^2}, \qquad W \longmapsto \frac{XY}{X^2 + uY^2}.$$

(b) $H = \mathbb{G}_m$, $U = \mathbb{Z}/2\mathbb{Z}$, and k is nonperfect of characteristic 2.

Choose k and u as in (a), and consider the subgroup E of GL_2 generated by the element

$$X = \begin{pmatrix} 0 & 1 \\ u & 0 \end{pmatrix} = \begin{pmatrix} u^{1/2} & 0 \\ 0 & u^{1/2} \end{pmatrix} \begin{pmatrix} 0 & u^{-1/2} \\ u^{1/2} & 0 \end{pmatrix}.$$

The group E is an extension of $\mathbb{Z}/2\mathbb{Z}$ by $\mathbb{G}_m$, but this extension is not trivial because the unipotent part of X is not defined over k.

(c) $H = \boldsymbol{\mu}_p$, $U = \boldsymbol{\alpha}_p$, and k is nonperfect.

Let $\mathfrak{e}$ be the commutative restricted Lie algebra generated by two elements X and Y such that

$$X^{(p)} = X, \qquad Y^{(p)} = aX.$$

By Appendix II, 2.2, $\mathfrak{e}$ is the restricted Lie algebra of an algebraic group E which is an extension of $\boldsymbol{\alpha}_p$ by $\boldsymbol{\mu}_p$. This extension is trivial if and only if there exists $b \in k$ such that $b^p = a$, for then

$$(Y - bX)^{(p)} = 0.$$

(d) $H = \boldsymbol{\mu}_2$, $U = \boldsymbol{\alpha}_2 \times \boldsymbol{\alpha}_2$, E noncommutative, and k a field of characteristic 2.

Consider the special linear group $\mathrm{SL}_{2,k}$, and put $E = F(\mathrm{SL}_{2,k})$. The group E is a radicial group of height 1, whose Lie algebra is generated by three elements X, Y, Z satisfying

$$[X, Y] = Z, \qquad [X, Z] = [Y, Z] = 0,$$
$$X^{(p)} = Y^{(p)} = 0, \qquad Z^{(p)} = Z.$$

Consequently, E is a central extension of $U \simeq \boldsymbol{\alpha}_2 \times \boldsymbol{\alpha}_2$ by $\boldsymbol{\mu}_2$. Each factor $\boldsymbol{\alpha}_2$ of U lifts uniquely in E, but U itself does not lift in E, since $[X, Y] \neq 0$.

7. Nilpotent Affine Algebraic Groups

7.1. Extensions of Groups of Multiplicative Type

Proposition 7.1.1.— *Let S be a prescheme, let H and K be two S-group preschemes of multiplicative type and finite type, and let E be an S-group prescheme which is an extension of K by H (that is, K is the quotient of E by H for the fpqc topology). Then E is of multiplicative type in either of the following two cases:*

(a) E is commutative.

(b) The fibres of K are connected.

Proof.

(i) Suppose first that S is the spectrum of a field k. The assertion is local for the fpqc topology, so we may suppose that k is algebraically closed and hence that H and K are diagonalizable. Notice that K acts trivially on H by inner automorphisms: this is clear in case (a), and follows from 6.2.1 in case (b). Induction on the length of a suitable composition series of K reduces us to the case where K is of one of the following three types:

$$\text{(a) } K = \mathbb{G}_m, \qquad \text{(b) } K = \boldsymbol{\mu}_p, \qquad \text{(c) } K = \boldsymbol{\mu}_q, \quad (q,p) = 1,$$

and in case (c), E is commutative. Now embed H in $\mathbb{G}_m^r$. It follows that E embeds in an extension of K by $\mathbb{G}_m^r$, and we may therefore suppose that $H = \mathbb{G}_m^r$.

(a) If $K = \mathbb{G}_m$, then E is a smooth, connected, affine algebraic group (Exposé $\mathrm{VI_B}$, 9.2) of unipotent rank zero, and is therefore a torus.

(b) Let $K = \boldsymbol{\mu}_p$. In this case E is a trivial extension. Indeed, since H is smooth, the canonical morphism

$$\operatorname{Lie} E \longrightarrow \operatorname{Lie} \boldsymbol{\mu}_p$$

is surjective (Appendix II, 3.2), and the assertion follows from 5.8.2(i), in view of Appendix II, 2.2.

(c) Let $K = \boldsymbol{\mu}_q$, where $(q,p) = 1$, and suppose that E is commutative. Again the extension E is trivial. Indeed, let $x \in E(\bar{k})$ lift a generator $\bar{x}$ of $\boldsymbol{\mu}_q(\bar{k})$. The element x^q belongs to $H(\bar{k})$, and hence has the form y^q, with $y \in H(\bar{k})$ (notice that $\mathbb{G}_m^r(\bar{k})$ is q-divisible). Since E is commutative, $y^{-1}x$ is a lift of $\bar{x}$ having order q.

(ii) Consider now the general case. The groups H and K are flat, affine, and of finite presentation over S (Exposé IX, 2.1), and consequently so is E (Exposé $\mathrm{VI_B}$, 9.2). Using the general technique of Exposé $\mathrm{VI_B}$, §10, we reduce to the case where S is noetherian. To prove that E is of multiplicative type, it then suffices to prove that E is commutative and that ${}_nE$ is finite over S for every n (Exposé X, 4.8(b)).

(a) E is commutative. We must verify that the morphism

$$E \times_S E \longrightarrow E, \qquad (x,y) \longmapsto [x,y] = xyx^{-1}y^{-1}$$

factors through the unit section of E, and it is enough to verify this when S is the spectrum of an artinian local ring. But then E is of multiplicative type by (i) and Exposé X, 2.3.

(b) The group ${}_nE$ is finite over S. Indeed, there is an exact sequence

$$0 \longrightarrow {}_nH \longrightarrow {}_nE \longrightarrow {}_nK \xrightarrow{u} H_n,$$

where H_n is the cokernel of the n-th power map on H. We know that H_n is of multiplicative type (Exposé IX, 2.7), and hence separated, and that ${}_nH$ and ${}_nK$ are finite over S. The group $\ker u$ is a closed subgroup of ${}_nK$, and hence is finite over S; consequently ${}_nE$ is finite over S, as an extension of one finite group by another (Exposé $\mathrm{VI_B}$, 9.2).

7.2. Structure of Commutative Affine Algebraic Groups

Theorem 7.2.1.— *Let k be a field and G a commutative affine algebraic k-group. Then:*

(a) G contains a largest subgroup M of multiplicative type. The group M is characteristic in G, the quotient G/M is unipotent, and the formation of M commutes with extension of the ground field k.

(b) If k is perfect, G is the direct product of M and a unipotent algebraic subgroup U, uniquely so.

Proof.

(i) Suppose that k is algebraically closed. When G is smooth, 7.2.1(b) is well known (BIBLE 4, Theorem 4). If G is radicial of height 1, the decomposition of $\mathrm{Lie}\, G$ described in 4.2.2 corresponds, in view of 4.3.1(v) and Appendix II, 2.2, to a decomposition of G of the kind asserted in 7.2.1(b). In general, G has a composition series whose successive quotients are smooth or radicial of height 1 (Appendix II, 3.1). To prove 7.2.1(b), it therefore suffices to observe the following. Suppose there is an exact sequence of commutative algebraic groups

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0,$$

in which G', respectively G'', is the product of a group of multiplicative type and a unipotent group,

$$G' = M' \cdot U', \qquad G'' = M'' \cdot U''.$$

Then the same is true of G. Indeed, consider the exact sequence

$$0 \longrightarrow G'/U' \longrightarrow G/U' \longrightarrow G'' \longrightarrow 0.$$

By 7.1.1(a), the inverse image of M'' in G/U' is a subgroup M_1 of multiplicative type. The group M_1 lifts to a subgroup M of G by 5.1.1(i)(a). Likewise, now using 6.1.1(C), one proves that there is a unipotent subgroup U of G which is an extension of U'' by U'. Clearly $G = M \cdot U$.

(ii) Let k now be arbitrary. By (i), $G_{\bar{k}}$ is the direct product of a group $M_{\bar{k}}$ of multiplicative type and a unipotent group $U_{\bar{k}}$. Put

$$S = \bar{k} \times_k \bar{k},$$

and let M_1 and M_2 be the two inverse images of $M_{\bar{k}}$ under the two projections $S \rightrightarrows \bar{k}$. The group

$$G_S/M_2 = (G/M_{\bar{k}})_S$$

has unipotent fibres; hence the image of M_1 in G_S/M_2 is trivial by 2.4(i), and $M_1 \subset M_2$. The converse inclusion follows in the same way, so $M_1 = M_2$. By fpqc descent, $M_{\bar{k}}$ therefore descends to an algebraic subgroup M of G. It is clear that M is of multiplicative type, that G/M is unipotent, and that the formation of M is compatible with every extension of k. For every k-prescheme S, every subgroup H of multiplicative type in G_S is contained in M_S. Indeed, its image in the group with unipotent fibres

$$(G/M)_S = G_S/M_S$$

is trivial by 2.5.

Taking H, in particular, to be the transform of M_S under an automorphism of G_S, we conclude that M is characteristic in G.

Finally, if k is perfect, G/M lifts in G to a unipotent group, uniquely so by 6.1.1(C).

Remark 7.2.2.—

(i) If k is not perfect, the unipotent component of $G_{\bar{k}}$ need not be defined over k, as Example 6.4(a) shows.

(ii) In contrast to the multiplicative-type component M, the unipotent component U is not in general characteristic in G, whatever the characteristic of k. Of course, the uniqueness of the decomposition in 7.2.1(b) implies that U is invariant under every k-automorphism of G. But if U and M are such that there exist a k-prescheme S and a nonzero S-homomorphism

$$h : U_S \longrightarrow M_S$$

(cf. 2.6), then one obtains an S-automorphism of G_S,

$$(u, m) \longmapsto (u, h(u) + m),$$

which does not leave U_S invariant.

(iii) If G is finite over k, then under Cartier duality (2.6) the quotient G/M corresponds to the connected component of the identity in the dual $D(G)$ of G.

7.3. Structure of Nilpotent Affine Algebraic Groups

Theorem 7.3.1.— *Let k be a field and G a nilpotent (Exposé VI$_B$, §8), affine, connected algebraic k-group. Then G has a largest subgroup M of multiplicative type. The group M is central and characteristic, and G/M is a unipotent algebraic group.*

Let Z be the centre of G, and let M be the largest subgroup of multiplicative type in Z, as supplied by 7.2.1. Since Z is characteristic in G, and M is characteristic in Z, the group M is characteristic in G. It remains to show that G/M is unipotent. By induction on the length of the ascending central series of G, this follows from the following more general lemma.

Lemma 7.3.2 (Rosenlicht).— *Let G be a connected algebraic k-group and let Z be its centre. Then the centre Z' of $G' = G/Z$ is unipotent.*

Proof. We may suppose that k is algebraically closed. It is then enough to show that Z' contains no subgroup isomorphic to $\boldsymbol{\mu}_\ell$, for any prime number ℓ, by 4.6.1(vi). Let $\boldsymbol{\mu}_\ell$ be such a subgroup of Z', and let N be its inverse image in G. Since $\boldsymbol{\mu}_\ell$ is central in G', the group N is invariant in G.

(i) Suppose that $(\ell, p) = 1$. We can find an element $x \in N(k)$ and an integer n with the following properties:

(a) x lifts a generator $\bar{x}$ of $\boldsymbol{\mu}_\ell$;
(b) $x^{\ell^n} \in Z^0(k)$.

Indeed, it suffices to choose a lift of $\bar{x}$ whose image in $(N/Z^0)(k)$ belongs to the ℓ-Sylow subgroup. Raising to the ℓ-th power in the commutative group Z^0 is an étale morphism, so $Z^0(k)$ is ℓ-divisible. Consequently, after multiplying x by an element of $Z^0(k)$, we may suppose that $x^{\ell^n} = 0$. The group N is then generated by two commuting commutative groups, namely Z and the group generated by x, and is therefore commutative. The group ${}_{\ell^n}N$ is of multiplicative type and is characteristic in N; hence it is invariant in G, and consequently central because G is connected (Exposé IX, 5.5). Thus ${}_{\ell^n}N$ is contained in Z, contradicting the fact that its image in G' contains $\boldsymbol{\mu}_\ell$.

(ii) Suppose that $\ell = p$. There is then an integer n such that the image of F_nN in G' contains $\boldsymbol{\mu}_p$, and hence an exact sequence

$$(*) \qquad 1 \longrightarrow K \longrightarrow F_nN \longrightarrow \boldsymbol{\mu}_p \longrightarrow 1.$$

The group K is contained in Z, and is therefore commutative. It follows from 7.2.1, 7.1.1(b), and 5.5.1 that there is a subgroup of multiplicative type in F_nN whose image in the quotient by K is $\boldsymbol{\mu}_p$. As in (i), we deduce that F_nN is commutative. Its multiplicative-type component (7.2.1) is characteristic in F_nN, hence invariant in G, and therefore central (Exposé IX, 5.5). Since its image in G' contains $\boldsymbol{\mu}_p$, this is a contradiction.

A. Appendix I. Hochschild Cohomology and Extensions of Algebraic Groups

A.1. Definition of the Cohomology Groups

Let k be a field, let G be an algebraic k-group, and let $(\mathrm{Ab})_G$ be the abelian category of commutative algebraic k-groups on which G acts. If $A \in \mathrm{Ob}(\mathrm{Ab})_G$, the functor

$$h_A : (\mathrm{Sch}\,/k)^\circ \longrightarrow \mathrm{Ens}$$

canonically defined by A is a G-$\mathbb{Z}$-module in the sense of Exposé I, 3.2. We may therefore consider the standard complex $C^\bullet(G, A)$ of algebraic cochains of G with values in A (Exposé I, 5.1), together with the group $Z^i(G, A)$ of i-cocycles, the group $B^i(G, A)$ of i-coboundaries, and the i-th cohomology group $H^i(G, A)$. As usual, $H^0(G, A)$ is identified with the group $A^G(k)$, where A^G is the k-functor of invariants of A under G. The group $H^1(G, A)$ classifies the trivial principal homogeneous spaces under A on which G acts.

In general, the functor

$$A \longmapsto H^\bullet(G, A)$$

from $(\mathrm{Ab})_G$ to Ab is not a cohomological functor. Nevertheless, we have the following proposition.

Proposition A.1.1.— *Let*

$$0 \longrightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \longrightarrow 0$$

be an exact sequence in $(\mathrm{Ab})_G$. *Then:*

(a) *if v has a section—that is, if there is a k-morphism of preschemes $s : A'' \to A$ such that $vs = 1_{A''}$—then there is the usual exact cohomology sequence*

$$\cdots \longrightarrow H^i(G, A) \longrightarrow H^i(G, A'') \xrightarrow{d} H^{i+1}(G, A') \longrightarrow \cdots ;$$

(b) *if $A(k) \to A''(k)$ is surjective, then the sequence*

(1)

$$0 \longrightarrow A'^G(k) \longrightarrow A^G(k) \longrightarrow A''^G(k) \xrightarrow{d} H^1(G, A') \longrightarrow H^1(G, A) \longrightarrow H^1(G, A'')$$

is exact.

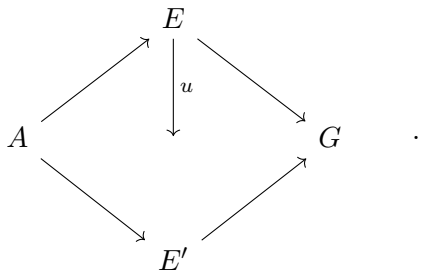
Proof. For (a), observe that the existence of a section implies the exactness of the sequence of complexes

$$0 \longrightarrow C^\bullet(G, A') \longrightarrow C^\bullet(G, A) \longrightarrow C^\bullet(G, A'') \longrightarrow 0.$$

For (b), if $x'' \in A''^G(k)$, its inverse image in A is a trivial principal homogeneous space under A' (because $A(k) \to A''(k)$ is assumed to be surjective) on which G acts, and therefore defines an element $d(x'') \in H^1(G, A')$. The exactness of the sequence (1) is then immediate.

A.2. The Group $\mathrm{Ext}_{\mathrm{alg}}(G, A)$

Let A and G be two algebraic k-groups, and let E and E' be two algebraic k-groups that are extensions of G by A. These two extensions are said to be *isomorphic* if there is a k-morphism of groups $u : E \to E'$ that makes the following diagram commutative:



The group E acts on A by inner automorphisms. If A is commutative, this action factors through G, so that G acts on A. Conversely, if $A \in \mathrm{Ob}(\mathrm{Ab})_G$, we denote by $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ the set of classes of algebraic extensions E of G by A for which the action of G on A defined by E agrees with the action coming from the structure of A as an object of $(\mathrm{Ab})_G$.

The set $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ is naturally a bifunctor, covariant in A and contravariant in G. More precisely:

(a) if $f : A \to B$ is a morphism in $(\mathrm{Ab})_G$, and if E represents an element of $\mathrm{Ext}_{\mathrm{alg}}(G, A)$, then $f_*(E) \in \mathrm{Ext}_{\mathrm{alg}}(G, B)$ is defined to be the class of the extension of G by B equal to the quotient of the semidirect product $B \cdot E$ (where E acts on B through G) by the algebraic subgroup that is the image of A under the morphism (f, i). This quotient is representable by Exposé $\mathrm{VI_A}$, §5, and there is therefore a commutative diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & A & \xrightarrow{i} & E & \longrightarrow & G & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle f} & & \downarrow & & \downarrow{\scriptstyle 1_G} & & \\ 1 & \longrightarrow & B & \longrightarrow & f_*(E) & \longrightarrow & G & \longrightarrow & 1 \end{array} .$$

(b) if $g : H \to G$ is a k-morphism of algebraic k-groups, and if E is an extension of G by A, the fiber product $E \times_G H$ is naturally an extension of H by A, denoted by $g^*(E)$. Thus there is a commutative diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & A & \longrightarrow & g^*(E) & \longrightarrow & H & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle 1_A} & & \downarrow & & \downarrow{\scriptstyle g} & & \\ 1 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & G & \longrightarrow & 1 \end{array} .$$

By adapting the proofs given in J.-P. Serre, *Groupes algébriques et corps de classe*, Chapter VII, one endows $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ with a natural abelian-group structure, functorial in A and G.

Proposition A.2.1.— *Let*

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

be an exact sequence in $(\mathrm{Ab})_G$. *Then:*

(a) there is a canonical exact sequence of abelian groups

$$Z^1(G, A) \longrightarrow Z^1(G, A'') \xrightarrow{d} \mathrm{Ext}_{\mathrm{alg}}(G, A') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A'');$$

(b) if $A(k) \to A''(k)$ is surjective, one deduces from (a) the exact sequence

$$(2)\quad H^1(G, A) \longrightarrow H^1(G, A'') \xrightarrow{d} \mathrm{Ext}_{\mathrm{alg}}(G, A') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A'').$$

The exact sequence in (a) generalizes the usual exact $\mathrm{Hom}(\,\cdot\,,\,\cdot\,)$ sequence valid for commutative extensions (*loc. cit.*), and is proved in the same way. We recall only the definition of the coboundary

$$d : Z^1(G, A'') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A').$$

For this purpose, consider the extension

$$1 \longrightarrow A' \longrightarrow A \cdot G \longrightarrow A'' \cdot G \longrightarrow 1,$$

obtained in the evident way from the exact sequence $0 \to A' \to A \to A'' \to 0$. If $u \in Z^1(G, A'')$, then u defines in the usual way a section homomorphism

$$u : G \longrightarrow A'' \cdot G.$$

One then has

$$d(u) = u^*(A \cdot G).$$

A.3. Comparison of $H^2(G, A)$ and $\mathrm{Ext}_{\mathrm{alg}}(G, A)$

It is well known that, for abstract groups, there is a functorial isomorphism between the abelian groups $H^2(G, A)$ and $\mathrm{Ext}(G, A)$. Likewise in the present setting, if A is an object of $(\mathrm{Ab})_G$, one can associate with every 2-cocycle $u \in Z^2(G, A)$ an algebraic-group structure on the prescheme $A \times_k G$, making it an element of $\mathrm{Ext}_{\mathrm{alg}}(G, A)$. Moreover, this extension

is trivial if and only if $u \in B^2(G, A)$ (see Exposé III, 1.2.2). Recall that the composition law on $A \times G$ is defined by

$$(a, g)(a', g') = a + {}^g a' + u(g, g').$$

It is clear that the extensions of G by A obtained in this way are not arbitrary, since they possess a section. Conversely, however, if $E \in \mathrm{Ext}_{\mathrm{alg}}(G, A)$ possesses a section s, then E is isomorphic to the extension of G by A associated with the 2-cocycle u such that

$$u(g, g') = s(g)s(g')s(gg')^{-1}.$$

We thus obtain the following proposition.

Proposition A.3.1.— *There is a functorial isomorphism between the bifunctors with values in abelian groups*

$$(G, A) \longmapsto H^2(G, A) \qquad \textit{and} \qquad (G, A) \longmapsto \mathrm{Ext}_s(G, A),$$

where $\mathrm{Ext}_s(G, A)$ *denotes the subgroup of* $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ *formed by the classes of extensions of* G *by* A *that possess a section.*

B. Appendix II. Review and Supplements on Radicial Groups

Let $p > 1$ be a prime number, and let S be an $\mathbb{F}_p$-prescheme.

B.1. The Frobenius morphism

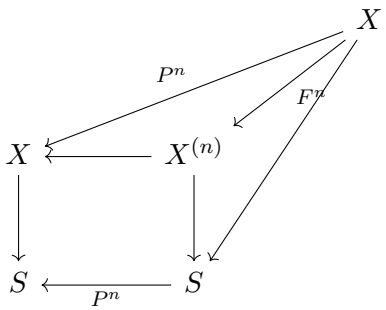
For every S-prescheme X and every integer $n > 0$, let

$$P^n : X \longrightarrow X$$

denote the $\mathbb{F}_p$-endomorphism corresponding to raising sections of $\mathcal{O}_X$ to the p^n-th power, and let $X^{(n)}$ denote the S-prescheme obtained from X by base change along $P^n : S \to S$. There is then a unique S-morphism

$$F^n : X \longrightarrow X^{(n)}$$

that makes the following diagram commutative:



It is clear that F^n identifies with the "n-th iterate" of $F^1 = F$, called the Frobenius morphism of X/S.

If G is an S-group prescheme, then $G^{(n)}$ is an S-group prescheme and

$$F^n : G \longrightarrow G^{(n)}$$

is an S-group morphism. Its kernel ${}_{F^n}(G)$ is a characteristic subgroup prescheme of G (that is, it is stable under the functor $\mathrm{Aut}_{S\text{-gr}}(G)$), and is radicial over S. If G is a radicial S-group prescheme, we say that G has height at most h if ${}_{F^h}(G) = G$.

B.2. Groups and restricted p-Lie algebras

If G is an S-group prescheme, then $\mathrm{Lie}(G)$ (Exposé II) is naturally a restricted p-Lie algebra (Exposé $\mathrm{VII_A}$, §§5 and 6). In particular, one has the following result (cf. Exposé $\mathrm{VII_A}$).

Proposition B.2.1.—

(i) $\mathrm{Lie}(\boldsymbol{\alpha}_p)_S = \mathrm{Lie}(\mathbb{G}_a)_S$[11] *is a sheaf of $\mathcal{O}_S$-modules, free of rank* 1 *over $\mathcal{O}_S$, generated by an element X such that $X^{(p)} = 0$.*

(ii) $\mathrm{Lie}(\boldsymbol{\mu}_p)_S = \mathrm{Lie}(\mathbb{G}_m)_S$ *is a sheaf of $\mathcal{O}_S$-modules, free of rank* 1 *over $\mathcal{O}_S$, with a canonical basis X such that $X^{(p)} = X$.*

We now recall the fundamental result proved in Exposé $\mathrm{VII_A}$, §7.

Theorem B.2.2.— *Suppose that S is affine, with ring A. Then the functor*

$$G \longmapsto \mathrm{Lie}\, G$$

defines an equivalence between, on the one hand, the category of S-group preschemes G that are of finite presentation and flat over S, of height 1*, and whose Lie algebra is locally free over $\mathcal{O}_S$, and, on the other hand, the category of restricted p-A-Lie algebras that are locally free of finite rank.*

Moreover, if G is as above and H is any S-group prescheme of finite presentation, the canonical morphism

$$\mathrm{Hom}_{S\text{-gr}}(G, H) \longrightarrow \mathrm{Hom}_{p\text{-}A\text{-Lie}}\big(\mathrm{Lie}\, G(S), \mathrm{Lie}\, H(S)\big)$$

is an isomorphism.

B.3. Radicial groups and smooth groups

We now suppose that S is the spectrum of a field k of characteristic p. Recall that Exposé $\mathrm{VI_A}$, §5 proves that, if G is an algebraic k-group and H an algebraic subgroup of G, then the sheaf G/H (for the fpqc topology) is representable. We also recall ($\mathrm{VII_A}$, 8.3):

Proposition B.3.1.— *Let G be an algebraic k-group. There exists an integer m such that, for every $n \geq m$, the algebraic group*

$$G/_{F^n}(G)$$

is smooth over k.

Proposition B.3.2.— *Consider an exact sequence of algebraic k-groups*

$$1 \longrightarrow G' \longrightarrow G \xrightarrow{u} G'' \longrightarrow 1$$

and the following assertions:

(i) the morphism u is smooth;

(ii) G' is smooth over k;

[11] Editor's note: the definition of $\boldsymbol{\alpha}_p$, already given at the beginning of this Exposé, has been omitted here.

(iii) for every integer $n > 0$, the sequence

$$1 \longrightarrow {}_{F^n}(G') \longrightarrow {}_{F^n}(G) \longrightarrow {}_{F^n}(G'') \longrightarrow 1$$

is exact;

(iv) the morphism ${}_FG \to {}_FG''$ is an epimorphism;

(v) the morphism

$$(\mathrm{Lie}\, G)(k) \longrightarrow (\mathrm{Lie}\, G'')(k)$$

is surjective.

Then

$$\text{(i)} \Longleftrightarrow \text{(ii)} \Longrightarrow \text{(iii)} \Longrightarrow \text{(iv)} \Longleftrightarrow \text{(v)}.$$

Moreover, if G is smooth, all five assertions are equivalent.

The equivalence (i)$\Longleftrightarrow$(ii) follows from Exposé $\mathrm{VI_B}$, 9.2(vii).
For (ii)$\Longrightarrow$(iii), if G' is smooth, then

$$F^n : G' \longrightarrow G'^{(n)}$$

is an epimorphism, and (iii) follows from the snake diagram. The implication (iii)$\Longrightarrow$(iv) is clear, while (iv)$\Longleftrightarrow$(v) follows from Theorem B.2.2. Finally, suppose that G is smooth and that (v) holds. Then G'' is smooth, and (v) implies

$$\dim G' = \dim_k (\mathrm{Lie}\, G')(k),$$

so G' is smooth.

C. Appendix III. Remarks and Supplements Concerning Exposés XV, XVI, and XVII

C.1.

It may be that Propositions 1.2 and 1.2 bis of Exposé XV remain valid if the hypothesis that H_0 is smooth over S_0 is removed. This is the case, in particular, if G is finite, flat, and commutative.

C.2. Complement to Exposé XV, 4.8

The following proposition, as well as Theorems C.3.1 and C.4.1 below, will appear in a forthcoming article by M. Raynaud on group schemes over a discrete valuation ring.[12]

Proposition C.2.1.— *Let S be the spectrum of a discrete valuation ring, let t be its generic point, let G be a flat S-group prescheme of finite type, and put*

$$\widetilde{G} = \mathrm{Spec}\, \Gamma(G, \mathcal{O}_G).$$

Let $u : G \to \widetilde{G}$ be the canonical morphism. Then:

(1) $\widetilde{G}$ is naturally an S-group scheme, and u is a homomorphism.

[12]Editor's note: comments are to be added here, including references to Exposé $\mathrm{VI_B}$, ...

(2) $\ker(u)$ *is flat over* S*, and* $(\widetilde{G}, u)$ *is the fpqc quotient of* G *by* $\ker(u)$*. Thus* $\widetilde{G}$ *is the largest affine quotient of* G*.*

(3) If G_t *is affine, then* $\ker(u)$ *is an étale group over* S*, and it is the trivial group if and only if* G *is separated over* S*. In particular, a flat, separated* S*-group scheme of finite type whose generic fiber is affine is affine.*

C.3.

In the statement of Exposé XV, 6.6, the hypothesis that H be equal to its connected normalizer is unnecessary in order that the set of points s of S for which H_s is a parabolic subgroup of G_s be open. Indeed, repeat the proof given in paragraph c) preceding Lemma 6.9, and let N denote the scheme-theoretic closure of N_t in G. Then N is a closed subgroup scheme of G, flat over S, which contains H and is contained in $\mathcal{N}$. The theorem below shows that G/N is representable. Since G_s/N_s is proper and connected, the proof concludes as in *loc. cit.*

Theorem C.3.1[13]**.**— *Let* S *be the spectrum of a henselian discrete valuation ring, let* G *be an* S*-group prescheme locally of finite type, and let* H *be a closed subgroup prescheme of* G*, flat over* S*. Then* G/H *is representable.*

C.4. Complement to Exposé XVI, 1.1

The following theorem refines Exposé VIII, 7.9.

Theorem C.4.1.— *Let* S *be the spectrum of a discrete valuation ring with residue characteristic* $p > 0$*, and let* G *be a commutative, smooth, separated* S*-group scheme of finite type. Then the following conditions are equivalent:*

(i) ${}_pG$ *is finite over* S*.*

(ii) For every S*-prescheme* S'*, every separated* S'*-group prescheme* H' *of finite presentation over* S'*, and every* S'*-monomorphism*

$$u : G_{S'} \longrightarrow H',$$

the morphism u *is an immersion.*

C.5.

Example XVII, 6.4(a), gives a smooth group over a field k whose unipotent radical is not defined over k. The following proposition gives a general method for obtaining such groups.

Proposition C.5.1.— *Let* k *be a field, let* K *be a finite radicial extension of* k *of degree greater than* 1*, and let* H *be a smooth, connected* K*-algebraic group of dimension* r*. Put*

$$G = \prod_{K/k} H/K,$$

[13]Editor's note: see Theorem 4C in S. Anantharaman, "Schémas en groupes, espaces homogènes et espaces algébriques sur une base de dimension 1," *Bull. Soc. Math. France, Mém.* 33 (1976), 5–79.

which is a smooth, connected k-algebraic group of dimension $r[K:k]$. Let

$$u : G_K \longrightarrow H$$

be the canonical homomorphism, and put $R = \ker(u)$. Then:

(i) u is an epimorphism, and R is a smooth, connected, unipotent algebraic group.

(ii) If U is a smooth algebraic subgroup of G such that U_K contains R, then $U = G$.

Corollary.— *Retain the notation of the preceding proposition.*

(a) If H is not unipotent, then the unipotent radical of $G_{\bar{k}}$ is not defined over k.

(b) If H is a nonzero abelian variety, then G is not an extension of an abelian variety by a smooth linear group.

(c) If H is not solvable, then the solvable radical of $G_{\bar{k}}$ is not defined over k, and G has no Borel subgroup defined over k.

Let us first show how the corollary follows from the proposition.

(a) Let U be a unipotent radical of G. Then U_K is the unipotent radical of G_K, and hence contains R, since R is smooth, connected, and unipotent by (i). Thus $U = G$ by (ii). But G_K has H as a quotient, and H is not unipotent by hypothesis. Therefore G is not unipotent, a contradiction.

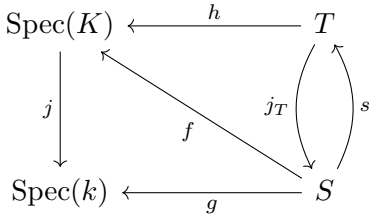
(b) Suppose that G is an extension of an abelian variety by a linear group L that is smooth over k. Then L_K necessarily contains R, so $L = G$. But G cannot be a linear group, because G_K has a nonzero abelian variety as a quotient.

(c) Let S be a solvable radical of G, and let B be a Borel subgroup of G. Then S_K and B_K contain R, so $S = G$ and $B = G$. But then G is solvable, contradicting the fact that G_K has a quotient that is not solvable.

Proof of Proposition C.5.1. *(i)* We begin by describing the morphism u. A K-scheme

$$f : S \longrightarrow \mathrm{Spec}(K)$$

gives rise to the diagram



where

$$g = j \circ f, \qquad T = \mathrm{Spec}(K) \times_{\mathrm{Spec}(k)} S,$$

h and j_T are the two projections, and s is the section of T over S such that $h \circ s = f$. The map

$$u(S) : G_K(S) \longrightarrow H(S)$$

is simply the composite

$$G_K(S) \xrightarrow{\sim} H(T) \longrightarrow H(S),$$

where the last arrow is defined by the section s.

Take, in particular, S to be the spectrum of an algebraic closure $\bar{k}$ of k, and let f be the unique k-morphism induced by $K \to \bar{k}$, so that T is a local Artinian scheme. To prove (i), it is enough to do so after extending the base field from K to $\bar{k}$. It is clear that

$$G_S = \prod_{T/S} H_T/T$$

represents the Greenberg functor of H_T relative to S (M. J. Greenberg, "Schemata over local rings," *Ann. of Math.* 73 (1961), 624–648). Under this identification, the description above shows that u_S is the canonical transition morphism

$$\mathrm{Green}(H_T) \longrightarrow H_S = H_T \times_T S.$$

Assertion (i) now follows from the smoothness of H over K and M. J. Greenberg, "Schemata over local rings II," *Ann. of Math.* 78 (1963), 256–266.

(ii) To prove (ii), we may suppose that k is separably closed. Let U be a smooth algebraic subgroup of G such that $U_K \supset R$, and let us prove that $U = G$. Since G is connected, we may suppose that U is connected. Let $V = u(U_K)$, a smooth, connected algebraic subgroup of H, and put

$$V' = \prod_{K/k} V/K,$$

a smooth, connected algebraic subgroup of G. The group V'_K is an algebraic subgroup of G_K, and the canonical morphism $V'_K \to V$ is the restriction of u to V'_K. By hypothesis, U_K contains R, hence *a fortiori* contains $\ker(V'_K \to V)$; by construction, moreover, the image of U_K in H is V. It follows that U_K contains V'_K, and hence that U contains V'.

On the other hand, the canonical isomorphism

$$G(k) \hookrightarrow G(K) \xrightarrow{u(K)} H(K)$$

plainly sends $U(k)$ into $V(K) = V'(k)$. Thus $U(k) \subset V'(k)$. Since U is smooth and k is separably closed, it follows that $U \subset V'$, whence $U = V'$. We then have

$$([K : k] - 1) \dim V = \dim \ker(V'_K \to V) = \dim R = ([K : k] - 1) \dim H.$$

Since $K \neq k$, we conclude that $\dim V = \dim H$, hence that $V = H$, and finally that $U = V' = G$.

Exposé XVIII
Weil's Theorem on the Construction of a Group from a Rational Law

by Michael Artin

0. Introduction

This Exposé is devoted to the well-known theorem of Weil [1], which constructs a group variety from a birational law. It seems that the generalization of this result to the case where the base is the spectrum of a discrete valuation ring was already known to several people; see, for example, [2]. Here we shall prove the theorem for a flat group of finite presentation over an arbitrary base prescheme S. Since the hypotheses must be formulated somewhat more carefully, the statement is not in the form given by Weil; but when S is the spectrum of a field, the form given here is seen to be essentially equivalent to Weil's. We use the following suggestion of Grothendieck: given a "group germ" X over a prescheme S, one should construct the group as the quotient of $X \times_S X$ by a suitable equivalence relation.

1. "Recollections" on Rational Maps

Let X/S be a relative prescheme and let $U \subseteq X$ be an open subset. We say that U is *schematically dense in X relative to S*, or that $U \hookrightarrow X$ is *relatively schematically dense*, if, for every base change $S' \to S$, the open subset $U \times_S S'$ is schematically dense in $X \times_S S'$. For the definition and properties of this notion, we refer to Exposé IX, §4.*

Proposition 1.1. *(i) A finite intersection, as well as a union of a nonempty family, of open subsets schematically dense relative to S, is schematically dense relative to S.*

(ii) If $U \subseteq X$ is schematically dense relative to S, and if $S \to T$ is a morphism, then $U_T \subseteq X_T$ is schematically dense relative to T.

(iii) Let $U \subseteq V \subseteq X$ be open immersions. For U to be schematically dense in X relative to S, it is necessary and sufficient that it be so in V, and that V be so in X.

[0] Version of May 8, 2009.

* See also EGA IV$_3$, 11.9 and 11.10 (especially 11.10.8), where the term "universally schematically dense relative to S" is used.

(iv) If $U \subseteq X$ and $V \subseteq Y$ are relatively schematically dense, then $U \times_S V$ is schematically dense in $X \times_S Y$ relative to S.

In Exposé IX one finds the following criterion:

Proposition 1.2. *Let X/S be locally of finite presentation and flat, and let U be an open subset of X. If, for every $s \in S$, the fiber U_s is schematically dense in X_s, then U is schematically dense in X relative to S.*

In particular:

Corollary 1.3. *Suppose that X/S is locally of finite presentation and flat, and that every fiber X_s is "without embedded components." Then U is schematically dense in X relative to S if and only if, for every $s \in S$, U_s is dense in X_s in the topological sense.*

One easily deduces from the definition the following proposition.

Proposition 1.4. *Let $U \subseteq X$ be schematically dense relative to S, and let $f, g : X \to Y$ be two morphisms of functors over S, where Y satisfies one of the following conditions:*

(i) *Y is a prescheme over S, and every fiber Y_s is separated.*

(ii) *Y is a presheaf[1] over S, such that the diagonal morphism $Y \to Y \times_S Y$ is representable by a closed immersion.*[2]

Then, if $f = g$ on U, one has $f = g$.

Proof. In case (i), the standard argument following IX, 4.1 shows that f and g are equal on every fiber; hence the subprescheme $\mathrm{Ker}(f, g)$ of X is set-theoretically equal to X, and is therefore closed. Since it contains U, it is equal to X, that is, $f = g$. Case (ii) is proved as in *loc. cit.*[3]

Definition 1.5. [*] *Let X be a prescheme over S, and let Y be a presheaf over S. A* rational map *$f : X \dashrightarrow Y$ over S is an equivalence class of morphisms $f : U \to Y$ over S, where U is an open subset of X that is schematically dense relative to S. One sets*

$$(f : U \to Y) \sim (f' : U' \to Y)$$

if and only if there exists an open subset $U'' \subseteq U' \cap U$, schematically dense relative to S, such that $f = f'$ on U''.

This definition is made in such a way that, for every base change $S' \to S$, the rational map $f \times_S S'$ may be defined in the obvious way.

If U is an open subset of X, we say that the rational map f is *defined on U* if there exists a morphism representing f whose set of definition contains U.

[1] Editor's note: specify for which topology; a priori, the fpqc topology.

[2] Editor's note: Let us recall here the definition of a closed immersion. A morphism of S-functors $F \to G$ is *relatively representable* if, for every S-morphism $T \to G$, where T is an S-prescheme, the T-functor $F_T = F \times_G T$ is representable by a T-prescheme. A morphism of S-functors $F \to G$ is an open (respectively, closed) immersion if it is relatively representable and if, for every S-morphism $T \to G$, where T is an S-prescheme, the T-morphism $F_T \to T$ is an open (respectively, closed) immersion.

[3] Editor's note: Give details on this point...

[*] See EGA IV_4, 20.5, where the term "pseudo-morphism from X to Y relative to S" is used in order not to conflict with EGA I, 7.12.

Definition 1.5.1. [4] *If Y satisfies one of conditions (i), (ii) of 1.4, it is clear that there exists a largest open subset U of X on which f is defined, and this U is schematically dense relative to S. It is called the* domain of definition of f over S, *and is denoted* $\mathrm{Dom}(f)$.

This notion does not in general commute with base change, but one has:

Proposition 1.6. *Let X be an S-prescheme and Y an S-functor satisfying one of hypotheses (i), (ii) of 1.4. Let $f : X \dashrightarrow Y$ be a rational map over S, let $S' \to S$ be a flat morphism locally of finite presentation, and put $f' = f \times_S S'$. Then*

$$\mathrm{Dom}(f') = \mathrm{Dom}(f) \times_S S'.$$

Proof. Put $U = \mathrm{Dom}(f)$. It is clear that $V' = \mathrm{Dom}(f')$ contains $U \times_S S'$. Let V be the image of V'; it is an open subset of X, because $X' \to X$ is open. We must prove that $V = U$, that is, we must find a morphism $V \to Y$ representing f. Put $S'' = S' \times_S S'$, and

$$X'' = X \times_S S'', \qquad U'' = U \times_S S'', \qquad V'' = V' \times_V V'.$$

Then U'' is schematically dense in X'' relative to S''; hence U'' is schematically dense in V'' relative to S'', since $U'' \subseteq V'' \subseteq X''$. The restriction of $f' : V' \to Y'$ to U' is obtained from $f : U \to Y$ by base change. The two morphisms $V'' \to Y$ obtained from f' by the two projections agree on U'', and hence agree. Since $V' \to V$ is flat and locally of finite presentation, it is an fppf covering (Exposé IV, 6.3), and the morphism $V \to Y$ is obtained by descent.

We shall frequently use the following triviality:

Proposition 1.7. *Let X/S be faithfully flat and locally of finite presentation, and let $U \subseteq X$ be a relatively schematically dense open subset. Then there exists an fppf covering base change $S' \to S$, and a section $x \in X(S')$ contained in $U(S')$.*

Indeed, U/S is an fppf covering. Take $S' = U$, and as the section take the S-graph of the inclusion $U \hookrightarrow X$.

2. Local Determination of a Group Homomorphism

Let G and H be groups, and let $U \subseteq G$ be a subset such that $U \cdot U = G$. If f and g are homomorphisms from G to H such that $f = g$ on U, then $f = g$. Likewise:

Proposition 2.1. *Let S be a site (cf. Exposé IV[4]), let G be a sheaf of groups on S, and let $U \subseteq G$ be a subsheaf of sets such that the morphism $U \times U \to G$ induced by multiplication is an epimorphism. If f and g are homomorphisms from G to a sheaf of groups H and are equal on U, then $f = g$.*

Corollary 2.2. *Let G be an S-group prescheme that is locally of finite presentation and flat, let $U \subseteq G$ be a relatively schematically dense open subset, and let H be a sheaf of groups on S for the fppf topology. Then every homomorphism $f : G \to H$ is determined by its restriction to U.*

Indeed, since G is flat over S, the composition law on G is a flat morphism ($\mathrm{VI_B}$, 9.2(xi)); it follows that $U \times_S U \to G$ is faithfully flat and locally of finite presentation, hence an fppf covering, and therefore an epimorphism.

[4] Editor's note: The number 1.5.1 has been added.

[4] Editor's note: that is, a category endowed with a topology; cf. IV, §4.2.

Proposition 2.3. *Let G be a group prescheme locally of finite presentation and flat over S, let $U \subseteq G$ be a relatively schematically dense open subset, and let H be a sheaf of groups for the fppf topology. Write m_G for the multiplication on G and m_H for that on H.*

Let $\bar{f} : U \to H$ be an S-morphism, and suppose that there exists a relatively schematically dense open subset V of $m_G^{-1}(U) \cap (U \times_S U)$ such that the following diagram is commutative:

$$\begin{array}{ccc} V & \xrightarrow{(\bar{f}\times\bar{f})|_V} & H \times_S H \\ \downarrow{\scriptstyle m_G} & & \downarrow{\scriptstyle m_H} \\ U & \xrightarrow{\bar{f}} & H \end{array}$$

Then there exists a morphism of S-groups $f : G \to H$, necessarily unique, extending $\bar{f}$, in each of the following situations:

(i) *H is representable.*

(ii) *The diagonal morphism $H \to H \times_S H$ is representable by a closed immersion.*[5]

(iii) *For every section $a \in U(S)$, the open subset $\mathrm{pr}_2((a \times_S U) \cap V)$ is relatively schematically dense in U*[*] *, and this assertion remains true after every base change $S' \to S$.*

Proof. First note that V is relatively schematically dense in $G \times_S G$. Indeed, V is relatively schematically dense in

$$m_G^{-1}(U) \cap (U \times_S G) \cap (G \times_S U),$$

and each of the three factors is obtained from $U \subseteq G$ by an evident base change $G \times_S G \to G$. The assertion therefore follows from 1.1.

To construct $f : G \to H$, since $U \times_S U \to G$ is an fppf covering, it is enough to construct a morphism $U \times_S U \to H$ whose two pullbacks to

$$(U \times_S U) \times_G (U \times_S U)$$

are equal. Take the morphism

$$m_H \circ (\bar{f} \times \bar{f}) : U \times_S U \longrightarrow H.$$

For sections $a, b, c, d \in U(S')$, where $S' \to S$ is arbitrary, such that $ab = cd$, we must prove

$$\bar{f}(a)\bar{f}(b) = \bar{f}(c)\bar{f}(d).$$

If (a, b) and (c, d) belong to $V(S')$, the desired equality follows at once. Thus it holds on $(V \times_G V)(S')$.

I claim that $V \times_G V$ is a relatively schematically dense open subset of

$$(U \times_S U) \times_G (U \times_S U).$$

This will complete the proof in cases (i) and (ii) by 1.4, because the morphism f constructed in this way extends $\bar{f}$, and its being a homomorphism is then evident, again by 1.4. Now

$$V \times_G V = (V \times_G (G \times_S G)) \cap ((G \times_S G) \times_G V).$$

[5] Editor's note: See the note at 1.4.

[*] Or in G, which is equivalent by 1.1(iv).

By symmetry and 1.1, it is enough to check that $V \times_G (G \times_S G)$ is relatively schematically dense in $(G \times_S G) \times_G (G \times_S G)$. The latter prescheme is isomorphic over S to $G \times_S G \times_S G$, the isomorphism being given by

$$(a, b, c, d) \longmapsto (a, b, c).$$

What remains is therefore the relative schematic density of $V \times_S G$ in $G \times_S G \times_S G$, which follows from that of V in $G \times_S G$.

It remains to treat case (iii). To prove $\bar{f}(a)\bar{f}(b) = \bar{f}(c)\bar{f}(d)$, we may make an fppf covering base change. Suppose that, after such a base change $S' \to S$, there is a section $x \in G(S')$ such that

$$(b, x), \quad (a, bx), \quad (d, x), \quad (c, dx)$$

all belong to V. Then

$$\begin{aligned}\bar{f}(a)\bar{f}(b)\bar{f}(x) &= \bar{f}(a)\bar{f}(bx) = \bar{f}(abx) = \bar{f}(cdx)\\ &= \bar{f}(c)\bar{f}(dx) = \bar{f}(c)\bar{f}(d)\bar{f}(x),\end{aligned}$$

which gives the desired equality.

To find such an x[5], set, for every $z \in U(S')$,

$$V_z = \mathrm{pr}_{2/S'}((z \times_{S'} U_{S'}) \cap V_{S'}).$$

The conditions imposed on x say exactly that $x \in W$, where

$$W = V_b(S') \cap V_d(S') \cap b^{-1}V_a(S') \cap d^{-1}V_c(S').$$

By hypothesis (iii), W is relatively schematically dense in G. Proposition 1.7 therefore supplies such a section x after an fppf covering base change.

Let us verify that the morphism thus constructed is multiplicative. Let $a, b \in G(S')$, where $S' \to S$ is arbitrary; after this base change, rename S' as S. Choose an fppf covering base extension $S' \to S$ and a section $x \in G(S')$ such that

$$x \in U(S') \qquad \text{and} \qquad ax^{-1} \in U(S'),$$

that is, $x \in (a^{-1}U)^{-1}(S')$. Choose in addition another fppf covering extension $S'' \to S'$ and a section $y \in G(S'')$ such that

$$y, \quad by^{-1}, \quad aby^{-1} \in U(S''),$$

and

$$by^{-1} \in V_x, \qquad xby^{-1} \in V_{ax^{-1}}.$$

Then, by the definition of f, over S'' we have

$$f(a) = \bar{f}(ax^{-1})\bar{f}(x), \qquad f(b) = \bar{f}(by^{-1})\bar{f}(y), \qquad f(ab) = \bar{f}(aby^{-1})\bar{f}(y).$$

Furthermore,

$$\begin{aligned}f(a)f(b) &= \bar{f}(ax^{-1})\bar{f}(x)\bar{f}(by^{-1})\bar{f}(y)\\ &= \bar{f}(ax^{-1})\bar{f}(xby^{-1})\bar{f}(y)\\ &= \bar{f}(aby^{-1})\bar{f}(y) = f(ab),\end{aligned}$$

[5]Editor's note: The beginning of the following sentence has been added.

which proves multiplicativity.

It is now easy to see that f extends $\bar{f}$. Let $a \in U(S')$; after base change, rename S' as S, and choose an fppf covering $S' \to S$ and sections $x, y \in G(S')$ such that

$$(x, y), \qquad (ax, y), \qquad (a, xy)$$

belong to $V(S')$. Then

$$f(xy) = \bar{f}(x)\bar{f}(y) = \bar{f}(xy), \qquad f(axy) = \bar{f}(ax)\bar{f}(y) = \bar{f}(axy).$$

Consequently,

$$\begin{aligned} f(a) &= f(axy)f\big((xy)^{-1}\big) = f(axy)f(xy)^{-1} \\ &= \bar{f}(axy)\bar{f}(xy)^{-1} = \bar{f}(a). \end{aligned}$$

□

Remark 2.3.1. In many cases hypothesis (iii) holds, for it will still hold if U is replaced by a smaller open subset U', still relatively schematically dense in G. For example:

Proposition 2.4. *In the situation of 2.3, suppose that every geometric fiber of G/S is irreducible. Put*

$$U' = \mathrm{pr}_1 V \qquad \textit{and} \qquad V' = V \cap m_G^{-1}(U') \cap (U' \times_S U').$$

Then $U' \subseteq U$ is an open subset relatively schematically dense in G, and U', $\bar{f}|_{U'}$, and V' satisfy hypothesis (iii).

Proof. The subset U' is open because $G \times_S G \to G$ is flat and locally of finite presentation. All the other verifications are immediate except for hypothesis (iii).

Let $a \in U(S')$; after base change, rename S' as S. To verify that

$$\mathrm{pr}_2\big((a \times_S U') \cap V'\big)$$

is relatively schematically dense in U', it suffices by Corollary 1.3 to do so fiber by fiber. Thus it is enough to treat the case where S is the spectrum of a field, and in that case it is enough to prove that U' is nonempty, since G is irreducible and has no embedded components (cf. $\mathrm{VI_A}$, 1.1.1). Now

$$\begin{aligned} \mathrm{pr}_2\big((a \times_S U') \cap V'\big) = \mathrm{pr}_2\big((a \times_S U') \cap V\big) \\ \cap\, \mathrm{pr}_2\big((a \times_S U') \cap m_G^{-1}(U')\big). \end{aligned}$$

The second factor on the right is dense in G. It therefore remains only to prove that

$$\mathrm{pr}_2\big((a \times_S U') \cap V\big)$$

is dense in G, that is, nonempty; and this is clear because $a \in U' = \mathrm{pr}_1 V$. □

3. Construction of a Group from a Rational Law

3.0. Suppose given an S-prescheme X/S and a rational map

$$X \times_S X \dashrightarrow X$$

over S. We seek an S-group G/S and a birational map relative to S[6]

$$X \dashrightarrow G$$

that commutes with the composition laws. We treat only the case in which X/S satisfies the following hypothesis:

($\diamond$) X/S is faithfully flat and of finite presentation, with separated fibers and without embedded components.

(The last two conditions hold for a group prescheme.[7])

We shall often omit the symbol S from fiber products.

Let X/S be a prescheme having the properties ($\diamond$) above, and let W be a subprescheme of finite presentation of $X \times X \times X$ having the following property:

(*) the three morphisms $W \to X \times X$ induced by the projections from X^3 onto X^2 are relatively schematically dense open immersions over S.

Notation. We shall use the following terminology. Given sections $a, b, c \in X(S')$, where $S' \to S$ is arbitrary, such that $(a, b, c) \in W(S')$, we write

$$c = ab, \qquad b = a^{-1}c, \qquad a = cb^{-1}.$$

Definition 3.0.1. [8] Given a section $(a, b) \in X^2(S')$, we say that ab is *defined* if, and only if, there exists a section $c \in X(S')$ such that $(a, b, c) \in W(S')$; equivalently, if and only if $(a, b) \in \mathrm{pr}_{12} W(S')$. Similarly, to say that $a^{-1}b$ or ab^{-1} is defined has the analogous meaning, and we extend this terminology also to products of several factors.

Remark 3.0.2. Let us note at once the following fact. By (*), W defines a rational map $X^2 \dashrightarrow X$ over S, namely the one given by $(a, b) \mapsto ab$. This rational map may well have a domain of definition larger than $\mathrm{pr}_{12} W$. Nevertheless, we say that ab is defined only if $(a, b) \in \mathrm{pr}_{12} W(S')$.

Definition 3.1. A *group germ* is an S-prescheme X/S having the properties ($\diamond$) above, together with a subprescheme W of finite presentation of $X \times X \times X$, having the following properties:

(i) W satisfies property (*) above.

[6] Editor's note: The notion of a birational map (relative to S) should be made explicit, perhaps in an addition to 1.5.2.

[7] Editor's note: Cf. VI_A; the reference is to be specified.

[8] Editor's note: The numbering 3.0.1 and 3.0.2 has been introduced.

(ii) For every section $a \in X(S)$, the sets

$$\begin{aligned} &\mathrm{pr}_i\big((a \times X \times X) \cap W\big), \quad i = 2, 3, \\ &\mathrm{pr}_i\big((X \times a \times X) \cap W\big), \quad i = 3, 1, \\ &\mathrm{pr}_i\big((X \times X \times a) \cap W\big), \quad i = 1, 2, \end{aligned}$$

are schematically dense in X relative to S, and this assertion remains true after every base change $S' \to S$. (Condition (i) implies that these sets are open in X.) Intuitively, this says that ax is defined for sufficiently general x, and similarly for the other products.

(iii) The law is associative: if $(a, b, c) \in X^3(S')$, for an arbitrary $S' \to S$, and both $(ab)c$ and $a(bc)$ are defined, then

$$(ab)c = a(bc).$$

Remarks. (a) In (i), the condition that W be schematically dense in X^2 relative to S may be replaced by the condition that, for every $s \in S$, the fiber W_s be topologically dense in X_s^2, by the hypotheses on X and by 1.2.

(b) Condition (iii) is equivalent to the following. Let V_1 (respectively V_2) be the open subset of X^3 on which the rational map

$$(a, b, c) \longmapsto a(bc) \qquad \big(\text{respectively } (a, b, c) \longmapsto (ab)c\big)$$

is defined* . Then there exists an open subset $V \subseteq V_1 \cap V_2$, schematically dense in X^3 relative to S, on which the two maps agree. This follows from 1.4, since V_1 and V_2 are schematically dense in X^3 relative to S.

(c) Hypothesis (ii) will be used below to ensure that X is a subobject of the group prescheme that it defines. In many cases one can deduce (ii) from (i), provided that X is replaced by a relatively schematically dense open subset X' and W by a relatively schematically dense open subset of $W \cap X'^3$. More precisely:

Proposition 3.2. *Suppose that every geometric fiber of X/S is irreducible, and let W be a subprescheme of X^3 satisfying condition (i) of 3.1. Then there exist a relatively schematically dense open subset $X' \subseteq X$ and a relatively schematically dense open subset $W' \subseteq W \cap X'^3$ such that the pair (X', W') satisfies conditions (i) and (ii). If (iii) holds for (X, W), then it holds for (X', W').*

Proof. Set

$$X' = \bigcap_{i=1}^{3} \mathrm{pr}_i\, W.$$

Each $\mathrm{pr}_i\, W$ is open in X, because $W \to X^2$ is an open immersion and the projections $X^2 \to X$ are flat and of finite presentation. Moreover, $\mathrm{pr}_i\, W$ is schematically dense in X relative to S, because W is so in X^2 and the projections $X^2 \to X$ are surjective; it is enough here to check topological density. Put

$$W' = W \cap X'^3.$$

*In the sense explained in 3.0.1; one could also replace these open subsets by the domains of definition (cf. 1.5) of the rational maps under consideration.

To verify (i), note that

$$W' = (W \cap (X' \times X' \times X)) \cap (W \cap (X \times X' \times X')) \\ \cap (W \cap (X' \times X \times X')).$$

Now

$$W \cap (X' \times X' \times X) \simeq \mathrm{pr}_{12} W \cap (X' \times X'),$$

so this term is schematically dense in W relative to S. The other two terms on the right are likewise schematically dense in W, and hence W' is schematically dense in W. Consequently $\mathrm{pr}_{ij} W'$ is schematically dense in $\mathrm{pr}_{ij} W$, hence in $X \times X$, and therefore in $X' \times X'$.

To verify (ii), let $a \in X'(S')$. After this base change, rename S' as S. We must show, for example, that

$$\mathrm{pr}_2((a \times X' \times X') \cap W')$$

is schematically dense in X' relative to S. By 1.3 it is enough to check this fiber by fiber. Thus it is enough to treat the case in which S is the spectrum of a field, and in that case it is enough to show that the open subset is nonempty, since the fibers of X/S are irreducible and have no embedded components. Since

$$\mathrm{pr}_2((a \times X \times X) \cap W)$$

is nonempty, because a is a section of X', this open subset is dense in X. We have

$$\mathrm{pr}_2((a \times X \times X) \cap W) = \mathrm{pr}_2((a \times X) \cap \mathrm{pr}_{12} W).$$

Therefore

$$\mathrm{pr}_2((a \times X) \cap \mathrm{pr}_{12} W) \cap X' \\ = \mathrm{pr}_2((a \times X') \cap \mathrm{pr}_{12} W) = \mathrm{pr}_2((a \times X' \times X) \cap W)$$

is dense in X, and hence in X'.

Similarly, $\mathrm{pr}_3((a \times X \times X') \cap W)$ is dense in X, hence dense in $\mathrm{pr}_3((a \times X \times X) \cap W)$. Equivalently, $(a \times X \times X') \cap W$ is dense in $(a \times X \times X) \cap W$, and therefore

$$\mathrm{pr}_2((a \times X \times X') \cap W)$$

is dense in $\mathrm{pr}_2((a \times X \times X) \cap W)$, hence in X, and thus in X'.

Finally,

$$\mathrm{pr}_2((a \times X' \times X') \cap W') = \mathrm{pr}_2((a \times X' \times X') \cap W) \\ = \mathrm{pr}_2((a \times X' \times X) \cap W) \cap \mathrm{pr}_2((a \times X \times X') \cap W),$$

which is indeed dense in X'. This is what was required. The other assertions in (ii) follow by symmetry, and preservation of condition (iii) is immediate.[8] This proves Proposition 3.2. □

[8]Editor's note: The following sentence has been added.

3.2.1. [9] Fix a group germ (X, W) over S. We first make some preliminary observations about the situation, collected below as rules. These rules will often be used without explicit mention in what follows.

Let $a \in X(S')$, and after base change rename S' as S. There is then a birational map φ over S from X to itself, sending a section x to ax whenever the latter is defined. By 3.1(ii), the domain of definition of φ contains the relatively schematically dense open subset

$$\mathrm{pr}_2\big((a \times X \times X) \cap W\big),$$

and φ defines an isomorphism from this open subset onto the open subset

$$\mathrm{pr}_3\big((a \times X \times X) \cap W\big),$$

on which $a^{-1}x$ is defined. This observation is generalized in the evident way in Rule 1 below.

Rule 1. Let $P = P(x, t_1, \ldots, t_n)$ be a "product" of the symbols $x, t_1, \ldots, t_n$, obtained recursively as follows: $P_0 = x$, and P_{i+1} is one of the expressions

$$P_i t, \qquad t P_i, \qquad P_i^{-1} t, \qquad t P_i^{-1}, \qquad P_i t^{-1}, \qquad t^{-1} P_i,$$

where t is one of the t_j; and $P_r = P$. Let $a_1, \ldots, a_n \in X(S')$. Then there exists a relatively schematically dense open subset U of $X_{S'}$ such that the product $P(x, a_1, \ldots, a_n)$ is defined (in the sense of Remark 3.0.2) for a section $x \in X(S'')$ if and only if $x \in U(S'')$, and the map

$$x \longmapsto P(x, (a))$$

gives an isomorphism from U onto another relatively schematically dense open subset of $X_{S'}$, denoted $P(U, (a))$.

Rule 2. Let $a, b \in X(S')$. Then:

if ab is defined, so is $a^{-1}(ab)$, and $a^{-1}(ab) = b$;
if $a^{-1}b$ is defined, so is $a(a^{-1}b)$, and $a(a^{-1}b) = b$;
if ba^{-1} is defined, so is $(ba^{-1})a$, and $(ba^{-1})a = b$.

Rule 3. Let $a, b, b' \in X(S')$. If

$$ab = ab', \qquad ba = b'a, \qquad a^{-1}b = a^{-1}b', \qquad \text{or} \qquad ba^{-1} = b'a^{-1},$$

then $b = b'$. It is understood here that an equality entails that both of its sides are defined.

Rule 4. Let $a, b, c \in X(S')$. Whenever both sides are defined, one has

$$a\big((ba)^{-1}c\big) = b^{-1}c.$$

Similarly,

$$(c(ab)^{-1})a = cb^{-1}, \qquad a^{-1}\big((ab^{-1})c\big) = b^{-1}c, \qquad \big(c(b^{-1}a)\big)a^{-1} = cb^{-1}.$$

[9] Editor's note: The numbering 3.2.1, as well as 3.2.2, has been introduced.

Rule 5. All the following associativity laws hold whenever both sides are defined:

$$(a^{-1}b)c = a^{-1}(bc), \qquad (ab^{-1})c = a(b^{-1}c), \qquad (ab)c^{-1} = a(bc^{-1}),$$
$$(ab)^{-1}c = b^{-1}(a^{-1}c), \quad (a^{-1}b)c^{-1} = a^{-1}(bc^{-1}), \quad (ab^{-1})c^{-1} = a(cb)^{-1}.$$

3.2.2. Verification of the rules. (1) This follows by an evident induction on the length r of P, the case $r = 1$ being a direct consequence of 3.1(ii).

(2) This is immediate from the definition.

(3) Indeed, by Rule 2, in the first case for example,

$$b = a^{-1}(ab) = a^{-1}(ab') = b'.$$

(4) Let us verify the first relation, for example. On the right-hand side, left multiplication by b is defined and gives c, by Rule 2. Suppose that it is also defined on the left-hand side. Then

$$b(a((ba)^{-1}c)) = (ba)((ba)^{-1}c) = c.$$

Indeed, ba is defined by hypothesis, since it occurs in the expression. The middle term is therefore defined and equal to c by Rule 2, and it is equal to the left-hand term by associativity (cf. 3.1(iii)). Rule 3 thus implies that the desired equality holds whenever this multiplication by b is defined.

Now fix a and b. Rule 1 implies that $b(a((ba)^{-1}c))$ is defined for c "in" a relatively schematically dense open subset U of X. Hence, on this relatively schematically dense open subset, the two rational maps

$$c \longmapsto b^{-1}c \qquad \text{and} \qquad c \longmapsto a((ba)^{-1}c)$$

are equal. By 1.4, they are equal on every common domain of definition, which proves the desired result.

(5) The argument is of the same kind as the preceding one. For example, let us verify

$$(ab^{-1})c^{-1} = a(cb)^{-1}.$$

If right multiplication by c is defined on the right-hand side, equality follows from Rule 4. Since it is enough to verify such a formula on a relatively schematically dense open subset, we are reduced to the case in which this multiplication is defined.

3.2.3. Consider the relation R on $X \times X$ defined as follows. For $a, b, a', b' \in X(S')$, one has

$$(a, b) \sim (a', b') \pmod{R(S')}$$

if and only if there exist an fppf covering $S'' \to S'$ and a section $x \in X(S'')$ such that $(xa)b$ and $(xa')b'$ are defined and equal. Then R is an equivalence relation.

Indeed, this relation is evidently symmetric. By Rule 1, the product $(xa)b$ is defined when x is "in" a suitable relatively schematically dense open subset. Thus 1.7 asserts that there exist an fppf covering $S'' \to S'$ and an $x \in X(S'')$ for which $(xa)b$ is defined. The relation is therefore reflexive. Its transitivity follows from the next lemma.

Lemma 3.3. *Let x, y, a, b, a', b' be sections of $X(S')$ such that $(xa)b$, $(xa')b'$, $(ya)b$, and $(ya')b'$ are defined. If*

$$(xa)b = (xa')b',$$

then

$$(ya)b = (ya')b'.$$

Indeed, the lemma says that one may test $(a, b) \sim (a', b')$ using any $x \in X(S')$, with $S' \to S$ an fppf covering, for which both products are defined. Given

$$a, b, a', b', a'', b'' \in X(S),$$

Rules 1, 1.1, and 1.7 allow us to find an fppf covering $S' \to S$ and a section $x \in X(S')$ such that the three products in question are defined; transitivity follows.

Proof of Lemma 3.3. Let us begin by writing formally

$$(za)b = \big(((zx^{-1})x)a\big)b = \big((zx^{-1})(xa)\big)b = (zx^{-1})\big((xa)b\big),$$
$$(za')b' = \big(((zx^{-1})x)a'\big)b' = \big((zx^{-1})(xa')\big)b' = (zx^{-1})\big((xa')b'\big).$$

The appropriate rules show that these equalities do hold whenever their terms are defined.[10] It follows that

$$(za)b = (za')b'$$

whenever all these expressions are defined. Moreover, by Rule 1 and the hypotheses already made, these expressions are defined when $z \in X(S'')$ belongs to $V(S'')$, where V is a certain relatively schematically dense open subset of X (we have taken $S' = S$). Therefore the two rational self-maps of X given by

$$z \longmapsto (za)b \qquad \text{and} \qquad z \longmapsto (za')b'$$

are equal, and consequently

$$(ya)b = (ya')b'.$$

□

Lemma 3.4. *Consider the rational map*

$$\varphi\colon X^3 \dashrightarrow X$$

over S, defined by

$$(a, b, c) \longmapsto c^{-1}(ab).$$

Let U be the domain of definition of φ, and let Γ be the graph of the morphism $f\colon U \to X$ induced by φ, regarded as a subprescheme of X^4. Then a section $(a, b, c, d) \in X^4(S)$ belongs to $\Gamma(S)$ if and only if

$$(a, b) \sim (c, d).$$

[10] Editor's note: To be made precise...

Proof. First observe that the rational map φ is also given by the formula

$$(a,b,c) \longmapsto (xc)^{-1}\big((xa)b\big)$$

for an arbitrary section $x \in X(S)$. Equivalently,

$$c^{-1}(ab) = (xc)^{-1}\big((xa)b\big)$$

whenever both sides are defined. We leave the verification to the reader.

We now show that φ is defined at a section $(a,b,c) \in X^3(S)$ if and only if there exists a d such that $(a,b) \sim (c,d)$, and that in this case

$$d = \varphi(a,b,c).$$

Suppose first that $(a,b) \sim (c,d)$. To verify that φ is defined, we may make an fppf-covering base change (Proposition 1.6), and may therefore suppose that there exists a section $x \in X(S')$ such that $(xa)b$ and $(xc)d$ are defined and equal. It follows from Rule 2 that

$$(xc)^{-1}\big((xa)b\big)$$

is defined and equal to d.

Conversely, suppose that φ is defined at (a,b,c), and put $d = \varphi(a,b,c)$. Choose an fppf covering $S' \to S$ and a section $x \in X(S')$ such that $(xa)b$ and $(xc)d$ are defined. We wish to show that they are equal. It is enough to show that the two rational self-maps of X over S, given by

$$b \longmapsto (xa)b \qquad \text{and} \qquad b \longmapsto (xc)\varphi(a,b,c),$$

are equal; this follows from the observation in the first paragraph. □

Proposition 3.5. *The equivalence relation*

$$R \rightrightarrows X^2$$

is representable, and it is a flat relation of finite presentation; that is, the projections of R onto X^2 are flat morphisms of finite presentation.

Proof. We may suppose that S is affine. Since (X,W) is of finite presentation over S, we may descend the entire situation to an S of finite type over $\operatorname{Spec}\mathbb{Z}$, hence to a noetherian S. We may therefore suppose that S is noetherian. It is then immediate that the graph Γ of 3.4 is of finite presentation over X^2. The projection $\mathrm{pr}_{12}|_\Gamma$ is flat because

$$\Gamma \xrightarrow{\sim} \mathrm{pr}_{123}\,\Gamma,$$

the latter being an open subset of X^3, and because the projection

$$\mathrm{pr}_{12}\colon X^3 \longrightarrow X^2$$

is flat, since X is flat over S.

I claim that Γ represents R. There is something to prove here, because the domain of definition of a rational map does not in general commute with base change. Let $S'' \to S$. What is clear from 3.4 applied over S'' is that

$$R(S'') \supseteq \Gamma(S''),$$

because $\varphi \times_S S''$ is certainly defined on $U \times_S S''$. Now let

$$(a, b, c, d) \in R(S'').$$

We must prove that $(a, b, c, d) \in \Gamma(S'')$. This can be checked locally, say for the étale topology. We may therefore suppose that S'' is strictly local, that is, the spectrum of a henselian ring with separably closed residue field. Moreover, by applying 1.6 and the usual passage-to-the-limit arguments, we are reduced to the case in which S is strictly local and $S'' \to S$ is local.

Suppose that there is a section $x \in X(S)$ such that over S'' the products $(xa)b$ and $(xc)d$ are defined. This implies that

$$(xc)^{-1}((xa)b)$$

is defined and equal to d. Now there is a relatively schematically dense open subset V of X^3 such that $(xc)^{-1}((xa)b)$ is defined if and only if $(a, b, c) \in V(S'')$, and $V \subseteq U$. Consequently, $(a, b, c, d) \in \Gamma(S'')$ if such an x exists. By 1.6, we may make an fppf-covering base change $S' \to S$ in order to find such an x. Since S is strictly local, we can find* a faithfully flat, local, finite morphism $S' \to S$ and a section $x \in X(S')$ that passes through an arbitrary closed point of the closed fiber of X/S. To say that $(xa)b$ and $(xc)d$ are defined means that, over S'', x lies in a certain relatively schematically dense open subset; this can be checked on the closed fiber of $X_{S''}$. Thus the required section exists. □

Let G now be the quotient of X^2 by R, as a sheaf for the fppf topology. We define a composition law on G as follows. Let

$$(g, g') \in G(S')$$

be represented by a section

$$((a, b), (c, d)) \in (X^2 \times_S X^2)(S''),$$

where $S'' \to S'$ is an fppf covering. Suppose moreover that there is a section $x \in X(S'')$ such that $a(b(cx))$ and $x^{-1}d$ are defined; this is permitted by Rule 1 and 1.7. We define gg' to be the class in $G(S')$ represented by the section

$$(a(b(cx)), x^{-1}d) \in X^2(S'').$$

Let us verify that gg' is independent of the choice of the section x and of the representative $((a, b), (c, d))$. Let

$$(a', b') \sim (a, b), \qquad (c', d') \sim (c, d),$$

and let x' be such that $a'(b'(c'x'))$ and ${x'}^{-1}d'$ are defined. We may suppose that all these are sections over S''. We must prove that

$$(a(b(cx)), x^{-1}d) \sim (a'(b'(c'x')), {x'}^{-1}d');$$

that is, for a suitable fppf covering $S''' \to S''$ and a suitable section $z \in X(S''')$, we must have

$$(z(a(b(cx))))(x^{-1}d) = (z(a'(b'(c'x'))))({x'}^{-1}d').$$

*Combining Exposé $\mathrm{VI_A}$, 1.1.1 and EGA $\mathrm{IV_4}$, 17.16.2.

Whenever all the products involved are defined,

$$\begin{aligned}(z(a(b(cx))))(x^{-1}d) &= ((za)(b(cx)))(x^{-1}d)\\ &= (((za)b)(cx))(x^{-1}d)\\ &= ((((za)b)c)x)(x^{-1}d)\\ &= (((za)b)c)(x(x^{-1}d))\\ &= (((za)b)c)d,\end{aligned}$$

and the same identities hold with primes. By Rule 1 and 1.7, such a z exists. It therefore remains to prove that

$$(((za)b)c)d = (((za')b')c')d'.$$

But $(za)b = (za')b'$, since $(a,b) \sim (a',b')$ by 3.3; the desired equality now follows from $(c,d) \sim (c',d')$.

Consider the natural morphism

$$i\colon X \longrightarrow G$$

defined as follows. For $a \in X(S')$, choose an fppf covering $S'' \to S'$ and a section $b \in X(S'')$ such that ab^{-1} is defined, and put

$$i(a) = \text{the class in } G \text{ of } (ab^{-1}, b).$$

It is readily verified that this class, which a priori lies in $G(S'')$, is independent of the choice of b, and therefore defines a well-determined element of $G(S')$.

The reader may enjoy verifying the following proposition.

Proposition 3.6. *The morphism i commutes with the composition laws on X and G. In other words, if $a, b \in X(S')$ are sections for which ab is defined, then*

$$i(a)i(b) = i(ab).$$

The goal of this section is the following theorem.

Theorem 3.7. [11] *Let (X, W) be a group germ over S, with X/S faithfully flat and of finite presentation, and with separated fibers having no embedded component. With the notation above:*

(i) *G is a sheaf of groups.*

(ii) *$i\colon X \to G$ is representable by an open immersion.*

(iii) *G is representable fppf-locally over S.*

(iv) *If G/S is representable, then it is a flat group of finite presentation, and $i\colon X \to G$ is schematically dense relative to S.*

Observe that G is plainly characterized by properties (i)–(iv); one may therefore forget its explicit construction.

The proof proceeds in several stages.

[11] Editor's note: If X is smooth and separated over S, faithfully flat and of finite presentation over S, then G/S is representable by a smooth S-group scheme of finite type over S. This is Theorem 6.6.1 of *Néron Models*, by Bosch–Lütkebohmert–Raynaud, Springer (1990).

Lemma 3.8.

(i) *Let c be a section of $X(S)$. Then the morphism from the prescheme $c \times X$, which is S-isomorphic to X, into G, given by*

$$c \times X \hookrightarrow X \times X \longrightarrow G,$$

is a monomorphism.

(ii) *Let $\{c_i\}_{i\in I}$ be sections of $X(S)$, put*

$$Z = \coprod_i c_i \times X,$$

and let $R' \rightrightarrows Z$ be the equivalence relation induced on Z by the evident morphism $Z \to X^2$. Then R' is a "gluing" of the $c_i \times X \simeq X$.

Proof. (i) For two sections (c, a) and (c, a') of $(c \times X)(S')$ to have the same image in G, one must have $(c, a) \sim (c, a')$; that is, for a suitable section $x \in X(S'')$, where $S'' \to S'$ is an fppf covering,

$$(xc)a = (xc)a'$$

. It follows from Rule 3 that $a = a'$.

(ii) Let c_i, c_j be two sections, and consider the birational self-map ψ_{ji} of X over S, given by

$$x \longmapsto c_j^{-1}(c_i x).$$

For $y \in X(S)$, this is the same map as

$$x \longmapsto (yc_j)^{-1}\big((yc_i)x\big),$$

as is readily seen. Moreover, ψ_{ji} is defined at a section $b \in X(S')$ if and only if there exists a section b' such that

$$(c_i, b) \sim (c_j, b'),$$

and then $b' = \psi_{ji}(b)$. Let U_{ji} be the domain of definition of ψ_{ji} over S. It remains to prove that this domain of definition is universal: if $b \in X(S'')$, for an arbitrary $S'' \to S$, and ψ_{ji} is defined at b, then $b \in U_{ji}(S'')$. Equivalently, one must prove that if $b, b' \in X(S'')$ satisfy

$$(c_i, b) \sim (c_j, b'),$$

then $b \in U_{ji}(S'')$. We leave to the reader the verification of this fact, which is analogous to that in 3.5. □

Lemma 3.9. *Suppose that $\{c_i\}_{i\in I}$ are sections of $X(S)$ such that*

$$\coprod_i c_i \times X \longrightarrow G$$

is surjective as a morphism of sheaves. Then G is representable, flat, and of finite presentation over S, and the structural morphism

$$X^2 \longrightarrow G$$

is flat and of finite presentation.

Proof. The representability of G is an immediate consequence of Lemma 3.8, and it follows that

$$\coprod_i c_i \times X \longrightarrow G$$

is an open covering. To prove that $X^2 \to G$ is flat, it is enough to do so locally, hence to prove that the rational map

$$X^2 \dashrightarrow c_i \times X$$

induces a flat morphism on its domain of definition. This rational map is given by

$$(a, b) \longmapsto (c_i, ((xc_i)^{-1}(xa))b),$$

where x is an arbitrary section. If it is defined at $(a, b) \in X^2(S')$, one can find an fppf covering $S'' \to S'$ and a section $x \in X(S'')$ such that $((xc_i)^{-1}(xa))b$ is defined. It is then readily seen that this morphism is flat. The same argument shows that it is locally of finite presentation, and hence is fppf-covering.

By construction the relation R is effective. Proposition 3.5 therefore shows that $X^2 \to G$ becomes a morphism of finite presentation after the fppf-covering base change $G \leftarrow X^2$. Thus $X^2 \to G$ is of finite presentation.

Let us prove that $G \to S$ is flat and of finite presentation. It is flat and locally of finite presentation because G is covered by the $c_i \times X \simeq X$. Moreover, X^2/S is quasi-compact and $X^2 \to G$ is surjective, so G/S is quasi-compact. To prove that $G \to S$ is quasi-separated, note the following cartesian diagram:

$$\begin{array}{ccc} R & \xrightarrow{\alpha} & X^2 \times_S X^2 \\ {\scriptstyle\gamma}\downarrow & & \downarrow{\scriptstyle\beta} \\ G & \xrightarrow{\Delta} & G \times_S G. \end{array}$$

Here γ is surjective and α, β are quasi-compact. Hence $\beta\alpha$ is quasi-compact, and therefore so is Δ.[10] □

Lemma 3.10. *Let $\{c_i\}_{i\in I}$ be sections of $X(S)$. For*

$$\coprod_i c_i \times X \longrightarrow G$$

to be surjective as a morphism of fppf sheaves, it is enough that the following condition hold: for every $S' \to S$ and every $(a, b) \in X^2(S')$, there exist an open covering $\{S'_\nu\}_{\nu\in N}$ of S' and a function

$$N \longrightarrow I, \qquad \nu \longmapsto i(\nu),$$

such that $(c_{i(\nu)}^{-1}a)b$ is defined over S'_ν.

Proof. Let $S'' \to S$ be arbitrary, and let $g \in G(S'')$. Choose an fppf covering $S' \to S''$ and a section $(a, b) \in X^2(S')$ representing g. Take the open covering $\{S'_\nu\}$ supplied by the hypothesis. Over each S'_ν one has

$$(a, b) \sim (c_{i(\nu)}, (c_{i(\nu)}^{-1}a)b).$$

Thus, locally over S'_ν, the element g is represented by a section of $c_{i(\nu)} \times X$. Since the family $\{S'_\nu \to S''\}$ is fppf-covering, the desired surjectivity follows. □

[10] Editor's note: Another way to conclude here is by faithfully flat descent (EGA IV$_2$, 2.7.1), since β is an fppf covering.

Lemma 3.11. *Let Y/S be a prescheme of finite presentation, and let $\{a_i\}_{i\in I}$ be sections of $Y(S)$. Let s_0, s_1 be points of S, with s_0 a specialization of s_1, and let Y_j be the fiber of Y/S at s_j. Let C_j be the closure in Y_j of the set of points $\{a_i(s_j)\}_{i\in I}$. Then*

$$\dim C_1 \geq \dim C_0.$$

Proof. It is enough to verify the assertion after a base change $S' \to S$ together with chosen points s_0', s_1' such that $s_j' \mapsto s_j$ and s_0' is a specialization of s_1'. By EGA II, 7.1.4, we are therefore reduced to the case in which S is the spectrum of a valuation ring A, s_0 is its closed point, and s_1 is its generic point.

Let V be the closure of C_1 in Y. Clearly $C_0 \subseteq V$. The lemma is consequently a special case of the "well-known" fact that an irreducible closed subprescheme V of a prescheme Y/S of finite presentation satisfies

$$\dim(V \times_S s_1) \geq \dim(V \times_S s_0)$$

when S is the spectrum of a valuation ring and $V \times_S s_1 \neq \varnothing$ (EGA IV, 13.1.6). □

Lemma 3.12. *Suppose that S is the spectrum of a local ring, with closed point s_0, and let $\{c_i\}_{i\in I}$ be sections such that the closure C_0, in the closed fiber X_0, of the set $\{c_i(s_0)\}_{i\in I}$ has dimension*

$$\dim C_0 = \dim X_0 = n.$$

Then the condition of Lemma 3.9 is satisfied.

Proof. First note that all the fibers of X/S have the same dimension. This follows from EGA IV, 12.1.1(i), and from the fact that X has a rational composition law. Lemma 3.11 therefore implies that, for every morphism $S_1 \to S$, with S_1 the spectrum of a field, the closure of $\{c_i \times_S S_1\}_{i\in I}$ in X_{S_1} has dimension n.

Let us verify the condition of Lemma 3.10. Let $(a, b) \in X^2(S')$. For $(c_i^{-1}a)b$ to be defined, it is necessary and sufficient that c_i lie in a certain open subset $U \subseteq X_{S'}$ that is schematically dense relative to S' (Rule 1). We must show that this holds for a suitable i, locally on S'. It is therefore enough to treat the case in which S' is the spectrum of a local ring; membership of c_i in $U(S')$ can then be checked on the closed fiber. We are thus reduced to $S' = \operatorname{Spec} k$, where k is a field.

With the notation above, take $S' = S_1$. Then

$$\dim C_1 = \dim X_{S_1},$$

and U is relatively schematically dense in X_{S_1}. Hence $U \cap C_1 \neq \varnothing$, and therefore

$$U \cap \{c_i \times_S S_1\}_{i\in I} \neq \varnothing.$$

This proves the assertion. □

Proof of Theorem 3.7. The proof is now easy. First note the following consequence of the finiteness assertion in Lemma 3.9. Let $\{A_i\}$ be a directed system of rings over S, put

$$\widetilde{A} = \varinjlim_i A_i,$$

and suppose that the hypotheses of Lemma 3.9 hold over $S = \operatorname{Spec} \widetilde{A}$. Then the object representing the quotient G of

$$R \rightrightarrows X^2$$

descends to some $S_i = \operatorname{Spec} A_i$, together with the finiteness and flatness properties stated in Lemma 3.9. This is the usual passage to the limit (EGA IV, 8 and 11).

It follows that, in proving Theorem 3.7(iii) and (iv), we may restrict ourselves to the case

$$S = \operatorname{Spec} A,$$

where A is a strictly local ring. Let x_0 be a closed point of the closed fiber X_0 of X/S. There exists* a local, finite, free extension A' of A, and a section of

$$X' = X \times_S S'$$

passing through the unique point x'_0 of X' lying over x_0. Notice that $X'_0 \to X_0$ is radicial because S is strictly local, and hence the residue field of A is separably closed.

It follows that there exists a directed system $\{A_i\}$ of local rings, flat and finite over A, such that, on putting

$$\widetilde{A} = \varinjlim_i A_i, \qquad \widetilde{X} = X \times_S \operatorname{Spec} \widetilde{A},$$

the prescheme $\widetilde{X}$ has a set of sections inducing a dense set on its closed fiber $\widetilde{X}_0$. Indeed, for each closed point x_0 of X_0, choose an extension $A(x_0)$ such that the corresponding base change of X admits a section "passing through x_0", and take for the directed system the finite tensor products of the $A(x_0)$. Notice that $\widetilde{A}$ is local, being a direct limit of local rings. Thus Lemmas 3.12 and 3.10 give the quotient G over $\operatorname{Spec} \widetilde{A}$, hence, by the preceding remarks, over one of the $\operatorname{Spec} A_i$, and therefore fppf-locally.

In fact, it follows from the constructions that fppf-locally one can find a finite set of sections $\{c_i\}$, $i = 1, \ldots, r$, such that G is covered by the $c_i \times X$. Assertions (ii) and (iv) follow easily from this fact, and we leave the verification of (i) to the reader.

Corollary 3.13. *The group sheaf G determined by a group germ (X, W) over S is representable in each of the following situations:*

(i) *S is Artinian.*

(ii) *For every local scheme $S' \to S$ at a closed point s of S, the scheme $X_{S'}$ has a set of sections whose restrictions to the closed fiber form a set with closure of dimension $\dim X_s$.*

(iii) *S is strictly local and X/S is smooth.*

(iv) *There exists an fppf covering $S' \to S$ such that $G_{S'}$ is representable and affine over S.*

Indeed, (ii) follows from Lemmas 3.10 and 3.12; (iii) follows directly from (ii) and Hensel's lemma; (iv) follows from descent for affine schemes; and (i) follows from descent for group schemes, which is possible here because every finite subset of a group over a field is known to be contained in an affine open subset (Exposé VI).

*Cf. the note at the foot of page 15 in Exposé VII.

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EXPOSÉ XIX

REDUCTIVE GROUPS—GENERALITIES

by M. Demazure

The remainder of this Seminar is devoted to the study of reductive groups. Its principal aim is to generalize Chevalley's classical results (the *Bible* and *Tôhoku*)[1] to arbitrary base schemes. The two central results are the uniqueness theorem (Exposé XXIII, Theorem 4.1 and Corollaries 5.1–5.10) and the existence theorem (Exposé XXV, Theorem 1.1) for pinned reductive group schemes corresponding to prescribed root data. The proofs used here are inspired by Chevalley's; the language of schemes makes it possible to give them greater effectiveness.

The results of the first volume of the *Bible* (Exposés 1–13) will be used systematically. By contrast, we shall prove directly over an arbitrary scheme the results of the second volume (in particular the isomorphism theorem); knowledge of the proofs over an algebraically closed field is therefore not absolutely indispensable.

In proving these two fundamental results, we shall use only the most elementary results from the theory of groups of multiplicative type, found essentially in Exposés VIII and IX; on the other hand, we shall make essential use of the results of Exposé XVIII.[2] The reader interested specifically in the existence and uniqueness theorems may, on a first reading, skip Exposés X–XVII.

1. Review of groups over an algebraically closed field

1.1. Throughout this subsection, k will denote an algebraically closed field. As announced above, the only results from the *Bible* used below are contained in its first volume (Exposés 1–13).

All the results of the *Bible* (*loc. cit.*) concerning semisimple groups hold more generally for reductive groups (defined below), and their proofs are identical after the following harmless modifications:

- Exposé 9, §4, Definition 3: see 1.6.1 below;

[0] Editor's note: version of 13 October 2024.

[1] Editor's note: see the bibliographical references at the end of this Exposé. In particular, the 2005 re-edition of the Chevalley Seminar 1956–1958, cited as *Bible*, was revised by P. Cartier; see [Ch05].

[2] Editor's note: more precisely, Proposition XVIII 2.3 (extension of a "generic homomorphism" between groups) is used in XXII 4.1.11 and then in the proof of uniqueness in XXIII, as well as in XXIV. Theorem XVIII 3.7 (construction of a group from a group germ) is used only in XXV.

- Exposé 12, §4, Theorem 2(e): omit "finite";
- Exposé 13, §3, Theorem 2: replace "rank" by "semisimple rank";
- Exposé 13, §4, Corollary 2 to Theorem 3: replace "rank" by "reductive rank".

1.2. Let G be a smooth affine connected k-group. The *radical* of G (*Bible*, §9.4, Proposition 2)[3] is the reduced subgroup associated with the identity component of the intersection of the Borel subgroups of G; it is also the largest smooth connected solvable normal subgroup of G. We denote it by $\mathrm{rad}(G)$.

The *unipotent radical* of G is the unipotent part of the radical of G; it is also the largest smooth connected unipotent normal subgroup of G. We denote it by $\mathrm{rad}^u(G)$.

1.3. Let G be a smooth affine connected k-group, and let Q be a torus in G. Then the centralizer $\underline{\mathrm{Centr}}_G(Q)$ of Q in G is a closed subgroup of G (Exposé VIII, 6.7), smooth (Exposé XI, 2.4), and connected (cf. *Bible*, §6.6, Theorem 6(a), or [Ch05], §6.7, Theorem 7(a)).

We have the fundamental relation

$$\mathrm{rad}^u(\underline{\mathrm{Centr}}_G(Q)) = \mathrm{rad}^u(G) \cap \underline{\mathrm{Centr}}_G(Q). \tag{1.3.1}$$

First,[4]

$$U = \mathrm{rad}^u(G) \cap \underline{\mathrm{Centr}}_G(Q)$$

is a normal unipotent subgroup of $\underline{\mathrm{Centr}}_G(Q)$; let us show that it is smooth and connected. Let Q act on $\mathrm{rad}^u(G)$ by inner automorphisms. Then U is precisely the scheme of fixed points of this action. This scheme is smooth and connected by Lemma 1.4 below.

Consequently, U is a closed subgroup of $\mathrm{rad}^u(\underline{\mathrm{Centr}}_G(Q))$. On the other hand, by the *Bible*, Exposé 12, §3, Corollary to Theorem 1,

$$\mathrm{rad}^u(\underline{\mathrm{Centr}}_G(Q))(k) \subset \mathrm{rad}^u(G)(k).$$

Equality (1.3.1) follows.

Let us record one special case of the preceding result. If G is a smooth affine connected k-group and T is a maximal torus of G, then

$$\underline{\mathrm{Centr}}_G(T) = T \cdot (\underline{\mathrm{Centr}}_G(T) \cap \mathrm{rad}^u(G)). \tag{1.3.2}$$

Indeed (cf. *Bible*, §6.6, Theorem 6(c), or [Ch05], §6.7, Theorem 7(c)), $\underline{\mathrm{Centr}}_G(T)$ is solvable and hence is the semidirect product of its maximal torus T and its unipotent radical.

By the density theorem (cf. *Bible*, §6.5, Theorem 5(a), or [Ch05], §6.6, Theorem 6(a)), the union of $\underline{\mathrm{Centr}}_G(T)$, as T ranges over the maximal tori of G, contains a dense open subset of G. It therefore follows from (1.3.2) that:

Corollary 1.3.3.— *If G is a smooth affine connected k-group, then the union of $T \cdot \mathrm{rad}^u(G)$, where T ranges over the maximal tori of G, contains a dense open subset of G.*

Notation 1.4.0.—[5] *We shall systematically use the following notation. If S is a scheme and s is a point of S, then $\bar{s}$ denotes the spectrum of an algebraic closure $\overline{\kappa(s)}$ of $\kappa(s)$.*

Lemma 1.4.—[5] *Let S be a scheme and let Q be an S-torus acting on an S-group scheme H, separated and smooth over S.*

[3] Editor's note: the reference 9-06 (that is, Exposé 9, p. 6) has been replaced by §9.4 (that is, Exposé 9, §4), which applies both to the *Bible* and to [Ch05]. When the numbering in [Ch05] differs from that in the *Bible*, as happens in Exposé 6, both references will be given explicitly.

[4] Editor's note: the argument that follows has been expanded.

[5] Editor's note: this notation, which appeared in 2.3 of the original, has been inserted here; moreover, the statement and proof of 1.4 have been expanded.

(i) The fixed-point functor H^Q is representable by a closed subscheme of H, smooth over S.

(ii) If, moreover, H is affine over S and has connected fibers, then the same holds for H^Q.

Proof. First, by VIII 6.5(d),[6] since H is separated over S and Q is essentially free over S, H^Q is representable by a closed subscheme of H. In particular, if H is affine over S, then so is H^Q.

Now consider the semidirect product $G = H \cdot Q$; it is smooth and separated over S. The centralizer $\underline{\mathrm{Centr}}_G(Q)$ is then representable by a closed subscheme of G, of finite presentation over G, by Exposé XI 6.11(a),[6] and it is smooth over S by Exposé XI 2.4.

Since

$$\underline{\mathrm{Centr}}_G(Q) = H^Q \cdot Q,$$

it follows that H^Q is smooth over S. Indeed, using the section

$$\sigma : H^Q \longrightarrow \underline{\mathrm{Centr}}_G(Q)$$

deduced from the identity section ϵ_Q of Q, one sees at once that H^Q is formally smooth over S; and since ϵ_Q is of finite presentation, H^Q is locally of finite presentation over S. This proves (i).

Finally, suppose that H is affine over S and has connected fibers. Then every geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa(\bar{s})$-group. Hence, by the first assertion of 1.3, the group

$$\underline{\mathrm{Centr}}_{G_{\bar{s}}}(Q_{\bar{s}}) = \left(\underline{\mathrm{Centr}}_G(Q)\right)_{\bar{s}} = (H^Q)_{\bar{s}} \cdot Q_{\bar{s}}$$

is also connected, and this implies that $(H^Q)_{\bar{s}}$ is connected. □

1.5. Recall that the *reductive rank* of a smooth affine k-group G is the common dimension of its maximal tori. We denote it by $\mathrm{rgred}(G/k)$ or $\mathrm{rgred}(G)$. The equality $\mathrm{rgred}(G/k) = 0$ holds if and only if G is unipotent (that is, $G = \mathrm{rad}^u(G)$), by the *Bible*, §6.4, Corollary 1 to Theorem 4, or [Ch05], §6.5, Corollary 1 to Theorem 5.

If H is a normal subgroup of the smooth affine k-group G, then the quotient G/H is affine and smooth (Exposé $\mathrm{VI_B}$, 11.17 and $\mathrm{VI_A}$, 3.3.2(iii)). Moreover (*Bible*, §7.3, Theorem 3(a),(c)),

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/H) + \mathrm{rgred}(H).$$

Definition 1.6.1.—[7] *A k-group G is said to be* reductive *if it is smooth, affine, and connected, and if* $\mathrm{rad}(G)$ *is a torus; that is, if*

$$\mathrm{rad}^u(G) = \{e\}.$$

Lemma 1.6.2.— *Let G be a reductive k-group.*

(i) If Q is a torus of G, then $\underline{\mathrm{Centr}}_G(Q)$ is reductive.

(ii) In particular, if T is a maximal torus of G, then $\underline{\mathrm{Centr}}_G(T) = T$.

[6] Editor's note: see also the additions made in $\mathrm{VI_B}$, 6.2.4(d) and 6.5.5(a).

[7] Editor's note: the former number 1.6 has been replaced by numbers 1.6.1 through 1.6.3. Note that in this Seminar every reductive k-group (respectively, every semisimple k-group; cf. 1.8) is connected by definition.

(iii) The center of G is contained in every maximal torus and is therefore diagonalizable.

(iv) The radical of G is the unique maximal torus of $\underline{\mathrm{Centr}}(G)$.

Proof. Indeed, (i) follows from (1.3.1), (ii) from (1.3.2), and (iii) follows from (ii). Finally, the maximal torus of $\underline{\mathrm{Centr}}(G)$, that is, the neutral component $\underline{\mathrm{Centr}}(G)^0$, is a smooth connected solvable subgroup normal in G, and hence is contained in $\mathrm{rad}(G)$. Conversely, $\mathrm{rad}(G)$ is a torus normal in G, and hence central (*Bible*, §4.3, Corollary to Proposition 2); this proves (iv). □

Remark 1.6.3.— *If G is reductive and* $\dim(G) \neq \mathrm{rgred}(G)$*, then*

$$\dim(G) - \mathrm{rgred}(G) \geq 2.$$

Indeed, this difference is always even (cf. 1.10 below).

1.7. Let G be a smooth affine connected k-group, and let H be a smooth connected normal subgroup. Then

$$\mathrm{rad}(H) = \big(\mathrm{rad}(G) \cap H\big)^0_{\mathrm{red}}, \qquad \mathrm{rad}^u(H) = \big(\mathrm{rad}^u(G) \cap H\big)^0_{\mathrm{red}}, \tag{1.7.1}$$

as is seen immediately. In particular, if G is reductive, then so is H.

Let $f : G \longrightarrow G'$ be a surjective morphism of smooth affine connected k-groups. Then

$$f\big(\mathrm{rad}^u(G)\big) = \mathrm{rad}^u(G'). \tag{1.7.2}$$

In particular, if G is reductive, then so is G'.

Let us prove (1.7.2). First, $f\big(\mathrm{rad}^u(G)\big)$ is contained in $\mathrm{rad}^u(G')$. On introducing

$$H = \big(f^{-1}(\mathrm{rad}^u(G'))\big)^0_{\mathrm{red}},$$

which contains $\mathrm{rad}^u(G)$, we have $\mathrm{rad}^u(H) = \mathrm{rad}^u(G)$; thus we are reduced to the case $G = H$, that is, to the case in which G' is unipotent. As T runs over the maximal tori of G, the union of the groups $T\,\mathrm{rad}^u(G)$ is dense in G. Hence the union of the groups $f(T)\,f\big(\mathrm{rad}^u(G)\big)$ is dense in G'. But $f(T)$ consists of semisimple elements, so $f(T) = \{e\}$, since G' is unipotent. It follows that $f\big(\mathrm{rad}^u(G)\big)$ is dense in G'. Therefore, by *Bible*, §5.4, Lemma 4, or [Ch05], §6.1, Lemma 1,[8] $f\big(\mathrm{rad}^u(G)\big)$ is an open subgroup of G'. Since G' is connected, it follows that

$$f\big(\mathrm{rad}^u(G)\big) = G'.$$

N.B. Under the same hypotheses one can prove that

$$f\big(\mathrm{rad}(G)\big) = \mathrm{rad}(G').$$

1.8. A k-group G is said to be *semisimple* if it is smooth, affine, and connected, and if

$$\mathrm{rad}(G) = \{e\},$$

that is, if G is reductive and $\underline{\mathrm{Centr}}(G)$ is finite. If G is an arbitrary smooth affine connected k-group, then $G/\mathrm{rad}(G)$ is semisimple (*Bible*, §9.4, Proposition 2), and $G/\mathrm{rad}^u(G)$ is reductive. The *semisimple rank of G*, denoted by $\mathrm{rgss}(G/k)$, or simply by $\mathrm{rgss}(G)$, is the reductive rank of $G/\mathrm{rad}(G)$.

[8] Editor's note: the following argument has been supplied in detail.

Thus, if G is reductive,

$$\operatorname{rgred}(G) = \operatorname{rgss}(G) + \dim\bigl(\operatorname{rad}(G)\bigr).$$

If G is a smooth affine connected k-group and Q is a central subtorus of G, then G/Q is semisimple if and only if G is reductive and $Q = \operatorname{rad}(G)$. Indeed, if G/Q is semisimple, then $Q \supset \operatorname{rad}(G)$; hence $\operatorname{rad}(G)$ is a torus, so G is reductive, and $\operatorname{rad}(G)$ is the maximal torus of $\underline{\operatorname{Centr}}(G)$, whence $\operatorname{rad}(G) = Q$. If G is reductive and Q is a central subgroup, then G/Q is reductive and

$$\operatorname{rgss}(G) = \operatorname{rgss}(G/Q).$$

1.9. Let K be an algebraically closed extension of k, and let G be a smooth affine connected k-group. Then G is reductive (respectively, semisimple) if and only if G_K is, and

$$\operatorname{rgred}(G/k) = \operatorname{rgred}(G_K/K), \qquad \operatorname{rgss}(G/k) = \operatorname{rgss}(G_K/K).$$

1.10. Let G be a smooth connected k-group and let T be a torus of G.[9] Write $\mathfrak{g}$ for the k-Lie algebra of G, that is, $\mathfrak{g} = \operatorname{Lie}(G) = \operatorname{Lie}(G/k)(k)$, and similarly write $\mathfrak{t} = \operatorname{Lie}(T)$. Under the action of T, through $T \longrightarrow G$ and the adjoint representation of G, the Lie algebra $\mathfrak{g}$ decomposes as

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha, \tag{1.10.1}$$

where the elements $\alpha \in R$ are nontrivial characters of T and the spaces $\mathfrak{g}^\alpha$ are nonzero. The characters $\alpha \in R$ are called the *roots of G relative to T*. By Exposé II, 5.2.3(i),

$$\mathfrak{g}^0 = \operatorname{Lie}\bigl(\underline{\operatorname{Centr}}_G(T)\bigr). \tag{1.10.2}$$

In particular,[10] since $\underline{\operatorname{Centr}}_G(T)$ is connected (by 1.3 when G is affine, and by Exposé XII, 6.6(b), in general), T is its own centralizer if and only if $\mathfrak{g}^0 = \operatorname{Lie}(T)$. This condition implies that T is maximal and that $\underline{\operatorname{Centr}}(G) \subset T$; hence, by Exposé XII, 8.8(d),

$$\underline{\operatorname{Centr}}(G) = \operatorname{Ker}(T \longrightarrow \operatorname{GL}(\mathfrak{g})) = \bigcap_{\alpha \in R} \operatorname{Ker}(\alpha). \tag{1.10.3}$$

Moreover, $\underline{\operatorname{Centr}}(G)$ is then affine, and therefore the morphism $G \longrightarrow G/\underline{\operatorname{Centr}}(G)$ is affine. Since the latter group is affine (Exposé XII, 6.1),[11] G is affine.

When G is reductive and T is maximal, the roots in the preceding sense coincide with the roots in the sense of *Bible*, §12.2, Definition 1. The latter are indeed roots in the sense of this number (*Bible*, §13.2, Theorem 1(c)), and their number is $\dim(G) - \dim(T)$ (*Bible*, §13.4, Corollary 2 to Theorem 3). Moreover, if α is a root, then so is $-\alpha$ (*Bible*, §12.2, Corollary to Proposition 1). As usual, we write the group law of $\operatorname{Hom}_{k\text{-gr}}(T, \mathbb{G}_{m,k})$ indifferently additively or multiplicatively. It follows that, for reductive G,

$$\dim(G) - \operatorname{rgred}(G) = \operatorname{Card}(R) \tag{1.10.4}$$

is always even.

[9] Editor's note: the following sentence has been added.
[10] Editor's note: the following argument has been given in detail.
[11] Editor's note: see also Addition $\mathrm{VI_B}$, 6.2.6.

The semisimple rank of the reductive group G is the rank of the subset R of the $\mathbb{Q}$-vector space

$$\mathrm{Hom}_{k\text{-gr}}(T, \mathbb{G}_{m,k}) \otimes_{\mathbb{Z}} \mathbb{Q}$$

(*Bible*, §13.3, Theorem 2).

Lemma 1.11.— *Let k be an algebraically closed field, G a smooth affine connected k-group, and T a torus of G. Put*

$$W(T) = \underline{\mathrm{Norm}}_G(T)/\underline{\mathrm{Centr}}_G(T),$$

the Weyl group of G relative to T. The following conditions are equivalent:

(i) *G is reductive, T is maximal, and* $\mathrm{rgss}(G) = 1$.

(ii) *G is reductive, T is maximal, and $G \neq T$; there exists a subtorus Q of T, of codimension 1 in T, that is central in G.*

(iii) *G is not solvable and* $\dim(G) - \dim(T) \leq 2$.

(iv) *$W(T) \neq \{e\}$ and* $\dim(G) - \dim(T) \leq 2$.

(v) *$W(T) = (\mathbb{Z}/2\mathbb{Z})_k$ and* $\dim(G) - \dim(T) = 2$.

Moreover, under these conditions there are exactly two roots of G relative to T, and they are opposite. Under the conditions of (ii), *$Q = \mathrm{rad}(G)$.*

Proof. The implication (v)⇒(iv) is clear, and (iv)⇒(iii) follows from *Bible*, §6.1, Corollary 3 to Theorem 1, or [Ch05], §6.2, Corollary 3 to Theorem 2.

Let us prove (iii)⇒(ii). Let U be the unipotent radical of G. The group G/U is reductive and is not a torus (otherwise G would be solvable); hence, by (1.10.4),

$$\dim(G/U) - \mathrm{rgred}(G/U) = \mathrm{Card}(R) \geq 2.$$

On the other hand,

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/U) + \mathrm{rgred}(U) = \mathrm{rgred}(G/U),$$

and therefore

$$\begin{aligned}\dim(G) - \dim(U) - \mathrm{rgred}(G) &= \dim(G/U) - \mathrm{rgred}(G/U)\\ &\geq 2 \geq \dim(G) - \dim(T) \geq \dim(G) - \mathrm{rgred}(G).\end{aligned}$$

It follows that $\dim(U) = 0$, so G is reductive, and that $\mathrm{rgred}(G) = \dim(T)$; thus T is maximal and $\dim(G) - \dim(T) = 2$. By *Bible*, §10.4, Proposition 8, there exists a singular torus Q of codimension 1 in T. Then $\underline{\mathrm{Centr}}_G(Q)$ is reductive (1.6.2(i)) and not solvable (by the definition of a singular torus), and hence

$$\dim(\underline{\mathrm{Centr}}_G(Q)) - \mathrm{rgred}(G) \geq 2.$$

This proves that $G = \underline{\mathrm{Centr}}_G(Q)$, and that Q is central in G.

Let us prove (ii)⇒(i). We know that G/Q is reductive by 1.7 and that $\mathrm{rgred}(G/Q) = 1$; hence $\mathrm{rgss}(G/Q)$ is either 0 or 1. The first case is impossible, since it would imply $\mathrm{rgss}(G) = 0$, and therefore $G = T$. Thus

$$\mathrm{rgred}(G/Q) = \mathrm{rgss}(G/Q) = 1,$$

which proves that G/Q is semisimple. Consequently Q is the radical of G and $\mathrm{rgss}(G) = 1$. Let us finally prove (i)⇒(v). If Q is the radical of G, then $\dim(T) - \dim(Q) = 1$, and Q is central in G. Hence $G = \underline{\mathrm{Centr}}_G(Q)$, which proves that Q is a singular torus. By *Bible*, §11.3, Theorem 2,

$$W_G(T) = (\mathbb{Z}/2\mathbb{Z})_k,$$

and, by *Bible*, §12.1, Lemma 1, $\dim(G) - \dim(T) = 2$. Thus G has at most two roots relative to T, while it has at least two, opposite to one another, by 1.10.

Proposition 1.12.— *Let k be an algebraically closed field, G a smooth connected k-group, T a torus of G, and R the set of roots of G relative to T. Write*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha\in R} \mathfrak{g}^\alpha, \qquad \mathfrak{g}^\alpha \neq 0,$$

for the decomposition of $\mathfrak{g}$ under $\mathrm{Ad}(T)$. For every $\alpha \in R$, put

$$T_\alpha = \mathrm{Ker}(\alpha)^0, \qquad Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha).$$

[12] *The following conditions are equivalent:*

(i) G is affine and reductive, and T is maximal.

(ii) $\mathfrak{g}^0 = \mathfrak{t}$, and every Z_α, for $\alpha \in R$, is reductive.

(iii) $\mathfrak{g}^0 = \mathfrak{t}$; every $\mathfrak{g}^\alpha$, for $\alpha \in R$, has dimension 1; if $\alpha, \beta \in R$ and $\beta = q\alpha$ for some $q \in \mathbb{Q}$, then $q = \pm 1$; and, for every $\alpha \in R$, there exists $w_\alpha \in G(k)$ which normalizes T, centralizes T_α, but does not centralize T.

Moreover, under these conditions every Z_α has semisimple rank 1, and

$$\mathrm{Lie}(Z_\alpha) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}.$$

Proof. Suppose first that G is affine and reductive and that T is maximal. Every Z_α is then affine and reductive by 1.6.2(i), with maximal torus T; furthermore, T is its own centralizer by 1.6.2(ii). Thus $\mathfrak{g}^0 = \mathfrak{t}$, proving (i)⇒(ii).

Conversely, the equality $\mathfrak{g}^0 = \mathfrak{t}$ implies that T is maximal and that G is affine, by 1.10. Hence every Z_α is affine, smooth, and connected, by 1.3. In all cases, Exposé II, 5.2.2 gives

$$\mathrm{Lie}(Z_\alpha) = \mathfrak{g}^{T_\alpha} = \mathfrak{g}^0 \oplus \bigoplus_{\beta\in R\cap\mathbb{Q}\alpha} \mathfrak{g}^\beta.$$

Consequently,

$$\mathrm{Lie}(Z_\alpha) \supset \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

so $Z_\alpha \neq T$. Since T_α is a central subtorus of Z_α, of codimension 1 in T, Lemma 1.11 applied to Z_α proves the equivalence of (ii) and (iii), together with the additional assertions.

It remains to prove (ii)⇒(i). We already know that (ii) implies that T is maximal and that G is affine. Let U be the unipotent radical of G. It is invariant in G, and its Lie algebra is invariant under $\mathrm{Ad}(T)$; hence

$$\mathrm{Lie}(U) = \big(\mathrm{Lie}(U) \cap \mathfrak{g}^0\big) \oplus \bigoplus_{\alpha\in R} \big(\mathrm{Lie}(U) \cap \mathfrak{g}^\alpha\big).$$

[12] Editor's note: thus T_α is the identity component of $\mathrm{Ker}(\alpha)$.

By (1.3.1),

$$U \cap T = U \cap \underline{\mathrm{Centr}}_G(T) = \mathrm{rad}^u\big(\underline{\mathrm{Centr}}_G(T)\big) = \mathrm{rad}^u(T) = \{e\},$$
$$U \cap Z_\alpha = U \cap \underline{\mathrm{Centr}}_G(T_\alpha) = \mathrm{rad}^u\big(\underline{\mathrm{Centr}}_G(T_\alpha)\big) = \mathrm{rad}^u(Z_\alpha) = \{e\}.$$

It follows[13] that

$$\mathrm{Lie}(U) \cap \mathfrak{g}^0 = \mathrm{Lie}(U) \cap \mathfrak{t} = \mathrm{Lie}(U \cap T) = 0,$$
$$\mathrm{Lie}(U) \cap \mathfrak{g}^\alpha \subset \mathrm{Lie}(U) \cap \mathrm{Lie}(Z_\alpha) = \mathrm{Lie}(U \cap Z_\alpha) = 0.$$

Thus $\mathrm{Lie}(U) = 0$, and therefore $U = \{e\}$; in other words, G is reductive.

Corollary 1.13.— *Let k, G, T, R, and*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha$$

be as in Proposition 1.12. For every $\alpha \in R$, let T_α be the maximal torus of $\mathrm{Ker}(\alpha)$ and put $Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha)$. The following conditions are equivalent:

(i) G is affine and semisimple, and T is maximal.

(ii) $\mathfrak{g}^0 = \mathfrak{t}$, every Z_α is reductive, and $\bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$ is finite.

[14]

2. Reductive group schemes: definitions and first properties

Scholium 2.1.— *For an S-group scheme G, the following properties are equivalent:*

(i) G is affine and smooth over S, with connected fibers.

(ii) G is affine and flat over S, of finite presentation over S, with geometrically integral fibers.

These properties are stable under base change and are local for the fpqc topology.

[15] Indeed, suppose that (i) holds. Since G is affine and smooth over S, it is of finite presentation over S; and since its fibers are smooth and connected, they are geometrically integral, by Exposé $\mathrm{VI_A}$, 2.4.

Conversely, if (ii) holds, the fibers of G are connected and geometrically reduced, and hence smooth (Exposé $\mathrm{VI_A}$, 1.3.1). It follows that G is smooth over S, by EGA IV_4, 17.5.2.

These properties are plainly stable under base change: see EGA II, 1.5.1 for "affine", EGA IV_1, 1.6.2(iii) for "of finite presentation", EGA IV_2, 2.1.4 for "flat", and EGA IV_2, 4.6.5(i) for "with geometrically integral fibers".

They are also clearly local for the Zariski topology. It is therefore enough to check that, if $S' \longrightarrow S$ is faithfully flat and quasi-compact and $G_{S'} \longrightarrow S'$ has the indicated properties, then so does $G \longrightarrow S$. This follows from EGA IV_2, 2.5.1 for "flat", 2.7.1(vi) and (xiii) for "of finite presentation" and "affine", and 4.6.5(i) for "with geometrically integral fibers"; for "smooth", see also EGA IV_4, 17.7.3(ii).

[13] Editor's note: if U and L are two subgroup schemes of G, then $\mathrm{Lie}(U) \cap \mathrm{Lie}(L) = \mathrm{Lie}(U \cap L)$.

[14] Editor's note: this follows from the equalities $\mathrm{rad}(G) = \underline{\mathrm{Centr}}(G)^0$ and $\underline{\mathrm{Centr}}(G) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$, established in 1.6.2(iv) and (1.10.3).

[15] Editor's note: the following details have been added.

2.2. Let G be an S-group scheme satisfying the preceding conditions, and let Q be a torus of G in the sense of Exposé IX, Definition 1.3.[16] By Exposé XI, 6.11(a) and 2.4, $\underline{\mathrm{Centr}}_G(Q)$ is represented by a closed subgroup scheme of G; it is therefore affine over S, and it is of finite presentation and smooth over S. Moreover, every geometric fiber $G_{\bar{s}}$ is a smooth affine connected $\kappa(\bar{s})$-group. The first assertion of 1.3 therefore shows that the same is true of

$$\underline{\mathrm{Centr}}_{G_{\bar{s}}}(Q_{\bar{s}}) = \left(\underline{\mathrm{Centr}}_G(Q)\right)_{\bar{s}}.$$

Lemma 2.3.— *Let S be a scheme, let G be an S-group scheme that is smooth and affine over S, with connected fibers, and let T be a torus of G. The set of points $s \in S$ for which $G_{\bar{s}}$ is reductive, of semisimple rank* 1*, with maximal torus $T_{\bar{s}}$, is an open subset U of S.*

Proof. Since G and T are flat over S, the function

$$s \longmapsto \dim(G_{\bar{s}}/\bar{s}) - \dim(T_{\bar{s}}/\bar{s})$$

is locally constant. Let U_1 be the open subset of S on which this function has the value 2.

[17] By 6.3 the Weyl group

$$W_G(T) = \underline{\mathrm{Norm}}_G(T)/\underline{\mathrm{Centr}}_G(T)$$

is represented by an S-group scheme that is étale and separated over S, and the function

$$s \longmapsto \text{the number of points of } W_G(T)_{\bar{s}}$$

is lower semicontinuous. Let U_2 be the locus on which this number is greater than 1; it is open. By Lemma 1.11, the locus on which $G_{\bar{s}}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}}$, is

$$U = U_1 \cap U_2.$$

Furthermore, $W_G(T)_{\bar{s}}$ has exactly two points for every $s \in U$. Consequently, by SGA 1, I, 10.9 and EGA IV_3, 15.5.1 and IV_4, 18.12.4, $W_G(T)_U$ is finite and étale over U.

Remark 2.4.— *The group $W_G(T)_U$ is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_U$.*

Indeed, the preceding argument shows that it is finite and étale over U. Since the automorphism functor of $(\mathbb{Z}/2\mathbb{Z})_U$ is the trivial group, it is enough to verify the assertion locally. It is immediate whenever the algebra $\mathscr{A}$ defining $W_G(T)_U$ is a free $\mathcal{O}_U$-module.[18]

Notations and reminders 2.5.0.—[19] Let S be a scheme, let G be an S-group scheme, and let $\epsilon\colon S \to G$ be the unit section of G. We saw in II, §4.11 that the functor $\underline{\mathrm{Lie}}(G/S)$ is represented by the vector bundle $\mathbb{V}(\omega^1_{G/S})$, where $\omega^1_{G/S} = \epsilon^*(\Omega^1_{G/S})$, and we write

$$\mathfrak{g} = \mathrm{Lie}(G/S) = \left(\omega^1_{G/S}\right)^\vee$$

[16] Editor's note: the following argument has been given in detail.

[17] Editor's note: the following details have been added. Note also that §6 is independent of the rest of this Exposé.

[18] Editor's note: the hypothesis implies that $\mathscr{A}$ is locally free of rank 2. Since the augmentation ideal $\mathscr{I}$ is a direct summand of $\mathscr{A}$, after replacing U by a sufficiently small affine open $\mathrm{Spec}(R)$, one may suppose that $I = \Gamma(U, \mathscr{I})$ is a free R-module of rank 1. If x is a generator of I, then $x^2 = ax$ for some $a \in R$; the assumption that $A = \Gamma(U, \mathscr{A})$ is étale implies that a is invertible, which gives the asserted local identification.

[19] Editor's note: this paragraph of notations and reminders has been added.

for the sheaf of sections of this vector bundle. Suppose in addition that G is smooth over S. Then $\omega^1_{G/S}$, and hence $\mathrm{Lie}(G/S)$, are finite locally free $\mathcal{O}_S$-modules, and (cf. I, 4.6.5.1)

$$\underline{\mathrm{Lie}}(G/S) = \mathbb{W}(\mathrm{Lie}(G/S));$$

that is, for every S-scheme S',

$$\underline{\mathrm{Lie}}(G/S)(S') = \Gamma\big(S', \mathrm{Lie}(G/S) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}\big)\,.$$

By II, 4.1.1.1, the adjoint action of G gives $\underline{\mathrm{Lie}}(G/S) = \mathbb{W}(\mathrm{Lie}(G/S))$ the structure of a G-$\mathcal{O}_S$-module; thus $\mathrm{Lie}(G/S)$ is a G-$\mathcal{O}_S$-module (cf. I, 4.7.1). If G is moreover affine over S, this says, by I, 4.7.2, that $\mathrm{Lie}(G/S)$ is an $A(G)$-comodule.

If T is a torus over S (IX, Definition 1.3), we say that T is *split* ("trivial" in the original) if it is isomorphic to $\mathbb{G}^r_{m,S}$ for some integer $r \geq 0$, and that T is *trivialized* if such an isomorphism has been fixed, or more generally if an isomorphism

$$T \simeq D_S(M)$$

has been fixed, where M is a free abelian group of rank r.

Finally, recall (XII, Definition 1.3) that a torus T of G is called *maximal* if, for every $s \in S$, $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Theorem 2.5.— *Let S be a scheme, let G be an S-group scheme that is affine and of finite presentation over S, with connected fibers, and let s_0 be a point of S. Suppose that G is flat over S at $\epsilon(s_0)$, and that the geometric fiber $G_{\bar{s}_0}$ is (reduced and) reductive* (resp. semisimple)*. Then there exist an open subset $U \subset S$, containing s_0, and an étale surjective morphism $S' \to U$ such that:*

(i) G_U is smooth, with reductive (resp. semisimple) *geometric fibers, and its reductive rank and semisimple rank are constant.*

(ii) $G_{S'}$ has a split[20] *maximal torus T, and the Weyl group* (cf. 6.3)

$$W_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T)/\underline{\mathrm{Centr}}_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T)/T$$

is finite over S'.

Proof. [21] Write $\pi\colon G \to S$ for the structure morphism and $\epsilon\colon S \to G$ for the unit section. Since G is flat over S at $\epsilon(s_0)$, and since $G_{\bar{s}_0}$ is reduced (hence smooth over $\bar{s}_0$, by $\mathrm{VI_A}$, 1.3.1), G is smooth over S at $\epsilon(s_0)$ (EGA IV_4, 17.5.1). Thus there is an open neighborhood V of $\epsilon(s_0)$ on which π is smooth. Then $S' = \epsilon^{-1}(V)$ is open in S, and $G_{S'}$ is smooth over S' at the points of $\epsilon(S')$. Since G has connected fibers, $\mathrm{VI_B}$, 3.10 shows that $G_{S'}$ is smooth over S'. Replacing S by S', we may therefore suppose that G is smooth over S.

By Exposé XI, Theorem 4.1, the functor of subgroups of multiplicative type of G is represented by an S-scheme $\mathscr{M}$, smooth and separated over S. Let r_0 be the rank of $G_{\bar{s}_0}$, and consider the open subscheme $\mathscr{M}_{r_0} \subset \mathscr{M}$ representing the functor of subtori of rank r_0 of G (IX, 1.4). Smoothness implies that $\mathscr{M}_{r_0}$ has a rational point over a finite separable extension of $\kappa(s_0)$ (EGA IV_4, 17.15.10(iii)). Consequently there is an étale morphism $S' \to S$, together with a point s'_0 above s_0, such that $G_{s'_0}$ has a torus of rank r_0. Replacing S by S', we may therefore suppose that G_{s_0} has a torus of rank r_0.

[20] Editor's note: the word "split" has been added.

[21] Editor's note: the proof has been expanded.

By "Hensel's lemma" (XI, 1.10), the smoothness of $\mathscr{M}_{r_0}$ allows this torus to be lifted to an S'-torus T of G, where $S' \to S$ is étale and is equipped with a point s'_0 above s_0. By Exposé X, 4.5 (see also 6.1[22]), there is an étale morphism $f\colon S'' \to S'$ that splits T and for which $f^{-1}(s'_0) \neq \varnothing$. Because an étale morphism is open, and the assertions in (i) are local for the étale topology, we may thus suppose that G has a split torus T,[23] maximal at s_0. Write $T = D_S(M)$, and let

$$\mathfrak{g} = \coprod_{m\in M} \mathfrak{g}^m$$

be the decomposition of $\mathfrak{g} = \mathrm{Lie}(G/S)$ under $\mathrm{Ad}(T)$ (Exposé I, 4.7.3).

Put $\mathfrak{t} = \mathrm{Lie}(T)$, and, for every $m \in M$, write

$$\mathfrak{g}^m(s_0) = \mathfrak{g}^m \otimes_{\mathcal{O}_S} \kappa(s_0).$$

[24] Let R be the set of $m \in M$, $m \neq 0$, such that $\mathfrak{g}^m(s_0) \neq 0$.[25] Since $G_{\bar{s}_0}$ is reductive, $\mathfrak{g}^0(s_0) = \mathfrak{t}(s_0)$, and hence

$$\mathfrak{g}(s_0) = \mathfrak{t}(s_0) \oplus \coprod_{\alpha\in R} \mathfrak{g}^\alpha(s_0).$$

Since the modules in question are locally free, after shrinking S we may suppose that the $\mathfrak{g}^\alpha$ are free and that

$$\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha\in R} \mathfrak{g}^\alpha.$$

Recall from Exposé XII, 1.12 that a group of multiplicative type has a unique maximal torus (this also follows almost immediately by descent, the diagonalizable case being evident). Let T_α be the maximal torus of $\ker(\alpha)$, and put

$$Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha).$$

[26] By 2.2, Z_α is affine and smooth over S, with connected fibers; and by 1.12, its fiber $(Z_\alpha)_{\bar{s}_0}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}_0}$. By 2.3 there is therefore an open subset $U_\alpha \subset S$, containing s_0, on which Z_α has reductive fibers.

Set

$$U = \bigcap_{\alpha\in R} U_\alpha.$$

By 1.12, for every $s \in U$, $G_{\bar{s}}$ is reductive, with maximal torus $T_{\bar{s}}$, and the set of roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$ is identified with R. Consequently

$$\mathrm{rgred}(G_{\bar{s}}) = \dim(T) = \mathrm{rg}(M), \qquad \mathrm{rgss}(G_{\bar{s}}) = \mathrm{rg}(R) \quad \text{(cf. 1.10)}.$$

This proves (i) and the first assertion of (ii). It remains to prove that $W_{G_U}(T_U)$ is finite over U, that is, that it has the same number of points in every geometric fiber (SGA 1, I, 10.9; EGA IV$_3$, 15.5.1; and EGA IV$_4$, 8.12.4).

For this it is enough to observe that, for $s \in U$, the geometric fiber of this group is determined by the configuration $R \subset M$: it is the constant group associated with the abstract Weyl group of this root system. In particular it is independent of s; see the *Bible*, §11.3, Theorem 2 (and also Exposé XXII, no. 3). □

[22] Editor's note: §6.1 is independent of the rest of this Exposé.
[23] Editor's note: throughout this volume, "trivial torus" has been replaced by "split torus".
[24] Editor's note: the preceding sentence has been added.
[25] Editor's note: since $\mathfrak{g}$ is a finite locally free $\mathcal{O}_S$-module, R is finite.
[26] Editor's note: the following argument has been expanded.

Corollary 2.6.— *Let G be an S-group scheme that is affine and smooth over S, with connected fibers. The set of points $s \in S$ for which $G_{\bar{s}}$ is reductive* (resp. semisimple) *is an open subset U of S, and the functions*

$$s \longmapsto \mathrm{rgred}(G_{\bar{s}}/\bar{s}), \qquad s \longmapsto \mathrm{rgss}(G_{\bar{s}}/\bar{s})$$

are locally constant on U.

Definition 2.7.— *An S-group* (that is, an S-group scheme) *G is called* reductive (resp. *semisimple*) *if it is affine and smooth over S, with connected reductive* (resp. semisimple) *geometric fibers.*

For an S-group scheme G, the property of being reductive (resp. semisimple) is stable under base change and local for the fpqc topology.

2.8.— Let G be a reductive S-group. For every torus (resp. maximal torus) Q of G,

$$Z(Q) = \underline{\mathrm{Centr}}_G(Q)$$

is reductive (resp. is equal to Q). This follows at once from 2.2 and 1.6. Applying 2.5 to $\underline{\mathrm{Centr}}_G(Q)$, we deduce that Q is contained, locally for the étale topology, in a maximal torus.

Remark 2.9.— *Using, as usual, the technique of EGA IV$_3$, §8, the finite-presentation hypotheses, and Theorem 2.5, one sees that if G is a reductive group over S, there is an open cover $\{U_i\}$ of S such that each G_{U_i} is obtained by base change from a reductive group over a noetherian ring* (in fact, over a finitely generated $\mathbb{Z}$-algebra). *Likewise, if G has a split maximal torus T, one may further suppose that T_{U_i} comes from a split maximal torus of the preceding group, ...*

3. Roots and root systems of reductive group schemes

3.1.— Let S be a scheme, and let T be an S-torus acting linearly on a finite locally free $\mathcal{O}_S$-module F (cf. I, §4.7). For every character α of T, that is,

$$\alpha \in \mathrm{Hom}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}),$$

define a subfunctor of $\mathbb{W}(F)$ by

$$\mathbb{W}(F)^\alpha(S') = \left\{x \in \mathbb{W}(F)(S') \mid t \cdot x = \alpha(t)x \text{ for every } t \in T(S''),\ S'' \to S'\right\}.$$

Lemma.—[27] *Then $\mathbb{W}(F)^\alpha = \mathbb{W}(F^\alpha)$, where F^α is a submodule of F, locally a direct summand of F, and hence also locally free.*

Indeed, the assertion is local for the fpqc topology, and we may suppose that $T = D_S(M)$, where M is a finite-type (free) abelian group. Then α is identified with a locally constant function from S to M (Exposé VIII, 1.3), and, after restricting S if necessary, we may suppose that this function is constant. We are then reduced to Exposé I, 4.7.3.

Definition 3.2.— *Let S be a scheme, let G be an S-group scheme smooth over S and with connected fibers,*[28] *and let T be a torus of G. Put $\mathfrak{g} = \mathrm{Lie}(G/S)$, and let T act on $\mathfrak{g}$ through the adjoint representation of G.*

[27] Editor's note: this statement has been made into a lemma in order to bring it out.

[28] Editor's note: the original also assumed that G was of finite presentation over S, an assumption that does not appear to be used below. In any case, since G is smooth over S and has connected fibers, it is quasi-compact and separated over S (VI$_B$, 5.5), and hence of finite presentation over S.

A character α of T is called a root *of G with respect to T if the following equivalent conditions are satisfied:*

(i) For every $s \in S$, α_s is a root of G_s with respect to T_s (1.10).

(ii) α is nontrivial on every fiber, and $\mathfrak{g}^\alpha(s) \neq 0$ for every $s \in S$.

Lemma 3.3.— *Let S be a scheme, let T be an S-torus, and let α be a character of T. The following conditions are equivalent:*

(i) α is nontrivial on every fiber; that is, for every $s \in S$, α_s is distinct from the unit character of T_s.

(ii) For every $S' \to S$ with $S' \neq \varnothing$, $\alpha_{S'}$ is distinct from the unit character of $T_{S'}$.

(iii) The morphism α is faithfully flat.

[29] It is clear that (ii) implies (i), and it is easy to see that (iii) implies (i). Condition (i) implies (ii): if $s' \in S'$ lies over s, and $\alpha_{s'}$ is the unit character, then so is α_s. Finally, to prove that (i) implies (iii), we reduce to the case in which T is diagonalizable and conclude by Exposé VIII, 3.2(a).

3.4.—[30] Let G be a reductive S-group scheme, let T be a maximal torus of G, and let α be a root of G with respect to T. Then, by 2.5.0 and 1.12, $\mathfrak{g}^\alpha$ is a locally free $\mathcal{O}_S$-module of rank one. Moreover, by 1.10, $-\alpha$ is also a root of G with respect to T. In particular, if G has semisimple rank one, 1.11 and 1.12 give the following statement.

Lemma 3.5.— *Let S be a scheme, let G be a reductive S-group of semisimple rank one, and let T be a maximal torus of G. If α is a root of G with respect to T, then so is $-\alpha$, and*

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank one.

Definition 3.6.— *Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let R be a set of roots of G with respect to T. We say that R is a* root system *of G with respect to T if the following equivalent conditions are satisfied:*

(i) For every $s \in S$, the map $\alpha \mapsto \alpha_s$ is a bijection from R onto the set of roots of G_s with respect to T_s.

(ii) The elements of R are distinct on every fiber (that is, if $\alpha, \alpha' \in R$ and $\alpha \neq \alpha'$, then $\alpha - \alpha' = \alpha\alpha'^{-1}$ is nontrivial on every fiber)*, and, for every $s \in S$,*

$$\dim(G_s/\kappa(s)) - \dim(T_s/\kappa(s)) = \mathrm{Card}(R).$$

(iii) One has

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha.$$

[29] Editor's note: the original has been simplified; it invoked Exposé IX, 5.2. Note, however, that loc. cit., 5.3 shows that, if S is connected, the conditions of the lemma hold as soon as (i) holds at one point s of S.

[30] Editor's note: what follows has been modified so as to recall the hypotheses of 3.2 and to add that T is a maximal torus.

The equivalence of these conditions is immediate.

Lemma 3.7.— *Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let R be a root system of G with respect to T. Every root of G with respect to T is locally on S equal to an element of R.*

Proof. This follows from condition (iii).

Put

$$\mathcal{M} = \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S});$$

this is a twisted constant S-group scheme (Exposé X, 5.6). If G has a root system R with respect to T, then the inclusion $R \hookrightarrow \mathcal{M}(S)$ defines a morphism

$$R_S \longrightarrow \mathcal{M},$$

where R_S is the constant S-scheme defined by R. By condition (ii), this morphism is an open and closed immersion whose image is precisely

$$\bigcup_{\alpha \in R} \alpha(S),$$

where each $\alpha \in R$ is regarded as a section $S \to \mathcal{M}$.

Let $\mathcal{R}$ be the functor of roots of G with respect to T. By definition, $\mathcal{R}(S')$ is the set of roots of $G_{S'}$ with respect to $T_{S'}$, for every $S' \to S$. If $S' = \varnothing$, set $\mathcal{R}(\varnothing) = \{e\}$; if $S' \neq \varnothing$, the inclusion $R \hookrightarrow \mathcal{M}(S')$ identifies R with a root system of $G_{S'}$ with respect to $T_{S'}$. Thus, by 3.7,

$$\mathcal{R}(S') = \mathrm{Hom}_{\text{loc. const.}}(S', R),$$

which shows that $\mathcal{R}$ is represented by R_S.

Now drop the assumption that G necessarily has a root system with respect to T. The functor $\mathcal{R}$ is in any case an fpqc subsheaf of $\mathcal{M}$. Locally for this topology, G does have a root system with respect to T (for example, make T split and use the proof of Theorem 2.5). By Exposé IV, 4.6.8 and the descent theory of open (respectively closed) subschemes,[31] we obtain the following proposition.

Proposition 3.8.— *Let S be a scheme, let G be an S-group, and let T be a maximal torus of G. The functor $\mathcal{R}$ of roots of G with respect to T is represented by a finite twisted constant S-scheme* (Exposé X, 5.1) *which is an open and closed subscheme of*

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}).$$

A subset $R \subset \mathrm{Hom}_{S\text{-}gr.}(T, \mathbb{G}_{m,S})$ is a root system of G with respect to T if and only if the morphism

$$R_S \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}),$$

defined by the preceding inclusion, induces an isomorphism

$$R_S \xrightarrow{\sim} \mathcal{R}.$$

3.9.— Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let α be a root of G with respect to T, that is, a section of $\mathcal{R}$. Consider the kernel $\mathrm{Ker}(\alpha)$ of α, its unique maximal torus T_α, and the centralizer of the latter,

$$Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha).$$

[31] Editor's note: see SGA 1, VIII, 4.4 or EGA IV_2, 2.7.1.

This is a closed S-subgroup of G, reductive (2.8), and of semisimple rank one (1.12). Moreover,

$$\mathrm{Lie}(Z_\alpha/S) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha};$$

hence $\{\alpha, -\alpha\}$ is a root system of Z_α with respect to T.

3.10.— Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let α be a root of G with respect to T. If $q \in \mathbb{Q}$ and $q\alpha$ is a root of G with respect to T, then $q = 1$ or $q = -1$. This follows at once from 1.12.

4. Roots and vector group schemes

4.1.— Let S be a scheme and let F be a finite locally free $\mathcal{O}_S$-module. The S-scheme $\mathbb{W}(F)$ is smooth over S. Its Lie algebra is canonically isomorphic to F. Indeed, there is a canonical isomorphism

$$\mathbb{W}(F) \xrightarrow{\sim} \mathrm{Lie}(\mathbb{W}(F)/S) = \mathbb{W}\big(\mathrm{Lie}(\mathbb{W}(F)/S)\big)$$

(Exposé II, 4.4.1 and 4.4.2). We shall always identify F with $\mathrm{Lie}(\mathbb{W}(F)/S)$.

Lemma 4.2.— *Let S be a scheme and let V be a vector bundle over S, smooth over S. There is then a unique isomorphism of $\mathcal{O}_S$-modules*

$$\exp\colon \mathbb{W}\big(\mathrm{Lie}(V/S)\big) \xrightarrow{\sim} V$$

which induces the identity on Lie algebras. [32]

Proof. We have $V = \mathbb{V}(\mathscr{F})$ for some quasi-coherent $\mathcal{O}_S$-module $\mathscr{F}$. Since V is smooth over S, the module $\mathscr{F} \simeq \omega^1_{V/S}$ is finite locally free, and hence

$$\mathrm{Lie}(V/S) = \mathbb{V}(\omega^1_{V/S}) \simeq \mathbb{W}\big(\mathrm{Lie}(V/S)\big).$$

[32] Editor's note: the proof in the original edition has been retained; it may also be expanded as follows.

Let $\mathscr{F}$ be a quasi-coherent $\mathcal{O}_S$-module and put $V = \mathbb{V}(\mathscr{F})$. Write $\pi\colon V \to S$ for the projection and $\epsilon\colon S \to V$ for the zero section. Then $\Omega^1_{V/S} = \pi^*\mathscr{F}$, whence

$$\omega^1_{V/S} = \epsilon^*\Omega^1_{V/S} \simeq \epsilon^*\pi^*\mathscr{F} \simeq \mathscr{F}, \tag{1}$$

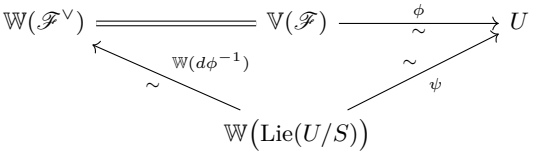
and therefore $\mathrm{Lie}(V/S) = (\omega^1_{V/S})^\vee \simeq \mathscr{F}^\vee$. If V is smooth over S, then by (1) $\mathscr{F}$ is finite locally free, and hence

$$V = \mathbb{V}(\mathscr{F}) \simeq \mathbb{W}(\mathscr{F}^\vee) \simeq \mathbb{W}\big(\mathrm{Lie}(V/S)\big). \tag{2}$$

Now let U be an S-scheme equipped with a section $\tau\colon S \to U$, and suppose that an isomorphism $\phi\colon V \xrightarrow{\sim} U$ of pointed S-schemes has been given, that is, $\phi \circ \epsilon = \tau$ (for example, when U is an S-group). Then ϕ induces an isomorphism

$$d\phi\colon \mathscr{F}^\vee \xrightarrow{\sim} \mathrm{Lie}(U/S),$$

and therefore an isomorphism ψ displayed by



which identifies U with $\mathbb{W}(\mathrm{Lie}(U/S))$. Since the functor $\mathbb{W}$ is fully faithful, there is a unique isomorphism

$$\exp\colon \mathbb{W}\big(\mathrm{Lie}(U/S)\big) \xrightarrow{\sim} U$$

such that $\mathrm{Lie}(\psi^{-1} \circ \exp) = \mathrm{id}$. In fact, one sees directly that $\mathrm{Lie}(\psi) = \mathrm{id}$, so that $\exp = \psi$.

Moreover, by Exposé II, loc. cit., there is a canonical isomorphism

$$V \xrightarrow{\sim} \mathrm{Lie}(V/S) \simeq \mathbb{W}(\mathrm{Lie}(V/S)),$$

and the uniqueness of exp follows at once because $\mathbb{W}$ is fully faithful. □

4.3. Notation.— If V is a vector bundle over S, let $V^\times$ denote the open subset obtained from V by removing the zero section. Write the group law of V multiplicatively. The action of $\mathcal{O}_S$ on V which defines its module structure will then be written exponentially:

$$\mathcal{O}_S \times_S V \longrightarrow V, \qquad (x, v) \longmapsto v^x.$$

Thus

$$(vv')^x = v^x (v')^x, \qquad v^{x+x'} = v^x v^{x'}, \qquad v^{xx'} = (v^x)^{x'}.$$

In particular, upon restricting the operator action to $\mathbb{G}_{m,S}$, the open subset $V^\times$ is stable and is therefore an object with operator group $\mathbb{G}_{m,S}$. We shall continue to write this action as $(z, v) \longmapsto v^z$.

Definition 4.4.1.[33]— *Let L be an invertible module on S, and let $\mathbb{W}(L)$ be the associated vector bundle. Then $\mathbb{W}(L)^\times$ is a* (locally trivial) *principal homogeneous bundle under $\mathbb{G}_{m,S}$. We write*

$$\Gamma(S, L)^\times = \mathbb{W}(L)^\times(S).$$

Scholium 4.4.2.— *There are natural bijections among the isomorphisms of $\mathcal{O}_S$-modules $\mathcal{O}_S \simeq L$, the isomorphisms of $\mathcal{O}_S$-modules $\mathcal{O}_S \simeq \mathbb{W}(L)$,*[34] *and the sections $S \to \mathbb{W}(L)^\times$.*

The correspondence is

$$f \longmapsto \mathbb{W}(f) \longmapsto \mathbb{W}(f)(1).$$

It is compatible with base change. Thus $\mathbb{W}(L)^\times$ may be regarded as the "scheme of trivializations of $\mathbb{W}(L)$".

4.5.— Let S be a scheme, let T be an S-torus, let P be an S-group with operator group $\mathbb{G}_{m,S}$ (for example, a vector bundle), and let α be a character of T. We write $T \cdot_\alpha P$ for the semidirect product of P by T, where T acts on P through the composite morphism

$$T \xrightarrow{\alpha} \mathbb{G}_{m,S} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(P).$$

Definition 4.6.— *Let S be a scheme, let G be an S-group scheme, let T be a subgroup of G, let α be a character of T, and let L be an $\mathcal{O}_S$-module. Let*

$$p\colon \mathbb{W}(L) \longrightarrow G$$

be a morphism of S-group functors.[35] *We say that p is* normalized by T with multiplier α *if it satisfies the following equivalent conditions:*

[33] Editor's note: for purposes of later references, paragraph 4.4 has been divided into 4.4.1 and 4.4.2.

[34] Editor's note: here the vector bundle $\mathbb{W}(L)$ is identified with the $\mathcal{O}_S$-module functor which it represents.

[35] Editor's note: the edition uses a bold $\mathbf{W}(L)$ here because, for an arbitrary $\mathcal{O}_S$-module L, the functor need not be representable. Below, however, L is assumed finite locally free (indeed, invertible), in which case the functor is represented by the vector bundle $\mathbb{V}(L^\vee)$, and the notation $\mathbb{W}(L)$ is used without this distinction.

(i) p is a morphism of objects with operator group T, where T acts on $\mathbb{W}(L)$ through α, and on G by inner automorphisms. In other words, for every $S' \to S$, every $t \in T(S')$, and every

$$x \in \mathbb{W}(L)(S') = \Gamma(S', L \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

one has

$$\mathrm{int}(t)p(x) = p(\alpha(t)x).$$

(ii) The morphism

$$T \cdot_\alpha \mathbb{W}(L) \longrightarrow G, \qquad (t, x) \longmapsto t\,p(x),$$

defined by multiplication in G, is a group morphism.

Lemma 4.7.— *Under the hypotheses of 4.6, suppose in addition that G is smooth with connected fibers,*[36] *that T is a maximal torus of G, and that L is invertible. If p is a monomorphism and α is nonzero on every fiber, then α is a root of G with respect to T.*

Indeed,

$$\mathrm{Lie}(p)\colon L \longrightarrow \mathfrak{g}$$

is a monomorphism which factors through $\mathfrak{g}^\alpha$.

Proposition 4.8.— *Under the hypotheses of 4.7, suppose that G is reductive and that p is a monomorphism. Then α is a root of G with respect to T, and* $\mathrm{Lie}(p)$ *induces an isomorphism*

$$\mathrm{Lie}(p)\colon L \xrightarrow{\sim} \mathfrak{g}^\alpha.$$

Indeed, by 4.7 and the fact that $\mathfrak{g}^\alpha$ is invertible, it is enough to prove that α is nonzero on every fiber. Let $s \in S$ be such that $\alpha_{\bar{s}} = 0$ ($= 1$ in multiplicative notation). If X is a nonzero section of $L \otimes_S \kappa(\bar{s})$, then $p(X)$ is a unipotent element different from the identity in $G(\bar{s})$ which centralizes $T_{\bar{s}}$. This is impossible, since $T_{\bar{s}}$ is its own centralizer.

Corollary 4.9.— *Under the hypotheses of 4.8, there exists a monomorphism of groups with operator group T* (that is, normalized by T with multiplier α)

$$\mathbb{W}(\mathfrak{g}^\alpha) \longrightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^\alpha \to \mathfrak{g}$.

We shall soon see that 4.9 in fact holds whenever G is a reductive group and α is a root of G with respect to T, and that such a morphism is unique.

Reminder 4.10.— *Let k be an algebraically closed field, let G be a reductive k-group, let T be a maximal torus of G, and let α be a root of G with respect to T. There exists a monomorphism*

$$p\colon \mathbb{G}_{a,k} \longrightarrow G$$

normalized by T with multiplier α.

See the *Bible*, §13.1, Theorem 1.

4.11.— We finish this section with a technical result which will be useful to us. Let S be a scheme, and let L be an invertible $\mathcal{O}_S$-module. Let $q > 0$ be an integer such that $x \longmapsto x^q$ is an endomorphism of the S-group $\mathbb{G}_{a,S}$. If $S \neq \varnothing$, then $q = 1$, or $q = p^n$, where

[36] Editor's note: compare editor's note (28).

p is a prime number which vanishes on S. This follows at once from the elementary fact that the greatest common divisor of the binomial coefficients $\binom{q}{i}$, for $i \neq 0, q$, is p if $q = p^n$ with p prime, and is 1 otherwise.

The morphism defined by taking the q-th power,

$$L \longrightarrow L^{\otimes q},$$

is a morphism of sheaves of abelian groups. By base change it defines a morphism of S-group schemes

$$\mathbb{W}(L) \longrightarrow \mathbb{W}(L^{\otimes q}).$$

In particular, if L' is another invertible module and an $\mathcal{O}_S$-linear morphism

$$h\colon L^{\otimes q} \longrightarrow L'$$

is given, one obtains a morphism of S-group schemes

$$\mathbb{W}(L) \longrightarrow \mathbb{W}(L'), \qquad x \longmapsto h(x^q).$$

With this notation, we have the following result.

Proposition 4.12.— *Let S be a scheme, let T* (resp. T') *be an S-torus, let L* (resp. L') *be an invertible $\mathcal{O}_S$-module, and let α* (resp. α') *be a character of T* (resp. T'). *Let $f\colon T \to T'$ be a group morphism and let*

$$g\colon \mathbb{W}(L) \longrightarrow \mathbb{W}(L')$$

be a morphism of S-schemes (not necessarily a morphism of S-groups) *satisfying*

$$(\star) \qquad g\big(\alpha(t)x\big) = \alpha'\big(f(t)\big)g(x)$$

for every $S' \to S$, every $x \in \mathbb{W}(L)(S')$, and every $t \in T(S')$. Let $s_0 \in S$ be such that $\alpha_{\bar{s}_0} \neq 0$. [37]

(a) *Suppose that g sends the zero section to the zero section and that, for every integer $n > 0$,*

$$(\alpha' \circ f)_{\bar{s}_0} \neq n\alpha_{\bar{s}_0}.$$

Then $g = 0$ in a neighborhood of s_0.

(b) *Suppose that g is a group morphism and that $g_{\bar{s}_0} \neq 0$. There then exist an open subset $U \subset S$ containing s_0 and an integer $q > 0$ such that $x \longmapsto x^q$ is an endomorphism of $\mathbb{G}_{a,U}$ and*

$$(\alpha' \circ f)_U = q\alpha_U.$$

[37] Editor's note: the phrases "consider the semidirect product $T \cdot_\alpha \mathbb{W}(L)$, respectively $T' \cdot_{\alpha'} \mathbb{W}(L')$" have been deleted. On the other hand, in part (b) the hypothesis that g be a group morphism, together with $(\star)$, is equivalent to requiring

$$(t, x) \longmapsto \big(f(t), g(x)\big)$$

to be a group morphism from $T \cdot_\alpha \mathbb{W}(L)$ to $T' \cdot_{\alpha'} \mathbb{W}(L')$.

(c) *Suppose that*

$$(\alpha' \circ f)_{\bar{s}_0} = q\alpha_{\bar{s}_0},$$

where $q > 0$ is an integer such that $x \longmapsto x^q$ is an endomorphism of $\mathbb{G}_{a,S}$. There then exist an open subset $U \subset S$ containing s_0 and a unique morphism of $\mathcal{O}_S$-modules

$$h\colon L^{\otimes q}|_U \longrightarrow L'|_U$$

such that g_U is the composite

$$\mathbb{W}(L)_U \xrightarrow{x\mapsto x^q} \mathbb{W}(L^{\otimes q})_U \xrightarrow{\mathbb{W}(h)} \mathbb{W}(L')_U.$$

Proof of Proposition 4.12.. Let us prove (a). Since the conclusion is local on S, we may suppose that $\mathbb{W}(L) = \mathbb{W}(L') = \mathbb{G}_{a,S}$, and hence that g is given by a polynomial

$$g(X) = \sum_{n\geq 0} a_n X^n, \qquad a_n \in \Gamma(S, \mathcal{O}_S).$$

Condition $(\star)$, which relates f and g, is the identity in $\Gamma(S', \mathcal{O}_{S'})[X]$

$$\sum_{n\geq 0} a_n \alpha'(f(t)) X^n = \sum_{n\geq 0} a_n \alpha(t)^n X^n.$$

Thus, for every $n \geq 0$, every $S' \to S$, and every $t \in T(S')$,

$$a_n(\alpha'(f(t)) - \alpha(t)^n) = 0.$$

For every $n \geq 0$, let S_n be the set of points $s \in S$ such that

$$(\alpha' \circ f)_{\bar{s}} = n\alpha_{\bar{s}}.$$

It is known (Exposé IX, 5.3) that the S_n are both open and closed and that

$$(\alpha' \circ f)_{S_n} = n\alpha_{S_n}.$$

Moreover, since $\alpha_{\bar{s}_0} \neq 0$, after shrinking S we may suppose that α is nonzero on every fiber (same reference); it follows that the S_n are pairwise disjoint. After shrinking S once more, we may therefore suppose that one of the following two cases holds: $S = S_n$ for some n, or every S_n is empty.

Let $m \geq 0$ be such that $S_m = \varnothing$. We claim that $a_m = 0$. Indeed, $\alpha' \circ f$ and $m\alpha$ are distinct on every fiber of S, and we have:

Lemma 4.13.— *Let S be a scheme, let T be an S-torus, and let α, α' be two characters of T which are distinct on every fiber. There exist an fpqc covering family $\{S_i \to S\}$ and, for every i, an element $t_i \in T(S_i)$ such that*

$$\alpha(t_i) - \alpha'(t_i) = 1.$$

The lemma is immediate by reduction first to the diagonalizable case, and then to $T = \mathbb{G}_{m,S}$.

We return to the proof of Proposition 4.12. We have just proved (a). In cases (b) and (c), there is an integer n such that $S = S_n$ ($n = q$ in (c)). The preceding argument gives $a_m = 0$ for $m \neq n$, and therefore

$$g(X) = a_n X^n, \qquad a_n \in \Gamma(S, \mathcal{O}_S).$$

This proves (c) at once. In case (b), we know that $a_n(s_0) \neq 0$; we may therefore suppose that a_n is invertible on S. It follows that $x \longmapsto x^n$ is an endomorphism of $\mathbb{G}_{a,S}$, by the hypothesis in (b), and this completes the proof.

5. An instructive example

5.1.— Let k be an algebraically closed field of characteristic 0. Put $A = k[t]$, the polynomial ring in one variable over k, and let $S = \mathrm{Spec}(A)$. Consider the following Lie algebra $\mathfrak{g}$ over $\mathcal{O}_S$. As a module it is free of rank 3, with basis $\{X, Y, H\}$, and its multiplication table is

$$[X, Y] = 2tH, \qquad [H, X] = X, \qquad [H, Y] = -Y.$$

For $s \in S$, $s \neq s_0$, where s_0 is the point defined by $t = 0$, the fiber

$$\mathfrak{g}(s) = \mathfrak{g} \otimes_A \kappa(s)$$

is isomorphic to the Lie algebra of $\mathrm{PGL}_{2,\kappa(s)}$. For $s = s_0$, it is a solvable Lie algebra.

5.2.— Let G_1 be the group scheme of automorphisms of $\mathfrak{g}$: for every $S' \to S$, $G_1(S')$ is the group of automorphisms of the $\mathcal{O}_{S'}$-Lie algebra $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$. It is a closed subscheme of the group $\mathrm{GL}(\mathfrak{g})$ of automorphisms of the $\mathcal{O}_S$-module $\mathfrak{g}$.

Let $S' \to S$, and let

$$u \in M_3(\Gamma(S', \mathcal{O}_{S'}))$$

be viewed as an endomorphism of the $\mathcal{O}_{S'}$-module $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$:

$$\begin{aligned} u(X) &= aX + bY + eH, \\ u(Y) &= b'X + a'Y + e'H, \\ u(H) &= cX + c'Y + dH. \end{aligned}$$

It is immediate that u is a section of G_1 if and only if $\det(u)$ is invertible and the following relations hold:

(1)	$a(d-1) = ec$,	(1')	$a'(d-1) = e'c'$,
(2)	$b(d+1) = ec'$,	(2')	$b'(d+1) = e'c$,
(3)	$e = 2t(bc - ac')$,	(3')	$e' = 2t(b'c' - a'c)$,
(4)	$2tc = eb' - ae'$,	(4')	$2tc' = be' - ea'$,
(5)	$2t(aa' - bb') = 2td$.		

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Lemma 5.3.— *Relations* (1), (1'), (2), *and* (2') *imply*

$$\det(u) = aa'(2-d) + bb'(d+2), \qquad aa' - bb' = d\det(u).$$

Indeed, the first assertion follows at once by substituting relations (1), (1'), (2), and (2') in the expansion of $\det(u)$:

$$\begin{aligned} \det(u) &= aa'd + be'c + b'c'e - a'ec - ae'c' - bb'd \\ &= aa'd + bb'(d+1) + bb'(d+1) - aa'(d-1) - bb'd - aa'(d-1) \\ &= aa'(d - d + 1 - d + 1) + bb'(d + 1 + d + 1 - d) \\ &= aa'(2-d) + bb'(d+2). \end{aligned}$$

[38] Editor's note: the equality $[u(X), u(Y)] = 2t\,u(H)$ gives relations (4), (4'), and (5); the equality $[u(H), u(X)] = u(X)$ gives (1)–(3); and the equality $[u(H), u(Y)] = -u(Y)$ gives (1')–(3').

Multiplying this relation by d, we obtain

$$d\det(u) = aa'(2d - d^2) + bb'(d^2 + 2d).$$

On the other hand, the relation (1) × (1') = (2) × (2') gives at once

$$aa'(d-1)^2 = bb'(d+1)^2.$$

Combining the last two relations immediately yields the second formula of Lemma 5.3.

5.4.— Now consider

$$G = G_1 \cap \mathrm{SL}(\mathfrak{g}).$$

This is the closed subgroup of G_1 defined by the equation $\det(u) = 1$; hence it is an affine group over S.

Proposition 5.5.— *The group G is smooth over S.*

To prove the proposition, we shall need the following lemmas.

Lemma 5.6.— *Let U be the open subset of*

$$\mathrm{End}_A(\mathfrak{g}) \simeq \mathbb{W}(M_3(\mathcal{O}_S))$$

defined by the condition that ad be invertible; that is, the principal open subset $\mathrm{End}_A(\mathfrak{g})_f$, where $f(u) = ad$. Then $U \cap G$ is the closed subscheme of U defined by the six equations

$$(1), \quad (2), \quad (2'), \quad (3), \quad (3'), \quad (\mathrm{D}): aa' - bb' = d.$$

It is clear that these six relations are satisfied by every point of G, by Lemma 5.3, and therefore by every point of $U \cap G$. Conversely, suppose that $u \in U(S')$, for some $S' \to S$, satisfies the six conditions in the statement. We must show that $\det(u) = 1$, and that u also satisfies (1'), (4), (4'), and (5).

First, (D) implies (5). Relations (2) and (2') give

$$bb'(d+1) = bce' = b'c'e.$$

On the other hand, from (3) and (3'), writing $2t(bc - ac')(b'c' - a'c)$ in two ways, we obtain

$$(bc - ac')e' = (b'c' - a'c)e.$$

Together with the preceding relation, this gives

$$ac'e' = a'ce.$$

By (1), however,

$$a'ce = a'a(d-1).$$

Thus

$$a(a'(d-1) - e'c') = 0,$$

and this implies (1'), since a is invertible.

Consequently, (1), (2), (2'), and (1') hold. Lemma 5.3 and (D) now give

$$d\big(\det(u) - 1\big) = 0.$$

Since d is invertible, it follows that $\det(u) = 1$.

It remains to prove (4) and (4'). We prove (4'); the other calculation follows by symmetry. From (3), (3'), and (D), we have

$$a'e + be' = -2t(aa' - bb')c' = -2tdc'.$$

Combining (1') and (2), we also have

$$a'e + be' = -d(be' - ea'),$$

[39] which proves (4'), since d is invertible.

Lemma 5.7.— *The group G is smooth over S along the identity section.*

Proof. By 5.6 and SGA 1, II 4.10, it is enough to prove that the differentials of the functions

$$a(d-1) - ec, \qquad b(d+1) - ec', \qquad b'(d+1) - e'c,$$
$$e - 2t(bc - ac'), \qquad e' - 2t(b'c' - a'c), \qquad aa' - bb' - d$$

at the points of the identity section of G are linearly independent. If the differential of a lower-case letter is denoted by the corresponding capital letter, these differentials are

$$D, \qquad 2B, \qquad 2B', \qquad E + 2tC', \qquad E' + 2tC, \qquad A + A' - D,$$

[40] and they are indeed linearly independent modulo every $(t - \lambda)$, $\lambda \in k$. [41]

Lemma 5.8.— *For $s \in S$, $s \neq s_0$, the fiber G_s is connected and semisimple.*

Proof. Indeed, since $s \neq s_0$, $\mathfrak{g}(s)$ is isomorphic to the Lie algebra of $\mathrm{PGL}_{2,\kappa(s)}$, and, on the other hand, $G_s = (G_1)_s$. The automorphism group of the Lie algebra of PGL_2 over a field of characteristic 0 is known to be PGL_2 itself, which is connected and semisimple. [42]

Lemma 5.9.— *The fiber G_{s_0} is solvable and has two connected components, of the following form:*

$$G^0_{s_0} = \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & a^{-1} & 0 \\ c & c' & 1 \end{pmatrix} \right\}, \quad G^-_{s_0} = \left\{ \begin{pmatrix} 0 & b & 0 \\ b^{-1} & 0 & 0 \\ c & c' & -1 \end{pmatrix} \right\}.$$

Proof. We have $e = e' = 0$, since $t = 0$ at s_0. The equations (1), (1'), ..., (5), and (D) can then be solved immediately.

Lemma 5.10.— *The matrix*

$$w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

is a section of G over S, such that $w(s_0) \in G^-_{s_0}$.

Let us now prove Proposition 5.5. Denote by G^0 the union of the identity components of the fibers of G, that is, the complement of $G^-_{s_0}$. Since G is smooth over S along the identity section by Lemma 5.7, G^0 is an open subgroup of G, smooth over S, by Exposé $\mathrm{VI_B}$, 3.10. By translation, G is also plainly smooth at the points of $w(S)$; hence G is smooth over S. [43]

[39] Editor's note: the sign has been corrected.
[40] Editor's note: $A + A' + B$ has been corrected to $A + A' - D$.
[41] Editor's note: "$(t - a)$, $a \in A$" has been corrected to "$(t - \lambda)$, $\lambda \in k$".
[42] Editor's note: the preceding sentence has been slightly modified.
[43] Editor's note: the preceding sentence has been modified, and the reference to Exposé $\mathrm{VI_B}$, 3.10 has been added.

5.11.— Consider the morphism

$$\mathbb{G}_{m,S} \longrightarrow G^0, \qquad z \longmapsto \begin{pmatrix} z & 0 & 0 \\ 0 & z^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is a monomorphism defining a torus T of G^0. We claim that

$$T = \mathrm{Centr}_G(T) = \mathrm{Centr}_{G^0}(T).$$

It is enough to verify the first equality. Since the two sides are smooth S-subgroups of G, it is enough to check that they have the same geometric points. In the fibers over points $s \neq s_0$, this follows because $\mathrm{PGL}_{2,\kappa(s)}$ is reductive and T_s is a maximal torus in it for dimensional reasons (cf. 1.11). In the fiber over s_0, the verification is immediate. In particular, T is a maximal torus of both G and G^0.

5.12.— The section w of G defined in Lemma 5.10 normalizes T. It follows immediately (cf. 2.4) that the Weyl group of G is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_S$, and is therefore finite over S:

$$W_G(T) = \mathrm{Norm}_G(T)/T = (\mathbb{Z}/2\mathbb{Z})_S.$$

By contrast, $W_{G^0}(T)$ is not finite over S: it "lacks one point" above s_0.

5.13.— The open immersion $G^0 \to G$ is not a closed immersion, since G^0 is dense in G; it is nevertheless an affine morphism (and hence G^0 is affine over S). Indeed, $G^0_{s_0}$ is closed in G_{s_0}, which is closed in G, so its complement U in G is open. The open subsets G^0 and U cover G, and it is enough to verify that the immersions

$$G^0 \longrightarrow G^0 \qquad \text{and} \qquad G^0 \cap U \longrightarrow U$$

are affine. This is trivial for the first; for the second, observe that $U \cap G^0$ is defined in U by the condition $b \neq 0$.

We have thus constructed a smooth affine S-group G^0, with connected fibers, possessing a self-centralizing maximal torus T whose Weyl group $W_{G^0}(T)$ is not finite (compare Theorem 2.5).

6. Local existence of maximal tori. The Weyl group

In the proof of Theorem 2.5 we used a result from Exposé XI on the local existence, for the étale topology, of maximal tori. The proof in Exposé XI uses a deep representability result (XI 4.1). In the particular case under consideration, one can give another, far more elementary proof, based on the ideas of Exposé XII, no. 7.

Proposition 6.1.— *Let S be a scheme, let G be a smooth affine S-group scheme with connected fibers over S, and let $s_0 \in S$ be a point such that the maximal tori of the geometric fiber $G_{\bar{s}_0}$ are self-centralizing. There exist an étale morphism $S' \to S$ whose image contains s_0 and a split maximal torus T of $G_{S'}$.*

Proof. First, we may suppose that S is affine. [44] Since G is of finite presentation over S, we may successively suppose that S is noetherian, local, and then henselian with separably closed residue field (cf. EGA IV, 8.12, §§8.8 and 18.8). Thus let

$$S = \mathrm{Spec}(A), \qquad k = \kappa(s_0),$$

[44] Editor's note: the preceding sentence and the reference to EGA IV below have been added.

where A is henselian and k is separably closed. Choose a maximal torus T_0 of $G_0 = G_k$. Such a torus exists, for example because the scheme of maximal tori of G_0 is smooth over k (Exposé XII, 7.1(c)). Since k is separably closed, T_0 is split (cf. Exposé X, 1.4), and is therefore given by a group monomorphism

$$f_0 \colon \mathbb{G}_{m,k}^r \longrightarrow G_0.$$

Let $m > 1$ be an integer prime to the characteristic of k. By Exposé VIII, 6.7, for every $h > 0$, $\mathrm{Centr}_{G_0}({}_{m^h}T_0)$ is represented by a closed subscheme of G_0. The subgroup schemes ${}_{m^h}T_0$ are schematically dense in T_0 (cf. Exposé IX, 4.10), and G_0 is noetherian. Consequently there is an h such that

$$\mathrm{Centr}_{G_0}({}_{m^h}T_0) = \mathrm{Centr}_{G_0}(T_0) = T_0.$$

Put $n = m^h$. Since n is invertible on S, ${}_n\mathbb{G}_{m,S}$ is isomorphic to $(\mathbb{Z}/n\mathbb{Z})_S$, so f_0 defines a group monomorphism

$$u_0 \colon (\mathbb{Z}/n\mathbb{Z})_k^r \longrightarrow G_0$$

such that $\mathrm{Centr}_{G_0}(u_0) = T_0$.

Now the S-functor

$$P = \mathrm{Hom}_{S\text{-gr}}\big((\mathbb{Z}/n\mathbb{Z})_S^r, G\big)$$

is represented by an S-scheme of finite type: it is a closed subscheme of r copies of

$${}_nG = \mathrm{Hom}_{S\text{-gr}}\big((\mathbb{Z}/n\mathbb{Z})_S, G\big) = \mathrm{Ker}\big(G \xrightarrow{n} G\big).$$

But P is smooth over S (Exposé IX, 3.6). Therefore $u_0 \in P(k)$ lifts to a section $u \in P(S)$, by Hensel's lemma (Exposé XI, 1.11):

$$u \colon (\mathbb{Z}/n\mathbb{Z})_S^r \longrightarrow G.$$

Consider $H = \mathrm{Centr}_G(u)$. It is a closed subgroup scheme of G, by Exposé VIII, 6.5(e), and its special fiber is $H_0 = T_0$. Moreover, H is smooth over S. Indeed, let $S' = \mathrm{Spec}(A')$ be an affine S-scheme, let

$$u' \colon ({}_n\mathbb{G}_{m,S'})^r \longrightarrow G_{S'}$$

be obtained from u by base change, put $S'_J = \mathrm{Spec}(A'/J)$, where J is a square-zero ideal, and let $x \in H(S'_J)$. Since G is smooth, x lifts to an element $g \in G(S')$. Then

$$v = \mathrm{int}(g)(u')$$

satisfies $v_J = u'_J$. By Exposé IX, 3.2, there is therefore an element $g' \in G(S')$ such that $g'_J = e$ and $\mathrm{int}(g')(v) = u'$. It follows that $h = g'g$ belongs to $H(S')$ and satisfies $h_J = x$.[45]

Let H^0 be the identity component of H. It is a subgroup scheme of G, smooth and with connected fibers, whose special fiber is a torus. By Exposé X, 8.1, it is a torus, necessarily split by Exposé X, 4.6. Put $T = H^0$, and let

$$C = \mathrm{Centr}_G(T).$$

This is a closed subgroup of G (Exposé VIII, 6.5(e)) and is smooth (Exposé XI, 2.4). Consider C^0 (in fact $C^0 = C$, but we do not need this). Then $C^0 \supset T$, and both are

[45] Editor's note: the reference to Exposé VIII, 6.5(e) was added above, and the preceding smoothness argument has been expanded.

smooth groups with connected fibers. They coincide at s_0, hence in a neighborhood of s_0. After shrinking S, we may therefore suppose that $C^0 = T$; then, a fortiori, T is maximal.

Remark 6.2.— *The proof shows in particular that the reductive rank of $G_{\bar{s}}$ is constant in a neighborhood of $s = s_0$.*

Proposition 6.3.— *Let S be a scheme, let G be an S-group scheme that is smooth and of finite presentation over S, and let Q be a subtorus of G.*

(i) $\mathrm{Centr}_G(Q)$ *and* $\mathrm{Norm}_G(Q)$ *are represented by closed subgroup schemes, smooth* (and hence of finite presentation) *over S.*

(ii) $\mathrm{Centr}_G(Q)$ *is an open and closed subscheme of* $\mathrm{Norm}_G(Q)$. *The quotient*

$$W_G(Q) = \mathrm{Norm}_G(Q)/\mathrm{Centr}_G(Q)$$

is represented by an open subgroup scheme of $\mathrm{Aut}_{S\text{-gr}}(Q)$*; it is therefore a quasi-finite, étale, separated S-group scheme.*

(iii) *For every $s \in S$, put*

$$w(s) = \left|\mathrm{Norm}_{G(\bar{s})}(Q(\bar{s}))/\mathrm{Centr}_{G(\bar{s})}(Q(\bar{s}))\right|.$$

Then $s \longmapsto w(s)$ is lower semicontinuous, and is constant in a neighborhood of s if and only if $W_G(Q)$ is finite over S in a neighborhood of s.

Proof. By Exposé XI, 6.11, $\mathrm{Centr}_G(Q)$ and $\mathrm{Norm}_G(Q)$ are represented by closed subschemes of G of finite presentation. They are smooth by Exposé XI, 2.4 and 2.4 bis, which proves (i). Assertions (ii) and (iii) are then proved as in Exposé XI, 5.9 and 5.10; those proofs in fact use only (i), and not the deeper theorems of Exposé XI, 4.1 and 4.2.

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Exposé XX
Reductive Groups of Semisimple Rank 1

by M. Demazure

0.1 Elementary systems. The groups U_α and $U_{-\alpha}$

Recollection 1.1. *Let $S = \mathrm{Spec}(k)$, where k is an algebraically closed field, and let G be a reductive S-group of semisimple rank one and T a maximal torus of G, necessarily split. Then*

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha},$$

where α and $-\alpha$ are the roots of G with respect to T. Moreover, there exist two group monomorphisms

$$p_\alpha \colon \mathbb{G}_{a,S} \longrightarrow G, \qquad p_{-\alpha} \colon \mathbb{G}_{a,S} \longrightarrow G$$

such that

$$t\, p_\alpha(x)\, t^{-1} = p_\alpha(\alpha(t)x), \qquad t\, p_{-\alpha}(x)\, t^{-1} = p_{-\alpha}(\alpha(t)^{-1}x)$$

for every $S' \to S$, $t \in T(S')$, and $x \in \mathbb{G}_a(S')$, and such that the morphism

$$\mathbb{G}_{a,S} \times_S T \times_S \mathbb{G}_{a,S} \longrightarrow G, \qquad (y,t,x) \longmapsto p_{-\alpha}(y)\, t\, p_\alpha(x),$$

is radicial and dominant (*Bible*, Section 13.4, Corollary 2 to Theorem 3).

Since its tangent map at the identity is bijective, this morphism is also separable, hence birational. Zariski's Main Theorem (EGA III_1, 4.4.9) therefore shows that it is an open immersion.

Lemma 1.2. *Let S be a scheme, G an S-group scheme, T a torus of G, Q a subtorus of T, and α a character of T whose restriction to Q is nontrivial on every fiber. Let*

$$p_\alpha \colon \mathbb{G}_{a,S} \longrightarrow G \quad \text{and} \quad p_{-\alpha} \colon \mathbb{G}_{a,S} \longrightarrow G$$

be group morphisms normalized by T, with multipliers α and $-\alpha$, respectively. Suppose that the morphism

$$u \colon \mathbb{G}_{a,S} \times_S T \times_S \mathbb{G}_{a,S} \longrightarrow G, \qquad u(y,t,x) = p_{-\alpha}(y)\, t\, p_\alpha(x),$$

[0]Editors' note: version of 13 October 2024.

defined on points by the displayed formula, is an open immersion. Finally, let $q \geq 0$ be an integer and let

$$p\colon \mathbb{G}_{a,S} \longrightarrow G$$

be a group morphism such that, for every $S' \to S$, $t \in Q(S')$, and $x \in \mathbb{G}_a(S')$,

$$\operatorname{int}(t)^q\big(p(x)\big) = p\big(\alpha(t)x\big).$$

Then there is a unique $\nu \in \mathbb{G}_a(S)$ such that

$$p(x) = p_\alpha(\nu x^q).$$

Indeed, let Ω be the image of u, and put $U = p^{-1}(\Omega)$. This is an open subset of $\mathbb{G}_{a,S}$ containing the zero section. For every section t of Q, multiplication by $\alpha(t)$ leaves U globally invariant. We have $U = \mathbb{G}_{a,S}$. It is enough to check this when S is the spectrum of an algebraically closed field k; then

$$\alpha\colon Q(k) \longrightarrow k^\times$$

is surjective, so $U(k) \supset k^\times$, and hence $U = \mathbb{G}_{a,k}$.

There are consequently three morphisms

$$a\colon \mathbb{G}_{a,S} \longrightarrow \mathbb{G}_{a,S}, \qquad b\colon \mathbb{G}_{a,S} \longrightarrow T, \qquad c\colon \mathbb{G}_{a,S} \longrightarrow \mathbb{G}_{a,S}$$

such that

$$p(x) = p_{-\alpha}\big(a(x)\big)\, b(x)\, p_\alpha\big(c(x)\big).$$

The condition on p becomes

$$a\big(\alpha(t)x\big) = \alpha(t)^{-q}a(x), \qquad b\big(\alpha(t)x\big) = b(x), \qquad c\big(\alpha(t)x\big) = \alpha(t)^q c(x).$$

For the same reason, for every $S' \to S$ and $z \in \mathbb{G}_m(S')$,

$$a(zx) = z^{-q}a(x), \qquad b(zx) = b(x), \qquad c(zx) = z^q c(x),$$

and therefore

$$z^q a(z) = a(1), \qquad b(z) = b(1), \qquad c(z) = z^q c(1).$$

Since $\mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}$, for every $S' \to S$ and $x \in \mathbb{G}_a(S')$ we obtain

$$\begin{aligned} x^q a(x) = a(1) = a(0) = 0, &\quad \text{hence } a = 0,\\ c(x) = x^q c(1) = \nu x^q, &\quad \text{for some } \nu \in \mathbb{G}_a(S),\\ b(x) = b(1) = b(0) = e, &\quad \text{hence } b = e. \end{aligned}$$

This completes the proof.

Definition 1.3. *An S-elementary system is a triple (G, T, α) in which*

(i) *G is a reductive S-group of semisimple rank one* (Exposé XIX, 2.7)*;*

(ii) *T is a maximal torus of G;*

(iii) *α is a root of G with respect to T* (Exposé XIX, 3.2).

Thus there is a direct-sum decomposition (Exposé XIX, 3.5)

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank one.[1]

[1] As $\mathcal{O}_S$-modules.

1.4. If (G, T, α) is an S-elementary system, then $(G_{S'}, T_{S'}, \alpha_{S'})$ is an S'-elementary system for every $S' \to S$. If (G, T, α) is an S-elementary system, then so is $(G, T, -\alpha)$.

If S is a scheme, G a reductive S-group, T a maximal torus of G, and α a root of G with respect to T, then (Z_α, T, α) is an S-elementary system (Exposé XIX, 3.9).

Let (G, T, α) be an S-elementary system. The invertible module $\mathfrak{g}^\alpha$ is canonically endowed with a T-module structure. Hence the vector bundle $W(\mathfrak{g}^\alpha)$ is also a T-module. On the other hand, the inner automorphisms by elements of T make G a group equipped with a group of operators T.

Theorem 1.5. *Let (G, T, α) be an S-elementary system.*

(i) *There exists a unique morphism of groups with operator group T*

$$\exp\colon W(\mathfrak{g}^\alpha) \longrightarrow G$$

whose induced morphism on Lie algebras is the canonical map $\mathfrak{g}^\alpha \to \mathfrak{g}$.[2] *Equivalently,* exp *is the unique morphism such that, for every $S' \to S$, $t \in T(S')$, and $X, X' \in W(\mathfrak{g}^\alpha)(S')$,*

$$\begin{aligned} \exp(X + X') &= \exp(X)\exp(X'), \\ \operatorname{int}(t)\big(\exp(X)\big) &= \exp\big(\alpha(t)X\big), \\ \operatorname{Lie}(\exp)(X) &= X. \end{aligned}$$

(ii) *Define analogously, in the S-elementary system $(G, T, -\alpha)$,*

$$\exp\colon W(\mathfrak{g}^{-\alpha}) \longrightarrow G.$$

Then the morphism

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^\alpha) \longrightarrow G, \qquad (Y, t, X) \longmapsto \exp(Y)\, t\, \exp(X),$$

defined on points by the displayed formula, is an open immersion.

Suppose for the moment that the required exponential morphisms exist, and prove the other assertions. We first prove (ii). Both sides are of finite presentation and flat over S, so it is enough to work when S is the spectrum of an algebraically closed field (SGA 1, I 5.7 and VIII 5.5). Put $S = \operatorname{Spec}(k)$, and choose

$$X \in \Gamma(S, \mathfrak{g}^\alpha)^\times, \qquad Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times.$$

It is enough to prove that

$$\mathbb{G}_{a,k} \times_k T \times_k \mathbb{G}_{a,k} \longrightarrow G, \qquad (y, t, x) \longmapsto \exp(yY)\, t\, \exp(xX),$$

is an open immersion. By 1.1 and 1.2, there exist $a, b \in k$ such that

$$\exp(yY) = p_{-\alpha}(ay), \qquad \exp(xX) = p_\alpha(bx).$$

Since $\exp\colon W(\mathfrak{g}^{-\alpha}) \to G$ induces a monomorphism on Lie algebras, $a \neq 0$; likewise $b \neq 0$. The assertion is therefore reduced to 1.1.

The uniqueness of exp may be proved locally on S. We reduce to the case in which $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free, and then apply 1.2 with $Q = T$ and $q = 1$.

[2] It will be seen below, in Corollary 5.9, that exp is an isomorphism from $W(\mathfrak{g}^\alpha)$ onto a closed subgroup of G.

It remains to prove existence. Faithfully flat descent, together with the uniqueness just proved, shows that it is enough to prove existence locally on S for the fpqc topology. The usual finite-presentation arguments reduce first to S noetherian and then to S noetherian local. We may therefore assume that

$$S = \mathrm{Spec}(A),$$

where A is a complete noetherian local ring with algebraically closed residue field k. Let

$$p_0 \colon \mathbb{G}_{a,k} \longrightarrow G_k$$

be a monomorphism of k-groups normalized by T_k, with multiplier $\alpha_0 = \alpha \otimes_A k$; such a morphism exists by 1.1. By 1.1 and 1.2, the associated morphism

$$T_k \cdot_{\alpha_0} \mathbb{G}_{a,k} \longrightarrow G_k$$

is an immersion and hence, in particular, a monomorphism. We provisionally admit the following two lemmas.

Lemma 1.6. *Let S be a scheme, G an S-group of finite presentation, T an S-torus, α a character of T that is nontrivial on every fiber, and $s_0 \in S$. Let*

$$f \colon T \cdot_\alpha \mathbb{G}_{a,S} \longrightarrow G$$

be an S-group morphism such that f_{s_0} is a monomorphism and the restriction of f to T is a monomorphism. There exists an open neighborhood U of s_0 such that $f|_U$ is a monomorphism.

Lemma 1.7. *Let A be a complete noetherian local ring with algebraically closed residue field k, let (G, T, α) be an A-elementary system, and let*

$$p_0 \colon \mathbb{G}_{a,k} \longrightarrow G_k$$

be a morphism of k-groups normalized by T_k, with multiplier $\alpha \otimes_A k$. There exists a group morphism

$$p \colon \mathbb{G}_{a,A} \longrightarrow G$$

normalized by T, with multiplier α.

Let p be the morphism asserted by 1.7, and let

$$f \colon T \cdot_\alpha \mathbb{G}_{a,S} \longrightarrow G$$

be the corresponding morphism. It satisfies the hypotheses of 1.6, so it is a monomorphism; in particular, p is a monomorphism. The theorem now follows from Exposé XIX, 4.9.

Proof of 1.6. Let

$$\epsilon \colon S \longrightarrow T \cdot_\alpha \mathbb{G}_{a,S}$$

be the identity section. Since f is unramified at $\epsilon(s_0)$, it is unramified at $\epsilon(s)$ for all s in an open neighborhood U of s_0. Hence $f|_U$ is unramified (Exposé X, 3.5),[3] and its kernel $\mathrm{Ker}(f)_U$ is unramified over U. To prove that $f|_U$ is a monomorphism, it is therefore enough[4] to prove that $\mathrm{Ker}(f)_U$ is radicial over U, which is a set-theoretic question. We are reduced to the following assertion.

[3] See also Exposé VI_B, 2.5.

[4] By EGA IV_4, 17.9.1.

Lemma 1.8. *Let k be an algebraically closed field, and let N be an invariant subgroup of $T \cdot_\alpha \mathbb{G}_{a,k}$, where α is a nontrivial character of the torus T. Suppose that N is étale over k and that $N \cap T = \{e\}$. Then $N = \{e\}$.*

We have

$$\mathrm{int}(t')(t,x) = (t, \alpha(t')x).$$

If (t,x) is a point of N with $x \neq 0$, then (t,zx) is also a point of N for every $z \in k^\times$. Thus (t,x) is not isolated, contradicting the quasi-finiteness of N. Set-theoretically, therefore, $N \subset T$, and the hypothesis $N \cap T = \{e\}$ completes the proof.

Proof of 1.7. Let $\mathfrak{m}$ be the radical of A, and put

$$S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1}), \qquad n \geq 0.$$

We first show, by induction on n, that p_0 can be extended, for every n, to a morphism of S_n-groups

$$p_n \colon \mathbb{G}_{a,S_n} \longrightarrow G_{S_n},$$

normalized by T_{S_n} with multiplier α_{S_n}, in such a way that

$$p_{n+1} \times_{S_{n+1}} S_n = p_n.$$

Let $H = T \cdot_\alpha \mathbb{G}_{a,S}$. Denote by f_n the morphism $H_{S_n} \to G_{S_n}$ defined by p_n. We provisionally admit the following lemma.

Lemma 1.9. *Let (G,T,α) be a k-elementary system, where k is algebraically closed, and let*

$$p \colon \mathbb{G}_{a,k} \longrightarrow G$$

be a monomorphism normalized by T, with multiplier α. Then

$$\mathrm{H}^2\big(T \cdot_\alpha \mathbb{G}_{a,k}, \mathfrak{g}\big) = 0.$$

Here $T \cdot_\alpha \mathbb{G}_{a,k}$ acts on $\mathfrak{g}$ through the morphism $T \cdot_\alpha \mathbb{G}_{a,k} \to G$ defined by p and through the adjoint representation of G.

By Exposé III, 2.8, f_n therefore extends to a morphism of S_{n+1}-groups

$$f'_{n+1} \colon H_{S_{n+1}} \longrightarrow G_{S_{n+1}}.$$

The morphism f'_{n+1} and the canonical immersion of $T_{S_{n+1}}$ into $G_{S_{n+1}}$ have the same restriction to T_{S_n}. By Exposé III, 2.5, there is an element $g \in G(S_{n+1})$ such that

$$g \times_{S_{n+1}} S_n = e$$

and such that

$$f_{n+1} = \mathrm{int}(g) \circ f'_{n+1}$$

restricts to $T_{S_{n+1}}$ by the canonical immersion. Let p_{n+1} be the restriction of f_{n+1} to $\mathbb{G}_{a,S_{n+1}}$. It is normalized by $T_{S_{n+1}}$, has multiplier $\alpha_{S_{n+1}}$, and extends p_n.

We have thus constructed a coherent system (f_n); it remains to algebraize it. For this we use:

Lemma 1.10. *Let A be a complete noetherian local ring, $\mathfrak{m}$ its maximal ideal,*

$$S = \mathrm{Spec}(A), \qquad S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1}),$$

let T be an S-torus, let α be a nonzero character of T, and let X be an affine S-scheme on which T acts. Let T act on $\mathbb{G}_{a,S}$ through α. Let $q \geq 0$, and let $(f_n)_{n\geq 0}$ be a coherent system of morphisms

$$f_n \colon \mathbb{G}_{a,S_n}^{q} \longrightarrow X_{S_n}$$

of objects with operators T_{S_n}. There exists a unique morphism of objects with operators T

$$f \colon \mathbb{G}_{a,S}^{q} \longrightarrow X$$

that induces the f_n (compare Exposé IX, 7.1).

Corollary 1.11. *If X is a group with operator group T, and if the f_n are group morphisms, then f is also a group morphism.*

Indeed, apply the uniqueness assertion of the lemma to the two morphisms $\mathbb{G}_{a,S}^{2q} \to X$ deduced from f in the usual way.

Proof of 1.10. Suppose that T is split, as it is in the application of 1.10 to the proof of 1.5. By Exposé I, 4.7.3.1, the functor

$$X \longmapsto \mathscr{A}(X)$$

defines an equivalence between the category of affine S-schemes equipped with an action of T and the category opposite to that of M-graded S-algebras, where

$$M = \mathrm{Hom}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}).$$

Thus there are gradings

$$B = \mathscr{A}(X) = \bigoplus_{m\in M} B_m, \qquad C = \mathscr{A}(\mathbb{G}_{a,S}^{q}) = \bigoplus_{m\in M} C_m.$$

Each C_m is visibly a finite free A-module. Indeed, $C_m = 0$ if m is not a multiple of α, whereas, if $m = d\alpha$, then C_m is isomorphic to the A-module of homogeneous polynomials of degree d in q variables. Put

$$\widehat{B}_m = \varprojlim_n B_m \otimes_A (A/\mathfrak{m}^{n+1}),$$
$$\widehat{C}_m = \varprojlim_n C_m \otimes_A (A/\mathfrak{m}^{n+1}),$$
$$\widehat{B} = \bigoplus_{m\in M} \widehat{B}_m, \qquad \widehat{C} = \bigoplus_{m\in M} \widehat{C}_m.$$

There are canonical morphisms of M-graded algebras

$$g_B \colon B \longrightarrow \widehat{B}, \qquad g_C \colon C \longrightarrow \widehat{C}.$$

The preceding observation shows that g_C is an isomorphism. Giving a coherent system (f_n) as in the statement is equivalent to giving a morphism of graded A-algebras

$$\widehat{F} \colon \widehat{B} \longrightarrow \widehat{C}.$$

Finding a morphism f as in the statement is equivalent to finding a morphism of graded A-algebras $F\colon B \to C$ making the diagram

$$\begin{array}{ccc} B & \xrightarrow{F} & C \\ {\scriptstyle g_B}\downarrow & & \downarrow{\scriptstyle g_C} \\ \widehat{B} & \xrightarrow{\widehat{F}} & \widehat{C} \end{array}$$

commute. Since g_C is an isomorphism, the existence and uniqueness of F are immediate. This proves 1.10.

To complete the proof of 1.5, it remains only to prove 1.9.

1.12. Proof of 1.9. We have

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha}.$$

As explained in 1.9, regard $\mathfrak{g}$ as a $T \cdot_{\alpha} \mathbb{G}_{a,k}$-module. Clearly $\mathfrak{t} \oplus \mathfrak{g}^{\alpha}$ is a submodule of $\mathfrak{g}$, and the quotient is isomorphic to $\mathfrak{g}^{-\alpha}$ as a k-vector space and even as a T-module. The group $\mathbb{G}_{a,k}$ acts trivially on this one-dimensional quotient, since every group morphism $\mathbb{G}_{a,k} \to \mathbb{G}_{m,k}$ is trivial. Likewise, $\mathfrak{g}^{\alpha}$ is a submodule of $\mathfrak{t} \oplus \mathfrak{g}^{\alpha}$; the quotient is isomorphic to $\mathfrak{t}$ as a T-module, and $\mathbb{G}_{a,k}$ acts trivially on it. In summary:

Lemma 1.13. *Under the hypotheses of 1.9, $\mathfrak{g}$, as a $T \cdot_{\alpha} \mathbb{G}_{a,k}$-module, has a composition series whose successive quotients are*

$$\mathfrak{g}^{-\alpha}, \qquad \mathfrak{t}, \qquad \mathfrak{g}^{\alpha},$$

regarded as $T \cdot_{\alpha} \mathbb{G}_{a,k}$-modules through the projection $T \cdot_{\alpha} \mathbb{G}_{a,k} \to T$.

We are thus reduced to computing the cohomology of $T \cdot_{\alpha} \mathbb{G}_{a,k}$ acting on $W(k)$ through the projection $T \cdot_{\alpha} \mathbb{G}_{a,k} \to T$ and a character β of T (here $\beta = 0, \alpha$, or $-\alpha$). [5] Write $k[x_1, \ldots, x_n]$ for the polynomial algebra over k in n variables, and $k_q[x_1, \ldots, x_n]$ for its subspace of homogeneous polynomials of degree q.

Lemma 1.14. *With the preceding notation,*

$$\mathrm{H}^n(T \cdot_{\alpha} \mathbb{G}_{a,k}, k) = \mathrm{H}^n(C^*_{\alpha,\beta}),$$

where the complex $C^*_{\alpha,\beta}$ is defined by

$$C^n_{\alpha,\beta} = \begin{cases} k_q[x_1, \ldots, x_n], & \text{if } \beta = q\alpha, \text{ with } q \in \mathbb{N}^*, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\begin{aligned} (\delta f)(x_1, x_2, \ldots, x_{n+1}) = {} & f(x_2, \ldots, x_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(x_1, \ldots, x_i x_{i+1}, \ldots, x_{n+1}) \\ & + (-1)^{n+1} f(x_1, \ldots, x_n). \end{aligned}$$

[5] The editors added the sentence that follows.

Indeed, by Exposé I, 5.3.2, the functor $M \mapsto \mathrm{H}^0(T, M)$ is exact on the category of T-modules, and the $\mathrm{H}^q(T, -)$ vanish. It follows, as in the usual theory of group cohomology, that $\mathrm{H}^n(T \cdot_\alpha \mathbb{G}_{a,k}, k)$ may be computed as the n-th cohomology group of the complex of cochains of $\mathbb{G}_{a,k}$ in k that are invariant under T, that is, those satisfying

$$f(\alpha(t)x_1, \ldots, \alpha(t)x_n) = \beta(t)f(x_1, \ldots, x_n).$$

This is precisely the complex just displayed.

To prove 1.9, it is therefore enough to show that $\mathrm{H}^2(C^*_{\alpha,\beta}) = 0$ for $\beta = 0, \alpha, -\alpha$, which is immediate.

Remark 1.15. The groups $\mathrm{H}^n(C^*_{\alpha,\beta})$ can be computed explicitly when $\beta = q\alpha$; see M. Lazard, *Lois de groupes et analyseurs*, Annales E.N.S., 1955. In particular,

$$\mathrm{H}^n(C^*_{\alpha,q\alpha}) = 0 \qquad \text{for } n > q.$$

Notations 1.16. The image of the canonical immersion

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^{\alpha}) \longrightarrow G$$

will be denoted by Ω. It is an open subset of G containing the identity section. The images of

$$W(\mathfrak{g}^{-\alpha}), \qquad W(\mathfrak{g}^{\alpha}), \qquad W(\mathfrak{g}^{-\alpha}) \times_S T, \qquad T \times_S W(\mathfrak{g}^{\alpha})$$

will be denoted, respectively,[6]

$$U_{-\alpha}, \qquad U_{\alpha}, \qquad U_{-\alpha} \cdot T, \qquad T \cdot U_{\alpha}.$$

Then U_α (respectively $U_{-\alpha}$) is a subgroup of G, canonically endowed with a vector-bundle structure, and

$$\mathrm{int}(t)(x) = x^{\alpha(t)} \qquad (\text{respectively } x^{-\alpha(t)})$$

for every $S' \to S$, $t \in T(S')$, and $x \in U_\alpha(S')$ (respectively $x \in U_{-\alpha}(S')$).

There are canonical isomorphisms

$$T \cdot U_\alpha \simeq T \cdot_\alpha U_\alpha, \qquad T \cdot U_{-\alpha} \simeq T \cdot_{-\alpha} U_{-\alpha}.$$

The open subset Ω is stable under $\mathrm{int}(T)$, and

$$\mathrm{int}(t')(y \cdot t \cdot x) = y^{-\alpha(t')} \cdot t \cdot x^{\alpha(t')}.$$

Corollary 1.17. *We have*

$$\mathrm{Lie}(U_\alpha/S) = \mathfrak{g}^{\alpha}, \qquad \mathrm{Lie}(U_{-\alpha}/S) = \mathfrak{g}^{-\alpha}.$$

The isomorphisms

$$W(\mathfrak{g}^{\alpha}) \xrightarrow[\sim]{\exp} U_\alpha, \qquad W(\mathfrak{g}^{-\alpha}) \xrightarrow[\sim]{\exp} U_{-\alpha}$$

are those of Exposé XIX, 4.2.

Corollary 1.18. *The open subset Ω is relatively schematically dense in G* (cf. Exposé XVIII, §1).

This follows immediately from Exposé XVIII, 1.3.[7]

[6] The editors replaced $P_{-\alpha}$ and P_α by $U_{-\alpha}$ and U_α.

[7] See also EGA IV$_3$, 11.10.10.

Corollary 1.19. *The center of G is*

$$\mathrm{Centr}(G) = \mathrm{Ker}(\alpha).$$

It is therefore a closed subgroup of G, of multiplicative type and of finite type.

The second assertion follows from the first by Exposé IX, 2.7. Let us prove the first. The inner automorphism defined by a section of $\mathrm{Ker}(\alpha)$ acts trivially on Ω, by the last formula of 1.16, and hence on G, by 1.18. Conversely, if $g \in G(S)$ centralizes G, then it centralizes T and U_α. It is therefore a section of T, by Exposé XIX, 2.8, and annihilates α. Since the same argument applies after every base change, we indeed have $\mathrm{Centr}(G) = \mathrm{Ker}(\alpha)$.

Corollary 1.20. *A monomorphism*

$$p_\alpha \colon \mathbb{G}_{a,S} \longrightarrow G$$

normalized by T, with multiplier α, exists if and only if the $\mathscr{O}_S$-module $\mathfrak{g}^\alpha$ is free. More precisely, the assignments

$$X_\alpha \longmapsto (x \longmapsto \exp(xX_\alpha)), \qquad p_\alpha \longmapsto \mathrm{Lie}(p_\alpha)$$

give a bijection between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of such monomorphisms p_α, which is also the set of isomorphisms of vector group schemes

$$\mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha.$$

8

Corollary 1.21. *The subgroups U_α and $U_{-\alpha}$ of G commute on no fiber.*

Indeed, if $(U_\alpha)_s$ and $(U_{-\alpha})_s$ commute, then Ω_s is a subgroup of G_s, and hence $\Omega_s = G_s$.[9] It follows that G_s is solvable, contrary to the hypothesis that G_s is reductive of semisimple rank 1.

0.2 Structure of elementary systems

Theorem 2.1. *Let S be a scheme and (G, T, α) an S-elementary system. There exist a morphism of $\mathscr{O}_S$-modules*

$$\mathfrak{g}^\alpha \otimes_{\mathscr{O}_S} \mathfrak{g}^{-\alpha} \longrightarrow \mathscr{O}_S, \qquad (X, Y) \longmapsto \langle X, Y \rangle,$$

and a morphism of S-groups

$$\alpha^* \colon \mathbb{G}_{m,S} \longrightarrow T$$

such that, for every $S' \to S$ and all

$$X \in \Gamma\big(S', \mathfrak{g}^\alpha \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}\big), \qquad Y \in \Gamma\big(S', \mathfrak{g}^{-\alpha} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}\big),$$

one has

$$\exp(X) \cdot \exp(Y) \in \Omega(S') \quad \Longleftrightarrow \quad 1 + \langle X, Y \rangle \in \mathbb{G}_m(S').$$

[8] Indeed, $\mathrm{Lie}(\mathbb{G}_{a,S}) = \mathscr{O}_S$, and $\mathrm{Lie}(p_\alpha)$ is an element of $\mathrm{Hom}_{\mathscr{O}_S}(\mathscr{O}_S, \mathfrak{g}^\alpha) = \Gamma(S, \mathfrak{g}^\alpha)$.

[9] By 1.18.

Under these conditions,

$$\text{(F)} \qquad \exp(X)\cdot\exp(Y) = \exp\left(\frac{Y}{1+\langle X,Y\rangle}\right)\alpha^*(1+\langle X,Y\rangle)\exp\left(\frac{X}{1+\langle X,Y\rangle}\right).$$

Moreover, the morphisms $(X,Y)\mapsto\langle X,Y\rangle$ and α^* are uniquely determined. The first is an isomorphism, and therefore places $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ in duality; furthermore,

$$\alpha\circ\alpha^* = 2,$$

where 2 denotes squaring on $\mathbb{G}_{m,S}$.

In view of the uniqueness assertions, it is enough to prove the theorem locally on S. We may therefore assume that $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free on S. Choose

$$X\in\Gamma(S,\mathfrak{g}^\alpha)^\times, \qquad Y\in\Gamma(S,\mathfrak{g}^{-\alpha})^\times,$$

and put

$$p_\alpha(x)=\exp(xX), \qquad p_{-\alpha}(y)=\exp(yY)$$

for $x,y\in\mathbb{G}_a(S')$, $S'\to S$. By 1.5 and 1.21, it is enough to prove the following proposition.

Proposition 2.2. *Let S be a scheme, G an S-group, T a torus of G, and α a character of T that is nontrivial on every fiber. Let*

$$p_\alpha\colon\mathbb{G}_{a,S}\longrightarrow G, \qquad p_{-\alpha}\colon\mathbb{G}_{a,S}\longrightarrow G$$

be group monomorphisms normalized by T, with multipliers α and $-\alpha$, respectively. Suppose that:

(i) *the morphism*

$$\mathbb{G}_{a,S}\times_S T\times_S\mathbb{G}_{a,S}\longrightarrow G, \qquad (y,t,x)\longmapsto p_{-\alpha}(y)\,t\,p_\alpha(x),$$

is an open immersion; denote its image by Ω;

(ii) *for every $s\in S$, the groups*

$$(p_\alpha)_s(\mathbb{G}_{a,\kappa(s)}) \quad\text{and}\quad (p_{-\alpha})_s(\mathbb{G}_{a,\kappa(s)})$$

do not commute.

Then there exist unique elements

$$a\in\mathbb{G}_a(S), \qquad \alpha^*\in\mathrm{Hom}_{S\text{-gr.}}(\mathbb{G}_{m,S},T)$$

with the following properties: for every $S'\to S$ and all $x,y\in\mathbb{G}_a(S')$,

$$p_\alpha(x)p_{-\alpha}(y)\in\Omega(S') \iff 1+axy\in\mathbb{G}_m(S'),$$

and, under this condition,

$$p_\alpha(x)p_{-\alpha}(y)=p_{-\alpha}\left(\frac{y}{1+axy}\right)\alpha^*(1+axy)p_\alpha\left(\frac{x}{1+axy}\right).$$

Moreover, a is invertible, that is, $a \in \mathbb{G}_m(S)$, and $\alpha \circ \alpha^* = 2$.

Proof. A) Consider the morphism

$$\mathbb{G}_{a,S}^2 \longrightarrow G, \qquad (x, y) \longmapsto p_\alpha(x)p_{-\alpha}(y).$$

Let U be the inverse image of Ω. It is an open subset of $\mathbb{G}_{a,S}^2$ containing $0 \times_S \mathbb{G}_{a,S}$ and $\mathbb{G}_{a,S} \times_S 0$. There are therefore unique morphisms of S-schemes

$$A\colon U \longrightarrow \mathbb{G}_{a,S}, \qquad C\colon U \longrightarrow \mathbb{G}_{a,S}, \qquad B\colon U \longrightarrow T$$

satisfying the set-theoretic identity

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(A(u, v))\, B(u, v)\, p_\alpha(C(u, v)).$$

Immediately,

$$A(0, v) = v, \qquad A(u, 0) = 0, \qquad C(u, 0) = u, \qquad C(0, v) = 0,$$
$$B(u, 0) = B(0, v) = e.$$

Let S' be a separated S-scheme and let $t \in T(S')$. Since $\Omega_{S'}$ is stable under $\mathrm{int}(t)$, the last formula of 1.16 shows that $U_{S'}$ is stable under the automorphism

$$(x, y) \longmapsto (\alpha(t)x, \alpha(t)^{-1}y)$$

of $\mathbb{G}_{a,S'}^2$, and

$$A(\alpha(t)u, \alpha(t)^{-1}v) = \alpha(t)^{-1}A(u, v),$$
$$C(\alpha(t)u, \alpha(t)^{-1}v) = \alpha(t)C(u, v),$$
$$B(\alpha(t)u, \alpha(t)^{-1}v) = B(u, v).$$

Because α is faithfully flat, it follows that for every $S' \to S$ and every $z \in \mathbb{G}_m(S')$, the open subset $U_{S'}$ is stable under $(x, y) \mapsto (zx, z^{-1}y)$, and

$$A(zu, z^{-1}v) = z^{-1}A(u, v),$$
$$C(zu, z^{-1}v) = zC(u, v),$$
$$B(zu, z^{-1}v) = B(u, v).$$

Suppose first that v is invertible. Taking $z = v$, we see that if (u, v) is a section of U, then $(uv, 1)$ is also a section of U, and

$$A(uv, 1) = v^{-1}A(u, v), \qquad B(uv, 1) = B(u, v).$$

Let V be the open subset of $\mathbb{G}_{a,S}^2$ defined by[10]

$$(u, v) \in V(S') \iff (u, v),\ (uv, 1),\ \text{and } (1, uv) \text{ belong to } U(S').$$

Since U contains $0 \times_S \mathbb{G}_{a,S}$ and $\mathbb{G}_{a,S} \times_S 0$, the open subset V is a neighborhood of the zero section of $\mathbb{G}_{a,S}^2$. The preceding calculation shows that, on $V \cap (\mathbb{G}_{a,S} \times_S \mathbb{G}_{m,S})$, the two morphisms

$$(u, v) \longmapsto A(u, v), \qquad (u, v) \longmapsto vA(uv, 1)$$

[10]The editors added the condition "$(1, uv) \in U(S')$".

coincide, as do

$$(u,v) \longmapsto B(u,v), \qquad (u,v) \longmapsto B(uv,1).$$

Since $\mathbb{G}_{a,S} \times_S \mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}^2$, these morphisms coincide on V.

We have $A(0,1) = 1$. Hence there is an open subset W_1 of $\mathbb{G}_{a,S}$, containing the zero section, such that $A(x,1)$ is invertible for every section x of W_1. Put

$$A(x,1)^{-1} = F(x).$$

If $(u,v) \in V(S')$[11] and $uv \in W_1(S')$, then

$$A(u,v) = vA(uv,1) = vF(uv)^{-1}.$$

The same argument applied to C gives an open subset W_2 of $\mathbb{G}_{a,S}$, containing the zero section, and an element $E \in \mathscr{O}(W_2)^{\times}$[12] such that

$$C(u,v) = uC(1,uv) = uE(uv)^{-1}$$

whenever $(u,v) \in V(S')$ and $uv \in W_2(S')$.

Consequently, with $W = W_1 \cap W_2$, there are S-morphisms

$$\begin{aligned} &F\colon W \to \mathbb{G}_{m,S}, && F(0) = 1,\\ &H\colon W \to T, && H(0) = e,\\ &E\colon W \to \mathbb{G}_{m,S}, && E(0) = 1, \end{aligned}$$

such that, if $(u,v) \in V(S')$ and $uv \in W(S')$, then

$$(+) \qquad p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\big(vF(uv)^{-1}\big)\, H(uv)\, p_\alpha\big(uE(uv)^{-1}\big).$$

B) We now use the associativity of G to write

$$p_\alpha(u)p_{-\alpha}(v)p_{-\alpha}(w) = p_\alpha(u)p_{-\alpha}(v+w).$$

There is an open subset L of $\mathbb{G}_{a,S}^3$, containing the zero section, such that $(u,v,w) \in L(S')$ is equivalent to

$$\begin{aligned} &(u,v) \in V(S'), \qquad \big(uE(uv)^{-1}, w\big) \in V(S'), \qquad (u, v+w) \in V(S'),\\ &uv \in W(S'), \qquad uwE(uv)^{-1} \in W(S'), \qquad u(v+w) \in W(S'). \end{aligned}$$

Using formula $(+)$, we immediately obtain, for $(u,v,w) \in L(S')$,

$$(1) \qquad E(uv+uw) = E\big(uwE(uv)^{-1}\big)E(uv),$$

$$(2) \qquad H(uv+uw) = H\big(uwE(uv)^{-1}\big)H(uv),$$

and

$$(3) \qquad \begin{aligned} (v+w)F(uv+uw)^{-1} = {}& \alpha\big(H(uv)^{-1}\big)\, wF\big(uwE(uv)^{-1}\big)^{-1}\\ &+ vF(uv)^{-1}. \end{aligned}$$

[11] Here and below, the editors replaced $U(S')$ by $V(S')$.

[12] The editors use E for the element denoted G in the original, since G already denotes the S-group under consideration.

It follows immediately from the definition of L that $(1,0,0) \in L(S)$. Consider therefore

$$L \cap_T \left(1 \times_S \mathbb{G}_{a,S} \times_S \mathbb{G}_{a,S}\right) = 1 \times_S M.$$

The scheme M is an open subset of $\mathbb{G}_{a,S}^2$ containing the section $(0,0)$. For $(v,w) \in M(S')$, the elements

$$v, \qquad wE(v)^{-1}, \qquad v+w$$

belong to $W(S')$, and

(1') $$E(v+w) = E\big(wE(v)^{-1}\big)E(v),$$

(2') $$H(v+w) = H\big(wE(v)^{-1}\big)H(v),$$

(3') $$\begin{aligned}(v+w)F(v+w)^{-1} = \alpha\big(H(v)\big)^{-1} wF\big(wE(v)^{-1}\big)^{-1} \\ + vF(v)^{-1}.\end{aligned}$$

Finally, consider the morphism from M to $\mathbb{G}_{a,S}^2$ defined set-theoretically by

$$(v,w) \longmapsto \big(v, wE(v)^{-1}\big).$$

[13] It preserves the section $(0,0)$ and induces an isomorphism of M onto an open subset N of $\mathbb{G}_{a,S}^2$ containing the zero section; the inverse isomorphism is

$$(x,y) \longmapsto \big(x, yE(x)\big).$$

[14] We have therefore proved the following assertion:

There is an open subset N of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, if $(x,y) \in N(S')$, then x, y, and $x + yE(x)$[15] belong to $W(S')$, and

(1") $$E\big(x+yE(x)\big) = E(x)E(y),$$

(2") $$H\big(x+yE(x)\big) = H(x)H(y),$$

(3") $$\begin{aligned}\big(x+yE(x)\big)F\big(x+yE(x)\big)^{-1} = xF(x)^{-1} \\ + \alpha\big(H(x)\big)^{-1} yE(x)F(y)^{-1}.\end{aligned}$$

C) Arguing in the same way with left associativity, we obtain the following assertion.[16]

There is an open subset N' of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, if $(x,y) \in N'(S')$, then x, y, and $x + yF(x)$[17] belong to $W(S')$, and

(4") $$F\big(x+yF(x)\big) = F(x)F(y),$$

[13] The editors corrected what follows.

[14] Thus the editors made the change of variables $x = v,\ y = wE(v)^{-1}$, or equivalently $v = x,\ w = yE(x)$.

[15] The editors corrected $yE(x)$ to $x + yE(x)$.

[16] That is, one writes the equalities resulting from $p_\alpha(t)p_\alpha(u)p_{-\alpha}(v) = p_\alpha(t+u)p_{-\alpha}(v)$, and sets $v = 1,\ x = u,\ t = yF(u)$, or equivalently $y = tF(u)^{-1}$.

[17] The editors corrected $yF(x)$ to $x + yF(x)$.

(5") $$H\big(x+yF(x)\big) = H(x)H(y),$$

(6") $$\big(x+yF(x)\big)E\big(x+yF(x)\big)^{-1} = xE(x)^{-1} + \alpha\big(H(x)\big)^{-1}yF(x)E(y)^{-1}.$$

We are thus led to solve the "functional equation" $(1'')$.

Lemma 2.3. *Let S be a scheme, let W be an open subset of $\mathbb{G}_{a,S}$ containing the zero section, and let*

$$F\colon W \longrightarrow \mathbb{G}_{m,S}$$

be an S-morphism. Suppose that $F(0) = 1$ and that there is an open subset N of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, for $(x,y) \in N(S')$, the elements x, y, and $x+yF(x)$ belong to $W(S')$, and

(†) $$F\big(x+yF(x)\big) = F(x)F(y).$$

(i) *If $S = \mathrm{Spec}(k)$ for a field k, there is an $a \in k$ such that $F(x) = 1+ax$.*

(ii) *If $a = F'(0) \in \Gamma(S, \mathscr{O}_S)$ is invertible, then $F(x) = 1+ax$.*

Proof. Under the hypotheses, we may differentiate the given equation at $x = 0$, respectively at $y = 0$, and obtain

(∗) $$F'(y)\big(1+yF'(0)\big) = F'(0)F(y) \qquad \text{for } (0,y) \in N(S'),$$

respectively

$$F'(x)F(x) = F(x)F'(0) \qquad \text{for } (x,0) \in N(S').$$

Since F takes its values in $\mathbb{G}_m$, the second relation gives

(∗′) $$F'(x) = F'(0) \qquad \text{for } (x,0) \in N(S').$$

The first relation therefore gives

$$F'(0)\big(1+yF'(0)\big) = F'(0)F(y) \qquad \text{for } (y,0),(0,y) \in N(S').$$

If $a = F'(0)$ is invertible, it follows that

$$F(y) = 1+ay$$

for y a section of an open subset of W containing the zero section, hence schematically dense in W. This proves (ii), and also proves (i) when $F'(0) \neq 0$.

Suppose $F'(0) = 0$. Then (∗′) shows that $F'(x) = 0$ for x near 0, and hence $F' = 0$ by schematic density. If k has characteristic zero, F is a rational function with zero derivative, and is therefore constant and equal to $F(0) = 1$.

Suppose now that k has characteristic p, and that F is not constant.[18] There are an integer $n > 0$ and a rational function $F_1 \in k(X)$ such that $F_1'(X) \neq 0$ and

$$F(X) = F_1\big(X^{p^n}\big) = F_1(X)^{p^n}.$$

[18]The editors corrected what follows.

Substitution in the functional equation gives

$$(\dagger_1) \qquad F_1(x + yF_1(x)^{p^n}) = F_1(x)F_1(y).$$

Differentiating at $x = 0$ gives

$$(*_1) \qquad F_1'(y) = F_1'(0)F_1(y),$$

whereas differentiation of $(\dagger_1)$ at $y = 0$ gives

$$(*_1') \qquad F_1'(x)F_1(x)^{p^n} = F_1(x)F_1'(0).$$

By hypothesis, $F_1'(X)$ is invertible in $k(X)$; these two equalities therefore imply

$$F_1(X)^{p^n} = 1.$$

Thus F_1 is constant, contrary to the choice of F_1. Consequently F is constant and equal to $1 = F(0)$. □

D) Suppose first that S is the spectrum of a field. If $F'(0) = 0$, then $F = 1$. Formula $(5'')$ then gives

$$H(x + y) = H(x)H(y),$$

which shows that H extends to a group morphism

$$\mathbb{G}_{a,S} \longrightarrow T$$

(Exposé XVIII, 2.3), necessarily the constant morphism with value e. On the other hand, Lemma 2.3 also gives

$$E(x) = 1 + bx$$

for some $b \in k$. Formula $(6'')$ therefore gives, for $(x, y) \in N(S')$,

$$(x + y)E(x + y)^{-1} = xE(x)^{-1} + yE(y)^{-1}.$$

Thus,[19] again by Exposé XVIII, 2.3, the map

$$x \longmapsto xE(x)^{-1}$$

extends to a morphism of k-groups $\mathbb{G}_{a,k} \to \mathbb{G}_{a,k}$. Consequently

$$\frac{x}{1 + bx} = cx$$

for some $c \in k$, whence $b = 0$ and $c = 1$.

Thus F, H, E are constant with values $(1, e, 1)$ in a neighborhood of the zero section, and hence everywhere. Formula $(+)$ then shows that U_α and $U_{-\alpha}$ commute, contrary to hypothesis (ii).

For arbitrary S, we have therefore proved that $F'(0)$ is nonzero on every fiber, and hence invertible. The same is plainly true of $E'(0)$. Lemma 2.3 now shows that there are $a, b \in \mathbb{G}_m(S)$ such that

$$(\diamondsuit_1) \qquad F(x) = 1 + ax, \qquad E(x) = 1 + bx \qquad \text{for } x \in W(S').$$

[19] The editors added the sentence that follows. One may also see by direct calculation that the preceding equality implies $0 = xyb\big(2 + (x + y)b\big)$, hence $0 = b\big(2 + (x + y)b\big)$, and finally $b = 0$.

E) The rest is now easy. Substitution of the preceding results in $(3'')$ gives

$$y\,\alpha\big(H(x)\big)(1+ay) = y\big(1+ax+ay(1+bx)\big)(1+ax).$$

This formula holds for every section (x,y) of N. Since $\mathbb{G}_{a,S}\times_S\mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}^2$, it follows that

$$(1+ay)\alpha\big(H(x)\big) = \big(1+ax+ay(1+bx)\big)(1+ax).$$

Setting $y=0$ gives

$$\alpha\big(H(x)\big) = (1+ax)^2.$$

Substitution in the preceding equality gives[20]

$$a^2xy = abxy.$$

Since $\mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}$, we obtain $a^2=ab$. As a is invertible,

$$(\diamondsuit_2) \qquad a=b, \qquad \alpha\big(H(x)\big) = (1+ax)^2.$$

Because a is invertible, the map $x\mapsto 1+ax$ is an automorphism of $\mathbb{G}_{a,S}$. There are therefore an open subset W' of $\mathbb{G}_{a,S}$, containing the section 1, and a morphism

$$P\colon W' \longrightarrow T$$

such that

$$P(1+ax) = H(x).$$

[21] Substitution in $(2')$ gives immediately, for $(x,y)\in N(S')$,

$$P(1+ax+ay) = P\left(\frac{1+ax+ay}{1+ax}\right)P(1+ax).$$

Thus there is an open neighborhood of 1 in $\mathbb{G}_{m,S}$ on which

$$P(x)P(y) = P(xy).$$

By Exposé XVIII, 2.3, there is a group morphism

$$(\diamondsuit_3) \qquad \alpha^*\colon \mathbb{G}_{m,S} \longrightarrow T$$

extending P. Since $\alpha(H(x)) = (1+ax)^2$ near the zero section, we have $\alpha(\alpha^*(z)) = z^2$ near the section 1, and hence

$$(\diamondsuit_4) \qquad \alpha\circ\alpha^* = 2.$$

F)[22] Combining $(+)$ and $(\diamondsuit_1)$–$(\diamondsuit_4)$, we see that there are

$$a\in\mathbb{G}_m(S), \qquad \alpha^*\in \mathrm{Hom}_{S\text{-gr.}}(\mathbb{G}_{m,S},T)$$

[20] In the three preceding equalities, the editors corrected the original by replacing the factor $1+bx$ on the right by $1+ax$. Substitution of the third equality in the second, and use of the invertibility of $1+ax$, gives precisely $a^2xy=abxy$.

[21] This is the change of variables $x'=1+ax$, or $x=(x'-1)/a$.

[22] The editors slightly modified what follows, since an open subset V had already been introduced in part A.

such that $\alpha \circ \alpha^* = 2$, and such that, whenever $(u, v) \in V(S')$ and $uv \in W(S')$, the element $1 + auv$ is invertible and

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\left(\frac{v}{1+auv}\right)\alpha^*(1+auv)p_\alpha\left(\frac{u}{1+auv}\right).$$

Let V' be the open subset of $\mathbb{G}_{a,S}^2$ defined by the invertibility of $1 + auv$; thus

$$V' = \left(\mathbb{G}_{a,S}^2\right)_f, \qquad f(u,v) = 1 + auv.$$

The two sides of the preceding formula define morphisms $V' \to G$ which agree in a neighborhood of the zero section, and hence agree on V'. Thus the formula holds for every section (u, v) of V', and in particular $V' \subset U$, where U is the open subset introduced at the beginning of part A.

We prove that $U = V'$. Returning to the notation of part A, there is a morphism

$$A\colon U \longrightarrow \mathbb{G}_{a,S}$$

whose restriction to V' is

$$A(u,v) = v(1+auv)^{-1}.$$

The equality $U = V'$ is a set-theoretic question and may therefore be checked when $S = \mathrm{Spec}(k)$ for a field k. It then reduces to the evident assertion that the domain of definition of the rational map

$$\mathbb{G}_{a,k}^2 \dashrightarrow \mathbb{G}_{a,k}, \qquad (X,Y) \longmapsto \frac{Y}{1+aXY},$$

is the open subset defined by $1 + aXY$.

G) We have proved the existence of a and α^*, as well as the two asserted supplementary properties. It remains to prove uniqueness. Let a' and $\alpha^{*\prime}$ also satisfy the required conditions. For $u, v \in \mathbb{G}_a(S')$, we immediately have

$$1 + auv \text{ invertible} \implies 1 + a'uv \text{ invertible},$$
$$\frac{v}{1+auv} = \frac{v}{1+a'uv}.$$

Hence, for every $u \in \mathbb{G}_a(S')$,

$$1 + au \text{ invertible} \implies 1 + au = 1 + a'u,$$

which proves $a = a'$. With the same notation,

$$1 + au \text{ invertible} \implies \alpha^*(1+au) = \alpha^{*\prime}(1+au),$$

and therefore $\alpha^* = \alpha^{*\prime}$. □

Corollary 2.4. *Let*

$$\exp(Y)t\exp(X) \quad \text{and} \quad \exp(Y')t'\exp(X')$$

be two elements of $\Omega(S')$. *Their product belongs to* $\Omega(S')$ *if and only if*

$$u = 1 + \langle X, Y' \rangle$$

is invertible; in that case,

(F') $$\begin{aligned}&\exp(Y)t\exp(X)\cdot\exp(Y')t'\exp(X')\\ &\quad= \exp\bigl(Y + u^{-1}\alpha(t)^{-1}Y'\bigr)\,tt'\alpha^*(u)\,\exp\bigl(u^{-1}\alpha(t')^{-1}X + X'\bigr).\end{aligned}$$

Remark 2.5. *Formula* (F) *of Theorem 2.1 may also be written without using the morphisms* exp. *Transporting the duality* $\mathfrak{g}_\alpha \otimes \mathfrak{g}_{-\alpha} \to \mathscr{O}_S$ *through those morphisms gives a canonical pairing of vector bundles*

$$U_\alpha \times_S U_{-\alpha} \longrightarrow \mathbb{G}_{a,S},$$

which we continue to denote by $(x, y) \mapsto \langle x, y\rangle$. *Thus*

$$\langle \exp X, \exp Y\rangle = \langle X, Y\rangle.$$

If $x \in U_\alpha(S')$, $y \in U_{-\alpha}(S')$, *and* $1 + \langle x, y\rangle \in \mathbb{G}_m(S')$, *then*

$$x \cdot y = y^{(1+\langle x,y\rangle)^{-1}}\, \alpha^*\big(1 + \langle x, y\rangle\big)\, x^{(1+\langle x,y\rangle)^{-1}}.$$

Corollary 2.6. *The pairing*

$$W(\mathfrak{g}_\alpha) \times_S W(\mathfrak{g}_{-\alpha}) \longrightarrow \mathbb{G}_{a,S}$$

induces a pairing of principal $\mathbb{G}_{m,S}$*-bundles*

$$W(\mathfrak{g}_\alpha)^\times \times_S W(\mathfrak{g}_{-\alpha})^\times \longrightarrow \mathbb{G}_{m,S}.$$

We denote this pairing by $(X, Y) \mapsto \langle X, Y\rangle$, *or simply* $(X, Y) \mapsto XY$.

For every $X \in \Gamma(S, \mathfrak{g}_\alpha)^\times$, there is therefore a unique

$$X^{-1} \in \Gamma(S, \mathfrak{g}_{-\alpha})^\times$$

such that $XX^{-1} = 1$. Moreover,

$$(zX)^{-1} = z^{-1}X^{-1}.$$

The morphism

$$s\colon W(\mathfrak{g}_\alpha)^\times \longrightarrow W(\mathfrak{g}_{-\alpha})^\times$$

so defined is an isomorphism of schemes, compatible with the isomorphism $s\colon z \mapsto z^{-1}$ of their operator groups.

Definition 2.6.1. *The elements* X *and* $s(X) = X^{-1}$ *are said to be paired.*

Apply Corollary 2.4 with $Y = 0 = X'$ and

$$Y' = aX^{-1}, \qquad a \in \mathscr{O}_S(S).$$

Then $u = 1 + a$, and

$$u^{-1}Y' = u^{-1}(u-1)X^{-1} = (1 - u^{-1})X^{-1}.$$

Hence:

Corollary 2.7. *Let* $X \in \Gamma(S, \mathfrak{g}_\alpha)^\times$ *and* $u \in \Gamma(S, \mathscr{O}_S)^\times$. *Then*

$$\alpha^*(u) = \exp\big((u^{-1} - 1)X^{-1}\big)\, \exp(X)\, \exp\big((u-1)X^{-1}\big)\, \exp(-u^{-1}X).$$

Definition 2.8. *The morphism* α^* *is called the coroot associated with the root* α.

Remark 2.9. *If (G,T,α) is an S-elementary system, then so is $(G,T,-\alpha)$. Theorem 2.1 therefore gives a duality between $\mathfrak{g}_{-\alpha}$ and $\mathfrak{g}_\alpha$, and a coroot $(-\alpha)^*$. Taking the inverse of formula* (F) *gives immediately*

$$\langle X,Y\rangle = \langle Y,X\rangle, \qquad (-\alpha)^* = -\alpha^*.$$

We now pass to the Lie algebra of G. The root α and the coroot α^* define linear forms

$$\mathscr{O}_S \xrightarrow{\overline{\alpha^*}} \mathfrak{t} \xrightarrow{\bar\alpha} \mathscr{O}_S.$$

Put

$$H_\alpha = \overline{\alpha^*}(1).$$

We call $\bar\alpha$ the infinitesimal root associated with α, and H_α the corresponding infinitesimal coroot.

Lemma 2.10. *Let $S' \to S$, and let*

$$X, X' \in W(\mathfrak{g}_\alpha)(S'), \qquad H \in W(\mathfrak{t})(S'), \qquad Y, Y' \in W(\mathfrak{g}_{-\alpha})(S'), \qquad t \in T(S').$$

Then

(1) $$\mathrm{Ad}(t)H = H, \qquad \mathrm{Ad}(t)X = \alpha(t)X, \qquad \mathrm{Ad}(t)Y = \alpha(t)^{-1}Y;$$

(2) $$\begin{aligned} &\mathrm{Ad}(\exp X)H = H - \bar\alpha(H)X, && \mathrm{Ad}(\exp X)X' = X', \\ &\mathrm{Ad}(\exp X)Y = Y + \langle X,Y\rangle H_\alpha - \langle X,Y\rangle X; \end{aligned}$$

(2') $$\begin{aligned} &\mathrm{Ad}(\exp Y)H = H + \bar\alpha(H)Y, && \mathrm{Ad}(\exp Y)Y' = Y', \\ &\mathrm{Ad}(\exp Y)X = X + \langle X,Y\rangle H_{-\alpha} - \langle X,Y\rangle Y; \end{aligned}$$

(3) $$\begin{aligned} [H,X] = \bar\alpha(H)X, \qquad [H,Y] = -\bar\alpha(H)Y, \\ [X,Y] = \langle X,Y\rangle H_\alpha; \end{aligned}$$

(4) $$H_{-\alpha} = -H_\alpha,$$

(5) $$\bar\alpha(H_\alpha) = 2.$$

The proof of these formulas is either immediate or an immediate consequence of formula (F) of 2.1.

Corollary 2.11. *Suppose that H_α is nonzero on every fiber; by* (5)*, this holds in particular if* 2 *is invertible on S. Then*

$$X_\alpha \in \Gamma(S,\mathfrak{g}_\alpha)^\times, \qquad X_{-\alpha} \in \Gamma(S,\mathfrak{g}_{-\alpha})^\times$$

are paired if and only if

$$[X_\alpha, X_{-\alpha}] = H_\alpha.$$

2.12. Let (G, T, α) be an S-elementary system. By 1.19,

$$\mathrm{Centr}(G) = \ker(\alpha)$$

is a group of multiplicative type and finite type. If Q is a subgroup of multiplicative type of $\mathrm{Centr}(G)$, then G/Q is affine over S (Exposé IX, 2.5), smooth over S (Exposé $\mathrm{VI_B}$, 9.2), and has connected reductive fibers of semisimple rank 1 (Exposé XIX, 1.8).

Put $G' = G/Q$. It is a reductive S-group of semisimple rank 1, and $T' = T/Q$ is a maximal torus of G'. The open subset

$$U_{-\alpha} \cdot T \cdot U_\alpha$$

of G is stable under Q, and its quotient is isomorphic to

$$U_{-\alpha} \times_S (T/Q) \times_S U_\alpha.$$

If α' denotes the character of T' induced by α, the canonical morphism $G \to G'$ induces isomorphisms

$$\mathfrak{g}_\alpha \xrightarrow{\sim} \mathfrak{g}'_{\alpha'}, \qquad \mathfrak{g}_{-\alpha} \xrightarrow{\sim} \mathfrak{g}'_{-\alpha'}.$$

In particular, α' is a root of G' relative to T'. Writing α/Q for the character

$$T/Q \longrightarrow \mathbb{G}_{m,S}$$

induced by α, we obtain:

Lemma 2.13. *If Q is a subgroup of multiplicative type of* $\ker(\alpha)$*, then*

$$(G/Q, T/Q, \alpha/Q)$$

is an elementary system.

Lemma 2.14. *Under the preceding conditions, the following diagrams are commutative:*

$$\begin{array}{ccccc}
\mathrm{W}(\mathfrak{g}^\alpha) & \xrightarrow{\exp} & G & \xleftarrow[\exp]{} & \mathrm{W}(\mathfrak{g}^{-\alpha}) \\
{\scriptstyle \mathrm{can}}\downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \mathrm{can}} & & {\scriptstyle \mathrm{can}}\downarrow{\scriptstyle \sim} \\
\mathrm{W}(\mathfrak{g}'^{\alpha'}) & \xrightarrow{\exp} & G' & \xleftarrow[\exp]{} & \mathrm{W}(\mathfrak{g}'^{-\alpha'})
\end{array}$$

$$\begin{array}{ccc}
\mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} & \xrightarrow{\sim} & \mathcal{O}_S \\
{\scriptstyle \mathrm{can}}\downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \mathrm{id}} \\
\mathfrak{g}'^{\alpha'} \otimes \mathfrak{g}'^{-\alpha'} & \xrightarrow{\sim} & \mathcal{O}_S
\end{array}$$

$$\begin{array}{ccccc}
 & & T & & \\
 & {\scriptstyle \alpha^*}\nearrow & \downarrow{\scriptstyle \mathrm{can}} & \searrow{\scriptstyle \alpha} & \\
\mathbb{G}_{m,S} & & & & \mathbb{G}_{m,S} \\
 & {\scriptstyle \alpha'^*}\searrow & & \nearrow{\scriptstyle \alpha'} & \\
 & & T' & &
\end{array}$$

0.3 The Weyl group

Notations 3.0. [23] If (G, T, α) is an S-elementary system, write

$$N = \mathrm{Norm}_G(T), \qquad W = \mathrm{Norm}_G(T)/T$$

(cf. Exposé XIX, 6.3). The group N is a closed subgroup of G, smooth over S. Write

$$N^\times = N - T$$

for the open subscheme of N induced on the complement of T.[24]

Let R be the unique maximal torus of $\mathrm{Ker}(\alpha)$, and let T' be the image of

$$\alpha^* \colon \mathbb{G}_{m,S} \longrightarrow T;$$

it is a one-dimensional subtorus of T. The morphism

$$T' \times_S R \longrightarrow T$$

induced by multiplication in T is surjective, hence faithfully flat. Indeed, it is enough to check this on the geometric fibers, where it follows at once from the formula

$$\alpha \circ \alpha^* = 2.$$

Theorem 3.1. *With the preceding notation:*

(i) *W is isomorphic to the constant group $(\mathbb{Z}/2\mathbb{Z})_S$.*

(ii) *$N^\times$ is a locally trivial principal homogeneous bundle under T, on the left by the law $(t, q) \mapsto tq$, and on the right by the law $(q, t) \mapsto qt$.*

(iii) *One has*

$$\mathrm{int}(w)t = t \cdot \alpha^*(\alpha(t)^{-1})$$

for $w \in N^\times(S')$, $t \in T(S')$, and $S' \to S$. In the decomposition $T_{S'} = T'_{S'} \cdot R_{S'}$, $\mathrm{int}(w)$ induces the identity on $R_{S'}$ and inversion on $T'_{S'}$. One has the relations

$$\alpha \circ \mathrm{int}(w) = \alpha^{-1}, \qquad \mathrm{int}(w) \circ \alpha^* = (\alpha^*)^{-1}.$$

(iv) *For $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \to S$, set*

$$w_\alpha(X) = \exp(X)\exp(-X^{-1})\exp(X).$$

Then $w_\alpha(X) \in N^\times(S')$, and the morphism $w_\alpha \colon W(\mathfrak{g}^\alpha)^\times \to N^\times$ so defined satisfies

$$w_\alpha(zX) = \alpha^*(z)w_\alpha(X) = w_\alpha(X)\alpha^*(z)^{-1}$$

for $z \in \mathbb{G}_m(S')$, $X \in W(\mathfrak{g}^\alpha)^\times(S')$, and $S' \to S$.

(v) *For $X, Y \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \to S$, one has, with the notation of 2.6,*

$$w_\alpha(X)w_\alpha(Y) = \alpha^\vee(-XY^{-1}).$$

[25] *In particular,*

$$w_\alpha(X)^2 = \alpha^*(-1) \in {}_2T(S) \cap \mathrm{Centr}(G)(S),$$

[23] The editors added the number 3.0 for later references.

[24] The editors replaced Q by the more suggestive notation $N^\times$.

[25] The editors corrected the first formula of the original.

$$w_\alpha(X)^{-1} = w_\alpha(-X) = \alpha^*(-1)w_\alpha(X).$$

(vi) *If, similarly, one defines for* $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$

$$w_{-\alpha}(Y) = \exp(Y)\exp(-Y^{-1})\exp(Y),$$

then, in addition to the formulas analogous to the preceding ones,

$$w_{-\alpha}(X^{-1}) = w_\alpha(X)^{-1} = w_\alpha(-X), \qquad w_\alpha(X)w_{-\alpha}(Y) = \alpha^*(XY).$$

Proof. Assertion (i) was already proved in Exposé XIX, 2.4. It follows at once that $N^\times$ is indeed a principal homogeneous bundle under T for the laws defined in (ii); its local triviality for the Zariski topology follows in particular from (iv).

Let us prove (iii). If $w \in N^\times(S)$, it is clear that $\alpha \circ \mathrm{int}(w)$ is a root of G relative to T, and hence is locally equal to α or $-\alpha$. Since it is $-\alpha$ on every fiber (Bible, 12-05, proof of the corollary to Proposition 1), one has

$$\alpha \circ \mathrm{int}(w) = -\alpha.$$

By transport of structure, it follows that

$$-\alpha^* = \mathrm{int}(w)^{-1} \circ \alpha^* = \mathrm{int}(w) \circ \alpha^*,$$

since $\mathrm{int}(w)^2 = \mathrm{int}(w^2)$ and w^2 is a section of T. Thus $\mathrm{int}(w)$ induces inversion on T'; since R is central, it induces the identity on R. The formula in (iii) defines a morphism $T \to T$ having the same properties, and therefore coincides with $\mathrm{int}(w)$.

Let us prove (iv). Successively,

$$\begin{aligned} w_\alpha(X)tw_\alpha(X)^{-1} &= \exp(X)\exp(-X^{-1})\exp(X)t\exp(-X)\exp(X^{-1})\exp(-X) \\ &= \exp(X)\exp(-X^{-1})\exp\big(X - \alpha(t)X\big)\exp\big(\alpha(t)^{-1}X^{-1}\big)\exp\big(-\alpha(t)X\big)t. \end{aligned}$$

Applying formula (F) gives

$$\begin{aligned} \exp(-X^{-1})\exp\big((1-\alpha(t))X\big) = {} & \exp\big((\alpha(t)^{-1}-1)X\big)\alpha^*\big(\alpha(t)^{-1}\big) \\ & \cdot \exp\big(-\alpha(t)^{-1}X^{-1}\big). \end{aligned}$$

Substitution in the preceding relation yields

$$\begin{aligned} \mathrm{int}(w_\alpha(X))t &= \exp\big(\alpha(t)^{-1}X\big)\alpha^*\big(\alpha(t)^{-1}\big)\exp\big(-\alpha(t)X\big)t \\ &= \exp(aX)\alpha^*\big(\alpha(t)^{-1}\big)t, \end{aligned}$$

where

$$a = \alpha(t)^{-1} - (\alpha \circ \alpha^*)\big(\alpha(t)^{-1}\big)\alpha(t).$$

But $\alpha \circ \alpha^* = 2$, which gives $a = 0$ at once and shows that $w_\alpha(X) \in N^\times(S')$.

We now prove the second assertion of (iv). One has[26]

$$\begin{aligned} \alpha^*(z)w_\alpha(X) &= \exp(z^2X)\exp(-z^{-2}X^{-1})\exp(z^2X)\alpha^*(z) \\ &= \exp(zX)\exp\big((z^2-z)X\big)\exp(-z^{-2}X^{-1})\exp(z^2X)\alpha^*(z) \\ &= \exp(zX)\exp(-z^{-1}X^{-1})\alpha^*(z)^{-1}\exp\big((z^3-z^2)X\big)\exp(z^2X)\alpha^*(z) \\ &= \exp(zX)\exp(-z^{-1}X^{-1})\exp(zX) = w_\alpha(zX). \end{aligned}$$

[26]The first equality follows from 1.5 (i), which, together with $\alpha \circ \alpha^* = 2$, gives $\alpha^*(z)\exp(X)\alpha^*(z)^{-1} = \exp(z^2X)$ and $\alpha^*(z)\exp(X^{-1})\alpha^*(z)^{-1} = \exp(z^{-2}X)$. The third equality follows from formula (F), and the fourth again from these formulas. Finally, an analogous calculation gives $w_\alpha(X)\alpha^*(z^{-1}) = w_\alpha(zX)$.

Let us prove (v). By the preceding result, the first formula in (v) follows at once from the second; we prove the latter:

$$\begin{aligned} w_\alpha(X)^2 &= \exp(X)\exp(-X^{-1})\exp(2X)\exp(-X^{-1})\exp(X) \\ &= \exp(X)\exp(-X^{-1})\exp(X^{-1})\alpha^*(-1)\exp(-2X)\exp(X) \\ &= \exp(X)\alpha^*(-1)\exp(-X) = \alpha^*(-1), \end{aligned}$$

since

$$\alpha(\alpha^*(-1)) = (-1)^2 = 1.$$

This also proves that $\alpha^*(-1) \in \mathrm{Centr}(G)(S)$.

Finally, let us prove (vi). The first assertion is a special case of the second, so we prove the latter. Its two sides define morphisms

$$W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times \longrightarrow G.$$

To prove that they coincide, it suffices to do so on a nonempty open subset of every fiber (Exposé XVIII, 1.4). We may therefore suppose that $1 + XY$ is invertible. Then, successively,

$$\begin{aligned} w_\alpha(X)w_{-\alpha}(Y) &= \exp(X)\exp(-X^{-1})\exp(X)\exp(Y) \\ &\qquad \exp(-Y^{-1})\exp(Y) \\ &= \exp(X)\exp(-X^{-1})\exp\left(\frac{Y}{1+XY}\right)\alpha^*(1+XY) \\ &\qquad \exp\left(\frac{X}{1+XY}\right)\exp(-Y^{-1})\exp(Y) \\ &= \exp(X)\exp\left(\frac{-X^{-1}}{1+XY}\right)\alpha^*(1+XY)\exp\left(\frac{-Y^{-1}}{1+XY}\right)\exp(Y) \\ &= \exp(-X^{-2}Y^{-1})\alpha^*\left(\frac{XY}{1+XY}\right)\exp(X+Y^{-1})\alpha^*(1+XY) \\ &\qquad \cdot \exp\left(\frac{-Y^{-1}}{1+XY}\right)\exp(Y) \\ &= \exp(-X^{-2}Y^{-1})\alpha^*(XY)\exp\left(\frac{Y^{-1}+X}{(1+XY)^2}\right) \\ &\qquad \exp\left(\frac{-Y^{-1}}{1+XY}\right)\exp(Y) \\ &= \alpha^*(XY)\exp(-Y)\exp(Y) = \alpha^*(XY). \end{aligned}$$

Corollary 3.2. *Let $n \in \mathbb{Z}$, $n \neq 0$. For every $w \in G(S)$, the following conditions are equivalent:*

(i) *$w \in N^\times(S)$;*

(ii) *$\mathrm{int}(w) \circ n\alpha^* = -n\alpha^*$, where $(n\alpha^*)(z) = \alpha^*(z)^n$.*

One has (i)⇒(ii) by Theorem 3.1 (iii). Conversely, one may suppose that $N^\times$ has a section, and one is reduced to proving:

Lemma 3.3. *For $n \neq 0$, one has*

$$\mathrm{Centr}_G(n\alpha^*) = T.$$

Indeed, the image T' of $n\alpha^*$ is a subtorus of G. It follows from Exposé XIX, 2.8 that $\mathrm{Centr}_G(n\alpha^*)$ is a reductive subgroup of G containing T. On every fiber,

$$\mathrm{Centr}_{G_s}(n\alpha_s^*) \neq G_s,$$

and hence

$$\mathrm{Centr}_{G_s}(n\alpha_s^*) = T_s$$

by Exposé XIX, 1.6.3.[27] Therefore

$$\mathrm{Centr}_G(n\alpha^*) = T,$$

since these are smooth subgroups of G.

Remark 3.4. *The construction of w_α, and the fact that $w_\alpha(X)$ normalizes T, depend only on formula* (F)*. In particular, if G is an S-group satisfying the conditions of 2.2, then* $\mathrm{Norm}_G(T)$ *differs from T on every fiber. Consequently, if G is an affine S-group with connected fibers satisfying the conditions of 2.2, then G is reductive of semisimple rank* 1*. Indeed, it is smooth in a neighborhood of the identity section, hence smooth, and the criterion of Exposé XIX, 1.11 applies.*

3.5. Before stating the following theorem, let us make a few remarks. As usual, we identify $\mathfrak{g}^{-\alpha}$ with $(\mathfrak{g}^\alpha)^{\otimes -1}$. Similarly, we identify

$$\mathrm{Hom}_{\mathscr{O}_S}(\mathfrak{g}^{-\alpha}, \mathfrak{g}^\alpha) \quad \text{with} \quad (\mathfrak{g}^\alpha)^{\otimes 2},$$

and hence

$$\mathrm{Isom}_{\mathscr{O}_S\text{-mod.}}\big(W(\mathfrak{g}^{-\alpha}), W(\mathfrak{g}^\alpha)\big) \simeq W\big((\mathfrak{g}^\alpha)^{\otimes 2}\big)^\times.$$

If $w \in N^\times(S)$, then $\mathrm{Ad}(w)$ permutes $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ (3.1 (iii)), and therefore defines an isomorphism

$$a_\alpha(w)\colon \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}^\alpha.$$

We thus identify it with a section

$$a_\alpha(w) \in \Gamma\big(S, (\mathfrak{g}^\alpha)^{\otimes 2}\big)^\times.$$

This construction is compatible with base change and therefore defines a morphism

$$a_\alpha\colon N^\times \longrightarrow W\big((\mathfrak{g}^\alpha)^{\otimes 2}\big)^\times$$

such that

$$a_\alpha(w)Y = \mathrm{Ad}(w)Y$$

for every $w \in N^\times(S')$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})^\times$, and $S' \to S$.

[27] The hypothesis $\mathrm{Centr}_{G_s}(n\alpha_s^*) \neq G_s$ implies

$$\dim \mathrm{Centr}_{G_s}(n\alpha_s^*) - \dim T_s < 2;$$

this difference is even by the cited result.

Theorem 3.6. (i) *For every $S' \to S$, $w \in N^\times(S')$, and $Y \in W(\mathfrak{g}^{-\alpha})(S')$, one has*

$$\operatorname{int}(w)\exp(Y) = \exp\bigl(a_\alpha(w)Y\bigr).$$

(ii) *One has*

$$a_\alpha(tw) = \alpha(t)a_\alpha(w), \qquad a_\alpha(wt) = \alpha(t)^{-1}a_\alpha(w).$$

(iii) *If one similarly defines*

$$a_{-\alpha}\colon N^\times \longrightarrow W\bigl((\mathfrak{g}^{-\alpha})^{\otimes 2}\bigr)^\times,$$

then

$$a_{-\alpha}(w) = a_\alpha(w)^{-1}.$$

[28]

(iv) *For every $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \to S$, one has*

$$a_\alpha(w_\alpha(X)) = -X^2.$$

Proof. Assertion (i) is immediate from the characterization of the morphisms exp given in 1.5. Assertion (ii) is immediate, as is (iii). Let us prove (iv). Let

$$X \in \Gamma(S', \mathfrak{g}^\alpha)^\times, \qquad Z \in \Gamma(S', \mathfrak{g}^\alpha).$$

By definition,[29]

$$\begin{aligned} a_\alpha(w_\alpha(X))^{-1}(Z) &= \operatorname{Ad}(w_\alpha(X))(Z) \\ &= \operatorname{Ad}(\exp(X))\operatorname{Ad}(\exp(-X^{-1}))\operatorname{Ad}(\exp(X))(Z). \end{aligned}$$

Applying formulas (2′) and (2) of Lemma 2.10, together with

$$H_{-\alpha} = -H_\alpha, \qquad \bar{\alpha}(H_\alpha) = 2$$

(loc. cit., (4) and (5)), and $\langle X, X^{-1}\rangle = 1$ (2.6), the expression on the right is, successively,

$$\begin{aligned} &\operatorname{Ad}(\exp(X))\operatorname{Ad}(\exp(-X^{-1}))(Z) \\ &\quad= \operatorname{Ad}(\exp(X))(Z + \langle X^{-1}, Z\rangle(H_\alpha - X^{-1})) \\ &\quad= Z + \langle X^{-1}, Z\rangle(H_\alpha - 2X - X^{-1} - H_\alpha + X) \\ &\quad= Z - \langle X^{-1}, Z\rangle X - \langle X^{-1}, Z\rangle X^{-1}. \end{aligned}$$

But

$$Z = \langle X^{-1}, Z\rangle X, \qquad \langle X^{-1}, Z\rangle X^{-1} = X^{-2}Z.$$

Thus

$$a_\alpha(w_\alpha(X))^{-1} = -X^{-2},$$

and consequently

$$a_\alpha(w_\alpha(X)) = -X^2.$$

[28] That is, $a_{-\alpha}(w)$ and $a_\alpha(w)$ are paired; cf. 2.6.1.

[29] The editors supplied the details that follow.

Corollary 3.7. *In particular, one has*

$$\operatorname{int}\bigl(w_\alpha(X)\bigr)\exp(X) = \exp(-X^{-1}).$$

Hence, by the definition of $w_\alpha(X)$,

$$w_\alpha(X)\exp(X)w_\alpha(X)^{-1} = \exp(-X)w_\alpha(X)\exp(-X),$$

and therefore, by an immediate calculation,

$$\bigl(w_\alpha(X)\exp(X)\bigr)^3 = e.$$

Corollary 3.8. *Let* $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, *and let* $n \in \mathbb{Z}$, $n \neq 0$. *Then* $w_\alpha(X)$ *is the unique section* $w \in G(S)$ *satisfying*

(i) $\operatorname{int}(w) \circ n\alpha^* = -n\alpha^*$;

(ii) $\bigl(w\exp(X)\bigr)^3 = e$.

We know that $w_\alpha(X)$ satisfies these conditions. Conversely, let $w \in G(S)$ satisfy (i) and (ii). By 3.2 and 3.1 (ii), there exists $t \in T(S)$ such that

$$w = w_\alpha(X)t.$$

Set $u = \exp(X)$. Then

$$\begin{aligned} wuw^{-1} &= w_\alpha(X)t\exp(X)t^{-1}w_\alpha(X)^{-1} \\ &= \exp\bigl(-\alpha(t)X^{-1}\bigr), \end{aligned}$$

whereas

$$\begin{aligned} u^{-1}wu^{-1} &= \exp(-X)w_\alpha(X)t\exp(-X) \\ &= \exp(-X)w_\alpha(X)\exp(-X)\exp\bigl(X - \alpha(t)X\bigr)t \\ &= \exp(-X^{-1})\exp\bigl(X - \alpha(t)X\bigr)t \\ &= \exp(-X^{-1})t\exp(H). \end{aligned}$$

Now

$$(wu)^3 = e \quad\Longleftrightarrow\quad wuw^{-1} = u^{-1}wu^{-1}.$$

Comparing the two decompositions of this element in

$$U_{-\alpha} \cdot T \cdot U_\alpha$$

gives $t = e$.

Remark 3.9. We may summarize a number of the results of this section by the following diagram of principal homogeneous bundles (on the left):

$$\begin{array}{ccccc} \mathrm{W}(\mathfrak{g}^\alpha)^\times & \xrightarrow{\quad w_\alpha \quad} & N^\times & \xrightarrow{\quad a_\alpha \quad} & \mathrm{W}\bigl((\mathfrak{g}^\alpha)^{\otimes 2}\bigr)^\times \\ \mathbb{G}_{m,S} & \xrightarrow{\quad \alpha^* \quad} & T & \xrightarrow{\quad \alpha \quad} & \mathbb{G}_{m,S} \end{array}$$

Notice that a_α is faithfully flat (since α is), and that w_α is a monomorphism if and only if α^* is a monomorphism. We leave to the reader the task of writing down the corresponding diagrams for right principal-bundle structures, analogous diagrams for the root $-\alpha$, and of studying the relations among these various diagrams.

Lemma 3.10. *Let S be a scheme, and let $q > 0$ be an integer such that $x \mapsto x^q$ defines an endomorphism of $\mathbb{G}_{a,S}$. Let (G, T, α) and (G', T', α') be two elementary S-systems, and let $f\colon G \to G'$ be a morphism of S-groups. Let*

$$h\colon (\mathfrak{g}^{\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

be an isomorphism of $\mathcal{O}_S$-modules, and let

$$h^{\vee}\colon (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}$$

be the contragredient isomorphism. Suppose that, for every $S' \to S$ and every $X \in W(\mathfrak{g}^{\alpha})(S')$,

$$f\big(\exp(X)\big) = \exp\big(h(X^q)\big).$$

Then the following conditions are equivalent:

(i) $f\big(\alpha^*(z)\big) = \alpha'^*(z)^q$.

(ii) $f\big(w_{\alpha}(Z)\big) = w_{\alpha'}\big(h(Z^q)\big)$.

(iii) $f\big(\exp(Y)\big) = \exp\big(h^{\vee}(Y^q)\big)$.

Each condition is to be read as holding for every $S' \to S$ and every

$$z \in \mathbb{G}_m(S'), \qquad Z \in W(\mathfrak{g}^{\alpha})^{\times}(S'), \qquad Y \in W(\mathfrak{g}^{-\alpha})(S'),$$

respectively. Indeed, (i) implies (ii) by 3.8, (ii) implies (iii) by 3.7, and (iii) implies (i) by 2.7.

Proposition 3.11. *Let S be a scheme, and let $q \in \mathbb{Z}$, $q > 0$, be such that $x \mapsto x^q$ defines an endomorphism of $\mathbb{G}_{a,S}$. Let (G, T, α) and (G', T', α') be two elementary S-systems, and let $f\colon G \to G'$ be a morphism of S-groups. The following conditions on f are equivalent:*

(i) *The restriction of f to T factors through a morphism $f_T\colon T \to T'$ making the diagram*

$$\begin{array}{ccccc}
\mathbb{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbb{G}_{m,S} \\
\Big\downarrow{\scriptstyle q} & & \Big\downarrow{\scriptstyle f_T} & & \Big\downarrow{\scriptstyle q} \\
\mathbb{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbb{G}_{m,S}
\end{array}$$

commutative.

(ii) *There exists a unique isomorphism of $\mathcal{O}_S$-modules*

$$h\colon (\mathfrak{g}^{\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

such that

$$f\big(\exp(X)\big) = \exp\big(h(X^q)\big), \qquad f\big(\exp(Y)\big) = \exp\big(h^{\vee}(Y^q)\big)$$

for every $S' \to S$, $X \in W(\mathfrak{g}^{\alpha})(S')$, and $Y \in W(\mathfrak{g}^{-\alpha})(S')$. Consequently, f also satisfies the equivalent conditions of 3.10.

We first prove that (ii) implies (i). By 3.10, condition (ii) gives

$$f \circ \alpha^* = q\alpha'^*;$$

hence, by 3.3, $f|_T$ factors through T'. It remains to prove

$$\alpha'\big(f(t)\big) = \alpha(t)^q,$$

which follows at once from the fact that f induces a morphism of groups

$$T \cdot U_\alpha \longrightarrow T' \cdot U_{\alpha'}.$$

We now prove that (i) implies (ii). Let

$$X \in \Gamma(S, \mathfrak{g}^\alpha), \qquad Y \in \Gamma(S, \mathfrak{g}^{-\alpha}),$$

and set

$$p_+(x) = f\big(\exp(xX)\big), \qquad p_-(y) = f\big(\exp(yY)\big).$$

These are morphisms of groups

$$p_+, p_- \colon \mathbb{G}_{a,S} \longrightarrow G.$$

Now

$$\begin{aligned} \mathrm{int}\big(\alpha'^*(z)\big)^q\big(p_+(x)\big) &= \mathrm{int}\big(f_T(\alpha^*(z))\big)\big(f(\exp(xX))\big) \\ &= f\Big(\mathrm{int}\big(\alpha^*(z)\big)\big(\exp(xX)\big)\Big) \\ &= f\big(\exp(z^2 xX)\big) = p_+(z^2 x). \end{aligned}$$

Applying Lemma 1.2 with $Q = \alpha'^*(\mathbb{G}_{m,S})$, we obtain a section

$$X' \in \Gamma(S, \mathfrak{g}'^{\alpha'})$$

such that

$$f\big(\exp(xX)\big) = p_+(x) = \exp(x^q X').$$

Similarly, there exists a section

$$Y' \in \Gamma(S, \mathfrak{g}'^{-\alpha'})$$

such that

$$f\big(\exp(yY)\big) = \exp(y^q Y').$$

Since f is a group morphism, it respects formula (F); hence

$$X^q Y^q = (XY)^q = X'Y'.$$

It follows readily that $X^q \mapsto X'$ and $Y^q \mapsto Y'$ define the announced isomorphisms h and $h^\vee$.

Proposition 3.12. *Let (G, T, α) be an elementary S-system and let $w \in N^\times(S)$. Put*

$$\Omega_0 = \Omega \cap \mathrm{int}(w^{-1})(\Omega).$$

Let d be the function on Ω defined by

$$d\big(\exp(Y) \cdot t \cdot \exp(X)\big) = \alpha(t)^{-1} + XY.$$

Then $\Omega_0 = \Omega_d$, and, for $\exp(Y) \cdot t \cdot \exp(X) \in \Omega_0(S')$, one has the following formula, where $z = d(\exp(Y) \cdot t \cdot \exp(X))$:

$$(\star) \qquad \mathrm{int}(w)\big(\exp(Y) \cdot t \cdot \exp(X)\big) = \exp\big(z^{-1} a_\alpha(w)^{-1} X\big) \cdot t\alpha^*(z) \cdot \exp\big(z^{-1} a_\alpha(w) Y\big).$$

Moreover,

$$d \circ \mathrm{int}(w) = d^{-1}.$$

Indeed, one immediately has[30]

$$\begin{aligned}\operatorname{int}(w)\big(\exp(Y)\cdot t\cdot\exp(X)\big) &= \exp\big(a_\alpha(w)Y\big)\cdot t\alpha^*\big(\alpha(t)^{-1}\big)\cdot\exp\big(a_\alpha(w)^{-1}X\big)\\ &= \exp\big(a_\alpha(w)Y\big)\cdot\exp\big(\alpha(t)a_\alpha(w)^{-1}X\big)\cdot t\alpha^*\big(\alpha(t)^{-1}\big).\end{aligned}$$

By 2.1, this is a section of Ω if and only if $1+\alpha(t)XY$ is invertible, which proves $\Omega_0=\Omega_d$. Applying formula (F) of the same paragraph then gives $(\star)$ by an immediate computation. Finally, $(\star)$ gives

$$\begin{aligned}(d\circ\operatorname{int}(w))\big(\exp(Y)\cdot t\cdot\exp(X)\big) &= \alpha\big(t\alpha^*(z)\big)^{-1}+z^{-2}XY\\ &= z^{-2}\big(\alpha(t)^{-1}+XY\big)=z^{-1},\end{aligned}$$

which proves the last assertion.

N.B. The function d is independent of the choice of w.

0.4 The Isomorphism Theorem

Theorem 4.1. *Let S be a scheme and let $q\in\mathbb{Z}$, $q>0$, be such that $x\mapsto x^q$ is an endomorphism of $\mathbb{G}_{a,S}$. Let (G,T,α) and (G',T',α') be two elementary S-systems. Let*

$$h\colon(\mathfrak{g}^\alpha)^{\otimes q}\xrightarrow{\sim}\mathfrak{g}'^{\alpha'},\qquad h^\vee\colon(\mathfrak{g}^{-\alpha})^{\otimes q}\xrightarrow{\sim}\mathfrak{g}'^{-\alpha'}$$

be mutually contragredient isomorphisms. Let $f_T\colon T\to T'$ be a morphism of S-groups making the following diagram commute:

$$\begin{array}{ccc}\mathbb{G}_{m,S} & \xrightarrow{\ q\ } & \mathbb{G}_{m,S}\\ {\scriptstyle\alpha^*}\downarrow & & \downarrow{\scriptstyle\alpha'^*}\\ T & \xrightarrow{\ f_T\ } & T'\\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\alpha'}\\ \mathbb{G}_{m,S} & \xrightarrow{\ q\ } & \mathbb{G}_{m,S}\end{array}$$

There exists a unique morphism of S-groups $f\colon G\to G'$ extending f_T and satisfying

$$f\big(\exp(X)\big)=\exp\big(h(X^q)\big)$$

for every $S'\to S$ and $X\in W(\mathfrak{g}^\alpha)(S')$. Moreover, this morphism also satisfies

$$f\big(\exp(Y)\big)=\exp\big(h^\vee(Y^q)\big),\qquad f\big(w_\alpha(Z)\big)=w_{\alpha'}\big(h(Z^q)\big)$$

for every $S'\to S$, $Y\in\Gamma(S',\mathfrak{g}^{-\alpha})$, and $Z\in\Gamma(S',\mathfrak{g}^\alpha)^\times$.

If $f\colon G\to G'$ extends f_T, then

$$f\circ\alpha^*=(\alpha'^*)^q.$$

If, in addition, f satisfies the second displayed condition in the statement, then it satisfies the other two conditions as well by 3.10. It follows that f is determined on Ω by

$$f\big(\exp(Y)t\exp(X)\big)=\exp\big(h^\vee(Y^q)\big)\,f_T(t)\,\exp\big(h(X^q)\big).$$

[30] The editors corrected the original by interchanging $a_\alpha(w)$ and $a_\alpha(w)^{-1}$, and supplied the details of the proof of $d\circ\operatorname{int}(w)=d^{-1}$.

Since Ω is schematically dense in G, this already proves the uniqueness of f. To prove existence, it is enough, by Exposé XVIII, 2.3, to show that the preceding formula defines a "generically multiplicative" morphism from Ω to G'. By 2.4, this amounts to checking

$$\alpha' \circ f = \alpha^q,$$

which follows from the fact that f extends f_T.

Scholium 4.2. Theorem 4.1 may also be interpreted as follows. Let $\mathcal{E}$ be the category of elementary S-systems, and let $\mathcal{D}$ be the category of tuples

$$(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\ L),$$

where T is a torus, α and α^* are group morphisms such that $\alpha \circ \alpha^* = 2$, and L is an invertible $\mathcal{O}_S$-module. The reader may specify the morphisms in the two categories under consideration. Define a functor

$$\mathcal{E} \longrightarrow \mathcal{D},$$
$$(G, T, \alpha) \longmapsto (\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\ \mathfrak{g}^\alpha).$$

The preceding theorem says that this functor is fully faithful. In fact, it is an equivalence of categories, as will be seen in the next section. We already have:

Corollary 4.3. *If $q = 1$ and f_T is an isomorphism, then f is an isomorphism.*

Corollary 4.4. *If $q = 1$ and f_T is faithfully flat with kernel Q* (cf. Exposé IX, 2.7)*, then f is faithfully flat and quasi-compact with kernel Q, and hence identifies G' with G/Q.*

Indeed, if f_T is faithfully flat with kernel Q, then

$$Q = \mathrm{Ker}(f_T) \subset \mathrm{Ker}(\alpha' \circ f_T) = \mathrm{Ker}(\alpha).$$

Introduce the elementary S-system $(G/Q, T/Q, \alpha/Q)$ of 2.13. By 2.14, we are reduced to proving that f/Q induces an isomorphism from G/Q onto G', which follows immediately from 4.3.

0.5 Examples of Elementary Systems; Applications

5.1. Let S be a scheme and L an invertible $\mathcal{O}_S$-module. Consider the group GL_L over S defined by

$$\mathrm{GL}_L(S') = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; \begin{array}{ll} a, d \in \mathbb{G}_a(S'), & b \in W(L)(S'), \\ c \in W(L^{-1})(S'), & ad - bc \in \mathbb{G}_m(S') \end{array} \right\},$$

equipped with the usual matrix-multiplication law. It is locally isomorphic to $\mathrm{GL}_{2,S}$. It is therefore an affine smooth S-group scheme with connected fibers.

Remark. Let L' and L'' be invertible sheaves on S such that

$$L = L' \otimes L''^{-1}.$$

Then there is an isomorphism of S-groups

$$\mathrm{GL}_L \xrightarrow{\sim} \mathrm{GL}(L' \oplus L'')$$

defined as follows: if x (respectively y) is a section of L' (respectively L'') on an open subset V of S, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}.$$

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5.2. Denote by SL_L the closed subgroup of GL_L defined by $ad-bc=1$. It too is an affine smooth S-group scheme with connected fibers; under the preceding isomorphism it is isomorphic to $\mathrm{SL}(L'\oplus L'')$.

Likewise, consider the morphism

$$\mathbb{G}_{m,S} \longrightarrow \mathrm{GL}_L, \qquad z \longmapsto \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}.$$

It is a central monomorphism. Passage to the quotient gives a smooth affine S-group P_L with connected fibers (cf. Exposé VIII, 5.7). By passing to the quotient in the isomorphism of the preceding remark, one sees that P_L identifies with the automorphism group of the projective bundle $\mathbb{P}(L'\oplus L'')$ (cf. EGA II, 4.2.7). Write i and p for the canonical morphisms

$$\mathrm{SL}_L \xrightarrow{i} \mathrm{GL}_L \xrightarrow{p} \mathrm{P}_L.$$

Here i is a closed immersion, while p is faithfully flat and affine.

5.3. Consider the group morphisms

$$t_G\colon \mathbb{G}_{m,S}^2 \longrightarrow \mathrm{GL}_L, \quad t_G(z,z') = \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix},$$

$$t_S\colon \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_L, \qquad t_S(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix},$$

$$t_P\colon \mathbb{G}_{m,S} \longrightarrow \mathrm{P}_L, \qquad t_P(z) = p(t_G(z,1)).$$

They are group monomorphisms and define in each group a split torus of relative codimension 2. For $s\in S$, let

$$X \in \Gamma(s, L\otimes s)^{\times}.$$

The section $\left(\begin{smallmatrix} 0 & X \\ -X^{-1} & 0 \end{smallmatrix}\right)$ of $\mathrm{GL}_{L,s}$ normalizes $t_G(\mathbb{G}_{m,s}^2)$ and does not centralize it. Exposé XIX, 1.6 therefore shows that GL_L is reductive of semisimple rank 1, with maximal torus $t_G(\mathbb{G}_{m,S}^2)$. The same argument shows that SL_L and P_L are reductive of semisimple rank 1, with respective maximal tori $t_S(\mathbb{G}_{m,S})$ and $t_P(\mathbb{G}_{m,S})$.

5.4. The Lie algebras of these groups, and the adjoint actions of the chosen maximal tori, are determined in the usual way. For GL_L, Exposé II, 4.8 immediately gives

$$\mathrm{Lie}(\mathrm{GL}_L/S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; a,d\in\mathcal{O}_S,\ b\in L,\ c\in L^{-1} \right\},$$

with the usual bracket, and

$$\mathrm{Ad}(t_G(z,z')) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & zz'^{-1}b \\ z'z^{-1}c & d \end{pmatrix}.$$

[31] The editors corrected $L''\otimes L'^{-1}$ to $L'\otimes L''^{-1}$ and expanded the sentence that follows.

Put $\mathfrak{g} = \mathrm{Lie}(\mathrm{GL}_L/S)$, and define

$$\alpha_G \colon t_G(\mathbb{G}_{m,S}^2) \longrightarrow \mathbb{G}_{m,S}, \qquad \alpha_G(t_G(z,z')) = zz'^{-1}.$$

The preceding identity shows that α_G is a root of GL_L relative to $t_G(\mathbb{G}_{m,S}^2)$, and that

$$u \colon L \longrightarrow \mathfrak{g}, \quad X \longmapsto \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \qquad u^- \colon L^{-1} \longrightarrow \mathfrak{g}, \quad Y \longmapsto \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix}$$

identify L with $\mathfrak{g}^{\alpha_G}$ and L^{-1} with $\mathfrak{g}^{-\alpha_G}$, respectively. Thus

$$(\mathrm{GL}_L, t_G(\mathbb{G}_{m,S}^2), \alpha_G)$$

is an elementary S-system.

Similarly, setting

$$\alpha_S(t_S(z)) = z^2, \qquad \alpha_P(t_P(z)) = z,$$

one finds elementary S-systems

$$(\mathrm{SL}_L, t_S(\mathbb{G}_{m,S}), \alpha_S), \qquad (\mathrm{P}_L, t_P(\mathbb{G}_{m,S}), \alpha_P),$$

and corresponding identifications of L and L^{-1} with the root summands of their Lie algebras.

5.5. Set

$$\exp \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

This defines a morphism

$$W(\mathfrak{g}^{\alpha_G}) \longrightarrow \mathrm{GL}_L$$

whose tangent map is the canonical inclusion; it is therefore the unique morphism of this kind by 1.5. Likewise, set

$$\exp \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}.$$

Direct calculation of formula (F) gives

$$\left\langle \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} \right\rangle = XY, \qquad \alpha_G^*(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = t_G(z, z^{-1}).$$

[32]

The open subset $N^\times = N_G^\times$ defined before 3.1 is

$$N_G^\times(S') = \left\{ \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \;\middle|\; P \in W(\mathfrak{g}^\alpha)^\times(S'),\ Q \in W(\mathfrak{g}^{-\alpha})^\times(S') \right\}.$$

For $X \in W(\mathfrak{g}^\alpha)^\times(S')$,

$$w_{\alpha_G}(X) = \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

[32] The editors expanded the calculations that follow.

Moreover, if

$$w = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \in N_G^{\times}(S'),$$

then

$$a_{\alpha_G}(w) = PQ^{-1} \in W\big((\mathfrak{g}^{\alpha})^{\otimes 2}\big)^{\times}(S');$$

that is, for $Y \in W(\mathfrak{g}^{-\alpha})^{\times}(S')$,

$$a_{\alpha_G}(w)(Y) = PQ^{-1}Y \in W(\mathfrak{g}^{\alpha})^{\times}(S').$$

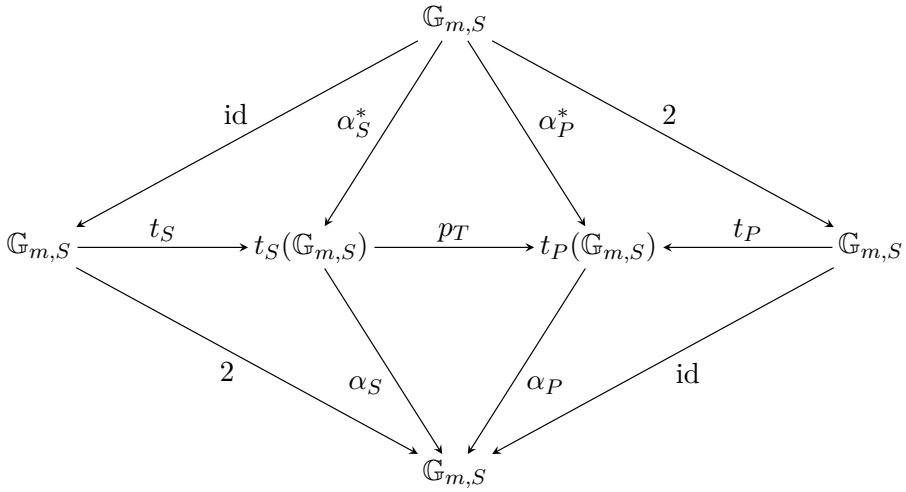
5.6. We leave the analogous calculations for SL_L and P_L to the reader. One obtains the same duality formula and the coroots

$$\alpha_S^*(z) = t_S(z), \qquad \alpha_P^*(z) = t_P(z^2).$$

Let p_T be the morphism induced by $p\colon \mathrm{GL}_L \to \mathrm{P}_L$ on $t_S(\mathbb{G}_{m,S})$, namely

$$p_T(t_S(z)) = t_P(z^2).$$

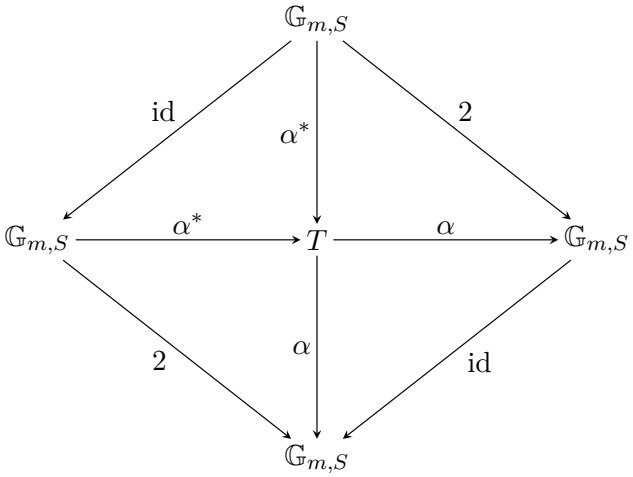
Thus the following diagram is commutative: [33]



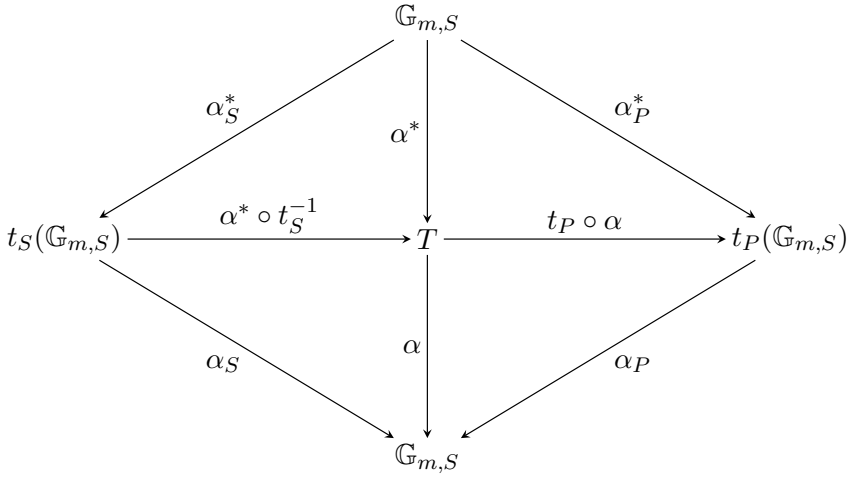
Its central portion is the commutative diagram of 4.1 [34] associated with the canonical morphism $p \circ i\colon \mathrm{SL}_L \to \mathrm{P}_L$, which induces a morphism between the preceding elementary S-systems.

5.7. Let (G, T, α) now be an arbitrary elementary S-system. Consider the commutative diagram

[33] Here t_S and t_P are isomorphisms.
[34] With $q = 1$.



Combining this diagram with that of 5.6 gives a commutative diagram



Using 4.1, one obtains:

Proposition 5.8. *Let S be a scheme and let (G, T, α) be an elementary S-system. Put $L = \mathfrak{g}^\alpha$, so that $L^{-1} = \mathfrak{g}^{-\alpha}$.*

(i) *There exists a unique group morphism $f\colon \mathrm{SL}_L \to G$ satisfying the following equivalent conditions:*

$$\text{(a)}\quad f\begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = \alpha^*(z), \qquad f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X);$$

$$\text{(b)}\quad f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), \qquad f\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \exp(Y);$$

$$\text{(c)}\quad f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), \qquad f\begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix} = w_\alpha(X).$$

(ii) *There exists a unique group morphism $g\colon G \to \mathrm{P}_L$ satisfying*

$$g(t) = \begin{pmatrix} \alpha(t) & 0 \\ 0 & 1 \end{pmatrix}, \qquad g(\exp(X)) = p\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

Moreover,

$$g(\exp(Y)) = p\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \qquad g(w_\alpha(X)) = p\begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

The morphism g is faithfully flat and quasi-compact, with

$$\mathrm{Ker}(g) = \mathrm{Ker}(\alpha) = \mathrm{Centr}(G),$$

and $g \circ f$ is the canonical morphism $\mathrm{SL}_L \to \mathrm{P}_L$.

Notice that condition (b) in (i) gives an explicit description of the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$.

Corollary 5.9. *Let (G, T, α) be an elementary S-system. The subgroups*

$$T \cdot U_\alpha, \qquad T \cdot U_{-\alpha}, \qquad U_\alpha, \qquad U_{-\alpha}$$

are closed.

Since U_α is a closed subgroup scheme of $T \cdot U_\alpha$, it suffices to prove the assertion for the latter. By Noether's theorem (Exposé IV, 5.3.1 and 6.4.1), it is enough to show that

$$(T \cdot U_\alpha)/\mathrm{Ker}(\alpha)$$

is a closed subgroup of $G/\mathrm{Ker}(\alpha)$. By 5.8, this reduces to the evident fact that the subgroup of P_L defined by $c = 0$ is closed; one may equivalently work in GL_L, by another application of Noether's theorem. Consequently, the exponential morphisms of Theorem 1.5(i) are closed immersions.

N.B. The corollary also follows from the fact that $T \cdot U_\alpha$ and $T \cdot U_{-\alpha}$ are "Borel subgroups" of G (cf. Exposé XII, 7.10).

5.10. Let L be an invertible $\mathcal{O}_S$-module, and let

$$\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S}$$

be a diagram of groups, where T is a torus,[35] such that $\alpha \circ \alpha^* = 2$. Let R be the maximal torus of $\mathrm{Ker}(\alpha)$, and put $K = \alpha^{*-1}(R)$. Then K is a subgroup of multiplicative type of $\mathbb{G}_{m,S}$; since $\alpha \circ \alpha^* = 2$, it is even a subgroup of $\mu_{2,S}$. In particular,

$$K \longrightarrow \mathrm{SL}_L, \qquad z \longmapsto \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}$$

is central. Hence there is a central group monomorphism

$$K \longrightarrow R \times \mathrm{SL}_L, \qquad z \longmapsto \left(\alpha^*(z), \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}\right).$$

Let

$$G = (R \times \mathrm{SL}_L)/K.$$

This is an affine smooth group over S, with connected fibers. The sequence

$$1 \longrightarrow K \longrightarrow R \times t_S(\mathbb{G}_{m,S}) \xrightarrow{u} T \longrightarrow 1, \qquad u(x, t_S(z)) = x\alpha^*(z),$$

[35] This qualification is supplied by the editors.

is exact. The image of $R \times t_S(\mathbb{G}_{m,S})$ in G is therefore a torus T' isomorphic to T. If α' is the character of T' transported from α, one readily checks that (G, T', α') is an elementary S-system, that $\mathfrak{g}^{\alpha'}$ is isomorphic to L, and that α'^* is obtained from α^* under the isomorphism $T \xrightarrow{\sim} T'$. Thus the object

$$\left(\mathbb{G}_{m,S} \xrightarrow{\alpha'^*} T' \xrightarrow{\alpha'} \mathbb{G}_{m,S},\, \mathfrak{g}^{\alpha'}\right)$$

of the category $\mathcal{D}$ defined in 4.2 is isomorphic to

$$\left(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\, L\right).$$

We have therefore proved:

Theorem 5.11. *With the notation of 4.2, the functor*

$$(G, T, \alpha) \longmapsto \left(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\, \mathfrak{g}^{\alpha}\right)$$

is an equivalence of categories between $\mathcal{E}$ and $\mathcal{D}$.

0.6 Generators and Relations for an Elementary System

6.1. Let S be a scheme and let (G, T, α) be an elementary S-system. Let

$$X \in W(\mathfrak{g}^{\alpha})^{\times}(S), \qquad u = \exp(X).$$

As observed in 3.8, the element $w = w_\alpha(X)$ satisfies, in particular,

$$(wu)^3 = e.$$

[36] Let s_α denote the automorphism of T induced by $\mathrm{int}(w)$. By Theorem 3.1(iii), for every $S' \to S$ and every $t \in T(S')$,

$$s_\alpha(t) = \mathrm{int}(w)(t) = t\,\alpha^*(\alpha(t)^{-1}).$$

Theorem 6.2. *Let H be a sheaf of groups over S for the fppf topology. Let*

$$f_T \colon T \longrightarrow H, \qquad f_\alpha \colon U_\alpha \longrightarrow H$$

be group morphisms, and let $h \in H(S)$. There exists a (necessarily unique) *group morphism*

$$f \colon G \longrightarrow H$$

extending f_T and f_α and satisfying $f(w) = h$ if and only if the following conditions hold:

(i) *For every $S' \to S$, $t \in T(S')$, and $x \in U_\alpha(S')$,*

$$f_T(t) f_\alpha(x) f_T(t)^{-1} = f_\alpha(txt^{-1}) = f_\alpha\big(x^{\alpha(t)}\big). \tag{1}$$

Equivalently, f_T and f_α extend to a group morphism $T \cdot U_\alpha \to H$.

(ii) *For every $S' \to S$ and $t \in T(S')$,*

$$h f_T(t) h^{-1} = f_T(s_\alpha(t)) = f_T\Big(t\,\alpha^*(\alpha(t)^{-1})\Big). \tag{2}$$

[36] The editors added the sentence that follows.

(iii) *The following two relations hold in $H(S)$:*

$$h^2 = f_T(\alpha^*(-1)), \tag{3}$$

$$\big(hf_\alpha(u)\big)^3 = e. \tag{4}$$

Proof. Write the laws on U_α and $U_{-\alpha}$ additively, while writing their vector-space structures multiplicatively. If f has the asserted properties, then for $y \in U_{-\alpha}(S')$ one necessarily has

$$f(y) = f(w^{-1}wyw^{-1}w) = hf_\alpha(w^{-1}yw)h^{-1}.$$

Define a morphism $f_{-\alpha}\colon U_{-\alpha} \to H$ by

$$f_{-\alpha}(y) = hf_\alpha(w^{-1}yw)h^{-1}. \tag{$*_1$}$$

It is a group morphism. On the big cell Ω, the morphism f is forced to satisfy

$$f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x).$$

This proves uniqueness. The conditions in the statement are manifestly necessary; let us prove that they are sufficient.

Relation (4) gives

$$hf_\alpha(u)h^{-1}h^2 = f_\alpha(-u)h^{-1}f_\alpha(-u).$$

By (3) and (1), $h^2 = f_T(\alpha^*(-1))$ commutes with $f_\alpha(-u)$, and hence

$$hf_\alpha(u)h^{-1} = f_\alpha(-u)hf_\alpha(-u).$$

By definition,

$$hf_\alpha(u)h^{-1} = f_{-\alpha}(wuw^{-1}),$$

whereas 3.7 gives

$$wuw^{-1} = -\widetilde{u}, \tag{$*_2$}$$

with $\widetilde{u}$ the element paired with u. Therefore

$$f_{-\alpha}(-\widetilde{u}) = f_\alpha(-u)hf_\alpha(-u). \tag{$*_3$}$$

Let t be a section of T over a variable $S' \to S$. Apply $\mathrm{int}(f_T(t))$ to this identity. On the left one obtains[37]

$$\begin{aligned} f_T(t)f_{-\alpha}(-\widetilde{u})f_T(t)^{-1} &= f_T(t)hf_\alpha(u)h^{-1}f_T(t)^{-1} \\ &= h\big(h^{-1}f_T(t)h\big)f_\alpha(u)\big(h^{-1}f_T(t)^{-1}h\big)h^{-1} \\ &= hf_T(s_\alpha(t))f_\alpha(u)f_T(s_\alpha(t))^{-1}h^{-1} \\ &= hf_\alpha(\alpha(s_\alpha(t))u)h^{-1} \\ &= hf_\alpha(\alpha(t)^{-1}u)h^{-1} \\ &= f_{-\alpha}(-\alpha(t)^{-1}\widetilde{u}). \end{aligned}$$

Here we used (2), (1), $s_\alpha(t) = t\alpha^*(\alpha(t)^{-1})$, and $\alpha \circ \alpha^* = 2$.

[37] The editors corrected the calculation that follows.

The right-hand side of $(*_3)$, after conjugation, becomes

$$f_\alpha(-\alpha(t)u)\, f_T(t) h f_T(t)^{-1} h^{-1}\, h f_\alpha(-\alpha(t)u).$$

Since

$$h f_T(t)^{-1} h^{-1} = f_T(s_\alpha(t^{-1})) = f_T\big(t\,\alpha^*(\alpha(t))\big),$$

this equals

$$f_\alpha(-\alpha(t)u)\, f_T\big(\alpha^*(\alpha(t))\big)\, h f_\alpha(-\alpha(t)u).$$

Comparison gives

$$f_{-\alpha}(-\alpha(t)^{-1}\widetilde{u}) = f_\alpha(-\alpha(t)u)\, f_T\big(\alpha^*(\alpha(t))\big)\, h f_\alpha(-\alpha(t)u).$$

Since $\alpha\colon T \to \mathbb{G}_{m,S}$ is faithfully flat and H is a separated presheaf, $-\alpha(t)^{-1}$ may be replaced by an arbitrary section of $\mathbb{G}_{m,S}$. Thus:

Lemma 6.2.1. *For every $S' \to S$ and every $z \in \mathbb{G}_m(S')$,*

$$f_{-\alpha}(z\widetilde{u}) = f_\alpha(z^{-1}u)\, f_T\big(\alpha^*(-z^{-1})\big)\, h f_\alpha(z^{-1}u).$$

Let $x, y \in \mathbb{G}_a(S')$, where $S' \to S$, and suppose that y and $1 + xy$ are invertible. Applying Lemma 6.2.1 first with $z = y$, one obtains[38]

$$f_\alpha(xu) f_{-\alpha}(y\widetilde{u}) = f_\alpha\big((x + y^{-1})u\big)\, f_T\big(\alpha^*(-y^{-1})\big)\, h f_\alpha(y^{-1}u).$$

Now $x + y^{-1} = y^{-1}(1 + xy)$. Applying the lemma with $z = y/(1 + xy)$ gives

$$\begin{aligned} f_\alpha\big((x + y^{-1})u\big) = f_{-\alpha}\left(\frac{y}{1+xy}\widetilde{u}\right) & f_\alpha\big(-(x + y^{-1})u\big) \\ & \cdot h\, f_T\left(\alpha^*\left(-\frac{y}{1+xy}\right)\right). \end{aligned}$$

Substitution in the preceding identity yields

$$\begin{aligned} f_\alpha(xu) f_{-\alpha}(y\widetilde{u}) = f_{-\alpha}\left(\frac{y}{1+xy}\widetilde{u}\right) & f_\alpha\big(-(x + y^{-1})u\big) h^{-1} \\ & \cdot f_T\big(\alpha^*(1 + xy)^{-1}\big) h f_\alpha(y^{-1}u). \end{aligned}$$

Since $h^{-1} f_T(t) h = f_T(s_\alpha(t))$ by (2) (recall that $s_\alpha^2 = \mathrm{id}$), and $s_\alpha \circ \alpha^* = -\alpha^*$, this is

$$f_{-\alpha}\left(\frac{y}{1+xy}\widetilde{u}\right) f_\alpha\big(-(x + y^{-1})u\big) f_T\big(\alpha^*(1 + xy)\big) f_\alpha(y^{-1}u).$$

For $x' \in U_\alpha(S')$ and $z \in \mathbb{G}_m(S')$, one has

$$f_\alpha(x') f_T(\alpha^*(z)) = f_T(\alpha^*(z)) f_\alpha(z^{-2}x').$$

Consequently,

$$\begin{aligned} f_\alpha(xu) f_{-\alpha}(y\widetilde{u}) &= f_{-\alpha}\left(\frac{y}{1+xy}\widetilde{u}\right) f_T\big(\alpha^*(1 + xy)\big) \\ &\quad \cdot f_\alpha\left(\left[-\frac{y^{-1}(1+xy)}{(1+xy)^2} + y^{-1}\right] u\right) \\ &= f_{-\alpha}\left(\frac{y}{1+xy}\widetilde{u}\right) f_T\big(\alpha^*(1 + xy)\big) f_\alpha\left(\frac{x}{1+xy}u\right). \end{aligned}$$

We have proved:

[38] The editors expanded the calculations that follow.

Lemma 6.2.2. *Let $S' \to S$. If $a \in U_\alpha(S')$, $b \in U^\times_{-\alpha}(S')$, and $1 + ab \in \mathbb{G}_m(S')$, then*

$$f_\alpha(a) f_{-\alpha}(b) = f_{-\alpha}\left(\frac{b}{1+ab}\right) f_T(\alpha^*(1+ab)) f_\alpha\left(\frac{a}{1+ab}\right).$$

By schematic density, this formula remains valid for $b \in U_{-\alpha}(S')$, with $1+ab$ invertible. Define

$$f\colon \Omega \longrightarrow H, \qquad f(ytx) = f_{-\alpha}(y) f_T(t) f_\alpha(x).$$

It follows at once from Lemma 6.2.2, condition 6.2(i), and formula (F') of 2.4 that whenever $g, g' \in \Omega(S')$ and $gg' \in \Omega(S')$, one has

$$f(gg') = f(g) f(g').$$

By Exposé XVIII, 2.3(iii) and 2.4,[39] there is therefore a group morphism $G \to H$ extending f. Denote it again by f. It remains only to prove that $f(w_\alpha) = h$. Since

$$w_\alpha = u \cdot (-\widetilde{u}) \cdot u,$$

formula $(*_3)$ gives[40]

$$f(w_\alpha) = f_\alpha(u) f_{-\alpha}(-\widetilde{u}) f_\alpha(u) = h.$$

This completes the proof of Theorem 6.2.

Remark 6.3. *These results will be completed in Exposé XXIII, 3.5.*

[39] Each geometric fiber of G is connected, for example by 1.1.

[40] The editors simplified the original argument by invoking $(*_3)$.

Exposé XXI. Root Data

This Exposé[1] collects, in the absence of a suitable reference,[2] known results on root data (that is, "abstract" root systems), most of which will be used below.

Notation. We denote by $\mathbb{Q}_+$ the set of positive (or zero) rational numbers; $\mathbb{Z} \cap \mathbb{Q}_+ = \mathbb{N}$. Let V be a $\mathbb{Q}$-vector space. If A (respectively B) is a subset of $\mathbb{Q}$ (respectively of V), we denote by $A \cdot B$ the image of $A \otimes B$ under the morphism $\mathbb{Q} \otimes V \to V$, in other words the set of linear combinations of elements of B with coefficients in A. We write $-B = \{-1\}B$. We denote by $E - F$ the set of elements of E that do not belong to F.

0.1 Generalities

0.1.1 Definitions and first properties

Definition 0.1.1.1. Let M and M^* be two free finite-type $\mathbb{Z}$-modules in duality. Write

$$V = M \otimes \mathbb{Q}, \qquad V^* = M^* \otimes \mathbb{Q};$$

these are two $\mathbb{Q}$-vector spaces in duality. We identify M (respectively M^*) with a subset of V (respectively V^*). The canonical bilinear form on $M^* \times M$ (respectively $V^* \times V$) is denoted $(\ ,\ )$.

Let R be a finite subset of M. Suppose given a map $\alpha \mapsto \alpha^*$ from R to M^*; the set of α^*, for $\alpha \in R$, is denoted R^*. To every $\alpha \in R$ associate the endomorphism s_α (respectively s_α^*) of M and V (respectively of M^* and V^*) given by

$$(1) \qquad s_\alpha(x) = x - (\alpha^*, x)\alpha, \qquad s_\alpha = \mathrm{id} - \alpha^* \otimes \alpha,$$
$$(1^*) \qquad s_\alpha^*(u) = u - (u, \alpha)\alpha^*, \qquad s_\alpha^* = \mathrm{id} - \alpha \otimes \alpha^*.$$

The pair (R, R^*) (more precisely, the pair $(R, R \to M^*)$) is called a *root datum* in (M, M^*), and (M, M^*, R, R^*) is called a root datum, if the following axioms hold:

$$\text{(DR I)} \quad (\alpha^*, \alpha) = 2 \qquad \text{for every } \alpha \in R,$$
$$\text{(DR II)} \quad s_\alpha(R) \subset R, \qquad s_\alpha^*(R^*) \subset R^* \qquad \text{for every } \alpha \in R.$$

The set R is called the *root system* of the root datum $\mathcal{R} = (M, M^*, R, R^*)$. The elements of R (respectively R^*) are called the *roots* (respectively *coroots*) of the root datum.

Remark 0.1.1.2. Axiom (DR I) is equivalent to any one of the following properties:

$$(2,2^*) \qquad s_\alpha s_\alpha = \mathrm{id}, \qquad s_\alpha^* s_\alpha^* = \mathrm{id},$$
$$(3,3^*) \qquad s_\alpha(\alpha) = -\alpha, \qquad s_\alpha^*(\alpha^*) = -\alpha^*.$$

[1] Editors' note: Version of 13 October 2024.

[2] Editors' note: For results on root systems (Sections 1–5), one may consult *Groupes et algèbres de Lie*, Chapter VI.

Remark 0.1.1.3. Axioms (DR I) and (DR II) imply

$$R = -R, \qquad R^* = -R^*, \qquad 0 \notin R, \qquad 0 \notin R^*.$$

Lemma 0.1.1.4. *The map $R \to R^*$ is a bijection. More generally, if $\alpha, \beta \in R$ and $(\alpha^*, x) = (\beta^*, x)$ for every $x \in R$, then $\alpha = \beta$.*

Proof. Indeed, one then has

$$s_\beta(\alpha) = \alpha - 2\beta, \qquad s_\alpha(\beta) = \beta - 2\alpha.$$

It follows at once that

$$s_\beta s_\alpha(\alpha) = 2\beta - \alpha = \alpha + 2(\beta - \alpha), \qquad s_\beta s_\alpha(\beta - \alpha) = s_\beta(\beta - \alpha) = \beta - \alpha.$$

Hence, by (DR II),

$$(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha) \in R.$$

Since R is finite, $\beta - \alpha = 0$. □

Corollary 0.1.1.5. *The inverse map $R^* \to R$ defines a root datum*

$$\mathcal{R}^* = (M^*, M, R^*, R),$$

called the dual *of $\mathcal{R}$.*[3]

Definition 0.1.1.6. We denote by $\Gamma_0(R)$ the subgroup of M generated by R. We denote by $\mathcal{V}(R)$ the vector subspace of V generated by R, that is,

$$\mathcal{V}(R) = \Gamma_0(R) \otimes \mathbb{Q}.$$

Applying these definitions to $\mathcal{R}^*$, one likewise obtains $\Gamma_0(R^*)$ and $\mathcal{V}(R^*)$.

The *reductive rank* of $\mathcal{R}$ is the number

$$\mathrm{rgred}(\mathcal{R}) = \mathrm{rank}(M) = \dim(V) = \dim(V^*) = \mathrm{rank}(M^*) = \mathrm{rgred}(\mathcal{R}^*).$$

The *semisimple rank* of $\mathcal{R}$ is the number

$$\mathrm{rgss}(\mathcal{R}) = \mathrm{rank}(R) = \mathrm{rank}(\Gamma_0(R)) = \dim(\mathcal{V}(R)).$$

Thus $\mathrm{rgss}(\mathcal{R}) \leq \mathrm{rgred}(\mathcal{R})$.

We shall see below that $\mathrm{rgss}(\mathcal{R}) = \mathrm{rgss}(\mathcal{R}^*)$, that is, that $\mathcal{V}(R)$ and $\mathcal{V}(R^*)$ have the same dimension.

Definition 0.1.1.7.

The root datum $\mathcal{R}$ is called *semisimple* (respectively *trivial*) if $\mathrm{rgss}(\mathcal{R}) = \mathrm{rgred}(\mathcal{R})$ (respectively $\mathrm{rgss}(\mathcal{R}) = 0$). Thus $\mathcal{R}$ is trivial if and only if R is empty. The trivial root datum of reductive rank zero is denoted

$$0 = (\{0\}, \{0\}, \varnothing, \varnothing).$$

[3] Editors' note: $(\alpha^*)^* = \alpha$.

Definition 0.1.1.8. We denote by $W(\mathcal{R})$ the group of transformations of M generated by the s_α, $\alpha \in R$. It is called the *Weyl group* of $\mathcal{R}$. We write

$$W^*(\mathcal{R}) = W(\mathcal{R}^*).$$

Then $W(\mathcal{R})$ acts on R, $\Gamma_0(R)$, $\mathcal{V}(R)$, M, and V. If $w \in W(\mathcal{R})$ and $x \in M$ (respectively $x \in V$), then

$$wx - x \in \Gamma_0(R) \quad \text{(respectively } wx - x \in \mathcal{V}(R)\text{)};$$

this follows immediately from formula (1). Likewise for $W^*(\mathcal{R})$.

Lemma 0.1.1.9. *For every $\alpha \in R$, $x \in V$, and $u \in V^*$, one has*

$$(s_\alpha^*(u), s_\alpha(x)) = (u, x). \tag{4}$$

Proof. Taking the scalar product of (1) and (1*), the left-hand side is

$$(u,x) + (u,\alpha)(\alpha^*,x)\big((\alpha^*,\alpha) - 2\big) = (u,x).$$

□

Remark 0.1.1.10. If one assumes $0 \notin R$ and $0 \notin R^*$, then Lemma 0.1.1.9 is equivalent to (DR I).

Corollary 0.1.1.11. *Let $h \mapsto h^\vee$ denote the isomorphism from $\mathrm{GL}(M)$ onto $\mathrm{GL}(M^*)$ that associates to h its contragredient. Formula (4) may then also be written*

$$(s_\alpha)^\vee = s_\alpha^*. \tag{5}$$

Corollary 0.1.1.12. *The preceding isomorphism induces an isomorphism from $W(\mathcal{R})$ onto $W^*(\mathcal{R})$.*

Scholium 0.1.1.13. *By the preceding result, we shall identify W and W^*, and regard W as a group of transformations of*

$$R,\ R^*,\ M,\ M^*,\ \Gamma_0(R),\ \Gamma_0(R^*),\ V,\ V^*,\ \mathcal{V}(R),\ \mathcal{V}(R^*).$$

We shall write s_α for s_α^*.

0.1.2 The map p

Lemma 0.1.2.1. *Let $p : M \to M^*$ (respectively $V \to V^*$) be the linear map defined by*

$$p(x) = \sum_{u \in R^*} (u,x)u. \tag{6}$$

Write $\ell(x) = (p(x), x)$. Then:

$$\ell(x) \geq 0, \qquad \ell(\alpha) > 0 \qquad \text{for } \alpha \in R, \tag{7}$$

$$\ell(wx) = \ell(x) \qquad \text{for } w \in W, \tag{8}$$

$$(p(x), y) = (p(y), x) \qquad \text{for } x, y \in V, \tag{9}$$

$$\ell(\alpha)\alpha^* = 2p(\alpha) \qquad \text{for } \alpha \in R. \tag{10}$$

Proof. The first three relations are immediate.[4] We prove the last one. By (1), for $u \in V^*$,

$$\begin{aligned}(u,\alpha)^2\alpha^* &= (u,\alpha)u - (u,\alpha)s_\alpha(u)\\ &= (u,\alpha)u + (u,s_\alpha(\alpha))s_\alpha(u)\\ &= (u,\alpha)u + (s_\alpha(u),\alpha)s_\alpha(u) \qquad (s_\alpha^2 = \mathrm{id}).\end{aligned}$$

Since s_α permutes R^* by (DR II), summing over $u \in R^*$ gives the result. □

Scholium 0.1.2.2. *Relation* (10) *says that the map* $\alpha \mapsto \ell(\alpha)\alpha^*$ *is the restriction to* R *of a linear map from* M *to* M^**. In particular,*

$$(-\alpha)^* = -\alpha^*.$$

Corollary 0.1.2.3. *The map* p *induces an isomorphism from* $\mathcal{V}(R)$ *onto* $\mathcal{V}(R^*)$*.*

Proof. The map p sends $\mathcal{V}(R)$ onto $\mathcal{V}(R^*)$. Hence

$$\dim \mathcal{V}(R) \geq \dim \mathcal{V}(R^*).$$

Applying this inequality to the dual root datum gives

$$\dim \mathcal{V}(R) = \dim \mathcal{V}(R^*),$$

so the surjective map p is also bijective. □

Corollary 0.1.2.4. *One has* $\dim \mathcal{V}(R) = \dim \mathcal{V}(R^*)$*, and therefore*

$$\mathrm{rgss}(\mathcal{R}) = \mathrm{rgss}(\mathcal{R}^*).$$

Corollary 0.1.2.5. *The bilinear form* (u,x) *is nondegenerate on* $\mathcal{V}(R^*) \times \mathcal{V}(R)$*, and thus puts these* $\mathbb{Q}$*-vector spaces in duality.*

Proof. If $(u,x) = 0$ for every $u \in R^*$, then $p(x) = 0$. □

Corollary 0.1.2.6. *The symmetric bilinear form* $(p(x),y)$ *is positive and nondegenerate on* $\mathcal{V}(R)$*.*

Corollary 0.1.2.7. *The group* W *acts faithfully on* R *(and hence on the other sets listed in Scholium 0.1.1.13).*

Proof. Let $u \in V^*$, and suppose $w \in W$ satisfies $w(\alpha) = \alpha$ for every $\alpha \in R$. We prove $w(u) = u$. For every $\alpha \in R$,

$$(w(u) - u, \alpha) = (w(u),\alpha) - (u,\alpha) = (u, w^{-1}(\alpha)) - (u,\alpha) = 0.$$

But $w(u) - u \in \mathcal{V}(R^*)$. Since it is orthogonal to every root, it vanishes by Corollary 0.1.2.5. □

Corollary 0.1.2.8. *The group* W *is finite.*

Proposition 0.1.2.9. *The operations of* W *respect the correspondence between roots and coroots. In other words, for* $\alpha \in R$ *and* $w \in W$*,*

$$w(\alpha^*) = w(\alpha)^*.$$

[4] Editors' note: Formula (8) follows from (4), (2,2*), and (DR II).

Proof. It suffices to verify this for $w = s_\beta$, $\beta \in R$, that is, to verify

$$s_\beta(\alpha^*) = s_\beta(\alpha)^*.$$

Now $\ell(s_\beta(\alpha))s_\beta(\alpha)^*/2$ equals

$$\begin{aligned} p(s_\beta(\alpha)) &= \sum_{u \in R^*} (u, s_\beta(\alpha))u \\ &= \sum_{u \in R^*} (s_\beta(u), \alpha)u \\ &= \sum_{u \in R^*} (u, \alpha)s_\beta(u) = s_\beta(p(\alpha)). \end{aligned}$$

Since $\ell(s_\beta(\alpha)) = \ell(\alpha)$, we obtain

$$s_\beta(\alpha)^* = s_\beta\left(\frac{2p(\alpha)}{\ell(\alpha)}\right) = s_\beta(\alpha^*).$$

□

Corollary 0.1.2.10. *If $\alpha \in R$ and $w \in W$, then*

$$ws_\alpha w^{-1} = s_{w(\alpha)}.$$

Proof. For $x \in V$,

$$ws_\alpha w^{-1}(x) = x - (\alpha^*, w^{-1}(x))w(\alpha) = x - (w(\alpha^*), x)w(\alpha).$$

By Proposition 0.1.2.9, the last expression is

$$x - (w(\alpha)^*, x)w(\alpha) = s_{w(\alpha)}(x).$$

□

Corollary 0.1.2.11.
Let $R' \subset R$. The quadruple (M, M^*, R', R'^*) is a root datum if and only if $\alpha, \beta \in R'$ implies $s_\alpha(\beta) \in R'$.

0.2 Relations between two roots

0.2.1 Proportional roots

Proposition 0.2.1.1. *Let α and β be two roots. The following conditions are equivalent:*

(i) there exists $k \in \mathbb{Q}$ such that $\alpha = k\beta$;

(ii) $s_\alpha = s_\beta$.

Under these conditions, $\alpha^* = k^{-1}\beta^*$, and k is one of

$$1,\ -1,\ 2,\ -2,\ \tfrac{1}{2},\ -\tfrac{1}{2}.$$

Proof. Assume first (i). Then

$$\alpha^* = \ell(\alpha)^{-1} 2p(\alpha) = k^{-2}\ell(\beta)^{-1} 2k\, p(\beta) = k^{-1}\beta^*.$$

This immediately gives $s_\alpha = s_\beta$. Conversely, if $s_\alpha = s_\beta$, then

$$\alpha = s_\alpha(-\alpha) = s_\beta(-\alpha) = (\beta^*, \alpha)\beta - \alpha,$$

so

$$2\alpha = (\beta^*, \alpha)\beta, \qquad (\beta^*, \alpha) \in \mathbb{Z};$$

thus (ii) implies (i). Finally, if $\alpha = k\beta$, then $\alpha^* = k^{-1}\beta^*$, whence

$$(\alpha^*, \beta) = 2k^{-1}, \qquad (\beta^*, \alpha) = 2k.$$

Thus $2k$ and $2k^{-1}$ are integers, which gives the stated list. □

Application 0.2.1.2. *Root data $\mathcal{R}$ with* $\mathrm{rgss}(\mathcal{R}) = 1$ *are of one of the following two types:*

(i) Type A_1: there exists $\alpha \in M$ such that the roots are α and $-\alpha$. The coroots are then α^* and $-\alpha^*$.

(ii) Type A_1': there exists $\alpha \in M$ such that the roots are $\alpha, -\alpha, 2\alpha, -2\alpha$. The coroots are then $\alpha^*, -\alpha^*, \alpha^*/2, -\alpha^*/2$.

Definition 0.2.1.3. A root $\alpha \in R$ is called *indivisible* if $\alpha/2 \notin R$. The root datum $\mathcal{R}$ is called *reduced* if every root is indivisible.

The root datum $\mathcal{R}$ is reduced if and only if $\mathcal{R}^*$ is reduced. If α is indivisible and $w \in W$, then $w(\alpha)$ is indivisible.

Definition 0.2.1.4. Let $\alpha \in R$. If α is indivisible, set $\mathrm{ind}(\alpha) = \alpha$; otherwise set $\mathrm{ind}(\alpha) = \alpha/2$.

Corollary 0.2.1.5. *If $\alpha \in R$ is indivisible and $k\alpha \in R$, where $k \in \mathbb{Q}$, then $k \in \mathbb{Z}$.*

Proposition 0.2.1.6. *Let $\mathcal{R}$ be a root datum. Then*

$$\mathrm{ind}(\mathcal{R}) = \big(M, M^*, \mathrm{ind}(R), \mathrm{ind}(R)^*\big)$$

is a reduced root datum, and

$$W\big(\mathrm{ind}(\mathcal{R})\big) = W(\mathcal{R}).$$

Proof. The quadruple $\mathrm{ind}(\mathcal{R})$ is a root datum by Corollary 0.1.2.11, since the Weyl group permutes the indivisible roots. The second assertion follows from Proposition 0.2.1.1. □

Remark 0.2.1.7. If $\mathcal{R}$ is not reduced, then

$$\mathrm{ind}(R)^* \neq \mathrm{ind}(R^*), \qquad \mathrm{ind}(\mathcal{R}^*) \neq \mathrm{ind}(\mathcal{R})^*.$$

0.2.2 Orthogonal roots

Lemma 0.2.2.1. *Let α and β be two roots. Then*

$$\ell(\alpha)(\alpha^*, \beta) = \ell(\beta)(\beta^*, \alpha). \tag{11}$$

Proof. This follows immediately from Lemma 0.1.2.1, formulas (9) and (10). □

Corollary 0.2.2.2. *Let $\alpha, \beta \in R$. The following conditions are equivalent:*

(i) $(\alpha^*, \beta) = 0$;

(i bis) $(\beta^*, \alpha) = 0$;

(ii) $(p(\alpha), \beta) = 0$;

(iii) $s_\alpha(\beta) = \beta$;

(iii bis) $s_\beta(\alpha) = \alpha$;

(iv) $s_\alpha(\beta^*) = \beta^*$;

(iv bis) $s_\beta(\alpha^*) = \alpha^*$;

(v) $s_\alpha \neq s_\beta$, *and* s_α *and* s_β *commute.*

Proof. All the equivalences are immediate except those involving (v). We first show that (i), together with its equivalent condition (i bis), implies (v). One has

$$s_\alpha s_\beta(x) = x - (\beta^*, x)\beta - (\alpha^*, x)\alpha + (\beta^*, x)(\alpha^*, \beta)\alpha.$$

If $(\alpha^*, \beta) = 0$, then $(\beta^*, \alpha) = 0$, and consequently

$$s_\alpha s_\beta(x) = s_\beta s_\alpha(x).$$

Conversely, suppose (v). Then

$$s_\alpha = s_\beta s_\alpha s_\beta = s_{s_\beta(\alpha)} \qquad \text{by Corollary 0.1.2.10.}$$

By Proposition 0.2.1.1, there is a $k \in \mathbb{Q}^*$ such that

$$\alpha = k\, s_\beta(\alpha) = k\alpha - k(\beta^*, \alpha)\beta.$$

Since $s_\alpha \neq s_\beta$, the roots α and β are not proportional, again by Proposition 0.2.1.1. It follows that $(\beta^*, \alpha) = 0$. □

Definition 0.2.2.3. Two roots satisfying the equivalent conditions of Corollary 0.2.2.2 are called *orthogonal*.

Remark 0.2.2.4. The roots α and β are orthogonal if and only if the coroots α^* and β^* are orthogonal.

Lemma 0.2.2.5. *If α and β are two orthogonal roots, then $\alpha + \beta \in R$ if and only if $\alpha - \beta \in R$.*

Proof. Indeed,

$$s_\beta(\alpha + \beta) = s_\beta(\alpha) + s_\beta(\beta) = \alpha - \beta.$$

□

Lemma 0.2.2.6. *Let α and β be two nonorthogonal roots. For a coroot γ^*, define $\ell(\gamma^*)$ to be the value of ℓ on the root γ^* of $\mathcal{R}^*$. Then*

$$\ell(\alpha)\ell(\alpha^*) = \ell(\beta)\ell(\beta^*). \tag{12}$$

Proof. Multiply formula (11) by the corresponding formula for $\mathcal{R}^*$, and use $(\gamma^*)^* = \gamma$ for every $\gamma \in R$. This gives

$$(\beta^*, \alpha)(\alpha^*, \beta)\ell(\alpha)\ell(\alpha^*) = (\beta^*, \alpha)(\alpha^*, \beta)\ell(\beta)\ell(\beta^*).$$

The common integer factor is nonzero because the roots are not orthogonal, so it may be cancelled. □

0.2.3 General case

Proposition 0.2.3.1. *If α and β are any two roots, then*

$$0 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 4. \tag{13}$$

If α and β are neither proportional nor orthogonal, then

$$1 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 3.$$

Proof. One has

$$4(p(\alpha), \beta)^2 = \ell(\alpha)\ell(\beta)(\alpha^*, \beta)(\beta^*, \alpha).$$

On the other hand, by Corollary 0.1.2.6, the symmetric bilinear form $(p(x), y)$ is positive and nondegenerate on $\mathcal{V}(R)$. Hence

$$(p(x), y)^2 \leq \ell(x)\ell(y).$$

[5] The first inequality follows. In the nonproportional, nonorthogonal case, the product is a nonzero integer and cannot equal 4, which gives the second inequality. □

Corollary 0.2.3.2. *Let α and β be two nonorthogonal roots. If $\ell(\alpha) = \ell(\beta)$, then there is a $w \in W$ such that $w(\alpha) = \beta$.*

Proof. If α and β are proportional, equality of their ℓ-values gives $\alpha = \beta$ or $\alpha = -\beta$; take $w = 1$ or $w = s_\alpha$, respectively. If they are neither proportional nor orthogonal, formula (11) and Proposition 0.2.3.1 give

$$(\beta^*, \alpha) = (\alpha^*, \beta) = \pm 1.$$

If both integers are 1, take $w = s_\beta s_\alpha s_\beta$. If both are -1, take $w = s_\alpha s_\beta$. □

Corollary 0.2.3.3. *Let α and β be roots. If $\alpha \neq \beta$ and $(\beta^*, \alpha) > 0$, then $\alpha - \beta$ is a root. Respectively, if $\alpha \neq -\beta$ and $(\beta^*, \alpha) < 0$, then $\alpha + \beta$ is a root.*

Proof. The second case follows from the first by replacing β by $-\beta$. Suppose first that α and β are proportional and that $(\beta^*, \alpha) > 0$. Then

$$\beta = \alpha, \qquad 2\beta = \alpha, \qquad \text{or} \qquad \beta = 2\alpha.$$

[5] Editors' note: equality holds if and only if x and y are proportional.

The first case is excluded. In the other two cases, $\alpha - \beta = \beta \in R$ and $\alpha - \beta = -\alpha \in R$, respectively.

If α and β are not proportional, then (α^*, β) and (β^*, α) are strictly positive integers whose product is at most 3. Thus one of them equals 1. If $(\beta^*, \alpha) = 1$, then

$$\alpha - \beta = s_\beta(\alpha) \in R;$$

if $(\alpha^*, \beta) = 1$, then

$$\alpha - \beta = -s_\alpha(\beta) \in R.$$

□

Lemma 0.2.3.4. *Let α and β be two nonproportional roots. If $\beta - \alpha$ is not a root, then $\beta + k\alpha \in R$ for $k = -(\alpha^*, \beta)$, but not for $k = -(\alpha^*, \beta) + 1$.*

Proof. Indeed,

$$\beta - (\alpha^*, \beta)\alpha = s_\alpha(\beta) \in R,$$

whereas

$$\beta + (-(\alpha^*, \beta) + 1)\alpha = s_\alpha(\beta - \alpha) \notin R.$$

□

Proposition 0.2.3.5. *Let α and β be two nonproportional roots. The set of integers k for which $\beta + k\alpha \in R$ is an interval $[p, q]$, with $p \leq 0$ and $q \geq 0$, and*

$$p + q = -(\alpha^*, \beta).$$

Proof. For the first assertion it suffices, for example, to show that if $\beta + k\alpha \in R$ and $k > 0$ is an integer, then $\beta + (k-1)\alpha \in R$. This is trivial if $k = 1$. If $k \geq 2$, then

$$(\alpha^*, \beta + k\alpha) = (\alpha^*, \beta) + 2k \geq -3 + 4 > 0,$$

and the assertion follows from Corollary 0.2.3.3.

Let $[p, q]$ be the interval in question. Applying Lemma 0.2.3.4 to $\beta + p\alpha$, one obtains

$$q - p = -(\alpha^*, \beta + p\alpha) = -(\alpha^*, \beta) - 2p.$$

[6] The asserted formula follows. □

Remark 0.2.3.6. The preceding formula contains the qualitative statements of Lemma 0.2.2.5 and Corollary 0.2.3.3.

Complements 0.2.3.7. [7] *By formulas* (9) *and* (10)*, and by Corollary 0.1.2.6, the bilinear form on $\mathcal{V}(R)$ defined by*

$$\langle x, y \rangle = (p(x), y)$$

is symmetric and positive definite. For every $\alpha \in R$ and $y \in \mathcal{V}(R)$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y), \qquad \ell(\alpha) = \langle \alpha, \alpha \rangle.$$

Consequently, Corollary 0.2.3.3 gives the following corollary.

Corollary. *Let $\alpha \neq \beta$ in R. If $\alpha - \beta \notin R$, then*

$$\langle \alpha, \beta \rangle \leq 0.$$

[6] Editors' note: in this equality, $+2p$ was corrected to $-2p$.

[7] Editors' note: these complements, useful for the proof of Proposition 0.3.1.5, were added by the editors.

0.3 Simple roots, positive roots

0.3.1 Systems of simple roots

Lemma 0.3.1.1. *Let α and α_i, $i = 1, \dots, n$, be roots, and suppose that α is distinct from the α_i. If*

$$\alpha = \sum_{i=1}^{n} q_i \alpha_i, \qquad q_i \in \mathbb{Q}_+,$$

then there is an index i such that

$$q_i \neq 0, \qquad (\alpha^*, \alpha_i) > 0, \qquad \alpha - \alpha_i \in R.$$

Proof. Write

$$2 = (\alpha^*, \alpha) = \sum_{i=1}^{n} q_i (\alpha^*, \alpha_i).$$

This proves the first two assertions. The third then follows from Corollary 0.2.3.3. □

Proposition 0.3.1.2. *Let α and α_i, $i = 1, \dots, n$, be roots. If*

$$\alpha = \sum_{i=1}^{n} m_i \alpha_i, \qquad m_i \in \mathbb{N},$$

then there is a sequence $\beta_1, \dots, \beta_m$ of roots chosen from among the α_i, not necessarily pairwise distinct, such that, on setting

$$\gamma_p = \sum_{i=1}^{p} \beta_i, \qquad p = 1, \dots, m,$$

one has $\gamma_p \in R$ and $\gamma_m = \alpha$.

Proof. We argue by induction on the integer $m' = \sum_i m_i$. If α equals one of the α_i, say α_{i_0}, which necessarily happens when $m' = 1$, take

$$m = 1, \qquad \beta_1 = \alpha_{i_0}.$$

Otherwise, Lemma 0.3.1.1 gives an index i for which $m_i \neq 0$ and $\alpha - \alpha_i$ is a root. Then $m_i - 1 \in \mathbb{N}$ and

$$\alpha - \alpha_i = (m_i - 1)\alpha_i + \sum_{j \neq i} m_j \alpha_j.$$

Apply the induction hypothesis to $\alpha - \alpha_i$, and append α_i to the resulting sequence. □

Corollary 0.3.1.3. *Let $R' \subset R$. The following conditions are equivalent:*

(i) if $\alpha, \beta \in R'$ and $\alpha + \beta \in R$, then $\alpha + \beta \in R'$;

(ii) $(\mathbb{N} \cdot R') \cap R \subset R'$.

Proof. Clearly (ii) implies (i). The converse follows immediately from Proposition 0.3.1.2. □

Definition 0.3.1.4. A set of roots satisfying the equivalent conditions of Corollary 0.3.1.3 is called *closed*.

Proposition 0.3.1.5. *Let $\Delta \subset R$ be a set of roots. The following conditions are equivalent:*

(i) the elements of Δ are indivisible and linearly independent, and

$$R \subset (\mathbb{Q}_+ \cdot \Delta) \cup (-\mathbb{Q}_+ \cdot \Delta);$$

(ii) the elements of Δ are linearly independent, and

$$R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta);$$

(iii) every root can be written uniquely as a linear combination of the elements of Δ, with integer coefficients all of the same sign.

Proof. [8] It is clear that (ii) implies (iii). Write $\alpha_1, \ldots, \alpha_n$ for the distinct elements of Δ.

Assume (i), and let $\alpha \in R$. There is a unique expression

$$\varepsilon\alpha = \sum_i q_i\alpha_i, \qquad q_i \in \mathbb{Q}_+, \qquad \varepsilon = \pm 1.$$

We show that the q_i are integers. This is immediate when all but one of them vanish, by Corollary 0.2.1.5. Otherwise, $\varepsilon\alpha$ is distinct from the α_i, and Lemma 0.3.1.1 gives an i_0 such that

$$\alpha' = \varepsilon\alpha - \alpha_{i_0} \in R, \qquad q_{i_0} \neq 0.$$

Thus

$$\alpha' = (q_{i_0} - 1)\alpha_{i_0} + \sum_{i \neq i_0} q_i\alpha_i.$$

At least one q_i, with $i \neq i_0$, is nonzero, so (i) forces $q_{i_0} - 1 \geq 0$. Repeating this operation with α', after finitely many steps we find that all q_i are integers. This proves (ii).

Assume now (iii). We first prove that $\alpha_1, \ldots, \alpha_n$ are linearly independent over $\mathbb{Q}$. Otherwise there would be an equality

$$x = \sum_{i \in I} a_i\alpha_i = \sum_{j \in J} b_j\alpha_j,$$

where I and J are disjoint subsets of $\{1, \ldots, n\}$, at least one of them, say I, is nonempty, and $a_i, b_j \in \mathbb{N}^*$. By the corollary in Complements 0.2.3.7, one has $\langle \alpha_i, \alpha_j \rangle \leq 0$ for $i \in I$ and $j \in J$. Therefore $\langle x, x \rangle \leq 0$, and hence $x = 0$.

Choose $i_0 \in I$. The two expressions

$$\alpha_{i_0} = \alpha_{i_0} + \sum_{i \in I} a_i\alpha_i = \alpha_{i_0} + \sum_{j \in J} b_j\alpha_j$$

then imply, by (iii), that $a_i = 0 = b_j$ for all i, j, a contradiction. Thus the elements of Δ are linearly independent over $\mathbb{Q}$.

It remains to show that they are indivisible. Suppose that $\beta \in \Delta$ is divisible. Then $\beta/2 \in R$, so

$$\frac{\beta}{2} = \varepsilon \sum_{\alpha \in \Delta} m_\alpha\alpha, \qquad m_\alpha \in \mathbb{N}, \qquad \varepsilon = \pm 1.$$

Consequently,

$$\beta = 2\varepsilon \sum_{\alpha \in \Delta} m_\alpha\alpha.$$

Uniqueness gives $m_\alpha = 0$ for $\alpha \neq \beta$ and $2\varepsilon m_\beta = 1$, a contradiction. □

[8] Editors' note: in the argument that follows, the order was changed and the proof of the implication (iii)⇒(i) was spelled out.

Definition 0.3.1.6. A set Δ of roots satisfying the conditions of Proposition 0.3.1.5 is called a *system of simple roots*, or a *base* of $\mathcal{R}$.

If $w \in W$ and Δ is a system of simple roots, then $w(\Delta)$ is a system of simple roots.

Remark 0.3.1.7. This definition involves only R, not $\mathcal{R}$; indeed, it involves only $\mathrm{ind}(R)$.

Remark 0.3.1.8. If Δ is a system of simple roots, then Δ is a basis of the free abelian group $\Gamma_0(R)$. Hence

$$\mathrm{Card}(\Delta) = \mathrm{rgss}(\mathcal{R}).$$

Remark 0.3.1.9. The conditions of Proposition 0.3.1.5 are therefore equivalent to the following:

(i′) the elements of Δ are indivisible, their number is at most $\mathrm{rgss}(\mathcal{R})$, and

$$R \subset (\mathbb{Q}_+ \cdot \Delta) \cup (-\mathbb{Q}_+ \cdot \Delta);$$

(ii′) $\mathrm{Card}(\Delta) \leq \mathrm{rgss}(\mathcal{R})$, and

$$R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta).$$

Corollary 0.3.1.10. *If Δ is a system of simple roots, then* $\mathrm{ind}(\Delta^*)$ *is a system of simple coroots* (i.e. *a system of simple roots of $\mathcal{R}^*$*).

Proof. If

$$\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha, \qquad a_\alpha \in \mathbb{N},$$

then

$$p(\beta) = \sum_{\alpha \in \Delta} a_\alpha p(\alpha).$$

Consequently, by (10),

$$\beta^* = \sum_{\alpha \in \Delta} a_\alpha \frac{\ell(\alpha)}{\ell(\beta)} \alpha^*,$$

which shows that $\mathrm{ind}(\Delta^*)$ satisfies condition (i′) of Remark 0.3.1.9. □

By Proposition 0.2.1.1, if α^* is not indivisible, then

$$\alpha^* = 2u^*, \qquad u^* \in \mathrm{ind}(\Delta^*) \quad \text{and} \quad u = 2\alpha.$$

It follows that:[9]

Corollary 0.3.1.11. *Let Δ be a system of simple roots, and let*

$$\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha, \qquad a_\alpha \in \mathbb{Z},$$

be the expression of β relative to Δ. Then $\ell(\beta)$ divides $2a_\alpha \ell(\alpha)$; it even divides $a_\alpha \ell(\alpha)$ if α^* is indivisible.

Before continuing to state the properties of systems of simple roots, we show that such systems exist.

[9] Editors' note: The preceding sentence was added, and in Corollary 3.1.11 the expression $4a_\alpha \ell(\alpha)$ was replaced by $2a_\alpha \ell(\alpha)$.

0.3.2 Systems of positive roots

Definition 0.3.2.1. A subset $R_+ \subset R$ is called a *system of positive roots* of $\mathcal{R}$ *(or of R; see Remark 0.3.2.2)* if it satisfies:

(P 1) $R_+ \cap (-R_+) = \varnothing$;

(P 2) $R_+ \cup (-R_+) = R$;

(P 3) $(\mathbb{Q}_+ \cdot R_+) \cap R \subset R_+$.

In particular, such a set is closed. We shall see later, in Corollary 0.3.3.8, that a closed set satisfying (P 1) and (P 2) also satisfies (P 3), and hence is a system of positive roots. If $w \in W$ and R_+ is a system of positive roots, then $w(R_+)$ is a system of positive roots.

Remark 0.3.2.2. This definition involves only R. We shall also say that R_+ is a system of positive roots of R.

Remark 0.3.2.3. Conditions (P 1) and (P 2) immediately give

$$\mathrm{Card}(R_+) = \frac{\mathrm{Card}(R)}{2}.$$

It follows that if R_+ and R'_+ are two systems of positive roots and $R_+ \subset R'_+$, then $R_+ = R'_+$.

Remark 0.3.2.4. If R_+ is a system of positive roots, then R^*_+ is a system of positive coroots *(i.e.* a system of positive roots of R^**)*.

This follows immediately from Lemma 0.1.1.4 and Scholium 0.1.2.2.

Definition 0.3.2.5. Let Δ be a system of simple roots. Set

$$\mathscr{P}(\Delta) = (\mathbb{Q}_+ \cdot \Delta) \cap R = (\mathbb{N}_+ \cdot \Delta) \cap R.$$

Proposition 0.3.2.6. *If Δ is a system of simple roots, then $\mathscr{P}(\Delta)$ is a system of positive roots. If Δ is a system of simple roots and R_+ is a system of positive roots, then*

$$\Delta \subset R_+ \quad \Longleftrightarrow \quad R_+ = \mathscr{P}(\Delta).$$

Proof. The first assertion is immediate. If $\Delta \subset R_+$, then $\mathscr{P}(\Delta) \subset R_+$ by (P 3), and therefore $\mathscr{P}(\Delta) = R_+$ by Remark 0.3.2.3. The rest is trivial. □

Remark 0.3.2.7. Systems of positive roots exist: endow $\mathcal{V}(R)$ with the structure of a totally ordered vector space. The set of roots that are ≥ 0 for this order is a system of positive roots.

Theorem 0.3.2.8. *Let R_+ be a system of positive roots. There exists a unique system of simple roots $\mathscr{S}(R_+)$ such that*

$$\mathscr{S}(R_+) \subset R_+, \qquad \textit{i.e.} \qquad R_+ = \mathscr{P}(\mathscr{S}(R_+)).$$

Uniqueness follows immediately from:

Lemma 0.3.2.9. *Let Δ be a system of simple roots. An element $\alpha \in \mathscr{P}(\Delta)$ belongs to Δ if and only if α is not the sum of two elements of $\mathscr{P}(\Delta)$.*

Proof. This follows immediately from the definitions and Proposition 0.3.1.2. □

We now prove the existence of $\mathscr{S}(R_+)$. Consider the set of subsets T of R_+ such that

$$T = \mathrm{ind}(T) \qquad \text{and} \qquad \mathbb{Q}_+ \cdot T \supset R_+.$$

This set is nonempty, since it contains $\mathrm{ind}(R_+)$. Let Δ be a minimal element of this set under inclusion. We show that Δ is a system of simple roots; by Proposition 0.3.1.5(i), it is enough to show that Δ is a linearly independent subset of $M \otimes \mathbb{Q}$.

Lemma 0.3.2.10. *If $\alpha, \beta \in \Delta$ and*

$$q\alpha + q'\beta \in R, \qquad q, q' \in \mathbb{Q},$$

then $qq' \geq 0$.

Proof. If $qq' < 0$, then, after interchanging α and β if necessary, we may write

$$a\alpha - b\beta \in R_+, \qquad a, b \in \mathbb{Q}_+^*.$$

By the defining property of Δ, there is a relation

$$a\alpha - b\beta = \sum_{\gamma \in \Delta} c(\gamma)\gamma, \qquad c(\gamma) \in \mathbb{Q}_+.$$

If $a \leq c(\alpha)$, this becomes

$$-\beta = \frac{c(\alpha) - a}{b}\alpha + \sum_{\gamma \neq \alpha} \frac{c(\gamma)}{b}\gamma.$$

Thus $-\beta \in (\mathbb{Q}_+ \cdot \Delta) \cap R$, which is contained in R_+ by (P 3). This gives $\beta \in R_+ \cap (-R_+)$, contradicting (P 1).

If instead $a > c(\alpha)$, then

$$(a - c(\alpha))\alpha = b\beta + \sum_{\gamma \neq \alpha} c(\gamma)\gamma,$$

which shows that

$$\Delta \subset \mathbb{Q}_+ \cdot (\Delta \setminus \{\alpha\}),$$

contrary to the minimality of Δ. □

Recall from Complements 0.2.3.7 that the bilinear form on $\mathcal{V}(R)$ defined by

$$\langle x, y \rangle = (p(x), y)$$

is a Euclidean inner product. Moreover, for $\alpha \in R$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y),$$

so $\langle \alpha, y \rangle$ and (α^*, y) have the same sign.

Lemma 0.3.2.11. *If $\alpha, \beta \in \Delta$, then*

$$(\beta^*, \alpha) \leq 0 \qquad \textit{and hence} \qquad \langle \beta, \alpha \rangle \leq 0.$$

Proof. Indeed,

$$s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta \in R,$$

and therefore $(\beta^*, \alpha) \leq 0$ by Lemma 0.3.2.10. □

We now show that Δ is linearly independent. Otherwise, Lemma 0.3.2.11 would imply, exactly as in the proof of Proposition 0.3.1.5, the existence of a nontrivial relation

$$\sum_{\alpha\in\Delta} m(\alpha)\alpha = 0, \qquad m(\alpha)\in\mathbb{N}.$$

If $m(\alpha_0)\geq 1$, then

$$-\alpha_0 = \big(m(\alpha_0)-1\big)\alpha_0 + \sum_{\alpha\neq\alpha_0} m(\alpha)\alpha.$$

By (P 3), this gives $\alpha_0 \in R_+ \cap (-R_+)$, contradicting (P 1). Thus Δ is a base of $\mathcal{R}$, completing the proof of Theorem 0.3.2.8.

Corollary 0.3.2.12. *Let R_+ be a system of positive roots, let Δ be a base of $\mathcal{R}$, and let $w\in W$. Then*

$$\Delta\subset R_+ \iff R_+ = \mathscr{P}(\Delta) \iff \Delta = \mathscr{S}(R_+),$$

$$\mathscr{P}\big(\mathrm{ind}(\Delta^*)\big) = \mathscr{P}(\Delta)^*, \qquad \mathscr{S}(R_+^*) = \mathrm{ind}\big(\mathscr{S}(R_+)^*\big),$$

and

$$\mathscr{S}\big(w(R_+)\big) = w\big(\mathscr{S}(R_+)\big), \qquad \mathscr{P}\big(w(\Delta)\big) = w\big(\mathscr{P}(\Delta)\big).$$

Definition 0.3.2.13. Once a system of simple roots Δ has been chosen, the elements of $\mathscr{P}(\Delta)$ are called *positive*. Once a system of positive roots R_+ has been chosen, the elements of $\mathscr{S}(R_+)$ are called *simple*.

Corollary 0.3.2.14. *Let R_+ be a system of positive roots, and let $\alpha\in R_+$. The following conditions are equivalent:*

(i) α is simple, that is, $\alpha\in\mathscr{S}(R_+)$;

(ii) α is not the sum of two elements of R_+;

(iii) $R_+\setminus\{\alpha\}$ is closed.

Proof. The equivalence of (i) and (ii) follows immediately from Lemma 0.3.2.9; the equivalence of (ii) and (iii) is immediate from Definition 0.3.1.4. □

Definition 0.3.2.15. Let Δ be a system of simple roots. The sum of the coefficients in the expression of a root α relative to Δ is called the *order of α relative to Δ* and is denoted $\mathrm{ord}_\Delta(\alpha)$.

One has

$$\alpha\in\mathscr{P}(\Delta) \iff \mathrm{ord}_\Delta(\alpha)>0, \qquad \alpha\in\Delta \iff \mathrm{ord}_\Delta(\alpha)=1.$$

Lemma 0.3.2.16. *Let Δ be a system of simple roots and $\alpha\in\mathscr{P}(\Delta)$. There is a sequence $\alpha_i\in\Delta$, $i=1,\dots,m$, such that*

$$\alpha_1+\alpha_2+\cdots+\alpha_p\in\mathscr{P}(\Delta) \quad \textit{for } p=1,\dots,m,$$

and

$$\alpha_1+\alpha_2+\cdots+\alpha_m=\alpha.$$

Moreover, for every sequence satisfying these conditions,

$$m=\mathrm{ord}_\Delta(\alpha).$$

Proof. This is immediate from Proposition 0.3.1.2. □

0.3.3 Characterization and conjugacy of systems of positive roots

Lemma 0.3.3.1. *If $\alpha \in \mathscr{P}(\Delta)$, $\beta \in \Delta$, and α and β are not proportional, then*

$$s_\beta(\alpha) \in \mathscr{P}(\Delta).$$

Proof. One has

$$s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta.$$

At least one simple root other than β occurs with a nonzero, and hence strictly positive, coefficient in the expression of α. It occurs with the same coefficient in the expression of $s_\beta(\alpha)$; therefore $s_\beta(\alpha)$ is also positive. □

Corollary 0.3.3.2. *If $\beta \in \Delta$, then the reflection s_β permutes the elements of $\mathscr{P}(\Delta)$ that are not proportional to β.*

Lemma 0.3.3.3. *Let*

$$\alpha \in \mathscr{P}(\Delta) \setminus \Delta$$

be indivisible. There exists $\beta \in \Delta$ such that

$$s_\beta(\alpha) \in \mathscr{P}(\Delta) \qquad \textit{and} \qquad \mathrm{ord}_\Delta\big(s_\beta(\alpha)\big) < \mathrm{ord}_\Delta(\alpha).$$

Proof. By Lemma 0.3.1.1, there is a $\beta \in \Delta$ such that $(\beta^*, \alpha) > 0$. Since $\alpha \notin \Delta$ and α is indivisible, it cannot be proportional to β. Thus $s_\beta(\alpha) \in \mathscr{P}(\Delta)$ by Lemma 0.3.3.1, and

$$\mathrm{ord}_\Delta\big(s_\beta(\alpha)\big) = \mathrm{ord}_\Delta(\alpha) - (\beta^*, \alpha) < \mathrm{ord}_\Delta(\alpha).$$

□

Corollary 0.3.3.4. *If $\alpha \in \mathscr{P}(\Delta)$ is indivisible, then there is a sequence $\beta_p \in \Delta$, $p = 1, \ldots, q$, such that*

$$\alpha_p = s_{\beta_p} \cdots s_{\beta_1}(\alpha) \in \mathscr{P}(\Delta) \quad \textit{for } p = 1, \ldots, q,$$

and $\alpha_q \in \Delta$.

Proof. Apply Lemma 0.3.3.3 repeatedly, inducting on $\mathrm{ord}_\Delta(\alpha)$. □

Proposition 0.3.3.5. *The Weyl group is generated by the s_α, for $\alpha \in \Delta$. Every indivisible root is conjugate under an element of the Weyl group to a simple root.*

Proof. The second assertion follows immediately from Corollary 0.3.3.4. The first then follows from Corollary 0.1.2.10 and Proposition 0.2.1.1. □

Proposition 0.3.3.6. *Let R_+ be a system of positive roots, and let $P' \subset R$ be closed and satisfy (P 2). Then there exists $w \in W$ such that*

$$w(R_+) \subset P'.$$

We first state the corollaries.

Corollary 0.3.3.7. *The Weyl group acts transitively on the set of systems of positive roots, and on the set of systems of simple roots.*

Corollary 0.3.3.8. *A subset of R is a system of positive roots if and only if it is closed and satisfies (P 1) and (P 2).*

Corollary 0.3.3.9. *Suppose that $\Gamma_0(R)$ is endowed with an ordered-group structure for which every root is either > 0 or < 0. Then the set of positive roots for this order is a system of positive roots.*

Proof of Proposition 0.3.3.6. Choose $w \in W$ such that $\mathrm{Card}(w(R_+) \cap P')$ is maximal. Replacing R_+ by $w(R_+)$, we may assume that

$$(*) \qquad \mathrm{Card}(R_+ \cap P') \geq \mathrm{Card}(R_+ \cap s_\alpha(P')) \quad \text{for every } \alpha \in \mathscr{S}(R_+) = \Delta.$$

We show that $R_+ \subset P'$. Otherwise, since P' is closed, there is an $\alpha \in \Delta$ such that $\alpha \notin P'$. As P' satisfies (P 2), one then has $-\alpha \in P'$, and also $-2\alpha \in P'$ if 2α is a root.

For every $\beta \in R_+ \cap P'$, one has $\beta \neq \alpha$. If 2α is not a root, then $s_\alpha(\beta) \in R_+$ by Lemma 0.3.3.1; hence

$$s_\alpha(R_+ \cap P') \subset R_+ \cap s_\alpha(P').$$

But $R_+ \cap s_\alpha(P')$ also contains $\alpha = s_\alpha(-\alpha)$, contradicting $(*)$. If 2α is a root, the same argument applies. □

Thus, to study closed sets of roots satisfying (P 2), it is enough to consider those that contain a system of positive roots.

Proposition 0.3.3.10. *Let R_+ be a system of positive roots, and let P' be a closed subset of R containing R_+. Put*

$$\Delta = \mathscr{S}(R_+), \qquad \Delta' = \Delta \cap (-P').$$

Then P' is the union of R_+ and the set of roots that are linear combinations of the elements of Δ' with negative coefficients.

Proof. We argue by induction on the order of a root $\gamma \in P' \setminus R_+$. If $\mathrm{ord}_\Delta(\gamma) = -1$, then $-\gamma \in \Delta$, and hence $\gamma \in -\Delta'$.

Suppose that $\mathrm{ord}_\Delta(\gamma) < -1$. By Proposition 0.3.1.2, there exists $\beta \in \Delta$ such that $-\gamma - \beta \in R$. Then

$$0 < \mathrm{ord}_\Delta(-\gamma - \beta) = \mathrm{ord}_\Delta(-\gamma) - 1 < \mathrm{ord}_\Delta(-\gamma).$$

The first inequality shows that $-\gamma - \beta \in R_+$. Consequently, since $R_+ \cap (-R_+) = \varnothing$, and since $\gamma + \beta$ is the sum of two roots of P', the root $\gamma + \beta$ belongs to $P' \setminus R_+$ and satisfies

$$\mathrm{ord}_\Delta(\gamma + \beta) > \mathrm{ord}_\Delta(\gamma).$$

By induction, $\gamma + \beta$ is a linear combination of the elements of Δ' with negative coefficients. Since $\gamma = (\gamma + \beta) - \beta$, it remains only to show that $\beta \in -P'$. But $\gamma \in P'$ and $-(\gamma + \beta) \in R_+ \subset P'$, so

$$-\beta = \gamma - (\gamma + \beta) \in P'.$$

□

Definition 3.3.10.1. [10] A subset R' of R is called *symmetric* if

$$-R' = R'.$$

[10] Editors' note: This definition was inserted here.

Scholium 0.3.3.11. *Let P' be a set of roots satisfying (P 2) and closed. There exist a system of simple roots Δ and a subset Δ' of Δ such that*

$$P' = R \cap (\mathbb{N} \cdot \Delta \cup -\mathbb{N} \cdot \Delta').$$

If

$$R_{\Delta'} = (\mathbb{Z} \cdot \Delta') \cap R,$$

which is a closed and symmetric subset of R, then P' is the disjoint union of $R_{\Delta'}$ and the closed subset of $\mathscr{P}(\Delta)$ formed by the

$$\alpha \in \mathscr{P}(\Delta) \setminus \mathbb{N} \cdot \Delta',$$

that is, the positive roots whose decomposition with respect to Δ "contains at least one root of $\Delta \setminus \Delta'$".

0.3.4 Closed and symmetric sets of roots

Proposition 0.3.4.1. *Let*

$$\mathcal{R} = (M, M^*, R, R^*)$$

be a root datum, and let R' be a closed and symmetric subset of R. Then:

(i) $\mathcal{R}' = (M, M^*, R', R'^*)$ is a root datum;

(ii) for every system of positive roots R_+ of R, $R'_+ = R_+ \cap R'$ is a system of positive roots of R';

(iii) the Weyl group $W(\mathcal{R}')$ of $\mathcal{R}'$ is the subgroup of $W(\mathcal{R})$ generated by the s_α, for $\alpha \in R'$.

Proof. The first assertion is immediate from Corollary 0.1.2.11; the second follows from Corollary 0.3.3.8; and the third is evident. □

Corollary 0.3.4.2. *Let $w \in W(\mathcal{R}')$. The order of w is the smallest integer $n > 0$ such that*

$$w^n(\alpha') = \alpha' \qquad \text{for every } \alpha' \in R'.$$

Proof. Apply Corollary 0.1.2.7 to the root datum $\mathcal{R}'$. □

Corollary 0.3.4.3. *Let α and β be two nonproportional roots. Let n be the smallest integer > 0 such that*

$$(s_\alpha s_\beta)^n(\alpha) = \alpha, \qquad (s_\alpha s_\beta)^n(\beta) = \beta.$$

Then the subgroup $W_{\alpha,\beta}$ of W generated by s_α and s_β is defined by the relations

$$s_\alpha^2 = 1, \qquad s_\beta^2 = 1, \qquad (s_\alpha s_\beta)^n = 1.$$

Proof. In view of Corollary 0.3.4.2, it remains only to verify the following lemma. □

Lemma 0.3.4.4. *Let G be the group generated by two elements x and y subject to the relations $x^2 = y^2 = 1$. Every normal subgroup of G containing neither x nor y is generated, as a normal subgroup, by an element of the form $(xy)^n$, where $n > 0$.*

Proof. [11] Every element of G can be written as $(xy)^n$, as $(yx)^n = (xy)^{-n}$, or in one of the forms

$$(a) \qquad x(yx)^{2n} \qquad \text{or } y(xy)^{2n+1},$$
$$(b) \qquad y(xy)^{2n} \qquad \text{or } x(yx)^{2n+1},$$

where $n \in \mathbb{N}$. The elements of type (a), respectively of type (b), are conjugate to x, respectively to y. □

Remark 0.3.4.5. The integer n is computed immediately. Put

$$(\alpha^*, \beta) = p, \qquad (\beta^*, \alpha) = q.$$

Then[12]

$$(s_\alpha s_\beta)(\alpha) = (pq-1)\alpha - q\beta, \qquad (s_\alpha s_\beta)(\beta) = p\alpha - \beta.$$

Thus n is the order of the matrix

$$\begin{pmatrix} pq-1 & p \\ -q & -1 \end{pmatrix}.$$

If $pq = 0$, then $p = q = 0$ by Corollary 0.2.2.2, whence $n = 2$. Otherwise, Proposition 0.2.3.1 shows that pq equals 1, 2, or 3, and one finds respectively $n = 3$, 4, or 6.

N.B. Expressing the finiteness of the order of the preceding matrix recovers inequality (13) of Proposition 0.2.3.1.

Definition 0.3.4.6. Let Δ be a system of simple roots and let $\Delta' \subset \Delta$. We write

$$R_{\Delta'} = R \cap (\mathbb{Q} \cdot \Delta') = R \cap (\mathbb{Z} \cdot \Delta').$$

Lemma 0.3.4.7. *The set $R_{\Delta'}$ is closed and symmetric, and Δ' is a system of simple roots of the root datum*

$$(M, M^*, R_{\Delta'}, R^*_{\Delta'}).$$

Its Weyl group is the subgroup $W_{\Delta'}$ of W generated by the s_α, for $\alpha \in \Delta'$. Moreover,

$$\Delta \cap R_{\Delta'} = \Delta'.$$

Proof. Trivial. □

Proposition 0.3.4.8. *Let $R' \subset R$. The following conditions are equivalent:*

(i) There exists a vector subspace V' of V, or of $\mathcal{V}(R)$, such that $R' = R \cap V'$.

(ii) There exist a system of simple roots Δ of R and a subset Δ' of Δ such that $R' = R_{\Delta'}$.

More precisely, under these conditions, every system of simple roots Δ' of (M, M^*, R', R'^*) is contained in a system of simple roots Δ of R, and one has $R' = R_{\Delta'}$.

Proof. The implication (ii) ⇒ (i) is clear. Suppose (i). Then R' is closed and symmetric, so (M, M^*, R', R'^*) is a root datum. Let Δ' be a system of simple roots of this datum, and put

$$R'_+ = \mathscr{P}(\Delta').$$

[11] Editors' note: The original has been corrected in what follows.
[12] Editors' note: The original has been corrected in what follows.

If $V' = V$, then Δ' is a system of simple roots of $\mathcal{R}$, and there is nothing to prove. Otherwise, there exists $x \in V^*$ such that

$$(x, V') = \{0\}, \qquad (x, \alpha) \neq 0 \quad \text{for every } \alpha \in R \setminus R'.$$

Put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}, \qquad R_+ = R_+(x) \cup R'_+.$$

For every $\alpha \in R$, one has

$$(x, \alpha) > 0 \Longleftrightarrow \alpha \in R_+(x),$$
$$(x, \alpha) < 0 \Longleftrightarrow \alpha \in -R_+(x),$$
$$(x, \alpha) = 0 \Longleftrightarrow \alpha \in R'.$$

It follows at once from Corollary 0.3.3.8 that R_+ is a system of positive roots of R. Put $\Delta = \mathscr{S}(R_+)$. It is enough to prove that $\Delta' \subset \Delta$. Otherwise, let $\alpha \in \Delta' \setminus \Delta$. By Corollary 0.3.2.14, there exist $\beta, \gamma \in R_+$ such that

$$\alpha = \beta + \gamma.$$

If $\beta, \gamma \in R_+(x)$, then $\alpha \in R_+(x)$, which is impossible because $(x, \Delta') = 0$. If one of β, γ, say β, belongs to R'_+, then, since R' is symmetric and closed,

$$\gamma = \alpha - \beta \in R_+ \cap R' = R'_+.$$

But then α is not simple in R'_+, again a contradiction. □

Lemma 0.3.4.9. *Under the preceding conditions, let*

$$\alpha \in \mathscr{P}(\Delta) \setminus R'.$$

For every $w \in W_{\Delta'}$, one has

$$w(\alpha) \in \mathscr{P}(\Delta) \setminus R'.$$

Proof. It suffices to verify the assertion for $w = s_\beta$, $\beta \in \Delta'$. It then follows from Lemma 0.3.3.1 and from $s_\beta(R') = R'$. □

Lemma 0.3.4.10. *Let $w \in W$. Under the conditions of Proposition 0.3.4.8, the following conditions are equivalent:*

(i) $w \in W_{\Delta'}$.

(ii) For every $m \in M$, one has $w(m) - m \in V'$.

(iii) For every $\alpha \in R$, one has $w(\alpha) - \alpha \in V'$.

Proof. Clearly (i) implies (ii), and (ii) implies (iii). Suppose (iii). Write

$$w = s_{\alpha_n} \cdots s_{\alpha_1}, \qquad \alpha_i \in \Delta,$$

and argue by induction on n that every α_i belongs to Δ'. We may assume that

$$w' = s_{\alpha_{n-1}} \cdots s_{\alpha_1} \in W_{\Delta'}.$$

Then, for every $\alpha \in R$,

$$w(\alpha) - \alpha = w'(\alpha) - \alpha - (\alpha_n^*, w'(\alpha))\alpha_n.$$

Taking $\alpha = {w'}^{-1}(\alpha_n)$, one obtains $2\alpha_n \in V'$, hence $\alpha_n \in \Delta'$. Therefore

$$w = s_{\alpha_n} w' \in W_{\Delta'}.$$

□

0.3.5 Miscellaneous remarks

Proposition 0.3.5.1. *Let R_+ be a system of positive roots. Put*

$$\rho_{R_+} = \frac{1}{2} \sum_{\alpha \in \mathrm{ind}(R_+)} \alpha.$$

Then

$$(\beta^*, \rho_{R_+}) = 1 \qquad \text{for every } \beta \in \mathscr{S}(R_+).$$

Proof. One can write

$$2\rho_{R_+} = \beta + \sum_{\substack{\alpha \in \mathrm{ind}(R_+) \\ \alpha \neq \beta}} \alpha.$$

Hence

$$s_\beta(2\rho_{R_+}) = 2\rho_{R_+} - 2\beta$$

by Lemma 0.3.3.2. □

Corollary 0.3.5.2. *Put*

$$\rho^*_{R_+} = \frac{1}{2} \sum_{\alpha^* \in \mathrm{ind}(R^*_+)} \alpha^*.$$

Then:

(i) $(\rho^*_{R_+}, \alpha) > 0$ *for every* $\alpha \in R_+$*; equivalently,* $\rho^*_{R_+} \in \mathcal{C}(R_+)$*, in the notation of Definition 0.3.6.8.*

(ii) For every $\alpha \in R$*,*

$$\mathrm{ord}_{\mathscr{S}(R_+)}(\alpha) = (\rho^*_{R_+}, \alpha).$$

[13]

Remark 0.3.5.3. If $w \in W$, then

$$\rho_{w(R_+)} = w(\rho_{R_+}), \qquad \rho^*_{w(R_+)} = w(\rho^*_{R_+}).$$

Proposition 0.3.5.4. *Let α and γ be two nonproportional roots, with α indivisible. There exist a system of simple roots containing α and a root β such that*

$$\gamma = a\alpha + b\beta, \qquad a, b \in \mathbb{N}.$$

Proof. Choose a basis of the vector space $\mathcal{V}(R)$ containing $\alpha_1 = \alpha$ and $\alpha_2 = \gamma$, and consider the lexicographic order with respect to this basis. Let R_+ be the set of roots > 0. It is clear that R_+ is a system of positive roots and[14] that the least element of R_+ not proportional to α is simple. This element has the form

$$\beta = p\alpha + q\gamma, \qquad 0 < q \leq 1.$$

Consequently

$$\gamma = q^{-1}\beta - q^{-1}p\alpha.$$

Since $q^{-1} > 0$, one has

$$q^{-1} \in \mathbb{N}^*, \qquad -q^{-1}p \in \mathbb{N}.$$

□

[13] Editors' note: By Lemma 0.3.1.5, it is enough to verify the formula for $\alpha \in \Delta$. Proposition 0.3.5.1, applied to R^*, then gives $(\rho^*_{R_+}, \alpha) = 1 = \mathrm{ord}_{\mathscr{S}(R_+)}(\alpha)$.

[14] Editors' note: The original was modified here to take account of the case in which 2α is a root.

We conclude with two remarks on the group $\Gamma_0(R)$.

Proposition 0.3.5.5. *Let G be an abelian group, and let $f : R \to G$ be a map satisfying:*

(i) if $\alpha \in R$, then $f(-\alpha) = -f(\alpha)$;

(ii) if $\alpha, \beta, \alpha + \beta \in R$, then

$$f(\alpha + \beta) = f(\alpha) + f(\beta).$$

There exists a unique group homomorphism

$$\overline{f} : \Gamma_0(R) \longrightarrow G$$

such that $\overline{f}(\alpha) = f(\alpha)$ for every $\alpha \in R$.

Proof. Let Δ be a system of simple roots of R, and write

$$\beta = \sum_{\alpha \in \Delta} a(\alpha)\alpha.$$

Lemma 0.3.2.16 gives at once

$$f(\beta) = \sum_{\alpha \in \Delta} a(\alpha) f(\alpha).$$

Now Δ is a basis of $\Gamma_0(R)$. □

Proposition 0.3.5.6. *Let Δ be a system of simple roots. There exists on $\Gamma_0(R)$ a totally ordered group structure such that the roots > 0 are the elements of $\mathscr{P}(\Delta)$, and such that*

$$\alpha \longmapsto \mathrm{ord}_\Delta(\alpha)$$

is increasing.

Proof. Let

$$\alpha_1, \alpha_2, \ldots, \alpha_n, \qquad n = \mathrm{rgss}(\mathcal{R}),$$

be the elements of Δ. Every $x \in \Gamma_0(R)$ has a decomposition

$$x = \sum_{i=1}^{n} m_i(x)\alpha_i.$$

Take the lexicographic order with respect to the functions

$$\sum_i m_i, \quad m_n, \quad m_{n-1}, \quad \ldots, \quad m_2.$$

□

Remark 0.3.5.7. The first roots, in order, are

$$\alpha_1, \alpha_2, \ldots, \alpha_n;$$

then come, when they are roots,

$$2\alpha_1, \quad \alpha_1 + \alpha_2, \quad \alpha_1 + \alpha_3, \quad \ldots.$$

0.3.6 Weyl chambers

Lemma 0.3.6.1. *Let V be a finite-dimensional real vector space.*[15] *Let f_i be independent linear forms. Put*

$$C = \{x \in V \mid f_i(x) > 0\}.$$

Then C is a maximal convex subset of

$$X = V \setminus \bigcup_i f_i^{-1}(0).$$

Proof. Trivial. □

Definition 0.3.6.2. A subset $C \subset V$ described by the construction of Lemma 0.3.6.1 is called, here, a *chamber* of V.

Definition 0.3.6.3. A hyperplane H of V is called a *wall* of C if

$$H \cap (\overline{C} \setminus C)$$

contains a nonempty open subset of H.

Remark 0.3.6.4. For a convex subset, the closure can be described without reference to the topology of V: it is the set of endpoints of all open segments contained in the given subset.

Lemma 0.3.6.5. *Under the conditions of Lemma 0.3.6.1,*

$$\overline{C} = \{x \in V \mid f_i(x) \geq 0\}.$$

The walls of C are the hyperplanes $f_i^{-1}(0)$.

Proof. The first assertion is clear. The second follows from

$$\overline{C} \setminus C \subset \bigcup_i f_i^{-1}(0)$$

and from the fact that the $f_i^{-1}(0)$ are visibly walls of C. □

Proposition 0.3.6.6. *Let C be a chamber of V. If*

$$H_i, \qquad i = 1, 2, \ldots, n,$$

are the distinct walls of C, then, for every system of linear forms $\{u_i\}$ such that $H_i = u_i^{-1}(0)$, there exist $\varepsilon_i \in \{-1, +1\}$ for which

$$C = \{x \in V \mid \varepsilon_i u_i(x) > 0\}.$$

For every wall H of C,

$$H \cap C = \varnothing, \qquad H \cap \overline{C} = \overline{C_H},$$

where C_H is a chamber in H. The walls of C_{H_i} are the $H_j \cap H_i$, for $j \neq i$.

Proof. This follows immediately from the lemma. □

Definition 0.3.6.7. The C_{H_i} are called the *faces* of C.

[15] Editors' note: The original $\mathbb{Q}$ has been replaced by $\mathbb{R}$.

Let now

$$\mathcal{R} = (M, M^*, R, R^*)$$

be a root datum, and put

$$V_{\mathbb{R}}^* = M^* \otimes \mathbb{R}.$$

Definition 0.3.6.8. [16] For every $\alpha \in R$, put

$$\mathcal{H}_\alpha = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) = 0\}.$$

Write

$$X = V_{\mathbb{R}}^* \setminus \bigcup_{\alpha \in R} \mathcal{H}_\alpha.$$

For every $x \in X$, put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}.$$

For every system of positive roots R_+, put

$$\mathcal{C}(R_+) = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in R_+\}.$$

Proposition 0.3.6.9. *(i) For every $x \in X$, $R_+(x)$ is a system of positive roots. For every system of positive roots R_+, $\mathcal{C}(R_+)$ is a chamber in $V_{\mathbb{R}}^*$. The $\mathcal{C}(R_+)$ are the maximal convex subsets of X.*

(ii) Let Δ be a system of simple roots. One has

$$\mathcal{C}(\mathscr{P}(\Delta)) = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in \Delta\}.$$

The walls of $\mathcal{C}(\mathscr{P}(\Delta))$ are the hyperplanes

$$\mathcal{H}_\alpha = \alpha^{-1}(0), \qquad \alpha \in \Delta;$$

its faces are

$$C_\alpha = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) = 0,\ (x, \beta) > 0 \text{ for } \beta \in \Delta,\ \beta \neq \alpha\}.$$

(iii) One has the equivalence

$$R_+(x) = R_+ \quad \Longleftrightarrow \quad x \in \mathcal{C}(R_+).$$

Proof. It is first clear that $R_+(x)$ is a system of positive roots. Since $x \in \mathcal{C}(R_+(x))$, the union of the $\mathcal{C}(R_+(x))$ is X. Property (iii) is immediate; it follows that the $\mathcal{C}(R_+)$ form a partition of X. The first assertion of (ii) is evident. It follows at once that $\mathcal{C}(R_+)$ is a chamber in $V_{\mathbb{R}}^*$, which proves the remainder of (i). Put

$$C = \mathcal{C}(\mathscr{P}(\Delta)).$$

It remains only to note that

$$\overline{C} \cap \left(V_{\mathbb{R}}^* \setminus \bigcup_{\alpha \in R} \alpha^{-1}(0)\right) = \overline{C} \cap \left(V_{\mathbb{R}}^* \setminus \bigcup_{\alpha \in \Delta} \alpha^{-1}(0)\right)$$

to complete the proof of (ii) by Lemma 0.3.6.1. □

[16] Editors' note: The definition of the hyperplanes $\mathcal{H}_\alpha$ was added here.

Definition 0.3.6.10. The $\mathcal{C}(R_+)$ are called the *Weyl chambers* of the root datum. Conversely, for every Weyl chamber C, put

$$R_+(C) = R_+(x)$$

for any $x \in C$.

Corollary 0.3.6.11. *The maps*

$$R_+ \longmapsto \mathcal{C}(R_+), \qquad C \longmapsto R_+(C)$$

give a bijective correspondence between systems of positive roots and Weyl chambers.

This correspondence is invariant under the Weyl group:

Lemma 0.3.6.12. *If $w \in W(\mathcal{R})$, then*

$$\mathcal{C}(w(R_+)) = w(\mathcal{C}(R_+)).$$

Corollary 0.3.6.13. *The correspondences*

$$\Delta \longleftrightarrow R_+ \longleftrightarrow C$$

are isomorphisms of homogeneous spaces under $W(\mathcal{R})$.

We shall see later that these homogeneous spaces are principal (5.5).

Remark 0.3.6.14. If C is a Weyl chamber, then $-C$ is also one; it is called the *opposite* of C. Consequently there exists $w_0 \in W$—and in fact exactly one, cf. 5.5—such that

$$w_0(C) = -C.$$

It is called the *symmetry* of the root datum relative to the Weyl chamber C (or relative to $R_+(C)$, or to $\mathscr{S}(R_+(C))$, and so on).

0.4 Reduced root data of semisimple rank 2

4.0.[17] Let $\mathcal{R}$ be a root datum of semisimple rank 2. Let $\{\alpha, \beta\}$ be a system of simple roots. Suppose that $\ell(\alpha) \leq \ell(\beta)$. By 0.2.3.1 and 0.3.2.1, there are then four possibilities:

Type	$\ell(\beta)/\ell(\alpha)$	$\ell(\beta^*)/\ell(\alpha^*)$	(β^*, α)	(α^*, β)
$A_1 \times A_1$	–	–	0	0
A_2	1	1	-1	-1
B_2	2	1/2	-1	-2
G_2	3	1/3	-1	-3

It follows from Remark 0.3.4.5 that the order of $s_\alpha s_\beta$ is respectively $2, 3, 4, 6$.

We study each of these systems separately and give the list of indivisible roots.

Type $A_1 \times A_1$. The indivisible roots are

$$\alpha, \quad \beta, \quad -\alpha, \quad -\beta.$$

[17] Editors' note: The numbering 4.0 was added for later references.

The corresponding coroots are

$$\alpha^*, \quad \beta^*, \quad -\alpha^*, \quad -\beta^*.$$

Type A_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha+\beta$
$\ell(\gamma)/\ell(\alpha)$	1	1	1
coroot γ^*	α^*	β^*	$\alpha^*+\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	1	1	1

The half-sum of the indivisible positive roots is

$$\rho = \alpha + \beta.$$

Type B_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha+\beta$	$2\alpha+\beta$
$\ell(\gamma)/\ell(\alpha)$	1	2	1	2
coroot γ^*	α^*	β^*	$2\alpha^*+\beta^*$	$\alpha^*+\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	2	1	2	1

The half-sum of the indivisible positive roots is

$$\rho = \frac{4\alpha + 3\beta}{2}.$$

Type G_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha+\beta$	$2\alpha+\beta$	$3\alpha+\beta$	$3\alpha+2\beta$
$\ell(\gamma)/\ell(\alpha)$	1	3	1	1	3	3
coroot γ^*	α^*	β^*	$\alpha^*+3\beta^*$	$2\alpha^*+3\beta^*$	$\alpha^*+\beta^*$	$\alpha^*+2\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	3	1	3	3	1	1

The half-sum of the indivisible positive roots is

$$\rho = 5\alpha + 3\beta.$$

Proposition 0.4.1. [18] *Let n be the order of $s_\alpha s_\beta$. Put $u_0 = 0$, and, for $p \in \mathbb{N}$, define*

$$\begin{cases} u_{2p+1} = u_{2p} + (s_\alpha s_\beta)^p(\alpha), \\ u_{2p+2} = u_{2p+1} + (s_\alpha s_\beta)^p s_\alpha(\beta). \end{cases}$$

Then: [19]

[18] The following numbers 4.1 to 4.4 are used in the proof of Theorem 0.5.1. There are now simpler proofs of that theorem; see *Groupes et algèbres de Lie*, Chapter V, §3, Theorem 1. Editors' note: The reference was made explicit and this note, which occurred at 4.2 in the original, was moved here.

[19] Editors' note: Let ρ be the half-sum of the positive roots, cf. Proposition 0.3.5.1. If, as below, one puts $w_0 = 1$, $w_1 = s_\alpha$, $w_2 = s_\beta s_\alpha$, $w_3 = s_\alpha s_\beta s_\alpha$, and so on, then $u_k = \rho - w_k^{-1}(\rho)$. This proves (i), since $w_{2n} = 1$.

(i) $u_{k+2n} = u_k$, *for every* $k \in \mathbb{N}$.

(ii) $u_0 = 0$, $u_1 = \alpha$, $u_{2n-1} = \beta$, *and* $u_{2n} = 0$.

(iii) If $1 < k < 2n-1$, *then*

$$u_k = a_k\alpha + b_k\beta, \qquad a_k, b_k \in \mathbb{N}^*.$$

Proof. Assertion (i) follows from $(s_\alpha s_\beta)^n = 1$ and $u_{2n} = 0$.

We prove (ii) and (iii). Direct calculation gives, in the four cases, the following sequences of values: [20]

$A_1 \times A_1$	$0, \alpha, \beta+\alpha, \beta, 0,$
A_2	$0, \alpha, \beta+2\alpha, 2\beta+2\alpha, 2\beta+\alpha, \beta, 0,$
B_2	$0, \alpha, \beta+3\alpha, 2\beta+4\alpha, 3\beta+4\alpha, 3\beta+3\alpha, 2\beta+\alpha, \beta, 0,$
G_2	$0, \alpha, \beta+4\alpha, 2\beta+6\alpha, 4\beta+9\alpha, 5\beta+10\alpha, 6\beta+10\alpha,$ $6\beta+9\alpha, 5\beta+6\alpha, 4\beta+4\alpha, 2\beta+\alpha, \beta, 0.$

□

Lemma 0.4.2. *Put*

$$w_{2p} = (s_\beta s_\alpha)^p, \qquad w_{2p+1} = s_\alpha(s_\beta s_\alpha)^p,$$

so that $w_0, \ldots, w_{2n-1}$ *are the distinct elements of* W. *Let* $u_0, \ldots, u_{2n-1}$ *be the elements of* V *defined in Proposition 0.4.1. For every* $x \in V^*$, *one has*

$$x - w_k(x) = n_k\alpha^* + m_k\beta^*,$$

where $n_k, m_k \in \mathbb{Q}$ *and*

$$n_k + m_k = (x, u_k).$$

[21]

Proof. The proof is by induction on k. If $k = 0$, the formula is trivial. We carry out, for example, the passage from w_{2p} to w_{2p+1}. Since $w_{2p+1} = s_\alpha w_{2p}$,

$$\begin{aligned} x - w_{2p+1}(x) &= x - w_{2p}(x) + w_{2p}(x) - s_\alpha w_{2p}(x) \\ &= n_{2p}\alpha^* + m_{2p}\beta^* + (w_{2p}(x), \alpha)\alpha^*. \end{aligned}$$

Consequently

$$\begin{aligned} n_{2p+1} + m_{2p+1} &= n_{2p} + m_{2p} + \big((s_\beta s_\alpha)^p(x), \alpha\big) \\ &= (x, u_{2p}) + \big(x, (s_\alpha s_\beta)^p(\alpha)\big) \\ &= (x, u_{2p+1}). \end{aligned}$$

□

[20] Editors' note: The original was corrected here to agree with the convention $\ell(\alpha) \le \ell(\beta)$ of 4.0.

[21] Editors' note: In view of the preceding editor's note, this follows immediately from Propositions 0.3.5.1 and 0.1.1.9: one has

$$n_k + m_k = (x - w_k(x), \rho) = (x, \rho - w_k^{-1}(\rho)) = (x, u_k).$$

Corollary 0.4.3. [22] *Let $x \in V^*$. For every $w \in W$, put*

$$x - w(x) = a_w\alpha^* + b_w\beta^*.$$

If $(x, \alpha) \geq 0$ and $(x, \beta) \geq 0$, then

$$a_w + b_w \geq 0.$$

If moreover $(x, \alpha) > 0$ (respectively $(x, \beta) > 0$), then

$$a_w + b_w > 0$$

for $w \neq 1, s_\beta$ (respectively for $w \neq 1, s_\alpha$).

Proof. This follows at once from Proposition 0.4.1 and Lemma 0.4.2. □

Corollary 0.4.4. *Let $\mathcal{R}$ be any root datum, and let Δ be a system of simple roots. Let γ be a positive root, let α, β be two simple roots, and let $W_{\alpha,\beta}$ be the subgroup of W generated by s_α and s_β. If*

$$\mathrm{ord}_\Delta(s_\alpha(\gamma)) < \mathrm{ord}_\Delta(\gamma), \qquad \mathrm{ord}_\Delta(s_\beta(\gamma)) \leq \mathrm{ord}_\Delta(\gamma),$$

then, for every $w \in W_{\alpha,\beta}$,

$$\mathrm{ord}_\Delta(w(\gamma)) \leq \mathrm{ord}_\Delta(\gamma).$$

Moreover,

$$\mathrm{ord}_\Delta(w(\gamma)) < \mathrm{ord}_\Delta(\gamma)$$

if $w \neq 1, s_\beta$.

Proof. Consider the dual root datum $\mathcal{R}^*$, and then the datum

$$(M^*, M, R'^*, R'),$$

where R' is the set of roots that are rational linear combinations of α and β. Applying Corollary 0.4.3 to this datum gives the stated result. [23] □

0.5 The Weyl group: generators and relations

Let $\mathcal{R}$ be a root datum. Since the Weyl group is the same for this datum and for the corresponding reduced datum, we may assume $\mathcal{R}$ reduced when studying the Weyl group.

Let

$$\Delta = \{\alpha_1, \ldots, \alpha_n\}$$

be a system of simple roots, where $n = \mathrm{rgss}(\mathcal{R})$. Let n_{ij} be the order of $s_{\alpha_i}s_{\alpha_j}$ in W. In particular $n_{ii} = 1$, and Remarks 0.3.4.2 and 0.3.4.3 show that the subgroup W_{ij} of W generated by s_{α_i} and s_{α_j} is defined by

$$s_{\alpha_i}^2 = s_{\alpha_j}^2 = (s_{\alpha_i}s_{\alpha_j})^{n_{ij}} = 1.$$

Theorem 0.5.1. *The group W is the group generated by the elements*

$$s_{\alpha_i}, \qquad i = 1, 2, \ldots, n,$$

subject to the relations

$$(s_{\alpha_i}s_{\alpha_j})^{n_{ij}} = 1.$$

[22] Editors' note: The statement was corrected.

[23] Editors' note: The hypotheses say that $(\gamma, \alpha^*) > 0$ and $(\gamma, \beta^*) \geq 0$. Consequently $\gamma - w(\gamma) \in \mathbb{N}\alpha + \mathbb{N}\beta$ for every $w \in W_{\alpha,\beta}$, and it is nonzero if $w \notin \{1, s_\beta\}$.

Proof. We have already seen that the theorem is true when $n = 2$; we shall use this fact in the proof. Introduce the group $\widetilde{W}$ generated by elements

$$T_i, \qquad i = 1, 2, \ldots, n,$$

subject to the relations

$$(T_i T_j)^{n_{ij}} = 1.$$

In particular $n_{ii} = 1$, whence $T_i^2 = 1$. Let

$$p : \widetilde{W} \longrightarrow W$$

be the group homomorphism sending T_i to s_{α_i}. We know that p is surjective; we shall prove that it is injective. □

Lemma 0.5.2. *For every $\alpha \in \mathscr{P}(\Delta)$, one can define uniquely an element $T_\alpha \in \widetilde{W}$ with the following properties:*

(i) $p(T_\alpha) = s_\alpha$;

(ii) $T_{\alpha_i} = T_i$;

(iii) if β and α are positive roots such that $s_{\alpha_i}(\alpha) = \beta$, then

$$T_i T_\alpha T_i = T_\beta.$$

Proof. First observe that it follows from 0.1.2.10 and 0.3.3.6 that (i) is a consequence of (ii) and (iii), and that (ii) and (iii) determine the T_α completely. We construct them by induction on $\mathrm{ord}_\Delta(\alpha)$. If $\mathrm{ord}_\Delta(\alpha) = 1$, then $\alpha \in \Delta$, and we put $T_\alpha = T_i$ if $\alpha = \alpha_i$.

Consider the hypothesis

(H_p) there exist T_α, for $\alpha \in \mathscr{P}(\Delta)$ with $\mathrm{ord}_\Delta(\alpha) \leq p$, satisfying (ii), and satisfying (iii) whenever $\mathrm{ord}_\Delta(\alpha) \leq p$ and $\mathrm{ord}_\Delta(\beta) \leq p$.

It holds for $p = 1$. Indeed, if α and $s_{\alpha_i}(\alpha) = \beta$ are simple, then α_i and α are orthogonal. Thus, if $\alpha_j = \alpha = \beta$, one has $n_{ij} = 2$, whence

$$T_i T_j T_i = T_j.$$

Suppose $p > 1$ and (H_{p-1}) holds.

A) Construction of the T_α for $\mathrm{ord}_\Delta(\alpha) \leq p$. It is enough to do this when $\mathrm{ord}_\Delta(\alpha) = p$. There exists $\alpha_i \in \Delta$ such that

$$s_{\alpha_i}(\alpha) \in \mathscr{P}(\Delta), \qquad \mathrm{ord}_\Delta(s_{\alpha_i}(\alpha)) < p$$

by Lemma 0.3.3.3. Put

$$(\star) \qquad T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

We verify that T_α depends only on α. Let $\alpha_j \in \Delta$ also satisfy

$$s_{\alpha_j}(\alpha) \in \mathscr{P}(\Delta), \qquad \mathrm{ord}_\Delta(s_{\alpha_j}(\alpha)) < p.$$

We must prove

$$(+) \qquad T_i T_{s_{\alpha_i}(\alpha)} T_i = T_j T_{s_{\alpha_j}(\alpha)} T_j.$$

We distinguish two cases.

(1) Suppose that α is a linear combination of α_i and α_j. The same is then true of $s_{\alpha_i}(\alpha)$ and $s_{\alpha_j}(\alpha)$; by (H_{p-1}), the elements $T_{s_{\alpha_i}(\alpha)}$ and $T_{s_{\alpha_j}(\alpha)}$ are words in T_i, T_j. Since the projection of $(+)$ to W holds and the theorem is true for $n = 2$, the restriction of p to the subgroup of $\widetilde{W}$ generated by T_i, T_j is injective. Thus $(+)$ holds.

(2) Suppose that α is not a linear combination of α_i and α_j. Then, for $w \in W_{\alpha_i,\alpha_j}$, all the $w(\alpha)$ are positive (cf. Lemma 0.3.4.9). The relation to be verified can also be written

$$(++) \qquad (T_iT_j)^{n_{ij}-1}T_{s_{\alpha_i}(\alpha)}(T_jT_i)^{n_{ij}-1} = T_{s_{\alpha_j}(\alpha)}.$$

Corollary 0.4.4 shows that all the $w(\alpha)$ have order $< p$ for $w \in W_{\alpha_i,\alpha_j}$, $w \neq 1$. We may therefore apply (H_{p-1}) $2(n_{ij}-1)$ times, which proves the relation.

B) Verification of (H_p)**.** [24] We must verify that, if $\alpha_j \in \Delta$ and $\beta = s_{\alpha_j}(\alpha)$ satisfies $\mathrm{ord}_\Delta(\beta) \leq p$, then

$$T_jT_\alpha T_j = T_\beta.$$

If $\mathrm{ord}_\Delta(\beta) < p$, this follows from the preceding construction, since $T_j^2 = 1$. We may therefore suppose

$$\mathrm{ord}_\Delta(\beta) = p = \mathrm{ord}_\Delta(\alpha).$$

In this case α and α_j are orthogonal, so $\beta = \alpha$, and it remains to prove

$$(\dagger) \qquad T_jT_\alpha T_j = T_\alpha.$$

By $(\star)$,

$$T_\alpha = T_iT_{s_{\alpha_i}(\alpha)}T_i,$$

so it remains to verify

$$(+++) \qquad T_jT_iT_{s_{\alpha_i}(\alpha)}T_iT_j = T_\alpha = T_iT_{s_{\alpha_i}(\alpha)}T_i.$$

□

Proof of Lemma 0.5.2, continued. Put $m = n_{ij}$, $s_i = s_{\alpha_i}$, and $s_j = s_{\alpha_j}$. We have

$$T_jT_i = (T_iT_j)^{m-1}.$$

By Corollary 0.4.4, $\mathrm{ord}_\Delta(w(\alpha)) < p$ for every $w \in W_{ij}$ distinct from

$$1 = (s_is_j)^m \quad \text{and} \quad s_j = s_i(s_is_j)^{m-1}.$$

The induction hypothesis therefore gives

$$\begin{aligned} &T_j(T_iT_j)^{m-2}T_{s_i(\alpha)}(T_jT_i)^{m-2}T_j \\ &\quad = T_{s_j(s_is_j)^{m-2}s_i(\alpha)} = T_{s_is_j(\alpha)} = T_{s_i(\alpha)}. \end{aligned}$$

The last equality holds because $s_j(\alpha) = \alpha$. Hence

$$(T_iT_j)^{m-1}T_{s_i(\alpha)}(T_jT_i)^{m-1} = T_iT_{s_i(\alpha)}T_i,$$

which proves $(+++)$. □

[24] Editors' note: The argument in the original has been expanded in what follows.

Lemma 0.5.3. *Let $h \in \widetilde{W}$. Write*

$$h = T_{\alpha_1} \cdots T_{\alpha_m},$$

where the $\alpha_i \in \Delta$, not necessarily distinct, are chosen so that m is minimal. Then

$$p(h)(\alpha_m) \in -\mathscr{P}(\Delta).$$

Proof. Since

$$p(T_{\alpha_m})(\alpha_m) = s_{\alpha_m}(\alpha_m) = -\alpha_m,$$

if $p(h)(\alpha_m)$ were positive, there would be an index k, $1 \leq k \leq m-1$, such that

$$u = s_{\alpha_{k+1}} \cdots s_{\alpha_m}(\alpha_m) = -s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) \in -\mathscr{P}(\Delta)$$

and $s_{\alpha_k}(u) \in \mathscr{P}(\Delta)$. Necessarily $u = -\alpha_k$, by Lemma 0.3.3.1, whence

$$s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) = \alpha_k.$$

Property (iii) of Lemma 0.5.2 then gives

$$T_{\alpha_k} T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}} T_{\alpha_m} = T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}},$$

contradicting the minimality of m. □

Let now $h \in \widetilde{W}$ satisfy

$$p(h)(\mathscr{P}(\Delta)) \subset \mathscr{P}(\Delta).$$

Lemma 0.5.3 gives $p(h) = 1$. This proves Theorem 0.5.1 and also the following two corollaries.

Corollary 0.5.4. *If R_+ is a system of positive roots and $w \in W$ satisfies $w(R_+) = R_+$, then $w = 1$.*

Corollary 0.5.5. *The Weyl group acts freely and transitively on the set of systems of positive roots, respectively on the set of systems of simple roots, respectively on the set of Weyl chambers.*

Choose a system of simple roots Δ, and put

$$R^+ = \mathscr{P}(\Delta).$$

[25] For every pair of simple roots $(\alpha, \beta) \in \Delta \times \Delta$, let $R_{\alpha,\beta}$ be the set of roots that are linear combinations of α and β. Put

$$R^+_{\alpha,\beta} = R^+ \cap R_{\alpha,\beta},$$

and let $W_{\alpha,\beta}$ be the Weyl group of $R_{\alpha,\beta}$, that is, the subgroup of W generated by s_α and s_β.

Theorem 0.5.6 (Tits)**.** *Let α and β be two simple roots, and let $w \in W$ satisfy $w(\alpha) = \beta$. There exist a sequence of simple roots $\alpha_0, \ldots, \alpha_m$ and a sequence $w_0, \ldots, w_{m-1}$ of elements of W satisfying:*

(i) $\alpha_0 = \alpha$ and $\alpha_m = \beta$;

[25] Editors' note: The superscript notation R^+, rather than R_+, is used here so that $R^+_{\alpha,\beta} = R_{\alpha,\beta} \cap R^+$.

(ii) $w = w_{m-1}w_{m-2}\cdots w_0$;

(iii) $w_i(\alpha_i) = \alpha_{i+1}$ *for* $0 \leq i \leq m-1$;

(iv) for every i, $0 \leq i \leq m-1$, *such that* $\alpha_i \neq \alpha_{i+1}$, *one has* $w_i \in W_{\alpha_i,\alpha_{i+1}}$;

(v) for every i, $0 \leq i \leq m-1$, *such that* $\alpha_i = \alpha_{i+1}$, *there exists a simple root* β_i *such that* $w_i \in W_{\alpha_i,\beta_i}$.

Proof. Put[26]

$$M(w) = \mathrm{Card}(R^+ \cap w^{-1}(-R^+)) = \mathrm{Card}\{\gamma \in R^+ \mid w(\gamma) \in -R^+\}.$$

If $M(w) = 0$, then $w(R^+) = R^+$, so $w = 1$ by Corollary 0.5.4; the theorem is then trivial: $m = 0$, and assertions (iii)–(v) are empty.

We argue by induction on $M(w)$. Suppose $M(w) > 0$. There exists $\beta_0 \in \Delta$ such that $w(\beta_0) \in -R^+$. Put $\alpha_0 = \alpha$, and consider

$$A = w^{-1}(R^+) \cap R_{\alpha_0,\beta_0}.$$

This is a system of positive roots of R_{α_0,β_0}. Hence there exists $w_0 \in W_{\alpha_0,\beta_0}$ such that

$$w_0^{-1}(R^+_{\alpha_0,\beta_0}) = A.$$

Put $w' = ww_0^{-1}$. Lemma 0.3.4.9 gives

$$R^+ - R^+_{\alpha_0,\beta_0} = w_0(R^+ - R^+_{\alpha_0,\beta_0}),$$

and therefore

$$\begin{aligned}(R^+ - R^+_{\alpha_0,\beta_0}) \cap w^{-1}(-R^+) \\ = w_0\Big((R^+ - R^+_{\alpha_0,\beta_0}) \cap w'^{-1}(-R^+)\Big).\end{aligned} \tag{1}$$

On the other hand,

$$\beta_0 \in R^+_{\alpha_0,\beta_0} \cap w^{-1}(-R^+). \tag{2}$$

Since $w_0(R_{\alpha_0,\beta_0}) = R_{\alpha_0,\beta_0}$, one has

$$w_0(-A) = R_{\alpha_0,\beta_0} \cap w'^{-1}(-R^+),$$

and hence

$$\begin{aligned}R^+_{\alpha_0,\beta_0} \cap w'^{-1}(-R^+) &= R^+_{\alpha_0,\beta_0} \cap w_0(-A) \\ &= R^+_{\alpha_0,\beta_0} \cap (-R^+_{\alpha_0,\beta_0}) = \varnothing.\end{aligned} \tag{2'}$$

It follows from (1), (2), and (2′) that

$$M(w') < M(w).$$

Put $\alpha_1 = w_0(\alpha_0)$. We show that $\alpha_1 \in \Delta$, or equivalently that $\alpha_0 \in w_0^{-1}(\Delta)$. Since $w(\alpha_0) \in \Delta$, we have

$$\alpha_0 \in w^{-1}(\Delta) \cap R_{\alpha_0,\beta_0}.$$

[26]Editors' note: In what follows, the original argument has been expanded, and equality (1) has been corrected.

Thus α_0 is a simple root of

$$A = w^{-1}(R^+) \cap R_{\alpha_0,\beta_0} = w_0^{-1}(R^+_{\alpha_0,\beta_0}),$$

and consequently belongs to

$$w_0^{-1}(\Delta \cap R^+_{\alpha_0,\beta_0}) = w_0^{-1}\{\alpha_0, \beta_0\}$$

by Lemma 0.3.4.8. Hence $\alpha_1 = w_0(\alpha_0)$ equals α_0 or β_0. If $\alpha_1 \neq \alpha_0$, then $\alpha_1 = \beta_0$ and $w_0 \in W_{\alpha_0,\alpha_1}$.

Finally,

$$\beta = w'(\alpha_1) \qquad \text{and} \qquad M(w') < M(w),$$

so the induction hypothesis completes the proof. □

0.6 Morphisms of root data

0.6.1 Definition

Let

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two root data. Let $f : M' \to M$ be a linear map and ${}^t f : M^* \to M'^*$ its transpose.

Definition 0.6.1.1. The map f is called a *morphism* from $\mathcal{R}'$ to $\mathcal{R}$, and is denoted

$$f : \mathcal{R}' \longrightarrow \mathcal{R},$$

if f induces a bijection from R' onto R, and ${}^t f$ induces a bijection from R^* onto R'^*.

Then ${}^t f$ is a morphism of the dual root data:

$${}^t f : \mathcal{R}^* \longrightarrow \mathcal{R}'^*.$$

If $f : \mathcal{R}' \to \mathcal{R}$ is a morphism and $\alpha = f(\alpha')$ for $\alpha' \in R'$, then

$$\alpha'^* = {}^t f(\alpha^*).$$

Indeed, if p and p' denote the maps of 0.1.2.1 for $\mathcal{R}$ and $\mathcal{R}'$, respectively, then

$$p' = {}^t f \circ p \circ f,$$

which gives the assertion at once. The reader may verify the statements below, almost all of which are immediate.

Proposition 0.6.1.2. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism of root data. If $\alpha' \in R'$ and $\alpha = f(\alpha')$, then*

$$\alpha'^* = {}^t f(\alpha^*).$$

Moreover, f induces isomorphisms

$$R' \xrightarrow{\sim} R, \qquad \Gamma_0(R') \xrightarrow{\sim} \Gamma_0(R), \qquad \mathcal{V}(R') \xrightarrow{\sim} \mathcal{V}(R),$$

and ${}^t f$ induces isomorphisms

$$R^* \xrightarrow{\sim} R'^*, \qquad \Gamma_0(R^*) \xrightarrow{\sim} \Gamma_0(R'^*), \qquad \mathcal{V}(R^*) \xrightarrow{\sim} \mathcal{V}(R'^*).$$

The last is the transpose of the corresponding morphism induced by f. The map

$$s_{\alpha'} \longmapsto s_{f(\alpha')}$$

extends to an isomorphism

$$W(\mathcal{R}') \xrightarrow{\sim} W(\mathcal{R})$$

compatible with the actions of these two groups on the sets of 0.1.1.13.

Proposition 0.6.1.3. *The maps*

$$\Delta' \longmapsto f(\Delta'), \qquad R'_+ \longmapsto f(R'_+), \qquad C' \longmapsto ({}^t f \otimes \mathbb{R})^{-1}(C')$$

define bijective correspondences between the systems of simple roots, the systems of positive roots, and the Weyl chambers for $\mathcal{R}'$ and $\mathcal{R}$. These correspondences are compatible with the actions of the Weyl groups and with the correspondences

$$\mathscr{S}(R_+) \longleftrightarrow R_+ \longleftrightarrow \mathscr{C}(R_+).$$

Lemma 0.6.1.4. *Morphisms compose. A morphism $f : \mathcal{R}' \to \mathcal{R}$ is an isomorphism if and only if the linear map $f : M' \to M$ is bijective.*

0.6.2 Isogenies

Definition 0.6.2.1. A morphism $f : \mathcal{R}' \to \mathcal{R}$ of root data is called an *isogeny* if $f : M' \to M$ is injective with finite cokernel.

If f is an isogeny, then ${}^t f$ is an isogeny.

Definition 0.6.2.2. Let $f : \mathcal{R}' \to \mathcal{R}$ be an isogeny. Put

$$K(f) = \mathrm{Coker}(M' \xrightarrow{f} M).$$

Lemma 0.6.2.3. *There is a natural pairing*

$$K(f) \times K({}^t f) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

that puts these two finite groups in duality.

Proof. This is classical. □

Lemma 0.6.2.4. *If $f : \mathcal{R}' \to \mathcal{R}$ is a morphism, then*

$$\mathrm{rgss}(\mathcal{R}') = \mathrm{rgss}(\mathcal{R}).$$

If f is moreover an isogeny, then

$$\mathrm{rgred}(\mathcal{R}') = \mathrm{rgred}(\mathcal{R}).$$

Proof. This is immediate. □

Lemma 0.6.2.5. *Every morphism of semisimple root data is an isogeny.*

Proof. The map f must induce an isomorphism from $V' = \mathcal{V}(R')$ onto $V = \mathcal{V}(R)$. □

If $\mathcal{R}'$ and $\mathcal{R}$ are semisimple, every isogeny $f : \mathcal{R}' \to \mathcal{R}$ defines a commutative diagram:

$$\begin{array}{ccc} \Gamma_0(R') & \xrightarrow{\sim} & \Gamma_0(R) \\ \downarrow & & \downarrow \\ M' & \xrightarrow[f]{} & M \end{array}$$

If $M = \Gamma_0(R)$, then f is necessarily an isomorphism.

Definition 0.6.2.6. A root datum is called *adjoint*, respectively *simply connected*, if

$$M = \Gamma_0(R), \qquad \text{respectively} \qquad M^* = \Gamma_0(R^*).$$

An adjoint or simply connected root datum is therefore semisimple. Moreover, $\mathcal{R}$ is adjoint, respectively simply connected, if and only if $\mathcal{R}^*$ is simply connected, respectively adjoint. The preceding result gives the following proposition.

Proposition 0.6.2.7. *Let $\mathcal{R}$ be a semisimple root datum. The following conditions are equivalent:*

(i) $\mathcal{R}$ is adjoint, respectively simply connected;

(ii) every isogeny $\mathcal{R}' \to \mathcal{R}$, respectively $\mathcal{R} \to \mathcal{R}'$, is an isomorphism.

Proposition 0.6.2.8. *Let $\mathcal{R}$ be an adjoint, respectively simply connected, root datum. Every indivisible root, respectively coroot, is an indivisible element of M, respectively of M^*.*

Proof. Every indivisible root belongs to a basis of $\Gamma_0(R)$, by 0.3.3.5. □

0.6.3 Radical and coradical

Let $\mathcal{R}$ be a root datum. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha^* \in R^*\}, \qquad N^* = M^*/(\mathcal{V}(R^*) \cap M^*).$$

Lemma 0.6.3.1. *Consider the canonical morphisms*

$$N \longrightarrow M, \qquad M^* \longrightarrow N^*.$$

They are transposes of one another, and N^* is identified with the dual of N.

Proof. This follows immediately from 0.1.2.5. □

Definition 0.6.3.2. The *coradical* of $\mathcal{R}$, denoted $\mathrm{corad}(\mathcal{R})$, is the trivial root datum

$$\mathrm{corad}(\mathcal{R}) = (N, N^*, \varnothing, \varnothing).$$

If

$$\mathcal{R}^0 = (M, M^*, \varnothing, \varnothing),$$

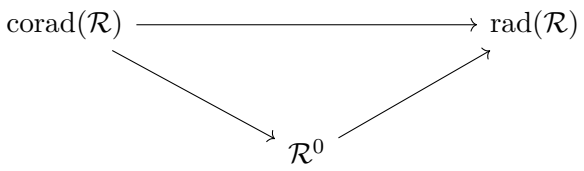
which is a trivial root datum, there is therefore a morphism

$$\mathrm{corad}(\mathcal{R}) \longrightarrow \mathcal{R}^0.$$

Definition 0.6.3.3. The *radical* of $\mathcal{R}$, denoted $\mathrm{rad}(\mathcal{R})$, is the trivial root datum [27]

$$\mathrm{rad}(\mathcal{R}) = \mathrm{corad}(\mathcal{R}^*)^*.$$

Thus there is a diagram



whose transpose is the corresponding diagram for $\mathcal{R}^*$.

Lemma 0.6.3.4. *The canonical morphism*

$$u : \mathrm{corad}(\mathcal{R}) \longrightarrow \mathrm{rad}(\mathcal{R})$$

is an isogeny.

Definition 0.6.3.5. Put

$$N(\mathcal{R}) = K(u) = M \Big/ \left(\big(\mathcal{V}(R) \cap M\big) + \bigcap_{\alpha \in R} \ker(\alpha^*) \right).$$

There is then a canonical pairing

$$N(\mathcal{R}) \times N(\mathcal{R}^*) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

Lemma 0.6.3.6. *One has*

$$\mathrm{rgred}\big(\mathrm{rad}(\mathcal{R})\big) = \mathrm{rgred}\big(\mathrm{corad}(\mathcal{R})\big) = \mathrm{rgred}(\mathcal{R}) - \mathrm{rgss}(\mathcal{R}),$$

and the following conditions are equivalent:

(i) $\mathcal{R}$ *is semisimple;*

(ii) $\mathrm{rad}(\mathcal{R}) = 0$*;*

(iii) $\mathrm{corad}(\mathcal{R}) = 0$.

0.6.4 Products of root data

Definition 0.6.4.1. Let

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two root data. The *product root datum* of $\mathcal{R}$ and $\mathcal{R}'$, denoted

$$\mathcal{R}'' = \mathcal{R} \times \mathcal{R}',$$

is the root datum (M'', M''^*, R'', R''^*), where

$$\begin{aligned} M'' &= M \times M', & M''^* &= M^* \times M'^*, \\ R'' &= (R \times 0) \cup (0 \times R'), & R''^* &= (R^* \times 0) \cup (0 \times R'^*), \end{aligned}$$

and the map $\alpha \mapsto \alpha^*$ is the evident one.

[27] Editors' note: Equivalently, if

$$P = \{y \in M^* \mid (y, \alpha) = 0 \text{ for every } \alpha \in R\}, \qquad P^* = M/\big(\mathcal{V}(R) \cap M\big),$$

then $\mathrm{rad}(\mathcal{R}) = (P^*, P, \varnothing, \varnothing)$.

Proposition 0.6.4.2. *Under the preceding hypotheses there are canonical isomorphisms*

$$\Gamma_0(R'') \simeq \Gamma_0(R) \times \Gamma_0(R'), \qquad \mathcal{V}(R'') \simeq \mathcal{V}(R) \times \mathcal{V}(R'),$$

$$W(\mathcal{R}'') \simeq W(\mathcal{R}) \times W(\mathcal{R}'),$$

and similarly for the other associated constructions. Moreover,

$$\mathrm{rgred}(\mathcal{R}'') = \mathrm{rgred}(\mathcal{R}) + \mathrm{rgred}(\mathcal{R}'), \qquad \mathrm{rgss}(\mathcal{R}'') = \mathrm{rgss}(\mathcal{R}) + \mathrm{rgss}(\mathcal{R}').$$

There is also a canonical isomorphism of root data

$$(\mathcal{R} \times \mathcal{R}')^* \simeq \mathcal{R}^* \times \mathcal{R}'^*.$$

The preceding definitions extend immediately to products of any finite number of factors.

Proposition 0.6.4.3. *Let*

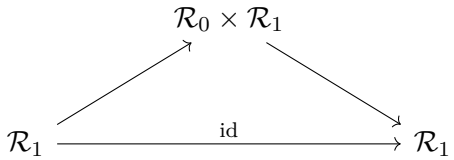
$$\mathcal{R} = \mathcal{R}_1 \times \mathcal{R}_2 \times \cdots \times \mathcal{R}_n$$

be a product of root data. The following conditions are equivalent:

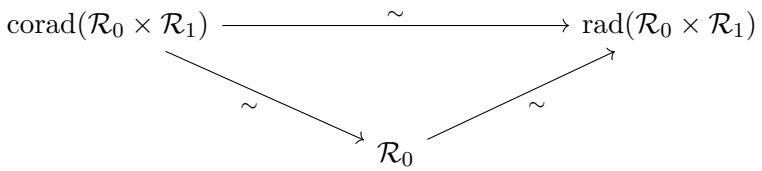
(i) $\mathcal{R}$ *is semisimple, respectively simply connected, respectively adjoint, respectively reduced;*

(ii) every $\mathcal{R}_i$ *is semisimple, respectively simply connected, respectively adjoint, respectively reduced.*

Consider the following special case. Let $\mathcal{R}_0$ be a trivial root datum and $\mathcal{R}_1$ a semisimple root datum. There is then a commutative diagram



Lemma 0.6.4.4. *There are canonical isomorphisms*



In particular,

$$N(\mathcal{R}_0 \times \mathcal{R}_1) = 0.$$

We shall see later that, conversely, if $N(\mathcal{R}) = 0$, then $\mathcal{R}$ is the product of a semisimple root datum and a trivial root datum.

0.6.5 Induced and coinduced root data

Let $\mathcal{R} = (M, M^*, R, R^*)$ be a root datum, and let $N \subset M$ be a subgroup containing the roots, that is,

$$\Gamma_0(R) \subset N \subset M.$$

The canonical linear map $i_N : N \to M$ gives, by transposition, a linear map

$${}^t i_N : M^* \longrightarrow N^*.$$

Put

$$R_N = R, \qquad R_N^* = {}^t i_N(R^*).$$

Lemma 0.6.5.1. *The quadruple*

$$\mathcal{R}_N = (N, N^*, R_N, R_N^*)$$

is a root datum, and i_N is a morphism.

Proof. First, ${}^t i_N$ induces an isomorphism from R^* onto R_N^*. Indeed, if $\alpha, \beta \in R$ and

$${}^t i_N(\alpha^*) = {}^t i_N(\beta^*),$$

then

$$(\alpha^*, x) = (\beta^*, x)$$

for every $x \in N$, in particular for $x \in R$. Hence $\alpha = \beta$ by 0.1.1.4. The rest follows without difficulty. □

Definition 0.6.5.2. The root datum $\mathcal{R}_N$ is called the root datum *induced by* $\mathcal{R}$ *on* N.

Lemma 0.6.5.3. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism, and put*

$$N = f(M') \subset M.$$

Then f factors uniquely through i_N.

In particular, up to isomorphism, the isogenies $\mathcal{R}' \to \mathcal{R}$ correspond bijectively to the finite-index subgroups of

$$M/\Gamma_0(R).$$

This sharpens 0.6.2.7.

Now let $N^* \subset M^*$ be a subgroup containing R^*. Define the root datum *coinduced by* $\mathcal{R}$ *on* N^* by

$$\mathcal{R}^{N^*} = (\mathcal{R}_{N^*}^*)^*.$$

There is a canonical morphism

$$p^{N^*} : \mathcal{R} \longrightarrow \mathcal{R}^{N^*}.$$

Lemma 0.6.5.4. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism. There exist subgroups $N \subset M$ and $N'^* \subset M'^*$ such that f factors as*

$$\begin{array}{ccc} \mathcal{R}' & \xrightarrow{\quad f \quad} & \mathcal{R} \\ {\scriptstyle p^{N'^*}}\downarrow & & \uparrow{\scriptstyle i_N} \\ \mathcal{R}'^{N'^*} & \xrightarrow[\sim]{f_0} & \mathcal{R}_N \end{array}$$

where f_0 is an isomorphism.

Proof. Take $N = f(M')$, as in 0.6.5.3. The resulting morphism $M' \to N$ is surjective, so its transpose is injective. Take its image for N'^*. □

We now treat some special cases. If $N = \Gamma_0(R)$, write

$$\mathcal{R}_N = \mathrm{ad}(\mathcal{R}).$$

If $N = \mathcal{V}(R) \cap M$, write

$$\mathcal{R}_N = \mathrm{ss}(\mathcal{R}).$$

Thus there is a diagram

$$\mathrm{ad}(\mathcal{R}) \longrightarrow \mathrm{ss}(\mathcal{R}) \longrightarrow \mathcal{R}.$$

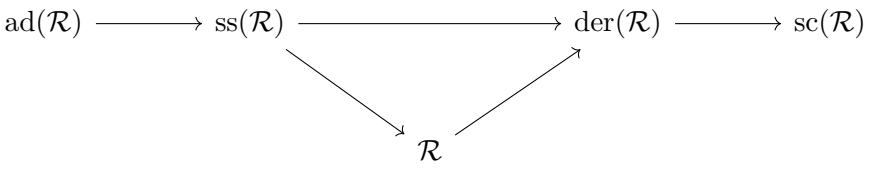
Put

$$\mathrm{der}(\mathcal{R}) = \mathrm{ss}(\mathcal{R}^*)^*, \qquad \mathrm{sc}(\mathcal{R}) = \mathrm{ad}(\mathcal{R}^*)^*.$$

By duality there is a diagram

$$\mathcal{R} \longrightarrow \mathrm{der}(\mathcal{R}) \longrightarrow \mathrm{sc}(\mathcal{R}).$$

Proposition 0.6.5.5. *(i) In the first row of the diagram*



the four root data are semisimple and the three morphisms are isogenies.

(ii) The datum $\mathrm{ad}(\mathcal{R})$ *is adjoint, and* $\mathcal{R}$ *is adjoint if and only if* $\mathrm{ad}(\mathcal{R}) \to \mathcal{R}$ *is an isomorphism.*

(iii) The datum $\mathrm{sc}(\mathcal{R})$ *is simply connected, and* $\mathcal{R}$ *is simply connected if and only if* $\mathcal{R} \to \mathrm{sc}(\mathcal{R})$ *is an isomorphism.*

(iv) The following conditions are equivalent:

- *(a)* $\mathcal{R}$ *is semisimple;*
- *(b)* $\mathrm{ss}(\mathcal{R}) \to \mathcal{R}$ *is an isomorphism;*
- *(c)* $\mathcal{R} \to \mathrm{der}(\mathcal{R})$ *is an isomorphism.*

Consider the morphism

$$\mathrm{ss}(\mathcal{R}) \longrightarrow \mathrm{der}(\mathcal{R}).$$

The definitions immediately give the following result.

Lemma 0.6.5.6. *Let*

$$h : \mathrm{ss}(\mathcal{R}) \longrightarrow \mathrm{der}(\mathcal{R})$$

be the canonical isogeny. Then

$$K(h) \simeq N(\mathcal{R}).$$

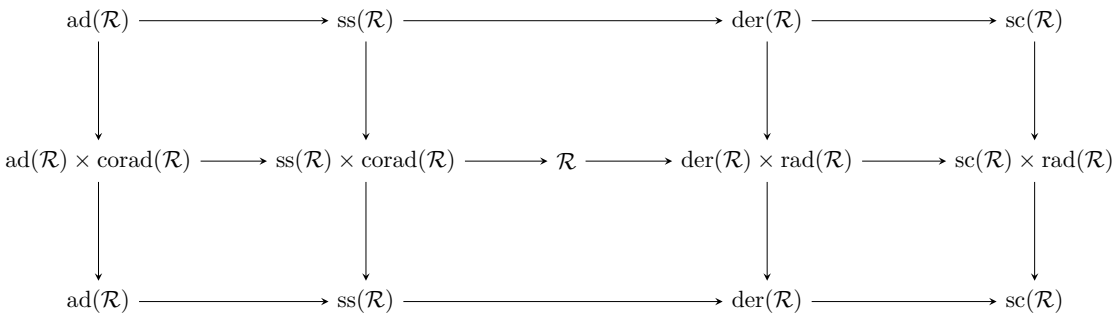
Consider other special cases of induced root data. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha \in R\} \times \Gamma_0(R).$$

The sum is direct by 0.1.2.5. It follows that $\mathcal{R}_N$ is identified with

$$\mathrm{ad}(\mathcal{R}) \times \mathrm{corad}(\mathcal{R}).$$

One may replace $\Gamma_0(R)$ by $\mathcal{V}(R) \cap M$, and then dualize both constructions. This gives the following diagram of root data:



It is commutative, as is checked at once, and is self-dual in the evident sense. The horizontal morphisms are isogenies, and the composites of the vertical arrows are the identity.

Lemma 0.6.5.7. *Let h_1 and h_2 be the canonical isogenies*

$$\mathrm{ss}(\mathcal{R}) \times \mathrm{corad}(\mathcal{R}) \xrightarrow{h_1} \mathcal{R} \xrightarrow{h_2} \mathrm{der}(\mathcal{R}) \times \mathrm{rad}(\mathcal{R}).$$

Then

$$K(h_1) \simeq K(h_2) \simeq N(\mathcal{R}).$$

Proof. This is immediate from the definitions. □

Corollary 0.6.5.8. *For a root datum $\mathcal{R}$, the following conditions are equivalent:*

(i) $N(\mathcal{R}) = 0$, *that is,* $\mathrm{corad}(\mathcal{R}) \to \mathrm{rad}(\mathcal{R})$ *is an isomorphism;*

(ii) the canonical map $\mathrm{ss}(\mathcal{R}) \to \mathrm{der}(\mathcal{R})$ *is an isomorphism;*

(iii) h_1 *is an isomorphism;*

(iv) h_2 *is an isomorphism;*

(v) $\mathcal{R}$ *is the product of a semisimple root datum and a trivial root datum.*

Corollary 0.6.5.9. *For every root datum $\mathcal{R}$, there are isogenies*

$$\mathrm{ad}(\mathcal{R}) \times \mathcal{R}_0 \longrightarrow \mathcal{R} \longrightarrow \mathrm{sc}(\mathcal{R}) \times \mathcal{R}_0,$$

where $\mathcal{R}_0$ is the trivial root datum of rank $\mathrm{rgred}(\mathcal{R}) - \mathrm{rgss}(\mathcal{R})$.

Lemma 0.6.5.10. *Let $\mathcal{R}$ be a root datum, let Δ be a system of simple roots, and let $\Delta' \subset \Delta$. Consider the root datum*

$$\mathcal{R}_{\Delta'} = (M, M^*, R_{\Delta'}, R^*_{\Delta'})$$

of 0.3.4.7.

(i) If $\mathcal{R}$ is simply connected, then $\mathrm{der}(\mathcal{R}_{\Delta'})$ *is simply connected.*

(ii) If $\mathcal{R}$ is adjoint, then $\mathrm{ss}(\mathcal{R}_{\Delta'})$ *is adjoint.*

Proof. The two assertions are equivalent by duality. The second reduces to

$$M \cap \mathcal{V}(R_{\Delta'}) = \Gamma_0(R_{\Delta'}).$$

When $M = \Gamma_0(R)$, both sides are the subgroup of M generated by Δ'. □

0.6.6 Weights

Definition 0.6.6.1. For a root datum $\mathcal{R}$, put [28]

$$\Lambda(\mathcal{R}) = \{x \in \mathcal{V}(R) \mid (\alpha^*, x) \in \mathbb{Z} \text{ for every } \alpha^* \in R^*\}.$$

The elements of $\Lambda(\mathcal{R})$ are called the *weights* of $\mathcal{R}$. The weights of $\mathcal{R}^*$ are called the *coweights* of $\mathcal{R}$.

One has

$$\Gamma_0(R) \subset \Lambda(\mathcal{R}),$$

and $\Lambda(\mathcal{R})$ is stable under $W(\mathcal{R})$.

Lemma 0.6.6.2. *The bilinear map*

$$V^* \times V \longrightarrow \mathbb{Q}$$

induces a duality

$$\Gamma_0(R^*) \times \Lambda(\mathcal{R}) \longrightarrow \mathbb{Z}.$$

Proof. Immediate. □

Corollary 0.6.6.3. *Let*

$$\Delta^* = (\alpha_1^*, \ldots, \alpha_n^*)$$

be a system of simple coroots, and let $p_i \in \mathcal{V}(R)$, $i = 1, \ldots, n$, be defined by

$$(\alpha_i^*, p_j) = \delta_{ij}.$$

Thus

$$s_{\alpha_i}(p_i) = p_i - \alpha_i, \qquad s_{\alpha_i}(p_j) = p_j \quad (i \neq j).$$ [29]

Then $\Lambda(\mathcal{R})$ is the free abelian group generated by the p_i.

The p_i are called the *fundamental weights* corresponding to the system of simple coroots Δ^*.

Corollary 0.6.6.4. *For every $\alpha^* \in \Delta^*$,*

$$\left(\alpha^*, \sum_i p_i\right) = 1.$$

Consequently,

$$\sum_i p_i = \rho_{R_+}, \qquad R_+ = \mathscr{P}(\mathrm{ind}(\Delta)),$$

in the notation of 0.3.5.1.

[28] Editors' note: The notation $\Gamma(\mathcal{R})$ has been replaced by $\Lambda(\mathcal{R})$ to avoid any possible confusion with $\Gamma_0(R)$.

[29] Caution: when the datum is not reduced, the α_i need not form a system of simple roots.

Corollary 0.6.6.5. *For every $x \in \mathcal{V}(R)$,*

$$x = \sum_i (\alpha_i^*, x) p_i.$$

Since

$$R^* \subset \Gamma_0(R^*), \qquad R \subset \Lambda(\mathcal{R}),$$

the quadruple

$$(\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*)$$

is a root datum.

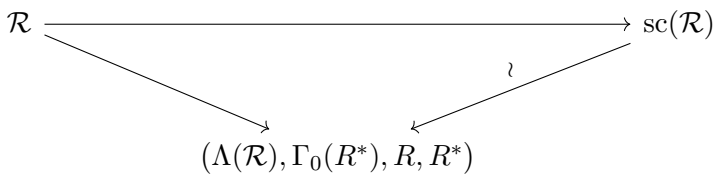
Corollary 0.6.6.6. *The canonical morphism*

$$\Gamma_0(R^*) \longrightarrow M^*$$

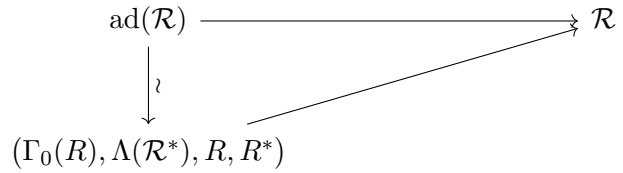
is the transpose of

$$x \longmapsto \sum_i (\alpha_i^*, x) p_i.$$

This map defines a morphism of root data, and there is a commutative diagram



Thus we have an explicit description of $\mathrm{sc}(\mathcal{R})$ in terms of the weights of $\mathcal{R}$. Likewise, one obtains a commutative diagram



Corollary 0.6.6.7. *The datum $\mathcal{R}$ is simply connected if and only if*

$$M = \Lambda(\mathcal{R}).$$

Remark 0.6.6.8. One has

$$\Lambda(\mathcal{R}) \cap M = \mathcal{V}(R) \cap M.$$

Consequently, $\mathcal{R}$ is semisimple if and only if

$$M \subset \Lambda(\mathcal{R}).$$

The results of §0.6.5 also give:

Corollary 0.6.6.9. *The datum $\mathcal{R}$ is the product of a simply connected datum and a trivial datum if and only if*

$$M \supset \Lambda(\mathcal{R}).$$

Consider now the canonical isogeny

$$f : \mathrm{ad}(\mathcal{R}) \longrightarrow \mathrm{sc}(\mathcal{R}),$$

and put $Z(\mathcal{R}) = K(f)$. One has

$$Z(\mathcal{R}) \simeq Z(\mathrm{sc}\,\mathcal{R}) \simeq Z(\mathrm{ad}\,\mathcal{R}).$$

Corollary 0.6.6.10. *There is a canonical isomorphism*

$$Z(\mathcal{R}) = \Lambda(\mathcal{R})/\Gamma_0(R).$$

More precisely, there is an exact sequence of $W(\mathcal{R})$-modules

$$0 \longrightarrow \Gamma_0(R) \longrightarrow \Lambda(\mathcal{R}) \longrightarrow Z(\mathcal{R}) \longrightarrow 0.$$

Corollary 0.6.6.11. *There is a canonical pairing*

$$Z(\mathcal{R}^*) \times Z(\mathcal{R}) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

which puts these groups in duality.

Remark 0.6.6.12. One has

$$Z(\mathcal{R} \times \mathcal{R}') \simeq Z(\mathcal{R}) \times Z(\mathcal{R}').$$

In particular, consider simply connected data $\mathcal{R}_i$, $i = 1, 2, \ldots, n$, and a trivial datum $\mathcal{R}_0$. Put

$$\mathcal{R} = \mathcal{R}_0 \times \mathcal{R}_1 \times \cdots \times \mathcal{R}_n.$$

If

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}_0 = (M_0, M_0^*, \varnothing, \varnothing),$$

then

$$M/\Gamma_0(R) \simeq M_0 \times Z(\mathcal{R}_1) \times \cdots \times Z(\mathcal{R}_n).$$

0.6.7 Automorphisms

By 0.6.1.4, an automorphism of $\mathcal{R}$ is an automorphism u of M such that

$$u(R) = R, \qquad {}^t u(R^*) = R^*.$$

In particular, every $w \in W(\mathcal{R})$ defines an automorphism of $\mathcal{R}$.

Lemma 0.6.7.1. *The group $W(\mathcal{R})$ is a normal subgroup of $\mathrm{Aut}(\mathcal{R})$. More precisely, if $u \in \mathrm{Aut}(\mathcal{R})$ and $\alpha \in R$, then*

$$u s_\alpha u^{-1} = s_{u(\alpha)}.$$

Proof. The proof is the same as that of 0.1.2.10. □

Proposition 0.6.7.2. *Let Δ be a system of simple roots, and put*

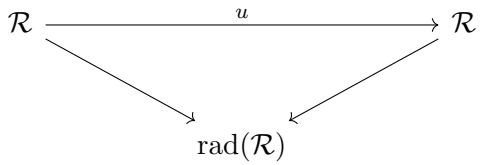
$$E_\Delta(\mathcal{R}) = \{u \in \mathrm{Aut}(\mathcal{R}) \mid u(\Delta) = \Delta\}.$$

Then $\mathrm{Aut}(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta(\mathcal{R})$.

Proof. This follows at once from the fact that $W(\mathcal{R})$ acts simply transitively on the systems of simple roots and from the fact that, if Δ is a system of simple roots of $\mathcal{R}$, then $u(\Delta)$ is a system of simple roots for every automorphism u of $\mathcal{R}$. □

We shall later give a simpler description of $E_\Delta(\mathcal{R})$ in the case of reduced irreducible root data.[30]

Definition 0.6.7.3. We denote by $\mathrm{Aut}^s(\mathcal{R})$[31] the set of $u \in \mathrm{Aut}(\mathcal{R})$ for which the following diagram is commutative:



We write

$$E^s_\Delta(\mathcal{R}) = E_\Delta(\mathcal{R}) \cap \mathrm{Aut}^s(\mathcal{R}).$$

Remark 0.6.7.4. If $u \in \mathrm{Aut}(\mathcal{R})$, then $u \in \mathrm{Aut}^s(\mathcal{R})$ if and only if

$$(u - \mathrm{id})(M) \subset \mathcal{V}(R).$$

In particular,

$$W(\mathcal{R}) \subset \mathrm{Aut}^s(\mathcal{R}).$$

Proposition 0.6.7.5. *For every system of simple roots Δ, the group $\mathrm{Aut}^s(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E^s_\Delta(\mathcal{R})$.*

By functoriality, every automorphism of $\mathcal{R}$ induces an automorphism of $\mathrm{ad}(\mathcal{R})$. Thus there is a canonical homomorphism

$$\mathrm{Aut}(\mathcal{R}) \longrightarrow \mathrm{Aut}(\mathrm{ad}(\mathcal{R})).$$

Lemma 0.6.7.6. *The homomorphism*

$$\mathrm{Aut}^s(\mathcal{R}) \longrightarrow \mathrm{Aut}(\mathrm{ad}(\mathcal{R}))$$

is injective.

Proof. Let u be an automorphism of M such that

$$(u - \mathrm{id})(M) \subset \mathcal{V}(R)$$

and

$$^t u(\alpha^*) = \alpha^* \qquad (\alpha^* \in R^*).$$

For every $x \in M$, one has

$$(\alpha^*, u(x) - x) = (^t u(\alpha^*) - \alpha^*, x) = 0.$$

Hence $u(x) - x = 0$, by 0.1.2.5. □

Lemma 0.6.7.7. *The group $\mathrm{Aut}^s(\mathcal{R})$ is finite.*

Proof. It suffices to prove that $\mathrm{Aut}(\mathcal{R})$ is finite when $\mathcal{R}$ is adjoint. In that case M is generated by R, so every automorphism of $\mathcal{R}$ is determined by the permutation it induces on R. □

Remark 0.6.7.8. It is immediate that $\mathrm{Aut}(\mathcal{R})$ (respectively $E_\Delta(\mathcal{R})$) is finite if and only if

$$\mathrm{rgred}(\mathcal{R}) - \mathrm{rgss}(\mathcal{R}) \leq 1.$$

[30] Editors' note: When $\mathcal{R}$ is simply connected or adjoint, see 7.4.5.

[31] Editors' note: The superscript s is intended to suggest "semisimple."

0.6.8 p-morphisms of reduced root data

Throughout this subsection, p is a fixed positive integer.

Definition 0.6.8.1. Let

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two reduced root data. A group homomorphism

$$f : M' \longrightarrow M$$

is called a *p-morphism from $\mathcal{R}'$ to $\mathcal{R}$* if there exist a bijection

$$u : R \xrightarrow{\sim} R'$$

and a map

$$q : R \longrightarrow \{p^n \mid n \in \mathbb{N}\}$$

such that:

(i) $f(u(\alpha)) = q(\alpha)\alpha$ for every $\alpha \in R$;

(ii) ${}^t f(\alpha^*) = q(\alpha)u(\alpha)^*$ for every $\alpha \in R$.[32]

Corollary 0.6.8.2. *A 1-morphism is precisely a morphism.*

Corollary 0.6.8.3. *The transpose of a p-morphism is a p-morphism.*

Lemma 0.6.8.4. *If $w \in W(\mathcal{R})$ and $\alpha \in R$, then*

$$q(w(\alpha)) = q(\alpha).$$

The map

$$s_\alpha \longmapsto s_{u(\alpha)}$$

extends to an isomorphism

$$u : W(\mathcal{R}) \xrightarrow{\sim} W(\mathcal{R}')$$

such that

$$u(w(\alpha)) = u(w)\big(u(\alpha)\big).$$

Proof. It suffices to prove, for $\alpha, \beta \in R$, that

$$u(s_\alpha(\beta)) = s_{u(\alpha)}u(\beta), \qquad q(s_\alpha(\beta)) = q(\beta).$$

Successively, one has

$$\begin{aligned} f\big(s_{u(\alpha)}u(\beta)\big) &= f(u(\beta)) - \big(u(\alpha)^*, u(\beta)\big)f(u(\alpha)) \\ &= q(\beta)\beta - q(\beta)q(\alpha)^{-1}(\alpha^*, \beta)q(\alpha)\alpha \\ &= q(\beta)\big(\beta - (\alpha^*, \beta)\alpha\big) \\ &= q(\beta)s_\alpha(\beta). \end{aligned}$$

[32] Editors' note: These two conditions imply

$$q(\alpha)\big(u(\alpha)^*, u(\beta)\big) = q(\beta)(\alpha^*, \beta)$$

for all $\alpha, \beta \in R$.

If

$$\gamma = u^{-1}\big(s_{u(\alpha)}u(\beta)\big),$$

then

$$q(\gamma)\gamma = f(u(\gamma)) = q(\beta)s_\alpha(\beta).$$

The roots γ and $s_\alpha(\beta)$ are proportional over $\mathbb{Q}$, hence equal or opposite. Since $q(\gamma)$ and $q(\beta)$ are positive, it follows that

$$q(\gamma) = q(\beta), \qquad \gamma = s_\alpha(\beta).$$

□

Definition 0.6.8.5. The numbers $q(\alpha)$ are called the *root exponents* of f.

Example 0.6.8.6. Let $\mathcal{R}$ be a reduced root datum and let $q = p^n$, $n \in \mathbb{N}$. Multiplication by q,

$$M \longrightarrow M, \qquad x \longmapsto qx,$$

is a p-morphism whose root exponents are all equal to q (and for which $u = \mathrm{id}$). It is denoted

$$q : \mathcal{R} \longrightarrow \mathcal{R}.$$

Proposition 0.6.8.7. *In the notation of 0.6.8.1, u induces an isomorphism from the set of systems of simple roots (respectively positive roots) of R onto the corresponding set for R'.*

Proof. This follows immediately from 0.3.1.5 (respectively 0.3.2.1). □

0.7 Structure

0.7.1 Decomposition of a root datum

Proposition 0.7.1.1. *Let $\mathcal{R}$ be a root datum and let Δ be a system of simple roots.*

(i) Let R' and R'' be two closed symmetric sets of roots forming a partition of R. If

$$\Delta' = \Delta \cap R', \qquad \Delta'' = \Delta \cap R'',$$

then

$$R' = R_{\Delta'}, \qquad R'' = R_{\Delta''},$$

and every root of Δ' is orthogonal to every root of Δ''.

(ii) Let Δ' and Δ'' be two orthogonal subsets forming a partition of Δ. Then $R_{\Delta'}$ and $R_{\Delta''}$ form a partition of R.

Proof. We first prove (i).

Lemma 0.7.1.2. *Under the hypotheses of* (i)*, if α, β, and $\alpha+\beta$ are roots, then all three belong to R', or all three belong to R''.*

Suppose, for example, that $\alpha+\beta \in R'$. One cannot have $\alpha, \beta \in R''$, since R'' is closed. Suppose therefore that $\alpha \in R'$. Then $-\alpha \in R'$, and

$$\beta = (\beta+\alpha) - \alpha \in R'.$$

We now prove $R' = R_{\Delta'}$ by induction on the order of a positive root $\alpha \in R' \cap \mathscr{P}(\Delta)$. If $\mathrm{ord}_\Delta(\alpha) = 1$, then

$$\alpha \in R' \cap \Delta = \Delta'.$$

If $\mathrm{ord}_\Delta(\alpha) > 1$, there is $\beta \in \Delta$ such that $\alpha - \beta \in R$. By the lemma,

$$\beta \in \Delta', \qquad \alpha - \beta \in R'.$$

Hence $\alpha - \beta \in R_{\Delta'}$ by induction, and finally

$$\alpha = (\alpha - \beta) + \beta \in R_{\Delta'}.$$

We next prove that Δ' and Δ'' are orthogonal. If $\alpha \in \Delta'$ and $\beta \in \Delta''$, then

$$(\beta^*, \alpha) \leq 0.$$

If $(\beta^*, \alpha) \neq 0$, then $\beta + \alpha$ is a root, contrary to the lemma.

We now prove (ii). If Δ' or Δ'' is empty, the assertion is immediate. Otherwise, if $R_{\Delta'}$ and $R_{\Delta''}$ do not form a partition of R, there is a root α of the form

$$\alpha = \sum_i m'_i \alpha'_i + \sum_j m''_j \alpha''_j, \qquad m'_i, m''_j \in \mathbb{Z}_+,$$

where the α'_i (respectively α''_j) belong to Δ' (respectively Δ''). Applying 0.3.1.2, and interchanging Δ' and Δ'' if necessary, yields a relation

$$\delta = \gamma + \beta, \qquad \gamma \in R_{\Delta'}, \quad \beta \in \Delta'', \quad \delta \in R.$$

But $(\beta^*, \gamma) = 0$, so $\gamma - \beta$ is also a root by 0.2.2.5, which is impossible. □

Proposition 0.7.1.3. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

(i) There is no nontrivial partition of R into two closed symmetric subsets.

(ii) For some (respectively every) system of simple roots Δ of R, there is no partition of Δ into two nonempty orthogonal subsets.

(iii) The natural representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible.

(iv) For every pair (α, β) of roots, there is a sequence of roots

$$\alpha_0, \alpha_1, \ldots, \alpha_n, \qquad \alpha_0 = \alpha, \quad \alpha_n = \beta,$$

such that α_i and α_{i+1} are nonorthogonal for $i = 0, \ldots, n-1$.

Proof. The equivalence (i)⇔(ii) follows from 0.7.1.1, and (iv)⇒(ii) is immediate. Conversely, if (ii) holds for Δ, then (iv) holds when $\alpha, \beta \in \Delta$. Every root is nonorthogonal to some simple root, by 0.3.1.1, for example. Moreover, (iii)⇒(i): under the hypotheses of 0.7.1.1, the space $\mathcal{V}(R')$ is stable under $W(\mathcal{R})$. It remains to prove (i)⇒(iii).

Let H be a vector subspace of $\mathcal{V}(R)$, stable under $W(\mathcal{R})$. For every $\alpha \in R$, the equality $s_\alpha(H) = H$ immediately gives

$$\alpha \in H \qquad \text{or} \qquad \alpha^* \in H^\perp,$$

where $H^\perp$ denotes the orthogonal complement of H in $\mathcal{V}(R^*)$, which is in duality with $\mathcal{V}(R)$. Put

$$R' = \{\alpha \in R \mid \alpha \in H\}, \qquad R'' = \{\alpha \in R \mid \alpha^* \in H^\perp\}.$$

This gives a partition of R into two closed symmetric subsets. □

Definition 0.7.1.4. A root datum (respectively root system) satisfying the equivalent conditions of 0.7.1.3 and having nonzero semisimple rank is called *irreducible*.

Corollary 0.7.1.5. *For every root datum $\mathcal{R}$, there is a unique partition, up to order, of R into closed symmetric irreducible subsets.*

Corollary 0.7.1.6. *Every adjoint (respectively simply connected) root datum is a product of irreducible adjoint (respectively simply connected) root data.*

Proof. It suffices to consider the adjoint case. The assertion then follows from the fact that, under the hypotheses of 0.7.1.1,

$$\Gamma_0(R) = \Gamma_0(R') \times \Gamma_0(R'').$$

□

Corollary 0.7.1.7. *For every root datum (respectively reduced root datum) $\mathcal{R}$, there is an isogeny*

$$\mathcal{R} \longrightarrow \mathcal{R}',$$

where $\mathcal{R}'$ is a product of a trivial root datum and irreducible simply connected root data (respectively irreducible reduced simply connected root data).

0.7.2 Properties of irreducible root data

Definition 0.7.2.1. Let $\mathcal{R}$ be an irreducible root datum. For every $\alpha \in R$, put

$$\mathrm{long}(\alpha) = \ell(\alpha)/\ell(\alpha_0),$$

where $\alpha_0 \in R$ is such that $\ell(\alpha_0)$ is minimal. The number $\mathrm{long}(\alpha)$ is called the *length* of α.

Lemma 0.7.2.2. *Let $\mathcal{R}$ be an irreducible root datum. The Weyl group acts transitively on the set of roots of a given length.*

Proof. Let $\alpha, \beta \in R$. Since the representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible, α cannot be orthogonal to all $w(\beta)$, $w \in W(\mathcal{R})$. Hence some $w(\beta)$ is nonorthogonal to α. Since

$$\ell(w(\beta)) = \ell(\beta),$$

the assertion follows from 0.2.3.2. □

Lemma 0.7.2.3. *If $\mathcal{R}$ is irreducible and reduced, then*

$$\mathrm{long}(R) \in \{\{1\}, \{1,2\}, \{1,3\}\}.$$

Proof. By the observation used above, for every $\alpha, \beta \in R$ there is $w \in W(\mathcal{R})$ such that $w(\beta)$ is nonorthogonal to α. Therefore, by 0.2.3.1,

$$\frac{\ell(\alpha)}{\ell(\beta)} \in \left\{1, 2, 3, \frac{1}{2}, \frac{1}{3}\right\}.$$

Thus $\mathrm{long}(\alpha)$ is 1, 2, or 3. But lengths 2 and 3 cannot both occur, since their ratio 2/3 is not in the displayed set. □

Remark 0.7.2.4. Reasoning similarly gives the following result. If $\mathcal{R}$ is irreducible and nonreduced, with $\mathrm{rgss}(\mathcal{R}) > 1$, then

$$\mathrm{long}(R) = \{1, 2, 4\}.$$

If $R_i = \mathrm{long}^{-1}(i)$, then

$$\mathrm{ind}(R) = R_1 \cup R_2, \qquad R_4 = 2R_1,$$

and two nonproportional roots of R_1 are orthogonal. Conversely, suppose that R is an irreducible reduced root system such that $\mathrm{long}(R) = \{1, 2\}$. Put $R_i = \mathrm{long}^{-1}(i)$, and suppose that two nonproportional roots of R_1 are orthogonal. Then $R \cup 2R_1$ is irreducible and nonreduced, and

$$\mathrm{ind}(R \cup 2R_1) = R.$$

Lemma 0.7.2.5. *If $\mathcal{R}$ is an irreducible root datum, then $\mathcal{R}^*$ is irreducible as well, and the product*

$$\mathrm{long}(\alpha)\,\mathrm{long}(\alpha^*)$$

is constant as α runs through R.

Proof. This follows immediately from 0.7.1.3 (iv) and 0.2.2.6. □

Definition 0.7.2.6. Let $\mathcal{R}$ be any root datum. The *length* of $\alpha \in R$, denoted $\mathrm{long}(\alpha)$, is the length of α in its irreducible component.

Lemma 0.7.2.7. *There is a unique group homomorphism*

$$u : \Gamma_0(R) \longrightarrow \Gamma_0(R^*)$$

such that

$$u(\alpha) = \mathrm{long}(\alpha)\alpha^* \qquad (\alpha \in R).$$

Proof. By 0.3.5.5, it is enough to verify that, whenever $\alpha, \beta, \alpha + \beta \in R$,

$$\mathrm{long}(\alpha)\alpha^* + \mathrm{long}(\beta)\beta^* = \mathrm{long}(\alpha + \beta)(\alpha + \beta)^*.$$

But $\alpha, \beta, \alpha + \beta$ belong to the same irreducible component of R, by 0.7.1.2; the assertion is therefore reduced to 0.1.2.2. □

Remark 0.7.2.8. Let u be as in 0.7.2.7. For $\alpha, \beta \in R$,

$$(u(\alpha), \beta) = (u(\beta), \alpha).$$

Indeed, this amounts to

$$\mathrm{long}(\alpha)(\alpha^*, \beta) = \mathrm{long}(\beta)(\beta^*, \alpha),$$

which is clear when α and β are orthogonal. If they are not orthogonal, they lie in the same irreducible component of R, and the assertion reduces to 0.1.2.1, formula (9).

Remark 0.7.2.9. The symmetric bilinear form

$$(x, y) \longmapsto (u(x), y)$$

is positive and nondegenerate on $\Gamma_0(R)$.

Proof. Let R_i be the irreducible components of R. Then

$$\Gamma_0(R) = \prod_i \Gamma_0(R_i),$$

and the bilinear form $(u(x), y)$ is the product, on the groups $\Gamma_0(R_i)$, of the forms

$$2^{-1}\ell(\alpha_i)^{-1}(p(x), y),$$

where $\ell(\alpha_i)$ is the minimum of $\ell(\alpha)$ for $\alpha \in R_i$. These symmetric bilinear forms are positive and nondegenerate by 0.1.2.6. □

0.7.3 Cartan matrix

Let $\mathcal{R}$ be a root datum. If Δ is a system of simple roots, the *Cartan matrix* of $\mathcal{R}$ relative to Δ is the square matrix indexed by Δ and defined by

$$a_{\alpha,\beta} = (\alpha^*, \beta), \qquad \alpha, \beta \in \Delta.$$

If Δ' is another system of simple roots and $w \in W(\mathcal{R})$ satisfies $w(\Delta) = \Delta'$, then

$$(w(\alpha)^*, w(\beta)) = (\alpha^*, \beta).$$

Thus the Cartan matrix relative to Δ' is obtained from that relative to Δ by the isomorphism $\Delta \to \Delta'$ of index sets defined by w. Consequently, up to isomorphism of its index set, the Cartan matrix depends only on $\mathcal{R}$.

Proposition 0.7.3.1. *The Cartan matrix has the following properties.*

(i) $a_{\alpha,\alpha} = 2$, *and* $a_{\alpha,\beta} \leq 0$ *for* $\alpha \neq \beta$.

(ii) $a_{\alpha,\beta} = 0$ *implies* $a_{\beta,\alpha} = 0$.

(iii) There are strictly positive integers $m_\alpha = \mathrm{long}(\alpha)$ *such that the matrix*

$$(m_\alpha a_{\alpha,\beta})$$

is symmetric, positive, and nondegenerate.

(iv) The principal minors of $(a_{\alpha,\beta})_{\alpha,\beta\in\Delta}$, *namely*

$$\det(a_{\alpha,\beta})_{\alpha,\beta\in\Delta'} \qquad (\Delta' \subset \Delta),$$

are strictly positive.

(v) One has

$$s_\alpha(\beta) = \beta - a_{\alpha,\beta}\alpha, \qquad s_\alpha(\beta^*) = \beta^* - a_{\beta,\alpha}\alpha^*.$$

Proof. Part (v) is a definition; (i) follows from 0.3.2.11, (ii) from 0.2.2.2, and (iii) from 0.7.2.9. Part (iv) follows immediately from (iii) and

$$\det(m_\alpha a_{\alpha,\beta})_{\alpha,\beta\in\Delta'} = \left(\prod_{\alpha\in\Delta'} m_\alpha\right) \det(a_{\alpha,\beta})_{\alpha,\beta\in\Delta'}.$$

□

Proposition 0.7.3.2. *Let $\mathcal{R}$ and $\mathcal{R}'$ be two reduced simply connected (respectively adjoint) root data, let Δ and Δ' be systems of simple roots of $\mathcal{R}$ and $\mathcal{R}'$, and let $u : \Delta \to \Delta'$ be an isomorphism. Denote the corresponding Cartan matrices by $(a_{\alpha,\beta})$ and $(a'_{\alpha',\beta'})$, and suppose that*

$$a'_{u(\alpha),u(\beta)} = a_{\alpha,\beta}.$$

Then there is a unique isomorphism from $\mathcal{R}$ onto $\mathcal{R}'$ that induces u on Δ.

Proof. It is enough to give the proof in the adjoint case. Then $M = \Gamma_0(R)$ and $M' = \Gamma_0(R')$ are the free abelian groups generated by Δ and Δ'. Hence there is a unique group isomorphism $M \to M'$ inducing u on Δ; denote it again by u.

We show that $u(R) \subset R'$. Every root α of R can be written

$$\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}), \qquad \alpha_i \in \Delta.$$

By the hypothesis on u and the relations 0.7.3.1 (v),

$$u(\alpha) = s_{u(\alpha_1)} \cdots s_{u(\alpha_n)}(u(\alpha_{n+1})).$$

It remains to prove that ${}^t u(R'^*) \subset R^*$. This follows from the fact that the elements of M^* (respectively M'^*) are determined by their duality with R or Δ (respectively R' or Δ'), by 0.1.2.5. □

Corollary 0.7.3.3. *A reduced simply connected or adjoint root datum is determined up to isomorphism by its Cartan matrix.*

Corollary 0.7.3.4. *Let $\mathcal{R}$ be a reduced simply connected (respectively adjoint) root datum and let Δ be a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the set Δ that preserve the Cartan matrix.*

Remark 0.7.3.5. The existence of a root datum corresponding to a given Cartan matrix satisfying, for example, (i), (ii), and (iv) is not easily settled directly without using the classification.

0.7.4 Dynkin diagrams

Definition 0.7.4.1. A *Dynkin-diagram structure* on a finite set Δ (the word "scheme" being banished for obvious reasons) consists of a set of pairs of distinct elements of Δ, called *linked pairs*, together with a map

$$\Delta \longrightarrow \{1, 2, 3\}.$$

The notion of isomorphism between such structures is evident.

Definition 0.7.4.2. Let $\mathcal{R}$ be a root datum and let Δ be a system of simple roots. The *Dynkin diagram* of $\mathcal{R}$ relative to Δ is the set Δ, in which two simple roots are linked if and only if they are nonorthogonal, and each root is assigned its length.

Proposition 0.7.4.3. *The Dynkin diagram and the Cartan matrix determine one another uniquely.*

Proof. The equivalence

$$\alpha \text{ and } \beta \text{ are not linked} \iff a_{\alpha,\beta} = 0$$

and the relation

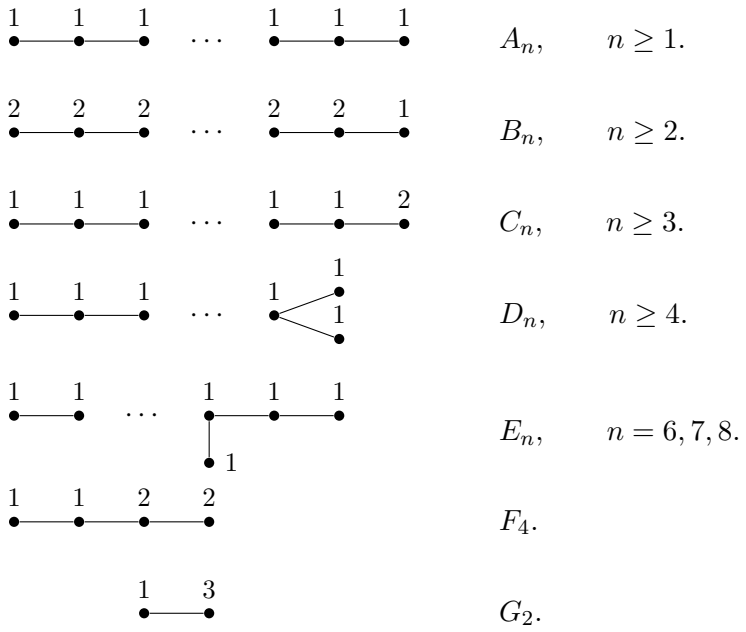
$$\operatorname{long}(\alpha) a_{\alpha,\beta} = \operatorname{long}(\beta) a_{\beta,\alpha},$$

with $\inf \operatorname{long}(\alpha) = 1$ in each connected component of the diagram, determine the $a_{\alpha,\beta}$ from the links and lengths, and conversely. The details are left to the reader. □

Corollary 0.7.4.4. *A reduced simply connected or adjoint root datum is determined by its Dynkin diagram.*

Corollary 0.7.4.5. *Let $\mathcal{R}$ be a reduced simply connected or adjoint root datum and let Δ be a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the Dynkin diagram of $\mathcal{R}$ relative to Δ; that is, with the group of permutations of Δ preserving lengths and links.*

Remark 0.7.4.6. The usual method[33] [34] classifies the connected Dynkin diagrams and shows that each actually corresponds to an irreducible reduced simply connected root datum. One obtains the familiar types:



By 0.7.4.5, the corresponding group $E_\Delta(\mathcal{R})$ is obtained at once:

$$\begin{aligned} E_\Delta(\mathcal{R}) &= \{e\}, && A_1,\ B_n,\ C_n,\ E_7,\ E_8,\ F_4,\ G_2, \\ E_\Delta(\mathcal{R}) &= \mathbb{Z}/2\mathbb{Z}, && A_n\ (n \geq 2),\ D_n\ (n \geq 5),\ E_6, \\ E_\Delta(\mathcal{R}) &= \mathfrak{S}_3, && D_4. \end{aligned}$$

[33] See Bourbaki, *Groupes et algèbres de Lie*, Chapter VI, no. 4.2, or the Sophus Lie Seminar.

[34] Editors' note: for a slightly different proof, see also [De80].

0.7.5 Complements on p-morphisms

Let $f : \mathcal{R} \to \mathcal{R}'$ be a p-morphism (cf. 0.6.8). It is clear from the definitions that the bijection $u : R \xrightarrow{\sim} R'$ associated with f makes the systems of simple roots, the systems of positive roots, the irreducible components, and so forth of R and R' correspond. We may therefore suppose, for simplicity, that R and R' are irreducible.

Lemma 0.7.5.1. *If R and R' are irreducible, there is a $k \in \mathbb{Q}$ such that, for every $\alpha \in R$,*

$$k \operatorname{long}(u(\alpha)) = q(\alpha)^2 \operatorname{long}(\alpha).$$

Proof. One has

$$\operatorname{long}(\alpha)(\alpha^*, \beta) = \operatorname{long}(\beta)(\beta^*, \alpha)$$

and, similarly,

$$\operatorname{long}(u(\alpha))(u(\alpha)^*, u(\beta)) = \operatorname{long}(u(\beta))(u(\beta)^*, u(\alpha)).$$

It follows immediately that, for nonorthogonal α, β,

$$\frac{q(\alpha)^2 \operatorname{long}(\alpha)}{\operatorname{long}(u(\alpha))} = \frac{q(\beta)^2 \operatorname{long}(\beta)}{\operatorname{long}(u(\beta))}.$$

The conclusion then follows from 0.7.1.3 (iv). □

Remark 0.7.5.2. It follows from 0.7.2.2 and 0.6.8.4 that $q(\alpha)$ depends only on $\operatorname{long}(\alpha)$. One then sees readily that, if $q(\alpha)$ is not constant, the product $q(\alpha)\operatorname{long}(\alpha)$ is constant; hence $p = 2$ or 3. Inspection of the preceding diagrams gives the four possible cases (the same letter denotes a Dynkin diagram and the corresponding reduced simply connected root datum):

$$\begin{aligned} &p = 2, \quad B_n \xrightarrow{f_1} C_n, \quad C_n \xrightarrow{f_2} B_n, \quad (C_2 = B_2), \\ &p = 2, \quad F_4 \xrightarrow{g} F_4, \\ &p = 3, \quad G_2 \xrightarrow{h} G_2. \end{aligned}$$

The reader will observe that $f_1 \circ f_2$, $f_2 \circ f_1$, $g \circ g$, and $h \circ h$ are p-morphisms of the form described in 0.6.8.6.

7.5.3. The preceding description shows immediately that, if $\mathcal{R}$ and $\mathcal{R}'$ are two reduced root data of semisimple rank at most 2 and there is a p-morphism from $\mathcal{R}'$ to $\mathcal{R}$, then $\mathcal{R}$ and $\mathcal{R}'$ have the same type. More precisely, one has the following table.

Notation. Let $f : \mathcal{R}' \to \mathcal{R}$ be a p-morphism. The symbols q and q_1 denote arbitrary positive powers of p. For rank-two systems we use the notation of Section 0.4, with α, β the simple roots and $\ell(\alpha) \leq \ell(\beta)$.

Type	p	**values of** f	**values of** ${}^t f$
Trivial	arbitrary	—	—
A_1	arbitrary	$f(\alpha') = q\alpha$	${}^t f(\alpha^*) = q\alpha'^*$
$A_1 \times A_1$	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q_1\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q_1\beta'^*$
A_2, B_2, G_2	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q\beta'^*$
B_2	$p = 2$	$f(\alpha') = q\beta$ $f(\beta') = 2q\alpha$	${}^t f(\alpha^*) = 2q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$
G_2	$p = 3$	$f(\alpha') = q\beta$ $f(\beta') = 3q\alpha$	${}^t f(\alpha^*) = 3q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$

Bibliography

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[De80] M. Demazure, "A, B, C, D, E, F, etc.," pp. 221–227 in *Séminaire sur les singularités des surfaces* (Palaiseau, 1976–1977), eds. M. Demazure, H. C. Pinkham, and B. Teissier, Lecture Notes in Mathematics **777**, Springer-Verlag, 1980.

Exposé XXII
Reductive Groups: Splittings, Subgroups, Quotient Groups

by Michel Demazure

[0] This Exposé consists of two parts. The first (§§1–5.5) gathers the technical results needed for the proofs of the uniqueness and existence theorems. The second (§5.6 to the end) will not be used in that proof; the end of §5 will be used in particular in Exposé XXVI, which is devoted to parabolic subgroups; §6 establishes, in the scheme-theoretic setting, the classical results concerning the derived group of a reductive group.

1. Roots and Coroots. Split Groups and Root Data

Theorem 1.1. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and α a root of G relative to T.*

(i) *There exists a unique T-equivariant homomorphism of group schemes*

$$\exp_\alpha : W(\mathfrak{g}^\alpha) \longrightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^\alpha \to \mathfrak{g}$. This homomorphism is a closed immersion. The corresponding homomorphism

$$T \cdot_\alpha W(\mathfrak{g}^\alpha) \longrightarrow G$$

is likewise a closed immersion.

If $p_\alpha : \mathbb{G}_{a,S} \to G$ is a monomorphism normalized by T, with multiplier α, there exists a unique $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^{\times 1}$ such that $p_\alpha(x) = \exp_\alpha(xX_\alpha)$; one has $\mathrm{Lie}(p_\alpha)(1) = X_\alpha$, and the preceding two formulas establish a bijective correspondence between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of monomorphisms $\mathbb{G}_{a,S} \to G$ normalized by T, with multiplier α.

(ii) *There exists a unique duality, denoted $(X, Y) \longmapsto XY$,*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathcal{O}_S,$$

and a unique homomorphism of group schemes

$$\alpha^* : \mathbb{G}_{m,S} \longrightarrow T,$$

[0] Editor's note: Version dated October 13, 2024.

[1] Editor's note: The set $\Gamma(S, \mathfrak{g}^\alpha)^\times$ is defined in Exposé XIX, 4.4.1.

such that formula (F) *of Exposé XX, 2.1 holds. One has*

$$\alpha \circ \alpha^* = 2, \qquad (-\alpha)^* = -\alpha^*,$$

and α^* *is given by the formula of Exposé XX, 2.7.*

Indeed, a homomorphism normalized by T, with multiplier α, necessarily factors through the closed subgroup

$$Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha)$$

of G (cf. Exposé XIX, 3.9). Now (Z_α, T, α) is an elementary S-system (Exposé XX, 1.4), and one is reduced to the results of Exposé XX (1.5, 2.1, and 5.9).

Remark 1.2. Part (i) of Theorem 1.1 remains valid if one assumes only that α is a character of T, nontrivial on every fiber. Indeed, there is then a decomposition

$$S = S' \amalg S'',$$

such that $\alpha|_{S'}$ is a root of $G_{S'}$ relative to $T_{S'}$, and $\mathfrak{g}^\alpha|_{S''} = 0$. If $S = S'$, one is reduced to 1.1; if $S = S''$, the result is trivial; the general case follows at once.

Notations 1.3. As in Exposé XX, let U_α denote the image of $W(\mathfrak{g}^\alpha)$; it is a closed subgroup of G, canonically endowed with a vector-group structure. We call it the *vector group associated with the root* α. We say that α^* is the *coroot associated with* α. Sections

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha) \quad \text{and} \quad X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})$$

are said to be *paired* if $X_\alpha X_{-\alpha} = 1$. Then $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, and likewise for $X_{-\alpha}$. The corresponding homomorphisms p_α and $p_{-\alpha}$ are contragredient to one another in the sense of Exposé XX, 1.5[2], and one has

$$p_\alpha(x)p_{-\alpha}(y) = p_{-\alpha}\left(\frac{y}{1+xy}\right)\alpha^*(1+xy)p_\alpha\left(\frac{x}{1+xy}\right).$$

Proposition 1.4. *Under the hypotheses of 1.1, let* $w \in \underline{\mathrm{Norm}}_G(T)(S)$. *Then*

$$\beta = \alpha \circ \mathrm{int}(w)^{-1} : T \longrightarrow \mathbb{G}_{m,S}$$

is a root of G *relative to* T, $\beta^* = \mathrm{int}(w) \circ \alpha^*$ *is the corresponding coroot, and the following diagram is commutative:*

$$\begin{array}{ccc} W(\mathfrak{g}^\alpha) & \xrightarrow{\exp_\alpha} & G \\ {\scriptstyle \mathrm{Ad}(w)}\downarrow & & \downarrow{\scriptstyle \mathrm{int}(w)} \\ W(\mathfrak{g}^\beta) & \xrightarrow{\exp_\beta} & G. \end{array}$$

Trivial: transport of structure.

[2]Editor's note: That is, if $\langle\ ,\ \rangle$ denotes the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$, and if $\mathfrak{g}^\alpha$ is identified with U_α via $p_\alpha(x) = \exp(xX_\alpha)$, and likewise for $-\alpha$, then $\langle p_\alpha(x), p_{-\alpha}(y)\rangle = xy$.

Definitions 1.5. (a) Under the hypotheses of 1.1, let s_α denote the automorphism of T defined by

$$s_\alpha(t) = t \cdot \alpha^*(\alpha(t))^{-1}.$$

Let (,) denote the canonical pairing

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T) \times \underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S}) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, \mathbb{G}_{m,S}) = \mathbb{Z}_S.$$

Then s_α acts on $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S})$, respectively on $\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T)$, by the following formulas, where χ, respectively u, denotes an arbitrary section of the corresponding scheme:

$$s_\alpha(\chi) = \chi - (\alpha^*, \chi)\alpha, \qquad s_\alpha(u) = u - (u, \alpha)\alpha^*.$$

One has

$$s_\alpha \circ s_\alpha = \mathrm{id}, \qquad s_{-\alpha} = s_\alpha.$$

(b) If $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then the inner automorphism $w_\alpha(X)$ of T, defined by

$$w_\alpha(X) = \exp_\alpha(X) \exp_{-\alpha}(-X^{-1}) \exp_\alpha(X)$$

(cf. Exposé XX, 3.1), coincides with s_α (loc. cit.). Proposition 1.4 therefore gives the following.

Corollary 1.6. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and α, β two roots of G relative to T. Then*

$$s_\alpha(\beta) = \beta - (\alpha^*, \beta)\alpha$$

is a root of G relative to T, and its corresponding coroot is

$$s_\alpha(\beta)^* = s_\alpha(\beta^*) = \beta^* - (\beta^*, \alpha)\alpha^*.$$

Corollary 1.7. *Under the preceding conditions, $\alpha^* = \beta^*$ implies $\alpha = \beta$.*

Indeed, if $\alpha^* = \beta^*$, one has (cf. Exposé XXI, 1.4)

$$s_\beta(\alpha) = \alpha - 2\beta, \qquad s_\alpha(\beta) = \beta - 2\alpha,$$

and hence immediately

$$(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha).$$

If $\beta \neq \alpha$, there exists $s \in S$ such that $\alpha_s \neq \beta_s$. The preceding formula would then give infinitely many distinct roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$, which is impossible.

Definitions 1.8.0. [3] If $u : \mathbb{G}_{m,S} \to T$ is a homomorphism of groups, we say that u is a *coroot* of G relative to T if there exists a root α of G relative to T such that $\alpha^* = u$. Consider the functor $\mathcal{R}^*$ of coroots of G relative to T, defined by

$$\mathcal{R}^*(S') = \{\text{coroots of } G_{S'} \text{ relative to } T_{S'}\}.$$

If $\mathcal{R}$ is the functor of roots of G relative to T (Exposé XIX, 3.8), there is a canonical morphism $\mathcal{R} \to \mathcal{R}^*$. By 1.7 and Exposé XIX, 3.8, one has:

[3] Editor's note: We added the numbering 1.8.0 to highlight these definitions.

Corollary 1.8. *The canonical morphism $\mathcal{R} \to \mathcal{R}^*$ is an isomorphism. In particular, $\mathcal{R}^*$ is represented by a twisted finite constant S-scheme that is an open and closed subscheme of*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T).$$

This leads to the following definition.

Definition 1.9. *Let S be a scheme and T an S-torus. A* twisted root datum *in T consists of*

(i) *a finite subscheme $\mathcal{R}$ of*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S});$$

(ii) *a finite subscheme $\mathcal{R}^*$ of*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T);$$

(iii) *an isomorphism $\mathcal{R} \xrightarrow{\sim} \mathcal{R}^*$, denoted $\alpha \mapsto \alpha^*$,*

satisfying the following conditions:

(DR 1) *For every $S' \to S$ and every $\alpha \in \mathcal{R}(S')$, one has $\alpha \circ \alpha^* = 2$.*

(DR 2) *For every $S' \to S$ and all $\alpha, \beta \in \mathcal{R}(S')$, one has*

$$\alpha - (\beta^*, \alpha)\beta \in \mathcal{R}(S'), \qquad \alpha^* - (\alpha^*, \beta)\beta^* \in \mathcal{R}^*(S').$$

Moreover, the twisted root datum is said to be reduced *if, for every nonempty $S' \to S$ and every $\alpha \in \mathcal{R}(S')$, one has $2\alpha \notin \mathcal{R}(S')$.*

Proposition 1.10. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and let $\mathcal{R}$, respectively $\mathcal{R}^*$, be the scheme of roots, respectively coroots, of G relative to T. Then $(\mathcal{R}, \mathcal{R}^*)$ is a reduced twisted root datum in T.*

It remains only to verify that this twisted root datum is reduced. This was done in Exposé XIX, 3.10.

1.11. Let $T = \mathrm{D}_S(M)$ be a *trivialized* torus. If M^* denotes the abelian group dual to M, there are canonical isomorphisms (cf. Exposé VIII, 1.5)

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S}) \xrightarrow{\sim} M_S, \qquad \underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T) \xrightarrow{\sim} M_S^*,$$

and hence group isomorphisms

$$\mathrm{Hom}_{S\text{-gr}}(T, \mathbb{G}_{m,S}) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{loc.const.}}(S, M),$$

$$\mathrm{Hom}_{S\text{-gr}}(\mathbb{G}_{m,S}, T) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{loc.const.}}(S, M^*).$$

A character of T, respectively a group homomorphism $\mathbb{G}_{m,S} \to T$, is said to be *constant* relative to the given trivialization if the preceding isomorphism carries it to a constant map from S to M, respectively to M^*.

1.12. With the same notation, let (M, M^*, R, R^*) be a root datum (Exposé XXI). Then (R_S, R_S^*) is a twisted root datum in T. Conversely, if $(\mathcal{R}, \mathcal{R}^*)$ is a twisted root datum in a torus T, a *splitting* of this root datum means the data of an ordinary root datum (M, M^*, R, R^*) and an isomorphism

$$T \simeq \mathrm{D}_S(M)$$

that carries $(\mathcal{R}, \mathcal{R}^*)$ to (R_S, R_S^*).

Definition 1.13. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. A* splitting of G relative to T *consists of*

(i) *an abelian group M and an isomorphism $T \simeq \mathrm{D}_S(M)$;*

(ii) *a system of roots R of G relative to T* (Exposé XIX, 3.6),

satisfying the following two conditions:

(D 1) *S is nonempty, and the roots $\alpha \in R$* (respectively, the corresponding coroots) *identify with constant functions from S to M* (respectively, to M^*).

(D 2) *The $\mathfrak{g}^\alpha$, $\alpha \in R$, are free $\mathcal{O}_S$-modules.*

We say that G is *splittable relative to T* if a splitting of G relative to T exists. A *splitting of G* consists of a maximal torus T of G together with a splitting of G relative to T. We say that G is *splittable* if it admits a splitting. A *split S-group* is a reductive S-group equipped with a splitting; we denote it by a symbol of the form (G, T, M, R), or simply by G when there is no risk of confusion.

Condition (D 1) canonically identifies R, respectively R^*, with a subset of M, respectively of M^*.

Proposition 1.14. *Let S be a nonempty scheme and let (G, T, M, R) be a split S-group. Then*

$$\mathcal{R}(G, T, M, R) = (M, M^*, R, R^*)$$

is a reduced root datum (Exposé XXI, 1.1 and 2.1.3)*; it is a splitting of the twisted root datum of 1.10.*

This follows trivially from 1.10 and Exposé XIX, 3.7.

For brevity we shall sometimes write $\mathcal{R}(G, T, M, R) = \mathcal{R}(G)$. We shall systematically use the notation V, $V(\mathcal{R})$, $W, \ldots$ of Exposé XXI.

Remark 1.15. (a) If S is nonempty and connected, respectively if $\mathrm{Pic}(S) = 0$, condition (D 1), respectively condition (D 2), is automatically satisfied.

(b) If (G, T, M, R) is a split S-group, then for every nonempty $S' \to S$, $(G_{S'}, T_{S'}, M, R)$ is a split S'-group and

$$\mathcal{R}(G, T, M, R) = \mathcal{R}(G_{S'}, T_{S'}, M, R).$$

1.16. Let $T = \mathrm{D}_S(M)$ be a trivialized torus. The Lie algebra $\mathfrak{t}$ of T identifies canonically (Exposé II, 5.1.1) with

$$\mathfrak{t} \simeq M^* \otimes \mathcal{O}_S.$$

For every homomorphism of groups $u : T \to \mathbb{G}_{m,S}$, $\mathrm{Lie}(u)$ is a linear form

$$\mathrm{Lie}(u) : \mathfrak{t} \longrightarrow \mathcal{O}_S = \mathrm{Lie}(\mathbb{G}_{m,S}/S).$$

In particular, if u is defined by an element $\alpha \in M$, then $\mathrm{Lie}(u)$ is the linear form $\bar{\alpha}$ on $M^* \otimes \mathcal{O}_S$ defined by

$$\bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Symmetrically, for every homomorphism of groups $h : \mathbb{G}_{m,S} \to T$, $\mathrm{Lie}(h)$ is an $\mathcal{O}_S$-linear morphism

$$\mathcal{O}_S = \mathrm{Lie}(\mathbb{G}_{m,S}/S) \longrightarrow \mathfrak{t},$$

canonically determined by the section

$$H = \mathrm{Lie}(h)(1) \in \Gamma(S, \mathfrak{t}).$$

In particular, if h is defined by an element $m \in M^*$, then

$$H = \mathrm{Lie}(h)(1) = m \otimes 1.$$

Comparing the two definitions, we find in particular

$$\bar{\alpha}(H) = (h, \alpha) \cdot 1 \in \Gamma(S, \mathcal{O}_S).$$

1.17. These definitions apply in particular when T is the maximal torus of a split group. Every root $\alpha \in R$ defines an *infinitesimal root*

$$\bar{\alpha} \in \mathrm{Hom}_{\mathcal{O}_S}(\mathfrak{t}, \mathcal{O}_S), \qquad \bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Every coroot α^*, $\alpha \in R$, defines an *infinitesimal coroot*

$$H_\alpha \in \Gamma(S, \mathfrak{t}), \qquad H_\alpha = \alpha^* \otimes 1.$$

For $\alpha, \beta \in R$, we have

$$\bar{\alpha}(H_\beta) = (\beta^*, \alpha) \cdot 1,$$

and in particular

$$\bar{\alpha}(H_\alpha) = 2.$$

Consequently, if 2 is invertible on S, then $\bar{\alpha}$ and H_α are nonzero on every fiber.

2. Existence of a Splitting. Type of a Reductive Group

Proposition 2.1. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. Suppose that T is split. Then G is locally splittable relative to T: for every $s_0 \in S$, there exists an open neighborhood U of s_0 such that the U-group G_U is splittable relative to T_U.*

Indeed, write $T \simeq \mathrm{D}_S(M)$ and

$$\mathfrak{g} = \coprod_{m \in M} \mathfrak{g}^m.$$

Let

$$R = \{m \in M \mid m \neq 0,\ \mathfrak{g}^m(s_0) \neq 0\}.$$

After restricting S and replacing it by an open neighborhood of s_0, we may suppose that the $\mathfrak{g}^\alpha$, $\alpha \in R$, are free and that the $\mathfrak{g}^m$, for $m \neq 0$ and $m \notin R$, vanish. We then have

$$\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha,$$

where the $\mathfrak{g}^\alpha$ are free of rank 1. It follows that R is a system of roots of G relative to T (Exposé XIX, 3.6). The coroots α^* corresponding to the $\alpha \in R$ then identify with locally constant functions on S with values in M^*. Shrinking S once more, we may suppose that they are constant, and the proof is complete.

Observe that the proof immediately gives:

Proposition 2.2. *Let S be a nonempty connected scheme such that* $\mathrm{Pic}(S) = 0$*, for example* $\mathrm{Spec}(\mathbb{Z})$ *or a local scheme* (in particular, the spectrum of a field)*. If G is a reductive S-group possessing a split maximal torus T, then G is splittable relative to T.*

We deduce immediately from 2.1 and from the fact that a reductive group locally possesses maximal tori for the étale topology (Exposé XIX, 2.5):

Corollary 2.3. *Let S be a scheme and G a reductive S-group* (respectively, let T also be a maximal torus of G)*. Then G is locally splittable* (respectively, locally splittable relative to T) *for the étale topology on S.*

Corollary 2.4. *Let k be a field and G a reductive k-group. There exists a finite separable extension K/k such that G_K is splittable.*

Remark 2.5. Using 2.1 and Exposé XIX, Remark 2.9, one immediately proves the following result: let $G = (G, T, M, R)$ be a split S-group. There exists an open covering (U_i) of S such that each split group G_{U_i} is obtained by base change from a split group over a noetherian ring (and in fact over a finitely generated $\mathbb{Z}$-algebra). We shall moreover prove that every split group over S is already obtained from a split $\mathbb{Z}$-group (Exposé XXV).

2.6. Let k be an algebraically closed field and G a reductive k-group. We know (for example by 2.4) that splittings of G exist. Let (G, T, M, R) and (G, T', M', R') be two splittings of G. Then the root data

$$\mathcal{R}(G, T, M, R) \qquad \text{and} \qquad \mathcal{R}(G, T', M', R')$$

are isomorphic.

Indeed, we first see that it is enough to reduce to the case $T = T'$: there exists $g \in G(k)$ such that $T' = \mathrm{int}(g)T$, and one readily verifies that transporting a splitting by an automorphism of G produces a root datum isomorphic to the original one. But since $S = \mathrm{Spec}(k)$ is connected, the isomorphism

$$\mathrm{D}_k(M) \xrightarrow{\sim} T \xrightarrow{\sim} \mathrm{D}_k(M')$$

comes from a unique isomorphism $M \simeq M'$. For the same reason, there exists at most one root system of G relative to T.

Definition 2.6.1. [4] If G is a reductive k-group, where k is algebraically closed, the *type of* G is the isomorphism class of the root datum defined by any splitting of G. If G is a torus of type M in the sense of Exposé IX, 1.4, then the type of G as a reductive group is given by the trivial root datum

$$(M, M^*, \varnothing, \varnothing).$$

By 1.15(b)[5], the type is invariant under an (algebraically closed) extension of the base field.

Definition 2.7. If G is a reductive S-group and $s \in S$, the *type of* G *at* s is the type of the reductive $\bar{s}$-group $G_{\bar{s}}$.

For every $S' \to S$ and every $s' \in S'$ mapping to $s \in S$, the type of $G_{S'}$ at s' is equal to the type of G at s.

If G is splittable and (G, T, M, R) is a splitting of G, then the type of G at s is the isomorphism class of $\mathcal{R}(G, T, M, R)$, by 1.15(b)[5]. The following then follows immediately from 2.3:

Proposition 2.8. *Let G be a reductive S-group, where $S \neq \varnothing$. The function*

$$s \longmapsto \textit{the type of } G \textit{ at } s$$

is locally constant on S. In particular, there exists a partition of S into nonempty open subschemes on each of which G has constant type. More precisely, let E be the set of types of the fibers of G, and, for every $t \in E$, let S_t be the set of points $s \in S$ where G has type t. Then $(S_t)_{t \in E}$ is a partition of S, and each S_t is open and closed (and nonempty).

3. The Weyl Group

3.1. Let S be a scheme, G a reductive S-group, and T a maximal torus of G. Then

$$W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$$

is a finite étale S-group (Exposé XIX, 2.5). The morphism $n \mapsto \mathrm{int}(n)$ induces, upon passage to the quotient, a canonical monomorphism (which is moreover an open immersion)

$$W_G(T) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(T).$$

3.2. Suppose now that G is splittable relative to T. Choose a splitting, say (G, T, M, R). We then have a canonical isomorphism (Exposé VIII, 1.5)

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(T) \simeq \left(\mathrm{Aut}_{\mathrm{gr.}}(M)\right)_S.$$

In particular, if W is the Weyl group of the root datum $\mathcal{R}(G)$ (Exposé XXI, 1.1.8), we have a monomorphism

$$W_S \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(T).$$

[4] Editor's note: The number 2.6.1 was added for later references.

[5] Editor's note: We have corrected the original, which referred to 1.17.

3.3. For every root $\alpha \in R$, the reflection $s_\alpha \in W$ acts on M by

$$s_\alpha(x) = x - (\alpha^*, x)\alpha,$$

and hence acts on T, by the preceding morphism, according to

$$s_\alpha(t) = t \cdot \alpha^*\big(\alpha(t)\big)^{-1}.$$

On the other hand, since $\mathfrak{g}^\alpha$ is assumed to be free, there exists

$$X \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

Consider

$$w_\alpha(X) \in \underline{\mathrm{Norm}}_G(T)(S) \qquad \text{(Exposé XX, 3.1)}.$$

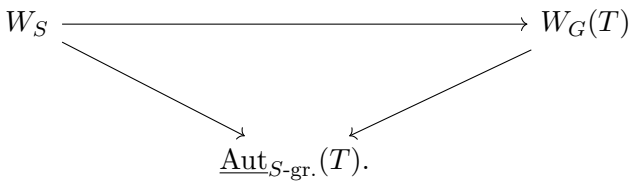
We have (loc. cit.)

$$\mathrm{int}\big(w_\alpha(X)\big)(t) = s_\alpha(t).$$

Since W is generated by the s_α, $\alpha \in R$, it follows from the preceding remarks that, if W and $\underline{\mathrm{Norm}}_G(T)(S)/T(S)$ are regarded as groups of automorphisms of T, then

$$W \subset \underline{\mathrm{Norm}}_G(T)(S)/T(S) \subset W_G(T)(S).$$

By definition of the constant group W_S associated with W (cf. Exposé I, 1.8), we therefore have a commutative diagram:



Proposition 3.4. *Let S be a scheme, let (G, T, M, R) be a split S-group, and let W be the Weyl group of the root datum $\mathcal{R}(G)$. Then the canonical monomorphism*

$$W_S \longrightarrow W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$$

is an isomorphism.

Indeed, these are étale groups over S; it therefore suffices to verify that, for every $s \in S$, the morphism

$$W_S(\bar{s}) \longrightarrow W_G(T)(\bar{s})$$

is an isomorphism.[6] The latter assertion follows, for example, from *Bible*, §11.3, Theorem 2.

Remark 3.5. Using 2.3, the preceding proposition gives a new proof of the fact that the Weyl group of a maximal torus of a reductive S-group G is finite over S (Exposé XIX, 2.5(ii)). [7]

[6] Editor's note: Indeed, since W_S and $W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$ are étale over S, the morphism $f : W_S \to W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$ is étale (EGA IV_4, 17.3.4). If, moreover, each f_s is an isomorphism, then, by loc. cit., 17.9.1, f is a surjective open immersion, hence an isomorphism.

[7] Editor's note: Indeed, let T be a maximal torus of G. The fact that $W_G(T)$ is finite over S is local for the fpqc topology (EGA IV_2, 2.7.1), hence *a fortiori* for the étale topology. By 2.3, we may therefore suppose that G is split relative to T, in which case the assertion follows from 3.4.

3.6. Under the hypotheses of 3.1, for every $w \in W_G(T)(S)$, let N_w[8] denote the fiber product in the following diagram:

$$\begin{array}{ccc} N_w & \longrightarrow & \underline{\mathrm{Norm}}_G(T) \\ \downarrow & & \downarrow \\ S & \xrightarrow{w} & W_G(T). \end{array}$$

It is an open and closed subscheme of $\underline{\mathrm{Norm}}_G(T)$, and is a principal homogeneous bundle under T on the left (respectively, on the right) for the law

$$(t, q) \longmapsto tq \qquad \text{(respectively, } (q, t) \longmapsto qt).$$

If $n \in N_w(S)$, then

$$N_{ww'} = n \cdot N_{w'}, \qquad N_{w'w} = N_{w'} \cdot n.$$

3.7. In particular, if α is a root of G with respect to T, then N_{s_α} is none other than what was denoted by $N^\times$ in Exposé XX, 3.0. If $\mathfrak{g}^\alpha$ is free over S, we therefore have

$$N_{s_\alpha}(S) \neq \varnothing.$$

By 3.4 and condition (D 2) of the splitting, we deduce:

Corollary 3.8. *Under the hypotheses of 3.4, the morphism*

$$\underline{\mathrm{Norm}}_G(T)(S) \longrightarrow W_G(T)(S) = \mathrm{Hom}_{\mathrm{loc.const.}}(S, W)$$

is surjective. In particular, for every $w \in W$, there exists $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ such that

$$\mathrm{int}(n_w)|_T = w.$$

4. Homomorphisms of Split Groups

4.1. The "Big Cell"

4.1.1. Let (G, T, M, R) be a split reductive S-group. Choose a system of positive roots R_+ of the root datum $\mathcal{R}(G)$ (Exposé XXI, 3.2.1). Put $R_- = -R_+$.

Choose a total ordering on R_+ (respectively, R_-) and consider the morphism induced by multiplication in G:

$$u : \left(\prod_{\alpha \in R_-} U_\alpha\right) \times_S T \times_S \left(\prod_{\alpha \in R_+} U_\alpha\right) \longrightarrow G.$$

This is an open immersion. Indeed, since both sides are flat and of finite presentation over S, it suffices to verify this on every geometric fiber (SGA 1, I, 5.7 and VIII, 5.4). We are thus reduced to the case in which S is the spectrum of an algebraically closed field. But by *Bible*, §13.4, Corollary 2 to Theorem 3, u is radicial and dominant. Since the tangent

[8]Editor's note: We have replaced Q_w by N_w, just as in Exposé XX, 3.0, Q was replaced by $N^\times$.

map of u at the origin is an isomorphism (definition of a root system), u is birational. As G is normal, Zariski's "Main Theorem" (EGA III_1, 4.4.9) applies, and u is an open immersion.

Let us show that the image Ω of this open immersion is independent of the ordering chosen on R_+ (respectively, R_-). Since this amounts to comparing open subsets of G, we are reduced to proving that they have the same geometric points, and may therefore again suppose that S is the spectrum of an algebraically closed field. The assertion is then precisely *Bible*, §13, Proposition 1(c) and Theorem 1(a).

We have therefore proved:

Proposition 4.1.2. *Let (G, T, M, R) be a split S-group, and let R_+ be a system of positive roots of R. There exists an open subset Ω_{R_+} of G such that, for every total ordering on R_+* (respectively, R_-)*, the morphism induced by multiplication in G*

$$\left(\prod_{\alpha\in R_-} U_\alpha\right) \times_S T \times_S \left(\prod_{\alpha\in R_+} U_\alpha\right) \longrightarrow G$$

is an open immersion with image Ω_{R_+}.

Remark 4.1.3. Proposition 4.1.2 may be restated as follows: for every $\alpha \in R$, choose an isomorphism of vector groups

$$p_\alpha : \mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha \qquad \text{(cf. 1.19)}.$$

Then the morphism (where $N = \mathrm{Card}(R_+) = \mathrm{Card}(R_-)$)

$$\mathbb{G}_{a,S}^N \times_S T \times_S \mathbb{G}_{a,S}^N \longrightarrow G$$

defined on points by

$$\left((x_\alpha)_{\alpha\in R_-}, t, (x_\alpha)_{\alpha\in R_+}\right) \longmapsto \prod_{\alpha\in R_-} p_\alpha(x_\alpha) \cdot t \cdot \prod_{\alpha\in R_+} p_\alpha(x_\alpha)$$

is an open immersion whose image depends only on R_+, and not on the choice of the p_α or of the orderings on R_+ and R_-.

Notation 4.1.4. We write[9]

$$\Omega_{R_+} = \prod_{\alpha\in R_-} U_\alpha \cdot T \cdot \prod_{\alpha\in R_+} U_\alpha.$$

Proposition 4.1.5. *The scheme Ω_{R_+} is of finite presentation over S* (hence quasi-compact in G) *and is universally schematically dense in G relative to S* (cf. Exposé XVIII, 1).

The first assertion is trivial. Thus,[10] Ω_{R_+} is flat and of finite presentation over S, and contains the identity section; it therefore meets every fiber of G in a nonempty, hence dense, open subset. The second assertion thus follows from Exposé XVIII, 1.3.[11]

[9] Editor's note: It is called the "big cell" corresponding to R_+.
[10] Editor's note: We have expanded the original in what follows.
[11] Editor's note: See also EGA IV_3, 11.10.10.

Corollary 4.1.6. *Let (G, T, M, R) be a split reductive S-group. Then*

$$\underline{\mathrm{Centr}}(G) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha).$$

Consequently, $\underline{\mathrm{Centr}}(G)$ is representable by a closed diagonalizable subgroup of G. [12]

The second assertion follows immediately from the first. To prove the first, one may invoke Exposé XII, 4.8 and 4.11; alternatively, one may argue directly as follows.[13] Let $S' \to S$. If $t \in T(S')$ and $\alpha(t) = 1$ for every $\alpha \in R$, then $\mathrm{int}(t)$ induces the identity on $T_{S'}$ and on every $(U_\alpha)_{S'}$, $\alpha \in R$, and therefore also on $(\Omega_{R_+})_{S'}$. By schematic density it consequently induces the identity on $G_{S'}$, whence

$$t \in \underline{\mathrm{Centr}}(G)(S').$$

Conversely, since

$$\underline{\mathrm{Centr}}_{G_{S'}}(T_{S'}) = T_{S'} \qquad \text{(cf. Exposé XIX, 2.8)},$$

if $g \in G(S')$ centralizes $T_{S'}$ and the $(U_\alpha)_{S'}$, then g is a section of $T_{S'}$ on which every $\alpha \in R$ vanishes.

Corollary 4.1.7. *Let S be a scheme and G a reductive S-group. Then the center of G is representable by a closed subgroup of G of multiplicative type. It is also "the intersection of the maximal tori of G" in the following sense: for every $S' \to S$, $\underline{\mathrm{Centr}}(G)(S')$ is the set of $g \in G(S')$ such that, for every $S'' \to S'$, the inverse image of g in $G(S'')$ belongs to $T(S'')$ for every maximal torus T of $G_{S''}$.*

In view of 2.3, the first assertion follows from 4.1.6 by descent.[14] Let us prove the second assertion. Let H denote "the intersection of the maximal tori of G" in the preceding sense. Clearly,[15]

$$\underline{\mathrm{Centr}}(G) \subset H.$$

By descent, it is therefore enough to prove $\underline{\mathrm{Centr}}(G) = H$ when G is split. Since H is contained in the intersection of the maximal tori of G in the usual sense, this follows from the following remark: if (G, T, M, R) is a splitting, $\alpha \in R$, and $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then

$$\mathrm{int}\big(\exp_\alpha(X)\big)(T) \cap T = \mathrm{Ker}(\alpha),$$

as a trivial calculation shows. See also Exposé XII, 8.6 and 8.8 for a more general statement.

Remark 4.1.8. In what follows, in the split case we shall systematically identify T with $D_S(M)$. Then

$$\underline{\mathrm{Centr}}(G) = D_S(M/\Gamma_0(R)),$$

where $\Gamma_0(R)$ is the subgroup of M generated by R (cf. Exposé XXI, 1.1.6).
If $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ is a system of simple roots of R, then one immediately has (cf. Exposé XX, 1.19)

$$\underline{\mathrm{Centr}}(G) = \bigcap_i \mathrm{Ker}(\alpha_i) = \bigcap_i \underline{\mathrm{Centr}}(Z_{\alpha_i}).$$

[12] Editor's note: We have added the preceding sentence.
[13] Editor's note: We have expanded the original in what follows.
[14] Editor's note: Indeed, representability of the center by a closed subscheme of G is local for the fpqc topology (SGA 1, VIII, 5.2 and 5.4), hence *a fortiori* for the étale topology; the same is true of the property of being of multiplicative type.
[15] Editor's note: We have expanded the original in what follows.

Proposition 4.1.9. *Let S be a scheme, let (G, T, M, R) be a split S-group, let Q be an S-torus, let α_0 be a character of Q, let $\mathcal{L}$ be an invertible $\mathcal{O}_S$-module, and let*

$$f : Q \longrightarrow T, \qquad p : W(\mathcal{L}) \longrightarrow G$$

be group homomorphisms satisfying the following set-theoretic relation:

$$p(\alpha_0(q)x) = \mathrm{int}\big(f(q)\big) \cdot p(x)$$

for every $q \in Q(S')$, $x \in W(\mathcal{L})(S')$, and $S' \to S$. Suppose that f separates the elements of R in the following sense: if $\alpha, \alpha' \in R$ and $m, m' \in \mathbb{Z}$, then[16]

$$m\alpha \circ f = m'\alpha' \circ f \quad \Longrightarrow \quad m\alpha = m'\alpha'.$$

Finally, let $s \in S$ be such that $(\alpha_0)_{\bar{s}} \neq e$ and $p_{\bar{s}} \neq e$.

There then exist an open subset U of S containing s, an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $\mathbb{G}_{a,U}$, a root $\alpha \in R$, and an isomorphism of $\mathcal{O}_U$-modules

$$h : (\mathcal{L}|_U)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}^{\alpha}|_U$$

such that

$$\text{(i)} \quad (\alpha \circ f)|_U = (q\alpha_0)|_U,$$

$$\text{(ii)} \quad p(X) = \exp_{\alpha}\big(h(X^q)\big) \quad \text{for every } X \in W(\mathcal{L})(S'),\ S' \to U.$$

Moreover, once U has been chosen, q, α, and h are uniquely determined.

After shrinking S, we may suppose that α_0 is nonzero on every fiber of S. Choose a system of positive roots R_+ of R, and put

$$V = p^{-1}(\Omega_{R_+}).$$

This is an open subset of $W(\mathcal{L})$ containing the zero section and stable under multiplication by every $\alpha_0(q)$, where $q \in Q(S')$ and $S' \to S$. Since α_0 is nontrivial on every fiber, it follows immediately that $V = W(\mathcal{L})$, and hence that p factors through Ω_{R_+}. Choose arbitrary orderings of R_+ and R_-; every product below is understood to be taken in the chosen ordering. There are therefore unique morphisms

$$a_\alpha : W(\mathcal{L}) \longrightarrow U_\alpha, \qquad \alpha \in R, \qquad b : W(\mathcal{L}) \longrightarrow T$$

such that

$$p(x) = \prod_{\alpha \in R_-} a_\alpha(x) \cdot b(x) \cdot \prod_{\alpha \in R_+} a_\alpha(x).$$

Writing out the covariance condition under Q, one immediately obtains

$$a_\alpha\big(\alpha_0(q)x\big) = \alpha\big(f(q)\big)a_\alpha(x), \qquad \alpha \in R,$$

$$b\big(\alpha_0(q)x\big) = b(x)$$

for every $x \in W(\mathcal{L})(S')$, $q \in Q(S')$, and $S' \to S$. The second condition immediately gives $b = e$.

[16] Editor's note: This is equivalent to the following hypothesis: if $\alpha, \alpha' \in R$, $m, m' \in \mathbb{Z}$, and

$$(m\alpha \circ f)_{\bar{s}} = (m'\alpha' \circ f)_{\bar{s}}$$

for every geometric point $\bar{s}$ of S, then $m\alpha = m'\alpha'$. In particular, this separation hypothesis is stable under base change.

Now let $\alpha \in R$ be such that $(a_\alpha)_{\bar{s}} \neq e$; such an α exists because $p_{\bar{s}} \neq e$ by hypothesis. Applying Exposé XIX, 4.12(a), we find an integer $n > 0$ such that

$$(\alpha \circ f)_{\bar{s}} = (n\alpha_0)_{\bar{s}}.$$

After shrinking S, we may suppose that $\alpha \circ f = n\alpha_0$ (Exposé IX, 5.3). But then, for every $\alpha' \in R$ with $\alpha' \neq \alpha$, one has

$$(\alpha' \circ f)_{\bar{s}} \neq m\alpha_0 \qquad \text{for every integer } m > 0,$$

by the hypothesis on f and the fact that the only roots proportional to α are α and $-\alpha$. Applying Exposé XIX, 4.12(a), now to $a_{\alpha'}$, we find that $a_{\alpha'}$ vanishes in a neighborhood of s. Since R is finite, after shrinking S once more we may suppose that $a_{\alpha'} = 0$ for every $\alpha' \in R$, $\alpha' \neq \alpha$. Thus $p = a_\alpha$, and we may apply Exposé XIX, 4.12(b), followed by 4.12(c), which gives the asserted result. The uniqueness assertions are clear.

Remark 4.1.10. The condition imposed on f in 4.1.9 is satisfied in particular when f is surjective (that is, faithfully flat).

Proposition 4.1.11. *Let (G, T, M, R) be a split S-group, let R_+ be a system of positive roots of R, and let Ω_{R_+} be the corresponding "big cell."*

(i) *Let H be a group-valued S-functor that is separated*[17] *for the fppf topology. If $f, g : G \rightrightarrows H$ are two group homomorphisms that coincide on Ω_{R_+}, then $f = g$.*

(ii) *Let H be an S-sheaf of groups for the fppf topology, and let $f : \Omega_{R_+} \to H$ be an S-morphism with the following property: for every $S' \to S$ and every $x, y \in \Omega_{R_+}(S')$ such that $xy \in \Omega_{R_+}(S')$, one has $f(xy) = f(x)f(y)$. Then there exists a group homomorphism $\bar{f} : G \to H$, unique by* (i)*, extending f.*

Indeed, by 4.1.5, (i) (respectively (ii)) follows immediately from Exposé XVIII, 2.2 (respectively 2.3 and 2.4).

Remark 4.1.12. If $\alpha \in R_+$, then

$$(\dagger) \qquad \Omega_{R_+} \cap Z_\alpha = U_{-\alpha} \cdot T \cdot U_\alpha.$$

Indeed,[18] for every $S' \to S$, if

$$g = \prod_{\beta \in R_-} p_\beta(x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(x_\beta)$$

is an element of $\Omega_{R_+}(S')$, and if $t' \in T_\alpha(S')$, then

$$t'gt'^{-1} = \prod_{\beta \in R_-} p_\beta(\beta(t')x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(\beta(t')x_\beta).$$

Since α and $-\alpha$ are the only elements of R whose restriction to T_α is 1, it follows that g belongs to $Z_\alpha = \mathrm{Centr}(T_\alpha)$ if and only if $x_\beta = 0$ for $\beta \neq \pm\alpha$.

From (†) and Exposé XX, 2.1, one deduces that if $X \in \Gamma(S, \mathfrak{g}^\alpha)$ and $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$, then

$$\exp_\alpha(X) \exp_{-\alpha}(Y) \in \Omega_{R_+}(S) \quad \Longleftrightarrow \quad 1 + XY \text{ is invertible.}$$

[17] Editor's note: Recall (cf. Exposé IV, 4.3.5) that an S-presheaf H is separated for a topology $\mathcal{T}$ if, for every $S' \to S$ and every family of S-morphisms $(S'_i \to S')_{i \in I}$ covering for $\mathcal{T}$, the map

$$H(S') \longrightarrow \prod_i H(S'_i)$$

is injective.

[18] Editor's note: We have expanded the original in what follows.

4.2. Morphisms of Split Groups

Definition 4.2.1. Let S be a nonempty scheme, and let (G, T, M, R) and (G', T', M', R') be two split S-groups. An S-group homomorphism $f : G \to G'$ is said to be *compatible with the splittings*, or to define a *morphism of split groups*, if the restriction of f to T factors as a morphism $f_T : T \to T'$ of the form $f_T = D_S(h)$, where $h : M' \to M$ is a group homomorphism satisfying the following condition: there exist a bijection $d : R \xrightarrow{\sim} R'$[19] and, for every $\alpha \in R$, an integer $q(\alpha) > 0$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbb{G}_{a,S}$ and

$$h(d(\alpha)) = q(\alpha)\alpha, \qquad {}^t h(\alpha^*) = q(\alpha)d(\alpha)^*.$$

Remark 4.2.2. It is immediate that h, d, and the $q(\alpha)$, for $\alpha \in R$, are uniquely determined by f. We write $h = \mathscr{R}(f)$. The $q(\alpha)$ are the radicial exponents of f (or of h).

Let p be the prime number, if one exists, that vanishes on S; put $p = 1$ if no prime number vanishes on S. Then $\mathscr{R}(f)$ is a p-morphism of reduced root data in the sense of Exposé XXI, 6.8. We have thus defined a functor $\mathscr{R}$ from the category of split S-groups to the category of reduced root data (equipped with p-morphisms).

Proposition 4.2.3. *Under the hypotheses of 4.2.1, the following properties hold:*

(i) *For every $\alpha \in R$, there exists a unique isomorphism of $\mathcal{O}_S$-modules*

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)} \xrightarrow{\sim} \mathfrak{g}'^{d(\alpha)}$$

such that

$$f\big(\exp_\alpha(X)\big) = \exp_{d(\alpha)}\big(f_\alpha(X^{q(\alpha)})\big)$$

for every $X \in W(\mathfrak{g}^\alpha)(S')$ and every $S' \to S$.

(ii) *For every $\alpha \in R$, one has $q(-\alpha) = q(\alpha)$, and f_α and $f_{-\alpha}$ are contragredient to one another.*

(iii) *For every $\alpha \in R$, every $Z \in W(\mathfrak{g}^\alpha)^*(S')$, and every $S' \to S$, one has*

$$f\big(w_\alpha(Z)\big) = w_{d(\alpha)}\big(Z^{q(\alpha)}\big).$$

By hypothesis, the following diagram is commutative:

$$\begin{array}{ccccc}
\mathbb{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbb{G}_{m,S} \\
\Big\downarrow{\scriptstyle q(\alpha)} & & \Big\downarrow{\scriptstyle f_T} & & \Big\downarrow{\scriptstyle q(\alpha)} \\
\mathbb{G}_{m,S} & \xrightarrow{d(\alpha)^*} & T' & \xrightarrow{d(\alpha)} & \mathbb{G}_{m,S}
\end{array}$$

It follows that f maps $\mathrm{Ker}(\alpha)$ into $\mathrm{Ker}(d(\alpha))$, hence T_α into $T'_{d(\alpha)}$, and hence Z_α into $Z'_{d(\alpha)}$. It remains only to apply Exposé XX, 3.10 and 3.11, to the groups Z_α and $Z'_{d(\alpha)}$.

[19] Editor's note: We have replaced the notation $u : R \to R'$ by $d : R \to R'$.

Proposition 4.2.4. *The morphism f induces a morphism f_N from $\underline{\mathrm{Norm}}_G(T)$ to $\underline{\mathrm{Norm}}_{G'}(T')$, hence a morphism f_W from $W_G(T)$ to $W_{G'}(T')$; the latter is an isomorphism. More precisely, let*

$$\bar{d} : W\big(\mathscr{R}(G)\big) = W \longrightarrow W' = W\big(\mathscr{R}(G')\big)$$

be the isomorphism extending $s_\alpha \mapsto s_{d(\alpha)}$ (Exposé XXI, 6.8.4)*. Then there is a commutative diagram of isomorphisms:*

$$\begin{array}{ccc} W_G(T) & \xrightarrow[\sim]{f_W} & W_{G'}(T') \\ \uparrow \sim & & \uparrow \sim \\ W_S & \xrightarrow[\sim]{\bar{d}_S} & W'_S. \end{array}$$

This follows immediately from 3.4, Exposé XXI, 6.8.4, and (iii) above.

Remark 4.2.5. With the notation of 4.2.3, the restriction of f to Ω_{R_+}, for a system of positive roots R_+, can be written explicitly: it maps Ω_{R_+} into $\Omega'_{d(R_+)}$ ($d(R_+)$ is a system of positive roots of R' by Exposé XXI, 6.8.7), and is given by the set-theoretic formula

$$f\Big(\prod_{\alpha\in R_-} \exp_\alpha(X_\alpha)\cdot t \cdot \prod_{\alpha\in R_+} \exp_\alpha(X_\alpha)\Big) = \prod_{\alpha\in R_-} \exp_{d(\alpha)}\Big(f_\alpha(X_\alpha^{q(\alpha)})\Big)\cdot f_T(t)\cdot \prod_{\alpha\in R_+} \exp_{d(\alpha)}\Big(f_\alpha(X_\alpha^{q(\alpha)})\Big).$$

Proposition 4.2.6. (i) *The morphism f is surjective* (= faithfully flat in the present case; cf. Exposé $\mathrm{VI_B}$, 3.11) *if and only if f_T is surjective.*

(ii) *One has* $\mathrm{Ker}(f) \subset \Omega_{R_+}$.

We prove (i). If f is surjective, then $f_T(T) = f(T)$ is a maximal torus of G'. Indeed, $f(T)$ is a subtorus of a maximal torus T' (Exposé IX, 6.8); to verify that $f(T) = T'$, one reduces to the case of an algebraically closed field, where this is *Bible*, §7.3, Theorem 3(a).

If f_T is surjective, the preceding formula shows that f induces a surjection from $\Omega = \Omega_{R_+}$ onto $\Omega' = \Omega'_{d(R_+)}$.[20] Since the fibers of G' are connected, it follows (cf. Exposé $\mathrm{VI_A}$, 0.5) that f is surjective.

We prove (ii), and for this purpose admit a result that will be proved below (5.7.4). For each $w \in W$, choose an $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ representing it. Then the open subschemes $n_w\Omega$, for $w \in W$, cover G. It is therefore enough to prove that

$$\mathrm{Ker}(f) \cap n_w\Omega \neq \varnothing \quad \Longrightarrow \quad w = 1.$$

If $x \in \Omega(S')$, where $S' \to S$, and $f(n_w x) = 1$, then $f(x) = f(n_w)^{-1}$. But $f(x) \in \Omega'(S')$, whereas

$$f(n_w)^{-1} \in \underline{\mathrm{Norm}}_{G'}(T')(S').$$

By 4.2.4, we are reduced to proving the following.

[20] Editor's note: We have given the reference to Exposé VI in greater detail below.

Lemma 4.2.7. *Under the hypotheses of 4.1.2, one has*

$$\Omega \cap \underline{\mathrm{Norm}}_G(T) = T.$$

Let

$$x = \prod_{\alpha\in R_-} p_\alpha(x_\alpha)\cdot t\cdot \prod_{\alpha\in R_+} p_\alpha(x_\alpha) = vtu \in \Omega(S').$$

If x normalizes $T_{S'}$, then, for every $t' \in T(S')$,

$$xt'x^{-1} = t'' \in T(S'),$$

that is, $xt' = t''x$, which may be written

$$v(tt')(t'^{-1}ut') = (t''vt''^{-1})(t''t)u.$$

This gives $t'^{-1}ut' = u$, hence

$$u \in \mathrm{Centr}_G(T)(S') = T(S'),$$

and therefore $u = 1$. Similarly, $v = 1$.

Corollary 4.2.8. *One has*

$$\mathrm{Ker}(f) = \prod_{\alpha\in R_-} K_\alpha \cdot \mathrm{Ker}(f_T)\cdot \prod_{\alpha\in R_+} K_\alpha,$$

where, for each $\alpha \in R$, K_α denotes the finite S-group

$$K_\alpha = \mathrm{Ker}\Big(U_\alpha \longrightarrow U_\alpha^{\otimes q(\alpha)}\Big) \simeq \boldsymbol{\alpha}_{q(\alpha),S}.$$

To apply this corollary, we make the following definition.

Definition 4.2.9. Let S be a scheme and let G and G' be two reductive S-groups. A faithfully flat and finite morphism of S-groups $f : G \to G'$ (i.e. one that is surjective and has finite kernel over S) is called an *isogeny*. If, moreover, $\mathrm{Ker}(f)$ is a central subgroup of G, then f is called a *central isogeny*.

Proposition 4.2.10. *Let $f : G \to G'$ be a morphism of split groups. For f to be an isogeny* (respectively, a central isogeny)*, it is necessary and sufficient that f_T be an isogeny, i.e. that $\mathscr{R}(f)$ be injective with finite cokernel* (respectively, and that $q(\alpha) = 1$ for every $\alpha \in R$).

Indeed, by 4.2.8, $\mathrm{Ker}(f)$ is finite over S if and only if $\mathrm{Ker}(f_T)$ is finite over S, and $\mathrm{Ker}(f) \subset T$ if and only if every $q(\alpha)$ is equal to 1. In that case $\mathrm{Ker}(f)$ is central, since it is of multiplicative type and invariant; cf. Exposé IX, 5.5.

Remark 4.2.11. (a) Thus $f : G \to G'$ is a central isogeny if and only if

$$\mathscr{R}(f) : \mathscr{R}(G') \longrightarrow \mathscr{R}(G)$$

is an isogeny in the sense of Exposé XXI, 6.2. Moreover, in this case, with the notation of *loc. cit.*, one has

$$\mathrm{Ker}(f) = D_S\big(K(\mathscr{R}(f))\big), \qquad K\big(\mathscr{R}(f)\big) = \mathrm{Coker}\big(\mathscr{R}(f)\big).$$

(b) If G and G' are semisimple, every morphism of split groups $f : G \to G'$ is an isogeny.[21]

(c) If $f : G \to G'$ is faithfully flat and finite and G is reductive (respectively, semisimple), then G' is so as well. Indeed, G' is of finite presentation over S (Exposé V, 9.1), affine over S (EGA II, 6.7.1), smooth over S (Exposé VI, 9.2), and has connected reductive (respectively, semisimple) geometric fibers by Exposé XIX, 1.7.

Definition 4.2.1 may seem arbitrary. It is justified by the following proposition, which, for simplicity, we shall state for semisimple groups.

A morphism $f : G \to G'$ of reductive S-groups is said to be *splittable* if there exist splittings of G and G' with which f is compatible. We then have the following.

Proposition 4.2.12. *Let S be a scheme, let G and G' be two semisimple S-groups, and let $f : G \to G'$ be a morphism of groups. Let $s \in S$. The following conditions are equivalent:*

(i) $\mathrm{Ker}(f_{\bar{s}})$ is finite ($\iff e$ is isolated in $\mathrm{Ker}(f)(\bar{s})$), and $f_{\bar{s}}$ is surjective, i.e. $f_{\bar{s}}$ is an isogeny.

(ii) $f_{\bar{s}}$ is splittable.

(iii) There exists an étale morphism $S' \to S$ covering s such that $f_{S'} : G_{S'} \to G'_{S'}$ is splittable.

Clearly, (iii) is equivalent to (ii), and (ii) implies (i) by 4.2.11(b). This is where the hypothesis that G and G' are semisimple enters; the other implications are valid for reductive groups.

We now prove that (i) implies (iii). We may suppose that G and G' are split in such a way that f induces a morphism

$$f_T : T \longrightarrow T'$$

(2.3 and Exposé XIX, 2.8). After restricting S, we may suppose that $f_T = D_S(h)$, where $h : M' \to M$ is a morphism of groups. Let $\alpha \in R$, and consider the composite morphism

$$p : W(\mathfrak{g}^\alpha) \xrightarrow{\exp_\alpha} G \xrightarrow{f} G'.$$

Since $\mathrm{Ker}(p_{\bar{s}})$ is finite, $p_{\bar{s}} \neq e$. On the other hand, $f_{T_{\bar{s}}}$ is surjective. We may therefore apply 4.1.9: there exist an open subset V_α of S containing s, a root $\alpha' \in R'$, and an integer $q(\alpha)$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbb{G}_{a,V_\alpha}$, and an isomorphism of $\mathcal{O}_{V_\alpha}$-modules

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)}\Big|_{V_\alpha} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}\Big|_{V_\alpha},$$

such that

$$f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)}))$$

and

$$\alpha' \circ f_T = h(\alpha') = q(\alpha)\alpha.$$

We may replace S by the intersection of the V_α, for $\alpha \in R$. Put $\alpha' = d(\alpha)$. It is clear that

$$d : R \longrightarrow R'$$

[21] Editor's note: Indeed, since G and G' are semisimple, f_T is surjective with finite kernel; hence f is faithfully flat with finite kernel by 4.2.6(i) and 4.2.8.

is a bijection, since the kernel of h is finite ($f_{T_{\bar{s}}}$ being surjective). It remains only to prove that

$$f_T \circ \alpha^* = q(\alpha)\alpha'^*,$$

which follows by a trivial modification of the argument used in Exposé XX, 3.11.

In any event, as we have seen in the course of the proof, (i) implies (iii). We have therefore proved the following.

Corollary 4.2.13. *Let S be a scheme and let $f : G \to G'$ be an isogeny of reductive groups. Then f is locally splittable for the étale topology.*

4.3. Central Quotients of Reductive Groups

We first consider a special case.

Proposition 4.3.1. *Let S be a scheme, let (G, T, M, R) be a split S-group, let N be a subgroup of M containing R, and put*

$$Q = D_S(M/N) \subset \mathrm{Centr}(G).$$

Then:

(i) *$G' = G/Q$ is a reductive S-group, and $T' = T/Q$ is a maximal torus of G';*

(ii) *if T' is identified with $D_S(N)$, then $R \subset N$ is a root system of G' relative to T', (G', T', N, R) is a splitting of G', and $\mathscr{R}(G')$ is canonically identified with the induced root datum* (Exposé XXI, 6.5) *$\mathscr{R}(G)_N$;*

(iii) *the canonical morphism $G \to G'$ is compatible with the splittings, has root exponents equal to* 1*, and, by functoriality, induces the canonical morphism* (loc. cit.)

$$\mathscr{R}(G)_N \longrightarrow \mathscr{R}(G).$$

We know that $G' = G/Q$ is representable by an affine group scheme over S (Exposé VIII, 5.7), smooth over S (Exposé $\mathrm{VI_B}$, 9.2), and with connected reductive geometric fibers (as quotients of reductive groups; cf. Exposé XIX, 1.7). Thus G' is a reductive S-group.

It is clear that

$$T' = T/Q \simeq D_S(N)$$

is a maximal torus of G'. Next, choose a system R_+ of positive roots of R. The open subset Ω_{R_+} of 4.2 is stable under Q, and there is a canonical isomorphism

$$\Omega_{R_+}/Q \simeq \left(\prod_{\alpha\in R_-} U_\alpha\right) \underset{S}{\times} (T/Q) \underset{S}{\times} \left(\prod_{\alpha\in R_+} U_\alpha\right).$$

Moreover, Ω_{R_+}/Q is an open subset of G' containing the identity section (cf. Exposé IV, 4.7.2 and 6.4.1).

It follows that, if $\mathfrak{g}'$ denotes the Lie algebra of G' and $\bar{\alpha}$ the character of T/Q induced by α (or, equivalently, defined by $\alpha \in N$ under the identification $T/Q = D_S(N)$), then the canonical morphism $\mathfrak{g} \to \mathfrak{g}'$ induces, for every $\alpha \in R$, an isomorphism

$$\mathfrak{g}^\alpha \xrightarrow{\sim} \mathfrak{g}'^{\bar{\alpha}}.$$

We have thus proved that R is a root system of G' relative to T', and the proof is completed without difficulty.

Corollary 4.3.2. *Let S be a scheme, let G be a reductive S-group, and let Q be a normal subgroup of G of multiplicative type. Then Q is central in G, the quotient G/Q is representable by a reductive S-group, and the canonical morphism $G \to G/Q$ is locally splittable for the étale topology* (with root exponents equal to 1).

The first assertion follows from Exposé IX, 5.5; the others are local for the étale topology and reduce to 4.3.1.

Definition 4.3.3. Let G be a reductive S-group. We say that G is *adjoint* (respectively, *simply connected*) if, for every $s \in S$, the type of G at s is given by an adjoint (respectively, simply connected) root datum, i.e. (Exposé XXI, 6.2.6) one for which M is generated by R (respectively, M^* is generated by R^*).

Proposition 4.3.4. (i) *An adjoint* (respectively, simply connected) *reductive group is semisimple.*

(ii) *If T is a maximal torus of the adjoint* (respectively, simply connected) *reductive group G, and if α is a root of G relative to T, then the infinitesimal root $\bar{\alpha}$ is nonzero on every fiber* (respectively, α^* is a monomorphism and the infinitesimal coroot H_α is nonzero on every fiber).

Indeed, (i) is trivial; (ii) may be checked on the geometric fibers and follows at once from Exposé XXI, 6.2.8.

Proposition 4.3.5. (i) *The reductive group G is adjoint if and only if*

$$\mathrm{Centr}(G) = \{e\}_S.$$

(ii) *For every reductive group G, the quotient group $G/\mathrm{Centr}(G)$ is an adjoint reductive group.*

Indeed, we may suppose that G is split. Then (i) is immediate, since

$$\mathrm{Centr}(G) = D_S(M/\Gamma_0(R)),$$

and (ii) follows from 4.3.1.

Definition 4.3.6. Let G be a reductive S-group. The *adjoint group* of G, denoted by $\mathrm{ad}(G)$, is the group $G/\mathrm{Centr}(G)$. The *radical* of G, denoted by $\mathrm{rad}(G)$, is the maximal torus (unique by Exposé XII, 1.12) of $\mathrm{Centr}(G)$. The *semisimple group associated with G* is the quotient $G/\mathrm{rad}(G)$.

The preceding definitions are compatible with base change. If $s \in S$, then $\mathrm{rad}(G)_{\bar{s}}$ is indeed the radical of $G_{\bar{s}}$ in the usual sense (Exposé XIX, 1.6).

4.3.7. If (G, T, M, R) is a split group, then

$$\mathrm{rad}(G) = D_S(M/N), \qquad N = M \cap V(R).$$

Thus the semisimple group associated with G (as also the adjoint group of G) is equipped with a canonical splitting (4.3.1), and we have a diagram of split groups

$$G \longrightarrow \mathrm{ss}(G) \longrightarrow \mathrm{ad}(G)$$

corresponding to the canonical diagram of root data (Exposé XXI, 6.5.5)

$$\mathrm{ad}(\mathscr{R}(G)) \longrightarrow \mathrm{ss}(\mathscr{R}(G)) \longrightarrow \mathscr{R}(G).$$

Remark 4.3.8. *Let (G, T, M, R) be a split adjoint* (respectively, simply connected) *S-group, and let Δ be a system of simple roots of R. Then the family $\{\alpha\}_{\alpha\in\Delta}$, respectively $\{\alpha^*\}_{\alpha\in\Delta}$, induces an isomorphism*

$$T \xrightarrow{\sim} (\mathbb{G}_{m,S})^{\Delta}, \qquad \textit{respectively} \qquad (\mathbb{G}_{m,S})^{\Delta} \xrightarrow{\sim} T.$$

Indeed,

$$M = \Gamma_0(R) \qquad (\text{respectively, } M^* = \Gamma_0(R^*)),$$

and Δ (respectively, $\{\alpha^*\}_{\alpha\in\Delta}$) is a basis of the free abelian group $\Gamma_0(R)$ (respectively, $\Gamma_0(R^*)$); see Exposé XXI, 3.1.8.

Remark 4.3.9. The radical of a reductive group is a *characteristic* subgroup (i.e. stable under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$), by its definition.

5. Subgroups of Type (R)

We are especially interested in reductive groups, but some of the results that we shall establish hold more generally for a wider class of groups: groups of type (RR).

5.1. Groups of Type (RR)

Definition 5.1.1. Let S be a scheme and G an S-group scheme. We say that G is *of type* (RR) if it satisfies the following conditions:

(i) G is smooth and of finite presentation over S, with connected fibers.

(ii) G locally possesses maximal tori for the fpqc topology.

(iii) For every $s \in S$, every maximal torus T of $G_{\bar{s}}$, and every root α of $G_{\bar{s}}$ relative to $T_{\bar{s}}$ (Exposé XIX, 1.10),

$$\mathrm{Lie}(G_{\bar{s}})^{\alpha}$$

has rank 1 (as a vector space over $\overline{\kappa(s)}$).

(iv) For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, let R denote the set of roots of $G_{\bar{s}}$ relative to T, and let $\Gamma_0(R)$ be the subgroup of

$$\mathrm{Hom}_{\bar{s}\text{-gr.}}(T, \mathbb{G}_{m,\bar{s}})$$

generated by R. Then the content[22] of every root $\alpha \in R$ in the free abelian group $\Gamma_0(R)$ (and hence a positive integer) is invertible on S.

Reminder 5.1.1.1. [23] Let us recall that, if G is a smooth connected algebraic group over an algebraically closed field k, a *Cartan subgroup* of G is the centralizer of a maximal torus of G (Exposé XII, 1.0), and such a subgroup is smooth and connected. For this, as well as for other characterizations of Cartan subgroups, see *Bible*, §7.1, Theorem 1 in the case where G is affine, and Exposé XII, Theorem 6.6 in the general case.

If S is an arbitrary scheme and G is a smooth S-group of finite type, a *Cartan subgroup* of G is a smooth S-subgroup C of G such that, for every $s \in S$, $C_{\bar{s}}$ is a Cartan subgroup of $G_{\bar{s}}$ (Exposé XII, Definition 3.1).

[22] Editor's note: The content of the root α is the positive generator of the ideal $\{f(\alpha) \mid f \in \Gamma_0(R)^*\}$ of $\mathbb{Z}$; it is the greatest integer $c > 0$ such that $\alpha/c \in \Gamma_0(R)$.

[23] Editor's note: This reminder was added.

Remark 5.1.2. (a) By Exposé XII, 7.1 (where the hypothesis that G is separated is satisfied here because G has connected fibers; cf. Exposé $\mathrm{VI_B}$, 5.5), conditions (i) and (ii) imply that G locally possesses, for the étale topology, maximal tori (respectively, Cartan subgroups), which are locally conjugate for the étale topology.

(b) The Cartan subgroups of G have connected fibers (Exposé XII, 6.6).

(c) If G is affine over S, conditions (i) and (ii) are respectively equivalent to:

(i′) G is smooth over S and has connected fibers;

(ii′) the reductive rank of the fibers of G is locally constant (Exposé XII, 1.7).

(d) By Exposé XII, 8.8 (c) and (d), G has a reductive center Z, and for every $s \in S$, in the notation of (iv),[24]

$$Z_{\bar{s}} = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha),$$

whence

$$\mathrm{Hom}_{\bar{s}\text{-gr.}}\big((T/Z)_{\bar{s}}, \mathbb{G}_{m,\bar{s}}\big) \simeq \Gamma_0(R).$$

(e) Condition (iv) holds in particular in the following two cases:

(1) S has characteristic 0;

(2) every root is an indivisible element of the group generated by the roots.

(f) The property of being of type (RR) is stable under base change and is local for the fpqc topology.

Remarks (c) and (e) immediately give the following proposition.

Proposition 5.1.3. *A reductive group is of type* (RR).

Proposition 5.1.4. *Let S be a scheme, let G be an S-group of type* (RR)*, and let Q be a central subgroup of G, of finite presentation over S, such that the quotient G/Q is representable* (for example, G affine over S and Q of multiplicative type (Exposé IX, 2.3), or S artinian (Exposé $\mathrm{VI_A}$, 3.3.2)). *Then G/Q is an S-group of type* (RR).

Indeed, G/Q is smooth over S (Exposé $\mathrm{VI_B}$, 9.2), of finite presentation over S (Exposé V, 9.1), and has connected fibers, so condition (i) holds. On the other hand, condition (ii) follows from Exposé XII, 7.6; it remains to verify conditions (iii) and (iv).

Put $G' = G/Q$, let

$$u : G \longrightarrow G'$$

be the canonical morphism, and let $T' = u(T)$ be the maximal torus of G' that is the image of T (cf. Exposé XII, 7.1 (e)). For each $\alpha \in R$, again denote by α the character of T' induced by α; indeed,

$$Q \cap T \subset \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$$

by 5.1.2 (d). We first prove:

[24] Editor's note: The original said "under the conditions of (iv)," but the last condition of (iv) does not appear to be used here.

Lemma 5.1.5. *Under the hypotheses of 5.1.4, let $T = D_S(M)$ be a trivialized maximal torus of G, and suppose that the decomposition of $\mathfrak{g} = \mathrm{Lie}(G)$ under $\mathrm{Ad}(T)$ has the form*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha, \qquad R \subset M - \{0\},$$

where, for every $s \in S$, $\mathfrak{g}^\alpha(s) \neq 0$ for $\alpha \in R$ (thus $\mathfrak{g}^\alpha$ is an invertible $\mathcal{O}_S$-module for every $\alpha \in R$, and R is the set of roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$, for every $s \in S$). *Then the Lie algebra $\mathfrak{g}'$ of G' decomposes under $\mathrm{Ad}(T')$ as follows:*

$$\mathfrak{g}' = \mathfrak{g}'^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}'^\alpha,$$

and $\mathrm{Lie}(u)$ induces an isomorphism from $\mathfrak{g}^\alpha$ onto $\mathfrak{g}'^\alpha$.

Indeed, let

$$p = \mathrm{Lie}(u) : \mathfrak{g} \longrightarrow \mathfrak{g}'.$$

One immediately has $p(\mathfrak{g}^\alpha) \subset \mathfrak{g}'^\alpha$ for every $\alpha \in R$, and $p(\mathfrak{g}^0) \subset \mathfrak{g}'^0$. Since

$$\mathrm{Ker}(p) = \mathrm{Lie}(Q) \subset \mathrm{Lie}\big(\mathrm{Centr}_G(T)\big) = \mathfrak{g}^0,$$

p induces a monomorphism from $\mathfrak{g}^\alpha$ into $\mathfrak{g}'^\alpha$, for every $\alpha \in R$.

To prove the lemma, it is enough to do so when S is the spectrum of an algebraically closed field; in view of the preceding remarks, it is then enough to prove that

$$\mathrm{rk}(\mathfrak{g}') = \mathrm{rk}(\mathfrak{g}'^0) + \mathrm{Card}(R).$$

Set

$$C = \mathrm{Centr}_G(T), \qquad C' = \mathrm{Centr}_{G'}(T').$$

By Exposé XII, 7.1 (e), u induces a faithfully flat morphism $C \to C'$ with kernel Q. Hence

$$\dim C' + \dim Q = \dim C.$$

But G, G', C, and C' are smooth, and therefore

$$\begin{aligned} \dim G = \mathrm{rk}(\mathfrak{g}) &= \mathrm{rk}(\mathfrak{g}^0) + \mathrm{Card}(R) = \dim C + \mathrm{Card}(R) \\ &= \dim Q + \dim C' + \mathrm{Card}(R) = \dim Q + \mathrm{rk}(\mathfrak{g}'^0) + \mathrm{Card}(R), \\ \mathrm{rk}(\mathfrak{g}') &= \dim G' = \dim G - \dim Q. \end{aligned}$$

It follows that

$$\mathrm{rk}(\mathfrak{g}') = \mathrm{rk}(\mathfrak{g}'^0) + \mathrm{Card}(R),$$

which is the desired relation.

We return to the proof of 5.1.4. We may suppose that S is the spectrum of an algebraically closed field. Let T be a maximal torus of G. Applying 5.1.5, we immediately obtain (iii) and (iv) for G/Q.

To use the preceding proposition, we introduce a definition.

Definition 5.1.6. We say that the S-group scheme G is *of type* (RA) if it is of type (RR) and also satisfies condition (iv′) below, which is stronger than (iv):

(iv′) For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, the content of every root of $G_{\bar{s}}$ relative to T, in

$$\mathrm{Hom}_{\bar{s}\text{-gr.}}(T, \mathbb{G}_{m,\bar{s}}),$$

is invertible on S.

Remarks 5.1.7. (a) An adjoint reductive S-group is of type (RA).

(b) If S has characteristic 0, every group of type (RR) is of type (RA).

(c) The property of being of type (RA) is stable under base change and is local for the fpqc topology.

Remark (a) above admits the following generalization.

Proposition 5.1.8. *Let S be a scheme, let G be an S-group of type* (RR)*, and let Z be its reductive center. Suppose that the quotient G/Z is representable* (for example, if G is affine over S, or if S is artinian)*. Then G/Z is of type* (RA)*.*

Indeed, this follows immediately from 5.1.4, 5.1.5, and 5.1.2 (d).

Remark 5.1.9. If G is of type (RR) and T is a maximal torus of G, Exposé XIX, 6.3 applies. In particular, $W_G(T)$ is étale, quasi-finite, and separated over S.

5.2. Subgroups of Type (R)

Definition 5.2.1. Let S be a scheme, let G be a smooth S-group scheme of finite presentation with connected fibers,[25] and let H be a group subscheme of G. We say that H is *of type* (R) if:

(i) H is smooth and of finite presentation over S, and has connected fibers;[25]

(ii) for every $s \in S$, $H_{\bar{s}}$ contains a Cartan subgroup of $G_{\bar{s}}$.

This notion is stable under base change and is local for the fpqc topology.

Recall 5.2.2. (cf. Exposé XII, 7.9) Under the preceding conditions:

(a)
$$H = \underline{\mathrm{Norm}}_G(H)^0.$$

(b) If G locally possesses Cartan subgroups (respectively maximal tori) for the étale topology, then so does H, and the Cartan subgroups (respectively maximal tori) of H are Cartan subgroups (respectively maximal tori) of G.

Examples 5.2.3. (a) *Borel subgroups.* A Borel subgroup of G is a subgroup H of type (R) whose geometric fibers are Borel subgroups of the corresponding fibers of G.[26]

(b) *Parabolic subgroups.* A parabolic subgroup of G is a subgroup of type (R) whose geometric fibers contain Borel subgroups.

Other examples are given by the following propositions.

Proposition 5.2.4. *Let G be as in 5.2.1, let $K \subset H$ be two group subschemes of G, and suppose that H is of type* (R)*. Then K is a subgroup of type* (R) *of H if and only if it is a subgroup of type* (R) *of G.*

Indeed,[27] let $s \in S$. Since H is of type (R), every maximal torus of $H_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$, and the same therefore holds for Cartan subgroups.

[25] Editor's note: The hypothesis that G (respectively H) is of finite presentation over S is automatically satisfied: since G (respectively H) is smooth over S with connected fibers, it is quasi-compact and separated over S (Exposé $\mathrm{VI_B}$, 5.5), and hence of finite presentation over S.

[26] Editor's note: Equivalently, H is a smooth subgroup of G such that every geometric fiber $H_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$ (since every Borel subgroup of $G_{\bar{s}}$ is connected and contains a Cartan subgroup of $G_{\bar{s}}$).

[27] Editor's note: We have expanded the argument that follows.

Proposition 5.2.5. *Let G be as in 5.2.1, let T be a maximal torus of G, let Q be a subtorus of T, and put $Z = \underline{\mathrm{Centr}}_G(Q)$. If H is a subgroup of type* (R) *of G containing T, then $H \cap Z$ is a subgroup of type* (R) *of Z.*

First recall that Z is a closed group subscheme of G, of finite presentation (Exposé XI, 6.11), with connected fibers (Exposé XII, 6.6), and smooth over S (Exposé XI, 2.4), so it satisfies the conditions imposed in Definition 5.2.1. Likewise, $H \cap Z$ is of finite presentation, smooth, and has connected fibers (since $H \cap Z = \underline{\mathrm{Centr}}_H(Q)$); moreover,

$$H \cap Z \supset \underline{\mathrm{Centr}}_G(T).$$

Proposition 5.2.6. *Let S be a scheme, let G be an S-group of type* (RR) (respectively (RA))*, and let H be a subgroup of type* (R) *of G. Then H is an S-group of type* (RR) (respectively (RA))*.*

Indeed, (i) is clear, (ii) follows from 5.2.2 (b), and (iii) and (iv) (respectively (iv$'$)) are to be verified when S is the spectrum of an algebraically closed field. Then H contains a maximal torus T of G, and therefore also

$$C = \underline{\mathrm{Centr}}_G(T),^{28}$$

and the assertions to be proved follow immediately from:

Lemma 5.2.7. *Let S be a scheme, let G be an S-group of type* (RR)*, and let T be a maximal torus of G equipped with a trivialization $T \simeq D_S(M)$. Suppose that*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha$$

(the $\mathfrak{g}^\alpha$ are then invertible $\mathcal{O}_S$-modules)*. Let H be a subgroup of type* (R) *containing $C = \underline{\mathrm{Centr}}_G(T)$* (that is, containing T)*. Then $\mathfrak{h} = \mathrm{Lie}(H/S)$ is locally on S of the form*

$$\mathfrak{g}^0 + \coprod_{\alpha \in R'} \mathfrak{g}^\alpha = \mathfrak{g}_{R'}.$$

More precisely, for each $s \in S$, put

$$R'(s) = \{\alpha \in R \mid \mathfrak{g}^\alpha(s) \subset \mathfrak{h}(s)\}.$$

Then $R'(s)$ is a locally constant function of s. If U is an open subset of S on which $R'(s) = R'$, then

$$\mathfrak{h}_U = \mathfrak{g}^0_U \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha_U.$$

Indeed, $\mathfrak{h}$ is a submodule of $\mathfrak{g}$, locally a direct factor, containing $\mathfrak{g}^0$, and stable under T.

[28] Editor's note: By hypothesis, H contains $\underline{\mathrm{Centr}}_G(T')$ for some maximal torus T' of G. The tori T and T' are then conjugate in H, and hence H also contains $C = \underline{\mathrm{Centr}}_G(T)$.

5.3. Strict Transporter of Two Subgroups of Type (R). Applications

Recall 5.3.0. [29] Let S be a scheme, let G be a smooth S-group, put $\mathfrak{g} = \mathrm{Lie}(G/S)$, and let $\mathfrak{h}$ be an $\mathcal{O}_S$-submodule of $\mathfrak{g}$ that is locally a direct factor. The $\mathcal{O}_S$-algebra

$$\mathscr{A} = \mathrm{Sym}(\omega^1_{G/S})$$

is locally free; hence the S-scheme

$$\underline{\mathrm{Lie}}(G/S) = W(\mathfrak{g}) = \mathrm{Spec}(\mathscr{A})$$

is essentially free in the sense of Exposé VIII, 6.1. Since $W(\mathfrak{h})$ is a closed subscheme of $\underline{\mathrm{Lie}}(G/S)$, of finite presentation over $\underline{\mathrm{Lie}}(G/S)$, it follows from Exposé VIII, 6.5 (a) that

$$N = \underline{\mathrm{Norm}}_G(\mathfrak{h})$$

is represented by a closed group subscheme of G, of finite presentation over G. (See also the additions 6.2.3 and 6.2.4 (a) in Exposé $\mathrm{VI_B}$.) On the other hand, by Exposé II, 5.3.1,

$$\underline{\mathrm{Lie}}(N/S) = \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(\mathfrak{h}).$$

Finally, by Exposé $\mathrm{VI_B}$, 3.10, if N is smooth over S at the points of the unit section, then the group subfunctor N^0 (defined in Exposé $\mathrm{VI_B}$, 3.1) is represented by an open group subscheme of N, smooth over S.

Proposition 5.3.1. *Let S be a scheme, let G be an S-group of type* (RA) (5.1.6)*, let H be a subgroup of type* (R) *of G, and let $\mathfrak{g} \supset \mathfrak{h}$ be their Lie algebras. Then* $\underline{\mathrm{Norm}}_G(\mathfrak{h})$ (which, by 5.3.0, is representable by a closed subscheme of G, of finite presentation over S) *is smooth over S at every point of the unit section, and*

$$\underline{\mathrm{Norm}}_G(\mathfrak{h})^0 = H.$$

[30] *Proof.* Put

$$N = \underline{\mathrm{Norm}}_G(\mathfrak{h}), \qquad \mathfrak{n} = \mathrm{Lie}(N/S).$$

We have $H \subset N$, and by Exposé II, 5.3.1, for every $s \in S$,

$$\mathfrak{h}(s) \subset \mathfrak{n}(s) = \mathrm{Norm}_{\mathfrak{g}(s)}(\mathfrak{h}(s)).$$

Now, by 5.3.2 below,

$$\mathfrak{h}(s) = \mathrm{Norm}_{\mathfrak{g}(s)}(\mathfrak{h}(s)),$$

and since H is smooth over S,

$$\dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s$$

(cf. [DG70], §II.5, Theorem 2.1). Hence

$$\dim_{\kappa(s)} \mathfrak{n}(s) = \dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s \leq \dim N_s.$$

It follows that

$$N^0_s = H^0_s = H_s,$$

since H has connected fibers. Consequently the group subfunctor N^0 (defined in Exposé $\mathrm{VI_B}$, 3.1) is represented by the smooth S-group H. This proves 5.3.1, modulo the following lemma.

[29] Editor's note: This reminder was added; it is used in the proofs of 5.3.1 and 5.3.4.
[30] Editor's note: We have expanded the original in the argument that follows.

Lemma 5.3.2. *Under the conditions of 5.2.7, if G is of type* (RA)*, then, for every $s \in S$,*

$$\mathrm{Norm}_{\mathfrak{g}(s)}(\mathfrak{h}(s)) = \mathfrak{h}(s).$$

Indeed, we are reduced to the case where S is the spectrum of a field, and hence where

$$\mathfrak{h} = \mathfrak{g}_{R'}$$

for some $R' \subset R$. But we already have

$$\mathrm{Transp}_{\mathfrak{g}}(\mathfrak{t}, \mathfrak{h}) = \mathfrak{h}.$$

Indeed, if $H \in \mathfrak{t}$ and $X \in \mathfrak{g}^{\alpha}$, then

$$[H, X] = \bar{\alpha}(H)X,$$

where $\bar{\alpha} : \mathfrak{t} \to \mathcal{O}_S$ is the morphism derived from α. Condition (iv′) says precisely that $\bar{\alpha} \neq 0$ for every $\alpha \in R$.

Corollary 5.3.3. *Let S be a scheme, let G be an S-group of type* (RA)*, let H and H' be two subgroups of type* (R) *of G, and let $\mathfrak{h}$ and $\mathfrak{h}'$ be their Lie algebras. Then*

$$H = H' \quad \Longleftrightarrow \quad \mathfrak{h} = \mathfrak{h}'.$$

Corollary 5.3.4. *Under the conditions of 5.2.7, with G of type* (RA)*, the maps*

$$H \longmapsto \mathrm{Lie}(H/S), \qquad \mathfrak{h} \longmapsto \underline{\mathrm{Norm}}_G(\mathfrak{h})^0$$

give a bijective correspondence between the set of subgroups of type (R) *of G containing T, and the set of Lie subalgebras of $\mathfrak{g}$ containing $\mathfrak{g}^0$, stable under T, and whose normalizer in G is smooth over S at every point of the unit section.*

Indeed,[31] let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ satisfying the properties above. By 5.3.0,

$$H = \underline{\mathrm{Norm}}_G(\mathfrak{h})^0$$

is a smooth S-group scheme. Moreover, since

$$C = \underline{\mathrm{Centr}}_G(T)$$

stabilizes every $\mathfrak{g}^{\alpha}$ and has connected fibers (Exposé XII, 6.6), we have $C \subset H$. Therefore H is a subgroup of G of type (R). By Exposé II, 5.3.1,

$$\underline{\mathrm{Lie}}(H) = \underline{\mathrm{Norm}}_{\mathfrak{g}}(\mathfrak{h}).$$

Finally, by the proof of 5.3.2,

$$\underline{\mathrm{Norm}}_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

[31] Editor's note: The proof that follows was added.

Corollary 5.3.5. *Let S be a scheme, let G be an S-group of type* (RR) (5.1.1)*, let T be a maximal torus of G, and let H and H' be two subgroups of type* (R) *of G, both containing T. Then*

$$H = H' \iff \mathfrak{h} = \mathfrak{h}'.$$

By virtue of the finite-presentation hypotheses, we reduce as usual (cf. EGA IV_3, §8, and Exposé VI_B, §10) to the case where S is noetherian. It is then enough to verify that $\mathfrak{h} = \mathfrak{h}'$ implies

$$H_{S'} = H'_{S'}$$

for every S' that is the spectrum of an artinian quotient of a local ring of S;[32] we are therefore reduced to the case where S is artinian, in which case we may apply 5.1.8.

Let

$$u : G \longrightarrow G' = G/Z$$

be the canonical morphism, and let $T' = T/Z$ be the maximal torus of G' corresponding to T. By Exposé XII, 7.12, there exist subgroups H_1 and H'_1 of type (R) of G', containing T', such that

$$H = u^{-1}(H_1), \qquad H' = u^{-1}(H'_1).$$

It is enough to prove that $H_1 = H'_1$. But by 5.2.7 and 5.1.5,

$$\mathrm{Lie}(H_1) = \mathrm{Lie}(H'_1),$$

and we are reduced to 5.3.3.

Remark 5.3.6. The fact that H and H' contain the same maximal torus is essential for the validity of 5.3.5 when G is not of type (RA). For example, consider the maximal tori of $\mathrm{SL}_{2,k}$, where k has characteristic 2.[33]

Corollary 5.3.7. *Let S be a scheme, let G be an S-group of type* (RR)*, let T be a maximal torus of G, and let H and H' be two subgroups of type* (R) *of G, both containing T. The set U of $s \in S$ such that $H_s = H'_s$ is open and closed in S, and*

$$H_U = H'_U.$$

Indeed, this follows immediately from 5.3.5 and 5.2.7.

Corollary 5.3.8. *The "functor of subgroups of type* (R) *containing T", where T is a given maximal torus in a group G of type* (RR)*, is formally unramified* (Exposé XI, 1.1).

[32] Editor's note: Indeed, let $g \in G$, let s be its image in S, let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{G,g}$, let $\mathfrak{n}$ be that of $\mathcal{O}_{S,s}$, and let I (respectively I') be the kernel of the morphism from $\mathcal{O}_{G,g}$ to $\mathcal{O}_{H,g}$ (respectively $\mathcal{O}_{H',g}$), the latter being the zero ring if $g \notin H$ (respectively $g \notin H'$). Since $\mathcal{O}_{G,g}$ is noetherian, I and I' are closed for the $\mathfrak{m}$-adic topology, hence *a fortiori* for the $\mathfrak{n}$-adic topology. It is therefore enough to prove

$$I + \mathfrak{n}^n \mathcal{O}_{G,g} = I' + \mathfrak{n}^n \mathcal{O}_{G,g}$$

for every $n \in \mathbb{N}$.

[33] Editor's note: Indeed, if k is algebraically closed, all maximal tori of $G = \mathrm{SL}_{2,k}$ are conjugate under $G(k)$, and they all have as Lie algebra the line $k\,\mathrm{id} \subset M_2(k)$, which is invariant under the adjoint action.

Theorem 5.3.9. *Let G be an S-group of type* (RR) (5.1.1)*, and let H and H' be two subgroups of type* (R) (5.2.1)*. Let $\underline{\mathrm{Transt}}_G(H, H')$ be the strict transporter of H into H', defined by*

$$\underline{\mathrm{Transt}}_G(H, H')(S') = \{g \in G(S') \mid \mathrm{int}(g)H_{S'} = H'_{S'}\}.$$

Then $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G, smooth and of finite presentation over S.

The fact that $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G, of finite presentation over S, follows from Exposé XI, 6.11 (a). To prove that it is smooth over S, it is necessary to prove that if S is affine, if S_0 is the closed subscheme defined by a nilpotent ideal J, and if

$$g_0 \in G(S_0), \qquad \mathrm{int}(g_0)H_0 = H'_0,$$

then there exists $g \in G(S)$, lifting g_0, such that

$$\mathrm{int}(g)H = H'.$$

Since smoothness is local for the étale topology, we may suppose that H contains a maximal torus T of G.

Then T_0 is a maximal torus of H_0, and hence $\mathrm{int}(g_0)T_0$ is a maximal torus of H'_0. By Exposé IX, 3.6 bis, there exists a torus T' of H' such that

$$T'_0 = \mathrm{int}(g_0)T_0;$$

by Exposé IX, 3.3 bis, there therefore exists $g \in G(S)$, lifting g_0, such that

$$\mathrm{int}(g)T = T'.$$

After replacing H by $\mathrm{int}(g)H$, we may therefore suppose that H and H' contain the same maximal torus T and that $H_0 = H'_0$. But then $H = H'$ by 5.3.7. □

Corollary 5.3.10. *Let G be an S-group of type* (RR)*, and let H be a subgroup of type* (R) *of G. Then $\underline{\mathrm{Norm}}_G(H)$ is representable by a closed group subscheme of G, of finite presentation and smooth over S.*

Using now the reasoning employed in Exposé XI to deduce 5.4 bis from 5.2 bis, we obtain:

Corollary 5.3.11. *Under the hypotheses of 5.3.9, the following conditions are equivalent:*

(i) *H and H' are locally conjugate in G for the étale topology;*

(i bis) *the same holds for the fpqc topology;*

(ii) *for every $s \in S$, $H_{\bar{s}}$ and $H'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$;*

(ii bis) *the structural morphism*

$$\underline{\mathrm{Transt}}_G(H, H') \longrightarrow S$$

is surjective;

(iii) *$\underline{\mathrm{Transt}}_G(H, H')$ is a principal homogeneous bundle under the action of the smooth S-group scheme of finite presentation $\underline{\mathrm{Norm}}_G(H)$.*

We merely note that the nontrivial implication (iii)⇒(i) is Hensel's lemma.

Using now *Bible*, §6.4, Theorem 4 (= [Ch05], §6.5, Theorem 5) and §9.3, Theorem 1, we obtain from 5.3.10 and 5.3.11:

Corollary 5.3.12. *Let G be an S-group of type* (RR)*. The Borel subgroups of G are closed in G, equal to their normalizers, and locally conjugate for the étale topology.*

Definition 5.3.13. *Let S be a scheme, and let G be a smooth S-group scheme of finite presentation with connected fibers.*[34] *A* Killing pair *of G is a pair $T \subset B$, where T is a maximal torus of G and B is a Borel subgroup of G containing T.*

Using now the conjugacy of maximal tori in B (cf. 5.1.2 (a) and 5.2.6, for example), we have:

Corollary 5.3.14. *Let G be an S-group of type* (RR)*. The Killing pairs of G are locally conjugate for the étale topology.*

Corollary 5.3.15. *Let G be an S-group of type* (RR)*. Let T be a maximal torus of G, and let*

$$W_G(T) = \underline{\mathrm{Norm}}_G(T)/\underline{\mathrm{Centr}}_G(T)$$

be the corresponding Weyl group (Exposé XIX, 6.3)*. The "functor of Borel subgroups of G containing T" is formally principal homogeneous under $W_G(T)$.*

This follows immediately from 5.3.14 and from the fact that, if B is a Borel subgroup of G containing T, then

$$\underline{\mathrm{Norm}}_G(T) \cap B = \underline{\mathrm{Centr}}_G(T),$$

cf. Exposé XIV, 4.4.

Proposition 5.3.16. *Let G be an S-group of type* (RR)*, let H be a subgroup of type* (R)*, and let $N = \underline{\mathrm{Norm}}_G(H)$ be its normalizer* (5.3.10)*. Let T be a maximal torus of H, and let $W_H(T)$ and $W_N(T)$ be the corresponding Weyl groups* (étale, quasi-finite, and separated by Exposé XIX, 6.3)*. There is the following exact sequence of sheaves* (for the étale topology)*, whose morphisms are induced by*

$$\underline{\mathrm{Norm}}_H(T) \longrightarrow \underline{\mathrm{Norm}}_N(T) \longrightarrow N/H :$$

$$1 \longrightarrow W_H(T) \longrightarrow W_N(T) \longrightarrow N/H \longrightarrow 1.$$

The only nontrivial point is that the last arrow is an epimorphism. Let $n \in N(S')$, where $S' \to S$. The two maximal tori T and $\mathrm{int}(n)T$ of H are locally conjugate in H for the étale topology. Thus there is a covering family $\{S'_i \to S'\}$ and, for every i, an $h_i \in H(S'_i)$ such that

$$\mathrm{int}(h_i)T = \mathrm{int}(n)T.$$

Consequently

$$nh_i^{-1} \in \underline{\mathrm{Norm}}_N(T),$$

which gives the desired result.

Remark 5.3.17. The group $W_N(T)$ may be described as follows. Suppose that we have reduced to the situation of 5.2.7, with $\mathfrak{h} = \mathfrak{g}_{R'}$. Then $W_N(T)$ is equal to $\underline{\mathrm{Norm}}_W(R')$, the sheaf of sections of $W = W_G(T)$ which, acting on R, normalize R'. Indeed, by 5.3.5,

$$\underline{\mathrm{Norm}}_N(T) = \underline{\mathrm{Norm}}_G(H) \cap \underline{\mathrm{Norm}}_G(T) = \underline{\mathrm{Norm}}_G(\mathfrak{h}) \cap \underline{\mathrm{Norm}}_G(T).$$

[34] Editor's note: See note 25.

Corollary 5.3.18. *Let G be an S-group of type* (RR)*, and let H be a subgroup of type* (R)*. Suppose that "the Weyl groups of G are finite", that is, that for every $S' \to S$ and every maximal torus T of $G_{S'}$, the étale S'-scheme*

$$\underline{\mathrm{Norm}}_{G_{S'}}(T)/\underline{\mathrm{Centr}}_{G_{S'}}(T)$$

is finite (cf. Exposé XIX, 6.3 (iii))*. Then the following conditions are equivalent:*

(i) *H is closed in G;*

(ii) *$\underline{\mathrm{Norm}}_G(H)/H$ is representable by a finite étale S-scheme;*

(iii) *"the Weyl groups of H are finite".*

Indeed, we may suppose that H has a maximal torus T. By 5.3.10, $N = \underline{\mathrm{Norm}}_G(H)$ is closed in G; therefore $W_N(T)$ is closed in $W_G(T)$, and hence is finite over S. Clearly (i) implies (iii), and (iii) implies (ii) by the exact sequence of 5.3.16. Finally, (ii) implies (i), since if N/H is finite, it is separated, so H is closed in N, and hence in G.

Remark 5.3.19. When G is reductive, the preceding conditions on H seem always to be satisfied. We prove them below in most cases.

5.4. Subgroups of Type (R) of a Split Reductive Group (Generalities)

5.4.1. If H is a subgroup of type (R) of the reductive group G, then H locally for the étale topology contains a maximal torus of G (5.2.2). By 2.3, locally for the étale topology we may suppose that G is split relative to this torus. Thus let (G, T, M, R) be a split S-group, and let H be a subgroup of type (R) of G containing T. By 5.3.5, such a subgroup is characterized by its Lie algebra, which (5.2.7) is locally on S of the form $\mathfrak{g}_{R'}$:

$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \prod_{\alpha \in R'} \mathfrak{g}^{\alpha}.$$

Definition 5.4.2. *Let (G, T, M, R) be a split S-group. A subset R' of R is said to be of type* (R) *if $\mathfrak{g}_{R'}$ is the Lie algebra of a subgroup of type* (R) *of G containing T. This subgroup, uniquely determined by R', is denoted $H_{R'}$.*

Lemma 5.4.3. *Under the preceding conditions, the following equivalences hold:*

$$\begin{aligned} H \cap Z_\alpha = T &\iff \alpha \notin R',\ -\alpha \notin R', \\ H \supset U_\alpha &\iff \alpha \in R', \\ H \cap U_\alpha = e &\iff \alpha \notin R', \\ H \supset Z_\alpha &\iff \alpha \in R',\ -\alpha \in R'. \end{aligned}$$

Indeed, $H \cap Z_\alpha$ is a subgroup of type (R) of Z_α, by 5.2.5; but a subgroup of type (R) of Z_α containing T is locally equal to one of the following subgroups:

$$T, \qquad T \cdot U_\alpha, \qquad T \cdot U_{-\alpha}, \qquad Z_\alpha,$$

by 5.3.5.

Lemma 5.4.4. *Under the conditions of 5.4.2, let R_+ be a system of positive roots; choose orderings on $R' \cap R_+$ and $R' \cap -R_+$. The morphism*

$$\Omega_{R_+,R'} = \prod_{\alpha \in R' \cap -R_+} U_\alpha \underset{S}{\times} T \underset{S}{\times} \prod_{\alpha \in R' \cap R_+} U_\alpha \longrightarrow G,$$

induced by multiplication in G, induces an open immersion

$$\Omega_{R_+,R'} \longrightarrow H_{R'}.$$

Indeed, by 5.4.3 this morphism factors through $H_{R'}$, and therefore induces an immersion $\Omega_{R_+,R'} \to H_{R'}$. We then argue as in 4.1.1.

Proposition 5.4.5. *Let (G, T, M, R) be a split S-group. Let R' and R'' be two subsets of R of type* (R).

(i) *$H_{R'} \cap H_{R''}$ is smooth at every point of the unit section, $R' \cap R''$ is of type* (R)*, and*

$$\left(H_{R'} \cap H_{R''}\right)^0 = H_{R' \cap R''}.$$

(ii) *The following equivalence holds:*

$$H_{R'} \subset H_{R''} \iff R' \subset R''.$$

Indeed, (ii) follows immediately from (i). To prove (i), it suffices to prove that $H_{R'} \cap H_{R''}$ is smooth at every point of the unit section: its neutral component (cf. Exposé $\mathrm{VI_B}$, 3.10) will then be a subgroup of type (R) containing T, and hence equal to $H_{R' \cap R''}$. But

$$\Omega_{R_+,R'} \cap \Omega_{R_+,R''} = \Omega_{R_+,R' \cap R''}$$

is an open subset of $H_{R'} \cap H_{R''}$, contains the unit section, and is smooth over S.

Corollary 5.4.6. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and s a point of S. If H and H' are two subgroups of type* (R) *containing T, and if $H_s \subset H'_s$, then there is an open subset U of S containing s such that $H_U \subset H'_U$.*

Indeed, we may suppose that G is split relative to T, and the assertion then follows immediately from 5.4.5 (ii).

We are thus led to ask which subsets R' of R are of type (R). We may suppose that the group is adjoint; one must then verify that $\mathfrak{g}_{R'}$ is a Lie algebra and that its normalizer is smooth at every point of the unit section. The most important case is given by the following theorem.

Theorem 5.4.7. *Every closed subset R' of R is of type* (R).

Recall (Exposé XXI, 3.1.4) that a subset R' of R is said to be *closed* if

$$\alpha, \beta \in R', \quad \alpha + \beta \in R \quad \Longrightarrow \quad \alpha + \beta \in R'.$$

Remark 5.4.8. We shall see later (Exposé XXIII, 6.6) that, if $6 \cdot 1_S \neq 0$[35]—for example, if S has residue characteristics different from 2 and 3—the fact that $\mathfrak{g}_{R'}$ is a Lie algebra already implies that R' is closed. In that case a subset R' is of type (R) if and only if it is closed. Thus Theorem 5.4.7 gives all the subsets of type (R) that are "independent of the characteristic."

Let us first prove the following lemma.

[35]Editor's note: In fact, it suffices (cf. loc. cit.) that 2 and 3 be nonzero on S.

Lemma 5.4.9. *For each $\alpha \in R$, choose an element*

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

Let $\alpha, \beta \in R$, with $\alpha + \beta \neq 0$, and let q be the largest integer i such that $\beta + i\alpha \in R$. There exist uniquely determined sections

$$M_{\alpha,\beta,i} \in \Gamma(S, \mathcal{O}_S)$$

such that

$$\mathrm{Ad}\big(\exp_\alpha(xX_\alpha)\big)(X_\beta) = X_\beta + \sum_{i=1}^{q} M_{\alpha,\beta,i} x^i X_{\beta+i\alpha}$$

for every $x \in \mathbb{G}_a(S')$, where $S' \to S$.

The map

$$x \longmapsto \mathrm{Ad}\big(\exp_\alpha(xX_\alpha)\big)(X_\beta)$$

defines a morphism

$$\mathbb{G}_{a,S} \longrightarrow \mathbf{W}(\mathfrak{g}) \simeq \mathbb{G}_{a,S}^m.$$

There are therefore uniquely determined sections $Y_n \in \Gamma(S, \mathfrak{g})$ such that

$$\mathrm{Ad}\big(\exp_\alpha(xX_\alpha)\big)(X_\beta) = \sum_{n \geq 0} x^n Y_n.$$

Applying the inner automorphism defined by a section t of T, we immediately find

$$\mathrm{Ad}(t)(Y_n) = \beta(t)\alpha(t)^n Y_n,$$

and hence $Y_n \in \Gamma(S, \mathfrak{g}^{\beta+n\alpha})$. Since α and β are not proportional, none of the $\beta + n\alpha$ is zero. Thus $Y_n = 0$ for $n > q$, and

$$Y_n = M_{\alpha,\beta,n} X_{\beta+n\alpha}$$

for $0 \leq n \leq q$, where $M_{\alpha,\beta,n} \in \mathbb{G}_a(S)$ is uniquely determined. Setting $x = 0$ in the formula obtained gives $Y_0 = X_\beta$, which completes the proof.

Remark 5.4.10. Differentiating the preceding formula at $x = 0$, we find

$$[X_\alpha, X_\beta] = \begin{cases} N_{\alpha,\beta} X_{\alpha+\beta}, & \text{where } N_{\alpha,\beta} = M_{\alpha,\beta,1}, \text{ if } \alpha + \beta \in R, \\ 0, & \text{if } \alpha + \beta \notin R \text{ and } \alpha + \beta \neq 0. \end{cases}$$

Let us now prove 5.4.7. If R' is a closed subset of R, then $\mathfrak{g}_{R'}$ is a Lie subalgebra of $\mathfrak{g}$, by 5.4.10 and Exposé XX, 2.10, formula (3). By 5.4.9 and Exposé XX, 2.10, formula (2), U_α normalizes $\mathfrak{g}_{R'}$ for every $\alpha \in R'$. Choose a system of positive roots R_+, and consider the open subset Ω_{R_+} of 4.1.2. Let $\Omega_{R_+,R'}$ be the closed subscheme of Ω_{R_+} defined by

$$\Omega_{R_+,R'} = \left(\prod_{\alpha \in R' \cap -R_+} U_\alpha \right) \cdot T \cdot \left(\prod_{\alpha \in R' \cap R_+} U_\alpha \right).$$

The canonical immersion $\Omega_{R_+,R'} \to G$ factors through

$$i : \Omega_{R_+,R'} \longrightarrow \underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'}).$$

Suppose that G is adjoint. The tangent map of i at the points of the unit section is bijective by 5.3.2. In particular, i is étale at every point of the unit section, and hence is a local immersion[36] at every point of the unit section. Therefore $\underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the unit section, as was to be proved.

[36] Editor's note: By EGA IV$_4$, 17.11.2, i is étale at every point of the unit section (and $\underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the unit section). Moreover, let V be the largest open subset of $\Omega_{R_+,R'}$ on which i is étale. Since i is a monomorphism, i_V is an open immersion (ibid., 17.9.1).

5.5. Borel Subgroups of a Split Reductive Group

Proposition 5.5.1. *Let (G, T, M, R) be a split S-group. For every system of positive roots R_+ of R, the subgroup H_{R_+}* (which exists by 5.4.7) *is a Borel subgroup of G. Moreover, for every ordering on R_+, the morphism induced by multiplication in G,*

$$T \underset{S}{\times} \prod_{\alpha \in R_+} U_\alpha \longrightarrow G,$$

is a closed immersion with image H_{R_+}. We write $B_{R_+} = H_{R_+}$.

By the definition of Borel subgroups, the first assertion may be verified after replacing S by the spectrum of an algebraically closed field. Let B be the Borel subgroup of G containing T and corresponding to the system of positive roots R_+ (*Bible*, §10.4, Proposition 9). The Lie algebra of B is $\mathfrak{g}_{R_+}$; therefore $B = H_{R_+}$ by 5.3.5.

For the second assertion, the morphism in the statement induces an open immersion

$$i : T \underset{S}{\times} \prod_{\alpha \in R_+} U_\alpha \longrightarrow H_{R_+}$$

by 5.4.4. But i is surjective (*Bible*, §15.1, Corollary 1 to Proposition 1).

Corollary 5.5.2. *Choose an arbitrary ordering on R_+, and for each $\alpha \in R_+$ choose an*

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

Let $\alpha, \beta \in R_+$. For every pair $(i, j) \in \mathbb{N}^* \times \mathbb{N}^*$ such that $i\alpha + j\beta \in R$, there is a unique section

$$C_{i,j,\alpha,\beta} \in \Gamma(S, \mathcal{O}_S)$$

such that, for all $x, y \in \mathbb{G}_a(S')$, where $S' \to S$, one has

$$\exp_\alpha(xX_\alpha) \exp_\beta(yX_\beta) \exp_\alpha(xX_\alpha)^{-1} = \exp_\beta(yX_\beta) \prod_{\substack{i,j \in \mathbb{N}^* \\ i\alpha + j\beta \in R}} \exp_{i\alpha+j\beta}(C_{i,j,\alpha,\beta} x^i y^j X_{i\alpha+j\beta}).$$

If $\alpha = \beta$, the assertion is trivial. Suppose therefore that $\alpha \neq \beta$. By the proposition there are unique morphisms

$$F_0 : \mathbb{G}_{a,S}^2 \longrightarrow T, \qquad F_\gamma : \mathbb{G}_{a,S}^2 \longrightarrow \mathbb{G}_{a,S} \quad (\gamma \in R_+)$$

such that

$$\exp(xX_\alpha) \exp(yX_\beta) \exp(xX_\alpha)^{-1} = F_0(x, y) \prod_{\gamma \in R_+} \exp(F_\gamma(x, y) X_\gamma).$$

Let $t \in T(S')$, where $S' \to S$. Applying $\mathrm{int}(t)$ to this formula, we immediately obtain

$$F_0(\alpha(t)x, \beta(t)y) = F_0(x, y), \tag{1}$$

$$F_\gamma(\alpha(t)x, \beta(t)y) = \gamma(t) F_\gamma(x, y). \tag{2}$$

Since α and β are two linearly independent characters of T over $\mathbb{Q}$, the first relation implies, as usual, that F_0 is constant, and hence $F_0(x, y) = e$. Write next

$$F_\gamma(x, y) = \sum a_{ij} x^i y^j, \qquad a_{ij} \in \Gamma(S, \mathcal{O}_S).$$

Substituting in relation (2) and comparing the polynomials on both sides gives

$$a_{ij}(\alpha(t)^i\beta(t)^j - \gamma(t)) = 0.$$

If $\gamma \neq i\alpha + j\beta$, we know (Exposé XIX, 4.13) that there exist a faithfully flat quasi-compact morphism $S' \to S$ and an element $t \in T(S')$ such that

$$\alpha(t)^i\beta(t)^j - \gamma(t) = 1.$$

Thus $a_{ij} = 0$ over S', and hence over S. If $\gamma = i\alpha + j\beta$, set $a_{ij} = C_{i,j,\alpha,\beta}$. Setting $x = 0$ (respectively, $y = 0$), one obtains

$$C_{0,1,\alpha,\beta} = 1 \qquad \text{(respectively, } C_{1,0,\alpha,\beta} = 0\text{)}.$$

Remark 5.5.3. Differentiating with respect to y at $y = 0$ and comparing with 5.4.9, one finds

$$C_{i,1,\alpha,\beta} = M_{\alpha,\beta,i}.$$

Corollary 5.5.4. *Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let $\alpha \neq \beta$ be two roots of G relative to T such that $\alpha + \beta$ is nontrivial on every fiber. Order arbitrarily the set of roots $i\alpha + j\beta$, where $(i, j) \in \mathbb{N}^* \times \mathbb{N}^*$. For all $i, j \in \mathbb{N}^*$ such that $i\alpha + j\beta \in R$, there is a unique morphism of $\mathcal{O}_S$-modules*

$$f_{\alpha,\beta,i,j} : (\mathfrak{g}^\alpha)^{\otimes i} \otimes (\mathfrak{g}^\beta)^{\otimes j} \longrightarrow \mathfrak{g}^{i\alpha+j\beta}.$$

such that, for every $S' \to S$ and all $X \in \mathbf{W}(\mathfrak{g}^\alpha)(S')$, $Y \in \mathbf{W}(\mathfrak{g}^\beta)(S')$, one has (the exponentials in the right-hand side being understood in the sense of 1.2)[37]

$$\exp_\alpha(X)\exp_\beta(Y)\exp_\alpha(-X) = \exp_\beta(Y)\prod_{(i,j)}\exp_{i\alpha+j\beta}\big(f_{\alpha,\beta,i,j}(X^{\otimes i} \otimes Y^{\otimes j})\big).$$

The assertion is local for the fpqc topology. By §2 we may therefore suppose that G is split relative to T, with α and β constant in the splitting. Since $\alpha + \beta \neq 0$, there is a system of positive roots R_+ containing α and β (Exposé XXI, 3.5.4), and we are reduced to 5.5.2.

Corollary 5.5.5. *Let S be a scheme and G a reductive S-group.*

(i) *G has Borel subgroups locally for the étale topology. If T is a maximal torus of G, then, locally for the étale topology, G also has Borel subgroups containing T.*

(ii) *If T is a maximal torus of G, the "functor of Borel subgroups of G containing T" is represented by a principal homogeneous bundle under $W_G(T)$.*

(iii) *If (G, T, M, R) is split, every Borel subgroup B of G containing T is locally on S of the form B_{R_+}, where R_+ is a system of positive roots of R.*

(iv) *If $T \subset B$ is a Killing pair of G, there is an étale covering family $\{S_i \to S\}$, and for every i a splitting $(G_{S_i}, T_{S_i}, M_i, R_i)$ and a system of positive roots R_{i+} of R_i, such that $B_{S_i} = B_{R_{i+}}$.*

Indeed, (i) follows from 2.3 and 5.5.1, (ii) from (i) and 5.3.15, (iii) from (ii) and 5.5.1, and (iv) from (iii) and 2.3.

[37] Editor's note: That is, if $\mathfrak{g}^{i\alpha+j\beta} = 0$ on a connected component of S, the corresponding exponential is 1 on that component.

Lemma 5.5.6. *Choose on the group $\Gamma_0(R)$ generated by the roots a totally ordered group structure such that the roots > 0 are the elements of R_+* (cf. Exposé XXI, 3.5.6)[38] *Let $\alpha_1 < \cdots < \alpha_N$ be the elements of R_+. Consider the isomorphism*

$$f : T \underset{S}{\times} U_{\alpha_1} \underset{S}{\times} \cdots \underset{S}{\times} U_{\alpha_N} \longrightarrow B_{R_+}$$

induced by multiplication in G. For $i = 1, \ldots, N$, put

$$U_{\geq i} = f\left(U_{\alpha_i} \underset{S}{\times} \cdots \underset{S}{\times} U_{\alpha_N}\right).$$

(i) *Each $U_{\geq i}$ is an invariant subgroup of B_{R_+}.*

(ii) *For $1 \leq i \leq N-1$, $U_{\geq i}$ identifies with the semidirect product*

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1}.$$

(iii) *B_{R_+} identifies with the semidirect product*

$$B_{R_+} = T \cdot U_{\geq 1}.$$

(iv) *For $1 \leq i \leq N-1$, the inner automorphisms of $U_{\geq 1}$ act trivially on $U_{\geq i}/U_{\geq i+1}$, identified with U_{α_i} by* (ii).

We first prove, by descending induction on i, the following assertion:

$U_{\geq i}$ is an invariant subgroup of B_{R_+}, the semidirect product of U_{α_i} and $U_{\geq i+1}$.

This is true for $i = N$; suppose it has been proved for $i+1$, and let us prove it for i. As schemes, we have

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1}.$$

We must show that $U_{\geq i}$ is invariant under the inner automorphisms of B_{R_+}. This is clear for $\mathrm{int}(t)$, where $t \in T(S)$; it is therefore enough to consider $\mathrm{int}(x)$, where $x \in U_\alpha(S)$ and $\alpha \in R_+$. Since $U_{\geq i+1}$ is invariant, it is enough to prove that

$$\mathrm{int}(x)(U_{\alpha_i}) \subset U_{\geq i}.$$

By 5.5.2, if $y \in U_{\alpha_i}(S')$, then

$$y^{-1}xyx^{-1} \in U_{\geq i+1}(S'),$$

which implies that

$$\mathrm{int}(x)(y) \in U_{\geq i}(S').$$

It remains to prove that $U_{\geq i}$ is a subgroup. Let $x, y \in U_{\geq i}(S)$, and write

$$x = px', \qquad y = qy',$$

where $p, q \in U_{\alpha_i}(S)$ and $x', y' \in U_{\geq i+1}(S)$. We then have

$$xy = px'qy' = pq(q^{-1}x'q)y' \in U_{\alpha_i}(S')U_{\geq i+1}(S'),$$

and likewise

$$x^{-1} = p^{-1}(px'^{-1}p^{-1}) \in U_{\alpha_i}(S')U_{\geq i+1}(S').$$

We have thus proved (i) and (ii), as well as (iv) along the way. Assertion (iii) is a trivial consequence of 5.5.1.

[38] Editor's note: Such an ordering is necessarily compatible with ord_Δ, where $\Delta = \mathscr{S}(R_+)$ (cf. Exposé XXI, 3.2.15).

Lemma 5.5.7. *With the preceding notation, choose, for each $1 \leq i \leq N$, an element*

$$X_i \in \Gamma(S, \mathfrak{g}^{\alpha_i})^{\times},$$

and consider the isomorphism

$$a : \mathbb{G}_{a,S}^{N} \longrightarrow U_{\geq 1}$$

defined on underlying sets by

$$a(x_1, \ldots, x_N) = \exp_{\alpha_1}(x_1 X_1) \cdots \exp_{\alpha_N}(x_N X_N).$$

There is a unique family $(Q_i)_{1 \leq i \leq N}$ of polynomials

$$Q_i = Q_i(x_1, \ldots, x_N, y_1, \ldots, y_N)$$

with coefficients in $\Gamma(S, \mathcal{O}_S)$, such that

$$\begin{aligned} &a(x_1, \ldots, x_N) a(y_1, \ldots, y_N) \\ &\quad = a\big(Q_1(x_1, \ldots, x_N, y_1, \ldots, y_N), \ldots, Q_N(x_1, \ldots, x_N, y_1, \ldots, y_N)\big). \end{aligned}$$

Moreover, the coefficients of these polynomials belong to the subring of $\Gamma(S, \mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$ of 5.5.2 $(\alpha, \beta \in R_+,\ i, j \in \mathbb{N}^*)$, and each Q_i is of the form

$$\begin{aligned} Q_i(x_1, \ldots, x_N, y_1, \ldots, y_N) = x_i + y_i \\ + Q_i'(x_1, \ldots, x_{i-1}, y_1, \ldots, y_{i-1}). \end{aligned}$$

The existence and uniqueness of the Q_i are immediate from the fact that a is an isomorphism of schemes. Let z, z', z'' denote sections of $U_{\geq i+1}$. We have

$$\begin{aligned} &a(x_1, \ldots, x_N) a(y_1, \ldots, y_N) \\ &\quad = a(x_1, \ldots, x_{i-1}, 0, \ldots, 0) \exp_{\alpha_i}(x_i X_i) z \\ &\qquad \cdot a(y_1, \ldots, y_{i-1}, 0, \ldots, 0) \exp_{\alpha_i}(y_i X_i) z'. \end{aligned}$$

Using 5.5.6 (i) and (iv), the right-hand side can be written

$$\begin{aligned} &a(x_1, \ldots, x_{i-1}, 0, \ldots, 0) a(y_1, \ldots, y_{i-1}, 0, \ldots, 0) \\ &\qquad \cdot \exp_{\alpha_i}((x_i + y_i) X_i) z'', \end{aligned}$$

and, applying 5.5.6 once again, this gives

$$Q_i(x_1, \ldots, x_N, y_1, \ldots, y_N) = x_i + y_i + Q_i'(x_1, \ldots, x_{i-1}, y_1, \ldots, y_{i-1}),$$

where

$$Q_i'(x_1, \ldots, x_{i-1}, y_1, \ldots, y_{i-1}) = Q_i(x_1, \ldots, x_{i-1}, 0, \ldots, 0, y_1, \ldots, y_{i-1}, 0, \ldots, 0);$$

this is precisely the required form.

It remains to prove the assertion concerning the coefficients of the polynomials Q_i. Let A be the subring of $\Gamma(S, \mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$, where $\alpha, \beta \in R_+$ and $i, j \in \mathbb{N}^*$. We prove by descending induction on i that, if

$$x_1 = \cdots = x_{i-1} = 0 \qquad \text{and} \qquad y_1 = \cdots = y_{i-1} = 0,$$

that is, if $a(x_1, \dots, x_N)$ and $a(y_1, \dots, y_N)$ are sections of $U_{\geq i}$, then the polynomials

$$R_j(x_i, \dots, x_N, y_i, \dots, y_N) = Q_j(x_1, \dots, x_N, y_1, \dots, y_N)$$

have coefficients in A. This is trivial for $i = N$, and also for $j < i$, since $R_j = 0$ when $j < i$. Let $i < N$; suppose the assertion proved for $i + 1$, and let us prove it for i and $j \geq i$. We have

$$\begin{aligned} a(0, \dots, 0, x_i, \dots, x_N) &= \exp(x_i X_i) a(0, \dots, 0, x_{i+1}, \dots, x_N) \\ &= \exp(x_i X_i) Z_i. \end{aligned}$$

Similarly, write

$$a(0, \dots, y_i, \dots, y_N) = \exp(y_i X_i) T_i.$$

Then

$$\begin{aligned} &a(0, \dots, x_i, \dots, x_N) a(0, \dots, y_i, \dots, y_N) \\ &\quad = \exp\big((x_i + y_i) X_i\big) \operatorname{int}\big(\exp(-y_i X_i)\big)(Z_i) \cdot T_i. \end{aligned}$$

Now

$$\operatorname{int}\big(\exp(-y_i X_i)\big)(Z_i) = \operatorname{int}\big(\exp(-y_i X_i)\big)\big(\exp(x_{i+1} X_{i+1}) \cdots \exp(x_N X_N)\big)$$

is a product of $N - i - 1$ sections of $U_{\geq i+1}$, whose coefficients in the decomposition

$$U_{\geq i+1} = U_{\alpha_{i+1}} \cdots U_{\alpha_N}$$

are polynomials in $y_i, x_{i+1}, \dots, x_N$ with coefficients in A, by 5.5.2. Applying the induction hypothesis, we deduce that the coefficients of

$$\operatorname{int}\big(\exp(-y_i X_i)\big)(Z_i) \cdot T_i$$

are likewise polynomials with coefficients in A, which completes the proof.

The preceding induction immediately gives a proof of the following.

Lemma 5.5.8. *With the notation of 5.5.6, let, for each $i = 1, \dots, N$,*

$$f_i : U_{\alpha_i} \longrightarrow H$$

be a group morphism, where H is an S-group functor. For the morphism

$$f : U_{\geq 1} \longrightarrow H$$

defined by

$$f\big(\exp(x_1 X_1) \cdots \exp(x_N X_N)\big) = f_1\big(\exp(x_1 X_1)\big) \cdots f_N\big(\exp(x_N X_N)\big)$$

to be a group morphism, it is necessary and sufficient that, for every pair $i < j$,

$$\begin{aligned} &f_j\big(\exp(x_j X_j)\big) f_i\big(\exp(x_i X_i)\big) f_j\big(\exp(-x_j X_j)\big) \\ &\quad = f\big(\exp(x_j X_j) \exp(x_i X_i) \exp(-x_j X_j)\big). \end{aligned}$$

5.6. Subgroups of Type (R) with Solvable Fibers

Proposition 5.6.1. *Let (G, T, M, R) be a split S-group, let R' be a subset of R of type* (R) (5.4.2)*, and let $H_{R'}$ be the corresponding subgroup of G. The following conditions are equivalent:*

(i) *$H_{R'}$ has solvable geometric fibers.*

(ii) *There exists a system of positive roots R_+ such that $R' \subset R_+$, and hence $H_{R'} \subset B_{R_+}$* (cf. 5.4.5)*.*

(iii) $R' \cap (-R') = \varnothing$.

(iv) *For every ordering of R', the morphism induced by multiplication in G*

$$T \times_S \prod_{\alpha \in R'} U_\alpha \longrightarrow H_{R'}$$

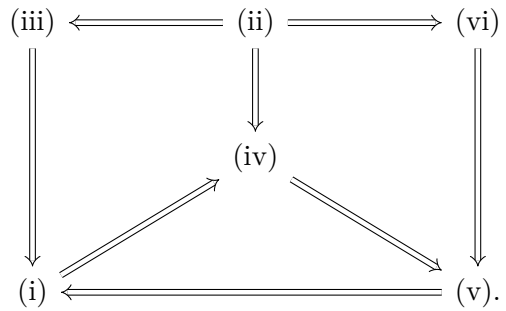
is an isomorphism.

(v) $H_{R'} \cap \underline{\mathrm{Norm}}_G(T) = T$.

(vi) *For every subset R'' of R of type* (R)*, one has* (cf. 5.4.5)

$$H_{R'} \cap \underline{\mathrm{Norm}}_G(H_{R''}) = H_{R' \cap R''}.$$

We shall prove these equivalences according to the following logical scheme.



Clearly (ii) implies (iii), and (vi) implies (v) (take $R'' = \varnothing$). By 5.4.6, it is enough to verify that (i) implies (ii) on the geometric fibers. If S is the spectrum of an algebraically closed field, then $H_{R'}$ is contained in a Borel subgroup containing T, and this subgroup is therefore of the form H_{R_+} (5.5.5(iii)).

Likewise, the implication from (iii) to (i) may be checked on the geometric fibers. Suppose that (iii) holds. If $H_{R'}$ were not solvable, there would exist a subtorus Q of T, of codimension 1 in T, such that

$$\underline{\mathrm{Centr}}_{H_{R'}}(Q)$$

is not solvable (*Bible*, §10.4, Proposition 8). But $\underline{\mathrm{Centr}}_{H_{R'}}(Q)$ has Lie algebra $\mathfrak{g}_{R''}$, where R'' is the set of roots in R' which vanish on Q. Thus, by (iii),

$$R'' = \varnothing \qquad \text{or} \qquad R'' = \{\alpha\}.$$

Consequently, $\underline{\mathrm{Centr}}_{H_{R'}}(Q)$, which is a subgroup of G of type (R), is T or $T \cdot U_\alpha$, and is therefore solvable, contrary to the hypothesis.

Likewise, the implication from (ii) to (iv) may be checked on the geometric fibers (since the schemes in question are flat S-schemes of finite presentation). By *Bible*, §13.2, Theorem 1(d), the morphism in question is bijective. It induces an isomorphism on the tangent spaces at the origin, and the conclusion follows as usual (cf. 4.1.1).

The implication from (iv) to (v) follows from 4.2.7. To prove that (v) implies (i), we may again reduce to the case in which S is the spectrum of an algebraically closed field; the conclusion then follows from *Bible*, §10.3, the corollary to Proposition 6, and §9.3, Corollary 3 to Theorem 1.

It remains only to prove the implication from (ii) to (vi). We may reduce to the case in which G is adjoint. Then, by 5.3.3,

$$\underline{\mathrm{Norm}}_G(H_{R''}) = \underline{\mathrm{Norm}}_G(\mathfrak{g}_{R''}) \subset \underline{\mathrm{Transp}}_G(\mathfrak{t}, \mathfrak{g}_{R''}). \tag{39}$$

[39] By 5.4.5, it is enough to prove

$$H_{R'}(S) \cap \underline{\mathrm{Transp}}_{G(S)}(\mathfrak{t}, \mathfrak{g}_{R''}) \subset H_{R'\cap R''}(S). \tag{x}$$

We first prove a lemma.

Lemma 5.6.2. *With the notation of 5.5.7, let*

$$u = \exp(x_1X_1)\cdots\exp(x_NX_N),$$

where $x_i \in \mathbb{G}_a(S)$. Let m, with $1 \le m \le N$, be such that $x_i = 0$ for $i < m$.

(a) *If $H \in \Gamma(S, \mathfrak{t})$, the component of $\mathrm{Ad}(u)H$ in $\mathfrak{g}^{\alpha_m}$ is*

$$-\overline{\alpha_m}(H)x_mX_m.$$

(b) *If $Y \in \Gamma(S, \mathfrak{g}^{-\alpha_m})$, the component of $\mathrm{Ad}(u)Y$ in $\mathfrak{t}$ is, with the notation of Exposé XX, 2.6,*

$$x_m\langle X_m, Y\rangle H_{\alpha_m}.$$

Proof. Set

$$\mathfrak{u} = \mathfrak{g}^{\alpha_{m+1}} + \cdots + \mathfrak{g}^{\alpha_N}.$$

By 5.4.9, one has

$$\mathrm{Ad}\big(\exp(x_iX_i)\big)\mathfrak{u} \subset \mathfrak{u} \qquad \text{for } i > m.$$

By Exposé XX, 2.10,

$$\mathrm{Ad}\big(\exp(x_iX_i)\big)H = H - \overline{\alpha_i}(H)x_iX_i.$$

It follows immediately that, modulo $\mathfrak{u}$,

$$\mathrm{Ad}(u)H = \mathrm{Ad}\big(\exp(x_mX_m)\big)H = H - \overline{\alpha_m}(H)x_mX_m,$$

which proves the first assertion.

Likewise, set[40]

$$\mathfrak{n} = \mathfrak{g}^{\alpha_1} + \cdots + \mathfrak{g}^{\alpha_N}, \qquad u_1 = \exp(x_{m+1}X_{m+1})\cdots\exp(x_NX_N).$$

[39] Editor's note: The inclusion follows from $\mathfrak{t} \subset \mathfrak{g}_{R''}$.

[40] Editor's note: The original has been corrected in what follows.

For $i > m$, one has $\alpha_i > \alpha_m$; hence, by 5.4.9, modulo $\mathfrak{n}$,

$$\mathrm{Ad}(u_1)Y \equiv Y, \qquad \text{and therefore} \qquad \mathrm{Ad}(u)Y \equiv \mathrm{Ad}\big(\exp(x_m X_m)\big)Y.$$

Applying Exposé XX, 2.10, one therefore obtains, modulo $\mathfrak{n}$,

$$\mathrm{Ad}(u)Y - Y \equiv x_m \langle X_m, Y\rangle H_{\alpha_m},$$

which proves the second assertion.

Let us return to the proof of inclusion (x). Suppose that there exists

$$h \in H_{R'}(S), \qquad h \notin H_{R'\cap R''}(S),$$

such that

$$\mathrm{Ad}(h)\mathfrak{t} \subset \mathfrak{g}_{R''}.$$

We may write

$$h = t\exp(x_1 X_1)\cdots\exp(x_N X_N).$$

Since $h \notin H_{R'\cap R''}(S)$, there is a least m such that

$$t\exp(x_1 X_1)\cdots\exp(x_{m-1}X_{m-1}) \in H_{R'\cap R''}(S), \qquad \alpha_m \notin R'', \qquad x_m \neq 0.$$

Then

$$h' = \exp(x_m X_m)\cdots\exp(x_N X_N)$$

still satisfies the conditions imposed above on h. By Lemma 5.6.2, for every $H \in \Gamma(S, \mathfrak{t})$, the component of $\mathrm{Ad}(h')H$ in $\mathfrak{g}^{\alpha_m}$ is

$$-\overline{\alpha_m}(H)x_m X_m.$$

It therefore follows from the hypothesis on h and from the definition of m that

$$\overline{\alpha_m}(H) = 0 \qquad \text{for every } H \in \Gamma(S, \mathfrak{t}).$$

This is impossible, since G is adjoint and $\overline{\alpha_m}$ is nonzero on every fiber.

Remark 5.6.2 bis. With the notation of Lemma 5.6.2, suppose that $\mathrm{Ad}(u)$ induces the identity on $\mathfrak{t}$ and on $\mathfrak{g}^{-\alpha_m}$. Then $x_m = 0$. Indeed,

$$x_m\overline{\alpha_m} = 0, \qquad x_m H_{\alpha_m} = 0.$$

If $\alpha_m \notin 2M$, then $\overline{\alpha_m}$ is nonzero on every fiber; if $\alpha_m \in 2M$, then $\alpha_m^* \notin 2M^*$, and H_{α_m} is nonzero on every fiber. Thus $x_m = 0$. It follows that $u = e$ if $\mathrm{Ad}(u)$ acts trivially on $\mathfrak{g}$.

Remark 5.6.3. Let H be a subgroup of type (R) of a reductive S-group G, and suppose that its geometric fibers are solvable. Then H is closed in G, and

$$\underline{\mathrm{Norm}}_G(H)/H$$

is represented by a separated finite étale S-scheme. This follows from 5.3.18 and either 3.5 or Exposé XIX, 2.5(ii).

Corollary 5.6.4. *Let (G, T, M, R) be a split reductive S-group. If $R' \subset R$ is closed and $R' \cap (-R') = \varnothing$, then R' is contained in a system of positive roots.* [41]

Proof. By 5.4.7, R' is of type (R), so the result follows from Proposition 5.6.1.

Corollary 5.6.5. *Under the hypotheses of Proposition 5.6.1, multiplication in G induces an isomorphism*

$$\prod_{\alpha \in R'} U_\alpha \xrightarrow{\sim} U_{R'},$$

where $U_{R'}$ is a closed subgroup scheme of G, smooth over S, with connected unipotent geometric fibers, and is independent of the chosen ordering of R'. Moreover, $H_{R'}$ is the semidirect product $T \cdot U_{R'}$, with $U_{R'}$ invariant in $H_{R'}$.

Proof. If $R' \subset R_+$, then

$$U_{R'} = H_{R'} \cap U_{\geq 1},$$

with the notation of 5.5.6, is a closed subgroup of G, of finite presentation, and invariant in $H_{R'}$. By Proposition 5.6.1(iv), one has $H_{R'} = T \cdot U_{R'}$, from which all the assertions follow.

Remark 5.6.6. In particular, U_{R_+} is the group $U_{\geq 1}$ of 5.5.6.

Let us draw some corollaries from the preceding results concerning groups of type $U_{R'}$.

Corollary 5.6.7. *Let (G, T, M, R) be a split reductive S-group, and let R' and R'' be two subsets of R of type* (R)*, with $R' \cap (-R') = \varnothing$.*

(i) *One has*

$$U_{R'} \cap \underline{\mathrm{Norm}}_G(H_{R''}) = U_{R' \cap R''}.$$

(ii) *Suppose that R' is closed. If the conditions $\alpha \in R'$, $\beta \in R''$, and $\alpha + \beta \in R$ imply that $\alpha + \beta \in R'$, then $H_{R''}$ normalizes $U_{R'}$.*

Proof. Assertion (i) follows at once from 5.6.5 and 5.6.1(vi). To prove (ii), it is enough, by 5.4.4, to show that T and each U_β, for $\beta \in R''$, normalize $U_{R'}$.[42] For T this is trivial; for U_β it follows from 5.5.2 and Exposé XXI, 3.1.2.[43]

Corollary 5.6.8. *Let (G, T, M, R) be a split S-group, let R_+ be a system of positive roots, and let α be a simple root of R_+* (that is, an element of R_+ such that $R_+ \setminus \{\alpha\}$ is closed)*. Set*

$$U_{\widehat{\alpha}} = U_{R_+ \setminus \{\alpha\}}.$$

Then:

[41] Editor's note: This is in fact a statement about root systems, supplementing Exposé XXI (cf. *Bible*, VI, §1.7, Proposition 22), and it is proved here indirectly.

[42] Editor's note: Indeed, since G has connected fibers, it is separated over S by Exposé $\mathrm{VI_B}$, 5.5; hence, by Exposé XI, 6.11 (see also Addendum 6.5.5 in Exposé $\mathrm{VI_B}$), $\underline{\mathrm{Norm}}_G(U_{R'})$ is represented by a closed subgroup scheme N of G. If N contains T and the U_β, for $\beta \in R''$, it contains the big cell $\Omega_{R_+, R''}$. This cell is schematically dense in $H_{R''}$ by 5.4.4 and EGA $\mathrm{IV_3}$, 11.10.10 (the fibers of $H_{R''}$ are integral, and $\Omega_{R_+, R''}$ contains a nonempty open subset of every fiber). It follows that $H_{R''} \subset N$, and therefore that $H_{R''}$ normalizes $U_{R'}$.

[43] Editor's note: One must show that, under the hypotheses of (ii), if $\alpha \in R'$ and $\beta \in R''$, then every root of the form $i\alpha + j\beta$, with $i, j \in \mathbb{N}^*$, belongs to R'. For this purpose the reference Exposé XXI, 2.3.5, has been replaced by Exposé XXI, 3.1.2. This may also be seen directly by inspecting the root systems of rank 2.

(i) *$U_{\widehat{\alpha}}$ is a normal subgroup of B_{R_+};*

(ii) *U_{R_+} is the semidirect product of $U_{\widehat{\alpha}}$ by U_α;*

(iii) *$U_{-\alpha}$ normalizes $U_{\widehat{\alpha}}$;*

(iv) *Z_α normalizes $U_{\widehat{\alpha}}$.*

If one similarly defines

$$U_{-\widehat{\alpha}} = U_{R_- \setminus \{-\alpha\}}, \qquad R_- = -R_+,$$

then

$$\Omega_{R_+} = U_{-\widehat{\alpha}} \cdot U_{-\alpha} \cdot T \cdot U_\alpha \cdot U_{\widehat{\alpha}}.$$

Proof. Assertion (ii) follows from 5.6.5, and (i) from 5.6.7(ii). Likewise, (iii) follows from 5.5.2: indeed, if $\beta \in R_+$, with $\beta \neq \alpha$, no combination

$$i(-\alpha) + j\beta, \qquad i, j > 0,$$

can be negative, since β contains at least one simple root different from α. Assertion (iv) then follows from (i) and (iii), because

$$U_{-\alpha} \cdot T \cdot U_\alpha$$

is schematically dense in Z_α. Finally, the last assertion follows from (ii) and its analogue for U_{R_-}.

Let us return to the general situation.

Proposition 5.6.9. *Let S be a scheme, let G be a reductive S-group, and let H be a subgroup of type* (R) *with solvable geometric fibers.*

(i) *The group*

$$\mathrm{D}_S(H) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(H, \mathbb{G}_{m,S})$$

is represented by a twisted constant S-group whose type at $s \in S$ is

$$\mathbb{Z}^{\mathrm{rgred}(G_s)}.$$

The biduality morphism (Exposé VIII, §1)

$$f : H \longrightarrow \mathrm{D}_S\big(\mathrm{D}_S(H)\big)$$

is smooth and surjective.

(ii) *The kernel H^u of f is the largest closed normal subgroup scheme of H, smooth over S, with connected unipotent geometric fibers. It is called the unipotent part of H, and is also denoted*

$$H^u = \mathrm{rad}^u(H).$$

Moreover, H^u is the sheaf of commutators of H: every morphism of groups from H to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on H^u, and therefore factors through

$$H/H^u = \mathrm{D}_S\big(\mathrm{D}_S(H)\big).$$

(iii) *If T is a maximal torus of H, the morphism $T \longrightarrow H$ induces isomorphisms*

$$\mathrm{D}_S(H) \xrightarrow{\sim} \mathrm{D}_S(T), \qquad T \xrightarrow{\sim} \mathrm{D}_S(\mathrm{D}_S(H)).$$

Moreover, H is identified with the semidirect product of H^u by T.

(iv) *In the situation of 5.6.1, if $H = H_{R'}$, then*

$$H^u = U_{R'}.$$

Proof. The assertions of the proposition are local for the étale topology (Exposé X, 5.5). We may therefore reduce to the situation of 5.6.1. We then know from 5.6.5 that $H_{R'}$ is the semidirect product of $U_{R'}$ by T. Let us show that $U_{R'}$ is the sheaf of commutators of $H_{R'}$. Since $H_{R'}/U_{R'} = T$ is commutative, it suffices to prove that every morphism of groups

$$\phi : H_{R'} \longrightarrow V$$

as in (ii) vanishes on $U_{R'}$. It is enough to prove that ϕ vanishes on each U_α, for $\alpha \in R'$. If $t \in T(S')$ and $X \in W(\mathfrak{g}^\alpha)(S')$, then

$$1 = \phi\big(t \exp_\alpha(X) t^{-1} \exp_\alpha(-X)\big) = \phi\big(\exp_\alpha((\alpha(t) - 1)X)\big).$$

Since $\alpha : T \longrightarrow \mathbb{G}_{m,S}$ is faithfully flat, it follows at once that ϕ vanishes on $U_\alpha^\times$. But every section of U_α is locally the sum of two sections of $U_\alpha^\times$. Thus

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, V) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(H/U_{R'}, V)$$

for every V as above. Applying this result to $V = \mathbb{G}_{m,S}$, we immediately obtain (i) and (iii), then (iv) and the second assertion of (ii).

It remains to prove the first assertion of (ii). The only nontrivial point is that every closed normal subgroup U of H, smooth over S, with connected unipotent geometric fibers, is a subgroup of H^u. We first have:

Lemma 5.6.10. *Let G be a reductive S-group, T a maximal torus, and U a subgroup scheme of G, smooth over S, with unipotent geometric fibers, and normalized by T. Then*

$$U \cap T = e.$$

Proof. Indeed, since

$$T = \underline{\mathrm{Centr}}_G(T),$$

one has $U \cap T = U^T$, where the right-hand side denotes the invariants under $\mathrm{int}(T)$. By Exposé XIX, 1.4, $U \cap T$ is therefore smooth over S. It is also radicial over S: for every $s \in S$, the group

$$U(\overline{s}) \cap T(\overline{s})$$

consists of elements that are both unipotent and semisimple. This proves the lemma.

We return to the proof of 5.6.9(ii). If U is a normal subgroup of H as above, then the semidirect product $T \cdot U$ is a subgroup of G of type (R), with solvable geometric fibers. We may therefore suppose it to be of the form $H_{R''}$, with $R'' \subset R'$. It is enough to prove that $U = U_{R''}$, and we are thus reduced to the case $H = T \cdot U$. Since H/U is commutative, the French text says that U is a subsheaf of the sheaf of commutators of H, which is H^u.[44] □

Let us note that we have in fact just proved:

[44] Editor's note: The implication in the French is stated in the wrong direction: commutativity of H/U gives $H^u \subset U$. The desired inclusion follows because H/H^u is a torus whereas U has connected unipotent geometric fibers.

Proposition 5.6.11. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. The maps*

$$H \longmapsto H^u, \qquad U \longmapsto T \cdot U$$

are mutually inverse bijections between the set of subgroups H of G of type (R)*, containing T, and having solvable geometric fibers, and the set of subgroups U of G, smooth over S, normalized by T, and having connected unipotent geometric fibers.*[45]

In particular, when (G, T, M, R) is split, the groups $H_{R'}$ and $U_{R'}$ correspond.

Corollary 5.6.12. *Let S be a scheme, let (G, T, M, R) be a split S-group* (respectively, and let R_+ be a system of positive roots of R defining the Borel subgroup B)*. Every smooth group subscheme of G with connected unipotent geometric fibers* (respectively, every smooth group subscheme of B^u)*, normalized by T, is locally on S of the form $U_{R'}$, where R' is a subset of R contained in a system of positive roots* (respectively, a subset of R_+) *and is of type* (R)*.*

Proof. For the "respectively" case, it is enough to observe that the geometric fibers of the given group are unipotent and connected by the *Bible*, §13.2, Theorem 1(d). □

Proposition 5.6.9 also has the following corollary:

Corollary 5.6.13. *Let S be a scheme, G a reductive S-group, and H a subgroup of type* (R) *with solvable geometric fibers. Let $\underline{\mathrm{Tor}}(H)$ be the functor of maximal tori of H:*

$$\underline{\mathrm{Tor}}(H)(S') = \{\, \text{maximal tori of } H_{S'} \,\}.$$

Then $\underline{\mathrm{Tor}}(H)$ is represented by an affine smooth S-scheme, which is a principal homogeneous bundle under H^u for the action

$$(h, T) \longmapsto \mathrm{int}(h)T.$$

[46]

Proof. Indeed, if T and T' are two maximal tori of $H_{S'}$, there exists a unique section $h \in H^u(S')$ such that $\mathrm{int}(h)T = T'$. The uniqueness of h follows at once from the equality

$$\underline{\mathrm{Norm}}_G(T) \cap H^u = e$$

(cf., for example, 5.6.1); it is therefore enough to prove the existence of h locally for the étale topology. By 5.2.6 and 5.1.2(a), we may suppose that T and T' are conjugate by a section of H, whence the desired conclusion because

$$H = H^u \cdot T$$

by 5.6.9(iii). It follows that $\underline{\mathrm{Tor}}(H)$ is a principal homogeneous sheaf under H^u, which is affine and smooth over S; the assertion follows at once.[47] □

[45] Editor's note: The hypothesis that U be closed has been omitted, since it is automatic. Indeed, for such a U, one has $U \cap T = e$ by 5.6.10, so the semidirect product $H = T \cdot U$ is a subgroup of G of type (R) (cf. 5.2.1), with solvable geometric fibers. Thus, by 5.6.3, H is closed in G; and since U is closed in H, it is closed in G.

[46] Editor's note: This reproves and sharpens Exposé XII, 1.10 (for G reductive).

[47] Editor's note: By SGA 1, VIII, 2.1, and EGA IV_4, 17.7.1.

5.7. Bruhat's Theorem

Recall 5.7.1. *Let k be an algebraically closed field, G a reductive k-group, B a Borel subgroup of G, T a maximal torus of B, and $N = \underline{\mathrm{Norm}}_G(T)$. Then*

$$G(k) = B(k)N(k)B(k);$$

this is Bruhat's theorem (*Bible*, §13.4, Corollary 1 to Theorem 3). *More precisely, with the notation of 3.6, the sets*

$$B(k)N_w(k)B(k) = B^u(k)N_w(k)B^u(k)$$

form a partition of $G(k)$ as w ranges over $(N/T)(k)$. If B' is another Borel subgroup of G containing T, the sets $B'(k)N_w(k)B(k)$ likewise form a partition of $G(k)$. Indeed, if $y \in N(k)$ is such that $\mathrm{int}(y)B = B'$, *then*

$$yB(k)N_w(k)B(k) = B'(k)N_{yw}(k)B(k).$$

Definition 5.7.2. *Let (G, T, M, R) be a split S-group, let R_- be a system of positive roots of R, and let $B' = B_{R_-}$ be the Borel subgroup it defines. For $w \in W$, put* (cf. 5.6.5)

$$R_-^w = R_- \cap w(R_-), \qquad B'^u_w = U_{R_-^w} = \prod_{\alpha \in R_-^w} U_\alpha.$$

If $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ is a representative of w (3.8), *one may also write*

$$B'^u_w = B'^u \cap \mathrm{int}(n_w)B'^u.$$

Lemma 5.7.3. *Let (G, T, M, R) be a split S-group, let R_+ be a system of positive roots of R, let $R_- = -R_+$, and let B* (respectively B') *be the Borel subgroup of G defined by R_+* (respectively R_-). *Let $w \in W$, and let N_w and B'^u_w be the corresponding subschemes of G* (3.8 and 5.7.2).

(i) *The sheaf $B' \cdot N_w \cdot B$, the image of the morphism*

$$B' \underset{S}{\times} N_w \underset{S}{\times} B \longrightarrow G$$

induced by multiplication in G, is represented by a subscheme of G (in fact, by a closed subscheme of the open subscheme $n_w\Omega_{R_+}$).

(ii) *The morphism*

$$B'^u_w \underset{S}{\times} N_w \underset{S}{\times} B^u \longrightarrow G,$$

induced by multiplication in G, is an immersion whose image is the preceding subscheme.

Proof. Let us first show that the morphism in (ii) is an immersion. By definition, $\mathrm{int}(n_w)^{-1}$ induces a closed immersion of B'^u_w into B'^u; hence the morphism

$$(u, b) \longmapsto n_w^{-1} u\, n_w b$$

induces a closed immersion

$$B'^u_w \underset{S}{\times} B \longrightarrow \Omega_{R_+}.$$

It follows immediately that the morphism in (ii) induces a closed immersion of its source into the open subscheme $n_w\Omega_{R_+}$. To prove (i), it is enough to see that

$$B'(S)N_w(S)B(S) = B'^u_w(S)N_w(S)B^u(S).$$

Now, if $\alpha \in R$, then $\mathrm{int}(n_w)U_\alpha(S) = U_{w(\alpha)}(S)$. Thus, if $w^{-1}(\alpha) \in R_+$,

$$\begin{aligned} U_\alpha(S)N_w(S)B(S) &= U_\alpha(S)n_wT(S)B^u(S) \\ &= n_wU_{w^{-1}(\alpha)}(S)T(S)B^u(S) \\ &= n_wB(S) = N_w(S)B^u(S). \end{aligned}$$

In view of the definition of R^w_-, this immediately gives the desired assertion. □

Theorem 5.7.4. *Let S be a scheme, let (G, T, M, R) be a split S-group, let B be the Borel subgroup defined by the system of positive roots R_+, and let B' be the Borel subgroup defined by $R_- = -R_+$.*

(i) (Bruhat's theorem) *The schemes $B'^u_w \cdot N_w \cdot B$, as w ranges over W, form a partition of the underlying set of G.*

(ii) *For each $w \in W$, let n_w be a representative of w in $\underline{\mathrm{Norm}}_G(T)(S)$ (3.8). Then the open subschemes*

$$n_w\Omega = n_wB'^u \cdot T \cdot B^u$$

as w ranges over W, form an open covering of G.

Proof. Both assertions may be checked on the geometric fibers, where they follow from 5.7.1 and 5.7.3. □

Remark 5.7.5. Assertion (i) implies that if S is the spectrum of a field, then $G(S)$ is the disjoint union of the sets

$$B'^u_w(S) \cdot T(S) \cdot B^u(S).$$

The corresponding assertion for an arbitrary S, even a local or Artinian one, is plainly false. Note, however, that (ii) implies that if S is local, then $G(S)$ is indeed the union of the $n_w\Omega(S)$. In fact:

Corollary 5.7.6. *Let Δ be a system of simple roots of the split group G over the local scheme S.*

(i) *Then $G(S)$ is generated by $T(S)$ and the groups $U_\alpha(S)$, for $\alpha \in \Delta \cup -\Delta$.*

(ii) *If G is simply connected* (4.3.3)*, then $G(S)$ is already generated by the groups $U_\alpha(S)$, for $\alpha \in \Delta \cup -\Delta$.*

Proof. Let H be the subgroup of $G(S)$ generated by the $U_\alpha(S)$, for $\alpha \in \Delta \cup -\Delta$. First observe that H contains a representative of every s_α, for $\alpha \in \Delta$, in $\underline{\mathrm{Norm}}_G(T)(S)$ (Exposé XX, 3.1), and hence a representative n_w of every $w \in W$. Since every $\alpha \in R$ can be written as $w(\alpha_0)$, with $w \in W$ and $\alpha_0 \in \Delta$, we have

$$U_\alpha(S) = \mathrm{int}(n_w)U_{\alpha_0}(S) \subset H.$$

The subgroup generated by H and $T(S)$ therefore contains $\Omega(S)$, and hence, by the preceding observation, is all of $G(S)$.

Now suppose that G is simply connected. Let us prove that $H \supset T(S)$. By Exposé XX, 2.7, H contains $\alpha^*(\mathbb{G}_m(S))$ for every $\alpha \in \Delta$, and it remains only to apply 4.3.8. □

Remark 5.7.6.1. Instead of taking, for each $\alpha \in \Delta$, both $U_\alpha(S)$ and $U_{-\alpha}(S)$, it is enough to take $U_\alpha(S)$ together with a representative w_α of the reflection s_α, or else $U_\alpha(S)$ together with a section of $U_{-\alpha}^\times$, ...

Corollary 5.7.7. *Suppose that G has semisimple rank 1, and choose $u_\alpha \in U_\alpha^\times(S)$. Then Ω and $u_\alpha\Omega$ form an open covering of G.*

Proof. If $u_{-\alpha}$ is the section of $U_{-\alpha}$ paired with u_α (cf. 1.3), then 5.7.4 (ii) gives

$$G = \Omega \cup u_{-\alpha}^{-1} u_\alpha u_{-\alpha}^{-1} \Omega,$$

and hence

$$G = u_{-\alpha} G = u_{-\alpha}\Omega \cup u_\alpha u_{-\alpha}^{-1}\Omega = \Omega \cup u_\alpha \Omega.$$

□

Corollary 5.7.8. *Let S be a scheme and let G be a reductive S-group. Then G is essentially free over S in the sense of Exposé VIII, 6.1.*[48]

Proof. The assertion is local for the fpqc topology, so we may assume that G is split. It then admits an open covering by open subschemes isomorphic to

$$\mathbb{G}_{a,S}^N \underset{S}{\times} \mathbb{G}_{m,S}^n,$$

and these are essentially free. □

Lemma 5.7.9. *Under the hypotheses of 5.7.4, let α be a simple root of R_+, and let $u_\alpha \in U_\alpha^\times(S)$. For every $v \in U_{-\alpha}(S)$, one has*

$$\Omega \cdot v \subset \Omega \cup u_\alpha \cdot \Omega.$$

Proof. We are comparing two open subschemes of G, so it is enough to do so when S is the spectrum of a field k. We must therefore prove

$$\Omega(k)v \subset \Omega(k) \cup u_\alpha \Omega(k).$$

Now

$$\begin{aligned} \Omega(k)v &= B'^u(k)T(k)B^u(k)v \\ &= U_{-\widehat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\widehat{\alpha}}(k)v \\ &\subset U_{-\widehat{\alpha}}(k)Z_\alpha(k)U_{\widehat{\alpha}}(k)v. \end{aligned}$$

Here we use the decomposition of 5.6.8. Applying 5.6.8 (iii) and then 5.7.7 to the group Z_α, we obtain

$$\begin{aligned} \Omega(k)v &\subset U_{-\widehat{\alpha}}(k)Z_\alpha(k)vU_{\widehat{\alpha}}(k) \\ &\subset U_{-\widehat{\alpha}}(k)Z_\alpha(k)U_{\widehat{\alpha}}(k) \\ &\subset U_{-\widehat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\widehat{\alpha}}(k) \\ &\quad \cup U_{-\widehat{\alpha}}(k)u_\alpha U_{-\alpha}(k)T(k)U_\alpha(k)U_{\widehat{\alpha}}(k). \end{aligned}$$

Applying 5.6.8 (iii) once more, now with R_- in place of R_+, gives the result. □

[48] Editor's note: See also the additions made in Exposé $\mathrm{VI_B}$, 6.2.1–6.2.6 and 6.5.2–6.5.5.

Proposition 5.7.10. *Under the hypotheses of 5.7.4, choose, for each simple root α, an element $u_\alpha \in U_\alpha^\times(S)$. Let U_1 be the submonoid of $B^u(S)$ generated by the u_α. Then the open subschemes $u\Omega$, for $u \in U_1$, form an open covering of G.*

Proof. Once again, we may assume that S is the spectrum of a field k. In view of 5.7.6, it is enough to prove that

$$\bigcup_{u\in U_1} u\Omega(k)$$

is stable under right multiplication by $T(k)$, $U_\alpha(k)$, and $U_{-\alpha}(k)$, for every simple root α. This is trivial in the first two cases, and in the last it follows from the lemma. □

Remark 5.7.11. Let us point out a special case of 5.7.2. If $w = s_\alpha$ is the reflection with respect to the simple root α, then

$$R_- \cap s_\alpha(R_-) = R_- \setminus \{-\alpha\}$$

(Exposé XXI, 3.3.1), and hence, in the notation of 5.6.8,

$$B'^u_{s_\alpha} = U_{-\widehat{\alpha}}.$$

Remark 5.7.12. In fact, the proof of 5.7.10 immediately gives the following statement. Under the hypotheses of 5.7.10, let Γ be a submonoid of $G(S)$. The open subschemes $g\Omega$, for $g \in \Gamma$, form an open covering of G if and only if, for every $s \in S$ and every simple root α, one has

$$(u_\alpha)_{\bar{s}} B'^u(\bar{s}) \subset \Gamma \cdot B'^u(\bar{s}) \cdot T(\bar{s}) \cdot B^u(\bar{s}).$$

Remark 5.7.13. By 5.5.5 (iii), arguing as in 5.7.1, one immediately obtains the following variant of 5.7.4. Let (G, T, M, R) be a split S-group, and let B and B' be two Borel subgroups of G containing T. For every $w \in W$, the sheaf $B' \cdot N_w \cdot B$ is representable by a subscheme of G; these subschemes, as w ranges over W, form a partition of the underlying set of G. One can also give the analogue of 5.7.3 (ii): one must put

$$B'^u_w = B'^u \cap \mathrm{int}(n_w)\widetilde{B}^u,$$

where $\widetilde{B}$ is the Borel subgroup opposite to B with respect to T (cf. 5.9.2).

Proposition 5.7.14. *Let S be a scheme, let G be a reductive S-group, and let*

$$\mathrm{Ad}\colon G \longrightarrow GL_{\mathcal{O}_S}(\mathfrak{g})$$

be its adjoint representation. Then

$$\mathrm{Ker}(\mathrm{Ad}) = \underline{\mathrm{Centr}}(G).$$

Equivalently, the canonical homomorphism induced by Ad *upon passage to the quotient,*

$$\overline{\mathrm{Ad}}\colon G/\underline{\mathrm{Centr}}(G) = \mathrm{ad}(G) \longrightarrow GL_{\mathcal{O}_S}(\mathfrak{g}),$$

is a monomorphism.[49]

[49] Editor's note: It is even a closed immersion, by Exposé XVI, 1.5(a).

Proof. We may assume that G is split. Choose a total order on $\Gamma_0(R)$ compatible with its group structure, and let R_+ be the set of positive roots. By 5.7.4 (ii) and 4.1.6, it is enough to prove that, if n_w represents $w \in W$, if

$$u \in U(S), \qquad t \in T(S), \qquad v \in U_-(S),$$

and if $\mathrm{Ad}(n_w vtu) = \mathrm{id}$, then $w = e$, $v = e$, and $u = e$. For each $m \in R \cup \{0\}$, put

$$\mathfrak{g}^{>m} = \bigoplus_{n>m} \mathfrak{g}^n, \qquad \mathfrak{g}^{<m} = \bigoplus_{n<m} \mathfrak{g}^n.$$

Let $X \in \Gamma(S, \mathfrak{g}^m)$; write

$$\mathrm{Ad}(tu)X = \mathrm{Ad}(v^{-1}n_w^{-1})X.$$

Now

$$\mathrm{Ad}(t)\,\mathrm{Ad}(u)X - m(t)X \in \Gamma(S, \mathfrak{g}^{>m}),$$
$$\mathrm{Ad}(v^{-1}n_w^{-1})X - \mathrm{Ad}(n_w^{-1})X \in \Gamma\big(S, \mathfrak{g}^{<w^{-1}(m)}\big).$$

If $w \neq e$, there is an $\alpha \in R$ such that $w^{-1}(\alpha) < \alpha$. On taking $m = \alpha$, this gives a contradiction, since

$$\begin{aligned}\mathrm{Ad}(tu)X &\in \Gamma\big(S, \mathfrak{g}^{\alpha} + \mathfrak{g}^{>\alpha}\big) \cap \Gamma\big(S, \mathfrak{g}^{w^{-1}(\alpha)} + \mathfrak{g}^{<w^{-1}(\alpha)}\big)\\ &= 0.\end{aligned}$$

Thus $w = e$, and we may choose $n_w = e$. We then have

$$\mathrm{Ad}(v^{-1})X - X \in \Gamma\big(S, \mathfrak{g}^{<m} \cap (\mathfrak{g}^m + \mathfrak{g}^{>m})\big) = 0,$$

whence $\mathrm{Ad}(v)X = X$ for every $X \in \Gamma(S, \mathfrak{g}^m)$, and therefore $\mathrm{Ad}(v) = \mathrm{id}$. Similarly, $\mathrm{Ad}(u) = \mathrm{id}$. The result now follows from 5.6.2 bis. □

5.8. Schemes Associated with a Reductive Group

Theorem 5.8.1. *Let S be a scheme and let G be a reductive S-group. Let $\mathcal{H}$ be the functor of subgroups of type* (R) *of G: for every $S' \to S$, $\mathcal{H}(S')$ is the set of subgroups of type* (R) *of $G_{S'}$* (cf. 5.2.1). *Then $\mathcal{H}$ is representable by a quasi-projective S-scheme of finite presentation over S.*

Proof.[50] Let

$$G' = G/\underline{\mathrm{Centr}}(G)$$

be the adjoint group of G (4.3.6), and let $u\colon G \to G'$ be the quotient morphism. By Exposé XII, 7.12, the map

$$H' \longmapsto u^{-1}(H')$$

is a bijection from the set of subgroups of type (R) of G' onto the set of subgroups of type (R) of G; this remains true after every base change. Thus, replacing G by G', we may assume that G is adjoint. Consider the morphism

$$u\colon \mathcal{H} \longrightarrow \underline{\mathrm{Grass}}(\mathfrak{g})$$

that assigns to every subgroup of type (R) its Lie algebra (a submodule of $\mathfrak{g}$ that is locally a direct summand[51]). Then u is a monomorphism by 5.3.3. It is enough to prove that it is

[50] Editor's note: The original is expanded in what follows.
[51] Editor's note: Cf. Exposé II, 4.11.8.

representable by an immersion of finite presentation; in other words, it is enough to prove the following assertion. For every $S_1 \to S$, given a locally direct-summand submodule $\mathfrak{h}$ of $\mathfrak{g}_{S_1}$, the morphisms $S' \to S_1$ for which $\mathfrak{h}_{S'}$ is the Lie algebra of a subgroup of type (R) of $G_{S'}$ are exactly those that factor through a certain subscheme Σ of finite presentation over S_1. Replacing S_1 by S, we reduce to $S_1 = S$, and we may moreover assume that S is affine; then there exists an affine Noetherian scheme S_0 such that G (respectively, $\mathfrak{h}$) is obtained by base change from an adjoint reductive S_0-group G_0 (respectively, from a locally direct-summand submodule $\mathfrak{h}_0$ of $\mathfrak{g}_0 = \mathrm{Lie}(G_0/S_0)$). It is enough to show that there exists a subscheme Σ_0 of S_0 having the required properties, for then $\Sigma = \Sigma_0 \times_{S_0} S$. Replacing S by S_0, we may therefore suppose that S is affine and Noetherian (note that then every subscheme of S is of finite presentation over S). Finally, after replacing S by a sufficiently small open subscheme, we may suppose that $\mathfrak{g}$ is free of rank n and that $\mathfrak{h}$ is a direct summand, free of rank r.

We must first express the condition that $\mathfrak{h}_{S'}$ be a Lie subalgebra of $\mathfrak{g}_{S'}$, that is, that the morphism induced by the Lie bracket

$$\varphi\colon \mathfrak{h} \otimes \mathfrak{h} \xrightarrow{[,]} \mathfrak{g}$$

factor through $\mathfrak{h}$. If $(e_1, \ldots, e_n)$ is a basis of $\mathfrak{g}$ such that $(e_1, \ldots, e_r)$ is a basis of $\mathfrak{h}$, then φ is given by sections a_k^{ij} of $\mathcal{O}_S$, where $i, j = 1, \ldots, r$ and $k = 1, \ldots, n$. The preceding condition is equivalent to requiring that $S' \to S$ factor through the closed subscheme of S defined by the equations

$$a_k^{ij} = 0 \qquad \text{for } k = r+1, \ldots, n \text{ and } i, j = 1, \ldots, r.$$

We replace S by this closed subscheme.

Then, by 5.3.0,

$$N = \underline{\mathrm{Norm}}_G(\mathfrak{h})$$

is a closed subgroup scheme of G, of finite presentation over G. We must now express (cf. 5.3.1) the condition that $N_{S'}$ be smooth at every point of the unit section, of relative dimension $r = \mathrm{rank}(\mathfrak{h})$, and that the inclusion of $\mathfrak{h}_{S'}$ in $\mathrm{Lie}(N_{S'}/S')$ be an equality.

Since N is affine over S[52] (being closed in G), the unit section $\varepsilon\colon S \to N$ is a closed immersion. Thus $\varepsilon(S)$ is defined by a quasi-coherent ideal $\mathcal{J}$ of $\mathcal{A}(N)$. Note that

$$\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2 \simeq \varepsilon^*(\Omega^1_{N/S}),$$

so its formation commutes with every base change $S' \to S$. By the equivalence $(c') \Longleftrightarrow (a)$ in EGA IV_4, 17.12.1 (applied to $f\colon N \to S$ and $j = \varepsilon$), the morphism $N_{S'} \to S'$ is smooth, of relative dimension r, at every point of $\varepsilon(S')$ if and only if

$$\mathfrak{n}_{S'/N'} = \mathfrak{n}_{S/N} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$$

is locally free of rank r, and the morphism

$$\varphi_n(S')\colon \mathrm{Sym}^n(\mathfrak{n}_{S'/N'}) \longrightarrow \mathcal{J}^n_{S'}/\mathcal{J}^{n+1}_{S'}$$

[52]Editor's note: The original then indicated that, denoting by $\mathfrak{n}_{N/G}$ the inverse image of the conormal sheaf $\mathcal{N}_{N/G}$ under the unit section $S \to N$, the condition that $\mathfrak{n}_{N_{S'}/G_{S'}} \to \omega^1_{G_{S'}/S'}$ be universally injective is equivalent to the condition that $S' \to S$ factor through a certain open subscheme of S. We were unable to justify this point, since formation of $\mathcal{N}_{N/G}$ does not commute with base change, and we have replaced that argument by the one that follows, suggested by O. Gabber.

is an isomorphism for every $n \geq 1$. Put

$$K_n(S') = \ker \varphi_n(S').$$

By TDTE I, Lemma 3.6, $\mathfrak{n}_{S/N} \otimes \mathcal{O}_{S'}$ is locally free of rank r if and only if $S' \to S$ factors through a certain subscheme Z of S. Replacing S by Z, we may therefore suppose that $\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2$ is locally free of rank r. Then, for every $S' \to S$,

$$(\mathcal{J}^2/\mathcal{J}^3) \otimes \mathcal{O}_{S'} = \mathcal{J}_{S'}^2/\mathcal{J}_{S'}^3,$$

and consequently

$$K_2(S') = K_2(S) \otimes \mathcal{O}_{S'}.$$

It follows that $\varphi_2(S')$ is an isomorphism if and only if $S' \to S$ factors through the closed subscheme S_2 of S defined by the ideal generated by the image of

$$\mathrm{Sym}^2(\mathfrak{n}_{S/N})^* \otimes K_2(S) \longrightarrow \mathcal{O}_S.$$

Over S_2, the module $\mathcal{J}^2/\mathcal{J}^3$ is then isomorphic to $\mathrm{Sym}^2(\mathfrak{n}_{S/N})$, hence is locally free; the same argument shows that $\varphi_3(S')$ is an isomorphism if and only if $S' \to S$ factors through a certain closed subscheme S_3 of S, and so on. We thus obtain that $N_{S'} \to S'$ is smooth, of relative dimension r, at every point of $\varepsilon(S')$ if and only if $S' \to S$ factors through the closed subscheme

$$Z = \bigcap_n S_n.$$

But then, for every $S' \to Z$, $\mathrm{Lie}(N_{S'}/S')$ is locally a direct summand of rank r of $\mathfrak{g}_{S'}$, and hence the inclusion

$$\mathfrak{h}_{S'} \subset \mathrm{Lie}(N_{S'}/S')$$

is an equality. We then put $H = N^0$.

Replacing S by Z, it remains only to express the condition that $H_{s'}$ have the same reductive rank as $G_{s'}$ at every point $s' \in S'$, or, equivalently, that H_s have the same reductive rank as G_s at every point s in the set-theoretic image of S' in S. This condition defines an open subset of S (Exposé XIX, 6.2). □

Remark. In general, the scheme $\mathcal{H}$ is not smooth over S. It is nevertheless smooth if 6 is invertible on S, or if there exists a prime number p such that $p \cdot 1_S = 0$ (that is, if S has characteristic $p > 0$).

Corollary 5.8.2. *Let S be a scheme, let G be a reductive S-group, and let H be a subgroup of type* (R) *of G. Recall from 5.3.10 that $\underline{\mathrm{Norm}}_G(H)$ is representable by a closed subgroup scheme of G, smooth over S. Then the quotient sheaf*

$$G/\underline{\mathrm{Norm}}_G(H)$$

is representable by a quasi-projective S-scheme, smooth and of finite presentation over S (in fact, by an open subscheme of $\mathcal{H}$).

Indeed, consider the morphism

$$f : G \longrightarrow \mathcal{H}$$

defined set-theoretically by

$$f(g) = \mathrm{int}(g)H.$$

By 5.3.9, this morphism is smooth and of finite presentation, hence open. Let $V = f(G)$, equipped with its open-subscheme structure in $\mathcal{H}$. The morphism $f : G \to V$ is a covering morphism with kernel $\underline{\mathrm{Norm}}_G(H)$. Consequently $G/\underline{\mathrm{Norm}}_G(H)$ is representable by V (cf. Exposé IV, 4.6.5). □

Corollary 5.8.3. *Let S be a scheme and let G be a reductive S-group. Consider the functors $\underline{\mathrm{Tor}}(G)$, $\underline{\mathrm{Bor}}(G)$, and $\underline{\mathrm{Kil}}(G)$, defined by*

$$\underline{\mathrm{Tor}}(G)(S') = \{\textit{maximal tori of } G_{S'}\},$$
$$\underline{\mathrm{Bor}}(G)(S') = \{\textit{Borel subgroups of } G_{S'}\},$$
$$\underline{\mathrm{Kil}}(G)(S') = \{\textit{Killing pairs of } G_{S'} \quad (\text{cf. } 5.3.13)\}.$$

(i) *$\underline{\mathrm{Tor}}(G)$, $\underline{\mathrm{Bor}}(G)$, and $\underline{\mathrm{Kil}}(G)$ are representable by smooth S-schemes of finite presentation, with geometrically integral fibres, and are respectively affine, projective, and affine over S.*

(ii) *The canonical morphism*

$$\underline{\mathrm{Kil}}(G) \longrightarrow \underline{\mathrm{Tor}}(G)$$

(respectively, $\underline{\mathrm{Kil}}(G) \to \underline{\mathrm{Bor}}(G)$) *is finite étale and surjective* (respectively, affine, smooth, and surjective).

(iii) *Let T be a maximal torus of G* (respectively, let B be a Borel subgroup of G, or let $B \supset T$ be a Killing pair of G). *The morphism*

$$G \longrightarrow \underline{\mathrm{Tor}}(G), \quad \textit{resp.} \quad G \longrightarrow \underline{\mathrm{Bor}}(G), \quad \textit{resp.} \quad G \longrightarrow \underline{\mathrm{Kil}}(G),$$

defined by

$$g \longmapsto \mathrm{int}(g)T, \quad \textit{respectively} \quad g \longmapsto \mathrm{int}(g)B, \quad \textit{respectively} \quad g \longmapsto (\mathrm{int}(g)B, \mathrm{int}(g)T)$$

induces an isomorphism

$$G/\underline{\mathrm{Norm}}_G(T) \xrightarrow{\sim} \underline{\mathrm{Tor}}(G), \quad \text{resp.} \quad G/B \xrightarrow{\sim} \underline{\mathrm{Bor}}(G), \quad \text{resp.} \quad G/T \xrightarrow{\sim} \underline{\mathrm{Kil}}(G).$$

One first sees that (iii) follows from the conjugacy theorem for maximal tori (respectively, Borel subgroups, respectively, Killing pairs) and from the facts that

$$\underline{\mathrm{Norm}}_G(B) = B, \qquad \underline{\mathrm{Norm}}_G(B) \cap \underline{\mathrm{Norm}}_G(T) = T,$$

all of which were established above (5.1.2, 5.3.12, 5.3.14, 5.6.1).

It follows first that the canonical morphisms

$$\underline{\mathrm{Tor}}(G) \longrightarrow \mathcal{H}, \qquad \underline{\mathrm{Bor}}(G) \longrightarrow \mathcal{H}$$

are representable, locally for the étale topology, by open immersions (5.8.2 and 5.1.2, respectively 5.5.5). Hence, by descent, $\underline{\mathrm{Tor}}(G)$ and $\underline{\mathrm{Bor}}(G)$ are representable by open subschemes of $\mathcal{H}$. Likewise, $\underline{\mathrm{Kil}}(G)$ is locally, for the étale topology, representable by a scheme affine over the base (Exposé IX, 2.3), and hence is representable by an affine S-scheme, by descent for affine schemes.[53]

The assertions of (ii) follow at once from 5.5.5(ii) and 5.6.13. It already follows that $\underline{\mathrm{Tor}}(G)$ is affine over S (EGA II, 6.7.1). It remains only to prove that $\underline{\mathrm{Bor}}(G)$ is projective over S. We already know that it is quasi-projective, so it remains to prove that it is proper;[54] but it has connected fibres, and therefore, by EGA IV_3, 15.7.10, one is reduced to proving this on geometric fibres. If S is the spectrum of an algebraically closed field, then $\underline{\mathrm{Bor}}(G) = G/B$ by (iii), and the conclusion follows from *Bible*, §6.4, Theorem 4 (or [Ch05], §6.5, Theorem 5). □

[53] Editor's note: Cf. SGA 1, VIII, 2.1.

[54] Editor's note: We may suppose that S is affine and, since $\underline{\mathrm{Bor}}(G)$ is of finite presentation over S by (i), reduce to the case in which S is Noetherian; we are then under the hypotheses of EGA IV_3, 15.7.10.

Remark 5.8.4. Under the hypotheses of 5.8.3, let Q be a central subgroup of multiplicative type of G. The evident morphisms define isomorphisms

$$\underline{\mathrm{Tor}}(G) \simeq \underline{\mathrm{Tor}}(G/Q), \qquad \underline{\mathrm{Bor}}(G) \simeq \underline{\mathrm{Bor}}(G/Q), \qquad \underline{\mathrm{Kil}}(G) \simeq \underline{\mathrm{Kil}}(G/Q).$$

Corollary 5.8.5. *Let S be a scheme, G a reductive S-group, and P a subgroup scheme of G, smooth and of finite presentation over S. The following conditions are equivalent:*

(i) *For every $s \in S$, $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$* (that is, the quotient scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper over $\bar{s}$, or equivalently $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$; cf. *Bible*, §6.4, Theorem 4, or [Ch05], §6.5, Theorem 5).

(ii) *The quotient sheaf G/P is representable by an S-scheme smooth and projective over S.*

Moreover, under these conditions, P is closed in G, has connected fibres, and

$$P = \underline{\mathrm{Norm}}_G(P).$$

Clearly (ii) implies (i). If (i) holds, then

$$P(\bar{s}) = \underline{\mathrm{Norm}}_{G(\bar{s})}(P_{\bar{s}})$$

and $P_{\bar{s}}$ is connected (for the first assertion, cf. *Bible*, §12.3, Lemma 4;[55] the second assertion follows from this, since $P' = P^0_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$ normalized by $P_{\bar{s}}$, whence $P'(\bar{s}) = P(\bar{s})$ and therefore $P' = P_{\bar{s}}$). It follows that P is of type (R) and that

$$P = \underline{\mathrm{Norm}}_G(P),$$

so that P is closed in G. By 5.8.2,

$$G/P = G/\underline{\mathrm{Norm}}_G(P)$$

is representable by a quasi-projective S-scheme. Its fibres are connected and proper, and it is therefore projective by the argument of 5.8.3. □

Remark 5.8.6. The statements 5.8.1, 5.8.2, and 5.8.5 are valid for an S-group of type (RA), or for an S-group of type (RR) satisfying 5.1.8.[56]

Remark 5.8.7. Through the inner automorphisms of G, there are canonical actions

$$G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Tor}}(G)), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Bor}}(G)), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G)),$$

which, in the situation of 5.8.3(iii), are identified with the canonical actions

$$G \longrightarrow \underline{\mathrm{Aut}}_S(G/\underline{\mathrm{Norm}}_G(T)), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(G/B), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(G/T).$$

In particular, one obtains

$$\begin{aligned} \ker(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Tor}}(G))) &= \ker(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Bor}}(G))) \\ &= \ker(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G))) \\ &= \underline{\mathrm{Centr}}(G). \end{aligned}$$

[55] Editor's note: We have supplied the details that follow.

[56] Editor's note: Indeed, the proof of 5.8.1 uses only 5.3.3 (which is valid for a group of type (RA)) and Exposé XIX, 6.2, which, by Exposé XII, 1.7(b), is also valid for groups of type (RR).

Indeed, it is clear that $\underline{\mathrm{Centr}}(G)$ acts trivially on each of the three schemes. Conversely, the kernel of

$$G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G))$$

is "the intersection of the maximal tori of G" in the sense of 4.1.7, and hence equals $\underline{\mathrm{Centr}}(G)$ (loc. cit.). For $\underline{\mathrm{Bor}}(G)$, one observes that "the intersection of the Borel subgroups of G" is also "the intersection of its maximal tori" (see the following section). For $\underline{\mathrm{Tor}}(G)$, one uses Exposé XII, 4.11.

5.9. Properties Peculiar to Borel Subgroups

Most of these properties will be generalized in Exposé XXVI to parabolic subgroups.

Definition 5.9.1. Let S be a scheme, G a reductive S-group, and B, B' two Borel subgroups of G. We say that B and B' are *in general position* (or that B' is in general position relative to B) if $B \cap B'$ is a torus (necessarily maximal) of G.

If T is a maximal torus of G contained in B and B', we say that B and B' are *opposite* (relative to T) if

$$B \cap B' = T.$$

Proposition 5.9.2. *Let S be a scheme, G a reductive S-group, B a Borel subgroup of G, and T a maximal torus of B. There exists a unique Borel subgroup B' of G, opposite to B relative to T.*

If (G, T, M, R) is a splitting of G relative to T, and if $B = B_{R_+}$ (5.5.1), then

$$B' = B_{-R_+}.$$

By faithfully flat descent, it is enough to prove the proposition in the split case, with $B = B_{R_+}$ (5.5.5(iv)). Then B_{-R_+} is indeed opposite to B (4.1.2). Let us show that it is the only Borel subgroup of G containing T that is opposite to B. If B' is a Borel subgroup of G containing T, then locally on S it has the form $B_{R'_+}$, where R'_+ is another system of positive roots of R (5.5.5(iii)). If $R'_+ \neq -R_+$, there exists $\alpha \in R'_+ \cap R_+$, and hence

$$U_\alpha \subset B_{R_+} \cap B_{R'_+}.$$

□

Proposition 5.9.3. *Let S be a scheme, G a reductive S-group, and B a Borel subgroup of G.*

(i) *If B' is a Borel subgroup of G, the following conditions are equivalent:*

(a) *B' is in general position relative to B* (5.9.1).

(b) *$B'^u \cap B^u = e$.*

(b′) *$B'^u \cap B = e$.*

(c) *Multiplication in G induces an open immersion*

$$B'^u \times_S B \longrightarrow G.$$

(c′) *The canonical morphism*

$$B'^u \longrightarrow G/B$$

is an open immersion.

(ii) *The functor* $\underline{\mathrm{Opp}}(B)$,

$$S' \longmapsto \left\{ \begin{array}{c} \textit{Borel subgroups of } G_{S'} \textit{ in general position} \\ \textit{relative to } B_{S'} \end{array} \right\},$$

is representable by an open subscheme of $\underline{\mathrm{Bor}}(G)$ (5.8.3). *The morphism*

$$\underline{\mathrm{Opp}}(B) \longrightarrow \underline{\mathrm{Tor}}(B)$$

defined by $B' \mapsto B \cap B'$ *is an isomorphism. In particular* (5.6.13), *the inner automorphisms of* B^u *equip* $\underline{\mathrm{Opp}}(B)$ *with the structure of a principal homogeneous bundle under* B^u.

Let us first examine (i). We have (a) ⇒ (c). Indeed, (c) is local for the étale topology; by 5.5.5(iv), one reduces to the case in which G is split relative to $B \cap B'$ and B has the form B_{R_+}. By 5.9.2,

$$B'^u = U_{-R_+},$$

and one is reduced to 4.1.2.

Trivially,

$$(\mathrm{c}') \Longleftrightarrow (\mathrm{c}) \Longrightarrow (\mathrm{b}') \Longrightarrow (\mathrm{b}).$$

It remains to prove (b) ⇒ (a). Let us first prove (ii). The second assertion is a formal consequence of 5.9.2, and the third follows at once from it by 5.6.13. Let us prove the first. It is local for the étale topology, and we may therefore suppose that B contains a maximal torus T. Let B'_0 be the Borel subgroup opposite to B relative to T (5.9.2).

By what precedes, the morphism

$$B^u \longrightarrow \underline{\mathrm{Bor}}(G)$$

induced by the canonical morphism

$$G \longrightarrow G/B'_0 \longrightarrow \underline{\mathrm{Bor}}(G)$$

(5.8.3) induces an isomorphism

$$B^u \xrightarrow{\sim} \underline{\mathrm{Opp}}(B).$$

Thus there is a commutative diagram

$$\begin{array}{ccc} B^u & \longrightarrow & G/B'_0 \\ \downarrow \sim & & \downarrow \sim \\ \underline{\mathrm{Opp}}(B) & \longrightarrow & \underline{\mathrm{Bor}}(G). \end{array}$$

Now the morphism $B^u \to G/B'_0$ is an open immersion (by (i), (a) ⇒ (c′)), which completes the proof of (ii). We record at once the following corollary.

Corollary 5.9.4. *Let G be a reductive S-group and let B, B' be two Borel subgroups of G. If $s \in S$ is such that $B_{\bar{s}}$ and $B'_{\bar{s}}$ are in general position, there exists an open subset V of S, containing s, such that B_V and B'_V are in general position.*

It therefore remains only to prove (b) ⇒ (a). By the preceding corollary, it is enough to do so when S is the spectrum of an algebraically closed field k. We may suppose that G is split relative to a maximal torus T of B. Let B'_0 be the Borel subgroup opposite to B. Since the Borel subgroups of G are conjugate under $G(k)$, there exists $g \in G(k)$ such that $\mathrm{int}(g)B'_0 = B'$. By the Bruhat theorem (5.7.4), we may write

$$g = bnb', \qquad b \in B(k), \quad b' \in B'_0(k), \quad n \in \underline{\mathrm{Norm}}_G(T)(k).$$

Thus

$$B' = \mathrm{int}(b)\,\mathrm{int}(n)B'_0$$

and

$$B' \cap B = \mathrm{int}(b)(\mathrm{int}(n)B'_0 \cap B).$$

If $n \notin T(k)$, then

$$\mathrm{int}(n)(B'_0)^u \cap B^u \neq e$$

(cf. the proof of 5.9.2); it follows that (b) implies

$$B' \cap B = \mathrm{int}(b)(B'_0 \cap B) = \mathrm{int}(b)T.$$

□

Proposition 5.9.5. *Let S be a scheme, G a reductive S-group, B a Borel subgroup of G, and B^u its unipotent part. There exists a sequence of subgroups of B,*

$$U_0 = B^u \supset U_1 \supset \cdots \supset U_n \supset \cdots,$$

having the following properties:

(i) *Each U_i is smooth, has connected fibers, and is characteristic in B; the inner automorphisms of B^u act trivially on the* (sheaf) *quotients U_i/U_{i+1}.*

(ii) *For each $i \geq 0$, there exists a locally free $\mathcal{O}_S$-module $\mathcal{E}_i$ and an isomorphism of sheaves of groups on S,*

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i).$$

(iii) *For every $s \in S$,*

$$(U_n)_s = e \qquad \textit{for} \qquad n \geq \dim(B^u_s).$$

Suppose first that there exists a splitting (G, T, M, R) of G and a system R_+ of positive roots of R such that $B = B_{R_+}$. Denote by Δ the set of simple roots of R_+. For each $\alpha \in R_+$, denote by $\mathrm{ord}(\alpha)$ the sum of the coefficients of α in the basis Δ of $\Gamma_0(R)$; this is the *order* of α relative to R_+. We have

$$\mathrm{ord}(\alpha) \leq \mathrm{Card}(R_+).$$

For every $i > 0$, let $R^{(i)}$ be the set of roots of order $> i$. This is a closed set of positive roots, and we may therefore construct (5.6.5)

$$U_i = U_{R^{(i)}}.$$

If $\alpha \in R_+$ and $\beta \in R^{(i)}$, then $\alpha + \beta \in R^{(i+1)}$. It follows from 5.5.2 that each U_i is an invariant subgroup of B, and that the inner automorphisms of B^u act trivially on U_i/U_{i+1}. Moreover, this group identifies with

$$\prod_{\mathrm{ord}(\alpha)=i+1} U_\alpha$$

and is therefore endowed with a vector-group structure.

If B is of the form $B_{R'_+}$ for another splitting (G, T', M', R') of G, let us show that the groups U'_i constructed as above using the new splitting coincide with the U_i, and that the vector-group structures on the successive quotients likewise coincide. By 5.6.13, there exists $b \in B^u(S)$ such that $T' = \mathrm{int}(b)T$; the assertion to be proved is local on S, and we may therefore suppose that the isomorphism

$$T \xrightarrow{\sim} T'$$

induced by $\mathrm{int}(b)$ arises by duality from an isomorphism of root data

$$h : (M', M'^*, R', R'^*) \xrightarrow{\sim} (M, M^*, R, R^*).$$

The roots in R'_+ are then the roots

$$\alpha \circ \mathrm{int}(b) = h(\alpha), \qquad \alpha \in R_+,$$

and its simple roots are the $h(\alpha)$, with $\alpha \in \Delta$. Consequently

$$\mathrm{ord}\big(h(\alpha)\big) = \mathrm{ord}(\alpha).$$

By transport of structure, the vector groups $U'_{h(\alpha)}$ are the groups $\mathrm{int}(b)U_\alpha$. It follows that

$$\mathrm{int}(b)U_i = U'_i.$$

But U_i is invariant in B, so that $U_i = U'_i$. Likewise, the isomorphism

$$\mathrm{int}(b) : U_i/U_{i+1} \xrightarrow{\sim} U'_i/U'_{i+1}$$

is the identity.

We now pass to the general case. Let $\{S_i \to S\}$ be an étale covering family such that each (G_i, T_i, M_i, R_i) is a splitting, and choose a system R_{i+} of positive roots such that

$$B \times_S S_i = B_{R_{i+}}$$

(5.5.5(iii)). For every i, the preceding construction gives a sequence

$$B^u_{S_i} = U_{i,0} \supset U_{i,1} \supset \cdots \supset U_{i,j} \supset \cdots$$

and vector-group structures on its successive quotients. By descent, it is enough to verify, for every i, i' and every j, that

$$U_{i,j} \times_{S_i} S_{ii'} = U_{i',j} \times_{S_{i'}} S_{ii'}, \qquad S_{ii'} = S_i \times_S S_{i'},$$

and that the vector-group structures on

$$(U_{i,j}/U_{i,j+1}) \times_{S_i} S_{ii'} \quad \text{and} \quad (U_{i',j}/U_{i',j+1}) \times_{S_{i'}} S_{ii'}$$

coincide. If $S_{ii'} = \varnothing$, there is nothing to prove. Otherwise we are in the situation considered above: $B \times_S S_{ii'}$ is defined by the system of positive roots R_{i+}, respectively $R_{i'+}$, in the splittings

$$(G_{S_{ii'}}, T_i \times_{S_i} S_{ii'}, M_i, R_i), \qquad (G_{S_{ii'}}, T_{i'} \times_{S_{i'}} S_{ii'}, M_{i'}, R_{i'}).$$

This proves the proposition. □

Corollary 5.9.6. *If S is affine, then*

$$H^1(S, B^u) = e;$$

that is, every principal homogeneous bundle under B^u has a section.

Indeed, S is a disjoint union of subschemes on each of which B^u has constant relative dimension. In view of 5.9.5(iii), we may therefore suppose that $U_n = e$. By TDTE I, B.1.1,[56] one has

$$H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0.$$

It follows by induction that $H^1(S, B^u) = e$.

Corollary 5.9.7. *If S is affine, then B has maximal tori. If T is a maximal torus of B, then*

$$H^1(S, T) = H^1(S, B).$$

The first assertion follows at once from 5.9.6 and 5.6.13; the second follows from it in the standard way.[57]

Corollary 5.9.8. *If G is a reductive S-group, the canonical morphism* (cf. 5.8.3)

$$\underline{\mathrm{Kil}}(G) \longrightarrow \underline{\mathrm{Bor}}(G)$$

has sections over every affine open subset.

Corollary 5.9.9. *Under the hypotheses of 5.9.5, suppose that S is affine. Then there exists a locally free $\mathcal{O}_S$-module $\mathcal{E}$ such that B^u, as a scheme, is S-isomorphic to $W(\mathcal{E})$.*

We prove by induction on i that B^u/U_i is S-isomorphic to

$$W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

This is clear for $i = 0$; suppose that $i \geq 1$. Then B^u/U_i is a principal homogeneous bundle with base

$$X = B^u/U_{i-1}$$

under the group $(U_{i-1}/U_i)_X$. Since B^u/U_{i-1} is affine by the induction hypothesis, and since

$$U_{i-1}/U_i = W(\mathcal{E}_{i-1}),$$

this bundle is trivial. We therefore have, at least, an isomorphism of S-schemes

$$B^u/U_i \xrightarrow{\sim} (B^u/U_{i-1}) \times_S W(\mathcal{E}_{i-1}) = W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

The conclusion now follows at once from condition 5.9.5(iii).

[56] Editor's note: This is the corollary on page 18 of TDTE I.

[57] Editor's note: Indeed, there is an exact sequence

$$H^1(S, B^u) \longrightarrow H^1(S, B) \xrightarrow{\pi} H^1(S, B/B^u) = H^1(S, T);$$

see [Se64], I §5.5, Proposition 38, or [Gi71], III Proposition 3.3.1. Now $H^1(S, T) \to H^1(S, B)$ is a section of π, so that π is surjective; on the other hand, $H^1(S, B^u) = 0$ by 5.9.6.

Corollary 5.9.10. *Let S be a semilocal scheme, let $\{s_i\}$ be its closed points, and let B be a Borel subgroup of the reductive S-group G. The canonical map*

$$B^u(S) \longrightarrow \prod_i B^u(\operatorname{Spec} \kappa(s_i))$$

is surjective.

Indeed, if $S = \operatorname{Spec}(A)$, $\kappa(s_i) = A/\mathfrak{p}_i$, and $\mathcal{E}$ is defined by the A-module E, then

$$B^u(S) = E \otimes A, \qquad B^u(\operatorname{Spec} \kappa(s_i)) = E \otimes_A (A/\mathfrak{p}_i).$$

The assertion follows at once from the surjectivity of

$$A \longrightarrow \prod_i A/\mathfrak{p}_i.$$

5.10. Subgroups of Type (R) with Reductive Fibers

Proposition 5.10.1. *Let (G, T, M, R) be a split S-group, let R' be a subset of R of type* (R) (5.4.2)*, and let $H_{R'}$ be the corresponding subgroup of G. The following conditions are equivalent:*

(i) *$H_{R'}$ is reductive* (that is, its geometric fibers are reductive)*.*

(ii) *$R' = -R'$, that is, R' is symmetric.*

Moreover, under these conditions, $(H_{R'}, T, M, R')$ is a splitting of $H_{R'}$; for every system R_+ of positive roots of R,

$$R'_+ = R' \cap R_+$$

is a system of positive roots of R', and

$$B_{R_+} \cap H_{R'} = H_{R'_+}.$$

This is a Borel subgroup of $H_{R'}$, whose unipotent part is

$$U_{R_+} \cap H_{R'} = U_{R'_+}.$$

Clearly (i) implies (ii): it is enough to check this fiber by fiber, and R' is the root system of $H_{R'}$ relative to T. To prove that (ii) implies (i), observe, by 5.4.3, that

$$H_{R'} \cap Z_\alpha = \underline{\operatorname{Centr}}_{H_{R'}}(T_\alpha) = Z_\alpha$$

for every $\alpha \in R'$, and apply the criterion of Exposé XIX, 1.12.

If R_+ is a system of positive roots of R, then $R'_+ = R_+ \cap R'$ is plainly a closed subset of R' such that

$$R'_+ \cup (-R'_+) = R', \qquad R'_+ \cap (-R'_+) = \varnothing;$$

it is therefore a system of positive roots of R'. The other two assertions follow respectively from 5.6.1(vi) and 5.6.7(i). □

Corollary 5.10.2. *Let S be a scheme, G a reductive S-group, and H a reductive S-subgroup such that, for every $s \in S$, $G_{\bar{s}}$ and $H_{\bar{s}}$ have the same reductive rank. Then H is closed in G, $\underline{\mathrm{Norm}}_G(H)$ is smooth over S, and*

$$\underline{\mathrm{Norm}}_G(H)/H$$

is representable by a finite étale S-scheme.

If T is a maximal torus of H, and B a Borel subgroup of G containing T, then $B \cap H$ is a Borel subgroup of H, whose unipotent part is

$$(B \cap H)^u = B^u \cap H.$$

The first assertions follow at once from 5.3.10 and 5.3.18, using the fact that the Weyl groups of G and H are finite (Exposé XIX, 2.5). The other assertions are local for the étale topology and reduce to the case studied in 5.10.1. □

Proposition 5.10.3. *Let S be a scheme and G a reductive S-group.*

(a) *If Q is a torus of G, then $\underline{\mathrm{Centr}}_G(Q)$ is a subgroup of G of type* (R) *with reductive fibers. If $Q \subset Q'$ are two tori of G, then*

$$\underline{\mathrm{Centr}}_G(Q) \supset \underline{\mathrm{Centr}}_G(Q').$$

(b) *If H is a subgroup of G of type* (R) *with reductive fibers, then* $\mathrm{rad}(H)$ (4.3.6) *is a torus of G. If $H \subset H'$ are two subgroups of G of type* (R) *with reductive fibers, then*

$$\mathrm{rad}(H) \supset \mathrm{rad}(H').$$

(c) *If Q is a torus of G, then*

$$\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q)) \supset Q \quad \textit{and} \quad \underline{\mathrm{Centr}}_G(\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q))) = \underline{\mathrm{Centr}}_G(Q).$$

(d) *If H is a subgroup of G of type* (R) *with reductive fibers, then*

$$\underline{\mathrm{Centr}}_G(\mathrm{rad}(H)) \supset H \quad \textit{and} \quad \mathrm{rad}(\underline{\mathrm{Centr}}_G(\mathrm{rad}(H))) = \mathrm{rad}(H).$$

Indeed, (a) follows at once from Exposé XIX, 2.8. To prove (b), it is enough to observe that $\mathrm{rad}(H') \subset H$, since H contains, fpqc-locally, a maximal torus of G, hence of H'. The first assertion of (c) (respectively, of (d)) is trivial, and the second follows by the usual argument. □

This proposition leads to the following definition:

Definition 5.10.4. Let S be a scheme, G a reductive S-group, H a reductive subgroup of G of type (R), and Q a subtorus of G.[58]

1) H is called a *critical subgroup* if it is the centralizer of its radical.

2) Q is called a *C-critical torus* if it is the radical of its centralizer.

Proposition 5.10.3 then gives:

[58] Editor's note: We have modified the original by introducing the term "C-critical torus" in place of "critical torus", in order to avoid confusion in later references (cf. Exposé XXVI, 3.9). We have also expanded the statement of 5.10.5 and added Remark 5.10.5.1.

Corollary 5.10.5.

(i) For every subtorus Q of G, $\underline{\mathrm{Centr}}_G(Q)$ is critical.

(ii) For every subgroup H of G of type (R) with reductive fibers, $\mathrm{rad}(H)$ is a C-critical torus of G.

(iii) The maps
$$Q \longmapsto \underline{\mathrm{Centr}}_G(Q), \qquad H \longmapsto \mathrm{rad}(H)$$
are mutually inverse bijections between the set of C-critical tori of G and the set of its critical reductive subgroups of type (R).

(iv) If Q is a torus of G, then
$$\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q))$$
is the smallest C-critical torus of G containing Q.

(v) If H is a reductive subgroup of G of type (R), then
$$\underline{\mathrm{Centr}}_G(\mathrm{rad}(H))$$
is the smallest critical reductive subgroup of G of type (R) containing H.

Remark 5.10.5.1. (58)

1) A torus T of G is a critical subgroup of G if and only if it is a maximal torus.

2) Henceforth, "critical torus" means "C-critical torus".

Proposition 5.10.6. *Let (G, T, M, R) be a split S-group, and let R' be a subset of R. The following conditions are equivalent:*

(i) *R' is of type* (R)*, and $H_{R'}$ is reductive and critical.*

(ii) *There exist a system Δ of simple roots of R and a subset Δ' of Δ such that R' is the set of those elements of R that are linear combinations of elements of Δ'.*

(iii) *R' is closed and symmetric, and every system of simple roots of R' is the intersection with R' of a system of simple roots of R.*

Indeed, by Exposé XXI, 3.4.8, conditions (ii) and (iii) are equivalent; they are also equivalent to the condition that R' be the intersection of R with a $\mathbf{Q}$-vector subspace of $M \otimes \mathbf{Q}$. This last condition follows from (i): if
$$H_{R'} = \underline{\mathrm{Centr}}_G(Q),$$
then R' is the set of elements of R that vanish on Q (Exposé II, 5.2.3(ii)). Conversely, this condition implies (i), because $\mathrm{rad}(H_{R'})$ is the maximal torus of
$$\bigcap_{\alpha \in R'} \mathrm{Ker}(\alpha).$$
Consequently,
$$\underline{\mathrm{Centr}}_G(\mathrm{rad}(H_{R'})) = H_{R''},$$
where R'' is the intersection of R with the vector subspace spanned by R'; hence $R'' = R'$. □

5.10.7. Let us summarize some of the preceding results. Let (G, T, M, R) be a split S-group, let Δ be a system of simple roots of R, and let R_+ be the corresponding system of positive roots. Choose a subset Δ' of Δ, denote by R' the set of those elements of R which are linear combinations of elements of Δ', and put

$$R'_+ = R' \cap R_+.$$

Let $T_{\Delta'}$ be the maximal torus of

$$\bigcap_{\alpha \in \Delta'} \mathrm{Ker}(\alpha)$$

and put

$$Z_{\Delta'} = \underline{\mathrm{Centr}}_G(T_{\Delta'}).$$

Then $Z_{\Delta'}$ is a reductive subgroup of G, with radical $T_{\Delta'}$; $(Z_{\Delta'}, T, M, R')$ is a split S-group; $B_{R_+} \cap Z_{\Delta'}$ is the Borel subgroup of $Z_{\Delta'}$ defined by the system R'_+ of positive roots (or, equivalently, by the system Δ' of simple roots), and its unipotent part is

$$U_{R_+} \cap Z_{\Delta'} = U_{R'_+}.$$

Remark 5.10.8. Under the conditions of 5.10.4, let Q be a critical torus of G, and let

$$L = \underline{\mathrm{Centr}}_G(Q)$$

be its centralizer. Since $Q = \mathrm{rad}(L)$, Q is a characteristic subgroup of L; it follows at once that

$$\underline{\mathrm{Norm}}_G(L) = \underline{\mathrm{Norm}}_G(Q),$$

and hence also that

$$\underline{\mathrm{Norm}}_G(L)/L = \underline{\mathrm{Norm}}_G(Q)/\underline{\mathrm{Centr}}_G(Q) = W_G(Q).$$

By 5.10.2, we deduce:

Proposition 5.10.9. *Let S be a scheme, let G be a reductive S-group, and let Q be a critical torus of G. The Weyl group $W_G(Q)$ is finite* (étale) *over S.*

Remark 5.10.10. Under the conditions of 5.10.7, one may describe explicitly

$$W_G(T_{\Delta'}) = \underline{\mathrm{Norm}}_G(Z_{\Delta'})/Z_{\Delta'}.$$

This is the constant group associated with the quotient W_1/W_2, where W_1 is the subgroup of W consisting of those elements which normalize the subgroup of M generated by Δ', and W_2 is the subgroup of W generated by the s_α, for $\alpha \in \Delta'$.

5.11. Subgroups of Type (RC)

Definition 5.11.1. Let S be a scheme and let G be a reductive S-group. A group subscheme H of G is said to be *of type* (RC) if it is of type (R), that is, if it satisfies the following two conditions from 5.2.1:

(i) H is smooth over S, with connected fibers;

(ii) for every $s \in S$, $H_{\bar{s}}$ contains a maximal torus of $G_{\bar{s}}$;

and if it also satisfies the following condition:

(iii) for every $s \in S$ and every maximal torus T of $H_{\bar{s}}$, the set of roots of $H_{\bar{s}}$ relative to T is a closed subset of the set of all roots of $G_{\bar{s}}$ relative to T.

Remark 5.11.2. As we already observed in 5.4.8, condition (iii) follows from the other two when 6 is invertible on S.[59]

Lemma 5.11.3. *Let (G, T, M, R) be a split S-group, and let R' be a closed subset of R. Put*

$$R_1 = \{\alpha \in R' \mid -\alpha \in R'\}, \qquad R_2 = \{\alpha \in R' \mid -\alpha \notin R'\}.$$

Then R_1 and R_2 are closed. Consider the groups $H_{R'}$, H_{R_1}, and U_{R_2} (5.4.7 and 5.6.5), which are smooth and have connected fibers.

(i) *U_{R_2} is normal in $H_{R'}$, and $H_{R'}$ is the semidirect product of U_{R_2} by H_{R_1}.*

(ii) *The group H_{R_1} is reductive, and U_{R_2} has connected unipotent geometric fibers. Every normal subgroup scheme of $H_{R'}$ that is smooth over S and has connected unipotent geometric fibers is contained in U_{R_2}. Every reductive subgroup of $H_{R'}$ that contains T is contained in H_{R_1}.*

(iii)
$$U_{R_2} \cap \underline{\mathrm{Norm}}_G(H_{R_1}) = e.$$

Let us first prove (iii), which follows from 5.6.7(i). The first assertion of (i) follows from 5.6.7(ii). Since

$$U_{R_2} \cap H_{R_1} = e$$

by (iii), the semidirect product $H_{R_1} \cdot U_{R_2}$ is a subgroup of $H_{R'}$. But these are two subgroups of G of type (R), containing T, and having the same Lie algebra $\mathfrak{g}_{R'}$; they coincide by 5.3.5, which proves (i).

For (ii), the first two assertions are merely 5.10.1 and 5.6.5. Let U be a subgroup scheme of $H_{R'}$, smooth and of finite presentation over S, normal in $H_{R'}$ (and hence normalized by T), and with connected unipotent geometric fibers. By 5.6.12, locally one has $U = U_{R''}$, where $R'' \subset R'$ and

$$R'' \cap (-R'') = \varnothing.$$

If U were not contained in U_{R_2}, then R'' would not be contained in R_2; there would therefore exist $\alpha \in R''$ such that $-\alpha \in R'$. Then $Z_\alpha \subset H_{R'}$ by 5.4.3, so Z_α normalizes U. But U contains U_α, and Z_α has a section w such that

$$\mathrm{int}(w)U_\alpha = U_{-\alpha};$$

thus $-\alpha \in R''$, a contradiction.

Finally, if L is a reductive subgroup of $H_{R'}$ containing T, locally one has $L = H_{R'''}$, where R''' is symmetric and contained in R'; hence $R''' \subset R_1$. □

Proposition 5.11.4. *Let S be a scheme, let G be a reductive S-group, and let H be a subgroup scheme of G of type* (RC).

(i) *H is closed in G, and $\underline{\mathrm{Norm}}_G(H)/H$ is representable by a finite étale S-group scheme.*

(ii) *H has a largest normal subgroup scheme that is smooth and of finite presentation over S, with connected unipotent geometric fibers. It is called the* unipotent radical *of H and is denoted $\mathrm{rad}^u(H)$. The quotient sheaf*

$$H/\mathrm{rad}^u(H)$$

is representable by a reductive S-group.

[59] Editor's note: That is, when every residue characteristic of S is greater than 3.

(iii) *If T is a maximal torus of H, then H has a reductive subgroup L, containing T, of type* (RC)*, with the following properties:*

(a) *every reductive subgroup of H containing T is contained in L;*

(b) *H is the semidirect product*

$$H = L \cdot \mathrm{rad}^u(H);$$

in other words, the canonical morphism

$$L \longrightarrow H/\mathrm{rad}^u(H)$$

is an isomorphism.

Moreover, L is the unique reductive subgroup of H that contains T and satisfies either one of the two preceding conditions. Finally,

$$\underline{\mathrm{Norm}}_H(L) = L, \qquad \underline{\mathrm{Norm}}_H(T) = \underline{\mathrm{Norm}}_L(T), \qquad W_H(T) = W_L(T);$$

in particular, $W_H(T)$ is finite over S.

Proof. First observe that (i) is local for the étale topology. Thus, by Corollary 5.3.18, (i) follows from the last assertion of (iii).

The assertions of (ii) are local for the étale topology. We may therefore suppose ourselves in the situation of 5.11.3, where they follow immediately from (i) and (ii).

By virtue of the uniqueness assertions contained there, (iii) is likewise local for the étale topology, and we may again reduce to the situation of 5.11.3, where properties (a) and (b) have been verified. The uniqueness of an L satisfying (a) is immediate; in view of (a), the uniqueness of an L satisfying (b) is clear. The equality

$$\underline{\mathrm{Norm}}_H(L) = L$$

is precisely 5.11.3(iii). If a section of H normalizes T, then it normalizes L, by the uniqueness of L, and hence is a section of L by what we have just proved. This proves the second equality; the third is then immediate.

Proposition 5.11.5. *Let S be a scheme, let G be a reductive S-group, and let $\mathcal{H}_c$ be the functor of subgroups of G of type* (RC)*, which is a subfunctor of the functor $\mathcal{H}$ of 5.8.1.*

(i) *$\mathcal{H}_c$ is representable by an open subscheme of $\mathcal{H}$, smooth, quasi-projective, and of finite presentation over S.*

(ii) *There exist a finite étale S-scheme $\mathcal{C}\ell_c$ and a morphism*

$$\mathrm{cl}\colon \mathcal{H}_c \longrightarrow \mathcal{C}\ell_c$$

that is smooth, quasi-projective, of finite presentation, surjective, and has connected geometric fibers, with the following property:

For every $S' \to S$ and all $H, H' \in \mathcal{H}_c(S')$, one has $\mathrm{cl}(H) = \mathrm{cl}(H')$ if and only if H and H' are conjugate in G locally for the étale topology (or, equivalently by 5.3.11, if for every $s \in S$, $H_{\bar{s}}$ and $H'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$).

(iii) *$\mathcal{C}\ell_c$ and* cl *are determined* (up to a unique isomorphism) *by the preceding conditions.*

(iv) *If (G, T, M, R) is a splitting of G, let E be the set of conjugacy classes modulo W of closed subsets of R. Then there is an isomorphism*

$$\mathcal{C}\ell_c \xrightarrow{\sim} E_S$$

such that, for every closed subset R' of R, $\mathrm{cl}(H_{R'})$ corresponds to the canonical image of R' in

$$E_S(S) = \mathrm{Hom}_{\mathrm{loc.const.}}(S, E).$$

It is first of all clear that $\mathcal{H}_c$ is a sheaf for the étale topology, and that (ii) implies that $\mathcal{C}\ell_c$ is simply the quotient sheaf of $\mathcal{H}_c$ by the equivalence relation defined by conjugation.

This first implies (iii), as well as the fact that it suffices to verify (i) and (ii) locally for the étale topology. We are therefore reduced to the situation of (iv); let us first construct a morphism

$$f\colon \mathcal{H}_c \longrightarrow E_S.$$

It suffices to construct a map $\mathcal{H}_c(S) \to E_S(S)$, functorial in S; let H be a subgroup of G of type (RC). Since H locally for the étale topology has maximal tori, and since the maximal tori of G are locally conjugate for the étale topology, there exist a covering family $(S_i \to S)$, elements $g_i \in G(S_i)$, and closed subsets R_i of R such that

$$\mathrm{int}(g_i)(H \times_S S_i) = H_{R_i} \times_S S_i.$$

Each R_i defines a section η_i of E_{S_i}, that is, an element of $E_S(S_i)$. It remains to prove that the family (η_i) comes from a section $\eta = f(H)$ of E_S over S, and that it depends only on H.

For this it is enough to prove that $H_{R'}$ and $H_{R''}$ are locally conjugate for the étale topology if and only if R' and R'' are conjugate by an element of the Weyl group W, which is clear.

For every $\eta \in E$, there exists $H_0 \in \mathcal{H}_c(S)$ such that $f(H_0) = \eta$: take $H_0 = H_{R'}$, where R' is a closed subset whose image is η. If $H \in \mathcal{H}_c(S')$, for $S' \to S$, then H is locally conjugate to H_0 if and only if $f(H) = \eta$. Consequently,

$$f^{-1}(\eta) \simeq G/\underline{\mathrm{Norm}}_G(H_0),$$

which, by 5.8.2, is an open subscheme of $\mathcal{H}$, smooth, quasi-projective, and of finite presentation over S, with nonempty connected fibers. Since E_S is the disjoint union of the open subschemes that are the images of the sections corresponding to the elements $\eta \in E$, $\mathcal{H}_c$ identifies with the disjoint union of the $f^{-1}(\eta)$. This proves (i) and (ii); assertion (iv) follows from the construction.

Corollary 5.11.6. *Let $u \in \mathcal{C}\ell_c(S')$, where $S' \to S$. Then $\mathrm{cl}^{-1}(u)$ is a smooth, quasi-projective S'-scheme of finite presentation, with nonempty connected fibers. It is an open subscheme of $\mathcal{H}_c$ and a "homogeneous" scheme under $G_{S'}$* (acting by inner automorphisms).

In particular, if $H \in \mathrm{cl}^{-1}(u)(S')$, the morphism

$$G_{S'} \longrightarrow (\mathcal{H}_c)_{S'}, \qquad g \longmapsto \mathrm{int}(g)H$$

identifies

$$G_{S'}/\underline{\mathrm{Norm}}_{G_{S'}}(H) \quad \textit{with} \quad \mathrm{cl}^{-1}(u).$$

Examples 5.11.7. There are two canonical sections u_t and u_b of $\mathcal{C}\ell_c$. The section u_t corresponds to maximal tori, for which $R' = \varnothing$. The section u_b corresponds to Borel subgroups, for which R' is a system of positive roots. The schemes $\mathrm{cl}^{-1}(u_t)$ and $\mathrm{cl}^{-1}(u_b)$ are respectively $\underline{\mathrm{Tor}}(G)$ and $\underline{\mathrm{Bor}}(G)$, introduced in 5.8.3. Other examples will be given in Exposé XXVI.

Remark 5.11.8. One can construct an S-scheme $\mathcal{C}\ell$, of finite presentation and unramified over S, and a smooth surjective morphism

$$\mathcal{H} \longrightarrow \mathcal{C}\ell$$

with connected geometric fibers, having properties analogous to those of 5.11.5(ii) and (iii).

6. The Derived Group

6.1. Preliminaries

In this section, we fix a scheme S, a split S-group (G, T, M, R), and a system R_+ of positive roots of R. We write

$$B = B_{R_+}, \qquad B^- = B_{R_-}, \qquad U = B^u, \qquad U^- = (B^-)^u,$$

and

$$\Omega = \Omega_{R_+} = U^- \cdot T \cdot U.$$

6.1.1. Let T' denote the subtorus of T that is the "image of the family α^*, $\alpha \in R$"; in other words, T' is the image of the homomorphism

$$(\mathbb{G}_{m,S})^R \longrightarrow T$$

defined by

$$(z_\alpha)_{\alpha\in R} \longmapsto \prod_{\alpha\in R} \alpha^*(z_\alpha).$$

It follows at once that, if Δ denotes the set of simple roots of R_+, the homomorphism

$$(\mathbb{G}_{m,S})^\Delta \longrightarrow T'$$

defined in the same way is surjective and has finite kernel. If we identify T with $D_S(M)$, then T' identifies with $D_S(M/N)$, where

$$N = M \cap V(R^*)^\perp$$

(we write $V(R^*)^\perp$ for the orthogonal complement of $V(R^*)$ under the duality between V and V^*).

Lemma 6.1.2. *The homomorphism defined by multiplication in T,*

$$\mathrm{rad}(G) \times_S T' \longrightarrow T,$$

is an isogeny (cf. 4.2.9).

Indeed, the canonical homomorphism

$$\mathrm{rad}(T) \longrightarrow T/T'$$

comes by duality from the homomorphism of commutative groups

$$M \cap V(R^*)^\perp \longrightarrow M/(M \cap V(R)),$$

which is immediately seen to be injective with finite cokernel (cf. Exposé XXI, 6.3).

Definition 6.1.3. We put

$$\Omega' = U^- \cdot T' \cdot U;$$

this is a closed subscheme of $\Omega = U^- \cdot T \cdot U$.

Lemma 6.1.4. *Let α be a simple root and let $w_\alpha \in \underline{\mathrm{Norm}}_G(T)(S)$ lift s_α. Then*

$$\mathrm{int}(w_\alpha)\Omega' \cap \Omega \subset \Omega'.$$

It is enough to prove that, if $g \in \Omega'(S)$ and $\mathrm{int}(w_\alpha)g \in \Omega(S)$, then $\mathrm{int}(w_\alpha)g \in \Omega'(S)$. By 5.6.8, write

$$g = a \exp_{-\alpha}(Y) t \exp_\alpha(X) b,$$

where

$$a \in U_{-\widehat{\alpha}}(S), \qquad Y \in \Gamma(S, \mathfrak{g}^{-\alpha}), \qquad t \in T'(S), \qquad X \in \Gamma(S, \mathfrak{g}^{\alpha}), \qquad b \in U_{\widehat{\alpha}}(S).$$

We then have

$$\mathrm{int}(w_\alpha)g = \mathrm{int}(w_\alpha)a \cdot \mathrm{int}(w_\alpha)\big(\exp_{-\alpha}(Y) t \exp_\alpha(X)\big) \cdot \mathrm{int}(w_\alpha)b.$$

By 5.6.8(iv),

$$\mathrm{int}(w_\alpha)a \in U_{-\widehat{\alpha}}(S), \qquad \mathrm{int}(w_\alpha)b \in U_{\widehat{\alpha}}(S).$$

It follows that, on putting

$$h = \exp_{-\alpha}(Y) t \exp_\alpha(X),$$

we have the equivalences

$$\mathrm{int}(w_\alpha)g \in \Omega(S) \quad \Longleftrightarrow \quad \mathrm{int}(w_\alpha)h \in \Omega(S),$$

and

$$\mathrm{int}(w_\alpha)g \in \Omega'(S) \quad \Longleftrightarrow \quad \mathrm{int}(w_\alpha)h \in \Omega'(S).$$

We are therefore reduced to the case $g = h$. Since, by (4.1.12),

$$Z_\alpha \cap \Omega = U_{-\alpha} \cdot T \cdot U_\alpha, \qquad Z_\alpha \cap \Omega' = U_{-\alpha} \cdot T' \cdot U_\alpha,$$

we are reduced to proving the following assertion:

$$\mathrm{int}(w_\alpha)h \in \big(U_{-\alpha} \cdot T \cdot U_\alpha\big)(S) \quad \Longrightarrow \quad \mathrm{int}(w_\alpha)h \in \big(U_{-\alpha} \cdot T' \cdot U_\alpha\big)(S).$$

But this follows at once from Exposé XX, 3.12, which shows that the T-component of $\mathrm{int}(w_\alpha)h$ is of the form

$$t \cdot \alpha^*(z) \in T'(S).$$

Lemma 6.1.5. *For every $w \in \underline{\mathrm{Norm}}_G(T)(S)$, there exists an open subscheme V_w of G, containing the unit section, such that*

$$\mathrm{int}(w)\Omega' \cap V_w \subset \Omega'.$$

For each simple root α, choose an element $n_\alpha \in \underline{\mathrm{Norm}}_G(T)(S)$ lifting s_α. For every point $s \in S$, there exist an open subscheme V of S containing s, an element $t \in T(V)$, and, over V, a relation

$$w = n_{\alpha_1} \cdots n_{\alpha_p} t,$$

where the α_i are simple. We may plainly restrict ourselves to proving the result when $V = S$; we argue by induction on p. If $p = 0$, then $w \in T(S)$, and we take $V_w = G$. Suppose, therefore, that $w = n_\alpha w'$, where w' satisfies the conclusion of the lemma. There is then an open subscheme $V_{w'}$ of G, containing the unit section, such that

$$\mathrm{int}(w')\Omega' \cap V_{w'} \subset \Omega'.$$

We may then write

$$\begin{aligned}\mathrm{int}(w)\Omega' \cap \mathrm{int}(n_\alpha)V_{w'} \cap \Omega &= \mathrm{int}(n_\alpha)\big(\mathrm{int}(w')\Omega' \cap V_{w'}\big) \cap \Omega \\ &\subset \mathrm{int}(n_\alpha)\Omega' \cap \Omega \subset \Omega',\end{aligned}$$

by 6.1.4. We now take

$$V_w = \mathrm{int}(n_\alpha)V_{w'} \cap \Omega,$$

and the proof is complete.

Lemma 6.1.6. *There exists an open subscheme V_0 of G, containing the unit section, such that for every $S' \to S$,*

$$U(S')U^-(S') \cap V_0(S') \subset \Omega'(S').$$

Let n_0 be an element of $\underline{\mathrm{Norm}}_G(T)(S)$ lifting the symmetry w_0 of the Weyl group,[60] that is, such that

$$\mathrm{int}(n_0)U = U^- \qquad \text{(cf. Exposé XXI, 3.6.14).}$$

Then $n_0^2 \in T(S)$. We show that the open subscheme $V_0 = V_{n_0}$ of 6.1.5 has the required property. Indeed,

$$\begin{aligned}U(S')U^-(S') &= \mathrm{int}(n_0)\big(\mathrm{int}(n_0)^{-1}U(S') \cdot \mathrm{int}(n_0)^{-1}U^-(S')\big) \\ &= \mathrm{int}(n_0)\big(U^-(S') \cdot U(S')\big) \subset \mathrm{int}(n_0)\Omega'(S').\end{aligned}$$

Consequently,

$$U(S')U^-(S') \cap V_0(S') \subset \mathrm{int}(n_0)\Omega'(S') \cap V_0(S') \subset \Omega'(S').$$

[60] Editor's note: The symmetry w_0 is defined in Exposé XXI, 3.6.14.

Lemma 6.1.7. *Consider the morphism*

$$f : \Omega = U^- \cdot T \cdot U \longrightarrow T/T'$$

obtained by composing the second projection with the canonical morphism from T to T/T'. Then f is "generically multiplicative": there exists an open subscheme V of $\Omega \times_S \Omega$, containing the unit section (and hence relatively schematically dense, Exposé XVIII, 1.3)*, such that for every $S' \to S$ and every $(x, y) \in V(S')$, one has $xy \in \Omega(S')$ and*

$$f(xy) = f(x)f(y).$$

Let x, y be sections of Ω over S'. Write

$$x = utv, \qquad y = u't'v',$$

where

$$u, u' \in U^-(S'), \qquad t, t' \in T(S'), \qquad v, v' \in U(S').$$

Let V_0 be the open subscheme of 6.1.6, and let V be the open subscheme of $\Omega \times_S \Omega$ defined by the condition

$$vu' \in V_0(S').$$

This open subscheme has the required property. Indeed, if $(x, y) \in V(S')$, then

$$xy = (utv)(u't'v') = (ut)(vu')(t'v').$$

Since $vu' \in \Omega'(S')$, we obtain

$$xy \in U^-(S')\, t\Omega'(S')t'\, U(S') \subset U^-(S')\, tt'T'(S')\, U(S'),$$

and therefore $xy \in \Omega(S')$ and

$$f(xy) = f(tt') = f(t)f(t') = f(x)f(y).$$

Proposition 6.1.8. *There exists a group morphism*

$$f : G \longrightarrow T/T'$$

inducing the canonical projection on T. The kernel $\mathrm{Ker}(f)$ *of f is a closed subgroup scheme of G, smooth over S, with connected fibers. Every group morphism from G to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on* $\mathrm{Ker}(f)$.

The first assertion follows at once from 4.1.11. We immediately have

$$\mathrm{Ker}(f) \cap \Omega = \Omega',$$

which proves that $\mathrm{Ker}(f)$ is smooth over S at every point of the unit section.[61] By 5.6.9(ii), every morphism ϕ from G to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on U and on U^-. By Exposé XX, 2.7, ϕ also vanishes on T', and hence on Ω'. Using the notation of 5.7.10, we see that the monoid U_1 is contained in $\mathrm{Ker}(f)(S)$, which shows that

$$\mathrm{Ker}(f) = \bigcup_{u \in U_1} u\Omega'.$$

It follows, on the one hand, that every ϕ as above vanishes on $\mathrm{Ker}(f)$, and, on the other hand, that $\mathrm{Ker}(f)$ has connected fibers and is therefore smooth over S, by Exposé $\mathrm{VI_B}$, 3.10.

[61] Editor's note: We have added "at every point of the unit section," as well as the reference to Exposé $\mathrm{VI_B}$, 3.10, at the end of the proof. Moreover, in the following sentence "prescheme" has been replaced by "presheaf."

6.2. The Derived Group of a Reductive Group

Theorem 6.2.1. *Let S be a scheme and G a reductive S-group.*

(i) *The group*

$$\mathrm{D}_S(G) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, \mathbb{G}_{m,S})$$

is represented by a twisted constant S-group whose type at $s \in S$ is

$$\mathbb{Z}^{\mathrm{rgred}(G_s)-\mathrm{rgss}(G_s)}.$$

(ii) *Write*

$$\mathrm{corad}(G) = \mathrm{D}_S\big(\mathrm{D}_S(G)\big),$$

which is therefore an S-torus. The biduality morphism (cf. Exposé VIII, §1)

$$f_0 : G \longrightarrow \mathrm{corad}(G)$$

is smooth and surjective.

(iii) *The composite morphism*

$$\mathrm{rad}(G) \longrightarrow G \longrightarrow \mathrm{corad}(G)$$

is an isogeny (cf. 4.2.9).

(iv) *The kernel of f_0, denoted*

$$\mathrm{der}(G) = \mathrm{Ker}(f_0),$$

is a closed subgroup scheme of G, semisimple over S, called the derived group of G. If G is semisimple, then $\mathrm{der}(G) = G$.

(v) *Every group morphism from G to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on* $\mathrm{der}(G)$*, and therefore factors through f_0.*

Proof. All the assertions of the theorem are local for the étale topology; we may therefore reduce to the case in which G is split over S. Consider the morphism f of 6.1.8. By the last assertion of 6.1.8, we immediately have an isomorphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, \mathbb{G}_{m,S}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(T/T', \mathbb{G}_{m,S}),$$

which proves (i), then (ii), and yields a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{f_0} & \mathrm{corad}(G) \\ & \searrow^{f} & \downarrow \sim \\ & & T/T'. \end{array}$$

Assertion (v) now follows from 6.1.8, and (iii) from 6.1.2. We also have $\mathrm{Ker}(f) = \mathrm{Ker}(f_0)$, which, by 6.1.8, implies that $\mathrm{der}(G)$ is smooth over S with connected fibers. It remains to verify that its fibers are semisimple. They are reductive by Exposé XIX, 1.7, as invariant subgroups of reductive groups. By (iii), $\mathrm{rad}(G) \cap \mathrm{der}(G)$ is finite, which indeed implies that the fibers of $\mathrm{der}(G)$ are semisimple.

Remark 6.2.2. a) By construction, when G is split, $\mathrm{der}(G)$ is the fppf subsheaf of G generated by the U_α, $\alpha \in R$. It even suffices to take the U_α, $\alpha \in \Delta \cup -\Delta$, where Δ is a base of R.

b)[62] Let C be the presheaf of commutators of G, that is, the S-functor in groups that assigns to every $S' \to S$ the commutator subgroup of $G(S')$ (that is, the subgroup of $G(S')$ generated by the elements $xyx^{-1}y^{-1}$, for $x, y \in G(S')$), and let $\widetilde{C}$ be the associated fppf sheaf. Since the quotient $G/\mathrm{der}(G) = T/T'$ is commutative, $\mathrm{der}(G)$ contains C, and hence $\widetilde{C}$ (cf. Exposé IV, 4.3.12).

On the other hand, the quotient presheaf $G/\widetilde{C}$ is separated (Exposé IV, 4.4.8.1), and therefore, by (v), we have $\mathrm{der}(G) \subset \widetilde{C}$. Thus $\mathrm{der}(G) = \widetilde{C}$; in other words, $\mathrm{der}(G)$ is the fppf sheaf of commutators of G.

Let us finally note that, since C is a subpresheaf of G, it is separated, but it is not equal to $\mathrm{der}(G)$ in general: for example, $\mathrm{der}(\mathrm{SL}_2) = \mathrm{SL}_2$, whereas $\mathrm{SL}_2(\mathbb{F}_2) \simeq \mathfrak{S}_3$ is not equal to its commutator subgroup.

c) When S is the spectrum of an algebraically closed field k, $\mathrm{der}(G)(k)$ is the commutator subgroup of $G(k)$ (Exposé $\mathrm{VI_B}$, 7.10).

6.2.3. Consider now the two exact sequences

$$1 \longrightarrow \mathrm{rad}(G) \longrightarrow G \longrightarrow \mathrm{ss}(G) \longrightarrow 1,$$
$$1 \longrightarrow \mathrm{der}(G) \longrightarrow G \longrightarrow \mathrm{corad}(G) \longrightarrow 1.$$

Since $\mathrm{rad}(G)$ is central in G, multiplication in G defines a group morphism

$$u : \mathrm{rad}(G) \times_S \mathrm{der}(G) \longrightarrow G,$$

which is covering by 6.2.1(iii), and hence surjective and flat (Exposé $\mathrm{VI_B}$, 9.2(xi)). [63] Its kernel is isomorphic to $\mathrm{rad}(G) \cap \mathrm{der}(G)$, which is also the kernel of

$$\mathrm{rad}(G) \longrightarrow \mathrm{corad}(G);$$

it is a finite subgroup of multiplicative type of $\mathrm{rad}(G)$. Similarly, the kernel of the morphism

$$G \longrightarrow \mathrm{corad}(G) \times_S \mathrm{ss}(G)$$

is isomorphic to $\mathrm{der}(G) \cap \mathrm{rad}(G)$.

Proposition 6.2.4. *Let G be a reductive S-group. The three morphisms*

$$\mathrm{rad}(G) \times_S \mathrm{der}(G) \longrightarrow G,$$
$$G \longrightarrow \mathrm{corad}(G) \times_S \mathrm{ss}(G),$$
$$\mathrm{rad}(G) \longrightarrow \mathrm{corad}(G)$$

are central isogenies, and their kernels are isomorphic.

[62] Editor's note: In what follows, we have expanded the original and removed the assertion that "$\mathrm{der}(G)$ is the fppf-separated presheaf of commutators of G."

[63] Editor's note: Indeed, G is the fppf quotient of $\mathrm{rad}(G) \times_S \mathrm{der}(G)$ by $\mathrm{Ker}(u)$, which is a group of multiplicative type and hence is flat over S. Therefore, by Exposé $\mathrm{VI_B}$, 9.2(xi), u is flat.

Corollary 6.2.5. *The following conditions are equivalent:*

(i) *G is the product of a semisimple group and a torus.*

(ii) *The morphism*
$$\mathrm{rad}(G) \times_S \mathrm{der}(G) \xrightarrow{\sim} G$$
is an isomorphism.

(iii) *The morphism*
$$G \xrightarrow{\sim} \mathrm{corad}(G) \times_S \mathrm{ss}(G)$$
is an isomorphism.

(iv) $\mathrm{rad}(G) \cap \mathrm{der}(G) = e$.

6.2.6. Let us return temporarily to the case of a split group. We retain the notation of 6.1. Put
$$N = M \cap V(R^*)^\perp, \qquad \text{so that} \qquad T' = \mathrm{D}_S(M/N).$$
Then $U^- \cdot T' \cdot U$ is an open neighborhood of the unit section in $\mathrm{der}(G)$. Consequently,
$$\mathrm{Lie}\big(\mathrm{der}(G)/S\big) = \mathfrak{t}' \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha.$$
Since the characters induced on T' by the $\alpha \in R$ are nonzero and distinct (Exposé XXI, 1.2.5—which we already used in 6.1.2), R is a root system of $\mathrm{der}(G)$ relative to T'. It is then immediate (since $U_\alpha \subset \mathrm{der}(G)$) that the exp-morphisms of $\mathrm{der}(G)$ "are" those of G, and likewise for the coroots.

It follows that:

Proposition 6.2.7. *With the preceding notation,*
$$\big(\mathrm{der}(G), T', M/N, R\big)$$
is a split group with root datum $\mathrm{der}\big(\mathcal{R}(G)\big)$. *The canonical morphism* $\mathrm{der}(G) \to G$ *induces, by functoriality, the canonical morphism of root data*
$$\mathcal{R}(G) \longrightarrow \mathrm{der}\big(\mathcal{R}(G)\big)$$
of Exposé XXI, 6.5.

N.B. As an exercise, the reader may construct the diagram of split groups corresponding to the three left-hand columns of the diagram of root data in Exposé XXI, 6.5.7.

Proposition 6.2.8. *Let S be a scheme, G a reductive S-group, and* $\mathrm{der}(G)$ *its derived group.*

(i) *For every maximal torus T of G, $T \cap \mathrm{der}(G)$ is a maximal torus of* $\mathrm{der}(G)$. *For every maximal torus T' of* $\mathrm{der}(G)$,
$$\underline{\mathrm{Centr}}_G(T') = \mathrm{rad}(G) \cdot T'$$
is a maximal torus of G. These two constructions are inverse to one another and establish a bijective correspondence between the maximal tori of G and those of $\mathrm{der}(G)$.

(ii) *For every Borel subgroup B of G, $B \cap \mathrm{der}(G)$ is a Borel subgroup B' of $\mathrm{der}(G)$. We have $B'^u = B^u$. For every Borel subgroup B' of $\mathrm{der}(G)$,*

$$\underline{\mathrm{Norm}}_G(B') = \mathrm{rad}(G) \cdot B'$$

is a Borel subgroup of G. These constructions are inverse to one another and establish a bijective correspondence between the Borel subgroups of G and those of $\mathrm{der}(G)$.

By the local conjugacy theorem for maximal tori and the construction of the derived group, the only assertion that remains to be proved in (i) is the following: if T is a maximal torus of G, then

$$T = \big(T \cap \mathrm{der}(G)\big) \cdot \mathrm{rad}(G) = \underline{\mathrm{Centr}}_G\big(T \cap \mathrm{der}(G)\big).$$

The first equality is immediate (since we may reduce to the split case); the second follows at once, because $\mathrm{rad}(G)$ is central in G, and hence

$$T = \underline{\mathrm{Centr}}_G(T) = \underline{\mathrm{Centr}}_G\big(T \cap \mathrm{der}(G)\big).$$

The argument for (ii) is the same.

6.3. Subgroups with Commutative Quotients

6.3.1. Let G be a reductive S-group. If H is a subsheaf in groups of G, the following conditions are equivalent:

- H contains $\mathrm{der}(G)$;
- H is normal and G/H is commutative.

In this case, the canonical morphism

$$f_0 : G \longrightarrow \mathrm{corad}(G)$$

maps H onto a subsheaf $f_0(H)$ of $\mathrm{corad}(G)$, and we have

$$G/H \simeq \mathrm{corad}(G)/f_0(H), \qquad H/\mathrm{der}(H) \simeq f_0(H),$$
$$\mathrm{der}(G) = \mathrm{der}(H), \qquad H = f_0^{-1}(f_0(H)).$$

Since $\mathrm{der}(G)$ is smooth over S with connected fibers,[64] it follows from Exposé IV, 5.3.1 and 6.3.1, and Exposé IV_B, 9.2, that the assignment

$$H \longmapsto f_0(H)$$

establishes a bijective correspondence between the group subschemes (respectively, the closed group subschemes) of G that contain $\mathrm{der}(G)$, are smooth over S, and have connected fibers, and the group subschemes (respectively, the closed group subschemes) of $\mathrm{corad}(G)$ that are smooth over S and have connected fibers.

Now, if H' is a group subscheme of $\mathrm{corad}(G)$, smooth over S with connected fibers, then H' is of finite presentation over S (Exposé VI_B, 5.5), and its fibers are tori (because those of $\mathrm{corad}(G)$ are). Hence, by Exposé X, 8.2, H' is a subtorus of $\mathrm{corad}(G)$, and is therefore closed in $\mathrm{corad}(G)$ (Exposé IX, 2.6).

Consequently, every subgroup of G that is smooth with connected fibers and contains $\mathrm{der}(G)$ is closed in G.

[64] Editor's note: We have expanded the original in what follows. In particular, we have added the conclusion, implicit in the original, that every subgroup of G that is smooth with connected fibers and contains $\mathrm{der}(G)$ is closed in G.

6.3.2. If H is a closed group subscheme of G, smooth over S, with connected fibers, and normal in G, then H is reductive. If, moreover, $H \supset \mathrm{der}(G)$, then

$$\mathrm{der}(H) = \mathrm{der}(G) \qquad \text{and} \qquad f_0(H) \simeq \mathrm{corad}(H).$$

We have therefore proved the following proposition.

Proposition 6.3.3. *Let G be a reductive S-group. Every group subscheme H of G that is normal in G, has commutative quotient* (that is, contains $\mathrm{der}(G)$)*, is smooth over S, and has connected fibers*[65] *is closed and reductive. We have*

$$\mathrm{der}(H) = \mathrm{der}(G), \qquad f_0(H) \simeq \mathrm{corad}(H),$$

and

$$G/H \simeq \mathrm{corad}(G)/\mathrm{corad}(H), \qquad H = \big(H \cap \mathrm{rad}(G)\big) \cdot \mathrm{der}(G).$$

Moreover, the assignment $H \mapsto f_0(H)$ defines a bijection between the set of subgroups H of G having the preceding properties and the set of subtori of $\mathrm{corad}(G)$.

By another application of Noether's theorem (Exposé IV, 5.3.1 and 6.3.1), we deduce the following.

Proposition 6.3.4. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. For every subgroup H of G as above, $T \cap H$ is a maximal torus of H, and*

$$G/H \simeq T/(T \cap H), \qquad H = (T \cap H) \cdot \mathrm{der}(G).$$

Moreover, the assignment $H \mapsto T \cap H$ is a bijection between the set of subgroups H of G as above and the set of subtori of T containing $T \cap \mathrm{der}(G)$.

[65] Editor's note: We have deleted the hypothesis that H be retrocompact in G, which is automatic: by Exposé $\mathrm{VI_B}$, 5.5, both G and H are separated and quasi-compact over S; hence $H \hookrightarrow G$ is quasi-compact by EGA $\mathrm{IV_1}$, 1.1.2(v).

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Exposé XXIII
Reductive Groups: Uniqueness of Pinned Groups

by M. Demazure

[1] The aim of this Exposé is to prove the uniqueness theorem (Theorem 4.1). It was proved by Chevalley in the case of an algebraically closed field; the reduction-to-rank-two method used here is likewise due to Chevalley (see *Bible*, Exposés 23 and 24). Along the way, we obtain an explicit description of reductive groups by generators and relations (3.5).

1. Pinnings

Definition 1.1.— *Let S be a scheme and let (G, T, M, R) be an S-split group (XXII, 1.13). A* pinning [2] *of this split group is the datum of a system Δ of simple roots of R, together with, for each $\alpha \in \Delta$, a section*

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

In other words, a pinning of the reductive group G over the nonempty scheme S is the datum of:

(i) a maximal torus T;

(ii) an abelian group M and an isomorphism $T \simeq D_S(M)$;

(iii) a system of roots R of G with respect to T;

(iv) a system Δ of simple roots of R;

(v) an element $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ for every $\alpha \in \Delta$, that is, an element

$$u_\alpha = \exp_\alpha(X_\alpha) \in U_\alpha^\times(S) \qquad \text{for every } \alpha \in \Delta,$$

[1] *Editors' note.* Version of 13 October 2024.

[2] *Editors' note.* Demazure tells us that behind this terminology lies the image of the butterfly (supplied to him by Grothendieck): the body is a maximal torus T, the wings are two Borel subgroups opposite with respect to T; one spreads the butterfly by laying out the wings and then fixes elements in the additive groups (pins) to rigidify the situation, that is, to eliminate the automorphisms.

satisfying condition (D1) of Exposé XXII, 1.13 (indeed, condition (D2) of *loc. cit.* is automatic[3]).

Every split group admits a pinning; in particular, every reductive group is locally pinnable for the étale topology.

1.2.— If G is a pinned S-group, that is, an S-split group equipped with a pinning, it is canonically endowed with the system of positive roots R^+ defined by Δ, the corresponding Borel subgroup $B = B_{R^+}$, the opposite Borel subgroup $B^- = B_{R^-}$, the unipotent groups $U = B_u$ and $U^- = (B^-)_u$, the open subscheme $U^- \cdot T \cdot U$, and so forth. Likewise, for each $\alpha \in \Delta$, there is a canonical isomorphism of vector groups

$$p_\alpha : \mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha, \qquad x \longmapsto \exp_\alpha(xX_\alpha) = u_\alpha^x,$$

normalized by T with multiplier α, and whose datum is equivalent to that of X_α (Exposé XXII, 1.1).

By duality one obtains an $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ and an isomorphism

$$p_{-\alpha} : \mathbb{G}_{a,S} \xrightarrow{\sim} U_{-\alpha}$$

which is contragredient to the preceding one (Exposé XXII, 1.3). We set (Exposé XX, 3.1)

$$w_\alpha = w_\alpha(X_\alpha) = p_\alpha(1)p_{-\alpha}(-1)p_\alpha(1) = p_{-\alpha}(-1)p_\alpha(1)p_{-\alpha}(-1).$$

Then (*loc. cit.*, 3.1 and 3.7)

$$w_\alpha^2 = \alpha^*(-1), \qquad \mathrm{int}(w_\alpha)t = s_\alpha(t) = t \cdot \alpha^*\big(\alpha(t)^{-1}\big),$$

$$\mathrm{int}(w_\alpha)p_\alpha(x) = p_{-\alpha}(-x) = p_{-\alpha}(x)^{-1}, \quad \mathrm{Ad}(w_\alpha)X_\alpha = -X_{-\alpha},$$
$$\mathrm{int}(w_\alpha)p_{-\alpha}(x) = p_\alpha(-x) = p_\alpha(x)^{-1}, \quad \mathrm{Ad}(w_\alpha)X_{-\alpha} = -X_\alpha.$$

We shall systematically use the preceding notation in what follows.

Definition 1.3.— *Let S be a scheme, and suppose that*

$$(G, T, M, R, \Delta, (X_\alpha)) \quad \textit{and} \quad (G', T', M', R', \Delta', (X'_{\alpha'}))$$

are two pinned S-groups. The morphism of split groups (Exposé XXII, 4.2.1)

$$f : G \longrightarrow G'$$

is said to be compatible with the pinnings*, or to define a* morphism of pinned groups*, if the associated bijection $d : R \to R'$ (cf. loc. cit.) satisfies $d(\Delta) = \Delta'$ and if, for every $\alpha \in \Delta$,*

$$f\big(\exp_\alpha(X_\alpha)\big) = \exp_{d(\alpha)}\big(X'_{d(\alpha)}\big), \qquad \textit{i.e.} \qquad f(u_\alpha) = u'_{d(\alpha)}.$$

[4]

1.4.— If $q(\alpha)$ denotes the integer of *loc. cit.*, then

$$f\big(p_\alpha(x)\big) = p'_{d(\alpha)}\big(x^{q(\alpha)}\big) \qquad (\alpha \in \Delta),$$

[3] *Editors' note.* It is implied by condition (v), namely the existence of a section $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

[4] *Editors' note.* The bijection $R \xrightarrow{\sim} R'$ is denoted by d (rather than by u) in order to avoid the notation $u'_{u(\alpha)}$.

and hence

$$f(p_{-\alpha}(x)) = p'_{-d(\alpha)}(x^{q(\alpha)}), \qquad f(w_\alpha) = w'_{d(\alpha)}.$$

Recall (Exposé XXII, 4.2) that, for every $\alpha \in R$, every $z \in \mathbb{G}_m(S')$, and every $t \in T(S')$,

$$f(\alpha^*(z)) = d(\alpha)^*(z)^{q(\alpha)} = d(\alpha)^*(z^{q(\alpha)}), \qquad d(\alpha)(f(t)) = \alpha(t)^{q(\alpha)}.$$

1.5.— We call a root datum equipped with a system of simple roots a *pinned root datum*, and a p-morphism of root data (Exposé XXI, 6.8) that sends simple roots to simple roots a *p-morphism of pinned root data*.

If G is a pinned S-group, we denote by $\mathscr{R}(G)$ the corresponding pinned root datum (the root datum of Exposé XXII, 1.14 equipped with Δ). Let p be the integer defined in Exposé XXII, 4.2.2. Then:

Scholium 1.6.— *The correspondence $G \longmapsto \mathscr{R}(G)$ defines a functor from the category of pinned reductive S-groups to the category of pinned root data (whose morphisms are the p-morphisms).*

1.7. The pinned groups Z_{Δ_1}

Let Δ_1 be a subset of the system Δ of simple roots of the pinned group G. Let T_{Δ_1} be the maximal torus of $\bigcap_{\alpha\in\Delta_1} \ker(\alpha)$, and set

$$Z_{\Delta_1} = \mathrm{Centr}_G(T_{\Delta_1}).$$

Put $R' = \mathbb{Z}\Delta_1 \cap R$. It is known (Exposé XXII, 5.10.7) that Z_{Δ_1} is a reductive S-group with radical T_{Δ_1}, that (T, M, R') is a splitting of this group, and that Δ_1 is a system of simple roots. Thus

$$(Z_{\Delta_1}, T, M, R', \Delta_1, (X_\alpha)_{\alpha\in\Delta_1})$$

is a pinned S-group. We shall always give Z_{Δ_1} this pinning. In particular, we shall consider

$$Z_\alpha = Z_{\{\alpha\}}, \qquad Z_{\alpha\beta} = Z_{\{\alpha,\beta\}}.$$

Write $B_{\Delta_1} = B \cap Z_{\Delta_1}$. It is known (*loc. cit.*) that this is the canonical Borel subgroup [5] of Z_{Δ_1}, and that its unipotent part is $U_{\Delta_1} = U \cap Z_{\Delta_1}$. In particular,

$$B_\alpha = T \cdot U_\alpha.$$

We write

$$U_{\alpha\beta} = U_{\{\alpha,\beta\}} = U \cap Z_{\alpha\beta} = \prod_{\gamma\in R^+_{\alpha\beta}} U_\gamma,$$

where $R^+_{\alpha\beta}$ is the set of positive roots that are linear combinations of α and β.

Now let $f : G \to G'$ be a morphism of pinned S-groups. If $d : R \to R'$ is the corresponding bijection and $\Delta_1 \subset \Delta$, then $d(\Delta_1) = \Delta'_1 \subset \Delta'$. Clearly f maps T_{Δ_1} into $T'_{\Delta'_1}$, and hence Z_{Δ_1} into $Z'_{\Delta'_1}$. The corresponding S-morphism

$$f_{\Delta_1} : Z_{\Delta_1} \longrightarrow Z'_{\Delta'_1}$$

is compatible with the canonical pinnings. It defines a morphism of pinned root data

$$\mathscr{R}(f_{\Delta_1}) : \mathscr{R}(Z_{\Delta_1}, T, M, \ldots) \longrightarrow \mathscr{R}(Z'_{\Delta'_1}, T', M', \ldots),$$

and the underlying morphisms $M' \to M$ of $\mathscr{R}(f)$ and $\mathscr{R}(f_{\Delta_1})$ coincide.

[5] *Editors' note.* That is, the Borel subgroup of Z_{Δ_1} corresponding to $R'^+ = R' \cap R^+$.

1.8. Study of the group $\mathrm{Norm}_G(T)$

For every pair (α, β) of simple roots, let $n_{\alpha\beta}$ be the order of $s_\alpha s_\beta$ in the Weyl group W. In particular $n_{\alpha\alpha} = 1$, and therefore $(w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S)$.

Definition 1.8.1.— *For every $(\alpha, \beta) \in \Delta \times \Delta$, set*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S).$$

Moreover, set (Exposé XX, 3.1)

$$t_\alpha = t_{\alpha\alpha} = w_\alpha^2 = \alpha^*(-1) \in T(S).$$

Proposition 1.8.2.— *Let H be an S-functor in groups that carries direct sums of schemes to products (for example, a sheaf for the Zariski topology). Let*

$$f_T : T \longrightarrow H$$

be a morphism of groups, and let $h_\alpha \in H(S)$ $(\alpha \in \Delta)$. There exists a necessarily unique morphism of groups

$$f_N : \mathrm{Norm}_G(T) \longrightarrow H$$

inducing f_T on T and satisfying $f_N(w_\alpha) = h_\alpha$ for $\alpha \in \Delta$ if and only if the following two conditions hold:

(i) for every $\alpha \in \Delta$,

$$f_T(s_\alpha(t)) = h_\alpha f_T(t) h_\alpha^{-1}$$

for every $t \in T(S')$, $S' \to S$; equivalently, $f_T \circ s_\alpha = \mathrm{int}(h_\alpha) \circ f_T$;

(ii) for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$f_T(t_{\alpha\beta}) = (h_\alpha h_\beta)^{n_{\alpha\beta}}.$$

Proof. Equip (Sch) with the topology $\mathcal{T}$ generated by the pretopology whose covering families are direct sums. The hypothesis says that H is a $\mathcal{T}$-sheaf. Let L be the free group on the generators $(m_\alpha)_{\alpha \in R}$, and let L_1 be the invariant subgroup generated by

$$(m_\alpha m_\beta)^{n_{\alpha\beta}}, \qquad (\alpha, \beta) \in \Delta \times \Delta.$$

Let $g : L \to W$ be defined by $g(m_\alpha) = s_\alpha$. It is known (Exposé XXI, 5.1) that g induces an isomorphism $\overline{g} : L/L_1 \xrightarrow{\sim} W$. Let L act on T through g, equivalently by $m_\alpha \cdot t = s_\alpha(t)$. Writing L_S for the constant group defined by L, form the semidirect product $N = T \cdot L_S$. There is a unique morphism of S-groups

$$h : N = T \cdot L_S \longrightarrow \mathrm{Norm}_G(T)$$

such that $h(m_\alpha) = w_\alpha$ and $h(t) = t$ for every $t \in T(S')$, $S' \to S$.

Let N_1 be the invariant subgroup sheaf of N generated by

$$t_{\alpha\beta}^{-1}(m_\alpha m_\beta)^{n_{\alpha\beta}}, \qquad (\alpha, \beta) \in \Delta \times \Delta.$$

Clearly $N_1 \subset \ker h$, so h induces

$$h_1 : N/N_1 \longrightarrow \mathrm{Norm}_G(T).$$

We prove that h_1 is an isomorphism. Since $h|_T$ is the canonical immersion, hence a monomorphism, the canonical morphism $T \to N/N_1$ is also a monomorphism and induces an isomorphism $T \xrightarrow{\sim} TN_1/N_1$. Thus h_1 induces an isomorphism $TN_1/N_1 \xrightarrow{\sim} T$. It remains only to show that

$$h_2 : N/TN_1 \longrightarrow \mathrm{Norm}_G(T)/T$$

is an isomorphism. But TN_1 is the invariant $\mathcal{T}$-subgroup sheaf of N generated by T and the $(m_\alpha m_\beta)^{n_{\alpha\beta}}$; hence it is the subgroup sheaf generated by T and L_1, namely $T \cdot (L_1)_S$. Therefore h_2 identifies with

$$\overline{g} : L_S/(L_1)_S \longrightarrow W_S,$$

which is an isomorphism by construction.

The proof of the proposition is now immediate. Necessity is clear. Conversely, condition (i) gives a morphism

$$u : N \longrightarrow H$$

such that $u(m_\alpha) = h_\alpha$ for $\alpha \in \Delta$ and $u|_T = f_T$. Condition (ii) says that u vanishes on N_1, and the result follows.

1.9. Faithfulness of the functor $\mathscr{R}$

Proposition 1.9.1.— *The functor $\mathscr{R}$ of 1.6 is faithful: if*

$$f, g : G \rightrightarrows G'$$

are two morphisms of pinned groups such that $\mathscr{R}(f) = \mathscr{R}(g)$, then $f = g$.

Indeed, f and g agree on T, on every U_α ($\alpha \in \Delta$), and on every $U_{-\alpha}$ ($\alpha \in \Delta$); it therefore suffices to apply:

Lemma 1.9.2.— *Let S be a scheme, let G be a pinned reductive S-group, and let H be an S-presheaf in groups separated for the fppf topology. Let*

$$f, g : G \rightrightarrows H$$

be two morphisms of S-groups. The following conditions are equivalent:

(i) $f = g$;

(ii) f and g agree on T, on every U_α ($\alpha \in \Delta$), and on every $U_{-\alpha}$ ($\alpha \in \Delta$);

(iii) f and g agree on T and on every U_α ($\alpha \in \Delta$), and $f(w_\alpha) = g(w_\alpha)$ for every $\alpha \in \Delta$.

Proof. The implication (i)⇒(ii) is trivial, while (ii)⇒(iii) follows immediately from the definition of w_α in 1.2. It remains to prove (iii)⇒(i). If $\alpha \in R$, there is a sequence $\{\alpha_i\} \subset \Delta$ such that

$$\alpha = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_n}(\alpha_{n+1}),$$

and hence

$$U_\alpha = \mathrm{int}(w_{\alpha_1} \cdots w_{\alpha_n}) U_{\alpha_{n+1}}.$$

Thus f and g agree on every U_α. They consequently agree on Ω, and therefore agree everywhere (Exposé XXII, 4.1.11).

Remark 1.9.3.— *If G is semisimple, the hypothesis that f and g agree on T may be omitted in (ii) and (iii). Indeed, G, as an fppf sheaf, is generated by the U_α, $\alpha \in R$ (Exposé XXII, 6.2.2(a)).*

2. Generators and relations for a pinned group

Throughout this section G is a fixed pinned S-group. For $\alpha, \beta \in \Delta$, we use the notation $Z_\alpha, Z_{\alpha\beta}, U_{\alpha\beta}, R^+_{\alpha\beta}$ of 1.7.

Theorem 2.1.— *Let S be a scheme, let G be a pinned S-group, and let H be an fppf S-sheaf in groups. Suppose given morphisms of groups*

$$f_N : \mathrm{Norm}_G(T) \longrightarrow H, \qquad f_\alpha : U_\alpha \longrightarrow H \quad (\alpha \in R).$$

There exists a necessarily unique morphism of groups

$$f : G \longrightarrow H$$

extending f_N and all the f_α, $\alpha \in R$, if and only if the following conditions hold:

(i) for every $\alpha \in \Delta$ and $\beta \in R$,

$$\mathrm{int}(f_N(w_\alpha)) \circ f_\beta = f_{s_\alpha(\beta)} \circ \mathrm{int}(w_\alpha);$$

(ii) for every $\alpha \in \Delta$, there exists a morphism of groups

$$F_\alpha : Z_\alpha \longrightarrow H$$

extending f_α, $f_{-\alpha}$, and $f_N|_{\mathrm{Norm}_{Z_\alpha}(T)}$;

(iii) for every $(\alpha, \beta) \in \Delta \times \Delta$, there exists a morphism of groups $U_{\alpha\beta} \to H$ inducing f_γ on U_γ for every $\gamma \in R^+_{\alpha\beta}$ (so $U_\gamma \subset U_{\alpha\beta}$).

2.1.1. Proof

The conditions are plainly necessary. Choose a totally ordered group structure on $\Gamma_0(R)$ for which the positive elements of R are precisely R^+ (Exposé XXI, 3.5.6); every product indexed by a subset of R will be taken in this order. Let f_T denote the restriction of f_N to T, and define a morphism of sets

$$f : \Omega \longrightarrow H$$

by

$$f\left(\prod_{\alpha\in R^-} y_\alpha \cdot t \cdot \prod_{\alpha\in R^+} x_\alpha\right) = \prod_{\alpha\in R^-} f_\alpha(y_\alpha) \cdot f_T(t) \cdot \prod_{\alpha\in R^+} f_\alpha(x_\alpha).$$

Every morphism satisfying the theorem must extend this map. Conversely, any morphism of groups $f : G \to H$ extending it also extends f_N: since it extends f_α and $f_{-\alpha}$, one has

$$f(w_\alpha) = F_\alpha(w_\alpha) = f_N(w_\alpha),$$

and it extends f_T by hypothesis. By Exposé XXII, 4.1.11(ii), it therefore remains to prove:

Proposition 2.1.2.— *The morphism $f : \Omega \to H$ defined above is* generically multiplicative*: for every $S' \to S$ and every $x, y \in \Omega(S')$ such that $xy \in \Omega(S')$, one has $f(xy) = f(x)f(y)$.*

Lemma 2.1.3.— *Let $n \in \mathrm{Norm}_G(T)(S')$, $S' \to S$, and let $\alpha, \beta \in R$ satisfy $\mathrm{int}(n)U_\alpha = U_\beta$, equivalently $n(\alpha) = \beta$. Then*

$$\mathrm{int}\big(f_N(n)\big) \circ f_\alpha = f_\beta \circ \mathrm{int}(n).$$

It suffices to verify this formula for a system of generators of $\mathrm{Norm}_G(T)$. It holds for every w_α, $\alpha \in \Delta$, by 2.1(i), so it remains to treat $n \in T(S')$. This is immediate from 2.1(ii) when α is simple. Otherwise choose $w \in W$ with $w^{-1}(\alpha) \in \Delta$. Writing w as a product of simple reflections[6], one reduces to showing that if the formula holds for α and every n, it also holds for $w_{\alpha_0}(\alpha)$ and $t \in T(S')$, where $\alpha_0 \in \Delta$. By 2.1(i),

$$\begin{aligned}\mathrm{int}\big(f_N(t)\big) \circ f_{w_{\alpha_0}(\alpha)} &= \mathrm{int}\big(f_N(tw_{\alpha_0})\big) \circ f_\alpha \circ \mathrm{int}(w_{\alpha_0}^{-1}) \\ &= f_{w_{\alpha_0}(\alpha)} \circ \mathrm{int}(tw_{\alpha_0}) \circ \mathrm{int}(w_{\alpha_0}^{-1}).\end{aligned}$$

Lemma 2.1.4.— *The restriction of f to U (respectively U^-) is a morphism of groups. In particular, this restriction is independent of the chosen ordering of the roots.*

It suffices to prove the assertion for U. By Exposé XXII, 5.5.8, it is enough to show that, for every pair $\alpha < \beta$ of positive roots and every $x_\alpha \in U_\alpha(S')$, $x_\beta \in U_\beta(S')$, $S' \to S$,

$$f_\beta(x_\beta) f_\alpha(x_\alpha) f_\beta(x_\beta^{-1}) = f(x_\beta x_\alpha x_\beta^{-1}).$$

By Exposé XXII, 5.5.2, there are elements $x_\gamma \in U_\gamma(S')$, where $\gamma = i\alpha + j\beta \in R$, $i > 0$, $j \geq 0$ [7], such that

$$x_\beta x_\alpha x_\beta^{-1} = \prod_\gamma x_\gamma.$$

It remains to verify

$$f_\beta(x_\beta) f_\alpha(x_\alpha) f_\beta(x_\beta^{-1}) = \prod_{\substack{\gamma = i\alpha + j\beta \\ i>0,\ j\geq 0}} f_\gamma(x_\gamma).$$

By Exposé XXI, 3.5.4, there is a $w \in W$ such that $w(\alpha) = \alpha_0 \in \Delta$ and $w(\beta) \in A_{\alpha_0\beta_0}$, in the notation of 1.7, for some $\beta_0 \in \Delta$. Lift w to $n \in \mathrm{Norm}_G(T)(S)$ by Exposé XXII, 3.8. It suffices to verify the relation after conjugation by $f_N(n)$; by 2.1.3, this reduces to $\alpha, \beta \in R^+_{\alpha_0\beta_0}$, where condition 2.1(iii) applies.

Lemma 2.1.5.— *Let $n \in \mathrm{Norm}_G(T)(S)$ satisfy $\mathrm{int}(n)U = U^-$, so that n represents the symmetry of the Weyl group[8] (Exposé XXI, 3.6.14). For every $u \in U(S')$, $S' \to S$ (respectively $u \in U^-(S')$), one has*

$$f(nun^{-1}) = f_N(n) f(u) f_N(n^{-1}).$$

This follows immediately from 2.1.3 and 2.1.4.

Lemma 2.1.6.— *Let $u \in B(S')$, $v \in B^-(S')$, and $g \in \Omega(S')$, where $S' \to S$. Then*

$$f(vgu) = f(v) f(g) f(u).$$

Write

$$v = v_1 t_1, \qquad g = v_2 t_2 u_2, \qquad u = t_3 u_3,$$

[6] *Editors' note.* "Fundamental symmetries" was replaced by "simple reflections."

[7] *Editors' note.* The printed $i, j > 0$ was corrected to $i > 0, j \geq 0$, since x_α occurs in the product on the right.

[8] *Editors' note.* Relative to Δ.

with $v_i \in U^-(S')$, $t_i \in T(S')$, and $u_i \in U(S')$. On the one hand,

$$f(v)f(g)f(u) = f(v_1)f_T(t_1)f(v_2)f_T(t_2)f(u_2)f_T(t_3)f(u_3).$$

On the other hand,

$$\begin{aligned} f(vgu) &= f(v_1 t_1 v_2 t_1^{-1}\, t_1 t_2 t_3\, t_3^{-1} u_2 t_3 u_3) \\ &= f(v_1 \cdot t_1 v_2 t_1^{-1}) f_T(t_1 t_2 t_3) f(t_3^{-1} u_2 t_3 \cdot u_3). \end{aligned}$$

Using 2.1.4 on the two outer factors gives

$$f(vgu) = f(v_1)f(t_1 v_2 t_1^{-1})f_T(t_1 t_2 t_3)f(t_3^{-1} u_2 t_3)f(u_3).$$

It remains only to use

$$\begin{aligned} f(t_1 v_2 t_1^{-1}) &= f_T(t_1) f(v_2) f_T(t_1)^{-1}, \\ f(t_3^{-1} u_2 t_3) &= f_T(t_3)^{-1} f(u_2) f_T(t_3), \end{aligned}$$

which follow from the definition of f and 2.1.3.

Lemma 2.1.7.— *Let $\alpha \in \Delta$ and*

$$g \in \Omega(S') \cap \mathrm{int}(w_\alpha)^{-1}\Omega(S'), \qquad S' \to S.$$

Then

$$f(w_\alpha g w_\alpha^{-1}) = f_N(w_\alpha) f(g) f_N(w_\alpha)^{-1}.$$

By Exposé XXII, 5.6.8, write

$$g = a x_{-\alpha} t x_\alpha b,$$

where $a \in U^-_{-\widehat{\alpha}}(S')$, $x_{-\alpha} \in U_{-\alpha}(S')$, $t \in T(S')$, $x_\alpha \in U_\alpha(S')$, and $b \in U_{\widehat{\alpha}}(S')$. By 2.1.3 and 2.1.4, and because s_α permutes the positive roots other than α (Exposé XXI, 3.3.2),

$$\mathrm{int}(w_\alpha) a \in U^-_{-\widehat{\alpha}}(S'), \qquad \mathrm{int}(w_\alpha) b \in U_{\widehat{\alpha}}(S').$$

The desired formula holds for $g = a$ and $g = b$. By 2.1.6 it therefore suffices to treat $g = x_{-\alpha} t x_\alpha \in Z_\alpha(S')$; in that case everything takes place in Z_α, and condition 2.1(ii) applies.

Lemma 2.1.8.— *Let $n \in \mathrm{Norm}_G(T)(S)$. For every $S' \to S$ and every $g \in \Omega(S')$ such that* $\mathrm{int}(n)g \in \Omega(S')$,

$$f(ngn^{-1}) = f_N(n) f(g) f_N(n)^{-1}.$$

This is immediate for $n \in T(S)$, by 2.1.3. The two sides define morphisms from $\Omega \cap \mathrm{int}(n)^{-1}\Omega$ to H. It is enough to find an open $V_n \subset \Omega$, containing the unit section, such that $\mathrm{int}(n)V_n \subset \Omega$, and on which $f \circ \mathrm{int}(n)$ and $\mathrm{int}(f_N(n)) \circ f$ agree. By the structure of $\mathrm{Norm}_G(T)$, it suffices to show that if this is true for n', then it is true for $n = n' w_\alpha$, $\alpha \in \Delta$. Put

$$V_n = \Omega \cap \mathrm{int}(w_\alpha)^{-1} V_{n'}.$$

Then $\mathrm{int}(n)V_n \subset \mathrm{int}(n')V_{n'} \subset \Omega$. If $g \in V_n(S')$, then

$$\mathrm{int}(n)g = \mathrm{int}(n')\,\mathrm{int}(w_\alpha)g.$$

Now $\mathrm{int}(w_\alpha)g \in V_{n'}(S')$, so the induction hypothesis gives

$$f\big(\mathrm{int}(n')\,\mathrm{int}(w_\alpha)g\big) = \mathrm{int}(f_N(n'))f(\mathrm{int}(w_\alpha)g).$$

Lemma 2.1.7 gives

$$f(\mathrm{int}(w_\alpha)g) = \mathrm{int}(f_N(w_\alpha))f(g),$$

and the conclusion follows.

We now prove Proposition 2.1.2. Let $x, x' \in \Omega(S')$, and write

$$x = vtu, \qquad x' = v't'u',$$

so $xx' = vt(uv')t'u'$. By 2.1.6 and $U^-(S')\Omega(S')U(S') = \Omega(S')$, it suffices to show that $f(uv') = f(u)f(v')$ whenever $uv' \in \Omega(S')$. Choose $n \in \mathrm{Norm}_G(T)(S')$ as in 2.1.5. Then

$$f(u) = f_N(n)^{-1} f(nun^{-1}) f_N(n),$$
$$f(v') = f_N(n)^{-1} f(nv'n^{-1}) f_N(n),$$

and hence

$$f(u)f(v') = f_N(n)^{-1} f(nun^{-1}) f(nv'n^{-1}) f_N(n).$$

Since $nun^{-1} \in U^-(S')$ and $nv'n^{-1} \in U(S')$,

$$f(nun^{-1})f(nv'n^{-1}) = f\big((nun^{-1})(nv'n^{-1})\big) = f(nuv'n^{-1}).$$

Thus

$$f(u)f(v') = f_N(n)^{-1} f(nuv'n^{-1}) f_N(n).$$

If $uv' \in \Omega(S')$, Lemma 2.1.8 finishes the proof.

Remark 2.2.— *Instead of prescribing f_N, one may prescribe a morphism $f_T : T \to H$ and sections $h_\alpha \in H(S)$, $\alpha \in \Delta$, satisfying the conditions of 1.8.2. Condition (ii) of Theorem 2.1 must then be replaced by:*

> (ii′) There exists a morphism of groups $F_\alpha : Z_\alpha \to H$ extending $f_\alpha, f_{-\alpha}, f_T$ and satisfying $F_\alpha(w_\alpha) = h_\alpha$.

We now give the preceding theorem a more explicit form, retaining its notation.

Theorem 2.3.— *Let S be a scheme, let G be a pinned S-group, and let H be an fppf S-sheaf in groups. Suppose given morphisms of groups*

$$f_T : T \longrightarrow H, \qquad f_\alpha : U_\alpha \longrightarrow H \quad (\alpha \in \Delta),$$

and sections $h_\alpha \in H(S)$, $\alpha \in \Delta$. There exists a necessarily unique morphism of groups $f : G \to H$ inducing f_T and the f_α and satisfying $f(w_\alpha) = h_\alpha$ for all $\alpha \in \Delta$ if and only if:

(i) for every $S' \to S$, $t \in T(S')$, $\alpha \in \Delta$, and $x \in U_\alpha(S')$,

$$f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(\mathrm{int}(t)x); \tag{1}$$

(ii) for every $\alpha \in \Delta$, $S' \to S$, and $t \in T(S')$,

$$h_\alpha f_T(t) h_\alpha^{-1} = f_T(s_\alpha(t)) = f_T\big(t\alpha^*(\alpha(t)^{-1})\big); \tag{2}$$

(iii) for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$(h_\alpha h_\beta)^{n_{\alpha\beta}} = f_T(t_{\alpha\beta}); \tag{3}$$

(iv) for every $\alpha \in \Delta$, recalling that $u_\alpha = \exp_\alpha(X_\alpha)$,

$$(h_\alpha f_\alpha(u_\alpha))^3 = e; \tag{4}$$

(v) for every distinct $\alpha, \beta \in \Delta$, there exists a morphism $f_{\alpha\beta} : U_{\alpha\beta} \to H$ such that

$$f_{\alpha\beta}|_{U_\alpha} = f_\alpha, \qquad f_{\alpha\beta}|_{U_\beta} = f_\beta, \tag{5}$$

and, for every $\gamma \in R^+_{\alpha\beta}$ with $\gamma \neq \alpha$ (respectively $\gamma \neq \beta$) and every $x \in U_\gamma(S')$, $S' \to S$,

$$\mathrm{int}(h_\alpha) f_{\alpha\beta}(x) = f_{\alpha\beta}(\mathrm{int}(w_\alpha)x) \tag{6}$$

(respectively $\mathrm{int}(h_\beta) f_{\alpha\beta}(x) = f_{\alpha\beta}(\mathrm{int}(w_\beta)x)$).

Proof. Uniqueness follows from 1.9.2. We prove existence.

Lemma 2.3.1.— *There exists a morphism*

$$f_N : \mathrm{Norm}_G(T) \longrightarrow H$$

extending f_T and satisfying $f_N(w_\alpha) = h_\alpha$.

This is precisely 1.8.2, in view of conditions (2) and (3).

Lemma 2.3.2.— *For every $\alpha \in \Delta$, there is a morphism of groups $F_\alpha : Z_\alpha \to H$ extending f_T and f_α, satisfying $F_\alpha(w_\alpha) = h_\alpha$, and hence extending $f_N|_{\mathrm{Norm}_{Z_\alpha}(T)}$.*

This follows from Exposé XX, 6.2 and conditions (1), (3), and (4).

Lemma 2.3.3.— *Let $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$. Let $n \in \mathrm{Norm}_{Z_{\alpha\beta}}(T)(S)$ satisfy $n(\alpha) = \alpha$ (respectively $n(\alpha) = \beta$), equivalently $\mathrm{int}(n)U_\alpha = U_\alpha$ (respectively U_β). Then, for every $S' \to S$ and $x \in U_\alpha(S')$,*

$$\mathrm{int}(f_N(n)) f_\alpha(x) = f_\alpha(\mathrm{int}(n)x)$$

(respectively the right-hand side is $f_\beta(\mathrm{int}(n)x)$).

There exist $t \in T(S)$ and a sequence $\{\alpha_i\} \subset \{\alpha, \beta\}$ such that

$$n = t w_{\alpha_k} \cdots w_{\alpha_1}.$$

Condition (1) treats $n = t$, so we may omit t. Proceed by induction on k, and set

$$\gamma_i = s_{\alpha_i} \cdots s_{\alpha_1}(\alpha).$$

If every γ_i is positive, condition (v) yields

$$\mathrm{int}\big(f_N(w_{\alpha_i} \cdots w_{\alpha_1})\big) f_\alpha(x) = f_{\alpha\beta}\big(\mathrm{int}(w_{\alpha_i} \cdots w_{\alpha_1})x\big),$$

and $i = k$ gives the result. Otherwise there is a $j < k$ such that $\gamma_j \in R^+$ but $\gamma_{j+1} \notin R^+$. Since $\gamma_{j+1} = s_{\alpha_j}(\gamma_j)$, Exposé XXI, 3.3.2 gives $\gamma_j = \alpha_j$, hence γ_j is α or β. Decompose

$$n = n' n'', \qquad n' = w_{\alpha_k} \cdots w_{\alpha_{j+1}}, \qquad n'' = w_{\alpha_j} \cdots w_{\alpha_1}.$$

Both n' and n'' satisfy the lemma's hypotheses and involve fewer than k reflections, so induction applies.

Lemma 2.3.4.— *Let $\alpha \in R$. Suppose $n, n' \in \mathrm{Norm}_G(T)(S)$ and $\beta, \beta' \in \Delta$ satisfy $n(\alpha) = \beta$ and $n'(\alpha) = \beta'$. Then, for every $x \in U_\alpha(S')$, $S' \to S$,*

$$\mathrm{int}(f_N(n)^{-1}) f_\beta(nxn^{-1}) = \mathrm{int}(f_N(n')^{-1}) f_{\beta'}(n'xn'^{-1}).$$

It suffices to prove that if $n(\alpha) = \beta$, with $\alpha, \beta \in \Delta$, then

$$\mathrm{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \mathrm{int}(n).$$

By Tits's lemma (Exposé XXI, 5.6), there are simple roots

$$\alpha_0 = \alpha, \alpha_1, \ldots, \alpha_m = \beta$$

and elements $w_i \in W$, $0 \leq i < m$, such that

$$n = w_{m-1} \cdots w_0, \qquad w_i(\alpha_i) = \alpha_{i+1},$$

and for each i there is a simple root β_i such that

$$w_i \in W_{\alpha_i \beta_i}, \qquad \alpha_{i+1} = \alpha_i \text{ or } \beta_i.$$

Induction reduces the assertion to Lemma 2.3.3.

Lemma 2.3.5.— *There is a family of morphisms of groups $f'_\gamma : U_\gamma \to H$, $\gamma \in R$, such that:*

(i) for $\alpha \in \Delta$, $f'_\alpha = f_\alpha$ and $f'_{-\alpha} = F_\alpha|_{U_{-\alpha}}$;

(ii) if $\alpha, \beta \in \Delta$ and $\gamma \in R^+_{\alpha\beta}$, then $f'_\gamma = f_{\alpha\beta}|_{U_\gamma}$;

(iii) if $n \in \mathrm{Norm}_G(T)(S)$ and $\alpha, \beta \in R$ satisfy $n(\alpha) = \beta$, then

$$\mathrm{int}(f_N(n)) f'_\alpha(x) = f'_\beta(\mathrm{int}(n)x)$$

for every $x \in U_\alpha(S')$, $S' \to S$.

For every root α, choose $n \in \mathrm{Norm}_G(T)(S)$ with $n(\alpha) \in \Delta$, and define $f'_\alpha(x)$ by the expression in Lemma 2.3.4. This is independent of n, and f'_α is a morphism of groups. Property (iii) holds by construction. The first part of (i) follows by taking $n = e$, and the second by taking $n = w_\alpha$ and using 2.3.2. If $\gamma \in R^+_{\alpha\beta}$, choose $n \in \mathrm{Norm}_{Z_{\alpha\beta}}(T)(S)$ with $n(\gamma) = \alpha$ or β; then (iii), together with conditions (5) and (6), proves (ii).

2.3.6.— We finish the proof by verifying the conditions of Theorem 2.1 for f_N and f'_α, $\alpha \in R$:

- 2.1(i) follows from 2.3.5(iii);
- 2.1(ii) follows from 2.3.5(i) and 2.3.2;
- 2.1(iii) follows from 2.3.5(ii) and the fact that $f_{\alpha\beta}$ is a morphism of groups.

An immediate corollary is:

Corollary 2.4.— *Let S be a scheme, let G be a pinned S-group of semisimple rank at least 1, and let H be an fppf S-sheaf in groups. For each $(\alpha, \beta) \in \Delta \times \Delta$, let*

$$F_{\alpha\beta} : Z_{\alpha\beta} \longrightarrow H$$

be a morphism of groups. There exists a morphism $f : G \to H$ inducing the $F_{\alpha\beta}$ if and only if, for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$F_{\alpha\beta}|_{Z_\alpha} = F_{\alpha\alpha}.$$

Necessity is clear. Conversely, put

$$f_T = F_{\alpha\alpha}|_T;$$

this is independent of α, since

$$F_{\alpha\alpha}|_T = F_{\alpha\beta}|_T = F_{\beta\beta}|_T.$$

Set

$$f_\alpha = F_{\alpha\alpha}|_{U_\alpha}, \qquad h_\alpha = F_{\alpha\alpha}(w_\alpha), \qquad f_{\alpha\beta} = F_{\alpha\beta}|_{U_{\alpha\beta}}.$$

The conditions of Theorem 2.3 hold: (1), (2), and (4) hold in Z_α, while (3), (5), and (6) hold in $Z_{\alpha\beta}$. Thus there is a morphism $f : G \to H$ extending f_T and all f_α, and satisfying $f(w_\alpha) = h_\alpha$. It agrees with $F_{\alpha\beta}$ on the generators of $Z_{\alpha\beta}$; hence $f|_{Z_{\alpha\beta}} = F_{\alpha\beta}$.

Corollary 2.5.— *Let S be a scheme; let G and G' be two pinned S-groups of semisimple rank 2; let $q > 0$ be an integer such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$; and let*

$$h : R(G') \longrightarrow R(G)$$

be a q-morphism of pinned root data. For each $\alpha \in R^+$, choose

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times, \qquad X'_{d(\alpha)} \in \Gamma(S, \mathfrak{g}'^{d(\alpha)})^\times,$$

extending the canonical choices for $\alpha \in \Delta$. Suppose:

(i) if $\Delta = \{\alpha, \beta\}$, then

$$D_S(h)(t_{\alpha\beta}) = t'_{d(\alpha)d(\beta)};$$

(ii) for every $\alpha \in \Delta$ and every $\beta \in R^+$, $\beta \neq \alpha$, if $z \in \mathbf{G}_m(S)$ is defined by

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{s_\alpha(\beta)},$$

then

$$\mathrm{Ad}(w'_{d(\alpha)})X'_{d(\beta)} = z^{q(\beta)}X'_{d(s_\alpha(\beta))};$$

(iii) there is a morphism of groups $f : U \to U'$ such that, for every $\alpha \in R^+$, every $S' \to S$, and every $x \in \mathbf{G}_a(S')$,

$$f(\exp(xX_\alpha)) = \exp\bigl(x^{q(\alpha)}X'_{d(\alpha)}\bigr).$$

Then there is a morphism of pinned groups $g : G \to G'$ such that $R(g) = h$.

Indeed, define $f_\alpha : U_\alpha \to G'$ by

$$f_\alpha(\exp(xX_\alpha)) = \exp\bigl(x^{q(\alpha)}X'_{d(\alpha)}\bigr),$$

and put

$$f_T = D_S(h), \qquad h_\alpha = w'_{d(\alpha)}.$$

The conditions of Theorem 2.3 hold: note that

$$q(s_\alpha(\beta)) = q(\beta)$$

by Exposé XXI, 6.8.4, and that

$$D_S(h)(t_\alpha) = t'_{d(\alpha)}$$

always holds. The assertion follows at once.

Remark 2.6.— *Condition (v) of Theorem 2.3 may be made more explicit as follows. One must first verify:*

(a) for every word in w_α, w_β whose corresponding word carries α or β to α or β, the corresponding relation of the type occurring in 2.3.3 holds.

The proof of 2.3.3 shows that it suffices to check the words that are minimal, in the sense that no nontrivial initial subword satisfies the imposed conditions.

If (a) holds, define $f_\gamma : U_\gamma \to H$ for every $\gamma \in R^+_{\alpha\beta}$ as in 2.3.5. One must then require:

(b) the map defined by the f_γ,

$$U_{\alpha\beta} = \prod_{\gamma \in R^+_{\alpha\beta}} U_\gamma \longrightarrow H,$$

is a morphism of groups.

By Exposé XXII, 5.5.8, condition (b) follows from:

(b′) the preceding map respects the commutation relations between U_γ and U_δ, for $\gamma, \delta \in R^+_{\alpha\beta}$, $\gamma > \delta$; namely, the relations involving $C_{i,j,\gamma,\delta}$ in Exposé XXII, 5.5.2.

Conversely, the conjunction of (a) and (b′) is equivalent to condition (v).

These conditions may even be reduced to conditions involving only

$$h_\alpha, \quad h_\beta, \quad f_\alpha(u_\alpha), \quad f_\beta(u_\beta)$$

in H. For example, if $s_\alpha s_\beta s_\alpha(\beta) = \alpha$, a condition of type (a) reads

$$\text{int}(h_\alpha h_\beta h_\alpha) f_\beta(x) = f_\alpha(\text{int}(w_\alpha w_\beta w_\alpha)x)\,. \tag{1}$$

Here $x \in U_\beta(S')$, $S' \to S$. In particular, for $x = u_\beta$,

$$\text{int}(w_\alpha w_\beta w_\alpha) u_\beta = u_\alpha^z$$

for a certain $z \in \mathbf{G}_m(S)$, and

$$\text{int}(h_\alpha h_\beta h_\alpha) f_\beta(u_\beta) = f_\alpha(u_\alpha)^z. \tag{1′}$$

Conversely, taking conditions (i)–(iv) of 2.3 into account, (1′) implies (1). If $t \in T(S')$, conjugating (1′) by t, and using (i) and (ii), gives (1) for

$$x = \text{int}(t) u_\beta = u_\beta^{\beta(t)}.$$

The morphism $\beta : T \to \mathbf{G}_{m,S}$ is faithfully flat, so (1) holds for $x \in U_\beta(S')^\times$. It is additive in x, and every section of U_β is locally the sum of two invertible sections. Thus

$$(1') + (\text{i}) + (\text{ii}) \quad \Longrightarrow \quad (1).$$

The same argument applies to conditions of type (b). One uses the fact that, for two distinct positive roots γ, γ', hence linearly independent over $\mathbf{Q}$, the morphism

$$T \longrightarrow \mathbf{G}_{m,S}^2$$

with components γ, γ' is faithfully flat. We leave the details of this transposition to the reader.

3. Groups of Semisimple Rank 2

3.1. Generalities

Lemma 3.1.1.— *Let S be a scheme, let G be a pinned S-group, and let α, β be two roots of G such that $\alpha + \beta \neq 0$.*

(i) If $\alpha + \beta \notin R$, then

$$\exp(X_\alpha)\exp(X_\beta) = \exp(X_\beta)\exp(X_\alpha)$$

for every $S' \to S$, $X_\alpha \in W(\mathfrak{g}^\alpha)(S')$, and $X_\beta \in W(\mathfrak{g}^\beta)(S')$.

(ii) If neither $\alpha + \beta$ nor $\alpha - \beta$ is a root, then

$$w_\alpha(z_\alpha)\exp(X_\beta)w_\alpha(z_\alpha)^{-1} = \exp(X_\beta)$$

for every $X_\beta \in W(\mathfrak{g}^\beta)(S')$ and $z_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$, and

$$w_\alpha(z_\alpha)w_\beta(z_\beta) = w_\beta(z_\beta)w_\alpha(z_\alpha)$$

for every $z_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$ and $z_\beta \in W(\mathfrak{g}^\beta)^\times(S')$.

(iii) Let $X_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$, $X_\beta \in W(\mathfrak{g}^\beta)^\times(S')$, and $w \in \mathrm{Norm}_G(T)(S')$ satisfy $w(\alpha) = \beta$. Define $z \in \mathbf{G}_m(S')$ by

$$\mathrm{Ad}(w)X_\alpha = zX_\beta.$$

Then

$$\begin{aligned}\mathrm{int}(w)\exp(xX_\alpha) &= \exp(xzX_\beta),\\ \mathrm{int}(w)\exp(yX_\alpha^{-1}) &= \exp(yz^{-1}X_\beta^{-1}),\\ \mathrm{int}(w)w_\alpha(X_\alpha) &= \beta^*(z)w_\beta(X_\beta).\end{aligned}$$

In particular, if $z = \pm 1$, then

$$\mathrm{int}(w)w_\alpha(X_\alpha) = w_\beta(X_\beta)^z.$$

(iv) If $t_\alpha = \alpha^*(-1)$ and $t_\beta = \beta^*(-1)$, then

$$s_\alpha(t_\beta) = t_\beta t_\alpha^{(\beta^*,\alpha)}, \qquad \beta(t_\alpha) = (-1)^{(\alpha^*,\beta)}.$$

Proof. Part (i) follows immediately from Exposé XXII, 5.5.2. Part (ii) follows from Exposé XX, 3.1 and from (i) applied to

$$(\beta,\alpha), \quad (\beta,-\alpha), \quad (-\beta,\alpha), \quad (-\beta,-\alpha).$$

Part (iii) is immediate from the definitions; for its last assertion, observe that

$$\beta^*(-1) = w_\beta(X_\beta)^{-2}.$$

Finally, (iv) is tautological.

Proposition 3.1.2 (Groups of type $A_1 \times A_1$).— *Let S be a scheme and G a pinned S-group of type $A_1 \times A_1$. Write $\Delta = R^+ = \{\alpha, \beta\}$.*

(i)
$$t_{\alpha\beta} = (w_\alpha w_\beta)^2 = t_\alpha t_\beta = (w_\beta w_\alpha)^2 = t_{\beta\alpha}.$$

(ii)
$$\mathrm{Ad}(w_\alpha)X_\beta = X_\beta, \qquad \mathrm{Ad}(w_\beta)X_\alpha = X_\alpha.$$

(iii) The groups U_α and U_β commute; in other words, U is commutative.

Proof. Lemma 3.1.1(ii) gives $w_\alpha w_\beta = w_\beta w_\alpha$, whence
$$(w_\alpha w_\beta)^2 = w_\alpha^2 w_\beta^2 = t_\alpha t_\beta,$$
which proves (i). The same lemma gives (ii), and (iii) is Lemma 3.1.1(i).

3.1.3.— Condition (v) of Theorem 2.3 takes the following explicit form. Put
$$v_\alpha = f_\alpha(u_\alpha) \qquad (\alpha \in \Delta).$$
Then the two groups of conditions are
$$\text{(A)} \quad \begin{cases} h_\alpha v_\beta h_\alpha^{-1} = v_\beta, \\ h_\beta v_\alpha h_\beta^{-1} = v_\alpha, \end{cases} \qquad \text{(B)} \quad v_\alpha v_\beta = v_\beta v_\alpha.$$

3.2. Groups of Type A_2

Proposition 3.2.1.— *Let S be a scheme and G a pinned S-group of type A_2. Write*
$$\Delta = \{\alpha, \beta\}, \qquad R^+ = \{\alpha, \beta, \alpha+\beta\}.$$

(i)
$$t_{\alpha\beta} = (w_\alpha w_\beta)^3 = e = (w_\beta w_\alpha)^3 = t_{\beta\alpha}.$$

(ii) Put $\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}$. Then
$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_\beta &= -X_{\alpha+\beta}, \\ \mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_\beta, \\ \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha. \end{aligned}$$

(iii) Put
$$p_{\alpha+\beta}(x) = \exp(xX_{\alpha+\beta}) = \mathrm{int}(w_\beta)p_\alpha(x).$$
Then
$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy).$$

3.2.2.— The proof occupies 3.2.2–3.2.7. First,
$$(\beta^*, \alpha) = (\alpha^*, \beta) = -1,$$
so $\alpha(t_\beta) = \beta(t_\alpha) = -1$. Put
$$\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}.$$
Then
$$\mathrm{Ad}(w_\beta)X_{\alpha+\beta} = \alpha(t_\beta)X_\alpha = -X_\alpha.$$

Write

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{\alpha+\beta}, \qquad z \in \mathbf{G}_m(S).$$

It follows that

$$\mathrm{Ad}(w_\alpha)X_{\alpha+\beta} = \beta(t_\alpha)z^{-1}X_\beta = -z^{-1}X_\beta.$$

By Exposé XXII, 5.5.2, there is a unique $A \in \mathbf{G}_a(S)$ such that

$$(+) \qquad p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy).$$

Thus (ii) and (iii) reduce to proving $A = -z = 1$.

3.2.3.— Conjugating (+) by w_β gives

$$(++) \qquad p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy).$$

3.2.4.— By definition,

$$w_\beta p_\alpha(x)w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Expanding w_β, this is

$$p_\beta(1)p_{-\beta}(-1)p_\beta(1)p_\alpha(x)p_\beta(-1)p_{-\beta}(1)p_\beta(-1) = p_{\alpha+\beta}(x).$$

Since p_β commutes with $p_{\alpha+\beta}$, this may be written

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (+) on the left and (++) on the right gives

$$p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_\beta(-1) = p_{\alpha+\beta}(x)p_{-\beta}(1) \cdot p_\alpha(Ax)p_{-\beta}(-1).$$

Because $\alpha + 2\beta$, $\alpha - \beta$, and $2\alpha + \beta$ are not roots, this reduces to

$$p_\alpha(x)p_{\alpha+\beta}(Ax) = p_\alpha(Ax)p_{\alpha+\beta}(x).$$

Exposé XXII, 4.1.3 now gives $A = 1$.

3.2.5.— Conjugating (+) by w_α, and using $A = 1$, gives

$$(+++) \qquad p_{\alpha+\beta}(zy)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(zy)p_\beta(-z^{-1}xy).$$

3.2.6.— As before,

$$w_\alpha p_\beta(y)w_\alpha^{-1} = p_{\alpha+\beta}(zy).$$

Since p_α and $p_{\alpha+\beta}$ commute,

$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{\alpha+\beta}(zy)p_{-\alpha}(-1).$$

Using (+) and (+++), this becomes

$$p_\beta(y)p_{\alpha+\beta}(-y) = p_{\alpha+\beta}(zy)p_\beta(-z^{-1}y).$$

The two factors commute, and hence $z = -1$.

3.2.7.— It remains to prove (i). One has

$$w_\alpha w_\beta w_\alpha = w_\alpha w_\beta w_\alpha^{-1} w_\alpha^2 = w_{\alpha+\beta}^{-1} t_\alpha,$$

and therefore

$$\begin{aligned} w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{\alpha+\beta}^{-1} t_\alpha w_\beta \\ &= w_\beta w_{\alpha+\beta}^{-1} w_\beta^{-1}\, s_\beta(t_\alpha) t_\beta \\ &= w_\alpha t_\alpha t_\beta t_\beta = w_\alpha t_\alpha = w_\alpha^{-1}. \end{aligned}$$

Thus

$$(w_\alpha w_\beta)^3 = (w_\beta w_\alpha)^3 = e.$$

Remark 3.2.8.— *Put $v_\alpha = f_\alpha(u_\alpha)$. Condition (v) of 2.3 becomes*

$$(A) \quad \begin{cases} \mathrm{int}(h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \mathrm{int}(h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases}$$

and

$$(B) \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta\, (h_\beta v_\alpha h_\beta^{-1}), \\ v_\beta (h_\beta v_\alpha h_\beta^{-1}) = (h_\beta v_\alpha h_\beta^{-1}) v_\beta, \\ v_\alpha (h_\beta v_\alpha h_\beta^{-1}) = (h_\beta v_\alpha h_\beta^{-1}) v_\alpha. \end{cases}$$

Writing $v_{\alpha+\beta} = \mathrm{int}(h_\beta) v_\alpha$, the latter conditions are equivalently

$$(B) \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta}, \\ v_\alpha v_{\alpha+\beta} = v_{\alpha+\beta} v_\alpha, \\ v_\beta v_{\alpha+\beta} = v_{\alpha+\beta} v_\beta. \end{cases}$$

3.3. Groups of Type B_2

Proposition 3.3.1.— *Let S be a scheme and G a pinned S-group of type B_2. Write*

$$\Delta = \{\alpha, \beta\}, \qquad R^+ = \{\alpha, \beta, \alpha+\beta, 2\alpha+\beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^4 = t_\alpha = (w_\beta w_\alpha)^4 = t_{\beta\alpha}.$$

(ii) If

$$\mathrm{Ad}(w_\beta) X_\alpha = X_{\alpha+\beta}, \qquad \mathrm{Ad}(w_\alpha) X_\beta = X_{2\alpha+\beta},$$

then

$$\begin{aligned} &\mathrm{Ad}(w_\alpha) X_{\alpha+\beta} = -X_{\alpha+\beta}, \quad \mathrm{Ad}(w_\alpha) X_{2\alpha+\beta} = X_\beta, \\ &\mathrm{Ad}(w_\beta) X_{\alpha+\beta} = -X_\alpha, \qquad \mathrm{Ad}(w_\beta) X_{2\alpha+\beta} = X_{2\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(x X_{\alpha+\beta}) = \mathrm{int}(w_\beta) p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(x X_{2\alpha+\beta}) = \mathrm{int}(w_\alpha) p_\beta(x). \end{aligned}$$

Then

$$\begin{aligned} p_\beta(y) p_\alpha(x) &= p_\alpha(x) p_\beta(y) p_{\alpha+\beta}(xy) p_{2\alpha+\beta}(x^2 y), \\ p_{\alpha+\beta}(y) p_\alpha(x) &= p_\alpha(x) p_{\alpha+\beta}(y) p_{2\alpha+\beta}(2xy). \end{aligned}$$

3.3.2.— The proof occupies 3.3.2–3.3.6. We have

$$(\beta^*, \alpha) = -1, \qquad (\alpha^*, \beta) = -2,$$

so $\alpha(t_\beta) = -1$ and $\beta(t_\alpha) = 1$. In particular, $\beta(t_\alpha) = \alpha(t_\alpha) = 1$, so t_α is a section of $\mathrm{Centr}(G)$. Put

$$\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}, \qquad \mathrm{Ad}(w_\alpha)X_\beta = X_{2\alpha+\beta}.$$

Then

$$\begin{aligned}\mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha,\\ \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= \beta(t_\alpha)X_\beta = X_\beta,\\ \mathrm{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}.\end{aligned}$$

For some $k \in \mathbf{G}_m(S)$,

$$\mathrm{Ad}(w_\alpha)X_{\alpha+\beta} = kX_{\alpha+\beta}.$$

By Exposé XXII, 5.5.2, there are $A, B, C \in \mathbf{G}_a(S)$ such that

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y), \tag{1}$$

$$p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Cxy). \tag{2}$$

It remains to prove

$$A = B = 1, \qquad C = 2, \qquad k = -1.$$

3.3.3.— Conjugating (1) by w_α gives

$$\begin{aligned}p_{2\alpha+\beta}(y)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{2\alpha+\beta}(y)\\ \cdot p_{\alpha+\beta}(Akxy)p_\beta(Bx^2y).\end{aligned} \tag{3}$$

Transforming (2) in the same way gives

$$p_{\alpha+\beta}(ky)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(ky)p_\beta(Cxy). \tag{4}$$

Conjugating (1) by w_β gives

$$\begin{aligned}p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)\\ \cdot p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y).\end{aligned} \tag{5}$$

3.3.4.— Write

$$w_\beta p_\alpha(x) w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Because $\alpha + 2\beta$ is not a root,

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (1) on the left and (5) on the right yields

$$\begin{aligned}&p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_\beta(-1)\\ &\quad = p_{\alpha+\beta}(x)p_{-\beta}(1)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{-\beta}(-1).\end{aligned}$$

After commuting the indicated factors and applying (2), this becomes

$$\begin{aligned}p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) = p_\alpha(Ax)p_{\alpha+\beta}(x)\\ \cdot p_{2\alpha+\beta}((AC-B)x^2).\end{aligned}$$

Hence

$$A = 1, \qquad C = 2B.$$

3.3.5.— Now write

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{2\alpha+\beta}(y).$$

Since $3\alpha + \beta$ is not a root,

$$p_\alpha(1) p_\beta(y) p_\alpha(-1) = p_{-\alpha}(1) p_{2\alpha+\beta}(y) p_{-\alpha}(-1).$$

Using (1) on the left and (3) on the right gives

$$p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) = p_{2\alpha+\beta}(y) p_{\alpha+\beta}(Aky) p_\beta(By).$$

The three groups commute, so

$$B = 1, \qquad -A = Ak.$$

Consequently

$$A = B = 1, \qquad C = 2, \qquad k = -1.$$

3.3.6.— We have proved (ii) and (iii). Since

$$s_\beta(t_\alpha) = t_\alpha t_\beta^{(\alpha^*,\beta)} = t_\alpha,$$

one obtains successively

$$\begin{aligned} w_\alpha w_\beta w_\alpha &= w_{2\alpha+\beta} t_\alpha, \\ w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_{2\alpha+\beta} t_\alpha t_\beta, \\ w_\alpha w_\beta w_\alpha w_\beta w_\alpha w_\beta w_\alpha &= w_\beta^{-1} t_\alpha. \end{aligned}$$

[9] It follows that

$$(w_\beta w_\alpha)^4 = t_\alpha, \qquad (w_\alpha w_\beta)^4 = s_\beta(t_\alpha) = t_\alpha.$$

Remark 3.3.7.— *Put*

$$v_{\alpha+\beta} = \mathrm{int}(h_\beta) v_\alpha, \qquad v_{2\alpha+\beta} = \mathrm{int}(h_\alpha) v_\beta.$$

Condition (v) of 2.3 becomes

$$(A) \quad \begin{cases} \mathrm{int}(h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \mathrm{int}(h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases}$$

and

$$(B) \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta}, \\ v_{\alpha+\beta} v_\alpha = v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2, \\ v_{\alpha+\beta} v_\beta = v_\beta v_{\alpha+\beta}, \\ v_{2\alpha+\beta} v_\alpha = v_\alpha v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_\beta = v_\beta v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_{\alpha+\beta} = v_{\alpha+\beta} v_{2\alpha+\beta}. \end{cases}$$

[9] *Editors' note.* The French prints $(\alpha^*, \beta) = 2$; the convention used immediately above gives -2. The conclusion is unchanged because $t_\beta^2 = e$.

3.4. Groups of Type G_2

Proposition 3.4.1.— *Let S be a scheme and G a pinned S-group of type G_2. Write*

$$\Delta = \{\alpha, \beta\}, \qquad R^+ = \{\alpha, \beta, \alpha+\beta, 2\alpha+\beta, 3\alpha+\beta, 3\alpha+2\beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^6 = e = (w_\beta w_\alpha)^6 = t_{\beta\alpha}.$$

(ii) If

$$\begin{aligned} \mathrm{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta},\\ \mathrm{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \mathrm{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, \end{aligned}$$

then

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, & \mathrm{Ad}(w_\alpha)X_{3\alpha+\beta} &= X_\beta,\\ \mathrm{Ad}(w_\alpha)X_{3\alpha+2\beta} &= X_{3\alpha+2\beta}, & \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha,\\ \mathrm{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \mathrm{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(xX_{\alpha+\beta}) = \mathrm{int}(w_\beta)p_\alpha(x),\\ p_{2\alpha+\beta}(x) &= \exp(xX_{2\alpha+\beta}) = \mathrm{int}(w_\alpha w_\beta)p_\alpha(x),\\ p_{3\alpha+\beta}(x) &= \exp(xX_{3\alpha+\beta}) = \mathrm{int}(w_\alpha)p_\beta(-x),\\ p_{3\alpha+2\beta}(x) &= \exp(xX_{3\alpha+2\beta}) = \mathrm{int}(w_\beta w_\alpha)p_\beta(-x). \end{aligned}$$

Then

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2),\\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(2xy)p_{3\alpha+\beta}(3x^2y)p_{3\alpha+2\beta}(3xy^2),\\ p_{2\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(3xy),\\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy),\\ p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) &= p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(3xy). \end{aligned}$$

3.4.2.— The proof occupies 3.4.2–3.4.9. Since

$$(\beta^*, \alpha) = -1, \qquad (\alpha^*, \beta) = -3,$$

we have $\beta(t_\alpha) = \alpha(t_\beta) = -1$. Define the four nonsimple root vectors as in (ii). Then

$$\begin{aligned} \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha,\\ \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= \alpha(t_\alpha)\beta(t_\alpha)X_{\alpha+\beta} = -X_{\alpha+\beta},\\ \mathrm{Ad}(w_\alpha)X_{3\alpha+\beta} &= -\beta(t_\alpha)X_\beta = X_\beta,\\ \mathrm{Ad}(w_\beta)X_{3\alpha+2\beta} &= \alpha(t_\beta)^3\beta(t_\beta)X_{3\alpha+\beta} = -X_{3\alpha+\beta}. \end{aligned}$$

Since $(3\alpha+2\beta) \pm \alpha$ and $(2\alpha+\beta) \pm \beta$ are not roots,

$$\mathrm{Ad}(w_\alpha)X_{3\alpha+2\beta} = X_{3\alpha+2\beta}, \qquad \mathrm{Ad}(w_\beta)X_{2\alpha+\beta} = X_{2\alpha+\beta}.$$

3.4.3.— By Exposé XXII, 5.5.2, there are sections

$$A, B, C, D, E, F, G, H, J \in \mathbf{G}_a(S)$$

such that

$$(1)\quad p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y)p_{3\alpha+\beta}(Cx^3y)p_{3\alpha+2\beta}(Dx^3y^2),$$
$$(2)\quad p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Exy)p_{3\alpha+\beta}(Fx^2y)p_{3\alpha+2\beta}(Gxy^2),$$
$$(3)\quad p_{2\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hxy),$$
$$(4)\quad p_{3\alpha+\beta}(y)p_\beta(x) = p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(Jxy).$$

3.4.4.— Conjugating (1), (3), and (4) by w_β gives

$$p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y)$$
$$(5)\qquad \cdot p_{3\alpha+2\beta}(Cx^3y)p_{3\alpha+\beta}(-Dx^3y^2),$$
$$(6)\qquad p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(Hxy),$$
$$(7)\qquad p_{3\alpha+2\beta}(y)p_{-\beta}(-x) = p_{-\beta}(-x)p_{3\alpha+2\beta}(y)p_{3\alpha+\beta}(-Jxy).$$

Conjugating (1) by w_α gives

$$p_{3\alpha+\beta}(-y)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{3\alpha+\beta}(-y)p_{2\alpha+\beta}(Axy)p_{\alpha+\beta}(-Bx^2y)$$
$$(8)\qquad \cdot p_\beta(Cx^3y)p_{3\alpha+\beta}(Dx^3y^2).$$

3.4.5.— The identity

$$w_\beta p_\alpha(x)w_\beta^{-1} = p_{\alpha+\beta}(x)$$

may be written, since $\alpha + 2\beta$ is not a root,

$$(9)\qquad p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (1) and then (4), its left side becomes

$$(10)\qquad p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}((D-CJ)x^3).$$

Using (5) and then (7), its right side becomes

$$p_{\alpha+\beta}(x)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}((CJ-D)x^3).$$

Applying (2) transforms this into

$$p_\alpha(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AE-B)x^2)p_{3\alpha+\beta}((A^2F+CJ-D)x^3)$$
$$(12)\qquad \cdot p_{3\alpha+2\beta}((AG-C)x^3).$$

Comparing (10) and (12) gives

$$A = 1, \qquad E = 2B, \qquad C + D = F + CJ, \qquad F = G.$$

3.4.6.— Write

$$w_\alpha p_\beta(y)w_\alpha^{-1} = p_{3\alpha+\beta}(-y).$$

Since $4\alpha + \beta$ is not a root,

$$(13)\qquad p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{3\alpha+\beta}(-y)p_{-\alpha}(-1).$$

Using (1), the left side is

$$p_\beta(y)p_{\alpha+\beta}(-Ay)p_{2\alpha+\beta}(By)p_{3\alpha+\beta}(-Cy)p_{3\alpha+2\beta}(-Dy^2).$$

Using (8), (6), and (4), the right side is

$$p_\beta(Cy)p_{\alpha+\beta}(-By)p_{2\alpha+\beta}(Ay)p_{3\alpha+\beta}(-y)p_{3\alpha+2\beta}((D-CJ-ABH)y^2).$$

Thus

$$C=1, \qquad A=B, \qquad D-CJ-ABH=-D.$$

Together with the preceding identities, this yields

$$A=B=C=1, \quad E=2, \quad F=G, \quad D+1=F+J, \quad 2D=H+J.$$

3.4.7.— From

$$w_\beta p_{3\alpha+\beta}(x)w_\beta^{-1}=p_{3\alpha+2\beta}(x)$$

one obtains

$$p_{3\alpha+\beta}(x)p_{3\alpha+2\beta}(-Jx)=p_{3\alpha+2\beta}(x)p_{3\alpha+\beta}(-Jx),$$

and therefore $J=-1$.

3.4.8.— Finally,

$$w_\alpha p_{\alpha+\beta}(y)w_\alpha^{-1}=p_{2\alpha+\beta}(y),$$

or equivalently

$$p_\alpha(1)p_{\alpha+\beta}(y)p_\alpha(-1)=p_{-\alpha}(1)p_\alpha(-1)p_{2\alpha+\beta}(y)p_\alpha(1)p_{-\alpha}(-1).$$

Using (2) on the left and (3) on the right gives

$$\begin{aligned}&p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-Ey)p_{3\alpha+\beta}(Fy)p_{3\alpha+2\beta}(-Gy^2)\\&\quad=p_{-\alpha}(1)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hy)p_{-\alpha}(-1).\end{aligned}$$

Commuting $p_{-\alpha}(-1)$ first past $p_{3\alpha+\beta}(Hy)$, and then past $p_{2\alpha+\beta}(y)$, introduces no new $p_{3\alpha+\beta}$-factor. Comparing those factors gives $F=H$. Hence

$$2D=D+1, \qquad D=1, \qquad F=G=H=2D-J=3.$$

This determines all the coefficients and proves (iii).

3.4.9.— For (i), one obtains successively

$$\begin{aligned}w_\alpha w_\beta w_\alpha&=w_{3\alpha+\beta}^{-1}t_\alpha,\\w_\beta(w_\alpha w_\beta)^2&=w_{3\alpha+2\beta}^{-1}t_\alpha,\\w_\alpha(w_\beta w_\alpha)^3&=w_{3\alpha+2\beta}^{-1},\\w_\beta(w_\alpha w_\beta)^4&=w_{3\alpha+\beta}t_\beta,\\w_\alpha(w_\beta w_\alpha)^5&=w_\beta^{-1}.\end{aligned}$$

Consequently

$$(w_\alpha w_\beta)^6=(w_\beta w_\alpha)^6=e.$$

Remark 3.4.10.— *Condition (v) of 2.3 consists first of*

$$(A)\quad\begin{cases}\mathrm{int}(h_\alpha h_\beta h_\alpha h_\beta h_\alpha)v_\beta=v_\beta,\\ \mathrm{int}(h_\beta h_\alpha h_\beta h_\alpha h_\beta)v_\alpha=v_\alpha,\end{cases}$$

and, after setting

$$v_{\alpha+\beta} = \mathrm{int}(h_\beta)v_\alpha, \qquad v_{2\alpha+\beta} = \mathrm{int}(h_\alpha h_\beta)v_\alpha,$$
$$v_{3\alpha+\beta} = \mathrm{int}(h_\alpha)v_\beta^{-1}, \quad v_{3\alpha+2\beta} = \mathrm{int}(h_\beta h_\alpha)v_\beta^{-1},$$

of the commutation relations

$$\begin{aligned}
v_\beta v_\alpha &= v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta} v_{3\alpha+\beta} v_{3\alpha+2\beta},\\
v_{\alpha+\beta} v_\alpha &= v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2 v_{3\alpha+\beta}^3 v_{3\alpha+2\beta}^3,\\
v_{2\alpha+\beta} v_\alpha &= v_\alpha v_{2\alpha+\beta} v_{3\alpha+\beta}^3,\\
v_{3\alpha+\beta} v_\alpha &= v_\alpha v_{3\alpha+\beta}, \qquad & v_{3\alpha+2\beta} v_\alpha &= v_\alpha v_{3\alpha+2\beta},\\
v_{\alpha+\beta} v_\beta &= v_\beta v_{\alpha+\beta}, & v_{2\alpha+\beta} v_\beta &= v_\beta v_{2\alpha+\beta},\\
v_{3\alpha+\beta} v_\beta &= v_\beta v_{3\alpha+\beta} v_{3\alpha+2\beta}^{-1},\\
v_{3\alpha+2\beta} v_\beta &= v_\beta v_{3\alpha+2\beta},\\
v_{2\alpha+\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{2\alpha+\beta} v_{3\alpha+2\beta}^3,\\
v_{3\alpha+\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{3\alpha+\beta},\\
v_{3\alpha+2\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{3\alpha+2\beta},\\
v_{3\alpha+\beta} v_{2\alpha+\beta} &= v_{2\alpha+\beta} v_{3\alpha+\beta},\\
v_{3\alpha+2\beta} v_{2\alpha+\beta} &= v_{2\alpha+\beta} v_{3\alpha+2\beta},\\
v_{3\alpha+2\beta} v_{3\alpha+\beta} &= v_{3\alpha+\beta} v_{3\alpha+2\beta}.
\end{aligned} \tag{B}$$

3.5. Explicit Form of the Generators-and-Relations Theorem

Theorem 3.5.1.— *Let S be a scheme and G a pinned S-group, with maximal torus T, system of simple roots Δ, and elements*

$$u_\alpha \in U_\alpha(S)^\times, \qquad w_\alpha \in \mathrm{Norm}_G(T)(S) \qquad (\alpha \in \Delta)$$

defined by the pinning. Let H be an fppf S-sheaf in groups, let

$$f_T : T \to H, \qquad f_\alpha : U_\alpha \to H \quad (\alpha \in \Delta)$$

be morphisms of groups, and let $h_\alpha \in H(S)$. Put

$$v_\alpha = f_\alpha(u_\alpha).$$

There exists a morphism of groups $f : G \to H$ extending f_T and all the f_α, and satisfying $f(w_\alpha) = h_\alpha$, if and only if:

(i) for every $S' \to S$, $\alpha \in \Delta$, $t \in T(S')$, and $x \in U_\alpha(S')$,

$$\mathrm{int}(f_T(t))f_\alpha(x) = f_\alpha(\mathrm{int}(t)x) = f_\alpha(x^{\alpha(t)}); \tag{1}$$

(ii) for every $\alpha \in \Delta$, every $S' \to S$, and every $t \in T(S')$,

$$\mathrm{int}(h_\alpha)f_T(t) = f_T(s_\alpha(t)) = f_T\Big(t\alpha^*(\alpha(t)^{-1})\Big); \tag{2}$$

(iii) for every $\alpha \in \Delta$,

$$h_\alpha^2 = f_T(\alpha^*(-1)), \tag{3_1}$$

and

$$(h_\alpha v_\alpha)^3 = e; \tag{4}$$

(iv) for distinct $\alpha, \beta \in \Delta$:

(a) the appropriate braid relation is

$$\begin{cases} (h_\alpha h_\beta)^2 = f_T(\alpha^*(-1)\beta^*(-1)), & (\alpha^*, \beta) = 0, \\ (h_\alpha h_\beta)^3 = e, & (\alpha^*, \beta) = -1, \\ (h_\alpha h_\beta)^4 = f_T(\alpha^*(-1)), & (\alpha^*, \beta) = -2, \\ (h_\alpha h_\beta)^6 = e, & (\alpha^*, \beta) = -3; \end{cases} \tag{3_2}$$

(b) the corresponding relations (A) and (B) of 3.1.3, 3.2.8, 3.3.7, or 3.4.10 hold.

When it exists, f is necessarily unique.

This follows immediately from Theorem 2.3, Remark 2.6, and the calculations in each rank-two case.

Remark 3.5.2.— *The result may equivalently be presented by giving morphisms*

$$a_\alpha : T \cdot U_\alpha \longrightarrow H, \qquad b_\alpha : \mathrm{Norm}_{Z_\alpha}(T) \longrightarrow H \qquad (\alpha \in \Delta),$$

and putting

$$h_\alpha = b_\alpha(w_\alpha), \qquad v_\alpha = a_\alpha(u_\alpha).$$

The required conditions are:

(1) all the a_α and b_α have the same restriction to T;

(2) condition (4) and all the conditions in (iv) of Theorem 3.5.1 hold.

3.5.3.— Further applications will be given in the next Exposé. Here is one. Theorem 3.5.1 describes G by generators and relations in the category of fppf S-sheaves. More explicitly, for every $S' \to S$, let $H(S')$ be the group generated by

$$T(S'), \qquad U_\alpha(S') \ (\alpha \in R), \qquad w_\alpha \ (\alpha \in R),$$

subject to relations analogous to

$$(1),\ (2),\ (3_1),\ (4),\ (3_2),\ (A),\ (B).$$

Then G is the sheaf associated with the presheaf $S' \mapsto H(S')$.

In particular, if S' is the spectrum of an algebraically closed field k, then $G(S') = H(S')$. This follows immediately from the Nullstellensatz in the form: an fppf-covering sieve over an algebraically closed field is trivial. Thus Theorem 3.5.1 gives an explicit presentation by generators and relations of the abstract group $G(k)$.

4. Uniqueness of Pinned Groups: Fundamental Theorem

Theorem 4.1.— *Let S be a nonempty scheme. The functor R of 1.6 is fully faithful. More precisely, let G, G' be pinned S-groups, let $p > 0$ be an integer such that $x \mapsto x^p$ is an endomorphism of $\mathbf{G}_{a,S}$, and let*

$$h : R(G') \longrightarrow R(G)$$

be a p-morphism of pinned root data. There is a unique morphism of pinned groups $f : G \to G'$ such that $R(f) = h$.

Uniqueness was proved in 1.9, so only existence remains. By hypothesis there are a bijection $d : R \xrightarrow{\sim} R'$ and a map

$$q : R \longrightarrow \{p^n \mid n \in \mathbf{N}\}$$

such that, for every $\alpha \in R$,

$$h(d(\alpha)) = q(\alpha)\alpha, \qquad {}^t h(\alpha^*) = q(\alpha)d(\alpha)^*.$$

In particular, G and G' have the same semisimple rank.

4.1.1.— Suppose that both semisimple ranks are 0. Then G, G' are tori:

$$G = T = D_S(M), \qquad G' = T' = D_S(M'),$$

and $h : M' \to M$ is an ordinary homomorphism. Take $f = D_S(h)$.

4.1.2.— Suppose that both semisimple ranks are 1, and put

$$f_T = D_S(h) : T \longrightarrow T'.$$

Writing $\alpha' = d(\alpha)$, the diagram

$$\begin{array}{ccccc}
\mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\
\downarrow{\scriptstyle q(\alpha)} & & \downarrow{\scriptstyle f_T} & & \downarrow{\scriptstyle q(\alpha)} \\
\mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S}
\end{array}$$

commutes. Apply Exposé XX, 4.1.

4.1.3.— Suppose that both semisimple ranks are 2. Exposé XXI, 7.5.3 lists every possible h; we verify the conditions of Corollary 2.5 in each case. Write

$$\Delta = \{\alpha, \beta\}, \qquad \Delta' = \{\alpha', \beta'\}, \qquad d(\alpha) = \alpha', \quad d(\beta) = \beta'.$$

4.1.4. G and G' of type $A_1 \times A_1$.— Here

$$h(\alpha') = q\alpha, \qquad h(\beta') = q_1\beta.$$

Conditions 2.5(ii) and (iii) follow from Proposition 3.1.2(ii),(iii). For (i),

$$\begin{aligned}
D_S(h)t_{\alpha\beta} &= D_S(h)(t_\alpha t_\beta) \\
&= {}^t h(\alpha^*)(-1)\, {}^t h(\beta^*)(-1) \\
&= \alpha'^*((-1)^q)\, \beta'^*((-1)^{q_1}).
\end{aligned}$$

The assumption on $x \mapsto x^p$, with $S \neq \varnothing$, implies that $p = 1$ or S has characteristic p. Thus $(-1)^q = -1$: if q is even, then $p = 2$ and $1 = -1$. Consequently

$$D_S(h)t_{\alpha\beta} = \alpha'^*(-1)\beta'^*(-1) = t'_{\alpha'\beta'}.$$

4.1.5. G and G' of type A_2.— Here

$$h(\alpha') = q\alpha, \qquad h(\beta') = q\beta.$$

Put

$$X_{\alpha+\beta} = \mathrm{Ad}(w_\beta)X_\alpha, \qquad X'_{\alpha'+\beta'} = \mathrm{Ad}(w'_{\beta'})X'_{\alpha'}.$$

Condition (i) follows as above from Proposition 3.2.1(i), and (ii) follows immediately from Proposition 3.2.1(ii). For (iii), the map

$$p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z) \longmapsto p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(z^q)$$

must be a homomorphism. The only nontrivial relation is

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy),$$

which maps to

$$p'_{\beta'}(y^q)p'_{\alpha'}(x^q) = p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(x^q y^q).$$

4.1.6.— The same argument treats types B_2 and G_2 when the radicial exponents are equal, using Propositions 3.3.1 and 3.4.1. It remains to treat the two exceptional cases in Exposé XXI, 7.5.3.

4.1.7. Type B_2, characteristic 2, $q(\alpha) = 2q$, $q(\beta) = q$.— The positive roots are

$$\{\alpha, \beta, \alpha+\beta, 2\alpha+\beta\}$$

and

$$\{\alpha', \beta', \alpha'+\beta', \alpha'+2\beta'\},$$

where α and β' are the short simple roots. We have

$$\begin{aligned} h(\alpha') &= 2q\alpha, & h(\beta') &= q\beta, \\ h(\alpha'+\beta') &= q(2\alpha+\beta), & h(\alpha'+2\beta') &= 2q(\alpha+\beta), \end{aligned}$$

so

$$\begin{aligned} d(\alpha+\beta) &= \alpha'+2\beta', & q(\alpha+\beta) &= 2q, \\ d(2\alpha+\beta) &= \alpha'+\beta', & q(2\alpha+\beta) &= q. \end{aligned}$$

Put

$$\begin{aligned} X_{\alpha+\beta} &= \mathrm{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \mathrm{Ad}(w_\alpha)X_\beta, \\ X'_{\alpha'+\beta'} &= \mathrm{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \mathrm{Ad}(w'_{\beta'})X'_{\alpha'}. \end{aligned}$$

For (i), characteristic 2 gives $-1 = 1$, hence

$$t_{\alpha\beta} = t_\alpha = \alpha^*(-1) = e = \beta'^*(-1) = t'_{\beta'} = t'_{\alpha'\beta'}.$$

For (ii), the two defining identities are

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_\beta &= X_{2\alpha+\beta}, & \mathrm{Ad}(w'_{\alpha'})X'_{\beta'} &= X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)}, \\ \mathrm{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \mathrm{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}. \end{aligned}$$

Proposition 3.3.1(ii), together with $-1 = 1$, gives the four remaining identities:

$$\begin{aligned}
\mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{\alpha+\beta}, & \mathrm{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= X'_{d(\alpha+\beta)},\\
\mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= X_\beta, & \mathrm{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= X'_{d(\beta)},\\
\mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= X_\alpha, & \mathrm{Ad}(w'_{\beta'})X'_{d(\alpha+\beta)} &= X'_{d(\alpha)},\\
\mathrm{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \mathrm{Ad}(w'_{\beta'})X'_{d(2\alpha+\beta)} &= X'_{d(2\alpha+\beta)}.
\end{aligned}$$

For (iii), the sole nontrivial relations are

$$\begin{aligned}
p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y),\\
p'_{\alpha'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(x'y')p'_{\alpha'+2\beta'}(x'y'^2).
\end{aligned}$$

The proposed map is

$$\begin{aligned}
&p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z)p_{2\alpha+\beta}(t)\\
&\quad\longmapsto p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q)p'_{\alpha'+2\beta'}(z^{2q})p'_{\alpha'+\beta'}(t^q).
\end{aligned}$$

It respects the first relation because the second, with $y' = x^{2q}$ and $x' = y^q$, reads

$$\begin{aligned}
p'_{\beta'}(y^q)p'_{\alpha'}(x^{2q}) = p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q)p'_{\alpha'+2\beta'}((xy)^{2q})\\
\cdot p'_{\alpha'+\beta'}((x^2y)^q).
\end{aligned}$$

4.1.8. Type G_2, characteristic 3, $q(\alpha) = 3q$, $q(\beta) = q$.— The positive roots are

$$\{\alpha, \beta, \alpha+\beta, 2\alpha+\beta, 3\alpha+\beta, 3\alpha+2\beta\}$$

and

$$\{\alpha', \beta', \alpha'+\beta', \alpha'+2\beta', \alpha'+3\beta', 2\alpha'+3\beta'\},$$

where again α and β' are short. We have

$$\begin{aligned}
h(\alpha') &= 3q\alpha, & h(\beta') &= q\beta,\\
h(\alpha'+\beta') &= q(3\alpha+\beta), & h(\alpha'+2\beta') &= q(3\alpha+2\beta),\\
h(\alpha'+3\beta') &= 3q(\alpha+\beta), & h(2\alpha'+3\beta') &= 3q(2\alpha+\beta),
\end{aligned}$$

and hence

$$\begin{aligned}
d(\alpha+\beta) &= \alpha'+3\beta', & q(\alpha+\beta) &= 3q,\\
d(2\alpha+\beta) &= 2\alpha'+3\beta', & q(2\alpha+\beta) &= 3q,\\
d(3\alpha+\beta) &= \alpha'+\beta', & q(3\alpha+\beta) &= q,\\
d(3\alpha+2\beta) &= \alpha'+2\beta', & q(3\alpha+2\beta) &= q.
\end{aligned}$$

Define

$$\begin{aligned}
X_{\alpha+\beta} &= \mathrm{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \mathrm{Ad}(w_\alpha)X_{\alpha+\beta},\\
X_{3\alpha+\beta} &= -\mathrm{Ad}(w_\alpha)X_\beta, & X_{3\alpha+2\beta} &= \mathrm{Ad}(w_\beta)X_{3\alpha+\beta},\\
X'_{\alpha'+\beta'} &= -\mathrm{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \mathrm{Ad}(w'_{\beta'})X'_{\alpha'+\beta'},\\
X'_{\alpha'+3\beta'} &= \mathrm{Ad}(w'_{\beta'})X'_{\alpha'}, & X'_{2\alpha'+3\beta'} &= \mathrm{Ad}(w'_{\alpha'})X'_{\alpha'+3\beta'}.
\end{aligned}$$

Condition (i) follows from Proposition 3.4.1(i). Four instances of (ii) hold by definition:

$$\begin{aligned}
\mathrm{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \mathrm{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{d(\alpha+\beta)},\\
\mathrm{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \mathrm{Ad}(w'_{\alpha'})X'_{\beta'} &= -X'_{d(3\alpha+\beta)},\\
\mathrm{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, & \mathrm{Ad}(w'_{\beta'})X'_{d(3\alpha+\beta)} &= X'_{d(3\alpha+2\beta)},\\
\mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, & \mathrm{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= X'_{d(2\alpha+\beta)}.
\end{aligned}$$

The remaining instances follow from Proposition 3.4.1(ii); for example,

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, \qquad \mathrm{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} = -X'_{d(\alpha+\beta)}, \\ \mathrm{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}, \quad \mathrm{Ad}(w'_{\beta'})X'_{d(3\alpha+2\beta)} = -X'_{d(3\alpha+\beta)}. \end{aligned}$$

For (iii), in characteristic 3, the nontrivial relations in U are

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy). \end{aligned}$$

The corresponding relations in U' are

$$\begin{aligned} p'_{\alpha'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(-x'y')p'_{\alpha'+2\beta'}(-x'^2y') \\ &\quad \cdot p'_{\alpha'+3\beta'}(-x'^3y')p'_{2\alpha'+3\beta'}(-x'^3y'^2), \\ p'_{\alpha'+\beta'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'+\beta'}(y')p'_{\alpha'+2\beta'}(x'y'), \\ p'_{\alpha'+3\beta'}(y')p'_{\alpha'}(x') &= p'_{\alpha'}(x')p'_{\alpha'+3\beta'}(y')p'_{2\alpha'+3\beta'}(x'y'). \end{aligned}$$

Define $\phi : U \to U'$ by

$$\begin{aligned} &\phi(p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(t)p_{2\alpha+\beta}(u)p_{3\alpha+\beta}(v)p_{3\alpha+2\beta}(w)) \\ &\quad = p'_{\alpha'}(x^{3q})p'_{\beta'}(y^q)p'_{\alpha'+3\beta'}(t^{3q})p'_{2\alpha'+3\beta'}(u^{3q})p'_{\alpha'+\beta'}(v^q)p'_{\alpha'+2\beta'}(w^q). \end{aligned}$$

Substituting x^{3q}, y^q in the three displayed relations for U' yields exactly the images of the three relations for U; thus ϕ is a homomorphism.

4.1.9. Semisimple rank greater than 2.— For $\alpha \in \Delta$, put $\alpha' = d(\alpha)$. For every (α, β), consider the pinned rank-at-most-two groups

$$Z_{\alpha\beta}, \qquad Z'_{\alpha'\beta'}.$$

The homomorphism $M' \to M$ underlying h induces a p-morphism

$$h_{\alpha\beta} : R(Z'_{\alpha'\beta'}) \longrightarrow R(Z_{\alpha\beta}).$$

By the preceding cases, there is a morphism of pinned groups

$$f_{\alpha\beta} : Z_{\alpha\beta} \longrightarrow Z'_{\alpha'\beta'}$$

with $R(f_{\alpha\beta}) = h_{\alpha\beta}$. The restrictions

$$f_{\alpha\beta}|_{Z_\alpha}, \qquad f_{\alpha\alpha} : Z_\alpha \to Z'_{\alpha'}$$

correspond to the same morphism of pinned root data, and therefore coincide by uniqueness. Corollary 2.4 now glues the $f_{\alpha\beta}$ to a morphism $f : G \to G'$. It is a morphism of pinned groups and satisfies $R(f) = h$, completing the proof.

5. Corollaries of the Fundamental Theorem

The most important is the following.

Corollary 5.1.— *Let S be a nonempty scheme, let G, G' be pinned S-groups, and let*

$$h : R(G') \xrightarrow{\sim} R(G)$$

be an isomorphism of pinned root data. There is a unique isomorphism of pinned S-groups $f : G \xrightarrow{\sim} G'$ such that $R(f) = h$.

This also follows from Theorem 3.5.1, because its relations can be written using only $R(G)$. It also follows from the elementary part of the proof of Theorem 4.1, without the exceptional isogenies of 4.1.7 and 4.1.8.

Corollary 5.2 (Uniqueness theorem).— *Let S be a scheme and let G, G' be two splittable S-groups. If they have the same type, then they are isomorphic.*

Corollary 5.3.— *Let S be a scheme and G, G' two splittable S-groups. The following are equivalent:*

(i) G and G' are isomorphic;

(ii) they are locally isomorphic for the fpqc topology;

(iii) there is $s \in S$ such that G_s and G'_s have the same type.

Clearly (i) implies (ii), which implies (iii). Conversely, (iii) gives

$$R(G) = R(G_s) = R(G'_s) = R(G'),$$

so Corollary 5.2 applies.

Corollary 5.4 (Uniqueness of Chevalley schemes).— *Let G, G' be reductive groups over $\mathbf{Z}$, each possessing a split maximal torus. The following are equivalent:*

(i) G and G' are isomorphic;

(ii) there is $s \in \operatorname{Spec} \mathbf{Z}$ such that G_s and G'_s have the same type;

(iii) $G_{\mathbf{C}}$ and $G'_{\mathbf{C}}$ have the same type.

Indeed, G and G' are splittable by Exposé XXII, 2.2. [10]

Corollary 5.5 (Existence of outer automorphisms).— *Let S be a scheme, G a pinned S-group, and h an automorphism of the pinned root datum $R(G)$. There is a unique automorphism u of G preserving the pinning and satisfying $R(u) = h$.*

Corollary 5.5 bis.— *Let S be a scheme, G a split reductive S-group, and R^+ a system of positive roots. For every simple root α, choose an isomorphism of vector groups*

$$p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha.$$

[10] *Editors' note.* In fact every $\mathbf{Z}$-torus is split: every $\mathbf{Z}$-torus is isotrivial, and every étale cover of $\operatorname{Spec} \mathbf{Z}$ is trivial.

Let h be an automorphism of M permuting the positive roots and the corresponding coroots: if $\alpha \in R^+$, then

$$h(\alpha) \in R^+, \qquad h^\vee(\alpha^*) = h(\alpha)^*.$$

There is a unique automorphism u of G inducing $D_S(h)$ on T and satisfying $u \circ p_\alpha = p_{h(\alpha)}$ for every simple root α.

Corollary 5.6.— *Let G, G' be reductive S-groups. The following are equivalent:*

(i) G, G' are locally isomorphic for the fpqc topology;

(i bis) they are locally isomorphic for the étale topology;

(ii) G_s, G'_s are isomorphic for every $s \in S$;

(iii) the functions $s \mapsto \mathrm{type}(G_s)$ and $s \mapsto \mathrm{type}(G'_s)$ are equal.

The implication (i bis)⇒(i) is immediate; (i)⇒(ii) follows from the principle of finite extension (EGA IV_3, 9.1.4); and (ii)⇒(iii) is immediate. For (iii)⇒(i bis), one may suppose G, G' splittable (Exposé XXII, 2.3), and then apply Corollary 5.3.

Corollary 5.7.— *Let S be a scheme, G a reductive S-group, and G' an affine smooth S-group with connected fibers. If $s \in S$ and $G_s \simeq G'_s$, then there is an étale morphism $S' \to S$, covering s, such that $G_{S'} \simeq G'_{S'}$.*

By Exposés XIX, 2.5 and XXII, 2.1, one may suppose G, G' reductive and splittable, and Corollary 5.3 applies.

Corollary 5.8.— *Let k be a field and G, G' reductive k-groups. The following are equivalent:*

(i) G, G' have the same type;

(ii) $G \otimes_k \bar{k}$ and $G' \otimes_k \bar{k}$ are isomorphic;

(iii) for some finite separable extension K/k, $G \otimes_k K$ and $G' \otimes_k K$ are isomorphic.

Corollary 5.9.— *Let S be a nonempty scheme and R a root datum. The following are equivalent:*

(i) there is a pinned S-group of type R;

(ii) there is an S-group of type R;

(iii) fpqc-locally, there is a reductive S-group of type R.

Only (iii)⇒(i) needs proof. Suppose, for simplicity, that there are a faithfully flat quasi-compact morphism $S' \to S$ and a reductive S'-group G' of type R. We may suppose G' splittable and choose a pinning E', writing $R = R(G', E')$. The two pullbacks of (G', E') to

$$S'' = S' \times_S S'$$

are pinned groups (G''_1, E''_1), (G''_2, E''_2), with canonical isomorphisms

$$p_i : R(G''_i, E''_i) \xrightarrow{\sim} R.$$

Thus

$$p = p_2^{-1} \circ p_1 : R(G_1'', E_1'') \xrightarrow{\sim} R(G_2'', E_2'').$$

The uniqueness theorem gives a unique isomorphism

$$f : (G_1'', E_1'') \xrightarrow{\sim} (G_2'', E_2'')$$

with $R(f) = p$. This is a descent datum on G'. Its cocycle condition may be checked after applying R, since R is fully faithful, and then follows from the construction of p. Because G' is affine, the datum is effective. The pinning is stable under descent, so it descends to a pinned S-group (G, E) of type R.

N.B. In the language of fibered categories, the preceding proof becomes both shorter and more transparent.

Corollary 5.10.— *Let S be a nonempty scheme and R a pinned root datum for which there exists a reductive S-group of type R. Then there is a pinned S-group of type R, unique up to unique isomorphism.*

Definition 5.11.— *Under these hypotheses, denote the unique pinned S-group of type R by $\acute{\mathrm{E}}\mathrm{p}_S(R)$, its canonical maximal torus by $T_S(R)$, its canonical Borel subgroup by $B_S(R)$, and so on.*

For every morphism $S' \to S$, with $S' \neq \varnothing$, one may identify

$$\acute{\mathrm{E}}\mathrm{p}_{S'}(R) = \acute{\mathrm{E}}\mathrm{p}_S(R) \times_S S'.$$

If $\acute{\mathrm{E}}\mathrm{p}_{\mathrm{Spec}\,\mathbf{Z}}(R)$ exists, write it $\acute{\mathrm{E}}\mathrm{p}(R)$; then

$$\acute{\mathrm{E}}\mathrm{p}_S(R) = \acute{\mathrm{E}}\mathrm{p}(R) \times_{\mathrm{Spec}\,\mathbf{Z}} S.$$

The group $\acute{\mathrm{E}}\mathrm{p}(R)$ is called the *Chevalley group scheme of type R.*

5.12.— Thus an S-sheaf in groups G is a reductive S-group of type R if and only if it is locally isomorphic, for the étale or fpqc topology, to $\acute{\mathrm{E}}\mathrm{p}_S(R)$. Similarly, by the conjugacy theorems, giving a reductive S-group of type R with a maximal torus is locally equivalent to

$$(\acute{\mathrm{E}}\mathrm{p}_S(R), T_S(R)).$$

The analogous statements hold for Borel subgroups and Killing pairs.

6. Chevalley Systems

The explicit calculations of Section 3 have important numerical consequences.

Definition 6.1.— *Let S be a scheme and (G, T, M, R) a split S-group. A* Chevalley system *of G is a family*

$$(X_\alpha)_{\alpha \in R}, \qquad X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times,$$

satisfying:

(SC) *for every pair $\alpha, \beta \in R$,*

$$\mathrm{Ad}(w_\alpha(X_\alpha))X_\beta = \pm X_{s_\alpha(\beta)}.$$

Recall that

$$w_\alpha(X_\alpha) = \exp(X_\alpha)\exp(-X_\alpha^{-1})\exp(X_\alpha).$$

Condition (SC) implies, in particular,

$$X_{-\alpha} = \pm X_\alpha$$

by the identity

$$\mathrm{Ad}(w_\alpha(X_\alpha))X_\alpha = -X_\alpha^{-1}.$$

Proposition 6.2.— *Every split group admits a Chevalley system. More precisely, if*

$$(\Delta, (X'_\alpha)_{\alpha\in\Delta})$$

is a pinning of (G, T, M, R)*, there is a Chevalley system* $(X_\alpha)_{\alpha\in R}$ *with* $X_\alpha = X'_\alpha$ *for* $\alpha \in \Delta$*.*

First, it suffices to verify (SC) for $\alpha \in \Delta$. Indeed, for every $\alpha \in R$, choose a sequence $\alpha_i \in \Delta$ such that

$$\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}).$$

Repeated use of (SC) gives

$$X_\alpha = \pm\,\mathrm{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))\, X_{\alpha_{n+1}}.$$

By Lemma 3.1.1(iii),

$$\begin{aligned} w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}) w_{\alpha_{n+1}}(X_{\alpha_{n+1}}) w_{\alpha_n}(X_{\alpha_n})^{-1} \cdots w_{\alpha_1}(X_{\alpha_1})^{-1} \\ = \alpha^*(\pm 1) w_\alpha(X_\alpha). \end{aligned}$$

Now

$$w_{\alpha_i}(X_{\alpha_i})^{-1} = \alpha_i^*(-1) w_{\alpha_i}(X_{\alpha_i}),$$

and

$$\beta(\gamma^*(-1)) = (-1)^{(\gamma^*,\beta)} = \pm 1.$$

It follows that (SC) for every pair (α_i, γ) implies (SC) for every (α, β).

For each $\alpha \in R$, choose such a sequence and define

$$X_\alpha = \mathrm{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))\, X'_{\alpha_{n+1}}.$$

It remains to prove the following lemma.

Lemma 6.3.— *Let*

$$(G, T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$$

be a pinned S-group. If α_i*,* $0 \le i \le n+1$*, are simple roots such that*

$$s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}) = \alpha_0,$$

then

$$\mathrm{Ad}(w_{\alpha_1} \cdots w_{\alpha_n}) X_{\alpha_{n+1}} = \pm X_{\alpha_0}.$$

Arguing as in 2.3.4, using Tits's lemma, reduces the proof to:

(a) G has semisimple rank at most 2; or

(b) $w_{\alpha_1} \cdots w_{\alpha_n}$ is a section of T.

Case (a) is exactly the calculation in 3.1.2(i), 3.2.1(i), 3.3.1(i), or 3.4.1(i).

In case (b), it suffices to show that if

$$s_{\alpha_1} \cdots s_{\alpha_n} = 1,$$

then $t = w_{\alpha_1} \cdots w_{\alpha_n}$ satisfies $\alpha(t) = \pm 1$ for every root α. By the structure of the Weyl group, the word in the free group generated by the s_α lies in the normal subgroup generated by

$$(s_\alpha s_\beta)^{n_{\alpha\beta}}.$$

Thus one reduces to a word

$$s_{\alpha_1} \cdots s_{\alpha_i} (s_{\alpha_{i+1}} s_{\alpha_{i+2}})^{n_{\alpha_{i+1}\alpha_{i+2}}} s_{\alpha_i} \cdots s_{\alpha_1},$$

for which

$$t = s_{\alpha_1} \cdots s_{\alpha_i} (t_{\alpha_{i+1}\alpha_{i+2}}).$$

The assertion is now immediate from the values of $t_{\alpha_1\alpha_2}$ computed in 3.1.2(i), 3.2.1(i), 3.3.1(i), and 3.4.1(i).

Proposition 6.4.— *Let (G, T, M, R) be a split S-group with Chevalley system $(X_\alpha)_{\alpha\in R}$. Let α, β be nonproportional roots, with*

$$\text{length}(\alpha) \leq \text{length}(\beta), \qquad |(\beta^*, \alpha)| \leq |(\alpha^*, \beta)|.$$

Let q and $p-1$ be the nonnegative integers for which the roots of the form $\beta + k\alpha$ are

$$\{\beta - (p-1)\alpha, \ldots, \beta, \ldots, \beta + q\alpha\}.$$

Thus $-(\alpha^*, \beta) = q - p + 1$. Writing $p_\gamma(x) = \exp(xX_\gamma)$, the commutation relation between U_α and U_β is given by:

(p, q)	$p_\beta(y)p_\alpha(x)p_\beta(-y)p_\alpha(-x)$
$(-, 0)$	e
$(1, 1)$	$p_{\alpha+\beta}(\pm xy)$
$(1, 2)$	$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)$
$(1, 3)$	$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)p_{3\alpha+\beta}(\pm x^3y)p_{3\alpha+2\beta}(\pm x^3y^2)$
$(2, 1)$	$p_{\alpha+\beta}(\pm 2xy)$
$(2, 2)$	$p_{\alpha+\beta}(\pm 2xy)p_{2\alpha+\beta}(\pm 3x^2y)p_{\alpha+2\beta}(\pm 3xy^2)$
$(3, 1)$	$p_{\alpha+\beta}(\pm 3xy)$

These are all possibilities, since the root string has length at most 3.

Proof. By Exposé XXI, 3.5.4, there is a system of simple roots Δ such that $\alpha \in \Delta$, and there are $\alpha' \in \Delta$, $a, b \in \mathbf{Q}_+$, with

$$\beta = a\alpha + b\alpha'.$$

The desired relation lies in $U_{\alpha\alpha'}$, so it follows from the rank-two calculations of Section 3 and condition (SC).

Corollary 6.5 (Chevalley's rule).— *Let $(X_\alpha)_{\alpha\in R}$ be a Chevalley system of the split S-group G. If $\alpha, \beta, \alpha + \beta \in R$, then*

$$[X_\alpha, X_\beta] = \pm pX_{\alpha+\beta},$$

where p is the least positive integer for which $\beta - p\alpha$ is not a root.

The assertion is symmetric in α, β; after arranging $\text{length}(\alpha) \leq \text{length}(\beta)$, it follows from Proposition 6.4.

Corollary 6.6.— *Let S be a scheme such that $6 \cdot 1_S \neq 0$, and let (G, T, M, R) be a split S-group. If $R' \subset R$ is such that*

$$(*) \qquad \mathfrak{g}_{R'} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R'} \mathfrak{g}^\alpha$$

is a Lie subalgebra of $\mathfrak{g}$, then R' is a closed subset of R.

Choose a Chevalley system and let $\alpha, \beta \in R'$ with $\alpha + \beta \in R$. By Corollary 6.5 and Proposition 6.4,

$$[X_\alpha, X_\beta] = \pm p X_{\alpha+\beta}, \qquad p \in \{1, 2, 3\}.$$

Since neither 2 nor 3 is zero on S, (*) implies $\mathfrak{g}^{\alpha+\beta} \subset \mathfrak{g}_{R'}$, and hence $\alpha + \beta \in R'$. [11]

6.7.— The exact signs in this section can be determined by studying $\text{Norm}_G(T)$, and more precisely the extended Weyl group

$$W^* = N(\mathbf{Z}), \qquad N = \text{Norm}_{\text{Ép}_{\mathbf{Z}}(R)}(T_{\mathbf{Z}}(R)).$$

It is an extension of $W(R)$ by an abelian group of type $(2, 2, \ldots, 2)$, which accounts for the signs.

Remark 6.7.1.— *By 3.1.2(i) and 3.n.1, $n = 2, 3, 4$, the elements $w_\alpha, w_\beta \in N(\mathbf{Z})$ satisfy the braid relations*

$$w_\alpha w_\beta \cdots = w_\beta w_\alpha \cdots$$

with $n_{\alpha\beta}$ factors on each side.

See also Tits [Ti66], Bourbaki [BLie], IX.4, Ex. 12, and Springer [Sp98, 9.3.2].

[11] *Editors' note.* It suffices that 2 and 3 are nonzero on S; for example the result applies to $S = \text{Spec}(\mathbf{Z}/6\mathbf{Z})$. In the excluded characteristics, short-root subsystems furnish standard counterexamples.

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Exposé XXIV
Automorphisms of Reductive Groups

by Michel Demazure

[0] The first part of this Exposé (nos. 1 to 5) is a direct consequence of the existence, for a reductive group, of "sufficiently many outer automorphisms", a result which is itself a consequence of the weakest form of the isomorphism theorem for pinned groups. The second part (nos. 6 and 7) sets out two applications of the more precise results of the preceding Exposé; in particular, no. 7 uses the generators-and- relations theorem in its explicit form. Finally, we have included in an appendix (no. 8) results on "Galois" cohomology used in the text.

Let us specify our cohomological notation. If S is a scheme and G an S-group scheme, we write $H^1(S,G)$ for the first cohomology set of S with coefficients in G, computed for the (fpqc) topology; this is also the set of isomorphism classes of principal homogeneous (fpqc) sheaves under G. We write $H^1_{\text{ét}}(S,G)$ for the corresponding set for the étale topology; it is therefore the part of $H^1(S,G)$ formed by classes of homogeneous sheaves under G which are quasi-isotrivial (that is, locally trivial for the étale topology). We write $\mathrm{Fib}(S,G)$ for the part of $H^1(S,G)$ formed by classes of representable sheaves (principal homogeneous bundles). Thus we have the inclusions

$$H^1_{\text{ét}}(S,G) \subset H^1(S,G), \qquad \mathrm{Fib}(S,G) \subset H^1(S,G).$$

If every principal homogeneous sheaf under G is representable (for example, if G is quasi-affine over S; cf. SGA 1, VIII, 7.9), then

$$\mathrm{Fib}(S,G) = H^1(S,G).$$

If $S' \to S$ is a covering morphism for the (fpqc) topology, we write $H^1(S'/S,G)$ for the kernel of the canonical map

$$H^1(S,G) \longrightarrow H^1(S', G_{S'}).$$

It is known that $H^1(S'/S,G)$ can be computed simplicially (TDTE I, §A.4), which implies that, when $S' \to S$ is a covering for the étale topology, $H^1(S'/S,G)$ is also the kernel of

$$H^1_{\text{ét}}(S,G) \longrightarrow H^1_{\text{ét}}(S', G_{S'}).$$

[0]Editor's note: Version dated October 13, 2024.

Finally, following Exp. VIII, 4.5, the name "Theorem 90" is given to the following assertion: "every principal homogeneous sheaf under $G_{m,S}$ is representable and locally trivial." This assertion is equivalent to

$$H^1(S, G_{m,S}) = \mathrm{Pic}(S),$$

or again to

$$H^1(S, G_{m,S}) = 0$$

when S is local (or, more generally, semilocal).

1. Scheme of Automorphisms of a Reductive Group

1.0.

We must first make precise certain definitions from the preceding Exposé. Let S be a nonempty scheme, let G be a reductive S-group, and let

$$\mathscr{R} = (M, M^*, R, R^*, \Delta)$$

be a pinned reduced root datum (Exp. XXIII, 1.5). A *pinning of G of type $\mathscr{R}$*, or an *$\mathscr{R}$-pinning of G*, consists of:

(i) an isomorphism of $D_S(M)$ onto a maximal torus T of G (or, equivalently, a monomorphism $D_S(M) \to G$ whose image is a maximal torus T of G), identifying R with a root system of G relative to T (Exp. XIX, 3.6) and R^* with the corresponding set of coroots;

(ii) for each $\alpha \in \Delta$, an element

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

The group G possesses an $\mathscr{R}$-pinning if and only if it is split and of type $\mathscr{R}$ (Exp. XXII, 2.7).

If $u : G \to G'$ is an isomorphism of reductive S-groups, then every $\mathscr{R}$-pinning $\mathcal{E}$ of G determines, by "transport of structure", an $\mathscr{R}$-pinning $u(\mathcal{E})$ of G'. If $v : \mathscr{R}' \to \mathscr{R}$ is an isomorphism of pinned root data, then every $\mathscr{R}'$-pinning $\mathcal{E}'$ of G' determines, by transport of structure, an $\mathscr{R}$-pinning $v(\mathcal{E}')$ of G.

Call a *pinned group* a triple $(G, \mathscr{R}, \mathcal{E})$, where G is a reductive S-group, $\mathscr{R}$ is a pinned reduced root datum, and $\mathcal{E}$ is a pinning of G of type $\mathscr{R}$. An isomorphism from the pinned group $(G, \mathscr{R}, \mathcal{E})$ onto the pinned group $(G', \mathscr{R}', \mathcal{E}')$ is a pair (u, v), where $u : G \to G'$ is an isomorphism and $v : \mathscr{R}' \to \mathscr{R}$ is an isomorphism of pinned root data, such that

$$u(\mathcal{E}) = v(\mathcal{E}').$$

[1]

N.B. If S is nonempty, v is uniquely determined by u, and by an abuse of language u will also be called an isomorphism of the pinned groups. In particular, if $(G, \mathscr{R}, \mathcal{E})$ is a pinned group, an automorphism of $(G, \mathscr{R}, \mathcal{E})$ is therefore an automorphism u of G for which there exists an automorphism v of $\mathscr{R}$ such that $u(\mathcal{E}) = v(\mathcal{E})$. It is thus an automorphism of G which normalizes T, induces on T an automorphism of the form $D_S(h)$, where h is

[1] Editor's note: Thus $\mathrm{Lie}(u)(X_{v(\alpha)}) = X'_\alpha$ for every $\alpha \in R'$.

an automorphism of M, and permutes the elements X_α, $\alpha \in \Delta$. (As is readily seen, the preceding conditions in fact characterize the automorphisms of $(G, \mathscr{R}, \mathcal{E})$.)

There is an evident contravariant functor

$$\Phi : \quad (G, \mathscr{R}, \mathcal{E}) \longmapsto \mathscr{R}, \qquad (u, v) \longmapsto v,$$

and the principal result of the preceding Exposé (Exp. XXIII, 4.1) shows that this functor is fully faithful (we shall moreover see in the following Exposé that it is an equivalence of categories). In particular, it follows that the automorphism group of $(G, \mathscr{R}, \mathcal{E})$ is canonically isomorphic to the automorphism group of the pinned root datum $\mathscr{R}$ (cf. Exp. XXIII, 5.5).

1.1.

Let S be a scheme; endow $(\mathbf{Sch})/S$ with the (fpqc) topology, and consider the S-sheaf of groups $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$, where G is an S-group scheme. There is an exact sequence of S-sheaves of groups

$$1 \longrightarrow \underline{\mathrm{Centr}}(G) \longrightarrow G \xrightarrow{\mathrm{int}} \underline{\mathrm{Aut}}_{S\text{-gr.}}(G),$$

which defines a monomorphism

$$j : G/\underline{\mathrm{Centr}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G).$$

The image sheaf of j is the sheaf of inner automorphisms of G. An automorphism u of G is inner if and only if there is a covering family $\{S_i \to S\}$ and, for every i, an element $g_i \in G(S_i)$ such that

$$\mathrm{int}(g_i) = u_{S_i}.$$

In this case, if v is another automorphism of G, one sees at once that

$$\mathrm{int}(v)u = vuv^{-1}$$

is the inner automorphism defined by the family $g_i' = v(g_i)$. It follows that the image of j is normal in $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$. The quotient sheaf of groups, denoted $\underline{\mathrm{Autext}}(G)$, is the sheaf of outer automorphisms of G. Thus there is an exact sequence

$$1 \longrightarrow G/\underline{\mathrm{Centr}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

All the preceding definitions are compatible with base change. They are naturally valid in any site.

1.2.

Let S be a scheme and let $(G, \mathscr{R}, \mathcal{E})$ be a pinned reductive S-group. Let E be the (abstract) automorphism group of the pinned root datum $\mathscr{R}$, that is, the group of automorphisms of $\mathscr{R}$ which normalize Δ. By Exp. XXIII, 5.5, there is a canonical monomorphism

$$E \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$$

which associates with $h \in E$ the unique automorphism u of the pinned group G such that $\Phi(u) = h$. This monomorphism canonically defines a monomorphism of sheaves

$$(*) \qquad a : E_S \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G).$$

An automorphism u of G is a section of the image sheaf of a if and only if the following conditions hold:

(i) u normalizes T. It is then known that u permutes the roots of G relative to T. If $\alpha \in R$, then

$$u(\alpha) : t \longmapsto \alpha(u^{-1}(t))$$

is therefore locally on S of the form $t \longmapsto \beta(t)$, with $\beta \in R$. The second condition may then be written as follows:

(ii) If $\alpha \in \Delta$ and U is an open subset of S such that $u(\alpha)_U \in R$, then $u(\alpha)_U \in \Delta$ and

$$\mathrm{Lie}(u_U)(X_\alpha)_U = (X_{u(\alpha)})_U.$$

It follows at once from the definitions that the sections of $a(E_S)$ normalize the subgroups of G defined by the pinning:

$$T, \quad B, \quad B^-, \quad U, \quad U^-.$$

With these definitions in place, we have:

Theorem 1.3. *Let S be a scheme and G a reductive S-group. Consider the canonical exact sequence of S-sheaves of groups* [2]

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \xrightarrow{p} \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

Then:

(i) $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable by a smooth separated S-scheme;

(ii) $\underline{\mathrm{Autext}}(G)$ is representable by a twisted constant S-scheme of finite generation (Exp. X, 5.1);

(iii) if G is split, the preceding exact sequence is split. More precisely, for every pinning of G, the morphism (cf. 1.2 $(*)$)

$$p \circ a : E_S \longrightarrow \underline{\mathrm{Autext}}(G)$$

is an isomorphism.

Let us first show how the theorem follows from the following lemma.

Lemma 1.4. *Under the hypotheses of* (iii), $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ *is the semidirect product*

$$a(E_S) \cdot \mathrm{ad}(G).$$

[3]

The lemma immediately implies the theorem when G is split. Since G is locally split for the étale topology (Exp. XXII, 2.3), hence also for the (fppf) topology, and since the latter is "of effective descent" for the fibred category of twisted constant morphisms (Exp. X, 5.5), assertion (ii) follows in the general case (cf. Exp. IV, 4.6.8). To deduce (i), observe that $\mathrm{ad}(G)$ is affine over S; thus p is affine whenever $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable, and the result follows by descent for affine schemes. [4]

It remains only to prove 1.4. For this, it suffices to prove:

[2] Editor's note: Recall that $\mathrm{ad}(G) = G/\underline{\mathrm{Centr}}(G)$ denotes the adjoint group of G; cf. Exp. XXII, 4.3.6.
[3] Editor's note: Here and below, the notation $\mathrm{int}(G)$ has been replaced by $\mathrm{ad}(G)$.
[4] Editor's note: Cf. SGA 1, VIII, 2.1.

Lemma 1.5. *If $(\mathscr{R}, \mathcal{E})$ and $(\mathscr{R}', \mathcal{E}')$ are two pinnings of the reductive S-group G, there exists a unique inner automorphism u of G over S carrying one pinning to the other; that is, such that there exists an isomorphism*

$$v : \mathscr{R}' \xrightarrow{\sim} \mathscr{R}$$

for which $u(\mathcal{E}) = v(\mathcal{E}')$; cf. 1.0.

1.5.1. Uniqueness.

It suffices to prove that, if G is a pinned S-group and $\mathrm{int}(g)$ is an automorphism of the pinned group ($g \in G(S)$), then $\mathrm{int}(g) = \mathrm{id}$. First,

$$\mathrm{int}(g)T = T, \qquad \mathrm{int}(g)B = B,$$

and hence

$$g \in \underline{\mathrm{Norm}}_G(T)(S) \cap \underline{\mathrm{Norm}}_G(B)(S) = T(S)$$

(cf., for example, Exp. XXII, 5.6.1). It follows that $\mathrm{int}(g)$ normalizes every U_α and that

$$\mathrm{Lie}(\mathrm{int}(g))X_\alpha = \mathrm{Ad}(g)X_\alpha = \alpha(g)X_\alpha$$

for every $\alpha \in \Delta$. Thus $\alpha(g) = 1$ for $\alpha \in \Delta$, and therefore

$$g \in \bigcap_{\alpha \in \Delta} (\ker \alpha)(S) = \underline{\mathrm{Centr}}(G)(S)$$

(Exp. XXII, 4.1.8). □

1.5.2. Existence.

It suffices to prove existence locally for the (fpqc) topology. Let

$$(T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta}) \quad \text{and} \quad (T', M', R', \Delta', (X'_{\alpha'})_{\alpha' \in \Delta'})$$

be the two pinnings. By conjugacy of maximal tori, we may assume $T = T'$. After restricting S, we may assume that the isomorphism

$$D_S(M) \simeq D_S(M')$$

comes from an isomorphism $M \simeq M'$ carrying R onto R'; we are thus reduced to the situation

$$T = T', \qquad M = M', \qquad R = R'.$$

Since the systems of simple roots are conjugate under the Weyl group (Exp. XXI, 3.3.7), we may also assume $\Delta = \Delta'$. For every $\alpha \in \Delta$, there is then a scalar $z_\alpha \in \mathbb{G}_m(S)$ such that

$$X'_\alpha = z_\alpha X_\alpha,$$

and it is enough to construct, locally for the (fpqc) topology, a section t of T such that $\alpha(t) = z_\alpha$ for every $\alpha \in \Delta$. But the morphism

$$T \longrightarrow (\mathbb{G}_{m,S})^\Delta$$

whose components are the α, $\alpha \in \Delta$, is dual to an injection $\mathbb{Z}^\Delta \to M$, and is therefore faithfully flat. This completes the proof of 1.5.2, and hence of 1.3.

Corollary 1.6. *Let S be a scheme and G a reductive S-group. The following conditions are equivalent:*

(i) $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is affine (respectively of finite type, respectively of finite presentation, respectively quasi-compact) over S;

(ii) $\underline{\mathrm{Autext}}(G)$ is finite over S;

(iii) for every $s \in S$,
$$\mathrm{rg}_{\mathrm{red}}(G_s) - \mathrm{rg}_{\mathrm{ss}}(G_s) \leq 1.$$

Indeed, since $\mathrm{ad}(G)$ is affine, flat, and of finite presentation over S, the morphism
$$p : \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\mathrm{Autext}}(G)$$
is affine, faithfully flat, and of finite presentation.

If $\underline{\mathrm{Autext}}(G)$ is finite over S, it is affine over S, and hence so is $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$; this proves (ii)⇒(i). If $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is quasi-compact over S, it is of finite presentation over S, since it is in any case locally of finite presentation and separated over S. By Exp. V, 9.1, $\underline{\mathrm{Autext}}(G)$ is then of finite presentation over S, and hence finite; this proves (i)⇒(ii). Finally, to prove the equivalence of (ii) and (iii), one may suppose that G is split, and the assertion reduces to Exp. XXI, 6.7.8.

Corollary 1.7. *Let S be a scheme and G a reductive S-group. Then*
$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)^0 \simeq \mathrm{ad}(G).$$

Corollary 1.8. *Let S be a scheme, G a reductive S-group, and H an S-group scheme that is smooth, affine, and has connected fibres over S. Then the S-functor*
$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H)$$
is representable by a smooth separated S-scheme, which is affine over S if G is semisimple.

Indeed, let U be the set of points $s \in S$ for which H_s is reductive; this is an open subset (Exp. XIX, 2.6). If S' is an S-scheme, $H_{S'}$ is reductive if and only if $S' \to S$ factors through U. It follows that the canonical morphism
$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H) \longrightarrow S$$
factors through U. We may therefore assume $S = U$, and we are reduced to the following corollary.

Corollary 1.9. *Let S be a scheme, and let G and G' be two reductive S-groups. Then*
$$F = \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$$
is representable by a smooth separated S-scheme, affine if G or G' is semisimple. Moreover, S is the disjoint union of two open subschemes S_1 and S_2 such that $F_{S_1} = \varnothing$, while F_{S_2} is a principal homogeneous bundle on the left (respectively, on the right) *under* $\underline{\mathrm{Aut}}_{S_2\text{-gr.}}(G'_{S_2})$ (respectively, $\underline{\mathrm{Aut}}_{S_2\text{-gr.}}(G_{S_2})$).

Indeed, let S_2 be the set of points of S at which G and G' have the same type, and let S_1 be its complement. Since the type of a reductive group is a locally constant function,

S_1 and S_2 are open. Clearly $F_{S_1} = \varnothing$, and we may assume $S = S_2$. By Exp. XXIII, 5.6, F is a principal homogeneous sheaf under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$, locally trivial for the étale topology. It follows that

$$F_0 = F/\operatorname{ad}(G)$$

is a principal homogeneous sheaf under $\underline{\mathrm{Autext}}(G)$, locally trivial for the étale topology, and hence representable (Exp. X, 5.5). Since $\operatorname{ad}(G)$ is affine, F is therefore representable as well.

We note that the proof also gives the following result.

Corollary 1.10. *Let S be a scheme, and let G and G' be two reductive S-groups having the same type at every $s \in S$. Then $\operatorname{ad}(G)$ acts freely on the right on $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$. The quotient sheaf*

$$\underline{\mathrm{Isomext}}(G, G') = \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')/\operatorname{ad}(G)$$

is representable by a twisted constant S-scheme; it is a principal homogeneous bundle under $\underline{\mathrm{Autext}}(G)$, and is therefore finite over S if G is semisimple. Moreover, the isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \simeq \underline{\mathrm{Isom}}_{S\text{-gr.}}(G', G), \qquad u \longmapsto u^{-1},$$

induces an isomorphism

$$\underline{\mathrm{Isomext}}(G, G') \simeq \underline{\mathrm{Isomext}}(G', G).$$

Remark 1.11. If $\underline{\mathrm{Isomext}}(G, G')(S) \neq \varnothing$, one says that G is an *inner twisted form* of G'. Then G' is an inner twisted form of G, and the structure group of $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ can be reduced to $\operatorname{ad}(G)$. More precisely, let

$$u \in \underline{\mathrm{Isomext}}(G, G')(S)$$

and regard u as a section $u : S \to \underline{\mathrm{Isomext}}(G, G')$. Denote by

$$\underline{\mathrm{Isomint}}_u(G, G')$$

the inverse image, under the canonical morphism

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \longrightarrow \underline{\mathrm{Isomext}}(G, G'),$$

of the closed subscheme of $\underline{\mathrm{Isomext}}(G, G')$ that is the image of u. The natural action of $\operatorname{ad}(G)$ on $\underline{\mathrm{Isomint}}_u(G, G')$ makes this scheme a principal homogeneous bundle. By extension of the structure group[5]

$$\operatorname{ad}(G) \longrightarrow \underline{\mathrm{Aut}}(G),$$

one recovers $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ from $\underline{\mathrm{Isomint}}_u(G, G')$.

By Hensel's lemma (Exp. XI, 1.11), Corollary 1.8 immediately gives:

[5]The source's editorial note corrects $\operatorname{ad}(G) \to \underline{\mathrm{Autext}}(G)$ to $\operatorname{ad}(G) \to \underline{\mathrm{Aut}}(G)$.

Corollary 1.12. *Let S be a henselian local scheme, let G be a reductive S-group, let G' be a smooth affine S-group with connected fibres, and let s be the closed point of S. If G_s and G'_s are isomorphic algebraic groups over $\kappa(s)$, then G and G' are isomorphic. More precisely, every $\kappa(s)$-isomorphism*

$$G_s \simeq G'_s$$

comes from an S-isomorphism $G \simeq G'$.

Applying Corollary 1.7 (respectively, Corollary 1.12) to the scheme of dual numbers over a field, Exp. III, 2.10 (respectively, 3.10) yields part (i) (respectively, part (ii)) of the next corollary.

Corollary 1.13. *Let k be a field and G a reductive k-group.*

(i) *If G is adjoint, then*

$$H^1(G, \mathrm{Lie}(G/k)) = 0.$$

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(ii) *One has*

$$H^2(G, \mathrm{Lie}(G/k)) = 0.$$

Remark 1.14.

(i) The assertion concerning H^1 was known to Chevalley; the assertion concerning H^2 had been proved in most cases of the classification by Chevalley.

(ii) In fact, the conjunction of Corollary 1.13 and the uniqueness theorem over an algebraically closed field is essentially equivalent to the uniqueness theorem. A direct proof of Corollary 1.13 would therefore give a way to deduce the general uniqueness theorem from Chevalley's uniqueness theorem over a field.

The existence of reductive groups of every type over every scheme (Exp. XXV) shows that the obstructions to lifting a reductive k-group G over artinian rings with residue field k, which by Exp. III, 3.8 are elements of

$$H^3(G, \mathrm{Lie}(G/k) \otimes V) \simeq H^3(G, \mathrm{Lie}(G/k)) \otimes V,$$

where V is a certain k-vector space endowed with the trivial action of G, vanish. This suggests that

$$H^3(G, \mathrm{Lie}(G/k)) = 0.$$

A direct proof, if the assertion is true, would likewise doubtless give a way to deduce the general existence theorem from the existence theorem over a field (Chevalley's Tôhoku).

[6] The source's editorial note corrects the original, which stated (i) without assuming G adjoint. By the description of $H^i(G, -)$ as the derived functors of $\mathrm{Hom}_G(k, -)$ (Exp. I, 5.3.1; see also [Ja87], I, 4.16), one has $H^i(G, V) = \mathrm{Ext}^i_G(k, V)$ for every G-module V. If $\mathrm{char}(k) = p > 0$ and $G = \mathrm{SL}_{p,k}$, then $\mathrm{Lie}(\mathrm{GL}_p/k)$ is a nontrivial extension of k by $\mathfrak{g} = \mathrm{Lie}(\mathrm{SL}_p/k)$, so $H^1(G, \mathfrak{g}) \neq 0$. See also the addition 1.15.1 below.

Corollary 1.15. *Let k be a field,*[7] *and let G be a reductive k-group. Regard k as a trivial G-module. Then*

$$H^1(G,k) = H^2(G,k) = 0.$$

[8] Since

$$H^i(G_{\bar{k}},\bar{k}) = H^i(G,k)\otimes_k \bar{k},$$

we may assume k algebraically closed. An element of $H^1(G,k)$ is simply a morphism of k-groups

$$\varphi : G \longrightarrow \mathbb{G}_{a,k}.$$

Then

$$\varphi(G) = G/\ker(\varphi)$$

is a smooth connected reductive subgroup of $\mathbb{G}_{a,k}$ (cf. XIX, 1.7), and is therefore trivial. Thus $H^1(G,k) = 0$. Now consider the reductive k-group

$$H = G \times_k \mathbb{G}_{m,k}.$$

There is an H-stable decomposition

$$\mathrm{Lie}(H/k) = \mathrm{Lie}(G/k) \oplus k.$$

For every H-module V,

$$H^i(H,V) = H^i\big(G, H^0(\mathbb{G}_{m,k},V)\big).$$

Indeed, the $H^i(H,-)$ are the derived functors of

$$H^0(H,-) = H^0\big(G, H^0(\mathbb{G}_{m,k},-)\big),$$

and the functor $H^0(\mathbb{G}_{m,k},-)$ is exact (cf. Exp. I, 5.3.1 and 5.3.3). In particular, for every i,

$$H^i(H,\mathrm{Lie}(H/k)) = H^i(G,\mathrm{Lie}(G/k)) \oplus H^i(G,k).$$

Applying 1.13(ii) to the reductive group H gives $H^2(G,k) = 0$.

Corollary 1.15.1. — [9] *Let k be a field, G a reductive k-group, Z its center, and $\mathscr{R} = (M, M^*, R, R^*)$ the type of G. Then*

$$H^1(G,\mathrm{Lie}(G/k)) \simeq \mathrm{Ext}^1_{\mathbb{Z}}(M/\Gamma_0(R),k).$$

In particular, $H^1(G,\mathrm{Lie}(G/k)) = 0$ if and only if Z is smooth over k.

Let $\mathfrak{g}$, $\mathfrak{z}$, and $\mathfrak{g}_{\mathrm{ad}}$ be the Lie algebras of G, Z, and $G_{\mathrm{ad}} = G/Z$, respectively. It follows from 1.15 and its proof that

$$H^1(G,\mathfrak{g}) = H^1(G_{\mathrm{ad}},\mathfrak{g}) \simeq H^1(G_{\mathrm{ad}},\mathfrak{g}/\mathfrak{z}).$$

Set $C = \mathrm{Coker}(\mathfrak{g} \to \mathfrak{g}_{\mathrm{ad}})$. There is an exact sequence

$$0 \longrightarrow \mathfrak{g}/\mathfrak{z} \longrightarrow \mathfrak{g}_{\mathrm{ad}} \longrightarrow C \longrightarrow 0.$$

[7] The source's editorial note replaces "scheme" with "field."

[8] Because of the correction made in 1.13, the source's editors supply another proof for H^1. Moreover, a theorem of G. Kempf gives $H^i(G,k) = 0$ for every $i > 0$; cf. [Ja87], II, 4.5 and 4.11.

[9] Editor's note: This corollary, drawn from remarks of O. Gabber, has been added to make 1.13(i) more precise.

Since $H^1(G_{\mathrm{ad}}, \mathfrak{g}_{\mathrm{ad}}) = 0$ by 1.13 and

$$H^0(G_{\mathrm{ad}}, \mathfrak{g}_{\mathrm{ad}}) = 0$$

(cf. Exp. II, 5.2.3), we obtain

$$H^1(G_{\mathrm{ad}}, \mathfrak{g}/\mathfrak{z}) \simeq H^0(G_{\mathrm{ad}}, C).$$

To calculate the right-hand side, we may assume k algebraically closed. Let (G, T, M, R) be a splitting of G. Then

$$Z = D_k(M/\Gamma_0(R))$$

and C is identified with

$$\mathrm{Coker}(\mathrm{Hom}_{\mathbb{Z}}(M, k) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\Gamma_0(R), k)) \simeq \mathrm{Ext}^1_{\mathbb{Z}}(M/\Gamma_0(R), k),$$

with trivial G_{ad}-action. The corollary follows, since Z is not smooth over k exactly when $\mathrm{char}(k) = p > 0$ and $M/\Gamma_0(R)$ has p-torsion (Exp. IX, 2.1).

Definition 1.16. — *Let S be a scheme and G a reductive S-group. A* form of G over S *is an S-group scheme G' locally isomorphic to G for the* (fpqc) *topology. It is equivalent to require that G' be locally isomorphic to G for the étale topology (cf. Exp. XXIII, 5.6), or that G' be a reductive S-group of the same type as G at every point of S.*

Corollary 1.17. — *Let S be a scheme and G a reductive S-group.*

(i) The functor

$$G' \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$$

is an equivalence between the category of forms of G over S and the category of principal homogeneous bundles under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$.

(ii) If $S' \to S$ is a covering morphism, forms of G trivialized by S' correspond to bundles trivialized by S'.

(iii) Every principal homogeneous sheaf under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$ is representable and quasi-isotrivial, that is, locally trivial for the étale topology.

The first assertion is formal in the category of sheaves, for example for the (fpqc) topology. Every sheaf locally isomorphic to G for the (fpqc) topology is representable, since G is affine over S, and is locally isomorphic to G for the étale topology. Finally, for every form G' of G, the S-sheaf $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ is representable by 1.8. The corollary follows.

Corollary 1.18. — *The set of isomorphism classes of forms of the reductive group G over S is isomorphic to*

$$H^1\Big(S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big) = H^1_{\text{ét}}\Big(S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big) = \mathrm{Fib}\Big(S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big).$$

If $S' \to S$ is a covering morphism, the subset consisting of forms trivialized by S' is isomorphic to

$$H^1\Big(S'/S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big).$$

Corollary 1.19. — *Let S be a scheme and $\mathscr{R}$ a pinned reduced root datum such that $\mathrm{Ép}_S(\mathscr{R})$ exists,*[10] *a condition which is automatically satisfied by Exp. XXV. Put*

$$A_S(\mathscr{R}) = \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{Ép}_S(\mathscr{R})) = \mathrm{ad}(\mathrm{Ép}_S(\mathscr{R})) \cdot E(\mathscr{R})_S.$$

[10] Editor's note: Cf. XXIII, Definition 5.11.

(i) The set of isomorphism classes of reductive S-groups of type $\mathscr{R}$ (Exp. XXII, 2.7) is isomorphic, by Exp. XXIII, 5.12, to

$$H^1(S, A_S(\mathscr{R})) = H^1_{\text{ét}}(S, A_S(\mathscr{R})) = \mathrm{Fib}(S, A_S(\mathscr{R})).$$

(ii) If $S' \to S$ is a covering morphism, the subset consisting of classes of groups splittable over S' is isomorphic to $H^1(S'/S, A_S(\mathscr{R}))$.

Remark 1.20. — With the preceding notation, every reductive S-group G of type $\mathscr{R}$ has a canonically associated right principal homogeneous bundle under $A_S(\mathscr{R})$:

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}), G) = P.$$

The scheme P may be interpreted as the "scheme of pinnings of G of type $\mathscr{R}$" (cf. 1.0). It is also a principal homogeneous bundle on the left under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$, a structure visible immediately from the description above.

Proposition 1.21. — *Let S be a henselian local scheme and s its closed point. The functor*

$$G \longmapsto G_s$$

induces a bijection from the set of isomorphism classes of reductive S-groups onto the set of isomorphism classes of reductive $\kappa(s)$-groups. In particular, for every reductive S-group G, there exists a finite étale surjective morphism $S' \to S$ such that $G_{S'}$ is splittable.

Using the existence of $\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})$ (Exp. XXV), it is enough to prove that, with $H = A_S(\mathscr{R})$, the canonical map

$$\mathrm{Fib}(S, H) \longrightarrow \mathrm{Fib}(\kappa(s), H_s)$$

is bijective, and that every element of $\mathrm{Fib}(S, H)$ has the property just stated. Every finite subset of H is contained in an affine open subscheme: this is clear for a constant group, and H is affine over a constant group. The result proved in the Appendix, 8.1, therefore applies.

2. Automorphisms and Subgroups

Introduce the following notation. If

$$\mathcal{H} = \underline{\mathrm{Aut}}_{S\text{-gr}}(G)$$

and X is a subfunctor of G, set

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G, X) = \mathrm{Norm}_{\mathcal{H}}(X), \qquad \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_X) = \underline{\mathrm{Centr}}_{\mathcal{H}}(X).$$

If Y is a second subfunctor of G, define

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G, X, Y) = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, X) \cap \underline{\mathrm{Aut}}_{S\text{-gr}}(G, Y).$$

If G' is a second S-group and X' a subfunctor of G', write

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, X; G', X')$$

for the subfunctor of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ defined, for every $S' \to S$, by

$$\left\{ u \in \underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')(S') \,\middle|\, u(X_{S'}) = X'_{S'} \right\}.$$

The notation $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, X, Y; G', X', Y')$, and so on, is defined similarly.[11]

Proposition 2.1. — *Let S be a scheme, G a reductive S-group, and T a maximal torus of G (respectively, B a Borel subgroup of G, or $B \supset T$ a Killing couple of G). Let T_{ad} (respectively, B_{ad}) denote the corresponding maximal torus (respectively, Borel subgroup) of* $\mathrm{ad}(G)$*:*

$$B_{\mathrm{ad}} \simeq B/\underline{\mathrm{Centr}}(G) = B/\underline{\mathrm{Centr}}(B), \qquad T_{\mathrm{ad}} \simeq T/\underline{\mathrm{Centr}}(G).$$

Then $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$ (respectively, $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B)$, or $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T)$) is represented by a closed subscheme of $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$, smooth over S, and the exact sequence of Theorem 1.3 induces the exact sequences

$$1 \longrightarrow \mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}}) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, T) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1,$$

$$1 \longrightarrow B_{\mathrm{ad}} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1,$$

$$1 \longrightarrow T_{\mathrm{ad}} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

By descent of closed subschemes,[12] we reduce immediately to the case in which G is pinned and $B \supset T$ is its canonical Killing couple (cf. Exp. XXII, 5.5.5(iv)). Since the group E of 1.2 normalizes B and T, the result follows at once from the normalization theorems in $\mathrm{ad}(G)$ (Exp. XXII, 5.3.12 and 5.6.1).

Using now the conjugacy theorems (cf. Exp. XXIII, 5.12), and arguing as in §1, one obtains:

Corollary 2.2. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, and let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G').*

(i) *The S-functor*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')$$

is represented by a closed smooth subscheme of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ which is a principal homogeneous space under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$. Moreover, $\mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}})$ acts freely on this scheme, and there is a canonical isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')/\mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}}) \simeq \underline{\mathrm{Isomext}}(G, G').$$

(ii) *The S-functor*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B; G', B')$$

is represented by a closed smooth subscheme of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ which is a principal homogeneous space under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B)$. Moreover, B_{ad} acts freely on this scheme, and there is a canonical isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B; G', B')/B_{\mathrm{ad}} \simeq \underline{\mathrm{Isomext}}(G, G').$$

(iii) *The S-functor*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B, T; G', B', T')$$

is represented by a closed smooth subscheme of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$, which is a principal homogeneous space under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T)$. Moreover, T_{ad} acts freely on this scheme, and there is a canonical isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B, T; G', B', T')/T_{\mathrm{ad}} \simeq \underline{\mathrm{Isomext}}(G, G').$$

[11] Editor's note: The preceding definitions have been made explicit. The original only said that $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, X, Y), \ldots$ and $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, X; G', X'), \ldots$ were defined similarly.

[12] Editor's note: Cf. SGA 1, VIII, 1.9.

Arguing once again as in §1, one obtains:

Corollary 2.3. — *Let S be a scheme, let G be a reductive S-group, and let $B \supset T$ be a Killing couple of G. The functor*

$$(G', T') \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T'),$$

respectively

$$(G', B') \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, B; G', B'),$$

respectively

$$(G', B', T') \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, B, T; G', B', T'),$$

is an equivalence between the category of pairs (G', T') (respectively, pairs (G', B'), respectively, triples (G', B', T')), where G' is a form of G and T' is a maximal torus of G' (respectively, B' is a Borel subgroup of G', respectively, $B' \supset T'$ is a Killing couple of G'), and the category of principal homogeneous bundles under the S-group H, where $H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$ (respectively, $H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B)$, respectively, $H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T)$). Moreover, every principal homogeneous sheaf under H is representable and quasi-isotrivial, so that

$$H^1(S, H) = H^1_{\text{ét}}(S, H) = \mathrm{Fib}(S, H).$$

Remark 2.4. — Under the hypotheses of 2.2, the morphism denoted set-theoretically by $u \mapsto u(T)$ (respectively, $u \mapsto u(B)$, respectively, $u \mapsto (u(B), u(T))$) induces an isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')/\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T) \simeq \mathrm{Tor}(G'),$$

respectively,

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')/\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B) \simeq \mathrm{Bor}(G'),$$

respectively,

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')/\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T) \simeq \mathrm{Kil}(G').$$

The proof is immediate: it suffices to prove the assertion locally for (fpqc), so one may assume $G \simeq G'$, and the result then reduces to Exp. XXII, 5.8.3(iii).

Remark 2.5. — The preceding results are immediately interpreted in terms of reduction of the structure group. If G' is a form of G, corresponding to the principal bundle $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$, to give a reduction of the structure group of this bundle to $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$ amounts to giving a maximal torus T' of G'. The indicated bijections are, on the one hand, that of 2.4 and, on the other hand, the map

$$T' \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T').$$

The same applies to Borel subgroups and Killing couples.

Proposition 2.6. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, and let T (respectively, T') be a maximal torus of G (respectively, G'). Then T_{ad} acts freely on $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')$, and the quotient*

$$P = \underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')/T_{\mathrm{ad}}$$

is representable; it is a principal homogeneous bundle under

$$A = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)/T_{\mathrm{ad}},$$

where A is represented by a twisted constant S-scheme and is an extension of $\underline{\mathrm{Autext}}(G)$ *by*

$$W_{\mathrm{ad}(G)}(T_{\mathrm{ad}}) = \mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}})/T_{\mathrm{ad}}.$$

Moreover, if A is made to act on T in the evident way, the bundle associated with P is precisely T'.[13]

The first part of the proposition follows immediately from the preceding results. To prove the second, observe that there is an evident morphism

$$P \times_S T \longrightarrow T', \qquad (u,t) \longmapsto u(t).$$

To show that, after passage to the quotient by A, it induces an isomorphism, one may once again assume $(G,T) \simeq (G',T')$, in which case the assertion is immediate.

In an entirely analogous way, one has:

Proposition 2.7. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, and let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G'). If*

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G,B,T)/T_{\mathrm{ad}} \simeq \underline{\mathrm{Autext}}(G)$$

is made to act on T in the evident way, the bundle associated with $\underline{\mathrm{Isomext}}(G,G')$ *is precisely T'.*

Corollary 2.8. — *Let G and G' be two reductive S-groups which are inner twisted forms of one another, and let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G'). Then T and T' are isomorphic.*[14]

Remark 2.9. — It is not true in general that B and B' are isomorphic; they are nevertheless inner twisted forms of one another (cf. §5).

One may develop "Isomint" variants[15] of the preceding results. We mention one:

Proposition 2.10. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G'), and let $u \in \underline{\mathrm{Isomext}}(G,G')(S)$. Consider*

$$\begin{aligned}&\underline{\mathrm{Isomint}}_u(G,B,T;G',B',T')\\ &\quad = \underline{\mathrm{Isom}}_{S\text{-gr}}(G,B,T;G',B',T') \cap \underline{\mathrm{Isomint}}_u(G,G').\end{aligned}$$

It is a smooth affine S-scheme which is a principal homogeneous bundle under T_{ad}. In particular,

$$\underline{\mathrm{Isomint}}_u(G,B,T;G',B',T')(S) \neq \varnothing$$

if and only if the corresponding element of $H^1(S,T_{\mathrm{ad}})$ is zero.

To conclude this section, we prove two results that will be useful later.

Proposition 2.11. — *Let S be a scheme, let G be a reductive S-group, and let T be a maximal torus of G. The evident morphism*

$$T_{\mathrm{ad}} \xrightarrow{\sim} \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_T)$$

[13] Editor's note: Consequently, if P has a section, then T and T' are isomorphic; in particular, T_P and T'_P are isomorphic. See the addition 4.5.1, where this observation is developed in the case $G = \underline{\mathrm{Ép}}_S(\mathscr{R})$.

[14] Editor's note: This follows from 2.7, since $\underline{\mathrm{Isomext}}(G,G')$ has a section (cf. the preceding editor's note).

[15] Editor's note: Recall that $\underline{\mathrm{Isomint}}_u(G,G')$ is defined in 1.11.

is an isomorphism.

This follows from the preceding statements. A direct proof was also given in the course of the proof of 1.5.2.

Corollary 2.12. — *Under the preceding hypotheses, there is an equivalence between the category of pairs (G', f), where G' is a form of G and f is an isomorphism from T onto a maximal torus of G', and the category of principal homogeneous bundles under T_{ad}.*

Corollary 2.13. — *If $H^1(S, T_{\mathrm{ad}}) = 0$, and if G' is a form of G having a maximal torus isomorphic to T, then G' is isomorphic to G.*

Corollary 2.14. — *Let S be a scheme such that $\mathrm{Pic}(S) = 0$, and let G be a reductive S-group of constant type. Then G is splittable if and only if G has a split maximal torus.*

Let G be a reductive group and let $\mathrm{rad}(G)$ be its radical (Exp. XXII, 4.3.9). Since $\mathrm{rad}(G)$ is central and characteristic in G, there is a canonical morphism

$$q : \underline{\mathrm{Autext}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{rad}(G)).$$

Proposition 2.15. — *Let G be a reductive S-group. The following sequence is exact:*

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_{\mathrm{rad}(G)}) \xrightarrow{p} \underline{\mathrm{Autext}}(G) \xrightarrow{q} \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{rad}(G)),$$

and $\ker(q) = \mathrm{im}(p)$ is an open and closed subscheme of $\underline{\mathrm{Autext}}(G)$, finite over S.

One may assume that G is split. The first assertion is immediate, and the second follows from Exp. XXI, 6.7.5 and 6.7.7.

Writing

$$H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_{\mathrm{rad}(G)}),$$

one obtains:

Corollary 2.16. — *There is an equivalence between the category of pairs (G', f), where G' is a form of G and f is an isomorphism from $\mathrm{rad}(G)$ onto the radical of G', and the category of principal bundles under a certain S-group scheme H, where H is such that there is an exact sequence*

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow H \longrightarrow F \longrightarrow 1,$$

in which the S-group F is étale and finite over S.

3. Dynkin Scheme of a Reductive Group. Quasi-Split Groups

3.1. — Recall (Exp. XXI, 7.4.1) that a Dynkin diagram is a finite set endowed with the structure defined by a set of pairs of distinct elements (bonds) and a map to $\{1, 2, 3\}$ (lengths). To every pinned reduced root datum $\mathscr{R}$ there is associated a Dynkin diagram $\Delta(\mathscr{R})$, whose underlying set is the set of simple roots.

3.2. — Let S be a scheme. A Dynkin S-scheme is a finite twisted constant S-scheme Δ, endowed with the structure defined by a subscheme L of $\Delta \times_S \Delta$ having empty intersection with the diagonal, and by a morphism

$$\Delta \longrightarrow \{1, 2, 3\}_S.$$

If S' is a connected S-scheme,[16] $\Delta(S')$ is naturally endowed with the structure of a Dynkin diagram.

[16]Editor's note: The hypothesis that S' be connected has been added.

The following notions are defined immediately: an isomorphism of two Dynkin schemes, the base extension of a Dynkin scheme, and the constant Dynkin scheme associated with a Dynkin diagram.

Every étale descent datum on a Dynkin scheme is effective.

3.3. — We shall associate a Dynkin S-scheme with every reductive S-group G. First suppose that G is splittable over S. For every pinning E of G, let $\Delta(E)$ be the constant Dynkin scheme associated with the pinned root datum defined by E. If E and E' are two pinnings of G, then by 1.5 there is a unique inner automorphism of G over S carrying E to E'. This automorphism of G defines an isomorphism

$$a_{EE'} : \Delta(E) \xrightarrow{\sim} \Delta(E').$$

The $a_{EE'}$ plainly form a transitive system, so the $\Delta(E)$ may be identified (that is, one takes their inductive limit); the result is a constant Dynkin scheme denoted by $\mathrm{Dyn}(G)$. If now G is an arbitrary reductive S-group, there is an étale covering family $\{S_i \to S\}$ such that

G_{S_i} is splittable. Arguing as above, one therefore obtains a canonical descent datum on the $\mathrm{Dyn}(G_{S_i})$, and by descent this constructs a Dynkin S-scheme $\mathrm{Dyn}(G)$.

3.4. — This construction has the following properties (which, moreover, essentially characterize it):

(i) With every reductive S-group G is associated a Dynkin scheme $\mathrm{Dyn}(G)$; with every isomorphism

$$u : G \xrightarrow{\sim} G'$$

is functorially associated an isomorphism

$$\mathrm{Dyn}(u) : \mathrm{Dyn}(G) \xrightarrow{\sim} \mathrm{Dyn}(G').$$

(ii) If S' is an S-scheme and G is a reductive S-group, then

$$\mathrm{Dyn}(G \times_S S') \simeq \mathrm{Dyn}(G) \times_S S'.$$

(iii) If E is a pinning of G defining the pinned root datum with Dynkin diagram Δ, then

$$\mathrm{Dyn}(G) \simeq \Delta_S.$$

(iv) If u is an inner automorphism of G, then $\mathrm{Dyn}(u)$ is the identity automorphism of $\mathrm{Dyn}(G)$.

3.5. — Let S be a scheme and G a reductive S-group. It is clear that the functor

$$\underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G))$$

of automorphisms of $\mathrm{Dyn}(G)$ as a Dynkin scheme is represented by a finite twisted constant S-scheme. By 3.4(i) and (ii), there is a canonical morphism

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G)),$$

which, by (iv), factors through a morphism

$$\underline{\mathrm{Autext}}(G) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G)).$$

More generally, if G and G' are two reductive S-groups, there is a canonical morphism

$$\underline{\mathrm{Isomext}}(G, G') \longrightarrow \underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G), \mathrm{Dyn}(G')).$$

In particular, if G' is an inner twisted form of G (1.11), the Dynkin schemes $\mathrm{Dyn}(G)$ and $\mathrm{Dyn}(G')$ are isomorphic.

3.6. — If G is semisimple (respectively, adjoint or simply connected), the morphism

$$\underline{\mathrm{Autext}}(G) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G))$$

is a monomorphism (respectively, an isomorphism). Indeed, one may assume that G is pinned, and the assertion reduces to the corresponding result for pinned reduced root data (cf. Exp. XXI, 7.4.5).

There is an analogous result for the Isom functors. It follows in particular that two semisimple adjoint (respectively, simply connected) S-groups are inner twisted forms of one another if and only if their Dynkin schemes are isomorphic.

3.7. — One can give a different construction of the Dynkin scheme associated with a reductive group. Let $\mathscr{R}$ be a pinned reduced root datum and G a reductive S-group of type $\mathscr{R}$, and write $\Delta(\mathscr{R})$ for the Dynkin diagram defined by $\mathscr{R}$. There is, by 3.5, a canonical morphism

$$A_S(\mathscr{R}) = \underline{\mathrm{Aut}}_{S\text{-gr}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\Delta(\mathscr{R})_S).$$

The reductive S-group G corresponds (1.17) to a bundle

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}), G),$$

principal homogeneous under $A_S(\mathscr{R})$. The associated bundle under $\underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\Delta(\mathscr{R})_S)$ corresponds to a form over S of $\Delta(\mathscr{R})_S$: it is $\mathrm{Dyn}(G)$. In other words, this associated bundle is precisely

$$\underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\Delta(\mathscr{R})_S, \mathrm{Dyn}(G)).$$

In this form the proof is immediate.

3.8. Dynkin Scheme and Killing Couples. — Let S be a scheme, G a reductive S-group, and $B \supset T$ a Killing couple of G. There is a canonical morphism

$$i : \mathrm{Dyn}(G) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S})$$

which identifies $\mathrm{Dyn}(G)$ with the "scheme of simple roots of B relative to T." This morphism is defined immediately by descent from the pinned case. Observe also that the data of T and i allow one to reconstruct B (the "one-to-one correspondence between systems of simple roots and systems of positive roots").

It follows from the preceding description of $D = \mathrm{Dyn}(G)$ that there is a canonical root of B_D relative to T_D: this root α_D is the image, under $i(D)$, of the identity morphism of D. One obtains from it a canonical invertible $\mathcal{O}_D$-module $\mathfrak{g}_D$:

$$\mathfrak{g}_D = (\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_D)^{\alpha_D}.$$

In the pinned case,

$$D = \coprod_{\alpha \in \Delta} S_\alpha,$$

where every S_α is a copy of S, and $\mathfrak{g}_D$ is then the $\mathcal{O}_D$-module inducing $\mathfrak{g}_\alpha$ on S_α, for every $\alpha \in \Delta$.

3.9. Quasi-Pinnings. Quasi-Pinned Groups. — If G is a reductive S-group, a *quasi-pinning of* G means the data

(i) of a Killing couple $B \supset T$ of G;

(ii) of a section

$$X \in \Gamma\big(\mathrm{Dyn}(G), \mathfrak{g}_D\big)^{\times}.$$

A reductive S-group is called *quasi-splittable* if it has a quasi-pinning. A *quasi-pinned group* is a reductive group equipped with a quasi-pinning.

Let $B \supset T$ be a Killing couple of the reductive S-group G. Then G is quasi-pinnable relative to this Killing couple if and only if $\mathfrak{g}_D$ has a section nonzero at every point, that is, if the element of $\mathrm{Pic}(\mathrm{Dyn}(G))$ defined by $\mathfrak{g}_D$ is zero. Suppose in particular that S is semilocal. Then $\mathrm{Dyn}(G)$ is also semilocal, and hence $\mathrm{Pic}(\mathrm{Dyn}(G)) = 0$. Thus:

Proposition 3.9.1. — *Let S be a semilocal scheme and G a reductive S-group. Then G is quasi-splittable if and only if it has a Borel subgroup.*

[17] Indeed, S is affine, so by the first assertion of Exp. XXII, 5.9.7, if G has a Borel subgroup B, it also has a Killing couple $B \supset T$. Then, because $\mathrm{Pic}(D) = 0$, the module $\mathfrak{g}_D$ has a nowhere-zero section X.

Continue to let A be a semilocal ring and $S = \mathrm{Spec}(A)$. For every reductive S-group G, the morphism

$$\mathrm{Bor}(G) \longrightarrow S$$

is surjective (because G_s has Borel subgroups for every $s \in S$), smooth, and projective (Exp. XXII, 5.8.3). It therefore has sections after a finite, surjective, étale extension of the base.[18] It follows that:

[17] Editor's note: The reference to Exp. XXII, 5.9.7 has been given in detail.

[18] Editor's note: The original referred to EGA IV, §24, which did not appear. The point is to use "Bertini's theorem." We give the details of the argument, which were indicated to us by O. Gabber. Let $X \to S$ be a surjective smooth projective morphism. Replacing S by a connected component, one may assume that X/S has relative dimension d at every point (cf. EGA IV_4, 17.10.2). One may also assume that X is a closed subscheme of a projective space $\mathbb{P}^n_S = \mathbb{P}(E)$, where E is a free A-module of rank $n+1$ (cf. EGA II, 5.3.3). Let $s_1, \ldots, s_r$ be the closed points of S. By Bertini's theorem (see, for example, [Jou83], I, 6.10), there is an open subset U of

$$P = \mathbb{P}(E^*)^d$$

with nonempty fibers such that, for every point $u = (f_1, \ldots, f_d)$ of U_{s_i}, the intersection of $X_{\kappa(u)}$ with the d hyperplanes of $\mathbb{P}^n_{\kappa(u)}$ defined by u is étale over $\kappa(u)$. After shrinking U, one may assume that U is the complement in affine space $\mathbb{A}^{nd}_S$ of the zero locus $V(Q)$ of a polynomial Q of degree $m > 0$. An easy induction on the number of variables then shows that $V(Q)$ cannot contain every rational point of $\mathbb{A}^{nd}_{\kappa(s_i)}$ if $|\kappa(s_i)| > m$. Since the morphism

$$A \longrightarrow \prod_i \kappa(s_i)$$

is surjective, one can find a monic polynomial $R \in A[X]$ of degree m whose image in $\kappa(s_i)[X]$ is irreducible when $\kappa(s_i)$ is finite and has m distinct roots when $\kappa(s_i)$ is infinite. Put

$$A' = A[X]/(R), \qquad S' = \mathrm{Spec}(A').$$

Then $S' \to S$ is étale, finite, and surjective, and the residue field at each of its closed points $s'_1, \ldots, s'_t$ has cardinality at least $2m$. Thus U has a rational point u_i above every closed point of S', and, since

$$A' \longrightarrow \prod_i \kappa(s'_i)$$

is surjective , these points lift to a section u of $P_{S'}$. Let Z be the intersection of $X_{S'}$ with the d hyperplanes of $\mathbb{P}^n_{S'}$ defined by u, and let V be the open subset of Z consisting of the points at which Z is étale over S'. By EGA IV_3, 11.3.8, V contains the fibers $Z_{s'_i}$ for every i. Since $\pi : Z \to S'$ is proper, it follows that the closed set $\pi(Z - V)$ is empty, whence $V = Z$. Thus $Z \to S'$ is surjective, étale, and proper, hence finite, as is the composite $Z \to S$; this gives the desired section of $X \to S$.

Corollary 3.9.2. — *Let S be a semilocal scheme and G a reductive S-group. There is an étale, finite, and surjective morphism $S' \to S$ such that $G_{S'}$ is quasi-splittable.*

Remark 3.9.3. — Under the preceding hypotheses, let T be a maximal torus of G (cf. Exp. XIV, 3.20). One may moreover assume that $G_{S'}$ is quasi-splittable relative to $T_{S'}$: it suffices to apply the preceding argument to the "scheme of Borel subgroups containing T," which is finite and étale over S (Exp. XXII, 5.5.5(ii)).

3.10. — Let E and E' be two quasi-pinnings of the reductive S-group G. There is a unique inner automorphism of G carrying E to E'. Indeed, one reduces immediately to the split case, where the assertion has already been proved (1.5; it suffices to observe that for an inner automorphism of G, preserving a pinning is equivalent to preserving the underlying quasi-pinning). One concludes, as in §1, that a

quasi-pinning of the reductive S-group G defines a splitting h of the exact sequence:

$$1 \longrightarrow \underline{\mathrm{ad}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \underset{p}{\overset{h}{\leftrightarrows}} \underline{\mathrm{Autext}}(G) \longrightarrow 1$$

the image of h being the subgroup of $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$ that preserves the quasi-pinning.

Likewise, if G and G' are two quasi-pinned S-groups, one defines naturally the subfunctor

$$\underline{\mathrm{Isom}}_{S\text{-gr},\, q\text{-}ép}(G, G')$$

of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$. The projection from $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ to $\underline{\mathrm{Isomext}}(G, G')$ induces an isomorphism

$$(*) \qquad \underline{\mathrm{Isom}}_{S\text{-gr},\, q\text{-}ép}(G, G') \xrightarrow{\sim} \underline{\mathrm{Isomext}}(G, G').$$

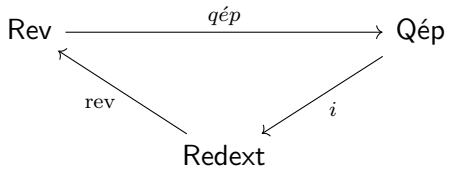
Theorem 3.11. — *Let S be a scheme, let $\mathscr{R}$ be a pinned reduced root datum such that $\mathrm{Ép}_S(\mathscr{R})$ exists (cf. Exp. XXV), and let E be its group of automorphisms. Consider the following three categories:*

(i) the category Rev *of principal Galois coverings of S with group E (the morphisms are isomorphisms of E-bundles);*

(ii) the category Redext*, whose objects are the reductive S-groups of type $\mathscr{R}$ (Exp. XXII, 2.7), the morphisms from G to G' being the elements of $\underline{\mathrm{Isomext}}(G, G')(S)$;*

(iii) the category Qép *of quasi-pinned reductive S-groups of type $\mathscr{R}$ (the morphisms are the isomorphisms preserving the quasi-pinnings).*

These three categories are equivalent. More precisely, there is a diagram of functors, commutative up to functorial isomorphisms:



We describe these three functors below, leaving to the reader the verification that the diagram commutes.[19]

[19] Editor's note: In 3.11.4 we have sketched the verification that the functors i, rev, and *qép* are fully faithful and that the composite rev $\circ i \circ$ *qép* is isomorphic to the identity functor of Rev.

3.11.1. The functor i. — This is the evident functor:

$$i(G) = G, \qquad i(f) = \text{the image of } f \text{ under the isomorphism 3.10 } (*).$$

3.11.2. The functor *qép*. — Let S' be a principal Galois covering of S with group E. Let

$$(G, T, M, \mathscr{R}, \Delta, (X_\alpha)_{\alpha\in\Delta})$$

be a pinning of $G = \text{Ép}_S(\mathscr{R})$, and let $(X'_\alpha)_{\alpha\in\Delta}$ be the corresponding pinning of

$$G' = G_{S'} = \text{Ép}_{S'}(\mathscr{R}).$$

By Exp. XXIII, 5.5 bis, there is a unique group homomorphism

$$\theta : E \longrightarrow \underline{\text{Aut}}_{S\text{-gr}}(\text{Ép}_S(\mathscr{R})) = A(\mathscr{R})(S)$$

such that, for every $h \in E$, the automorphism $\theta(h)$ induces $D_S(h)$ on T, and

$$\text{Lie}(\theta(h))(X_\alpha) = X_{h(\alpha)} \qquad (\alpha \in \Delta).$$

Taking into account the action of E on S', this yields an action of E on G', compatible with the action of E on S' and preserving the canonical quasi-pinning of G'.[20] Since $S' \to S$ is a covering for the (fpqc) topology, it is an effective descent morphism for the fibered category of affine morphisms. Write

$$qép(S') = Q\text{-}\,\text{Ép}_{S'/S}(\mathscr{R})$$

for the quasi-pinned S-group obtained by Galois descent.[21]

3.11.3. The functor rev**.** — Let G be a reductive S-group of type $\mathscr{R}$. Write

$$\text{rev}(G) = \underline{\text{Isomext}}(\text{Ép}_S(\mathscr{R}), G).$$

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[20] Editor's note: The preceding discussion expands the original in order to show that the action of E on G' is obtained by combining the natural action of E on $\text{Ép}_{S'}(\mathscr{R})$ with the given action on S'. The canonical Killing couple of G' is preserved by this action, as is the quasi-pinning given by the element $\widetilde{X}'$ of

$$\Gamma\left(\Delta_{S'}, \text{Lie}(G'/S')_{\Delta_{S'}}\right),$$

equal to X'_α on the copy of S' indexed by α. Indeed, since E acts on $\Delta_{S'}$ by permuting the copies of S', one has $h(\widetilde{X}') = \widetilde{X}'$ for every $h \in E$.

[21] Editor's note: See, for example, TDTE I, p. 22, Example 1. One may also describe $qép(S')$ as the twist of $G = \text{Ép}_S(\mathscr{R})$ by the E-torsor S'/S, that is, the (fpqc) sheaf quotient of $S' \times_S G$ by the right action of E defined by

$$(s', g) \cdot h = (s'h, h^{-1}(g)).$$

Since G is affine over S, and since E acts on G by group automorphisms, this sheaf is represented by an S-group $G^\sharp$, which is a "twisted" form of G, and $D^\sharp = \text{Dyn}(G^\sharp)$ is the twist of the constant Dynkin scheme Δ_S by the torsor S'/S. Since E normalizes B and T, one obtains likewise a Killing couple $(B^\sharp, T^\sharp)$ of $G^\sharp$. On the other hand, put $\mathfrak{g} = \text{Lie}(G/S)$ and $\mathfrak{g}^\sharp = \text{Lie}(G^\sharp/S)$. For every $U \to S$, the sections of $\mathfrak{g}^\sharp$ over U are the E-equivariant S-morphisms

$$U \times_S S' \longrightarrow W(\mathfrak{g}).$$

Since $D^\sharp \times_S S'$ is E-isomorphic to $\Delta \times S'$, endowed with the action

$$(\alpha, s') \cdot h = (h^{-1}(\alpha), s'h),$$

the morphism $(\alpha, s') \mapsto X_\alpha$ gives a section of $\mathfrak{g}^\sharp$ over $D^\sharp$ which is a quasi-pinning, that is, a nowhere-zero section of

$$\left(\mathfrak{g}^\sharp \otimes_{\mathcal{O}_S} \mathcal{O}_{D^\sharp}\right)^{D^\sharp}$$

(cf. 3.8).

This is a principal homogeneous bundle under E_S (cf. 1.10 and 1.19), that is, an object of Rev.

3.11.4. — [22] It is clear from the isomorphism 3.10 $(*)$ that the functor i is fully faithful. On the other hand, if G_0, G, G' are reductive S-groups, the natural morphism

$$\underline{\mathrm{Isom}}(G_0, G) \times_S \underline{\mathrm{Isom}}(G, G') \longrightarrow \underline{\mathrm{Isom}}(G_0, G')$$

induces a morphism

$$(\star) \qquad \underline{\mathrm{Isomext}}(G_0, G) \times_S \underline{\mathrm{Isomext}}(G, G') \longrightarrow \underline{\mathrm{Isomext}}(G_0, G'),$$

since, for every $S' \to S$, if

$$g_0 \in G_0(S'), \quad g \in G(S'), \quad u \in \underline{\mathrm{Isom}}(G_0, G)(S'), \quad v \in \underline{\mathrm{Isom}}(G, G')(S'),$$

one has

$$v \circ \mathrm{int}(g) \circ u \circ \mathrm{int}(g_0) = v \circ u \circ \mathrm{int}\big(u^{-1}(g) g_0\big).$$

Taking $G_0 = \acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})$, one obtains a map

$$\underline{\mathrm{Isomext}}(G, G')(S) \longrightarrow \mathrm{Hom}_{\mathsf{Rev}}\big(\underline{\mathrm{Isomext}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}), G), \underline{\mathrm{Isomext}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}), G')\big),$$

and, to show that this map is a bijection, one may assume by (fpqc) descent that G and G' are pinned, in which case it is evident.

Similarly, if S_1, S_2 are two objects of Rev, and if one writes

$$S_i \times^E \acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})$$

for the S-group $q\acute{e}p(S_i)$ obtained by twisting $\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})$ by the E-torsor S_i (cf. editor's note (21)), there is a natural map

$$\mathrm{Hom}_{\mathsf{Rev}}(S_1, S_2) \longrightarrow \mathrm{Hom}_{\mathsf{Q\acute{e}p}}(S_1 \times^E \acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}), S_2 \times^E \acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})).$$

To show that this map is a bijection, one may assume by (fpqc) descent that S_1 and S_2 are trivial E-bundles, in which case the assertion is evident.

Thus the three functors in the diagram are fully faithful. Moreover, for every object S' of Rev, there is a natural morphism

$$S' \longrightarrow \underline{\mathrm{Isomext}}\Big(\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}), S' \times^E \acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})\Big),$$

and to see that it is an isomorphism, one may assume by (fpqc) descent that S' is a trivial E-bundle, in which case the assertion is evident.

There is therefore an isomorphism of functors

$$\mathrm{Id}_{\mathsf{Rev}} \longrightarrow \mathrm{rev} \circ i \circ q\acute{e}p.$$

Since rev is fully faithful, for every reductive S-group H (not necessarily quasi-split) one obtains a bijection

$$\underline{\mathrm{Isomext}}\Big(Q\text{-}\acute{\mathrm{E}}\mathrm{p}_{\mathrm{rev}(H)/S}(\mathscr{R}), H\Big)(S) \simeq \mathrm{Aut}_{\mathsf{Rev}}(\mathrm{rev}(H)),$$

[22] Editor's note: This subsection number has been added.

functorial in H. In particular, the identity morphism of $\mathrm{rev}(H)$ corresponds to a "canonical" element

$$u_0 \in \underline{\mathrm{Isomext}}\Big(Q\text{-}\,\acute{\mathrm{E}}\mathrm{p}_{\mathrm{rev}(H)/S}(\mathscr{R}), H\Big)\,(S).$$

Moreover, every element u of this same set corresponds to an automorphism φ_u of $\mathrm{rev}(H)$, and $q\acute{e}p(\varphi_u)$ is the unique automorphism of the pinned S-group

$$Q\text{-}\,\acute{\mathrm{E}}\mathrm{p}_{\mathrm{rev}(H)/S}(\mathscr{R})$$

such that

$$u_0 \circ q\acute{e}p(\varphi_u) = u.$$

This proves Theorem 3.11.

We develop one of the corollaries of 3.11.

Corollary 3.12. — *For every reductive S-group G, there is a quasi-pinned S-group $G_{q\text{-}\acute{e}p}$ and an "outer isomorphism"*

$$u \in \underline{\mathrm{Isomext}}(G_{q\text{-}\acute{e}p}, G)(S).$$

The pair $(G_{q\text{-}\acute{e}p}, u)$ is unique up to a unique isomorphism.

Indeed, one may assume that G has constant type $\mathscr{R}$, and one takes

$$G_{q\text{-}\acute{e}p} = Q\text{-}\,\acute{\mathrm{E}}\mathrm{p}_{\mathrm{rev}(G)/S}(\mathscr{R}).$$

3.12.1. — Observe that the data of the pair $(G_{q\text{-}\acute{e}p}, u)$ canonically define the S-scheme (cf. 1.11)

$$Q = \underline{\mathrm{Isomint}}(G_{q\text{-}\acute{e}p}, G),^{23}$$

which is a principal homogeneous bundle under $\mathrm{ad}(G_{q\text{-}\acute{e}p})$, and whose data are "equivalent" to those of the inner twisted form G of $G_{q\text{-}\acute{e}p}$. Moreover, Q is precisely the "scheme of quasi-pinnings of G," a definition that explains its structure as a principal homogeneous bundle (on the left) under $\mathrm{ad}(G)$ (3.10); compare 1.20.

Proposition 3.13. *Let S be a scheme, G a semisimple adjoint (resp. simply connected) S-group, $B \supset T$ a Killing couple of G,* $\mathrm{Dyn}(G)$ *the Dynkin S-scheme of G. There exists a canonical isomorphism of S-group schemes*

$$T \xrightarrow{\sim} \prod_{\mathrm{Dyn}(G)/S} \mathbb{G}_{m,\mathrm{Dyn}(G)}.$$

(Recall (cf. Exp. II, § 1) that the right-hand side is by definition the S-functor which to $S' \to S$ associates $G_m(\mathrm{Dyn}(G) \times_S S')$, or, what amounts to the same, $G_m(Dyn(G_{S'}))$.)

First proof. Let us do it for simplicity in the adjoint case. Consider the composite morphism

$$T \longrightarrow \prod_{D/S} T_D \longrightarrow \prod_{D/S} \mathbb{G}_{m,D}.$$

where the first morphism is the canonical morphism, the second is $\prod_{D/S} \alpha_D$ (we have written $D = \mathrm{Dyn}(G)$). To verify that this morphism is an isomorphism, one may assume G split; now, in this case, this is none other than the morphism $T \to (G_{m,S})^{\Delta}$ with components α, for $\alpha \in \Delta$, and the latter is an isomorphism (Exp. XXII, 4.3.8).

[23] Editor's note: We write $\underline{\mathrm{Isomint}}(G_{q\text{-}\acute{e}p}, G)$ instead of $\underline{\mathrm{Isomint}}_u(G_{q\text{-}\acute{e}p}, G)$ (cf. 1.11), since the pair $(G_{q\text{-}\acute{e}p}, u)$ is unique up to a unique isomorphism.

Second proof. By 2.8, 3.5 and 3.11, one may assume that $G = Q - \acute{E}p_{S'/S}(R)$, T being the canonical maximal torus. On S', one has by Exp. XXII 4.3.8, an isomorphism $T_{S'} \xrightarrow{\sim} (G_{m,S'})^{\Delta}$, defined by the simple roots (resp. the simple coroots). The group E operates on the right-hand side by permutation of Δ. Now the bundle associated to S'/S by $E \to \mathrm{Aut}(\Delta)$ is $\mathrm{Dyn}(G)$ (3.7), and one concludes at once.

Using Prop. 8.4 of the appendix, one deduces:

Corollary 3.14. *Under the preceding conditions, one has*

$$H^1(S,T) \xrightarrow{\sim} H^1(\mathrm{Dyn}(G), \mathbb{G}_m) \xrightarrow{\sim} \mathrm{Pic}\big(\mathrm{Dyn}(G)\big).$$

In particular, $H^1(S,T) = 0$ when S is semi-local.

Remark 3.15. Let S be a scheme, G (resp. G') an S-reductive group, $B \supset T$ (resp. $B' \supset T'$) a Killing couple of G (resp. G'), $u \in Isomext(G,G')(S)$. Set (cf. 2.10)

$$P = \underline{\mathrm{Isomint}}_u(G,B,T;G',B',T').$$

this is a principal homogeneous bundle under T_{ad} (by $(f,t) \mapsto f \circ int(t)$).

Let on the other hand $D = \mathrm{Dyn}(G) = \mathrm{Dyn}(G')$ (identified via u (3.5)), and let $L = Isom_D(g_D, g'_D)$ be the principal homogeneous bundle under $G_{m,D}$ defined by the invertible $\mathcal{O}_D$-module

$$\mathrm{Hom}_{\mathcal{O}_D}(\mathfrak{g}_D, \mathfrak{g}'_D) = \mathfrak{g}'_D \otimes (\mathfrak{g}_D)^{\vee}.$$

Each $f \in P(S')$ defines an isomorphism of $g_D \otimes_{O_S} O_{S'}$ onto $g'_D \otimes_{O_S} O_{S'}$, whence a canonical morphism

$$P \to \prod_{D/S} L.$$

This morphism is an isomorphism, compatible with the isomorphism of operator groups

$$T_{\mathrm{ad}} \xrightarrow{\sim} \prod_{D/S} \mathbb{G}_{m,D}.$$

defined above. Indeed, it suffices to verify it in the case where the two groups are pinned, where it is easy.

It follows in particular that in the isomorphism

$$H^1(S, T_{ad}) \xrightarrow{\sim} \mathrm{Pic}(\mathrm{Dyn}(G))$$

of 3.14, the class of the bundle P is transformed into $c\ell(g'^D) - c\ell(g_D)$. The bundle P is therefore trivial if and only if g'^D and g_D are isomorphic.

If (G,B,T) is quasi-splittable, for example if one takes for G the group $G'_{q-\acute{e}p.}$, with its canonical Killing couple, it follows that the image of the class of P is none other than the obstruction to the quasi-splitting of G' defined in 3.9.

3.16. Symmetries

3.16.1.

Let S be a scheme, G an S-reductive group, $B \supset T$ a Killing couple of G. Write $D = \mathrm{Dyn}(G)$. Recall (Exp. XXII, 5.9.1) that there exists a unique Borel subgroup B^- of G such that $B \cap B^- = T$. If $X \in \Gamma(D, g_D)^{\times}$ defines a quasi-pinning of G relative to (B,T)

(3.9), then $Y = -X^{-1} \in \Gamma(D, (g_D)^\vee)^\times$ defines a quasi-pinning of G relative to (B^-, T); one says that this is the *opposite quasi-pinning*.

If R is a pinned reduced root datum and if E is an R-pinning of the S-reductive group G, one defines an R-pinning E^- said to be *opposite to* E in the following way: one keeps the same maximal torus T, one takes the opposite of the isomorphism $D_S(M) \xrightarrow{\sim} T$, and one "pins" by $Y_\alpha = -X_\alpha^{-1} \in \Gamma(S, g^{-\alpha})^\times$, for $\alpha \in \Delta$. The quasi-pinning underlying E^- is the quasi-pinning opposite to the quasi-pinning underlying E.

Remark. In the notation of Exp. XIX, 3.1, if one sets

$$w_\alpha(X_\alpha) = \exp(X_\alpha)\exp(-X_\alpha^{-1})\exp(X_\alpha).$$

one has $w_\alpha(X_\alpha) = w_{-\alpha}(Y_\alpha)$ (loc. cit., 3.1 (vi)).

Proposition 3.16.2. *Let S be a scheme, G an S-reductive group.*

(i) Let T be a maximal torus of G; there exists a unique

$$i_T \in \left(\underline{\mathrm{Aut}}_{S\text{-gr.}}(G,T)/T_{\mathrm{ad}}\right)(S) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(T).$$

such that $i_T(t) = t^{-1}$ for every section t of T.

(ii) Let (B,T) be a Killing couple of G; there exists a unique section

$$w_{B,T} \in \big(\mathrm{Norm}_G(T)/T\big)(S) = W_G(T)(S).$$

such that $int(w_{B,T})(B) = B^-$ (with the obvious abuse of language).

(iii) Let $Q = (B,T,X)$ be a quasi-pinning of G, $Q^- = (B^-,T,Y)$ the opposite quasi-pinning; there exists a unique inner automorphism of G

$$n_Q \in Norm_{ad(G)}(T_{ad})(S) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$$

such that $n_Q(Q) = Q^-$, that is, $n_Q(T) = T$, $n_Q(B) = B^-$, $n_Q(X) = Y$.

(iv) Let (R,E) be a pinning of G, (R,E^-) the opposite pinning; there exists a unique automorphism of G

$$u_E \in \underline{\mathrm{Aut}}_{S\text{-gr.}}(G,T) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(G).$$

such that $u_E(E) = E^-$, i.e. $u_E(t) = t^{-1}$ for every section t of T, and $Ad(u_E)X_\alpha = Y_\alpha$ for every $\alpha \in \Delta$.

Proof. (ii) follows from Exp. XXII, 5.5.5 (ii), then (iii) follows from 3.10, and (iv) follows from Exp. XXIII, 4.1. Finally to prove (i), one may assume G pinned. Existence follows from (iv) for example, uniqueness from the fact that an automorphism of G inducing the identity on T is given by a section of T_{ad} (2.11).

Corollary 3.16.3. *One has*

$$i_T^2 = w_{B,T}^2 = n_Q^2 = u_E^2 = e.$$

Moreover, i_T (resp. u_E) is $\neq e$ if $G \neq e$, and $w_{B,T}$ (resp. n_Q) is $\neq e$ if G is not a torus.

Corollary 3.16.4. *In the situation of (iii) (resp. (iv)), n_Q projects onto $w_{B,T}$ (resp. u_E projects onto i_T) by the canonical morphism*

$$Norm_{ad(G)}(T_{ad}) \to W_{ad(G)}(T_{ad}) \simeq W_G(T),$$

resp.

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(G,T) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G,T)/T_{\mathrm{ad}}.$$

Corollary 3.16.5. *The preceding definitions are compatible with base change, and are functorial under isomorphism (in an obvious sense).*

Proposition 3.16.6. *(i) One can define uniquely for each reductive group G over a scheme S an element*

$$s_G \in Autext(G)(S)$$

in such a way that this construction is functorial in G under isomorphism, is compatible with base change, and is such that whenever T is a maximal torus of the S-reductive group G, s_G is the image of i_T under the canonical morphism

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(G,T)/T_{\mathrm{ad}} \longrightarrow \underline{\underline{\mathrm{Autext}}}(G).$$

(ii) One has $s_G^2 = e$, and s_G is a central element of $Autext(G)$.

(iii) Under the conditions of 3.16.2 (ii), if one identifies $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G,B,T)/T_{ad}$ with $Autext(G)$ (2.2), one has

$$w_{B,T} i_T = i_T w_{B,T} = s_G.$$

(iv) Under the conditions of 3.16.2 (iv), if one identifies $\mathrm{Aut}_{S-gr.-ép}(G,R,E)$ with $Autext(G)$ (1.3 (iii)), one has

$$n_E u_E = u_E n_E = s_G.$$

Proof. (i) is proved without difficulty by descent. On the other hand, since i_T is evidently a central section of square e in $\underline{\mathrm{Aut}}_{S\text{-gr.}}(T)$, (ii) follows immediately; (iii) is a consequence of (iv) by descent. Finally, under the conditions of (iv), it is clear that $n_E u_E = u_E n_E$ and that this automorphism of G respects the pinning; modulo the identification made, it is therefore equal to its image in $Autext(G)$; but n_E is inner and u_E projects onto s_G.

Remark 3.16.7. (i) One determines s_G explicitly in each case of the classification thanks to (iii): for each irreducible pinned root datum it suffices to compose the symmetry through the origin with the symmetry in the Weyl group (i.e. the element w_0 of the Weyl group such that $w_0(\Delta) = -\Delta$). One finds the following results: one has $s_G = e$ except for A_n ($n \geqslant 2$), D_n (n odd) and E_6, in which case s_G is the unique non-trivial "outer automorphism".

(ii) The automorphism u_E is the one used to construct "the compact real forms" in the theory of semisimple Lie algebras.

Remark 3.16.8. We defined in 3.16.1 an involution in the S-scheme

$$Q = Isomint(G_{q-ép.}, G)$$

of quasi-pinnings of G (cf. 3.12.1); by transport of structure from $G_{q-ép.}$ to G, one sees at once that this involution is given by the action of an element of $ad(G_{q-ép.})(S)$: the element n_0 defined (3.16.2 (iii)) by the canonical quasi-pinning of $G_{q-ép.}$.

In the same way, one has defined an involution in the S-scheme

$$P = Isom_{S-gr.}(\acute{E}p_S(R), G)$$

of R-pinnings of G (cf. 1.20). Arguing as before, one sees that this involution is given by the action of the automorphism u_0 of $\acute{E}p_S(R)$ defined (3.16.2 (iv)) by the canonical pinning of $\acute{E}p_S(R)$.

4. Isotriviality of reductive groups and of principal bundles under reductive groups

4.1. Definitions. Isotriviality theorem

Definition 4.1.1. *Let S be a scheme, G an S-group scheme, P a principal homogeneous bundle under G. One says that P is* locally isotrivial *(resp.* semi-locally isotrivial*)* if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open set) there exist an open set U of S containing s (resp. F) and a finite surjective étale morphism $S' \to U$ such that $P_{S'}$ is trivial.

Definition 4.1.2. *Let S be a scheme, G an S-reductive group. One says that G is* locally isotrivial *(resp.* semi-locally isotrivial*)* if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open set) there exist an open set U of S containing s (resp. F) and a finite surjective étale morphism $S' \to U$ such that $G_{S'}$ is splittable.*

Remark 4.1.3. (i) The equivalence of categories 4.1.1 of 1.17 respects by definition local (resp. semi-local) isotriviality.

(ii) Add to the conditions of 4.1.1: G locally of finite presentation over S. Then the principal bundle P (or the reductive group G) is locally isotrivial (resp. semi-locally isotrivial) if and only if for every $S' \to S$, S' local (resp. semi-local), $P_{S'}$ is isotrivial (or $G_{S'}$ isotrivial), that is to say, if there exists $S'' \to S'$ finite surjective étale such that $P_{S''}$ is trivial (or $G_{S''}$ split).

(iii) In the case of tori, definition 4.1.2 coincides with that of Exp. IX, 1.1.

4.1.4.

Recall (Exp. XXII, 4.3 and 6.2) that for every reductive group G, we have introduced the groups $rad(G)$, $corad(G)$ and $dér(G)$. The groups $rad(G)$ and $corad(G)$ are tori, which are split when G is split; moreover, there exists an isogeny $rad(G) \to corad(G)$. The S-group $dér(G)$ is semisimple, one has $G/dér(G) = corad(G)$; it follows that for every principal homogeneous bundle P under G, $P/dér(G)$ is a principal homogeneous bundle under $corad(G)$. This said, one has:

Theorem 4.1.5. *Let S be a scheme, G an S-reductive group of constant type.*

(i) The following conditions are equivalent:

(a) G is locally (resp. semi-locally) isotrivial.

(b) The torus $rad(G)$ is.

(b') The torus $corad(G)$ is.

(c) The Galois covering of S associated to G (3.11) is.

If T is a maximal torus of G, these conditions are also equivalent to

(d) The torus T is locally (resp. semi-locally) isotrivial.

(ii) Let P be a principal homogeneous bundle under G; for P to be locally (resp. semi-locally) isotrivial, it is necessary and sufficient that the $corad(G)$-principal bundle $P/dér(G)$ be so.

Corollary 4.1.6. *The conditions of (i) are satisfied when G is semisimple or when S is locally noetherian and normal (or more generally geometrically unibranch). The conditions of (ii) are satisfied when G is locally (resp. semi-locally) isotrivial.*

For (i), the first assertion is trivial from (b), the second follows from (c) and Exp. X, 5.14 and 5.15. For (ii), it suffices to note that by virtue of theorem 90, a principal bundle under a split torus is semi-locally isotrivial.

4.2. Proof: the semisimple case

Let us first prove, for later reference:

Proposition 4.2.1. *Let S be a scheme, G an S-reductive group, T a maximal torus of G (resp. B a Borel subgroup, resp. $B \supset T$ a Killing couple of G), P a principal homogeneous bundle under G, G' the twisted form of G associated to P (via the morphism $int : G \to \mathrm{Aut}_{S-gr.}(G)$). One has canonical isomorphisms*

$$P/\operatorname{Norm}_G(T) \xrightarrow{\sim} \operatorname{Tor}(G'), \qquad P/B \xrightarrow{\sim} \operatorname{Bor}(G'), \qquad P/T \xrightarrow{\sim} \operatorname{Kil}(G').$$

By construction, G' is the quotient of $P \times_S G$ by a certain action of G ($((p, g'))g = (pg, g^{-1}g'g)$); one therefore has a morphism $P \times_S G \to G'$, that is, a morphism

$$P \to \operatorname{Hom}_S(G, G'),$$

which, as one sees at once, factors through a morphism

$$f : P \longrightarrow \operatorname{Isom}_{S\text{-gr.}}(G, G').$$

(to verify this assertion, one may assume $G = P$, in which case one has $G' = G$, $f = int$). The set-theoretic map $p \mapsto f(p)(T)$ defines a morphism

$$P \to Tor(G').$$

To verify that this morphism induces an isomorphism $P/Norm_G(T) \xrightarrow{\sim} Tor(G')$ as announced, one may again assume $P = G$ in which case one is reduced to Exp. XXII, 5.8.3 (iii). (In fact, *loc. cit.* should be replaced by the statement above.) One argues similarly for Bor and Kil.

Proposition 4.2.2. *Let S be a semi-local scheme, G an S-semisimple group of constant type.*

(i) G is isotrivial.

(ii) Every principal bundle under G is isotrivial.

Let us prove (i). Up to making a finite surjective étale extension of the base, one may, by 3.9.2, assume G quasi-split. But then G is isotrivial by construction (3.11, the group E being finite). To prove (ii), one may, by (i), assume G split; one is then reduced to:

Lemma 4.2.3. *Let S be a semi-local scheme. Every principal bundle under a split reductive group is isotrivial.*

Indeed, with the notation of 4.2.1, where $B \supset T$ denotes the canonical Killing couple of the split G, the S-scheme $Kil(G')$ possesses a section, after finite surjective étale extension of the base, by 3.9.2. One may therefore reduce the structure group of G to T; but T is split, so $H^1(S, T) = 0$ (theorem 90).

Corollary 4.2.4. *Let S be a scheme and*

$$1 \to G \to G' \to G'' \to 1$$

an exact sequence of S-group schemes (for (fpqc)), G being semisimple of constant type. Let P be a principal homogeneous bundle under G', suppose the sheaf[24] *P/G associated*

[24] N.D.E.: We have replaced "bundle" by "sheaf", and then G' by G''.

representable (for example G'' affine over S). For P to be locally isotrivial (resp. semi-locally isotrivial), it is necessary and sufficient that P/G be so (as bundle under G'').

If P is trivial, P/G is too, which shows that the condition is necessary. Conversely, suppose S local (resp. semi-local) and P/G isotrivial, so trivialized by a finite surjective étale extension S' of S. Extending the base to S', one may reduce the structure group of $P_{S'}$ to $G_{S'}$. But S' is still semi-local and $G_{S'}$ semisimple of constant type, so the obtained bundle is isotrivial (4.2.2).

4.3. Proof: general case

Let us first note that 4.1.5 (ii) follows at once from the application of 4.2.4 to the exact sequence

$$1 \to dér(G) \to G \to corad(G) \to 1.$$

Let us therefore prove (i). If G is split, $rad(G)$ and $corad(G)$ are split, as is $rev(G)$; so (a) implies (b), (b'), and (c).

4.3.1.

One has (c) $\Rightarrow$ (a). Let R be the type of G; one has an exact sequence

$$1 \to ad(\acute{E}p_S(R)) \to A_S(R) \to E_S \to 1.$$

Applying 4.2.4 to the bundle $P = Isom_{S-gr.}(\acute{E}p_S(R), G)$ and to the associated bundle $rev(G) = P/ad(\acute{E}p_S(R))$, one has (c) $\Rightarrow$ (a).

4.3.2.

One has (b) $\Rightarrow$ (a). It suffices to prove that if $rad(G)$ is split, G is semi-locally isotrivial. Now let $G_0 = \acute{E}p_S(R)$; consider the category of pairs (G', f), where G' is a form of G_0 and f is an isomorphism of $rad(G_0)$ onto $rad(G)$.

One knows (2.16) that this category is equivalent to the category of principal homogeneous bundles under a certain S-group H extension of a finite twisted constant group by a semisimple group. It suffices to prove that every principal bundle under H is semi-locally isotrivial; this follows at once from 4.2.4.

4.3.3.

One has (b') $\Rightarrow$ (a). One can argue as previously (it will moreover be the same group H that arises). One can also see that (b) and (b') are equivalent: a torus isogenous to a locally split torus is also locally split (cf. Exp. IX, 2.11 (iii)).

4.3.4.

One has (d) $\Rightarrow$ (a). It suffices to prove that a group of constant type possessing a split maximal torus is semi-locally isotrivial; this follows at once from 2.14.

4.3.5.

One has (a) ⇒ (d). It suffices to prove that a maximal torus of a split group is semi-locally isotrivial. More precisely:

Lemma 4.3.6. *Let S be a semi-local scheme, G an S-reductive group, T_0 a split maximal torus of G, $W_0 = Norm_G(T_0)/T_0$ (this is a locally constant S-group, constant if G is of constant type, by 2.14), T a maximal torus of G.*

There exists a morphism $S' \to S$ which is principal homogeneous under W_0 (hence étale finite and surjective, and even principal Galois if G is of constant type) such that $T_{S'}$ is conjugate to $(T_0)_{S'}$ by an element of $G(S')$ (and so in particular split).

Indeed, one knows that $Transp_G(T_0, T)$ is a principal homogeneous bundle under $Norm_G(T_0)$ (cf. for example Exp. XI, 5.4 bis). Set $S' = Transp_G(T_0, T)/T_0$; this is a principal homogeneous bundle under W_0. Extending the base from S to S', one can reduce the structure group of $Transp_G(T_0, T)$ to T_0. Now S' is semi-local and T_0 split, so $Transp_G(T_0, T)$ possesses a section over S'.[25] QED

4.4. Use of the existence of maximal tori

Using Grothendieck's existence theorem for maximal tori (Exp. XIV, 3.20), one can considerably sharpen the preceding results. Let us state at once:

Theorem 4.4.1. *Let S be a semi-local scheme, R a pinned reduced root datum, W its Weyl group, E the group of its automorphisms. (Recall that E operates naturally on W and that the semi-direct product $A = W \cdot E$ is identified with the group of automorphisms of R non-pinned, cf. Exp. XXI, 6.7.2).*

(i) Every principal homogeneous bundle under $\acute{E}p_S(R)$ is trivialized by a principal Galois covering $S' \to S$ of group W.

(ii) Let G be an S-reductive group of type R; let $rev(G) = Isomext(\acute{E}p_S(R), G)$ be the associated Galois covering of S with group E. Let W_0 be the form of W_S associated to $rev(G)$. There exists a morphism $S' \to S$, which is a principal homogeneous bundle under W_0, such that $G_{S'}$ is quasi-splittable (i.e. possesses a Borel subgroup).

(iii) Every S-reductive group of type R is split by a principal Galois covering $\tilde{S} \to S$ with group A.

Let us first state:

Proposition 4.4.2. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, P a principal homogeneous bundle under G, G' the twisted form of G associated to P (one then has, cf. 4.2.1, a canonical isomorphism $P/Norm_G(T) \xrightarrow{\sim} Tor(G')$). Let T' be a maximal torus of G', defining a composite morphism*

$$S \longrightarrow \mathrm{Tor}(G') \xrightarrow{\sim} P/\mathrm{Norm}_G(T).$$

Consider the canonical morphisms

$$P \longrightarrow P/T \longrightarrow P/\mathrm{Norm}_G(T).$$

and take their inverse images by the preceding morphism:

$$\begin{array}{ccccc} P & \longrightarrow & P/T & \longrightarrow & P/\mathrm{Norm}_G(T) \\ \uparrow & & \uparrow & & \uparrow \\ H & \longrightarrow & S' & \longrightarrow & S \end{array}$$

[25] N.D.E.: by "theorem 90".

Then S' (resp. H) is a principal homogeneous bundle over S (resp. S') under $W_G(T)$ (resp. $T_{S'}$). Moreover, if one makes $W_G(T)$ operate on T in the obvious way, the bundle associated to S' is isomorphic to T'.

The first two assertions are trivial, the last is proved like the corresponding assertion of 2.6.

Corollary 4.4.3. *Let S be a semi-local scheme, G an S-reductive group, T a maximal torus of G. Suppose one of the two following conditions is satisfied:*

(i) T is split.

(ii) T is contained in a Borel subgroup of G, and G is either adjoint, or simply connected.

Let moreover P be a principal homogeneous bundle under G. There exists an S-scheme S', which is a principal homogeneous bundle under $W_G(T)$, such that $P_{S'}$ is trivial.

Indeed, if G' is the form of G associated to P, then G' possesses a maximal torus T' (Exp. XIV, 3.20). Resuming the notation of the preceding proposition, one sees that $H^1(S', T_{S'}) = 0$ (by theorem 90 for (i), by 3.14 for (ii)). The morphism $H \to S'$ possesses a section, so $P_{S'}$ also possesses a section over S'. QED

Let us now prove the theorem. Assertion (i) is a particular case of the preceding corollary (take $G = \acute{E}p_S(R)$, endowed with its canonical split torus). Let us prove (ii). One knows (3.12) that G is an inner twisted form of

$$G_0 = Q - \acute{E}p_{rev(G)/S}(R).$$

If T_0 is the canonical maximal torus of G_0, $W_{G_0}(T_0)$ is the group W_0 described in the statement. The form G of G_0 corresponds to a principal homogeneous bundle P under $ad(G_0)$ ($P = Isomint(G_0, G)$). The group $W_{ad(G_0)}(T_0^{ad})$ is canonically isomorphic to W_0, and one obtains the wanted result by applying 4.4.3 to the situation $(ad(G_0), T_0^{ad}, P)$, hypothesis (ii) of 4.4.3 being well verified. Let us finally prove (iii). Resume the notation of (ii); one has a diagram

$$\begin{array}{ccc} & & \mathrm{rev}(G) \\ & & \Big\downarrow{\scriptstyle E_S} \\ S' & \xrightarrow[W_0]{} & S \end{array}$$

One knows that $G_{S'}$ is isomorphic to $(G_0)_{S'}$ and that $(G_0)_{rev(G)}$ is splittable. If one sets $\tilde{S} = S' \times_S rev(G)$, $G_{\tilde{S}}$ is splittable, and it only remains to verify that $\tilde{S}$ is indeed a principal Galois covering of S with group A, which follows from the following more general lemma (naturally valid in any site):

Lemma 4.4.4. *Let S be a scheme, G and H two S-group schemes, $G \to \mathrm{Aut}_{S-gr.}(H)$ an action of G on H, E a G-principal homogeneous bundle, F an H_0-principal homogeneous bundle, where H_0 is the form of H associated to E. Then $E \times_S F$ is endowed naturally with a structure of principal homogeneous bundle under the semi-direct product $H \cdot G$.*

Write $(e, g) \mapsto eg$ (resp. $(f, u) \mapsto fu$) for the action (on the right) of G on E (resp. of H_0 on F). Write

$$p : E \times_S H \longrightarrow H_0.$$

for the canonical projection (H_0 is by definition the quotient of $E \times_S H$ by G acting on it by the formula $(e, h)g = (eg, g^{-1}hg)$). Consider the morphism

$$r : E \times_S F \times_S H \times_S G \longrightarrow E \times_S F.$$

defined set-theoretically by

$$r(e, f, h, g) = (eg, f \cdot p(e, h)).$$

The morphism r indeed defines an action of the semi-direct product $H \cdot G$ on $E \times_S F$. Indeed, one has set-theoretically

$$r(r(e, f, h, g), h', g') = (egg', f\, p(e, h)\, p(eg, h')).$$

but

$$p(e, h)p(eg, h') = p(e, h)p(e, gh'g^{-1}) = p(e, hgh'g^{-1}).$$

whence

$$r(r(e, f, h, g), h', g') = r(e, f, hgh'g^{-1}, gg').$$

which had to be proved.

To prove now that this law is indeed a law of principal homogeneous bundle, one may assume that E and F are trivial, in which case one sees at once that $E \times_S F$ is also a trivial bundle.

4.5. Isotriviality of maximal tori of semisimple groups

[26]

Let us bring out here the following result, contained implicitly in 2.6 (cf. the N.D.E. (13)) and which was mentioned at the end of Exp. IX 1.2. Recall (XXIII 5.11) that, for every scheme S and every reduced root datum R, one writes $\acute{E}p_S(R)$ for the unique pinned S-group of type R (which exists by Exp. XXV) and $T_S(R)$ its canonical maximal torus.

Proposition 4.5.1. *Let G be an S-semisimple group of constant type R; assume that G possesses a maximal torus T (which is the case if S is semi-local, by XIV 3.20).*

(i) Then $P = Isom_{S-gr.}(\acute{E}p_S(R), T_S(R); G, T)/T_S(R)_{ad}$ is a principal S-bundle under the finite group $\mathrm{Aut}(R)$, and T_P is a split maximal torus of G_P. Consequently, T is isotrivial.

(ii) Moreover, for every finite set F of points of P contained in an affine open set, there exists an open set U of P containing F such that G_U is split.

Proof. The exact sequence

$$1 \longrightarrow T_S(\mathscr{R})_{\mathrm{ad}} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}), T_S(\mathscr{R})) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(T_S(\mathscr{R})).$$

induces for every $S' \to S$ a morphism of bundles $P_{S'} \to Isom_{S'-gr.}(T_{S'}(R), T_{S'})$, so if $P_{S'}$ possesses a section, then $T_{S'}(R)$ and $T_{S'}$ are isomorphic. This is in particular the case for $S' = P$.

Note moreover that $\mathrm{Aut}_{S-gr.}(\acute{E}p_S(R), T_S(R))/T_S(R)_{ad}$ is none other than the constant S-group $\mathrm{Aut}(R)_S$, and that P is identified with the $\mathrm{Aut}(R)_S$-torsor $Isom(R, \tilde{R}(G, T))$, where $\tilde{R}(G, T)$ denotes the twisted root datum associated to (G, T) (cf. XXII 1.10).

Then (ii) follows from (i) and from 2.14, taking into account the fact that the g_α (α ranging over the roots of R) are free $\mathcal{O}_U$-modules of rank 1 on a sufficiently small affine open set U containing F.

[26] *Editors' note.* This no. 4.5 has been added.

5. Canonical decomposition of an adjoint or simply connected group

In this section, we shall use the results of no 1 to generalize to the case of schemes a classical decomposition of adjoint (resp. simply connected) groups. So as not to overburden the exposition indefinitely, the proofs are sketched and the details are left to the reader; in fact it is always a matter of absolutely standard proofs in the theory of principal bundles: reduction of the structure group, twisting, etc.

5.1.

Recall (Exp. XXI, 7.4) that a Dynkin diagram is a disjoint union of its connected components, which are Dynkin diagrams. Moreover, every non-empty connected Dynkin diagram corresponding to a root datum is isomorphic to one of the standard diagrams (A_n, B_n, ..., G_2) which were exhibited in Exp. XXI, 7.4.6. In the sequel, we shall be interested only in Dynkin diagrams whose connected components are of one of the preceding types. Let T be the set of these standard diagrams. For every Dynkin diagram D, let $n(t) = n_D(t)$ be the number of connected components of D isomorphic to t, where $t \in T$. The *type* of D is by definition $\sum_{t\in T} n_D(t) \cdot t$.

A Dynkin diagram of type t is said to be *simple of type* t, a Dynkin diagram of type $n \cdot t$ is said to be *isotypic of type* t. Let D_0 be the set of connected components of D and let

$$a : D_0 \to T$$

be the obvious map. The type of D is none other than $\sum_{x\in D_0} a(x)$.

5.2.

Let S be a scheme, D an S-Dynkin scheme (verifying the restrictive condition stated above). The cokernel of the pair of morphisms $L \Rightarrow D$ ($L =^{27}$ scheme of bonds of D) is written D_0. This is the "scheme of connected components" of D (it

exists trivially when D is constant; the general case is deduced from this by descent); it is a finite twisted constant S-scheme. One has a canonical morphism

$$a : D_0 \to T_S.$$

For $t \in T$, set $a^{-1}(t) = D_{0,t}$; this is a subscheme of D_0, whose inverse image in D, written D_t, is the *isotypic component of type* t of the Dynkin scheme D. Each D_t is a subscheme of D, and one has

$$D = \coprod_{t\in T} D_t.$$

Note that the degree of $D_{0,t}$ at $s \in S$ is $n(s,t)$, if the type of D at s is $\sum_t n(s,t) \cdot t$.

5.3.

In what follows, we shall consider only semisimple adjoint (resp. simply connected) groups. To simplify the language, we shall state the results for adjoint groups; all statements will remain valid if one substitutes everywhere "simply connected" for "adjoint".

[27] N.D.E.: We have replaced R by L, as in 3.2.

Recall that an adjoint reduced root datum is determined up to isomorphism by the type of its Dynkin diagrams. We shall therefore say that an adjoint root datum R (resp. a semisimple adjoint group G) is *of type* $\sum n(t)\cdot t$ if its Dynkin diagrams are (resp. if its type is given by an adjoint root datum of type $\sum n(t)\cdot t$). We shall say that R or G is *simple of type* t (resp. *isotypic of type* t) if its type is t (resp. $n\cdot t$, $n>0$).

If G is a semisimple adjoint group, we shall use the symbols $Dyn_0(G)$ and $Dyn_{0,t}(G)$ in the sense defined in 5.2.

5.4.

Let t_i, $i=1,2,\dots,n$, be distinct elements of T, and let R_i be a pinned adjoint root datum isotypic of type t_i. Consider the product $R=R_1\times\cdots\times R_n$ (Exp. XXI, 6.4.1). Let S be a scheme such that the different $\acute{E}p_S(R_i)$ exist (cf. Exp. XXV). One verifies at once that there exists a canonical isomorphism

$$(*)\qquad \acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})=\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}_1)\times_S\cdots\times_S\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}_n).$$

Proposition 5.5.— *Let S be a scheme and G an adjoint semisimple S-group. There is a unique decomposition*

$$G\simeq\prod_{t\in\mathrm{T}}G_t,$$

where G_t is an adjoint semisimple S-group, isotypic of type t. Moreover, the preceding decomposition induces isomorphisms

$$\mathrm{Dyn}_t(G)\simeq\mathrm{Dyn}(G_t),\qquad \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)\simeq\prod_{t\in\mathrm{T}}\underline{\mathrm{Aut}}_{S\text{-gr.}}(G_t).$$

This was proved above when G is split. In the general case one may suppose that G has constant type $\mathscr{R}$. Using the preceding decomposition of $A_S(\mathscr{R})$ and 1.17, one obtains the desired decomposition of G. The other assertions then follow by descent.

Remark 5.6.— More generally, if G and H are two adjoint semisimple groups, there are canonical isomorphisms in the following commutative diagram:

$$\begin{array}{ccc}
\underline{\mathrm{Isom}}_{S\text{-gr.}}(G,H) & \xrightarrow{\sim} & \prod_{t\in\mathrm{T}}\underline{\mathrm{Isom}}_{S\text{-gr.}}(G_t,H_t)\\
\downarrow & & \downarrow\\
\underline{\mathrm{Isomext}}(G,H) & \xrightarrow{\sim} & \prod_{t\in\mathrm{T}}\underline{\mathrm{Isomext}}(G_t,H_t)\\
\wr\downarrow & & \downarrow\wr\\
\underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G),\mathrm{Dyn}(H)) & \xrightarrow{\sim} & \prod_{t\in\mathrm{T}}\underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G_t),\mathrm{Dyn}(H_t)).
\end{array}$$

Remark 5.7.— One can give the following intrinsic characterization of G_t, which we state without proof: it is the largest reductive subgroup of G that is invariant (and indeed characteristic) and isotypic of type t.

The preceding proposition reduces the study of adjoint semisimple groups to that of isotypic adjoint semisimple groups. We now undertake this study.

5.8.

Let $\mathscr{R}$ be a pinned reduced adjoint simple root datum of type t, let I be a nonempty finite set, and let $\mathscr{R}^I$ be the product root datum of copies $\mathscr{R}_i$ of $\mathscr{R}$, for $i \in I$. Denote by E the group of automorphisms of $\mathscr{R}$, identified with the group of automorphisms of its Dynkin diagram D. The Dynkin diagram of $\mathscr{R}^I$ is identified with D^I, and hence there is an exact sequence

$$1 \longrightarrow \mathrm{Aut}(D)^I \longrightarrow \mathrm{Aut}(D^I) \longrightarrow \mathrm{Aut}(I) \longrightarrow 1,$$

where $\mathrm{Aut}(I)$ is the group of permutations of I. It follows that the canonical isomorphism

$$\acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R})^I \simeq \acute{\mathrm{E}}\mathrm{p}_S(\mathscr{R}^I)$$

induces an exact sequence

$$1 \longrightarrow A_S(\mathscr{R})^I \longrightarrow A_S(\mathscr{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S) \longrightarrow 1.$$

The last group is the constant S-group associated with $\mathrm{Aut}(I)$. Notice also that I is canonically identified with the set of connected components of D^I.

Let G be an S-semisimple group of type $\mathscr{R}^I$, thus defining (cf. 1.17) a principal homogeneous bundle P under $A_S(\mathscr{R}^I)$. The definitions by descent of $\mathrm{Dyn}(G)$ (3.7) and of $\mathrm{Dyn}_0(G)$ (5.2) show that the bundle associated with P by

$$A_S(\mathscr{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S)$$

is precisely $\underline{\mathrm{Isom}}_S(I_S, \mathrm{Dyn}_0(G))$, corresponding to the form $\mathrm{Dyn}_0(G)$ of I_S. Using again the equivalence of categories 1.17 and the preceding exact sequence, one deduces formally that there is a reductive $\mathrm{Dyn}_0(G)$-group G_0 of type $\mathscr{R}$, and an S-isomorphism

$$G \simeq \prod_{\mathrm{Dyn}_0(G)/S} G_0.$$

Proposition 5.9.— *Let S be a scheme and G an adjoint semisimple S-group isotypic of type t. There exist an adjoint semisimple $\mathrm{Dyn}_0(G)$-group G_0, simple of type t, and a unique S-isomorphism*

$$G \simeq \prod_{\mathrm{Dyn}_0(G)/S} G_0.$$

Moreover, this isomorphism induces an exact sequence

$$1 \longrightarrow \prod_{\mathrm{Dyn}_0(G)/S} \underline{\mathrm{Aut}}_{\mathrm{gr.}}(G_0) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\mathrm{Aut}}_S(\mathrm{Dyn}_0(G)) \longrightarrow 1.$$

One may plainly suppose that G has constant type $n\,t$. The first assertion has already been proved (and uniqueness is clear). The second then follows from the split case by descent.

Combining 5.5 and 5.9 gives:

Proposition 5.10.— *Let S be a scheme, let G be an adjoint semisimple S-group, and put $\mathfrak{D} = \mathrm{Dyn}(G)$.*

(i) There is a canonical decomposition

$$G \simeq \prod_{t \in \mathrm{T}} \prod_{\mathfrak{D}_{0,t}/S} G_{0,t} \simeq \prod_{\mathrm{Dyn}_0(G)/S} G_0,$$

where each $G_{0,t}$ is a simple adjoint $\mathrm{Dyn}_{0,t}(G)$-group of type t; equivalently, G_0 is an adjoint semisimple $\mathrm{Dyn}_0(G)$-group whose type at $x \in \mathrm{Dyn}_0(G)$ is $a(x) \in \mathrm{T}$, where $a : \mathrm{Dyn}_0(G) \to \mathrm{T}_S$ was defined in 5.2.

(ii) The preceding decompositions induce isomorphic exact sequences, written vertically below. Here $\underline{\mathrm{Aut}}_{S,a}(\mathfrak{D}_0)$ denotes the scheme of automorphisms of $\mathrm{Dyn}_0(G)$ commuting with $a : \mathrm{Dyn}_0(G) \to \mathrm{T}_S$:

$$\begin{array}{ccc}
1 & & 1 \\
\downarrow & & \downarrow \\
\prod_{t\in \mathrm{T}} \prod_{\mathfrak{D}_{0,t}/S} \underline{\mathrm{Aut}}_{\mathrm{gr.}}(G_{0,t}) & \xrightarrow{\sim} & \prod_{\mathrm{Dyn}_0(G)/S} \underline{\mathrm{Aut}}_{\mathrm{gr.}}(G_0) \\
\downarrow & & \downarrow \\
\underline{\mathrm{Aut}}_{S\text{-gr.}}(G) & = & \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \\
\downarrow & & \downarrow \\
\prod_{t\in \mathrm{T}} \underline{\mathrm{Aut}}_S(\mathrm{Dyn}_{0,t}(G)) & \xrightarrow{\sim} & \underline{\mathrm{Aut}}_{S,a}(\mathrm{Dyn}_0(G)) \\
\downarrow & & \downarrow \\
1 & & 1
\end{array}$$

Corollary 5.11.— *Under the preceding hypotheses, the following three categories are equivalent:*

(i) the category of principal homogeneous bundles under G;

(ii) the category of principal homogeneous bundles under G_0;

(iii) the product, for $t \in \mathrm{T}$, of the categories of principal homogeneous bundles under the $G_{0,t}$.

This follows formally from the preceding decompositions and 8.4.

Corollary 5.12.— *There are canonical isomorphisms*

$$\underline{\mathrm{Tor}}(G) \simeq \prod_{t\in \mathrm{T}} \prod_{\mathfrak{D}_{0,t}/S} \underline{\mathrm{Tor}}(G_{0,t}) \simeq \prod_{\mathrm{Dyn}_0(G)/S} \underline{\mathrm{Tor}}(G_0),$$

and likewise with Tor *replaced by* Bor*, respectively by* Kil*.*

This is immediate from the split case.

Remark 5.13.— The canonical morphism

$$\mathrm{Dyn}(G) \longrightarrow \mathrm{Dyn}_0(G)$$

allows one to regard $\mathrm{Dyn}(G)$ as a Dynkin scheme over $\mathrm{Dyn}_0(G)$; in fact, it is the Dynkin scheme $\mathrm{Dyn}(G_0)$ of G_0.

Likewise, if $T \subset B$ is a Killing couple of G, corresponding canonically to a Killing couple $T_0 \subset B_0$ of G_0, then the obstructions to the quasi-splitting of G and G_0, which lie by 3.9 in

$$\mathrm{Pic}\big(\mathrm{Dyn}(G)\big) = \mathrm{Pic}\big(\mathrm{Dyn}(G_0)\big),$$

coincide. Hence:

Corollary 5.14.— *The following conditions are equivalent:*

(i) G is quasi-splittable;

(ii) G_0 is quasi-splittable;

(iii) each $G_{0,t}$, $t \in \mathrm{T}$, is quasi-splittable.

6. Automorphisms of Borel Subgroups of Reductive Groups

Lemma 6.1.— *Let S be a scheme, let (G, T, M, R) be a split S-group, let Δ be a system of simple roots of R, and let B be the corresponding Borel subgroup. Then B^u is generated, as an* (fppf) *sheaf of groups, by the U_α, $\alpha \in \Delta$.*

Let H be the subgroup sheaf of B^u generated by the U_α, $\alpha \in \Delta$. We prove $H \supset U_\beta$, for $\beta \in R_+$, by induction on $\mathrm{ord}(\beta) = \mathrm{ord}_\Delta(\beta) > 0$ (cf. Exp. XXI, 3.2.15). The assertion holds by hypothesis when $\mathrm{ord}(\beta) = 1$. Suppose $\mathrm{ord}(\beta) > 1$, and suppose $U_\gamma \subset H$ whenever $\mathrm{ord}(\gamma) < \mathrm{ord}(\beta)$. There is an $\alpha \in \Delta$ such that $\beta - \alpha \in R_+$ (Exp. XXI, 3.1.1). Let p be the largest integer such that $\beta - p\alpha = \beta' \in R_+$. Then $U_\alpha, U_{\beta'} \subset H$ and $\beta' - \alpha \notin R$. Thus it remains to prove:

Lemma 6.2.— *Keep the notation of Exp. XXIII, 6.4. Suppose $p = 1$, that is, $\beta - \alpha$ is not a root. If H is a subgroup sheaf of B^u such that $U_\alpha, U_\beta \subset H$, then $U_{i\alpha+j\beta} \subset H$ whenever $i\alpha + j\beta \in R$, $i > 0$, $j > 0$.*

Distinguish four cases according as $q = 0, 1, 2, 3$. In what follows, x and y denote arbitrary sections of $\mathbb{G}_{a,S'}$, where $S' \to S$.

For $q = 0$, there is nothing to prove.

For $q = 1$, one has

$$p_{\alpha+\beta}(\pm x) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S'),$$

and hence $U_{\alpha+\beta} \subset H$.

For $q = 2$, one similarly has

$$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2 y) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S').$$

Thus, after changing some signs,

$$F(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2 y) \in H(S').$$

Since $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ commute,

$$F(x, 1)F(1, -x) = p_{2\alpha+\beta}(x^2 - x).$$

[28] If $a \in \mathbb{G}_a(S)$, the equation $X^2 - X = a$ defines a free surjective extension of S, namely $\mathrm{Spec}\, \mathcal{O}_S[X]/(X^2 - X - a)$. Consequently $p_{2\alpha+\beta}(a) \in H(S')$, so $U_{2\alpha+\beta} \subset H$, and then $U_{\alpha+\beta} \subset H$, since $F(1, a) \in H(S')$.

For $q = 3$, one similarly has

$$F(x, y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2 y)p_{3\alpha+\beta}(x^3 y)p_{3\alpha+2\beta}(x^3 y^2) \in H(S'),$$

[28] *Editors' note.* We corrected $F(-1, x)$ to $F(1, -x)$.

and, since

$$p_{3\alpha+2\beta}(\pm x) = F(1,1)^{-1}p_\beta(-x)F(1,1)p_\beta(x) \in H(S'),$$

one obtains $U_{3\alpha+2\beta} \subset H$, and therefore

$$K(x,y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y) \in H(S').$$

Computing

$$K(x+y,1)K(-x,1)K(1,y)^{-1}$$

modulo $U_{3\alpha+2\beta}$, one finds

$$p_{2\alpha+\beta}(2x^2+y^2+2xy-y)p_{3\alpha+\beta}(y^3+3xy^2+3x^2y-y) \in H(S').$$

If $a \in \mathbb{G}_a(S)$, the "equations"

$$x^2 = -xy-y+1-a, \qquad y^2 = 3y-2+3a$$

define a free surjective extension of S. One then has $y^3+3xy^2+3x^2y-y=0$, and the preceding expression gives $p_{2\alpha+\beta}(a) \in H(S')$. [29] Thus H contains $U_{2\alpha+\beta}$ and $U_{3\alpha+2\beta}$, and

$$E(x,y) = p_{\alpha+\beta}(xy)p_{3\alpha+\beta}(x^3y) \in H(S').$$

Since $p_{\alpha+\beta}(xy)$ and $p_{3\alpha+\beta}$ commute, the preceding computation applies:

$$E(1,x)E(1,-x) = p_{3\alpha+\beta}(x^3-x).$$

Hence $U_{3\alpha+\beta} \subset H$, and then $U_{\alpha+\beta} \subset H$.

Remark 6.2.1.— The preceding proof shows that the hypothesis "H contains U_α and U_β" could have been replaced by: "H contains U_α or U_β, and H is invariant under $\mathrm{int}(U_\alpha)$ and $\mathrm{int}(U_\beta)$."

Theorem 6.3.— *Let S be a scheme, let G and G' be two semisimple S-groups, and let B (respectively B') be a Borel subgroup of G (respectively G'). Every isomorphism $B \xrightarrow{\sim} B'$ is induced by a unique isomorphism $G \xrightarrow{\sim} G'$.*

The assertion is local for the étale topology, so we may suppose that G is split: $G = (G,T,M,R)$, with B defined by the system R_+ of positive roots of R. The given isomorphism $u : B \xrightarrow{\sim} B'$ induces an isomorphism from T onto a maximal torus T' of B', hence of G'. The given isomorphism $T \simeq D_S(M)$ gives an isomorphism $T' \simeq D_S(M)$ in which the elements of R_+ become the constant roots of B' relative to T', and hence the elements of R become the constant roots of G' relative to T'. Since G and G' are semisimple, the coroots are determined by duality (Exp. XXI, 1.2.5); this proves that (T',M,R) is a splitting of G' such that $\mathscr{R}(G) = \mathscr{R}(G')$.

Applying Exp. XXIII, 5.1 (the uniqueness theorem), one obtains a unique isomorphism $G \xrightarrow{\sim} G'$ agreeing with u on T and on the U_α, $\alpha \in \Delta$. By 5.1, its restriction to B is u.

Remark 6.3.1.— (i) Using Exp. XXII, 4.1.9 and arguing as in *loc. cit.*, 4.2.12, one may replace "isomorphism" in the theorem by "isogeny" (respectively, "central isogeny").

(ii) The theorem is false for reductive groups. For example, take $G = G' = \mathrm{SL}_2 \times \mathbb{G}_m$, identified with the group

$$\left\{ \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & h \end{pmatrix} \;\middle|\; ad-bc=1,\ h \text{ invertible} \right\}.$$

[29] *Editors' note.* We corrected $p_{3\alpha+\beta}(a)$ to $p_{2\alpha+\beta}(a)$ and supplied the end of the argument.

take $B = B'$ to be the Borel subgroup defined by $c = 0$, and let u be the following automorphism of B:

$$u(a, b, d, h) = (a, b, d, ah).$$

Corollary 6.4.— *The functor*

$$(G, B) \longmapsto B$$

from the category of pairs (G, B), where G is an S-semisimple group and B is a Borel subgroup of G, to the category of S-group schemes, with isomorphisms as morphisms, is fully faithful.

Corollary 6.5.— *Let S be a scheme, G an S-semisimple group, B a Borel subgroup of G, and B' an S-group locally isomorphic to B for the* (fpqc) *topology. Then B' is locally isomorphic to B for the local finite étale topology,*[30] *and there exists an S-semisimple group G' of which B' is a Borel subgroup. Moreover, G' is unique up to a unique isomorphism inducing the identity on B'.*

Corollary 6.6.— *Let S be a scheme, G an S-semisimple group, and B a Borel subgroup of G. Then $\underline{\mathrm{Aut}}_{S\text{-gr.}}(B)$ is representable by an affine smooth S-scheme, $\underline{\mathrm{Autext}}(B)$ is representable by an étale finite S-scheme, and the evident morphisms induce an isomorphism of exact sequences.*

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \underline{B}_{\mathrm{ad}}(G) & \longrightarrow & \underline{\mathrm{Aut}}_{S\text{-gr.}}(G, B) & \longrightarrow & \underline{\mathrm{Autext}}(G) & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \underline{B}_{\mathrm{ad}} & \longrightarrow & \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) & \longrightarrow & \underline{\mathrm{Autext}}(B) & \longrightarrow & 1.
\end{array}$$

This follows immediately from 2.1 and the preceding results. We leave to the reader the task of developing the analogues of the results of nos. 1, 2, 3, and 4 in the foregoing setting.

Remark 6.7.— If S is a scheme and B an S-group, then B can be a Borel subgroup of only one semisimple group, uniquely determined by B. It remains to characterize those S-groups B which are Borel subgroups of semisimple groups.[31]

7. Representability of the functors $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$, for G reductive

7.1. The split case

7.1.1.

Let S be a scheme, G a pinned S-group, T its maximal torus, and Δ the set of simple roots. For $\alpha \in \Delta$, let

$$u_\alpha \in U_\alpha^\times(S), \qquad w_\alpha \in \mathrm{Norm}_G(T)(S)$$

[30] *Editors' note.* This topology is denoted (étf) in Exposé IV, 6.3. In other words, for every $s \in S$ there exist an open neighborhood U of s and a finite surjective étale morphism $V \to U$ such that $B' \times_S V \simeq B \times_S V$.

[31] *Editors' note.* One may ask whether every smooth affine S-group whose geometric fibers are Borel subgroups of semisimple groups is itself a Borel subgroup of a semisimple S-group. We have suppressed the counterexample in the original, with S the scheme of dual numbers over a field k, because it was erroneous, as M. Demazure pointed out.

be the elements defined by the pinning.

Let H be an S-group scheme. We are interested in the functor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H)$, and more precisely in the morphism

$$q : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T,H).$$

Let

$$f_T : T \longrightarrow H$$

be an S-group homomorphism and put $q^{-1}(f_T) = \mathscr{F}$. Thus

$$\mathscr{F}(S') = \left\{ f \in \mathrm{Hom}_{S'\text{-gr.}}(G_{S'}, H_{S'}) \;\middle|\; f|_{T_{S'}} = (f_T)_{S'} \right\}.$$

There is a morphism of S-functors

$$i : \mathscr{F} \longrightarrow H^{2n},$$

where $n = \mathrm{Card}(\Delta)$, defined on points by

$$i(f) = \big(f(u_\alpha), f(w_\alpha)\big)_{\alpha\in\Delta}.$$

By Exposé XXIII, 1.9, i is a monomorphism.

Proposition 7.1.2.— *If H is separated over S, then $\mathscr{F}$ is representable and i is a closed immersion.*

The usual technique of relative representability[32] shows that it is enough to prove the following. Given sections

$$v_\alpha, h_\alpha \in H(S), \qquad \alpha \in \Delta,$$

the S-schemes S' for which there exists an S'-homomorphism

$$f : G_{S'} \longrightarrow H_{S'}$$

extending $(f_T)_{S'}$ and satisfying

$$f(u_\alpha) = v_\alpha, \qquad f(w_\alpha) = h_\alpha$$

are exactly those which factor through a certain closed subscheme of S. We may plainly suppose S affine.

For convenience, say that a morphism $Y \to X$[33] of affine schemes satisfies condition (L) if Y is the spectrum of an $\mathcal{O}(X)$-algebra which is a free $\mathcal{O}(X)$-module. Within the category of affine schemes, products and composites of morphisms satisfying (L) again satisfy (L), and (L) is stable under base change.

Lemma 7.1.3.— *Suppose that S is affine, and let $\alpha \in \Delta$. Consider the morphism*

$$a : T \times_S T \longrightarrow G_{a,S}, \qquad a(t,t') = \alpha(t) + \alpha(t').$$

Then:

(i) a is faithfully flat and of finite presentation;

(ii) if R is the fiber square of a, the structure morphism $R \to S$ satisfies (L).

[32] *Editors' note.* See, for example, the proof of Exposé XXII, 5.8.1.

[33] *Editors' note.* We have corrected $X \to Y$ to $Y \to X$.

The morphism a is surjective, since $\alpha : T \to G_{m,S}$ is surjective, and it is plainly of finite presentation. Since α satisfies (L), it is enough, for both assertions, to show that

$$u : G_{m,S}^2 \longrightarrow G_{a,S}, \qquad u(x,y) = x + y$$

satisfies (L). Equivalently, for every ring A, the ring

$$A[X, Y, X^{-1}, Y^{-1}]$$

must be free over its subring $A[X+Y]$.[34] Now $A[X,Y]$ is free with basis $\{1, X\}$ over

$$B = A[X+Y, XY],$$

because $X^2 - (X+Y)X + XY = 0$. Hence

$$A[X^{\pm 1}, Y^{\pm 1}] = A[X,Y]_{XY}$$

is free over

$$B_{XY} = A[X+Y, (XY)^{\pm 1}]$$

with basis $\{1, X\}$, and free over $A[X+Y]$ with basis

$$(XY)^p, \quad X(XY)^p, \qquad p \in \mathbb{Z}.$$

Lemma 7.1.4.— *Let $\alpha \in \Delta$, and let*

$$b : T \times_S T \longrightarrow H$$

be the morphism defined on points by

$$b(t,t') = \mathrm{int}(f_T(t))(v_\alpha)\ \mathrm{int}(f_T(t'))(v_\alpha).$$

Let $f_\alpha : U_\alpha \to H$ be an S-morphism. The following conditions are equivalent:[35]

(i) f_α is a group homomorphism, $f_\alpha(u_\alpha) = v_\alpha$, and

$$f_T(t) f_\alpha(x) f_T(t)^{-1} = f_\alpha(\alpha(t)x)$$

for all $t \in T(S')$, $x \in U_\alpha(S')$, and $S' \to S$;

(ii) $f_\alpha(u_\alpha) = v_\alpha$, and

$$f_\alpha(a(t,t')u_\alpha) = b(t,t')$$

for all $t, t' \in T(S')$ and $S' \to S$.

If f_α satisfies (i), then

$$f_\alpha(\alpha(t)u_\alpha) = \mathrm{int}(f_T(t))(v_\alpha),$$

which immediately gives (ii). Conversely, suppose (ii). Since a is an (fpqc) covering, to prove the last condition in (i) it suffices to show that, for $t, t', t'' \in T(S')$,

$$f_T(t)\, f_\alpha(a(t',t'')u_\alpha)\, f_T(t)^{-1} = f_\alpha(\alpha(t)a(t',t'')u_\alpha).$$

[34] *Editors' note.* We have simplified the proof in the original.

[35] *Editors' note.* In what follows the group law of U_α is written additively. If $p_\alpha : G_{a,S} \xrightarrow{\sim} U_\alpha$ is the isomorphism with $p_\alpha(1) = u_\alpha$, and $x = p_\alpha(z)$, then $\alpha(t)x$ denotes $p_\alpha(\alpha(t)z)$, while $a(t,t')u_\alpha$ denotes $p_\alpha(a(t,t')) = \alpha(t)u_\alpha + \alpha(t')u_\alpha$.

This is the same as

$$f_T(t)b(t',t'')f_T(t)^{-1} = b(tt',tt''),$$

which follows directly from the definition of b.

It remains to prove that f_α is a group homomorphism. The preceding property gives

$$\begin{aligned} f_\alpha(\alpha(t)u_\alpha)f_\alpha(\alpha(t')u_\alpha) &= \big(f_T(t)f_\alpha(u_\alpha)f_T(t)^{-1}\big)\big(f_T(t')f_\alpha(u_\alpha)f_T(t')^{-1}\big) \\ &= b(t,t') \\ &= f_\alpha(\alpha(t)u_\alpha + \alpha(t)\alpha(t')u_\alpha). \end{aligned}$$

Thus $f_\alpha(x+x') = f_\alpha(x)f_\alpha(x')$ whenever x, x' are sections of the schematically dense open subset $U_\alpha^\times \subset U_\alpha$. Exposé XVIII, 1.4 then gives the result.

7.1.5.

Fix provisionally an $\alpha \in \Delta$. Since a is faithfully flat and quasi-compact, $G_{a,S}$ is identified with the quotient of $T \times_S T$ by the equivalence relation R defined by a. Write $i_1, i_2 : R \to T \times_S T$ for its two projections.

$$R \underset{i_2}{\overset{i_1}{\rightrightarrows}} T \times_S T$$

For b to factor through a,[36] it is necessary and sufficient that

$$b \circ i_1 = b \circ i_2.$$

Let K be the equalizer of $b \circ i_1$ and $b \circ i_2$.

$$R \underset{b\circ i_2}{\overset{b\circ i_1}{\rightrightarrows}} H$$

Then the required condition is $K = R$. Since H is separated over S, K is a closed subscheme of R. On the other hand, R is essentially free over S, by 7.1.3.

7.1.6.

It follows that, for an S-scheme S', an f_α satisfying 7.1.4(i) exists over S', and is then unique, if and only if

$$K_{S'} = R_{S'}$$

and the resulting morphism satisfies

$$f_\alpha(u_\alpha) = v_\alpha.$$

The first condition is equivalent to requiring $S' \to S$ to factor through a certain closed subscheme of S (Exposé VIII, 6.4[37]). After replacing S by this closed subscheme, the second condition again defines a closed subscheme of S, since it is the equality of two sections of the separated S-scheme H.

After replacing S by the resulting closed subscheme, we may thus suppose that a morphism $f_\alpha : U_\alpha \to H$ satisfying 7.1.4(i) exists. Intersecting the closed subschemes obtained for all $\alpha \in \Delta$, we may suppose this condition holds for every $\alpha \in \Delta$.

[36] *Editors' note.* That is, for condition 7.1.4(ii) to hold.

[37] *Editors' note.* See also the addition in Exposé $\mathrm{VI_B}$, 6.2.3.

7.1.7.

Similarly, for each $\alpha \in \Delta$, consider the two S-group homomorphisms

$$f_T \circ \mathrm{int}(w_\alpha), \qquad \mathrm{int}(h_\alpha) \circ f_T : T \longrightarrow H.$$

Since H is separated over S and T is essentially free over S, the same reasoning shows that, after replacing S by a closed subscheme, one may suppose that

$$f_T \circ \mathrm{int}(w_\alpha) = \mathrm{int}(h_\alpha) \circ f_T$$

for every $\alpha \in \Delta$.

7.1.8.

The generators-and-relations theorem (Exposé XXIII, 3.5) now shows that a group homomorphism $f : G \to H$ with the required properties exists if and only if a certain finite set of algebraic relations among the sections

$$h_\alpha, \qquad v_\alpha, \qquad f_T(\alpha^*(-1)), \qquad \alpha \in \Delta,$$

is satisfied:

$$R_i\Big((h_\alpha)_\alpha, (v_\alpha)_\alpha, \big(f_T(\alpha^*(-1))\big)_\alpha\Big) = e, \qquad i = 1, \ldots, n.$$

Because H is separated over S, these relations again define a closed subscheme of S, proving Proposition 7.1.2.

Corollary 7.1.9.— *Let S be a scheme, G a split reductive S-group, T its maximal torus, and H a separated S-group scheme. Let $\mathcal{P}$ be a property of morphisms satisfying:*

(i) every closed immersion has $\mathcal{P}$;

(ii) a composite of morphisms having $\mathcal{P}$ has $\mathcal{P}$;

(iii) $\mathcal{P}$ is stable under base change;

(iv) the structure morphism $H \to S$ has $\mathcal{P}$.

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$$

is relatively representable by a separated morphism having $\mathcal{P}$.[38]

Indeed, we may suppose G pinned. The structure morphism $H^{2n} \to S$ has $\mathcal{P}$, and 7.1.2 gives the assertion.

Corollary 7.1.10.— *Let S be a scheme, G a split reductive S-group, and H a smooth S-group scheme with affine fibers.*[39] *Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S-scheme locally of finite presentation and separated over S.*

Since H is smooth over S, its neutral component H^0 is defined; it has affine fibers and is smooth, separated, and of finite presentation over S (Exposé $\mathrm{VI_B}$, 3.10 and 5.5). Since G has connected fibers, H may be replaced by H^0. As G and H are then of finite presentation, we may suppose S noetherian and apply 7.1.9 with $\mathcal{P}$ equal to the property "of finite type". By Exposé XV, 8.9, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$ is representable by a separated S-scheme locally of finite type.

Remark 7.1.11.— If H is affine over S, the reference to Exposé XV may be replaced by Exposé XI, 4.2.

[38] *Editors' note.* More explicitly, for every representable $S' \to \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$, the fiber product $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \times S'$ is representable, and its morphism to S' is separated and has $\mathcal{P}$.

[39] *Editors' note.* We have suppressed the superfluous hypothesis that H be quasi-separated.

7.2. General case

Proposition 7.2.1.— *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. Let H be an S-group scheme whose structure morphism $H \to S$ satisfies:*

> (+) *Every point $s \in S$ has an open neighborhood U such that $H_U \to U$ is quasi-projective.*

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H)$$

is relatively representable by a morphism satisfying (+).

When G is split relative to T, apply 7.1.9 with $\mathcal{P} = (+)$. When G is locally isotrivial, for example when G is semisimple (4.2.2), note that the assertion is local for the local finite étale topology. Indeed, quasi-projectivity is local for the global finite étale topology and ensures effectiveness of descent for that topology (SGA 1, VIII, 7.7). The general case follows from the next lemma.

Lemma 7.2.2.— *Let S be a scheme, G a reductive S-group, T a maximal torus of G, G' the derived group of G (Exposé XXII, 6.2), and $T' = T \cap G'$ the maximal torus of G' corresponding to T (Exposé XXII, 6.2.8). The natural diagram involving G, G', T, T' is a pushout in the category of S-group sheaves. Equivalently, for every S-group sheaf H, the corresponding square of Hom-functors is cartesian.*

$$\begin{array}{ccc} G & \longleftarrow & G' \\ \uparrow & & \uparrow \\ T & \longleftarrow & T' \end{array}$$

$$\begin{array}{ccc} \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) & \longrightarrow & \underline{\mathrm{Hom}}_{S\text{-gr.}}(G', H) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H) & \longrightarrow & \underline{\mathrm{Hom}}_{S\text{-gr.}}(T', H) \end{array}$$

If $\mathrm{rad}(G)$ is the radical of G, then $\mathrm{rad}(G) \subset T$, so

$$\mathrm{rad}(G) \cap G' = \mathrm{rad}(G) \cap T' = K.$$

Multiplication in G induces isogenies (Exposé XXII, 6.2)

$$i : G' \times_S \mathrm{rad}(G) \longrightarrow G, \qquad j : T' \times_S \mathrm{rad}(G) \longrightarrow T.$$

Let

$$f_{G'} : G' \to H, \qquad f_T : T \to H, \qquad f_{T'} : T' \to H$$

be S-group homomorphisms such that

$$f_{G'}|_{T'} = f_T|_{T'} = f_{T'}.$$

We show that there is a unique S-group homomorphism $f : G \to H$ inducing $f_{G'}$ and f_T. Put

$$f_{\mathrm{rad}} = f_T|_{\mathrm{rad}(G)}$$

[40] and define

$$f_1 = f_{G'} \cdot f_{\mathrm{rad}} : G' \times_S \mathrm{rad}(G) \longrightarrow H.$$

For f to exist, and it is plainly then unique, it is necessary and sufficient that f_1 be trivial on the kernel of i, that is, that $f_{G'}$ and f_{rad} coincide on K. The existence of f_T similarly shows that $f_{T'}$ and f_{rad} coincide on K. Finally, $f_{G'}$ and $f_{T'}$ coincide on $K \subset T'$, proving the lemma.

Arguing as in 7.1.10, Proposition 7.2.1 gives:

Corollary 7.2.3.— *Let S be a scheme, G a reductive S-group, and H a smooth, quasi-projective S-group scheme with affine fibers. Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable by an S-scheme locally of finite presentation and separated over S.*

7.3. Phenomena specific to characteristic zero

If G and H are smooth S-groups, with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, there is a canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie}}(\mathfrak{g}, \mathfrak{h}),$$

where the functor on the right has the evident meaning.

Proposition 7.3.1.— *Let S be a scheme of characteristic zero, that is, a $\mathbb{Q}$-scheme; let G be a reductive S-group, and H a smooth, quasi-projective S-group scheme with affine fibers.*

(i) $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ *is representable by a smooth separated S-scheme.*

(ii) If G is semisimple, this scheme is affine and of finite presentation over S.

(iii) If G is simply connected (Exposé XXII, 4.3.3), the canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie}}(\mathfrak{g}, \mathfrak{h})$$

is an isomorphism.

Lemma 7.3.2.— *Let k be a field of characteristic zero, G a reductive k-group, V a finite-dimensional k-vector space, and $G \to \mathrm{GL}(V)$ a linear representation. Then*

$$H^0(G, V) = H^0(\mathfrak{g}, V), \qquad H^i(G, V) = 0 \quad (i > 0).$$

The first equality is valid in general for a smooth connected group.[41] For a reductive group it can also be proved as follows. We may suppose k algebraically closed, hence G split and generated by subgroups isomorphic to $G_{m,k}$,[42] and the assertion for this group is immediate.

The first equality shows that the functor $H^0(G, V)$ is exact in V when G is semisimple: $\mathfrak{g}$ is then a semisimple Lie algebra, and one applies [BLie], §I.6, Exercise 1(b). The

[40] *Editors' note.* We have replaced f_r by f_{rad}.

[41] *Editors' note.* The inclusion $V^G \subset V^{\mathfrak{g}}$ is clear. Conversely, $C = \mathrm{Centr}_G(V^{\mathfrak{g}})$ is a closed subgroup of G, smooth because $\mathrm{char}(k) = 0$. By Exposé II, 5.3.1, $\mathrm{Lie}(C) = \mathrm{Centr}_{\mathfrak{g}}(V^{\mathfrak{g}}) = \mathfrak{g}$; because C is smooth and G connected, $C = G$. Thus $V^{\mathfrak{g}} \subset V^G$. See also [DG70], §II.6, Proposition 2.1(c).

[42] *Editors' note.* The union of the maximal tori of G is dense in G; cf. *Bible*, §6.5, Theorem 5, or [Ch05], §6.6, Theorem 6.

assertion remains true when G is reductive. Indeed, introducing the radical R of G[43] and the semisimple quotient G/R, one has

$$H^0(G,V) = H^0(G/R, H^0(R,V)),$$

so $H^0(G,-)$ is a composite of two exact functors. Exposé I, 5.3.1 then gives $H^i(G,V) = 0$ for $i > 0$.

Remark 7.3.3.— Under the preceding hypotheses, if G is semisimple, then

$$H^1(\mathfrak{g},V) = H^2(\mathfrak{g},V) = 0;$$

see [BLie], *loc. cit.*, (b) and (d).

7.3.4.

We prove the proposition. By 7.2.3, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H)$ is already representable by an S-scheme locally of finite presentation and separated over S. To show it smooth, it suffices to show it infinitesimally smooth (Exposé XI, 1.8),[44] and this follows from Exposé III, 2.8(i) and 7.3.2. This proves (i).

We next show that (ii) follows from (iii). First it is enough to prove that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H)$ is affine over S, since it is then of finite presentation, being smooth by (i). In any case,

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie}}(\mathfrak{g},\mathfrak{h})$$

is represented by a closed subscheme of

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod}}(\mathfrak{g},\mathfrak{h}) \simeq \mathrm{W}\left(\mathfrak{g}^\vee \otimes_{\mathcal{O}_S} \mathfrak{h}\right),$$

which is an affine S-scheme. Thus the desired conclusion follows when G is simply connected.

In general, we may suppose G split, so that $G \simeq \mathrm{Ép}_S(\mathscr{R})$. Introduce the simply connected root datum $\mathrm{sc}(\mathscr{R})$ (Exposé XXI, 6.5.5). The existence theorem (Exposé XXV, 1.1) gives a simply connected S-group $\widetilde{G}$ and a central isogeny

$$\widetilde{G} \longrightarrow G.$$

Its kernel K is a finite diagonalizable S-group, hence a twisted constant S-group because S has characteristic zero. The functor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(K,H)$ is plainly represented by a separated S-scheme. For example, if $K \simeq (\mathbb{Z}/n\mathbb{Z})_S$, then it is isomorphic to

$$\ker\Big(H \xrightarrow{n} H\Big).$$

There is an exact sequence of pointed S-schemes

$$1 \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H) \xrightarrow{u} \underline{\mathrm{Hom}}_{S\text{-gr.}}(\widetilde{G},H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(K,H).$$

Thus u is a closed immersion, and $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H)$ is affine over S.

[43] *Editors' note.* Observe that R is a torus.

[44] *Editors' note.* This means that for every local artinian S-scheme S', with closed point s', every homomorphism of $\kappa(s')$-groups $G_{s'} \to H_{s'}$ lifts to an S'-group homomorphism $G_{S'} \to H_{S'}$. By Exposé III, 2.8, this follows from the vanishing of $H^2(G_{s'},V)$, where $V = \mathrm{Lie}(H_{s'}/\kappa(s'))$.

7.3.5.

It remains to prove (iii). As in the proof of (i), and using 7.3.3, one shows that

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie}}(\mathfrak{g},\mathfrak{h})$$

is smooth over S. We may therefore suppose $S = \mathrm{Spec}(k)$, where k is algebraically closed of characteristic zero. It is enough to show that Lie is bijective on k-points and induces a bijection on tangent spaces at corresponding points. First we show that

$$\mathrm{Hom}_{k\text{-gr.}}(G,H) \longrightarrow \mathrm{Hom}_{k\text{-Lie}}(\mathfrak{g},\mathfrak{h})$$

is bijective.

If $u : G \to H$ is a k-group homomorphism, its graph is a connected subgroup of $G \times_k H$, determined by its Lie algebra, which is the graph of $\mathrm{Lie}(u)$. Hence the map is injective. Conversely, if $v : \mathfrak{g} \to \mathfrak{h}$ is a Lie algebra homomorphism, its graph $\mathfrak{k}$ is a Lie subalgebra of $\mathfrak{g}\times\mathfrak{h}$, isomorphic to $\mathfrak{g}$. Because $\mathfrak{g}$ is its own derived algebra, so is $\mathfrak{k}$. A theorem of Chevalley [Ch51], §14, Theorem 15, therefore shows that $\mathfrak{k}$ is the Lie algebra of a connected subgroup K of $G \times_k H$.[45]

Since $\mathfrak{k} \simeq \mathfrak{g}$ is semisimple, K is a semisimple k-group . Moreover,

$$\dim K = \dim \mathfrak{k} = \dim \mathfrak{g} = \dim G,$$

and $K \cap H$ is finite because its Lie algebra is zero. Thus the canonical morphism

$$\mathrm{pr}_1 : K \longrightarrow G$$

is finite and dominant; because G is connected, it is surjective, and hence an isogeny. Since G is simply connected and K is semisimple, Exposé XXI, 6.2.7 shows that pr_1 is an isomorphism. Consequently K is the graph of a k-group homomorphism $u : G \to H$ with $\mathrm{Lie}(u) = v$.

Finally, let $u : G \to H$ be a k-group homomorphism. The tangent space at u to

$$\underline{\mathrm{Hom}}_{k\text{-gr.}}(G,H)$$

identifies with $Z^1(G,\mathfrak{h})$ (Exposé II, 4.2), where G acts on $\mathfrak{h}$ through $\mathrm{Ad}_H \circ u$. Similarly, the tangent space to

$$\underline{\mathrm{Hom}}_{k\text{-Lie}}(\mathfrak{g},\mathfrak{h})$$

at $\mathrm{Lie}(u)$ identifies with $Z^1(\mathfrak{g},\mathfrak{h})$. We must show that

$$Z^1(G,\mathfrak{h}) \longrightarrow Z^1(\mathfrak{g},\mathfrak{h})$$

is bijective.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathfrak{h}^G & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(G,\mathfrak{h}) & \longrightarrow & H^1(G,\mathfrak{h}) \\
 & & \downarrow & & \| & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathfrak{h}^{\mathfrak{g}} & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(\mathfrak{g},\mathfrak{h}) & \longrightarrow & H^1(\mathfrak{g},\mathfrak{h})
\end{array}$$

By 7.3.2 and 7.3.3,

$$\mathfrak{h}^G = \mathfrak{h}^{\mathfrak{g}}, \qquad H^1(G,\mathfrak{h}) = 0 = H^1(\mathfrak{g},\mathfrak{h}),$$

[45] *Editors' note.* See also [Bo91], II 7.9. We have also added the following sentence.

which gives the desired conclusion.

Remark 7.3.6.— It is presumably true that, if S is locally noetherian, G a simply connected semisimple S-group, H a smooth S-group scheme, and $\widehat{G}, \widehat{H}$ their formal completions along the identity section, then every homomorphism $\widehat{G} \to \widehat{H}$ comes from a unique homomorphism $G \to H$. This would generalize 7.3.1(iii). When $S = \mathrm{Spec}(k)$ and H is affine and of finite type, this follows from a theorem of Dieudonné [Di57], §15, Theorem 4.[46]

7.4. An example

As an example, we shall determine

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S}).$$

7.4.1.

Recall (Exposé XX, no. 5) that $\mathrm{SL}_{2,S}$ is the S-group scheme of matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

over S satisfying $ad - bc = 1$. A maximal torus T is defined by the monomorphism

$$\alpha^* : \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}, \qquad \alpha^*(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}.$$

A root of G with respect to T is defined by

$$\alpha(\alpha^*(z)) = z^2,$$

and a corresponding monomorphism is

$$p : \mathbb{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}, \qquad p(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Finally, putting $u = p(1)$, the corresponding representative of the Weyl group is

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Recall also (Exposé XX, 6.2) that $\mathrm{SL}_{2,S}$ is generated by T, $p(\mathbb{G}_{a,S})$, and w, subject to the relations

$$\alpha^*(z)p(x)\alpha^*(z^{-1}) = p(z^2x),$$
$$w\alpha^*(z)w^{-1} = \alpha^*(z^{-1}), \qquad w^2 = \alpha^*(-1), \qquad (wu)^3 = 1.$$

[46] *Editors' note.* See Exposé $\mathrm{VII_B}$ for the definition of the formal groups $\widehat{G}$ and $\widehat{H}$. Suppose that $S = \mathrm{Spec}(k)$, where k is a field. By loc. cit., 2.2.1, giving a morphism of formal k-groups $v : \widehat{G} \to \widehat{H}$ is equivalent to giving a morphism of Hopf k-algebras $\phi : \mathrm{Dist}(G) \to \mathrm{Dist}(H)$, where $\mathrm{Dist}(G)$ denotes the algebra of distributions (at the origin) of G; see Exposé $\mathrm{VII_A}$, 2.1, [DG70], §II.4, 6.1, or [Ja87], I 7.7. It is called the hyperalgebra of G in [Di57] and [Ta75]. Theorem 4 of [Di57] (see also [Ta75], 0.3.4 (f) and (g)) generalizes in this context the theorem of Chevalley used in 7.3.5: one obtains a closed connected k-subgroup $K \subset G \times H$ such that $\widehat{K}$ equals the graph of v. Since $\mathrm{Dist}(K) \to \mathrm{Dist}(G)$ is an isomorphism, $K \to G$ is finite étale. It follows that K is semisimple and then, by Exposé XXI, 6.2.7, that $K \to G$ is an isomorphism (because G is simply connected; see also [Ta75], 1.8 and 2.2). More generally, loc. cit. studies the k-groups G having property (SC): every finite étale morphism of k-groups $G' \to G$, with G' connected, is an isomorphism. Finally, the preceding argument shows that every finite-dimensional $\mathrm{Dist}(G)$-module is, uniquely, a G-module; for an extension to a split reductive k-group G (or even to a Borel subgroup of G), see [Ja87], II 1.20 (and the references given there).

7.4.2.

Let f be an endomorphism of $\mathrm{SL}_{2,S}$. It first defines a homomorphism

$$f \circ \alpha^* : \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}.$$

The closed subgroup $\ker(f \circ \alpha^*) \subset \mathbb{G}_{m,S}$ is, locally on S, either $\mu_{n,S}$, with $n \geq 1$, or $\mathbb{G}_{m,S}$. After restricting S, we may therefore assume that there are an integer $n \in \mathbb{N}$ and a monomorphism

$$f' : \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$$

such that

$$f \circ \alpha^*(z) = f'(z^n).$$

By conjugacy of monomorphisms $\mathbb{G}_{m,S} \to \mathrm{SL}_{2,S}$, after an étale covering extension of the base there is a section g of G such that

$$f \circ \alpha^*(z) = \mathrm{int}(g) \circ \alpha^*(z^n).$$

Replacing f by an inner conjugate, we are thus reduced to the case

$$f \circ \alpha^*(z) = \alpha^*(z^n)$$

for some $n \in \mathbb{N}$.

7.4.3.

Consider now

$$f \circ p : \mathbb{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}.$$

It satisfies

$$\alpha^*(z^n)(f \circ p)(x)\alpha^*(z^n)^{-1} = (f \circ p)(z^2 x).$$

If $n = 0$, it follows at once that $f \circ p$ is invariant under the homotheties of $\mathbb{G}_{a,S}$, and hence is constant. If $n \neq 0$, Exposé XXII, 4.1.9 applies: the map $x \mapsto x^n$ is an endomorphism of $\mathbb{G}_{a,S}$, and there is a $\lambda \in \mathbb{G}_{m,S}$ such that

$$f \circ p(x) = p(\lambda x^n).$$

This relation is also valid for $n = 0$, on taking $\lambda = 1$. After extending S once more, choose $z \in \mathbb{G}_{m,S}$ with $z^2 = \lambda$. Replacing f by $\mathrm{int}(\alpha^*(z)) \circ f$, we are reduced to

$$f \circ \alpha^*(z) = \alpha^*(z^n), \qquad f \circ p(x) = p(x^n).$$

7.4.4.

Finally, one must have

$$f(w)\alpha^*(z^n)f(w)^{-1} = \alpha^*(z^n)^{-1}, \qquad (f(w)u)^3 = 1.$$

By Exposé XX, 3.8, this implies $f(w) = w^n$. Since G is generated by T, $p(\mathbb{G}_{a,S})$, and w, it follows that, for every $S' \to S$ and every $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(S')$,

$$f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$

7.4.5.

Summarizing, locally on S for the étale topology, every endomorphism f of $\mathrm{SL}_{2,S}$ can be written

$$f = \phi \circ F_n,$$

where ϕ is an inner automorphism, $n \geq 0$, and

$$F_n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$

If $f = \phi \circ F_n$, the integer n is determined by any fiber f_s, for example by $\ker(f_s)$. Consequently n is a locally constant function on S.

7.4.6.

It follows that, for every endomorphism f of $\mathrm{SL}_{2,S}$, the scheme S has a canonical decomposition into open and closed subschemes

$$S_0, \qquad S_1, \qquad S_{p^n},$$

where p^n runs through the positive powers of prime numbers, such that:

(i) f_{S_0} is the zero morphism;

(ii) f_{S_1} is an isomorphism, necessarily an inner automorphism;

(iii) S_{p^n} has characteristic p, and $f_{S_{p^n}}$ has a unique decomposition

$$f_{S_{p^n}} = \phi \circ F_p^n,$$

where ϕ is an inner automorphism and F_p is the Frobenius endomorphism of $\mathrm{SL}_{2,\mathbb{F}_p}$.

7.4.7.

Equivalently,

$$\underline{\mathrm{Hom}}_{\mathbb{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbb{Z}}, \mathrm{SL}_{2,\mathbb{Z}})$$

is the disjoint-union scheme consisting of:

(i) one scheme isomorphic to $\mathrm{Spec}(\mathbb{Z})$;

(ii) one scheme isomorphic to

$$\underline{\mathrm{Aut}}_{\mathbb{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbb{Z}}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbb{Z}});$$

(iii) for each prime p and each integer $n > 0$, one scheme isomorphic to

$$\underline{\mathrm{Aut}}_{\mathbb{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbb{F}_p}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbb{F}_p}).$$

7.4.8.

In particular:

(i) $\underline{\mathrm{Hom}}_{\mathbb{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbb{F}_p}, \mathrm{SL}_{2,\mathbb{F}_p})$ has infinitely many connected components and is therefore not quasi-compact.

(ii) If S is a scheme of unequal characteristics, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S})$ is not flat over S.

8. Appendix: Cohomology of a smooth group over a henselian ring. Cohomology and the functor $\prod$

Proposition 8.1.— *Let S be a henselian local scheme, s its closed point, and G a smooth S-group scheme such that every finite subset of G is contained in an affine open subset.*[*]

(i) If P is a principal homogeneous bundle under G, there is a finite surjective étale morphism $S' \to S$ which trivializes P. Moreover,

$$\mathrm{Fib}(S, G) \simeq H^1_{ét}(S, G).$$

(ii) For every finite surjective étale morphism $S' \to S$, the canonical map

$$H^1(S'/S, G) \longrightarrow H^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is bijective.

(iii) The canonical map

$$\mathrm{Fib}(S, G) \longrightarrow \mathrm{Fib}(\kappa(s), G_s)$$

is bijective.

8.1.2.

If K is a finite separable extension of $\kappa(s)$, there is a finite surjective étale morphism $S' \to S$ such that

$$K \simeq S' \otimes_S \kappa(s).$$

[47] If P is a principal homogeneous bundle under G, then P is smooth over S, hence P_s is smooth over $\kappa(s)$. Consequently there is a finite separable extension $K/\kappa(s)$ such that P_K has a section (EGA IV_4, 17.15.10). Representing K as above, Hensel's lemma (EGA IV_4, 18.5.17) shows that $P_{S'}$ has a section. This proves the first assertion of (i).

Conversely, if P is a principal homogeneous sheaf under G for the étale topology, then a finite surjective étale morphism $S' \to S$ trivializes P: every étale covering family of a henselian local scheme is dominated by a family consisting of one morphism of that form. By the descent hypothesis imposed on G, P is representable (SGA 1, VIII, 7.6). This completes the proof of (i), and shows that (ii) implies (iii). It remains to prove (ii).

8.1.3.

The map in (ii) is injective. Let P and Q be two principal homogeneous bundles under G, trivialized by S', and consider the S-group sheaf

$$\mathcal{H} = \underline{\mathrm{Isom}}_{G\text{-bundles}}(P, Q).$$

Since $\mathcal{H}_{S'}$ is isomorphic to $G_{S'}$, $\mathcal{H}$ is representable by the second hypothesis on G, as above. If $\mathcal{H}(\kappa(s)) \neq \varnothing$, Hensel's lemma gives $\mathcal{H}(S) \neq \varnothing$, and hence $P \simeq Q$.

[*] This last condition is in fact unnecessary; see A. Grothendieck, *Le groupe de Brauer III*, in *Dix Exposés sur la cohomologie des schémas*, North-Holland, 1968, Theorem 11.7 and Remarks 11.8.3.

[47] *Editors' note.* The scheme S' is still local and henselian. We have also retained the numbering of the original: there is no no. 8.1.1.

8.1.4.

To prove surjectivity in (ii), it is equivalent to prove that

$$Z^1(S'/S, G) \longrightarrow Z^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is surjective. Let Z^1 be the S-functor defined by

$$Z^1(T) = Z^1(S' \times_S T/T, G_T).$$

The preceding map is then

$$Z^1(S) \longrightarrow Z^1(\kappa(s)).$$

Another application of Hensel's lemma reduces the assertion to showing that Z^1 is representable by a smooth S-scheme.

8.1.5.

We first show that Z^1 is representable by an S-scheme locally of finite presentation. For $i = 0, 1, \ldots$, define the S-functor C_i by

$$C_i(T) = C_i(S' \times_S T/T, G_T).$$

Equivalently,

$$C_i(T) = G((S'/S)^{i+1} \times_S T), \qquad C_i = \underline{\mathrm{Hom}}_S((S'/S)^{i+1}, G).$$

Since Z^1 is obtained from C_1 and C_2 by fiber products, it is enough to prove that C_i, for $i = 1, 2$, is representable by an S-scheme locally of finite presentation.

8.1.6.

Let $T \to S$ be a finite surjective étale morphism which splits S'. Then

$$C_i \times_S T = \underline{\mathrm{Hom}}_T((S' \times_S T/T)^{i+1}, G_T)$$

is represented by a product of n copies of G_T, where n is the degree of $(S'/S)^{i+1}$. The hypothesis on G, applied once again, shows that C_i is represented by an S-scheme locally of finite presentation (SGA 1, VIII, *loc. cit.*).

8.1.7.

It remains to prove that Z^1 is smooth. By definition, if T is an affine S-scheme and T_0 is the closed subscheme defined by a square-zero ideal J, we must show that

$$Z^1(T) \longrightarrow Z^1(T_0)$$

is surjective. Since G is smooth, the canonical map $G(T) \to G(T_0)$ is surjective, and it is enough to prove that

$$H^1(S' \times_S T/T, G_T) \longrightarrow H^1(S' \times_S T_0/T_0, G_{T_0})$$

is bijective. After a slight change of notation and a harmless generalization of the hypotheses, this reduces to the following lemma.

Lemma 8.1.8.— *Let S and S' be affine schemes, let $S' \to S$ be faithfully flat, let J be a square-zero ideal on S, and let S_0 be the closed subscheme which it defines. Let G be a smooth S-group. Put*

$$S'_0 = S' \times_S S_0, \qquad G_0 = G \times_S S_0.$$

Then the canonical map

$$H^1(S'/S, G) \longrightarrow H^1(S'_0/S_0, G_0)$$

is bijective.

Remark 8.1.9.— If G is commutative, the same assertion holds for every H^i, $i > 0$, with an analogous proof.

Proof. Let M_0 be the quasi-coherent $\mathcal{O}_{S_0}$-module

$$M_0 = \mathrm{Lie}(G_0/S_0) \otimes_{\mathcal{O}_{S_0}} J.$$

For every S_0-scheme T_0, put

$$M_0(T_0) = H^0\Big(T_0, M_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}\Big),$$

and define the S-group functors

$$M = \prod_{S_0/S} M_0, \qquad \overline{G} = \prod_{S_0/S} G_0$$

by

$$M(T) = M_0(T_0), \qquad \overline{G}(T) = G_0(T_0), \qquad T_0 = T \times_S S_0.$$

By Exposé III, 0.9 and (0.6.2), for every affine S-scheme T there is a functorial exact sequence

$$1 \longrightarrow M(T) \longrightarrow G(T) \longrightarrow \overline{G}(T) \longrightarrow 1.$$

We must study the map

$$H^1(S'/S, G) \longrightarrow H^1(S'_0/S_0, G_0) = H^1(S'/S, \overline{G}).$$

Suppose first that G is commutative. There is then an exact cohomology sequence

$$H^1(S'/S, M) \longrightarrow H^1(S'/S, G) \longrightarrow H^1(S'/S, \overline{G}) \longrightarrow H^2(S'/S, M).$$

Moreover,

$$H^i(S'/S, M) = H^i(S'_0/S_0, M_0) = H^i(S'_0/S_0, \mathcal{M}_0) = 0 \qquad (i \neq 0),$$

the last equality following from TDTE I, B, Lemma 1.1.

Now suppose that G is not commutative. We use the exact sequence of nonabelian cohomology. If $u \in Z^1(S'/S, \overline{G})$, the elements of $H^1(S'/S, G)$ whose image in $H^1(S'/S, \overline{G})$ equals the class of u lie in the image of the corresponding coboundary map

$$H^1(S'/S, M_u) \longrightarrow H^1(S'/S, G),$$

where M_u is M twisted by u. Similarly, for $v \in Z^1(S'/S, \overline{G})$, there is an obstruction

$$\Delta(v) \in H^2(S'/S, M_v)$$

which vanishes exactly when the class of v is in the image of $H^1(S'/S, G)$. It therefore suffices to prove

$$H^1(S'/S, M_u) = 0, \qquad H^2(S'/S, M_v) = 0.$$

Recall (Exposé III, 0.8) that the action of G on M defined by the exact sequence above is induced functorially by the adjoint representation

$$\mathrm{ad} : G_0 \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_{S_0}}(\mathrm{Lie}(G_0/S_0)).$$

The cocycle u, respectively v, therefore defines a descent datum on $\mathrm{Lie}(G_0/S_0)$; write L_u, respectively L_v, for the resulting quasi-coherent $\mathcal{O}_{S_0}$-module. For $T \to S$,

$$M_u(T) = H^0\Big(T_0, L_u \otimes_{\mathcal{O}_{S_0}} J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}\Big),$$

and similarly with v in place of u. Hence

$$H^1(S'/S, M_u) = H^1(S'_0/S_0, L_u \otimes J),$$

$$H^2(S'/S, M_v) = H^2(S'_0/S_0, L_v \otimes J),$$

and both groups vanish by the result already used. □

Proposition 8.2.— *Let $\mathcal{C}$ be a category with fiber products, endowed with a topology coarser than the canonical topology. Let $S' \to S$ be a morphism of $\mathcal{C}$, let G' be an S'-group sheaf, and let*

$$G = \prod_{S'/S} G'$$

be the corresponding S-group sheaf. Let

$$H^1_S(S', G') \subset H^1(S', G')$$

be the subset of classes of principal homogeneous sheaves under G' which are trivialized by a sieve on S' obtained by base change from a suitable covering sieve on S. The canonical map

$$H^1(S, G) \longrightarrow H^1(S', G'), \qquad P \longmapsto P \times_S S',$$

induces a bijection

$$H^1(S, G) \xrightarrow{\sim} H^1_S(S', G').$$

Its inverse is induced by

$$P' \longmapsto \prod_{S'/S} P'.$$

For every object X of $\mathcal{C}/S$, the definition gives an isomorphism, functorial in X,

$$\mathrm{Hom}_S(X, G) \xrightarrow{\sim} \mathrm{Hom}_{S'}(X \times_S S', G').$$

Thus, for every S-object T, there is a bijection functorial in T,

$$H^1(T/S, G) \xrightarrow{\sim} H^1(T'/S', G'), \qquad T' = T \times_S S'.$$

Replacing the single morphism $T \to S$ by an arbitrary covering family of S, and then passing to the inductive limit, gives the first part of the proposition. The second follows immediately.

Lemma 8.3.— *Under the hypotheses of 8.2, the equality*

$$H^1_S(S', G') = H^1(S', G')$$

is local on S. More precisely, suppose that there is a covering family $\{S_i \to S\}$ such that, for every i,

$$H^1_{S_i}(S' \times_S S_i, G') = H^1(S' \times_S S_i, G').$$

Then $H^1_S(S', G') = H^1(S', G')$.

Indeed, let P' be a principal homogeneous sheaf under G', and put

$$P'_i = P' \times_{S'} (S' \times_S S_i).$$

By hypothesis, there is a covering family $\{S_{ij} \to S_i\}$ such that, for each j,

$$P' \times_{S'} (S' \times_S S_{ij})$$

has a section. But $\{S_{ij} \to S\}$ is a covering family of S, and P' is therefore trivialized by the covering family of S' obtained from it by base change.

Proposition 8.4.— *Let $S' \to S$ be a finite étale morphism of schemes. Let G' be an S'-group sheaf and let*

$$G = \prod_{S'/S} G'$$

be the corresponding S-group sheaf. For the étale topology, respectively the local finite étale topology, respectively the (fpqc) *topology, the following functors induce mutually inverse bijections.*

$$\mathcal{P} \longmapsto \mathcal{P} \times_S S'$$

$$\prod_{S'/S} \mathcal{P}' \longleftarrow\!\shortmid \mathcal{P}'$$

In particular,

$$H^1(S, G) \simeq H^1(S', G').$$

By 8.2, it is enough to show

$$H^1_S(S', G') = H^1(S', G').$$

By 8.3, this may be checked locally for the local finite étale topology. We may therefore suppose that S' is a finite disjoint union of copies of S, say I_S, where I is a finite set. Then G' is given by a family $(G_i)_{i \in I}$ of sheaves on S, and

$$H^1(S', G') \simeq \prod_{i \in I} H^1(S, G_i).$$

On the other hand,

$$H^1(S, G) \simeq \prod_{i \in I} H^1(S, G_i).$$

It follows from 8.2 that $H^1_S(S', G') = H^1(S', G')$. □

Remarks 8.5.— Propositions 8.2 and 8.4 may be interpreted through the following exact sequence, where $f : S' \to S$ is the given morphism:

$$1 \longrightarrow H^1(S, f_*G') \longrightarrow H^1(S', G') \longrightarrow H^0(S, R^1 f_* G').$$

In the commutative case, this exact sequence follows from the Leray spectral sequence. It remains valid in the noncommutative case.[48]

In this form one sees that the result remains valid when f is assumed only finite, or even merely integral, provided that the topology is the étale topology. Indeed, for every G',

$$R^1 f_*(G') = \text{the final sheaf}$$

by SGA 4, VIII, 5.3.[49]

The result becomes false for a topology such as (fpqc) or (fppf), even if

$$S = \mathrm{Spec}(k), \qquad S' = \mathrm{Spec}(k[t]/t^2),$$

where k is an algebraically closed field of characteristic $p \neq 0$, and $G' = \mu_p$ or α_p.

Likewise, 8.2 becomes false, even for the étale topology, if the finiteness hypothesis on f is omitted. For example, take f to be an open immersion with

$$S = \mathrm{Spec}(V),$$

where V is a complete discrete valuation ring with algebraically closed residue field, take S' to be the open subscheme consisting of the generic point, and take G' to be the constant group $(\mathbb{Z}/n\mathbb{Z})_{S'}$, where n is prime to the residue characteristic of V. Then

$$H^1(S, G) = 0, \qquad H^1(S', G') \neq 0.$$

Replacing S' by $S \amalg S'$ gives an analogous example in which $S' \to S$ is étale and surjective, hence a cover for the topology under consideration.

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[48] *Editors' note.* See §V.3.1.4 of [Gi71].

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Exposé XXV
The Existence Theorem

by M. Demazure

[1] For completeness, in this Exposé we give a proof of the existence theorem for split groups. Like Chevalley's original proof ("Sur certains schémas de groupes semi-simples", *Séminaire Bourbaki*, May 1961, no. 219), it rests on the existence of complex semisimple algebraic groups of every possible type. Cartier gave the principle of a more satisfactory proof, establishing directly the existence of a simply connected split semisimple $\mathbb{Z}$-group associated with a given Cartan matrix (unpublished).[2] Let us point out, however, that the difficulty is not to give an explicit construction of a group scheme, but to verify that the group so constructed really satisfies the required conditions—essentially, that its fibers are indeed smooth and reductive.

1. Statement of the theorem

Theorem 1.1.— *Let S be a nonempty scheme. The functor*

$$G \longmapsto R(G)$$

is an equivalence from the category of pinned reductive S-groups to the category of pinned reduced root data.

By the uniqueness theorem (Exposé XXIII, 4.1), the preceding statement is equivalent to the following.

Corollary 1.2 (Existence of split groups).— *For every reduced root datum R, there exists a split reductive $\mathbb{Z}$-group G such that $R(G) \simeq R$.*[3]

In particular:

Corollary 1.3.— *Let k be a field. For every splittable reductive k-group G_k, there exists a reductive $\mathbb{Z}$-group G such that*

$$G \otimes_{\mathbb{Z}} k \simeq G_k.$$

[1] Editors' note: Version of 13 October 2024.

[2] Editors' note: Such a principle was also sketched by B. Kostant *[Ko66]*; a complete proof, using quantum groups, was given more recently by G. Lusztig *[Lu09]*.

[3] Editors' note: The work *[BT84]* of F. Bruhat and J. Tits also contains a variant of Chevalley's construction (*loc. cit.*, 2.2.3–2.2.5 and §3.2), which in particular gives a smooth affine $\mathbb{Z}$-group G with connected fibers, possessing a split maximal torus, and whose generic fiber is a reductive $\mathbb{Q}$-group of type R. The reductivity of the geometric fibers of G follows from the study of the unipotent radical of a special fiber in *loc. cit.*, 4.6.12 and 4.6.15 (valid for more general G, associated with a valued root datum); but it is simpler to deduce it from the description of $\mathfrak{g} = \mathrm{Lie}(G)$, from Exposé XIX, 1.12(iii), and from the existence of the elements $w_\alpha(X)$ of Exposé XX, 3.1(iv); see *[BT84]*, 3.2.1.

Conversely, first observe that to prove 1.2 it is enough, by Exposé XXII, 4.3.1, and Exposé XXI, 6.5.10, to consider the case in which the root datum R is simply connected (and even irreducible, if desired, by Exposé XXI, 7.1.6).

On the other hand, under the hypotheses of 1.3 the group scheme G has constant type (since $\mathrm{Spec}(\mathbb{Z})$ is connected), hence type $R(G_k)$. By Exposé XXIII, 5.9, the validity of 1.3 for a given group G_k therefore entails the existence of a split $\mathbb{Z}$-group of type $R(G_k)$.[4]

Thus, to prove 1.2 and hence 1.1, it is enough to prove 1.3 when k has characteristic zero (for example, $k = \mathbb{C}$) and G_k is simply connected (and in particular semisimple), together with the following proposition.

Proposition 1.4.— *For every simply connected reduced root datum R, there exists a semisimple algebraic $\mathbb{C}$-group of type R.*

One may prove 1.4 as follows. First, there exists a complex semisimple Lie algebra $\mathfrak{g}$ of type R; see, for example, N. Jacobson, *Lie Algebras*, Chapter VII, Theorem 5.[5] Then $G = \mathrm{Aut}(\mathfrak{g})^0$ is a semisimple algebraic $\mathbb{C}$-group of type $\mathrm{ad}(R)$.[6] By *Bible*, §23.1, Proposition 1, it follows that there exists a semisimple $\mathbb{C}$-group of type R.

The remainder of this Exposé is devoted to the proof of 1.3 when k has characteristic zero and G_k is semisimple. The proof proceeds in two stages: the construction of a "piece of $\mathbb{Z}$-group scheme" (§2), and the study of the group obtained by applying "Weil's theorem" (§3).

To avoid confusion, in §2 we shall not use the abbreviated notation X_k for the k-scheme $X \otimes_{\mathbb{Z}} k$, where X is a $\mathbb{Z}$-scheme.

2. Existence theorem: construction of a piece of group

2.1.

Choose once and for all a splitting of G_k, denoted

$$(G_k, T_k, M, R)$$

(see Exposé XXII, 1.13), a system Δ of simple roots of R (defining the system R^+ of positive roots), and a Chevalley system $(X_{\alpha,k})_{\alpha\in R}$ of G_k (Exposé XXIII, 6.1 and 6.2) satisfying the additional condition (see Exposé XX, 2.6)

$$X_{\alpha,k}X_{-\alpha,k} = 1 \qquad (\alpha \in R).$$

[4]Editors' note: In fact, Exposé XXIII, 5.9 is unnecessary: the present Exposé constructs, for every semisimple $\mathbb{Q}$-group $G_{\mathbb{Q}}$, a semisimple $\mathbb{Z}$-group G of the same type as $G_{\mathbb{Q}}$, equipped with a split maximal torus T; hence G is split by Exposé XXII, 2.2.

[5]Editors' note: Let Δ be a base of R, and let $\widetilde{\mathfrak{g}}$ be the complex Lie algebra generated by $(e_\alpha, f_\alpha, h_\alpha)_{\alpha\in\Delta}$, subject to the relations $[h_\alpha, h_\beta] = 0$, $[e_\alpha, f_\beta] = h_\alpha$ if $\beta = \alpha$ and $= 0$ otherwise, $[h_\alpha, e_\beta] = (\alpha^*, \beta)e_\beta$, and $[h_\alpha, f_\beta] = -(\alpha^*, \beta)f_\beta$. In *loc. cit.*, $\mathfrak{g}$ is defined as the quotient of $\widetilde{\mathfrak{g}}$ by the intersection of the kernels of its finite-dimensional irreducible representations. In *[Se66]*, §VI.5, Theorem 9 (see also *[BLie]*, VIII, §4.3, Theorem 1), it is shown that $\mathfrak{g}$ is the quotient of $\widetilde{\mathfrak{g}}$ by the relations

$$\mathrm{ad}(e_\alpha)^{1-(\alpha^*,\beta)}(e_\beta) = 0, \qquad \mathrm{ad}(f_\alpha)^{1-(\alpha^*,\beta)}(f_\beta) = 0.$$

For an explicit description of the structure constants, and in particular the choice of signs, see Exposé XXIII, 6.5 and 6.7, as well as *[Ti66]*, §4, Theorem 1, and, for types A, D, E, *[Sp98]*, 10.2.5.

[6]Editors' note: One may suppose R irreducible, hence $\mathfrak{g}$ simple. By a general argument, $\mathrm{Lie}(G)$ is the complex Lie algebra of derivations of $\mathfrak{g}$; see *[DG70]*, §II.4, Proposition 2.3. These derivations are all inner; see *[BLie]*, I, §6.1, Corollary 3 to Proposition 1. Thus $\mathrm{Lie}(G) = \mathfrak{g}$, so G has no invariant subgroup of positive dimension and is semisimple. Its root system is consequently the same as that of $\mathfrak{g}$. Moreover, the center of G acts both trivially and faithfully on $\mathfrak{g}$, and is therefore trivial; hence G is adjoint.

Finally, choose on the subgroup of M generated by R a total order compatible with the group structure and such that the roots > 0 are the elements of R^+. We then write the roots as

$$-\alpha_n < -\alpha_{n-1} < \cdots < -\alpha_1 < \alpha_1 < \alpha_2 < \cdots < \alpha_n.$$

For $\alpha \in R$, let $U_{\alpha,k}$ denote the vector group corresponding to the root α, and let

$$p_{\alpha,k} \colon \mathbb{G}_{a,k} \xrightarrow{\sim} U_{\alpha,k}$$

be the isomorphism of vector groups defined by $X_{\alpha,k}$.

2.2.

The splitting of G_k includes, in particular, an isomorphism of k-groups

$$T_k \simeq D_k(M).$$

Put $T = D(M)$. This is a $\mathbb{Z}$-torus, and the preceding isomorphism may be regarded as an isomorphism

$$T_k \simeq T \otimes_{\mathbb{Z}} k.$$

We have

$$\mathrm{Hom}_{\mathbb{Z}\text{-gr.}}(T, \mathbb{G}_m) = M,$$

and shall regard the elements of $R \subset M$ as characters of T. Likewise, the elements of R^* will be regarded as morphisms of $\mathbb{Z}$-groups $\mathbb{G}_m \to T$.

2.3.

For each $\alpha \in R^+$, let $\mathbb{G}_a(\alpha)$ be a copy of $\mathbb{G}_a$, and consider the $\mathbb{Z}$-scheme

$$U = \mathbb{G}_a(\alpha_1) \times \cdots \times \mathbb{G}_a(\alpha_n).$$

If U_k denotes the unipotent part of the Borel subgroup B_k of G_k defined by R^+, let

$$a \colon U \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U_k$$

be the isomorphism of k-schemes defined by

$$a(x_1, \ldots, x_n) = p_{\alpha_1,k}(x_1) \cdots p_{\alpha_n,k}(x_n).$$

2.4.

The group law of U_k is expressed by relations of the form

$$a(x_1, \ldots, x_n)a(y_1, \ldots, y_n) = a(z_1, \ldots, z_n),$$

where, for $h = 1, \ldots, n$, each z_h is a polynomial

$$z_h = x_h + y_h + Q_h(x_1, \ldots, x_{h-1}, y_1, \ldots, y_{h-1}),$$

whose coefficients are integers (Exposé XXII, 5.5.8, and Exposé XXIII, 6.4). Moreover,

$$Q_h(x_1, \ldots, x_{h-1}, 0, \ldots, 0) = 0.$$

Equip U with the composition law defined by these formulas, which are indeed "defined over $\mathbb{Z}$". Since this law induces the group law on U_k, it is associative, and (0) is a unit element. Indeed, these two assertions are expressed by relations among the polynomials Q_h, and $\mathbb{Z} \to k$ is injective. This is a group law: if (x_i) is a section of U over an arbitrary scheme S, its inverse (y_i) is computed recursively by

$$y_i = -x_i - Q_i(x_1, \ldots, x_{i-1}, y_1, \ldots, y_{i-1}),$$

formulas which are again "defined over $\mathbb{Z}$".

In summary, we have constructed a group law on U for which the isomorphism a above is an isomorphism of groups.

For each $\alpha \in R^+$, consider the morphism

$$p_\alpha \colon \mathbb{G}_a \longrightarrow U$$

defined by $p_\alpha(x) = (x_i)$, where

$$x_i = \begin{cases} x, & \alpha_i = \alpha, \\ 0, & \alpha_i \neq \alpha. \end{cases}$$

It is a closed immersion and a group homomorphism; denote its image by U_α. Since

$$(x_i) = p_{\alpha_1}(x_1) \cdots p_{\alpha_n}(x_n),$$

U is identified with the product

$$U = U_{\alpha_1} U_{\alpha_2} \cdots U_{\alpha_n}.$$

2.5.

Let $T = D(M)$ act on each U_α through the character α. One verifies immediately that this defines an action of T on the group U, and one may form the semidirect product $B = T \ltimes U$. There is a canonical isomorphism of k-groups

$$B \otimes_{\mathbb{Z}} k \xrightarrow{\sim} B_k.$$

If an arbitrary order is now placed on R^+, the morphism

$$\prod_{\alpha \in R^+} U_\alpha \longrightarrow U$$

defined by multiplication in U is still an isomorphism. Indeed, both sides are flat $\mathbb{Z}$-schemes of finite presentation, so it is enough to verify the assertion on geometric fibers. We are then reduced to Lazard's theory (*Bible*, §13.1, Proposition 1): regard U as a group with operators T, and use the fact that the U_α are pairwise nonisomorphic as groups with operators, since the characters $\alpha \in R^+$ of T are pairwise distinct on every fiber.

2.6.

Replacing R^+ by $R^- = -R^+$, one similarly constructs the groups U^-, B^-, and U_α for $\alpha \in R^-$, together with isomorphisms

$$p_\alpha \colon \mathbb{G}_a \xrightarrow{\sim} U_\alpha, \qquad \alpha \in R^-.$$

Finally introduce the product scheme

$$\Omega = U^- \times T \times U.$$

There is a canonical isomorphism of k-schemes

$$\Omega \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U_k^- \times_k T_k \times_k U_k \simeq \Omega_k,$$

where Ω_k is the "big cell" of G_k (Exposé XXII, 4.1.2).

From now on we identify $\Omega \otimes_{\mathbb{Z}} k$ with Ω_k by this isomorphism, and regard U^-, T, and U as subschemes of Ω through their unit sections. Write

$$e = ((0), e, (0))$$

for the "unit section" of Ω. Our aim is now to put a piece-of-group law on Ω.

Lemma 2.7.— *Let $\alpha \in \Delta$, and let $w_{\alpha,k}$ be the element of $\mathrm{Norm}_{G_k}(T_k)(k)$ defined by $X_{\alpha,k}$; recall that, by definition,*

$$w_{\alpha,k} = p_{-\alpha,k}(-1)p_{\alpha,k}(1)p_{-\alpha,k}(-1).$$

There exist an open subset V_α of Ω, containing the section e, and a morphism

$$h_\alpha \colon V_\alpha \longrightarrow \Omega$$

such that:

(i) $h_\alpha(e) = e$;

(ii) $(h_\alpha) \otimes_{\mathbb{Z}} k$ *coincides with the restriction of* $\mathrm{int}(w_{\alpha,k})$ *to* $V_\alpha \otimes_{\mathbb{Z}} k \subset G_k$;

(iii) $T \subset V_\alpha$, *and h_α sends T into T. For every $\beta \in R$, one has $U_\beta \subset V_\alpha$, and h_α sends U_β into $U_{s_\alpha(\beta)}$.*

By the definition of a Chevalley system (Exposé XXIII, 6.1), for every $\beta \in R$ there exists an integer $e_\beta = \pm 1$ such that

$$\mathrm{int}(w_{\alpha,k})p_{\beta,k}(x) = p_{s_\alpha(\beta),k}(e_\beta x)$$

for every $x \in \mathbb{G}_a(S)$ and every $S \to \mathrm{Spec}(k)$.

Let S be any scheme. Write an arbitrary element of $\Omega(S)$ in the form

$$\left(\left(\prod_{\substack{\beta \in R^- \\ \beta \neq -\alpha}} p_\beta(x_\beta) \right) p_{-\alpha}(x_{-\alpha}),\ t,\ p_\alpha(x_\alpha) \left(\prod_{\substack{\beta \in R^+ \\ \beta \neq \alpha}} p_\beta(x_\beta) \right) \right),$$

where arbitrary orders have been chosen on $R^- \setminus \{-\alpha\}$ and $R^+ \setminus \{\alpha\}$, as permitted by 2.5. Define a morphism $d \colon \Omega \to \mathrm{Spec}(\mathbb{Z})$ by

$$d(u) = \alpha(t) + x_\alpha x_{-\alpha}.$$

Let $V_\alpha = \Omega_d$, the open subset of Ω defined by the invertibility of $d(u)$. It contains e, T, and every U_β, $\beta \in R$. Define

$$h_\alpha \colon V_\alpha \longrightarrow \Omega, \qquad h_\alpha(u) = (a(u), b(u), c(u)),$$

where

$$a(u) = \left(\prod_{\substack{\beta \in R^- \\ \beta \neq -\alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right) p_{-\alpha}\Big(-x_\alpha d(u)^{-1}\Big),$$
$$b(u) = t \cdot \alpha^*(d(u)),$$
$$c(u) = p_\alpha\Big(-x_{-\alpha} d(u)^{-1}\Big) \left(\prod_{\substack{\beta \in R^+ \\ \beta \neq \alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right).$$

Since s_α permutes the positive roots distinct from α (respectively the negative roots distinct from $-\alpha$), $c(u)$ (respectively $a(u)$) is a section of U (respectively U^-), so the preceding morphism is well defined. It plainly satisfies (i) and (iii). Assertion (ii) follows immediately from the definition of the e_β, $\beta \in R$, and Exposé XX, 3.12.

Lemma 2.8.— *There exist open subsets V, V' of Ω and morphisms*

$$h\colon V \longrightarrow \Omega, \qquad h'\colon V' \longrightarrow \Omega$$

with the following properties:

(i) *V and V' contain e, and $h(e) = h'(e) = e$;*

(ii) *the morphism induced by*
$$h' \circ h\colon h^{-1}(V') \longrightarrow \Omega$$
is the restriction of the identity morphism;

(iii) *the k-morphisms $h \otimes_{\mathbb{Z}} k$ and $h' \otimes_{\mathbb{Z}} k$ are the restrictions, to $V \otimes_{\mathbb{Z}} k$ and $V' \otimes_{\mathbb{Z}} k$, of automorphisms of the group G_k;*

(iv) *V and V' contain U, T, U^-; the morphisms h, h' send U into U^-, U^- into U, and T into T.*

Let w^0 be the element of the Weyl group of G_k that sends R^+ to R^-. Write

$$w^0 = s_{\alpha_n} \cdots s_{\alpha_1}, \qquad \alpha_i \in \Delta$$

(with no relation to the numbering in 2.1), and put

$$w_0 = w_{\alpha_n, k} \cdots w_{\alpha_1, k} \in \mathrm{Norm}_{G_k}(T_k)(k).$$

Define recursively, for $i \leq n$, an open subset V_i of Ω and a morphism $h_i\colon V_i \to \Omega$ by

$$V_0 = \Omega, \qquad h_0 = \mathrm{id},$$

and, for $i = 0, \ldots, n-1$,

$$V_{i+1} = h_i^{-1}(V_{\alpha_{i+1}}), \qquad h_{i+1} = h_{\alpha_{i+1}} \circ h_i,$$

where V_{α_j} and h_{α_j} are those of 2.7.

Take $V = V_n$ and $h = h_n$. The conditions (i), (iii), and (iv) concerning V and h hold: for (i) and (iii) this follows immediately from 2.8, and for (iv) from the fact that $h \otimes_{\mathbb{Z}} k$ is the restriction of $\mathrm{int}(w_0)$ to $V \otimes_{\mathbb{Z}} k$.

Since $(w^0)^2 = 1$, we also have

$$w^0 = s_{\alpha_1} \cdots s_{\alpha_n} = (s^3_{\alpha_1}) \cdots (s^3_{\alpha_n}).$$

Put

$$w'_0 = (w_{\alpha_1,k})^3 \cdots (w_{\alpha_n,k})^3.$$

Repeating the construction above gives an open subset V' and a morphism h' that also satisfy (i), (iii), and (iv). Moreover, $h' \otimes_{\mathbb{Z}} k$ is the restriction of $\mathrm{int}(w'_0)$ to $V' \otimes_{\mathbb{Z}} k$. For every simple root $\alpha \in \Delta$, however, $(w_{\alpha,k})^4 = e$ (Exposé XX, 3.1), hence $w'_0 w_0 = e$. It follows that $h' \circ h$ induces the identity morphism on a nonempty open subset of $\Omega \otimes_{\mathbb{Z}} k$. Since Ω is smooth and of finite presentation over $\mathbb{Z}$, $\Omega \otimes_{\mathbb{Z}} k$ is schematically dense in Ω; this proves (ii).

Proposition 2.9.— *There exist an open subset V_1 of $\Omega \times \Omega$, an open subset V_2 of Ω, and morphisms*

$$\pi\colon V_1 \longrightarrow \Omega, \qquad \sigma\colon V_2 \longrightarrow \Omega$$

having the following properties:

(i) *if $x \in \Omega(S)$, then (e, x) and (x, e) are sections of V_1, and*

$$\pi(e, x) = \pi(x, e) = x;$$

(ii) *V_2 contains e, and $\sigma(e) = e$;*

(iii) *π_k and σ_k are the restrictions of the morphisms $G_k \times_k G_k \to G_k$ and $G_k \to G_k$ defined respectively by multiplication and inversion.*

Proof. Let (v, t, u) and (v', t', u') be two sections of Ω. By 2.8(iv), $h(u)$ is a section of U^- and $h(v')$ a section of U; thus $(h(u), e, h(v'))$ may be regarded as a section of Ω. Define V_1 as the open subset of $\Omega \times \Omega$ specified by

$$\big((v, t, u), (v', t', u')\big) \in V_1(S) \quad \Longleftrightarrow \quad (h(u), e, h(v')) \in V'(S),$$

using the notation of 2.8. If the pair on the left is a section of V_1, then $h'(h(u), e, h(v'))$ is defined and is a section of Ω; write

$$h'(h(u), e, h(v')) = (v'', t'', u'').$$

Set

$$\pi\big((v, t, u), (v', t', u')\big) = \big(v \cdot tv''t^{-1},\ tt''t',\ t'^{-1}u''t' \cdot u'\big).$$

Assertion (i) follows immediately from 2.8(ii). To verify the assertion about π in (iii), observe that

$$h'(h(u), e, h(v')) = uv' = v''t''u''$$

whenever $u \in U(S)$, $v' \in U^-(S)$, and $S \to k$, by 2.8(iii) and (ii).

The morphism σ is constructed similarly. If (v, t, u) is a section of Ω, then $h(u^{-1})$ is a section of U^-, $h(v^{-1})$ a section of U, and $h(t^{-1})$ a section of T. Thus

$$(h(u^{-1}), h(t^{-1}), h(v^{-1}))$$

is a section of Ω. Define an open subset V_2 of Ω by

$$(v,t,u) \in V_2(S) \quad \Longleftrightarrow \quad (h(u^{-1}), h(t^{-1}), h(v^{-1})) \in V'(S),$$

and define

$$\sigma(v,t,u) = h'(h(u^{-1}), h(t^{-1}), h(v^{-1})).$$

The required properties of σ are verified in the same way.

Corollary 2.10.— *π is "generically associative" and σ is a "generic inverse": if $x, y, z \in \Omega(S)$, and if the expressions below are defined (as they always are over an open subset of Ω containing the unit section), then*

$$\pi(x, \pi(y,z)) = \pi(\pi(x,y), z), \qquad \pi(x, \sigma(x)) = e = \pi(\sigma(x), x).$$

Indeed, the two sides of each formula define morphisms between smooth $\mathbb{Z}$-schemes of finite presentation, and the morphisms agree on the generic fibers by 2.9(iii).

Corollary 2.11.— *Let $\alpha \in R$, and let S be a scheme. Suppose that $x, y \in \mathbb{G}_a(S)$ satisfy*

$$(p_\alpha(x), p_{-\alpha}(y)) \in V_1(S), \qquad 1 + xy \in \mathbb{G}_m(S).$$

These conditions define an open subset of $\mathbb{G}_{a,S}^2$ containing the section $(0,0)$. Then

$$\pi(p_\alpha(x), p_{-\alpha}(y)) = \left(p_{-\alpha}\left(\frac{y}{1+xy}\right), \alpha^*(1+xy), p_\alpha\left(\frac{x}{1+xy}\right)\right).$$

The proof is the same as above, by Exposé XX, 2.1.

3. Existence theorem: conclusion of the proof

For convenience, make the following definition.

Definition 3.1.— *Let S be a scheme and G an S-group scheme. The group G is called* admissible *if there is an open immersion of S-schemes*

$$i \colon \Omega_S = \Omega \times S \longrightarrow G$$

such that:

(i) *the diagram*

$$\begin{array}{ccccc} (V_1)_S & \longrightarrow & \Omega_S \times_S \Omega_S & \xrightarrow{i \times_S i} & G \times_S G \\ \downarrow{\scriptstyle \pi} & & & & \downarrow{\scriptstyle \pi_G} \\ \Omega_S & & \xrightarrow{\qquad i_S \qquad} & & G \end{array}$$

is commutative, where π_G denotes the multiplication morphism of G;

(ii) *there is a finite set of sections $a_j \in \Omega(S)$ such that the open subsets $i(a_j)\, i(\Omega_S)$ cover G.*

By "Weil's theorem" (Exposé XVIII, 3.13(iii) and (iv)), we have:

Lemma 3.2.— *If every admissible S-group is affine whenever S is étale and of finite type over $\mathbb{Z}$, then there exists an admissible affine $\mathbb{Z}$-group.*

We now have:

Lemma 3.3.— *Let S be a scheme and G an admissible S-group. Then G is smooth and of finite presentation over S, with affine, connected, semisimple fibers.*

Since Ω_S is smooth and of finite presentation over S, with connected fibers, the same is true of G by condition (ii). To prove 3.3, we may therefore suppose that S is the spectrum of a field K. Identify Ω_K with its image in G. Plainly Ω_K is the product

$$\left(\prod_{\alpha\in R^-} U_{\alpha,K}\right) T_K \left(\prod_{\alpha\in R^+} U_{\alpha,K}\right)$$

of the subgroups T_K and $U_{\alpha,K}$, $\alpha \in R$, of G. Thus the Lie algebra of G is identified with the direct sum

$$\mathrm{Lie}(T_K) \oplus \bigoplus_{\alpha\in R} \mathrm{Lie}(U_{\alpha,K}).$$

The inner automorphism defined by a section of T_K acts on $U_{\alpha,K}$, and hence on $\mathrm{Lie}(U_{\alpha,K})$, through the character

$$\alpha \in R \subset M \simeq \mathrm{Hom}_{K\text{-gr.}}(T_K, \mathbb{G}_{m,K}).$$

The preceding decomposition of $\mathrm{Lie}(G)$ is therefore exactly its decomposition under the adjoint action of T. Thus the roots of G_K with respect to T_K are the elements $\alpha \in R$.

Apply Exposé XIX, 1.13. Let T_α be the maximal torus of $\ker(\alpha) \subset T_K$, and put

$$Z_\alpha = \mathrm{Centr}_G(T_\alpha).$$

It is enough to prove that each Z_α is reductive. But

$$Z_\alpha \cap \Omega_K = \left(\prod_{\substack{\beta\in R^-\\ \beta|_{T_\alpha}=e}} U_{\beta,K}\right) T_K \left(\prod_{\substack{\beta\in R^+\\ \beta|_{T_\alpha}=e}} U_{\beta,K}\right).$$

The roots vanishing on T_α are the rational multiples of α, hence precisely α and $-\alpha$. Consequently,

$$Z_\alpha \cap \Omega_K = U_{-\alpha,K}\, T_K\, U_{\alpha,K}.$$

By Exposé XX, 3.4, to prove Z_α reductive it is enough to show that $U_{\alpha,K}$ and $U_{-\alpha,K}$ do not commute, and this follows immediately from 2.11.

It follows from 3.2 and 3.3 that the proof will be complete once the following lemma is proved.

Lemma 3.4.— *Let S be a locally noetherian scheme of dimension at most 1, and let G be a smooth S-group of finite type, with affine, connected, semisimple fibers. Then G is affine (and hence semisimple).*

Nota. In Exposé XVI it was shown that 3.4 holds without any hypothesis on S, but the proof is comparatively delicate. Since only the special case 3.4 is needed here, we give a direct proof.

Consider the Lie algebra $\mathfrak{g}$ of G, a locally free $\mathcal{O}_S$-module, and the adjoint representation

$$\mathrm{Ad}\colon G \longrightarrow \mathrm{GL}_{\mathcal{O}_S}(\mathfrak{g}).$$

To prove that G is affine over S, it is enough to prove that Ad is affine. Since G is smooth with connected fibers, it is separated over S (Exposé VI_B, 5.5), so Ad is separated. By the result proved in the Appendix (see 4.1), it suffices to show that Ad is quasi-finite. We are thus reduced to the case in which S is the spectrum of a field. In that case G is affine and therefore semisimple, and the assertion is Exposé XXII, 5.7.14.

4. Appendix

We used the following proposition in the course of the proof.

Proposition 4.1.— *Let S be a locally noetherian scheme of dimension at most 1; let G and H be two S-group schemes of finite type; and let $f\colon G \to H$ be a quasi-finite separated group homomorphism. If G is flat over S,*[7] *then f is affine.*

We shall give the proof only when G is smooth over S, which is indeed the hypothesis satisfied in the application above.

4.2.

By EGA II, 1.6.4, we may suppose that S is reduced. By the usual techniques of passage to the limit,[8] we may suppose S local. If $\dim(S) = 0$, the assertion is trivial.[9] Suppose therefore that $\dim(S) = 1$. By faithfully flat descent, we may suppose that S is complete with algebraically closed residue field. After replacing S by its normalization $\widetilde{S}$, if necessary, we may suppose S normal (EGA II, 6.7.1, and EGA 0_IV, 23.1.5).[10] We are thus reduced to the case in which S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field.

4.3.

Let η (respectively s) be the generic (respectively closed) point of S. Consider the image $f_\eta(G_\eta)$ of G_η in H_η. It is a closed subgroup scheme of H_η. Let H' be the schematic closure of $f_\eta(G_\eta)$ in H. Since $H' \to H$ is affine (it is a closed immersion), we may replace H by H', and thus suppose that H is flat over S and that f_η is surjective. Since f_η is finite and G, H are flat over S, we have

$$\dim(G_s) = \dim(G_\eta) = \dim(H_\eta) = \dim(H_s).$$

4.4.

Let $H_s^0, \ldots, H_s^n$ be the irreducible components of H_s, where H_s^0 denotes the identity component, and let $z_0, \ldots, z_n$ be their generic points. Since each local ring $\mathcal{O}_{H,z_i}$ has dimension

[7] Editors' note: We have added the flatness hypothesis, which had been omitted.

[8] Editors' note: See EGA IV_3, 8.10.5(viii).

[9] Editors' note: Indeed, if $S = \mathrm{Spec}(k)$, with k a field, then f is the composite of the projection $p\colon G \to G/\ker(f)$ and a closed immersion i. Since $\ker(f)$ is finite over k, p is finite (Exposé VI_B, 9.2).

[10] Editors' note: Indeed, the irreducible components $S_1, \ldots, S_r$ of S are complete integral noetherian local schemes. By a theorem of Nagata (EGA 0_IV, 23.1.5), the normalization $\widetilde{S}_i$ is finite over S_i, and hence $\widetilde{S}$ is finite over S. A theorem of Chevalley (EGA II, 6.7.1) then shows that if $f \times_S \widetilde{S}$ is affine, so is f.

at most 1, the morphism

$$G \times_H \mathcal{O}_{H,z_i} \longrightarrow \mathcal{O}_{H,z_i}$$

is affine, being quasi-finite and separated (see Exposé XVI, Lemma 4.2). Thus G is affine over H in a neighborhood of z_i. Let V be the largest open subset of H over which $G|_V$ is affine. Then V contains every z_i,[11] and therefore contains at least one closed point $y_i \in H_s^i$. Moreover, $\kappa(y_i) = \kappa(s)$, since $\kappa(s)$ is algebraically closed.

On the other hand, V is plainly stable under translation by any element $g \in G(S)$. We have $\dim(G_s) = \dim(H_s)$, and f_s is finite, so

$$f(s)\colon G_s^0(s) \longrightarrow H_s^0(s)$$

is surjective. Since A is complete and G is smooth over S, the canonical map $G^0(S) \to G_s^0(s)$ is surjective. Since $H_s^0(s)$ acts transitively on each $H_s^i(s)$, it follows that $V \supset H_s(s)$, and therefore, since $\kappa(s)$ is algebraically closed, $V \supset H_s$.[12]

Clearly $V \supset H_\eta$, since f_η is finite. Consequently, $V = H$. □

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[11] Editors' note: The following sentence has been added.

[12] Editors' note: When G is not assumed smooth over S, one may argue as follows. Let h_0 be a rational point of H_s^0, and let g_0 be a rational point of G_s^0 such that $f(g_0) = h_0$. By Exposé $\mathrm{VI_B}$, 5.6.1, there is a commutative diagram

$$\begin{array}{ccc} S'' & \xrightarrow{g} & G \\ {\scriptstyle\pi}\downarrow & \overset{\phi}{\searrow} & \downarrow \\ S' & \xrightarrow{w} & S \end{array}$$

in which w is étale and surjective, π is finite and surjective, and $\phi^{-1}(s)$ consists of a single point s'' such that $g(s'') = g_0$. Then $G_{S''} \to H_{S''}$ is affine above a neighborhood of $h_0 y_i$, and the same is true of $G_{S'} \to H_{S'}$ (EGA II, 6.7.1), and hence of $G \to H$ by faithfully flat descent (EGA $\mathrm{IV_2}$, 2.7.1(xiii)). Thus $h_0 y_i \in V$, from which it follows that V contains $H_s(s)$, and therefore H_s.

[13] Editors' note: Additional references cited in this Exposé.

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Exposé XXVI
Parabolic Subgroups of Reductive Groups

M. Demazure

This Exposé studies the parabolic subgroups of an S-reductive group G. Its essential result is the conjugacy theorem (5.4). Its essential tool is the notion of the transversal position of two parabolic subgroups, a notion studied systematically in §4. Another fact plays an important role: the decomposition of the unipotent radical $\mathrm{rad}^u(P)$ of a parabolic subgroup P into successive extensions of vector groups (2.1).[1]

Various schemes associated with G are studied in §3; §6 deals with the split subtori[2] of G and their relations with parabolic subgroups.

Finally, in §7, we explain briefly how, over a semi-local base, one formulates the "relative theory" of reductive groups as presented over fields in the article of A. Borel and J. Tits, *Groupes réductifs*, Publications Mathématiques de l'IHÉS, no. 27. In that article, cited as [BT65] below, the reader will also find, for a base field, other results not touched upon here. [3]

1. Recollections. Levi subgroups

Definition 1.1.— *Let S be a scheme, G an S-reductive group, and P a sub-S-group-scheme of G. One says that P is a* parabolic subgroup *of G if:*

(i) *P is smooth over S;*

(ii) *for every $s \in S$, the quotient $\bar{s}$-scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper (that is, by the Bible, §6.4, Theorem 4 = [Ch05], §6.5, Theorem 5, $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$).*

Proposition 1.2 (Exp. XXII, 5.8.5).— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G. Then P is closed in G, has connected fibers, and*

$$P = \underline{\mathrm{Norm}}_G(P).$$

Moreover, the quotient sheaf G/P is representable by an S-scheme that is smooth and projective over S.

[1] *Editor's note.* Since the Levi subgroups of P form a torsor under $\mathrm{rad}^u(P)$ (1.9), this entails, when S is semi-local, that P has a Levi subgroup and hence a maximal torus (2.4).

[2] *Editor's note.* The terminology "trivial torus" has been replaced by "split torus".

[3] *Editor's note.* Version of 13 October 2024.

Proposition 1.3 (Exp. XXII, 5.3.9 and 5.3.11).— *Let S be a scheme, G an S-reductive group, and P, P' two parabolic subgroups of G. The following conditions are equivalent:*

(i) *P and P' are conjugate in G, locally for the étale* (respectively, fpqc) *topology.*

(ii) *For every $s \in S$, $P_{\bar{s}}$ and $P'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$.*

(iii) *The strict transporter $\underline{\mathrm{Transt}}_G(P, P')$ of P into P', defined for every $S' \to S$ by*

$$\underline{\mathrm{Transt}}_G(P, P')(S') = \{g \in G(S') \mid \mathrm{int}(g)P_{S'} = P'_{S'}\},$$

is a closed subscheme of G, smooth and of finite presentation over S, and is a principal homogeneous bundle on the right under P and on the left under P'.

Proposition 1.4.— *Let S be a nonempty scheme, let (G, T, M, R) be a split S-reductive group, and let R' be a subset of R. The following conditions on R' are equivalent:*

(i)
$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R'} \mathfrak{g}^\alpha$$
is the Lie algebra of a parabolic subgroup of G containing T (necessarily unique, Exp. XXII, 5.3.5).

(ii) *R' is of type* (R) (Exp. XXII, 5.4.2) *and contains a system of positive roots.*

(iii) *R' is a closed subset of R and satisfies: if $\alpha \in R - R'$, then $-\alpha \in R'$; equivalently, $R = R' \cup (-R')$.*

(iv) *There exist a system Δ of simple roots and a subset $A \subset \Delta$ such that R' is the union of the positive roots and the negative roots that are linear combinations of elements of A.*

(v) *R' contains a system of simple roots of R; moreover, if $\Delta \subset R'$ is a system of simple roots of R and*
$$A = (-R') \cap \Delta,$$
then R' is the union of the positive roots and the negative roots that are linear combinations of elements of A.

One has (i)⇔(ii) by Exp. XXII, 5.4.5(ii) and 5.5.1, and (iii)⇒(ii) by Exp. XXI, 3.3.6 and Exp. XXII, 5.4.7. Clearly (v)⇒(iv)⇒(iii), while (iii)⇒(v) follows from Exp. XXI, 3.3.6 and 3.3.10.

It remains to prove that (i) implies that R' is closed in R. This last assertion may be checked on any geometric fiber, so we may suppose that S is the spectrum of an algebraically closed field.

Let P be the parabolic subgroup of G containing T whose Lie algebra is $\mathfrak{g}_{R'}$. Since the Borel subgroups of P are the Borel subgroups of G contained in P, the Bible, §12.3, Theorem 1, and Exp. XXII, 5.4.5(i), show that, if U denotes the unipotent radical of P, then $T \cdot U$ is the subgroup of G containing T whose Lie algebra is

$$\mathfrak{g}_{R''} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R''} \mathfrak{g}^\alpha,$$

where R'' is the intersection of the systems of positive roots of R contained in R'. In particular, R'' is closed and $R'' \cap (-R'') = \varnothing$.

On the other hand, $H = P/U$ is reductive, the canonical image $\bar{T}$ of T is a maximal torus of H (and $T \to \bar{T}$ is an isomorphism), and there is an isomorphism of $\bar{T}$-modules, or equivalently of M-graded vector spaces,

$$\mathrm{Lie}(H) \simeq \mathfrak{t} \oplus \bigoplus_{\alpha \in R_s} \mathfrak{g}^\alpha,$$

where R_s is the complement of R'' in R'. Thus R_s identifies naturally with the roots of H relative to $\bar{T}$, and in particular $R_s = -R_s$. Consequently,

$$R'' = \{\alpha \in R' \mid -\alpha \notin R'\}, \qquad R_s = \{\alpha \in R' \mid -\alpha \in R'\}.$$

We now prove that R' is closed. Let $\alpha, \beta \in R'$ with $\alpha + \beta \in R$. If $\alpha, \beta \in R''$, then $\alpha + \beta \in R''$ because R'' is closed. If $\alpha \in R_s$, $\beta \in R''$, and $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$, whence

$$-\alpha = -(\alpha + \beta) + \beta \in R'',$$

because R'' is closed. This gives $-\alpha \in R'' \cap R_s$, a contradiction. It remains to consider $\alpha, \beta \in R_s$. If $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$. Since $\alpha + \beta \neq 0$, some system of positive roots of the root system R_s contains both α and β, and hence some Borel subgroup of $H = P/U$ contains the canonical images of U_α and U_β. Its inverse image in P is a Borel subgroup containing U_α, U_β, U, and therefore $U_{-(\alpha+\beta)}$, which is impossible.

Corollary 1.5.— *A parabolic subgroup of a reductive group is of type* (RC) (Exp. XXII, 5.11.1).

Proposition 1.6 (Exp. XXII, 5.11.4).— *Let S be a scheme, G an S-reductive group, and P a subgroup of G of type* (RC).[4]

(i) *P has a largest invariant sub-group-scheme that is smooth and of finite presentation over S, with connected unipotent geometric fibers. It is a characteristic subgroup of P, called the* unipotent radical *of P and denoted* $\mathrm{rad}^u(P)$. *The quotient sheaf* $P/\mathrm{rad}^u(P)$ *is representable by an S-reductive group.*

(ii) *If T is a maximal torus of P, then P has a reductive subgroup L containing T such that:*

  (a) *every reductive subgroup of P containing T is contained in L;*

  (b) *P is the semidirect product $L \cdot \mathrm{rad}^u(P)$, that is, the canonical morphism $L \to P/\mathrm{rad}^u(P)$ is an isomorphism.*

  Moreover, L is the unique subgroup (respectively, reductive subgroup) *of P containing T and satisfying (b)* (respectively, (a)). *Finally,*

$$\underline{\mathrm{Norm}}_P(L) = L, \qquad \underline{\mathrm{Norm}}_P(T) = \underline{\mathrm{Norm}}_L(T).$$

[4] *Editor's note.* "Parabolic subgroup" has been replaced by "subgroup of type (RC)", as in Exp. XXII, 5.11.4, because the statement will later be applied (4.5.1, 6.17) to $P \cap P'$ when P, P' are parabolic and $P \cap P'$ is of type (RC).

1.7. A subgroup L of P satisfying condition 1.6(ii)(b) is called a *Levi subgroup* of P. It is a maximal reductive subgroup of P. Indeed, it is reductive because it is isomorphic to $P/\operatorname{rad}^u(P)$. If L' is a reductive subgroup of P containing L, then, locally for the fpqc topology, L has a maximal torus T, and 1.6(ii) gives $L' = L$.

If L and L' are two Levi subgroups of P, they are conjugate in P, locally for the fpqc topology. Indeed, locally one may choose maximal tori $T \subset L$ and $T' \subset L'$; after a further fpqc localization they are conjugate in P, so one may assume $T = T'$, and then 1.6(ii) gives $L = L'$. Since $P = L \cdot \operatorname{rad}^u(P)$ and $\underline{\operatorname{Norm}}_P(L) = L$, one obtains:

Corollary 1.8.— *Let P be a subgroup of type* (RC) *of the S-reductive group G. If L, L' are two Levi subgroups of P, there is a unique $u \in \operatorname{rad}^u(P)(S)$ such that* $\operatorname{int}(u)L = L'$.

Let $\underline{\operatorname{Lev}}(P)$ denote the functor of Levi subgroups of P: for $S' \to S$, $\underline{\operatorname{Lev}}(P)(S')$ is the set of Levi subgroups of $P_{S'}$. Corollary 1.8 gives:

Corollary 1.9.— *Let P be a subgroup of type* (RC) *of the S-reductive group G. Then $\underline{\operatorname{Lev}}(P)$ is a principal homogeneous bundle under the S-group $\operatorname{rad}^u(P)$; in particular, it is representable by an S-scheme that is smooth and affine over S, with integral geometric fibers.*

Likewise, 1.6 immediately yields:

Corollary 1.10.— *Let P be a subgroup of type* (RC) *of the S-reductive group G. The functor $\underline{\operatorname{Tor}}(P)$ of maximal tori of P is representable by a smooth affine S-scheme. The relation $L \supset T$ defines a morphism*

$$\underline{\operatorname{Tor}}(P) \longrightarrow \underline{\operatorname{Lev}}(P),$$

whose fiber above $L \in \underline{\operatorname{Lev}}(P)(S)$ identifies with $\underline{\operatorname{Tor}}(L)$ (Exp. XXII, 5.8.3).

The first assertion of 1.10 follows from the other two and Exp. XXII, 5.8.3.

Definition 1.11.— *Let S be a nonempty scheme, G an S-reductive group, P a parabolic subgroup of G, and*

$$\mathcal{E} = (T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$$

a pinning of G. One says that $\mathcal{E}$ is adapted *to P, or is a* pinning of the pair (G, P), *if $P \supset T$ and the Lie algebra of P is of the form*

$$\mathfrak{t} \oplus \bigoplus_{\alpha\in R'} \mathfrak{g}^\alpha,$$

where $R' \subset R$ contains R^+.

In particular, if $T \subset B$ is the Killing pair defined by the pinning, then $T \subset B \subset P$. Under these conditions set $\Delta(P) = \Delta \cap (-R')$. Exp. XXII, 5.4.3 gives

$$\alpha \in \Delta(P) \iff \alpha \in \Delta \text{ and } U_{-\alpha} \subset P \iff \alpha \in \Delta \text{ and } U_{-\alpha} \cap P \neq e.$$

Proposition 1.4(v) and Exp. XXII, 5.11.3 and 5.10.6 now immediately give:

Proposition 1.12.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and $(T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$ a pinning of G adapted to P. Let $\Delta(P) \subset \Delta$ be as above.*

(i) *The unipotent radical of P is*

$$\operatorname{rad}^u(P) = U_{R''} = \prod_{\alpha\in R''} U_\alpha,$$

where R'' is the set of positive roots whose expansion in Δ contains at least one element of $\Delta - \Delta(P)$[5] *with a nonzero coefficient.*

(ii) *The unique Levi subgroup L of P containing T is*

$$Z_{\Delta(P)} = \underline{\mathrm{Centr}}_G(T_{\Delta(P)}),$$

where $T_{\Delta(P)}$ is the maximal torus of $\bigcap_{\alpha\in\Delta(P)} \ker(\alpha)$. Moreover, $T_{\Delta(P)} = \mathrm{rad}(L)$.

Corollary 1.13.— *Every Levi subgroup L of a parabolic subgroup P of a reductive group G is a* critical subgroup *of G; that is* (Exp. XXII, 5.10.4)*,*

$$L = \underline{\mathrm{Centr}}_G(\mathrm{rad}(L)).$$

This follows immediately from 1.12 and the following lemma, which is contained in 1.4 and Exp. XXII, 5.4.1:

Lemma 1.14.— *Locally for the étale topology, every pair (G, P), where P is a parabolic subgroup of the reductive group G, admits a pinning in the sense of 1.11.*

We also record:

Proposition 1.15.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and $\mathcal{E}, \mathcal{E}'$ two pinnings of G adapted to P. The unique inner automorphism of G over S that transforms $\mathcal{E}$ into $\mathcal{E}'$* (Exp. XXIV, 1.5) *comes from P, via*

$$P \longrightarrow P/\underline{\mathrm{Centr}}(P) = P/\underline{\mathrm{Centr}}(G) \longrightarrow G/\underline{\mathrm{Centr}}(G).$$

Indeed, it suffices to argue as in Exp. XXIV, 1.5, using:

Lemma 1.16 (Exp. XXII, 5.3.14 and 5.2.6).— *The maximal tori* (respectively, Borel subgroups, respectively, Killing pairs) *of a parabolic subgroup P of an S-reductive group G are conjugate in P, locally for the étale topology.*

Proposition 1.17.— *Let S be a scheme, G an S-reductive group, P, P' two parabolic subgroups of G, and B a Borel subgroup contained in both P and P'. If P and P' are conjugate in G, locally for the étale topology, then $P = P'$.*

Indeed, we may assume that $g \in G(S)$ satisfies $\mathrm{int}(g)P = P'$. The groups B and $\mathrm{int}(g)^{-1}B$ are then two Borel subgroups of P. After extending S, Lemma 1.16 lets us choose $p \in P(S)$ with $\mathrm{int}(p)\,\mathrm{int}(g^{-1})B = B$. Hence

$$pg^{-1} \in \underline{\mathrm{Norm}}_G(B)(S) = B(S),$$

so $g \in B(S) \cdot p \subset P(S)$, and therefore $P' = \mathrm{int}(g)P = P$.

Remark 1.18.— *If P, P' are two parabolic subgroups of G containing a common Borel subgroup, then $P \cap P'$ is again a parabolic subgroup of G. Indeed, it is smooth along the unit section* (Exp. XXII, 5.4.5) *and contains a Borel subgroup.*

Proposition 1.19.— *Let S be a scheme, G an S-reductive group, and G' its derived group* (Exp. XXII, 6.2.1)*.*

[5] *Editor's note.* $\Delta(P)$ has been corrected to $\Delta - \Delta(P)$.

(i) *The assignments*

$$P \longmapsto P' = P \cap G', \qquad P' \longmapsto P' \cdot \mathrm{rad}(G) = \underline{\mathrm{Norm}}_G(P')$$

are mutually inverse bijections between the parabolic subgroups of G and those of G'. Moreover, $\mathrm{rad}^u(P) = \mathrm{rad}^u(P')$.

(ii) *Let P be a parabolic subgroup of G and $P' = P \cap G'$. The assignments*

$$L \longmapsto L' = L \cap G' = L \cap P', \qquad L' \longmapsto L' \cdot \mathrm{rad}(G) = \underline{\mathrm{Centr}}_G(\mathrm{rad}(L'))$$

are mutually inverse bijections between the Levi subgroups of P and those of P'. Moreover,

$$\mathrm{rad}(L') = (\mathrm{rad}(L) \cap G')^0.$$

The proof, for example by reduction to the split case, is straightforward and is left to the reader, together with the immediate proof of:

Proposition 1.20.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and L a Levi subgroup of P. The assignments*

$$Q \longmapsto Q \cap L = Q', \qquad Q' \longmapsto Q' \cdot \mathrm{rad}^u(P)$$

are mutually inverse bijections between the parabolic subgroups of G contained in P and the parabolic subgroups of L. Moreover, the Levi subgroups of Q' are precisely the Levi subgroups of Q contained in L.

Proposition 1.6 may be completed as follows.

Proposition 1.21.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G.*

(i) *P has a largest invariant subgroup that is smooth and of finite presentation over S, with connected solvable geometric fibers. It is a characteristic subgroup of P, called the* radical *of P and denoted* $\mathrm{rad}(P)$. *The quotient sheaf $P/\mathrm{rad}(P)$ is representable by an S-semisimple group.*

(ii) *If L is a Levi subgroup of P, then* $\mathrm{rad}(P)$ *is the semidirect product of* $\mathrm{rad}^u(P)$ *and* $\mathrm{rad}(L)$*; one has*

$$\mathrm{rad}(L) = L \cap \mathrm{rad}(P), \qquad L = \underline{\mathrm{Centr}}_G(L \cap \mathrm{rad}(P)), \qquad P/\mathrm{rad}(P) \simeq L/\mathrm{rad}(L).$$

Indeed, assertion (i) being local, one may suppose that P has a Levi subgroup L. It is then immediate that

$$R = \mathrm{rad}^u(P) \cdot \mathrm{rad}(L)$$

has the properties asserted in (i). For (ii), it only remains to prove that $\mathrm{rad}(L) = L \cap \mathrm{rad}(P)$; this follows at once because $L \cap \mathrm{rad}(P)$ is smooth with connected fibers, L being the centralizer of a torus.[6]

[6] *Editor's note.* See Exposé XIX, 1.4.

2. Structure of the unipotent radical of a parabolic subgroup

Proposition 2.1.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and $\mathrm{rad}^u(P)$ its unipotent radical. There is a sequence of sub-group-schemes of $\mathrm{rad}^u(P)$*

$$\mathrm{rad}^u(P) = U_0 \supset U_1 \supset U_2 \supset \cdots \supset U_n \supset \cdots$$

having the following properties:

(i) *Every U_i is smooth, with connected fibers, characteristic and closed in P. The commutator of a section of U_i and a section of U_j is a section of U_{i+j+1}, over a varying $S' \to S$.*

(ii) *For every $i \geq 0$, there is a locally free $\mathcal{O}_S$-module $\mathcal{E}_i$ and an isomorphism of sheaves of S-groups*

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i).$$

Moreover, the automorphisms of P, over a varying $S' \to S$, act linearly on $W(\mathcal{E}_i)$.

(iii) *For every $s \in S$, one has $U_{n,s} = e$ whenever*

$$n > \dim\big(\mathrm{rad}^u(P)_s\big).$$

2.1.1. Suppose first that the pair (G, P) admits a pinning. Let $(T, M, R, \Delta, \ldots)$ be a pinning of G adapted to P, and let $\Delta(P) \subset \Delta$ be determined by P. Write $\alpha_1, \ldots, \alpha_p$ for the elements of $\Delta(P)$, and $\beta_1, \ldots, \beta_q$ for the elements of $\Delta - \Delta(P)$. Every root $\gamma \in R$ has a unique expression

$$\gamma = a_1\alpha_1 + \cdots + a_p\alpha_p + b_1\beta_1 + \cdots + b_q\beta_q.$$

Set

$$a(\gamma) = b_1 + \cdots + b_q. \tag{7}$$

[7] The definitions immediately give (cf. 1.12):

$$\begin{aligned} U_\gamma \subset P &\iff a(\gamma) \geq 0, \\ U_\gamma \subset \mathrm{rad}^u(P) &\iff a(\gamma) > 0, \\ a(n\gamma + m\gamma') &= n\,a(\gamma) + m\,a(\gamma') \qquad (n, m \in \mathbb{Z}). \end{aligned}$$

For $i \geq 0$, let R_i be the set of roots $\gamma \in R$ such that $a(\gamma) > i$. Each R_i is a closed set of roots satisfying $R_i \cap (-R_i) = \varnothing$. Consider (Exp. XXII, 5.6.5) the S-group

$$U_i = U_{R_i} = \prod_{\gamma \in R_i} U_\gamma.$$

It is a closed sub-group-scheme of G, smooth over S, with connected fibers.

Let $\alpha, \beta \in R$. Consider the commutation relation of Exp. XXII, 5.5.2:

$$p_\alpha(x)p_\beta(y)p_\alpha(-x) = p_\beta(y) \prod_{n,m \in \mathbb{N}^*} p_{n\alpha+m\beta}(C_{n,m,\alpha,\beta}x^n y^m),$$

[7] *Editor's note.* The original $a_1 + \cdots + a_p$ has been corrected to $b_1 + \cdots + b_q$.

where every

$$p_\gamma : \mathbb{G}_{a,S} \xrightarrow{\sim} U_\gamma$$

is an isomorphism of vector groups. If $a(\alpha) > i$ and $a(\beta) > j$, then, for $n, m > 0$,

$$\begin{aligned} a(n\alpha + m\beta) &= n\, a(\alpha) + m\, a(\beta) \\ &\geq n(i+1) + m(j+1) > i + j + 1. \end{aligned}$$

Hence the commutator of a section of U_α and a section of U_β belongs to U_{i+j+1}; therefore

$$(U_i, U_j) \subset U_{i+j+1}.$$

In particular U_i/U_{i+1} is commutative, and identifies naturally with

$$U_i/U_{i+1} \simeq \prod_{a(\gamma)=i+1} U_\gamma \simeq W(\mathcal{E}_i),$$

where $\mathcal{E}_i$ is the direct sum of the $\mathfrak{g}^\gamma$ for which $a(\gamma) = i + 1$.

Return to the commutation formula and suppose $a(\alpha) \geq 0$, $a(\beta) > i$. If $n, m \in \mathbb{N}^*$, then either

$$a(n\alpha + m\beta) > i + 1,$$

or $a(n\alpha + m\beta) = i + 1$, in which case necessarily $m = 1$. This first proves that $\mathrm{int}(p_\alpha(x))$ preserves U_i, and hence also U_{i+1}. It then shows that, modulo U_{i+1}, the expression for $\mathrm{int}(p_\alpha(x))p_\beta(y)$ contains only terms

$$p_{n\alpha+\beta}(C_{n,1,\alpha,\beta} x^n y),$$

which are linear in y. Thus the inner automorphisms defined by sections of U_α act linearly on $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$. The same is clear for inner automorphisms defined by sections of T. Since P is generated by T and by the U_α with $a(\alpha) \geq 0$, it follows that:

(i) every U_i is invariant in P;

(ii) the inner automorphisms defined by sections of P act linearly on $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$.

2.1.2. Let $(T', M', R', \Delta', \ldots)$ now be another pinning of G adapted to P. By 1.15 there is an inner automorphism of G, coming from P, that carries the first pinning to the second. After extending S, it may be written $\mathrm{int}(p)$ for some $p \in P(S)$. Repeating the construction with the second pinning produces groups U'_i and vector-group isomorphisms

$$U'_i/U'_{i+1} \simeq W(\mathcal{E}'_i)$$

obtained from U_i and $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$ by transport of structure under $\mathrm{int}(p)$. The preceding invariance and linearity statements therefore give $U'_i = U_i$, and the two vector-space structures on

$$U_i/U_{i+1} = U'_i/U'_{i+1}$$

coincide.

Thus the U_i and the vector structures on their successive quotients are independent of the chosen pinning and, in particular, invariant under every automorphism of P. This proves the proposition when (G, P) admits a pinning. Assertion (iii) is immediate: by Exp. XXI, 3.1.2, the set $\{a(\gamma) \mid \gamma \in R\}$ is an interval in $\mathbb{Z}$, so $\dim U_{i,s} = \dim U_{i+1,s}$ cannot occur unless $U_{i,s} = e$.

2.1.3. In general there is an fpqc covering family $\{S_j \to S\}$ such that every pair (G_{S_j}, P_{S_j}) admits a pinning (1.14). What precedes gives descent data on the $\mathrm{rad}^u(P_{S_j})_i$, compatible with the vector structures on the quotients. The result follows by descent for closed subschemes and for locally free modules.[8]

Corollary 2.2.— *Let S be affine, G an S-reductive group, and P a parabolic subgroup of G. Then*

$$H^1(S, \mathrm{rad}^u(P)) = 0;$$

that is, every principal homogeneous bundle under $\mathrm{rad}^u(P)$ *is trivial.*

Indeed, S is the disjoint union of subschemes on each of which $\mathrm{rad}^u(P)$ has constant relative dimension. By 2.1(iii) one may thus assume that $U_n = e$ for some n. Since

$$H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0$$

by TDTE I, B, 1.1 (or SGA 1, Exposé XI, 5.1), the conclusion follows.

Corollary 2.3.— *Under the preceding hypotheses, P has a Levi subgroup L. If L is any Levi subgroup of P, the canonical map*

$$H^1(S, L) \longrightarrow H^1(S, P)$$

is bijective (see the introduction to Exp. XXIV for the definition of $H^1(S, -)$).

The first assertion follows from 2.2 and 1.9. The map $H^1(S, L) \to H^1(S, P)$ is surjective because $P = L \cdot \mathrm{rad}^u(P)$. For injectivity it is enough to show that

$$H^1(S, \mathrm{rad}^u(P)_Q) = 0$$

for every principal homogeneous L-bundle Q, where the subscript denotes twisting by Q. One may either repeat the proof of 2.2, using the L-invariance of the vector structures on U_i/U_{i+1}, or observe that $\mathrm{rad}^u(P)_Q$ is the unipotent radical of the parabolic subgroup P_Q of G_Q and apply 2.2 to P_Q.

Corollary 2.4.— *Let S be a semi-local scheme, G an S-reductive group, and P a parabolic subgroup of G. There is a maximal torus T of G contained in P.*

Indeed, by 2.3 the group P has a Levi subgroup L, and it suffices to know that L has a maximal torus, which follows from Exp. XIV, 3.20.

Corollary 2.5.— *Let S be affine, G an S-reductive group, and P a parabolic subgroup of G. There is a locally free $\mathcal{O}_S$-module $\mathcal{E}$ such that* $\mathrm{rad}^u(P)$*, as an S-scheme, is isomorphic to $W(\mathcal{E})$.*

We prove by induction on i that there is an isomorphism of S-schemes

$$\mathrm{rad}^u(P)/U_i \simeq W(\mathcal{E}_0 \oplus \mathcal{E}_1 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

This is clear for $i = 0$. If $i > 0$, then $\mathrm{rad}^u(P)/U_i$ is a principal homogeneous bundle over $\mathrm{rad}^u(P)/U_{i-1}$ under

$$(U_{i-1}/U_i)_{\mathrm{rad}^u(P)/U_{i-1}} \simeq W\Big(\mathcal{E}_{i-1} \otimes \mathcal{O}_{\mathrm{rad}^u(P)/U_{i-1}}\Big).$$

[8] *Editor's note.* See SGA 1, Exposé VIII, 1.3, 1.9, and 1.10.

The base is affine, for example by induction, so the bundle is trivial (TDTE I or SGA 1, Exposé XI, *loc. cit.*). Hence

$$\mathrm{rad}^u(P)/U_i \simeq \big(\mathrm{rad}^u(P)/U_{i-1}\big) \times_S (U_{i-1}/U_i),$$

which completes the induction.

Corollary 2.6.— *Let S be semi-local, let $\{s_i\}$ be its closed points, let G be an S-reductive group, and let P be a parabolic subgroup of G. The canonical map*

$$\mathrm{rad}^u(P)(S) \longrightarrow \prod_i \mathrm{rad}^u(P)(\mathrm{Spec}\,\kappa(s_i))$$

is surjective.

Indeed, write $S = \mathrm{Spec}\,A$ and $\kappa(s_i) = A/\mathfrak{p}_i$. If $\mathcal{E}$ corresponds to the projective, hence flat, A-module E, it remains to prove that

$$E \longrightarrow \prod_i E \otimes_A A/\mathfrak{p}_i$$

is surjective. It suffices to do so for $E = A$, when it is well known (Bourbaki, *Algèbre commutative*, Chapter II, §1, no. 2, Proposition 5).

Corollary 2.7.— *Let k be an infinite field, G a k-reductive group, and P a parabolic subgroup of G. Then* $\mathrm{rad}^u(P)(k)$ *is dense in* $\mathrm{rad}^u(P)$.

Corollary 2.8.— *Let S be semi-local, let $\{s_i\}$ be its closed points, let G be an S-reductive group, let P be a parabolic subgroup of G, and, for every i, let L_i be a Levi subgroup of P_{s_i}. There is a Levi subgroup L of P whose fiber at every s_i is L_i.*

Indeed, let L_0 be a Levi subgroup of P (2.3). For every i, choose

$$u_i \in \mathrm{rad}^u(P)(\mathrm{Spec}\,\kappa(s_i)) \quad \text{with} \quad \mathrm{int}(u_i)L_{0,s_i} = L_i$$

by 1.8. If $u \in \mathrm{rad}^u(P)(S)$ induces the u_i, as 2.6 permits, then $L = \mathrm{int}(u)L_0$ has the required fibers.

Corollary 2.9.— *In the setting of 2.1, let H also be a sub-group-scheme of G, smooth and of finite presentation over S, with connected fibers, such that $P \cap H$ contains, locally for the fpqc topology, a maximal torus of G. For every $i \geq 0$ there is a locally direct-factor submodule $\mathcal{F}_i \subset \mathcal{E}_i$ for which the isomorphism*

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i)$$

induces an isomorphism

$$(U_i \cap H)/(U_{i+1} \cap H) \xrightarrow{\sim} W(\mathcal{F}_i).$$

Indeed, H is a subgroup of G of type (R) (Exp. XXII, 5.2.1). The assertion is fpqc-local, so we may suppose that G is split relative to a maximal torus of $P \cap H$, and even reduce to 2.1.1 with H defined by a subset $R' \subset R$. In the notation used there, Exp. XXII, 5.6.7(ii) gives

$$U_i \cap H = \prod_{\alpha \in R_i \cap R'} U_\alpha.$$

Thus $(U_i \cap H)/(U_{i+1} \cap H)$ identifies with

$$\prod_{\substack{\alpha \in R' \\ a(\alpha)=i+1}} U_\alpha,$$

which proves the assertion.

Corollary 2.10.— *In the setting of 2.9, the conclusions of 2.2, 2.5, 2.6, and 2.7 remain valid with* $\mathrm{rad}^u(P)$ *replaced by* $\mathrm{rad}^u(P) \cap H$.

Corollary 2.11.— *Let*[9] G *be an* S*-reductive group,* P *a parabolic subgroup, and* H *a subgroup of* P *of type* (RC) *such that*

$$\mathrm{rad}^u(H) = \mathrm{rad}^u(P) \cap H.$$

Then statements 2.2 through 2.8 remain valid with P *replaced by* H.

3. Scheme of parabolic subgroups of a reductive group

3.1. Let E be a finite twisted constant S-scheme (Exp. X, 5.1). Consider the S-functor $\mathrm{Of}(E)$, where $\mathrm{Of}(E)(S')$ is the set of open-and-closed subschemes of $E_{S'}$, or equivalently the set of open-and-closed subsets of $E_{S'}$. Then $\mathrm{Of}(E)$ is representable by a finite twisted constant S-scheme. Indeed, if $E = A_S$, for a finite set A, then

$$\mathrm{Of}(E) \simeq \mathcal{P}(A)_S,$$

where $\mathcal{P}(A)$ denotes the power set of A; the general case follows by descent of open-and-closed subschemes. Plainly,

$$\mathrm{Of}(E_{S'}) = \mathrm{Of}(E)_{S'}, \qquad \mathrm{Of}(E \times_S E') = \mathrm{Of}(E) \times_S \mathrm{Of}(E').$$

3.2. Let S be a scheme and G an S-reductive group. Define the functor $\underline{\mathrm{Par}}(G)$ of parabolic subgroups of G by

$$\underline{\mathrm{Par}}(G)(S') = \{\text{parabolic subgroups of } G_{S'}\}.$$

In particular $G \in \underline{\mathrm{Par}}(G)(S)$, and $\underline{\mathrm{Bor}}(G) \subset \underline{\mathrm{Par}}(G)$. We shall define a morphism

$$t : \underline{\mathrm{Par}}(G) \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G))$$

with the following properties:

(i) t is functorial in G, with respect to isomorphisms, and commutes with base change.

(ii) If $(T, M, R, \Delta, \ldots)$ is a pinning of G adapted to a parabolic subgroup P, then the canonical isomorphism

$$\underline{\mathrm{Dyn}}(G) \simeq \Delta_S \qquad \text{(Exp. XXIV, 3.4(iii))}$$

carries $t(P)$ onto $\Delta(P)_S$, in the notation of 1.11 and 1.12.

[9] *Editor's note.* Corollary 2.11 has been added; it is used in 4.5.1 and 6.17.

First suppose that P is a parabolic subgroup of G and that $(T, M, R, \Delta, \ldots)$ is a pinning of G adapted to P. Define $t(P)$ by (ii). The resulting subscheme of $\underline{\mathrm{Dyn}}(G)$ is independent of the pinning. Indeed, if $(T', M', R', \Delta', \ldots)$ is another pinning adapted to P, then the unique inner automorphism of G carrying the first pinning to the second comes from P, by 1.15. The canonical isomorphism $\Delta \xrightarrow{\sim} \Delta'$ therefore carries $\Delta(P)$ onto $\Delta'(P)$.

Without assuming that (G, P) admits a pinning, 1.14 and the definition of $\underline{\mathrm{Dyn}}(G)$ (Exp. XXIV, 3.3) allow one to descend a unique open-and-closed subscheme $t(P)$ such that

$$t(P)_{S'} = t(P_{S'})$$

whenever $(G, P)_{S'}$ admits a pinning.

Theorem 3.3.— *Let S be a scheme, G an S-reductive group, and*

$$t : \underline{\mathrm{Par}}(G) \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G))$$

the morphism just defined.

(i) *Two parabolic subgroups P, P' of G are conjugate locally for the fpqc topology* (cf. 1.3) *if and only if $t(P) = t(P')$.*

(ii) $\underline{\mathrm{Par}}(G)$ *is representable, and t is smooth and projective, with integral geometric fibers.*

By 3.2(i), together with the fact that inner automorphisms of G act trivially on $\underline{\mathrm{Dyn}}(G)$ (Exp. XXIV, 3.4(iv)), conjugate parabolics have the same type. Conversely, suppose $t(P) = t(P')$. Locally one may first assume that both (G, P) and (G, P') admit pinnings (1.14). By the conjugacy of pinnings in G (Exp. XXIV, 1.5), one may then choose a single pinning (T, M, R, Δ) adapted to both P and P'. Equality of types gives $\Delta(P) = \Delta(P')$, hence $P = P'$ by 1.4(v). This proves (i).

For (ii), use the notation of Exp. XXII, 5.11.5.[10] There is a canonical morphism $\underline{\mathrm{Par}}(G) \to \mathscr{H}_c$, which fits into the cartesian square

$$\begin{array}{ccc} \underline{\mathrm{Par}}(G) & \xrightarrow{\ t\ } & \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \\ \downarrow & & \downarrow \\ \mathscr{H}_c & \xrightarrow{\ c\ell\ } & \mathcal{C}\ell_c. \end{array}$$

The vertical arrows are monomorphisms. By *loc. cit.*, $\mathscr{H}_c$ is representable and $c\ell$ is smooth, quasi-projective, and of finite presentation, with integral geometric fibers; hence the same is true of t.

It remains to prove that t is proper. This is fpqc-local, so one may reduce to a pinned group $G = (G, T, M, R, \Delta, \ldots)$. Then $\underline{\mathrm{Dyn}}(G) \simeq \Delta_S$, and it is enough to show that

$$t^{-1}((\Delta_1)_S)$$

is proper over S for every $\Delta_1 \subset \Delta$. Let P_1 be the parabolic containing T with $\Delta(P_1) = \Delta_1$. Part (i) shows that $g \mapsto \mathrm{int}(g)P_1$ induces an isomorphism

$$G/\underline{\mathrm{Norm}}_G(P_1) \xrightarrow{\sim} t^{-1}((\Delta_1)_S).$$

[10] *Editor's note.* Recall that $\mathscr{H}_c = \mathscr{H}_c(G)$ denotes the functor of subgroups of G of type (RC), that $\mathcal{C}\ell_c = \mathcal{C}\ell_c(G)$ denotes the functor of their "conjugacy classes", and that $c\ell : \mathscr{H}_c \to \mathcal{C}\ell_c$ is the canonical projection.

By 1.2, $G/\underline{\mathrm{Norm}}_G(P_1) = G/P_1$, which is projective over S.

Definition 3.4.— $\mathrm{Of}(\underline{\mathrm{Dyn}}(G))$ *is called the* scheme of types of parabolics *of* G, *and* $t(P)$ *is called the* type *of* P.

Corollary 3.5.— *The* S*-functor* $\underline{\mathrm{Par}}(G)$ *is representable by an* S*-scheme that is smooth and projective over* S. *The factorization*

$$\underline{\mathrm{Par}}(G) \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \longrightarrow S$$

is the Stein factorization (EGA III, 4.3.3) *of the structural morphism* $\underline{\mathrm{Par}}(G) \to S$.

Corollary 3.6.— *For every* $t \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S)$, *the* S*-scheme*

$$\underline{\mathrm{Par}}_t(G) = t^{-1}(t)$$

of parabolic subgroups of type t *is smooth and projective over* S *and homogeneous under* G. *If* P *is a parabolic subgroup of* G, *there is a canonical isomorphism*

$$G/P \xrightarrow{\sim} \underline{\mathrm{Par}}_{t(P)}(G).$$

Moreover,

$$\underline{\mathrm{Par}}_{\varnothing}(G) = \underline{\mathrm{Bor}}(G), \qquad \underline{\mathrm{Par}}_{\underline{\mathrm{Dyn}}(G)}(G) \xrightarrow{\sim} S.$$

Remark 3.7.— *The* S*-scheme* $\mathrm{Of}(\underline{\mathrm{Dyn}}(G))$ *has a natural order structure, the domination relation being set-theoretic inclusion of subschemes. It is a lattice; in particular, the infimum of two open-and-closed subschemes of* $\underline{\mathrm{Dyn}}(G)_{S'}$ *is their intersection.*

If B *is a Borel subgroup of* G, *let* X *be the functor of parabolic subgroups containing* B. *The morphism*

$$X \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G))$$

induced by t *is an isomorphism of ordered schemes. This follows from* $P \subset Q \Rightarrow t(P) \subset t(Q)$ *and the next lemma.*

Lemma 3.8.— *Let* S *be a scheme,* G *an* S*-reductive group,* P *a parabolic subgroup of* G, *and*

$$t' \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S) \quad \textit{with} \quad t(P) \subset t'.$$

There is a unique parabolic subgroup P' *of* G *containing* P *and satisfying* $t(P') = t'$.

After extending the base, one may suppose that P contains a Borel subgroup B of G. Uniqueness then follows from 1.17. For existence one may work in the split case, where the assertion is immediate from §1.

Remarks 3.8.1.—

(i) *The analogue of 3.8 obtained by reversing the inclusions is false. It would imply, for example, that every group of type* A_1 *has a Borel subgroup, which is not so* (Exp. XX, §5).

(ii) *The preceding discussion shows that* $t(P) \subset t(Q)$ *means that locally for the fpqc, or étale, topology,* P *is conjugate to a subgroup of* Q. *It is enough to check this on geometric fibers. Section 5 will show that the étale topology may be replaced by the Zariski topology.*

3.9. The preceding discussion can be repeated for critical subgroups. Recall (Exp. XXII, 5.10.4 and 5.10.5) that a reductive subgroup H of a reductive group G is *critical* if

$$H = \underline{\mathrm{Centr}}_G(\mathrm{rad}(H)),$$

that a subtorus $Q \subset G$ is a *C-critical torus* if

$$Q = \mathrm{rad}(\underline{\mathrm{Centr}}_G(Q)),$$

and that critical subgroups and C-critical tori are in bijection via

$$H \longmapsto \mathrm{rad}(H), \qquad Q \longmapsto \underline{\mathrm{Centr}}_G(Q).$$

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If (G, T, M, R) is split, the subgroup of G containing T and corresponding to $R' \subset R$ (Exp. XXII, 5.4.2) is critical if and only if R' is *vectorial*, meaning that it is the intersection of R with a vector subspace of $M \otimes \mathbb{Q}$ (Exp. XXII, 5.10.6).

For arbitrary G, Exp. XXII, 5.11.5 defines a finite étale S-scheme $\mathcal{C}\ell_{\mathrm{crit}}$; in the split case it is the constant scheme associated with the vectorial subsets of R, modulo the action of the Weyl group. If $\underline{\mathrm{Crit}}(G)$ denotes the functor of critical subgroups, there is a canonical morphism

$$c\ell : \underline{\mathrm{Crit}}(G) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

in the cartesian square

$$\begin{array}{ccc} \underline{\mathrm{Crit}}(G) & \xrightarrow{c\ell} & \mathcal{C}\ell_{\mathrm{crit}} \\ \downarrow & & \downarrow \\ \mathscr{H}_c & \xrightarrow{c\ell} & \mathcal{C}\ell_c. \end{array}$$

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Proposition 3.10.— *Let S be a scheme, G an S-reductive group, $\underline{\mathrm{Crit}}(G)$ the functor of its critical subgroups, and let $\mathcal{C}\ell_{\mathrm{crit}}$ and*

$$c\ell : \underline{\mathrm{Crit}}(G) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

be as above.

(i) *Critical subgroups H, H' of G are conjugate, locally for the fpqc topology, if and only if $c\ell(H) = c\ell(H')$.*

(ii) *$\underline{\mathrm{Crit}}(G)$ is representable, and $c\ell$ is smooth and affine, with integral geometric fibers.*

The proof is the same as for 3.3 except for affineness of $c\ell$. It is enough to prove that $\underline{\mathrm{Crit}}(G)$ is affine over S. The functor $\underline{\mathrm{Crit}}(G)$ identifies naturally with the functor of critical tori of G, giving a canonical monomorphism

$$\underline{\mathrm{Crit}}(G) \longrightarrow \mathcal{M},$$

where $\mathcal{M}$ is the scheme of subgroups of multiplicative type of G (Exp. XI, 4.1). By Exp. XII, 5.3, it is enough to show that this is an open-and-closed immersion. After the base change $\mathcal{M} \to S$, this reduces to the following: if Q is a subgroup of multiplicative type in G, the $S' \to S$ for which $Q_{S'}$ is a critical torus of $G_{S'}$ are precisely those factoring through an open-and-closed subscheme of S. The condition is the conjunction of:

[11] *Editor's note.* "Critical torus" has here been replaced by "C-critical torus". Below, "critical torus" is again used for "C-critical torus".

[12] *Editor's note.* See 3.3, editor's note, for the notation $\mathscr{H}_c, \mathcal{C}\ell_c$.

(1) Q is a torus;

(2) once Q is a torus, the torus $\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q))$ has the same relative dimension as Q.

Both conditions are of the required kind.

Corollary 3.11.— *The S-functor $\underline{\mathrm{Crit}}(G)$ is representable by an S-scheme that is smooth and affine over S.*

Corollary 3.12.— *Let H be a critical subgroup of the S-reductive group G. Then $G/\underline{\mathrm{Norm}}_G(H)$ and G/H are representable by S-schemes that are affine and smooth over S.*

The first assertion follows from 3.11, and the second from the first and Exp. XXII, 5.10.2.

Corollary 3.13.— *Let Q be a subtorus of the S-reductive group G. Then $G/\underline{\mathrm{Centr}}_G(Q)$ is representable by an S-scheme that is smooth and affine over S. The same holds for $G/\underline{\mathrm{Norm}}_G(Q)$ when Q is a critical subtorus of G.*

Indeed, $H = \underline{\mathrm{Centr}}_G(Q)$ is critical (Exp. XXII, 5.10.5), and when Q is critical one has $\underline{\mathrm{Norm}}_G(H) = \underline{\mathrm{Norm}}_G(Q)$ (*loc. cit.*, 5.10.8).

3.14. The conjugacy of Levi subgroups of parabolic subgroups of G gives a unique morphism

$$u : \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

such that, for every parabolic subgroup P and every Levi subgroup $L \subset P$,

$$c\ell(L) = u(t(P)),$$

compatibly with every base change.

3.15. Let S be a scheme and G an S-reductive group. Consider the S-functors[13]

$$\begin{aligned}
PL(S') &= \{(P, L) \mid P \supset L,\ P \text{ parabolic in } G_{S'},\ L \text{ a Levi subgroup of } P\},\\
PT(S') &= \{(P, T) \mid P \supset T,\ P \text{ parabolic in } G_{S'},\ T \text{ a maximal torus of } P\},\\
CT(S') &= \{(C, T) \mid C \supset T,\ C \text{ critical in } G_{S'},\ T \text{ a maximal torus of } C\},\\
PLT(S') &= \{(P, L, T) \mid P \supset L \supset T,\ (P, T) \in PT(S'),\ L \text{ a Levi subgroup of } P\}.
\end{aligned}$$

There are evident morphisms between these functors and $\underline{\mathrm{Par}}(G), \underline{\mathrm{Crit}}(G)$, and $\underline{\mathrm{Tor}}(G)$. They form the following commutative truncated cube.

Theorem 3.16 (cf. Figure 3.15.1).—

(i) *Every morphism in the diagram is smooth, surjective, and of finite presentation.*

(ii) *Every morphism except t is affine; t is projective.*

(iii) *Every morphism is either finite étale or has integral geometric fibers: f, g, h, k, p, q, u are finite étale; $a, b, c, d, r, c\ell$ have integral geometric fibers; and e is an isomorphism.*

(iv) *The square (a, b, f, g) is cartesian.*

[13] *Editor's note.* The original LT has been changed to CT.

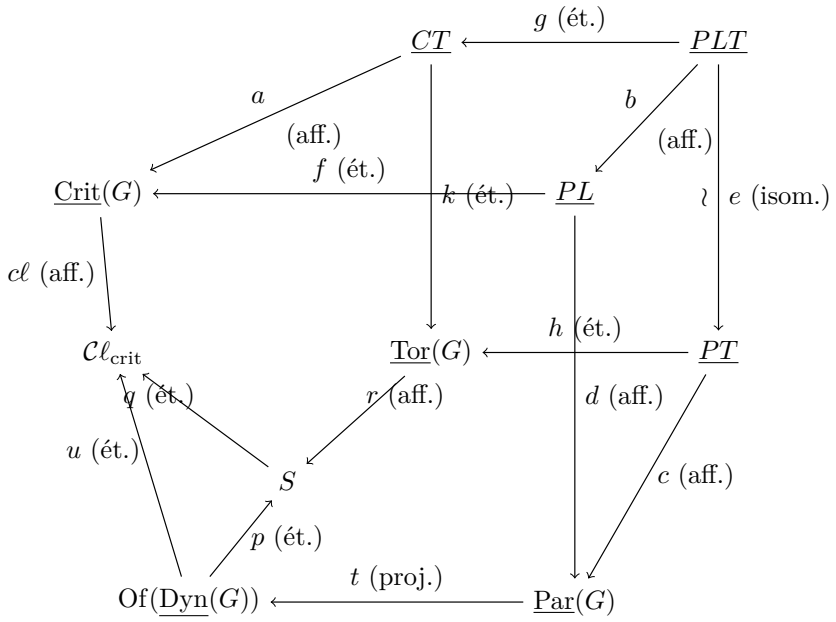

Figure 3.15.1

Proof. It is clear from 1.6(ii) that e is an isomorphism, and (iv) is immediate. The morphism a is smooth and affine with integral geometric fibers. Indeed, after the base change $\underline{\mathrm{Crit}}(G) \to S$, this reduces to the same statement for

$$\underline{\mathrm{Tor}}(L_0) \longrightarrow S,$$

where L_0 is the universal critical subgroup; L_0 is reductive by definition, so Exp. XXII, 5.8.3 applies. By (iv), b has the same properties.

The morphism d is likewise smooth and affine with integral geometric fibers, by 1.9; the same is therefore true of

$$c = dbe^{-1}.$$

The morphism r has these properties by Exp. XXII, 5.8.3, and $c\ell$ by 3.10.

We already know that p and q are finite étale and surjective (3.1 and 3.9). If f and k are finite étale and surjective, then so is g, by (iv), and so is $h = kge^{-1}$. Since the assertions about t were proved in 3.3(ii), it remains only to prove the claim for f and k. We do so for k, the proof for f being the same.

If T is a maximal torus of G, let $\mathcal{C}$ be the functor of critical subgroups containing T. We must show that $\mathcal{C}$ is represented by a finite étale S-scheme with nonempty fibers. We may suppose that G is split relative to T. If E is the set of vectorial subsets of the root system R, then $\mathcal{C}$ is represented by E_S, by 3.9. □

Corollary 3.17.— *All the functors in the preceding diagram are representable by S-schemes that are smooth over S, and all are affine over S except* $\underline{\mathrm{Par}}(G)$.

Remark 3.18.—

(i) *The fact that $f : PL \to \underline{\mathrm{Crit}}(G)$ is étale and surjective says that a subgroup of G is critical if and only if, locally for the étale topology, it is a Levi subgroup of a parabolic subgroup.*

In general the map $PL(S) \to \underline{\mathrm{Crit}}(G)(S)$ need not be surjective: a critical subgroup C may fail to arise over S from a parabolic subgroup. For example, a maximal torus

need not be contained in a Borel subgroup, as for a nonsplit form of SL_2 (Exp. XX, §5).

(ii) *Likewise,*

$$u : \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

need not be an isomorphism. Two parabolic subgroups of different types may have Levi subgroups of the same type. In a group of type A_2*, for instance, the two types of parabolics whose Levi subgroups have semisimple rank* 1 *correspond to the two vertices of the diagram, whereas there is only one type of critical subgroup of rank* 1.

The analogous example for type A_3 *shows that, even over an algebraically closed field, nonisomorphic parabolic subgroups may have Levi subgroups of the same type.*[14]

We end this section with an application to principal bundles.

Lemma 3.20.— *Let*[15] S *be a scheme,* G *an* S*-reductive group,* F *a principal homogeneous bundle under* G*, and* G_F *the corresponding twisted form. Identify* $\underline{\mathrm{Dyn}}(G)$ *with* $\underline{\mathrm{Dyn}}(G_F)$ (Exp. XXIV, 3.5)*. For every parabolic subgroup* P *of* G*, there is a canonical isomorphism*

$$F/P \xrightarrow{\sim} \underline{\mathrm{Par}}_{t(P)}(G_F).$$

In particular, the structural group of F *reduces to* P *if and only if* G_F *has a parabolic subgroup of type* $t(P)$*.*

The proof is exactly the same as that of Exp. XXIV, 4.2.1.

3.21. If S is a scheme, G an S-reductive group, and

$$t \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S),$$

write $H^1_t(S,G)$ for the subset of $H^1(S,G)$ consisting of the classes of principal G-bundles F for which the associated group G_F has a parabolic subgroup of type t. If G itself has a parabolic subgroup P of type t, then $H^1_t(S,G)$ is precisely the image of $H^1(S,P)$ in $H^1(S,G)$, and hence does not depend on the chosen P.

[14] *Editor's note.* Concretely, in GL_4, the parabolics

$$P = \left\{ \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \qquad P' = \left\{ \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

are not isomorphic (their unipotent radicals are not isomorphic), while their Levi subgroups

$$L = \left\{ \begin{pmatrix} * & * & 0 & 0 \\ * & * & 0 & 0 \\ 0 & 0 & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \quad L' = \left\{ \begin{pmatrix} * & 0 & 0 & 0 \\ 0 & * & * & 0 \\ 0 & * & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

are conjugate by $\left(\begin{smallmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$.

[15] *Editor's note.* The numbering of the original is retained: there is no §3.19.

4. Relative position of two parabolic subgroups

4.1. A preliminary result

Lemma 4.1.1.— *Let k be a field, G a reductive k-group, and P, P' two parabolic subgroups of G.*

(i) *The intersection $P \cap P'$ is smooth, has the same reductive rank as G, and contains a maximal torus T of G.*

(ii) *The set R' of roots of $P \cap P'$ relative to T is a closed subset of the set R of roots of G relative to T.*

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Proof. Suppose first that k is algebraically closed. Let B (respectively B') be a Borel subgroup of P (respectively P'). There is an element $g \in G(k)$ such that $\mathrm{int}(g)B = B'$. On the other hand, if T_0 is a maximal torus of B and $N = \underline{\mathrm{Norm}}_G(T_0)$, Bruhat's theorem (*Bible*, §13.4, Corollary 1 to Theorem 3) gives

$$G(k) = B(k)N(k)B(k).$$

Thus there are $b, b_1 \in B(k)$ and $n \in N(k)$ such that $g = bnb_1$, and hence $\mathrm{int}(b)\,\mathrm{int}(n)B = B'$. Consequently

$$P \cap P' \supset B \cap B' = \mathrm{int}(b)\big(B \cap \mathrm{int}(n)B\big) \supset \mathrm{int}(b)T_0.$$

Now let k be arbitrary. The preceding result shows that $P_{\bar{k}} \cap P'_{\bar{k}}$ contains a maximal torus of $G_{\bar{k}}$. By Exp. XXII, 5.4.5, $(P \cap P')_{\bar{k}}$ is smooth along the identity section; since the base is a field, it is therefore smooth (Exp. $\mathrm{VI_A}$, 1.3.1), and so $P \cap P'$ is smooth. By Exp. $\mathrm{VI_A}$, 2.3.1, the identity component $(P \cap P')^0$ is an open subgroup of $P \cap P'$, smooth over k. Applying Exp. XIV, 1.1 to it proves (i).

Finally, the set R_P (respectively $R_{P'}$) of roots of P (respectively P') relative to T is closed, and

$$R' = R_P \cap R_{P'}.$$

This proves (ii). □

Remark 4.1.2.— *One can prove* ([BT65], 4.5) *that $P \cap P'$ is connected; this fact will be used in 4.5.1.*

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Remark 4.1.3.— *The preceding lemma is false over an arbitrary scheme. For example, let G be a reductive group over an algebraically closed field k, and let B be a Borel subgroup of G. Take $S = G$ as base and consider the Borel subgroups B_1, B_2 of G_G, where*

$$B_1 = B_G, \qquad B_2 = \mathrm{int}(g_0)B_1,$$

and g_0 is the canonical (diagonal) section of G_G. For every $g \in G(k)$, the fiber of $B_1 \cap B_2$ at g is $B \cap \mathrm{int}(g)B$. If $B \neq G$, the dimension of this fiber varies with g, so $B_1 \cap B_2$ cannot be smooth over G.

[16] *Editor's note.* Part (ii), which will be useful in 4.5.1, was added by the editors.

[17] *Editor's note.* In view of the details added in 4.5.1, the original wording "we shall not use this fact" was changed.

4.2. Transversal position

Theorem 4.2.1.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G. The following conditions on the pair (P, Q) are equivalent:*

(i) $\mathrm{Lie}(P/S) + \mathrm{Lie}(Q/S) = \mathrm{Lie}(G/S)$.

(ii) *The canonical morphism $P \times_S Q \to G$ is smooth.*

(ii′) *The canonical morphism $P \to G/Q$ is smooth.*

(iii) *The canonical morphism $P \times_S Q \to G$ is open.*

(iii′) *The canonical morphism $P \to G/Q$ is open.*

(iv) *The canonical morphism $P \times_S Q \to G$ is fiberwise dominant.*

(iv′) *The canonical morphism $P \to G/Q$ is fiberwise dominant.*

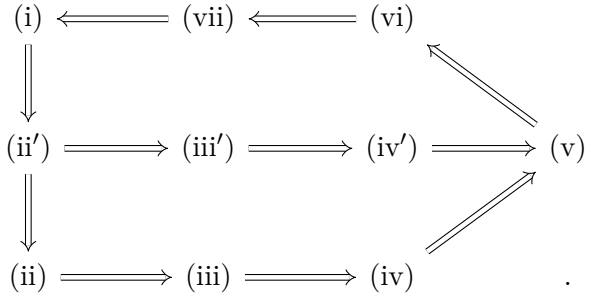
(v) *For every $s \in S$, the group $P_s \cap Q_s$ has minimum dimension, that is,*

$$\dim(P_s \cap Q_s) = \dim P_s + \dim Q_s - \dim G_s.$$

(vi) *There is an étale covering family $\{S_i \to S\}$ and, for every i, a Borel subgroup B_i of P_{S_i} and a Borel subgroup B_i' of Q_{S_i}, such that $B_i \cap B_i'$ is a maximal torus of G_{S_i}.*

(vii) *There is an étale covering family $\{S_i \to S\}$ and, for every i, a splitting $(T_i, \dots, R_i)$ of G_{S_i} and a system R_i^+ of positive roots of R_i, such that P_{S_i} (respectively Q_{S_i}) is the subgroup of type* (R) *of G_{S_i} containing T_i and defined by a subset $R_i^{(1)}$ (respectively $R_i^{(2)}$) of R_i containing R_i^+ (respectively $-R_i^+$); see Exp. XXII, 5.4.2 and 5.2.1.*

Proof. We prove the theorem according to the following logical diagram:



The implications (ii) ⇒ (iii) and (ii′) ⇒ (iii′) are immediate. If (iii) holds, the set-theoretic image of $P \times_S Q \to G$ is an open subset of G containing the identity section. Since the fibers of G are connected, this image is dense in every fiber; thus (iv) holds. The implication (iii′) ⇒ (iv′) is identical.

Condition (ii′) implies (ii), by the cartesian diagram

$$\begin{array}{ccc} P \times_S Q & \longrightarrow & G \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow \\ P & \longrightarrow & G/Q. \end{array}$$

On the other hand, (iv) or (iv′) implies (v) by dimension theory (EGA IV_2, 5.6.6). Indeed, one may assume $S = \mathrm{Spec}(k)$, with k algebraically closed; every nonempty fiber of the morphism in (iv) at a point of $G(k)$, or of the morphism in (iv′) at a point of $(G/Q)(k)$, is isomorphic to $P \cap Q$.

The implication (vi) ⇒ (vii) follows from Exp. XXII, 5.5.1 (iv) and 5.9.2. Condition (vii) implies (i), because the latter may be checked fpqc-locally. Under (vii) we may therefore suppose that G is split, $P \supset B_{R^+}$, and $Q \supset B_{-R^+}$; then already

$$\mathrm{Lie}(B_{R^+}) + \mathrm{Lie}(B_{-R^+}) = \mathrm{Lie}(G).$$

Let us prove that (i) implies (ii′). Let $u : P \to G/Q$ be the canonical morphism. Smoothness can be checked on geometric fibers, since both P and G/Q are smooth over S; hence we may suppose that S is the spectrum of an algebraically closed field. The morphism u is compatible with the evident action of P: $u(pp') = p\,u(p')$. A translation argument (compare the proof of Exp. $\mathrm{VI_B}$, 1.3) therefore reduces smoothness to the point $e \in P(k)$. By SGA 1, II, 4.7, it is enough that the tangent map at e be surjective. This map is naturally identified with

$$\mathrm{Lie}(P) \longrightarrow \mathrm{Lie}(G)/\mathrm{Lie}(Q),$$

which is surjective under (i).

It remains to prove (v) ⇒ (vi). Suppose first that S is the spectrum of an algebraically closed field. By 4.1.1 there is a maximal torus T contained in P and Q. Let R, R_1, and R_2 be the respective sets of roots of G, P, and Q relative to T. By Exp. XXII, 5.4.4 and 5.4.5,

$$\dim G = \dim T + \mathrm{Card}(R), \qquad \dim P = \dim T + \mathrm{Card}(R_1),$$
$$\dim Q = \dim T + \mathrm{Card}(R_2), \quad \dim(P \cap Q) = \dim T + \mathrm{Card}(R_1 \cap R_2).$$

Condition (v) is therefore equivalent to

$$\mathrm{Card}(R_1 \cap R_2) = \mathrm{Card}(R_1) + \mathrm{Card}(R_2) - \mathrm{Card}(R),$$

that is, to $R_1 \cup R_2 = R$. By Exp. XXII, 5.9.2 and 5.4.5, it is enough to prove that $R_1 \cap (-R_2)$ contains a system of positive roots of R. We are thus reduced to the following lemma.

Lemma 4.2.2.— *Let R be a "root system"—for example, the set of roots of a root datum in the sense of Exposé XXI. Let R_1, R_2 be two closed subsets of R, each containing a system of positive roots. If $R_1 \cup R_2 = R$, then $R_1 \cap (-R_2)$ contains a system of positive roots.*

Indeed, $R_3 = R_1 \cap (-R_2)$ is closed, so by Exp. XXI, 3.3.6 it is enough to show that $R_3 \cup (-R_3) = R$. Now

$$R_1 \cup (-R_1) = R = R_2 \cup (-R_2),$$

and the conclusion follows from the following elementary fact. If A, A', B, B' are four subsets of a set E, and

$$A \cup A' = B \cup B' = A \cup B = A' \cup B' = E,$$

then

$$(A \cap B') \cup (A' \cap B) = E.$$

This proves (v) ⇒ (vi) over the spectrum of an algebraically closed field. Return now to the general case and suppose (v) holds. Let $s \in S$. By the preceding argument, there

is a Borel subgroup $\bar{B}$ of $P_{\bar{s}}$ and a Borel subgroup $\bar{B}'$ of $Q_{\bar{s}}$ such that $\bar{B} \cap \bar{B}'$ is a maximal torus of $G_{\bar{s}}$.

The S-scheme

$$\underline{\mathrm{Bor}}(P) \simeq \underline{\mathrm{Bor}}(P/\operatorname{rad}^u(P))$$

of Borel subgroups of P is smooth. Applying "Hensel's lemma" (Exp. XI, 1.10) and working étale-locally around s, we may suppose that there is a Borel subgroup B of P specializing to $\bar{B}$. Similarly, there is a Borel subgroup B' of Q specializing to $\bar{B}'$. Since $\bar{B} \cap \bar{B}'$ is a maximal torus of $G_{\bar{s}}$, some open neighborhood U of s has the property that $B_U \cap B'_U$ is a maximal torus of G_U (Exp. XXII, 5.9.4). This proves (vi). □

Definition 4.2.3.— *A pair (P, Q) satisfying the equivalent conditions* (i)–(vii) *of Theorem 4.2.1 is said to be in* transversal position. *One also says that P is in transversal position relative to Q, or, by an abuse of language, that P and Q are in (mutual) transversal position.*

In view of (vi), this definition agrees, for Borel subgroups, with that of Exp. XXII, 5.9.1.

Corollary 4.2.4.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G.*

(i) *The pair (P, Q) is in transversal position if and only if, for every $s \in S$, the pair $(P_{\bar{s}}, Q_{\bar{s}})$ is in transversal position. If $S' \to S$ is surjective and $(P_{S'}, Q_{S'})$ is in transversal position, then so is (P, Q).*

(ii) *There is an open subscheme $U \subset S$ with the following property: a morphism $S' \to S$ factors through U if and only if $(P_{S'}, Q_{S'})$ is in transversal position.*

(iii) *Consider the subfunctors*

$$\underline{\mathrm{Gen}}(G) \subset \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G), \qquad \underline{\mathrm{Gen}}(/Q) \subset \underline{\mathrm{Par}}(G), \qquad \underline{\mathrm{Gen}}(P/Q) \subset G,$$

defined as follows. For $S' \to S$, the set $\underline{\mathrm{Gen}}(G)(S')$ consists of pairs of parabolic subgroups of $G_{S'}$ in transversal position; $\underline{\mathrm{Gen}}(/Q)(S')$ consists of parabolic subgroups of $G_{S'}$ in transversal position relative to $Q_{S'}$; and $\underline{\mathrm{Gen}}(P/Q)(S')$ consists of the $g \in G(S')$ such that $\mathrm{int}(g)P_{S'}$ is in transversal position relative to $Q_{S'}$.

Each of these functors is represented by an open subscheme, universally schematically dense over S, of the corresponding S-scheme

$$\underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G), \qquad \underline{\mathrm{Par}}(G), \qquad G,$$

respectively.

[18]

Assertion (i) follows immediately from condition 4.2.1(i). For (ii), take

$$U = S - \mathrm{Supp}(\mathrm{Coker}\, u),$$

where

$$u : \mathrm{Lie}(P) \oplus \mathrm{Lie}(Q) \longrightarrow \mathrm{Lie}(G)$$

[18] *Editor's note.* The original expression "relatively dense" was replaced by "universally schematically dense over S"; compare EGA IV$_3$, Definition 11.10.8.

is the canonical morphism.[19]

There are cartesian diagrams

$$\begin{array}{ccc} G & \xrightarrow{f} & \underline{\mathrm{Par}}(G) \\ \uparrow & & \uparrow \\ \underline{\mathrm{Gen}}(P/Q) & \longrightarrow & \underline{\mathrm{Gen}}(/Q) \end{array} \qquad \begin{array}{ccc} \underline{\mathrm{Par}}(G) & \xrightarrow{f'} & \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G) \\ \uparrow & & \uparrow \\ \underline{\mathrm{Gen}}(/Q) & \longrightarrow & \underline{\mathrm{Gen}}(G), \end{array}$$

where $f(g) = \mathrm{int}(g)P$ and $f'(R) = (R, Q)$. It therefore suffices to prove (iii) for $\underline{\mathrm{Gen}}(G)$.

Let P_0 be the canonical parabolic subgroup of $G_{\underline{\mathrm{Par}}(G)}$, and set

$$X = \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G), \qquad P = \mathrm{pr}_1^*(P_0), \qquad Q = \mathrm{pr}_2^*(P_0).$$

Applying (ii) to the parabolic subgroups P, Q of G_X constructs an open subscheme $U \subset X$, which is immediately identified with $\underline{\mathrm{Gen}}(G)$. The density assertion may be checked on geometric fibers,[20] so assume $S = \mathrm{Spec}(k)$, where k is algebraically closed. Since $\underline{\mathrm{Par}}(G)$ is smooth, it is enough to show that $\underline{\mathrm{Gen}}(G)$ meets every irreducible component of $\underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G)$. By 3.3, this amounts to the following: given $t, t' \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S)$, find a pair (P, P') in transversal position such that $t(P) = t$ and $t(P') = t'$.

Choose Borel subgroups B, B' of G such that $B \cap B'$ is a maximal torus (split G and apply Exp. XXII, 5.9.2). By 3.8 there are parabolic subgroups $P \supset B$ and $P' \supset B'$ with $t(P) = t$ and $t(P') = t'$. They have the required types and are in transversal position by 4.2.1(vi). □

Corollary 4.2.5.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G, with (P, Q) in transversal position.*

(i) *Let P', Q' be parabolic subgroups of G of the same types as P, Q, respectively. The pair (P', Q') is in transversal position if and only if, étale-locally, it is conjugate to (P, Q).* (In §5 we shall see that the étale topology can be replaced by the Zariski topology.)

(ii) *The canonical morphism $P \times_S Q \to G$ induces a smooth surjective morphism*

$$P \times_S Q \longrightarrow \underline{\mathrm{Gen}}(Q/P)$$

and an isomorphism

$$(P \times_S Q)/(P \cap Q) \xrightarrow{\sim} \underline{\mathrm{Gen}}(Q/P),$$

where $R = P \cap Q$ acts on $P \times_S Q$ by

$$(p, q)r = (pr, r^{-1}q).$$

(iii) *The canonical morphism*

$$P \longrightarrow \underline{\mathrm{Par}}_{t(Q)}(G), \qquad p \longmapsto \mathrm{int}(p)Q,$$

induces a smooth surjective morphism

$$P \longrightarrow \underline{\mathrm{Gen}}(/P) \cap \underline{\mathrm{Par}}_{t(Q)}(G)$$

and an isomorphism

$$P/(P \cap Q) \xrightarrow{\sim} \underline{\mathrm{Gen}}(/P) \cap \underline{\mathrm{Par}}_{t(Q)}(G).$$

[19] *Editor's note.* The second summand, printed as $\mathrm{Lie}(G)$, has been corrected to $\mathrm{Lie}(Q)$; this is the map used in Theorem 4.2.1(i).

[20] *Editor's note.* Compare EGA IV_3, 11.10.10.

(iv) *The canonical morphism*

$$G \longrightarrow \underline{\mathrm{Par}}_{t(P)}(G) \times_S \underline{\mathrm{Par}}_{t(Q)}(G), \qquad g \longmapsto (\mathrm{int}(g)P, \mathrm{int}(g)Q),$$

induces a smooth surjective morphism

$$G \longrightarrow \underline{\mathrm{Gen}}(G) \cap (\underline{\mathrm{Par}}_{t(P)}(G) \times_S \underline{\mathrm{Par}}_{t(Q)}(G))$$

and an isomorphism

$$G/(P \cap Q) \xrightarrow{\sim} \underline{\mathrm{Gen}}(G) \cap (\underline{\mathrm{Par}}_{t(P)}(G) \times_S \underline{\mathrm{Par}}_{t(Q)}(G)).$$

Proof. Let us prove (i). The stated condition is plainly sufficient. Conversely, suppose (P', Q') is in transversal position. Since P and P' are conjugate étale-locally, we may assume $P = P'$. It is then enough to show that two parabolic subgroups Q, Q', of the same type and both in transversal position relative to P, are étale-locally conjugate by a section of P.

By 4.2.1(vi), after another étale localization there are Borel subgroups $B, B' \subset P$, $B_1 \subset Q$, and $B'_1 \subset Q'$ such that

$$B \cap B_1 = T, \qquad B' \cap B'_1 = T'$$

are maximal tori of G. The Killing pairs (B, T) and (B', T') of P are étale-locally conjugate in P (1.16). We may therefore assume $B = B'$ and $T = T'$. Then $B_1 = B'_1$ by Exp. XXII, 5.9.2, and hence $Q = Q'$ by 3.8.

Parts (ii), (iii), and (iv) are proved in parallel. We prove (ii). Let $g \in \underline{\mathrm{Gen}}(Q/P)(S)$; thus $g \in G(S)$ and $\mathrm{int}(g)Q$ is in transversal position relative to P. The preceding argument shows that Q and $\mathrm{int}(g)Q$ are, étale-locally, conjugate by a section of P. Working locally for this topology, choose $p \in P(S)$ such that $\mathrm{int}(g)Q = \mathrm{int}(p)Q$. Then

$$p^{-1}g \in \underline{\mathrm{Norm}}_G(Q)(S) = Q(S),$$

so $g = pq$ for some $q \in Q(S)$. Thus the morphism in (ii) is an étale covering. Comparing with 4.2.1(ii), it is smooth and surjective.

Finally, the equivalence relation on $P \times_S Q$ defined by $P \times_S Q \to G$ is exactly the relation associated with the action of $R = P \cap Q$,

$$(p, q)r = (pr, r^{-1}q).$$

This proves the quotient assertion, since a smooth surjective morphism is, in particular, an effective epimorphism. □

Remark 4.2.6.— *If P, Q are in transversal position, the open subset $\underline{\mathrm{Gen}}(Q/P)$ of G will often be denoted $P \cdot Q$; this notation is justified by 4.2.5*(ii).

Proposition 4.2.7.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G, with (P, Q) in transversal position.*

(i) *The group $P \cap Q$ is smooth over S (and in fact has connected fibers by 4.1.2). Its identity component $(P \cap Q)^0$* (Exp. $\mathrm{VI_B}$, 3.10) *is a subgroup of type* (RC) *of G* (Exp. XXII, 5.11.1)*, and its unipotent radical* (loc. cit., 5.11.4) *decomposes as the direct product*

$$\mathrm{rad}^u((P \cap Q)^0) = (\mathrm{rad}^u(P) \cap Q) \times_S (P \cap \mathrm{rad}^u(Q)).$$

(ii) *If S is affine, then*

$$H^1(S, \mathrm{rad}^u((P \cap Q)^0)) = 0.$$

If S is semilocal, $P \cap Q$ contains a maximal torus of G.

Proof. The group $P \cap Q$ is smooth by 4.2.1(ii) and the cartesian diagram

$$\begin{array}{ccc} P \times_S Q & \longrightarrow & G \\ \uparrow & & \uparrow \\ P \cap Q & \longrightarrow & S \end{array}.$$

The remaining assertions about $(P \cap Q)^0$ may be checked étale-locally. By 4.2.1(vii), choose a splitting (G, T, M, R) of G such that P, Q contain T and are defined by subsets R_1, R_2 of R, where R_1 contains a system R^+ of positive roots and R_2 contains the opposite system R^-. Let Δ be the set of simple roots of R^+, and put

$$A_1 = \Delta \cap (-R_1), \qquad A_2 = \Delta \cap R_2, \qquad A = A_1 \cap A_2.$$

By 1.4(v) and 1.12,

$$R_1 = R^+ \cup (R^- \cap (-\mathbb{N}A_1)), \qquad R_2 = (R^+ \cap \mathbb{N}A_2) \cup R^-,$$
$$\mathrm{rad}^u(P) = \prod_{\substack{\alpha \in R_1 \\ \alpha \notin -R_1}} U_\alpha, \qquad \mathrm{rad}^u(Q) = \prod_{\substack{\alpha \in R_2 \\ \alpha \notin -R_2}} U_\alpha.$$

Exp. XXII, 5.6.7 therefore gives

$$\mathrm{rad}^u(P) \cap Q = \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_1}} U_\alpha = \prod_{\alpha \in K_2} U_\alpha,$$
$$\mathrm{rad}^u(Q) \cap P = \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_2}} U_\alpha = \prod_{\alpha \in K_1} U_\alpha,$$

where K_2 consists of the positive roots that are linear combinations of elements of A_2, but not of elements of A, and K_1 consists of the negative roots that are linear combinations of elements of A_1, but not of elements of A. If $\alpha \in K_2$ and $\beta \in K_1$, then $\alpha + \beta$ is neither a root nor zero. Hence the two groups displayed above commute.

By Exp. XXII, 5.4.5, the group

$$H = (P \cap Q)^0$$

is defined by the set of roots

$$R_1 \cap R_2 = (R^+ \cap \mathbb{N}A_2) \cup (R^- \cap (-\mathbb{N}A_1)).$$

This set is closed, so H is of type (RC) by definition (Exp. XXII, 5.11.1). By loc. cit., 5.11.3 and 5.11.4,

$$\mathrm{rad}^u(H) = \prod_{\alpha \in K} U_\alpha,$$

where K consists of the $\alpha \in R_1 \cap R_2$ such that $\alpha \notin -(R_1 \cap R_2)$. The symmetric part of $R_1 \cap R_2$ is $R \cap \mathbb{Z}A$, and therefore $K = K_1 \cup K_2$. This proves (i).

The first assertion of (ii) follows from (i) and 2.10. For the second, the reductive group

$$(P \cap Q)^0 / \operatorname{rad}^u((P \cap Q)^0)$$

has a maximal torus $\bar{T}$ when the base is semilocal. The inverse image N of $\bar{T}$ in $(P \cap Q)^0$ is a subgroup of type (R) of G, with solvable fibers, and

$$N^u = \operatorname{rad}^u((P \cap Q)^0)$$

(Exp. XXII, 5.6.9). The scheme of maximal tori of N is a principal homogeneous bundle under N^u (loc. cit., 5.6.13), and therefore has a section because $H^1(S, N^u) = 0$. □

4.3. Opposite parabolic subgroups

4.3.1. Let G be an S-reductive group. Exp. XXIV, 3.16.6 defines a canonical "outer automorphism" s_G of G, of order at most 2.[21] It induces a canonical automorphism, again of order at most 2, of $\underline{\mathrm{Dyn}}(G)$, and hence of $\mathrm{Of}(\underline{\mathrm{Dyn}}(G))$. We denote all of these automorphisms by s_G, or simply by s. Two types of parabolic subgroups

$$t, t' \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S)$$

are said to be *opposite* if $t = s_G(t')$.

Theorem 4.3.2.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G.*

(a) *If L is a Levi subgroup of P, there is a unique parabolic subgroup P' of G such that $P \cap P' = L$.*

(b) *For every parabolic subgroup Q of G, the following conditions are equivalent:*

(i) *For every $s \in S$, the identity component $\big((P \cap Q)_{\bar{s}}\big)^0$, which is smooth by 4.1.1, is reductive.*

(ii) *The group $P \cap Q$ is a Levi subgroup of both P and Q.*

(iii) *The groups P, Q have opposite types and the pair (P, Q) is in transversal position.*

(iv) *The groups P, Q have opposite types and* $\operatorname{rad}^u(P) \cap Q = e$.

(v)
$$\operatorname{rad}^u(P) \cap Q = \operatorname{rad}^u(Q) \cap P = e.$$

(vi) *The canonical morphism* $\operatorname{rad}^u(P) \times_S Q \to G$ *is an open immersion.*

(vi′) *The canonical morphism* $\operatorname{rad}^u(P) \to G/Q$ *is an open immersion.*

(vii) *There is an étale covering family $\{S_i \to S\}$ and, for every i, a splitting (T_i, M_i, R_i) of G_{S_i} and a subset $R_i^{(1)} \subset R_i$, such that P_{S_i} is the subgroup of type* (R) *containing T_i and defined by $R_i^{(1)}$, while Q_{S_i} is the subgroup defined by*

$$R_i^{(2)} = -R_i^{(1)}.$$

[21] *Editor's note.* "Of order 2" was replaced by "of order at most 2", since s_G may be trivial, for example when G has type $A_1, B_n, C_n, \ldots$.

Proof. We first prove part (b). Conditions (iii) and (vii) are equivalent by 4.2.1(vii) and the definition of s_G in the split case (Exp. XXIV, 3.16.2(iv)). Plainly (ii) implies (i), and (vi′) implies (vi) by base change along $G \to G/Q$.

Suppose now that (vii) holds. Since every assertion to be proved is étale-local, assume that $G = (G, T, M, R)$ is split, that P is defined by $R' \subset R$, and that Q is defined by $-R'$. Let L be the subgroup of type (R) containing T and defined by $R' \cap (-R')$. By Exp. XXII, 5.11.3, L is a Levi subgroup of both P and Q. Moreover,

$$P = L \cdot \mathrm{rad}^u(P), \qquad Q = L \cdot \mathrm{rad}^u(Q),$$

and Exp. XXII, 5.6.7 gives

$$\mathrm{rad}^u(P) \cap Q = Q \cap \mathrm{rad}^u(P) = e.$$

Thus $P \cap Q = L$, proving (ii) and (v).

Since P, Q are in transversal position, the canonical morphism $P \to G/Q$ induces an open immersion

$$P/(P \cap Q) \longrightarrow G/Q$$

by 4.2.1. The canonical morphism

$$\mathrm{rad}^u(P) \longrightarrow P/(P \cap Q) = P/L$$

is an isomorphism, so (vi′) follows. Hence all the conditions follow from (vii).

It remains to show that each of (i), (iv), (v), and (vi) implies (vii). Since the equivalence of (ii) and (iii) is already known, this may be checked on geometric fibers. We may suppose that S is the spectrum of an algebraically closed field. By 4.1.1, a maximal torus T of G is contained in $P \cap Q$. Let R, R_1, R_2 be the sets of roots of G, P, Q, respectively, relative to T. Write

$$R_1^a = \{\alpha \in R_1 \mid -\alpha \notin R_1\},$$

and define R_2^a similarly. We must prove $R_1 = -R_2$.

- Condition (i) says that $R_1 \cap R_2$ is symmetric. If $\alpha \in R_1$ and $\alpha \notin R_2$, then $\alpha \in -R_2$; if $\alpha \in R_2$, then
$$\alpha \in R_1 \cap R_2 = -(R_1 \cap R_2) \subset -R_2.$$
Thus $R_1 \subset -R_2$, and symmetry gives $R_1 = -R_2$.

- Condition (iv) gives $\mathrm{Card}(R_1) = \mathrm{Card}(R_2)$ and $R_1^a \cap R_2 = \varnothing$. The latter condition is equivalent to $R_2 \subset -R_1$, and equality of cardinalities then gives $R_2 = -R_1$.

- Condition (v) gives
$$R_1^a \cap R_2 = R_2^a \cap R_1 = \varnothing.$$
Hence $R_2 \subset -R_1$ and $R_1 \subset -R_2$, so again $R_2 = -R_1$.

- Condition (vi) gives
$$\mathrm{Lie}(\mathrm{rad}^u(P)) \oplus \mathrm{Lie}(Q) = \mathrm{Lie}(G).$$
Thus R is the disjoint union of R_2 and R_1^a, which implies $R_2 = -R_1$.

This proves part (b). For part (a), the implication (vii) ⇒ (ii) already gives the desired parabolic subgroup P' étale-locally. It remains only to prove uniqueness, and this too is étale-local. We may suppose that G is split relative to a maximal torus T of L, with P and P' defined by subsets R_1 and R'_1 of R. By hypothesis $R_1 \cap R'_1$ is symmetric. The preceding argument gives $R'_1 = -R_1$; hence P' is determined by P and L. □

Definition 4.3.3.— *Two parabolic subgroups of G satisfying the equivalent conditions* (i)–(vii) *of Theorem 4.3.2 are said to be* opposite. *If P is a parabolic subgroup of G and L is a Levi subgroup of P, the* parabolic subgroup opposite to P relative to L *is the unique parabolic subgroup Q of G such that $P \cap Q = L$.*

If T is a maximal torus of P, the parabolic subgroup opposite to P relative to T *is the unique Q such that $P \cap Q$ is the unique Levi subgroup of P containing T* (1.6)*; equivalently, it is the unique Q such that $P \cap Q$ contains T and P, Q are opposite.*

By 4.3.2(iii), Corollaries 4.2.4 and 4.2.5 immediately yield parallel results. We record one sample.

Corollary 4.3.4.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G.*

(i) *The groups P, Q are opposite if and only if $P_{\bar{s}}, Q_{\bar{s}}$ are opposite for every $s \in S$. If $S' \to S$ is surjective and $P_{S'}, Q_{S'}$ are opposite, then P, Q are opposite.*

(ii) *Let $\underline{\mathrm{Opp}}(G)(S')$ be the set of pairs of opposite parabolic subgroups of $G_{S'}$, for $S' \to S$. This functor is represented by an open subscheme of $\underline{\mathrm{Par}}(G)^2$. Let $\underline{\mathrm{Opp}}(/P)(S')$ be the set of parabolic subgroups of $G_{S'}$ opposite to $P_{S'}$. This functor is represented by an open subscheme, universally schematically dense over S, of*

$$\underline{\mathrm{Par}}_{s(t(P))}(G).$$

(iii) *Suppose P, Q are opposite, and let P', Q' be parabolic subgroups of G, with P' of the same type as P. The groups P', Q' are opposite if and only if, étale-locally, the pair (P', Q') is conjugate to (P, Q).* (In §5 the étale topology will be replaced by the Zariski topology.)

[22]

Corollary 4.3.5.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G.*

(i) *The morphism*

$$\underline{\mathrm{Opp}}(/P) \longrightarrow \underline{\mathrm{Lev}}(P), \qquad Q \longmapsto P \cap Q,$$

is an isomorphism. The scheme $\underline{\mathrm{Opp}}(/P)$ is a principal homogeneous bundle under $\mathrm{rad}^u(P)$, acting by inner automorphisms. If S is affine, G has a parabolic subgroup opposite to P.

(ii) *Suppose S is semilocal, with closed points $\{s_i\}$. For each i, let Q_i be a parabolic subgroup of G_{s_i} opposite to P_{s_i}. There is a parabolic subgroup Q of G, opposite to P, such that $Q_{s_i} = Q_i$ for every i.*

[22] *Editor's note.* Here again, "relatively dense" was replaced by "universally schematically dense over S"; compare EGA IV$_3$, Definition 11.10.8.

(iii) *The morphism*

$$\underline{\mathrm{Opp}}(G) \longrightarrow \mathrm{PL}, \qquad (P,Q) \longmapsto (P, P \cap Q),$$

is an isomorphism.

These assertions follow from Theorem 4.3.2(a) and from 1.9, 2.3, and 2.8.

Remark 4.3.6.— *Let P, Q be opposite parabolic subgroups of G, and let $P \cdot Q$ be the open subscheme of G, introduced in 4.2.6, which is the sheaf-theoretic image of $P \times_S Q$. The product morphism $G \times_S G \to G$ induces isomorphisms*

$$\mathrm{rad}^u(P) \times_S Q \xrightarrow{\sim} P \cdot Q \xleftarrow{\sim} P \times_S \mathrm{rad}^u(Q).$$

Indeed, this follows from 4.3.2 (or 4.2.5(ii)) and from the fact that $P \cap Q$ is a Levi subgroup of both P and Q:

$$P = \mathrm{rad}^u(P) \cdot (P \cap Q), \qquad Q = \mathrm{rad}^u(Q) \cdot (P \cap Q).$$

Likewise, there is a commutative diagram

$$\begin{array}{ccc} \mathrm{rad}^u(P) & \hookrightarrow & G/Q \\ \downarrow \sim & & \downarrow \sim \\ \underline{\mathrm{Opp}}(/P) & \hookrightarrow & \underline{\mathrm{Par}}_{t(Q)}(G), \end{array}$$

in which the vertical arrows are induced by $g \mapsto \mathrm{int}(g)Q$.

4.4. Osculating position

Proposition 4.4.1.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G. The following conditions are equivalent:*

(i) *$P \cap Q$ is a parabolic subgroup of G.*

(ii) *Étale-locally, $P \cap Q$ contains a Borel subgroup of G.*

(iii) *Étale-locally, $P \cap Q$ contains a maximal torus of G; moreover, for every $S' \to S$ and every maximal torus T of $G_{S'}$ contained in both $P_{S'}$ and $Q_{S'}$, the parabolic subgroup opposite to $P_{S'}$ relative to T is in transverse position relative to $Q_{S'}$.*

(iv) *There is an étale covering family $\{S_i \to S\}$ and, for each i, a maximal torus T_i of G_{S_i}, contained in both P_{S_i} and Q_{S_i}, such that the parabolic subgroup opposite to P_{S_i} relative to T_i is in transverse position relative to Q_{S_i}.*

(v) *There is an étale covering family $\{S_i \to S\}$ and, for each i, a splitting (T_i, M_i, R_i) of G_{S_i} such that P_{S_i}, respectively Q_{S_i}, is the subgroup of type* (R) *of G_{S_i} containing T_i and defined by a set of roots $R_i^{(1)}$, respectively $R_i^{(2)}$, with $R_i^{(1)} \cap R_i^{(2)}$ containing a system of positive roots of R_i.*

Moreover, when these conditions hold,

$$t(P \cap Q) = t(P) \cap t(Q)$$

in the notation of 3.2.

The implications (v)⇒(ii) and (iii)⇒(iv) are immediate. Moreover, (ii)⇒(i) follows from 1.18. To prove (iv)⇒(v), one may suppose that G is split and that P, respectively Q, is defined by a set of roots R_1, respectively R_2. The parabolic subgroup opposite to P is then defined by $-R_1$, and the assertion reduces to Lemma 4.2.2. The implication (i)⇒(iii) is proved in the same way after splitting. Finally, the last assertion may be proved étale-locally: one may suppose that $P \cap Q$ contains a Borel subgroup B of G, and the assertion then reduces to 3.7.

Definition 4.4.2.— *Two parabolic subgroups of G satisfying the equivalent conditions* (i)–(v) *of 4.4.1 are said to be in* osculating position.

Corollary 4.4.3.— *Let P, Q be parabolic subgroups in osculating position, and let P', Q' be parabolic subgroups of G of the same types as P, Q, respectively. The subgroups P', Q' are in osculating position if and only if, étale-locally, the pair (P', Q') is conjugate to the pair (P, Q).*

It is enough to show that, if both P and P' are in osculating position relative to Q, then they are conjugate, étale-locally, by a section of Q. Now $P \cap Q$ and $P' \cap Q$ are parabolic subgroups of the same type contained in Q. By part (ii) of the lemma below, they are therefore conjugate étale-locally by a section of Q. We may consequently suppose $P \cap Q = P' \cap Q$; part (i) of that lemma then gives $P = P'$.

Lemma 4.4.4.— *Let P, P', Q be three parabolic subgroups of the S-reductive group G.*

(i) *The equality $P = P'$ holds if and only if P and P' are in osculating position and have the same type.*

(ii) *Suppose $P \subset Q$ and $P' \subset Q$. If $g \in G(S)$ is such that* $\mathrm{int}(g)P$ *and P' are in osculating position, then $g \in Q(S)$.*

Part (i) follows immediately from the last assertion of 4.4.1. To prove (ii), observe that Q and $\mathrm{int}(g)Q$ both contain $P' \cap \mathrm{int}(g)P$, and hence are in osculating position. They coincide by (i), so that

$$g \in \underline{\mathrm{Norm}}_G(Q)(S) = Q(S).$$

Assertions (iii) and (iv) of Proposition 4.4.1 give at once:

Corollary 4.4.5.— *Let P, P', Q be three parabolic subgroups of the S-reductive group G, containing the same maximal torus T of G. Suppose that P and P' are opposite relative to T. Then Q is in osculating position relative to P if and only if it is in transverse position relative to P'. Under these conditions, $P \cap Q$ is also in transverse position relative to P'.*

Corollary 4.4.6.— *Let P, Q be two parabolic subgroups of G containing the same maximal torus T. The subgroups P, Q are in transverse position if and only if there are parabolic subgroups $P' \subset P$ and $Q' \subset Q$ of G, opposite relative to T. One may even choose*

$$t(P') = t(P) \cap s(t(Q)).$$

[23]

The condition is plainly sufficient, by 4.2.1(i) and 4.3.2(iii). Conversely, let P^-, respectively Q^-, be the parabolic subgroup opposite to P, respectively Q, relative to T. By 4.4.5, $P^- \cap Q$ is in transverse position relative to both P and Q^-, and therefore also relative to $P \cap Q^-$, by another application of 4.4.5. Moreover,

$$\begin{aligned} t(P^- \cap Q) &= t(P^-) \cap t(Q) \\ &= s(t(P)) \cap s(t(Q^-)) \\ &= s\big(t(P) \cap t(Q^-)\big) \\ &= s\big(t(P \cap Q^-)\big). \end{aligned}$$

It follows that

$$P^- \cap Q = P', \qquad P \cap Q^- = Q'$$

are opposite, by 4.3.2(iii). Since $P' \cap Q' \supset T$, they are indeed opposite relative to T.

4.5. Standard position

In this subsection we briefly indicate how some of the preceding results generalize.

Proposition 4.5.1.— *Let P_1, P_2 be two parabolic subgroups of the S-reductive group G. The following conditions are equivalent:*

(i) *$P_1 \cap P_2$ is smooth.*

(ii) *$P_1 \cap P_2$ is a subgroup of type* (R)*, or of type* (RC)*, of G.*

(iii) *Fpqc-locally, $P_1 \cap P_2$ contains a maximal torus of G.*

(iv) *Zariski-locally, $P_1 \cap P_2$ contains a maximal torus of G.*

When S is semilocal, these conditions are also equivalent to:

(v) *$P_1 \cap P_2$ contains a maximal torus of G.*

Proof. Clearly, (iv) implies (iii), and (v) implies (iv) when S is semilocal; condition (ii) implies (i) by Exp. XXII, Definition 5.2.1. We shall prove

$$\text{(i)} \Rightarrow \text{(ii)} \Rightarrow \text{(iii)},$$

and then show that (iii) implies (i) and (iv), as well as (v) when S is semilocal. Put $K = P_1 \cap P_2$. By 4.1.1 and [BT65], 4.5, every geometric fiber $K_{\bar{s}}$ contains a maximal torus $T_{\bar{s}}$ of $G_{\bar{s}}$, is connected, and has, relative to $T_{\bar{s}}$, a closed subset of the roots of $G_{\bar{s}}$. Consequently, if K is smooth over S, then it is a subgroup of type (RC), hence a fortiori of type (R); see Exp. XXII, 5.2.1 and 5.11.1. Thus (i) and (ii) are equivalent.

If K is of type (R), it contains a maximal torus of G étale-locally, by Exp. XXII, 2.2 and Exp. XIX, 6.1. Hence (ii) implies (iii).

Assume (iii), and let us prove that K is smooth. By fpqc descent we may suppose that

$$G = (G, T, M, R)$$

[23] *Editor's note.* The involution s was defined in 4.3.1.

is split, where T is a maximal torus contained in $P_1 \cap P_2$, and that there are closed subsets R_1, R_2 of R such that

$$P_1 = H_{R_1}, \qquad P_2 = H_{R_2}.$$

Since K has connected fibers, Exp. XXII, 5.4.5 gives

$$K = H_{R_1 \cap R_2};$$

therefore K is smooth over S and of type (RC).

Let R_1^s be the symmetric part of R_1, and put $R_1^a = R_1 - R_1^s$; then

$$\mathrm{rad}^u(P_1) = H_{R_1^a}.$$

As observed in the proof of [BT65], 4.4,

$$R' = (R_1^s \cap R_2) \cup R_1^a$$

is a closed subset of R such that $R' \cup (-R') = R$. Hence, by Exp. XXI, 3.3.6, R' contains a system of positive roots of R. Moreover, the symmetric part of R' is $R_1^s \cap R_2^s$, which is contained in $R_1 \cap R_2$.

It follows that

$$P' = K \cdot \mathrm{rad}^u(P_1)$$

is a parabolic subgroup of G, and that K is a subgroup of type (RC) of P' satisfying

$$\mathrm{rad}^u(K) = K \cap \mathrm{rad}^u(P').$$

By 2.11, K therefore has a Levi subgroup L. Exp. XIV, 3.20 and 3.21 then show that K satisfies (iv), and also (v) when S is semilocal. This proves Proposition 4.5.1. □

[24]

Definition 4.5.1.1.— *When the preceding conditions hold, P_1 and P_2 are said to be in* mutual standard position*. This is the case, for example, when they are in transverse position, when they are in osculating position, or when the base is the spectrum of a field. The notion is stable under base extension and is fpqc-local.*

[25]

4.5.2. Let (P, Q) and (P', Q') be two pairs of parabolic subgroups of G in standard position. Define the subfunctor H of G by letting $H(S')$ be the set of $g \in G(S')$ such that

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)Q = Q'.$$

It is a closed subscheme of G, smooth over S and formally a principal homogeneous space under $P \cap Q$. It follows that the following conditions are equivalent:

(i) (P, Q) and (P', Q') are conjugate fpqc-locally;

(ii) (P, Q) and (P', Q') are conjugate étale-locally;

(iii) (P, Q) and (P', Q') are conjugate on every geometric fiber.

The pairs (P, Q) and (P', Q') are then said to have the same *type of mutual position*. This notion is stable under base change and is fpqc-local.

[24] *Editor's note.* The proof of the equivalence, together with condition (v), which was used implicitly in 6.17 of the original edition, was added in the Polo–Gille edition. The original 4.5.1 was consequently divided into Proposition 4.5.1 and Definition 4.5.1.1.

[25] *Editor's note.* See the preceding editorial note. For an example of parabolic subgroups that are not in standard position, see 7.11 below.

4.5.3. Let $\underline{\mathrm{Stand}}(G)$ be the subfunctor of $\underline{\mathrm{Par}}(G)\times_S\underline{\mathrm{Par}}(G)$ formed by the pairs in mutual standard position. The functor $\underline{\mathrm{Stand}}(G)$ is representable. There is a finite étale S-scheme $\underline{\mathrm{TypeStand}}$, the *scheme of types of mutual standard position*, and a smooth morphism of finite presentation with irreducible geometric fibers, hence in particular faithfully flat,

$$t_2 : \underline{\mathrm{Stand}}(G) \longrightarrow \underline{\mathrm{TypeStand}}.$$

This morphism is a quotient of $\underline{\mathrm{Stand}}(G)$ by the action of G: two sections of $\underline{\mathrm{Stand}}(G)$, over any $S' \to S$, have the same type of mutual position if and only if they have the same image under t_2. There is a commutative diagram

$$\begin{array}{ccc} \underline{\mathrm{Stand}}(G) & \xrightarrow{\quad t_2 \quad} & \underline{\mathrm{TypeStand}} \\ \downarrow & & \downarrow q \\ \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G) & \xrightarrow{t\times t} & \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \times_S \mathrm{Of}(\underline{\mathrm{Dyn}}(G)). \end{array}$$

The morphism q may be described by descent as follows. Let (P, Q) be a pair of parabolic subgroups of G in mutual standard position, and let T be a maximal torus of $P \cap Q$. The morphism

$$\underline{\mathrm{Norm}}_G(T) \longrightarrow \underline{\mathrm{Stand}}(G), \qquad n \longmapsto (P, \mathrm{int}(n)Q),$$

defined set-theoretically as displayed, induces an isomorphism

$$W_P(T)\backslash W_G(T)/W_Q(T) \xrightarrow{\sim} q^{-1}\big(t(P), t(Q)\big).$$

The left-hand side denotes the sheaf of double cosets. These assertions are immediate; note in particular that

$$t_2^{-1}\big(t_2(P, Q)\big) \simeq G/(P \cap Q).$$

4.5.4. Fix a parabolic subgroup P of G, and let $\underline{\mathrm{Par}}(G; P)$ be the functor of parabolic subgroups of G in standard position relative to P. For every

$$t \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S),$$

put

$$\underline{\mathrm{Par}}_t(G; P) = \underline{\mathrm{Par}}(G; P) \cap \underline{\mathrm{Par}}_t(G).$$

Both functors are obtained from $\underline{\mathrm{Stand}}(G)$ by fiber products, and are therefore represented by S-schemes, smooth and of finite presentation over S, with nonempty fibers. There is a canonical morphism induced by t_2,

$$t_P : \underline{\mathrm{Par}}_t(G; P) \longrightarrow q^{-1}\big(t(P), t\big), \qquad t_P(Q) = t_2(P, Q),$$

which is smooth and of finite presentation, with irreducible geometric fibers. The canonical morphism

$$\underline{\mathrm{Par}}_t(G; P) \longrightarrow \underline{\mathrm{Par}}_t(G)$$

is a surjective monomorphism, and may therefore be viewed as a *cell decomposition* of $\underline{\mathrm{Par}}_t(G)$, indexed by the set of connected components of $q^{-1}(t(P), t)$.

4.5.5. Suppose that the type t is of the form $t(Q)$, where Q is a parabolic subgroup of G in standard position relative to P, and suppose that $P \cap Q$ contains a maximal torus T. Then

$$\underline{\mathrm{Par}}_t(G) \simeq G/Q, \qquad q^{-1}(t(P), t) \simeq W_P(T)\backslash W_G(T)/W_Q(T),$$

and there is a diagram

$$\begin{array}{ccc} \underline{\mathrm{Par}}_{t(Q)}(G;P) & \xrightarrow{\quad f \quad} & W_P(T)\backslash W_G(T)/W_Q(T) \\ \Big\downarrow{\scriptstyle i} & & \\ \underline{\mathrm{Par}}_{t(Q)}(G) & \xrightarrow{\quad\sim\quad} & G/Q. \end{array}$$

Here i is a surjective monomorphism, and f is smooth and of finite presentation, with irreducible geometric fibers. Moreover, if Q_1, Q_2 are two sections of $\underline{\mathrm{Par}}_{t(Q)}(G;P)$, over some $S' \to S$, that is, two parabolic subgroups of $G_{S'}$ which are fpqc-locally conjugate to Q and are in standard position relative to $P_{S'}$, then Q_1, Q_2 are fpqc-locally conjugate by a section of P if and only if

$$f(Q_1) = f(Q_2).$$

If S is the spectrum of an algebraically closed field k, this gives

$$P(k)\backslash G(k)/Q(k) \simeq W_P(T)(k)\backslash W_G(T)(k)/W_Q(T)(k).$$

More generally, suppose that the scheme $W_P(T)\backslash W_G(T)/W_Q(T)$ is constant, of the form E_S, as happens for example when G is split relative to T and S is connected. The subschemes $f^{-1}(e)$, for $e \in E$, then decompose $\underline{\mathrm{Par}}_{t(Q)}(G;P)$ into open and closed subschemes. They are homogeneous spaces under P, smooth and of finite presentation over S, with irreducible geometric fibers.

4.5.6. Return to the general situation of 4.5.4. The scheme $q^{-1}(t(P), t)$ always has two distinguished sections, corresponding respectively to the types "transverse position" and "osculating position". The inverse image of the first section is a relatively dense open subscheme of $\underline{\mathrm{Par}}_t(G;P)$, as seen above; it is the cell of greatest relative dimension in the decomposition.

The inverse image of the second section is presumably a closed subscheme of $\underline{\mathrm{Par}}_t(G;P)$; it is the cell of least relative dimension in the decomposition. [26]

5. Conjugacy theorem

Theorem 5.1.— *Let S be a semilocal scheme, G an S-reductive group, and P, P' two opposite parabolic subgroups of G* (4.3.3)*. Then*

$$\mathrm{rad}^u(P)(S) \cdot P' \cdot P = G.$$

In other words, as u runs through $\mathrm{rad}^u(P)(S)$*, the open subsets $uP' \cdot P$ of G* (4.3.6) *cover all of G.*

The proof is divided into several steps.

[26] *Editor's note.* In this paragraph the Polo–Gille edition corrects $t^{-1}(t(P), t)$ to $q^{-1}(t(P), t)$, and twice corrects $\underline{\mathrm{Par}}_t(G)$ to $\underline{\mathrm{Par}}_t(G;P)$.

5.1.1. It is enough to prove the theorem when S is the spectrum of a field k; this follows at once from 2.6.

5.1.2. Put $L = P \cap P'$. Suppose that L has a Borel subgroup B_L, and let T be a maximal torus of B_L, as in Exp. XXII, 5.9.7. One checks immediately that

$$B = B_L \cdot \mathrm{rad}^u(P)$$

is a Borel subgroup of P. Let B' be the Borel subgroup of G opposite to B relative to T, so that $B \cap B' = T$. After splitting G relative to T, one immediately sees that $B' \subset P'$. We claim that

$$B^u(S) \cdot B' \cdot B \subset \mathrm{rad}^u(P)(S) \cdot P' \cdot P. \tag{x}$$

Indeed,

$$B^u(S) \subset P(S) = \mathrm{rad}^u(P)(S) \cdot L(S) \subset \mathrm{rad}^u(P)(S) \cdot P'(S),$$

so it is enough to observe that $B' \cdot B \subset P' \cdot P$. Thus (x) reduces Theorem 5.1 to the pair (B, B').

5.1.3. The theorem holds when k is algebraically closed: the hypothesis of 5.1.2 is then satisfied, and Exp. XXII, 5.7.10 applies.

5.1.4. The theorem also holds when k is infinite. Indeed, by 2.7,

$$\mathrm{rad}^u(P)(k) \text{ is dense in } \mathrm{rad}^u(P)(\bar{k}),$$

and the theorem holds over $\bar{k}$.

5.1.5. We are therefore reduced to the case of a finite field k. The scheme $\underline{\mathrm{Bor}}(L)$ is a smooth homogeneous space under L; Lang's theorem (Amer. J. Math. **78** (1956)) therefore shows that L has a Borel subgroup B_L. [27] By 5.1.2, we may consequently suppose that $P = B$ and $P' = B'$ are Borel subgroups. We write $T = B \cap B'$.

5.1.6. Let K be an algebraic closure of k. Choose a pinning of (G_K, B_K, T_K), and let R^+, respectively Δ, be the set of positive, respectively simple, roots. By Exp. XXII, 5.7.2, it is enough to prove, for every $\alpha \in \Delta$, that

$$u_\alpha B'^u(K) \subset B^u(k) B'^u(K) B(K). \tag{1}$$

Let α_i be the distinct conjugates of α over k. They belong to Δ, since B is defined over k. Let R' be the set of roots which are linear combinations of the α_i, and put $R'_- = R_- \cap R'$. Since R' is defined over k, there is a subtorus Q of T such that Q_K is the maximal torus of the common kernel of the α_i.

Put

$$Z = \underline{\mathrm{Centr}}_G(Q), \qquad B_Z = B \cap Z, \qquad B'_Z = B' \cap Z$$

(Exp. XXII, 5.10.2). It is enough to verify the desired assertion in Z, namely

$$u_\alpha B'^u_Z(K) \subset B^u_Z(k) B'^u_Z(K) B_Z(K). \tag{2}$$

[27] *Editor's note.* See also [DG70], §III.5, 7.4.

Indeed,

$$(B'^u_Z)_K = \prod_{\beta \in R'_-} U_\beta.$$

Let $R'' = R - R'$, and set

$$V = \prod_{\beta \in R'' \cap R_-} U_\beta.$$

Then

$$(B'^u)_K = (B'^u_Z)_K \cdot V,$$

and $(B_Z)_K$ normalizes V, by Exp. XXII, 5.6.7. From (2) we therefore obtain successively

$$\begin{aligned} u_\alpha B'^u(K) &= u_\alpha B'^u_Z(K) V(K) \\ &\subset B^u_Z(k) B'^u_Z(K) B_Z(K) V(K) \\ &\subset B^u_Z(k) B'^u_Z(K) V(K) B_Z(K), \end{aligned}$$

which immediately gives (1). We are thus reduced to the case $G = Z$, that is, to the case in which $\mathrm{Gal}(K/k)$ acts transitively on the simple roots.

5.1.7. The assertion to be proved says equivalently that G/B is the union of the translates, under $B^u(k)$, of the open image of B'^u. This assertion is unchanged when G is replaced by its adjoint group, or indeed by any group having the same adjoint group. We may therefore suppose that G is adjoint.

5.1.8. Consider the Dynkin diagram of G_K. The Galois group acts transitively on it. But this Galois group has only cyclic quotients, whereas the Dynkin diagram has no cycles. The diagram is consequently of type nA_1, with $n \geq 0$, or mA_2, with $m \geq 0$. Using the canonical decomposition of Exp. XXIV, 5.9, we may write

$$G = \prod_{D/K} G_0,$$

where D is a finite K-scheme and G_0 is either a torus or is of type A_1 or A_2.

By Exp. XXIV, 5.12, the group B comes from a Borel subgroup B_0 of G_0, and T comes from a maximal torus T_0 of G_0; the group B' comes from the Borel subgroup B'_0 of G_0 opposite to B_0 relative to T_0. We have

$$B'^u(k) = B^u_0(D), \qquad B'^u \cdot T \cdot B^u = \prod_{D/k} B'^u_0 \cdot T_0 \cdot B^u_0.$$

It is therefore enough to prove the assertion for the triple (G_0, B_0, T_0).

5.1.9. We may consequently suppose that G has type $\varnothing$, A_1, or A_2. Since G has a Borel subgroup B, it is quasi-split relative to B, by Exp. XXIV, 3.9.1, and hence split if it has type $\varnothing$ or A_1. The theorem is already known in the split case, by Exp. XXII, 5.7.10, so only type A_2 remains.

By Exp. XXIV, 3.11, there is a morphism $E \to \mathrm{Spec}(k)$, a principal Galois bundle under the group $\mathbb{Z}/2\mathbb{Z}$ of automorphisms of the Dynkin diagram of type A_2, such that

$$G = \text{Q-}\acute{\mathrm{E}}\mathrm{p}_{E/\,\mathrm{Spec}(k)}(A_2).$$

If E has a section, G is split and the theorem follows. Otherwise $E = \mathrm{Spec}(k')$, where k'/k is a quadratic extension. Finally, as observed in 5.1.7, we may suppose that G is simply connected, that is, that G is a form of $\mathrm{SL}_{3,k}$.

5.1.10. We are now in the following situation: k is a finite field and k'/k is a quadratic extension. Pin $\mathrm{SL}_{3,k'}$, the group of 3×3 matrices of determinant 1, as follows. Its maximal torus is the group of diagonal matrices, its Borel subgroup is the group of upper triangular matrices, and its root vectors are

$$u_\alpha = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad u_\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

The big cell Ω is given by

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in \Omega(S) \quad \Longleftrightarrow \quad a \text{ and } ae - bd \text{ are invertible,}$$

and

$$B^u(k) = \left\{ \begin{pmatrix} 1 & \bar{x} & z \\ 0 & 1 & x \\ 0 & 0 & 1 \end{pmatrix} \;\middle|\; x, z \in k',\ z + \bar{z} = x\bar{x} \right\}.$$

We must prove inclusion (1) of 5.1.6. Thus, for every $a, b, c \in K$, we must find $x, z \in k'$ such that $z + \bar{z} = x\bar{x}$ and

$$\begin{pmatrix} 1 & \bar{x} & z \\ 0 & 1 & x \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{pmatrix} \in \Omega(K). \tag{3}$$

If $a \neq -1$, take $x = z = 0$. If $a = -1$, the required conditions are

$$\begin{cases} z + \bar{z} = x\bar{x}, \\ bz - \bar{x} \neq 0, \\ (b + c)\bar{z} + bx - 1 \neq 0. \end{cases} \tag{4}$$

Let $q \geq 2$ be the number of elements of k. For every $m \in k$, the equation $z + \bar{z} = m$, with $z \in k'$, has q solutions.

If $b = 0$, take $x = 1$. It remains to solve

$$z + \bar{z} = 1, \qquad c\bar{z} \neq 1,$$

which is always possible by the preceding observation.

If $b \neq 0$, take $x = 0$. It remains to solve

$$z + \bar{z} = 0, \qquad z \neq 0, \qquad (b + c)z \neq 1.$$

This is always possible when $q \geq 3$. When $k = \mathbb{F}_2$, it is possible if $b + c \neq 1$, by taking $z = 1$.

It remains to consider $k = \mathbb{F}_2$, $b + c = 1$, and $b \neq 0$. The system becomes

$$z + \bar{z} = x\bar{x}, \qquad bz \neq \bar{x}, \qquad \bar{z} + bx \neq 1.$$

If $b = 1$, or respectively $b \notin k'$, take $x = 1$; the last two conditions become

$$z \neq b^{-1}, \qquad z \neq 1 - b,$$

and follow from any solution of $z + \bar{z} = 1$. Finally, if $b \in k' - k$, take $x = b$ and $z = \bar{b}$. This completes the proof. □

Corollary 5.2.— *Let S be a semilocal scheme, G an S-reductive group, and P, P' two opposite parabolic subgroups of G. The canonical map*

$$\mathrm{rad}^u(P)(S) \cdot \mathrm{rad}^u(P')(S) \longrightarrow (G/P)(S)$$

is surjective. In particular,

$$(G/P)(S) = G(S)/P(S).$$

Every parabolic subgroup Q of G having the same type as P is of the form

$$Q = \mathrm{int}(uu')P$$

with $u \in \mathrm{rad}^u(P)(S)$ and $u' \in \mathrm{rad}^u(P')(S)$.

The second assertion is plainly equivalent to the first, so let us prove the latter. Let s_i be the closed points of S, let V be the open subscheme of G/P which is the image of P' (and is isomorphic to $\mathrm{rad}^u(P')$, by 4.3.6), and take $x \in (G/P)(S)$. By 5.1, for every i there is

$$u_i \in \mathrm{rad}^u(P)\big(\kappa(s_i)\big)$$

such that $u_i x_{s_i}$ is a section of V_{s_i}. Choose $u \in \mathrm{rad}^u(P)(S)$ lifting all the u_i, as permitted by 2.6. Then ux is a section of V, since this may be checked on the closed fibers. Finally,

$$\mathrm{rad}^u(P')(S) \xrightarrow{\sim} V(S)$$

is bijective, and the result follows.

Corollary 5.3.— *Let S be a semilocal scheme, G an S-reductive group, and P, P', Q three parabolic subgroups of G. There is a $g \in G(S)$ such that* $\mathrm{int}(g)Q$ *is in transverse position relative to both P and P'.*

In the notation of 4.2.4(ii), we must show that the universally schematically dense open subscheme

$$\underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P')$$

of G has a section over S. [28] Choose a parabolic subgroup Q_1 opposite to Q, by 4.3.5(i), and put

$$U = \mathrm{rad}^u(Q), \qquad U' = \mathrm{rad}^u(Q_1).$$

We shall find the desired g in $U(S)U'(S)$. In this form, 4.2.4(i) and 2.6 show that it suffices to work on the fibers over the closed points of S; we may therefore suppose $S = \mathrm{Spec}(k)$.

If k is algebraically closed, there is a $g \in G(k)$ with the required property. By 5.2 it may be written

$$g = uu'q, \qquad u \in U(k), \quad u' \in U'(k), \quad q \in Q(k),$$

and $\mathrm{int}(uu')Q = \mathrm{int}(g)Q$.

Suppose k is infinite. Let V be the open subset of $U \times_k U'$ defined by the cartesian diagram

$$\begin{array}{ccc} U \times_k U' & \longrightarrow & G \\ \uparrow & & \uparrow \\ V & \longrightarrow & \underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P'). \end{array}$$

[28] *Editor's note.* The Polo–Gille edition replaces "relatively dense" by "universally schematically dense over S"; compare EGA IV$_3$, Definition 11.10.8.

Since $V(\bar{k}) \neq \varnothing$, as just shown, V is dense in $U \times_k U'$; it therefore has a k-point by 2.7.

Suppose finally that k is finite. Lang's theorem, as used in 5.1.5, shows that P, respectively P', has a Borel subgroup B, respectively B', since

$$\underline{\mathrm{Bor}}(P) \simeq \underline{\mathrm{Bor}}(P/\operatorname{rad}^u(P)), \qquad \underline{\mathrm{Bor}}(P') \simeq \underline{\mathrm{Bor}}(P'/\operatorname{rad}^u(P'))$$

are smooth. Let B_1 be a Borel subgroup opposite to B. By 5.2 there are

$$a \in B^u(k), \qquad a_1 \in B_1^u(k)$$

such that $\operatorname{int}(aa_1)B = B'$. Then

$$B_0 = \operatorname{int}(aa_1)B_1 = \operatorname{int}(a)B_1$$

is opposite to both B' and B. Let Q_0 be the unique parabolic subgroup of G containing B_0 and having the same type as Q, as in 3.8. It is in transverse position relative to P and P', by 4.2.1(vi). On the other hand, 5.2 gives

$$Q_0 = \operatorname{int}(uu')Q, \qquad u \in U(k), \quad u' \in U'(k),$$

which proves the assertion.

Corollary 5.4.— *Let S be a semilocal scheme, G an S-reductive group, and P, Q two parabolic subgroups of G. There is a $g \in G(S)$ such that* $\operatorname{int}(g)P$ *is in osculating position relative to Q; equivalently,* $\operatorname{int}(g)P \cap Q$ *is a parabolic subgroup of G.*

Choose a parabolic subgroup P' opposite to P, by 4.3.5(i). By 5.3, there is a parabolic subgroup P_1' of the same type as P', in transverse position relative to both P and Q. Let T be a maximal torus of $P_1' \cap Q$, as in 4.2.7(ii), and let P_1 be the parabolic subgroup opposite to P_1' relative to T. Then P_1 and Q are in osculating position, by 4.4.5. Since P and P_1 are both opposite to P_1', there is a

$$g \in \operatorname{rad}^u(P_1')(S)$$

such that $\operatorname{int}(g)P = P_1$, by 4.3.5(i). □

For the same reason there is a $u \in \operatorname{rad}^u(P)(S)$ such that $\operatorname{int}(u)P' = P_1'$. Hence $g = \operatorname{int}(u)u'$ for some $u' \in \operatorname{rad}^u(P')(S)$, and

$$P_1 = \operatorname{int}(uu'u^{-1})P = \operatorname{int}(uu')P.$$

This also gives another proof of 5.2.

Corollaries 5.3 and 5.4 are the essential results of this section. We first record some consequences of 5.4.

Corollary 5.5.— *Let S be a semilocal scheme and G an S-reductive group.*

(i) *If P, Q are parabolic subgroups of G and $t(P) \subset t(Q)$, then there is a $g \in G(S)$ such that* $\operatorname{int}(g)P \subset Q$.

(ii) *Let*

$$P_1 \supset P_2 \supset \cdots \supset P_n, \qquad P_1' \supset P_2' \supset \cdots \supset P_n'$$

be two chains of parabolic subgroups of G, with $t(P_i) = t(P_i')$ for every i. There is a $g \in G(S)$ such that $\operatorname{int}(g)P_i = P_i'$ *for every i.*

(iii) *Let P, Q, P', Q' be four parabolic subgroups of G, with $t(P) = t(P')$ and $t(Q) = t(Q')$. If the pairs (P, P') and (Q, Q') are both in transverse position, respectively both in osculating position, then there is a $g \in G(S)$ such that*

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)Q = Q'.$$

(iv) *Let P, P' be parabolic subgroups of the same type, and let L, respectively L', be a Levi subgroup of P, respectively P'. There is a $g \in G(S)$ such that*

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)L = L'.$$

Proof. Assertion (i) follows immediately from 5.4. Assertion (ii) is proved by induction on n, the case $n = 0$ being trivial. We may suppose that $P_i = P'_i$ for $i = 1, \ldots, n-1$. By 5.2 there is a $g \in G(S)$ such that $\mathrm{int}(g)P_n = P'_n$. But P_n and $\mathrm{int}(g)P_n$ are both contained in $P_{n-1} = P'_{n-1}$, and hence

$$g \in P_{n-1}(S)$$

by 4.4.4(ii). It follows that $\mathrm{int}(g)P_i = P'_i$ for every i.

Assertion (iv) follows at once from 5.2 and 1.8. Let us prove (iii) first in the transverse case. When the types of P and Q are opposite, the assertion follows from (iv) and 4.3.3(iii). In general, by 4.2.7(iii) and 4.4.6 one can find parabolic subgroups

$$P_1 \subset P, \quad P'_1 \subset P', \quad Q_1 \subset Q, \quad Q'_1 \subset Q'$$

such that P_1, P'_1 are opposite, Q_1, Q'_1 are opposite, and $t(P_1) = t(P'_1)$. There is consequently a $g \in G(S)$ such that

$$\mathrm{int}(g)P_1 = P'_1, \qquad \mathrm{int}(g)Q_1 = Q'_1.$$

After conjugating, we may suppose $P_1 = P'_1$ and $Q_1 = Q'_1$. Then P, P' are in osculating position and have the same type, so $P = P'$ by 4.4.4(i). Likewise $Q = Q'$.

It remains to prove (iii) in the osculating case. By 5.2 we may suppose $P = P'$. Again by 5.2 there is a $g \in G(S)$ such that $\mathrm{int}(g)Q = Q'$. But then $g \in P(S)$, by 4.4.4(ii), and therefore

$$\mathrm{int}(g)P = P = P'.$$

□

Definition 5.6.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G. The subgroup P is said to be* minimal *if every parabolic subgroup Q of G contained in P is equal to P.*

Notice that this notion is not, in general, stable under passage to the fibers.

Corollary 5.7.— *Let S be a semilocal scheme and G an S-reductive group.*

(i) *Let*

$$t, t' \in \mathrm{Of}\big(\mathrm{Dyn}(G)\big)(S).$$

If G has a parabolic subgroup of type t and one of type t', then it has a parabolic subgroup of type $t \cap t'$. In particular, the set of $t(P)$, as P ranges over the parabolic subgroups of G, has a least element $t_{\min}$.

(ii) *Every parabolic subgroup of G contains a minimal parabolic subgroup. A parabolic subgroup of G is minimal if and only if it has type $t_{\min}$. Any two minimal parabolic subgroups of G are conjugate by an element of $G(S)$.*

This follows immediately from 5.4 and 5.5(i).

Remark 5.8.— *A parabolic subgroup opposite to a minimal parabolic subgroup is also minimal. Consequently,*

$$s(t_{\min}) = t_{\min}.$$

Corollary 5.9.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G. The canonical morphism*

$$G \longrightarrow G/P = X$$

makes G a Zariski-locally trivial P_X-bundle over X. If L is a Levi subgroup of P, the canonical morphism (cf. 3.12)

$$G \longrightarrow G/L = Y$$

makes G a Zariski-locally trivial L_Y-bundle over Y.

It is enough to prove the following lifting assertion. Given a morphism $S' \to S$, with S' local, every morphism $S' \to X$ (respectively $S' \to Y$) lifts to a morphism $S' \to G$. In other words, we may assume that S is local, and must show that

$$G(S) \longrightarrow X(S), \qquad \text{respectively} \qquad G(S) \longrightarrow Y(S),$$

is surjective.

The first assertion was proved in 5.2. For the second, let $y \in Y(S)$. Its canonical image in $X(S)$ is the image of some $g \in G(S)$. The image y' of g in $Y(S)$ therefore has the same image in $X(S)$ as y. There is consequently a unique

$$u \in \mathrm{rad}^u(P)(S)$$

such that $y'u = y$, and the image of gu in $Y(S)$ is indeed y.

Corollary 5.10.— *Let S be a semilocal scheme and G an S-reductive group.*

(i) *Let P be a parabolic subgroup of G, and L a Levi subgroup of P. The canonical maps* (cf. 3.21) *induce bijections*

$$H^1(S, L) \xrightarrow{\sim} H^1(S, P) \xrightarrow{\sim} H^1_{t(P)}(S, G).$$

(ii) *If*

$$t, t' \in \mathrm{Of}(\mathrm{Dyn}(G))(S),$$

then (cf. 3.21)

$$H^1_t(S, G) \cap H^1_{t'}(S, G) = H^1_{t \cap t'}(S, G).$$

(iii) *If P and Q are two parabolic subgroups of G in osculating position, the following canonical diagram is cartesian and all its arrows are injective:*

$$\begin{array}{ccc} H^1(S, P) & \longrightarrow & H^1(S, G) \\ \uparrow & & \uparrow \\ H^1(S, P \cap Q) & \longrightarrow & H^1(S, Q). \end{array}$$

Let us prove (i). The map

$$H^1(S, L) \longrightarrow H^1(S, P)$$

is bijective by 2.3, while

$$H^1(S, P) \longrightarrow H^1_{t(P)}(S, G)$$

is surjective by 3.21. We must prove that it is injective, or equivalently that the canonical map

$$H^1(S, P) \longrightarrow H^1(S, G)$$

is injective. Let Q be a principal homogeneous bundle under P, let Q_1 be the associated principal homogeneous bundle under G, and let P' and G' be the corresponding twisted forms of P and G. Clearly G' is an S-reductive group and P' is a parabolic subgroup of G'.

The elements of $H^1(S, P)$ having the same image in $H^1(S, G)$ as the class of Q are naturally identified with the kernel of

$$H^1(S, P') \longrightarrow H^1(S, G').$$

By the exact sequence in cohomology, this kernel is in turn identified with the set of $G'(S)$-orbits in $(G'/P')(S)$.[29] But $G'(S)$ acts transitively on $(G'/P')(S)$, by 5.2.

For (ii), let Q be a principal homogeneous bundle under G, and let G_Q be the corresponding twisted form of G. By definition (3.21), we must prove that G_Q has a parabolic subgroup of type $t \cap t'$ if and only if it has parabolic subgroups of types t and t'. This is precisely the combination of 3.8 and 5.7(i). Finally, (iii) follows immediately from (i) and (ii).

We now state a consequence of 5.3.

Corollary 5.11.— *Let S be a semilocal scheme, G an S-reductive group, and P a parabolic subgroup of G. If $P \neq G$, there exist at least three distinct parabolic subgroups of G having the same type as P. In other words,*

$$P \neq G \qquad \Longrightarrow \qquad (G(S) : P(S)) \geq 3.$$

Indeed, let P' be a parabolic subgroup opposite to P, as in 4.3.5(i). Since $P \neq G$, one has $\mathrm{rad}^u(P') \neq e$, for example by 4.3.2. By 2.1, $\mathrm{rad}^u(P')(S) \neq e$; choose

$$u \in \mathrm{rad}^u(P')(S), \qquad u \neq e.$$

Then $\mathrm{int}(u)P \neq P$. By 5.3 there is a parabolic subgroup P_1 of the same type as P, opposite to both P and $\mathrm{int}(u)P$. Thus P_1, P, and $\mathrm{int}(u)P$ are three distinct parabolic subgroups of G, all of the same type. □

6. Parabolic subgroups and split tori

Proposition 6.1.— *Let S be a scheme, G an S-reductive group, $\mathfrak{g} = \mathrm{Lie}(G/S)$, and Q a split subtorus of G. Write $Q = D_S(M)$, and let*

$$\mathfrak{g} = \bigoplus_{\alpha \in M} \mathfrak{g}^\alpha$$

be the decomposition of $\mathfrak{g}$ under the action of Q. Let $M_1 \subset M$ satisfy

$$0 \in M_1, \qquad \alpha, \beta \in M_1 \Longrightarrow \alpha + \beta \in M_1.$$

[29] For these arguments in nonabelian cohomology, see Giraud's thesis, [Gi71], § III.3.

(i) *There is a unique smooth subgroup H_{M_1} of G, with connected fibers, which contains $\underline{\mathrm{Centr}}_G(Q)$ and whose Lie algebra is*

$$\bigoplus_{\alpha\in M_1} \mathfrak{g}^\alpha.$$

(ii) *The following implications hold:*

$$\begin{aligned} M_1 = \{0\} &\Longrightarrow H_{M_1} = \underline{\mathrm{Centr}}_G(Q), \\ M_1 = -M_1 &\Longrightarrow H_{M_1} \text{ is reductive}, \\ M_1 \cup (-M_1) = M &\Longrightarrow \begin{array}{c} H_{M_1} \text{ and } H_{-M_1} \text{ are opposite parabolic} \\ \text{subgroups of } G, \text{ with common Levi subgroup } H_{M_1\cap(-M_1)}. \end{array} \end{aligned}$$

Assertions (i) and (ii) are local for the fpqc topology. We may therefore suppose that Q is contained in a maximal torus T of G, and may moreover split G relative to T. Assertion (i) then follows immediately from Exposé XXII, 5.3.5, 5.4.5, and 5.4.7. The assertions in (ii) follow from Exposé XXII, 5.3.5, 5.10.1, and 5.11.3, together with 1.4 and 4.3.2 of the present Exposé.

Corollary 6.2.— *Let S be a scheme, G an S-reductive group, and Q a split subtorus of G. There is a parabolic subgroup of G having $\underline{\mathrm{Centr}}_G(Q)$ as a Levi subgroup.*

Indeed, write $Q = D_S(M)$, choose a total order on the group M, and let M_1 be the set of its positive elements. The subgroup H_{M_1} has the required property.

Corollary 6.3.— *If the S-reductive group G has a noncentral split subtorus, then it has a proper parabolic subgroup* (that is, one different from G).

Combining 6.2 with 5.9 and 5.10 gives the following result.

Corollary 6.4.— *Let S be a scheme, G an S-reductive group, and Q a split subtorus of G. The canonical morphism*

$$G \longrightarrow G/\underline{\mathrm{Centr}}_G(Q)$$

is a locally trivial fibration.

If S is semilocal, the map

$$G(S) \longrightarrow (G/\underline{\mathrm{Centr}}_G(Q))(S)$$

is surjective, and the map

$$H^1(S, \underline{\mathrm{Centr}}_G(Q)) \longrightarrow H^1(S, G)$$

is injective.

6.5.

Suppose that S is connected. If T is an S-torus and T', T'' are two split subtori of T, their product

$$T' \cdot T''$$

is again a split subtorus of T.[30] Indeed, it is identified with the quotient of $T' \times_S T''$ by $T' \cap T''$, and this quotient is split by Exposé IX, 2.11. It follows that T has a largest split subtorus, which we denote by T_{spl}.

[30] *Editor's note.* The Polo–Gille edition adopts multiplicative notation here, replacing the source's "sum $T' + T''$" by "product $T' \cdot T''$".

Lemma 6.6.— *Let S be a connected scheme, T an isotrivial S-torus, and T_{spl} its largest split subtorus. The following conditions are equivalent:*

(i) *There is a homomorphism*
$$T \longrightarrow \mathbf{G}_{m,S}$$
different from the trivial homomorphism.

(ii) $T_{\mathrm{spl}} \neq e$.

Since T is isotrivial, there are a finite group Γ, a connected principal Galois covering $S' \to S$ with group Γ, and an isomorphism
$$T_{S'} \simeq D_{S'}(M).$$
The group M is then a Γ-module, and there is a natural isomorphism
$$\mathrm{Hom}_S(T, \mathbf{G}_{m,S}) = H^0(\Gamma, M).$$

Let V be the vector subspace of $M \otimes \mathbf{Q}$ generated by the elements
$$\gamma(m) - m, \qquad \gamma \in \Gamma, \quad m \in M.$$
One checks immediately that
$$(T_{\mathrm{spl}})_{S'} \simeq D_{S'}\big(M/(M \cap V)\big).$$
Thus (i) is equivalent to
$$H^0(\Gamma, M) \neq 0,$$
or equivalently to
$$H^0(\Gamma, M \otimes \mathbf{Q}) \neq 0,$$
whereas (ii) is equivalent to $M \neq M \cap V$, or to $M \otimes \mathbf{Q} \neq V$. But
$$M \otimes \mathbf{Q} = H^0(\Gamma, M \otimes \mathbf{Q}) \oplus V.$$
This follows at once by considering the projector from $M \otimes \mathbf{Q}$ onto its invariants which sends x to the average of its transforms under Γ.

Lemma 6.7.— *Let S be a scheme, G an S-reductive group, $P \neq G$ a parabolic subgroup of G, L a Levi subgroup of P, and $Q = \mathrm{rad}(L)$. There is a homomorphism*
$$Q \longrightarrow \mathbf{G}_{m,S}$$
different from the trivial homomorphism.

Let $U = \mathrm{rad}^u(P)$. It is invariant under $\mathrm{int}(P)$, hence under $\mathrm{int}(Q)$. Consider the invertible $\mathcal{O}_S$-module
$$\det\big(\mathrm{Lie}(U)\big),$$
the highest exterior power of the locally free $\mathcal{O}_S$-module $\mathrm{Lie}(U)$. The adjoint representation defines a homomorphism
$$f : Q \longrightarrow \mathrm{Aut}\big(\det\big(\mathrm{Lie}(U)\big)\big) = \mathbf{G}_{m,S}.$$

Since $P \neq G$, one has $U \neq e$. Choose $s \in S$ such that $U_s \neq e$. After splitting G_s relative to a maximal torus containing Q_s, it is immediate that $f_s \neq e$.

Proposition 6.8.— *Let S be a connected semilocal scheme, G an S-reductive group, P a parabolic subgroup of G, L a Levi subgroup of P, and $Q = \mathrm{rad}(L)$. Let Q_{spl} be the largest split subtorus of Q, equivalently the largest central split subtorus of L. Then*

$$L = \underline{\mathrm{Centr}}_G(Q_{\mathrm{spl}}).$$

Put $L' = \underline{\mathrm{Centr}}_G(Q_{\mathrm{spl}})$. This is a reductive subgroup of G containing L, and

$$P' = P \cap L'$$

is a parabolic subgroup of L' with Levi subgroup L, by 1.20. If $L' \neq L$, then $L' \not\subset P$, since L is a maximal reductive subgroup of P by 1.7. Hence $P' \neq L'$.

Let G_1 be the derived group of G, and put $P_1 = P' \cap G_1$. By 1.19, P_1 is a parabolic subgroup of the semisimple group G_1, the group $L_1 = L \cap G_1$ is a Levi subgroup of P_1, and

$$Q_1 = \mathrm{rad}(L_1) = \big(\mathrm{rad}(L) \cap G_1\big)^0 = \big(Q \cap G_1\big)^0.$$

The group L_1 has a maximal torus T_1, by Exposé XIV, 3.20, and T_1 is isotrivial, by Exposé XXIV, 4.1.5. Therefore its subtorus Q_1 is also isotrivial, by Exposé IX, 2.11. Since $P_1 \neq G_1$, Lemmas 6.7 and 6.6 give

$$(Q_1)_{\mathrm{spl}} \neq e.$$

It follows that

$$\big(Q_{\mathrm{spl}} \cap G_1\big)^0 \neq e,$$

and hence that $Q_{\mathrm{spl}} \not\subset \mathrm{rad}(L')$, because $\mathrm{rad}(L') \cap G_1$ is finite. This contradicts the definition of L'. □

Corollary 6.9.— *Let S be a connected semilocal scheme, G an S-reductive group, and Q a critical subtorus of G, that is, a subtorus such that*

$$\mathrm{rad}\big(\underline{\mathrm{Centr}}_G(Q)\big) = Q.$$

The group $\underline{\mathrm{Centr}}_G(Q)$ is a Levi subgroup of a parabolic subgroup of G if and only if

$$\underline{\mathrm{Centr}}_G(Q) = \underline{\mathrm{Centr}}_G(Q_{\mathrm{spl}}),$$

or equivalently,

$$\mathrm{Lie}(G)^Q = \mathrm{Lie}(G)^{Q_{\mathrm{spl}}}.$$

This follows from 6.2 and 6.8.

Corollary 6.10.— *Let S be a connected semilocal scheme, G an S-reductive group, and L a subgroup of G. The following conditions are equivalent:*

(i) *There is a parabolic subgroup of G having L as a Levi subgroup.*

(ii) *There is a split subtorus of G whose centralizer is L.*

(iii) *There is a homomorphism*

$$\mathbf{G}_{m,S} \longrightarrow G$$

whose centralizer is L.

The implication (i)⇒(ii) follows from 6.8, and (iii)⇒(i) from 6.1. It remains to prove (ii)⇒(iii). Suppose therefore that

$$L = \underline{\mathrm{Centr}}_G(Q), \qquad Q = D_S(M).$$

Write $\mathfrak{g} = \mathrm{Lie}(G/S)$ and

$$\mathfrak{g} = \bigoplus_{\alpha \in M} \mathfrak{g}^\alpha,$$

and let R be the set of $\alpha \in M \setminus \{0\}$ for which $\mathfrak{g}^\alpha \neq 0$.

Since R is finite and does not contain 0, there is a homomorphism

$$u : M \longrightarrow \mathbf{Z}$$

such that $u(\alpha) \neq 0$ for every $\alpha \in R$. By duality, u yields a homomorphism $\mathbf{G}_{m,S} \to Q$, and hence a homomorphism

$$f : \mathbf{G}_{m,S} \longrightarrow G.$$

One has

$$\underline{\mathrm{Centr}}_G(f) \supset \underline{\mathrm{Centr}}_G(Q).$$

These are two smooth subgroups of G, with connected fibers, and their Lie algebras coincide, since both are equal to $\mathfrak{g}^0$. They are therefore equal, by the usual argument.

Corollary 6.11.— *Let S be a connected semilocal scheme and G an S-reductive group. The maps*

$$L \longmapsto \mathrm{rad}(L)_{\mathrm{spl}}, \qquad Q \longmapsto \underline{\mathrm{Centr}}_G(Q)$$

are mutually inverse, order-reversing bijections between:

- *the set of subgroups L of G which are Levi subgroups of parabolic subgroups of G; and*
- *the set of split subtori Q of G such that*

$$\mathrm{rad}\big(\underline{\mathrm{Centr}}_G(Q)\big)_{\mathrm{spl}} = Q.$$

Corollary 6.12.— *Let S be a connected semilocal scheme and G an S-reductive group. Consider the following assertions:*

(i) *G has a parabolic subgroup different from G.*

(ii) *G has a noncentral split subtorus.*

(ii bis) *G has a noncentral split subtorus of relative dimension 1.*

(iii) *There is a group homomorphism*

$$\mathbf{G}_{a,S} \longrightarrow G$$

which is a closed immersion.

Then

$$\text{(i)} \iff \text{(ii)} \iff \text{(ii bis)} \implies \text{(iii)}.$$

The only new assertion is (i)⇒(iii). Let $P \neq G$ be a parabolic subgroup of G. Then $U = \mathrm{rad}^u(P) \neq e$. Consider the last nontrivial subgroup U_n in the composition series of U, as in 2.1. There is an isomorphism

$$U_n \simeq W(E_n),$$

where E_n is a locally free $\mathcal{O}_S$-module, and hence is free because S is semilocal and connected. Since $E_n \neq 0$, there is a monomorphism

$$\mathcal{O}_S \longrightarrow E_n$$

whose image is locally a direct summand. It induces a closed immersion

$$\mathbf{G}_{a,S} = W(\mathcal{O}_S) \hookrightarrow W(E_n) \simeq U_n,$$

which proves (iii).

Remark 6.12.1.— *When S is the spectrum of a field of characteristic* 0*, the Jacobson–Morozov theorem gives* (iii)⇒(ii bis)*. The four preceding conditions are then equivalent* ("Godement's criterion"; cf. [BT65], 8.5).[31]

Definition 6.13.— *Let S be a connected semilocal scheme and G an S-reductive group. The group G is said to be* anisotropic *if it contains no nontrivial split subtorus.*

Corollary 6.14.— *Let S be a connected semilocal scheme. An S-reductive group G is anisotropic if and only if it has no parabolic subgroup $P \neq G$ and its radical is anisotropic.*

Combining 6.6 with Exposé XXIV, 4.1.5, and Exposé XXII, 6.2 gives:

Corollary 6.15.— *Let S be a connected semilocal scheme and G an isotrivial S-reductive group* (for example, G semisimple, or S normal; cf. Exposé X, 5.16)*. The group G is anisotropic if and only if it has no parabolic subgroup $P \neq G$ and*

$$\mathrm{Hom}_{S\text{-gr}}(G, \mathbf{G}_{m,S}) = e.$$

Proposition 6.16.— *Let S be a connected semilocal scheme and G an S-reductive group. The maximal split subtori of G are precisely the largest central split subtori of the Levi subgroups of the minimal parabolic subgroups of G. Any two such tori are conjugate by an element of $G(S)$.*

Let Q be a maximal split subtorus of G. By 6.2,

$$L = \underline{\mathrm{Centr}}_G(Q)$$

is a Levi subgroup of a parabolic subgroup P of G. Since

$$Q \subset \mathrm{rad}\big(\underline{\mathrm{Centr}}_G(Q)\big)_{\mathrm{spl}},$$

the maximality of Q gives

$$Q = \mathrm{rad}\big(\underline{\mathrm{Centr}}_G(Q)\big)_{\mathrm{spl}}.$$

[31] More generally, this remains true when S is the spectrum of a perfect field (Tits); see [BT71], Corollary 3.7.

By 6.11, L is minimal among the Levi subgroups of parabolic subgroups of G. Hence P is a minimal parabolic subgroup of G, by 1.20. It follows from 5.7 and 5.5(iv) that any two such subtori Q are conjugate by an element of $G(S)$. The conjugacy of the tori Q and of the pairs (P, L) also proves the first assertion.

Corollary 6.17.— *Let S be a connected semilocal scheme, and let P, P' be two minimal parabolic subgroups in standard position* (4.5.1.1). *Then $P \cap P'$ contains a common Levi subgroup of P and P'.*

Indeed, $P \cap P'$ contains a maximal torus T of G, by 4.5.1(v). Let L be the unique Levi subgroup of P which contains T. By 1.21,

$$\mathrm{rad}(P) \cap T = \mathrm{rad}(P) \cap L = \mathrm{rad}(L).$$

Thus $\mathrm{rad}(P) \cap T$ contains $\mathrm{rad}(L)_{\mathrm{spl}}$, which is a maximal split subtorus of G, and is therefore necessarily equal to T_{spl}. Hence

$$L = \underline{\mathrm{Centr}}_G(T_{\mathrm{spl}}),$$

and, by symmetry, L is also a Levi subgroup of P'.

Remark 6.18.— *It follows from 1.21 that a parabolic subgroup P of G is minimal if and only if* $\mathrm{rad}(P)$ *contains a maximal split subtorus of G. In that case, if T is a maximal torus of P, the torus T_{spl} is a maximal split subtorus both of G and of* $\mathrm{rad}(P)$, *and* $\underline{\mathrm{Centr}}_G(T_{\mathrm{spl}})$ *is a Levi subgroup of P. Moreover, every Levi subgroup of P is obtained in this way.*

7. Relative root datum

Throughout this section, S denotes a nonempty connected semilocal scheme, G an S-reductive group, Q a maximal split subtorus of G, and L the centralizer of Q in G:

$$L = \underline{\mathrm{Centr}}_G(Q).$$

7.1.

Since Q is the largest central split subtorus of L, every element of $G(S)$ which normalizes L also normalizes Q. Thus (cf. 7.1.1)

$$\underline{\mathrm{Norm}}_G(L)(S) = \underline{\mathrm{Norm}}_G(Q)(S).$$

On the other hand, 6.4 shows that

$$G(S) \longrightarrow (G/L)(S)$$

is surjective. Consequently there is a canonical identification

$$W_G(Q)(S) = (\underline{\mathrm{Norm}}_G(Q)/\underline{\mathrm{Centr}}_G(Q))(S) \simeq \underline{\mathrm{Norm}}_G(L)(S)/L(S).$$

Let

$$M = \mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S}),$$

so that $Q \simeq D_S(M)$ canonically. Denote by W the group of automorphisms of M defined by $W_G(Q)(S)$. Thus

$$W \simeq W_G(Q)(S) \simeq \underline{\mathrm{Norm}}_G(L)(S)/L(S).$$

7.1.1.

In general one does not have $\underline{\mathrm{Norm}}_G(L) = \underline{\mathrm{Norm}}_G(Q)$. For example, let S be the spectrum of a field k having a quadratic extension k', and take for G the unitary group

$$^2A_{2,k'/k}$$

(cf. Exposé XXIV, 3.11.2). The minimal parabolic subgroups of G are its Borel subgroups, and their Levi subgroups are maximal tori; hence

$$(\underline{\mathrm{Norm}}_G(L)/L)(k) = \mathfrak{S}_3.$$

Since G is not split, its maximal split tori have dimension at most 1, and are therefore isomorphic to $\mathbf{G}_{m,k}$. The group $\underline{\mathrm{Norm}}_G(Q)/L$ acts faithfully on Q, and

$$(\underline{\mathrm{Norm}}_G(Q)/L)(k) = \mathbf{Z}/2\mathbf{Z}.$$

[32]

7.2.

Let P be a parabolic subgroup of G with Levi subgroup L; such a subgroup exists by 6.2. It is necessarily minimal, by 6.18. Using the conjugacy of the minimal parabolic subgroups of G (5.7), the conjugacy of the Levi subgroups of a parabolic subgroup (1.8), and the equalities

$$P = \underline{\mathrm{Norm}}_G(P), \qquad \underline{\mathrm{Norm}}_G(L) \cap P = L$$

from 1.6, we obtain the following conclusion: the set of (minimal) parabolic subgroups of G having Levi subgroup L is a principal homogeneous space under W.

7.3.

The Lie algebra of G decomposes under the action of Q as

$$\mathrm{Lie}(G) = \mathrm{Lie}(L) \oplus \bigoplus_{\alpha \in R} \mathrm{Lie}(G)^\alpha,$$

where R is the set of nonzero characters α of Q such that $\mathrm{Lie}(G)^\alpha \neq 0$. These are the roots of G relative to Q.

Let

$$M^* = \mathrm{Hom}_{S\text{-gr}}(\mathbf{G}_{m,S}, Q).$$

It is in duality with M, and W acts on it naturally by transport of structure.

Theorem 7.4.— *With the notation of 7.3, there is a unique map*

$$R \longrightarrow M^*, \qquad \alpha \longmapsto \alpha^*,$$

which makes (M, M^*) *into a root datum* (Exposé XXI, 1.1) *with Weyl group* W.

Moreover, parabolic subgroups P *of* G *having Levi subgroup* L *correspond bijectively to systems of positive roots* R^+ *in* R *by the relation*

$$\mathrm{Lie}(P) = \mathrm{Lie}(L) \oplus \bigoplus_{\alpha \in R^+} \mathrm{Lie}(G)^\alpha.$$

[32] *Editor's note.* The French edition prints $\underline{\mathrm{Norm}}_G(Q)/Q$ in this final equality, although the immediately preceding action and the definition of the relative Weyl group use $\underline{\mathrm{Norm}}_G(Q)/L$.

7.4.1.

Suppose first that the existence of the desired map $\alpha \mapsto \alpha^*$ has been proved. By Exposé XXI, 3.4.10, s_α is the unique element of W such that

$$s_\alpha(m) - m$$

is a rational multiple of α for every $m \in M$. Consequently s_α is determined by α. Since

$$\alpha^*(m)\alpha = m - s_\alpha(m),$$

[33] the coroot α^* is also determined by α, proving the uniqueness of the map.

7.4.2.

Let $\alpha \in R$. Denote by L_α, H_α, and $H_{-\alpha}$, respectively, the unique smooth subgroups of G with connected fibers which contain L and satisfy (cf. 6.1(i))

$$\begin{aligned}
\mathrm{Lie}(L_\alpha) &= \mathrm{Lie}(L) \oplus \bigoplus_{\gamma \in \mathbf{Z}\alpha \cap R} \mathrm{Lie}(G)^\gamma, \\
\mathrm{Lie}(H_\alpha) &= \mathrm{Lie}(L) \oplus \bigoplus_{\gamma \in \mathbf{N}\alpha \cap R} \mathrm{Lie}(G)^\gamma, \\
\mathrm{Lie}(H_{-\alpha}) &= \mathrm{Lie}(L) \oplus \bigoplus_{\gamma \in -\mathbf{N}\alpha \cap R} \mathrm{Lie}(G)^\gamma.
\end{aligned}$$

The group L_α is a reductive subgroup of G, and $H_\alpha, H_{-\alpha}$ are opposite parabolic subgroups of L_α, with common Levi subgroup L, by 6.1(ii).

By 7.2 there is therefore an element

$$s_\alpha \in \underline{\mathrm{Norm}}_{L_\alpha}(Q)(S)/L(S) \subset W$$

such that $s_\alpha(H_\alpha) = H_{-\alpha}$. One has

$$s_\alpha(\alpha) = -\alpha,$$

because α, respectively $-\alpha$, is the common divisor of the elements of R occurring in H_α, respectively $H_{-\alpha}$. Moreover $s_\alpha^2 = \mathrm{id}$, because $s_\alpha^2(H_\alpha)$ and H_α are both opposite to $H_{-\alpha}$ relative to L. We have thus constructed an element $s_\alpha \in W$ with the properties

(x) $$s_\alpha(\alpha) = -\alpha, \qquad s_\alpha^2 = \mathrm{id},$$

and

(xx) s_α can be represented by an element of $L_\alpha(S)$.

The construction is canonical in α; in particular,

(xxx) $$w s_\alpha w^{-1} = s_{w(\alpha)} \qquad \text{for every } w \in W.$$

[33] *Editor's note.* Here and again in 7.4.4, the Polo–Gille edition restores the factor α omitted after $\alpha^*(m)$ in the original text.

7.4.3.

We now prove

(xxxx) $$s_\alpha(m) - m \in \mathbf{Z}\alpha \qquad \text{for every } m \in M.$$

Since S is connected, this assertion is local for the fpqc topology. We may therefore suppose that

$$L_\alpha = G_1$$

is split relative to a maximal torus T_1 of L. Let (G_1, T_1, M_1, R_1) be such a splitting. The monomorphism $Q \to T_1$ identifies M with a quotient of M_1; denote the canonical map by

$$p : M_1 \longrightarrow M.$$

The image of R_1 under p, possibly including zero, consists of the roots of G_1 relative to Q, hence of the elements of R which are integral multiples of α. Thus

$$p(R_1) \subset \mathbf{Z}\alpha.$$

By (xx), an element of $\underline{\mathrm{Norm}}_{G_1}(L)(S)$ induces s_α on Q. Exposé XXII, 5.10.10, then gives a section

$$w \in (W_1)_S(S)$$

which induces s_α on Q, where W_1 is the Weyl group of the root datum $(M_1, R_1, \ldots)$. After restricting S if necessary, we may suppose that $w \in W_1$ and that

$$p\big(w(m_1)\big) = s_\alpha\big(p(m_1)\big) \qquad (m_1 \in M_1).$$

By definition, w is a product of reflections associated with elements of R_1. Hence $w(m_1) - m_1$ is an integral linear combination of elements of R_1. It follows that

$$s_\alpha\big(p(m_1)\big) - p(m_1)$$

is an integral linear combination of elements of $p(R_1) \subset \mathbf{Z}\alpha$, and is therefore an integral multiple of α, as required.

7.4.4.

We may now define $\alpha^* \in M^*$ by

$$\alpha^*(m)\alpha = m - s_\alpha(m).$$

By (x),

$$\langle \alpha^*, \alpha \rangle = 2.$$

Moreover, (xxx) implies that for every $(\alpha, \alpha') \in R \times R$,

$$s_\alpha(\alpha') \in R, \qquad s_\alpha(\alpha'^*) = s_\alpha(\alpha')^*.$$

By Exposé XXI, 1.1, the map $\alpha \mapsto \alpha^*$ therefore defines a root datum on (M, M^*).

7.4.5.

Let W' be the Weyl group of this root datum, that is, the group of transformations of M generated by the s_α. Then $W' \subset W$.

Choose a total order on the free abelian group M, and put

$$R^+ = \{\alpha \in R \mid \alpha \geq 0\}.$$

This is a system of positive roots in R. Let $w \in W$, and represent it by an element

$$n \in \underline{\mathrm{Norm}}_G(L)(S) = \underline{\mathrm{Norm}}_G(Q)(S).$$

Put $P = H_{R^+}$, in the notation of 6.1. Then P is a parabolic subgroup of G with Levi subgroup L, and

$$\mathrm{int}(n)P = H_{w(R^+)}.$$

It follows from 7.3 that $w(R^+) = R^+$ implies $w = e$. The group W' acts transitively on the systems of positive roots in R, by Exposé XXI, 3.3.7, while the stabilizer of R^+ in W is trivial. Therefore $W = W'$.

It follows as well that $W = W'$ acts simply transitively both on the systems of positive roots in R and on the parabolic subgroups of G having Levi subgroup L. This proves the last assertion of 7.4. □

7.5.

Let P, P_1 be two minimal parabolic subgroups of G, with Levi subgroups L, L_1, and let Q, Q_1 be the largest central split subtori of L, L_1, respectively. Then the pairs (P, Q) and (P_1, Q_1) are conjugate: there is a $g \in G(S)$ such that

$$\mathrm{int}(g)P = P_1, \qquad \mathrm{int}(g)Q = Q_1.$$

Indeed, P and P_1 are conjugate by 5.7, so we may suppose $P = P_1$. Then L and L_1 are conjugate by an element of $P(S)$, by 1.8. Moreover, if g, g' are two elements of $G(S)$ which conjugate (P, Q) to (P_1, Q_1), then $g'^{-1}g$ normalizes P and Q, hence also P and L. But

$$\underline{\mathrm{Norm}}_G(P) \cap \underline{\mathrm{Norm}}_G(L) = P \cap \underline{\mathrm{Norm}}_G(L) = L = \underline{\mathrm{Centr}}_G(Q).$$

Consequently the isomorphism

$$Q \xrightarrow{\sim} Q_1$$

induced by $\mathrm{int}(g)$ is independent of g.

Let $\mathcal{R}, \mathcal{R}_1$ be the root data defined by 7.4 in

$$\mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S}) \quad \text{and} \quad \mathrm{Hom}_{S\text{-gr}}(Q_1, \mathbf{G}_{m,S}),$$

and let R^+, R_1^+ be the systems of positive roots corresponding to P, P_1. The canonical isomorphism $Q \simeq Q_1$ just constructed carries $(\mathcal{R}, R^+)$ to $(\mathcal{R}_1, R_1^+)$. We may therefore define the *based relative root datum* of G over S, by identifying the various pairs $(\mathcal{R}, R^+)$ through the transitive system of isomorphisms above.[34]

From now on, denote this based root datum by

$$\mathcal{R}(G/S) = (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}^+).$$

[34] A based root datum is a root datum together with the choice of a system of positive roots, equivalently of simple roots; cf. Exposé XXIII, 1.5.

For every pair $P \supset Q$ as above there is therefore a canonical isomorphism

$$\underline{M} \xrightarrow{\sim} \mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S})$$

which carries $\underline{R}$, respectively $\underline{R}^+$, to the roots of G, respectively of P, relative to Q, and carries $W(\mathcal{R})$ to $W_G(Q)(S)$.

7.6.

Let again Q be a maximal split torus of G, let P be a minimal parabolic subgroup of G with Levi subgroup $\underline{\mathrm{Centr}}_G(Q)$, let

$$(M, M^*, R, R^*, R^+)$$

be the corresponding based root datum of 7.4, and let Δ be the set of simple roots in R^+. For every $A \subset \Delta$, let

$$R_A = R^+ \cup (\mathbf{Z}A \cap R^-).$$

Thus R_A consists of the positive roots together with the negative roots which are integral linear combinations of elements of A. It is a closed set of roots, by Exposé XXI, 3.1.4, and every closed set containing R^+ is uniquely of this form, by Exposé XXI, 3.3.10.

By 6.1 there is a unique smooth subgroup P_A of G, with connected fibers, which contains $\underline{\mathrm{Centr}}_G(Q)$ and satisfies

$$\mathrm{Lie}(P_A) = \mathrm{Lie}(G)^0 \oplus \bigoplus_{\alpha \in R_A} \mathrm{Lie}(G)^\alpha.$$

The following result now follows immediately from 6.1, the conjugacy of minimal parabolic subgroups, and the fact that the set of roots of a parabolic subgroup of G containing Q is closed (which follows from 1.4 after splitting).

Proposition 7.7.—

(i) *The map*

$$A \longmapsto P_A$$

is an inclusion-preserving bijection from the set of subsets of Δ onto the set of parabolic subgroups of G containing P.

(ii) *Every parabolic subgroup of G is conjugate by an element of $G(S)$ to a unique P_A.*

7.8.

Let $P \supset Q$ be as above. Consider the relative root datum of G over S defined in 7.5 and the canonical isomorphism

$$f : (M, M^*, R, R^*, R^+) \xrightarrow{\sim} (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}^+).$$

The set Δ of simple roots in R^+ is carried to the set $\underline{\Delta}$ of simple roots in $\underline{R}^+$. Thus every subset $A \subset \Delta$ is carried to a subset

$$f(A) \subset \underline{\Delta}.$$

Definition 7.9.0.— *Let H be any parabolic subgroup of G. By 7.7*(ii)*, it is conjugate to a unique P_A. Put*

$$t_r(H) = f(A) \subset \underline{\Delta}.$$

The conjugacy theorems show immediately that $t_r(H)$ is independent of the choice of the pair $P \supset Q$. It is called the relative type *of H.*

Proposition 7.9.—

(i) *The map $H \mapsto t_r(H)$ induces a bijection from the set of $G(S)$-conjugacy classes of parabolic subgroups of G onto the set of subsets of $\underline{\Delta}$.*

(ii) *Let H be a parabolic subgroup of G, P a minimal parabolic subgroup contained in H, Q the largest central split torus of a Levi subgroup of P, Δ the set of simple roots of P relative to Q, and*

$$f : \Delta \xrightarrow{\sim} \underline{\Delta}$$

the canonical isomorphism. For every $\alpha \in \Delta$,

$$f(\alpha) \in t_r(H) \quad \Longleftrightarrow \quad \mathrm{Lie}(H)^{-\alpha} \neq 0,$$

and

$$H = P_A, \qquad A = f^{-1}(t_r(H)).$$

(iii) *If H, H' are two parabolic subgroups of G containing P, then* (see 3.8.1(ii) and 5.5(i) for other equivalent conditions)

$$t_r(H) \subset t_r(H') \quad \Longleftrightarrow \quad t(H) \subset t(H').$$

7.10.

The relative positions of two minimal parabolic subgroups can be described as follows. We use the notation $t_2(P, P_1)$ of 4.5.2.

(1) Let P, P_1, P', P_1' be four minimal parabolic subgroups of G. Then

$$t_2(P, P_1) = t_2(P', P_1'),$$

that is, the pairs (P, P_1) and (P', P_1') are fpqc-locally conjugate, if and only if there is a $g \in G(S)$ such that

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)P_1 = P_1'.$$

(2) Fix a minimal parabolic subgroup P of G with Levi subgroup L, and let T be a maximal torus of L. Consider the scheme

$$\underline{\mathrm{Par}}_{t_{\min}}(G; P)$$

of minimal parabolic subgroups of G in standard position relative to P. There is a morphism (cf. 4.5.5)

$$f : \underline{\mathrm{Par}}_{t_{\min}}(G; P) \longrightarrow W_P(T) \backslash W_G(T) / W_P(T)$$

whose fibers are the orbits of P in $\underline{\mathrm{Par}}_{t_{\min}}(G;P)$. By (1), f therefore induces a monomorphism

$$P(S)\backslash\underline{\mathrm{Par}}_{t_{\min}}(G;P)(S) \hookrightarrow (W_P(T)\backslash W_G(T)/W_P(T))(S).$$

Its image is identified with W. This is the Bruhat theorem: each orbit of $P(S)$ in $\underline{\mathrm{Par}}_{t_{\min}}(G;P)(S)$ contains exactly one parabolic subgroup of G with Levi subgroup L, that is, one of the form

$$\mathrm{int}(n)P, \qquad n \in \underline{\mathrm{Norm}}_G(L)(S).$$

(3) Equivalently, let $E(S)$ be the set of elements $g \in G(S)$ such that $\mathrm{int}(g)P$ and P are in mutual standard position. Then $E(S)$ is partitioned into double cosets modulo $P(S)$, indexed by W:

$$E(S) = P(S)\, W\, P(S).$$

Putting $U = \mathrm{rad}^u(P)$, one may also write

$$E(S) = U(S) \cdot \underline{\mathrm{Norm}}_G(L)(S) \cdot U(S).$$

This exhibits a partition of $E(S)$ into double cosets modulo $U(S)$, indexed by $\underline{\mathrm{Norm}}_G(L)(S)$.

(4) If S is the spectrum of a field, then $E(S) = G(S)$, and one recovers [BT65], 5.15.

Counterexamples 7.11.— *Let*

$$S = \mathrm{Spec}(\mathbf{Z}/4\mathbf{Z}), \qquad G = \mathrm{SL}_{2,S}\,.$$

Let B be the usual Borel subgroup consisting of the matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \textit{with } c = 0,$$

and put

$$g = \begin{pmatrix} -1 & 1 \\ 2 & 1 \end{pmatrix} \in G(S), \qquad B' = \mathrm{int}(g)B.$$

Then

$$B(S) = B'(S),$$

whereas $B \cap B'$ contains no maximal torus. [35]

This shows both that two distinct minimal parabolic subgroups may have the same group of sections and that, in general, one cannot decide from the groups $P(S)$ and $P'(S)$ alone whether two minimal parabolic subgroups P, P' are in standard position. In particular, the subset $E(S)$ of $G(S)$ does not appear to be definable solely from the configuration

$$\{G(S), P(S), \underline{\mathrm{Norm}}_G(L)(S)\}.$$

In the example above, this subset is defined by $c \neq 2$. [36]

[35] Indeed, B' is the subgroup of G defined by $c = 2(a+b)$. Thus $B \cap B'$ is not flat over S, and hence contains no maximal torus, by 4.5.1.

[36] More generally, for every S-scheme S', the set $E(S')$ is defined by the condition that c is either zero or invertible.

7.12.

We now study how $\mathcal{R}(G/S)$ varies with S. Let S' be an S-scheme which is likewise nonempty, connected, and semilocal. If Q is a maximal split torus of G, then $Q_{S'}$ is a split torus of $G_{S'}$. Choose a maximal split torus Q' of $G_{S'}$ containing $Q_{S'}$, and put

$$M = \mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S}) \simeq \mathrm{Hom}_{S'\text{-gr}}(Q_{S'}, \mathbf{G}_{m,S'}),$$
$$M' = \mathrm{Hom}_{S'\text{-gr}}(Q', \mathbf{G}_{m,S'}).$$

The monomorphism $Q_{S'} \to Q'$ induces an epimorphism

$$u : M' \longrightarrow M.$$

Write

$$L = \underline{\mathrm{Centr}}_G(Q), \qquad L' = \underline{\mathrm{Centr}}_{G_{S'}}(Q').$$

Then $L' \subset L_{S'}$.

If H is a subgroup of G containing L, then $H_{S'}$ contains L', and

$$\mathrm{Lie}(H) = \mathrm{Lie}(L) \oplus \bigoplus_{\alpha \in R_H} \mathrm{Lie}(H)^{\alpha},$$
$$\mathrm{Lie}(H_{S'}) = \mathrm{Lie}(L') \oplus \bigoplus_{\alpha' \in R'_{H_{S'}}} \mathrm{Lie}(H_{S'})^{\alpha'}.$$

Here R_H, respectively $R'_{H_{S'}}$, denotes the set of roots of H, respectively $H_{S'}$, relative to Q, respectively Q'. It follows immediately that

$$R_H \subset u(R'_{H_{S'}}) \subset R_H \cup \{0\}.$$

Taking $H = G$, we first obtain

$$R \subset u(R') \subset R \cup \{0\}.$$

Next take for H a minimal parabolic subgroup P with Levi subgroup L. Then $R'_{H_{S'}}$ contains a system of positive roots in R'. By 7.4, there is therefore a minimal parabolic subgroup P' of $G_{S'}$, with Levi subgroup L', contained in $P_{S'}$. We have constructed the diagram of inclusions

$$\begin{array}{ccccc} Q_{S'} & \subset & L_{S'} & \subset & P_{S'} \\ \cap & & \cup & & \cup \\ Q' & \subset & L' & \subset & P'. \end{array}$$

If R^+, respectively Δ, is the system of positive, respectively simple, roots in R defined by P, and if R'^+, Δ' are defined similarly, then

$$R^+ \subset u(R'^+) \subset R^+ \cup \{0\}, \qquad \Delta \subset u(\Delta') \subset \Delta \cup \{0\}.$$

Now let

$$w \in W \simeq \underline{\mathrm{Norm}}_G(Q)(S)/\underline{\mathrm{Centr}}_G(Q)(S),$$

and choose a representative $n \in \underline{\mathrm{Norm}}_G(Q)(S)$. Since $\mathrm{int}(n)Q = Q$, one has $\mathrm{int}(n)L = L$, and hence $\mathrm{int}(n)L_{S'} = L_{S'}$. The tori Q' and $\mathrm{int}(n)Q'$ are therefore two maximal split tori of $L_{S'}$, so they are conjugate by an element $x \in L(S')$. We may choose x so that

$$\mathrm{int}(nx)Q' = Q'.$$

Thus $n' = nx$ belongs to $\underline{\mathrm{Norm}}_{G_{S'}}(Q')(S')$. Let w' be the image of n' in

$$W' \simeq \underline{\mathrm{Norm}}_{G_{S'}}(Q')(S')/\underline{\mathrm{Centr}}_{G_{S'}}(Q')(S').$$

The action of w' on M' is compatible with $u : M' \to M$, and the induced action on M is the action of w.

Using the definition of relative root data together with the conjugacy theorems gives the following result.

Theorem 7.13.— *Let S, S' be nonempty connected semilocal schemes, let $S' \to S$ be a morphism of schemes, let G be an S-reductive group, and write*

$$\mathcal{R}(G/S) = (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}^+),$$
$$\mathcal{R}(G_{S'}/S') = (\underline{M}', \underline{M}'^*, \underline{R}', \underline{R}'^*, \underline{R}'^+)$$

for the based relative root data. There is a canonical homomorphism

$$u : \underline{M}' \longrightarrow \underline{M}$$

with the following properties:

(i) *The homomorphism u is surjective.*

(ii) *For every $w \in W$, there is an element $w' \in W'$ compatible with u and inducing w on $\underline{M}$.*

(iii) *For every subset $X \subset \underline{M}$, put*

$$X^\wedge = X \cap (\underline{M} \setminus \{0\}).$$

Then

$$u(\underline{R}'^+)^\wedge = \underline{R}^+, \qquad u(\underline{\Delta}')^\wedge = \underline{\Delta}.$$

(iv) *For every parabolic subgroup H, consider*

$$t_r(H) \subset \underline{\Delta}, \qquad t_r(H_{S'}) \subset \underline{\Delta}'.$$

Then

$$\begin{aligned} t_r(H_{S'}) &= u^{-1}\big(t_r(H) \cup \{0\}\big) \cap \underline{\Delta}' \\ &= \left\{\alpha' \in \underline{\Delta}' \;\middle|\; u(\alpha') \in t_r(H) \text{ or } u(\alpha') = 0\right\}. \end{aligned}$$

Remark 7.14.— *If G is splittable over S, its maximal split tori are maximal tori, and the relative notions introduced here coincide with the previously defined absolute notions. The preceding theorem therefore describes the relative root datum $\mathcal{R}(G/S)$ and the relative type t_r in terms of the absolute root datum and absolute type of $G_{S'}$, where S' is chosen so that $G_{S'}$ is splittable* (cf. Exposé XXIV, 4.4.1). *See* [BT65], *6.12 and following, for this description.*

7.15.

Let S be a Henselian local scheme, s_0 its closed point, and S_0 the spectrum of the residue field at s_0, identified with a closed subscheme of S. For every object X over S, write X_0 for its base change to S_0. Let G be an S-reductive group.

For every parabolic subgroup $\overline{P}$ of G, the group $\overline{P}_0$ is a parabolic subgroup of G_0. Conversely, for every parabolic subgroup P of G_0, there is a parabolic subgroup $\overline{P}$ of G such that

$$\overline{P}_0 = P.$$

This follows from Hensel's lemma and from the fact that $\underline{\mathrm{Par}}(G)$ is a smooth S-scheme. In particular, by 5.7, $\overline{P}$ is minimal if and only if $\overline{P}_0$ is minimal.

Once such a subgroup $\overline{P}$ has been chosen, the same argument shows that every maximal split subtorus of $\overline{P}_0$ is of the form T_0, where T is a maximal split subtorus of $\overline{P}$. It follows readily that the relative root data of G over S and of G_0 over S_0 are canonically isomorphic. Thus the theory of parabolic subgroups of G reduces to the corresponding theory for G_0.

Conversely, every S_0-reductive group is of the form G_0, by Exposé XXIV, Proposition 1.21. This makes it possible to reduce the study of the parabolic subgroups of an S_0-reductive group to the corresponding study over S.

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Theory of Topoi and Étale Cohomology of Schemes. Volumes 1 to 3

M. Artin, A. Grothendieck, J.-L. Verdier

With the participation of N. Bourbaki, P. Deligne, B. Saint-Donat

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EXPOSÉ I

Presheaves

A. Grothendieck and J.-L. Verdier

In numbers 0 through 5 of this exposé, we present the elementary, and most often well-known, properties of categories of presheaves[i]. These properties are used constantly throughout the rest of the seminar, and knowledge of them is essential for understanding the subsequent exposés. The proofs are immediate; most often, they are omitted. In numbers 6 through 9, we take up a few themes used on various occasions later. The reader in a hurry may omit them on a first reading. Number 10 fixes the terminology used. Appendix 11 is due to N. Bourbaki. 1

0. Universes

A *universe* is a nonempty set[1] $\mathcal{U}$ having the following properties:

($\mathcal{U}$ 1) If $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$.
($\mathcal{U}$ 2) If $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$.
($\mathcal{U}$ 3) If $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$.
($\mathcal{U}$ 4) If $(x_i, i \in I)$ is a family of elements of $\mathcal{U}$ and $I \in \mathcal{U}$, then $\cup_{i\in I} x_i \in \mathcal{U}$.

From the preceding axioms one readily deduces the following properties: 2

- If $x \in \mathcal{U}$, the set $\{x\}$ belongs to $\mathcal{U}$.
- If x is a subset of $y \in \mathcal{U}$, then $x \in \mathcal{U}$.
- If $x, y \in \mathcal{U}$, the ordered pair $(x, y) = \{\{x, y\}, x\}$ (Kuratowski's definition) is an element of $\mathcal{U}$.
- If $x, y \in \mathcal{U}$, the union $x \cup y$ and the product $x \times y$ are elements of $\mathcal{U}$.
- If $(x_i, i \in I \in \mathcal{U})$ is a family of elements of $\mathcal{U}$, the product $\prod_{i\in I} x_i$ is an element of $\mathcal{U}$.
- If $x \in \mathcal{U}$, then $\mathrm{card}(x) < \mathrm{card}(\mathcal{U})$ (strictly). In particular, the relation $\mathcal{U} \in \mathcal{U}$ does not hold.

We can therefore carry out all the usual operations of set theory starting from the elements of a universe without thereby causing the final result to cease to be an element of the universe.

The first advantage of the notion of a universe is that it provides a definition of the usual categories: the category of sets belonging to the universe $\mathcal{U}$ ($\mathcal{U}$-Ens), the category of topological spaces belonging to the universe $\mathcal{U}$, the category of commutative groups belonging to the universe $\mathcal{U}$ ($\mathcal{U}$-Ab), the category of categories belonging to the universe $\mathcal{U}$....

However, the only known universe is the set of symbols of the form $\{\{\emptyset\}, \{\{\emptyset\}, \emptyset\}\}$ etc.. (all the elements of this universe are finite sets, and this universe is countable). In particular, no universe is known that contains an element of infinite cardinality. We are

[i] The reader may also consult SGA 3 I § § 1 to 3.

[1] N.D.E.: This definition contradicts Bourbaki's appendix *below*. However, the usefulness of empty universes seems debatable...

therefore led to add to the axioms of set theory the axiom: ($\mathcal{U}\,A$) *For every set x there exists a universe $\mathcal{U}$ such that $x \in \mathcal{U}$.*

3 Since the intersection of a family of universes is a universe, it follows immediately that every set is an element of a smallest universe. One can show that axiom ($\mathcal{U}\,A$) is independent of the axioms of set theory.

We shall also add the axiom:

($\mathcal{U}\,B$) *Let $R\{x\}$ be a relation and let $\mathcal{U}$ be a universe. If there exists an element $y \in \mathcal{U}$ such that $R\{y\}$, then $\tau_x R\{x\} \in \mathcal{U}$.*

The consistency of axioms ($\mathcal{U}\,A$) and ($\mathcal{U}\,B$) relative to the other axioms of set theory has not been proved and, it seems, cannot be proved.

Let $\mathcal{U}$ be a universe and let $c(\mathcal{U})$ be the supremum of the cardinalities of the elements of $\mathcal{U}$ ($c(\mathcal{U}) \leq \mathrm{card}(\mathcal{U})$). The cardinal $c(\mathcal{U})$ has the following properties:

($F\,I$) If $a < c(\mathcal{U})$, then $2^a < c(\mathcal{U})$.

($F\,I\,I$) If $(a_i, i \in I)$ is a family of cardinals strictly less than $c(\mathcal{U})$ and if $\mathrm{card}(I)$ is strictly less than $c(\mathcal{U})$, then $\Sigma_{i\in I} a_i < c(\mathcal{U})$.

The cardinals that have properties ($F\,I$) and ($F\,I\,I$) are called *strongly inaccessible cardinals.*

The cardinal 0 and the countably infinite cardinal are strongly inaccessible.

Axiom ($\mathcal{U}\,A$) implies:

($\mathcal{U}\,A'$) *Every cardinal is strictly less than a strongly inaccessible cardinal.*

Conversely, one can show (11) that the consistency of ($U\,A'$) implies the consistency of ($U\,A$), and that the consistency of these axioms entails that of axiom ($\mathcal{U}\,B$).

4 Call an *Artinian set* any set E for which there is no infinite family $(x_n, n \in N)$ such that $x_\circ \in E$, $x_{n+1} \in x_n$. One can then show [*loc. cit.*] that there is a one-to-one correspondence between the strongly inaccessible cardinals and the Artinian universes, defined as follows: to every strongly inaccessible cardinal c we associate the unique Artinian universe U_c such that

$$\mathrm{card}(\mathcal{U}_c) = c.$$

Let $c < c'$ be two strongly inaccessible cardinals. We have:

$$\mathcal{U}_c \in \mathcal{U}'_{c'}.$$

In particular, the Artinian universes of cardinality less than a given cardinal form a well-ordered set (with respect to the membership relation). Axiom ($\mathcal{U}\,A'$) is equivalent to the axiom:

($\mathcal{U}\,A''$). *Every Artinian set is an element of an Artinian universe.*

Observe that all the usual sets ($\mathbf{Z}, \mathbf{Q}, \ldots$) are Artinian sets.

1. $\mathcal{U}$-categories. Presheaves of sets

1.0. *Throughout the remainder of the seminar and unless expressly stated otherwise, the universes under consideration will contain an element of infinite cardinality.* Let $\mathcal{U}$ be a universe. A set is said to be *$\mathcal{U}$-small* (or, when no confusion can result, *small*) if it is *isomorphic* to an element of $\mathcal{U}$. We shall also use the terminology: small group, small ring, small category[2] ...We shall often assume, without explicit mention, that the
5 schemes, topological spaces, indexing sets...with which we work are $\in \mathcal{U}$, or at least have cardinality $\in \mathcal{U}$; nevertheless, many of the categories with which we shall work will not be $\in \mathcal{U}$.

[2] N.D.E.: A category C is regarded as a set of arrows (equipped with the subset of objects, regarded as the set of identity arrows, and with the "source, target" maps). Thus, the expressions "C is an element of $\mathcal{U}$" or "C is $\mathcal{U}$-small" make sense.

DEFINITION 1.1. *Let $\mathcal{U}$ be a universe and C a category. We say that C is a $\mathcal{U}$-category if, for every pair (x, y) of objects of C, the set $\mathrm{Hom}_C(x, y)$ is $\mathcal{U}$-small.*

1.1.1. Let C and D be two categories and let $\mathrm{Fonct}(C, D)$ be the category of functors from C to D. The following assertions are immediate:

a) If C and D are elements of a universe $\mathcal{U}$ (resp. are $\mathcal{U}$-small), the category $\mathrm{Fonct}(C, D)$ is an element of $\mathcal{U}$ (resp. is $\mathcal{U}$-small).
b) If C is $\mathcal{U}$-small and D is a $\mathcal{U}$-category, $\mathrm{Fonct}(C, D)$ is a $\mathcal{U}$-category.

REMARK 1.1.2. Let D be a category having the following properties:

(C1) The set $\mathrm{ob}(D)$ is contained in the universe $\mathcal{U}$.
(C2) For every pair (x, y) of objects of D, the set $\mathrm{Hom}_D(x, y)$ is an element of $\mathcal{U}$.

(The usual categories constructed from a universe $\mathcal{U}$ have these two properties: $\mathcal{U}$-Ens, $\mathcal{U}$-Ab,...). Let C be a category belonging to $\mathcal{U}$. Then the category $\mathrm{Fonct}(C, D)$ does not in general have properties (C1) and (C2). For example, the category $\mathrm{Fonct}(C, \mathcal{U}\text{-Ens})$ has neither property (C1) nor property (C2). This justifies the adopted definition of $\mathcal{U}$-category, in preference to the more restrictive notion given by conditions (C1) and (C2) above.

DEFINITION 1.2. *Let C be a category. The category of presheaves of sets on C relative to the universe $\mathcal{U}$ (or, when no confusion can result, the category of presheaves on C) is the category of contravariant functors on C with values in the category of $\mathcal{U}$-sets.* 6

We denote by $\widehat{C}_{\mathcal{U}}$ (or more simply, when no confusion can result, by $\widehat{C}$) the category of presheaves of sets on C relative to the universe $\mathcal{U}$. The objects of $\widehat{C}_{\mathcal{U}}$ are called $\mathcal{U}$-presheaves (or simply presheaves) on C. When C is $\mathcal{U}$-small, the category $\widehat{C}_{\mathcal{U}}$ is a $\mathcal{U}$-category. When C is a $\mathcal{U}$-category, $\widehat{C}_{\mathcal{U}}$ is not necessarily a $\mathcal{U}$-category.

CONSTRUCTION-DEFINITION 1.3. Let x be an object of a $\mathcal{U}$-category C. The *$\mathcal{U}$-functor represented by x* is the functor $h_{\mathcal{U}}(x) : C^\circ \to \mathcal{U}\text{-Ens}$ whose construction is as follows [2]. Let y be an object of C.

a) If $\mathrm{Hom}_C(y, x)$ is an element of $\mathcal{U}$, then:

$$h_{\mathcal{U}}(x)(y) = \mathrm{Hom}_C(y, x)$$

b) Suppose that $\mathrm{Hom}_C(y, x)$ is not an element of $\mathcal{U}$ and let $R(Z, x, y)$ be the relation: "The set Z is the codomain of an isomorphism $\mathrm{Hom}_C(y, x) \xrightarrow{\sim} Z$". We then set:

$$h_{\mathcal{U}}(x)(y) = \tau_Z R(Z).$$

(By axiom ($\mathcal{U}B$), $h_{\mathcal{U}}(x)(y)$ is an element of $\mathcal{U}$). Let $R'(u, x, y)$ be the relation: « u is a bijection

$$u : \mathrm{Hom}_C(y, x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y) ».$$

We then set: 7

$$\varphi(y, x) = \tau_u R'(u).$$

It should be noted that, in cases (a) and (b), we have *a* canonical isomorphism:

$$\varphi(y, x) : \mathrm{Hom}_C(y, x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y).$$

[2] C° will always denote the category opposite to C.

(In case a) $\varphi(y,x)$ is the identity). Now let $u : y \to y'$ be an arrow of C. By composition of morphisms, the morphism u defines a map:

$$\mathrm{Hom}_C(u,x) : \mathrm{Hom}_C(y',x) \longrightarrow \mathrm{Hom}_C(y,x).$$

We then set

$$h_{\mathscr{U}}(x)(u) = \varphi(y,x)\,\mathrm{Hom}_C(x,u)\varphi(y,x)^{-1}.$$

It is immediate that $h_{\mathscr{U}}(x)$, so defined, is a functor $\widehat{C} \to \mathscr{U}$-Ens.

1.3.1. Similarly, using the isomorphisms φ, for every morphism $y : x \to x'$ we define a morphism of functors:

$$h_{\mathscr{U}}(y) : h_{\mathscr{U}}(x) \longrightarrow h_{\mathscr{U}}(x')$$

and it is immediate that this defines a functor

$$h_{\mathscr{U}} : C \longrightarrow \widehat{C}_{\mathscr{U}}.$$

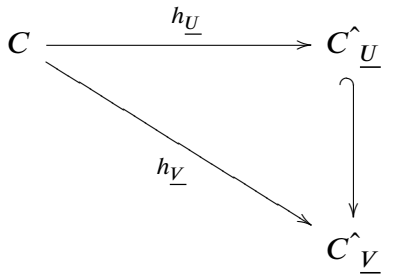
1.3.2. Now let $\mathscr{V}$ be a universe such that $\mathscr{U} \subset \mathscr{V}$. There is then a canonical fully faithful functor:

$$\mathscr{U}\text{-Ens} \hookrightarrow \mathscr{V}\text{-Ens}$$

whence a fully faithful functor:

$$\widehat{C}_{\mathscr{U}} \hookrightarrow \widehat{C}_{\mathscr{V}}$$

8 and the diagram:



commutes up to canonical isomorphism. When C is an element of $\mathscr{U}$, this canonical isomorphism is the identity.

1.3.3. *In practice, the universe $\mathscr{U}$ is fixed once and for all and is not mentioned.* We then use the notations $\widehat{C}$ (for the category of $\mathscr{U}$-presheaves of sets) and

$$h : C \longrightarrow \widehat{C}.$$

For every object x of C, the presheaf $h(x)$ is called the *presheaf represented by x*, and we shall always identify the value at $y \in \mathrm{ob}(C)$ of the presheaf $h(x)$ with $\mathrm{Hom}_C(y,x)$.

Proposition 1.4. *Let C be a $\mathscr{U}$-category, F a presheaf on C, and X an object of C. There is an isomorphism functorial in X and in F:*

$$i : \mathrm{Hom}_{\widehat{C}}(h(X),F) \xleftarrow{\sim} F(X),$$

where the map i, when F is of the form $h(Y)$, is none other than the map:

$$h : \mathrm{Hom}(h(X),h(Y)) \longleftarrow \mathrm{Hom}(X,Y).$$

In particular, the functor h is fully faithful.

1.4.1. This proposition justifies the customary abuse of language identifying an ob-
9 ject of C with the corresponding contravariant functor. A presheaf isomorphic to an object in the image of h (or, using the abuse of language noted above, isomorphic to an object of C) is called a *representable presheaf.*

1.4.2. Let $\mathscr{V}$ be a universe containing the universe $\mathscr{U}$. The category of $\mathscr{U}$-presheaves is a *full* subcategory of the category of $\mathscr{V}$-presheaves, and consequently a $\mathscr{U}$-presheaf is representable if and only if its image in the category of $\mathscr{V}$-presheaves is a representable $\mathscr{V}$-presheaf.

2. Projective and inductive limits

Let C be a $\mathscr{U}$-category and I a small category. Denote by $\mathrm{Fonct}(I, C)$ the category of functors from I to C. The category $\mathrm{Fonct}(I, C)$ is a $\mathscr{U}$-category. To every object X of C, associate the subcategory $\mathscr{X}$ of C whose only object is X and whose only arrow is the identity of X. Let $i_X : \mathscr{X} \to C$ denote the inclusion functor. There is a unique functor $e_X : I \to \mathscr{X}$, and we denote by $k_X : I \to C$ the functor $i_X \circ e_X$. (We say that k_X is the constant functor with value X.) The correspondence $X \mapsto k_X$ is clearly functorial in X, which allows us to define, for every functor $G : I \to C$, the presheaf $X \mapsto \mathrm{Hom}_{\mathrm{Fonct}(I,C)}(k_X, G)$.

Definition 2.1. *The projective limit of G, denoted by $\varprojlim_I G$, is the presheaf:*

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fonct}(I,C)}(k_X, G).$$

When the presheaf $\varprojlim_I G$ is representable, we also denote by $\varprojlim_I G$ an object of C rep- 10
resenting it. Thus the object $\varprojlim_I G$ is defined only up to isomorphism. When no confusion can result, we use the abbreviated notation $\varprojlim G$.

As G varies, $\varprojlim G$ is a functor from the category $\mathrm{Fonct}(I, C)$ to $\widehat{C}$.

2.1.1. By symmetry (reversing the direction of the arrows in C), one similarly defines the inductive limit of a functor: it is a covariant functor on C with values in the category of $\mathscr{U}$-sets. We shall use the notations $\varinjlim_I G$ or $\varinjlim G$.

Observe that products, fiber products, and kernels are projective limits. Similarly, sums, amalgamated sums, and cokernels are inductive limits.

Definition 2.2. *Let I and C be two categories and $G : I \to C$ a functor. The projective limit of G is said to be representable if there exists a universe $\mathscr{U}$ such that:*

1) *The category I is $\mathscr{U}$-small.*
2) *The category C is a $\mathscr{U}$-category.*
3) *The presheaf $\varprojlim G$, with values in the category of $\mathscr{U}$-sets, is representable.*

It follows from n° 1 that the object $\varprojlim G$ representing the presheaf $\varprojlim G$ is independent, up to isomorphism, of the universe $\mathscr{U}$. Note also that there exists a smallest universe $\mathscr{U}'$ having properties (1) and (2), and that the presheaf $\varprojlim G$ necessarily takes values in the category of $\mathscr{U}'$-sets.

2.2.1. Let I and C be two categories. We say that *the I-projective limits in C are* 11
representable if, for every functor $G : I \to C$, the projective limit of G is representable. Finally, let C be a category and $\mathscr{U}$ a universe. We say that the *$\mathscr{U}$-projective limits in C are representable* if, for every category I that is $\mathscr{U}$-small and every functor $G : I \to C$, the projective limit of G is representable.

Proposition 2.3. *Let C be a category and $\mathscr{U}$ a universe. The following assertions are equivalent:*

i) *The $\mathcal{U}$-projective limits in C are representable.*
ii) *Products indexed by a small set are representable, and kernels of pairs of arrows are representable.*
iii) *Products indexed by a small set are representable, and fiber products are representable.*

PROOF. It suffices to observe that there is an isomorphism, functorial in G,

$$\varprojlim G = \mathrm{Ker}\Big(\prod_{i\in \mathrm{ob}(I)} G(i) \rightrightarrows \prod_{u\in \mathrm{F\ell(I)}} G(\mathrm{but}(u))\Big),$$

where the pair of arrows is defined by the morphisms

$$\prod_{i\in \mathrm{Ob}(I)} G(i) \xrightarrow{\mathrm{pr}_{\mathrm{but}(u)}} G(\mathrm{but}(u))$$

$$\prod_{i\in \mathrm{Ob}(I)} G(i) \xrightarrow{\mathrm{pr}_{\mathrm{source}(u)}} G(\mathrm{source}\ (u)) \xrightarrow{G(u)} G(\mathrm{but}(u)).$$

Moreover, it is clear that $\mathrm{Ker}(X \overset{u}{\underset{v}{\rightrightarrows}} Y) = X \prod_{X\times Y} X$, where the two morphisms from X to $X \times Y$ are $\mathrm{id}_X \times u$ and $\mathrm{id}_X \times v$.

2.3.1. There are of course analogous definitions and assertions for inductive limits,
12 which we shall not spell out. The same applies to finite projective and inductive limits (i.e. those relative to a finite category I).

COROLLARY 2.3.2. *Denote by $\mathcal{U}$-*Ens *the category of $\mathcal{U}$-sets and by $\mathcal{U}$-*Ab *the category of abelian group objects in $\mathcal{U}$-*Ens*. The $\mathcal{U}$-projective and inductive limits in $\mathcal{U}$-*Ens *and in $\mathcal{U}$-*Ab *are representable.*

PROPOSITION 2.3.3. *Let I be a small category, $F : I \to \mathcal{U}$-*Ens *a functor, and $G \in \mathrm{ob}\ I$ a small set of objects of I such that, for every object X of I, there exists an object $Y \in G$ and a morphism $Y \to X$ (resp. $X \to Y$). Then $\varprojlim_I F$ (resp. $\varinjlim_I F$) is representable, and $\mathrm{card}(\varprojlim_I F) \leq \Pi_{Y\in G} \mathrm{card}(F(Y))$ (resp. $\mathrm{card}(\varinjlim_I F) \leq \Sigma_{Y\in G} \mathrm{card}(F(Y))$).*

PROOF. One immediately verifies that $\varprojlim_I F'$ (resp. $\varinjlim_I F'$) is isomorphic to a subobject (resp. a quotient) of $\Pi_{Y\in G} F(Y)$ (resp. $\amalg_{Y\in G} F(Y)$); hence the proposition.

DEFINITION 2.4.1. *Let C be a category in which finite projective (resp. inductive) limits are representable, and let $u : C \to C'$ be a functor. We say that u is left exact (resp. right exact) if it "commutes" with finite projective (resp. inductive) limits. A functor that is both left and right exact is called an exact functor.*

2.4.2. It follows from 2.3 that, for a functor to be exact on the left, it is necessary and sufficient that it take the terminal object (= empty product) to the terminal object, the product of two objects to the product of their images, and the kernel of a pair of arrows to the kernel of the image pair; equivalently, that it take the terminal object to the terminal object and fiber products to fiber products (we assume that in C finite $\varprojlim$ are representable).

13 2.5.0. Let I, J, and C be three categories, and let $G : I \times J \to C$ be a functor (i.e. a functor from I with values in the category of functors from J to C). Suppose that the projective limits of the functors

$$G_i : J \longrightarrow C \qquad i \in \mathrm{ob}(I)$$
$$j \longmapsto G(i \times j)$$

are representable, and that the functor

$$i \longmapsto \varprojlim G_i$$

admits a representable projective limit. It is then clear that the functor G admits a representable projective limit and that there is a canonical isomorphism

$$\varprojlim_{I \times J} G \xrightarrow{\sim} \varprojlim_I \varprojlim_J G_i,$$

and consequently a canonical isomorphism

$$\varprojlim_I \varprojlim_J G_i \xrightarrow{\sim} \varprojlim_J \varprojlim_I G_j.$$

Hereafter we shall say more briefly that projective limits commute with projective limits. Similarly, inductive limits commute with inductive limits. But in general it is not true that inductive limits commute with projective limits.

Definition 2.5. *Let C be a category with representable fiber products, I a category, $G : I \to C$ a functor, $g : G \to X$ a morphism from G to an object of C (i.e. a morphism from G to the constant functor associated with X), and $m : Y \to X$ a morphism in C. Let* 14
$G \times_X Y : I \to C$ *be the functor*

$$i \longmapsto G(i) \times_X Y.$$

We say that the inductive limit of G is universal if, for every object X, every morphism $g : G \to X$, and every morphism $m : Y \to X$,

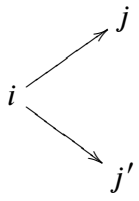
a) *the inductive limit of the functor $G \times_X Y$ is representable,*
b) *the canonical morphism $\varinjlim(G \times_X Y) \to (\varinjlim G) \times_X Y$ is an isomorphism.*

Proposition 2.6. *Let $\mathcal{U}$ be a universe. The inductive limits in $\mathcal{U}$-*Ens *that are representable (in particular, the $\mathcal{U}$-inductive limits (2.2.1) in $\mathcal{U}$-*Ens*) are universal.*

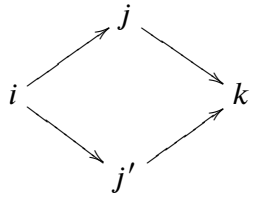
We shall also use another commutation result for projective and inductive limits, which we now present.

Definition 2.7. *A category I is pseudo-filtered if it has the following properties:*

PS 1) *Every diagram of the form*



can be embedded in a commutative diagram



PS 2) *Every diagram of the form*

$$i \underset{v}{\overset{u}{\rightrightarrows}} j$$

15 *can be embedded in a diagram*

$$i \underset{v}{\overset{u}{\rightrightarrows}} j \xrightarrow{w} k$$

such that

$$w \circ u = w \circ v.$$

A category I is said to be filtered if it is pseudo-filtered, nonempty, and connected, i.e. if any two objects of I can be joined by a sequence of arrows (with no condition imposed on the directions of the arrows); in the presence of PS 2*, this also means that $I \neq \emptyset$ and that, for any two objects a, b of I, there is always an object c of I and arrows $a \to c$ and $b \to c$. We also say that a category I is cofiltered if I° is filtered.*

Example 2.7.1. *If, in I, amalgamated sums (resp. sums of two objects) and cokernels of pairs of parallel arrows are representable, then I is pseudo-filtered (resp. filtered).*

Proposition 2.8. *Let $\mathscr{U}$ be a universe. Filtered $\mathscr{U}$-inductive limits in $\mathscr{U}$-*Ens *commute with finite projective limits.*

We are immediately reduced to proving that filtered $\mathscr{U}$-inductive limits commute with fiber products. The proof is left to the reader. One may use the description of the limit given by the following

Lemma 2.8.1. *Let I be a small filtered category and $i \mapsto X_i$ a functor from I to $\mathscr{U}$-*Ens*. On the disjoint union $\coprod_{i \in \mathrm{ob}\, I} X_i$, let R be the relation:*

(R) *Two elements $x_i \in X_i$ and $x_j \in X_j$ are related if there exists an object $k \in \mathrm{ob}\, I$ and two morphisms $u : i \to k$ and $v : j \to k$ such that the images in X_k of x_i and x_j under the transition maps u and v, respectively, are equal.*

16 *Then:*

1) *R is an equivalence relation.*
2) *The quotient $\coprod_{i \in \mathrm{ob}\, I} X_i / R$ is canonically isomorphic to $\varinjlim_I X_i$.*
3) *Two elements $x_i \in X_i$ and $x_j \in X_j$ are respectively equivalent under R to two elements of the same X_k.*
4) *For every $i \in \mathrm{ob}\, I$, two elements α and β of X_i are equivalent under R if and only if there is a morphism $u : i \to j$ such that $u(\alpha) = u(\beta)$.*

Assertion 1) follows from (PS 1) (2.7). To prove 2), one checks that $\coprod_{i \in \mathrm{ob}\, I} X_i / R$ has the universal property of the inductive limit (only (PS 1) is used). Assertion 3) follows from the fact that I is connected. Assertion 4) follows from (PS 2).

COROLLARY 2.9. *Let γ be a kind of algebraic structure "defined by finite projective limits." (The reader is asked to give a mathematical meaning to the preceding phrase. We merely note that the structures of groups, abelian groups, rings, modules, etc. are structures of this kind.) Denote by $\mathscr{U}$-γ the category of γ-objects in $\mathscr{U}$-*Ens. *The functor that associates with each object of $\mathscr{U}$-γ its underlying set commutes with filtered $\mathscr{U}$-inductive limits. Consequently, filtered $\mathscr{U}$-inductive limits in $\mathscr{U}$-γ commute with finite projective limits.*

COROLLARY 2.10. *Pseudo-filtered $\mathscr{U}$-inductive limits in $\mathscr{U}$-*Ab *commute with finite projective limits.*

(One reduces to filtered limits by decomposing the indexing category into connected components.)

We now record a result that we shall use constantly: let C and C' be two $\mathscr{U}$-categories, and let u and v be two functors $C \underset{u}{\overset{v}{\leftrightarrows}} C'$, with u *left adjoint* to v. Recall that this means that there is an isomorphism between the bifunctors with values in $\mathscr{U}$-Ens: 17

$$\mathrm{Hom}_{C'}(u(X), X') \xrightarrow{\sim} \mathrm{Hom}_{C}(X, v(X')).$$

PROPOSITION 2.11. *The functor u commutes with representable inductive limits; the functor v commutes with representable projective limits.*

This assertion means that, for every category I and every functor $G : I \to C'$ such that the projective limit of G is representable, the functor $v \circ G$ has a representable projective limit and there is a canonical isomorphism

$$v(\varprojlim G) \longrightarrow \varprojlim(v \circ G).$$

The reader may spell out the assertion concerning the functor u.

Finally, we record a computation of projective limits in the case where the indexing category has products.

PROPOSITION 2.12. *Let I and C be two categories, and $\mathscr{V}$ a universe. Assume that the $\mathscr{V}$-projective limits in C are representable and that there is in I a family of objects $(i_\alpha)_{\alpha \in A}$, where A is an element of $\mathscr{V}$, such that:*

1) *the products $i_\alpha \times i_\beta$ are representable for every pair $(\alpha, \beta) \in A \times A$,*
2) *every object of I admits a morphism to at least one of the i_α.*

Then, for every contravariant functor F from I to C,

$$F : I^\circ \longrightarrow C,$$

the projective limit of F exists, and there is an isomorphism functorial in F:

$$\varprojlim F \xrightarrow{\sim} \mathrm{Ker}\left(\prod_{\alpha \in A} F(i_\alpha) \rightrightarrows \prod_{(\alpha,\beta) \in A \times A} F(i_\alpha \times i_\beta)\right),$$

where the two arrows are defined by the projections of the products $i_\alpha \times i_\beta$ onto their factors. 18

3. Exactness properties of the category of presheaves

Let C be a $\mathscr{U}$-category and let $\widehat{C}$ be the category of presheaves on C. All the exactness properties of $\widehat{C}$ follow from the following proposition:

PROPOSITION 3.1. *The $\mathcal{U}$-projective and inductive limits in $\widehat{C}$ are representable. For every object X of C, the functor on $\widehat{C}$*

$$F \longmapsto F(X) \qquad F \in \widehat{C}$$

commutes with inductive and projective limits.

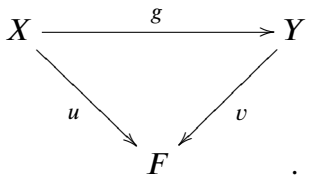
In other words, "inductive and projective limits in $\widehat{C}$ are computed argument by argument." We mention a few corollaries.

COROLLARY 3.2. *Let γ be an algebraic structure defined by finite projective limits. The category of contravariant functors with values in $\mathcal{U}$-γ is equivalent to the category of γ-objects of $\widehat{C}$.*

COROLLARY 3.3. *A morphism of $\widehat{C}$ that is both a monomorphism and an epimorphism is an isomorphism. A morphism of $\widehat{C}$ factors uniquely as an epimorphism followed by a monomorphism. The representable inductive limits in $\widehat{C}$ are universal. The filtered $\mathcal{U}$-inductive limits in $\widehat{C}$ commute with finite projective limits. The canonical functor $C \to \widehat{C}$ commutes with representable projective limits.*

19 More generally, one may say that the category $\widehat{C}$ inherits all the properties of the category of $\mathcal{U}$-sets that involve inductive and projective limits.

3.4.0. Let F be an object of $\widehat{C}$. We denote by C/F the following category. The objects of C/F are pairs consisting of an object X of C and a morphism u from X to F. Let (X,u) and (Y,v) be two objects. A morphism from (X,u) to (Y,v) is a morphism g from X to Y such that the following diagram is commutative:



PROPOSITION 3.4. *With the notation of (3.4.0), the source functor $C/F \to \widehat{C}$ admits a representable inductive limit in $\widehat{C}$. The canonical morphism*

$$\varinjlim_{C/F} \text{source}\,(.) \longrightarrow F$$

is an isomorphism.

COROLLARY 3.5. *Let F and H be two objects of $\widehat{C}$. There is a canonical isomorphism*

$$\mathrm{Hom}(F,H) \xrightarrow{\sim} \varprojlim_{(X,u)\in \mathrm{ob}(C/F)} H(X).$$

PROOF. The corollary follows immediately from 3.4 and 1.2.

20 PROPOSITION 3.6. *Let C be a $\mathcal{U}$-category, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, and let $i : \widehat{C}_{\mathcal{U}} \to \widehat{C}_{\mathcal{V}}$ be the natural inclusion functor of the corresponding categories of presheaves (1.3). The functor i commutes with inductive and projective limits. Moreover, for every object F of $\widehat{C}_{\mathcal{U}}$, every subobject of $i(F)$ is isomorphic to the image under i of a unique subobject of F (thus giving a bijection from the set of subobjects of F to the set of subobjects of $i(F)$).*

4. Sieves

DEFINITION 4.1. *Let C be a category. A* sieve *in the category C is a full subcategory D of C with the following property: every object of C from which there exists a morphism to an object of D belongs to D. If X is an object of C, the sieves of X are (by an abuse of language) the sieves in the category C/X.*

Let $\mathscr{U}$ be a universe such that C is a $\mathscr{U}$-category. Let $\widehat{C}$ be the corresponding category of presheaves. To every sieve of X one associates a subobject of X in $\widehat{C}$ as follows: to every object Y of C, one assigns the set of morphisms $f : Y \to X$ such that the object (Y, f) belongs to the sieve.

PROPOSITION 4.2. *The map defined above establishes a bijection between the set of sieves of X and the set of subobjects of X in $\widehat{C}$.*

PROOF. We shall only describe what is the inverse map. To every subfunctor R of X we associate the category C/R of objects of C over R (3.4). One immediately checks that C/R is a sieve of X.

REMARK 4.2.1. One sees likewise that the sieves of C are in canonical bijective correspondence with the set of subfunctors of the "terminal functor" on C (the "terminal" object of $\widehat{C}$). 21

4.3. Let C be a $\mathscr{U}$-category. By an abuse of language, we shall also call the subobjects of X in the category $\widehat{C}$ the sieves of X. This abuse of language allows us, for every presheaf F and every sieve R of X, to define $\mathrm{Hom}_{\widehat{C}}(R, F)$ as the set of morphisms from the functor R to F. There is moreover a canonical isomorphism, functorial in F (3.5):

$$\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_{C/R} F(.),$$

which also gives a direct definition[3]. Likewise, Proposition 4.2 allows us to transfer the usual operations on functors to sieves. We mention the following:

4.3.1. *Base change.* Let R be a sieve of X and let $f : Y \to X$ be a morphism of objects of C. The fiber product $R \times_X Y$ is a sieve of Y, called the *sieve deduced from R by base change.* The corresponding subcategory of C/Y is the inverse image of the subcategory of C/X defined by R under the canonical functor $C/Y \to C/X$ defined by f.

4.3.2. *Order relation, intersection, union.* The inclusion relation on the subfunctors of X is an order relation. The union and intersection of a family of sieves indexed by an arbitrary set may be defined as the least upper bound and the greatest lower bound of the family of corresponding subpresheaves.

4.3.3. *Image, generated sieve.* Let $(F_\alpha)_{\alpha \in A}$ be a family of presheaves and, for every $\alpha \in A$, let $f_\alpha : F_\alpha \to X$ be a morphism, where X is an object of C. The *image* of this family of morphisms is the union of the images of the f_α. The image of this family is therefore a sieve of X. In particular, if all the F_α are objects of C, the image sieve will be called the sieve generated by the morphisms f_α. The category C/R is the full subcategory of C/X formed by the objects $X' \to X$ over X for which there exists an X-morphism from X' to one of the F_α. 22

[3]N.D.E.: More simply, C/R is identified with R, so that one has the formula $\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_R F$, where, by an abuse of notation, F denotes the composite of F and the tautological "source" functor $R \to C/X \to C$.

The reader may, as an exercise, translate the relations and operations defined here on subfunctors into the language of the categories C/R. One will then observe that these relations and operations do not depend on the universe $\mathscr{U}$ for which C is a $\mathscr{U}$-category, and that they are therefore defined for every category, without having to specify the universe to which the morphism sets belong; this could moreover have been foreseen *a priori* from 3.6.

5. Functoriality of categories of presheaves

5.0. Let C, C', and D be three categories, and let $u : C \to C'$ be a functor. We denote by u^* the functor

$$u^* : \begin{cases} \mathbf{Hom}(C'^\circ, D) & \longrightarrow & \mathbf{Hom}(C^\circ, D) \\ G & \longmapsto & G \circ u \end{cases}$$

obtained by composition with the functor u. The functor u^* commutes with inductive and projective limits.

23 PROPOSITION 5.1. *Suppose that C is small and that the $\mathscr{U}$-inductive (resp. projective) limits in D are representable. The functor u^* has a left adjoint $u_!$ (resp. a right adjoint u_*). Thus there is an isomorphism*

$$\mathrm{Hom}_{\mathbf{Hom}(C^\circ,D)}(F, u^*G) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Hom}(C'^\circ,D)}(u_!F, G)$$
$$(\textit{resp.}\ \mathrm{Hom}_{\mathbf{Hom}(C^\circ,D)}(u^*G, F) \simeq \mathrm{Hom}_{\mathbf{Hom}(C'^\circ,D)}(G, u_*F)).$$

PROOF. We shall give only the proof of the existence of the left adjoint functor. The resp. part of the proposition will then follow formally from the isomorphisms

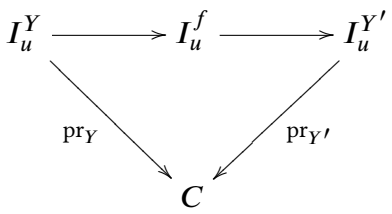
$$\mathbf{Hom}(C^\circ, D) \xrightarrow{\sim} \mathbf{Hom}(C, D^\circ)^\circ \xrightarrow{\sim} \mathbf{Hom}((C^\circ)^\circ, D^\circ)^\circ.$$

Let Y be an object of C'. Denote by I_u^Y the following category. The objects of I_u^Y are pairs (X, m), where X is an object of C and m is a morphism $Y \to u(X)$. Let (X, m) and (X', m') be two objects of I_u^Y. A morphism from (X, m) to (X', m') is a morphism $\xi : X \to X'$ such that $m' = u(\xi)m$. Composition of morphisms is defined in the evident way.

Let $f : Y \to Y'$ be a morphism of C'. By composition, the morphism f defines a functor $I_u^f : I_u^Y \to I_u^{Y'}$.

There is moreover a functor $\mathrm{pr}_Y : I_u^Y \to C$ that assigns the object X to the object (X, m). Notice that the diagram

(∗)



is then commutative.

24 Now let F be a presheaf on C, and set

(5.1.1) $$u_!F(Y) = \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y(.).$$

The commutativity of diagram (*) and the functoriality of the inductive limit make $u_!F$ a presheaf on C'. Let us show that the functor $u_!$ is a left adjoint of the functor u^*. For this, we show that for every presheaf G on C', there is a functorial isomorphism

$$\mathrm{Hom}(u_!F, G) \xrightarrow{\sim} \mathrm{Hom}(F, u^*G).$$

Let $\xi \in \mathrm{Hom}(u_!F, G)$. Thus, for every object X of C, there is a morphism

$$\xi_X : u_!(u(X)) \longrightarrow G(u(X)).$$

But $(u(X), \mathrm{id}_{u(X)})$ is an object of $I_u^{u(X)}$, and by the definition of the inductive limit there is a canonical morphism

$$F(X) \longrightarrow u_!F(u(X)).$$

For every object X of C, we therefore obtain a morphism

$$\eta_X : F(X) \longrightarrow G(u(X))$$

which is visibly functorial in X. Hence a morphism

$$\eta : F \longrightarrow u^*G.$$

Conversely, let $\eta \in \mathrm{Hom}(F, u^*G)$. For every object Y of C, this gives a morphism of functors

$$\eta_Y : F \circ \mathrm{pr}_Y \longrightarrow (u^*G)\,\mathrm{pr}_Y\,,$$

and hence, by composition with the evident morphism from the functor $(u^*G)\,\mathrm{pr}_Y$ to the constant functor $G(Y)$, a morphism

$$\xi_Y : u_!F(Y) \longrightarrow G(Y)$$

which is functorial in Y. Hence a morphism

$$\xi : u_!F \longrightarrow G.$$

The reader will check that the two maps thus defined are inverse to one another, thereby completing the proof. 25

Proposition 5.2. *Suppose that in D the $\mathscr{U}$-inductive limits are representable, finite projective limits are representable, and filtered $\mathscr{U}$-inductive limits commute with finite projective limits. Suppose also that finite projective limits are representable in C and that the functor u is left exact* (2.3.2). *Then finite projective limits are representable in* $\mathbf{Hom}(C^\circ, D)$ *and in* $\mathbf{Hom}(C'^\circ, D)$, *and the functor $u_!$ is left exact.*

Proof. The first assertion is trivial. Let us prove the second. By the proof of 5.1, for every presheaf F on C and every object Y of C', one has

$$u_!F(Y) \xrightarrow{\sim} \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y\,.$$

It therefore suffices to show that the category $(I_u^Y)^\circ$ satisfies axioms (PS 1) and (PS 2) (2.7), and that this category is connected. The verification is left to the reader.

5.3. Specializing these results to the case where D is the category of $\mathscr{U}$-sets, one obtains a sequence of three functors:

$$u_!, u^*, u_*,$$

which is a "sequence of adjoint functors" in the sense that for any two consecutive functors in the sequence, the one on the right is right adjoint to the other. Their essential 26
properties are summarized in the following proposition:

PROPOSITION 5.4. *Let C be a small category, C' a $\mathscr{U}$-category, and $u : C \to C'$ a functor.*

1) *The functor $u^* : \widehat{C'} \to \widehat{C}$ commutes with inductive and projective limits.*
2) *The functor $u_* : \widehat{C} \to \widehat{C'}$ commutes with projective limits. For every presheaf F on C and every object Y of C', one has:*

$$u_*F(Y) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{C}}(u^*(Y), F).$$

3) *The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with inductive limits. The functor $u_!$ is defined only up to isomorphism, but it can always be chosen so that the diagram*

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h'} \\ \widehat{C} & \xrightarrow{u_!} & \widehat{C'} \end{array}$$

(h and h' are the canonical inclusion functors) is commutative.
For every presheaf F on C, one has:

$$u_!F \xrightarrow{\sim} \varinjlim_{C/F} h' \circ u.$$

4) *If finite projective limits are representable in C and if u is left exact (2.3.2), the functor $u_!$ is left exact.*

PROOF. Assertion (1) is trivial. Assertion (2) follows from the fact that u_* is a right adjoint functor, by (2.11) and (1.4). The same is true of assertion (3), but one also applies (3.4). Finally, assertion (4) is simply (5.2), which can be applied thanks to (2.7).

27 PROPOSITION 5.5. *Let C and C' be two small categories and let $C \underset{u}{\overset{v}{\leftrightarrows}} C'$ be a pair of functors, where v is left adjoint to u. There are then isomorphisms, compatible with the adjunction isomorphisms:*

$$v^* \xrightarrow{\sim} u_!$$

$$v_* \xrightarrow{\sim} u^* \quad .$$

PROOF. It suffices to exhibit an isomorphism $v^* \xrightarrow{\sim} u_!$; the other isomorphism follows from it by adjunction. Let F be a presheaf on C and let Y be an object of C'. Then:

$$v^*F(Y) \xrightarrow{\sim} \mathrm{Hom}(v(Y), F).$$

Next, using (3.4):

$$\mathrm{Hom}(v(Y), F) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(v(Y), .).$$

But v is left adjoint to u, and hence:

$$\varinjlim_{C/F} \mathrm{Hom}(v(Y), .) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(Y, u(.)).$$

Now using (5.4.3)), one obtains:

$$\varinjlim_{C/F} \mathrm{Hom}(Y, u(.)) \xrightarrow{\sim} \mathrm{Hom}(Y, u_! F) \xrightarrow{\sim} u_! F(Y).$$

Thus, for every object Y of C', we have determined an isomorphism $v^* F(Y) \xrightarrow{\sim} u_! F(Y)$ that is clearly functorial in Y and in F, C.Q.F.D.

Corollary 5.5.1. *Let $u : C \to C'$ be a functor that admits a left adjoint. The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with projective limits (recall that it commutes with inductive limits by (1.4.3)).*

Remark 5.5.2. One thus obtains a "sequence of *four* adjoint functors" (cf. 5.3): 28

$$v_!, v^* = u_!, v_* = u^*, u_*,$$

of which the first three (resp. the last three) therefore commute with $\varinjlim$ (resp. $\varprojlim$).

Proposition 5.6. *The hypotheses are those of 5.4. The following conditions are equivalent:*

i) *The functor u is fully faithful.*
ii) *The functor $u_!$ is fully faithful.*
iii) *The adjunction morphism $\mathrm{id}_{\widehat{C}} \to u^* u_!$ is an isomorphism.*
iv) *The functor u_* is fully faithful.*
v) *The adjunction morphism $u^* u_* \to \mathrm{id}_{\widehat{C}}$ is an isomorphism.*

Proof. It is clear that ii) $\Leftrightarrow$ iii) and iv) $\Leftrightarrow$ v) (general properties of adjoint functors), and that ii) $\Rightarrow$ i) (5.4.3)). Let us show that i) $\Rightarrow$ iii). The functors $\mathrm{id}_{\widehat{C}}$, u^*, and $u_!$ commute with inductive limits. By (3.4), it therefore suffices to prove that $H \to u^* u_! H$ is an isomorphism when H is representable, which is evident. Let us show that iii) is equivalent to v). For every object H (resp. K) of $\widehat{C}$, denote by $\Phi(H) : H \to u^* u_! H$ (resp. by $\Psi(K) : u^* u_* \to K$) the adjunction morphism. We then have a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Hom}_{\widehat{C}}(H, u^* u_* K) & \xrightarrow{\mathrm{Hom}(H, \Psi(K))} & \mathrm{Hom}_{\widehat{C}}(H, K) \\
\wr \downarrow & \nearrow_{\mathrm{Hom}(\Phi(H), K)} & \\
\mathrm{Hom}_{\widehat{C}}(u_! H, u_* K) & & \\
\wr \downarrow & & \\
\mathrm{Hom}_{\widehat{C}}(u^* u_! H, K) & &
\end{array} .$$

It follows that $\Phi(H)$ is an isomorphism for every H if and only if $\Psi(\mathscr{K})$ is an isomorphism for every K, C.Q.F.D. 29

Remark 5.7. a) The equivalences ii) $\Leftrightarrow$ iii) $\Leftrightarrow$ iv) $\Leftrightarrow$ v) are general results on adjoint functors.

b) The explicit form of $u_!$ resp. u_* given in the proof of 1.4 immediately shows that, under the general hypotheses of 1.1, if u is fully faithful, then the adjunction morphism $\mathrm{id} \to u^* u_!$ (resp. $u^* u_* \to \mathrm{id}$) is an isomorphism, i.e. that $u_!$ (resp. u_*) is fully faithful.

5.8.0. Let γ be a type of algebraic structure defined by finite projective limits, let $\mathscr{U}$-γ-Ens be the category of γ-objects of $\mathscr{U}$-Ens, and let esj: $\mathscr{U}$-γ-Ens $\to$ $\mathscr{U}$-Ens be the underlying-set functor (for simplicity, we suppose that the type of structure under consideration has only one underlying set). Let C be a category. Composition with esj gives a functor denoted by

$$\hat{\text{esj}} : \mathbf{Hom}(C^\circ, \mathscr{U}\text{-}\gamma\text{-Ens}) \longrightarrow \widehat{C}.$$

Since projective limits in $\widehat{C}$ are computed argument by argument, the functor $\hat{\text{esj}}$ factors as an equivalence

$$\mathbf{Hom}(C^\circ, \mathscr{U}\text{-}\gamma\text{-Ens}) \xrightarrow{\approx} \widehat{C}_\gamma, \tag{5.8.1}$$

where $\widehat{C}_\gamma$ is the category of γ-objects of $\widehat{C}$, followed by a functor, still denoted by

$$\hat{\text{esj}} : \widehat{C}_\gamma \longrightarrow \widehat{C},$$

and called the "*underlying presheaf of sets*" functor.

30 5.8.2. Suppose that the functor esj : $\mathscr{U}$-γ-Ens $\to$ $\mathscr{U}$-Ens has a left adjoint Lib : $\mathscr{U}$-Ens $\to$ $\mathscr{U}$-γ-Ens [3] (in fact, one can show that this condition is always satisfied). Composition with Lib gives a functor

$$\hat{\text{Lib}} : \widehat{C} \longrightarrow \mathbf{Hom}(C^\circ, \mathscr{U}\text{-}\gamma\text{-Ens})$$

and, after composition with the equivalence (5.8.1), a functor still denoted by

$$\hat{\text{Lib}} : \widehat{C} \longrightarrow \widehat{C}_\gamma$$

and called the "freely generated presheaf of γ-objects" functor. The functor $\hat{\text{Lib}}$ is left adjoint to the functor $\hat{\text{esj}}$.

Proposition 5.8.3. *Let γ be a type of algebraic structure defined by finite projective limits such that, in the category of γ-objects of $\mathscr{U}$-*Ens*, the $\mathscr{U}$-inductive limits are representable [3]; let C be a category belonging to $\mathscr{U}$, let C' be a $\mathscr{U}$-category, and let $u : C \to C'$ be a functor. Denote by $\widehat{C}_\gamma$ (resp. $\widehat{C'}_\gamma$) the category of γ-objects of $\widehat{C}$ (resp. $\widehat{C'}$), and by $u^{*\gamma}$ the functor on γ-objects induced by the functor u^*. It follows from 5.1 and the equivalence 5.8.1 that the functor $u^{*\gamma}$ has a left (resp. right) adjoint. This functor is denoted by $u_{!_\gamma}$ (resp. u_{*_γ}).*

(1) *The functor $u^{*\gamma}$ commutes with inductive and projective limits. The diagram*

(∗)
$$\begin{array}{ccc} \widehat{C'}_\gamma & \xrightarrow{u^{*\gamma}} & \widehat{C}_\gamma \\ {\scriptstyle \hat{\text{esj}}'}\downarrow & & \downarrow{\scriptstyle \hat{\text{esj}}} \\ \widehat{C'} & \xrightarrow{u_*} & \widehat{C} \end{array}$$

31 *is commutative ($\hat{\text{esj}}'$ and $\hat{\text{esj}}$ denote the "underlying-set" functors).*

[3] The functor Lib is the "free generated γ-object" functor. Examples: free group, free abelian group, free A-module, etc.

[3] One can show that this condition is always satisfied.

(2) *The functor u_{*_γ} commutes with projective limits. The diagram*

(**)
$$\begin{array}{ccc} \widehat{C}_\gamma & \xrightarrow{u_{*_\gamma}} & \widehat{C'}_\gamma \\ {\scriptstyle \mathrm{esj}\,\hat{}}\downarrow & & \downarrow{\scriptstyle \mathrm{esj}'\,\hat{}} \\ \widehat{C} & \xrightarrow{u_*} & \widehat{C'} \end{array}$$

commutes up to isomorphism.

(3) *The functor $u_{!_\gamma}$ commutes with inductive limits. Suppose that* $\mathrm{esj}\hat{}$ *(resp.* $\mathrm{esj}'\hat{}$*) has a left adjoint* $\mathrm{Lib}\hat{}$ *(resp.* $\mathrm{Lib}'\hat{}$*) (5.8.2). The diagram*

(***)
$$\begin{array}{ccc} \widehat{C} & \xrightarrow{u_!} & \widehat{C'} \\ {\scriptstyle \mathrm{Lib}\,\hat{}}\downarrow & & \downarrow{\scriptstyle \mathrm{Lib}'\,\hat{}} \\ \widehat{C}_\gamma & \xrightarrow{u_{!\gamma}} & \widehat{C'}_\gamma \end{array}$$

commutes up to isomorphism.

Suppose that $u_!$ commutes with finite projective limits (5.4) and (5.6). Then the diagram

(****)
$$\begin{array}{ccc} \widehat{C}_\gamma & \xrightarrow{u_{!\gamma}} & \widehat{C'}_\gamma \\ {\scriptstyle \mathrm{esj}\,\hat{}}\downarrow & & \downarrow{\scriptstyle \mathrm{esj}'\,\hat{}} \\ \widehat{C} & \xrightarrow{u_!} & \widehat{C'} \end{array}$$

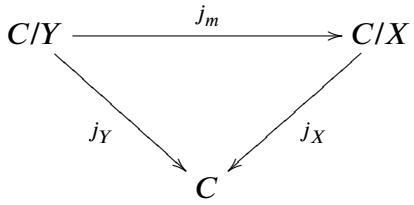
commutes up to isomorphism, and $u_{!\gamma}$ commutes with finite projective limits.

Proof. Assertion (1) is evident. So is assertion (2), since u_* is a right adjoint and therefore (2.11) commutes with projective limits and, in particular, with finite projective limits; this gives the commutativity of diagram (**). The commutativity of diagram (***) follows from the uniqueness, up to isomorphism, of the left adjoint functor, and 32
finally the commutativity of diagram (****) follows immediately from the fact that $u_!$ commutes with finite projective limits.

Notation 5.9. By an abuse of notation, the functors $u^{*\gamma}$ and $u_{*\gamma}$ will often be denoted by u^* and u_*. In view of the commutativity of diagrams (*) and (**), this cannot cause confusion. On the other hand, when $u_!$ does not commute with finite projective limits, diagram (****) does not commute up to isomorphism, and the notations $u_!$ and $u_{!\gamma}$ must be used to avoid confusion.

5.10. Let C be a small category. For every object X of $\widehat{C}$, let C/X denote the category of arrows with target X and source an object of C. The source functor defines a functor $j_X : C/X \to C$. Let $m : Y \to X$ be a morphism of $\widehat{C}$. Composition of

morphisms defines a functor $j_m : C/Y \to C/X$. The diagram



is commutative. It follows from 5.1 that, for every object F of $(C/X)\hat{}$ and every object Y of C, one has

$$j_{X!}F(Y) = \coprod_{u \in \mathrm{Hom}_{\widehat{C}}(Y,X)} F(u). \tag{5.10.1}$$

Formula (5.10.1) defines $j_{X!}$ when C is a $\mathscr{U}$-category, and one checks that the functor
33 $j_{X!}$ thus defined is always right adjoint to the functor $j_X^* : \widehat{C} \to (C/X)\hat{}$ (5.0).

Proposition 5.11. *Let C be a $\mathscr{U}$-category and let X be a presheaf on C.*

1) *The functor*

$$j_{X!} : (C/X)\hat{} \longrightarrow \widehat{C}$$

factors through the category $\widehat{C}/X$:

$$(C/X)\hat{} \xrightarrow{e_X} \widehat{C}/X \longrightarrow \widehat{C}.$$

The functor e_X is an equivalence of categories.

2) *The functor $e_X \circ j_X^* : \widehat{C} \to \widehat{C}/X$ is canonically isomorphic to the functor $H \mapsto (H \times X \xrightarrow{\mathrm{pr}_2} X)$.*

Proof. 1) Let f be the terminal object of $(C/X)\hat{}$. There is a canonical isomorphism $f \xrightarrow{\sim} \varinjlim_{Y \in \mathrm{ob}\, C/X} Y$, and hence $j_{X!}(f) = \varinjlim_{Y \in \mathrm{ob}\, C/X} j_X(Y)$ (5.4). But $j_{X!}(f) \simeq X$, which gives the factorization. To show that e_X is an equivalence, it is enough to exhibit a quasi-inverse functor. To every object $H \to X$ of $\widehat{C}/X$, associate the presheaf on C/X

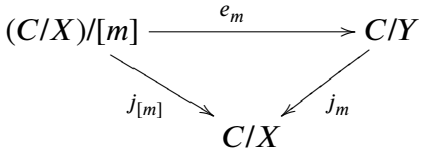
$$(Y \to X) \longmapsto \mathrm{Hom}_{\widehat{C}/X}((Y \to X), (H \to X)).$$

2) The functor $e_X \circ j_X^*$ is right adjoint to the forgetful functor, and therefore the functor $e_X \circ j^*$ is canonically isomorphic to the functor $H \mapsto (H \times H \xrightarrow{\mathrm{pr}_2} X)$.

5.12. Let $m : Y \to X$ be a morphism of $\widehat{C}$. By (5.11), the morphism m is canonically isomorphic to the image under e_X of an object of $(C/X)\hat{}$, which we denote by $[m]$. By restriction to the subcategories, the functor e_X defines an equivalence

$$(C/X)/[m] \xrightarrow{e_m} C/Y.$$

34 The diagram



commutes up to canonical isomorphism.

5.13. We record a result that will be useful in Exp. VI. Let $u : C \to C'$ be a functor between small categories. For every object H of $\widehat{C}$, denote by $u/H : C/H \to C'/u_!H$ the functor that assigns to every morphism $m : X \to H$ the morphism $u_!X \xrightarrow{u_!m} u_!H$ (we know (5.4) that we may always put $u_!X = uX$). The following diagram is commutative:

$$\begin{array}{ccc} C/H & \xrightarrow{u/H} & C'/u_!H \\ {\scriptstyle J_H}\downarrow & & \downarrow{\scriptstyle j_{u_!H}} \\ C & \xrightarrow{u} & C' \end{array} \quad . \tag{5.13.1}$$

We therefore have a diagram commutative up to isomorphism: and since $(u/H)_!$ takes the terminal object of $(C/H)\hat{}$ to the terminal object of $(C'/u_!H)\hat{}$, the diagram

$$\begin{array}{ccc} (C/H)\hat{} & \xrightarrow{(u/H)_!} & (C'/u_!H)\hat{} \\ {\scriptstyle e_H}\downarrow & & \downarrow{\scriptstyle e_{u_!H}} \\ C\hat{}/H & \xrightarrow{u_!/H} & C'\hat{}/u_!H \end{array} \tag{5.13.3}$$

commutes up to isomorphism (6.1). 35

Proposition 5.14. *Let $u : C \to C'$ be a functor between small categories. Suppose that u has the following property:*

(PPF) *For every object X of C, the functor $u/X : C/X \to C'/uX$ is fully faithful.*

Then:

1) *Let f be the terminal object of $\widehat{C}$. The functor u factors as*

$$C = C/f \xrightarrow{u/f} C'/u_!f \xrightarrow{j_{u_!f}} C'.$$

The functor u/f is fully faithful.

2) *The functor $u_! : \widehat{C} \to \widehat{C'}$ factors as*

$$\widehat{C} = (C/f)\hat{} \xrightarrow{(u/f)_!} (C'/u_!f)\hat{} \xrightarrow{e_{u_!f}} \widehat{C'}/u_!f \longrightarrow \widehat{C'},$$

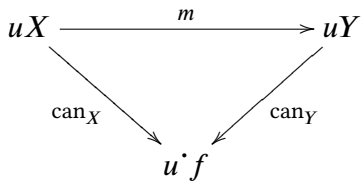
where the functor $(u/f)_!$ is fully faithful, the functor $e_{u_!f}$ is an equivalence, and the functor $\widehat{C'}/u_!f \to \widehat{C'}$ is the forgetful functor.

3) *In particular, the functor $u_!$ is faithful, and therefore the unit morphism $\mathrm{id} \xrightarrow{\Phi} u^*u_!$ is a monomorphism. Moreover, for every morphism $\alpha : H \to K$ of $\widehat{C}$, the diagram*

$$\begin{array}{ccc} H & \xrightarrow{\Phi(H)} & u^*u_!H \\ {\scriptstyle \alpha}\downarrow & & \downarrow{\scriptstyle u.u^\bullet(\alpha)} \\ K & \xrightarrow{\Phi(K)} & u^*u_!K \end{array}$$

is cartesian.

PROOF. 1) The factorization comes from diagram (5.13.1). The functor u is faithful. Hence u/f is faithful. Let us show that it is fully faithful. Let X and Y be two objects of C, let $\mathrm{can}_X : uX \to u_!f$ (resp. $\mathrm{can}_Y : uY \to u_!f$) be the canonical morphisms, and let 36



be a morphism of $C'/u_!f$. We have $u_!f = \varinjlim_{Z\in \mathrm{ob}\, C} uZ$, and therefore (3.1)

$$\mathrm{Hom}_{\widehat{C'}}(uX, u_!f) = \varinjlim_{Z\in \mathrm{ob}\, C} \mathrm{Hom}_{C'}(uX, uZ).$$

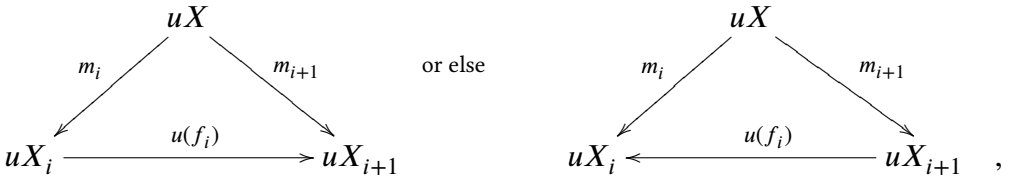
By the definition of the inductive limit, to say that $\mathrm{can}_Y\, m = \mathrm{can}_X$ is equivalent to saying that there exist

a) a finite sequence of objects X_i of C, $i \in [0, n]$, with $X_0 = X$ and $X_n = Y$,

b) for every i, a morphism $m_i : uX \to uX_i$ ($m_0 = \mathrm{id}_X, m_n = m$),

c) for every pair $(i, i+1)$, a morphism $f_i : X_i \to X_{i+1}$ or else $f_i : X_{i+1} \to X_i$,

such that the diagrams



are commutative.

It then follows immediately, by induction on i and using property (PPF), that m_i is of the form $u(p_i)$. In particular $m = u(p)$, and hence u/f is fully faithful.

37 2) The factorization is immediate. The functor $(u/f)_!$ is fully faithful by 5.6. The other assertions follow from 5.11.

3) The functor $u_!$ is the composite of the faithful forgetful functor and fully faithful functors. It is therefore faithful. It follows from the general properties of adjoint functors that the unit morphism $\mathrm{id} \to u^*u_!$ is a monomorphism. By 2), the functor $u_!$ appears as the composite of a fully faithful functor $v : \widehat{C} \to \widehat{C'}/u_!f$ and the forgetful functor. The functor u^*, right adjoint to $u_!$, is therefore the composite of the "product with $u_!f$" functor, right adjoint to the forgetful functor, and a functor w right adjoint to v. Moreover, since v is fully faithful, the unit morphism $\mathrm{id} \to wv$ is an isomorphism. The last assertion follows readily.

6. Faithful functors and conservative functors

38

DEFINITION 6.1. *Let E be a category and let $(\varphi_i : E \to F_i)_{i\in I}$ be a family of functors*

$$\varphi_i : E \longrightarrow F_i.$$

The family of functors (φ_i) is said to be faithful if, for every pair of objects X, Y of E and every pair of arrows u, $v : X \rightrightarrows Y$, the relation $\varphi_i(u) = \varphi_i(v)$ for every $i \in I$ implies $u = v$ (in other words, if the map $\mathrm{Hom}(X, Y) \to \Pi_i \mathrm{Hom}(\varphi_i(X), \varphi_i(Y))$ defined by (φ_i) is injective). The family of functors (φ_i) is said to be conservative if every arrow u of E for which $\varphi_i(u)$ is an isomorphism for every $i \in I$ is an isomorphism. We say that

(φ_i) is conservative for monomorphisms (resp. for epimorphisms, resp. ...) if the preceding condition holds whenever u is a monomorphism (resp. an epimorphism, resp. ...).

6.1.1. If we introduce the unique functor

$$\varphi : E \longrightarrow F = \prod_{i \in I} F_i$$

defined by the family of functors (φ_i), it is clear that this family is faithful (resp. conservative, resp. conservative for monomorphisms, resp. ...) if and only if the functor φ is faithful (resp. conservative, resp. ...) (by which we mean that the family consisting only of φ is faithful, resp. conservative, resp. ...). Thus, without major inconvenience, we could restrict ourselves below to a family consisting of a single functor. Nevertheless, for ease of future reference, we shall state the following results for families.

The notions of 6.1 are especially useful when the φ_i satisfy suitable exactness properties, and in that case they tend to coincide: 39

PROPOSITION 6.2. *The notation is that of 5.1.*

(i) *If the kernels of double arrows, or the cokernels of double arrows, are representable in E, and if the φ_i commute with them, then one has the implication*

$$(\varphi_i) \text{ conservative} \implies (\varphi_i) \text{ faithful.}$$

(ii) *Suppose that fiber products (resp. amalgamated sums) are representable in E, and that the φ_i commute with them. Suppose that (φ_i) is faithful or conservative; then for every arrow u of E, u is a monomorphism (resp. an epimorphism) if and only if the same holds for $\varphi_i(u)$ for every $i \in I$.*

(iii) *Suppose that fiber products and amalgamated sums are representable in E and that the φ_i commute with them, and that every arrow in E which is a bimorphism (i.e. both a monomorphism and an epimorphism) is an isomorphism. Then one has the implication*

$$(\varphi_i) \text{ faithful} \implies (\varphi_i) \text{ conservative.}$$

(iv) *Suppose that fiber products (resp. amalgamated sums) are representable in E, and that the φ_i commute with them. Then, if (φ_i) is conservative for monomorphisms (resp. for epimorphisms), it is in fact conservative.*

(v) *Let D be a type of diagram, let $F : d \mapsto F(d)$ be a diagram of type D in E, let X be an object of E, and let $u = (u_d)_{d \in D}$ be a family of arrows*

$$X \longrightarrow F(d) \quad (\text{resp. } F(d) \longrightarrow X).$$

Suppose that (φ_i) is conservative, that projective (resp. inductive) limits of type D are representable in E, and that the φ_i commute with them. Then u makes X a projective (resp. inductive) limit of F in E if and only if, for every $i \in I$, $\varphi_i(u)$ makes $\varphi_i(X)$ a projective (resp. inductive) limit of $\varphi_i(D)$ in E_i. 40

Proof.

(i) For the non-resp. statement, given a double arrow $u, v : X \rightrightarrows Y$, it suffices to express the equality $u = v$ by the condition that the inclusion $\mathrm{Ker}(u, v) \to X$ is an isomorphism. Here and below, we omit repetition of the dual argument for the dual statement.

(ii) If (φ_i) is faithful, the condition that $u : X \to Y$ be a monomorphism is expressed by the equality $\mathrm{pr}_1 = \mathrm{pr}_2$ for the fiber product $X \times_Y X$. If (φ_i) is conservative, it is expressed by the condition that the diagonal morphism $\delta : X \to X \times_Y X$ be an isomorphism.

(iv) Since in the latter argument the morphism δ is a monomorphism, we see that it would in fact have been enough to suppose that (φ_i) is conservative for monomorphisms. But this then implies that (φ_i) is conservative without qualification. Indeed, if $u \in F\ell\, E$ is such that the $\varphi_i(u)$ are isomorphisms, the preceding argument shows that these are monomorphisms, and hence they are isomorphisms by the hypothesis on (φ_i).

(iii) This is a trivial consequence of (ii).

(v) This is a trivial consequence of the definitions.

We record the following consequence of (i), (ii), and (iv):

41 COROLLARY 6.3. *Suppose that fiber products and amalgamated sums are representable in E and that the φ_i commute with them, and that kernels of double arrows or cokernels of double arrows are representable and that the φ_i commute with them. (It suffices, for example, that finite projective limits and finite inductive limits be representable in E, and that the φ_i be exact functors.) Then the following conditions are equivalent:*

a) *(φ_i) is faithful.*
b) *(φ_i) is conservative.*
c) *(φ_i) is conservative for monomorphisms.*
c′) *(φ_i) is conservative for epimorphisms.*

We also record, for reference:

PROPOSITION 6.4. *Let $\varphi : E \to F$ be a functor with a right adjoint ψ (thus $\mathrm{Hom}(\varphi(X), Y) \simeq \mathrm{Hom}(X, \psi(Y))$). The functor φ (resp. ψ) is faithful if and only if, for every element X of E (resp. every element Y of F), the unit morphism $X \to \psi\varphi(X)$ is a monomorphism. The functor φ (resp. ψ) is fully faithful if and only if the preceding unit morphism is an isomorphism.*

Indeed, if X', X are two objects of E, the map

$$(*) \qquad \mathrm{Hom}(X', X) \longrightarrow \mathrm{Hom}(\varphi(X'), \varphi(X))$$

is identified with the map obtained from

$$(**) \qquad X \longrightarrow \psi\varphi(X)$$

42 by applying the functor $\mathrm{Hom}(X', -)$. Thus, for $(*)$ to be a monomorphism (resp. an isomorphism) for every X', with X fixed, it is necessary and sufficient that the unit morphism $(**)$ be a monomorphism (resp. an isomorphism).

PROPOSITION 6.5. *Let $p : E' \to E$ be a functor. The following conditions are equivalent:*

(i) *p is faithful, conservative, and fibered* (SGA 1 VI 6.1).

(ii) *p is a fibered functor whose fibers are discrete categories.*

(iii) *For every $X' \in \mathrm{ob}\, E'$, the functor $E'_{/X'} \to E_{/p(X)}$ induced by p is an equivalence of categories that is surjective on objects.*

(iv) *(When E is a $\mathcal{U}$-category.) There exists an element $F \in \mathrm{ob}\, \widehat{C}$ and an equivalence of categories over E* (SGA 1 VI 4.3) *$E' \xrightarrow{\approx} E_{/F}$ (where $E_{/F}$ is the full subcategory of $\widehat{E}_{/F}$ formed by the arrows $X \to F$ whose source belongs to E).*

6.5.1. Recall that a category C is said to be *discrete* if it is a *groupoid* (i.e. every arrow in it is invertible) and if it is *rigid* (i.e. the automorphism group of every object is reduced to the trivial group); equivalently, the category is equivalent to the category C' defined by a set I (with $\mathrm{ob}C' = I$, and with the identity arrows as its only arrows). When C is already assumed to be a groupoid, saying that C is discrete amounts to saying that for two objects X, Y of C, there exists at most one arrow from X to Y, i.e. that C is isomorphic to the category defined by a preordered set.

The equivalence of conditions (i) and (ii) of 6.5 is an immediate consequence of the 43
preceding observations and of the following

LEMMA 6.5.2. *Let $p : E' \to E$ be a fibered functor. Then:*

(i) *For p to be conservative, it is necessary and sufficient that its fiber categories be groupoids.*
(ii) *For p to be faithful, it is necessary and sufficient that its fiber categories be ordered categories.*

Proof of 6.5.2.

(i) Suppose p is conservative. For every arrow u in a fiber E'_X, $p(u) = \mathrm{id}_X$ is an isomorphism, so u is an isomorphism in E', hence also in E'_X (since an inverse of u in E' will clearly be an inverse in E'_X). Thus E'_X is a groupoid. Conversely, suppose that the E'_X are groupoids, and let u' be an arrow of E' such that $p(u')$ is an isomorphism; let us prove that u is an isomorphism. To this end, note that, since p is fibered, we can factor $u' : X' \to Y'$ as a composite $X' \to u^*(Y') \to Y'$, where the first arrow is an X-morphism (N.B. $X = p(X')$, $u = p(u')$), and the second is a cartesian morphism over u. The first arrow is an isomorphism since E'_X is a groupoid, and the second is as well, since a *cartesian* morphism of a fibered category is clearly an isomorphism whenever its projection is an isomorphism.
(ii) Suppose p is faithful, and let X', Y' be two objects of a fiber category E_X. Then two arrows from X' to Y' lie over the same arrow id_X of E, and hence are identical, so E_X is ordered. Conversely, suppose that the fiber categories are 44
ordered, and let us prove that p is faithful. Thus let $u', v' : X' \rightrightarrows Y'$ be arrows of E' lying over the same arrow $u : X \to Y$ of E. They then factor as $X' \rightrightarrows u^*(Y) \to Y'$, where the two arrows $X' \rightrightarrows u^*(Y')$ are arrows of E'_X with the same source and target; they are therefore equal, and hence $u' = v'$, C.Q.F.D.

Let us return to the proof of 6.5. We have proved (i) $\Leftrightarrow$ (ii). On the other hand, (ii) $\Leftrightarrow$ (iv) is fairly clear: indeed, on the one hand, the category $E_{/F}$ is fibered over E, with fiber categories the discrete categories defined by the sets $F(X)$, as follows immediately from the definitions; on the other hand, if p is as in (ii), then by the sorites of SGA 1 VI 8 the fibered category E' over E is E-equivalent to the split category over E defined by the functor $E \to (\mathrm{Cat})$ determined by the functor $F : E \to (\mathrm{Ens})$, which assigns to every $X \in \mathrm{ob}\, E$ the set of isomorphism classes of objects of E'_X. Now this split category is E-isomorphic to the category $E_{/F}$. Since (iv) $\Rightarrow$ (iii) is clear, it remains to prove (iii) $\Rightarrow$ (i). Now it is clear that for p to be faithful (resp. conservative), it is necessary and sufficient that the induced functors $E'_{/X'} \to E_{/p(X)}$ be so; *a fortiori*, it suffices that these be fully faithful; therefore it remains only to prove that (iii) implies that p is fibered. But one again sees that a functor p is fibered if and only if the induced functors $E'_{/X'} \to E_{/p(X')}$ are fibered. This is in particular the case if they are equivalences of categories that are surjective on objects.

7. Generating and cogenerating subcategories

45

DEFINITION 7.1. *Let E be a category and C a full subcategory of E. We say that C is a subcategory of E* generating by strict epimorphisms (resp. by epimorphisms) *if, for every object X of E, the family of arrows of E with target X and source $X' \in \mathrm{ob}\, C$ is epimorphic (resp. strictly epimorphic) (1.3). We say that C is a subcategory of E* generating (resp. generating for monomorphisms, resp. generating for strict monomorphisms) *if, for every arrow $u : Y \to X$ (resp. every monomorphism of E, resp. every strict monomorphism of E (10.5)) such that, for every $X' \in \mathrm{ob}\, C$, the corresponding map $\mathrm{Hom}(X', Y) \to \mathrm{Hom}(X', X)$ is bijective, u is an isomorphism. Finally, we say that a family (X_i) of objects of E is generating by strict epimorphisms (resp. ...) if the full subcategory C of E generated by this family is generating by strict epimorphisms (resp. ...).*

7.1.1. Observe that, in terms of the family $(h_{X'})_{X' \in \mathrm{ob}\, C}$ of functors

$$h_{X'} : E \longrightarrow (\mathrm{Ens}) \ (X' \in \mathrm{ob}\, C)$$

represented by the $X' \in \mathrm{ob}\, C$, the condition that C be generating (resp. generating for monomorphisms, resp. generating for strict monomorphisms) can be expressed by saying that the family $(h_{X'})$ is *conservative* (resp. conservative for monomorphisms, resp. conservative for strict monomorphisms) (6.1). It also follows immediately from the
46 definitions that C is generating by epimorphisms if and only if the family $(h_{X'})_{X' \in \mathrm{ob}\, C}$ is *faithful* (6.1). We shall also give below (7.2 (i)) an analogous interpretation of the condition that C be generating by strict epimorphisms.

7.1.2. As with the notions introduced in 6.1, the notions of 7.1 are especially useful when E has suitable exactness properties, in which case the various notions introduced have a marked tendency to be all equivalent (7.3). Thus the question of which of these notions in 7.1 should be regarded as the most important scarcely arises; in the most important cases, these notions coincide, and the term "generating subcategory" may therefore be interpreted indifferently as referring to any one of the properties considered in 7.1 (for example the first, which is the strongest of all, as we shall see (7.2 (ii))).

7.1.3. Suppose that E is a $\mathscr{U}$-category, and consider the canonical functor

$$\varphi : E \longrightarrow \widehat{C} = \mathbf{Hom}(C, \mathscr{U}\text{-Ens}) \tag{7.1.3.1}$$

which is the composite of the functors $E \to \widehat{E} \to \widehat{C}$, where the first functor is the canonical functor (1.3.3), and the second is the restriction functor to C. Observe that it is plainly equivalent to say that the preceding functor φ is conservative (resp. faithful), or to say that the family of functors $h_{X'} : X \to \mathrm{Hom}(X', X) = \varphi(X)(X')$, for variable $X' \in \mathrm{ob}\, C$, is a conservative (resp. faithful) family, that is, also (7.1.2), that C is gener-
47 ating (resp. generating by epimorphisms). Similarly, φ is conservative for monomorphisms (resp. for strict monomorphisms) if and only if the family of the $h_{X'}(X' \in \mathrm{ob}\, C)$ is conservative for monomorphisms (resp. for strict monomorphisms), i.e. if and only if the subcategory C of E is generating for monomorphisms (resp. for strict monomorphisms).

PROPOSITION 7.2. *Let E be a $\mathscr{U}$-category and C a full subcategory.*

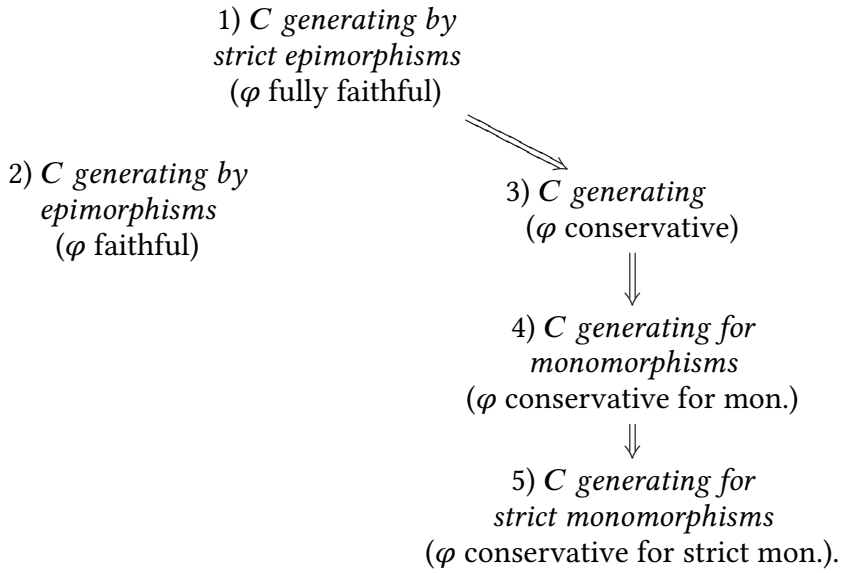
(i) *The following conditions are equivalent:*
   a) *C is a subcategory generating by strict epimorphisms.*
   b) *For every $X \in \mathrm{ob}\, E$, let $C_{/X}$ denote the full subcategory of $E_{/X}$ consisting of arrows $X' \to X$ with source $X' \in \mathrm{ob}\, C$. The natural arrow from the inclusion functor $j : C_{/X} \to E$ to the constant functor on $C_{/X}$ with value X makes X*

an inductive limit of j:

$$X \xleftarrow{\sim} \varinjlim_{C_{/X}} X'.$$

c) *The canonical functor φ of (7.1.3.1) is fully faithful.*

(ii) *Among the notions of 7.1 there are the following implications :*



(iii) *There are the following conditional implications:* 48

a) *If in E epimorphic families of arrows are strictly epimorphic, then* 2) ⇒ 1). *If in E monomorphisms are strict, then* 5) ⇒ 4).

b) *If in E the kernels of pairs of arrows (resp. fiber products) are representable, then* 3) ⇒ 2) *(resp.* 4) ⇒ 3)*).*

c) *If in E every family of morphisms $X_i \to X$ with common target X factors as a strictly epimorphic (resp. epimorphic) family $X_i \to Y$ followed by a monomorphism (resp. by a strict monomorphism) $Y \to X$, then* 4) ⇒ 1) *(resp.* 5) ⇒ 2)*).*

Let us immediately record the following.

Corollary 7.3. *All the notions considered in 6.1 (and repeated in the implication diagram in* (ii) *above) are equivalent in each of the following two cases:*

(i) *In E, the kernels of pairs of arrows and the fiber products are representable, the monomorphisms are strict, and the epimorphic families of arrows are strictly epimorphic.*

(ii) *In E, every family $(X_i \to X)_{i\in I}$ of arrows with common target X factors as an epimorphic family $(X_i \to Y)$ followed by a monomorphism $Y \to X$, every monomorphism of E is strict, and every epimorphic family of arrows of E is strict.*

Indeed, in case (i) we have 4) ⇒ 3) and 3) ⇒ 2) by b), and 5) ⇒ 4) and 2) ⇒ 1) by a). 49
In case (ii), by c), we have the implications 4) ⇒ 1) and 5) ⇒ 2). We therefore conclude by the implication diagram 6.2 (ii).

Proof of 7.2.

(i) The implication b) ⇒ a) follows at once from the definitions. Let us prove a) ⇒ b). Thus, under hypothesis a), we must prove that, for all X, $Y \in \mathrm{ob}\, E$, every

system of arrows

$$u_{X'} : X' \longrightarrow Y$$

indexed by the $X' \in \mathrm{ob}\, C_{/X}$, such that $u_{X'} f = u_{X''}$ for every arrow $f : X'' \to X'$ in $C_{/X}$, factors through an arrow (necessarily unique by hypothesis a)) $X \to Y$. By hypothesis a), it suffices to check that, for every object Z of $E_{/X}$ and every pair of morphisms $v' : Z \to X'$, $v'' : Z \to X''$ in $E_{/X}$, with X' and X'' in $C_{/X}$, we have $u_{X'} v' = u_{X''} v''$. But, by hypothesis a), the family of arrows $w : X''' \to Z$, with $X''' \in \mathrm{ob}\, C$, is epimorphic, and it therefore suffices to check that, for every such w, we have $(u_{X'} v')w = (u_{X''} v'')w$, which can also be written $u_{X'}(v'w) = u_{X''}(v''w)$ and follows at once from the hypothesis made on the family of the u.

Let us now prove the equivalence of conditions b) and c. For this, observe that, for every $X \in \mathrm{ob}\, E$, the object $\varphi(X)$ of $\widehat{C}$ is the inductive limit in $\widehat{C}$ of the canonical functor $\widehat{C}_{/\varphi(X)} \to \widehat{C}$ (3.4); but $\widehat{C}_{/\varphi(X)}$ is canonically isomorphic to $C_{/X}$, so that in $\widehat{C}$ we have

$$\varphi(X) = \varinjlim_{C_{/X}} X',$$

50 where the object X' of C is identified with the functor $\in \widehat{C}$ that it represents. Consequently, for a second object Y of E, we have a canonical isomorphism

$$\begin{aligned} \mathrm{Hom}(\varphi(X), \varphi(Y)) \simeq \varprojlim_{C_{/X}} \mathrm{Hom}(X', \varphi(Y)) &\simeq \varprojlim_{C_{/X}} \varphi(Y)(X') \text{(1.4)} \\ (*) \qquad &\simeq \varprojlim_{C_{/C}} \mathrm{Hom}(X', Y). \end{aligned}$$

This being so, the map

$$\mathrm{Hom}(X, Y) \longrightarrow \mathrm{Hom}(\varphi(X), \varphi(Y))$$

defined by φ is none other, via the isomorphism between the extreme terms of (∗), than the map

$$\mathrm{Hom}(X, Y) \longrightarrow \varprojlim_{C_{/X}} \mathrm{Hom}(X', Y)$$

deduced from the inductive system of arrows $X' \to X$ indexed by $C_{/X}$ considered in 7.2 (i) b). Thus the first map is bijective for every Y (with X fixed) if and only if X is an inductive limit of the inclusion functor $j : C_{/X} \to E$, which proves the equivalence of b) and c).

(ii) The implications 1) ⇒ 2) and 3) ⇒ 4) ⇒ 5) are trivial by virtue of the definitions. The implication 1) ⇒ 3) is obtained by interpreting property 1) as the full faithfulness of φ by (i), and by observing that a fully faithful functor is conservative. But we have already observed (7.1.1) that 3) means that φ is conservative.

51 (iii) Assertion a) is a tautology. Assertion b) follows from 6.2 (i) (resp. 6.2 (iv)) applied to the functor φ, taking into account that the latter is left exact. Finally, let us prove c). For $X \in \mathrm{ob}\, E$, consider the family of all the morphisms $X_i \to X$, with $X_i \in \mathrm{ob}\, C$; by hypothesis on E it factors as a family $X_i \to Y$ that is effective epimorphic (resp. epimorphic), followed by a monomorphism (resp. by a strict monomorphism) $Y \to X$. Then hypothesis 4) (resp. 5)) implies that $Y \to X$ is an isomorphism, so the family considered is strictly epimorphic (resp. epimorphic), C.Q.F.D.

PROPOSITION 7.4. *Let E be a category, C a full subcategory of E, and X an object of E. Suppose that C is generating in E (resp. C is generating for monomorphisms, and that the fiber product of two subobjects of X over X is representable in E). Then a strict subobject (10.11) (resp. a subobject) X' of X is determined when, for every $T \in \mathrm{ob}\, C$, one knows the subset of $\mathrm{Hom}(T, X)$ which is the image of $\mathrm{Hom}(T, X')$. Consequently, the cardinality of the set of strict subobjects (resp. of the set of subobjects) of X is bounded above by $\prod_{T \in \mathrm{ob}\, C} 2^{\mathrm{card}\, \mathrm{Hom}(T,X)}$, and if $X \in \mathrm{ob}\, C$, it is bounded above by $2^{\mathrm{card}\, F\ell\, C}$.*

Let us first prove the parenthetical assertion. Let X' and X'' be two subobjects of X such that, for every $T \in \mathrm{ob}\, C$, the images of $\mathrm{Hom}(T, X')$ and $\mathrm{Hom}(T, X'')$ in $\mathrm{Hom}(T, X)$ are equal. They are therefore also equal to the image of $\mathrm{Hom}(T, X''')$, where X''' is the fiber product of X' and X'' over X. Since C is generating for monomorphisms, it follows that the monomorphisms $X''' \to X'$ and $X''' \to X''$ are isomorphisms, so X' and X'' 52
are equal, each being equal to the subobject X''' of X.

Let us prove the nonparenthetical assertion. By definition of the notion of strict subobject, it suffices to check that knowledge of the subset $\mathrm{Hom}(T, X')$ of $\mathrm{Hom}(T, X)$ for every $T \in \mathrm{ob}\, C$ implies knowledge of those pairs of arrows $X \overset{u,v}{\rightrightarrows} T$ such that $ui = vi$, where $i : X' \to X$ is the canonical injection. But since C is generating, the relation $ui = vi$ is equivalent to the relation $(ui)f = (vi)f$ for every $f \in \mathrm{Hom}(T, X')$, i.e. to $ug = vg$ for every $g \in \mathrm{Hom}(T, X)$ coming from $\mathrm{Hom}(T, X')$ (i.e. of the form if, with $f \in \mathrm{Hom}(T, X')$), which proves our assertion.

COROLLARY 7.5. *Let E be a category, C a full generating subcategory, and X an object of C. Then a strict quotient (10.8) X' of X is determined when, for every $T \in \mathrm{ob}\, C$, one knows the subset of $\mathrm{Hom}(T, X)^2$ consisting of pairs (u, v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. Thus the cardinality of the set of strict quotients of X is bounded above by $\prod_{T \in \mathrm{ob}\, C} 2^{\mathrm{card}\, \mathrm{Hom}(T,X)^2}$.*

Indeed, by definition, a strict quotient X' of X is determined when one knows, for every object Y of E, the subset of $\mathrm{Hom}(Y, X) \times \mathrm{Hom}(Y, X)$ consisting of pairs (u, v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. But the relation $qu = qv$ is equivalent to the relation $(qu)f = (qv)f$ for every morphism $f : T \to Y$ with source T in C, since C is generating. This relation may also be written $q(uf) = q(vf)$, which proves the first assertion of 7.5. The second follows at once.

7.5.1. We can generalize 7.5 by introducing, for a family of objects $(X_i)_{i \in I}$ of E, 53
the notion of a *strict quotient in E of the family*, by which we mean a strictly epimorphic family $(p_i : X_i \to X')_{i \in I}$ of morphisms of E, — with the understanding, as for ordinary quotients, that we identify two such families $(p_i : X_i \to X')$ and $(q_i : X_i \to X'')$ if one can find an isomorphism $v : X' \to X''$ (necessarily unique) such that $f p_i = q_i$ for every $i \in I$. With this terminology, the proof of 7.5 applies *ne varietur* to give the

VARIANT 7.5.2. Let E, C be as in 7.5, and let $(X_i)_{i \in I}$ be a family of objects of E. Then a strict quotient X' of $(X_i)_{i \in I}$ in E (7.5.1) is determined by the following data: for every $T \in \mathrm{ob}\, C$, and every pair $(i, j) \in I \times I$, the subset of $\mathrm{Hom}(T, X_i) \times \mathrm{Hom}(T, X_j)$ consisting of the pairs (u, v) such that $p_i u = p_j v$, where for every $i \in I$, $p_i : X_i \to X'$ denotes the canonical morphism. Consequently, the cardinality of the set of strict quotients of $(X_i)_{i \in I}$ in E is bounded above by $\prod_{T \in \mathrm{ob}\, C} \prod_{i,j \in I} 2^{\mathrm{card}\, \mathrm{Hom}(T,X_i) \times \mathrm{card}\, \mathrm{Hom}(T,X_j)}$.

7.5.3. It is immediately clear that the analogous conclusion holds if one assumes only that C is generating for strict monomorphisms, provided that the products $X_i \times X_j$ are representable and that one restricts to the *effective* quotients of the family $(X_i)_{i \in I}$,

i.e. to the strict quotients X' such that the fiber products $X_i \times_{X'} X_j$ are representable in E (which is no restriction if E is stable under fiber products).

54 PROPOSITION 7.6. *Let E be a category, C a full subcategory of E generating by epimorphisms (7.1), $\Pi_\circ$ = an infinite cardinal $\geq \operatorname{card} F\ell C$, $\Pi \geq \Pi_\circ$ a cardinal, and $(u_i : X_i \to X)_{i\in I}$ a universally epimorphic family (10.3) in E consisting of pullbackable morphisms (10.7) with sources $X_i \in \operatorname{ob} C$, and such that $\operatorname{card} I \leq \Pi$. Then $\operatorname{card} F\ell C_{/X} \leq \Pi^{\Pi_\circ}$.*

Let $T \in \operatorname{ob} C$; it will suffice to prove that

$$\operatorname{card} \operatorname{Hom}(T, X) \leq \Pi^{\Pi_\circ}, \tag{7.6.1}$$

for one then obtains successively

$$\operatorname{card} \operatorname{ob} C_{/X} \leq (\Pi^{\Pi_\circ}) \times \operatorname{card} \operatorname{ob} C_{/X} \leq (\Pi^{\Pi_\circ}) \times \Pi_\circ = \Pi^{\Pi_\circ},$$

and finally $\operatorname{card} F\ell C_{/X} \leq ((\Pi^{\Pi_\circ})^2) \times \Pi_\circ = \Pi^{\Pi_\circ}$, since for two objects S, T of $C_{/X}$, one has $\operatorname{card} \operatorname{Hom}_{C_{/X}}(S, T) \leq \operatorname{car} \operatorname{Hom}_C(S, T) \leq \Pi_\circ$. To prove 7.5.3, first note the

LEMMA 7.6.2. *Let J be a set such that $\operatorname{card} J = \Pi_\circ$. For every object T of C, and every morphism $f : T \to X$, there exists an epimorphic family $(v_j : S_j \to T)_{j\in J}$, with sources $S_j \in \operatorname{ob} C$, and, for every $j \in J$, an $i(j) \in I$ and a $g_j : S_j \to X_{i(j)}$ such that $u_{i(j)} g_j = f v_j$.*

Indeed, the family of $X_i \times_X T \to$ is epimorphic by hypothesis; on the other hand, since C is generating by strict epimorphisms, for every i the family of arrows $S \to X_i \times_X T$ with source $S \in \operatorname{ob} C$ is epimorphic, and hence, by transitivity, the family of arrows $v : S \to T$ with source in $\operatorname{ob} C$ which factor through one of the $X_i \times_X T$ is epimorphic. These are precisely the arrows $v : S \to T$ with source in C for which there
55 exist an $i \in I$ and a $g : S \to X_i$ such that $u_i g = f v$. Since the set of these arrows $v : S \to T$ is contained in $F\ell\, C$, and therefore has cardinality $\leq \Pi_\circ$, it can be indexed by the index set J, whence the lemma.

Now note that a morphism $f : T \to X$ is determined once the $f v_j$ are known, and these are determined once the g_j are known. Thus $\operatorname{card} \operatorname{Hom}(T, X)$ is bounded above by the cardinality of the set of families $(i(j), S_j, v_j, g_j)_{j\in J}$; since the cardinality of the set of maps $j \to I$ is $\leq \Pi^{\Pi_\circ}$, and since, for a fixed such map $j \mapsto i(j)$, the cardinality of the set of corresponding families S_j, f_j, v_j is bounded above by that of the set of maps from J to $F\ell C \times F\ell C$, which is $\leq \Pi_\circ^{\Pi_\circ}$ because $\operatorname{card}(F\ell C \times F\ell C) = \Pi_\circ^2 = \Pi_\circ$, we find that the left-hand side of (7.6.1) is bounded above by $\Pi^{\Pi_\circ} \times \Pi_\circ^{\Pi_\circ} = \Pi^{\Pi_\circ}$, which completes the proof of 7.6.

PROPOSITION 7.7. *Let E be a category in which fiber products are representable, $(X_\alpha)_{\alpha\in A}$ a generating family of objects of E, and $(\varphi_i)_{i\in I}$ a family of functors $\varphi_i : E \to E_i$ commuting with fiber products. The family (φ_i) is conservative (5.1) if and only if, for every $\alpha \in A$ and every subobject X' of X_α distinct from X_α, there exists an $i \in I$ such that $\varphi_i(X') \to \varphi_i(X_\alpha)$ is not an isomorphism. In this case, if E is a $\mathcal{U}$-category and if A is $\mathcal{U}$-small, there exists a $\mathcal{U}$-small subset J of I such that $(\varphi_j)_{j\in J}$ is already a conservative family of functors.*

The necessity of the stated conservativity condition being obvious, let us prove its sufficiency. By 6.1 (iv), it suffices to prove that (φ_i) is conservative for monomorphisms.
56 Let $u : Y' \to Y$ be a monomorphism in E which is not an isomorphism; we must prove that there exists $i \in I$ such that $\varphi_i(u)$ is not an isomorphism. By the hypothesis on (X_α), there exist an α and a morphism $X_\alpha \to Y$ which does not factor through Y', in

other words, such that the inverse image X' of Y' is a subobject of X_α distinct from X_α. By hypothesis, there exists an $i \in I$ such that $\varphi_i(X') \to \varphi(X)$ is not an isomorphism. Since $\varphi_i(X') \simeq \varphi(X_\alpha) \times_{\varphi_{i(Y)}} \varphi_i(Y')$, it indeed follows that $\varphi_i(Y') \to \varphi_i(Y)$ is not an isomorphism.

The second assertion of 7.7 follows immediately from the first, in view of 7.4.

Corollary 7.7.1. *Let E be a $\mathscr{U}$-category in which fiber products are representable, and which admits a $\mathscr{U}$-small generating family of objects. Then every generating family $(Y_i)_{i\in I}$ of E contains a $\mathscr{U}$-small generating subfamily $(Y_j)_{j\in J}$ (card $J \in \mathscr{U}$).*

Indeed, it suffices to apply 7.7 to a $\mathscr{U}$-small generating family $(X_\alpha)_{\alpha\in A}$ of E and to the family of functors φ_i ($i \in I$) represented by the Y_i.

Proposition 7.8. *Let E be a category, C a full subcategory generating by strict epimorphisms, D a category, and $\mathbf{Hom}'(E, D)$ the full subcategory of $\mathbf{Hom}(E, D)$ consisting of the functors which commute with inductive limits of type $C_{/X}$, where X is an arbitrary object of E. Then the functor $F \to F|C$ induces a fully faithful functor*

$$\mathbf{Hom}'(E, D) \longrightarrow \mathbf{Hom}(C, D).$$

Consequently, if $\mathbf{Hom}(C, D)$ is a $\mathscr{U}$-category (1.1.), for example (1.1.1. b)) if C is $\mathscr{U}$-small 57
and D is a $\mathscr{U}$-category, then $\mathbf{Hom}'(E, D)$ is also a $\mathscr{U}$-category.

Let $F, G : E \to D$ be two functors in the first member, and let $u : F \to G$ be a natural transformation. If $X = \varinjlim X_i$ in E and if F and G commute with the inductive limit under consideration, then $u(X) : F(X) \to G(X)$ is identified with the inductive limit of the morphisms $u(X_i) : F(X_i) \to G(X_i)$, and is therefore determined once the $u(X_i)$ are known. This shows that the functor considered in 7.8 is faithful, in view of the implication a) $\Rightarrow$ b) in 7.2 (i). Conversely, let $v : F|C \to G|C$ be a natural transformation; let us prove that it comes from a natural transformation $u : F \to G$. For every $X \in \mathrm{ob}\, E$, define

$$u(X) : F(X) = \varinjlim_{C_{/X}} F(X_i) \longrightarrow G(X) = \varinjlim_{C_{/X}} G(X_i)$$

as the inductive limit of the $v(X_i)$. It is immediate that one thus obtains maps functorial in X, hence a natural transformation $u : F \to G$, and finally that the morphism induced by u from $F|C$ to $G|C$ is v, which completes the proof.

7.9. **Cogenerating families and subcategories.** Let E be a category and C a full subcategory of E. We say that C is cogenerating by strict monomorphisms (resp. cogenerating by monomorphisms, resp. cogenerating, resp. cogenerating for epimorphisms, resp. cogenerating for strict epimorphisms) if the full subcategory C° of E° is generating by strict epimorphisms (resp. etc.). The terminology for families is analogous. All the results of this section concerning the notion of a generating subcategory and its variants (7.1) therefore yield corresponding results for the dual notions, which we leave to the reader to formulate for their personal satisfaction. 58

Proposition 7.10. *Let E be a category, C a full generating subcategory (7.1), and D a full subcategory of E. The subcategory D is cogenerating (7.9) if and only if, for every pair of parallel arrows $T \overset{u,v}{\rightrightarrows} X$ in E with source $T \in \mathrm{ob}\, C$ and $u \neq v$, there exists an arrow $w : X \to I$ with target $I \in \mathrm{ob}\, D$ such that $wu \neq wv$.*

By definition, to say that D is cogenerating means that for every pair of parallel arrows $Y \overset{u,v}{\rightrightarrows} X$ in E such that $u \neq v$, there exists an arrow $w : X \to I$, with $I \in \mathrm{ob}\, D$, such that $wu \neq wv$. Thus 7.10 simply means that it suffices to test this property when $Y \in \mathrm{ob}\, C$. Now, since C is generating, the hypothesis $u \neq v$ implies that there exists $f : T \to Y$ such that $uf \neq vf$, whence, by hypothesis, there exists a $w : X \to I$ with target $I \in \mathrm{ob}\, D$ such that $w(uf) \neq w(vf)$, and hence $wu = wv$, C.Q.F.D.

Corollary 7.11. *With the notation of 7.10, suppose that the objects I of D are injective objects of E, i.e. such that for every monomorphism $X \to Y$ in E, the corresponding map $\mathrm{Hom}(Y, I) \to \mathrm{Hom}(X, I)$ is surjective. Suppose moreover that every pair of parallel arrows $T \overset{u,v}{\rightrightarrows} X$ in E factors as an epimorphic (resp. effective epimorphic) pair of parallel arrows $Y \overset{u',v'}{\rightrightarrows} X$ followed by a monomorphism $i : X' \to X$. Then, in the criterion of 7.10 for D to be cogenerating, one may restrict to the pairs (u, v) which are epimorphic (resp. effective epimorphic).*

59 We conclude:

Corollary 7.12. *Let E be a $\mathcal{U}$-category admitting a small generating subcategory C, and such that every object of E is the source of a monomorphism into an injective object of E. Suppose moreover that, for every $T \in \mathrm{ob}\, C$, the coproduct $T \amalg T$ in E is representable, and that every morphism with source $T \amalg T$ factors as an effective epimorphism followed by a monomorphism. Then E admits a small full cogenerating subcategory D. More precisely, one may take D such that* $\mathrm{card}\,\mathrm{ob}\, D \leq \prod_{T,T' \in \mathrm{ob}\, C} 2^{\mathrm{card}(\mathrm{Hom}(T', T \amalg T))^2}$.

Indeed, by 7.11, it suffices, for every $T \in \mathrm{ob}\, C$ and every effective quotient X of $T \amalg T$, to choose an embedding of X into an injective object I of E, and to take for D the full subcategory of E spanned by these I. The conclusion then follows from 7.5.

Examples 7.13. To construct small cogenerating subcategories in terms of small generating subcategories, we are thus led to seek conditions under which a $\mathcal{U}$-category E has "enough injectives", i.e. every object embeds into an injective object (by a monomorphism). It is well known [Tohoku] that this condition holds in a abelian $\mathcal{U}$-category with exact (small) filtered inductive limits (axiom AB 5 of *loc. cit.*) which admits a small generating family. Moreover, the construction of *loc. cit.* is not essentially tied to abelian categories, and also works in the category of sheaves of $\mathcal{U}$-sets on a topological space $X \in \mathcal{U}$. We shall not state here the exactness properties which make the construction
60 in question work, and shall merely point out that, in the special case of the category of sheaves of sets on X, one immediately reduces to the case of *loc. cit.* as follows. Note that if $\mathcal{O}_X$ is a ring on X, then every injective object I of the category of $\mathcal{O}_X$-modules is also injective as an object of the category of sheaves of sets. Indeed, if F is a sheaf of sets, and $\mathcal{O}_X[F]$ the "free $\mathcal{O}_X$-module generated by F", there is by definition a homomorphism of sheaves of sets

$$(*) \qquad F \longrightarrow \mathcal{O}_X[F]$$

giving rise to an isomorphism, functorial in F,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X[F], I) \xrightarrow{\sim} \mathrm{Hom}(F, I).$$

Since the functor $F \mapsto \mathcal{O}_X[F]$ manifestly takes monomorphisms to monomorphisms, it follows immediately that if I is an injective $\mathcal{O}_X$-module, it is also an injective sheaf of sets. It follows that if the morphism $(*)$ is a monomorphism (as is the case if one

takes for $\mathcal{O}_X$ a constant ring with value a ring $A \neq 0$), then an embedding of $\mathcal{O}_X[F^*]$ into an injective $\mathcal{O}_X$-module I gives an embedding of F into the injective sheaf of sets underlying I.

The same argument applies, without change, to the category of sheaves of $\mathcal{U}$-sets on a $\mathcal{U}$-site, which will be introduced in the following exposé.

The chief interest of the existence of a small cogenerating subcategory lies in the representability criterion 8.12.7 below.

8. Ind-objects and pro-objects

61

8.1. Cofinal functors and cofinal subcategories.

Definition 8.1.1. *Let*

$$\varphi : I \longrightarrow I' \tag{8.1.1.1}$$

be a functor. The functor φ is said to be cofinal if, for every category C and every functor $u : I' \to C$, regarding $\varinjlim u$ and $\varinjlim u \circ \varphi$ as objects of $\mathbf{Hom}(G(\mathcal{V}\text{-Ens}))^\circ$ (2.1, 2.3.1) (where $\mathcal{V}$ is a universe such that $I, I' \in \mathcal{V}$ and C is a $\mathcal{V}$-category), the canonical morphism

$$\varinjlim u \circ \varphi \longrightarrow \varinjlim u \tag{8.1.1.2}$$

is an isomorphism. When φ is the inclusion functor of a subcategory of I', the subcategory I is said to be cofinal if φ is cofinal.

8.1.2. Notice that for given φ and u, the bijectivity of (8.1.1.2) does not depend on the choice of the universe $\mathcal{V}$.

Returning to the definition of the terms occurring in (8.1.1.2) (2.1, 2.3.1), we see that to say that φ is cofinal also means that, for every universe $\mathcal{V}$ such that I and I' are $\mathcal{V}$-small, and every functor

$$v : I'^\circ \longrightarrow (\mathcal{V}\text{-Ens}),$$

the canonical homomorphism

$$\varprojlim v \longrightarrow \varprojlim v \circ \varphi \tag{8.1.2.1}$$

is an isomorphism, i.e. a bijection. To see that this condition is indeed necessary, observe, on putting $v = v^\circ$, that it also says that the condition of Definition 8.1.1 is satisfied when $C = (\mathcal{V}\text{-Ens})^\circ$ (which implies that the inductive limits considered in (8.1.1.2) are 62
representable in C).

. It follows immediately from the definition that *the composite of two cofinal functors is cofinal.*

Proposition 8.1.3. *Let $\varphi : I \to I'$ be a functor.*

a) *For φ to be cofinal, it is necessary that φ satisfy the condition:*
   F 1) *For every object i' of I', there exists an object i of I such that $\varphi(i)$ is an upper bound of i' (i.e. such that $\mathrm{Hom}(i', \varphi(i)) \neq \phi$).*

b) *Suppose that I is filtered. For φ to be cofinal, it is necessary and sufficient that φ satisfy condition* (F 1) *and the following condition:*
   F 2) *For every object i of I and every double arrow $i' \overset{f',g'}{\rightrightarrows} \varphi(i)$ in I' with target $\varphi(i)$, there exists an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$.*
   Moreover, if φ is cofinal, I' is filtered.

c) *Suppose that I' is filtered and φ is fully faithful. For φ to be cofinal, it is necessary and sufficient that it satisfy condition* F 1) *of* a). *This implies that I is filtered.*

Proof. For necessity in a) and b), use Definition 8.1.1 only in the case where u is the canonical inclusion functor (1.3.3)

$$u : I' \hookrightarrow \widehat{I'}$$

(having chosen a universe $\mathscr{U}$ such that I and I' are $\mathscr{U}$-small). We may then interpret
63 (8.1.1.2) as an arrow of $\widehat{I'}$ (3.1) whose target is the *terminal* functor on I' (3.4). It is immediately clear that condition F 1) says that this arrow is an epimorphism of $\widehat{I'}$, i.e. that it is surjective at each argument (taking into account that inductive limits in $\widehat{I'}$ are computed "argument by argument"). This proves a). Now suppose that I is filtered and φ is cofinal. Then I' is filtered: indeed, the condition that two objects of I' be bounded above by a third follows at once from the same condition on I and from F 1), and it remains only to prove condition PS 2) of 2.7, i.e. that for every double arrow

$$f', g' : i' \rightrightarrows j'$$

in I', there exists an arrow $h' : j' \to k'$ of I' such that $h'f' = h'g'$. By F 1), we may suppose that k' is of the form $\varphi(i)$. But since φ is cofinal, for every object i' of I' the set $\varinjlim_i \mathrm{Hom}(i', \varphi(i))$ consists of a single element. The standard description of *filtered* inductive limits in (Ens) (2.8.1) therefore implies that there exists an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$. This completes the proof that I' is filtered and at the same time proves condition F 2).

To prove that the conditions in b) are sufficient for φ to be cofinal, use the form 8.1.2 of the definition. One sees at once that condition F 1) implies that (8.1.2.1) is a monomorphism, i.e. is injective at each argument, while condition F 2) (together with F 1)) ensures that it is bijective (taking into account the argument-by-argument computation of projective limits in $\widehat{I'^{\circ}}$). This proves b).

64 Finally, if I' is filtered and φ is fully faithful, it is immediate that condition F 1) implies that I is filtered and implies condition F 2). Thus c) follows from a) and b).

8.1.4. When I' is a filtered category and I is a full subcategory of I', we see from 8.1.3 c) that the condition that I be cofinal in I' depends only on the subset $\mathrm{Ob}\, I$ of the preordered set $OB\, I'$ (for the preorder relation "$x \leq y$" $\Leftrightarrow$ "$\mathrm{Hom}(x, y) \neq \emptyset$"). If J' is a preordered set and J a subset of J', we shall sometimes say that J is a *cofinal subset of* J' when every element of J' is bounded above by an element of J. When J' is filtered, this therefore means that the inclusion functor $\mathscr{J} \hookrightarrow \mathscr{J}'$ for the associated categories is cofinal.

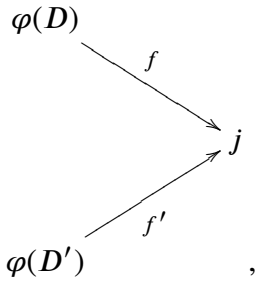
8.1.5. In what follows, we shall use the notion of cofinal functor only when the categories I and I' are filtered. Classically, one even restricted oneself to categories associated with preordered sets (i.e. categories in which there is at most one arrow with given source and target). It turns out, however, that this restriction is inconvenient in applications, since the "natural" filtered categories arising in many applications are *not* ordered categories. The following result, due to P. Deligne, shows nevertheless that there is no essential difference between the two points of view:

65 Proposition 8.1.6. *Let I be a small filtered category. Then there exist a small ordered set E and a cofinal functor $\varphi : \mathscr{E} \to I$, where $\mathscr{E}$ denotes the category associated with E.*

First suppose that the preordered set $\mathrm{Ob}\, I$ has no greatest element. A subdiagram of I is a pair $D = (\mathscr{O}, F)$ consisting of a subset F of $F\ell(I)$ and a subset $\mathscr{O}$ of $\mathrm{Ob}\, I$, such that for every $f \in F$, the source and target of f belong to $\mathscr{O}$. An element e of $\mathscr{O}$ is called a *terminal object* of the subdiagram D if, for every $x \in D'$, the set $\mathrm{Hom}(x, e) \cap F$ of arrows of D with source x and target y has exactly one element f_x, if $f_x = f_y \circ g$ for every arrow

$g : x \to y$ of D, and if $f_e = \mathrm{id}_e$. Let E be the set of *finite* subdiagrams D of I having a *unique* terminal element $\varphi(D)$, and order E by inclusion. If D, D' are two elements of E and $D \subset D'$, there is a unique arrow $\varphi(D', D)$ of D' with source $\varphi(D)$ and target $\varphi(D')$. If $D \subset D' \subset D''$ are inclusions in E, one clearly has $\varphi(D'', D) = \varphi(D'', D')\varphi(D', D)$; one also has $\varphi(D, D) = \mathrm{id}_{\varphi(D)}$. Thus we obtain a functor $\varphi : \mathscr{E} \to I$, and it remains to prove that E is filtered and φ is cofinal. In view of 8.1.3 b), three verifications are required:

1) Condition F 1) of 8.1.3: for every $i \in I$, take for D the subdiagram of I consisting only of the object i and its identity arrow; then $i \leq \varphi(D) = i$. This condition F 1) also shows that $E \neq \emptyset$.
2) Condition F 2): For every $i \in \mathrm{Ob}\, I$, every $D \in E$, and every double arrow $i \rightrightarrows \varphi(D)$, find a diagram $D' \in E$ containing D such that $\varphi(D', D) : \varphi(D) \to \varphi(D')$ *equalizes* the double arrow. Since I is filtered, there is an object j of I and an arrow $f : \varphi(D) \to j$ that equalizes the double arrow. Since I has no greatest element, we may suppose that j is a *strict* upper bound of $\varphi(D)$, which implies that it is distinct from every other object of D. Let D' be the subdiagram of I whose set of objects is $\mathcal{O} \cup \{j\}$ and whose set of arrows consists of the arrows of D, together with the composites $x \xrightarrow{f_x} \varphi(D) \xrightarrow{f} j$, where x is an object of D and f_x is the unique arrow of D with source x and target $\varphi(D)$, and id_j. It is then clear that D' is a finite subdiagram of I, that it has j as its unique terminal object, hence that $D' \in E$, and that D' satisfies the required condition. 66
3° Two subdiagrams D, $D' \subset E$ are contained in the same $D'' \in E$. Indeed, one may find a strict upper bound j of $\varphi(D)$ and $\varphi(D')$,



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and take for D'' the subdiagram whose set of objects is the union of the sets of objects of D and D' with $\{j\}$, and whose set of arrows is the union of the sets of arrows of D and D', the set of composites $f \circ f_x$ (x an object of D) and $f' \circ f'_{x'}$ (x' an object of D'), and $\{\mathrm{id}_j\}$. This indeed gives a finite subdiagram of E. Let us show that, after replacing j, f, f' by j', gf, gf' for a suitable $g : j \to j'$, we may arrange that j is a terminal object of D'. Equivalently, for every x that is an object of both D and D', one must have $f f_x = f' f'_{x'}$. This amounts to a *finite* set of double arrows to be equalized by a $g : j \to j'$, which is possible since I is filtered. 67

This completes the proof in the case considered. The general case is reduced to it by introducing the filtered category $\mathbf{N}$ associated with the ordered set $\mathbf{N}$ of natural integers, and observing that the category $\mathbf{N} \times I$ is filtered, has no greatest element, and that the projection $\mathbf{N} \times I \to I$ is a cofinal functor.

Corollary 8.1.7. *Let I be a $\mathscr{U}$-category. There exist a small filtered ordered set E and a cofinal functor $\mathscr{E} \to I$ if and only if I is filtered and* Ob I *has a small cofinal subset I' (8.1.4).*

Necessity follows from 8.1.3 a), and sufficiency follows from 8.1.6 applied to the full subcategory of I defined by I'.

DEFINITION 8.1.8. *Let I be a filtered category. We say that I is essentially small if I is a $\mathscr{U}$-category and satisfies the equivalent conditions of 8.1.7.*

8.2. Ind-objects and ind-representable functors.

8.2.1. In what follows, we fix a $\mathscr{U}$-category C, which we always regard as embedded in the category $\widehat{C}$ of $\mathscr{U}$-presheaves by means of the canonical functor (1.3.1)

$$h : C \hookrightarrow \widehat{C}. \tag{8.2.1.1}$$

An *ind-object of* C (or an *inductive system* of C) is any functor

$$\varphi : I \longrightarrow C, \tag{8.2.1.2}$$

68 where I is a *filtered* category (2.7), called the *index category* of the ind-object. (Apart from this, I is arbitrary; in particular, we do not require I to be a $\mathscr{U}$-category.) It will often be convenient to denote an ind-object φ by the indexed notation $(X_i)_{i\in \mathrm{ob}\, I}$, or even (by another abuse of notation) $(X_i)_{i\in I}$, or $(X_i)_i$, or (X_i), where $X_i = \varphi(i)$ for $i \in \mathrm{ob}\, I$. This notation is no more and no less abusive than the notation classically used for inductive systems indexed by filtered ordered sets (where the transition morphisms are likewise not mentioned). The X_i are called the *component objects* of the ind-object $\mathscr{X}$. Note that, in the general case considered here, the "*transition morphisms*" $\varphi(f)$ of the inductive system are indexed by the set $F\ell I$ of arrows of I, which in general is not identified with a subset of $\mathrm{Ob}\, I \times \mathrm{Ob}\, I$; in other words, for given i and j, with j an upper bound of i, there may exist more than one transition morphism from X_i to X_j.

8.2.2. The most useful ind-objects $\mathscr{X} = (X_i)_{i\in I}$ of C are those for which the index category I is essentially small (8.1.8); such an ind-object is called *essentially small*. If $(X_i)_{i\in I}$ is such an object, then, using the fact that small inductive limits are representable in $\widehat{C}$ (3.1), we may consider

$$L(\mathscr{X}) \overset{\text{def}}{=} \varinjlim h \cdot \mathscr{X} = \varinjlim_i X_i \in \mathrm{ob}\, \widehat{C}, \tag{8.2.2.1}$$

where the last limit is taken *in* $\widehat{C}$. Thus $L(\mathscr{X})$ is the presheaf

$$L(\mathscr{X}) : Y \longmapsto \varinjlim_i \mathrm{Hom}(Y, X_i) \tag{8.2.2.2}$$

69 on C. This presheaf is called the *presheaf ind-represented by the ind-object $\mathscr{X} = (X_i)_{i\in I}$ of C*. A presheaf F on C is called an *ind-representable presheaf* if it is isomorphic to a presheaf $L(\mathscr{X})$ ind-represented by an essentially small ind-object. It follows moreover from 8.1.6 that, in this definition, we may suppose that I is a small category, and even that I is associated with an ordered set $\in \mathscr{U}$.

8.2.3. Consider an ind-object $\mathscr{X} = (X_i)_{i\in I}$ given by a functor $\varphi : I \to C$, and let

$$u : I' \longrightarrow I$$

be a functor, where I' is a second filtered category. Then the functor $\varphi \circ u$ is an ind-object $\mathscr{X}'$ of C with index category I'; in indexed notation it is $(X_{u(i')})_{i'\in I'}$. It is called the ind-object of C *deduced from* $(X_i)_{i\in I}$ *by a change of index categories* using the functor u. If I and I' (i.e. $\mathscr{X}$ and $\mathscr{X}'$) are essentially small, one obtains a canonical morphism

$$L(\mathscr{X}') = \varinjlim_{i'} X_{u(i')} \longrightarrow L(\mathscr{X}) = \varinjlim_i X_i \tag{8.2.3.1}$$

between the presheaves ind-represented by $\mathscr{X}'$ and $\mathscr{X}$, respectively. When u is a cofinal functor (8.1.1), $\mathscr{X}$ is essentially small if and only if $\mathscr{X}'$ is (8.1.7), and the preceding homomorphism (8.2.3.1) is an isomorphism

$$L(\mathscr{X}') \simeq L(\mathscr{X}). \tag{8.2.3.2}$$

8.2.4. Let 70

$$\mathscr{X} = (X_i)_{i\in I}, \mathscr{Y} = (Y_j)_{j\in I} \tag{8.2.4.1}$$

be two essentially small ind-objects of C, indexed by the (filtered and essentially small) categories I and J. A *morphism* from the ind-object $\mathscr{X}$ to the ind-object $\mathscr{Y}$ is any morphism

$$L(\mathscr{X}) \longrightarrow L(\mathscr{Y}) \tag{8.2.4.2}$$

between the presheaves that they ind-represent. We put

$$\mathrm{Hom}_{\mathrm{ind-ob}}(\mathscr{X}, \mathscr{Y}) = \mathrm{Hom}_{\widehat{C}}(L(\mathscr{X}), L(\mathscr{Y})). \tag{8.2.4.3}$$

(The subscript ind-ob on Hom is omitted when there is no risk of confusion.) Composition of morphisms of ind-objects of $\mathscr{C}$ is defined as composition of the morphisms of the presheaves that they ind-represent. If $\mathscr{V} \supset \mathscr{U}$ is a universe containing $\mathscr{U}$, and if we restrict to the essentially small ind-objects that belong to $\mathscr{V}$, these form the set of objects of a category whose set of arrows consists of triples $(\mathscr{X}, \mathscr{Y}, u)$, where $\mathscr{X}$ and $\mathscr{Y}$ are essentially small ind-objects $\in \mathscr{V}$ and $u : \mathscr{X} \to \mathscr{Y}$ is a morphism of ind-objects, i.e. a morphism $L(\mathscr{X}) \to L(\mathscr{Y})$. The category thus obtained is denoted by

$$\mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U}). \tag{8.2.4.4}$$

When $\mathscr{V} = \mathscr{U}$, it is denoted simply by

$$\mathrm{Ind}_{\mathscr{U}}(C) \text{ or } \mathrm{Ind}(C), \tag{8.2.4.5}$$

and is called the *category of ind-objects of* C relative to $\mathscr{U}$, with the mention of $\mathscr{U}$ omitted when no confusion is likely.

Clearly, if $\mathscr{V} \subset \mathscr{V}'$ are two universes containing $\mathscr{U}$, then $\mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U})$ is a full subcategory of $\mathrm{Ind}_{\mathscr{V}'}(, \mathscr{U})$; it follows from 8.2.3 that the inclusion functor is an *equivalence of categories*: 71

$$\mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U}) \xrightarrow{\approx} \mathrm{Ind}_{\mathscr{V}'}(C, \mathscr{U}). \tag{8.2.4.6}$$

This shows in particular that *every essentially small ind-object of* C *defines an element of* $\mathrm{Ind}(C)$*, up to unique isomorphism.* This observation justifies the fairly common abuse of language of identifying every essentially small ind-object of C with an object of $\mathrm{Ind}(C)$.

It follows at once from the definitions that, for every universe $\mathscr{V}$, there is a canonical functor

$$L : \mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U}) \longrightarrow \widehat{C} \tag{8.2.4.7}$$

which is fully faithful. By (8.2.4.6), these functors are clearly determined up to unique isomorphism once one of them is known, and in particular once the canonical functor

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}. \tag{8.2.4.8}$$

is known. Since this functor is fully faithful, it is often used to identify an object of the first member with the functor that it ind-represents, or even to identify $\mathrm{Ind}(C)$ with its essential image in $\widehat{C}$, consisting of the ind-representable presheaves. Note, however, that this identification has distinctly more disadvantages than the analogous identification

of C with a full subcategory of $\widehat{C}$ by the functor h (8.2.1.1), since, unlike the latter, the functor (8.2.4.8) is not in general injective on objects.

72 8.2.5. Let us express Definition (8.2.4.3) explicitly in terms of the indexed expressions (8.2.4.1) of the ind-objects under consideration. Taking Definition (8.2.2.1) into account, one obtains

$$\mathrm{Hom}(\mathscr{X},\mathscr{Y}) \simeq \varprojlim_i \mathrm{Hom}(X_i,\mathscr{Y}) \overset{\mathrm{dfn}}{=} \varprojlim_i \mathrm{Hom}(X_i, L(\mathscr{Y})) = \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i,Y_j),$$

where the last equality follows from (8.2.2.2) applied to $\mathscr{Y}$. Thus one obtains a canonical bijection

$$\mathrm{Hom}((X_i)_{i\in I},(Y_j)_{j\in J}) \simeq \varprojlim_{i\in I} \varinjlim_{j\in J} \mathrm{Hom}_C(X_i,Y_j). \tag{8.2.5.1}$$

We leave to the reader the task of expressing composition of morphisms of ind-objects explicitly in terms of this formula. Notice that it follows at once from this formula that the set $\mathrm{Hom}(\mathscr{X},\mathscr{Y})$ of homomorphisms of ind-objects is *$\mathscr{U}$-small*, i.e. that the categories (8.2.4.4) are $\mathscr{U}$-categories. Equivalently (by (8.2.4.6)), the *category* $\mathrm{Ind}(C)$ *is a $\mathscr{U}$-category*, i.e. the second member of (8.2.5.1) is small when $I, J \in \mathscr{U}$. This follows at once from the fact that C is a $\mathscr{U}$-category, so the sets $\mathrm{Hom}_C(X_i,Y_j)$ are small.

8.2.6. Let I be an essentially small filtered category. It follows from (8.2.5.1), or from the functoriality of the inductive limit of a functor $I \to \widehat{C}$ with respect to that functor, that there is a canonical functor

$$\mathbf{Hom}(I,C) \longrightarrow \mathrm{Ind}(C). \tag{8.2.6.1}$$

Note that this functor is in general *not* fully faithful, nor even faithful.

73 Remark 8.2.7. The definitions of 8.2.4 make clear the notion of *isomorphism* of two (essentially small) ind-objects, which also means the isomorphism of the presheaves that they ind-represent. Note that this is a very weak notion of isomorphism when compared with the notion of isomorphism in categories of the form $\mathbf{Hom}(I,C)$. Thus, many fairly natural conditions that may be imposed on an ind-object (such as having its components in a given strictly full subcategory, or being strict (8.12 below)) *are not* invariant under isomorphism. Hence, if one wishes to work with notions invariant under isomorphism of ind-objects, the notions in question must be "saturated" by passing to the strictly full subcategory of $\mathrm{Ind}(C)$ generated by the subcategory of $\mathrm{Ind}(C)$ formed by the objects satisfying the condition under consideration. See, for example, the notion of essentially constant ind-object introduced below (8.4).

Exercise 8.2.8. Let $\mathscr{U} \subset \mathscr{V}$ be two universes and let C be a $\mathscr{U}$-category belonging to $\mathscr{V}$. Denote by $\mathrm{SysInd}(C)$ the following category:

a) The objects of $\mathrm{SysInd}(C)$ are the functors $\varphi : I \to C$ where I is an essentially $\mathscr{U}$-small filtered category belonging to $\mathscr{V}$.
b) Let (I,φ) and (J,ψ) be two objects of $\mathrm{SysInd}(C)$. A morphism in $\mathrm{SysInd}(C)$ from (I,φ) to (J,ψ) is a pair (m,u), where $m : I \to J$ is a functor and $u : \varphi \to \psi \circ m$ is a morphism of functors. Composition of morphisms in $\mathrm{SysInd}(C)$ is defined in the evident way.

74 Let S be the set of morphisms $(m,u) : (I,\varphi) \to (J,\Psi)$ of $\mathrm{SysInd}(C)$ such that m is cofinal and u is an isomorphism. Let (I,φ) be an object of $\mathrm{SysInd}(C)$. The category $F\ell(I)$ of arrows of I maps to I by two natural functors: source and target. These two functors are moreover linked by the canonical morphism of functors v : source $\to$ target. Hence

there are two morphisms in SysInd(C) from $(F\ell(I), \varphi \circ \text{source})$ to (I, φ): $p_1(I,\varphi) =$ (source, id) and $p_2(I,\varphi) =$ (target, $\varphi^* v : \varphi\circ\text{source} \to \varphi\circ$ target). Let p : SysInd(C) $\to$ Ind(C) be the evident functor. Let B be a category. Show that the functor $F \mapsto F \circ p$

$$\mathbf{Hom}(\mathrm{Ind}(C), B) \longrightarrow \mathbf{Hom}(\mathrm{SysInd}(C), B)$$

is fully faithful, and that a functor G : SysInd(C) $\to B$ belongs to its essential image if and only if it has the following two properties:

1) For every $s \in S$, $F(s)$ is an isomorphism of B.
2) For every object (I, φ) of SysInd(C), $F(p_1(I,\varphi)) = F(p_2(I,\varphi))$.

8.3. Characterization of ind-representable functors.

Proposition 8.3.1. *Let F be an ind-representable presheaf on C. Then F is left exact, i.e. (2.3.2) for every finite category J and every functor $\varphi : J \to C$ such that $\varinjlim \varphi$ is representable (i.e. $\varprojlim \varphi^\circ$ is representable), the canonical map*

$$F(\varinjlim \varphi) \longrightarrow \varprojlim F \circ \varphi$$

is bijective.

Indeed, since representable presheaves are clearly left exact, it follows at once from 2.8 that the same is true of every filtered inductive limit of such functors, C.Q.F.D. 75

8.3.2. Given a presheaf F on C, we shall often have occasion to work with the category

$$C_{/F} \hookrightarrow \widehat{C}_{/F}, \tag{8.3.2.1}$$

which denotes the full subcategory of the latter category formed by arrows $X \to F$ *whose source lies in* C. It follows from 1.4 that the objects of this category may be identified with pairs (X, u), where X is an object of C and $u \in F(X)$. A morphism from (X,u) to (X',u') is then interpreted as an arrow $f : X \to X'$ in C such that $F(f)(u') = u$. The functor from $\widehat{C}_{/F}$ to $\widehat{C}$ that "forgets the target" induces a functor (called *canonical*).

$$C_{/F} \longrightarrow C, \tag{8.3.2.2}$$

already considered in 3.4, where it is proved that the inductive limit of this functor exists in $\widehat{C}$ (without any smallness condition on $C_{/F}$!) and is canonically isomorphic to F.

. Note that if finite inductive limits are representable in C, and if F is left exact (i.e. takes them to finite projective limits in (Ens)), then the same is true in $C_{/F}$, and a fortiori (2.7.1) $C_{/F}$ is filtered. If, more generally, sums of two objects and cokernels of pairs of parallel arrows are representable in C, and if F takes them to products resp. kernels, then $C_{/F}$ is closed under the same types of finite inductive limits, hence it is filtered if and only if it is nonempty, i.e. if and only if the functor F is not the constant functor with value $\varnothing$. 76

Theorem 8.3.3. *Let C be a $\mathscr{U}$-category, and let $F \in \mathrm{ob}\,\widehat{C}$, $F : C^\circ \to (\mathscr{U}\text{-Ens})$ be a $\mathscr{U}$-presheaf on C. The following conditions are equivalent:*

(i) *F is ind-representable (8.2.2).*
(ii) *The category $C_{/F}$ (8.3.2) is filtered and essentially small.*
(iii) *(if finite inductive limits are representable in C.) The functor F is left exact, and* ob $C_{/F}$ *has a small cofinal subset (8.1.4).*

(iii bis) *(If sums of two objects and cokernels of pairs of parallel arrows are representable in C.) The functor F takes sums of two objects of C to products, and cokernels of pairs of parallel arrows of C to kernels; the category $C_{/F}$ is nonempty, i.e. the functor F is not the constant functor with value $\emptyset$; finally, there exists a small family of objects of $C_{/F}$ such that every object X of $C_{/F}$ is dominated by an object X_i, i.e. admits an F-morphism $X_i \to X$.*

(iv) *(if the category C is equivalent to a small category.) The category $C_{/F}$ is filtered.*

(v) *(If the category C is equivalent to a small category, and if filtered inductive limits are representable in it.) The functor F is left exact.*

77 *Proof.* (i) $\Rightarrow$ (ii). Suppose that F is ind-represented by $(X_i)_{i\in I}$, with I small. Let us prove that $C_{/F}$ is filtered. Let X, X' be two objects of $C_{/F}$, i.e. objects of C equipped with morphisms $X \to F$, $X' \to F$; let us prove that they have a common upper bound in $C_{/F}$. The morphisms $X \to F$, $X' \to F$ come from morphisms $X \to X_i$, $X' \to X'_i$, and, after replacing i, i' by a common upper bound in $\mathrm{Ob}\, I$, we may assume that $i = i'$, and take as a common upper bound of X, X' the object X_i equipped with the canonical morphism $X_i \to F$. Now let $X \overset{f,g}{\rightrightarrows} X' \xrightarrow{h} F$ be a pair of parallel arrows in $C_{/F}$; let us prove that it is equalized by an arrow $X' \to X''$ in $C_{/F}$. Now $h : X' \to F$ is given by a morphism $h_i : X' \to X_i$, and the condition $hf = hg$ means that there exists $\alpha : i \to j$ in I such that $\varphi(\alpha)(h_i f) = \varphi(\alpha)(h_i g)$, i.e. after replacing h_i by $\varphi(\alpha)h_i$, this morphism equalizes f and g. This proves that $C_{/F}$ is filtered. It is then immediate that it is essentially small, since the objects $(X_i \to F)_{\in \mathrm{Ob}\, I}$ form a small family in $\mathrm{Ob}\, C_{/F}$ that is cofinal.

(ii) $\Rightarrow$ (i). Put $I = C_{/F}$, and consider the canonical functor (8.3.2.1)

$$\varphi : I = C_{/F} \longrightarrow F.$$

We know that the presheaf ind-represented by this ind-object of C is F (3.4), so F is ind-representable by definition (8.2.2).

(ii) $\Leftrightarrow$ (iii). Indeed, (ii) $\Rightarrow$ (iii) because an ind-representable functor is left exact (8.3.1),
78 and (iii) $\Rightarrow$ (ii) because we noted (8.3.1) that if F is left exact, then $C_{/F}$ is filtered, and we apply Definition 8.1.8 to conclude that this category is essentially small.

(ii) $\Leftrightarrow$ (iii bis) is proved in the same way as (ii) $\Leftrightarrow$ (iii).

The equivalences (ii) $\Leftrightarrow$ (iv) and (iii) $\Leftrightarrow$ (v) are immediate, since if C is equivalent to a small category, then $C_{/F}$ is clearly likewise equivalent to a small category and *a fortiori* it is automatically essentially small as soon as it is filtered. This completes the proof of 8.3.3.

Remark 8.3.4. Let $F : C' \to C''$ be a functor between $\mathscr{U}$-categories. It appears that the notion of left exactness (2.3.2) is of little practical interest unless finite projective limits are representable in C'. In the case where $C'' = (\mathscr{U}\text{-Ens})$, and where C' is $\mathscr{U}$-small, writing C' in the form C° (thus $C = C'^\circ$), one should regard the "right notion" that replaces, in the general case, that of left exactness as the ind-representability of F (regarded as a presheaf on C°; i.e. the pro-representability of F, regarded as a functor on C', cf. 8.10). This notion does indeed coincide with that of left exactness when finite projective limits are representable in C' *i.e.* finite inductive limits in C are representable (8.3.3). When C' is no longer assumed $\mathscr{U}$-small, or at least equivalent to a $\mathscr{U}$-small category, the two notions for a functor F, left exactness and pro-representability, no longer necessarily coincide, even if finite projective limits in C' are representable (cf. 8.12.9). In
79 fact, it seems that in this case, left exact functors that are not ind-representable should

be regarded as pathological, while the "good" objects remain the ind-representable functors; compare 8.13.3. For the case of functors $F : C' \to C''$, with C' and C'' once again arbitrary $\mathscr{U}$-categories, we shall develop below (8.11.5), more generally, a notion that "improves" upon left exactness (namely, that of a functor admitting a pro-adjoint).

8.4. Constant and essentially constant ind-objects. Choose a terminal category e, i.e. one such that $\mathrm{Ob}\, e$ and $F\ell\, e$ each consist of a single element (for example, one may take for e the full subcategory of (Ens) on the empty set, if a canonical choice is desired). This is obviously a small filtered category. If X is an object of C, we associate with it the constant ind-object indexed by e, with value X, denoted by $c(X)$. It is clear that, as X varies, this gives a fully faithful functor

$$c : C \longrightarrow \mathrm{Ind}(C), \tag{8.4.1}$$

which is moreover injective on objects, and by which we shall identify C with a full subcategory of $\mathrm{Ind}(C)$. Note that the composite functor

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C} \tag{8.4.2}$$

is the canonical functor h (8.2.1.1).

8.4.3. More generally, let I be a filtered category. The constant functor on I whose value is an object X of C is also called *the constant ind-object with value X indexed by I*. It is clear that, if I is essentially small, this ind-object ind-represents the functor $h(X)$ represented by X, hence this ind-object is isomorphic to $c(X)$. 80

8.4.4. An ind-object $\mathscr{X}$ of C is said to be *essentially constant* if it is isomorphic (in a category $\mathrm{Ind}_{\mathscr{V}}(C, \mathscr{V}')$, for suitable $\mathscr{V}$, $\mathscr{V}'$) to an ind-object of the form $c(X)$, $X \in \mathrm{ob}\, C$. The object X, which is then determined up to canonical isomorphism, is called the *value* of the essentially constant ind-object under consideration. Thus, by definition, the functor c of (8.4.1) induces an equivalence of C with the full subcategory of $\mathrm{Ind}(C)$ consisting of the ind-objects that are essentially constant, this subcategory being the essential image of the functor c.

Clearly a constant ind-object is essentially constant, whereas the converse holds only in the trivial case in which C is empty or a one-point category.

8.5. Filtered inductive limits in $\mathrm{Ind}(C)$.

Proposition 8.5.1. *Let C be a $\mathscr{U}$-category. In* $\mathrm{Ind}(C)$*, small filtered inductive limits are representable, and the canonical functor* (8.2.4.8)

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}$$

commutes with them.

Since L is fully faithful, the assertions of the proposition are equivalent to the following: 81

Corollary 8.5.2. *Every small filtered inductive limit (in $\widehat{C}$) of ind-representable presheaves is ind-representable.*

This follows easily from criterion 8.3.3 (ii). The details of the verification are left to the reader.

8.5.3. The "tautological" case of filtered inductive limits in $\mathrm{Ind}(C)$ is that in which one starts with an essentially small ind-object $\mathscr{X} = (X_i)_{i\in I}$ of C. Then, in $\widehat{C}$ and hence also in $\mathrm{Ind}(C)$ (or in $\widehat{C}$), one has:

$$\mathscr{X} = \varinjlim_I X_i. \tag{8.5.3.1}$$

When writing this formula, care must be taken that it is not an inductive limit in C, and that even when the latter exists, it is not isomorphic in $\mathrm{Ind}(C)$ to $\mathscr{X}$: indeed, the canonical functor (8.4.1) $c : C \to \mathrm{Ind}(C)$ *does not in general commute with filtered inductive limits.* (Cf. 8.5.5 below.)

To avoid this possible confusion in writing (8.5.3.1), "some authors" (= P. Deligne) prefer to write it as

$$\mathscr{X} = \text{«}\varinjlim\text{»}\, X_i, \tag{8.5.3.2}$$

the quotation marks indicating that the inductive limit is taken in a category of ind-objects $\mathrm{Ind}(C)$. By extension, one should then denote by «lim» every inductive-limit operation in $\mathrm{Ind}(C)$ (without the components of the inductive system under consideration in $\mathrm{Ind}(C)$ necessarily lying in the image, or the essential image, of $c : C \to \mathrm{Ind}(C)$).

82 8.5.4. To compute the inductive or projective limit of a functor

$$\varphi : J \longrightarrow \mathrm{Ind}(C), \tag{8.5.4.1}$$

where J is now an arbitrary category, it is often convenient to make φ explicit by means of a functor

$$\Phi : J \times I \longrightarrow C, \tag{8.5.4.2}$$

where I is an essentially small filtered category, or preferably a small one. Such a Φ indeed defines a functor

$$J \longmapsto (i \longmapsto \Phi(j,i)) : J \longmapsto \mathbf{Hom}(I, C),$$

whence, by composing with the canonical arrow (8.2.6.1) $\mathbf{Hom}(I, C) \to \mathrm{Ind}(C)$, a functor

$$J \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \mathrm{Ind}(C). \tag{8.5.4.3}$$

We shall say that Φ is an *indexed expression of φ, indexed by the (filtered) category I*, if the corresponding functor (8.5.4.3) is isomorphic to φ. We shall study below (8.8) general conditions for the existence of an indexed expression for a given functor φ, and here we shall restrict ourselves to starting from a functor φ given in indexed form (8.5.4.3), in the case where J is a small filtered category, in order to indicate in this case how to compute $\text{«}\varinjlim\text{»}\,\varphi = \text{«}\varinjlim_j\text{»}\,\varphi(j)$. Formula (8.5.3.2), together with the associativity formula for inductive limits (2.5.0), then immediately gives the "*tautological computation*" of $\text{«}\varinjlim\text{»}\,\varphi$:

$$\text{«}\varinjlim_J\text{»}\,\varphi = \text{«}\varinjlim_{J\times I}\text{»}\,\Phi, \tag{8.5.4.4}$$

83 i.e.

$$\text{«}\varinjlim_j\text{»}\,(i \longmapsto \Phi(j,i)) = \text{«}\varinjlim_{j,i}\text{»}\,\Phi(j,i), \tag{8.5.4.5}$$

Thus, the sought-for inductive system is none other than Φ itself (the indexing category being $J \times I$, which is indeed filtered since J and I are).

Exercise 8.5.5. Let C be a $\mathscr{U}$-category

a) Prove that the following conditions are equivalent:

(i) In C, small inductive limits are representable, and the functor $c : C \to \mathrm{Ind}(C)$ commutes with them.
(iii) The functor c is an equivalence of categories.
(ii) In C, small inductive limits are representable, and for every $X \in \mathrm{Ob}\, C$, the functor $Y \mapsto \mathrm{Hom}(X, Y)$ commutes with these limits.
(iv) Small filtered inductive limits are representable in C, and the functor $\lim c : \mathrm{Ind}\, C \to C$ (cf. 8.7.1.5) is fully faithful (or equivalently, an equivalence).

b) If C is a finite category, prove that the preceding conditions are equivalent to the following: for every projector in C (10.6), its image is representable in C.

8.6. Extension of a functor to ind-objects.

8.6.1. Let

$$f : C \longrightarrow C' \tag{8.6.1.1}$$

be a functor between $\mathscr{U}$-categories. For every ind-object

$$\varphi : I \longrightarrow C$$

of C (I a filtered category), the composite functor $f \circ \varphi$ is an ind-object of C', also denoted by $\mathrm{Ind}(f)(\varphi)$. In indexed notation, if φ is written as $\mathscr{X} = (X_i)_{i\in I}$, one obtains 84

$$\mathrm{Ind}(f)(X_i)_{i\in I} = (f(X_i))_{i\in I}. \tag{8.6.1.2}$$

Let $\mathscr{Y} = (Y_j)_{j\in J}$ be a second inductive system of C. One then obtains an evident map, also denoted by $u \to \mathrm{Ind}(f)(u)$:

$$\mathrm{Hom}(\mathscr{X}, \mathscr{Y}) = \varprojlim_i \varinjlim_j \mathrm{Hom}(X_i, Y_j) \to \tag{8.6.1.3}$$

$$\mathrm{Hom}(\mathrm{Ind}(f)(\mathscr{X}), \mathrm{Ind}(f)(\mathscr{Y})) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}(f(X_i), f(Y_j)).$$

Referring to the unwritten formula for composition of morphisms of ind-objects (8.2.5), one immediately sees that these maps are compatible with composition of morphisms of ind-objects. Thus, for every universe $\mathscr{V}$, we have obtained a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U}) \longrightarrow \mathrm{Ind}_{\mathscr{V}}(C', \mathscr{U}) \tag{8.6.1.4}$$

(notation of (8.2.4.5)), and in particular a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C'). \tag{8.6.1.5}$$

If there is a second functor

$$g : C' \longrightarrow C'',$$

then one clearly has an identity of functors between Ind-categories (or $\mathrm{Ind}_{\mathscr{V}}$-categories, as desired):

$$\mathrm{Ind}(gf) = \mathrm{Ind}(g) \circ \mathrm{Ind}(f), \tag{8.6.1.6}$$

and, when $C' = C$, the perennial formula 85

$$\mathrm{Ind}(\mathrm{id}_C) = \mathrm{id}_{\mathrm{Ind}(C)}. \tag{8.6.1.7}$$

One may therefore say, heuristically, that the category $\mathrm{Ind}(C)$ *depends functorially on* C (covariantly) [3]. We leave to the reader the task of making explicit even a 2-*functorial* dependence by defining, for every pair of $\mathscr{U}$-categories C, C', a canonical functor

$$\mathbf{Ind}(f) : \mathbf{Hom}(C, C') \longrightarrow \mathbf{Hom}(\mathrm{Ind}(C), \mathrm{Ind}(C')), \tag{8.6.1.8}$$

[3] For a contravariant dependence, under certain conditions, cf. 8.11.

and specifying the functorial nature of the identical isomorphisms (8.6.1.6) and (8.6.1.7).

8.6.2. Return to the functor (8.6.1.1), and consider the corresponding functor (5.1)

$$f_! : \widehat{C} \longrightarrow \widehat{C'}, \tag{8.6.2.1}$$

and the diagram of functors

$$\begin{array}{ccc} \mathrm{Ind}(C) & \xrightarrow{\mathrm{ind}(f)} & \mathrm{Ind}(C') \\ {\scriptstyle L_C}\downarrow & & \downarrow{\scriptstyle L_{C'}} \\ \widehat{C} & \xrightarrow{f_!} & \widehat{C'} \end{array} \tag{8.6.2.2}$$

where the vertical arrows denote the canonical functors L of (8.2.4.8). Since the functor $f_!$ "extends f" and commutes with inductive limits (5.4.3), and the functors L_C and $L_{C'}$ commute with small filtered inductive limits (8.5.1), it follows from (8.5.3.2) that, for every ind-object $(X_i)_{i\in I}$ of C, there is a canonical isomorphism in $\widehat{C'}$

$$f_!(L_C((X_i)_{i\in I})) \simeq \varinjlim_i f_! L_C(X_i) \simeq \varinjlim_i L_{C'} f(X_i) = L_{C'}(f(X_i)_{i\in I})$$

86 i.e. a canonical isomorphism

$$f_! L_C(\mathscr{X}) \simeq L_{C'}(\mathrm{Ind}(f)(\mathscr{X})).$$

We leave to the reader the verification that the latter is functorial, i.e. that it corresponds to a canonical isomorphism

$$f_! L_C \simeq L_{C'} \circ \mathrm{Ind}(f). \tag{8.6.2.3}$$

In other words, the *square (8.6.2.2) commutes up to canonical isomorphism.* Taking into account the fact that $f_!$ commutes with small inductive limits, and using 8.5.1, we conclude:

Proposition 8.6.3. *Let $f : C \to C'$ be a functor between $\mathscr{U}$-categories. Then the functor*

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

commutes with filtered inductive limits. Moreover, it makes the following diagram commutative:

$$\begin{array}{ccc} C & \xrightarrow{f} & C' \\ {\scriptstyle c_C}\downarrow & & \downarrow{\scriptstyle c_{C'}} \\ \mathrm{Ind}(C) & \xrightarrow{\mathrm{Ind}(f)} & \mathrm{Ind}(C') \end{array} ,$$

where the vertical arrows c_C, $c_{C'}$ are the canonical functors (8.4.1).

The last assertion is trivial from the definitions and has been included for ease of reference. Notice moreover that the properties stated in 8.6.3 *characterize the functor* 87 $\mathrm{Ind}(f)$ up to unique isomorphism, as the functor induced by $f_!$ (8.6.2.1), as follows from the proof of (8.6.2.3) just given.

Proposition 8.6.4. *The notation is that of 8.6.3.*

a) *The functor* $\mathrm{Ind}(f)$ *is faithful (resp. fully faithful) if and only if f is.*

b) *The functor* $\mathrm{Ind}(f)$ *is an equivalence of categories if and only if* f *is fully faithful and every object of* C' *is isomorphic to a direct factor (10.6) of an object in the image of* f.

Proof.

a) Necessity clearly follows from the fact that f is induced by $\mathrm{Ind}(f)$. For sufficiency, it is enough to use the form (8.6.1.3) of $\mathrm{Ind}(f)$ on the sets $\mathrm{Hom}(\mathscr{X}, \mathscr{Y})$, recalling that filtered inductive limits of sets and arbitrary projective limits take monomorphisms to monomorphisms and isomorphisms to isomorphisms.

b) We may already suppose that f, hence $\mathrm{Ind}(f)$, is fully faithful. Since every object of $\mathrm{Ind}(C')$ is a small filtered inductive limit of objects of C', the full faithfulness of $\mathrm{Ind}(f)$ implies that, for this functor to be essentially surjective, it is equivalent that every object X' of C' belong to the essential image. But if there is an isomorphism

$$X' \xrightarrow{\sim} \text{«}\varinjlim\text{»}\, f(X_i),$$

this isomorphism factors through one of the $f(X_i)$, showing that X' is a direct factor of $f(X_i)$ and proving necessity in b). For sufficiency, using the full faithfulness of $\mathrm{Ind}(f)$, the assertion follows immediately.

Corollary 8.6.5. *In* $\mathrm{Ind}(E)$ *the images of projectors (10.6) are representable, and the functor* $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$ *commutes with the formation of these images.* 88

Indeed, this follows from the fact that small filtered inductive limits are representable in $\mathrm{Ind}(C)$ (8.5.1) and that $\mathrm{Ind}(f)$ commutes with them (8.6.3), taking into account that the image of a projector $p : X \to X$ may be interpreted as the inductive limit of a filtered inductive system indexed by $\mathbf{N}$

$$X \xrightarrow{P} X \xrightarrow{P} X \longrightarrow \dots$$

or, equivalently, as the inductive limit of the evident functor on the filtered category P having one object and one nonidentity arrow Π such that $\Pi^2 = \Pi$.

8.7. *The functor* $\varinjlim_C : \mathrm{Ind}(C) \to C$. *Universal characterizations of the category* $\mathrm{Ind}(C)$.

8.7.1. Let us return to a $\mathscr{U}$-category C and the canonical functor

$$c : C \longrightarrow \mathrm{Ind}(C). \tag{8.7.1.1}$$

Let $\mathscr{X} = (X_i)_{i\in I}$ be an object of $\mathrm{Ind}(C)$. It follows immediately from the definition of representable inductive limits (2.1, 2.1.1) that $\varinjlim_i X_i$ *is representable in* C *if and only if the left adjoint of* c *is defined at* $\mathscr{X}$, and that in this case $\varinjlim_i X_i$ (computed in C) is precisely the value at $\mathscr{X}$ of that left adjoint:

$$\mathrm{Hom}_C(\varinjlim_i X_i, Y) \simeq \mathrm{Hom}_{\mathrm{Ind}(C)}((X_i)_{i\in I}, c(Y)). \tag{8.7.1.2}$$

If we denote by 89

$$\mathrm{Ind}(C)' \subset \mathrm{Ind}(C) \tag{8.7.1.3}$$

the full subcategory of $\mathrm{Ind}(C)$ formed by the ind-objects of C that admit an inductive limit *in* C, the preceding observation shows that this subcategory is *strictly full*, and that there is a canonical functor

$$\varinjlim_C : \mathrm{Ind}(C)' \longrightarrow C, \tag{8.7.1.4}$$

whose value at each object $(X_i)_{i\in I}$ of $\mathrm{Ind}(C)'$ is its inductive limit in C. In the special case where small filtered inductive limits are representable in C, one therefore obtains a natural functor

$$\varinjlim_C : \mathrm{Ind}(C) \longrightarrow C. \tag{8.7.1.5}$$

. Of course, the preceding functors may also be defined on categories of the form $\mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U})'$ and $\mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U})$, but in view of (8.2.4.6) they are already determined (up to unique isomorphism) by the preceding functors corresponding to the case $\mathscr{V} = \mathscr{U}$.

. The preceding construction of $\varinjlim_C$ as a left adjoint functor immediately implies that this functor *commutes with arbitrary inductive limits*, and in particular with small filtered inductive limits (the latter being representable in $\mathrm{Ind}(C)$ (8.5.1)). Notice also that $\mathrm{Ind}(C)'$ always contains the essential image of c (formed by the essentially constant ind-objects), and that there is a canonical isomorphism, functorial in $X \in \mathrm{ob}\,\mathrm{Ind}(C)'$:

$$\varinjlim_C c(X) \simeq X, \tag{8.7.1.8}$$

90 i.e. , in the favorable case where C is stable under small inductive limits, there is a canonical isomorphism

$$\varinjlim_C \circ\, c \simeq \mathrm{id}_C\,. \tag{8.7.1.9}$$

8.7.2. Now consider a functor

$$f : C \longrightarrow E, \tag{8.7.2.1}$$

where E is a $\mathscr{U}$-category in which small inductive limits are representable, so that there is a functor

$$\varinjlim_E : \mathrm{Ind}(E) \longrightarrow E.$$

Since we have also defined (8.6)

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(E),$$

we may consider the composite

$$\overline{f} = \varinjlim_E \circ\, \mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow E, \tag{8.7.2.2}$$

sometimes called the *canonical extension of f to ind-objects* (but not to be confused with $\mathrm{Ind}(f)$!). As a composite of two functors that commute with small filtered inductive limits (8.6.3, 8.7.1.7), this functor itself commutes with small filtered inductive limits. Moreover, the isomorphism (8.7.1.9) applied to E, together with (8.6.2.3), shows that $\overline{f}$ *extends* f up to isomorphism, i.e. there is a canonical isomorphism

$$\overline{f} \circ c_C \simeq f\,. \tag{8.7.2.3}$$

91 It is moreover fairly clear, in view of the fact that every object of $\mathrm{Ind}(C)$ is a small filtered inductive limit of objects of C, that the two preceding properties still *characterize* $\overline{f}$ up to unique isomorphism. This can be sharpened to obtain at the same time a universal characterization, up to equivalence, of $\mathrm{Ind}(C)$ among the $\mathscr{U}$-categories E in which small filtered inductive limits are representable. If E and F are two such $\mathscr{U}$-categories, denote by

$$\mathbf{Hom}(E, F)' \subset \mathbf{Hom}(E, F) \tag{8.7.2.4}$$

the full subcategory of $\mathbf{Hom}(E, F)$ formed by the functors that commute with small filtered inductive limits. Then:

PROPOSITION 8.7.3. *Let C be a $\mathscr{U}$-category, and use the notation (8.7.2.4) above. Then the canonical functor*

$$c_C : C \longrightarrow \mathrm{Ind}(C)$$

is 2-universal among the functors with source C and target a $\mathscr{U}$-category having representable small filtered inductive limits. In other words, $\mathrm{Ind}(C)$ *is such a category (8.2.5, 8.5.1), and if E is such a category, the functor*

$$g \longmapsto g \circ c_C : \mathbf{Hom}(\mathrm{Ind}(C), E)' \longrightarrow \mathbf{Hom}(C, E) \tag{8.7.3.1}$$

is an equivalence of categories.

To prove the latter point, it is enough to verify that the map $f \mapsto \overline{f}$ defined by (8.7.2.2) can be refined to a *functor*

$$f \longmapsto \overline{f} = \varinjlim_E \cdot \mathrm{Ind}(f) : \mathbf{Hom}(C, E) \longrightarrow \mathbf{Hom}(\mathrm{Ind}(C), E)', \tag{8.7.3.2}$$

and that the latter is a quasi-inverse of (8.7.3.1). The details of the essentially trivial verification are left to the reader. Alternatively, one may invoke 7.8 to conclude first that (8.7.3.1) is fully faithful, and then (8.7.2.3) to conclude that it is essentially surjective. 92

8.7.4. Return to a functor between $\mathscr{U}$-categories

$$f : C \longrightarrow E, \tag{8.7.4.1}$$

where E is a $\mathscr{U}$-category in which small filtered inductive limits are representable, giving a functor

$$\overline{f} : (X_i)_{i\in I} \longmapsto \varinjlim_i f(X_i) : \mathrm{Ind}(C) \longrightarrow E. \tag{8.7.4.2}$$

We propose to study conditions on f ensuring that $\overline{f}$ is fully faithful, resp. an equivalence of categories. Since f is isomorphic to the composite $\overline{f} \circ c_C$, and $c_C : C \to \mathrm{Ind}(C)$ is fully faithful, we see that for $\overline{f}$ to be fully faithful, it is *necessary* that f be so. Replacing C by its essential image in E, we therefore lose essentially no generality by supposing that f is the inclusion functor of a subcategory C of E, as we shall do below to simplify the notation.

PROPOSITION 8.7.5. *The notation is that of 8.7.4.*

a) *The functor $\overline{f}$ is fully faithful if and only if:*
   (i) *Every object X of C satisfies the following condition:*
   (PF) *For every small filtered inductive system $(Y_i)_{i\in I}$ in C whose inductive limit $\varinjlim_i Y_i$ exists in E, the canonical map* 93

$$\varinjlim_i \mathrm{Hom}(X, Y_i) \longrightarrow \mathrm{Hom}(X, \varinjlim_i Y_i) \tag{8.7.5.1}$$

   is bijective.

b) *The functor $\overline{f}$ is an equivalence of categories if and only if C satisfies condition* (i) *and the following two conditions:*
   (ii) *C is a generating subcategory of E by strict epimorphisms (7.1).*
   (iii) *For every object X of E, the full subcategory $C_{/X}$ of $E_{/X}$ formed by the objects X' over X whose source is in* $\mathrm{Ob}\, C$ *is filtered and essentially small.*

Condition (iii) *is satisfied in particular if C is equivalent to a small category, the finite inductive limits in C are representable, and the inclusion functor $f : C \to E$ commutes with them (i.e. for every finite category J and every functor $\varphi : J \to C$, the inductive limit of $f\varphi$ is representable and is isomorphic in E to an object of C).*

Proof.

a) With the notation of condition (i), let $\mathscr{Y}$ denote the ind-object (Y_i) and let $\mathscr{X}$ denote the constant ind-object defined by X. Then (8.7.5.1) is precisely the canonical map

$$\mathrm{Hom}(\mathscr{X}, \mathscr{Y}) \longrightarrow \mathrm{Hom}(\overline{f}(\mathscr{X}), \overline{f}(\mathscr{Y})),$$

so if $\overline{f}$ is fully faithful, this map is indeed bijective. Conversely, if $\mathscr{X} = (X_j)_{j\in J}$ and $\mathscr{Y} = (Y_i)_{i\in I}$ are arbitrary ind-objects of E, then the map $u : \mathrm{Hom}(\mathscr{X}, \mathscr{Y}) \to$
94 $\mathrm{Hom}(\overline{f}(\mathscr{X}), \overline{f}(\mathscr{Y}))$ is the projective limit over j of the maps $u_j : \mathrm{Hom}(\mathscr{X}_j, \mathscr{Y}) \to \mathrm{Hom}(\overline{f}(\mathscr{X}_j), \mathscr{Y})$, where $\mathscr{X}_j$ is the constant ind-object with value X_j. Thus u is bijective if the u_j are, and (i) therefore implies that $\overline{f}$ is fully faithful.

b) Suppose that (i), (ii), and (iii) hold, and let us prove that $\overline{f}$ is an equivalence. It remains to prove that it is essentially surjective, hence that every $X \in \mathrm{ob}\, E$ belongs to the essential image. But hypothesis (ii) means that $\varinjlim_{C_{/X}} Z \to X$ is an isomorphism, while (iii) says that the category $C_{/X}$ is filtered and essentially small. Thus X is the image of the ind-object of E defined by the natural functor $C_{/X} \to E$. Conversely, suppose that $\overline{f}$ is an equivalence. In view of a), it remains to verify (ii) and (iii). We may suppose that $E = \mathrm{Ind}(C)$, with C identified with the subcategory $c(C)$ of E. But we know (8.3.3 (ii)) that for every $X \in \mathrm{Ind}(C)$, identified if desired with the functor F on C° that it ind-represents, $C_{/X} = C_{/F}$ is an essentially small filtered category (which proves (iii)), and that the inductive limit in $\widehat{C}$ of the functor $C_{/F} \to \widehat{C}$ is F. It is therefore also so in the full subcategory E of $\widehat{C}$, since $F = X$ belongs to E, which proves (ii). It remains to prove the last assertion of b). But the hypothesis on C clearly implies that finite inductive limits are representable in the category $C_{/X}$; *a fortiori*, the category $C_{/X}$ is filtered.

95 Remark 8.7.5.2. The preceding argument shows more generally that, when condition (i) is satisfied, the essential image of the fully faithful functor $\overline{f}$ (8.7.4.2) consists of the $X \in \mathrm{ob}\, E$ such that the category $C_{/X}$ is filtered and essentially small (a condition automatically satisfied if C satisfies the conditions stated at the end of b)), *and* such that the inductive limit of the canonical functor $C_{/X} \to E$ is X.

Corollary 8.7.6. *Let E_{PF} be the full subcategory of E formed by the objects satisfying condition (PF) of 8.7.5 a). Then, for every functor $J \to E_{PF}$, where J is a finite category, that admits an inductive limit X in E, one has $X \in \mathrm{ob}\, E_{PF}$. The canonical functor $(X_i)_{i\in I} \to \varinjlim_i {}_E X_i$ deduced from the inclusion $g : E_{PF} \hookrightarrow E$*

$$\overline{g} : \mathrm{Ind}(E_{PF}) \longrightarrow E \tag{8.7.6.1}$$

is fully faithful. If finite inductive limits are representable in E and E_{PF} is equivalent to a small category, then the essential image of the functor $\overline{f}$ consists of the objects X of E such that the canonical morphism $\varinjlim_{(E_{PF})_{/X}} Y \to X$ is an isomorphism. If, moreover, the subcategory E_{PF} of E is generating by strict epimorphisms (7.1), then the functor (8.7.6.1) is an equivalence of categories.

All the assertions are evident, in the order in which they are stated, in view of 8.7.5, using 2.8 for the first assertion.

96 COROLLARY 8.7.7. *Suppose that finite inductive limits are representable in E. Let C be a full subcategory of E, equivalent to a small category, and consider the functor $(X_i)_{i\in I} \to \varinjlim_{i,E} X_i$:*

$$\overline{f} : \mathrm{Ind}(C) \to E. \tag{8.7.7.1}$$

Let E_{PF} be as in 8.7.6. The following conditions are equivalent:

(i) *The functor $\overline{f} : \mathrm{Ind}(C) \to E$ is an equivalence.*

(ii) *The category C is generating in E by strict epimorphisms, is contained in E_{PF}, and every object of E_{PF} is isomorphic in E (or in E_{PF}, which is equivalent (8.7.6)) to a direct factor of an object of C.*

When the subcategory C of E is stable under direct factors (10.6), the preceding conditions are also equivalent to the following:

(ii bis) *$C = E_{PF}$, and C is generating in E by strict epimorphisms.*

(iii) *The subcategory C of E is generating by strict epimorphisms, is contained in E_{PF}, and finite inductive limits are representable in it.*

When, moreover, finite inductive limits are representable in E, these conditions are also equivalent to

(iii bis) *The subcategory C of E is generating by strict epimorphisms, is contained in E_{PF}, and is stable in E under finite inductive limits (or, equivalently, under finite sums and cokernels of double arrows).*

We already know (8.7.5) that condition (i) implies $C \subset E_{PF}$ and that C is generating 97
by strict epimorphisms. Let us also prove that every object X of E_{PF} is then isomorphic to a direct factor of an object of C. Indeed, we know that X is a filtered inductive limit $\varinjlim_i X_i$ in E of objects of C, and by the definition of E_{PF}, for a suitable i the canonical homomorphism $X_i \to X$ has a left inverse, so X is indeed a direct factor of the object X_i of C. This proves (i) $\Rightarrow$ (ii) (without any hypothesis on C or E, moreover). Let us prove (ii) $\Rightarrow$ (i). Since C is equivalent to a small category and every object of E_{PF} is a direct factor of an object of C, we see that E_{PF} is also equivalent to a small category. Thus, by 8.7.6, the functor (8.7.6.1) is an equivalence of categories, and we conclude by 8.6.4 b) applied to the inclusion $E \hookrightarrow E_{PF}$.

When every direct factor in E of an object of C belongs to C, it is clear that (ii) $\Leftrightarrow$ (ii bis). On the other hand, (ii bis) implies (iii) resp. (iii bis) by 8.7.6. Finally, (iii) (and *a fortiori* (iii bis)) implies (i) by 8.7.5. This completes the proof.

EXERCISE 8.7.8. (*Karoubi envelopes*). Let C be a $\mathscr{U}$-category, let $E = \mathrm{Ind}(C)$, and let $E_{PF} \supset C$ be the full subcategory of E defined in 8.7.6.

a) Prove that images of projectors are representable in E_{PF} and that every object of E_{PF} is a direct factor of an object of C.

b) Call a category F *Karoubian* if images of projectors are representable in it. Let

$$\varphi : C \longrightarrow K$$

be a fully faithful functor such that K is Karoubian and every object of K is a 98
direct factor of an object in the image of C. Prove that φ is 2-*universal* among functors from C with values in Karoubian categories, more precisely: for every Karoubian category F, the functor

$$g \longmapsto g \circ \varphi : \mathbf{Hom}(K, F) \longrightarrow \mathbf{Hom}(K, F)$$

is an equivalence of categories. (Use the fact that every functor commutes with images of projectors.) The category K equipped with φ, determined up to equivalence (itself determined up to unique isomorphism) by the preceding properties, is called the *Karoubi envelope* $\widetilde{C}$ of C. (Compare also IV 7.5.)

c) Deduce from a) and b) that $\mathrm{Ind}(C)_{PF}$, equipped with the functor $C \to \mathrm{Ind}(C)_{PF}$ induced by $c : C \to \mathrm{Ind}(C)$, is a Karoubi envelope of C.

d) Show that every functor $f : C \to C'$ extends in an essentially unique way to a functor $\widetilde{f} : \widetilde{C} \to \widetilde{C'}$ of Karoubi envelopes, and that if these Karoubi envelopes are taken as in c), $\widetilde{f}$ is the functor induced by $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$. Show that the conditions of 8.5.4 b) are also equivalent to the following: $\widetilde{f}$ is an equivalence of categories.

Exercise 8.7.9. Let E be a $\mathscr{U}$-category.

a) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, where C is a $\mathscr{U}$-category.

99 (i bis) As in (i), with C moreover Karoubian (8.7.8 b)).

(ii) The subcategory E_{PF} of E (8.7.6) is generating by strict epimorphisms, and for every object X of E, $(E_{PF})_{/X}$ is an essentially small filtered category.

Show that if C is a *Karoubian* $\mathscr{U}$-category such that E is equivalent to $\mathrm{Ind}(C)$, then C is equivalent to E_{PF} (by an equivalence determined up to unique isomorphism). Establish a 2-equivalence between the 2-category formed by the categories E satisfying the preceding conditions and the functors between them that *commute with small filtered inductive limits*, and the 2-category formed by *Karoubian* $\mathscr{U}$-categories and arbitrary functors between them.

b) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, where C is a $\mathscr{U}$-category in which finite inductive limits are representable.

(i bis) As in (i), but with C moreover Karoubian.

(ii) Finite inductive limits in E are representable, the subcategory E_{PF} of E is generating by strict epimorphisms, and for every object X of E, there exists a small subset I of $\mathrm{Ob}\, E_{PF/X}$ such that every object of $E_{PF/X}$ is bounded above by an object of I. (Use 8.9.5 b).)

100

8.8. Indexed representation of a functor $J \to \mathrm{Ind}(C)$.

8.8.1. Let

$$\varphi : J \longrightarrow \mathrm{Ind}(C) \text{ i.e. } \varphi \in \mathrm{Ob}\, \mathbf{Hom}(J, \mathrm{Ind}(C))$$

be a functor, where C is a $\mathscr{U}$-category. Recall (8.5.4) that an *indexed representation of* φ is, by definition, an ind-object of $\mathbf{Hom}(J, C)$

$$\psi : I \longrightarrow \mathbf{Hom}(J, C),$$

with I an essentially small filtered category, together with a specified isomorphism from its inductive limit in $\mathbf{Hom}(J, \mathrm{Ind}(C))$ to φ. Thus φ admits an indexed representation if and only if it is isomorphic to an essentially small filtered inductive limit in $\mathbf{Hom}(J, \mathrm{Ind}(C))$ of objects of the full subcategory $\mathbf{Hom}(J, C)$. We shall say that the category J is *admissible for* C if every functor $\varphi : J \to \mathrm{Ind}(C)$ admits an indexed representation.

Since in $\mathrm{Ind}(C)$ small filtered inductive limits are representable (8.5.1), the same is true in $\mathbf{Hom}(J, \mathrm{Ind}(C))$, and they are computed "argument by argument". Consequently, the inclusion functor

$$\mathbf{Hom}(J, C) \hookrightarrow \mathbf{Hom}(J, \mathrm{Ind}(C)) \tag{8.8.1.1}$$

extends canonically to a functor

$$\mathrm{Ind}(\mathbf{Hom}(J,C)) \longrightarrow \mathbf{Hom}(J,\mathrm{Ind}(C)) \tag{8.8.1.2}$$

(8.7.2). The objects of the essential image of this functor are then precisely the φ that admit an indexed representation; thus, to say that J is admissible for C also means that the functor (8.8.1.2) is essentially surjective. In this connection, we note:

PROPOSITION 8.8.2. *If the category J is equivalent to a finite category, the canonical functor (8.8.1.2) is fully faithful; thus J is admissible for C if and only if this functor is an equivalence of categories.* 101

By 8.7.5 a), it suffices to prove that for every small filtered inductive system $(\varphi_i)_{i\in I}$ in $\mathbf{Hom}(J,C)$ and every object φ of $\mathbf{Hom}(C,J)$, the canonical map

$$\varinjlim_i \mathrm{Hom}(\varphi,\varphi_i) \longrightarrow \mathrm{Hom}(\varphi,\varinjlim_i \varphi_i) \tag{8.8.2.1}$$

is bijective, where $\varinjlim_i \varphi_i$ denotes the inductive limit taken in $\mathrm{Hom}(J,\mathrm{Ind}(C))$. Now, if φ, ψ are two functors $J \to C$, there is an exact diagram of sets, functorial in φ and ψ:

$$\mathrm{Hom}(\varphi,\psi) \longrightarrow \prod_{X\in \mathrm{Ob}\, J} \mathrm{Hom}(\varphi(X),\psi(X)) \rightrightarrows \prod_{\substack{f\in Fl J\\ f:X\to Y}} \mathrm{Hom}(\varphi(X),\psi(Y)).$$

Since filtered inductive limits commute with kernels of double arrows and with finite products, the bijectivity of (8.8.2.1) follows when J is finite. The case in which J is equivalent to a finite category reduces immediately to the preceding case.

PROPOSITION 8.8.3. *Let $\varphi : J \to \mathrm{Ind}(C)$ be a functor, with J equivalent to a small category. For φ to admit an indexed representation, the following two conditions are sufficient, and they are also necessary when J is equivalent to a finite category:*

a) *The category $\mathbf{Hom}(J,C)_{/\varphi}$ (formed by the arrows in $\mathbf{Hom}(J,\mathrm{Ind}(C))$ with target φ and source in $\mathrm{Hom}(J,C)$) is filtered.*

b) *For every object j of J, the functor* 102

$$\psi \longmapsto \psi(j) : \mathbf{Hom}(J,C)_{/\varphi} \longrightarrow C_{/\varphi(j)} \tag{8.8.3.1}$$

is cofinal, i.e. satisfies conditions F 1), F 2) *of* 8.1.3.

Proof. Suppose a) and b) hold; let us prove that φ admits an indexed representation. We may clearly suppose that J is small. Let

$$I = \mathbf{Hom}(J,C)_{/\varphi}, \tag{8.8.3.2}$$

which is a filtered category by hypothesis, and first suppose that I is essentially small, so that the "inclusion" of I in $\mathbf{Hom}(J,C)$ defines an ind-object of $\mathbf{Hom}(J,C)$, $i \mapsto \psi_i$. Moreover, there is a canonical homomorphism

$$\varinjlim_i \psi_i \longrightarrow \varphi, \tag{8.8.3.3}$$

given argument by argument by the homomorphism

$$\varinjlim_I \psi_i(j) \longrightarrow \varphi(j) = \varinjlim_{C_{/\varphi(j)}} X$$

induced by the functor (8.8.3.1). Since the latter is cofinal, we conclude that (8.8.3.3) is an isomorphism, which proves the assertion. In the case where we do not suppose I essentially small, it is enough to construct an essentially small full subcategory I'

such that (8.8.3.3) remains an isomorphism when $\varinjlim_{I'}$ is used in place of $\varinjlim_{I}$. For this purpose, it is enough that the composite functors

$$I' \longrightarrow I \longrightarrow C_{/\varphi(j)}$$

103 induced by the functors (8.8.3.1) all be cofinal. We then conclude by the following lemma, whose proof is immediate from criterion (8.1.3 b)) and is left to the reader:

LEMMA 8.8.3.4. *Let I be a filtered $\mathscr{U}$-category, and let*

$$f_j : I \longrightarrow I_j \quad (j \in J)$$

be a small family of cofinal functors from I to essentially small filtered categories. Then there exists a full subcategory I' of I which is filtered and essentially small, and such that the functors induced by the f_j are cofinal.

Finally, let us prove necessity in 8.8.3 when J is assumed equivalent to a finite category. An indexed representation of φ by means of an essentially small filtered indexing category I defines a functor ψ

(8.8.3.5)
$$\begin{array}{ccc} I & \xrightarrow{\psi} & \mathbf{Hom}(J, C)_{/\varphi} \\ & \searrow^{\psi_j} & \downarrow \\ & & C_{/\varphi(j)} \end{array} ,$$

and it is enough to prove that ψ and each of the functors ψ_j induced by it are cofinal: by 8.1.3 b), it will then indeed follow that $\mathbf{Hom}(J, C)_{/\varphi}$ is filtered, and by definition 8.1.1 that the vertical arrow in (8.8.3.5) is also cofinal. By 8.8.2, we may identify φ with an object of $\mathrm{Ind}(E)$, where $E = \mathbf{Hom}(J, C)$, and the fact that the functor

$$\psi : I \longrightarrow E_{/\varphi},$$

104 arising from the ind-object φ of E indexed by I is cofinal is a general fact, which follows immediately from criteria (F 1) and (F 2) of 8.1.3. The same argument (with E, φ replaced by C, $\varphi(j)$) shows that ψ_j is cofinal, which completes the proof.

REMARK 8.8.4.
a) In the case where J is equivalent to a finite category, if φ admits an indexed representation, we have seen that (8.8.3.5) is a cofinal functor, which implies that the category $\mathbf{Hom}(J, C)_{/\varphi}$ is itself essentially small.
b) Suppose that in C finite inductive limits are representable. Then the same is true in $E = \mathbf{Hom}(J, C)$, and these limits are computed argument by argument and are also inductive limits in $\mathbf{Hom}(J, \widehat{C})$ and *a fortiori* in $\mathbf{Hom}(J, \mathrm{Ind}(C))$. It follows immediately that condition a) of 8.8.3 is then automatically satisfied, and it remains only to consider condition b). I do not know whether it is automatically satisfied when J is in addition assumed small.

We now come to the main result of this section:

PROPOSITION 8.8.5. *Let J be a category equivalent to a finite category, and suppose in addition that J is rigid, i.e. that for every $j \in \mathrm{Ob}\, J$, every endomorphism of j is the identity. Then J is admissible for C (for any $\mathscr{U}$-category C), i.e. every functor $J \to \mathrm{Ind}(C)$ admits an indexed representation.*

105 Upon replacing J by an equivalent category if necessary, we may suppose that J is *skeletal*, i.e. that two isomorphic objects of J are equal. Then J is in fact finite. We proceed by induction on card Ob J; the case where this number is ≤ 0 is trivial, so we shall suppose it is ≥ 1. Let $j_\circ$ be a *maximal* object of J, i.e. such that for every arrow $j_\circ \to j$, there exists an arrow $j \to j_\circ$; in view of the fact that J is rigid, this implies that $j_\circ \to j$ is an isomorphism, and since J is skeletal, it implies that $j_\circ = j$. Let J' therefore be the full subcategory obtained from J by removing the object $j_\circ$, and let J'' be the full subcategory consisting of the object $j_\circ$. The proposition then follows from the

LEMMA 8.8.5.1. *Let J be a finite category, and let J' and J'' be two full subcategories such that* Ob J = Ob J' ∪ Ob J'', *and such that for every $j' \in$ Ob J' and $J'' \in$ Ob J'', one has* $\mathrm{Hom}(j'', j') = \emptyset$. *If J' and J'' are admissible for C, then so is J.*

It is enough to prove criteria a) and b) of 8.8.3, which amounts to carrying out six elementary verifications, namely the last two filteredness conditions for $\mathbf{Hom}(J, C)_{/\varphi}$, and conditions F 1) and F 2) for the functors (8.8.3.1), successively in the cases $j \in$ Ob J' and $j \in$ Ob J''; the latter conditions moreover imply that $\mathbf{Hom}(J, C)_{/\varphi}$ is nonempty. The rather tedious verification presents no difficulty and is left to the reader. (The editor sought in vain for an elegant proof that would circumvent these unpleasant verifications.)

EXERCISE 8.8.6. a) Let J, J' be two categories admissible for C, one of them 106
being finite. Prove that $J \times J'$ is admissible for C. (Use 8.8.2.)

b) Prove that a small discrete category is admissible for every category C.

c) Let J be a category satisfying the following conditions: (i) J is rigid, (ii) Ob J is countable, (iii) For any two objects j, j' of J, $\mathrm{Hom}(j, j')$ is finite, (iv) For every object j of J, the set of objects of J that are bounded above by j, i.e. that are sources of an arrow with target j, is finite. Show that J is admissible for every $\mathscr{U}$-category C. (First show that J can be written as a filtered union of finite full subcategories J_n, such that every object of J bounded above by an object of J_n belongs to J_n. Then verify criteria a), b) of 8.8.3.)

EXERCISE 8.8.7. In our notation we identify an ordered set with the category it defines.

a) Let C be an ordered set. Let $\tilde{C}$ be the set of subsets A of C that are filtered and such that $x \in A$ and $y \leq x$ imply $y \in A$. Let $L_\circ : C \to \tilde{C}$ be the map that associates to every $x \in C$ the set $L_\circ(x)$ consisting of those $y \in A$ such that $y \leq x$. Show that L is injective and that the order on C is induced by that on $\tilde{C}$. For every $A \in \tilde{C}$, consider the inclusion $f(A) : A \to C$, which is an ind-object of C; for every ind-object $\varphi : I \to C$ of C, let $g(\varphi) \in \tilde{C}$ be the subset of C consisting of the elements of C bounded above by an element of the form $\varphi(i)$. Show that one thus obtains two mutually quasi-inverse equivalences 107

$$\mathrm{Ind}(C) \underset{\approx}{\overset{f,g}{\leftrightarrows}} \tilde{C},$$

transforming the functor $L_\circ$ into the canonical functor $L : C \to \mathrm{Ind}(C)$.

b) Suppose that for every $x \in C$, the set of elements bounding x from above (resp. from below) is finite. Show that every ind-object (resp. pro-object) of C is essentially constant, and indeed that for every such $\varphi : I \to C$ (resp. $\varphi : I^\circ \to C$) there exists an $i_\circ \in$ Ob I such that the functor induced on $I_{/i_\circ}$ is constant (i.e. takes every arrow to an isomorphism).

c) Let $C \subset \mathbf{N}^\circ \times \mathbf{N}$ be formed by the pairs of natural numbers (j, i) such that $i \geq j$. Show that $\tilde{\mathbf{N}}$ is isomorphic to the ordered set obtained from $\mathbf{N}$ (identified with an ordered subset of $\tilde{\mathbf{N}}$ as in a)) by adjoining a greatest element ∞. Show that $\tilde{C}$ is isomorphic to $\mathbf{N}^\circ \times \tilde{\mathbf{N}}$. Consider the functor

$$\varphi : \mathbf{N}^\circ \longrightarrow \tilde{C} \quad , \quad \varphi(j) = (j, \infty).$$

Show that:

1) the projective limit of φ is not representable in $\tilde{C}$, although filtered projective limits in C are representable (and are essentially constant, by b)).
2) For every functor $\psi : \mathbf{N}^\circ \to C$, one has $\mathrm{Hom}(\psi, \varphi) = \varnothing$, and *a fortiori* the functor φ admits no indexed representation.

108 Exercise 8.8.8. Let $X = \mathbf{N} \amalg \mathbf{N}$ be the disjoint union of two copies of $\mathbf{N}$, let s be the symmetry of X, and for every $i \in \mathbf{N}$, let X_i be the subset $\Delta_{i+1} \amalg \Delta_i$ of $\mathbf{N}$ (where Δ_i denotes the subset of $\mathbf{N}$ formed by the j such that $j \leq i$). Let C be the subcategory of the category of subsets of X whose objects are the X_i and whose arrows are the maps between the X_i induced by the identity of X or by its symmetry s, and let $\tilde{C}$ be the category defined analogously, but with the additional object X.

a) Show that $\mathrm{Ind}(C)$ is equivalent to $\tilde{C}$, with the canonical functor $C \to \mathrm{Ind}(C)$ corresponding to the inclusion $C \to \tilde{C}$. Show that there exists no pair (Y, f) consisting of an object Y of C and an endomorphism f of Y, together with a morphism of objects with endomorphism in $\mathrm{Ind}(C)$ from (Y, f) to (X, s).
b) Deduce that if J is the category having a single object and, in addition to the identity arrow, a single arrow of order 2, then the functor $J \to \mathrm{Ind}(C)$ defined by (X, s) admits no indexed representation.

8.9. Exactness properties of $\mathrm{Ind}(C)$.

Proposition 8.9.1. *Let C be a $\mathscr{U}$-category.*

a) *The canonical functors L (8.2.4.8) and c (8.4.1)*

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C}$$

commute with projective limits.
b) *Suppose that finite inductive limits are representable in C and that C is equivalent to a small category. Then small projective limits are representable in* $\mathrm{Ind}(C)$.
109 c) *If small projective limits (resp. finite projective limits) are representable in C, then the same is true in* $\mathrm{Ind}(C)$. *Let J be a category that is finite and rigid, or discrete; if projective limits of type J are representable in C, projective limits of the same type are representable in* $\mathrm{Ind}(C)$.
d) *Small filtered inductive limits in* $\mathrm{Ind}(C)$ *(which are representable by virtue of 8.5.1) are left exact, i.e. commute with finite projective limits.*

Proof.

a) The fact that L commutes with projective limits follows from the fact that it is fully faithful and from the computation of projective limits in $\widehat{C}$ "argument by argument," which implies that, to verify that a projective system of morphisms $F \to F_i$ $(i \in I)$ in $\widehat{C}$ exhibits F as a projective limit of the F_i, it suffices to verify that the analogous assertion holds for the projective systems of maps of sets $\mathrm{Hom}(X, F) \to \mathrm{Hom}(X, F_i)$, for every $X \in \mathrm{Ob}\, C$. But $C \subset \mathrm{Ind}(C)$.

The fact that c commutes with projective limits follows formally from the fact that L does so, as does the composite $L \circ c : C \to \widehat{C}$.

b) In view of a), the assertion amounts to saying that every projective limit of ind-representable presheaves is ind-representable, which follows at once from the criterion 8.3.3 (v), since a projective limit of left exact functors is left exact (which follows from the fact that "projective limits commute with projective limits" (2.5.0)).

c) This follows formally from the analogous property of $\widehat{C}$ (3.3), and from the fact that L is conservative (being fully faithful) and commutes with limits of the type under consideration (a) and 8.5.1). 110

d) It is well known (cf. 2.3) that the representability of finite projective limits is equivalent to that of projective limits of the types

empty , … ,

(corresponding to certain finite ordered sets), and the representability of small projective limits amounts to that of products and finite projective limits. Thus the second assertion made in (c) implies the first.

Suppose first that I is finite. Using the result 8.8.5 on the representability of functors $\varphi : J \to \mathrm{Ind}(C)$ in indexed form (8.5.4)

$$\phi : J \times I \longrightarrow C, \tag{8.9.1.1}$$

the existence of $\varinjlim \varphi$ is therefore a special case of the following more precise and more general result:

Corollary 8.9.2. *Consider a functor $\varphi : J \to \mathrm{Ind}(C)$ given in indexed form Φ (8.9.1.1), where J is a finite category. If for every $i \in \mathrm{Ob}\, I$, the partial functor $j \to \Phi(j,i)$ has a projective (resp. inductive) limit representable in C, then φ has a projective (resp. inductive) limit representable in* $\mathrm{Ind}(C)$*, and there is a canonical isomorphism*

$$\varinjlim \varphi \simeq \text{«}\varinjlim_i\text{»} \varprojlim_j \Phi(j,i) \tag{8.9.2.1}$$

(resp. 111

$$\varinjlim \varphi \simeq \text{«}\varinjlim_i\text{»} \varprojlim_j \Phi(j,i)), \tag{8.9.2.2}$$

where $\varinjlim_j$ (resp. . $\varprojlim_j$) is computed in C.

For the first formula, one simply uses the fact that in $\mathrm{Ind}(C)$ filtered inductive limits commute with $\varprojlim_j$, and that c commutes with $\varprojlim_J$ (8.9.1 d) and a)):

$$\varinjlim \varphi = \varprojlim_j \text{«}\varinjlim_i\text{»}\, \Phi(j,i) \overset{\sim}{\leftarrow} \text{«}\varinjlim_i\text{»} \varprojlim_j{}^{\mathrm{Ind}(C)} \Phi(j,i) \simeq \text{«}\varinjlim_i\text{»} \varprojlim_j{}^{C}\, \Phi(j,i).$$

The second is proved in the same way, using the commutation of inductive limits of type I with inductive limits of type J (2.5.0) and the fact that c commutes with inductive limits of type J, proved in 8.9.4 a) below.

The case where J is discrete. This case is contained in the following more precise assertion:

Corollary 8.9.3. *Let the I_α be essentially small filtered categories indexed by a small set A, and for every $\alpha \in A$, let $X(\alpha) = (X(\alpha)_{i_\alpha})_{i_\alpha \in I_\alpha}$ be an ind-object of C indexed by I_α. Let I be the product category of the I_α, and suppose that for every $\imath = (i_\alpha)_{\alpha\in A} \in I$, the product $\prod_{\alpha\in A} X(\alpha)_{i_\alpha}$ is representable in C. Then the product $\prod_{\alpha\in A} X(\alpha)$ is representable in* Ind(C)*, and it is canonically isomorphic to the ind-object indexed by I given by the formula*

$$\prod_{\alpha\in A} \text{« }\varinjlim_{i_\alpha\in I_\alpha}\text{ »}\ X(\alpha)_{i_\alpha} \simeq \text{« }\varinjlim_{\imath=(i_\alpha)_{\alpha\in A}\in I}\text{ »} \left(\prod_{\alpha\in A} A(\alpha)_{i_\alpha}\right), \tag{8.9.3.1}$$

112 *where the product on the right-hand side is the product computed in C. (Note that I is filtered and essentially small, since the I_α are.)*

Indeed, it is well known (and immediate by reduction to the case in which one works in the category of sets) that the formula in question holds when the limits are computed in $\widehat{C}$. The conclusion then follows from the fact that L commutes with the limits in question (8.9.1 a) and 8.5.1).

Remarks 8.9.4. The proof given for 8.9.2 and 8.9.3 shows, more generally, that if for every $i \in \mathrm{Ob}\, I$, $\varinjlim_j \Phi(j,i)$ (resp. $\varprojlim_j \Phi(j,i)$) *computed in* Ind(C) is representable, then the same is true of $\varprojlim_i \varphi$ (resp. of $\varinjlim_j \varphi$). This, together with the argument of c), shows that for a given category J arising from a finite or discrete ordered set (or more generally, one that is C-admissible (8.3.1)), projective limits (resp. inductive limits) of type J are representable in Ind(C) if (and only if) for every functor $\varphi : J \to C$, the projective (resp. inductive) limit of φ *computed in* Ind(C) is representable. In the nonparenthetical case, this also means, by virtue of (a), that every projective limit of type J of representable presheaves on C is ind-representable.

Proposition 8.9.5. *Let C be a $\mathscr{U}$-category.*

a) *The canonical functor*
$$c : C \longrightarrow \mathrm{Ind}(C)$$
is right exact (hence exact, in view of 8.9.1 a)*).*

b) *If finite inductive limits (resp. finite sums) are representable in C, then small inductive limits (resp. small sums) are representable in* Ind(C)*. Let J be a preordered* 113 *set that is finite or discrete; if inductive limits of type J are representable in C, the same is true in* Ind(C)*.*

Proof.

a) Suppose that in C one has $X \simeq \varinjlim_J X_j$, where J is a finite category and X and the X_j belong to C. Then, for every ind-object $\mathscr{Y} = (Y_i)_{i\in I}$ of C, one has:
$$\mathrm{Hom}(X,\mathscr{Y}) \simeq \varinjlim_i \mathrm{Hom}(X,Y_i) \simeq \varinjlim_i \varprojlim_j \mathrm{Hom}(X_j,Y_i)$$
$$\overset{2.8}{\simeq} \varprojlim_j \varinjlim_i \mathrm{Hom}(X_j,Y_i) \simeq \varprojlim_j \mathrm{Hom}(X_j,\mathscr{Y}),$$
and the desired conclusion follows by comparing the outer terms.

b) We already noted in 8.8.4 that the assertions made follow from a) and 8.9.2, at least for finite inductive limits. To prove the stated conclusions in the infinite cases, we are reduced to the case of sums, which can be interpreted as a filtered inductive limit of finite sums, and we therefore conclude by 8.5.1.

Remark 8.9.6. We have already observed that the functor c does not in general commute with filtered inductive limits, and hence does not commute with infinite sums either, in contrast to what happens for $L : \mathrm{Ind}(C) \to \widehat{C}$. On the other hand, apart from its commutation with *filtered* inductive limits (8.5.1), L has practically no commutation properties with any other type of inductive limit (initial object, sum of two objects, cokernels of double arrows). Thus, if C admits an initial object $\varnothing_C$, which is therefore an initial object of $\mathrm{Ind}(C)$ by a), $\varnothing_C$ is never an initial object of $\widehat{C}$ (i.e. identical to the constant presheaf with value $\varnothing$), since $\mathrm{Hom}(\varnothing_C, \varnothing_C) \neq \varnothing$!

Proposition 8.9.7. *Let* 114

$$f : C \longrightarrow C'$$

be a functor between $\mathscr{U}$-categories, and let J be a finite rigid category (8.8.5). If inductive (resp. projective) limits of type J are representable in C and f commutes with them, then $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$ *also commutes with limits of this type.*

This follows at once from 8.8.5 and from the calculation 8.9.2 of finite limits in a category of ind-objects, for a functor represented in indexed form.

Corollary 8.9.8. *If finite inductive (resp. projective) limits are representable in C, and if f is right exact (resp. left exact), then the same is true of* $\mathrm{Ind}(f)$*; in the unparenthesized case,* $\mathrm{Ind}(f)$ *even commutes with all small inductive limits.*

The first assertion follows from 8.9.7. The second follows from the first, taking into account that $\mathrm{Ind}(f)$ commutes with filtered inductive limits (8.6.3).

Exercise 8.9.9. Let C be a $\mathscr{U}$-category.

a) Suppose that finite sums are representable in C. Show that if *finite* sums in C are disjoint (resp. universal) (cf. II 4.5), then in $\mathrm{Ind}(C)$ *small* sums are disjoint (resp. universal).

b) Suppose that every morphism in C factors as an epimorphism followed by a monomorphism (resp. as an effective epimorphism followed by a monomorphism, resp. as an epimorphism followed by an effective monomorphism, resp. as an effective epimorphism followed by an effective monomorphism). Show that $\mathrm{Ind}(C)$ satisfies the same property. In the last case, it follows that every epimorphism of $\mathrm{Ind}(C)$ is effective, every monomorphism of $\mathrm{Ind}(C)$ is effective, and every bimorphism of $\mathrm{Ind}(C)$ is an isomorphism; show that in this case every epimorphism (resp. monomorphism) of $\mathrm{Ind}(C)$ can be represented by an inductive system of epimorphisms (resp. monomorphisms) of C. If, in addition, every epimorphism (resp. every monomorphism) in C is assumed to be *universal*, the same property holds in $\mathrm{Ind}(C)$. 115

c) Suppose that C is additive (resp. abelian); then so is $\mathrm{Ind}(C)$.

d) Let $X = \text{« } \varinjlim_I \text{ » } X_i$ be an ind-object of C, and let $R \rightrightarrows X$ be an equivalence relation on X. For every $i \in \mathrm{Ob}\, I$, let $R_i \rightrightarrows X_i$ be the equivalence relation on X_i (regarded as an object of $\mathrm{Ind}(C)$) induced by R via $X_i \to X$. (We assume that the fiber product $R_i = R \times_{X \times X} X_i \times X_i$ in $(\mathrm{Ind}(C))$ is representable by an object of $\mathrm{Ind}(C)$, as is the case if finite projective limits are representable in C.) Show that, for R to be effective, it suffices that the R_i be effective.

e) Let X be an object of C, and let R be an equivalence relation on $c(X)$, i.e. on X regarded as an object of $\mathrm{Ind}(C)$. Suppose that fiber products are representable in C. For R to be effective, it is necessary that R be of the form $\text{« } \varinjlim_i \text{ » } R_i$, where

the R_i are equivalence relations on X (regarded as an object of C), and this
116 condition is sufficient if equivalence relations in C are assumed to be effective.

f) Suppose that every morphism in C factors as an effective epimorphism followed by an effective monomorphism, and that finite projective limits are representable. Let X be an object of C, and let R be a subobject of $X \times X$, regarded as an object of $\mathrm{Ind}(C)$. Write R in the form « $\varinjlim$ » Z_i, where $(Z_i)_{i\in I}$ is a filtered increasing family of subobjects of $X \times X$ *in* C (this is possible by b)). Show that, for R to be an equivalence relation on X, it is necessary and sufficient that, for every $i \in \mathrm{Ob}\, I$, there exists an index $j \in \mathrm{Ob}\, I$ whose corresponding subobject contains $s(Z_i)$ and $Z_i \circ Z_i$, where s is the symmetry of $X \times X$.

g) Suppose that in C finite projective limits and finite sums are representable, that every equivalence relation there is effective, and that every morphism there factors as an effective epimorphism followed by an effective monomorphism. Show that equivalence relations in $\mathrm{Ind}(C)$ are universal if and only if C satisfies the following condition:

ST) For every object X of C and every subobject Z of $X \times X$ in C, if the sequence of subobjects Z_n ($n \geq 0$) of $X \times X$ in C is defined recursively by $Z_0 = Z$, $Z_n = \mathrm{Sup}(Z_n, s(Z_n), Z_n \circ Z_n)$ (where Sup is taken in the set of subobjects of $X \times X$, which exists by the hypotheses made on C), then the sequence $(Z_n)_{n\geq 0}$ is *stationary.*

k) Show that equivalence relations in Ind((Ens)) are not necessarily effective.

* NB. For criteria ensuring that $\mathrm{Ind}(C)$ is a topos, cf. VI 8.9.9.$_*$

117 Exercise 8.9.10. Let C be a $\mathscr{U}$-category. Put $\mathrm{Pro}(C) = (\mathrm{Ind}(C^\circ))^\circ$ (cf. 8.11).

a) Suppose that finite sums (resp. small sums) are representable in C. Show that if they are disjoint (cf. II 4.5), then $\mathrm{Pro}(C)$ satisfies the same condition.

b) Let J be a small set such that sums indexed by J are representable in C, and hence also in $\mathrm{Pro}(C)$. Let $(X(\alpha))_{\alpha\in J}$ be a family of objects of $\mathrm{Pro}(C)$, with $X(\alpha) = $ « $\varprojlim_{I_\alpha}$ » X_{i_α}. Show that, for the sum of the $X(\alpha)$ in $\mathrm{Ind}(C)$ to be universal, it suffices that the same hold for each of the families $\varprojlim_J X_{i_\alpha}$, where $(i_\alpha)_{\alpha\in I} \in \prod_{\alpha\in J} I_\alpha$. Deduce that, if in C sums of type J are universal, then for the same to hold in $\mathrm{Pro}(C)$ it is necessary and sufficient that, for every family $(X(\alpha))_{\alpha\in J}$ as above, with $I_\alpha = I$ for every $\alpha \in J$, the canonical homomorphism in $\mathrm{Pro}(C)$

$$« \varprojlim_I » \amalg_\alpha X(\alpha)_i \longleftarrow « \varprojlim_{I^J} » \amalg_{\alpha\in J} X(\alpha)_{i_\alpha}$$

be an isomorphism. Deduce that in Pro((Ens)), sums of type J are universal if and only if J is finite.

Exercise 8.9.11. Let C be a $\mathscr{U}$-category.

a) Let $(X_i \to X)_{i\in I}$ be a family of morphisms in C. Show that, for it to be epimorphic in $\mathrm{Ind}(C)$, it suffices that there exists a *finite* subfamily that is epimorphic in C. Prove that this condition is also necessary when it is assumed that in C every finite family of morphisms with target X_i factors as an epimorphic family,
118 followed by an effective monomorphism (10.5). (For this last assertion, let P be the set of finite subsets of I, ordered by inclusion, and let $X' = (X'_J)_{J\in P}$ be the ind-object formed by the images of the finite subfamilies of the given family;

finally, let $X'' = X \amalg_{X'} X = \text{« lim »}\, X \amalg_{X'_J} X$. Show that the two canonical morphisms $X \rightrightarrows X''$ are distinct, but coincide on the X'_J.)

b) Suppose that every finite family of morphisms with the same target in C factors as an epimorphic family followed by an effective monomorphism. Prove that $\mathrm{Ind}(C)$ admits a small subcategory generating by epimorphisms (7.1) if and only if C admits a small subcategory C' such that every object of C is the target of a *finite* epimorphic family with source in C'. If C is assumed to admit a small generating subcategory (7.1), then $\mathrm{Ind}(C)$ admits a small subcategory generating by epimorphisms if and only if C is equivalent to a small category. (For this last statement, use 7.5.2).

c) $\mathrm{Ind}((\mathrm{Ens}))$ does not admit a small generating subcategory.

Exercise 8.9.12. Let C be a $\mathscr{U}$-category, and consider $\mathrm{Pro}(C) = \mathrm{Ind}(C^\circ)^\circ$.

a) Show that if $\mathrm{Pro}(C)$ admits a small subcategory P' generating by epimorphisms (7.1), then there exists a small subcategory C' of C that is generating by epimorphisms in $\mathrm{Pro}(C)$. (Take the category of components of the objects of P'.)

b) Show that the category $\mathrm{Pro}((\mathrm{Ens}))$ has no small subcategory generating by epimorphisms. (If C' is as in a), choose a set X whose cardinality strictly exceeds the cardinalities of the objects of C', and consider the pro-set $\mathscr{X}$ formed by the complements in X of subsets of cardinality $< \mathrm{card}(X)$. Show that, for every nonempty set Y of cardinality $< \mathrm{card}(X)$, one has $\mathrm{Hom}(Y, \mathscr{X}) = \emptyset$, but that $\mathscr{X}$ is not isomorphic to the constant pro-set $\emptyset$.) 119

8.10. Dual notions: pro-objects, pro-representable functors. Let C be a $\mathscr{U}$-category. A *pro-object* of C is a functor

$$\varphi : I^\circ \longrightarrow C, \tag{8.10.1}$$

where I is an essentially small filtered category (called the index category), and, as usual, I° denotes the opposite category. Let $\mathscr{V}$ be a universe such that $\mathscr{V} \supset \mathscr{U}$. One should note that the pro-objects of C indexed by $I \in \mathscr{V}$ are in one-to-one correspondence with the ind-objects of C° indexed by $I \in \mathscr{V}$: to every such ind-object $\psi : I \to C^\circ$ one associates the pro-object $\varphi = \psi^\circ : I^\circ \to C$. Following this correspondence, one defines "by transport of structure and reversal of arrows" the notion of a morphism between pro-objects of C from the analogous notion (8.2.4.2), (8.2.5.1) for ind-objects, and consequently defines the category of pro-objects of C indexed by $I \in \mathscr{V}$, together with a canonical isomorphism

$$\mathrm{Pro}_{\mathscr{V}}(C, \mathscr{U}) \simeq \mathrm{Ind}_{\mathscr{V}}(C^\circ, \mathscr{U})^\circ; \tag{8.10.2}$$

for $\mathscr{V} = \mathscr{U}$ one simply speaks of the *category of pro-objects of* C, denoted $\mathrm{Pro}_{\mathscr{U}}(C)$ or simply $\mathrm{Pro}(C)$, which is thus defined by 120

$$\mathrm{Pro}(C) = \mathrm{Pro}_{\mathscr{U}}(C, \mathscr{U}) \simeq \mathrm{Ind}(C^\circ)^\circ. \tag{8.10.3}$$

If two pro-objects are given in *indexed form*

$$\mathscr{X} = (X_i)_{i \in I} \quad , \quad \mathscr{Y} = (Y_j)_{j \in J}, \tag{8.10.4}$$

formula (8.2.5.1) takes here the form

$$\mathrm{Hom}(\mathscr{X}, \mathscr{Y}) \simeq \varprojlim_j \varinjlim_i \mathrm{Hom}(X_i, Y_j). \tag{8.10.5}$$

The functor (8.4.1), applied to C°, gives, upon passing to opposite categories, a canonical functor (denoted by the same letter c when there is no risk of confusion) that identifies C with a full subcategory of $\mathrm{Pro}(C)$:

$$c : C \longrightarrow \mathrm{Pro}(C); \tag{8.10.6}$$

it is in order to have such a functor, rather than $C \to \mathrm{Pro}(C)^\circ$, that we "reversed the arrows" in the definition of morphisms of pro-objects starting from the analogous definition for ind-objects. Pro-objects in the essential image of (8.10.6) are again called *essentially constant pro-objects.*

Put

$$\check{C} = (C^\circ)\hat{} = \mathbf{Hom}(C, (\mathrm{Ens})), \tag{8.10.7}$$

then the functor (8.2.4.7) may be regarded as a canonical fully faithful functor

$$L : \mathrm{Pro}_{\mathscr{V}}(C, \mathscr{U}) \longrightarrow \check{C}^\circ, \tag{8.10.8}$$

121 and in particular one obtains

$$L : \mathrm{Pro}(C) \hookrightarrow \check{C}^\circ. \tag{8.10.9}$$

We leave it to the reader to translate, as the need arises, the results of the preceding sections and of those that follow from the language of ind-objects into that of pro-objects. Depending on the mathematical context, one language or the other is more useful, not to mention the cases in which both notions arise simultaneously, for example when one must consider complex categories such as $\mathrm{Pro}(\mathrm{Ind}(C))$ or $\mathrm{Ind}(\mathrm{Pro}(C))$. We confine ourselves to giving a few additional indications, in order to fix the terminology and notation and to make the "yoga" precise.

8.10.10. A *covariant functor* $F : C \to (\mathrm{Ens})$ is called a *pro-representable functor* if it belongs to the essential image of (8.10.9) (or of 8.10.8, which amounts to the same thing); the full subcategory of $\check{C}$ formed by these functors is therefore equivalent to $\mathrm{Pro}(C)^\circ$. In general one will prefer to work in the opposite category, the essential image *as a subcategory of* (8.10.9), which is therefore equivalent to $\mathrm{Pro}(C)$ itself. In accordance with this usage, it is often preferable to regard covariant functors

$$F : C \longrightarrow (\mathrm{Ens})$$

as objects of $\mathbf{Hom}(C, (\mathrm{Ens}))^\circ = \check{C}^\circ$, and consequently to write the category of morphisms

$$X \longrightarrow F \text{ in } \check{C}^\circ,$$

122 i.e. of pairs (X, u) consisting of an object X of C and an element $u \in F(X)$, as

$$_F\backslash C\ . \tag{8.10.11}$$

With this convention, one thus obtains a *covariant* functor (source functor)

$$_F\backslash C \to C, \tag{8.10.12}$$

and 3.4 is rewritten in the form

$$F \xrightarrow{\sim} \text{“}\varinjlim_{(_F\backslash C)^\circ}\text{”}\, X \tag{8.10.13}$$

8.10.14. Criterion 8.3.3, applied to C°, becomes a pro-representability criterion for F: it is necessary and sufficient that $(_F\backslash C)^\circ$ be filtered and essentially small, the latter condition being superfluous if C is equivalent to a small category; if, moreover, finite projective limits are representable in C, this is equivalent to saying that F commutes with them, i.e. is *left exact.*

8.10.15. In $\mathrm{Pro}(C)$ small filtered projective limits are representable and the functor (8.10.9) commutes with them; in other words, this functor carries projective limits in $\mathrm{Pro}(C)$ to *inductive* limits in $\check{C} = \mathbf{Hom}(C, (\mathrm{Ens}))$, just like its composite $C \to \check{C}^\circ$ with (8.10.6), which shares with it the unfortunate property of being contravariant when regarded as taking values in $\check{C}$. One should note that pro-representable functors from C to (Ens) are the functors that are small filtered *inductive* limits of representable functors (and not projective limits, as the terminology might perhaps suggest). 123

8.11. Ind-adjoints and pro-adjoints.

8.11.1. Let

$$f : C \longrightarrow C' \tag{8.11.1.1}$$

be a functor between $\mathscr{U}$-categories, whence a functor $F \mapsto F \circ f$

$$f^* : \widehat{C}' \longrightarrow \widehat{C}. \tag{8.11.1.2}$$

We say that f *admits an ind-adjoint* if the preceding functor carries ind-representable functors to ind-representable functors. Since f^* commutes with inductive limits, and since the full subcategory of $\widehat{C}'$ formed by the ind-representable functors is stable under small filtered inductive limits, it is equivalent to say that f admits an ind-adjoint, or that f^* carries objects of C' (identified with a full subcategory of $\widehat{C}'$) to ind-representable functors on C, i.e.

$$X \longmapsto \mathrm{Hom}(f(X), Y') \text{ is ind-representable for every } Y' \in \mathrm{Ob}\, C'. \tag{8.11.1.3}$$

Another way to express the condition that f admit an ind-adjoint is to say that there exists a functor

$$g : \mathrm{Ind}(C') \longrightarrow \mathrm{Ind}(C) \tag{8.11.1.4}$$

essentially induced by f^*, i.e. such that there is an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(X, g(\mathscr{Y}')) \simeq \mathrm{Hom}_{\mathrm{Ind}(C')}(f(X), \mathscr{Y}'), \tag{8.11.1.5}$$

where $X \in \mathrm{Ob}\, C$ and $\mathscr{Y}' \in \mathrm{Ob}\, \mathrm{Ind}(C')$. In fact, it even suffices to have a functor 124

$$g_\circ : C' \longrightarrow \mathrm{Ind}(C) \tag{8.11.1.6}$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(X, g_\circ(Y')) \simeq \mathrm{Hom}_{C'}(f(X), Y') \tag{8.11.1.7}$$

in $X \in \mathrm{Ob}\, C$, $Y' \in \mathrm{Ob}\, C'$. The functor g (respectively $g_\circ$) is evidently determined up to unique isomorphism by f, and conversely f can be reconstructed up to canonical isomorphism from the knowledge of this g (respectively $g_\circ$). It is clear from (8.11.1.5) that the functor g commutes with inductive limits and that it "extends" the functor $g_\circ$; this therefore determines it up to unique isomorphism in terms of $g_\circ$ (8.7.2). The functor g, and sometimes also the functor $g_\circ$ that it extends, is called the *ind-adjoint functor of* f (there is no need here to specify: on the right, since the other one, if it exists, will be called the pro-adjoint; cf. 8.11.5 below).

Of course, when f admits a right adjoint

$$f^{\mathrm{ad}} : C' \longrightarrow C,$$

it admits an ind-adjoint g, and the latter is canonically isomorphic to the canonical extension of f^{ad} to ind-objects:

$$g \simeq \mathrm{Ind}(f^{\mathrm{ad}}). \tag{8.11.1.8}$$

The notion of an ind-adjoint is thus a natural generalization of the notion of a right adjoint.

125 Now consider the canonical extension

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C'). \tag{8.11.1.9}$$

We then have:

Proposition 8.11.2. *For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary and sufficient that the functor* $\mathrm{Ind}(f)$ (8.11.1.9) *admit a right adjoint. The latter is canonically isomorphic to the ind-adjoint* (8.11.1.4).

Sufficiency and the last assertion are immediate from the ind-adjunction formula (8.11.1.5). For necessity, note that by passing to the projective limit of this formula over the X_i, one obtains an adjunction isomorphism for the ind-objects $\mathscr{X} = (X_i)_{i\in I}$ and $\mathscr{Y}'$:

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(\mathscr{X}, g(\mathscr{Y}')) \simeq \mathrm{Hom}_{\mathrm{Ind}(C')}(\mathrm{Ind}(f)(\mathscr{X}), \mathscr{Y}'), \tag{8.11.2.1}$$

C.Q.F.D.

Corollary 8.11.3. *If f admits an ind-adjoint, then* $\mathrm{Ind}(f)$ *commutes with inductive limits, and the ind-adjoint g commutes with projective limits.*

Proposition 8.11.4. *For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary that f be right exact, and this condition is sufficient when C is equivalent to a small category.*

This follows from criterion (8.11.1.3), and from 8.3.1 and 8.3.3 (iv).

For another criterion in terms of the notion of an accessible functor, cf. 8.13.3.

126 8.11.5. Now consider the functor $F \mapsto F \circ f$

$$f^{\circ *} : \check{C}' \longrightarrow \check{C} \tag{8.11.5.1}$$

induced by f. We say that f *admits a pro-adjoint* if the preceding functor carries pro-representable functors to pro-representable functors, i.e. if there exists a functor (called the *pro-adjoint functor of* f)

$$g : \mathrm{Pro}(C') \longrightarrow \mathrm{Pro}(C), \tag{8.11.5.2}$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Pro}(C)}(g(\mathscr{Y}'), \mathscr{X}) \simeq \mathrm{Hom}_{\mathrm{Pro}(C')}(\mathscr{Y}', f(\mathscr{X})). \tag{8.11.5.3}$$

Of course, to say that f admits a pro-adjoint means that f° admits an ind-adjoint, so that the notions and results for ind-adjoints translate immediately into terms of pro-adjoints. We merely point out that f admits a pro-adjoint if and only if $\mathrm{Pro}(f) : \mathrm{Pro}(C) \to \mathrm{Pro}(C')$ admits a *left* adjoint, and that in this case the preceding functor g is such a left adjoint of $\mathrm{Pro}(f)$; and that for this it is necessary that f be *left* exact, this condition also being sufficient when C is equivalent to a small category. In this case, f is therefore exact if and only if it admits both an ind-adjoint and a pro-adjoint.

Example 8.11.6. Consider the case of a functor

$$f : C \longrightarrow (\mathrm{Ens}).$$

If this functor admits a pro-adjoint, it is pro-representable, and the converse is true if and only if the full subcategory of $\check{C}$ formed by the pro-representable functors is stable
127 under small products; this is the case in particular if C is equivalent to a small category (8.10.14) or if small products are representable in C (8.9.5 b), applied to C°). Up to minor

qualifications, one may thus say that for a functor $f : C \to C'$, the existence of a pro-adjoint is the *natural generalization of the notion of pro-representability of* f, which is defined when $C' = (\mathrm{Ens})$.

8.12. Strict ind-objects and pro-objects. Application to a representability criterion.

8.12.1. Let

$$\mathscr{X} = (X_i)_{i\in I}$$

be an ind-object of the $\mathscr{U}$-category C, and let F be the presheaf that it ind-represents. It is then immediately clear that it is equivalent to say that the canonical morphisms

$$X_i \longrightarrow F = \text{“}\varinjlim_i\text{”}\, X_i$$

are monomorphisms in $\mathrm{Ind}(C)$ (or, equivalently, in $\widehat{C}$, i.e. monomorphisms of functors “argument by argument”), or to say that for every arrow $i \to j$ of I, the corresponding transition arrow

$$X_i \longrightarrow X_j$$

is a monomorphism. When these conditions are satisfied, and if moreover I is an ordered category, we say that $\mathscr{X}$ is a *strict ind-object*. One should note that this condition 128
is not invariant under isomorphism of ind-objects; an ind-object will be called *essentially strict* if it is isomorphic to a strict ind-object. A presheaf will be called *strictly ind-representable* if it is ind-representable by a strict ind-object; thus $F = \text{“}\varinjlim_i\text{”}\, X_i$ is strictly ind-representable if and only if $\mathscr{X} = (X_i)_{i\in I}$ is essentially strict.

. Let F be a presheaf on C, and consider the full subcategory of $C_{/F}$ formed by the arrows $X \to F$ with source in C that are monomorphisms. We shall call it the *category of representable subfunctors of* F; it is the category associated with the ordered set of representable subfunctors of F, ordered by the order induced from that of the set of subobjects of F. The following then follows immediately from the definitions:

Proposition 8.12.2. *For the presheaf F on C to be strictly ind-representable, it is necessary and sufficient that the (ordered) category I of representable subfunctors of F be filtered and essentially small, and that one have*

$$\varinjlim_I X_i \xrightarrow{\sim} F. \tag{8.12.2.1}$$

When, for every object X of C, the set of subobjects of X in C is small (for example, if C admits a small generating subcategory (7.4)), this implies that the category of representable subfunctors is itself small.

. If F is strictly ind-representable, there is thus a preferred way of ind-representing 129
it by an ind-object, and the latter is even a strict ind-object: one takes the representation (8.12.2.1). A strict ind-object $\mathscr{Y}$ is called *saturated* if it is isomorphic to an ind-object of the form (X_i) occurring in (8.12.2.1), for a suitable F; thus, up to unique isomorphism, there exists a single saturated strict ind-object isomorphic to the given strict ind-object, namely the one considered in 8.12.2, taking $F = \varinjlim \mathscr{Y}$ (the limit in $\widehat{C}$).

. Suppose that it is already known that one can find a small cofinal subset in the set $\mathrm{Ob}\, C_{/F}$, as is the case in particular if F is ind-representable; it then follows that the same condition is satisfied in the full subcategory I considered in 8.12.2, so in criterion 8.12.2 one may omit the condition that I be essentially small.

8.12.3. Let F be a presheaf on C, and consider an object

$$u : X \longrightarrow F$$

of $C_{/F}$. We say that u (or the pair (X, u)) is *minimal* if, for every factorization of u as

$$X \xrightarrow{p} X' \xrightarrow{u'} F, \tag{8.12.3.1}$$

with p a strict epimorphism (10.2), p is an isomorphism. Regarding u as an element of
130 $F(X)$ (1.4), to say that u is minimal thus means that every strict epimorphism $p : X \to X'$ such that $u \in \mathrm{Im}(F(p) : F(X') \to F(X))$ is an isomorphism. This notion is clarified by part a) of the following lemma:

LEMMA 8.12.4. *Let F be a presheaf on C.*

a) *Suppose that F carries cokernels to kernels, and consider a morphism*

$$u : X \longrightarrow F,$$

with $X \in \mathrm{Ob}\, C$. For u to be a monomorphism, it is sufficient, when cokernels of pairs of arrows are representable in C, that u be minimal; this condition is also necessary if fiber products are representable in C.

b) *Suppose that finite $\varinjlim$ are representable in C. For the subcategory of representable subfunctors of F (8.12.1.1) to be filtered (not necessarily small) and to have F as its inductive limit, it is sufficient that F be left exact and that every morphism $u : X \to F$, with $X \in \mathrm{Ob}\, C$, factor as*

$$X \to X' \xrightarrow{u'} F,$$

with u' minimal; this condition is also necessary if fiber products are representable in C.

c) *Suppose that C admits a small generating subcategory (7.1), and that I, the category of representable subfunctors of F, is filtered; then I is small.*

131 *Proof.*

a) Suppose u is minimal; let us prove that u is a monomorphism, i.e. that for every pair of arrows $v, v' : Y \rightrightarrows X$ such that $uv = uv'$, one has $v = v'$. Indeed, if $X' = \mathrm{Coker}(v, v')$, then u factors as $X \xrightarrow{p} X' \xrightarrow{u'} F$ (since F carries cokernels to kernels), and since p is a strict epimorphism by construction, it follows that p is an isomorphism, i.e. $v = v'$. Suppose that u is a monomorphism; let us prove that it is minimal. Consider a factorization (8.12.3.1); since p is a strict epimorphism, it is the cokernel of the canonical pair of arrows $v, v' : X \times_{X'} X \rightrightarrows X$, and since $(u'p)v = (u'p)v'$ and $u'p = u$ is a monomorphism, one has $v = v'$, hence p is an isomorphism.

b) Sufficiency: Since F is left exact, $C_{/F}$ is filtered. Since we know that $F = \varinjlim_{C_{/F}} X$, by 8.1.3 c) it remains to prove that the full subcategory I of $C_{/F}$ consisting of the subobjects of F is cofinal in $C_{/F}$; by the "sufficient" part of a), this is precisely what is ensured by the hypothesis that every object of $C_{/F}$ is dominated by a "minimal" object. Necessity: Since every filtered inductive limit of right exact functors is again right exact, the first condition is trivially necessary. The second then follows from the "necessary" part of a).

c) A representable subfunctor $X \hookrightarrow F$ of F is known once one knows the full subcategory $C'_{/X}$ of $C'_{/F}$, where C' is a fixed small generating subcategory of C. (Use the hypothesis that I is filtered.) Since $C'_{/F}$ is small, the set of its full subcategories is small, whence the conclusion.

PROPOSITION 8.12.5. *Let C be a $\mathscr{U}$-category in which finite inductive limits are representable, and which admits a small generating subcategory (7.1). Let F be a presheaf on C. For F to be strictly ind-representable, it is sufficient that it satisfy the following two conditions, and these conditions are also necessary if fiber products are representable in C:* 132

a) *F is left exact.*
b) *Every pair (X,u), with $X \in \mathrm{Ob}\, C$ and $u \in F(X)$, is dominated in $C_{/F}$ by a minimal pair (8.12.3), i.e. there exists a minimal pair (X',u') ($X' \in \mathrm{Ob}\, C, u' \in F(X')$) and a morphism $f : X \to X'$ such that $u = F(f)(u')$.*

Sufficiency follows from 8.12.2 and 8.12.4 b), c), and necessity from 8.3.1 and 8.12.4 a).

REMARK 8.12.6. When F carries amalgamated sums in C to fiber products, it is immediately clear that for a given pair (X,u) as in b), the set of strict quotients X' of X such that $u \in \mathrm{Im}(F(X') \to F(X))$ is filtered decreasingly; this implies that if the set of strict quotients of X is Artinian, then condition b) of 8.12.5 is automatically satisfied. Thus, if the preceding condition on X is satisfied for every object X of C (one then also says, sometimes, that the objects of C° are *Artinian*), it follows from 8.12.5 that F is strictly ind-representable if and only if F is left exact; in this case, F is therefore strictly ind-representable as soon as it is ind-representable (8.3.1).

COROLLARY 8.12.7. *Let C be a $\mathscr{U}$-category in which small projective limits are representable, and which admits a small cogenerating subcategory (7.9, 7.13). Then a functor $F : C \to (\mathrm{Ens})$ is representable if and only if F commutes with small projective limits.* 133

Necessity is clear. For sufficiency, apply 8.12.5 to the opposite category C°. To prove first that F is (strictly) pro-representable, it remains to show that every pair (X,u), $X \in \mathrm{Ob}\, C$ and $u \in F(X)$, is dominated by a "minimal" pair. Now the family $(X_i)_{i\in I}$ of *strict* subobjects of X such that $u \in \mathrm{Im}(F(X_i) \to F(X))$ is small (7.5 in dual form). Since F is left exact, this family is cofiltered (8.12.6), and since F commutes with small projective limits, one sees that $X' = \varprojlim_i X_i$ is a least object of this family. (Use the fact that, being left exact, F carries monomorphisms to monomorphisms.) If $u' \in F(X')$ is the unique element whose image in $F(X)$ is u, one then sees that (X',u') is a minimal pair dominating (X,u). This proves that F is pro-representable, and the conclusion then follows from the

LEMMA 8.12.8.1. *Let $F : C \to (\mathrm{Ens})$ be a functor, where C is a $\mathscr{U}$-category in which small projective limits are representable. For F to be representable, it is necessary and sufficient that it be pro-representable and commute with small projective limits.*

Necessity is clear; let us prove sufficiency. To say that F is representable evidently means that ${}_F\backslash C$ admits an initial object (in fact, the initial objects of ${}_F\backslash C$ are precisely the isomorphisms $F \to X$, i.e. the representation data for F). Now, since F is pro-representable, $({}_F\backslash C)^\circ$ is filtered and equivalent to a small category, so $X = \varprojlim_{({}_F\backslash C)^\circ} X$ is representable in C, and since F commutes with the limit under consideration, it follows that X is an initial object of ${}_F\backslash C$, C.Q.F.D. 134

Corollary 8.12.8. *Under the hypotheses on C of 8.12.7, let $f : C \to C'$ be a functor from C to a $\mathscr{U}$-category C'. For f to admit a left adjoint, it is necessary and sufficient that f commute with small projective limits.*

Indeed, this reduces immediately to 8.12.7, by applying that statement to composite functors of the form $X \mapsto \mathrm{Hom}(Y', f(X))$.

Examples 8.12.9. As was pointed out in 7.13, the hypotheses on C in 8.12.7 and 8.12.8 are satisfied if C is the category of $\mathscr{U}$-sheaves of sets on a topological space $X \in \mathscr{U}$ (or, more generally, on a $\mathscr{U}$-site (II (3.0.2)). We give an instructive example[3] showing that the hypothesis of the existence of a small cogenerating subcategory D of C is not superfluous in 8.12.7. Take for C the category of groups that are elements of $\mathscr{U}$. Let J be the set of isomorphism classes of simple groups in C; for every $j \in J$ choose a simple group G_j in the class j, and let I be the filtered ordered set of $\mathscr{U}$-small subsets of J, and, for $i \in I$,
135 let $X_i = \amalg_{j\in i} G_j$. The X_i then form a projective system $(X_i)_{i\in I}$ in C, whose transition morphisms are epimorphisms, and the corresponding functor

$$(*) \qquad F(X) = \varinjlim_i \mathrm{Hom}(X_i, X)$$

takes its values in ($\mathscr{U}$-Ens) (although the index set I clearly does not have cardinality in $\mathscr{U}$); more precisely, let us show that for every small full subcategory $C_\circ$ of C, there exists an $i_\circ \in I$ such that the restriction of F to $C_\circ$ is represented by $X_{i_\circ}$, which will prove both that F takes values in $\mathscr{U}$-Ens and that it commutes with small projective limits. To prove our assertion, it suffices to note that for every $X \in \mathrm{Ob}\, C_\circ$, the cardinality of the set $J(X)$ of $j \in J$ for which there exists a nontrivial homomorphism from G_j to X is necessarily small, since such a morphism is necessarily a monomorphism (G_j being simple); consequently, if $i_\circ$ is the subset of J given by the union of the $J(X)$ for $X \in \mathrm{ob}\, C_\circ$, then $i_\circ$ is small, i.e. $i_\circ \in I$, and it has the required property. On the other hand, it is clear that F is not representable, since $\mathrm{card}\, I \notin \mathscr{U}$. From this and 8.12.7 one therefore concludes that *the category C of groups in $\mathscr{U}$ admits no small full cogenerating subcategory.* Since, on the other hand, the object $\mathbf{Z}$ of C is a generator, it then follows from the proof of 7.12 that there exists a two-generator group G that cannot be embedded in an injective object of the category C of groups in $\mathscr{U}$. It moreover seems plausible that C has no injective objects other than the trivial groups.
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8.13. Pro-representable functors and accessible functors.

8.13.1. In this section, we use some notions and results from the following section, in particular 9.11 and 9.13, to obtain a pro-representability criterion that we shall use (incidentally) in IV 9.16. In what follows, C denotes a $\mathscr{U}$-category satisfying condition L of 9.1 b), this condition being satisfied, for example, if small filtered inductive limits are representable in C.

Proposition 8.13.2. *Let C be as above, and let*

$$f : C \longrightarrow (\mathrm{Ens})$$

be a functor.

a) *Suppose that every object of C is accessible (9.3). If f is pro-representable, then f is accessible (9.2) and left exact.*

b) *Suppose that finite projective limits are representable in C, and that C admits a cardinal filtration (9.12). If f is accessible and left exact, then f is pro-representable.*

[3] (due to H. BASS).

Proof.

a) The hypothesis on C means that the covariant representable functors from C to (Ens) are accessible. The same is therefore true of every small inductive limit of such functors (9.6 (i)), and hence also of every pro-representable functor.

b) By 8.3.3 (iii), it remains to prove that $\mathrm{Ob}({}_F\backslash C)^\circ$ contains a small cofinal subcategory. By hypothesis, there exists a cardinal Π such that f is Π-accessible. Let (X, u), $u \in F(X)$, be an object of ${}_F\backslash C$. With the notation of 9.12 c), we then have 137
$X = \varinjlim_i X_i$, with I filtered and large relative to Π and the X_i in $C' = \mathrm{Filt}^\Pi(C)$, whence $F(X) \xleftarrow{\sim} \varinjlim_i F(X_i)$. This shows that the small subcategory $({}_F\backslash C')^\circ$ is cofinal in $({}_F\backslash C)^\circ$, and completes the proof.

Corollary 8.13.3. *Let C be a $\mathscr{U}$-category satisfying the following conditions:*

a) *Finite projective limits are representable in C.*
b) *Small filtered inductive limits are representable in C.*
c) *The functor* Ker *on the category of double arrows of C is accessible (for example, it commutes with small filtered* $\varinjlim$*).*
d) *Every strict epimorphism of C is a universal strict epimorphism* (10.2).
e) *There exists a small subcategory of C generating by strict epimorphisms* (7.1).

Under these conditions, a functor $f : C \to$ (Ens) is pro-representable if and only if it is left exact and accessible (9.2).

Indeed, the conditions of 8.13.2 a) and b) on C are satisfied, by 9.11 and 9.13 respectively.

Corollary 8.13.4. *Let $f : C \to C'$ be a functor between $\mathscr{U}$-categories satisfying conditions* a) *through* e) *of 8.13.3. For f to admit a pro-adjoint, it is necessary and sufficient that f be left exact and accessible.*

Since the representable functors $h_{Y'} : X' \to \mathrm{Hom}(Y', X') : C' \to$ (Ens) are 138
left exact and accessible (9.11), if f has the same properties, so do its composites with the preceding functors; these are therefore pro-representable by 8.13.3, i.e. f admits a pro-adjoint. Conversely, suppose that f admits a pro-adjoint, and let C'_1 be a small generating subcategory of C, and Π a cardinal such that the $Y' \in \mathrm{Ob}\, C'$ are Π-accessible; to prove that f is Π-accessible, it therefore suffices to prove the same of its composites with the $h_{Y'}$, $Y' \in \mathrm{Ob}\, C'$; by hypothesis, these composites are pro-representable, hence accessible (8.13.3), and hence Π-accessible provided Π is taken sufficiently large, C.Q.F.D.

9. Accessible functors, cardinal filtrations, and the construction of small generating subcategories

The present section, which is more technical in nature than the other sections of this exposé, will be used in this seminar only in IV 9 and VI 4, which are not used elsewhere in the Seminar. The reader should therefore omit the present section, at least on a first reading!

9.0. All categories considered in this section are assumed to be $\mathscr{U}$-categories. Except for the small index categories $I, J \ldots$ that we shall have occasion to use, the developments that follow will apply primarily to "large" categories $E, F \ldots$ that are stable under small filtered inductive limits. Most often, however, it will suffice that a slightly weaker condition be satisfied (condition L in 9.1 below). All cardinals considered in the 139

present section are assumed to belong to $\mathscr{U}$.

Following a suggestion of P. DELIGNE, we shall study, for a functor $f : E \to F$ between large categories, a condition that f commute with certain types of filtered inductive limits. This condition is remarkably stable, and is satisfied by the most important functors occurring in nature. The applications we have in mind for our seminar are 9.13.3, 9.13.4 (used in VI 4), and especially 9.25, which gives, in a nontrivial case, the existence of a small generating family in a category of sections of a fibered category; this result will be used in IV 9.16.

DEFINITION 9.1. a) *Let I be a preordered set and π a cardinal. We say that I is large relative to π if I is filtered and every subset of I of cardinality $\leq \pi$ has an upper bound in I.*

b) *Let E be a category. If π is a cardinal, we say that E satisfies condition L_π if, for every small ordered set I large relative to π, E is stable under inductive limits of type I. We say that E satisfies condition L if there exists a cardinal $\pi \in \mathscr{U}$ such that E satisfies condition L_π.*

9.1.1. When, in 9.1 a), one has $\pi \geq 2$, the second stated condition already implies that I is filtered, and if π is finite, for I to be large relative to π simply means that I is filtered. In what follows we shall be concerned almost exclusively with the case in which π is infinite. Note that if π, π' are two cardinals such that $\pi' \geq \pi$, then I being large relative to π' evidently implies that it is large relative to π.

140 9.1.2. As announced in 9.0, conditions L_π and L should be regarded as technical variants of the stronger condition of stability under small filtered inductive limits. It is clear that if π, π' are cardinals such that $\pi' \geq \pi$, then condition L_π implies condition $L_{\pi'}$.

DEFINITION 9.2. *Let $f : E \to F$ be a functor. If Π is a cardinal, we say that f is Π-preaccessible (respectively Π-accessible) if E satisfies L_Π (9.1) and if, for every ordered set $I \in \mathscr{U}$ large relative to Π, and every inductive system $(X_i)_{i\in I}$ in E of type I, the canonical morphism*

$$\varinjlim f(X_i) \longrightarrow f(\varinjlim X_i)$$

is a monomorphism (respectively an isomorphism). We say that f is preaccessible (respectively accessible) (relative to the universe $\mathscr{U}$) if there exists a cardinal $\Pi \in \mathscr{U}$ such that f is Π-preaccessible (respectively Π-accessible).

The category of Π-accessible (respectively accessible) functors from E to F will be denoted $\mathbf{Hom}(E, F)_\Pi$, respectively $\mathbf{Hom}(E, F)_{\mathrm{acc}}$.

9.2.1. Evidently, a functor commuting with small filtered $\varinjlim$ (for example, a functor admitting a right adjoint) is Π-accessible for every cardinal $\Pi \geq 2$.

DEFINITION 9.3. *Let E be a category, X an object of E,*

$$h^\circ_X : E \longrightarrow (\mathscr{U}\text{-Ens})$$

the covariant functor it represents, and $\Pi \in \mathscr{U}$ a cardinal. We say that X is a Π-
141 *preaccessible (respectively Π-accessible) object of E if the functor h°_X is Π-preaccessible (respectively Π-accessible); we say that X is* preaccessible (respectively accessible) *if there exists a cardinal $\Pi \in \mathscr{U}$ such that X is a Π-preaccessible (respectively Π-accessible) object of E.*

9.3.1. We denote by E_Π the full subcategory of E formed by the Π-accessible objects of E.

When Definition 9.3 is applied to a category of the form $\mathbf{Hom}(C, F)$, the terminology just introduced presents *a priori* an ambiguity with the analogous terminology introduced in 9.2, when the objects of $\mathbf{Hom}(C, F)$ are interpreted as functors; there does not, however, seem to be any serious risk of confusion.

DEFINITION 9.4. *Let E be a category and let $\pi \in \mathscr{U}$ be a cardinal. We say that E is a π-preaccessible (respectively π-accessible) category if there exists in E a small full subcategory C that is generating (7.1) and whose objects are π-preaccessible (respectively π-accessible) (9.3). We say that E is a preaccessible (respectively accessible) category if there exists a cardinal $\pi \in \mathscr{U}$ such that E is π-preaccessible (respectively π-accessible).*

For important examples, cf. 9.11.3 below.

PROPOSITION 9.5. *Let $f : E \to F$ be a functor between categories such that F is accessible and every object of E is accessible. Then, if f admits a left adjoint, f is accessible.*

Indeed, by hypothesis, F admits a small conservative family of representable functors $F \to (\mathscr{U}\text{-Ens})$ that are accessible, so we are immediately reduced to showing that the composite of f with each of the preceding functors is accessible. But since f admits a left adjoint, these composites are functors $E \to (\mathscr{U}\text{-Ens})$ that are *representable*, and hence accessible by hypothesis on E. 142

PROPOSITION 9.6. *Let E and F be two categories, let $\pi \in \mathscr{U}$ be a cardinal, and consider the full subcategory $\mathbf{Hom}(E, F)_\Pi$ of $\mathbf{Hom}(E, F)$ formed by the Π-accessible functors (9.2) from E to F.*

(i) *This subcategory is stable under every type of $\varinjlim$ that is representable in F.*

(ii) *Suppose that F is π-accessible (9.4), and let J be a category such that $\operatorname{card} F\ell\, J \leq \pi$ and that the $\varinjlim$ of type J are representable in F, so the $\varinjlim$ of type J are representable in $\mathbf{Hom}(E, F)$. Then the subcategory $\mathbf{Hom}(E, F)_\Pi$ is stable under the $\varinjlim$ of type J.*

COROLLARY 9.7. *Let E and F be two categories, with F accessible (9.4). Then the full subcategory $\mathbf{Hom}(E, F)_{\mathrm{acc}}$ of $\mathbf{Hom}(E, F)$ formed by the accessible functors is stable under every type of inductive or projective limit, indexed by a small category J, that is representable in F (and hence in $\mathbf{Hom}(E, F)$).*

Proof of 9.6. Assertion (i) follows trivially from the commutation of the functor $\varinjlim$ with arbitrary inductive limits. For (ii), let $(f_j)_{j\in J}$ be a projective system of Π-accessible functors $E \to F$, and let $f = \varprojlim f_j$ be its projective limit, computed "argument by argument." We prove that f is Π-accessible, i.e. that for every ordered set $I \in \mathscr{U}$ large relative to Π, and every inductive system $(X_i)_{i\in I}$ in E, the canonical morphism 143

$$\varinjlim_i((\varprojlim_j f_j)(X_i)) \longrightarrow (\varprojlim_j f_j)(\varinjlim_i X_i)$$

is an isomorphism. Now the "argument-by-argument" computation of the functor $\varprojlim_j f_j$ allows us to identify the preceding canonical morphism with the canonical morphism

$$\varinjlim_i \varprojlim_j f_j(X_i) \longrightarrow \varprojlim_j \varinjlim_i f_j(X_i)$$

associated with the bifunctor

$$(i, j) \longmapsto f_j(X_i) : I \times J \longrightarrow F.$$

Thus 9.6 is a consequence of the following more general assertion:

COROLLARY 9.8. *Let F be a Π-accessible category, I an ordered set large relative to Π, J a small category such that* card $F\ell \leq \Pi$ *and such that projective limits of type J are representable in F, and $h : I \times J \to F$ a functor; then the canonical morphism*

$$\varinjlim_i \varprojlim_j h(i,j) \longrightarrow \varprojlim_j \varinjlim_i h(i,j) \tag{9.8.1}$$

is an isomorphism. In other words, the functor

$$\varinjlim_i : \mathbf{Hom}(I,F) \longrightarrow F \tag{9.8.2}$$

144 *commutes with projective limits of type J (for every small category J such that* card $F\ell(J) \leq \Pi$ *and such that projective limits of type J are representable in F). Equivalently, for every J as above, the functor*

$$\varinjlim_j : \mathbf{Hom}(J,F) \longrightarrow F \tag{9.8.3}$$

is Π-accessible.

By hypothesis, F admits a conservative family of covariant representable Π-accessible functors $g : F \to (\mathscr{U}\text{-Ens})$, so it is immediately clear that it suffices to prove 9.8 in the case $F = \mathscr{U}$-Ens. We shall then prove the assertion in the form that (9.8.2) commutes with projective limits of type J, where card $F\ell\, J \leq \Pi$. Since I is filtered, we know that (9.8.2) commutes with finite projective limits (2.8). By a standard argument (cf. 2.3), it therefore remains to prove that it commutes with products indexed by a set J such that card $J \leq \Pi$. This reduces us to proving the bijectivity of (9.8.1) when J is discrete. Let us prove injectivity: consider two elements a, b of the first member; they therefore come from $\prod_j h(i_\circ, j)$ for a suitable $i_\circ \in I$, say $\prod_j a(i_\circ, j)$ and $\prod_j b(i_\circ, j)$; suppose that the elements $\prod_j h(\infty, j)$ of the second member that they define are equal, i.e. that for every j, there exists $i(j) \geq i_\circ$ such that $a(i_\circ, j)$ and $b(i_\circ, j)$ have the same image in $h(i,j)$. Since I is large relative to card J, it follows that there exists a common upper bound $i_1 \in I$ of all the $i(j)$, which implies that the two elements under consideration of $\prod_j h(i_\circ, j)$ have the same image in $\prod_j h(i_1, j)$ and therefore define the same element of the first member
145 of (9.8.1), establishing injectivity. For surjectivity, let $\prod_j a(\infty, j)$ be an element of the second member; thus, for every $j \in J$, $a(\infty, j)$ comes from an element of $h(i(j), j)$ for a suitable $i(j) \in I$. As before, one can find a common upper bound $i \in I$ of the $i(j)$, so $\prod_j a(\infty, j)$ comes from an element $\prod_j a(i, j)$ of $\prod_j h(i, j)$, and hence lies in the image of (9.8.1). This completes the proof.

COROLLARY 9.9. *Let F be a Π-accessible (respectively accessible) category. Then, for every category J such that* card $F\ell\, J \leq \Pi$ *(respectively every small category J), and for every functor $(X_j)_{j\in J} : J \to F$ whose projective limit in F is representable, if the X_j are Π-accessible (respectively accessible), then so is $\varprojlim X_j$.*

Indeed, the functor $F \to (\mathscr{U}\text{-Ens})$ represented by $\varprojlim X_j$ is the projective limit of the functors represented by the X_j, and one applies 9.6 (ii) to the projective system formed by these functors.

REMARK 9.10. In the statements 9.6, 9.7, 9.8, and 9.9, one may everywhere replace the words "Π-accessible", "accessible" by "Π-preaccessible", "preaccessible". Indeed, the proof given also proves this variant of the preceding statements.

PROPOSITION 9.11. *Let E be a category satisfying condition L (9.1), and let C be a small full generating subcategory (7.1). Suppose that the following condition is satisfied:*

a) *E is stable under kernels of pairs of arrows, and the functor* Ker *on the category of pairs of arrows in E is accessible (for example, it commutes with small filtered inductive limits).* 146

Then every object of E is preaccessible (9.3), a fortiori E is preaccessible. Suppose that C is even generating by strict epimorphisms (7.1), and suppose that E satisfies the following conditions:

b) *E is stable under fiber products.*

c) *Every small strictly epimorphic family in E is universally strictly epimorphic (10.3), or small direct sums are representable in E and every strict epimorphism of E is a universal strict epimorphism.*

Then every object of E is accessible, a fortiori E is accessible.

The fact that, under (a), every object of E is preaccessible follows from the fact that for every object X of E, the set of strict subobjects of X is small (7.4), and from the

LEMMA 9.11.1. *Under the conditions of (9.11a), let $X \in \mathrm{ob}\, E$ and let Π be a cardinal such that E satisfies L_Π (9.1), the functor* Ker *in (9.11a) is Π-accessible, and the set of strict subobjects of X has cardinality bounded by Π. Then X is Π-preaccessible.*

Indeed, let I be an ordered set large relative to Π, let $(Y_i)_{i\in I}$ be an inductive system in E with inductive limit Y, and let $(u_i, v_i : X \rightrightarrows Y_i)$ be a pair of arrows such that the composite pair $X \overset{u,v}{\rightrightarrows} Y$ satisfies $u = v$; let us prove that there exists $j \geq i$ in I such that $u_j = v_j$. To this end, consider the inductive system of pairs of arrows $(u_j, v_j : X \rightrightarrows Y_j)_{j\geq i}$, whose inductive limit is (u, v). By hypothesis one has $X = \mathrm{Ker}(u, v) = \varinjlim_j \mathrm{Ker}(u_j, v_j) = \varinjlim_j X_j$, where $X_j = \mathrm{Ker}(u_j, v_j)$. Now the X_j are strict subobjects of X, so there exists a subset I' of I, formed by indices $j \geq i$, such that $\mathrm{card}\, I' \leq \Pi$ and every X_j is equal to one of the $X_{i'}$ $(i' \in I')$. Since I is large relative to Π, there exists an upper bound j of I' in I. Then X_j contains all the $X_{j'}$ for $j' \geq i$, so $X_j \to X$ is an epimorphism (since the family of the $X_{j'} \to X$ is epimorphic), and hence an isomorphism because it is a strict monomorphism. Thus $u_j = v_j$, which proves 9.11.1. 147

The second assertion of 9.11 follows from the first and from the

LEMMA 9.11.2. *Under the conditions of (9.11a), b), c), let X be an object of E, and let Π be an infinite cardinal such that E satisfies L_Π (9.1), that* card ob $C_{/X} \leq \Pi$, *that for two objects $X' \to X$ and $X'' \to X$ of $C_{/X}$, $X' \times_X X''$ is Π-preaccessible, and finally that X is Π-preaccessible. Then X is Π-accessible.*

With the notation of the proof of 9.11.1, it suffices to prove that every morphism $u : X \to Y$ comes from a morphism $u_i : X \to Y_i$. Consider the family of morphisms $Y_i \to Y$, which is strictly epimorphic; by b) and c), the family of morphisms $Y_i \times_Y X \to X$ is also strictly epimorphic; on the other hand, for every i, since C is generating by strict epimorphisms, the family of arrows

$$T \longrightarrow Y_i \times_Y X$$

with source in C is strictly epimorphic. In the second alternative considered in c), one can find such a strictly epimorphic arrow whose source is a (small) sum of objects of C. By the transitivity of the notion of a universally strictly epimorphic family (II 2.5), it then follows that the family of arrows $T_\alpha \overset{f_\alpha}{\longrightarrow} X$ with source in C that factor through one of 148

the $Y_i \times_Y X$ is strictly epimorphic. Let J be the index set of this family; its cardinality is bounded by $\operatorname{card} \operatorname{ob} C_{/X} \leq \Pi$. For every $\alpha \in J$, choose an $i = i(\alpha) \in I$ and an X-morphism $T_\alpha \to Y_i \times_Y X$, or, equivalently, a $v_\alpha : T_\alpha \to Y_i$ such that $p_i v_\alpha = u f_\alpha$, where $p_i : Y_i \to Y$ is the canonical morphism. Since I is large relative to Π, one can choose $i(\alpha)$ independently of α, say equal to i. For every pair of indices $\alpha, \beta \in J$, consider the composites

$$\operatorname{pr}_1 v_\alpha, \operatorname{pr}_2 v_\beta : T_\alpha \times_X T_\beta \rightrightarrows Y_i.$$

Their composites with $Y_i \to Y$ are equal, so, since $T_\alpha \times_X T_\beta$ is Π-preaccessible by hypothesis, there exists an index $i' = i(\alpha, \beta) \geq i$ such that the composites of the arrows under consideration with $Y_i \to Y_{i'}$ are equal. Since the set of pairs α, β has cardinality $\leq \Pi^2 = \Pi$ (Π being infinite), it follows once again that i' can be chosen independently of α, β. We may evidently suppose that $i' = i$. But then, since the family $(f_\alpha : T_\alpha \to X)$ is strictly epimorphic, one can find a morphism $u_i : X \to Y_i$ such that $u_i f_\alpha = v_\alpha$. One then has $p_i u_i = u$, since for every α one has $(p_i u_i) f_\alpha = p_i(u_i f_\alpha) = p_i v_\alpha = u f_\alpha$, and the family of the f_α is epimorphic. This completes the proof of 9.11.2.

149 Remark 9.11.3. As a special case of 9.11, one sees that every object in the category E is accessible in each of the following two cases: 1) E is a $\mathscr{U}$-abelian category with exact filtered inductive limits and admitting a small generating subcategory (here it is the *second* alternative of c) that applies). 2) E is the category of sheaves of sets on a topological space $X \in \mathscr{U}$. More generally, it suffices that E be the category of sheaves of sets on a $\mathscr{U}$-site (II 2.1), or that E be a $\mathscr{U}$-topos (IV 1.1).

Definition 9.12. *Let E be a category. A cardinal filtration of E is an increasing filtration $(\operatorname{Filt}^\Pi(E))_{\Pi \geq \Pi_\circ}$ of E by strictly full subcategories $\operatorname{Filt}^\Pi(E)$, indexed by the cardinals $\Pi \in \mathscr{U}$ such that $\Pi \geq \Pi_\circ$ (where $\Pi_\circ$ is a fixed infinite cardinal, depending on the cardinal filtration under consideration), and satisfying the following conditions:*

a) *For every $\Pi \geq \Pi_\circ$, $\operatorname{Filt}^\Pi(E)$ is equivalent to a small category.*

b) *E satisfies $L_{\Pi_\circ}$ (9.3), and for every $\Pi \geq \Pi_\circ$, $\operatorname{Filt}^\Pi(E)$ is stable in E under filtered inductive limits indexed by ordered sets I large relative to $\Pi_\circ$ such that $\operatorname{card} I \leq \Pi$.*

c) *For every $\Pi \geq \Pi_\circ$, and every $X \in \operatorname{ob} E$, one can find an isomorphism*

$$X \simeq \varinjlim_I X_i,$$

150 *where $(X_i)_{i \in I}$ is an inductive system in E indexed by an ordered set I large relative to Π (9.1), and where the X_i belong to $\operatorname{Filt}^\Pi(E)$. Moreover, if $X \in \operatorname{ob} \operatorname{Filt}^\Pi(E)$, $\Pi' \geq \Pi$, one may choose I such that $\operatorname{card} I \leq \Pi'^{\Pi}$.*

9.12.1. Note that it follows from b) and c) that every $X \in \operatorname{ob} E$ belongs to some $\operatorname{Filt}^\Pi(E)$, for Π sufficiently large.

Example 9.12.2. Take $E = (\mathscr{U}\text{-Ens})$, $\Phi_\circ = \aleph_\circ$, $\operatorname{Filt}^\Pi(E)$ = the full subcategory of E consisting of the sets such that $\operatorname{card} X \leq \Pi$. More generally:

Proposition 9.13. *Let E be a $\mathscr{U}$-category. Suppose that E is stable under small filtered $\varinjlim$, under sums of two objects, and under cokernels of pairs of arrows, that E is stable under fiber products, and that the epimorphisms in E are universal strict epimorphisms. Let C be a small full subcategory of E that is generating by strict epimorphisms. Let $\Pi_\circ$ be an infinite cardinal $\geq \operatorname{card} Fl\, C$. For every cardinal $\Pi \geq \Pi_\circ$, let $\operatorname{Filt}^\Pi(E)$ be the strictly full subcategory of E consisting of the objects X of E for which there exists a strictly epimorphic*

family $(X_i \to X)_{i\in I}$ *with target* X, *such that* $\operatorname{card} I \leq \Pi$ *and* $X_i \in \operatorname{ob} C$ *for every* $i \in I$. *Then* $(\operatorname{Filt}^{\Pi}(E))_{\Pi\geq\Pi_\circ}$ *is a cardinal filtration of* E. *Moreover, for every* $X \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$, *the cardinality of the set of arrows of* $C_{/X}$ *(and a fortiori, of the set of objects of* $C_{/X}$*) is bounded by* $\Pi^{\Pi_\circ}$.

This last assertion is precisely 7.6.

To show that this is a cardinal filtration, one must prove conditions a), b), and c) of 9.12. Condition a) follows at once from 7.5.2. For b), suppose that $X = \varinjlim_I X_i$, with $\operatorname{card} I \leq \Pi$ and the $X_i \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$. Thus the family of canonical morphisms $X_i \to X$ 151
is strictly epimorphic, and by the hypothesis on the X_i, for every $i \in I$ there exists a strictly epimorphic family $(X_{ij} \to X_i)_{j\in J_i}$, with $\operatorname{card} J_i \leq \Pi$. Taking the sums $\amalg_i X_i$ and $\amalg_{i,j} X_{ij}$, one sees by the transitivity of universal strict epimorphisms (II 2.5) that the family of composites $X_{ij} \to X_i \to X$ (indexed by the disjoint union of the J_i, for $i \in I$) is strictly epimorphic. But $\operatorname{card} J \leq \Pi^2 = \Pi$, whence the conclusion. (NB. we have used only the fact that $\varinjlim_I X_i$ exists, and not the fact that I is large relative to $\Pi_\circ$, or even that it is filtered.) Finally, let us prove c). Note that for every $X \in \operatorname{ob} E$, the family of arrows $X_i \to X$ with source in $\operatorname{ob} C$ is strictly epimorphic, since C is generating by strict epimorphisms, and is plainly small, so there exists a cardinal $\Pi \geq \Pi_\circ$ such that $X \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$. It remains to prove that if $\Pi' \geq \Pi \geq \Pi_\circ$ are cardinals, and if $X \in \operatorname{ob}\operatorname{Filt}^{\Pi'}(E)$, then there is an isomorphism

$$X \simeq \varinjlim_I X_i,$$

with I large relative to Π, $\operatorname{card} I \leq \Pi'^{\Pi}$, and the X_i in $\operatorname{ob}\operatorname{Filt}^{\Pi}(E)$. Now, since C is generating by strict epimorphisms, one has

$$(*) \qquad X \simeq \varinjlim_{C_{/X}} T.$$

Let us also note that by successively adjoining to C sums and cokernels of pairs of arrows in C, and doing the same for the category thus obtained, and so on, one reduces to the case where C is stable under sums of two objects in E and under cokernels of pairs of arrows in E, without losing the hypothesis $\Pi_\circ \geq \operatorname{card} F\ell\, C$. We may suppose moreover that, if E contains a chosen initial object $\varnothing_E$, then $\varnothing_E \in \operatorname{ob} C$. This implies that the 152
category $C_{/X}$ is stable under sums of two objects and cokernels of pairs of arrows. It is then filtered, since it is nonempty: if it were empty, the relation $(*)$ would show that X is an initial object, and hence $C_{/X}$ would contain the arrow $\varnothing_E \to X$, a contradiction. Let I be the set, ordered by inclusion, of full subcategories i of $C_{/X}$ that are filtered and such that $\operatorname{card}\operatorname{ob} i \leq \Pi$. For every $i \in I$, let $X_i \in \operatorname{ob} E$ be the inductive limit of the composite functor $i \to C_{/X} \to E$, which exists by hypothesis. Then plainly

$$\varinjlim_I X_i \simeq \varinjlim_{C/X} T \simeq X.$$

On the other hand, we have already noted that $\operatorname{card}\operatorname{ob} C_{/X} \leq \Pi'^{\Pi_\circ}$. It follows that I is large relative to Π, and that $\operatorname{card} I \leq (\Pi'^{\Pi_\circ})^{\Pi} = \Pi'^{\Pi\Pi_\circ} = \Pi'^{\Pi}$ by the following lemma, whose proof is left to the reader (where we take $I = C_{/X}$, $c = \Pi'^{\Pi_\circ}$):

Lemma 9.13.1. *Let* J *be a filtered category, and let* c *and* Π *be two cardinals such that* $c \geq \operatorname{Sup}(\operatorname{card} F\ell\, J, \Pi)$, *and suppose that* $\operatorname{card}\operatorname{Hom}(j, J') \leq \Pi_\circ$ *for every pair of objects* j, j' *of* J. *Let* I *be the set of full filtered subcategories* i *of* J *such that* $\operatorname{card}\operatorname{ob} i \leq \Pi$. *Then, ordered by inclusion,* I *is large relative to* Π, *and one has* $\operatorname{card} I \leq c^{\Pi}$.

This completes the proof of 9.13.

153 Remark 9.13.2. A slight additional effort should make it possible to replace in 9.13 the hypothesis that E is stable under small filtered inductive limits by the hypothesis that E satisfies L (9.1), provided E is stable under small sums, or that in E every epimorphic *family* is universally epimorphic. In the proof of c), one must then choose $\Pi_\circ$ such that E satisfies $L_{\Pi_\circ}$, and restrict oneself to those full subcategories i of $C_{/X}$ that are not only filtered, but such that the preordered set $\mathrm{ob}\, i$ is large relative to $\Pi_\circ$. Using Section 8 and the hypothesis that E satisfies $L_{\Pi_\circ}$, one then finds that the X_i exist, and one should be able to conclude by a suitable variant of 9.13.1, which the editor has not verified.

Proposition 9.14. *Let $f : E \to F$ be an accessible functor (9.2) between two categories equipped with cardinal filtrations $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_\circ}$ and $(\mathrm{Filt}^{\Pi}(F))_{\Pi \geq \Pi'_\circ}$. Then there exists a cardinal $\Pi_1 \geq \mathrm{Sup}(\Pi_\circ, \Pi'_\circ)$ such that for every cardinal $\Pi \geq \Pi_1$, one has*

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi^{\Pi_1}}(F). \tag{9.14.1}$$

In particular, if $\Pi = 2^c$ with $c \geq \Pi_1$, whence $\Pi^{\Pi_1} = 2^{c\Pi_1} = 2^c = \Pi$, one has

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi}(F). \tag{9.14.2}$$

We immediately point out the

154 Corollary 9.15. *Let E be a category equipped with two cardinal filtrations $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_\circ}$ and $(\mathrm{Filt}'^{\Pi}(E))_{\Pi \geq \Pi'_\circ}$. Then there exists a cardinal $\Pi_1 \geq \mathrm{Sup}(\Pi_\circ, \Pi'_\circ)$ such that, for every cardinal $c \geq \Pi_1$, putting $\Pi = 2^c$, one has $\mathrm{Filt}^{\Pi}(E) = \mathrm{Filt}'^{\Pi}(E)$.*

Proof of 9.14. Let c be a cardinal such that $c \geq \mathrm{Sup}(\Pi_\circ, \Pi'_\circ)$ and such that f is c-accessible. Let, moreover, $\Pi_1 \geq c$ be such that

$$f(\mathrm{Filt}^{c}(E)) \subset \mathrm{Filt}^{\Pi_1}(F). \tag{$*$}$$

Such a Π_1 exists because $\mathrm{Filt}^{c}(E)$ is equivalent to a small category (9.12 a)), and hence so is $f(\mathrm{Filt}^{c}(E))$, so that one can apply 9.12.1 to the objects of the latter to find a $\mathrm{Filt}^{\Pi}(F)$ containing them all (taking into account that the $\mathrm{Filt}^{\Pi}(F)$ are full subcategories). Let then $\Pi \geq \Pi_1$ and $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$; let us prove that $f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F)$. Indeed, write

$$X = \varinjlim_{I} X_i,$$

where the $X_i \in \mathrm{ob}\,\mathrm{Filt}^{c}(E)$, I is large relative to c, and $\mathrm{card}\, I \leq \Pi^{c} \leq \Pi^{\Pi_1}$ (9.12c)). Since f is c-accessible and I is large relative to c, one has

$$f(X) \simeq \varinjlim_{I} f(X_i),$$

and since $\mathrm{card}\, I \leq \Pi^{\Pi_1}$ and $f(X_i) \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_1}(F) \subset \mathrm{ob}\,\mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by $(*)$, one has $f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by 9.12 b), C.Q.F.D.

9.15.1. The notion of cardinal filtration 9.12 is of interest mainly when the objects of E are accessible. We point out that it follows from 9.11 that this condition is satisfied
155 when, in addition to the hypotheses of 9.13, one supposes that the functor Ker on the category of pairs of arrows in E is accessible. We also point out:

Proposition 9.16. *Let E be a category equipped with a cardinal filtration $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_\circ}$. Suppose that the objects of $\mathrm{Filt}^{\Pi_\circ}(E)$ are accessible objects of E; then there exists a cardinal*

$\Pi_1 \geq \Pi_\circ$ in $\mathscr{U}$ such that the objects of $\mathrm{Filt}^{\Pi_\circ}(E)$ *are Π_1-accessible; if Π_1 is chosen in this way, then for every cardinal $\Pi \geq \Pi_1$, we have (with the notation of 9.3.1):*

$$E_\Pi \subset \mathrm{Filt}^{\Pi}(E) \subset E_{(\Pi^{\Pi_\circ})}. \tag{9.16.1}$$

The existence of Π_1 follows immediately from the fact that $\mathrm{Filt}^{\Pi_\circ}(E)$ is equivalent to a small category (9.12a)). Now let $\Pi \geq \Pi_1$. If X belongs to $\mathrm{Filt}^{\Pi}(E)$, write $X \simeq \varinjlim_I X_i$ with $\mathrm{card}\, I \leq \Pi^{\Pi_\circ}$, $X_i \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_\circ}(E)$ for every i (9.12c)). It then follows from 9.9 that X is $\Pi^{\Pi_\circ}$-accessible, whence the second inclusion (9.16.1). Suppose that X is Π-accessible, and write $X \simeq \varinjlim_I X_i$, with I large relative to Π and the $X_i \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$ (9.12c)). By the hypothesis on X, the given isomorphism $X \xrightarrow{\sim} \varinjlim_I X_i$ factors through one of the X_i, so X is isomorphic to a direct factor of that X_i. We conclude that $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$, and hence obtain the first inclusion (9.16.1), by the following lemma.

Lemma 9.16.2. *Every object of E that is a direct factor of an object of* $\mathrm{Filt}^{\Pi}(E)$ *belongs to* $\mathrm{Filt}^{\Pi}(E)$.

Indeed, if X is the image of a projector p on an object Y of E (i.e. an endomorphism p such that $p^2 = p$), and if I is an ordered filtered set, X is the inductive limit of the 156
filtered inductive system $(Y_i)_{i\in I}$ defined by $Y_i = Y$ for every $i \in I$, $p : Y_i \to Y_j$ if $i < j$. Taking I large relative to $\Pi_\circ$ and $\mathrm{card}\, I \leq \Pi$, we thus see that if $Y \in \mathrm{Filt}^{\Pi}(E)$, then the same holds for X, by 9.12 b).

Corollary 9.17. *Under the conditions of 9.16, for every cardinal $c \geq \Pi_1$, setting $\Pi = 2^c$,* $\mathrm{Filt}^{\Pi}(E)$ *coincides with the strictly full subcategory E_Π of E consisting of the Π-accessible objects.*

Corollary 9.18. *Let E be a category satisfying condition L_Π (9.1), where Π is a cardinal, and let C be a full subcategory of E. Denote by* $\mathrm{Ind}(C)_\Pi$ *the full subcategory of the category* $\mathrm{Ind}(C)$ *of ind-objects of C (8.2) consisting of ind-objects of the form $(X_i)_{i\in I}$, where I is an ordered set large relative to Π. Consider the canonical functor*

$$(X_i)_{i\in I} \longmapsto \varinjlim_I X_i : \mathrm{Ind}(C)_\Pi \longrightarrow E. \tag{9.18.1}$$

a) *This functor is fully faithful if and only if every object X of C is a Π-accessible object (9.3) of E.*
b) *Assume the conditions of 9.17, in particular $\Pi = 2^c$, and take $C = \mathrm{Filt}^{\Pi}(E)$. Then the functor (9.18.1) is an equivalence of categories.*

Assertion a) is an immediate generalization of 8.7.5 a), and is proved in the same way. Part b) then follows from 9.17 and condition 9.12 c) on cardinal filtrations, which implies that the functor under consideration is essentially surjective.

Corollary 9.19. *Under the conditions of 9.17, let $C = \mathrm{Filt}^{\Pi}(E)$, let F be a $\mathscr{U}$-category,* 157
and consider the functor

$$\mathbf{Hom}(E, F)_\Pi \longrightarrow \mathbf{Hom}(C, F) \tag{9.19.1}$$

induced by the "restriction to C" functor $f \mapsto f|C$, where the source of (9.19.1) is the category of Π-accessible functors from E to F (9.2). The preceding functor is fully faithful. If F satisfies condition L_Π (9.1), then the functor (9.19.1) is an equivalence of categories.

The first assertion is proved as in 7.8, using 9.18 b). The second is obtained by constructing a quasi-inverse functor to (9.19.1), associating to every functor $f : C \to F$ the functor $(X_i)_{i\in I} \to \varinjlim_i f(X_i)$ from $\mathrm{Ind}(E_\Pi)$ to F, and using 9.18 b) to deduce from it a functor $\overline{f} : E \to F$. It remains only to show that thelatter is Π-accessible. But this is proved in the same way as the analogous assertion 8.7.3.

Corollary 9.20. *Let E be a category admitting a cardinal filtration and such that every object of E is accessible (cf. 9.16), and let F be a category. Then:*

a) *The category $\mathbf{Hom}(E,F)_{acc}$ of accessible functors from E to F (9.2) is a $\mathscr{U}$-category. (*NB *we recall (9.0) that the given categories E, F are assumed to be $\mathscr{U}$-categories.)*

b) *Suppose that F is stable under small filtered inductive limits. For every full subcategory C of E equivalent to a small category, with inclusion functor $i : C \to E$, consider the corresponding functor*

$$i_! : \mathbf{Hom}(C,F) \longrightarrow \mathbf{Hom}(E,F)$$

158 *(5.1). A functor $f : E \to F$ is accessible if and only if there exists a small subcategory C of E such that f belongs to the essential image of the preceding functor $i_!$.*

Proof.

a) It suffices to prove that for every cardinal Π such that E satisfies L_Π, the full subcategory $\mathbf{Hom}(E,F)_\Pi$ of $\mathbf{Hom}(E,F)_{acc}$ is a $\mathscr{U}$-category. It is clearly enough to verify this for cardinals of the form 2^c, with c sufficiently large. But then this follows from 9.19, since ($C = \mathrm{Filt}^\Pi(E)$ being essentially small) $\mathbf{Hom}(C,F)$ is obviously a $\mathscr{U}$-category.

b) By transitivity of the formation of the functors $i_!$, one may restrict in the statement to subcategories C of the form $\mathrm{Filt}^\Pi(E)$, where Π is as in 9.17. One then sees readily that the composite functor $\mathbf{Hom}(\mathrm{Filt}^\Pi(E),F) \to \mathbf{Hom}(E,F)_\Pi \to \mathbf{Hom}(E,F)$, where the first arrow is quasi-inverse to (9.19.1), and the second is the inclusion, is none other than the functor $i_!$, up to isomorphism. Thus assertion b) follows from 9.19.

*Exercise 9.20.1. (The present exercise uses the notions of site and topos, developed in Exposés II and IV.) Let E be a $\mathscr{U}$-topos, let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$, let C be a small full generating subcategory of the $\mathscr{U}$-topos E, and let $\Pi_\circ \in \mathscr{U}$ be an infinite cardinal such that $\Pi_\circ \in \mathrm{card}\, F\ell\, C$. For every cardinal $\Pi \geq \Pi_\circ$, $\Pi \in \mathscr{U}$, let $\mathrm{Filt}^\Pi(E)$ be the strictly full subcategory of E consisting of the objects X such that there exists
159 a strictly epimorphic family $(X_i \to X)_{i\in I}$ with target X, such that $\mathrm{card}\, I \leq \Pi$ and $X_i \in \mathrm{Ob}\, C$ for every $i \in I$.

a) Show that $(\mathrm{Filt}^\Pi(E))_{\Pi\geq\Pi_\circ}$ is a cardinal filtration of the $\mathscr{U}$-category E, and that one can choose $\Pi_\circ$ such that for every $\Pi \geq \Pi_\circ$, one has the inclusions

$$E_\Pi \subset \mathrm{Filt}^\Pi(E) \subset E_{\Pi^{\Pi_\circ}},$$

where, for every cardinal c, E_c denotes the strictly full subcategory of E consisting of the c-accessible objects.

b) Choosing a cardinal $\Pi \geq \Pi_\circ$ such that $\Pi = \Pi^{\Pi_\circ}$ (for example, Π of the form 2^c, with $c > \Pi_\circ$), and putting $C = \mathrm{Filt}^\Pi(E)$, show that there is an equivalence of categories

$$\mathrm{Ind}(C)_\Pi \xrightarrow{\sim} E,$$

(notation $\mathrm{Ind}(C)_\Pi$ of 9.18).

c) Retain the notation of b), and let $\mathscr{V}$ be a universe such that $\mathscr{U} \subset \mathscr{V}$, let $E^{\sim}_{V}$ be the $\mathscr{V}$-topos of $\mathscr{V}$-sheaves on the site E (endowed with its canonical topology). Denote by $\mathrm{Ind}(C, \mathscr{V})_\Pi$ the category of $\mathscr{V}$-Ind-objects of C indexed bypreordered index sets *large relative to* Π. Show that there is an equivalence of categories

$$\mathrm{Ind}(C, \mathscr{V})_\Pi \xrightarrow{\approx} \tilde{E}_{\mathscr{V}}.$$

d) Let $c \in \mathscr{V}$ be a cardinal, and let $\mathrm{Ind}(E, \mathscr{V})'_c$ be the full subcategory of $\mathrm{Ind}(E, \mathscr{V})$ consisting of the $\mathscr{V}$-ind-objects of E indexed by a preordered set that is large relative to every cardinal $< c$. Taking $c = \mathrm{card}\,\mathscr{U}$, show that there is an equivalence of categories 160

$$\mathrm{Ind}(E, \mathscr{V})'_c \xrightarrow{\approx} \tilde{E}_{\mathscr{V}}. \quad *$$

9.21. The present section 9.21 develops technical preliminaries for the proof of Theorem 9.22 below, which is the main result of the present section 9. Let

$$p : E \to B \tag{9.21.1}$$

be a fibered functor, where B is a small category, and where the inverse-image functors

$$f^* : E_\beta \longrightarrow E_\alpha$$

associated with the arrows $f : \alpha \to \beta$ of B are accessible (9.2). In particular, the fiber categories E_α ($\alpha \in \mathrm{ob}\, B$) satisfy condition L (9.1), and therefore, since B is small, there exists a cardinal $\Pi'_\circ \in \mathscr{U}$ such that all the categories E_α satisfy condition $L_{\Pi'_\circ}$. Let $\Pi_\circ \in \mathscr{U}$ be an infinite cardinal $> \Pi'_\circ$, so that one has

$$\Pi_\circ > \Pi'_\circ, E_\alpha \text{ satisfies } L_{\Pi'_\circ} \text{ for every } \alpha \in \mathrm{ob}\, B. \tag{9.21.2}$$

Moreover, the hypotheses made immediately imply the existence of a cardinal $c \in \mathscr{U}$ satisfying the following conditions:

$$\begin{cases} \text{a)} & c \text{ is infinite,} \\ \text{b)} & c \geq \mathrm{card}\, F\ell\, B, \\ \text{c)} & \text{for every arrow } f : \alpha \to \beta \text{ in } B, \\ & \text{the functor } f^* : E_\beta \to E_\alpha \text{ is } c\text{-} \\ & \text{accessible.} \end{cases} \tag{9.21.3}$$

Suppose moreover that, for every $\alpha \in \mathrm{ob}\, B$, one can find a full subcategory C_α of E_α satisfying the following conditions: 161

$$\begin{cases} \text{a)} & \text{For every } X \in \mathrm{ob}\, C_\alpha, \ X \text{ is } c\text{-} \\ & \text{accessible in } E_\alpha \text{ (9.3).} \\ \text{b)} & \text{For every } X \in \mathrm{ob}\, E_\alpha, \text{ one can find} \\ & \text{an isomorphism } X \simeq \varinjlim_I X_i \text{ in} \\ & E_\alpha, \text{ where the } X_i \text{ belong to } C_\alpha \text{ and} \\ & I \text{ is an ordered set large relative to} \\ & c. \end{cases} \tag{9.21.4}$$

Let Π be a cardinal such that

$$\Pi \geq \Pi_\circ \quad , \quad \text{hence } \Pi > \Pi'_\circ, \tag{9.21.5}$$

and, for every $\alpha \in \mathrm{ob}\, B$, let

$$C_\alpha^\Pi \subset E_\alpha \tag{9.21.6}$$

be formed by the objects that can be represented in the form $\varinjlim_I X_i$, where I is an ordered set large relative to $\Pi'_\circ$, such that card $I \leq \Pi$, and where the X_i belong to C_α. It is then clear, by 7.5.2, that C_α^Π is essentially small, i.e. is equivalent to a small category.

Let

$$F = \mathbf{Hom}_B(B, E) \tag{9.21.7}$$

be the category of sections of E over B, and let

$$F^\Pi \subset F \tag{9.21.8}$$

be the strictly full subcategory of F formed by the sections $X : \alpha \mapsto X(\alpha)$ such that for every $\alpha \in \mathrm{ob}\, B$ one has

$$X(\alpha) \in C_\alpha^\Pi .$$

162 Since the C_α^Π are essentially small, it is clear that the same is true of the category F^Π. We shall show that this category is generating, and more precisely:

LEMMA 9.21.9. *Under the preceding conditions and with the preceding notation, the following statements hold:*

(i) *Every object of F is accessible (9.3). If d is a cardinal $\geq \Pi'_\circ$, and if $\alpha \mapsto X(\alpha)$ is an element of F such that, for every α, $X(\alpha)$ is d-accessible in E_α, then X is d-accessible in F.*

(ii) *Suppose that $\Pi \leq c$, or that $\Pi'_\circ$ is finite (i.e. the E_α are stable under small filtered $\varinjlim$). Then every object X of F is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i belong to F^Π and I is large relative to Π.*

To prove (i), note that by (9.21.4) and 9.9, every object of E_α is accessible. On the other hand, F satisfies condition $L_{\Pi'_\circ}$ by the

LEMMA 9.21.10. *Let $p : E \to B$ be a fibered functor, let $F = \mathbf{Hom}_B(B, E)$, let I be a category, and let $i \mapsto X_i$ be a functor from I to F. For $\varinjlim_I X_i$ to be representable in F, it suffices that, for every $\alpha \in \mathrm{ob}\, B$, $\varinjlim_I X_i(\alpha)$ be representable in the fiber category E_α; when this is so, $\varinjlim_I X_i$ is "computed argument by argument". In particular, if the fiber categories satisfy condition $L_{\Pi'_\circ}$ (where $\Pi'_\circ$ is a given cardinal), then so does F.*

163 We leave the details of the (easy) proof of 9.21.10 to the reader, merely noting that it is convenient to use the following result, whose proof is immediate:

COROLLARY 9.21.10.1. *With the hypotheses and notation of 9.21.10 for $p : E \to B$ and for I, for every $\alpha \in \mathrm{ob}\, B$, the inclusion functor $E_\alpha \to E$ commutes with inductive limits of type I.*

Returning now to the general conditions of 9.21.9, let X be an object of F. Since, for every $\alpha \in \mathrm{ob}\, B$, $X(\alpha)$ is accessible in E_α, there exists a cardinal $d \in \mathscr{U}$ such that, for every α, $X(\alpha)$ is d-accessible in E_α. We may choose $d \geq \Pi'_\circ$, so that F satisfies L_d by 9.21.10. Using 9.21.10 once more to compute the inductive limits $\varinjlim_I Y_i$ in F, with I large relative to d, one immediately sees that X is d-accessible. This proves (i).

We shall now prove 9.21.9 (ii) in several steps ((9.21.11) to (9.21.16)).

Let then

$$X : \alpha \longmapsto X(\alpha)$$

be an object of F. By (9.21.4 b)), for every $\alpha \in \mathrm{ob}\, B$ one can find an ordered set I_α large relative to c, and an isomorphism $X(\alpha) = \varinjlim_{i \in I_\alpha} X(\alpha)_i$, where $i \mapsto X(\alpha)_i$ is an inductive system of type I_α in E_α, with the $X(\alpha)_i$ in C_α. Consider the product ordered

set $I = \Pi_\alpha I_\alpha$; it is clear that it is large relative to c, and that the preceding inductive systems give rise to inductive systems in the E_α, 164

$$i \longmapsto X(\alpha)_i \in \mathrm{ob}\, C_\alpha, i \in \mathrm{ob}\, I, \tag{9.21.11}$$

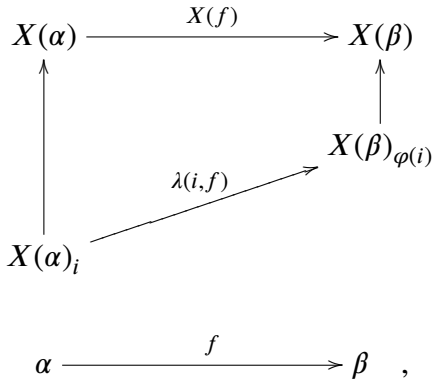
indexed by the same ordered set I, and to isomorphisms

$$X(\alpha) \xrightarrow{\sim} \varinjlim_I X(\alpha)_i \text{ in } E_\alpha, \tag{9.21.12}$$

for every $\alpha \in \mathrm{ob}\, B$. In what follows we suppose that data (9.21.11) and (9.21.12) have been fixed.

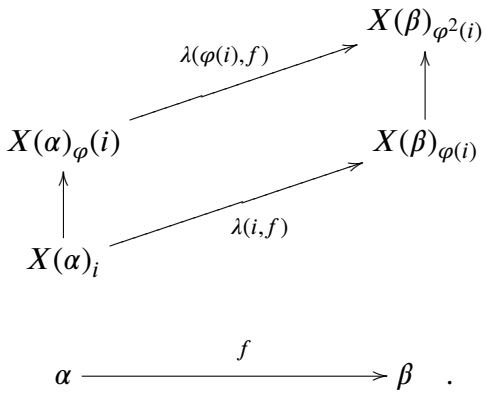
LEMMA 9.21.13. *Under the preceding conditions, one can find a map $\varphi : I \to I$ such that $\varphi(i) \geq i$ for every $i \in I$, and a map $(i, f) \mapsto \lambda(i, f)$ from $I \times \mathrm{Fl}\, B$ to $\mathrm{Fl}\, E$, satisfying the following conditions:*

a) *For $i \in I$ and $(f : \alpha \to \beta) \in F\ell B$, $\lambda(i, f)$ is an f-morphism from $X(\alpha)_i$ to $X(\beta)_{\varphi(i)}$, making the diagram*



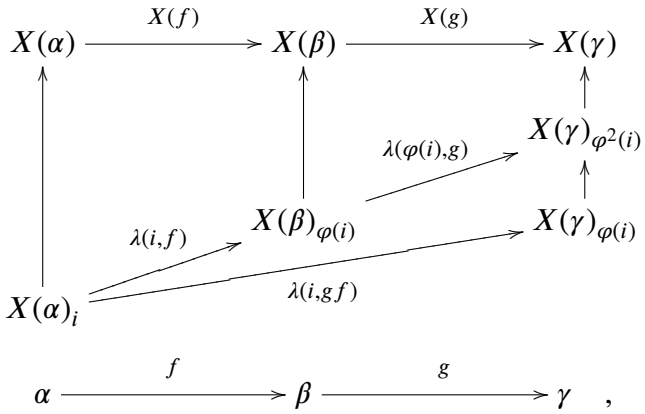
commutative, where the vertical arrows are the canonical morphisms deduced from (9.21.12).

b) *For every $i \in I$, and every $(f : \alpha \to \beta) \in \mathrm{Fl}\, B$, the diagram* 165



is commutative.

c) *For every pair of consecutive arrows* $\alpha \xrightarrow{f} \beta \xrightarrow{g} \gamma$ *of* $\boldsymbol{B}$*, the following diagram is commutative:*



where the vertical arrows are those deduced from (9.21.12).

166 One should moreover note that the commutativity of the two upper trapezoids in c) is already contained in a), so c) in fact asserts the commutativity of the lower triangle of the diagram under consideration.

Let us prove 9.21.13. Let $i \in I$, and let $f : \alpha \to \beta$ be an arrow in $\boldsymbol{B}$. Consider the composite $X(\alpha)_i \to X(\alpha) \xrightarrow{X(f)} X(\beta) \simeq \varinjlim_j X(\beta)_j$; it defines a morphism in $\boldsymbol{E}$:

$$X(\alpha)_i \to f^*(X(\beta)) \simeq f^*(\varinjlim_j X(\beta)_j) \simeq \varinjlim_j f^*(X(\beta)_j),$$

where the last equality follows from the fact that f^* is c-accessible (9.21.3 c)and that I is large relative to c. By (9.21.4 a)), since $X(\alpha)_i \in \mathrm{ob}\, C_\alpha$, the morphism under consideration factors through some $f^*(X(\beta)_j)$, where j *a priori* depends on i and on $f \in \mathrm{Fl}\,(\boldsymbol{B})$. But by (9.21.3 b)), and since I is large relative to c, one may choose j independently of f, say $j = \varphi(i)$. Since I is filtered, one may suppose $\varphi(i) \geq i$. One thus obtains, for $i \in I$ and $f \in \mathrm{Fl}\, \boldsymbol{B}$, a morphism

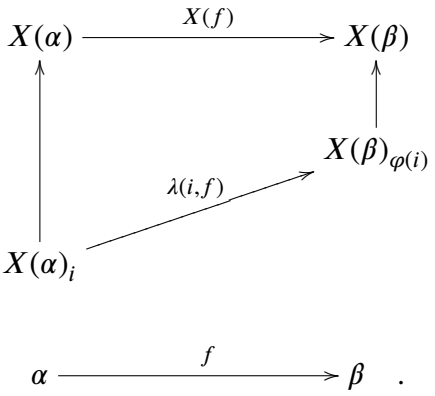
$$X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)}),$$

or, equivalently, an f-morphism

$$\lambda(i, f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)},$$

which by construction makes the diagram

$D(f)$



167 commutative. Now , for given i, f, g, consider the lower triangle of the diagram in 9.21.13

c); it is not clear that it is commutative, but the two composites $X(\alpha)_i \rightrightarrows X(\gamma)_{\varphi^2(i)}$ become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)$, as follows from the commutativity of the two upper trapezoids and of the outer trapezoid, which are respectively the diagrams $D(f)$, $D(g)$, and $D(gf)$. Thus, since $X(\alpha)_i$ is c-accessible (9.21.4 a)) and $X(\gamma) \simeq \varinjlim_{j\in I} X(\gamma)_j$, with I large relative to c, it follows that one can find an element $j \geq \varphi^2(i)$ of I such that the two arrows under consideration become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)_j$. A priori, j depends on i, f, g. But, for fixed i, the set of possible pairs f, g has cardinality $\leq c^2 = c$, by (9.21.3 a) and b)), and hence, since I is large relative to c, one may choose j independently of f and g, say $j = \varphi'(i) \geq \varphi^2(i) \geq \varphi(i)$. For every $i \in I$ and $f : \alpha \to \beta$, let then

$$\lambda'(i, f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi'(i)}$$

be the composite f-morphism $X(\alpha)_i \to X(\beta)_{\varphi(i)} \to X(\beta)_{\varphi'(i)}$, where the second arrow is the transition morphism. It is then immediate from the construction that (φ', λ') satisfies conditions 9.21.13 a) and c) for (φ, λ). Proceeding in the same way for condition b), one sees that φ' may be chosen so that this condition is also satisfied for (φ', λ'). This completes the proof of (9.21.13).

9.21.14. Now let 168

$$J \subset I$$

be a subset of I satisfying the following conditions:

a) J is filtered,
b) for every $j \in J$, one has $\varphi(j) \in J$,
c) For every $\alpha \in \mathrm{ob}\, B$, $X_J(\alpha) = \varinjlim_{i\in J} X(\alpha)_i$ is representable in E_α.

Conditions a) and c) are satisfied in particular if J is large relative to $\Pi'_\circ$ (9.21.2). It then follows from 9.21.10.1 that, for every $\alpha \in \mathrm{ob}\, B$, $X_J(\alpha)$ is the inductive limit $\varinjlim_{i\in J} X(\alpha)_i$ in E. Now, for an arrow $f : \alpha \to \beta$ of B, the arrows $\lambda(i, f)$, with i varying in J, define, by (9.21.13 b)), a morphism of ind-objects from $(X(\alpha)_i)_{i\in J}$ to $(X(\beta)_i)_{i\in J}$, whence a morphism

$$X_J(f) : X_J(\alpha) \to X_J(\beta)$$

on the inductive limits, which is manifestly an f-morphism. Using (9.21.13 (c)), one obtains the transitivity relations

$$X_J(gf) = X_J(g)X_J(f),$$

so that a section $X_J \in \mathrm{ob}\, F$ of E over B has been defined. Finally, (9.21.13 a)) shows us that the canonical morphisms

$$X_J(\alpha) \longrightarrow X(\alpha) \quad , \quad \alpha \in \mathrm{ob}\, B,$$

are functorial in α, so there is a canonical morphism

$$u_J : X_J \longrightarrow X.$$

On the other hand, if 169

$$J' \supset J$$

is another subset of I satisfying conditions a), b), c) above, one obtains a canonical morphism

$$u_{J',J} : X_J \longrightarrow X_{J'},$$

so that the X_J form an inductive system in F, parametrized by the set (ordered by inclusion) of subsets J of I satisfying the conditions under consideration. Finally, the

morphisms u_J above define a morphism from this inductive system to (the constant inductive system defined by) X.

9.21.15. Now let K be a set of subsets J of I satisfying conditions a), b), c) considered in 9.21.14, and suppose that K is filtered and has union I. It is then clear that one has

$$\varinjlim_{J\in K} X_J \xrightarrow{\sim} X,$$

using 9.21.10.1, which reduces us to checking that there is an isomorphism argument by argument.

Take, for example, for K the set of all subsets J of I that are large relative to $\Pi'_\circ$, stable under φ, and such that $\operatorname{card} J \leq \Pi$. Then, by the definition (9.21.6) of C_α^Π, for every $\alpha \in \operatorname{ob} B$ one has $X_J(\alpha) \in \operatorname{ob} C_\alpha^\Pi$, hence $X_J \in \operatorname{ob} F^\Pi$. Consequently, 9.21.9 (ii) will be proved if we establish that K is large relative to Π (and hence filtered) and has union I. For this, it will evidently suffice to prove that every subset S of I such that
170 $\operatorname{card} S \leq \Pi$ is contained in a $J \in K$. This will indeed follow from the preliminary hypothesis stated in 9.21.9 (ii), and from the

LEMMA 9.21.16. *Let I be an ordered set, let Π be an infinite cardinal, and let $\varphi : I \to I$ be a map from I to itself.*

a) *Suppose I is filtered. Then, for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a filtered subset $J \supset S$ of I such that $\varphi(J) \subset J$ and $\operatorname{card}(J) \leq \Pi$.*

b) *Let $\Pi'_\circ$ be a cardinal $< \Pi$, and suppose that I is large relative to Π. Then, for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a subset $J \supset S$ of I large relative to $\Pi'_\circ$, such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.*

Let us prove b), for example (the proof of a) being similar and simpler). Let P be the set of subsets of I of cardinality $\leq \Pi$, and let $T \mapsto i_T : P \to I$ be a map such that, for $T \in P$, i_T is an upper bound of T in I; the existence of this map merely expresses the hypothesis that I is large relative to Π. Let A be a well-ordered set such that $\operatorname{card} A = \Pi$, and such that, for every $a \in A$, the set of $a' \leq a$ has cardinality $< \Pi$. It follows in particular, since $\Pi'_\circ < \Pi$, that A *is large relative to* $\Pi'_\circ$. Define, by transfinite recursion, maps

$$S \longmapsto S_a : P \longrightarrow P \quad (a \in A),$$

by the following formulas:

$S_\circ = S$, $S_{a+1} = S_a \cup \varphi(S_a) \cup \{i_{S_a}\}$, $S_a = \bigcup_{a'<a} S_{a'}$ if a is a limit ordinal.

171 Then put, for $S \in P$,

$$J(S) = \bigcup_{a\in A} S_a.$$

It is then clear that $J(S) \supset S$, that $J(S)$ is stable under φ, that $\operatorname{card} J(S) \leq \Pi$ (since $\operatorname{card} A = \Pi$), and finally that $J(S)$ is large relative to $\Pi'_\circ$ (using the fact that A is large relative to $\Pi'_\circ$).

This proves 9.21.16 and completes the proof of 9.21.9.

We also point out a variant of 9.21.9:

LEMMA 9.21.17. *The notation is that of 9.21.9. Let F' denote the full subcategory* $\mathbf{Homcart}_B(B, E)$ *of $F = \mathbf{Hom}_B(B, E)$ formed by the cartesian sections of E over B. Then:*

(i) *Every object of F' is accessible. More precisely, let $\alpha \mapsto X(\alpha)$ be an element of F', and let d be a cardinal such that $d \geq \Pi'_\circ$, $d \geq c$, and such that, for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is a d-accessible object of E_α. Then X is a d-accessible object of F'.*

(ii) *Suppose $\Pi \leq c$ and that the functors $f^* : E_\beta \to E_\alpha$ are $\Pi'_\circ$-accessible, or that the categories E_α are stable under small filtered inductive limits and the functors $f^* : E_\beta \to E_\alpha$ commute with these inductive limits. Then every object X of F' is isomorphic to an object of the form $\varinjlim_I X_i$, where, for every $i \in I$, X_i is an object of the full subcategory $F'^{\Pi} = F' \cap F^{\Pi}$ of F', and where I is large relative to Π.*

The proof being entirely analogous to that of 9.21.9, we shall merely indicate the 172
points at which the latter must be modified. First note:

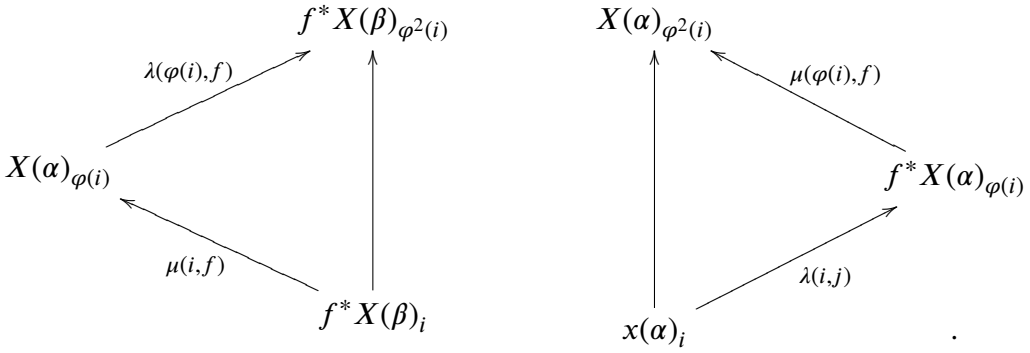
LEMMA 9.21.18. *The statement of 9.21.10 remains valid when $F = \mathbf{Hom}_B(B, E)$ is replaced there by $F' = \mathbf{Homcart}_B(B, E)$, provided that the inverse image functors $f^* : E_\beta \to E_\alpha$ commute with inductive limits of type I.*

Thus, under the conditions of 9.21.17, if d is a cardinal such that $d \geq c$ and $d \geq \Pi'_\circ$, it follows from the above and from hypothesis (9.21.3 c)) that, for every ordered set I large relative to d, inductive limits of type I are representable in F' and are computed argument by argument, whence the conclusion 9.21.17 (i) follows by an immediate argument.

To prove (ii), one must supplement 9.21.13:

LEMMA 9.21.19. *Under the conditions of 9.21.13, suppose that $X \in \mathrm{ob}\, F'$ i.e. that X is a cartesian section of E over B. Then the conclusion can be strengthened by asserting that there exists a function μ which assigns to every $i \in I$ and every arrow $f : \alpha \to \beta$ of B a morphism $\mu(i, f) : f^*X(\beta)_i \to X(\alpha)_{\varphi(i)}$ in E_α, such that:*

d) *For every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, the following two diagrams are commutative:*



To prove this, one proceeds as in 9.21.13, by considering the composite morphism 173
$f^*(X(\beta)_i) \to f^*(X(\beta)) \xrightarrow{X(f)^{-1}} X(\alpha)$, where (as already in the formulation of condition d)) we identify notationally an f-morphism $R \to S$ of E with the corresponding morphism $R \to f^*(S)$ of E_α. Since $X(\alpha) = \varinjlim_j X(\alpha)_j$, and I is large relative to c, one can factor the preceding morphism through one of the $X(\alpha)_j$, where j *a priori* depends on i *and* on f, but can be chosen independently of f, say $\psi(i)$. After enlarging the function φ of 9.21.13, one may suppose that $\psi = \varphi$. Proceeding as for conditions b) and c) of 9.21.13, one sees that, after further enlarging the map φ, one may suppose that the two diagrams of 9.21.19 d) are commutative.

9.21.20. With the map φ chosen as in 9.21.19, let us repeat the argument of 9.21.14, except that we must assume that J satisfies, in addition to the stated conditions a), b), c), the following condition:

d) For every arrow $f : \alpha \to \beta$ in B, the functor $f^* : E_\beta \to E_\alpha$ commutes with inductive limits of type J.

174 We claim that, under this additional condition, the section X_J of E over B is cartesian, i.e. for every arrow $f : \alpha \to \beta$ of B, the morphism

$$(*) \qquad X_J(\alpha) \longrightarrow f^* X_J(\beta)$$

is an isomorphism. Indeed, by condition d) above, the target of the morphism under consideration is identified with $\varinjlim_{i\in J} f^*(X(\beta)_i)$, and the morphism is obtained by taking $\varinjlim_J$ of the morphism of ind-objects in E_α

$$(X(\alpha)_i)_{i\in J} \longrightarrow (f^*)(X(\beta)_i)_{i\in J},$$

induced by the morphisms $\lambda(i, f) : X(\alpha)_i \to f^*(X(\beta)_{\varphi(i)})$ for $i \in J$. Now the conditions set out in 9.21.19 ensure that the preceding homomorphism of ind-objects is in fact an isomorphism, an inverse being obtained from the homomorphism induced by the system of $\mu(i, f)$, for $i \in J$. This proves our assertion that $(*)$ is an isomorphism, and hence that $X \in \mathrm{ob}\, F'$.

9.21.21. We shall now assume that the functors $f^* : E_\beta \to E_\alpha$ are $\Pi'_\circ$-*accessible*. Then the conditions on J considered in 9.21.20 are satisfied if J is large relative to $\Pi'_\circ$, stable under φ, and such that $\mathrm{card}\, J \leq \Pi$, and we complete the proof of 9.21.17 just as in 9.21.15.

175 Theorem 9.22. *Let $p : E \to B$ be a fibered functor, where B is a category essentially equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ in B the inverse image functor $f^* : E_\beta \to E_\alpha$ is accessible (9.2). Suppose moreover that for every $\alpha \in \mathrm{ob}\, E$, the fiber category E_α admits a cardinal filtration (9.12) and that every object of E_α is accessible (9.3). Then each of the categories $F = \mathbf{Hom}_B(B, E)$ and $F' = \mathbf{Homcart}_B(B, E)$ has a small full subcategory generating by strict epimorphisms (7.1), and every object of either category is accessible.*

It is immediate that the statement is not essentially changed when B is replaced by a full subcategory $B_\circ$ such that the inclusion functor $B_\circ \to B$ is an equivalence, and E by $E_\circ = E \times_B B_\circ$. This allows us to assume that B is a small category.

For every $\alpha \in \mathrm{ob}\, B$, let $(\mathrm{Filt}^\Pi(E))_{\Pi\geq\Pi_\alpha}$ be a cardinal filtration of E_α. Let $c_\circ \in \mathscr{U}$ be a cardinal satisfying the following conditions:

$$(9.22.1) \qquad \begin{cases} c_\circ \geq \operatorname{Sup}_{\alpha\in ob B} \Pi_\alpha \quad , \quad c_\circ \geq \mathrm{card}\, \mathrm{Fl}\, B; \\ \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_\beta \to E_\alpha \text{ is } c_\circ\text{-accessible}; \\ \text{for every } \alpha \in \mathrm{ob}\, B, \text{ the objects of } \mathrm{Filt}^{\Pi_\alpha}(E_\alpha) \text{ are } c_\circ\text{-accessible}. \end{cases}$$

Set

$$c = 2^{c_\circ},$$

so that $c > c_\circ$, and, for every $\alpha \in \mathrm{ob}\, B$, let

$$C_\alpha = \mathrm{Filt}^c(E_\alpha).$$

176 We claim that the preliminary conditions of 9.21.9 and 9.21.17 are satisfied on taking $\Pi = \Pi_\circ = c$. This is clear for (9.21.2) and (9.21.3). For (9.21.4 a)), this follows from 9.17, and (9.21.4 b)) follows from condition 9.12 c) for a cardinal filtration. Let us also note that condition 9.12 b) on cardinal filtrations implies that for every $\alpha \in \mathrm{ob}\, B$, we have $C_\alpha^\Pi = C_\alpha = \mathrm{Filt}^C(E_\alpha)$, where C_α^Π is defined in (9.21.6). Consequently, 9.22 follows from 9.21.9 and 9.21.17, and more precisely, we have proved the

COROLLARY 9.23. *Under the conditions of 9.22, if $c_\circ \in \mathcal{U}$ is a cardinal satisfying conditions (9.22.1), and if $c = 2^{c_\circ}$, then the subcategory F^c of F (respectively F'^c of F') formed by the X such that $X(\alpha) \in \operatorname{ob} \operatorname{Filt}^c(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$ is essentially small and generating by strict epimorphisms; more precisely, every object X of F (respectively F') is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i belong to F^c (respectively F'^c), and where I is an ordered set large relative to c.*

Under slightly stronger hypotheses in 9.22, one can moreover make 9.22 and 9.23 considerably more precise:

COROLLARY 9.24. *Under the conditions of 9.22, suppose that each of the categories E_α ($\alpha \in \operatorname{ob} B$) is stable under small filtered inductive limits, and that, for every $\Pi \geq \Pi_\alpha$, $\operatorname{Filt}^\Pi(E_\alpha)$ is stable under filtered inductive limits indexed by filtered ordered sets I such that $\operatorname{card} I \leq \Pi$ (which slightly strengthens condition 9.12 b) for cardinal filtrations). In* 177
the case where F' is being considered, suppose moreover that, for every arrow $f : \alpha \to \beta$ of B, the functor $f^* : E_\beta \to E_\alpha$ commutes with small filtered inductive limits. Finally, let $c_\circ \in \mathcal{U}$ be a cardinal satisfying conditions (9.22.1), and, for every cardinal $\Pi \geq c = 2^{c_\circ}$, consider the strictly full subcategory F^Π (respectively F'^Π) of F (respectively F') formed by the X such that $X(\alpha) \in \operatorname{Filt}^\Pi(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$. Then the subcategories under consideration define a cardinal filtration (9.12) of F (respectively F').

Conditions a), b), c) of 9.12 must be checked. Conditions a) and b) follow from the analogous conditions for the given cardinal filtrations of the E_α, and from 9.21.10 (respectively 9.21.18, taking account of the second of conditions (9.22.1)). It remains to prove c), whose first part follows immediately from 9.21.9 (respectively 9.21.17), by taking $\Pi'_\circ = 0$ and $\Pi_\circ = c$. It remains to prove that if $\Pi' \geq \Pi \geq c$ are two cardinals, then for every X in $F^{\Pi'}$ (respectively $F'^{\Pi'}$) one has $X \simeq \varinjlim_{i \in K} X_i$, with the X_i in F^Π (respectively F'^Π), K large relative to Π, and $\operatorname{card} K \leq \Pi'^\Pi$. For this, return to the proofs of 9.21.9 (ii) and 9.21.17 (ii). By condition 9.12 c) on the cardinal filtration of E_α, one may suppose that each I_α has cardinality $\leq \Pi'$, and therefore one will have

$$\operatorname{card} I \leq (\Pi'^\Pi)^{\operatorname{card} \operatorname{ob} B} \leq (\Pi'^\Pi)^c = \Pi'^{\Pi c} = \Pi'^\Pi.$$

The index set K used in 9.21.15 (respectively 9.21.21) is the set of subsets J of I that are filtered (i.e. large relative to $\Pi'_\circ = 0$), stable under φ, and such that $\operatorname{card} J \leq \Pi$. It is then immediate that

$$\operatorname{card} K \leq (\operatorname{card} I)^\Pi \leq (\Pi'^\Pi)^\Pi = \Pi'^{\Pi^2} = \Pi'^\Pi,$$

which completes the proof of 9.24. 178

To conclude, it is appropriate to give a statement freed from the somewhat technical hypotheses of 9.22, replacing them by the conjunction, for the E_α, of the hypotheses that occur in 9.11 (which ensure the accessibility of the objects of the E_α) and in 9.13 (which ensure the existence of a cardinal filtration in E_α):

COROLLARY 9.25. *Let $p : E \to B$ be a fibered functor, where B is a category equivalent to a small category, and where, for every arrow $f : \alpha \to \beta$ of B, the functor $f^* : E_\beta \to E_\alpha$ is accessible (9.2). Suppose moreover that, for every $\alpha \in \operatorname{ob} B$, the fiber category E_α satisfies the following conditions:*

a) *E_α admits a small full subcategory generating by strict epimorphisms (7.1).*

b) *E_α is stable under fiber products, kernels of pairs of arrows, sums of two objects, and cokernels of pairs of arrows, and finally under small filtered inductive limits* [3].
c) *The functor* $\mathrm{Ker}(u, v)$ *on the category of pairs of arrows of E_α is accessible (for example, it commutes with small filtered inductive limits).*
d) *Every strict epimorphism of E_α (10.2) is a universal strict epimorphism.*

Under these conditions, each of the categories $F = \mathbf{Hom}_B(B, E)$ and $F' = \mathbf{Homcart}_B(B, E)$ admits a small subcategory generating by strict epimorphisms, and all its objects are accessible (9.3).

179 Moreover, 9.24 gives us:

Corollary 9.26. *Under the conditions of 9.25, and in the case where one considers $F' = \mathbf{Homcart}_B(B, E)$, suppose that the functors $f^* : E_\beta \to E_\alpha$ commute with small filtered inductive limits. Then F (respectively F') admits a cardinal filtration, which can be made explicit by the procedure of 9.24.*

10. Glossary

For the reader's convenience, we collect here the definitions of some terms used in the preceding sections. We denote a category by C.

10.1. Cartesian, cocartesian. A diagram in C

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

is called *cartesian* if it is commutative and the canonical morphism from A to the fiber product $C \times_D B$ is an isomorphism. It is called *cocartesian* if the corresponding diagram in the category opposite to C is cartesian.

180 **10.2. Epimorphism, strict epimorphism.** etc. Cf. 10.3.

10.3. Epimorphic family, strictly epimorphic family. etc.: A family

$$f_i : A_i \longrightarrow B, i \in I,$$

of arrows with the same target in C is called an *epimorphic family* if, for every object C of C, the map induced by the f_i,

$$\mathrm{Hom}_C(B, C) \longrightarrow \prod_i \mathrm{Hom}_C(A_i, C), \tag{10.3.1}$$

is injective.

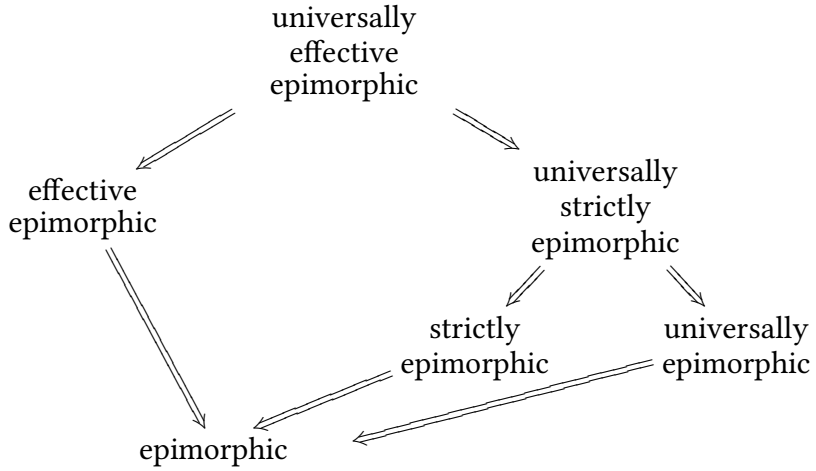
The family under consideration is called *strictly epimorphic* if the image of the map (10.3.1) consists of the families (g_i) such that, for every object R of C, every pair of indices $i, j \in I$, and every pair of arrows $u : R \to A_i$, $v : R \to A_j$ with $f_i u = f_j v$, one also has $g_i u = g_j v$. When the fiber products $A_i \times_B A_j$ are representable for $i, j \in I$, this is equivalent to saying that the image of (10.3.1) consists of the (g_i) such that, for every pair of indices $i, j \in I$, one has $g_i \,\mathrm{pr}_1 = g_j \,\mathrm{pr}_2$, where $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections of $A_i \times_B A_j$. The family $(f_i)_{i \in I}$ is called an *effective epimorphic family* if the preceding

[3] This last condition is probably unnecessary; cf. 9.13.2.

condition is satisfied, i.e. if it is strictly epimorphic and the fiber products $A_i \times_B A_j$ are representable.

The family $(f_i)_{i\in I}$ is called a *universally epimorphic* (respectively *universally effective epimorphic*) family if the morphisms f_i are pullbackable (10.3), and if, for every arrow $B' \to B$, the family of arrows $f'_i : A'_i = A_i \times_B B' \to B'$, $i \in I$, deduced from the family $(f_i)_{i\in I}$ by the base change $B' \to B$, is epimorphic (respectively effective epimorphic).

The notion of a *universally strictly epimorphic family* will be defined in (II 2.5, 2.6). 181
Note that when the morphisms f_i are pullbackable (for example, if fiber products are representable in C), this notion coincides with that of a universally effective epimorphic family (use II 2.4). In the general case, the six variants of the notion of epimorphicity under consideration are related by the logical implications:



A morphism $f : A \to B$ of C is called an *epimorphism (respectively a strict epimorphism, respectively an effective epimorphism, respectively a universal epimorphism, respectively a universal effective epimorphism, respectively a universal strict epimorphism)* if the family of morphisms consisting of the single element f is epimorphic (respectively ...).

10.4. Monomorphic family, strictly monomorphic family. etc. A family of arrows $f_i : A_i \to B$ of C is called *monomorphic (respectively strictly monomorphic,...)* if, regarded as a family of arrows of the opposite category C°, it is epimorphic (respectively strictly epimorphic,...). An arrow $f : A \to B$ of C is called a *monomorphism (respectively a strict monomorphism,...)* if the family consisting only of f is monomorphic (respectively strictly monomorphic,...), i.e. if f, regarded as an arrow of the opposite 182
category C°, is an epimorphism (respectively a strict epimorphism,...).

10.5. Monomorphism, strict monomorphism. etc.Cf. 10.4.

10.6. Projector, image of a projector, direct factor of an object. Let $f : X \to X$ be an endomorphism of an object of C. The endomorphism f is called a *projector* if $f^2 = f$. The pair (f, id_X) then has a representable cokernel X' if and only if it has a representable kernel X''; when this is the case, there exists a unique isomorphism $u : X' \to X''$ in C such that $iup = f$, where $p : X \to X'$ and $i : X'' \to X$ are the canonical morphisms.[4] One then generally identifies X' with X'' and calls it the *image of the projector* f; the projector f is said to *admit an image* if $\mathrm{Ker}(f, \mathrm{id}_X)$ is representable, i.e. if $\mathrm{Coker}(f, \mathrm{id}_X)$

[4]N.D.E.: Representability of the kernel, or of the cokernel, is also equivalent to the existence of morphisms i, p, and an isomorphism u such that $iup = f$ and $pi = u^{-1}$. In particular, this property is preserved by every functor.

is representable. A subobject (respectively a quotient) Y of X is called a *direct factor* of X if there exists a projector f on X that factors through Y and has Y as its image as a subobject (respectively as a quotient) of X. Sometimes, by abuse of language, an object of C is called a *direct factor of* X if it is isomorphic to the image of a projector on X.

10.7. Pullbackable. An arrow $f : A \to B$ of C is called *pullbackable* if, for every arrow $B' \to B$ of C, the fiber product $A \times_B B'$ is representable in C.

183 **10.8. Quotient, strict quotient.** etc.

Let A be an object of C. Two epimorphisms $f : A \to B$ and $f' : A \to B'$ with source A are called *equivalent* if there exists an isomorphism $u : B \xrightarrow{\sim} B'$ such that $uf = f'$. This defines an equivalence relation on the set of epimorphisms with source A, whose classes are called the *quotients, or quotient objects,* of A. For each quotient of A, one generally chooses an element of this class, say $f : A \to B$, and one often speaks (by abuse of language) of the quotient B of A (or of the quotient $f : A \to B$ of A). The quotient B is called a *strict quotient* (respectively an *effective quotient, respectively a universal quotient,...*) if the morphism $f : A \to B$ is a strict epimorphism (respectively an effective epimorphism, respectively a universal epimorphism,...) (10.3).

10.9. Equivalence relations. Let F and G be two presheaves of sets on C. A diagram $p_1, p_2 : F \rightrightarrows G$ is called an *equivalence relation on* G if, for every object A of C, the map

$$(p_1(A), p_2(A)) : F(A) \longrightarrow G(A) \times G(A)$$

induces a bijection from $F(A)$ onto the graph of an equivalence relation on the set $G(A)$.

A diagram $p_1, p_2 : B \rightrightarrows C$ in C is called an equivalence relation on C if the corresponding diagram of presheaves is an equivalence relation. When finite products and fiber products are representable in C, a functor $\Phi : C \to C'$ commuting with finite products and fiber products carries equivalence relations on C to equivalence relations on $\Phi(C)$.

184 Let $\pi : C \to D$ be a morphism of C such that the fiber product $C \times_D C$ is representable. Then the diagram $\mathrm{pr}_1, \mathrm{pr}_2 : C \times_D C \rightrightarrows C$ is an equivalence relation on C.

10.10. Effective equivalence relation, universally effective equivalence relation. An equivalence relation $p_1, p_2 : R \rightrightarrows A$ on an object A of C is called an *effective equivalence relation* if there exists a morphism $\pi : A \to B$ such that the square

$$\begin{array}{ccc} R & \xrightarrow{p_2} & A \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle \pi} \\ A & \longrightarrow & B \end{array}$$

is cartesian and cocartesian. (The morphism π is the cokernel of the pair (p_1, p_2) and is therefore determined up to unique isomorphism.) The morphism π is then an effective epimorphism; if it is a universal effective epimorphism (10.3), the *equivalence relation* is said to be *universally effective.*

10.11. Subobject, strict subobject. etc. These are the notions dual to those of quotient and strict quotient etc. (10.8).

Bibliography

[1] B. Mitchell, "Theory of Categories," Academic Press (1965).

II. Appendix: Universes (by N. Bourbaki(*))

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1. Definition and first properties of universes

Definition 1. *A set U is called a universe if it satisfies the following conditions:*

(U.I) *if $x \in U$ and $y \in x$, then $y \in U$;*
(U.II) *if $x, y \in U$, then $\{x, y\} \in U$;*
(U.III) *if $x \in U$, then $\mathscr{P}(x) \in U$;*
(U.IV) *if $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then the union $\bigcup_{\alpha \in I} x_\alpha$ belongs to U;*
(U.?) *if $x, y \in U$, then the ordered pair $(x, y) \in U$.*

N.B. Since it has, I believe, been decided that in future editions the ordered pair will be defined à la Kuratowski by $(x, y) = \{\{x, y\}, \{x\}\}$, condition (U.?) is unnecessary, since it follows from (U.II).

Examples. 1) The *empty* set is a universe, denoted by $U_\circ$.

2) Consider the nonempty finite words formed with the four symbols $''\{'', ''\}'', ''$ and $''\varnothing''$ (cf. Alg. I). Define, by induction on the length n of such a word, the notion of a *meaningful* word:

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(a) For $n = 1$, only the word $\varnothing$ is meaningful;
(b) for a word A of length n to be meaningful, it is necessary and sufficient that there exist p distinct meaningful words $A_1, \dots, A_p$ ($p \geq 1$) of lengths $< n$ such that

$$A = \{A_1, A_2, \dots, A_p\}.$$

For example, $\varnothing$, $\{\varnothing\}$, $\{\{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\}$ are meaningful words. It is clear, by induction on the length, that every meaningful word denotes a term of Set Theory; for example, if $A = \{A_1, A_2, \dots, A_p\}$, A denotes the set whose elements are $A_1, A_2, \dots,$ and A_p; these terms are plainly *finite sets.* Let U_1 be the set of sets thus obtained. It is readily checked that U_1 satisfies conditions (U.I), (U.II), (U.III), and (U.IV) of Definition 1 (but not that idiotic condition (U.?), which is no problem if one is willing to decanulate the ordered pair). Thus U_1 is a *universe.* Note that the elements of U_1 are finite, and that U_1 is countable.

In the statements that follow, *U denotes a universe.*

Proposition 1. *If $X \in U$ and if $y \subset x$, then $y \in U$.*

Indeed, $\mathscr{P}(x) \in U$ by (U.III), whence $y \in \mathscr{P}(x)$ and $y \in U$ by (U.I).

Corollary. *If $x \in U$, every quotient set y of x is an element of U.*

Indeed, y is a subset of $\mathscr{P}(x)$. Hence $y \in U$ by (U.III) and Proposition 1.

187 Proposition 2. *If $x \in U$, then $\{x\} \in U$.*

This follows from (U.II), applied with $y = x$.

Proposition 3. *Every ordered pair, ordered triple, and ordered quadruple of elements of U is an element of U.*

This is true for ordered pairs by the N.B. or (U.?). The case of ordered triples follows, since $(x, y, z) = ((x, y), z)$, and then that of ordered quadruples, since $(x, y, z, t) = ((x, y, z), t)$.

[4] We reproduce here, with his permission, secret papers of N. Bourbaki. The references in this text are to his learned work.

PROPOSITION 4. *If $X, Y \in U$, then $X \times Y \in U$.*

Indeed, for $x \in X$, $\{x\} \times Y$ is the union of the family $\{(x, y)\}_{y \in Y}$; since $\{(x, y)\} \in U$ by (U.I), (U.II), and Proposition 3, we have $\{x\} \times Y \in U$ by (U.IV). Finally, $X \times Y$ is the union of the family $(\{x\} \times Y)_{x \in X}$; it is therefore an element of U, again by (U.IV).

COROLLARY 1. *If $X, Y, Z, \ldots$ are elements of U, all the sets of the scale constructed on $X, Y, Z, \ldots$ are elements of U* (cf. Chapter IV, §).

This indeed follows by successive applications of Proposition 4 and of (U.III).

COROLLARY 2. *If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U and if $I \in U$, the sum set $\Sigma_{\alpha \in I} x_\alpha$ is an element of U.*

Indeed, this sum set is a subset of the product $(\bigcup_{\alpha \in I} x_\alpha) \times I$ (Chapter II), a product both of whose factors are elements of U (by (U.IV) and the hypothesis). We then apply Propositions 4 and 1.

PROPOSITION 5. *If X and Y are elements of U, every correspondence between X and Y (in particular, every map from X to Y) is an element of U.* 188

Indeed, such a correspondence C is an ordered triple (X, Y, Γ), where Γ is a subset of $X \times Y$ (the graph of C) (Chapter II, §). We have $\Gamma \in U$ by Propositions 4 and 1. Hence $C \in U$ by Proposition 3.

PROPOSITION 6. *If $X, Y \in U$, every set Z of correspondences between X and Y (in particular, of maps from X to Y) is an element of U.*

Indeed, Z is a subset of $\{X\} \times \{Y\} \times \mathscr{P}(X \times Y)$. Now this product is an element of U by Proposition 2 and the corollary to Proposition 4. Therefore $Z \in U$ by Proposition 1.

COROLLARY. *If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then $\Pi_{\alpha \in I} x_\alpha \in U$.*

Indeed, this product is a set of maps from I to $\bigcup_{\alpha \in I} x_\alpha$, and this union is an element of U by (U.IV).

PROPOSITION 7. *If X is a subset of U whose cardinality is at most that of an element of U, then X is an element of U.*

Let I be an element of U such that $\mathrm{card}(X) \leq \mathrm{card}(I)$. There is a surjection $i \mapsto x_i$ from a subset I' of I onto X. Then X is the union of the family $(\{x_i\})_{i \in I'}$; now this union is an element of U by Proposition 1, Proposition 2, and (U.IV).

COROLLARY. *If U is nonempty, every finite subset of U is an element of U, and U has elements of arbitrarily large finite cardinality.* 189

On the other hand, if $U = \varnothing$, then $\varnothing$ is a finite subset of $\varnothing$ but not an element of $\varnothing$.

Indeed, if U is nonempty, Proposition 2 shows that a one-element set x_0 belongs to U. By induction on n, put $x_{n+1} = \mathscr{P}(x_n)$. We have $x_n \in U$ by (U.III), and $\mathrm{card}(x_{n+1}) = 2^{\mathrm{card}(x_n)}$, so that U has elements of arbitrarily large finite cardinality.

REMARK. It follows from Proposition 2 and the corollary to Proposition 7 that every nonempty universe U contains the universe U_1 of Example 2; indeed, $\varnothing \in U$ by Proposition 1. Thus U_1 is the intersection of all nonempty universes. More generally:

PROPOSITION 8. *If $(U_\lambda)_{\lambda \in L}$ is a nonempty family of universes, then $U = \bigcap_{\lambda \in L} U_\lambda$ is a universe.*

This follows at once from Definition 1.

2. Universes and species of structures

Let $\mathscr{E}$ be a species of structure; suppose, to fix ideas and simplify the exposition, that every structure of species $\mathscr{E}$ is defined on an underlying set. Let (X, S) be a structure of species $\mathscr{E}$ (where X is the underlying set and S the structure), and let U be a *universe*. If $X \in U$, then all the constituent objects of the structure S on X are elements of U (by Cor. 1 of Prop. 4, n° 1, and by (U.I)), so the structure (X, S) is an element of U.

190 Suppose now that $\mathscr{E}$ is a species of structure with *morphisms*. If X and X' are elements of U endowed with structures of species $\mathscr{E}$, then the set of morphisms from X to X' is again an element of U (n° 1, Prop. 6).

Now consider the *category* $(U\text{-}\mathscr{E})$ defined in § 1, n° 2, Ex. c). Since the objects and arrows of $(U\text{-}\mathscr{E})$ are elements of U, $(U\text{-}\mathscr{E})$ is a *pair of subsets* of U, endowed with a quadruple of maps; in general, it is not an element of U. The notation $X \in (U\text{-}\mathscr{E})$ will mean that X is an element of U endowed with a structure of species $\mathscr{E}$.

The category $(U\text{-}\mathscr{E})$ is *stable* under many operations:

a) If $X \in (U\text{-}\mathscr{E})$, and if X' is a subset of X that admits an *induced structure*, then $X' \in (U\text{-}\mathscr{E})$. This follows from Prop. 1, n° 1.

b) If $X \in (U\text{-}\mathscr{E})$, and if X'' is a quotient set of X that admits a *quotient structure*, then $X'' \in (U\text{-}\mathscr{E})$ (Cor. of Prop. 1, n° 1).

c) If $X, Y \in (U\text{-}\mathscr{E})$, and if $X \times Y$ admits a *product structure*, then $X \times Y \in (U\text{-}\mathscr{E})$ (Prop. 4, n° 1). More generally, if $(X_i)_{i\in I}$ is a family of elements of $(U\text{-}\mathscr{E})$, *if* $I \in U$, and if $\Pi_{i\in I} X_i$ admits a *product structure*, then $\Pi_{i\in I} X_i \in (U\text{-}\mathscr{E})$ (Cor. of Prop. 6). The analogous assertions hold for *sum structures* (Cor. 2 of Prop. 4).

d) Let I be a preordered set, and let $(X_i, f_{ij})_{i,j\in I}$ be a *projective system* (resp. *inductive system*) of sets endowed with structures of species $\mathscr{E}$ and morphisms; let L be the limit of this system, in the sense of Chap. III. If $X_i \in U$ for every i, if $I \in U$, and if L admits a projective-limit structure (resp. an inductive-limit
191 structure), then $L \in (U\text{-}\mathscr{E})$: indeed, L is a subset of $\Pi_{i\in I} X_i$ (resp. a quotient set of $\Sigma_{i\in I} X_i$), and is therefore an element of U by n° 1.

Here are a few more particular examples:

* 1) Let $X, Y \in (U\text{-}\mathrm{Top})$ be two *locally compact* spaces. Then the set $\mathscr{C}(X, Y)$ of continuous maps from X to Y, endowed with the topology of compact convergence (General Topology X), is an element of $(U\text{-}\mathrm{Top})$. Indeed, $\mathscr{C}(X, Y) \in U$ by Prop. 6 of n° 1.*

* 2) Let $X, Y \in (U\text{-}\mathrm{Ab})$. Then the group $\mathrm{Hom}_{\mathbf{Z}}(X, Y)$ is an element of $(U\text{-}\mathrm{Ab})$ by Prop. 6 of n° 1. If, in addition, $\mathbf{Z} \in U$, the tensor product $X \otimes_{\mathbf{Z}} Y$ is an element of $(U\text{-}\mathrm{Ab})$; indeed, this tensor product is a quotient of $\mathbf{Z}^{X\times Y}$, which is a subset of $\mathbf{Z}^{X\times Y}$, which in turn is an element of U by Props. 4 and 6 of n° 1.*

* 3) Let $X \in (U\text{-}\mathrm{Unif})$ be a uniform space. Then its *completion* $\widehat{X}$ is an element of $(U\text{-}\mathrm{Unif})$. Indeed, $\widehat{X}$ is a set of equivalence classes of Cauchy filters on X; now a filter on X is an element of $\mathscr{P}(\mathscr{P}(X))$, so an equivalence class of filters is an element of $\mathscr{P}(\mathscr{P}(\mathscr{P}(X)))$, whence $\widehat{X}$ is an element of $\mathscr{P}(\mathscr{P}(\mathscr{P}(\mathscr{P}(X))))$, and hence an element of U by (U.III) and (U.I).*

N.B. Moral: Bourbaki will have to take care to "canonicalize" its constructions properly. For example, in constructions in which an element ∞ is adjoined (Alexandroff
192 compactification, projective field), it will be advantageous to take $\infty = \varnothing$ (since $\varnothing$ is an element of every nonempty universe) and to form the disjoint union of $\{\varnothing\}$ and the

given set. Of course, one should also "canonicalize" the two-element set used to construct disjoint unions; thus $\{\emptyset, \{\emptyset\}\}$ seems to me a good candidate, since it belongs to every nonempty universe.

3. Universes and categories

Let U be a universe and C a category. We write $C \in U$ if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$. This notation is justified by the fact that C is the sextuple consisting of $\mathrm{Ob}(C)$, $\mathrm{Fl}(C)$, and the four structure maps; thus, if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$, these four maps are elements of U (n° 1, Corollary 1 of Proposition 4), and hence so is the sextuple C (n° 1, Proposition 3, *cum grano salis*). This is moreover a special case of n° 2, if one considers the species of structure "cat" of a category. With the notation of n° 2, the relation $C \in U$ is also written $C \in (U\text{-cat})$.

One should note that a category such as $(U\text{-Ens})$, $(U\text{-Ord})$, $(U\text{-Top})$, or $(U\text{-Gr})$ is *not*, in general, an element of U.

Let C, D be two categories and U a universe such that $C, D \in U$. If C' is a *subcategory* of C, then $C' \in U$; the *product category* $C \times D$, the sum category of C and D, and the opposite categories C° and D° are also elements of U (cf. n° 2). The *functor category* $E = \mathbf{Hom}(C, D)$ is likewise an element of U: indeed, $\mathrm{Ob}(E)$ is a set of pairs of maps $\mathrm{Ob}(C) \to \mathrm{Ob}(D)$, $F\ell(C) \to F\ell(D)$, and hence $\mathrm{Ob}(E) \in U$ by n° 1; as for $F\ell(E)$, it is a set of natural transformations, that is, of maps $\mathrm{Ob}(C) \to F\ell(D)$, whence $F\ell(E) \in U$ by Proposition 6 of n° 1. 193

PROPOSITION 9. *Let $\mathscr{E}$ be a species of structure with morphisms, and let U, U' be two universes such that $U' \subset U$. Then $(U'\text{-}\mathscr{E})$ is a full subcategory of $(U\text{-}\mathscr{E})$.*

Indeed, if x and y are objects of $(U'\text{-}\mathscr{E})$, they are by definition objects of $(U\text{-}\mathscr{E})$. As for $\mathrm{Hom}(x, y)$, it is the same set in both categories (cf. § 1, n° 2, Example c)).

N.B. One should note that the hypothesis that U and U' are universes is unnecessary, so that Proposition 9 would be more appropriately placed in § 1, n° 4. But the Tribe asked that it be put here.

Remark. It may happen that the axioms of the species of structure $\mathscr{E}$ imply that the *cardinals* of sets endowed with structures of species $\mathscr{E}$ are *bounded* by a fixed cardinal, say c (* for example, the species of structure of finite group, finitely generated group, finitely generated module over a fixed ring A, or finitely generated algebra over A *). Suppose then that there exists an element z of U' such that $c \leq \mathrm{card}(z)$. Then, for every set x endowed with a structure of species $\mathscr{E}$, there exists an element x' of U' equipotent to x, for example a subset of z; endow x' with the structure deduced from that of x by transport of structure; this gives an element of $(U'\text{-}\mathscr{E})$. It follows that *the inclusion functor from $(U'\text{-}\mathscr{E})$ to $(U\text{-}\mathscr{E})$ is then essentially surjective* (§ 4, n° 2, Definition 2) *and is therefore an equivalence of categories* (§ 4, n° 3, Theorem 1).

4. The universe axiom

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The very pleasant stability results of n° 2 are of interest only if they can be applied to something other than the two small universes U_0 and U_1 described in n° 1. We shall therefore add the following axiom to the axioms of Set Theory:

(A.6) *For every set x, there exists a universe U such that $x \in U$.*

This axiom implies that, if $(x_\alpha)_{\alpha \in I}$ is a *family* of sets, there exists a universe U such that $x_\alpha \in U$ for every $\alpha \in I$: indeed, it suffices to apply (A.6) to $x = \bigcup_{\alpha \in I} x_\alpha$ and then apply Proposition 1 of n° 1.

In particular, given a *category* C, there exists a universe U such that $C \in U$ in the sense of n° 3: apply the preceding result to the family $(\mathrm{Ob}(C), F\ell(C))$. This applies to categories of the form $(V\text{-}\mathscr{E})$, where $\mathscr{E}$ is a species of structure with morphisms and V is a set, for example a universe; in general one has $V \neq U$.

For example, if V is a universe, then $\mathrm{Ob}(V\text{-}\mathrm{Ens}) = V$ and $F\ell(V\text{-}\mathrm{Ens}) \subset V$ (n° 1, Proposition 5). The universes U such that $(V\text{-}\mathrm{Ens}) \in U$ are therefore those such that $V \in U$. But the relation $V \in V$ is impossible for a universe: indeed, for every subset A of V, one would have $A \in V$ (n° 1, Proposition 1), whence $\mathrm{card}(\mathscr{P}(V)) \leq \mathrm{card}(V)$, which is impossible.

5. Universes and strongly inaccessible cardinals

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Let U be a universe. By condition (U.I) of Definition 1, every element x of U is a subset of U; hence $\mathrm{card}(x) \leq \mathrm{card}(U)$. Since the cardinals of the subsets of U form a well-ordered set, the cardinal

$$\mathfrak{c}(U) = \sup_{x \in U} \mathrm{card}(x) \tag{1}$$

exists. Note that, for every cardinal $\mathfrak{c} < \mathfrak{c}(U)$, there exists an element x of U such that $\mathrm{card}(x) = \mathfrak{c}$; indeed, by definition there exists $y \in U$ such that $\mathfrak{c} \leq \mathrm{card}(y) \leq \mathfrak{c}(U)$, and one takes for x a suitable subset of y. Conversely, if $x \in U$, then $\mathrm{card}(x) < \mathfrak{c}(U)$; indeed, $\mathscr{P}(x) \in U$ by (U.III), whence $2^{\mathrm{card}(x)} \leq \mathfrak{c}(U)$. Thus the cardinal $\mathfrak{c}(U)$ has the following properties:

1) If $\mathfrak{c}$ is a cardinal $< \mathfrak{c}(U)$, then $2^{\mathfrak{c}} < \mathfrak{c}(U)$; indeed, if x is an element of U of cardinality $\mathfrak{c}$, then $\mathscr{P}(x) \in U$ by (U.III), whence $2^{\mathfrak{c}} < \mathfrak{c}(U)$.
2) If $(\mathfrak{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{c}(U)$ for every $\lambda \in I$ and $\mathrm{card}(I) < \mathfrak{c}(U)$, then the sum cardinal $\Sigma_{\lambda \in I} \mathfrak{c}_\lambda$ is $< \mathfrak{c}(U)$. Indeed, let $x_\lambda \in U$ be such that $\mathrm{card}(x_\lambda) = \mathfrak{c}_\lambda$; after replacing I by an equipotent index set if necessary, one may suppose that $I \in U$; then the sum set of the x_λ is an element of U (Corollary 2 of Proposition 4 of n° 1), which proves our assertion.

We make the following definition:

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DEFINITION 2. *A cardinal $\mathfrak{d}$ is called strongly inaccessible if:*

(FI.1) *If $\mathfrak{c}$ is a cardinal such that $\mathfrak{c} < \mathfrak{d}$, then $2^{\mathfrak{c}} < \mathfrak{d}$.*
(FI.2) *If $(\mathfrak{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{d}$ for every $\lambda \in I$, and if $\mathrm{card}(I) < \mathfrak{d}$, then $\Sigma_{\lambda \in I} \mathfrak{c}_\lambda < \mathfrak{d}$.*

Examples. The cardinal 0 and the countably infinite cardinal are strongly inaccessible. No nonzero finite cardinal is strongly inaccessible. Thus we have just proved that the universe axiom (A.6) implies the statement:

(A′.6) *Every cardinal is strictly bounded above by a strongly inaccessible cardinal.*

Conversely:

THEOREM 1. *Statement (A′.6) implies the universe axiom (A.6).*

Indeed, let A be a set. We must construct a universe U of which A is an element. Define recursively a sequence $(A_n)_{n \geq 0}$ of sets by:

(2) $A_0 = A$, A_{n+1} = the union of the elements of A_n = the set of elements of the elements of A_n.

Put $B = \bigcup_{n=0}^{\infty} A_n$. By (A′.6), let $\mathfrak{c}$ be a strongly inaccessible cardinal such that $\mathrm{card}(B) < \mathfrak{c}$.

There exists a *well-ordered* set I such that $\mathrm{card}(I) = \mathfrak{c}$. After replacing I by its smallest segment of cardinality $\mathfrak{c}$ if necessary, we may suppose that *every segment of I distinct from I has cardinality $< \mathfrak{c}$*. We denote by $\mathscr{E}$ the *least element of I*. It follows 197
from the hypothesis on segments that I has no greatest element (otherwise we would remove it), and hence that every element of I has a successor; for $\alpha \in I$, we denote the *successor* of α by $s(\alpha)$.

Now define, by *transfinite recursion*, a family $(B_\alpha)_{\alpha\in I}$ of sets by:

$$\text{(3)}\qquad \begin{cases} B_{\mathscr{E}} = B \\ B_{s(\alpha)} = B_\alpha \cup \mathscr{P}(B_\alpha) \\ B_\alpha = \bigcup_{\beta<\alpha} B_\beta \text{ if } \alpha \text{ has no predecessor.} \end{cases}$$

Put $U = \bigcup_{\alpha\in I} B_\alpha$. We shall show that U is the desired universe.

First, $A \in U$, since $A = A_0$ is a subset of $B = B_{\mathscr{E}}$, and therefore an element of $\mathscr{P}(B_{\mathscr{E}}) \subset B_{s(\mathscr{E})}$.

Next note that, for every $\alpha \in I$, one has:

$$\text{(4)}\qquad \mathrm{card}(B_\alpha) < \mathfrak{c}.$$

Indeed, proceed by transfinite recursion on α. This is true for $\alpha = \mathscr{E}$ by hypothesis. From (4) one obtains $\mathrm{card}(B_{s(\alpha)}) \leq \mathrm{card}(B_\alpha) + 2^{\mathrm{card}(B_\alpha)} < \mathfrak{c} + \mathfrak{c}$ (by (FI.1)) $= \mathfrak{c}$ (since $\mathfrak{c}$ is infinite). Finally, if α has no predecessor, the set I' of $\beta < \alpha$ has cardinality $< \mathfrak{c}$ by construction; hence (if $\mathrm{card}(B_\beta) < \mathfrak{c}$ for every $\beta < \alpha$), $\mathrm{card}(B_\alpha) = \mathrm{card}(\bigcup_{\beta\in I'} B_\beta) \leq \Sigma_{\beta\in I'} \mathrm{card}(B_\beta) < \mathfrak{c}$ (by (FI.2)). This proves (4).

Let us now show that 198

$$\text{(5)}\qquad \mathrm{card}(x) < \mathfrak{c} \text{ for every } x \in U.$$

Indeed, it suffices to show that, for every $\alpha \in I$, one has "$\mathrm{card}(x) < \mathfrak{c}$ for every $x \in B_\alpha$". Proceed again by transfinite recursion on α. This is true for $\alpha = \mathscr{E}$, since, if $x \in B_{\mathscr{E}} = B$, there exists $n \geq 0$ such that $x \in A_n$; hence $x \subset A_{n+1}$ by (2), $x \subset B$, and $\mathrm{card}(x) \leq \mathrm{card}(B) < \mathfrak{c}$. The case where α has no predecessor is evident. Finally, if $x \in B_\alpha$ and $\alpha = s(\beta)$, either $x \in B_\beta$, in which case the assertion "$\mathrm{card}(x) < \mathfrak{c}$" is true by recursion, or $x \in \mathscr{P}(B_\beta)$, whence $x \subset B_\beta$ and the assertion "$\mathrm{card}(x) < \mathfrak{c}$" is true by (4).

We are now in a position to prove that U is indeed a *universe*:

(U.I) Let $x \in U$ and $y \in x$. We must show that $y \in U$. In other words, we must show that, for every $\alpha \in I$, one has:

$$\text{“}x \in B_\alpha \text{ and } y \in x \Longrightarrow y \in B_\alpha\text{”}.$$

Proceed again by transfinite recursion on α. This is true for $\alpha = \mathscr{E}$, since $x \in B_{\mathscr{E}} = B$ implies $x \in A_n$ for some n; hence, if $y \in x$, then $y \in A_{n+1}$, whence $y \in B = B_{\mathscr{E}}$. Passage to an element α with no predecessor is evident. Finally, pass from α to $s(\alpha)$: if $x \in B_{s(\alpha)}$ and $y \in x$, either $x \in B_\alpha$, whence $y \in B_\alpha \subset B_{s(\alpha)}$ by recursion, or $x \in \mathscr{P}(B_\alpha)$, whence again $y \in B_\alpha \subset B_{s(\alpha)}$.

(U.II) Let $x, y \in U$. Since the family $(B_\alpha)_{\alpha\in I}$ is increasing by (3), there exists $\alpha \in I$ such that $x, y \in B_\alpha$. Then $\{x, y\} \in \mathscr{P}(B_\alpha) \subset B_{s(\alpha)} \subset U$.

(U.III) Let $x \in U$. Then there exists $\alpha \in I$ such that $x \in B_\alpha$. It therefore suffices to 199
show that, for every $\alpha \in I$, one has:

$$\text{“}x \in B_\alpha \Longrightarrow \mathscr{P}(x) \in B_{s(s(\alpha))}\text{”}.$$

Proceed again by transfinite recursion on α. If $\alpha = \mathscr{E}$, then $x \in A_n$ for some n, whence $y \subset A_{n+1} \subset B = B_{\mathscr{E}}$ for every subset y of x; thus $y \in \mathscr{P}(B_{\mathscr{E}}) \subset B_{s(\mathscr{E})}$ for every $y \in \mathscr{P}(x)$, whence $\mathscr{P}(x) \subset B_{s(\mathscr{E})}$ and $\mathscr{P}(x) \in \mathscr{P}(B_{s(\mathscr{E})}) \subset B_{s(s(\mathscr{E}))}$, so our assertion is true for $\alpha = \mathscr{E}$. Passage to an element α with no predecessor is immediate, since $\beta < \alpha$ then implies $s(s(\beta)) < \alpha$. Finally, pass from α to $s(\alpha)$; let $x \in B_{s(\alpha)}$; if $x \in B_\alpha$, then $\mathscr{P}(x) \in B_{s(s(\alpha))} \subset B_{s(s(s(\alpha)))}$ by recursion; if $x \in \mathscr{P}(B_\alpha)$, then $x \subset B_\alpha$, whence $\mathscr{P}(x) \subset \mathscr{P}(B_\alpha)$ and $\mathscr{P}(x) \in \mathscr{P}(\mathscr{P}(B_\alpha)) \in B_{s(s(s(\alpha)))}$.

(U.IV) Let $(x_\lambda)_{\lambda \in K}$ be a family of elements of U such that $K \in U$. We must show that the union $x = \bigcup_{\lambda \in K} x_\lambda$ is an element of U. For every $\lambda \in K$, choose an $\alpha(\lambda) \in I$ such that $x_\lambda \in B_{\alpha(\lambda)}$. Let us show that the set $\alpha(K)$ of the $\alpha(\lambda)$ is *bounded above* in I: indeed, if it were not, one would have

$$I = \bigcup_{\lambda \in K} [\mathscr{E}, \alpha(\lambda)],$$

which would contradict (FI.2), since $\mathrm{card}(K) < \mathscr{c}$ and $\mathrm{card}([\mathscr{E}, \alpha(\lambda)]) < \mathscr{c}$ for every $\lambda \in K$. Let then β be an upper bound of $\alpha(K)$; one has $x_\lambda \in B_\beta$ for every $\lambda \in K$, since the family $(B_\alpha)_{\alpha \in I}$ is increasing. Every element of x_λ then belongs to B_β by what was proved in (U.I), whence $x = \bigcup_{\lambda \in K} x_\lambda \subset B_\beta$. By (3), it follows that $x \in B_{s(\beta)}$, whence $x \in U$. C.Q.F.D.

200 DEFINITION 3. *Let x, y be two sets and let n be an integer ≥ 0. We say that y is a component of order n of x if there exists a sequence $(x_j)_{j=0,\dots,n}$ such that $x_0 = x$, $x_n = y$, and $x_{j+1} \in x_j$ for $j = 0, \dots, n-1$.*

Thus x is the only component of order 0 of x. The components of order 1 (resp. 2) of x are the elements of x (resp. the elements of the elements of x). We say that y is a *component* of x if there exists $n \geq 0$ such that y is a component of order n of x. The relation "y is a component of x" is a *preorder relation*. By the selection–union scheme (Chapter II, ...), the relation "y is a component of x" is collectivizing with respect to y, so that the components of x form a *set*.

DEFINITION 4. *Let $\mathscr{c}$ be a cardinal. A set x is said to be of type $\mathscr{c}$ (resp. of strict type $\mathscr{c}$, of finite type) if all the components of x have cardinalities $\leq \mathscr{c}$ (resp. $< \mathscr{c}$, finite cardinalities).*

EXAMPLES. The elements of the universe U_1 (n° 1, Example 2) are all of finite type. If $\mathscr{c}$ is a cardinal and x is of type $\mathscr{c}$ (resp. of strict type $\mathscr{c}$, of finite type), then every component of x and every subset of x is of type $\mathscr{c}$ (resp. of strict type $\mathscr{c}$, of finite type); likewise $\mathscr{P}(x)$ is of type $2^{\mathscr{c}}$ (resp. of strict type $2^{\mathscr{c}}$, of finite type). If x is of type $\mathscr{c}$ and if $\mathscr{c} \leq \mathscr{c}'$, then x is of type $\mathscr{c}'$.

201 LEMMA. *Let $\mathscr{c}$ be an uncountable strongly inaccessible cardinal, and let X be a set of strict type $\mathscr{c}$. Then there exists a cardinal $\mathscr{d} < \mathscr{c}$ such that X is of type $\mathscr{d}$.*

Indeed, for every integer n, let $\mathscr{d}_n$ denote the cardinality of the set X_n of components of order n of X. We have $\mathscr{d}_0 = 1$, $\mathscr{d}_1 = \mathrm{card}(X) < \mathscr{c}$, and, since $X_n = \bigcup_{y \in X_{n-1}} y$, induction on n using (FI.2) gives $\mathscr{d}_n < \mathscr{c}$ for every n. The $\mathscr{d}_n$ are therefore bounded above by the cardinal $\mathscr{d} = \Sigma_{j \geq 0} \mathscr{d}_j$, which is $< \mathscr{c}$ by (FI.2) and the hypothesis that $\mathscr{c}$ is uncountable. If Y is a component of order n of X, then $\mathrm{card}(Y) \leq \mathrm{card}(X_{n+1}) \leq \mathscr{d}$. C.Q.F.D.

PROPOSITION 10. *Let U be a universe and let $\mathscr{c}$ be a strongly inaccessible cardinal. Then the set U' of those $x \in U$ that are of strict type $\mathscr{c}$ is a universe.*

Indeed, if $x, y \in U'$ and if $z \in x$, then plainly $z \in U'$ and $\{x, y\} \in U'$. If $x \in U'$, then $\mathrm{card}(x) < \mathfrak{c}$, whence $\mathrm{card}(\mathscr{P}(x)) < \mathfrak{c}$ by (FI.1); since a component of $\mathscr{P}(x)$ is either $\mathscr{P}(x)$ itself, a subset of x, or a component of x, we have $\mathscr{P}(x) \in U'$. Finally, if $(x_\alpha)_{\alpha\in I}$ is a family of elements of U' such that $I \in U'$, then $\mathrm{card}(\bigcup_\alpha x_\alpha) < \mathfrak{c}$ by (FI.2); since every component of order > 0 of $\bigcup_\alpha x_\alpha$ is a component of some x_α, we indeed have $\bigcup_\alpha x_\alpha \in U'$. C.Q.F.D.

Remark on the cardinal $\mathfrak{c}(U)$. Let U be a universe and let $\mathfrak{c}(U)$ be the cardinal defined by (1), i.e.

$$\mathfrak{c}(U) = \sup_{x\in U} \mathrm{card}(x).$$

We plainly have $\mathfrak{c}(U) \leq \mathrm{card}(U)$, since, as we recall, $x \in U$ implies $x \subset U$. But 202
the equality $\mathfrak{c}(U) = \mathrm{card}(U)$ does not always hold. For example, let $(x_\alpha)_{\alpha\in I}$ be an *uncountable* family of symbols subject to the axioms:

$$\text{«}x_\alpha = \{x_\alpha\} \text{ for every } \alpha \in I\text{»} \quad , \quad \text{«}x_\alpha \neq x_\beta \text{ if } \alpha \neq \beta\text{»}$$

(in other words, x_α is a one-element set, namely itself). As in Example 2 of n° 1, form the "significant words" constructed from the symbols $\emptyset$, x_α, $\{$, $\}$ and so forth. Each denotes a *finite* set. Let U be the set of these words. One readily verifies that the conditions of Definition 1 are satisfied, so that U is a *universe*. Since the significant words are finite sequences of elements of a set of cardinality $\mathrm{card}(I) + 4 = \mathrm{card}(I)$, we have $\mathrm{card}(U) = \mathrm{card}(I)$ (Chapter III; in fact, $\mathrm{card}(U) \geq \mathrm{card}(I)$ suffices, and this is obvious). On the other hand, $\mathfrak{c}(U)$ is the countably infinite cardinal, and hence $\mathfrak{c}(U) < \mathrm{card}(U)$.

The strict inequality $\mathfrak{c}(U) < \mathrm{card}(U)$ is due to our having introduced sets x_α such that $x_\alpha \in x_\alpha$. If one rules out horrors of this kind, one obtains $\mathrm{card}(U) = \mathfrak{c}(U)$ for every universe U, as well as other very pretty results "which cannot possibly be useful." This is what we shall do in the next section.

6. Artinian sets and universes

DEFINITION 5. *A set x is said to be Artinian if there is no infinite sequence $(x_n)_{n\geq 0}$ such that $x_0 = x$ and $x_{n+1} \in x_n$ for every $n \geq 0$.*

Examples. The sets $\emptyset$, $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$, and, more generally, the elements of the universe U_1 of n° 1 are Artinian.

In pictorial terms, if x is Artinian, and if one takes an element x_1 of x, then an 203
element x_2 of x_1, etc., the process must stop, and one arrives at a component x_n of x that is *empty*. In other words, Artinian sets are "built from $\emptyset$"; this will be made more precise later (cf. (*) in the proof of Theorem 2).

Artinian sets plainly have the following properties:

(AR.I) If x is Artinian, every subset of x and every component of x is Artinian. For x to be Artinian, it is necessary and sufficient that every element of x be Artinian.
(AR.II) If x and y are Artinian, then $\{x, y\}$ is Artinian.
(AR.III) If x is Artinian, $\mathscr{P}(x)$ is Artinian.
(AR.IV) Every union of Artinian sets is an Artinian set.

These properties immediately yield the following result:

PROPOSITION 11. *If U is a universe, the set of Artinian elements of U is a universe, necessarily Artinian.*

COROLLARY. *If x is an Artinian set, there exists an Artinian universe U such that $x \in U$.*

Indeed, by axiom (A.6), x is an element of a universe V; take for U the set of Artinian elements of V.

The following proposition is even less useful than the rest of n° 6:

204 PROPOSITION 12. *Let A be an Artinian set. Then:*

a) *For every $x \in A$, one has $x \notin x$;*
b) *If $x, y \in A$, one cannot have both $x \in y$ and $y \in x$;*
c) *For every $x \in A$, the relation "y is a component of z" on the components y, z of x is an order relation;*
d) *For every nonempty element x of A, there exists $y \in x$ such that $x \cap y = \emptyset$.*

Indeed, the negation of a) (resp. of b)) entails the existence of an infinite sequence $(x_n)_{n\geq 0}$ contradicting Definition 5, namely $(x, x, x, \dots)$ (resp. $(x, y, x, y, x, y, \dots)$). The negation of c) means that there exist $x \in A$, distinct components y, z of x, and membership chains:

$$y \in y_1 \in \cdots \in y_q \in z \quad , \quad z \in z_1 \in \cdots \in z_r \in y;$$

whence, as in b), an infinite sequence $(x_n)_{n\geq 0}$ contradicting Definition 5. Finally, if d) is false, there exists a nonempty element x of A such that $x \cap y \neq \emptyset$ for every $y \in x$; set $x_\circ = x$, and take for x_1 an element of x; since $x \cap x_1 \neq \emptyset$, take for x_2 an element of $x \cap x_1$, and so on; more formally, define by induction an infinite sequence $(x_n)_{n\geq 1}$ of elements of x by means of $x_1 \in x$, $x_{n+1} \in x_n \cap x$ for $n \geq 1$; then the sequence $(x_n)_{n\geq 0}$ contradicts Definition 5.

REMARK. Let B be a set. Then B is Artinian if and only if every set A whose elements are sets of components of B satisfies condition d) of Proposition 12. Indeed, necessity
205 follows from (AR.I) and Proposition 12. Conversely, if B is not Artinian, there exists an infinite sequence $(x_n)_{n\geq 0}$ with $x_\circ = B$ and $x_{n+1} \in x_n$ for every $n \geq 0$; then take for A the set whose sole element is the set X of the x_n; thus A contains a nonempty element X such that, for every element y of X, one has $y \cap X \neq \emptyset$ (indeed, y is of the form x_n, and one has $x_{n+1} \in y \cap X$).

THEOREM 2. *Let $\mathfrak{c}$ be an infinite cardinal. Then:*

a) *The relation "x is an Artinian set of type $\mathfrak{c}$ (resp. of strict type $\mathfrak{c}$)" is collectivizing with respect to x; the set $A_\mathfrak{c}$ of Artinian sets of type $\mathfrak{c}$ has cardinality $2^\mathfrak{c}$.*
b) *If $\mathfrak{c}$ is strongly inaccessible, the set $U_\mathfrak{c}$ of Artinian sets of strict type $\mathfrak{c}$ is a universe of cardinality $\mathfrak{c}$; the cardinal $\mathfrak{c}(U_\mathfrak{c}) = \sup_{x\in U_\mathfrak{c}} \mathrm{card}(x)$ is $\mathfrak{c}$.*
c) *If a universe U contains an element of cardinality $\mathfrak{c}$, every Artinian set of type $\mathfrak{c}$ belongs to U (in other words, $A_\mathfrak{c} \subset U$).*
c′) *If a universe U is nonempty, every Artinian set of finite type is an element of U.*

Before proving Theorem 2, let us deduce from it a few illuminating corollaries:

COROLLARY 1. *If a universe U is Artinian, then* $\mathrm{card}(U)$ *is strongly inaccessible, and U is the set of Artinian sets of strict type* $\mathrm{card}(U)$.

206 Indeed, set $\mathfrak{c}(U) = \sup_{x\in U} \mathrm{card}(x)$. This is a strongly inaccessible cardinal (beginning of n° 5). First suppose that it is uncountable; then, for every infinite cardinal $\mathfrak{c} < \mathfrak{c}(U)$, every Artinian set of type $\mathfrak{c}$ is an element of U by c); hence every Artinian set of strict type $\mathfrak{c}(U)$ is an element of U by the lemma of n° 5. This last assertion remains valid if $\mathfrak{c}(U)$ is countable, by c′). Conversely, by (U.I), every set in U is of strict type $\mathfrak{c}(U)$. Thus U is the universe $U_{\mathfrak{c}(U)}$ of b), whence $\mathfrak{c}(U) = \mathrm{card}(U)$ by b).

It follows from Corollary 1 that an Artinian universe is uniquely *determined* by its cardinality (which is moreover strongly inaccessible). Thus there is a "one-to-one correspondence" between Artinian universes and strongly inaccessible cardinals. In particular:

COROLLARY 2. *The inclusion relation $U \subset U'$ on Artinian universes is a well-ordering relation.*

Indeed, the relation $\mathfrak{c} \leq \mathfrak{c}'$ on cardinals is a well-ordering relation (Chapter III), and, with the notation of b) of Theorem 2, the relations $\mathfrak{c} \leq \mathfrak{c}'$ and $U_{\mathfrak{c}} \subset U_{\mathfrak{c}'}$ are equivalent.

Observe that b) of Theorem 2 gives a second proof of Theorem 1 (n° 5).

We now turn to the proof of Theorem 2. Given a set A, we shall call a *chain* of A any finite sequence $(x_j)_{j=0,\ldots,n}$ such that $x_0 = A$ and $x_{j+1} \in x_j$ for $j = 0, \ldots, n-1$. The chains of A form a *set*, by the selection–union scheme; denote it by $G(A)$. Given a 207
chain $X = (x_j)_{j=0,\ldots,n}$, the chains of the form $(x_i)_{i=0,\ldots,q}$ with $q \leq n$ will be called less than X; this endows $G(A)$ with the structure of an *ordered set*. For A to be Artinian, it is necessary and sufficient that $G(A)$ be a "*Noetherian*" ordered set (i.e. one satisfying the equivalent conditions of Chapter III, § 6, n° 5).

We shall prove that:

(*) *If A is Artinian, it is uniquely determined by the isomorphism class of the ordered set $G(A)$.*

Indeed, given an ordered set G and an element $g \in G$, let $S(g)$ denote the set of all $g' \in G$ such that $g < g'$ and such that $g \leq h \leq g'$ implies $h = g$ or $h = g'$ (in other words, the set of "immediate successors" of g). Consider the map θ that assigns to every chain $X = (x_0, \ldots, x_n)$ in $G(A)$ the set x_n; then, for every $X \in G(A)$, we have:

$$\theta(X) = \{\theta(X') | X' \in S(X)\}.$$

Since A is the image under θ of the least element $X_0 = (A)$ of $G(A)$, it will suffice to show that θ is uniquely determined by the isomorphism class of the ordered set $G(A)$. This follows from the following lemma:

LEMMA 1. *Let G be a Noetherian ordered set and let φ be a function with domain G* 208
such that, for every $g \in G$, one has:

$$\varphi(g) = \{\varphi(g') | g' \in S(g)\}. \tag{1}$$

Then φ is uniquely determined. Moreover, if U is a universe containing an element equipotent to G, then φ takes its values in U.

The hypothesis implies that $\varphi(g) = \varnothing$ if g is a maximal element of G.

Indeed, let φ and φ' be two functions satisfying (1). If $\varphi \neq \varphi'$, the set of all $g \in G$ such that $\varphi(g) \neq \varphi'(g)$ is nonempty and hence, since G is Noetherian, has a *maximal* element h. We then have $\varphi(g) = \varphi'(g)$ for every $g > h$, in particular for every immediate successor $g \in S(h)$; hence $\varphi(h) = \varphi'(h)$ by (1), which is a contradiction. Thus indeed $\varphi = \varphi'$. Now let U be a universe containing an element x equipotent to G; let us show that φ takes its values in U. Otherwise, let h be a maximal element among those $g \in G$ such that $\varphi(g) \notin U$. Then $\varphi(g') \in U$ for every $g' \in S(h)$, so that

$$\varphi(h) = \{\varphi(g') | g' \in S(h)\}$$

is a subset of U. Since its cardinality does not exceed $\mathrm{card}(G)$, and hence does not exceed the cardinality of an element of U, we have $\varphi(h) \in U$ (n° 1, Proposition 7). This contradiction shows that φ takes its values in U.

With this established, let us prove a) of Theorem 2. We may confine ourselves to the 209

assertion without the resp. clause, since the other follows at once. Let $\mathfrak{c}$ be an *infinite* cardinal. If A is a set of type $\mathfrak{c}$, then:

$$\text{card}(G(A)) \leq \mathfrak{c} \ . \tag{2}$$

Indeed, if A_n denotes the set of components of order n of A, then $\text{card}(A_1) \leq \mathfrak{c}$ and $\text{card}(A_{n+1}) \leq \mathfrak{c} \cdot \text{card}(A_n)$, whence $\text{card}(A_n) \leq \mathfrak{c}^n$ by induction. But $\mathfrak{c}^n = \mathfrak{c}$, since $\mathfrak{c}$ is infinite (Chapter III); hence $\text{card}(\bigcup_{n=0}^{\infty} A_n) \leq \text{card}(\mathbf{N}) \cdot \mathfrak{c} = \mathfrak{c}$ (Chapter III). Now $G(A)$ is a set of *finite* sequences of elements of $\bigcup_n A_n$, so inequality (2) indeed holds (Chapter III). Now, if E is a set of cardinality $\mathfrak{c}$, specifying an order structure on a subset E' of E is equivalent to specifying the subset of $E \times E$ consisting of the (x, y) such that $x \in E'$, $y \in E'$, and $x \leq y$. Consequently, by (2), the isomorphism classes of the ordered sets $G(A)$ (where A is of type $\mathfrak{c}$) form a set $\mathfrak{F}_{\mathfrak{c}}$, and $\text{card}(\mathfrak{F}_{\mathfrak{c}}) \leq \text{card}(\mathscr{P}(E \times E)) = 2^{\mathfrak{c}}$. Let $\mathfrak{F}'_{\mathfrak{c}}$ be the subset of $\mathfrak{F}_{\mathfrak{c}}$ consisting of the classes of Noetherian ordered sets having a least element. If, to every $G \in \mathfrak{F}'_{\mathfrak{c}}$, one assigns the value of the function φ of Lemma 1 at the least element of G, one obtains a map θ on $\mathfrak{F}'_{\mathfrak{c}}$ whose image contains *all* Artinian sets of type $\mathfrak{c}$. The latter therefore do indeed form a set $A_{\mathfrak{c}}$, and we have $\text{card}(A_{\mathfrak{c}}) \leq \text{card}(\mathfrak{F}'_{\mathfrak{c}}) \leq \text{card}(\mathfrak{F}_{\mathfrak{c}}) \leq 2^{\mathfrak{c}}$.

210 It remains to show that $\text{card}(A_{\mathfrak{c}}) = 2^{\mathfrak{c}}$. For this it suffices to show that there exists an Artinian set B of type $\mathfrak{c}$ and cardinality $\mathfrak{c}$, since the subsets of B will then be elements of $A_{\mathfrak{c}}$. This existence follows from the following lemma:

LEMMA 2. *For every cardinal $\mathfrak{c}$, there exists an Artinian set B of type $\mathfrak{c}$ and cardinality $\mathfrak{c}$.*

Proceed by transfinite induction on $\mathfrak{c}$. For $\mathfrak{c} = 0$ there is no choice, and one takes $B = \emptyset$. If $\mathfrak{c}$ has a predecessor $\mathfrak{c}'$, let B' be an Artinian set of type $\mathfrak{c}'$ and cardinality $\mathfrak{c}'$; one then has $\mathfrak{c} \leq 2^{\mathfrak{c}'}$, so there exists a subset B of $\mathscr{P}(B')$ of cardinality $\mathfrak{c}$; the elements of B are subsets of B' and therefore have cardinalities $\leq \mathfrak{c}' \leq \mathfrak{c}$; the higher-order components of B are components of B', and hence also have cardinalities $\leq \mathfrak{c}' \leq \mathfrak{c}$. Finally, if $\mathfrak{c}$ has no predecessor, choose, for every cardinal $\mathfrak{c}_\lambda < \mathfrak{c}$, an Artinian set B_λ of type $\mathfrak{c}_\lambda$ and cardinality $\mathfrak{c}_\lambda$; then $B = \bigcup_\lambda B_\lambda$ has the required property. This proves Lemma 2 and completes the proof of part a).

We turn to b). Let $\mathfrak{c}$ be a strongly inaccessible cardinal. By a), we already know that the Artinian sets of strict type $\mathfrak{c}$ form a set $U_{\mathfrak{c}}$. The fact that $U_{\mathfrak{c}}$ is a universe follows immediately from properties (AR.I) to (AR.IV) of Artinian sets (beginning of n°),

211 and from cardinal estimates analogous to those of Proposition 10 of n° 5. The relation $\sup_{x \in U_{\mathfrak{c}}} \text{card}(x) = \mathfrak{c}$ followsfrom Lemma 2, applied to cardinals $< \mathfrak{c}$. Finally, to show that $\text{card}(U_{\mathfrak{c}}) = \mathfrak{c}$, first suppose that $\mathfrak{c}$ is *uncountable*; bythe lemma of n° 5, $U_{\mathfrak{c}}$ is the union $\bigcup_{\mathfrak{d} < \mathfrak{c}} A_{\mathfrak{d}}$, where $A_{\mathfrak{d}}$ denotes the set of Artinian sets of type $\mathfrak{d}$; now $\text{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}} < \mathfrak{c}$ (by a)); on the other hand, the set of cardinals $\mathfrak{d} < \mathfrak{c}$ has cardinality $\leq \mathfrak{c}$ (Chapter III); whence $\text{card}(U_{\mathfrak{c}}) \leq \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$, and also $\text{card}(U_{\mathfrak{c}}) \geq \mathfrak{c}$, since $\text{card}(U_{\mathfrak{c}}) \geq \text{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}}$ for every $\mathfrak{d} < \mathfrak{c}$.

The case $\mathfrak{c} = 0$ being trivial, there remains the case where $\mathfrak{c}$ is the *countably infinite cardinal.* In this case $U_{\mathfrak{c}}$ is the set of Artinian sets of finite type (i.e. finite together with all their components), and one uses a pleasant result of a combinatorial nature.

LEMMA 3. (D. K"onig?). *Consider two infinite sequences $(E_n)_{n \geq 1}$, $(f_n)_{n \geq 1}$ of finite sets E_n and maps $f_n : E_{n+1} \to E_n$. If there exists no infinite sequence $(x_n)_{n \geq 1}$ such that $x_n \in E_n$ and $f_n(x_{n+1}) = x_n$ for every n, then E_n is empty for all sufficiently large n.*

In other words, if all sequences (x_n) such that $f_n(x_{n+1}) = x_n$ are *finite*, their lengths are *bounded*. This can be expressed in terms of projective limits: a projective limit of nonempty finite sets is nonempty (cf. General Topology, Chapter I, 2nd ed., § 9, n° 6, Proposition 8, 2°)).

Indeed, argue by contradiction. If there exist nonempty E_n for arbitrarily large n, 212
no E_n is empty (since $E_n = \varnothing$ implies $E_{n+1} = \varnothing$ in view of the existence of $f_n : E_{n+1} \to E_n$). Call a finite sequence $(x_j)_{1\leq j\leq n}$ *coherent* if $f_j(x_{j+1}) = x_j$ for $j = 1, \dots, n-1$. We prove by induction on n the existence of a coherent sequence $(a_1, \dots, a_n)$ $(a_i \in E_i)$ that, for every $q \geq n$, can be extended to a coherent sequence $(a_1, \dots, a_n, x_{n+1}, \dots, x_q)$ of length q. This is evident for $n = 0$, since no E_q is empty. Pass from n to $n+1$. If, for every $x \in f_n^{-1}(\{a_n\}) \subset E_{n+1}$, all coherent sequences extending $(a_1, \dots, a_n, x)$ had lengths bounded by an integer $q(x)$, then all coherent sequences extending $(a_1, \dots, a_n)$ would have bounded lengths (by $\sup_x q(x)$), since E_{n+1} is finite; hence there exists $a_{n+1} \in f_n^{-1}(\{a_n\})$ such that the coherent sequence $(a_1, \dots, a_n, a_{n+1})$ admits extensions of arbitrary length. We then obtain an infinite sequence $(a_n)_{n\geq 1}$ contradicting the hypothesis.

It follows from Lemma 3 that if A is an Artinian set of finite type, then the ordered set $G(A)$ of its chains is *finite*: indeed, take for E_n the set of chains $(x_n \in x_{n-1} \in \cdots \in x_1 \in A)$ with $n+1$ terms (which is finite because the components of A of order $\leq n+1$ are finite in number and are all finite), and for f_n the map $(x_{n+1} \in x_n \in x_{n-1} \in \dots) \mapsto (x_n \in x_{n-1} \in \dots)$. Now the set of isomorphism classes of finite ordered sets is *countable*: 213
indeed, giving an order structure on a finite subset of $\mathbf{N}$ is equivalent to giving its graph, which is a finite subset of $\mathbf{N}\times\mathbf{N}$; on the other hand, the set of finite subsets of a countable set is countable (Chapter III). It follows from (*) and Lemma 1 that the set $\mathfrak{a}$ of Artinian sets of finite type is countable. It is infinite by Lemma 2 (or, more simply, by using the sequence $(z_n)_{n\geq 0}$ defined by $z_0 = \varnothing$, $z_{n+1} = \{z_n\}$). This completes the proof of b).

Remark. We have just proved that if A is an Artinian set of finite type, it has only finitely many components. There therefore exists an integer n such that A is of type n.

We turn to the proof of c). Let $\mathfrak{c}$ be an infinite cardinal, let U be a universe admitting an element of cardinality $\mathfrak{c}$, and let A be an Artinian set of type $\mathfrak{c}$. Then the ordered set $G(A)$ has cardinality $\leq \mathfrak{c}$ (formula (2) above). The second assertion of Lemma 1, applied to $G(A)$, shows that the map $(x_0, \dots, x_n) \rightsquigarrow x_n$ from $G(A)$ takes its values in U. In other words, all the components of A are elements of U. This proves c). The proof of c′) is analogous: if A is an Artinian set of finite type, we have just seen that $G(A)$ is finite; since U is nonempty, it contains an element equipotent to $G(A)$ by the corollary to Proposition 7 (n° 1). C.Q.F.D.

7. Vague metamathematical remarks

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a) The axiom "*every set is Artinian*" is *harmless*. Indeed, if one has a model M of Set Theory, the set M' of the Artinian elements of M is also a model (cf. Proposition 11).

b) The universe axiom (A.6) (and the equivalent axiom (A′.6) on strongly inaccessible cardinals) is *independent* of the rest of Set Theory. Indeed, let $\mathfrak{c}$ be the first uncountable strongly inaccessible cardinal. The universe $U_{\mathfrak{c}}$ of Artinian sets of strict type $\mathfrak{c}$ (Theorem 2, b)) is a model of Set Theory without (A.6): the elements of $U_{\mathfrak{c}}$ are called "sets," the "membership relation" is the restriction to $U_{\mathfrak{c}}$ of the usual one, etc. The "universes" of the model are therefore the ordinary universes that are elements of $U_{\mathfrak{c}}$. But we have seen that the only universes

which are elements of U_c are the two pequeños $U_0 = \varnothing$ and U_1. Thus U_1 is a "set" which is not an element of any "universe." We therefore have a model of Set Theory in which (A.6) is false.

c) Bourbaki was too cautious in merely "presuming" that the *axiom of infinity* (A.5) is *independent* of the preceding axioms and schemes. It is indeed independent, since the countable universe U_1 of Artinian sets of finite type is a model in which (A.5) is false and the preceding axioms and schemes are true.

d) It would be very interesting to prove that the universe axiom (A.6) is harmless. That looks difficult—and in fact it is unprovable, says Paul Cohen.

215 The adjective "vague" in the title means that, when constructing models, we did not take the trouble to fiddle with the symbol τ so that it could not take us outside the model. The trick, if one is considering a model M, is to replace the ordinary $\tau_x(\mathscr{R}(x))$ by:

$$\tau_x(\mathscr{R}(x) \text{ and } x \in M).$$

One still has to check that this really turns the ordinary quantifiers into the quantifiers formerly called "typical":

$$(\forall x \in M) \quad , \quad (\exists x \in M).$$

Exercises. 1) Let n be an integer ≥ 1. Show that the set of Artinian sets of type n is infinite (the sets $z_0 = \varnothing$, $z_1 = \{\varnothing\}$, $z_{q+1} = \{z_q\}$ are of type 1).

2) Define the *height* of a set A to be the supremum (finite or infinite) of the integers n for which there exists a sequence $x_n \in x_{n-1} \in \cdots \in x_0 = A$. Show that the sets of height $\leq n$ are finite and form a finite set, whose cardinality p_n can be calculated from $p_0 = 1$, $p_{n+1} = 2^{p_n}$ (proceed by induction on n, noting that the elements of a set A of height $\leq n$ are sets of height $\leq n-1$).

3) Let G be a Noetherian ordered set having a least element g_0, and suppose that, for every $h \in G$, the set of $g \in G$ such that $g \leq h$ is finite and totally ordered.

216 a) Show that every element $h \neq g_0$ of G has a unique predecessor.

b) Let φ be the map on G defined in Lemma 1 (i.e. $\varphi(g)$ is the set of all $\varphi(g')$ as g' ranges over the successors of g). Put $A = \varphi(g_0)$. Show that A is an Artinian set. For $g \in G$, let $g_0 < g_1 < \cdots < g_n = g$ be the sequence of elements $\leq g$; put $f(g) = (\varphi(g_0), \varphi(g_1), \ldots, \varphi(g_n))$; show that f is an increasing surjection from G onto the ordered set $G(A)$ of the text. Show that, if f is *injective*, it is an *isomorphism* from G onto $G(A)$.

c) For $g \in G$, let M_g be the set of upper bounds of g. Suppose that, for every pair of distinct elements g, g' having the same predecessor p, the ordered sets M_g and $M_{g'}$ are not isomorphic. Show that the map f of b) is then an *isomorphism* from G onto $G(A)$. [If $f(g) = f(g')$ with $g \neq g'$, and if $g_0 < g_1 < \cdots < g_n = g$ and $g'_0 < g'_1 < \cdots < g'_{n'} = g'$ are the sequences of elements $\leq g$ and $\leq g'$, respectively, show that $n = n'$, and that one may assume that g and g' have the same predecessor p; then consider the set of $p \in G$ for which there exist two distinct successors g, g' of p such that $f(g) = f(g')$, a maximal element q of this set, and two distinct successors h, h' of q such that $f(h) = f(h')$; note that the restrictions of f to M_h and $M_{h'}$ are injective and hence (by b)) are isomorphisms from M_h onto $G(\varphi(h))$ and from $M_{h'}$ onto $G(\varphi(h'))$; deduce from the hypothesis that M_h and $M_{h'}$ are not isomorphic that $\varphi(h) \neq \varphi(h')$, which contradicts $f(h) = f(h')$].

217 N.B. Exercise 3) gives very precise information on how Artinian sets are "made."

We already knew that such a set A is determined by the isomorphism class of the ordered set $G(A)$ (Lemma 1). We now know how to *characterize* the ordered sets G isomorphic to some $G(A)$:

- α) G is Noetherian and has a least element;
- β) For every $g \in G$, the set of $h \in G$ such that $h \leq g$ is totally ordered and finite (whence the existence and uniqueness of the predecessor of every nonleast g);
- γ) If g, g' are distinct elements having the same predecessor, the set M_g of upper bounds of g and the set $M_{g'}$ of upper bounds of g' are not isomorphic.

4) Let $(A_n)_{n\geq 0}$ be the sequence of finite *ordinals* defined by $A_0 = \varnothing$, $A_{n+1} = A_n \cup \{A_n\}$. Show that the ordered set $G(A_n)$ has 2^n elements, and that the number of its elements of height q (in the sense of Exercise 2) is $\binom{n}{q}$.

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EXPOSÉ II

Topologies and sheaves

J.-L. Verdier

After defining topologies and pretopologies (Section 1) and sheaves of sets (Section 2), in Section 3 we turn to the central theorem of this exposé (3.4): the existence of the sheaf associated with a presheaf. In order to cover all the cases encountered in practice, the existence of this functor is proved for presheaves on a $\mathscr{U}$-site (3.0.2). The consequences of this theorem for the behavior of inductive and projective limits in categories of sheaves are drawn in Sections 4 and 5. In Section 6 we define and study sheaves with values in arbitrary categories, immediately focusing attention on sheaves of rings, groups, modules, etc. 218

1. Topologies, covering families, pretopologies

DEFINITION 1.1. *A topology on a category C consists, for every object X of C, of a set $J(X)$ of sieves on X, subject to the following axioms:*

T 1) (Stability under base change). *For every object X of C, every sieve $R \in J(X)$, and every morphism $f : Y \to X$ ($Y \in \mathrm{ob}(C)$), the sieve $R \times_X Y$ on Y belongs to $J(Y)$.*

T 2) (Local character). *If R and R' are two sieves on X, if $R \in J(X)$, and if, for every $Y \in \mathrm{ob}(C)$ and every morphism $Y \to R$, the sieve $R' \times_X Y$ belongs to $J(Y)$, then R' belongs to $J(X)$.* 219

T 3) *For every object X of C, X belongs to $J(X)$.*

1.1.1. The sieves belonging to $J(X)$ will be called the *sieves covering* X, or the *refinements of* X. It follows immediately from axioms (T1), (T2), and (T3) that the set of sieves covering X is stable under finite intersections, and that every sieve containing a covering sieve is itself a covering sieve. Thus the set $J(X)$, ordered by inclusion, is cofiltered (I 8).

1.1.2. Let C be a category, and let T and T' be two topologies on C. The topology T is said to be *finer than the topology* T' if, for every object X of C, every refinement of X for the topology T' is a refinement of X for the topology T. This defines an order structure on the set of topologies.

1.1.3. Let $(T_i)_{i\in I}$ be a family of topologies on C. For every object X of C, the set of sieves on X that are covering for all the topologies T_i satisfies axioms (T 1), (T 2), and (T 3), and hence defines a topology: the *intersection topology* of the T_i, i.e. the infimum of the T_i. It is the finest topology that is coarser than all the topologies T_i. Consequently, the family $(T_i)_{i\in I}$ also admits a *supremum*: the intersection topology of the topologies finer than each of the T_i.

1.1.4. In particular, taking $J(X)$ to be the set of all sieves on X gives a topology finer than every topology on C; we call it the *discrete topology*. 220

There is a topology coarser than every topology on C: the topology for which $J(X) = \{X\}$ for every X of C. This topology is called the *coarse* or *chaotic* topology.

1.1.5. A category C equipped with a topology is called a *site*. The category C is called the *category underlying the site*.

DEFINITION 1.2. *Let C be a site and X an object of C. A family of morphisms $(f_\alpha : X_\alpha \to X)$, $\alpha \in A$, is called a covering family if the sieve generated by the family f_α (I 4.3.3) is a sieve covering X.*

1.1.6. Let C be a category. Suppose that, for each object X of C, we are given a set of families of morphisms with target X. There is then a coarsest topology T for which the given families are covering, namely the intersection (1.1.3) of all the topologies in question. This topology is called the *topology generated by the given sets of families of morphisms*. In general, it is difficult to describe all the covering sieves of this topology. The situation is more favorable, however, in the following case:

DEFINITION 1.3. *Let C be a category. A pretopology on C consists, for each object X of C, of a set* $\mathrm{Cov}(X)$ *of families of morphisms with target X, subject to the following axioms:*

221 PT 0) *For every object X of C, the morphisms in the families belonging to* $\mathrm{Cov}(X)$ *are pullbackable. (Recall that a morphism $Y \to X$ is called pullbackable if, for every morphism $Z \to X$, the fiber product $Z \times_X Y$ is representable.)*

PT 1) *For every object X of C, every family $(X_\alpha \to X)_{\alpha\in A}$ belonging to* $\mathrm{Cov}(X)$*, and every morphism $Y \to X$ ($Y \in \mathrm{ob}(C)$), the family $(X_\alpha \times_X Y \to Y)_{\alpha\in A}$ belongs to* $\mathrm{Cov}(Y)$*. (Stability under base change.)*

PT 2) *If $(X_\alpha \to X)_{\alpha\in A}$ belongs to* $\mathrm{Cov}(X)$ *and if, for every $\alpha \in A$, $(X_{\beta_\alpha} \to X_\alpha)_{\beta_\alpha\in B_\alpha}$ belongs to* $\mathrm{Cov}(X_\alpha)$*, then the family $(X_\gamma \to X)_{\gamma\in\coprod_{\alpha\in A} B_\alpha}$ (where, for every $\gamma = (\alpha, \beta_\alpha)$, $\alpha \in A$, $\beta_\alpha \in B_\alpha$, the morphism $X_\gamma \to X$ is the composite morphism $X_\gamma = X_{\beta_\alpha} \to X_\alpha \to X$) belongs to* $\mathrm{Cov}(X)$*. (Stability under composition.)*

PT 3) *The family $X \xrightarrow{\mathrm{id}_X} X$ belongs to* $\mathrm{Cov}(X)$.

1.3.1. The definition in 1.1.6 allows us to consider, for a given *pretopology* on C, the *topology* on C generated by that pretopology. Note that if C is a category in which fiber products are representable, then every topology T on C can be defined by a pretopology, namely the one for which $\mathrm{Cov}(X)$ consists of *all* the families covering X for the topology T.

PROPOSITION 1.4. *Let C be a category, E a pretopology on C, T the topology defined by the pretopology E (1.1.6), and X an object of C. Denote by $J_E(X)$ the set of sieves on X generated by families of the pretopology, and by $J_T(X)$ the set of refinements of X for*
222 *the topology T. Then $J_E(X)$ is cofinal in $J_T(X)$. In other words, a sieve R on X belongs to $J_T(X)$ if and only if there exists a sieve $R' \in J_E(X)$ such that $R' \subset R$.*

PROOF. For every object X of C, let $J'(X)$ be the set of sieves on X that contain a sieve belonging to $J_E(X)$. Clearly, $J'(X) \subset J_T(X)$. To prove that $J'(X) = J_T(X)$, it is enough to show that the data $J'(X)$ define a topology on C. Now the $J'(X)$ clearly satisfy axioms (T 1) and (T 3). It therefore remains to verify (T 2). For this, it is enough to prove that if R' is a subsieve of $R \in J_E(X)$ such that, for every $Y \to R$, the sieve $R' \times_X Y$ on Y belongs to $J'(Y)$, then the sieve R' belongs to $J'(X)$. Now the sieve R is generated by a family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, and for every α the sieve $R' \times_X X_\alpha$ contains a sieve generated by a family $(X_{\beta_\alpha} \to X_\alpha)$ belonging to $\mathrm{Cov}(X_\alpha)$. It follows that the sieve R' contains a sieve generated by the family $(X_{\beta_\alpha} \to X)$. Thus, by axiom (PT 2), R' contains a sieve belonging to $J_E(X)$ and consequently belongs to $J'(X)$, C.Q.F.D.

2. Sheaves of sets

DEFINITION 2.1. *Let C be a site whose underlying category is a $\mathcal{U}$-category. A presheaf F with values in $\mathcal{U}$ – Ens is said to be separated (resp. to be a sheaf) if, for every sieve R covering an object X of C, the map*

$$\mathrm{Hom}_{\hat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\hat{C}}(R, F)$$

is injective (resp. bijective). The full subcategory of $\hat{C}$ whose objects are the sheaves is 223
called the category of sheaves of sets on C, and is most often denoted by $\widehat{C}$ [4]. When no ambiguity can result, we shall simply say the category of sheaves on C; on the other hand, we shall sometimes be more precise and say the category of sheaves with values in $\mathcal{U}$ – Ens, or the category of $\mathcal{U}$-sheaves.

PROPOSITION 2.2. *Let C be a $\mathcal{U}$-category, and let $\mathcal{F} = (F_i)_{i\in I}$ be a family of $\mathcal{U}$-presheaves on C. For each object X of C, denote by $J_{\mathcal{F}}(X)$ the set of sieves $R \to X$ such that, for every morphism $Y \to X$ in C with target X, the sieve $R \times_X Y$ has the following property: the map*

$$\mathrm{Hom}_{\hat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\hat{C}}(R \times_X Y, F_i)$$

is bijective (resp. injective) for every $i \in I$. Then the sets $J_{\mathcal{F}}(X)$ define a topology on C, which is the finest topology for which each F_i is a sheaf (resp. separated presheaf).

PROOF. The $J_{\mathcal{F}}(X)$ clearly satisfy axioms (T 1) and (T 3). It remains to show that they satisfy (T 2). For this it suffices to show the following:

1) If $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R \in J_{\mathcal{F}}(X)$ and such that, for every $Y \to R$ (where Y is an object of C), the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$, then the sieve R' belongs to $J_{\mathcal{F}}(X)$.
2) If $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R' \in J_{\mathcal{F}}(X)$, then the sieve R belongs to $J_{\mathcal{F}}(X)$.

Note that, in case 2), for every $Y \to R$ (where Y is an object of C), the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. Now, in cases 1) and 2), R is the inductive limit of the objects Y of C over R (I 3.4). The inductive limits in $\hat{C}$ are universal (I 3.3). Consequently, in cases 1) 224
and 2), R' is the inductive limit of the $R' \times_X Y$. Now, in cases 1) and 2), $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. Passing to the inductive limit over the objects Y of C over R, we deduce that the map

$$\mathrm{Hom}_{\hat{C}}(R, F_i) \longrightarrow \mathrm{Hom}_{\hat{C}}(R', F_i)$$

is a bijection (resp. an injection) for every $i \in I$. It follows that the maps

$$\mathrm{Hom}_{\hat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\hat{C}}(R', F_i) \quad , \quad \mathrm{Hom}_{\hat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\hat{C}}(R, F_i)$$

are, in cases 1) and 2), bijections (resp. injections). Moreover, hypotheses 1) and 2) are plainly stable under arbitrary base change $Y \to X$ (Y an object of C). It then follows that, for every object Y of C over X and every $i \in I$, the maps

$$\mathrm{Hom}_{\hat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\hat{C}}(R \times_X Y, F_i)$$
$$\mathrm{Hom}_{\hat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\hat{C}}(R' \times_X Y, F_i)$$

are in both cases bijections (resp. injections); this shows that R' and R belong to $J_{\mathcal{F}}(X)$, C.Q.F.D.

[4] or $\tilde{C}$, depending on the mood of the typewriter.

COROLLARY 2.3. *Let C be a $\mathscr{U}$-category, and for every X of C let $K(X)$ be a set of sieves on X. Suppose that the $K(X)$ satisfy axiom* (T 1) *of* (1.1). *A presheaf F is a sheaf (resp. a separated presheaf) for the topology generated (1.1.6) by the $K(X)$ if and only if, for every object X of C and every sieve $R \in K(X)$, the map*

$$\mathrm{Hom}_{C^\wedge}(X, F) \longrightarrow \mathrm{Hom}_{C^\wedge}(R, F)$$

is a bijection (resp. an injection).

225 COROLLARY 2.4. *In particular, let C be a $\mathscr{U}$-category equipped with a pretopology. A presheaf F is a sheaf (resp. a separated presheaf) if and only if, for every object X of C and every covering family $(X_\alpha \to X)$ belonging to* $\mathrm{Cov}(X)$, *the diagram of sets*

$$F(X) \to \prod_{\alpha \in A} F(X_\alpha) \rightrightarrows \prod_{(\alpha,\beta)\in A\times A} F(X_\alpha \times_X X_\beta)$$

is exact (resp. the map

$$F(X) \longrightarrow \prod_{\alpha \in A} F(X_\alpha)$$

is injective).

PROOF. Apply 2.3, then (I 3.5) and (I 2.12).
Corollary 2.4 recovers the definition given in [1].

DEFINITION 2.5. *Let C be a $\mathscr{U}$-category. The canonical topology of C is the finest topology for which the representable functors are sheaves (2.2). A sieve covering X for the canonical topology will be called a universally strictly epimorphic sieve. A covering family for the canonical topology will be called a universally strictly epimorphic family. If, in addition, the morphisms of the covering family are pullbackable, the family will be called universally effective epimorphic* [2].

REMARK 2.5.1. For almost all the sites that have been used up to now, the topology is *coarser* than the canonical topology; in other words, the representable functors on C
226 are sheaves, i.e. the covering families of C are universally strictly epimorphic. The only exception to this rule is the site $\widehat{C}$ studied in n° 5 (whose topology is *finer,* and most often strictly finer, than the canonical topology).

PROPOSITION 2.6. *A sieve $R \hookrightarrow X$ is universally strictly epimorphic if and only if, for every object $Y \to X$ of C over X, and for every object Z of C, the map*

$$\mathrm{Hom}_C(Y, Z) \longrightarrow \varprojlim_{C/(Y\times_X R)} \mathrm{Hom}_C(., Z)$$

is a bijection.

PROOF. Immediate upon applying 2.2 and then (I 5.3).

REMARKS 2.7.
1) Proposition 2.6 gives a characterization of the universally strictly epimorphic sieves of a category C that is independent of the universe in which the presheaves take their values, provided only that the morphism sets $\mathrm{Hom}(X, Y)$ of C belong to that universe. It thus makes it possible to define the canonical topology for any category C, without having to specify the universes.
2) Let C be a site whose underlying category is a $\mathscr{U}$-category, let F be a $\mathscr{U}$-presheaf, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. The category underlying C is a $\mathscr{V}$-category, and F may be regarded as a $\mathscr{V}$-presheaf. The $\mathscr{U}$-presheaf F is a sheaf (resp. a separated presheaf) if and only if the $\mathscr{V}$-presheaf F is a sheaf

(resp. a separated presheaf). In other words, the condition for a presheaf F to be a sheaf (resp. a separated presheaf) does not depend (in the sense just specified) on the universe in which the presheaf F takes its values. In particular, let C be a site and let F be a $\mathscr{U}$-presheaf; we say that F is a sheaf (resp. a separated presheaf) if there exists a universe $\mathscr{V}$ containing $\mathscr{U}$ such that the category C is a $\mathscr{V}$-category and such that F is a $\mathscr{V}$-sheaf (resp. a separated $\mathscr{V}$-presheaf). This property does not depend on the universe $\mathscr{V}$. 227

3. The sheaf associated with a presheaf

DEFINITION 3.0.1. *Let C be a site. A topologically generating family (or, when no confusion can result, a generating family) of C is a set G of objects of C such that every object of C is the target of a covering family of morphisms of C (1.2) whose sources are elements of G.*

DEFINITION 3.0.2. *Let $\mathscr{U}$ be a universe. A $\mathscr{U}$-site is a site C whose underlying category is a $\mathscr{U}$-category* (I 1.1)*, and which possesses a small topologically generating family. Let C be a $\mathscr{U}$-category; a $\mathscr{U}$-topology on C is a topology on C making C a $\mathscr{U}$-site. A site C is called $\mathscr{U}$-small, or, by abuse of language, small, if its underlying category is small* (I 1.0).

3.0.3. It follows immediately from the definitions that every topology finer than a $\mathscr{U}$-topology is a $\mathscr{U}$-topology, and that a small site is a $\mathscr{U}$-site.

PROPOSITION 3.0.4. *Let C be a $\mathscr{U}$-site and G a small topologically generating family of C. For every $X \in \operatorname{ob} C$, denote by $J_G(X)$ the set of sieves covering X generated by a family of morphisms* 228

$$(Y_\alpha \xrightarrow{u_\alpha} X), \alpha \in A, \text{ where } Y_\alpha \in G.$$

1) *The set $J_G(X)$ is small.*
2) *The set $J_G(X)$ is cofinal in the set $J(X)$ of all sieves covering X, ordered by inclusion.*
3) *For every $R \in J_G(X)$, there exists a small epimorphic family* (I 10.2) *$(u_\alpha : Y \to R)$, $\alpha \in A$, with $Y \in G$.*

PROOF. 1) Put $A(X) = \coprod_{Y \in G} \hom(Y, X)$. The set $A(X)$ is small (I 0) and $\operatorname{card}(J_G(X)) \leq 2^{\operatorname{card}(A(X))}$. Consequently $J_G(X)$ is small.

2) Let $R \in J(X)$. Put $A(R) = \coprod_{Y \in G} \operatorname{Hom}(Y, R)$, and let R' be the sieve of X generated by the family $(Y \xrightarrow{u} R \hookrightarrow X)$, $u \in A(R)$. One has $R' \subset R$, and it suffices to show that R' is covering. By axiom (T 2) of 1.1, it suffices to show that for every morphism $Z \to R$, $Z \in \operatorname{ob} C$, the sieve $R' \times_X Z$ of Z is covering. Now the sieve $R' \times_X Z$ contains the sieve generated by the family of all morphisms $Y \to Z$ with $Y \in G$, which is covering by hypothesis. Therefore the sieve $R' \times_X Z$ of Z is covering (axiom T2 of 1.1).

3) Let $R \in J_G(X)$. The family $(Y \xrightarrow{u} R)$, $u \in A(R) = \coprod_{Y \in G} \operatorname{Hom}(Y, R)$, is epimorphic by hypothesis. Now, for every $Y \in \operatorname{ob} C$, $\operatorname{Hom}(Y, R)$ is contained in $\operatorname{Hom}(Y, X)$, which is small. Consequently $A(R)$ is small.

3.0.5. Let C be a $\mathscr{U}$-site, $\mathscr{V}$ a universe such that $C \in \mathscr{V}$ and $\mathscr{U} \subset \mathscr{V}$, and G a $\mathscr{U}$-small topologically generating family of C. The category $\hat{C}$ of presheaves of $\mathscr{U}$-sets on C is a $\mathscr{V}$-category (I 1.1.1). Let X be an object of C. The set $J(X)$ of sieves covering X is $\mathscr{V}$-small and, ordered by inclusion, is cofiltered (1.1.1). For every $\mathscr{U}$-presheaf F, the inductive limit $\varinjlim_{J(X)} \operatorname{Hom}_{\hat{C}}(R, F)$ is therefore representable by an element of $\mathscr{V}$ (I 229

2.4.1). Moreover, it follows from (3.0.4 3)) that, for every $R \in J_G(X)$, $\mathrm{Hom}_{C^\wedge}(R, F)$ is $\mathscr{U}$-small, and since $J_G(X)$ is a $\mathscr{U}$-small set cofinal in $J(X)$ (*loc. cit.*), it follows from (I 2.4.2) that $\varinjlim_{J(X)} \mathrm{Hom}_{C^\wedge}(R, F)$ is $\mathscr{U}$-small. Choose, for every F and every X, an element of $\mathscr{U}$ representing this inductive limit, and put

$$LF(X) = \varinjlim_{J(X)} \mathrm{Hom}_{C^\wedge}(., F).$$

Let $g : Y \to X$ be a morphism of C. The base-change functor $g^* : J(X) \to J(Y)$ defines a map

$$LF(g) : LF(X) \longrightarrow LF(Y),$$

which makes $X \mapsto LF(X)$ a presheaf on C.

Since the morphism $\mathrm{id}_X : X \to X$ is an element of $J(X)$, for every object X of C one has a map

$$\ell(F)(X) : F(X) \longrightarrow LF(X),$$

which visibly defines a morphism of functors

$$\ell(F) : F \longrightarrow LF.$$

It is moreover clear that $F \mapsto LF$ is a functor in F and that the morphisms $\ell(F)$ define a morphism

$$\ell : \mathrm{Id} \longrightarrow L.$$

Finally, let $R \hookrightarrow X$ be a covering sieve of X. The definition of $LF(X)$ and (I 1.4) provide a map

$$Z_R : \mathrm{Hom}_{C^\wedge}(R, F) \longrightarrow \mathrm{Hom}_{C^\wedge}(X, LF),$$

230 and, for every morphism $Y \xrightarrow{g} X$ of C, the definition of the functor LF shows that the following diagram is commutative:

(∗)
$$\begin{array}{ccc} Z_R : & \mathrm{Hom}_{C^\wedge}(R, F) & \longrightarrow \mathrm{Hom}_{C^\wedge}(X, LF) \\ & \downarrow & \downarrow \\ Z_{R\times_X Y} : & \mathrm{Hom}_{C^\wedge}(R \times_X Y, F) & \longrightarrow \mathrm{Hom}_{C^\wedge}(Y, LF) \end{array}$$

(the vertical arrows are the evident arrows).
Let us now copy [2].

Lemma 3.1. 1) *For every covering sieve $i_R : R \hookrightarrow X$ and every $u : R \to F$, the diagram*

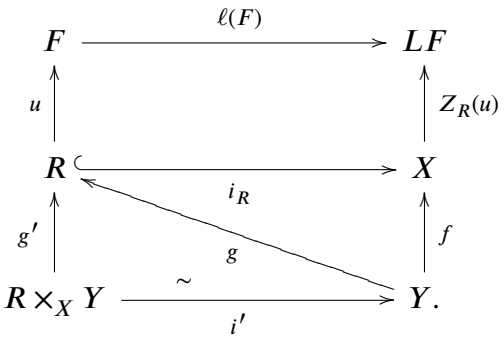
(∗∗)
$$\begin{array}{ccc} F & \xrightarrow{\ell(F)} & LF \\ {\scriptstyle u}\uparrow & & \uparrow{\scriptstyle Z_R(u)} \\ R & \xrightarrow[i_R]{} & X \end{array}$$

is commutative.

2) *For every morphism $v : X \to LF$, there exists a covering sieve R of X and a morphism $u : R \to F$ such that $Z_R(u) = v$.*
3) *Let Y be an object of C, and let $u, v : Y \rightrightarrows F$ be two morphisms such that $\ell(F) \circ u = \ell(F) \circ v$. Then the kernel of the pair (u, v) is a covering sieve of Y.*

4) *Let R, R' be two covering sieves of X, and let $u : R \to F$, $u' : R' \to F$ be two morphisms. For $Z_R(u) = Z_{R'}(u')$, it is necessary and sufficient that u and u' coincide on a covering sieve $R'' \hookrightarrow R \times_X R'$.*

PROOF. The only nontrivial assertion is assertion 1). We must show that $Z_R(u) \circ i_R = \ell(F) \circ u$. For this it suffices to show (I 3.4) that the composites of these morphisms with every morphism $Y \xrightarrow{g} R$ (Y an object of C) are equal. Consider $f = i_R \circ g$ and the fiber product $R \times_X Y$: 231



In the diagram above, i' is a monomorphism admitting a section. Consequently (1.3 3) i' is an isomorphism. By the definition of the morphism $\ell(F)$, the morphism $Z_{R\times_X Y}(u \circ g')$ is equal to $\ell(F) \circ u \circ g$. On the other hand, the commutativity of diagram (*) gives the equality $Z_{R\times_X Y}(u \circ g') = Z_R(u) \circ f$, and hence one indeed has $\ell(F) \circ u \circ g = Z_R(u) \circ i_R \circ g$.

PROPOSITION 3.2. 1) *The functor L is left exact* (I 2.3.2).

2) *For every presheaf F, LF is a separated presheaf.*

3) *The presheaf F is separated if and only if the morphism $\ell(F) : F \to LF$ is a monomorphism. The presheaf LF is then a sheaf.*

4) *The following properties are equivalent:*
   (i) *$\ell(F) : F \to LF$ is an isomorphism.*
   (ii) *F is a sheaf.*

PROOF. 1) It suffices to show (I 3.1) that, for every object X of C, the functor $F \mapsto LF(X)$ commutes with finite projective limits. By the definition of a projective limit, the functor

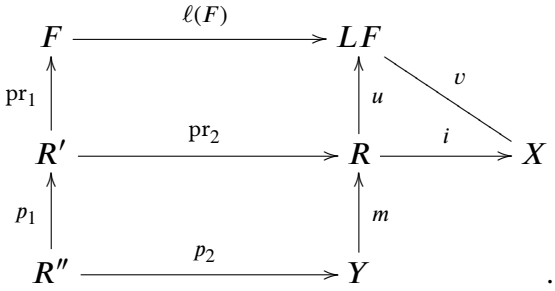
$$F \longmapsto \mathrm{Hom}_{\hat{C}}(R, F) \quad R \hookrightarrow X \in \mathrm{ob}(J(X))$$

commutes with projective limits, and the inductive limit $\varinjlim_{J(X)}$ commutes with finite projective limits because $J(X)$ is a cofiltered set (I 2.7). 232

2) Let X be an object of C, and let $f, g : X \rightrightarrows LF$ be two morphisms that coincide on a covering sieve $R \hookrightarrow X$ of X. By 3.1 2), there exists a covering sieve $R' \hookrightarrow X$, which may always be supposed contained in R, and two morphisms $u, v : R' \rightrightarrows F$ such that $Z_{R'}(u) = f$ and $Z_{R'}(v) = g$. By 3.1 1), one then has $\ell(F) \circ u = \ell(F) \circ v$. Consequently (3.1 4)) u and v coincide on a covering sieve $R'' \hookrightarrow R'$. Let w be the restriction of u to R''. One has $Z_{R''}(w) = Z_{R'}(u) = Z_{R'}(v)$, and consequently $f = g$. Thus the presheaf LF is separated.

3) If F is separated, the morphism $\ell(F)$ is a monomorphism because a filtered inductive limit of monomorphisms is a monomorphism. If $\ell(F)$ is a monomorphism, the presheaf F is a subpresheaf of a separated presheaf, and therefore is separated. Let us show that LF is then a sheaf. Let $i : R \hookrightarrow X$ be a sieve

covering an object X of C, and let $u : R \to LF$ be a morphism. It suffices to show that u factors through X. Put $R' = F \times_{LF} R$ and $v = Z_{R'}(\mathrm{pr}_1)$:



233 To show that $u = v \circ i$, it suffices to show (I 3.4) that for every morphism $m : Y \to R$ (Y an object of C), $v \circ i \circ m = u \circ m$. Put $R'' = Y \times_R R'$, and let p_1, p_2 be the projections. The projection $R'' \overset{p_2}{\hookrightarrow} Y$ is a monomorphism and makes R'' a covering sieve of Y (3.1 2)). One has $v \circ i \circ \mathrm{pr}_2 = \ell(F) \circ \mathrm{pr}_1$ (3.1 1)), and hence $v \circ i \circ \mathrm{pr}_2 = u \circ m \circ p_2$. It follows that $v \circ i \circ \mathrm{pr}_2 \circ p_1 = u \circ \mathrm{pr}_2 \circ p_1$, i.e. $v \circ i \circ m \circ p_2 = u \circ m \circ p_2$. Since the presheaf LF is separated, one has $v \circ i \circ m = u \circ m$, C.Q.F.D.

4) Clear.

Remark 3.3. Let $J'(X)$ be a cofinal subset of $J(X)$. One has

$$LF(X) = \varinjlim_{J'(X)} \mathrm{Hom}_{\hat{C}}(., F).$$

In particular, if the topology of C is defined by a pretopology $X \mapsto \mathrm{Cov}(X)$ (1.1.3), the functor L can be described using the covering families of $\mathrm{Cov}(X)$ (I 2.12 and I 3.5). Writing the formulas explicitly, one recovers the construction of [

Theorem 3.4. *Let C be a $\mathcal{U}$-site. The inclusion functor $i : \tilde{C} \hookrightarrow \hat{C}$ from sheaves into presheaves admits a left adjoint $\underline{a}$, which is left exact (I 2.3.2):*

$$\tilde{C} \underset{i}{\overset{\underline{a}}{\rightleftarrows}} \hat{C} \quad .$$

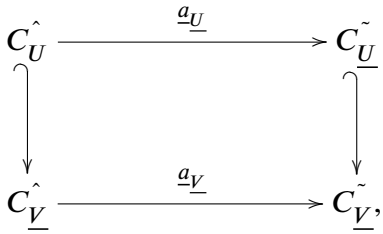
The functor $i \circ \underline{a}$ is canonically isomorphic to the functor $L \circ L$ (cf. 3.0.5). For every presheaf F, the adjunction morphism $F \to i \circ \underline{a}(F)$ is deduced, via the preceding isomorphism, from the morphism $\ell(LF)\ell(F) : F \to L \circ L(F)$.

234 Definition 3.5. *The sheaf $\underline{a}F$ is called the sheaf associated with the presheaf F.*

Theorem 3.4 follows immediately from Proposition 3.2.

Proposition 3.6. *Let C be a $\mathcal{U}$-site and let $\mathcal{V} \supset \mathcal{U}$ be a universe. Denote by $\hat{C}_{\mathcal{U}}$ and $\tilde{C}_{\mathcal{U}}$ (respectively $\hat{C}_{\mathcal{V}}$ and $\tilde{C}_{\mathcal{V}}$) the categories of $\mathcal{U}$-presheaves and $\mathcal{U}$-sheaves (respectively $\mathcal{V}$-presheaves and $\mathcal{V}$-sheaves), and $\underline{a}_{\mathcal{U}} : \hat{C}_{\mathcal{U}} \to \tilde{C}_{\mathcal{U}}$ (respectively $\underline{a}_{\mathcal{V}} : \hat{C}_{\mathcal{V}} \to \tilde{C}_{\mathcal{V}}$) the*

corresponding "associated sheaf" functors. The diagram



where the vertical functors are the canonical inclusions, is commutative up to canonical isomorphism.

PROOF. This follows from the construction of the functors $\underline{a}_U$ and $\underline{a}_V$ (3.4 and 3.0.5).

4. Exactness properties of the category of sheaves

The exactness properties of the category of sheaves follow from the exactness properties of the category of presheaves via Theorem 3.4. This section makes this philosophy explicit through some of the most useful standard statements.

THEOREM 4.1. *Let C be a $\mathscr{U}$-site, let $\widetilde{C}$ be the category of sheaves, let $\underline{a} : \hat{C} \to \widetilde{C}$ be the associated-sheaf functor, and let $i : \widetilde{C} \to \hat{C}$ be the inclusion functor.*

1) *The functor $\underline{a}$ commutes with inductive limits and is exact.*
2) *The $\mathscr{U}$-inductive limits in $\widetilde{C}$ are representable. For every small category I and* 235
   every functor $E : I \to \widetilde{C}$, the canonical morphism
$$\varinjlim_I E \longrightarrow \underline{a}(\varinjlim_I i \circ E)$$
   is an isomorphism.
3) *The $\mathscr{U}$-projective limits in $\widetilde{C}$ are representable. For every object X of C, the functor on $\widetilde{C}$: $F \mapsto F(X)$ commutes with projective limits, i.e. the inclusion functor $i : \widetilde{C} \to \hat{C}$ commutes with projective limits.*

PROOF. These properties follow essentially from Theorem 3.4 and from (I 2.11).

Thus, in the category of sheaves, products indexed by an element of $\mathscr{U}$, fiber products, sums indexed by an element of $\mathscr{U}$, amalgamated sums, kernels, cokernels, images, and coimages are representable.

COROLLARY 4.1.1. *Let C be a $\mathscr{U}$-site and let F be a sheaf of sets on C. The canonical homomorphism*
$$\varinjlim_{C/F} \underline{a}X \longrightarrow F$$
is an isomorphism.

PROOF. This follows from (I 3.4) and the fact that $\underline{a}$ commutes with inductive limits.

PROPOSITION 4.2. *Every morphism of $\widetilde{C}$ that is both an epimorphism and a monomorphism is an isomorphism.*

PROOF. Let $u : G \to H$ be a morphism of $\widetilde{C}$ that is both an epimorphism and a monomorphism. First observe that the morphism u is a monomorphism of presheaves 236
(4.1 1)). Construct the amalgamated sum $H \amalg_G H = K$ and the two canonical morphisms

$i_1, i_2 : H \rightrightarrows K$ in the category of presheaves. Since u is a monomorphism of presheaves, one immediately verifies that the following diagram is cartesian and cocartesian (I 10.1):

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle i_2} \\ H & \xrightarrow{i_1} & K. \end{array}$$

Applying the "associated sheaf" functor, one therefore obtains a cartesian and cocartesian diagram in the category of sheaves (4.1 1)):

$$(*) \qquad \begin{array}{ccc} G & \xrightarrow{u} & H \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \underline{a}i_2} \\ H & \xrightarrow{\underline{a}i_1} & \underline{a}K. \end{array}$$

Since u is an epimorphism of sheaves, the morphism $\underline{a}i_1$ is an isomorphism, and since diagram (*) is cartesian, the morphism u is an isomorphism.

PROPOSITION 4.3. 1) *The inductive limits in $\tilde{C}$ that are representable are universal* (I 2.5).
2) *Every epimorphic family* (I 10.2) *of morphisms is universally effective epimorphic (2.6).*
3) *Every equivalence relation is universally effective* (I 10.6).
4) *Filtered $\mathscr{U}$-inductive limits commute with finite projective limits* (I 2.6).

237 PROOF. Assertions 1) through 4) hold in the category of sets, hence in the category of presheaves (I 3.1). Assertions 1) and 4) then follow immediately from the corresponding assertions for presheaves and from 4.1. Let us prove 3). Let $p_1, p_2 : R \rightrightarrows X$ be an equivalence relation in $\tilde{C}$. The diagram $R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X$ is then an equivalence relation of presheaves (4.1.1). There is therefore a morphism of presheaves $u : X \to Y$ such that the diagram

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle u} \\ X & \xrightarrow{u} & Y \end{array}$$

is cartesian and cocartesian in the category of presheaves. Applying the "associated sheaf" functor, one obtains a cartesian and cocartesian diagram in the category of sheaves (4.1 1)). It follows that $R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X$ is an effective equivalence relation. Since every equivalence relation is effective by what precedes, every equivalence relation is universally effective. Let us prove 2). Let $(u_i : X_i \to X)$, $i \in I$, be an epimorphic family in $\tilde{C}$. By (II 2.6) and (I 2.12), it suffices to prove the following assertions:

a) For every morphism of sheaves $v : Y \to X$, the family of morphisms $\mathrm{pr}_{2,i} : X_i \times_X Y \to Y$, $i \in I$, is an epimorphic family in $\tilde{C}$.

b) For every sheaf Z, the following diagram of sets is exact:

$$\mathrm{Hom}(X,Z) \to \prod_{i\in I} \mathrm{Hom}(X_i,Z) \rightrightarrows \prod_{(i,k)\in I\times I} \mathrm{Hom}(X_i \times_X X_k, Z)$$

Let X' be the union of the images of the morphisms u_i, formed in the category of presheaves, and let $j : X' \hookrightarrow X$ be the canonical injection. For every $i \in I$, the morphism $u_i : X_i \to X$ factors as a morphism $u'_i : X_i \to X'$ followed by j, and the family $u'_i : X_i \to X'$, $i \in I$, is an epimorphic family in $\hat{C}$. Since j is a monomorphism, the morphism $\underline{a}j : \underline{a}X' \to X$ is a monomorphism of sheaves (4.1). Since, for every $i \in I$, one has $u_i = \underline{a}j\underline{a}u'_i$, $\underline{a}j$ is an epimorphism of sheaves. It follows that $\underline{a}j$ is an isomorphism (4.2). Let $v : Y \to X$ be a morphism of sheaves. Base change then yields a morphism $j_v : X' \times_X Y \to Y$ and morphisms $u'_{i,v} : X_i \times_X Y \to X' \times_X Y$. Since $\underline{a}$ commutes with fiber products, $\underline{a}j_v$ is an isomorphism. Since epimorphic families in $\hat{C}$ remain epimorphic under base change, the family $u'_{i,v}$, $i \in I$, is epimorphic in $\hat{C}$. Since $\underline{a}$ commutes with inductive limits, the family $\underline{a}u'_{i,v} : X_i \times_X Y \to \underline{a}(X' \times_X Y)$, $i \in I$, is epimorphic in $\tilde{C}$, proving a). Let us prove b). Let Z be a sheaf. Since $\underline{a}j$ is an isomorphism, the map

$$\mathrm{Hom}(X,Z) \longrightarrow \mathrm{Hom}(X',Z)$$

is a bijection. Since the u'_i, $i \in I$, form an epimorphic family in $\hat{C}$, the following diagram of sets is exact:

$$\mathrm{Hom}(X',Z) \to \prod_{i\in I} \mathrm{Hom}(X_i,Z) \rightrightarrows \prod_{(i,k)\in I\times I} \mathrm{Hom}(X_i \times_{X'} X_k, Z)$$

Finally, since X' is a subpresheaf of X, the presheaf $X_i \times_{X'} X_k$, $(i,k) \in I \times I$, is canonically isomorphic to $X_i \times_X X_k$, proving b).

Remarks 4.3.1. 1) Proposition 4.3 2) formally gives us a second proof of 4.2: it is clear that, in any category, a morphism that is both a universal effective epimorphism and a monomorphism is in fact an isomorphism.

2) It follows from the preceding propositions that every morphism in the category of sheaves factors uniquely as an effective epimorphism followed by an effective monomorphism. This property naturally generalizes, to nonadditive categories, axiom (AB2) for abelian categories (coim $\simeq$ im) [Tohoku].

4.4.0. Let C be a $\mathscr{U}$-site. Composing the functor $h : C \to \hat{C}$ (I 1.3.1) with the associated sheaf functor yields a functor

$$\epsilon_C : C \longrightarrow \tilde{C},$$

called the *canonical functor* from C to $\tilde{C}$, which will be used constantly in what follows. The functor ϵ_C commutes with finite projective limits. When the topology on C is coarser than the canonical topology, ϵ_C is fully faithful and commutes with projective limits; moreover, it is then defined even when C is not assumed to possess a small topologically generating family. We shall not study in detail the behavior of ϵ_C with respect to inductive limits (cf. [SGA 3 IV]). We shall, however, need the results below.

Theorem 4.4. *Let C be a $\mathscr{U}$-site and let $(u_i : X_i \to X)$, $i \in I$, be a family of morphisms of C with target X. The following conditions are equivalent* [4]*:*

i) *The family $(\epsilon_C u_i : \epsilon_C X_i \to \epsilon_C X)$, $i \in I$, is an epimorphic family in $\tilde{C}$* (I 10.2).

[4] When the site C does not necessarily possess a small topologically generating family and when the topology on C is coarser than the canonical topology, one has ii) $\Rightarrow$ i) (cf. the proof of 4.4).

ii) *The family* $(u_i : X_i \to X)$, $i \in I$, *is a covering family of* C *(1.2).*

PROOF. ii) $\Rightarrow$ i). Let $R \hookrightarrow X$ be the sieve generated by the u_i, $i \in I$ (I 4.3.3), and let $u_i' : X_i \to R$ be the morphisms induced by the u_i. The morphisms u_i', $i \in I$, form an epimorphic family in $\hat{C}$, and R is a covering sieve of X. Consequently, for every sheaf F, the map

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(R, F)$$

is bijective, and the map

$$\mathrm{Hom}(R, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$

is injective. Thus the map

$$\mathrm{Hom}(X, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$

is injective, and consequently the map

$$\mathrm{Hom}(\underline{a}X, F) \longrightarrow \prod_i \mathrm{Hom}(\underline{a}X_i, F)$$

is injective, which proves i).
i) $\Rightarrow$ ii). With the notation introduced above, let $J_R : R \hookrightarrow X$ be the canonical inclusion into X of the sieve generated by the u_i, $i \in I$. It follows from i) that the morphism of sheaves $\underline{a}J_R : \underline{a}R \to \underline{a}X$ is both an epimorphism and a monomorphism in the category of sheaves. It is therefore an isomorphism of sheaves (4.2). With the notation of n° 3, we have a commutative diagram:

$$\begin{array}{ccccc} X & \xrightarrow{\ell(X)} & LX & \xrightarrow{\ell(LX)} & \underline{a}X \\ \uparrow{\scriptstyle J_R} & & \uparrow{\scriptstyle LJ_R} & & \uparrow{\scriptstyle \wr}{\scriptstyle \underline{a}J_R} \\ R & \xrightarrow{\ell(R)} & LR & \xrightarrow{\ell(LR)} & \underline{a}R. \end{array}$$

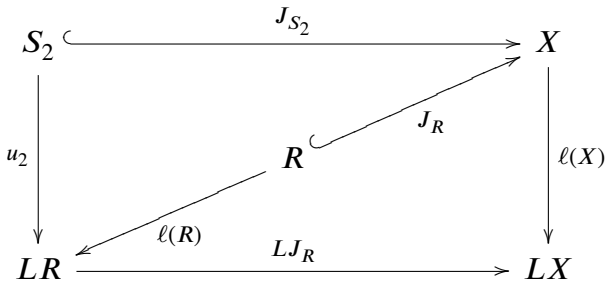
241 First of all, by 3.1 2), there exist a covering sieve $J_{S_1} : S_1 \to X$ and a morphism $u_1 : S_1 \to LR$ such that

$$\ell(LX) \circ \ell(X) \circ J_{S_1} = \underline{a}J_R \circ \ell(LR) \circ u_1 = \ell(LX) \circ LJ_R \circ u_1. \tag{4.6.1}$$

Put $m = \ell(X) \circ J_{S_1}$ and $n = LJ_R \circ u_1$. The morphisms m and n are morphisms from S_1 to LX. Let $S_2 \hookrightarrow S_1$ be the kernel of the pair of arrows (m, n). For every object Y of C and every morphism $\alpha : Y \to S_1$, one has, by 4.6.1, $\ell(LX) \circ m \circ \alpha = \ell(LX) \circ n \circ \alpha$. It then follows from (3.1 3)) that the kernel $S_2 \times_{S_1} Y$ of the pair of arrows $(m \circ \alpha, n \circ \alpha)$ is a covering sieve of Y. Consequently, by axiom (T 2) for topologies, $J_{S_2} : S_2 \hookrightarrow X$ is a covering sieve of X. Thus there is a morphism $u_2 : S_2 \to LR$ and a commutative

diagram:

(4.6.2)



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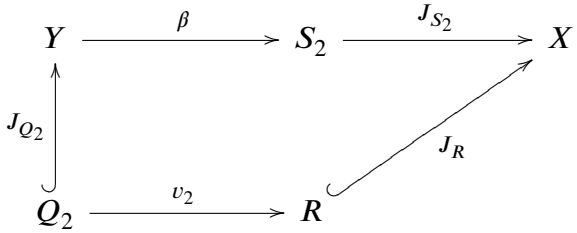
Let Y be an object of C and let $\beta : Y \to S_2$ be a morphism. By 3.1 2) and 1), there exist a covering sieve $J_{Q_1} : Q_1 \hookrightarrow Y$ and a morphism $v_1 : Q_1 \to R$ such that $u_2 \circ \beta \circ J_{Q_1} = \ell(R) \circ v_1$. Put $f = J_{S_2} \circ \beta \circ J_{Q_1}$ and $g = J_R \circ v_1$. Let Q_2 be the kernel of the pair $(f, g) : Q_1 \rightrightarrows X$, and let $J_{Q_2} : Q_2 \hookrightarrow Y$ be the canonical inclusion. For every object Z of C and every morphism $\gamma : Z \to Q_1$, the commutativity of (4.6.2) gives:

$$\ell(X) \circ f \circ \gamma = \ell(X) \circ J_{S_2} \circ \beta \circ J_{Q_1} \circ \gamma = LJ_R \circ u_2 \circ \beta \circ J_{Q_1} \circ \gamma =$$
$$= LJ_R \circ \ell(R) \circ v_1 \circ \gamma = \ell(X) \circ J_R \circ v_1 \circ \gamma = \ell(X) \circ g \circ \gamma.$$

It then follows from 3.1 3) that the kernel $Q_2 \times_{Q_1} Z$ of the pair $(f \circ \gamma, g \circ \gamma)$ is a covering sieve of Z. By axiom (T 2) for topologies, $J_{Q_2} : Q_2 \hookrightarrow Y$ is a covering sieve of Y. Thus, for every morphism $\beta : Y \to S_2$, there exist a covering sieve $J_{Q_2} : Q_2 \hookrightarrow Y$ and a morphism $v_2 : Q_2 \to R$ such that the following diagram is commutative: 242



Denote by $S_2 \cap R$ the fiber product of S_2 and R over X. The sieve (of Y) $(S_2 \cap R) \times_{S_2} Y$ contains the sieve Q_2, and hence $(S_2 \cap R) \times_{S_2} Y$ is a covering sieve of Y. It then follows from axiom (T 2) for topologies that $S_2 \cap R$ is a covering sieve of X, and consequently the sieve R, which contains $S_2 \cap R$, is a covering sieve of X, C.Q.F.D.

Corollary 4.4.4. *Let T be the topology of the $\mathcal{U}$-site C, and let T' (resp. T'') be the finest topology on C among those for which the objects of $\widetilde{C}$ are sheaves* (resp. separated presheaves) *(2.2). Then $T = T' = T''$.*

Indeed, one trivially has $T \leq T' \leq T''$, and 4.4 and 2.2 imply that $T'' \leq T$, C.Q.F.D.

Definition 4.5. *An initial object of a category A is an object ϕ_A that represents the empty inductive limit, i.e. such that, for every $X \in$ ob A, there exists a unique arrow $\phi_A \to X$. An object ϕ_A is called* strict initial *if it is initial and every morphism with target ϕ_A is an isomorphism. Let (S_i), $i \in I$, be a family of objects of a category A. Suppose that the sum $S = \coprod_{i \in I} S_i$ is representable. The sum S is called disjoint if the structure morphisms $S_i \to S$ are pullbackable, if they are monomorphisms, and if, for every pair i, j with $i \neq j$, the fiber product $S_i \times_S S_j$ is an initial object of A. The sum S is called disjoint and universal if it is disjoint and remains a disjoint sum after every base change $T \to S$; it follows that, for every pair i, j with $i \neq j$, the objects $S_i \times_S S_j$ are strict initial objects of A.* 243

EXAMPLE 4.5.1. In the category of sets, sums are disjoint and universal. The same is therefore true in every category of presheaves of sets (I 3.1), and hence in every category of sheaves of sets on a $\mathscr{U}$-site (4.1); in particular, the initial object of $\tilde{C}$ is strict.

PROPOSITION 4.6. *Let C be a $\mathscr{U}$-category and let $(s_i : X_i \to X)$, $i \in I$, be a family of morphisms of C. For every $\mathscr{U}$-topology $\mathscr{T}$ on C (3.0.2), denote by $\tilde{C}_{\mathscr{T}}$ the corresponding category of sheaves and by $\epsilon_{\mathscr{T}} : C \to \tilde{C}_{\mathscr{T}}$ the corresponding functor.*

1) *Let $\mathscr{T}$ be a $\mathscr{U}$-topology such that*

$$\coprod_{i\in I} \epsilon_{\mathscr{T}}(X_i) \xrightarrow{(\epsilon_{\mathscr{T}}(s_i))} \epsilon_{\mathscr{T}}(X)$$

is an isomorphism. Then, for every topology $\mathscr{T}'$ finer than $\mathscr{T}$, the morphism

$$\coprod_{i\in I} \epsilon_{\mathscr{T}'}(X_i) \xrightarrow{(\epsilon_{\mathscr{T}'}(s_i))} \epsilon_{\mathscr{T}'}(X)$$

is an isomorphism.

2) *Let $\mathscr{T}$ be a $\mathscr{U}$-topology on C. The following properties* (i) *and* (ii) *are equivalent:*

i) a) *The family $(s_i : X_i \to X)$, $i \in I$, is covering for $\mathscr{T}$.*

b) *For every $i \in I$, the diagonal morphism of presheaves $\Delta_i : X_i \hookrightarrow X_i \times_X X_i$ is sent by the associated sheaf functor (for $\mathscr{T}$) to an isomorphism (as* 244 *is the case if the s_i are monomorphisms).*

c) *For every pair (i, j) of elements of I with $i \neq j$, the presheaf $X_i \times_X X_j$ is sent by the associated sheaf functor (for $\mathscr{T}$) to the initial object of $\tilde{C}_{\mathscr{T}}$.*

ii) *$\epsilon_{\mathscr{T}}(X)$ is the sum of the $\epsilon_{\mathscr{T}}(X_i)$.*

PROOF. 1) A sheaf for $\mathscr{T}'$ is a sheaf for $\mathscr{T}$. Consequently, for every sheaf F for $\mathscr{T}'$, the morphism

$$\mathrm{Hom}_{\hat{C}}(X, F) \longrightarrow \prod_{i\in I} \mathrm{Hom}_{\hat{C}}(X_i, F)$$

is an isomorphism, which proves the assertion.

2) By definition, property ii) holds if and only if the morphism of presheaves

$$\Phi = (h(s_i), i \in I) : \coprod_i h(X_i) \longrightarrow h(X)$$

is sent by the associated sheaf functor to an isomorphism, i.e. (4.2) if and only if $\underline{a}(\Phi)$ is both an epimorphism and a monomorphism of sheaves. By (4.4), the morphism $\underline{a}(\Phi)$ is an epimorphism if and only if property a) holds. The diagonal morphism

$$\Delta : \coprod_i h(X_i) \longrightarrow (\coprod_i h(X_i)) \times_{h(X)} (\coprod_i h(X_i))$$

is the direct sum of a family $\Delta_{i,j}$, $(i, j) \in I \times I$, of morphisms defined as follows:

β) When $i = j$, $\Delta_{i,i}$ is the diagonal morphism $h(X_i) \to h(X_i) \times_{h(X)} h(X_i)$.

γ) When $i \neq j$, $\Delta_{i,j}$ is the morphism $\phi_{\hat{C}} \to h(X_i) \times_{h(X)} h(X_j)$ ($\phi_{\hat{C}}$ denotes the initial object of $\hat{C}$).

245 The morphism $\underline{a}(\Phi)$ is a monomorphism if and only if $\underline{a}(\Delta)$ is an isomorphism, and by (4.1) $\underline{a}(\Delta)$ is an isomorphism if and only if the $\underline{a}(\Delta_{i,j})$ are isomorphisms, i.e. if and only if properties b) and c) hold.

COROLLARY 4.6.1. *Let C be a $\mathcal{U}$-category and X an object of C.*

1) *Let $\mathcal{T}$ be a $\mathcal{U}$-topology on C. The object X of C is sent by the associated sheaf functor (for the topology $\mathcal{T}$) to the initial object of $\tilde{C}_{\mathcal{T}}$ if and only if the empty sieve covers X.*
2) *When X is a strict initial object of C, the sheaf associated with X, for every $\mathcal{U}$-topology finer than the canonical topology, is an initial object.*

PROOF. 1) Take the indexing set I in 4.6 2) to be the empty set.
2) By 1) and 4.6 1), it is enough to show that the empty sieve covers X for the canonical topology; i.e. by (2.6), it is enough to show that, for every object $Y \to X$ over X, Y is an initial object of C, which follows from the definition (4.5).

COROLLARY 4.6.2. *Let C be a $\mathcal{U}$-site and let $(s_i : X_i \to X)$, $i \in I$, be a family of pullbackable morphisms of C with the same target and having the following properties:*

α) The family of the s_i, $i \in I$, is covering.
β) For every $i \in I$, $s_i : X_i \to X$ is a monomorphism.
γ) For every pair (i, j) of elements of I with $i \neq j$, $X_i \times_X X_j$ is covered by the empty sieve, i.e. for every sheaf F on C, $F(X_i \times_X X_j)$ is a singleton set.

Then $\epsilon_C(X)$ is the sum of the $\epsilon_C(X_i)$, $i \in I$.

PROOF. This follows immediately from 4.6 2) and 4.6.1 1). 246

COROLLARY 4.6.3. *Let C be a $\mathcal{U}$-category and let $(s_i : X_i \to X)$, $i \in I$, be a family of pullbackable morphisms with the same target. Suppose that the canonical topology of C is a $\mathcal{U}$-topology. The following conditions are equivalent:*

i) *There exists a $\mathcal{U}$-topology $\mathcal{T}$ on C, coarser than the canonical topology, such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$, $i \in I$.*
ii) *The object X is the disjoint and universal sum (4.5) (in C) of the X_i.*

PROOF. ii) $\Rightarrow$ i). Since X is the universal sum of the X_i, $i \in I$, the family of the s_i, $i \in I$, is covering for the canonical topology of C (2.6). Condition α) of 4.6.2 is therefore satisfied when C is endowed with the canonical topology. Condition β) is evidently satisfied, and property γ) follows from 4.6 1) 2), whence i).
i) $\Rightarrow$ ii). Let $\mathcal{T}$ be a topology on C coarser than the canonical topology such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$. For every sheaf F for $\mathcal{T}$, the canonical map $\mathrm{Hom}(X, F) \to \prod_i \mathrm{Hom}(X_i, F)$ is a bijection. Moreover, every representable presheaf is a sheaf. It follows immediately that X is the sum in C of the X_i, $i \in I$. Applying this to the case where the set I is empty, one sees that if an object of C is carried to the initial object of $\tilde{C}$, that object is an initial object of C. The functor $\epsilon_{\mathcal{T}} : C \to \tilde{C}$ is fully faithful and commutes with finite projective limits. Consequently, condition b) of 4.6.2 implies that the s_i, $i \in I$, are monomorphisms of C, and condition c) implies, by what precedes, that the $X_i \times_X X_j$, $i \neq j$, are initial objects of C. Consequently X is the disjoint sum of the X_i. Since $\epsilon_{\mathcal{T}}$ commutes with fiber products, this last property is stable under every base 247
change $Y \to X$, and therefore X is the disjoint and universal sum of the X_i, $i \in I$.

COROLLARY 4.6.4. *Let C be a $\mathcal{U}$-category and X an object of C. Suppose that the canonical topology on C is a $\mathcal{U}$-topology. The following properties are equivalent:*

i) *There exists on C a topology coarser than the canonical topology such that $\epsilon_{\mathcal{T}}(X)$ is an initial object of $\tilde{C}_{\mathcal{T}}$.*
ii) *X is a strict initial object of C.*

PROOF. Take the empty set for the set I in 4.6.3.

PROPOSITION 4.7. *Let C be a $\mathscr{U}$-category, let $\tilde{C}$ be the category of sheaves on C for the canonical topology, let $\epsilon_C : C \to \tilde{C}$ be the canonical functor (4.4.0), and let $R \rightrightarrows X$ be an equivalence relation in C admitting a cokernel Y in C. Let $\Pi : X \to Y$ be the canonical morphism and suppose it is pullbackable. Consider the following properties:*

i) *$\epsilon_C(Y)$ is the quotient of the equivalence relation $\epsilon_C(R) \rightrightarrows \epsilon_C(X)$.*
ii) *The equivalence relation $R \rightrightarrows X$ is universally effective* (cf. n° 7).

One has ii) ⇒ i). *When C possesses a $\mathscr{U}$-topology coarser than the canonical topology, one has* i) ⇒ ii).

PROOF. ii) ⇒ i). The canonical morphism $R \to X \times_Y X$ is an isomorphism. Let $S \to Y$ be the sieve generated by $\pi : X \to Y$. For every presheaf F, there is an exact diagram (I 2.12):

$$\mathrm{Hom}(S, F) \to \mathrm{Hom}(X, F) \rightrightarrows \mathrm{Hom}(R, F).$$

248 It therefore suffices to show that, for every sheaf F for the canonical topology, the map

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(S, F)$$

is a bijection, i.e. it suffices to show that S is covering for the canonical topology, or equivalently that the sieve generated by $\pi : X \to Y$ is covering, which follows from Definition 2.5.
i) ⇒ ii). The functor $\epsilon_C : C \to \tilde{C}$ commutes with finite projective limits and is fully faithful. The canonical morphism $\epsilon_C(R) \to J_C(X) \times_{J_C(Y)} J_C(X)$ is an isomorphism (4.3). Consequently the canonical morphism $R \to X \times_Y X$ is an isomorphism. The morphism $\epsilon_C(X) \to \epsilon_C(Y)$ is an epimorphism, which implies (4.4) that $\pi : X \to Y$ is a covering morphism of C for the canonical topology, C.Q.F.D.

PROPOSITION 4.8. *Let C be a $\mathscr{U}$-site. The category of sheaves on C has the following properties:*

a) *Finite projective limits are representable.*
b) *Direct sums indexed by an element of $\mathscr{U}$ are representable. They are disjoint and universal (4.5).*
c) *Equivalence relations are universally effective.*

This was proved in 4.1, 2) and 3), 4.3, 3), and 4.5.1.

These properties have been singled out because they will later allow us to characterize the $\mathscr{U}$-categories equivalent to categories of sheaves on categories belonging to $\mathscr{U}$ (IV 1.2).

REMARK. Recall that property a) is equivalent to the property:
There exists a terminal object, and fiber products are representable (I 2.3.1).

249 **4.9.** Let C be a site whose underlying category is a $\mathscr{U}$-category. Then the category $\widetilde{C}$ of $\mathscr{U}$-sheaves on C satisfies the conditions considered in I 7.3 ((i) or (ii), as desired), and therefore, by *loc. cit.*, the various variants of the notion of a *generating family* in $\widetilde{C}$ considered in I 7.1 coincide. This being understood:

PROPOSITION 4.10. *With the preceding notation, consider the canonical functor $\epsilon : C \to \widetilde{C}$ (4.4.0), and let $G \subset \mathrm{ob}\, C$ be a topologically generating family in C (3.0.1). Then the family $(\epsilon(X))_{X \in G}$ of objects of $\widetilde{C}$ is a generating family. In particular, the family $(\epsilon(X))_{X \in \mathrm{ob}\, C}$ of objects of $\widetilde{C}$ is generating.*

We must prove that every morphism $u : F \to F'$ in $\widetilde{C}$ such that

$$\mathrm{Hom}(\epsilon(X), F) \longrightarrow \mathrm{Hom}(\epsilon(X), F') \tag{4.10.1}$$

is a bijection for every $X \in G$, is an isomorphism. The preceding map identifies with the map

$$u(X) : F(X) \longrightarrow F'(X), \tag{4.10.2}$$

and we must show that if the latter is bijective for every $X \in G$, then it is so for every $X \in \mathrm{ob}\, C$. Let $(X_i \to X)_{i\in I}$ be a covering family of X by objects of G, and, for every pair of indices (i, j) in I, consider the set of all morphisms $X_{ijk} \to X_i \times_X X_j$ in $\hat{C}$, with $X_{ijk} \in G$ (k varying in an index set $I_{i,j}$). One then obtains a morphism of exact diagrams of sets

$$\begin{array}{ccccc} F(X) & \longrightarrow & \prod_i F(X_i) & \rightrightarrows & \prod_{ijk} F(X_{ijk}) \\ \downarrow & & \downarrow\wr & & \downarrow\wr \\ F'(X) & \longrightarrow & \prod_i F'(X_i) & \rightrightarrows & \prod_{ijk} F'(X_{ijk}) \end{array} ,$$

where , by hypothesis, the last two vertical arrows are bijections. The same is therefore true of the first vertical arrow, C.Q.F.D. 250

Corollary 4.11. *Suppose that C is a $\mathcal{U}$-site (3.0.2). Then the category $\widetilde{C}$ is a $\mathcal{U}$-category and admits a $\mathcal{U}$-small generating family.*

Indeed, by hypothesis one may take for G in 4.10 a small generating family, which proves the existence of a small generating family. On the other hand, if $F, F' \in \mathrm{ob}\, \widetilde{C}$, a morphism from F to F' is known once one knows the morphism (4.10.1), i.e. (4.10.2), for every $X \in \mathrm{ob}\, G$ (I 7.1.1), whence the map

$$\mathrm{Hom}(F, F') \longrightarrow \prod_{X\in G} \mathrm{Hom}(F(X), F'(X))$$

is injective. Since the second member is $\mathcal{U}$-small, so is the first, which proves that $\widetilde{C}$ is a $\mathcal{U}$-category.

Corollary 4.12. *Let C be a $\mathcal{U}$-site. Then, for every object X of $\widetilde{C}$, the set of subobjects of X and the set of quotient objects of X are $\mathcal{U}$-small.*

This follows from I 7.4, respectively I 7.5, which apply by 4.11.

5. Extension of a topology from C to $\hat{C}$

Proposition 5.1. *Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $\hat{C}$. The following conditions are equivalent:*

i) *For every $X \to K$, with $X \in \mathrm{ob}\, C$, the image of the corresponding morphism $H \times_K X \to X$ is a covering sieve of X.*

ii) *The morphism $\underline{a}(f) : \underline{a}H \to \underline{a}K$ between the associated sheaves is an epimorphism in $\tilde{C}$.* 251

ii bis) *For every sheaf F on C, the map $\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$ induced by f is injective.*

Proof. It is clear that ii) is equivalent to ii bis).

Let us prove that (ii) ⇒ (i). By 4.4, (i) means that $u : \underline{a}(H \times_K X) \to \underline{a}(X)$ is an epimorphism. Now, since $\underline{a}(H \times_K X) \simeq \underline{a}(H) \times_{\underline{a}(K)} \underline{a}(X)$, because $\underline{a}$ commutes with finite $\varprojlim$ (4.1), the morphism in question is obtained from $\underline{a}(f) : \underline{a}(H) \to \underline{a}(K)$ by base change. Since the epimorphisms of $\widetilde{C}$ are universal, this proves (ii) ⇒ (i). Conversely, since the family of all $X_i \to K$, with $X_i \in \mathrm{ob}\, C$, is epimorphic, the same is true of the family of $\underline{a}(X_i) \to \underline{a}(K)$ in $\widetilde{C}$ (4.1). Hence, to verify that $\underline{a}(f) : \underline{a}(H) \to \underline{a}(K)$ is epimorphic, it suffices to do so after every base change of the preceding kind $\underline{a}(X_i) \to \underline{a}(K)$, which proves i) ⇒ ii), C.Q.F.D.

Definition 5.2. 1) *A morphism $f : H \to K$ satisfying the three equivalent conditions of 5.1 is called a* covering morphism. *A family of morphisms $f_i : H_i \to K$, $i \in I$, with the same target is called* covering *if the corresponding morphism $f : \coprod_i H_i \to K$ is covering.*

2) *A morphism $f : H \to K$ is called* bicovering *if it is covering and if the diagonal morphism $H \to H \times_K H$ is covering. A family $f_i : H_i \to K$, $i \in I$, with the same target is called bicovering if the corresponding morphism $f : \coprod_i H_i \to K$ is bicovering.*

By condition i) of 5.1, to say that a family $(f_i : H_i \to K)$, $i \in I$, is covering means that, for every morphism $X \to K$, with $X \in \mathrm{ob}\, C$, the image of the family $H_i \times_K X \to X$,
252 $i \in I$, is a covering sieve of X; or equivalently, by ii), that the family of morphisms $\underline{a}(f_i) : \underline{a}H_i \to \underline{a}K$ in $\tilde{C}$ is epimorphic (taking into account that the functor $\underline{a}$ commutes with direct sums (4.1)).[4]

Proposition 5.3. *Let C be a $\mathscr{U}$-site and let $f : H \to K$ be a morphism of $\hat{C}$. The following conditions are equivalent:*

i) *The morphism f is bicovering (5.2).*

i bis) *The morphism f is covering and, for every object X of C and every pair of morphisms $u, v : X \rightrightarrows H$ such that $fu = fv$, the kernel of (u, v) is a covering sieve of X.*

ii) *The morphism $\underline{a}(f) : \underline{a}H \to \underline{a}K$ is an isomorphism in $\tilde{C}$.*

ii bis) *For every sheaf F on C, the map $\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$ is a bijection.*

Proof. The equivalence i) ⇔ i bis) follows from condition i) of 5.1, applied to the diagonal morphism $H \to H \times_K H$; the equivalence ii) ⇔ ii bis) is trivial.
i) ⇒ ii). The morphism $\underline{a}(f) : \underline{a}H \to \underline{a}K$ is an epimorphism (5.1). Since the functor $\underline{a}$ commutes with the formation of fiber products (4.1), the diagonal morphism $\underline{a}H \to \underline{a}H \times_{\underline{a}K} \underline{a}H$ is an epimorphism (5.1). Since a diagonal morphism is always a monomorphism, it is an isomorphism (4.2). It follows that the morphism $\underline{a}(f)$ is a monomorphism. It is therefore an isomorphism (4.2).
ii) ⇒ i). Since the functor $\underline{a}$ commutes with the formation of fiber products, the morphism f and the diagonal morphism $H \to H \times_K H$ are carried by $\underline{a}$ to isomorphisms. In particular, they are carried by $\underline{a}$ to epimorphisms. They are therefore covering, C.Q.F.D.

253 5.3.1. It follows from 5.3, and from the fact that $\underline{a}$ commutes with direct sums, that a family $(f_i : H_i \to K)$, $i \in I$, of morphisms of $\hat{C}$ is bicovering if and only if the f_i induce an isomorphism from the direct sum $\coprod_i \underline{a}H_i$ onto $\underline{a}K$ on the associated sheaves,

[4] Note that the property of a morphism or a family of morphisms of $\hat{C}$ being covering (resp. bicovering) is stable under base change.

or equivalently if and only if, for every sheaf F, the map

$$\mathrm{Hom}(K, F) \longrightarrow \prod_i \mathrm{Hom}(H_i, F)$$

is bijective.

PROPOSITION 5.4. *Let C be a $\mathscr{U}$-site. There exists a (necessarily unique) topology on $\hat{C}$ such that a family $H_i \to K$ of arrows of $\hat{C}$ with the same target is covering for this topology if and only if it is covering in the sense of 5.2. It is also the coarsest topology on $\hat{C}$ among the topologies T having the following properties:*

a) *T is finer than the canonical topology of $\hat{C}$ (i.e. every epimorphic family in $\hat{C}$ is covering for T).*
b) *Every covering family in C is covering in $\hat{C}$.*

PROOF. We shall confine ourselves to giving indications. We leave it to the reader to show that the families which are covering in the sense of 5.2 are the covering families of a topology T_C on $\hat{C}$ (one uses 4.1). The covering families of the canonical topology on $\hat{C}$ are the epimorphic families in $\hat{C}$ (2.6 and I 3.1). Since $\underline{a}$ commutes with inductive limits (4.1), the topology T_C is finer than the canonical topology of $\hat{C}$. Moreover, the covering families of C are covering families for T_C (5.1). Thus, if T' denotes the coarsest topology on $\hat{C}$ having properties a) and b), T_C is finer than T'. Let $(f_i : H_i \to K)$, $i \in I$, be a covering family for T_C. We show that $(f_i, i \in I)$ is a covering family for T'. 254
Let $s_i : H_i \to H = \coprod_i H_i$ be the canonical monomorphisms, and let $f = (f_i, i \in I) : H = \coprod_i H_i \to K$ be the morphism defined by the f_i. The family of the s_i is covering for T'. Thus, by axiom (T 2) for topologies, to show that $(f_i, i \in I)$ is a covering family for T', it suffices to show that the morphism $f : H \to K$ is covering for T'. There exists an epimorphic family in $\hat{C}$, $u_\lambda : X_\lambda \to K$, $\lambda \in \Lambda$, with $X_\lambda \in \mathrm{ob}\, C$ (I 3.4). Therefore, again by axiom (T 2) for topologies, to show that $f : H \to K$ is covering for T', it suffices to show that, for every $\lambda \in \Lambda$, the morphism $\mathrm{pr}_2 : H \times_K X_\lambda \to X_\lambda$ is covering for T'. Let $v_j : Y_j \to H \times_K X_\lambda$, $j \in J$, with $Y_j \in \mathrm{ob}\, C$, be an epimorphic family in $\hat{C}$. The family $(\mathrm{pr}_2 \circ v_j, j \in J)$ is covering for T_C. It is therefore a covering family in C (5.1), and hence a covering family for T'. Consequently, the sieve generated by $\mathrm{pr}_2 : H \times_K X_\lambda \to X_\lambda$ contains a covering sieve for T'. It is therefore covering for T', and hence $\mathrm{pr}_2 : H \times_K X_\lambda \to X_\lambda$ is covering, C.Q.F.D.

REMARK 5.4.1. The proof of 5.4 in fact shows that every topology T' on $\hat{C}$ which is finer than the canonical topology of $\hat{C}$ is the coarsest of the topologies T on $\hat{C}$ having the following properties:

a) T is finer than the canonical topology of $\hat{C}$.
b) Every family covering for T' of the form $u_i : X_i \to X$, $i \in I$, where X and the X_i are objects of C, is covering for T.

In particular, every topology T' on $\hat{C}$ which is finer than the canonical topology is uniquely determined by the families $u_i : X_i \to X$, $i \in I$, with X_i and X objects of C, which are covering for T'.

REMARK 5.4.2. It is easy to show that, for every topology T' on $\hat{C}$ which is finer 255
than the canonical topology, the set of families of morphisms $(X_i \to X)$, $i \in I$, with the same target in C, which are covering for T' is the set of covering families of a topology on C. Thus 5.4 and 5.4.1 establish a one-to-one correspondence between the topologies on C and the topologies on $\hat{C}$ finer than the canonical topology.

5.5.0. Let C be a small category. Denote by Caf the set of strictly full subcategories (every object isomorphic to an object of the subcategory is an object of the subcategory) of $\hat{C}$ whose inclusion functor admits a left adjoint which commutes with finite projective limits. Also denote by $\mathscr{T}$ the set of topologies on C. Theorem 3.4 defines a map

$$\Phi : \mathscr{T} \longrightarrow \text{Caf}\, quoi.$$

We shall define a map in the opposite direction. For this purpose, to every element $e = i' : C' \underset{\longrightarrow}{\overset{a'}{\leftarrow}} \hat{C}$ of Caf, with $\underline{a}'$ left adjoint to i', we must associate a topology T_e on C. For every object X of C, define $J_e(X)$ to be the set of subobjects of X whose inclusion morphism is carried by $\underline{a}'$ to an isomorphism. Using the hypotheses made on $\underline{a}'$, one verifies immediately that this defines a topology T_e on C. We have therefore defined a map

$$\Psi : \text{Caf} \longrightarrow \mathscr{T}\, quoi.$$

We then have the following result (due to J. GIRAUD):

256 THEOREM 5.5. *The map Φ is a bijection, and Ψ is its inverse.*

PROOF. The map $\Psi \circ \Phi$ is the identity. Indeed, this follows immediately from 5.1. The map $\Phi \circ \Psi$ is the identity. Indeed, let $i' : C' \underset{\longrightarrow}{\overset{\underline{a}'}{\leftarrow}} \hat{C}$ be an element of Caf, let T_e be the topology corresponding to it under Ψ, and let C_e be the category of sheaves for T_e. Using the definition of the topology T_e and the definition of bicovering morphisms (5.2), one proves that the following assertions are equivalent:

i) The morphism u of $\hat{C}$ is bicovering for the topology T_e.
ii) The morphism u of $\hat{C}$ is carried by $\underline{a}'$ to an isomorphism.

It is clear that C' is a full subcategory of C_e. It therefore suffices to show that every sheaf F for the topology T_e is an object of C'. But, by the equivalence above, the morphism $F \to i' \circ a'(F)$ is bicovering, and since its source and target are sheaves, it follows from 5.3 (ii) that $F \to i' \circ a'(F)$ is an isomorphism, C.Q.F.D.

6. Sheaves with values in a category

6.0. Let C and D be two categories. A contravariant functor from C to D, $F : C^\circ \to D$, is called a *presheaf on C with values in D*.

DEFINITION 6.1. *Let C be a site and D a category. A presheaf $F : C^\circ \to D$ is called a* sheaf on C with values in D *(or, more briefly, a sheaf with values in D) if, for every object S of D, the presheaf of sets*

$$X \longmapsto \text{Hom}_D(S, F(X)) \quad , \quad X \in \text{ob}(C)$$

is a sheaf. The full subcategory of $\mathbf{Hom}(C^\circ, D)$ *formed by the sheaves on C with values in D will be denoted by* $\mathbf{Hom}^\sim(C^\circ, D)$.

257 REMARKS 6.2. 1) The condition of 6.1 means that for every object X of C, every covering sieve R of X, and every object S of D, one has an isomorphism (I 3.5 and 2.1):

$$\text{Hom}_D(S, F(X)) \xrightarrow{\sim} \varprojlim_{C/R} F(.).$$

Thus, when D is the category of $\mathscr{U}$-sets, one recovers the definition of sheaves of sets (*loc. cit.*).

2) Let D' be a $\mathcal{U}$-category and let $G : D \to D'$ be a functor commuting with projective limits. By composition, the functor G carries sheaves with values in D to sheaves with values in D'.

6.3.0. Let γ be a type of algebraic structure defined by finite projective limits (I 2.9), and let $\mathcal{U} - \gamma$ be the category of γ-sets belonging to $\mathcal{U}$. Denote by $\mathrm{esj} : \mathcal{U} - \gamma \to \mathcal{U} - \mathrm{Ens}$ the "underlying set" functor. The functor esj commutes with projective limits and therefore, by the preceding remark, defines by composition the functors

$$\hat{\mathrm{esj}} : \mathbf{Hom}(C^\circ, \mathcal{U} - \gamma) \longrightarrow \hat{C}$$
$$\tilde{\mathrm{esj}} : \mathbf{Hom}^\sim(C^\circ, \mathcal{U} - \gamma) \longrightarrow \tilde{C}.$$

The functor $\tilde{\mathrm{esj}}$ is called the "underlying sheaf of sets" functor. In fact, it factors canonically through the category $\gamma - \widetilde{C}$ of γ-objects of $\tilde{C}$. Denoting again by $\widetilde{\mathrm{esj}}$ the resulting functor

$$\mathbf{Hom}^\sim(C^\circ, \mathcal{U} - \gamma) \longrightarrow \gamma - \widetilde{C},$$

we may state:

Proposition 6.3.1. *The functor* esj$\tilde{}$ *establishes an equivalence among the category of sheaves on C with values in $\mathcal{U} - \gamma$, the category of γ-objects in the category of sheaves of sets, and the category of γ-objects in the category of presheaves of sets whose underlying presheaf is a sheaf.*

Proof. The proof essentially uses (I 3.2) and the fact that a finite projective limit of 258
sheaves in the category of presheaves is a sheaf (4.1).

6.3.2. The preceding proposition justifies the abuse of language whereby a sheaf with values in $\mathcal{U} - \gamma$ is identified with the corresponding γ-object of $\tilde{C}$. We shall henceforth make this abuse of language systematically.

We shall use the classical terminology: *sheaves of groups, sheaves of commutative groups* (which we shall most often call *abelian sheaves*), *sheaves of rings, sheaves of A-modules* etc.

6.3.3. We denote by

$$\tilde{C}_\gamma (resp.\ \hat{C}_\gamma)$$

the category of γ-objects of $\tilde{C}$ (resp. of $\hat{C}$). When γ is the type of structure "A-module", we also write $\tilde{C}_A$, or simply $\widetilde{C}_{\mathrm{ab}}$ when $A = \mathbf{Z}$.

Suppose that C is a $\mathcal{U}$-site. The "associated sheaf" functor is left exact (4.1), and therefore:

Proposition 6.4. *The inclusion functor $i_\gamma : \tilde{C}_\gamma \to \hat{C}_\gamma$ admits a left adjoint a_γ which is left exact; that is, one has an isomorphism*

$$\mathrm{Hom}_{\tilde{C}_\gamma}(a_\gamma, \ ) \xrightarrow{\sim} \mathrm{Hom}_{\hat{C}_\gamma}(\ , i_\gamma).$$

Let X be an object of $\hat{C}_\gamma$. The underlying sheaf of sets of $a_\gamma(X)$ is canonically isomorphic to the sheaf of sets associated with the underlying presheaf of sets of X. The morphism of underlying presheaves of sets of the adjunction morphism $\mathrm{id} \to i_\gamma a_\gamma$ *is identified with the adjunction morphism applied to the underlying presheaf of sets.*

Proposition 6.5. *Suppose that the functor* esj $: \gamma - \mathcal{U} \to \mathcal{U} - \mathrm{Ens}$ *admits a left* 259

adjoint [4]*:*

$$\mathrm{Hom}_{\mathscr{U}-\mathrm{Ens}}(.,\mathrm{esj}(.)) \xrightarrow{\sim} \mathrm{Hom}_{\gamma-\mathscr{U}}(\mathrm{Lib}(.),.).$$

The functor $\mathrm{esj}^{\sim} : C_{\gamma}^{\sim} \to C^{\sim}$ *(resp.* $\mathrm{esj}^{\wedge} : C_{\gamma}^{\wedge} \to C^{\wedge}$*) admits a left adjoint* $\mathrm{Lib}^{\sim}$ *(resp.* $\mathrm{Lib}^{\wedge}$*), and one has a canonical isomorphism*

$$\mathrm{Lib}^{\sim} = a_{\gamma}\,\mathrm{Lib}^{\wedge}\,i.$$

Proof. The proof is formal and is left to the reader.

Corollary 6.6. *Let* (X_i)*,* $i \in I$*, be a generating family of* $C^{\sim}$ *(4.9). Then the family* $\mathrm{Lib}^{\sim}(X_i)$ *is a generating family of* $C_{\gamma}^{\sim}$*.*

Proof. The proof is formal once one observes that the functor $\mathrm{esj}^{\sim}$ commutes with projective limits and is conservative ($\mathrm{esj}^{\sim}(u)$ is an isomorphism $\Leftrightarrow$ u is an isomorphism).

Proposition 6.7. *Let* A *be a* $\mathscr{U}$*-sheaf of rings, or a small ring, and let* $C_A^{\sim}$ *(resp.* $C_A^{\wedge}$*) be the category of sheaves (resp. of presheaves) of unital* A*-modules* (6.3.3) *on a* $\mathscr{U}$*-site* C*. Then* $C_A^{\sim}$ *is an abelian* $\mathscr{U}$*-category satisfying the axioms* (AB 5) *("existence of left exact filtered inductive limits") and* (AB 3)* *("existence of infinite products") of* [Tohoku]*. It has a family of generators indexed by an element of* $\mathscr{U}$*.*

Proof. It is clear that the category $C_A^{\wedge}$ is an abelian category satisfying the axioms
260 (AB 3), (AB 4), and (AB 5) (I 2.8 and I 3.3). It then follows from 6.4 that the category $C_A^{\sim}$ is an additive category in which kernels and cokernels are representable. More precisely, let i_A and a_A be the inclusion and associated-sheaf functors, respectively, and let $u : X \to Y$ be a morphism of $C_A^{\sim}$. The functor a_A is left exact and commutes with inductive limits. The functor i_A commutes with projective limits. Consequently the functor $i_A a_A : C_A^{\wedge} \to C_A^{\wedge}$ is an additive left exact functor. One deduces canonical isomorphisms

$$(*) \qquad i_A(\mathrm{Ker}(u)) \xrightarrow{\sim} \mathrm{Ker}(i_A(u)),$$

$$(**) \qquad \mathrm{coker}(u) \xrightarrow{\sim} a_A\,\mathrm{coker}(i_A(u)).$$

Consequently there is a canonical isomorphism

$$(***) \qquad \mathrm{coim}(u) \xrightarrow{\sim} a_A\,\mathrm{coim}(i_A(u)).$$

Let us show that the canonical morphism $\mathrm{coim}(u) \to \mathrm{im}(u)$ is an isomorphism. For this, by (*), it suffices to show that the sequence

$$0 \longrightarrow i_A(\mathrm{coim}(u)) \longrightarrow i_A(Y) \longrightarrow i_A(\mathrm{coker}(u))$$

is exact. Using the isomorphisms (**) and (***), and observing that $a_A i_A(Y)$ is isomorphic to Y, one sees that this sequence is isomorphic to the sequence

$$0 \longrightarrow i_A a_A(\mathrm{coim}(i_A(u))) \longrightarrow i_A a_A i_A(Y) \longrightarrow i_A a_A(\mathrm{coker}(i_A(u))),$$

which is precisely the image under the functor $i_A a_A$ of the sequence

$$0 \longrightarrow \mathrm{coim}(i_A(u)) \longrightarrow i_A(Y) \longrightarrow \mathrm{coker}(i_A(u)).$$

This sequence is exact, and the functor $i_A a_A$ is left exact. Thus the category $C_A^{\sim}$ is an
261 abelian category. The functor $a_A : C_A^{\wedge} \to C_A^{\sim}$ is left exact and commutes with arbitrary

[4] One proves that such an adjoint always exists [C.F.]: free group, free abelian group, free A-module etc.

inductive limits. Consequently it is additive, exact, and commutes with inductive limits. Since the category $\hat{C}_A$ satisfies axiom (AB 5), the same is true of the category $\tilde{C}_A$. Finally, it is clear that in the category $\tilde{C}_A$, products indexed by an element of $\mathscr{U}$ are representable, and hence axiom (AB 3)* is satisfied. This completes the proof of the first assertion.

For every ring A which is an element of $\mathscr{U}$, the functor

$$\text{esj} : \mathscr{U} - A - \text{module} \longrightarrow \mathscr{U} - \text{Ens}$$

admits a left adjoint $X \longmapsto X_A$ (the free A-module generated by X). It therefore suffices to apply 4.10 and 6.6 to complete the proof.

NOTATION 6.8. Let H be a sheaf of sets. The sheaf of A-modules associated with the presheaf (cf. notation 6.5)

$$X \longmapsto \text{Lib}_A\, H(X) \qquad X \in \text{ob}\, C$$

is denoted by A_H. When $H = \underline{a}(X)$ (the sheaf associated with the presheaf represented by X), one sometimes simply writes A_X (by abuse of notation). The proof of 6.7 shows that the family of sheaves of A-modules A_X, $(X \in \text{ob}(C))$, is a generating family, indexed by an element of $\mathscr{U}$, of the category of sheaves of A-modules.

REMARK 6.9. According to [TÔHOKU], 6.7 shows that the category $\tilde{C}_A$, when A is an element of $\mathscr{U}$, has enough injectives. It is also known that infinite products are not necessarily exact in $\tilde{C}_A$, and therefore that, in general, the category $\tilde{C}_A$ does not have enough projectives [ROOS].

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EXPOSÉ III

Functoriality of categories of sheaves

J.-L. Verdier

In I 5, we studied the behavior of categories of presheaves with respect to functors between the categories of arguments. In this exposé, we extend this study to the case of sites and categories of sheaves (n° 1 and 2). After introducing the induced topology (n° 3), we turn in n° 4 to the comparison lemma, which will play an important role in Exp. IV. In number 5, for the patient reader, we study certain commutative diagrams related to localized categories (of the form C/X). In Theorem I 5.1, we impose smallness hypotheses on the categories under consideration. In general, these hypotheses are not satisfied by the sites encountered in practice ($\mathscr{U}$-sites). This necessitates a few contortions (4.2, 4.3, 4.4). 263

1. Continuous functors

DEFINITION 1.1. *Let C and C' be two $\mathscr{U}$-sites and let $u : C \to C'$ be a functor between their underlying categories. We say that u is continuous if, for every sheaf of sets F on C', the presheaf $X \mapsto F \circ u(X)$ on C is a sheaf.*

This notion of continuity depends a priori on the universe $\mathscr{U}$ with respect to which the two sites are $\mathscr{U}$-sites. Proposition 1.5 shows that it is independent of it.

1.11. Denote by $i : C^\sim \to \hat{C}$ (resp. by $i' : C'^\sim \to \hat{C'}$) the canonical inclusion functor from sheaves into presheaves. By Definition 1.1, the functor u is continuous if and only if there exists a functor $u_s : C'^\sim \to C^\sim$ such that $iu_s = u^* i'$ (where $u^* F = F \circ u$ (I 5.0)). 264

PROPOSITION 1.2. *Let C be a small site, C' a $\mathscr{U}$-site, and $u : C \to C'$ a functor between their underlying categories. The following properties are equivalent:*

i) *The functor u is continuous.*
ii) *For every object X of C and every sieve $R \hookrightarrow X$ covering X, the morphism $u_! R \hookrightarrow u(X)$ is bicovering in $\hat{C'}$* (II 5.2 and I 5.1). [4]
iii) *For every bicovering family $H_i \to K$, $i \in I$, in $\hat{C}$, the family of morphisms $u_! H_i \to u_! K$, $i \in I$, is bicovering in $\hat{C'}$.*
iv) *There exists a functor $u^s : C^\sim \to C'^\sim$, commuting with inductive limits and "extending u", i.e. such that the diagram of canonical functors*

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle \epsilon_C}\downarrow & & \downarrow{\scriptstyle \epsilon_{C'}} \\ C^\sim & \xrightarrow{u^s} & C'^\sim, \end{array}$$

commutes up to isomorphism.

[4] "Covering" in place of "bicovering" did not necessarily suffice; cf. Example 1.9.3 below.

Moreover, when C is no longer necessarily a small site but is a $\mathscr{U}$-site, one still has i) $\Leftrightarrow$ iv), *and the functor u^s of* iv) *is necessarily a left adjoint of the functor $u_s : \widetilde{C'} \to \widetilde{C}$, and is consequently determined up to a unique isomorphism.*

265 PROOF. The proof of i) $\Leftrightarrow$ iv) when C is a $\mathscr{U}$-site will be given in 4.2. Assume that C is small. It is clear that iii) $\Rightarrow$ ii).

i) $\Rightarrow$ iii): Let $H_i \to K$, $i \in I$, be a bicovering family in $\hat{C}$. For every sheaf F on C', the presheaf u^*F (I 5.0) is a sheaf on C. Hence (II 5.3) the canonical map

$$\mathrm{Hom}(K, u^*F) \longrightarrow \prod_i \mathrm{Hom}(H_i, u^*F),$$

is bijective. By adjunction (I 5.1), it follows that the canonical map

$$\mathrm{Hom}(u_!K, F) \longrightarrow \prod_i \mathrm{Hom}(u_!H_i, F)$$

is bijective. Consequently (II 5.3) the family $u_!H_i \to u_!K$, $i \in I$, is bicovering.

ii) $\Rightarrow$ i): Let X be an object of C, let $R \hookrightarrow X$ be a covering sieve, and let F be a sheaf on C'. By (II 5.3), the map

$$\mathrm{Hom}(u_!X, F) \longrightarrow \mathrm{Hom}(u_!R, F)$$

is bijective. By adjunction (I 5.1), it follows that the map

$$\mathrm{Hom}(X, u^*F) \longrightarrow \mathrm{Hom}(R, u^*F)$$

is bijective. Consequently u^*F is a sheaf.

i) $\Rightarrow$ iv): We use the customary notation $\underline{a}$ and i (resp. $\underline{a}'$ and i') for the "associated sheaf" and "inclusion into presheaves" functors. For every sheaf G on C, set $u^sG = \underline{a}'u_!iG$. For every sheaf F on C', we have the following sequence of natural isomorphisms:

$$\begin{aligned}\mathrm{Hom}(G, u_sF) &\simeq \mathrm{Hom}(iG, iu_sF) \simeq \mathrm{Hom}(iG, u^*i'F)\\ &\simeq \mathrm{Hom}(u_!iG, i'F) \simeq \mathrm{Hom}(\underline{a}'u_!iG, F).\end{aligned}$$

266 (The first isomorphism follows from the full faithfulness of i; the second follows from the definition of u_s; the third and fourth are obtained by adjunction.) Consequently u^s is left adjoint to u_s. In particular, u^s commutes with inductive limits. For every presheaf K on C and every sheaf F on C', we have the following sequence of natural isomorphisms: $\mathrm{Hom}(u^s\underline{a}K, F) \simeq \mathrm{Hom}(\underline{a}K, u_sF) \simeq \mathrm{Hom}(K, iu_sF) \simeq \mathrm{Hom}(K, u^*i'F) \simeq \mathrm{Hom}(u_!K, i'F) \simeq \mathrm{Hom}(\underline{a}'u_!K, F)$ (the third isomorphism follows from the definition of u_s; the others are obtained by adjunction); whence a functorial isomorphism $u^s\underline{a}K \simeq \underline{a}'u_!K$. In particular, using this isomorphism when K is representable, and taking I 5.4 3) into account, we obtain a diagram commutative up to isomorphism:

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle \epsilon_C}\downarrow & & \downarrow{\scriptstyle \epsilon_{C'}} \\ \widetilde{C} & \xrightarrow{u^S} & \widetilde{C'} \end{array}$$

iv) ⇒ ii): Let $h : C \to \hat{C}$ (resp. $h' : C' \to \hat{C'}$) be the functor that assigns to an object the presheaf it represents. Consider the diagram:

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h'} \\ \hat{C} & \xrightarrow{u_!} & \hat{C'} \\ {\scriptstyle \underline{a}}\downarrow & & \downarrow{\scriptstyle \underline{a}'} \\ \tilde{C} & \xrightarrow{u^S} & \tilde{C'}. \end{array}$$

The upper square is commutative (1.4 3)), and we have $\underline{a}h = \epsilon_C$, $\underline{a}'h' = \epsilon_{C'}$. 267
For every object K of $\hat{C}$, we have a canonical isomorphism (I 3.4):

$$\varinjlim_{(h(X)\to K)\in \mathrm{ob}(C/K)} h(X) \xrightarrow{\sim} K.$$

Since the functors $u_!$, u^s, $\underline{a}$, and $\underline{a}'$ commute with inductive limits, we obtain isomorphisms:

$$\varinjlim_{(h(X)\to K)\in \mathrm{ob}(C/K)} u^s\underline{a}h(X) \xrightarrow{\sim} u^s\underline{a}K,$$

$$\varinjlim_{(h(X)\to K)\in \mathrm{ob}(C/K)} \underline{a}'u_!h(X) \xrightarrow{\sim} \underline{a}'u_!K.$$

We have $u_!h = h'u$ and, by hypothesis, an isomorphism $\underline{a}'h'u \simeq u^s\underline{a}h$, whence an isomorphism

$$u^s\underline{a}K \simeq \underline{a}'u_!K,$$

which is immediately seen to be functorial in K. The diagram

$$\begin{array}{ccc} \hat{C} & \xrightarrow{u_!} & \hat{C'} \\ {\scriptstyle \underline{a}}\downarrow & & \downarrow{\scriptstyle \underline{a}'} \\ \tilde{C} & \xrightarrow{u^S} & C', \end{array}$$

is therefore commutative up to isomorphism. Since $\underline{a}i$ is isomorphic to the identity functor, we have an isomorphism $u^s \simeq \underline{a}'u_!i$, whence the uniqueness of u^s. Let $v : H \to K$ be a bicovering morphism in $\hat{C}$. Then $\underline{a}(v)$ is an isomorphism (II 5.3), and consequently $\underline{a}'u_!(v)$ (isomorphic to $u^s\underline{a}(v)$) is an isomorphism. Thus $u_!(v)$ is carried by $\underline{a}'$ to an isomorphism, and consequently $u_!(v)$ is bicovering (II 5.3), whence ii).

The additional assertion, in the case where C is small, was proved in the course of 268
the proof, C.Q.F.D.

1.2.1. The functor $u^s : \tilde{C} \to \tilde{C'}$ of 2.2 iv) may often be interpreted as an "inverse image" functor for a "morphism of topoi" $\tilde{C'} \to \tilde{C}$, cf. IV 4.9. Its properties are summarized in the following proposition:

Proposition 1.3. *Let $u : C \to C'$ be a continuous functor from a small site C to a U-site C'.*

1) *The functor u^s is left adjoint to the functor u_s.*

2) *There is a canonical isomorphism* $u^s \simeq \underline{a}' u_! i$.
3) *There is a canonical isomorphism* $u^s \underline{a} \simeq \underline{a}' u_!$.
4) *The functor* u^s *commutes with inductive limits.*
5) *When the functor* $u_!$ *is left exact (cf. I 5.4 4)), the functor* u^s *is also left exact. More generally, the functor* u^s *commutes with those types of finite projective limits with which the functor* $u_!$ *commutes.*

Proof. Assertions 1), 2), 3), and 4) were proved in the course of the proof of 1.2. Assertion 5) follows from the isomorphism in 2), since i commutes with projective limits and $\underline{a}'$ commutes with finite projective limits (II 4.1).

Remark 1.4. Combining I 5.4 3) and 1.3 3), we obtain a diagram:

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h'} \\ C\hat{} & \xrightarrow{u_!} & C'\hat{} \\ {\scriptstyle \underline{a}}\downarrow & & \downarrow{\scriptstyle \underline{a}'} \\ C\tilde{} & \longrightarrow & C'\tilde{}, \end{array}$$

269 where the upper square commutes and the lower square commutes up to isomorphism. But the functors $\underline{a}$, $\underline{a}'$, $u_!$, and u^s are defined only up to isomorphism. The reader may verify that the functors $\underline{a}$, $\underline{a}'$, and u^s can be chosen so that:

1) The composite functors $\underline{a}h$ and $\underline{a}'h'$ are injective on the sets of objects;
2) the diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle \underline{a}h}\downarrow & & \downarrow{\scriptstyle \underline{a}'h'} \\ C\tilde{} & \xrightarrow{u^s} & C'\tilde{} \end{array}$$

commutes.

More precisely, the functors $\underline{a}$ and $\underline{a}'$ can be chosen so as to satisfy condition 1). Once the functors $\underline{a}$ and $\underline{a}'$ have been chosen, the functor u^s can be chosen so as to satisfy condition 2).

Proposition 1.5. *Let* $\mathscr{U} \subset \mathscr{V}$ *be two universes, let* C *and* C' *be two* $\mathscr{U}$*-sites, and let* $u : C \to C'$ *be a functor between the underlying categories. Then* u *is continuous relative to* $\mathscr{U}$ *if and only if it is continuous relative to* $\mathscr{V}$*. Moreover, when* u *is continuous, denote by* $u^s_{\mathscr{U}}$ *(resp.* $u^s_{\mathscr{V}}$*) the functor between the categories of* $\mathscr{U}$*-sheaves (resp.* $\mathscr{V}$*-sheaves) introduced in 1.2* iv)*. Then the diagram*

(1.5.1)
$$\begin{array}{ccc} \widetilde{C}_{\mathscr{U}} & \xrightarrow{u^s_{\underline{U}}} & \widetilde{C'}_{\underline{U}} \\ \downarrow & & \downarrow \\ C\tilde{}_{\underline{V}} & \xrightarrow{u^s_{\underline{V}}} & C'\tilde{}_{\underline{V}}, \end{array}$$

270 *where the vertical functors are the canonical inclusion functors, commutes up to canonical isomorphism.*

PROOF. We give the proof only in the case where C is $\mathscr{U}$-small. The general case will be proved in 4.3. It is clear that if the functor u^* carries every $\mathscr{V}$-sheaf on C' to a $\mathscr{V}$-sheaf on C, then it carries every $\mathscr{U}$-sheaf on C' to a $\mathscr{U}$-sheaf on C. Suppose that u^* carries every $\mathscr{U}$-sheaf on C' to a $\mathscr{U}$-sheaf on C. Denote by $\hat{C}_{\mathscr{U}} \hookrightarrow \hat{C}_{\mathscr{V}}$ (resp. $\hat{C'}_{\mathscr{U}} \hookrightarrow \hat{C'}_{\mathscr{V}}$) the categories of $\mathscr{U}$-presheaves and $\mathscr{V}$-presheaves, and by $u_{\mathscr{U}!}$ and $u_{\mathscr{V}!}$ the "inverse image" functors for $\mathscr{U}$-presheaves and $\mathscr{V}$-presheaves, respectively. We have a commutative diagram (cf. the explicit construction of $u_!$ in I 5.1):

(∗)
$$\begin{array}{ccc} \hat{C}_{\underline{U}} & \xrightarrow{u_{\underline{U}!}} & \hat{C'}_{\underline{V}} \\ \downarrow & & \downarrow \\ \hat{C}_{\underline{V}} & \xrightarrow{u_{\underline{V}!}} & \hat{C'}_{\underline{V}}. \end{array}$$

The inclusion functors from $\mathscr{U}$-presheaves into $\mathscr{V}$-presheaves are fully faithful and commute with projective limits. It follows from (II 5.1 i)) that a bicovering morphism of $\mathscr{U}$-presheaves is a bicovering morphism of $\mathscr{V}$-presheaves. Now let $R \hookrightarrow X$ be a sieve covering an object X of C. By 1.2 ii), the morphism $u_{\mathscr{U}!}R \to u_{\mathscr{U}!}X$ is bicovering. Using the commutativity of (∗) and what precedes, we deduce that the morphism $u_{\mathscr{V}!}R \to u_{\mathscr{V}!}X$ is bicovering. Consequently (1.2), the functor u^* carries $\mathscr{V}$-sheaves on C' to $\mathscr{V}$-sheaves on C. The commutativity of (1.5.1) then follows from the commutativity of (∗), from II 3.6, and from 1.3 2).

PROPOSITION 1.6. *Let C and C' be two $\mathscr{U}$-sites, and let $u : C \to C'$ be a functor* 271 *between the underlying categories. Consider the following conditions:*

i) *u is continuous* (cf. . 1.1 and 1.2).

ii) *For every covering family $(X_i \to X)$, $i \in I$, in C, the family $(uX_i \to uX)$, $i \in I$, in C' is covering.*

We have the implication i) ⇒ ii).

Suppose that the topology of C can be defined by a pretopology T (II 1.3) *and that the functor u commutes with fiber products.*[4] *Then conditions* i) *and* ii) *are equivalent and are also equivalent to the following condition:*

iii) *The functor u carries the covering families of T to covering families in C'.*

In particular, suppose that fiber products in C are representable and that the functor u commutes with fiber products. Then conditions i) *and* ii) *are equivalent.*

PROOF. Let us prove that i) ⇒ ii). Let $X_i \to X$, $i \in I$, be a covering family in C. For every sheaf F on C', u^*F is a sheaf. Consequently, the map

$$\mathrm{Hom}(X, u^*F) \longrightarrow \prod_i \mathrm{Hom}(X_i, u^*F)$$

is injective (II 5.1). Hence there is an injective map

$$\mathrm{Hom}(u_!X, F) \longrightarrow \prod_i \mathrm{Hom}(u_!X_i, F).$$

Therefore the family $u_!X_i \to u_!X$, $i \in I$, is covering in C' (II 5.1).

Every covering family of the pretopology T is a covering family in C, whence ii) ⇒ iii). Let us prove that iii) ⇒ i). Let X be an object of C and let $X_i \to X$, $i \in I$, be a 272

[4] In fact, it suffices that u commute with the fiber products occurring in base changes of morphisms belonging to covering families for T.

covering family in $\mathrm{Cov}(X)$ (II 1.3). The morphisms $X_i \to X$ are pullbackable, and the functor u commutes with fiber products. Consequently, the fiber products $u(X_i) \times_{u(X)} u(X_j)$ are representable and canonically isomorphic to $u(X_i \times_X X_j)$. Let F be a sheaf on C'. Then the diagram of sets

$$F(u(X)) \to \prod_i F(u(X_i)) \rightrightarrows \prod_{i,j} F(u(X_i \times_X X_j))$$

is exact (II 2.1, I 3.5, and I 2.12). Consequently, the diagram of sets

$$u^* F(X) \to \prod_i u^* F(X_i) \rightrightarrows \prod_{i,j} u^* F(X_i \times_X X_j)$$

is exact. Therefore $u^* F$ is a sheaf (II 2.4), whence i). The final assertion of 1.6 follows from what precedes and from the fact that, when fiber products are representable in C, all covering families in C[4] define a pretopology on C whose associated topology is the topology of C.

PROPOSITION 1.7. *Let C be a small site, C' a $\mathscr{U}$-site, and $u : C \to C'$ a continuous functor. Let γ be an algebraic structure defined by finite projective limits, such that in the category of γ-objects of $\mathscr{U}$ − Ens the $\mathscr{U}$-indexed inductive limits are representable* [4]. *We use the notation of Proposition* II 6.4. *The functor u_s commutes with projective limits and consequently defines a functor $u_s^\gamma : C\tilde{}_\gamma \to C'\tilde{}_\gamma$. The functor u_s^γ has a left adjoint u_γ^s with the following properties:*

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1) *There is a canonical isomorphism $u_\gamma^s \xrightarrow{\sim} \underline{a}'_\gamma u_{\gamma!} i_\gamma$, and u_γ^s commutes with every finite projective limit with which $u_{\gamma!}$ commutes.*
2) *There is a canonical isomorphism $u_\gamma^s \underline{a}_\gamma \xrightarrow{\sim} \underline{a}'_\gamma u'_{\gamma!}$.*
3) *The functor u_γ^s commutes with inductive limits.*
4) *If u^s is left exact, the diagram* (in the notation of II 6.3.0)

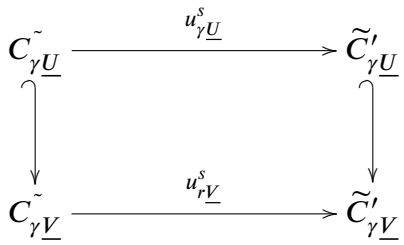
$$\begin{array}{ccc} C\tilde{}_\gamma & \xrightarrow{u_Y^s} & C'\tilde{}_\gamma \\ \downarrow{\scriptstyle \mathrm{esj}\tilde{}} & & \downarrow{\scriptstyle \mathrm{esj}'\tilde{}} \\ C\tilde{} & \xrightarrow{u^s} & C'\tilde{} \end{array}$$

   is commutative up to isomorphism, and the functor u_γ^s is exact.
5) *Suppose that the functor* esj$\tilde{}$ *(resp.* esj$\tilde{}$*) has a left adjoint* Lib$\tilde{}$ *(resp.* Lib$'\tilde{}$*)* (II 6.5). *There is a canonical isomorphism $u_\gamma^s \mathrm{Lib}\tilde{} \xrightarrow{\sim} \mathrm{Lib}'\tilde{} u^s$.*

Moreover, when C is not necessarily a small site but is a $\mathscr{U}$-site, the functor u_s^γ has a left adjoint u_γ^s. Finally, if V is a universe containing $\mathscr{U}$ and if $u_{\gamma\mathscr{U}}^s$ (resp. $u_{\gamma\mathscr{V}}^s$) denotes the left adjoint of u_s relative to the universe $\mathscr{U}$ (resp. $\mathscr{V}$), the following diagram, analogous to diagram (1.5.1):

[4] indexed by sets belonging to a suitable universe.

[4] In fact, one can show that this condition is always satisfied.

(1.7.1)


is commutative up to canonical isomorphism.

Proof. The proof, in the case where C is a small site, is left to the reader. It will be completed in 4.3 in the case where C is a $\mathscr{U}$-site.

Notation 1.8. Henceforth we shall use the notation u_s for the functor u^{γ}_s. This notation is unlikely to cause confusion, because the functor u_s (defined on sheaves of sets) commutes with finite projective limits. 274

Likewise, when u^s is left exact, we shall use the notation u^s for the functor u^s_{γ}. This abuse of notation is justified by 1.7 4).

Example 1.9.1. Let X be a small topological space. Denote by $\mathrm{Ouv}(X)$ the category of open subsets of X (the objects of $\mathrm{Ouv}(X)$ are the open subsets of X, and the morphisms of $\mathrm{Ouv}(X)$ are inclusions), equipped with the canonical topology. A family $(U_i \subset U)$, $i \in I$, is covering if and only if the open subsets U_i cover U (II 2.6). The sheaves of sets on $\mathrm{Ouv}(X)$ are therefore the sheaves of sets in the sense of [TF]. Let $f : X \to Y$ be a continuous map. It induces a functor $f^{-1} : \mathrm{Ouv}(Y) \to \mathrm{Ouv}(X)$: for every open subset U of Y, $f^{-1}(U)$ is the inverse image of U under the map f. The functor f^{-1} is continuous (1.6). The functor $f^{-1}_s : \mathrm{Ouv}(X)^{\sim} \to \mathrm{Ouv}(Y)^{\sim}$ is the direct-image functor for sheaves, in the sense of [TF]. The functor $(f^{-1})^s : \mathrm{Ouv}(Y)^{\sim} \to \mathrm{Ouv}(X)^{\sim}$ is the inverse-image functor for sheaves in the sense of [TF]. The functor $(f^{-1})^s$ commutes with finite projective limits, by 1.3 5).

Example 1.9.2. Let X be a topological space and let $Y \hookrightarrow X$ be an open subset of X. Every open subset of Y is an open subset of X, and thus we obtain a functor $\Phi : \mathrm{Ouv}(Y) \to \mathrm{Ouv}(X)$. The functor Φ is continuous (1.6). The functor $\Phi_s : \mathrm{Ouv}(X)^{\sim} \to \mathrm{Ouv}(Y)^{\sim}$ is the "restriction to Y" functor. The functor $\Phi^s_{\mathrm{ab}} : \mathrm{Ouv}(Y)^{\sim}_{\mathrm{ab}} \to \mathrm{Ouv}(X)^{\sim}_{\mathrm{ab}}$ (II 6.3.3, I 1.7) is the "extension by zero outside Y" functor introduced in [TF]. The functor $\Phi^s : \mathrm{Ouv}(Y)^{\sim} \to \mathrm{Ouv}(X)^{\sim}$ (1.3) is analogous to the "extension by zero" functor: for every sheaf H on Y and every point $y \in Y$, the stalk of $\Phi^s H$ at y is canonically isomorphic to the stalk of H at y; for every point $x \in X$, $x \notin Y$, the stalk of $\Phi^s H$ at x is empty. The functor Φ^s is called the "extension by the empty set outside Y" functor. The functor Φ^s commutes with fiber products and with finite products over a nonempty index set, but it does not carry the terminal object of $\mathrm{Ouv}(Y)^{\sim}$ to the terminal object of $\mathrm{Ouv}(X)^{\sim}$. 275

Example 1.9.3. Let C and C' be two small sites, and let $u : C \to C'$ be a functor between the underlying categories that carries every covering family of C to a covering family of C'. The functor u is not continuous in general (cf. nevertheless 1.6). Here is a counterexample. Let S be a set of infinite cardinality in the universe $\mathscr{U}$, and let C be the full subcategory of the category of sets whose objects are the finite, nonempty subsets of S. Equip C with the canonical topology. Observe that the covering families of C are the surjective families of maps and that covering families are nonempty. Observe also that, up to isomorphism, there are only two constant sheaves on C: the sheaf with value the

empty set and the sheaf with value a one-element set. Set $C = C'$ and let $u : C \to C'$ be a constant functor. The functor u carries the covering families of C to covering families of C'. The functor u is not continuous. Indeed, since the representable presheaves of
276 C' are sheaves, for every object X of C' there exists a sheaf F on C' such that the cardinality of $F(X)$ is ≥ 2, and consequently the presheaf u^*F is not a sheaf.

2. Cocontinuous functors

DEFINITION 2.1. *Let C and C' be two $\mathscr{U}$-sites, and let $u : C \to C'$ be a functor between the underlying categories. We say that u is cocontinuous if it has the following property:*

(COC) For every object Y of C and every covering sieve $R \hookrightarrow u(Y)$, the sieve on Y generated by the arrows $Z \to Y$ such that $u(Z) \to u(Y)$ factors through R covers Y.

PROPOSITION 2.2. *Let C be a small site, let C' be a $\mathscr{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. Denote by $\hat{u}_* : \hat{C} \to \hat{C'}$ the functor right adjoint to the functor $F \mapsto F \circ u = \hat{u}^* F$ (I 5.1). The functor u is cocontinuous if and only if, for every sheaf G on C, $\hat{u}_* G$ is a sheaf on C'.*

PROOF. *The condition is sufficient.* Suppose that for every sheaf G on C, the presheaf $\hat{u}_* G$ is a sheaf. Let Y be an object of C, and let $R \hookrightarrow u(Y)$ be a covering sieve. There is an isomorphism $\mathrm{Hom}(u(Y), u_* G) \xrightarrow{\sim} \mathrm{Hom}(R, u_* G)$, hence, by adjunction (I 5.1), an isomorphism $\mathrm{Hom}(\hat{u}^* \circ u(Y), G) \xrightarrow{\sim} \mathrm{Hom}(\hat{u}^* R, G)$. It follows (II 5.3) that the morphism $\hat{u}^* R \to \hat{u}^* u(Y)$ is bicovering. But the functor $\hat{u}^*$ commutes with projective limits, and therefore the morphism $\hat{u}^* R \to \hat{u}^* u(Y)$ is a monomorphism. It is thus a covering monomorphism. On the other hand, there is a canonical morphism $Y \xrightarrow{p} \hat{u}^* u(Y)$ deduced
277 from the adjunction morphism $\mathrm{id}_C \to \hat{u}^* u_!$ (I 5.4 3)). We then make the base change $Y \xrightarrow{p} \hat{u}^* u(Y)$. This gives a covering sieve on Y:

$$\hat{u}^* R \times_{\hat{u}^* u(Y)} Y \longrightarrow Y,$$

and, on determining the arrows $Z \to Y$ that factor through this sieve, one obtains the sieve described by property (COC).

The condition is necessary. Let S be an object of C' and let $R \hookrightarrow S$ be a covering sieve on S. We must prove that, for every sheaf G on C, there is an isomorphism

$$\mathrm{Hom}(S, u_* G) \xrightarrow{\sim} \mathrm{Hom}(R, u_* G).$$

Using Proposition II 5.3 and the adjunction isomorphisms, we see that it suffices to prove that the monomorphism $\hat{u}^* R \to \hat{u}^* S$ is covering. For this, it suffices to prove (II 5.1) that for every base change $Y \to \hat{u}^* S$, the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y$ is covering, where Y is an object of C. But, by the properties of adjoint functors and I 5.4 3), the morphism $Y \to \hat{u}^* S$ factors through $Y \xrightarrow{p} \hat{u}^* u(Y) \to \hat{u}^* S$, and consequently the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y \hookrightarrow Y$ is obtained, by the base change $Y \xrightarrow{p} \hat{u}^* u(Y)$, from the monomorphism $\hat{u}^* R \times_{\hat{u}^* S} \hat{u}^* u(Y) \xrightarrow{q} \hat{u}^* u(Y)$. Now the functor $\hat{u}^*$ commutes with projective limits, and consequently the monomorphism q is the image under $\hat{u}^*$ of the sieve $R \times_S u(Y) \hookrightarrow u(Y)$, which is covering. Hypothesis (COC) then allows us to conclude that the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y \hookrightarrow Y$ is covering, C.Q.F.D.

PROPOSITION 2.3. *Let C and C' be two $\mathscr{U}$-sites, and let $u : C \to C'$ be a cocontinuous functor. Denote by $u^* : \hat{C'} \to \hat{C}$ the functor $u^* = \underline{a} \hat{u}^* i'$, where $\underline{a}$ denotes the "associated*
278 *sheaf" functor for $\hat{C}$, $\hat{u}^*$ the functor $F \mapsto F \circ u$, and i' the inclusion functor $\widetilde{C'} \to \hat{C'}$.*

1) *The functor u^* commutes with inductive limits and is exact.*
2) *There is a canonical isomorphism $u^*\underline{a}' \simeq \underline{a}\hat{u}^*$, where $\underline{a}'$ denotes the "associated sheaf" functor for $C'^{\wedge}$.*
3) *The functor u^* has a right adjoint $u_* : \widetilde{C} \to \widetilde{C}'$. When C is small, the diagram*

(2.3.1)
$$\begin{array}{ccc} C^{\sim} & \xrightarrow{u_*} & \widetilde{C}' \\ \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i'} \\ C^{\wedge} & \xrightarrow{\hat{u}_*} & C'^{\wedge}, \end{array}$$

in which the lower functor is right adjoint to the functor $\hat{u}^*$ (I 5.1), commutes up to canonical isomorphism.

4) *Let $\mathscr{V} \supset \mathscr{U}$ be a universe. Denote by $u_{*\mathscr{U}} : C^{\sim}_{\mathscr{U}} \to \widetilde{C}'_{\mathscr{U}}$ (resp. $u_{*\mathscr{V}} : C^{\sim}_{\mathscr{V}} \to \widetilde{C}'_{\mathscr{V}}$) the right adjoint relative to the universe $\mathscr{U}$ (resp. $\mathscr{V}$) whose existence is asserted in 3). The diagram*

(2.3.2)
$$\begin{array}{ccc} C^{\sim}{}_{\underline{U}} & \xrightarrow{u_{*\underline{U}}} & \widetilde{C}'_{\underline{U}} \\ \downarrow & & \downarrow \\ C^{\sim}{}_{\underline{V}} & \xrightarrow{u_{*\underline{V}}} & \widetilde{C}'_{\underline{V}} \end{array}$$

commutes up to isomorphism.

Proof. Assertions 1) and 2) follow immediately from II 3.4. Assertion 3), when C is small, follows from 2.2. When C is a $\mathscr{U}$-site, it will be proved in 4.4. Assertion 4), when C is small, follows from the analogous commutativity assertion when the topologies on C and C' are chaotic and from I 3.5. When the topologies on C and C' are chaotic, the commutativity of (2.3.2) follows immediately from the explicit description of u_* (I 5.1). 279

2.4. We shall not develop here the considerations concerning γ-objects of categories of sheaves. It will suffice to note that the functors u^* and u_* introduced in this section always commute with finite projective limits (unlike what happened in the preceding section for the functor u^s). Consequently, they extend naturally to functors defined on γ-objects; these functors are adjoint to one another and commute with the "underlying sheaf of sets" functors. We shall make the notational abuses mentioned in 1.8, using u^* and u_* also for the extensions to γ-objects.

Proposition 2.5. *Let C and C' be two $\mathscr{U}$-sites, and let $C \underset{u}{\overset{v}{\rightleftarrows}} C'$ be a pair of functors, where v is left adjoint to u. The following properties are equivalent:*

i) *The functor u is continuous.*
ii) *The functor v is cocontinuous.*

Moreover, under these equivalent conditions, there are canonical isomorphisms $v_* \simeq u_s$ and $v^* \simeq u^s$. In particular, the functor u^s commutes with finite projective limits.

Proof. The property of a functor being continuous or cocontinuous does not depend on the universe $\mathscr{U}$ for which C and C' are $\mathscr{U}$-sites (this is immediate for a cocontinuous

functor, and follows from 1.5 for a continuous functor). We may therefore, after enlarging the universe, suppose that C and C' are small. The proposition then follows from 2.2 and I 5.6.

280 PROPOSITION 2.6. *Let $u : C \to C'$ be a continuous and cocontinuous functor between two $\mathscr{U}$-sites. Then the functor $u^* : C'^{\sim} \to C^{\sim}$ (2.3) commutes with $\mathscr{U}$-inductive and projective limits. Denote by $u_! : C^{\sim} \to C'^{\sim}$ and $u_* : C^{\sim} \to C'^{\sim}$ the functors left and right adjoint to u^*, respectively. The functor $u_!$ is fully faithful if and only if the functor u_* is fully faithful. When u is fully faithful, $u_!$ is fully faithful, and the converse holds when the topologies of C and C' are coarser than the canonical topology.*

The functor u^* has a right adjoint u_* (2.3) and a left adjoint $u_!$ (1.2). The functor u^* therefore commutes with inductive and projective limits (I 2.11). The fact that $u_!$ is fully faithful if and only if u_* is fully faithful is a general property of adjoint functors (I 5.7 1). When u is fully faithful, the functor $\hat{u}_* : \hat{C}_{\mathscr{V}} \to \hat{C'}_{\mathscr{V}}$ on $\mathscr{V}$-presheaves, for a sufficiently large universe $\mathscr{V}$, is fully faithful (I 5.7). Consequently, by the commutativity of (2.3.1) and (2.3.2), the functor $u_* : C^{\sim} \to \widetilde{C'}$ is fully faithful, and hence so is $u_!$ by what precedes. The converse follows from the existence of the commutative diagram in 1.2 iv), taking into account the fact that the functors ϵ_C and $\epsilon_{C'}$ are fully faithful when the topologies are coarser than the canonical topology, C.Q.F.D.

EXAMPLE 2.7. With the notation of 1.9.2, the functor

$$\boldsymbol{\Phi} : \mathrm{Ouv}(Y) \longrightarrow \mathrm{Ouv}(X)$$

is continuous (1.9.2) and cocontinuous (2.2). Moreover, let $i : Y \to X$ be the canonical
281 inclusion and let $i^{-1} : \mathrm{Ouv}(X) \to \mathrm{Ouv}(Y)$ be the functor that it defines (1.9.1). The functor i^{-1} is right adjoint to the functor $\boldsymbol{\Phi}$. There is therefore a sequence of three adjoint functors:

$$\begin{array}{ccc} \boldsymbol{\Phi}^s & , & \boldsymbol{\Phi}_s \\ & & \| \\ & & \boldsymbol{\Phi}^* \quad , \quad \boldsymbol{\Phi}_*, \end{array}$$

and canonical isomorphisms $\boldsymbol{\Phi}^* \simeq (i^{-1})^s$ and $\boldsymbol{\Phi}_* \simeq i_s^{-1}$.

3. Induced topology

3.1. Let C' be a site, C a category, and $u : C \to C'$ a functor. For every universe $\mathscr{U}$ such that C' is a $\mathscr{U}$-site and C is a $\mathscr{U}$-small category, denote by $\mathscr{T}_{\mathscr{U}}$ the finest among the topologies T on C that make u continuous (1.1). (Such a topology exists by II 2.2.) The topology $\mathscr{T}_{\mathscr{U}}$ does not depend on the universe $\mathscr{U}$. Indeed, if $\mathscr{V} \supset \mathscr{U}$ is a universe, then $\mathscr{T}_{\mathscr{U}} = \mathscr{T}_{\mathscr{V}}$ (1.1 and 1.5). The topology $\mathscr{T}_{\mathscr{U}}$ is called *the topology induced on C by the topology of C' by means of the functor u* [4].

PROPOSITION 3.2. *Let C be a small category, C' a $\mathscr{U}$-site, $u : C \to C'$ a functor, and $\mathscr{T}$ the topology on C induced by u. Let X be an object of C, and let $R \to X$ be a sieve on X. The following properties are equivalent:*

i) *The sieve $R \hookrightarrow X$ is covering for $\mathscr{T}$.*
ii) *For every base change $Y \to X$, where Y is an object of C, the morphism $u_!(R \times_X Y) \to u(Y)$ is bicovering in $\hat{C'}$* (II 5.2).

[4] When no confusion can result, this topology is called the *topology induced on C by the topology of C'*.

PROOF. i) ⇒ ii): This follows from axiom (T 1) for topologies and from 1.2.
ii) ⇒ i): For every sheaf F on C', the map

$$\mathrm{Hom}(Y, u^*F) \longrightarrow \mathrm{Hom}(R \times_X Y, u^*F)$$

is isomorphic, by adjunction (I 5.1), to the map 282

$$\mathrm{Hom}(u(Y), F) \longrightarrow \mathrm{Hom}(u_!(R \times_X Y), F),$$

which is bijective (II 5.3). Consequently (II 2.2), the sieve $R \hookrightarrow X$ is covering for $\mathscr{T}$.

COROLLARY 3.3. *Let C' be a site, C a category, $u : C \to C'$ a functor, and $\mathscr{T}$ the topology on C induced by the topology of C'. Let $(X_i \to X)$, $i \in I$, be a family of pullbackable morphisms in C, and suppose that u commutes with fiber products* [4]. *The following properties are equivalent:*

i) *The family $X_i \to X$, $i \in I$, is covering for $\mathscr{T}$.*
ii) *The family $(u(X_i) \to u(X))$, $i \in I$, is covering.*

PROOF. i) ⇒ ii): This follows from 1.6.
ii) ⇒ i): Let $\mathscr{U}$ be a universe such that C is $\mathscr{U}$-small and C' is a $\mathscr{U}$-site. Let $R \hookrightarrow X$ be the sieve generated by the $X_i \to X$. The presheaf R is the coequalizer of the pair of arrows (I 2.12 and I 3.5)

$$\coprod_{i,j} X_i \times_X X_j \rightrightarrows \coprod_i X_i$$

(the coproduct is taken here in $C^\wedge$). Since the functor $u_!$ commutes with inductive limits (I 5.4), the presheaf $u_!R$ is the coequalizer of the pair of arrows

$$\coprod_{i,j} u(X_i \times_X X_j) \rightrightarrows \coprod_i u(X_i).$$

Since the functor u commutes with fiber products, the presheaf $u_!R$ is the coequalizer of the pair of arrows

$$\coprod_{i,j} u(X_i) \times_{u(X)} u(X_j) \rightrightarrows \coprod_i u(X_i),$$

and consequently $u_!R \to u(X)$ is the sieve on $u(X)$ generated by the $u(X_i) \to$ 283
$u(X)$, $i \in I$. Using once again the fact that u commutes with fiber products, one shows that for every base change $Y \to X$, where Y is an object of C, $u_!(R \times_X Y) \to u(Y)$ is the sieve on $u(Y)$ generated by the $u(X_i) \times_{u(X)} u(Y) \to u(Y)$, $i \in I$. Since the family $u(X_i) \to u(X)$, $i \in I$, is covering, the family $u(X_i) \times_{u(X)} u(Y) \to u(Y)$, $i \in I$, is covering. Thus $u_!(R \times_X Y) \to u(Y)$ is a covering sieve, and consequently (3.2) $R \to X$ is covering for $\mathscr{T}$, C.Q.F.D.

COROLLARY 3.4. *Let C' be a $\mathscr{U}$-site, let C be a full subcategory of C', and let $u : C \to C'$ be the inclusion functor. Suppose that fiber products are representable in C and that u commutes with fiber products. The following conditions are equivalent:*

i) a) *For every object X of C, every covering family $(Y_j \to X)$, $j \in J$, in C' is refined by a covering family $X_i \to X$, $i \in I$, where the X_i are objects of C.*
   b) *There exists a small set G of objects of C such that every object of C is the target of a covering family in C' of morphisms whose sources belong to G.*

[4] In fact, it suffices that u commute with fiber products of the form $X_i \times_X X'$.

ii) *The topology induced by the topology of C' is a $\mathcal{U}$-topology* (I 3.0.2), *and the functor u is continuous and cocontinuous for this topology.*

PROOF. This follows immediately from 3.3 and 2.1.

284 PROPOSITION 3.5. *Let C be a $\mathcal{U}$-site and let $\epsilon_C : C \to C^\sim$ be the canonical functor* (II 4.4.0). *Equip $C^\sim$ with the canonical topology. Then the topology of the site C is the topology induced by the topology of $C^\sim$.*

PROOF. The covering families of $C^\sim$ for the canonical topology are the universally effective epimorphic families (II 2.5), that is (II 4.3), the epimorphic families. Let T be the topology of the site C, and let $\mathcal{T}_\mathcal{U}$ be the finest among the topologies T' on C such that, for every sheaf F on $C^\sim$, $F \circ \epsilon_C$ is a sheaf for T'. It suffices to show that $T = \mathcal{T}_\mathcal{U}$, for then $\mathcal{T}_\mathcal{U}$ is a $\mathcal{U}$-topology and consequently $\mathcal{T}_\mathcal{U}$ is the induced topology.

A) *The topology T is finer than $\mathcal{T}_\mathcal{U}$*: Let $(X_i \to X)$, $i \in I$, be a covering family for $\mathcal{T}_\mathcal{U}$. Then, for every sheaf F on $C^\sim$, the map $F(\epsilon_C(X)) \to \prod_i F(\epsilon_C(X_i))$ is injective, and consequently (II 5.2) $(\epsilon_C X_i \to \epsilon_C X)$, $i \in I$, is covering for the canonical topology of $C^\sim$, hence (II 5.2) $(X_i \to X)$, $i \in I$, is covering for T.

B) *The topology T is coarser than $\mathcal{T}_\mathcal{U}$*: It suffices to show that $\epsilon_C : C \to C^\sim$ is continuous (1.1). Now we shall prove in IV n° 1 that every sheaf on $C^\sim$ is representable, and consequently, for every sheaf F on $C^\sim$, $F \circ \epsilon_C$ is a sheaf on C. (We shall not use 3.5 before IV 1.)

We record two results that will be used in VI 7.

PROPOSITION 3.6. *Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, and, for every $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathcal{T}_\mathcal{U}$ on C that is the coarsest among the topologies for which the u_i are continuous. The topology $\mathcal{T}_\mathcal{U}$ does not depend on the universe $\mathcal{U}$ relative to which the categories under consideration are small.*

The last assertion of 3.6 follows from 1.5.

Let T be a topology on C. The functors u_i are continuous if and only if, for every
285 $i \in I$ and every covering sieve $R \hookrightarrow X$ on an object X of C_i, the morphism

$$u_{i!}(R) \longrightarrow u_i(X)$$

in $\widehat{C}$ is bicovering (1.2 (ii)). Thus 3.6 is a consequence of the following lemma.

LEMMA 3.6.1. *Let C be a small category and let $(u_i : F_i \to G_i)_{i\in I}$ be a family of arrows in $\widehat{C}$. There exists on C a topology that is coarsest among those making the morphisms u_i covering (resp. bicovering)* (II 5.2).

To say that $u : F \to G$ is covering for a given topology T means that, for every arrow $X \to G$, with $X \in \mathrm{Ob}\, C$, the corresponding arrow $F \times_G X \to X$ is covering; equivalently, the family of arrows $X' \to X$ in C that factor through the preceding arrow is covering. The existence of a coarsest topology among those for which the $u_i : F_i \to G_i$ are covering thus follows from I 1.1.6, proving the assertion of 3.6.1 without the parenthetical alternative. The parenthetical assertion follows by recalling that a morphism $u : F \to G$ is bicovering if and only if the morphisms $u : F \to G$ and $\mathrm{diag}_u : F \to F \times_G F$ are covering.

PROPOSITION 3.7. *Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, for every $i \in I$ let $u_i : C_i \to C$ be a functor, and let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathfrak{T}_\mathcal{U}$ that is the finest for which the u_i are cocontinuous.*

The topology $\mathfrak{T}_{\mathscr{U}}$ does not depend on the universe $\mathscr{U}$ relative to which the categories under consideration are small.

Let $\mathscr{U}$ be a universe relative to which the categories under consideration are small. The functors u_i are cocontinuous for a topology $\mathfrak{T}$ on C if and only if, for every $i \in I$ and every sheaf F on C_i, the presheaf $\hat{u}_{i*}F$ is a sheaf for $\mathfrak{T}$ (2.2). It follows that the topology $\mathfrak{T}_{\mathscr{U}}$ is the finest topology for which the presheaves $\hat{u}_{i*}F$, $i \in I$, $F \in C_i^{\sim}$, are sheaves (II 2.2). The last assertion follows from 2.2.

4. Comparison lemma

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THEOREM 4.1. (comparison lemma). *Let C be a small category, let C' be a site whose underlying category is a $\mathscr{U}$-category, and let $u : C \to C'$ be a fully faithful functor. Endow C with the topology induced by u (3.1). Consider the following properties:*

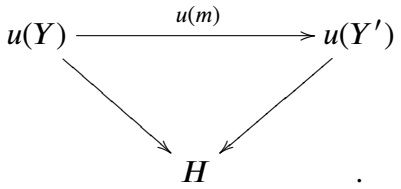
i) *Every object of C' can be covered by objects coming from C.*
ii) *The functor $F \mapsto F \circ u$ induces an equivalence from the category of sheaves on C' to the category of sheaves on C.*

We always have i) ⇒ ii). *When C' is a $\mathscr{U}$-site and the topology of C' is coarser than the canonical topology, we have* ii) ⇒ i).

4.1.1. Let us first prove i) ⇒ ii). The proof proceeds in two steps.

First step. For every presheaf H on C', the adjunction morphism $\varphi : u_!u^*H \to H$ (I 5.1) is bicovering (II 5.3), and the functor $u_s : \widetilde{C'} \to \widetilde{C}$ (1.1.1) is fully faithful.

Let C/H be the small category whose objects are the objects Y of C equipped with a morphism $u(Y) \to H$, and whose morphisms are the commutative diagrams



We have $u^*H = \varinjlim_{Y \in \mathrm{ob}\, C/H} Y$ (I 3.4), and hence (I 5.4) $u_!u^*H = \varinjlim_{Y \in \mathrm{ob}\, C/H} u(Y)$. The adjunction morphism is the evident morphism arising from the description of $u_!u^*H$ as an inductive limit. The morphism φ has the following property: (∗) For every object Y 287
of C, every morphism $m : u(Y) \to H$ factors uniquely as the canonical morphism $\alpha(m) : u(Y) \to u_!u^*H$ followed by φ. It follows immediately from i) that φ is covering (II 5.1); let us show that it is bicovering. Let $p, q : Z \rightrightarrows u_!u^*H$ be two morphisms from an object Z of C' such that $\varphi p = \varphi q$. For every object Y of C and every morphism $n : u(Y) \to Z$, we have $\varphi pn = \varphi qn$. Property (∗) then implies that $pn = qn$, and since the $u(Y)$ cover Z, the kernel of (p, q) is a covering sieve of Z. Thus φ is bicovering (II 5.3). For every sheaf H on C', u^*H is a sheaf on C, denoted by u_sH (1.1.1). Moreover, $u^su_sH = \underline{a}'u_!u_sH$ (1.3), and the adjunction morphism $u^su_sH \to H$ is obtained by applying the "associated sheaf" functor to $\varphi : u_!u^*H \to H$. Consequently (II 5.3), the adjunction morphism $u^su_sH \to H$ is an isomorphism. Hence $u_s : \widetilde{C'} \to \widetilde{C}$ is fully faithful.

Second step. The functor u is cocontinuous and the functor $u^s : C^{\sim} \to C'^{\sim}$ (1.2) is fully faithful. Consequently $u_s : C'^{\sim} \to C^{\sim}$ is an equivalence. Let Y be an object of C and let $i : R \hookrightarrow u(Y)$ be a covering sieve. Since the functor u^* commutes with projective limits (I 5.5), the morphism $u^*(i) : u^*(R) \hookrightarrow u^*u(Y)$ is a monomorphism. Since u is fully faithful, we have $u^*u(Y) \simeq Y$, and thus obtain a sieve of Y, $u^*(i) : u^*(R) \to Y$. To show that u is cocontinuous, it suffices to show that $u^*(i) : u^*(R) \hookrightarrow Y$ is a covering sieve for the

induced topology on C (2.1). For this, it suffices to show (3.2) that for every base change $m : X \to Y$, the morphism $u_!(u^*(R)\times_Y X) \to u(X)$ is bicovering. But since u^* commutes
288 with projective limits, we have $u^*(R) \times_Y X = u^*(R \times_{u(Y)} u(X))$, and $R \times_{u(Y)} u(X)$ is a covering sieve of $u(X)$. It therefore suffices to show that for every object Y of C and every covering sieve $i : R \hookrightarrow u(Y)$, the morphism in $\hat{C'}$, $u_!u^*(R) \xrightarrow{u_!u^*(i)} u(Y)$, is bicovering. Now this morphism factors as the adjunction morphism $u_!u^*(R) \to R$, which is bicovering (first step), followed by the covering monomorphism $R \hookrightarrow u(Y)$. It is therefore bicovering (II 5.3). This shows that u is cocontinuous. Since u is continuous and fully faithful, it follows from 2.6 (used in the case where C is small) that u^s (denoted by $u_!$ in 2.6) is fully faithful. Since u^s and u_s are adjoint to one another and are fully faithful, they are quasi-inverse functors, and hence u_s is an equivalence.

4.1.2. Let us now prove ii) ⇒ i). The objects Y of C form a generating family of $C^\sim$ (II 4.10), and consequently, for every object X of C', there exists an epimorphic family (in $C^\sim$) $v_i : Y_i \to u_sX$, $i \in I$. We have $u_su(Y_i) = Y_i$, and, since u_s is an equivalence of categories, $v_i = u_s(w_i)$, where the $w_i : u(Y_i) \to X$ form an epimorphic family in $C'^\sim$. It then follows from II 5.1 that the family $w_i : u(Y_i) \to X$, $i \in I$, is covering.

4.2. Completion of the proof of 1.2. Let G be a full subcategory of C whose objects form a small topologically generating family of C (II 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$ (3.1). The functor $(u \circ i)_s = i_s \circ u_s$ (1.1.1) has
289 a left adjoint by the first part of the proof of 1.2. Since i_s is an equivalence of categories (4.1), the functor u_s has a left adjoint u^s, and there is a functorial isomorphism $u^s \simeq (u \circ i)^s \circ i_s$. Let us show that the diagram

(4.2.1)
$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle \epsilon_C}\downarrow & & \downarrow{\scriptstyle \epsilon_{C'}} \\ C^\sim & \xrightarrow{u^s} & \widetilde{C'} \end{array}$$

commutes up to isomorphism. For every sheaf H on C', we have $\mathrm{Hom}_{\widetilde{C'}}(\epsilon_{C'}uX, H) \simeq u_sH(X)$ for every $X \in \mathrm{ob}\, C$. Moreover, $\mathrm{Hom}_{\widetilde{C'}}(u^s\epsilon_C X, H) \simeq \mathrm{Hom}_{\widetilde{C}}(\epsilon_C X, u_sH)$ by adjunction, and then $\mathrm{Hom}_{\widetilde{C}}(\epsilon_C X, u_sH) \simeq u_sH(X)$ by the definition of ϵ_C. Hence there is an isomorphism $\epsilon_{C'}uX \simeq u^s\epsilon_C X$ for every $X \in \mathrm{ob}\, C$. This proves i) ⇒ iv). Let us prove iv) ⇒ i). We have a commutative diagram

(4.2.2)
$$\begin{array}{ccccc} G & \xrightarrow{i} & C & \xrightarrow{u} & C' \\ {\scriptstyle \epsilon_G}\downarrow & & \downarrow{\scriptstyle \epsilon_C} & & \downarrow{\scriptstyle \epsilon_{C'}} \\ \widetilde{G} & \xrightarrow{i^s} & \widetilde{C} & \xrightarrow{u^s} & \widetilde{C'}, \end{array}$$

and consequently, by the first part of the proof, the functor $u \circ i : G \to C'$ is continuous. It follows at once from 4.1 and 1.1 that u is continuous, whence iv) ⇒ i). This also gives the uniqueness of u^s, since the first part of the proof gives uniqueness when C is small.

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4.3. Completion of the proofs of 1.5 and 1.7. Let $u : C \to C'$ be a functor between two $\mathscr{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (II 3.0.1). Endow G with the topology induced by the

inclusion functor $i : G \to C$. It follows immediately from 4.1 and 1.1 that u is continuous if and only if $u \circ i$ is continuous. This observation and the first part of the proof of 1.5 give the general case. The same observation also reduces the proof of 1.7 to the case where the category C is small.

4.4. Completion of the proof of 2.3. Let $u : C \to C'$ be a functor between two $\mathscr{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (II 3.0.1), endowed with the topology induced by the inclusion functor $i : G \to C$. It follows from the proof of 4.1 (4.1.1, second step) that i is cocontinuous. Consequently, when u is cocontinuous, the functor $u \circ i : G \to C'$ is cocontinuous. Moreover, $i^* : \widetilde{C} \to \widetilde{G}$ is an equivalence of categories (4.1). Therefore the functor $u^* : \widetilde{C'} \to \widetilde{C}$ has a right adjoint. This proves 3). To prove 4), it suffices to observe that $(u \circ i)_* \simeq u_* \circ i_*$ and that i_* is an equivalence (4.1). The commutativity of (2.3.2) then follows from the commutativity of these diagrams when C is small.

5. Localization

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5.1. Now let C be a $\mathscr{U}$-site and X an object of $\hat{C}$. Unless expressly stated otherwise, the category C/X will be endowed with the topology $\mathscr{T}$ induced by the functor $j_X : C/X \to C$ (3.1). The notation C/X will denote the category C/X endowed with the topology $\mathscr{T}$. Proposition I 5.11 shows that the functor $j_{X!}$ commutes with fiber products and consequently sends every monomorphism to a monomorphism. In particular, for every object $(Y \to X)$ of C/X, the functor $j_{X!}$ establishes a one-to-one correspondence between the sieves, in the category C/X, on the object $(Y \to X)$ and the sieves, in C, on the object Y.

PROPOSITION 5.2. *Let C be a $\mathscr{U}$-site, X an object of $\hat{C}$, and $j_X : C/X \to C$ the corresponding continuous functor.*

1) *A sieve R on an object $(Z \to X)$ is covering in C/X if and only if the sieve $j_{X!}(R) \hookrightarrow Z$ is covering in C.*
2) *The functor $j_X : C/X \to C$ is cocontinuous and continuous.*
3) *Let $m : Y \to X$ be a morphism of $\hat{C}$. We then have the commutative diagram*

$$\begin{array}{ccc} C/Y & \xrightarrow{j_m} & C/X \\ & {\scriptstyle j_Y} \searrow \quad \swarrow {\scriptstyle j_X} & \\ & C & \end{array} .$$

The topology induced by the functor j_Y on C/Y is equal to the topology induced by j_m on C/Y.

4) *The topology induced by $j_X : C/X \to C$ is a $\mathscr{U}$-topology (II 3.0.2).*

PROOF. 1) If the sieve R on $(Z \to X)$ is covering, the sieve $j_{X!}(R) \hookrightarrow Z$ is covering (1.6). 292

Conversely, if the sieve $j_{X!}(R) \to Z$ is covering in C, one sees that the same is true for every sieve obtained by a base change in C/X. The sieve R is therefore covering (3.2).

2) This follows immediately from 1) by applying 2.1.
3) This follows immediately from the description of covering sieves given in 1).

4) Let $(G_i)_{i\in I}$ be a small topologically generating family of C. One verifies immediately that the small family $(u : G_i \to X)$, $u \in \coprod_{i\in I} \mathrm{Hom}_{C^\sim}(G_i, X)$, is topologically generating in C/X.

Terminology and notation 5.3. By the preceding proposition, the functor j_X is both continuous and cocontinuous. It therefore defines a sequence of three adjoint functors (4.3.2) between the categories of sheaves of sets (1.3 and 2.3):

$$j_X^s : (C/X)^\sim \longrightarrow C^\sim$$
$$j_X^* = j_{X,s} : C^\sim \longrightarrow (C/X)^\sim$$
$$j_{X*} : (C/X)^\sim \longrightarrow C^\sim.$$

In the particular situation of Proposition 5.2, we shall use the following terminology and notation:

1) The functor j_{X*} will be called the *direct image functor.*
2) The functor $j_{X,s} = j_X^*$ will be denoted by j_X^* and will be called the *restriction functor to* C/X.
293 3) The functor j_X^s on sheaves of sets will be called the "extension by the empty set to C" functor and will be denoted by $j_{X!}$ (cf. 2.9.2).

Thus we have a sequence of three adjoint functors between $(C/X)^\sim$ and $C^\sim$:

$$j_{X!},\ j_X^*,\ j_{X*}.$$

Proposition 5.4. *The functor $j_{X!} : (C/X)^\sim \to C^\sim$ factors through the category $C^\sim/\underline{a}X$ ($\underline{a}$ is the associated sheaf functor):*

$(C/X)^\sim \xrightarrow{e_X^\sim} C^\sim/\underline{a}X \to C^\sim$. *The functor*

$$e_X^\sim : (C/X)^\sim \longrightarrow \widetilde{C}/\underline{a}X$$

is an equivalence of categories. The restriction functor to C/X, followed by the equivalence $e_X^\sim$, is isomorphic to the "base change by $\underline{a}X \to \epsilon$" functor ($\epsilon$ being the terminal object of $C^\sim$): $F \mapsto (F \times \underline{a}X \xrightarrow{\mathrm{pr}_2} \underline{a}X)$.

Proof. The image under $j_{X!}$ of the terminal object of $(C/X)^\sim$ is the object $\underline{a}X$, which gives the factorization. To show that the functor $e_X^\sim$ is an equivalence, we shall confine ourselves to a few indications. By I 5.11, a presheaf on C/X is defined by a presheaf F on C equipped with a morphism $F \to X$. One then proves that the following two properties are equivalent:

i) The presheaf on C/X defined by $F \to X$ is a sheaf.
ii) The following diagram is cartesian (where i and $\underline{a}$ denote the inclusion in the category of presheaves and the associated sheaf functor, respectively):

$$\begin{array}{ccc} F & \longrightarrow & i\underline{a}F \\ \downarrow & & \downarrow \\ X & \longrightarrow & i\underline{a}X \end{array} .$$

294 It follows immediately that $e_X^\sim$ is an equivalence. The last assertion is trivial.

PROPOSITION 5.5. 1) *Let C be a $\mathscr{U}$-site and X an object of $\hat{C}$. The diagram below of categories and functors commutes up to canonical isomorphisms:*

$$\begin{array}{ccccccccc}
C/X & \xrightarrow{h_X} & (C/X)\hat{} & \xrightarrow{\underline{a}_X} & (C/X)\tilde{} & \xrightarrow{\quad i_X \quad} & (C/X)\hat{} \\
\downarrow & & \downarrow{\scriptstyle e\hat{}_X} & & \downarrow{\scriptstyle e\tilde{}_X} & & \downarrow{\scriptstyle e\hat{}/X} \\
C/X & \xrightarrow{h/X} & C\hat{}/X & \xrightarrow{\underline{a}/X} & C\tilde{}/\underline{a}X & \xrightarrow{i/\underline{a}X} & C\hat{}/i\underline{a}X & \xrightarrow{\pi} & C\hat{}/X.
\end{array}$$

The notations $\underline{a}_X$ and i_X denote the "associated sheaf" and inclusion into presheaves functors for the site C/X. The notation $\underline{a}/X$ denotes the natural extension of the functor $\underline{a}$ (the associated sheaf functor for the site C) to the category of arrows with target X; similarly for the notation $i/\underline{a}X$. Finally, the notation π denotes the base-change functor along the canonical arrow $X \to i\underline{a}X$.

2) *Let, in addition, $m : Y \to X$ be a morphism of $\hat{C}$. The following diagram commutes up to canonical isomorphism:*

$$\begin{array}{ccccc}
C\tilde{}/\underline{a}X/\underline{a}Y & \xleftarrow{\quad f \quad} & (C/X)\tilde{}/\underline{a}_X Y & & \\
\| & & & \overset{g}{\nwarrow} & \\
 & & & & (C/X/Y)\tilde{} \\
 & & & \overset{e\tilde{}_m}{\swarrow} & \\
C\tilde{}/\underline{a}Y & \xleftarrow{e\tilde{}_Y} & (C/Y)\tilde{} & &
\end{array} .$$

The arrow f is the natural extension of the equivalence $e\tilde{}/X$ to the category of arrows with target $\underline{a}_X Y$, and is consequently an equivalence of categories. The arrow g is precisely the equivalence of 5.4 applied to the situation $(C/X)/[m]$. 295

3) *Denote by $j_{\underline{a}X!} : C\tilde{}/\underline{a}X \to C\tilde{}$ the "forget $\underline{a}X$" functor, and by $j^*_{\underline{a}X} : C\tilde{} \to C\tilde{}/\underline{a}X$ the "product with $\underline{a}X$" functor. The following diagrams commute up to canonical isomorphism:*

$$\begin{array}{ccc}
(C/X)\tilde{} & \xrightarrow{j_{X!}} & C\tilde{} \\
\downarrow & & \| \\
C\tilde{}/\underline{a}X & \xrightarrow{j_{\underline{a}X!}} & C\tilde{}
\end{array} \quad , \quad
\begin{array}{ccc}
C\tilde{} & \xrightarrow{j^*_X} & (C/X)\tilde{} \\
\| & & \downarrow{\scriptstyle e\tilde{}/X} \\
C\tilde{} & \xrightarrow{j^*_{\underline{a}X}} & C\tilde{}/\underline{a}X.
\end{array}$$

PROOF. The proof of these assertions follows immediately from the definition of the equivalences $e\tilde{}_X$ (5.4) and $e\hat{}_X$ (I 5.11).

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EXPOSÉ IV

Topoi

A. Grothendieck and J.-L. Verdier

0. Introduction

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0.1. We saw in II various exactness properties of categories of the form $\widetilde{\mathscr{C}}$ = the category of sheaves of sets on $\mathscr{C}$, where $\mathscr{C}$ is a small site. These properties can be expressed by saying that in many respects such categories (which we shall call *topoi*) inherit the familiar properties of the category (Ens) of (small) sets. On the other hand, experience has shown that various situations in Mathematics should be regarded *above all as a technical means of constructing the corresponding categories of sheaves* (of sets), i.e. *the corresponding "topoi."* It appears that all the genuinely important notions attached to a site (for example its cohomological invariants, studied in V; various other "topological" invariants, such as its homotopy invariants recently studied by M. Artin and B. Mazur [1]; and the notions studied in J. Giraud's book on noncommutative cohomology) are in fact expressed directly in terms of the associated topos. From this point of view, two sites should be regarded as essentially equivalent when their associated topoi are equivalent categories. The datum of a site (at least in the case, especially important in practice, where its topology is coarser than its canonical topology) should be regarded as amounting to the datum of a topos $\mathscr{E}$ (namely the associated topos formed by the sheaves of sets on the site) together with a generating family of objects of $\mathscr{E}$ (cf. II 4.9, and 1.2.1 below). This viewpoint is analogous to associating a group with a system of generators and a system of relations among those generators, and attaching interest rather to the structure of that group than to the system of generators and relations used to generate it (which are regarded as auxiliary data of the situation). Moreover, the "comparison lemma" III 4.1 provides numerous examples of pairs of sites $\mathscr{C}$, $\mathscr{C}'$ that are not isomorphic, and are not even equivalent as categories, but give rise to equivalent topoi; thus $\mathscr{C}$ and $\mathscr{C}'$ should be regarded as essentially equivalent.

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0.2. In the present exposé we give a characterization of topoi by simple exactness properties (due to J. Giraud); we study the natural notion of a morphism of topoi, inspired by the notion of a continuous map from one topological space to another; and we develop in the setting of topoi certain constructions familiar from ordinary sheaf theory (sheaves **Hom**, tensor-product sheaves, supports). Finally, following M. Artin, we show how a topos can be reconstructed from an "open" part, its complementary "closed" part, and a certain left exact functor connecting them, which may in fact be chosen almost arbitrarily.

0.3. This gives a procedure for gluing topoi which, when applied to topoi arising from ordinary topological spaces, will in general produce a topos no longer of the same type. This is a first indication of the remarkable stability of the notion of a topos under various natural constructions, a stability lacking for the notion of a topological space (from which the notion of a topos is inspired). As a second remarkable example, let

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us also mention the relative classifying topos of a group in a topos (cf. [1] or 5.9 below), inspired by the classical notion of the classifying space of a topological group, and the notion of a "moduli topos" associated with various "moduli problems" in Algebraic Geometry or Analytic Geometry [10] [13].

Other topoi, such as the *étale topos* of a scheme (systematically studied in the present Seminar beginning with Exp. VII) or the *crystalline topos* of a relative scheme [6], arise naturally when one wishes to develop usable cohomology theories for abstract algebraic varieties (and, more generally, schemes), replacing the classical Betti cohomology of algebraic varieties over the field of complex numbers.

0.4. Thus one may say that the notion of a topos, a natural outgrowth of the *sheaf-theoretic viewpoint* in Topology, in turn constitutes a substantial enlargement of the notion of a topological space[4], embracing a large number of situations which formerly were not regarded as belonging to topological intuition. The characteristic feature of such situations is the availability of a notion of "localization," a notion formalized precisely by that of a site and, ultimately, by that of a topos (through the topos associated with the site). As the term "topos" itself is intended precisely to suggest, it seems reasonable and legitimate to the authors of the present Seminar to regard the subject of Topology as the study of *topoi* (and not merely of topological spaces).

0.5. It seemed useful to include in this general exposé on topoi a fairly large number
302 of examples, many of which are only remotely related to the original aim of this seminar (namely, the study of étale cohomology). The reader in a hurry, interested exclusively in étale cohomology, may of course omit these examples without inconvenience, as well as most of the present exposé, consulting it only when the need arises.

1. Definition and characterization of topoi

DEFINITION 1.1. *A category E is called a $\mathcal{U}$-topos, or simply a topos when no confusion is likely, if there exists a site $C \in \mathcal{U}$ such that E is equivalent to the category $\tilde{C}$ of $\mathcal{U}$-sheaves of sets on C.*

1.1.1. Let E be a $\mathcal{U}$-topos. We always regard E as endowed with its canonical topology (II 2.5), which therefore makes it a *site*, and even, by 1.1.2 d) below, a $\mathcal{U}$-site (II 3.0.2). Unless expressly stated otherwise, we shall consider no topology on E other than the one just specified.

1.1.2. We saw in II 4.8, II 4.11 that a $\mathcal{U}$-topos E is a $\mathcal{U}$-category (I 1.1) satisfying the following conditions:

a) Finite projective limits are representable in E.
b) Direct sums indexed by an element of $\mathcal{U}$ are representable in E. They are disjoint and universal (II 4.5).
c) Equivalence relations in E are universally effective (I 10.10).
d) E admits a generating family (II 4.9) indexed by an element of $\mathcal{U}$.

303 In fact, we shall see that these intrinsic properties *characterize* $\mathcal{U}$-topoi:

THEOREM 1.2. (J. Giraud). *Let E be a $\mathcal{U}$-category. The following properties are equivalent:*

i) *E is a $\mathcal{U}$-topos* (1.1).
ii) *E satisfies conditions* a), b), c), *and* d) of 1.1.2.

[4] Cf. [9], or 4.1 and 4.2 below, for the precise relations between the notion of a topos and that of a topological space.

iii) *The $\mathscr{U}$-sheaves on E for the canonical topology are representable, and E has a small generating family (condition* 1.1.2 d)*).*

iv) *There exist a category $C \in \mathscr{U}$ and a fully faithful functor $i : E \to \hat{C}$ (where $\hat{C}$ denotes the category of $\mathscr{U}$-presheaves on C) having a left adjoint a which is left exact.*

i′) *There exists a site $C \in \mathscr{U}$ such that projective limits are representable in C and the topology of C is coarser than its canonical topology* (II 2.5)*, and such that E is equivalent to the category $\tilde{C}$ of $\mathscr{U}$-sheaves of sets on C.*

Moreover:

COROLLARY 1.2.1. *Let E be a $\mathscr{U}$-topos, and let C be a full subcategory of E, endowed with the topology induced* (III 3.1) *by that of E (1.1.1). Consider the functor*

$$E \longrightarrow \tilde{C}$$

which associates to every $X \in \mathrm{ob}\, E$ the restriction to C of the functor represented by X. This functor is an equivalence of categories if and only if ob C *is a generating family of E.*

The equivalence i) ⇔ iv) follows immediately from II 5.5, and we have already re- 304
called above that i) ⇒ ii). Since i′) ⇒ i) trivially, it remains to prove ii) ⇒ iii) and iii) ⇒ i′), which will be done in 1.2.4 and 1.2.3 below. The corollary is then obtained by observing that the functor in question factors as $E \to \tilde{E} \to \tilde{C}$; since $E \to \tilde{E}$ is an equivalence by virtue of (iii), the question reduces to determining when $\tilde{E} \to \tilde{C}$ is an equivalence. The conclusion then follows from the "comparison lemma" III 4.1; cf. the proof in 1.2.3 below.

1.2.3. *Proof of* iii) ⇒ i′). Let $\mathfrak{X} = (X_i)_{i \in I}$, $I \in \mathscr{U}$, be a small generating family of E. Since E is a $\mathscr{U}$-category, the set of isomorphism classes of finite diagrams in E whose objects are members X_i is $\mathscr{U}$-small. Consequently, the smallest set $\mathfrak{X}'$ of objects of E containing the finite projective limits of objects of $\mathfrak{X}$ is a countable union of small sets and is therefore small [4]. Setting $\mathfrak{X}^{(n+1)} = (\mathfrak{X}^{(n)})'$, and $\mathfrak{X} = \bigcup_n \mathfrak{X}^{(n)}$, we see that, after enlarging the generating family, we may suppose that $\mathfrak{X}$ is stable under finite projective limits. Let $\mathscr{V}$ be a universe containing $\mathscr{U}$ such that E is $\mathscr{V}$-small. For every object H of E, denote by $I(H)$ the set $\coprod_{i \in I} \mathrm{Hom}(X_i, H)$. Let H be an object of E and let $H' \hookrightarrow H$ be the subsheaf of H, for the canonical topology, which is the "union" of the images of the morphisms $u : X_i \to H$, $(u, i) \in I(H)$ (II 4.1). Since H' is a subsheaf of a $\mathscr{U}$-sheaf, H' is a $\mathscr{U}$-sheaf. It is therefore representable. Since the family $\mathfrak{X}$ is generating and, for every $i \in I$, the map $\mathrm{Hom}(X_i, H') \to \mathrm{Hom}(X_i, H)$ is bijective, the morphism $H' \hookrightarrow H$ is an isomorphism (II 4.9). Consequently (II 5.2), the family $(u : X_i \to H), (u, i) \in I(H)$, is covering for the canonical topology of E. Thus every object of E can be covered, 305
for the canonical topology of E, by objects of $\mathfrak{X}$. Let C be the full subcategory of E defined by the objects of $\mathfrak{X}$, and let $u : C \hookrightarrow E$ be the inclusion functor. The topology $\mathscr{C}$ induced on C by the canonical topology of E is coarser than the canonical topology of C, and when $\mathscr{U}$ contains an element of infinite cardinality, finite projective limits are representable in C. It follows from III 4.1 that the functor $F \mapsto F \circ u$ is an equivalence from E to the category of $\mathscr{U}$-sheaves on C for the topology $\mathscr{C}$, C.Q.F.D.

1.2.4. *Proof of* ii) ⇒ iii). The proof consists of four steps.

Let $\mathscr{V}$ be a universe containing $\mathscr{U}$, such that E is an element of $\mathscr{V}$. Let $\widetilde{E}$ be the category of $\mathscr{V}$-sheaves on E for the canonical topology, and let $J_E : E \to \widetilde{E}$ be the canonical functor.

[4] Here we reason with classes of objects of E up to isomorphism.

. Let $(g_i : G_i \to H)$, $i \in I \in \mathscr{U}$, be an epimorphic family in $\widetilde{E}$. If the G_i and the fiber products $G_i \times_H G_j$ are representable, then the sheaf H is representable.

Indeed, the direct sum $\coprod_{i\in I} G_i$ is represented in $\widetilde{E}$ (II 4.1) by a representable sheaf G (property b) and II 4.6). The same holds for the direct sum $K = \coprod_{(i,j)\in I\times I} G_i \times_H G_j$. Moreover, the diagram $K \rightrightarrows G \to H$ is exact and K is the fiber square of G over H. (This latter property holds in the category of presheaves, and hence in the category of sheaves (II 4.1).) It follows from c) and II 4.7 that the sheaf H is representable.

. Let X_α, $\alpha \in A \in \mathscr{U}$, be a generating family of E. For every sheaf H, denote by $I(H)$
306 the set $\coprod_{\alpha\in A} \mathrm{Hom}_E(X_\alpha, H)$. The family $(u : X_\alpha \to H, (u,\alpha) \in I(H))$ is epimorphic in $\widetilde{E}$. When H is a $\mathscr{U}$-sheaf, $I(H)$ is an element of $\mathscr{U}$.

Indeed, since every sheaf H is the target of an epimorphic family of morphisms whose sources are representable sheaves (I 3.4 and II 4.1), it is enough to prove the first assertion when the sheaf H is representable. Let G then be the image of the family $(u : X_\alpha \to H, (u,\alpha) \in I(H))$. The morphism $G \to H$ is a monomorphism. We are therefore in the situation of the first step, since $X_\alpha \times_G X_\beta$ is isomorphic to $X_\alpha \times_H X_\beta$, which is representable, and moreover $I(H)$ is an element of $\mathscr{U}$. It follows that G is representable. But since X_α is a generating family, $G \to H$ is an isomorphism. The last assertion is immediate from the definition of $\mathscr{U}$-sheaves.

. Every subsheaf of a representable sheaf is representable.

Indeed, let $G \to H$ be a subsheaf of a representable sheaf. Then G is a $\mathscr{U}$-sheaf. The family $(u : X_\alpha \to G, (u,\alpha) \in I(G))$ is therefore epimorphic in $\widetilde{E}$ and indexed by an element of $\mathscr{U}$. Moreover, the fiber products $X_\alpha \times_G X_\beta$ are isomorphic to the fiber products $X_\alpha \times_H X_\beta$, which are representable. Thus we are in the situation of the first step and G is representable.

. Every $\mathscr{U}$-sheaf is representable.

Indeed, by the first and second steps, it is enough to show that the fiber products $X_\alpha \times_H X_\beta$ are representable. But these fiber products are subobjects of the products $X_\alpha \times X_\beta$. The conclusion therefore follows from the third step. This completes the proof of Theorem 1.2.

307 Remark 1.3. Of course, for a given $\mathscr{U}$-topos E, there is in general no preferred way of representing it, up to equivalence, in the form $\tilde{C}$, with C a *small* site; or, what amounts essentially to the same thing by virtue of 1.2.1 when one restricts to C whose topology is coarser than the canonical topology, there is no preferred *small* generating family in E. When one ceases to impose on C the condition that C be small, there is, on the other hand (by virtue of 1.2 iii)), an entirely canonical choice of a $\mathscr{U}$-site C such that E is equivalent to $\tilde{C}$, namely E itself! This is one of the technical reasons why in practice it is inconvenient to work only with small sites: in fact, the most important sites of all, namely $\mathscr{U}$-topoi, are not small! Moreover, the sites generating topoi that arise in many questions in algebraic geometry (and even in topology, cf. 2.5) are not small either; examples include the étale site of a scheme (VII.1).

Proposition 1.4. *Let E be a $\mathscr{U}$-topos, let E' be a category (not necessarily a $\mathscr{U}$-category), and let F be a presheaf on E with values in E'. In order that F be a sheaf with values in E'* (II 6.1)*, it is necessary and sufficient that F take $\mathscr{U}$-inductive limits in E to projective limits in E'.*

Let $\mathscr{V}$ be a universe containing $\mathscr{U}$ and such that E' is a $\mathscr{V}$-category. By composing F with the functors $\mathrm{Hom}(X', -) : E' \to \mathscr{V}$-Ens defined by the objects X' of E', we are

reduced to the case where $E' = \mathscr{V}\text{-Ens}$, with $\mathscr{V}$ a universe. Suppose that F takes $\mathscr{U}$-inductive limits to projective limits. It then follows at once from the definitions that F is a sheaf, since a covering family $X_i \to X$ in $\mathscr{E}$, by the definition of the canonical topology 308
of $\mathscr{E}$, allows one to regard X as an inductive limit of the diagram $\left(X_i \times_X X_j \overset{\nearrow X_i}{\searrow X_j}\right)$.
Suppose that F is a sheaf, and let us prove that it takes $\mathscr{U}$-inductive limits to projective limits. If $\mathscr{V} \subset \mathscr{U}$, we may suppose that $\mathscr{V} = \mathscr{U}$, and it suffices to apply criterion 1.2 iii). To treat the general case, a little more work is needed. Let C be a full subcategory of E generated by a generating family indexed by an element of $\mathscr{U}$, and let $u : C \to E$ be the inclusion functor. Endow C with the topology induced by the canonical topology of E (III 3.5). Denote by $\tilde{C}_{\mathscr{U}}$ and $\tilde{C}_{\mathscr{V}}$ the categories of sheaves on C, by $\tilde{E}_{\mathscr{V}}$ the category of V-sheaves on E for the canonical topology, by $J_E : E \to E_{\mathscr{V}}$ the canonical functor, and by $i_{\mathscr{U},\mathscr{V}} : \tilde{C}_{\mathscr{U}} \to \tilde{C}_{\mathscr{V}}$ the inclusion functor. The diagram

(∗)
$$\begin{array}{ccc} \tilde{C}_{\underline{U}} & \xrightarrow{i_{\underline{U},\underline{V}}} & \tilde{C}_{\underline{V}} \\ \uparrow & & \uparrow \\ E & \xrightarrow{J_E} & \tilde{E}_{\underline{V}} \end{array},$$

where the vertical arrows are induced by the functor $F \mapsto F \circ u$, is commutative. It follows from the explicit construction of the associated sheaf functor (II 3) and from II 4.1 that the functor $i_{\mathscr{U},\mathscr{V}}$ commutes with $\mathscr{U}$-inductive limits. Moreover, it follows from III 5.1 and 1.5 that the vertical arrows in (∗) are equivalences of categories. Consequently, $J_E : E \to \tilde{E}_{\mathscr{V}}$ commutes with $\mathscr{U}$-inductive limits. For every object X of E, we have

$$F(X) \simeq \mathrm{Hom}_{\tilde{E}_{\mathscr{V}}}(J_E(X), F).$$

Thus F takes $\mathscr{U}$-inductive limits in E to projective limits. 309

Corollary 1.5. *Let E be a $\mathscr{U}$-topos, let E' be a $\mathscr{U}$-category, and let $f : E \to E'$ be a functor. In order that f admit a right adjoint, it is necessary and sufficient that f commute with $\mathscr{U}$-inductive limits.*

This is an immediate consequence of 1.4 in the case $E' = \mathscr{U}\text{-Ens}$.

Corollary 1.6. *Let E and E' be two $\mathscr{U}$-topoi, and let $f : E \to E'$ be a functor. The following conditions are equivalent:*

i) *f commutes with $\mathscr{U}$-inductive limits.*
ii) *f admits a right adjoint.*
iii) *f is continuous* (III 1.1).

The equivalence i) ⇔ ii) was proved in 1.5. To prove i) ⇔ iii), apply the definition of continuous functors, choosing a universe $\mathscr{V}$ such that $\mathscr{U} \in \mathscr{V}$ (so E and E' are $\mathscr{V}$-small sites). We must express that for every $\mathscr{V}$-sheaf F on E', the composite $F \circ f$ is a sheaf on E, that is (1.4), that it takes $\mathscr{U}$-inductive limits to projective limits. For this it is sufficient that i) hold, since by 1.4 F itself takes $\mathscr{U}$-inductive limits to projective limits; it is also necessary, as one sees by taking F to be a representable functor.

Corollary 1.7. *With the notation of 1.6, in order that f be the inverse-image functor u^* for a morphism of topoi* (3.1) *$u : E' \to E$, it is necessary and sufficient that f be left exact and take epimorphic families to epimorphic families.*

310 The necessity is trivial by definition (NB every right exact functor takes epimorphisms to epimorphisms). The sufficiency follows from III 1.6, which implies that f is continuous under the stated conditions, and hence commutes with $\mathscr{U}$-inductive limits by 1.6.

Remark 1.8. With the notation of 1.5, one can show that f admits a left adjoint if and only if it commutes with $\mathscr{U}$-projective limits. In other words, a *covariant* functor $E \to (\mathscr{U}\text{-Ens})$ is representable if and only if it commutes with $\mathscr{U}$-projective limits [4]. We indicate only the principle of the proof, which proceeds in three steps:

a) Standard arguments [5, n° 195, § 3] show that if F commutes with $\varprojlim$, it is prorepresentable by a strict projective system $(T_i)_{i\in I}$, where I is a filtered ordered set, *not necessarily small.* We may suppose that if $i > j$, then $T_i \to T_j$ is not an isomorphism, and under this hypothesis F is representable if and only if I is small (which in fact implies that the projective system is essentially constant).

b) To prove that I is small, knowing that for every object X of E the set $F(X) = \varinjlim_i \mathrm{Hom}(T_i, X)$ is small, it suffices to have a *small cogenerating family* $(X_j)_{j\in J}$ (i.e. one that is generating for the opposite category E°). Now one shows that every $\mathscr{U}$-topos E has a small cogenerating family.

c) To prove this last point, one observes by standard arguments [Toh] that every object X of E admits a monomorphism into an "injective" object; and then that
311 for every generating family (L_α) of E, if each L_α is embedded in this way into an injective object I_α, the family (I_α) is cogenerating.

2. Examples of topoi

2.0. We have brought together in this section a fairly large number of typical examples of topoi, which the reader will already have had occasion to encounter elsewhere, and which are intended to facilitate his access to the "yoga" of topoi. For other examples (drawn from algebraic geometry) of topologies on sites, giving rise to as many topoi, he may consult SGA 3 IV 6, and (for the étale topos) Exposé VII of the present seminar. Since later on we shall hardly refer to the present section except for questions of notation or terminology, we leave it to the reader, as an exercise, to verify the statements with which we have supplemented these examples for his general instruction. All the examples of the present section will be made more precise in 4, where their functorial dependence on the data will be examined, and in the following sections by way of illustration of the general notions concerning topoi.

2.1. Topos associated with a topological space. Let X be a small topological space, and let $\mathrm{Ouv}(X)$ be the category of open subsets of X, endowed with the canonical topology (III 1.9.1). We shall denote by $\mathrm{Top}(X)$ the topos of $\mathscr{U}$-sheaves on $\mathrm{Ouv}(X)$. This topos is equivalent to the category of étale topological spaces over X, by associating with every such space X' over X the sheaf $U \mapsto \Gamma(X'/U) = \mathrm{Hom}_X(U, X')$ on $\mathrm{Ouv}(X)$ [TF][5]
312 By an abuse of language, we shall sometimes be unable to refrain from denoting by the same letter X the topos $\mathrm{Top}(X)$ defined by the topological space X.

It is of course the preceding example that served chiefly as a guide and as an intuitive basis for the development of the theory of topoi. One should note, however, that the topoi obtained from topological spaces are of a very special nature, owing in particular

[4] A more general statement is found in I 8.12.8, I 8.12.9; the outline of the proof following a) through c) corresponds to the proof given in *loc. cit.*

[5] N.D.E.: Here [TF] denotes reference [8].

to the fact that they have been described by sites $C = \mathrm{Ouv}(X)$ in which *all morphisms are monomorphisms* (and hence whose underlying category reduces to a preordered set). It follows in particular that the sheaves represented by the objects of C are subsheaves of the terminal sheaf, and consequently that *the subsheaves of the terminal sheaf form a generating family of the topos under consideration.* This property is not shared by most topoi that arise naturally in algebraic geometry or in algebra, cf. the examples below. It is, up to minor qualifications, characteristic of topoi of the form $\mathrm{Top}(X)$ (cf. 7.1.9 below).

It is easily verified that the map $\mathrm{Ouv}(X) \to \mathrm{Top}(X)$, which associates with every open subset of X the sheaf it represents, is a *bijection* from $\mathrm{Ouv}(X)$ onto the set of subobjects of the terminal object of $\mathrm{Top}(X)$; this bijection is even an isomorphism for the natural order structures, i.e. it induces an isomorphism of the corresponding categories. This suggests that it should be possible to reconstruct the topological space X, up to homeomorphism, from $\mathrm{Top}(X)$ known up to equivalence. We shall see below (4.2) that this is indeed the case, subject to a slight restriction on X.

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2.2. **Point topos or terminal topos, and empty topos or initial topos.** When X is a topological space consisting of a single point, the functor

$$F \longrightarrow F(X) : \mathrm{Top}(X) \longrightarrow \mathscr{U}\text{-Ens}$$

is an equivalence of categories. This shows in particular that the *category* $\mathscr{U}$-Ens is a *$\mathscr{U}$-topos.* We saw by examples in Exp. II that this $\mathscr{U}$-topos is typical from the point of view of exactness properties, in that the verification of many properties (especially exactness properties) of general topoi reduces to this particular topos. The interpretation given here in terms of the one-point topological space justifies the abuse of language by which a topos equivalent to the category $\mathscr{U}$-Ens is called a *point topos* (although, as a category, it is not at all equivalent to the one-point category!). This is the terminology that corresponds to the correct geometric intuition of the role played by these topoi. By another abuse of language, a point topos is sometimes called a *terminal topos*, cf. 4.3; we shall say "*the* terminal topos" for the topos ($\mathscr{U}$-Ens).

When X is the empty topological space, so that $\mathrm{Ouv}(X)$ is the one-point category, a presheaf F on $\mathrm{Ouv}(X)$ is a sheaf if and only if its value at the unique object ϕ of $\mathrm{Ouv}(X)$ is a singleton set. It follows that $\mathrm{Top}(\varnothing)$ is isomorphic to the category of singleton $\mathscr{U}$-sets, a category equivalent to the one-point category. From this one concludes in particular that the one-point category (as well as any $\mathscr{U}$-category equivalent to it) is a $\mathscr{U}$-topos. By an abuse of language, it is sometimes called the *empty topos* or the *initial topos* (cf. 4.4); 314
one should take care that it is not equivalent to the empty category.

2.3. **Topos associated with a space with operators.** Let X be a topological space and G a discrete group acting on X by homeomorphisms. In [Toh][6] 5.1 the category of *G-sheaves on X*, or, as we shall also say, sheaves on (X, G), was defined; these are sheaves (of sets) on X *equipped* with actions of G compatible with those of G on X. One sees at once, using Giraud's criterion 1.2 iii), that this category is a topos (NB $\mathscr{U}$ is understood throughout), which we shall simply denote by $\mathrm{Top}(X, G)$. When G is the trivial group, we recover the example of 2.1; when X is the one-point space, we obtain the topos of sets on which G acts on the left (or *G-sets*), also called the *classifying topos of the discrete group G*, and denoted by B_G. It is easily verified that the only subobjects of the terminal object e of the topos B_G are e and the empty sheaf ϕ; in particular, if G is not the trivial group, the subobjects of the terminal object of the classifying topos B_G

[6] N.D.E.: Here [Toh] denotes reference [4].

do not form a generating family of B_G. Thus B_G is then not equivalent to a topos of the type Top(X) considered in 2.1.

The notion of a G-sheaf was introduced in *loc. cit.* in order to develop the cohomological theory of abelian G-sheaves. Interpreting the latter as the abelian sheaves of the topos (X, G), that theory becomes part of the theory of Exp. V, developed in the setting of general topoi.

315 More generally, one may seek to associate an appropriate topos with a topological space X equipped with a topological group G (not necessarily discrete) of automorphisms, which would give rise to an adequate cohomological theory. The same may be done in the context of differentiable manifolds, real or complex analytic manifolds, or schemes. This is indeed possible, cf. 2.5 below.

2.4. Classifying topos of a group object. Let E be a topos, and let G be a group object of E. Let (E, G) be the category of objects of E on which G acts. It follows at once from Giraud's criterion that this is a topos. It is called the *classifying topos* of the group object G, and is denoted by B_G. When E is the point topos (2.2), i.e. when G is an ordinary group, we recover the classifying topos of 2.3.

The terminology adopted here is justified by the fact that the topos B_G plays a universal role in the classification of "torsors" (or homogeneous principal bundles) under G, or more generally under $G_{E'} = f^*(G)$, where E' is a topos "over E", i.e. endowed with a morphism $f : E' \to E$ (cf. 3.1 below). This role, made explicit in [3 Chap V] or in 5.9 below, shows that B_G plays, in the context of topoi, the same role as the classical classifying spaces of topological groups in the homotopy theory of topological spaces. The latter may be regarded (cf. 2.5) as a weakened version of the former, obtained by retaining from the classifying topos only the "homotopy type" of that topos, in a suitable sense which need not be specified here.

2.5. "Big site" and "big topos" of a topological space. Classifying topos of a topo-
316 **logical group.** [6]

Let U – Esp, or simply (Esp), be the category of topological spaces $\in \mathcal{U}$. It is known that finite projective limits are representable in (Esp). Consider on (Esp) the pretopology (II 1.3) for which Cov(X) is the set of surjective families of open embeddings $u_i : X_i \to X$. We regard (Esp) as a site by means of the topology generated by the preceding pretopology. For every object X of (Esp), regard the category

$$(\mathrm{Esp})_{/X}$$

of objects of (Esp) over X, i.e. topological spaces over X, as a site, using the topology induced by that of (Esp) via the forgetful functor $(\mathrm{Esp})_{/X} \to (\mathrm{Esp})$ (III 5.2 4)). This site is called the *big site* associated with X. One should note that it is not $\in \mathcal{U}$; nor is it a $\mathcal{U}$-site in the sense of II 3.0.2, so precautions are necessary when applying the usual results to it. To remedy this inconvenience, one may choose a universe $\mathcal{V}$ such that $\mathcal{U} \in \mathcal{V}$, so that $(\mathrm{Esp})_{/X}$ becomes a $\mathcal{V}$-site, and work with the associated $\mathcal{V}$-topos $(\mathrm{Esp})_{/X}$, which may be denoted by TOP(X) and will be called the *big topos of* X. If one is reluctant to enlarge $\mathcal{U}$, one may choose a cardinal c bounding the cardinalities of X and of all the topological spaces one intends to use in the arguments (most often, Sup(card X, card $\mathbb{R}$) will suffice!), and replace $(\mathrm{Esp})_{/X}$ by the subcategory $(\mathrm{Esp})'_{/X}$ consisting of those X' over X such that card $X' \leq c$, endowed with the induced topology; one then denotes by
317 TOP(X) the topos of sheaves on this site. To fix ideas, suppose that the first definition

[6] The introduction of these sites and topoi is due to M. Giraud, who also brought out their advantages over the traditional "small" site.

has been adopted.

The advantage of the big topos of X over the small one is that the site defining it contains $(\mathrm{Esp})_{/X}$ as a full subcategory; since the topology of this site is manifestly coarser than the canonical topology, the canonical functor from $(\mathrm{Esp})_{/X}$ to $\mathrm{TOP}(X)$, associating with every space X' over X the sheaf it represents, is seen to be *fully faithful.* Consequently, *a space X' over X is determined up to X-isomorphism by the sheaf* ($\in \mathrm{Top}(X)$) *it defines*; thus the notion of a sheaf on (the big site of) X may be regarded as a *generalization* of the notion of a topological space over X, by means of which all constructions of sheaf theory acquire a meaning for topological spaces over X.

Thus, when G is a group object of the category $(\mathrm{Esp})_{/X}$ of topological spaces over X, one may associate with it the classifying topos B_G (2.4), hence classifying cohomology groups, classifying homotopy groups, etc. (defined as the corresponding invariants of the $\mathscr{V}$-topos B_G). In particular, when X is a one-point space, G is identified with an ordinary topological group. Subject to the usual local conditions ensuring that the singular cohomology of the Cartesian products G^n agrees with cohomology in the sense of sheaves (for constant coefficients, say), for example if G is locally contractible, one may verify that the cohomology of the classifying topos of G is canonically isomorphic to that of the classifying space of G in the sense of topologists.

The introduction of classifying topoi (via "big sites") has over classifying spaces 318
the advantage of providing a richer theory, since in particular they provide useful cohomological invariants for coefficients more general than constant or locally constant coefficients. Moreover, the definition considered here adapts in an evident way to the other customary contexts: differentiable manifolds, real or complex analytic manifolds or spaces, and schemes. This point of view makes it possible in particular to connect the study of characteristic classes from the traditional and the "arithmetic" points of view, by regarding the "classical groups" as arising from schemes defined over the ring of integers; cf. [7] for indications in this direction. Likewise, the general results of J. Giraud [3] on the classification of extensions of group objects, developed in the very general and flexible setting of topoi, may, by means of "big topoi", be specialized to results on the classification of extensions of topological groups, or of real or complex Lie groups, which seem to have been scarcely known to topologists except in the case of extensions with abelian kernel [11].

2.6. Topoi of the form $\widehat{C}$. Let C be a small category. Then the category $\widehat{C}$ of U-presheaves on C is plainly a $\mathscr{U}$-topos, since it is of the form $\tilde{C}$, where C is endowed with the chaotic topology. We shall give below some details on the relations between C and $\widehat{C}$. Let us note here only that a topos E equivalent to a topos of the form $\widehat{C}$ is of a rather special nature, because *it has a small generating family (X_i) consisting of connected projective objects*, i.e. objects X such that the functor $Y \mapsto \mathrm{Hom}(X, Y)$ takes 319
epimorphisms to epimorphisms and coproducts to coproducts: indeed, it suffices to take in $\widehat{C}$ the generating family consisting of the functors represented by the $X \in \mathrm{ob}\, C$. Moreover, note that if in a topos E one has a covering family $X_i \to X$, where the X_i are projective and connected, then every other covering family of X is refined by the former (I 4.3.2, I 4.3.3). Consequently, in a topos E of the form $\widehat{C}$ every object X admits a covering family refining all the others. A topos of the form $\mathrm{Top}(X)$ (2.1), where X is a topological space whose points are closed, has the preceding property only if X is discrete.

When the category C has a single object, C is identified with a monoid G. A presheaf on C is then identified with a set on which G acts *on the right* (since it is a functor

$G^\circ \to$ (Ens)), and $\widehat{C}$ is the topos of sets with a monoid of right operators, which may also be denoted by B_{G° in view of 2.3: it is the *topos of sets with monoid of operators* G° *(the monoid opposite to* G*).* When G is a group, using the isomorphism $g \mapsto g^{-1}$ from G onto G°, one recovers the classifying topos B_G of 2.3.

2.7. Classifying topos of a pro-group.

2.7.1. Let $\mathscr{G} = (G_i)_{i\in I}$ be a projective system of groups, with G_i, $I \in \mathscr{U}$. Assume that the projective system is *strict*, i.e. that the transition morphisms $G_j \to G_i$ are surjective. If E is a set, by a (left, say) *action of* $\mathscr{G}$ *on* E we mean the following structure: a) a family $(E_i)_{i\in I}$ of subsets of E, whose union is E; b) for every $i \in I$, an action of
320 the group G_i on the set E_i; these data are furthermore required to satisfy the following condition: for $j \geq i$, E_i is the subset of E_j consisting of the elements fixed under the kernel of $G_j \to G_i$. We also say that $\mathscr{G}$ *acts on* E (on the left) if a (left) action of $\mathscr{G}$ on E has been given. The sets $\in \mathscr{U}$ endowed with an action of $\mathscr{G}$ form a category in the evident way. By Giraud's criterion, one sees at once that this category is a $\mathscr{U}$-topos. It is denoted by $B_{\mathscr{G}}$ and called the *classifying topos of* $\mathscr{G}$. When I has an initial object $i_\circ$, on putting $G = G_{i_\circ}$ one recovers the classifying topos of 2.3.

2.7.2. Another important example is that in which the groups G_i are finite, so that

$$G = \varprojlim G_i$$

is a compact totally disconnected topological group, or *profinite group.* An action of $\mathscr{G}$ on E then amounts to a continuous action of G on E, or, equivalently, to one for which the stabilizer of every point of E is an open subgroup of G. The classifying topos $B_{\mathscr{G}}$ will also be denoted by B_G, where, of course, G is to be regarded as endowed with its profinite topology.

2.7.3. Using the remarks of 2.6, it is easy to verify that the topos $B_{\mathscr{G}}$ defined by a strict projective system $\mathscr{G} = (G_i)_{i\in I}$ of groups is equivalent to a topos of the form $\widehat{C}$ only if this projective system is essentially constant; in the case of a profinite group, this means that the group is in fact finite.

2.7.4. The following geometric interpretation of the classifying topos B_G of a discrete group G is useful for giving a correct geometric intuition for these topoi. (Cf. also, in the same vein, 4.5, 5.8, 5.9, and 7.2 below.) Let X be a connected, locally connected,
321 and locally simply connected topological space, let x be a point of X, and let G be its fundamental group at x. (NB It is known that, up to isomorphism, every discrete group G may be obtained in this way.) Then the Galois theory of coverings of X gives an equivalence between the category B_G of G-sets and the category of *étale coverings* of X, i.e. spaces X' over X that are locally X-isomorphic to X-spaces of the form $X \times I$, where I is a discrete space. (Compare SGA 1 V 4,5.) The latter can moreover be interpreted as the category of locally constant sheaves on X, i.e. the category of locally constant objects (IX 2.0) of the topos Top(X).

When G is a profinite group, there is an analogous geometric interpretation of B_G as the category of X-schemes that are coproducts of finite étale coverings of a connected scheme X, equipped with a geometric point x and an isomorphism $G \simeq \pi_1(X, x)$. It is again known that every profinite group may be obtained as the fundamental group of a suitable connected scheme (the spectrum of a field, if desired). Finally, pro-groups (not necessarily profinite or essentially constant) also occur in the classification of coverings of connected and locally connected spaces that are not locally simply connected, or in the classification of étale coverings, not necessarily finite or ind-finite, of connected nonnormal schemes. For the latter case, cf. SGA 3 X 6.

EXERCISE 2.7.5. For a topos E, define the notions of connectedness, local connectedness [6], simple connectedness, and local simple connectedness. Define the notion of a constant and locally constant object of E (cf. IX 2.0). Let $f : E' \to E$ be a morphism of 322
topoi (3.1), with E' simply connected (for example, E' the point topos (2.2)), and E connected and locally connected. Define a strict pro-group $\pi_1(E, f) = (G_i)_{i\in I} = \mathscr{G}$ (called the *fundamental pro-group of E at f*) and an equivalence of categories between $B_{\mathscr{G}}$ and the category of locally constant objects of E.[6] Show that, when E is locally simply connected, $\pi_1(E, f)$ is essentially constant and hence is identified with an ordinary discrete group $\pi_1(E, f)$, called the *fundamental group of E at f*. Show that every strict pro-group $\mathscr{G}$ is isomorphic (as a pro-group) to the fundamental pro-group of a suitable connected and locally connected topos at a suitable f, with E' the point topos (take $E = B_{\mathscr{G}}$, and $f : E' \to E$ defined by the forgetful functor $f^* : E \to E' = (\mathrm{Ens})$). When $\mathscr{G}$ is essentially constant, i.e. isomorphic (as a pro-group) to an ordinary discrete group, prove that one may take E above to be locally simply connected (again take $E = B_G$).

2.8. Example of a false topos. Let $\mathscr{G} = (G_i)_{i\in I}$ be a strict pro-group, where I is a filtered ordered set and where $i > j$ implies that $G_i \to G_j$ is not an isomorphism. Suppose that $\mathrm{card}(I) \notin \mathscr{U}$. Consider the category of sets $E \in \mathscr{U}$ on which $\mathscr{G}$ acts on the left (2.7.1). It is a $\mathscr{U}$-category, and one sees as in 2.7.1 that this category satisfies conditions a), b), and c) of 1.1.2. Nevertheless, it is not a $\mathscr{U}$-topos, since it has no generating family that is $\mathscr{U}$-small. Similarly, it is not a $\mathscr{V}$-topos for any universe $\mathscr{V}$.

3. Morphisms of topoi

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DEFINITION 3.1. *Let E and E' be two $\mathscr{U}$-topoi. A morphism[7] from E to E', or sometimes (by abuse of language) a continuous map from E to E', is a triple $u = (u_*, u^*, \varphi)$, consisting of functors*

$$u_* : E \longrightarrow E' \quad , \quad u^* : E' \longrightarrow E$$

and an "adjunction" isomorphism φ of bifunctors in $X' \in \mathrm{Ob}\, E'$, $Y \in \mathrm{Ob}\, E$:

$$\varphi : \mathrm{Hom}_E(u^*(X'), Y) \xrightarrow{\sim} \mathrm{Hom}_{E'}(X', u_*(Y)),$$

where the functor u^* is moreover required to be left exact, i.e. to commute with finite projective limits. The functor u_* is called the direct-image functor for the morphism of topoi u, the functor u^* is called the inverse-image functor for the morphism of topoi u, and the isomorphism φ is called the adjunction isomorphism for u.

3.1.1. Unless expressly stated otherwise, for a morphism of topoi $u : E \to E'$, we shall henceforth denote the corresponding direct- and inverse-image functors by u_* and u^* [7]. Note that, since u_* is right adjoint to u^* and u^* is left adjoint to u_* by the adjunction isomorphism φ, each of the two functors u_*, u^* determines the other up to unique isomorphism, by the well-known general facts about adjoint functors [14]. In practice, depending on the circumstances, it may be more convenient to define a morphism of topoi $u : E \to E'$ either by specifying $u^* : E' \to E$, or by specifying $u_* : E \to E'$; in the first case, one need only verify that the given functor u^* has a 324
right adjoint and is left exact. In the second case, one must verify that the given functor u_* has a left adjoint which is left exact. In either case, from the partial data, by *choosing*

[6] cf. 8.7 l).

[6] One may take inspiration from SGA 3 X 6.

[7] N.D.E.: Some authors speak of a *geometric morphism*.

[7] It is sometimes appropriate also to write u^{-1} instead of u^*, cf. 13.2.3.

an adjoint functor and an adjunction isomorphism, one obtains a morphism of topoi $u : E \to E'$, and the latter will be "unique up to unique isomorphism" in terms of the datum u^* resp. u_*, in a sufficiently clear sense which will moreover be made completely explicit below (3.2.1).

3.1.2. If $u : E \to E'$ is a morphism of topoi, it follows from the properties of adjoint functors (I 2.11) that the direct-image functor $u_* : E \to E'$ commutes with projective limits, and the functor $u^* : E' \to E$ commutes with inductive limits [7]. Since the latter is assumed in addition to be left exact, i.e. to commute with finite projective limits, we see in particular that u^* *is exact.* We may therefore say that *it is the inverse-image functor* u^*, in the pair (u_*, u^*), *which has the most remarkable exactness properties.* These properties ensure that, for every type of algebraic structure Σ whose data can be described in terms of arrows between the underlying sets and sets deduced from them by repeated application of operations of finite projective limits and arbitrary inductive limits, and for every "object of $\mathscr{E}'$ endowed with a structure of type Σ" (a notion which makes sense by virtue of the internal exactness properties of the topos $\mathscr{E}'$ (II 4.1)), its image under u^* is endowed with the same structures. Rather than undertake the unrewarding task
325 of giving a precise meaning to this statement and formally justifying it, we advise the reader to spell it out and convince himself of its validity for types of structure such as groups, rings, modules over a ring, comodules over a coring, bialgebras over a ring, and torsors under a group. (In these examples, the first three types of structure are defined exclusively in terms of finite projective limits, while the other notions implicitly involve constructions which also use inductive limits.) Furthermore, the functor u^* "*commutes*" with all the usual functorial operations performed in terms of such structures, more precisely with all operations which can be expressed in terms of $\varinjlim$ and finite $\varprojlim$: constructions of free objects (free groups or modules, for example) generated by an object, tensor products (cf. § 12 below), etc.

As for the direct-image functor $u_* : E \to E'$, since it commutes with projective limits, it consequently "preserves" every algebraic structure on an object (or a family of objects) of E which can be defined exclusively in terms of projective limits (such as the structures of group, ring, or module over a ring among the preceding examples). On the other hand, the functor u_* is not in general right exact, i.e. it does not in general commute with finite inductive limits, and it does not even in general take epimorphisms to epimorphisms (it is, moreover, this failure of exactness of u_* which is the source of its cohomological properties; these will be studied, from the viewpoint of commutative homological algebra, in the following exposé). Consequently it does not, in general,
326 extend to a functor on objects of the type comodule, bialgebra, or torsor under a group, and does not in general commute with operations such as "free module generated by", tensor product of modules, etc.

3.1.3. In practice, when one is given a functor $f : E \to F$ from one $\mathscr{U}$-topos to another, one should make all of its exactness properties explicit, including the possible existence of left or right adjoint functors (cf. the footnote on page 162), in order to understand the "geometric nature" of f, an understanding which will generally be an indispensable guide to a correct geometric intuition of the situation. Thus, if it turns out that f commutes with arbitrary inductive limits and finite projective limits, one should write f in the form

$$f = u^*,$$

[7] Moreover (1.5 and 1.8), for a given functor $u_* : E \to E'$ resp. $u^* : E' \to E$, this functor has a left adjoint (resp. a right adjoint) if and only if it commutes with $\mathscr{U}$-projective limits (resp. $\mathscr{U}$-inductive limits).

where

$$u : F \longrightarrow E$$

is a morphism of topoi, i.e. interpret f as an "inverse-image" functor for a "continuous map" of topoi. When f commutes with arbitrary projective limits, so that it has a left adjoint, and if the latter (which a priori commutes with arbitrary inductive limits) *also* commutes with finite projective limits, one should write f in the form

$$f = v_*,$$

where

$$v : E \longrightarrow F$$

is a morphism of topoi. In certain cases it may happen that f satisfies both of the properties under consideration (cf. 4.10 for an example). In that case one should introduce simultaneously the morphisms of topoi 327

$$u : F \longrightarrow E \quad , \quad v : E \longrightarrow F,$$

which give rise to a sequence of three adjoint functors (I 5.3):

$$v^* \quad , \quad v_* = u^* = f \quad , \quad u_*.$$

One must then distinguish carefully between the morphisms of topoi u and v, lest one lose the geometric intuition of the situation.

In this connection, note that when between two topoi E, F one has a sequence of three adjoint functors

$$e, f, g \ (e, g : F \rightrightarrows E, f : E \to F),$$

then f commutes with arbitrary inductive and projective limits, and therefore can always be written in the form u^*, where $u : F \to E$ is a morphism of topoi. Thus g can be written $g = u_*$. On the other hand, of course, e can be written in the form v^* (and then f in the form v_*) only if it also commutes with finite projective limits. This will evidently be the case if e is itself the right adjoint of a fourth functor d. Without a condition of this kind, one will often write

$$e = u_!$$

for the left adjoint of an inverse-image functor $f = u^*$ when such a left adjoint exists, this notation being suggested by the example of an open immersion $u : X \to Y$ of topological spaces. Similarly, if e is of the form v^*, i.e. if f is of the form v_*, one sometimes writes $g = v^!$ for the right adjoint of a direct-image functor v_*, when this adjoint exists [7].

3.2. Let E, E' be two $\mathscr{U}$-topoi, and let 328

$$u = (u_*, u^*, \varphi) \quad , \quad v = (v_*, v^*, \Psi) : E \rightrightarrows E'$$

be two morphisms of topoi from E to E'. A *morphism from u to v* is any morphism from u_* to v_* (in the sense of the category $\mathbf{Hom}(E, E')$ of functors from E to E'). Morphisms of morphisms of topoi compose in the evident way, thereby defining a category, which is in fact a *$\mathscr{U}$-category* (I 7.8), denoted

$$\mathbf{Homtop}(E, E'),$$

and called the *category of morphisms* (or *continuous maps*) *from E to E'*. One then has an evident functor

$$u \longmapsto u_* : \mathbf{Homtop}(E, E') \longrightarrow \mathbf{Hom}(E, E'),$$

[7] Cf. below 14.4, in the case of abelian sheaves.

which is fully faithful by the definition of arrows in the first member (but is not injective on objects).

3.2.1. Care must be taken that, if u and v are given as above, the theory of adjoint functors supplies a canonical bijection

$$\mathrm{Hom}(u_*, v_*) \xrightarrow{\sim} \mathrm{Hom}(v^*, u^*);$$

in particular, one obtains a *contravariant functor* on $\mathbf{Homtop}(E, E')$:

$$u \longmapsto u^* \,:\, \mathbf{Homtop}(E, E')^\circ \longrightarrow \mathbf{Hom}(E, E').$$

In accordance with geometric intuition, according to which the direct-image functor u_* "goes in the same direction" as the continuous map which gives rise to it, one should therefore define the direction of arrows for morphisms between morphisms of topoi *in terms of the direct-image functors*, and not in terms of the inverse-image functors (although the latter, as we have seen, possess the exactness properties characteristic of the notion of morphism of topoi).

329 3.2.2. Having defined the category $\mathbf{Homtop}(E, E')$ of morphisms from the topos E to the topos E', the notion of an *isomorphism* between two morphisms u, v from E to E' is also defined. In practice there is no need to distinguish essentially between two isomorphic morphisms of topoi, at least when one has a *canonical* isomorphism between the two (just as there is often no need to distinguish between two objects of a category when a canonical isomorphism between them is given). Let us point out in this connection that, most often, when one deals with diagrams of morphisms of topoi and questions of commutativity of such diagrams (a notion which makes sense by (3.3)), one means only commutativity up to ("canonical") isomorphism; by abuse of language, one then treats these diagrams as actually commutative diagrams.

One might try to justify this abuse of language by introducing the set $\mathrm{Homtop}(E, E')/\mathrm{Isom}$ of morphisms from E to E' up to isomorphism, and calling such an isomorphism class a morphism, instead of following Definition 3.2. But this runs into the very serious difficulties which arise whenever one tries to identify two isomorphic objects of a category without having a canonical isomorphism between them. Experience shows that such a viewpoint is impracticable, and that one must retain the "fine" notion 3.1 of a morphism between morphisms of topoi, even if one is sometimes obliged to struggle with compatibilities between canonical isomorphisms [7].

330 3.3.1. Let E, E', E'' be three $\mathscr{U}$-topoi, and consider morphisms of topoi

$$u \,:\, E \longrightarrow E' \quad , \quad v \,:\, E' \longrightarrow E''.$$

The theory of adjoint functors then gives us an adjunction isomorphism between the composite functors v_*u_* and u^*v^*, in terms of the adjunction isomorphisms for the pairs (u_*, u^*) and (v_*, v^*). On the other hand, the functor u^*v^* is left exact, as the composite of two left exact functors. Consequently one obtains a morphism from E to E'', called the *composite of the morphisms u and v*, and denoted vu:

$$vu \,:\, E \longrightarrow E''.$$

One verifies at once that composition of morphisms is associative, and that for every $\mathscr{U}$-topos E there is a morphism from E to itself which is a two-sided identity for composition: it is the morphism $(\mathrm{id}_E, \mathrm{id}_E, \varphi)$, where φ is the evident adjunction isomorphism

[7] For examples of such struggles (apparently successful), we refer the reader to Mme M. Hakim's book on relative schemes [9].

of id_E with itself. Let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$. One defines a category

$$(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Top}),$$

whose objects are the $\mathscr{U}$-topoi which belong to $\mathscr{V}$, whose arrows are morphisms between such $\mathscr{U}$-topoi, and whose composition of arrows is the one just described.

3.3.2. In fact, the composition map

$$\mathrm{Homtop}(E, E') \times \mathrm{Homtop}(E', E'') \longrightarrow \mathrm{Homtop}(E, E'')$$

is the map induced on objects by a "functor of composition of morphisms":

$$\mathbf{Homtop}(E, E') \times \mathbf{Homtop}(E', E'') \longrightarrow \mathbf{Homtop}(E, E''),$$

whose effect on arrows is the usual "convolution product" operation for morphisms between functors (here, direct-image functors). These composition functors satisfy a (strict) associativity property, for four topoi E, E', E'', E''', which refines the associativity of composition of morphisms of topoi. One may also say, in language which is beginning to become familiar [2], [9], that the $\mathscr{U}$-topoi are the objects (or 0-arrows) of a 2-*category*, whose 1-arrows are the morphisms of topoi and whose 2-arrows are morphisms of morphisms of topoi. 331

It is the fact that the $\mathscr{U}$-topoi (elements of a universe $\mathscr{V}$) form a 2-category, rather than merely an ordinary category like ordinary topological spaces, which from the technical viewpoint constitutes the most important difference between the theory of topoi and that of topological spaces. This fact is the source of certain technical complications already alluded to, but also of facts essentially new in comparison with traditional topology.

3.4. The fact that the $\mathscr{U}$-topoi (elements of a universe $\mathscr{V}$) form a 2-category (3.3.2) makes it possible in particular to define the notion of *equivalence of two $\mathscr{U}$-topoi E, E'*: one says that E and E' are *equivalent* if there exist morphisms of topoi $u : E \to E'$ and $v : E' \to E$ such that the composites uv and vu are isomorphic respectively to the identity morphisms of E and E'; the morphisms u and v are then said to be *quasi-inverse equivalences of one another.* 332

One sees at once that for the morphism of topoi $u : E \to E'$ to be an equivalence, it is necessary and sufficient that u_* be an equivalence, or, what amounts to the same thing, that u^* be an equivalence. (Use the fact that a functor $f : E \to E'$ between two topoi which is an equivalence is both of the form u_* and of the form v^*, with u and v morphisms of topoi); and for E and E' to be equivalent in the sense of the preceding paragraph, it is necessary and sufficient that they be equivalent as categories (i.e. as objects of the 2-category $(\mathscr{V}\text{-}\mathrm{cat})$). As it should, this shows that the notions of equivalence introduced do not depend on the choice of the universe $\mathscr{V}$ in 3.3.1.

3.4.1. In practice, there is most often no reason to distinguish essentially between equivalent $\mathscr{U}$-topoi, just as there is often no reason to distinguish essentially between two equivalent categories—provided, however, that one has an explicit equivalence from one to the other, or at least an equivalence defined up to unique isomorphism. Here it is the notion of equivalence of topoi which replaces the traditional notion of homeomorphism between two topological spaces. See the example in 4.2 below for the precise relations between these two notions.

4. Examples of morphisms of topoi

Here we return to the examples of 2, using the notion of a morphism of topoi. The comments of 2.0 apply equally to the present section. The universe $\mathscr{U}$ will generally be understood.

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4.1. The topos $\mathrm{Top}(X)$ for a varying topological space X.

4.1.1. Let

$$f : X \longrightarrow Y$$

be a continuous map of topological spaces. We shall associate with it canonically a morphism of topoi

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X) \to \mathrm{Top}(Y),$$

with the notation of 2.1. When $\mathrm{Top}(X)$ is defined as $\mathrm{Ouv}(X)^{\sim}$, the most convenient description of $\mathrm{Top}(f)$ is by means of the direct-image functor on sheaves

$$f_* : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(Y),$$

defined by the formula

$$f_*(F) = F \circ f^{-1},$$

where

$$f^{-1} : \mathrm{Ouv}(Y) \longrightarrow \mathrm{Ouv}(X)$$

is the evident functor $U \to f^{-1}(U)$. We have already noted that this functor is continuous and left exact, and therefore indeed defines the functor f_* above, which admits a left adjoint f^* that is left exact (III 1.9.1). Of course, strictly speaking, the morphism of topoi $\mathrm{Top}(f)$ depends on the choice of the left adjoint f^* of f_* and hence is defined only up to canonical isomorphism. In what follows we shall refrain from explicitly pointing out phenomena of this kind.

When one adopts the "étale spaces" point of view in defining $\mathrm{Top}(X)$, it is the inverse-image functor

$$f^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X)$$

334 that is the most convenient for defining the morphism of topoi $\mathrm{Top}(f)$, by simply setting

$$f^*(Y') = X \times_Y Y'$$

for every étale space Y' over Y; it is clear that the fiber product is indeed an étale space over X, and that the functor f^* so obtained is left exact and commutes with arbitrary $\varinjlim$, and therefore defines a morphism of topoi $\mathrm{Top}(f)$. For the compatibility of the two definitions obtained, we refer to [TF].

For two composable continuous maps

$$X \xrightarrow{f} Y \xrightarrow{g} Z,$$

there is a canonical isomorphism $\mathrm{Top}(gf) \simeq \mathrm{Top}(g)\,\mathrm{Top}(f)$ of morphisms of topoi. These *transitivity isomorphisms*, for three composable continuous maps f, g, h, satisfy a compatibility relation which we shall refrain from writing here, and which is precisely the one considered in SGA 1 VI 7.4 B) (for $\epsilon = (\mathrm{Esp})^\circ$). This may be expressed by saying that, as X varies in the category (Esp),

$$X \longmapsto \mathrm{Top}(X)$$

is a "pseudofunctor"

$$(\mathscr{U}\text{-esp}) \longrightarrow (\mathscr{V}\text{-}\mathscr{U}\text{-}\mathsf{T}), \tag{4.1.1.1}$$

or, in the terminology of 2-categories, that one has a *non-strict* functor of 2-categories [9]. In practice, we shall most often permit ourselves the abuse of language consisting in identifying $\mathrm{Top}(gf)$ and $\mathrm{Top}(g)\,\mathrm{Top}(f)$, that is, reasoning as if (4.1.1.1) were an actual functor between ordinary categories. We shall permit ourselves analogous abuses of language in the other examples treated below. 335

4.1.2. The preceding considerations extend immediately to topoi associated with topological spaces equipped with operator groups (2.3). If $f = (f^{\mathrm{es}}, f^{\mathrm{gr}})$,

$$f : (X, G) \longrightarrow (Y, H)$$

is a morphism of spaces with operators (where

$$f^{\mathrm{es}} : X \longrightarrow Y \quad , \quad f^{\mathrm{gr}} : G \longrightarrow H$$

are respectively continuous maps and group homomorphisms, compatible in the evident sense), one associates with it a morphism of topoi

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X, G) \longrightarrow \mathrm{Top}(Y, H),$$

whose definition is left to the reader. When the groups G and H are trivial, one recovers the definition of 4.1.1; when, on the other hand, the spaces X and Y are each reduced to a point, the inverse-image functor f^* is the "restriction of the operator group" functor

$$f^* : B_H \longrightarrow B_G,$$

which associates with every H-set the G-set that it defines by means of $f : G \to H$. We shall encounter this example again in other forms in 4.5 and 4.6.1.

4.1.3. Given a continuous map $f : X \to Y$ of topological spaces, one also associates with it a morphism between the corresponding "large topoi" (2.5)

$$\mathrm{TOP}(f) \text{ or } f : \mathrm{TOP}(X) \longrightarrow \mathrm{TOP}(Y),$$

most conveniently defined by the inverse-image functor 336

$$f^* : \mathrm{TOP}(Y) \longrightarrow \mathrm{TOP}(X),$$

which is simply the *restriction functor*. This morphism $\mathrm{TOP}(f)$ is a special case of the morphism called an "inclusion" for an induced topos, which will be studied in 5. Thus, subject to the same qualifications as in 4.1.1, the topos $\mathrm{TOP}(X)$ may be regarded as a functor of X, as X varies in (Esp).

4.2. Fidelity properties of $X \mapsto \mathrm{Top}(X)$. We propose to specify to what extent a topological space X can be reconstructed in terms of the topos $\mathrm{Top}(X)$. For this purpose it is appropriate, for two spaces X and Y, to describe the category of morphisms from $\mathrm{Top}(X)$ to $\mathrm{Top}(Y)$ (3.2), so as to determine the fidelity properties of the "functor" $X \mapsto \mathrm{Top}(X)$. We shall merely state the results obtained, referring the reader to [9] for details. The reader who wishes to carry out the verification himself may consult Exercise 7.8.

4.2.1. Recall that a topological space X is called *sober* if every irreducible closed subset of X has exactly one generic point. Almost all spaces used in practice are sober; this is true in particular of a Hausdorff space, more generally of a space all of whose points are closed, or of the underlying space of a scheme. With a topological space X one associates (*loc. cit.* or EGA 0_I, new edition) a sober topological space X_{sob} and a continuous map 337

(4.2.1.1) $$\varphi : X \longrightarrow X_{\mathrm{sob}}$$

which is *universal* for continuous maps from X to sober spaces; in other words, one constructs a left adjoint $X \mapsto X_{\mathrm{sob}}$ to the inclusion functor (Espsob) $\to$ (Esp) from the

category of sober spaces to that of "arbitrary" topological spaces (the quotation marks serving as a reminder that a universe is involved!). The explicit construction takes as the points of X_{sob} the irreducible closed subsets of X, and as its open subsets the sets of the form U', where U is an open subset of X and $U' \subset X_{\text{sob}}$ denotes the set of irreducible closed subsets of X that meet U. The map (4.2.1.1) is obtained by associating with each $x \in X$ the closure of $\{x\}$. The space X is sober if and only if the preceding map is bijective, and hence a homeomorphism.

One checks that the functor

$$\varphi^{-1} : \text{Ouv}(X_{\text{sob}}) \longrightarrow \text{Ouv}(X)$$

induced by φ is an isomorphism, which implies that the morphism of topoi

$$\text{Top}(\varphi) : \text{Top}(X) \longrightarrow \text{Top}(X_{\text{sob}})$$

defined by φ is also an isomorphism. This explains a priori why X_{sob} must necessarily enter into the question of reconstructing X from $\text{Top}(X)$: since the latter depends only on X_{sob} up to isomorphism, the question can have an affirmative answer only if X is
338 sober. We shall specify below (7.1) how X_{sob} can in fact be reconstructed in terms of $\text{Top}(X)$, by interpreting its points as "points" of the topos $\text{Top}(X)$ (or, equivalently, as fiber functors).

4.2.2. On every topological space one should introduce the order relation $\leq$ for which

$$x \leq y \Longleftrightarrow \overline{\{x\}} \subset \overline{\{y\}} \text{ i.e. } x \in \overline{\{y\}}$$

(one also expresses this by saying that x is a *specialization* of y, or that y is a *generization* of x). For a space of the form X_{sob}, this is simply the inclusion relation among the irreducible closed subsets of X.

With this understood, on the set of maps from a space X to another space Y one should introduce the order relation called "specialization", deduced from that of Y, namely

$$f \leq g \Longleftrightarrow f(x) \leq g(x) \text{ for every } x \in X.$$

With these conventions, one has the following result:

4.2.3. *Let X, Y be two topological spaces, with Y sober, and let f and g be two continuous maps from X to Y. Then there is at most one morphism from* $\text{Top}(f)$ *to* $\text{Top}(g)$*, and such a morphism exists if and only if g is a specialization of f. Finally, every morphism of topoi* $\text{Top}(X) \to \text{Top}(Y)$ *is isomorphic to a morphism of the form* $\text{Top}(f)$*, where $f : X \to Y$ is a continuous map (uniquely determined by the first assertion).*

339 If , for an ordered set I, one defines the associated category $\text{cat}(I)$ by declaring that, for $i \geq j$, there is exactly one arrow from i to j, the preceding result may be summarized by saying that there is a canonical equivalence of categories

$$\text{cat}(\text{Hom}_{(\text{esp})}(X, Y)) \xrightarrow{\approx} \textbf{Homtop}(\text{Top}(X), \text{Top}(Y))$$

(where the second member is defined in 3.2).

The following is a formal consequence of these results:

Corollary 4.2.4. a) *Let $f : X \to Y$ be a continuous map. Then* $\text{Top}(f) : \text{Top}(X) \to \text{Top}(Y)$ *is an equivalence of topoi if and only if $f_{\text{sob}} : X_{\text{sob}} \to Y_{\text{sob}}$ is a homeomorphism (and hence, when X and Y are sober, if and only if f is a homeomorphism).*

b) *Let X and Y be topological spaces. The topoi* $\text{Top}(X)$ *and* $\text{Top}(Y)$ *are equivalent if and only if X_{sob} and Y_{sob} are homeomorphic (and hence, if X and Y are sober, if and only if X and Y are homeomorphic).*

4.3. Morphisms into the terminal topos: constant objects of a topos, section functors. Let P (the initial letter of "point") denote the standard terminal topos, i.e. $P = (\mathrm{Ens})$ (2.2). Let E be any topos. We shall see that *up to unique isomorphism, there exists a unique morphism of topoi*

$$f : E \to P$$

or, more precisely, that *the category* $\mathbf{Homtop}(E, P)$ *is equivalent to the one-point category*: between any two objects of this category there is therefore a unique arrow, and it is an isomorphism. (This justifies to some extent the name "terminal topos".) For this, recall (3.2.1) that $\mathbf{Homtop}(E, P)$ is equivalent to the full subcategory of $\mathbf{Hom}(P, E)^\circ$ formed by the functors 340

$$f^* : P = (\mathrm{Ens}) \longrightarrow E$$

that commute with $\varinjlim$ and are left exact. Let e be a singleton set. Every set X can then be written canonically as the "coproduct of X copies of e", from which it follows that the category of functors $g : (\mathrm{Ens}) \to E$ that commute with $\varinjlim$ is equivalent to the category E, by associating with each g the object $g(e)$ of E. In terms of $T = g(e)$, the functor g is reconstructed, up to unique isomorphism, by $g(I) = T \times I$, where one sets $T \times I = \coprod_{i \in I} T_i$, with $T_i = T$ for every $i \in I$. The fact that the functor of I so defined by T does indeed commute with $\varinjlim$ follows from the fact that it is manifestly left adjoint to the functor $X \mapsto \mathrm{Hom}(T, X)$ from E to (Ens). This said, for g to be left exact, it is plainly necessary that $g(e) = T$ be a terminal object of E (since e is a terminal object of (Ens)), and it follows easily from the fact that in E "coproducts are universal" (1.1.2 b)) that this condition is also sufficient. Thus the category of inverse-image functors f^* is equivalent to the category of terminal objects of E, which is itself plainly equivalent to the terminal category.

It follows from the foregoing that the choice of a morphism (4.3.1) is essentially equivalent to the choice of a terminal object of E, say e_E. In terms of this object, one then has canonical isomorphisms of functors

$$f^*(I) \simeq e_E \times I = \text{ coproduct of } I \text{ copies of } e_E \quad \text{for } I \in \mathrm{ob}(\mathrm{Ens}), \tag{4.3.2}$$

and 341

$$f_*(X) = \mathrm{Hom}(e_E, X) \qquad \text{for } X \in \mathrm{Ob}\, E. \tag{4.3.3}$$

4.3.4. The preceding two functors will play an important role below. For a set I, the object $f^*(I) = e_E \times I$ of (4.3.2) is called the *constant object with value* I in E (or, when E is presented as a category $\tilde{C}$ in terms of a site C, the *constant sheaf with value* I *on* C). It will also often be denoted by I_E, or by I_C when E is defined by the site C. The fact that $I \mapsto I_E$ is the inverse-image functor of a morphism of topoi specifies its exactness properties, which imply in particular that this functor respects all the usual kinds of algebraic structure, transforming a group into a group object of E, and so on (3.1.2). If, for example, one has a group G of E, one says that it is a *constant group* (or, where appropriate, a *constant sheaf of groups*) if it is isomorphic to a group of the form $G_{\circ E}$, where $G_\circ$ is an ordinary group. The same terminology is used for every other kind of "algebraic" structure, in the sense specified (more or less) in 3.1.2.

4.3.5. Notice that the functor $I \mapsto I_E$ is not necessarily fully faithful (or even faithful: take E to be the "empty topos" (2.2)); thus a constant object of E does not in general determine, up to unique isomorphism, the set I from which it arises. We say that E is 0-*acyclic*, or *connected and nonempty*, if the functor $I \mapsto I_E$ is fully faithful. By the general

properties of adjoint functors, this amounts to saying that the adjunction morphism

$$I \longrightarrow f_*(f^*(I)) = f_*(I_E) = \mathrm{Hom}(e_E, I_E)$$

342 is an isomorphism (so that the functor (4.3.3) recovers the "value" of a constant object of E). It is easily checked that this holds if and only if e_E is not the initial object ϕ_E of E, i.e. if E is not an "empty topos" (which expresses the *faithfulness* of the functor $I \mapsto I_E$[7]), *and* if e_E is a *connected* object of E, i.e. is not the sum of two objects of E which are not "empty" (i.e. which are not initial objects of E).

4.3.6. The functor (4.3.3) is also often called the *sections functor* and is denoted by Γ_E, $\Gamma(E, -)$, or simply Γ:

$$\mathrm{Hom}(e_E, X) = \Gamma_E(X) = \Gamma(E, X) = \Gamma(X). \tag{4.3.6.1}$$

It commutes with arbitrary projective limits, but is not in general right exact; its derived functors (on abelian group objects) will be studied in the next exposé.

4.4. Morphisms from the "empty topos". Let ϕ_{top} be an empty topos, corresponding to a category of sheaves Φ equivalent to the terminal category (2.2). Let E be a topos. The category of functors from E to Φ is clearly equivalent to the one-point category, and every such functor commutes with inductive and projective limits (without any merit in this), and hence may be regarded as an inverse-image functor for a morphism of topoi $\phi_{\mathrm{top}} \to E$. It follows that the *category* $\mathbf{Homtop}(\phi_{top}, E)$ *is equivalent to the one-point cat-*
343 *egory*, and in particular that *there exists, up to unique isomorphism, exactly one morphism of topoi*

$$\phi_{\mathrm{top}} \longrightarrow E. \tag{4.4.1}$$

This justifies to some extent the terminology "initial topos" introduced in 2.2.

One may also determine the morphisms of topoi

$$E \longrightarrow \phi_{\mathrm{top}}\ ; \tag{4.4.2}$$

one checks at once that such a morphism exists if and only if the initial object of E is also a terminal object, i.e. if and only if E itself is an "empty topos," and that in this case the category $\mathbf{Homtop}(E, \phi_{\mathrm{top}})$ is again equivalent to the one-point category. The unique morphism (4.4.2) (up to isomorphism) is then an equivalence of topoi.

4.5. The classifying topos B_G for a varying group G.

4.5.1. Let E be a topos, and let

$$f : G \longrightarrow H$$

be a morphism of group objects in E. It induces a "restriction of the group of operators" functor

$$f^* : B_H \longrightarrow B_G,$$

where the notation is that of 2.4. It is immediate that this functor commutes with inductive and projective limits; a fortiori, it may be interpreted as an inverse-image functor associated with a morphism of topoi

$$B_f \text{ or } f : B_G \longrightarrow B_H.$$

344 The corresponding direct-image functor

$$f_* : B_G \longrightarrow B_H$$

[7] or equivalently the fact that this functor is conservative (I 6.3).

is readily made explicit by the formula

$$f_*(X) = \mathbf{Hom}_G(H_s, X),$$

where X is an object of E with a left action of G, where H_s is H regarded as endowed with the left actions by G induced by f, and where $\mathbf{Hom}_G$ denotes the evident subobject of the object **Hom** defined below (10.2); the left action of H on $\mathbf{Hom}_G(H_s, X)$ is induced by the right actions of H on H_s by *right* translations.

Since the inverse-image functor f^* commutes with arbitrary $\varprojlim$ (and not only with finite $\varprojlim$), it is itself the right adjoint of a functor

$$f_! : B_G \longrightarrow B_H,$$

so that there is a sequence of three adjoint functors as in 3.1.3:

$$f_!, f^*, f_*.$$

The functor $f_!$ is readily made explicit by the formula

$$f_!(X) = H \times^G X,$$

where the right-hand side denotes the "contracted product" induced by the actions of G on X (on the left) and on H (on the right, via right translations and f), defined as the quotient of $H \times X$ by the action of G given by

$$g \circ (h, x) = (hg^{-1}, gx).$$

The functor $f_!$, being a left adjoint, clearly commutes with inductive limits, but is not in general left exact (i.e. it cannot in turn be regarded as an inverse-image functor 345
for a morphism of topoi $B_H \to B_G$). In fact, one checks easily that it can be left exact only if $f : G \to H$ is an isomorphism. Likewise, the functor f_*, which commutes with projective limits, is not in general right exact and, a fortiori, does not in general admit a right adjoint. At least when E is the punctual topos, i.e. when G and H are ordinary groups, f_* is right exact only if f is an isomorphism.

Given a second morphism of groups $g : H \to K$, one obtains, as in 4.1.1, transitivity up to canonical isomorphism, so that, subject to the usual reservation, the classifying topos B_G may be regarded as depending *functorially* on the group G. We leave it to the reader to generalize this functorial behavior to the case where both G *and* the topos E vary.

4.5.2. *The topos $B_{\mathscr{G}}$ for a varying pro-group $\mathscr{G}$.* We leave it to the reader to make precise the covariant character of the topos $B_{\mathscr{G}}$ (2.7) with respect to $\mathscr{G}$, by following the account given in 4.5.1. One should note, however, that in the case of a morphism $f : \mathscr{G} \to \mathscr{H}$ of pro-groups which are not essentially constant, the corresponding morphism of topoi $B_{\mathscr{G}} \to B_{\mathscr{H}}$ does not in general permit the definition of a functor $f_!$ (whose right adjoint would be the functor f^* restricting the pro-group of operators). Suppose, to simplify the statement, that $\mathscr{G}$ and $\mathscr{H}$ are defined by *profinite* groups G and H. It is then easily seen that $f_!$ exists if and only if the image of the morphism in question $f : G \to H$ has finite index in H, and in that case $f_!$ is given by the same formula as 346
in 4.5.

4.6. The topos $\widehat{C}$ for a varying category C.

4.6.1. Let

$$f : C \longrightarrow C'$$

be a functor from a category $C \in \mathscr{U}$ to another category C', and hence let

$$f^* : \widehat{C}' \longrightarrow \widehat{C} \quad , \quad f^*(F) = F \circ f .$$

It is immediate that this functor commutes with projective and inductive limits; a fortiori, it may be regarded as the inverse-image functor for a morphism of topoi

$$\widehat{f} \text{ or } f : \widehat{C} \to \widehat{C}'.$$

The corresponding direct-image functor

$$f_* : \widehat{C} \longrightarrow \widehat{C}'$$

is none other than the functor likewise denoted f_* in I 5.1. Moreover (as was to be expected from the fact that f^* also commutes with arbitrary $\varprojlim$), f^* also admits a left adjoint

$$f_! : \widehat{C} \longrightarrow \widehat{C}',$$

(which was likewise denoted $f_!$ in I 5.1). Thus one obtains a sequence of three adjoint functors

$$f_!, f^*, f_*,$$

the first of which is moreover an extension of $f : C \to C'$ to the categories of presheaves (for the usual embeddings of C and C' in the categories $\widehat{C}$ and $\widehat{C}'$). One should note that $f_!$ is not in general left exact, nor is f_* in general right exact; this removes any ambiguity
347 about the direction of variance of the topos $\widehat{C}$ associated with the varying category C: this topos is a *covariant* functor of C, subject to the usual reservation arising from the transitivity isomorphisms (cf. 4.1.1). When the sets of objects of C and C' each consist of a single element, so that C and C' identify with monoids G and G', the corresponding topoi are the classifying topoi B_{G° and $B_{G'^\circ}$, and one recovers their variance with respect to the varying G, already encountered from other viewpoints in 4.1.2 and 4.5 (where the restriction to groups rather than monoids was inessential).

4.6.2. The 2-functorial dependence of the topos $\widehat{C}$ on C can be made more precise by introducing, for two categories $C, C' \in \mathscr{U}$, a canonical functor

$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Homtop}(\widehat{C}, \widehat{C}'). \tag{4.6.2.1}$$

It remains to define this functor on arrows. For this, note that $f \mapsto f^*$ identifies (up to equivalence of categories) the right-hand side with a full subcategory of $\mathbf{Hom}(\widehat{C}', \widehat{C})^\circ$ (3.2.1). If $f, g : C \to C'$ are two functors, every morphism of functors $u : f \to g$ defines, for $F \in \mathrm{ob}\,\widehat{C}'$, a morphism of functors $F \circ g \to F \circ f$ (since F is a *contra*variant functor), hence a morphism $g^* \to f^*$ and therefore a morphism $\widehat{f} \to \hat{g}$, as asserted.

When C is the one-point category, $\widehat{C}$ is the punctual topos (2.2), denoted P, and (4.6.2.1) is interpreted as a natural functor

$$C' \longrightarrow \mathbf{Homtop}(P, \widehat{C}') \overset{\text{dfn}}{=} \mathbf{Points}(\widehat{C}') \tag{4.6.2.2}$$

348 from C' to the "*category of points*" of $\widehat{C}'$, which will be studied in 6. This functor is not necessarily an equivalence of categories (7.3.3); a fortiori, (4.6.2.1) is not necessarily an equivalence of categories.

On the other hand, the functor (4.6.2.1) is always *fully faithful.* To see this, note that if $f, g : E \to E'$ are two morphisms of topoi for which $f_!$ and $g_!$ are defined (3.1.3), then the theory of adjoint functors gives canonical isomorphisms

$$\mathrm{Hom}(f_!, g_!) \simeq \mathrm{Hom}(g^*, f^*) \simeq \mathrm{Hom}(f_*, g_*) \overset{\text{dfn}}{=} \mathrm{Hom}(f, g).$$

Applying this to functors of the form $\widehat{f}$ and $\hat{g}$ associated with $f, g : C \to C'$, one obtains the asserted result, taking into account that the natural extension map $\mathrm{Hom}(f, g) \to \mathrm{Hom}(f_!, g_!)$ is bijective (I 7.8).

4.6.3. One may ask when the functor (4.6.2.1) is an equivalence of categories, i.e. when it is essentially surjective. This is a special case of the problem of determining all morphisms of topoi $\widehat{C} \to \widehat{C}'$. More generally, if C is a category in $\mathscr{U}$ and E is a topos, one may seek to determine the morphisms of topoi

$$f : \widehat{C} \longrightarrow E.$$

The category of these morphisms is equivalent to the opposite of the category of functors $f^* : E \to \widehat{C}$ commuting with arbitrary inductive limits and finite projective limits. A functor $E \to \widehat{C}$ may be interpreted as a functor $E \times C^\circ \to (\mathscr{U}\text{-Ens}) = (\mathrm{Ens})$, or equivalently as a functor $F : C^\circ \to \mathbf{Hom}(E, (\mathrm{Ens}))$. The exactness property in question is then expressed by requiring that, for every object X of C, the functor $F(X) : E \to (\mathrm{Ens})$ commute with arbitrary inductive limits and finite projective limits (or, as we shall say in 6, that $F(X)$ is a "*fiber functor*" on E). This amounts to saying that $F(X)$ is the 349
inverse-image functor for a morphism of topoi $P \to E$, and one therefore obtains finally a canonical equivalence of categories

$$\mathbf{Homtop}(\widehat{C}, E) \xrightarrow{\approx} \mathbf{Hom}(C, \mathbf{Points}(E)), \tag{4.6.3.1}$$

where, for brevity, we write

$$\mathbf{Points}(E) = \mathbf{Homtop}(P, E)$$

for the category of points of the topos E.

When E is of the form $\widehat{C}'$, it follows immediately from the definitions that the composite of (4.6.2.1) and the preceding equivalence (4.6.3.1) is the functor

$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Hom}(C, \mathbf{Points}(\widehat{C}')) \tag{4.6.3.2}$$

defined by $F \mapsto i \circ F$, where $i : C' \to \mathbf{Points}(\widehat{C}')$ is the canonical embedding (4.6.2.2). It follows at once that *for* (4.6.2.1) *to be an equivalence, it is necessary and sufficient that* C *be empty or that* (4.6.2.2) *be essentially surjective* (and hence an equivalence of categories). This last condition on C' is satisfied in certain interesting cases, notably when C' is the one-object category defined by a group G (7.2.5).

Remark 4.6.4. [8] The association of a topos $\widehat{C}$ with an arbitrary category C suggests that a category C has invariants of a topological nature (cohomology groups, homotopy groups, etc.), just as a topos does. The cohomology groups of $\widehat{C}$ with coefficients in an abelian group object F (in the general sense studied in V) are precisely the values of the derived functors $\varprojlim^{(n)}$ of the functor $\varprojlim$, already familiar to topologists. J. L. 350
Verdier and (independently) D. G. Quillen verified that, when one restricts to constant, or more generally locally constant, coefficients, these cohomology groups identify with the cohomology groups of the semi-simplicial set $\mathrm{Nerf}(C)$ canonically associated with C

[8] N.D.E.: For a recent perspective on the notion of the homotopy type of a topos, see [15] § 7.1.6.

[5, n° 212, prop. 4.1], and moreover that, up to isomorphism in the "homotopy category" of [2], every semi-simplicial set can be obtained from a category C. With a suitable notion of homotopy type for topoi, which we do not specify here, one may say that the semi-simplicial homotopy types of topologists are precisely the homotopy types of topoi of the special form $\widehat{C}$, and more generally of topoi E in which every object admits a covering that refines every other one (2.6) [8]. In the absence of this condition on E, its homotopy type can at best be expressed by a suitable *projective system* of semi-simplicial sets [1].

4.7. The topos $C^\sim$ for a variable site C (cocontinuous functors). Let

$$f : C \longrightarrow C'$$

be a cocontinuous functor (III 2.1) between sites in $\mathscr{U}$, i.e. a functor such that the functor $\widehat{f}_*$ from $\widehat{C}$ to $\widehat{C}'$ (4.6.1) maps $C^\sim$ into $C'^\sim$, that is, induces a functor

$$\widetilde{f}_* : \widetilde{C} \longrightarrow C'^\sim \tag{4.7.1}$$

making the following diagram of functors commutative:

$$\begin{array}{ccc} C^\sim & \xrightarrow{\widetilde{f}_*} & C'^\sim \\ {\scriptstyle i_*}\downarrow & & \downarrow{\scriptstyle i'_*} \\ \widehat{C} & \xrightarrow{\widehat{f}_*} & \widehat{C}' \end{array} , \tag{4.7.2}$$

351 where i_* and i'_* are the inclusion functors. We saw (III 2.3) that the functor $\widetilde{f}_*$ then has a left adjoint $\widetilde{f}^*$, and that the latter is left exact. In other words, $\widetilde{f}_*$ is the direct-image functor associated with a morphism of topoi

$$\widetilde{f} \text{ or } f : C^\sim \longrightarrow C'^\sim,$$

the corresponding inverse-image functor being, of course, $\widetilde{f}^*$. Taking left adjoints of the functors involved, the commutative diagram (4.7.2) also gives a diagram commutative up to isomorphism:

$$\begin{array}{ccc} C^\sim & \xleftarrow{\widetilde{f}^*} & C'^\sim \\ {\scriptstyle \underline{a}}\uparrow & & \uparrow{\scriptstyle \underline{a}'} \\ \widehat{C} & \xleftarrow{\widehat{f}^*} & \widehat{C}' \end{array} , \tag{4.7.3}$$

where $\underline{a}$ and $\underline{a}'$ are the "associated sheaf" functors. This diagram immediately recovers the formula (III 2.3)

$$\widetilde{f}^* = \underline{a}\widehat{f}^* i'.$$

The transitivity property for the morphisms of topoi $\widehat{f} : \widehat{C} \to \widehat{C}'$ implies the same property for the morphisms of topoi $\widetilde{f} : C^\sim \to C'^\sim$ associated with cocontinuous functors. Thus one may say that *the topos $C^\sim$ varies functorially with C* in a *covariant* manner, when cocontinuous functors are taken as the "morphisms" of sites.

352 In all the cases encountered thus far, the cocontinuous functor f used is also con-

[8] and, more generally still, of topoi that are "locally ∞-connected" in an evident sense which we leave it to the reader to make precise.

tinuous, that is (III 1.1), it "extends" to a functor

$$\widetilde{f}_! : \widetilde{C} \longrightarrow \widetilde{C'}$$

commuting with inductive limits, which is left adjoint to $\widetilde{f}^*$, so that there is a sequence of three adjoint functors

$$\widetilde{f}_!, \widetilde{f}^*, \widetilde{f}_*.$$

One should note that the functor $\widetilde{f}_!$ is not in general left exact, nor is $\widetilde{f}_*$ right exact. This removes any ambiguity concerning the direction of the variance of the topos $\widetilde{C}$, when C varies through functors assumed only to be cocontinuous, or even continuous and cocontinuous.

Remark 4.7.4. Given a morphism of topoi

$$F : E \longrightarrow E',$$

for there to exist a functor $F_!$ left adjoint to F^*, it is necessary and sufficient that one be able to "realize" (up to equivalence) E and E' in the form $\widetilde{C}$ and $\widetilde{C'}$, for two sites $C, C' \in \mathscr{U}$, and that one be able to find a *continuous and cocontinuous* functor $f : C \to C'$ such that F identifies with $\widetilde{f}$. This is clearly sufficient; for necessity, it is enough to take for C and C' small full generating subcategories of E and E', respectively, such that

$$F_!(\mathrm{ob}\, C) \subset \mathrm{ob}\, C',$$

endowed with the topologies induced by those of E and E', and to take for f the functor induced by $F_!$ (cf. 1.2.1). Recall (1.8) that the existence of $F_!$ also means that F^* commutes with projective limits, or, equivalently here since this functor is left exact, that it 353
commutes with products.

4.7.5. If one asks which morphisms of topoi $F : E \to E'$ can be realized by a cocontinuous (not necessarily continuous) functor of sites, one likewise finds the following answer: the set of objects X of E such that the functor $X' \mapsto \mathrm{Hom}(X, F^*(X'))$ from E' to (Ens) commutes with projective limits (or, equivalently, with products) must be a *generating family of* E.[9]

4.8. The morphism of topoi $\widetilde{C} \to \widehat{C}$ for a site C. Let C be a small site, with which the two topoi $\widetilde{C}$ and $\widehat{C}$ are therefore associated (the latter does not depend on the topology placed on C). In II 3.4 we defined the "associated sheaf" functor

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C},$$

and proved that it is left exact and commutes with inductive limits. It is therefore the inverse-image functor associated with a morphism of topoi

$$p : \widetilde{C} \longrightarrow \widehat{C},\ p^* = \underline{a},$$

the corresponding direct-image functor being the canonical inclusion

$$i = p_* : \widetilde{C} \longrightarrow \widehat{C}.$$

[9] N.D.E.: This statement does not appear to be correct: the condition expressed is equivalent to the assertion that F^* commutes with arbitrary products, or equivalently that F^* has a left adjoint $F_!$. For every site C, the identity functor $F = id : C \longrightarrow C$ is cocontinuous when the term on the right is endowed with the trivial topology. The functor $F^* : \widehat{C} \longrightarrow \widetilde{C}$ is then the associated sheaf functor, and does not in general commute with arbitrary products.

It is well known that the latter functor is not in general right exact (its derived functors on abelian objects give rise to the cohomology presheaves $\mathscr{X}^n(F)$ of V 2), and that $\underline{a}$ does not in general commute with arbitrary $\varprojlim$. This removes any ambiguity concerning the direction of the "natural" morphism of topoi between $\widehat{C}$ and $C\tilde{}$.

Given a cocontinuous functor

$$f : C \longrightarrow C'$$

354 of sites, one obtains a diagram of morphisms of topoi

$$\begin{array}{ccc} C\tilde{} & \xrightarrow{\tilde{f}} & C'\tilde{} \\ {\scriptstyle p_*}\downarrow & & \downarrow{\scriptstyle p'_*} \\ \widehat{C} & \xrightarrow{\hat{f}} & \widehat{C}' \end{array}$$

which is commutative up to canonical isomorphism: this is precisely what the commutativity of the diagram of functors (4.7.2) expresses. Thus one may say that the morphism of topoi $p : C\tilde{} \to \widehat{C}$ is functorial in C, when C is varied through *cocontinuous* functors between sites.

4.9. Effect of a continuous functor of sites. Morphisms of sites.

4.9.1. Let

$$f : C \longrightarrow C'$$

be a continuous functor from C to C', i.e. such that the functor $\widehat{f}^* : \widehat{C}' \to \widehat{C}$ maps $C'\tilde{}$ into $C\tilde{}$, and hence induces a functor

$$f_s : C'\tilde{} \longrightarrow C\tilde{}.$$

We saw (III 1.2) that the latter admits a left adjoint

$$f^s : C\tilde{} \longrightarrow C'\tilde{},$$

which moreover "extends" f in the evident sense. One should note that in general f^s is not left exact (even if f is also cocontinuous), nor does f_s commute with inductive limits, so that the datum of f does not, without further hypotheses, describe a morphism of topoi in either direction between $C\tilde{}$ and $C'\tilde{}$. The case where f is cocontinu-
355 ous, i.e. where f_s commutes with inductive limits and may therefore be regarded as an inverse-image functor for a morphism of topoi $\tilde{f} : C\tilde{} \to C'\tilde{}$, was examined in 4.7. We shall examine the case where the functor f^s is left exact, and may therefore be regarded as an inverse-image functor for a morphism of topoi in the *opposite direction*:

$$\text{(4.9.1.1)} \qquad \mathrm{Top}(f) = g : C'\tilde{} \longrightarrow C\tilde{}.$$

Notice that *we have deliberately not denoted this morphism by the letter* $\tilde{f}$ *or* f*, so as to avoid confusion with the situation of 4.7, in accordance with the general recommendations of 3.1.3.* A continuous functor $f : C \to C'$ is sometimes called a *morphism of sites from* C' *to* C (note: *not from* C *to* C') if the functor f^s is left exact; in other words, if there exists a morphism of topoi (4.9.1.1) such that the corresponding inverse-image functor

$$g^* : C\tilde{} \longrightarrow C'\tilde{}$$

extends the functor f, i.e. makes the following diagram of functors commute up to isomorphism:

$$\begin{array}{ccc} C & \xrightarrow{f} & C' \\ {\scriptstyle\epsilon}\downarrow & & \downarrow{\scriptstyle\epsilon'} \\ \tilde{C} & \xrightarrow{g^*} & \tilde{C'} \end{array} ,$$

where ϵ and ϵ' are the canonical functors of II 4.4.0.

4.9.2. In practice, one recognizes that a continuous functor $f : C \to C'$ is a morphism of sites from C' to C from the fact that finite projective limits are representable in C and that f commutes with them (III 1.3 5)). Subject to the indicated condition 356
on C (which is almost always satisfied in practice), and assuming in addition that the topology of C' is coarser than the canonical topology (also almost always satisfied), the preceding sufficient condition (f left exact) for f to be a morphism of sites from C' to C is also necessary.

4.9.3. In the spirit of the preceding discussion, if C and C' are two $\mathscr{U}$-sites, the category of morphisms of sites $\mathbf{Morsite}(C', C)$ from C to C' should be defined as the full subcategory of the *opposite category* $\mathbf{Hom}(C, C')^\circ$ of the category of functors from C to C', formed by those functors which are morphisms of sites (from C' to C). In this way one obtains a canonical functor (defined up to unique isomorphism)

$$\mathbf{Morsite}(C', C) \longrightarrow \mathbf{Homtop}(\tilde{C'}, \tilde{C}).$$

Proposition 4.9.4. *Let E be a $\mathscr{U}$-topos and C a $\mathscr{U}$-site. The functor $f \mapsto f^*|C = f^* \circ \epsilon_C$, associating with every morphism of topoi $f : E \to \tilde{C}$ the "restriction" to C of the associated inverse-image functor $f^* : \tilde{C} \to E$, induces an equivalence of categories*

$$\mathbf{Homtop}(E, \tilde{C}) \xrightarrow{\approx} \mathbf{Morsite}(E, C)(\hookrightarrow \mathbf{Hom}(C, E)^\circ).$$

When finite $\varprojlim$ are representable in C, the corresponding fully faithful functor

$$\mathbf{Homtop}(E, \tilde{C}) \longrightarrow \mathbf{Hom}(C, E)^\circ$$

has as its essential image the functors $g : C \to E$ which are left exact and continuous, or equivalently those which are left exact and carry covering families to covering families. 357

The last assertion follows from the first by 4.9.2. On the other hand, it follows from III 1.2 iv) and IV 1.2 iii) that the functor $G \to G \circ \epsilon$ induces an equivalence between the category of *continuous* functors from $\tilde{C}$ to E and the category of *continuous* functors from C to E, a quasi-inverse being given by $g \to g^s$ in the notation of *loc. cit.* On the other hand, by definition, g is a morphism of sites if and only if g^s is the inverse-image functor associated with a morphism of topoi $E \to \tilde{C}$, whence the conclusion by 3.2.1; the "or equivalently" follows from III 1.6.

The main point to retain from 4.9.4 is that (when finite $\varprojlim$ are representable in C) "it amounts to the same thing" to give a morphism of topoi $f : E \to \tilde{C}$ or to give a functor $g : C \to E$ which is left exact and carries covering families to covering families.

4.9.5. Using the developments of 4.9.1 and 4.9.3, one sees as usual that, for a varying $\mathscr{U}$-site C, via the notions of a morphism of sites and a morphism between morphisms of sites just made explicit, the topos $\tilde{C}$ depends functorially (or more exactly, 2-functorially) on the site C. Note that, with the terminology just introduced, $\tilde{C}$ depends *covariantly* on the site C.

4.9.6. It is immediate that (unlike what happens for the notion of a cocontinuous functor, cf. 4.7 4) *every* morphism of topoi $F : E \to E'$ can be realized by means of a morphism of sites $f : C \to C'$ (i.e. a continuous functor $f : C' \to C$ such that
358 ...): it suffices to choose small full generating subcategories C and C' of E and E', respectively, endowed with the topologies induced by those of E and E', such that

$$F^*(\mathrm{ob}\, C') \subset \mathrm{ob}\, C,$$

and to take for f the functor induced by F^*. This explains why most morphisms of topoi encountered in practice are indeed described by morphisms of sites (rather than by cocontinuous functors as in 4.7).

4.10. Relations between the small and large topoi associated with a topological space X. We retain the notation of 2.5. In particular, $\mathscr{V}$ is a universe such that $\mathscr{U} \in \mathscr{V}$. We depart from the convention of 2.1 by using $\mathrm{Top}(X)$ to denote the topos of $\mathscr{V}$-sheaves (rather than $\mathscr{U}$-sheaves) on X. Thus, we shall work with $\mathscr{V}$-topoi and not $\mathscr{U}$-topoi. (NB it would be possible to retain $\mathscr{U}$-topoi by adopting the appropriate convention for the definition of $\mathrm{TOP}(X)$, so that the latter is a $\mathscr{U}$-topos, cf. 2.5). With this understood, we shall define TWO morphisms of topoi

$$\text{(4.10.1)} \qquad \begin{cases} f : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X) \\ g : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X) \end{cases}, \quad gf \simeq \mathrm{id}_{\mathrm{Top}(X)},$$

giving rise to a sequence of three adjoint functors

$$\text{(4.10.1.1)} \qquad g^* = f_! \quad , \quad g_* = f^* \quad , \quad g^! = f_*,$$

359 with the notation of 3.1.3. We shall successively define the first two functors in this sequence, the third then being defined as the right adjoint of the second.

4.10.2. The functor

$$g_* = f^* : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X)$$

is defined simply as the *restriction functor* from $(\mathrm{Esp})_{/X}$ to the site $\mathrm{Ouv}(X)$ of open subsets of X; it does indeed take sheaves to sheaves, as follows immediately from the definition. Using underlined letters for sheaves on the large site of X, for such a sheaf $\underline{F}$ we denote by $\underline{F}_X$ its restriction to the site $\mathrm{Ouv}(X)$, thereby obtaining a functor

$$\text{(4.10.2.1)} \qquad \mathrm{Restr} : \underline{F} \longmapsto \underline{F}_X : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X).$$

It is clear that this functor commutes with $\mathscr{V}$-projective limits, since these are computed pointwise (one could also invoke the existence of the left adjoint constructed in 4.10.4 below). I claim that it also commutes with $\mathscr{V}$-inductive limits. To see this, we shall give a very convenient interpretation of "large" sheaves on X, i.e. of objects of $\mathrm{TOP}(X)$, in terms of ordinary sheaves (NB these are $\mathscr{V}$-sheaves) on topological spaces X' over X.

4.10.3. For a large sheaf $\underline{F}$ on X, and for every space X' over X (understood that $X' \in \mathscr{U}$), one defines in the obvious way, as in 4.10.2, the "small" sheaf $\underline{F}_{X'}$, the restriction of $\underline{F}$ to X'. If

$$u : X'' \longrightarrow X'$$

360 is a morphism in $(\mathrm{Esp})_{/X}$, one then defines in the obvious way a morphism $u_*(\underline{F}_{X''}) \to \underline{F}_{X'}$, or, what amounts to the same thing, a "*transition morphism*"

$$\text{(4.10.3.1)} \qquad \varphi_u : u^*(\underline{F}_{X'}) \longrightarrow \underline{F}_{X''}.$$

As u varies, these morphisms satisfy an evident transitivity condition for a composite

$$X''' \overset{u}{\to} X'' \overset{v}{\to} X'$$

of morphisms in $(\mathrm{Esp})_{/X}$, whose explicit formulation is left to the reader. In this way one obtains a natural functor from the category $\mathrm{TOP}(X)$ of large sheaves on X to the category of systems

$$(\underline{F}_{X'}) \quad (X' \in \mathrm{ob}(\mathrm{Esp})_{/X}), (\varphi_u) \quad (u \in Fl(\mathrm{Esp})_{/X}),$$

satisfying the transitivity condition just described and, in addition, such that for every morphism $u : X'' \to X'$ which is an *open immersion* (or, more generally, a local homeomorphism), the transition morphism φ_u is an isomorphism. Since the functors $u^* : \mathrm{Top}(X') \to \mathrm{Top}(X'')$ used to describe the φ_u commute with $\mathscr{V}$-inductive limits, it follows at once that, in the preceding description of the objects of $\mathrm{TOP}(X)$ in terms of "small" sheaves $F_{X'}$, $\mathscr{V}$-inductive limits are computed pointwise, i.e. the functors of the form $\underline{F} \to \underline{F}_{X'}$ commute with $\mathscr{V}$-inductive limits.

In particular, this holds for the functor $\underline{F} \mapsto \underline{F}_X$ considered in 4.10.2. It therefore does have a right adjoint (1.5). It remains to construct a left adjoint (whose existence also follows a priori from 1.8) and to verify that the latter is left exact. This will complete 361
the definition of the morphisms of topoi announced in (4.10.1) in terms of the functors (4.10.1.1).

4.10.4. The functor

$$g^* = f_! : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X)$$

is obtained by associating with every small sheaf F on X the system of its inverse images $(F_{X'})$, $X' \in \mathrm{ob}(\mathrm{Esp})_{/X}$, under the structural morphisms $X' \to X$, the transition morphism φ_u for $u : X'' \to X'$ being the transitivity isomorphism for inverse images of sheaves (4.1.1). One sees at once that this gives a functor

$$\mathrm{Prol} : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X),$$

which is *fully faithful*, and whose essential image consists of the large sheaves $\mathscr{F}$ on X for which all the transition morphisms φ_u of (4.10.3.1) are isomorphisms. We leave to the reader the task of defining an adjunction isomorphism between this canonical prolongation functor and the restriction functor of 4.10.2, proving that the latter is right adjoint to the former. This is immediate from the description in 4.10.3 of the category $\mathrm{TOP}(X)$.

Remarks 4.10.5. a) The construction in 4.10.4 also shows that $g^* = f_!$ is fully faithful (or, what amounts to the same thing, that $g_* = f^*$ is a localization functor, or again that $g^! = f_*$ is fully faithful). The large sheaves on X belonging to the essential image of the functor $g^* = f_! = \mathrm{Prol}$ deserve the name of large *étale sheaves* on X, since they form a category equivalent to the category of ordinary sheaves on X, or again to that of étale spaces over X. It is moreover 362
easy to see that if X' is a topological space over X, then the large sheaf on X represented by it is étale in the preceding sense if and only if X' is an étale space over X.

b) The full faithfulness of f_* may also be expressed by saying that the adjunction morphism $f^* f_* \to \mathrm{id}_{\mathrm{Top}(X)}$ is an isomorphism, i.e. that $g_* f_* \simeq \mathrm{id}$, i.e. that $g f \simeq \mathrm{id}$. Thus g makes $\mathrm{TOP}(X)$ into a *topos over* $\mathrm{Top}(X)$, admitting a "section" f over $\mathrm{Top}(X)$.

c) The fact that the functor $g^* = f_!$ is fully faithful, exact, and commutes with $\mathscr{V}$-inductive limits partially justifies the very convenient point of view (due to J. Giraud) according to which, in practically all questions in the theory of sheaves on X, it is harmless to replace the usual sheaves, or "small" sheaves, by the associated "large" sheaves. This applies in particular to cohomological questions, since g_* is exact, the functors $R^i g_*$ (V 5) vanish for $i > 0$, and hence (V 5.4) for every *large* sheaf $\mathscr{F}$ on X there are canonical isomorphisms

$$H^i(\mathrm{TOP}(X), \mathscr{F}) \simeq H^i(\mathrm{Top}(X), \mathscr{F}_X) = H^i(X, F).$$

Applying this to a sheaf of the form $\mathrm{Prol}(F)$, where F is a small sheaf on X, we obtain (since $\mathrm{Prol}(F)_X \simeq F$ canonically) a canonical isomorphism

$$H^i(X, F) = H^i(\mathrm{TOP}(X), \mathrm{Prol}(F)).$$

Thus, *the cohomological invariants of X, computed by means of the small or the large topos of X, are essentially identical.* The same result is moreover valid in noncommutative cohomology.

363 Exercise 4.10.6. Let S be a $\mathscr{U}$-site whose topology is coarser than the canonical topology, and let $\underline{M}$ be a subset of $Fl\ S$ satisfying the following conditions:

a) The morphisms of $\mathscr{M}$ are pullbackable (I 10.7), and $\underline{M}$ is stable under base change.
b) $\underline{M}$ contains the identity arrows and is stable under composition.
c) An arrow $u : X \to Y$ such that there exists a covering family $Y_i \to Y$, with the arrows $X \times_Y Y_i \to Y_i$ in $\underline{M}$, itself belongs to $\underline{M}$.
d) For every $X \in \mathrm{ob}\, S$, every covering family of X is refined by a covering family $(f_i : X_i \to X)$, with $f_i \in \underline{M}$.

For every $X \in \mathrm{ob}\, S$, consider the site $S(X)$ (the "small site of X"), whose underlying category is the full subcategory of $S_{/X}$ consisting of the objects whose structural morphism belongs to $\underline{M}$, endowed with the topology induced (III 3.1) by that of S.

1°) For every arrow $u : X \to Y$ of S, show that base change by u from $S(Y)$ to $S(X)$ is a continuous functor, hence gives a functor commuting with inductive limits (III 1.2 iv))

$$S(u)^x : S(Y)^\sim \longrightarrow S(X)^\sim.$$

2°) Define an equivalence between the topos $S^\sim$ and the category of systems

$$(F_X) \quad (X \in \mathrm{ob}\, S) \quad , \quad (\varphi_u) \quad (u \in Fl S)$$

364 consisting of objects $F_X \in \mathrm{ob}\, S(X)^\sim$ and, for every arrow $u : X \to Y$ in S, of a morphism $\varphi_u : S(u)^x(F_Y) \to F_X$, these systems being subject to a transitivity condition for a composite $v \circ u$ of arrows of S, and to the condition that $u \in \underline{M}$ imply that φ_u is an isomorphism.

3°) Define "restriction" and "prolongation" functors

$$\mathrm{Res}_X : (S_{/X})^\sim \longrightarrow S(X)^\sim, \mathrm{Prol}_X : S(X)^\sim \longrightarrow (S_{/X})^\sim$$

Show that Res_X commutes with small inductive and projective limits and that Prol_X is *fully faithful*, its essential image consisting of the sheaves F such that φ_u is an isomorphism for every arrow u of $S_{/X}$.

4°) Define an adjunction morphism making Res_X the right adjoint of Prol_X. Deduce that there exists a morphism of topoi

$$f : S(X)^\sim \to (S_{/X})^\sim$$

making $S(X)^\sim$ a subtopos of $(S_{/X})^\sim$ and such that

$$f_! = \mathrm{Prol}_X$$
$$f^x = \mathrm{Res}_X .$$

Show that Prol_X takes abelian sheaves to abelian sheaves.

5°) Show that if $S(X)$ admits fiber products, then Prol_X is exact. Deduce that there then exists a morphism of topoi $g : (S_{/X})^\sim \to S(X)^\sim$ which is a left retraction of f, i.e. such that

$$g^x = \mathrm{Prol}_X$$
$$g_x = \mathrm{Res}_X .$$

(For an example in which $S(X)$ does not admit fiber products, take for S the category of schemes endowed with the étale topology, for M the smooth morphisms, and for X a Noetherian scheme of dimension > 0.) 365

6°) Show that for every abelian sheaf F of $(S_{/X})^\sim$ there is an isomorphism

$$H^q(X, F) \simeq H^q(X, \mathrm{Res}_X F) \qquad \forall q.$$

Show, using hypercoverings for example, that for every abelian sheaf G of $S(X)^\sim$ there is an isomorphism

$$H^q(X, G) \simeq H^q(X, \mathrm{Prol}_X G) \qquad \forall q.$$

7°) Buy a chocolate medal for the editor.

5. Induced topos

5.1. Let E be a topos and X an object of E. Then the category $E_{/X}$ of objects of E over X is a topos, as follows, for example, immediately from Giraud's criterion 1.2 ii). Alternatively, by 1.2.1, one may realize E as a category of sheaves $C^\sim$, where C is a generating subcategory of E that may be chosen so that $X \in \mathrm{ob}\, C$; one then knows 366
that $E_{/X} = C^\sim_{/X}$ is equivalent to $(C_{/X})^\sim$ (III 5.4), and hence is a topos.

The topos $E_{/X}$ is called the *topos induced* on the object X of E.

5.2. We shall define a canonical morphism of topoi

$$j_X : E_{/X} \longrightarrow E, \tag{5.2.1}$$

which is called the *inclusion morphism* of the induced topos $E_{/X}$ into the ambient topos E, or, better (cf. 5.7), the *localization morphism of E at X*. This morphism corresponds to a sequence of three adjoint functors (cf. 3.1.3)

$$j_{X!} \quad , \quad j_X^* \quad , \quad j_{X*}, \tag{5.2.2}$$

which may be described explicitly as follows:

a) The functor

$$j_{X!} : E_{/X} \longrightarrow E$$

is the functor that *forgets the structural arrow* from $E_{/X}$ to E.

b) The functor

$$j_X^* : E \longrightarrow E_{/X}$$

is defined by

$$j_X^*(Z) = (X \times Z, \mathrm{pr}_1),$$

where $\mathrm{pr}_1 : X \times Z \to X$ is the first projection. It may also be interpreted as the *base-change functor* relative to the morphism

$$X \longrightarrow e_E,$$

where e_E is the terminal object of E. By the definition of the base-change functor, it is immediate that j_X^* is indeed right adjoint to $j_{X!}$. In particular, it commutes with projective limits. It also commutes with inductive limits, by virtue of the special exactness properties of topoi (II 4).

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c) It follows from this last fact (1.6) that j_X^* has a right adjoint

$$j_{X*} : E_{/X} \longrightarrow E.$$

For an object X' over X, the object $j_{X*}(X')$ is also sometimes denoted by one of the symbols $\prod_{X/e_E}(X'/X)$, $\mathbf{Hom}_{X/e_E}(X, X')$, or $\mathrm{Res}_{X/e_E}(X'/X)$ ("Weil restriction"), one or another of which is doubtless already familiar to the learned reader of our modest work.

5.3. One should note that the functor $j_{X!}$ commutes with fiber products and takes monomorphisms to monomorphisms, but it is not in general left exact for all that. It is left exact only if X is a terminal object of E (since $X = j_{X!}(X, \mathrm{id}_X)$ and (X, id_X) is a terminal object of $E_{/X}$), that is, if and only if j_X is in fact an equivalence of topoi (and hence all the functors in (5.2.2) are equivalences). [9]

Likewise, the functor j_{X*} is not in general right exact and does not necessarily take epimorphisms to epimorphisms.

It follows from these observations that the direction of the morphism of topoi relating $E_{/X}$ to E, for an object X of the topos E, is determined without any possible ambiguity.

5.4. The functor j_X^* is often called the *localization functor* or the *restriction functor*. The latter terminology is justified by identifying the objects of E resp. of $E_{/X}$ with the sheaves on E resp. on $E_{/X}$ (1.2 iii)), and observing that, under this identification, the adjunction formula between $j_{X!}$ (the forgetful functor) and j_X^* may be interpreted as saying that $j_X^*(F)$ *is the restriction to* $E_{/X}$ *of the sheaf* F *on* E (where "restriction" is understood in the generalized sense: composition with the "inclusion" functor $j_{X!} : E_{/X} \to E$). In accordance with notation familiar from other contexts, we shall also often write, interchangeably:

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$$j_X^*(F) = F|X = F_X = \text{ restriction of } F \text{ to } X. \tag{5.4.1}$$

Likewise, when E is of the form $C^\sim$, where C is a (small) site, and X is of the form $\epsilon(S)$ with $S \in \mathrm{ob}\, C$, so that one has the equivalence of categories recalled in 5.1

$$E_{/X} = C^\sim_{/\epsilon(S)} \xrightarrow{\approx} (C_{/S})^\sim$$

(where $C_{/S}$ is endowed with the topology induced by that of C), the functor j_X^* is identified simply with the functor that "restricts" a variable sheaf F on C to the category $C_{/S}$.

These considerations also make it possible to foresee the important role that induced topoi and the localization morphisms (5.2.1) will play in all questions in which one is led to argue by "localization on E" (cf. 8), that is, in practically all questions involving sites or topoi.

[9] For a commutation property of $j_{X!}$ under change of topos, cf. XVII 5.1.2.

5.5. Let

$$f : X \longrightarrow Y$$

be an arrow of E, which thus allows us to regard X (or, more correctly, (X, f)) as an object of $E_{/Y}$. It is clear that there is a canonical isomorphism

$$(E_{/Y})_{/X} \xrightarrow{\sim} E_{/X} \tag{5.5.1}$$

(*transitivity of induced topoi*). Applying the construction of 5.1 to $E_{/Y}$ in place of E, one therefore obtains a canonical morphism of topoi

$$\mathrm{loc}(f) \text{ or } f : E_{/X} \longrightarrow E_{/Y}, \tag{5.5.2}$$

also called the *localization morphism (associated with f).* When Y is the terminal object of E, one essentially recovers (5.2.1). The localization morphism for f is associated with a sequence of three adjoint functors 369

$$f_! \quad , \quad f^* \quad , \quad f_*, \tag{5.5.3}$$

which may be interpreted respectively as a forgetful functor, a restriction functor, and a functor denoted, according to preference (with X' as argument), by $\prod_{X/Y}(X'/X)$, $\mathbf{Hom}_{X/Y}(X, X')$, or $\mathrm{Res}_{X/Y}(X'/X)$.

5.6. The transitivity of the forgetful functors implies that, for two composable morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

of E, there is a canonical isomorphism of morphisms of topoi

$$\mathrm{loc}(gf) \simeq \mathrm{loc}(g)\,\mathrm{loc}(f) : E_{/X} \longrightarrow E_{/Y} \longrightarrow E_{/Z}.$$

Moreover, for three composable morphisms, the preceding transitivity isomorphisms satisfy the usual compatibility relation. Thus, subject to the customary abuse of language, this allows one to regard

$$X \longmapsto E_{/X}$$

as a covariant functor from E to the category ($\mathscr{V}$-$\mathscr{U}$-top) (3.3.1). More precisely, one obtains a 2-functor (not strict in general) from E to the 2-category ($\mathscr{V}$-$\mathscr{U}$-top).

Before continuing with the general theory of induced topoi (5.10), let us give a few instructive examples.

5.7. Let X be a topological space, and hence let $\mathrm{Top}(X)$ be a topos (2.1). Let X' be an object of $\mathrm{Top}(X)$, which it will be convenient to interpret as an étale space over X, $p : X' \to X$. The category $\mathrm{Top}(X)_{/X'}$ is then identified with the category of étale spaces X'' over X, equipped with an X-morphism $X'' \to X'$. It is known that such a morphism makes X'' an étale space over X', and in this way one obtains a canonical equivalence of topoi 370

$$\mathrm{Top}(X)_{/X'} \xrightarrow{\approx} \mathrm{Top}(X').$$

The localization morphism is therefore identified with a morphism of topoi $\mathrm{Top}(X') \to \mathrm{Top}(X)$, and one sees immediately that the latter is none other than the morphism

$$\mathrm{Top}(p) : \mathrm{Top}(X') \longrightarrow \mathrm{Top}(X)$$

associated with the structural continuous map $p : X' \to X$. Intuitively, the localization morphism is thus simply the translation, into the language of topoi, of the étale map $p : X' \to X$. This both explains the appropriateness of the term "localization morphism" and counsels caution in using the term "inclusion morphism", which seems

chiefly appropriate when p is an open immersion. It will consequently be prudent in 5.1 to reserve the term "inclusion morphism" for the case where the morphism $X \to e_E$ is a monomorphism, i.e. where X is identified with a subobject of the terminal object of E.

5.8. Let E be a topos, G a group in E, and H a subgroup of G, and let

$$X = G/H$$

be the quotient homogeneous space, regarded as an object of E with a left action by G,
371 i.e. as an object of the classifying topos B_G (2.4). We shall determine the induced topos

$$(B_G)_{/X} \quad (X = G/H),$$

by defining an equivalence of topoi

$$c : B_H \xrightarrow{\approx} (B_G)_{/X}, \tag{5.8.1}$$

such that the following diagram commutes up to canonical isomorphism:

$$B_H \underset{B_i}{\overset{c}{\underset{\longrightarrow}{\hookrightarrow}}} (B_G)_{/X} \xrightarrow{j_X} B_G \tag{5.8.2}$$

where j_X is the localization morphism in B_G, and where

$$B_i : B_H \longrightarrow B_G$$

is the morphism deduced from the inclusion $i : H \to G$ (4.5).

To define (5.8.1), we shall simply give the descriptions of the inverse-image functor c^* and the direct-image functor c_*, leaving it to the reader to verify that these are indeed quasi-inverse equivalences and give rise to the commutative diagram (5.8.2) of morphisms of topoi. Let e be the subobject of E underlying X, the image of the section of X (over a chosen terminal object e_E of E) deduced from the unit section of G. If X' is an object of B_G over X, then $c^*(X')$ denotes the inverse image of e in X',

$$c^*(X') = X' \times_X e,$$

which is stable under the left action of H on X', obtained by restricting the given action of G on X'. This indeed defines a functor

$$c^* : (B_G)_{/X} \longrightarrow B_H.$$

372 On the other hand, if X' is an object of B_H, then the morphism from X' to the terminal object e_H of B_H (i.e. the terminal object e_E of E with the trivial action of H) gives, upon application of the functor $i_!$ (4.5), a morphism in B_G

$$i_!(X') \longrightarrow i_!(e_H) \simeq X,$$

which allows $i_!(X')$ to be regarded as an object of $(B_G)_{/X}$, and hence gives the desired functor

$$c_* : X' \longmapsto (i_!(X') \longrightarrow X) : B_H \longrightarrow B_G.$$

5.8.3. Thus, by (5.8.2), one sees that if G is a group in a topos and H a subgroup, *the functor "restriction of the operators from G to H" may be interpreted as a localization functor.* In particular, taking H to be the unit subgroup, one sees that the functor that *forgets the actions of G* may be interpreted as a localization functor. More generally, it follows that if (X, G) is an object of E with an action of G (i.e. an object of the classifying topos B_G), then the functor that "forgets the actions of G",

$$E' = (B_G)_{/(X,G)} \longrightarrow E_{/X}$$

may be interpreted as a localization functor with respect to a suitable object $E_{G,(X,G)}$ of E' covering the terminal object. It suffices to take $E_{G,(X,G)} = E_G \times (X, G)$, giving the cartesian diagram (where $E_G = G_S = G$ with the action of G by left translation)

(5.8.3.1)
$$\begin{array}{ccc} E_G & \longleftarrow & E_{G,(X,G)} = Z \\ \downarrow & & \downarrow \\ e_G & \longleftarrow & (X, G) \end{array} ,$$

and to observe that, by transitivity of induced topoi, the induced topos $E' = ((B_G)_{X/G)/Z}$ is equivalent to the topos $((B_G)_{/E_G})_{/Z}$; since $(B_G)_{/E_G}$ is equivalent to E by the functor c^* of (5.8.1) (with $H = e$), it suffices to verify that this equivalence takes Z to the object X of E (which is immediate) in order to deduce the desired equivalence of categories 373

$$E'_{/Z} \xrightarrow{\approx} E_{/X}.$$

We leave it to the reader to verify that the composite of this equivalence with the localization functor

$$E' = (B_G)_{/(X,G)} \longrightarrow E'_{/Z}$$

is the functor that "forgets the actions of G". Thus one sees that, upon passing to induced topoi, the cartesian diagram above gives a diagram of morphisms of topoi, commutative up to canonical isomorphism (5.6):

(5.8.3.2)
$$\begin{array}{ccccc} (B_G)_{/E_G} \approx E & \xleftarrow{\ i\ } & (B_G)_{/Z} \approx E'_{/Z} & \approx & E_{/X} \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle j'} & & \\ B_G & \xleftarrow{\ f\ } & (B_G)_{/(X,G)} = E' & & \end{array} ,$$

where the inverse-image functors associated with the vertical arrows j, j' are the functors that "forget the actions of G", and where the inverse-image functor i^* is identified with the localization functor $E \to E_{/X}$. This diagram is, moreover, "2-cartesian" in the sense of 5.11 below. 374

5.8.4. We leave it to the reader to extend the preceding considerations to the case of a pro-group $\mathscr{G} = (G_i)$, in the particular case where H is defined as the "kernel" of $\mathscr{G} \to G_{i_0}$. (In the case of profinite groups, this means restricting attention to *open* subgroups H of G, i.e. subgroups that are closed and of *finite index*.)

Exercise 5.9. Let E be a topos, G a group of E, B_G the classifying topos of G (2.4), and

$$\pi : B_G \longrightarrow E$$

the morphism of topoi deduced from the homomorphism from G to the unit group e of E (taking into account that $B_e \approx E$).

a) Let

$$E_G = G_s$$

be the object of B_G whose underlying E-object is G, with the actions of G defined by left translations. Regard $\pi^*(G)$ as a group of B_G (it is the group G of E with the trivial action of G on it). Show that the morphism of E

$$G_s \times G \longrightarrow G_s$$

defined by the right translations of G on G_s is compatible with the actions of G on G_s and on $G = \pi^*(G)$, and defines a morphism of B_G

$$E_G \times \pi^*(G) \longrightarrow E_G.$$

Show that this morphism equips E_G with a right action of the B_G-group $\pi^*(G)$,
375 and that this action makes E_G a *torsor* under $\pi^*(G)$ (that is, the terminal object of B_G can be covered by objects X_i such that, for every i, the restriction of E_G to X_i is isomorphic to the trivial bundle with operators $(\pi^*(G)_{X_i})_d$).

b) Let

$$q : E' \longrightarrow E$$

be a topos over E. For every topos B over E, define the category $\mathbf{Homtop}_E(E', B)$ of morphisms of topoi from E' to B compatible, up to a specified isomorphism, with the structural morphisms $E' \to E$ and $B \to E$. Taking $B = B_G$, define, by the formula $f \mapsto f^*(E_G)$, an equivalence of categories

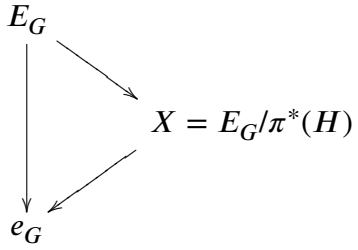
$$\mathbf{Homtop}_E(E', B_G) \longrightarrow \mathbf{Tors}(E, q^*(G)),$$

where the right-hand side denotes the category of (right) $q^*(G)$-torsors on E.

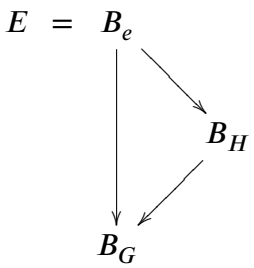
c) Let H be a subgroup of G, so that $\pi^*(H)$ is a subgroup of $\pi^*(G)$ and hence acts on E_G by restriction of the operator group, giving a quotient object

$$X = E_G/\pi^*(H),$$

which is simply the object G/H endowed with the usual left action of G. Let e_G be the terminal object of B_G (that is, the terminal object e_E of E with the necessarily trivial action of G), and consider in B_G the diagram of morphisms



376 which, upon passing to the topoi induced by B_G on these objects, gives a diagram of morphisms of topoi, commutative up to canonical isomorphism (5.6). Show that this diagram is equivalent to the diagram



,

whose three arrows are the arrows associated by 4.5 with the group morphisms $e \to H \to G$ in E. (Thus, *the classifying topos B_H may intuitively be interpreted as a homogeneous space over B_G, with group $\pi^*(G)$, associated with the universal torsor* (= homogeneous principal bundle) *E_G over B_G.*)

d) Reconsider the example of 5.7 in the case where X' is an *étale covering* (locally trivial, cf. 2.7.4) of X, interpreting it in terms of the considerations of the present exercise.

5.10. Inverse images of induced topoi. Let

$$f : E' \to E$$

be a morphism of $\mathscr{U}$-topoi, let X be an object of E, and let $X' = f^*(X)$ be its inverse image in E'. We shall define a diagram of morphisms of topoi

(5.10.1)
$$\begin{array}{ccc} E_{/X} & \xleftarrow{f_{/X}} & E'_{/X'} \\ {\scriptstyle j_X}\downarrow & & \downarrow{\scriptstyle j_{X'}} \\ E & \xleftarrow{f} & E' \end{array},$$

where j_X and $j_{X'}$ are the localization morphisms (5.2), and where $f_{/X}$ is defined below; 377
this diagram will be commutative up to canonical isomorphism. We shall define $f_{/X}$ here by means of its inverse image functor

(5.10.2)
$$(f_{/X})^* : E_{/X} \longrightarrow E'_{/X'},$$

which we simply take to be the functor "induced" by f^*, in the evident sense. Since the inclusion functors $j_{X!} : E_{/X} \to E$ and $j_{X'!} : E'_{/X'} \to E'$ commute with $\mathscr{U}$-inductive limits and fiber products, and are conservative, it follows at once that the functor $(f_{/X})^*$ commutes with $\mathscr{U}$-inductive limits and fiber products, just as does the functor f^* which induces it. Moreover, since $(f_{/X})^*$ plainly takes the terminal object X to the terminal object X', it is left exact and hence is associated with a morphism of topoi

(5.10.3)
$$f_{/X} : E'_{/X'} \longrightarrow E_{/X},$$

defined up to unique isomorphism (3.1.1). On the other hand, the diagram deduced from (5.10.1) by passing to the inverse-image functors is commutative (up to canonical isomorphism) by construction and by the fact that f^* commutes with the products $X \times Y$. Thus (5.10.1) itself is commutative, up to a unique isomorphism inducing the canonical isomorphism on the inverse-image functors of the two composites $f \circ j_{X'}$ and $j_X \circ f_{/X}$ (3.2.1).

5.10.4. When $f : E' \to E$ is associated with a continuous map $Y' \to Y$ of topological spaces (4.1), so that X is identified with an étale space over Y, and X' with the fiber product $X \times_Y Y'$, then, identifying the induced topoi $\mathrm{Top}(Y)_{/X}$ and $\mathrm{Top}(Y')_{/X'}$ with $\mathrm{Top}(X)$ and $\mathrm{Top}(X')$ respectively (5.7), one sees that the morphism $f_{/X}$ just constructed is (up to unique isomorphism) the morphism $\mathrm{Top}(X') \to \mathrm{Top}(X)$ deduced from the 378
canonical continuous map $X' \to X$. In the light of this example, one may therefore say that the topos $E'_{/X'}$ plays the role of a fiber product of $E_{/X}$ and E' over E. This intuition is made precise by the following result:

Proposition 5.11. *With the notation of 5.10, the diagram (5.10.1), equipped with the compatibility isomorphism $\alpha : f \circ j_{X'} \to j_X \circ f_{/X}$, is "2-cartesian." More precisely, for every $\mathscr{U}$-topos F, if to each morphism of topoi $g : F \to E'_{/X'}$ one associates the triple*

$(f_{/X} \circ g, j_{X'} \circ g, \alpha * g) = (g_1, g_2, \beta)$, where $g_1 : F \to E_{/X}$ and $g_2 : F \to E'$ are morphisms of topoi and $\beta : j_X \circ g_1 \to f \circ g_2$ is an isomorphism of morphisms of topoi from F to E, then one obtains an equivalence between the category $\mathbf{Homtop}(F, E'_{/X'})$ *and the category* $\mathbf{Homtop}(F, E_{/X}) \times^2_{\mathbf{Homtop}(F,E)} \mathbf{Homtop}(F, E')$ *of all triples* (g_1, g_2, β) *as above.*

(NB. A 2 has been placed above the cartesian-product sign to recall that this is not an ordinary fiber product of categories, but a "2-fiber product," in the sense made explicit in the statement.)

To prove 5.11, we shall describe morphisms of topoi explicitly by means of their associated inverse-image functors. For this we shall need the auxiliary results given in 5.11.1 through 5.12 below.

5.11.1. Consider, in general, the situation in which two categories E and E' are given, together with an object X of E which is assumed to be pullbackable, that is, such that the functor

$$j : A \longmapsto A \times X : E \longrightarrow E_{/X}$$

379 is defined, and finally a left exact functor

$$(*) \qquad \varphi : E_{/X} \longrightarrow E'.$$

Since $E_{/X}$ has a terminal object, namely $(X, \mathrm{id}_X) = X$, the same is therefore true of E', which has the terminal object

$$e' \simeq \varphi(X).$$

Having chosen the terminal object e' of E', we shall associate with every φ as above a pair

$$(**) \qquad (\Psi, u), \text{ with } \Psi = \varphi \circ j : E \to E', \text{ and } u \in \mathrm{Hom}(e', \Psi(X)).$$

To define u, note that

$$j(X) = X \times X$$

has a canonical section δ_X over the terminal object X of $E_{/X}$, namely the diagonal section, and define

$$u = \varphi(\delta_X) : \varphi(X) = e' \longrightarrow \varphi(X \times X) = \Psi(X).$$

I claim that knowledge of the pair $(**)$ makes it possible to reconstruct the functor φ of $(*)$ up to unique isomorphism. For this, for every object

$$p : X' \longrightarrow X$$

of $E_{/X}$, consider the following cartesian diagram in $E_{/X}$:

$$\begin{array}{ccc} X' & \longrightarrow & j(X') = X' \times X \\ \downarrow & & \downarrow{\scriptstyle j(p)} \\ X & \xrightarrow{\delta_X} & j(X) = X \times X \end{array} ,$$

where the first horizontal arrow is the graph morphism $\Gamma_p = (\mathrm{id}_X, p)$. Applying the left exact functor φ to this diagram, and using the definition of Ψ as $\varphi \circ j$, gives a cartesian

diagram

$$\begin{array}{ccc} \varphi(X',p) & \longrightarrow & \Psi(X') \\ \downarrow & & \downarrow{\scriptstyle \Psi(p)} \\ e' & \longrightarrow & \Psi(X) \end{array} ,$$

or, in other words, a canonical isomorphism 380

$$\varphi(X',p) \simeq \Psi(X') \times_{\Psi(X)} e' \tag{5.11.1.1}$$

It is immediately seen to be functorial in the object (X',p) of $E_{/X}$, which describes explicitly how the functor $\varphi : E_{/X} \to E'$ is reconstructed, up to isomorphism, from the pair (Ψ,u) of $(**)$. Notice also that the functor Ψ is left exact. More generally, Ψ commutes with every type of projective limit with which φ commutes, because j commutes with projective limits.

Conversely, suppose now that finite projective limits are representable in E and E', and start with a pair

$$(\Psi,u), \Psi : E \longrightarrow E', u \in \mathrm{Hom}(e', \Psi(X)),$$

where the functor Ψ is left exact. Formula (5.11.1.1) then defines a functor $\varphi : E_{/X} \to E'$. One checks at once that this functor is left exact and, more generally, that it commutes with every type of projective limit representable in E with which Ψ commutes. This follows immediately from the cartesian diagram

$$\begin{array}{ccc} \varprojlim_I{}^{E} X'_i & \longleftarrow & \varprojlim_I{}^{E_{/X}} X'_i \\ \downarrow & & \downarrow \\ \varprojlim_I{}^{E} X & \xleftarrow{\delta} & X \end{array}$$

relating the projective limits computed in E and in $E_{/X}$ (where δ denotes the diagonal 381
morphism into the projective limit of the constant functor with value X): apply the functor Ψ to it, obtaining a cartesian diagram in E', and compose the latter with the cartesian square deduced from $\Psi(X) \leftarrow e'$. The resulting composite cartesian square expresses the compatibility of φ with the projective limit under consideration.

The pair (Ψ,u) is reconstructed up to isomorphism from φ by observing that there is a canonical isomorphism

$$\Psi(X') = \varphi(j(X')), \tag{5.11.1.2}$$

deduced from (5.11.1.1) by replacing (X',p) there with $j(X') = (X' \times X, \mathrm{pr}_2)$ and observing that $\Psi(X' \times X) \simeq \Psi(X') \times \Psi(X)$. The preceding isomorphism is manifestly functorial in X', that is, it gives an isomorphism

$$\Psi \simeq \varphi \circ j,$$

and one likewise checks that, under this isomorphism, $u : e' \to \Psi(X)$ is identified with $\varphi(\delta_X)$. We have thus obtained the essential part of the following result.

LEMMA 5.11.2. *Let E and E' be two categories in which finite projective limits are representable, let X be an object of E, let* $\mathbf{Sex}(E, E')$ *be the category of left exact functors Ψ from E to E', and let* $\mathbf{Sex}(E_{/X}, E')$ *be the analogous category of left exact functors φ from $E_{/X}$ to E'. There is then an equivalence between the category* $\mathbf{Sex}(E_{/X}, E')$ *and the category* $\mathbf{Sex}(E, E')_{/\Gamma}$ *of pairs (Ψ, u), where $\Psi \in \mathrm{ob}\,\mathbf{Sex}(E, E')$ and $u : e' \to \Psi(X)$ (with e' the terminal object of E'). This equivalence, as well as a quasi-inverse functor, is described explicitly in 5.11.1. If φ and (Ψ, u) correspond, then φ commutes with a given type*
382 *of projective limit if and only if Ψ does. If the base-change functors in E and E' commute with a given type of inductive limit, then φ commutes with inductive limits of that type if and only if Ψ does.*

We leave it to the reader to make explicit the category structure on $\mathbf{Sex}(E, E')_{/\Gamma}$ corresponding to the pairs (Ψ, u). If the functor $\Gamma : \mathbf{Sex}(E, E') \to (\mathrm{Ens})$ is defined by $\Gamma(\Psi) = \mathrm{Hom}(e', \Psi(X))$, then u in the pair (Ψ, u) may be interpreted as an element of $\Gamma(\Psi)$, that is, as a homomorphism $\Psi \to \Gamma$ in the category of presheaves on $\mathbf{Sex}(E, E')^\circ$. This justifies the notation $_{/\Gamma}$ used in the statement of 5.11.1. To finish the proof of 5.11.2, it remains to make explicit in 5.11.1 the functorial nature of the constructions considered when φ, respectively (Ψ, u), varies. This is essentially trivial and is left to the reader. It also remains to verify the assertion concerning commutation with inductive limits; this too follows immediately from the explicit formulas (5.11.1.1) and (5.11.1.2) relating these functors.

In particular, when E and E' are $\mathscr{U}$-topoi, and morphisms of topoi are interpreted by means of their associated inverse-image functors, one obtains:

PROPOSITION 5.12. *Let E and E' be two $\mathscr{U}$-topoi, let X be an object of E, and consider on* $\mathbf{Homtop}(E', E)$ *the contravariant functor*

$$\gamma : f \longmapsto \Gamma(E', f^*(X)) = \mathrm{Hom}(e', f^*(X)) : \mathbf{Homtop}(E', E)^\circ \longrightarrow (\mathrm{Ens}).$$

There is then an equivalence of categories

$$\mathbf{Homtop}(E', E_{/X}) \xrightarrow{\approx} \mathbf{Homtop}(E', E)_{/\gamma}, \tag{5.12.1}$$

383 *where the right-hand side is the full subcategory of* $\mathbf{Homtop}(E', E)\hat{}_{/\gamma}$ *consisting of pairs (f, u) with $f \in \mathrm{Homtop}(E', E)$ and $u \in \gamma(f)$, that is, $u \in \Gamma(E', f^*(X))$. This functor is obtained by associating with every morphism of topoi $h : E' \to E_{/X}$ the pair (f, u) consisting of the composite*

$$f = j_X \circ h : E' \to E_{/X} \to E,$$

and the morphism

$$u : e' = h^*(X, \mathrm{id}_X) \longrightarrow f^*(X) = h^*(j_X^*(X)) = h^*(X \times X)$$

deduced from the diagonal morphism $\delta_X : X \to X \times X$ by applying h^* to it.

5.12.2. Using 5.12, the proof of 5.11 becomes almost evident, and essentially reduces to the following (which will serve in place of a formal proof that would be no more instructive): to give a morphism of topoi $g : F \to E'_{/X'}$, "amounts to the same thing" as to give a morphism of topoi $g_2 : F \to E'$ and a section u of $g_2^*(X')$. But $g_2^*(X') = g_2^*(f^*(X)) = (f g_2)^*(X)$, so giving u is also equivalent to giving a lifting of the morphism of topoi $f g_2 : F \to E$ to a morphism of topoi $g_1 : F \to E_{/X}$. This expresses precisely that giving a morphism of topoi $g : F \to E'_{/X'}$ is essentially equivalent to giving a triple (g_1, g_2, α) as in 5.11.

REMARK 5.13. We shall see (§ 15[10]) that for every diagram

$$\begin{array}{ccc} E_1 & & \\ \downarrow & & \\ E & \longleftarrow & E_2 \end{array}$$

of topoi, there exists a 2-fiber product in the sense of the 2-category ($\mathscr{V}$-$\mathscr{U}$-Top) of 384
$\mathscr{U}$-topoi belonging to $\mathscr{V}$ (independent, up to equivalence, of the choice of a universe $\mathscr{V}$ such that $\mathscr{U} \in \mathscr{V}$). For natural examples of "fiber products of topoi" other than 5.11, see the final chapter of GIRAUD's book [3].

EXERCISE 5.14.
a) Let F and G be two topoi over a topos E. Define the category $\mathbf{Homtop}_E(F,G)$ of pairs (f,α), where $f : F \to G$ is a morphism of topoi and $\alpha : p \to q \circ f$ is an isomorphism of morphisms of topoi from F to E (where $p : F \to E$ and $q : G \to E$ are the structural morphisms of topoi). Define pairing functors $\mathbf{Homtop}_E(F,G) \times \mathbf{Homtop}_E(G,H) \to \mathbf{Homtop}_E(F,H)$, and associativity isomorphisms.
b) Let X and Y be two objects of a topos E. Define a natural functor from the discrete category defined by the set $\mathrm{Hom}(X,Y)$ to the category $\mathbf{Homtop}_E(E_{/X}, E_{/Y})$. Establish compatibility with the composition of morphisms in E and with the pairings considered in a).
c) Prove that the functor considered in b) is an equivalence of categories. In particular, an object X of a topos E can be reconstructed up to unique isomorphism when one knows, "up to E-equivalence," the induced topos $E_{/X}$ as a topos *over* E. [10]

6. Points of a topos and fiber functors

6.0. Let P be the standard *$\mathscr{U}$-punctual topos* (2.2)

$$P = (\mathscr{U}\text{-Ens}).$$

In view of the rather different intuitions attached on the one hand to the symbol P, rep- 385
resenting a geometric object of the nature of a point, and on the other hand to ($\mathscr{U}$-Ens), representing a "large category" to be interpreted as "the category of sheaves on P," one should use one or the other symbol exclusively according to context, although from a strictly logical viewpoint they denote one and the same object.

DEFINITION 6.1. *Let E be a $\mathscr{U}$-topos. A point of E is a morphism of topoi $P \to E$ from the punctual topos P (6.0) to E. The category* $\mathbf{Homtop}(P,E)$ *(3.2) is called the category of points of E and is denoted by* $\mathbf{Points}(E)$ *or* $\mathrm{Pt}(E)$. *If $p : P \to E$ is a point of E, with associated inverse-image functor $p^* : E \to (\mathscr{U}\text{-Ens})$, and if F is an object of E, the set $p^*(F)$ is called the fiber of F at p and is denoted by F_p.*

6.1.1. Of course, when F is a group object (resp. ring object, resp. ...) of E, its fiber F_p at p is a group (resp. a ring, resp. ...) (3.1.2).

[10] N.D.E.: This section does not exist. See, however, [16], Proposition 4.47, for a general existence statement for fiber products of topoi (noting that all geometric morphisms between Grothendieck topoi are bounded).

[10] This result is due to P. DELIGNE. The case where E is the punctual topos had previously been treated by Mme HAKIM.

6.1.2. By 3.2.1, the category of points of E is equivalent, via the functor $p \mapsto p^*$, to the *opposite* of the category of functors

$$\varphi : E \longrightarrow (\mathscr{U}\text{-Ens})$$

which commute with $\mathscr{U}$-inductive limits and are left exact. Thus it amounts essentially to the same thing to give a point of E or to give such a functor φ.

DEFINITION 6.2. *A fiber functor of the $\mathscr{U}$-topos E is a functor $\varphi : E \to (\mathscr{U}\text{-Ens})$ which commutes with $\mathscr{U}$-inductive limits and is left exact. The category of fiber functors*
386 *on E, denoted by* $\mathbf{Fib}(E)$*, is the full subcategory of* $\mathbf{Hom}(E, (\mathscr{U}\text{-Ens}))$ *formed by the fiber functors on E.*

6.2.3. By 6.1.2, the category $\mathbf{Fib}(E)$ of fiber functors on E is therefore equivalent to the *opposite* of the category $\mathbf{Point}(E)$ of points of E, via the functor $p \mapsto p^*$ from the latter to the former:

$$\mathbf{Point}(E) \xrightarrow{\approx} \mathbf{Fib}(E)^\circ. \tag{6.2.1.1}$$

A functor

$$\varphi : E \longrightarrow (\mathscr{U}\text{-Ens})$$

is a fiber functor if and only if it is the fiber functor $F \mapsto F_p$ associated with a point p of E (6.1). In other words, the functor (6.2.1.1) is surjective. Let us also note that a functor φ is a fiber functor if and only if it is left exact and carries covering families to surjective families (1.7). This last condition may also be expressed by saying that φ commutes with $\mathscr{U}$-sums and carries epimorphisms to epimorphisms.

6.3. Let $C \in \mathscr{U}$ be a site. A *point of the site C* is a point of the topos $C^\sim$, and the *category of points of the site C* is the category

$$\mathbf{Point}(C) = \mathbf{Point}(C^\sim).$$

Consider, on the other hand, the functor

$$\varphi \longrightarrow \varphi|C = \varphi \circ \epsilon_C : \mathbf{Fib}(C^\sim) \longrightarrow \mathbf{Hom}(C, (\mathscr{U}\text{-Ens})), \tag{6.3.1}$$

which is fully faithful and whose essential image is the full subcategory $\mathbf{Morsite}((\mathscr{U}\text{-Ens}), C)^\circ$ (4.9.4), which we shall also denote by $\mathbf{Fib}(C)$. A functor

$$\Psi : C \longrightarrow (\mathscr{U}\text{-Ens})$$

387 is called a *fiber functor on the site C* if it belongs to the essential image of (6.3.1), just described. Such a functor is therefore *continuous* (a fortiori (III 1.6), it carries covering families to covering families, i.e. to surjective families), and it is left exact. When finite projective limits are representable in C (a condition satisfied in almost every case encountered in practice), the preceding properties *characterize* the fiber functors on C: they are then precisely the functors $\Psi : C \to (\mathscr{U}\text{-Ens})$ which are left exact and carry covering families to surjective families (4.9.4).

One should retain that the functor $\varphi \to \varphi|C$ is an equivalence of categories

$$\mathbf{Fib}(C^\sim) \xrightarrow{\approx} \mathbf{Fib}(C),$$

so that *it amounts essentially to the same thing to give a fiber functor on the topos $C^\sim$* (or equivalently a point of that topos) *or to give a fiber functor on the site C*. Given a fiber

functor Ψ on C, arising up to isomorphism from a fiber functor φ on $C^{\sim}$, the latter is reconstructed from Ψ, up to canonical isomorphism, by the formula

$$\varphi(F) \simeq \varinjlim_{C_{/F}} \Psi(X), \tag{6.3.2}$$

where $C_{/F}$ is the category of objects X of C endowed with a morphism $X \to F$ (in $\widehat{C}$), i.e. an element of $F(X)$. Formula (6.3.2) is an immediate consequence of the fact that φ commutes with inductive limits and of the canonical isomorphism, functorial in F (II 4.1.1):

$$F \simeq \varinjlim_{C_{/F}} \epsilon_C(X).$$

More generally, when G is a presheaf on C, there is a canonical isomorphism, functorial in G: 388

$$\varphi(\underline{a}G) \simeq \varinjlim_{C_{/G}} \Psi(X), \tag{6.3.3}$$

where $\underline{a}G$ is the sheaf associated with G (II 4.1.1). Indeed, one has the formula

$$\underline{a}G \simeq \varinjlim_{C_{/G}} \epsilon_C(X).$$

6.4.0. We refer to I 6.1 for the notion of a *conservative family of functors*

$$\varphi_i : E \longrightarrow E_i \qquad i \in I.$$

Note that when E and the E_i are $\mathscr{U}$-topoi and the functors φ_i are inverse-image functors f_i^* associated with morphisms of topoi

$$f_i : E_i \longrightarrow E,$$

all the exactness properties assumed in I 6.2, I 6.3, and I 6.4 are satisfied, provided that in I 6.2 (v) D is assumed finite. The following conditions are then equivalent: $(f_i^*)_{i\in I}$ is conservative; it is conservative for *monomorphisms* (I 6.1); it is conservative for *epimorphisms*; and it is *faithful*.

When the family of functors f_i^* is conservative, one also sometimes says that *the family of morphisms of topoi* $(f_i)_{i\in I}$ *is conservative*. In particular: 389

Definition 6.4.1. *Let E be a $\mathscr{U}$-topos and let $(p_i : P \to E)_{i\in I}$ be a family of points of E. This family of points is called conservative if the family of associated fiber functors $F \mapsto F_{p_i}$ from E to ($\mathscr{U}$-Ens) is conservative (6.4.0). The topos E is said to have enough points if it admits a conservative family of points (i.e. if the family of all points of the topos is conservative).*

6.4.2. Let us note that all $\mathscr{U}$-topoi used thus far have enough points. It is nevertheless possible, "on purpose," to construct topoi which do not have enough points (7.2.6 e) and 7.4); such a topos is necessarily not "empty" in the geometric sense of 2.2. A topos having enough points admits a conservative family of points indexed by some $I \in \mathscr{U}$ (6.5). Note, however, that if E is a $\mathscr{U}$-topos, the set of isomorphism classes of points of E is not necessarily $\mathscr{U}$-small (7.3), although it is so in many cases encountered in practice. Finally, for an interesting existence theorem (due to P. Deligne) providing enough points and covering all cases arising in algebraic geometry, see VI 9.

6.4.3. When $(p_i)_{i\in I}$ is a conservative family of points of the topos E, the remarks of I 6.2 apply. In particular, they imply that two arrows $u, v : F \rightrightarrows G$ in E are equal if and only if, for every $i \in I$, the induced arrows on fibers $u_{p_i}, v_{p_i} : F_{p_i} \rightrightarrows G_{p_i}$ are
390 equal; that an arrow $u : F \to G$ in E is a monomorphism (resp. an epimorphism) if and only if, for every $i \in I$, the map $u_{p_i} : F_{p_i} \to G_{p_i}$ is so; that an object F of E is initial (resp. terminal) if and only if, for every $i \in I$, F_{p_i} is empty (resp. consists of one point); and that an object F is a subobject of the terminal object e_E if and only if, for every $i \in I$, F_{p_i} has at most one point.

PROPOSITION 6.5. a) *Let C be a $\mathcal{U}$-site and let $(\varphi_i)_{i\in I}$ be a family of fiber functors on C (6.3). The family $(p_i)_{i\in I}$ of corresponding points of the topos $E = C^\sim$ is conservative if and only if every family $(X_j \to X)_{j\in J}$ in C for which, for every $i \in I$, the corresponding family $(\varphi_i(X_j) \to \varphi_i(X))_{j\in J}$ is surjective is a covering family.*

b) *Let E be a $\mathcal{U}$-topos. If E has enough points (6.4.1), then E admits a $\mathcal{U}$-small conservative family of points.*

Assertion b) is a special case of I 7.7. To prove a), apply I 7.7 to the generating family in E formed by the $\epsilon(X)$, $X \in \mathrm{ob}\, C$. Recall (II 4.4) that the family $X_j \to X$ is covering if and only if the family $\epsilon(X_j) \to \epsilon(X)$ is epimorphic. If the family of points $(p_i)_{i\in I}$ is conservative, this amounts (6.4.3) to saying that, for every $i \in I$, the family $(X_j)_{p_i} \to (X)_{p_i}$ is surjective, i.e. that the family $\varphi_i(X_j) \to \varphi_i(X)$ is surjective. This proves the "only if" part of a). For the "if" part, apply the criterion I 7.7, which reduces
391 us to verifying that every monomorphism $F \to \epsilon(X)$ for which $F_{p_i} \to \epsilon(X)_{p_i}$ is an isomorphism for every $i \in I$ is an isomorphism, or equivalently an epimorphism. Now one can find a covering family of morphisms $\epsilon(X_j) \to F$, and, after refining further, one may suppose that the morphisms $\epsilon(X_j) \to \epsilon(X)$ are induced by morphisms $X_j \to X$. By hypothesis, for every $i \in I$, the composite morphisms $(X_j)_{p_i} \to F_{p_i} \to (X)_{p_i}$ form a surjective family (as the composite of two surjective families), i.e. the family $\varphi_i(X_j) \to \varphi_i(X)$ is surjective. It follows from the hypothesis that the family $X_j \to X$ is covering, hence that the family $\epsilon(X_j) \to \epsilon(X)$ is epimorphic, and a fortiori that $F \to \epsilon(X)$ is an epimorphism.

COROLLARY 6.5.1. *Let C be a category. A topology on C making C a $\mathcal{U}$-site with enough fiber functors (i.e. such that the associated topos has enough points) is completely determined by the full subcategory Φ of* $\mathbf{Hom}(C, \mathrm{Ens})$ *formed by the fiber functors on C.*

Indeed, by 6.5 a), one can then describe the covering families in terms of the family of fiber functors.

In fact, we shall see below (6.8.5) that the subcategory Φ is contained in the category of pro-representable functors on C, so that $\Phi^\circ \simeq \mathbf{Point}(C^\sim)$ identifies, up to equivalence, with a full subcategory of $\mathrm{Pro}-C$.

Compare 6.5.1 with GIRAUD's theorem II 5.5, which establishes a one-to-one correspondence between the set of topologies on C and a certain set of full subcategories of $\mathbf{Hom}(C^\circ, \mathrm{Ens})$.

392 EXERCISE 6.5.2. Let C be a category equivalent to a category in $\mathcal{U}$, and let P be a subcategory of $\mathrm{Pro}(C)$. For every $p \in P$, denote by $X \mapsto X_p$ the functor pro-represented by p. A family $(X_i \to X)_{i\in I}$ in C is called *P-covering* if, for every $p \in P$, the family $(X_{ip} \to X_p)_{i\in I}$ is surjective. Show that there is a topology T_p on C whose covering families are precisely the P-covering families, and that T_p is the finest topology on C

for which the functors $X \mapsto X_p$, $p \in P$, are fiber functors. Show that the topology T_p makes C a site with *enough* fiber functors. Show that, for every topology T on C, among the topologies T' finer than T which have enough fiber functors there is a coarsest one: it is the topology T_p, where P is the strictly full subcategory of $\mathrm{Pro}(C)$, equivalent to $\mathbf{Point}(C)$, described in (6.8.5) below.

PROBLEM 6.5.3. Characterize the subcategories (strictly full, stable under $\mathscr{U}$-filtered projective limits, ...) P of $\mathrm{Pro}(C)$ which can arise from a topology on C as the essential image of $\mathbf{Point}(C^{\sim})$ (6.8.5). [10]

6.6. Let

$$f : E \longrightarrow E'$$

be a morphism of $\mathscr{U}$-topoi. It induces a canonical functor

$$\mathbf{Point}(f) = (p \longmapsto f \circ p) : \mathbf{Point}(E) \longrightarrow \mathrm{Point}(E').$$

Given two composable morphisms of topoi

$$E \xrightarrow{f} E' \xrightarrow{g} E'',$$

one checks immediately that 393

$$\mathbf{Point}(gf) = \mathbf{Point}(g) \circ \mathbf{Point}(f) : \mathbf{Point}(E) \longrightarrow \mathbf{Point}(E') \longrightarrow \mathbf{Point}(E'').$$

This makes precise the functorial dependence (without any abuse of language for once) of $\mathbf{Point}(E)$ on the topos E: if $\mathscr{V}$ is a universe such that $\mathscr{U} \in \mathscr{V}$, one obtains a (genuine!) functor

$$\mathbf{Point} : (\mathscr{V}\text{-}\mathscr{U}\text{-top}) \longrightarrow (\mathscr{V}\text{-cat})$$

from the category of $\mathscr{U}$-topoi which are elements of $\mathscr{V}$ to the category of categories which are elements of $\mathscr{V}$.

6.7. Let E be a $\mathscr{U}$-topos. Every object F of E then defines a contravariant functor

$$p \longmapsto F_p = p^*(F) : \mathbf{Point}(E)^\circ \longrightarrow (\mathscr{U}\text{-Ens}),$$

and hence, for varying F, a canonical functor

$$E \longrightarrow \mathbf{Point}(E)\hat{} = \mathbf{Hom}(\mathbf{Point}(E)^\circ, (\mathscr{U}\text{-Ens})), \tag{6.7.1}$$

which, by definition (6.5), is faithful if and only if E has enough points. (The functor (6.7.1) bears a certain formal analogy to a Fourier transform...)

Let X be an object of E, which therefore defines a presheaf $\widehat{X}$ on

$$C = \mathbf{Point}(E).$$

We may therefore define the category

$$C_{/\widehat{X}} = \mathbf{Point}(E)_{/\widehat{X}}$$

of morphisms $p \to \widehat{X}$ in the category $\widehat{C}$ of presheaves on C, i.e. the category of pairs (p, ξ) where p is a point of E and ξ is an element of $\widehat{X}(p) = X_p$, the fiber of X at p. With this understood, I claim that *there is a canonical equivalence of categories*

$$C' = \mathbf{Point}(E_{/X}) \xrightarrow{\approx} C_{/\widehat{X}} \quad , \quad \text{where} \quad C = \mathbf{Point}(E), \tag{6.7.2}$$

whose composite with the inclusion functor $C_{/\widehat{X}} \to C$ is none other than the functor 394

[10] cf. 9.1.8 a) and e) for necessary conditions, and 9.1.12 b) for necessary and sufficient conditions in the cases of $\mathrm{Top}(X)$, with X a sober locally Noetherian space.

$$\mathbf{Point}(j_X) : C' = \mathbf{Point}(E_{/X}) \longrightarrow C = \mathbf{Point}(E)$$

induced functorially (6.6) by the localization morphism (5.2.1)

$$j_X : E_{/X} \longrightarrow E.$$

This is simply the special case of 5.12 obtained by taking $E' = P = (\mathrm{Ens})$.

Corollary 6.7.3. *Let E be a topos. If E has enough points, then so does every induced topos $E_{/X}$, $X \in \mathrm{ob}\, E$. Conversely, if $(X_i)_{i\in I}$ is a family of objects of E covering the terminal object and, for every $i \in I$, $E_{/X_i}$ has enough points, then E has enough points.*

For the first assertion, let $f : X' \to X''$ be a morphism in $E_{/X}$ which is not an isomorphism. We prove that there exists a point q of $E_{/X}$ such that f_q is not an isomorphism. Since E has enough points, there is a point p of E such that $f_p : X'_p \to X''_p$ is not an isomorphism. This implies that there exists $q \in X_p$ such that the map induced by f_p on the fibers X'_q and X''_q of X'_p and X''_p over X_p at q is not an isomorphism. But by the equivalence (6.7.2), q may be identified with a point of $E_{/X}$, and the map $X'_q \to X''_q$ just considered is none other than the map on fibers at q induced by $f : X' \to X''$, which proves the assertion.

Conversely, suppose that the X_i cover the terminal object of E and that the $E_{/X_i}$
395 have enough points. We prove the same for E. Let $f : X' \to X''$ be a morphism in E which is not an isomorphism. There is then some $i \in I$ such that $f_i : X'_i \to X''_i$ is not an isomorphism, where the subscript i denotes restriction to X_i. Hence there is a point p_i of $E_{/X_i}$ such that $f_{ip_i} : X'_{ip_i} \to X''_{ip_i}$ is not an isomorphism. If p denotes the image of p_i in E under the localization morphism $E_{/X_i} \to E$, this says that $X'_p \to X''_p$ is not an isomorphism, which proves that E has enough points.

Remark 6.7.4. The preceding argument shows that if (p_α) is a conservative family of points of E, then the family of points of $E_{/X}$ lying over one of the p_α (a family indexed by the disjoint union of the X_{p_α}) is conservative. Likewise, if the X_i cover the terminal object of E and, for every $i \in I$, P_i is a conservative set of points of $E_{/X_i}$, then the set of points of E which are the *images* of the $p \in P_i$ under the localization morphisms $E_{/X_i} \to E$ (with $i \in I$ and p varying) is conservative.

6.8. Let E be a topos and p a point of E. A *neighborhood of the point p of the topos E* is a pair (X, u), where $X \in \mathrm{ob}\, E$ and $u \in X_p$. By (6.7.2), one may also interpret u as a lift of p to a point of the induced topos $E_{/X}$. Writing φ_p for the fiber functor associated with the point p, one may also interpret a neighborhood of the point p as an object of the category $E_{/\varphi_p}$ (the full subcategory of $\hat{(E^\circ)}_{/\varphi_p}$ formed by the objects whose source lies in $E \subset \hat{E}$). This latter category, i.e. the category of pairs (X, u) as above, will naturally be called the *category of neighborhoods* of the point p of the topos E; it is also denoted by
396 $\mathbf{Vois}(p)$. Since φ_p is left exact and finite projective limits are representable in E, finite projective limits are representable in the category $\mathbf{Vois}(p)$, and the canonical functor $\mathbf{Vois}(p) \to E$ commutes with them (I 8.10.10). A fortiori, the opposite category $\mathbf{Vois}(p)^\circ$ is filtered. Moreover, it is clear (I 3.4) that there is a canonical isomorphism, functorial in $F \in \mathrm{ob}\, E$,

$$\varphi_p(F) = F_p \simeq \varinjlim_{\mathbf{Vois}(p)^\circ} F(X). \tag{6.8.1}$$

In other words, *the functor φ_p, "fiber at the point p," is isomorphic to the filtered inductive limit of the functors represented in the topos E by the neighborhoods of the point p of E.* In

fact, *the fiber functor is pro-representable* (I 8), which also means that it is represented by a pro-object $(X_i)_{i\in I}$ of E, where I is a filtered ordered set such that $I \in \mathscr{U}$. Indeed, by *loc. cit.*, this is equivalent to saying that the category $\mathbf{Vois}(p)$ admits a small full cofinal subcategory. Now let C be a small full generating subcategory of E; I claim that the full subcategory $C_{/\varphi_p}$ of $\mathbf{Vois}(p) = E_{/\varphi_p}$, formed by the neighborhoods (X,u) of p such that $X \in \mathrm{ob}\, C$ (a category that is obviously small), is cofinal in $\mathbf{Vois}(p)$. Indeed, let (X,u) be a neighborhood of p; one must find a neighborhood (X',u') of p lying over (X,u), with $X' \in \mathrm{ob}\, C$. There is a covering family $X'_i \to X$, with the X'_i in C, from which it follows that the X'_{ip} cover X_p; hence there exist an $X' = X'_{i_\circ}$ and a $u' \in X'_p$ such that (X',u') lies over (X,u), as claimed.

6.8.2. Let C be a $\mathscr{U}$-site and p a point of C, i.e. a point of the topos $C^\sim$. By an abuse of notation, we again denote by φ_p the "restriction" of the fiber functor φ_p to C, i.e. the composite $\varphi_p \circ \epsilon$; for an object X of C, we shall also often write 397

$$X_p = \varphi_p(X) = \epsilon(X)_p.$$

A *neighborhood of the point p in the site C* is a pair (X,u), where $X \in \mathrm{ob}\, C$ and $u \in X_p = \varphi_p(X)$. These neighborhoods again form a category $C_{/\varphi_p}$, which may be denoted by $\mathbf{Vois}_C(p)$. Let us show that *the category $\mathbf{Vois}_C(p)^\circ$ is again filtered, that it admits a $\mathscr{U}$-small full cofinal subcategory, and that there is an isomorphism, functorial in the sheaf F on C:*

$$F_p \simeq \varinjlim_{\mathbf{Vois}_C(p)^\circ} F(X). \tag{6.8.3}$$

First note, by repeating the argument of (6.8.1), that if C' is a full subcategory of C that is topologically generating (II 3.0.1), then $\mathbf{Vois}_{C'}(p)^\circ$ is cofinal in $\mathbf{Vois}_C(p)^\circ$. Taking $C' \in \mathscr{U}$ to be small, we are thus reduced, in proving the assertion, to the case where $C \in \mathscr{U}$. In this case $\widehat{C}$ is a $\mathscr{U}$-topos, and there is an associated-sheaf functor $\underline{a} : \widehat{C} \to C^\sim$, which allows us to construct on $\widehat{C}$ the composite fiber functor $\varphi_p\underline{a}$. Applying (6.8.1) to the topos $\widehat{C}$ and to its full generating subcategory C, one finds that the category $C^\circ_{/\varphi_p\underline{a}}$, which is manifestly isomorphic to $\mathbf{Vois}_C(p)^\circ$, is filtered, and that there is an isomorphism functorial in the presheaf P on C:

$$\varphi_p\underline{a}(P) \simeq \varinjlim_X P(X).$$

If F is a sheaf on C, then $F = \underline{a}F$, and applying the preceding isomorphism to F regarded as a presheaf gives (6.8.3). At the same time, this argument establishes the more general formula 398

$$(\underline{a}P)_p \simeq \varinjlim_{\mathbf{Vois}_C(p)^\circ} P(X), \tag{6.8.4}$$

an isomorphism functorial in an arbitrary presheaf P on C, at least when C is small. The general case follows by introducing a universe $\mathscr{V}$ such that $C \in \mathscr{V}$, by regarding φ_p on C as a fiber functor with values in ($\mathscr{V}$-Ens), and using the compatibility of the formation of the associated sheaf with enlargement of universes (II 3.6).

6.8.5. To every point p of the topos $C^\sim$ we have associated a pro-object of the $\mathscr{U}$-site C, defined by the canonical functor $\mathbf{Vois}_C(p) \to C$, taking into account the fact that the category of neighborhoods of p in C is cofiltered and admits a small full cofinal subcategory. One immediately verifies that, as p varies, this gives a functor

$$\mathbf{Point}(C^\sim) \longrightarrow \mathrm{Pro}(C). \tag{6.8.5.1}$$

This functor is *fully faithful.* Via the associated fiber functors, it may be interpreted as a functor $\mathrm{Fib}(C) \to \mathrm{Fib}(C^\sim) \to \mathrm{Pro}(C)^\circ$. The first functor is an equivalence by 4.9.4 (recalled for fiber functors in 6.3), and the second is fully faithful by the general formalism (I 8) of pro-objects.

6.8.6. Consider the special case where C is $\mathscr{U}$-small and endowed with the chaotic topology, so that $C^\sim = \widehat{C}$. I claim that in this case the preceding functor is even an equivalence of categories

$$\mathbf{Point}(\widehat{C}) \xrightarrow{\approx} \mathrm{Pro}(C). \tag{6.8.6.1}$$

399 Indeed, it remains to prove that this functor is essentially surjective, which follows from the fact that the functors of the form $P \mapsto P(X)$ on $\widehat{C}$ are fiber functors, and from the evident fact that every filtered inductive limit of fiber functors on a topos is again a fiber functor.

Identifying C with a full subcategory of $\mathrm{Pro}(C)$ (I 8), the functor

$$C \longrightarrow \mathbf{Point}(\widehat{C})$$

induced by a quasi-inverse to (6.8.6.1) is manifestly isomorphic to the canonical functor already considered in (4.6.2.2); the quasi-inverse in question may be regarded as its canonical extension (I 8), taking into account the fact that $\mathbf{Point}(\widehat{C})$ is stable under small projective limits.

6.8.7. More generally, let $(X_i)_{i\in I}$ be a pro-object of the $\mathscr{U}$-site C, and hence a functor on $C^\sim$

$$\varphi(F) = \varinjlim_I F(X_i), \tag{6.8.7.1}$$

and, in view of (6.8.5), it is natural to ask when this functor is a fiber functor. The following conditions are equivalent:

(i) The preceding functor φ is a fiber functor on $C^\sim$.

(ii) The "restriction" of φ to C is a fiber functor on the site C (6.3).

(iii) For every covering family $Y_\alpha \to Y$ of an object Y of C, every $i_\circ \in I$, and every morphism $X_{i_\circ} \to Y$, there exist an $i \geq i_\circ$, an α, and a morphism $X_i \to Y_\alpha$ making the diagram

$$\begin{array}{ccc} X_i & \longrightarrow & Y_\alpha \\ \downarrow & & \downarrow \\ X_{i_o} & \longrightarrow & Y \end{array} .$$

commutative.

400 Indeed, condition (iii) simply expresses the fact that $\varphi|C$ carries covering families to covering families. The implications (i) ⇒ (ii) ⇒ (iii) are trivial, and it remains to prove (iii) ⇒ (i). Since the functor φ is manifestly left exact, it remains to prove (4.9.4) that it carries covering families $F_\alpha \to F$ in $C^\sim$ to covering families, i.e. that for every $i_\circ$ and every $x_\circ \in F(X_{i_\circ})$, there exist an $i \geq i_\circ$, an α, and a $y \in F_\alpha(X_i)$ such that $x_\circ$ and y have the same image in $F(X_i)$. Since the family $F_\alpha \to F$ is covering, there is a covering family $Z_\gamma \to X_{i_\circ}$ of $X_{i_\circ}$ such that, for every γ, the inverse image of $x_\circ$ in $F(Z_\gamma)$ lifts to an element of some $F_\alpha(Z_\gamma)$. By hypothesis (iii), there exist an $i \geq i_\circ$ and an $X_{i_\circ}$-morphism $X_i \to Z_\gamma$. The image of $x_\circ$ in $F(X_i)$ therefore lifts to an element y of some $F_\alpha(X_i)$, C.Q.F.D.

EXERCISE 6.9. Let $C \in \mathscr{U}$ be a site, and let φ be a fiber functor on C, pro-represented by an object $(X_i)_{i\in I}$ of $\mathrm{Pro}(C)$. For every index $i \in I$, every object Y of C, every morphism $u : X_i \to Y$, and every covering sieve R on Y, choose an index $j \geq i$ such that the composite $X_j \to X_i \xrightarrow{u} Y$ factors through R. For fixed i, let $J(i)$ be the set formed by i and by the j obtained as Y, u, and R vary; for a subset I' of I, likewise let $J(I')$ be the union of the $J(i)$ for $i \in I'$. On the other hand, if I' is a finite subset of I, let $m(I') \in I$ be an upper bound of I', and for an arbitrary subset I' of I, let $M(I')$ be the union of the $m(I'')$, where I'' ranges over the set of finite subsets of I'. We assume 401
that the function $I' \mapsto m(I')$ is chosen so that, when I' consists of a single element i, one has $m(I') = i$; this implies that, for every subset I' of I, one has $I' \subset M(I')$. Consider the iterates $(MJ)^n$ of the map MJ from the set of subsets of I to itself, and, for every subset I' of I, let $P(I')$ be the union of the $(MJ)^n(I')$.

a) Show that, for every subset I' of I, $P(I')$ is a filtered subset of I for the induced order, and that I is the increasing filtered union of the $P(I')$ as I' ranges over the set of *finite* subsets of I.
b) Show that there is a cardinal $c \in \mathscr{U}$, depending only on the cardinal $\mathrm{card}\,\mathrm{fl}(C) = a$, such that, for every finite subset I' of I, one has $\mathrm{card}(P(I')) \leq c$. (If a is infinite, one may take $c = 2^a$.)
c) Show, using 6.8.7, that for every subset I' of I the functor $\varphi_{I'}$ on C, pro-represented by the pro-object $(X_i)_{i\in P(I')}$, is a fiber functor on C, and that the fiber functor φ is the filtered inductive limit of the functors $\varphi_{I'}$ as I' ranges over the set of finite subsets of I.
d) Deduce that there is a small full subcategory $\boldsymbol{\Phi}$ of $\mathbf{Fib}(C) \approx \mathbf{Fib}(C)$ such that every object of $\mathbf{Fib}(C)$ is a filtered inductive limit of objects of $\boldsymbol{\Phi}$. (If a is infinite, one may take $\boldsymbol{\Phi}$ to have cardinal $\leq 2^{2^a}$; if a is finite, one may of course take $\boldsymbol{\Phi} = C$.)

EXERCISE 6.10. Let C be a $\mathscr{U}$-site, and let D be a $\mathscr{U}$-category in which small filtered $\varinjlim$ are representable; let $\mathrm{Fais}(C, D)$ be the category of sheaves on C with values in D (II 6.1). By extending (6.8.3), define a "fiber functor"

$$F \longrightarrow F_p : \mathrm{Fais}(C, D) \longrightarrow D. \tag{6.10.1}$$

If finite $\varprojlim$ are representable in D and commute with filtered $\varinjlim$, then the preceding functor is left exact.

7. Examples of fiber functors and points of topoi

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7.1. The case of $\mathrm{Top}(X)$ for a topological space X. Let x be a point of X. Regard the singleton $\{x\}$ as a topological space, and consider the inclusion

$$i_x : \{x\} \longrightarrow X.$$

By 4.1.1, it defines a morphism of topoi (defined up to unique isomorphism)

$$\mathrm{Top}(i_x) : \mathrm{Top}(\{x\}) \longrightarrow \mathrm{Top}(X). \tag{7.1.1}$$

On the other hand, one may identify P with $\mathrm{Top}(\{x\})$ (2.2), and one thereby obtains a point p_x of $\mathrm{Top}(X)$:

$$p_x : P \longrightarrow \mathrm{Top}(X), \tag{7.1.2}$$

defined by x up to unique isomorphism. The associated fiber functor is given by the familiar formula [TF], which moreover follows immediately from I 5.1:

$$p_x^*(F) = F_x = F_{p_x} = \varinjlim_{U \ni x} F(U), \tag{7.1.3}$$

the limit being taken over the open neighborhoods U of x in X.

Since $\mathrm{Top}(X)$ is known to be isomorphic to $\mathrm{Top}(X_{\mathrm{sob}})$ (4.2.1), one sees more generally that every point of X_{sob}, i.e. every irreducible closed subset Z of X, defines a point of $\mathrm{Top}(X)$, or equivalently a fiber functor on $\mathrm{Top}(X)$, which we shall still denote by $F \mapsto F_Z$. Returning to its definition by formula (7.1.3) on X_{sob}, one obtains

$$F_Z = \varinjlim_{U \cap Z \neq \phi} F(U), \tag{7.1.4}$$

403 the inductive limit being taken over the open subsets U of X which meet Z.

7.1.5. It is well known that the fiber functors $F \mapsto F_x$ ($x \in X$) already form a conservative family. This is especially clear from the interpretation of sheaves on X in terms of étale spaces, since the fiber of F at x is then simply its fiber in the sense of spaces fibered over X. Note that the index set of the conservative family of fiber functors under consideration is X, and hence is $\mathscr{U}$-small.

7.1.6. By 4.2.3, the category **Point**($\mathrm{Top}(X)$) is equivalent to the category associated with the set X_{sob} ordered by the specialization relation. In other words, every fiber functor on $\mathrm{Top}(X)$ is isomorphic to a fiber functor $F \mapsto F_Z$, where Z is a uniquely determined element of X_{sob}, i.e. a uniquely determined irreducible closed subset of X; moreover, if Z, $Z' \in X_{\mathrm{sob}}$, then the set $\mathrm{Hom}(F_Z, F_{Z'})$ is either empty or consists of a single point, the latter case occurring if and only if Z is a specialization of Z' in X_{sob}, i.e. if and only if $Z \subset Z'$ as a subset of X. When X itself is sober, one may replace X_{sob} by X itself in these statements.

Note that the automorphism group of a fiber functor of $\mathrm{Top}(X)$ (opposite to the automorphism group of the corresponding point of $\mathrm{Top}(X)$) is always the trivial group (more generally, 4.2.3 tells us the same thing about the automorphism group of every morphism of topoi associated with topological spaces). This is a phenomenon very special to the
404 case under consideration: see the example in 7.2, as well as VIII 7. In the case of topoi associated with "moduli problems" (cf. for example [10]), the automorphism groups of fiber functors moreover have a remarkable interpretation as the automorphism groups of the algebraic structures (over algebraically closed fields) which one proposes to classify.

7.1.7. We have just seen how one can reconstruct a sober space X (or the sober space X_{sob} associated with an arbitrary topological space), at least as a set ordered by the specialization relation, from the topos $\mathrm{Top}(X)$ (which it even suffices to know up to equivalence), as the set of isomorphism classes of "points" of $\mathrm{Top}(X)$. From what was said in 2.1, one also reconstructs the topology of X, i.e. the family of its open subsets, as follows: for every subobject U of the terminal object e_E of E, let $\mathrm{Pt}(U)$ be the set of $x \in X$ such that $U_x \neq \phi$ (which moreover identifies, by (6.7.2), with the set of isomorphism classes of points of the induced topos $E_{/U}$). Then $U \mapsto \mathrm{Pt}(U)$ is an isomorphism of ordered sets from the set of subobjects of e_E onto the set of open subsets of X.

7.1.8. The determination in 7.1.6 of the category of points of the topos $\mathrm{Top}(X)$ defined by a topological space X leads to the following terminology for points p, p' of an arbitrary topos E: one says that p is a *specialization* of p', or that p' is a *generization* of p, when there exists a morphism from p' to p (in the sense of the category **Point**(E) of 6.1). These relations are still transitive, but, using 6.8.6, one easily finds examples in
405 which p and p' are each a specialization of the other, without p and p' being isomorphic

(much less equal). The category of generizations of a point p of the topos E, by which we mean the category $\mathbf{Point}(E)_{/p}$, plays in some respects a role analogous to that of passing to the localization of a scheme at a point. This role will be made precise in VI 8.4 by the construction of the topos obtained by localizing E at the point p, whose category of points is canonically equivalent to the category of generizations of p.

Exercise 7.1.9. Let E be a $\mathscr{U}$-topos. Prove that E is equivalent to a topos of the form $\mathrm{Top}(X)$, where X is a topological space $\in \mathscr{U}$, if and only if it satisfies the following two conditions:

a) The family of subobjects of the terminal object e_E is generating for the topos E.
b) E has enough points (6.5). (This condition is not superfluous: cf. 7.4.)

Compare with 7.8 c).

Exercise 7.1.10. Let X be a topological space endowed with an operator group G ($X, G \in \mathscr{U}$), and let $E = \mathrm{Top}(X, G)$ (2.3).

a) Show that, for every object (X', G) of E, the induced topos $E_{/(X',G)}$ is canonically equivalent to the topos $\mathrm{Top}(X', G)$ (compare 5.7).
b) When G acts properly and freely on X' (Bourbaki, Gen. Top. Chap. III 4), show that the morphism of spaces with operators $(X', G) \to (X'/G, e)$ induces (4.1.2) an equivalence of topoi

$$\mathrm{Top}(X', G) \longrightarrow \mathrm{Top}(X'/G).$$

c) Deduce from b) that there exists an object Z of E such that the induced topos $E_{/Z}$ is equivalent to $\mathrm{Top}(X)$, the localization functor $E \to E_{/Z}$ being isomorphic to the functor that "forgets the actions of G". (Imitate the argument of 5.8.3.) 406
d) Deduce from c) that every fiber functor on E is induced, via the functor that "forgets the actions of G", by a fiber functor on $\mathrm{Top}(X)$, and can therefore be defined by a point of X_{sob}.
e) Determine the structure of the category $\mathbf{Point}(E)$ (up to equivalence) in terms of the space with operators (X_{sob}, G). Conclude in particular that two points of X_{sob} define isomorphic fiber functors on E if and only if they are conjugate under the action of G.

Exercise 7.1.11. Let $X \in \mathscr{U}$ be a topological space. Prove that the $\mathscr{V}$-topos $\mathrm{TOP}(X)$ (2.5) has enough points. More precisely, for every object X' of $\mathrm{TOP}(X)$ and every $x' \in X'$, define a fiber functor $F \mapsto F_{X',x'}$ on $\mathrm{TOP}(X)$ which depends only on the "germ" of the space (X', x') over X, and prove that this family of fiber functors is conservative (and indexed by a $\mathscr{V}$-small index set). Show, using a suitable filtered projective system $(X'_i, x'_i)_{i\in I}$, with I not $\mathscr{U}$-small, that one does not obtain in this way all the fiber functors of the $\mathscr{V}$-topos $\mathrm{TOP}(X)$. Show that the set of isomorphism classes of such fiber functors does not have cardinality $\in \mathscr{V}$.

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7.2. **Points of a classifying topos B_G.** Let E be a $\mathscr{U}$-topos and G a group of E, whence a classifying topos B_G (2.4). If e is the unit group, the morphisms $e \to G \to e$ define (4.5) morphisms of topoi (taking into account that $B_e \approx E$)

$$E \longrightarrow B_G \longrightarrow E, \tag{7.2.1}$$

whose composite is isomorphic to id_E; passage to the categories of points therefore gives functors

$$\mathbf{Point}(E) \longrightarrow \mathbf{Point}(B_G) \longrightarrow \mathbf{Point}(E), \tag{7.2.2}$$

whose composite is isomorphic to the identity. Use the fact that the first morphism of topoi in (7.2.1) identifies with a localization morphism relative to the object $E_G = G$ of B_G (5.8); by (6.7.2), it follows that the first functor in (7.2.2) identifies with the natural "inclusion" functor

$$\mathbf{Point}(B_G)_{/\widehat{E}_G} \longrightarrow \mathbf{Point}(B_G), \tag{7.2.3}$$

where $\widehat{E}_G$ denotes the presheaf $p' \mapsto (E_G)_{p'}$ on $\mathbf{Point}(B_G)$. Since E_G plainly covers the terminal object of B_G, its fibers $(E_G)_p$ are nonempty, from which it follows that (7.2.3) is essentially surjective; this means that *every fiber functor on B_G is induced (up to isomorphism) by a fiber functor on E, via the functor that "forgets the actions of G" $B_G \to E$*. One concludes in particular that *if E has enough points (resp. a small conservative family of points),* then the same is true of B_G.

Remark 7.2.4. One can go further and determine (up to equivalence) the structure of the category $C' = \mathbf{Point}(B_G)$ in terms of the category $C = \mathbf{Point}(E)$ and the presheaf of groups

$$\widehat{G} : p \longmapsto G_p : C^\circ \longrightarrow (\text{Groups})$$

408 on it. Indeed, define a new category C_1, having the same objects as C, and such that for two objects p, q of C, one has

$$\mathrm{Hom}_{C_1}(p,q) = \mathrm{Hom}_C(p,q) \times \widehat{G}(p),$$

composition of arrows in C_1 being induced by that in C and by the group law of the presheaf $\widehat{G}$. The datum of a functor $\varphi_1 : C_1 \to D$ into an arbitrary category D is then equivalent to the datum of a functor $\varphi : C \to D$ endowed with an action of the presheaf $\widehat{G}$ on the latter (i.e. the datum, for every $p \in \mathrm{ob}\, C$, of an action of $\widehat{G}(p)$ on $\varphi(p)$, satisfying an evident functoriality condition as p varies). Using this observation, one defines a canonical functor

$$\varphi_1 : C_1 \longrightarrow C', \tag{7.2.4.1}$$

corresponding to the functor $\varphi : C \to C'$ of (7.2.2), which associates with every point p of E the point $\varphi(p)$ induced on B_G, together with the natural actions of $\widehat{G}(p) = G_p$ on the latter, arising from the actions of G_p on the X_p as X ranges over B_G. We leave to the reader the proof that (7.2.4.1) is an equivalence of categories, using the structure (7.2.3) of the first functor in (7.2.2), and noting that the natural inclusion functor $C \to C_1$ has an analogous structure, relative to the inverse image P_1 of the presheaf $P' = \widehat{E}_G$ on C'.

409 7.2.5. One concludes in particular that the *isomorphism classes of points of B_G correspond exactly to the isomorphism classes of points of E: every point of the classifying topos B_G is induced, up to a* non-unique *isomorphism, by a point of E*. In particular, when E is the punctual topos P, and hence G is an ordinary group, the category $\mathbf{Point}(B_G)$ is a connected groupoid with fundamental group G: every fiber functor on B_G is isomorphic (noncanonically) to the functor ω that "forgets the actions of G", and the monoid of endomorphisms of the latter functor is the group G (hence every endomorphism of ω is an automorphism).

Exercise 7.2.6. a) Let (Tors) be the category of pairs $(\Gamma, P) \in \mathcal{U}$, where Γ is a group and P is a right torsor under Γ. The functor $(\Gamma, P) \mapsto \Gamma$

$$(\text{Tors}) \longrightarrow (\text{Groupes}) \tag{7.2.6.1}$$

is a cofibered functor. With the notation of 7.2.4, consider the functor

$$\widehat{G} : C^\circ \longrightarrow (\text{Groupes}),$$

and let D' be the cofibered category over C° obtained by inverse image from the cofibered category (7.2.6.1), and let D be the corresponding category fibered over C, having the same fiber categories as D', so that for $p \in \operatorname{ob} C$ one has

$$D_p = \text{ category of right torsors under } G_p.$$

Construct canonical functors

$$D \longrightarrow C' = \mathbf{Point}(B_G) \quad , \quad C' \longrightarrow D, \tag{7.2.6.2}$$

quasi-inverse to one another. In particular, conclude that the category of points of B_G above a given point p of E is canonically equivalent to the category of right torsors under the group G_p.

b) Let $\mathscr{G} = (G_i)_{i\in I}$ be a strict projective system of groups of the $\mathscr{U}$-topos E ($I \in \mathscr{U}$ a filtered preordered set), whence a classifying topos $B_{\mathscr{G}}$, by imitating the definition of 2.7.1. Determine the category of points of $B_{\mathscr{G}}$ by imitating a). (NB. Replace the categories (Tors) and (Groupes) by the categories of projective systems indexed by I in these categories.) Conclude that if I admits a countable cofinal subset, then every fiber functor on $B_{\mathscr{G}}$ is isomorphic to a functor induced by a fiber functor on E (via the functor that "forgets the actions of G"), the latter being determined up to non-unique isomorphism. 410

c) Taking E to be the punctual topos, so that $\mathscr{G}$ is an ordinary pro-group, show that the conclusion of b) is likewise valid if I is arbitrary but $\mathscr{G}$ is profinite (i.e. the G_i are finite): every fiber functor is isomorphic to the functor that "forgets the actions of G".

d) Still taking the case where E is the punctual topos, give an example of a fiber functor on $B_{\mathscr{G}}$, for a suitable projective system $\mathscr{G} = (G_i)_{i\in I}$, which is not isomorphic to the functor that "forgets the actions of G".

e) Regard $\mathscr{G} = (G_i)$ as a group of the topos $\widehat{I}$, and let $\mathscr{T}$ be a *gerbe* on I with *band* G [3]. Show how one can "twist" the classifying topos B_G by means of the gerbe T, to obtain a topos $B^T_{\mathscr{G}}$, an inductive limit of subcategories $B^T(i)$ equivalent (noncanonically) to the classifying topoi B_{G_i}. Show that the topos $B^T_{\mathscr{G}}$ admits a fiber functor if and only if the gerbe T is "neutral", by establishing an equivalence of categories between the category of fiber functors on $B^T_{\mathscr{G}}$ and 411
the category of sections of T over I. Conclude, if the G_i are commutative, so that gerbes on I with band $\mathscr{G}$ are classified by $\varprojlim^{(2)}_I G_i = H^2(I, \mathscr{G})$ (*loc. cit.*), that there is an example of a topos (not "empty" (2.2)) of the form $B^T_{\mathscr{G}}$ which has no points. (Take an example in which $\varprojlim^{(2)}_I G_i \neq 0$.)

7.3. **Points of presheaf topoi $\widehat{C}$; examples of $\mathscr{U}$-presheaf topoi $\widehat{C}$ whose category of points is not equivalent to a small category.** Let E be a $\mathscr{U}$-topos, and let $(\varphi_i)_{i\in I}$ be a filtered family of fiber functors on E. It follows from the exactness properties of the functors $\varinjlim$ (I 2.8) that the functor $\varphi = \varinjlim \varphi_i$ is also a fiber functor: *every filtered inductive limit of fiber functors is a fiber functor.* Consequently, the category $\mathbf{Fib}(E)$ admits $\mathscr{U}$-filtered inductive limits (and the inclusion functor $\mathbf{Fib}(E) \to \mathbf{Hom}(E, (\mathscr{U}\text{-Ens}))$

commutes with them); in other words, the category **Point**(E) admits $\mathscr{U}$-filtered projective limits. This fact was already used in Exercise 7.1.11 to give an example of "large" categories of fiber functors.

Considerably simpler examples can be constructed, with presheaf topoi of the form $E = \widehat{C}$, $C \in \mathscr{U}$, by using (6.8.6.1). This immediately gives examples in which **Fib**($\widehat{C}$) is not equivalent to a category $\in \mathscr{U}$, i.e. in which the cardinal of the set of isomorphism classes of objects of this $\mathscr{U}$-category is not $\in \mathscr{U}$. This means that the category of pro-objects Pro$-(C)$ is not equivalent to a category $\in \mathscr{U}$. For example, it suffices to take
412 $D = C^\circ$ to be the category of finite sets of the form $[0, n]$, where $n \geq 0$ is an integer; in this case Ind(D) $\simeq$ Pro(C)$^\circ$ is equivalent to the category of nonempty sets. The topos E is then the familiar category of *cosimplicial sets*. One could also take C to be the category of nonempty finite sets; one then finds that the category of points of the presheaf topos $\widehat{C}$ of simplicial sets is equivalent to the category of profinite sets, or equivalently to the category of totally disconnected compact spaces.

7.4. **Nonempty topoi without points**. The following example is due to *P*. Deligne. (For another example, cf. 7.2.6 e).) Let K be a compact space equipped with a measure μ, and let U be the ordered set of measurable subsets of K, modulo sets of measure zero. Make U into a category $\underline{U}$ with ob$\,\underline{U} = U$, the morphisms of $\underline{U}$ being the "inclusion morphisms" between elements of U. Make $\underline{U}$ into a site by taking the pretopology for which Cov(E) (for $E \in U$) consists of the *countable* families of elements E_i of U bounded above by E, such that E is the union of the E_i modulo a set of measure zero. This gives a topos Top(μ) $= \underline{U}^\sim$ whose set of subobjects of the terminal object is a generating family (a topos which appears to have escaped the attention of probabilists). This topos is an "empty topos" (2.2) if and only if $\mu = 0$. On the other hand, the category of points of this topos is equivalent to the discrete category defined by the set of points $x \in K$ such that $\mu(\{x\}) \neq 0$ (proof left to the reader). It is therefore empty if K has no such points, for example if K is the unit interval in the real line, with the measure induced by Lebesgue measure.

413 Exercise 7.5. (*karoubian categories and morphisms of topoi*)[10]

a) Let C be a category, let X be an object of C, and let p be an endomorphism of X. We say that p is a *projector* if $p^2 = p$. Prove that, if p is a projector, Ker(id$_X$, p) is representable if and only if Coker(id$_X$, p) is representable, and that the two objects of C so obtained are canonically isomorphic. We then say that the projector p *admits an image,* and Ker(id$_X$, p) $\simeq$ Coker(id$_X$, p) is called the *image of the projector p*; according to the context, it is identified with either a subobject or a quotient object of C. An object isomorphic to Im p, for a suitable projector p in X, is called a *direct factor of the object X of C*. We say that C is a *category with direct factors*, or a *karoubian category,* if every projector in an object of C admits an image. If $F : C \to C'$ is a functor that commutes with kernels or cokernels, then F sends a projector admitting an image to a projector admitting an image[11].

b) Show that, for every category C, one can find a functor $\varphi : C \to \mathrm{kar}(C)$ from C to a Karoubian category (determined up to equivalence) such that, for every Karoubian category C', the functor

$$f \mapsto f \circ \varphi : \mathbf{Hom}(\mathrm{kar}(C), C') \longrightarrow \mathbf{Hom}(C, C')$$

[10] Compare I 8.7.8.

[11] N.D.E.: No condition on F is needed here: every functor preserves projectors admitting images.

is an equivalence of categories; $\mathrm{kar}(C)$ is called the *Karoubi envelope* of the category C. (Hint: take $\mathrm{ob}\,\mathrm{kar}(C)$ to be the set of pairs (X, p), with $X \in \mathrm{ob}\,C$ and p a projector in X, and take $\mathrm{Hom}((X,p),(Y,q))$ to be the subset of $\mathrm{Hom}(X,Y)$ consisting of those $f : X \to Y$ such that $f = qfp$.) Show that φ is fully faithful.

c) Let $F : E \to E'$ be a functor from one category to another. Show that if F commutes with a given type of inductive or projective limits, then the same is 414
true of every direct factor of F. In particular, if E and E' are $\mathscr{U}$-topoi and F is an inverse-image functor for a morphism of topoi $E' \to E$, then the same is true of every direct factor of F; consequently, the category $\mathbf{Homtop}(E', E)$ is Karoubian, and in particular the category $\mathbf{Point}(E)$ is Karoubian.

d) Show that, for every $\mathscr{U}$-category C, the category $\mathrm{Ind}(C)$ is Karoubian (where $\mathrm{Ind}(C)$ is formed using inductive systems in C indexed by a filtered preordered set $I \in \mathscr{U}$). (If $C \in \mathscr{U}$, use, for example, the fact (7.3) that $\mathrm{Ind}(C)$ is equivalent to the category $\mathbf{Fib}(\widehat{(C^\circ)}) = \mathbf{Point}(\widehat{(C^\circ)}^\circ)$.) Deduce another construction of $\mathrm{kar}(C)$ as the full subcategory of $\mathrm{Ind}(C)$ formed by the images in $\mathrm{Ind}(C)$ of the projectors of objects X of C.

Exercise 7.6. (*Essential morphisms of topoi, essential points*).

a) Let $f : E \to F$ be a morphism of topoi. Show that the following conditions are equivalent:
   i) $f_!$ exists, i.e. f^* admits a left adjoint.
   ii) f^* commutes with $\mathscr{U}$-projective limits.
   ii′) f^* commutes with $\mathscr{U}$-products.

   We then say that f is an *essential morphism* from the topos E to the topos F.

   Show that if f satisfies these conditions, then the same is true of every direct factor of f. (Use 1.8 and 7.5 c).)

b) Let E be a topos. An *essential point of the topos* E is a point $p : P \to E$ of E such that $p_!$ exists. Show that if $E = \mathrm{Top}(X)$, where X is a topological space, and if p is the point of E defined by an $x \in X$, then p is essential if and only if x has 415
a smallest open neighborhood U (which means, if the points of X are closed, that x is an *isolated point* of X). For a morphism of topoi $f : E \to F$ to be essential (a)), it is necessary that f send essential points to essential points, and this condition is also sufficient when E has enough essential points (i.e. when the family of these points is conservative) (cf. d)).

c) A point p of the topos E is essential if and only if the associated fiber functor is representable by an object X of E. For the covariant functor $\varphi : E \to (\mathrm{Ens})$ represented by a given object X of E to be a fiber functor (i.e. to define a point, necessarily an essential point, of E), it is necessary and sufficient that X be nonempty and connected (4.3.5), and projective (i.e. that the functor φ send epimorphisms to epimorphisms). (Hint: first show that φ commutes with coproducts if and only if X is nonempty and connected, and then use the criterion 4.9.4.) Deduce an equivalence between the category $\mathbf{Pointess}(E)$ and the full subcategory P of E formed by the nonempty connected projective objects.

d) Show that the topology on P induced by that of E is the chaotic topology. Deduce the equivalence of the following conditions on E (due to J. E. ROOS [12 c), prop. 1]): (i) The family of essential points of E is conservative; (ii) The full subcategory P of E formed by the nonempty connected projective objects is generating; (iii) E is equivalent to a presheaf topos of the form $\widehat{C}$, where C is a category equivalent to a category $\in \mathscr{U}$ (or, if one prefers, $C \in \mathscr{U}$). (Use c) and

the fact that, if P is generating, the natural functor $E \to P^{\sim}$ is an equivalence
416 of categories.)

e) The category of essential points of a topos E is Karoubian (cf. 7.5 c)). Let C be a category equivalent to a category $\in \mathscr{U}$. Prove that the canonical fully faithful functor (4.6.2)

$$C \longrightarrow \mathbf{Point}(\widehat{C})$$

factors through a functor

$$C \longrightarrow \mathbf{Pointess}(\widehat{C}),$$

which makes $\mathbf{Pointess}(\widehat{C})$ a *Karoubi envelope* (7.5 b)) *of* C. In particular, this functor is an equivalence of categories if and only if the category C is Karoubian. (Hint: using c), prove that every essential point of $\widehat{C}$ is isomorphic to a direct factor of some $X \in \mathrm{ob}\, C$.)

f) Let C and C' be two categories equivalent to categories $\in \mathscr{U}$. Prove that the canonical fully faithful functor (4.6.2)

$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Homtop}(\widehat{C}, \widehat{C}')$$

takes its values in the full subcategory $\mathbf{Homtopess}(\widehat{C}, \widehat{C}')$ of essential morphisms *of topoi* (a)), and that the induced functor

$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Homtopess}(\widehat{C}, \widehat{C}')$$

is an equivalence of categories if and only if C is empty or C' is Karoubian. (Use g) below.)

g) Let C be a category equivalent to a category $\in \mathscr{U}$, and let E be a topos. Define an equivalence between the category $\mathbf{Homtopess}(\widehat{C}, E)$ of essential morphisms
417 of topoi $\widehat{C} \to E$, and the category $\mathbf{Hom}(C, \mathbf{Pointess}(E))$. (Use b) and the fact that $\widehat{C}$ has enough essential points.)

h) Let C be a finite category. Prove that every point of the presheaf topos $\widehat{C}$ is essential, and that $\mathbf{Point}(\widehat{C})$ is equivalent to a finite category. (Note that the Karoubi envelope of C is finite, and, using (6.8.6.1), reduce to proving that if C is a finite Karoubian category, then the canonical functor $C \to \mathrm{Pro}(C)$ is an equivalence of categories.) Using b), deduce that *every* morphism of topoi $\widehat{C}' \to \widehat{C}$ is essential, and that $\mathbf{Hom}(C, C') \to \mathrm{Hom}(\widehat{C}, \widehat{C}')$ is an equivalence if and only if C is empty or C' is Karoubian.

i) Let X be a topological space, and let $f : \mathrm{Top}(X) \to P$ be the morphism of topoi induced by the continuous map from X to the one-point space. Show that f is essential if and only if the space X is *locally connected*, i.e. if it satisfies the following condition: for every $x \in X$ and every open neighborhood U of x, the connected component of x in U is a neighborhood of x; or equivalently: for every open subset U of X, the connected components of U are open. (cf. 8.7 l) for a generalization).

Exercise 7.7. (*Unusual points of a classifying topos*)[11].

a) Let G be a monoid. The monoid G contains an element g such that $g \neq e$ and $g^2 = g$ if and only if the classifying topos B_G admits an essential point that is not isomorphic to the trivial point (corresponding to the fiber functor "forget the G-action"). (Use 7.6 e)).

[11] cf. also 7.2.6 d).

b) Let G be the additive monoid of the integers ≥ 0. Show that every essential point of the classifying topos B_G is isomorphic to the trivial point. Construct a point of B_G that is not isomorphic to the trivial point. Show that the category **Point**(B_G) is not equivalent to a category $\in \mathscr{U}$.

EXERCISE 7.8. (*Topology on* Point(E)*, and topoi associated with ordered sets*).

a) Let E be a topos. Denote by Point(E) the set of isomorphism classes of points of E. For every open U of E, let U' be the set of classes of points p of E such 418
that $U_p \neq \phi$. Show that $U \mapsto U'$ is an order-preserving map from the ordered set Ouv(E) of opens of E to the set of subsets of $X = \text{Point}(E)$, and that this map commutes with finite Inf and arbitrary Sup. Deduce that the subsets of X of the form U' define a topology on X (called the *canonical topology on the set of classes of points of the topos* E). Show that if E has enough points, the map $U \mapsto U'$ is an isomorphism from the ordered set Ouv(E) onto the ordered set Ouv(X).

b) Let $\mathscr{O} \in \mathscr{U}$ be an ordered set admitting finite Inf and arbitrary Sup, and hence defining a category cat$(\mathscr{O})$ having finite products and arbitrary coproducts. We say that a family of morphisms $U_i \to U$ with common target U is covering if U is the Sup of the U_i. Suppose that coproducts in cat$(\mathscr{O})$ are universal, i.e. that in $\mathscr{O}$ the arbitrary Sup commute with Inf. This defines a topology on cat$(\mathscr{O})$, making it a site, and hence a topos cat$(\mathscr{O})^\sim$, also denoted simply by $\mathscr{O}^\sim$. Show that $\mathscr{O}$ is canonically isomorphic to Ouv$(\mathscr{O}^\sim)$.

Show that the category **Homtop**$(E, \mathscr{O}^\sim)$ is equivalent to the category associated with the ordered set of morphisms of ordered sets $\mathscr{O} \to$ Ouv(E) that commute with finite Inf and arbitrary Sup. (Use the criterion 4.9.4.) If $\mathscr{O}'$ is an ordered set satisfying the same conditions as $\mathscr{O}$, deduce that **Homtop**$(\mathscr{O}^\sim, \mathscr{O}'^\sim)$ is equivalent to the category associated with the ordered set of all morphisms of ordered sets $\mathscr{O} \to \mathscr{O}'$ that commute with finite Inf and arbitrary Sup.

c) Show that the topos E is equivalent to a topos of the form $\mathscr{O}^\sim$, with $\mathscr{O}$ as in 419
b), if and only if the family of opens of E is generating in E. It is equivalent to some Top(X) with $X \in \mathscr{U}$ if and only if it satisfies the preceding condition and has enough points. Assuming the first condition, express the second in terms of the structure of the ordered set $\mathscr{O} =$ Ouv(E), using the last assertion of b).

d) Define a canonical morphism of topoi

$$E \longrightarrow \text{Ouv}(E)^\sim,$$

which is universal (up to equivalence) for morphisms from E to topoi generated by their opens (cf. c)).

e) Let X' be a *small* subset of $X = \text{Point}(E)$, equipped with the topology induced by that of X. Define a canonical morphism of topoi

$$\text{Ouv}(E)^\sim \longrightarrow \text{Top}(X'),$$

which is an equivalence when the family X' of points of E is conservative. Then deduce a canonical morphism of topoi

$$E \longrightarrow \text{Top}(X').$$

Give a universal characterization (up to equivalence) of this morphism of topoi with respect to morphisms from the topos E to topoi of the form Top(Y). (Note in this connection that, up to equivalence, Top(X') does not depend on the

chosen small conservative family of fiber functors, or equivalently (4.2), that X'_{sob} is independent, up to homeomorphism, of the choice of such a family.)

420 f) Suppose that the category **Point**(E) is not equivalent to a small category (cf. 7.3), and that E admits a small conservative family of fiber functors. Show that, if such a family is taken sufficiently large, the subset X' of **Point**(E) that it defines is not a sober topological space for the induced topology.

g) For the relations with Exercise 7.6, cf. 8.

8. Localization. Opens of a topos

8.1. Let C be a category, and let $T(X)$ be a relation involving an object X of C. The relation $T(X)$ is said to be *stable under base change* (in X) if, for every morphism $X' \to X$ of C, the relation $T(X)$ implies $T(X')$. This amounts to saying that the full subcategory R of C formed by the objects X for which $T(X)$ holds is a *sieve* (I 4.1).

Remark 8.1.1. Suppose that C is a $\mathscr{U}$-category (I 1.1), so that C may be identified with a full subcategory of $\widehat{C}$ (I 1.4). It is sometimes convenient to extend the definition of the relation T on $\mathrm{ob}\,C$ to a relation $\widehat{T}$ on an object F of $\widehat{C}$, by defining $\widehat{T}(F)$ to mean that $T(X)$ holds for every arrow $X \to F$ in $\widehat{C}$ with $X \in \mathrm{ob}\,C$. The relation $\widehat{T}(F)$ is clearly still stable under base change in F. Again denoting by R the subobject of the terminal object e of $\widehat{C}$ defined by the sieve R of C (I 4.2.1), the relation $\widehat{T}(F)$ may also be expressed as $\widehat{\mathrm{Hom}}_C(F, R) \neq \varnothing$, i.e. it means that the unique morphism $F \to e$ factors through the subobject R of e.

8.2. Now suppose that C is a *site*. A relation $T(X)$ in an argument $X \in \mathrm{ob}\,C$ is

421 said to be *stable under descent* if, for every covering family $(X_i : X_i \to X)_{i\in I}$ of an object X of C, the relation "$T(X_i)$ for every $i \in I$" implies $T(X)$. We say that T is a *relation of local nature* if it is stable under base change (8.1) and stable under descent. When C is a $\mathscr{U}$-category and T is already assumed stable under base change, i.e. is defined by a sieve R of C, which we identify with a subpresheaf of the terminal presheaf on C, one sees immediately that T is of local nature if and only if R is a *sheaf*. Thus *relations of local nature in an argument $X \in \mathrm{ob}\,C$ correspond exactly to the subsheaves of the terminal sheaf of C, i.e. to the subobjects of the terminal object of the topos $C^\sim$.* They therefore depend essentially only on the topos $C^\sim$ defined by the site C (like all important notions associated with a site!). In terms of the subsheaf R of the terminal sheaf, the relation $T(X)$ is expressed as $\mathrm{Hom}(\epsilon(X), R) \neq \varnothing$. It extends canonically to the relation $T^\sim(F) : \mathrm{Hom}(F, R) \neq \varnothing$ on the topos $C^\sim$.

Remark 8.2.1. With the notation introduced in 8.1.1, for a presheaf F on C, since $\mathrm{Hom}(F, R) \simeq \mathrm{Hom}(\underline{a}(F), R)$, the relation $\widehat{T}(F)$ is equivalent to the relation $\widehat{T}(\underline{a}(F))$.

Definition 8.3. *An open of a $\mathscr{U}$-topos E is a subobject of the terminal object e_E of E; if X is an object of E, an open of X sometimes means an open of the induced topos $E_{/X}$, i.e. a subobject of X.*

An open of a $\mathscr{U}$-site C is an open of the associated $\mathscr{U}$-topos $C^\sim$.

Since the terminal objects of E are canonically isomorphic, the ambiguity introduced

422 in 8.3 by the choice of e_E is harmless. One may also, in a more intrinsic but practically less convenient way, define the opens of E as the full subcategories R of E for which the relation $F \in \mathrm{ob}\,R$, for an object F of E, is of local nature. Likewise, the opens of a $\mathscr{U}$-site C correspond bijectively to the full subcategories of C for which the relation $F \in \mathrm{ob}\,R$, for an object of C, is of local nature; this notion is therefore essentially independent of

the chosen universe $\mathscr{U}$. Finally, by what was said in 8.2, a relation of local nature on a topos (or on a site) is essentially the same thing as an open of that topos (or site).

8.4. Examples of opens.

8.4.1. Let X be a topological space, and interpret $\mathrm{Top}(X)$ (2.1) as the category of étale spaces over X. Since the monomorphisms in $\mathrm{Top}(X)$ are plainly the injective maps, it follows that the opens of the topos $\mathrm{Top}(X)$ identify with the opens of the space X. More generally, the opens of a topos $\mathrm{Top}(X, G)$ (2.4) identify with the opens of X which are *invariant* under the action of G.

8.4.2. The opens of a $\mathscr{U}$-topos E form a $\mathscr{U}$-small set (I 7.4), ordered by inclusion, admitting arbitrary suprema and finite infima. Its least element is the initial object (or "empty object") ϕ_E, and its greatest element is the terminal object e_E of E. These two elements are identical only if E is an "empty topos" (2.2).

8.4.3. Let G be a monoid object in E, giving a topos B_G (2.6), the category of objects of E on which G acts on the left. The terminal object e_G of B_G is the terminal object 423
e_E of E with the trivial action of G. It follows that the ordered set of opens of B_G is canonically isomorphic to that of the opens of E. In particular, if E is the punctual topos, so that G is an ordinary monoid, the set of opens of B_G consists of exactly two elements, namely ϕ_{B_G} and e_G.

8.4.4. If E is a topos of the form $\widehat{C}$, where C is a $\mathscr{U}$-category, then (8.1) the ordered set of opens of E is isomorphic to the ordered set of sieves of C.

8.4.5. Let X be a sober nondiscrete topological space. Show that $\mathrm{TOP}(X)$ (2.5) has opens which do not arise from opens of X, i.e. from opens of $\mathrm{Top}(X)$, by inverse image under the canonical morphism (4.10.1) $\mathrm{TOP}(X) \to \mathrm{Top}(X)$.

8.5. Examples of relations of local nature. For every object X of the topos E, denote by $F \to F_X$ the *localization functor* $j_X^* : E \to E_{/X}$, $Y \mapsto Y \times X$ (5.2).

8.5.1. Let $u : F \to G$ be a morphism of E. The following relation in the argument $X \in \mathrm{ob}\, E$:

$u_X : F_X \to G_X$ is a monomorphism (resp. an epimorphism, resp. an isomorphism)

is a relation of local nature. In particular, if G is a group object of E, the relation "G_X is the trivial group over X" is of local nature; the corresponding open of E is called the *cosupport of the group* G.

8.5.2. Let $u, v : F \rightrightarrows G$ be two morphisms of E. The relation in the argument 424
$X \in \mathrm{ob}\, E$

$$u_X = v_X : F_X \longrightarrow G_X$$

is a relation of local nature. In particular, if G is a group object of E, and u is a *section* of G (4.3.6), the relation in X

$$\text{the section } u_X \text{ of the group } G_X \text{ of } E_{/X} \text{ is zero}$$

is of local nature; the corresponding open of E is called the *cosupport of the section* u *of the group* G.

Remark 8.5.3. When $E = \mathrm{Top}(X)$, the cosupport of a sheaf of groups G, resp. of a section of such a sheaf G, is simply the open of X complementary to the *support of* G, resp. to the *support of* u, in the classical terminology [TF].

8.5.4. Let $P(f)$ be a relation in the argument $f \in \mathrm{fl}\, C$, where C is a site. When fiber products are representable in C, the relation $P(f)$ is said to be *stable under base change (resp. stable under descent, resp. of local nature) on the target* (or *on the base*, or

"*downstairs*") if, for every morphism $f : X \to Y$ of C, the following relation in the argument $Y' \in \mathrm{ob}\, C_{/Y}$:

"the morphism $f' : X' = X \times_Y Y' \to Y'$ obtained from f by base change along the structural morphism $Y' \to Y$ satisfies $P(f')$"

is stable under base change, resp. stable under descent, resp. is of local nature. When the topology of E is defined by means of a pretopology, the relation $P(f)$ is said to be *of local nature on the source* (or *"upstairs"*) if, for every morphism $f : X \to Y$ of C
425 and every family $(g_i : X_i \to X)_{i\in I}$ in $\mathrm{Cov}(X)$, the relation $P(f)$ is equivalent to the relation "$P(fg_i)$ for every $i \in I$." When $\mathrm{Cov}(X)$ consists of all covering families of C, this condition also means that the following relation in the object X' of the induced site $C_{/X}$

$$P(fg), \text{ where } g : X' \to X \text{ is the structural morphism}$$

is of local nature. Note that if the relation $P(f)$ is of local nature upstairs, it is a fortiori of local nature downstairs.

8.5.5. Take, for example, $C = (\mathrm{Sch})$, the category of schemes in $\mathcal{U}$, with the faithfully flat quasi-compact topology (SGA 3 IV 6.3) or a coarser topology. Each of the following properties of a morphism is then of local nature downstairs:

surjective, radicial, universally open, universally closed, proper, quasi-compact, quasi-compact and dominant, universal homeomorphism, separated, quasi-separated, locally of finite type, locally of finite presentation, of finite type, of finite presentation, an isomorphism, a monomorphism, an open immersion, a closed immersion, a quasi-compact immersion, affine, quasi-affine, finite, quasi-finite, integral, flat, faithfully flat, neat, smooth, étale

(EGA IV 2.6.4, 2.7.1 and EGA IV 17.7.1). On the other hand, endow (Sch) with the pretopology for which, for every scheme $X \in \mathcal{U}$, $\mathrm{Cov}(X)$ consists of the families of morphisms $(f_i : X_i \to X)_{i\in I}$ which are surjective and for which the f_i are flat and locally of finite presentation. Each of the following properties is then *of local nature upstairs*:

locally of finite type, locally of finite presentation, of finite type, flat, neat, smooth, étale.

(EGA IV 17.7.5, 17.7.7).

426 Exercise 8.6. Let $f : E \to F$ be a morphism of $\mathcal{U}$-topoi. Consider the following relation in the argument $X \in \mathrm{ob}\, F$: the induced morphism $E_{/f^*(X)} \to F_{/X}$ is an equivalence of topoi. Prove that this relation is of local nature.

Exercise 8.7. (*Partitions of a topos, sum of topoi*). Let E be a $\mathcal{U}$-topos. A family $(e_i)_{i\in I}$ of opens of E, i.e. of subobjects of the terminal object e of E, is called a *partition of* e (or of the topos E) if the canonical morphism $\coprod_{i\in I} e_i \to e$ is an isomorphism, i.e. (II 4.6.2) if e is the supremum of the e_i and $i \neq j$ implies $e_i \cap e_j = \emptyset_E$.

a) Show that a family $(e_i)_{i\in I}$ of objects of E is a partition of E if and only if the functor

(8.7.1) $$E \longrightarrow \prod_i E_{/e_i}$$

defined by the localization functors $E \to E_{/e_i}$ is an equivalence of categories.

b) Conversely, let $(E_i)_{i\in I}$ be a family of $\mathcal{U}$-topoi, with $I \in \mathcal{U}$. Prove that $E = \Pi_{i\in I} E_i$ is a $\mathcal{U}$-topos (called the *sum topos* of the family $(E_i)_{i\in I}$). Define a partition $(e_i)_{i\in I}$ of E and equivalences of categories $E_{/e_i} \approx E_i$ in such a way that the projection functors $E \to E_i$ identify with the localization functors $E \to E_{/e_i}$.

c) Let E be the sum topos of the family $(E_i)_{i\in I}$. Show that, for every $i \in I$, the projection functor $E \to E_i$ is of the form u_i^*, where

$$u_i : E_i \longrightarrow E \tag{8.7.2}$$

is a morphism of topoi. Prove that, for every $\mathscr{U}$-topos F, the functor $v \mapsto (v \circ u_i)_{i\in I}$ 427

$$\mathbf{Homtop}(E, F) \longrightarrow \prod_i \mathbf{Homtop}(E_i, F) \tag{8.7.3}$$

is an equivalence of categories.

d) Let $(e_i)_{i\in I}$ be a partition of the topos E. For every morphism of topoi $g : F \to E$, the family $(f_i)_{i\in I}$, where $f_i = g^*(e_i)$, is a partition of F, and the induced morphisms $g_i : E_{/e_i} \to F_{/f_i}$ make it possible to reconstruct g (up to unique isomorphism), the functor g^* identifying with the Cartesian product of the functors g_i^* (taking into account the equivalences of the type in 8.7 1)). Deduce a complete description of the category $\mathbf{Homtop}(F, E)$ in terms of categories of the form $\mathbf{Homtop}(F', E_i)$, where F' is a topos of the form $F_{/f}$ (with f a direct summand of the terminal object of F) and $E_i = E_{/e_i}$.

e) In particular, if F is connected and nonempty (cf. 4.3.5), i.e. if, for every partition $(f_i)_{i\in I}$ of F, there exists a unique $i \in I$ such that $f_i \neq \phi_F$, prove that the canonical functor

$$\coprod_{i\in I} \mathbf{Homtop}(F, E_i) \longrightarrow \mathbf{Homtop}(F, E) \tag{8.7.4}$$

induced by the morphisms of topoi (8.7.2) is an equivalence of categories. More particularly, the family of morphisms of topoi (8.7.2) induces an equivalence of categories

$$\coprod_{i\in I} \mathbf{Point}(E_i) \xrightarrow{\approx} \mathbf{Point}(E).$$

f) A subset I of the set of opens of E is called a *reduced partition* of E if the identity family $(e_i)_{i\in I}$ indexed by I is a partition and the e_i are different from ϕ_E. Show that the refinement relation (I 4.3.2) between reduced partitions makes the set P of reduced partitions into a $\mathscr{U}$-small ordered set in which the supremum of any two elements exists. This gives a strict projective system $(I_p)_{p\in P}$, regarded as a pro-set and denoted by $\pi_\circ(E)$. It is called the *pro-$\pi_\circ$ of the topos E*. 428

g) Prove that $\pi_\circ(E)$ pro-represents the functor

$$T \longmapsto f_* f^*(T) = \Gamma(E, T_E) : (\mathscr{U}\text{-Ens}) \longrightarrow (\mathscr{U}\text{-Ens})$$

associated with the canonical morphism $f : E \to P$ from E to the punctual topos P (4.3). In particular, this functor is representable if and only if $\pi_\circ(E)$ is essentially constant, i.e. isomorphic (as a pro-set) to an "ordinary" set, again denoted $\pi_\circ(E)$. The pro-set $\pi_\circ(E)$ consists of one point if and only if E is "connected and nonempty." It is empty if and only if E is the "empty topos" (2.2).

h) Suppose that E is quasi-compact, i.e. every covering of its terminal object e admits a finite subcovering. Show that $\pi_\circ(E)$ is then a profinite set and hence may be identified (using the familiar equivalence between pro-objects of the category of finite sets and compact totally disconnected spaces) with a compact totally disconnected space, also denoted $\pi_\circ(E)$.

i) Let X be a topological space, and let $\pi_\circ(X)$ be the set of connected components of X, regarded as a constant pro-set. Define a canonical morphism of pro-sets 429

$$\pi_\circ(X) \longrightarrow \pi_\circ(\mathrm{Top}(X)), \tag{8.7.5}$$

or equivalently a canonical map

$$\pi_\circ(X) \longrightarrow \varprojlim \pi_\circ(\mathrm{Top}(X)). \tag{8.7.6}$$

Show that the first morphism is an isomorphism if and only if X is homeomorphic to a sum space of connected spaces (which is the case, in particular, when X is locally connected).

j) Suppose that X is a quasi-compact scheme. Prove that (8.7.6) is bijective.

k) Establish the "functorial character" in X of the morphisms (8.7.5) and (8.7.6), first making precise the meaning of this expression.

l) (Compare 7.6 i)). Show that the following two conditions on the $\mathscr{U}$-topos E are equivalent: (i) The canonical morphism from E to the punctual topos is "essential" (Exercise 7.6 a)), i.e. the functor $I \mapsto I_E$ from ($\mathscr{U}$-Ens) to E commutes with products indexed by sets in $\mathscr{U}$. (ii) For every object X of E, the induced topos $E_{/X}$ has a $\pi_\circ(E_{/X})$ which is an ordinary set, i.e. X is isomorphic to the sum of a family of connected objects of E. In this case E is called a *locally connected topos* (compare 2.7.5).

Exercise 8.8. (*Dominant morphisms of topoi*).

a) Let $f : E' \to E$ be a morphism of topoi. Show that the following conditions are equivalent:

(i) $f_*(\phi_{E'}) = \phi_E$ (where ϕ_E and $\phi_{E'}$ denote the initial objects of E and E', respectively).

430 (ii) For every object X of E which is "nonempty," i.e. not isomorphic to ϕ_E, the object $f^*(X)$ is nonempty in E'.

(ii′) As in (ii), with X a subobject of the terminal object e_E of E, i.e. with X an open of the topos E.

The morphism of topoi f is then called *dominant*.

b) Let $f : X' \to X$ be a continuous map of topological spaces, giving a morphism of topoi $\mathrm{Top}(f) : \mathrm{Top}(X') \to \mathrm{Top}(X)$ (4.1). Show that $\mathrm{Top}(f)$ is dominant if and only if f is dominant, i.e. if $f(X')$ is dense in X.

c) Show that if $f : E' \to E$ is a "conservative" morphism of topoi (6.4.0), i.e. if f^* is faithful, then f is dominant. Show that the converse need not hold, even for a localization morphism $E_{/U} \to E$, where U is an open of the topos E. (Show that in this case f is conservative only if U is the terminal object of E (and hence only if f is an equivalence of topoi), and use the example in b).)

d) Let $f : X' \to X$ be an arrow in a topos E. We call f a *dominant morphism* in the topos E if the corresponding morphism of induced topoi $E_{/X'} \to E_{/X}$ (5.5.2) is dominant. Show that this property depends only on the image $f(X')$ of X' in X, as an element of the ordered set $\mathrm{ouv}(X)$ of subobjects of X. Show that the morphism $E_{/X'} \to E_{/X}$ is conservative if and only if f is covering.

9. Subtopoi and gluing of topoi

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9.1. Subtopoi and embeddings of topoi.

DEFINITION 9.1.1. a) *A morphism of topoi $f : F \to E$ is called an embedding if the functor $f_* : F \to E$ is fully faithful (i.e. if the adjunction morphism $f^* f_* \to \mathrm{id}_F$ is an isomorphism).*

b) *A subtopos of a topos E is a strictly full subcategory E' of E which is a topos and for which the inclusion functor $\alpha : E' \to E$ is of the form i_*, where $i : E' \to E$ is a morphism of topoi (i.e. (3.1.1) α admits a left adjoint i^* which is left exact). The morphism i (which is determined up to unique isomorphism) is then called the inclusion morphism of the subtopos E' in E.*

REMARK 9.1.2. a) It is clear that the inclusion morphism of a subtopos is an embedding, and that a subcategory E' of a topos E is a subtopos if and only if it is the essential image of a direct-image functor f_* associated with a morphism of topoi $g : F \to E$ which is an embedding.

b) It follows immediately from the definitions that a morphism of topoi $f : F \to E$ is an embedding if and only if it is isomorphic to a composite morphism

$$F \xrightarrow{g} E' \xrightarrow{i} E,$$

where g is an equivalence of topoi and i is the inclusion morphism of a subtopos E' of E. The latter is uniquely determined by the preceding conditions as the essential image of f_*, and the preceding factorization is likewise unique up to unique isomorphism. In practice, of course, one should use g to identify F with the subtopos E' of E (compare 3.4.1).

c) An equivalence of topoi (3.4) $u : E \xrightarrow{\approx} E_1$ defines in the evident way a bijection between the set of subtopoi of E and the set of subtopoi of E_1. Indeed, since u_* is an equivalence of categories, it establishes a bijection between the set of 432
strictly full subcategories of E and the set of strictly full subcategories of E_1 (to $E' \subset E$ corresponds the essential image E'_1 of $u_* | E'$), and one sees immediately that E' is a subtopos of E if and only if E'_1 is a subtopos of E_1. Thus, in studying subtopoi, one may reduce to the case where E is of the form $\widetilde{C}$, with C a small site. In this case, II 5.5 shows that a strictly full subcategory E' of E is a subtopos if and only if its inclusion functor $\alpha : E' \to E$ admits a left adjoint which is left exact (which already implies that E' is a topos), and that these subtopoi are in one-to-one correspondence with the topologies T' on C finer than the given topology T of C: to such a T' one associates the strictly full subcategory $(C, T')^\sim$ of $C^\sim$ formed by the sheaves for the topology T'. This also shows, for *every* $\mathcal{U}$-topos E, that a strictly full subcategory E' of E is a subtopos if and only if its inclusion functor $\alpha : E' \to E$ admits a left adjoint which is left exact; *the ordered set $\mathfrak{E}(E)$ of subtopoi of E is $\mathcal{U}$-small, and every subset of $\mathfrak{E}(E)$ admits a supremum and an infimum.*

d) If E' and E'' are two subtopoi of a topos E, their intersection is a subtopos of E. Indeed, since this intersection is strictly full, one reduces to the case where E is of the form $C^\sim$. With the notation of c), E' and E'' then correspond to two topologies T' and T'' on C finer than T, and it suffices to see that *$E' \cap E''$ is the category of sheaves for the topology T''' which is the supremum of T' and T''*. We indicate the proof of this fact (which should have appeared as a corollary after II 4.4 or II 5.5!). Let E''' be the category of sheaves for T''', plainly contained in E' and E'', and hence in $E' \cap E''$. For every full subcategory D of E, let $t(D)$ be the finest topology among those for which the objects of D are sheaves (II 2.2). The inclusions $E''' \subset E' \cap E'' \subset E', E''$ plainly imply $t(E'), t(E'') \leq$ 433

$t(E' \cap E'') \leq t(E''')$. On the other hand, (II 4.4.4) $t(E') = T'$, $t(E'') = T''$, and $t(E''') = T'''$, so the preceding inequalities and the definition of T''' as the supremum imply $T''' = t(E' \cap E'')$, and hence $E' \cap E'' \subset E'''$, C.Q.F.D.
More generally, this argument shows that if T' is the supremum of an arbitrary family of topologies (T_i), the category E' of sheaves for T' is the intersection of the categories E_i of sheaves for the T_i. In particular, one obtains:

PROPOSITION 9.1.3. Let E be a topos. For every family of subtopoi $(E_i)_{i\in I}$ of E, the intersection $\bigcap_{i\in I} E_i$ is a subtopos of E.

PROPOSITION 9.1.4. *Let $i : E' \to E$ be an embedding of topoi, and let F be a topos. The functor*

$$f \longmapsto i \circ f : \mathbf{Homtop}(F, E') \longrightarrow \mathbf{Homtop}(F, E) \tag{9.1.4.1}$$

is fully faithful, and its essential image consists of the morphisms of topoi $g : F \to E$ for which the essential image of g_* is contained in that of i_*.

Indeed, consider the evident commutative diagram

$$\begin{array}{ccc} \mathbf{Homtop}(F, E') & \longrightarrow & \mathbf{Homtop}(F, E) \\ \downarrow & & \downarrow \\ \mathbf{Hom}(F, E') & \longrightarrow & \mathbf{Hom}(F, E), \end{array}$$

where the vertical arrows are the functors $h \mapsto h_*$, which are fully faithful by the definition of **Homtop** (3.2). The second horizontal arrow $G \mapsto i_*G$ is also fully faithful, since i_* is, and therefore so is the first horizontal arrow. It remains to prove the characterization of the essential image of (9.1.4.1). The necessity of the condition is clear from the formula $(if)_* = i_*f_*$. Conversely, suppose that the essential image of g_* is contained in that of i_*, i.e. that one may write $g_* = i_*f_*$, with $f_* : F \to E'$ a functor. We must show that g belongs to the essential image of (9.1.4.1). This amounts to saying that f_* admits a left
434 adjoint which is left exact. Now it is clear that f_* admits g^*i_* as a left adjoint, and this functor is left exact, being the composite of the left exact functors g^* and i_*, C.Q.F.D.

For a converse to 9.1.4, cf. Exercise 9.1.6.

Taking F to be the punctual topos (2.2), one obtains in particular from 9.1.4:

COROLLARY 9.1.5. *If $f : E' \to E$ is an embedding morphism of topoi, the corresponding functor*

$$\mathrm{Point}(f) : \mathbf{Point}(E') \longrightarrow \mathbf{Point}(E) \tag{9.1.5.1}$$

is fully faithful.

EXERCISE 9.1.6. (*Embeddings of topoi and 2-monomorphisms*).
Let $i : E' \to E$ be a morphism of topoi.

a) If F and G are two topoi *over* E', define (with the notation of 5.14 a)) a canonical functor

$$f \longmapsto f * i : \mathbf{Homtop}_{E'}(F, G) \longrightarrow \mathbf{Homtop}_{E}(F, G). \tag{9.1.6.1}$$

b) Suppose that i, as a 1-arrow of the 2-category of $\mathscr{V}$-$\mathscr{U}$-topoi (3.3.2), is a 2-monomorphism, i.e. that for every $\mathscr{U}$-topos F which is an element of $\mathscr{V}$, the functor (9.1.4.1) is fully faithful. Show that, for every pair of $\mathscr{U}$-topoi F, G (not necessarily in $\mathscr{V}$), the functor (9.1.6.1) is fully faithful.

c) Show that i is an embedding if and only if i is a 2-monomorphism. (For sufficiency, apply b) to induced topoi $E'_{/X'}$ and $E'_{/Y'}$, and use 5.14 c)).

d) Let $g : F \to E$ be a morphism of topoi, and suppose that the 2-fiber product (5.11) $F' = F \times^2_E E'$ exists (a condition which is always satisfied, as P. DELIGNE has shown[12]). Let $j : F' \to F$ be the projection morphism. Prove that if i is an embedding, then so is j. (Use c), and note that the stability of the notion of a 2-monomorphism under 2-base change is formal.) When E' is a subtopos of E and $i : E' \to E$ is the canonical inclusion morphism, the essential image of F' under j_* (which is a subtopos of F independent of the indeterminacy in the construction of a 2-fiber product) is called the *subtopos of F which is the inverse image under $g : F \to E$ of the subtopos E' of E*. 435

e) With the notation at the end of d), choose as 2-fiber product the subtopos of F which is the inverse image of the subtopos E' of E. Let X be an object of F. Show that, for X to belong to F', it is necessary that it satisfy the following condition: for every object U of F, writing $g_U : F_{/U} \to E$ for the morphism of topoi induced by g, one has $g_{U*}(X|U) \in \mathrm{Ob}\, E'$, where $X|U = (X \times U \xrightarrow{\mathrm{pr}_2} U)$ is the "restriction of X to U." (NB: Note that if $i_U : F_{/U} \to F$ is the canonical inclusion, then $g_U = g \circ i_U$, whence $g_{U*}(X|U) = g_*(i_{U*}(X|U)) = g_*(\mathbf{Hom}(U, X))$, where $\mathbf{Hom}(U, X)$ is the object defined in 10.1 below.)
Problem. Conversely, is the preceding condition on X sufficient for X to belong to F'? This amounts to asking whether the strictly full subcategory of F formed by these objects X is a *subtopos* of F.

f) Let X be an object of E, let $X' = i^*(X)$, and let $i_{/X} : E'_{/X'} \to E_{/X}$ be the morphism of topoi induced by i ((5.10.1)). Show that if i is an embedding, then so is $i_{/X}$. (Use d) and 5.11.) Conversely, if $(X_\alpha)_\alpha$ is a family of objects covering the terminal object, and if $i_{/X_\alpha}$ is an embedding for every α, then so is i.

Exercise 9.1.7. (*Subtopoi and multiplicative systems of arrows*).

a) Let $i : E' \to E$ be a morphism of topoi, let S be the set of arrows u of E such that $i^*(u)$ is an isomorphism, and consider the canonical functor induced by i^* (cf. [2, Chap. I] or the proof of VI 6.2):

$$\rho : E[S^{-1}] \longrightarrow E'. \tag{9.1.7.1}$$

Show that i is an embedding if and only if the preceding functor ρ is an equivalence of categories, and that in this case S admits both a calculus of left fractions and a calculus of right fractions. (Use [2, Chap. I, 1.3].) In this case, the essential image of i_* can be recovered from S as consisting of those $X \in \mathrm{Ob}(E)$ such that for every $u : Y \to Y'$ in S, the map $\mathrm{Hom}(u, X) : \mathrm{Hom}(Y', X) \to \mathrm{Hom}(Y, X)$ is bijective. 436

b) We henceforth assume that i is an embedding. Show that, for every topos F, the functor

$$f^* \longmapsto f^* i^* : \mathbf{Hom}(E', F) \longrightarrow \mathbf{Hom}(E, F)$$

induces an equivalence between the category of the f^* arising from morphisms of topoi $f : F \to E'$ (i.e. which are left exact and admit a right adjoint) and the category of functors $g^* : E \to F$ arising from morphisms of topoi $g : F \to E$ and such that g^* takes arrows of S to isomorphisms.

[12] N.D.E.: See also [16], Proposition 4.47.

c) Let S be a subset of $Fl\ E$, where E is a $\mathscr{U}$-topos. Show that S is associated with a subtopos E' of E by the procedure of a) if and only if S satisfies the following conditions:

ST 1) S contains the invertible arrows and is stable under composition and base change.

ST 2) If a composite vu belongs to S, then $u \in S$ if and only if $v \in S$.

ST 3) If $u : X \to Y$ is such that there exists a covering family $Y_i \to Y$, $i \in I$, for which $u_i : X_i = X \times_Y Y_i \to Y_i$ belongs to S for every $i \in I$, then $u \in S$.

Show that this gives a one-to-one correspondence between the set of subtopoi E' of E and the set of subsets S of $Fl\ E$ satisfying conditions ST 1), ST 2), and ST 3). (Given S, associate with it a $\mathscr{U}$-topology T', finer than the canonical topology T of E, whose covering families $(X_i \to X)_{i\in I}$ are those for which the inclusion $X' \hookrightarrow X$ of the image of the X_i is $\in S$. Show that the category of sheaves for T' is equivalent to a subtopos E' of E, and use II 4.4 to prove that it recovers S.)

d) Show that S is determined by the subset S_0 of S consisting of the *monomorphisms*. Show that the map $S \mapsto S_0$ establishes a bijection between the set of subsets S of $Fl\ E$ satisfying conditions ST 1) through ST 3) of c) and the
437 set of subsets of $Fl\ E$ consisting of monomorphisms and satisfying the same conditions ST 1) through ST 3).

e) Let $(E'_i)_{i\in I}$ be a family of subtopoi of E, and let $(S_i)_{i\in I}$ be the corresponding family of subsets of $Fl\ E$. Show that $\bigcap_{i\in I} S_i$ satisfies conditions ST 1) through ST 3) of c), and hence corresponds to a subtopos E' of E, and that the latter is the *subtopos generated by the* E'_i, i.e. the *supremum* (cf. 9.1.2 c)) of the subtopoi E'_i.

f) Let A be a subset of $Fl\ E$. Show that there is a smallest subset S of $Fl\ E$ among those containing A and satisfying conditions ST 1) through ST 3), and that S corresponds to the largest subtopos E' of E such that the corresponding multiplicative system of $Fl\ E$ contains S. Show that S is equal to the union of the subsets A_n ($n \geq 0$) of $Fl\ E$, constructed inductively as follows: $A_0 = A$, and A_{n+1} consists of the arrows $u : X \to Y$ which are of one of the following four types (i) through (iv):

(i) u is invertible, or is the composite of two arrows of A_n, or is obtained from an arrow of A_n by base change.

(ii) u is one of the factors of a composite arrow whose other factor and whose composite both belong to A_n.

(iii) u is the coproduct of a small family of arrows $\in A_n$.

(iv) There exists an epimorphism $Y' \to Y$ such that $u' = X' = X \times_Y Y' \to Y'$, obtained from u by base change, belongs to A_n.

g) With the notation of e), let A be the union of the S_i, let S be the subset of $Fl\ E$ obtained from A by the procedure of f), and let E' be the corresponding subtopos of E. Show that E' is the intersection (= infimum) of the subtopoi E'_i.

Exercise 9.1.7.2. (*Canonical factorization of a morphism of topoi*).

a) Let $f : F \longrightarrow E$ be a morphism of topoi. Show that, up to isomorphism, one can find a factorization of f as

$$F \xrightarrow{f'} E' \xrightarrow{i} E, \tag{9.1.7.3}$$

where f' is *conservative* (i.e. (6.4.0) f'^* is conservative, or equivalently faithful) and i is an embedding, and that this factorization is unique up to equivalence 438
(cf. 9.1.4). To do this, introduce the set $S_f \subset Fl\ E$ of arrows u of E such that $f^*(u)$ is an isomorphism, prove that S_f satisfies the conditions of 9.1.7 c), take for E' the corresponding subtopos of E, and define f'^* by (9.1.7.1), which corresponds to a morphism of topoi f' by 9.1.7 b).

b) The subtopos of E defined by the embedding i is called the subtopos of E which is the *image of the morphism of topoi* f. Show that it is the smallest of the subtopoi E_1 of E such that f factors, up to isomorphism, through E_1, i.e. (9.1.4) the smallest of the subtopoi of E containing the image of f_* (or equivalently the essential image of f_*). For f to be an embedding, it is necessary and sufficient that f induce an equivalence f' of F with its image; for f to be conservative, it is necessary and sufficient that its image be all of E.

c) Suppose that $f = \mathrm{Top}(f_0)$ is associated with a continuous map of sober topological spaces $f_0 : Y \to X$ (4.1). Consider the canonical factorization of f_0 as

$$Y \xrightarrow{f'_0} X' = f_0(Y) \xrightarrow{i_0} X,$$

where i_0 is the inclusion. Show that the canonical factorization of f then identifies with the corresponding factorization

$$\mathrm{Top}(Y) \xrightarrow{\mathrm{Top}(f'_0)} \mathrm{Top}(X') \xhookrightarrow{\mathrm{Top}(i_0)} \mathrm{Top}(X).$$

Let Z be the smallest sober subspace of X containing $X' = f_0(Y)$, i.e. the set of points x of X such that $\{\overline{x}\} \cap X'$ is dense in $\{\overline{x}\}$. (Note that $Z = X'$ if the points of X are closed, or if X' is a *constructible* subset of the locally Noetherian space X.) Then f is conservative, i.e. the subtopos of $E = \mathrm{Top}(X)$ which is the image of $F = \mathrm{Top}(Y)$ under $f = \mathrm{Top}(f_0)$ is equal to E itself, if and only if $Z = X$, i.e. if and only if $f_0(Y)$ is *very dense* (EGA IV 10.13) in X. This makes precise the extent to which the notion of a conservative morphism of topoi may legitimately be regarded as the natural generalization of the notion of a *surjective* continuous map of topological spaces.

d) Let 439

$$\begin{array}{ccc} Y & \longleftarrow & Y' \\ \downarrow & & \downarrow \\ X & \longleftarrow & X' \end{array}$$

be a Cartesian diagram of topological spaces. If Y and X' are discrete spaces, and hence Y' is discrete, then the corresponding diagram of topoi is a 2-fiber product diagram (5.11). (NB. For a counterexample when Y and X' are no longer assumed discrete, with Y and X' subspaces of the Hausdorff space X, cf. 9.1.10 c)). Deduce an example of a 2-fiber product diagram of topoi

$$\begin{array}{ccc} F & \longleftarrow & F' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ E & \longleftarrow & E' \end{array}$$

such that f is conservative but f' is not conservative, more precisely with $E' =$ the punctual topos and $F' =$ the empty topos (2.2). (Take Y such that the image

of Y in X is very dense and $\neq X$, and take X' to consist of a point not belonging to the image.)

e) Let $(E_i)_{i\in I}$ be a family of subtopoi of a topos E. Prove that E is the supremum of the E_i if and only if the family of embedding morphisms $E_i \to E$ is conservative (6.4.0).

Exercise 9.1.8. (*Subtopoi and points of E. The case of topological spaces*).
Let E be a topos.

a) Let Z be a full subcategory of **Point**(E) (or, what amounts to the same thing, a subset of Ob **Point**(E)). Show that the set $S(Z)$ of arrows of E which are taken to bijections by the fiber functors corresponding to the $z \in \mathrm{Ob}\, Z$ satisfies the conditions of 9.1.7 c), and hence defines a subtopos E_Z of E. Show that this subtopos has enough points, and that Z is equivalent to a full subcategory of **Point**(E_Z). Show that every subtopos E' of E which has enough points can be obtained by the preceding procedure. (Take for Z the essential image
440 of the functor (9.1.5.1).) Show that one obtains a bijection between the set of subtopoi E' of E having enough points and a subset of the set of strictly full subcategories Z of **Point**(E) which are stable under direct factors (I 10.6) and under small filtered projective limits (cf. 7.3). (Problem: Which subcategories of **Point**(E) arise in this way? This is essentially, in another form, the problem already raised in 6.5.3, in view of the dictionary in 9.1.2 c). Note that, besides the necessary conditions just indicated, there are also subtler conditions on the topological nature of Point(E), such as sobriety; cf. e)).

b) Let $(Z_i)_{i\in I}$ be a family of full subcategories of **Point**(E), and let Z be the full subcategory which is the union of the Z_i. Show that the corresponding subtopos E_Z defined in a) is the subtopos of E generated by the E_{Z_i}. (Use 9.1.7 e).) Conclude that the subtopos of E generated by a family of subtopoi having enough points also has enough points.

c) Let $f : E' \to E$ be an embedding of topoi. Show that the map $U \mapsto f^*(U)$

$$\mathrm{Ouv}(E) \longrightarrow \mathrm{Ouv}(E') \tag{9.1.8.1}$$

is surjective, and conclude that the map

$$\mathrm{Point}(E') \longrightarrow \mathrm{Point}(E) \tag{9.1.8.2}$$

induced by f on the isomorphism classes of points is a homeomorphism from Point(E') onto a subspace of Point(E), for the topologies defined in 7.8 a). (For injectivity, use 9.1.5.)

d) Let $f : X' \to X$ be a continuous map, giving a morphism of topoi (4.1)

$$\mathrm{Top}(f) : \mathrm{Top}(X') \longrightarrow \mathrm{Top}(X).$$

Prove that the latter is an embedding if and only if the topology of X' is the inverse image under f of that of X (i.e. , when X' is a Kolmogorov space, for example a sober space, if and only if f induces a homeomorphism of X' onto a
441 subspace of X). (For the necessity, use c), and for the sufficiency use 9.1.7.2 c)).

e) Let X be a sober topological space, and let $E = \mathrm{Top}(X)$. Using a) and d), establish an isomorphism of ordered sets $X' \mapsto E_{X'}$ between the set of sober subspaces X' of X (i.e. , when X is Hausdorff, the set of *all* subsets of X) and the set of subtopoi of E having enough points. Show that $E_{X'}$ is equivalent to Top(X'). If $(X'_i)_{i\in I}$ is a family of sober subspaces of X, show that there is a smallest sober subspace X' of X containing the X'_i, equal to the union of the

X'_i if I is finite or X is Hausdorff, and that $E_{X'}$ is the subtopos of E generated by the $E_{X'_i}$. (Use b)).

f) Give an example of a topos E of the form $\mathrm{Top}(X)$ and a subtopos E' which does not have enough points, and which is even nonempty (2.2) but has no points. (Take E of the form $\widehat{U}$ and $E' = U^{\sim}$, where U is the ordered set defined in 7.4. Or better, take the example in 9.1.10 b), which shows that one may take for X any nonempty compact space without isolated points.)

Exercise 9.1.9. (*The case of topoi defined by an ordered set*).

a) We use the yoga of 7.8 b) and c), which gives a dictionary between, on the one hand, topoi generated by their opens and morphisms of such topoi, and, on the other hand, ordered sets admitting arbitrary Sup, finite infima, and in which the infimum of two elements distributes over arbitrary Sup; morphisms between such ordered sets are increasing maps commuting with finite infima and arbitrary Sup.

b) Let E be a topos and $f : E' \to E$ an embedding of topoi. Show that, for every generating family $(X_i)_{i\in I}$ in E, the family $(f^*(X_i))_{i\in I}$ is generating. Deduce that if E is generated by its opens, then the same is true of E'. (Use 9.1.8 c).)

c) Let $f : E' \to E$ be a morphism of topoi generated by their opens, associated with a morphism $g : \mathcal{O} \to \mathcal{O}'$ of the corresponding ordered sets. Show that f is an embedding if and only if g is surjective, *and* g is surjective on the graphs of the order relations, i.e. the order on $\mathcal{O}'$ is the quotient of that on $\mathcal{O}$. 442

d) Let E be a topos of the form $\mathcal{O}^{\sim}$ as in a). Deduce from b) and c) a bijective correspondence between the set of subtopoi of E and the set of equivalence relations R on $\mathcal{O}$ having the following properties, analogous to conditions ST 1) to ST 3) of 9.1.7 c):

Q0 1) The relation R is compatible with finite infima; in other words, if U and U' are equivalent under R, then so are $U \cap V$ and $U' \cap V$ for every $V \in \mathcal{O}$.

Q0 2) The relation R is compatible with arbitrary Sup; in other words, if U_i is equivalent to U'_i for every $i \in I$, then $\mathrm{Sup}_i U_i$ is equivalent to $\mathrm{Sup}_i U'_i$.

Show that if subtopoi E'_i of E correspond to the equivalence relations R_i, then the subtopos generated by the E'_i corresponds to the intersection of the relations R_i.

Exercise 9.1.10. (*Intersections of subspaces; new topoi without points; nonexistence of a complement to a subtopos*).

Let X be a topological space, $E = \mathrm{Top}(X) = \mathrm{Ouv}(X)^{\sim}$ (2.1).

a) Let R be the following relation on $\mathrm{Ouv}(X)$: $(U, U') \in R \iff U \cap U'$ is dense in both U and U'. Show that R is an equivalence relation satisfying conditions Q0 1) and Q0 2) of 9.1.9 d), and therefore defines a subtopos E' of E. Show that, for a point $x \in X$, the corresponding point belongs to the essential image of $\mathbf{Point}(E')$ if and only if x is *thick*, i.e. belongs to the closure of the interior of $\overline{x}$. Show that E' is the empty topos (2.2) if and only if X is empty.

b) If X is sober, show that the category $\mathbf{Point}(E')$ is equivalent to the category defined by the ordered subset of X (for the specialization relation) consisting of the thick points of X. Show that if the points of X are closed (resp. if X is Noetherian), then the thick points of X are the isolated points of X, i.e. those for which $\{x\}$ is open (resp. are the maximal points of X). Conclude that if X is a nonempty Hausdorff topological space without isolated points, then the 443

subtopos E' of $E = \mathrm{Top}(X)$ is a non-"empty" topos having no points.

c) Let X' be a subset of X, and let $E_{X'}$ be the subtopos of E associated with X' by the construction of 9.1.8 a). Show that if X' is dense, then $E_{X'}$ contains E'. Show that if X is sober and X' and X'' are two sober subsets of X, then their intersection $X' \cap X''$ is sober and $E_{X' \cap X''} \subset E_{X'} \cap E_{X''}$, and that the inclusion can be strict [12]. (Take a nonempty Hausdorff space X having two disjoint dense subsets X', X''.) Show that the preceding inclusion is an equality if and only if the subtopos $E_{X'} \cap E_{X''}$ has enough points. Show that this is the case if X' or X'' is a *locally closed* subset of X.

d) Let X' be a subset of X, and let X'' be the complement of X'. A subtopos F of E contains the $E_{\{x\}}$ for $x \in X''$ if and only if it contains $E_{X''}$ (cf. 9.1.8 b)). Conclude that if X'' is a union of locally closed subsets of X (for example, if the points of X'' are closed), then the set of subtopoi F of E whose intersection with $E_{X'}$ is the empty subtopos $\mathrm{Top}(\emptyset)$ has a greatest element (which then deserves the name of *weak complement of* $E_{X'}$ in E; cf. 9.1.13) *only if* $E_{X'} \cap E_{X''}$ is the empty subtopos (a condition that is not satisfied if $X \neq \emptyset$ and if X' and X'' are both dense in X, with X, X', and X'' sober; cf. c)). Show that when this condition fails, the infimum of two elements in the ordered set of subtopoi of E is not distributive with respect to arbitrary Sup. (The editor does not know whether it is distributive with respect to finite Sup; cf. , however, 9.1.11 e) and 9.1.12 d)).

Exercise 9.1.11. (*Subtopoi of locally Noetherian topoi. The case of Noetherian spaces*).

a) Let $f : E' \to E$ be an embedding of topoi. Show that if X is a pre-Noetherian
444 object of E (i.e. (VI 1.30) every increasing sequence of subobjects of X is stationary), then $X' = f^*(X)$ is a pre-Noetherian object of E'. (By introducing the induced topoi $E'_{/X'}$ and $E_{/X}$ and using 9.1.6 e), first reduce to the case where X is the final object e of E, so that X' is the final object e' of E'. Then observe that every increasing sequence of subobjects of e' defines, upon applying f_*, an increasing sequence of subobjects of $f_*(e') = e$.) Conclude that if E admits a generating family consisting of pre-Noetherian objects (resp. if E is a Noetherian topos, resp. locally Noetherian (VI 2.11)), then E' satisfies the same condition. (Use 9.1.9 b).)

b) Let E be a locally Noetherian topos (VI 2.11). Show that every subtopos of E is of the form E_Z (9.1.8 a)), for a suitable full subcategory Z of $\mathbf{Point}(E)$. (Use a) and Deligne's theorem that every locally Noetherian topos has enough points.) Conclude that if E is of the form $\mathrm{Top}(X)$, where X is a sober locally Noetherian topological space, then $Z \mapsto E_Z$ is an isomorphism from the ordered set of sober subspaces of X to the ordered set of subtopoi of E. (Use 9.1.8 e).) The intersection of sober subspaces Z_i (is necessarily sober and) defines the intersection of the subtopoi E_{Z_i} (in contrast with what can occur in the general case (9.1.10 c)).

c) Let X be a sober locally Noetherian space, let X' be a subset of X, and let X'' be its complement. Show that the following conditions are equivalent:
   (i) X' is constructible.
   (ii) X' and X'' are sober.
   (iii) X' is sober, and the subtopos $E_{X'} \cong \mathrm{Top}(X')$ of $E = \mathrm{Top}(X)$ admits a "complementary" subtopos (cf. 9.1.13).

[12] See, however, 9.1.11 b) below for the case where X is locally Noetherian.

(iv) The intersection of the subtopoi $E_{X'} \approx \mathrm{Top}(X')$ and $E_{X''} \approx \mathrm{Top}(X'')$ of $E = \mathrm{Top}(X)$ is the "empty" subtopos of E.

Show that if E' and E'' are two subtopoi of E, they are weak complements if and only if they are complements (9.1.13).

d) Let X be a sober Noetherian space. Show that the following conditions are equivalent: 445

(i) Every sober subspace X' of X is constructible, i.e. , according to c), its complement is sober.

(i bis) For every maximal point x of X, $X - \{x\}$ is sober, i.e. $\{x\}$ is constructible.

(ii) X is finite.

(iii) Every subtopos of $\mathrm{Top}(X)$ admits a complementary subtopos (cf. 9.1.13).

(iv) In the ordered set of subtopoi of $\mathrm{Top}(X)$, the infimum of two elements is distributive with respect to arbitrary Sup.

(Use the argument of 9.1.10 d), together with the intersection formula proved in b).)

e) Let X be a locally Noetherian space. Show that in the ordered set of subtopoi of $\mathrm{Top}(X)$, the infimum of two elements is distributive with respect to *finite* Sup. (Reduce to the case where X is sober, and use b).)

f) For every sober subspace Z of the sober locally Noetherian space X, let T_Z be the topology on the category $\mathrm{Ouv}(X)$ of open subsets of X for which a family of inclusions $U_i \to U$ is covering if and only if $U \cap Z = \bigcup_i U_i \cap Z$. Show that $Z \mapsto T_Z$ establishes a bijection (reversing inclusion relations) between the set of sober subspaces Z of X and the set of topologies on $\mathrm{Ouv}(X)$ finer than the canonical topology T_X. (Use b) and 9.1.2 c).)

g) Let X be a sober topological space. A subset X' of X is sober if and only if its complement is a union of locally closed subsets; if X is locally Noetherian, this also means that X' is "proconstructible" (EGA IV 1.9.4). Show that there is a topology $T_{\mathrm{cons}'}$ on X such that the closed subsets of $X_{\mathrm{cons}'} = (X, T_{\mathrm{cons}'})$ are the sober subsets of X. Assume henceforth that X is locally Noetherian, so that $X_{\mathrm{cons}'}$ is none other than the space X_{cons} of EGA IV 9.1.13. Deduce 446 from b) and c) that the ordered set Σ of subtopoi of $\mathrm{Top}(X)$ is isomorphic to the ordered set of closed subsets of X_{cons}, and that, under this correspondence, the subtopoi admitting a complement correspond to the subsets that are both open and closed in X_{cons}. Show that X_{cons} is locally compact, and that it is compact if and only if X is Noetherian, i.e. the topos $\mathrm{Top}(X)$ is Noetherian.

h) Let E be a locally Noetherian topos. Suppose that it has the property considered in 9.1.14 b) (a condition that may perhaps always hold...). Show that the essential images of the canonical functors $\mathbf{Point}(E') \to \mathbf{Point}(E)$, where E' ranges over the set Σ of subtopoi of E, define a collection of subsets of the set $X = \mathrm{Point}(E)$ of isomorphism classes of points of E which is the collection of closed subsets for a *topology* T_{cons} *on* X, and that in this way one obtains an isomorphism of ordered sets between the set Σ of subtopoi of E and the set of closed subsets of X_{cons}. Conclude that in Σ, the Sup of two elements is distributive with respect to *arbitrary* infima. Show that X_{cons}, which is not necessarily $\mathscr{U}$-small, has an associated sober space (4.2.1) which is $\mathscr{U}$-small.

Exercise 9.1.12. (*Finite topoi*). A topos E is said to be *finite* if it is equivalent to a topos of the form $\widehat{C}$, with C a finite category.

a) (Dictionary). Let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$. Let (Karfiness) be the 2-category defined as the full 2-subcategory of the category $(\mathrm{Cat})_{\mathscr{V}}$ consisting of categories that are elements of $\mathscr{V}$, are *Karoubian and equivalent to a finite category*, and let (Topfin) be the 2-category defined as the full 2-subcategory of ($\mathscr{V}$-$\mathscr{U}$-Top) (3.3.1) consisting of the *finite* $\mathscr{U}$-topoi that are $\in \mathscr{V}$. Show that there are mutually quasi-inverse 2-equivalences:

(9.1.12.1) $\qquad C \longmapsto \widehat{C} : (\text{Karfiness}) \longrightarrow (\text{Topfin})$

(9.1.12.2.) $\qquad E \longmapsto \mathbf{Point}(E) : (\text{Topfin}) \longrightarrow (\text{Karfiness}).$

(Use 7.6 h), e).)

447 b) Show that a finite topos is Noetherian VI 2.11.

c) Show that every subtopos of a finite topos E is a finite topos. More precisely, if E is equivalent to $\widehat{C}$, where C is a finite Karoubian category (or, more generally, is equivalent to a finite category), show that one obtains an order isomorphism between the set of subtopoi E' of E and the set of strictly full subcategories C' of C that are Karoubian (or, equivalently, stable under taking direct factors in C), by associating with every C' the essential image of $f_* : \widehat{C'} \to \widehat{C}$, where $f : C' \to C$ is the inclusion functor. (Use 5.6 to verify that f_* is fully faithful, and b) and 9.1.11 b) for the fact that every subtopos E' of E is obtained in the indicated way.)

d) Let E be a finite topos. Construct on the finite set $X = \mathrm{Point}(E)$ of isomorphism classes of points of E a topology making it a sober topological space and such that the ordered set of subtopoi of E is canonically isomorphic to the set of closed subsets of X. (Take the "constructible topology" $\mathscr{C}_{\mathrm{cons}}$ on X, defined by the order relation $(x \leq y) \Leftrightarrow (x$ is a direct factor of $y)$, for which the closure of a point x consists of the y such that $y \leq x$.) In particular, the set Σ of subtopoi of E is finite, and the binary infimum operation $\wedge$ is distributive with respect to arbitrary Sup.

e) Let C be a category equivalent to a finite category. Show that, for every full subcategory C' of C, there exists a topology $T_{C'}$ on C for which a family $(X_i \to X)_{i \in I}$ is covering if and only if, for every $Y \in \mathrm{Ob}\, C'$, the family of maps $\mathrm{Hom}(Y, X_i) \to \mathrm{Hom}(Y, X)$ $(i \in I)$ is surjective. Show that the category of sheaves on $(C, T_{C'})$ is equivalent to $\widehat{C'}$, and that $C' \mapsto T_{C'}$ is a bijection (reversing the order relations) between the set of strictly full subcategories C' of C that are Karoubian (or, equivalently, stable under taking direct factors in C), and the set of topologies (II 1.1) on the category C. (Use c) and 9.1.2 c).)

448 In particular, if C is a site whose underlying category is equivalent to a finite category, then the topos $C^{\sim}$ is a finite topos.

f) Let $f : C' \to C$ be a functor of $\mathscr{U}$-categories, with C equivalent to a finite category. Show that the morphism of topoi $\widehat{f} : \widehat{C'} \to \widehat{C}$ (4.6) is conservative if and only if every object of C is isomorphic to a direct factor of an object in the image of f. (Reduce to the case where f is an inclusion $f : C' \hookrightarrow C$, with C' as in c).)

g) Let C be the category having two objects e (the final object) and a, and, besides the identity arrows, three arrows $f : a \to e$, $g : e \to a$, $p = gf$ (thus $p^2 = p$). Show that $E = \widehat{C}$ has, besides the subtopoi E and $\mathrm{Top}(\varnothing)$, exactly one subtopos E', namely $E' = \widehat{\{e\}}$; consequently, for every subtopos E'' of E with $E'' \neq \mathrm{Top}(\varnothing)$, one has $E' \cap E'' \neq \mathrm{Top}(\varnothing)$.

Exercise 9.1.13. (*Complementary subtopoi of a topos*).

Let E be a topos and let E' and E'' be two subtopoi. We say that E' and E'' are *complementary* if $E' \cap E'' = \mathrm{Top}(\varnothing)$ and $E' \vee E'' = E$ (where $\vee$ denotes the Sup in the ordered set of subtopoi of E).

Suppose that in the set Σ of subtopoi of E, the infimum of two elements is distributive over finite suprema.

a) If E' and E'' are complementary, then E'' is the largest among the subtopoi E''_1 such that $E' \cap E''_1 = \mathrm{Top}(\varnothing)$ (i.e. E'' is a weak complement of E', in the terminology of 9.1.10 d)).

b) Show that the following conditions on E are equivalent:
   (i) If E' and E'' are two subtopoi of E such that E'' is a weak complement of E', then E'' is a complement of E'.
   (ii) For every subtopos E' of E such that $E' \neq E$, there exists a subtopos E'' of E such that $E'' \neq \mathrm{Top}(\varnothing)$ and $E' \cap E'' = \mathrm{Top}(\varnothing)$.
   (i bis) If E' and E'' are two subtopoi of E such that E'' is maximal among the subtopoi E''_1 satisfying $E' \cap E''_1 = \mathrm{Top}(\varnothing)$, then E'' is a complement of E'.

   Show that even if E is a finite topos (9.1.12), these conditions need not hold (9.1.12 g)). They do hold, however, if E is of the form $\mathrm{Top}(X)$, with X a locally Noetherian space. (Use 9.1.11 b)). 449

c) A subtopos E' of E is called *complemented* if it admits a complement (compare 9.1.12 c)). Prove that the set of complemented subtopoi is stable under finite infima and suprema and under complementation, and that complementation transforms infima into suprema and suprema into infima.

d) State the duals of observations a), b), and c) (which were in fact trivial general statements about an abstract ordered set Σ; note that the hypothesis made on Σ is itself self-dual).

9.1.14. *Open questions*[13]. Let E be a topos.

a) In the set of subtopoi of E, is the infimum of two elements distributive over finite suprema?

b) Let E' and E'' be two subtopoi of E such that $E = E' \vee E''$. Is every point of E then isomorphic to the image of a point of E' or a point of E''?

   Note that if the answer to b) is affirmative for E and all its subtopoi, then the answer to a) is affirmative at least for subtopoi having enough points (and hence for all subtopoi if E is locally Noetherian (9.1.11 b)). The answer to a) (and a fortiori to b)) is nevertheless not known for all Noetherian topoi. The answers to a) and b) are affirmative, however, if E is of the form $\mathrm{Top}(X)$, with X a locally Noetherian space (9.1.11), or if E is a finite topos (9.1.12).

   Note that a) can be reformulated as follows (applied to all subtopoi F of E):

a′) If E' and E'' are two subtopoi of a topos F such that the canonical morphism (cf. 8.7 b))

$$E' \amalg E'' \longrightarrow F \tag{9.1.14.1}$$

is conservative (which also means that $F = E' \vee E''$ (9.1.7.2 e)), then the same is true after 2-base change by $F_1 \hookrightarrow F$, where F_1 is a subtopos of F. 450

   There is an analogous reformulation b′) of b), taking a base change $F_1 \to F$ by a *punctual* topos F_1 (2.2). Thus affirmative answers to a) and b) would follow

[13] N.D.E.: Positive answers to questions a) and d) were given in [17].

from an affirmative answer to the following question (where F is any subtopos of E):

c) If E' and E'' are two subtopoi of the topos F whose supremum is F, i.e. such that the morphism of topoi (9.1.14.1) is conservative, is this morphism even *universally conservative*, i.e. does it remain conservative after every 2-base change $F_1 \to F$?

This statement makes sense by a result of DELIGNE (see [16], Proposition 4.47) asserting the existence of 2-fiber products of topoi. See, however, the example in 9.1.7.2 d).

On the other hand, 9.1.11 h) shows that an affirmative answer to b) would, in the locally Noetherian case, provide an affirmative answer to the following question:

d) In the ordered set Σ of subtopoi of E, is the infimum of two elements distributive over *arbitrary* (not necessarily finite) suprema?

This also means that Σ may be interpreted as the ordered set of "closed subtopoi" of a suitable topos $E_{\mathrm{cons}'}$ (namely $\Sigma^{0\tilde{}}$, where the opposite ordered set Σ^0 of Σ, regarded as a category, is endowed with its canonical topology, cf. 7.8 b)), which is generated by the subobjects of the terminal object and is therefore very close to an ordinary topological space. It would remain to study this topos $E_{\mathrm{cons}'}$, taking inspiration from the examples 9.1.11 g) and 9.1.12 d), and in particular to determine whether it is Noetherian when E is. This amounts to the following question for subtopoi F of E, which makes sense independently of d):

e) Let F be a Noetherian topos, and let $(E_i)_{i\in I}$ be a family of subtopoi of F whose intersection is $\mathrm{Top}(\emptyset)$. Is there a finite subfamily with the same property?

451 In a rather different direction, recall also the question raised in 9.1.6 e), which deserves clarification.

9.2. The case of open subtopoi.

9.2.1. Let U be an open of a topos E, i.e. a subobject of the terminal object e of E (8.3). Consider the localization morphism (5.2)

$$j : E_{/U} \longrightarrow E. \tag{9.2.2}$$

Since $U \to e$ is a monomorphism, the functor $j_!$ (which is interpreted as the forgetful functor) is fully faithful, and hence so is its biadjoint j_* (I 5.7). In other words, *the localization morphism j in (9.2.2) associated with an open U of the topos E is an embedding of topoi* (9.1.1 a)).

An embedding of topoi $F \to E$ is called an *open embedding* if it is isomorphic to the embedding of topoi defined by an open U of E. A subtopos E' (9.1.1 b)) of E is called an *open subtopos* if it is defined by an open of E, i.e. if the canonical inclusion morphism $E' \to E$ is an open embedding. Note that the open U of E associated with an open embedding $j : E' \to E$ is uniquely determined as the subobject $j_!(e')$ of e, where e' denotes the terminal object of E'. Thus one obtains a one-to-one correspondence between opens U of E and open subtopoi of E.

REMARK 9.2.3. In accordance with the preceding definitions, the open subtopos of E associated with the open U of E is the strictly full subcategory of E which is the essential image of the direct-image functor

$$j_* : E_{/U} \longrightarrow E.$$

One should not confuse this subcategory of E with the strictly full subcategory which is the essential image of $j_!$, formed by the objects X of E for which the structural morphism $X \to e$ factors through U. These two subcategories plainly determine one another and are canonically equivalent. This is why some authors may have been tempted to call (and indeed have called) the strictly full subcategory which is the essential image of $j_!$ the "open subtopos defined by U," since its description in terms of U is simpler than that of the essential image of j_*. This terminology causes no difficulty so long as one does not intend to study kinds of subtopoi other than open subtopoi. Since that is not our situation, we shall not follow the authors just mentioned.

For the reader's convenience, we summarize the most important special properties of open embeddings:

PROPOSITION 9.2.4. *Let*

$$j : \mathcal{U} \longrightarrow E$$

be an open embedding of topoi (9.2.1). Then the functor $j_!$ left adjoint to j^* exists, so that there is a sequence of three adjoint functors

$$j_!, \quad j^*, \quad j_* :$$

$$\mathcal{U} \;\underset{j^*}{\overset{j_!}{\rightleftarrows}}\; E \;\xrightarrow{j_*}$$

Moreover:

a) *The functors $j_!$ and j_* are fully faithful.*
b) *The functor $j_!$ commutes with fiber products, with products indexed by small sets $I \neq \phi$, and with projective limits indexed by small cofiltered categories (I 2.7).*
c) *For every object X of E, the adjunction morphism*

$$j_! j^*(X) \longrightarrow X$$

is a monomorphism.

Proof. We may suppose that j is the localization morphism defined by an open U of E. Then a) is recalled for reference, b) follows from the interpretation of $j_!$ as a forgetful functor and from the fact that $U \to e$ is a monomorphism. Finally, c) is immediate on observing that the morphism in question is obtained from the inclusion $U \to e$ by base change along $X \to e$, since a monomorphism is carried to a monomorphism by base change.

REMARK 9.2.4.1. There exist embeddings of topoi $j : E' \to E$ for which $j_!$ exists but does not commute with products of two objects or with cofiltered projective limits (Exercise 9.5.9 c)).

9.2.5. Now let

$$g : E' \longrightarrow E$$

be an arbitrary morphism of topoi, and let $U' = g^*(U)$ be the inverse image of the open U of E. We saw in 5.11 that the diagram of topoi

$$\begin{array}{ccc} E'/_{U'} & \xrightarrow{j'} & E' \\ {\scriptstyle g_U}\downarrow & & \downarrow{\scriptstyle g} \\ E/_U & \xrightarrow{j} & E \end{array}$$

is 2-Cartesian. It follows in particular that the inverse image under g of an open subtopos of E (cf. 9.1.6 d)) is an open subtopos of E', or equivalently that the notion of an open embedding of topoi (9.2.1) is stable under 2-base changes in the 2-category of $\mathscr{U}$-topoi (which are elements of a given universe $\mathscr{V}$).

Also note that the morphism of topoi $g : E' \to E$ factors, up to isomorphism, through $j : E_{/U} \to E$ if and only if $j' : E'_{/U'} \to E'$ is an equivalence of categories, which also means that the inclusion $U' \hookrightarrow e'$ (where e' is the terminal object of E') is an isomorphism. This makes 9.1.4 more precise by giving a simple characterization of the essential image of the functor considered in *loc. cit.*

9.2.6. Apply 9.2.5 when E' is of the form $E_{/V}$, where V is a second open of E, and g is the localization morphism. Then $U' = U \cap V$, and the 2-Cartesian product of $E_{/U}$ and $E_{/V}$ over E identifies with $E_{/U\cap V}$. It follows that *a finite intersection of open subtopoi of E is an open subtopos of E*; more precisely, the map (9.2.6.1)

$$U \longmapsto \text{ the subtopos of } E \text{ defined by } U$$

commutes with finite intersections.

Exercise 9.2.7. Prove that the map (9.2.6.1) also commutes with *arbitrary* Sup. (Use 9.1.7.2 e)).

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9.3. Construction of the closed subtopos complementary to an open subtopos.

9.3.1. Let T be a topological space and U an open subset of T, so that $\mathrm{Top}(U)$ identifies with an open subtopos of $\mathrm{Top}(\mathsf{T})$ (notation of 2.1). Let Y be the closed topological subspace of T complementary to U. Up to equivalence, $\mathrm{Top}(Y)$ may then be regarded as a subtopos of $\mathrm{Top}(\mathsf{T})$, namely the subtopos formed by the objects F of $\mathrm{Top}(\mathsf{T})$ whose restriction to U is the terminal object of U. This description of a subcategory of $E = \mathrm{Top}(\mathsf{T})$ makes sense whenever one has a topos and an open U of it, and we shall see that it always gives a subtopos of E. Moreover, in the particular case first considered, it is immediate (with the terminology introduced in Exercise 9.1.13) that $\mathrm{Top}(Y)$ and $\mathrm{Top}(U)$ are complementary subtopoi (use 9.2.5 and 9.1.7.2 e)). The same will hold in the general case, and we shall see that this property uniquely characterizes the subtopos in question. It therefore deserves in every respect the name of the closed subtopos complementary to the given open U, or to the open subtopos defined by U. The details of the construction of this topos $\mathscr{F}$ and of the inverse-image functor $i^* : E \to \mathscr{F}$ will be given in the present section, while Section 9.4 develops its first properties.

9.3.2. Thus let E be a topos and U an open of E, as in 9.2.1, and consider the localization morphism

$$j : \mathscr{U} = E_{/U} \longrightarrow E, \tag{9.3.2.1}$$

which is an open embedding.

For every object X of E, put

$$X_{\complement U} = U \coprod_{U \times X} X, \tag{9.3.2.2}$$

where the amalgamation is formed with respect to the canonical projections

$$\mathrm{pr}_1 : U \times X \longrightarrow U, \quad \mathrm{pr}_2 : U \times X \longrightarrow X.$$

Thus there is a *cocartesian* diagram (I 10) in E, depending functorially on X: 455

$$\begin{array}{ccc} U \times X & \xrightarrow{\mathrm{pr}_2} & X \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow \\ U & \longrightarrow & X_{\complement U} \end{array} \tag{9.3.2.3}$$

Note that in the example considered in 9.3.1, $X_{\complement U}$ is canonically isomorphic to $i_* i^*(X)$ (where $i : Y \to T$ is the inclusion), and $X \to X_{\complement U}$ identifies with the adjunction morphism.

Proposition 9.3.3. *With the preceding notation, the following statements hold:*

a) *For every object X of E, the following conditions are equivalent:*
   (i) *There exists an object Y of E and an isomorphism $X \simeq Y_{\complement U}$.*
   (ii) *The canonical morphism $X \xrightarrow{m} X_{\complement U}$ is an isomorphism.*
   (iii) *The canonical morphism $\mathrm{pr}_2 : X \times U \to U$ is an isomorphism (i.e. $j^*(X)$ is a terminal object of $E_{/U}$).*

b) *For every morphism*

$$m : X \longrightarrow X'$$

in E, the following conditions are equivalent:
   (i′) *The diagram*

$$\begin{array}{ccc} X & \xrightarrow{m} & X' \\ \downarrow & & \downarrow \\ X_{\complement U} & \longrightarrow & X'_{\complement U} \end{array}$$

   is Cartesian.
   (ii′) *The morphism $m \times \mathrm{id}_U : X \times U \to X' \times U$ is an isomorphism, i.e. $j^*(m)$ is an isomorphism.*

c) *The functor*

$$X \longrightarrow X_{\complement U}$$

from E to E is left exact, carries epimorphic families to epimorphic families, and commutes with filtered inductive limits and amalgamated sums. 456

Proof. First prove the proposition when E is of the form $\widehat{C}$, where C is a small category. By I.3.1 this reduces to the case where E is the "punctual" topos (Ens), where the proposition is evident. One then passes to the general case by the standard procedures of II 4, using the "associated sheaf" functor.

PROPOSITION 9.3.4. *With the notation of 9.3.2, the strictly full subcategory $\mathscr{F}$ of E formed by the objects X of E such that $j^*(X)$ is a terminal object of $\mathscr{U}$* (cf. 9.3.3 a)) *is a subtopos of E: i.e. (9.1.1) it is a topos, and the inclusion functor $i_* : \mathscr{F} \to E$ is the direct-image functor of a morphism of topoi $i : \mathscr{F} \to E$, whose inverse-image functor is $X \mapsto i^*X = X_{\complement U}$. The adjunction morphism $X \to i_*i^*X$ is the canonical morphism $X \to X_{\complement U}$ of (9.3.2.3).*

Proof: For every object X of E, denote by $u(X)$ the canonical morphism $X \to X_{\complement U}$. The morphism $u(X)$ is functorial in X, and for every object X, $u(X_{\complement U})$ is an isomorphism. It follows formally and immediately from these two properties that the functor i^* is left adjoint to the functor i_* and that the adjunction morphism is $u(X)$. Since the functor $X \to X_{\complement U}$ is left exact, the functor i^* is left exact. It then follows from II 5.5 that $\mathscr{F}$ is a topos, and from Definition 3.1 that the pair (i^*, i_*) is a morphism of topoi.

9.3.5. The subtopos $\mathscr{F}$ of E described in 9.3.4 is called the *closed subtopos of E complementary to the open U of E*, or, equivalently, to the open subtopos of E corresponding to U. In accordance with 9.1.1 b), the morphism $i : \mathscr{F} \to E$ constructed in 9.3.4 is called the *inclusion morphism*. A full subcategory $\mathscr{F}$ of E is called a *closed subtopos* of E if there exists an open U of E for which $\mathscr{F}$ is the closed subtopos complementary to U. Note that this U is uniquely determined as $i_*(\phi_{\mathscr{F}})$, where $\phi_{\mathscr{F}}$ is the initial object of $\mathscr{F}$, equal to $i^*(\phi_E)$; it is also called *the open of E complementary to the closed subtopos $\mathscr{F}$*,
457 and the subtopos of E defined by U is also called the *open subtopos of E complementary to $\mathscr{F}$*. Finally, an embedding of topoi (9.1.1 a)) $i : F \to E$ is called a *closed embedding* if the corresponding subtopos of E is closed.

If G is a group object on E and $s \in \mathrm{Hom}(e_E, G)$, the *support* of G (resp. of s) is the closed subtopos of E complementary to the cosupport open of G (resp. of s) (8.5.1, 8.5.2).

9.3.6. With the notation of 9.3.2 and 9.3.4, one therefore has two morphisms of topoi

$$\mathscr{U} \xrightarrow{j} E \xleftarrow{i} \mathscr{F}, \tag{9.3.6.1}$$

defining five functors which form two sequences of adjoint functors, $(j_!, j^*, j_*)$ and (i^*, i_*):

$$\mathscr{U} \underset{\longleftarrow j^* \longrightarrow}{\overset{j_!}{\longrightarrow}} E \underset{\longleftarrow i_* \longrightarrow}{\overset{i^*}{\longrightarrow}} \mathscr{F}. \tag{9.3.6.2}$$

The functor

$$\rho = i^*j_* : \mathscr{U} \longrightarrow \mathscr{F} \tag{9.3.6.3}$$

is called the *gluing functor* relative to the open U of E. More generally, if $\mathscr{U}$ and $\mathscr{F}$ are arbitrary topoi, a functor $\rho : \mathscr{U} \to \mathscr{F}$ is called a *gluing functor* if it is left exact and accessible (I.9.2), or equivalently (I 8) if it admits a pro-adjoint. The reasons for this terminology will appear in 9.4.1 d) and 9.5.4 below.

9.4. First properties of the closed subtopos $\mathscr{F}$ and of $i : \mathscr{F} \to E$.

PROPOSITION 9.4.1. *The notation is that of 9.3.2 and 9.3.4.*

a) *The functor $i_* : \mathscr{F} \to E$ carries epimorphic families to epimorphic families. It commutes with the formation of amalgamated sums and filtered inductive limits.*
b) *For every object X of E such that the projection morphism $U \times X \to U$ is an epimorphism, the adjunction morphism $X \to i_*i^*X$ is an epimorphism.*

458 c) *The pair of functors (i^*, j^*) from E to $\mathscr{F}$ and $\mathscr{U}$, respectively, is conservative* (I 6.1), *i.e. a morphism m of E is an isomorphism if and only if $i^*(m)$ and $j^*(m)$ are isomorphisms (equivalently, the pair (i^*, j^*) is faithful (6.4.0)).*

d) *The gluing functor $\rho = i^* j_* : \mathscr{U} \to \mathscr{F}$ is left exact and accessible* (I 9.2), *or equivalently* (I 8) *it admits a pro-adjoint $\sigma : \mathrm{Pro}\,\mathscr{F} \to \mathrm{Pro}\,\mathscr{U}$.*

Proof: a) follows from 9.3.3 c). For b), it suffices to use the definition (9.3.2.2) of $X_{\complement U}$, taking into account that $i_* i^* X = X_{\complement U}$ (9.3.4). Assertion c) follows from 9.3.3 b). The functors i^* and j_* are left exact, and hence so is ρ. The functor j_* is a direct-image functor for a morphism of topoi and is therefore accessible (I 9.5); note that a topos is an accessible category (I 9.11.3). Since the functor i^* commutes with inductive limits (it is an inverse-image functor), the functor ρ, being a composite of two accessible functors, is accessible.

PROPOSITION 9.4.2. *Let E' be a topos. Then the functor $h \mapsto i \circ h$,*

$$\mathrm{Homtop}(E', \mathscr{F}) \longrightarrow \mathrm{Homtop}(E', E),$$

is fully faithful, and its essential image consists of the morphisms of topoi $g : E' \to E$ such that $g^*(U)$ is an initial object $\phi_{E'}$ of E'.

By 9.1.4, the functor in question is fully faithful, and g belongs to its essential image if and only if, for every object X' of E', the object $g_*(X')$ of E belongs to $\mathscr{F}$, i.e. if

(*) $$j^*(g_*(X')) \simeq e_{\mathscr{U}} \text{ for every } X' \in \mathrm{Ob}\, E'.$$

Put $U' = g^*(U)$ and consider the diagram of topoi

$$\begin{array}{ccc} E'_{/U'} & \xrightarrow{j'} & E' \\ {\scriptstyle g_U}\downarrow & & \downarrow{\scriptstyle g} \\ E_{/U} & \xrightarrow{j} & E \end{array},$$

There is plainly a canonical isomorphism 459

$$j^*(g_*(X')) \simeq g_{U*}(j'^*(X')).$$

The functor j'^* is also plainly essentially surjective, so condition (*) is equivalent to

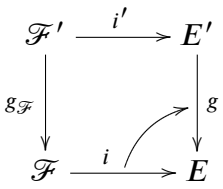
(**) $$g_{U*}(Y') \simeq e_{\mathscr{U}} \text{ for every } Y' \in \mathrm{Ob}\, E'_{/U'}.$$

Using the adjunction between g_{U*} and g_U^*, one sees immediately that this means that $g_U^*(Y)$ is an initial object of $E'_{/U'}$, for every object Y of $E_{/U}$; or, using the fact that the initial object of $E'_{/U'}$ is strict (II 4.5), that it suffices for this to hold for $Y = e_{\mathscr{U}}$, the terminal object of $\mathscr{U}$. But this also means that the terminal object U' of $E'_{/U'}$ is initial, which is manifestly equivalent to U' being an initial object $\phi_{E'}$ of E', C.Q.F.D.

COROLLARY 9.4.3. *Let $g : E' \to E$ be a morphism of topoi, let $U' = g^*(U)$, let $\mathscr{U}' = E'_{/U'}$, and let $\mathscr{F}'$ be the closed subtopos of E' complementary to the open U'; finally, let $i' : \mathscr{F}' \to E'$ be the inclusion morphism. Then the morphism of topoi $g \circ i' : \mathscr{F}' \to E$ factors, up to isomorphism, through a morphism of topoi $g_{\mathscr{F}} : \mathscr{F}' \to \mathscr{F}$, and the*

corresponding diagram of topoi

(9.4.3.1)



is 2*-Cartesian* (cf. 5.11).

This follows formally from 9.4.2 and the definitions.

With the terminology introduced in 9.1.6 d), one may therefore say in particular that the inverse image under a morphism of topoi $g : E' \to E$ of a *closed* subtopos $\mathscr{F}$ of E is a *closed* subtopos $\mathscr{F}'$ of E' (namely the closed subtopos complementary to the open $U' = g^*(U)$, where U is the open of E complementary to $\mathscr{F}$).

Corollary 9.4.4. *Let E be a topos, $\mathscr{U}$ an open subtopos of E, and $\mathscr{F}$ the complementary closed subtopos.*

460 a) *One has*

$$\mathscr{U} \cap \mathscr{F} \cong \mathrm{Top}(\varnothing) \quad , \quad \mathscr{U} \vee \mathscr{F} = E,$$

where $\vee$ denotes the Sup *in the set of all subtopoi of E* (9.1.2 c)). *(In the terminology introduced in 9.1.13, $\mathscr{U}$ and $\mathscr{F}$ are complementary subtopoi of E.)*

b) *$\mathscr{U}$ (resp. $\mathscr{F}$) is the largest among the subtopoi E' of E such that $E' \cap \mathscr{F} \cong \mathrm{Top}(\varnothing)$ (resp. $\mathscr{U} \cap E' \cong \mathrm{Top}(\varnothing)$).*

Proof.

a) The first relation amounts to saying that if E' is a topos and $g : E' \to E$ is a morphism of topoi factoring, up to isomorphism, through both $\mathscr{U}$ and $\mathscr{F}$, then $E' \cong \mathrm{Top}(\varnothing)$. By 9.2.5 and 9.4.2, the hypothesis means that $g^*(U)$ is both an initial and a terminal object of E', which proves the assertion. For the second relation, note that by 9.1.7.2 e) it is equivalent to 9.4.1 c).

b) Suppose that $E' \cap \mathscr{F} \cong \mathrm{Top}(\varnothing)$. By 9.4.3, $E' \cap \mathscr{F}$ is equivalent to the subtopos of E' complementary to $g^*(U)$, where $g : E' \to E$ is the inclusion morphism. Thus the condition means that $U' = g^*(U)$ is a terminal object of E', i.e. (9.2.5) that $E' \subset \mathscr{U}$. Suppose that $\mathscr{U} \cap E' \cong \mathrm{Top}(\varnothing)$. By 5.11, $\mathscr{U} \cap E'$ is equivalent to $E'_{/U'}$, so the condition means that U' is an initial object of E', i.e. (9.4.2) that $E' \subset \mathscr{F}$.

Corollary 9.4.5. *Let $(\mathscr{F}_i)_{i \in I}$ be a family of closed subtopoi of the topos E, and for every $i \in I$ let U_i be the open of E complementary to $\mathscr{F}_i$. Then the intersection of the $\mathscr{F}_i$ as subtopoi of E (9.1.3) is a closed subtopos whose complementary open is $U = \mathrm{Sup}_i U_i$.*

This follows formally from the universal property of the intersection as a 2-product and from 9.4.2, taking into account that for a morphism of topoi $g : E' \to E$, $g^*(U) = \mathrm{Sup}_i g^*(U_i)$ is an initial object if and only if all the $g^*(U_i)$ are initial.

Similarly one proves:

Corollary 9.4.6. *With the notation of 9.4.5, suppose that I is finite. Then the supre-*
461 *mum of the subtopoi $\mathscr{F}_i$ of E* (9.1.2 c)) *is a closed subtopos of E, whose complementary open is $\bigcap_i U_i$.*

Exercise 9.4.7. (*Gluing of subtopoi*). Let E be a topos.

a) Let E' be a subtopos of E, let $(S_i)_{i\in I}$ be a family of objects of E covering the terminal object, and, for every $i \in I$, let S'_i be the inverse image of S_i in E', and let $E'_i \approx E'_{/S'_i}$ be the subtopos of $E_i = E_{/S_i}$ which is the inverse image, under the localization morphism $E_i \to E$, of the subtopos E' of E (9.1.6 d) and 5.11). Show that, for an object X of E, X belongs to E' if and only if, for every $i \in I$, the inverse image $X|S_i$ of X in E_i belongs to E'_i.

b) Suppose that E is of the form $C^\sim$, where C is a $\mathscr{U}$-site. Show that the presheaf on C

$$S \mapsto \text{set of subtopoi of the topos } (C_{/S})^\sim \approx E_{/\varepsilon(S)}$$

is a sheaf. For arbitrary E, deduce that there exists an object of E which represents the functor

$$S \mapsto \text{set of subtopoi of the topos } E_{/S};$$

c) With the notation of a), show that the subtopos E' of E is an open subtopos (resp. a closed subtopos) if and only if, for every $i \in I$, the same is true of the induced subtopos E'_i of E_i.

Exercise 9.4.8. (*Interior, exterior, closure, and boundary of a subtopos.*) Let E be a topos, E' a subtopos, and $i : E' \to E$ the inclusion morphism.

a) Show that $i_*(\phi_{E'})$ is the largest open U of E such that $E'|\mathscr{U} \approx \mathrm{Top}(\varnothing)$, where $\mathscr{U}$ is the open subtopos of E defined by U. This U, or $\mathscr{U}$, is called the *exterior* of the subtopos E' of E, and the closed subtopos of E complementary to U is called the *closure* of E'.

b) Show that there exists a largest open V of E such that the corresponding open subtopos $\mathscr{V}$ of E is contained in E'. (Use 9.2.7.) This V, or $\mathscr{V}$, is called the *interior* of the subtopos E' of E. The opens U and V of E are disjoint ($U \cap V = \phi_E$); the closed subtopos of E complementary to $\mathrm{Sup}(U, V) = \mathscr{U} \amalg \mathscr{V}$ is called the *boundary* of the subtopos E' of E. 462

c) The subtopos E' is an open subtopos (resp. a closed subtopos) of E if and only if it is equal to its interior (resp. if it is equal to its closure). The boundary of E' is empty if and only if E' is both an open and a closed subtopos of E, or equivalently, if it is the open subtopos of E defined by an open V of E which is a *direct summand* of the terminal object e, i.e. such that there exists a subobject U of e with $e \simeq U \amalg V$. The exterior of E' is "empty" i.e. the closure is equal to E, if and only if $i : E' \to E$ is dominant (8.8).

d) Let S be an object of E, let $F = E_{/S}$ be the induced topos, and let $F' \simeq F_{/S'}$ be the subtopos of F induced by the subtopos E' of E. Show that the exterior (resp. the closure, resp. the interior, resp. the boundary) of F' is induced by the exterior (resp. ...) of E'.

Exercise 9.4.9. (*Locally closed subtopoi of a topos.*) A subtopos F of a topos E is called a *locally closed subtopos* of E if it is the intersection of an open subtopos and a closed subtopos.

a) Prove that the intersection of a finite family of locally closed subtopoi of E is locally closed. If $g : E' \to E$ is a morphism of topoi, prove that the inverse image (9.1.6 d)) of a locally closed subtopos of E is a locally closed subtopos of E'.

b) Let X be a topological space, and let $E = \mathrm{Top}(X)$. For every subset X' of X, consider the subtopos $T(X')$ of E which is the essential image of $\mathrm{Top}(X')$ in

Top(X). Show that if X' is a locally closed subset, $T(X')$ is a locally closed subtopos of E, the converse being true if, in addition, X and X' are assumed sober. Prove that $X' \mapsto T(X')$ defines an order isomorphism between the set of locally closed subsets X' of X and the set of locally closed subtopoi of E.

463 c) Let F be a subtopos of E. Show that every open subtopos of F is induced by an open subtopos of E. (Use 9.1.8 c).) Show that F is a locally closed subtopos of E if and only if it is an open subtopos of its closure (9.4.8 a)). Deduce, using 9.4.8 d), that the property for F of being a locally closed subtopos of E is a local property on E.

d) Let F be a locally closed subtopos of E. Show that, among all the ways of writing F as the intersection of an open subtopos $\mathscr{U}$ and a closed subtopos $\mathscr{F}$ of E, there is one for which $\mathscr{U}$ is the largest possible, and $\mathscr{F}$ the smallest possible. (Take $\mathscr{F}$ = the closure of F, and $\mathscr{U}$ = the largest open subtopos inducing on $\mathscr{F}$ the subtopos F.) Show that the formation of $\mathscr{U}$ and $\mathscr{F}$ is compatible with localization on E.

Exercise 9.4.10. Develop the notions of a *constructible, locally constructible, quasi-constructible, and locally quasi-constructible subtopos*, following the model of the familiar notions for topological spaces (EGA 0_{III} 9.1, EGA IV 10.1). For a topos of the form Top(X), one recovers a bijection between the set of constructible subsets (resp. ...) of X, and the set of constructible subtopoi (resp. ...) of Top(X).

9.5. The gluing theorem.

9.5.1. Let $f : A \to B$ be a functor between two categories. Denote by (B, A, f) the following category: the objects of (B, A, f) are the triples

$$(X, Y, u)$$

where X is an object of B, Y an object of A, and u a morphism from X to $f(Y)$; the morphisms between two objects (X, Y, u) and (X', Y', u') are the pairs (m, m'), where m is a morphism from X to X' and m' a morphism from Y to Y', such that the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{u} & f(Y) \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle f(m')} \\ X' & \xrightarrow{u'} & f(Y') \end{array}$$

464 the composition of morphisms is defined in the evident way.

9.5.2. Note that the category (B, A, f) depends functorially on the system of data A, B, f, in a sense whose precise formulation we leave to the reader. In particular, if f' is a second functor from A to B, then every morphism of functors

$$f \overset{u}{\to} f'$$

defines a functor $(B, A, f) \to (B, A, f')$, which is an isomorphism if $u : f \to f'$ is one.

9.5.3. Return to the situation of 9.3, and consider the category $(\mathscr{F}, \mathscr{U}, \rho)$ defined in 9.5.1. To each object X of E there corresponds an object $(i^*X, j^*X, h(X) : i^*X \to \rho j^*X)$ of $(\mathscr{F}, \mathscr{U}, \rho)$, where $h(X)$ is obtained by applying the functor i^* to the adjunction morphism $X \to j_*j^*(X)$ (one has $\rho = i^*j_*$ 9.3.6.3). Thus this defines a functor

$$\Phi : E \longrightarrow (\mathscr{F}, \mathscr{U}, \rho). \tag{9.5.3.1}$$

THEOREM 9.5.4. a) *Let E be a topos, let $\mathcal{U}$ be an open subtopos of E, and let $\mathcal{F}$ be the complementary closed subtopos. The functor*

$$\rho = i^* j_* : \mathcal{U} \longrightarrow \mathcal{F}$$

is left exact and accessible, and the functor Φ (9.5.3.1) is an equivalence of categories.

b) *Conversely, let $\mathcal{U}$ and $\mathcal{F}$ be two topoi, and let $\rho : \mathcal{U} \to \mathcal{F}$ be a gluing functor (9.3.6) (i.e. accessible (I 9.2) and left exact). The category $E = (\mathcal{F}, \mathcal{U}, \rho)$ is a topos. Let $\phi_{\mathcal{F}}$ be an initial object of $\mathcal{F}$, let $e_{\mathcal{U}}$ be a terminal object of $\mathcal{U}$, and let $U = (\phi_{\mathcal{F}}, e_{\mathcal{U}}, \phi_{\mathcal{F}} \to \rho(e_{\mathcal{U}})) \in \mathrm{Ob}\, E$. The functor $X \mapsto (\phi_{\mathcal{F}}, X, \phi_{\mathcal{F}} \to \rho(X))$ induces an equivalence of $\mathcal{U}$ with $E_{/U}$, hence with the open subtopos $\mathcal{U}'$ of E defined by U. The functor $Y \mapsto (Y, e, Y \to \rho(e))$ is a closed embedding (9.3.5), i.e. it induces an equivalence of $\mathcal{F}$ with a closed subtopos $\mathcal{F}'$ of E. The subtopoi $\mathcal{U}'$ and $\mathcal{F}'$ are complementary. Let $\rho' : \mathcal{U}' \to \mathcal{F}'$ be the gluing functor. The* 465
functors ρ and ρ' are compatible with the equivalences described above.

DEFINITION 9.5.5. *Let $\mathcal{U}$ and $\mathcal{F}$ be two topoi, and let $\rho : \mathcal{U} \to \mathcal{F}$ be a gluing functor (9.3.6). The topos $(\mathcal{F}, \mathcal{U}, \rho)$ is called the topos constructed from $\mathcal{U}$ and $\mathcal{F}$ by gluing by means of the functor ρ.*

9.5.6. *Proof of* 9.5.4 a) [13] The first assertion merely recalls 9.4.1 d). The pair of functors (i^*, j^*) is conservative, or equivalently, faithful (9.4.1 c)). A fortiori, the functor Φ is faithful. Let us show that it is fully faithful. It follows from 9.3.3 b) that, for every object X of E, the following diagram is cartesian:

$$(*) \qquad \begin{array}{ccc} X & \longrightarrow & j_* j^* X \\ \downarrow & & \downarrow \\ i_* i^* X & \longrightarrow & i_* i^* j_* j^* X; \end{array}$$

Let X and Y be two objects of E, and suppose that (m, m'):

$$\begin{array}{ccc} i^* X & \longrightarrow & i^* j_* j^* X \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle i^* j_* m'} \\ i^* Y & \longrightarrow & i^* j_* j^* Y; \end{array}$$

is a morphism from $\Phi(X)$ to $\Phi(Y)$. Using the functoriality of the fiber product and the diagrams $(*)$, one obtains a morphism $w : X \to Y$ which makes the following diagrams commutative:

$$\begin{array}{ccc} X & \xrightarrow{w} & Y \\ \downarrow & & \downarrow \\ j_* j^* X & \xrightarrow{j_* m'} & j_* j^* Y, \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{w} & Y \\ \downarrow & & \downarrow \\ i_* i^* X & \xrightarrow{i_* m} & i_* i^* Y \end{array}$$

By applying the functors j^* and i^* respectively, one obtains $j^* w = m'$ and $i^* w = m$. Let 466

[13] For a gluing statement more general than 9.5.5, cf. Exercise[14] [N.D.E.: This exercise does not seem to exist.].

us now show that the functor Φ is essentially surjective. Let $(Y, X, u : Y \to i^* j_*(X))$ be an object of $(\mathscr{F}, \mathscr{U}, i^* j_*)$. From it one obtains a diagram:

$$\begin{array}{ccc} & & j_*X \\ & & \downarrow \\ i_*Y & \xrightarrow{i_*u} & i_*i^*j_*X \end{array}$$

Hence, on taking the fiber product, one obtains a cartesian diagram:

$$(**) \qquad \begin{array}{ccc} W & \longrightarrow & j_*X \\ \downarrow & & \downarrow \\ i_*Y & \xrightarrow{i_*u} & i_*i^*j_*X \end{array}$$

We leave it to the reader to show that $\Phi(W)$ is isomorphic to $(Y, X, u : Y \to i^* j_*(X))$ and that the diagram $(**)$ is isomorphic to the diagram $(*)$ associated with W.

9.5.7. *Proof* of 9.5.4 b). We shall confine ourselves to showing that $(\mathscr{F}, \mathscr{U}, \rho)$ is a topos. The proof of the other assertions is left to the reader. We verify properties a), b), c), d) of 1.1.2.

a) *Finite projective limits are representable*: This is clear because the categories $\mathscr{U}$ and $\mathscr{F}$ are topoi and ρ commutes with finite projective limits.

b) *Direct sums are representable, disjoint, and universal*: Let $(Y_i \xrightarrow{u_i} \rho(X_i))$, $i \in I$, be a family of objects of $(\mathscr{F}, \mathscr{U}, \rho)$. By the definition of inductive limits, there is a canonical morphism

$$\coprod_{i \in I} \rho(X_i) \longrightarrow \rho(\coprod_{i \in I} X_i),$$

467 and hence, by composing with $\coprod u_i$, a morphism

$$n : \coprod_{i \in I} Y_i \longrightarrow \rho(\coprod_{i \in I} X_i).$$

One verifies immediately that this last object is the direct sum of the given family and that this direct sum is disjoint and universal.

c) *Equivalence relations are effective and universal*:

Let

$$\begin{array}{ccc} R_1 & \rightrightarrows & Y \\ \downarrow & & \downarrow w \\ \rho(R_2) & \underset{f(v)}{\overset{f(u)}{\rightrightarrows}} & \rho(X) \end{array}$$

be an equivalence relation on the object $Y \xrightarrow{w} \rho(X)$. Then $R_1 \rightrightarrows Y$ is an equivalence relation on Y in $\mathscr{F}$, and $R_2 \underset{v}{\overset{u}{\rightrightarrows}} X$ is an equivalence relation on X in $\mathscr{U}$. The quotients Y/R_1, $\rho(X)/\rho(R_2)$, and X/R_2 are representable in $\mathscr{F}$ and $\mathscr{U}$ because these categories are topoi. Moreover, by the definition of inductive limits, the canonical morphism $\rho(X) \to \rho(X/R_2)$ factors through $\rho(X)/\rho(R_2)$. By functoriality of inductive limits, one obtains a canonical morphism $Y/R_1 \xrightarrow{t}$

$\rho(X/R_2)$. One verifies that this last object is the quotient of $Y \overset{w}{\longrightarrow} \rho(X)$ by the equivalence relation in question, that this quotient is effective (I 10), and that all these properties are preserved under base change.

d) $(\mathscr{F}, \mathscr{U}, \rho)$ *admits a small generating family.* This property follows from I 9.25 (taking into account that ρ is accessible), by taking $B = \Delta_1 = (0 \overset{f}{\longrightarrow} 1)$, $E_0 = \mathscr{F}$, $E_1 = \mathscr{U}$, and $f^* = \rho : E_1 \to E_0$, whence $\mathbf{Hom}_B(B, E) = (\mathscr{F}, \mathscr{U}, \rho)$.

One may also avoid using I 9.25 (whose proof as given was rather arduous!) and use the hypothesis on ρ in the form that ρ admits a pro-adjoint

$$\sigma : \mathrm{Pro}(\mathscr{F}) \longrightarrow \mathrm{Pro}(\mathscr{U}),$$

(this interpretation (I 8) being based on I 9, whose proof is distinctly more comprehensible). It suffices to note that if C' (resp. C'') is a generating subcategory of $\mathscr{U}$ (resp. $\mathscr{F}$), and if, for every $X'' \in \mathrm{Ob}\, C''$, one represents $\sigma(X'')$ as a projective system $(\sigma(X'')_i)_{i \in I(X'')}$, where $I(X'')$ is a small filtered ordered set, then the family of objects of E which are either of the form $(\phi_{\mathscr{F}}, X', \phi_{\mathscr{F}} \to \rho(X'))$ with $X' \in \mathrm{Ob}\, C'$, or of the form $(X'', \sigma(X'')_i, u_{X''} : X'' \to \rho(\sigma(X'')_i))$, where $X'' \in \mathrm{Ob}\, C''$, $i \in I(X'')$, and $u_{X''}$ corresponds under adjunction to the canonical morphism $\sigma(X'') \to \sigma(X'')_i$, is a generating family (plainly a small one) when C' and C'' are chosen small. The verification is indeed immediate and is left to the reader. 468

. Use the notation of 9.5.7 b), and let us explain how, up to canonical isomorphisms of functors, the system of functors (9.3.6.2) can be made explicit when $E = (\mathscr{F}, \mathscr{U}, \rho)$. The details verifying the assertions below (which are essentially mechanical applications of 9.5.4) are left to the reader.

. The functor

$$i^* : (\mathscr{F}, \mathscr{U}, \rho) \longrightarrow \mathscr{F}$$

is given by

$$i^*(X, Y, u : X \longrightarrow \rho(Y)) = X.$$

. The functor

$$i_* : \mathscr{F} \longrightarrow (\mathscr{F}, \mathscr{U}, \rho)$$

is given by $i_*(X) = (X, e, X \to \rho(e))$, where e is the terminal object of $\mathscr{U}$.

. The adjunction morphism

$$\mathrm{id} \longrightarrow i_* i^*$$

is isomorphic to the functorial morphism

$$\begin{array}{ccc} X & \xrightarrow{u} & \rho(Y) \\ {\scriptstyle id}\downarrow & & \downarrow \\ X & \longrightarrow & \rho(e) \end{array}$$

. The functor 469

$$j^* : (\mathscr{F}, \mathscr{U}, \rho) \longrightarrow \mathscr{U}$$

is given by $j^*(X, Y, u : X \longrightarrow \rho(Y)) = Y$.

. The functor

$$j_! : \mathscr{U} \longrightarrow (\mathscr{F}, \mathscr{U}, \rho)$$

is given by $j_!(Y) = (\phi_{\mathscr{F}}, Y, \phi_{\mathscr{F}} \longrightarrow \rho(Y))$
(where $\phi_{\mathscr{F}}$ is the initial object of $\mathscr{F}$).

. The functor

$$j_* : \mathscr{U} \longrightarrow (\mathscr{F}, \mathscr{U}, \rho)$$

is given by $j_*(Y) = (\rho(Y), Y, \mathrm{id} : \rho(Y) \longrightarrow \rho(Y))$.

. The adjunction morphism

$$j_! j^* \longrightarrow \mathrm{id}$$

is isomorphic to the functorial morphism

$$\begin{array}{ccc} \phi_{\mathscr{F}} & \longrightarrow & \rho(Y) \\ \downarrow & & \downarrow{\scriptstyle \rho(\mathrm{id})} \\ X & \xrightarrow{u} & \rho(Y) \end{array} \quad .$$

. The adjunction morphism

$$\mathrm{id} \longrightarrow j_* j^*$$

is isomorphic to the functorial morphism

$$\begin{array}{ccc} X & \xrightarrow{u} & \rho(Y) \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \rho(\mathrm{id})} \\ \rho(Y) & \xrightarrow{\mathrm{id}} & \rho(Y) \end{array}$$

Exercise 9.5.9. (*Exactness properties of a glued topos.*) Let $\mathscr{U}$, $\mathscr{F}$ be two topoi, let $\rho : \mathscr{U} \to \mathscr{F}$ be a gluing functor, let $E = (\mathscr{F}, \mathscr{U}, \rho)$ be the glued topos, and let $j : \mathscr{U} \to E$ and $i : \mathscr{F} \to E$ be the canonical morphisms of topoi, so that $\rho = i^* j_*$.

a) Prove that i^* commutes with (small) products (i.e. (1.8 or I 8) the functor i^* admits a left adjoint $i_!$) if and only if the same holds for ρ, i.e. (*loc. cit.*) if and only if ρ admits a left adjoint $\sigma : \mathscr{F} \to \mathscr{U}$ (or equivalently if and only if the
470 pro-adjoint of ρ, restricted to $\mathscr{F}$ (9.6.3), factors, up to isomorphism, through $\mathscr{U}$): one then has $\sigma = j^* i_!$. Thus the gluing functors ρ having this property correspond, up to isomorphism, to the functors $\sigma : \mathscr{F} \to \mathscr{U}$ which have a right adjoint, i.e. (1.6) which commute with $\mathscr{U}$-small inductive limits, or equivalently which are continuous. When $\mathscr{F}$ is equivalent to a topos $C^\sim$, where C is a $\mathscr{U}$-site, these functors therefore correspond, up to isomorphism, to continuous functors $C \to \mathscr{U}$ (III 1.2 and III 1.7).

b) Suppose that the functor $i_!$ is defined. Show that $i_!$ commutes with a specified type of small projective limits if and only if $\sigma : \mathscr{F} \to \mathscr{U}$ commutes with them. (For necessity, use the fact that j^* commutes with small projective limits; for sufficiency, use the equivalent description of E in terms of σ as consisting of the triples (Y, X, u) with $Y \in \mathrm{Ob}\,\mathscr{F}$, $X \in \mathrm{Ob}\,\mathscr{U}$, $u : \sigma(Y) \to X$.)

c) Deduce from a) and b) an example of a closed embedding of topoi $i : \mathscr{F} \to E$ such that $i_!$ exists, but commutes neither with fiber products nor with filtered projective limits. (It suffices to find a continuous functor between two topoi which has neither of these two exactness properties.)

Exercise 9.5.10. (*Gluing of topoi of the form $\widehat{C}$.*) Let C be a small category, and let $E = \widehat{C}$, which is a $\mathscr{U}$-topos.

a) Recall that the opens U of E correspond to *sieves* C' of C (I 4.1, 8.4.4), with $E_{/U}$ then being equivalent to $\widehat{C'}$, and the functor $j^* : E \to E_{/U}$ identifying with the restriction functor 5.4, i.e. the inclusion morphism $j : E_{/U} \to E$ being $\widehat{f} : \widehat{C'} \to \widehat{C}$ (4.6.1), where $f : C' \to C$ is the inclusion. Let C'' be the full

subcategory of C complementary to C' (i.e. $\mathrm{Ob}\, C''$ is the complement of $\mathrm{Ob}\, C'$ in $\mathrm{Ob}\, C$), and let $g : C'' \to C$ be the inclusion. Then the closed subtopos $\mathscr{F}$ of $E = \widehat{C}$, complementary to the open subtopos defined by C', is canonically equivalent to $\widehat{C''}$, with the inclusion morphism $i : \mathscr{F} \to E$ identifying with $\hat{g} : \widehat{C''} \to \widehat{C}$.

b) Consider the functor 471

$$h : C'^{0} \times C'' \longrightarrow (\mathrm{Ens}), \quad h(X', X'') = \mathrm{Hom}_C(X', X''), \tag{9.5.10.1}$$

and note that C, together with its full subcategory C', can be reconstructed up to isomorphism from the categories C', C'' and the bifunctor h (the latter may be chosen arbitrarily). (Compare 9.7.1 below.) The datum of h is equivalent to that of a functor $C'' \to \widehat{C'} = \mathbf{Hom}(C'^{0}, (\mathrm{Ens}))$, or equivalently (III 1.2 and III 1.7) to that of a functor

$$\sigma : \widehat{C''} \longrightarrow \widehat{C'} \tag{9.5.10.2}$$

which is continuous, i.e. (1.6) commutes with small inductive limits, or equivalently, admits a right adjoint

$$\rho : \widehat{C'} \longrightarrow \widehat{C''}. \tag{9.5.10.3}$$

Show that ρ is precisely the gluing functor associated with the open subtopos $\widehat{C'}$ of $E = \widehat{C}$.

c) Deduce from a) and b) that if one is given a gluing functor (9.5.10.3) between two presheaf categories, and hence a glued topos E, the following conditions are equivalent:

(i) E is equivalent to a topos of the form $\widehat{C}$.

(ii) The functor ρ commutes with small products (or equivalently, it admits a left adjoint (9.5.10.2)).

(iii) The inclusion morphism $i : \widehat{C''} \to E$ is "essential," i.e. i^* commutes with products, or equivalently i^* admits a left adjoint $i_!$.

When this is so, describe explicitly a category C such that $E \approx \widehat{C}$ using the bifunctor (9.5.10.1) defined by the restriction of (9.5.10.2) to C''.

d) Show that a topos obtained by gluing two punctual topoi need not be equivalent to a topos of the form $\widehat{C}$.

Exercise 9.5.11. (Morphism from a glued topos to a topos.)

a) Let $\rho : \mathscr{U} \to \mathscr{F}$ be a functor between two categories, let $E = (\mathscr{F}, \mathscr{U}, \rho)$ be the "glued" category of 9.5.1, and let F be an arbitrary category. Consider the functor $f \mapsto \rho \circ f$: 472

$$\overline{\rho} : \overline{\mathscr{U}} = \mathbf{Hom}(F, \mathscr{U}) \longrightarrow \overline{\mathscr{F}} = \mathbf{Hom}(F, \mathscr{F}),$$

and define an equivalence of categories

$$\mathbf{Hom}(F, E) \xrightarrow{\approx} (\overline{\mathscr{U}}, \overline{\mathscr{F}}, \overline{\rho}).$$

b) Let $f : F \to E$ be a functor, and consider its composites with the canonical functors $j^* : E \to \mathscr{U}$ and $i^* : E \to \mathscr{F}$. Show that f commutes with a specified type of inductive limits (assumed to be representable in $\mathscr{U}$ and $\mathscr{F}$, and hence in E) if and only if $j^* f$ and $i^* f$ commute with them. Show that if ρ commutes with a specified type of projective limits (which is then representable in E), then f commutes with them if and only if the same holds for $j^* f$ and $i^* f$.

c) Now suppose that $\mathscr{U}$ and $\mathscr{F}$ are topoi, and that ρ is a gluing functor. Let F be a topos and $f : F \to E$ a functor. Conclude from b) that f is an inverse-image functor associated with a morphism of topoi $E \to F$ if and only if the same holds for the functors $j^*f : F \to \mathscr{U}$ and $i^*f : F \to \mathscr{F}$. Conclude from this and from a) that there is an equivalence between the category **Homtop**(E, F) and the category of triples (f', f'', u), where f' (resp. f'') is an object of **Homtop**$(\mathscr{U}, F)$ (resp. **Homtop**$(\mathscr{F}, F)$), and where $u : f''^* \to \rho \circ f'^*$ is a homomorphism of functors $F \to \mathscr{F}$: if f is an object of **Homtop**(E, F) and (f', f'', u) is the corresponding triple, then $f' = f \circ j$, $f'' = f \circ i$.

d) Specify in what sense c) may be said to provide a characterization, up to equivalence (in a 2-category of topoi), of the topos E in terms of the triple $(\mathscr{U}, \mathscr{F}, \rho)$.

Exercise 9.5.12. (*Gluing of two topological spaces.*)

a) Let X be a topological space, let $\rho : \mathrm{Top}(X) \to (\mathrm{Ens}) = \mathrm{Top}(pt)$ be a gluing functor, i.e. (I 8) a pro-representable functor, let $A \in \mathrm{Ob}\,\mathrm{Pro}(\mathrm{Top}(X))$ be the object which pro-represents it, and let E be the topos obtained from ρ by gluing. Show that E is equivalent to a topos of the form $\mathrm{Top}(Y)$ if and only if A is
473 isomorphic to an object of $\mathrm{Pro}(\mathrm{Ouv}(X))$. Conclude that if X is nonempty, one can find ρ such that E is not equivalent to a topos of the form $\mathrm{Top}(Y)$.

b) More generally, consider a gluing functor $\rho : \mathrm{Top}(X) \to \mathrm{Top}(Y)$, where X, Y are two topological spaces. Deduce from it a map

$$\phi : X \longrightarrow \mathrm{Ob}\,\mathrm{Pro}(\mathrm{Top}(Y))$$

(the composite of $X \to \mathrm{Ob}\,\mathbf{Point}(\mathrm{Top}(X)) \to \mathrm{Ob}\,\mathrm{Pro}(\mathrm{Top}(X))$ (6.8.5) and the pro-adjoint σ of ρ). (N.B. for $x \in X$, $\phi(x)$ identifies with the *fiber at x of the gluing sheaf* (9.6.4) *defined by ρ*.) Show that if E is equivalent to a topos of the form $\mathrm{Top}(Z)$, then ϕ takes its values in the essential image of $\mathrm{Pro}(\mathrm{Ouv}(Y))$.

Problem. Is the converse true?

9.6. Gluing sheaf. We saw in 9.5.4 that the datum of a topos E equipped with an open U amounts essentially to that of two topoi $\mathscr{U}$ and $\mathscr{F}$ and a gluing functor

$$\rho : \mathscr{U} \longrightarrow \mathscr{F} \tag{9.6.1}$$

i.e. a functor admitting a pro-adjoint

$$\sigma : \mathrm{Pro}(\mathscr{F}) \longrightarrow \mathrm{Pro}(\mathscr{U}). \tag{9.6.2}$$

By the sorites on pro-adjoint functors (I 8), the functor ρ is determined up to unique isomorphism (and consequently the glued topos $E = (\mathscr{F}, \mathscr{U}, \rho)$ is determined up to equivalence of topoi) once the functor σ is known, or, equivalently, once the functor induced by σ is known:

$$\sigma_\circ : \mathscr{F} \longrightarrow \mathrm{Pro}(\mathscr{U}). \tag{9.6.3}$$

The functor σ is determined, up to unique isomorphism, by the property of commuting with small filtered projective limits and of extending the functor $\sigma_\circ$. On the other hand, for a given functor (9.6.3) to be isomorphic to a pro-adjoint, it is plainly necessary and sufficient that, for every object X of $\mathscr{U}$, the contravariant functor

$$Y \longmapsto \mathrm{Hom}_{\mathrm{Pro}(\mathscr{U})}(\sigma_\circ(Y), X)$$

474 on $\mathscr{F}$ be representable, or equivalently be a sheaf on $\mathscr{F}$ for the canonical topology (1.2

iii)). It follows at once that this also means that the opposite functor

$$\sigma_\circ^\circ : \mathscr{F}^\circ \longrightarrow \mathrm{Pro}(\mathscr{U})^\circ \subset \check{\mathscr{U}} = \mathbf{Hom}(\mathscr{U}, (\mathrm{Ens})) \tag{9.6.4}$$

is a *sheaf* on $\mathscr{F}$ with values in $\mathrm{Pro}(\mathscr{U})^\circ$ (II 6.1). Thus, *the datum of a gluing functor* (9.6.1) *is also equivalent* (up to isomorphism) *to that of a sheaf on* $\mathscr{F}$ *with values in* $\mathrm{Pro}(\mathscr{U})^\circ$. The sheaf thus associated with a gluing functor (or with a situation $(E, U, \mathscr{F})$ as in (9.3)) is called the *gluing sheaf* associated with the gluing functor ρ under consideration (resp. with the open U of the topos E).

Exercise 9.6.5. Refine 9.5.4 into a statement of 2-equivalence of 2-categories between the 2-category consisting of pairs (E, U), where E is a topos and U an open of the latter, and the 2-category consisting of triples $(\mathscr{U}, \mathscr{F}, \rho)$, where $\mathscr{U}$ and $\mathscr{F}$ are topoi and $\rho : \mathscr{U} \to \mathscr{F}$ is a gluing functor (the explicit description of the 2-categories in question being left to the reader). Give a variant of this statement involving the 2-category of triples $(\mathscr{U}, \mathscr{F}, \phi)$, where ϕ is now a sheaf on $\mathscr{F}$ with values in $\mathrm{Pro}(\mathscr{U})^\circ$.

9.7. Points of a glued topos.

9.7.1. Let C be a category, and let

$$u : D' \longrightarrow C \quad , \quad v : D'' \longrightarrow C$$

be two fully faithful functors such that the essential images C' and C'' of u and v satisfy

$$\mathrm{Ob}\, C' \cap \mathrm{Ob}\, C'' = \phi \quad , \quad \mathrm{Ob}\, C' \cup \mathrm{Ob}\, C'' = \mathrm{Ob}\, C.$$

Suppose, in addition, that

$$X' \in \mathrm{Ob}\, C' \quad , \quad X'' \in \mathrm{Ob}\, C'' \Rightarrow \mathrm{Hom}(X'', X') = \phi,$$

or, equivalently, that

$$\mathrm{Hom}(v(A''), u(A')) = \phi \text{ for } A' \in \mathrm{Ob}\, D', A'' \in \mathrm{Ob}\, D''.$$

It is then immediate that the category C can be reconstructed, up to canonical isomorphism, from the categories C' and C'' and the functor 475

$$(X', X'') \longmapsto \mathrm{Hom}(X', X'') : C'^\circ \times C'' \longrightarrow (\mathrm{Ens});$$

similarly, C can be reconstructed up to equivalence (this equivalence being defined up to unique isomorphism) from D', D'', and the functor

$$(A', A'') \longmapsto \mathrm{Hom}(u(A'), v(A'')) : D'^\circ \times D'' \longrightarrow (\mathrm{Ens}).$$

This observation applies in particular to the situation described in the following proposition, and shows that the category $\mathbf{Point}(E)$ is determined, up to equivalence, by the categories $\mathbf{Point}(\mathscr{U})$ and $\mathbf{Point}(\mathscr{F})$ and by a certain functor

$$\mathbf{Point}(\mathscr{U})^\circ \times \mathbf{Point}(\mathscr{F}) \longrightarrow (\mathrm{Ens})$$

derived from the gluing functor $\rho : \mathscr{U} \to \mathscr{F}$.

Proposition 9.7.2. *Let E be a topos, let U be an open of E, let $\mathscr{U} = E_{/U}$, and let $\mathscr{F}$ be the closed subtopos of E complementary to U. Consider the functors $p \mapsto p \circ j$ and $q \mapsto q \circ i$ induced by the inclusion morphisms $j : \mathscr{U} \to E$ and $i : \mathscr{F} \to E$:*

$$u : \mathbf{Point}(\mathscr{U}) \longrightarrow \mathbf{Point}(E), \quad v : \mathbf{Point}(\mathscr{F}) \longrightarrow \mathbf{Point}(E). \tag{9.7.2.1}$$

Then the following assertions hold:

a) *The preceding functors u and v are fully faithful, and every point of E belongs to the essential image of exactly one of the functors u and v.*

b) *Let p and q be two points of E such that p (resp. q) belongs to the essential image of u (resp. of v). Then*

(9.7.2.2) $$\mathrm{Hom}(q, p) = \phi.$$

c) *Let p (resp. q) be a point of $\mathscr{U}$ (resp. of $\mathscr{F}$). There is then an isomorphism, functorial in (q, p):*

(9.7.2.3) $$\mathrm{Hom}(u(p), v(q)) \simeq \mathrm{Hom}(\mathrm{Pro}(\rho)(p), q) \simeq \mathrm{Hom}(p, \sigma(q)),$$

where

$$\sigma : \mathrm{Pro}(\mathscr{F}) \longrightarrow \mathrm{Pro}(\mathscr{U})$$

is the functor (9.6.2), pro-adjoint to the gluing functor $\rho : \mathscr{U} \to \mathscr{F}$, and where, in writing

$$\rho : \mathscr{U} \longrightarrow \mathscr{F},$$

476 *we have identified (up to equivalence) the categories* **Point**$(\mathscr{U})$ *and* **Point**$(\mathscr{F})$ *with full subcategories of* $\mathrm{Pro}(\mathscr{U})$ *and* $\mathrm{Pro}(\mathscr{F})$*, respectively (6.8.5).*

Proof

a) The full faithfulness of u and v is known (9.1.4). The assertion concerning the essential images of u and v follows from the criteria 9.2.5 and 9.2.4 for a point $P \to E$ to factor, up to isomorphism, through $\mathscr{U}$ resp. $\mathscr{F}$, taking into account the fact that the punctual topos P has only two opens, namely the initial object and the terminal object of P.

b) In view of a), the assertion means that if the point $p : P \to E$ factors through $\mathscr{U}$, then the same holds for every point $p' : P \to E$ such that $\mathrm{Hom}(p', p) \neq \phi$. Now, if $\alpha : p' \to p$, it induces $p^*(e_{\mathscr{U}}) \to p'^*(e_{\mathscr{U}})$, and then $p^*(e_{\mathscr{U}}) = e_p$ implies $p'^*(e_{\mathscr{U}}) = e_p$, C.Q.F.D.

c) By the lemma 9.7.2.4 below, applied to the inclusions $j : \mathscr{U} \to E$ and $i : \mathscr{F} \to E$, there is a diagram of functors commuting up to canonical isomorphisms:

$$\begin{array}{ccccc} \mathrm{Point}(\mathscr{U}) & \xrightarrow{u} & \mathrm{Point}(E) & \xleftarrow{v} & \mathrm{Point}(\mathscr{G}) \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Pro}(\mathscr{U}) & \xrightarrow{\mathrm{Pro}(j_*)} & \mathrm{Pro}(E) & \xleftarrow{\mathrm{Pro}(i_*)} & \mathrm{Pro}(\mathscr{G}), \end{array}$$

where the vertical arrows denote the canonical fully faithful functors (6.8.5). Moreover, since i_* admits a left adjoint i^*, $\mathrm{Pro}(i_*)$ admits the left adjoint $\mathrm{Pro}(i^*)$ I 8.11.5, and one finally obtains the functorial isomorphisms

$$\begin{aligned} \mathrm{Hom}(u(p), v(q)) &\simeq \mathrm{Hom}_{\mathrm{Pro}(E)}(\mathrm{Pro}(j_*)(p), \mathrm{Pro}(i_*)(q)) \\ &\simeq \mathrm{Hom}_{\mathrm{Pro}(\mathscr{F})}(\mathrm{Pro}(i^*)(\mathrm{Pro}(j_*)(p)), q) \\ &\simeq \mathrm{Hom}_{\mathrm{Pro}(\mathscr{F})}(\mathrm{Pro}(i^* j_*)(p), q), \end{aligned}$$

which is precisely the first formula in (9.7.2.3), the second then being immediate from the definition of σ as a pro-adjoint of ρ. This proves 9.7.2, subject to the following lemma.

Lemma 9.7.2.4. *Let $f : F \to E$ be a morphism of topoi. Then the diagram of functors*

(9.7.2.4.1)
$$\begin{array}{ccc} \mathrm{Point}(F) & \xrightarrow{p \mapsto f \circ p} & \mathrm{Point}(E) \\ \downarrow & & \downarrow \\ \mathrm{Pro}(F) & \xrightarrow{\mathrm{Pro}(f_*)} & \mathrm{Pro}(E) \end{array}$$

commutes up to canonical isomorphism (the vertical arrows denoting the fully faithful functors 6.8.5). 477

The verification is immediate from the definitions and is left to the reader.

COROLLARY 9.7.3. *Let E be a topos having enough points. Then every closed subtopos $\mathscr{F}$ of E also has enough points. More precisely, if $(p_i)_{i\in I}$ is a conservative family of points of E, then the family of points of $\mathscr{F}$ defined by the subfamily $(p_i)_{i\in J}$ consisting of those p_i which come from points of $\mathscr{F}$ is conservative. (Compare 6.7.3, 6.7.4.)*

Indeed, let $f : X \to Y$ be an arrow of $\mathscr{F}$ which is carried to an isomorphism by the fiber functors associated with the points under consideration of $\mathscr{F}$. Since $i^*i_* \simeq \mathrm{id}_{\mathscr{F}}$, it follows that $i_*(f) : i_*(X) \to i_*(Y)$ is carried to an isomorphism by the fiber functors associated with the p_i for $i \in J$; the same is true for the p_i with $i \in I - J$, i.e. those coming from points of U, since j^*i_* is the constant functor with value the terminal object. Therefore $i_*(f)$ is an isomorphism, and hence so is $f \simeq i^*i_*(f)$, C.Q.F.D.

REMARK 9.7.4. If E is a topos having enough points, it is clear that an open U of E is determined once the full subcategory of $\mathbf{Point}(E)$ which is the essential image of $\mathbf{Point}(E_{/U})$ is known; in fact, if C is a full subcategory of $\mathbf{Point}(E)$ defining a conservative family of points of E, it even suffices to know the subcategory $C_{/U}$ consisting of those elements of C which belong to the essential image of $\mathbf{Point}(U)$. It follows from this and from 9.7.2 a) that a closed subtopos $\mathscr{F}$ of E is determined once the essential image of $\mathbf{Point}(\mathscr{F})$ in $\mathbf{Point}(E)$ is known, or even merely its intersection with C.

EXERCISE 9.7.5. Let E be a topos. For every subtopos F of E, let $P(F)$ be the subset of $\mathrm{Point}(E)$ formed by the isomorphism classes of points of E which factor through F. Thus $P(F)$ is homeomorphic to $\mathrm{Point}(F)$ (9.1.8 c)). Show that if F is an open (resp. closed, resp. locally closed (9.4.9)) subtopos of E, then $P(F)$ is an open (resp. closed, resp. locally closed) subset of $\mathrm{Point}(E)$. Show that when E has enough points, the same is true of every locally closed subtopos of E, and the map $F \mapsto P(F)$ induces a bijection between the set of open (resp. closed, resp. locally closed) subtopoi of E and the set of open (resp. closed, resp. locally closed) subsets of $\mathrm{Point}(E)$. Extend the preceding considerations to the case where $\mathrm{Point}(E)$ is replaced by any subspace X of $\mathrm{Point}(E)$ corresponding to a conservative family of points of E. 478

9.8. Complements concerning certain topoi associated with topological spaces. In this section we give a few complements in the form of exercises. The reader is advised to look through them in order to become accustomed to the viewpoint of topoi in various classical situations.

EXERCISE 9.8.1. (*Descent of sheaves on a topological space.*)
Let $f : X \to Y$ be a continuous map of topological spaces in $\mathscr{U}$, giving a functor

$$f^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X).$$

a) The following conditions are equivalent:
   (i) f^* is faithful.
   (ii) f^* is conservative.
   (iii) $f(X)$ is a very dense subset (EGA IV 10.1.3) of Y.
   It suffices for f or f_{sob} to be surjective. Give an example where f^* is faithful and $f = f_{\mathrm{sob}}$ is not surjective (take X discrete, f injective, $f(X)$ the set of closed points of Y, and Y a nondiscrete Noetherian topological space). Give an example where f_{sob} is surjective but f is not, and another where f is surjective but f_{sob} is not.

b) Suppose that $f(X)$ is very dense in Y and that its topology is the quotient topology induced from X. Prove that the arrow f of (Esp) is a descent morphism for the fibered category of sheaves of sets on varying spaces, i.e. that the functor
479 f^* induces a *fully faithful* functor from $\mathrm{Top}(Y)$ to the category of objects of $\mathrm{Top}(X)$ equipped with descent data relative to $f : X \to Y$. Reduce to the case where f is surjective and follow the argument of VIII 9.1 given in the setting of étale topologies of schemes. Give a converse.

c) Suppose that $f(X)$ is very dense in Y, that its topology is the quotient topology induced from X, and that the fibers of f are connected. Prove that the functor f^* is fully faithful. (Follow the argument of SGA 1 IX 3.4.) Give a converse, at least if the points of Y are closed.

d) Suppose that Y is the union of open subsets Y_i over which X admits sections. Prove that f is then an *effective descent morphism* for the fibered category of sheaves of sets on varying topological spaces, i.e. that the functor considered in c) is in fact an equivalence of categories.

Exercise 9.8.2. (*Quotient topoi of topological spaces. Topological étendues.*)

a) Let

(9.8.5.1) $$X_2 \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; X_1 \;\underset{p_2}{\overset{p_1}{\rightrightarrows}}\; X_0$$

be a semi-simplicial object truncated in degree 2 in the category (Esp) of topological spaces $\in \mathscr{U}$; the inverse-image functors then give a diagram of categories of sheaves

$$\mathrm{Top}(X_0) \;\underset{p_2^*}{\overset{p_1^*}{\rightrightarrows}}\; \mathrm{Top}(X_1) \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \mathrm{Top}(X_2).$$

(More precisely, one has a category cofibered over the category of standard simplices Δ_n, $0 \leq n \leq 2$.) Show that the category $\varprojlim$ of this diagram (i.e. the category of sheaves of sets on X_0 *equipped* with a descent datum relative to the diagram 9.8.2.1, i.e. with an isomorphism $p_1^*(F) \simeq p_2^*(F)$ satisfying...), is a $\mathscr{U}$-topos T. (For a more general statement on the stability of topoi under $\varprojlim$ operations, cf. VI 8.) Define a morphism of topoi $q : \mathrm{Top}(X_0) \to T$, and prove
480 that for every $\mathscr{U}$-topos E, $\mathbf{Homtop}(T, E)$ is equivalent, via q^*, to the category $\varprojlim$ of the following diagram of categories, deduced from (9.8.2.1):

$$\mathbf{Homtop}(\mathrm{Top}(X_0), E) \;\substack{\longrightarrow\\ \longrightarrow}\; \mathbf{Homtop}(\mathrm{Top}(X_1), E) \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \mathbf{Homtop}(\mathrm{Top}(X_2), E).$$

b) In particular, let R be an equivalence relation in an object X of (Esp), i.e. (I 10.9) a subobject R of $X \times X$ (N.B. the inclusion $R \hookrightarrow X \times X$ is continuous, but the topology of R is not necessarily induced by that of $X \times X$), such that, for every object Y of (Esp), $\mathrm{Hom}(Y, R)$ is the graph of an equivalence relation in $\mathrm{Hom}(Y, X)$. This gives a diagram of the form (9.8.2.1), with $X_0 = X$, $X_1 = R$, $X_2 = (R, p_2) \times_X R, p_1$, where p_1, p_2 are the two projections from R to X, and hence a topos, which we denote by $\mathrm{Top}(X)/R$, or even $\mathrm{Top}(X/R)$, or indeed X/R, by abuse of notation, and which plays the role of a quotient of X by R in the sense made precise in a). Suppose that the topology of R is induced by that of $X \times X$ and that, denoting by X/R the ordinary quotient topological space, X

locally admits sections over X/R. Prove that T is then equivalent to $\mathrm{Top}(X/R)$. (Use 9.8.1 d).)

c) Let $H \to X$ be an injective morphism of topological groups, i.e. a monomorphism of group objects in (Esp), and let R be the equivalence relation that it defines in X, i.e. $R = H \times X$, with $p_1 = \mathrm{pr}_1$ and p_2 defined by the action of H on X by left translations. Show that the topology of R is induced by that of $X \times X$ if and only if the topology of H is induced by that of X. Give examples in which this condition is not satisfied and in which the ordinary quotient topology on X/H is the indiscrete topology, with a) $H = \mathbb{Z} \times \mathbb{Z}$, $X = \mathbb{R}$, and b) $H = \mathbb{R}$, $X = \mathbb{T} \times \mathbb{T}$, where $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ ("geodesics on the torus"). Prove that the two topoi so obtained are equivalent and are not equivalent to $\mathrm{Top}(X/H)$ (which is a final topos (2.2)), nor to any topos of the form $\mathrm{Top}(Y)$.

d) Let G be a topological group acting on a topological space X, whence, in the familiar way, a semi-simplicial object

$$\ldots G \times G \times X \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; G \times X \;\substack{\longrightarrow\\ \longrightarrow}\; X.$$

Show that the topos T deduced from it by a) identifies with the category of spaces X' over X having G as a group of operators, such that $X' \to X$ is an étale map (compatible with the action of G). In particular, when G is discrete, one recovers the topos $\mathrm{Top}(X, G)$ of 2.3. 481

The preceding considerations justify the notation $\mathrm{Top}(X)/G$, or indeed $\mathrm{Top}(X/G)$ or even simply X/G, for the preceding topos T. One should bear in mind, however, that when G is a discrete group (to fix ideas) acting *properly* on X, so that the ordinary quotient space X/G has reasonably good properties, the natural morphism of topoi $T \to \mathrm{Top}(X/G)$ deduced from the universal characterization a) of T is an equivalence of topoi only if G acts freely, i.e. without fixed points, i.e. when $G \times X \to X \times X$ is a monomorphism. Thus, in the case of a "pre-equivalence relation" (or "groupoid" in the sense of (SGA 3 V 1)) which is not an equivalence relation, passage to the quotient in the sense of topoi (or "passage to the fine quotient") does not in general correspond (via the correspondence $Y \mapsto \mathrm{Top}(Y)$) to the usual topological quotient.

e) Let T be a $\mathscr{U}$-topos. Show that the following conditions are equivalent:

(i) There exists a family $(S_i)_{i \in I}$ of objects of T covering the final object of T, such that for every $i \in I$ the induced topos $T_{/S_i}$ is equivalent to a topos of the form $\mathrm{Top}(X)$, with $X \in$ (Esp).

(i′) There exists an $S \in \mathrm{Ob}\, T$ covering the final object such that the induced topos $T_{/S}$ is equivalent to a topos of the form $\mathrm{Top}(X)$.

(ii) There exist a topological space X and an *étale* pre-equivalence relation R in X (in the sense of the category (Esp)) (i.e. one for which $p_1 : R \to X$ is an étale map), such that T is equivalent to $\mathrm{Top}(X/R)$, with the notation of b).

The topos T is then said to be *locally a topological space*, or alternatively a *topological étendue* (or simply an *étendue*, when no confusion is likely). 482

f) Show that the topos T constructed in a) has enough points and, more precisely, that it admits a conservative family of points parametrized by the (small) set X_0. In particular, every étendue has a small conservative family of points, and hence has enough points. When an étendue T is represented in the form $\mathrm{Top}(X/R)$ as in e) ii), prove that every point of T is isomorphic to the image of a point of

Top(X), and is therefore defined by an ordinary point of X_{sob} (or of X itself, when X is sober).

g) Show that the topos Top(X/G) of 2.3 is an étendue. (Use its description in d) or 7.1.10 c).) Show that, for every $x \in X$, the monoid of endomorphisms of the point of Top$(X, G) =$ Top$(X)/G$ defined by x is canonically isomorphic to the stabilizer group G_x of x. In particular, the points of an étendue may have nontrivial automorphism groups. Determine **Point**(T) for the étendue T defined by an étale pre-equivalence relation in a space X, and show that every endomorphism of a point of T is an automorphism.

h) Let T be a topos. Show that the following conditions are equivalent:

  i) For every point p of T, every endomorphism of p (resp. every automorphism of p) is the identity.

  ii) For every topos S having enough points and every morphism of topoi $f : S \to T$, every endomorphism of f (resp. every automorphism of f) is the identity.

Likewise, show that the following conditions are equivalent:

  i′) For two points p, q of T, there exists at most one morphism from p to q.

483 ii′) For two morphisms $f, g : S \rightrightarrows T$ from a topos S having enough points to T, there exists at most one morphism from f to g.

When these latter conditions hold, the topos T is said to be *pointwise rigid*, or simply *rigid*. Show that if X is a topological space equipped with an étale pre-equivalence relation R, the quotient étendue $T =$ Top$(X)/R$ is rigid if and only if R is an equivalence relation.

i) Let (X, G) be a topological space with a discrete group of operators, let H be a normal subgroup of G such that the induced action of H on X is proper and free, and let $X' = X/H$, $G' = G/H$, so that G' acts on X' in the evident way. Prove that the morphism of topoi

$$\mathrm{Top}(X, G) \longrightarrow \mathrm{Top}(X', G')$$

induced by the canonical morphism $(X, G) \to (X', G')$ of spaces with operators (4.1.2) is an equivalence of topoi.

Show that the set of open subtopoi of Top(X/G) identifies with the set of open subsets of X stable under the action of G, or equivalently with the set of open subsets of the quotient topological space X/G. Deduce that Top(X/G) is connected if and only if every subset of X which is both open and closed and stable under the action of G is either empty or equal to X. When the connected components of X are open, Top(X/G) is connected if and only if G acts transitively on the set $\pi_0(X)$ of connected components of X; more generally, π_0(Top(X/G)) (8.7 g)) is an essentially constant pro-set isomorphic to the quotient set $\pi_0(X)/G$.

Show that when X is locally connected and locally simply connected, and $T =$ Top(X, G) is connected, i.e. G is transitive on $\pi_0(X)$, then among all the ways of representing T by means of a space with operators (X', G') (cf. above for some such representations), there is one, unique up to nonunique isomorphism, for which X' is connected and simply connected. Show that choosing such an (X', G') amounts to choosing a "universal covering" of the final object of T, and that G' is isomorphic to the group $\pi_1(T)$ relative to this universal covering (cf. 2.7.5).

484 (Hint: show that giving an equivalence of T with a Top(X, G) amounts to

giving a G_T-torsor P in T, and that, if (X, G) and P correspond, there is a canonical equivalence $\mathrm{Top}(X) \simeq T/P$.)

Deduce from this a trivial proof of the last assertion of c). Characterize the topoi T equivalent to a $\mathrm{Top}(X, G)$, where G is a discrete group acting on a locally connected and locally simply connected topological space while transitively permuting its connected components, as the connected, locally connected, and locally simply connected étendues whose universal covering P of the final object e_T of T is such that the induced topos $T_{/P}$ is equivalent to a topos of the form $\mathrm{Top}(X)$ (or, as we shall simply say by abuse of language, $T_{/P}$ "is" a topological space).

Give an example of an étendue, locally isomorphic to $\mathbb{R}$, which is connected and simply connected but is not a topological space. (Take the universal covering of the line $\mathbb{R}$ with doubled origin.)

j) A ringed topos $(T, \mathcal{O}_T)$ (cf. § 11) is called a *differentiable étendue* (resp. a *real-analytic étendue*, resp. a *complex-analytic étendue*) if there exist objects S_i of T covering the final object such that the induced ringed topos $(T_{/S_i}, \mathcal{O}_T/S_i)$ is equivalent to the ringed topos defined by a differentiable manifold with its sheaf of real C^∞ functions (resp. to the ringed topos defined by a real-analytic space, resp. to the ringed topos defined by a complex-analytic space). Give a constructive description of these ringed topoi of the type in e) ii) (taking for X, respectively, a differentiable manifold, a complex-analytic space, or a real-analytic space). Give examples of such ringed topoi which are not equivalent to ringed topoi associated with topological spaces, by returning to the examples considered in c).

EXERCISE 9.8.3. (*Topoi associated with local equivalence relations and with foliations*) 485
[14].

Let X be a topological space.

a) For every open subset U of X, let $\mathrm{Quot}(U)$ be the set of equivalence relations on U, and, for an open subset $V \subset U$, consider the natural map $\mathrm{Quot}(U) \to \mathrm{Quot}(V)$. This gives a presheaf Quot_X on X. Let Q_X, or simply Q, be the associated sheaf, and let r be a section of Q (a "*local equivalence relation on X*"). From the order relation on the set $\mathrm{Quot}(U)$ of equivalence relations on an open subset U of X, deduce on the presheaf Quot, and on the associated sheaf Q, an order structure, i.e. a structure of a presheaf resp. a sheaf with values in the category of ordered sets.

b) For every equivalence relation R on X, consider the section $\mathrm{loc}(R)$ of Q that it defines. Let $E(r)$ be the set of equivalence relations on X such that $\mathrm{loc}(R)$ is less fine than r, and let $\mathrm{glob}(r)$ be the least upper bound of the equivalence relations in $E(r)$ (its graph is the intersection of the graphs of the $R \in E(r)$). Let $(U_x)_{x\in X}$ be a family of open neighborhoods of the points $x \in X$, and, for every $x \in X$, let R_x be an equivalence relation on U_x whose germ at x is r_x. For every family $\mathscr{V}$ of open subsets V_x ($x \in X, x \in V_x \subset U_x$), let $R_{\mathscr{V}}$ be the equivalence relation on X generated by the family of relations $R_x|V_x$. Show that if $\mathscr{V}' \leq \mathscr{V}$ (in the obvious sense), then $R_{\mathscr{V}'} \geq R_{\mathscr{V}}$, and that $\mathrm{glob}(r)$ is the least upper bound of the increasing filtered family of equivalence relations $R_{\mathscr{V}}$. Give an example in

[14] This exercise is given subject to every reservation, since it was written hastily and insufficiently checked. Mr. N. SAAVEDRA checked parts a) through e). The reader is requested to communicate any observations to us.

which glob(r) does not belong to $E(r)$, i.e. in which there is an $a \in X$ such that,
486 for every open neighborhood $W \subset U_a$ of a, there exist a $\mathscr{V} = (V_x)$ and two points y, z of W which are equivalent under R_a but not under $R_{\mathscr{V}}$. Show that, for every $R \in \mathrm{Quot}(X)$, one has $\mathrm{glob}\,\mathrm{loc}(R) \geq R$.

c) The local equivalence relation r is said to be *precoherent (resp. coherent)* if $\mathrm{glob}(r) \in E(r)$, i.e. if $\mathrm{loc}(\mathrm{glob}(r)) \leq r$ (resp. if, for every open subset U of X, the restriction of r to U is precoherent). It is said to be globally coherent if it is coherent and if, in addition, $r = \mathrm{loc}\,\mathrm{glob}(r)$. An equivalence relation R on X is said to be *locally coherent* if $r = \mathrm{loc}(R)$ is coherent, and *coherent* if, in addition, $R = \mathrm{glob}(r)$, i.e. if $R = \mathrm{glob}(\mathrm{loc}(R))$. Show that globally coherent local equivalence relations r on X correspond bijectively to coherent equivalence relations R on X, by $r \mapsto \mathrm{glob}(r)$ and $R \mapsto \mathrm{loc}(R)$.

d) Show that, in order for r to be coherent, it is enough that, for every $a \in X$ and every open neighborhood $W' \subset U_a$ of a, there exist an open neighborhood $W \subset W'$ of a such that every equivalence class of $R_a|W$ is contained in a connected component of an equivalence class of $R_a|W'$ (a fortiori, it is enough that there exist a fundamental system of open neighborhoods $W_a \subset U_a$ of a such that the fibers of the induced equivalence relation $R_a|W_a$ are connected). (Hint: reduce to proving precoherence in the case where r is defined by an $R \in \mathrm{Quot}(X)$, and observe in that case that the fibers of the equivalence relations $R_{\mathscr{V}}$ are relatively open—and hence also relatively closed—subsets of the fibers of R.) Show that, in order for a global equivalence relation R to be coherent, it is enough that it be locally coherent and that its fibers be connected; prove that this condition is necessary if the fibers are closed. Give an example of an equivalence relation with connected and locally connected fibers which is not locally coherent (?).

487 e) Let $f : X' \to X$ be a continuous map. Define a natural homomorphism of presheaves $f_*(\mathrm{Quot}_{X'}) \leftarrow \mathrm{Quot}_X$, inducing a homomorphism of sheaves $f_*(Q_{X'}) \leftarrow Q_X$, and hence a homomorphism $f^*(Q_X) \to Q_{X'}$. If r is a local equivalence relation on X, then f is said to be a "*fiber map*" for the local equivalence relation r on X if the inverse image of r is the coarse local equivalence relation on X'. For every topological space X', let $\mathrm{Homfib}_r(X', X)$ be the set of continuous maps from X' to X which are fiber maps relative to r. Show that, as X' varies, this defines a contravariant functor in X'.

f) Suppose that r is coherent. Show that the preceding functor is representable by a topological space X^r over X. Show that the universal fiber map $\Phi : X^r \to X$ is bijective, and that every $x \in X^r$ has an open neighborhood U such that the map $U \to X$ induced by Φ is a homeomorphism of U onto its image. Show that the connected components of X^r are open and correspond under the bijection Φ to the fibers of the equivalence relation $R = \mathrm{glob}(r)$. Give an example in which the restriction of Φ to the connected components of X^r does not induce homeomorphisms from those spaces onto their images in X.

g) The local equivalence relation r is said to be *open* if it is a section of the subsheaf Qouv_X of Q arising from the subpresheaf $\mathrm{Quotouv}_X$ of Quot_X whose value on every open subset U of X is the set of *open* equivalence relations on U. The local equivalence relation r is said to be *strictly open* if it is coherent and open or, equivalently, if it is coherent and if, for every open subset U of X, $\mathrm{glob}(r|U)$ is an open equivalence relation on U. An equivalence relation R on X is said to be *strictly open* if it is coherent and if $\mathrm{loc}(R)$ is a strictly open local equivalence

relation. Prove that, for the open relation r to be strictly open, it is enough that it satisfy the condition considered in d); if r is locally defined by an R with closed fibers, then that condition is also necessary. 488

h) Let F be a sheaf of sets on X. For every open subset U of X, let $\mathrm{Quot}(U, F)$ be the set of pairs consisting of an equivalence relation R_U on U and an equivalence relation $R_{F|U}$ on $F|U$ (F being regarded as an étale space over X), such that the structure map $p : F|U \to U$ is compatible with the equivalence relations R_U and $R_{F|U}$, and such that the corresponding diagram of topological spaces

$$\begin{array}{ccc} F|U & \longrightarrow & (F|U)/R_{F|U} \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U & \longrightarrow & U/R_U \end{array}$$

is Cartesian, with q an étale map (so that $F|U$ is identified with the inverse image of the sheaf $(F|U)/R_{F|U}$ on U/R_U). Show that the $\mathrm{Quot}(U, F)$, for varying U, define a presheaf on X, whose associated sheaf will be denoted by $Q(F/X)$. Define a homomorphism of sheaves $Q(F/X) \to Q_X$. For a given section r of Q_X, an *r-structure* on the sheaf F is a section of $Q(F/X)$ lying above the given section r of Q_X. Define the category $\mathrm{Top}(X/r)$ of r-sheaves on X, and define a conservative and faithful functor, "forgetting the r-structure", $\mathrm{Top}(X/r) \to \mathrm{Top}(X)$. Prove that finite projective limits and finite inductive limits are representable in $\mathrm{Top}(X/r)$, and that the preceding functor commutes with those limits. Deduce that finite sums in $\mathrm{Top}(X/r)$ are disjoint and universal and that equivalence relations are universally effective. Prove that $\mathrm{Top}(X/r)$ has a small generating family. Give an example, with r coherent, in which $\mathrm{Top}(X/r)$ does not have infinite coproducts (?), and hence is not a topos.

i) Let $f : X \to Y$ be a continuous map compatible with $R = \mathrm{glob}(r)$. Define a functor

$$f_r^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X/r)$$

whose composite with the inclusion functor $\mathrm{Top}(X/r) \to \mathrm{Top}(X)$ is f^*. Show that, if r is globally coherent and strictly open (i.e. if R is strictly open and $r = \mathrm{loc}(R)$), and if f is the canonical map $X \to Y = X/R$, then f_r^* is an equivalence of categories. Consequently, $\mathrm{Top}(X/r)$ is a $\mathscr{U}$-topos, and $\mathrm{Top}(X/r) \to \mathrm{Top}(X)$ is the inverse-image functor of a morphism of topoi. 489

j) Deduce from i) that, if r is strictly open, then $\mathrm{Top}(X/r)$ is a topos, and that the inclusion functor $\mathrm{Top}(X/r) \to \mathrm{Top}(X)$ is the inverse-image functor associated with a morphism of topoi

$$p : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(X/r).$$

k) Define a functorial construction of the topos $\mathrm{Top}(X/r)$ and of the preceding morphism of topoi p with respect to the pair (X, r) consisting of a topological space X equipped with a strictly open local equivalence relation. Show that, if $X' \to X$ is an étale map and r' is the inverse image of r on X', then the induced morphism $\mathrm{Top}(X'/r') \to \mathrm{Top}(X/r)$ is equivalent to a localization morphism $\mathrm{Top}(X/r)_{/E} \to \mathrm{Top}(X/r)$, where E is an object of $\mathrm{Top}(X/r)$ (determined up to a unique isomorphism). The object E covers the final object of $\mathrm{Top}(X/r)$ if and only if the saturation under $R = \mathrm{glob}(r)$ of the image of X' in X is all of X.

l) Deduce from k) that $\mathrm{Top}(X/r)$ is a *rigid étendue* (9.8.2 h)). Show that, if X is sober, the set of isomorphism classes of points of $\mathrm{Top}(X/r)$, endowed with its canonical topology (7.8), is homeomorphic to the quotient topological space $X/\mathrm{glob}(r)$, and that, for any two points of $\mathrm{Top}(X/r)$ arising from points x and x' of X, there is at most one morphism from one to the other. Such a morphism exists if and only if x' and x are equivalent modulo $\mathrm{glob}(r)$ to points x'_1 and x_1 such that x_1 is a generalization of x'_1.

490 m) Study the canonical morphism of topoi $\mathrm{Top}(X) \to \mathrm{Top}(X/r)$, observing that, for every object U of $\mathrm{Top}(X/r)$ such that the topos induced on U "is" an ordinary topological space, the induced morphism $\mathrm{Top}(X)_{/p^*(U)} \to \mathrm{Top}(X/r)_{/U}$ "is" a continuous map of ordinary topological spaces $X_U \to U$. Show that the fibers of the map $X_U \to U$ are homeomorphic to connected components of the space X^r of f).

n) Let X be a differentiable manifold equipped with a foliation, i.e. with a subsheaf F, locally a direct factor of the tangent sheaf of X, which is stable under brackets. Define on X a strictly open local equivalence relation associated with the foliation, and define on the quotient topos $T = \mathrm{Top}(X/r)$ a ring which makes it a *differentiable étendue* (9.8.2 j)). Define an equivalence between the category of locally free modules on the differentiable étendue T (also called differentiable vector bundles on T) and the category of locally free modules on X equipped with a connection relative to the given subbundle F of the tangent bundle. Give variants in the real-analytic and complex-analytic cases.

10. Sheaves of morphisms

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Proposition 10.1. *Let E be a topos and let X and Y be two objects of E; the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y) \simeq \mathrm{Hom}_{E_{/Z}}(X_Z, Y_Z)$ is representable.*

Indeed, inductive limits are universal in E (II 4.3). Hence the functor $Z \mapsto Z \times X$ commutes with inductive limits. Consequently, the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ carries inductive limits in the argument Z to projective limits. It is therefore representable (1.2.1).

10.2. The object representing the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ is denoted by $\mathbf{Hom}_E(X, Y)$ (or more simply by $\mathbf{Hom}(X, Y)$) and is called the *sheaf of morphisms from X to Y*. It is a bifunctor in X and Y. We thus have a trifunctorial isomorphism

$$\mathrm{Hom}_E(Z, \mathbf{Hom}_E(X, Y)) \simeq \mathrm{Hom}_E(Z \times X, Y). \tag{10.2.1}$$

It then follows from formula (10.2.1) that the bifunctor $(X, Y) \mapsto \mathbf{Hom}(X, Y)$ carries inductive limits in the argument X (resp. projective limits in the argument Y) to projective limits in E.

Proposition 10.3. *Let $u : E \to E'$ be a morphism of topoi. For a variable object X of E and a variable object Y of E', there is a bifunctorial isomorphism*

$$u_* \mathbf{Hom}_E(u^*Y, X) \simeq \mathbf{Hom}_{E'}(Y, u_*X). \tag{10.3.1}$$

Proof. For every object Z of E', we have a sequence of isomorphisms, functorial in all the arguments:

$$\begin{aligned}
&\mathrm{Hom}_{E'}(Z, v_*\,\mathbf{Hom}_E(v^*Y, X)) \simeq \mathrm{Hom}_E(v^*Z, \mathbf{Hom}_E(v^*Y, X)) && \text{(adjunction)}\\
&\mathrm{Hom}_E(v^*Z, \mathbf{Hom}_E(v^*Y, X)) \simeq \mathrm{Hom}_E(v^*Z \times v^*Y, X) && \text{(10.2.1)}\\
&\mathrm{Hom}_E(v^*Z \times v^*Y, X) \simeq \mathrm{Hom}_E(v^*(Z \times Y), X) && (v^* \text{ is left exact})\\
&\mathrm{Hom}_E(v^*(Z \times Y), X) \simeq \mathrm{Hom}_{E'}(Z \times Y, v_*X) && \text{(adjunction)}\\
&\mathrm{Hom}_{E'}(Z \times Y, v_*X) \simeq \mathrm{Hom}_{E'}(Z, \mathbf{Hom}_{E'}(Y, v_*X)) && \text{(10.2.1)}
\end{aligned}$$

The two sides of (10.3.1) therefore represent isomorphic functors, C.Q.F.D.

COROLLARY 10.4. *Let C be a $\mathscr{U}$-site, X a $\mathscr{U}$-presheaf on C, Y a $\mathscr{U}$-sheaf on C, and $\mathscr{V}$* 492
a universe containing $\mathscr{U}$ such that C is $\mathscr{V}$-small. Let $\hat{C}_{\mathscr{V}}$ be the topos of $\mathscr{V}$-presheaves on C, and let $\tilde{C}_{\mathscr{U}}$ be the topos of $\mathscr{U}$-sheaves on C. The presheaf $\mathbf{Hom}_{\hat{C}_{\mathscr{V}}}(X, Y)$ is a $\mathscr{U}$-sheaf. Let $X \to \underline{a}X$ be the canonical morphism from X to its associated sheaf. The corresponding morphism $\mathbf{Hom}_{\hat{C}_{\mathscr{V}}}(\underline{a}X, Y) \to \mathbf{Hom}_{\hat{C}_{\mathscr{V}}}(X, Y)$ is an isomorphism. There is a canonical isomorphism $\mathbf{Hom}_{\hat{C}_{\mathscr{V}}}(\underline{a}X, Y) \simeq \mathbf{Hom}_{\tilde{C}_{\mathscr{U}}}(\underline{a}X, Y)$.

10.4.1. Suppose first that C is a small site and that $\mathscr{V} = \mathscr{U}$. We then have a morphism of topoi $\tilde{C}_{\mathscr{U}} \to \hat{C}_{\mathscr{U}}$ (4.8), from which (10.3.1) gives the assertions in this case. To pass from this to the general case, note first that the "associated sheaf" functor is independent of the universe (II 3.6), and that the topos of $\mathscr{V}$-sheaves on C is equivalent to the topos of $\mathscr{V}$-sheaves on $\tilde{C}_{\mathscr{U}}$ (III 4 and 1). It therefore suffices to prove the following lemma:

LEMMA 10.4.2. *Let E be a $\mathscr{U}$-topos, let $\mathscr{V}$ be a universe containing $\mathscr{U}$, let $\tilde{E}_{\mathscr{V}}$ be the topos of $\mathscr{V}$-sheaves on E, and let X and Y be two objects of E. There exists a canonical isomorphism $\mathbf{Hom}_E(X, Y) \simeq \mathbf{Hom}_{\tilde{E}_{\mathscr{V}}}(X, Y)$.*

10.4.3. The functor $\epsilon_E : E \to \tilde{E}_{\mathscr{V}}$ is fully faithful and left exact. Consequently, for every object Z of E, there is an isomorphism

$$\mathrm{Hom}_{\tilde{E}_{\mathscr{V}}}(\epsilon_E Z, \epsilon_E\,\mathbf{Hom}_E(X, Y)) \simeq \mathrm{Hom}_{\tilde{E}_{\mathscr{V}}}(\epsilon_E Z \times \epsilon_E X, \epsilon_E Y).$$

But every object of $\tilde{E}_{\mathscr{V}}$ is an inductive limit of objects coming from E; the assertion therefore follows from (10.2.1).

10.5. Let $v : E \to E'$ be a morphism of topoi. We shall define four bifunctorial morphisms:

$$\begin{aligned}
&\Phi_v : v^*\,\mathbf{Hom}(X, Y) \longrightarrow \mathbf{Hom}(v^*X, v^*Y) \quad , \quad X \text{ and } Y \text{ objects of } E',\\
&\Psi_v : v_*\,\mathbf{Hom}(X, Y) \longrightarrow \mathbf{Hom}(v_*X, v_*Y) \quad , \quad X \text{ and } Y \text{ objects of } E,\\
&\Xi_v : \mathbf{Hom}(v_!X, Y) \longrightarrow v_*\,\mathbf{Hom}(X, v^*Y) \quad , \quad X \in \mathrm{ob}\,E, Y \in \mathrm{ob}\,E',\\
&\Lambda_v : v_!(X \times v^*Y) \longrightarrow v_!(X) \times Y \qquad\qquad X \in \mathrm{ob}\,E, Y \in \mathrm{ob}\,E',
\end{aligned}$$

where, in order to define the morphisms Ξ_v and Λ_v, we suppose that the inverse-image 493
functor v^* of v admits a left adjoint $v_!$.

10.5.1. *Definition of Ψ_v.* Let X be an object of E. There is an adjunction morphism $v^*v_*X \to X$, and hence, for every object Z of E', a bifunctorial morphism $v^*(Z \times v_*X) \to (v^*Z) \times X$. Thus, for every object Y of E, there is a trifunctorial morphism

$$\mathrm{Hom}_E((v^*Z) \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*(Z \times v_*X), Y).$$

By adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, v_* \mathbf{Hom}(X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, \mathbf{Hom}(v_*X, v_*Y));$$

and hence the morphism Ψ_v.

10.5.2. *Definition of Λ_v.* For every object X of E, there is an adjunction morphism $X \to v^*v_!X$, and hence, for every object Y of E', a bifunctorial morphism $X \times v^*Y \to v^*v_!(X) \times v^*Y \simeq v^*(v_!(X) \times Y)$; by adjunction this gives the morphism Λ_v.

10.5.3. *Definition of Ξ_v.* Let X be an object of E, and let Z and Y be two objects of E'. The morphism $\Lambda_v : v_!(X \times v^*Z) \to v_!(X) \times Z$ gives a trifunctorial morphism $\mathrm{Hom}_{E'}(v_!(X) \times Z, Y) \to \mathrm{Hom}_{E'}(v_!(X \times v^*Z), Y)$; by adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, \mathbf{Hom}(v_!X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_* \mathbf{Hom}(X, v^*Y));$$

and hence the morphism Ξ_v.

10.5.4. *Definition of Φ_v.* The trifunctorial morphism of (10.5.3)

$$\mathrm{Hom}_{E'}(v_!(Z) \times X, Y) \longrightarrow \mathrm{Hom}_{E'}(v_!(Z \times v^*X), Y),$$

where Z is an object of E and X, Y are objects of E', gives, by adjunction, a trifunctorial morphism

$$\mathrm{Hom}_E(Z, v^* \mathbf{Hom}(X, Y)) \longrightarrow \mathrm{Hom}_E(Z, \mathbf{Hom}(v^*X, v^*Y));$$

and hence a definition of the morphism Φ_v. There is a second way to define Φ_v. Let X, Y, and Z be three objects of E'. The functor $v^* : E' \to E$ gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*(Z) \times v^*(X), v^*Y);$$

and hence, by adjunction, a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, \mathbf{Hom}(X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_* \mathbf{Hom}(v^*X, v^*Y)).$$

494 We therefore have a bifunctorial morphism

$$\mathbf{Hom}(X, Y) \longrightarrow v_* \mathbf{Hom}(v^*X, v^*Y);$$

and hence, by adjunction, a morphism which the reader may verify is indeed the morphism Φ_v defined above.

PROPOSITION 10.6. *Let $v : E \to E'$ be a morphism of topoi.*

1) *The morphism Ψ_v (10.5.1) is an isomorphism for every object Y of E if and only if the adjunction morphism $v^*v_*X \to X$ is an isomorphism. In particular, Ψ_v is an isomorphism for every pair of objects (X, Y) if and only if the functor $v_* : E \to E'$ is fully faithful, i.e. if v is an embedding of topoi.*
2) *The following conditions are equivalent:*
   (i) *For every pair (X, Y) of objects of E', the morphism Φ_v (10.5.4) is an isomorphism.*
   (ii) *The functor v^* admits a left adjoint $v_!$, and for every object X of E and every object Y of E', the morphism Ξ_v (10.5.3) is an isomorphism.*
   (iii) *The functor v^* admits a left adjoint $v_!$, and for every object X of E and every object Y of E', the morphism Λ_v (10.5.2) is an isomorphism.*

10.6.1. The first assertion follows immediately from (10.5.1). It also follows immediately from (10.5.2), (10.5.3), and (10.5.4) that (iii) $\Leftrightarrow$ (ii) $\Leftrightarrow$ ((i) and the functor v^* admits a left adjoint $v_!$). It remains to show that (i) implies that v^* admits a left adjoint, i.e. (1.8) that (i) implies that v^* commutes with small products. Let e' be the final object of E' and let I be a small set. Since v^* is left exact, $v^*(e')$ is a final object of E, and since v^* commutes with inductive limits, $v^*(\amalg_I e') \simeq \amalg_I v^*(e')$. For every object Y of E', we have $\mathbf{Hom}(\amalg_I e', Y) \simeq \Pi_I \mathbf{Hom}(e', Y) \simeq \Pi_I Y$ and $\mathbf{Hom}(\amalg_I v^*(e'), v^*Y) \simeq \Pi_I \mathbf{Hom}(v^*(e'), v^*Y) \simeq \Pi_I v^*Y$. The functorial morphism Φ_v therefore induces an isomorphism $v^*\Pi_I Y \simeq \Pi_I v^*Y$, which one checks is the canonical isomorphism. C.Q.F.D. 495

Corollary 10.7. *Let E be a topos, let Z be an object of E, let $j_Z : E_{/Z} \to E$ be the localization morphism (5.2), and let X, Y be two objects of E.*

1) *There is a bifunctorial isomorphism*

$$j_Z^* \mathbf{Hom}(X, Y) \simeq \mathbf{Hom}(j_Z^* X, j_Z^* Y).$$

2) *Let X' be an object of $E_{/Z}$. There is a bifunctorial isomorphism*

$$\mathbf{Hom}_E(j_{Z!}X', Y) \simeq j_{Z*} \mathbf{Hom}_{E/Z}(X', j_Z^* Y).$$

3) *Let Y' be an object of E/Z. When Z is open in E, there is a bifunctorial isomorphism*

$$j_{Z*} \mathbf{Hom}(X', Y') \simeq \mathbf{Hom}(j_{Z*}X', j_{Z*}Y').$$

Assertions 1) and 2) are proved by noting that the morphism Λ_{j_Z} is an isomorphism. For assertion 3), it suffices to note that j_{Z*} is fully faithful, since $j_{Z!}$ is fully faithful (I 5.7. a)).

Corollary 10.8. *Let E be a topos, let X, Y, and Z be three objects of E, and let $j_Z : E_{/Z} \to E$ be the localization morphism.*

1) *There is a canonical isomorphism*

$$j_{Z*} j_Z^* Y \simeq \mathbf{Hom}(Z, Y).$$

2) *There is a canonical isomorphism*

$$\mathrm{Hom}_E(Z, \mathbf{Hom}(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

Let e_Z be the final object of $E_{/Z}$ (i.e. the object $\mathrm{id}_Z : Z \to Z$). It follows from (10.2.1) that there is a canonical isomorphism

$$\mathbf{Hom}_{E/Z}(e_Z, j_Z^* Y) \simeq j_Z^* Y;$$

hence, by (10.7 2), an isomorphism

$$j_{Z*} j_Z^* Y \simeq \mathbf{Hom}(j_{Z!} e_Z, Y).$$

Since $j_{Z!}e_Z = Z$, this proves 1). Let us prove 2). From (10.7) we obtain a canonical isomorphism 496

$$\mathrm{Hom}_{E_{/Z}}(e_Z, j_Z^* \mathbf{Hom}(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y);$$

hence, by adjunction on the first member,

$$\mathrm{Hom}_E(Z, \mathbf{Hom}(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

11. Ringed topoi, localization in ringed topoi

11.1.1. Let $\mathscr{U}$ be a universe. A *$\mathscr{U}$-ringed topos* is a pair (E, A) where E is a $\mathscr{U}$-topos and A is an object endowed with a ring structure. A $\mathscr{U}$-ringed site is a pair (C, A) where C is a $\mathscr{U}$-site and A is a $\mathscr{U}$-sheaf of rings on C. We omit mention of the universe when the context leaves no room for confusion. To a ringed site (C, A) is associated the ringed topos $(C^{\sim}, A)$. To a ringed topos (E, A) is associated the ringed site consisting of the site E and the sheaf of rings represented by A.

11.1.2. Let (E, A) be a ringed topos. We denote by ${}_AE$ (resp. E_A) the category of sheaves of unital left (resp. right) A-modules (we will also write A-Modules). The category ${}_AE$ (resp. E_A) is an abelian category (II 6.7). Let M and N be two sheaves of left (resp. right) A-modules. The commutative group of A-Module morphisms from M to N is denoted by $\mathrm{Hom}_A(M, N)$.

11.1.3. Let A, B, and C be three rings in a topos E, let M be a sheaf of $A-B$-bimodules, and let N be a sheaf of $A-C$-bimodules. Let e be a final object of E, and put $\mathrm{Hom}(e, B) = \Gamma(B)$ and $\mathrm{Hom}(e, C) = \Gamma(C)$. The $A-B$-bimodule structure on M gives a ring homomorphism from $\Gamma(B)^{\circ}$ to the ring of endomorphisms of the A-module M; similarly, the $A-C$-bimodule structure on N gives a ring homomorphism from $\Gamma(C)^{\circ}$ to the ring of endomorphisms of the A-module N. By functoriality, we deduce a $\Gamma(B)-\Gamma(C)$-bimodule structure on the commutative group $\mathrm{Hom}_A(M, N)$. In particular, when A is a sheaf of commutative rings, the group $\mathrm{Hom}_A(M, N)$ is canonically endowed with a $\Gamma(A)$-module structure.

497 11.1.4. Let E be a topos, let A and B be two rings in E, let $u : A \to B$ be a morphism of sheaves of rings, and let M be a B-module (a left module, to fix ideas). The sheaf M may be regarded as a sheaf of A-modules by means of u: for every object X of E, $M(X)$ is endowed with the $A(X)$-module structure obtained from its $B(X)$-module structure and the homomorphism $u(X) : A(X) \to B(X)$ by restriction of scalars. We thus obtain a functor "*restriction of scalars along u*"

$$\mathrm{Res}(u) : {}_BE \longrightarrow {}_AE.$$

11.1.5. In particular, when $A = \mathbf{Z}$ (the constant sheaf $\underline{\mathbf{Z}}$, that is, the sheaf associated to the constant presheaf $\mathbf{Z}$) and when $u : \underline{\mathbf{Z}} \to B$ is the unique canonical morphism, we obtain a functor from ${}_BF$ to the category ${}_{\underline{\mathbf{Z}}}E$, which is none other than the category of abelian sheaves, denoted by E_{Ab}. This functor is called the *underlying abelian-sheaf functor*.

Proposition 11.1.6. *The restriction-of-scalars functor along u commutes with inductive and projective limits. It is conservative.*

The proposition is true for the restriction-of-scalars functor for ordinary modules, that is, when E is the punctual topos. It is therefore true when E is the topos of presheaves on a small site. It then follows from the determination of inductive and projective limits by means of the associated-sheaf functor (II 6.4) that the proposition is true in general.

11.2.1. Let (E, A) be a ringed topos, let X be an object of E, and let $j_X : E_{/X} \to E$ be the localization morphism 5.2. By III 1.7 or 3.1.2, the sheaf j_X^*A is canonically endowed with a ring structure. The sheaf of rings j_X^*A is most often denoted by $A|X$, or, by abuse, simply by A. Unless otherwise stated, the topos $E_{/X}$ will be ringed by $A|X$. The sheaf $j_{X*}j_X^*A$ is canonically endowed with a ring structure. The adjunction morphism $A \to j_{X*}j_X^*A$ is a morphism of sheaves of rings.

498 11.2.2. Let , in addition, M be an A-module (a left module, to fix ideas). The sheaf $j_X^* M$ is endowed with an $A|X$-module structure (III 1.7; or 3.1.2); hence there is a functor

$$j_X^* : {}_A E \longrightarrow {}_{A|X} E_{/X}$$

called the restriction functor to $E_{/X}$, or also the restriction functor to X. The functor j_X^* commutes with inductive and projective limits (*loc. cit.*). In particular, it is exact. Now let N be an $A|X$-Module. The sheaf $j_{X*}N$ is a sheaf of $j_{X*}A|X$-Modules (III 1.7 or 3.1.2); hence, by restriction of scalars along the adjunction morphism $A \to j_{X*}(A|X)$ (11.1.4), it is an A-Module, again denoted by $j_{X*}N$. We have therefore defined a functor

$$j_{X*} : {}_{A|X} E_{/X} \longrightarrow {}_A E$$

which commutes with projective limits (11.1.6 and III 1.7). For every A-Module M, the adjunction morphism $M \to j_{X*}j_X^* M$ is a morphism of A-Modules. For every $A|X$-Module N and every A-Module M, the adjunction morphism $M \to j_{X*}j_X^* M$ defines a morphism, bifunctorial in M and N,

$$\mathrm{Hom}_{A|X}(j_X^* M, N) \longrightarrow \mathrm{Hom}_A(M, j_{X*}N). \tag{11.2.2.1}$$

The latter morphism is an isomorphism; that is, the functors j_X^* and j_{X*} for A-Modules are adjoint (III 1.7).

11.2.3. The functors j_X^* and j_{X*} for Modules commute with the "underlying set" functor (III 1.7 4)). They therefore commute with the restriction-of-scalars functors and, in particular, with the underlying abelian-sheaf functor.

PROPOSITION 11.3.1. *Let (E, A) be a ringed topos and let X be an object of E. The functor*

$$j_X^* : {}_A E \longrightarrow {}_{A|X} E_{/X}$$

admits a left adjoint, denoted by $j_{X!} : {}_{A|X} E_{/X} \to {}_A E$ and called extension by zero. The extension-by-zero functor is exact and faithful, and commutes with inductive limits. The functors $j_{X!}$ for Modules commute with restriction-of-scalars functors and, in particular, 499 *they commute with the underlying abelian-sheaf functor.*

The existence of the functor $j_{X!}$ follows from III 1.7. Since $j_{X!}$ is a left adjoint, it commutes with inductive limits (I 2). To prove the other assertions, first suppose that E is the topos of presheaves of sets on a small category C containing the object X. We then know that $E_{/X}$ is equivalent to the category of presheaves on $C_{/X}$ and that, under this equivalence, the functor $j_X^* : E \to E_{/X}$ is simply composition with the forgetful functor $C_{/X} \to C$ (I 5.11). It then follows immediately from the explicit construction of $j_{X!}$ (I 5.1) that, for every $A|X$-Module N and every object Y of C, one has

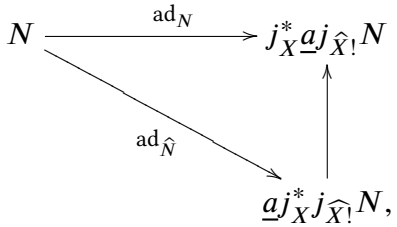
$$j_{X!}N(Y) = \bigoplus_{u \in \mathrm{Hom}_C(Y,X)} N(u);$$

hence $j_{X!}$ is exact, and extension by zero commutes with restriction-of-scalars functors in this case. In the general case, we may suppose that E is the topos of sheaves on a small site C and that X comes from an object of C (1.2.1). The topos $E_{/X}$ is then equivalent to the topos of sheaves on $C_{/X}$ (endowed with the induced topology) (III 5.4), and the topos morphism $j_X : E_{/X} \to E$ comes from the forgetful functor $C_{/X} \to C$, which is continuous and cocontinuous (III 5.2). It then follows from III 1.7 that extension by zero for sheaves is obtained by composing extension by zero for presheaves with the associated-sheaf functor; hence the assertion of exactness and the compatibility with restrictions of scalars.

To prove faithfulness, it is equivalent to show that, for every $A|X$-Module N, the adjunction morphism

$$\mathrm{ad}_N : N \longrightarrow j_X^* j_{X!} N$$

is a monomorphism. Now, denoting by $j_{\widehat{X}!}$ the extension-by-zero functor for presheaves and by $\underline{a}$ the "associated sheaf" functor, we have a commutative diagram



500 where $\mathrm{ad}_{\widehat{N}}$ is the adjunction morphism for presheaves. The morphism $\mathrm{ad}_{\widehat{N}}$ is a monomorphism by what precedes, and therefore $\underline{a}(\mathrm{ad}_{\widehat{N}})$ is a monomorphism. Moreover, since the forgetful functor $C_{/X} \to C$ is continuous and cocontinuous, the canonical morphism $\underline{a} j_X^* \to j_X^* \underline{a}$ is an isomorphism (III 2.3). Consequently, ad_N is a monomorphism.

Remark 11.3.2. Let N be an $A|X$-Module. The sheaf of sets underlying $j_{X!}N$ is not, in general, isomorphic to the sheaf obtained by extending the sheaf of sets underlying N by the empty set (III 5.3 and 5.2). One must therefore take care not to confuse the *extension-by-zero* functor denoted by $j_{X!} : {}_{A|X}E_{/X} \to {}_A E$ in 11.3.1 with the extension-by-the-empty-set functor, also denoted by $j_{X!} : E_{/X} \to E$, in III 5.3 and IV 5.2. In most cases encountered in practice, the abuse of notation just noted causes no confusion. When confusion is nevertheless possible, we will use the notations $j_{X!}^{\mathrm{ab}}$ and $j_{X!}^{\mathrm{ens}}$ for the extension-by-zero and extension-by-the-empty-set functors, respectively.

Proposition 11.3.3. *Let (E, A) be a ringed topos and let X be an object of E. The A-Module $j_{X!}(A|X)$, most often denoted by A_X or $A_{X,E}$, is the free A-Module generated by X (II 6.5); that is, for every A-Module M, there is a canonical isomorphism, functorial in M:*

$$\mathrm{Hom}_E(X, M) \xleftarrow{\sim} \mathrm{Hom}_A(A_X, M).$$

Let $(X_i)_{i\in I}$ be a topologically generating family of E (II 3.0.1). The family $(A_{X_i})_{i\in I}$ is a generating family of the category of A-Modules.

Let e_x be the final object of the topos $E_{/X}$ ($e_X = X \xrightarrow{\mathrm{id}} X$). There is an isomorphism

$$\mathrm{Hom}_{A|X}(A|X, j_X^* M) \xrightarrow{\sim} \mathrm{Hom}_{E_{/X}}(e_X, j_X^* M);$$

501 deduced from the unit section $e_X \to A/X$; hence, by adjunction, there is an isomorphism

$$\mathrm{Hom}_A(j_{X!}^{\mathrm{ab}}(A|X), M) \xrightarrow{\sim} \mathrm{Hom}_E(j_{X!}^{\mathrm{ens}}(e_X), M).$$

Since $j_{X!}^{\mathrm{ens}}(e_X) = X$, we obtain the asserted isomorphism. The last assertion follows from II 6.6.

Remark 11.3.4. Let A and B be two sheaves of rings on a topos E, let X be an object of E, and let N be an $A|X$-$B|X$-bimodule. The abelian sheaf obtained by extending N by zero is canonically endowed with an A-B-bimodule structure, as follows immediately from its explicit description (11.3.1). In particular, the free A-module so obtained, A_X, is an A-bimodule.

EXERCISE 11.3.5. Let E be a topos, let X be an object of E, let $x : P \to E$ be a point of E (6.1), and let $(x_i)_{i\in I}$ be the family of points of $E_{/X}$ above x (in bijective correspondence with the fiber X_x (6.7.2)). Show that, for every abelian sheaf M on $E_{/X}$, the fiber $(j_{X!}M)_x$ is canonically isomorphic to $\oplus_i M_{x_i}$.

12. Operations on modules

PROPOSITION 12.1. *Let (E, A) be a ringed topos, and let M and N be two left A-modules (resp. right A-modules). The functor on E which assigns to every object X of E the commutative group $\mathrm{Hom}_A(M, \mathbf{Hom}_E(X, N))$ is representable by an abelian sheaf denoted by $\mathbf{Hom}_A(M, N)$ (or sometimes $\mathbf{Hom}(M, N)$ when no confusion can result). For every object X of E, there is a canonical isomorphism*

$$\mathbf{Hom}_A(M, N)(X) \simeq \mathrm{Hom}_{A|X}(j_X^* M, j_X^* N). \tag{12.1.1}$$

There is a canonical isomorphism $\mathbf{Hom}_E(X, N) = j_{X*}j_X^* N$ (10.8); consequently, $\mathbf{Hom}_E(X, N)$ is endowed, functorially in X, with a structure of left A-module (resp. right A-module). The functor $X \longmapsto \mathbf{Hom}_E(X, N)$ transforms inductive limits in the argument X into projective limits of A-modules (10.2 and 11.2.3); consequently, the functor

$$X \longmapsto \mathrm{Hom}_Z(M, \mathbf{Hom}_E(X, N))$$

transforms inductive limits in the argument X into projective limits of commutative groups, and hence into projective limits of the underlying sets. It is therefore representable (1.4 and 1.2), and the object representing it is endowed with the structure of an abelian sheaf. By definition, for every object X of E, there is a canonical isomorphism

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$$\mathbf{Hom}_A(M, N)(X) = \mathrm{Hom}(X, \mathbf{Hom}_A(M, N)) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}(X, N)) \tag{12.1.2}$$

and the isomorphism (12.1.1) follows from (12.1.2), from the isomorphism $\mathbf{Hom}(X, N) \approx j_{X*}j_X^* N$ (10.8), and from the adjunction formulas of 10.

12.2. The abelian sheaf $\mathbf{Hom}_A(M, N)$ is called the *sheaf of A-module homomorphisms from M to N*. It is a bifunctor in M and N. It follows from its definition and from (10.2) that it transforms projective limits in the argument N (resp. inductive limits in M) into projective limits of abelian sheaves. In particular, it is left exact in both arguments.

PROPOSITION 12.3. *Let (E, A) be a ringed topos, let M and N be two right A-modules (resp. left A-modules), and let X be an object of E:*

a) *There are canonical isomorphisms:*

$$\mathbf{Hom}_A(M, N)(X) \approx \mathrm{Hom}_A(M, j_{X*}j_X^* N) \simeq \mathrm{Hom}_{A|X}(j_X^* M, j_X^* N) \simeq \mathrm{Hom}_A(j_{X!}j_X^* M, N).$$

b) *There is a canonical isomorphism:*

$$\Phi_X : j_X^* \mathbf{Hom}_A(M, N) \simeq \mathbf{Hom}_{A|X}(j_X^* M, j_X^* N).$$

Let, in addition, P be a right $A|X$-module (resp. a left $A|X$-module):

c) *There is a canonical isomorphism:*

$$j_{X*}\mathbf{Hom}_{A|X}(j_X^* M, P) \simeq \mathbf{Hom}_A(M, j_{X*}P).$$

d) *There is a canonical isomorphism:*

$$\Xi_X : \mathbf{Hom}_A(j_{X!}P, N) \simeq j_{X*}\mathbf{Hom}_{A|X}(P, j_X^* N).$$

The isomorphisms in a) follow from (12.1.2), from the isomorphism $\mathbf{Hom}_E(X, N) \simeq j_{X*}j_X^* N$ (10.8), and from the adjunction formulas of 10.

Proof of b). For every object Y of $E_{/X}$, there is the following sequence of isomorphisms:

$$\mathrm{Hom}(Y, j_X^* \mathbf{Hom}_A(M, N)) \simeq \mathrm{Hom}(j_{X!}Y, \mathbf{Hom}_A(M, N)) \text{ (adjunction for sheaves of sets)}$$

$$\mathrm{Hom}(j_{X!}Y, \mathbf{Hom}_A(M, N)) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}_E(j_{X!}Y, N)) \quad (12.2.1)$$

$$\mathrm{Hom}_A(M, \mathbf{Hom}_E(j_{X!}Y, N)) \simeq \mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(Y, j_X^* N)) \quad (10.7)$$

$$\mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(Y, j_X^* N)) \simeq \mathrm{Hom}_{A|X}(j_X^* M, \mathbf{Hom}_E(Y, j_X^* N)) \quad (11.2.2.1)$$

$$\mathrm{Hom}_{A|X}(j_X^* M, \mathbf{Hom}_E(Y, j_X^* N)) \simeq \mathrm{Hom}(Y, \mathbf{Hom}_{A|X}(j_X^* M, j_X^* N)) \quad (12.2.1)$$

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Proof of c). For every object Y of E, there is the following sequence of isomorphisms:

$$\mathrm{Hom}(Y, j_{X*}\mathbf{Hom}_{A|X}(j_X^* M, P)) \simeq \mathrm{Hom}(j_X^* Y, \mathbf{Hom}_{A|X}(j_X^* M, P)) \quad \text{(adjunction)}$$

$$\mathrm{Hom}(j_X^* Y, \mathbf{Hom}_{A|X}(j_X^* M, P)) \simeq \mathrm{Hom}_{A/X}(j_X^* M, \mathbf{Hom}_{E_{/X}}(j_X^* Y, P)) \quad (12.2.1)$$

$$\mathrm{Hom}_{A|X}(j_X^* M, \mathbf{Hom}_{E_{/X}}(j_X^* Y, P)) \simeq \mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(j_X^* Y, P)) \quad (11.2.2.1)$$

$$\mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(j_X^* Y, P)) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}_E(Y, j_{X*}P)) \quad (10.3)$$

$$\mathrm{Hom}_A(M, \mathbf{Hom}_E(Y, j_{X*}P)) \simeq \mathrm{Hom}(Y, \mathbf{Hom}_A(M, j_{X*}P)) \quad (12.2.1)$$

Proof of d). For every object Y of E, there is the following sequence of isomorphisms:

$$\mathrm{Hom}(Y, \mathbf{Hom}_A(j_{X!}P, N)) \simeq \mathrm{Hom}_A(j_{X!}P, \mathbf{Hom}_E(Y, N)) \quad (12.2.1)$$

$$\mathrm{Hom}_A(j_{X!}P, \mathbf{Hom}_E(Y, N)) \simeq \mathrm{Hom}_{A|X}(P, j_X^* \mathbf{Hom}_E(Y, N)) \quad (11.3.1)$$

$$\mathrm{Hom}_{A/X}(P, j_X^* \mathbf{Hom}_E(Y, N)) \simeq \mathrm{Hom}_{A|X}(P, \mathbf{Hom}_{E_{/X}}(j_X^* Y, j_X^* N)) \quad (10.7)$$

$$\mathrm{Hom}_{A/X}(P, \mathbf{Hom}_{E_{/X}}(j_X^* Y, j_X^* N)) \simeq \mathrm{Hom}(j_X^* Y, \mathbf{Hom}_{A/X}(P, j_X^* N)) \quad (12.2.1)$$

$$\mathrm{Hom}(j_X^* Y, \mathbf{Hom}_{A|X}(P, j_X^* N)) \simeq \mathrm{Hom}(Y, j_{X*}\mathbf{Hom}_{A|X}(P, j_X^* N)) \quad \text{(adjunction)}$$

Corollary 12.4. *Let C be a $\mathscr{U}$-site, let A' be a $\mathscr{U}$-presheaf of rings on C, let M be a $\mathscr{U}$-presheaf of A'-modules (left modules, to fix ideas), and let N be a $\mathscr{U}$-sheaf of A'-modules (left modules). Denote by A the sheaf associated with A', so that the sheaves N and $\underline{a}M$ (the sheaf associated with M) are A-modules. The presheaf $X \longmapsto \mathrm{Hom}_{A'|X}(j_X^* M, j_X^* N)$ is a $\mathscr{U}$-abelian sheaf canonically isomorphic to $\mathbf{Hom}_A(\underline{a}M, N)$.*

Indeed, it follows from III 5.5 and III 2.3 that the localization functor to $C_{/X}$ commutes with the associated-sheaf functors (for C and $C_{/X}$). Consequently, there are isomorphisms, functorial in the variable object X of C:

$$\begin{aligned}\mathrm{Hom}_{A'|X}(j_X^* M, j_X^* N) &\simeq \mathrm{Hom}_{A|X}(\underline{a}j_X^* M, j_X^* N)\\ &\simeq \mathrm{Hom}_{A|X}(j_X^* \underline{a}M, j_X^* N) \simeq \mathbf{Hom}_A(\underline{a}M, N)(X).\end{aligned}$$

504 Corollary 12.5. *Let E be a topos, let A, B, and C be three sheaves of rings on E, let M be an A-B-bimodule, and let N be an A-C-bimodule. The abelian sheaf $\mathbf{Hom}_A(M, N)$ is canonically endowed with a B-C-bimodule structure. In particular, when A is a sheaf of commutative rings and M and N are A-modules, $\mathbf{Hom}_A(M, N)$ is canonically endowed with an A-module structure. The canonical isomorphisms b), c), and d) of 12.3 are isomorphisms of bimodules.*

We shall confine ourselves to giving indications. For every object X of E, one has $\mathbf{Hom}_A(M, N)(X) \simeq \mathrm{Hom}_{A|X}(j_X^* M, j_X^* N)$ (12.3). Consequently, the commutative group

$\mathbf{Hom}_A(M,N)(X)$ is canonically endowed with a $B(X)$-$C(X)$-bimodule structure (11.1.3), which is readily checked to be functorial in X. The other assertions are left to the reader.

COROLLARY 12.6. *Let (E,A) be a ringed topos, let X be an object of E, and let N be an A-module (a left module, to fix ideas). There are canonical isomorphisms of A-modules:*

$$\mathbf{Hom}_E(X,N) \simeq j_{X*}j_X^*N \simeq \mathbf{Hom}_A(A_X,N);$$

the A-module structure on $\mathbf{Hom}_A(A_X,N)$ comes from the bimodule structure on A_X (11.3.4).

The first isomorphism follows from 10.7, and the second from the isomorphism of A-modules $N \simeq \mathbf{Hom}_A(A,N)$ and from 12.3.

PROPOSITION 12.7. *Let (E,A) be a ringed topos, let M be a right A-module, and let N be a left A-module. The functor which assigns to every abelian sheaf P the commutative group $\mathrm{Hom}_A(M,\mathbf{Hom}_{\underline{Z}}(N,P))$ is representable by an abelian sheaf denoted by $M\otimes_A N$ and called the tensor product over A of M and N.*

There is a canonical injection of $\mathrm{Hom}_A(M,\mathbf{Hom}_{\underline{Z}}(N,P))$ into the set $\mathrm{Hom}(M,\mathbf{Hom}(N,P)) \simeq \mathrm{Hom}(M\times N,P)$; returning to the definitions, one sees immediately that a morphism f of sheaves of sets from $M\times N$ to P comes from an element of $\mathrm{Hom}_A(M,\mathbf{Hom}_Z(N,P))$ if and only if, for every object X of E, the map $f(X) : M(X)\times N(X) \to P(X)$ is an $A(X)$-bilinear map from $M(X)\times N(X)$ to $P(X)$. We call morphisms of sheaves of 505
sets from $M\times N$ to P having this property A-bilinear morphisms. Thus, the problem is to represent the functor of A-bilinear morphisms from $M\times N$ to P. To do this, one proceeds as in the ordinary case, i.e. the case where E is the punctual topos. Consider the eight morphisms:

(i): $M\times M\times N \longrightarrow M\times N \qquad 1\le i\le 3$
(i): $M\times N\times N \longrightarrow M\times N \qquad 4\le i\le 6$
(i): $M\times A\times N \longrightarrow M\times N \qquad 7\le i\le 8$

defined by the formulas:

(1) $(m_1,m_2,n) \longmapsto (m_1+m_2,n)$
(2) $(m_1,m_2,n) \longmapsto (m_1,n)$
(3) $(m_1,m_2,n) \longmapsto (m_2,n)$
(4) $(m,n_1,n_2) \longmapsto (m,n_1+n_2)$
(5) $(m,n_1,n_2) \longmapsto (m,n_1)$
(6) $(m,n_1,n_2) \longmapsto (m,n_2)$
(7) $(m,a,n) \longmapsto (ma,n)$
(8) $(m,a,n) \longmapsto (m,an)$;

here formula (i) describes the map (i)(X) for the variable objects X of E. Let L denote the "free generated abelian sheaf" functor. It is clear that the largest quotient (in the sense of abelian sheaves) of $L(M\times N)$ which equalizes $L(1)$ with $L(2)+L(3)$, $L(4)$ with $L(5)+L(6)$, and $L(7)$ with $L(8)$ represents the functor of A-bilinear morphisms from $M\times N$ to P.

12.8. One observes that the functor $P \longmapsto \mathrm{Hom}_A(N,\mathbf{Hom}_{\underline{Z}}(M,P))$ is also canonically isomorphic to the functor of A-bilinear maps from $M\times N$ to P. Consequently, the tensor product $M\otimes_A N$ also represents the functor $\mathrm{Hom}_A(N,\mathbf{Hom}_{\underline{Z}}(M,P))$. Thus there are isomorphisms, functorial in all the arguments:

(12.8.1) $\qquad \mathrm{Hom}_Z(M\otimes_A N,P) \simeq \mathrm{Hom}_A(M,\mathbf{Hom}_{\underline{Z}}(N,P)),$

(12.8.2) $\qquad \mathrm{Hom}_Z(M\otimes_A N,P) \simeq \mathrm{Hom}_A(N,\mathbf{Hom}_{\underline{Z}}(M,P)).$

506 **12.9.** It follows from (12.8), or alternatively from (12.8.1) and (12.2), that the functor $\otimes_A$ commutes with inductive limits in both arguments.

PROPOSITION 12.10. *Let C be a $\mathscr{U}$-site, let A' be a presheaf of rings on C, let M' (resp. N') be a right (resp. left) A'-module, and let A, M, and N be the sheaves associated with A', M', and N' respectively. The sheaf associated with the presheaf $X \longmapsto M'(X) \otimes_{A'(X)} N'(X)$ $(X \in \operatorname{ob} C)$ is canonically isomorphic to the sheaf $M \otimes_A N$.*

The presheaf $X \longmapsto M'(X) \otimes_{A'(X)} N'(X)$ can be constructed from the presheaves M', N', and A' by the operations specified in the proof of 12.7. Since the "associated sheaf" functor commutes with finite projective and inductive limits and with the "free generated abelian object" functors, formation of the tensor product commutes with the "associated sheaf" functor.

PROPOSITION 12.11. *Let (E, A) be a ringed topos, let M be a right A-module, let N be a left A-module, and let X be an object of E.*

a) *There is a canonical isomorphism*

$$j_X^*(M \otimes_A N) \simeq (j_X^* M) \otimes_{A|X} (j_X^* N).$$

Let, in addition, P be a right $A|X$-module and Q a left $A|X$-module.

b) *There are canonical isomorphisms (projection formulas):*

$$j_{X!}(P \otimes_{A|X} j_X^* N) \simeq (j_{X!} P) \otimes_A N$$
$$j_{X!}(j_X^* M \otimes_{A|X} Q) \simeq M \otimes_A (j_{X!} Q).$$

Proof of a). For every abelian sheaf R on $E_{/X}$, there is the following sequence of isomorphisms, functorial in R:

$$\operatorname{Hom}_{\underline{Z}}(j_X^*(M \otimes_A N), R) \simeq \operatorname{Hom}_{\underline{Z}}(M \otimes_A N, j_{X*} R) \qquad (11.2.2.1)$$
$$\operatorname{Hom}_{\underline{Z}}(M \otimes_A N, j_{X*} R) \simeq \operatorname{Hom}_A(M, \mathbf{Hom}_Z(N, j_{X*} R)) \qquad (12.8.1)$$
$$\operatorname{Hom}_A(M, \mathbf{Hom}_Z(N, j_{X*} R)) \simeq \operatorname{Hom}_A(M, j_{X*} \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \qquad (12.3)$$
$$\operatorname{Hom}_A(M, j_{X*} \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \simeq \operatorname{Hom}_{A|X}(j_X^* M, \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \qquad (11.2.2.1)$$
$$\operatorname{Hom}_{A/X}(j_X^* M, \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \simeq \operatorname{Hom}_{\underline{Z}}(j_X^* M \otimes_A j_X^* N, R) \qquad (12.8.1)$$

507 Let us exhibit the first isomorphism in b). For every abelian sheaf R, there is a sequence of isomorphisms, functorial in R:

$$\operatorname{Hom}_Z(j_{X!}(P \otimes_{A|X} j_X^* N), R) \simeq \operatorname{Hom}_Z(P \otimes_{A|X} j_X^* N, j_X^* R) \qquad (11.3.1)$$
$$\operatorname{Hom}_Z(P \otimes_{A|X} j_X^* N, j_X^* R) \simeq \operatorname{Hom}_{A/X}(P, \mathbf{Hom}_Z(j_X^* N, j_X^* R)) \qquad (12.8.1)$$
$$\operatorname{Hom}_{A|X}(P, \mathbf{Hom}_Z(j_X^* N, j_X^* R)) \simeq \operatorname{Hom}_{A|X}(P, j_X^* \mathbf{Hom}_Z(N, R)) \qquad (12.3)$$
$$\operatorname{Hom}_{A|X}(P, j_X^* \mathbf{Hom}_Z(N, R)) \simeq \operatorname{Hom}_A(j_{X!} P, \mathbf{Hom}_Z(N, R)) \qquad (11.3.1)$$
$$\operatorname{Hom}_A(j_{X!} P, \mathbf{Hom}_Z(N, R)) \simeq \operatorname{Hom}_Z(j_{X!} P \otimes_A N, R) \qquad (12.8.1)$$

The second isomorphism is obtained similarly using (12.8.2).

COROLLARY 12.12. *Let E be a topos, let A, B, and C be three sheaves of rings on E, let M be a B-A-bimodule, and let N be an A-C-bimodule. The abelian sheaf $M \otimes_A N$ is canonically endowed with a B-C-bimodule structure. In particular, when A is a sheaf of commutative rings and M and N are A-modules, $M \otimes_A N$ is canonically endowed with an A-module structure. The isomorphisms of 12.11 are isomorphisms of bimodules.*

For every object X of E, $j_X^* M$ is a right $A|X$-module on which the ring $B(X)$ acts on the left, and likewise $j_X^* N$ is a left A/X-module on which $C(X)$ acts on the right. By functoriality, the abelian sheaf $j_X^* M \otimes_{A/X} j_X^* N \simeq j_X^*(M \otimes_A N)$ is endowed with the structure of a $B(X)$-$C(X)$-"object", and this structure varies functorially in X. Consequently, $M \otimes_A N$ is endowed with a B-C-bimodule structure. This $B(X)$-$C(X)$-object structure on $j_X^*(M \otimes_A N)$ is reflected in the evident way on the functor of "A-bilinear morphisms". One then observes that, when A is commutative and M and N are A-modules, the right and left A-module structures so obtained are "equal"; consequently, $M \otimes_A N$ is an A-module in this case. The last assertion is left to the reader.

COROLLARY 12.13. *Let (E, A) be a ringed topos, let M be a left A-module (resp. a right A-module), and let X be an object of E. With the notation A_X of 11.3.3, there is a canonical isomorphism* 508

$$A_X \otimes_A M \simeq j_{X!} j_X^* M \quad \textit{(resp. } M \otimes_A A_X \simeq j_{X!} j_X^* M\textit{)}.$$

This follows from the projection formulas (12.11).

PROPOSITION 12.14. *Let E be a topos, let A and B be two sheaves of rings on E, let M be a right A-module, let N be an A-B-bimodule, and let P be a right B-module. There is a canonical isomorphism*

$$\mathrm{Hom}_B(M \otimes_A N, P) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}_B(N, P)).$$

From (12.8.1) one obtains a canonical A-bilinear morphism $\mathbf{Hom}_Z(N, P) \times N \to P$; hence, by restricting it to $\mathbf{Hom}_B(N, P) \times N$ (which is a subsheaf), one obtains an A-bilinear morphism $\mathbf{Hom}_B(N, P) \times N \to P$, which is immediately seen to be B-linear in the second factor. Thus there is a canonical morphism of B-modules $\mathbf{Hom}_B(N, P) \otimes_A N \to P$; hence a canonical map, functorial in M:

$$\mathrm{Hom}_A(M, \mathbf{Hom}_B(N, P)) \longrightarrow \mathrm{Hom}_B(M \otimes_A N, P). \tag{12.14.1}$$

Let us show that this morphism of functors is an isomorphism. Since both sides transform inductive limits in M into projective limits, it is enough to show that (12.14.1) is an isomorphism when M ranges over a generating family of the category E_A. It is therefore enough to show that (12.14.1) is an isomorphism when $M = A_X$, where X is an object of E. The verification is then immediate.

13. Morphisms of ringed topoi

DEFINITION 13.1. *Let (E, A) and (E', A') be two ringed topoi. A morphism of ringed topoi $u : (E, A) \to (E', A')$ is a pair (m, θ), where $m : E \to E'$ is a morphism of topoi (3.1) and $\theta : m^* A' \to A$ is a morphism of Rings.*

13.1.1. Since the functor $m^* : E' \to E$ is left adjoint to the functor $m_* : E \to E'$, giving a morphism of ringed topoi $(m, \theta) : (E, A) \to (E', A')$ amounts to giving a morphism of topoi $m : E \to E'$ and a morphism of sheaves of rings $\Theta : A' \to m_* A$.

13.2. To a morphism of ringed topoi $u = (m, \theta) : (E, A) \to (E', A')$, we associate two notable functors between the categories of Modules: 509

13.2.1. *The direct-image functor for Modules:* Let M be a left (resp. right) A-Module. The object $m_* M$ is canonically endowed with an $m_* A$-Module structure; restriction of scalars along the canonical morphism $\Theta : A' \to m_* A$ therefore gives an A'-Module, denoted by $u_*(M)$ and called the *direct image of M under the morphism u.*

13.2.2. *The inverse-image functor for Modules:* Let N be a left (resp. right) A'-Module. The object m^*N of E is canonically endowed with the structure of a left (resp. right) m^*A'-Module. The left A-Module $A \otimes_{m^*A'} m^*N$ (resp. the right A-Module $m^*N \otimes_{m^*A'} A$), where A is endowed with the m^*A'-Module structure defined by $\theta : m^*A' \to A$, is denoted by u^*N and is called the *inverse image of the Module N under the morphism of ringed topoi u*.

13.2.3. Note that for every A-Module M, the sheaf of sets and the abelian sheaf underlying u_*M is the sheaf m_*M. Thus the notation u_* is most often also used for the functor m_*, the direct image for sheaves of sets. On the other hand, for an A'-Module N, the sheaf of sets or abelian sheaf underlying u^*N is in general neither equal nor isomorphic to m^*N (except when θ is an isomorphism). One must therefore distinguish between the inverse image for Modules and the inverse image for sheaves of sets or abelian sheaves. The notation $u^{-1} : E' \to E$ is most often used for the functor m^*, the inverse image for sheaves of sets. The functor u^{-1} is called the *"set-theoretic" inverse-image functor* under the morphism of ringed topoi u, as opposed to the inverse image "in the sense of Modules" u^*.

Definition 13.3. *Let (C, A) and (C', A') be two $\mathscr{U}$-ringed sites. A morphism of ringed sites $u : (C, A) \to (C', A')$ is a pair (m, θ), where m is a morphism from the site C to the*
510 *site C' (4.9) and $\theta : m^*A' \to A$ is a morphism of Rings.*

13.3.1. A morphism of ringed topoi is a morphism of $\mathscr{U}$-ringed sites. A morphism of ringed sites gives rise to a morphism between the corresponding ringed topoi (11.1.1 and 4.9.1).

Proposition 13.4. *Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi.*

a) *For a variable left (resp. right) A-Module M and a variable left (resp. right) A'-Module N, there are canonical bifunctorial isomorphisms (called adjunction isomorphisms):*

$$\mathrm{Hom}_A(u^*N, M) \simeq \mathrm{Hom}_{A'}(N, u_*M), \tag{13.4.1}$$

$$u_* \mathbf{Hom}_A(u^*N, M) \simeq \mathbf{Hom}_{A'}(N, u_*M). \tag{13.4.2}$$

b) *For a variable object X of E', there is a canonical isomorphism, functorial in X:*

$$u^*A'_X \xrightarrow{\sim} A_{u^{-1}(X)}, \tag{13.4.3}$$

where A'_X and $A_{u^{-1}(X)}$ denote the generated free Modules (11.3.3).

c) *When A' is commutative and the canonical morphism $u^{-1}A' \to A$ is central (resp. when the canonical morphism $u^{-1}A' \to A$ is an isomorphism), there is, for a variable right A'-Module M and a variable left A'-Module N, a canonical bifunctorial isomorphism:*

$$u^*M \otimes_A u^*N \simeq u^*(M \otimes_{A'} N) \tag{13.4.4}$$

$$\text{(resp. (13.4.5))} \qquad u^*M \otimes_A u^*N \simeq u^{-1}(M \otimes_{A'} N).$$

First, there is a canonical isomorphism $\mathrm{Hom}_A(u^*N, M) \simeq \mathrm{Hom}_{u^{-1}A'}(u^{-1}N, M)$ (12.14), and then an isomorphism $\mathrm{Hom}_{u^{-1}A'}(u^{-1}N, M) \simeq \mathrm{Hom}_{A'}(N, u_*M)$ (III 1.7); this

gives (13.4.1). Let us exhibit the isomorphism (13.4.2). For every object X of E', there is the following sequence of functorial isomorphisms:

$$\begin{aligned}
&\mathrm{Hom}_{E'}(X, u_*\,\mathbf{Hom}_A(u^*N, M)) \simeq \mathrm{Hom}(u^{-1}X, \mathbf{Hom}_A(u^*N, M)) && \text{(adjunction)},\\
&\mathrm{Hom}_A(u^*N, \mathbf{Hom}_E(u^{-1}X, M)) \simeq \quad '' \quad '' \quad '' \quad '' && (12.1),\\
&\mathrm{Hom}_A(u^*N, \mathbf{Hom}_E(u^{-1}X, M)) \simeq \mathrm{Hom}_{A'}(N, u_*\,\mathbf{Hom}_E(u^{-1}X, M)) && (13.4.1),\\
&\mathrm{Hom}_{A'}(N, \mathbf{Hom}_{E'}(X, u_*M)) \simeq \quad '' \quad '' \quad '' \quad '' && (10.3.1),\\
&\quad '' \quad '' \quad '' \quad '' \simeq \quad \mathrm{Hom}_{E'}(X, \mathbf{Hom}_{A'}(N, u_*M)) && (12.1).
\end{aligned}$$

Formula (13.4.5) then follows from formula (13.4.2) by adjunction (the definition of the tensor product (12.7)). Formula (13.4.4) follows from (13.4.5), using the commutativity of the tensor product when the base ring is commutative (12.8). Finally, to prove (13.4.3), consider the sequence of isomorphisms

$$\begin{aligned}
&\mathrm{Hom}_A(u^*A'_X, M) \simeq \mathrm{Hom}_{A'}(A'_X, u_*M) && (13.4.1),\\
&\mathrm{Hom}_{E'}(X, u_*M) \simeq \quad '' \quad '' \quad '' && (11.3.3),\\
&\quad '' \quad '' \quad '' \simeq \mathrm{Hom}_E(u^{-1}X, M) && \text{(adjunction)},\\
&\mathrm{Hom}_A(A_{u^{-1}X}, M) \simeq \quad '' \quad '' \quad '' && (11.3.3).
\end{aligned}$$

Corollary 13.5. *Let (E, A) be a ringed topos, let $x : P \to E$ be a point of E, and let $F \mapsto F_x$ be the associated fiber functor (6.1). The fiber functor at x carries the tensor product of A-Modules to the (ordinary) tensor product of A_x-modules.*

The tensor product in P is the ordinary tensor product of modules (P, the punctual topos, is the category of sets). The assertion therefore follows from (13.4.4).

Corollary 13.6. *Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi. The direct-image functor u_* for right or left Modules commutes with projective limits and, in particular, is left exact. The inverse-image functor u^* for right or left Modules commutes with inductive limits and, in particular, is right exact.*

This follows from formula (13.4.1) (I 2.11).

13.7. Note that the inverse-image functor u^* for Modules is not, in general, exact, whereas the set-theoretic inverse-image functor u^{-1} is exact. The exactness of u^* can nevertheless be asserted when the canonical morphism $u^{-1}A' \to A$ is flat (on the right or on the left) (V 1.8). This is so, in particular, when the canonical morphism $u^{-1}A' \to A$ is an isomorphism.

13.8. Let C be a $\mathscr{U}$-site and let A' be a presheaf of rings on C; denote its associated sheaf by A. The category of presheaves of A'-modules which are sheaves is equivalent to the category of A-Modules. The notation $\mathrm{Hom}_{A'}(M, N)$ is sometimes used for the group $\mathrm{Hom}_A(M, N)$ (11.1.2). Likewise, and by abuse, the notations $\mathbf{Hom}_{A'}(M, N)$ and $M \otimes_{A'} N$ are used for the sheaves $\mathbf{Hom}_A(M, N)$ and $M \otimes_A N$. When $A' = k_E$ is the constant presheaf defined by an ordinary ring k, we also write $\mathbf{Hom}_k$, $\otimes_k$ instead of $\mathbf{Hom}_{k_E}$, $\otimes_{k_E}$.

Exercise 13.9. *Locally ringed topoi* (cf. [9] for further information in the direction considered below).

Let $(E, \mathscr{O}_E)$ be a commutatively ringed topos. For $X \in \mathrm{ob}\, E$ and $f \in \mathscr{O}_E(X)$, let X_f be the largest subobject of X on which f is invertible.

a) Show that, for $X' \in \mathrm{ob}\, E_{/X}$, $f_{X'}$ is invertible if and only if the structural morphism $X' \to X$ factors through X_f, and that, for $f, g \in \mathcal{O}_E(X)$, one has

$$X_{fg} = X_f \cap X_g.$$

b) Show that the following conditions (i) through (iii) are equivalent:

(i) For $X \in \mathrm{ob}\, E$ and $f, g \in \mathcal{O}_E(X)$, one has

$$X_{f+g} \subset \mathrm{Sup}(X_f, X_g).$$

(ii) For $X \in \mathrm{ob}\, E$ and $f, g \in \mathcal{O}_E(X)$ such that $f + g$ is invertible, one has $X = \mathrm{Sup}(X_f, X_g)$.

(iii) For $X \in \mathrm{ob}\, E$ and $f \in \mathcal{O}_E(X)$, one has

$$X = \mathrm{Sup}(X_f, X_{1-f}).$$

513 Show that these conditions imply the following condition (iv), and are equivalent to it if E has enough points:

(iv) For every point p of E, the fiber ring $\mathcal{O}_{E,p}$ is a *local ring*.

When the equivalent conditions (i) through (iii) hold, $(E, \mathcal{O}_E)$ is said to be a *locally ringed topos*; likewise, a ringed site is said to be a *locally ringed site* if the corresponding ringed topos is a locally ringed topos.

c) Let $(E, \mathcal{O}_E)$ and $(E', \mathcal{O}_{E'})$ be two locally ringed topoi, and let f be a morphism of ringed topoi from the first to the second. Show that the following condition (i) implies condition (ii), and is equivalent to it if E has enough points:

(i) For every $X' \in \mathrm{ob}\, E'$ and $s' \in \mathcal{O}_{E'}(X')$, setting $X = f^{-1}(X')$ and $s = f^*(s')$, one has

$$X_s = f^{-1}(X'_{s'}).$$

(ii) For every point p of E, setting $p' = f(p)$, the natural homomorphism on the fibers

$$\mathcal{O}_{E',p'} \longrightarrow \mathcal{O}_{E,p}$$

is a *local* homomorphism of local rings.

When condition (i) holds, f is said to be a *morphism of locally ringed topoi*, or also an *admissible morphism of locally ringed topoi* if there is a risk of confusion. We denote by **Homtoplocan**(E, E') the full subcategory of the category **Homtopan**(E, E') of all morphisms of ringed topoi from E to E' defined by the admissible morphisms from E to E'.

d) Suppose that $(E, \mathcal{O}_E)$ is the ringed topos associated with a ringed topological space $(X, \mathcal{O}_X)$ (2.1). Show that $(E, \mathcal{O}_E)$ is locally ringed if and only if $(X, \mathcal{O}_X)$ is locally ringed, i.e. if and only if, for every $x \in X$, the fiber $\mathcal{O}_{X,x}$ is a local ring. (Note that in criterion (iv) of b), it suffices to take p in a *conservative* family of
514 points of E.) Let $f : (X, \mathcal{O}_X) \to (X', \mathcal{O}_{X'})$ be a morphism of ringed spaces, where $\mathcal{O}_X$ and $\mathcal{O}_{X'}$ are sheaves of local rings. Show that the corresponding morphism of ringed topoi $\mathrm{Top}(X, \mathcal{O}_X) \to \mathrm{Top}(X', \mathcal{O}_{X'})$ is admissible if and only if the same is true of the morphism f, i.e. if and only if, for every $x \in X$,

$$\mathcal{O}_{X',f(x)} \longrightarrow \mathcal{O}_{X,x}$$

is a *local* homomorphism of local rings.

e) Let $(E, \mathcal{O}_E)$ and $(E', \mathcal{O}_{E'})$ be two locally ringed topoi such that E has enough points and $(E', \mathcal{O}_{E'})$ is equivalent to the ringed topos defined by a *scheme* $(X', \mathcal{O}_{X'})$;

prove that the category **Homtoplocan**(E, E') is equivalent to a *discrete* category, i.e. that it is a rigid groupoid (every morphism is an isomorphism, and the automorphism groups of the objects are the trivial groups).
(Hint: reduce to the case where $(E, \mathcal{O}_E)$ is the punctual topos ringed by a field.)
Recall, on the other hand, that **Homtop**(E, E') is not in general equivalent to a discrete category, even if E and E' are topoi defined by schemes (and even if E is the punctual topos), cf. 4.2.3.

f) Let E be a locally ringed topos, and let P be the ringed topos defined by the ringed space Spec k, where k is a field. Show that the admissible morphisms of locally ringed topoi $P \to E$ correspond to pairs (p, u) consisting of a point p of E and an injection of $k(p)$ into k, where $k(p)$ is the residue field of the local ring $\mathcal{O}_{E,p}$. Generalize this to a statement exhibiting the admissible morphisms from a punctual locally ringed topos to E (generalizing EGA I 2.4.4).

A *geometric point of a locally ringed topos* E is any admissible morphism into E from the locally ringed topos defined by a ringed space of the form Spec(k), where k is an *algebraically closed* field. Define the *category of geometric points of* E (with the understanding that the corresponding fields k belong to $\mathcal{U}$), denoted by **Ptgéom**(E), and a canonical functor **Ptgéom**$(E) \to$ **Point**(E). Show that this functor is faithful if and only if E has no point, i.e. **Point**(E) is the empty category. 515

g) Let E be a $\mathcal{U}$-topos and let $\mathcal{V}$ be a universe such that $\mathcal{U} \in \mathcal{V}$. Define a 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-Top})_{/E}$ in terms of the object E of the 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-Top})$ (3.3.1), following 5.14 a). When E is ringed by a Ring A, define similarly a 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-Topan})_{/E}$, and when E is locally ringed, use the notion of admissible morphism (cf. c)) to define a 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-Topanloc})_{/E}$ having **Ptgéom**(E) (cf. f)) as a full subcategory.

14. Modules on a topos defined by gluing

14.1. Let (E, A) be a ringed topos, let U be an open subtopos of E, let F be the complementary closed subtopos, and let $j : U \to E$ and $i : F \to E$ be the canonical morphisms (9.3). The topos U is ringed by j^*A (11.2.1), and the morphism of topoi i, completed in the evident way, becomes a morphism of ringed topoi. In this section, the topos F will be ringed by the sheaf i^*A, denoted by $A|F$, and the morphism j, completed in the evident way, is then a morphism of ringed topoi.

14.2. We therefore have two morphisms of ringed topoi:

$$j : (U, A|U) \longrightarrow (E, A) \text{ and } i : (F, A|F) \longrightarrow (E, A);$$

whence five functors between the corresponding categories of modules (left modules, to fix ideas):

$$(14.2.1) \qquad ({}_{A|U}U) \underset{\longleftarrow}{\overset{j_!}{\underset{j_*}{\overset{j^*}{\rightrightarrows}}}} ({}_AE) \underset{\longleftarrow}{\overset{i^*}{\rightarrow}} ({}_{A|F}F) \quad .$$

Each functor in diagram (14.2.1) is left adjoint to the functor lying below it.

14.3. We know that the functor $X \mapsto (j^*X, i^*X; i^*X \to i^*j_*j^*X)$ is an equivalence from E to the topos (U, F, i^*j_*) (9.5.4). This equivalence induces an equivalence between the corresponding categories of modules, and hence the functor $P \mapsto (j^*P, i^*P; i^*P \to i^*j_*j^*P)$ is an equivalence, denoted by Φ, from the category ${}_AE$ to 516

the category $({}_{A|U}U, {}_{A|F}F, i^*j_*)$. The functors in (14.2.1), composed with Φ or Φ^{-1}, are then the functors:

$$\begin{aligned}
\Phi \circ j_* &: M \mapsto (M, i^*j_*M; i^*j_*M \xrightarrow{\mathrm{id}} i^*j_*M),\\
j^* \circ \Phi^{-1} &: (M, N; N \to i^*j_*M) \mapsto M,\\
\Phi \circ j_! &: M \mapsto (M, 0; 0 \to i^*j_*M),\\
\Phi \circ i_* &: N \mapsto (0, N; N \to 0),\\
i^* \circ \Phi^{-1} &: (M, N; N \to i^*j_*M) \mapsto N.
\end{aligned}$$

The reader may, as an exercise, make the various adjunction morphisms explicit.

14.4. Denote by

$$i^! : {}_AE \longrightarrow {}_{A|F}F \tag{14.4.1}$$

the functor defined by the formula

$$i^! \circ \Phi^{-1}(M, N; N \xrightarrow{u} i^*j_*M) = \mathrm{Ker}(u). \tag{14.4.2}$$

PROPOSITION 14.5. *The functor $i^!$ is right adjoint to the functor i_*. The adjunction morphism $i_*i^! \to \mathrm{id}$ is a monomorphism. The functors $i^!$ (for varying rings) commute with restriction-of-scalars functors.*

It is clear that every morphism

$$(0, N; 0) \longrightarrow (M, N; N \xrightarrow{u} i^*j_*M)$$

factors uniquely through $(0, \ker(u); 0)$, which proves the adjunction property. The other assertions are trivial.

517 PROPOSITION 14.6. *For every object P of ${}_AE$, there are exact sequences, functorial in P:*

$$0 \longrightarrow j_!j^*P \longrightarrow P \longrightarrow i_*i^*P \longrightarrow 0; \tag{14.6.1}$$

$$0 \longrightarrow i_*i^!P \longrightarrow P \longrightarrow j_*j^*P, \tag{14.6.2}$$

where the nontrivial arrows are the adjunction arrows.

First note that a sequence

$$(M', N'; u') \longrightarrow (M, N; u) \longrightarrow (M'', N''; u'')$$

in $({}_{A/U}U, {}_{A/F}F; i^*j_*)$ is exact if and only if the corresponding sequences

$$\begin{aligned}
M' &\longrightarrow M \longrightarrow M''\\
N' &\longrightarrow N \longrightarrow N''
\end{aligned}$$

are exact.

The sequences (14.6.1) and (14.6.2) are transformed by the equivalence Φ into sequences of the type in (14.3):

$$\begin{aligned}
0 &\longrightarrow (M, 0; 0) \longrightarrow (M, N; u) \longrightarrow (0, N; 0) \longrightarrow 0\\
0 &\longrightarrow (0, \ker(u); 0) \longrightarrow (M, N, u) \longrightarrow (M, i^*j_*M; \mathrm{id}).
\end{aligned}$$

The exactness of these sequences is immediate.

14.7. The exact sequence (14.6.2) gives a new interpretation of the functor $i_*i^!$. Indeed, let X be an object of E. Sequence (14.6.2) gives the exact sequence of abelian groups

$$0 \longrightarrow \mathrm{Hom}_E(X, i_*i^!P) \longrightarrow \mathrm{Hom}_E(X, P) \longrightarrow \mathrm{Hom}_E(X, j_*j^*P).$$

Also denote by U the open subobject of E which is the final object of the open subtopos U. It follows from the adjunction properties of the functors j_*, j^*, and $j_!^{\mathrm{ens}}$ that the group $\mathrm{Hom}_E(X, j_*j^*P)$ is canonically isomorphic to $\mathrm{Hom}_E(X \times U, P)$, and that the morphism $\mathrm{Hom}_E(X, P) \to \mathrm{Hom}_E(X, j_*j^*P)$ is precisely the morphism defined by the monomorphism $X \times U \to X$ (5). In other words (8.5.2):

PROPOSITION 14.8. *The sections of $i_*i^!P$ on $E_{/X}$ are the sections of P on $E_{/X}$ whose support* (9.3.5) *is contained in F . In other words, $i_*i^!P$ is the largest subsheaf of P whose support is contained in F.* 518

14.9. By abuse of language, one sometimes says that $i_*i^!P$ is the submodule of P defined by the sections of P supported in F. It follows from 8.5.3 that this terminology merely extends to general topoi terminology used for topoi of sheaves on topological spaces.

PROPOSITION 14.10. 1) *The functor $P \mapsto i_*i^*P$ is isomorphic to the functor $P \mapsto i_*i^*A \otimes_A P$.*

2) *The functor $P \mapsto i_*i^!P$ is isomorphic to the functor $P \mapsto \mathbf{Hom}_A(i_*i^*A, P)$.*

The exact sequence (14.6.1), in the particular case where P is the A-module A, reads

$$0 \longrightarrow A_U \longrightarrow A \longrightarrow i_*i^*A \longrightarrow 0. \tag{14.10.1}$$

Here A_U is the free A-module generated by the open subobject U corresponding to the open subtopos U (9 and 11.3.3). For every A-module P, the A-modules $j_!j^*P$ and j_*j^*P are canonically isomorphic respectively to $A_U \otimes_A P$ and $\mathbf{Hom}_A(A_U, P)$ (12.3 and 12.6). Moreover, under these isomorphisms, the canonical morphisms $j_!j^*P \to P$ and $P \to j_*j^*P$ arise from the monomorphism $A_U \to A$. The exact sequence 14.10.1 therefore gives two exact sequences (12.2 and 12.11):

$$j_!j^*P \longrightarrow P \longrightarrow i_*i^*A \otimes_A P \longrightarrow 0,$$
$$0 \longrightarrow \mathbf{Hom}_A(i_*i^*A, P) \longrightarrow P \longrightarrow j_*j^*P;$$

and the stated isomorphisms follow by comparison with (14.6.1) and (14.6.2).

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EXPOSÉ V

Cohomology in Topoi

J.-L. Verdier

Introduction

In this exposé we present the commutative and elementary cohomological invariants of topoi. In n° 1, we study flat modules and flat morphisms of ringed topoi. The proofs are given using the hypothesis, most often satisfied in practice, that the topoi have enough points (IV 6). These proofs are repeated in the general case in Appendix n° 8, where Deligne, by means of the technique of local inductive limits, also generalizes to topoi D. Lazard's theorem on the structure of flat modules. The theorems in this exposé are existence theorems for spectral sequences relating the various cohomological invariants (n° 3, 5, 6). It is known that, even for topological spaces, Čech cohomology does not in general coincide with sheaf cohomology [11]. In Appendix n° 7, we introduce a modified Čech computation which makes it possible to obtain the sheaf cohomology of an arbitrary topos by means of coverings. In that appendix we are led to use simplicial coverings (hypercoverings), whose homotopy invariants were studied in [1] (cf. also [17]). 2

The cohomological invariants introduced here are elementary in the sense that we do not use derived categories [12]. A reader familiar with this language will immediately translate the various statements of this exposé and will then be able to generalize them to complexes and hypercohomology. The language of derived categories is, moreover, used later in this seminar.

Here we restrict ourselves to commutative cohomology. For noncommutative H^1, which is used in this seminar, and for noncommutative H^2, we refer to [9]. The functors that we derive are additive; we say nothing about multiplicative structures (cf. [7]).

0. Generalities on abelian categories

In this section we recall several lemmas, most of which can be found in [11]. 3

PROPOSITION 0.1. *Let $\mathfrak{A}$ be an abelian category having a generator. The following conditions are equivalent:*

i) *The category $\mathfrak{A}$ satisfies axiom AB 5): Small direct sums are representable, and if (X_i), $i \in I$, is a small increasing filtered family of subobjects of an object X of $\mathfrak{A}$ and Y is a subobject of X, then*

$$(\sup_i X_i) \cap Y = \sup_i (X_i \cap Y).$$

ii) *Small pseudo-filtered inductive limits* (I.2.7) *are representable and commute with finite projective limits.*

Moreover, if the preceding conditions hold, small filtered inductive limits are universal (I 2.6).

PROOF. It is clear that (ii) ⇒ (i). To show that (i) ⇒ (ii), and to prove the additional assertion, it suffices to use the fact that $\mathfrak{A}$ is a full subcategory of a category of modules $\mathscr{M}$ over a suitable ring, such that the inclusion functor $u : \mathfrak{A} \to \mathscr{M}$ admits an exact left adjoint v [5]. The verification is then immediate.

0.1.1. It is known [11] that an abelian category $\mathfrak{A}$ having a generator and satisfying axiom *AB* 5) has enough injectives, that is, every object embeds in an injective object. Moreover, by the result already cited [5], small products are representable in $\mathfrak{A}$ (axiom *AB* 3) *).

PROPOSITION 0.2. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories and let $\mathfrak{A} \underset{u}{\overset{v}{\rightleftarrows}} \mathfrak{B}$ be two adjoint functors (u is left adjoint to v). Consider the following two properties:*

i) *The functor u is exact.*

4 ii) *The functor v sends injective objects of $\mathfrak{B}$ to injective objects of $\mathfrak{A}$.*

1) *The implication* (i) ⇒ (ii) *always holds.*
If, in addition, every nonzero object of $\mathfrak{B}$ is the source of a nonzero morphism into an injective object, then (ii) ⇔ (i).

2) *Suppose that:*

a) *the category $\mathfrak{B}$ has enough injectives;*

b) *one of the two equivalent conditions* (i) *and* (ii) *above holds;*

c) *the functor u is faithful.*

Then the category $\mathfrak{A}$ has enough injectives.

PROOF. The proof is left to the reader.

REMARK 0.2.1. The category of commutative groups has enough injectives. Applying Proposition 0.2, we deduce that every category of unital modules over a ring with identity has enough injectives. Applying the result of [5] (used in the proof of 0.1) together with 0.2, we then deduce that every abelian category having a generator and exact small filtered inductive limits has enough injectives; this gives a new proof of that fact.

PROPOSITION 0.3. *Let $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}$ be three abelian categories, and let $u : \mathfrak{A} \to \mathfrak{B}$ and $v : \mathfrak{B} \to \mathfrak{C}$ be two left exact additive functors. Suppose that $\mathfrak{A}$ and $\mathfrak{B}$ have enough injective objects. The following two properties are equivalent:*

i) *There exists a spectral functor whose $E_2^{p,q}$ term is*

$$R^p v R^q u$$

and which converges to $R^{p+q}vu$ (suitably filtered).

ii) *The functor u sends injective objects of $\mathfrak{A}$ to objects that are acyclic for the functor v.*

PROOF. (i) ⇒ (ii) is immediate, since it suffices to apply the spectral functor to an

5 injective object. The implication (ii) ⇒ (i) is proved in [11].

PROPOSITION 0.4. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories and let $u : \mathfrak{A} \to \mathfrak{B}$ be a left exact additive functor. Let M be a subset of the set of objects of $\mathfrak{A}$ having the following properties:*

1) *Every object of $\mathfrak{A}$ embeds in an element of M.*

2) *If $X \oplus Y$ belongs to M, then the object X belongs to M.*

3) *If*

$$0 \longrightarrow X' \longrightarrow X \longrightarrow X'' \longrightarrow 0$$

is an exact sequence and X' and X belong to M, then X'' belongs to M, and the sequence

$$0 \longrightarrow u(X') \longrightarrow u(X) \longrightarrow u(X'') \longrightarrow 0$$

is exact. Zero objects belong to M.

Then every injective belongs to M, and the objects of M are acyclic for u, that is, for every $q \neq 0$ and every object X of M, one has $R^q u(X) = 0$. (In particular, resolutions by objects of M can be used to compute the derived functors of u.)

For the proof, see [11] 3.3.1.

PROPOSITION 0.5. *Let $\mathscr{U} \subset \mathscr{V}$ be two universes, let $\mathfrak{A}$ (resp. $\mathfrak{B}$) be a $\mathscr{U}$-category (resp. $\mathscr{V}$-category) which is abelian, satisfies axiom AB 5) relative to $\mathscr{U}$ (resp. to $\mathscr{V}$), and has a $\mathscr{U}$-small (resp. $\mathscr{V}$-small) generating family. Let $\epsilon : \mathfrak{A} \to \mathfrak{B}$ be an exact fully faithful functor. The following conditions are equivalent:*

1) *There exists a generating family $(X_i)_{i\in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i\in I}$ is generating in $\mathfrak{B}$.*

1′) *Every object of $\mathfrak{B}$ is isomorphic to a quotient of an object of the form $\bigoplus_{\alpha\in A} \epsilon(Y_\alpha)$, where $A \in \mathscr{V}$.*

Under these conditions, the following property holds:

2) *ϵ sends $\mathscr{U}$-small products to products (and thus commutes with $\mathscr{U}$-small projective limits).*

Moreover, under the equivalent conditions 1) or 1′), the following conditions are equivalent : 6

a) *For every object Y of $\mathfrak{A}$, every subobject of $\epsilon(Y)$ is isomorphic to the image under ϵ of a subobject of Y.*

a′) *There exists a generating family $(X_i)_{i\in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i\in I}$ is generating in $\mathfrak{B}$ and such that, for every $i \in I$, every subobject of $\epsilon(X_i)$ is isomorphic to the image under ϵ of a subobject of X_i.*

b) *Every object of $\mathfrak{B}$ is isomorphic to a subobject of an object of the form $\prod_{\alpha\in A} \epsilon(Y_\alpha)$, where $A \in \mathscr{V}$.*

c) *ϵ commutes with $\mathscr{U}$-small direct sums (and thus commutes with $\mathscr{U}$-small inductive limits).*

Moreover, under conditions 1) and a′), one has:

d) *ϵ sends injective objects to injective objects.*

REMARK 0.5.1. When in 0.5 one has $\mathscr{U} = \mathscr{V}$, conditions 1) and a′) imply that ϵ is an equivalence of categories (since b), 2), and a) then hold).

0.5.2. We will confine ourselves to giving indications for the proof. The implications 1) $\Leftrightarrow$ 1′) and 1) $\Rightarrow$ 2) are left to the reader. It is clear that a) $\Rightarrow$ a′). We show that a′) implies d). Let M be injective in $\mathfrak{A}$. For every $i \in I$ and every subobject $Y \hookrightarrow X_i$ of X_i, the homomorphism $\mathrm{Hom}(X_i, M) \to \mathrm{Hom}(Y, M)$ is surjective. Consequently, by a′) and the full faithfulness of ϵ, for every subobject $U \hookrightarrow \epsilon(X_i)$, the homomorphism $\mathrm{Hom}(\epsilon(X_i), \epsilon(M)) \to \mathrm{Hom}(U, \epsilon(M))$ is surjective. Since the $\epsilon(X_i)$ form a generating family of $\mathfrak{B}$, $\epsilon(M)$ is injective [11].

We show that a′) $\Rightarrow$ b). By enlarging the family of the X_i if necessary, we may suppose that, for every $i \in I$, every quotient of X_i is isomorphic to an X_j for a suitable j. For every $i \in I$, let $X_i \hookrightarrow M_i$ then be a monomorphism into an injective object. The family $(\epsilon(X_i))_{i\in I}$ is stable under quotients, and for $i \in I$, the morphism $\epsilon(X_i) \hookrightarrow \epsilon(M_i)$

is, by d), a monomorphism into an injective object. It is then immediately verified that, since the family $\epsilon(X_i)$ is generating, the family $(\epsilon(M_i))_{i\in I}$ is cogenerating, whence b).

We show that b) ⇒ c). Let $(Z_\alpha)_{\alpha\in A}$ be a $\mathscr{U}$-small family of objects of $\mathfrak{A}$, and let us show that the canonical morphism

$$\bigoplus_{\alpha\in A}\epsilon(Z_\alpha)\longrightarrow\epsilon(\bigoplus_{\alpha\in A}Z_\alpha)$$

7 is an isomorphism. Since the family of objects $\epsilon(Y)$ is cogenerating, it suffices to show that, for every object Y of $\mathfrak{A}$, the homomorphism

$$\mathrm{Hom}(\epsilon(\bigoplus_{a}Z_a),\epsilon(Y))\longrightarrow\mathrm{Hom}(\bigoplus_{a}\epsilon(Z_a),\epsilon(Y))$$

is an isomorphism, which follows from the full faithfulness of ϵ.

It remains to show that c) ⇒ a). Let Z be a subobject of $\epsilon(Y)$. By 1′), there exists a $\mathscr{U}$-small family Y_α of objects of $\mathfrak{A}$ and an epimorphism from $\bigoplus_\alpha\epsilon(Y_\alpha)$ onto Z. Since ϵ commutes with $\mathscr{U}$-small direct sums, there is therefore an epimorphism $\epsilon(Y')\to Z$. Denote by $u:\epsilon(Y')\to Z\to\epsilon(Y)$ the composite morphism. We have $Z=\mathrm{Im}(u)$. Since ϵ is fully faithful, we have $u=\epsilon(v)$, and consequently $Z=\mathrm{Im}(\epsilon(v))=\epsilon(\mathrm{Im}(v))$.

Exercise 0.5.2. Let $\mathrm{Sex}_{\mathscr{V}}(\mathfrak{A})$ be the category of contravariant functors from $\mathfrak{A}$ to the category of $\mathscr{V}$-commutative groups which commute with $\mathscr{U}$-small inductive limits. Show that, under conditions 1) and a′), the canonical functor from $\mathfrak{B}$ to $\mathrm{Sex}_{\mathscr{V}}(\mathfrak{A})$ is an equivalence.

1. Flat modules

Definition 1.1. *Let (E,A) be a ringed topos* (IV 11.1.1). *A right (resp. left) A-module M is said to be flat if the functor $M\otimes_A\cdot$ (resp. $\cdot\otimes_A M$) from the category of left (resp. right) A-modules to the category of abelian sheaves of E is exact.*

Proposition 1.2. *Let M be a B-A-bimodule.*

1) *The following properties are equivalent:*
   i) *The module M is flat as a right A-module.*
   ii) *For every injective B-module I, the left A-module $\mathbf{Hom}_B(M,I)$ is injective.*
2) *A module M which is a pseudo-filtered inductive limit* (I 2.7.1) *of flat modules is flat.*
3) *Finally, if $M_\bullet=\cdots\to M_{i+1}\to M_i\to\cdots$ is an acyclic complex of flat modules ($M_i=0$ for $i<i_0$), then, for every module F, the complex*

$$M_\bullet\otimes_A F=\cdots\longrightarrow M_{i+1}\otimes_A F\longrightarrow M_i\otimes_A F\longrightarrow\cdots$$

8 *is acyclic.*

Proof. By (IV 12.12), there is an adjunction isomorphism:

$$\mathrm{Hom}_B(M\otimes_A\cdot,\cdot)\xrightarrow{\sim}\mathrm{Hom}_A(\cdot,\mathbf{Hom}_B(M,\cdot)).$$

It is then enough to apply 0.2 to obtain the equivalence (i) ⇔ (ii). This adjunction isomorphism also shows that the tensor product commutes with inductive limits. The fact that pseudo-filtered inductive limits are exact (0.1) implies the second assertion. To show that the complex $M_\bullet\otimes_A F$ is acyclic, it is enough to show that, for every injective abelian sheaf I, the complex $\mathrm{Hom}^\bullet_Z(M_\bullet\otimes_A F,I)$ is acyclic. By the adjunction formulas, this

complex is isomorphic to $\mathrm{Hom}_A^{\bullet}(F, \mathbf{Hom}_Z(M_{\bullet}, I))$. But, by the equivalence (i) $\Leftrightarrow$ (ii), the complex of sheaves

$$\mathbf{Hom}_Z(M_{\bullet}, I)$$

is an acyclic complex whose objects are injective, whence the conclusion.

PROPOSITION 1.3.1. *Let (E, A) be a ringed topos, let H be an object of E, and let M be a flat $A_{/H}$-module. Then $j_{H!}M$ is a flat A-module. In particular, A_H is flat on both the right and the left.*

Suppose, to fix ideas, that M is a right $A_{/H}$-module. For every left A-module P, there is a canonical isomorphism (IV 12)

$$j_{H!}M \otimes_A P \simeq j_{H!}(M \otimes_{A_{/H}} j_H^* P).$$

The functors $j_{H!}$ and j_H^* are exact (IV 11.3.1 and 11.12.2), and, by hypothesis, the functor $M \otimes_{A_{/H}} -$ is exact. Consequently, the functor $P \longmapsto j_{H!}M \otimes_A P$ is exact, and $j_{H!}M$ is flat.

PROPOSITION 1.3.2. *(Projection formula for closed immersions):*
Let (E, A) be a ringed topos, and let $i : F \to E$ be a closed subtopos of E. Set $i^*A = A_{/F}$. For every right $A_{/F}$-module M and every left A-module P, there is a functorial 9
isomorphism

$$i_*(M \otimes_{A_{/F}} i^* P) \simeq (i_* M) \otimes_A P.$$

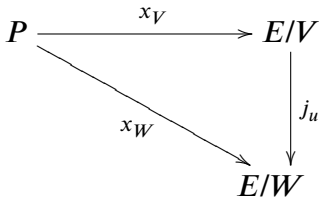
Let U be the open complement of F, and let $j : U \to E$ be the canonical immersion morphism. One has $j^*(i_*M \otimes_A P) \simeq 0 \otimes_{A_{/U}} j^*(P)$ (IV 12); consequently, $i_*M \otimes_A P$ has its support in F. Hence the adjunction morphism $i_*M \otimes_A P \to i_* i^*(i_*M \otimes_A P)$ is an isomorphism (IV 14). One has $i^*(i_*M \otimes_A P) \simeq i^* i_* M \otimes_{A_{/F}} i^* P$ (IV 12), and, since $i : F \to E$ is a closed immersion, $i^* i_* M \simeq M$ (IV 14); this gives the asserted isomorphism.

COROLLARY 1.3.3. *For every flat $A_{/F}$-module M, i_*M is flat.*

It follows from 1.3.2 and from (IV 14) that the functor $P \longmapsto (i_*M) \otimes_A P$ is exact.

1.4. Let (E, A) be a ringed topos, let $x : P \to E$ be a point of E (IV 6.1), and let $\mathrm{Vois}(x)$ be the category of neighborhoods of x (IV 6.8). To every object V of $\mathrm{Vois}(x)$ there corresponds an object of E, also denoted by V, and a point $x_V : P \to E/V$ of E/V. Moreover, to every morphism $u : V \to W$ in $\mathrm{Vois}(x)$ there corresponds an essentially commutative diagram of morphisms of topoi (IV 6.7)

(1.4.1)



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Let N be an A_x-module (resp. a set). From the diagram (1.4.1), one obtains a diagram of A_x-modules (resp. sets):

(1.4.2)
$$\begin{array}{ccc} x_W^* x_{W*} N & \xrightarrow{\mathrm{ad}_W} & N \\ {\scriptstyle \phi(u)}\downarrow & \nearrow_{\mathrm{ad}_V} & \\ x_V^* x_{V*} N & & \end{array}$$

10 where ad_W and ad_V are the adjunction morphisms. Moreover, one immediately checks that $\phi(uv) = \phi(v)\phi(u)$. Thus there is a functor from the filtered category $\mathrm{Vois}(x)^\circ$ to the category of A_x-modules (resp. sets), and a homomorphism

(1.4.3)
$$\Lambda(N) : \varinjlim_{V \in \mathrm{Vois}(x)^\circ} x_V^* x_{V*} N \longrightarrow N.$$

Proposition 1.5. *For every A_x-module (resp. set) N, $\Lambda(N)$ is an isomorphism.*

This proposition is a special case of a proposition due to Deligne (8.2.6). Let us give a direct proof. By passing to the underlying sets, it is enough to prove the proposition when N is a set (I 2.8). The cofiltered category $\mathrm{Fl}(\mathrm{Vois}(x))$ of arrows of $\mathrm{Vois}(x)$ is fibered over the category $\mathrm{Vois}(x)$ by the functor $p : \mathrm{Fl}(\mathrm{Vois}(x)) \to \mathrm{Vois}(x)$ which assigns to an arrow its target. Let D be a category and $F : \mathrm{Fl}(\mathrm{Vois}(x)) \to D$ a pro-object (I 8.10). For every object V of $\mathrm{Vois}(x)$, denote by F_V the pro-object obtained by restricting the functor F to the fiber category $\mathrm{Vois}(x)/V$. To every morphism $m : U \to V$ of $\mathrm{Vois}(x)$, the base-change functor by m associates a morphism from the pro-object F_U to the pro-object F_V, and the canonical morphisms of pro-objects $F \to F_V$ determine a morphism of pro-objects

(1.5.1)
$$F \longrightarrow \varprojlim_{V \in \mathrm{Vois}(x)^\circ} F_V$$

which is immediately checked to be an isomorphism. Apply this observation to the pro-object of pointed sets

(1.5.2)
$$(U, V, m) \longmapsto x_V^*(m) = F(U, V, m).$$

One obtains an isomorphism of pro-objects

(1.5.3)
$$F \xrightarrow{\sim} \varprojlim_{\mathrm{Vois}(x)} \varprojlim_{\mathrm{Vois}(x)/V} x_V^*(m);$$

hence, for every set N, a bijection

(1.5.4)
$$\varinjlim_{\mathrm{Fl}(\mathrm{Vois}(x))^\circ} \mathrm{Hom}(x_V^*(m), N) \simeq \varinjlim_{\mathrm{Vois}(x)^\circ} \varinjlim_{(\mathrm{Vois}(x)/V)^\circ} \mathrm{Hom}(x_V^*(m), N).$$

11 But, for fixed V, $\varinjlim_{(\mathrm{Vois}(x)/V)^\circ} \mathrm{Hom}(x_V^*(m), N)$ is identified with $x_V^* x_{V*} N$. Indeed, by IV 6.8.1 one has $x_V^* x_{V*} N \simeq \varinjlim_{\mathrm{Vois}(x_V)^\circ} \mathrm{Hom}_{E/V}(W, x_{V*} N)$, and hence, by adjunction, $x_V^* x_{V*} N \simeq \varinjlim_{\mathrm{Vois}(x_V)^\circ} \mathrm{Hom}(x_V^*(W), N)$; moreover, the category $\mathrm{Vois}(x_V)$ is equivalent to the category $\mathrm{Vois}(x)/V$ (IV 6.7.2). Finally, the canonical transition maps $\varinjlim_{(\mathrm{Vois}(x)/U)^\circ} \mathrm{Hom}(x_U^*(m), N) \to$

$\varinjlim_{(\mathrm{Vois}(x)/V)^\circ} \mathrm{Hom}(x_V^*(n), N)$ in (1.5.4) are identified with the canonical maps $x_U^* x_{U*} N \to x_V^* x_{V*} N$ of (1.4.3), as the reader may verify. Thus there is a bijection

$$\varinjlim_{\mathrm{Fl}(\mathrm{Vois}(x))^\circ} \mathrm{Hom}(x_V^*(m), N) \simeq \varinjlim_{\mathrm{Vois}(x)^\circ} x_V^* x_{V*} N \tag{1.5.5}$$

and the map $\Lambda(N) : \varinjlim_{\mathrm{Vois}(x)^\circ} x_V^* x_{V*} N \to N$ of (1.4.3), composed with the bijection (1.5.5), comes from the maps $\mathrm{Hom}(x_V^*(m), N) \to N$ which assign to a map $r : x_V^*(m) \to N$ the image under r of the distinguished point of $x_V^*(m)$. To prove the proposition, it is then enough to observe that every object (U, V, m) of $\mathrm{Fl}(\mathrm{Vois}(x))$ is bounded below by (U, U, id_U), and that $x_U^*(\mathrm{id}_U)$ consists of a single element or, in other words, that the canonical morphism from the constant pro-object consisting of a single element to F is an isomorphism of pro-objects.

PROPOSITION 1.6. *Let (E, A) be a ringed topos, and let M be an A-module.*

1) *If M is flat, then, for every point $x : P \to E$ of E, the A_x-module M_x is flat.*
2) *Let $(x_i)_{i\in I}$ be a conservative family of points of E such that, for every $i \in I$, M_{x_i} is a flat A_{x_i}-module.*

Then M is flat.

LEMMA 1.6.1. *Let M be a flat module and let V be an object of E. The A/V-module $j_V^* M$ is flat.*

We must show that the functor $P \longmapsto P \otimes_{A/V} j_V^* M$ is exact. Since the extension-by-zero functor $j_{V!}$ is exact and faithful (IV 11.3.1), it is enough to show that the functor $P \longmapsto j_{V!}(P \otimes_{A/V} j_V^* M)$ is exact. There is a functorial isomorphism $j_{V!}(P \otimes_{A/V} j_V^* M) \simeq$ 12
$j_{V!}(P) \otimes_A M$ (IV 12.11); consequently, the functor $P \longmapsto j_{V!}(P) \otimes_A M$ is exact (IV 11.3.1).

1.6.2. Let us prove 1). To fix ideas, suppose that M is a flat left A-Module, and let $x : P \to E$ be a point of E. We shall show that the functor $N \mapsto N \otimes_{A_x} M_x$ from the category of right A_x-modules to the category of commutative groups is exact. For this it suffices to show that this functor is left exact. With the notation of 1.4, let V be a neighborhood of x, and let $j_V : E/V \to E$ be the localization morphism. For every A_x-module N, we have $x_V^* x_{V*} N \otimes_{A_x} M_x \simeq x_V^* x_{V*} N \otimes_{A_x} x_V^* j_V^* M \simeq x_V^*(x_{V*} N \otimes_{A/V} j_V^* M)$ (IV 12.11). Consequently, the functor $N \mapsto x_V^* x_{V*} N \otimes_{A_x} M_x$ is left exact. It follows that the functor $N \mapsto N \otimes_{A_x} M_x$, being a filtered inductive limit of left exact functors (1.5), is left exact.

Let us prove 2). Let $0 \to P' \to P \to P'' \to 0$ be an exact sequence of A-Modules, and let us show that the sequence $0 \to P' \otimes_A M \to P \otimes_A M \to P'' \otimes_A M \to 0$ is exact. It suffices to show that, for every $i \in I$, the sequence obtained by passing to the fibers at x_i is exact (IV 6). But the sequence of fibers at x_i is the sequence (IV 13.5)

$$0 \longrightarrow P'_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow P_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow P''_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow 0,$$

and since M_{x_i} is flat, this sequence is exact.

PROPOSITION 1.7. *Let (E, A) be a ringed topos, let $u : E' \to E$ be a morphism of topoi, and put $A' = u^* A$. Let M be a right (resp. left) A'-Module. The following conditions are equivalent:*

i) *The functor $N \mapsto M \otimes_{A'} u^* N$ (resp. $N \mapsto u^* N \otimes_{A'} M$) from the category of left (resp. right) A-Modules to the category of abelian sheaves on E' is exact.*
ii) *M is a flat A'-Module.*

It is clear that ii) ⇒ i).

Let us prove that i) ⇒ ii). We give the proof only in the case where E' possesses a conservative family of points $(x_i)_{i\in I}$. The general case is treated in (8.2.7). In this
13 particular case, which covers most applications, we reduce to the case where E' is the punctual topos and hence where the morphism u, which we then denote by x, is a point of E (1.6 and IV 11.3.1). We restrict ourselves to the case where M is a right A'-Module. The case where M is a left A'-Module reduces to this one by passing to opposite rings. Let V be a neighborhood of x, and let V_x be its fiber at x. With the notation of 1.4, there is an essentially commutative diagram of morphisms of topoi:

$$\begin{array}{ccccc} P & \xrightarrow{j} & P/V_x & \xrightarrow{j_{V_x}} & P \\ & \searrow^{x_V} & \downarrow{\scriptstyle x/V} & & \downarrow{\scriptstyle x} \\ & & E/V & \longrightarrow & E \end{array}$$

where x/V is the morphism deduced from x by localization over V (IV 5.10), and j is the morphism deduced from the marked point of V_x. Let us show that the functor $N' \mapsto M \otimes_{A'} x_V^* N'$ is exact. We have $x_V^* = j^*(x/V)^*$ and $\mathrm{id} = j^* j_{V_x}^*$; consequently, there is a canonical isomorphism

$$M \otimes_{A'} x_V^* N' \simeq j^*\big(j_{V_x}^* M \otimes_{A'/V_x} (x/V)^* N'\big).$$

Since j^* is exact and the functor $j_{V_x!}$ is exact and faithful (IV 11.3.1), it suffices to show that the functor

$$N' \longmapsto j_{V_x!}\Big(j_{V_x}^* M \otimes_{A'/V_x} (x/V)^* N'\Big)$$

is exact; consequently (IV 12.11), it suffices to show that the functor $N' \mapsto M \otimes_{A'} x^* j_{V!} N'$ is exact. But it follows from the definition of the inverse image of an induced topos (IV 5.10.2) that $j_{V_x!}(x/V)^* N'$ is equal to $x^* j_{V!} N'$, and since the functor $j_{V!}$ is exact (IV 11.3.1), the functor $N' \mapsto M \otimes_{A'} j_{V_x!}(x/V)^* N'$ is exact by hypothesis.

For every A'-module Q, there is a functorial isomorphism (1.5):

$$Q \simeq \varinjlim_{\mathrm{Vois}(x)^\circ} x_V^* x_{V*} Q.$$

By what precedes, the functor $Q \mapsto M \otimes_{A'} Q$ is a filtered inductive limit of left exact functors. It is therefore left exact, and hence M is flat.

Corollary 1.7.1. *Let $u : (E', A') \to (E, A)$ be a morphism of ringed topoi, and let M be a flat A-Module. Then $u^* M$ is a flat A'-Module.*

This follows from the criterion given in 1.7 and formula (IV 13.4.5).

14 Corollary 1.7.2. *Let $u : (E', A') \to (E, A)$ be a morphism of ringed topoi. The following conditions are equivalent:*

i) *The right (resp. left) $u^{-1}(A)$-Module A' is flat.*
ii) *The functor u^* from the category of left (resp. right) A-Modules to the category of left (resp. right) A'-Modules is exact.*

The equivalence follows from 1.7 and the definition of u^* (IV 13.2.1).

Definition 1.8. *A morphism of ringed topoi having the equivalent properties of 1.7.2 is called a morphism of ringed topoi which is left (resp. right) flat.*

PROPOSITION 1.9. *Let $\mathscr{U} \subset \mathscr{V}$ be two universes, let C be a $\mathscr{U}$-site, let $C^\sim_\mathscr{U}$ and $C^\sim_\mathscr{V}$ be the categories of $\mathscr{U}$- and $\mathscr{V}$-sheaves of sets, respectively, let $\epsilon : C^\sim_\mathscr{U} \to C^\sim_\mathscr{V}$ be the canonical inclusion functor, and let A be a $\mathscr{U}$-sheaf of rings on C.*

1) *The functor ϵ commutes with $\mathscr{U}$-small inductive and projective limits of Modules. The functor ϵ is conservative and fully faithful on the categories of Modules.*
2) *For every object H of $C^\sim_\mathscr{U}$, there is a canonical isomorphism $\epsilon(A_H) \simeq \epsilon(A)_{\epsilon(H)}$.*
3) *Every injective left or right A-Module is carried by ϵ to an injective $\epsilon(A)$-Module.*
4) *Every flat right or left A-Module is carried by ϵ to a flat $\epsilon(A)$-Module.*
5) *The functor ϵ commutes with the formation of tensor products and sheaves of homomorphisms.*

The formation of the associated sheaf is independent of the universe (II 3.6). It then follows from the construction of inductive and projective limits in categories of sheaves (II 4.1 and 6.4) that the functor ϵ commutes with $\mathscr{U}$-small projective limits of sets or Modules, which gives 1). Let us prove 2). By taking a $\mathscr{U}$-small generating subcategory of $C^\sim_\mathscr{U}$, we may reduce to the case where C is $\mathscr{U}$-small (III 4.1). It then follows from 15
(II 6.5) that the free A-Module generated by H is obtained by forming the presheaf of free A-modules generated by H, a construction that commutes with enlargement of universes, and then forming the associated sheaf, an operation which likewise commutes with enlargement of universes (II 3.6). To prove 3), it suffices, by 0.5, to show that every $\mathscr{V}$-subsheaf of a $\mathscr{U}$-sheaf is a $\mathscr{U}$-sheaf, which is clear.

Let us prove 4). Let M be a flat A-Module. It suffices to show that, for every injective $\epsilon(A)$-Module J, the abelian sheaf $\mathbf{Hom}_{\epsilon(A)}(\epsilon(M), J)$ is injective (1.2). By 0.5, J is a subobject, hence a direct factor, of an object of the form $\prod_\alpha \epsilon(I_\alpha)$, where the I_α are injective. We may therefore suppose that $J = \epsilon(I)$, where I is an injective object. If the homomorphism

$$\epsilon(\mathbf{Hom}_A(M, I)) \longrightarrow \mathbf{Hom}_{\epsilon(A)}(\epsilon(M), \epsilon(I))$$

is an isomorphism, it follows from 3) that $\mathbf{Hom}_{\epsilon(A)}(\epsilon(M), \epsilon(I))$ is injective. It therefore remains to prove 5). The fact that formation of the tensor product commutes with the functor ϵ follows from IV 12.6 and the fact that formation of the associated sheaf commutes with enlargement of the universe (II 3.6). For every object X of $C^\sim_\mathscr{U}$ and every pair of A-Modules M, N of $C^\sim_\mathscr{U}$, we therefore have

$$\mathrm{Hom}_{C^\sim_\mathscr{V}}(\epsilon(X), \epsilon(\mathbf{Hom}_A(M, N))) \simeq \mathrm{Hom}_{C^\sim_\mathscr{U}}(X, \mathbf{Hom}_A(M, N)) \simeq \mathrm{Hom}_A(\mathbf{Z}_X \otimes_\mathbf{Z} M, N)$$

by the full faithfulness of ϵ, and then

$$\mathrm{Hom}_A(\mathbf{Z}_X \otimes_\mathbf{Z} M, N) \simeq \mathrm{Hom}_{\epsilon(A)}(\epsilon(\mathbf{Z}_X \otimes_\mathbf{Z} M), \epsilon(N)) \simeq \mathrm{Hom}_{\epsilon(A)}(\epsilon(\mathbf{Z}_X) \otimes_\mathbf{Z} \epsilon(M), \epsilon(N))$$

by the full faithfulness of ϵ and what precedes. Using 2) and IV 12.14, we finally obtain an isomorphism

$$\mathrm{Hom}_{C^\sim_\mathscr{V}}(\epsilon(X), \epsilon(\mathbf{Hom}_A(M, N))) \simeq \mathrm{Hom}_{C^\sim_\mathscr{V}}(\epsilon(X), \mathbf{Hom}_{\epsilon(A)}(\epsilon(M), \epsilon(N)));$$

hence the asserted isomorphism.

1.10. Let E be a topos and let $\mathscr{U} = (U_i \to X)_{i \in I}$ be a small family of morphisms with the same target. For every finite set $\Delta_n = \{0, \dots, n\}$, put

$$S_n(\mathscr{U}) = \coprod_{f : \Delta_n \to I} U_{f(0)} \times_X \cdots \times_X U_{f(n)},$$

the coproduct being taken over the set of maps from Δ_n to I. Let m, n be two integers, and let $g : \Delta_m \to \Delta_n$ be a map. We define a morphism 16

(1.10.1) $$s(g) : S_n(\mathcal{U}) \longrightarrow S_m(\mathcal{U})$$

as follows: for every $f : \Delta_n \to I$, the restriction of $s(g)$ to $U_{f(0)} \times_X \cdots \times_X U_{f(n)}$ is the composite of the canonical inclusion

$$U_{f(g(0))} \times_X \cdots \times_X U_{f(g(m))} \hookrightarrow S_m(\mathcal{U}),$$

and the unique morphism

$$s_f(g) : U_{f(0)} \times_X \cdots \times_X U_{f(n)} \longrightarrow U_{f(g(0))} \times_X \cdots \times_X U_{f(g(m))}$$

such that, for every $i \in \Delta_m$,

$$\mathrm{pr}_{f(g(i))}\, s_f(g) = \mathrm{pr}_{f(g(i))}$$

(pr_a denotes the a-th projection). We thus obtain a contravariant functor $\Delta_n \mapsto S_n$ from the category of finite sets to E; in other words, a semi-simplicial complex $S_\bullet(\mathcal{U})$ of objects of E. Note that this complex is canonically augmented toward X. Every functor from E to a category C carries $S_\bullet(\mathcal{U})$ to a simplicial object of C. In particular, if A is a Ring of E, the "free A-Module generated by" functor carries $S_\bullet(\mathcal{U})$ to a simplicial complex of A-biModules, augmented toward A_X and denoted by $A_\bullet(\mathcal{U})$. We have

(1.10.2) $$A_n(\mathcal{U}) = \bigoplus_{f:\Delta_n \to I} A_{U_{f(0)} \times_X \cdots \times_X U_{f(n)}}.$$

This complex will often be written

(1.10.3) $$\cdots \quad \overset{\overset{s_0}{\rightarrow}\ \overset{s_1}{\rightarrow}\ \overset{s_2}{\rightarrow}}{} \quad \bigoplus_{i,j} A_{U_i \times_X U_j} \quad \overset{\overset{s_0}{\rightarrow}\ \overset{s_1}{\rightarrow}}{} \quad \bigoplus_i A_{U_i} \longrightarrow A_X$$

where the arrows in (1.10.3) (except the last, which is the augmentation) represent the face operators of the simplicial complex $A_\bullet(\mathcal{U})$, that is, the morphisms corresponding to the increasing injective maps from Δ_n into Δ_{n+1} (s_i omits the integer i). To the complex $A_\bullet(\mathcal{U})$ we associate a differential complex augmented toward A_X:

(1.10.4) $$\cdots \xrightarrow{d} \bigoplus_{i,j} A_{U_i \times_X U_j} \xrightarrow{d} \bigoplus_i A_{U_i} \longrightarrow A_X,$$

17 by putting

(1.10.5) $$d = \sum (-1)^i s_i.$$

Proposition 1.11. *When the family $\mathcal{U} = (U_i \to X)_{i \in I}$ is epimorphic, the differential complex (1.10.4) is acyclic and gives a resolution of A_X.*

Denote by $\mathbf{Z}$ the constant sheaf with value $\mathbf{Z}$. By the definition of the "free A-Module generated by" functor, for every object H of E we have

$$A_H \simeq \mathbf{Z}_H \otimes_{\mathbf{Z}} A,$$

hence

$$A_\bullet(\mathcal{U}) \simeq \mathbf{Z}_\bullet(\mathcal{U}) \otimes_{\mathbf{Z}} A.$$

Since the components of $\mathbf{Z}_\bullet(\mathcal{U})$ are flat $\mathbf{Z}$-Modules (1.3.1), it suffices to prove the proposition when $A = \mathbf{Z}$.

Suppose first that E is the topos of sets. The augmented complex $S_\bullet(\mathscr{U})$ is then a direct sum of augmented complexes of the form

$$\cdots \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; S\times S\times S \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; S\times S \;\rightrightarrows\; S \longrightarrow \{e\} \qquad (\{e\} \text{ a one-element set}).$$

Each of these augmented complexes is homotopically trivial. Thus $S_\bullet(\mathscr{U})$ is a homotopically trivial augmented complex, and its homology is therefore trivial, proving the proposition in this case.

Now let $p : \mathrm{Ens} \to E$ be a point of E. Since formation of the complex $\mathbf{Z}_\bullet(\mathscr{U})$ commutes with the inverse-image functors of morphisms of topoi, $p^*(\mathbf{Z}_\bullet(\mathscr{U})) \simeq \mathbf{Z}_\bullet(p^*(\mathscr{U}))$ is a resolution of $\mathbf{Z}_{p^*X} \simeq p^*(\mathbf{Z}_X)$; this proves the proposition when E has enough fiber functors (IV 4.6). This holds, in particular, when E is a topos of presheaves $C^\wedge$, since for every object X of C, $\Gamma(X,-)$ is a fiber functor. In the general case, E is equivalent to a topos of sheaves on a small site C (IV 1), and the epimorphic family $\mathscr{U} = (U_i \to X)_{i\in I}$ is the image, under the "associated sheaf" functor, of an epimorphic family $\mathscr{U}' = (U'_i \to X')_{i\in I}$. Consequently, $\mathbf{Z}_\bullet(\mathscr{U}) \simeq \underline{a}\mathbf{Z}_\bullet(\mathscr{U}')$ is a resolution of $\underline{a}\mathbf{Z}_{X'} \simeq \mathbf{Z}_X$. 18

2. Čech cohomology. Cohomological notation

2.1. General notation.

2.1.1. Let (E, A) be a ringed topos, and let M, N be two A-Modules (left modules, to fix ideas). We denote by $\mathrm{Ext}^q_A(E; M, N)$ the value at N of the q-th right derived functor of the functor $\mathrm{Hom}_A(M,\cdot)$ [11]. The functors $\mathrm{Ext}^q_A(E; M, N)$, $q \in \mathbf{N}$, form a δ-functor in the variable N. They also form a contravariant δ-functor in the variable M. By definition, $\mathrm{Ext}^0_A(E; M, N) = \mathrm{Hom}_A(M, N)$.

2.1.2. Let X be an object of E. When $M = A_X$, the free A-Module generated by X (IV 12), we put $\mathrm{Ext}^q_A(E; A_X, N) = H^q(X, N)$. Notice that the ring A no longer appears in this new notation. This cannot cause confusion, since we will show (3.5) that the formation of $H^q(X,\cdot)$ commutes with restriction of scalars. The functor $H^q(X,\cdot)$ is the q-th right derived functor of $\mathrm{Hom}_A(A_X,\cdot) = \mathrm{Hom}_E(X,\cdot)$, also denoted by $\Gamma(X,\cdot)$. When $M = A$, we put $\mathrm{Ext}^q_A(E; A, N) = H^q(E, N)$.

2.2. Localization.

Let X be an object of E, and let $j : E_{/X} \to E$ be the localization morphism (IV 8). The functor j_X^* for Modules is exact (4.11) and admits an exact left adjoint $j_{X!}$ (4.11). Consequently, it sends injective Modules to injective Modules. Thus, for a variable A-Module N of E and a variable $A|X$-Module M of $E_{/X}$, there are functorial isomorphisms

$$\mathrm{Ext}^q_{A|X}(E_{/X}; M, j_X^* N) \simeq \mathrm{Ext}^q_A(E; j_{X!}M, N). \tag{2.2.1}$$

In particular, there are canonical isomorphisms

$$H^q(E_{/X}, j_X^* N) \simeq H^q(X, N).$$

For every object X of E and every pair M, N of A-Modules, we put

$$\mathrm{Ext}^q_A(X; M, N) \simeq \mathrm{Ext}^q_{A|X}(E_{/X}; M|X, N|X). \tag{2.2.2}$$

By what precedes, the functors $\mathrm{Ext}^q_A(X; M,\cdot)$ are the derived functors of the functors $N \mapsto \mathrm{Hom}_{A|X}(M|X, N|X)$. The functors $(M, N) \mapsto \mathrm{Ext}^q_A(X; M, N)$, $q > 0$, form a δ-functor with respect to each variable. 19

2.3. The case of presheaf topoi. Cohomology of a covering.

2.3.1. Let C be a small category endowed with a presheaf of rings A, let $C^\wedge$ be the topos of presheaves on C, and let X be a representable object of $C^\wedge$. The functor which associates to an A-Module M the group $\Gamma(X, M) = M(X)$ is exact (I 3). Consequently, $H^q(X, M) = 0$ for every $q > 0$ and every A-Module M, or, equivalently, A_X is a projective A-Module.

2.3.2. Let S be a presheaf on C. For every A-Module M, there is a canonical isomorphism (I 2):

$$\Gamma(S, M) \simeq \varprojlim_{C/S} M|S$$

Moreover, for every injective A-Module M, the $A|S$-Module $M|S$ is injective (2.2). Consequently, the groups $H^q(S, M)$ are the values at $M|S$ of the right derived functors of $\varprojlim_{C/S}$. Denoting these derived functors by $\varprojlim^q_{C/S}$, there are canonical isomorphisms

$$H^q(S, M) \simeq \varprojlim\nolimits^q_{C/S} M|S.$$

In particular, there are canonical isomorphisms

$$H^q(C^\wedge, M) \simeq \varprojlim\nolimits^q_{C} M.$$

2.3.3. Let X be an object of C, and let $\mathscr{U} = (U_i \to X)$, $i \in I$, be a family of morphisms of C such that, for every $i \in I$, $U_i \to X$ is pullbackable (I 10). Denote by $A_\bullet$ the simplicial complex studied in 1.10:

$$A_\bullet = \cdots \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \coprod_{(i,j)\in I\times I} A_{U_i\times_X U_j} \;\substack{\longrightarrow\\ \longrightarrow}\; \coprod_{i\in I} A_{U_i}.$$

For every A-Module M, put $C^\bullet(\mathscr{U}, M) = \mathrm{Hom}_A(A_\bullet, M)$:

$$C^\bullet(\mathscr{U}, M) : \prod_{i\in I} M(U_i) \;\substack{\longrightarrow\\ \longrightarrow}\; \prod_{(i,j)\in I\times I} M(U_i \times_X U_j) \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots. \tag{2.3.3.1}$$

Put $H^q(\mathscr{U}, M) = H^q(C^\bullet(\mathscr{U}, M))$.

Proposition 2.3.4.

20 1) *With the notation of 2.3.3, let $R \hookrightarrow X$ be the sieve generated by the family $(U_i \to X)$, $i \in I$. There is a canonical isomorphism*

$$H^q(\mathscr{U}, M) \simeq H^q(R, M).$$

2) *The functors $H^q(\mathscr{U}, \cdot)$ commute with restrictions of scalars.*

Since R is a subobject of X in $C^\wedge$, the fiber products $U_{i_1} \times_R U_{i_2} \times_R \cdots \times_R U_{i_n}$ and $U_{i_1} \times_X U_{i_2} \times_X \cdots \times_X U_{i_n}$ are canonically isomorphic. It follows from 1.11 that the complex $A_{\mathscr{U}\bullet}$ is a resolution of A_R, and from 2.3.1 that the components of $A_{\mathscr{U}\bullet}$ are projective Modules. The cohomology groups of $C^\bullet(\mathscr{U}, M)$ are therefore canonically isomorphic to $\mathrm{Ext}^q_A(C^\wedge; A_R, M)$, whence the isomorphism. Assertion 2) follows immediately from the description of $C^\bullet(\mathscr{U}, M)$ (2.3.3.1).

Corollary 2.3.5. *Let $\mathscr{U} = (U_i \to X, i \in I)$ and $\mathscr{U}' = (U'_j \to X, j \in J)$ be two families of morphisms with target X; let $\phi = (\phi : I \to J, m_i : U_i \to U'_{\phi(i)})$ and $\phi' = (\phi' : I \to J, m'_i : U_i \to U'_{\phi(i)})$ be two morphisms (over X) from $\mathscr{U}$ to $\mathscr{U}'$. The morphisms ϕ and ϕ' induce the morphisms ϕ^q and $\phi'^q : H^q(\mathscr{U}, M) \to H^q(\mathscr{U}', M)$. The*

morphisms ϕ^q and ϕ'^q are equal. In particular, if the families $\mathcal{U}$ and $\mathcal{U}'$ are equivalent (i.e. if there is a morphism from $\mathcal{U}$ to $\mathcal{U}'$ and a morphism from $\mathcal{U}'$ to $\mathcal{U}$), the A-Modules $H^q(\mathcal{U}, M)$ and $H^q(\mathcal{U}', M)$ are canonically isomorphic.

Proof. Indeed, let R and R' be the sieves of X generated by the families $\mathcal{U}$ and $\mathcal{U}'$. The morphisms ϕ and ϕ' define the same morphism from R to R', and therefore induce two homotopic morphisms between the projective resolutions $A_{\mathcal{U}\bullet}$ and $A_{\mathcal{U}'\bullet}$.

Exercise 2.3.6. (Standard resolution)

a) Let C be a small category. For every integer $n > 0$, denote by $\mathrm{Fl}^n(C)$ the set of sequences of morphisms of C : $(u_1, \dots, u_n)$ such that, for every i, $0 < i < n$, the target of u_i equals the source of u_{i+1}, so that the morphisms u_i and u_{i+1} are composable. Define a semisimplicial set 21

$$ES(C) = \mathrm{ob}\ C \;\leftleftarrows\; \mathrm{Fl}^1(C) \;\lefttriplearrows\; \mathrm{Fl}^2(C) \;\leftleftarrows\!\!\!\!\!\!\equiv\; \mathrm{Fl}^3(C) \quad \cdots,$$

whose face operators $s_i : \mathrm{Fl}^n(C) \to \mathrm{Fl}^{n-1}(C)$ are as follows:

$$\begin{aligned} s_0(u_1, \dots, u_n) &= (u_2, \dots, u_n), \\ s_i(u_1, \dots, u_n) &= (u_1, \dots, u_{i+1}u_i, \dots, u_n), \quad 0 < i < n, \\ s_n(u_1, \dots, u_n) &= (u_1, \dots, u_{n-1}). \end{aligned}$$

b) Show that when C has an initial object or a final object, the complex $ES(C)$ is homotopically trivial.

c) For every object X of C, denote by ${}_{X\backslash}C$ the category of arrows with source X. Define a semisimplicial presheaf $ES(C)$ whose value at every object X of C is $ES({}_{X\backslash}C)$. Denote by $\mathbf{Z}_{ES(C)}$ the free abelian semisimplicial presheaf generated by $ES(C)$. It has a canonical augmentation $\mathbf{Z}_{ES(C)} \to \mathbf{Z}$ to the constant presheaf with value $\mathbf{Z}$. Show that, upon passing to the associated differential complex, the complex $\mathbf{Z}_{ES(C)}$ is a resolution of $\mathbf{Z}$ and that its components are projective abelian presheaves.

d) For every abelian presheaf M, put $ST^\bullet(M) = \mathrm{Hom}(\mathbf{Z}_{ES(C)}, M)$. Describe the components of $ST^\bullet(M)$ explicitly. Show that, for every integer $q \geq 0$, $H^q(ST^\bullet(M)) = H^q(C^\wedge, M)$.

e) Show that, for every sheaf S on C, the functors $H^q(S, \cdot)$ commute with restrictions of scalars.

f) Observe that when C is a group, $\mathbf{Z}_{ES(C)}$ is the standard resolution of the trivial module $\mathbf{Z}$, [3].

g) Show that, for every abelian presheaf M on C, $\mathbf{Z}_{ES(C)} \otimes M$ is a (left) resolution of M. Define the functor $\varinjlim_C$ on presheaves. Show that it is right exact. Denote its left derived functors by $H_q(C, M)$. Show that the components of the complex $\mathbf{Z}_{ES(C)} \otimes M$ are acyclic for $\varinjlim_C$. Deduce, denoting by $ST_\bullet(M)$ the complex $\varinjlim_C(\mathbf{Z}_{ES(C)} \otimes M)$, isomorphisms $H_q(ST_\bullet(M)) \simeq H_q(C, M)$.

h) Let $u : C^\wedge \to \mathrm{Ens}$ be the unique morphism from the topos $C^\wedge$ to the punctual topos. Denote by $u_! : C^\wedge_{\mathbf{Z}} \to \mathrm{Ab}$ the left adjoint of the functor $u^* : \mathrm{Ab} \to C^\wedge_{\mathbf{Z}}$. Show that for every abelian presheaf M, $u_!M = \varinjlim_C M$. 22

i) Henceforth denote by $ST^\bullet(C, M)$ and $ST_\bullet(C, M)$ the complexes denoted above by $ST^\bullet(M)$ and $ST_\bullet(M)$. A presheaf M on C is called locally constant if it sends all morphisms of C to isomorphisms. Associate to every locally constant presheaf M a locally constant presheaf $\check{M}$ on the category C° (the category

opposite to C) such that, for every morphism u of C, $\check{M}(u) = M(u)^{-1}$. Find a canonical isomorphism between the complexes $ST^{\bullet}(C, M)$ and $ST^{\bullet}(C^{\circ}, \check{M})$ (resp. $ST_{\bullet}(C, M)$ and $ST_{\bullet}(C^{\circ}, \check{M})$).

j) Let C be a small category having an initial object (resp. a small filtered category). Show that, for every constant presheaf M, $H^q(C^{\wedge}, M) = 0$ for $q > 0$ (resp. $H_q(C, M) = 0$ for $q > 0$).

2.4. The case of small sites. Čech cohomology.

2.4.1. Let (C, A) be a ringed $\mathscr{U}$-site, let $C^{\sim}$ be the topos of sheaves on C, and let $\epsilon : C \to C^{\sim}$ be the canonical functor which associates to an object of C the sheaf associated to the presheaf represented by that object. By abuse of notation, for every object X of C and every sheaf of A-modules M, we put $H^q(X, M) = H^q(\epsilon(X), M)$ (cf. 2.3.1). Recall that when the topology of C is coarser than the canonical topology, as is always the case in practice, the functor ϵ is fully faithful and allows us to identify C with its image under ϵ.

2.4.2. Denote by $\mathscr{H}^0 : C_A^{\sim} \to C_A^{\wedge}$ the inclusion functor from sheaves of A-modules to the category of presheaves of A-modules. Thus, for every sheaf of A-modules M, one has by definition

$$\mathscr{H}^0(M)(X) = H^0(X, M) = M(X), \tag{2.4.2.1}$$

for every object X of C. The functor $\mathscr{H}^0$ is left exact. Its right derived functors are denoted by $\mathscr{H}^q$. Since, for every object X of C, the "sections over X" functor is exact in the category of presheaves, one has

$$\mathscr{H}^q(M)(X) = H^q(X, M), \tag{2.4.2.2}$$

23 for every object X of C and every sheaf of A-modules M, so that the presheaf $\mathscr{H}^q(M)$ is simply the presheaf $X \mapsto H^q(X, M)$.

2.4.3. Suppose that (C, A) is a ringed *small site*, so that $C^{\wedge}$ is a *topos* to which the results of 2.3 may be applied. Let X be an object of C, and let $R \hookrightarrow X$ be a covering sieve. For every *presheaf* of A-modules G, the groups $H^q(R, G)$ (which are thus computed in the topos $C^{\wedge}$) are called the *Čech cohomology groups of the presheaf G relative to the covering sieve R*. When $R \hookrightarrow X$ is the sieve generated by a covering family $\mathscr{U} = (U_i \to X)_{i \in I}$ of pullbackable morphisms, these groups can be computed by means of the complex $C^{\bullet}(\mathscr{U}, G)$ (2.3.3), called the *Čech complex of G relative to the covering family $\mathscr{U}$* (or relative to the covering $\mathscr{U}$). The groups $H^q(\mathscr{U}, G) = H^q(R, G)$ (2.3.4) are then called the *Čech cohomology groups of G relative to the covering family $\mathscr{U}$*.

2.4.4. Let M be a sheaf of A-modules on C. The groups $H^q(\mathscr{U}, \mathscr{H}^0(M))$ are most often denoted, by abuse, by $H^q(\mathscr{U}, M)$ and called the Čech cohomology groups of the sheaf M relative to the covering family $\mathscr{U}$.

2.4.5. Denote by $\check{\mathscr{H}}^0 : C_A^{\wedge} \to C_A^{\wedge}$ the natural extension to presheaves of A-modules of the functor L described in II. Thus, by definition, for a presheaf G and an object X of C,

$$\check{\mathscr{H}}^0(G)(X) = \varinjlim_{R \hookrightarrow X} G(R), \tag{2.4.5.1}$$

where the inductive limit is taken over the covering sieves of X. It follows from (2.4.5.1) that the functor $\check{\mathscr{H}}^0$ is left exact. The right derived functors of $\check{\mathscr{H}}^0$ are denoted by $\check{\mathscr{H}}^q$. Since the "sections over X" and "filtered inductive limit" functors are exact, it follows

from (2.4.5.1) that

$$\check{\mathscr{H}}^q(G)(X) = \varinjlim_{R \hookrightarrow X} H^q(R, G), \tag{2.4.5.2}$$

where the limit is taken over the covering sieves of X.

The presheaves $\check{\mathscr{H}}^q$ are called the *Čech cohomology presheaves*. For every object X of C, put 24

$$\check{H}^q(X, G) = \check{\mathscr{H}}^q(G)(X). \tag{2.4.5.3}$$

The groups $\check{H}^q(X, G)$ are called the *Čech cohomology groups*. When the topology of C is defined by a pretopology, as is most often the case in practice, it follows from 2.3.4 that

$$\check{H}^q(X, G) \simeq \varinjlim_{\mathscr{U}} H^q(\mathscr{U}, G), \tag{2.4.5.4}$$

where the inductive limit is taken over the pullbackable covering families preordered by the natural order relation on their corresponding sieves (2.3.5).

2.4.6. Let M be a sheaf of A-modules. By abuse of notation, put

$$\check{H}^q(X, M) = \check{H}^q(X, \check{\mathscr{H}}^0(M)), \qquad \check{\mathscr{H}}^q(M) = \check{\mathscr{H}}^q(\check{\mathscr{H}}^0(M)), \tag{2.4.6.1}$$

and the groups $\check{H}^q(X, M)$ are called the Čech cohomology groups of the sheaf M. Note that, although the functors $\check{H}^q$ are derived functors on the category of presheaves, they do not in general form a δ-functor on the category of sheaves.

2.5. Change of universe. Čech cohomology for $\mathscr{U}$-sites.

2.5.1. Let (C, A) be a ringed $\mathscr{U}$-site, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. The site (C, A) is then a $\mathscr{V}$-site, and there are a $\mathscr{U}$-topos $C^\sim_{\mathscr{U}}$, a $\mathscr{V}$-topos $C^\sim_{\mathscr{V}}$, and a canonical inclusion functor $\epsilon : C^\sim_{\mathscr{U}} \hookrightarrow C^\sim_{\mathscr{V}}$. The functor ϵ is exact and fully faithful on the categories of Modules, and sends injective Modules to injective Modules (1.9). Thus, for every pair of $\mathscr{U}$-sheaves of A-modules, there are canonical isomorphisms

$$\mathrm{Ext}^q_A(C^\sim_{\mathscr{U}}; M, N) \simeq \mathrm{Ext}^q_{\epsilon A}(C^\sim_{\mathscr{V}}; \epsilon M, \epsilon N), \quad q \geq 0. \tag{2.5.1.1}$$

In particular, for every $\mathscr{U}$-sheaf of sets R on C, there are canonical isomorphisms (2.1.1)

$$H^q(R, M) \simeq H^q(\epsilon R, \epsilon M), \quad q \geq 0, \tag{2.5.1.2}$$

and, even more particularly, for every object X of C, there are canonical isomorphisms 25 (2.4.1)

$$H^q(X, M) \simeq H^q(X, \epsilon M). \tag{2.5.1.3}$$

Loosely speaking, we may therefore say that sheaf cohomology does not depend on the choice of universes, and, when required for a proof or a construction, one may always enlarge the universe in order to compute the cohomology of a sheaf.

2.5.2. Let (C, A) be a ringed $\mathscr{U}$-site, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. Denote by $C^\sim_A$ the category of $\mathscr{U}$-sheaves of modules, and let $\hat{\epsilon} : C^\wedge_{A,\mathscr{U}} \to C^\wedge_{A,\mathscr{V}}$ be the inclusion functor from $\mathscr{U}$-presheaves to $\mathscr{V}$-presheaves of A-modules. The functor $\hat{\epsilon}$ is exact, and therefore the derived functors of the functor $\hat{\epsilon}\mathscr{H}^0 : C^\sim_A \to C^\wedge_{A,\mathscr{V}}(\quad)$ are the functors $\hat{\epsilon}\mathscr{H}^q$, $q \geq 0$. By abuse of notation, we will again denote by $\mathscr{H}^q : C^\sim_A \to C^\wedge_{A,\mathscr{V}}$, $q \geq 0$, the functors $\hat{\epsilon}\mathscr{H}^q$. When C is $\mathscr{V}$-small, this enlargement of the universe has the following advantage: the category $C^\wedge_{\mathscr{U}}$ is not in general a $\mathscr{U}$-topos, and $\mathscr{U}$-presheaves of A-modules are not necessarily submodules of injective $\mathscr{U}$-presheaves, whereas the category of $\mathscr{V}$-presheaves is a topos (a $\mathscr{V}$-topos), and consequently every $\mathscr{V}$-presheaf of

A-modules is a subobject of an injective $\mathscr{V}$-presheaf (0.1.1). Thus, for every $\mathscr{V}$-presheaf of sets R (and in particular when R is a $\mathscr{U}$-presheaf) and every $\mathscr{U}$-sheaf of A-modules M, the groups $H^q(R, \mathscr{H}^q(M))$ are defined by 2.1.1, and it follows from (2.5.1.2) that these groups do not depend on the universe $\mathscr{V}$ under consideration. Likewise, for every pair of integers ≥ 0, denoted by p and q, the presheaves $\check{\mathscr{H}}^p(\mathscr{H}^q(M))$ are defined by 2.4.5, and it follows from (2.5.1.2) and (2.4.5.4) that these presheaves do not depend on the universe $\mathscr{V}$ used to define them.

3. The Cartan–Leray spectral sequence associated with a covering

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PROPOSITION 3.1. *Let (C, A) be a ringed $\mathscr{U}$-site, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. The functor $\mathscr{H}^0 : C_A^\sim \to C_{A,\mathscr{V}}^\wedge$ (2.5.2) sends injective A-modules to injective presheaves. For every integer $q > 0$ and every A-module M, the sheaf associated with the presheaf $\mathscr{H}^q(M)$ is zero.*

Denote by $\underline{a}_{\mathscr{V}}$ the "associated sheaf" functor for $\mathscr{V}$-presheaves, and by $\epsilon : C_A^\sim \hookrightarrow C_{A,\mathscr{V}}^\sim$ the inclusion functor from $\mathscr{U}$-sheaves to $\mathscr{V}$-sheaves. One has $\underline{a}_{\mathscr{V}}\mathscr{H}^0 = \epsilon$ (II 3.6), and, since the functors $\underline{a}_{\mathscr{V}}$ and ϵ are exact, one has $\underline{a}_{\mathscr{V}}\mathscr{H}^q = 0$ for every integer $q > 0$. For every $\mathscr{U}$-sheaf M and every $\mathscr{V}$-presheaf N, there is a functorial isomorphism $\mathrm{Hom}_{C_{A,\mathscr{V}}^\wedge}(N, \mathscr{H}^0(M)) \simeq \mathrm{Hom}_{C_{A,\mathscr{V}}^\sim}(\underline{a}_{\mathscr{V}}N, \epsilon M)$. When M is injective, ϵM is injective (1.9), and, since the functor $\underline{a}_{\mathscr{V}}$ is exact, the functor $N \longmapsto \mathrm{Hom}_{C_{A,\mathscr{V}}^\wedge}(N, \mathscr{H}^0(M))$ is exact. Consequently, $\mathscr{H}^0(M)$ is injective.

THEOREM 3.2. *Let (C, A) be a ringed $\mathscr{U}$-site, let R be a $\mathscr{U}$-presheaf of sets on C, and let M be a sheaf of A-modules. There is a spectral sequence, functorial in R and M:*

$$H^{p+q}(\underline{a}R, M) \Longleftarrow E_2^{p,q} = H^p(R, \mathscr{H}^q(M)). \tag{3.2.1}$$

(When C is not $\mathscr{U}$-small, the term $H^p(R, \mathscr{H}^q(M))$ must be interpreted as the cohomology of the presheaf $\mathscr{H}^q(M)$ in the topos $C_{\mathscr{V}}^\wedge$, where $\mathscr{V}$ is a universe containing $\mathscr{U}$ such that C is $\mathscr{V}$-small (2.5.2).)

By the definition of the "associated sheaf" functor (cf. II), there is an isomorphism of functors $H^0(\underline{a}R, M) \simeq H^0(R, \mathscr{H}^0(M))$. The functor $\mathscr{H}^0$ sends injective objects to injective objects. The spectral sequence of composite functors (0.3) is the desired spectral sequence.

COROLLARY 3.3. *Let X be an object of C, and let $\mathscr{U} = (U_i \to X)$, $i \in I$, be a covering family such that, for every $i \in I$, the morphism $U_i \to X$ is pullbackable. Then there is a spectral sequence (called the Cartan–Leray spectral sequence):*

$$H^{p+q}(X, M) \Longleftarrow E_2^{p,q} = H^p(\mathscr{U}, \mathscr{H}^q(M)). \tag{3.3.1}$$

27 Let $R \hookrightarrow X$ be the sieve generated by $\mathscr{U}$. Since this sieve is covering, the sheaf associated with R is the sheaf associated with X (II 5.2). Taking 2.4.1 into account, one has $H^{p+q}(\underline{a}R, M) = H^{p+q}(X, M)$. The corollary then follows from 2.3.4.

COROLLARY 3.4. *There is a spectral sequence, functorial in the sheaf M and in the object X of C,*

$$H^{p+q}(X, M) \Longleftarrow E_2^{p,q} = \check{H}^p(X, \mathscr{H}^q(M)). \tag{3.4.1}$$

As X varies in $\mathbf{C}$, this spectral sequence gives the spectral sequence of presheaves

$$\mathscr{H}^{p+q}(M) \Longleftarrow E_2^{p,q} = \check{\mathscr{H}}^p(\mathscr{H}^q(M)). \tag{3.4.2}$$

These spectral sequences give functorial morphisms (edge morphisms):

$$\Phi^q(M) : \check{\mathscr{H}}^q(M) \longrightarrow \mathscr{H}^q(M), \tag{3.4.3}$$

$$\Phi_X^q(M) : \check{H}^q(X, M) \longrightarrow H^q(X, M). \tag{3.4.4}$$

The morphisms Φ^q and Φ_X^q are isomorphisms when q is 0 or 1. The morphisms Φ^2 and Φ_X^2 are monomorphisms. More generally, when the presheaves $\mathscr{H}^i(M)$ vanish for $0 < i < n$, the morphisms $\Phi^q(M)$ and $\Phi_X^q(M)$ are isomorphisms for $0 \leq q \leq n$ and monomorphisms for $q = n + 1$.

The first spectral sequence is obtained by taking the inductive limit in the spectral sequence (3.2.1) over the covering sieves $R \hookrightarrow X$ of X (2.4.5.2). The second spectral sequence follows immediately from it (2.4.5.3). The sheaves associated with the presheaves $\mathscr{H}^q(M)$ are zero when $q > 0$ (3.1). Thus $\check{\mathscr{H}}^0\check{\mathscr{H}}^0\check{\mathscr{H}}^q(M) = 0$ for every $q > 0$ (II.3.4). Consequently, $\check{\mathscr{H}}^0\check{\mathscr{H}}^q(M) = 0$ for $q > 0$ (II 3.2). The assertions concerning the morphisms Φ^q and Φ_X^q follow immediately.

Corollary 3.5. *Let (E, A) be a ringed topos, and let M be an A-module (a left module, to fix ideas). Denote by $\mathscr{M}$ the underlying abelian sheaf of M. The functor $M \longmapsto \mathscr{M}$ is exact; consequently, for every object X of E, the canonical isomorphism*

$$H^0(X, M) \xrightarrow{\sim} H^0(X, \mathscr{M})$$

extends to morphisms 28

$$H^q(X, M) \longrightarrow H^q(X, \mathscr{M}) \qquad q \geq 0.$$

These morphisms are isomorphisms.

For every object Y of E, one has (2.4.5.4):

$$H^p(Y, M) = \varinjlim_{\mathscr{U}} H^p(\mathscr{U}, M) \quad , \quad H^p(Y, \mathscr{M}) = \varinjlim_{\mathscr{U}} H^p(\mathscr{U}, \mathscr{M}),$$

where the inductive limit is taken over the epimorphic families $\mathscr{U} = (Y_i \to Y)$, $i \in I$.

Consequently, the canonical homomorphism $\check{H}^p(Y, M) \to \check{H}^p(Y, \mathscr{M})$ is an isomorphism (2.3.4). Hence, by (2.4.5.3), the canonical homomorphism $\check{\mathscr{H}}^p(M) \to \check{\mathscr{H}}^p(\mathscr{M})$ is an isomorphism. If M is an injective A-module, then $\check{\mathscr{H}}^p(\mathscr{M}) = 0$ for $p > 0$; hence $\mathscr{H}^1(\mathscr{M}) = 0$ and, by induction on p, $\mathscr{H}^p(\mathscr{M}) = 0$ for $p > 0$ (3.4). Thus $H^p(X, \mathscr{M}) = 0$ for $p > 0$ (2.4.2.2) and 2.5. Consequently, the functor $M \longmapsto \mathscr{M}$ sends injective objects to objects acyclic for the functor $H^0(X, \cdot)$, which gives the asserted isomorphism.

Exercise 3.6. Let G be a topological group and let B_G be its classifying topos (IV 2.5). Denote by E_G the object of B_G consisting of the underlying topological space of G, endowed with the action of G by left translations. The canonical morphism from E_G to the final object e_G of B_G is an epimorphism; hence there is a covering $\mathscr{U} = (E_G \to e_G)$ and, for every abelian sheaf F of B_G, a spectral sequence

$$E_2^{p,q} = H^p(\mathscr{U}, \mathscr{H}^q(F)) \Longrightarrow H^{p+q}(B_G, F), \tag{3.6.1}$$

which we propose to study.

1) Show that, for every integer n, the topos

$$B_{G/E_G\times E_G\times\cdots\times E_G} \quad (n \quad \text{factors})$$

is canonically equivalent to the topos (IV 2.5)

$$\mathrm{TOP}(G\times G\times G\times\cdots\times G) \quad (n-1 \quad \text{factors}).$$

29 For every integer n, denote by F_n the sheaf on the topological space $G\times\cdots\times G$ (n factors) induced by F.

2) Deduce that the term $E_2^{p,q}$ of (3.6.1) is canonically isomorphic to the p-th cohomology group of a complex of the form

$$H^q(e,F_0)\longrightarrow H^q(G,F_1)\longrightarrow H^q(G\times G,F_2)\longrightarrow\cdots,$$

which is to be made explicit (e denotes the one-point topological space).

3) Deduce that the cohomology of the classifying topos B_G with values in locally constant sheaves is isomorphic to the corresponding singular cohomology of a classifying space, provided that, on the spaces of the form $G\times G\times\cdots\times G$, cohomology with coefficients in locally constant sheaves coincides with the corresponding singular cohomology (as is the case, for example, when G is locally contractible). (For classifying spaces, one may consult [15].)

4. Acyclic sheaves

Definition 4.1. Let (E,A) be a ringed topos, let F be an A-Module, and let S be a topologically generating family of E. The sheaf F is said to be S-acyclic if, for every object X of S and every integer $q>0$, one has $H^q(X,F)=0$. When S is equal to ob E, the S-acyclic sheaves are called *flasque sheaves.*

4.2. Let (C,A) be a $\mathscr{U}$-ringed site, and let F be a sheaf of A-modules on C. The sheaf F is said to be *C-acyclic* if it is S-acyclic, where S is the family of sheaves associated with the objects of C.

Proposition 4.3. *Let (C,A) be a $\mathscr{U}$-ringed site, and let F be a sheaf of A-modules. Denote by $\mathscr{H}^0 : C_A^\sim \to C_A^\wedge$ the inclusion functor from the A-Modules into the category of presheaves of A-modules (here these are $\mathscr{V}$-presheaves, where $\mathscr{V}$ is a universe such that C is $\mathscr{V}$-small). The following conditions are equivalent:*

30 i) *F is C-acyclic.*

ii) *For every object X of C and every covering sieve $R\hookrightarrow X$, one has*

$$H^q(R,\mathscr{H}^0(F))=0 \quad \textit{for every } q>0.$$

iii) *For every object X of C, one has*

$$\check{H}^q(X,F)=0 \quad \textit{for every } q>0.$$

i ⇒ ii) : Since F is C-acyclic, the presheaves $\mathscr{H}^q(F)$ vanish for $q>0$. The spectral sequence (3.2.1) then gives an isomorphism $H^q(R,\mathscr{H}^0(F))\simeq H^q(X,F)$ for every q. Consequently, $H^q(R,\mathscr{H}^0(F))=0$ for $q>0$.

ii ⇒ iii) : This is clear upon passing to the inductive limit.

iii ⇒ i) : By induction on q, one proves that the presheaves $\mathscr{H}^q(F)$ vanish for $q>0$. For this, use (3.4.2).

4.4. It follows from criterion 4.3 ii) and 3.5 that the property of being S-acyclic depends only on the underlying abelian sheaf. In particular, a sheaf of A-modules is flasque if and only if its underlying abelian sheaf is flasque.

COROLLARY 4.5. *Let (E, A) be a ringed topos, and let F be an A-Module. The following properties are equivalent:*

i) *F is flasque;*
ii) *For every epimorphic family $\mathscr{X} = (X_i \to X)_{i\in I}$,*

$$H^q(\mathscr{X}, F) = 0 \quad \text{for every } q > 0.$$

In 4.2, take C to be the topos E itself. The corollary then follows from the equivalence i) $\Leftrightarrow$ ii) of 4.3.

4.6. Injective sheaves are flasque. Flasque sheaves are S-acyclic for every topologically generating family S. Flasque sheaves are not necessarily injective (take for E the topos of sets). S-acyclic sheaves are not necessarily flasque (Exercise 4.1.3).

PROPOSITION 4.7. *Let (E, A) be a ringed topos, let F be a flasque A-Module, and let X be an object of E. For every subobject Y of X, the canonical homomorphism $H^0(X, F) \to H^0(Y, F)$ is surjective.* 31

Suppose that Y is a subobject of X such that the morphism $H^0(X, F) \to H^0(Y, F)$ is not surjective. Let Z be the object obtained by gluing two copies X_1 and X_2 of X along Y. The object Z is covered by the two subobjects X_1 and X_2, and one has $X_1 \times_Z X_2 = Y$. Denote the resulting covering by $\mathscr{X} = (X_1, X_2)$. One sees immediately that $H^1(\mathscr{X}, F) \neq 0$, a contradiction.

4.8. For general topoi, the property described in 4.7 does not characterize flasque sheaves (Exercise 4.15). It does, however, characterize them for topoi generated by their open subobjects, and in particular for topoi associated with topological spaces (Exercise 4.16). In the case of topological spaces, the terminology "flasque" adopted here agrees with that of [7] (Exercise 4.16).

PROPOSITION 4.9. *Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi.*

1) *The functor u_* carries flasque A-Modules to flasque A'-Modules.*
2) *Let S and S' be topologically generating families of E and E', respectively, such that $u^*(S') \subset S$. The functor u_* carries S-acyclic A-Modules to S'-acyclic A'-Modules.*
3) *When u is a flat morphism (1.8), the functor u_* carries injective A-Modules to injective A'-Modules.*

Let X be an object of E', let $\mathscr{X} = (X_i \to X)_{i\in I}$ be an epimorphic family, let F be a flasque A-Module, and let $C^\bullet(\mathscr{X}, u_*F)$ be the Čech complex of the covering $\mathscr{X}$. Using the adjunction between u_* and u^*, and the fact that u^* commutes with fiber products, one obtains a canonical isomorphism

$$C^\bullet(\mathscr{X}, u_*F) \simeq C^\bullet(u^*(\mathscr{X}), F).$$

Moreover, u^* commutes with inductive limits, and hence $u^*(\mathscr{X}) = (u^*X_i \to u^*X)_{i\in I}$ is an epimorphic family. Since F is flasque, $H^q(u^*(\mathscr{X}), F) = 0$ for $q > 0$, and consequently

$$H^q(\mathscr{X}, u_*F) = H^q(C^\bullet(\mathscr{X}, u_*F)) = H^q(C^\bullet(u^*(\mathscr{X}), F)) = H^q(u^*(\mathscr{X}), F) = 0$$

for $q > 0$. Thus u_*F is flasque.

Let us prove 2). When S' is stable under fiber products, an argument analogous to the preceding one proves 2). In the general case, we use the spectral sequence of
32 the morphism u (5.3.2). Assertion 2) will not be used before (5.3.2). Let F be an S-acyclic sheaf. The sheaves $R^q u_* F$ are the sheaves associated with the presheaves $X \mapsto H^q(u^*X, F)$. Since S' is a topologically generating family and F is S-acyclic, one has $R^q u_* F = 0$ for every $q > 0$. The spectral sequence (5.3.2) then gives, for every object X of E', an isomorphism $H^q(X, u_* F) \simeq H^q(u^*X, F)$; consequently, for every X in S', $H^q(X, u_* F) = 0$.

Let us prove 3). When u is flat, the inverse-image functor u^* for Modules is exact (1.8). Therefore its right adjoint u_* carries injective objects to injective objects (0.2).

Proposition 4.10. *Let F be an A-Module of the ringed topos (E, A), and let G be an injective A-Module.*

1) *The functor $F \mapsto \mathbf{Hom}_A(F, G)$ is exact.*
2) *The abelian sheaf $\mathbf{Hom}_A(F, G)$ is flasque.*

Proof. Let us prove the assertions in turn.

1) Let
$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$
be an exact sequence. We must show that the sequence
$$0 \longrightarrow \mathbf{Hom}_A(F'', G) \longrightarrow \mathbf{Hom}_A(F, G) \longrightarrow \mathbf{Hom}_A(F', G) \longrightarrow 0$$
is exact; for this it suffices to show that, for every object H of E, the sequence
$$0 \longrightarrow \mathrm{Hom}_E(H, \mathbf{Hom}_A(F'', G)) \longrightarrow \mathrm{Hom}_E(H, \mathbf{Hom}_A(F, G)) \longrightarrow \mathrm{Hom}_E(H, \mathbf{Hom}_A(F', G)) \longrightarrow 0$$
is exact. By (IV 6.12), the latter sequence is isomorphic to
$$0 \longrightarrow \mathrm{Hom}_A(A_H \otimes_A F'', G) \longrightarrow \mathrm{Hom}_A(A_H \otimes_A F, G) \longrightarrow \mathrm{Hom}_A(A_H \otimes_A F', G) \longrightarrow 0,$$
and the A-Module A_H is flat (1.3.1).

2) Let us show that $\mathbf{Hom}_A(F, G)$ is flasque. Let
$$\mathscr{X} = (X_i \longrightarrow X)_{i \in I}$$
be an epimorphic family of E, and let
$$C_\bullet(\mathscr{X}) : \cdots \rightrightarrows \coprod_{i,j} \mathbf{Z}_{X_i \times_X X_j} \rightrightarrows \coprod_i \mathbf{Z}_{X_i}$$
33 be the complex defined in 1.5. This complex is a flat resolution of the object $\mathbf{Z}_X$, which is itself flat (1.4 and 1.3).
We then compute the groups:
$$H^q(\mathscr{X}, \mathbf{Hom}_A(F, G)) \xrightarrow{\sim} H^q\big(\mathrm{Hom}^\bullet_{\mathbf{Z}}(C_\bullet(\mathscr{X}), \mathbf{Hom}_A(F, G))\big) \xrightarrow{\sim} H^q\big(\mathrm{Hom}_A(C_\bullet(\mathscr{X}) \otimes_{\mathbf{Z}} F, G)\big).$$
The complex
$$C_\bullet(\mathscr{X}) \otimes_{\mathbf{Z}} F$$
is acyclic in degrees different from 0, because $C_\bullet(\mathscr{X})$ is a flat resolution of a flat module. Consequently, $\mathrm{Hom}_A(C_\bullet(\mathscr{X}) \otimes_{\mathbf{Z}} F, G)$ is acyclic in degrees different from 0.

Proposition 4.11. *Let (E, A) be a ringed topos, and let F be a flasque (resp. injective) sheaf of A-modules.*

1) *For every object X of E, the $A|X$-Module $j_X^* F$ is flasque (resp. injective).*
2) *For every closed subtopos Z of E, the sheaf of sections of F with support in Z* (IV 14) *is flasque (resp. injective).*

Assertion 1) follows from 4.5 when F is flasque. The functor j_X^* admits an exact left adjoint $j_{X!}$ (IV 11). Consequently, when F is injective, $j_X^* F$ is injective (0.2). Let us prove 2). Denote the inclusion morphism by $i : Z \to E$. The sheaf of sections of F with support in Z is the sheaf $i_* i^! F$ (IV 14). The functor $i_* i^!$ is right adjoint to the exact functor $i_* i^*$ (IV 14). It therefore carries injective sheaves to injective sheaves. Let U be the open complement of Z, let $j : U \to E$ be the inclusion morphism, and let F be a flasque sheaf. There is an exact sequence (IV 14):

$$0 \longrightarrow i_* i^! F \longrightarrow F \longrightarrow j_* j^* F. \tag{4.11.1}$$

For every object X of E, one has $j_* j^* F(X) = F(X \times U)$, and the morphism $F(X) \to j_* j^* F(X)$ induced by the last arrow of (4.11.1) comes from the canonical injection $X \times U \hookrightarrow X$. Since F is flasque, this morphism is surjective (4.7), and hence the last arrow of (4.11.1) is an epimorphism of presheaves. For every object X of E, the exact cohomology sequence deduced from (4.11.1) gives $H^q(X, i_* i^! F) = 0$ for $q > 0$, and hence $i_* i^! F$ is flasque.

4.12. The property of a sheaf being flasque (resp. injective) localizes (4.11.1). It 34
is not, in general, a property of local nature (Exercise 4.15). It is, however, a property of local nature for topoi generated by their open subobjects, and in particular for topoi associated with topological spaces (Exercise 4.16).

Exercise 4.13. Let X be a locally compact space, and let F be a c-soft sheaf [7]. Using the local nature of softness (*loc. cit.*), show that the restriction of F to every paracompact open subset of X is a soft sheaf. Show that the paracompact open subsets form a basis for the topology of X. Deduce that, if S denotes the family of paracompact open subsets of X, then F is S-acyclic. Show that the sheaf of continuous functions on X is S-acyclic but is not flasque when X is not discrete.

Problem 4.14. Study totally acyclic topoi, i.e. topoi such that $H^q(X, F) = 0$ for every $q > 0$, every object X, and every abelian sheaf F.

Exercise 4.15. Let G be a discrete group, and let B_G be its classifying topos (IV 2.4). Show that for every abelian sheaf F and every monomorphism $X \hookrightarrow Y$, the homomorphism $F(Y) \to F(X)$ is surjective. Let $E(G)$ be the group G regarded as a homogeneous space under itself. It is an object of B_G. The topos $B_{G/E(G)}$ is equivalent to the punctual topos (IV 8). The morphism $E(G) \to e$ (e the final object of B_G) is an epimorphism. For every abelian sheaf F on B_G, the sheaf $F|E(G)$ is flasque. Deduce that the property of being flasque or injective is not of local nature.

Exercise 4.16. A topos E is said to be generated by its open subobjects if the open subobjects of E (i.e. the subobjects of the final object e of E) form a generating family (I 7). Such a topos has the following property:
(P) Every epimorphic family $X_i \to X$, $i \in I$, admits a refinement by an epimorphic family $U_j \to X$, $j \in J$, in which the $U_j \to X$ are monomorphisms.

a) Are there topoi which have property (P) but are not generated by their open 35
subobjects? Topoi associated with topological spaces are generated by their open subobjects. If E is generated by its open subobjects (resp. has (P)), then for every object X of E, $E_{/X}$ is generated by its open subobjects (resp. has (P)).

Every subtopos of a topos generated by its open subobjects is generated by its open subobjects. Property (P) is not a property of local nature.

b) Let E be a topos, and let $\mathrm{Ouv}(E)$ be the category of open subobjects of E, endowed with the induced topology. The inclusion functor $\mathrm{Ouv}(E) \to E$ is a morphism of sites, and hence gives a morphism of topoi $\Pi : E \to \mathrm{Ouv}(E)^\sim$. The functor Π^* is fully faithful. With respect to the 2-category of topoi generated by their open subobjects, the morphism Π has a universal property which the reader is invited to make explicit.

c) Let E be a topos generated by its open subobjects, and let $\mathrm{Point}(E)$ be the set of points of E up to isomorphism. Show that $\mathrm{Point}(E)$ is small. Put a topology on $\mathrm{Point}(E)$ and define a morphism $\mathrm{Top}(\mathrm{Point}(E)) \to \mathrm{Ouv}(E)^\sim$ making $\mathrm{Top}(\mathrm{Point}(E))$ a subtopos of $\mathrm{Ouv}(E)^\sim$. For every topological space X, show that every morphism from $\mathrm{Top}(X)$ to E gives a morphism from $\mathrm{Top}(X)$ to $\mathrm{Top}(\mathrm{Point}(E))$. Show that a topos generated by its open subobjects is equivalent to a topos $\mathrm{Top}(X)$ (X a topological space) if and only if it has enough points. Show that there exist topoi generated by their open subobjects which are not equivalent to any topos $\mathrm{Top}(X)$, where X is a topological space.

d) Let E be a topos having property (P). An abelian sheaf F on E is flasque if and only if, for every monomorphism $X \hookrightarrow Y$, the homomorphism $F(Y) \to F(X)$ is surjective.

e) Let E be a topos having property (P). Show that the property of a sheaf F being flasque is a property of local nature.

f) Let E be a topos generated by its open subobjects, and let e be the final object of E. An abelian sheaf F is flasque if and only if, for every open subobject U of e, the canonical morphism $F(e) \to F(U)$ is surjective.

36 Exercise 4.17. (Flasque sheaves and change of universe)

Let C be a $\mathscr{U}$-site (for example, a $\mathscr{U}$-topos), and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. Denote the corresponding categories of sheaves by $C^\sim_{\mathscr{U}}$ and $C^\sim_{\mathscr{V}}$, and denote the canonical inclusion by $\epsilon : C^\sim_{\mathscr{U}} \to C^\sim_{\mathscr{V}}$. Let F be a flasque abelian $\mathscr{U}$-sheaf on C. We shall show that ϵF is flasque. First note that, for every object X of $C^\sim_{\mathscr{U}}$, one has $H^q(\epsilon X, \epsilon F) = 0$ for $q > 0$. Every object Y of $C^\sim_{\mathscr{V}}$ admits an epimorphic family $\epsilon X_i \to Y$, $i \in I$, in which the $\epsilon X_i \to Y$ are monomorphisms. It follows that

$$H^q(Y, \epsilon F) = displaystyle \varinjlim_{\epsilon X \hookrightarrow Y}{}^{q} F(X).$$

To show that these $\varinjlim^q$ vanish for $q \neq 0$, one may reduce to the topos $\mathrm{Ouv}(C_{\mathscr{V}/Y})^\sim$ and use the fact that, for this topos, a sheaf is flasque if it is so locally.

5. The $R^q u_*$ and the Cartan–Leray spectral sequence for a morphism of topoi

5.0. Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi, and let $u_* : E \to E'$ be the direct-image functor. The notation u_* will also denote the extension of the direct-image functor to modules. The functor u_* on modules (left modules, to fix ideas) is left exact. Its right derived functors are denoted by $R^q u_*$, $q \geq 0$.

Proposition 5.1. 1) *For every A-module M, the sheaf $R^q u_* M$ is the sheaf associated with the presheaf $X' \mapsto H^q(u^* X', M)$ ($X' \in \mathrm{ob}\, E'$).*

2) *Formation of the functors $R^q u_*$ commutes with restriction of scalars.*

3) *Formation of the functors $R^q u_*$ commutes with localization. More precisely, for every object X' of E', if $u_{/X'} : E_{/u^*X'} \to E'_{/X'}$ denotes the morphism deduced from u by localization* (IV 8)*, then, for every A-module M, there is a canonical isomorphism*

$$R^q(u_{/X'})_*(M|u^*X') \simeq R^q u_*(M)|X' \qquad q \geq 0. \tag{5.1.1}$$

Let $u_*^\wedge : E^\wedge \to E'^\wedge$ denote the direct-image functor for $\mathscr{U}$-presheaves ($u_*^\wedge M = M \circ u^*$). Since u^* and u_* are adjoint, there is an isomorphism

$$u_* \simeq \underline{a} u_*^{\wedge 0},$$

where $\underline{a}$ is the associated-sheaf functor for E'. Since the functors $\underline{a}$ and $u_*^\wedge$ are exact, 37
we have

$$R^q u_* \simeq \underline{a} u_*^\wedge \mathscr{H}^q,$$

which is another way of stating 1). Assertion 2) then follows from 1) and 3.5. By definition of the morphism $u_{/X'}$, there is a canonical isomorphism (5.1.1) for $q = 0$. The general case follows by noting that the localization functors (IV 8) are exact and take injective objects to injective objects (4.11).

PROPOSITION 5.2. *Let $u : E \to E'$ be a morphism of topoi and let S' be a generating family of E'. The sheaves M which are acyclic for the functors $H^0(u^*X', \cdot)$, $X' \in S'$, are acyclic for the functor u_*. In particular, flabby sheaves are acyclic for u_*.*

This follows from 5.1, 1).

PROPOSITION 5.3. *Let $u : E \to E'$ be a morphism of topoi and let M be an abelian group of E. There is a spectral sequence*

$$E_2^{p,q} = H^p(E', R^q u_* M) \Longrightarrow H^{p+q}(E, M). \tag{5.3.1}$$

More generally, for every object X' of E', there is a spectral sequence

$$E_2^{p,q} = H^p(X', R^q u_* M) \Rightarrow H^{p+q}(u^*X', M). \tag{5.3.2}$$

By definition of the direct- and inverse-image functors, there is an isomorphism $H^0(X', u_* M) \simeq H^0(u^*X', M)$. The functor u_* takes injective objects to flabby sheaves (4.6 and 4.9). The asserted spectral sequences are therefore spectral sequences of composite functors (0.3).

PROPOSITION 5.4. *Let $u : E \to E'$ and $v : E' \to E''$ be two morphisms of topoi. There is a spectral sequence*

$$R^p v_* R^q u_* \Longrightarrow R^{p+q}(v \circ u)_*.$$

We have $v_* u_* \simeq (vu)_*$, and the functor u_* takes injective objects to flabby sheaves 38
(4.9), hence to objects acyclic for v_* (5.2). Thus there is a spectral sequence of composite functors (0.3).

6. Local Ext and cohomology with supports

6.0. Let (E, A) be a ringed topos, and let M be an A-module, a left module to fix ideas. The functor $N \longmapsto \mathbf{Hom}_A(M, N)$, from the category of left A-modules to the category of abelian groups, is left exact (IV 12). Its right derived functors are denoted by

$$\mathbf{Ext}^q_A(M, N). \tag{6.0.1}$$

In particular, one has

$$\mathbf{Ext}^0_A(M, N) = \mathbf{Hom}_A(M, N). \tag{6.0.2}$$

By definition, there are canonical isomorphisms (2.1 and IV 12)

$$H^0(E, \mathbf{Ext}^0_A(M, N)) = \mathrm{Ext}^0_A(E; M, N) = \mathrm{Hom}_A(M, N). \tag{6.0.3}$$

More generally, for every object X of E, one has (2.2)

$$H^0(X, \mathbf{Ext}^0_A(M, N)) = \mathrm{Ext}^0_A(X; M, N) = \mathrm{Hom}_{A|X}(M|X, N|X). \tag{6.0.4}$$

Proposition 6.1. 1) *Formation of the functors* $\mathbf{Ext}^q_A$ *commutes with localization. More precisely, for every object* X *of* E*, there are functorial isomorphisms*

$$\mathbf{Ext}^q_A(M, N)|X \simeq \mathbf{Ext}^q_{A|X}(M|X, N|X). \tag{6.1.1}$$

2) *The sheaf* $\mathbf{Ext}^q_A(M, N)$ *is isomorphic to the sheaf associated with the presheaf* $X \longmapsto \mathrm{Ext}^q_A(X; M, N)$.

3) *There is a spectral sequence*

$$\mathrm{Ext}^{p+q}_A(E; M, N) \Longleftarrow E_2^{p,q} = H^p(E, \mathbf{Ext}^q_A(M, N)). \tag{6.1.2}$$

39 *More generally, for every object* X *of* E*, there is a spectral sequence functorial in* X *and in the arguments* M *and* N*:*

$$\mathrm{Ext}^{p+q}_A(X; M, N) \Longleftarrow E_2^{p,q} = H^p(X, \mathbf{Ext}^q_A(M, N)). \tag{6.1.3}$$

By definition, (6.1.1) is an isomorphism when $q = 0$ (IV 12). The general case follows, taking into account the fact that the localization functors are exact and send injective modules to injective modules (2.2). The functor $N \longmapsto \mathbf{Hom}_A(M, N) = \mathbf{Ext}^0_A(M, N)$ sends injective modules to flasque sheaves, and hence to sheaves acyclic for $H^0(X, \cdot)$ (4.10). The spectral sequences (6.1.2) and (6.1.3) are the spectral sequences of composite functors, in view of (6.0.3) and (6.0.4). As X varies in E, the spectral sequence (6.1.3) gives a spectral sequence of presheaves and hence, upon passing to the associated sheaves, a spectral sequence of sheaves. Since the sheaves associated with the presheaves $X \longmapsto H^p(X, \cdot)$ are zero when $p \neq 0$ (3.1), this spectral sequence degenerates and gives the isomorphism asserted in 2).

Proposition 6.2. *The functors* $(M, N) \longmapsto \mathbf{Ext}^q_A(M, N)$, $q \geq 0$, *form a* δ*-functor in the variable* M *and in the variable* N*. Explicitly, for every exact sequence* $0 \to N' \to N \to N'' \to 0$ *(resp.* $0 \to M' \to M \to M'' \to 0$*), there are long exact sequences*

(6.2.1)

$$\cdots \longrightarrow \mathbf{Ext}^q_A(M, N') \longrightarrow \mathbf{Ext}^q_A(M, N) \longrightarrow \mathbf{Ext}^q_A(M, N'') \xrightarrow{\delta} \mathbf{Ext}^{q+1}_A(M, N') \longrightarrow \cdots$$

(resp.

(6.2.2)

$$\cdots \longrightarrow \mathbf{Ext}^q_A(M'', N) \longrightarrow \mathbf{Ext}^q_A(M, N) \longrightarrow \mathbf{Ext}^q_A(M', N) \xrightarrow{\delta} \mathbf{Ext}^{q+1}_A(M'', N) \longrightarrow \cdots).$$

It follows from the general properties of the functors Ext^q that, for every object X of E, the functors $(M, N) \longmapsto \mathrm{Ext}^q_A(X; M, N)$, $q \geq 0$, form a δ-functor in each argument. The assertion follows by allowing X to vary and taking the associated sheaf (6.1).

6.3. Let (E, A) be a ringed topos, let M be an A-module, let Z be a closed subtopos of E (IV 9), and let U be its open complement. Denote by $H^0_Z(E, M)$ the group of sections of M whose support is contained in Z (IV 14), and by $\mathscr{H}^0_Z(M)$ the subsheaf of 40
M defined by the sections of M "with supports in Z" (IV 14). The functors $H^0_Z(E, \cdot)$ and $\mathscr{H}^0_Z(\cdot)$ are left exact (IV 14). Their derived functors are denoted by $H^q_Z(E, \cdot)$ and $\mathscr{H}^q_Z(\cdot)$ respectively, and are called the *cohomology groups (*resp. *sheaves) of M with supports in Z*[14].

There are canonical isomorphisms (IV 14)

$$H^0_Z(E, M) \simeq \mathrm{Hom}_A(A_Z, M) \simeq \mathrm{Ext}^0_A(E; A_Z, M), \tag{6.3.1}$$

$$\mathscr{H}^0_Z(M) \simeq \mathbf{Hom}_A(A_Z, M) \simeq \mathbf{Ext}^0_A(A_Z, M), \tag{6.3.2}$$

and hence, for every $q \geq 0$, there are isomorphisms

$$H^q_Z(E, M) \simeq \mathrm{Ext}^q_A(E; A_Z, M), \tag{6.3.3}$$

$$\mathscr{H}^q_Z(M) \simeq \mathbf{Ext}^q_A(A_Z, M). \tag{6.3.4}$$

Note that, since A_Z is a bimodule, the sheaves $\mathbf{Ext}^q_A(A_Z, M) \simeq \mathscr{H}^q_Z(M)$ are canonically endowed with A-module structures.

For every object X of E, denote by $Z_{/X}$ the closed subtopos of $E_{/X}$ obtained from Z by localization (it is the complement of $U \times X$). By definition, there are canonical isomorphisms

$$H^0(X, \mathscr{H}^0_Z(M)) \simeq H^0_{Z_{/X}}(E_{/X}, M|X). \tag{6.3.5}$$

Set

$$H^q_Z(X, M) \simeq H^q_{Z_{/X}}(E_{/X}, M|X). \tag{6.3.6}$$

Taking (6.3.2) into account, there are canonical isomorphisms

$$H^q_Z(X, M) \simeq \mathrm{Ext}^q_A(X; A_Z, M). \tag{6.3.7}$$

PROPOSITION 6.4. 1) *Formation of the functors $\mathscr{H}^q_Z$ commutes with localization.* 41
More precisely, for every object X of E and every A-module M, there are canonical isomorphisms

$$\mathscr{H}^q_Z(M)|X \simeq \mathscr{H}^q_{Z_{/X}}(M|X). \tag{6.4.1}$$

2) *The sheaf $\mathscr{H}^q_Z(M)$ is the sheaf associated with the presheaf $X \longmapsto H^q_Z(X, M)$.*

[14] Compare with SGA 2 I for the case of ordinary topological spaces, as well as with the exposé of HARTSHORNE cited on p. 80 below.

3) *There is a spectral sequence*

$$H_Z^{p+q}(E,M) \Longleftarrow H^p(E,\mathscr{H}_Z^q(M)). \tag{6.4.2}$$

More generally, for every object X of E, there is a spectral sequence

$$H_Z^{p+q}(X,M) \Longleftarrow H^p(X,\mathscr{H}_Z^q(M)). \tag{6.4.3}$$

This is simply Proposition 6.1 translated by means of the dictionary in 6.3.

PROPOSITION 6.5. *With the notation of 6.3, denote by $j : U \to E$ the canonical morphism. For every A-module M, there is an exact sequence of sheaves*

$$0 \longrightarrow \mathscr{H}_Z^0(M) \longrightarrow M \longrightarrow j_*(M|U) \longrightarrow \mathscr{H}_Z^1(M) \longrightarrow 0, \tag{6.5.1}$$

and, for $q \geq 2$, there are isomorphisms

$$\mathscr{H}_Z^q(M) \simeq \mathbf{Ext}_A^{q-1}(A_U, M) \simeq R^{q-1}j_*(M|U). \tag{6.5.2}$$

There is, in addition, a long exact sequence

$$\cdots \longrightarrow H_Z^q(E,M) \longrightarrow H^q(E,M) \longrightarrow H^q(U,M) \longrightarrow H_Z^{q+1}(E,M) \longrightarrow \cdots \text{ [14]} \tag{6.5.3}$$

and, more generally, for every object X of E, there is a long exact sequence

$$\cdots \longrightarrow H_Z^q(X,M) \longrightarrow H^q(X,M) \longrightarrow H^q(X\times U,M) \longrightarrow H_Z^{q+1}(X,M) \longrightarrow \cdots. \tag{6.5.4}$$

By the definition of A_Z, there is an exact sequence (IV 14)

$$0 \longrightarrow A_U \longrightarrow A \longrightarrow A_Z \longrightarrow 0. \tag{6.5.5}$$

42 The functors $\mathbf{Ext}_A^q(A,\cdot)$ vanish for $q>0$. One has $\mathbf{Hom}_A(A_U,M) \simeq j_*(M|U)$ (IV 14), and hence (2.2) $\mathbf{Ext}_A^q(A_U,M) \simeq R^q j_*(M|U)$. Finally, $\mathbf{Ext}_A^q(A_Z,M) \simeq \mathscr{H}_Z^q(M)$ (6.3.4). The long exact sequence (6.2.2) gives, in this case, (6.5.1) and (6.5.2). The exact sequences (6.5.3) and (6.5.4) follow from the dictionary in 6.3 and from the long exact sequence of the δ-functor $\mathrm{Ext}_A^q(X;\cdot,M)$, $q\geq 0$, associated with (6.5.5).

PROPOSITION 6.6. 1) *Flasque sheaves are acyclic for the functors $\mathscr{H}_Z^0$ and $H_Z^0(X,\cdot)$.*

2) *For every $q \geq 0$, the functors $\mathscr{H}_Z^q$ and $H_Z^q(X,\cdot)$ commute with restriction of scalars.*

Let M be a flasque sheaf. The exact sequence (6.5.4) gives the equalities $H_Z^q(X,M) = 0$ for every X and every $q\geq 2$, together with an exact sequence

$$0 \longrightarrow H_Z^0(X,M) \longrightarrow H^0(X,M) \longrightarrow H^0(X\times U,M) \longrightarrow H_Z^1(X,M) \longrightarrow 0.$$

But, since the sheaf M is flasque, the morphism $H^0(X,M) \to H^0(X\times U,M)$ is surjective (4). Consequently, $H_Z^1(X,M)=0$, and M is acyclic for $H_Z^0(X,\cdot)$. Passing to the associated sheaves, one deduces that M is acyclic for the functor $\mathscr{H}_Z^0$ (6.4). It is clear that the functors $\mathscr{H}_Z^0$ and $H_Z^0(X,\cdot)$ commute with restriction of scalars; since flasque sheaves are acyclic for both functors, assertion 2) follows.

[14] This exact sequence makes precise the role of the global cohomological invariants $H_Z^q(E,M)$ as "cohomology groups of E *modulo* the open subtopos U, with coefficients in M."

PROPOSITION 6.7. *Let (E, A) be a ringed topos, let Z be a closed subtopos of E, let U be its open complement, and let M and N be two A-modules. There are isomorphisms, functorial in M and N and compatible with changes of closed subtopos:*

$$\mathscr{H}^0_Z(\mathbf{Hom}_A(M,N)) \simeq \mathbf{Hom}_A(M, \mathscr{H}^0_Z(N)) \simeq \mathbf{Hom}_A(M \otimes_A A_Z, N).$$

This follows from (IV 14), taking (6.3.2) into account.

6.8. Set

$$\mathbf{Hom}_{A,Z}(M,N) = \mathscr{H}^0_Z(\mathbf{Hom}_A(M,N)). \tag{6.8.1}$$

The functor $N \longmapsto \mathbf{Hom}_{A,Z}(M,N)$ is left exact. Its derived functors are denoted by $\mathbf{Ext}^q_{A,Z}(M,N)$: these are the **Ext** *sheaves with supports in* Z. Thus

$$\mathbf{Ext}^0_{A,Z}(M,N) = \mathbf{Hom}_{A,Z}(M,N). \tag{6.8.2}$$

Set $M_Z = M \otimes_A A_Z$. It follows from IV 14 that M_Z is the direct image on E of the 43
inverse image of M on Z. Thus, taking 6.7 into account, there are isomorphisms

$$\mathbf{Ext}^q_{A,Z}(M,N) \simeq \mathbf{Ext}^q_A(M_Z,N). \tag{6.8.3}$$

We now pass to the global invariants. Set

$$\mathrm{Hom}_{A,Z}(M,N) = \mathrm{Ext}^0_{A,Z}(E;M,N) = H^0_Z(E, \mathbf{Hom}_A(M,N)). \tag{6.8.4}$$

The group $\mathrm{Hom}_{A,Z}(M,N)$ is the subgroup of the group of morphisms from M to N whose support is in Z (IV 14), i.e. those which vanish on U. More generally, for every object X of E, set

$$\mathrm{Ext}^0_{A,Z}(X;M,N) = H^0_Z(X, \mathbf{Hom}_A(M,N)). \tag{6.8.5}$$

The functors $N \longmapsto \mathrm{Ext}^0_{A,Z}(X;M,N)$ are left exact. Their derived functors are denoted by $\mathrm{Ext}^q_{A,Z}(X;M,N)$. These are the Ext *groups with supports in* Z. Definitions (6.3.6), (6.8.1), and (6.8.5), together with the isomorphisms of 6.7, give isomorphisms

$$\mathrm{Ext}^0_{A,Z}(X;M,N) \simeq \begin{cases} H^0(X, \mathbf{Hom}_{A,Z}(M,N)), \\ \mathrm{Ext}^0_A(X;M, \mathscr{H}^0_Z(N)), \\ \mathrm{Ext}^0_A(X;M_Z,N). \end{cases} \tag{6.8.6}$$

The last isomorphism in (6.8.6) gives isomorphisms

$$\mathrm{Ext}^q_{A,Z}(X;M,N) \simeq \mathrm{Ext}^q_A(X;M_Z,N). \tag{6.8.7}$$

PROPOSITION 6.9. 1) *There are two spectral sequences, functorial in M and N and compatible with changes of closed subtopos:*

$$\mathbf{Ext}^{p+q}_{A,Z}(M,N) \Longleftarrow \begin{cases} E_2^{p,q} = \mathscr{H}^p_Z(\mathbf{Ext}^q_A(M,N)), \\ {}'E_2^{p,q} = \mathbf{Ext}^p_A(M, \mathscr{H}^q_Z(N)). \end{cases} \tag{6.9.1}$$

2) *There are three spectral sequences, functorial in X, M, and N and compatible* 44
with changes of closed subtopos:

$$\mathrm{Ext}^{p+q}_{A,Z}(X;M,N) \Longleftarrow \begin{cases} E_2^{p,q} = H^p_Z(X, \mathbf{Ext}^q_A(M,N)), \\ {}'E_2^{p,q} = H^p(X, \mathbf{Ext}^q_{A,Z}(M,N)), \\ {}''E_2^{p,q} = \mathrm{Ext}^p_A(X;M, \mathscr{H}^q_Z(N)). \end{cases} \tag{6.9.2}$$

3) *The sheaves* $\mathbf{Ext}^q_{A,Z}(M,N)$ *are canonically isomorphic to the sheaves associated with the presheaves* $X \longmapsto \mathrm{Ext}^q_{A,Z}(X;M,N)$.

When N is injective, the sheaf $\mathbf{Hom}_A(M,N)$ is flasque (4.10), and hence acyclic for $\mathscr{H}^0_Z$ (6.6). The first spectral sequence in (6.9.1) is a spectral sequence of composite functors (6.8.1). Likewise, when N is injective, $\mathscr{H}^0_Z(N)$ is injective (4.11), and the second spectral sequence in (6.9.1) is a spectral sequence of composite functors deduced from 6.7. The spectral sequences in (6.9.2) are spectral sequences of composite functors deduced from definition (6.8.5) and the first two isomorphisms in (6.8.6). Finally, by allowing X to vary in the second spectral sequence in (6.9.2) and taking the associated sheaves, one obtains a spectral sequence of sheaves which degenerates by 3.1 and gives the isomorphisms in 3).

PROPOSITION 6.10. *The functors* $(M,N) \longmapsto \mathbf{Ext}^q_{A,Z}(M,N)$, $q \geq 0$, *and* $(M,N) \longmapsto \mathrm{Ext}^q_{A,Z}(X;M,N)$, $q \geq 0$, *are* δ*-functors in each of the variables* M *and* N. *Denoting by* M_U *the sheaf* $M \otimes_A A_U$ *(cf. IV 11), there is a long exact sequence*

(6.10.1)

$$\ldots \xrightarrow{\delta} \mathbf{Ext}^q_{A,Z}(M,N) \longrightarrow \mathbf{Ext}^q_A(M,N) \longrightarrow \mathbf{Ext}^q_A(M_U,N) \xrightarrow{\delta} \mathbf{Ext}^{q+1}_{A,Z}(M,N) \longrightarrow \ldots$$

and an analogous exact sequence for the Ext *groups.*

The functors $\mathbf{Ext}^q_A(\cdot,\cdot)$ and $\mathrm{Ext}^q_A(X;\cdot,\cdot)$ form δ-functors with respect to each variable (6.2), and the functor $M \longmapsto M_Z$ is exact because A_Z is flat (1.3.3). The first assertion therefore follows from the isomorphisms (6.8.3) and (6.8.7). The exact sequence (6.10.1), and the analogous exact sequence for the Ext groups, is the long exact sequence of the δ-functor

$$\mathbf{Ext}^q_A(\cdot,N),\ q \geq 0 (resp.\ \ \mathrm{Ext}^q_A(X;\cdot,N),\ q \geq 0)$$

45 associated with the exact sequence $0 \to M_U \to M \to M_Z \to 0$, taking (6.8.3) and (6.8.7) into account.

6.11. Let us indicate briefly how these results may be extended to the case of families of supports. Let E be a topos. For every object X of E, denote by $\mathrm{Fer}(X)$ the set of closed subtopoi of the topos $E_{/X}$. By passage to complements, this set is in bijective correspondence with the set of open subtopoi of $E_{/X}$, i.e. with the set of subobjects of X (IV 8). The presheaf $\mathrm{Fer} : X \longmapsto \mathrm{Fer}(X)$ is in fact a $\mathscr{U}$-sheaf, as is immediately verified, and is therefore representable. In other words, there is an object of E, again denoted by Fer, and a closed subtopos Z_{Fer} of $E_{/\mathrm{Fer}}$ such that, for every X, every element of $\mathrm{Fer}(X)$ is obtained from Z_{Fer} by base change along a morphism $u_Z : X \to \mathrm{Fer}$ uniquely determined by Z.

DEFINITION 6.12. *A family of supports of* E *is a subset* Φ *of the set of closed subtopoi of* E *having the following properties:*

(S1) *The union of a finite family of elements of* Φ *belongs to* Φ.

(S2) *Every closed subtopos of* E *contained in an element of* Φ *is an element of* Φ.

6.12.1. Let E be a topos, let Φ be a family of supports of E, and let X be an object of E. Denote by $\Phi(X)$ the smallest family of supports of $E_{/X}$ containing the closed subtopoi of $E_{/X}$ obtained from closed subtopoi in Φ by the base change $X \to e$ (e being the final object of E). The functor $X \longmapsto \Phi(X)$ is a subpresheaf of the sheaf Fer. It is therefore separated.

6.12.2. A family Φ of supports of E is said to be *of local character* if it has the following property:
(LC) For every epimorphic family $(X_i \to e)$, $i \in I$ (where e is the final object of E), the sequence of sets

$$\Phi \longrightarrow \prod_i \Phi(X_i) \rightrightarrows \prod_{i,j} \Phi(X_i \times X_j)$$

is exact.

Let $a\Phi$ be the sheaf associated with the presheaf $X \longmapsto \Phi(X)$ (6.12.1). Condition (LC) is equivalent to the condition that the canonical morphism $\Phi \to a\Phi(e)$ be a bijection (cf. the construction of the associated sheaf in II in the case of a separated presheaf). To verify (LC), one may therefore restrict to a final family of epimorphic families $(X_i \to e)$, $i \in I$. 46

Example 6.12.3. 1) Let Z be a closed subtopos of E. The set of closed subtopoi of E contained in Z is a family of supports of E. It is of local character.
2) Let T be a topological space and let Φ be a paracompactifying family of supports of T [7]. In general, the family Φ is not of local character.
3) Let T be a topological space and let p be an integer. The family Φ_p of closed subsets of T of Krull codimension $\geq p$ is a family of supports of T. It is of local character.
4) Let E be a topos with the following property: every epimorphic family $(X_i \to e)$, $i \in I$, is dominated by a finite epimorphic family. Then every family of supports of E is of local character. Indeed, using example 1), it is enough to show that a filtered inductive limit Φ_λ of families of local character is a family of local character. This follows immediately by taking the inductive limit in the sequence of sets

$$\Phi_\lambda \longrightarrow \prod_i \Phi_\lambda(X_i) \rightrightarrows \prod_{i,j} \Phi_\lambda(X_i \times X_j),$$

where $(X_i \to e)$, $i \in I$, is a finite epimorphic family. Indeed, by 6.12.2 and the hypothesis on E, it is enough to consider finite epimorphic families in verifying condition (LC); and, since filtered inductive limits commute with finite projective limits (I 2), exactness of these sequences of sets is preserved upon passage to the inductive limit.
5) Let E be a coherent topos (VI 2.3). Then, by the preceding example, every family of supports of E is of local character.

6.13. Let (E, A) be a ringed topos, let Φ be a family of supports of E, let N be an A-module (a left module, to fix ideas), and let X be an object of E. Set

$$\mathscr{H}^0_\Phi(N) = \varinjlim_{Z \in \Phi} \mathscr{H}^0_Z(N), \tag{6.13.1}$$

$$H^0_\Phi(E, N) = \varinjlim_{Z \in \Phi} H^0_Z(E, N). \tag{6.13.2}$$

Let M be a left A-module, and set 47

$$\mathbf{Ext}^0_{A,\Phi}(M, N) = \mathbf{Hom}_{A,\Phi}(M, N) = \varinjlim_{Z \in \Phi} \mathbf{Ext}^0_{A,Z}(M, N), \tag{6.13.3}$$

$$\mathrm{Ext}^0_{A,\Phi}(X; M, N) = \mathrm{Hom}_{A|X,\Phi(X)}(M|X, N|X) = \varinjlim_{Z \in \Phi} \mathrm{Ext}^0_{A,Z}(X; M, N). \tag{6.13.4}$$

There are canonical isomorphisms

$$\mathbf{Ext}^0_{A,\Phi}(M,N) \simeq \mathscr{H}^0_\Phi(\mathbf{Ext}^0_A(M,N)), \tag{6.13.5}$$

$$\mathrm{Ext}^0_{A,\Phi}(X;M,N) \simeq H^0_\Phi(X,\mathbf{Ext}^0_A(M,N)). \tag{6.13.6}$$

When Φ is the family of closed subtopoi contained in a fixed closed subtopos Z, there are isomorphisms

$$\begin{cases} \mathscr{H}^0_\Phi \simeq \mathscr{H}^0_Z, \\ H^0_\Phi \simeq H^0_Z, \\ \mathbf{Ext}^0_{A,\Phi}(\cdot,\cdot) \simeq \mathbf{Ext}^0_{A,Z}(\cdot,\cdot), \\ \mathrm{Ext}^0_{A,\Phi}(X;\cdot,\cdot) \simeq \mathrm{Ext}^0_{A,Z}(X;\cdot,\cdot). \end{cases} \tag{6.13.7}$$

The inductive limits defining the preceding functors are filtered. Consequently, these functors are left exact. Their right derived functors are denoted by

$$\mathscr{H}^q_\Phi,\ H^q_\Phi,\ \mathbf{Ext}^q_{A,\Phi}(\cdot,\cdot),\ \mathrm{Ext}^q_{A,\Phi}(X;\cdot,\cdot). \tag{6.13.8}$$

They too are computed by inductive limits of the functors $\mathscr{H}^q_Z$, H^q_Z, etc. as Z ranges through Φ.

From the isomorphisms (6.13.5) and (6.13.6), one obtains two spectral sequences by taking inductive limits in the first spectral sequence in (6.9.1) and the first spectral sequence in (6.9.2):

$$\mathbf{Ext}^{p+q}_{A,\Phi}(M,N) \Leftarrow E_2^{p,q} = \mathscr{H}^p_\Phi(\mathbf{Ext}^q_A(M,N)), \tag{6.13.9}$$

$$\mathrm{Ext}^{p+q}_{A,\Phi}(X;M,N) \Leftarrow E_2^{p,q} = H^p_\Phi(X,\mathbf{Ext}^q_A(M,N)). \tag{6.13.10}$$

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6.14. With the notation of 6.13, and without any hypothesis on Φ, there is a functorial morphism

$$\theta : H^0_\Phi(N) \longrightarrow H^0(E,\mathscr{H}^0_\Phi(N)). \tag{6.14.1}$$

This morphism is always injective, but is not in general an isomorphism. However, if Φ is *of local character*, the morphism θ is an isomorphism, as is immediately verified. Likewise, if Φ is *of local character*, there is an isomorphism

$$\theta' : \mathrm{Ext}^0_{A,\Phi}(E;M,N) \simeq H^0(E,\mathbf{Ext}^0_{A,\Phi}(M,N)). \tag{6.14.2}$$

PROPOSITION 6.15. *Let (E,A) be a ringed topos such that E is coherent* (VI)*, let Φ be a family of supports of E, and let M and N be two A-modules. There are two spectral sequences:*

$$H^{p+q}_\Phi(E,N) \Leftarrow E_2^{p,q} = H^p(E,\mathscr{H}^q_\Phi(N)), \tag{6.15.1}$$

$$\mathrm{Ext}^{p+q}_{A,\Phi}(E;M,N) \Leftarrow E_2^{p,q} = H^p(E,\mathbf{Ext}^q_{A,\Phi}(M,N)). \tag{6.15.2}$$

These spectral sequences are deduced from the spectral sequence (6.4.2) and the second spectral sequence in (6.9.2) by taking the inductive limit over the closed subtopoi Z in Φ, taking into account that the cohomology of a coherent topos commutes with inductive limits of sheaves (IV 5).

6.16. We mention, without proof, that the second spectral sequence in (6.9.1) and the third spectral sequence in (6.9.2) (with X equal to the final object of E) can be extended to the case of families of supports, provided that the topos E is coherent and the module M is perfect [13].

Finally, one can also generalize the notion of families of supports by introducing presheaves of families of supports, sheaves of families of supports, and the corresponding cohomology groups and sheaves[14]. 49

7. Appendix: Čech cohomology

Developing an idea due to P. Cartier, we show in this section how, in an arbitrary topos, one can compute the cohomology of a sheaf by means of coverings. For the classical theory of paracompact spaces, we refer to [7]; for another method which applies to certain topoi, and in particular to the étale topos, see [14].

7.1. Skeleton and coskeleton.

7.1.0. Let $\mathbf{\Delta}$ be the category of standard simplices (the objects of $\mathbf{\Delta}$ are the ordered sets $\Delta_n = [0, \dots, n]$, and the morphisms are the increasing maps). Let E be a $\mathscr{U}$-topos, and let $\mathbf{\Delta}(E)$ be the category of presheaves on $\mathbf{\Delta}$ with values in E; in other words, the category of *semi-simplicial objects* of E. Denote by $\mathbf{\Delta}[n]$ the full subcategory of $\mathbf{\Delta}$ defined by the objects Δ_p, $p \leq n$, by $i_n : \mathbf{\Delta}[n] \to \mathbf{\Delta}$ the inclusion functor, and by $\mathbf{\Delta}E[n]$ the category of presheaves on $\mathbf{\Delta}[n]$ with values in E; in other words, the category of *semi-simplicial objects of E truncated at order n*. By I 5.1, there is a sequence of three adjoint functors (I 5.3)

$$\mathbf{\Delta}E[n] \quad \begin{array}{c} \xrightarrow{i_{n!}} \\ \xleftarrow{i_n^*} \\ \xrightarrow{i_{n*}} \end{array} \quad \mathbf{\Delta}E$$

where i_n^* is the restriction functor to the category $\mathbf{\Delta}[n]$, and i_{n*} and $i_{n!}$ are its right and left adjoints, respectively. Note that $\mathbf{\Delta}E$ and $\mathbf{\Delta}E[n]$ are $\mathscr{U}$-topoi (IV 1.2), and that (i_n^*, i_{n*}) is a morphism of topoi (IV 3.1) which is an embedding of $\mathbf{\Delta}E[n]$ into $\mathbf{\Delta}E$ (I 5.6 and IV 9.1.1).

Definition 7.1.1. *The functor $i_{n!}i_n^*$ (resp. $i_{n*}i_n^*$) from $\mathbf{\Delta}_E$ to $\mathbf{\Delta}_E$ is denoted by* sk_n *(resp.* cosk_n*) and is called the* skeleton functor of order n *(resp. the* coskeleton functor of order n*).* 50

The skeleton and coskeleton functors are used constantly in the theory of semi-simplicial sets [3].

Proposition 7.1.2. 1) *The adjunction morphism* $\mathrm{sk}_n \to \mathrm{id}$ *is a monomorphism. The canonical morphisms* $\mathrm{sk}_n \mathrm{sk}_m \to \mathrm{sk}_n$ *(resp.* $\mathrm{cosk}_n \to \mathrm{cosk}_n \mathrm{cosk}_m$*) are isomorphisms when $n \leq m$ (resp. $n \geq m$).*

[14] cf. [12], Chap. IV, § 1 (Lecture Notes 20, Springer), for the development of these notions in the form of a fugue with variations.

2) *Let $u : E \to E'$ be a morphism of topoi. The functor u^*, extended to semi-simplicial and truncated semi-simplicial objects, commutes with the functors $i_{n!}$, i_n^*, sk_n, and cosk_n.*

The first assertion follows immediately from the definitions. To prove 2), it suffices to observe that, by I 5.1, the functors $i_{n!}, \ldots, \mathrm{cosk}_n$ are computed by means of inductive limits and finite projective limits.

7.2. An acyclicity lemma.

7.2.0. Let M be an abelian group of E. The *homology* of a semi-simplicial object K of E with coefficients in M is defined in the usual way. First consider $A(K)$, the free abelian semi-simplicial group generated by K, and then form the tensor product $M \otimes_{\mathbf{Z}} A(K)$; this gives an abelian semi-simplicial group of E. One then considers the associated complex of abelian groups (formed by taking the alternating sum of the boundary operators) and takes its homology. The homology objects are denoted by $H_i(K, M)$. They are abelian groups of E. When M is the group $\mathbb{Z}_E$ (the free abelian group generated by the final object of E), we write simply $H_i(K)$. The groups $H_i(K)$ are called the *homology groups* of K. A semi-simplicial object K is said to be *acyclic* if, for every abelian group M of E, the groups $H_i(K, M)$ vanish for $i > 0$.

51 Formation of homology commutes with inverse images under morphisms of topoi .

The purpose of this section is to prove the following lemma:

LEMMA 7.2.1. *Let E be a topos, let $K_\bullet$ and $L_\bullet$ be two semi-simplicial objects of E, let $v_\bullet : K_\bullet \to L_\bullet$ be a morphism of semi-simplicial objects, and let n be an integer ≥ 0. Suppose that v has the following properties:*

1) *For every integer $p < n$ ($p \geq 0$), the morphism $v_p : K_p \to L_p$ is an isomorphism.*
2) *The morphism $K_n \xrightarrow{v_n} L_n$ is an epimorphism.*
3) *The canonical morphisms $K_\bullet \to \mathrm{cosk}_n(K_\bullet)$ and $L_\bullet \to \mathrm{cosk}_n(L_\bullet)$ are isomorphisms.*

Then, for every integer p, the morphism v_p is an epimorphism, and the morphism $v_\bullet$ induces an isomorphism on homology objects.

REMARK. The proof in fact shows that, after adopting a suitable definition of homotopy in topoi, the morphism $v_\bullet$ is a homotopy equivalence.

PROOF. We may suppose that E is the topos of sheaves on a small site C (IV 1), and hence that E is a subtopos of a topos E' having enough fiber functors (for example, the topos $C^\wedge$). Denote by $a^* : E' \to E$ the inverse-image functor of the embedding morphism $E \hookrightarrow E'$. Let

$$v_\bullet[n] : K_\bullet[n] \longrightarrow L_\bullet[n]$$

be the morphism of truncated semi-simplicial objects obtained by truncating $v_\bullet$ at order n. Denote by $L_\bullet[n]'$ the image (in E') of $K_\bullet[n]$ under $v_\bullet[n]$, and by $v_\bullet[n]' : K_\bullet[n] \to L_\bullet[n]'$ the morphism induced by $v_\bullet[n]$. Put $L! = i_{n*}L_\bullet[n]'$, $K! = i_{n*}K_\bullet[n]$, and $v! = i_{n*}v_\bullet[n]'$ (here i_{n*} is relative to E'). By 1), the morphism $v! : K! \to L!$ has properties 1), 2), and 3) of the lemma. Moreover, by 2), 3), and 7.1.2, the morphism $a^*v!$ is none other than $v_\bullet$. It therefore suffices to prove the lemma for $v!$, and hence we may suppose that E has enough fiber functors. Using these fiber functors, we may then reduce to the case where E is the category of sets.

52 First observe that the hypotheses of the lemma on the morphism $v_\bullet$ are stable under every base change $M_\bullet \to L_\bullet$ for which $M_\bullet$ is a coskeleton of order n. Consequently, the

fiber $P_\bullet$ of $v_\bullet$ at any base point of $L_\bullet$ is of the form $i_{n*}P_\bullet[n]$, where $P_\bullet[n]$ is a nonempty complex truncated at order n of the form

$$P_n \longrightarrow e \longrightarrow e \longrightarrow e \longrightarrow \cdots \longrightarrow e. \tag{7.2.2}$$

For every integer i, denote by $\delta\Delta_i$ the semi-simplicial set forming the boundary of the simplex Δ_i. Every morphism from $\delta\Delta_i$ to $P_\bullet$ extends to a morphism from Δ_i to $P_\bullet$. Indeed, this property is immediate for $i \geq n$, because $P_\bullet$ is a coskeleton of order n, and it is clear for $i < n$ from the description (7.2.2). Since $P_\bullet$ is a Kan complex, it is homotopically trivial [6]. Note that the morphism $v_\bullet$ is a fibration in the sense of Kan [6]. Since the fibers of this fibration are homotopically trivial, $v_\bullet$ induces an isomorphism on the homotopy groups of $K_\bullet$ and $L_\bullet$ at all base points of $K_\bullet$. By the Hurewicz theorem, this implies that $v_\bullet$ induces an isomorphism on homology [6]. This proves the second assertion of the lemma. To prove the first, observe that every morphism from a complex truncated at order n to $L_\bullet[n]$ lifts to a morphism to $K_\bullet[n]$. Consequently, by 3), every morphism from a complex to $L_\bullet$ lifts to a morphism to $K_\bullet$. Thus $v_\bullet$ is surjective.

7.3. **Hypercoverings**.

7.3.0. Let C be a site belonging to the universe $\mathscr{U}$, in which fiber products and products of two objects are representable. Denote by $C^\wedge$ the topos of $\mathscr{U}$-presheaves on C, and by $C^\sim$ the topos of $\mathscr{U}$-sheaves on C. An object K of $C^\wedge$ is called *semi-representable* if it is isomorphic to a coproduct of representable presheaves.

Let $SR(C)$ be the full subcategory of $C^\wedge$ defined by the semi-representable objects. In the category $SR(C)$, projective limits indexed by finite nonempty categories are representable, and the inclusion functor $SR(C) \hookrightarrow C^\wedge$ commutes with these finite projective limits. By a remark already made, it follows that if $K_\bullet$ is a semi-simplicial object truncated in degree n in $C^\wedge$, all of whose objects are semi-representable, then the extension $i_n^+(K_\bullet)$ has the same property. Thus, if $K_\bullet$ is a semi-simplicial object of $C^\wedge$ all of whose objects are semi-representable, then, for every n, the object $\mathrm{cosk}_n(K_\bullet)$ has the same property. 53

7.3.1. *Definitions and notation.*

. Let $p > 0$ be an integer. A semi-simplicial object $K_\bullet$ of $C^\wedge$ is called a *hypercovering of type p* of C, or, when no confusion can result, a hypercovering of type p, if it has the following properties:

HR1) For every integer $n \geq 0$, K_n is semi-representable.
HR2) The canonical morphism $K_\bullet \to \mathrm{cosk}_p(K_\bullet)$ is an isomorphism.
HR3) For every integer $n \geq 0$, the canonical morphism of presheaves $K_{n+1} \to (\mathrm{cosk}_n(K_\bullet))_{n+1}$ is a covering morphism of presheaves (II 6.2). The canonical morphism of presheaves $K_0 \to e$, where e is the final object of $C^\wedge$, is a covering morphism of presheaves.

. A hypercovering of type p is a hypercovering of type q for every $q \geq p$. A semi-simplicial object $K_\bullet$ will be called a *hypercovering* (of C) if it has properties HR 1) and HR 3).

. Denote by HR_p (resp. HR), and call the category of hypercoverings of type p (resp. the category of hypercoverings), the following category:

a) The objects of HR_p (resp. HR) are the hypercoverings of type p (resp. the hypercoverings).
b) Let $K_\bullet$ and $L_\bullet$ be two objects of HR_p (resp. HR). A morphism of HR_p (resp. HR) with source $K_\bullet$ and target $L_\bullet$ is a morphism $v_\bullet : K_\bullet \to L_\bullet$ of semi-simplicial objects of $C^\wedge$.

. Let C be a site in which fiber products are representable, and let X be an object of C. A hypercovering of X (resp. a hypercovering of type p of X) means a hypercovering of the site C/X (resp. a hypercovering of type p of C/X). The category of hypercoverings
54 of X (resp. of hypercoverings of type p of X) is defined similarly . There is a sequence of inclusion functors

$$\cdots HR_p \hookrightarrow HR_{p+1} \hookrightarrow \cdots \hookrightarrow HR.$$

These functors are fully faithful. We denote by HR_∞ the inductive limit of the categories HR_p.

. For every integer $n \geq 0$, denote by Δ_n^c the standard semi-simplicial object of dimension n with values in the category of sets, i.e. the functor on $\mathbf{\Delta}$ represented by the ordered set $[0, n]$. Also denote by Δ_n^c the presheaf on C, with values in the category of semi-simplicial sets, that is constant with value Δ_n. Thus the presheaf Δ_n^c is a presheaf of semi-simplicial sets. The two canonical injections from Δ_0 into Δ_1 define two morphisms of semi-simplicial presheaves:

$$\underline{\Delta}_0^c \underset{e_1}{\overset{e_0}{\rightrightarrows}} \underline{\Delta}_1^c,$$

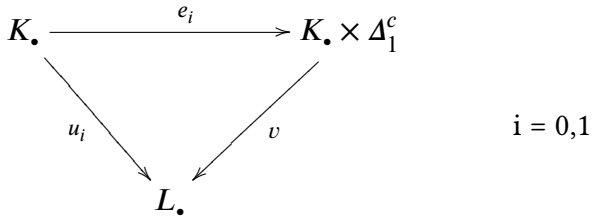
and consequently define, for every semi-simplicial presheaf $K_\bullet$, two canonical injections:

$$K_\bullet \underset{e_1}{\overset{e_0}{\rightrightarrows}} K_\bullet \times \underline{\Delta}_1^c.$$

Two morphisms $K_\bullet \underset{u_1}{\overset{u_0}{\rightrightarrows}} L_\bullet$ of semi-simplicial presheaves are called *homotopic morphisms* if there exists a morphism

$$v : K_\bullet \times \Delta_1^c \longrightarrow L_\bullet$$

such that the diagrams



i = 0,1

55 are commutative. The morphism v is called a *homotopy joining u_0 to u_1*. The relation "u_0 and u_1 are two homotopic morphisms from $K_\bullet$ to $L_\bullet$" is not, in general, an equivalence relation. Nevertheless, the equivalence relation generated by this relation is compatible with composition of morphisms. This therefore allows us to define the category of semi-simplicial presheaves up to homotopy by passing to the quotient by the equivalence relation generated by the homotopy relation.

. In this way we define the categories $\mathscr{H}\mathscr{R}_p$, $\mathscr{H}\mathscr{R}_\infty$, and $\mathscr{H}\mathscr{R}$, the *categories of hypercoverings up to homotopy*.

THEOREM 7.3.2. *Let C be a site satisfying the conditions of 7.3.0.*

1) *The category $\mathscr{H}\mathscr{R}_p^0$ (resp. $\mathscr{H}\mathscr{R}^0$) is filtered* (I 2.7).
2) *Let $K_\bullet$ be an object of HR_p (resp. of HR), let n be an integer such that $0 \leq n \leq p$ (resp. $0 \leq n$), and let*

$$u : X \longrightarrow K_n$$

be a covering morphism of presheaves. There exists an object $L_\bullet$ of HR_p (resp. of HR) and a morphism $f : L_\bullet \to K_\bullet$ such that the morphism

$$f_n : L_n \longrightarrow K_n$$

factors through u.

3) *The associated semi-simplicial sheaves* (II 3.5) *of hypercoverings are acyclic (7.2.0) in strictly positive degrees. The 0-th homology sheaf is isomorphic to the sheaf associated to the constant presheaf $\mathscr{Z}$.*

Proof. We prove 3). Let $K_\bullet^\sim$ be the semi-simplicial sheaf associated to $K_\bullet$. The "associated sheaf" functor commutes with the functor cosk_n (7.1.2. 2)). It follows that the semi-simplicial sheaf $K_\bullet^\sim$ has the following properties:

a) The canonical morphism $K_0 \to e$ is an epimorphism of sheaves.
b) For every integer $n \geq 0$, the canonical morphism $K_{n+1} \to (\mathrm{cosk}_n(K_\bullet))_{n+1}$ is an epimorphism of sheaves.

For $n \geq 0$, put $L_{\bullet n} = \mathrm{cosk}_n K_\bullet^\sim$. Then, for every n, there is a sequence of morphisms of semi-simplicial sheaves

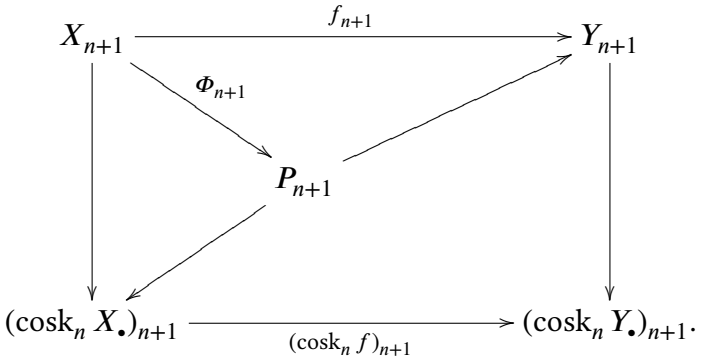
$$K_\bullet^\sim \xrightarrow{u_n} L_{\bullet n} \xrightarrow{v_n} L_{\bullet n-1} \cdots \longrightarrow L_{\bullet 0} \xrightarrow{v_0} e_\bullet,$$

and , for every j, $0 \leq j \leq n$, v_j has the properties of 7.2.1. Moreover, the morphism u_n 56
induces an isomorphism on the components of degree $\leq n$. It then follows from 7.2.1, by induction on n, that $K_\bullet$ is acyclic.

To prove 1) and 2), we introduce some terminology.

Definition 7.3.3. *Let $f : X_\bullet \to Y_\bullet$ be a morphism of semi-simplicial presheaves. We say that f is* special of type p *(resp.* special*) if:*

1) *The objects $X_\bullet$ and $Y_\bullet$ are canonically isomorphic to their coskeleta of order p, and $\mathrm{cosk}_p(f) = f$ (resp. no conditions are imposed on f, $X_\bullet$, or $Y_\bullet$).*
2) *For every integer n such that $0 \leq n \leq p$ (resp. $0 \leq n$), the morphism Φ_{n+1} appearing in the diagram below is covering:*



(The vertical arrows are defined by the canonical morphisms $X_\bullet \to \mathrm{cosk}_n X_\bullet$ and $Y_\bullet \to \mathrm{cosk}_n Y_\bullet$. The object P_{n+1} is the fiber product, and Φ_{n+1} is the unique arrow making the diagram commutative.)

3) *The morphism $f_0 : X_0 \to Y_0$ is covering.*

A semi-simplicial presheaf $K_\bullet$ is called special of type p (resp. special) if the canonical morphism

$$K_\bullet \longrightarrow e_\bullet \quad (e_\bullet \text{ is the final semi-simplicial object})$$

is special of type p (resp. special), i.e. if $K_\bullet$ satisfies conditions HR 2) *and* HR 3) *(resp.* HR 3)*) of 7.3.1.1.*

57 LEMMA 7.3.4. 1) *The composite of two special morphisms (resp. special morphisms of type p) is a special morphism (resp. a special morphism of type p).*

2) *Let $K_\bullet$ be a special semi-simplicial presheaf (resp. special of type p), let $X_\bullet \xrightarrow{f} Y_\bullet$ be a special morphism (resp. special of type p), and let $u : K_\bullet \to Y_\bullet$ be a morphism of complexes. Then the fiber product $P_\bullet = K_\bullet \times_{Y_\bullet} X_\bullet$ is special (resp. special of type p).*

The proof of this lemma is left to the reader.

7.3.5. *Proof of assertion* 2) *of* 7.3.2. We may first suppose that X is semi-representable. For every semi-simplicial object $L_\bullet$, denote by $\mathrm{Hom}_{(X)}(L_\bullet, K_\bullet)$ the set of morphisms of semi-simplicial objects equipped with a factorization $L_n \to X \xrightarrow{u} K_n$. The functor $\mathrm{Hom}_{(X)}(\cdot, K_\bullet)$ is representable. Indeed, this functor carries inductive limits to projective limits, and the category $\mathbf{\Delta}(C^\wedge)$ is a topos. Denote by $P_\bullet$ an object representing this functor. For an object Z of $C^\wedge$, likewise denote by $j_n^+(Z)$ the object representing the functor $L_\bullet \mapsto \mathrm{Hom}(L_n, Z)$ on $\mathbf{\Delta}C^\wedge$, whose construction by $\varprojlim$ is made explicit in III 1.1. Observe here that this construction involves only finite projective limits. It follows immediately that j_n^+ carries semi-representable presheaves to semi-simplicial presheaves having semi-representable components, and covering morphisms $Z' \to Z$ to special morphisms (7.3.3).

By the definition of $P_\bullet$, there is a cartesian square:

$$\begin{array}{ccc} P_\bullet & \longrightarrow & j_n^+(X) \\ \downarrow & & \downarrow{\scriptstyle j_n^+(u)} \\ K_\bullet & \longrightarrow & j_n^+(K_n) \end{array} .$$

By what has just been observed, the semi-simplicial objects $j_n^+(X)$ and $j_n^+(K_n)$ are semi-representable, and the morphism $j_n^+(u)$ is a special morphism of type n. The result now follows by applying 3.4 and the sorites 7.3.0.

58 . Let $M_\bullet$ and $N_\bullet$ be two semi-simplicial presheaves. The functor

$$L_\bullet \longmapsto \mathrm{Hom}(L_\bullet \times M_\bullet, N_\bullet)$$

is representable. The semi-simplicial presheaf representing it will be denoted by

$$\mathbf{Hom}_\bullet(M_\bullet, N_\bullet).$$

The degree-n component presheaf of this object will be denoted by $\mathbf{Hom}_n(M_\bullet, N_\bullet)$. There is a canonical isomorphism

$$\mathbf{Hom}_n(M_\bullet, N_\bullet) \xrightarrow{\sim} \mathbf{Hom}_0(M_\bullet \times \Delta_n^c, N_\bullet).$$

LEMMA 7.3.6. *Let*

$$\begin{array}{ccc} M_\bullet & \hookrightarrow & N_\bullet \\ \uparrow & & \uparrow \\ \mathrm{sk}_n \Delta_{n+1}^c & \xrightarrow{u} & \Delta_{n+1}^c \end{array} \qquad (u \text{ the canonical injection})$$

be a cocartesian diagram of semi-simplicial presheaves.

For every hypercovering $L_\bullet$, the morphism

$$\mathbf{Hom}_0(N_\bullet, L_\bullet) \longrightarrow \mathbf{Hom}_0(M_\bullet, L_\bullet)$$

is covering.

PROOF. There is a cartesian diagram:

$$\begin{array}{ccc} \mathbf{Hom}_0(N_\bullet, L_\bullet) & \longrightarrow & \mathbf{Hom}_0(M_\bullet, L_\bullet) \\ \downarrow & & \downarrow \\ \mathbf{Hom}_0(\Delta^c_{n+1}, L_\bullet) & \xrightarrow{(*)} & \mathbf{Hom}_0(\mathrm{sk}_n \Delta^c_{n+1}, L_\bullet). \end{array}$$

It therefore suffices to show that the morphism $(*)$ is covering. But $(*)$ is isomorphic to the morphism

$$L_{n+1} \longrightarrow (\mathrm{cosk}_n L_\bullet)_{n+1},$$

which is covering by hypothesis. C.Q.F.D.

7.3.7. Proof of assertion 1) of 7.3.2. The product of two objects of HR_p (resp. of HR) is again an object of HR_p (resp. of HR) (7.3.4). To prove that the category $\mathscr{H}\mathscr{R}^0_p$ (resp. $\mathscr{H}\mathscr{R}^0$) is filtered, it suffices to show that, given two morphisms in HR_p (resp. HR)

$$K_\bullet \underset{u_1}{\overset{u_0}{\rightrightarrows}} L_\bullet$$

there exists a morphism $v : M_\bullet \to K_\bullet$ in HR_p (resp. in HR) such that the morphisms

$$M_\bullet \underset{u_1 v}{\overset{u_0 v}{\rightrightarrows}} L_\bullet$$

are homotopic, i.e. such that there exists $w : M_\bullet \times \Delta^c_1 \to L_\bullet$ making the diagrams 59

(x)
$$\begin{array}{ccc} M_\bullet & \xrightarrow{e_i} & M_\bullet \times \underline{\Delta}^c_1 \\ {\scriptstyle v}\downarrow & & \downarrow{\scriptstyle w} \\ K_\bullet & \xrightarrow{u_i} & L_\bullet \end{array} \qquad i = 0, 1.$$

commute. For every semi-simplicial object $M_\bullet$ of $C^\wedge$ (not necessarily a hypercovering), let $F(M_\bullet)$ be the set of pairs (v, w), where $v : M_\bullet \to K_\bullet$ and $w : M_\bullet \times \Delta^c_1 \to L_\bullet$, such that the diagrams (x) commute. The functor $M_\bullet \mapsto F(M_\bullet)$ is contravariant in $M_\bullet$, carries inductive limits to projective limits, and is consequently represented by a semi-simplicial object $F_\bullet$. The object $F_\bullet$ is clearly the upper-left vertex of a cartesian diagram

$$\begin{array}{ccc} F_\bullet & \longrightarrow & \mathbf{Hom}_\bullet(\Delta^c_1, L_\bullet) \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ K_\bullet & \xrightarrow[u_0, u_1]{} & L_\bullet \times L_\bullet, \end{array}$$

where π is defined by the two inclusions of Δ^c_0 into Δ^c_1. To prove the assertion, it therefore suffices to show that $F_\bullet$ is an object of HR_p (resp. of HR). For this, by 7.3.4 2) and the sorites 7.3.0, it suffices to show that

60 a) $\mathbf{Hom}_\bullet(\Delta_1^c, L_\bullet)$ is semi-representable,
b) the morphism π is special of type p (resp. special).

First, it follows immediately that $\mathbf{Hom}_\bullet(\Delta_1^c, L_\bullet)$ is a semi-representable semi-simplicial object. Indeed, its components are computed by finite projective limits from the L_n, and are therefore semi-representable. Let us now verify b). First one shows that, when $L_\bullet$ is special of type p, the morphism

$$\mathbf{Hom}_\bullet(\Delta_1^c, L_\bullet) \longrightarrow \mathrm{cosk}_p \mathbf{Hom}_\bullet(\Delta_1^c, L_\bullet)$$

is an isomorphism; this verifies property 1) of 7.3.3. Next, the morphism

$$\pi_0 : \mathbf{Hom}_0(\Delta_1^c, L_\bullet) \longrightarrow L_0 \times L_0$$

is isomorphic to the canonical morphism

$$L_1 \xrightarrow{(d_0,d_1)} L_0 \times L_0,$$

which is covering by hypothesis ($L_\bullet$ is a hypercovering).

It remains to verify property 2) of 7.3.3, i.e. to verify that, for every n, in the diagram

$$\begin{array}{ccc}
\mathbf{Hom}_{n+1}(\Delta_1^c, L_\bullet) & \xrightarrow{\pi_{n+1}} & L_{n+1} \times L_{n+1} \\
\downarrow & \overset{\Phi_{n+1}}{\searrow} \quad \nearrow & \downarrow \\
 & P_{n+1} & \\
\downarrow & \swarrow & \downarrow \\
(\mathrm{cosk}_n \mathbf{Hom}_\bullet(\Delta_1^c, L_\bullet))_{n+1} & \xrightarrow{(\mathrm{cosk}_n \pi)_{n+1}} & (\mathrm{cosk}_n L_\bullet)_{n+1} \times (\mathrm{cosk}_n L_\bullet)_{n+1}
\end{array}$$

(P_{n+1} is the fiber product), the morphism Φ_{n+1} is covering. A simple "adjoint-functor chase" shows that

$$P_{n+1} \xrightarrow{\sim} \mathbf{Hom}_0\big(\mathrm{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c), L_\bullet\big),$$
$$\mathbf{Hom}_{n+1}(\Delta_1^c, L_\bullet) \xrightarrow{\sim} \mathbf{Hom}_0(\Delta_{n+1}^c \times \Delta_1^c, L_\bullet),$$

61 and that the morphism Φ_{n+1} is isomorphic to the morphism

$$\mathbf{Hom}_0(\Delta_{n+1}^c \times \Delta_1^c, L_\bullet) \longrightarrow \mathbf{Hom}_0\big(\mathrm{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c), L_\bullet\big)$$

arising from the injection

$$\mathrm{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c) \hookrightarrow \Delta_{n+1}^c \times \Delta_1^c.$$

There is a sequence of subobjects of $\Delta_{n+1}^c \times \Delta_1^c$:

$$\mathrm{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c) = M_0 \hookrightarrow M_1 \hookrightarrow \cdots \hookrightarrow M_k = \Delta_{n+1}^c \times \Delta_1^c,$$

such that, for every $0 \leq i < k$, the morphism

$$M_i \hookrightarrow M_{i+1}$$

fits into a cocartesian diagram:

$$\begin{array}{ccc} M_i & \hookrightarrow & M_{i+1} \\ \uparrow & & \uparrow \\ \mathrm{sk}_{n+1}\,\underline{\Delta}^c_{n+2} & \hookrightarrow & \underline{\Delta}^c_{n+2}\,. \end{array}$$

(The nondegenerate simplices of dimension $n+2$ in $\Delta_{n+1}\times\Delta_1$ are added one after another.) By 7.3.6, the morphism Φ_{n+1} is a composite of covering morphisms and is therefore itself covering. This completes the proof of 7.3.2.

7.4. The isomorphism theorem.

7.4.0. Let F be a presheaf of abelian groups on a site C satisfying condition 7.3.0. For every hypercovering $K_\bullet$, put

$$H^q(K_\bullet, F) = H^q(\mathrm{Hom}_{C^\wedge}(K_\bullet, F)).$$

The $H^q(K_\bullet, F)$ form a δ-functor on the category of abelian presheaves on C, but they are not, in general, the derived functors of the functor $H^0(K_\bullet, F)$.

Now put 62

$$\overset{r}{\vee}H^q\ (C, F) = \varinjlim_{HR_r} H^q(\cdot, F)\ (r \text{ may be infinite}), \tag{7.4.0.1}$$

and

$$\overset{\infty}{\vee}H^q\ (C, F) = \varinjlim_{HR} H^q(\cdot, F). \tag{7.4.0.2}$$

The $\overset{r}{\vee}H^q\ (C, F)$ (resp. $\overset{\infty}{\vee}H^q\ (C, F)$) again form a δ-functor, because the category $\mathscr{H}\mathscr{R}^0_r$ (resp. $\mathscr{H}\mathscr{R}^0$) is filtered (7.3.2). Since the categories $\mathscr{H}\mathscr{R}_r$ are not necessarily $\mathscr{U}$-categories, these δ-functors take values in the category of abelian $\mathscr{V}$-groups, for a suitable universe $\mathscr{V}$.

Suppose now that the presheaf F is a sheaf. Since the semi-simplicial sheaf associated with a hypercovering is acyclic (7.3.2), there is a spectral sequence, functorial in F and in $K_\bullet$:

$$H^p(K_\bullet, \mathscr{H}^q(F)) \Rightarrow H^{p+q}(C^\sim, F). \tag{7.4.0.3}$$

Passing to the limit gives the spectral sequences

$$\begin{aligned} &\overset{r}{\vee}H^p\ (C, \mathscr{H}^q(F)) \Rightarrow H^{p+q}(C^\sim, F), \\ &\overset{\infty}{\vee}H^p\ (C, \mathscr{H}^q(F)) \Rightarrow H^{p+q}(C^\sim, F). \end{aligned} \tag{7.4.0.4}$$

Theorem 7.4.1. *Let C be a site satisfying condition 7.3.0, and let F be an abelian sheaf on C.*

1) *The spectral sequences (7.4.0.4) define isomorphisms*

$$\overset{r}{\vee}H^q\ (C, F) \xrightarrow{\sim} H^q(C^\sim, F), \quad q \le r+1,$$

and a monomorphism

$$\overset{r}{\check{H}}{}^{r+2}(C,F) \longrightarrow H^{r+2}(C^{\sim},F).$$

63 2) *There is an isomorphism of δ-functors:*

$$\overset{\infty}{\check{H}}{}^{q}(C,F) \xrightarrow{\sim} \overset{\vee\vee}{\check{H}}{}^{q}(C,F) \xrightarrow{\sim} H^{q}(C^{\sim},F) \quad (\textit{for every } q).$$

3) *Let G be a presheaf of abelian groups, and let F be its associated sheaf. There are functorial isomorphisms*

$$\overset{r}{\check{H}}{}^{q}(C,G) \xrightarrow{\sim} H^{q}(C^{\sim},F), \quad q \leq r-1,$$

and a monomorphism

$$\overset{r}{\check{H}}{}^{r}(C,G) \longrightarrow H^{r}(C^{\sim},F).$$

When G is a separated presheaf, the latter morphism is an isomorphism, and there is a monomorphism

$$\overset{r}{\check{H}}{}^{r+1}(C,G) \longrightarrow H^{r+1}(C^{\sim},F).$$

4) *There are isomorphisms*

$$\overset{\infty}{\check{H}}{}^{q}(C,G) \xrightarrow{\sim} \overset{\vee\vee}{\check{H}}{}^{q}(C,G) \xrightarrow{\sim} H^{q}(C^{\sim},F) \quad (\textit{for every } q).$$

Proof. It suffices to show that, if N is an abelian presheaf whose associated sheaf is zero, then $\overset{r}{\check{H}}{}^{q}(C,N) = 0$ for $q \leq r$ (resp. $\overset{\vee\vee}{\check{H}}{}^{q}(C,N) = 0$ for every q); this follows immediately by using 3.2.

Remark 7.4.2. 1) Note that, for sites satisfying condition 7.3.0, hypercoverings of type 0 are ordinary coverings, and the functors $\overset{r}{\check{H}}{}^{q}$ for $r = 0$ are none other than the functors $\overset{\vee}{\check{H}}{}^{q}$ introduced in (2.4.5.1).

2) Let C be a site in which finite projective limits are representable, and suppose that for every object X of C and every covering family $(X_i \to X)$, $i \in I$, there exists an $i_0 \in I$ such that $X_{i_0} \to X$ is covering. One can then show that hypercoverings of type p whose components are representable are cofinal in $\mathscr{H}\mathscr{R}_p$. Likewise, hypercoverings $K_\bullet$ for which there exists an integer p such that
64 $K_\bullet$ is of type p, and whose components are representable, are cofinal in $\mathscr{H}\mathscr{R}$. (This may be proved by following the argument of 3.5.) Thus, in the case under consideration, these hypercoverings with representable components suffice to compute the cohomology of the sheaves associated with presheaves.

8. Appendix. Local direct limits (by P. Deligne)

The editor recommends that the reader should, as a rule, avoid reading this appendix. It presents a technique which sometimes makes it possible to extend to topoi without enough points assertions which the existence of points makes trivial. This technique yields a sheaf-theoretic variant of D. Lazard's theorem asserting that flat modules over a ring are direct limits of finite free modules.

8.1. Locally filtering categories.

8.1.0. If $\mathscr{B}$ is a category and $p : \mathscr{A} \to \mathscr{B}$ is a category over $\mathscr{B}$, we shall use the following notation:

- For $U \in \mathrm{Ob}\,\mathscr{B}$, $\mathscr{A}_U$ is the fiber category $p^{-1}(U)$.
- For $f : U \to V$ in $\mathscr{B}$ and $\lambda, \mu \in \mathrm{Ob}\,\mathscr{A}$ such that $p(\lambda) = U$ and $p(\mu) = V$, put

$$\mathrm{Hom}_f(\lambda, \mu) = p^{-1}(f), \quad \text{where } p : \mathrm{Hom}(\lambda, \mu) \longrightarrow \mathrm{Hom}(U, V).$$

Definition 8.1.1. *Let $\mathcal{S}$ be a site. A* locally cofiltering category (or locally left filtering category) *over $\mathcal{S}$ is a category $p : \mathscr{L} \to \mathcal{S}$ over $\mathcal{S}$ such that:*

($\mathscr{L}0$) *For every $f : U \to V$ in $\mathcal{S}$ and every $\mu \in \mathrm{Ob}\,\mathscr{L}_V$, there exists $\lambda \in \mathrm{Ob}\,\mathscr{L}_U$ such that $\mathrm{Hom}_f(\lambda, \mu) \neq \emptyset$;*

($\mathscr{L}1$) *For every $U \in \mathrm{Ob}\,\mathcal{S}$ and every finite family (λ_i) of objects of $\mathscr{L}_U$, there exist a covering $f_j : U_j \to U$ of U and objects $\mu_j \in \mathrm{Ob}\,\mathscr{L}_{U_j}$ such that $\mathrm{Hom}_{f_j}(\mu_j, \lambda_i) \neq \emptyset$ for all i and j;*

($\mathscr{L}2$) *For every $f : U \to V$ in $\mathcal{S}$ and every pair of parallel arrows $(\phi_0, \phi_1) : \lambda \rightrightarrows \mu$ over f, there exist a covering $f_j : U_j \to U$ of U and arrows ψ_j with target λ such that $p(\psi_j) = f_j$ and $\phi_0\psi_j = \phi_1\psi_j$.*

. This definition simplifies when $\mathscr{L}$ is fibered over $\mathcal{S}^*$: the axiom

($\mathscr{L}_*0$) is then satisfied, and ($\mathscr{L}1$), ($\mathscr{L}2$) may be stated as follows:

($\mathscr{L}'1$) For every $U \in \mathrm{Ob}\,\mathcal{S}$ and every finite family λ_i of objects of $\mathscr{L}_U$, locally on U there exists a μ mapping to all the λ_i.

($\mathscr{L}'2$) For every $U \in \mathrm{Ob}\,\mathcal{S}$, every pair of parallel arrows $(\phi_0, \phi_1) : \lambda \rightrightarrows \mu$ in $\mathscr{L}_U$ can, locally on U, be equalized by an arrow with target λ.

8.1.2. Let $\mathcal{S}$ be a site, let $\mathscr{L}$ be a category over $\mathcal{S}$, and let e be a stack over $\mathcal{S}$. Denote by $\Gamma\mathsf{e}$ the category of global sections $\mathbf{Hom}^{\mathrm{cart}}_{\mathcal{S}}(\mathcal{S}, \mathsf{e})$ of e. If $\mathcal{S}$ has a final object S, then $\Gamma\mathsf{e}$ "is none other than" e_S.

Definition 8.1.2.1. *The category of* local inverse systems of objects of e indexed by $\mathscr{L}$ *is the category* $\mathbf{Hom}_{\mathcal{S}}(\mathscr{L}, \mathsf{e})$.

Denote by c ("associated constant local inverse system") the composite functor

$$\Gamma\mathsf{e} = \mathbf{Hom}^{\mathrm{cart}}_{\mathcal{S}}(\mathcal{S}, \mathsf{e}) \hookrightarrow \mathbf{Hom}_{\mathcal{S}}(\mathcal{S}, \mathsf{e}) \xrightarrow{u \mapsto u \circ p} \mathbf{Hom}_{\mathcal{S}}(\mathscr{L}, \mathsf{e}).$$

When the dependence on $\mathscr{L}$ must be made explicit, we shall write $c_{\mathscr{L}}$ instead.

Definition 8.1.3. *The local inverse-limit functor, denoted by $\varprojlim_{\lambda \in \mathscr{L}}$, is the partially defined functor right adjoint to the functor c.*

Thus the functor

$$\varprojlim : \mathbf{Hom}_{\mathcal{S}}(\mathscr{L}, \mathsf{e}) \longrightarrow \Gamma\mathsf{e}$$

satisfies

$$\mathrm{Hom}(X, \varprojlim_{\lambda \in \mathscr{L}} X_\lambda) \simeq \mathrm{Hom}(c(X), (X_\lambda)_{\lambda \in \mathscr{L}}).$$

DEFINITION 8.1.4. *An $\mathcal{S}$-functor $F : \mathscr{L} \to \mathscr{M}$ between locally cofiltering categories over $\mathcal{S}$ is said to be cofinal if it satisfies the following two conditions:*

65 ($C\mathfrak{F}1$) *For every $U \in \mathrm{Ob}\,\mathcal{S}$ and $\mu \in \mathrm{Ob}\,\mathscr{M}_U$, there exist a covering $f_j : U_j \to U$ of U and objects $\lambda_j \in \mathrm{Ob}\,\mathscr{L}_{U_j}$ such that $\mathrm{Hom}_{f_j}(F(\lambda_j), \mu) \neq \emptyset$.*

($C\mathfrak{F}2$) *For every $f : U \to V$ in $\mathcal{S}$ and every pair of parallel arrows $(\phi_0, \phi_1) : F(\lambda) \rightrightarrows \mu$ over f, there exist a covering $f_j : U_j \to U$ of U and arrows ψ_j with target λ in $\mathscr{L}$, such that $p(\psi_j) = f_j$ and $\phi_0 F(\psi_j) = \phi_1 F(\psi_j)$.*

The composite of two cofinal functors is cofinal; equivalences of locally filtering categories are cofinal (the proof is left to the reader).

PROPOSITION 8.1.5. *Let $F : \mathscr{L} \to \mathscr{M}$ be a cofinal functor between locally cofiltering categories over $\mathcal{S}$, let e be a stack over $\mathcal{S}$, and let $(X_\mu)_{\mu \in \mathscr{M}}$ be a local inverse system indexed by $\mathscr{M}$. Then the morphism, functorial in $X \in \Gamma\mathsf{e}$,*

$$F^* : \mathrm{Hom}(c_{\mathscr{M}}(X), (X_\mu)_{\mu \in \mathscr{M}}) \longrightarrow \mathrm{Hom}(c_{\mathscr{L}}(X), (X_{F(\lambda)})_{\lambda \in \mathscr{L}})$$

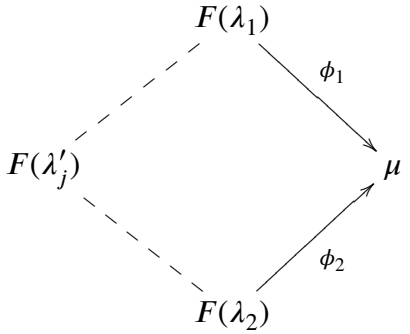
is an isomorphism, so that

$$F^* : \varprojlim_{\mathscr{M}} X_\mu \longrightarrow \varprojlim_{\mathscr{L}} X_{F(\lambda)}$$

is an isomorphism, the two sides being simultaneously defined or undefined.

The proof uses the following lemma, which is left to the reader:

LEMMA 8.1.6. *If $F : \mathscr{L} \to \mathscr{M}$ is cofinal, then for every $f : U \to V$ in $\mathcal{S}$ and arrows over f, $\phi_i : F(\lambda_i) \to \mu$ $(i = 1, 2)$, there exist a covering $f_j : U_j \to U$ of U, objects $\lambda'_j \in \mathrm{Ob}\,\mathscr{L}_{U_j}$, and commutative diagrams*



$$U_j \xrightarrow{f_j} U \longrightarrow V$$

whose projection in $\mathcal{S}$ is (f_j, f).

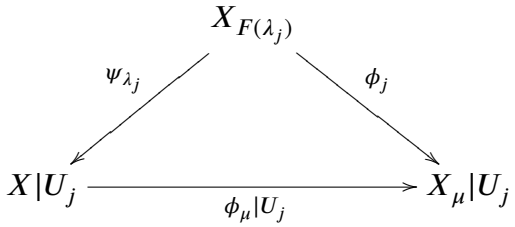
For every $U \in \mathrm{Ob}\,\mathcal{S}$ and $\mu \in \mathscr{M}_U$, condition ($C\mathfrak{F}1$) gives a covering $f_j : U_j \to U$ of U and arrows $\phi_j : F(\lambda_j) \to \mu$ over the f_j. If

$$\psi = (\psi_\lambda) \in \mathrm{Hom}(c_{\mathscr{L}} X, (X_{F(\lambda)})_{\lambda \in \mathscr{L}}) : \quad \psi_\lambda : X|p(\lambda) \longrightarrow X_{F(\lambda)}$$

is the image of

$$\phi = (\phi_\mu) \in \mathrm{Hom}(c_{\mathscr{M}} X, (X_\mu)_{\mu \in \mathscr{M}}) : \quad \phi_\mu : X|p(\mu) \longrightarrow X_\mu,$$

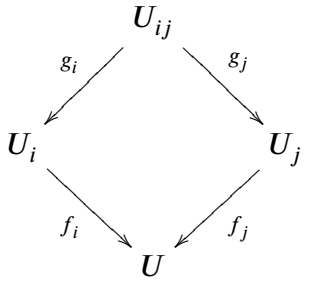
then the diagrams



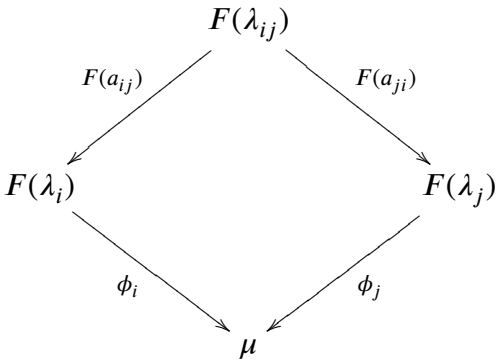
commute. Thus the ϕ_μ are locally determined by ψ; since e is a stack, it follows that F^* is injective. To prove that F^* is surjective, start with ψ and construct a ϕ such that $\psi = F^*\phi$. The preceding construction gives the composite arrows

$$\phi_{\mu,j} = \phi_j \circ \psi_{\lambda_j} : X|U_j \longrightarrow X_\mu|U_j.$$

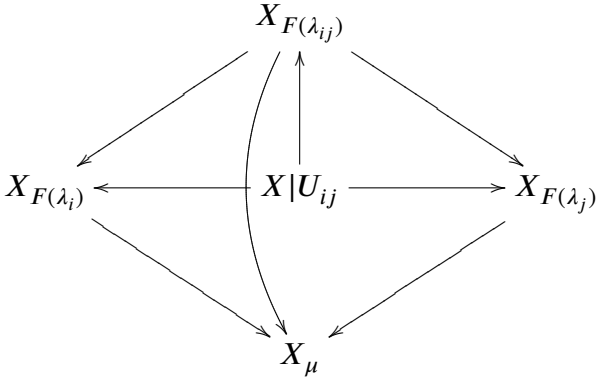
Let us verify that these arrows glue, so that there exists $\phi_\mu : X|U \to X_\mu$ such that $\phi_\mu|U_j = \phi_{\mu,j}$. Consider a commutative diagram in



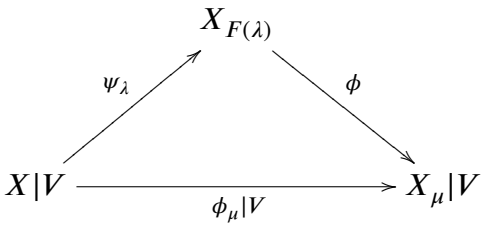
To verify that $\phi_{\mu,j}|U_{ij} = \phi_{\mu,i}|U_{ij}$, it is enough to do so locally on U_{ij}; by 8.1.6 and $(\mathscr{L}0)$ this reduces us to the case where there is a commutative diagram 66



over the preceding one. The diagram



is then commutative, and there exists a family (ϕ_μ) such that, for every arrow $\phi : F(\lambda) \to \mu$ over $f : V \to U$, the diagram



commutes. If $\sigma : \mu \to \mu'$ is an arrow of $\mathscr{M}$, then $(\sigma|V)(\phi_\mu|V) = (\sigma|V)\phi \cdot \psi_\lambda = (\sigma\phi|V) \cdot \psi_\lambda = \phi_{\mu'}|V$; hence $\phi = (\phi_\mu)$ is a morphism of functors such that $\psi = F^*\phi$, as required.

67 If E is an ordered set, we shall also denote by E the category whose set of objects is E and in which $\mathrm{Hom}(i, j)$ is a singleton or empty according as $i \leq j$ or not.

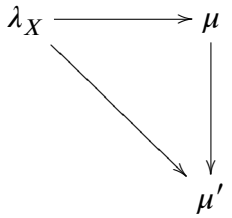
PROPOSITION 8.1.7. *Let $\mathscr{L}$ be a locally cofiltering category over a site $\mathscr{S}$. There exist an ordered set E and a functor F from E to $\mathscr{L}$ such that:*

(i) *E, regarded as a category over $\mathscr{S}$ by means of $pF : E \to \mathscr{S}$, is locally cofiltering.*
(ii) *The functor F is cofinal.*

Let $\mathscr{L}'$ be the locally cofiltering category which is the product of $\mathscr{L}$ with the category defined by the ordered set $\mathbf{Z}$. The projection from $\mathscr{L}'$ to $\mathscr{L}$ is cofinal, so it suffices to prove 8.1.7 for $\mathscr{L}'$.

If X is a finite set of arrows in $\mathscr{L}'$, denote by X^0 the set of endpoints of the arrows in X. Let E be the set of nonempty finite subsets X of $\mathrm{Fl}(\mathscr{L}')$ for which there exists $\lambda_X \in \mathrm{Ob}\,\mathscr{L}'$ satisfying:

a) no arrow of X, except 1_{λ_X}, has target λ_X;
b) if $\mu \in X^0$, then $1_\mu \in X$ and $\mathrm{Hom}(\lambda_X, \mu) \cap X \neq \emptyset$;
c) every diagram of the form



consisting of arrows of X is commutative.

Order E by the relation opposite to inclusion. If $X \in E$ and $\mu \in X^0$, there is a unique arrow from λ_X to μ which belongs to X. If $X, Y \in E$ and $X \supset Y$, let $\lambda_{X,Y}$ be the unique arrow in X from λ_X to λ_Y. The assignments $X \mapsto \lambda_X$ and $(X \supset Y) \mapsto \lambda_{X,Y}$ form a functor from E to $\mathscr{L}'$.

68 LEMMA 8.1.8. *The category E is locally cofiltering over $\mathscr{S}$, and the functor $\lambda : E \to \mathscr{L}'$ is cofinal.*

We must verify the following axioms successively:

(i) *axiom* $(\mathscr{L}0)$ Let $f : V \to U$ and $X \in E_U$. There exist $\lambda \in \mathrm{Ob}\,\mathscr{L}'_V$ and $\phi \in \mathrm{Hom}_f(\lambda, \lambda_X)$. Moreover, λ may be chosen so that $\lambda \notin X^0$ (this is where $\mathscr{L}' = \mathscr{L} \times \mathbf{Z}$ is used).

Let $X' = X \cup \{1_\lambda\} \cup Y$, where Y is the set of composite arrows $\lambda \xrightarrow{\phi} \lambda_X \xrightarrow{\psi} \mu$ $(\psi \in X)$. Then $X' \in E$, $\lambda_{X'} = \lambda$, and $\mathrm{Hom}(X', X) \neq \emptyset$.

(ii) *axiom* ($\mathscr{L}1$) Let (X_i) be a finite family of objects of E_U. There exist a covering $f_j : U_j \to U$, objects $\lambda_j \in \mathrm{Ob}\,\mathscr{L}'_{U_j}$, and arrows $\phi_{ji} \in \mathrm{Hom}_{f_j}(\lambda_j, \lambda_{X_i})$. If $x \in X_a^0 \cap X_b^0$, then for each j there exist a covering $g_{jk} : U_{jk} \to U_j$ and arrows $\psi_{jk} : \lambda_{jk} \to \lambda_j$ such that $p(\psi_{jk}) = g_{jk}$ and which equalize the pair of arrows $\lambda_j \rightrightarrows x$, one belonging to X_a and the other to X_b. We may arrange that the λ_{jk} belong to none of the X_i^0. Replace the $(f_j, \lambda_j, \phi_{ji})$ by the $(f_j g_{jk}, \lambda_{jk}, \phi_{ji}\psi_{jk})$, and repeat this construction for every x in an intersection of two of the X_i. We obtain a new system $(f_j, \lambda_j, \phi_{ji})$; put

$$X_j = \bigcup_i X_i \cup \{1_{\lambda_j}\} \cup Y_j,$$

where Y_j is the set of composite arrows

$$\lambda_j \xrightarrow{\phi_{ji}} \lambda_{X_i} \xrightarrow{\psi} x \quad (\psi \in X_i).$$

Then $X_j \in E_{U_j}$, and these X_j satisfy ($\mathscr{L}1$).

(iii) *axiom* ($\mathscr{L}2$) Trivial, since there are no nontrivial parallel arrows!

(iv) *axiom* ($C\mathfrak{F}1$) Trivial, since the functor λ is surjective.

(v) *axiom* ($C\mathfrak{F}2$) Let $f : U \to V$ and let $(\phi_0, \phi_1) : \lambda_X \rightrightarrows \lambda$ be a pair of parallel arrows over f. There exist a covering $f_j : U_j \to U$ of U and arrows $\psi_j : \lambda_j \to \lambda_X$ such that $p(\psi_j) = f_j$, $\phi_0\psi_j = \phi_1\psi_j$, and $\lambda_j \notin X^0$. Let $X_j = X \cup \{1_{\lambda_j}\} \cup Y_j$, where Y_j is the set of composite arrows

$$\lambda_j \xrightarrow{\psi_j} \lambda_X \xrightarrow{\psi} \mu \quad (\psi \in X).$$

Then $X_j \in E_{U_j}$, and $F(\lambda_{X_j,X}) = \psi_j$ equalizes (ϕ_0, ϕ_1). 69

. Suppose that $\mathscr{S}$ is (the site of nonempty open subsets of) the one-point topological space. A locally cofiltering category over $\mathscr{S}$ is then simply a left filtering category (i.e. one for which $\mathscr{S}^0$ is filtering in the sense of I 2.7), and cofinal functors correspond to cofinal functors between left filtering categories (I 8). Proposition 8.1.7 may then be restated as follows.

Proposition 8.1.9. *For every cofiltering category $\mathscr{L}$, there exist a cofiltering ordered set E and a cofinal functor from E to $\mathscr{L}$.*

In this special case, the proof of 8.1.7 reduces essentially to the proof of 8.1.9 given in I 8.

8.1.10. Let $q : \mathscr{T} \to \mathscr{S}$ be a morphism of sites, and let $p : \mathscr{L} \to \mathscr{S}$ be a locally cofiltered category over $\mathscr{S}$. Define the category $\mathscr{L}^*$ as follows:

(i) An object of $\mathscr{L}^*$ is a quadruple (V, U, ϕ, λ) such that $V \in \mathrm{Ob}\,\mathscr{T}$, $U \in \mathrm{Ob}\,\mathscr{S}$, $\phi \in \mathrm{Hom}(V, q^*U)$, and $\lambda \in \mathrm{Ob}\,\mathscr{L}_U$.

(ii) An arrow $f : (V, U, \phi, \lambda) \to (V', U', \phi', \lambda')$ is a triple (f_1, f_2, f_3), where $f_1 \in \mathrm{Hom}(V, V')$, $f_2 \in \mathrm{Hom}(U, U')$, and $f_3 \in \mathrm{Hom}(\lambda, \lambda')$, such that $q^*f_2 \circ \phi = \phi' \circ f_1$ and $p(f_3) = f_2$.

The functor $f \mapsto f_1$ makes $\mathscr{L}^*$ a category over $\mathscr{T}$.

Lemma 8.1.11. *The category $\mathscr{L}^*$ is locally cofiltered over $\mathscr{T}$.*

Axiom ($\mathscr{L}0$) is trivial. Let us verify ($\mathscr{L}1$).

Let $L_i = (V, U_i, \phi_i, \lambda_i)$ be a finite family. The ϕ_i define a morphism from V to the sheaf $q^*a\prod U_i = a\prod q^*U_i$. Consequently, there are commutative diagrams

$$\begin{array}{ccc} V_j & \xrightarrow{f_j} & V \\ \downarrow{\scriptstyle \psi_j} & & \downarrow{\scriptstyle \phi_i} \\ q^*U_j & \xrightarrow[q^*(p_{ij})]{} & q^*U_i \end{array}$$

70 such that the f_j cover V.

Choose λ_{ji} such that $\mathrm{Hom}_{p_{ij}}(\lambda_{ji}, \lambda_i) \neq \emptyset$. By ($\mathscr{L}1$), after refining U_j (and V_j), we may take $\lambda_{ji} = \lambda_j$ independently of i, and the $L_j = (V_j, U_j, \psi_j, \lambda_j)$ satisfy ($\mathscr{L}1$).

Let us verify ($\mathscr{L}2$). Let $(f_1, f_2) : L^1 \rightrightarrows L^2$, with $L^i = (V^i, U^i, \phi^i, \lambda^i)$ and $f_i = (f, g_i, h_i)$. The ϕ^i define a morphism from V^1 to the sheaf $q^*a\,\mathrm{Ker}(U^1 \rightrightarrows U^2) = a\,\mathrm{Ker}(q^*U^1 \rightrightarrows q^*U^2)$, so there are commutative diagrams

$$\begin{array}{ccccc} V_j & \xrightarrow{f_j} & V^1 & \longrightarrow & V^2 \\ \downarrow{\scriptstyle \psi_j} & & \downarrow{\scriptstyle \phi^1} & & \downarrow{\scriptstyle \phi^2} \\ q^*U_j & \xrightarrow[q^*p_j]{} & q^*U^1 & \underset{q^*g_2}{\overset{q^*g_1}{\rightrightarrows}} & q^*U^2 \end{array} \qquad (g_1 p_j = g_2 p_j),$$

such that the f_j cover V^1. Let $\chi_j : \lambda_j \to \lambda^1$ satisfy $p(\chi_j) = p_j$. Applying ($\mathscr{L}2$) to $(h_1\chi_j, h_2\chi_j)$, and refining U_j (and V_j) if necessary, we may arrange that $h_1\chi_j = h_2\chi_j$; the $L_j = (V_j, U_j, \psi_j, \lambda_j)$ then satisfy ($\mathscr{L}2$).

Proposition 8.1.12. *Let $t^* : \mathcal{S}^* \to \mathcal{S}$ be a locally cofiltered category over $\mathcal{S}$, and endow $\mathcal{S}^*$ with the topology induced from that of $\mathcal{S}$ by t^*. Then t^* is an equivalence of sites $t : \mathcal{S} \to \mathcal{S}^*$.*

By construction, the functor $t^* : \mathscr{F} \mapsto \mathscr{F} \circ t^*$ carries sheaves on $\mathcal{S}$ to sheaves on $\mathcal{S}^*$. By ($\mathscr{L}0$) and ($\mathscr{L}1$) applied to the empty family, every "sufficiently small" object of $\mathcal{S}$ belongs to the image of t^* (that is, the image of t^* is a *covering sieve* in $\mathcal{S}$). Hence, by the "comparison lemma" (III 4), the inclusion into $\mathcal{S}$ of the full subcategory defined by $t^*(\mathrm{Ob}\,\mathcal{S}^*)$ is an equivalence of sites. We may therefore suppose that t^* is surjective on objects.

Let $\mathscr{F}$ be a sheaf on $\mathcal{S}^*$.

(i) Let $(f_j : U_j \to U)$ be a covering in $\mathcal{S}$, let $\lambda^1, \lambda^2 \in \mathrm{Ob}\,\mathcal{S}^*_U$, and let $f_j^i \in \mathrm{Hom}_{f_j}(\mu_j, \lambda^i)$ for $i = 1, 2$. Then $\mathscr{F}(\lambda^1)$ and $\mathscr{F}(\lambda^2)$ have the same image in $\prod_j \mathscr{F}(\mu_j)$.

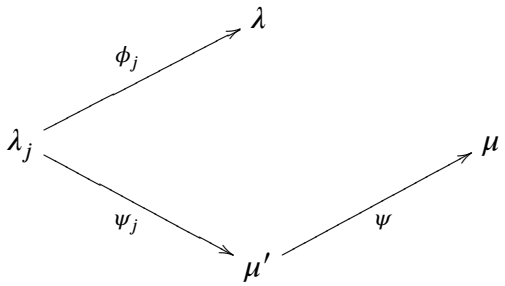
71 Indeed, let $x = (x_j)$ belong to the image of $\mathscr{F}(\lambda^1)$. For x to belong to the

image of $\mathscr{F}(\lambda^2)$, it suffices that, for every commutative diagram

$$\begin{array}{ccc} \mu & \longrightarrow & \mu_j \\ \downarrow & & \downarrow \\ \mu_k & \longrightarrow & \lambda^2, \end{array}$$

x_j and x_k have the same image in $\mathscr{F}(\mu)$. Since $\mathscr{F}$ is a sheaf, it suffices to verify this after replacing μ by the objects of one of its coverings. By ($\mathscr{L}2$), we may then suppose that the diagrams analogous to the preceding one, with λ^2 replaced by λ^1, also commute, in which case the assertion is clear.

(ii) Let $f : U \to V$ be an arrow of $\mathscr{S}$, let $\lambda \in \mathrm{Ob}\, \mathscr{S}_U^*$, and let $\mu \in \mathrm{Ob}\, \mathscr{S}_V^*$. There then exist a covering $f_j : U_j \to U$, an object $\mu' \in \mathrm{Ob}\, \mathscr{S}_U^*$, and arrows in the diagram



lying over

$$U_j \xrightarrow{f_j} U \xrightarrow{f} V.$$

By (i), there is a unique arrow $\overline{f}$ making the following diagram commute:

$$\begin{array}{ccc} \mathscr{F}(\mu) & \xrightarrow{\overline{f}} & \mathscr{F}(\lambda) \\ \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \phi_j} \\ \mathscr{F}(\mu') & \xrightarrow[\psi_j]{} & \mathscr{F}(\lambda_j). \end{array}$$

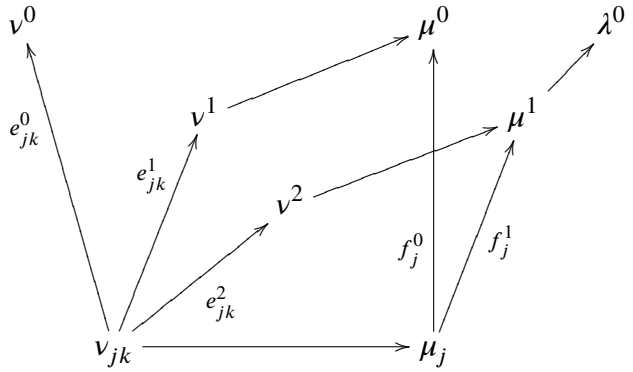
The arrow $\overline{f}$ does not change if (f_j) is replaced by a finer covering $f_k : U_k \to U_{j(k)}$, if λ_j is replaced by λ_k equipped with $\phi_k \in \mathrm{Hom}_{f_k}(\lambda_k, \lambda_{j(k)})$, and if ϕ_j (respectively ψ_j) is replaced by $\phi_{j(k)} \circ \phi_k$ (respectively $\psi_{j(k)} \circ \phi_k$). 72

The reader may deduce that $\overline{f}$ depends only on f.

(iii) Let us prove that $\overline{f}$ is functorial in f. Thus let

$$W \xrightarrow{h} V \xrightarrow{g} U$$

be arrows in $\mathcal{S}$, and let λ^0, μ^0, and ν^0 belong respectively to Ob $\mathcal{S}^*_U$, Ob $\mathcal{S}^*_V$, and Ob $\mathcal{S}^*_W$. There exists a commutative diagram



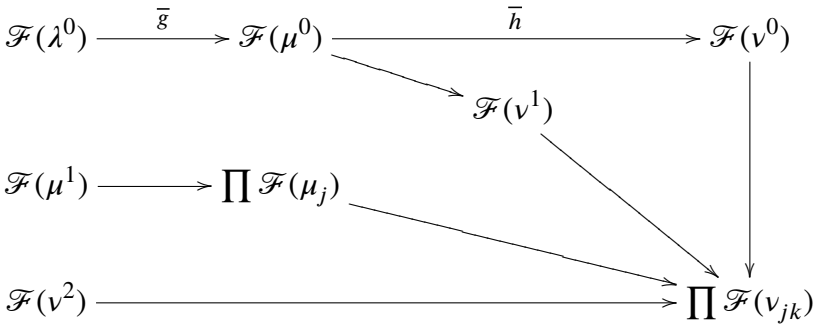
such that

a) $t^*(\nu^i)$, $t^*(\mu^i)$, $t^*(e^i_{jk})$, and $t^*(f^i_j)$ do not depend on i;

b) for fixed i, the e^i_{jk} cover W, and the f^i_j cover V.

First construct μ^1, ν^1, and ν^2 by ($\mathscr{L}0$), and the μ_j as in (ii). Let (W'_{jk}) be a covering of W mapping to the covering $V_j = t^*(\mu_j)$ as required by the diagram. After refining (W'_{jk}) and applying ($\mathscr{L}0$) and ($\mathscr{L}1$), one obtains a diagram of the required type, satisfying a) and b), but not necessarily commutative. Refining once more and applying ($\mathscr{L}2$) makes it commutative.

It remains only to contemplate the following commutative diagram:



73 (iv) It is clear that $\overline{f}$ is functorial in $\mathscr{F}$. Thus every sheaf $\mathscr{F}$ on $\mathcal{S}^*$ is the direct image under t of a uniquely determined presheaf $\mathscr{F}_0$ on $\mathcal{S}$, and every morphism of sheaves $f : \mathscr{F} \to \mathfrak{g}$ is the image of a unique morphism of presheaves $f_0 : \mathscr{F}_0 \to \mathfrak{g}_0$.

It already follows that t_* is fully faithful; it remains to prove that $\mathscr{F}_0$ is a sheaf. If $(f_j : U_j \to U)$ is a covering, then this covering is the image of a covering in $\mathcal{S}^*$, so $\mathscr{F}_0(U)$ injects into $\prod_j \mathscr{F}_0(U_j)$. If elements $x_j \in \mathscr{F}_0(U_j)$ glue, then a fortiori they glue in $\mathcal{S}^*$, and hence come from an element of $\mathscr{F}_0(U)$. This completes the proof of 8.1.12.

. Let $q : \mathcal{T} \to \mathcal{S}$ and $p : \mathscr{L} \to \mathcal{S}$ be as in (8.1.10), let $\mathscr{L}^*$ be the category defined there, and let e be a stack on $\mathcal{T}$. The stack $q_*\mathsf{e}$ on $\mathcal{S}$ is the stack defined by $(q_*\mathsf{e})_U = \mathsf{e}_{q^*U}$.

If $(X_\lambda)_{\lambda\in\mathscr{L}}$ is a local projective system indexed by $\mathscr{L}$ and taking values in $q_*\mathsf{e}$, it defines a local projective system $(X_L)_{L\in\mathscr{L}^*}$ taking values in e by the formula

$$X_L = \phi^* X_\lambda quad \text{for } L = (V, U, \phi, \lambda).$$

The reader may verify the following:

PROPOSITION 8.1.13.1. *With the preceding notation, one has*

$$\varprojlim_{\lambda\in\mathscr{L}} X_\lambda = \varprojlim_{L\in\mathscr{L}^*} X_L,$$

the two sides being simultaneously defined or undefined.

Let $\mathscr{S}$ be a site, and let $p^* : \mathscr{S}^* \to \mathscr{S}$ be a locally filtered category over $\mathscr{S}$. If e is a stack on $\mathscr{S}$, then $p_*\mathsf{e}$ is a stack on $\mathscr{S}^*$ (8.1.13.0). Moreover, the identity functor $\mathscr{S}^* \to \mathscr{S}^*$ is a locally filtered category over $\mathscr{S}^*$, endowed with the induced topology, and every local inductive system (X_λ) indexed by $\mathscr{S}^*$ defines a local inductive system over $\mathscr{S}^*$ indexed by $\mathscr{S}^*$.

One verifies easily:

PROPOSITION 8.1.14. *With the preceding notation, one has* 74

$$\varprojlim_{\lambda\in\mathscr{S}^*/\mathscr{S}} X_\lambda = \varprojlim_{\lambda\in\mathscr{S}^*/\mathscr{S}^*} X_\lambda,$$

the two sides being simultaneously defined or undefined.

8.2. Local inductive limits in categories of sheaves.

8.2.0. Let $p : \mathscr{T} \to \mathscr{S}$ be a morphism of sites. Let e be the following stack on $\mathscr{S}$:

(i) An object of e is a pair $(U, \mathscr{F})$ consisting of an object U of $\mathscr{S}$ and a sheaf $\mathscr{F}$ on p^*U.

(ii) An arrow $f : (U, \mathscr{F}) \to (V, \mathscr{G})$ is a pair consisting of an arrow $f_0 : U \to V$ and a morphism of sheaves on p^*U:

$$f_1 : p^*(f_0)^*\mathscr{G} \longrightarrow \mathscr{F}.$$

(iii) The functor from e to $\mathscr{S}$ is given by $f \longmapsto f_0$.

Care must be taken that e_U is the *opposite category* of the category of sheaves on p^*U.

Let $\mathscr{L}$ be a locally cofiltered category (on the left) over $\mathscr{S}$.

DEFINITION 8.2.1. *With the preceding notation, the category of local inductive systems, indexed by $\mathscr{L}$, of sheaves on $\mathscr{T}$ is the category* $\mathrm{Hom}_{\mathscr{S}}(\mathscr{L}, \mathsf{e})^\circ$.

The "projective limit" functor of 8.1.3 is called here the *"local inductive limit" functor.* It is a functor from $\mathrm{Hom}_{\mathscr{S}}(\mathscr{L}, \mathsf{e})^\circ$ to $(\Gamma\mathsf{e})^\circ = (\mathscr{T})^\sim$.

THEOREM 8.2.2. *The "local inductive limit" functor from the category of inductive systems of sheaves on $\mathscr{T}$, indexed by $\mathscr{L}$, to the category of sheaves on $\mathscr{T}$ is defined everywhere, commutes with finite projective limits, and commutes with arbitrary inductive limits.*

The propositions 8.1.13, 8.1.14, and 8.1.12 reduce the assertion to the case $\mathscr{S} = \mathscr{T} = \mathscr{L}$. A local inductive system then consists of a sheaf $\mathscr{F}_U$ on $\mathscr{S}/U$ for every $U \in \mathrm{Ob}\,\mathscr{S}$, together, for every arrow $f : V \to U$ in $\mathscr{S}$, with an arrow $f^*\mathscr{F}_U \to \mathscr{F}_V$, functorial in f. 75

For such inductive systems, the $\varinjlim$ are computed as follows:

LEMMA 8.2.3. *Let $U \longmapsto \mathscr{F}_U$ be a section of the category of presheaves on the objects of $\mathscr{S}$. Then*

$$a(U \longmapsto \Gamma(U, \mathscr{F}_U)) \simeq \varinjlim_U a\mathscr{F}_U.$$

By definition, the right-hand side represents the functor which assigns to every sheaf $\mathscr{F}$ on $\mathcal{S}$ the set of coherent systems of arrows $\phi_U : \mathscr{F}_U \to \mathscr{F}|U$. In turn, an arrow ϕ_U is a coherent system of arrows ϕ_g, one for every $g : V \to U$, where $\phi_g : \mathscr{F}_U(V) \to \mathscr{F}(V)$. Apply 8.1.11 to the identity functor of $\mathcal{S}$, thereby obtaining $\mathcal{S}^*$ locally filtered over $\mathcal{S}$. We have seen that

$$\varinjlim_{g \in \mathcal{S}^*} \mathscr{F}_U(V_g)^\sim \simeq \varinjlim_U a\mathscr{F}_U,$$

where V_g is the source of g and $^\sim$ denotes the "generated constant sheaf" functor. The functor $U \longmapsto 1_U$ from $\mathcal{S}$ to $\mathcal{S}^*$ is cofinal, and hence, again by 8.1.5,

$$\varinjlim_U \mathscr{F}_U(U)^\sim \simeq \varinjlim_U a\mathscr{F}_U.$$

Returning to the definitions, one finally finds

$$\varinjlim_U \mathscr{F}_U(U)^\sim = a(U \longmapsto \Gamma(U, \mathscr{F}_U)).$$

This shows that, in the case to which we have reduced, the functor $\varinjlim$ is defined everywhere and is computed as the composite of the "sheaf generated by a presheaf" functor and the functor $(\mathscr{F}_U) \longmapsto (U \longmapsto \Gamma(U, \mathscr{F}_U))$. Both of these functors commute with finite projective limits, and the functor $\varinjlim$ also commutes with arbitrary inductive limits by its definition as an adjoint functor.

76 Lemma 8.2.3 provides a systematic method for proving properties of the local-inductive-limit functor from the analogous properties of the "sheaf associated with a presheaf" functor. One obtains in this way:

PROPOSITION 8.2.4. *Let $\mathcal{T}$, $\mathcal{S}$, and $\mathscr{L}$ be as above, let $\mathscr{A}_\lambda$ be a local inductive system of sheaves of rings on $\mathcal{T}$, and let $\mathscr{L}_\lambda$ (resp. $\mathscr{M}_\lambda$) be a local inductive system of sheaves of right (resp. left) modules over $\mathscr{A}_\lambda$, all three indexed by $\mathscr{L}$.*

Then

$$\varinjlim(\mathscr{L}_\lambda \otimes_{\mathscr{A}_\lambda} \mathscr{M}_\lambda) \simeq \varinjlim \mathscr{L}_\lambda \underset{\varinjlim \mathscr{A}_\lambda}{\otimes} \varinjlim \mathscr{M}_\lambda.$$

(Reduce to the case $\mathcal{T} = \mathcal{S} = \mathscr{L}$, and note that both sides coincide with the sheaf generated by the presheaf

$$\mathscr{L}_U(U) \underset{\mathscr{A}_U(U)}{\otimes} \mathscr{M}_U(U).$$

)

PROPOSITION 8.2.5. *Let $\mathcal{T}_1 \overset{q}{\to} \mathcal{T}_2 \to \mathcal{S}$ be two morphisms of sites, and let (F_λ) be a local inductive system of sheaves on $\mathcal{T}_2$ indexed by a locally cofiltered category $\mathscr{L}$ over $\mathcal{S}$. Then*

$$q^* \varinjlim_{\lambda \in \mathscr{L}} F_\lambda \simeq \varinjlim_{\lambda \in \mathscr{L}} q^* F_\lambda.$$

This follows immediately from the fact that q^* has a right adjoint q_*.

Let $\mathcal{S}$ and $\mathcal{T}$ be two sites in which finite $\varprojlim$ are representable, let $p : \mathcal{T} \to \mathcal{S}$ be a morphism of sites such that $p^* : \mathcal{S} \to \mathcal{T}$ commutes with finite $\varprojlim$, and let $\mathscr{L}$ be the locally cofiltered category over $\mathcal{T}$ whose objects are triples (V, U, ϕ), where $V \in \mathrm{Ob}\,\mathcal{T}$, $U \in \mathrm{Ob}\,\mathcal{S}$, and $\phi \in \mathrm{Hom}(V, p^*U)$. It is obtained by applying 8.1.11 to $p : \mathcal{T} \to \mathcal{S}$.

PROPOSITION 8.2.6. *For every sheaf $\mathscr{F}$ on $\mathscr{T}$, one has*

$$\mathscr{F} = \varinjlim_{\lambda \in \mathscr{L}} \phi^* \phi_* \mathscr{F},$$

where, by abuse of notation, ϕ denotes the morphism induced by p from $\mathscr{T}/V$ to $\mathscr{S}/U$.

Let ϕ' be the inverse-image functor in the sense of presheaves. From 8.2.3 one 77
obtains

$$\varinjlim_{\lambda \in \mathscr{L}} \phi'^* \phi_* \mathscr{F} = \varinjlim_{\lambda \in \mathscr{L}} a\phi'^* \phi_* \mathscr{F} = \varinjlim_{\lambda \in \mathscr{L}} \mathscr{F}(V)^\sim = \varinjlim_{V} \mathscr{F}(V)^\sim = \mathscr{F}.$$

PROPOSITION 8.2.7. *Let $p : T \to S$ be a morphism of topoi, let $\mathscr{O}_S$ be a sheaf of rings on S, and let $\mathscr{O}_T$ be its inverse image on T. A right $\mathscr{O}_T$-module $\mathscr{M}$ is flat over $\mathscr{O}_T$ if and only if the functor*

$$\mathscr{N} \longmapsto \mathscr{M} \underset{\mathscr{O}_T}{\otimes} p^* \mathscr{N} \tag{8.2.7.1}$$

from the category of left $\mathscr{O}_S$-modules to the category of abelian sheaves on T is exact.

The "only if" assertion is trivial; suppose, therefore, that the functor (8.2.7.1) is exact, and let us prove that $\mathscr{M}$ is flat.

We may suppose that p is defined by a morphism of sites $p : \mathscr{T} \to \mathscr{S}$ satisfying the hypotheses made in 8.2.6. Let $V \in \operatorname{Ob} \mathscr{T}$, $U \in \operatorname{Ob} \mathscr{S}$, and $\phi : V \to p^*U$. We again denote by ϕ the induced morphism of sites $\phi : \mathscr{T}/V \to \mathscr{S}/U$.

LEMMA 8.2.8. *The functor $\mathscr{N} \longmapsto \mathscr{M}|V \otimes_{\mathscr{O}_V} \phi^* \mathscr{N}$, from the category of $\mathscr{O}_S|U$-modules on $\mathscr{S}/U$ to that of $\mathscr{O}_T|V$-modules on $\mathscr{T}/V$, is exact.*

We reduce to taking $V = p^*U$. Let j and j' be the "inclusion morphisms" $j : \mathscr{S}/U \to \mathscr{S}$ and $j' : \mathscr{T}/p^*U \to \mathscr{T}$. One has

$$j'_!(\mathscr{M}|V \underset{\mathscr{O}_V}{\otimes} \phi^* \mathscr{N}) = \mathscr{M} \underset{\mathscr{O}_T}{\otimes} p^* j_! \mathscr{N},$$

which gives 8.2.8, since $j_!$ and $j'_!$ are exact and faithful.

Let now Q be a left $\mathscr{O}_T$-module. By (8.2.6), one has

$$Q = \varinjlim_{\phi : V \to p^*U} \phi^* \phi_* (Q|V),$$

so that (8.2.4)

$$\mathscr{M} \underset{\mathscr{O}_T}{\otimes} Q = \varinjlim_{\phi : V \to p^*U} \mathscr{M}|V \underset{\mathscr{O}_V}{\otimes} \phi^* \phi_* (Q|V).$$

By 8.2.8, the functors $Q \longmapsto \mathscr{M}|V \otimes_{\mathscr{O}_V} \phi^* \phi_* (Q|V)$ are left exact; therefore $\mathscr{M} \otimes_{\mathscr{O}_T} Q$ 78
is left exact in Q, and hence exact.

COROLLARY 8.2.9. *Let $f : (T, \mathscr{O}_T) \to (S, \mathscr{O}_S)$ be a morphism of ringed topoi. Then the inverse image by f of a sheaf of flat right modules is a sheaf of flat modules.*

Factoring f through $(T, f^* \mathscr{O}_S)$ reduces us to the case $\mathscr{O}_T = f^* \mathscr{O}_S$. Let $\mathscr{M}$ then be a sheaf of flat modules on S. By 8.2.7, to verify that $f^* \mathscr{M}$ is flat, it is enough to verify the exactness of the functor

$$\mathscr{N} \longmapsto f^* \mathscr{M} \underset{\mathscr{O}_T}{\otimes} f^* \mathscr{N} = f^* (\mathscr{M} \underset{\mathscr{O}_S}{\otimes} \mathscr{N}),$$

which proves the assertion, since f^* is exact.

LEMMA 8.2.10. *Let $(S, \mathcal{O}_S)$ be a ringed topos, let $\mathcal{F}$ be a sheaf of left modules locally of finite presentation, let $\mathcal{G}$ be a sheaf of bimodules, and let $\mathcal{H}$ be a flat sheaf of left modules. Then the canonical arrow*

$$\mathbf{Hom}(\mathcal{F}, \mathcal{G}) \otimes \mathcal{H} \longrightarrow \mathbf{Hom}(\mathcal{F}, \mathcal{G} \otimes \mathcal{H})$$

is an isomorphism.

The question is local on S, which allows us to suppose that $\mathcal{F}$ has a finite presentation

$$\mathcal{F}_1 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F} \longrightarrow 0.$$

The first row of the following diagram is exact, since $\mathcal{H}$ is flat:

$$\begin{array}{ccccc} 0 \to \mathbf{Hom}(\mathcal{F}, \mathcal{G}) \otimes \mathcal{H} & \longrightarrow & \mathbf{Hom}(\mathcal{F}_0, \mathcal{G}) \otimes \mathcal{H} & \longrightarrow & \mathbf{Hom}(\mathcal{F}_1, \mathcal{G}) \otimes \mathcal{H} \\ \downarrow & & \downarrow s & & \downarrow s \\ 0 \to \mathbf{Hom}(\mathcal{F}, \mathcal{G} \otimes \mathcal{H}) & \longrightarrow & \mathbf{Hom}(\mathcal{F}_0, \mathcal{G} \otimes \mathcal{H}) & \longrightarrow & \mathbf{Hom}(\mathcal{F}_1, \mathcal{G} \otimes \mathcal{H}) \end{array}$$

which proves the assertion. Taking $\mathcal{G} = \mathcal{O}_S$, one deduces from 8.2.10:

79 LEMMA 8.2.11. *Every morphism from a sheaf of modules locally of finite presentation to a flat sheaf of modules factors locally on S through a finite free sheaf $\mathcal{O}_S^n$ ($n \in \mathbf{N}$).*

THEOREM 8.2.12. (D. Lazard). *A sheaf of modules $\mathcal{M}$ on a ringed site $(S, \mathcal{O}_S)$ is flat if and only if it is a local inductive limit of finite free modules.*

The "if" assertion follows from 8.2.2 and 8.2.4.

For every $U \in \mathrm{Ob}\, \mathcal{S}$, let $\mathcal{L}_0(U)$ (resp. $\mathcal{L}_1(U)$) be the category of finite free sheaves of modules (resp. sheaves of modules of finite presentation) on U, equipped with a linear map to $\mathcal{M}|U$. For every arrow $f : V \to U$ in $\mathcal{S}$, let f^* be the restriction functor from $\mathcal{L}_i(U)$ to $\mathcal{L}_i(V)$. These functors define a category $\mathcal{L}_i$, fibered over $\mathcal{S}$, whose fibers are the opposite categories of the categories $\mathcal{L}_i(U)$.

The reader will verify that the category $\mathcal{L}_1$ is locally filtered over $\mathcal{S}$ and that

$$\mathcal{M} = \varinjlim_{(M,f) \in \mathcal{L}_1} M.$$

If $\mathcal{M}$ is flat, the inclusion of $\mathcal{L}_0$ in $\mathcal{L}_1$ is fully faithful and, by 8.2.11, satisfies ($C\mathfrak{F}1$) of 8.1.4. It then automatically satisfies ($C\mathfrak{F}2$), and $\mathcal{L}_0$ is locally filtered. By 8.1.5, one has

$$\mathcal{M} = \varinjlim_{(M,f) \in \mathcal{L}_0} M,$$

which proves 8.2.12.

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EXPOSÉ Vbis

Cohomological Descent Techniques

B. Saint-Donat

Introduction

1. Let X be a topological space and let $\mathcal{U} = (U_i)_{i\in I}$ be a covering of X, assumed either open or closed and locally finite. If $\mathfrak{F}$ is an abelian sheaf on X, the Leray spectral sequence 84

$$\overset{\vee}{H}{}^p(\mathcal{U}, \mathscr{H}^q(\mathfrak{F})) \Rightarrow H^*(X, \mathfrak{F}) \tag{1.1}$$

defined by $\mathcal{U}$ [[5] II (5.2.4) and (5.4.1)] may be described as follows.

The covering $\mathcal{U}$ defines a "Čech" resolution $\mathscr{C}^*(\mathcal{U}, \mathfrak{F})$, functorial in $\mathfrak{F}$ (ibid. (5.2.1)). On the other hand, for every $\mathfrak{F}$ there is a "canonical flasque" resolution $C^*(\mathfrak{F})$, functorial in $\mathfrak{F}$ (ibid. (4.3)). With this notation, when $\mathcal{U}$ is open the spectral sequence (1.1) is obtained from the double complex

$$\Gamma(X, \mathscr{C}^*(\mathcal{U}, C^*(\mathfrak{F}))). \tag{1.2}$$

When $\mathcal{U}$ is closed and locally finite, one considers the double complex

$$\Gamma(X, C^*(\mathscr{C}^*(\mathcal{U}, \mathfrak{F}))). \tag{1.3}$$

2. Let us seek a unified description of these double complexes. Let X_0 be the topological disjoint union of the U_i, and for $n \geq 0$ let X_n be the $(n+1)$-fold fiber product of X_0 with itself over X:

$$X_n = \coprod_{i_0,\dots,i_n\in I} U_{i_0} \cap \cdots \cap U_{i_n} = \coprod_{\sigma\in \mathrm{Hom}([n],I)} \bigcap_{i=0}^{n} U_{\sigma(i)}. \tag{2.1}$$

The X_n form a simplicial system of topological spaces, and if j_n denotes the projection from X_n to X, then

$$\mathscr{C}^n(\mathcal{U}, \mathfrak{F}) = j_{n*}j_n^*(\mathfrak{F}). \tag{2.2}$$

Note that formation of the canonical flasque resolutions commutes with restriction to an open subset and with direct image by a closed immersion. Consequently: 85

(a) if $\mathcal{U}$ is open,

$$\mathscr{C}^q(\mathcal{U}, C^p(\mathfrak{F})) = j_{q*}j_q^*C^p(\mathfrak{F}) = j_{q*}C^p(j_q^*\mathfrak{F}); \tag{2.3}$$

(b) if $\mathcal{U}$ is closed and locally finite,

$$C^p(\mathscr{C}^q(\mathcal{U}, \mathfrak{F})) = C^p(j_{q*}j_q^*\mathfrak{F}) = j_{q*}C^p(j_q^*\mathfrak{F}). \tag{2.4}$$

Thus, to obtain a unified description of (1.2) and (1.3), it is enough to take the canonical flasque resolution of $j_q^*(\mathfrak{F})$ on X_q for every q, and then apply the functor j_{q*} to this resolution.

3. The preceding description remains meaningful for any simplicial system of topological spaces over X:

$$\begin{aligned} \Delta^0 &\longrightarrow \mathrm{Top}/X \\ [n] &\longmapsto X_n. \end{aligned} \tag{3.1}$$

Such a system need not be of the form (2.1). The double complex, coaugmented by $\mathfrak{F}$,

$$\mathfrak{F} \longrightarrow (j_{q*}C^p(j_q^*(\mathfrak{F})))_{p,q} \tag{3.2}$$

will not, however, in general define a resolution of $\mathfrak{F}$.

This work is devoted to finding sufficient conditions for (3.2) to define a resolution of $\mathfrak{F}$. In that case, the spectral sequence (1.1) generalizes to a spectral sequence

$$\overset{\vee}{H}{}^p\ (H^q(X_p, j_p^*(\mathfrak{F}))) \Rightarrow H^{p+q}(X, \mathfrak{F}), \tag{3.3}$$

called the "descent spectral sequence".

86 In the case of "constant coefficients", analogous spectral sequences were obtained by Segal (cf. [8]), by other methods and for other "cohomology theories", such as K-theory. Segal works in the category of CW-complexes: he uses a "geometric realization" functor which associates a new topological space with a semi-simplicial complex of topological spaces; this new space is to be compared with the topos associated with a simplicial topos [cf. (1.2.12)].

4. In Section 5 we illustrate the criteria obtained by using resolution of singularities to construct, for every analytic space X over $\mathbf{C}$, a simplicial system of nonsingular analytic spaces over X,

$$[n] \longmapsto X_n,$$

such that (3.2) defines a resolution of $\mathfrak{F}$. Taking the constant sheaf $\mathbf{C}$ for $\mathfrak{F}$ gives in particular a spectral sequence

$$H^q(X_p, \mathbf{C}) \Longrightarrow H^{p+q}(X, \mathbf{C}) \tag{4.1}$$

which expresses the complex cohomology of X in terms of the complex cohomology of nonsingular analytic spaces. Moreover, if X is projective, all the X_p may be assumed projective; this is the essential ingredient which yields a kind of "Hodge theory" for X (cf. [2]).

5. The preceding constructions extend verbatim when the sheaf $\mathfrak{F}$ is replaced by a bounded-below complex of sheaves. They lead to "localization" techniques in derived categories.

It is known that, for X_0 given by (2.1), the arrow

$$j_0^* : D^b(X) \longrightarrow D^b(X_0)$$

87 is not faithful in general. We shall show that data more precise than $j_0^*(K^\cdot)$ (for $K^\cdot \in D^+(X)$), involving the X_n, sometimes make it possible to reconstruct the complex $K^\cdot$.

The resulting statements will be used in the appendix to Exposé XVII to extend the definition of the functor $Rf_!$ (where f is a separated morphism of finite type between schemes) to the case in which f is not assumed compactifiable.

For this application it is not possible to consider only topological spaces, which are replaced here by the étale sites of schemes. Moreover, to carry out the proofs it will be necessary to consider both simplicial systems of spaces and multisimplicial systems. This explains, justifies, or excuses the degree of hypergenerality from which we shall begin.

1. Preliminaries

1.1. Notation.

1.1.1. Throughout what follows, $\mathcal{U}$ is a universe such that $\mathbf{Z} \in \mathcal{U}$: *all the topoi considered will be $\mathcal{U}$-topoi*.

Let T and T' be two topoi: a morphism $\varphi : T \to T'$ consists of a pair of functors $\varphi_* : T \to T'$ and $\varphi^* : T' \to T$, endowed with an adjunction

$$\mathrm{Hom}_{T'}(., \varphi_*.) \xrightarrow{\sim} \mathrm{Hom}_T(\varphi^*., .),$$

such that φ^* is left exact (i.e. preserves finite projective limits).

Let $(T, \mathcal{O}_T)$ and $(T', \mathcal{O}_{T'})$ be two ringed topoi: a morphism from $(T, \mathcal{O}_T)$ to $(T', \mathcal{O}_{T'})$ is a pair (φ, θ), where $\varphi : T \to T'$ is a morphism of topoi and $\theta : \mathcal{O}_{T'} \to \varphi_*(\mathcal{O}_T)$ is a morphism of rings.

1.1.2. We will make constant use of the language of fibered categories as presented in [SGA 1 VI]; the reader may also consult [4]. Let us merely fix some notation: if $E \to B$ is a fibered (resp. cofibered) functor, then, for a morphism $m : i \to j$ in B, we denote by $m^* : E_j \to E_i$ (resp. $m_* : E_i \to E_j$) the associated "inverse image" (resp. "direct image") functor; each of these functors is defined up to a unique functorial isomorphism. If $\varphi : E \to E'$ is a B-functor, then, for every object i of B, we denote by $\varphi_i : E_i \to E'_i$ the restriction of φ to E_i. 88

1.1.3. Finally, we denote by Δ the following category: the objects of Δ are the ordered sets $[n] = \{0, 1, \dots, n\}$, and the morphisms of Δ are all weakly increasing maps. Δ^+ (resp. Δ^-) denotes the category whose objects are those of Δ and whose arrows are the monomorphisms (resp. the epimorphisms) of Δ. We will freely use, as needed, the classical notation introduced in connection with Δ [cf. [3] II 2].

1.2. D-topoi.

Definition 1.2.1. *Let D be a category, and let $E \to D$ be a fibered and cofibered functor. We say that E is bifibered in topoi over D, or that E is a D-topos, if the following conditions hold:*

(a) *For every object i of D, the fiber category E_i is a topos.*
(b) *For every morphism $m : i \to j$ in D, there exists a morphism of topoi $f : E_j \to E_i$ such that $f_* = m^*$ and $f^* = m_*$.*

Remark. In view of (a), condition (b) can equivalently be expressed by saying that the functor m_* is left exact[15].

When $D = \Delta$ (resp. $\Delta \times \Delta$), we speak of a *simplicial topos* (resp. a *double simplicial topos*) to mean a Δ-topos (resp. a $\Delta \times \Delta$-topos).

[15] N.D.E.: Care must be taken here: in a fibered category, $m_\star$ is left adjoint to $m^\star$ and is therefore tautologically right exact; cf. [SGA 1 VI.10]. This explains the formulas $f_* = m^*$ and $f^* = m_*$.

In practice, we will encounter D-topoi by means of the following considerations.

89 DEFINITION 1.2.2. *Let $\mathscr{E}$ be a fibered and cofibered category over D. We say that $\mathscr{E}$ is* bifibered in duals of topoi over D *if $\mathscr{E}^\circ$ is a D°-topos.*

REMARK 1.2.3. Explicitly, $\mathscr{E}$ is bifibered in duals of topoi over D if and only if the following conditions hold:

(a) For every object i of D, the opposite category $\mathscr{E}_i{}^\circ$ of the fiber category $\mathscr{E}_i$ is a topos.
(b) For every morphism $m : i \to j$ in D, the functor m^* is left exact[16].

1.2.4. The reader will find examples of categories bifibered in duals of topoi in paragraph 4.

1.2.5. Let $\mathscr{E} \to B$ be a category bifibered in duals of topoi over B, and let $X : D^\circ \to B$ be a functor. We leave it to the reader to verify that $(D^\circ \times_B \mathscr{E})^\circ$ is a D-topos, which we denote by $\overline{X}$.

1.2.6. We will often say that a functor $X : D^\circ \to B$ is a *D-object of B*, and we denote by X_i the image under this functor of an object i of D; the D-objects of B form a category denoted by $D^\circ B$. If S is an object of B, a D-object of B/S will be called a *D-object of B augmented toward S*. The datum of a D-object of B augmented toward S is trivially equivalent to that of a functorial morphism $X \to C_S^D$, where X is a D-object of B and C_S^D is the constant D-object of B with value S.

When $D = \Delta$, we speak of a *simplicial object* (resp. an *augmented simplicial object*). We will also use *double simplicial objects* (taking $D = \Delta \times \Delta$).

90 1.2.7. Now suppose that the category B has finite fiber products.

Let $f : R \to S$ be an arrow in B: the bifunctor

$$([n], X) \rightsquigarrow \mathrm{Hom}_{\mathrm{Ens}}([n], \mathrm{Hom}_S(X, R))$$

from $\Delta^\circ \times (B/S)^\circ$ to Ens defines a functor

$$[n] \rightsquigarrow X_{[n]} = X_n$$

by taking as X_n a representative of the functor

$$Z \rightsquigarrow \mathrm{Hom}_{\mathrm{Ens}}([n], \mathrm{Hom}_S(Z, R))$$

$$(X_n \simeq \overbrace{R \times_S R \times \cdots \times_S R}^{n+1 \text{ times}}).$$

We denote by $[R|_f S]$ or $[R|S]$ the semi-simplicial object augmented toward S thus constructed.

Finally, if X and X' are two semi-simplicial objects of B (or of B/S), and if

$$u : X \longrightarrow X'$$

is a functorial morphism, then, for technical reasons, we introduce the object

$$[X|_u X']$$

computed in the category $\Delta^\circ B$. This object can be interpreted as a double simplicial object[17] of B, which we then denote by $[[X|_u X']]$: indeed, there is a canonical isomorphism

[16] N.D.E.: Cf. the preceding note. In other words, the family of functors $((m_*)^\circ, (m^*)^\circ)$ defines a compatible system of morphisms of topoi $\mathscr{E}_i^\circ \to \mathscr{E}_j^\circ$.

[17] N.D.E.: Explicitly, with the usual ordering of the indices, one has

$$[[X|_u X']]_{(n,m)} = \overbrace{X_m \times_{X'_m} X_m \times \cdots \times_{X'_m} X_m}^{n+1 \text{ times}}$$

of categories

$$\Delta^\circ(\Delta^\circ B) \xrightarrow{\sim} (\Delta \times \Delta)^\circ B.$$

We now return to the general notion of a D-topos.

1.2.8. Let E be a D-topos: we denote by $\Gamma(E)$ the category $\mathrm{Hom}_D(D, E)$. Let $f : D' \to D$ be a functor: the category $D' \times_D E$ is a D'-topos and, by composition with f, we obtain a functor

$$f^* : \Gamma(E) \longrightarrow \Gamma(D' \times_D E).$$

In the case where D' is reduced to an object i of D (with the identity of i as its only morphism), and where f is the canonical inclusion denoted by e_i, we may take $D' \times_D E$ to be the fiber category E_i, and e_i^* is then identified with the "*evaluation at i*" functor. 91

Proposition 1.2.9. *If D' is a $\mathcal{U}$-small category, the functor f^* has a right adjoint and a left adjoint (denoted by f_* and $f_!$, respectively).*

This follows from a slight generalization of Kan's lemma [III 1.1], which we will use repeatedly.

Lemma 1.2.10. *Let I, J, and A be three categories over the same category B. Suppose that I is $\mathcal{U}$-small and that A is fibered and cofibered over B. Let a B-functor $f : I \to J$ be given, and denote by f^* the functor*

$$\mathrm{Hom}_B(J, A) \longrightarrow \mathrm{Hom}_B(I, A)$$

defined by composition with f. If, in every fiber of A over B, $\mathcal{U}$-inductive limits (resp. projective limits) exist, then f^* has a left adjoint (resp. a right adjoint).

Proof. We give only the proof of the existence of the left adjoint, the resp. part of the lemma following by duality. We use the following fact, whose verification is left to the reader.

Lemma 1.2.10.1. *Let A be a bifibered category over a category B, let b be an object of B, and let $(F_\lambda)_{\lambda \in \Lambda}$ be a functor from a category Λ to the fiber A_b. For every G in $A_{b'}$, and every $u : b' \to b$ (resp. $u : b \to b'$) in B, there is a bijection*

$$\mathrm{Hom}_u(G, \varprojlim F_\lambda) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_u(G, F_\lambda)$$

(resp. $\mathrm{Hom}_u(\varinjlim F_\lambda, G) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_u(F_\lambda, G)$*) whenever the first member is defined.*

With this understood , let $I \amalg_f J$ be the category over B defined by 92

$$\mathrm{ob}(I \amalg_f J) = \mathrm{ob}(I) \amalg \mathrm{ob}(J)$$

$$\mathrm{Hom}(x, y) = \begin{cases} \mathrm{Hom}_I(x, y) & \text{if } x, y \in \mathrm{ob}(I) \\ \mathrm{Hom}_J(x, y) & \text{if } x, y \in \mathrm{ob}(J) \\ \mathrm{Hom}_J(f(x), y) & \text{if } x \in \mathrm{ob}(I) \text{ and } y \in \mathrm{ob}(J) \\ \varnothing & \text{if } x \in \mathrm{ob}(J) \text{ and } y \in \mathrm{ob}(I). \end{cases}$$

The category $\mathrm{Hom}_B(I \amalg_f J, A)$ is equivalent to the category of triples consisting of a B-functor F from I to A, a B-functor G from J to A, and a morphism φ of B-functors from F to $G \circ f = f(G)$.

For every object j of J, denote by I/j the category of objects of I "placed by f over j": the objects of I/j are the pairs (i, α), where i is an object of I and $\alpha : i \to j$ is an arrow in $I \amalg_f J/j$; the morphisms of I/j are those of $I \amalg_f J/j$. If p_I and p_J are the projections

with the functorialities deduced from those on the factors.

of I and J onto B, and if (i,α) is an object of I/j, denote by $\alpha_* : A_{p_I(i)} \to A_{p_J(j)}$ the functor $p_J(\alpha)_*$.

To give a B-functor from $I \amalg_f J$ to A is equivalently to give $F \in \mathrm{Hom}_B(I,A)$, $G \in \mathrm{Hom}_B(J,A)$, and a morphism, functorial in j,

$$\psi_j : \varinjlim_{(i,\alpha)\in I/j} \alpha_*(F(i)) \longrightarrow G(j).$$

(The functoriality in j of the left-hand member follows from 1.2.10.1.) The left adjoint of f^* is therefore given by the formula

$$f_!(F)(j) = \varinjlim_{(i,\alpha)\in I/j} \alpha_*(F(i)).$$

93 Corollary 1.2.11. *Let E be a D-topos and let i be an object of D; the functor e_i^* has a left adjoint (resp. a right adjoint), given by*

$$e_{i!}(a)(j) = \coprod_{\alpha\in\mathrm{Hom}(i,j)} \alpha_*(a) \left(\text{resp. } e_{i*}(a)(j) = \prod_{\alpha\in\mathrm{Hom}(j,i)} \alpha^*(a) \right)$$

where a is an object of E over i.

Proposition 1.2.12. *Let D be a $\mathcal{U}$-small category and let E be a D-topos. Then the category $\Gamma(E)$ is a $\mathcal{U}$-topos.*

One verifies "fiber by fiber," using 1.2.10.1, that the category $\Gamma(E)$ has the following properties:

a) Finite projective limits are representable.
b) Coproducts indexed by an element of $\mathcal{U}$ are representable. They are disjoint and universal.
c) Equivalence relations are universally effective.

It remains to show that $\Gamma(E)$ has a system of generators indexed by an element of $\mathcal{U}$. But if, for every object i of D, $(G_{i\lambda})_{\lambda\in\Lambda_i}$ is a system of generators of E_i (where Λ_i is a $\mathcal{U}$-small set), then the family $(e_{i!}(G_{i\lambda}))_{i,\lambda}$ is a system of generators of $\Gamma(E)$.

1.2.13. We now introduce the notion of a morphism between D-topoi. First, let us specify that if F and F' are two categories over the same category B and $T : F \to F'$ is a B-functor, a *B-left adjoint* to T means a functor $S : F' \to F$ left adjoint to T such that the canonical morphisms $1 \to T\circ S$ and $S\circ T \to 1$ are B-morphisms of functors. Under these conditions, one verifies immediately that T is cartesian and S is cocartesian.

94 Definition 1.2.14. *Let E and E' be two D-topoi. A morphism from E to E' is a pair of D-functors $\Phi_* : E \to E'$ and $\Phi^* : E' \to E$, equipped with a D-adjunction*

$$\mathrm{Hom}(\Phi^*(-),-) \underset{\xi}{\tilde{\to}} \mathrm{Hom}(-,\Phi_*(-)),$$

such that, for every object i of D, the pair (Φ_{*i},Φ_i^*), equipped with the adjunction induced by ξ, is a morphism of topoi from E_i to E_i'.

Proposition 1.2.15. *Let E and E' be two D-topoi, and let $(\Phi_*,\Phi^*) : E \to E'$ be a morphism. If D is a $\mathcal{U}$-small category, then the pair $(\Gamma(\Phi_*),\Gamma(\Phi^*)) : \Gamma(E) \to \Gamma(E')$ is a morphism of topoi.*

This follows immediately from the preceding definition.

The following lemma, whose proof is left to the reader, provides a way of constructing morphisms of D-topoi.

LEMMA 1.2.16. *Let E and E' be two bifibered categories over the same category D, and let Φ be a* cartesian *D-functor from E to E' such that, for every object i of D, $\Phi_i : E_i \to E'_i$ has a left adjoint. Then the choice, for every i, of a left adjoint to Φ_i canonically determines a D-functor $\Psi : E' \to E$ which is D-left adjoint to Φ.*

SCHOLIUM 1.2.17. Under the conditions of (1.2.16), suppose that E and E' are two D-topoi and that, for every object i of D, every left adjoint to Φ_i is left exact. Then, if $\Psi : E' \to E$ is a D-left adjoint to Φ, the pair $(\Phi, \Psi) : E \to E'$ is a morphism of D-topoi.

Now let $\mathscr{E} \to B$ be a category bifibered in duals of topoi over B, let X and X' be two D-objects of B, and let $\alpha : X \to X'$ be a functorial morphism. A choice of normalized cleavages for $\mathscr{E}$ and $\mathscr{E}^\circ$ then canonically determines a morphism $(\alpha_*, \alpha^*) : \overline{X} \to \overline{X}'$ of D-topoi such that, for every object i of D, $(\alpha_{*i}, \alpha_i^*) : \overline{X}_i \to \overline{X}'_i$ is equal to $(\alpha_{i*}^\circ, \alpha_i^{*\circ})$, 95
where $\alpha_i : X_i \to X'_i$ is the arrow of B given by α. For two different choices of cleavages for $\mathscr{E}$ and $\mathscr{E}^\circ$, there is a unique D-isomorphism between the resulting morphisms. (For the verification of these facts, the reader may consult [4], (1.17).)

1.3. Ringed D-topoi.

DEFINITION 1.3.1. *A ringed D-topos is a pair (E, A), where E is a D-topos and A is a ring of $\Gamma(E)$.*

One then verifies that, for every object i of D, A_i is a ring of the topos E_i, and that, for every morphism $m : i \to j$, the canonical arrow $A_i \to m^*(A_j)$ is a ring homomorphism.

DEFINITION 1.3.2. *Let (E, A) and (E', A') be two ringed D-topoi. A morphism from (E, A) to (E', A') is a pair (Φ, θ), where $\Phi : E \to E'$ is a morphism of D-topoi and $\theta : A' \to \Gamma(\Phi_*)(A)$ is a ring homomorphism.*

REMARK 1.3.3. A morphism $\Phi : (E, A) \to (E', A')$ of ringed D-topoi induces a morphism $(\Gamma(\Phi), \theta) : (\Gamma(E), A) \to (\Gamma(E'), A')$ of ringed $\mathscr{U}$-topoi when D is a $\mathscr{U}$-small category (cf. 1.2.15).

1.3.4. Let $\mathscr{E} \to B$ be a category bifibered in duals of topoi over B, and let $\mathscr{O}$ be a ring of $\Gamma(\mathscr{E}^\circ) = \mathrm{Hom}_{B^\circ}(B^\circ, \mathscr{E}^\circ)$. If $X : D^\circ \to B$ is a D-object of B, the D-topos $\overline{X}$ (cf. (1.2.5)) is naturally ringed by $(\mathscr{O} \,.\, X^\circ) : D \to \mathscr{E}$; the resulting ringed D-topos is denoted by $(\overline{X}, \mathscr{O})$. If $\alpha : X \to X'$ is a functorial morphism, the morphism $(\alpha_*, \alpha^*) : \overline{X} \to \overline{X}'$ (cf. 1.2.17) canonically induces a morphism $(\overline{X}, \mathscr{O}) \to (\overline{X}', \mathscr{O})$ of ringed D-topoi, again denoted by α.

1.3.5. A ringed D-topos (E, A) canonically defines a category $\mathrm{Mod}(E, A)$ bifibered in abelian categories over D, whose fiber at an object i of D is the category $\mathrm{Mod}(E_i, A_i)$ 96
of modules on the ringed topos (E_i, A_i). With this notation, the category of modules of $\Gamma(E)$ over A, denoted by $\mathrm{Mod}(\Gamma(E), A)$, identifies with the category $\mathrm{Hom}_D(D, \mathrm{Mod}(E, A))$.

1.3.6. Let $\varphi = (\Phi, \theta) : (E, A) \to (E', A')$ be a morphism of ringed D-topoi. It defines two functors $\varphi_* : \underline{\mathrm{Mod}}(E, A) \to \underline{\mathrm{Mod}}(E', A')$ and $\varphi^* : \underline{\mathrm{Mod}}(E', A') \to \mathrm{Mod}(E, A)$ between the corresponding categories of modules:

- Let M be an object of $\mathrm{Mod}(E, A)$ over an object i of D. Then $\Phi_*(M)$ is a module over $\Phi_*(A_i)$, and the morphism $\theta_i : A'_i \to \Phi_*(A_i)$ makes it an A'_i-module, denoted by $\varphi_*(M)$. The functor φ_* is called the *direct-image functor under the morphism φ*.

- Let M' be an object of $\mathrm{Mod}(E', A')$ over an object i of D. Then $\Phi^*(M')$ is a module over $\Phi^*(A'_i)$, and $\varphi^*(M') = \Phi^*(M') \otimes_{\Phi^*(A'_i)} A_i$ is canonically endowed with an A_i-module structure. By means of (1.2.16), this defines a functor φ^* left adjoint to φ_*, called the *inverse-image functor under the morphism* φ.

. We say that φ is *flat* if the functor $\Gamma(\varphi^*)$ is exact.

Proposition 1.3.7. *Let $f : D' \to D$ be a functor, and let (E, A) be a ringed D-topos. Then the canonical functor*

$$f^* : \mathrm{Mod}(\Gamma(E), A) \longrightarrow \mathrm{Mod}(\Gamma(E \times_D D'), A \circ f)$$

has both a right and a left adjoint if D' is a $\mathcal{U}$-small category. In particular, it is exact.

This follows immediately from (1.2.10) and the identification $\mathrm{Hom}_D(D, \mathrm{Mod}(E, A)) \simeq \mathrm{Mod}(\Gamma(E), A)$.

97 In accordance with the general notation, we denote the left (resp. right) adjoint to f^* by $f_!$ (resp. f_*).

1.3.8. The considerations below provide a convenient computational method for derived functors of the form

$$R\Gamma(\varphi_*) : D^+(\Gamma(E), A) \longrightarrow D^+(\Gamma(E'), A'),$$

where $\varphi : (E, A) \to (E', A')$ is a morphism of ringed D-topoi (cf. (1.3.6)). [If $(S, \mathcal{O}_S)$ is a ringed topos, we denote by $D^+(S, \mathcal{O}_S)$ the bounded-below derived category[18] of the category of $\mathcal{O}_S$-modules on S.]

1.3.9. This formal calculation may moreover be applied in other contexts, such as "descent in ℓ-adic cohomology" (cf. SGA 5).

Proposition 1.3.10. *Let D be a $\mathcal{U}$-small category. If (E, A) is a ringed D-topos, denote by $I_{(E,A)}$ the class of objects of $\mathrm{Mod}(\Gamma(E), A)$ isomorphic to an object of the form $\prod_{i \in \mathrm{ob}(D)} e_{i*}(Q_i)$, where, for every object i of D, Q_i is totally acyclic* (cf. V 4.1)[18]*; $I_{(E,A)}$ has the following properties:*

(i) *For every object F of $I_{(E,A)}$ and every object i of D, $e_i^*(F)$ is totally acyclic.*
(ii) *Every object of $\mathrm{Mod}(\Gamma(E), A)$ embeds into an object of $I_{(E,A)}$.*
(iii) *$I_{(E,A)}$ is stable under finite direct sums.*
(iv) *For every morphism $\varphi : (E, A) \to (E', A')$ of ringed D-topoi, the functor $\Gamma(\varphi_*)$ carries every acyclic complex in $C^+(\Gamma(E), A)$ formed of objects of $I_{(E,A)}$ to an acyclic complex formed of objects of $I_{(E',A')}$.*

Proof

(i) In view of the explicit expression for e_i^* (cf. (1.2.1.1)), it suffices to prove the following lemma:

98 Lemma 1.3.10.1. *Let $(S, \mathcal{O}_S)$ be a ringed topos, and let $(F_t)_{t \in T}$ be a family of totally acyclic $\mathcal{O}_S$-modules indexed by a $\mathcal{U}$-small set T. Then $\prod_{t \in T} F_t$ is totally acyclic.*

[18] N.D.E.: If $f : (S, \mathcal{O}_S) \to (T, \mathcal{O}_T)$ is a flat morphism of ringed topoi, we simply write f^* (and not $L^+(f^*)$) for the derived functor

$$f^* : D^+(T, \mathcal{O}_T) \to D^+(S, \mathcal{O}_S),$$

as in the original version.

[18] "Flasque" in the terminology of V.

Let X be an object of S. There is a spectral sequence

$$H^p\left(X, \prod_{t\in T}^{(q)} F_t\right) Rightarrow \prod_{t\in T} H^{p+q}(X, F_t) \tag{1.3.10.2}$$

where $\prod_{t\in T}^{(q)}$ denotes the q-th derived functor of the "product indexed by T" functor. Since $\prod_{t\in T}^{(q)} F_t$ is the sheaf associated with the presheaf $U \mapsto \prod_{t\in T} H^q(U, F_t)$, (1.3.10.2) degenerates and gives isomorphisms

$$H^n\left(X, \prod_{t\in T} F_t\right) \xrightarrow{\sim} \prod_{t\in T} H^n(X, F_t) \tag{1.3.10.3}$$

and hence, finally, $H^n(X, \prod_{t\in T} F_t) = 0$ for every $n > 0$.

(ii) Let F be an object of $\mathrm{Mod}(\Gamma(E), A)$, and put $F_i = e_i^*(F)$. For every i, choose a monomorphism $F_i \to Q_i$, where Q_i is totally acyclic in $\mathrm{Mod}(E_i, A_i)$. Then $e_{i*}(F_i) \to e_{i*}(Q_i)$ is a monomorphism (because e_{i*} has a left adjoint), and the canonical arrow

$$F \longrightarrow \prod_{i\in \mathrm{Ob}(D)} e_{i*}(F_i) \longrightarrow \prod_{i\in \mathrm{Ob}(D)} e_{i*}(Q_i) \tag{1.3.10.4}$$

is again a monomorphism, as is immediately verified.

(iii) The proof is left to the reader.

(iv) The reader may also verify that $\Gamma(\varphi_*)$ carries an object of $I_{(E,A)}$ to an object of $I_{(E',A')}$. Once this point is established, in view of (i), it suffices to verify the following lemma:

LEMMA 1.3.10.5. *If $0 \to F \to G \to H \to 0$ is a short exact sequence in* $\mathrm{Mod}(\Gamma(E), A)$, *with $F, G \in I_{(E,A)}$, then the sequence* 99

$$0 \longrightarrow \Gamma(\varphi_*)(F) \longrightarrow \Gamma(\varphi_*)(G) \longrightarrow \Gamma(\varphi_*)(H) \longrightarrow 0$$

is exact in $\mathrm{Mod}(\Gamma(E'), A')$.

It suffices to observe that $e_i^*(H)$ is acyclic for every object i of D, and that $\Gamma(\varphi_*)$ is computed fiber by fiber. This completes the proof of (1.3.10).

COROLLARY 1.3.11. *Let $\varphi : (E, A) \to (E', A')$ be a morphism of ringed D-topoi. The functor $R\Gamma(\varphi_*)$ can be computed by means of resolutions formed of objects of $I_{(E,A)}$.*

Let $L \subset K^+(\Gamma(E), A)$ be the full subcategory of closed complexes of objects of $I_{(E,A)}$. By ((1.3.10), (ii)), L is a triangulated subcategory, and the theorem (5.1) of [[7], Chapter 1] applies to it.

COROLLARY 1.3.12. *Let $\varphi : (E, A) \to (E', A')$ be a morphism of ringed D-topoi. For every object i of D, denote by $\varphi_i : (E_i, A_i) \to (E_i', A_i')$ the induced morphism of ringed topoi over i. Then the diagram*

$$\begin{array}{ccc} D^+(\Gamma(E), A) & \xrightarrow{e_i^*} & D^+(E_i, A_i) \\ \downarrow{\scriptstyle R\Gamma(\varphi_*)} & & \downarrow{\scriptstyle R\varphi_{i*}} \\ D^+(\Gamma(E'), A') & \xrightarrow[e_i^*]{} & D^+(E_i', A_i') \end{array}$$

is essentially commutative.

There is a morphism

$$e_i^* \circ R\Gamma(\varphi_*) \longrightarrow R\varphi_{i*} \circ e_i^*$$

which is verified to be an isomorphism by (1.3.10).

100 REMARK 1.3.13. Denote by $\mathscr{A}_{(E,A)}$ the class of objects F of $\mathrm{Mod}(\Gamma(E), A)$ such that $e_i^*(F)$ is totally acyclic for every object i of D. It follows from the proof of (1.3.10) that $R\Gamma(\varphi_*)$ can be computed by means of resolutions formed of objects of $\mathscr{A}_{(E,A)}$; this is what is done, in particular, in the introduction by using the "canonical flasque resolution"[18]. By [(1.3.10)(i)], one has $I_{(E,A)} \subset \mathscr{A}_{(E,A)}$, but we will use $I_{(E,A)}$ explicitly in Section 2.

2. The method of cohomological descent

In this section, D is a $\mathscr{U}$-small category.

2.1. Generalities. Notation.

2.1.1. Let $(S, \mathcal{O}_S)$ be a ringed topos. The category $S \times D$, equipped with the canonical projection $S \times D \to D$, is a D-topos; moreover, the constant section with value $\mathcal{O}_S$ defines a ringed D-topos $(S \times D, \mathcal{O}_S)$, called the *constant ringed D-topos with value* $(S, \mathcal{O}_S)$. With this notation, the category $\mathrm{Mod}(\Gamma(S \times D), \mathcal{O}_S)$ is identified with the category of covariant functors from D to $\mathrm{Mod}(S, \mathcal{O}_S)$.

Define an exact functor

$$\epsilon^* : \mathrm{Mod}(S, \mathcal{O}_S) \longrightarrow \mathrm{Mod}(\Gamma(S \times D), \mathcal{O}_S)$$

by associating with every module F on S the constant functor with value F. The functor ϵ^* has a right adjoint ϵ_*, which associates with every functor $H : D \to \mathrm{Mod}(\mathcal{O}_S)$ its projective limit; the adjunction morphism $F \to \epsilon_*\epsilon^*(F)$ is the morphism from F to the projective limit of the constant functor with value F. Thus ϵ^* is fully faithful if and only if D is connected.

101 DEFINITION 2.1.2. *Let (E, A) be a ringed D-topos. An augmentation of (E, A) is a morphism (of ringed D-topoi) from (E, A) to a constant ringed D-topos. A ringed D-topos equipped with an augmentation will be called an augmented ringed D-topos.*

2.1.3. Let $\mathscr{E} \to B$ be a category bifibered in duals of topoi, and let $\mathcal{O}$ be a ring in $\Gamma(\mathscr{E}^\circ) = \mathrm{Hom}_{B^\circ}(B^\circ, \mathscr{E}^\circ)$. Let S be an object of B. A functor $D^\circ \to B/S$, that is, a functorial morphism $X \xrightarrow{\theta} C_S^D$, where X is a functor $D^\circ \to B$ (cf. (1.2.6)), induces an augmentation

$$\theta : (\overline{X}, \mathcal{O}) \longrightarrow (\mathscr{E}_S^\circ \times D, \mathcal{O}_S) \qquad \text{(cf. (1.3.4))}.$$

which we shall denote by the same letter as the functorial morphism from which it arises.

2.2. Cohomological descent.

[18] See Exposé XVII for the generalization of this notion.

2.2.1. Let $\theta : (E, A) \to (S \times D, \mathcal{O}_S)$ be an augmented ringed D-topos. With the notation of (1.3.6) and (2.1.1), set

$$\overline{\theta}^* = \Gamma(\theta^*) \circ \epsilon^* : \mathrm{Mod}(S, \mathcal{O}_S) \longrightarrow \mathrm{Mod}(\Gamma(E), A)$$

and $\overline{\theta}_* = \epsilon_* \circ \Gamma(\theta_*) : \mathrm{Mod}(\Gamma(E), A) \to \mathrm{Mod}(S, \mathcal{O}_S)$.

The image of $\overline{\theta}^*$ lies in the full subcategory of $\mathrm{Mod}(\Gamma(E), A)$ consisting of the cocartesian sections of $\Gamma(D, \mathrm{Mod}(E, A))$. We shall denote this category by $\Gamma^{\mathrm{cocart}}(D, \mathrm{Mod}(E, A))$ [19]

Definition 2.2.2. *The morphism θ is said to be an* effective descent augmentation *if*

$$\overline{\theta}^* = \Gamma(\theta^*) \circ \epsilon^* : \mathrm{Mod}(S, \mathcal{O}_S) \longrightarrow \Gamma^{\mathrm{cocart}}(D, \mathrm{Mod}(E, A))$$

is an equivalence of categories.

2.2.3. With the notation of (2.2.1), suppose that θ is flat, so that $\overline{\theta}^*$ is exact and passes trivially to the derived categories; we again denote by $\overline{\theta}^*$ the functor thereby obtained. With this notation: 102

Lemma 2.2.3.1. *There are two functorial morphisms*

$$\alpha : \mathrm{id}_{D^+(S,\mathcal{O}_S)} \longrightarrow R\overline{\theta}_* \circ \overline{\theta}^* \quad \textit{and} \quad \beta : \overline{\theta}^* \circ R\overline{\theta}_* \longrightarrow \mathrm{id}_{D^+(\Gamma(E),A)}$$

which make these two functors adjoint.

(cf. [9] (3.3)).

Definition 2.2.4. *The morphism θ is said to be a* 1-cohomological-descent augmentation *if*

1°) *θ is flat;*

2°) *$\overline{\theta}^* : D^+(S, \mathcal{O}_S) \to D^+(\Gamma(E), A)$ is fully faithful.*

Remark 2.2.5. Condition 2°) of the preceding definition can also be expressed by saying that the morphism $\alpha : \mathrm{id}_{D^+(S,\mathcal{O}_S)} \to R\overline{\theta}_* \circ \overline{\theta}^*$ in (2.2.3.1) is an isomorphism. This is what one proves in practice and what is useful.

Definition 2.2.6. *The morphism θ is said to be a* 2-cohomological-descent augmentation *(or an* effective cohomological-descent augmentation*) if θ is both an effective descent augmentation and a 1-cohomological-descent augmentation.*

The preceding terminology is justified by the following result:

[19] N.D.E.: In other words, an augmented ringed D-topos is the datum of a compatible system of morphisms of S-topoi

$$(m^*, m_*) : E_j \longrightarrow E_i,$$

where $m \in \mathrm{Hom}_D(i, j)$. The category of cocartesian sections is the category of descent data; its objects are families $F_i \in \mathrm{Ob}(E_i)$ equipped with cocycles of isomorphisms

$$F_j \xrightarrow{\sim} m^* F_i.$$

The functor which associates with every object F on S the constant functor with value F is the inverse image of a morphism of topoi

$$\epsilon : \Gamma(S \times D) \longrightarrow S,$$

inducing the morphism of ringed topoi of 2.1.3. The pair $(\overline{\theta}_*, \overline{\theta}^*)$ is therefore simply induced by the composite of morphisms of topoi $\overline{\theta} = \epsilon \circ \Gamma(\theta) : \Gamma(E) \to S$ (cf. 1.2.15).

PROPOSITION 2.2.7. *Let θ be an effective cohomological-descent augmentation. Then the essential image of $\overline{\theta}^*$ is the full subcategory of $D^+(\Gamma(E), A)$ consisting of the complexes $F^\cdot$ such that, for every i, $H^i(F^\cdot)$ is a cocartesian section of* $\mathrm{Mod}(E, A)$.

For brevity, we shall call a complex $F^\cdot$ satisfying the preceding conditions a *cohomological descent datum.*

PROOF. It is clear that, if $K^\cdot \in D^+(S, \mathcal{O}_S)$, then $\overline{\theta}^*(K^\cdot)$ is a cohomological descent datum. It is therefore enough to show that, for every cohomological descent datum $F^\cdot$, the canonical morphism

$$\beta(F^\cdot) : \overline{\theta}^* \circ R\overline{\theta}_*(F^\cdot) \longrightarrow F^\cdot$$

103 is an isomorphism in $D^+(\Gamma(E), A)$.

a) *The case where $F^\cdot$ is bounded*: we argue by induction on the length ℓ of the interval of integers i for which $H^i(F^\cdot) \neq 0$:
   - for $\ell \leq 1$, the assertion is true because θ is of effective descent.
   - suppose $\ell > 1$, and let n be the greatest integer such that $H^n(F^\cdot) \neq 0$. There is a distinguished triangle[19]

$$\begin{array}{ccc} & H^n(F^\cdot)[-n] & \\ \swarrow & & \nwarrow \\ F'^\cdot & \longrightarrow & F^\cdot \end{array}$$

   such that the induction hypothesis applies to $F'^\cdot$. Since $\beta(H^n(F^\cdot)[-n])$ and $\beta(F'^\cdot)$ are isomorphisms, the same is true of $\beta(F^\cdot)$.

b) *The general case*: denote by $\tau_{\leq n}(F^\cdot)$ the complex

$$\ldots \longrightarrow F^{n-2} \longrightarrow F^{n-1} \longrightarrow \mathrm{Ker}\, d^n \longrightarrow 0 \longrightarrow 0 \ldots,$$

   (cf. [7] (7.1)), so that, for every n, there is a commutative diagram

$$\begin{array}{ccc} \overline{\theta}^* \circ R\overline{\theta}_*(\tau_{\leq n}(F^\cdot)) & \xrightarrow{\sim} & \tau_{\leq n}(F^\cdot) \\ \downarrow & & \downarrow \\ \overline{\theta}^* \circ R\overline{\theta}_*(F^\cdot) & \longrightarrow & F^\cdot \end{array}$$

   and it is enough to see that the morphism

$$R\overline{\theta}_*(\tau_{\leq n}(F^\cdot)) \longrightarrow R\overline{\theta}_*(F^\cdot)$$

104 induces an isomorphism on H^i for $i \leq n$. But there is a triangle

$$\begin{array}{ccc} & F''^\cdot & \\ \swarrow & & \nwarrow \\ \tau_{\leq n}(F^\cdot) & \longrightarrow & F^\cdot \end{array}$$

[19] For every complex $K^\cdot$, $K^\cdot[-n]$ denotes the complex $T^{-n}(K^\cdot)$, where T is the translation functor.

with $H^i(F''^{\cdot}) = 0$ for $i \leq n$. It follows that $H^i(R\overline{\theta}_*(F''^{\cdot})) = 0$ for $i \leq n$, and then that $H^i(R\overline{\theta}_*(\tau_{\leq n}(F^{\cdot}))) \to H^i(R\overline{\theta}_*(F^{\cdot}))$ is an isomorphism for $i \leq n$, which completes the proof.

C.Q.F.D.

We shall now present an explicit method for computing

$$R\overline{\theta}_* = R\epsilon_* \circ R\Gamma(\theta)_*,$$

which is very useful in proving certain criteria for cohomological descent and in exploiting that notion.

2.3. A method for computing $R\epsilon_*$. We shall begin with two sorites.

2.3.1. Let D be a $\mathscr{U}$-small category, and let Ab be the category of abelian groups belonging to $\mathscr{U}$. Denote by r_D the functor $D \to (\mathrm{Hom}(D, \mathrm{Ab}))^\circ$ defined by

$$r_D(i)(j) = \mathbf{Z}^{\mathrm{Hom}(i,j)}$$

for every pair (i, j) of objects of D. For every pair (i, X) consisting of an object of D and an object of $\mathrm{Hom}(D, \mathrm{Ab})$, one then constructs a canonical and functorial isomorphism

$$\mathrm{Hom}(r_D(i), X) \xrightarrow{\sim} X(i).$$

. Let A be an additive category (cf. Tôhoku), and let F be a functor $D \to A$. Define a functor $\overline{F} : (\mathrm{Hom}(D, \mathrm{Ab}))^\circ \to \mathrm{Hom}(A^\circ, \mathrm{Ens})$ by

$$\overline{F}(X)(a) = \mathrm{Hom}_{\mathrm{Hom}(D,\mathrm{Ab})}(X, \mathrm{Hom}_A(a, F(.))),$$

so that the following diagram, in which h_A denotes the canonical functor, is essentially commutative: 105

$$\begin{array}{ccc} D & \xrightarrow{r_D} & (\mathrm{Hom}(D, \mathrm{Ab}))^\circ \\ {\scriptstyle F}\downarrow & & \downarrow{\scriptstyle \overline{F}} \\ A & \xrightarrow[h_A]{} & \mathrm{Hom}(A^\circ, \mathrm{Ens}). \end{array}$$

We leave it to the reader to verify that $\overline{F}$ sends finite direct sums in $\mathrm{Hom}(D, \mathrm{Ab})$ to products and that, if $\mathbf{Z}$ denotes the constant functor $D \to \mathrm{Ab}$ with value $\mathbf{Z}$, then $\overline{F}(\mathbf{Z}) = \varprojlim F$.

Finally, the correspondence $F \rightsquigarrow \overline{F}$ defines a covariant functor

$$\mathrm{Hom}(D, A) \longrightarrow \mathrm{Hom}((\mathrm{Hom}(D, \mathrm{Ab}))^\circ, \mathrm{Hom}(A^\circ, \mathrm{Ens})).$$

. Denote by $\mathrm{Add}(D)$ the full subcategory of $(\mathrm{Hom}(D, \mathrm{Ab}))^\circ$ defined by the objects of the form $r_D(i)$, where i is an object of D, and their finite direct sums. The category $\mathrm{Add}(D)$ is additive, and the functor $r_D : D \to \mathrm{Add}(D)$ has the following universal property:

. *For every additive category A, the functor $G \longmapsto G \circ r_D$ induces an equivalence from the full subcategory of* $\mathrm{Hom}(\mathrm{Add}(D), A)$ *consisting of the additive functors* onto *the category* $\mathrm{Hom}(D, A)$.

The category $\mathrm{Add}(D)$, characterized up to equivalence by (2.3.1.3), will be called the *additive category generated by D*.

2.3.2. Let A be an abelian category, and let $K^{\cdot\cdot}$ be a double complex, regarded as an object of $C^+(C^+(A))$, with the first index corresponding to the first symbol C^+. Denote by $(K^{\cdot\cdot})_s$ the *associated simple complex* (cf. Tôhoku (2.4), whose notation we shall retain). This defines a functor 106

$$(2.3.2.1) \qquad (\quad)_s : C^+(C^+(A)) \longrightarrow C^+(A)$$

and we leave it to the reader to verify the following lemma:

LEMMA 2.3.2.2. *The functor* $(\quad)_s$ *defines a triangulated functor*

$$K^+(C^+(A)) \longrightarrow K^+(A)$$

which preserves quasi-isomorphisms; it therefore defines a triangulated functor

$$D^+(C^+(A)) \longrightarrow D^+(A),$$

again denoted by $(\quad)_s$.

REMARK. The same result holds for the functor $(\quad)_s : C(C^b(A)) \to C(A)$, since the spectral sequence under consideration is biregular by (EGA III (11.3.3)).

2.3.3. Now let $(S, \mathcal{O}_S)$ be a ringed topos, let D be a $\mathscr{U}$-small category, and let $Z^\cdot \in C^+(\mathrm{Add}(D))$ be a cochain complex such that $Z^n = 0$ for $n < 0$.

Every object F of $\mathrm{Mod}(\Gamma(S \times D), \mathcal{O}_S)$ (which is identified with a functor $D \to \mathrm{Mod}(S, \mathcal{O}_S)$ by (2.1.1)) defines, by the universal property of $\mathrm{Add}(D)$ (2.3.1.3), an additive functor

$$\widetilde{F} : \mathrm{Add}(D) \longrightarrow \mathrm{Mod}(S, \mathcal{O}_S).$$

The image complex

$$\varepsilon_{Z^\cdot *}(F) := \widetilde{F}(Z^\cdot)$$

is an object of $C^+(S, \mathcal{O}_S)$ which varies functorially with F. We have thus defined an additive functor

$$\varepsilon_{Z^\cdot *} : \mathrm{Mod}(\Gamma(S \times D), \mathcal{O}_S) \longrightarrow C^+(S, \mathcal{O}_S),$$

which is visibly exact.

This defines a triangulated functor [20]:

$$(2.3.3.1) \qquad K^+(\varepsilon_{Z^\cdot *}) : K^+(\Gamma(S \times D), \mathcal{O}_S) \longrightarrow K^+(C^+(S, \mathcal{O}_S)).$$

which preserves quasi-isomorphisms. Moreover, it follows from (2.3.2.2) that the composite functor

$$(\quad)_s \circ K^+(\varepsilon_{Z^\cdot *}) : K^+(\Gamma(S \times D), \mathcal{O}_S) \longrightarrow K^+(S, \mathcal{O}_S)$$

107 sends acyclic objects to acyclic objects. It therefore defines a triangulated functor

$$(2.3.3.2) \qquad R\varepsilon_{Z^\cdot *} : D^+(\Gamma(S \times D), \mathcal{O}_S) \longrightarrow D^+(S, \mathcal{O}_S)$$

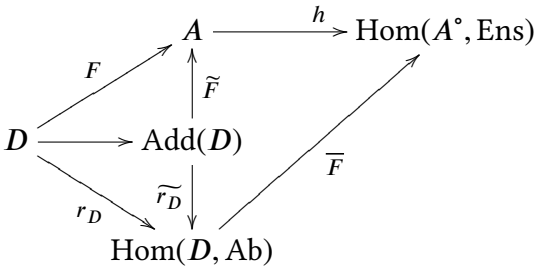
satisfying the relation [21] $Q \circ (\quad)_s \circ K^+(\varepsilon_{Z^\cdot *}) = R\varepsilon_{Z^\cdot *} \circ Q$ [21]

[20] N.D.E.: K^+ denotes the category of bounded-below complexes up to homotopy.

[21] N.D.E.: In other words, $\varepsilon_{Z^\cdot *}$ derives trivially as $F^\cdot \rightsquigarrow \varepsilon_{Z^\cdot *}(F^\cdot)$.

[21] For every abelian category A, the letter Q denotes the canonical functor $K^+(A) \to D^+(A)$.

2.3.4. Retain the preceding data and set $A = C^+(S, \mathcal{O}_S)$. First, $\widetilde{F}$ and $\overline{F}$ are compatible in the sense of commutativity of the Yoneda-type diagram



where h and r_D are respectively the Yoneda-type arrows defined by

$$h(a) = (\alpha \longmapsto \mathrm{Hom}_A(\alpha, a)) \quad \text{and} \quad r_D(d) \overset{2.3.1}{=} (\delta \longmapsto \mathrm{Hom}_D(d, \delta)),$$

that is, $h \circ \widetilde{F} = \overline{F} \circ \widetilde{r_D}$. It follows that

$$h(\varepsilon_{Z^\cdot *}(F)) = \overline{F}(\widetilde{r_D}(Z^\cdot)).$$

By the definition of a projective limit, $h(\epsilon_* F) = \varprojlim F$, and, as already observed, one has tautologically $\overline{F}(\mathbf{Z}) = \varprojlim F$; hence

$$\overline{F}(\mathbf{Z}) = h(\epsilon_*(F)).$$

Regard $Z^\cdot$ as a chain complex in $\mathrm{Hom}(D, \mathrm{Ab})$, that is, as a complex of the abelian category $\mathrm{Hom}(D, \mathrm{Ab})^\circ$ by means of the functor $\tilde{r}_D$. We may therefore speak of exact sequences and resolutions. Explicitly, a sequence in $\mathrm{Add}(D)$

$$Z \longrightarrow Z' \longrightarrow Z''$$

is exact if, for every object j of D, the sequence

$$\mathrm{Hom}_{\mathrm{Add}(D)}(Z'', [j]) \longrightarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z', [j]) \longrightarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z, [j])$$

is exact in Ab. Let $t : Z^0 \to \mathbf{Z}$ be a morphism from Z^0 to the constant functor with value $\mathbf{Z}$ such that the composite morphism

$$Z^1 \longrightarrow Z^0 \overset{t}{\to} \mathbf{Z} \tag{2.3.4.1}$$

is zero.

By functoriality (cf. (2.3.1.1)), one deduces a morphism $\varepsilon_*(F) \to \varepsilon_{Z^\cdot *}(F)^0$ such that the composite morphism

$$\varepsilon_*(F) \longrightarrow \varepsilon_{Z^\cdot *}(F)^0 \longrightarrow \varepsilon_{Z^\cdot *}(F)^1 \tag{2.3.4.2}$$

is zero.

There is then a canonical morphism of triangulated functors

$$Q \circ K^+(\varepsilon_*) \overset{j}{\to} Q \circ (\quad)_s \circ K^+(\varepsilon_{Z^\cdot *}) = R\varepsilon_{Z^\cdot *} \circ Q \tag{2.3.4.3}$$

Proposition 2.3.5. *Retain the conditions and notation of (2.3.3) and (2.3.4); suppose, in addition, that the augmented complex*

$$\ldots \longrightarrow Z^n \longrightarrow Z^{n-1} \longrightarrow \ldots \longrightarrow Z^1 \longrightarrow Z^0 \overset{t}{\to} \mathbf{Z}$$

defines a resolution of $\mathbf{Z}$.

Then, for every complex $F^\cdot$ consisting of objects of $I_{(S\times D,\mathcal{O}_S)}$ (cf. 1.3.10),

$$j(F^\cdot) : Q \circ K^+(\varepsilon_*)(F^\cdot) \longrightarrow (R\varepsilon_{Z^\cdot *} \circ Q)(F^\cdot)$$

108 *is an isomorphism.*

Let i be an object of D, and let Q_i be a module on $(S, \mathcal{O}_S)$. We show that $j(e_{i*}(Q_i))$ is an isomorphism. It is enough to see that the sequence

(2.3.5.1) $$0 \longrightarrow Q_i \longrightarrow \varepsilon_{Z^\cdot *}(e_{i*}(Q_i))^0 \longrightarrow \varepsilon_{Z^\cdot *}(e_{i*}(Q_i))^1 \longrightarrow \dots$$

is exact.

For this, it is enough to show that, for every object X of $\mathrm{Mod}(S, \mathcal{O}_S)$, the sequence (2.3.5.2)

$$0 \longrightarrow \mathrm{Hom}(X, Q_i) \longrightarrow \mathrm{Hom}(X, \varepsilon_{Z^\cdot *}(e_{i*}(Q_i))^0) \longrightarrow \mathrm{Hom}(X, \varepsilon_{Z^\cdot *}(e_{i*}(Q_i))^1) \longrightarrow \dots$$

is exact. An immediate calculation shows that (2.3.5.2) reduces to

(2.3.5.3) $$0 \longrightarrow \mathrm{Hom}(\mathbf{Z}, \mathrm{Hom}(X, Q_i)) \longrightarrow \mathrm{Hom}(\mathrm{Hom}_{\mathrm{Add}(D)}(Z^0, [i]), \mathrm{Hom}(X, Q_i)) \longrightarrow$$
$$\mathrm{Hom}(\mathrm{Hom}_{\mathrm{Add}(D)}(Z^1, [i]), \mathrm{Hom}(X, Q_i)) \longrightarrow \dots .$$

By hypothesis, the sequence

$$0 \leftarrow \mathbf{Z} \leftarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z^0, [i]) \leftarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z^1, [i]) \leftarrow \dots$$

is an exact sequence of free modules and is therefore null-homotopic. It follows that (2.3.5.3) is exact.

If, for every object i of D, Q_i is a totally acyclic module, then, by applying the functor $\prod_{i\in \mathrm{Ob}(D)}$ to the complexes (2.3.5.1), one sees that the resulting complex is still acyclic by (1.3.10.1). Thus $j(\prod_{i\in \mathrm{Ob}(D)} e_{i*}(Q_i))$ is an isomorphism.

We leave it to the reader to deduce that $j(F^\cdot)$ is an isomorphism when $F^n \in I_{(S\times D,\mathcal{O}_S)}$ for every n (for example, by spectral sequences).

109 Corollary 2.3.6. *Under the conditions of (2.3.5), the morphism j identifies $R\varepsilon_{Z^\cdot *}$ with the derived functor of ε_*.*

The subcategory of $K^+(\Gamma(S\times D), \mathcal{O}_S)$ defined by the complexes consisting of objects of $I_{(S\times D,\mathcal{O}_S)}$ satisfies the conditions of theorem (5.1) of [[7], I] for the functor $K^+(\varepsilon_*)$, by (1.3.10) and (2.3.5). The corollary follows immediately from this observation.

Corollary 2.3.7. *Under the conditions of (2.3.5), let (E, A) be a ringed D-topos and let $\theta : (E, A) \to (S \times D, \mathcal{O}_S)$ be an augmentation. There is a canonical isomorphism*

$$R\bar{\theta}_* \xrightarrow{\sim} R\varepsilon_{Z^\cdot *} \circ R\Gamma(\theta_*).$$

This follows from [[7], I, 5.4].

Remark 2.3.8. We knew a priori that $R\varepsilon_*$ exists and that there is an isomorphism $R\bar{\theta}_* \xrightarrow{\sim} R\varepsilon_* \circ R\Gamma(\theta_*)$, since $\Gamma(\theta_*)$ and ε_* are induced by morphisms of ringed topoi (cf. (1.3.6) and (2.1.1)). Nevertheless, the preceding calculation will prove very useful and applicable in other contexts, since in practice the conditions of (2.3.5) will always be satisfied.

We shall now give the fundamental examples to which (2.3.5) can be applied.

2.3.9. In $\mathrm{Add}(\Delta)$, put $Z^n = [n]$ and define $d^n : Z^n \to Z^{n+1}$ by the formula

$$d^n = \sum_{i=0}^{n+1} (-1)^i \partial^i. \tag{2.3.9.1}$$

Here $\partial^i : [n] \to [n+1]$ is the ith face [cf. [3] II 2]. For $t : Z^0 \to \mathbf{Z}$, take the evident natural augmentation[22].

The condition of 2.3.4 is trivially satisfied. The condition of 2.3.5 follows from [5] I (3.7.4).

2.3.10. In $\mathrm{Add}(\Delta \times \Delta)$, consider the simple complex associated with the double complex 110

$$\begin{array}{ccc} \big([n],[m+1]\big) & \xrightarrow{s_1^{n,m+1}} & \big([n+1],[m+1]\big) \\ \uparrow{\scriptstyle s_2^{n,m}} & & \uparrow{\scriptstyle s_2^{n+1,m}} \\ \big([n],[m]\big) & \xrightarrow{s_1^{n,m}} & \big([n+1],[m]\big) \end{array} \tag{2.3.10.1}$$

with

$$\begin{cases} S_1^{n,m} = \displaystyle\sum_{i=0}^{n+1} (-1)^i (\partial^i, \mathrm{id}_{[m]}), \\ S_2^{n,m} = \displaystyle\sum_{j=0}^{m+1} (-1)^j (\mathrm{id}_{[n]}, \partial^j). \end{cases} \tag{2.3.10.2}$$

The reader may verify (using 2.3.9) that the complex so defined, equipped with the evident natural augmentation, satisfies the conditions of 2.3.5.

2.3.11. The multisimplicial case is treated similarly (with $\underbrace{\Delta \times \Delta \times \cdots \times \Delta}_{p \text{ times}}$).

111

2.4. Relative cohomological descent.

2.4.0. Let $\mathscr{E} \to B$ be a category bifibered in duals of topoi, and let $\mathscr{O}$ be a ring of $\Gamma(\mathscr{E}^\circ)$.

These data canonically define a category fibered and cofibered in abelian categories over B°, denoted by $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O})$ (cf. (1.3.3) and (1.3.4)). Throughout what follows, the "direct-image" and "inverse-image" functors will always be taken with respect to the fibered and cofibered functor $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O}) \to B^\circ$.

Definition 2.4.0.1. *A morphism $f : T \to S$ in B is said to be* flat *(relative to $(\mathscr{E}, \mathscr{O})$) if $f_* : \mathrm{Mod}(\mathscr{E}_S^\circ, \mathscr{O}_S) \to \mathrm{Mod}(\mathscr{E}_T^\circ, \mathscr{O}_T)$ is an exact functor*[23].

[22] N.D.E.: Explicitly, if F is a simplicial complex with differential d_F, then $\varepsilon_{Z^{\bullet}*}(F)$ is the simple complex associated with the double complex $F^j(i)$ whose horizontal differential is the simplicial differential d^i and whose vertical differential is d_F^j.

[23] N.D.E.: This notation may cause confusion: one should probably write $(f^\circ)_\star$, which formally coincides with $(f^\star)^\circ$, where $f^\star$ is the fibered functor associated with $\mathscr{E} \to B$. For example, for the category of sheaves on topological spaces fibered over the category of topological spaces (cf. 4.1), one has $(f^\circ)_\star(S, F) = (T, f^{-1}F) = (f^\star)^\circ(S, F)$, where $f : T \to S$ is a continuous map of topological spaces, F is a sheaf on S, and f^{-1} is the usual inverse image. One then recognizes the usual flatness conditions.

2.4.1. Let G be a fibered and cofibered subcategory of $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O})$ such that, for every object S of B, G_S is a *thick* subcategory (cf. Tôhoku (1.11)) of $(\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O}))_S$. The reader will find examples of such situations in Section 4.

If S is an object of B, denote by $D^+(S)$ the derived category of $\mathrm{Mod}(\mathscr{E}^\circ_S, \mathscr{O}_S)$. Following [[7] (I § 4)], introduce the category $D^+_{G_S}(S)$, which identifies with the full subcategory of $D^+(S)$ formed by the objects $X^\cdot$ such that $H^n(X^\cdot) \in G_S$ for every n.

We assume throughout that the following condition holds.

. For every morphism $h : T \to S$ in B, the functor $Rh^* : D^+(T) \to D^+(S)$ has the property that $Rh^*(F) \in D^+_{G_S}(S)$ whenever $F \in G_T$[24].

By [[7] (I. (7.3). (ii))], this condition implies that $Rh^*(D^+_{G_T}(T)) \subset D^+_{G_S}(S)$.

112 2.4.2. Let D be a $\mathscr{U}$-small category. *Throughout what follows, we suppose given a complex $Z^\cdot$ of* $\mathrm{Add}(D)$ *satisfying the conditions of* (2.3.5) (in applications, one will in fact have $D = \Delta$ or $D = \Delta \times \Delta$).

If $X : D^\circ \to B$ is a D-object of B, denote by $\mathrm{Mod}_G(\Gamma(\overline{X}), \mathscr{O})$ the full subcategory of $\mathrm{Mod}(\Gamma(\overline{X}), \mathscr{O})$ formed by the objects F such that $e_i^*(F) \in G_i$ $(= G_{X_i})$ for every object i of D. The category $\mathrm{Mod}_G(\Gamma(\overline{X}), \mathscr{O})$ is a thick subcategory of $\mathrm{Mod}(\Gamma(\overline{X}), \mathscr{O})$. Denote by $K^+_G(\Gamma(\overline{X}), \mathscr{O})$ (resp. $D^+_G(\Gamma(\overline{X}), \mathscr{O})$) the full subcategory of $K^+(\Gamma(\overline{X}), \mathscr{O})$ (resp. $D^+(\Gamma(\overline{X}), \mathscr{O})$) formed by the objects $X^\cdot$ such that $H^n(X^\cdot) \in \mathrm{Mod}_G(\Gamma(\overline{X}), \mathscr{O})$ for every n.

2.4.3. Let X and X' be two D-objects of B, and let $\alpha : X \to X'$ be a functorial morphism. Recall (cf. (1.3.4)) that the corresponding morphism of ringed D-topoi is again denoted by $\alpha : (\overline{X}, \mathscr{O}) \to (\overline{X}', \mathscr{O})$.

Definition 2.4.3.1. *The morphism $\alpha : X \to X'$ is said to be* flat *if the corresponding morphism of ringed D-topoi is flat in the sense of (1.3.6.1).*

We then have the evident lemma:

Lemma 2.4.3.2. *The morphism $\alpha : X \to X'$ is flat if and only if, for every object i of D, $\alpha_i : X_i \to X'_i$ is a flat morphism* (cf. (2.4.0.1)).

2.4.4. A morphism $\alpha : X \to X'$ defines a functor

$$K^+(\Gamma(\alpha_*)) : K^+(\Gamma(\overline{X}), \mathscr{O}) \longrightarrow K^+(\Gamma(\overline{X}'), \mathscr{O}). \tag{2.4.4.1}$$

Lemma 2.4.4.2. *The functor*

$$K^+(\Gamma(\alpha_*))|_{K^+_G(\Gamma(\overline{X}),\mathscr{O})} : K^+_G(\Gamma(\overline{X}), \mathscr{O}) \longrightarrow K^+(\Gamma(\overline{X}'), \mathscr{O})$$

has a right derived functor, denoted by $R_G\Gamma(\alpha_*)$, and the canonical morphism

$$R_G\Gamma(\alpha_*) \longrightarrow R\Gamma(\alpha_*)|_{D^+_G(\Gamma(\overline{X}),\mathscr{O})}$$

113 *is an isomorphism.*

Moreover,

$$R_G\Gamma(\alpha_*)(D^+_G(\Gamma(\overline{X}), \mathscr{O})) \subset D^+_G(\Gamma(\overline{X}'), \mathscr{O}).$$

[24] N.D.E.: As above, it would be more correct to write $R(h^\circ)^* = R(h_*)^\circ$ rather than Rh^*, which would make the notation more consistent with the usual geometric intuition. One could equally well have taken the fibered and cofibered functors with respect to $\mathrm{Mod}(\mathscr{E}, \mathscr{O}) \to B$ rather than the opposite categories. The fibers, however, would then have been only duals of topoi.

The first part follows from (1.3.10) and [[7], (I. (5.2))]. To verify the last assertion, by [[7], (I. (7.3). (ii))], it suffices to show that $R\Gamma(\alpha_*)(F) \in D^+_G(\Gamma(\overline{X}'), \mathcal{O})$ whenever $F \in \mathrm{Mod}_G(\Gamma(\overline{X}), \mathcal{O})$. This follows from the explicit calculation of $R\Gamma(\alpha_*)$ given by (1.3.1.1) and from hypothesis (2.4.1.1).

2.4.5. Let S be an object of B. The ringed D-topos $(\overline{C^D_S}, \mathcal{O})$ (cf. (1.2.6)) identifies with the constant ringed D-topos $(\mathscr{E}^\circ_S \times D, \mathcal{O}_S)$, and we have the following lemma, which follows from [[7] (I. (7.3). (ii))]:

Lemma 2.4.5.1. *The functor $R\varepsilon_{Z\cdot *}|_{D^+_G(\Gamma(\overline{C^D_S}),\mathcal{O})}$, denoted by $R_G\varepsilon_{Z\cdot *}$, takes values in $D^+_{G_S}(S)$ and identifies with a right derived functor of*

$$K^+(\varepsilon_*)|_{K^+_G(\Gamma(\overline{C^D_S}),\mathcal{O})} : K^+_G(\Gamma(\overline{C^D_S}), \mathcal{O}) \longrightarrow K^+(S).$$

Proposition 2.4.6. *Let $\theta : X \to C^D_S$ be an augmented D-object of B. The functor $K^+(\overline{\theta}_*)|_{K^+_G(\Gamma(\overline{X}),\mathcal{O})}$ has a right derived functor, denoted by $R_G\overline{\theta}_*$, taking values in $D^+_{G_S}(S)$. Moreover, there is a canonical isomorphism*

$$R_G\overline{\theta}_* \xrightarrow{\sim} R_G\varepsilon_{Z\cdot *} \circ R_G\Gamma(\theta_*).$$

The proof, from what precedes, is left to the reader.

We are now in a position to give the definitions concerning cohomological descent.

2.4.7. Let $\theta : X \to C^D_S$ be an augmented D-object of B. Suppose that θ is flat, so that the restriction to $D^+_{G_S}(S)$ of the functor $\overline{\theta}^*$ takes values in $D^+_G(\Gamma(\overline{X}), \mathcal{O})$; denote this restriction by $\overline{\theta}^*_G$. With this notation: 114

Lemma 2.4.7.1. *There are two functorial morphisms $\alpha : \mathrm{id}_{D^+_{G_S}(S)} \to R_G\overline{\theta}_* \circ \overline{\theta}^*_G$ and $\beta : \overline{\theta}^*_G \circ R_G\overline{\theta}_* \to \mathrm{id}_{D^+_G(\Gamma(\overline{X}),\mathcal{O})}$ which make these two functors adjoint.*

This follows immediately from (2.2.3.1) and (2.4.6).

Definition 2.4.8. *The morphism θ is called an augmentation of cohomological* 1*-descent relative to G if*

1°) *θ is flat.*

2°) *The functor $\overline{\theta}^*_G$ is fully faithful.*

Remark 2.4.9. *Condition* 2°) *of the preceding definition may equivalently be expressed by saying that the morphism α in (2.4.7.1) is an isomorphism.*

Definition 2.4.10. *Let $\theta : X \to C^D_S$ be an augmented D-object of B. The morphism θ is called an* augmentation of effective descent relative to *G if the functor*

$$\Gamma(\theta^*) \circ \varepsilon^* : G_S \longrightarrow \Gamma^{\mathrm{Cocart}}(D, G)$$

is an equivalence of categories.

Definition 2.4.11. *The morphism θ is called an* augmentation of cohomological 2-descent *(or of* effective cohomological descent*)* relative to *G if θ is both an augmentation of cohomological* 1*-descent and an augmentation of effective descent relative to G.*

The proof of (2.2.7) then gives the following result:

THEOREM 2.4.12. *Let $\theta : X \to C_S^D$ be an augmentation of cohomological 2-descent* 115
relative to G. Then the essential image of $\overline{\theta}_G^*$ is the full subcategory of $D_G^+(\Gamma(\overline{X}), \mathcal{O})$ formed by the complexes $F^\cdot$ such that, for every n, $H^n(F^\cdot)$ is a cocartesian section of G.

The following proposition, whose verification is left to the reader, translates Definition (2.4.10) into the language of ([4], 6):

PROPOSITION 2.4.13. *The morphism $\theta : X \to C_S^D$ is an augmentation of effective descent relative to G if and only if the functor $D^\circ \to B/S$ over B defined by θ induces an equivalence between the category of cartesian sections of G° over B/S and the category of cartesian sections of G° over D°.*

COROLLARY 2.4.14. *Suppose that finite fiber products are representable in B, and let $f : R \to S$ be a morphism. The augmented semi-simplicial object $[R|_f S]$* (cf. (1.2.7)) *is of effective descent relative to G if and only if f is an effective descent morphism for the fibered category G° over B.*

This follows from (2.4.13) and ([4], 9).

DEFINITION 2.4.15. *Suppose that finite fiber products are representable in B, and let $f : R \to S$ be a morphism. We say that f is a morphism of cohomological i-descent relative to G $(i = 1, 2)$ if the augmented semi-simplicial object $[R|_f S]$ is of cohomological i-descent relative to G.*

DEFINITION 2.4.16. *Suppose that finite fiber products are representable in B, and let $\theta : X \to C_S^D$ be an augmentation. We say that θ is an* augmentation of universal cohomological i-descent $(i = 1, 2)$ relative to G *if, for every morphism $T \to S$, the augmenta-*
116 *tion $X_T \xrightarrow{\theta_T} C_T^D$ obtained by base change is an augmentation of cohomological i-descent relative to G.*

The notion of a *morphism of universal cohomological i-descent relative to G* follows immediately from the two preceding definitions.

We will often use the terminology "augmentation of G-cohomological i-descent" in place of "augmentation of cohomological i-descent relative to G".

2.5. The descent spectral sequence.

2.5.1. Let A be an abelian category. Denote by $F(A)$ the category whose objects are the objects of A equipped with a discrete and codiscrete filtration, and whose morphisms are the filtered morphisms of A: the category $F(A)$ is an additive category (but is not abelian).

The category $K(F(A))$ of filtered complexes of A, whose filtration is discrete and codiscrete degree by degree, is a triangulated category, and the canonical functors

$$\mathrm{Gr}_n : K(F(A)) \longrightarrow K(A) \tag{2.5.1.1}$$

are triangulated. The set Σ of morphisms f of $K(F(A))$ such that $\mathrm{Gr}_n(f)$ is a quasi-isomorphism for every n is therefore a saturated multiplicative system [cf. [9]. § 2. n° 1].

By inverting the arrows of this system, one obtains a new triangulated category, denoted by $DF(A)$, and the functors Gr_n extend to functors

$$\mathrm{Gr}_n : DF(A) \longrightarrow D(A). \tag{2.5.1.2}$$

In what follows, we will consider only complexes bounded below: this naturally gives rise to the notation $K^+(F(A))$ and $D^+F(A)$.

2.5.2. Let B be a thick subcategory of A, and denote by $K_B^+(F(A))$ the full subcategory of $K^+(F(A))$ whose objects are the complexes $X^\cdot$ such that $H^i(\mathrm{Gr}_j(X^\cdot)) \in B$ for every pair of integers (i,j) : $K_B^+(F(A))$ is a *localizing triangulated subcategory* of $K(F(A))$ [cf. [7]. (I. § 5)]. We denote by $D_B^+F(A)$ the full subcategory of $DF(A)$ whose objects are the complexes bounded below $X^\cdot$ such that $H^i(\mathrm{Gr}_j(X^\cdot)) \in B$ for every pair (i,j): by (*loc. cit.*), $D_B^+F(A)$ identifies with the category of fractions $K_B^+(F(A))_{\Sigma\cap K_B^+(F(A))}$. 117

The "forgetful functor for filtrations"

$$\iota : K^+(F(A)) \longrightarrow K^+(A)$$

is triangulated. Moreover, it follows from [[5]. I. (4.7)] that $\iota(f)$ is a quasi-isomorphism if $f \in \Sigma$ and that $\iota(K_B^+(F(A))) \subset K_B^+(A)$. The functor ι thus extends to a triangulated functor

$$\iota : D^+F(A) \longrightarrow D^+(A) \tag{2.5.2.1}$$

such that $\iota(D_B^+F(A)) \subset D_B^+(A)$.

Finally, note that there is a canonical *spectral functor* $D_B^+F(A) \to B$, converging to $H^n \circ \iota$, whose E_1 term is given by $H^{p+q} \circ \mathrm{Gr}_p$ [cf. [5] I. (4.2)].

We now return to the notation of (2.3), *assuming that* $D = \Delta$: $Z^\cdot$ will denote the complex of $\mathrm{Add}(\Delta)$ defined by (2.3.9.1).

Proposition 2.5.3. *Let* $(S, \mathcal{O}_S)$ *be a ringed topos. The functor*

$$R\varepsilon_{Z^\cdot *} : D^+(\Gamma(S\times D), \mathcal{O}_S) \longrightarrow D^+(S, \mathcal{O}_S)$$

admits a canonical factorization

$$D^+(\Gamma(S\times D), \mathcal{O}_S) \xrightarrow{F^{R\varepsilon}Z^\cdot *} D^+F(S, \mathcal{O}_S) \xrightarrow{\iota} D^+(S, \mathcal{O}_S).$$

Moreover, for every integer i, *the following diagram is commutative:*

$$\begin{array}{ccc} D^+(\Gamma(S\times D), \mathcal{O}_S) & \xrightarrow{Re_i^*} & D^+(S, \mathcal{O}_S) \\ \Big\downarrow{\scriptstyle F^{R\varepsilon}Z^\cdot *} & \nearrow{\scriptstyle \mathrm{Gr}_i} & \\ D^+F(S, \mathcal{O}_S) & & \end{array}$$

Consider the functor $C^+(C^+(S,\mathcal{O}_S)) \xrightarrow{(\ )_s} C^+(S,\mathcal{O}_S)$: if $X^{\cdot\cdot}$ is an object of $C^+(C^+(S,\mathcal{O}_S))$, we equip $(X^{\cdot\cdot})_s$ with its second canonical filtration [cf. [5]. I.4.8]; this gives a factorization 118

$$(\ )_s : C^+(C^+(S,\mathcal{O}_S)) \longrightarrow C^+(F(S,\mathcal{O}_S)) \xrightarrow{\iota} C^+(S,\mathcal{O}_S) \tag{2.5.3.1}$$

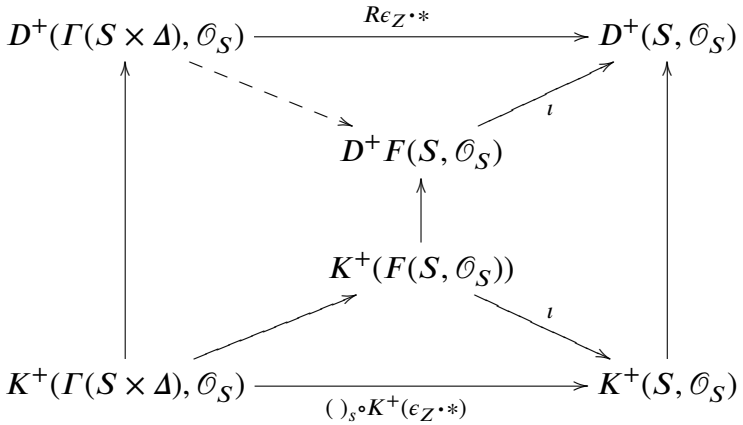
which passes to the categories K^+:

$$(\ )_s : K^+(C^+(S,\mathcal{O}_S)) \longrightarrow K^+(F(S,\mathcal{O}_S)) \xrightarrow{\iota} K^+(S,\mathcal{O}_S) \tag{2.5.3.2}$$

because a homotopy in $C^+(C^+(S,\mathcal{O}_S))$ induces a filtered homotopy.

We then have a diagram

(2.5.3.3)



and one verifies that there is a unique functor

$$ {}_F R\epsilon_{Z^\cdot *} : D^+(\Gamma(S\times\Delta),\mathcal{O}_S) \longrightarrow D^+F(S,\mathcal{O}_S)$$

119 making (2.5.3.3) commutative.

The rest of the proposition is evident.

Proposition 2.5.4. *Let (E,A) be a semi-simplicial ringed topos and*

$$\theta : (E,A) \longrightarrow (S\times\Delta,\mathcal{O}_S)$$

an augmentation. For every integer i, denote by

$$\theta_i : (E_i,A_i) \longrightarrow (S,\mathcal{O}_S)$$

the morphism of ringed topoi induced by θ over i.

The functor $R\bar{\theta}_* : D^+(\Gamma(E),A) \to D^+(S,\mathcal{O}_S)$ admits a canonical factorization

$$D^+(\Gamma(E),A) \xrightarrow{{}_F R\bar{\theta}_*} D^+F(S,\mathcal{O}_S) \xrightarrow{\iota} D^+(S,\mathcal{O}_S)$$

such that, for every integer i, the canonical diagram

$$\begin{array}{ccc} D^+(\Gamma(E),A) & \xrightarrow{Re_i^*} & D^+(E_i,A_i) \\ \downarrow{\scriptstyle {}_F R\bar{\theta}_*} & & \downarrow{\scriptstyle R\theta_{i*}} \\ D^+F(S,\mathcal{O}_S) & \xrightarrow[\mathrm{Gr}_i]{} & D^+(S,\mathcal{O}_S) \end{array}$$

is essentially commutative.

This follows from what precedes and from (1.3.1.2).

Proposition 2.5.5. *Let (E,A) be a semi-simplicial ringed topos and let*

$$\theta : (E,A) \longrightarrow (S\times\Delta,\mathcal{O}_S)$$

be an augmentation of cohomological 1-descent. Let

$$H : (S,\mathcal{O}_S) \longrightarrow (R,\mathcal{O}_R)$$

be a morphism of ringed topoi. Then there is a spectral functor from $D^+(S,\mathcal{O}_S)$ to $\mathrm{Mod}(R,\mathcal{O}_R)$:

$$E_1^{pq} = R^q(H\circ\theta_p)_*\circ\theta_p^* \longrightarrow R^{p+q}H_*$$

120 *called the* descent spectral functor.

Let $H \circ \theta : (E, A) \to (R \times \Delta, \mathcal{O}_R)$ be the augmentation canonically deduced from θ by composition with H; we have

$$\overline{(H \circ \theta)}_* = H_* \circ \overline{\theta}_* \tag{2.5.5.1}$$

so that the diagram

$$\begin{array}{ccc} D^+(S, \mathcal{O}_S) & \xrightarrow{\overline{\theta}^*} & D^+(\Gamma(E), A) \\ & \searrow^{RH_*} & \downarrow{\scriptstyle R(\overline{H\circ\theta})_*} \\ & & D^+(R, \mathcal{O}_R) \end{array} \tag{2.5.5.2}$$

is essentially commutative, since

$$R\overline{(H \circ \theta)}_* \xrightarrow{\sim} RH_* \circ R\overline{\theta}_* \text{ and } \mathrm{id}_{D^+(S,\mathcal{O}_S)} \xrightarrow{\sim} R\overline{\theta}_* \circ \overline{\theta}^*.$$

By (2.5.4), for every i we have an essentially commutative diagram

$$\begin{array}{ccccc} D^+(S, \mathcal{O}_S) & \xrightarrow{\overline{\theta}^*} & D^+(\Gamma(E), A) & \xrightarrow{Re_i^*} & D^+(E_i, A_i) \\ & \searrow^{RH_*} & \downarrow{\scriptstyle {}_F R(\overline{H\circ\theta})_*} & & \downarrow{\scriptstyle R(H\circ\theta_i)_*} \\ & & D^+F(R, \mathcal{O}_R) & \xrightarrow[\mathrm{Gr}_i]{} & D^+(R, \mathcal{O}_R) \\ & & \downarrow{\scriptstyle \iota} & & \\ & & D^+(R, \mathcal{O}_R) & & \end{array} \tag{2.5.5.3}$$

.

The asserted spectral functor is obtained by taking the canonical spectral functor on $D^+F(R, \mathcal{O}_R)$ and composing it with ${}_F R(\overline{H \circ \theta})_* \circ \overline{\theta}^*$. 121

We leave it to the reader to verify the following proposition:

Proposition 2.5.6. *With the preceding notation, the E_2^{pq} term of the preceding spectral functor is*

$$\check{H}^p \circ R^q\Gamma((H \circ \theta)_*) \circ \overline{\theta}^*$$

where $\check{H}^p : \mathrm{Mod}(\Gamma(R\times\Delta), \mathcal{O}_R) \to \mathrm{Mod}(R, \mathcal{O}_R)$ is the functor that assigns to every functor $\Delta \xrightarrow{F} \mathrm{Mod}(R, \mathcal{O}_R)$ the homology object in degree p of the associated complex (denoted by $\varepsilon_{Z\cdot *}(F)^\cdot$ in (2.3.3)).

We now return to the terminology introduced in (2.4): by the sorites (2.5.2), all the preceding considerations carry over word for word by placing the letter G as a subscript wherever this makes sense. The details are left to the reader; in particular, one obtains:

Proposition 2.5.7. *Let $\theta : X \to C_S^D$ be an augmentation of cohomological 1-descent relative to G, and let $h : S \to R$ be a morphism of B; there is a spectral functor from $D^+_{G_S}(S)$ to G_R:* 122

$$E_1^{pq} = R^q_G(h \circ \theta)_* \circ \theta^*_{p,G} \longrightarrow R^{p+q}_G h_*$$

with $E_2^{pq} = \check{H}^p \circ R_G^q \Gamma((h \circ \theta)_*) \circ \overline{\theta}_G^{\star}$.

3. Descent criteria

3.0. Notation. Throughout what follows, we retain the notation of (2.4). We shall always assume *that finite fiber products are representable in* B.

We shall also assume that B satisfies the following condition:

3.0.0. *Coproducts exist in* B*, are disjoint and universal, and are families of effective* $\mathscr{E}$*-descent and effective* G°*-descent* [cf. [4] (9.23), (9.25), and (9.27)].

3.0.1. We now recall some classical notation concerning simplicial objects of a category [cf. [3], Chap. II, and (V, Appendix)].

Let E be a category *having finite projective limits.* $\Delta^\circ E$ denotes the category of simplicial objects of E (cf. (1.2.6)).

For an integer n, denote by Δ_n (resp. ${\Delta_n}^+$, ${\Delta_n}^-$) the full subcategory of Δ (resp. Δ^+, Δ^-) consisting of the objects $[p]$ such that $p \leq n$.

The restriction functor

$$i_n^* : \Delta^\circ E \longrightarrow \underset{n}{\Delta}{}^\circ E \tag{3.0.1.1}$$

123 has a right adjoint i_{n*}, since finite projective limits exist in E (cf. (1.2.10)). The functor $i_{n*} \circ i_n^*$ is denoted by cosq_n and is called the *coskeleton functor of order n*; by abuse of notation, we shall also use cosq_n for the functor i_{n*}.

For every integer p, denote by $\Delta_{n[p]}$ (resp. ${\Delta_{n[p]}}^+$) the full subcategory of $\Delta_{[p]}$ (resp. $\Delta^+_{[p]}$), the category of objects of Δ (resp. Δ^+) over $[p]$, defined by the objects $[q]$ over $[p]$ such that $q \leq n$. We leave it to the reader to verify that

$$(\mathrm{cosq}_n(X))_p = \varprojlim_{\Delta_{n[p]}} X_q \xrightarrow{\sim} \varprojlim_{\Delta^+_{n[p]}} X_q. \tag{3.0.1.2}$$

3.0.2. Let X and X' be two objects of $\Delta^\circ E$, and let $f, g : X \rightrightarrows X'$ be two morphisms. A *homotopy from* f *to* g consists of a map $h_n : \mathrm{Hom}_\Delta([n],[1]) \to \mathrm{Hom}_E(X_n, X'_n)$ for every n, satisfying the following two conditions:

$$h_n(\partial^1) = f_n \quad \text{and} \quad h_n(\partial^\circ) = g_n \quad \text{for every } n. \tag{3.0.2.1}$$

. For every arrow $[n] \to [p]$ in Δ, the following diagram is commutative:

$$\begin{array}{ccc} \mathrm{Hom}_\Delta\big([p],[1]\big) & \longrightarrow & \mathrm{Hom}_E(X_p, X'_p) \\ \downarrow & & \searrow \\ & & \mathrm{Hom}_E(X_p, X'_p) \\ & & \uparrow \\ \mathrm{Hom}_\Delta\big([n],[1]\big) & \longrightarrow & \mathrm{Hom}_E(X_n, X'_n) \end{array} .$$

N.B.: The relation thus introduced on the set of morphisms from X to X' is not an equivalence relation. This inconvenience can be remedied by introducing the notion of *composite homotopy* [cf. [3], Chap. III], but we shall not use that notion.

124 Lemma 3.0.2.3. *Let* X *and* X' *be two objects of* $\Delta^\circ E$*, let* $f, g : X \rightrightarrows X'$ *be two morphisms, and let* $F : E \to C$ *be a functor, where* C *is an abelian category. A simplicial*

homotopy from f to g induces a homotopy between the morphisms of cochain complexes canonically associated with $F(f)$ and $F(g)$.

This reduces to the case where C is the category of modules over a ring, and one applies [[5], I, (3.7.1)].

LEMMA 3.0.2.4. *Let n be an integer, and let X and X' be two objects of $\Delta^\circ E$. Let f and g be two morphisms from X to X' such that $f_p = g_p$ for $p < n$. Then there exists a simplicial homotopy from $\mathrm{cosq}_n(f)$ to $\mathrm{cosq}_n(g)$.*

For $p < n$, define

$$h_p : \mathrm{Hom}_\Delta([p],[1]) \longrightarrow \mathrm{Hom}(X_p, X'_p)$$

to be the constant map with value $f_p = g_p$. Next define $h_n : \mathrm{Hom}_\Delta([n],[1]) \to \mathrm{Hom}(X_n, X'_n)$ by sending every element of $\mathrm{Hom}_\Delta([n],[1])$, except ∂°, to f_n, and ∂° to g_n. Finally, observe that

$$\mathrm{Hom}_\Delta([k],[1]) = \varprojlim_{\substack{[q]\to[k] \\ q\le n}} \mathrm{Hom}([q],[1])$$

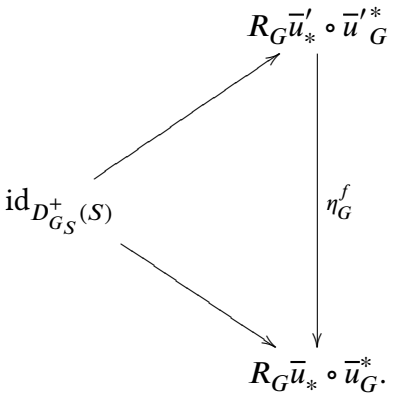
for $k > n$, which makes it possible to define h_k canonically.

3.1. Comparison of two augmentations from the viewpoint of 1-cohomological descent. Let S be an object of B. Denote by $\mathrm{Hom}_{\mathrm{plat}}(D^\circ, B/S)$ the full subcategory of $\mathrm{Hom}(D^\circ, B/S)$ whose objects are the flat augmentations [cf. (1.2.6) and (2.4.3.1)].

PROPOSITION 3.1.1. *Let $X \xrightarrow{u} C_S^D$ and $X' \xrightarrow{u'} C_S^D$ be two objects of $\mathrm{Hom}_{\mathrm{plat}}(D^\circ, B/S)$. Let $f : X \to X'$ be a morphism over S (i.e. a morphism in $\mathrm{Hom}(D^\circ, B/S)$). There is naturally associated with it a functorial morphism*

$$\eta_G^f : R_G\overline{u}'_* \circ \overline{u}'^*_G \longrightarrow R_G\overline{u}_* \circ \overline{u}^*_G$$

such that the following diagram is commutative:



Moreover, if $D = \Delta$, then η_G^f depends only on the homotopy class (cf. (3.0.2)) *of f in* $\mathrm{Hom}(\Delta^\circ, B/S)$. 125

It is clear that, for the construction of η_G^f, we may suppose that $G = \mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O})$.

Let I be the category defined by the graph

$$0 \longrightarrow 1. \tag{3.1.1.1}$$

Denote by $r_0 : D \to I \times D$ (resp. r_1) the fully faithful functor defined by $r_0(i) = (0, i)$ (resp. $r_1(i) = (1, i)$).

By virtue of the canonical isomorphism

$$\mathrm{Hom}((I\times D)^\circ, B/S) \xrightarrow{\sim} \mathrm{Hom}(I^\circ, \mathrm{Hom}(D^\circ, B/S)) \tag{3.1.1.2}$$

the data of (3.1.1) define a flat augmentation [25]

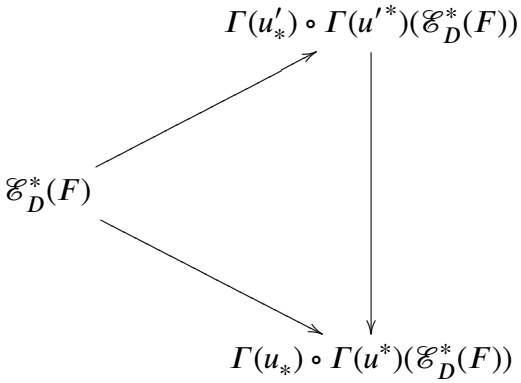
$$XfX' \xrightarrow{ufu'} C_S^{I\times D}. \tag{3.1.1.3}$$

Let F be an object of $\mathrm{Mod}(\mathscr{E}_S^\circ, \mathscr{O}_S)$. The canonical morphism

$$\mathscr{E}^*_{I\times D}(F) \longrightarrow \Gamma((ufu')_*)\circ\Gamma((ufu')^*)(\mathscr{E}^*_{I\times D}(F)) \tag{3.1.1.4}$$

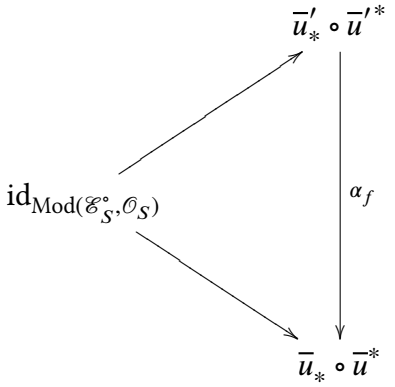
may be interpreted as a commutative triangle in $\mathrm{Mod}(\Gamma(\overline{C_S^D}), \mathscr{O})$:

(3.1.1.5)



126 and hence gives a functorial morphism $\overline{u}'_*\circ\overline{u}'^* \xrightarrow{\alpha_f} \overline{u}_*\circ\overline{u}^*$ such that the diagram

(3.1.1.6)

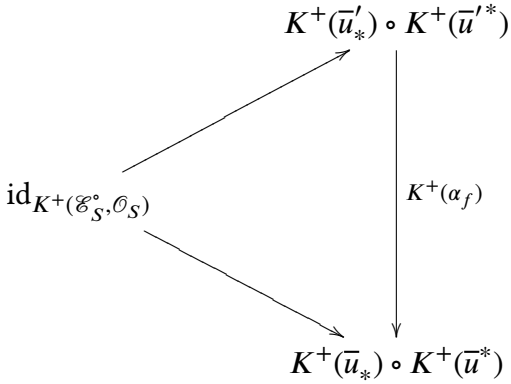


is commutative.

[25] N.D.E.: Explicitly, one has $(XfX')_{(0,i)} = X_i$ and $(XfX')_{(1,j)} = X'_j$; to $\alpha \in \mathrm{Hom}_D(i,j) = \mathrm{Hom}((0,i),(0,j))$ is associated $\alpha\circ f_i : (XfX')_{(0,i)} \to (XfX')_{(1,j)}$, and the endomorphisms of (ε, i), $\varepsilon = 0, 1$, act in the evident way.

One consequently obtains a functorial morphism such that the diagram

(3.1.1.7)



is commutative.
(Observe that $K^+(\alpha_f)$ can also be obtained formally by adjunction, using the isomorphism $K^+(\Gamma(f^*)) \circ K^+(\overline{u}'^*) \xrightarrow{\sim} K^+(\overline{u}^*)$.)

Let $F^\cdot$ be a complex in $K^+(S)$, and let $\xi : K^+(\overline{ufu'}^*)(F^\cdot) \to J^\cdot$ be a quasi-isomorphism such that, for every integer n, J^n is totally acyclic objectwise (cf. (1.3.13)). By (*loc. cit.*), the canonical arrows

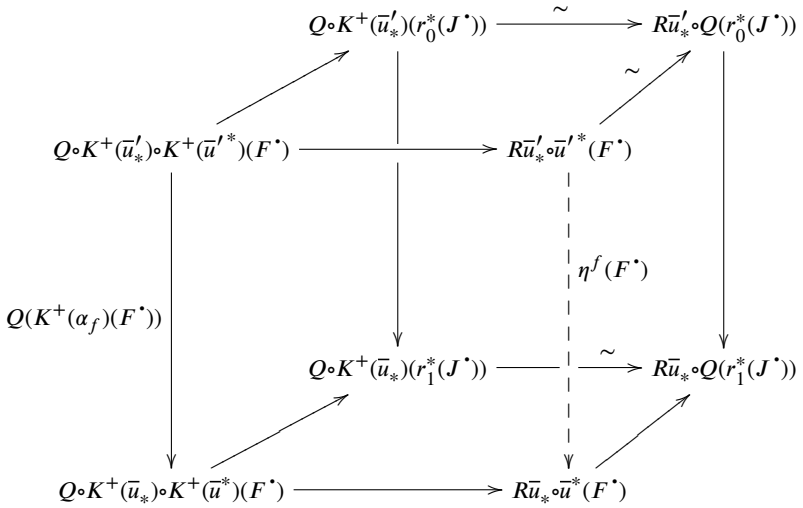
$$K^+(\overline{u}'^*)(F^\cdot) \longrightarrow r_1^*(J^\cdot) \quad \text{and} \quad K^+(\overline{u}^*)(F^\cdot) \longrightarrow r_0^*(J^\cdot)$$

are quasi-isomorphisms (cf. (1.2.8) for the notation). One therefore obtains a morphism 127

$$R\overline{u}'_* \circ \overline{u}'^*(F^\cdot) \xrightarrow{\eta^f(F^\cdot)} R\overline{u}_* \circ \overline{u}^*(F^\cdot)$$

such that the following diagram is commutative in $D^+(S)$:

(3.1.1.8)



It is clear that $\eta^f(F^\cdot)$ is independent of the chosen resolution, and one verifies that it is functorial in $F^\cdot$ as the latter varies in $D^+(S)$.

To complete the proof of (3.1.1), it is enough to prove the following lemma:

LEMMA 3.1.1.9. *Let* $X \xrightarrow{u} C_S^\Delta$ *and* $X' \xrightarrow{u'} C_S^\Delta$ *be two objects of* $\mathrm{Hom}_{\mathrm{plat}}(\Delta^\circ, B/S)$. *Let* 128 $f, g : X \to X'$ *be two morphisms in* $\mathrm{Hom}(\Delta^\circ, B/S)$. *If there exists a simplicial homotopy from* f *to* g, *then* $\eta^f = \eta^g$.

Let $h_n : \mathrm{Hom}_\Delta([n],[1]) \to \mathrm{Hom}_{B/S}(X_n, X'_n)$ be a simplicial homotopy from f to g. Let $\overline{I \times D}$ be the category obtained from $I \times D$ by adjoining the arrows $(1,n) \to (0,n)$ in bijective correspondence with $h_n(\mathrm{Hom}([n],[1]))$, together with the relations imposed by the compatibility of the h_n as n varies. The data of the lemma define a flat augmentation $\overline{X f X'} \xrightarrow{t} C_S^{\overline{I\times D}}$. Let $F^\cdot$ be a complex in $C^+(S)$, and let $J^\cdot$ be a resolution of $C^+(t^*)(F^\cdot)$ such that, for every integer n, J^n is totally acyclic objectwise. There exists a simplicial homotopy between

$$C^+(\Gamma(u'_*))(r_0^*(J^\cdot)) \rightrightarrows C^+(\Gamma(u_*))(r_1^*(J^\cdot)),$$

and hence a homotopy after passage to the total complexes (cf. (3.0.2.3)).

Definition 3.1.2. *With the notation of (3.1.1), we say that f is an* equivalence for G-1-cohomological descent *if η_G^f is an isomorphism.*

3.1.3. In what follows, denote by $L_i : \Delta \to \Delta \times \Delta$ (resp. $C_i : \Delta \to \Delta \times \Delta$) the canonical functor defined by $[n] \longmapsto [n] \times [i]$ (resp. $[n] \longmapsto [i] \times [n]$).

Proposition 3.1.4. *Let $X \xrightarrow{u} C_S^{\Delta\times\Delta}$ and $X' \xrightarrow{u'} C_S^{\Delta\times\Delta}$ be two objects of $\mathrm{Hom}_{\mathrm{plat}}((\Delta\times\Delta)^\circ, B/S)$. Let $f : X \to X'$ be a morphism over S. Suppose that $f * \mathrm{id}_{L_i} : X \circ L_i \to X' \circ L_i$ (resp. $f * \mathrm{id}_{C_i} : X \circ C_i \to X' \circ C_i$) is an equivalence for G-1-cohomological descent for every integer i. Then f is an equivalence for G-1-cohomological descent.*

By the description of η^f (cf. the proof of (3.1.1)), it is enough to show that a certain morphism of triple complexes induces an isomorphism on the cohomology of the total complexes. A standard spectral-sequence argument then gives the conclusion.

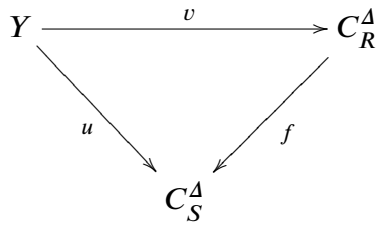
129 Proposition 3.1.5. *Let X and X' be two objects of $\mathrm{Hom}_{\mathrm{plat}}(\Delta^\circ, B/S)$. Let $f : X \to X'$ be a functorial morphism such that $f_n : X_n \to X'_n$ is a morphism of G-1-cohomological descent. Then the canonical morphism*

$$[[X|_f X']] \longrightarrow [[X'|_{\mathrm{id}} X']]$$

(cf. (1.2.7)) *is an equivalence for G-1-cohomological descent.*

This follows from (3.1.4) and the following lemma, whose proof is left to the reader:

Lemma 3.1.5.1. *Let $f : R \to S$ be a flat morphism, and let $Y \xrightarrow{u} C_S^\Delta$ and $Y \xrightarrow{v} C_R^\Delta$ be two* flat *augmentations such that the following diagram is commutative in $\mathrm{Hom}(\Delta^\circ, B)$:*



If v is an augmentation of G-1-cohomological descent, then it is an equivalence for G-1-cohomological descent.

In the applications, we shall combine (3.1.4) and (3.1.5) with the following result:

Proposition 3.1.6. *Let $X \xrightarrow{u} C_S^{\Delta\times\Delta}$ be a flat augmentation. For u to be an augmentation of G-1-cohomological descent, it is enough that*

$$X \circ L_i \xrightarrow{u*\mathrm{id}_{L_i}} C_S^\Delta \quad (\textit{resp.} \quad X \circ C_i \xrightarrow{u*\mathrm{id}_{C_i}} C_S^\Delta)$$

be so for every integer i.

This follows from the explicit construction of $R_G\bar{u}_*$ (cf. (2.3.10)).

COROLLARY 3.1.7. *Let $X \overset{u}{\to} C_S^\Delta$ be an augmentation. Then u is of G-1-cohomological descent if and only if $[[X|_{\mathrm{id}}X]] \to C_S^{\Delta\times\Delta}$ is so.*

130

3.2. Localization criteria.

PROPOSITION 3.2.1. *Let $Y \overset{v}{\to} C_S^\Delta$ be an augmentation of G-cohomological 1-descent. For a flat augmentation $X \overset{u}{\to} C_S^\Delta$ to be of G-cohomological 1-descent, it suffices that it become so after all base changes $Y_n \to S$.*

By means of the base changes $Y_n \to S$, one constructs a double semi-simplicial object $X \times_S Y$ augmented toward S. By (3.1.4) and (3.1.5.1), the canonical morphism

$$X \times_S Y \longrightarrow [[X|_{\mathrm{id}}X]]$$

is an equivalence for G-cohomological 1-descent. By (3.1.7), it therefore suffices to show that the augmentation of $X\times_S Y$ toward S is of G-cohomological 1-descent. This follows from the fact that the canonical morphism

$$X \times_S Y \longrightarrow [[Y|_{\mathrm{id}}Y]]$$

is an equivalence for G-cohomological 1-descent; again one uses (3.1.4), (3.1.5.1), and (3.1.7).

3.2.2. Let

(3.2.2.1)
$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow[g]{} & X \end{array}$$

be a commutative diagram in B, and let F be an object of $\mathrm{Mod}(\mathscr{E}_Y^\circ, \mathscr{O}_Y)$. There is a unique morphism

$$\varphi(F) : g_*(f^*(F)) \longrightarrow f'^*(g'_*(F))$$

such that the following diagram (in $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O})$) commutes[26]:

(3.2.2.2)
$$\begin{array}{ccccc} & & f^*(F) & \xrightarrow{f^\circ} & F \\ & {\scriptstyle g^\circ}\swarrow & & & \downarrow{\scriptstyle g'^\circ} \\ g_*(f^*(F)) & & & & g'_*(F) \\ & {\scriptstyle \mathrm{Id}}\searrow & & \nearrow{\scriptstyle f'^\circ} & \\ & & f'^*(g'_*(F)) & & \end{array}$$

[26]N.D.E.: As above, one must regard the commutative diagram as lying in B° and use the fibered and cofibered structure of $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O}) \to B^\circ$, which explains the unusual appearance of the base-change morphism. Under the (adjunction) arrows in the following diagram we have written their images in B°

131 If all the arrows of (3.2.2.1) are now assumed flat, one defines in the same way a morphism, called the *base-change* morphism[27]

$$\xi_G : g_{G*} \circ R_G f^* \longrightarrow R_G f'^* \circ g'_{G*}. \tag{3.2.2.3}$$

Definition 3.2.2.4. *The diagram* (3.2.2.1) is said to satisfy the base-change theorem relative to G if f, g, f', and g' *are flat morphisms and* ξ_G *is an isomorphism.*

3.2.3. Let $h : S \to S'$ be a *flat* morphism in B, let $X \xrightarrow{u} C_S^\Delta$ and $X' \xrightarrow{u'} C_{S'}^\Delta$ be two *flat* augmentations, and let $f : X \to X'$ be a functorial morphism such that the diagram (in $\mathrm{Hom}(\Delta^0, B)$)

$$\begin{array}{ccc} X & \xrightarrow{f} & X' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u'} \\ C_S^\Delta & \xrightarrow{h_c} & C_{S'}^\Delta \end{array} \tag{3.2.3.1}$$

commutes. (Here h_c denotes the constant functorial morphism defined by h.)

Let $K^\cdot$ be a complex in $D_G^+(S')$. The morphism

$$\eta_G^f(K^\cdot) : R\overline{u}'_* \circ \overline{u}'^*(K^\cdot) \longrightarrow R(\overline{h_c \circ u})_* \circ (\overline{h_c \circ u})^*(K^\cdot)$$

induces, in view of the identities

$$R(\overline{h_c \circ u})_* \simeq Rh_c^* \circ R\overline{u}_* \quad \text{and} \quad (\overline{h_c \circ u})^* \simeq \overline{u}^* \circ h_{c*},$$

a morphism $\mathrm{ch}_G^{f,h}(K^\cdot)$, functorial in $K^\cdot$, such that the following diagram commutes[28]:

$$\begin{array}{ccc} h_{c*} \circ R\overline{u}'_* \circ \overline{u}'^*(K^\cdot) & \xrightarrow{\mathrm{ch}_G^{f,h}(K^\cdot)} & R\overline{u}_* \circ \overline{u}^* \circ h_{c*}(K^\cdot) \\ & {\scriptstyle h_{c*}(\alpha(K^\cdot))}\nwarrow \quad \nearrow & \\ & h_{c*}(K^\cdot) & \end{array} \tag{3.2.3.2}$$

132 Lemma 3.2.3.3. *Suppose that, for every integer* i*, the diagram*

[27] N.D.E.: Here again, it would be more correct to write f°, g°, f'°, g'° instead of f, g, f', g'. If one considers the nonopposite fibered and cofibered structure, ξ_G becomes

$$g_G^* \circ R_G f_* \longrightarrow R_G f'_* \circ g_G'^*.$$

[28] N.D.E.: For readability, we have suppressed the reference to the thick category G in the derived functors. The same unusual derivations h_{c*} reappear, owing to the choice of the cofibered structure.

$$\begin{array}{ccc} X_i & \xrightarrow{f_i} & X'_i \\ {\scriptstyle u_i}\downarrow & & \downarrow{\scriptstyle u'_i} \\ S & \xrightarrow{h} & S' \end{array}$$

satisfies the base-change theorem relative to G. *Then* $\mathrm{ch}_G^{f,h}$ *is an isomorphism.*

The morphism $\mathrm{ch}_G^{f,h}(K^\cdot)$ may be computed as follows. Consider the augmentation

$$X f X' \xrightarrow{(h_c \circ u) f u'} C_S^{I \times \Delta}$$

(cf. (3.1.1.2)), and choose a resolution $J^\cdot$ of $K^+(\overline{(h_c \circ u) f u'}^*)(K^\cdot)$ such that, for every integer n, J^n is totally acyclic object by object.

One obtains a quasi-isomorphism

$$\theta : K^+(\Gamma(f^*))(r_0^* J^\cdot) \longrightarrow r_1^*(J^\cdot)$$

and a morphism

$$t : K^+(\Gamma(h_c^*)) \circ K^+(\Gamma(u'_*))(r_0^*(J^\cdot)) \longrightarrow K^+(\Gamma(u_*))(r_1^*(J^\cdot))$$

which has the factorization

$$K^+(\Gamma(h_c^*)) \circ K^+(\Gamma(u'_*))(r_0^*(J^\cdot)) \xrightarrow{\varphi} K^+(\Gamma(u_*)) \circ K^+(\Gamma(f^*))(r_0^*(J^\cdot)) \xrightarrow{K^+(\Gamma(u_*))(\theta)} \longrightarrow K^+(\Gamma(u_*))(r_1^*(J^\cdot)),$$

and $\mathrm{ch}^{f,h}(K^\cdot)$ is obtained by taking the image of t under the functor $R\epsilon_{Z^{\cdot *}}$. 133

The preceding factorization shows that t identifies "object by object" with the base-change morphisms associated with the diagrams

$$\begin{array}{ccc} X_i & \xrightarrow{f_i} & X'_i \\ {\scriptstyle u_i}\downarrow & & \downarrow{\scriptstyle u'_i} \\ S & \xrightarrow{h} & S' \end{array}$$

It follows, by spectral sequences, that $\mathrm{ch}_G^{f,h}(K^\cdot)$ is an isomorphism, completing the proof.

This leads to a *second localization criterion*:

Proposition 3.2.4. *Let* $X \xrightarrow{u} C_S^\Delta$ *be an object of* $\mathrm{Hom}_{\mathrm{plat}}(\Delta^\circ, B/S)$. *For* u *to be an augmentation of* G*-cohomological* 1*-descent, it suffices that it become so after "enough for* G*"*[28] *flat base changes* $h : S' \to S$ *for which the cartesian diagrams*

$$\begin{array}{ccc} X'_i & \longrightarrow & X_i \\ \downarrow & & \downarrow \\ S' & \xrightarrow{h} & S \end{array}$$

[28]The expression "enough for G" means that there exists a family $(S_\alpha \xrightarrow{h_\alpha} S)_{\alpha \in A}$ of flat morphisms satisfying the preceding conditions and such that the family of functors $(h_{\alpha *} : G_S \to G_{S_\alpha})_{\alpha \in A}$ is *conservative*.

satisfy the base-change theorem relative to G.

134 Proposition (3.2.4) follows immediately from (3.2.3.3), in view of diagram (3.2.3.2).

3.3. Properties of morphisms of cohomological descent. In this section, we assume that the *class of flat morphisms in B is stable under base change.*

PROPOSITION 3.3.1. *Let $i = 1, 2$.*

a) *Every flat morphism which has a section is a morphism of universal G-cohomological 2-descent.*
b) *Let $f : X \to S$ be a flat morphism, let $g : S' \to S$ be a morphism of G-cohomological i-descent, put $X' = X \times_S S'$, and let $f' = f_{(S')} : X' \to S'$. Then f is of universal G-cohomological i-descent if and only if f' is.*
c) *If the composite of two morphisms $f : X \to Y$ and $g : Y \to Z$ is of universal G-cohomological i-descent, then g is of universal G-cohomological i-descent.*
d) *The composite of two morphisms of universal G-cohomological i-descent is a morphism of universal G-cohomological i-descent.*
e) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms of universal G-cohomological i-descent, then $f \times_S g$ is of universal G-cohomological i-descent.*
f) *Let $(u_\alpha : X_\alpha \to Y_\alpha)_{\alpha \in A}$ be a family of morphisms of G-cohomological i-descent. Then $\amalg u_\alpha : \amalg_{\alpha \in A} X_\alpha \to \amalg_{\alpha \in A} Y_\alpha$ is a morphism of G-cohomological i-descent.*

135 *Proof.* For effective descent relative to G, we refer to [4].

a) Using a section s of f, compare the semi-simplicial objects augmented toward S, $[X|_f S]$ and $[S|_{\mathrm{id}} S]$. By (3.0.2.4), one obtains an equivalence for G-cohomological 1-descent[29].
b) This follows from (3.2.1).
c) This follows from a) and b).
d) Let $f : X \to Y$ and $g : Y \to S$ be two morphisms of universal G-cohomological 1-descent. Put $h = g \circ f$ and consider the fiber product

$$\begin{array}{ccc} Y & \xleftarrow{h'} & R \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g'} \\ S & \xleftarrow[h]{} & X \end{array}$$

The morphism g' has a section $s : X \to R$ such that $h' \circ s = f$. By c), h' is of universal G-cohomological 1-descent, and the same holds for h by b).
e) This follows formally from b) and d).
f) Let $(Z_\lambda)_{\lambda \in \Lambda}$ be a family of objects of B. It follows from hypotheses (3.0.0) that there is a canonical equivalence

$$\mathrm{Mod}\left(\mathscr{E}^\circ_{\amalg_\lambda Z_\lambda}, \mathcal{O}_{\amalg_\lambda Z_\lambda}\right) \xrightarrow{\sim} \prod_\lambda \mathrm{Mod}(\mathscr{E}^\circ_{Z_\lambda}, \mathcal{O}_{Z_\lambda})$$

which induces an equivalence

$$G_{\amalg_\lambda Z_\lambda} \xrightarrow{\sim} \prod_\lambda G_{Z_\lambda}.$$

[29] N.D.E.: For this, note that $[X|_f S]$ and $[S|_{\mathrm{id}} S]$ are simply the coskeleta of order 0 of f and id, respectively, so that s and f induce inverse isomorphisms in cohomology by 3.1.1.

Moreover, a family $(Q_\lambda)_\lambda \in \prod_\lambda \mathrm{Mod}(\mathscr{E}^\circ_{Z_\lambda}, \mathcal{O}_{Z_\lambda})$ corresponds to a totally acyclic object if and only if Q_λ is totally acyclic for every λ.

Now let $(u_\lambda : X_\lambda \to Y_\lambda)_{\lambda\in\Lambda}$ be a family of morphisms of G-cohomological 1-descent. Since direct sums in B are disjoint and universal, for every n there is an identification 136

$$\left[\coprod_\lambda X_\lambda \,|_{\coprod u_\lambda}\, \coprod Y_\lambda\right]_n \simeq \coprod_\lambda [X_\lambda|_{u_\lambda} Y_\lambda]_n$$

which is functorial in n. It then follows from the preceding remarks and the explicit calculation of $R_G\theta_*$ (cf. (2.3)) that $\coprod_\lambda u_\lambda$ is a morphism of G-cohomological 1-descent; the details are left to the reader. The reader may also verify that $\coprod_\lambda u_\lambda$ *remains of G-cohomological 1-descent after every base change if the same is true of every u_λ.*

3.3.2. Associate with each object X of B the set of families of morphisms $(X_\alpha \to X)_{\alpha\in A}$ such that $\coprod_\alpha X_\alpha \to X$ is a morphism of universal G-i-cohomological descent. By 3.3.1, this defines a pretopology on B.

The topology generated by this pretopology is called the *topology of G-i-cohomological descent.* It follows from 3.3.1 c) that the covering morphisms for this topology are precisely the morphisms of universal G-i-cohomological descent. Finally, note that the topology of G-2-cohomological descent is coarser than the topology of G°-descent [cf. [4] (6.23)].

Theorem 3.3.3. *Suppose that* all arrows of B are flat. *Let S be an object of B. Every hypercovering of S for the topology of G-1-cohomological descent whose objects are all representable* (cf. V, Appendix) *defines an augmentation of universal G-1-cohomological descent.*

The proof proceeds in two steps; we specify that all coskeleta are to be computed in the category B/S.

Lemma 3.3.3.1. *Let X be an object of* $\mathrm{Hom}(\Delta^\circ, B/S)$. *For X to be of* 1*-cohomological descent (resp. universal* 1*-cohomological descent), it is enough that* $\mathrm{cosq}_n(X)$ *have this property for all sufficiently large n.* 137

Indeed, let $\alpha_n : X \to \mathrm{cosq}_n(X)$ be the canonical morphism. One checks that if $K^\cdot$ is a complex of $D^+_{G_S}(S)$ such that $H^j(K^\cdot) = 0$ for $j < N$, then $H^i(\eta_G^{\alpha_n})$ is an isomorphism for $i < N + n$, whence the assertion.

Lemma 3.3.3.2. *Let $n \geq 0$, let X and X' be two simplicial objects of B/S, and let $f : X \to X'$ be a morphism. Suppose that:*

(i) *$X \to \mathrm{cosq}_{n+1}(X)$ is an isomorphism;*
(ii) *$X' \to \mathrm{cosq}_{n+1}(X')$ is an isomorphism;*
(iii) *$f_i : X_i \to X'_i$ is an isomorphism for $i \leq n$;*
(iv) *f_{n+1} is a morphism of universal G-1-cohomological descent.*

Then, if X is of universal G-1-cohomological descent, the same is true of X'.

Lemmas 3.3.3.1 and 3.3.3.2 clearly prove Theorem 3.3.3 by induction.

Lemma 3.3.3.3. *Under the hypotheses of 3.3.3.2, all the morphisms f_p are morphisms of universal G-1-cohomological descent.*

This is trivial for $p \leq n+1$. For $p > n+1$, X_p (resp. X'_p) may be written as $\varprojlim_{\substack{\Delta^+ \\ n+1[p]}} X_q$ (resp. $\varprojlim_{\substack{\Delta^+ \\ n+1[p]}} X'_q$).

We may suppose that $p > n+1$. Write

$$\mathrm{cosq}_{n+1}(X)_p = \varprojlim_{\substack{[q]\to[p] \\ q\le n+1}} X_q$$

as the kernel of the pair of arrows

$$\Pi X = \prod_{\substack{[q]\to[p] \\ q\le n+1}} X_q \rightrightarrows \prod_{\substack{\alpha \\ [i]\hookrightarrow[j] \\ \searrow\swarrow \\ [p] \\ j\le n+1}} X_i.$$

The component α_X indexed by $\alpha \in \mathrm{Hom}_{[p]}([i],[j])$ of this pair consists, on the one hand, of the projection $\Pi X \to X_i$ indexed by $[i] \to [p]$, and, on the other hand, of the composite

$$\Pi X \longrightarrow X_j \longrightarrow X_i$$

of the projection indexed by $[j] \to [p]$ with $\alpha \in \mathrm{Hom}(X_j, X_i)$.

It follows that the diagram

$$\begin{array}{ccc} \mathrm{cosq}_{n+1}(X)_p & \longrightarrow & \Pi X \\ \downarrow & & \downarrow \\ \mathrm{cosq}_{n+1}(X')_p & \longrightarrow & \Pi X' \end{array}$$

is Cartesian. Indeed, by the preceding description, it is enough to prove that for every *increasing injection* $\alpha \in \mathrm{Hom}_{[p]}([i],[j])$, the diagram

$$\begin{array}{ccc} \mathrm{Ker}(\alpha_X : \Pi X \rightrightarrows X_i) & \longrightarrow & \Pi X \\ \downarrow & & \downarrow \\ \mathrm{Ker}(\alpha_{X'} : \Pi X' \rightrightarrows X'_i) & \longrightarrow & \Pi X' \end{array}$$

is Cartesian. If $i = n+1$, then $\alpha : [i] \hookrightarrow [j]$ is the identity of $[n+1]$, so the pair of arrows imposes no condition. The respective kernels are ΠX and $\Pi X'$, which settles this case. Otherwise $i < n+1$ and $X_i \to X'_i$ is an isomorphism, so the diagram is Cartesian in this case as well (testing an equality in X_i or X'_i amounts to the same thing)[30].

Using 3.3.1, the relevant morphisms are therefore morphisms of universal G-1-cohomological descent. Let $[X'/X]^p$ be the $(p+1)$-fold fiber product of X' over X. By 3.3.3.3, it is enough to verify 3.3.3.2 after a base change $[X'/X]^p \to X$. After such a base change, the hypotheses of 3.3.3.2 still hold and, moreover, f_{n+1} admits a section. Lemma 3.0.2.4 then completes the proof.

For effective descent, we have the following result.

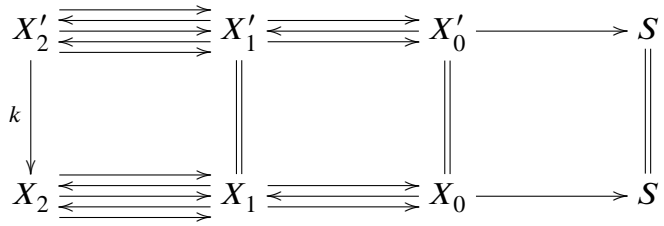
Proposition 3.3.4. *Let $X : \Delta^\circ \to B/S$ be a functor. Suppose that $X_0 \to S$ is a morphism of universal G°-2-descent, as are the morphisms $X_{n+1} \to (\mathrm{cosq}_n(X))_{n+1}$ for $n = 0, 1$. Then the functor X is G°-2-faithful and remains so after every base change (in other words, X defines an augmentation of universal effective descent in the sense of 2.4.11).*

By [4] (7.12), it is enough to show that $i_2^*(X)$ is G°-2-faithful; for this we use the following lemmas.

138 Lemma 3.3.4.1. *Let X and X' be two functors $\Delta_2^\circ \to B/S$, and consider a commutative*

[30] N.D.E.: The proof of Lemma 3.3.3.3 has been slightly rewritten.

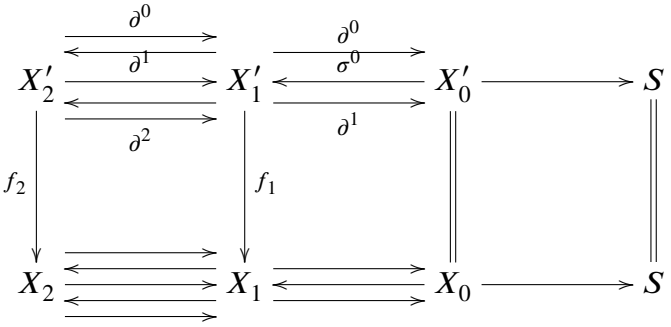
diagram



such that k is a morphism of G°-0-descent. If X is G°-2-faithful, then so is X'.

This is evident.

LEMMA 3.3.4.2. *Let X and X' be two functors $\Delta_2^\circ \to B/S$, and consider a commutative diagram*



such that f_1 is of G°-1-descent and f_2 is of G°-0-descent. Suppose in addition that $X'_2 \xrightarrow{\sim} (\mathrm{cosq}_1(X'))_2$. If X is G°-2-faithful, then so is X'.

Let $X''_1 = X'_1 \times_{X_1} X'_1$, and let $\partial_0 : X''_1 \to X'_1$ and $\partial_1 : X''_1 \to X'_1$ be the two projections.

The definition of a coskeleton gives an arrow $\varphi : X''_1 \to X'_2$ such that

$$\begin{cases} \partial_0 \circ \varphi = \partial_0, \\ \partial_1 \circ \varphi = \partial_1, \\ \partial_2 \circ \varphi = \sigma_0 \partial_1 \partial_0 = \sigma_0 \partial_1 \partial_1. \end{cases}$$

Now take a descent datum on X': it gives an object on X_0, two objects on X_1, and an isomorphism between them on X'_1. The arrow φ shows that the two inverse images of these isomorphisms on X''_1 are equal. Since f_1 is of 1-descent, we obtain an isomorphism between the two objects on X_1; this isomorphism is a descent datum (because $X'_2 \to X_2$ is of 0-descent), and the result follows.

COROLLARY 3.3.5. *Suppose that* all arrows of B are flat. *Every hypercovering of S for the topology of G-2-cohomological descent whose objects are representable defines an augmentation of universal G-2-cohomological descent.*

4. Examples

4.1. Sheaves of abelian groups on topological spaces.

4.1.0. In this section, Top denotes the category of topological spaces [elements of the fixed universe $\mathscr{U}$]. We define a category $\mathscr{E}$ bifibered in duals of topoi over Top by taking as objects the pairs (X, F), where X is a topological space and F is a sheaf of sets on X, and as morphisms the pairs $(f, \varphi) : (X, F) \to (Y, H)$, where $f : X \to Y$ is a continuous map and $\varphi : H \to f_*(F)$ is a morphism of sheaves. We take for $\mathscr{O}$ the section of $\mathscr{E}^0$ over Top^0 that associates with each topological space the constant sheaf

$\mathbf{Z}_X$: we set $G = \mathrm{Mod}(\mathscr{E}^0, \mathscr{O})$, so that, for every topological space X, G_X identifies with the category of sheaves of abelian groups on X.

Recall that a morphism $f : X \to Y$ is *separated* if the diagonal in $X \times_Y X$ is closed. A *proper* morphism is a separated and universally closed morphism (note that this definition is more restrictive than Bourbaki's).

139 The proof of the "base-change theorem" below is inspired by ([5] II(4.11.1)).

THEOREM 4.1.1. *Let $f : X \to Y$ be a proper morphism, let F be an abelian sheaf on X, and consider a Cartesian diagram*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y. \end{array}$$

Then the canonical map (XII 4.2) *or* (3.2.2.3)

$$g^* R^i f_* F \xrightarrow{\sim} R^i f'_* g'^* F$$

is an isomorphism[31].

This theorem is equivalent to the following corollary (the corollary is obtained by taking $Y' = (\mathrm{Point})$; the theorem is obtained by applying the corollary to f and f').

COROLLARY 4.1.2. *Let $f : X \to Y$ be a proper morphism, let F be an abelian sheaf on X, and let $y \in Y$. The canonical map*

$$(R^i f_* F)_y \longrightarrow H^i(f^{-1}(y), F)$$

is bijective.

By definition, as U ranges over the neighborhoods of y, we have

$$(R^i f_* F)_y = \varinjlim H^i(f^{-1}(U), F).$$

Since f is closed, the $f^{-1}(U)$ form a fundamental system of neighborhoods of $f^{-1}(y)$, and the result follows from the following lemma, (4.1.3).

140 LEMMA 4.1.3. *Let X be a topological space, let $K \subset X$, and let F be an abelian sheaf on X. Suppose that K is compact and that any two distinct points of K have disjoint neighborhoods in X. Then, as U ranges over the neighborhoods of K, we have*

$$\varinjlim H^i(U, F) \xrightarrow{\sim} H^i(K, F). \tag{4.1.3.1}$$

We first treat the case $i = 0$. In this case, it is clear that (4.1.3.1) is injective. For surjectivity, we use the following result.

LEMMA 4.1.4. *Under the hypotheses of* (4.1.3), *if A and B are two closed subsets of K and W is a neighborhood of $A \cap B$ in X, then there are neighborhoods U and V of A and B in X such that $U \cap V \subset W$.*

[31] N.D.E.: Unlike above, these are the derived functors of the usual functors, and not those arising from the fibered-category structure $G^\circ \to \mathrm{Top}^\circ$.

We first treat the case where A consists of a single point a. The assertion is trivial if $a \in B$ (take $U = W$). If $a \notin B$, then for every $b \in B$ there are disjoint open neighborhoods U_b and V_b of a and b in X. We take for V a finite union of the V_b that covers B, and for U the intersection of the corresponding U_b.

In the general case, for every $a \in A$, there are open neighborhoods U_a and V_a of a and B in X such that $U_a \cap V_a \subset W$. We take for U a finite union of the U_a that covers A, and for V the corresponding intersection of the V_a.

Return to (4.1.3). If $s \in H^*(K, F)$, there is an open covering U_i of K in X and there are sections $s_i \in H^*(U_i, F)$ such that $s_i = s$ on $U_i \cap K$. We may suppose that there are only finitely many U_i. Let (K_i) be a closed covering of K, with $K_i \subset K \cap U_i$. Let W_y be the open subset of $U_i \cap U_j$ on which $s_i = s_j$; we have $K_i \cap K_j \subset W_{ij}$. Apply (4.1.4) to all pairs (K_i, K_j) and to the W_{ij}: we find open subsets V_{ij} with $K_i \subset V_{ij} \subset U_i$ and $V_{ij} \cap V_{ji} \subset W_{ij}$. Let $U_i' = \bigcap_j V_{ij}$: then $K_i \subset U_i' \subset U_i$, and $s_i = s_j$ on $U_i' \cap U_j'$. The s_i therefore glue on the neighborhood of K given by the union of the U_i', and (4.1.3.1) is surjective (hence bijective) for $i = 0$[32]. 141

LEMMA 4.1.5. *If F is flabby, the sheaf $F|K$ on K is soft.*

Let $A \subset K$ be closed in K. Every section of F on A extends to a neighborhood, by what precedes. Since F is flabby, it extends to X, and hence in particular to K.

We now prove (4.1.3). Let F^* be a flabby resolution of F. We have

$$\varinjlim H^i(U, F) \overset{33}{=} \varinjlim H^i(\Gamma(U, F^*)) = H^i \varinjlim \Gamma(U, F^*) \overset{34}{=} H^i\Gamma(K, F^*) \overset{(4.1.5)}{=} H^i(K, F).$$

COROLLARY 4.1.6. *Let $f : X \to Y$ be a proper surjective morphism of topological spaces. Then f is of G-1-cohomological descent.*

This follows from (4.1.2) and (3.2.4).

COROLLARY 4.1.7. *Let $\mathcal{U} = (U_i)_{i\in I}$ be a locally finite closed covering of a topological space X. Then the canonical morphism $f : \coprod_{i\in I} U_i \to X$ is a morphism of G-1-cohomological descent.*

Indeed, f is proper and surjective. Applying (2.5.6), we recover the spectral sequence of [[5]. II (5.2.4)] (cf. the Introduction).

PROPOSITION 4.1.8. *Let $f : X \to Y$ be a surjective local homeomorphism of topological spaces. Then f is a morphism of universal G-1-cohomological descent.*

By localization on Y (cf. (3.2.4)), we are reduced to the case where f has a section.

COROLLARY 4.1.9. *Let $\mathcal{U} = (U_i)_{i\in I}$ be an open covering of a topological space X. Then the canonical morphism $f : \coprod_{i\in I} U_i \to X$ is a morphism of universal G-1-cohomological descent.* 142

The descent spectral sequence then gives [[5] II (5.4.1)] (cf. the Introduction).

REMARK 4.1.10. The proofs of (VIII 9) show that "1-cohomological descent" may be replaced by "2-cohomological descent" in all the preceding statements.

[32] N.D.E.: The two meanings of the symbol i should not be confused.

[33] N.D.E.: $H^i(\Gamma(U, F^*))$ denotes here the i-th cohomology group of the complex $\Gamma(U, F^*)$.

[34] N.D.E.: By (4.1.3), applied with $\Gamma = H^0$

4.2. Quasi-coherent modules on schemes.

4.2.0. Let $(\mathrm{Sch})_s$ be the category whose objects are the schemes that are elements of $\mathscr{U}$ and whose arrows are the quasi-compact and quasi-separated morphisms. We define a category $\mathscr{E}$ bifibered in duals of topoi over $(\mathrm{Sch})_s$ by taking as objects the pairs (X, F), where X is a scheme and F a sheaf on X (for the Zariski topology), and as morphisms the pairs $(f, \varphi) : (X, F) \to (Y, G)$, where $f : X \to Y$ is a continuous map and $\varphi : G \to f_*(F)$ is a morphism of sheaves. For $\mathscr{O}$ we take the section of $\mathscr{E}^0$ over $(\mathrm{Sch})_s^0$ that associates with every scheme its structure sheaf; for G we take the subcategory of $\mathrm{Mod}(\mathscr{E}^0, \mathscr{O})$ such that, for every scheme X, G_X is the category of quasi-coherent modules on X.

Proposition 4.2.1. *Every faithfully flat, quasi-compact, and quasi-separated morphism $f : X \to Y$ is a morphism of universal G-2-cohomological descent.*

By [SGA I (VIII.5.2)], f is a morphism of effective G^0-descent. By [EGA IV (1.7.21)], we may apply (3.2.4) to the Cartesian square

$$\begin{array}{ccc} X \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow f \\ X & \xrightarrow[f]{} & Y \end{array}$$

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4.3. Étale sheaves on schemes.

4.3.0. Let (Sch) be the category of all schemes belonging to $\mathscr{U}$. We thus define a category $\mathscr{E}$ bifibered in duals of topoi over (Sch) by taking as objects the pairs $(X, \mathscr{F})$, where X is a scheme and $\mathscr{F}$ an étale sheaf on X, and as morphisms the pairs $(f, \varphi) : (X, \mathscr{F}) \to (Y, \mathscr{G})$, where $f : X \to Y$ is a morphism of schemes and $\varphi : \mathscr{G} \to f_*(\mathscr{F})$ is a morphism of sheaves.

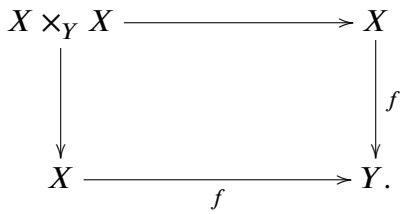
Let n be an integer ≥ 0. We denote by $\mathscr{O}_n$ the section of $\mathscr{E}^0$ over $(\mathrm{Sch})^0$ that associates with every scheme X the constant sheaf $(\mathbf{Z}/n\mathbf{Z})_X$. Put $F_n = \mathrm{Mod}(\mathscr{E}^0, \mathscr{O}_n)$, so that, for every scheme X, F_{n_X} identifies with the category of sheaves of $\mathbf{Z}/n\mathbf{Z}$-modules on X.

4.3.1. Now let $(\mathrm{Sch})_s$ be the subcategory of (Sch) whose objects are the schemes that are elements of $\mathscr{U}$ and whose arrows are the quasi-compact and quasi-separated morphisms. Denote by $\mathscr{E}_s$ the category bifibered in duals of topoi over $(\mathrm{Sch})_s$ deduced from $\mathscr{E}$ by the base change $(\mathrm{Sch})_s \to (\mathrm{Sch})$; we again denote by $\mathscr{O}_0$ the restriction of $\mathscr{O}_0$ to $(\mathrm{Sch})_s$. Let $G \subset \mathrm{Mod}(\mathscr{E}_s^0, \mathscr{O}_0)$ be the fibered and cofibered subcategory of $\mathrm{Mod}(\mathscr{E}_s^0, \mathscr{O}_0)$ such that, for every scheme X, G_X is the category of torsion abelian sheaves on X. The reader will note that condition (2.4.1.1) is satisfied.

Proposition 4.3.2. *Let $f : X \to Y$ be a* proper surjective *morphism. Then f is a morphism of universal F_n-2-cohomological descent for $n \geq 1$. Considered as a morphism of $(\mathrm{Sch})_s$, f is a morphism of universal G-2-cohomological descent.*

144 By (VIII 9.4), f is a morphism of effective F^0-descent, and by (XII.(5.1)) we may

apply (3.2.4) to the Cartesian diagram



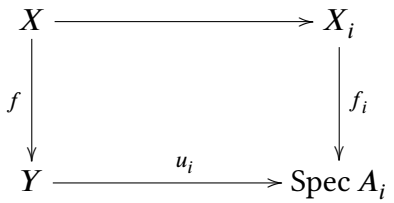
PROPOSITION 4.3.3. *Let $f : X \to Y$ be a* surjective *morphism of schemes. Suppose that one of the following hypotheses is satisfied:*

a) *f is integral.*
b) *Y is locally Noetherian, and f is universally open and locally of finite presentation.*
c) *f is flat and locally of finite presentation.*

Then f is a morphism of F_n-2-cohomological descent for every $n \geq 0$.

a) Repeat the proof of (4.3.2), replacing (XII.(5.1)) by ((VIII 5.6)).
b) Since the question is local on Y, we may suppose that Y is Noetherian. Apply (EGA IV (14.5.10)). By (3.3.1) and a), we reduce to the case in which every point y of Y has a neighborhood U such that $f^{-1}(U) \to U$ has a section, and the result then follows from (3.2.4).
c) We must show that f is a morphism of F-1-*cohomological descent* (we know from ((VIII 9.4)) that f is a morphism of effective F^0-descent).

By localization on Y (cf. (3.2.4)), we may suppose that Y is affine, with ring A. We may then write $\operatorname{Spec} A = \varprojlim_{i\in I} \operatorname{Spec} A_i$, where I is a filtered ordered set and A_i is a Noetherian ring for every i, with $X = \varprojlim X_i$, where, for every i, X_i is a scheme *flat* and *locally of finite presentation* over $\operatorname{Spec} A_i$ such that $f_i : X_i \to \operatorname{Spec} A_i$ is *surjective*; moreover, the diagrams



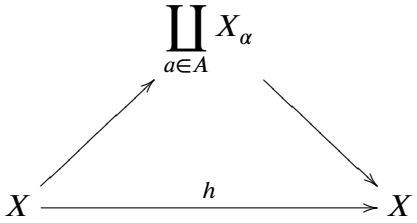
are Cartesian. 145

Let $\mathfrak{F}$ be an abelian sheaf on Y. We have $\mathfrak{F} = \varinjlim_i u_i^* u_{i*}(\mathfrak{F})$, so that, by a standard passage to the limit, we reduce to the case in which $\mathfrak{F}$ comes from a sheaf on $\operatorname{Spec} A_i$ for some i, and b) gives the conclusion.

REMARK 4.3.4. If $f : X \to Y$ is integral, then $R^q f_* = 0$ for $q > 0$. It follows readily that it is unnecessary to restrict to complexes bounded below when performing descent by means of f.

COROLLARY 4.3.5. *Let $(X_\alpha \xrightarrow{u_\alpha} X)_{\alpha\in A}$ be a covering family for the étale topology. Then $\coprod_{\alpha\in A} X_\alpha \to X$ is a morphism of universal F_n-2-cohomological descent for every $n \geq 0$.*

Indeed, there exists a factorization



where h is a surjective étale morphism.

5. Applications

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5.1. Construction of simplicial objects.

5.1.0. Up to and including 5.1.3, H is a category in which nonempty finite inverse limits exist and in which finite coproducts exist and are disjoint and universal. If $X : \Delta^0 \to H$ is a simplicial object of H, we denote by $d_i^k : X_k \to X_{k-1}$ (resp. by $s_i^k : X_k \to X_{k+1}$), for $i \in [k]$ and $0 \leq k$, the arrows corresponding to the face operators $\partial_k^i : [k-1] \to [k]$ (resp. to the degeneracy operators $\sigma_k^i : [k] \to [k+1]$) (cf. [3] II 2).

Definition 5.1.1. *A simplicial object $X : \Delta^0 \to H$ of H is σ-split if there exists a family of subobjects NX_j of X_j $(j \geq 0)$ such that the morphisms*

$$h_i : \coprod_{\substack{\mathrm{Hom}_{\Delta^-}([i],[j]) \\ j\leq i}} NX_j \longrightarrow X_i$$

are isomorphisms.

Likewise, a k-truncated simplicial object $X : \Delta_k^0 \to H$ is σ-split if there exist NX_j $(0 \leq j \leq k)$ satisfying the preceding condition for $i \leq k$.

The NX_j are uniquely determined by X. Indeed, if X is σ-split, the degeneracy operators $s_\ell^{k-1} : X_{k-1} \to X_k$ are isomorphisms from X_{k-1} onto direct summands of X_k, and

$$NX_k = \bigcap_\ell \quad \text{(complement of } s_\ell^{k-1}(X_{k-1})\text{)}.$$

For $k = 0$, $NX_0 = X_0$.

5.1.2. For a σ-split $(n+1)$-truncated simplicial object X of H, we denote by $\alpha_n(X)$ the triple consisting of

a) the restriction $i_n^*(X)$ of X to Δ_n^0;

147 b) NX_{n+1};

c) the evident map from NX_{n+1} to $(\mathrm{cosq}_n(X))_{n+1}$.

This triple (Y, N, β) satisfies the following condition:

$(*)$ Y is a σ-split n-truncated simplicial object of H, and β is a map from N to $(\mathrm{cosq}\, Y)_{n+1}$.

Proposition 5.1.3. *Let (Y, N, β) be a triple satisfying $(*)$ above.*

(i) *Up to unique isomorphism, there exists a unique σ-split $X : \Delta_{n+1}^0 \to H$ such that $\alpha_n(X) \simeq (Y, N, \beta)$.*

(ii) *To give a morphism f from X to a truncated simplicial object $Z : \Delta_{n+1}^0 \to H$ is equivalent to giving*

a) *a morphism $f' : Y \to i_n^*(Z)$;*

b) *a morphism* $f'' : N \to Z_{n+1}$ *such that the diagram*

$$\begin{array}{ccc} N & \longrightarrow & (\mathrm{cosq}_n Y)_{n+1} \\ \downarrow{\scriptstyle f''} & & \downarrow{\scriptstyle f'} \\ Z_{n+1} & \longrightarrow & (\mathrm{cosq}_n Z)_{n+1} \end{array}$$

commutes.

Let us construct X.

Set $Y'_{n+1} = (\mathrm{cosq}_n Y)_{n+1}$ and denote by

$$d'^{n+1}_i : Y'_{n+1} \longrightarrow Y_n \quad (\text{resp. } s'^{n}_i : Y_n \longrightarrow Y'_{n+1})$$

the face operators (resp. the degeneracy operators) obtained by functoriality.

Set $Y_{n+1} = N \coprod (\coprod_{\mathrm{Hom}_{\Delta^-_{\ell\leq n}}([n+1],[\ell])} NY_\ell)$, and let $\alpha : Y_{n+1} \to Y'_{n+1}$ be the arrow whose components are β and the arrows $NY_\ell \to Y_\ell \to Y'_{n+1}$ for every epimorphism $[n+1] \to [\ell]$ with $\ell \leq n$.

Let $s^n_i : Y_n \simeq \coprod_{\mathrm{Hom}_{\Delta^-_{\ell'\leq n}}([n],[\ell'])} NY_{\ell'} \to Y_{n+1}$ be the arrow induced by σ^i_n, so that 148

(I) $\alpha \circ s^n_i = s'^{n}_i$.

Set

(II) $d^{n+1}_i = d'^{n+1}_i \circ \alpha$.

To show that this defines an object of $\mathrm{Hom}(\Delta^0_{n+1}, H)$, it is enough, by [[3]. II (2.4)], to verify the formulas

a) $d^n_i \cdot d^{n+1}_j = d^n_{j-1} \cdot d^{n+1}_i \qquad i < j;$

b) $s^n_i \cdot s^{n-1}_j = s^n_{j+1} \cdot s^{n-1}_i \qquad i \leq j;$

c) $d^{n+1}_i \cdot s^n_j = \begin{cases} s^{n-1}_{j-1} \cdot d^n_i & i < j, \\ \mathrm{id} & i = j \text{ or } i = j+1, \\ s^{n-1}_j \cdot d^n_{i-1} & i > j+1. \end{cases}$

Formulas a) and c) are evident by (I) and (II), since they are known to hold if d^{n+1}_i is replaced by d'^{n+1}_i and s^n_i by s'^{n}_i.

Formula b) holds by construction.

We leave it to the reader to verify that the object X thus obtained satisfies 5.1.3.

Note that the functor constructed in the proof of 5.1.3, from the category of triples satisfying (5.1.2) $(*)$ to the category of $(n+1)$-truncated simplicial objects, is faithful but not full.

Proposition 5.1.3 provides a very convenient method for constructing simplicial objects of H by induction. In the remainder of this section, we formalize this observation in the form that will be used in 5.2 and 5.3.

5.1.4. Let $E \overset{\pi}{\to} B$ be a functor such that, for every object b of B, the fiber category E_b is nonempty and satisfies the conditions of 5.1.0. 149

Let Q be a property of an object of E, stable under isomorphism. Suppose that the following conditions hold:

a) *For every object b of B, there exists an object x of E_b satisfying Q.*

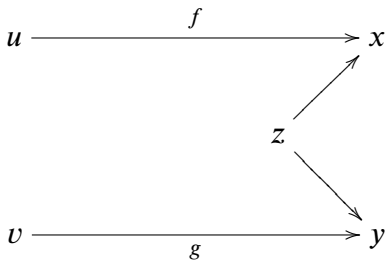
b) *For every object b of B and every pair (x, y) of objects of E_b satisfying Q, there exists an object z of E_b satisfying Q and morphisms $z \to x$, $z \to y$ in E_b.*

c) *For every morphism $b' \xrightarrow{t} b$ of B and every object x of E_b satisfying Q, there exists an object x' of $E_{b'}$ satisfying Q and a morphism $x' \to x$ over t.*

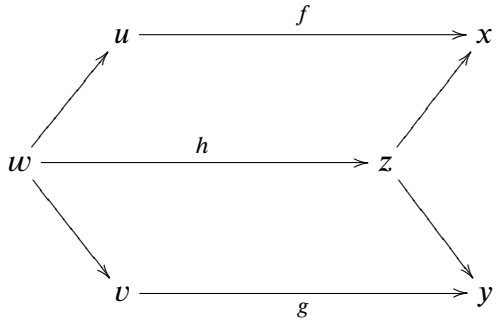
Let, on the other hand, P be a property of an arrow of E over an identity, stable under isomorphism. Suppose that the following conditions hold:

d) *For every object b of B and every object x of E_b, there exists an arrow $y \to x$ in E_b satisfying P.*

e) *For every object b of B, every diagram in E_b of the form*

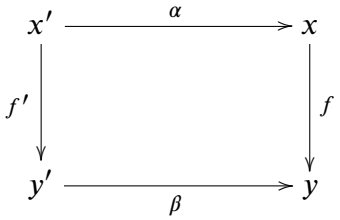


in which f and g satisfy P, there is a completion to a commutative diagram in E_b



150 *in which h satisfies P.*

f) *For every morphism $b' \xrightarrow{t} b$ of B and every morphism $x \xrightarrow{f} y$ of E_b satisfying P, there exists a morphism $x' \xrightarrow{f'} y'$ of $E_{b'}$ satisfying P and a commutative diagram*



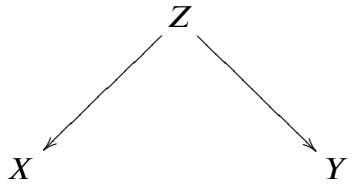
in which α and β are morphisms over t.

DEFINITION 5.1.5. *With the notation of 5.1.4, we say that an object X of $\mathrm{Hom}(\Delta^0, E)$ satisfies condition* (APQ) *if the following conditions hold:*

(i) *The image of $\pi \circ X$ is the constant simplicial object defined by an object b of B.*
(ii) *As an object of $\mathrm{Hom}(\Delta^0, E_b)$, X is σ-split.*
(iii) *X_0 satisfies condition Q.*
(iv) *For every integer n, the arrow $NX_{n+1} \to (\mathrm{cosq}_n X)_{n+1}$ satisfies condition P.*

5.1.6. Let $E_{(APQ)}$ denote the subcategory of $\mathrm{Hom}(\Delta^0, E)$ whose objects are the simplicial objects satisfying (APQ) and whose morphisms are the morphisms $X \xrightarrow{\alpha} Y$ of $\mathrm{Hom}(\Delta^0, E)$ such that $\pi(\alpha_n) : \pi(X_n) \to \pi(Y_n)$ is the same morphism of B for every n. There is then a canonical functor $\overline{\pi} : E_{(APQ)} \to B$.

PROPOSITION 5.1.7. *The functor $\overline{\pi}$ is surjective. Moreover, for every object b of B and every pair (X, Y) of objects of $E_{(APQ),b}$, there exists a diagram* 151



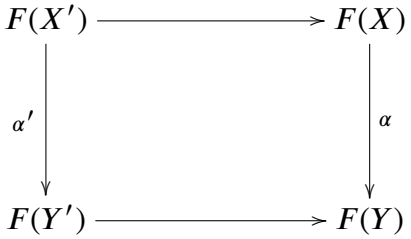
in $E_{(APQ),b}$.

The proof follows immediately from 5.1.3, by virtue of the hypotheses made in 5.1.4.

PROPOSITION 5.1.8. *Let C be a category and F a functor $E_{(APQ)} \to C$. We introduce the following conditions on F:*

(i) *For every object b of B, the restriction of F to $E_{(APQ),b}$ factors through the* coarse *category*[34] *defined by the objects of $E_{(APQ),b}$.*

(ii) *For every morphism $t : b' \to b$ of B and every pair $(X' \to X, Y' \to Y)$ of morphisms over t, the following diagram commutes:*



where α and α' are the arrows determined by condition (i).

Then the functor $G \mapsto G \circ \overline{\pi}$ induces an equivalence between the category $\mathrm{Hom}(B, C)$ *and the full subcategory of* $\mathrm{Hom}(E_{(APQ)}, C)$ *defined by the functors satisfying conditions* (i) and (ii).

The details of the proof are left to the reader. We indicate only how to prove that 152
the preceding functor is essentially surjective.

Let $F : E_{(APQ)} \to C$ be a functor satisfying conditions (i) and (ii). For every object b of B, choose an object X of $E_{(APQ),b}$ and set $\overline{F}(b) = F(X)$; one then verifies that $\overline{F}(b)$ varies functorially with b.

5.2. Singular cohomology of a scheme of finite type over a field of characteristic zero. In this section, k will denote a field of characteristic zero.

[34] A category is called *coarse* if, for every pair (x, y) of objects, $\mathrm{Hom}(x, y)$ consists of exactly one element.

5.2.0. We shall denote by Sch_f/k (resp. Ann) the category of schemes *locally of finite type* over k (resp. complex analytic spaces). All simplicial objects of Sch_f/k (resp. Ann) will be regarded as simplicial objects of Top by means of the canonical functor $\mathrm{Sch}_f/k \to \mathrm{Top}$ (resp. $\mathrm{Ann} \to \mathrm{Top}$). We retain the data of (4.1.0) concerning cohomological descent.

Following the usual notation, we write $S \to S^{\mathrm{an}}$ for the canonical functor $\mathrm{Sch}_f/\mathbf{C} \to \mathrm{Ann}$ defined in G.A.G.A.

5.2.1. Let X be a simplicial object of Sch/k (resp. Ann): the usual compatibilities in the construction of the de Rham complex of a morphism of schemes (resp. of analytic spaces) make it possible to construct a complex $\Omega^{\mathrm{Zar}\cdot}_{X/k}$ (resp. $\Omega^{\cdot}_{X/\mathbf{C}}$) of $D^+(\Gamma(\overline{X}, \mathbf{Z}))$ such that, for every integer i, $Re_i^*(\Omega^{\mathrm{Zar}\cdot}_{X/k})$ (resp. $Re_i^*(\Omega^{\cdot}_{X/\mathbf{C}})$) is the de Rham complex $\Omega^{\mathrm{Zar}\cdot}_{X/k}$ (resp. $\Omega^{\cdot}_{X_i/\mathbf{C}}$).

Let $X \xrightarrow{u} C_S^{\Delta}$ be an augmented simplicial object of Sch_f/k (resp. Ann). We define $H^{\mathrm{Zar}\cdot}_{DR}(u)$ (resp. $H^{\cdot}_{DR}(u)$ and $H^{\cdot}(u, \mathbf{C})$) to be the hypercohomology groups of the complex $R\overline{u}_+(\Omega^{\mathrm{Zar}\cdot}_{X/k})$ (resp. $R\overline{u}_*(\Omega^{\cdot}_{X/\mathbf{C}})$ and $R\overline{u}(\mathbf{C})$).

153 5.2.2. If $X \xrightarrow{u} C_S^{\Delta}$ is a semisimplicial object of $\mathrm{Sch}_f/\mathbf{C}$, there are three canonical morphisms:

$$H^{\mathrm{Zar}\cdot}_{DR}(u) \xrightarrow{\alpha} H^{\cdot}_{DR}(u^{\mathrm{an}})$$

$$H^{\cdot}(u^{\mathrm{an}}, \mathbf{C}) \xrightarrow{\beta} H^{\cdot}_{DR}(u^{\mathrm{an}})$$

$$H^{\cdot}(S^{\mathrm{an}}, \mathbf{C}) \xrightarrow{\gamma} H^{\cdot}(u^{\mathrm{an}}, \mathbf{C}).$$

Proposition 5.2.3. *With the notation of (5.2.2), suppose that X_n is smooth for every n: then α and β are isomorphisms. If, moreover, $X^{\mathrm{an}} \xrightarrow{u^{\mathrm{an}}} C^{\Delta}_{S^{\mathrm{an}}}$ is a hypercovering of S^{an} for the topology of universal 1-cohomological descent, then γ is an isomorphism.*

The assertion that α is an isomorphism follows from [[6], Th. 1′]. The Poincaré lemma shows that β is an isomorphism. The final part of the proposition follows from (3.3.3).

5.2.4. We shall now apply (5.1.4) in the case where $B = \mathrm{Sch}_f/k$ and E is the category over B such that, for every scheme S, the fiber category E_S is the category of schemes of Sch_f/k over S, the arrows over an arrow of B being the obvious ones. We shall say that a morphism $h : X \to Y$ over S satisfies property P if X is smooth over k and if h is a composite of finitely many morphisms satisfying one of the following conditions:

(i) it is the coproduct of a family of étale surjective morphisms;
(ii) it is the coproduct of a family of proper surjective morphisms.

We shall say that a scheme X of E_S satisfies property Q if the structural morphism $X \to S$ satisfies property P.

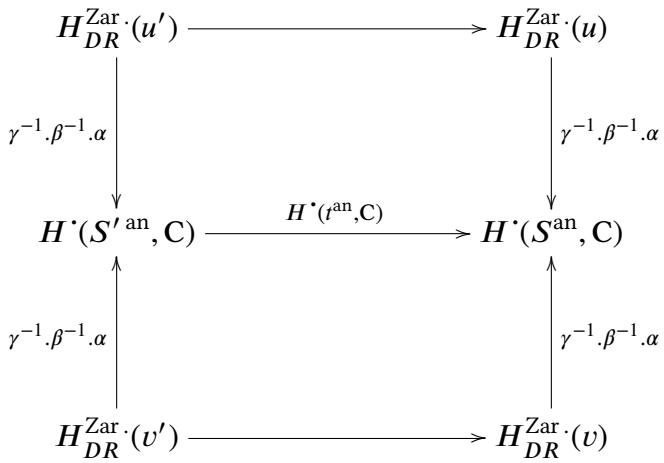
We leave it to the reader to verify, using resolution of singularities, that all the conditions of (5.1.4) are satisfied.

154 The groups $H^{\mathrm{Zar}\cdot}_{DR}(u)$ define a contravariant functor from the category $E_{(APQ)}$ to the category of graded abelian groups, denoted by $H^{\mathrm{Zar}\cdot}_{DR}$.

Proposition 5.2.5. *The functor $H^{\mathrm{Zar}\cdot}_{DR}$ satisfies the conditions of (5.1.8).*

By the Lefschetz principle, we are reduced to the case $k = \mathbf{C}$. In that case, for every morphism $S' \xrightarrow{t} S$ and every pair $(u' \to u, v' \to v)$ of morphisms over t, there is a

commutative diagram:



which completes the proof.

Definition 5.2.6. *The contravariant functor defined by (5.1.8) from $H^{\mathrm{Zar}\cdot}_{DR}$ will be denoted by $H^{\cdot}(*,k)$. For every scheme S locally of finite type over k, the graded abelian group $H^{\cdot}(S,k)$ will be called the* singular cohomology of S.

5.2.7. We conclude this section with an illustration of the general principles introduced in it. Recall first that if S is a complex analytic space, the *de Rham cohomology* of S, denoted by $\mathbf{H}^{\cdot}_{DR}(S)$, is by definition the hypercohomology of the de Rham complex $\Omega^{\cdot}_{S/\mathbf{C}}$.

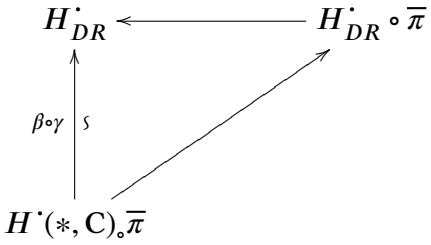
Proposition 5.2.8. (Blum–Herrera). *Let S be a scheme locally of finite type over* $\mathbf{C}$. 155
The canonical arrow

$$H^{\cdot}(S^{\mathrm{an}},\mathbf{C}) \longrightarrow \mathbf{H}^{\cdot}_{DR}(S^{\mathrm{an}})$$

has a retraction (in the category of graded abelian groups) that is functorial in S.

By (5.1.8), it is enough to consider the morphism $H^{\cdot}(+,\mathbf{C})\circ\overline{\pi} \to \mathbf{H}^{\cdot}_{DR}\circ\overline{\pi}$ obtained from the preceding one by composition with $\overline{\pi}$.

Using, for every augmentation $X \xrightarrow{u} C^{\Delta}_{S}$, the canonical morphism $\mathbf{L}\overline{u}^{+}(\Omega^{\cdot}_{S/C}) \to \Omega^{\cdot}_{X/C}$, one defines a functorial morphism $\mathbf{H}^{\cdot}_{DR}\circ\overline{\pi} \to \mathbf{H}^{\cdot}_{DR}$ so that the following diagram is commutative:



which gives the desired retraction.

Remark 5.2.9. Statement (5.2.8) remains valid for every analytic space once resolution of singularities is available in the analytic setting.

5.3. Mixed Hodge theories. We give here a technical result that plays a key role in [2].

k is a field of characteristic zero.

5.3.1. Let $\mathrm{Sch}_{\mathrm{fs}}/k$ be the category of separated schemes of finite type over k; denote by $E \xrightarrow{\pi} \mathrm{Sch}_{\mathrm{fs}}/k$ the category over $\mathrm{Sch}_{\mathrm{fs}}/k$ defined as follows:

156 If S is an object of $\mathrm{Sch}_{\mathrm{fs}}/k$, an object of E_S is a triple $(\overline{X}, X, i)$, where $\overline{X}$ is a reduced projective scheme over k, X is a proper scheme over S, and $i : X \to \overline{X}$ is an open immersion over k such that $i(X)$ is dense in $\overline{X}$.

If $t : S' \to S$ is a morphism of schemes, $(\overline{X}, X, i)$ is an object over S, and $(\overline{X}', X', i')$ is an object over S', a morphism from $(\overline{X}, X, i)$ to $(\overline{X}', X', i')$ over t is a pair $(\overline{h}, h)$, where $\overline{h} : \overline{X}' \to \overline{X}$ is a morphism over k and $h : X' \to X$ is a morphism over S, such that the following diagram is commutative.

$$\begin{array}{ccc} X' & \xrightarrow{i'} & \overline{X}' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \overline{h}} \\ X & \xrightarrow{i} & \overline{X}. \end{array}$$

N.B.: the properness hypotheses imply that $\overline{h}^{-1}(i(X)) = i(X')$.

Lemma 5.3.2. *For every object S of $\mathrm{Sch}_{\mathrm{fs}}/k$, the category E_S is nonempty, has nonempty projective limits, and has finite coproducts that are disjoint and universal.*

The fact that the category E_S is nonempty follows from Chow's lemma.

We shall describe only the construction of the direct product of two objects $(\overline{X}, X, i)$ and $(\overline{X}', X', i')$ of E_S.

The canonical arrow $j : X \times_S X' \to \overline{X} \times_k \overline{X}'$ is an immersion. Let T be the reduced closed subscheme of $\overline{X} \times_k \overline{X}'$ whose underlying space is the closure of the image of j. Then $(T, X \times_S X'_{\mathrm{red}}, j_{\mathrm{red}})$ has the required universal property.

157 5.3.3. Let S be an object of $\mathrm{Sch}_{\mathrm{fs}}/k$. We shall say that an object $(\overline{X}, X, i)$ of E_S satisfies property Q if $\overline{X}$ is smooth over k and $i(X)$ is the complement of a normal crossings divisor. We shall say that a morphism $(\overline{h}, h) : (\overline{X}', X', i') \to (\overline{X}, X, i)$ satisfies property P if $\overline{h}$ is surjective and $(\overline{X}', X', i')$ satisfies property Q.

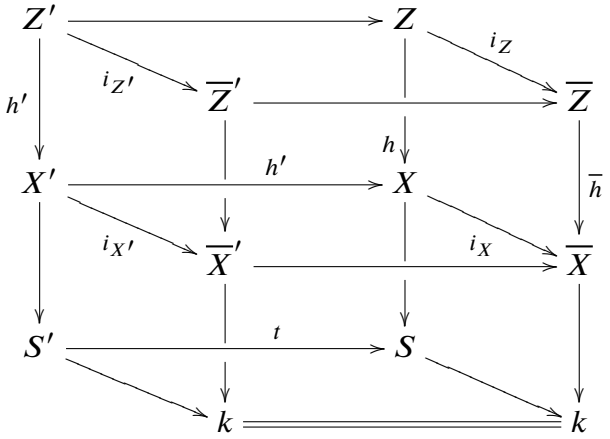
Proposition 5.3.4. *With the notation of (5.3.3), all the conditions of (5.1.4) are satisfied.*

Conditions a), b), d), and e) are readily verified using resolution of singularities. Moreover, c) follows from f), so it remains to verify f).

We must show that every commutative diagram

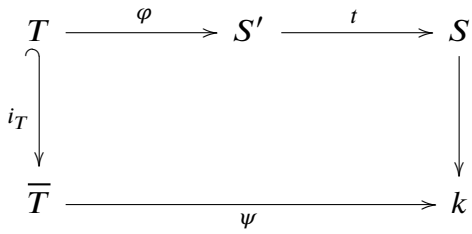
$$\begin{array}{ccccc} Z & \xrightarrow{i_Z} & \overline{Z} & & \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \overline{h}} & & \\ X & \xrightarrow{i_X} & \overline{X} & & \\ \downarrow & & \downarrow & & \\ S' & \xrightarrow{t} & S & \longrightarrow & k \end{array}$$

where $(\overline{X}, X, i_X)$ and $(\overline{Z}, Z, i_Z)$ are objects of E_S and $\overline{h}$ is surjective can be completed to a commutative diagram

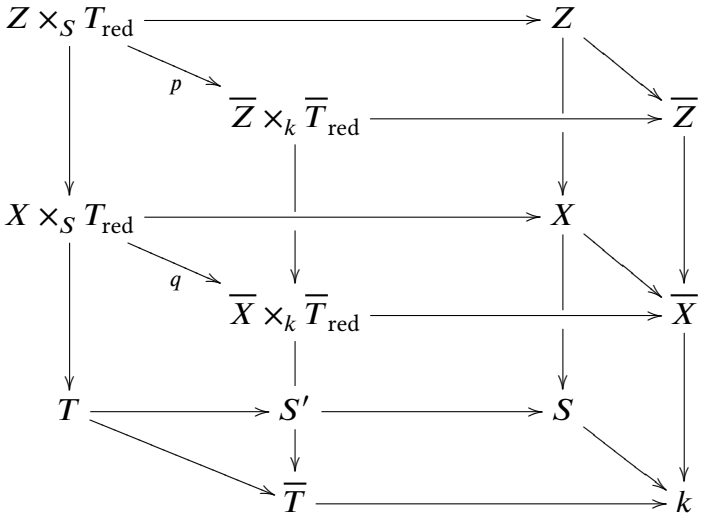


where $(\overline{X}', X', i_{X'})$ and $(\overline{Z}', Z', i_{Z'})$ are objects of $E_{S'}$ and $\overline{h'}$ is surjective. 158

Consider a diagram



where φ is proper and ψ is projective. By base change we obtain a diagram



where p and q are immersions. It then suffices to replace $\overline{Z} \times_k \overline{T}_{\mathrm{red}}$ and $\overline{X} \times_k \overline{T}_{\mathrm{red}}$ by 159
the closed images of p and q to obtain the required diagram. This completes the proof of (5.3.4).

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EXPOSÉ VI

Finiteness Conditions. Fibered Topoi and Sites. Applications to Questions of Passage to the Limit.

A. Grothendieck and J.-L. Verdier

0. Introduction

In the present exposé, we study two closely related types of questions. 165

First, we undertake a systematic study of finiteness conditions for topoi, successively examining the notions of *quasi-compactness*, *quasi-separatedness*, and *coherence* (= quasi-compactness + quasi-separatedness). These are inspired by the familiar analogous notions from scheme theory. We study them first for an object X of a topos and for an arrow $u : X \to Y$ of a topos (§ 1), then for the topos E itself (§ 2), and finally for a morphism of topoi $f : E \to E'$ (§ 3). Section 4 studies these notions in the special case of a topos obtained by gluing two topoi (IV 9). For the subsequent sections, the essential facts to retain from these somewhat technical developments are the following: a *coherent topos* is a topos equivalent to one of the form $C^{\sim}$, where C is a small site in which finite projective limits are representable and every covering family admits a finite covering subfamily; and a *coherent morphism* $f : E \to E'$ of coherent topoi is a morphism that can be described (up to equivalence) as the morphism associated with a morphism of sites $f : C \to C'$, where C and C' are as above and the corresponding functor $f^* : C' \to C$ is left exact.

It is no doubt useful to note that, with the finiteness notions used here, and especially that of a coherent topos, we resolutely depart (and for the first time) from the topological spaces familiar to analysts and "topologists"[34]. Thus the topos associated (IV 2.1) with a separated topological space X is coherent only if X is a finite set! In other words, the present exposé is directed entirely toward the types of topoi arising from algebraic geometry and algebra. In fact, all the topoi used thus far in these disciplines (except, of course, in algebraic geometry by transcendental methods over the field of complex numbers!) turn out to be locally coherent.

Second, we develop two *theorems on passage to the inductive limit* for the cohomology of topoi. One (5.2) tells us that the functors $H^i(E, -)$ on a coherent topos commute 166
with inductive limits of abelian sheaves. The other (8.7) says essentially that if a topos X is represented as a filtered projective limit of coherent topoi X_i, with coherent transition morphisms $f_{ij} : X_j \to X_i$, then the cohomology of X (with coefficients in an arbitrary abelian sheaf) is computed as the inductive limit of the cohomologies of the X_i. To give this statement a precise meaning, the main point is to specify the notion of a "projective limit of topoi," which is done in §§ 7, 8. This is a very useful geometric notion, undoubtedly also in contexts other than that of projective limits of schemes (which had served as the model for M.Artin in his initial definition of this notion). We show, for example, in (8.4) how the various notions of being "localized at a point" (for a Noetherian space,

[34] The quotation marks indicate that this is a "..." unfamiliar with the notion of a topos (cf. IV 0.4 for the meaning of the word "topology").

for a scheme with its usual topology, or even with its étale topology) can be unified by defining the localization as the projective limit of the "neighborhoods" of that point—in perfect agreement with the intuitive idea of localization at a point.

The two limit-passage theorems will be stated again explicitly in the next exposé in the special setting of étale cohomology of schemes (VII 3.3 and 5.7); in that case, they will moreover be used constantly throughout the remainder of this Seminar, and practically everywhere étale cohomology has been used up to the present. The reader willing to accept these two theorems in the special case in which they are stated in Exposé VII, and interested exclusively in applications of étale cohomology, may therefore omit the present exposé. It is true that the notion of a *constructible object* of a topos (i.e. one whose structural morphism $X \to e$ to the final object e is coherent) will likewise play an important role throughout the rest of the Seminar (whereas it plays only a very secondary role in the present exposé, if only because, for a coherent topos, it coincides with the notion of a coherent object, which takes center stage here). But this notion is developed independently in Exposé IX, in the special case of étale sheaves on schemes, without using the general study made in the present exposé; in that case it also exhibits very useful special phenomena that served as the basis for its treatment in IX (beginning
167 with Definition IX 2.3). Since the text of IX predates the present text of Exposé VI by five years, yet remains quite usable for the reader, we have confined ourselves at the end of Exposé IX to proving the equivalence between the notion of constructibility used in IX and the one introduced here.

1. Finiteness conditions for the objects and arrows of a topos

DEFINITION 1.1. *Let C be a site. An object X of C is called quasi-compact if, for every covering family $(X_i \to X)_{i\in I}$, there is a finite subset J of the index set I such that the subfamily $(X_i \to X)_{i\in J}$ is still covering.*

Since a $\mathcal{U}$-topos E is always regarded as a site (IV 1.1.1), the preceding definition applies in particular to an object of E. This case is moreover the general one, by the following proposition.

PROPOSITION 1.2. *Let C be a $\mathcal{U}$-site, let $\epsilon : C \to C^\sim$ be the canonical functor, and let X be an object of C. Then X is a quasi-compact object of the site C if and only if $\epsilon(X)$ is a quasi-compact object of the topos $C^\sim$.*

Suppose that X is quasi-compact; we prove that $\epsilon(X)$ is as well. Let $(F_i \to \epsilon(X))_{i\in I}$ be an epimorphic family. For every $i \in I$, choose an epimorphic family $\epsilon(X_{ij}) \to F_i$, $j \in J_i$. The composite $\epsilon(X_{ij}) \to \epsilon(X)$ identifies with an element α_{ij} of $\epsilon(X)(X_{ij})$. In general one cannot assert that it comes from a morphism $X_{ij} \to X$, but, by the construction of the associated sheaf $\epsilon(X)$, one knows that, for fixed i and j, a covering family $Y_k \to X_{ij}$ can be found such that the images of α_{ij} in the $\epsilon(X)(Y_k)$ come from elements of $X(Y_k)$, i.e. from morphisms $Y_k \to X$. Thus, after refining the epimorphic family $\epsilon(X_{ij}) \to F_i$, we may suppose that the composites $\epsilon(X_{ij}) \to \epsilon(X)$ come from
168 morphisms $X_{ij} \to X$. As i and j vary, the family of $\epsilon(X_{ij}) \to \epsilon(X)$ is epimorphic, so (II 4.4) the family of $X_{ij} \to X$ is covering. By the hypothesis on X, it admits a finite covering subfamily, and the corresponding finite family of morphisms $\epsilon(X_{ij}) \to \epsilon(X)$ is again epimorphic. Since these factor through finitely many of the morphisms $F_i \to \epsilon(X)$, that finite family is already epimorphic. This proves that $\epsilon(X)$ is quasi-compact.

Conversely, the fact that quasi-compactness of $\epsilon(X)$ implies quasi-compactness of X is trivial, by *loc. cit.*

The preceding result therefore shows that the notion of quasi-compactness on sites reduces to the corresponding notion in topoi.

PROPOSITION 1.3. *Let C be a site and let $(X_i \to X)_{i\in I}$ be a finite covering family, with the X_i quasi-compact. Then X is quasi-compact.*

To see this, we may assume, by 1.2, that C is a topos. Let $(X'_j \to X)$ be a covering, i.e. epimorphic, family in C. Then, for every $i \in I$, the family obtained by the base change $X_i \to X$ is epimorphic and hence (since X_i is quasi-compact) admits a finite epimorphic subfamily, corresponding to a finite subset $J_i \subset J$ of the index set J. Set $J' = \bigcup_{i\in I} J_i$. Then J' is a finite subset of J because I is finite, and the family $(X'_j \to X)_{j\in J'}$ is already epimorphic, since it is so locally, C.Q.F.D.

COROLLARY 1.4. *Let $(X_i)_{i\in I}$ be a family of objects of a topos E, and let X be their sum.* 169
Then X is quasi-compact if and only if all the X_i are quasi-compact and all but finitely many of the X_i are isomorphic to the "empty object" $\emptyset_E$ of E.

Sufficiency follows immediately from 1.3 (since the sum does not change when the empty summands are omitted). For necessity, note that a finite subfamily $(X_i)_{i\in J}$ must cover X, which implies that, for $i \in I - J$, one has $X_i = X_i \times_X \amalg_{j=J} X_j = \amalg_{j=J} X_i \times_X X_j = \emptyset$ (II 4.5.1). On the other hand, to show that every X_i is quasi-compact, note that $X \simeq X_i \amalg X'_i$, and that every covering family $Y_j \to X_i$ of X_i defines a covering family $Y_j \amalg X'_i \to X_j \amalg X'_i = X$ of X. The latter admits a finite covering subfamily, which plainly implies that the corresponding finite family of $Y_j \to X_i$ is also covering; thus X_i is quasi-compact.

REMARK 1.5.1. The property for an object X of a topos E of being quasi-compact depends only on the induced topos $E_{/X}$, and means that the final object of the latter is quasi-compact. Likewise, an object X of a site C is quasi-compact if and only if the final object of the induced site $C_{/X}$(III 5.1) is quasi-compact.

1.5.2. Let X be an object of a topos E. Then X is quasi-compact if and only if every filtered increasing family $(X_i)_{i\in I}$ of subobjects of X that covers X contains some X_i equal to X. This condition therefore depends only on the ordered set of subobjects of 170
X (i.e. the open subsets of the induced topos $E_{/X}$).

1.5.3. Consider a generating family $(X_i)_{i\in I}$ of E. It follows from the definitions and from 1.3 that, for an object X of E to be quasi-compact, it is necessary that it admit a covering family consisting of finitely many of the X_i, i.e. that X be isomorphic to a quotient object of a finite sum of objects of E; this condition is sufficient if the X_i are quasi-compact. Note also that, under the hypothesis made on E, after replacing the generating family under consideration by a subfamily if necessary, we may clearly suppose that $I \in \mathcal{U}$. Using the fact that the set of quotients of an object of E is small (I 7.5), we then find that the full subcategory E_{qc} of E formed by the quasi-compact objects of E is equivalent to a category $\in \mathcal{U}$, i.e. the cardinal of the set of isomorphism classes of objects of E_{qc} is $\in \mathcal{U}$.

EXAMPLE 1.6.1. Let X be a topological space. An open subset U of X is a quasi-compact object of the site $\mathrm{Ouv}(X)$ of open subsets of X if and only if U is a quasi-compact space for the topology induced by X. More generally, a sheaf F on X is a quasi-compact object of the topos $\mathrm{Top}(X)$ if and only if the étalé space X' over X associated with F is quasi-compact. (This follows from the preceding assertion, from 1.5, and from the fact that $\mathrm{Top}(X)_{/F}$ is equivalent to $\mathrm{Top}(X')$ (IV 5.7)).

1.6.2. Consider on the category (Sch) of schemes (elements of $\mathscr{U}$) one of the topologies T_i of [6] 6.3 (for example, the Zariski topology, the étale topology, the fppf topology, or the fpqc topology). A scheme is quasi-compact for this topology if and only if it is a quasi-compact scheme in the usual sense. (With the notation of *loc. cit.*, only the fact that every family $\in P'$ is finite is used.) 171

DEFINITION 1.7. *Let E be a topos and let $f : X \to Y$ be an arrow of E. We say that f is a quasi-compact morphism if, for every arrow $Y' \to Y$ with Y' quasi-compact, the object $X' = X \times_Y Y'$ is also quasi-compact. We say that f is quasi-separated if the diagonal morphism $X \to X \times_Y X$ is quasi-compact. We say that f is coherent if f is quasi-compact and quasi-separated.*

1.7.1. Unwinding the definition of a quasi-separated morphism $f : X \to Y$, we see that it also means that, for every pair of arrows $g_1, g_2 : X' \rightrightarrows X$ "over Y," i.e. such that $fg_1 = fg_2$, quasi-compactness of X' implies quasi-compactness of $\mathrm{Ker}(g_1, g_2)$.

PROPOSITION 1.8. *Let E be a topos.*

(i) *Every isomorphism in E is a coherent morphism, i.e. is quasi-compact and quasi-separated. The composite of two quasi-compact (resp. quasi-separated, resp. coherent) morphisms is quasi-compact (resp. quasi-separated, resp. coherent).*

172 (ii) *Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms in E, and let $f' : X' = X \times_Y Y' \to Y'$ be deduced from f by the base change g. If f is quasi-compact (resp. quasi-separated, resp. coherent), then so is f'.*

(iii) *Let $f : X \to Y$ and $g : Y \to Z$ (the arrow $Y \to Z$) be morphisms in E. If gf is quasi-separated, then f is quasi-separated.*

(iv) *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms in E, with g quasi-separated. If gf is quasi-compact (resp. coherent), then f is quasi-compact (resp. coherent).*

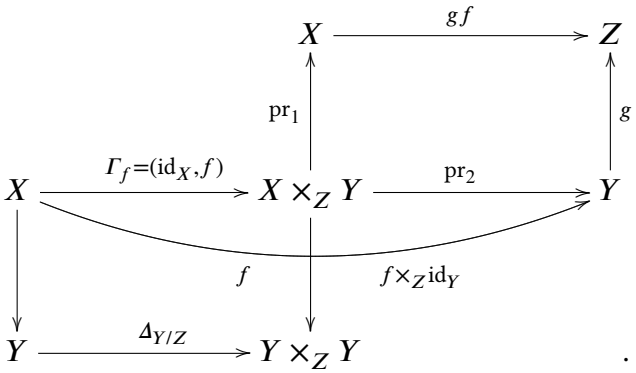
The proofs are well known (Cf. EGA IV 1 or EGA I, 2nd edition). Statements (i) and (ii) in the quasi-compact case follow trivially from the definitions. In the quasi-separated case, stability under base change in (ii) follows immediately from the definition and the preceding case, as does the fact that every isomorphism is quasi-separated. Stability under composition $X \to Y \to Z$ follows from the diagram

(∗)
$$\begin{array}{ccc} X & & \\ \downarrow{\scriptstyle \Delta_{X/Y}} \searrow{\scriptstyle \Delta_{X/Z}} & & \\ X \times_Y X & \longrightarrow & Y \\ \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle \Delta_{Y/Z}} \\ X \times_Z X & \longrightarrow & Y \times_Z Y \end{array}$$

173 whose square is Cartesian. By hypothesis, $\Delta_{Y/Z}$ is quasi-compact, and hence so is u, which is obtained from it by base change; moreover, $\Delta_{X/Y}$ is quasi-compact, and therefore so is the composite $u\Delta_{X/Y} = \Delta_{X/Z}$. This proves (i) and (ii), the coherent case following by combining the two preceding cases. To prove the first case considered in (iv),

consider the following diagram with Cartesian squares:

(**)



By hypothesis, g is quasi-separated, so $\Delta_{Y/Z}$ is quasi-compact and hence Γ_f is quasi-compact by base change. Likewise, by hypothesis gf is quasi-compact, so pr_2 is also quasi-compact by base change. Consequently, $f = \mathrm{pr}_2\,\Gamma_f$ is quasi-compact as a composite of two quasi-compact morphisms. The corresponding case of (iv) follows immediately from the preceding case and (iii), which remains to be proved. For this, return to diagram (∗). The hypothesis that gf is quasi-separated means that $\Delta_{X/Z} = u\Delta_{X/Y}$ is quasi-compact, while the conclusion that f is quasi-separated means that $\Delta_{X/Y}$ is quasi-compact. Equivalently, the conclusion that f is quasi-separated again amounts to the quasi-compactness of $\Delta_{X/Y}$. This follows from (iv) once we have proved that u is quasi-separated. But u is plainly a monomorphism, and indeed we have the following trivial result: 174

Corollary 1.8.1. *Every monomorphism is quasi-separated.*

Indeed, if $u : X \to Y$ is a monomorphism, then $\Delta_{X/Y}$ is an isomorphism and is therefore quasi-compact, C.Q.F.D.

Corollary 1.8.2. *Let $f : X \to Y$ and $f' : X' \to Y'$ be two S-morphisms in E. If f and f' are quasi-compact (resp. quasi-separated, resp. coherent), then so is $f \times_S f' : X \times_S X' \to Y \times_S Y'$.*

This follows in the usual way by combining (i) and (ii).

Remark 1.9.1. *Let E be any category, and let Q be a subset of* ob E *stable under isomorphism (thus, giving Q amounts to giving a strictly full subcategory of E), playing the role of the "quasi-compact" objects. Suppose that fiber products are representable in E. One can then paraphrase Definition 1.7 to define the notions of Q-quasi-compact, Q-quasi-separated, and Q-coherent morphisms. Proposition 1.8 and Corollary 1.8.1 then hold, as does 1.8.2 when products of two objects are representable in E.*

1.9.2. Let S be an object of E, and let $f : X \to Y$ be a morphism of $E_{/S}$. It follows immediately from Remark 1.5 that f is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the corresponding morphism in E (obtained from f by the "forget S" functor) has the same property. In particular, if $f : X \to Y$ is a morphism in E, then, by taking $S = Y$ above, we see that f is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the structural morphism of the object (X, f) of the induced topos $E_{/Y}$, with target the final object of that topos, is quasi-compact (resp. quasi-separated, resp. coherent). One also says that (by means of the structural morphism f) X *is quasi-compact (*resp. *quasi-separated,* resp. *coherent) over* Y, an expression that may equally well be interpreted as referring to the topos E or to the induced topos $E_{/Y}$ (in which Y is regarded as a final object). 175

1.9.3. An object X of a topos E is called *constructible* if it is coherent over the final object. Every morphism from one constructible object to another is coherent (1.8 (iv)). The full subcategory E_{cons} of E formed by the constructible objects of E is stable under finite inverse limits, as follows immediately from 1.8.2 and 1.8 (ii).

1.9.4. The notions of 1.7 are of real interest only in topoi E that admit a generating family formed by quasi-compact objects. In that case, these notions are "local on the base" (IV 8.5.4), as we now show.

176 PROPOSITION 1.10. *Suppose that the topos E admits a generating subcategory C formed by quasi-compact objects, and let $f : X \to Y$ be a morphism in E.*

(i) *The morphism f is quasi-compact if and only if, for every morphism $S \to Y$ with $S \in \mathrm{Ob}\, C$, the object $X \times_Y S$ is quasi-compact.*

(ii) *Let $Y_i \to Y$, $i \in I$, be a covering family. The morphism f is quasi-compact (*resp. *quasi-separated,* resp. *coherent) if and only if, for every $i \in I$, the morphism $f_i : X_i = X \times_Y Y_i \to Y_i$ has the same property.*

Let us prove (i). Necessity is immediate from the definition. For sufficiency, we must prove that, for every quasi-compact object Y' and every morphism $Y' \to Y$, the object $X' = X \times_Y Y'$ is quasi-compact. There is a covering family $S_i \to Y'$ with the S_i in $\mathrm{Ob}\, C$, and since Y' is quasi-compact, this covering family may be taken to be *finite*. By hypothesis the objects ${X'}_i = X \times_Y S_i \simeq X' \times_{Y'} S_i$ are quasi-compact, and since they cover X' (the S_i covering Y'), X' is quasi-compact by 1.3.

Let us prove (ii). It is enough to treat the quasi-compact case, since the quasi-separated case follows from Definition 1.7, and the coherent case likewise follows by combining the two preceding cases. The necessity has already been proved in 1.8 (ii). It remains to show that if the f_i are quasi-compact, then f is quasi-compact, i.e. that, for every quasi-compact Y' and every morphism $Y' \to Y$, the object $X' = X \times_Y Y'$ is
177 quasi-compact. Since $(Y_i \to Y)$ is a covering family, there exist objects S_α over Y that cover Y' and such that each composite $S_\alpha \to Y' \to Y$ factors through some Y_i. Since C is generating, the S_α may be taken in $\mathrm{Ob}\, C$, and since Y' is quasi-compact, the covering family may be taken finite. The $X' \times_{Y'} S_\alpha$ then form a finite covering family of X', and by 1.3 we are reduced to proving that each $X' \times_{Y'} S_\alpha = X \times_Y S_\alpha$ is quasi-compact. Choosing a factorization of $S_\alpha \to Y$ through some Y_i, the object in question is also isomorphic to $X_i \times_{Y_i} S_\alpha$, which is indeed quasi-compact since X_i is quasi-compact over Y_i and S_α is quasi-compact.

COROLLARY 1.11. *Let $g : Y \to Z$ be a morphism in the topos E, and let $(f_i : X_i \to Y)_{i \in I}$ be a covering family of morphisms with target Y. Then:*

(i) *Suppose that I is finite. If the $g f_i$ are quasi-compact, then so is g.*

(ii) *Suppose that the f_i are quasi-compact. If the $g f_i$ are quasi-separated, then so is g. If I is finite and the $g f_i$ are coherent, then g is coherent.*

Case (i) follows immediately from the definitions and 1.3, without any hypothesis on E. To prove (ii), it is enough, by (i), to treat the nonparenthetical case. Consider diagram 1.8 $(*)$ with X replaced by some X_i. Since the family of the f_i is covering, so
178 is the family of the $f_i \times_Z \mathrm{id}_Y$ obtained from it by base change. Hence, by 1.10 (ii), to prove that $\Delta_{Y/Z}$ is quasi-compact, it is enough to prove that the Γ_{f_i} are quasi-compact. Since $g f_i$ is quasi-separated by hypothesis, so is $\mathrm{pr}_2 : X_i \times_Z Y \to Y$, which is obtained from it by base change. On the other hand, $\mathrm{pr}_2 \Gamma_{f_i} = f_i$ is quasi-compact by hypothesis. Thus Γ_{f_i} is quasi-compact by 1.8 (iv), applied to the composite of Γ_{f_i} and pr_2, C.Q.F.D.

COROLLARY 1.12. *Let $(X_i)_{i\in I}$ be a finite family of objects of E, let X be the coproduct of the X_i, let $f_i : X_i \to Y$ be morphisms in E, and let $f : X \to Y$ be the corresponding morphism. The morphism f is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the f_i have the same property; in the quasi-separated case, the hypothesis that I be finite may be omitted.*

Sufficiency is a special case of 1.11, taking into account that the canonical morphisms $X_i \to X$ form a finite covering family and are quasi-compact morphisms, as follows immediately from the definition and 1.4. Necessity follows from transitivity, 1.8 (i), taking into account that the morphisms $X_i \to X$ are even coherent (for they are quasi-separated by 1.8.1).

DEFINITION 1.13. *Let X be an object of a topos E. We say that X is a* quasi-separated object of the topos E *if, for every quasi-compact object S of E, every morphism from S to X is quasi-compact, i.e. (1.7) if, for any two quasi-compact objects S, T of E and morphisms $S \to X, T \to X$, the fiber product $S \times_X T$ is always quasi-compact. We say that X is a coherent object of the topos E if X is quasi-compact and quasi-separated.*

PROPOSITION 1.14. *Let $f : X \to Y$ be a morphism in a topos E.* 179

(i) *If Y is quasi-compact, then f quasi-compact implies that X is quasi-compact. If Y is quasi-separated, then X quasi-compact implies that f is quasi-compact. If Y is coherent, f is quasi-compact if and only if X is quasi-compact.*

(ii) *If Y is quasi-compact (*resp. *quasi-separated,* resp. *coherent) and f is quasi-compact (*resp. *quasi-separated,* resp. *coherent), then X is quasi-compact (*resp. *quasi-separated,* resp. *coherent).*

The first two assertions of (i) are contained immediately in the definitions of a quasi-compact morphism (1.7) and, respectively, a quasi-separated object (1.13); the third follows by combining the first two. The first case of (ii) is merely a restatement, and the third again follows by combining the first two. It remains to treat the quasi-separated case. Let S be a quasi-compact object of E and $g : S \to X$ a morphism; we prove that g is quasi-compact. Since Y is quasi-separated, fg is quasi-compact, and since f is quasi-separated, it follows (1.8 (iii)) that g is quasi-compact, C.Q.F.D.

COROLLARY 1.15. *Every subobject of a quasi-separated object of E is quasi-separated. Let $(X_i)_{i\in I}$ be a family of objects of E, with $I \in U$; then $X = \amalg_i X_i$ is quasi-separated (*resp. *coherent) if and only if the X_i are.*

The first assertion follows from 1.14 (ii) and 1.8.1, but is also immediate from the 180
definition. For the second assertion in the nonparenthetical case, since the $X_i \to X$ are monomorphisms, it remains only to prove sufficiency, which follows easily from the definitions and 1.3; the parenthetical case follows from the case just treated and 1.4.

COROLLARY 1.16. *Under the conditions of 1.10 (ii), if the Y_i are coherent, then f is quasi-compact if and only if the $X_i = X \times_Y Y_i$ are quasi-compact objects of E.*

Indeed, by the criterion 1.10 (ii), one must express that the $f_i : X_i \to Y_i$ are quasi-compact; by 1.14 (i), this is equivalent to the X_i being quasi-compact.

COROLLARY 1.17. *Let E be a topos admitting a generating family formed by quasi-compact objects, and let $(g_i : Y_i \to Y)_{i\in I}$ be a covering (resp. finite covering) family in E, with the Y_i coherent. The object Y is quasi-separated (resp. coherent) if and only if, for every $i \in I$, g_i is quasi-compact, or equivalently if, for every pair of indices $i, j \in I$, $Y_i \times_Y Y_j$ is a quasi-compact object of E.*

The parenthetical case follows immediately from the nonparenthetical case and 1.3. Necessity of the first condition is immediate from the definition; let us prove sufficiency.
181 Let $f : X \to Y$ be a morphism, with X quasi-compact; we prove that f is quasi-compact. Since the g_i are quasi-compact, the $X_i = X \times_Y Y_i$ are quasi-compact, which implies by 1.16 that f is quasi-compact. Finally, expressing the quasi-compactness of the g_i by the same criterion 1.16 gives the second criterion of 1.17. In particular:

COROLLARY 1.17.1. *Let E be as in 1.17, let X be a coherent object of E, let $\mathscr{R} \rightrightarrows X$ be an equivalence relation in X, and let $Y = X/\mathscr{R}$. Then the following conditions are equivalent:*

(i) *Y is coherent;*
(ii) *$f : X \to Y$ is quasi-compact;*
(iii) *$\mathscr{R}$ is quasi-compact.*

COROLLARY 1.18. *Let E be a topos admitting a generating subcategory C formed by quasi-compact objects, and let Y be an object of E. The object Y is quasi-separated if and only if every morphism $X \to Y$ with $X \in \mathrm{ob}\, C$ is quasi-compact.*

Necessity is immediate from the definition, since the objects $X \in \mathrm{ob}\, C$ are quasi-compact by hypothesis. To prove sufficiency, let Z be a quasi-compact object of E; we prove that every morphism $Z \to Y$ is quasi-compact. By 1.10 (i), it is enough to verify that, for every $X \to Y$ with $X \in \mathrm{ob}\, C$, the object $Z \times_Y X$ is quasi-compact. This follows from the fact that the $X \to Y$ are quasi-compact by hypothesis and that Z is quasi-compact.

182 COROLLARY 1.19. *Let E be a topos admitting a generating subcategory formed by quasi-compact objects, and let $(g_i : Y_i \to Y)_{i \in I}$ be a covering family in E, with the Y_i quasi-separated and the g_i quasi-compact. Then Y is quasi-separated.*

Indeed, let X be a quasi-compact object of E; we show that every morphism $f : X \to Y$ is quasi-compact. By 1.10 (ii), it is enough to prove that the induced morphisms $f_i : X_i = X \times_Y Y_i \to Y_i$ are quasi-compact. Since g_i is quasi-compact, X_i is quasi-compact (because X is), and since Y_i is quasi-separated, it follows that f_i is quasi-compact.

REMARK 1.20.1. The reader accustomed to the use of the terms "quasi-compact" and "quasi-separated" in algebraic geometry will expect certain other relations between the "absolute" notions (concerning objects of the topos E) and the "relative" notions (concerning arrows of E) considered here; for example, if $f : X \to Y$ is a morphism in E such that X is quasi-compact and quasi-separated, then f is quasi-separated. Such properties hold only under restrictive finiteness conditions on the topos E, stronger than those considered in 1.10; we shall study these conditions in the next section.

1.20.2. It follows again from 1.5 that whether an object X of the topos E is quasi-separated depends only on the induced topos $E_{/X}$, and means that the final object of that topos is quasi-separated, i.e. that in $E_{/X}$ the product of two quasi-compact objects is quasi-compact.

183 **1.21. Case of sites.** Let C be a $\mathscr{U}$-site, E the topos of $\mathscr{U}$-sheaves on C, f an arrow of C, and X an object of C. We say that f is a *quasi-compact morphism* (resp. quasi-separated morphism) *of the site C* if $\epsilon(f)$ is a quasi-compact morphism (resp. quasi-separated morphism) of the topos E; likewise, we say that X is a *quasi-separated object of the site C* if $\epsilon(X)$ is a quasi-separated object of the topos E (compare 1.2). When C is a $\mathscr{U}$-topos endowed with its canonical topology, the canonical functor $\epsilon : C \to E$ is an

equivalence of categories, and consequently the preceding definition is compatible with the earlier definitions 1.7 and 1.13. Note also that the definition under consideration does not change when the universe $\mathscr{U}$ is replaced by another universe $\mathscr{V}$ such that C is also a $\mathscr{V}$-site, at least when C admits a topologically generating family consisting of quasi-compact objects. To see this, we are immediately reduced to the case $\mathscr{U} \subset \mathscr{V}$, so that E is a $\mathscr{V}$-site, and to proving that, for every arrow f of the $\mathscr{U}$-topos E, f is quasi-compact (resp. quasi-separated) if and only if it defines a quasi-compact arrow in the topos E' of $\mathscr{V}$-sheaves on E, and likewise that, for every object X of E, X is quasi-separated if and only if the corresponding object of E' is quasi-separated. We already know (without any hypothesis on E) the analogous statement for a quasi-compact object (1.2). The case "f quasi-compact" follows from this, by the criterion 1.10 (i), applied successively in E and in E' to the same generating family $(X_i)_{i\in I}$ of E consisting of 184
quasi-compact objects. This also gives the case "f quasi-separated", which means that the corresponding diagonal morphism is quasi-compact. Finally, in the case "X quasi-separated", we are again reduced to the case "X quasi-compact" by the criterion 1.18.

1.22. ***Examples.*** Return to the example 1.6.2 of the category (Sch) with a topology T_i. This is a $\mathscr{V}$-site for a suitable $\mathscr{V}$, and it admits the subcategory of affine schemes as a topologically generating category consisting of quasi-compact objects. We leave it to the reader to verify that an arrow $f : X \to Y$ of this site is quasi-compact (resp. quasi-separated) if and only if it is a quasi-compact (resp. quasi-separated) morphism of schemes in the usual sense ([3] or [4]); likewise, an object of the site is quasi-compact (resp. quasi-separated) if and only if it is a quasi-compact (resp. quasi-separated) scheme in the usual sense of *loc. cit.*. It is in fact appropriate to say, as we do here, *coherent morphism of schemes, coherent scheme*, instead of "quasi-compact and quasi-separated morphism of schemes", "quasi-compact and quasi-separated scheme", as is also done in EGA I, 2nd edition.

1.22.1. For a given scheme $X \in \mathscr{U}$, one may also consider the topos $\mathrm{Top}(X)$ associated with the underlying topological space, or the topos $\mathrm{Top}(X_{\mathrm{et}})$ (resp. $\mathrm{Top}(X_{\mathrm{fppf}})$) associated with the site of étale X-schemes (resp. X-schemes locally of finite presentation) 185
that are elements of $\mathscr{U}$, endowed with the étale topology (resp. fppf topology). It is a $\mathscr{U}$-topos admitting a generating family consisting of coherent objects, corresponding to the affine objects of the site of definition.

With this understood, the final object of this topos is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the scheme X is quasi-compact (resp. quasi-separated, resp. coherent). Likewise, a morphism of the site of étalé spaces over X (resp. of the site X_{et} of étale schemes over X, resp. of the site X_{fppf} of schemes locally of finite presentation over X) is quasi-compact (resp. quasi-separated, resp. coherent) if and only if it is a morphism of schemes that is, in the usual sense of *loc. cit.*, quasi-compact (resp. quasi-separated, resp. coherent).

THEOREM 1.23. *Let E be a $\mathscr{U}$-topos and X an object of E.*

(i) *For X to be quasi-compact, it is necessary and sufficient that, for every filtered inductive system $(Y_i)_{i\in I}$ of E, with $I \in \mathscr{U}$, the natural map*

$$\varinjlim \mathrm{Hom}(X, Y_i) \longrightarrow \mathrm{Hom}(X, \varinjlim Y_i) \tag{1.23.1}$$

be injective.

(ii) *Suppose that E admits a generating subcategory C consisting of quasi-compact objects. A necessary condition for X to be coherent is that, for every datum as in* (i)*, the map* (1.23.1) *be bijective.*

186 (i) Necessity. Suppose that X is quasi-compact. We must prove that if $i \in I$ and $f_i, g_i : X \rightrightarrows Y_i$ are such that the composites f, g of f_i, g_i with $Y_i \to Y = \varinjlim Y_j$ are equal, then there exists $j \geq i$ such that the composites f_j, g_j with $Y_i \to Y_j$ are already equal. Since filtered inductive limits in E commute with finite projective limits, and in particular with kernels, we have

$$\mathrm{Ker}(f, g) = \varinjlim_{j \geq l} \mathrm{Ker}(f_j, g_j).$$

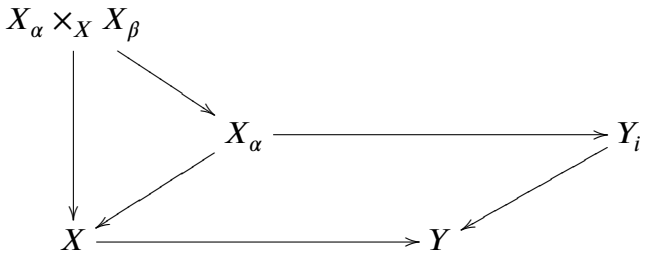
By hypothesis $\mathrm{Ker}(f, g) = X$, so X is the limit of the filtered increasing family of its subobjects $\mathrm{Ker}(f_j, g_j)$. Since X is quasi-compact, it is equal to one of these subobjects; hence there exists $j \geq i$ such that $f_j = g_j$.

Sufficiency. To prove that X is quasi-compact, it is equivalent (1.5.2) to prove that, for every filtered increasing family $(X_i)_{i \in I}$ of subobjects of X whose $\varinjlim$ is X, there exists $i \in I$ such that $X_i = X$. Since every monomorphism in E is effective (II 4.0), it follows that

$$X_i = \mathrm{Ker}(f_i, g_i : X \rightrightarrows X \amalg_{X_i} X).$$

Let $Y_i = X \amalg_{X_i} X$, and consider the inductive system formed by the Y_i. If Y is its inductive limit, then $Y \simeq X \amalg_{\varinjlim X_i} X = X \amalg_X X = X$. Consequently, for a fixed $i \in I$, f_i, g_i give the same element of $\mathrm{Hom}(X, Y)$, so by hypothesis there exists $j \geq i$ such that $f_j = g_j$, i.e. $X_j = X$, C.Q.F.D.

187 (ii) It remains to prove that, under the stated conditions, every morphism $X \to Y$ comes from a morphism $X \to Y_i$ for a suitable $i \in I$. Since C generates E and the Y_i cover Y, we may cover X by objects X_α of C in such a way that each composite $X_\alpha \to X \to Y$ factors through one of the Y_i. Since X is quasi-compact, we may suppose that the family of the X_α is finite, and hence that the morphisms $X_\alpha \to Y$ factor through a fixed Y_i as morphisms $g_{\alpha i} : X_\alpha \to Y_i$. Since X is quasi-separated, the $X_\alpha \times_X X_\beta$ are quasi-compact. Moreover, for every (α, β), the composites of pr_1 and pr_2 on $X_\alpha \times_X X_\beta$ with $X_\alpha, X_\beta \to Y_i$ are



such that their composites with $Y_i \to Y$ are equal. By (i), there therefore exists $j \geq i$ such that the composites with $Y_i \to Y_j$ are already equal. Since the set of pairs (α, β) is finite, j may be chosen independently of the pair. Consequently, the morphisms $g_{\alpha j}$ arise by composition from a morphism $g_j : X \to Y_j$, whose composite with $Y_j \to Y$ is moreover the given morphism $X \to Y$, since this is so after composition with the covering maps $X_\alpha \to X$. This completes the proof.

Remarks 1.23.2. The hypothesis on E in (ii) may be replaced by an additional hypothesis *on* X, namely that X admits a fundamental system of covering families by quasi-compact objects.

188 Corollary 1.24. *Let E be a topos, and let C be a strictly full subcategory of E consist-*

ing of coherent objects. Using the fact that the $\mathscr{U}$-inductive limits in E are representable, one obtains (I 8.6.1) *a canonical functor*

$$\mathrm{Ind}(C) \longrightarrow E. \tag{1.24.1}$$

This functor is fully faithful. If E admits a generating subcategory consisting of coherent objects, and if C is stable under direct factors, then the following conditions are equivalent:

(i) *The functor* (1.24.1) *is essentially surjective, i.e. it is an equivalence of categories.*
(ii) *C is a generating subcategory of E, stable under finite inductive limits.*
(iii) *The category C is equal to the full subcategory E_{coh} of E consisting of all coherent objects of E, and the necessary condition* 1.23 (ii) *for an object of E to be coherent is also sufficient.*

Let E_{PF} be the full subcategory of E defined in I 8.7.6. By I 8.7.5 a), statement 1.23 (ii) is equivalent to the second inclusion in

$$C \subset E_{\mathrm{coh}} \subset E_{PF},$$

and the composite inclusion $C \supset E_{PF}$ means that (1.24.1) is fully faithful. Since , by 189
hypothesis, the subcategory E_{coh} of E is generating, the same is a fortiori true of E_{PF}; moreover, finite projective limits are representable in E, so that we may apply I 8.7.7. Condition (iii) of 1.24 may also be written $C = E_{\mathrm{coh}} = E_{PF}$ and is equivalent to the condition $C = E_{PF}$, which is precisely condition (ii bis) of *loc. cit.* (taking into account that, by hypothesis, E_{coh} and hence E_{PF} is a generating subcategory of E). Likewise, condition (ii) of 1.24 is precisely condition (iii bis) of *loc. cit.*. Thus the equivalence of conditions (i), (ii bis), and (iii bis) in *loc. cit.* gives the equivalence of conditions (i), (ii), and (iii) of 1.24, C.Q.F.D.

The proof (or, more precisely, the invocation of I 8.7.7 (ii bis)) also gives part a) of the

Corollary 1.24.2. *Let E be a topos admitting a generating subfamily consisting of coherent objects, and let E_{PF} be the strictly full subcategory of objects X of E such that the covariant functor* $\mathrm{Hom}(X, -)$ *represented by X commutes with filtered inductive limits. Then:*

a) *The natural functor*
$$\mathrm{Ind}(E_{PF}) \longrightarrow E$$
is an equivalence of categories.
b) *For an object X of E to belong to E_{PF}, it is necessary and sufficient that it be isomorphic to a direct factor (= image of a projector (I, 10.6)) of the cokernel of a* 190
double arrow $X_1 \rightrightarrows X_0$, with X_1 and X_0 coherent.
c) *The subcategory E_{PF} of E is stable under finite inductive limits, and is equivalent to a small category.*

It remains to prove b) and c). The first assertion of c) is immediate from I 2.8, and implies the "if" part of b) by 1.23 (ii). For the "only if" part, use 1.23 (i), which shows that X is quasi-compact, whence the existence of an epimorphism $X_0 \to X$, with X_0 coherent, in view of the fact that E_{coh} generates E and is stable under finite sums. Let $R = X_0 \times_X X_0$, so that X identifies with the cokernel of $R \rightrightarrows X_0$. By a), we have $R = \varinjlim R_i$, the filtered inductive limit of the coker $(R_i \rightrightarrows X_0) = X_i$. By the hypothesis $X \in \mathrm{Ob}\, E_{PF}$, for sufficiently large i the morphism $u_i : X_i' \to X$ admits a section $v : X \to X_i'$. Let $w = vu_i : X_i' \to X_i'$, so that $w^2 = w$ and $X \simeq \mathrm{Coker}(w, \mathrm{id}_{X_i'} : X_i' \rightrightarrows X_i')$. Since R_i is again quasi-compact, cover it by $X_1 \to R_i$, with X_i coherent, whence $X_i \simeq \mathrm{Coker}(X_1 \rightrightarrows X_0)$, which proves b).

Finally, since $E_{PF} \subset E_{\text{qu.cpct}}$, the last assertion of c) follows from 1.5.3.

COROLLARY 1.25. *Let E be a topos such that the full subcategory C of E consisting of coherent objects is generating. Consider the canonical functor*

$$\varphi : \mathrm{Ind}(c) \longrightarrow E.$$

191 *The following conditions are equivalent:*

(i) *The functor φ is essentially surjective, i.e. every object of E is a filtered inductive limit of coherent objects.*
(i bis) *The functor φ is an equivalence of categories.*
(ii) *Every finite inductive limit in E of coherent objects is a coherent object.*
(ii bis) *The cokernel in E of a double arrow of coherent objects is a coherent object.*
(iii) *The converse of* 1.23 (ii) *holds.*

This is a special case of 1.24, taking into account that C is stable under finite sums (1.4), which implies the equivalence of (ii) and (ii bis), and taking into account the

LEMMA 1.25.1. *Let E be a topos. Every direct factor X' of a coherent object X of E is coherent.*

Indeed, X' is quasi-compact as a quotient of X (1.3), and quasi-separated as a subobject of X (1.15).

COROLLARY 1.26. *Let E be a topos satisfying the equivalent conditions* 1.25. *Then every induced topos does likewise.*

This follows immediately from criterion (ii), taking into account 1.20.2 and the fact that the existence of a generating subcategory of E consisting of coherent objects plainly implies the same property for an induced topos.

192 REMARK 1.27. Among the important examples of topoi satisfying the conditions of 1.25, let us mention Noetherian topoi (2.14), and the Zariski or étale topos (VII 1) of a scheme X (cf. IX, note p.42). But even when the topos E is *coherent* (2.3) (i.e. the full subcategory C of coherent objects is generating and stable under finite projective limits), it is not always true that C is *perfect* (2.9.1), i.e. satisfies the equivalent conditions of 1.25; cf. 1.28 below.

EXERCISE 1.28. Let E be a topos and $p_1, p_2 : X_1 \rightrightarrows X_0$ a double arrow in E. Define an increasing sequence of subobjects R_n ($n \geq 0$) of $X_0 \times X_0$ by the condition $R_0 = \mathrm{Sup}(\mathrm{Im}(p_1, p_2), \mathrm{Im}(p_2, p_1))$, and $R_n = \mathrm{pr}_{13}(\mathrm{pr}_{12}^{-1}(R_{n-1}), \mathrm{pr}_{23}^{-1}(R_{n-1}))$ for every $n \geq 1$, where pr_{ij} ($1 \leq i < j \leq 3$) denote the evident projections from the triple product of X_0 to the double product.

a) Show that one thus obtains an increasing sequence of subobjects of $X_0 \times X_0$, that $R = \varinjlim R_n$ is an equivalence graph (I 10.4) in X_0, and that the canonical morphism
$$X = \mathrm{Coker}(p_1, p_2) \longrightarrow X_0/R$$
is an isomorphism.

b) Deduce that if X_0 is quasi-compact and $X = \mathrm{Coker}(p_1, p_2)$ is quasi-separated
193 (hence coherent), then the sequence (R_n) is stationary. Conversely, if this sequence is stationary, and if X_0 is assumed coherent and X_1 quasi-compact, then X is coherent. (For this last point, write $R_n \simeq (R_{n-1}, \mathrm{pr}_2) \times_{X_0} (R_{n-1}, \mathrm{pr}_1)$ and deduce that R_n is quasi-compact over X_0 for every n, then use 1.19.)

c) Construct an example of two maps of sets $p_1, p_2 : X_1 \rightrightarrows X_0$ such that the sequence (R_n) is not stationary.

d) Let C be a small category in which finite inductive limits and finite projective limits are representable, and in which every morphism factors as an effective epimorphism followed by a monomorphism. Repeat the construction of the R_n for a double arrow (p_1, p_2) of C. Endowing C with the canonical topology, show that the canonical functor $\epsilon : C \to C^\sim$ is exact, and consequently commutes with the formation of the R_n.

e) Take for C a small full subcategory of (Ens), stable up to isomorphism under finite projective limits and finite inductive limits, and containing the sets X_0, X_1 of c). Deduce that the topos $C^\sim$ (which is a *coherent* topos (2.3), i.e. is generated by the subcategory of its coherent objects, which is stable under finite projective limits) does not satisfy the equivalent conditions of 1.25.

f) Let k be a field, and let C be the fppf site of schemes locally of finite presentation over k (SGA 3 IV 6.3). Show that there exists a reduced affine scheme X over k admitting an automorphism f that is not of finite order (take, for example, the affine plane E^2 endowed with the automorphism $f(x) = x$, $f(y) = x+y$). Show 194
that if $p_1, p_2 : X_1 \rightrightarrows X_0$ is a double arrow in C such that the corresponding double arrow in (Ens) $p_1(\overline{k}) : p_2(\overline{k}) : X_1(\overline{k}) \rightrightarrows X_0(\overline{k})$ gives a nonstationary sequence (R_n), then the same is true of the double arrow of $C^\sim$ defined by p_1, p_2. Thus, if X_0 is of finite type over k, the cokernel in $C^\sim$ of (p_1, p_2) is not coherent. Deduce that the cokernel in $C^\sim$ of the pair (f, id_X) is not coherent, and hence that the topos $C^\sim$ does not satisfy the equivalent conditions of 1.25. Deduce more generally that if S is a nonempty scheme and C is the fppf site of schemes locally of finite presentation over S, then the topos $C^\sim$ does not satisfy the conditions of 1.25. (Care must be taken that, contrary to appearances, $C^\sim$ is therefore Noetherian (2.11) only if S is empty, as follows from the preceding discussion and 2.14.)

DEFINITION 1.30. *Let E be a topos. An object X of E is called a* pre-Noetherian object[34] *of the topos E if it satisfies the following two equivalent conditions:*

(i) *Every subobject of X is quasi-compact.*

(ii) *Every increasing sequence of subobjects of X is stationary.*

1.30.1. The equivalence of conditions (i) and (ii) is clear. Note that these conditions again depend only on the induced topos $E_{/X}$, and indeed only on the ordered set of opens of $E_{/X}$, which is isomorphic to the ordered set of subobjects of X. They are stable under passage to a subobject of X.

1.31. One proves, as in 1.3 and 1.4, the following facts: if $(X_i \to X)$ is a finite 195
covering family, with the X_i pre-Noetherian, then X is pre-Noetherian; in particular, a quotient of a pre-Noetherian object of E is pre-Noetherian. If $(X_i)_{i\in I}$ is a family of objects of E, then their coproduct X is a pre-Noetherian object of E if and only if all but finitely many of the X_i are isomorphic to $\varnothing_E$, and all the X_i are pre-Noetherian.

1.32. Clearly, a pre-Noetherian object of a topos E is quasi-compact. When E admits a generating family formed by pre-Noetherian objects, the converse holds (as follows immediately from 1.31): the pre-Noetherian objects of E are then precisely its quasi-compact objects.

[34] The stronger notion of a *Noetherian* object is given in 2.11 below.

Suppose that E admits a generating family $(X_i)_{i\in I}$ formed by quasi-compact objects. Then the following three conditions are equivalent:

(i) The X_i are pre-Noetherian.
(ii) Every quasi-compact object of E is pre-Noetherian.
(iii) Every monomorphism in E is quasi-compact.

(We have just proved the equivalence of (i) and (ii); (ii) $\Rightarrow$ (iii) is immediate from the definitions, and (iii) $\Rightarrow$ (ii) follows from Definition 1.30 (i).)

196 Note that (iii) implies that *every morphism of E is quasi-separated*; hence (1.14 (ii)) every object of E lying over a quasi-separated object of E is itself quasi-separated.

EXAMPLE 1.33. (*Classifying topos of a group*). Let G be a group $\in \mathscr{U}$, and let B_G be its classifying topos (IV 2.4), i.e. the category of sets $\in \mathscr{U}$ on which G acts on the left. The nonempty connected objects of B_G (IV 4.3.5) are the nonempty G-sets on which G acts transitively, and every object E of B_G is the coproduct of its connected components, which are the minimal nonempty subobjects of E (namely, the orbits of G in E). Consequently, E is a quasi-compact object of B_G if and only if the set $\pi_0(E)$ of orbits of G in E is finite, and in that case E is even a pre-Noetherian object of B_G. These objects (and even the single object G_s, which is nonempty and connected) form a generating subcategory of B_G. The final object e of B_G is quasi-separated, i.e. the product of two quasi-compact objects of B_G is quasi-compact, if and only if the product $G_s \times G_s$ is quasi-compact, which is plainly equivalent to G being finite. More generally, one concludes that an object E of B_G is quasi-separated if and only if the stabilizers of its points are *finite* subgroups of G, i.e. if it is isomorphic to a coproduct of homogeneous spaces G/H, with H *finite*. Indeed, by 1.15, one may assume in this criterion that E is
197 connected; but if $x \in E$ has stabilizer group G_x, we know (IV 5.8) that B_G/E is equivalent to the topos B_{G_x}, and hence its final object is quasi-separated if and only if G_x is finite. This criterion shows that every object of B_G lying over a quasi-separated object is quasi-separated; a fortiori, the product of two quasi-separated objects of B_G is quasi-separated. It also shows that the coherent objects of B_G are the objects isomorphic to finite coproducts of homogeneous spaces G/H, with H *finite*. Since G_s is coherent, the coherent objects of B_G already form a generating subcategory of B_G. Moreover, this subcategory is stable under fiber products (as follows from the fact that every object lying over a quasi-separated object is quasi-separated). (In the terminology of 2.11 below, B_G is a *locally Noetherian* topos, but it is quasi-separated only when G is finite, in which case B_G is even Noetherian.)

It is immediate that a morphism $f : E' \to E$ is quasi-compact if and only if the induced map $\pi_0(f) : \pi_0(E') \to \pi_0(E)$ on the sets of orbits is proper, i.e. has finite fibers. We recover the fact that a monomorphism is quasi-compact (1.32), and hence that every morphism is quasi-separated. Thus the coherent morphisms in B_G are simply the quasi-compact morphisms, which have just been characterized.

Let E be an object of the topos B_G. One verifies easily that E is a filtered inductive limit of coherent objects of B_G if and only if the stabilizers of the points of E are ind-
198 finite groups (i.e. filtered inductive limits of their finite subgroups). In particular, if G is not ind-finite (for example, if $G = \mathbf{Z}$), the final object of the topos B_G is not an inductive limit of coherent objects. More precisely, the topos B_G satisfies the equivalent conditions of 1.25 if and only if its final object is a filtered inductive limit of coherent objects, or equivalently if the group G is ind-finite.

2. Finiteness conditions for a topos

PROPOSITION 2.1. *Let E be a $\mathscr{U}$-topos. The following conditions are equivalent:*

(i) *There exists a full generating subcategory C of E, formed by quasi-compact objects and stable under fiber products.*

(ii) *There exists a $\mathscr{U}$-site C such that every object of C is quasi-compact, fiber products are representable in C, and E is equivalent to the category of sheaves $C^{\sim}$.*

Suppose that these conditions hold. Then in (i) (resp. (ii)) *the category C may even be chosen to be $\mathscr{U}$-small. Moreover, for every $X \in \mathrm{ob}\, C$, the object X* (resp. $\epsilon(X)$) *of the topos E is* coherent.

The implication (i) ⇒ (ii) follows from IV 1.2.1, and the fact that in (i) (resp. (ii)) one may take C to be $\mathscr{U}$-small follows immediately from the argument of IV 1.2.3, applied to a small full generating subcategory in the topological sense (II 3.0.1) of C. It remains to prove (ii) ⇒ (i) and the last assertion of 2.1; these follow immediately from the following result.

COROLLARY 2.1.1. *Let C be a $\mathscr{U}$-site in which fiber products are representable and every* 199
object is quasi-compact, let $E = C^{\sim}$, and let $\varepsilon : C \to E$ be the canonical functor. Then, for every $X \in \mathrm{ob}\, C$, $\varepsilon(X)$ is a coherent object of E. The full subcategory of E formed by the coherent objects of E lying over some $\varepsilon(X)$ is stable under fiber products (and is plainly generating in E).

We already know that $\epsilon(X)$ is quasi-compact (1.2); let us prove that it is quasi-separated. Let F, G be two quasi-compact objects of E, and let $F \to \varepsilon(X)$, $G \to \epsilon(X)$ be two morphisms. We must prove that the fiber product $F \times_{\epsilon(X)} G$ is quasi-compact. Covering F and G by finitely many objects of the form $\epsilon(Y_i)$ and, respectively, $\epsilon(Z_j)$, and proceeding as in 1.2 to reduce to the case in which the composites $\epsilon(Y_i) \to \epsilon(X)$ and $\epsilon(Z_j) \to \epsilon(X)$ come from morphisms $Y_i \to X$ and, respectively, $Z_j \to X$, we are reduced by 1.3 to the case in which the morphisms $F \to \epsilon(X)$ and $G \to \epsilon(X)$ are obtained by applying ϵ to morphisms $Y \to X$ and $Z \to X$. But since fiber products are representable in C and ϵ commutes with them, we are reduced to proving that $\epsilon(Y \times_X Z)$ is quasi-compact, which was already noted above for every object of the form $\epsilon(U)$.

Now let $G \to F$, $H \to F$ be morphisms between coherent objects of E, and suppose that there exists a morphism $F \to \epsilon(X)$ for some $X \in \mathrm{ob}\, C$. We prove that $G \times_F H$ is coherent. Since G and H are quasi-compact and F is quasi-separated, it already follows from the definitions that the fiber product is quasi-compact. It remains to see that it is 200
quasi-separated, or, equivalently (1.15), that $G \times_{\epsilon(X)} H$ is quasi-separated. This follows from the more general fact:

COROLLARY 2.1.2. *Under the conditions of 2.1.1, for every quasi-separated object G of E, every morphism $G \to \epsilon(X)$ is quasi-separated. If $H \to \epsilon(X)$ is a second morphism, with H quasi-separated, then $G \times_{\epsilon(X)} H$ is quasi-separated.*

The first assertion implies the second: by base change, it implies that $G \times_{\epsilon(X)} H \to H$ is quasi-separated, and since H is quasi-separated, $G \times_{\epsilon(X)} H$ is likewise quasi-separated (1.14 (ii)). To prove that $G \to \epsilon(X)$ is quasi-separated, we shall use the following lemma.

LEMMA 2.1.3. *Let E be a topos and $f : X \to Y$ a morphism in E. Suppose that E admits a generating family of quasi-compact objects, and that there exists a covering family $(g_i : X_i \to X)_{i \in I}$ of X such that the X_i are quasi-compact and, for every pair of indices $i, j \in I$, $X_i \times_Y X_j$ is quasi-separated. Then, if X is a quasi-separated object of E, f is a quasi-separated morphism.*

Indeed, the family $(g_i \times_Y g_j : X_i \times_Y X_j \to X \times_Y X)_{(i,j)\in I\times I}$ is covering. Therefore, to see that the morphism $\Delta_{X/Y} : X \to X \times_Y X$ is quasi-compact, it is enough to see that,
201 for every $(i, j) \in I \times I$, the morphism obtained from it by base change along $g_i \times_Y g_j$ is quasi-compact (1.10 (ii)). The latter is precisely the canonical morphism $X_i \times_X X_j \to X_i \times_Y X_j$. By hypothesis X is quasi-separated and the X_i are quasi-compact, so $X_i \times_X X_j$ is quasi-compact; and since $X_i \times_Y X_j$ is quasi-separated by hypothesis, the morphism $X_i \times_X X_j \to X_i \times_Y X_j$ is indeed quasi-compact, C.Q.F.D.

We can now complete the proof of 2.1.2 by proving that every morphism $f : G \to \epsilon(X)$, with G quasi-separated, is quasi-separated. To do this, cover G by objects $\epsilon(X_i)$ such that the composite morphisms $\epsilon(X_i) \to \epsilon(X)$ come from morphisms $X_i \to X$. By 2.1.3, since the $\epsilon(X_i)$ are quasi-compact, it is enough to verify that the fiber products $\epsilon(X_i) \times_{\epsilon(X)} \epsilon(X_j) = \epsilon(X_i \times_X X_j)$ are quasi-separated. We have already seen that every object of E of the form $\epsilon(U)$ is coherent. This proves 2.1.2 and completes the proof of 2.1.

Proposition 2.2. *Let E be a topos containing a full generating subcategory C formed by coherent objects. The following conditions on E are equivalent:*

(i) *Every quasi-separated object of E is quasi-separated over the final object.*
(i bis) *For every morphism $f : X \to Y$ in E, if X is quasi-separated, then f is quasi-separated.*
(i ter) *Every object X of C is quasi-separated over the final object of E, i.e. the diagonal morphism $X \to X \times X$ is quasi-compact.*
202 (ii) *The product of two quasi-separated objects of E is quasi-separated.*
(ii bis) *The product in E of two objects of C is quasi-separated.*

These conditions imply that the full subcategory E_{coh} of E formed by the coherent objects of E is stable under fiber products (and, a fortiori, that E satisfies the equivalent conditions of 2.1).

Clearly (i bis) ⇒ (i), and the converse follows from 1.8 (iii). We have (i) ⇒ (ii) by an argument already used: if X and Y are quasi-separated, then by hypothesis (i), X is quasi-separated over e, so $X \times Y$ is quasi-separated over Y by base change, and hence $X \times Y$ is quasi-separated by 1.14 (ii). The same argument shows that (i ter) ⇒ (ii bis), while (i) ⇒ (i ter) and (ii) ⇒ (ii bis) are immediate. It therefore remains to prove (ii bis) ⇒ (i). If X is a quasi-separated object of E, cover X by objects X_i of C (which is possible since C is generating). Since the X_i are quasi-compact by the hypothesis on C, we may apply 2.1.3 to the morphism $X \to e$; thus, to prove that this morphism is quasi-separated, we are reduced precisely to verifying that the $X_i \times X_j$ are quasi-separated, which is hypothesis (ii bis).

Remark 2.2.1. Condition (i ter) is also equivalent to the following condition (i quater): for every double arrow $g_1, g_2 : Y \rightrightarrows X$ in C, its kernel in E is quasi-compact (or equivalently (1.15) coherent).

203 To see this, it is enough to apply criterion 1.10 (i) to the diagonal morphism $X \to X \times X$ considered in (i ter).

Definition 2.3. *A topos E satisfying the equivalent conditions of 2.2* (resp. 2.1) *is called an* algebraic topos (resp. *a* locally algebraic topos, *or* locally coherent). *An object X of a topos E is called an* algebraic object of the topos E *if the induced topos $E_{/X}$ is algebraic. A topos E is called* quasi-separated (resp. coherent) *if it is algebraic and its final object is quasi-separated (resp. coherent).*

2.3.1. It is in algebraic topoi that the finiteness notions developed in n° 1 have fully satisfactory properties, analogous to those of the notions bearing the same names in the category of schemes. All locally algebraic (= locally coherent) topoi encountered in practice are in fact algebraic, so the practical interest of this notion seems rather limited for the present. Nevertheless, one can construct locally algebraic topoi that are not algebraic (2.17 f).

2.4. Here are some general remarks concerning the notions introduced in 2.3, which will convince the reader (or so we hope) that the terminology introduced there is coherent.

2.4.1. Under the conditions of 2.1 (i) resp. (ii), for every $X \in C$, X (resp. $\epsilon(X)$) is an algebraic (coherent) object of E (2.1.2 and 1.19.2); in other words, the induced topos 204
$E_{/X}$ (resp. $E_{\epsilon(X)} \cong (C_{/X})^{\sim}$) is algebraic, hence *coherent*. Equivalently, *when C admits a final object, E itself is algebraic, hence coherent.*

2.4.2. If E is an algebraic topos, then every induced topos $E_{/X}$ is also algebraic (2.2 (i bis)). For a topos E to be locally algebraic = locally coherent, it is necessary and sufficient that there exist a family $(X_i)_{i\in I}$ of objects of E covering the final object, such that the X_i are algebraic (resp. coherent), i.e. such that the induced topoi $E_{/X_i}$ are algebraic (resp. coherent). Sufficiency follows immediately from the definitions, and necessity from the observations in 2.4.1. Of course, an algebraic topos is locally algebraic, and if E is a locally algebraic topos, every induced topos $E_{/X}$ is locally algebraic.

2.4.3. If X is an algebraic object of a topos E, then every object X' of E lying over X is again algebraic (cf. 2.4.2). In particular, every object of an algebraic topos is algebraic, and conversely, of course.

2.4.4. The full subcategory of E consisting of the *quasi-separated* objects that are *algebraic* is stable under fiber products and subobjects (1.15), and hence also under kernels. If E is algebraic, this category (which is then simply the category of quasi-separated 205
objects of E) is also stable under products of two objects. If E is quasi-separated (and only then), this category is stable under arbitrary finite projective limits, or equivalently contains the final object of E.

The full subcategory C of E consisting of the *coherent* objects that are *algebraic* is stable under fiber products. If E is locally algebraic, i.e. locally coherent, saying that E is algebraic amounts to saying that C is also stable under kernels of double arrows (2.2.1); and saying that E is quasi-separated amounts to saying that, in addition, C is stable under products of two objects (cf. 1.17). Finally, saying that E is coherent amounts to saying that C is stable under arbitrary finite projective limits, or equivalently that it contains the final object of E.

2.4.5. Let E be a topos. For E to be locally coherent (resp. algebraic, resp. quasi-separated, resp. coherent), it is necessary and sufficient that it admit a full generating subcategory C consisting of quasi-compact objects (which will automatically be coherent (2.1)) and *stable* under fiber products (resp. under fiber products and kernels, resp. under fiber products and products of two objects, resp. under arbitrary finite projective limits); or equivalently that E be equivalent to a topos $C^{\sim}$, where C is a $\mathscr{U}$-site every object of which is quasi-compact, and in which fiber products and products of two objects (resp. all finite projective limits) are representable. (For necessity, take for C the 206
category of objects of E that are coherent and *algebraic*, and apply 2.4; for sufficiency, apply 2.2 (i ter).) In this statement, one may of course also suppose that C is $\mathscr{U}$-small.

2.4.6. Let E be a topos. If E is algebraic, an object X of E is quasi-separated (resp. coherent) if and only if the induced topos $E_{/X}$ is so. When E is no longer assumed

algebraic, one can only say that $E_{/X}$ is quasi-separated (resp. coherent) if and only if X is quasi-separated (resp. coherent) and *algebraic*. This is of little consequence, since we will probably not have to use these notions outside the case in which E is algebraic.

2.4.7. For the notion of a coherent (resp. quasi-separated) object of a topos E to have satisfactory properties, one must restrict to objects that are also algebraic, a condition automatically satisfied if E is algebraic. Since we will probably never have to work with locally algebraic topoi E that are not algebraic, the question of revising the terminology introduced in 1.13, possibly reserving it for algebraic objects alone, therefore does not arise with any great urgency!

Moreover, we have refrained from defining the notion of a quasi-compact topos. In the spirit of the present paragraph, this name would naturally be given to *algebraic* topoi whose final object is quasi-compact.

207 We hesitated to introduce such a usage, since if X is a topological space, it would not be true that X is quasi-compact if and only if the associated topos $\mathrm{Top}(X)$ is so. In this connection, note that $\mathrm{Top}(X)$ is locally coherent if and only if X admits a basis of opens consisting of quasi-compact opens U such that, for $U_j, U_k \subset U_i$, $U_j \cap U_k$ is again quasi-compact (and then $\mathrm{Top}(X)$ is even algebraic). This condition is satisfied for the topological spaces underlying schemes, but rarely for spaces encountered by analysts, even when the spaces themselves are compact. For ease of reference, let us state explicitly:

PROPOSITION 2.5. *Let E be an algebraic topos and $f : X \to Y$ a morphism in E. If Y is a quasi-separated (resp. coherent) object of E, then, for X to be a quasi-separated (resp. coherent) object of E, it is necessary and sufficient that the morphism f be quasi-separated (resp. coherent).*

Sufficiency was already seen in 1.14 (ii), and holds without any hypothesis on E. Necessity in the non-parenthetical case holds even without supposing that Y is quasi-separated, and is simply Definition 2.3, via 2.2 i bis). It remains to prove that if Y is coherent and X is coherent, then f is coherent. Since we already know that f is quasi-separated, it is enough to see that f is quasi-compact, which follows from the fact that X is quasi-compact and Y quasi-separated (1.13).

208 COROLLARY 2.6. *Let E be an algebraic topos, $f : X \to Y$ a morphism in E, and $(Y_i \to Y)_{i\in I}$ a covering family, with the Y_i quasi-separated (resp. coherent). For f to be quasi-separated (resp. quasi-compact, resp. coherent), it is necessary and sufficient that, for every $i \in I$, $X_i = X \times_Y Y_i$ be a quasi-separated (resp. quasi-compact, resp. coherent) object of E.*

It is enough to combine 1.10 (ii) and 2.5 in the "quasi-separated" or "coherent" case; the "quasi-compact" case was already seen in 1.16. Note that the necessity of the condition stated in 2.6 also holds without any hypothesis on the topos E.

As a special case of 2.6, one obtains:

COROLLARY 2.7. *Let E be a coherent topos and X an object of E. For X to be a quasi-compact (resp. quasi-separated, resp. coherent) object of E, it is necessary and sufficient that it be quasi-compact (resp. quasi-separated, resp. coherent) over the final object (1.9.2).*

In particular, X is constructible (1.9.2) if and only if it is coherent.

COROLLARY 2.8. *Let E be a locally coherent topos and $f : X \to Y$ a morphism in E. For f to be quasi-separated, it is necessary and sufficient that, for every quasi-separated*

object Y' of $E_{/Y}$, $f^*(Y') = X \times_Y Y'$ be a quasi-separated object of $E_{/X}$ (or of E, which amounts to the same thing (1.20.2)).

The "only if" part holds without any condition on E, since if Y' is quasi-separated, 209
then $f' : X' = X \times_Y Y' \to Y'$ is quasi-separated by base change, and hence X' is quasi-separated (1.14 (ii)). For the converse, note that there exists a covering family $Y_i \to Y$ of Y by quasi-separated algebraic objects. By 1.10 (ii), to verify that f is quasi-separated it suffices to verify that the $f_i : X_i = X \times_Y X_i \to Y_i$ are so. But the hypothesis on f^* implies that the X_i are quasi-separated, and therefore so are the f_i, since $E_{/Y_i}$ is an algebraic topos.

Remark 2.8.1. Statements 2.5 and 2.6 remain valid if the hypothesis "E algebraic" is replaced by the more general hypothesis "Y is algebraic", as is seen immediately by applying the original statement to the induced topos $E_{/Y}$, which is algebraic. Likewise, in 2.8 it is enough to suppose that $E_{/Y}$ (rather than E) is locally coherent.

Proposition 2.9. *Let E be a topos and C a subcategory of E. The following conditions are equivalent:*

(i) *E is locally coherent (resp. coherent), satisfies the equivalent conditions of 1.25, and C is the full subcategory E_{coh} of E consisting of the coherent objects.*
(ii) *C is a strictly full generating subcategory of E, stable under finite inductive limits and fiber products (resp. and under finite projective limits), and consists of quasi-compact objects.*

The implication (i) ⇒ (ii) follows immediately from the definitions and from formu- 210
lation 1.25 (ii) of the conditions under consideration in 1.25. Let us prove the converse. The fact that C is a full generating subcategory consisting of quasi-compact objects and stable under fiber products (resp. under finite projective limits) implies, by definition, that E is locally coherent (resp. coherent). The additional fact that C is strictly full and stable under finite inductive limits then implies that $C = E_{\mathrm{coh}}$ (by 1.24 (ii) ⇒ (iii)), and that condition 1.25 (ii) is satisfied, C.Q.F.D.

Definition 2.9.1. *A $\mathscr{U}$-topos E is called perfect if it is coherent (2.3) and satisfies the equivalent conditions of 1.25, i.e. if the subcategory E_{coh} of E consisting of the coherent objects of E is stable under finite inductive limits. We say that E is locally perfect if there exists a family $(X_i)_{i\in I}$ of objects of E covering the final object such that, for every $i \in I$, the induced topos $E_{/X_i}$ is perfect.*

2.9.2. It follows immediately from this definition that a perfect (resp. locally perfect) topos is coherent (resp. locally coherent). As examples of perfect (resp. locally perfect) topoi, let us mention the Noetherian (resp. locally Noetherian) topoi introduced in 2.11 below (cf. 2.14), and the Zariski and étale topoi of a coherent scheme (resp. an arbitrary scheme) (IX, note page 42). As a counterexample, let us mention the fppf topos 211
of a scheme S, which is locally coherent (and even coherent if S is a coherent scheme, i.e. quasi-compact and quasi-separated) (VII 5.6), but is not locally perfect if $S \neq \emptyset$ (1.28 f)).

In a locally perfect topos, the notion of constructible object is particularly stable (2.9.3 below), which is one reason why the notion seems useful. Another reason is that, like the notion of a coherent or locally coherent topos, the notion of a perfect or locally perfect topos is stable under formation of the closed subtopos complementary to an open (4.6), and behaves particularly simply under the operation of gluing topoi (4.11, 4.14).

PROPOSITION 2.9.3. *Let E be a locally perfect topos (2.9.1). Then the full subcategory E_{cons} of E consisting of the constructible objects of E (1.9.3) is stable under finite inductive limits (and also, of course, under finite projective limits, by (1.9.3)).*

Since the property of an object of E being constructible is local on E (1.10 (ii)), we are reduced to the case in which E is a perfect topos. But then $E_{\mathrm{cons}} = E_{\mathrm{coh}}$ (2.7), and the conclusion follows from Definition 2.9.1.

212 COROLLARY 2.9.4. *Let E be a locally perfect topos. Then E is perfect if and only if E is coherent. For every object X of E, the induced topos $E_{/X}$ is locally perfect.*

PROBLEM 2.9.5. It is conceivable that perfect topoi are sufficiently special to admit a *structure theory* that is equally explicit (following a model proposed by M.ARTIN at the time of the oral seminar). Let us call a topos *finite* if it is equivalent to a topos of the form $\widehat{C}$, where C is a finite category (cf. Exercise 3.11), and call a topos *profinite* if it is a filtered projective limit of finite topoi (in the sense of 6 below, which applies by 3.12 c)). Since a finite topos is Noetherian (2.17 g)), hence perfect, and a filtered projective limit of perfect topoi is perfect (6), it follows that a profinite topos is perfect. The question is whether, conversely, every perfect topos is profinite. This is equivalent to asking whether every finite subcategory of E_{coh} is contained in a full subcategory F_0 of E_{coh} that is equivalent to a category F_{coh}, where F is a finite topos (cf. 3.11). (If the answer were negative, one would need to find manageable additional intrinsic conditions on a perfect topos that ensure that it is profinite.) Finally, it would remain to study the structure of profinite topoi in terms of a suitable notion of a "Karoubian profinite category", inspired by IV
213 7.6 h). This study has not yet been carried out even in the special case in which E is the Zariski or étale topos of a good coherent scheme X, and should then give invariants finer than the compact space X_{cons} (EGA IV 1.) associated with X.

In the same vein, let us mention the following question: Let E be a perfect topos, E' the topos induced on an open of E, and E'' the corresponding closed subtopos (IV 9.9). If E' and E'' are finite topoi, is the same true of E?

PROPOSITION 2.10. *Let E be a topos. The following conditions are equivalent:*

(i) *E admits a full generating subcategory, stable under fiber products, consisting of pre-Noetherian objects of E.*

(ii) *E admits a full generating subcategory consisting of quasi-separated pre-Noetherian objects (i.e. coherent pre-Noetherian objects).*

(iii) *E is locally coherent (2.3), and every quasi-compact object of E is pre-Noetherian.*

(iii bis) *E is locally coherent and admits a generating family consisting of pre-Noetherian objects of E.*

The equivalence of (iii) and (iii bis) follows from 1.32, and it is trivial that (i) ⇒ (iii bis) and (iii) ⇒ (i); hence (i) is equivalent to (iii) and (iii bis). Finally, (i) ⇒ (ii) follows from the
214 last assertion of 2.1, and it remains to prove (ii) ⇒ (i). This implication follows from the fact that the full subcategory of E formed by the quasi-separated Noetherian objects, if it is generating, is stable under fiber products: such a fiber product is quasi-compact by the definitions, hence Noetherian by the first assertion of 1.32; and it is quasi-separated by the last assertion of 1.32.

DEFINITION 2.11. *A topos E is called locally Noetherian if it satisfies the equivalent conditions of 2.10, and Noetherian if, in addition, its final object is coherent (or, equivalently (1.32), pre-Noetherian and quasi-separated). An object X of a topos E is called a Noetherian object of E if the induced topos $E_{/X}$ is a Noetherian topos.*

2.12. If E is a locally Noetherian topos, then so is every induced topos $E_{/X}$. Conversely, if one can find a family (X_i) covering the final object such that the $E_{/X_i}$ are locally Noetherian, then so is E. If C is a full generating subcategory of E consisting of pre-Noetherian objects and stable under fiber products (2.11 (i)), then for every $X \in C$, $E_{/X}$ is a Noetherian topos, i.e. the objects of C are even *Noetherian* (for, if E is locally Noetherian and $X \in \mathrm{Ob}\, E$, then $E_{/X}$ is Noetherian if and only if X is coherent). It follows that a topos E is locally Noetherian if and only if the final object of E can be covered by Noetherian objects X_i.

2.13. In a locally Noetherian topos E, every monomorphism is quasi-compact and every morphism is quasi-separated (1.32). A fortiori, E is an *algebraic* topos (2.3), and not merely locally coherent, i.e. locally algebraic. Thus a topos E is Noetherian if and only if it is locally Noetherian and *coherent* (2.3). In other words, if E is a locally Noetherian topos, then the Noetherian objects of E are precisely the *coherent* objects of E. 215

2.14. Let E be a locally Noetherian topos. If E is quasi-separated, i.e. if its final object is quasi-separated, then *every* object of E is quasi-separated (1.32); hence the pre-Noetherian objects of E are the same as the Noetherian (i.e. coherent) objects of E. Since a quotient object of a pre-Noetherian object is pre-Noetherian, it follows that E satisfies the equivalent conditions of 1.25 (because it satisfies condition 1.25 (ii bis)). In particular, if C is the full subcategory of E consisting of the Noetherian (coherent) objects of E, then the natural functor

$$\mathrm{Ind}(C) \longrightarrow E$$

is an equivalence of categories. We conclude that *a Noetherian topos is perfect* (2.9.1), and hence that a locally Noetherian topos is locally perfect.

If one assumes only that E is locally Noetherian, it remains true that every object of E is the inductive limit of its pre-Noetherian subobjects (for, in a topos admitting a generating subcategory consisting of quasi-compact objects, every object is the filtered inductive limit of its quasi-compact subobjects). But since a pre-Noetherian object of E is no longer necessarily coherent, this statement then has only very limited usefulness, and we know from (1.33) that the conditions of 1.25 are no longer necessarily satisfied. 216

Example 2.15. (*Topological spaces.*) Let X be a topological space. Then the following conditions are equivalent: (i) the topological space X is Noetherian (i.e. every open subset of X is quasi-compact, i.e. every increasing sequence of open subsets of X is stationary); (ii) the topos $\mathrm{Top}(X)$ is Noetherian; (iii) the final object X of $\mathrm{Top}(X)$ is Noetherian; (iv) the final object X of $\mathrm{Top}(X)$ is pre-Noetherian.

Indeed, trivially (ii) $\Leftrightarrow$ (iii) $\Rightarrow$ (iv) $\Leftrightarrow$ (i). On the other hand, (i) implies that the generating subcategory Ouv(X) of $\mathrm{Top}(X)$, which is stable under finite projective limits, consists of pre-Noetherian objects, whence (ii).

Likewise, one sees that the topological space X is locally Noetherian (i.e. is a union of Noetherian open subsets) if and only if the topos $\mathrm{Top}(X)$ is locally Noetherian.

2.15.1. Thus our Definitions 1.30 and 2.11 agree with the accepted terminology for topological spaces. On the other hand, in the general case we have chosen to give the term "Noetherian object" a stronger meaning than the more naive notion of 1.30, so that all the properties associated with this notion (rather than merely the grammatical structure of the definition) agree fully with the intuition attached to *Noetherian* topological spaces and schemes. 217

The reader will find further examples in the following exercises.

EXERCISE 2.16. (*Spaces with operators and classifying topoi.*)

Let X be a topological space on which a discrete group G acts, giving (IV 2.3) a topos $E = \mathrm{Top}(X, G)$. The objects X' of this topos are interpreted as étalé spaces over X with operator group G, and we note that the induced topos $E_{/X'}$ is canonically equivalent to $\mathrm{Top}(X', G)$.

a) The final object of the topos E is quasi-compact if and only if the quotient topological space X/G is quasi-compact (use IV 8.4.1).

b) Let P be the object of E defined by the trivial étalé space $X \times G$ with the diagonal action of G. Show that the induced topos $E_{/P}$ is equivalent to the topos $\mathrm{Top}(X)$ (compare IV 5.8.3). Conclude that E is locally coherent (resp. locally Noetherian) if and only if $\mathrm{Top}(X)$ is locally coherent (cf. 2.4.7) (resp. locally Noetherian (cf. 2.15)). Show that if E is locally coherent, i.e. locally algebraic, it is in fact
218 algebraic, using the generating family formed by the objects T_U, where U is a coherent open subset of X, $T_U = G \times U$ with the action $g \cdot (u, g') = (u, gg')$ of G, and the structural morphism $p_U : T_U \to X$ defined by $p_U(u, g) = g \cdot u$.

c) Suppose that E is algebraic, i.e. that $\mathrm{Top}(X)$ is algebraic (2.4.7). Show that the structural morphism $P \to e$ is quasi-separated. Show that E is quasi-separated if and only if $\mathrm{Top}(X)$ is quasi-separated (or, equivalently, the topological space X is quasi-separated, i.e. the intersection of two quasi-compact open subsets of X is a quasi-compact open subset), *and* G is finite or X is empty. (Compare 1.33.)

d) Show that the structural morphism $P \to e$ is quasi-compact if and only if G is finite. Show that E is coherent (resp. Noetherian) if and only if $\mathrm{Top}(X)$ is so, *and* G is finite or X is empty.

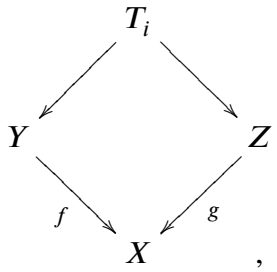
e) Let E be a topos and G a group of E, giving a classifying topos B_G (IV 2.4). Study the finiteness conditions in B_G, using what precedes as a guide. Ask the same question starting from a strict pro-group $\mathscr{G} = (G_i)_{i \in I}$ of E. In particular, show that if E is coherent (resp. Noetherian) and the G_i are coherent, then $B_{\mathscr{G}}$ is coherent (resp. Noetherian).

EXERCISE 2.17. (*Topoi of the form* $\widehat{C}$.) Let C be a category equivalent to a category $\in \mathscr{U}$, and let $E = \widehat{C}$ be the category of presheaves on C. By means of the canonical functor $\varepsilon : C \to \widehat{C}$, we identify C with a full subcategory of E.

219 (a) C is a generating subcategory of E consisting of quasi-compact objects. These objects are pre-Noetherian (cf. 1.32) if and only if, for every object X of C, every sieve on X is generated by a finite family of morphisms $X_i \to X$ of C. The final object of E is quasi-compact (resp. pre-Noetherian) if and only if there is a final subcategory of C whose set of objects is finite (resp. if every sieve on C is generated by a subcategory of C whose set of objects is finite).

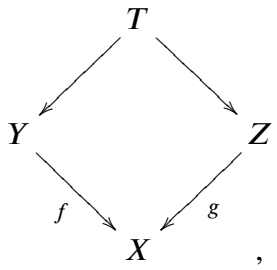
(b) The topos E admits a generating subcategory consisting of coherent (or, equivalently, quasi-separated) objects if and only if the objects of C are coherent in E, or equivalently if, for two arrows $Y \xrightarrow{f} X$, $Z \xrightarrow{g} X$ in C, the object $Y \times_X Z$ of E is quasi-compact, i.e. if one can find a finite family of commutative squares

in C



,

such that every other commutative square in C



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comes from a morphism $T \to T_i$. (Note that if $(F_i)_{i\in I}$ is a generating family of E, then every $X \in \mathrm{ob}\, C$ is isomorphic to a direct factor of one of the F_i.) 220

(c) The topos E is locally coherent, i.e. is generated by a family of coherent and *algebraic* objects, if and only if the objects X of C are coherent and algebraic objects of E, i.e. if condition b) and the following condition hold: for every pair of arrows $f, g : Y \rightrightarrows Z$ in C *over an object* X *of* C, the kernel of this pair of arrows in E is a quasi-compact object of E, i.e. there is a finite family of commutative diagrams in C

$$T_i \longrightarrow Y \overset{f,g}{\rightrightarrows} Z$$

such that every other commutative diagram in C

$$T \longrightarrow Y \overset{f,g}{\rightrightarrows} Z$$

comes from a morphism $T \to T_i$. The topos E is algebraic if and only if it satisfies condition b) and the preceding condition, but with an arbitrary pair of arrows (f, g) of C (not necessarily over an object X of C). The topos E is quasi-separated if and only if it satisfies the conditions of b), and if, in addition, for two objects X, $Y \in \mathrm{ob}\, C$, the product $X \times Y$ in E is quasi-compact. The topos E is coherent if and only if it satisfies the two preceding conditions and there is a final subcategory of C whose underlying set is finite.

(d) Give an example in which the objects of the generating subcategory C of E are pre-Noetherian, but E admits no generating family consisting of coherent objects (i.e. b) fails, or the objects of C are not all coherent). For this, take for C the category having as objects X, T_i $(i = 0, 1, \dots)$, and e (the final object), the only morphisms between these objects, besides the structural morphisms to e and the identity morphisms, being morphisms $u_i : T_i \to X$ and $v_i : X \to T_i$ subject to the conditions $u_i v_i = \mathrm{id}_X$, and finally set $p_i = v_i u_i$ (which necessarily satisfies $p_i^2 = p_i$). One verifies that the only sieves on X or on any T_i are the two trivial sieves, and that e admits exactly one nontrivial sieve. Thus, by a), the 221

objects of C are pre-Noetherian objects of E. Nevertheless, the product $X \times X$ in E is not quasi-compact, since it does not satisfy the criterion of b).

(e) Give an example in which the generating subcategory C of E consists of coherent objects of E, but E is not locally coherent. For this, take for C the category whose set of objects consists of the distinct objects e (the final object), Y, Z, and T_i ($i = 0, 1, \dots$), with the only morphisms between these objects, besides the morphisms to the final object e and the identity morphisms, being morphisms $f, g : Y \rightrightarrows Y$ subject to the conditions $fu_i = gu_i$. One verifies that C satisfies condition b) (with a little patience; one finds that all fiber products of objects
222 of C are isomorphic to $\varnothing_E$ or belong to C, with the exception of $Z \times_e Z$ and $Y \times_e Z$, which are covered by two elements of C), but clearly $\mathrm{Ker}(f, g)$ does not satisfy condition c).

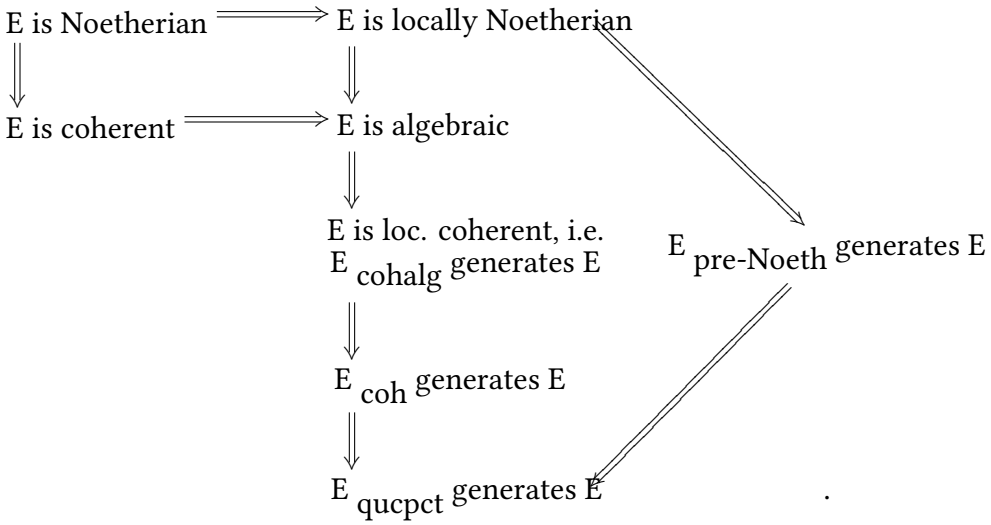
(f) Give an example in which C is stable under fiber products (a fortiori, E is a locally coherent, i.e. locally algebraic, topos), but E is not an algebraic topos. (Take the example given in d), and the full subcategory C' of E consisting of the objects Y, Z, T_i ($i > 0$), and $\varnothing_E$.)

(g) Suppose that the category C is finite. Prove that the topos E is Noetherian; that its Noetherian (i.e. quasi-compact) objects are the contravariant functors $F : C^\circ \to (\mathrm{Ens})$ such that, for every object X of C, $F(X)$ is a *finite* set; and that the fiber functors on E take Noetherian objects to finite sets (cf. IV 7.6. h)).

(h) Suppose that every morphism $f : X \to X$ of C factors as a composite ip, where i is a monomorphism and p has a right inverse. Let X be an object of C, let $I(X)$ be the set of its subobjects in the sense of C, regarded as an ordered set, and let $S(I(X))$ be the set of subsets U of $I(X)$ such that, for two elements X', $X'' \in I(X)$, the conditions $X' \in U$ and $X'' \leq X'$ imply $X'' \in U$. Show that the ordered set of subobjects in E of X is isomorphic to the set $S(I(X))$ (ordered by inclusion). Conclude that if $I(X)$ is finite, then X is a pre-Noetherian object of E. In particular, if (in addition to the factorization condition above) C is
223 stable under fiber products (resp. under finite projective limits), and if, for every $X \in \mathrm{ob}\, C$, the set $I(X)$ of subobjects of X in C is finite, then the topos E is locally Noetherian (resp. Noetherian).

(i) Let C be the category of finite sets (or, alternatively, of nonempty finite sets) $\in \mathscr{U}$, so that $E = \widehat{C}$ is the category of augmented simplicial sets (resp. of simplicial sets simpliciter). Show that E is a Noetherian topos. (Use h).) Taking a nonessentially constant pro-object $(X_i)_{i \in I}$ of C, show that E admits fiber functors that do not take Noetherian objects to Noetherian objects (i.e. to finite sets).

(j) Consider the following diagram of implications among properties of a topos E:



Show that every implication in this diagram is strict, and that among the notions considered there are no implications other than the composites of the implications in the preceding diagram. Here E_{coh}, E_{qucpct}, $E_{\mathrm{prénoeth}}$, E_{cohalg} denote 224
respectively the full subcategories of E consisting of the coherent, resp. quasi-compact, resp. pre-Noetherian, resp. coherent and algebraic objects of E. (One may restrict attention to topoi of the form $\widehat{C}$, using the results stated in d), e), and f).)

(k) Answer the following question (to which the author of these lines does not know the answer): can $E = \widehat{C}$ be locally Noetherian without the $\mathrm{Hom}(X, Y)$ (for X, $Y \in \mathrm{ob}\, C$) being finite?

Exercise 2.18. Let E be a topos.

a) Let X be a pre-Noetherian object of E. Show that X is isomorphic to a finite sum of connected objects (IV 4.3.5) of E. (Prove by contradiction that an increasing sequence of partitions of X (IV 8.7) is stationary.)

b) Let X be an object of E and let $(f_i : X_i \to X)_{i\in I}$ be a covering family of X. Show that if each X_i is isomorphic to a sum of connected objects of E, then so is X. (Reduce to the case in which the X_i are connected, then to the case in which they are subobjects of X, and consider on the index set I the equivalence relation R generated by the relation $X_i \cap X_j \neq \phi_E$. Show that if $J = I/R$, then X is the sum of the $X(i) = \underset{i\in I}{\mathrm{Sup}} X_i$.)

c) Conclude from b) that if $(X_i)_{i\in I}$ denotes a generating family of E, then E is locally connected (IV 8.7. ℓ)) if and only if every X_i ($i \in I$) is isomorphic to a 225
sum of connected objects of E.

d) Conclude from a) and c) that if E admits a generating subcategory consisting of pre-Noetherian objects, in particular if E is locally Noetherian, then E is locally connected, and a fortiori is isomorphic to the sum topos (IV 8.7 b)) of a family of connected topoi (IV 8.7 e)), i.e. topoi whose final object is connected. (In particular, one can associate with E, for every morphism $f : P \to E$, where P is a "nonempty connected and simply connected" topos (IV 2.7.5), a fundamental pro-group $\pi_1(E, f)$. Note that if E is locally Noetherian and "nonempty," such

an f can always be found, with P the punctual topos, by a result of DELIGNE ([9]).

3. Finiteness conditions for a morphism of topoi

DEFINITION 3.1. *Let $f : E' \to E$ be a morphism of topoi. We say that f is* quasi-compact (resp. quasi-separated) *if, for every quasi-compact (resp. quasi-separated) object X of E, $f^*(X)$ is quasi-compact (resp. quasi-separated). We say that f is coherent if f is quasi-compact and quasi-separated.*

3.1.1. Note that if f has any one of the preceding properties, then every morphism of topoi isomorphic to f has the same property. It is also immediate that the composite of two quasi-compact (resp. quasi-separated, resp. coherent) morphisms of topoi is again quasi-compact (resp. quasi-separated, resp. coherent).

226 PROPOSITION 3.2. *With the notation of 3.1, let $(X_i)_{i\in I}$ be a generating family of quasi-compact (resp. coherent) objects of E. In the parenthetical case, suppose that E' admits a generating family formed by quasi-compact objects. Then f is quasi-compact (resp. coherent) if and only if, for every $i \in I$, $f^*(X_i)$ is quasi-compact (resp. coherent).*

Necessity is immediate. For sufficiency in the first case, note that, for every quasi-compact object X of E, there is a finite epimorphic family of morphisms $X_i \to X$. Hence there is a finite epimorphic family $f^*(X_i) \to f^*(X)$, with the $f^*(X_i)$ quasi-compact by hypothesis, and therefore $f^*(X)$ is quasi-compact (1.3). In the second case, it remains to see that if X is a quasi-separated object of E, then $f^*(X)$ is quasi-separated. The hypothesis on X may be expressed (1.17) by the existence of an epimorphic family of morphisms $X_i \to X$ such that the $X_i \times_X X_j$ are quasi-compact. We then have a covering family $f^*(X_i) \to f^*(X)$ such that the fiber products $f^*(X_i) \times_{f^*(X)} f^*(X_j)$ are quasi-compact (since, by the first case, f^* carries quasi-compact objects to quasi-compact objects). As the $f^*(X_i)$ are coherent, the conclusion follows again from 1.17.

COROLLARY 3.3. *Let C and C' be two $\mathscr{U}$-sites in which fiber products are representable and every covering family admits a finite covering subfamily, and let $g : C \to C'$ be a morphism of sites* (IV 4.9.1). *Then the morphism of topoi $f : C^\sim \to C'^\sim$ defined by g is coherent.*

227 Indeed, $\epsilon_C(C)$ (resp. $\epsilon_{C'}(C')$) is a generating family of $C^\sim$ (resp. $C'^\sim$) satisfying the corresponding conditions of 3.2 (2.1.1).

PROPOSITION 3.4. *Let E be a locally coherent topos (2.3), and let $f : X \to Y$ be an arrow of E, whence a morphism of induced topoi* (IV (5.5.2)) $E_{/X} \to E_{/Y}$. *This morphism of topoi is quasi-compact (resp. quasi-separated, resp. coherent) if and only if f is a quasi-compact (resp. quasi-separated, resp. coherent) morphism.*

The quasi-compact case follows immediately from the definitions (without any condition on E); the quasi-separated case is precisely 2.8; and the coherent case follows by combining the two preceding cases.

PROPOSITION 3.5. *Let $f : E' \to E$ be a morphism of locally coherent topoi. Let $(X_i)_{i\in I}$ be a generating family in E formed by coherent algebraic objects. The following conditions are equivalent:*

(i) *For every quasi-compact arrow u of E, $f^*(u)$ is quasi-compact.*

(i bis) *For every arrow $u : X_i \to X_j$, $f^*(u)$ is quasi-compact.*

(ii) *For every coherent object Y' of E', every coherent algebraic object Y of E, and every arrow $v : Y' \to f^*(Y)$, the corresponding morphism of topoi*

$$f_v : E'_{/Y'} \longrightarrow E_{/Y}$$

obtained by composing the localization morphism $E'_{/Y'} \to E'_{/f^*(Y)}$ deduced from v (IV (5.5.2)) *with the morphism $E'_{/f^*(Y)} \to E_{/Y}$ induced by f* (IV (5.10.1)) *is coherent.* 228

(ii bis) *The same condition as in* (ii)*, but restricted to a collection of data $v_\alpha : Y'_\alpha \to f^*(Y_\alpha)$ such that the Y'_α are algebraic and cover the final object of E'.*

(ii ter) *(When a generating family $(X'_{i'})_{i' \in I'}$ in E' formed by coherent objects is given, together with, for every $i' \in I'$, an $i \in I$ and an arrow $v_{i'} : X'_{i'} \to f^*(X_i)$.) For every $i' \in I'$, the morphism of topoi $E'_{/X'_{i'}} \to E_{/X_i}$ defined by $v_{i'}$ is coherent.*

Since every arrow $X_i \to X_j$ is quasi-compact, it is clear that (i) ⇒ (i bis). Conversely, suppose (i bis) holds and let us prove (i). Let $u : X \to Y$ be a quasi-compact arrow of E. In terms of an epimorphic family $X_i \to Y$, the hypothesis on u says that the $X \times_Y X_i$ are quasi-compact (1.16), i.e. that for every i there exists a finite epimorphic family of morphisms $X_j \to X \times_Y X_i$ (1.3). We then have the corresponding epimorphic family $f^*(X_i) \to f^*(Y)$, such that for every i there is a finite epimorphic family $f^*(X_j) \to f^*(X \times_Y X_i) \simeq f^*(X) \times_{f^*(Y)} f^*(X_i)$, whose composite with the projection pr_2 to $f^*(X_i)$ is a quasi-compact morphism $f^*(X_j) \to f^*(X_i)$ by hypothesis (i bis). It follows (1.11 (i)) that, for every i, $\mathrm{pr}_2 : f^*(X) \times_{f^*(Y)} f^*(X_i)$ is quasi-compact, and hence (1.10 (ii)) that $f^*(u) : f^*(X) \to f^*(Y)$ is quasi-compact.

It remains to prove (i) ⇒ (ii) and (ii bis) ⇒ (i); the implication (ii) ⇒ (ii bis) is immedi- 229
ate, and (ii ter) is a special case of (ii bis). The implication (i) ⇒ (ii) is immediate: indeed, an object X of $E_{/Y}$ is quasi-compact (resp. quasi-separated) if and only if its structural morphism $X \to Y$ (resp. its diagonal morphism $X \to X \times_Y X$) is quasi-compact (2.7), and the same criterion of quasi-compactness and quasi-separatedness applies to $f_Y^*(X')$. Finally, let us prove (ii bis) ⇒ (i). Let $u : X \to Y$ be a quasi-compact arrow in E; we must show that $f^*(u)$ is quasi-compact. Since the Y'_α cover the final object of E', it is enough to prove that the $f^*(u)|Y'_\alpha$ are quasi-compact (1.10 (ii)). These morphisms are precisely the $f^*_{v_\alpha}(u|Y_\alpha)$, so we are reduced to proving the following result.

Corollary 3.6. *Let $f : E' \to E$ be a coherent morphism of locally coherent topoi. Then, for every quasi-compact arrow u of E, $f^*(u)$ is quasi-compact.*

By the implication (i bis) ⇒ (i) of 3.5 already proved, it is enough to show that, if $u : X \to Y$ is an arrow of E with X, Y coherent, then $f^*(u)$ is a quasi-compact morphism. This follows immediately from the hypothesis on f, which implies that $f^*(X)$ and $f^*(Y)$ are coherent as well.

Definition 3.7. *Let $f : E' \to E$ be a morphism of locally coherent topoi. We say that f is* locally coherent *if it satisfies the equivalent conditions of 3.5.* 230

Observe that "locally" means locally on the source.

3.7.1. This notion again depends only on the isomorphism class of the morphism of topoi f, and it is plainly stable under composition of morphisms of topoi. Moreover, by 3.5, if f is coherent then it is locally coherent; the converse holds if E' and E are coherent (by 3.5 (ii bis)). Criterion 3.5 (ii bis) immediately gives the following criterion:

Corollary 3.7. *Let $f : E' \to E$ be a morphism of locally coherent topoi, let $(Y_i)_{i \in I}$ and $(Y'_i)_{i \in I}$ be two families of objects of E and E', and, for every $i \in I$, let $v_i : Y'_i \to$*

$f^*(Y_i)$ be a morphism, whence a morphism of topoi

$$f_{v_i} : E'_{/Y'_i} \longrightarrow E_{/Y_i}.$$

Suppose that the Y'_i cover the final object of E'. Then f is locally coherent if and only if, for every $i \in I$, f_{v_i} is locally coherent.

In particular, applying this to the identity morphism of an induced topos gives:

COROLLARY 3.8. *Let E be a locally coherent topos, and let $v : X \to Y$ be an arrow of E. Then the morphism of induced topoi $E_{/X} \to E_{/Y}$ defined by v is locally coherent.*

231 REMARK 3.9. *It seems that all morphisms of algebraic topoi encountered in practice are locally coherent (which is why the term "locally coherent" is probably not destined to be used very often). Equivalently, the morphisms between coherent topoi encountered in practice are coherent. Nevertheless, one can construct noncoherent morphisms between coherent topoi and, more particularly, a noncoherent morphism from the punctual topos* (IV 2.2) *to a Noetherian topos E,* cf. 2.17 i).

EXAMPLES 3.10. (*Topoi associated with schemes.*) Return to the example 1.22 of the category (Sch) with the topologies T_i of SGA IV 6.3. For every scheme X, let X_{T_i} be the $\mathscr{V}$-topos defined by the induced site $(\mathrm{Sch})_{/X}$. A morphism of schemes $f : X \to Y$ then defines a morphism of topoi $f_{T_i} : X_{T_i} \to Y_{T_i}$, and it follows from 3.4 and 1.22 that this morphism of topoi is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the morphism of schemes f is quasi-compact (resp. quasi-separated, resp. coherent) in the usual sense of EGA IV 1. In any event, f_{T_i} is locally coherent by 3.8.

One may also associate with a scheme X the $\mathscr{U}$-topoi $\mathrm{Top}(X)$ and $\mathrm{Top}(X_{\text{ét}})$ as in 1.22.1, and to every morphism of schemes $f : X \to Y$ there are then associated morphisms $\mathrm{Top}(f)$ and $\mathrm{Top}(f_{\text{ét}})$ of the corresponding topoi. Let $T(f)$ denote either of
232 these two morphisms of topoi. One verifies immediately by 3.2 that this morphism of topoi is always locally coherent, and that it is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the morphism of schemes f is quasi-compact (resp. quasi-separated, resp. coherent) in the usual sense.

These observations show once again that the terminology introduced in the present section is compatible with the accepted terminology of scheme theory, and therefore deserves to be accepted by even the most recalcitrant reader.

EXERCISE 3.11. (*Coherent topoi and pretopoi.*)

a) A $\mathscr{U}$-pretopos (or simply a pretopos) is a category C satisfying the following conditions:
   1) Finite projective limits are representable in C.
   2) Finite coproducts are representable in C, and they are disjoint and universal.
   3) Equivalence relations in C are effective, and every epimorphism in C is universally effective.
   4) C is equivalent to a category $\in \mathscr{U}$.

   Show that if E is a coherent $\mathscr{U}$-topos, then the full subcategory E_{coh} of E formed by its coherent objects is a $\mathscr{U}$-pretopos, and that the inclusion functor $E_{\mathrm{coh}} \to E$
233 is left exact, commutes with finite coproducts, and commutes with passage to the quotient by an equivalence relation. (Use 1.5.3, 1.15, and 1.17.1.) Show that the topology induced by E on C is the "precanonical" topology, i.e. the topology whose covering families $X_i \to X$ are those that admit a *finite* subfamily

covering for the canonical topology of C. Consequently, E can be recovered up to equivalence from the pretopos $C = E_{\mathrm{coh}}$ as the topos $C^{\sim}$ (with C endowed with its precanonical topology).

b) Let C be a $\mathscr{U}$-pretopos. Endow C with the precanonical topology. Show that every morphism f of C factors as $f''f'$, with f' an epimorphism and f'' a monomorphism. Show that the topos $E = C^{\sim}$ is coherent and that the canonical functor $\varepsilon : C \to E$ induces an equivalence from C to the subcategory E_{coh} of E. (First show that ε is a fully faithful left exact functor commuting with finite coproducts and with passage to the quotient by an equivalence relation, and then that a coherent subobject in E of an object of C lies in the essential image of C.) Consequently, the $\mathscr{U}$-pretopos C can be recovered up to equivalence of categories from its associated $\mathscr{U}$-topos $E = C^{\sim}$.

c) Let C and C' be two $\mathscr{U}$-pretopoi. Show that a functor $\varphi : C' \to C$ is a morphism of sites from C to C' (for the precanonical topologies) if and only if φ is left exact and commutes with finite coproducts and with passage to a quotient by an equivalence relation.

d) With the notation of c), suppose that C and C' are associated with two coherent 234
$\mathscr{U}$-topoi E, E' as in a). Show that the canonical functor (IV 4.9.3)

$$\mathbf{Morsite}(C, C') \longrightarrow \mathbf{Homtop}(E, E')$$

induces an equivalence from its source (described explicitly in c)) to the full subcategory of its target formed by the *coherent* morphisms of topoi $E \to E'$. A quasi-inverse is obtained by associating with every coherent morphism of topoi $f : E \to E'$ the functor $E'_{\mathrm{coh}} = C' \to E_{\mathrm{coh}} = C$ induced by f^*.

e) Let C be a $\mathscr{U}$-pretopos and $E = C^{\sim}$. Express directly in terms of exactness properties of C the equivalent conditions of 1.25 (consult 1.28 b). Show that an object X of C is Noetherian in E if and only if every increasing sequence of subobjects of X in C is stationary, and hence that E is Noetherian if and only if every object of C satisfies the preceding condition.

Exercise 3.12. A topos E is called a *finite* topos if there exists a finite category C such that E is equivalent to $\widehat{C}$.

a) The topos E is finite if and only if E has enough essential points (IV 7.6 b)) and the category $\mathbf{Pointess}(E)$ of essential points of E is equivalent to a finite category. (Use IV 7.6 d) and h).)

b) Suppose that E is finite. Prove that every point of E is essential. Prove that 235
the 2-functors $E \mapsto \mathbf{Point}(E)$ and $C \mapsto \widehat{C}$ establish 2-equivalences between the 2-category of finite topoi E and the 2-category of Karoubian categories (IV 7.5 a)) C that are equivalent to finite categories, and that every morphism of finite topoi is essential (IV 7.6 a)). (Use a) and IV 7.6 h).)

c) Suppose that $E = \widehat{C}$, with C equivalent to a finite category, and let F be a coherent topos. Recall (IV (4.6.3.1)) that morphisms of topoi $f : E \to F$ correspond to functors $u : C \to \mathbf{Point}(F)$. Show that f is coherent if and only if f carries every coherent point of E (3.1) to a coherent point of F, or equivalently if, for every $X \in \mathrm{ob}\, C$, $u(X)$ is a *coherent* point of F. (Use 3.2 and 2.17 g).) Conclude that every morphism $f : E \to F$ from a finite topos to a finite topos is coherent.

d) Let X be a sober topological space (IV 4.2.1). The topos $\mathrm{Top}(X)$ (IV 2.1) is finite if and only if X is a finite set. (Use IV 7.1.6.)

4. Finiteness conditions in a topos obtained by gluing

The present section will not be used later in the Seminar.

4.1. Let E be a topos and U an open of E (i.e. a subobject of the final object of E),
236 and consider the corresponding open and closed subtopoi of E (IV 9)

$$E' = E_{/U}, E'' = E_{/U}.$$

We denote by

$$j : E' \longrightarrow E, i : E'' \longrightarrow E$$

the canonical morphisms of topoi. Recall (3.4) that for j to be a quasi-compact (resp. coherent) morphism of topoi, it is necessary and sufficient that the inclusion

$$j : U \longrightarrow e$$

of U into the final object of E (an inclusion that we shall also denote by j) be a quasi-compact (resp. coherent) morphism in E; moreover, $j : E' \to E$ is always quasi-separated (because $j : U \to e$ is so (1.8.1)). We propose to give criteria for the topos E'' to be coherent and for the morphism of topoi $i : E'' \to E$ to be coherent. First note:

PROPOSITION 4.2. *With the notation of 4.1, the inclusion morphism $i : E'' \to E$ is quasi-compact, i.e. for every quasi-compact object X of E, the object $i^*(X)$ of E'' is quasi-compact; moreover, if X is pre-Noetherian (1.30), then $i^*(X)$ is pre-Noetherian.*

This follows from Definition 1.1 and the

237 LEMMA 4.2.1. *The map $Y \mapsto i^*(Y)$ induces an isomorphism of ordered sets between the set of subobjects of X containing the subobject $X_U = X \times U$ of X, and the set of subobjects of $i^*(X)$.*

Note that since i_* is conservative (because it is fully faithful) and left exact (in particular, it commutes with fiber products), it follows immediately that a morphism $u : Y \to Z$ in E'' is a monomorphism if and only if $i_*(u) : i_*(Y) \to i_*(Z)$ is one. Consequently, on identifying (by i_*) E'' with the full subcategory of E consisting of objects Z satisfying the equivalent conditions of IV 9.7 1), the subobjects in E'' of the object Z of E'' identify with the subobjects Y of Z in E that belong to E'', or, equivalently, that contain $Z_U \xrightarrow{\sim} U$. For Z of the form $i_*i^*(X) = X_{\complement U}$ given by the amalgamated sum IV (9.5.1),

$$i_*i^*(X) = X_{\complement U} = X \amalg_{X_U} U,$$

the datum of a subobject Y of the latter in E is equivalent to the datum of a pair consisting of a subobject Y_1 of X and a subobject Y_2 of U, subject to the condition that the inverse images of these subobjects in X_U coincide. The condition that Y contain U then means that $Y_2 = U$, and the remaining condition on Y_1 is that it contain X_U. This therefore establishes a bijection between the set of subobjects of $i^*(X)$ in E'' and the set of
238 subobjects of X that contain X_U. It is clear that this is an isomorphism of ordered sets and that its inverse map is indeed the one asserted in 4.2.1.

COROLLARY 4.3. a) *Let X be an object of E. For X to be quasi-compact, it is sufficient that $i^*(X)$ and $j^*(X)$ be so, and this condition is also necessary if $j : U \to e$ is quasi-compact.*

b) *Let Y be an object of E''. For Y to be quasi-compact, it is sufficient that $i_*(Y)$ be so, and this condition is also necessary if U is quasi-compact.*

Proof

a) Sufficiency follows from Definition 1.1 and the fact that the pair (i^*, j^*) is conservative (IV 9.11 3)). Necessity follows from the fact that i and j are quasi-compact (by 3.14 and the hypothesis that j is quasi-compact).
b) Since $Y \simeq i^*i_*(Y)$, sufficiency follows from 4.2. Necessity follows from the sufficiency in a), taking into account that $i^*i_*(Y) \simeq Y$ and $j^*i_*(Y) \simeq U$.

COROLLARY 4.4. *Let Y be an object of E''. If U is quasi-compact or if E admits a generating family consisting of quasi-compact objects, then for Y to be quasi-separated (resp. coherent), it is* sufficient *that $i_*(Y)$ be so. If U is quasi-separated (resp. coherent) and if $j : U \to e$ is a quasi-compact morphism, then for Y to be quasi-separated (resp. coherent), it is necessary that $i_*(Y)$ be so.*

The parenthetical case follows from the non-parenthetical case, taking 4.3 b) into account. Suppose that $i_*(Y)$ is quasi-separated, and let us prove that Y is so under one of the two stated hypotheses. Thus we must prove that, for two quasi-compact objects Y' and Y'' over Y, the fiber product $Y' \times_Y Y''$ is quasi-compact. When U is assumed quasi-compact, note that it is enough to prove that $i_*(Y' \times_Y Y'')$ is quasi-compact (4.3 b)); this follows from the fact that $i_*(Y')$ and $i_*(Y'')$ are so (4.3 b)), using here the quasi-compactness of U, the fact that i_* commutes with fiber products, and the hypothesis that $i_*(Y)$ is quasi-separated. When E is assumed to admit a generating family consisting of quasi-compact objects, $i_*(Y')$ is a filtered inductive limit of its quasi-compact subobjects ${X'}_\alpha$; hence $Y' \simeq i^*i_*(Y')$ is a filtered inductive limit of the subobjects $i^*({X'}_\alpha)$, and since Y' is quasi-compact, it is equal to one of the $i^*({X'}_\alpha)$. Likewise, Y'' is of the form $i^*({X''}_\beta)$, where ${X''}_\beta$ is a quasi-compact subobject of $i_*(Y'')$. But then $Y' \times_Y Y'' = i^*(X'_\alpha \times_{i_*Y} X''_\beta)$, and since $X'_\alpha \times_{i_*(Y)} X''_\beta$ is quasi-compact by the hypothesis that $i_*(Y)$ is quasi-separated, it follows by 4.2 that $Y' \times_Y Y''$ is also quasi-compact. 239

Conversely, suppose that U is quasi-separated and $j : U \to e$ is quasi-compact, and assume that Y is quasi-separated. Let us show that $i_*(Y)$ is also quasi-separated, i.e. that for two quasi-compact objects X', X'' over $i_*(Y)$, the fiber product $X' \times_{i_*(Y)} X''$ is quasi-compact. By 4.3 a), it is enough to prove that its image under i^* is quasi-compact (which follows from 4.2 and the hypothesis that Y is quasi-separated) *and* that its image under j^* is quasi-compact. The latter is $j^*(X') \times j^*(X'')$, and is indeed quasi-compact because $j^*(X')$ and $j^*(X'')$ are so (j being quasi-compact) and U is quasi-separated. 240

COROLLARY 4.5. *Suppose that E admits a generating subcategory consisting of quasi-compact (*resp. *pre-Noetherian) objects. Then the same is true of E''.*

This follows from 4.2 and the

LEMMA 4.5.1. *If $(X_\alpha)_\alpha$ is a generating family in E, then $(i^*(X_\alpha))_\alpha$ is a generating family in E''.*

Indeed, this means that the family of functors

$$Y \longrightarrow \mathrm{Hom}(i^*(X_\alpha), Y)$$

on E'' is conservative. But $\mathrm{Hom}(i^*(X_\alpha), Y) \simeq \mathrm{Hom}(X_\alpha, i_*(Y))$, and it is enough to use the fact that the functor i_* is conservative.

PROPOSITION 4.6. *The notation is that of 4.1.*

a) *If E admits a generating family consisting of coherent objects, then the same is true of the closed subtopos E'', and the inclusion morphism $i : E'' \to E$ is coherent.*

b) *Suppose that the morphism $j : U \to e$ in E is quasi-compact. If E is locally coherent (*resp. *coherent,* resp. *quasi-separated,* resp. *locally Noetherian,* resp. *Noetherian,* resp. *locally perfect,* resp. *perfect), then the same is true of E'', and the inclusion morphism $i : E'' \to E$ is coherent.* 241

a) The first assertion will follow from the second and 4.5.1. In any event, 4.5 implies that E'' admits a generating family consisting of quasi-compact objects, so 3.2 applies and shows that it is enough to verify that, for every coherent object X of E, the object $i^*(X)$ of E'' is coherent. Using the compatibility of the formation of the topos complementary to an open with localization (IV 9.13 b)), we reduce to the case in which X is the final object of E, and hence to proving that the final object of E'' is coherent. Now 4.4 applies, so it is enough to prove that $i_*(e_{E''}) = e_E$ is coherent, which is indeed the case.

b) Under each of the hypotheses made in b), E admits a generating family consisting of coherent objects, so it already follows from a) that $i : E'' \to E$ is coherent. Moreover, 4.5.1 then implies that E'' admits as a generating subcategory the full subcategory ${E''}_{\mathrm{coh}}$ consisting of its coherent objects. Suppose that E is coherent, and let us show that E'' is also coherent, i.e. (2.4.5) that the subcategory ${E''}_{\mathrm{coh}}$ of E'' is stable under finite projective limits, knowing that this is so for the subcategory E_{coh} of E. This follows immediately from the parenthetical criterion 4.4 (which applies because U is now coherent, e being
242 coherent and $j : U \to e$ being quasi-compact), taking into account that i_* is left exact. By localization, again using IV 9.13 b), we conclude that if E is locally coherent, then the same is true of E''. If E is quasi-separated, then E'' is also quasi-separated: indeed, its final object is quasi-separated by 4.4, since that of E is, and it remains to prove that the product of two quasi-separated objects of E'' is quasi-separated (knowing that this is so in E); this again follows from the criterion 4.4 and the fact that i_* commutes with products (taking into account that the subobject U of e is quasi-separated, since e is, so that 4.4 applies).

Since a topos is locally Noetherian (resp. Noetherian) if and only if it is locally coherent (resp. coherent) and admits a generating family consisting of pre-Noetherian objects (2.10 (iii bis)), it follows from the preceding discussion and 4.5 that if E is locally Noetherian (resp. Noetherian), then the same is true of E''. Now suppose that E is perfect, and let us prove that E'' is perfect. Since we already know that it is coherent, it remains to verify that it satisfies criterion 1.25 (i), i.e. that every object Y of E'' is a filtered inductive limit of coherent objects, knowing that the analogous statement is true in E. Since $i_*(Y)$ is a filtered inductive limit of coherent objects X_α, $Y \simeq i^* i_*(Y)$ is a filtered inductive limit of the coherent objects $i^*(X_\alpha)$. By localization (IV 9.13 b)), we conclude that if E is locally perfect, then the same is true of E''. This completes the proof of 4.6.

243 Remark 4.6.1. In fact, in 4.6 b), the hypothesis that $j : U \to e$ is quasi-compact is unnecessary, except perhaps in the case "E quasi-separated". Indeed, by the preceding proof it is enough to see that if E is coherent, then E'' is coherent. Now U is the filtered inductive limit of its quasi-compact subobjects U_α, and in 7 we shall verify that E'' then identifies with the projective-limit topos (in the sense of 7) of the closed subtopoi E''_i complementary to the $E_{/U_i}$, which by 4.6 b) are coherent topoi with coherent transition morphisms. It then follows that E'' is a coherent topos (7).

Corollary 4.7. *Suppose that the subcategory E_{coh} of E consisting of coherent objects is generating, and let X be an object of E.*

a) *If X is quasi-separated, then $i^*(X)$ and $j^*(X)$ are quasi-separated; this condition is also sufficient if $j : U \to e$ is quasi-compact.*
b) *Suppose that $j : U \to e$ is quasi-compact. Then X is coherent if and only if $i^*(X)$ and $j^*(X)$ are coherent.*
c) *Suppose that E is coherent and $j : U \to e$ quasi-compact. Then E is perfect (2.9.1) if and only if E' and E'' are perfect.*

a) We noted in 4.1 that $j : E' \to E$ is quasi-separated, and we know from 4.6 a) that the same is true of i, which gives necessity. For sufficiency, let X' and X'' be quasi-compact objects over X. We must prove that $X' \times_X X''$ is quasi-compact, and for this it is enough to prove that its images under i^* and j^* are so (4.3 a)). Since i and j are quasi-compact, by 4.2 and the hypothesis that $j : U \to e$ is quasi-compact, the conclusion follows from the fact that i^* and j^* commute with fiber products and from the hypothesis that $i^*(X)$ and $j^*(X)$ are quasi-separated. 244
b) This follows by combining a) with 4.3 a).
c) Necessity was already seen (4.6 b)). Sufficiency follows from the fact that, since E is coherent, E' and E'' are coherent (4.6 b)), so that for any of the topoi under consideration, being perfect means that the category of its coherent objects is stable under finite $\varinjlim$. We therefore conclude by b).

Exercise 4.8. a) Solve the following question (to which the writer confesses, to his embarrassment, that he does not know the answer): with the notation of 4.1, if E is quasi-separated (resp. algebraic), is the same true of E''?
b) Suppose that E is equivalent to a topos of the form $\widehat{C}$, where C is a category equivalent to a category $\in \mathscr{U}$. Prove directly, using 2.17 c) and IV 9.24, that if E is locally coherent (resp. coherent, resp. algebraic, resp. quasi-separated, resp. locally Noetherian, resp. Noetherian), then the same is true of E''.

4.9. Keep the notation of 4.1, and recall (IV 9.16) that the datum of a situation (E, U) consisting of a topos E and an open U of E is essentially equivalent to that of a triple (E', E'', f), where E' and E'' are topoi and 245

$$f : E' \longrightarrow E'' \tag{4.9.1}$$

is a "gluing functor", i.e. a functor that is *left exact and accessible*; starting from (E, U) as in 4.1, the associated gluing functor is given by

$$f = i^* j_*. \tag{4.9.2}$$

We propose to express, in terms of the data E', E'', and f, the fact that E is a coherent (resp. perfect) topos and that U is a quasi-compact object, or equivalently, that E and E' are coherent. We already know that this implies that E'' is also coherent (4.6 b)), so the question becomes the following: given two *coherent* topoi E' and E'' and a gluing functor f (4.9.1), under what conditions on f is the glued topos E coherent (resp. perfect)? We also know (4.7 c)) that E is perfect if and only if E is coherent and E' and E'' are perfect; thus the parenthetical case of the problem reduces to the non-parenthetical case. Let us nevertheless note that the solution (4.10) of the latter is more elegant if E' is already assumed perfect.

4.9.3. First note that if E is coherent, then the full subcategory E_{coh} of E consisting of the objects $X = (X', X'', u : X'' \to f(X'))$ of E that are coherent is reconstructed as the subcategory E_0 of E consisting of the X for which $X' = j^*(X)$ and $X'' = i^*(X)$ 246

are coherent (4.7 b)). A necessary condition for E to be coherent is therefore that the preceding subcategory E_0 be generating. This condition is also sufficient, because by 4.3 a), E_0 consists of quasi-compact objects; moreover, it is clear that E_0 is stable under finite projective limits, and we conclude by 3.4.5.

Recall now (IV 9.18) that the datum of a gluing functor (4.9.1) is equivalent to that of a *sheaf* on E'' with values in $\mathrm{Pro}(E')^0$:

$$G \in \mathbf{Faisc}(E'', \mathrm{Pro}(E')^0), G : E''^0 \longrightarrow \mathrm{Pro}(E'^0), \tag{4.9.4}$$

or equivalently to that of a functor $g = G^0$,

$$g : E'' \longrightarrow \mathrm{Pro}(E') \tag{4.9.5}$$

that *commutes with inductive limits*. If C is a generating subcategory of E'', endowed with the induced topology, we know (II 6.10) that the datum of G is also equivalent (up to unique isomorphism) to that of a sheaf on C with values in $\mathrm{Pro}(E')^0$

$$G_C \in \mathbf{Faisc}(C, \mathrm{Pro}(E')^0), G_C : C^\circ \longrightarrow \mathrm{Pro}(E')^0, \tag{4.9.6}$$

or, equivalently, to that of a functor $g_C = G_C^0$,

$$g_C : C \longrightarrow \mathrm{Pro}(E'), \tag{4.9.7}$$

247 satisfying the familiar left-exactness conditions (II 6.2.1)). Of course, g_C is simply the *restriction of* g (4.9.5) *to* C. The case of greatest interest to us is that in which we take $C = E''_{\mathrm{coh}}$, giving the objects

$$G_0 \in \mathbf{Faisc}(E''_{\mathrm{coh}}, \mathrm{Pro}(E')^0), G_0 : {E''}^0_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E')^0, \tag{4.9.8}$$

$$g_0 = G_0^0 : E''_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'), \tag{4.9.9}$$

each of which is again equivalent to the datum of f.

4.9.10. Suppose that the full generating subcategory C of E'' consists of quasi-compact objects and is stable in E'' under finite sums, passage to the quotient by equivalence relations, and fiber products; this is the case, for example, for $C = E''_{\mathrm{coh}}$ (1.15 and 1.17.1). It is then immediate, if P is a category in which finite projective limits are representable (for example $P = \mathrm{Pro}(E')^0$), that a functor $G_C : C^\circ \to P$ is a sheaf with values in P if and only if the functor $g_C = G_C^0 : C \to P^0$ commutes with finite sums and with passage to the quotient by an equivalence relation. This specifies, in particular, which objects are the sheaves (4.9.6) (and thus expresses the gluing functors $f : E' \to E''$).

With these reminders in place, we can give the solution to the problem posed in 4.9:

248 Proposition 4.10. *Let E' and E'' be two coherent topoi, and let $f : E' \to E''$ be a gluing functor* (IV 9.10)*, i.e. a left exact and accessible functor. Let E be the topos deduced from it by gluing* (IV 9.16)*, and denote by E'_{coh} (resp. E'_{PF}) the strictly full subcategory of E' consisting of the coherent objects (resp. the objects X such that the covariant functor* $\mathrm{Hom}(X, -)$ *represented by X commutes with small filtered inductive limits). Consider the following conditions:*

(i) *E is coherent.*

(ii) *For every object X' of E', there exists a family of morphisms*

$$v_\alpha : Y'_\alpha \longrightarrow X'$$

with target X' and with coherent objects of E' as sources, such that the family of the

$$f(v_\alpha) : f(Y'_\alpha) \longrightarrow f(X')$$

in E'' is covering.

(ii bis) *f commutes with (small) filtered inductive limits, and* (ii) *holds for every* $X' \in \mathrm{Ob}\, E'_{PF}$.

(ii ter) *The canonical functor* (IV 9.20.1)

$$\pi : \mathbf{Point}(E'') \longrightarrow \mathrm{Pro}(E') \tag{4.10.1}$$

defined by the gluing functor f factors (up to isomorphism) through $\mathrm{Pro}(E'_{\mathrm{coh}})$ *(via the fully faithful functor* $\mathrm{Pro}(E'_{\mathrm{coh}}) \to \mathrm{Pro}(E')$ *arising from the inclusion* $E'_{\mathrm{coh}} \to E'$*):*

$$\pi_0 : \mathbf{Point}(E'') \longrightarrow \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.2}$$

(iii) *The functor f commutes with (small) filtered inductive limits.* 249

(iii bis) *The functor g_0 (4.9.9) factors (up to isomorphism) through a functor*

$$E''_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'_{PF}), \tag{4.10.3}$$

via the fully faithful functor $\mathrm{Pro}(E'_{PF}) \to \mathrm{Pro}(E')$ *deduced from the inclusion* $E'_{PF} \to E'$.

(iii ter) *The canonical functor π (4.10.1) factors (up to isomorphism) as*

$$\pi_1 : \mathbf{Point}(E'') \longrightarrow \mathrm{Pro}(E'_{PF}). \tag{4.10.4}$$

(iv) *The functor g_0 (4.9.9) factors (up to isomorphism) through a functor*

$$E''_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.5}$$

We then have the following diagram of implications:

$$\text{(iv)} \Rightarrow \text{(i)} \Leftrightarrow \text{(ii)} \Leftrightarrow \text{(ii bis)} \Leftrightarrow \text{(ii ter)} \Rightarrow \text{(iii)} \Leftrightarrow \text{(iii bis)} \Leftrightarrow \text{(iii ter)}. \tag{4.10.6}$$

Let us immediately record the

COROLLARY 4.11. a) *Suppose that E' is perfect. Then all the conditions considered in 4.10 are equivalent; in particular, E is coherent if and only if f commutes with filtered inductive limits, or equivalently if and only if the functor g_0 (4.9.9) factors (up to isomorphism) through* $\mathrm{Pro}(E'_{\mathrm{coh}})$. 250

b) *For E to be perfect, it is necessary and sufficient that E' and E'' be perfect and that f (resp. g_0) satisfy the condition stated in* a).

Indeed, a) follows from the fact that E' perfect means $E'_{\mathrm{coh}} = E'_{PF}$, so that (iii bis) implies (iv). Assertion b) follows, as is clear from the preliminary remarks of 4.9.

4.12. Proof of 4.10.

a) Let us spell out the condition equivalent to (i) obtained in 4.9.3, namely that the full subcategory E_0 of E is generating, i.e. that for every object $X = (X', X'', u : X'' \to f(X'))$ of E, the family of all morphisms

$$v = (v', v'') : Y = (Y', Y'', v : Y'' \longrightarrow f(Y')) \longrightarrow X,$$

with $Y \in \mathrm{Ob}\, E_0$, i.e. with Y' and Y'' coherent, is epimorphic. Since the pair of functors (i^*, j^*) is conservative, this also means that the family of corresponding morphisms $Y' \to X'$ is epimorphic in E', and that the family of corresponding morphisms $Y'' \to X''$ is epimorphic in E''. This is clear for the first family, as is seen by taking $Y \in \mathrm{Ob}\, E_0$ *over* U, i.e. such that $Y'' = \varnothing_{E''}$: one obtains

the family of all arrows in E' with target X' and coherent source, which is indeed epimorphic because E' is coherent and hence the subcategory ${E'}_{\mathrm{coh}}$ is generating. It therefore remains to express the condition that the family $Y'' \to X''$ be epimorphic for every given object X of E.

251 b) Now suppose that condition (ii) is satisfied for the object X' under consideration. It follows that one can find an epimorphic family of morphisms $Y''_\beta \to X''$ whose composites with $u : X'' \to f(X')$ each lift to a morphism $Y''_\beta \to f(Y'_\alpha)$ for a suitable $\alpha = \phi(\beta)$. Since E''_{coh} is a generating subcategory of E'', we may moreover suppose the Y''_β coherent. The commutative diagrams

$$\begin{array}{ccc} Y''_\beta & \longrightarrow & f(Y'_{\varphi(\beta)}) \\ \downarrow & & \downarrow \\ X'' & \longrightarrow & f(X') \end{array}$$

then provide the required family of morphisms in E, giving an epimorphic family $Y''_\beta \to X''$. Thus (ii) ⇒ (i).

Conversely, suppose (i), and let us prove (ii). Apply the condition made explicit in a) to the object

$$X = j_*(X') = (X', f(X'), \mathrm{id} = f(X') \xrightarrow{\sim} f(X')).$$

Since the $Y'' \to X'' \simeq f(X')$ form an epimorphic family, a fortiori so does the family of $f(v') : f(Y') \to f(X')$. The conclusion follows, since the Y' are coherent by hypothesis. Thus

$$\text{(i)} \Leftrightarrow \text{(ii)}.$$

252 c) Note now that condition (iv) also means that for every object X'' of E'', the pro-representable functor

$$g(X'') : X' \longmapsto \mathrm{Hom}(X'', f(X'))$$

on E' is pro-represented by a pro-object whose components belong to ${E'}_{\mathrm{coh}}$. Equivalently, in the category $E'_{/g(X'')}$ formed by pairs consisting of an object X' of E' and an element $u \in g(X'')(X') = \mathrm{Hom}(X'', f(X'))$, $u : X'' \to f(X')$, there is a cofinal set of objects (X'_α, u_α), with $u_\alpha : X'' \to f(X'_\alpha)$ and $X'_\alpha \in \mathrm{Ob}\, E'_{\mathrm{coh}}$. Since the category $E'_{/g(X'')}$ is filtered, the cofinality condition simply means that for every object (X', u), $u : X'' \to f(X')$, of $E'_{/g(X'')}$, there is a morphism from one of the (X'_α, u_α) to this object, i.e. a morphism $v'_\alpha : X'_\alpha \to X'$ making the triangle

(∗)
$$\begin{array}{ccc} & & f(X'_\alpha) \\ & \overset{u_\alpha}{\nearrow} & \downarrow{\scriptstyle f(v'_\alpha)} \\ X'' & \longrightarrow & f(X'). \end{array}$$

commutative. Thus condition (iv) is equivalent to the following:

(iv bis) For every arrow $u : X'' \to f(X')$, with $X' \in \mathrm{Ob}\, E'$ and $X'' \in \mathrm{Ob}\, E''_{\mathrm{coh}}$, there is an arrow $v' : Y' \to X'$ in E', with Y' coherent, such that u factors through $f(v') : f(Y') \to f(X')$.

It is clear that condition (iv bis) implies (ii), since for fixed X' the family of $u : X'' \to f(X')$ whose source is coherent is epimorphic. A fortiori, the same is therefore true of the family of all $f(v'_\alpha)$ in the corresponding diagrams $(*)$. Thus

$$\text{(iv)} \Rightarrow \text{(i)}.$$

d) Let us spell out condition (ii ter). Let p be a point of E''. By definition, $\pi(p)$ is the pro-object of E' which pro-represents the functor $h : X' \longmapsto f(X')_p$, or, equivalently, the pro-object defined by the canonical functor $E'_{/h} \to E'$, where $E'_{/h}$ denotes the category of pairs (X', u) with $X' \in \mathrm{Ob}\, E'$ and $u \in h(X')$, i.e. $u \in f(X')_p$. To say that this pro-object is isomorphic to a pro-object coming from $\mathrm{Pro}(E'_{\mathrm{coh}})$ means that the subcategory of $E'_{/h}$ formed by the (X', u) with $X' \in \mathrm{Ob}\, E'_{\mathrm{coh}}$ is cofinal. In other words, for every object (X', u), $u \in f(X')_p$, of $E'_{/h}$, there is an object (X'_0, u_0) mapping to it, with X'_0 coherent; that is, there is a morphism $X'_0 \to X'$ (X'_0 coherent) such that $u \in \mathrm{Im}(f(X'_0) \to f(X'))$. To say that this holds for every point p means that, for every object X' of E', the family of maps $f(X'_0) \to f(X')$, with X'_0 coherent, is surjective on every stalk. As we shall see in the appendix (10) that a coherent topos E'' has enough points, the resulting condition also says that, for every $X' \in \mathrm{Ob}\, E'$, the family $f(X'_0) \to f(X')$ is epimorphic. This is precisely condition (ii), and hence

$$\text{(ii)} \Leftrightarrow \text{(ii ter)}.$$

e) The condition that f commute with filtered inductive limits (condition (iii)) is equivalent, since the family of fiber functors of E'' is conservative, to the condition that the composite functors $h : X' \longmapsto f(X')_p$ do so. Let us prove that this means that the pro-object $\pi(p)$ which pro-represents this functor lies in the essential image of $\mathrm{Pro}(E'_{PF})$. This will prove the implications 253

$$\text{(iii)} \Leftrightarrow \text{(iii ter)} \Leftarrow \text{(ii ter)} \Rightarrow \text{(iii bis)}.$$

We are therefore led to prove the following lemma.

Lemma 4.12.1. *Let E' be a topos in which the subcategory E'_{coh} is generating, and let $X' \in \mathrm{Ob}\,\mathrm{Pro}(E')$. Then X' belongs to the essential image of $\mathrm{Pro}(E'_{PF})$ if and only if the functor $h : E' \to (\mathrm{Ens})$ which it pro-represents commutes with filtered inductive limits.*

Since a filtered inductive limit of functors commuting with filtered inductive limits has the same property, sufficiency follows from the definition of E'_{PF}. For necessity, we must prove that if the functor h commutes with filtered inductive limits, then in the filtered category $E'_{/h}$ of pairs (X', u) with $u \in h(X')$, the subcategory formed by the pairs (X', u) with $X' \in \mathrm{Ob}\, E'_{PF}$ is cofinal. That is, for every (X', u) in $E'_{/h}$, there is a morphism $X'_0 \to X'$, with $X'_0 \in \mathrm{Ob}\, E'_{PF}$, such that $u \in \mathrm{Im}(h(X'_0) \to h(X'))$. But X' is known to be a filtered inductive limit of objects X'_i of E'_{PF} (1.24.2 a)), so $h(X')$ is the filtered inductive limit of the $h(X'_i)$, which proves the assertion. 254

f) It remains only to prove the implications

$$\text{(ii bis)} \Rightarrow \text{(ii)} \quad \text{and} \quad \text{(iii)} \Leftrightarrow \text{(iii bis)}.$$

The first implication follows trivially from the fact that every object X' of E' is a filtered inductive limit of objects X'_i of E'_{PF}. For the second, it is enough to

note that the family of functors

$$Y'' \longmapsto \mathrm{Hom}(X'', Y''),$$

for $X'' \in \mathrm{Ob}\, E''_{\mathrm{coh}}$, is conservative and consists of functors commuting with filtered inductive limits (1.23 (ii)). Consequently, the functor $f : E' \to E''$ commutes with filtered inductive limits if and only if the composite functors $X' \longmapsto \mathrm{Hom}(X'', f(X')) = \mathrm{Hom}_{\mathrm{Pro}(E')}(g_0(X''), X')$ do so. By 4.12.1, this is equivalent to the fact that the $g_0(X'')$ lie in the essential image of $\mathrm{Pro}(E'_{PF})$. C.Q.F.D.

Remark 4.13. It is useful to state in sheaf-theoretic language the meaning of condition 4.10 (iv). Consider the commutative diagram of functors

$$\begin{array}{ccc} \mathrm{Pro}(E'_{\mathrm{coh}})^{\circ} & \xhookrightarrow{\alpha} & \mathrm{Pro}(E')^{\circ}_{\beta} \\ \downarrow{\scriptstyle \beta_0} & & \downarrow{\scriptstyle \beta} \\ \overset{\vee}{E}{}'_{\mathrm{coh}} & \xleftarrow[\gamma]{} & \overset{\vee}{E}{}' \end{array} \qquad (4.13.1)$$

where the superscript circles denote opposite categories, and the second horizontal arrow γ is restriction of functors, all the other arrows denoting the familiar fully faithful functors. Note that the inclusion arrow α *does not in general commute with projective limits*, even finite ones. Thus, in general, a sheaf F on a site C (such as E'') defines a presheaf $\alpha \circ F$ on the same site, with values in $\mathrm{Pro}(E'')^0$, which need not be a sheaf. Nevertheless, if a *presheaf* F on C with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ is such that the corresponding presheaf αF with values in $\mathrm{Pro}(E')^0$ is a sheaf, then F itself is a sheaf. Indeed, the sheaf condition is a left-exactness property (II 6.2.1)), and the functors β, β_0, and γ commute with small projective limits.

On the other hand, the characterization in 4.9.10 of sheaves on E''_{coh} with values in an arbitrary category P shows that, for every functor $\alpha : P \to Q$ such that $\alpha^0 : P^0 \to Q^0$ commutes with *finite coproducts* and with *passage to quotients by equivalence relations*, the functor $g \mapsto \alpha \circ g$ from $\mathrm{Hom}({E''}^0_{\mathrm{coh}}, P)$ to $\mathrm{Hom}({E''}^0_{\mathrm{coh}}, Q)$ takes sheaves to sheaves. We also know that in $\mathrm{Pro}(E'_{\mathrm{coh}})$ and in $\mathrm{Pro}(E')$, finite coproducts and quotients by equivalence relations are representable, and that the inclusion $E'_{\mathrm{coh}} \to E'$ commutes with them (1.15 and 1.17.1). The same is therefore true for the categories $\mathrm{Pro}(E'_{\mathrm{coh}})$ and $\mathrm{Pro}(E')$ and for the inclusion between them (I 8.9.5 b) and 8.9.7), which is precisely α^0, where α is the functor occurring in (4.13.1). It follows that if

$$\varphi : {E''}^0_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'_{\mathrm{coh}})^0 \qquad (4.13.2)$$

is a functor, then φ *is a sheaf on* E''_{coh} *with values in* $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ *if and only if* $\alpha \circ \varphi$ *is a sheaf on* E''_{coh} *with values in* $\mathrm{Pro}(E')^0$. Thus the category of gluing functors $f : E' \to E''$ satisfying condition 4.10 (iv) is canonically equivalent to the category of sheaves φ on E''_{coh} with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$. Taking 4.10 into account, we conclude that such a sheaf φ canonically defines a gluing functor (up to unique isomorphism), and that the topos E obtained from the latter is *coherent*. Moreover (4.11), *if* E' *is perfect*, one thereby obtains *all* coherent topoi that can be obtained from E' and E'' by gluing.

Note also that E'_{coh} is equivalent to a small category and is stable under finite projec-

tive limits. Consequently (I 8.10.14), by the fully faithful functor β_0 of (4.13.1), $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ is equivalent to the full subcategory of $\widehat{E'_{\mathrm{coh}}} = \mathrm{Hom}(E'_{\mathrm{coh}}, (\mathrm{Ens}))$ formed by the left exact functors. Thus the data of a sheaf (4.13.2) are equivalent to those of a functor

$$F : {E''}^0_{\mathrm{coh}} \times E'_{\mathrm{coh}} \longrightarrow (\mathrm{Ens}) \tag{4.13.3}$$

which is a sheaf in the first argument and left exact in the second; or, equivalently, to those of a functor

$$f_0 : E'_{\mathrm{coh}} \longrightarrow E'' \tag{4.13.4}$$

which is left exact. If f is the gluing functor satisfying condition 4.10 (iv) associated with φ, one immediately verifies that f_0 is canonically isomorphic to the restriction of $f : E' \to E''$. Thus *under the conditions being considered, f_0 determines f* (up to canonical isomorphism). To summarize the content of 4.11, we finally state:

Corollary 4.14. *When E' is a perfect topos, the category of gluing functors $f : E' \to E''$ which give a coherent topos E by gluing is equivalent, under $f \mapsto f_0 = f|E'_{\mathrm{coh}}$, to the category of left exact functors $f_0 : E'_{\mathrm{coh}} \to E''$, or equivalently to the category of sheaves* 258
φ on the site E''_{coh} (or, what amounts to the same thing, on the topos E'') with values in the category $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ (cf. 4.9.10).

The first assertion can also be deduced from 4.10 by observing that $E' \approx \mathrm{Ind}(E'_{\mathrm{coh}})$ (1.25). Indeed, restriction gives an equivalence between the category of functors $f : E' \to P$ commuting with filtered inductive limits (where P is a category in which small filtered inductive limits are representable) and the category of arbitrary functors $f : E'_{\mathrm{coh}} \to P$. It is immediate that, under these conditions, if finite projective limits are representable in P, then f is left exact if and only if f_0 is.

Remarks 4.15. a) The proof given for 4.10 shows that conditions equivalent to (ii ter) resp. (iii ter) are obtained by replacing the category **Point**(E'') by a subcategory which defines a *conservative* family of fiber functors.

b) Taking for E'' the point topos (IV 2.2), one sees immediately that conditions (ii ter) and (iii ter) are equivalent *only if* $E'_{\mathrm{coh}} = E'_{PF}$, i.e. if E' is perfect. Thus the condition "E is coherent" is equivalent to the condition "f commutes with filtered inductive limits" only if E' is perfect. On the other hand, taking more generally for E'' a coherent topos of the form $\widehat{C}$, one sees that *in this case condition* (i) *is equivalent to* (iv) *for every E'*. In fact, in the general case the editor has been unable to construct an example in which E is coherent without condition (iv) being satisfied. This is shameful!

5. Commutation of the functors $H^i(X,-)$ with filtered inductive limits

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Theorem 5.1. *Let E' be an algebraic topos (2.3), let E be a locally coherent topos (2.3), and let $f : E' \to E$ be a coherent morphism of topoi (3.1). Then, for every integer q, the functors $R^q f_*$* (V 5.0) *commute with filtered inductive limits of abelian sheaves.*

Corollary 5.2. *Let E' be a coherent topos (2.3). For every integer q, the functor $H^q(E',-)$ commutes with filtered inductive limits of abelian sheaves.*

Corollary 5.3. *Let E be a topos (resp. an algebraic topos) (2.3), and let X be an algebraic and coherent (resp. coherent) object of E (2.3). For every integer q, the functor $H^q(X,-)$ commutes with filtered inductive limits of abelian sheaves.*

Corollary 5.2 follows from 5.1 by taking for E the point topos and for $f : E' \to E$ the unique morphism (IV 2.2). Let us show that 5.2 implies 5.3. Let $j_X : E_{/X} \to E$ be the localization morphism. There is a canonical isomorphism, functorial in the abelian sheaf F (V 2.2.1), $H^q(X, F) \simeq H^q(E_{/X}, j_X^* F)$. The functor j_X^* commutes with inductive limits, and the topos $E_{/X}$ is coherent (2.4.6), whence 5.3. Let us show that 5.3 implies 5.1. Let E_{cohalg} be the full subcategory of E defined by the algebraic and coherent objects of E. It is a generating category (2.4.5). For every abelian sheaf F, $R^q f_* F$ is the sheaf associated with the presheaf $X \mapsto H^q(f^* X, F)$ ($X \in \mathrm{ob}\, C$) (V 5.1). Since f is coherent, $f^* X$ is coherent for every object X of C (3.1). By 5.3, the presheaf $X \mapsto H^q(f^* X, F)$ commutes with filtered inductive limits in the argument F, and 5.1 follows, since the associated-sheaf functor commutes with inductive limits. (Only the following property of the morphism f is used: there is a generating family S of E such that, for every X of S, $f^*(X)$ is algebraic and coherent.) It therefore remains to prove 5.2. We already know that the functor $H^\cdot(E', -)$ commutes with filtered inductive limits of abelian sheaves (1.2.3). By a classical result of [7], to prove 5.2 it is enough to show that a filtered inductive limit of sheaves acyclic for the functor $H^\cdot(E', -)$ is acyclic for $H^\cdot(E', -)$. This follows from the next lemma, applied with $C = E'_{\text{coh}}$.

260 LEMMA 5.4. *Let E be a locally coherent topos (2.3), and let C be a full generating subcategory of E, stable under fiber products, all of whose objects are quasi-compact (1.1). A filtered inductive limit of C-acyclic sheaves* (V 4.2) *is a C-acyclic sheaf.*

Let $(F_i)_{i \in I}$ be a filtered inductive system of C-acyclic sheaves, and denote its inductive limit by F. Every object Y of C is coherent (2.1), and consequently $F(Y) = \varinjlim_i F_i(Y)$ (1.2.3). To show that F is C-acyclic, it is enough to show that $\check{H}^q(Y, F) = 0$ for every $q > 0$ and every Y of C (V 4.3). Let X be an object of C, and let $\mathscr{X} = (X_\alpha \to X)_{\alpha \in A}$ be a finite covering family by objects of C. We have $H^q(\mathscr{X}, F) = H^q(C^\cdot(\mathscr{X}, F))$ (V 2.4.3). By the preceding and the fact that $C^\cdot(\mathscr{X}, F)$ involves only finite products of groups (V 2.3.3), we have $C^\cdot(\mathscr{X}, F) = \varinjlim_i C^\cdot(\mathscr{X}, F_i)$. Hence

$$H^q(\mathscr{X}, F) = H^q\big(\varinjlim_i C^\cdot(\mathscr{X}, F_i)\big) = \varinjlim_i H^q(C^\cdot(\mathscr{X}, F_i)) = 0$$

for $q > 0$ (V 4.3). Consequently $\check{H}^q(X, F) = \varinjlim_{\mathfrak{X}} H^q(\mathfrak{X}, F) = 0$.

COROLLARY 5.5. *Let E be a coherent topos (2.3), and let $(X_i)_{i \in I}$ be a decreasing filtered family of coherent subobjects of the final object. Denote by Φ the family of closed subtopoi of E* (IV 9) *contained in the closed complement of one of the X_i. Then Φ is a family of supports of E* (V 6.12)*, and for every integer q the functors $H^q_\Phi(E, -)$ and H^q_Φ* (V 6.13) *commute with filtered inductive limits.*

Let Z_i be the closed complement of X_i. For every abelian sheaf F, there is an exact sequence (V 6.5):

$$0 \longrightarrow H^0_{Z_i}(E, F) \longrightarrow H^0(E, F) \longrightarrow H^0(X_i, F) \longrightarrow H^1_{Z_i}(E, F) \longrightarrow \dots \quad .$$

The functors $H^q(E, F)$ and $H^q(X_i, F)$ commute with filtered inductive limits in F (5.2, 5.3). Consequently $H^q_{Z_i}(E, F)$ does so as well, whence the analogous property for $H^q_\Phi(E, F)$ after taking the inductive limit over the Z_i (V 6.13). The assertion concerning the functors $\mathscr{H}^q_\Phi$ follows by localization and passage to the associated sheaf.

261 DEFINITION 5.6. Let E be a ringed topos with ring A, and let q be an integer. An

A-module G is said to be of finite q-presentation if there is a family X_i of objects of E covering the final object of E such that, for every i, there is an exact sequence of modules on $E_{/X_i}$:

$$L_q \longrightarrow L_{q-1} \longrightarrow \cdots L_1 \longrightarrow L_0 \longrightarrow G_{/X_i} \longrightarrow 0 \tag{5.5.1}$$

where, for every p, $L_p = A_{/X_i}^{n_p}$, with n_p an integer.

Proposition 5.7. *Let G be a sheaf of finite q-presentation. For every $i \leq q-1$, the functor $\mathbf{Ext}_A^i(G,-)$ commutes with filtered inductive limits of A-modules.*

The problem is local on E (V 6.1). Thus, after localizing to the $E_{/X_i}$, we may suppose that there is an exact sequence of the form (5.5.1). Put $L^\cdot = L_q \to \cdots \to L_1 \to L_0$. We have $\mathbf{Ext}_A^i(G,F) \simeq \mathscr{H}^i(\mathbf{Hom}_A(L^\cdot, F))$, and the functor $F \mapsto \mathbf{Hom}_A(L^\cdot, F)$ commutes with inductive limits, which proves the proposition.

Corollary 5.8. *Let (E,A) be a ringed topos, let G be a sheaf of A-modules of finite q-presentation, and let X be an algebraic and coherent object of E. The functors $F \mapsto \mathrm{Ext}_A^i(X;G,F)$, for $\frac{i(i-1)}{2} \leq q-1$, commute with filtered inductive limits in F.*

This follows from 5.7 and 5.3 by the spectral sequence of V 6.1.3.

5.9. Let (E,A) be a ringed topos, let Φ be a family of supports of E (V 6.12), let G be an A-module of finite r-presentation (5.6), and let X be an algebraic and coherent object of E (2.3). Taking the inductive limit over the closed subtopoi in Φ in the last spectral sequences of V 6.9.1 and V 6.9.2, respectively, gives two spectral sequences (V 6.13):

$$\begin{cases} {}'E_2^{pq} = \varinjlim_{Z\in\Phi} \mathrm{Ext}_A^p(X;G,H_Z^qF) \Rightarrow \mathrm{Ext}_{A,\Phi}^{p+q}(X;G,F), \\ {}''E_2^{pq} = \varinjlim_{Z\in\Phi} \mathbf{Ext}_A^p(G,H_Z^qF) \Rightarrow \mathbf{Ext}_{A,\Phi}^{p+q}(G,F). \end{cases} \tag{5.9.1}$$

It follows from 5.7 and 5.8 that there are canonical isomorphisms

$$\begin{cases} {}'E_2^{pq} \simeq \mathrm{Ext}_A^p(X;G,H_\Phi^qF) & , \quad \frac{p(p-1)}{2} \leq r-1, \\ {}''E_2^{pq} \simeq \mathbf{Ext}_A^p(G,H_\Phi^qF) & , \quad p \leq r-1. \end{cases} \tag{5.9.2}$$

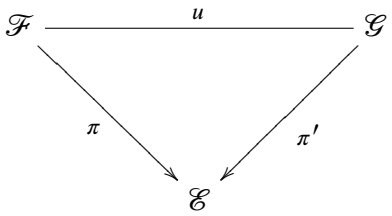
Corollary 5.10. *Use the hypotheses and notation of 5.5. Let A be a ring of E, and let* 262
G be an A-module of finite q-presentation. For every integer i such that $\dfrac{i(i-1)}{2} \leq q-1$, the functors $F \mapsto \mathrm{Ext}_{A,\Phi}^i(E;G,F)$ and $F \mapsto \mathbf{Ext}_{A,\Phi}^i(G,F)$ (V 6.13) commute with filtered inductive limits.

This follows from 5.5 and 5.7 by the first spectral sequences of V 6.9.1 and V 6.9.2, respectively.

6. Inductive and projective limits of a fibered category

6.0. We assume that the reader is familiar with the theory of fibered categories [5]. The essential purpose of this section is to fix the terminology and notation concerning fibered categories. Unless otherwise stated, all categories considered in this section belong to a fixed universe $\mathscr{U}$.

Let $\mathscr{F}$, $\mathscr{G}$, and $\mathscr{E}$ be three categories, and let $\pi : \mathscr{F} \to \mathscr{E}$ and $\pi' : \mathscr{G} \to \mathscr{E}$ be two functors. We denote by $\mathbf{Hom}_{\mathscr{E}}(\mathscr{F}, \mathscr{G})$ the category whose objects are the functors $u : \mathscr{F} \to \mathscr{G}$ such that the diagram



commutes, and whose morphisms are the $\mathscr{E}$-morphisms of functors, i.e. the morphisms of functors carried by π' to identity morphisms.

Let ξ be an object of $\mathscr{E}$. The objects X of $\mathscr{F}$ such that $\pi(X) = \xi$ are called *the objects of $\mathscr{F}$ over ξ*. Let $f : \xi \to \eta$ be a morphism of $\mathscr{E}$. The morphisms m of $\mathscr{F}$ such that
263 $\pi(m) = f$ are called the morphisms of $\mathscr{F}$ *over* f. Let $f : \xi \to \eta$ be a morphism of $\mathscr{E}$, let X be an object of $\mathscr{F}$ over ξ, and let Y be an object of $\mathscr{F}$ over η; we denote by $\mathrm{Hom}_f(X, Y)$ the set of morphisms from X to Y over f.

DEFINITION 6.1. 1) *A morphism $m : X \to Y$ of $\mathscr{F}$ is called* Cartesian *if, for every morphism $p : Z \to Y$ over $\pi(m)$, there exists a unique morphism $q : Z \to X$ over the identity of $\pi(X)$ such that $mq = p$.*

2) *The category $\mathscr{F}$ is called* prefibered *by π over $\mathscr{E}$ if, for every morphism $f : \xi \to \eta$ of $\mathscr{E}$, every object of $\mathscr{F}$ over η is the target of a Cartesian morphism over f.*

3) *The category $\mathscr{F}$ is called* fibered *by π over $\mathscr{E}$ if it is prefibered and the composite of any two composable Cartesian morphisms of $\mathscr{F}$ is Cartesian.*

6.1.1. Let ξ be an object of $\mathscr{E}$. The *fiber category* of $\mathscr{F}$ at ξ, denoted by $\mathscr{F}_\xi$, is the subcategory of $\mathscr{F}$ whose objects are the objects of $\mathscr{F}$ over ξ and whose morphisms are the morphisms of $\mathscr{F}$ over id_ξ. For two objects X and Y of $\mathscr{F}_\xi$, we denote by $\mathrm{Hom}_\xi(X, Y)$ the set of morphisms from X to Y in $\mathscr{F}_\xi$.

6.1.2. Let $f : \xi \to \eta$ be a morphism of $\mathscr{E}$. An object Y over η is the target of a Cartesian morphism over f if and only if the set-valued functor on the fiber $\mathscr{F}_\xi$

$$Z \mapsto \mathrm{Hom}_f(Z, Y) \quad Z \in \mathrm{ob}(\mathscr{F}_\xi),$$

is representable. When the category $\mathscr{F}$ is prefibered, the functor

$$Y \mapsto \mathrm{Hom}_f(., Y) \quad Y \in \mathrm{ob}(\mathscr{F}_\eta),$$

with values in the category of presheaves of sets on $\mathscr{F}_\xi$, in fact takes its values in the category of representable functors on $\mathscr{F}_\xi$, and therefore defines, up to unique isomorphism,
264 a functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$, called the *inverse image functor for f*. Suppose that $\mathscr{F}$ is prefibered and choose, for every morphism f of $\mathscr{E}$, a base-change functor f^*. The category $\mathscr{F}$ is then fibered over $\mathscr{E}$ if and only if, for every pair f, g of composable morphisms of $\mathscr{E}$, the composite functor f^*g^* is an inverse image functor. Let

$$C_{f,g} : f^*g^* \longrightarrow (gf)^* \tag{6.1.2.1}$$

be the canonical isomorphism. The isomorphisms $C_{f,g}$ satisfy a cocycle condition arising from the associativity of composition of morphisms in $\mathscr{E}$:

$$C_{f,hg}(f^* \circ C_{g,h}) = C_{gh,f}(C_{f,g} \circ h^*) \tag{6.1.2.2}$$

6.1.3. Conversely, suppose that for every object ξ of $\mathscr{E}$ one is given a category $\mathscr{F}_\xi$, for every morphism $f : \xi \to \eta$ a functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$, and for every pair (f, g) of morphisms of $\mathscr{E}$ an isomorphism of functors $C_{f,g} : f^*g^* \to (fg)^*$, such that the $C_{f,g}$ satisfy condition (6.1.2.2). One can then construct, in an essentially unique way, a category $\mathscr{F}$ fibered over $\mathscr{E}$ whose fibers "are" the categories $\mathscr{F}_\xi$ and whose base-change functors may be chosen to be "equal" to the functors given in advance [5].

6.1.4. Let $\mathscr{F}$ and $\mathscr{G}$ be two categories over $\mathscr{E}$. We denote by

$$\mathbf{Hom}_{\mathrm{cart}/\mathscr{E}}(\mathscr{F}, \mathscr{G})$$

the full subcategory of $\mathbf{Hom}_{\mathscr{E}}(\mathscr{F}, \mathscr{G})$ defined by the functors that carry Cartesian morphisms of $\mathscr{F}$ to Cartesian morphisms of $\mathscr{G}$. The objects of $\mathbf{Hom}_{\mathrm{cart}/\mathscr{E}}(\mathscr{F}, \mathscr{G})$ are called *Cartesian functors*.

Let C and C' be two categories, and let S be a set of morphisms of the category C. We denote by $\mathrm{Hom}_{S^{-1}}(C, C')$ the set of functors from C to C' that carry the morphisms of S to isomorphisms.

Let (Cat) denote the category whose objects are the categories belonging to the universe and whose morphisms are the functors between these categories (the category (Cat) does not belong to the universe).

Let $\mathscr{F}$ be a category over $\mathscr{E}$, and let S be the set of Cartesian morphisms of $\mathscr{F}$. The category $\mathscr{F}$ defines two functors on (Cat) with values in the category of sets belonging to the universe: 265

$$C \mapsto \mathrm{Hom}_{S^{-1}}(\mathscr{F}, C) \quad C \in \mathrm{ob}(\mathrm{Cat})$$

$C \mapsto \mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{F}, C \times \mathscr{E}) = \{\text{Cartesian functors from } \mathscr{F} \text{ to } C \times \mathscr{E}\}$

(the category $C \times \mathscr{E}$ is regarded as a category over $\mathscr{E}$ by the second projection functor).

Proposition 6.2. *The functors* (Cat) $\to$ (Ens) :

$$C \mapsto \mathrm{Hom}_{S^{-1}}(\mathscr{F}, C)$$
$$C \mapsto \mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{F}, C \times \mathscr{E})$$

are canonically isomorphic. They are representable.

Proof. To prove the first assertion, it is enough to observe that the Cartesian morphisms of $C \times \mathscr{E}$ are the morphisms of the form $m \times f$, where m is an isomorphism of C. To prove the second assertion, it is enough to prove that the functor $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, .)$ is representable. This follows from [1]. We indicate only the idea of the proof. One formally adjoins to the morphisms of $\mathscr{F}$ inverses of the morphisms of S. One considers the free category generated by the objects of $\mathscr{F}$, the morphisms of $\mathscr{F}$, and the formal inverses of the morphisms of S (the path category). One then takes the quotient by the relations arising from the relations among morphisms of $\mathscr{F}$ and relations of the form

$$s^{-1}s = \mathrm{id} \quad , \quad ss^{-1} = \mathrm{id} \quad s \in S.$$

The category thus obtained is denoted by $\mathscr{F}(S^{-1})$; equipped with the canonical functor $Q : \mathscr{F} \to \mathscr{F}(S^{-1})$, it represents the functor $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, .)$.

Definition 6.3. *When $\mathscr{F}$ is fibered over $\mathscr{E}$, the functor* $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, .)$ *is denoted by* $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ *and is called the inductive-limit functor of $\mathscr{F}$ over $\mathscr{E}^\circ$. The category representing it is likewise denoted by* $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ *and is called the inductive-limit category of $\mathscr{F}$ over $\mathscr{E}^\circ$.*

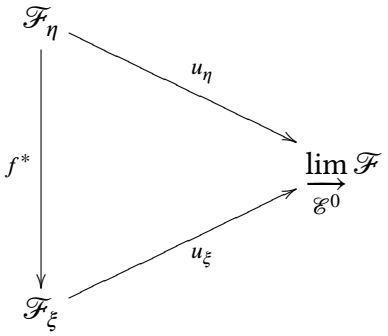
6.4.0. Suppose that $\mathscr{F}$ is fibered over $\mathscr{E}$. For every morphism f of $\mathscr{E}$, choose a base-change functor over f, and let

$$Q : \mathscr{F} \longrightarrow \varinjlim_{\mathscr{E}^\circ} \mathscr{F}$$

266 be the canonical functor. Composing the inclusion functors of the fibers of $\mathscr{F}$ into $\mathscr{F}$ with Q gives, for every object ξ of $\mathscr{E}$, a functor

$$u_\xi : \mathscr{F}_\xi \longrightarrow \varinjlim_{\mathscr{E}} \mathscr{F}$$

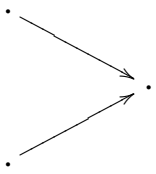
and, for every morphism $f : \xi \to \eta$ of $\mathscr{E}$, a diagram that commutes up to canonical isomorphism:



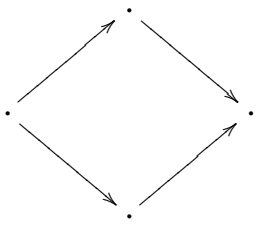
The category $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ then appears as the inductive limit *in the sense of pseudofunctors* [5] of the pseudofunctor $\mathscr{E}^\circ \to \mathrm{Cat}$ that associates $\mathscr{F}_\xi$ with every $\xi \in \mathrm{ob}$, and the functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$ with every $f : \xi \to \eta$. Note, however, that even when $\xi \mapsto \mathscr{F}_\xi$ is an actual functor, the category $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ is not in general the inductive limit in the sense of I 2 of the functor $\xi \mapsto \mathscr{F}_\xi$ (cf. 6.8).

PROPOSITION 6.4. *Let $\mathscr{F}$ be a category fibered over $\mathscr{E}$. Suppose that the category $\mathscr{E}$ has the following properties:*

L1) *Every diagram*



can be embedded in a commutative diagram

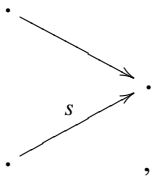


267 L2) *For every pair of morphisms $u, v : \cdot \rightrightarrows \cdot$ such that there exists a morphism t satisfying $tu = tv$, there exists a morphism w satisfying $uw = vw$.*

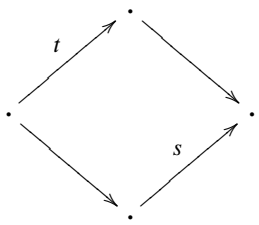
(Note that if the category $\mathscr{E}^\circ$ is pseudofiltered (I 2.7)*, it has properties* L1) *and* L2)*.) The set S of Cartesian morphisms of $\mathscr{F}$ then has the following properties:*

Fr1) *The set S is stable under composition. Isomorphisms belong to S.*

Fr2) *Every diagram*



where s belongs to S, can be completed to a commutative diagram



where t belongs to S.

Fr3) *For every pair of morphisms $u, v : \cdot \rightrightarrows \cdot$ such that there exists a morphism $s \in S$ satisfying $su = sv$, there exists a morphism $t \in S$ such that $ut = vt$.*

Proof. Left to the reader as an exercise.

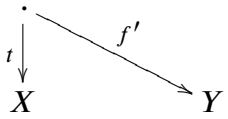
Proposition 6.5. *Let $\mathscr{F}$ be a category, and let S be a set of morphisms of $\mathscr{F}$ having properties* Fr1), Fr2), and Fr3) (6.4). *For every object X of $\mathscr{F}$, let $S(X)$ denote the category of morphisms of S with target X. The category $S(X)$ is cofiltered* (I 2.7). *The category $\mathscr{F}(S^{-1})$ (which represents the functor $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, \cdot)$) may then be described as follows:* 268

a) *The objects of $\mathscr{F}(S^{-1})$ are the objects of $\mathscr{F}$.*

b) *Let X and Y be two objects of $\mathscr{F}(S^{-1})$. We have*

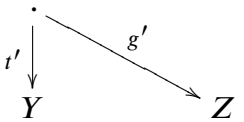
$$\mathrm{Hom}_{\mathscr{F}(S^{-1})}(X, Y) = \varinjlim_{S(X)} \mathrm{Hom}_{\mathscr{F}}(., Y).$$

c) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of $\mathscr{F}(S^{-1})$. Let*



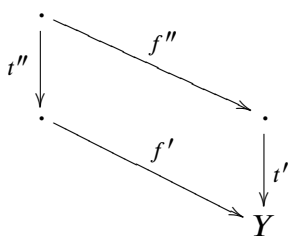
$t \in S$

(resp.



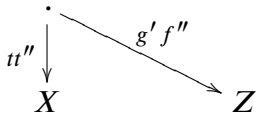
$t' \in S$)

be a diagram in $\mathscr{F}$ whose image is f (resp. g). Let



$t'' \in S$

be a commutative diagram whose existence is ensured by Fr2). The image in $\mathrm{Hom}_{\mathscr{F}(S^{-1})}(X, Z)$ *of the diagram*



is the composite morphism $g f$.

269 *Let*

$$Q : \mathscr{F} \longrightarrow \mathscr{F}(S^{-1})$$

be the canonical functor.

d) *The functor* $Q^* : \mathscr{F}(S^{-1})\hat{} \to \mathscr{F}\hat{}$ *(*$F \mapsto F \circ Q$ I 5.0*) is fully faithful and injective on objects.*

e) *The functor* $Q_! : \mathscr{F}\hat{} \to \mathscr{F}(S^{-1})\hat{}$ *(left adjoint to the functor* Q^* (I 5.1)*) is left exact. (Consequently, the same is true of* Q *when finite projective limits are representable in* $\mathscr{F}$*.)*

Proof. For the proof, we refer to [1].

Exercise 6.6. Let $\mathscr{F}$ be a category, and let S be a set of morphisms of $\mathscr{F}$ having properties Fr1), Fr2), and Fr3). For every object X of $\mathscr{F}$, let $S(X)$ denote the category of morphisms with target X.

a) The category $S(X)$ is cofiltered (I 2).

b) For every object X of $\mathscr{F}$, let $J_S(X)$ denote the set of sieves of X that contain a sieve generated by a morphism of S. The $J_S(X)$ define a topology T on $\mathscr{F}$.

c) For the topology T, the morphisms of S are bicovering (II 5.2).

d) For every object X of $\mathscr{F}$, let $\mathscr{J}^2(X)$ denote the category of bicovering morphisms with target X. The category $S(X)$ is cofinal in $\mathscr{J}^2(X)$ (I 8).

e) Let G be a presheaf of sets on $\mathscr{F}$, and let a be the associated-sheaf functor for the topology T. For every object X of $\mathscr{F}$, there is a canonical isomorphism

$$aG(X) \xrightarrow{\sim} \varinjlim_{S(X)} G(\cdot).$$

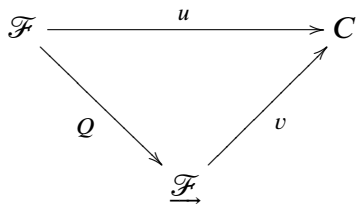
A presheaf is a sheaf if and only if it carries every morphism of S to an isomorphism.

f) Choose the associated-sheaf functor so that its restriction to $\mathscr{F}$ is injective on objects. Let $\underrightarrow{\mathscr{F}}$ denote the full subcategory of the category of sheaves of sets
270 on $\mathscr{F}$ defined by the sheaves associated with representable presheaves. The category $\underrightarrow{\mathscr{F}}$ is isomorphic to the category $\mathscr{F}(S^{-1})$ described in 6.5. The objects of $\underrightarrow{\mathscr{F}}$ form a generating family of the topos $\mathscr{F}\tilde{}$ of sheaves of sets on $\mathscr{F}$. The topology on $\underrightarrow{\mathscr{F}}$ induced by the canonical topology of $\mathscr{F}\tilde{}$ is the coarse topology (II 1.1.4).

g) Let $Q : \mathscr{F} \to \underrightarrow{\mathscr{F}}$ be the canonical functor. The functor Q is a morphism from the site $\underrightarrow{\mathscr{F}}$ (with the coarse topology) to the site $\mathscr{F}$ (with the topology T of b)) (IV 5.9), and induces an isomorphism on the corresponding topoi.

h) Let $u : \mathscr{F} \to C$ be a functor. The functor u is continuous if and only if it carries the morphisms of S to isomorphisms (III 1.1).

i) Let $u : \mathscr{F} \to C$ be a functor carrying the morphisms of S to isomorphisms. The functor u factors uniquely as



(Use III 1.4.)

j) Let $\mathrm{Pro}_S \mathscr{F}$ be the full subcategory of $\mathrm{Pro}\,\mathscr{F}$ (I 8) defined by the pro-objects whose transition morphisms belong to S. Show that the functor Q factors as

$$\mathscr{F} \xrightarrow{\ i\ } \mathrm{Pro}_S \mathscr{F} \xrightarrow{\ j\ } \mathscr{F}_{S^{-1}}$$

where j is the restriction of $\mathrm{Pro}\,Q$ to $\mathrm{Pro}_S \mathscr{F}$. Show that the functor j admits a fully faithful left adjoint

$$A : \mathscr{F}_{S^{-1}} \longrightarrow \mathrm{Pro}_S \mathscr{F}.$$

Show that, for every object X of $\mathscr{F}$, $AQ(X)$ is the pro-object

$$Y \longrightarrow X \in S(X) \mapsto Y.$$

PROPOSITION 6.7. *Let $\mathscr{E}$ be a cofiltered category (I 8.7) and $\mathscr{F} \to \mathscr{E}$ a fibered category* 271 *(6.1.3).*

1) *Let ξ be an object of $\mathscr{E}$, and let $\mathscr{F}/\xi$ be the category fibered over $\mathscr{E}/\xi$ obtained by the base change $\mathscr{E}/\xi \to \mathscr{E}$. The canonical functor $\varinjlim_{\mathscr{E}/\xi} \mathscr{F}/\xi \to \varinjlim_{\mathscr{E}} \mathscr{F}$ is an equivalence of categories.*
2) *More generally, let $\mathscr{E}' \to \mathscr{E}$ be a cofinal functor* (I 8). *Let $\mathscr{F}' \to \mathscr{E}'$ be the fibered category deduced from $\mathscr{F} \to \mathscr{E}$ by the base change $\mathscr{E}' \hookrightarrow \mathscr{E}$. The canonical functor*

$$\varinjlim_{\mathscr{E}'} \mathscr{F}' \longrightarrow \varinjlim_{\mathscr{E}} \mathscr{F}$$

is an equivalence of categories.

PROOF. Exercise.

EXERCISE 6.8. 1) Let $g : \mathscr{E}^\circ \to \mathrm{Cat}$ be a functor and $\mathscr{G} \to \mathscr{E}$ the corresponding fibered category [5]. Show that there exists a canonical functor $\varinjlim_{\mathscr{E}^\circ} \mathscr{G} \to \varinjlim_{\mathscr{E}^\circ} g$. Show that this functor is not necessarily an equivalence of categories. (One may take for g the constant functor whose value is the final object of Cat, and for $\mathscr{E}$ the category associated with a group.)

2) Suppose that the category $\mathscr{E}^\circ$ is pseudo-filtered. Show that the canonical functor $\varinjlim_{\mathscr{E}^\circ} \mathscr{G} \to \varinjlim_{\mathscr{E}^\circ} g$ is then an equivalence of categories.

PROPOSITION 6.9. *Let $\mathscr{F} \to \mathscr{E}$ be a fibered category (6.1). The functor* (Cat) $\to$ (Ens)

$$C \mapsto \mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(C \times \mathscr{E}, \mathscr{F})$$

is represented by the category $\mathbf{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{E}, \mathscr{F})$.

Let C be a category, let $F : C \times \mathscr{E} \to \mathscr{F}$ be a Cartesian $\mathscr{E}$-functor (6.1.4), and let X be an object of C. The functor $\xi \mapsto F((X, \xi))$ is a Cartesian functor from $\mathscr{E}$ to $\mathscr{F}$, denoted $F'(X)$, that depends functorially on X. Hence there is a map $F \mapsto F'$ from

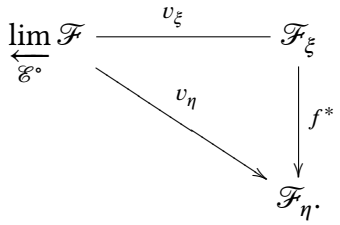
$\mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(C\mathscr{E}, \mathscr{F})$ to $\mathrm{Hom}(C, \mathbf{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{E}, \mathscr{F}))$, functorial in C, which is immediately verified to be a bijection.

272 **6.10.** The category $\mathbf{Hom}_{\mathrm{Cart}/}(\mathscr{E}, \mathscr{F})$ is called the category of *Cartesian sections* of the fibered category $\mathscr{F}$. It is also sometimes called the *projective-limit* category of $\mathscr{F}$ over $\mathscr{E}^\circ$. It is then denoted $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}$ [2].

6.11. Choose a splitting of the fibered category $\mathscr{F} \to \mathscr{E}$, i.e. for every arrow $f : \xi \to \eta$ of $\mathscr{E}$, choose a base-change functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$. For every object ξ of $\mathscr{E}$, denote by

$$v_\xi : \varprojlim_{\mathscr{E}^\circ} \mathscr{F} \longrightarrow \mathscr{F}_\xi$$

the evaluation functor at ξ. For every morphism $f : \xi \to \eta$, there is a diagram of functors commutative up to canonical isomorphism:



The category $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}$ thus appears as the projective limit, *in the sense of pseudofunctors*, of $\xi \mapsto \mathscr{F}_\xi$. Even when $\xi \mapsto \mathscr{F}_\xi$ is an actual functor (i.e. even when the chosen splitting is a cleavage [5]), the categories $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}$ (6.10) and $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}_\xi$ (I 2) are not equivalent in general.

7. Fibered topoi and sites

7.1. Fibered topoi.

Definition 7.1.1. *Let $\mathscr{U}$ be a universe. A $\mathscr{U}$-topos fibered over a category I is a category fibered over I (6.1)*

$$p : F \longrightarrow I$$

whose fibers are $\mathscr{U}$-topoi and whose inverse-image functors[34] *relative to the morphisms $f : i \to j$ of I are functors $f^* : F_j \to F_i$ that are the inverse-image functors of morphisms of topoi $f\cdot : F_i \to F_j$* (IV 3.1.2).

7.1.2. Equivalently, a $\mathscr{U}$-topos fibered over a category I is a fibered category $p :$
273 $F \to I$ whose fibers are $\mathscr{U}$-topoi and whose inverse-image functors are exact and commute with inductive limits (IV 1.6).

7.1.3. For every morphism $f : i \to j$ of I, choose an inverse-image functor $f^* : F_j \to F_i$. For every pair $i \xrightarrow{f} j \xrightarrow{g} k$ of composable morphisms of I, there is then a canonical isomorphism $c_{f,g} : f^*g^* \to (gf)^*$ (6.1.2.1), and the $c_{f,g}$ have the cocycle property (6.1.2.2). In addition, for every $f : i \to j$, choose a right adjoint $f_* : F_i \to F_j$ to the functor $f^* : F_j \to F_i$. The functor $f_* : F_i \to F_j$ is defined up to canonical isomorphism by the property of being right adjoint to f^*. The general results on adjoint functors then give canonical isomorphisms

(7.1.3.1) $$c'_{g,f} : g_*f_* \longrightarrow (gf)_*.$$

[34] For the fibered structure (6.1.2).

These have a cocycle property analogous to property 6.2.2.2, which we leave to the reader to make explicit. Applying the results of 6.1.3, we then obtain a category fibered over I°:

$$p' : F' \longrightarrow I^\circ. \tag{7.1.3.2}$$

It is readily verified that the fibered category $p' : F' \to I^\circ$ is independent, up to unique I°-isomorphism, of the various choices used to construct it[34]. The fiber at every object i of I° of the fibered category $p' : F' \to I^\circ$ is canonically isomorphic to the $\mathscr{U}$-topos F_i and is identified with it. The inverse-image functors of the fibered category $p' : F' \to I^\circ$ are *direct-image* functors of morphisms of topoi.

7.1.4. Let $p : F \to I$ be a fibered $\mathscr{U}$-topos. For every $f : i \to j$, choose a morphism of topoi (IV 3.1.2) $f\cdot : F_i \to F_j$ such that the inverse-image functor f^* (under $f\cdot$) is the inverse-image functor for the fibered structure (6.1). We then obtain cocycles $c_{f,g}$ relating these morphisms of topoi (7.1.3.1) and (7.1.3.2). The collection of the morphisms $f.$, $f \in \mathrm{Fl}(I)$, and the cocycles $c_{f,g}$ is called a *bisplitting* of the fibered topos $F \to I$. Choosing a bisplitting makes it possible to associate with the fibered topos $F \to I$ a pseudofunctor $i \mapsto F_i$ from the category I to the 2-category of $\mathscr{U}$-topoi. Conversely, one can associate with it in an essentially unique manner (cf. [5]) a fibered $\mathscr{U}$-topos $F \to I$ endowed with a bisplitting such that the corresponding pseudofunctor is canonically isomorphic to $i \mapsto F_i$. In practice, a fibered $\mathscr{U}$-topos $F \to I$ will most often be denoted by $(F_i)_{i\in I}$, with the implicit assumption that a bisplitting has been chosen. Note, however, that the constructions carried out on fibered topoi (such as the associated total topos (7.4) and the projective limit (8.1)) depend only on the fibered topoi and not on any bisplittings that may have been chosen. 274

Definition 7.1.5. *Let $p : F \to I$ and $q : G \to I$ be two $\mathscr{U}$-topoi fibered over I. A morphism m of fibered topoi from (F, p) to (G, q) consists of the datum of an I-functor $m^* : G \to F$ (6.1.4) and, for every object i of I, the datum of a right adjoint $m_{i*} : F_i \to G_i$ to the functor $m_i^* : G_i \to F_i$ obtained by restricting m^* to the fibers at i. These data must moreover be subject to the following condition: for every object i of I, the pair of adjoint functors (m_i^*, m_{i*}) is a morphism from the topos F_i to the topos G_i.*

7.1.6. Let $m : (F, p) \to (G, q)$ be a morphism of $\mathscr{U}$-topoi fibered over I, and let $p' : F' \to I^\circ$ and $q' : G' \to I^\circ$ be the fibered categories associated by the construction of 7.1.3 with the fibered topoi (F, p) and (G, q), respectively. Choose bisplittings of F and G (7.1.4), denoted in both cases by $(f : i \to j) \mapsto f. : (f^*, f_*)$, from F_i to F_j and from G_i to G_j, respectively. Since $f^* : G \to F$ is an I-functor, for every morphism $f : i \to j$ there is a morphism of functors (transition morphism)

$$b_f : f^* m_j^* \longrightarrow m_i^* f^*, \tag{7.1.6.1}$$

and for every pair $i \xrightarrow{f} j \xrightarrow{g} k$ of composable morphisms, we have

$$m_i^*(c_{f,g}) b_f^* f^*(b_g) = b_{gf} c_{f,g}. \tag{7.1.6.2}$$

Passing to adjoint functors gives morphisms

$$b'_f : f_* m_{i*} \longrightarrow m_{j*} f_* \tag{7.1.6.3}$$

that satisfy relations analogous to (7.1.6.2). Since the functors f_* are the inverse-image 275

[34] The writer apologizes for the nonintrinsic character of this construction.

functors of the fibered categories (F', p') and (G', q') (7.1.3), one can construct an I°-functor

$$m_* : F' \longrightarrow G', \tag{7.1.6.4}$$

whose restrictions to the fibers are the functors m_{i*} and whose transition isomorphisms are the b'_f. It is verified, with a little patience, that the functor m_* so constructed does not depend on the choice of the morphisms of topoi f.[34]

7.1.7. Denote by $\mathrm{Homtop}_I(F, G)$ the set of morphisms of fibered topoi between the fibered topos (F, p) and the fibered topos (G, q). The set $\mathrm{Homtop}_I(F, G)$ is the set of objects of a category denoted $\mathbf{Homtop}_I(F, G)$: if m and n are two morphisms of fibered topoi, a morphism from m to n is an I-morphism of functors from n^* to m^*. By adjunction, one also obtains an I°-morphism from the functor m_* to the functor n_* (7.1.6). Thus, in the end, there are two functors

$$\mathbf{Homtop}_I(F, G) \longrightarrow \mathbf{Hom}_I(G, F)^\circ \tag{7.1.7.1}$$

$$\mathbf{Homtop}_I(F, G) \longrightarrow \mathbf{Hom}_{I^\circ}(F', G') \tag{7.1.7.2}$$

that associate with every morphism $(m^*, (m_{i*})_{i\in \mathrm{Ob}\, I})$, on the one hand the functor $m^* \in \mathrm{Hom}_I(G, F)$ and, on the other hand, the functor $m_* \in \mathrm{Hom}_{I^\circ}(F', G')$ obtained by gluing the m_{i*} as in (7.1.6). The functor $m^* : G \to F$ (7.1.7.1) is called the *inverse-image functor* under the morphism of fibered topoi $m : F \to G$; the functor $m_* : F' \to G'$ (7.1.7.2) is called the *direct-image functor* under the morphism of fibered topoi $m : F \to G$. It follows immediately from the definitions that the functors $m \mapsto m^*$ and $m \mapsto m_*$ are *fully faithful*, and that their essential images are, on the one hand, the set of I-functors from G to F that, fiber by fiber, are exact and commute with inductive limits and, on the other hand, the set of I^0-functors from F' to G' that, fiber by fiber, are exact and commute with inductive limits and, on the other hand, the set of I^0-functors from F' to G' that, fiber by fiber, possess an exact left adjoint. This justifies the abuses of language in which a morphism of fibered topoi is defined by its direct image or its inverse image.
276 In fact, such a morphism of fibered topoi is then defined only up to unique isomorphism.

Denote by $\mathbf{Homtop}_{\mathrm{Cart}/I}(F, G)$ the full subcategory of $\mathbf{Homtop}_I(F, G)$ generated by the morphisms m such that m_* (or, equivalently, m^*) is a Cartesian functor (6.1.4). Such morphisms are called *Cartesian morphisms*. Thus there are fully faithful functors induced by (7.1.7.1) and (7.1.7.2):

$$\mathbf{Homtop}_{\mathrm{Cart}/I}(F, G) \longrightarrow \mathbf{Hom}_{\mathrm{cart}/I}(G, F)^\circ, \tag{7.1.7.3}$$

$$\mathbf{Homtop}_{\mathrm{Cart}/I}(F, G) \longrightarrow \mathbf{Hom}_{\mathrm{cart}/I^\circ}(F', G'). \tag{7.1.7.4}$$

7.1.8. Let $p : F \to I$, $q : G \to I$, and $r : H \to I$ be three $\mathscr{U}$-topoi fibered over I, and let $m : F \to G$ and $n : G \to H$ be two morphisms of fibered $\mathscr{U}$-topoi. As in IV 3.3, define the composite of the two morphisms m and n, denoted $nm : F \to H$, and if $\mathscr{V}$ is a universe such that $\mathscr{U} \in \mathscr{V}$, define a category $(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Top}_{/I})$ whose objects are the $\mathscr{U}$-topoi fibered over I with fiber categories that are elements of $\mathscr{V}$, and whose morphisms are the morphisms of $\mathscr{U}$-topoi fibered over I. Moreover, as in IV 3.3.2, the composition map

$$\mathrm{Homtop}_{/I}(F, G) \times \mathrm{Homtop}_{/I}(G, F) \longrightarrow \mathrm{Homtop}_{/I}(F, H)$$

[34] See the note at the bottom of page (page 112).

is the map induced on objects by a "functor of composition of morphisms":

$$\mathbf{Homtop}_{/I}(F,G)\,\mathbf{Homtop}_{/I}(G,H) \longrightarrow \mathbf{Homtop}_{/I}(F,H)$$

and the considerations of IV 3.3 and IV 3.4 extend without change to the case of fibered U-topoi. The reader will moreover have noticed that when I is a one-point category (one object and one identity morphism), a $\mathscr{U}$-topos fibered over I is "nothing but" a $\mathscr{U}$-topos, and a morphism of fibered $\mathscr{U}$-topoi is "nothing but" a morphism of $\mathscr{U}$-topoi; the preceding constructions then reduce to those of IV 3.

7.1.9. Let $p : F \to I$ be a fibered $\mathscr{U}$-topos and $\phi : J \to I$ a functor. The fibered category $p_J : F_J \to J$ deduced from (F,p) by the base change $\phi : J \to I$ is a $\mathscr{U}$-topos fibered over J. Let $p' : F' \to I^\circ$ and $(p_J)' : (F_J)' \to J^\circ$ be the fibered categories deduced respectively from (F,p) and (F_J,p_J) by the construction of 7.1.3. It is immediately verified that the category $((F_J)',(p_J)')$ is canonically isomorphic to the category $(p')_J : (F')_J \to J^\circ$ deduced from $p' : F' \to I^\circ$ by the base change $\phi : J^\circ \to I^\circ$. These 277
two categories fibered over J° will henceforth be identified and denoted $p'_j : F'_j \to J^\circ$. The base-change operation is functorial, and even 2-functorial, in its argument: for every pair $p : F \to I$ and $G \to I$ of $\mathscr{U}$-topoi fibered over I, there are two commutative diagrams of categories and functors:

$$\begin{array}{ccc} \mathbf{Homtop}_I(F,G) & \longrightarrow & \mathbf{Hom}_I(G,F)^\circ \\ \downarrow & & \downarrow \\ \mathbf{Homtop}_J(F_J,G_J) & \longrightarrow & \mathbf{Hom}_J(G_J,F_J)^\circ, \end{array} \tag{7.1.9.1}$$

$$\begin{array}{ccc} \mathbf{Homtop}_I(F,G) & \longrightarrow & \mathbf{Hom}_{I^\circ}(F',G') \\ \downarrow & & \downarrow \\ \mathbf{Homtop}_J(F_J,G_J) & \longrightarrow & \mathbf{Hom}_{J^\circ}(F'_{J^\circ},G_{J^\circ}), \end{array} \tag{7.1.9.2}$$

where the vertical functors are base-change functors, and the horizontal functors in (7.1.9.1) are respectively the functors "morphism $\mapsto$ inverse image" (7.1.7.1), while those in (7.1.9.2) are the functors "morphism $\to$ direct image" (7.1.7.2).

7.2. Fibered sites.

7.2.1. A *fibered $\mathscr{U}$-site* C over a category I is a fibered category $p : C \to I$ whose fibers are endowed with topologies making them $\mathscr{U}$-sites (IV 3.0.2), such that for every morphism $f : i \to j$ of I, the inverse-image functor $f^* : C_j \to C_i$ is a morphism from the site C_i to the site C_j (IV 4.9.1).

7.2.2. Let $p : C \to I$ and $q : D \to I$ be two fibered $\mathscr{U}$-sites over I. A *morphism of fibered $\mathscr{U}$-sites from C to D* is an I-functor (6.0) $m : D \to C$ such that, for every object i of I, the functor $m_i : D_i \to C_i$ is a morphism from the site C_i to the site D_i. Denote by $\mathrm{Morsite}_I(C,D)$ the set of morphisms of fibered sites from C to D. A *continuous functor from the fibered site D to the fibered site C* is an I-functor from D to C that induces continuous functors on the fibers (III 1). Denote by $\mathrm{Cont}_I(D,C)$ the set of continuous functors from the fibered site D to the fibered site C. Denote 278
by $\mathbf{Morsite}_I(C,D)$ the full subcategory of $\mathbf{Hom}_I(D,C)^\circ$ defined by the set of objects $\mathrm{Morsite}_I(C,D) \subset \mathrm{Hom}_I(D,C)$. Likewise, denote by $\mathscr{C}ont_I(D,C)$ the full subcategory of $\mathbf{Hom}_I(D,C)$ defined by the subset of objects $\mathscr{C}ont_I(D,C) \subset \mathrm{Hom}_I(D,C)$. Thus we have two fully faithful functors, injective on objects,

$$\mathbf{Morsite}_I(C,D) \hookrightarrow \mathscr{C}ont_I(D,C)^\circ \hookrightarrow \mathbf{Hom}_I(D,C)^\circ. \tag{7.2.2.1}$$

As in 7.1.7, one defines *cartesian* morphisms from the fibered site C to the fibered site D. We have a diagram of full subcategories

$$\begin{array}{ccc} \mathbf{Morsite}_{\mathrm{Cart}/I}(C,D) & \hookrightarrow & \mathbf{Hom}_{\mathrm{Cart}/I}(D,C)^{\circ} \\ \downarrow & & \downarrow \\ \mathbf{Morsite}_{I}(C,D) & \hookrightarrow & \mathbf{Hom}_{I}(D,C)^{\circ}. \end{array} \tag{7.2.2.2}$$

7.2.3. Morphisms of fibered $\mathscr{U}$-sites can be composed. This makes it possible to define the category ($\mathscr{V}$-$\mathscr{U}$-Site/I), whose objects are the $\mathscr{U}$-sites whose fibers belong to a universe $\mathscr{V}$ (of which $\mathscr{U}$ is an element). The category $\mathscr{V}$-$\mathscr{U}$-site is moreover endowed with a 2-category structure; in particular, composition of morphisms of fibered $\mathscr{U}$-sites is functorial with respect to morphisms between morphisms. For every functor $\phi : J \to I$ and every fibered site $p : C \to I$, the category $p_J : C_J \to J$ deduced from (C,p) by the base change $\phi : J \to I$ is canonically endowed with a fibered $\mathscr{U}$-site structure over J. The base-change operation is 2-functorial.

7.2.4. Let $p : C \to I$ be a fibered category whose fibers are endowed with topologies making them $\mathscr{U}$-sites. Suppose that projective limits are representable in the fiber categories. For $p : C \to I$ to be a fibered $\mathscr{U}$-site, it is necessary that the inverse-image functors be left exact and take covering families to covering families; this condition is also sufficient when the topologies of the fibers are coarser than the canonical topology (IV 4.9.2). All the fibered sites used in this Seminar will be of the type described above. Likewise, let $p : C \to I$ and $q : D \to I$ be two fibered sites such that finite projective
279 limits are representable in the fiber categories. A cartesian functor $m : D \to C$ that induces left exact functors on the fiber categories and takes covering families of the fiber categories to covering families is a morphism of fibered sites from C to D; the converse holds if the fiber topologies of C are coarser than the canonical topology (IV 4.9.2). All the morphisms of fibered sites used in this Seminar will be of the type described above.

7.2.5. A fibered $\mathscr{U}$-topos $p : F \to I$ is a fibered $\mathscr{U}$-site when its fibers are endowed with the canonical topology, as we shall always do from now on. Let $m : F \to G$ be a morphism of fibered $\mathscr{U}$-topoi. The inverse-image functor $m^* : G \to F$ (7.1.7) is a morphism from the fibered $\mathscr{U}$-site F to the fibered $\mathscr{U}$-site G. Thus we have a functor ($\mathscr{V}$-$\mathscr{U}$-Top$_{/I}$) $\to$ ($\mathscr{V}$-$\mathscr{U}$-Site$_{/I}$), which is not faithful in general. But this functor extends naturally to a 2-functor which, for two fibered $\mathscr{U}$-topoi F and G over I, induces a functor

$$\mathbf{Homtop}_I(F,G) \longrightarrow \mathbf{Morsite}_I(F,G) \tag{7.2.5.1}$$

that is an *equivalence of categories*, as follows immediately from the definitions. Moreover, the functor (7.2.5.1), composed with the canonical inclusion $\mathbf{Morsite}_I(F,G) \hookrightarrow \mathbf{Hom}_I(G,F)^{\circ}$, is precisely the functor (7.1.7.1) that associates its inverse image with a morphism of fibered topoi.

7.2.6. Let $p : C \to I$ be a fibered $\mathscr{U}$-site. For every morphism $f : i \to j$ of I, choose a base-change functor $f^* : C_j \to C_i$. Passing to the categories of $\mathscr{U}$-sheaves, the functor f^* extends to a functor $\mathrm{Top}(f)^* : C_j^{\sim} \to C_i^{\sim}$, where $\mathrm{Top}(f) : C_i^{\sim} \to C_j^{\sim}$ is a morphism of topoi (IV 4.9.1.1), making the following diagram commutative up to

isomorphism:

(7.2.6.1)
$$\begin{array}{ccc} C_j & \xrightarrow{f^*} & C_i \\ {\scriptstyle \epsilon_j}\downarrow & & \downarrow{\scriptstyle \epsilon_i} \\ \tilde{C_j} & \xrightarrow{\mathrm{Top}(f)^*} & \tilde{C_i} \end{array}$$

Let

(7.2.6.2) $$c_{f,g} : f^*g^* \longrightarrow (gf)^*$$

be the canonical isomorphisms (6.1.2.1), and let 280

(7.2.6.3) $$b_f : \mathrm{Top}(f)^*\epsilon_j \longrightarrow \epsilon_i f^*$$

be the isomorphisms that "make" the diagrams (7.2.6.1) commutative. Using the fact that the functors $\mathrm{Top}(f)^*$ commute with inductive limits, one verifies immediately that, for every pair of composable morphisms $i \xrightarrow{f} j \xrightarrow{g} k$, there is a unique isomorphism

(7.2.6.4) $$\mathrm{Top}(c_{f,g}) : \mathrm{Top}(f)^* \mathrm{Top}(g)^* \longrightarrow \mathrm{Top}(gf)^*$$

such that

(7.2.6.5) $$\epsilon_i(c_{f,g})b_f \,\mathrm{Top}(f)^*(b_g) = b_{gf}\,\mathrm{Top}(c_{f,g}).$$

Moreover, the isomorphisms (7.2.6.4) automatically satisfy the cocycle condition of 6.2.2.2. We can therefore construct a fibered category

(7.2.6.6) $$p^\sim : C^{\sim/I} \longrightarrow I$$

whose fibers are canonically isomorphic to the $\mathcal{U}$-topoi $C_i^\sim$ (6.1.3), and a cartesian functor

(7.2.6.7) $$\epsilon_{C/I} : C \longrightarrow C^{\sim/I}$$

that induces on the fibers the functors $\epsilon_i : C_i \to C_i^\sim$. It follows immediately from the definitions that $p^\sim : C^{\sim/I} \to I$ is a fibered $\mathcal{U}$-topos over I, that $\epsilon_{C/I}$ is a cartesian morphism of fibered $\mathcal{U}$-sites from $C^{\sim/I}$ to C, and that the fibered $\mathcal{U}$-topos $C^{\sim/I} \to I$ and the cartesian functor $\epsilon_{C/I}$ are independent, up to canonical isomorphism, of the various choices used to construct them (the choices of the base-change functors f^* and of the extensions $\mathrm{Top}(f)^*$). The fibered $\mathcal{U}$-topos $C^{\sim/I}$ is called the *associated fibered $\mathcal{U}$-topos* of the fibered $\mathcal{U}$-site.

. One verifies that formation of the fibered $\mathcal{U}$-topos associated with a fibered site "commutes" with base-category change operations.

Proposition 7.2.7. *Let E be a fibered $\mathcal{U}$-topos over I, and let C be a fibered $\mathcal{U}$-site over I. The functor $m \mapsto m^*\epsilon_{C/I}$, which associates with every morphism of fibered topoi $m : E \to C^{\sim/I}$ the composite with $\epsilon_{C/I}$ of the associated inverse-image functor $m^* : C^{\sim/I} \to E$, induces an equivalence of categories preserving cartesian objects* 281

$$\mathbf{Homtop}_I(E, C^{\sim/I}) \xrightarrow{\approx} \mathbf{Morsite}_I(E, C).$$

When finite projective limits are representable in the fiber categories of C, the corresponding fully faithful functor

$$\mathbf{Homtop}_I(E, C^{\sim/I}) \longrightarrow \mathbf{Hom}_I(C, E)^\circ$$

has as its essential image the set of I-functors $g : C \to E$ that are left exact fiber by fiber and take covering families of the fiber categories to epimorphic families of the corresponding fiber categories.

This proposition merely generalizes Proposition IV 4.9.4 to the fibered context, and the proof given in IV 4.9.4 generalizes immediately.

7.3. Example.

EXAMPLE 7.3.1. (*The fibered topos of arrows of a topos*)

Let E be a $\mathscr{U}$-topos, let $\mathrm{Fl}(E)$ be the category of arrows of E, and let $p : \mathrm{Fl}(E) \to E$ be the functor that assigns to an arrow its target. The category $(\mathrm{Fl}(E), p)$ is a fibered $\mathscr{U}$-topos over E (7.1.1 and IV 5.1). The fiber over every object X of E is the topos $E_{/X}$. The inverse-image functor $f^* : E_{/Y} \to E_{/X}$ for a morphism $f : X \to Y$ is the fiber-product functor. Thus, with every functor $\phi : I \to E$, one can associate the fibered $\mathscr{U}$-topos over I

$$p_I : \mathrm{Fl}(E)_I \longrightarrow I, \tag{7.3.1.1}$$

obtained by the base change $\phi : I \to E$, whose fiber over every object i of I is $E_{/\phi(i)}$ (7.1.9). If $\mathscr{V}$ is a universe of which E is an element, then, with the notation of 7.1.8, we have a map

$$\mathrm{Hom}(I, E) \longrightarrow \mathrm{ob}(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{top}_{/I}), \tag{7.3.1.2}$$

which associates with every functor $\phi \in \mathrm{Hom}(I, E)$ the fibered $\mathscr{U}$-topos over I deduced
282 from $(\mathrm{Fl}(E), p)$ by the base change $\phi : I \to E$. Let ϕ_1 and ϕ_2 be two functors from I to E, and let $m : \phi_1 \to \phi_2$ be a morphism of functors. Denote by $p_1 : \mathrm{Fl}(E)_1 \to I$ and $p_2 : \mathrm{Fl}(E)_2 \to I$ the fibered $\mathscr{U}$-topoi over I deduced from $p : \mathrm{Fl}(E) \to E$ by the base changes $\phi_1 : I \to E$ and $\phi_2 : I \to E$, respectively. For every object i of I, choose a morphism of topoi (IV 5.5.2)

$$\mathrm{loc}(m(i)) : E_{/\phi_1(i)} \longrightarrow E_{/\phi_2(i)}. \tag{7.3.1.3}$$

One then verifies easily that there is a unique[34] cartesian morphism of fibered topoi over I,

$$\mathrm{loc}(m) : \mathrm{Fl}(E)_1 \longrightarrow \mathrm{Fl}(E)_2, \tag{7.3.1.4}$$

that induces the morphisms $\mathrm{loc}(m(i))$ on the fiber topoi. The morphism $\mathrm{loc}(m) : \mathrm{Fl}(E)_1 \to \mathrm{Fl}(E)_2$ is determined up to canonical isomorphism by the morphism of functors $m : \phi_1 \to \phi_2$. If $n : \phi_2 \to \phi_3$ is a morphism of functors, we have an isomorphism

$$c(m, n) : \mathrm{loc}(n)\,\mathrm{loc}(m) \simeq \mathrm{loc}(nm), \tag{7.3.1.5}$$

and the morphisms $c(m, n)$ satisfy a cocycle property analogous to (6.1.2.2).

EXAMPLE 7.3.2. (The fibered site of arrows of a site).

Let C be a $\mathscr{U}$-site in which *fiber products are representable*, let $\mathrm{Fl}(C)$ be the category of arrows of C, and let $p : \mathrm{Fl}(C) \to C$ be the functor that assigns to an arrow its target. The fibered category $(\mathrm{Fl}(C), p)$ is a fibered $\mathscr{U}$-site over C when the fiber categories are endowed with the induced topologies (II 5.2). Let $\epsilon_C : C \to C^\sim$ be the canonical functor from C to the category of $\mathscr{U}$-sheaves on C. Denoting by $\mathrm{Fl}(\epsilon_C)$ the natural extension of

[34] Uniqueness comes from requiring, with the notation of (IV 5.5.3), the formula $(gf)_! = g_! f_!$.

the latter functor to the arrow categories, we have a commutative diagram of categories and functors:

$$\begin{array}{ccc} \mathrm{Fl}(C) & \xrightarrow{\mathrm{Fl}(\epsilon_C)} & \mathrm{Fl}(C^{\sim}) \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ C & \xrightarrow{\epsilon_C} & C^{\sim} \end{array} \tag{7.3.2.1}$$

and hence a cartesian functor between categories fibered over C: 283

$$\begin{array}{ccc} \mathrm{Fl}(C) & \longrightarrow & \mathrm{Fl}(C^{\sim})_C \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle p_{/C}} & \\ & C & \end{array} \tag{7.3.2.2}$$

where $p_{/C} : \mathrm{Fl}(C^{\sim})_C \to C$ is deduced from $\mathrm{Fl}(C^{\sim}) \to C^{\sim}$ by the base change $\epsilon_C : C \to C^{\sim}$. The functor $\mathrm{Fl}(C) \to \mathrm{Fl}(C^{\sim})_C$ of (7.3.2.2) is a morphism from the fibered site $\mathrm{Fl}(C^{\sim})_C$ to the fibered site $\mathrm{Fl}(C)$ (7.2.2); hence, by 7.2.7, there is a morphism of fibered topoi over C,

$$\mathrm{can} : \mathrm{Fl}(C^{\sim})_C \longrightarrow \mathrm{Fl}(C)^{\sim/C}, \tag{7.3.2.3}$$

where $\mathrm{Fl}(C)^{\sim/C}$ is the fibered topos over C associated with the fibered site $\mathrm{Fl}(C) \to C$ (7.2.6). It follows from the definition of the associated fibered topos (7.2.6) and from II 5.5 that the morphism can of (7.3.2.3) is a C-equivalence of fibered $\mathscr{U}$-topoi over C, i.e. there is a morphism $\Theta : \mathrm{Fl}(C)^{\sim/C} \to \mathrm{Fl}(C^{\sim})_C$ of fibered $\mathscr{U}$-topoi over C such that the composites $\Theta \circ \mathrm{can}$ and $\mathrm{can} \circ \Theta$ are C-isomorphic to the identity.

Let $\phi : I \to C$ be a functor. By base change, we deduce a fibered site over I,

$$p_I : \mathrm{Fl}(C)_I \longrightarrow I \tag{7.3.2.4}$$

and, since formation of the associated fibered topos commutes with base change (7.2.6), we have an I-equivalence of fibered $\mathscr{U}$-topoi

$$\mathrm{can}_I : \mathrm{Fl}(C^{\sim})_I \longrightarrow \mathrm{Fl}(C)_I^{\sim/I} \tag{7.3.2.5}$$

where $\mathrm{Fl}(C^{\sim})_I \to I$ is the fibered $\mathscr{U}$-topos obtained by the base change $\epsilon_C \circ \phi : I \to C^{\sim}$.

Finally, let $\phi_1 : I \to C$ and $\phi_2 : I \to C$ be two functors, and let $m : \phi_1 \to \phi_2$ be a morphism of functors. Denote by $\mathrm{Fl}(C)_1$ and $\mathrm{Fl}(C)_2$ the fibered sites over I deduced from $p : \mathrm{Fl}(C) \to C$ by the base changes ϕ_1 and ϕ_2, respectively. As in 7.3.1, one constructs a morphism of fibered sites over I,

$$\mathrm{loc}(m) : \mathrm{Fl}(C)_1 \longrightarrow \mathrm{Fl}(C)_2. \tag{7.3.2.6}$$

Denote by $\mathrm{Fl}(C^{\sim})_1 \to I$ and $\mathrm{Fl}(C^{\sim})_2 \to I$ the fibered $\mathscr{U}$-topoi over I obtained by the base changes $\epsilon_C \circ \phi_1$ and $\epsilon_C \circ \phi_2$, respectively. We have a diagram of morphisms over I that is commutative up to isomorphism: 284

$$\begin{array}{ccc} \mathrm{Fl}(C^{\sim})_1 & \xrightarrow{\mathrm{loc}(\epsilon_C * m)} & \mathrm{Fl}(C^{\sim})_2 \\ \downarrow & & \downarrow \\ \mathrm{Fl}(C)_1 & \xrightarrow{\mathrm{loc}(m)} & \mathrm{Fl}(C)_2 \end{array} \tag{7.3.2.7}$$

where the vertical morphisms are deduced by base change from the morphisms of (7.3.2.2).

EXERCISE 7.3.3. (*Small and big fibered sites*).

This exercise continues Exercise IV 4.10.6, whose hypotheses and notation we use. Denote by $\mathbf{M}$ the full subcategory of $\mathrm{Fl}(S)$ defined by the morphisms of $\mathscr{M}$, and let $p : \mathbf{M} \to S$ be the functor that assigns to a morphism its target.

6°) Show that $(\mathbf{M}, p)$ is a fibered site over S and that, when fiber products are representable in S, the inclusion functor $\mathbf{M} \to \mathrm{Fl}(S)$ is a cartesian morphism from the fibered site $\mathrm{Fl}(S) \to S$ to the fibered site $\mathbf{M} \to S$.

7°) Let $\mathbf{M}^{\sim/S} \to S$ be the fibered topos associated with $\mathbf{M} \to S$. Show that, with the notation of 7.3.2, there is a canonical morphism $g : \mathrm{Fl}(S^{\sim})_{/S} \to \mathbf{M}^{\sim/S}$ of fibered topoi over S which induces on each fiber over $X \in \mathrm{ob}(S)$ the morphism $(\mathrm{Prol}_X, \mathrm{Res}_X)$ (IV 4.10.6 4°)). Show that, although for every $X \in \mathrm{ob}(S)$ the morphism $g_X : S^{\sim}_{/X} \to S(X)^{\sim}$ admits a section (*loc. cit.*), the morphism g does not in general admit a section.

7.4. The total topology of a fibered topos or site.

DEFINITION 7.4.1. *Let $p : C \to I$ be a fibered $\mathscr{U}$-site and, for every $i \in \mathrm{ob}(I)$, denote by $\alpha_{i!} : C_i \to C$ the inclusion functor of the fiber category C_i into C. The* total topology *on C is the coarsest topology on C for which the functors $\alpha_{i!}$, $i \in \mathrm{ob}(I)$, are continuous* (III 3.6). *The category C equipped with the total topology is called the* total site.

285 PROPOSITION 7.4.2. *Let $p : C \to I$ be a fibered $\mathscr{U}$-site, let T be the total topology on C, and, for every $i \in \mathrm{ob}(I)$, let T_i be the topology of the fiber C_i.*

1) *Let X be an object of C over an object i of I. A family $(X_\beta \to X)_{\beta \in B}$ is covering for T if and only if it is refined by a covering family for T_i, $(Y_\gamma \to X)_{\gamma \in \Gamma}$, in C_i.*
2) *The topology T induces the topology T_i on each category C_i.*
3) *For every $i \in \mathrm{ob}(I)$, the functor $\alpha_{i!} : C_i \to C$ is cocontinuous.*

. Property 3) follows from 1) and III 2.1. Let us prove 1). For every object X of C, let $J(X)$ be the set of sieves of C/X described in 1). One verifies that the $J(X)$ have properties T 1), T 2), and T 3) of II 1.1, and hence that they define a topology T' on C. The topology T' is the coarsest topology on C for which the covering families for the topologies T_i are covering. Thus T' is coarser than T (III 1.6). To prove that $T' = T$, it is therefore enough to show that, for every object i of I, the functor $\alpha_{i!} : C_i \to C$ is continuous when C is equipped with the topology T'; equivalently, it is enough to show that the topology induced by T' on the category C_i is T_i, which will prove property 2) at the same time. Let T_i' be the topology on C_i induced by T', and let $u : R \hookrightarrow X$ be a covering sieve for T_i'. By III 3.2, the morphism $\alpha_{i!}(u) : \alpha_{i!}(R) \to \alpha_{i!}(X)$ is bicovering and in particular covering. By the definition of T', this means that the sieve $R \hookrightarrow X$ is covering for T_i. Thus T_i' is coarser than T_i. To prove that T_i' is finer than T_i, by the definition of the induced topology (III 3.1), it is enough to show that the functor $\alpha_{i!} : C_i \to C$ is continuous when C_i and C are equipped with T_i and T', respectively. In summary, by III 1.2 and 1.5, to prove properties 1) and 2), it is enough to prove the following lemma:

LEMMA 7.4.2.2. *Under the hypotheses and with the notation of 7.4.2 and 7.4.2.1, let X be an object of C over i, and let $R \xrightarrow{u} X$ be a morphism of $\hat{C_i}$. The following properties are equivalent:*

286 i) *The morphism $R \to X$ is bicovering for T_i.*
ii) *The morphism $\alpha_{i!}(u) : \alpha_{i!}(R) \to \alpha_{i!}(X)$ is bicovering for T'.*

i) $\Rightarrow$ ii): It is clear that if $R \to X$ is bicovering for T_i, then the morphism $\alpha_{i!}(u)$ is covering for T'. It remains to show that it is bicovering (II 5.2). Let Y be an object

of C over an object j of I, let $m, n : Y \rightrightarrows \alpha_{i!}(R)$ be two morphisms of $\hat{C}$ such that $\alpha_{i!}(u)m = \alpha_{i!}(u)n$, and set $f = p(\alpha_{i!}(u)m)$. Since the functor $\alpha_{i!}$ commutes with inductive limits (I 5.4.3), we have (I 3.4):

$$\mathrm{Hom}_{\hat{C}}(Y, \alpha_{i!}(R)) \simeq \varinjlim_{C_i/R} \mathrm{Hom}_C(Y, \alpha_{i!}(.)).$$

For every object Z of C_i, with the notation of 6.1.1, we have

$$\mathrm{Hom}_C(Y, Z) \simeq \coprod_{w \in \mathrm{Hom}(j,i)} \mathrm{Hom}_w(Y, Z),$$

and therefore

$$\varinjlim_{C_i/R} \mathrm{Hom}(Y, \alpha_{i!}(.)) \simeq \coprod_{w \in \mathrm{Hom}(j,i)} \varinjlim_{C_i/R} \mathrm{Hom}_w(Y, \alpha_{i!}(.)).$$

For every $w \in \mathrm{Hom}(j, i)$, however, there is an inverse image functor $w^* : C_i \to C_j$, and $\mathrm{Hom}_w(Y, \alpha_{i!}(.)) \simeq \mathrm{Hom}_{C_j}(Y, w^*(.))$. Thus, still denoting by $w^* : \hat{C_i} \to \hat{C_j}$ the extension of w^* to presheaves (which, according to I 5.4, should be denoted $(w^*)_!$), and observing that the functor $w^* : \hat{C_i} \to \hat{C_j}$ commutes with inductive limits (I 5.4.3), we obtain an isomorphism

$$\mathrm{Hom}_{\hat{C}}(Y, \alpha_{i!}(R)) = \coprod_{w \in \mathrm{Hom}(j,i)} \mathrm{Hom}_{\hat{C_j}}(Y, w^*(R)).$$

Under this isomorphism, the morphisms $m, n : Y \rightrightarrows \alpha_{i!}(R)$ correspond to two morphisms $m', n' : Y \rightrightarrows f^*R$ such that $f^*(u)m' = f^*(u)n'$. Since the inverse image functor is a morphism of topoi, it is in particular continuous, and consequently $f^*(u)$ is bicovering (III 3.2). There is therefore a covering family $(Y_\delta \xrightarrow{v_\delta} Y)_\delta$ in C_j such that $m'v_\delta = n'v_\delta$ for every δ. It follows that $mv_\delta = nv_\delta$ for every δ, and hence that $\alpha_{i!}(u)$ is bicovering for T'.

ii) ⇒ i): Suppose that $\alpha_{i!}(R) \xrightarrow{\alpha_{i!}(u)} \alpha_{i!}(X)$ is bicovering. The functor $\alpha_{i!} : C_i \to C$ 287
is cocontinuous (III 2.1). Hence $\alpha_i^*\alpha_{i!}(u)$ is bicovering (III 2.3.2 and II 5.3 ii)). The functor $\alpha_{i!} : C_i \to C$ has property (PPF) of I 5.14. We therefore have a Cartesian diagram in $\hat{C_i}$ (I 5.14 3)):

$$\begin{array}{ccc} R & \longrightarrow & \alpha_i^*\alpha_{i!}(R) \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \alpha_i^*\alpha_{i!}(u)} \\ X & \longrightarrow & \alpha_i^*\alpha_{i!}(X). \end{array}$$

It follows that u is bicovering (II 5.2).

Remark 7.4.3. 1) Proposition 7.4.2 can be stated and proved in a slightly more general setting than that of fibered sites, but apparently without useful application: instead of a fibered site $p : C \to I$, one considers a fibered category $p : C \to I$ whose fibers are equipped with topologies and whose inverse image functors are continuous for these topologies (whereas in the case of fibered sites one requires in addition that these inverse image functors be morphisms of sites (IV 4.9)).

2) One can prove that the total topology of a fibered site is the finest topology making the functors $\alpha_{i!} : C_i \to C$ cocontinuous (III 2).

3) Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a category equivalent to a $\mathscr{U}$-small category. Then the total site C is a $\mathscr{U}$-site (II 3.0.2). Indeed, if, for every object i of I, $(X_\beta)_{\beta\in B_i}$ is a small topologically generating family of C_i, then the family $(X_\beta)_{\beta\in \coprod_i B_i}$ is topologically generating for C and is $\mathscr{U}$-small. The topos of sheaves on the total site is called the *total topos* and is denoted by $\mathrm{Top}(C)$.
4) Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a small category. Since the functor $\alpha_{i!} : C_i \to C$ is cocontinuous, it gives rise to a morphism of topoi $\alpha_i = (\alpha_i^*, \alpha^{i*})$ from $C_i^\sim$ to $\mathrm{Top}(C)$ (IV 4.7). Since, moreover, the functor $\alpha_{i!} : C_i \to C$ is continuous, the functor $\alpha_i^* : \mathrm{Top}(C) \to C_i^\sim$ admits a left adjoint, again denoted
288 by $\alpha_{i!} : C_i^\sim \to \mathrm{Top}(C)$, which extends the functor $\alpha_{i!} : C_i \to C$ (III 1.2 iv)).

Proposition 7.4.4. *Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a small category I.*

1) *A presheaf F on C is a sheaf for the total topology if and only if, for every object i of I, the presheaf $F \circ \alpha_{i!} = F|C_i$ is a sheaf on C_i.*
2) *A functor m from C to a site D is a continuous functor from the total site C to the site D if and only if, for every object i of I, the functor $m|C_i = m \circ \alpha_{i!} : C_i \to D$ is continuous.*

If F is a sheaf for the total topology, the presheaf $F \circ \alpha_{i!}$ is a sheaf on C_i because $\alpha_{i!} : C_i \to C$ is continuous. Conversely, suppose that $F \circ \alpha_{i!}$ is a sheaf on C_i for every i. Let X be an object of C over an object i of I. For every covering sieve R of C_i/X, we have $F(X) \simeq F \circ \alpha_{i!}(R)$ and hence $F(X) \simeq F(\alpha_{i!}(R))$. Let $R' \hookrightarrow X$ be the sieve of C/X that is the image of the canonical morphism of presheaves $\alpha_{i!}(R) \to X$. The morphism $\alpha_{i!}(R) \to R'$ is an epimorphism of presheaves, and therefore $F(R') \to F(\alpha_{i!}(R))$ is injective. Since $F(R') \to F(\alpha_{i!}(R))$ is also surjective, it is bijective, and hence $F(X) \to F(R')$ is bijective. Since the covering sieves of X are described by (7.4.2.1), the presheaf F is a sheaf: returning to the construction of the associated sheaf (II 3), one sees that F is isomorphic to its associated sheaf (II 3.3). The second assertion follows immediately from the first and the definition of continuity (III 1.1).

7.4.5. Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a category I equivalent to a small category, and let

$$f : i \longrightarrow j$$

be an arrow of the index category I. Again denoting by $f : C_i^\sim \to C_j^\sim$ the morphism of topoi determined by the fibered-site structure, we have (with the notation of 7.4.3.4) a diagram of morphisms of $\mathscr{U}$-topoi

(7.4.5.1)
$$\begin{array}{ccc} C_i^\sim & \xrightarrow{\alpha_i} & \mathrm{Top}(C) \\ {\scriptstyle f}\downarrow & \nearrow_{\alpha_j} & \\ C_j^\sim & & \end{array}$$

289 where $\mathrm{Top}(C)$ is the total topos (7.4.3.3). The diagram (7.4.5.1) *is not commutative in general.* We shall define a canonical morphism of morphisms of topoi:

(7.4.5.2)
$$\rho_f : \alpha_i \longrightarrow \alpha_j \circ f.$$

It is enough to define such a morphism at the level of inverse images, i.e. to define a morphism of functors

(7.4.5.3)
$$\beta_f^* : f^* \circ \alpha_j^* \longrightarrow \alpha_i^*,$$

or, equivalently, using the adjunction between f^* and f_*, it is enough to define a morphism of functors

$$\gamma_f : \alpha_j^* \longrightarrow f_* \alpha_i^*. \tag{7.4.5.4}$$

Let Y be an object of C_j and F a sheaf on the total site C. By definition, $\alpha_j^* F(Y) = F(\alpha_{j!}(Y))$ and $f_* \alpha_i^* F(Y) = F(\alpha_{i!} f^*(Y))$, where in the second member $f^* : C_j \to C_i$ denotes the inverse image functor for the fibered structure. By the definition of this inverse image functor, however, there is a canonical Cartesian morphism

$$m_Y : \alpha_{i!} f^* \longrightarrow \alpha_{j!}; \tag{7.4.5.5}$$

hence, on applying the functor F, a map functorial in Y:

$$\gamma_f(F)(Y) = F(m_Y) : \alpha_j^* F(Y) \longrightarrow f_* \alpha_i^* F(Y), \tag{7.4.5.6}$$

which defines a morphism of sheaves on C_j:

$$\gamma_f(F) : \alpha_j^* F \longrightarrow f_* \alpha_i^* F; \tag{7.4.5.7}$$

whence the morphism of functors γ_f of (7.4.5.4).

Now let $f : i \to j$ and $g : j \to k$ be two composable morphisms of I. There is a canonical isomorphism 290

$$c'_{g,f} : g_* f_* \longrightarrow (gf)_*$$

and one immediately verifies the compatibility formula

$$c'_{g,f} \circ g_*(\gamma_f) \circ \gamma_g = \gamma_{gf}. \tag{7.4.5.8}$$

7.4.6. Introduce the fibered $\mathcal{U}$-topos $p^{\sim/I} : C^{\sim/I} \to I$ associated with $p : C \to I$ (7.2.6) and the corresponding category fibered over I°, $(p^{\sim/I})' : (C^{\sim/I})' \to I^\circ$ (7.1.3.2). The preceding considerations associate with every object F of $\mathrm{Top}(C)$ a family $F_i = (\alpha_i^* F)_{i \in \mathrm{ob}(I)}$ of objects of the $C_i^\sim$, together with a family of morphisms

$$\gamma_f F : F_j \longrightarrow f_* F_i, \quad f \in F1(I), \quad f : i \longrightarrow j,$$

subject to the compatibility conditions of (7.4.5.8). Thus with F we have associated an I°-functor from the category I° to the category $(C^{\sim/I})'$, i.e. an object $\Theta_C(F)$ of the category $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$. Since this construction is functorial in F, we obtain a functor

$$\Theta_C : \mathrm{Top}(C) \longrightarrow \mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})'). \tag{7.4.6.1}$$

Proposition 7.4.7. *Let $C \to I$ be a fibered $\mathcal{U}$-site, with I equivalent to a small category. The functor Θ_C (7.4.6.1) is an equivalence of categories.*

We shall content ourselves with describing a quasi-inverse to Θ_C. Let $\sigma \in \mathrm{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$, i.e. an I°-functor $i \mapsto \sigma(i)$ from I° to $(C^{\sim/I})'$. For every object X of C over $p(X) = i \in \mathrm{ob}(I)$, there is a sheaf $\sigma(p(X)) \in \mathrm{ob}\, C_i^\sim$, and hence a set $\sigma(p(X))(X)$. Let $m : X \to Y$ be a morphism of C over the morphism $f : i \to j$ of I. The morphism m factors uniquely as a morphism $m' : X \to f^*Y$ of C_i followed by the canonical Cartesian morphism $f^*Y \to Y$. Likewise, the morphism $\sigma(f)$ factors uniquely as a morphism $\gamma_f(\sigma) : \sigma(p(Y)) \to f_*\sigma(p(X))$ followed by the canonical I°-Cartesian morphism $f_*\sigma(p(X)) \to \sigma(p(X))$. We therefore obtain a map from $\sigma(p(Y))(Y)$ to $\sigma(p(X))(X)$, by composing the maps 291

$$\sigma(p(Y))(Y) \xrightarrow{\gamma_f(\sigma)(Y)} f_*\sigma(p(X))(Y) = \sigma(p(X))(f^*Y) \xrightarrow{\sigma(p(X))(m')} \sigma(p(X))(X).$$

Thus with every object X of C we have associated a set $\sigma(p(X))(X)$, and with every morphism $m : X \to Y$ a map $\sigma(p(Y))(Y) \to \sigma(p(X))(X)$. One verifies that this indeed defines a presheaf on C, and that this presheaf does not depend on the choice of inverse image functors used in its construction. It follows from 7.4.4 that the presheaf $X \mapsto \sigma(p(X))(X)$ is a sheaf on the total site C, and it is clear that this sheaf depends functorially on the object σ of $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$. Returning to the definition of Θ_C, it is also clear that the functor from $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ to $\mathrm{Top}(C)$ just constructed is a quasi-inverse to Θ_C.

Remark 7.4.8. It follows in particular from 7.4.7 that the category $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ is a $\mathscr{U}$-topos. This latter fact can be seen directly. Using I 9.21.10, one sees immediately that conditions a), b), and c) of IV 1.1.2 are satisfied, and the existence of a small generating category follows from I 9.25.

7.4.9. Let $p : C \to I$ and $q : D \to I$ be two $\mathscr{U}$-sites fibered over a small category, and let m be a morphism from the fibered site C to the fibered site D (7.2.2). Denote by $m^{\sim/I} : C^{\sim/I} \to D^{\sim/I}$ the corresponding morphism between the associated fibered $\mathscr{U}$-topoi. The morphism of fibered topoi $m^{\sim/I}$ is defined up to canonical isomorphism (7.2.7). It has a corresponding direct-image functor $m_*^{\sim/I} : (C^{\sim/I})' \to (D^{\sim/I})'$ (7.1.7), which, on passing to sections over I°, gives a functor

$$\mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I}) : \mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})') \longrightarrow \mathbf{Hom}_{I^\circ}(I^\circ, (D^{\sim/I})'). \tag{7.4.9.1}$$

Moreover, the functor m is a continuous functor between the total sites (7.4.4), and hence, by composition, gives a functor $m_* : \mathrm{Top}(C) \to \mathrm{Top}(F)$.

In summary, there is a diagram of functors

$$\begin{array}{ccc} \mathrm{Top}(C) & \xrightarrow{\theta_C} & \mathbf{Hom}_I^\circ(I^\circ, (C^{\sim/I})') \\ \Big\downarrow{\scriptstyle m_*} & & \Big\downarrow{\scriptstyle \mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I})} \\ \mathrm{Top}(D) & \xrightarrow{\theta_D} & \mathbf{Hom}_I^\circ(I^\circ, (D^{\sim/I})') \end{array} \tag{7.4.9.2}$$

292 Proposition 7.4.10. *The diagram (7.4.9.2) is commutative up to isomorphism. When m is Cartesian, m is a morphism from the total site C to the total site D.*

We leave it to the reader to verify that diagram (7.4.9.2) is commutative up to isomorphism. To show that m is a morphism between the total sites, it is enough, in view of 7.4.7, to show that the functor left adjoint to $\mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I})$ is left exact. We first describe the inverse-image functor of the morphism $m^{\sim/I}$ (7.2.7); for every object i of I, denote by $m_i^* : D_i^\sim \to C_i^\sim$ the functor induced by $(m^{\sim/i})^*$ on the fibers at i. For every morphism $f : i \to j$ of I, choose morphisms of topoi, denoted in both cases by $(f^*, f_*) : C_i^\sim \to C_j^\sim$ and $(f^*, f_*) : D_i^\sim \to D_j^\sim$. Since $(m^{\sim/I})^*$ is a Cartesian I°-functor, for every $f : i \to j$ there is a canonical isomorphism

$$m_i^* f_* \longrightarrow f^* m_j^*, \tag{7.4.10.1}$$

and since the functors f^* and f_* are adjoint, there are canonical morphisms (adjunction morphisms)

$$\phi : \mathrm{id} \longrightarrow f_* f^*$$
$$\psi : f^* f_* \longrightarrow \mathrm{id}\,. \tag{7.4.10.2}$$

Now consider the following sequence of functorial morphisms:

$$m_j^* f^* \xrightarrow{(1)} f_* f^* m_j^* f_* \xrightarrow{(2)} f_* m_i^* f^* f_* \xrightarrow{(3)} f_* m_i^*, \tag{7.4.10.3}$$

where morphism (1) is deduced from ϕ, morphism (2) is deduced from the isomorphism (7.4.10.1)), and morphism (3 is deduced from ψ. By composing the various morphisms in the sequence (7.4.10.3), we obtain a functorial morphism 293

$$\mathrm{can}_f : m_j^* f_* \longrightarrow f_* m_i^*. \tag{7.4.10.4}$$

Now let τ be an object of $\mathbf{Hom}_{I^\circ}(I^\circ(D^{\sim/I})')$. Thus, for every object i of I, there is an object $\tau(i)$ of $D_i^\sim$, and for every morphism $f : i \to j$, a morphism

$$\gamma_f(\tau) : \tau(j) \longrightarrow f_* \tau(i), \tag{7.4.10.5}$$

where the family of the $\gamma_f(\tau)$ has a compatibility property analogous to (7.4.5.8)). Put $\sigma(i = m_i^*(\tau(i))$ and denote by

$$\gamma_f(\sigma) : \sigma(j) \longrightarrow f_* \sigma(i) \tag{7.4.10.6}$$

the composite morphism $m_j^* \tau(j) \xrightarrow{m_j^* \gamma_f(\tau)} m_j^* f_* \tau(i) \xrightarrow{\mathrm{can}_f(\tau(i))} f_* m_i^* \tau(i)$. One verifies that the $\gamma_f(\sigma)$ have the compatibility property of (7.4.5.8), and hence that an object of $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ has thus been defined. An "adjoint functor chasing" then shows that the functor from $\mathbf{Hom}_{I^\circ}(I^\circ, (D^{\sim/I})')$ to $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ just constructed is left adjoint to the functor $\mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I})$. It remains to show that this left adjoint commutes with finite projective limits, which follows immediately from the fact that projective limits in the categories of sections are computed fiber by fiber, and that all the functors used to construct this left adjoint commute with finite projective limits.

Corollary 7.4.11. *Let $p : C \to I$ be a $\mathscr{U}$-site fibered over a small category, let $p^{\sim/I} : C^{\sim/I} \to I$ be the associated fibered $\mathscr{U}$-topos (7.2.6), and let $\epsilon_I : C \to C^{\sim/I}$ be the canonical morphism from the fibered $\mathscr{U}$-site $C^{\sim/I}$ to the fibered $\mathscr{U}$-site C. The morphism ϵ_I induces an equivalence $\mathrm{Top}(\epsilon_I) : \mathrm{Top}(C) \to \mathrm{Top}(C^{\sim/I})$ between the corresponding total topoi.*

This follows from the commutativity of diagram (7.4.9.2) and from the fact that the morphism ϵ_I gives an equivalence between the associated fibered $\mathscr{U}$-topoi.

Remark 7.4.11.1. It follows from 7.4.11 that in questions concerning the total topos one may replace fibered sites by the corresponding fibered topoi.

Lemma 7.4.12. *Let $p : C \to I$ be a site fibered over a small category I, and let i be a final object of I* (I 10). *The functor $\alpha_{i!} : C_i^\sim \to \mathrm{Top}(C)$ (7.4.3.4) is exact and fully faithful. The pairs of functors $\beta_i = (\alpha_{i!}, \alpha_i^*) : \mathrm{Top}(C) \to C_i^\sim$ and $\alpha_i = (\alpha_i^*, \alpha_{i*}) : C_i^\sim \to \mathrm{Top}(C)$ are morphisms of topoi. The composite morphism $\beta_i \alpha_i : C_i^\sim \to C_i^\sim$ is isomorphic to the identity morphism.* 294

For every object j of I, denote by $f_j : j \to i$ the unique morphism from j to i. Using the equivalence $\Theta_C : \mathrm{Top}(C) \xrightarrow{\sim} \mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ (7.4.7), one sees that for every object X of C_i, $\alpha_{i!(X)}$ is the section $j \to f_i^*(X)$, and that for every section $j \mapsto$

$\sigma(j) \in \mathrm{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$, $\alpha_i^*(j \mapsto \sigma(j))$ is the object $\sigma(i)$. Hence $\alpha_i^*\alpha_{i!}$ is isomorphic to the identity. Therefore $\alpha_{i!}$ is fully faithful. Since the functors f_j^* are left exact, the functor $\alpha_{i!}$ is left exact. Thus β_i is a morphism of topoi, and since $\alpha_i^*\alpha_{i!}$ is isomorphic to the identity, the morphism $\beta_i\alpha_i$ is isomorphic to the identity morphism.

THEOREM 7.4.13.1. *Let $p : C \to I$ and $q : D \to I$ be two $\mathscr{U}$-sites fibered over a small category I, and let $m : D \to C$ be an I-functor. For m to be a morphism from the fibered site C to the fibered site D (7.2.2), it is sufficient that m be a morphism from the total site C to the total site D.*

. We must show that if an I-functor $m : D \to C$ is a morphism from the total site C to the total site D, then m is a morphism from the fibered site C to the fibered site D, i.e. for every object i of I, the functor $m_i : D_i \to C_i$ induced by m on the fibers at i is a morphism from the site C_i to the site D_i. The proof of this last assertion occupies paragraphs 7.4.13.3 through 7.4.13.7.

. Let us show that, for every object i of I, the functor $m_i : D_i \to C_i$ is continuous. Indeed, there is a commutative diagram:

$$\begin{array}{ccc} D_i & \xrightarrow{m_i} & C_i \\ {\scriptstyle \alpha_{i!}}\downarrow & & \downarrow{\scriptstyle \alpha_{i!}} \\ D & \xrightarrow{m} & C \end{array}$$

295 and it is enough to show that, for every covering sieve $R \hookrightarrow X$ of C_i/X, the morphism $m_{i!}(R) \to m_i(X)$ is bicovering (III 1.2 and 1.5). But by the commutativity of the diagram on page (page 133) and the fact that the functors $\alpha_{i!}$ and m are continuous, the morphism $\alpha_{i!}m_{i!}(R) \to \alpha_{i!}m_i(X)$ is bicovering. The assertion therefore follows from 7.4.2.2.

. *Reduction to the case in which C and D are fibered $\mathscr{U}$-topoi.* Since the functors m_i are continuous, they admit natural extensions to the categories of sheaves, which we denote by $m_i^* : D_i^\sim \to C_i^\sim$ (III 1.2 iv)). Let $D^{\sim/I}$ and $C^{\sim/I}$ be the fibered $\mathscr{U}$-topoi associated with the fibered sites D and C, respectively, and for every morphism f of I, denote by f^* the inverse-image functors for the four fibered categories C, D, $C^{\sim/I}$, and $D^{\sim/I}$. Since the functor m is an I-functor, for every $f : i \to j$ there is a canonical morphism

$$f^*mj \leftarrow m_i f^*.$$

Since the functors f^*, m_i, and m_j are continuous, by passing to the categories of sheaves we deduce canonical isomorphisms

$$f^*m_j^* \longleftarrow m_i^* f^*.$$

These isomorphisms have compatibility properties that make it possible to construct a Cartesian functor $m^* : D^{\sim/I} \to C^{\sim/I}$ inducing the functors m_i^* on the fiber topoi. Moreover, the diagram

$$\begin{array}{ccc} D & \xrightarrow{m} & C \\ {\scriptstyle \varepsilon_I}\downarrow & & \downarrow{\scriptstyle \varepsilon_I} \\ D^{\sim/I} & \xrightarrow{m^*} & C^{\sim/I} \end{array}$$

is commutative up to isomorphism. Since the functors ϵ_I induce equivalences on the total topoi (7.4.11), and since m is a morphism between the total sites, the functor $m^* :$

$D^{\sim/I} \to C^{\sim/I}$ is a morphism between the total sites. We are therefore reduced to proving the assertion of 7.4.13.2 when D and C are fibered $\mathscr{U}$-topoi, which we assume henceforth.

. *For every object i of I, the functor $m_i : D_i \to C_i$ takes the final object of D_i to the final object of C_i.* Denote by $m : D^\sim \to C^\sim$ the natural extension of m to the total topoi (III 1.2). Using the equivalences $\Theta_D : D^\sim \to \mathbf{Hom}_{I^\circ}(I^\circ, D')$ (7.4.7), one immediately verifies that m associates with every section $\sigma \in \mathrm{Hom}_{I^\circ}(I^\circ, D')$ the section $(i \mapsto m_i\sigma(i)) \in \mathrm{Hom}_{I^\circ}(I^\circ, C')$. Since m is a morphism of sites, the functor m is left exact and in particular takes the final object of $\mathbf{Hom}_{I^\circ}(I^\circ, D)$ to the final object of $\mathbf{Hom}_{I^\circ}(I^\circ, C')$. For every object i of I, denote by e_i a final object of D_i. A final object of $\mathbf{Hom}_{I^\circ}(I^\circ, D')$ is the section $i \mapsto e_i$. Thus the section $i \mapsto m_i(e_i)$ is a final object of $\mathbf{Hom}_{I^\circ}(I^\circ, C')$, and consequently, for every object i of I, $m_i(e_i)$ is a final object of C_i. 296

. *Reduction to the case in which i is a final object of I.* Let i be an object of I. The functor $\alpha_{i!} : D_i \to D$ factors as

$$D_i \xrightarrow{\overline{\alpha}_{i!}} D_{/e_i} \xrightarrow{\mathrm{loc}(e_i)} D,$$

where $\overline{\alpha}_i$ is the evident functor and $\mathrm{loc}(e_i)$ the forgetful functor. Denote by $m/i : D_{/e_i} \to D_{m(e_i)}$ the functor that associates with every object $u : X \to e_i$ of $D_{/e_i}$ the object $m(u) : m(X) \to m(e_i)$ of $C_{/m(e_i)}$. There is a commutative diagram

$$\begin{array}{ccc} D_i & \xrightarrow{m_i} & C_i \\ {\scriptstyle \overline{\alpha}_{i!}}\downarrow & & \downarrow{\scriptstyle \overline{\alpha}_{i!}} \\ D_{/e_i} & \xrightarrow{m/i} & C_{/m(e_i)} \end{array}$$

Since e_i and $m(e_i)$ are final objects of D_i and C_i, respectively (7.4.13.5), the categories $D_{/e_i}$ and $C_{/m(e_i)}$ are $\mathscr{U}$-topoi fibered over $I_{/i}$, and the functor $m_{/i}$ is an $I_{/i}$-functor. Since the property of a functor being a morphism of sites is local (IV 4.9), the functor $m_{/i}$ is a morphism from the total site $C_{/m(e_i)}$ to the total site $D_{/(e_i)}$. We are therefore reduced to proving that m_i is a morphism of topoi when i is a final object of I, which we assume henceforth. 297

. *End of the proof.* Denote by $\alpha_{i!} : D_i \to D^\sim$, $m^\sim : D^\sim \to C^\sim$, and $\alpha_{i!} : C_i \to C^\sim$ the natural extensions to the categories of sheaves of the functors $\alpha_{i!}$ and m. There is a diagram commutative up to isomorphism:

$$\begin{array}{ccc} D_i & \xrightarrow{m_i} & C_i \\ {\scriptstyle \alpha_{i!}}\downarrow & & \downarrow{\scriptstyle \alpha_{i!}} \\ D^\sim & \xrightarrow{m^\sim} & C. \end{array}$$

Since i is a final object, the functor $\alpha_i^*\alpha_{i!} : C_i \to C_i$ is isomorphic to the identity (7.4.12), and consequently m_i is isomorphic to $\alpha_i^* m^\sim \alpha_{i!}$. Moreover, the functors α_i^*, $m^\sim$, and α_i^* are left exact (7.4.12). Consequently, m_i is left exact and is therefore a morphism from the site C_i to the site D_i.

Exercise 7.4.14. Let $X = (X_i)_{i\in I}$ be a topos fibered over I and let T be a topos. Show that the categories $\mathbf{Homtop}(X_i, T)$ are the fibers of a fibered category denoted

$\mathbf{Homtop}_I(X, T \times I)$. Show that the category of sections over I of $\mathbf{Homtop}_I(X, T \times I)$ is canonically equivalent to $\mathbf{Homtop}_I(\mathrm{Top}(X), T)$, where $\mathrm{Top}(X)$ is the total topos of X.

EXERCISE 7.4.15. Let $X = (X_i)_{i \in I}$ be a topos fibered over I. Define a Cartesian morphism m from X to the constant fibered topos with fiber (Ens). Show that the total topos of the constant fibered topos with fiber (Ens) is canonically equivalent to $\hat{I}$. Deduce a morphism of topoi

$$\mathrm{Top}(m) : \mathrm{Top}(X) \longrightarrow \hat{I}.$$

Let $F = (F_i)_{i \in I}$ (where $F_i = F|X_i$) be an object of $\mathrm{Top}(X)$ (7.4.7). Show that

$$\mathrm{Top}(m)(F) = (i \mapsto \Gamma(X_i, F_i)).$$

Deduce from 7.4.12 that if F is injective abelian, then F_i is injective abelian for every $i \in \mathrm{ob}\, I$. Deduce that

$$R^q \mathrm{Top}(m)(F) = (i \mapsto H^q(X_i, F_i)).$$

298 Show that the Cartan–Leray spectral sequence of the morphism m (V 5) is

$$H^{p+q}(\mathrm{Top}(X), F) \Leftarrow \varprojlim_I{}^{(p)} H^q(X_i, F|X_i).$$

8. Projective limits of fibered topoi

8.1. Generalities.

DEFINITION 8.1.1. *Let $\mathscr{U}$ be a universe and let $p : F \to I$ be a fibered $\mathscr{U}$-topos. A pair (C, m) consisting of a $\mathscr{U}$-topos C and a cartesian morphism $m : C \times I \to F$ of fibered $\mathscr{U}$-topoi over I is called a projective limit of the fibered topos F if, for every $\mathscr{U}$-topos D, the functor*

$$\mathbf{Homtop}(D, C) \longrightarrow \mathbf{Homtop}_{\mathrm{cart}/I}(D \times I, F) \tag{8.1.1.1}$$

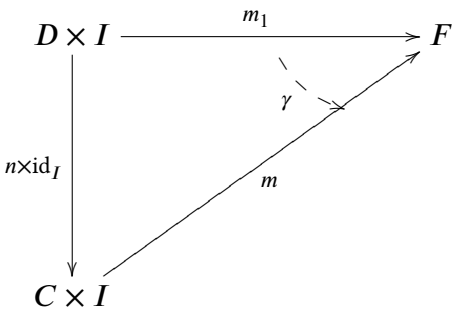
obtained by composing the base-change functor

$$\mathbf{Homtop}(D, C) \longrightarrow \mathbf{Homtop}_I(D \times I, C \times I)$$

with the functor of composition with m (7.1.8) is an equivalence of categories. (cf. 8.1.3.3 for a more "geometric" formulation).

8.1.2. We use the notation of 8.1.1. Let D be a $\mathscr{U}$-topos and let $m_1 : D \times I \to F$ be a morphism of fibered $\mathscr{U}$-topoi. Translating Definition 8.1.1, one immediately sees that there is a morphism of topoi $n : D \to C$ and an isomorphism of morphisms of fibered topoi over I, $\gamma : m_1 \xrightarrow{\sim} m \circ (n \times \mathrm{id}_I)$, making the diagram commutative:

(8.1.2.1)



Moreover, if (n', γ') is another pair with the property above, there is a *unique* isomorphism $\eta : n \xrightarrow{\sim} n'$ such that

$$\gamma \circ m(\eta \times \mathrm{id}) = \gamma'. \tag{8.1.2.2}$$

It follows that if (D, m_1) is a projective limit of F, then n is an equivalence of $\mathscr{U}$- 299
topoi (IV 3.4), and the additional datum of the isomorphism γ making diagram (8.1.2.1) commutative determines the pair (n, γ) up to canonical isomorphism. Existence of the projective limit will be proved in 8.2.3 when I is cofiltered.

8.1.3. We denote by

$$\varprojlim_I \text{top } F \text{ or } \varprojlim_I \text{top } F_i \text{ or also } \underleftarrow{F}$$

a projective limit of the fibered topos F, by $\mu : (\varprojlim_I \text{top } F) \times I \to F$ the canonical morphism, and, for every object i of I, by

$$\mu_i : \underleftarrow{F} \simeq \underleftarrow{F} \times i \longrightarrow F_i$$

the morphism of topoi induced by μ on the fiber over i.

Choose a bisplitting of F (7.1.4), i.e. for every $f : i \to j$ a morphism of topoi $f\cdot = (f^*, f_*) : F_i \to F_j$ such that f^* is an inverse-image functor for the fibered structure. Then, for every pair $i \xrightarrow{f} j \xrightarrow{g} k$ of composable morphisms of I, we have an isomorphism of morphisms of topoi (7.1.3)

$$c_{f,g} : g.f. \simeq (gf). \quad .$$

Since μ is a cartesian morphism of fibered topoi, for every morphism $f : i \to j$ of I we have an isomorphism of morphisms of topoi (7.1.6)

$$b_f : f.\mu_i \xrightarrow{\sim} \mu_j, \tag{8.1.3.1}$$

and the family of the b_f satisfies relations analogous to (7.1.6.2).

. Let D be a $\mathscr{U}$-topos, let $m : D \times I \to F$ be a morphism of fibered topoi, let $m_i : D = D \times \{i\} \to F_i$ be the morphisms induced by m on the fibers, and, for every $f : i \to j$, let

$$b'_f : f.m_i \xrightarrow{\sim} m_j,$$

be the transition isomorphism. It follows from 8.1.2 that there is a morphism of topoi $n : D \to \underleftarrow{F}$ and a family of isomorphisms 300

$$\gamma_i : m_i \xrightarrow{\sim} \mu_i n \quad i \in \text{ob } I,$$

such that, for every $f : i \to j$,

$$b_f \circ f.(\gamma_i) = \gamma_j \circ b'_f.$$

Moreover, the morphism n and the family of the γ_i, $i \in \text{ob}(I)$, are determined up to unique isomorphism (8.1.2).

Conversely, suppose we are given a topos $\underleftarrow{F}$, morphisms $\mu_i : \underleftarrow{F} \to F_i$, and isomorphisms b_f as in (8.1.3.1.), satisfying the usual compatibility conditions, and suppose all these data have the universal property described above. Then there is a unique morphism $\mu : \underleftarrow{F} \times I \to F$ of fibered topoi giving rise to the μ_i and the b_f, and $(\underleftarrow{F}, \mu)$ is a projective limit of the fibered topos F.

. Let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$. The preceding considerations therefore allow projective limits of fibered topoi to be interpreted as "projective limits," in the sense of 2-categories, of the pseudofunctors with values in the 2-category of fibered $\mathscr{U}$-topoi that are elements of $\mathscr{V}$, deduced from the fibered structure by choosing the morphisms f.. The reader should take care, however, not to confuse this notion of "projective limit" in the sense of 2-categories (with commutation up to isomorphism) with the strict (or naive) notion of projective limit (strict commutation of diagrams). Even when the strict notion makes sense (when the pseudofunctor is a genuine functor), generalized "projective limits" and strict projective limits do not coincide in general.

8.1.4. Let $p : F \to I$ and $q : G \to I$ be two fibered $\mathscr{U}$-topoi, and let $m : F \to G$ be a morphism of fibered topoi. Suppose that the fibered topoi F and G have projective limits (8.1.1). It follows immediately from Definition 8.1.1 that there is a morphism of topoi

$$\underleftarrow{m} : \underleftarrow{F} \longrightarrow \underleftarrow{G}, \tag{8.1.4.1}$$

and an isomorphism of morphisms of fibered topoi $\gamma : m \circ \mu \xrightarrow{\sim} \mu \circ \underleftarrow{m} \times \mathrm{id}_I$, making the
301 diagram commutative:

(8.1.4.2)
$$\begin{array}{ccc} \underleftarrow{F} \times I & \xrightarrow{\mu} & F \\ {\scriptstyle \underleftarrow{m}\times \mathrm{id}_I}\downarrow & \overset{\gamma}{\swarrow} & \downarrow{\scriptstyle m} \\ \underleftarrow{G} \times I & \xrightarrow{\mu} & G \end{array} .$$

Moreover, if $(\underleftarrow{m}', \gamma')$ is another pair with the property above, there is a *unique* isomorphism $\eta : \underleftarrow{m} \xrightarrow{\sim} \underleftarrow{m}'$ of morphisms of topoi such that

$$\gamma \circ \mu(\eta \times \mathrm{id}) = \gamma'.$$

8.1.5. Let $p : F \to I$ be a fibered $\mathscr{U}$-topos that is an element of $\mathscr{V}$, let $\phi : I' \to I$ be a functor that is an element of $\mathscr{V}$, and let $p_\phi : F_\phi \to I'$ be the fibered $\mathscr{U}$-topos deduced from (F, p) by the base change ϕ (7.1.9). Let $(\underleftarrow{F}, \mu)$ be a projective limit of F. By base change, we obtain a morphism of fibered topoi over I':

$$\mu' : \underleftarrow{F} \times I' \longrightarrow F_\phi.$$

Let $(\underleftarrow{F_\phi}, \mu_\phi)$ be a projective limit of F_ϕ. By the universal property of the projective limit (8.1.2), we then have a morphism of topoi

$$m_\phi : \underleftarrow{F} \longrightarrow \underleftarrow{F_\phi},$$

and an isomorphism of morphisms of fibered topoi over I':

$$\gamma : \mu' \xrightarrow{\sim} \mu_\phi \circ (m_\phi \times \mathrm{id}_{I'}).$$

The pair (m_ϕ, γ) is determined up to unique isomorphism.

8.1.6. Let $p : F \to I$ be a fibered $\mathscr{U}$-site over a small category I, let $F^{\sim/I} \to I$ be the associated fibered $\mathscr{U}$-topos (7.2.6), and let D be a $\mathscr{U}$-topos. First, we have a functor

$$\mathbf{Homtop}_{\mathrm{cart}/I}(D \times I, F^{\sim/I}) \longrightarrow \mathbf{Morsite}_{\mathrm{cart}/I}(D \times I, F), \tag{8.1.6.1}$$

which associates with every m the composite of m^* with $\epsilon_{F/I}$ (7.2.7). Moreover, by the definition of the category $\mathbf{Morsite}_{\mathrm{cart}/I}(D \times I, F)$ (7.2.2.2), we have a functor

$$\mathbf{Morsite}_{\mathrm{cart}/I}(D \times I, F) \longrightarrow \mathbf{Hom}_{\mathrm{cart}/I}(F, D \times I)^\circ. \tag{8.1.6.2}$$

Finally, by the definition of the inductive limit (6.3), we have a functor 302

$$\mathbf{Hom}_{\mathrm{cart}/I}(F, D\times I)^\circ \longrightarrow \mathbf{Hom}(\varinjlim_{I^\circ} F, D)^\circ. \tag{8.1.6.3}$$

Denote by

$$\Theta : \mathbf{Homtop}_{\mathrm{cart}/I}(D\times I, F^{\sim/I}) \longrightarrow \mathrm{Hom}(\varinjlim_{I^\circ} F, D)^\circ \tag{8.1.6.4}$$

the composite of these last three functors, and by

$$\Pi : F \longrightarrow \varinjlim_{I^\circ} F, \tag{8.1.6.5}$$

the canonical functor.

Proposition 8.1.7. *The functor Θ is fully faithful. Its essential image is the set of functors $m : \varinjlim_{I^\circ} F \to D$ such that the functor*

$$(m\circ\Pi, p) : F \longrightarrow D\times I$$

is a morphism from the total site $D\times I$ to the total site F.

It follows from 7.2.7 that the functor (8.1.6.1) is an equivalence of categories, from 7.2.2 that the functor (8.1.6.2) is fully faithful, and from 6.3 that the functor (8.1.6.3) is an equivalence of categories. Consequently, Θ is fully faithful. A functor $m : \varinjlim_{I^\circ} F \to D$ is isomorphic to the image, under $\mathbf{Hom}_{\mathrm{cart}/I}(F, D\times I)^\circ \to \mathbf{Hom}(\varinjlim_{I^\circ} F, D)^\circ$, of the functor $(m\circ\Pi, p)$, and such a functor belongs to the essential image of (8.1.6.2) if and only if it is a morphism from the total site $D\times I$ to the total site F (7.4.13.1); hence the proposition.

8.2. Construction of the projective limit when the index category is cofiltered.

Lemma 8.2.1. *Let I and I' be two cofiltered categories* (I 8.1)*, and let $\phi : I' \to I$ be a functor such that $\phi^\circ : I'^\circ \to I^\circ$ is cofinal* (I 8.1)*. Let, moreover, $p : F \to I$ be a fibered $\mathscr{U}$-topos, let D be a $\mathscr{U}$-topos, let $m : D\times I \to F$ be a Cartesian morphism of fibered topoi, and let $m' : D\times I' \to F'$ be the morphism of fibered topoi deduced from m by the base change ϕ (7.1.9). The pair (D, m) is a projective limit of F if and only if (D, m') is a projective limit of F'.*

The proof is merely a rather lengthy routine verification. It is left to the patient reader, who may use the viewpoint of projective limits of pseudofunctors (8.1.3.3). 303

Lemma 8.2.2. *Let $p : F \to I$ be a fibered $\mathscr{U}$-site over a small cofiltered category, let $\underrightarrow{F}$ be the inductive limit of the fibered category F (6.3.9), and let $\pi : F \to \underrightarrow{F}$ be the canonical functor. Equip $\underrightarrow{F}$ with the coarsest topology making π continuous* (III 1)*. Let D be a $\mathscr{U}$-site and $n : \underrightarrow{F} \to D$ a functor. The following properties are equivalent:*

i) *n is a morphism of sites $D \to \underrightarrow{F}$.*
ii) *$n\circ\pi$ is a morphism of sites (where F is equipped with the total topology).*
iii) *$(n\circ\pi, p) : F \to D\times I$ is a morphism from the total site $D\times I$ to the total site F.*
iv) *For every object i of I, the composite functor $F_i \xrightarrow{\alpha_{i!}} F \xrightarrow{n\circ\pi} D$ is a morphism of sites.*

The equivalence of i) and ii) follows from 6.6, and that of iii) and iv) follows from 7.4.13. Let us prove the equivalence of iii) and ii). For this it is enough to show that the first projection functor $\mathrm{pr}_1 : D\times I \to D$ is a morphism of sites. Denote by $D^\sim$ the

topos of sheaves on D. It follows from 7.4.7 that $(D \times I)^\sim$ is equivalent to the topos $\mathbf{Hom}_{I^\circ}(I^\circ, D^\sim \times I^\circ) = \mathbf{Hom}(I^\circ, D^\sim)$. Moreover, one sees immediately that the natural extension of $\mathrm{pr}_1 : D \times I \to D$ to sheaves (III 1.2) is the functor from $\mathbf{Hom}(I^\circ, D^\sim)$ to $D^\sim$ that associates with the functor $i \mapsto \sigma(i)$ the object $\varinjlim_{I^\circ} \sigma(i)$. Since I° is filtered, inductive limits over I° are exact (II 4), and hence pr_1 is a morphism of sites (IV 4.9). It remains to prove the equivalence of ii) and iv). First treat the case in which D is a $\mathscr{U}$-topos and F a fibered $\mathscr{U}$-topos over I.

Let i be an object of I. Arguing as in 7.4.13.6, one sees that the functor $\alpha_{i!} : F_i \to F$ factors as a morphism $\overline{\alpha}_{i!} : F_i \to F/\alpha_{i!}(e_i)$ followed by the localization functor $F/\alpha_{i!}(e_i) \to F$, where e_i is a final object of F_i. It is therefore enough to show that the composite functor $F/\alpha_{i!}(e_i) \xrightarrow{n\circ\pi} D$ is a morphism of sites. But there is a diagram commutative up to isomorphism:

$$\begin{array}{ccc} F/\alpha_{i!}(e_i) & \xrightarrow{n\circ\pi/\alpha_{i!}(e_i)} & D/n\circ\pi\alpha_{i!}(e_i) \\ \downarrow & & \downarrow \\ F & \xrightarrow{n\circ\pi} & D \end{array}$$

304 Since the property of being a morphism of sites is local (IV 5.10), it is enough to show that $n \circ \pi \circ \alpha_{i!}(e_i)$ is a final object of D. For every morphism $f : j \to k$ of I, there exists a unique morphism $e_f : \alpha_{j!}(e_j) \to \alpha_{k!}(e_k)$ over f, and this morphism is Cartesian. Thus $j \mapsto \alpha_{j!}(e_j)$ is a Cartesian section of F over I. Consequently, the canonical morphism $n \circ \pi \circ \alpha_{i!}(e_i) \to \varinjlim_I n \circ \pi \circ \alpha_{j!}(e_j)$ is an isomorphism, because π carries Cartesian morphisms to isomorphisms. Moreover, $\varinjlim_I \alpha_{j!}(e_j)$, where the inductive limit is taken in $\hat{F}$, is a final object of $\hat{F}$. Since $n \circ \pi$ is a morphism of sites, it carries the final object of $F^\sim$ to a final object of D. Hence $\varinjlim_I n \circ \pi \circ \alpha_{j!}(e_j)$ is a final object of D, completing the proof in this case.

In the general case, the functor $(n\pi, p) : F \to D \times I$ is Cartesian and continuous (7.4.4). Moreover, $n \circ \pi$ is the composite functor

$$F \xrightarrow{(n\pi,p)} D \times I \xrightarrow{\mathrm{pr}_1} D.$$

Passing to the associated fibered $\mathscr{U}$-topoi gives a diagram commutative up to isomorphism:

$$\begin{array}{ccccc} F & \xrightarrow{(n\pi,p)} & D \times I & \xrightarrow{\mathrm{pr}_1} & D \\ \downarrow{\scriptstyle \varepsilon_{F/I}} & & \downarrow{\scriptstyle \varepsilon_D\times\mathrm{id}} & & \downarrow{\scriptstyle \varepsilon_D} \\ F^{\sim/I} & \xrightarrow{(n\pi,p)^{\sim/I}} & D^\sim \times I & \xrightarrow{\mathrm{pr}_1} & D^\sim \end{array} .$$

Since $(n\pi, p)^{\sim/I}$ is Cartesian, it is of the form $(n'\pi', p')$, where $p' : F^{\sim/I} \to I$ and $\pi' : F^{\sim/I} \to \varinjlim_{I^\circ} F_i^\sim$ are the canonical functors. Thus $\mathrm{pr}_{1^\circ}(n\pi, p)^{\sim/I} = n'$. Since the functors $\epsilon_{F/I}$ and ϵ_D induce equivalences on the topoi of sheaves (7.4.11), the functors n and n' induce isomorphic functors after passage to associated sheaves (III 1.2), and hence
305 n' is a morphism of sites. By what precedes, the functor $(n\pi, p)^{\sim/I}$ induces morphisms of topoi on the fibers. It is therefore a morphism from the total site $F^{\sim/I}$ to the total site $D^\sim \times I$ (7.4.13). Since the functors $\epsilon_{F/I}$ and $\epsilon_D \times \mathrm{id}$ induce equivalences on the total

topoi (7.4.11), the functor $(n\pi, p)$ is a morphism of fibered sites (7.4.13), which proves the assertion.

THEOREM 8.2.3. *Let $p : F \to I$ be a fibered $\mathscr{U}$-site over an essentially small cofiltered category* (I 8.1).

1) *Let $\underrightarrow{F}$ denote the site whose underlying category is the inductive limit of the fibered category $F \to I$ (6.3) and whose topology is the coarsest topology making the canonical functor $\pi : F \to \underrightarrow{F}$ continuous. Then $\underrightarrow{F}$ is a $\mathscr{U}$-site and*

$$(\pi, p) : F \longrightarrow \underrightarrow{F} \times I$$

is a morphism from the fibered site $\underrightarrow{F} \times I$ to the fibered site F.

2) *Let $\mu : \underrightarrow{F}^{\sim} \times I \to F^{\sim/I}$ be the morphism of fibered $\mathscr{U}$-topoi deduced from (π, p) by passing to the associated fibered $\mathscr{U}$-topoi. The pair $(\underrightarrow{F}^{\sim}, \mu)$ is a projective limit of $F^{\sim/I}$.*

REMARK 8.2.4. 1) In view of the notation introduced in 8.1.3, one may set

$$\varprojlim_{I}\text{top}\ F^{\sim} = \underrightarrow{F}^{\sim}.$$

2) It follows from 7.4.4 that the topology on $\underrightarrow{F}$ is also the coarsest topology making the family of functors $\pi_j : F_j \to F \xrightarrow{\pi} \underrightarrow{F}$, $j \in \text{ob}\, I$, continuous.

DEFINITION 8.2.5. *The site $\underrightarrow{F}$ introduced in 8.2.3 is called the projective-limit site of the fibered site F.*

As an exercise, the reader may, as in 8.2.1, define the notion of a projective limit of a fibered site and verify that the site $\underrightarrow{F}$ is indeed a projective limit of the site F. The second assertion of the theorem may therefore be paraphrased as follows: the topos of sheaves on the projective-limit site is equivalent to the projective limit of the associated fibered topos.

8.2.6. *Proof of the theorem: Reduction to the case of a small index category.* We shall give only indications. Let I' be a full subcategory of I such that I'° is cofinal in I°. Denote by $F' \to I'$ the fibered category deduced from F by the base change $I' \to I$, and by $\pi' : F' \to \underrightarrow{F'}$ the canonical functor. One first observes that the categories $\underrightarrow{F}$ and $\underrightarrow{F'}$ are equivalent and that the topology on $\underrightarrow{F'}$ deduced from the topology of $\underrightarrow{F}$ by this equivalence is the coarsest topology making the functor $\pi' : F' \to \underrightarrow{F'}$ continuous. Assertion 1) of the theorem follows when it has been proved in the case where I is small. Assertion 2) in the general case then follows from assertion 2) in the case where I is small by Lemma 8.2.1. 306

8.2.7. *End of the proof of the theorem.* The first assertion follows from Lemma 8.2.2, applied with $D = \underrightarrow{F}$ and $n = \text{id}$. Let D be a $\mathscr{U}$-topos. There is a diagram commutative up to isomorphism:

$$\begin{array}{ccccc}
\mathbf{Homtop}(D,(\underrightarrow{F})^{\sim}) & \xrightarrow{(1)} & \mathbf{Morsite}(D,\underrightarrow{F}) & \xrightarrow{(2)} & \mathbf{Hom}(\underrightarrow{F},D)^{\circ} \\
\downarrow{\scriptstyle (3)} & & \downarrow{\scriptstyle (3)} & & \\
\mathbf{Homtop}_{\mathrm{Cart}/I}(D\times I,(\underrightarrow{F})^{\sim}\times I) & \xrightarrow{(1)} & \mathbf{Morsite}_{\mathrm{Cart}/I}(D\times I,(\underrightarrow{F})\times I) & & \downarrow{\scriptstyle (5)} \\
\downarrow{\scriptstyle (4)} & & \downarrow & & \\
\mathbf{Homtop}_{\mathrm{Cart}/I}(D\times I,F^{\sim/I}) & \xrightarrow{(1)} & \mathbf{Morsite}_{\mathrm{Cart}/I}(D\times I,F) & \xrightarrow{(2)} & \mathbf{Hom}_{\mathrm{Cart}/I}(F,D\times I)^{\circ},
\end{array}$$

where the functors of type (1) are equivalences (IV 4.9.4 and 7.2.7), the functors of type (2) are fully faithful (IV 4.9.1 and 7.2.2), the functors of type (3) are base-change functors, the functors of type (4) are composition functors (with μ, resp. (π, p)), and functor (5) is the equivalence deduced from the universal property of the inductive limit (6.3). To prove that the functor $\mathbf{Homtop}(D, (\underrightarrow{F})^{\sim}) \xrightarrow{(4)\circ(3)} \mathbf{Homtop}_{/I}(D\times I, F^{\sim/I})$ is an equivalence, it is enough to show that the functor

$$\mathbf{Morsite}(D, \underrightarrow{F}) \xrightarrow{(4)\circ(3)} \mathbf{Morsite}_{/I}(D\times I, F)$$

307 is an equivalence of categories. Equivalently, it is enough to show that the two fully faithful functors

$$\mathbf{Morsite}(D, \underrightarrow{F}) \xrightarrow{(5)\circ(2)} \mathbf{Homcart}_I(F, D\times I)^{\circ}$$

$$\mathbf{Morsite}_{/I}(D\times I, F) \xrightarrow{(2)} \mathbf{Homcart}_I(F, D\times I)^{\circ}$$

have the same essential images, which follows immediately from Lemma 8.2.2.

8.2.8. Let $p : F \to I$ be a fibered $\mathscr{U}$-topos over a category I, and let $(\varprojlim_I \text{top}\ F_i, \mu)$ be a projective limit of F. Thus there is a Cartesian morphism of fibered $\mathscr{U}$-topoi over I:

$$\mu : \varprojlim_I \text{top}\ F_i \times I \longrightarrow F,$$

whence, on passing to the categories fibered over I° by the direct image functors (7.1.3), a Cartesian functor

$$\mu_* : \varprojlim_I \text{top}\ F_i \times I^{\circ} \longrightarrow F'$$

which is the direct image functor for μ (7.1.7). By the definition of the projective limit of a fibered category (6.10), there is therefore a canonical functor

$$\Theta : \varprojlim_I \text{top}\ F_i \longrightarrow \varprojlim_{I, f_*} F_i = \mathbf{Hom}_{\text{cart}/I^{\circ}}(I^{\circ}, F'). \tag{8.2.8.1}$$

Theorem 8.2.9. *Use the notation of 8.2.8, and suppose that I is an essentially small cofiltered category. Then the functor Θ is an equivalence of categories. Suppose that I is small. Set $\varprojlim_I \text{top}\ F_i = (\underrightarrow{F})^{\sim}$ (which is permitted by 8.2.3 and 8.2.4), and interpret the category $\mathbf{Hom}_{I^{\circ}}(I^{\circ}, F')$ as the category of sheaves on the total site F (7.4.7). Then the composite functor*

$$(\underrightarrow{F})^{\sim} = \varprojlim_I \text{top}\ F_i \xrightarrow{\Theta} \mathbf{Hom}_{\text{cart}/I^{\circ}}(I^{\circ}, F') \hookrightarrow \mathbf{Hom}_{I^{\circ}}(I^{\circ}, F') \cong F^{\sim},$$

is precisely the direct image functor for the morphism of sites defined by the functor

$$\pi : F \longrightarrow \underrightarrow{F}.$$

308 8.2.10. Some comments before the proof of Theorem 8.2.9. Besides solving the problem of the existence of projective limits when the index category is cofiltered, a solution which is curiously nontrivial in this theory, Theorems 8.2.3 and 8.2.9 provide two further pieces of information essential for working with projective-limit topoi in practice. First, Theorem 8.2.3 interprets the projective-limit topos as the topos of sheaves on a site whose underlying category is the inductive-limit category of the categories F_i

under the system of inverse image functors. In applications, one will be able to interpret this category and its topology concretely. Second, Theorem 8.2.9 asserts that a sheaf on the projective limit is determined by the system of its direct images in the topoi F_i (which gives a Cartesian section over I° of the category $(F^{\sim/I})'$). The theorem further asserts conversely that, when one is given for every object i of I a sheaf X_i on F_i and, for every morphism $f : i \to j$ of I, an isomorphism $X_j \simeq f_* X_i$, these isomorphisms being subject to compatibility conditions, there exists essentially a unique sheaf on the projective limit whose direct images give the system of the X_i.

8.2.11. Assertion 2) of 8.2.3 and the first assertion of 8.2.9 may be summarized by the striking formula

$$\varprojlim_{I, f_*} F_i^\sim \approx \varprojlim_{I, f.} \mathrm{top}\ F_i^\sim \approx (\varinjlim_{I^\circ, f_*} F_i)^\sim. \tag{8.2.11.1}$$

8.2.12. *Proof of Theorem* 8.2.9. To prove the first assertion, reduce to the case where the category I is small by using 8.2.1 and an analogous lemma on projective limits of fibered categories. We leave the details to the reader. Suppose henceforth that I is a small category. We know that the functor

$$\pi : F \longrightarrow \varinjlim_{I^\circ} F_i$$

is a morphism of sites (8.2.2). The direct image functor

$$\pi_* : (\varinjlim_{I^\circ} F_i)^\sim \longrightarrow F^\sim$$

that it defines on the topoi is then fully faithful, and its essential image is the set of 309
sheaves on F that carry the Cartesian morphisms of F to isomorphisms (6.6). These sheaves correspond, under the equivalence $F^\sim \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$, to the Cartesian sections of F' over I° (7.4.7). Moreover, for every $j \in \mathrm{ob}(I)$, denote by $\alpha_{j!} : F_j \to F$ the inclusion functor, which gives rise to the three functors $(\alpha_{j!}, \alpha_j^*, \alpha_{j*}) : F_j \rightleftarrows F$ (7.4.3). It follows from the definition of the equivalence $F^\sim \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$ (7.4.7) that a sheaf X on $\varinjlim_{I^\circ} F_i$ is carried by the functor $(\varinjlim_{I^\circ} F_i)^\sim \xrightarrow{\pi_*} F^\sim \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$ to the Cartesian section $j \mapsto \alpha_j^* \pi_*(X)$. But the description in 8.2.3 of the functor $\mu : (\varinjlim_{I^\circ} F_i)^\sim \times I \to F$ shows that, for every object j of I, the direct image functor $\mu_{j*} : (\varinjlim_{I^\circ} F_i)^\sim \to F_j$ for the morphism $\mu_j : (\varinjlim_{I^\circ} F_i)^\sim \to F_j$ obtained from μ by passing to the fiber at j is precisely the functor $\alpha_j^* \pi_*$. Consequently, the functor $(\varinjlim_{I^\circ} F_i)^\sim \xrightarrow{\pi_*} F^\sim \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$ is precisely the functor $(\varinjlim_{I^\circ} F_i)^\sim \xrightarrow{\Theta} \mathbf{Hom}_{\mathrm{cart}/I^\circ}(I^\circ, F') \hookrightarrow \mathbf{Hom}_{I^\circ}(I^\circ, F')$, which proves the theorem.

8.3. Topology of the projective-limit site: the case of coherent topoi.

8.3.1. In this section, $p : F \to I$ is a fibered $\mathscr{U}$-site over an essentially small cofiltered category, whose inverse-image functors $f^* : F_i \to F_j$ commute with fiber products. In addition, for every object X of a fiber category F_i, we are given a set $\mathrm{Cov}(X)$ of covering families for the topology of F_i which have properties (PTO) and (PT1) of II 1.3 and generate the topology of F_i. Finally, suppose that for every $f : i \to j$, every $Y \in \mathrm{ob}\, F_j$, and every family $(Y_\alpha \to Y)_{\alpha \in A}$ in $\mathrm{Cov}(Y)$, the family $(f^*(Y_\alpha) \to f^*(Y))_{\alpha \in A}$ belongs to $\mathrm{Cov}(f^*(Y))$. These conditions on $\mathrm{Cov}(X)$ are satisfied, for example, if one takes $\mathrm{Cov}(X) = \{\text{all covering families of } X \text{ in } F_i\}$. In this section we shall study the 310

projective-limit site of F under suitable finiteness hypotheses (8.3.13).

8.3.2. Let $\pi : F \to \underrightarrow{F}$ be the canonical functor, and let Y be an object of $\underrightarrow{F}$. Denote by $\mathrm{Cov}(Y)$ the set of families $(Y_\alpha \xrightarrow{n_\alpha} Y)_{\alpha\in A}$ of the following form:

There exist $i \in I$, an object X of F_i, a family $(X_\alpha \xrightarrow{n'_\alpha} X)_{\alpha\in A} \in \mathrm{Cov}(X)$, an isomorphism $\phi : \pi(X) \simeq Y$, and a family of isomorphisms $\phi_\alpha : \pi(X_\alpha) \simeq Y_\alpha$, $\alpha \in A$, such that for every $\alpha \in A$ the diagram

$$\begin{array}{ccc} \pi(X_\alpha) & \overset{\phi_\alpha}{\sim} & Y_\alpha \\ \downarrow{\scriptstyle \pi(n'_\alpha)} & & \downarrow{\scriptstyle n_\alpha} \\ \pi(X) & \overset{\phi}{\sim} & Y \end{array}$$

is commutative.

Proposition 8.3.3. 1) *The families belonging to* $\mathrm{Cov}(Y)$, *for* $Y \in \mathrm{ob}\,\underrightarrow{F}$, *generate the topology of the projective-limit site (8.2.5).*

2) *The family* $Y \mapsto \mathrm{Cov}(Y)$, $Y \in \mathrm{ob}\,\underrightarrow{F}$, *has properties* (PTO) *and* (PT1) *of* II 1.3.

We shall need the following lemma.

Lemma 8.3.4. *Let i be an object of I, and let $n : X' \to X$ be a base-changeable morphism of F_i such that $f^*(n)$ is base-changeable for every $f : j \to i$. Then $\pi(n)$ is base-changeable. The functors*

$$\pi_j : F_j \hookrightarrow F \xrightarrow{\pi} \underrightarrow{F}, \quad j \in \mathrm{ob}\, I,$$

commute with fiber products.

Recall that every object Y of $\underrightarrow{F}$ is equal to an object $\pi(Z)$, $Z \in \mathrm{ob}\, F$, and that every morphism $m : Y \to \pi(X)$ is of the form

$$\pi(Z) \xrightarrow{\pi(s)^{-1}} \pi(Z') \xrightarrow{\pi(m')} \pi(X),$$

where s is a cartesian morphism of F. Consequently, to show that $\pi(n)$ is base-changeable,
311 it is enough to show that the fiber product of every diagram

$$\begin{array}{ccc} & & \pi(X') \\ & & \downarrow{\scriptstyle \pi(n)} \\ \pi(Z') & \xrightarrow{\pi(m')} & \pi(X) \end{array}$$

is representable. The morphism $m' : Z' \to X$ factors as $Z' \xrightarrow{m''} X'' \xrightarrow{s'} X$, where $p(m'')$ is the identity of $p(Z') = j$ and s' is cartesian over $p(s') = f$. Put $X''' = f^*(X')$

and $n' = f^*(n)$. We then have a commutative diagram

$$\begin{array}{ccc} X''' & \xrightarrow{s''} & X' \\ \downarrow{\scriptstyle n'} & & \downarrow{\scriptstyle n} \\ X'' & \xrightarrow{s'} & X, \end{array}$$

and hence a commutative diagram in $\underrightarrow{F}$:

$$\begin{array}{ccccc} & & \pi(X''') & \xrightarrow{\sim} & \pi(X') \\ & & \downarrow{\scriptstyle \pi(n')} & & \downarrow{\scriptstyle \pi(n)} \\ \pi(Z') & \xrightarrow{\pi(m'')} & \pi(X'') & \xrightarrow{\sim} & \pi(X) \end{array}$$

where n' is a base-changeable morphism of F_j. Thus, to show that $\pi(n)$ is base-changeable, it is enough to show that the product $\pi(Z') \times_{\pi(X'')} \pi(X''')$ is representable, and therefore it is enough to show that $\pi_j : F_j \hookrightarrow F \to \underrightarrow{F}$ commutes with fiber products.

Let

$$\begin{array}{ccc} Y' & \longrightarrow & X' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & X \end{array}$$

be a cartesian diagram in F_j, and let W be an object of F over $k \in \mathrm{ob}\, I$. We have (6.5):

$$\mathbf{Hom}(\pi(W), \pi(Y')) \simeq \varinjlim_{\substack{f \nearrow j \\ l \\ g \searrow k}} \mathbf{Hom}_{F_\ell}(g^*(W), f^*(Y'))$$

Since $f^* : F_j \to F_\ell$ commutes with fiber products, we have 312

$$\mathrm{Hom}(g^*(W), f^*(Y')) \simeq \mathrm{Hom}(g^*(W), f^*(Y)) \times_{\mathrm{Hom}(g^*(W), f^*(X))} \mathrm{Hom}(g^*(W), f^*(X')).$$

Using the commutation of filtered limits with fiber products (I 2) and formula (6.5), we obtain an isomorphism

$$\mathrm{Hom}(\pi(W), \pi(Y')) \simeq \mathrm{Hom}(\pi(W), \pi(Y)) \times_{\mathrm{Hom}(\pi(W), \pi(X))} \mathrm{Hom}(\pi(W), \pi(X')),$$

which shows that the diagram

$$\begin{array}{ccc} \pi(Y') & \longrightarrow & \pi(X') \\ \downarrow & & \downarrow \\ \pi(Y) & \longrightarrow & \pi(X) \end{array}$$

is cartesian in $\underrightarrow{F}$.

8.3.5. *Proof of* 8.3.3. We first prove the second assertion. It follows from 8.3.4 that the families in $\mathrm{Cov}(Y)$, $Y \in \underrightarrow{F}$, consist of base-changeable morphisms, whence property (PTO). To show that these families have property (PT1) (stability under base change), a series of reductions like those in the proof of 8.3.3 reduces us to proving the following assertion:

For every $i \in \mathrm{ob}\, I$, every $Y \in \mathrm{ob}\, F_i$, every $(Y_\alpha \to Y)_{\alpha\in A} \in \mathrm{Cov}(Y)$, and every morphism $u : Z \to Y$ of F_i, the family

$$(\pi(Z) \times_{\pi(Y)} \pi(Y_\alpha) \longrightarrow \pi(Z))_{\alpha\in A}$$

belongs to $\mathrm{Cov}(\pi(Z))$.

This assertion follows from the fact that the $\pi_i : F_i \hookrightarrow F \xrightarrow{\pi} \underrightarrow{F}$ commute with fiber products (8.3.4) and that the families $X \mapsto \mathrm{Cov}(X)$ of F_i are stable under base change. Let us prove the first assertion. It is clear that the families belonging to the $\mathrm{Cov}(Y)$, $Y \in \underrightarrow{F}$, are covering for the coarsest topology T making the functor π continuous (III 1). Thus the topology T' generated by these families is coarser than T. To show that it
313 is finer than T (and hence equal to T), it is enough to show that every sheaf M for T' is such that $M \circ \pi$ is a sheaf on F, or equivalently (8.2.4) that $M \circ \pi_i$ is a sheaf on F_i for every $i \in \mathrm{ob}\, I$. Since the functors $\pi_i : F_i \to \underrightarrow{F}$ commute with fiber products (8.3.4), M is a sheaf for T' if and only if (II 2.3 and I 2.12), for every $i \in \mathrm{ob}\, I$, every $Y \in \mathrm{ob}\, F_i$, and every $(Y_\alpha \to Y)_{\alpha\in A} \in \mathrm{Cov}(Y)$, the sequence of sets

$$M(\pi_i(Y)) \longrightarrow \prod_{\alpha\in A} M(\pi_i(Y_\alpha)) \rightrightarrows \prod_{(\alpha,\beta)\in A\times A} M\left(\pi_i(Y_\alpha \times_Y Y_\beta)\right)$$

is exact, i.e. (*loc. cit.*) if and only if $M \circ \pi_i$ is a sheaf on F_i for every $i \in \mathrm{ob}\, I$.

Proposition 8.3.6. *We use the notation and hypotheses of 8.3.1 and 8.3.2. If, for every $i \in \mathrm{ob}\, I$, $X \mapsto \mathrm{Cov}(X)$ is a pretopology on F_i (II 1.3), and if for every $X \in \mathrm{ob}\, F_i$ the covering families in $\mathrm{Cov}(X)$ are finite, then for every $Y \in \underrightarrow{F}$ the covering families in $\mathrm{Cov}(Y)$ are finite and $Y \mapsto \mathrm{Cov}(Y)$ is a pretopology on $\underrightarrow{F}$.*

8.3.7. The only thing to prove is that the families $Y \mapsto \mathrm{Cov}(Y)$ have property (PT2) of (II 1.3) (stability under composition). Let $i \in \mathrm{ob}\, I$, let Y be an object of F_i, and let $(Y_\alpha \to Y)_{\alpha\in A} \in \mathrm{Cov}(Y)$. In addition, let i_α, $\alpha \in A$, be a family of objects of I; for every α, let Z_α be an object of F_i; and let $(Z_{\alpha\beta} \xrightarrow{n_{\alpha\beta}} Z_\alpha)_{\alpha\beta\in B_\alpha}$ be a family in $\mathrm{Cov}(Z_\alpha)$. Finally, for every $\alpha \in A$, let an isomorphism $\phi : \pi(Z_\alpha) \xrightarrow{\sim} \pi(Y_\alpha)$ be given. Returning to the definition of the families in $\mathrm{Cov}(X)$ for $X \in \mathrm{ob}\, \underrightarrow{F}$ (8.3.2), we see that we must prove that the family

$$(\pi(Z_{\alpha\beta}) \xrightarrow{\pi(n_\alpha)\circ\phi_\alpha\circ\pi(n_{\alpha\beta})} \pi(Y))_{\alpha\beta\in\coprod_A B_\alpha}$$

belongs to $\mathrm{Cov}(\pi(Y))$. We shall do this after five reductions.

8.3.8. *Reduction to the case in which, for every $\alpha \in A$, we have $\phi_\alpha = \pi(m_\alpha)$, $m_\alpha : Z_\alpha \to Y_\alpha$. For every $\alpha \in A$, we have $\phi_\alpha = \pi(m_\alpha)\pi(s_\alpha)^{-1}$ (6.5), where s_α is a Cartesian*

morphism over $f_\alpha : i'_\alpha \to i_\alpha$. *Thus there are commutative diagrams*

$$\begin{array}{ccc} Z_{\alpha\beta} & \xleftarrow{s_{\alpha\beta}} & f^*(Z_{\alpha\beta}) \\ \downarrow{\scriptstyle n_{\alpha\beta}} & & \downarrow{\scriptstyle f_\alpha^*(n_{\alpha\beta})} \\ Z_\alpha & \xleftarrow{s_\alpha} & f_\alpha^*(Z_\alpha), \end{array} \qquad \alpha\beta \in \coprod_A B_\alpha,$$

which give commutative diagrams 314

$$\begin{array}{ccccc} \pi(Z_{\alpha\beta}) & & \xrightarrow{\sim} & & \pi(f_\alpha^*(Z_{\alpha\beta})) \\ \downarrow{\scriptstyle \pi(n_{\alpha\beta})} & & & & \downarrow{\scriptstyle \pi(f_\alpha^*(m_{\alpha\beta}))} \\ \pi(Z_\alpha) & \xrightarrow[\sim]{\pi(S_\alpha)^{-1}} & \pi(f_\alpha^*(Z_\alpha)) & \xrightarrow{\pi(m_\alpha)} & \pi(Y_\alpha), \end{array} \qquad \alpha\beta \in \coprod_A B_\alpha.$$

Since the families $(f_\alpha^*(Z_{\alpha\beta}) \xrightarrow{f^*(n_\alpha)} f^*(Z_\alpha))$ belong to $\mathrm{Cov}(f_\alpha^*(Z_\alpha))$, we may reduce to the case in which $\phi_\alpha = \pi(m_\alpha)$ for every α.

8.3.9. *Reduction to the case in which, for every* $\alpha \in A$, $i_\alpha = j$ *and* $p(m_\alpha) = k : j \to i$. Put $g_\alpha = p(m_\alpha)$. Since I is cofiltered and A is finite, there exist an object j of I and morphisms $h_\alpha : j \to i_\alpha$ such that, for every pair (α, α'), we have $g_\alpha h_\alpha = g_{\alpha'} h_{\alpha'} = k$. Using the inverse-image functors h_α^* as in 8.3.8, we reduce to the stated case.

8.3.10. *Reduction to the case in which* $i = j$ and $k : j \to i$ *is the identity.* For every α, there is a commutative diagram

$$\begin{array}{ccccc} Z_\alpha & \xrightarrow{m'_\alpha} & k^*(Y_\alpha) & \xrightarrow{t_\alpha} & Y_\alpha \\ & & \downarrow{\scriptstyle k^*(n_\alpha)} & & \downarrow{\scriptstyle n_\alpha} \\ & & k^*(Y) & \xrightarrow{t} & Y, \end{array}$$

where the morphisms t and t_α, $\alpha \in A$, are Cartesian, m'_α lies over the identity of j, and $t_\alpha \circ m'_\alpha = m_\alpha$. Replacing the family $(Y_\alpha \to Y)\alpha \in A$ by the family $(k^*(Y_\alpha) \to k^*(Y))_{\alpha\in A} \in$ 315
$\mathrm{Cov}(k^*(Y_\alpha))$, and the morphisms m_α by the morphisms m'_α, we reduce to the stated case.

LEMMA 8.3.11. *Let* j *be an object of* I *and let* $m : X' \to X$ *be a morphism of* F_j *such that* $\pi(m)$ *is an isomorphism. There exists a morphism* $\ell : j' \to j$ *such that* $\ell^*(m)$ *is an isomorphism.*

Let us show that there exist a morphism $f : i \to j$ and a morphism $q : f^*(X) \to f^*(X')$ such that $f^*(m)q = \mathrm{id}_{f^*(X)}$ and such that $\pi(q)$ is an isomorphism. Indeed, there exists an isomorphism $n : \pi(X) \to \pi(X')$ such that $\pi(m) \circ n = \mathrm{id}_{\pi(X)}$. There consequently exist (6.5) two morphisms $X \xleftarrow{s_1} Y_1 \xrightarrow{q_1} X'$, where s_1 is Cartesian, such that $n = \pi(q_1)\pi(s_1)^{-1}$. Thus $\pi(m) \circ \pi(q_1) = \pi(s_1)$. The morphisms mq_1 and s_1 from Y_1 to X have the same image in $\underline{F}$. Consequently (cf. the description of the morphisms in the inductive limit (6.5)), there exists a Cartesian morphism $s_2 : Y_2 \to Y_1$ such that $mq_1 s_2 = s_1 s_2$. Put $q_1 s_2 = q_2$ and denote by s_3 the Cartesian morphism $s_1 s_2$. Thus

$mq_2 = s_3$. The morphism $q_2 : Y_2 \to X'$ factors in an essentially unique manner as $Y_2 \xrightarrow{q} Y_3 \xrightarrow{s_4} X'$, where s_4 is Cartesian and q lies over the identity of $p(Y_2)$. Thus there is a commutative diagram

(8.3.11.1)
$$\begin{array}{ccc} Y_3 & \xleftarrow{\quad q \quad} & Y_2 \\ {\scriptstyle s_4}\downarrow & & \downarrow{\scriptstyle s_3} \\ X' & \xrightarrow{\quad m \quad} & X.. \end{array}$$

Put $p(s_4) = p(s_3) = f : i \to j$. We have $f^*(X') \simeq Y_3$, $f^*(X') \simeq Y_2$, and $f^*(m) : Y_2 \to Y_3$ makes the following diagram commutative:

$$\begin{array}{ccc} Y_3 & \xrightarrow{\quad f^*(m) \quad} & Y_2 \\ {\scriptstyle s_4}\downarrow & & \downarrow{\scriptstyle s_3} \\ X' & \xrightarrow{\quad m \quad} & X. \end{array}$$

316 Thus $s_3 = s_3(f^*(m) \circ q)$; since s_3 is Cartesian and $f^*(m) \circ q$ lies over the identity of $p(Y_2)$, we have $f^*(m) \circ q = \mathrm{id}_{Y_2}$. Moreover, the commutativity of diagram (8.3.11.1) shows that $\pi(q)$ is an isomorphism.

Now apply the preceding result to the morphism $q : f^*(X) \to f^*(X')$: there exist a morphism $f' : j' \to i$ and a morphism $q' : f'^*(X') \to f'^*(X)$ such that $f'^*(q)q' = \mathrm{id}_{f'^*(f'^*(X'))}$. But on putting $\ell^* = ff'$, there is an isomorphism $f'^*f^* \simeq \ell^*$. Thus there is a morphism $f'^*(q) : \ell^*(X) \to \ell^*(X')$ and an isomorphism $f'^*f^* \simeq \ell^*$. Thus there is a morphism $f'^*(q) : \ell^*(X) \to \ell^*(X')$ and a morphism $q' : \ell^*(X') \to \ell^*(X)$ such that $\ell^*(m)f'^*(q) = \mathrm{id}_{\ell^*(X)}$ and $f'^*(q)q' = \mathrm{id}_{\ell^*(X')}$. Consequently $\ell^*(m)$ is an isomorphism.

8.3.12. *End of the proof.* Since I is cofiltered and A finite, it follows from 8.3.11 that there exists an inverse-image functor ℓ^* that takes all the morphisms m_α, $\alpha \in A$, to isomorphisms. We are thus reduced to the case in which the m_α are isomorphisms. But then the covering families $(Z_{\alpha\beta} \to Z)_{\alpha\beta \in B_\alpha}$ are deduced by the base change $m_\alpha : Z_\alpha \to Y_\alpha$ from covering families $(Y_{\alpha\beta} \to Y_\alpha)_{\alpha\beta \in B_\alpha} \in \mathrm{Cov}(Y_\alpha)$. We are therefore reduced to the case in which $Z_\alpha = Y_\alpha$ and $\Phi_\alpha = \mathrm{id}_{\pi(Y_\alpha)}$. But in this case the family $(Y_{\alpha\beta} \xrightarrow{n_\alpha \circ n_{\alpha\beta}} Y)_{\alpha\beta \in \coprod_A B_\alpha}$ belongs to $\mathrm{Cov}(Y)$ (property (PT2)). Hence its image under π belongs to $\mathrm{Cov}(\pi(Y))$ (8.3.2).

THEOREM 8.3.13. *Let $p : F \to I$ be a $\mathcal{U}$-topos fibered over an essentially small cofiltered category. If, for every object i of I, the fiber topos F_i is coherent (2.3), and if, for every morphism $f : i \to j$, the morphism of topoi $f\cdot = (f^*, f_*) : F_i \to F_j$ is coherent (3.1), then the topos $\varprojlim_I \mathrm{top}\, F_i$ is coherent and, for every object i of I, the canonical morphism of topoi*

$$\mu_i : \varprojlim_I \mathrm{top}\, F_i \longrightarrow F_i$$

is coherent.

Moreover, if F_{coh} denotes the full subcategory of F defined by the objects of F that are coherent in their fiber (1.13), then the category F_{coh} is fibered over I, and $\varinjlim F_{\mathrm{coh}}$ is canonically equivalent to the category of coherent objects of $\varprojlim_{I}\mathrm{top}\ F_i$.

For every $i \in \mathrm{ob}\, I$, denote by $(F_i)_{\mathrm{coh}}$ the $\mathscr{U}$-site (endowed with the induced topology) of coherent objects of F_i (1.13). Since F_i is coherent, F_i is the topos of sheaves on $(F_i)_{\mathrm{coh}}$. For every morphism $f : i \to j$, the morphisms $f\cdot = (f^*, f_*)$ are coherent, and consequently $f^*(F_j)_{\mathrm{coh}} \subset (F_i)_{\mathrm{coh}}$. Thus there is a fibered $\mathscr{U}$-site $p_{\mathrm{coh}} : F_{\mathrm{coh}} \to I$ whose fiber sites are the sites $(F_i)_{\mathrm{coh}}$ and whose associated fibered topos is equivalent to $p : F \to I$. Consequently (8.2.3), 317

$$\varprojlim_{I,f.}\mathrm{top}\ F_i \cong (\varinjlim F_{\mathrm{coh}})^{\sim} \tag{8.3.13.1}$$

Since the topoi F_i are coherent, finite products and fiber products are representable in $(F_i)_{\mathrm{coh}}$ (2.2). Moreover, the functors $f^* : (F_i)_{\mathrm{coh}} \to (F_j)_{\mathrm{coh}}$ are left exact. Finally, since the objects of $(F_i)_{\mathrm{coh}}$ are quasi-compact, the finite covering families form a pretopology. It then follows from (8.3.4) that finite products and fiber products are representable in $\varprojlim_{I^\circ,F}(F_i)_{\mathrm{coh}}$ (addendum to Lemma 8.3.4: show that the canonical functors $(F_i)_{\mathrm{coh}} \to \varinjlim F_{\mathrm{coh}}$ take the final object to the final object), and from (8.3.6) that the objects of $\varinjlim F_{\mathrm{coh}}$ are quasi-compact. Consequently (2.4.5), $\varprojlim_{I}\mathrm{top}\ (F_i)$ is coherent. Finally, the canonical morphisms $\mu_i : \varprojlim_{I}\mathrm{top}\ (F_i) \to F_i$ are deduced, by passage to the topoi of sheaves, from the morphisms of sites (8.2.3)

$$(F_i)_{\mathrm{coh}} \longrightarrow \varinjlim F_{\mathrm{coh}}$$

and hence (3.3) the morphisms μ_i are coherent.

Let us prove the last assertion. We have (8.3.13.1)

$$\varprojlim_{\mathscr{I}}\mathrm{top}\ (F_i) \simeq (\varinjlim F_{\mathrm{coh}})^{\sim}.$$

Let X be a coherent object of $(\varinjlim F_{\mathrm{coh}})^{\sim}$. Since X is quasi-compact, there exists a finite covering family $(Y_\alpha \to X)$, where $Y_\alpha \in \mathrm{ob}\, \varinjlim F_{\mathrm{coh}}$ (1.1). On taking an index $i \in \mathrm{ob}\, I$ sufficiently small, we may suppose that the Y_α are the images under $\mu_i : F_{i\mathrm{coh}} \to \varinjlim F_{\mathrm{coh}}$ of a family Y'_α; hence, by taking the direct sum of the Y'_α (1.15), we see that there exist an index $i \in \mathrm{ob}\, I$, an object $Y' \in F_{i\mathrm{coh}}$, and a surjective morphism $\mu_i(Y') \to X$. The equivalence relation $R = \mu_i(Y') \times_X \mu_i(Y')$ is then a coherent object because $(\varinjlim F_{\mathrm{coh}})^{\sim}$ is a coherent topos (2.2). Let us first show that R is an object of $\varinjlim F_{\mathrm{coh}}$. By definition, R is a subobject of $\mu_i(Y' \times Y')$, and since R is coherent, we may, after changing the index i, find an arrow $m : Z \to (Y' \times Y')$ of $F_{i\mathrm{coh}}$ such that $\mathrm{Im}(\mu_i(m)) = R$. It follows that, since F_i is coherent, $\mathrm{Im}(m)$ is coherent in F_i (1.17.1), and hence belongs to $F_{i\mathrm{coh}}$. There consequently exist an index i and a diagram $R \underset{p_2}{\overset{p_1}{\rightrightarrows}} Y'$ of $F_{i\mathrm{coh}}$ such that $\mu_i(R) \underset{\mu_i(p_2)}{\overset{\mu_i(p_1)}{\rightrightarrows}} \mu_i(Y')$ is an equivalence relation and such that $X = \mu_i(Y')/\mu_i(R)$. Put $X_i = \mathrm{coker}(p_1, p_2)$ and $R' = Y' \times_{X'_i} Y'$. There is a canonical morphism in F_i, $u : R \to R'$. Moreover, $\mu_i^*(u)$ is an isomorphism. Thus, after changing the index i, we may suppose that $R = R'$, i.e. that R is an equivalence relation. But then X_i belongs to $F_{i\mathrm{coh}}$ (1.17.1), and consequently $X = \mu_i(X_i)$ belongs to $\varinjlim F_{\mathrm{coh}}$. 318

COROLLARY 8.3.14. *Let I be an essentially small cofiltered category, and let $p : F \to I$ and $q : G \to I$ be two $\mathscr{U}$-topoi fibered over I such that, for every object i of I, the topoi F_i and G_i are coherent, and such that, for every morphism $f : i \to j$, the morphisms $f. : F_i \to F_j$ and $G_i \to G_j$ are coherent. In addition, let $m : F \to G$ be a Cartesian morphism of fibered $\mathscr{U}$-topoi such that, for every $i \in \mathrm{ob}(I)$, $m_i : F_i \to G_i$ is coherent. Then the morphism $\underline{m}$ deduced from m by passage to the projective limit is coherent.*

For every object i of I, the morphism m_i induces a functor $m^*_{i,\mathrm{coh}} : G_{i,\mathrm{coh}} \to F_{i,\mathrm{coh}}$, whence a Cartesian functor $m^*_{\mathrm{coh}} = G_{\mathrm{coh}} \to F_{\mathrm{coh}}$ that is a morphism from the fibered site F_{coh} to the fibered site G_{coh} (7.4.13). The functor $\varinjlim_{I^\circ} m^*_{\mathrm{coh}} : \varinjlim_{I^\circ} G_{\mathrm{coh}} \to \varinjlim_{I^\circ} F_{\mathrm{coh}}$ is a morphism of sites that, on passage to the corresponding topoi, gives the morphism $\underline{m}$. The assertion then follows from 8.3.13 and 3.3.

EXERCISE 8.3.15. With the notation of 8.3.13, show that if the F_i, $i \in I$, are algebraic
319 (2.3) and if the morphisms $f. : F_i \to F_j$, $f \in \mathrm{Fl}(I)$, are coherent, then $\varprojlim_I \mathrm{top}\ F_i$ is algebraic and $(\varprojlim_I \mathrm{top}\ F_i)_{\mathrm{coh}} \xrightarrow{\sim} \underline{F}_{\mathrm{coh}}$.

8.4. Example: local topoi.

8.4.1. Let $X = (X_i)_{i\in I}$ be a topos fibered over a category I. One deduces from it a topos fibered over the category $\mathrm{Pro}(I)$ (I 8.10) by the formula

$$X_\alpha = \varprojlim_{I_\alpha} \mathrm{top}\ X_{i\alpha} \tag{8.4.1.1}$$

for every pro-object $\alpha = (i_\alpha)$ of I (to see this, one may generalize Exercise I 8.2.8 to pseudofunctors).

8.4.2. Let E be a topos, and let $p : \mathrm{Fl}(E) \to E$ be the fibered topos considered in 7.3.1. Extending it as in 8.4.1, one obtains a fibered topos $\overline{\mathrm{Fl}(E)} \to \mathrm{Pro}(E)$. The category $\mathrm{Point}(E)$ maps by a fully faithful functor into $\mathrm{Pro}(E)$ (IV 6.8.5), whence, by base change (7.1.9), a fibered topos $\mathrm{Loc}(E) \to \mathrm{Point}(E)$. The fiber $\mathrm{Loc}_p(E)$ at a point p of E is called the *localization of E at the point p*. By the preceding, this topos depends covariantly on the point p. By definition,

$$\mathrm{Loc}_p(E) = \varprojlim_{X\in \mathrm{Vois}(p)} \mathrm{top}\ E_{/X}, \tag{8.4.2.1}$$

where $\mathrm{Vois}(p)$ is the category of neighborhoods of p (IV 6.8).

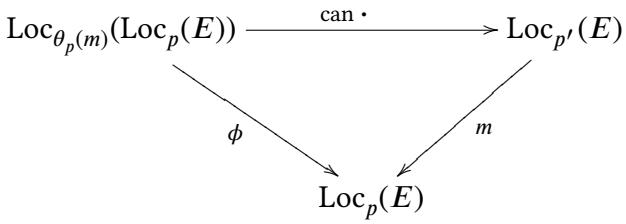
8.4.3. Let $m : p' \to p$ be a morphism of $\mathrm{Point}(E)$ (IV 6). For every $X \in \mathrm{Vois}(p)$ one thereby obtains a point p'_X of $E_{/X}$ and hence, by the universal property of $\varprojlim \mathrm{top}$ 8.1, a point $\theta_p(m)$ of $\mathrm{Loc}_p(E)$. This defines a functor

$$\theta_p : \mathrm{Point}(E)_{/p} \longrightarrow \mathrm{Point}(\mathrm{Loc}_p(E)), \tag{8.4.3.1}$$

which is immediately seen to be an equivalence of categories. Thus $\mathrm{Point}(\mathrm{Loc}_p(E))$ is canonically equivalent to the category of generalizations of p (IV 7.1.8). There is moreover a canonical equivalence of topoi

$$\mathrm{can}\cdot : \mathrm{Loc}_{\theta_p(m)}(\mathrm{Loc}_p(E)) \longrightarrow \mathrm{Loc}_{p'}(E),$$

which fits into a diagram commutative up to isomorphism



where ϕ is the canonical morphism of (8.4.2.1). 320

8.4.4. Localized topoi seem to be of interest only for topoi arising from algebraic geometry, or at least for topoi having suitable finiteness properties. Thus, when E is the topos of sheaves on a separated topological space, one sees immediately (using, for example, 8.2.9) that the localized topoi are point topoi. Moreover, in this case the category Point(E) is discrete and the situation described in 8.4.2 is trivial. On the other hand, let $E = \mathrm{Top}(X)$ be the topos of sheaves for the Zariski topology on a scheme X, let x be a point of X, let O_x be the local ring of X at x, and let $Y = \mathrm{spec}(O_x)$. One finds that

$$\mathrm{Loc}_x(E) = \mathrm{Top}(Y).$$

For other examples arising from the étale topology, we refer to Exposé VIII of this seminar.

8.4.5. When E is locally coherent (2.3), the localized topoi $\mathrm{Loc}_p(E)$ are coherent (in (8.4.2.1) one may restrict to the $X \in \mathrm{Vois}(p)$ which are algebraic and coherent, and then apply 8.3.13). For every morphism of points $m : p' \to p$, the morphism of topoi $m\cdot : \mathrm{Loc}_{p'}(E) \to \mathrm{Loc}_p(E)$ is coherent (8.3.14).

8.4.6. A *local topos* is a topos X such that the functor $\Gamma(X, -) =$ "sections over X" (IV 4.3) is a fiber functor (IV 6). The point corresponding to this fiber functor is called the *center* of the local topos. When E is locally coherent, its localized topoi are local topoi (1.2.3, 8.5.2, and 8.5.7). The center of $\mathrm{Loc}_p(E)$ is canonically isomorphic to $\theta_p(\mathrm{id}_p)$ ((8.4.3.1)). The image in E of the center of $\mathrm{Loc}_p(E)$ is isomorphic to p. The localized topos $\mathrm{Loc}_p(E)$, equipped with the canonical morphism $\mathrm{Loc}_p(E) \to E$, is the solution of 321
the universal (2-universal!) problem of mapping local topoi into E in such a way that the center maps "to" p.

8.4.7. A number of questions concerning local topoi have not been addressed by the editors. Thus, if X is a local topos, the final object e of X has a maximal open subtopos $U \neq e$. The complement Y of U is a local topos having no nontrivial open subtopos. Is such a topos a point topos? Let U be a topos and let $X = (U, \Phi : U \to (\mathrm{Ens}))$ be a topos obtained by gluing. What conditions must the gluing functor Φ satisfy for X to be a local topos?

8.5. Structure of the sheaves of a filtered projective limit of topoi.

8.5.1. In this section, $F = (F_i)_{i \in I}$ is a $\mathscr{U}$-topos fibered over a small cofiltered category I. Choose a bicleavage (7.1.4) of F, i.e. for every $f \in \mathrm{Fl}(I) : i \to j$, choose morphisms of topoi $f\cdot : F_i \to F_j$ such that $f^* : F_j \to F_i$ is the inverse-image functor for the fibered structure. There are then canonical isomorphisms $c_{f,g}$ satisfying a cocycle condition (7.1.3). Denote by $\varprojlim F$ the projective limit of the fibered topos F (8.2.3), and for every $i \in \mathrm{ob}\, I$, denote by

$$\mu_i : \varprojlim F \longrightarrow F_i, \tag{8.5.1.1}$$

the canonical morphism (8.1.3). Denote by $\mathrm{Top}(F)$ the total topos of F (7.4.3.3). By 8.2.9, the canonical functor $F \to \underset{\rightarrow}{F}$ defines a morphism of topoi

$$Q : \underset{\leftarrow}{F} \longrightarrow \mathrm{Top}(F). \tag{8.5.1.2}$$

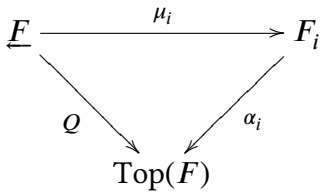
The topos $\underset{\leftarrow}{F}$ is identified with $\mathbf{Hom}_{\mathrm{Cart}/I^\circ}(I^\circ, F')$ (8.2.9), and the topos $\mathrm{Top}(F)$ is identified with $\mathbf{Hom}_{I^\circ}(I^\circ, F')$ (7.4.7). Under these identifications, Q is the embedding morphism of $\mathbf{Hom}_{\mathrm{cart}/I^\circ}(I^\circ, F')$ into $\mathbf{Hom}_{I^\circ}(I^\circ, F')$ (8.2.9), and for every object M of $\underset{\leftarrow}{F}$ we have

$$Q_*(M) = (i \longmapsto \mu_{i*}(M)). \tag{8.5.1.3}$$

By 7.4.3.4, for every $i \in \mathrm{ob}\, I$ there is a morphism of topoi

$$\alpha_i : F_i \longrightarrow \mathrm{Top}(F). \tag{8.5.1.4}$$

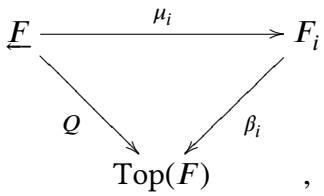
322 The diagram

(8.5.1.5)


is not commutative in general (even up to isomorphism). But the definition of the morphisms involved gives a canonical isomorphism

$$\alpha_i^* Q_* \simeq \mu_{i*}. \tag{8.5.1.6}$$

When i is a *final object* of I, α_i is an embedding having a retraction $\beta_i : \mathrm{Top}(F) \to F_i$ (7.4.12), and the diagram

(8.5.1.7)


is commutative up to canonical isomorphism.

PROPOSITION 8.5.2. *Let $j \mapsto M_j$ be an object of* $\mathrm{Top}(F)$. *There is a functorial isomorphism*

$$Q^*(j \mapsto M_j) \simeq \varinjlim_{I^\circ} \mu_j^* M_j. \tag{8.5.2.1}$$

An object of $\mathrm{Top}(F)$ (7.1.3) consists of the data of an assignment $j \mapsto M_j$, $M_j \in \mathrm{ob}\, F_j$, and, for every morphism $f : i \to j$, a morphism

$$\beta_f : M_j \longrightarrow f_* M_i \tag{8.5.2.2}$$

or, equivalently by adjunction, a morphism

$$\beta'_f : f^* M_j \longrightarrow M_i, \tag{8.5.2.3}$$

where the morphisms β'_f are subject to the condition that, for every pair of composable morphisms $i \xrightarrow{f} j \xrightarrow{g} k$, the following diagram be commutative:

$$\begin{array}{ccc} f^*g^*M_k & \xrightarrow{f^*\beta'_g} & f^*M_j \\ \downarrow{\scriptstyle c^*_{g,f}} & & \downarrow{\scriptstyle \beta'_f} \\ (gf)^*M_k & \xrightarrow{\beta'_{gf}} & M_i \end{array} \tag{8.5.2.4}$$

Recall moreover from (8.1.3.1) that, for every morphism $f : i \to j$, there is an isomorphism $b_f : f\cdot\mu_i \xrightarrow{\sim} \mu_j$, and that for every pair of composable morphisms $i \xrightarrow{f} j \xrightarrow{g} k$ there is a commutative diagram 323

$$\begin{array}{ccc} g\cdot f\cdot\mu_i & \xrightarrow{c_{g,f}} & (gf)\cdot\mu_i \\ \downarrow{\scriptstyle g\cdot(b_f)} & & \downarrow{\scriptstyle b_{gf}} \\ g\cdot\mu_j & \xrightarrow{b_g} & \mu_k \end{array} \tag{8.5.2.5}$$

Let $f : i \to j$ be a morphism of I. Denote by

$$t_f : \mu_j^*(M_j) \longrightarrow \mu_i^*(M_i) \tag{8.5.2.6}$$

the composite $\mu_j^*(M_j) \xrightarrow{b_f^*} \mu_i^*(f^*(M_j)) \xrightarrow{\mu_i^*(\beta'_f)} \mu_i^*(M_i)$. We leave it to the reader to verify that the commutativity of diagrams (8.5.2.4) and (8.5.2.5) implies that for every pair of composable morphisms $i \to j \to k$, one has

$$t_f t_g = t_{gf}. \tag{8.5.2.7}$$

We have therefore defined a functor $I^\circ \to \underleftarrow{F}$, whose inductive limit $\varinjlim_{I^\circ} \mu_i^*(M_i)$ can be considered. Let N be an object of $\underleftarrow{F}$. We have

$$\mathrm{Hom}(\varinjlim_{I^\circ} \mu_i^*(M_i), N) \simeq \varprojlim_{I^\circ} \mathrm{Hom}(\mu_i^*(M_i), N), \tag{8.5.2.8}$$

and hence, by adjunction, an isomorphism

$$\mathrm{Hom}(\varinjlim_{I^\circ} \mu_i^*(M_i), N) \simeq \varprojlim_{I^\circ} \mathrm{Hom}_{F_i}(M_i, \mu_{i*}(N)). \tag{8.5.2.9}$$

We now interpret the right-hand side of (8.5.2.9). For every morphism $f : i \to j$, 324 denote by

$$d_f : \mathrm{Hom}_{F_i}(M_i, \mu_{i*}(N)) \longrightarrow \mathrm{Hom}_{F_j}(M_j, \mu_{j*}(N))$$

the transition map of the projective system occurring in (8.5.2.9). It follows from the definition of the morphisms t_f ((8.5.2.6)) that d_f associates with a morphism

$$u_i : M_i \longrightarrow \mu_{i*}(N)$$

the morphism

$$d_f(u_i) : M_j \longrightarrow \mu_{j*}(N),$$

obtained by composing the morphisms

$$M_j \xrightarrow{\beta_f} f_*(M_i) \xrightarrow{f_*(u_i)} f_*\mu_{i*}(N) \xrightarrow{b_f} \mu_{j*}(N).$$

Thus an element of the right-hand side of (8.5.2.9) is a family of morphisms $u_i : M_i \to \mu_{i*}(N)$, $i \in \mathrm{ob}\, I$, such that for every $f : i \to j$ the diagram

$$\begin{array}{ccc} M_j & \longrightarrow & f_*(M_i) \\ {\scriptstyle u_j}\downarrow & & \downarrow{\scriptstyle f_*(u_i)} \\ \mu_{j*}(N) & \xrightarrow{\sim} & f_*\mu_{i*}(N) \end{array}$$

is commutative, that is, a morphism from the section $(i \mapsto M_i) \in \mathbf{Hom}_{I^\circ}(I^\circ, F')$ to the section $Q_*(N)$. We therefore have a functorial isomorphism

$$\mathrm{Hom}(\varinjlim_{I^\circ} \mu_i^*(M_i), N) \simeq \mathrm{Hom}((i \mapsto M_i), Q_*(N));$$

the proposition follows by adjunction.

Proposition 8.5.3. *If, for every $f : i \to j \in \mathrm{Fl}(I)$, the functors $f_* : F_i \to F_j$ commute with small filtered inductive limits, then for every section $(i \mapsto M_i) \in \mathrm{Hom}^\circ_I(I^\circ, F')$ we have*

$$Q_*Q^*(i \mapsto M_i) \simeq i \mapsto \varinjlim_{f : j \to i} f_* M_j. \tag{8.5.3.1}$$

325 We leave it to the reader to describe explicitly, in terms of the β_f (8.5.2.2) and the $c_{f,g}$ (8.5.1), the transition morphisms of the inductive systems $(f : j \to i) \mapsto f_*(M_j)$ and the transition morphisms of the section $i \mapsto \varinjlim_{f : j \to i} f_* M_i$. Since the functors f_* commute with filtered inductive limits, the section $i \mapsto \varinjlim_{f : j \to i} f_* M_j$ is Cartesian. Moreover, there is a natural morphism u from $(i \mapsto M_i)$ to the section $(i \mapsto \varinjlim_{f : j \to i} f_* M_j)$, and it is clear that every morphism from $(i \mapsto M_i)$ to a Cartesian section factors uniquely through u, giving the isomorphism (8.5.3.1).

Corollary 8.5.4. *Under the hypotheses of 8.5.3, the functor Q_* commutes with small filtered inductive limits.*

Let $\alpha \mapsto N^\alpha$ be a small filtered inductive system of $\underline{F}$. Put $M^\alpha = Q_*(N^\alpha)$, so that M^α is a section $i \mapsto M_i^\alpha$. We have $N^\alpha \simeq Q^*Q_*(N^\alpha)$, and since Q^* commutes with inductive limits, we have $\varinjlim_\alpha N^\alpha \simeq Q^*(\varinjlim_\alpha M^\alpha)$. Inductive limits in $\mathrm{Top}(F) = \mathrm{Hom}_{I^\circ}(I^\circ, F')$ are computed fiber by fiber. Thus $\varinjlim_\alpha M \simeq (i \mapsto \varinjlim_\alpha M_i^\alpha)$. By (8.5.3.1), we have

$$Q_*(\varinjlim_\alpha N^\alpha) \simeq Q_*Q^*(\varinjlim_\alpha M^\alpha) \simeq i \mapsto \varinjlim_{f : j \to i} f_*(\varinjlim_\alpha M_j^\alpha).$$

Since the functors f_* commute with filtered inductive limits, it follows that

$$Q_*(\varinjlim_\alpha N^\alpha) \simeq i \mapsto \varinjlim_{f : j \to i} \varinjlim_\alpha f_*(M_j^\alpha) \simeq i \mapsto \varinjlim_\alpha \varinjlim_{f : j \to i} f_*(M_j^\alpha) \simeq \varinjlim_\alpha Q_*(N^\alpha).$$

Corollary 8.5.5. *If the functors $f_* : F_i \to F_j$ commute with small filtered inductive limits, then for every object i of I and every $M_i \in \mathrm{ob}\, F_i$ we have*

$$\mu_{i*}\mu_i^*(M_i) \simeq \varinjlim_{f : j \to i} f_* f^*(M_i). \tag{8.5.5.1}$$

After making the base change $I/i \to I$, we may suppose that i is a final object of I (8.2.1).
326 The morphism $\mu_i : \underset{\leftarrow}{F} \to F_i$ is then the composite of the morphisms (8.2.9) and (7.4.12):

$$Q : \underset{\leftarrow}{F} \longrightarrow F^{\sim},$$
$$\beta_i : (\alpha_{i!}, \alpha_i^*) : F^{\sim} \longrightarrow F_i.$$

Thus $\mu_i^*(M_i) \simeq Q^*\beta_i^*(M_i)$, where $\beta_i^*(M_i)$ is the section $(f : i \to j) \mapsto f^*(M_i)$ whose transition morphisms are deduced from the $c_{f,g}$. The assertion then follows from 8.5.3.

Corollary 8.5.6. *Under the hypotheses of 8.5.5, the functors*

$$\mu_{i*} : \underset{\leftarrow}{F} \longrightarrow F_i$$

commute with filtered inductive limits.

The functor μ_{i*} is the composite of the functor Q_*, which commutes with filtered inductive limits, and the functor "restriction to the fiber at i", which commutes with inductive limits.

Corollary 8.5.7. *Under the hypotheses of 8.5.3, let i be an object of I and X an object of F_i such that the functor* $\mathrm{Hom}(X, -)$ *on F_i commutes with filtered inductive limits. Then the functor* $\mathrm{Hom}(\mu_i^*(X), -)$ *commutes with filtered inductive limits, and for every object M_i of F_i we have*

$$\mathrm{Hom}(\mu_i^*(X), \mu_i^*(M_i)) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^*(X), f^*(M_i)). \tag{8.5.7.1}$$

For every object $(j \mapsto M_j)$ of $\mathrm{Top}(F)$, we have

$$\mathrm{Hom}(\mu_i^*(X), Q^*(j \mapsto M_j)) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^*(X), M_j). \tag{8.5.7.2}$$

The functor $\mathrm{Hom}(\mu_i^*(X), -)$ is isomorphic to the functor $\mathrm{Hom}(X, \mu_{i*}(-))$, and the latter commutes with filtered inductive limits (8.5.4). By (8.5.5.1), we have

$$\mathrm{Hom}(\mu_i^*(X), \mu_i^*(M_i)) \simeq \mathrm{Hom}(X, \mu_{i*}, \mu_i^*(M_i)) \simeq \mathrm{Hom}(X, \varinjlim_{f : j \to i} f_* f^*(M_i)),$$

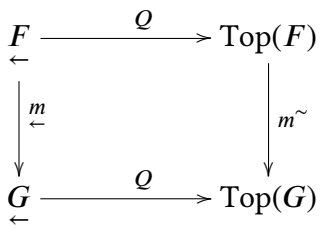
which gives formula (8.5.7.1)). Likewise, by ((8.5.3.1), we have

$$\mathrm{Hom}(\mu_i^*(X), Q^*(j \mapsto M_j)) \simeq \mathrm{Hom}(X, \mu_{i*}Q^*(j \mapsto M_j)) \simeq \mathrm{Hom}(X, \varinjlim_{f : j \to i} f_*(M_j)),$$

which gives formula (8.5.7.2). 327

Proposition 8.5.8. *Let $G \to I$ be a fibered $\mathscr{U}$-topos and $m : F \to G$ a Cartesian morphism of fibered topoi (7.1.15). For G we use the notation introduced in 8.1.1. Let $\underset{\leftarrow}{m} : \underset{\leftarrow}{F} \to \underset{\leftarrow}{G}$ be the morphism deduced from m by passage to the projective limit (8.1.4). The diagram of topoi and morphisms of topoi*

(8.5.8.1)



is commutative up to isomorphism. For every object N of $\underleftarrow{F}$, we have

$$\underleftarrow{m}_*(N) \simeq \varinjlim_{I^\circ} \mu_j^* m_{j*}\mu_{j*}(N). \tag{8.5.8.2}$$

The commutativity of diagram (8.5.8.1) follows immediately from the definitions (8.2.8 and 8.5.1). We have $\underleftarrow{m}_*(N) \simeq Q^*Q_*\underleftarrow{m}_*(N)$ and, by the commutativity of (8.5.8.1), $\underleftarrow{m}_*(N \simeq Q^*m_*^\sim Q_*(N)$. Consequently, there is an isomorphism $\underleftarrow{m}_*(N) \simeq Q^*m_*^\sim(j \mapsto \underleftarrow{\mu}_{j*}(N))$. The functor $m_*^\sim$ is simply the functor $\mathrm{Hom}_{I^\circ}(I^\circ, m_*)$ (7.4.10), and consequently there is a canonical isomorphism $\underleftarrow{m}_*(N) \simeq Q^*(j \mapsto m_{j*}\mu_{j*}(N))$. Formula (8.5.8.2) then follows from 8.5.2.

Proposition 8.5.9. *Under the hypotheses and with the notation of 8.5.8, suppose in addition that, for every object i of I, the functor $m_{i*} : F_i \to G_i$ commutes with filtered inductive limits, and that for every morphism $f : i \to j$ of I, the functor $f_* : F_i \to F_j$ commutes with filtered inductive limits. Then the functor $\underleftarrow{m}_* : \underleftarrow{F} \to \underleftarrow{G}$ commutes with filtered inductive limits, and for every object $(i \mapsto M_i)$ of $\mathrm{Top}(F)$ there is a canonical isomorphism*

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{I^\circ} \mu_j^* m_{j*}(M_j). \tag{8.5.9.1}$$

328 The first assertion follows from (8.5.8.2)) and (8.5.6. By (8.5.8.2)) and ((8.5.3.1), there is a functorial isomorphism

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \mu_i^* m_{i*}(\varinjlim_{f : j \to i} f_*(M_j)).$$

Using the commutation of the m_{i*} with filtered inductive limits and the isomorphisms $m_{i*}f_* \simeq f_* m_{j*}$ (7.1.6), we obtain

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \mu_i^*(\varinjlim_{f : j \to i} f_* m_{j*}(M_j)).$$

Since the functors μ_i^* commute with inductive limits, it follows that

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \varinjlim_{f : j \to i} \mu_i^* f_* m_{j*}(M_j).$$

This last object can be interpreted as an inductive limit over the category $\mathrm{Fl}(I)^\circ$, where $\mathrm{Fl}(I)$ is the category of morphisms of I. Let $\phi : I \to \mathrm{Fl}(I)$ be the functor that associates with every object i of I the identity morphism of I. The functor $\phi^\circ : I^\circ \to \mathrm{Fl}(I^\circ)$ is cofinal, and consequently (I 8.1) there is a canonical isomorphism

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{I^\circ} \mu_j^* m_{j*}(M_j).$$

Corollary 8.5.10. *Under the conditions of 8.5.9, for every object i of I and every object M_i of F_i, there is a canonical isomorphism*

$$\underleftarrow{m}_* \mu_i^*(M_i) \simeq \varinjlim_{f : j \to i} \mu_j^* m_{j*} f^*(M_i). \tag{8.5.10.1}$$

The proof is analogous to that of Corollary 8.5.5.

REMARK 8.5.11. Recall that the direct-image functor under a coherent morphism between coherent topoi commutes with filtered inductive limits (5.1). Consequently, Propositions 8.5.3 through 8.5.7, and 8.5.9, 8.5.10, apply when the fibered topoi under consideration are coherent and the morphisms of fibered topoi under consideration are coherent.

8.6. Ringed fibered $\mathscr{U}$-topoi.

8.6.1. A fibered $\mathscr{U}$-topos is called *ringed* if it is equipped with a sheaf of rings on the total site (7.4.1). Let $p : F \to I$ be a fibered $\mathscr{U}$-topos. Choose a bisplitting (7.1.4). 329
It follows from 7.4.7 that giving a sheaf of rings on F is equivalent to giving, for every object i of I, a ring A_i of F_i, and, for every morphism $f : i \to j$, a morphism $\phi_f : A_j \to f_* A_i$, the family of the ϕ_f being subject to the compatibility conditions made explicit in *loc. cit.*. The morphisms of topoi $f. : F_i \to F_j$ are therefore morphisms of topoi ringed respectively by A_i and A_j (IV 11), and the ringed structure on F, together with the choice of the morphisms f, gives a pseudofunctor from I to the 2-category of ringed topoi. Conversely, given such a pseudofunctor, one can reconstruct the ringed fibered $\mathscr{U}$-topos from which it arises.

8.6.2. As in 8.1, one can define the notion of the projective limit of a ringed $\mathscr{U}$-topos $(F_i, A_i, i \in I)$. We shall suppose that this immediate generalization has been made, and we shall freely apply to this situation the language introduced in 8.1. When it exists, this projective limit is a ringed $\mathscr{U}$-topos (H, B), and, for every object i of I, we have morphisms of ringed topoi $\mu_i : (H, B) \to (F_i, A_i)$. If the underlying (unringed) fibered topos F admits a projective limit $\varprojlim_I \text{top}\ F_i$, and if the inductive system of rings $i \mapsto \mu_i^*(A_i)$ admits an inductive limit $\varinjlim_{I^\circ} \mu_i^* A_i$ in the category of rings of $\varprojlim_I \text{top}\ F_i$ (as is always the case if I is small), then the ringed topos $(\varprojlim_I \text{top}\ F_i, \varinjlim_{I^\circ} \mu_i^* A_i)$ is evidently a projective limit of the ringed fibered topos under consideration.

8.6.3. The formulas of 8.5, established for sheaves of sets, are valid for abelian sheaves. They are also valid for sheaves of modules, provided the inverse-image functors occurring in them are interpreted as inverse-image functors in the sense of sheaves of modules (IV 1.1). We leave the verification to the reader.

8.6.4. Let $(p : F \to I, A)$ be a ringed fibered $\mathscr{U}$-topos over a cofiltered category. We call $(p : F \to I, A)$ a ringed fibered $\mathscr{U}$-topos flat on the right (resp. on the left) if, for every morphism $f : i \to j$, the morphism $f. : (F_i, A_i) \to (F_j, A_j)$ is a morphism of ringed topoi flat on the right (resp. on the left) (V 1.8). Suppose that I is essentially small. Then the morphism of ringed topoi $Q : (\varprojlim_I \text{top}\ F_i, \varinjlim_{I^\circ} \mu_i^*(A_i)) \to (\text{Hom}_{I^\circ}(I^\circ, F'), A)$ (8.5.1) is flat on the right (resp. on the left), as follows from the definition and from 330
8.5.2; this holds without any flatness hypothesis on the transition morphisms $f.$. If, in addition, $(p : F \to I, A)$ is a ringed fibered $\mathscr{U}$-topos flat on the right (resp. on the left), the morphisms $\mu_i : (\varprojlim_I \text{top}\ F_i, \varinjlim_{I^\circ} \mu_i^* A_i) \to (F_i, A_i)$ are flat on the right (resp. on the left). This follows from the fact that the inverse image of a flat module is flat (V 1.7.1), and that a filtered inductive limit of flat modules is flat. The same arguments prove the following fact: if $m : (F_i, A_i, i \in I) \to (G_i, A_i, i \in I)$ is a cartesian morphism of ringed fibered $\mathscr{U}$-topoi over an essentially small cofiltered category, and if, for every object $i \in \text{ob}(I)$, the morphism m_i is flat on the right (resp. on the left), then the morphism $\underline{m}$ deduced from m by passage to the projective limit is flat on the right (resp. on the left).

8.7. Cohomology of sheaves on a projective limit of topoi.

8.7.1. In this section, I denotes a small cofiltered category, $(p : F \to I, A)$ and $(q : G \to I, B)$ two ringed fibered $\mathscr{U}$-topoi, and $m : (p : F \to I, A) \to (q : G \to I, B)$ a morphism of ringed fibered $\mathscr{U}$-topoi. We suppose that, for every morphism $f : i \to j$ of I, the derived functors $R^n f_*$, $n \in \mathbf{N}$, of the functor $f_* : \mathrm{Mod}(F_i, A_i) \to \mathrm{Mod}(F_j, A_j)$ on modules commute with filtered inductive limits, and that, for every object $i \in I$, the derived functors $R^n m_{i*}$ of the functor $m_{i*} : (F_i, A_i) \to (G_i, B_i)$ on modules[34] commute with filtered inductive limits. Recall that these hypotheses hold when the categories F_i, G_i, $(i \in \mathrm{ob}(I))$, are *coherent* and when the morphisms $f : F_i \to F_j$, $(f : i \to j \in F1(I))$, and $m_i : F_i \to G_i$, $(i \in I)$, are *coherent* (5.2). We use the notation of 8.5.1. In addition, we denote by $\underline{A}$ (resp. $\underline{B}$) the sheaf $\varinjlim \mu_i^*(A_i)$ (resp. $\varinjlim \mu_i^*(B_i)$). Finally, for modules, we use the notation μ_i^{-1} for inverse image in the sense of abelian sheaves, reserving μ_i^* for inverse image in the sense of modules (IV 11).

331 LEMMA 8.7.2. *Let $j \mapsto M_j$ be an injective A-Module of* $\mathrm{Top}(F)$. *Then $Q^*(j \mapsto M_j)$ is an A-Module acyclic for $\underleftarrow{m}_*$, and, for every $i \in \mathrm{ob}(I)$, M_i is flasque.*

Let $i \in \mathrm{ob}(I)$, and let e_i be a final object of F_i. The fibered $\mathscr{U}$-topos $F_{/e_i} \to I_{/i}$ is deduced from $F \to I$ by the base change $I_{/i} \to I$. Since the sheaf $j \mapsto M_j$ is injective, it is flasque (V 4.6). Its restriction to the localized topos $F_{/e_i}$ is flasque (V 4.12). The restriction functor from $F_{/e_i}$ to F_i is a direct image functor for a morphism of topoi (7.4.12). It therefore carries flasque Modules to flasque Modules (V 5.2). Thus M_i is flasque.

Let us prove the first assertion of the lemma. Set $N' = Q^*(j \mapsto M_j)$. By (8.5.3.1), for every $i \in \mathrm{ob}(I)$ we have

$$\mu_{i*}(N') = \mu_{i*}Q^*(j \mapsto M_j) \simeq \varinjlim_{f : j \to i} f_*(M_j).$$

Using the hypothesis of 8.7.1, we see that the sheaf N' has the following property:

(P) For every object $i \in \mathrm{ob}(I)$, $\mu_{i*}(N')$ is m_{i*}-acyclic, and, for every morphism $f : i \to j$ of I, $\mu_{i*}(N')$ is f_*-acyclic.

It is enough to show that every object N' having property (P) is $\underleftarrow{m}_*$-acyclic. Since injectives have property (P), it is enough to verify the following two properties (V 0.4): for every exact sequence $0 \to N' \to N \to N'' \to 0$, where N' and N have property (P), (a) N'' has property (P), and (b) $m_*(N) \to m_*(N'')$ is an epimorphism. Verification of (a) is immediate. Let us verify (b). Set $K = \mathrm{coker}(Q_*(N') \to Q_*(N))$, so that K is the section $(i \mapsto K_i = \mathrm{coker}(\mu_{i*}(N') \to \mu_{i*}(N)))$. For every $f : i \to j$, there is a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & f_*\mu_{i*}(N') & \longrightarrow & f_*\mu_{i*}(N) & \longrightarrow & f_*(K_i) & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & \mu_{j*}(N') & \longrightarrow & \mu_{j*}(N) & \longrightarrow & K_j & \longrightarrow & 0.
\end{array}$$

332 Since $\mu_{i*}(N')$ is f_*-acyclic, the top row is exact. Since $i \mapsto \mu_{i*}(N')$ and $i \mapsto \mu_{i*}(N)$ are Cartesian sections, the first two vertical morphisms are isomorphisms. Consequently, the canonical morphism $K_j \to f_*(K_i)$ is an isomorphism, and $i \mapsto K_i$ is a Cartesian section. Thus K is of the form $Q_*(K')$, and the canonical morphisms $Q_*(N) \to K$ and

[34] To fix ideas, we shall take left modules.

$K \to Q_*(N'')$ are of the form $Q_*(u)$ and $Q_*(v)$, respectively, because Q_* is fully faithful. Since Q^* is exact and Q^*Q_* is isomorphic to the identity, v is an isomorphism, and hence $Q_*(v)$ is an isomorphism. Thus the sequence $0 \to Q_*(N') \to Q_*(N) \to Q_*(N'') \to 0$ is exact, and consequently, for every object i of I, the sequence $0 \to \mu_{i*}(N') \to \mu_{i*}(N) \to \mu_{i*}(N'') \to 0$ is exact. Since $\mu_{i*}(N')$ is m_{i*}-acyclic, the sequence $0 \to m_{i*}\mu_{i*}(N') \to m_{i*}\mu_{i*}(N) \to m_{i*}\mu_{i*}(N'') \to 0$ is exact. Moreover, there is an isomorphism $\underleftarrow{m}_* \simeq \varinjlim_{I^\circ} \mu_j^{-1} m_{j*}\mu_{j*}$ (8.5.8.2). The functor μ_j^{-1} is exact, and filtered inductive limits are exact. Therefore the sequence $0 \to \underleftarrow{m}_*(N') \to \underleftarrow{m}_*(N) \to \underleftarrow{m}_*(N'') \to 0$ is exact.

THEOREM 8.7.3. *The notation and hypotheses are those of 8.7.1. For every integer $n \in \mathbf{N}$, the derived functor $R^n\underleftarrow{m}_*$ commutes with filtered inductive limits, and for every A-Module $j \mapsto M_j$ there is a functorial isomorphism*

$$R^n\underleftarrow{m}_* Q^*(j \mapsto M_j) \simeq \varinjlim_{I^\circ} \mu_j^* R^n m_{j*}(M_j). \tag{8.7.3.1}$$

We have $Q^{-1}(A) = \underrightarrow{A}$, and hence inverse image for Modules is isomorphic to inverse image for abelian sheaves. It follows from 8.5.8.2 that, for every section $j \mapsto M_j$, there is a canonical isomorphism $\varinjlim_{I^\circ} \mu_j^{-1}(M_j) \simeq \varinjlim_{I^\circ} \mu_j^*(M_j)$. Moreover, under the hypotheses of 8.7.1, there is a canonical isomorphism (8.5.9.1) $\underleftarrow{m}_* Q^*(j \mapsto M_j) \simeq \varinjlim_{I^\circ} \mu_j^{-1} m_{j*}(M_j)$. Since, for every injective $j \mapsto M_j$, the M_j are flasque and $Q^*(j \mapsto M_j)$ is acyclic for $\underleftarrow{m}_*$ (8.7.2), there is an isomorphism $R^n\underleftarrow{m}_*(Q^*(j \mapsto M_j)) \simeq \varinjlim_{I^\circ} \mu_j^{-1} R^n m_{j*}(M_j)$, which gives formula (8.7.3.1) by what precedes. The first assertion follows immediately from formula (8.7.3.1).

COROLLARY 8.7.4. *For $n \in \mathbf{Z}$, there are functorial isomorphisms* 333

$$R^n\underleftarrow{m}_*(N) \simeq \varinjlim_{I^\circ} \mu_j^* R^n m_{j*}\mu_{j*}(N). \tag{8.7.4.1}$$

Indeed, N is isomorphic to $Q^*(j \mapsto \mu_{j*}(N))$.

COROLLARY 8.7.5. *For every object i of I, every integer n, and every A_i-Module M_i, there is a functorial isomorphism*

$$R^n\underleftarrow{m}_*\mu_i^*(M_i) \simeq \varinjlim_{f\,:\,j\to i} \mu_j^* R^n m_{j*} f^*(M_i). \tag{8.7.5.1}$$

After making the base change $I_{/i} \to I$, we may assume that i is a final object of I (8.2.1). The object $\mu_i^*(M_i)$ is then isomorphic to $Q^*((f : j \to i) \mapsto f^*(M_i))$ (cf. 8.5.5).

COROLLARY 8.7.6. *Let i be an object of I and n an integer. There is an isomorphism, functorial in the section $j \mapsto M_j$:*

$$R^n\mu_{i*}Q^*(j \mapsto M_j) \simeq \varinjlim_{f\,:\,j\to i} R^n f_*(M_j). \tag{8.7.6.1}$$

There is an isomorphism functorial in the $\underrightarrow{A}$-Module N:

$$R^n\mu_{i*}(N) \simeq \varinjlim_{f\,:\,j\to i} R^n f_*\mu_{j*}(N). \tag{8.7.6.2}$$

There is an isomorphism functorial in the A_i-Module M_i:

$$R^n\mu_{i*}\mu_i^*(M_i) \simeq \varinjlim_{f:j\to i} R^n f_* f^*(M_i). \tag{8.7.6.3}$$

The functors $R^n\mu_{i*}$ commute with filtered inductive limits.

Making the base change $I_{/i} \to I$ reduces us to the case where i is a final object of I. Formulas (8.7.6.1) and (8.7.6.2) are then special cases of formulas (8.7.3.1), (8.7.4.1), and (8.7.5.1), respectively, obtained by taking for G the constant fibered topos with fiber F_i and for m the morphism $(f : F_j \to F_i, f \in \mathrm{ob}(I_{/i}))$. The last assertion follows from (8.7.6.2), taking 8.5.4 into account.

334 Corollary 8.7.7. *Let i be an object of I, and let X be an object of F_i such that the functors $H^n(X,-)$, $n \in \mathbf{N}$, commute with filtered inductive limits. For every n, there is a canonical isomorphism, functorial in the A_i-Module M_i:*

$$H^n(\mu_i^*(X), \mu_i^*(M_i)) \simeq \varinjlim_{f:j\to i} H^n(f^*(X), f^*(M_i)) \tag{8.7.7.1}$$

and the functors $H^n(\mu_i^*(X),-)$ commute with filtered inductive limits. In particular, if the functors $H^n(F_i,-)$ commute with filtered inductive limits, then the functors $H^n(\underleftarrow{F},-)$ commute with filtered inductive limits, and there are canonical isomorphisms, functorial in the A_i-Modules M_i:

$$H^n(\underleftarrow{F}, \mu_i^*(M_i)) \simeq \varinjlim_{f:j\to i} H^n(F_j, f^*(M_i)). \tag{8.7.7.2}$$

It follows from V 5.3 that there are two spectral sequences

$$'E_2^{p,q} = \varinjlim_{f:j\to i} H^p(X, R^q f_* f^*(M_i)) \Rightarrow \varinjlim_{f:j\to i} H^{p+q}(f^*(X), f^*(M_i))$$

$$''E_2^{p,q} = H^p(X, R^q\mu_{i*}\mu_i^*(M_i)) \Rightarrow H^{p+q}(\mu_i^*(X), \mu_i^*(M_i)),$$

and from (8.5.7.1) that there is a morphism between these two spectral sequences. Since the $H^{p+q}(X,-)$ commute with filtered inductive limits, it follows from (8.7.6.2) that this morphism of spectral sequences is an isomorphism at the $E_2^{p,q}$ level. It therefore induces an isomorphism on the abutments, giving (8.7.7.1). For every $\underrightarrow{A}$-Module N, there is a spectral sequence (V 5.3)

$$E_2^{p,q} = H^p(X, R^q\mu_{i*}(N)) \Rightarrow H^{p+q}(\mu_i^*(X), N).$$

The assertion that the functors $H^n(\mu_i^*(X),-)$ commute with filtered inductive limits then follows from the analogous properties of the functors $H^n(X,-)$ and $R^n\mu_{i*}$ (8.7.6).

Remark 8.7.8. When the topoi F_j, $j \in \mathrm{ob}(I)$, are coherent and the transition morphisms are coherent, the hypothesis made on X (resp. F_i) in 8.7.7 holds when X (resp. F_i) is coherent (5.2). Moreover, we know that in this case $\mu_i^*(X)$ (resp. $\underleftarrow{F}$) is coherent (8.3.13).

335 Corollary 8.7.9. *Let i be an object of I, let M_i be a left A_i-Module, and let L_i be a right A_i-Module admitting a resolution of the form*

$$P_{i,k} \longrightarrow P_{i,k-1} \longrightarrow \cdots \longrightarrow P_{i,0} \longrightarrow L_i \longrightarrow 0$$

where, for every integer k, $P_{i,k}$ is isomorphic to a finite direct sum of objects of the form $A_{i|X}$ (IV 11.3.3), with X satisfying the hypotheses of 8.7.7. If, in addition to the hypotheses

of 8.7.1, the fibered topos F is right flat (8.6), then for $n \leq k-1$ there are canonical isomorphisms

$$\text{(8.7.9.1)} \qquad \operatorname{Ext}^n_{\underline{A}}(\underline{F}, \mu_i^*(L_i), \mu_i^*(M_i)) \simeq \varinjlim_{f:j\to i} \operatorname{Ext}^n_{A_j}(F_j, f^*(L_i), f^*(M_i)).$$

Denote by $P_{i,\cdot}$ the resolution of L_i. Since F is right flat, for every $f : j \to i$, $f^*(P_{i,\cdot})$ is a resolution of $f^*(L_i)$, and $\mu_i^*(P_{i,\cdot})$ is a resolution of $\mu_i^*(L_i)$ (8.6). These resolutions give two spectral sequences converging respectively to the two sides of (8.7.9.1), together with a morphism between the two spectral sequences. At the $E_1^{p,q}$ level this morphism is an isomorphism by (8.7.7.1). It therefore induces an isomorphism on the abutments.

9. Appendix. Criterion for the existence of points

by P. Deligne

Proposition 9.0. *Every locally coherent topos S has enough points.*

Since the question is local on S, we may suppose that the topos S is defined by a site $\mathcal{S}$ in which finite projective limits are representable and such that every covering $f_i : U_i \to U$ admits a finite subcovering. It is enough to prove that if $f : \mathcal{F} \to \mathcal{G}$ is not a monomorphism, then there is a point x of $\mathcal{S}$ such that f_x is not injective. By hypothesis, there are $U \in \operatorname{ob}\mathcal{S}$ and $s, s' \in \mathcal{F}(U)$ such that $s \neq s'$ and $f(s) = f(s')$. Replacing $\mathcal{S}$ by $\mathcal{S}/U$, we reduce to the following lemma.

Lemma 9.1. *If s and s' are two distinct global sections of a sheaf $\mathfrak{F}$ on a site $\mathcal{S}$ satisfying the preceding hypotheses, then there is a point x of $\mathcal{S}$ such that $s_x \neq s'_x$.* 336

Let $P = (U_i)_{i\in I}$ be a projective system in $\mathcal{S}$, indexed by a filtered ordered set I. For every sheaf $\mathcal{H}$ on $\mathcal{S}$, put $P(\mathcal{H}) = \varinjlim \mathcal{H}(U_i)$, and for every $V \in \operatorname{Ob}\mathcal{S}$, put $P(V) = \varinjlim \operatorname{Hom}(U_i, V)$. The functors $P(\mathcal{H})$ and $P(V)$ commute with fiber products. If the functor $P(V)$ takes coverings to surjective families of maps, it is a morphism of sites from the point topos to $\mathcal{S}$, and $P(\mathcal{H})$ is the corresponding fiber functor.

If $P = (U_i)_{i\in I}$ and $Q = (V_j)_{j\in J}$ are two projective systems in $\mathcal{S}$, we say that Q refines P if I is a subset of J with the induced order and P is the restriction of Q to I. There are then morphisms of functors from $P(\mathcal{H})$ to $Q(\mathcal{H})$ and from $P(V)$ to $Q(V)$.

For every P as above, denote by s_P and s'_P the images of s and s' in $P(\mathcal{F})$. Lemma 9.1 follows from the next lemma, in which P is taken to be the projective system indexed by $\{0\}$ and consisting of the final object of $\mathcal{S}$.

Lemma 9.2. *For every P as above satisfying $s_P \neq s'_P$, there is a refinement Q of P satisfying $s_Q \neq s'_Q$ and such that, for every covering $f_i : V_i \to V$ and every $f \in Q(V)$, f belongs to the image of one of the $Q(V_i)$.*

We first prove:

Lemma 9.3. *Let $P = (U_i)_{i\in I}$ satisfy the hypotheses of Lemma 9.2, let $f_i : V_i \to V$ be a finite covering in $\mathcal{S}$, and let $f \in P(V)$. There is a refinement Q of P satisfying $s_Q \neq s'_Q$ and such that the image of f in $Q(V)$ belongs to the image of one of the $Q(V_i)$.*

There are $i_\circ \in I$ and $f' \in \operatorname{Hom}(U_{i_\circ}, V)$ which define f. For $i \geq i_\circ$, put $U_{i,k} = U_i \times_V V_k$, this fiber product being defined by f'. Since (V_k) is a covering of V, $(U_{i,k})$ is a covering of U_i; the following arrow is injective (for $i > i_\circ$):

$$\mathcal{F}(U_i) \longrightarrow \prod_k \mathcal{F}(U_{i,k}).$$

337 Passing to the limit, and using the commutation of *finite* products with filtered inductive limits, we see that

$$P(\mathscr{F}) \longrightarrow \prod_k \varinjlim \mathscr{F}(U_{i,k})$$

is injective. Hence there is a k such that $\mathscr{S}_P$ and $\mathscr{S}'_P$ have distinct images in $\varinjlim_{i>i_\circ} \mathscr{F}(U_{i,k})$. Put $I_\circ = \{i \mid i \geq i_\circ \text{ and } i \in I\}$ and $J = I \coprod I_\circ$. Order J by declaring $j' \leq j''$ if the images of j' and j'' in I satisfy $j' \leq j''$ and it is not the case that $j' \in I_\circ$ and $j'' \in I$. The U_i and $U_{i,k}$, indexed by J, form a refinement Q of P such that $\mathscr{S}_Q \neq \mathscr{S}'_Q$ and the image of f in $Q(V)$ belongs to the image of $Q(V_k)$.

LEMMA 9.4. *Let $P = (U_i)_{i\in I}$ satisfy the hypotheses of Lemma 9.2. There is a refinement Q of P such that $\mathscr{S}_Q \neq \mathscr{S}'_Q$ and such that, for every finite covering (V_k) of an object V of $\mathscr{S}$ and every $f \in P(V)$, the image of f in $Q(V)$ belongs to the image of one of the $Q(V_k)$.*

Let E be the set of triples consisting of an object V of $\mathscr{S}$, a finite covering (V_k) of V, and an element $f \in P(V)$. Let $\leq$ be a well-ordering of E, and let $\overline{E}$ be the set obtained from E by adjoining a greatest element, denoted by ∞. By transfinite induction on $e \in \overline{E}$, we shall define refinements Q_e of P satisfying $\mathscr{S}_{Q_e} \neq \mathscr{S}'_{Q_e}$ and such that

(i) if $e' \leq e$, then Q_e refines $Q_{e'}$;
(ii) if $e = (V, (V_k), f)$, then the image of f in $Q_e(V)$ belongs to the image of one of the $Q_e(V_k)$.

Suppose the $Q_{e'}$ already defined for $e' < e$. If e is the first element of E, put $Q'_e = P$. If e has a predecessor $e-1$, put $Q'_e = Q_{e-1}$. Otherwise, let Q'_e be the projective system with index set $\bigcup_{e'<e} I_{e'}$ which refines the $Q_{e'}$ for $e' < e$. In this case,

$$Q'_e(\mathscr{F}) = \varinjlim_{e'<e} Q_{e'}(\mathscr{F}),$$

338 so that, in every case, Q'_e refines the $Q_{e'}$ for $e' < e$ and satisfies $\mathscr{S}_{Q'_e} \neq \mathscr{S}'_{Q'_e}$.

The required projective system Q_e is obtained by applying Lemma 9.3 to Q'_e and $e = (V, (V_k), f)$ (resp. by taking $Q_e = Q'_e$ if $e = \infty$).

The projective system Q_∞ satisfies Lemma 9.4.

Lemma 9.4 makes it possible to define, inductively on n, a sequence $Q_n = (U_i)_{i\in I_n}$ of projective systems in $\mathscr{S}$ such that

(i) $Q_0 = P$;
(ii) Q_{n+1} refines Q_n;
(iii) $\mathscr{S}_{Q_n} \neq \mathscr{S}'_{Q_n}$;
(iv) for every covering $f_k : V_k \to V$ and every $f \in Q_n(V)$, the image of f in $Q_{n+1}(V)$ belongs to the image of one of the $Q_{n+1}(V_k)$.

The projective system Q, whose index set is the union of the I_n and which extends the various Q_n, then satisfies Lemma 9.2. The proof also shows:

COROLLARY 9.5. *Let $\mathscr{S}$ be a site in which fiber products are representable and such that every covering in $\mathscr{S}$ admits a finite subcovering. Put $c = \sup(\aleph_0, \operatorname{card}(Fl\,\mathscr{S}))$. Then there is a very dense set X of points of $\mathscr{S}$ such that*

(i) $\operatorname{card}(X) \leq 2^c$;
(ii) *if $x \in X$ and $U \in \operatorname{Ob}\mathscr{S}$, then* $\operatorname{card}(U_x) \leq c$.

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EXPOSÉ VII

The étale site and topos of a scheme

A. Grothendieck

In the present exposé and the next, we develop the most elementary properties 342
concerning étale topology and cohomology. The developments of the present exposé concern certain properties that are, for the most part, valid for other, very different topologies, such as the "fppf topology". In the next exposé, we shall develop properties rather special to the étale topology, arising from the very particular nature of étale morphisms.

Here we partly follow three oral exposés by J.E. Roos (which he had not been able to write up), notably in the proof of VIII 6.3.

Unless expressly stated otherwise, it will be understood that the schemes considered in the present exposé and the following ones are elements of the fixed universe $\mathscr{U}$.

1. The étale topology

1.1. We denote by (Sch) the category of schemes (elements of the fixed universe $\mathscr{U}$). Recall that a morphism $f : X \to Y$ in (Sch) is called *étale* if it is locally of finite 343
presentation and flat, and if for every $y \in Y$, the fiber X_y is discrete and its local rings are finite *separable* extensions of $k(y)$. Equivalently, f is locally of finite presentation and, for every affine Y-scheme Y' (in the absolute sense) and every subscheme Y'_0 defined by a nilpotent ideal, the map

$$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X)$$

is *bijective*. For the most important properties of this notion, I refer to EGA IV §§ 17 and 18, and, in the meantime, to SGA 1 I and IV.

1.2. The *étale topology* on (Sch) is the topology generated by the pretopology for which, for every $X \in$ (Sch), the set Cov X consists of the families $(X_i \xrightarrow{u_i} X)_{i\in I}$ (indexed by an $I \in U$) such that the u_i are étale and $X = \bigcup_i u_i(X_i)$ (in the set-theoretic sense).

It is convenient to associate with every X the subcategory $\mathrm{Et}_{/X}$ of $(\mathrm{Sch})_{/X}$ consisting of the arrows $X' \to X$ that are étale. We endow it with the "induced topology" (III) from the étale topology of (Sch) (also called the "étale topology on X"), and denote the resulting site by $X_{\text{ét}}$ (the "étale site" of X) [34]. Note that every morphism of $\mathrm{Et}_{/X}$ is étale (SGA 1 I 4.8), whence a family $(X'_i \xrightarrow{u_i} X')_{i\in I}$ in $X_{\text{ét}}$ is covering if and only if it is *surjective*, i.e. $\bigcup_i u_i(X'_i) = X'$. One sees immediately (cf. 3.1) that $X_{\text{ét}}$ is a $\mathscr{U}$-site, so the results of Exposés I through VI apply. The topos $\widetilde{X_{\text{ét}}}$ of $\mathscr{U}$-sheaves on $X_{\text{ét}}$ is the *étale topos* of X.

1.3. In what follows, unless expressly stated otherwise, all the topological notions 344

[34] Note, however, that it would be preferable to denote by $X_{\text{ét}}$ the topos $\widetilde{X_{\text{ét}}}$ defined by the étale site of X. For practical reasons, in this Seminar we shall retain the notation introduced here (which is that of the original seminar).

considered in $\mathrm{Et}_{/X}$ or (Sch) are understood with respect to the étale topology. Moreover, in language and notation we shall commonly write X in place of $X_{\text{ét}}$ or $Et_{/X}$. Thus a "sheaf on X" (understood: for the étale topology) means a sheaf on the site $X_{\text{ét}}$. We denote by $\widetilde{X_{\text{ét}}}$ the category of these sheaves, which is a $\mathscr{U}$-topos. If F is an abelian sheaf on X, we denote its cohomology groups by $H^i(X, F)$; these would be denoted $H^i(X_{\text{ét}}, F)$ in Exposé V.

1.4. Let $f : X \to Y$ be a morphism of schemes. The inverse-image functor

$$f^* : Y' \mapsto Y' \times_Y X$$

induces a functor

$$(*) \qquad f^*_{\text{ét}} : \mathrm{Et}_{/Y} \longrightarrow \mathrm{Et}_{/X}$$

that commutes with finite $\varprojlim$ and takes covering families to covering families; it is therefore a morphism of sites

$$f^*_{\text{ét}} : X_{\text{ét}} \longrightarrow Y_{\text{ét}},$$

and thus induces a functor on the categories of sheaves:

$$f^{\text{ét}}_* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{Y_{\text{ét}}}$$

given by

$$f^{\text{ét}}_*(F) = F \circ f^*_{\text{ét}}.$$

Moreover, $f^{\text{ét}}_*$ admits a left adjoint

$$f^*_{\text{ét}} : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

345 extending the functor considered in $(*)$ and commuting with arbitrary $\varinjlim$ and finite $\varprojlim$, i.e. $f^*_{\text{ét}}$ defines a *morphism of topoi*

$$f_{\text{ét}} : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{Y_{\text{ét}}}.$$

Clearly, for a composite of two morphisms of schemes

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

there are canonical isomorphisms

$$\begin{cases} (gf)^{\text{ét}}_* \simeq g^{\text{ét}}_* f^{\text{ét}}_* & , \quad \text{whence} \\ (gf)^*_{\text{ét}} \simeq f^*_{\text{ét}} g^*_{\text{ét}} & , \quad \text{i.e. there is an isomorphism} \\ (gf)_{\text{ét}} \simeq g_{\text{ét}} f_{\text{ét}} & , \end{cases}$$

(so that one obtains a "pseudofunctor" (SGA 1 VI 8) from the category (Sch) to the category (Top) of topoi $\in \mathscr{U}'$, where $\mathscr{U}'$ is the smallest universe such that $\mathscr{U} \in \mathscr{U}'$; compare IV).

1.5. Notation. In what follows, we shall often write f^* and f_* in place of $f^*_{\text{ét}}$ and $f^{\text{ét}}_*$. If $f : X \to Y$ is a morphism of schemes, the functors $R^i f^{\text{ét}}_*$ will therefore simply be denoted $R^i f_*$. In this notation, recall the Leray spectral sequence for f and an abelian sheaf F on X (V 5):

$$H^*(X, F) \Longleftarrow E_2^{p.q} = H^p(Y, R^q f_*(F)).$$

Likewise, if there are two morphisms $f : X \to Y$ and $g : Y \to Z$, there is a Leray spectral sequence (the spectral sequence of composite functors)

$$R^*(gf)_*(F) \Longleftarrow E_2^{p.q} = R^p g_*(R^q f_*(F)).$$

1.6. When $f : X \to Y$ is itself an étale morphism, X may be regarded as an object 346
of $Y_{\text{ét}}$, and there is a canonical *isomorphism of sites*

$$X_{\text{ét}} \xrightarrow{\sim} (Y_{\text{ét}})_{/X}.$$

Under this isomorphism, the functor $f^*_{\text{ét}} : Y' \mapsto Y' \times_Y X$ of $(*)$ identifies with the internal base-change functor in the category $\mathrm{Et}_{/Y}$. It follows that the functor

$$f^* : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

is then isomorphic to the functor "*restriction* to Y"

$$f^*(F) \simeq F \circ i_{X/Y},$$

where F is a sheaf on $Y_{\text{ét}}$ and $i_{X/Y} = X_{\text{ét}} \to Y_{\text{ét}}$ is the evident functor (which regards an étale scheme over X, $u : X' \to X$, as an étale scheme over Y via $fu : X' \to Y$).

1.7. Questions of universes. Note that if $X \neq \emptyset$, then $X_{\text{ét}}$ *is not* an element of the chosen universe $\mathscr{U}$. Nevertheless, as we have noted, it is easy to see (3.1) that one can find a full subcategory C of $X_{\text{ét}}$, *an element of* $\mathscr{U}$, satisfying the conditions of the "comparison lemma" (III), so that the restriction functor induces an equivalence $\widetilde{C} \to \widetilde{X_{\text{ét}}}$. Thus the *étale topos* is equivalent to a topos defined in terms of a site $C \in \mathscr{U}$. In other words, $X_{\text{ét}}$ is a U-site, and hence by *loc. cit.* the results of Exposés I through VI apply to this site.

2. Examples of sheaves

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a) Let $F \in \mathrm{Ob}(\mathrm{Sch})_{/X}$ be a scheme over X, and for every X' over X put

$$F(X') = \mathrm{Hom}_X(X', F).$$

The functor $(\mathrm{Sch})^\circ_{/X} \to (\mathrm{Ens})$ thus defined is a sheaf for the étale topology (and even for the finer fpqc topology studied in SGA 3 IV). In other words, the étale topology on (Sch) is coarser than the canonical topology. This is, essentially, the content of SGA 1 VIII 5.1 (cf. also SGA 3 IV 6.3.1). A fortiori, the restriction of this sheaf to $X_{\text{ét}}$ is a sheaf. It will again be denoted by F when there is no risk of confusion[34]. Note also that the functor thus obtained

$(*)$ $$(\mathrm{Sch})_{/X} \longrightarrow \widetilde{X_{\text{ét}}}$$

commutes with finite $\varprojlim$ (this is immediate). This implies, for example, that when F is a group scheme (resp. ...) over X, the sheaf it defines is a sheaf of groups (resp. ...). Note that the functor $\mathrm{Et}_{/X} \to \widetilde{X_{\text{ét}}}$ induced by (*) is simply the canonical functor, associating with every $X' \in \mathrm{Ob}\, \widetilde{X_{\text{ét}}}$ the functor on $X_{\text{ét}}$ that it represents. It is therefore an isomorphism of the category $\mathrm{Et}_{/X}$ onto a full subcategory of the étale topos $\widetilde{X_{\text{ét}}}$, by which we generally identify an $X \in \mathrm{Ob}\, X_{\text{ét}}$ with the corresponding sheaf, denoted $\widetilde{X}$ or simply X.

Clearly, $H^\circ(X, F) = \mathrm{Hom}_{(\mathrm{Sch})_{/X}}(X, F)$.

Let us also give an interpretation of $H^1(X, F)$ when F is a group prescheme over X (commutative, if desired, so that the definition of $H^1(X, F)$ falls within Exposé V; for the general case one may consult J. Giraud's thesis[34]). Familiar ar- 348

[34] Care must be taken, however, that if $X \neq \emptyset$, the functor φ that associates with $F \in \mathrm{Ob}(\mathrm{Sch})_{/X}$ the corresponding sheaf $\varphi(F)$ is *not* fully faithful, nor even faithful, and that F cannot be reconstructed (up to isomorphism) from $\varphi(F)$. For example, if $S = \mathrm{Spec}\, k$, with k an algebraically closed field, knowledge of $\varphi(F)$ is equivalent merely to knowledge of the underlying set of F (by VIII 2.4). Compare IV

guments, which I shall not repeat here, then show that $H^1(X, F)$ is canonically isomorphic to the group of classes (up to isomorphism) of *sheaves of sets on $X_{ét}$ that are principal homogeneous under F*[34]. When F is *affine* over X, SGA 1 VIII 2.1 implies (again by standard arguments, cf. J. GIRAUD's thesis) that $H^1(X, F)$ is also the group of classes of *schemes* P over X on which F acts on the right and which are "principal homogeneous bundles over X for the étale topology", i.e. locally trivial for the étale topology[34].

REMARKS 2.1. Care must be taken that if, for every scheme Z over X, the associated sheaf on $X_{\text{ét}}$ is denoted by $\varphi_X(Z)$, and if $f : Y \to X$ is a morphism of schemes, there is an evident homomorphism (functorial in Z)

$$(*) \qquad f^*(\varphi_X(Z)) \longrightarrow \varphi_Y(Z_Y), \quad Z_y = Z \times_X Y,$$

i.e. an evident homomorphism

$$\varphi_X(Z) \longrightarrow f_*(\varphi_Y(Z_Y)),$$

namely the homomorphism, functorial in $X' \in \mathrm{Ob}\, X_{\text{ét}}$,

$$\mathrm{Hom}_X(X', Z) \longrightarrow \mathrm{Hom}_Y(X'_Y, Z_Y).$$

But note that *in general* $(*)$ *is not an isomorphism, i.e. the formation of the étale sheaf associated with a relative scheme does not commute with inverse-image functors.* Nevertheless, $(*)$ is an isomorphism in the special case in which Z is étale over X. Likewise,
349 the functor

$$\varphi_X : \mathrm{Sch}_{/X} \longrightarrow \widetilde{X_{\text{ét}}}$$

is not faithful if $X \neq \varnothing$[34]; nevertheless, its restriction to $X_{\text{ét}}$, which is simply the canonical functor $X_{\text{ét}} \to \widetilde{X_{\text{ét}}}$, is *fully faithful*, because by a) the topology of $X_{\text{ét}}$ is coarser than its canonical topology.

b) Note that $(*)$ also commutes with direct sums indexed by the $I \in \mathscr{U}$, as is readily verified. In particular, if for every set I we denote by I_X the constant X-scheme defined by I (SGA 3 I 1.8), the direct sum of I copies of X, then the associated sheaf is simply the constant sheaf $I_{X_{\text{ét}}}$ (the direct sum of I copies of the final sheaf on $X_{\text{ét}}$). Since the functor $I \mapsto I_X$ also commutes with finite $\varprojlim$, we see that it takes a group to a group, a commutative group to a commutative group, etc. If G is an ordinary commutative group, we shall simply write $H^i(X, G)$ in place of $H^i(X, G_X)$. Suppose, for example, that G is a finite group. Then G_X is finite, hence affine over X, and thus, using the final remark of a), we obtain an interpretation of $H^1(X, G)$ as the set *of classes of principal Galois coverings with group G* (SGA 1 V 2.7). When X is connected and endowed with a geometric point a, the "fundamental group" $\pi_1(X, a)$ (SGA 1 V 7) gives the canonical isomorphism

$$(**) \qquad H^1(X, G) \simeq \mathrm{Hom}(\pi_1(X, a), G),$$

(where G has been assumed commutative).

One may say that in passing from Zariski cohomology to the étale topology, "we have done what was needed" to obtain the "right" H^1 (which occurs in
350 the second member of $(*)$) for a finite constant coefficient group G. *It is a*

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[34] or F-torsors in the terminology of *loc. cit.*.

[34] or representable *F-torsors* in the now accepted terminology

[34] Cf. the note at the bottom of page (page 6)

remarkable fact, to be proved in the remainder of this seminar, that this also suffices to obtain the "right" $H^i(X, G)$ for every torsion coefficient group (at least if G is prime to the residual characteristics of X).

c) Let F be an $\mathcal{O}_X$-module on X, for the Zariski topology. Then (with the notation of SGA 3 I 4.6) define a functor

$$W(F) : (\mathrm{Sch})^{\circ}_{/X} \longrightarrow (\mathrm{Ens})$$

by

$$W(F)(X') = \Gamma(X', F \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}).$$

This functor is again a sheaf for the étale topology (and even for the fpqc topology), as follows again from SGA 1 VIII 1.6 and SGA 3 IV 6.3.1. A fortiori, the restriction of $W(F)$ to $X_{\text{ét}}$ is again a sheaf, which we also denote by $W(F)$. By definition, therefore, $H^\circ(X, W(F)) = \Gamma(F) = H^\circ(X, F)$. But there is a stronger result if F is quasi-coherent; cf. 4.3.

3. Generators of the étale topos. Cohomology of an $\varinjlim$ of sheaves

PROPOSITION 3.1. *Let X be a scheme, and let C be a full subcategory of $X_{\text{ét}}$ such that, for every affine scheme X' étale over X, C contains an object isomorphic to X'. Then C is a "family of topological generators" of the site $X_{\text{ét}}$, and hence a family of generators of the topos $\widetilde{X}_{\text{ét}}$. The restriction functor $\widetilde{X}_{\text{ét}} \to \widetilde{C}$ is an equivalence of categories (where C is equipped with the topology induced by that of $X_{\text{ét}}$).*

Immediate from the "comparison lemma." 351

COROLLARY 3.2. *Suppose that X is quasi-separated. Call the full subcategory C of $X_{\text{ét}}$ formed by the schemes étale over X that are of finite presentation over X, equipped with the topology induced by $X_{\text{ét}}$, the restricted étale site of X. Then:*

(i) *C is stable under fiber products and is a site of finite type if X is quasi-compact.*

(ii) *The restriction functor $\widetilde{X}_{\text{ét}} \to \widetilde{C}$ is an equivalence of categories.*

Assertion (i) is immediate, and (ii) follows from 3.1, since C satisfies the condition of 3.1 by the quasi-separatedness of X. Indeed, this implies that, if X' is quasi-compact, for example affine, then it is quasi-compact *over* X (EGA IV 1.2.4, with $Z = \operatorname{Spec} \mathbf{Z}$), and hence is of finite presentation over X if X' is locally of finite type over X.

PROPOSITION 3.3. *Suppose that X is quasi-compact and quasi-separated. Then the functors $H^q(X_{\text{ét}}, F)$ on the category of abelian sheaves on $X_{\text{ét}}$ commute with $\varinjlim$.*

Indeed, by 3.2 (ii), one may replace $X_{\text{ét}}$ by C. Since X is quasi-compact, $X \in \operatorname{Ob} C$, and the conclusion follows from 3.2 (i) and VI 6.1.2 (3).

4. Comparison with other topologies

4.0. First note that the examples of sheaves on $X_{\text{ét}}$ considered in n° 2 are in fact naturally *restrictions* of sheaves defined on $(\mathrm{Sch})_{/X}$ equipped with its étale topology (or even with the fpqc topology). More generally, let 352

$$u^* : X_{\text{ét}} \longrightarrow (\mathrm{Sch})_{/X}$$

be the inclusion functor. It is continuous (III) and commutes with finite $\varprojlim$, and hence defines a *morphism of sites*

$$u : (\mathrm{Sch})_{/X} \longrightarrow X_{\text{ét}},$$

whence a functor

$$u_* : \widetilde{(\mathrm{Sch})_{/X}} \longrightarrow \widetilde{X_{\text{ét}}}$$

on the associated categories of sheaves:

$$u_*(F) = F \circ u,$$

and a left adjoint[34] to the latter:

$$u^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{\mathrm{Sch}_{/X}}.$$

With this understood:

PROPOSITION 4.1. (i) *The functor u^* is fully faithful; hence, for every sheaf G on $X_{\text{ét}}$, the canonical homomorphism $G \to u_*u^*(G)$ is an isomorphism.*

(ii) *Let G (resp. F) be an abelian sheaf on $X_{\text{ét}}$ (resp. $(\mathrm{Sch})_{/X}$). Then the following canonical homomorphisms (defined, for example, as edge homomorphisms of the*
353 *corresponding Leray spectral sequences) are isomorphisms:*

$$H^*(X_{\text{ét}}, G) \xrightarrow{\sim} H^*((\mathrm{Sch})_{/X}, u^*G)$$

$$H^*(X_{\text{ét}}, F \circ u) \xrightarrow{\sim} H^*((\mathrm{Sch})_{/X}, F).$$

Indeed, the functor u is a comorphism, and one can apply (III).

Thus (i) shows that, for the study of the cohomology of sheaves, it is essentially equivalent to work with the "small" étale site $X_{\text{ét}}$ or with the "big" étale site $(\mathrm{Sch})_{/X}$.

4.2. One should also introduce on (Sch) (and hence on the $(\mathrm{Sch})_{/X}$) various topologies other than the étale topology, the most useful of which are defined in SGA 3 IV 6.3. The coarsest among them is the *Zariski topology* (Zar), defined by the pretopology whose covering families are the surjective families of open immersions; it is coarser than the étale topology. The finest of these topologies is the *"faithfully flat quasi-compact" topology*, abbreviated (fpqc), which is the coarsest topology for which the covering families in the Zariski sense, as well as faithfully flat quasi-compact morphisms, are covering; the fpqc topology is finer than the étale topology. As already observed, the various examples of sheaves on $(\mathrm{Sch})_{/X}$ considered in n° 2 are in fact already sheaves for the fpqc topology.

4.2.1. One must bear in mind, however, that for an abelian sheaf F on $(\mathrm{Sch})_{/X}$ for the fpqc topology (or for a topology such as fppf, considered in SGA 3), the cohomology groups of F for the fpqc topology (resp. ...) are not always isomorphic to the cohomology
354 groups for the étale topology. This is so even if X is the spectrum of a field k and F is representable by an algebraic group over k, even for H^1. In general, one can show that the étale topology gives the "correct" cohomology groups for coefficient groups that are *étale*, or more generally *smooth*, group schemes over X[34], but this is no longer true

[34] Strictly speaking, since $(\mathrm{Sch})_{/X}$ is a $\mathscr{U}$-category but *not* a $\mathscr{U}$-site, one cannot invoke III for the existence of u^*. Nevertheless, one constructs u^* easily by

$$u^*(F)(X') = (p_{X'})^*_{\text{ét}}(F)(X') \quad (X' \in \mathrm{Ob}\,\mathrm{Sch}_{/X}),$$

where $p_{X'}$ denotes the structural morphism $X' \to X$. Moreover, to give meaning to 4.1 (ii) and justify the indicated proof of 4.1, one must introduce a universe $\mathscr{V}$ such that $\mathscr{U} \in \mathscr{V}$, and regard the sheaves occurring in the formulas of 4.1 (i) as sheaves with values in $\mathscr{V}$-(Ens). Alternatively, if one wishes to work only with $\mathscr{U}$-sites, one may replace $(\mathrm{Sch})_{/X}$ by a full subcategory of $(\mathrm{Sch})_{/X}$ stable under finite $\varprojlim$, containing the C of 3.1, and $\mathscr{U}$-small.

[34] Cf. A. Grothendieck, Le groupe de Brauer III, Th. 11.7, in "Dix exposés sur la topologie des schémas," North-Holland Publishing Co.

for group schemes such as *radicial* groups over S, for which one must replace the étale topology, still too coarse, by the fpqc or fppf topology.

4.2.2. As an example of the relations between cohomologies for different topologies, consider the Zariski and étale topologies. Denote by X_{Zar} the site of Zariski open subsets of X, so that there is a canonical inclusion functor

$$u^* : X_{\mathrm{Zar}} \longrightarrow X_{\text{ét}}$$

which defines a morphism of sites

$$f : X_{\text{ét}} \longrightarrow X_{\mathrm{Zar}},$$

whence corresponding direct image functors

$$f_* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X_{\mathrm{Zar}}} \quad (f_*(F) = F \circ u),$$

and inverse image functors

$$f^* : \widetilde{X_{\mathrm{Zar}}} \longrightarrow \widetilde{X_{\text{ét}}},$$

adjoint to one another. Geometrically, the pair (f_*, f^*) should be interpreted as a morphism of topoi

$$f : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X_{\mathrm{Zar}}}.$$

One obtains a homomorphism of cohomological functors

$$H^*(X_{\mathrm{Zar}}, f_*(F)) \longrightarrow H^*(X_{\text{ét}}, F)$$

and a Leray spectral sequence 355

$$H^*(X_{\text{ét}}, F) \Longleftarrow H^p(X_{\mathrm{Zar}}, R^q f_*(F)),$$

where F is an abelian sheaf on $X_{\text{ét}}$. This spectral sequence summarizes the general relations between étale and Zariski cohomology. Of course, for coefficient sheaves F such as constant coefficient sheaves, this spectral sequence is in general far from trivial, i.e. in general $R^q f_*(F) \neq 0$ (notably, take the case where X is the spectrum of a field). Nevertheless:

PROPOSITION 4.3. *Let F be a sheaf of varying modules on the scheme X (a sheaf for the Zariski topology), whence a sheaf $F_{\text{ét}}$ on $X_{\text{ét}}$ (cf. 2, c) and a homomorphism of cohomological functors*

$$H^*(X, F) \longrightarrow H^*(X_{\text{ét}}, F_{\text{ét}}).$$

When F is quasi-coherent, the preceding homomorphism is an isomorphism.

Indeed, the first member is precisely $H^*(X_{\mathrm{Zar}}, f_*(F_{\text{ét}}))$, and with the preceding notation it is enough to prove that

$$R^q f_*(F_{\text{ét}}) = 0 \quad \text{for} \quad q > 0.$$

By the method for computing the $R^q f_*$ and the fact that the affine open subsets form a basis for the topology of X, this follows immediately from:

COROLLARY 4.4. *If X is affine, then*

$$H^q(X_{\text{ét}}, F_{\text{ét}}) = 0 \quad \textit{for} \quad q > 0.$$

To see this, let $C = X_{\text{ét}}$, and let C' be the full subcategory formed by the *affine* X'. Then, by 3.1,

$$H^q(C, F_{\text{ét}}) \simeq H^q(C', F|C').$$

It is therefore enough to prove that $F|C'$ is C'-acyclic (V 4.1), or equivalently that 356 it satisfies the condition of V 4.3, i.e. that, for every $X' \in \mathrm{Ob}\, C'$ and every covering

family $\mathscr{R} = (X'_i \to X')_i$ in C', one has $H^q(\mathscr{R}, F) = 0$ for $q > 0$. One may suppose the family $(X'_i)_i$ finite and then, after replacing the X'_i by their coproduct, that the covering family consists of a single covering morphism $X'_i \to X'$, i.e. an (étale and) surjective morphism. We are thus reduced to proving that, if $f : X \to Y$ is a surjective étale morphism of affine schemes and F a quasi-coherent module on Y, then

$$(*) \qquad H^q(X/Y, F) = 0 \quad \text{for} \quad q > 0.$$

This is proved in TDTE I, B, 1.1 (in FGA, cf. reference [4] of IX).

Remarks 4.5. In fact, by *loc. cit.*, for $(*)$ it is enough that $X \to Y$ be surjective and flat (rather than étale). The preceding proof then shows that one still has isomorphisms analogous to that of 4.3, with the étale site $X_{\text{ét}}$ replaced by $(\text{Sch})_{/X}$ equipped with any of the topologies finer than the Zariski topology considered in SGA 3 IV 6.3, for example the fpqc topology.

5. Cohomology of a projective limit of schemes

5.1. Let I be an increasing filtered preordered set,

$$\mathfrak{X} = (X)_{i \in I}$$

a projective system of schemes, the transition morphisms $u_i : X_i \to X_j$ $(i \geq j)$ being
357 *affine*. Recall (EGA IV 8...) that under these conditions, the projective limit

$$X = \varprojlim X_i$$

exists in the category of schemes (and even in the category of locally ringed spaces) and may be constructed as follows: one chooses $i_0 \in I$, so that for $i \geq i_0$ one has an X_{i_0}-isomorphism (EGA II 1.2 and 1.3)

$$X_i = \operatorname{Spec} \mathscr{A}_i,$$

where $\mathscr{A}_i$ is a quasi-coherent Algebra on X_i, the $\mathscr{A}_i$ $(i \geq i_0)$ thus forming an inductive system of such Algebras. Setting

$$\mathscr{A} = \varinjlim_i \mathscr{A}_i,$$

one may take

$$X = \operatorname{Spec} \mathscr{A}.$$

Moreover, denoting by $\operatorname{esp} S$ the underlying topological space of a scheme S, one proves (EGA IV 8...) that the canonical map

$$\operatorname{esp} X \longrightarrow \varprojlim \operatorname{esp}(X_i)$$

is a *homeomorphism* and that the canonical homomorphism of sheaves of rings

$$\varinjlim_i \gamma_i^{-1}(\mathscr{O}_{X_i}) \longrightarrow \mathscr{O}_X$$

is bijective, where $\gamma_i : X \to X_i$ is the canonical continuous map.

5.2. Informally, the content of EGA IV 8, 9 may be summarized by saying that, *when X_{i_0} is quasi-compact and quasi-separated*, then "every datum of a schematic nature
358 on X, of finite presentation over X", is equivalent to a datum of the same nature on one of the X_i, "for sufficiently large i". Thus one proves (EGA IV 8 …):

(a) if $i_1 \in I$ and X'_{i_1}, X''_{i_1} are two schemes of finite presentation over X_{i_1}, then, setting for every $i \geq i_1$

$$X'_i = X'_{i_1} \times_{X_{i_1}} X_i, \quad X' = X'_{i_1} \times_{X_{i_1}} X,$$

and defining X''_i, X'' in the same way, the canonical map

$$\varinjlim_{i \geq i_1} \mathrm{Hom}_{X_i}(X'_i, X''_i) \longrightarrow \mathrm{Hom}_X(X', X'')$$

is bijective.

(b) For every scheme X' of finite presentation over X, there exist an index $i_1 \in I$, a scheme X'_{i_1} of finite presentation over X_{i_1}, and an X-isomorphism

$$X' \simeq X'_{i_1} \times_{X_{i_1}} X.$$

5.3. The preceding result, and analogous results contained in EGA IV 8, may be expressed in the language of the $\varinjlim$ of fibered categories introduced in VI. For this purpose, consider the category $\mathscr{F}$ of morphisms $f : T \to S$ of finite presentation of schemes, and the "target functor"

$$\pi : \mathscr{F} \longrightarrow (\mathrm{Sch}),$$

which is plainly a *fibered functor*, its fiber categories being, moreover, equivalent to $\mathscr{U}$-small categories. Consider the inverse image of the fibered category $\mathscr{F}/(\mathrm{Sch})$ by the functor

$$i \mapsto X_i : \mathrm{cat}(I) \longrightarrow (\mathrm{Sch})$$

(where $\mathrm{cat}(I)$ is the category associated with I, with $\mathrm{Hom}(i, j) \neq \emptyset$ if and only if $i \geq j$), 359
giving a fibered category

$$\pi_{\mathscr{X}} : \mathscr{F}_{\mathscr{X}} \longrightarrow \mathrm{cat}(I),$$

whose fiber at each $i \in I$ is canonically isomorphic to the full subcategory $\mathscr{F}_{X_i}$ of $(\mathrm{Sch})_{/X_i}$ consisting of schemes of finite presentation over X_i, the base-change morphism $\mathscr{F}_i \to \mathscr{F}_j$ relative to $j \geq i$ being none other than $X'_i \mapsto X'_j = X'_i \times_{X_i} X_j$. This being so, by associating to every X'_i of finite presentation over X_i its inverse image on X

$$\varphi(X'_i) = X'_i \times_{X_i} X,$$

one obtains a natural functor

$$\varphi : \mathscr{F}_{\mathscr{X}} \longrightarrow \mathscr{F}_X,$$

which plainly takes every cartesian morphism of $\mathscr{F}_{\mathscr{X}}$ to an isomorphism, and hence, by definition, factors uniquely through a canonical functor

$$(*) \qquad \psi : \varinjlim_{\mathscr{F}_X/\mathrm{Cat}\, I} \mathscr{F}_{\mathscr{X}} \longrightarrow \mathscr{F}_X.$$

In view of the description of the first member, the result of EGA IV 8 recalled above may then be stated by saying that $(*)$ is an *equivalence of categories*.

5.4. Statement a) of 5.2. above may be refined in various ways, by introducing some class $(\mathscr{M})$ of morphisms of schemes, stable under base change, and stating that for a given X_{i_1}-morphism $u_{i_1} : X'_{i_1} \to X''_{i_1}$, the corresponding morphism $u' : X' \to X''$ 360
is $\mathscr{E}(\mathscr{M})$ if and only if there exists $i \geq i_1$ such that the X_i-morphism $u_i : X'_i \to X''_i$ is $\mathscr{E}(\mathscr{M})$. The statement obtained in this way is true, for example, when $(\mathscr{M})$ is one of the following classes of arrows: proper morphisms (respectively projective, resp. quasi-projective, resp. affine, resp. affine, resp. finite, resp. quasi-finite, resp. radicial, resp. surjective, resp. flat, resp. smooth, resp. étale, resp. unramified). The reader will find the corresponding statements in EGA IV, §8 (for the first nine), §11 (for the flat case), and §17 (for the last three cases).

5.5. Now replace $\mathscr{F} \to (\mathrm{Sch})$ by the fibered subcategory $\mathscr{G}$ consisting of *étale morphisms* of finite presentation, whose fiber category $\mathscr{G}_S$, for every scheme S, is none other than the category of schemes étale and of finite presentation over S, and similarly form the fibered category $\mathscr{G}_{\mathfrak{X}}$ over $\mathrm{cat}(I)$, whose fiber category at every $i \in I$ is the category $\mathscr{G}_{X_i}$. By 3.2 (ii), the étale topoi $\widetilde{X_{i_{\text{ét}}}}$, $\widetilde{X_{\text{ét}}}$ are also canonically equivalent to $\widetilde{\mathscr{G}_{X_i}}$, $\widetilde{\mathscr{G}_X}$, (where each $\mathscr{G}_S$ is endowed with the topology induced by $S_{\text{ét}}$). Moreover, by 3.2 (i), each $\mathscr{G}_{X_i}$ is a site of finite type, and is moreover equivalent to a $\mathscr{U}$-small site. In view of the topologies on the $\mathscr{G}_{X_i}$, $\mathscr{G}$ becomes a *fibered site* over $\mathrm{cat}(I)$. This being so, we may state the following

LEMMA 5.6. *The canonical functor*

$$\varinjlim_{\mathscr{G}_{\mathfrak{X}}/\,\mathrm{cat}(I)} \mathscr{G}_{\mathfrak{X}} \longrightarrow \mathscr{G}_X$$

361 *defines an equivalence, respecting the topologies, between the* $\varinjlim$ *site of the restricted étale sites (3.2) of the* X_i *and the restricted étale site of* $X = \varprojlim_i X_i$.

The fact that one obtains an equivalence of categories is one of the statements recalled in 5.4 (the one in which $(\mathscr{M})$ is the class of étale morphisms in (Sch)). The assertion concerning the topologies is obtained in the same way by taking $(\mathscr{M})$ = the class of surjective morphisms in (Sch).

The preceding result allows us to apply the results of VI to the present situation. In particular, we obtain

THEOREM 5.7. *Let* $\mathfrak{X} = (X_i)_{i\in I}$ *be a projective system of schemes, with* I *an increasing filtered preordered set. Suppose that the transition morphisms* $u_{ij} : X_j \to X_i$ *are affine (so that the scheme* $X = \varprojlim X_i$ *is defined (5.1)), and that the* X_i *are quasi-compact and quasi-separated. Consider the fibered site* $\mathscr{G}$ *over* $\mathrm{cat}(I)$ *described in 5.5, whose fiber* $\mathscr{G}_i$ *at* $i \in I$ *is canonically isomorphic to the "restricted étale site" of* X_i *(consisting of schemes étale and of finite presentation over* X_i*). Let* F *be an abelian sheaf on* $\mathscr{G}$ *(i.e.* (3) *a functor* $\mathscr{G}^0 \to (\mathrm{Ab})$ *whose restriction* F_i *to each* $\mathscr{G}_i$ *is a sheaf, i.e. a sheaf on* X_i*). Let* $F_\infty = \varinjlim_i u_i^*(F_i)$ *be the induced sheaf on* X*, where* $u_i : X \to X_i$ *is the canonical morphism. Then the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, F_i) \longrightarrow H^n(X, F_\infty)$$

are isomorphisms.

362 In view of 5.6, this follows by combining VI 8.7.3 and VI 8.5.2. We shall use 5.7 chiefly in the following particular case, which is also made explicit in

COROLLARY 5.8. *With the preceding hypotheses and notation for $(X_i)_{i\in I}$, X, u_{ij}, u_i, let $i_0 \in I$, and let F_{i_0} be an abelian sheaf on X_{i_0}. For every $i \geq i_0$, let $F_i = {u_{i_0 i}}^*(F_{i_0})$, and let moreover $F_\infty = u_i^*(F_{i_0})$. Under these conditions, the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, F_i) \longrightarrow H^n(X, F_\infty)$$

are isomorphisms.

Here is, however, a sometimes useful case that follows from 5.7 and not from 5.8:

COROLLARY 5.9. *With the hypotheses and notation of 5.7 for $(X_i)_{i\in I}$, X, u_{ij}, u_i, let G_{i_0} be a commutative group scheme locally of finite presentation over X_{i_0} (where $i_0 \in I$ is given). For every $i \geq i_0$, let $G_i = G_{i_0} \times_{X_{i_0}} X_i$, and similarly let $G_\infty = G_{i_0} \times_{X_{i_0}} X$. Then the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, G_i) \longrightarrow H^n(X, G_\infty)$$

are isomorphisms.

Indeed, we know that G_{i_0} defines a sheaf on $(\mathrm{Sch})_{/X_{i_0}}$ (cf. 2 (a)), and, supposing that i_0 is an initial object of I (which we may do), hence obtaining a canonical functor $\mathscr{G} \to (\mathrm{Sch})_{/X_{i_0}}$, composition gives a contravariant functor F on $\mathscr{G}$, whose restriction to $\mathscr{G}_i$ is canonically isomorphic to the sheaf on X_i defined by G_i. In particular, F is a sheaf on $\mathscr{G}$, and we may apply 5.7 to it. The hypothesis that G_{i_0} is locally of finite presentation over X_{i_0} serves to ensure that the sheaf F_∞ of 5.7 is isomorphic, by the natural homomorphism $F_\infty \to \mathrm{faisc}(G_\infty)$, to the sheaf $\mathrm{faisc}(G_\infty)$ defined by G_∞ (as follows immediately from EGA IV 8.8.2 (i), by considering the values of the sheaves in question on the objects of the restricted étale site of G_∞ and applying 5.6). This completes the proof of 5.9. 363

COROLLARY 5.10. *With the hypotheses and notation of 5.7 for $(X_i)_{i\in I}$, X, u_{ij}, u_i, let F be a sheaf of commutative groups on X; then the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, {u_i}_*(F)) \longrightarrow H^n(X, F)$$

are isomorphisms.

The statements 5.7 through 5.10 generalize to corresponding statements for the Ext^i of sheaves of Modules, by VI 8.7.9; we shall not repeat them in the present particular case. On the other hand, for later reference we shall make explicit the variants of 5.8 and 5.9 in terms of the functors $R^n f_*$.

COROLLARY 5.11. *Let I be an increasing filtered preordered set, let $(X_i)_{i\in I}$ and $(Y_i)_{i\in I}$ be two projective systems of schemes with affine transition morphisms, $X = \varprojlim_i X_i$, $Y = \varprojlim_i Y_i$, $(f_i)_{i\in I}$ a projective system of quasi-compact and quasi-separated morphisms $f_i : X_i \to Y_i$, $i_0 \in I$, and F_{i_0} a sheaf on X_{i_0}. For every $i \geq i_0$, let F_i be the sheaf on X_i that is the inverse image of F_{i_0}. Similarly, let $f : X \to Y$ be deduced from (f_i) and let F be the sheaf on X that is the inverse image of F_{i_0}. Under these conditions, the canonical homomorphism*

$$\varinjlim_i u_i^*(R^n {f_i}_*(F_i)) \longrightarrow R^n f_*(F)$$

is an isomorphism (where $u_i : X \to X_i$ is the canonical morphism). 364

We may suppose that i_0 is an initial object of I, and, since the question is local on Y_{i_0}, that Y_{i_0} is affine, so that the Y_i and the X_i are quasi-compact and quasi-separated. We are then under the hypotheses for applying VI 8.7.3 (where one reduces to 5.8, in view of the computation of the $R^n f_*$ as sheaves associated with presheaves). In the same way one proves:

COROLLARY 5.12. *The same statement as 5.11, except that now F_{i_0} denotes a commutative group scheme locally of finite presentation over X_{i_0}, and F_i, F its inverse images (in the sense of base change for schemes).*

Note that 5.9 (resp. 5.12) is not a particular case of 5.8 (resp. 5.11), cf. 2 a) remark 2.1.

COROLLARY 5.13. *With the notation of 5.11, let F be a sheaf of abelian groups on X. Then the canonical homomorphisms*

$$\varinjlim_i v_i^*(R^n f_{i*}(u_{i*}(F))) \longrightarrow R^n f_*(F)$$

are isomorphisms (where $u_i : X \to X_i$ and $v_i : Y \to Y_i$ are the canonical homomorphisms).

The proof is essentially the same as before.

365 REMARKS 5.14. a) The preceding limit-passage results remain valid for H^0 resp. f_*, for sheaves *of sets* (instead of abelian sheaves), and for H^1 resp. $R^1 f_*$ for sheaves of groups (not necessarily commutative), where the noncommutative H^1 and $R^1 f_*$ are defined as usual in terms of torsors (or principal homogeneous bundles) (cf. Giraud's thesis [34]). This can be checked directly in the general context of VI. It is certainly possible (and doubtless useful) also to give a variant for the noncommutative H^2 of "links" studied by Giraud.

b) The limit-passage results for fibered sites developed in VI assumed only that the base category was a filtered category, without it being necessary to suppose that it was associated with a filtered preordered set. The case of an arbitrary filtered index category is also the natural framework for the statements developed in the present n°. If we have worked in an overly restrictive framework, it was in order to be able to give correct references to EGA IV, where it is unfortunately assumed, in questions of passage to the limit (from §8 onward), that the index category is defined in terms of a filtered preordered set. We shall nevertheless assume henceforth that all the results used from EGA IV are valid for arbitrary filtered index categories (the proofs given in *loc. cit.* being valid, essentially without change, in this more general case). Thus we shall also use, without further comment, the results of the present n° in the case where I is replaced by an arbitrary filtered index category.

[34] cited p.(page 7) footnote (∗)

EXPOSÉ VIII

Fiber functors, supports, cohomological study of finite morphisms

A. Grothendieck

1. Topological invariance of the étale topos

THEOREM 1.1. *Let* $f : X' \to X$ *be an* integral surjective radicial *morphism. Then* 367
the base-change functor

$$f^* : X_{\text{ét}} \longrightarrow X'_{\text{ét}}$$

is an equivalence of sites (i.e. an equivalence of categories that is continuous, as is every quasi-inverse functor). Consequently, the functors

$$f_* : \widetilde{X'_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$
$$f^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

are equivalences of categories, quasi-inverse to one another.

The first assertion is well known (SGA 1 IX 4.10 and 4.11) when f is of finite presentation or when f is flat; the general case reduces to the case in which f is of finite presentation. Indeed, it is already known that the functor f^* is fully faithful (SGA 1 IX 3.4). It is enough to prove that f^* is essentially surjective, i.e. that every Z' étale over 368
X' "comes from" a Z étale over X. Using the fact that f is a universal homeomorphism and the full faithfulness of f^*, we are reduced to the case where X, X', and Z' are affine, the spectra of rings A, A', and B'. Writing A' as the inductive limit of its finitely generated A-subalgebras A'_i, B' comes from an étale algebra B'_i over some A'_i (cf. EGA IV 8), which reduces us to the case in which A' is integral and of *finite type* over A, hence finite over A. We then have an A-isomorphism $A' \simeq A''/J$, where $A'' \simeq A[t_1, \dots t_n]$ and J is an ideal of A''. Writing J as the inductive limit of its finitely generated subideals J_i, and setting $A'_i = A''/J_i$, we again have $A' = \varinjlim_i A'_i$, so B' comes from an étale algebra B'_i over some A'_i (*loco citato*). On the other hand, since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is surjective, the same holds for the $\operatorname{Spec} A'_i \to \operatorname{Spec} A$; moreover, one easily verifies that since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is radicial, the same holds for $\operatorname{Spec} A'_i \to \operatorname{Spec} A$ for sufficiently large i (apply EGA IV 1.9.9 to Spec A[34]). This reduces us to the case in which A' is one of the A'_i, hence is of finite presentation over A.

This proves the first assertion of 1.1. The assertion that f_* and f^* are mutually quasi-inverse equivalences follows at once.

COROLLARY 1.2. *Let* F *be an abelian sheaf on* X*, and let* F' *be its inverse image on* X'*. Then the canonical homomorphisms*

$$(*) \qquad H^i(X, F) \longrightarrow H^i(X', F')$$

[34] Let $S_i \subset S = \operatorname{Spec} A$ be the set of $s \in S$ such that the fiber of $X_i = \operatorname{Spec} A'_i$ at s is of separable rank 1. Then the S_i are *constructible* subsets of S (EGA IV 9.7.9) forming an increasing family; their union is S by the hypothesis on $S' = \operatorname{Spec} A' \to S$ and the fact that the fibers of S'_i are *noetherian* schemes. Thus $S = S_i$ for sufficiently large i, by EGA IV 1.9.9, C.Q.F.D..

are isomorphisms. Likewise, if F' is an abelian sheaf on X' and F its direct image on X, the canonical homomorphisms $(*)$ are isomorphisms.

369 EXAMPLES 1.3. Here are some frequently used applications of 1.1:

a) X' is a subscheme of X having the same underlying set as X, i.e. defined by a nil-ideal I.
b) X is a scheme over a separably closed field k, k' is an algebraic closure of k, and $X' = X \otimes_k k'$.
c) Let X be a geometrically unibranch scheme (for example, an algebraic curve over a field k whose only singularities are "cuspidal" singularities, thus excluding ordinary double points in particular). Then, if X' is the normalization of X_{red}, by definition $X' \to X$ is radicial, so 1.1 applies.

REMARKS 1.4. A morphism f as in 1.1 is a *universal homeomorphism*, i.e. it is a homeomorphism and remains so after every base extension. Conversely, if f is a universal homeomorphism, one proves that f satisfies the hypotheses of 1.1 [34], which explains the title of this section.

Let us point out explicitly, in connection with Example 1.3 c), that if X is an algebraic curve (over an algebraically closed field k, to fix ideas) having at least one singular point that is not "unibranch" (for example, an ordinary double point), then (with X' denoting the normalization of X_{red}) the conclusion of 1.2 already fails for H^1 with constant coefficients (compare SGA 1 I 11 a) and SGA 1 IX 5).

2. Sheaves on the spectrum of a field

370 PROPOSITION 2.1. *Let k be a field, $\overline{k}$ a separable closure of k, $\pi = \mathrm{Gal}(\overline{k}/k)$ its topological Galois group, $X = \operatorname{Spec} k$, and $\overline{X} = \operatorname{Spec} \overline{k}$ (on which π acts on the left). Consider the canonical functor*

$$i : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}}$$

(defined by $i(X')(X'') = \mathrm{Hom}_X(X', X'')$), and the canonical functor

$$j : \widetilde{X_{\text{ét}}} \longrightarrow \mathscr{B}_\pi$$

from the category of sheaves on $X_{\text{ét}}$ to the category $\mathscr{B}_\pi$ (cf. IV 2.7) of (discrete) sets on which π acts continuously on the left, defined by

$$j(F) = \varinjlim_{\alpha} F(\operatorname{Spec}(k_\alpha)),$$

where k_α ranges over the finite subextensions of $\overline{k}$. Then i and j are equivalences of categories. (Consequently, the topos $\widetilde{X_{\text{ét}}}$ is equivalent (IV 3.4) to the "classifying topos" $\mathscr{B}_\pi$..)

The composite functor

$$ji : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}} \longrightarrow (\pi\text{-Ens})$$

is an equivalence of categories by the fundamental theorem of Galois theory (in the form of SGA I V). Since $\mathscr{B}_\pi$ is plainly a topos (II 4.14), the same is therefore true of $X_{\text{ét}}$. Moreover, a family $(X_i \to X)$ of morphisms in $X_{\text{ét}}$ is covering if and only if it is surjective, i.e. its image in $\mathscr{B}_\pi$ is surjective, or, equivalently, covering for the canonical topology of $\mathscr{B}_\pi$. This shows that the topology of $X_{\text{ét}}$ is indeed its canonical topology. Consequently i is an equivalence, and hence so is j.

371 COROLLARY 2.2. *The functor j of 2.1 induces an equivalence between the category of abelian sheaves on $X = \operatorname{Spec} k$ and the category of Galois π-modules.*

Indeed, the latter are precisely the "abelian groups" of the topos $\mathscr{B}_\pi$. The same argument, by "transport of structure," gives:

COROLLARY 2.3. *Let F be an abelian sheaf on $X = \operatorname{Spec} k$, and let $M = j(F)$ be the associated Galois π-module. Then there is a canonical isomorphism of ∂-functors in F:*

$$H^*(X, F) \simeq H^*(\pi, M),$$

(where the right-hand side denotes Galois cohomology, studied, for example, in Serre's course C.G.*).*

COROLLARY 2.4. *Suppose that k is separably closed. Then the functor*

$$\Gamma : \widetilde{X_{\text{ét}}} \longrightarrow (\text{Ens})$$

is an equivalence of categories. If F is an abelian sheaf on $X = \operatorname{Spec} k$, then $H^i(X, F) = 0$ for $i \neq 0$. (In other words, the topos $\widetilde{X_{\text{ét}}}$ is a "punctual topos" (IV 2.2)*.)*

COROLLARY 2.5. *Let k' be a separably closed extension of a separably closed field k, let $X = \operatorname{Spec} k$, $X' = \operatorname{Spec} k'$, and let $u : X' \to X$ be the canonical morphism. Then the functor*

$$u^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

is an equivalence of categories.

3. Fiber functors relative to geometric points of a scheme

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3.1. Let X be a scheme. We shall call a *geometric point* of X any X-scheme ξ that is the spectrum of a *separably closed* field Ω. The datum of such a point is therefore equivalent to that of an ordinary point $x \in X$ together with a *separably closed* extension Ω of the residue field $k(x)$ of x. The case most frequent for us will be that in which Ω is a *separable closure* of $k(x)$, which we shall then generally denote by $\overline{k(x)}$; the corresponding geometric point of X is then denoted by $\overline{x}$. For a given $x \in X$, the $\overline{k(x)}$ (resp. $\overline{x}$) are determined up to *nonunique* k-isomorphism (resp. X-isomorphism).

REMARKS 3.2. In most geometric questions, it is more natural to restrict to the case in which Ω is *algebraically closed*. The different convention used here is specific to the study of the étale topology. It is the convention we adopted in the definition of the fundamental group in SGA 1 V 7. Note, however, that the developments in *loco citato* are equally valid with the convention adopted here (because the property of Ω used there is that every étale covering of the spectrum of Ω is trivial, i.e. precisely that Ω is separably closed).

DEFINITION 3.3. *Let X be a scheme, let ξ be a geometric point of X, and let $u : \xi \to X$ be its structural morphism. The (geometric)* fiber functor *relative to ξ, denoted $F \mapsto F_\xi$, is the composite functor*

$$\widetilde{X_{\text{ét}}} \longrightarrow (\text{Ens})$$

of

$$\widetilde{X_{\text{ét}}} \xrightarrow{u^*} \widetilde{\xi_{\text{ét}}} \xrightarrow{\Gamma_\xi} (\text{Ens}).$$

Equivalently, by 2.4, one may say that a geometric point ξ of X defines a point of the topos $X^\sim_{\text{ét}}$ (IV 6.1), whose associated fiber functor (*loc. cit.*) is simply being taken.

[34] Cf. EGA IV 8.11.6 if f is of finite presentation, and EGA IV 18.12.11 in the general case.

373 **3.4.** Since the functor Γ_ξ (the global-sections functor on ξ) is an *equivalence* of categories (12.4), knowing the fiber functors $F \mapsto F_\xi$ is essentially equivalent to knowing the inverse-image functors u^*. Moreover, it follows at once from 2.5 that if ξ is a geometric point of X for which there exists an X-morphism $\xi' \to \xi$ (i.e. ξ' corresponds to a separably closed extension Ω' of Ω), then the canonical homomorphism $F_\xi \to F_{\xi'}$ is a *functorial isomorphism*

$$F_\xi \simeq F_{\xi'}.$$

This shows that, for the study of fiber functors, one may (after replacing Ω by the separable algebraic closure of k in Ω) restrict oneself to the case where $\omega = k(\xi)$ is a separable closure $\overline{k(x)}$ of $k(x)$, and that the fiber functors corresponding to geometric points of X lying over the same $x \in X$ are isomorphic to one another (noncanonically).

In accordance with the conventions of 3.1, we shall denote by $F_{\overline{x}}$ a fiber functor corresponding to the choice of a separable closure $\overline{k(x)}$ of $k(x)$.

Let us point out an evident transitivity property (which shows the technical usefulness of not restricting oneself exclusively to geometric points defined by separable closures of residue fields): Let $f = X' \to X$ be a morphism of schemes, and let $u' : \xi' \to X'$ be a geometric point of X. Then $u = fu' : \xi' \to X$ is a geometric point of X, which we shall denote by ξ; relative to these geometric points, there is an isomorphism, functorial in $F \in \mathrm{Ob}\, \widetilde{X_{\text{ét}}}$:

$$f^*(F)_{\xi'} \simeq F_\xi.$$

374 Indeed, this follows from the transitivity formula for inverse-image functors

$$(fu)^* \xrightarrow{\sim} u^* f^*.$$

THEOREM 3.5. *Let X be a scheme.*

a) *For every geometric point ξ of X, the fiber functor $F \mapsto F_\xi$, $\widetilde{X_{\text{ét}}} \to (\mathrm{Ens})$ commutes with arbitrary $\varinjlim$ (indexed by categories $\in \mathcal{U}$) and with finite $\varprojlim$[34]. In particular, it takes sheaves of groups (resp. abelian sheaves, etc.) to groups (resp. abelian groups, etc.).*

b) *As x ranges over the points of X, the fiber functors $F_{\overline{x}}$ form a "conservative" family of functors, i.e. if $u : F \to G$ is a homomorphism of sheaves on X, then u is an isomorphism if and only if, for every $x \in X$, the corresponding homomorphism $u_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$ is an isomorphism*[34].

Taking into account the exactness properties in a), let us immediately note the following formal consequences of b):

COROLLARY 3.6. *Let $u : F \to G$ be a homomorphism of sheaves. Then u is a monomorphism (resp. an epimorphism) if and only if, for every $x \in X$, $u_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$ is injective (resp. surjective).*

COROLLARY 3.7. *Let $u, v : F \to G$ be two morphisms of sheaves on X. Then $u = v$ if and only if, for every $x \in X$, one has $u_{\overline{x}} = v_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$. In particular, if u, v are two*
375 *sections of F, one has $u = v$ if and only if $u_{\overline{x}} = v_{\overline{x}}$ in $F_{\overline{x}}$ for every $x \in X$.*

COROLLARY 3.8. *Let $F \to G \to H$ be a sequence of two homomorphisms of sheaves on X. Then the sequence is exact if and only if, for every $x \in X$, the corresponding sequence $F_{\overline{x}} \to G_{\overline{x}} \to H_{\overline{x}}$ is exact.*

[34] Thus it is indeed a "fiber functor" in the sense of Exposé IV.

[34] Thus there are "enough fiber functors" of the form $F \mapsto F_{\overline{x}}$, in the terminology of IV 6.5.

We leave to the reader the task of deducing the corollaries [34], which will moreover be proved in part in the proof that follows.

Assertion a) follows from the exactness properties already noted in (VII 1.4) for every inverse-image functor, such as $u^* : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$, and from the fact that $\widetilde{\xi_{\text{ét}}} \to (\text{Ens})$ is an equivalence. To prove b), let us note the following formula, which gives a procedure for *computing* fiber functors:

PROPOSITION 3.9[34]. *Let $\xi \xrightarrow{u} X$ be a geometric point of X, and let C_ξ be the category of "pointed étale X-schemes at ξ", i.e. of pairs (X', u'), where $X' \in \text{Ob}\, X_{\text{ét}}$ is a scheme étale over X, and u' is an X-morphism $\xi \to X'$. Then the opposite category C_ξ^0 is filtered and, for a varying sheaf F (resp. presheaf P) on X, there is a canonical functorial isomorphism*

$$F_\xi \simeq \varinjlim_{C_\xi^0} F(X'), (\text{resp. } (aP)_\xi \simeq \varinjlim_{C_\xi^0} P(X')),$$

(where aP = the sheaf associated with P).

Indeed, let us first note that for every presheaf P on $\xi_{\text{ét}}$, if aP is the associated sheaf, the natural homomorphism

$$P(\xi) \longrightarrow aP(\xi)$$

is an isomorphism, as follows, for example, immediately from the explicit computation of aP as $L(LP)$ (II 3.19). Let $\varphi : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ be the "inverse-image" functor. Then 376
$u^* = \varphi^*$ is nothing other (III 2.3. (d)) than the composite $a\varphi^\cdot i$, where $i : \widetilde{X_{\text{ét}}} \to \widehat{X_{\text{ét}}}$ is the inclusion functor, and $a : \widehat{\xi_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ is the "associated sheaf" functor. By the preceding remark we find

$$u^*(F)(\xi) = (\varphi^\cdot i(F))(\xi),$$

so it is enough to apply the explicit expression for $\varphi^\cdot$ given in the relevant proof. The corresponding case, where one starts with a presheaf P on X, is proved in the same way, by using the functorial isomorphism $\varphi^* a(P) \simeq a\varphi^*(P)$. The fact that the category C_ξ is filtered (which remains true for *every* X-scheme ξ, not necessarily reduced to a geometric point) follows immediately from the fact that finite projective limits exist in C_ξ, which follows from the fact that $X_{\text{ét}}$ is a subcategory of $(\text{Sch})/X$ stable under finite projective limits.

Now let $u : F \to G$ be a homomorphism of sheaves on X such that, for every $x \in X$, $u_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$ is a monomorphism; let us prove that u is a monomorphism. For this, we must prove that if $X' \in \text{Ob}\, X_{\text{ét}}$, $\varphi, \psi \in F(X')$ are such that $u(\varphi) = u(\psi)$, then $\varphi = \psi$. Replacing X by X' and using the transitivity of fiber functors, we may suppose $X = X'$. For every $x \in X$, we then have $u(\varphi)_{\overline{x}} = u(\psi)_{\overline{x}}$, i.e. $u_{\overline{x}}(\varphi_{\overline{x}}) = u_{\overline{x}}(\psi_{\overline{x}})$, hence $\varphi_{\overline{x}} = \psi_{\overline{x}}$ since $u_{\overline{x}}$ is a monomorphism. Using 3.9, it follows that there exists an $X'_x \in \text{Ob}\, X_{\text{ét}}$, whose image contains x, such that φ and ψ have the same inverse image on X'_x. Since, as x varies in X, the X'_x form a covering family of X, it follows that $\varphi = \psi$.

This proves that if the $u_{\overline{x}}$ are monomorphisms, then so is u. Suppose in addition that 377
the $u_{\overline{x}}$ are epimorphisms; let us prove that the same holds for u, and hence that u is an isomorphism. It remains to prove that for every $X' \in \text{Ob}\, X_{\text{ét}}$ and every $\psi \in G(X')$, there exists $\varphi \in F(X')$ such that $u(X')(\varphi) = \psi$. We may again suppose $X' = X$, and using 3.9, we see that, for every $x \in X$, there exists $X'_x \in \text{Ob}\, X_{\text{ét}}$ such that the image of X'_x in X contains x, and a $\varphi_x \in F(X'_x)$ whose image in $G(X'_x)$ is the inverse image of

[34] Cf. IV 6.5.2.

[34] Cf. IV 6.8.3

ψ. Using the fact that u is a monomorphism, we see that the φ_x agree on the $X'_x \times_X X'_x$ $(x, y \in X)$ and therefore come from a uniquely determined $\varphi \in F(X)$; one then has $u(\varphi) = \psi$, since the inverse images on the X'_x agree. C.Q.F.D..

Scholium 3.10. Theorem 3.5 implies that the collection of "fiber functors" for the étale topology enjoys the same essential properties as in the usual theory of sheaves on a topological space. We shall use 3.5 and its corollaries very frequently, and for this reason we shall generally dispense with referring to them explicitly.

3.11. For every $x \in X$ we shall also, by abuse of language, denote by x the $X-$scheme $\operatorname{Spec} k(x)$, for the usual structural morphism

$$i_x : x = \operatorname{Spec} k(x) \longrightarrow X.$$

It gives rise to a canonical functor

$$i_x^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{x_{\text{ét}}}.$$

If F is a sheaf on X, we shall put $F_x = i_x^*(F)$; thus it is a sheaf on $x = \operatorname{Spec} k(x)$ (and
378 as such may also be identified with an étale scheme over x, by virtue of 2.1). It depends functorially on F, and the functor $i_x^* : F \mapsto F_x$ still commutes with arbitrary $\varinjlim$ and finite $\varprojlim$, by virtue of VII 1.4.

If $\overline{x} = \operatorname{Spec} \overline{k(x)}$, the fiber functor $F \mapsto F_{\overline{x}}$ is canonically isomorphic to the composite of the functor $F \mapsto F_x$ and the functor j of 2.1 (taking into account that the latter is isomorphic to the fiber functor relative to the geometric point $\operatorname{Spec} \overline{k}$ of $\operatorname{Spec} k$). It follows immediately from this and from 3.5 that the system of functors $(F \mapsto F_x)$ $(x \in X)$ is likewise conservative. Moreover, if $k(x)$ is separably closed, i.e. $x = \overline{x}$, the notations F_x and $F_{\overline{x}}$ are, strictly speaking, in conflict (the first denoting an object of $\widetilde{x_{\text{ét}}}$, the second a set), but by virtue of 2.1 this conflict is inessential.

3.12. The preceding remarks show that if $\overline{x} = \operatorname{Spec} k(x)$, then, for every sheaf F on X, the group $\pi_x = \operatorname{Gal}(\overline{k(x)}/k(x))$ acts naturally (on the left) on the fiber set $F_{\overline{x}}$, so that $F \mapsto F_{\overline{x}}$ may in fact be regarded as a functor

$$\widetilde{X_{\text{ét}}} \longrightarrow \mathscr{B}_{\pi_x}$$

whose knowledge is essentially equivalent (thanks to 2.1) to that of the functor $F \mapsto F_x$.

This shows, in an important special case, that contrary to what happens for sheaves on topological spaces, "fiber functors" generally admit nontrivial automorphisms. In general, we shall determine in 7.9 all the homomorphisms from one fiber functor to another.

379 Remark 3.13. Here is a sometimes useful generalization of 3.5 b). Consider a subset E of X such that, for every étale morphism $f : X' \to X$, $E' = f^{-1}(E)$ is "very dense" in X' (EGA IV 10.1.3), i.e. for every f as above, every open subset U of X' which contains E' is equal to X'. Then the fiber functors $F_{\overline{x}}$ on $\widetilde{X_{\text{ét}}}$, for $x \in E$, already form a "conservative family" (and consequently Corollaries 3.6, 3.7, and 3.8 remain valid when one restricts oneself to choosing $x \in E$ in them). This follows immediately by repeating the proof given for 3.5 b). The preceding result applies in particular in the following situations:

a) X is a Jacobson scheme (EGA IV 10.4.1.), for example locally of finite type over a field, or over $\operatorname{Spec}(\mathscr{Z})$, and E is the set of closed points of X.
b) X is locally of finite type over a scheme S, and E is the set of points of X which are closed in their fiber.

c) X is locally noetherian, and E is the set of $x \in X$ such that $\overline{x}$ is a finite set (i.e. a semilocal scheme of dimension ≤ 1) cf. EGA IV 10.5.3 and 10.5.5.

4. Strictly local rings and schemes

4.1. Recall that, following Azumaya, a local ring A is said to be *henselian* if it satisfies the following equivalent conditions:

(i) Every finite A-algebra B is a direct product $\prod_i B_i$ of local rings (the B_i are then necessarily isomorphic to the $B_{\mathfrak{m}_i}$, where $\mathfrak{m}_i$ runs through the maximal ideals of B). 380

(ii) Every A-algebra B which is the localization at a prime ideal $\mathfrak{p}$ lying over the maximal ideal $\mathfrak{m}$ of A of an A-algebra C of finite type, and such that $B/\mathfrak{m}B$ is finite over $k = A/\mathfrak{m}$, is finite over A.

(iii) The same as (ii), but with C assumed étale over A at $\mathfrak{p}$ and $k \to B/\mathfrak{m}B$ an isomorphism.

(iv) "Hensel's lemma" in its classical form holds for A, that is, for every monic polynomial $F \in A[T]$, writing $\dot{F} \in k[T]$ for the corresponding reduced polynomial, and for every factorization

$$\dot{F} = \dot{F}_1 \dot{F}_2$$

of $\dot{F}$ as a product of two relatively prime monic polynomials, $\dot{F}_1$ and $\dot{F}_2$ lift (necessarily uniquely) to $F_1, F_2 \in A[T]$ such that $F = F_1 F_2$.

For the proof of these equivalences and the general properties of henselian rings, see EGA IV 18.5.11. Recall that if A is any local ring, one proves (Nagata) that there exists a local homomorphism

$$A \longrightarrow A^h$$

with A^h *henselian*, which is *universal* for local homomorphisms from A to local henselian rings. The ring A^h is called the henselization of A. The construction given in EGA IV 18.6 (different from Nagata's) consists in setting

$$A^h = \varinjlim_i A_i,$$

where A_i runs through the A-algebras essentially of finite type and essentially étale over A (that is, localizations of an étale A-algebra) such that $A \to A_i$ is a local homomorphism and the residue extension $k(A_i)/k(A)$ is trivial. This construction shows in particular that A^h is flat over A, and that if A is Noetherian, then so is its henselization. 381

Definition 4.2. *A local ring A is said to be strictly local* if it satisfies one of the following equivalent conditions*:*

(i) *A is henselian (4.1) and its residue field is separably closed.*

(ii) *Every local homomorphism $A \to B$, where B is a localization of an étale algebra over A, is an isomorphism.*

A scheme is called a strictly local scheme *if it is isomorphic to the spectrum of a strictly local ring.*

The equivalence of (i) and (ii) follows at once from 4.1. Note that, in geometric form, 4.2 (ii) (respectively 4.1 (iii)) says that for every étale Y-scheme X and every point x of

the fiber X_y (respectively every point of X_y *rational over* $k(y)$), where y is the closed point of Y, there exists a section of X over Y passing through x (and this section is necessarily unique).

DEFINITION 4.3. *Let X be a scheme, let $\mathcal{O}_{X_{\text{ét}}}$ be the sheaf on $X_{\text{ét}}$ defined by*

$$\mathcal{O}_{X_{\text{ét}}}(X') = \Gamma(X', \mathcal{O}_{X'})$$

(compare VII 2. c)*, and let $u : \xi = \operatorname{Spec}(\Omega) \to X$ be a geometric point of X. The* strictly
382 local ring of X at ξ*, denoted $\mathcal{O}_{X,\xi}$ or simply $\mathcal{O}_\xi$, is the fiber of the sheaf $\mathcal{O}_{X_{\text{ét}}}$ at the point ξ. Its spectrum is called the* strict localization of X at ξ.

By 3.9, the fiber $\mathcal{O}_{X,\xi}$ has the description

$$(*) \qquad \mathcal{O}_{X,\xi} = \varinjlim \Gamma(X', \mathcal{O}_{X'}),$$

where X' runs through the opposite filtered category of étale X-schemes pointed by ξ, that is, equipped with an X-morphism $\xi \to X'$. By transitivity of inductive limits, one may replace $\Gamma(X', \mathcal{O}_{X'})$ in the right-hand side by $\mathcal{O}_{X',x'}$, where x' is the image of ξ in X', and one obtains

$$(**) \qquad \mathcal{O}_{X,\xi} = \varinjlim_i A_i$$

where A_i runs through the algebras essentially of finite type and étale over $A = \mathcal{O}_{X,x}$, equipped with a k-homomorphism $k(A_i) \to \Omega$. (Here x denotes the image of ξ in X.) The transition homomorphisms between the A_i are, of course, the local homomorphisms of A-algebras $A_i \to A_j$ that make the corresponding triangle commute:

$$\begin{array}{ccc} k(A_i) & \longrightarrow & k(A_j) \\ & \searrow & \downarrow \\ & & \Omega. \end{array}$$

This implies that if there exists an admissible homomorphism from A_i to A_j, it is unique. Thus the inductive limit $(**)$ is a limit over an upward-filtered ordered index set.

383 **4.4.** The description $(**)$ shows that the strictly local ring of X at ξ depends essentially only on the usual local ring $A = \mathcal{O}_{X,x}$ $(x = u(\xi))$ and on the extension Ω of $k = k(A)$. It is also called the *strict henselization* of A (relative to the given separably closed extension Ω of k), and is denoted A^{hs}. We refer to EGA IV 18.8 for the general properties of A^{hs}. We merely point out that the construction again shows that A^{hs} is flat over A, and that it is Noetherian if A is. Moreover, one proves easily (*loc. cit.*) that the local homomorphism

$$A \longrightarrow A^{hs}$$

together with the k-homomorphism

$$k(A^{hs}) \longrightarrow \Omega$$

solves the universal problem, relative to the data of A and the extension Ω of k, of finding local homomorphisms

$$A \longrightarrow A',$$

where A' is *strictly local*, equipped with a k-homomorphism

$$k(A') \longrightarrow \Omega.$$

In particular, A^{hs} *is indeed a strictly local ring* (which justifies the terminology of 4.3).

4.5. Keep the notation of 4.3, and choose an affine open neighborhood U of x. Note that in 3.9 one may plainly restrict to those X' which are affine over U. These then form a pseudo-filtered inverse system of affine schemes, so that we are in the general situation 384
of VII 5 (see VI 5.12 b). By the expression $(*)$ for $\mathcal{O}_{X,\xi}$, there is an X-isomorphism

$$(***) \qquad \varprojlim_{\substack{X' \text{ étale} \\ \text{affine over } U, \\ X'_\xi\text{-pointed}}} X' \simeq \operatorname{Spec} \mathcal{O}_{X,\xi}$$

which makes precise the intuitive geometric meaning of $\operatorname{Spec}(\mathcal{O}_{X,\xi})$ as the "*limit of the étale neighborhoods of* ξ".

Proposition 4.6. *Let X be a strictly local scheme, and let x be its closed point, which is therefore a geometric point of X. Then the functor $F \mapsto \Gamma(X,F) : \widetilde{X_{\text{ét}}} \to (\text{Ens})$ is canonically isomorphic to the fiber functor $F \mapsto F_x$, and hence commutes with arbitrary* $\varinjlim$[34].

Indeed, by Definition 4.2 (ii), the final object X of the category of étale X-schemes pointed by x considered in 3.9 dominates every other object; hence $\varinjlim F(X')$ is canonically isomorphic to $F(X)$, as required.

In particular, the functor induced by Γ on the category of abelian sheaves is exact, and therefore:

Corollary 4.7. *Under the hypotheses of 4.6, for every abelian sheaf F on X one has*

$$H^q(X,F) = 0 \quad \text{for} \quad q \neq 0.$$

Corollary 4.8. *Let X be a scheme, let ξ be a geometric point of X, let $\overline{X} = \operatorname{Spec} \mathcal{O}_{X,\xi}$ be the corresponding strict localization, let F be a sheaf on X, and let $\overline{F}$ be its inverse image on $\overline{X}$. Then there is a functorial isomorphism*

$$F_\xi \simeq \Gamma(\overline{X}, \overline{F}).$$

Indeed, ξ may also be regarded as a geometric point of $\overline{X}$, and it suffices to apply 385
4.6 and the transitivity of fibers (3.4).

5. Application to the computation of the fibers of the $R^q f_*$

5.1. Let

$$f : X \longrightarrow Y$$

be a morphism of schemes and let F be an abelian sheaf on X. We wish to determine the $R^q f_*(F)$. By 3.5, this amounts, up to very little, to knowing the geometric fibers $R^q f_*(F)_{\overline{y}}$ for $y \in Y$. Now (V 5.1) $R^q f_*(F)$ is the sheaf associated with the presheaf

$$\mathscr{H}^q : Y' \longmapsto H^q(X \times_Y Y', F)$$

on Y, and hence (3.9)

$$R^q f_*(F)_{\overline{y}} = \varinjlim_{Y'} \mathscr{H}^q(Y'),$$

where the inductive limit is taken over the opposite filtered category of the étale Y-schemes pointed by $\overline{y}$. After choosing an affine open neighborhood U of y, one may

[34] not necessarily filtered!

restrict the limit on the right to the cofinal category of those Y' which are affine and lie over U. Thus one obtains

$$R^q f_*(F)_{\overline{y}} \simeq \varinjlim H^q(X \times_Y Y', F), \tag{5.1.1}$$

where Y' varies in the preceding category.

Put

$$\overline{Y} = \mathrm{Spec}(\mathcal{O}_{Y,\overline{y}}) = \varprojlim Y'$$

(see 4.5), and

$$\overline{X} = X \times_Y \overline{Y}.$$

386 Observe that in the inverse system $X \times_Y Y'$ (deduced from that of the Y' by the base change $X \to Y$), the transition morphisms are affine. We are therefore in the general situation of VII 5.1, and the construction of $\varprojlim$ in *loc. cit.* immediately gives a canonical isomorphism

$$\overline{X} = \varprojlim X \times_Y Y'.$$

Let $\overline{F}$ denote the sheaf on $\overline{X}$ that is the inverse image of F. We then obtain a canonical homomorphism

$$\varinjlim_{Y'} H^q(X \times_Y Y', F) \longrightarrow H^q(\overline{X}, \overline{F}),$$

and hence, by comparison with (5.1.1), a canonical homomorphism

$$R^q f_*(F)_{\overline{y}} \longrightarrow H^q(\overline{X}, \overline{F}), \tag{5.1.2}$$

which is plainly functorial in F.

Suppose now that $f : X \to Y$ is quasi-compact and quasi-separated. Then the same is true of $X \times_Y Y' \to Y'$, and since Y' is affine, the schemes $X \times_Y Y'$ are quasi-compact and quasi-separated. Applying (5.1.1) and the limit-passage theorem VII 5.8, one obtains:

THEOREM 5.2. *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes, let F be an abelian sheaf on X, let y be a point of Y, and let $\overline{y}$ be the geometric point over y associated with a separable closure $k(\overline{y})$ of $k(y)$. Let $\overline{Y} = \mathrm{Spec}(\mathcal{O}_{Y,\overline{y}})$ be the corresponding strict localization, let $\overline{X} = X \times_Y \overline{Y}$, and let $\overline{F}$ be the inverse image of F on $\overline{X}$. Then the canonical homomorphism (5.1.2) is an isomorphism:*

$$R^q f_*(F)_{\overline{y}} \simeq H^q(\overline{X}, \overline{F}).$$

387 This statement practically reduces the determination of the fibers of a sheaf $R^q f_*(F)$ to the determination of the cohomology of a scheme over a strictly local scheme, and it will be used constantly below. Technically, 5.2 explains the important role of henselian and strictly local rings in the study of étale cohomology.

REMARK 5.3. The statement remains valid for $R^0 f_*(F) = f_*(F)$, for an arbitrary sheaf of sets. Moreover, 5.2 also remains valid for $R^1 f_*(F)$ when F is a sheaf of groups (not necessarily commutative), with the usual definition of $R^1 f_*(F)$ (see Giraud's thesis).

5.4. Suppose now that f is a *finite* morphism. Then $\overline{f} : \overline{X} \to \overline{Y}$ is finite, and since $\overline{Y}$ is strictly local, it follows that $\overline{X}$ is a finite disjoint union of strictly local schemes $\overline{X}_i$. Hence, using 4.7, one finds

$$H^q(\overline{X}, \overline{F}) = 0 \quad \text{for} \quad q \neq 0. \tag{5.4.1}$$

Furthermore, the components $\overline{X}_i$ of $\overline{X}$ correspond to the points $\overline{x}_i$ of $\overline{X}$ above the closed point $\overline{y}$ of $\overline{Y}$, that is, to the points of $\overline{X}_{\overline{y}} = X_y \otimes_{k(y)} k(\overline{y})$. These points may also be regarded as geometric points of X, and $\overline{X}_i$ is then precisely the strict localization of X at $\overline{x}_i$. Thus (4.8)

$$H^0(\overline{X}_i, \overline{F}) \simeq F_{\overline{x}_i},$$

whence

$$H^0(\overline{X}, \overline{F}) = \prod_i F_{\overline{x}_i}. \tag{5.4.2}$$

This is an isomorphism functorial in the sheaf of sets F (there is no need here to restrict to abelian sheaves). Taking 5.2 and 5.3 into account, formulas (5.4.1) and (5.4.2) give: 388

Proposition 5.5. *Let $f : X \to Y$ be a finite morphism of schemes, and let y be a point of Y. Then for every sheaf F on X, there is an isomorphism, functorial in F,*

$$f_*(F)_{\overline{y}} \simeq \prod_{\overline{x} \in X_y \otimes_{k(y)} k(\overline{y})} F_{\overline{x}}.$$

Consequently, the formation of $f_*(F)$ commutes with every base change $Y' \to Y$; and if F is an abelian sheaf, then

$$R^q f_*(F) = 0 \quad \text{if} \quad q > 0.$$

Note that the first formula of 5.5 is in fact independent of 5.2 and of the general limit-passage theorem VII 5.7; together with 3.5, it implies that $F \mapsto f_*(F)$ is an *exact functor* on the category of abelian sheaves, and hence again that $R^q f_*(F) = 0$ for $q \neq 0$. Here is a slight variant of 5.5:

Corollary 5.6. *Let $f : X \to Y$ be an integral morphism. Then for every abelian sheaf F on X one has*

$$R^q f_*(F) = 0 \quad \text{for} \quad q \neq 0.$$

Moreover, the functor f_* on sheaves of sets commutes with every base change $Y' \to Y$.

Indeed, by 5.2 it is enough to prove that when Y is strictly local, $H^q(X, F) = 0$ for every abelian sheaf F on X. We then have $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, where B is an integral A-algebra; writing $B = \varinjlim B_i$, where B_i runs through the finite-type 389 subalgebras of B (which are in fact finite over A), one has $X = \varprojlim_i X_i$, where $X_i = \operatorname{Spec} B_i$. By VII 5.13, it is enough to prove that $R^q f_*(F_i) = 0$ for $q \neq 0$, where F_i is an abelian sheaf on X_i; this follows from 5.5. The last assertion of 5.6 is proved by the same reduction to 5.5.

Corollary 5.7. *Let $f : X \to Y$ be an immersion. Then for every sheaf F on X, the canonical morphism*

$$f^* f_*(F) \longrightarrow F$$

is an isomorphism.

Since f factors as the composite of an open immersion and a closed immersion, and 5.7 is trivial for an open immersion (IV no. 3), it remains to consider the case where f is a closed immersion. We must prove that for every $x \in X$, the corresponding homomorphism

$$(f^* f_*(F))_{\overline{x}} \longrightarrow F_{\overline{x}}$$

is bijective. By transitivity of fibers (3.4), the left-hand side is precisely the fiber $f_*(F)_{\overline{x}}$; it is therefore enough to verify that the canonical homomorphism

$$f_*(F)_{\overline{x}} \longrightarrow F_{\overline{x}}$$

is bijective, which follows at once from 5.5.

Remark 5.8. Using 5.5 and arguing again as in 5.6, one easily proves that if $f : X \to Y$ is an integral morphism and F is a sheaf of groups on X (not necessarily commutative), then $R^1 f_*(F)$ is the terminal sheaf on Y (compare Remark 5.3).

6. Supports

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Let U be a Zariski open subset of the scheme X. Then $I \in \mathrm{Ob}\, X_{\text{ét}}$, and in fact U is a subobject of the final object X of $X_{\text{ét}}$, hence defines a subobject $\widetilde{U}$ of the final object of $\widetilde{X_{\text{ét}}}$, i.e. an "open subset" of the étale topos $X_{\text{ét}}$ of X (IV 8.3).

Proposition 6.1. *The preceding map $U \mapsto \widetilde{U}$ is an isomorphism from the ordered set of open subsets (in the Zariski sense) of X onto the set of open subsets of the étale topos $\widetilde{X_{\text{ét}}}$.*

Since $X_{\text{ét}} \to \widetilde{X_{\text{ét}}}$ is fully faithful, it follows immediately that the map $U \mapsto \widetilde{U}$ preserves the order structures, i.e. $(U \subset V) \Leftrightarrow (\widetilde{U} \subset \widetilde{V})$; in particular, the preceding map is injective. To prove that it is surjective, consider a subsheaf F of the final sheaf $\widetilde{X}$, and consider the objects of $X_{\text{ét}/F}$, i.e. the schemes X' étale over X such that $F(X') \neq \emptyset$. Since $X' \to X$ is étale, it is an open map (EGA IV 2.4.6); in particular, its image (in the set-theoretic sense) $\mathrm{Im}(X')$ is open. Let U be the open subset which is the union of the $\mathrm{Im}(X')$ ($X' \in \mathrm{Ob}\, X_{\text{ét}/F}$). Since the family of the $X' \to U$ ($X' \in \mathrm{Ob}\, X_{\text{ét}/F}$) is surjective, hence covering, we conclude that $U \in \mathrm{Ob}\, X_{\text{ét}/F}$, hence $X_{\text{ét}/F'} = X_{\text{ét}/U}$, hence $F = U$, C.Q.F.D..

In view of 6.1, we may therefore speak unambiguously of an "open subset" of X, without specifying whether we mean this notion in the usual Zariski sense or in the sense of the étale topology.

Corollary 6.2. *Let U be an open subset of X, and let $j : U \to X$ be the canonical immersion. Then the functor*

$$j^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{U_{\text{ét}}}$$

391 *induces an equivalence of categories*

$$\widetilde{X_{\text{ét}}}/\widetilde{U} \longrightarrow \widetilde{U_{\text{ét}}}.$$

Indeed, one checks immediately that for every site C in which finite $\varprojlim$ exist, and for every subobject U of the final object e of C, if one considers the functor $\overleftarrow{j} : C \to C_{/U}$ defined by $j(S) = S \times U$, then j is a morphism of sites and $j^* : \widetilde{C} \to \widetilde{C/U}$ induces an equivalence $\widetilde{C}/\widetilde{U} \to \widetilde{C/U}$. It then suffices to combine this general fact with the fact that $U_{\text{ét}}$ is canonically isomorphic to $X_{\text{ét}/U}$.

In pictorial terms, one can express 6.2 by saying that the operations of "restriction to an open subset," in the usual sense for schemes on the one hand and in the sense of topoi on the other, are compatible. Here is an analogous compatibility for the operations of "restriction to a closed subset":

THEOREM 6.3. *Let X be a scheme, let Y be a closed subscheme of X, let $U = X - Y$ be endowed with the induced structure, and let $i : Y \to X$ and $j : U \to X$ be the canonical immersions. Then the functor*

$$i_* : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

is fully faithful, and if $F \in \mathrm{Ob}\, \widetilde{X_{\text{ét}}}$, then F is isomorphic to a sheaf of the form $i_*(G)$ sss $j^*(F)$ is isomorphic to the final sheaf $\widetilde{U}$ on U.

Proof. Since i_* and i^* are adjoint to one another, the fact that i_* is fully faithful is also equivalent to the fact that the functorial homomorphism

$$i^*(i_*(G)) \longrightarrow G$$

is an isomorphism, which is precisely 5.7. On the other hand, if $G \in \mathrm{Ob}\, \widetilde{Y_{\text{ét}}}$, then one 392
checks trivially, using 6.2, that $j^*(i_*(G))$ is the final sheaf on U. Conversely, if $F \in \mathrm{Ob}\, \widetilde{X_{\text{ét}}}$ is such that $j^*(F)$ is the final sheaf, let us prove that F is isomorphic to a sheaf of the form $i_*(G)$, or, what amounts to the same thing, that the canonical homomorphism

$$G \longrightarrow i_* i^* G$$

is an isomorphism. It is enough once again to check that for every $x \in X$, the homomorphism induced on the geometric fibers at $\overline{x}$ is an isomorphism. When $x \in U$, this is precisely the hypothesis made on G. When $x \in Y$, by transitivity of fibers we are reduced to checking that the homomorphism on the fibers at $\overline{x}$ induced by

$$i^*(G) \longrightarrow i^*(i_* i^*(G))$$

is an isomorphism; but the preceding homomorphism is an isomorphism by 5.7, applied to $F = i^*G$ and to i. This completes the proof of 6.3.

COROLLARY 6.4. *The functor i_* induces an equivalence from the category of abelian sheaves on Y to the category of abelian sheaves on X whose restriction to $U = X - Y$ is zero.*

In accordance with common usage, we shall therefore often identify an abelian sheaf on Y with an abelian sheaf on X which is zero on U.

6.5. By virtue of 6.1, we thus know (if X is a scheme) how to interpret the "open subsets" of the topos $\mathscr{E} = \widetilde{X_{\text{ét}}}$ as open subsets U of X in the usual sense, and by virtue of 6.3, under this identification the "residual topos" $\mathscr{E}_{\complement U}$ of IV 3 is canonically equivalent 393
to the topos $\widetilde{Y_{\text{ét}}}$, where $Y = X - U$ is endowed with any induced structure making Y a scheme. We may consequently apply to this situation the results of IV 3, in particular the description IV 3.3 of sheaves F on X in terms of triples (F', F'', u), where F' is a sheaf on Y, F'' is a sheaf on U, and u is a homomorphism from F' to $i^* j_*(F')$. In what follows we shall freely use the notation $i^!, j_!$ of IV.3, which denotes functors

$$j_! : \widetilde{(U_{\text{ét}})_{\text{ab}}} \longrightarrow \widetilde{(X_{\text{ét}})_{\text{ab}}},$$
$$i^! : \widetilde{(X_{\text{ét}})_{\text{ab}}} \longrightarrow \widetilde{(Y_{\text{ét}})_{\text{ab}}},$$

where the subscript "ab" denotes the category of abelian sheaves. These functors give rise to the two exact sequences IV 3.7.

6.6. In view of the preceding developments and of the general terminology introduced in IV 8.5, it is appropriate, for an abelian sheaf F on X, or a section φ of such a sheaf, to introduce the notion of the *support of* F, resp. of φ, as the closed subset (in the usual, i.e. Zariski, sense) complementary to the largest open subset on which the object in question vanishes.

In the present situation, one should also replace the general notation V 4.3 $H^q(\mathscr{C}_{\complement U} F)$, $\mathscr{H}^q_{\complement U}(F)$ by the notation $H^q_Y(X, F)$, $\mathscr{H}^q_Y(F)$, the first denoting an abelian group and the second an abelian sheaf on Y (or equivalently an abelian sheaf on X which is zero on U). Thus, one has $\mathscr{H}^q_Y = R^{q}{}_{y}{}^{!}$, etc.

We refer to V 6 for the general properties of the preceding functors.

7. Specialization morphisms of fiber functors

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7.1. We saw in N° 4 how one associates, with every geometric point ξ of the scheme X, a strictly local x-scheme

$$\overline{X}(\xi) = \operatorname{Spec} \mathscr{O}_{X,\xi},$$

which in fact depends only on the geometric point above the image x of ξ defined by the separable closure $\overline{k(x)}$ of $k(x)$ in $\Omega = k(\xi)$. In what follows we shall often restrict ourselves to geometric points $\xi = \operatorname{Spec} \Omega$ which are *algebraic separable* over X, i.e. such that Ω is a separable closure of $k(x)$, i.e. $\xi = \overline{x}$. Let us call an X-scheme Z a *strict localization of* X if it is X-isomorphic to a scheme of the form $\operatorname{Spec} \mathscr{O}_{X,\overline{x}}$; thus one sees that

$$\xi \mapsto \operatorname{Spec} \mathscr{O}_{X,\xi} = \overline{X}(\xi)$$

is an equivalence from the category of geometric points *algebraic separable* over X to the category of X-schemes which are strict localizations of X, when only isomorphisms are taken as morphisms in both categories.

The last assertion is no longer correct when all X-morphisms are taken as morphisms, since in the second category there may be X-morphisms which are not isomorphisms. We then make the following

Definition 7.2. *Let ξ, ξ' be two geometric points of the scheme X. A specialization arrow from ξ' to ξ is any X-morphism between the corresponding strict localizations*

$$\overline{X}(\xi') \longrightarrow \overline{X}(\xi).$$

395 *We say that ξ is a specialization of ξ', or that ξ' is a generization of ξ, if there exists a specialization arrow from ξ' to ξ.*

Note that specialization arrows compose in the evident way, so that, with specialization arrows as morphisms, *the geometric points of X form a category*, equivalent (as is the full subcategory formed by the geometric points algebraic and separable over X) to the full subcategory of $(\mathrm{Sch})_{/X}$ formed by the strict localizations of X.

Lemma 7.3. *Let X be a scheme, let Z be an X-scheme which is isomorphic to a pseudo-filtered projective limit of X-schemes X_i which are étale, with affine transition morphisms* (VII 5.1)*, let ξ' be a geometric point of X, and let $Z' = \overline{X}(\xi')$ be the corresponding strict localization. Then:*

a) *The restriction map*

$$\operatorname{Hom}_X(Z', Z) \longrightarrow \operatorname{Hom}_X(\xi', Z)$$

is bijective.

b) *In order that the two sides be nonempty, it is necessary and sufficient that the image x' of ξ' in X lie in the image of Z.*
c) *Let T be a Z-scheme. In order that T be a strict localization of Z, it is necessary and sufficient that it be a strict localization of X.*

Proof.

a) We are immediately reduced to the case where Z is one of the X_i, i.e. where Z is étale over X, and by the base change $Z' \to X$ we may suppose that $Z' \xrightarrow{\sim} X$, whence the conclusion follows from 4.2 (ii).
b) Note that if x' is the image of a point z of Z, then $k(z)$ is necessarily a separable algebraic extension of $k(x')$; hence there exists a $k(x')$-homomorphism from the latter into $k(\xi')$, whence the conclusion.
c) The proof is an easy exercise left to the reader. 396

From 7.2 and 7.3 we conclude:

PROPOSITION 7.4. *Let ξ, ξ' be two geometric points of the scheme X. Then the restriction map defines a bijection from the set $\mathrm{Hom}_X(\overline{X}(\xi'), \overline{X}(\xi))$ of specialization arrows from ξ' to ξ to the set of X-morphisms from ξ' to $\overline{X}(\xi)$.*

COROLLARY 7.5. *In order that ξ be a specialization of ξ', it is necessary and sufficient that the same hold for the images x, x' of ξ, ξ' in X, i.e. that x belong to the closure of $\{x'\}$.*

Since $\mathrm{Spec}\, \mathcal{O}_{X,\xi} \to \mathrm{Spec}\, \mathcal{O}_{X,x}$ is faithfully flat (4.4), it is surjective, and it is therefore enough to apply the second assertion of 7.3.

COROLLARY 7.6. *For every scheme X, let $\mathrm{Pt}(X)$ be the category of geometric points over X (or equivalently of geometric points algebraic separable over X), the morphisms being specialization arrows. Let ξ be a geometric point of a scheme X, and let $\overline{X}(\xi)$ be the corresponding strict localization. Then there is an equivalence of categories*

$$\mathrm{Pt}(\overline{X}(\xi)) \xrightarrow{\sim} \mathrm{Pt}(X)_{/\xi},$$

obtained by associating with every geometric point ξ' of $\overline{X}(\xi)$ the corresponding geometric point over X, together with the specialization morphism to ξ deduced from the structural morphism $\xi' \to \overline{X}(\xi)$ by 7.4.

In other words, *the datum of a generization ξ' of the geometric point ξ is essentially equivalent to the datum of a geometric point of $\overline{X}(\xi)$.* Note moreover that, by 7.3 c), the corresponding homomorphism $\overline{X}(\xi') \to \overline{X}(\xi)$ makes $\overline{X}(\xi')$ the strict localization of 397
$\overline{X}(\xi)$ relative to ξ'.

7.7. Now interpret, for every geometric point ξ of X and every sheaf F on X, the fiber F_ξ as $\Gamma(\overline{X}(\xi), \overline{F}(\xi))$, where $\overline{F}(\xi)$ is the inverse image of F on $\overline{X}(\xi)$ (4.8). Then one sees that every specialization arrow

$$u : \xi' \longrightarrow \xi$$

induces a homomorphism, functorial in F:

$$u^* : F_\xi \longrightarrow F_{\xi'},$$

called the *specialization homomorphism* associated with the specialization arrow u. It is evident from the transitivity of inverse images of sheaves that, for a composite specialization arrow, one has:

$$(wu)^* = u^* w^*.$$

7.8. If $\mathscr{E}$ is a topos, recall from Exposé IV that a "fiber functor," or (by abuse of language) a "point" of the topos $\mathscr{E}$, is any morphism from the "final topos" (Ens) (isomorphic to the category of sheaves on a space reduced to one point!) to $\mathscr{E}$, i.e. any functor

$$\varphi = \mathscr{E} \longrightarrow (\mathrm{Ens})$$

which commutes with arbitrary inductive limits, and with finite projective limits. The set of fiber functors of $\mathscr{E}$ should be regarded as the set of objects of a full subcategory of $\mathbf{Hom}(\mathscr{E}, (\mathrm{Ens}))$, called the *category of fiber functors* of the topos E, whose opposite is called the *category of points of* $\mathscr{E}$, and is denoted $\mathrm{Pt}(E)$, cf. (IV 6.1).

398 When E is of the form $\widetilde{X_{\text{ét}}}$, where X is a scheme, we have defined in 3.3 and 7.7 a functor

$$(*) \qquad \mathrm{Pt}(X) \longrightarrow \mathrm{Pt}(\widetilde{X_{\text{ét}}}),$$

where the first member is defined in 7.6. With this understood, we have the following

THEOREM 7.9. *Let X be a scheme. The preceding functor $(*)$ is an equivalence from the category of geometric points over X (with specialization arrows as morphisms) to the category of points of the étale topos $\widetilde{X_{\text{ét}}}$ (the* opposite *of that of fiber functors on $\widetilde{X_{\text{ét}}}$).*

Since this theorem will not be used later in the seminar, we confine ourselves here to a sketch of the proof, in which we shall allow ourselves certain liberties with questions of universes.

a) *The functor under consideration is fully faithful.* For every $X \in \mathrm{Ob}\, X_{\text{ét}}$, let $\overset{\vee}{X} : \widetilde{X_{\text{ét}}} \to (\mathrm{Ens})$ be defined by

$$\overset{\vee}{X}(F) = \mathrm{Hom}(X, F) = F(X).$$

Note that the fiber functor

$$\mathscr{E}_\xi : F \longrightarrow F_\xi$$

can be written

$$\mathscr{E}_\xi \simeq \varinjlim_i X_i^\vee,$$

where the X_i are affine schemes étale over X, indexed by a certain filtered category (cf. 4.3 and 4.5). Hence, for every functor $\varphi : \widetilde{X_{\text{ét}}} \to (\mathrm{Ens})$, one has:

$$\mathrm{Hom}(\mathscr{E}_\xi, \varphi) \simeq \varprojlim_i \mathrm{Hom}(X_i^\vee, \varphi),$$

and on the other hand it is well known that there is a bifunctorial isomorphism

$$\mathrm{Hom}(X_i^\vee, \varphi) \simeq \varphi(X_i).$$

399 When φ is of the form $\mathscr{E}_\xi$, hence

$$\varphi \simeq \varinjlim_j X'^\vee_j,$$

there is therefore a natural bijection

$$(*) \qquad \mathrm{Hom}(\mathscr{E}_\xi, \mathscr{E}_{\xi'}) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_X(X'_j, X_i).$$

On the other hand, one has (in the category of schemes)

$$\overline{X}(\xi) = \varprojlim_i X_i, \quad \overline{X}(\xi') = \varprojlim_j X'_j,$$

whence

$$\mathrm{Hom}_X(\overline{X}(\xi'), \overline{X}(\xi)) \simeq \varprojlim_i \mathrm{Hom}_X(\overline{X}(\xi'), X_i),$$

and on the other hand, since X_i is locally of finite presentation over X, one has

$$\mathrm{Hom}_X(\overline{X}(\xi'), X_i) \simeq \varinjlim_j \mathrm{Hom}_X(X_j, X_i),$$

whence

$$(**) \qquad \mathrm{Hom}_X(\overline{X}(\xi'), \overline{X}(\xi)) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_X(X'_j, X_i).$$

Comparison of $(*)$ and $(**)$ gives the desired conclusion (subject to checking compatibilities, which is left to the reader).

b) *The functor under consideration is essentially surjective.*

7.9.1. First note that if C is a site in which finite $\varprojlim$ are representable, and $\widetilde{C} = \mathscr{E}$ is the corresponding topos, then every fiber functor φ on $\mathscr{E}$ can be represented as a "filtered inductive limit" of functors of the form

$$\check{Y}(F) = F(Y) = \mathrm{Hom}(\widetilde{Y}, F), Y \in \mathrm{Ob}\, C,$$

(where $\widetilde{Y}$ is the sheaf associated with Y), as follows [34]. Consider the category $C_{/\varphi}$ of 400
pairs (Y, ξ), with $Y \in \mathrm{Ob}\, C$, $\xi \in \varphi(\widetilde{Y})$ (the morphisms being defined in the evident way), and note that there is an evident homomorphism

$$(*) \qquad \varinjlim_{(Y,\xi)\in C^0_{/\varphi}} \check{Y} \longrightarrow \varphi$$

by observing that

$$\mathrm{Hom}(\varinjlim_{C_\varphi} \check{Y},) \simeq \varprojlim_{C_{/\varphi}} \mathrm{Hom}(\check{Y}, \varphi) \simeq \varprojlim_{C_{/\varphi}} \varphi(\check{Y}).$$

There is a canonical element in $\varprojlim_{C_{/\varphi}} \varphi(\widetilde{Y})$, obtained by associating with every (Y, ξ) the element ξ of $\varphi(Y)$. Using the fact that finite $\varprojlim$ exist in C, and that the functor $Y \mapsto \varphi(\widetilde{Y})$ commutes with them, one sees that in $C_{/\varphi}$ finite $\varprojlim$ exist; a fortiori $C^0_{/\varphi}$ is filtered. Using the fact that every sheaf F is an inductive limit of sheaves of the form $\widetilde{Y}$, and using the fact that φ commutes with inductive limits, one easily concludes that the homomorphism of functors $(*)$ is bijective and thus gives the asserted representation.

[34] This is a special case of IV 6.8.3.

7.9.2. Note also that if $\mathscr{E}$ is a topos and Y is an object of $\mathscr{E}$, hence defines a topos $\mathscr{E}_{/Y}$, then the datum of a fiber functor φ for $\mathscr{E}_{/Y}$ is equivalent to the datum of a pair (φ, ξ), where φ is a fiber functor for $\mathscr{E}$, and $\xi \in (Y)$. With the fiber functor $\psi : \mathscr{E}_{/Y} \to (\mathrm{Ens})$ one associates the pair (φ, ξ), where φ is the composite $Z \mapsto \varphi(Z) = \psi(Z \times Y)$, and where $\xi \in \varphi(Y) = \psi(Y \times Y)$ is the image of the unique element of $\psi(Y)$ under the diagonal morphism of $Y \times Y$.

7.9.3. We now return to the case of the site $C = \widetilde{X_{\text{ét}}}$, and let $\varphi : \widetilde{X_{\text{ét}}} \to (\mathrm{Ens})$
401 be a fiber functor on the étale topos of X. Considering the restriction of φ to the full subcategory of $\widetilde{X_{\text{ét}}}$ formed by the open subsets of this topos, or, what amounts to the same thing (6.1), by the open subsets U of X, one obtains a map

$$\varphi_0 : \mathscr{U}(X) \longrightarrow \{0, 1\}$$

which commutes with arbitrary suprema and finite infima. (Note that for $u \in \mathscr{U}(X)$, $\varphi(U)$ is either empty or consists of a single point, and one takes $\varphi_0(U) = 0$ or 1 according as one is in the former or the latter case.)

One easily sees, using the fact that every irreducible closed subset of X has one and only one generic point [34], that φ_0 is defined by means of a unique $x \in X$, by the condition

$$\varphi_0(U) = 1 \quad \text{sss} \quad x \in U,$$

i.e.

$$\varphi(U) = \varnothing \Leftrightarrow x \in U.$$

Let $F \in \mathrm{Ob}\, \widetilde{X_{\text{ét}}}$. Then the image of F in the final sheaf X is an open subset; and since

$$\varphi(F) \longrightarrow \varphi(U)$$

is surjective, one sees that $\varphi(F) \neq \varnothing$ sss $\varphi(U) \neq \varnothing$, i.e. sss $x \in U$. Let $C_{/\varphi}$ be the category of pairs (X', ξ) where $X' \in \mathrm{Ob}\, X_{\text{ét}}$, and $\xi \in \varphi(X')$. It was pointed out in 7.9.1 that $C^0_{/\varphi}$ is filtered and connected. Moreover, by 7.9.2 and by the preceding discussion applied to X' instead of X, if $(X', \xi) \in \mathrm{Ob}\, X_{\text{ét}}$, there is associated with it a unique point $x' \in X'$, such that for an étale morphism $X'' \to X'$, ξ lies in the image of $\varphi(X'') \to \varphi(X')$ sss x' lies in the image of X''. It follows that the scheme $\varprojlim$ of the X' over the category C of the (X', ξ) exists and is a strict localization Z of X, thus corresponding
402 to a geometric point ξ and a fiber functor $\mathscr{E}_\xi$. For every sheaf F, one then has

$$\mathscr{E}_\xi(F) \simeq \varinjlim_{C_\varphi} (F)(X').$$

In view of 7.9.1, we conclude that φ is isomorphic to $\mathscr{E}_\xi$, which completes the proof of 7.9.

8. Two spectral sequences for integral morphisms

Proposition 8.1. *Let $f : X \to Y$ be a surjective integral morphism, and let F be an abelian sheaf on Y. For every integer i, let $\mathscr{H}^i(F)$ be the presheaf on* $(\mathrm{Sch})_{/Y}$ *defined by*

$$\mathscr{H}^i(F)(Z) = H^i(Z, F_Z),$$

where F_Z is the inverse image of F on Z. Then there exists a spectral sequence (functorial in F)

$$H^*(X, F) \Longleftarrow E_2^{pq} = H^p(X/Y, \mathscr{H}^q(F))$$

("descent spectral sequence").

[34] i.e. X is *sober* in the terminology of IV 4.2.1. The assertion made is a special case of IV 4.2.3.

Of course, the symbol $H^p(X/Y, G)$, for a presheaf G on $(\mathrm{Sch})_{/Y}$, denotes the p-th relative Čech cohomology group, defined by means of the complex of the $C^n(X/Y, G) = G(X^{n+1})$, where X^m denotes the m-th power in $(\mathrm{Sch})_{/Y}$. To prove 8.1, let $p_n : X^{n+1} \to Y$ be the projection, and set

$$A^n = p_{n*}(\mathscr{Z}_{X^{n+1}}),$$

where $\mathscr{Z}_{X^{n+1}}$ denotes the constant sheaf $\mathscr{Z}$ on X^{n+1}. Then the A^n are the components of a simplicial abelian sheaf on Y, hence of a complex of abelian sheaves A^* on Y. Note that there is an evident homomorphism

$$(*) \qquad \mathscr{Z}_Y \longrightarrow A^0.$$

With this understood, we have the following 403

Lemma 8.2. *The complex (A^*), endowed with the homomorphism $(*)$, is a resolution of $\mathscr{Z}_Y$. More generally, for every abelian sheaf F on Y, $A^* \otimes_{\mathscr{Z}} F$ is a resolution of F.*

Proof: We may suppose that Y is affine, hence $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, where B is an integral A-algebra. We have $B = \varinjlim_i B_i$, where B_i ranges over the finitely generated, hence finite, subalgebras of B, whence $X = \varprojlim X_i$, where $X_i = \operatorname{Spec} B_i$. Using VII 5.11, one sees that the augmented complex $A^*(X/Y)$ is then the inductive limit of the augmented complexes $A^*(X_i/Y)$. This reduces us to the case where f is *finite*. It suffices to prove that for every geometric point $\overline{y}$ of Y, the complex $A^*_{\overline{y}}$ is a resolution of $\mathscr{Z}_{\overline{y}}$, and likewise after tensoring with an F. But if $X_{\overline{y}}$ is the fiber of X at $\overline{y}$, it follows from 5.5 that the complex $A^*_{\overline{y}}$ is none other than the analogous complex $A^*(X_{\overline{y}}/\overline{y})$. This reduces us to the case where Y is the spectrum of a separably closed field k. Using 1.3 a) and b), we may even suppose that k is algebraically closed and X is reduced, hence that X is of the form I_Y, where I is a finite set. But then the complex $A^* \otimes F$ is identified with the trivial cochain complex of the index set I with coefficients in $F_{\overline{y}}$, and is therefore indeed a resolution of $F_{\overline{y}}$.

Thus, as a consequence of 8.2, we obtain a spectral sequence

$$H^*(Y, F) \Longleftarrow E_2^{pq} = H^p(H^q(Y, A^* \otimes F)),$$

and it remains to make the initial term explicit. There is a canonical homomorphism 404

$$A^n \otimes F \longrightarrow p_{n_*} p_n^*(F),$$

and the latter is an isomorphism, as one sees once again by reduction to the case where f is finite and passage to fibers. Using 5.5, one concludes that

$$H^q(Y, A^n \otimes F) \simeq H^q(X^{n+1}, p_n^*(F)) \Longrightarrow \mathscr{H}^q(F)(X^{n+1}),$$

which indeed gives the initial term stated in 8.1.

Remark 8.3. a) When one is given a *locally finite* covering of Y by closed subsets Y_i, and one sets $X = \coprod_i Y_i$, the canonical morphism $f : X \to Y$ is finite, and the initial term of the spectral sequence 8.1 can be made explicit as the cohomology of the complex defined by the $H^q(Y_{i_0 \dots i_p}, F|Y_{i_0 \dots i_p})$, where $Y_{i_0 \dots i_p} = Y_{i_0} \cap \dots \cap Y_{i_p}$. This is therefore the analogue of the Leray spectral sequence for a locally finite closed covering of an ordinary topological space (TF Chapter II 5.2.4). The latter can moreover also be generalized to a spectral sequence relative to a "finite" morphism, i.e. one which is proper with finite fibers, by proceeding as in 8.1.

b) Note the analogy between the spectral sequence of 8.1 and the Leray spectral sequence of a covering (X_i) of Y (in this case by X_i étale over Y); the latter would formally be obtained by writing the spectral sequence 8.1 for $X = \coprod_i X_i$. It is in fact likely that these two spectral sequences admit a common generalization, valid whenever one has a family of morphisms $(X_i \to Y)$ which is a "universally effective descent family" for the fibered category of étale sheaves over a
405 variable base scheme (cf. n° 9 below). The analogous question also arises in ordinary topology, concerning a common generalization of the two kinds of Leray spectral sequences of a covering, assumed either open or locally finite closed [34].

PROPOSITION 8.4. *Let Y be a scheme, let π be a profinite group, let $(\pi_i)_i$ be the projective system of finite discrete quotient groups of π, let $(X_i)_{i\in I}$ be a projective system of principal coverings of Y, with groups π_i, the transition homomorphisms $X_j \to X_i$ being compatible with the homomorphisms $\pi_j \to \pi_i$ on the groups of operators, and let $X = \varprojlim X_i$* (cf. VII 5.1)*, so that the group π acts on the Y-scheme X. Let F be an abelian sheaf on Y. Then there is a "*Hochschild–Serre*" spectral sequence (functorial in F)*

$$H^*(Y, F) \Longleftarrow E_2^{pq} = H^p(\pi, \varinjlim_i H^q(X_i, F_{X_i})),$$

where F_{X_i} is the inverse image of F on X_i, and $H^p(\pi, -)$ denotes Galois cohomology.

The term E_2^{pq} written here is also (by the definition of $H^p(\pi, -)$ and the transitivity of inductive limits) isomorphic to

$$E_2^{pq} = \varinjlim_i H^p(\pi_i, H^q(X_i, F_{X_i})),$$

which shows that it is enough to find an inductive system of spectral sequences (depending on the index i)

$$H^*(Y, F) \Longleftarrow {}^iE_2^{pq} = H^p(\pi_i, H^q(X_i, F_{X_i})).$$

This reduces us to defining the spectral sequence in the case where π is finite. Then
406 $f : X \to Y$ is a covering morphism in $Y_{\text{ét}}$, so one may write the Leray spectral sequence of this morphism. In view of the canonical isomorphisms

$$X^n \simeq X \overset{G}{\times} G^n,$$

a well-known calculation then shows that for every presheaf $\mathscr{H}$ on $X_{\text{ét}}$ which transforms sums into products, $C^*(X/Y, \mathscr{H})$ is none other than the complex of homogeneous cochains of G with coefficients in $\mathscr{H}(X)$, whence the stated form of the initial term.

A variant of this proof, avoiding any calculation, consists in regarding $E \mapsto X \times_\pi E$ as a morphism from the site of finite π-sets to the étale site of Y, and writing the Leray spectral sequence of this morphism. Finally, when π is finite, one may also regard the Hochschild–Serre spectral sequence as a special case of 8.1 relative to the morphism $f : X \to Y$.

COROLLARY 8.5. *Suppose that Y is quasi-compact and quasi-separated. Then the spectral sequence of 8.4 takes the form*

$$H^*(Y, F) \Longleftarrow E_2^{pq} = H^p(\pi, H^q(X, F_X)).$$

[34] Since these lines were written, P. Deligne has developed a general theory of "descent spectral sequences," cf. Exp. V bis.

Indeed, by VII 5.8, there are then canonical isomorphisms

$$\varinjlim_i H^q(X_i, F_{X_i}) \simeq H^q(X, F_X).$$

COROLLARY 8.6. *Let Y be a local henselian scheme with closed point y, let F be an abelian sheaf on Y, and let F_0 be the induced sheaf on $Y_0 = \mathrm{Spec}(k(y))$. Then the canonical homomorphisms*

$$H^n(Y, F) \longrightarrow H^n(Y_0, F_0)$$

are isomorphisms. 407

Indeed, let $\overline{y}$ be a geometric point over y, corresponding to a separable closure $k(y) = k(\overline{y})$ of $k(y)$; since Y is henselian, the strict localization X of Y at $\overline{y}$ is the projective limit of the connected étale Galois coverings pointed at $\overline{y}$, denoted X_i, so that the hypotheses of 8.5 apply. Since $H^q(X, F_X) = 0$ for $q > 0$ by 4.7, one obtains isomorphisms

$$H^n(Y, F) \xleftarrow{\sim} H^n(\pi, F(X)),$$

where π is the "Galois group" of X over Y, isomorphic to that of $\overline{k(y)}$ over $k = k(y)$. Similarly (or by 2.3) one finds

$$H^n(Y_0, F_0) \xleftarrow{\sim} H^n(\pi, F_0(X_0)),$$

where $X_0 = X \times_{Y_0} Y \simeq \mathrm{Spec}(k(\overline{y}))$. But the restriction homomorphism $F(X) \to F_0(X_0)$ is an isomorphism by 4.8, from which the conclusion 8.6 follows immediately.

9. Descent of étale sheaves

This section will not be used later in this Seminar, and may be omitted on a first reading.

PROPOSITION 9.1. *Let $f : X' \to X$ be a surjective (resp. universally submersive* (SGA I IX 2.1)*) morphism of schemes. Then the functor $f^* : X_{\text{ét}} \to X'_{\text{ét}}$ is faithful and "conservative"* (cf. 3.6) *(resp. induces a* fully faithful *functor from the category of sheaves on $X_{\text{ét}}$ to the category of sheaves on $X'_{\text{ét}}$ endowed with descent data relative to $f : X' \to X$).* 408

The first assertion follows immediately from the transitivity of fiber functors (3.4) and from (3.7); the second (which can also be stated by saying that f is a *descent morphism* relative to the fibered category of étale sheaves on variable schemes) also means that for two sheaves F, G on Y, the natural diagram

$$(*) \qquad \mathrm{Hom}(F, G) \to \mathrm{Hom}(F', G') \rightrightarrows \mathrm{Hom}(F'', G'')$$

is exact, where F' and G' (resp. F'' and G'') are the inverse images of F and G on X' (resp. on $X'' = X' \times_X X'$). Taking the final sheaf for F, the assertion gives the following

COROLLARY 9.2. *Let $f : X' \to X$ be a universally submersive morphism. Then for every sheaf F on X, the diagram*

$$\Gamma(X, F) \to \Gamma(X', F') \rightrightarrows \Gamma(X'', F'')$$

is exact, where $X'' = X' \times_X X'$, and where F', F'' are the inverse images of F on X', X''.

We shall prove 9.1 using the following

LEMMA 9.3. *Suppose that f is universally submersive. Let F be a sheaf on X, denote by $S(F)$ the set of subsheaves of F, and define $S(F')$, $S(F'')$ analogously. Then the diagram* 409
of natural maps

$$S(F) \to S(F') \rightrightarrows S(F'')$$

is exact.

Proof. The fact that $S(F) \to S(F')$ is injective follows immediately from the fact that the functor f^* is conservative; for if F_i $(i = 1, 2)$ are two subsheaves of F such that $f^*(F_1) = f^*(F_2)$, then the inclusions $F_3 = F_1 \cap F_2 \to F_i$ $(i = 1, 2)$ are isomorphisms, since they become so after applying the functor f^*, hence $F_1 = F_2$. It remains to prove that if G' is a subsheaf of F' such that $\mathrm{pr}_1^*(G') = \mathrm{pr}_2^*(G')$, then G' is the inverse image of a subsheaf of F. Note that one can find an epimorphism $F_1 \to F$ in $X_{\text{ét}}$, where F_1 is *representable* by a scheme étale over X. Introduce $F_2 = F_1 \times_F F_1$, and likewise F_1', F_1'', F_2', F_2'', sheaves giving rise to a diagram of sets

$$\begin{array}{ccccc}
S(F) & \longrightarrow & S(F') & \longrightarrow & S(F'') \\
\downarrow & & \downarrow & & \downarrow \\
S(F_1) & \longrightarrow & S(F_1') & \rightrightarrows & S(F_1'') \\
\downarrow\downarrow & & \downarrow\downarrow & & \downarrow\downarrow \\
S(F_2) & \longrightarrow & S(F_2') & \rightrightarrows & S(F_2''),
\end{array}$$

in which the columns are exact, by the exactness properties in the category of sheaves on X, X', X'', respectively. A standard diagram chase then shows that, to prove that the first row is exact, it is enough to prove this for rows 2 and 3. For row 2, this follows
410 from the fact that F_1 is representable, that $X' \to X$ is universally submersive, and from SGA 1 IX 2.3. On the other hand, F_2 is also representable, since it is a subsheaf of $F_1 \times_X F_1$, which is representable, and one applies 6.1. Thus row 3 is also exact, C.Q.F.D.

We now prove 9.1, i.e. that every morphism $u' : F' \to G'$ compatible with the descent data on F', G', the morphism u' comes from a morphism $u : F \to G$. Let $H = F \times G$, hence $H' = F' \times G'$, $H'' = F'' \times G''$; then the *graph* of u' is a subsheaf Γ' of H', whose two inverse images are equal to the graph of the same morphism $u'' : F'' \to G''$. Hence, by 9.3, Γ' comes from a subsheaf Γ of H. I claim that Γ is the graph of a morphism $u : F \to G$, i.e. that the morphism $p : \Gamma \to F$ induced by pr_1 is an isomorphism: indeed, it becomes an isomorphism after the base change $X' \to X$, and we apply the already proved part of 9.1. The morphism $u : F \to G$ then has the required property, C.Q.F.D.

THEOREM 9.4. *Let $f : X' \to X$ be a surjective morphism of schemes. Suppose that one of the following hypotheses holds:*

a) *f is integral.*
b) *f is proper.*
c) *f is flat and locally of finite presentation* [34].
d) *X is discrete (e.g. the spectrum of a field).*

Then f is an effective descent morphism *for the fibered category $\mathscr{F}$ over* (Sch) *of étale sheaves on variable schemes, i.e. the functor f^* induces an* equivalence *between the category of*
411 *sheaves on X and the category of sheaves on X' endowed with descent data relative to $f : X' \to X$.*

[34] It is likely that this second hypothesis is in fact superfluous.

9.4.1. Let F' be a sheaf on X' endowed with descent data relative to $f : X' \to X$. Define a functor

$$G : X^0_{\text{ét}} \longrightarrow (\text{Ens})$$

by the formula

$$G(Y) = \text{Ker}(F'(Y') \rightrightarrows F''(Y'')).$$

One sees immediately that G is a *sheaf* for the étale topology. On the other hand, there is a canonical injective homomorphism $G \to f_*(F')$, hence a homomorphism $f^*(G) \to F'$, and the latter is evidently compatible with the descent data. It remains to determine whether this homomorphism is an *isomorphism*. Note that G can also be defined by the exactness of

$$G \to f_*(F') \rightrightarrows g_*(F''),$$

so for every base change $Y \xrightarrow{h} X$ *which commutes with the formation of* $f_*(F')$ and $g_*(F'')$, $h^*(G) = G_Y$ is identified with the sheaf "descended" from $F'_{Y'}$ by $Y' \to Y$, and of course the homomorphism ${f_Y}^*(G_Y) \to (F_Y)'$ is the one deduced from $f^*(G) \to F'$ by base change. Now *suppose that the base change* $f : Y = X' \to X$ *commutes with the formation of* $f_*(F')$ and $g_*(F'')$, and note that since $Y' = X' \times_X X'$ has a section over $Y' = X'$, hence $Y' \to Y$ is an *effective* descent morphism for every fibered category, in particular for $\mathscr{F}$, it follows that $f_Y^*(G_Y) \to (F_Y)'$ is an isomorphism. Thus $f^*(G) \to F'$ becomes an isomorphism after the base change $Y' = X' \times_X X' \to X'$. Since the latter has a section, $f^*(G) \to F'$ is an isomorphism, and hence the descent datum under 412
consideration on F' is effective.

9.4.2. This proves Theorem 9.4 in case a), by 5.6. Case b) also follows from the fact that if f is proper, then f_* commutes with *every* base change $Y' \to Y$ (which will be proved in a later exposé (XII 5.1 (i))).

9.4.3. In case c), the question being local on X, we may suppose that X is affine and that there exists an affine scheme X'_1, of finite presentation, quasi-finite and faithfully flat over X, together with an X-morphism $X'_1 \to X'$, which reduces us to the case where f is quasi-finite. After localizing further on X, this time in the sense of the étale topology, one sees that there will exist an open subset X'_1 of X' such that $X'_1 \to X$ is finite and surjective, which reduces us to the case where f is finite and surjective, already treated in a).

9.4.4. In case d), we may suppose (by localizing on X) that X consists of a single point, and even, in view of 1.1, that it is the spectrum of a field k. Moreover, f is universally open (EGA IV 2.4.9), so the hypotheses of 9.1 hold, and the argument of 9.4.1 applies again, making a base change with $Y = \text{Spec}\, k$, where $k = k(x)$, for an $x \in X$. But then it is still true that f_* commutes with the base change $Y \to X$, as we shall see later (XVI 1.4 and 1.5).

This completes the proof of 9.4.

EXPOSÉ IX

Constructible sheaves; cohomology of an algebraic curve

M. Artin

0. Introduction

1

Note that, beginning with the present exposé, in contrast to the preceding exposés and to the classical cohomological theory of topological spaces, we are obliged, for the validity of the essential statements, to restrict ourselves to *torsion* sheaves (and often, more precisely, to torsion sheaves prime to the residue characteristics).

Let us point out that in fact there is no "reasonable" theory of cohomology with integral (or even real) coefficients for varieties over an algebraically closed field of characteristic $p \neq 0$, analogous to the classical theory for the case $k = \mathbf{C}$ (as Serre observed). More precisely, there is no contravariant functor H^1, defined, say, on the category of smooth projective schemes over an algebraically closed field K of characteristic $p > 0$, with values in the category of finite-dimensional vector spaces over $\mathbf{R}$ (or a subfield of $\mathbf{R}$), "commuting with products," i.e. such that $H^1(X \times Y) \xleftarrow{\sim} H^1(X) \times H^1(Y)$, and such that $\dim H^1(X) = 2$ if X is a connected curve of genus 1. Indeed, it would follow that if X is an abelian variety over k, then the map $u \mapsto u^1$ from $\mathrm{End}(X)$ to $\mathrm{End}(H^1(X))$ is *additive*, hence a *representation* of the opposite ring of $\mathrm{End}(X)$. But in characteristic p there exist elliptic curves X whose endomorphism ring is a maximal order A in a 2
definite quaternion algebra over $\mathbf{Z}$ ([3], p.198), and such an algebra plainly has no representation in a vector space of dimension 2 over $\mathbf{R}$, since $A \otimes_{\mathbf{Z}} \mathbf{R}$ is a division algebra (the quaternion algebra).

1. The formalism of torsion sheaves

Let T be a topos and F an abelian sheaf on T. One may multiply by n in F ($n \in \mathbf{Z}$ an integer), this multiplication being induced by the analogous operation in (Ab). Denote by ${}_nF$ the kernel of this multiplication; thus ${}_nF(X)$, for $X \in \mathrm{Ob}\, T$, is the group of sections of $F(X)$ whose order divides n.

We denote by $\mathbf{P}$ the set of all prime numbers.

Definition 1.1. *Let p be a set of prime numbers. We say that F is a p-torsion sheaf, or a p-sheaf, if the canonical morphism*

$$\varinjlim_n {}_nF \longrightarrow F \text{ is bijective,}$$

where n ranges over the set of integers such that $\mathrm{ass}\, n \subset p$, that is, such that the prime numbers dividing n belong to p. If $p = \mathbf{P}$ is the set of all prime numbers, we simply say that F is a torsion sheaf.

PROPOSITION 1.2. (i) *F is a p-torsion sheaf if and only if F is the sheaf associated with a presheaf taking values in p-torsion abelian groups; one may take*

$$P = \frac{\lim(\text{presheaf})}{\operatorname{ass} n \subset p}{}_nF.$$

3 (ii) *If F is a p-torsion sheaf and if $X \in \operatorname{Ob} T$ is a quasi-compact object*[34]*, then $F(X)$ is a p-torsion group.*

(iii) *If T is locally of finite type* (VI 1.1)*, then F is a p-torsion sheaf if and only if $F(X)$ is a p-torsion group for every quasi-compact $X \in \operatorname{Ob} T$. In this case, one has*

$$\varinjlim_{\operatorname{ass} n \subset p} H^q(T/X; {}_nF) \xrightarrow{\sim} H^q(T/X; F)$$

for every quasi-compact X and every q.

(iv) *If $u : T \to T'$ is a morphism of topoi and if F' is a p-torsion sheaf on T', then the inverse image u^*F is a p-torsion sheaf on T.*

(v) *If $u : T \to T'$ is a morphism of topoi, with T and T' locally of finite type and u quasi-compact*, and if F is a p-torsion sheaf on T, then the $R^q u_* F'$ are p-torsion sheaves on T for every q.*

Proof:

(i) Clearly, if a sheaf F is p-torsion, then F is the sheaf associated with the presheaf P whose group of sections $P(X)$ is the subgroup of p-torsion elements of $F(X)$. Conversely, let $F = \underline{a}P$ (II) be the sheaf associated with P, where P takes values in the category of abelian p-torsion groups. From the exact sequence

$$0 \to {}_nP \to P \xrightarrow{n} P$$

one deduces an exact sequence

$$0 \to \underline{a}({}_nP) \to F \xrightarrow{n} F,$$

4 whence $\underline{a}({}_nP) = {}_nF$. Since the functor $\underline{a}$ commutes with inductive limits, it follows from

$$\frac{\lim(\text{presheaf})}{\operatorname{ass} n \subset p}{}_nP \xrightarrow{\sim} P$$

that one also has

$$\varinjlim_{\operatorname{ass} n \subset p} {}_nF \xrightarrow{\sim} F.$$

(ii) Put $P = \varinjlim(\text{presheaf})_n F$. Then one has $P \subset F$, so P is a separated presheaf, and consequently

$$\underline{a}P(X) = \varinjlim(\ker(\prod_i P(X_i) \rightrightarrows)),$$

where, as usual, the limit is taken over the covering families $\{X_i \to X\}$. But since X is quasi-compact, it is enough to take finite covering families. For these families, $\prod_i P(X_i)$ is p-torsion; therefore so is the ker. It follows that $\underline{a}P(X) = F(X)$ is p-torsion.

[34] An object X of a topos is called quasi-compact if every covering family $\{X_i \to X\}$ can be refined by a finite covering family. A morphism u of topoi is quasi-compact if the inverse image of every quasi-compact object is again quasi-compact.

(iii) To verify that $\varinjlim_n F \to F$ is bijective, it is enough to do so on the values at $X \in \mathrm{Ob}\, T$ for every X in a family of generators. We are therefore reduced to (ii) for the first assertion. The second assertion is a consequence of

(iv) A consequence of (i), since $u^*(\underline{a}P') = \underline{a}(u \cdot P)$

(v) An easy consequence of (iii).

We shall also need a nonabelian variant. But there are technical problems in the definition (I do not know whether they are serious), and we shall therefore content ourselves with giving a definition for the étale topology of a scheme.

DEFINITION 1.3. *Let p be a set of prime numbers. A group G is called an* ind-p-group *if every finite subset of G generates a finite subgroup of order n, with $\mathrm{ass}\, n \subset p$. This is equivalent to saying that the finite subgroups G_α of order n_α, with $\mathrm{ass}\, n_\alpha \subset p$, form a filtered set, and that $G = \varinjlim G_\alpha$. If $p = \mathbf{P}$, one says* ind-finite group *instead of ind-p-group.* 5

One immediately verifies the following

PROPOSITION 1.4. (i) *A subgroup as well as a quotient of an ind p-group is again an ind p-group.*

(ii) *A filtered inductive limit of ind-p-groups is an ind-p-group.*

(iii) *A finite projective limit of ind-p-groups is an ind-p-group.*

DEFINITION 1.5. *Let X be a scheme. A sheaf F of groups on X is called a* sheaf of ind p-groups *(resp. of* ind-finite groups*) if, for every étale $U \to X$, with U quasi-compact, $F(U)$ is an ind p-group (resp. an ind-finite group).*

PROPOSITION 1.6. *Let F be a sheaf of groups on the scheme X.*

(i) *F is a sheaf of ind-p-groups if and only if, for every geometric point ξ of X, the fiber F_ξ is an ind-p-group.*

(ii) *If $f : X \to X'$ is a morphism and F' is a sheaf of ind-p-groups on X', then f^*F' is a sheaf of ind-p-groups.*

(iii) *If $f : X \to X'$ is a quasi-compact morphism and F is a sheaf of ind-p-groups on X, then f_*F is a sheaf of ind-p-groups.*

Proof. (i) Suppose that F is a sheaf of ind-p-groups, and let ξ be a geometric point of X. Then $F = \varinjlim F(X')$, where X' ranges over a pseudo-filtered system of schemes étale over X (VIII 3.9). It is evident that the affine X' (hence quasi-compact) form a cofinal set, and consequently that F is an inductive limit of ind p-groups, and hence is an ind-p-group by 1.4 (ii). Conversely, suppose that for every ξ the fiber F_ξ is an ind-p-group, and let $U \to X$ be étale, with U quasi-compact. Let $S \subset F(U)$ be a finite subset. For every geometric point ξ of U, the image of S in F_ξ generates a finite group. Since a finite group is finitely presented, one easily concludes that there exists a U_ξ étale over U, pointed at ξ, such that S generates a finite group in $F(U_\xi)$ (cf. VIII 4). A finite number of such U_ξ form a covering of U, say $\{U_1, \dots, U_n\}$. Then the image of S in $\prod F(U_i)$ generates a finite subgroup, and since $F(U) \subset \prod F(U_i)$, the result follows. 6

Assertion (ii) is trivial from (i), and (iii) is immediate.

2. Constructible sheaves

2.0. Let T be a topos. Recall that with every element $S \in \mathrm{Ob}(\mathrm{Ens})$ one associates the sheaf S_T associated with the presheaf P such that $P(X) = S$ for every $X \in \mathrm{Ob}(T)$. The object of T representing S_T is

$$\coprod_{s \in S} e = \text{“}\, S \times e\text{”},$$

where e is the final object of T. The sheaf S_T is called the *constant sheaf* with value S (IV). In the evident way one defines the notion of a *constant sheaf of groups, constant sheaf of A-modules* (A a ring), and *constant abelian sheaf*, with value M. A morphism $f_T : S_T \to S'_T$ of constant sheaves is said to be constant if it comes from a morphism of sets $f : S \to S'$.

A sheaf F is said to be *locally constant* if there exists a covering $\{e_i \to e\}$ of the final
7 object such that F becomes constant on each e_i. One defines analogously the notion of a locally constant *sheaf of groups* or sheaf of *A-modules*, and of a *locally constant morphism* $f : F \to G$, when F and G are locally constant sheaves. Finally, a locally constant sheaf of groups or of A-modules is said to be *of finite type* (resp. *of finite presentation*) if its local values are groups or A-modules of finite type (resp. of finite presentation).

LEMMA 2.1. (i) *Let $f : F \to G$ be a morphism of locally constant sheaves of sets on T, and suppose that the local values of F are finite sets. Then f is locally constant.*

(ii) *Let $f : F \to G$ be a morphism of locally constant sheaves of A-modules, with F of finite type. Then f is locally constant, and the kernel and cokernel of f are locally constant.*

(iii) *Let $0 \to F' \to F \to F'' \to 0$ be an exact sequence of sheaves of A-modules (resp. of groups), with F' and F'' locally constant, and F'' of finite presentation. Then F is locally constant.*

Proof of (i) and (ii). We reduce without difficulty to the case in which F and G are constant. Consider (ii): let $F = M_T$, $G = N_T$ (M, N being A-modules), and let $S \subset M$ be a finite set of generators of M. Then f is determined by its restriction to S_T, and f will be locally constant if $f|S_T$ is. Thus we are reduced to (i) for the first assertion of (ii), and the second assertion is a trivial consequence of the first.

It therefore remains to prove (i). Suppose that $F = S_T$, $G = {S'}_T$ are constant, and also denote by f the morphism of objects $f : S \times e \to S' \times e$ (notation as above). Let $f_s : e \to S' \times e$ ($s' \in S$) be the components of f: since S is finite, it plainly suffices to treat the case $f = f_s$, that is, the case in which S is a one-element set. Let $X_{s'} \to e$ ($s' \in S'$) be the family of morphisms that make the diagrams

$$\begin{array}{ccc} X_{s'} & \longrightarrow & e \\ \downarrow & & \downarrow{\scriptstyle i_{S'}} \\ e & \xrightarrow{\ f\ } & S' \times e \end{array}$$

8 cartesian, where $i_{s'} : e \to S' \times e$ is the "inclusion" into the s'-th component. The family $\{i_{s'}\}$ is trivially covering, and hence so is $\{X_{s'} \to e\}$. One checks immediately (since coproducts in a topos are disjoint) that f becomes constant on $X_{s'}$, and consequently f is locally constant.

Proof of (iii). Consider the case of sheaves of A-modules. We may suppose F' and F'' constant, with F'' having value M'', an A-module of finite presentation. If M'' is free, and hence has a finite basis $(e_i)_{1 \le i \le n}$, then the e_i define sections of F'' and hence lift locally to sections of F; this shows that the extension F of F'' by F' splits locally, which proves that F is locally isomorphic to $F' \times F''$, and is therefore locally constant. In the general case, choose a surjective homomorphism $\varphi : L'' \to M''$, with L'' free of finite type; the kernel R of φ is of finite type, and setting $L = L''_T \times_{M''_T} F = L''_T \times_{F''} F$, one sees on the one hand that L is an extension of L''_T by G, and is therefore locally

constant by what precedes, and on the other hand that there is an exact sequence $0 \to R_T \to L \to F \to 0$, which, by (ii), implies that F is locally constant.

From now on we restrict ourselves to the étale topoi of schemes.

LEMMA 2.2. *Let X be a scheme and let F be a locally constant sheaf on X. Then F is represented by an étale Y/X. If, moreover, the fibers of F are finite (resp. and nonempty), then Y is an étale covering (resp. and is surjective).*

Proof. Descent (SGA 1 IX 4.1 if F has finite fibers, SGA 3 X 5.4 in the general case).

DEFINITION 2.3. *A sheaf of sets (resp. of groups, resp. of A-modules) is said to be constructible if, for every affine open $U \subset X$, there exists a decomposition of U as a union of finitely many constructible* (EGA 0_{III} 9.1.2) *reduced locally closed subschemes U_i such that the sheaf induced by F on each U_i is locally constant, with finite value (resp. finite, resp. of finite presentation*[34]*).* 9

2.3.1. The finiteness hypothesis on the local values amounts to saying that for every geometric point ξ, the fiber $F_{\bar{\xi}}$ is finite (resp. finite, resp. of finite presentation). It follows immediately from 2.1 (i) that a sheaf of groups F is constructible if and only if the underlying sheaf of sets of F is constructible.

Care must be taken that if F is an abelian sheaf, it is constructible as a sheaf of groups (or equivalently, as a sheaf of sets) if and only if it is constructible as a sheaf of $\mathbf{Z}$-modules, and if *in addition* its fibers are finite. Thus, $\mathbf{Z}_X$ is constructible as a $\mathbf{Z}$-module, but not as a sheaf of groups.

PROPOSITION 2.4. (i) *Let X be quasi-compact and quasi-separated. Then a sheaf (resp. sheaf of groups, resp. sheaf of A-modules) is constructible if and only if there exists a decomposition of X as a union of constructible locally closed subsets such that F becomes locally constant and finite (resp. finite, resp. of finite presentation) on each X_i.*

(ii) *To verify that F is constructible, it is enough to verify that $F|U_i$ is constructible for the U_i of a given open covering of X.*

(iii) *Let $F : Y \to X$ be a morphism. Then f^*F is constructible if F is.*

(iv) *The set of points at which the fiber of a constructible sheaf is nonempty (resp. nonzero) is a locally constructible subset* (EGA 0_{III} 9.1.2). 10

(v) *Let X be locally noetherian. Then a sheaf of sets (resp. ...) F on X is constructible if and only if for every $x \in X$ there exists a nonempty open subset of the closure $\overline{x}$ of x such that F induces a locally constant sheaf with finite value (resp. ...) on U.*

Proof. Plainly, if $f : Y \to X$ is a morphism and if there exists a decomposition of X as a union of constructible locally closed subsets X_i such that F becomes locally constant on each X_i, the same is true of Y and f^*F (cf. EGA IV 1.8.2). This proves the implication $\Leftarrow$ of (i). Now suppose that X is quasi-compact and quasi-separated, and let $\{U_i, \dots, U_n\}$ be an affine open covering of X such that $F|U_i$ is constructible for every i. To find a decomposition of X as in (i), it is enough to do so for U_n and for $Y = X - U_n$ (with the induced reduced structure), since U_n and Y are constructible in X by virtue

[34] When A is not assumed noetherian, for the purposes of Homological Algebra it is preferable to require here, instead of finite presentation alone for the A-module M, that it have a left resolution by free A-modules of finite type. The notion introduced in (2.3) should then be called: F is 1-constructible (more generally, one would define n-constructibility in the evident way, for every integer $n \geq 0$). For the present Exposé we shall provisionally retain the terminology in the form (2.3), which is chiefly reasonable when A is noetherian, in which case it coincides with that just indicated. Compare SGA 6 I for finiteness notions in nonnoetherian cases.

of the hypothesis that "X is quasi-separated". Now a decomposition exists for U_n by the definition of constructibility, and Y is the union of $n-1$ affine opens $V_i = Y \cap U_i$ ($i = 1, \dots, n-1$) with $F|V_i$ constructible, whence result (i) follows by induction on n. The same reasoning proves (ii); indeed, the hypothesis implies that $F|U$ is constructible for every "sufficiently small" affine open U, and an arbitrary affine open V can be covered by finitely many such opens. Assertion (ii) shows that the notion of constructibility is local on X, and (iii), (iv) follow immediately. The implication $\Rightarrow$ of (v) is immediate and does not depend on the noetherian hypothesis. If X is noetherian, one proves the implication $\Leftarrow$ by the usual "noetherian induction".

PROPOSITION 2.5. *Let X be quasi-compact and quasi-separated, and let F be a constructible sheaf of groups or of A-modules on X. Then there exists a finite filtration of F whose successive quotients are of the form $i_!G$, where $i : U \to X$ is the inclusion of a constructible locally closed subset and where G is locally constant and constructible on U. If X is noetherian, there exists such a filtration with the U irreducible.*

11 Proof. By 2.4 (i), there exists a decomposition of X as a finite union of constructible locally closed subsets X_i such that each $F|X_i$ is locally constant and finite (resp. of finite presentation). Write $X_i = U_i \cap \complement V_i$, where U_i and V_i are constructible opens, and let N be the number of opens of X in the subtopology T generated by $\{U_i, V_i\}$. We argue by induction on N: if W is a minimal nonempty element of T, it is plain that $F|W$ is locally constant and of finite presentation. Let $i : W \to X$ be the inclusion morphism and let $Y = X - W$. We have the exact sequence

$$0 \longrightarrow i_! i^* F \longrightarrow F \longrightarrow F|Y \longrightarrow 0,$$

and are thereby reduced to the same assertion for Y and the sheaf $F|Y$, where the induction hypothesis applies.

PROPOSITION 2.6. *Suppose that A is a noetherian ring.*

(i) *A finite projective (resp. inductive) limit of constructible sheaves of A-modules or sets is constructible. In particular, the kernel, cokernel, and image of a morphism of constructible sheaves of A-modules are constructible.*

(i bis) *A finite projective limit of constructible sheaves of sets (resp. of groups) is constructible, and if $f : F \to G$ is a morphism of constructible sheaves of groups, the kernel, image, and cokernel (if $\mathrm{Im}(f)$ is a normal subgroup) are constructible. A finite inductive limit of constructible sheaves of sets is constructible.*

(ii) *Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of sheaves of A-modules (resp. of groups), with M', M'' constructible. Then M is constructible.*

(iii) *Let $M_1 \to M_2 \to M_3 \to M_4 \to M_5$ be an exact sequence of sheaves of A-modules. Then if M_i is constructible for $i = 1, 2, 4, 5$, the same holds for $i = 3$.*

12 *Proof.* (i) We leave the assertions for sets and groups to the reader. For sheaves of A-modules, it is enough to prove that the kernel and cokernel of a morphism $f : F \to G$ of constructible sheaves of A-modules are constructible. The assertion is local on X, and we may therefore suppose that X is affine, hence quasi-compact and quasi-separated. By 2.4. (i), we then reduce to the case in which F and G are locally constant, and conclude by using 2.1 (ii) and the noetherian hypothesis on A. Assertion (ii) is proved analogously using 2.1 (iii). Assertion (iii) follows immediately from (i) and (ii).

PROPOSITION 2.7. *Let X be a quasi-compact and quasi-separated scheme, and let F be a sheaf of sets (resp. of A-modules). In order that F be constructible, it is necessary and sufficient that it be isomorphic to the cokernel of a pair of morphisms $H \rightrightarrows G$, where H*

and G are sheaves of sets representable by schemes étale and of finite presentation over X (resp. that F be isomorphic to the cokernel of a homomorphism $\mathscr{A}_{V,X} \to \mathscr{A}_{U,X}$, where U and V are two schemes étale and of finite presentation over X).

Sufficiency follows easily from 2.6 (i bis).

Suppose that F is a constructible sheaf of sets. Using the existence of infinite coproducts in the étale site over X, one sees that one can find an epimorphism $G \to F$, with G representable by a scheme étale over X, which may moreover be supposed to be a coproduct of affine schemes G_i ($i \in I$), and hence separated over S. For every finite subset J of the index set I, let G_J be the coproduct of the G_i for $i \in J$, and let F_J be its image in F. Since G_J and F are constructible, so is F_J (2.6 (i bis)); hence the set X_J of $x \in X$ such that the fibers of F and F_J at a geometric point of X over x are equal is locally constructible (2.4 (iv)). Since the family of X_J is increasing and has union X, and since X is quasi-compact, one of the X_J equals X (EGA IV 1.9.9). This shows, after replacing G by G_J, that we may suppose G affine, and hence separated over X, and quasi-compact over X since X is quasi-separated (EGA IV 1.2.4). Let then $H = G \times_F G$. It is a subsheaf of $G \times_S G$, which is representable, and is therefore itself representable (VIII 6.1). Since G is moreover constructible (2.6 (i bis)), it follows again from the preceding argument that 13
it is quasi-compact, and therefore (being separated over X) of finite presentation over X. This proves the first assertion of 2.7, and the second is proved by the same method (NB it is not necessary that A be noetherian). Let us note in passing that the proof also proves the following corollary:

Corollary 2.7.1. *Let X be a scheme and let F be a scheme étale over X. In order that the corresponding étale sheaf on X be constructible, it is necessary and sufficient that F be of finite presentation over X.*

Proposition 2.7 will be useful here chiefly for deducing certain limit-passage properties for constructible sheaves:

Corollary 2.7.2. *Let X be a quasi-compact and quasi-separated scheme. Then every sheaf of sets (resp. of ind-p-groups, resp. of A-modules) on X is the inductive limit of a filtered inductive system of constructible sheaves of sets (resp. ...).*

If F is a sheaf of sets, retain the notation of the proof of 2.7, and let L be the set of pairs (J, H'), where J is a finite subset of I and H' is a quasi-compact open subset of $H_J = G_J \times_F G_J$. Order L in the evident way, and observe that, since F is the limit of the F_J, and each F_J is the limit of the $\mathrm{Coker}(H' \to G_J)$ for the quasi-compact opens H' of H_J, F is the inductive limit of the inductive system of the $F_\ell = \mathrm{Coker}(H' \to F_J)$ for $\ell = (H', J) \in L$, proving the case of sets. The case of sheaves of A-modules is treated in essentially the same way, beginning by representing F as the cokernel of a homomorphism $\mathscr{A}_{V,X} \to \mathscr{A}_{U,X}$, with U, V schemes étale and separated over X. The case of ind-$\mathscr{p}$-groups requires a little more attention. Retaining the preceding notation, denote by $L(G_J)$ the free group sheaf generated by the sheaf of sets G_J, by N_J the kernel of the homomorphism $L(G_J) \to F$ deduced from $G_J \to F$, and by L the set of pairs (J, N'), where N' is a subsheaf of *sets* of N_J which is "quasi-compact", i.e. such that there exists an epimorphism $P \to N'$, with P representable by a quasi-compact X-scheme. Order L by declaring that (J, N') is smaller than (J_1, N'_1) if $J \subset J_1$ and if the canonical homomorphism $L(G_J) \to L(G_{J_1})$ maps N_1 into N'_1. It is immediate that L is 14
filtered, and that one obtains an inductive system of sheaves of groups F_ℓ, indexed by L, by taking, for $\ell = (J, N')$, F_ℓ = the quotient sheaf of groups of $L(G_J)$ by the invariant subsheaf of groups generated by N'. It is moreover immediate that F is isomorphic to

the inductive limit of the F_ℓ, and it remains to prove that there exists a subset L' of L, cofinal in L, such that for $\ell \in L'$, F_ℓ is a constructible sheaf of $\mathcal{p}$-groups. For this, it is enough to prove that for every J one can find a subsheaf of sets N' of N_J which is quasi-compact, contains a given quasi-compact subsheaf of sets N'_0, and is such that the corresponding F_ℓ is a constructible sheaf of $\mathcal{p}$-groups. Note first that it is enough for this that the fibers of F_ℓ be finite $\mathcal{p}$-groups: then F_ℓ will automatically be constructible, as the reader will readily verify. On the other hand, the reader may also verify that the set $X_{N'}$ of points x of X such that the fiber of F_ℓ at a geometric point over x is a finite $\mathcal{p}$-group is locally constructible. Moreover, the subsets $X_{N'}$ of X are increasing functions of N', and their union is all of X: this last point follows from the fact that the inductive limit of the F_ℓ, for $\ell = (N', J)$ with J fixed, is the subsheaf of groups F_J of F generated by the subsheaf of sets F_J, which is quasi-compact, and hence $\overline{F_J}$ has fibers that are finite $\mathcal{p}$-groups. On the other hand, in a free group on a finite number of generators, if it is expressed as a filtered union of its finite subsets, one of these generates the whole subgroup. It follows readily that the union of the

$X_{N'}$ is indeed X. Using (EGA IV 1.9.9) once again, one finds that one of the $X_{N'}$ is equal to X, which completes the proof of 2.7.2.

. Note that the somewhat laborious proof in the case of group schemes brings to light the lack of a convenient sorites for constructibility conditions on sheaves of groups with fibers not necessarily finite; we apologize to the reader, inviting them to fill this gap by their own means if necessary, and point out by way of consolation that in 2.9 (iii) we obtain a simpler proof when X is Noetherian.

15 Corollary 2.7.3. *Let X be a quasi-compact and quasi-separated scheme, and let F be a constructible sheaf of sets (resp. of finite ind-groups, resp. of A-modules). Then the functor* $\mathrm{Hom}(F, G)$, *where G is a variable sheaf of sets (resp. ...), commutes with inductive limits in G. When F is a sheaf of A-modules, with A Noetherian, the functors* $\mathrm{Ext}^i(X; F, G)$ *in the A-module G also commute with inductive limits.*

This follows from 2.7 and (VII 3.3) by the usual arguments, which are left to the reader. Similarly, the conjunction of 2.7 and the results of (VII 5) easily gives:

Corollary 2.7.4. *Let I be a filtered category, and let $i \mapsto X_i : I^0 \to$ (Sch) be a functor which transforms arrows into affine morphisms and objects into quasi-compact and quasi-separated schemes. For every scheme Y, let $F_c(Y)$ be the category of constructible sheaves of sets (resp. of ind-$\mathcal{p}$-groups, resp. of A-modules) on Y. Then there is an equivalence of categories*

$$F_c(X) \overset{\approx}{\leftarrow} \varinjlim_i F_c(X_i),$$

where $X = \varprojlim_i X_i$ (VII 5) *and where the second member denotes the $\varinjlim$ category of fibered categories, defined in In particular, every constructible sheaf of sets (resp. ...) on X is isomorphic to the inverse image of a sheaf of the same kind on one of the X_i.*

Proposition 2.8. *Let $f : X \to Y$ be a surjective morphism locally of finite presentation, and let F be a sheaf of sets (resp. ...) on Y. Then F is constructible if and only if $f^*(F)$ is constructible.*

We shall use the following

Lemma 2.8.1. *Let $f : X \to Y$ be a surjective morphism locally of finite presentation, with Y quasi-compact and quasi-separated. Then there exists a finite partition of Y into*

16 *subschemes Y_i of finite presentation over Y (in particular, each Y_i is constructible in X), and, for every i, finite surjective morphisms $Y_i'' \xrightarrow{g_i} Y_i' \xrightarrow{h_i} Y_i$, with h_i étale and g_i locally free (i.e. flat and of finite presentation, in addition to the condition that g_i be finite) and radical, and finally a Y-morphism $Y_i'' \to X$* [34].

Using the fact that Y is quasi-compact and quasi-separated, we are easily reduced to the case where Y is quasi-affine, hence separated (by using a finite affine open covering $(U_i)_{1\leq i\leq n}$ of Y and considering the $Y_i = U_i - U_i \cap \bigcup_{j<i} U_j$), and then to the case where Y is affine (by the same argument). The scheme X is a union of affine open subsets X_i, and the image T_i of X_i in Y is constructible (EGA IV 1.8.4); the union of the T_i is X since f is surjective, whence it follows that X is the union of a finite number of the T_i (EGA IV 1.9.9), so that, replacing X by the sum scheme of the corresponding X_i, we may suppose that X is affine. The standard limit procedure (EGA IV 8 8.1.2 c)) then allows us to suppose in addition that Y is Noetherian. The usual Noetherian induction, together with the limit procedure (EGA IV 8.1.2 a)), reduces us to the case where Y is the spectrum of a field k. Since then $X \neq \emptyset$, there exists a closed point x in X, corresponding to a finite extension k' of k, itself a radicial extension of a separable extension k'_s of k. We then take $Y' = \mathrm{Spec}(k'_s)$, $Y'' = \mathrm{Spec}(k')$, which completes the proof of 2.8.1.

Necessity in 2.8 being trivial (2.4 (iii)), let us prove sufficiency. By the preceding lemma we are reduced to the case where f is of the form gh, where g and h satisfy the conditions stated for g_i, h_i in the lemma. We may therefore suppose successively either that f is finite radicial, a trivial case by (VIII 1.1), or that f is finite étale. Note moreover that, using the hypothesis that $f^*(F)$ is constructible and the definition of constructibility, we could suppose (after replacing X by an X-prescheme $X' = \coprod_i X_i$ of finite presentation and with surjective structural morphism) that $f^*(F)$ is already locally constant. But when f is étale and surjective, this implies that F is locally constant. Since in addition the hypotheses made on the fibers of $f^*(F)$ remain valid for those of F, with f surjective, it indeed follows that F is constructible, which completes the proof of 2.8.

Proposition 2.9. *Let X be a Noetherian scheme, let A be a Noetherian ring, and let p* 17
be a set of prime numbers.

(i) *The category of sheaves of sets (resp. of ind p-groups, resp. of A-modules) on X is locally Noetherian, that is, it has a set of generators formed of Noetherian objects.*
(ii) *A sheaf F of sets (resp. of ind p-groups, resp. of A-modules) on X is constructible if and only if it is Noetherian. If F is a sheaf of A-modules, then F is constructible if and only if it is a quotient of a finite sum of sheaves of the form $A_{U/X}$, where $U \to X$ is étale and of finite type (notation of* IV.2.4*: $A_{U/X} = A_U$).*
(iii) *The constructible subsheaves of a sheaf F of sets (resp. ...) form an inductive system, and F is the inductive limit of these subsheaves.*

Proof. We leave the assertion for sets to the reader. First consider the case of a sheaf of A-modules. For (i), one must find a set of generators whose elements are Noetherian objects. But the sheaves of the form $A_{U/X}$, with $U \to X$ étale and of finite type, form a set of generators which are easily seen to be constructible, and it therefore suffices to prove the following lemma:

Lemma 2.10. *Let X be Noetherian. Then every constructible A-module is Noetherian.*

Proof of the lemma.

[34] This assertion is also given in EGA IV 17.16.4.

By Noetherian induction, we suppose the lemma true for every subscheme X' of X distinct from X. Choose a nonempty open subset X_0 of X such that F induces a locally constant sheaf F_0 on X_0.

Now let $\{F_i\}$, $i \in \mathbf{N}$, be an increasing sequence of subsheaves of F_0. Since, for a geometric point P which is generic for an irreducible component of X_0, the fiber of F_0
18 at P is of finite type, the sequence of fibers $(F_i)_P$ is stationary, and we may suppose that it is constant. Now let Q be a geometric point of X_0 which is a *specialization* of P (VIII 7.2). Since F_0 is locally constant on X_0, one sees immediately that the specialization morphism $F_{0_Q} \to F_{0_P}$ (VIII.7) is bijective, hence $(F_i)_Q \to (F_i)_P$ is *injective* for every i.

Let V/X be étale and pointed at P, and let $s_1, \dots, s_n$ be sections $\in F_1(V)$ which generate $(F_1)_P$. Then the s_i generate $(F_i)_Q$ for every i and every geometric point Q which is a specialization of P above V. The sequence of sheaves F_i is thus constant in a neighborhood of the image of P, say in $X' \neq \emptyset$. Let $Y = X - X'$; it therefore suffices to prove that the sequence of the $F_i|Y$ is stationary, which follows from the Noetherian induction hypothesis.

The assertions (ii) for A-modules are now immediate: since the $A_{U/X}$ are Noetherian generators, one plainly has F Noetherian $\Leftrightarrow$ F is a quotient of a finite sum of sheaves $A_{U/X}$. But $A_{U/X}$ is constructible, and consequently, in view of 2.6, F Noetherian $\Rightarrow$ F constructible; the converse implication having already been proved (2.10), this establishes (ii). Finally, assertion (iii) for A-modules is a property of locally Noetherian categories (cf. [1], Ch. IV).

Now consider the case of ind-p-groups. First note that a constructible sheaf of groups F is certainly Noetherian. Indeed, since the fibers are finite, we may use the same argument as for A-modules. We shall now prove assertion (iii) directly, which will imply (i)—indeed, it will show that constructible sheaves form a set of generators. Moreover, any Noetherian sheaf will then necessarily be constructible, whence (ii).

To prove (iii), it plainly suffices to prove that a subsheaf (of groups) S of a sheaf F of ind p-groups generated by a finite number of sections $s_i \in F(U_i)$, with $U_i \to X$ étale
19 and of finite type, is constructible [34]. (Since a constructible sheaf is Noetherian, it is generated by a finite number of sections.) By Noetherian induction, we may suppose that this is true for every closed subscheme Y of X distinct from X. Now the notion of a subsheaf generated by sections commutes with the inverse image of sheaves; indeed, let $L_{U/X}$ be the sheaf which represents the functor $F \mapsto F(U)$ in the category of groups. Then if $f : Y \to X$, one has $f^*(L_{U/X}) = L_{L \times_X U/Y}$. Since S is the image of the sum of the $L_{U_i/X}$, and since f^* commutes with sums, it is clear that f^*S is the subsheaf of f^*F generated by the sections $f^*(S_i)$.

We may therefore suppose that S induces a constructible sheaf on every closed subset Y distinct from X, and it thus suffices (2.4 (i)) to prove that S induces a constructible sheaf on a suitable nonempty open subset of X. Replacing X by a sufficiently small open subset, we may suppose that every U_i is an étale covering of X. It suffices to prove that S is then locally constant, with finite fibers, on a suitable nonempty open subset. This assertion is local on X for the étale topology, so we may suppose that every U_i is completely split, say $U_i = \amalg_{n_i} X$ (the sum of n_i copies of X). Let s_{ij} $(j = 1, \dots, n_i)$ be the sections of $F(X)$ such that s_i $(s_{i1}, \dots, s_{in_i})$. Plainly, S is none other than the subsheaf of F generated by the $s_{ij} \in F(X)$. Since $F(X)$ is an ind p-group (1.5), the s_{ij}

[34] Recall the definition of this sheaf: let $L_{U/X}$ be the sheaf representing the functor $F \mapsto F(U)$ in the category of groups. The section $s_i \in F(U_i)$ induces a morphism $L_{U_i/X} \to F$, and S is the image of the sum morphism $\amalg_i L_{U_i/X} \to F$. It is the smallest subsheaf which "contains" the sections s_i.

generate a finite subgroup 1.3. Thus we may suppose that the set of the s_i is a finite subgroup of $F(X)$; but then S is identical to the sheaf *of sets* generated by the s_i, which is constructible by 2.6 (i) as a sheaf of sets, hence as a sheaf of groups, C.Q.F.D.

In the following two propositions, we give criteria for a sheaf to be locally constant or constructible, using specialization homomorphisms (VIII 7.7).

PROPOSITION 2.11. *Let X be a scheme, and let F be a constructible sheaf of sets (resp. of ind-finite groups, resp. of A-modules); in the case of sets, suppose in addition that F has* 20
finite fibers. Let $x \in X$, and let x be a geometric point over x. In order that F be locally constant in a neighborhood of x, it is necessary and sufficient that, for every geometric point $\overline{x}'$ which is a generization of $\overline{x}$, and every specialization arrow $\overline{x'} \to \overline{x}$ (VIII 7.2)*, the specialization homomorphism* (VIII 7.7) $F_{\overline{x}} \to F_{\overline{x'}}$ *be an isomorphism.*

The case of a sheaf of groups being a special case of that of a sheaf of sets, we confine ourselves to the latter; the case of a sheaf of modules is treated in essentially the same way. Necessity of the condition being trivial, we need only establish sufficiency. Since the fiber $I = F_{\overline{x}}$ is finite, after replacing X by a suitable scheme X' étale over X, we may suppose that one can find a homomorphism $u : I_X \to F$ from the constant sheaf I_X to F, inducing an isomorphism on the fibers at $\overline{x}$. Using the constructibility of both sheaves and 2.6, one easily sees that the set Z of points z of X such that the homomorphism induced on the fibers at a geometric point $\overline{z}$ over z is bijective is a locally constructible subset of X. The hypothesis immediately implies that it contains the generizations of x, hence (EGA IV 1.10.1) it is a neighborhood of x, which proves that F is locally constant (indeed, even constant in this case) in a neighborhood of x, C.Q.F.D.

DEFINITION 2.12. [34]*: A scheme X is said to be path-connected if for every pair P, Q of geometric points of X, there exist geometric points $P = P_0, \dots, P_n; Q_1, \dots, Q_n = Q$ and specializations* (VIII 7.2) *$P_i \to Q_i$ and $P_{i-1} \to Q_i$ $(i = 1, \dots, n)$. One says that X is locally path-connected (for the étale topology) if for every étale $U \to X$ there exists an étale covering $\{U_i \to U\}$ of U such that U_i is path-connected for every i.*

One checks immediately that a locally Noetherian prescheme is locally path-connected.

PROPOSITION 2.13. (i) *Let X be locally path-connected and let F be a sheaf of* 21
sets (resp. of groups, resp. of A-modules) on X. Suppose that the fibers of F are finite (resp. finite, resp. of finite presentation). Then F is locally constant if and only if, for every specialization $P \to Q$ of geometric points, the specialization morphism (VIII 7.7) *$F_Q \to F_P$ is bijective.*

(ii) *Suppose that X is locally Noetherian and that the fibers of F are finite (resp. finite, resp. of finite presentation). Then, if for every specialization $P \to Q$ the specialization morphism $F_Q \to F_P$ is injective, F is constructible.*

(iii) *Suppose that X is locally Noetherian and that the fibers of the sheaf of sets F are finite. Let $c : X \to \mathbf{N}$ be the function which associates with $x \in X$ the number of elements of the fiber $F_{\overline{x}}$ of F at a geometric point $\overline{x}$ above x. Then F is constructible if and only if c is a constructible function, i.e. if and only if $c^{-1}(n)$ is constructible for every $n \in \mathbf{N}$.*

Proof.

(i) Let P be a geometric point of X. We shall find an étale neighborhood of P, that is, an étale $U \to X$ pointed at P, such that F becomes constant on U. Let V' be

[34] The terminology is due to S. Lubkin.

an étale neighborhood of P such that $F(V) \to F_P$ is surjective. Such a V' exists by the finiteness hypothesis. Let $U \to V'$ be an étale neighborhood of P in V' such that U is path-connected. Then $F(U) \to F_P$ is still surjective. Choose a subset $S \subset F(U)$ which maps bijectively onto F_P. If F is a sheaf of groups (resp. of A-modules), one easily sees that, after changing U if necessary, one may find such a set which is moreover a subgroup (resp. submodule) of $F(U)$. This gives a morphism $S_U \to F|U$, where S_U is the constant sheaf with value S. One has

$$(S_U)_P \xrightarrow{\sim} F_P.$$

22 Since every specialization morphism on the fibers of F (resp. S_U) is bijective, and since U is path-connected, it follows that $S_U \xrightarrow{\sim} F|U$, whence the result.

(ii) The assertion is local, and we may therefore suppose that X is Noetherian. Let P be a geometric point above a maximal point of X. There exists an irreducible étale neighborhood, hence a path-connected one, $U \to X$ of P such that $F(U)$ maps surjectively onto F_P. It then maps surjectively onto F_Q for every geometric point Q of V, or, what amounts to the same thing, for every geometric point of the image U of V in X, since such a point is a specialization of P and $F_Q \to F_P$ is injective. Thus every specialization morphism in U is bijective, hence $F|U$ is locally constant. By Noetherian induction we may suppose that $F|X - U$ is constructible, whence the result.

(iii) It plainly suffices to treat the case where the function c is constant. Let $V \to X$ be an étale neighborhood of a geometric point P above a maximal point of X, such that $F(V)$ maps surjectively onto F_P. Let S be a subset of $F(V)$ such that $S \xrightarrow{\sim} F_P$. Then, for the constant sheaf S_V and every specialization $P \to Q$ of geometric points, one has a commutative diagram

$$\begin{array}{ccc} (S_V)_P & \xrightarrow{\sim} & F_P \\ \uparrow s & & \uparrow \\ (S_V)_Q & \longrightarrow & F_Q. \end{array}$$

Since $c(F_Q) = c(F_P)$, the arrow $F_Q \to F_P$ is bijective, so F is locally constant in a neighborhood of P by (i). By Noetherian induction, we are done.

The following result (ii) will be of technical use later; it will allow us to reduce certain verifications to the case of a constant sheaf.

23 PROPOSITION 2.14. *Let X be a scheme.*

(i) *For every finite morphism of finite presentation $f : X' \to X$, and every constructible sheaf of sets (resp. of ind-finite groups, resp. of A-modules) on X', $f_*(F)$ is constructible.*

(ii) *Suppose that X is quasi-compact and quasi-separated, and let F be a constructible sheaf of sets (resp. ...) on X. Then one can find a finite family $(p_i : X'_i \to X_i)$ of finite morphisms, for every i a constant constructible sheaf of sets (resp. ...) C_i on X'_i, and a monomorphism*

$$F \hookrightarrow \prod_i p_{i*}(C_i).$$

First prove (i). By definition, we may suppose that X is affine, and then, using 2.7.4, the standard limit procedure reduces us to the case where X is Noetherian. A further passage to the limit, again using 2.7.4 and also 2.4. (v), reduces us to the case where X is the spectrum of a field. The conclusion then reduces to the corresponding assertion concerning the fiber of $f_*(F)$ at a geometric point over X, which follows immediately from (VIII 5.5, 5.8).

Let us prove (ii). By 2.4 (i), there exists a partition of X into subschemes Y_i, with inclusion morphisms $u_i : Y_i \to X$ of finite presentation, such that the restriction F_i of F to every Y_i is a locally constant sheaf. It is then immediate that the canonical homomorphisms $F \to u_{i*}(F_i)$ define a homomorphism $F \to \prod_i u_{i*}(F_i)$ which is a monomorphism.

In the preceding construction, one could even arrange that every F_i be not only locally constant, but also *isotrivial*, i.e. that there exist a surjective *finite* étale morphism $q_i : Y_i' \to Y_i$ such that $q_i^*(F_i)$ is a constant sheaf on Y_i', say $C_{iY_i'}$. This follows easily, for example, from the definition of 2.8.1 and from (VIII 1.1). On the other hand, it follows 24
from the "Main Theorem" in the form (EGA IV 8.12.6) that there exists a finite morphism $p_i : Z_i \to X$ and an open immersion $v_i : Y_i' \to Z_i$ of X-schemes.

Since F_i embeds into $q_{i*}(q_i^*(F_i)) = q_{i*}(C_{iY_i'})$, F embeds into the product of the $u_{i*}(q_{i*}(C_{iY_i'})) = p_{i*}(v_{i*}(C_{iY_i'}))$. If every Z_i is normal, then Lemma (2.14.1) below implies that $v_{i*}(C_{iY_i'}) \simeq C_{iZ_i}$, hence F embeds into the product of the $p_{i*}(C_{iZ_i})$, and we are done. In the general case, introduce the normalization $\overline{Z}_i$ of Z_i, the inverse image $\overline{Y}_i'$ of Y_i' in $\overline{Z}_i$, and the immersion $\overline{v}_i : \overline{Y}_i' \to \overline{Z}_i$. One sees immediately that $\overline{Y}_i'$ contains the maximal points of $\overline{Z}_i$, and is therefore dense in $\overline{Z}_i$. If $r_i : \overline{Z}_i \to Z_i$ and $s_i : \overline{Y}_i' \to Y_i'$ are the projections, since the latter is surjective, it follows that $C_{iY_i'}$ embeds into $s_{i*}(C_{i\overline{Y}_i'})$, hence $v_{i*}(C_{iY_i'})$ embeds into $v_{i*}(s_{i*}(C_{i\overline{Y}_i'})) = r_{i*}(\overline{v}_{i*}(C_{i\overline{Y}_i'}))$, which is isomorphic to $r_{i*}(C_{i\overline{Z}_i})$ by (2.14.1) below. Thus F embeds into the product of the $p_{i*}(r_{i*}(C_{\overline{Z}_i})) = t_{i*}(C_{i\overline{Z}_i})$, where $t_i = p_i r_i$ is the canonical projection $\overline{Z}_i \to X$. When this morphism is finite for every i, we are done. In the general case, one can only say that $\overline{Z}_i$ is *integral* over X, hence of the form $\mathrm{Spec}(\mathscr{A}_i)$, where $\mathscr{A}_i$ is a quasi-coherent sheaf of $\mathcal{O}_X$-algebras which is integral over $\mathcal{O}_X$. Using (EGA IV 1.7.7), one sees that $\mathscr{A}_i$ is the inductive limit of its subalgebras $\mathscr{A}_{i,j}$ finite over X, hence $\overline{Z}_i$ is the projective limit (VII 5) of the preschemes $Z_{ij} = \mathrm{Spec}(A_{ij})$, finite over X. Applying (VII 5.11), one finds that $t_{i*}(C_{i\overline{Z}_i})$ is the inductive limit of the $t_{ij*}(C_{iZ_{ij}})$, where $t_{ij} : Z_{ij} \to X$ is the canonical projection. By (2.7.3), the homomorphism $F \to t_{i*}(C_{i\overline{Z}_i})$ therefore factors through a homomorphism $F \to t_{ij*}(C_{iZ_{ij}})$ for a suitable $j = j(i)$. We now set $X_i' = Z_{i,j(i)}$, and obtain an embedding as stated in 2.14. It remains to prove the following

Lemma 2.14.1. *Let X be a geometrically unibranch scheme (for example, a normal scheme), and let $i : U \to X$ be a quasi-compact open immersion. Then the closure Z of U, endowed with the induced reduced structure, is also geometrically unibranch; more* 25
precisely, for every $z \in Z$, one has $\mathcal{O}_{Z,z} \xleftarrow{\sim} \mathcal{O}_{X_{\mathrm{red}},z}$. Suppose on the other hand that i is dominant, and let C be a set, with C_U the constant sheaf on U defined by C. Then the natural homomorphism $C_X \to i_*(C_U)$ is an isomorphism. The same conclusions hold if the open immersion i is replaced by an inclusion $i : x \to X$, where x is a maximal point of X.

Using (EGA IV 2.3.11), we are reduced for the first assertion to the case where X is local, with closed point z; but then $X_{\text{réd}}$ is integral and U is nonempty, hence $Z = X_{\text{réd}}$,

and the assertion is evident (thus, instead of assuming X geometrically unibranch, it would have sufficed here to suppose that the local rings of $X_{\text{réd}}$ are integral domains). For the second assertion, looking fiber by fiber, we are reduced by (VIII 5.3) *loc. cit.* to the case where X is strictly local, and then to proving that $H^0(X, C_X) \to H^0(U, C_U)$ is bijective. Here again, X is irreducible, and U is a nonempty open subset of it and hence irreducible, and the two sides of the preceding map are identified with C, whence the desired conclusion. (Here the notion "geometrically unibranch" has been used in the form of (EGA IV 18.8.14), as meaning that the strict localizations of the rings of $X_{\text{réd}}$ are integral domains.) The case of the inclusion of a maximal point (stated here for ease of future reference) is treated in exactly the same way.

Remarks 2.14.2. When X is Noetherian, the proof given (or an immediate deduction) shows that in 2.14 (ii), the X_i' may be assumed integral; if in addition X is assumed universally Japanese, i.e. the rings of its affine open subsets are universally Japanese (EGA 0_{IV} 23.1.1, IV 7.7.2), the X_i' may moreover be assumed normal.

3. Kummer and Artin–Schreier theories

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3.0. We denote by $(\mathbf{G}_m)_X$ the sheaf represented by the multiplicative group $\operatorname{Spec} \mathcal{O}_X[t, t^{-1}]$ over X. Thus, for $U \to X$ étale, the sections of $(\mathbf{G}_m)_X$ over U are the invertible elements $\Gamma(U, \mathcal{O}_U^*)$ of $\Gamma(U, \mathcal{O}_U)$. Let $n \in \mathbf{N}$. The kernel of the n-th power map on $(\mathbf{G}_m)_X$ is the "sheaf of n-th roots of unity", denoted $(\boldsymbol{\mu}_n)_X$. Thus there is an exact sequence

$$0 \longrightarrow (\boldsymbol{\mu}_n)_X \longrightarrow (\mathbf{G}_m)_X \xrightarrow{n} (\mathbf{G}_m)_X. \tag{3.1}$$

If n is invertible on X (that is, if n is prime to every residue characteristic of X), the n-th power map is surjective *as a morphism of sheaves*:

Kummer theory 3.2. *If $n \in \mathbf{N}$ is invertible on X, there is an exact sequence*

$$0 \longrightarrow (\boldsymbol{\mu}_n)_X \longrightarrow (\mathbf{G}_m)_X \xrightarrow{n} (\mathbf{G}_m)_X \longrightarrow 0.$$

Indeed, let $u \in (\mathbf{G}_m)_X(U)$, with $U \to X$ étale. We must find an étale covering $\{U_i \to U\}$ such that u becomes an n-th power on every U_i. Since n is invertible on U, the equation

$$T^n - u = 0$$

is "separable" over U, that is,

$$U' = \operatorname{Spec} \mathcal{O}_U[T]/(T^n - u)$$

is étale over U. Since $U' \to U$ is surjective, it is a covering, and u becomes an n-th power on U'.

27 We point out that when n is invertible on X, $(\boldsymbol{\mu}_n)_X$ is a locally constant sheaf, in fact locally isomorphic to the sheaf $(\mathbf{Z}/n\mathbf{Z})_X$, and is constant in important cases. Kummer theory therefore gives information about the cohomology of X with values in certain constant sheaves. By definition,

$$H^0(X, (\mathbf{G}_m)_X) = \Gamma(X, \mathcal{O}_X^*),$$

and in dimension one we have:

Theorem 3.3. *("Hilbert's Theorem 90") There is an isomorphism*

$$H^1(X, (\mathbf{G}_m)_X) \xrightarrow{\sim} \operatorname{Pic} X,$$

where $\operatorname{Pic} X$ *is the group of isomorphism classes of invertible sheaves on X.*

Proof. This says that the canonical morphism

$$H^1(X_{\text{zar}}, (\mathbf{G}_m)_X) \longrightarrow H^1(X_{\text{ét}}, (\mathbf{G}_m)_X)$$

is bijective. Equivalently,

$$R^1\epsilon_*(\mathbf{G}_m)_X = 0,$$

where ϵ_* is direct image for the evident morphism of sites $\epsilon : X_{\text{et}} \to X_{\text{zar}}$ (VIII 4). We are therefore reduced to the case where X is affine. Since H^1 may be computed by Čech cohomology (V 2.5), and since for quasi-compact X it is enough to use quasi-compact coverings, this follows immediately from descent theory for sheaves (see FGA 190 or SGA VIII 1).

3.3.1. Suppose that X is Noetherian and has no embedded component, and let x_j be the maximal points of X. Let R_j be the local Artinian ring of X at x_j, and let i_j : Spec $R_j \to X$ be the inclusion. There is a canonical injection 28

$$(\mathbf{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbf{G}_m)_{\text{Spec } R_j}$$

of sheaves on $X_{\text{ét}}$ (respectively X_{zar}). Its cokernel D is called the *sheaf of Cartier divisors* on $X_{\text{ét}}$ (respectively X_{zar})[34].

Corollary 3.4. *Let X be Noetherian and without embedded components, let $\epsilon : X_{\text{ét}} \to X_{\text{zar}}$ be the canonical morphism of sites, and let D be the sheaf of Cartier divisors on $X_{\text{ét}}$. Then $\epsilon_* D$ is the sheaf D_{zar} of Cartier divisors on X_{zar}, and in particular*

$$H^0(X_{\text{ét}}, D) \simeq H^0(X_{\text{zar}}, D_{\text{zar}}).$$

Proof. Applying ϵ_* to the exact sequence

$$0 \longrightarrow (\mathbf{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbf{G}_m)_{\text{Spec } R_j} \longrightarrow D \longrightarrow 0$$

gives an exact sequence, since $R^1\epsilon_*(\mathbf{G}_m)_X = 0$ (3.3); the first assertion follows.

When X has characteristic $p \neq 0$, the following result sometimes replaces 3.2 in the study of p-torsion coefficients:

Artin–Schreier theory 3.5. *If X has characteristic $p > 0$, there is an exact sequence*

$$0 \longrightarrow (\mathbf{Z}/p\mathbf{Z})_X \longrightarrow \mathcal{O}_X \xrightarrow{\mathfrak{p}} \mathcal{O}_X \longrightarrow 0,$$

where $\mathfrak{p}$ is the morphism of additive groups defined by $\mathfrak{p}(f) = f^p - f$.

The kernel of $\mathfrak{p}$ is the sheaf of sections of $\mathcal{O}_X$ which locally belong to the prime field, and is therefore a constant sheaf. The morphism $\mathfrak{p}$ is surjective for the étale topology 29 because the equation $T^p - T - a$ ($a \in \Gamma(U, \mathcal{O}_U)$) is always "separable" in characteristic p, that is, defines an étale covering of U. In applying this exact sequence, we shall of course use VII 4.2.

Proposition 3.6. (i) *Let X be a scheme and let $i : x \to X$ be the inclusion of a point. Then*

$$H^1(X, i_*(\mathbf{Z})_x) = 0,$$

where $(\mathbf{Z})_x$ is the "constant sheaf" of value $\mathbf{Z}$ (see 2.0).

[34] In EGA IV 21, the word "divisors" is used instead of "Cartier divisors".

(ii) *If X is irreducible and if, for every geometric point $\overline{x}$ of X, the strict localization $X_{\overline{x}}$ of X at $\overline{x}$ is irreducible (one should say that "X is geometrically unibranch", see* EGA *IV 18.8.14), then*

$$H^1(X, (\mathbf{Z})_X) = 0.$$

Proof.

(i) The Leray spectral sequence

$$E_2^{pq} = H^p(X, R^q i_*(\mathbf{Z})_x) \Longrightarrow H^{p+q}(x, (\mathbf{Z})_x)$$

gives an injection $H^1(X, i_*(\mathbf{Z})_x) \to H^1(x, (\mathbf{Z})_x)$. Since $x = \operatorname{Spec} k(x)$ is the spectrum of a field, it is well known that $H^1(x, (\mathbf{Z})_x) = 0$. (This follows from the fact that $H^q(x, F)$ is torsion for $q > 0$ in any case (see VII 2.2 and CG), and from the exact sequences $0 \to \mathbf{Z} \xrightarrow{n} \mathbf{Z} \to \mathbf{Z}/n \to 0$.) This proves (i).

(ii) Let $i : x \to X$ be the inclusion of the generic point of X. Under the hypotheses of (ii), the canonical morphism $(\mathbf{Z})_X \to i_*(\mathbf{Z})_x$ is immediately seen to be bijective (and the converse is also true); thus (ii) is reduced to (i).

Remarks 3.7. In 3.6 (i) and (ii), $\mathbf{Z}$ may be replaced by any torsion-free abelian group. On the other hand, (ii) becomes false if the hypothesis that X is geometrically unibranch is omitted, as one sees, for example, by taking X to be an algebraic curve over a field K having an ordinary double point (compute $H^1(X, \mathbf{Z})$ in this case!).

4. The case of an algebraic curve

4.0. Let X be a Noetherian scheme of dimension 1. Let $R(X)$ be the ring of rational functions on X, and let $i : \operatorname{Spec} R(X) \to X$ be the inclusion. The ring $R(X)$ is Artinian, the product of the local rings of X at its maximal points, and the results of VIII 2 may be applied.

Let F be an abelian sheaf on $\operatorname{Spec} R(X)$, and consider the sheaves $R^q i_* F$, $q > 0$, on X. It follows at once from VIII 5.2 that the fiber of $R^q i_* F$ at a geometric point lying over a maximal point of X vanishes. Since X has dimension 1, such a sheaf is of a very special kind:

Lemma 4.1. *Let X be Noetherian and let F be a sheaf on X. The following conditions are equivalent (a sheaf satisfying them will be called a "skyscraper sheaf"):*

(i) *The fiber of F at every geometric point $\overline{x}$ of X lying over a nonclosed point $x \in X$ vanishes.*

(ii) *Every section $s \in F(X')$, where $X' \to X$ is étale of finite type, vanishes away from finitely many closed points of X.*

(iii) *There is an isomorphism $F \simeq \bigoplus_x i_{x*} i_x^* F$, where x runs through the closed points of X and $i_x : x \to X$ is the inclusion.*

30 *Proof.* Suppose (i) holds and let $z \in F(X')$, where $X' \to X$ is étale of finite type. The support of z (VIII 6.6) is then a closed subset of X' consisting of *closed* points, and is therefore finite because X' is Noetherian. Thus (i) implies (ii). The implication (ii) to (i) follows from the computation of fibers (VIII 3.9), and (iii) implies (i), since for every family (G_x) of sheaves on the closed points x of X, the sheaf $F = \bigoplus i_{x*}(G_x)$ on X satisfies (i): fiber functors commute with direct sums, and the support of $i_{x*}(G_x)$ is plainly contained in $\{x\}$.

It remains to prove that (ii) implies (iii). Consider the canonical morphism

$$F \longrightarrow \prod_x i_{x*} i_x^* F,$$

where x runs through the closed points of X. By (ii), it factors through

$$F \longrightarrow \bigoplus_x i_{x*} i_x^* F,$$

which gives the desired morphism. We leave to the reader the verification that it is bijective.

4.1.1. Apply the lemma to the sheaf $R^q i_* F$ introduced above. One has

$$R^q i_* F \xrightarrow{\sim} \bigoplus_x i_{x*} i_x^* (R^q i_* F),$$

where x runs through the closed points of X. By VIII 5, the fiber of $R^q i_* F$ at a geometric point $\overline{x}$ of X is

$$(R^q i_* F)_{\overline{x}} \simeq H^q(\operatorname{Spec} R(X_{\overline{x}}), F),$$

where $R(X_{\overline{x}}) \simeq R(X) \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\overline{x}}}$, $X_{\overline{x}}$ is the strict localization of X at $\overline{x}$ (VIII 4), and F also denotes the sheaf induced on $\operatorname{Spec} R(X_{\overline{x}})$. Suppose now that the residue field of every closed point x of X is separably closed. Then (see also VI 1)

$$\begin{aligned} H^p(X, R^q i_* F) &\simeq H^p\left(X, \bigoplus_x i_{x*} i_x^* (R^q i_* F)\right) \\ &\simeq \bigoplus_x H^p\left(X, i_{x*} i_x^* (R^q i_* F)\right) \\ &\simeq \bigoplus_x H^p\left(x, i_x^* (R^q i_* F)\right) \\ &\simeq \begin{cases} \bigoplus_x H^q(\operatorname{Spec} R(X_x), F) & \text{if } p = 0, \\ 0 & \text{if } p > 0. \end{cases} \end{aligned}$$

Thus:

Corollary 4.2. *Let X be a Noetherian scheme of dimension 1 such that, for every closed point x of X, the residue field $k(x)$ is separably closed. Let F be an abelian sheaf on $\operatorname{Spec} R(X)$, where $R(X)$ is the ring of rational functions on X. Consider the Leray spectral sequence*

$$E_2^{pq} = H^p(X, R^q i_* F) \Longrightarrow H^{p+q}(\operatorname{Spec} R(X), F).$$

One has

$$\begin{aligned} E_2^{pq} &= 0 \quad \text{if } p > 0 \text{ and } q > 0, \\ E_2^{0q} &= \bigoplus_x H^q(\operatorname{Spec} R(X_x), F) \quad \text{if } q > 0, \end{aligned}$$

where x runs through the closed points of X, and X_x is the strict localization of X at x.

The spectral sequence therefore gives an exact sequence

$$0 \longrightarrow H^1(X, i_*F) \longrightarrow H^1(\operatorname{Spec} R(X), F) \longrightarrow \bigoplus_x H^1(\operatorname{Spec} R(X_x), F)$$
$$\longrightarrow H^2(X, i_*F) \longrightarrow H^2(\operatorname{Spec} R(X), F) \longrightarrow \bigoplus_x H^2(\operatorname{Spec} R(X_x), F)$$
$$\longrightarrow \cdots .$$

4.3. Suppose moreover that the ℓ-cohomological dimension of $\operatorname{Spec} R(X)$, for a given prime number ℓ, is at most 1 (that is, $H^q(\operatorname{Spec} R(X), F) = 0$ for $q > 1$ and every ℓ-torsion sheaf F on $\operatorname{Spec} R(X)$). Let $K^1, \dots, K^m$ be the residue fields of the Artinian ring $R(X)$, that is, the residue fields of the maximal points of X. By the dictionary of VIII 2, the ℓ-cohomological dimension of $\operatorname{Spec} R(X)$ is

$$\operatorname{cd}_\ell(\operatorname{Spec} R(X)) = \sup_i \{\operatorname{cd}_\ell(G(\overline{K}^i/K^i))\},$$

where $\operatorname{cd}_\ell(G)$ denotes the ℓ-cohomological dimension of the profinite group G (see CG).

Every residue field K_x of $R(\overline{X}(x))$, for a strict localization $\overline{X}(x)$ of X, is plainly a separable (possibly infinite) extension of one of the K^i, and hence

$$\operatorname{cd}_\ell(G(\overline{K}_x/K_x)) \le \operatorname{cd}_\ell(G(\overline{K}^i/K^i))$$

by CG II 4.1, Proposition 10. Thus $\operatorname{cd}_\ell(\operatorname{Spec} R(\overline{X}(x))) \le 1$ as well.

Applying the exact sequence of 4.2 to an ℓ-torsion sheaf F, one obtains

$$\text{(4.3.1)} \qquad \begin{cases} 0 \to H^1(X, i_*F) \to H^1(\operatorname{Spec} R(X), F) \to \\ \qquad \bigoplus_x H^1(\operatorname{Spec} R(\overline{X}(x)), F) \longrightarrow H^2(X, i_*F) \longrightarrow 0, \\ H^q(X, i_*F) = 0 \quad \text{for } q > 2. \end{cases}$$

32 Note that the hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \le 1$ holds if X is an algebraic curve over a separably closed field (Tsen's theorem [6, Theorem 4.3]). Analogous exact sequences in this case were studied by Ogg[2] and Šafarevič.

4.4. Suppose that ℓ is invertible on X and take $F = (\mathbf{G}_m)_{R(X)}$. Applying 3.3 and [CG, Chapter I, Proposition 12], still under the hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \le 1$, one finds

$$\text{(4.4.1)} \qquad \begin{cases} H^1(\operatorname{Spec} R(X), \mathbf{G}_m) = 0, \\ H^q(\operatorname{Spec} R(X), \mathbf{G}_m) \underset{\ell}{=} 0 \quad (q \ge 2), \end{cases}$$

where the ℓ under the equality sign means that the group on the left, which is torsion in any case, has no ℓ-torsion.

The same result holds with $R(X)$ replaced by $R(\overline{X}(x))$. Since the sheaf $\mathbf{G}_m$ on $\operatorname{Spec} R(X)$ plainly induces the sheaf $\mathbf{G}_m$ on $\operatorname{Spec} R(\overline{X}(x))$, 4.2 gives

$$\text{(4.4.2)} \qquad \begin{cases} H^1(X, i_*(\mathbf{G}_m)_{R(X)}) = 0, \\ H^q(X, i_*(\mathbf{G}_m)_{R(X)}) \underset{\ell}{=} 0 \quad \text{if } q > 1. \end{cases}$$

For every sheaf F on X, 4.1 shows that the kernel P and cokernel Q of the canonical homomorphism $F \to i_*i^*F$ are skyscraper sheaves. Since the residue fields at the closed points of X are assumed separably closed, one therefore has $H^j(X, P) = H^j(X, Q) = 0$ for $j > 0$. The long exact cohomology sequence immediately implies that

$$H^j(X, F) \longrightarrow H^j(X, i_*i^*F)$$

is an isomorphism for $j \geq 2$. Taking (4.4.2) into account, one obtains 33

$$H^j(X, \mathbf{G}_m) \underset{\ell}{=} 0 \quad \text{for } j \geq 2, \tag{4.5}$$

valid when X is a Noetherian scheme of dimension 1 satisfying $\mathrm{cd}_\ell(R(X)) \leq 1$ and whose residue fields at closed points are separably closed. Kummer theory (3.2) and 3.3 now give:

THEOREM 4.6. *Let X be Noetherian of dimension 1, and let n be an integer invertible on X. Suppose that $\mathrm{cd}_\ell(R(X)) \leq 1$ for every prime number ℓ dividing n, and that the residue field $k(x)$ is separably closed for every closed point x of X. Then*

$$H^q(X, (\boldsymbol{\mu}_n)_X) = 0 \quad \text{if } q > 2,$$

and the cohomology for $q \leq 2$ is given by the exact sequence

$$0 \longrightarrow H^0(X, (\boldsymbol{\mu}_n)_X) \longrightarrow \Gamma(X, \mathcal{O}_X^*) \xrightarrow{n} \Gamma(X, \mathcal{O}_X^*) \longrightarrow$$
$$H^1(X, (\boldsymbol{\mu}_n)_X) \longrightarrow \mathrm{Pic}\, X \xrightarrow{n} \mathrm{Pic}\, X \longrightarrow H^2(X, (\boldsymbol{\mu}_n)_X) \longrightarrow 0.$$

COROLLARY 4.7. *Let X be a complete connected curve over a separably closed field k whose characteristic is prime to n. Then*

$$H^0(X, (\boldsymbol{\mu}_n)_X) \simeq \boldsymbol{\mu}_n(k),$$
$$H^1(X, (\boldsymbol{\mu}_n)_X) \simeq {}_n(\mathrm{Pic}\, X) = \{\textit{elements of } \mathrm{Pic}\, X \textit{ whose order divides } n\},$$
$$H^2(X, (\boldsymbol{\mu}_n)_X) \simeq (\mathrm{Pic}\, X)/n \simeq (\mathbf{Z}/n\mathbf{Z})^c,$$

where c is the number of irreducible components of X. 34

For the last assertion, recall that the Picard scheme [TDTE V] of such an $X/\mathrm{Spec}\, k$ has a decomposition

$$0 \longrightarrow A \longrightarrow \mathrm{Pic}_{X/k} \longrightarrow \mathbf{Z}^c \longrightarrow 0 \tag{4.8}$$

where A is a connected algebraic group over $\mathrm{Spec}\, k$. The "multiplication by n" map on such an A is étale, hence surjective on k-valued points. It follows that

$${}_n(\mathrm{Pic}\, X) \simeq {}_n(A(k))$$

and

$$(\mathrm{Pic}\, X)/n \simeq (\mathbf{Z}/n\mathbf{Z})^c.$$

If X is smooth over $\mathrm{Spec}\, k$, so that its Jacobian A is an abelian variety, the group ${}_n(\mathrm{Pic}\, X)$ is isomorphic to $(\mathbf{Z}/n)^{2g}$, where g is the genus of X. Of course, $\boldsymbol{\mu}_n(k) \simeq \mathbf{Z}/n\mathbf{Z}$ as well. Thus the ranks of the cohomology groups as $\mathbf{Z}/n$-modules are 1, $2g$, and 1, as expected.

5. The trace method

5.1. Let X be a scheme, let $f : X' \to X$ be an étale covering, and let F be a sheaf on X. Then the adjunction formulas give a *restriction* morphism

$$F \xrightarrow{\mathrm{res}} f_* f^* F.$$

Since f is finite, one has $H^q(X, f_* f^* F) \simeq H^q(X', f^* F)$, whence a morphism, also called restriction:

$$H^q(X, F) \xrightarrow{\mathrm{res}} H^q(X', f^* F), \tag{5.1.1}$$

which is moreover the evident morphism. But when f is étale, there is also another 35

morphism, called the *trace morphism*,

$$f_* f^* F \xrightarrow{\text{tr}} F, \tag{5.1.2}$$

which induces morphisms on cohomology

$$H^q(X', f^*F) \xrightarrow{\text{tr}} H^q(X, F). \tag{5.1.3}$$

The trace morphism is characterized by the following two properties:

(i) It is local for the étale topology.

(ii) If X' is the disjoint sum of d copies of X, so that $f_* f^* F \simeq F^d$, the trace is the summation map $F^d \to F$.

Uniqueness follows trivially from (i), (ii), since every étale covering is locally constant. Once uniqueness is known, existence follows by the usual descent argument.

From (ii) one deduces the following formula:

(5.1.4) $\quad \text{tr} \circ \text{res} : F \longrightarrow F$ is multiplication by the local degree of f.

If the degree is a constant d, it follows that the morphism $\text{tr} \circ \text{res} : H^q(X, F) \to H^q(X, F)$ is multiplication by d.

COROLLARY 5.2. *Let $f : X' \to X$ be an étale covering of constant degree d. If F is an r-torsion sheaf on X with $(r, d) = 1$, then $H^q(X', f^*F) = 0$ implies that $H^q(X, F) = 0$.*

36 Indeed, multiplication by d induces an isomorphism of F, hence an isomorphism on cohomology. If the latter isomorphism factors through zero, then $H^q(X, F) = 0$.

5.2.1. Now let $i : U \to X$ be an immersion, let $f : U' \to U$ be an étale covering of degree d, and suppose that f is induced by a finite morphism $g : X' \to X$ by base change, i.e. that one has the cartesian diagram

$$\begin{array}{ccc} U' & \xrightarrow{i'} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ U & \xrightarrow{i} & X \end{array}$$

Let F be a sheaf on U. One has

$$F \xrightarrow{\text{res}} f_* f^* F \xrightarrow{\text{tr}} F, \qquad \text{tr} \circ \text{res} = d$$

hence, since $i_* f_* = g_* i'_*$,

$$i_* F \xrightarrow{\text{res}} g_* i'_* f^* F \xrightarrow{\text{tr}} i_* F, \qquad \text{tr} \circ \text{res} = d \tag{5.3}$$

and since $i_! f_* \simeq g_* i'_!$ (which follows, for example, from (VIII 5.5)),

$$i_! F \xrightarrow{\text{res}} g_* i'_! f^* F \xrightarrow{\text{tr}} i_! F, \qquad \text{tr} \circ \text{res} = d \tag{5.4}$$

whence

$$(5.3') \qquad H^q(X, i_*F) \xrightarrow{\text{res}} H^q(X', i'_*f^*F) \xrightarrow{\text{tr}} H^q(X, i_*F)$$
$$\qquad\qquad\qquad

where Y is integral and finite over X, and F is of the form $i_!(\mathbf{Z}/\ell)_U$, where $i : U \to X$ is a nonempty open subscheme.

Now there is an exact sequence

$$0 \longrightarrow i_!(\mathbf{Z}/\ell)_U \longrightarrow (\mathbf{Z}/\ell)_Y \longrightarrow (\mathbf{Z}/\ell)_Z \longrightarrow 0$$

where $Z = Y - U$, hence $Z \neq Y$. Arguing by induction on the value of φ, we may
39 suppose that $H^q(\mathbf{Z}, \mathbf{Z}/\ell) = 0$ for $q > \varphi(Y) - 1$, whence $H^q(Y, i_!(\mathbf{Z}/\ell)) = 0$ for $q > \varphi(Y)$ by the exact cohomology sequence.

Corollary 5.7. *Let X be an algebraic curve over a separably closed field k of characteristic different from ℓ, with $\ell \in \mathbf{P}$. Then the cohomology of X with coefficients in a sheaf F of ℓ-torsion vanishes in degrees > 2. If X is affine, the cohomology vanishes in degrees > 1.*

For integral Y finite over X, put $\varphi(Y) = 2 \dim Y$ (resp. $\varphi(Y) = \dim Y$ if X, hence Y, is affine). By 7.6 it is enough to prove that $H^q(Y, \mathbf{Z}/\ell) = 0$ if $q > \varphi(Y)$, which is trivial if $\dim Y = 0$. It therefore remains to prove that $H^q(Y, \mathbf{Z}/\ell) = 0$ if $q > 2$ (resp. if $q > 1$ in the affine case) for an integral curve Y. Since $(\mathbf{Z}/\ell)_Y \simeq (\boldsymbol{\mu}_\ell)_Y$, this follows from 4.6 for $q > 2$.

Suppose that X is affine. Since radicial extensions do not affect cohomology (VIII 1.1), we are reduced to the case where k is algebraically closed. Let $f : X' \to X$ be the normalization of X, which is a finite morphism and an isomorphism outside a finite set of points, say S. There is an exact sequence of sheaves on X

$$0 \longrightarrow (\mathbf{Z}/n)_X \longrightarrow f_*(\mathbf{Z}/n)_{X'} \longrightarrow \epsilon \longrightarrow 0,$$

where ϵ is concentrated on S, so that $H^q(X, \epsilon) = 0$ for $q > 0$. By (VIII 5.6), we are reduced to proving that

$$H^q(X, f_*(\mathbf{Z}/n)_{X'}) \simeq H^q(X', \mathbf{Z}/n) = 0$$

if $q > 1$, that is, to the case where X is normal. We may also plainly suppose that X is connected. Let $i : X \to \overline{X}$ be an open immersion with $\overline{X}$ complete and nonsingular. Since X is affine, at least one point is missing from $\overline{X}$, and it follows immediately that the morphism

$$A(k) \longrightarrow \operatorname{Pic} X$$

40 is surjective, where A is the Jacobian of $\overline{X}$ (cf. (4.8)). Since $A(k)$ is divisible by $\ell \neq \operatorname{car} k$, the same is true of $\operatorname{Pic} X$, whence $H^2(X, \boldsymbol{\mu}_\ell) = 0$ by 4.6, C.Q.F.D.

Remark 5.8. Let us record, for later reference, that the argument of 5.5 in fact establishes the following result: Let X be a quasi-compact and quasi-separated scheme, let ℓ be a prime number, and let C be a strictly full subcategory of the category of constructible abelian ℓ-torsion sheaves, stable under direct factors and extensions. Suppose that for every finite morphism $f : Y \to X$ and every open subscheme $i : U \to Y$ of Y that is of finite presentation over X, one has $f_*(i_!(\mathbf{Z}/\ell\mathbf{Z})_U) \in C$. Then C contains all constructible ℓ-torsion sheaves.

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EXPOSÉ X

Cohomological dimension: first results

M. Artin

1. Introduction

43

In this Exposé, unless expressly stated otherwise, "sheaf" means "abelian sheaf".

Let X be a scheme and let $\ell \in \mathbf{P}$. "Recall" that the *ℓ-cohomological dimension* of X is the largest integer $\mathrm{cd}_\ell X = n$ (or $= \infty$ if no such number exists) such that $H^n(X, F) \neq 0$ for at least one ℓ-torsion sheaf F on X. If $X = \operatorname{Spec} A$, we write $\mathrm{cd}_\ell A = \mathrm{cd}_\ell \operatorname{Spec} A$. Taking into account the dictionary (VIII 2), we thus recover the "classical" notion of [CG] when $A = k$ is a field. Since we plainly have the formula

$$\mathrm{cd}_\ell(\coprod_i X_i) = \sup_i \mathrm{cd}_\ell X_i$$

the classical theory may, in the evident way, be extended to an arbitrary artinian ring.

This theory is set out in (CG) = "Galois cohomology" by Serre. We shall add a few refinements in n° 2. 44

Consider, for example, the case of an irreducible algebraic variety X, of dimension n, over a separably closed field k, and fix $\ell \neq \operatorname{car} k$. A theorem to be found in n° 4 gives the upper bound $2n$ for its cohomological dimension; this is the "true" dimension of X (suggested by the topological analogy when $k = \mathbf{C}$). This result is fairly elementary once the theory for fields is known, and in fact the upper bound $2n$ can be improved except when X is complete: there is another upper bound, $2n - 1$, which holds whenever X is "missing at least one point". The latter is the correct "stable upper bound", in the sense that every X of dimension n contains an open subset of cohomological dimension $2n - 1$. A third upper bound is obtained for an affine variety, where one obtains n as the cohomological dimension. This is a generalization of one of the Lefschetz theorems for hyperplane sections. These last two assertions are deeper and can only be proved later (XIV).

Note that cohomological dimension is not a local notion (in contrast to what happens for paracompact spaces). For example, a henselian ring with separably closed residue field has cohomological dimension zero, but if the closed point is removed, one may obtain an arbitrarily large number, namely $2n - 1$ if $n = \dim A$ in the most important cases; this also agrees with the topological analogy supplied by the cohomology of $U - \{x\}$, where U is a small open neighborhood of a point x of a complex analytic space.

2. Auxiliary results over a field

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We have the following

THEOREM 2.1. *Let k'/k be an extension of finite type of a field k, and let $\ell \in \mathbf{P}$. Then*

$$\mathrm{cd}_\ell k' \leqq \mathrm{cd}_\ell k + \deg . \operatorname{tr} .(k'/k),$$

If the inequality is strict and if $\ell \neq \operatorname{car} k$, then $\ell = 2$, k is orderable, and k' is not orderable (i.e. -1 is a sum of squares in k').

Recall that when $\ell = \operatorname{car} k$ one has $\operatorname{cd}_\ell k' \leqq 1$ (CG II 2.2), so the question of equality in 2.1 scarcely arises. Moreover, $\operatorname{cd}_2 k = \infty$ if k is orderable. Indeed, the algebraic closure $\overline{k}$ of k contains a subextension $\tilde{k}$ (a maximal ordered subfield of $\overline{k}$) with $[\overline{k} : \tilde{k}] = 2$. Plainly, $\operatorname{cd}_2 \tilde{k} = \infty$, and $\operatorname{cd}_2 \tilde{k} \leqq \operatorname{cd}_2 k$ by (CG I Prop. 14).

Proof. Almost all of this is well known. The inequality, as well as equality if $\operatorname{cd}_\ell k < \infty$, are results of Tate (CG II 4.1.2). It remains to prove that $\operatorname{cd}_\ell k' < \infty$ and $\operatorname{cd}_\ell k = \infty$ imply that k is orderable and $\ell = 2$, and it is enough to treat the following two cases:

a) $[k' : k] < \infty$,
b) $k' = k(x)$ with x transcendental over k.

In case a) we may suppose k'/k separable. Then the Galois group $H = G(\overline{k}/k')$ is an open subgroup of $G = G(\overline{k}/k)$, and by a result of Serre [5] one has $\operatorname{cd}_\ell H = \operatorname{cd}_\ell G$ unless G contains an element of order ℓ. Since an algebraically closed field $\overline{k}$ cannot have a subfield $\tilde{k}$ of finite index unless $\ell = 2$ and $\tilde{k}$ is orderable [7], the result follows.

46 Consider case b). Let R be the henselization of $\operatorname{Spec} k[x]$ at the point $x = 0$, which is a discrete valuation ring, and let K be the fraction field of R. Then $\operatorname{cd}_\ell k(x) < \infty$ implies $\operatorname{cd}_\ell K < \infty$. It is therefore enough to prove that $\operatorname{cd}_\ell K < \infty$ implies $\operatorname{cd}_\ell k < \infty$, which is a consequence of the following theorem:

THEOREM 2.2. [34]. *Let R be a henselian discrete valuation ring, with fraction field K and residue field k. Let $\ell \in \mathbf{P}$. We have*

$$\operatorname{cd}_\ell K = \operatorname{cd}_\ell k + 1$$

in the following two situations:

(i) $\ell \neq \operatorname{car} k$.
(ii) $\ell \neq \operatorname{car} K$, *$k$ perfect, and* $\operatorname{cd}_\ell k < \infty$.

REMARK. Naturally, if $\ell = \operatorname{car} K$ one has $\operatorname{cd}_\ell K \leq 1$ in every case (CG II 2.2). It therefore remains to study the relation of $\operatorname{cd}_\ell K$ with, perhaps, the module of absolute differentials Ω^1_k (EGA IV 20.4.3), in the unequal characteristic case when $\operatorname{car} k = \ell$, k imperfect.

Proof of the theorem. As for Theorem (2.1), most of it is due to Tate. Case (ii) is treated explicitly in (CG II Prop. 12) if R is complete, and we reduce to that case by means of the following lemma:

LEMMA 2.2.1. *Let R be a henselian discrete valuation ring with fraction field K, and let $\hat{K}$ be the fraction field of the completion $\hat{R}$ of R. Then*

$$G(\overline{\hat{K}}/\hat{K}) \simeq G(\overline{K}/K).$$

47 Proof. This means that the functor $L \mapsto \hat{L} = L \otimes_K \hat{K}$ from the category of étale K-algebras to the category of étale $\hat{K}$-algebras is an equivalence. Now every étale algebra $\hat{K}'$ over $\hat{K}$ is induced by a suitable étale algebra K' over K. Indeed, if $\hat{K}' = \hat{K}[\alpha]$ where α satisfies the separable irreducible equation $f(T) = 0$ with coefficients in $\hat{R}$, it is enough to take as K' the algebra $K[\alpha_0]$, where α_0 is a root of an equation $f_0(T) = 0$ with coefficients in R, with $f_0 \equiv f(\hat{m}^N)$, N large and $\hat{m}$ the maximal ideal of $\hat{R}$. This is an

[34] This theorem was proved recently by *Ax* [1].

application of the usual "Newton method" (Bourbaki, Alg. Comm. III 4.5). It remains to prove that the functor is fully faithful (it is plainly faithful), and this amounts to proving that if K' is a *field*, then so is $\widehat{K}'$. Let R' be the normalization of R in K', which is finite over R since K'/K is separable. The ring R' is a discrete valuation ring, and the completion of R' is $\widehat{R}' = R' \otimes_R \widehat{R}$. Hence $\widehat{R}'$ is a discrete valuation ring, and since K' is its total ring of fractions, it is indeed a field.

Consider assertion (i) of 2.2. If $\mathrm{cd}_\ell k < \infty$, we are again reduced by 2.2.1 to (CG II Prop. 12) (the hypothesis in the statement of that proposition that k be perfect is used in the proof only in the case of residue characteristic ℓ). It therefore remains to prove that $\mathrm{cd}_\ell K < \infty$ implies $\mathrm{cd}_\ell k < \infty$.

Let $\overline{G} = G(\overline{k}/k) = G(\widetilde{K}/K)$, where $\widetilde{K}$ is the maximal unramified extension of K (i.e. the fraction field of the discrete valuation ring $\widetilde{R}$ with separably closed residue field, the strict henselization of R), let $G = G(\overline{K}/K)$, $H = G(\overline{K}/\widetilde{K})$, so that $\overline{G} = G/H$. Let $\overline{G}' \subset \overline{G}$ be an ℓ-Sylow subgroup, and let k'/k, K'/K be the extensions corresponding to $\overline{G}'$. We have $\mathrm{cd}_\ell k < \infty$ if and only if $\mathrm{cd}_\ell k' < \infty$ (CG I Prop. 14), so by replacing R with its normalization in K', we reduce to the case in which $\overline{G}$ is an ℓ-group. It is then enough to prove that if $\mathrm{cd}_\ell G < \infty$, then $H^N(\overline{G}, \mathbf{Z}/\ell) = 0$ for sufficiently large N (CG I Prop. 20).

We have $\mathrm{cd}_\ell H = 1$ by the part of i) already proved, applied to $\widetilde{R}$. Moreover, we have 48

$$H^1(H, \boldsymbol{\mu}_\ell) = H^1(\operatorname{Spec} \widetilde{K}, \boldsymbol{\mu}_\ell) = \widetilde{K}^*/\widetilde{K}^{*\ell} = \mathbf{Z}/\ell. \tag{2.2.2}$$

Indeed, $H^1(\operatorname{Spec} \widetilde{K}, \boldsymbol{\mu}_\ell) = \widetilde{K}^*/\widetilde{K}^{*\ell}$ by Kummer theory IX 3.2. We have an exact sequence

$$0 \longrightarrow \widetilde{R}^* \longrightarrow \widetilde{K}^* \longrightarrow \mathbf{Z} \longrightarrow 0,$$

and $\widetilde{R}^*$ is divisible by ℓ since $\widetilde{R}$ is strictly local and ℓ is invertible in $\widetilde{R}$, whence (2.2.2).

Consider the Hochschild–Serre spectral sequence

$$E_2^{pq} = H^p(\overline{G}, H^q(H, \boldsymbol{\mu}_\ell)) \Longrightarrow H^{p+q}(G, \boldsymbol{\mu}_\ell).$$

We have $E_2^{pq} = 0$ if $q > 1$. If, moreover, $\mathrm{cd}_\ell G < \infty$, the "boundary" morphism

$$H^{n-1}(\overline{G}, H^1(H, \boldsymbol{\mu}_\ell)) \longrightarrow H^{n+1}(G, \boldsymbol{\mu}_\ell)$$

must be bijective for sufficiently large n, and it is necessary to prove that then

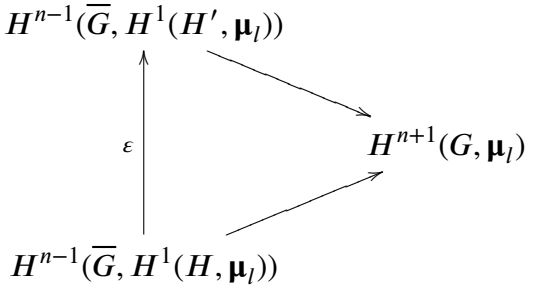
$$H^{n-1}(\overline{G}, H^1(H, \boldsymbol{\mu}_\ell)) = H^{n-1}(\overline{G}, \mathbf{Z}/\ell) = 0.$$

Set $R' = R[x']$ with $x'^\ell = x$, where x is the local parameter of R, and let K' be the fraction field of R', $G' = G(\overline{K}/K')$, etc. The residue field of R' is k, since we have supposed that $\overline{G}$ is an ℓ-group, and hence that the ℓ-th roots of unity are in K. We have a diagram

$$\begin{array}{ccccc} H' & \longrightarrow & G' & \longrightarrow & \overline{G} \\ \downarrow & & \downarrow & & \\ H & \longrightarrow & G & \longrightarrow & \overline{G}, \end{array}$$

whence a morphism of Hochschild–Serre spectral sequences, which gives a commuta- 49

tive diagram



Since $\mathrm{cd}_\ell G < \infty$ implies $\mathrm{cd}_\ell G' < \infty$, all the morphisms in this diagram are isomorphisms for large n. But the morphism

$$\mathbf{Z}/r = H^1(H, \mu_\ell) \longrightarrow H^1(H', \mu_\ell) = \mathbf{Z}/\ell$$

which induces the arrow ϵ is plainly zero, and hence $H^{n-1}(\overline{G}, \mathbf{Z}/\ell) = 0$, C.Q.F.D.

THEOREM 2.3. *Let A be a noetherian henselian ring of dimension 1, and let $\ell \in \mathbf{P}$ be invertible in A. Let k be the residue field of A and let $R(A)$ be the ring of rational functions on* Spec A. *We have*

$$\mathrm{cd}_\ell R(A) \leq \mathrm{cd}_\ell k + 1,$$

and equality holds if $\ell \neq 2$ or if k is not orderable.

Proof. Replacing A by A/P for a minimal prime ideal P of A and applying VIII 1, we see that we may suppose A integral. Let K be its fraction field, and let R be its normalization, which is local and henselian, and hence a discrete valuation ring (it is noetherian by Krull–Akizuki (Bourbaki, Alg. Comm. Chap. 7, § 2, prop. 5)), and let k' be the residue field of R. It is known that $[k' : k] < \infty$ (*loc. cit.*). Now K is the fraction field of R, and therefore

$$\mathrm{cd}_\ell K = \mathrm{cd}_\ell k' + 1 \leq \mathrm{cd}_\ell k + 1$$

50 by 2.2, with equality under the last hypothesis of 2.3, by 2.1.

EXAMPLE. Equality does not hold for $\ell = 2$, when A is the henselization at the origin of $\mathbf{R}[x, y]/(x^2 + y^2)$.

COROLLARY 2.4. *Let A be a noetherian local ring with residue field k, and let $\ell \in \mathbf{P}$ be invertible in A. Suppose that $\ell \neq 2$ or that no residue field of* Spec A *is orderable. Then*

$$\mathrm{cd}_\ell R(A) \geq \mathrm{cd}_\ell k + \dim A,$$

where $R(A)$ is the ring of rational functions on Spec A.

Proof. We may suppose A integral, with fraction field $R(A) = K$. Let $x \in \mathfrak{m}_A$. Then $\dim A/(x) = \dim A - 1$. By induction on $\dim A$, we have

$$\mathrm{cd}_\ell R(A/(x)) \geq \mathrm{cd}_\ell k + \dim A - 1.$$

Let k' be a residue field of $R(A/(x))$ such that $\mathrm{cd}_\ell k' = \mathrm{cd}_\ell R(A/(x))$ (cf. VIII 1), and let P be the prime ideal that is the kernel of $A \to k'$. The ring A_P has dimension 1. It is therefore enough to prove the corollary for the local ring A_P; that is, we are reduced to the case in which $\dim A = 1$. Let $\widetilde{A}$ be the henselization of A. Every residue field of $R(\widetilde{A})$ identifies with a separable extension of $R(\widetilde{A}) = K$, and therefore

$$\mathrm{cd}_\ell R(A) = \mathrm{cd}_\ell K \geq \mathrm{cd}_\ell R(\widetilde{A}).$$

We are thus reduced to the case in which A is henselian of dimension 1, which follows from 2.3.

COROLLARY 2.5. *Let X be a noetherian scheme and let $\ell \in \mathbf{P}$ be invertible on X. Then* 51

$$\mathrm{cd}_\ell R(X) \geq \dim X.$$

Indeed, let $x \in X$ be a point such that $\dim X = \dim \mathcal{O}_{X,x}$ (or such that $\dim \mathcal{O}_{X,x}$ is very large if $\dim X = \infty$). Let A be the strict henselization of $\mathcal{O}_{X,x}$. Then $\mathrm{cd}_\ell R(X) \geq \mathrm{cd}_\ell R(\mathcal{O}_{X,x}) \geq \mathrm{cd}_\ell R(A)$ by the usual argument. Moreover, $\dim A = \dim X$ (resp. is very large). If $\ell = 2$, and if ℓ is invertible on X, then the equation $T^2 + 1 = 0$ is separable over A, so -1 is a square in A because A is strictly local; hence the residue fields of A are not orderable. Thus the hypotheses of 2.4 are satisfied for A in every case, and $\mathrm{cd}_\ell R(A) \geq \dim A$, whence the result.

3. Field of fractions of a strictly local ring

3.0. In the following sections we shall have essential need of knowing $\mathrm{cd}_\ell R(A)$, where A is a strictly local ring, that is, a henselian ring with separably closed residue field. Unfortunately, $\mathrm{cd}_\ell R(A)$ is not known in general, for example if $A = \hat{\mathbf{Z}}_p[[x]]$($p \neq \ell$).

It is nevertheless conjectured that if A is noetherian and ℓ is invertible in A, then one has

$$\mathrm{cd}_\ell R(A) = \dim A \quad (\ell \text{ invertible in } A), \tag{3.1}$$

at least in the most important cases, for example if A is an *excellent* ring (EGA IV 7.8.2).

Note that if $\dim A = 1$, (3.1) is true by 2.3. Moreover, the inequality $\geq$ holds in any case by 2.5. We shall prove later (XIX), using resolution of singularities [3], that (3.1) is also true if A is excellent of residual characteristic zero. Here we shall prove (3.1) only 52 in the easy case of 3.2 below.

Naturally one has $\mathrm{cd}_\ell R(A) \leq 1$ if A has characteristic ℓ, so this case is trivial. It remains, moreover, to analyze the case of unequal characteristics with residual characteristic ℓ, and one does not expect (3.1) to be true in this case. It will certainly be necessary to take into account the rank of the module Ω^1_k (k the residue field), i.e. the degree $[k : k^\ell]$.

PROPOSITION 3.2. *Let X be a prescheme of finite type over the spectrum of a field k or over the spectrum of a discrete valuation ring R, and let $\ell \in \mathbf{P}$ be invertible in k (resp. R). Then 3.1 is true for every ring A that is a strict localization of X.*

Proof. Take X integral (one reduces to this case as usual), and let $n = \dim X$.

Suppose that X is defined over a *separably closed* field k. Then for every ring A that is a strict localization of X at a *closed* point, one has $\dim A = n$. Since every residue field K of $R(A)$ is identified with an algebraic extension of $R(X)$, and since

$$\deg . \mathrm{tr} .(R(X)/k) = \dim X = n,$$

one has $\mathrm{cd}_\ell K \leq n$ by 2.1, whence $\mathrm{cd}_\ell R(A) \leq n$. The opposite inequality follows from 2.5.

Now suppose that X is of finite type over Spec R, where R is a discrete valuation ring, and let A be the strict localization of X at a point x. If x does not lie above the closed point of Spec R, one reduces immediately to the case where X is of finite type over a field (by localizing R). For the same reason, one may also suppose that the morphism $X \to$ Spec R is dominant.

Suppose for the moment, in addition, that R is a *strictly local* discrete valuation ring 53 and that the point x is *closed* in X. Then one has $\dim A = \dim X = n$ (EGA IV 5.6.5).

Since every residue field K of $R(A)$ is identified with a separable extension of $R(X)$, one has

$$\mathrm{cd}_\ell\, R(A) \leq \mathrm{cd}_\ell\, R(X) \leq \mathrm{cd}_\ell\, L + \deg.\,\mathrm{tr}.(R(X)/L) = 1 + (n-1) = n$$

by 2.1 and 2.2, where L is the field of fractions of R; the result follows in this case, taking 2.5 into account.

The general case is deduced from these special cases by means of the following lemma:

LEMMA 3.3. (i) *Let X be of finite type over the spectrum of a field k, and let A be a strict localization ring of X at a point x. Then there exist a separably closed field k', a scheme X' of finite type over* Spec k', *and a point x' closed in X' such that A is isomorphic to a strict localization ring of X' at x'.*

(ii) *Let X be of finite type over* Spec R, *where R is a discrete valuation ring, let x be a point of X above the closed point of* Spec R, *and let A be a strict localization of X at x. There exist a strictly local discrete valuation ring R', a scheme X' of finite type over* Spec R', *and a closed point x' of X' above the closed point of* Spec R', *such that A is isomorphic to a strict localization of X' at x'.*

Proof.

(i) We may take X affine.

Let Y be the closure of x in X, with the induced reduced structure, and let $d = \dim Y$. By the normalization lemma ([2] Chap. V § 3 n° 1), there exists a morphism

$$X \longrightarrow \mathrm{Spec}\, k[y_1, \ldots, y_d] = T$$

54 inducing a *finite* morphism $Y \to T$. Consequently, the strict localization ring of $\mathcal{O}_{X,x}$ is isomorphic to the strict localization of $\mathcal{O}_{X',x'}$, where $X' = X \times_T \mathrm{Spec}\, \overline{R(T)}$, and where x' is a closed point of X'.

(ii) This case is treated similarly. We may suppose that X is affine, with ring A. Let Y be the closure of x in X with the induced reduced structure; it is a closed subscheme of the closed fiber X_0 of X over Spec R, and let $d = \dim Y$. Then, lifting the elements $y_1, \ldots, y_d$ of $A \otimes_R k$ considered in (i) to elements of A, one obtains a morphism

$$X \longrightarrow \mathrm{Spec}\, R[y_1, \ldots, y_d] = T$$

inducing a *finite* morphism $Y \to T_0$. Let R' be the strict localization of $R[y_1, \ldots, y_d] = B$ at the prime ideal $\mathfrak{m}B$. Then R' is a strictly local discrete valuation ring, and one may take $X' = X \times_T \mathrm{Spec}\, R'$, and x' a closed point of X' above x.

4. Cohomological dimension: the case where ℓ is invertible in $\mathcal{O}_X$

THEOREM 4.1. *Let X be a noetherian scheme, let $\ell \in \mathbf{P}$, and let φ be a function on X with values in* $\mathbf{N}$ *and satisfying the following conditions:*

(i) *For every $x \in X$, $\mathrm{cd}_\ell\, k(x) \leq \varphi(x)$.*

(ii) *Let $x \in X$ and let $y \in Y =$ the closure of x, with $y \neq x$. Let K be the ring of rational functions of a strict localization of Y above y. One has $\mathrm{cd}_\ell\, K < \varphi(x) - \varphi(y)$.*

Then for every ℓ-torsion sheaf F on X, one has

$$H^q(X, F) = 0$$

if

$$q > \varphi(F) \underset{\text{def}}{=} \mathrm{Sup}\{\varphi(x) \mid x \in X, F_{\overline{x}} \neq 0\}.$$

Proof. The hypotheses will also be satisfied by the restriction of φ to any subprescheme of X, so by noetherian induction we may suppose the theorem true for every closed subprescheme of X distinct from X. Let F be given, and write $F = \varinjlim F_i$, where the F_i are the constructible subsheaves of F (IX 2.9 (iii)). 55

For every F_i, the set of $x \in X$ such that $F_x \neq 0$ is constructible in X, and we may apply the induction hypothesis to every F_i whose support is not dense. We thus see that the induction hypothesis in fact implies that the theorem is true for every F that vanishes at at least one of the maximal points of X. We shall therefore suppose that F does not satisfy this condition.

Now let $i : \operatorname{Spec} R(X) \to X$ be the inclusion, where $R(X)$ is the ring of rational functions of X, and consider the following exact sequence, where K and C are defined as kernel and cokernel, respectively:

$$0 \longrightarrow K \longrightarrow F \longrightarrow i_* i^* F \longrightarrow C \longrightarrow 0.$$

Here K and C vanish at every maximal point of X, so by (ii), $\varphi(K)$ (resp. $\varphi(C)$) is strictly less than $n = \varphi(F)$, whence $H^q(X, K) = H^q(X, C) = 0$ for $q \geq n$. Thus, , in order to prove $H^q(X, F) = 0$ for $q > n$, it is enough, by the cohomology exact sequences, to prove $H^q(X, i_* i^* F) = 0$; hence we may suppose that F is of the form $i_* G$, for a suitable ℓ-sheaf G on $\operatorname{Spec} R(X)$.

Consider the Leray spectral sequence

$$E_2^{p,q} = H^p(X, R^q i_* G) \Longrightarrow H^{p+q}(\operatorname{Spec} R(X), G).$$

Now $R(X)_{\text{réd}}$ is the product of the fields $k(x)$ for x maximal in X, so $\operatorname{cd}_\ell R(X) \leq n$ by (i); therefore the abutment of the spectral sequence vanishes in dimension strictly greater than n. We wish to prove that $E_2^{p0} = 0$ for $p > n$. Examining the spectral sequence, 56 one sees that it is enough to prove that $E_2^{pq} = 0$ if $q > 0$ and if $p > n - q - 1$. Thus, by induction, it is enough to prove that $\varphi(R^q i_* G) < n - q$ for $q > 0$. But by VIII 5.2, the fiber of $R^q i_* G$ at a geometric point $\overline{y}$ above a point $y \in X$ is precisely $H^q(\operatorname{Spec} K, G_K)$, where K is the ring of rational functions of the strict localization of X at $\overline{y}$, and G_K is the induced sheaf on $\operatorname{Spec} K$. By (ii), one has $\operatorname{cd}_\ell K < n - \varphi(y)$, whence $(R^q i_* G)_{\overline{y}} = 0$ if $q \geq n - \varphi(y)$, which gives the desired result.

Corollary 4.2. *Let X be a noetherian scheme, let $\ell \in \mathbf{P}$, let $n \in \mathbf{N}$, and suppose that the following conditions are satisfied:*

(i) *For every point y of X of codimension c, one has $\operatorname{cd}_\ell k(y) \leq n - 2c$.*

(ii) *For every ring A that is a strict localization of an irreducible closed subscheme Y of X, one has $\operatorname{cd}_\ell R(A) \leq \dim A$.*

Then

$$\operatorname{cd}_\ell X \leq n.$$

More precisely, if F is an ℓ-sheaf that vanishes at every point y of codimension $< s$, one has

$$H^q(X, F) = 0 \text{ if } q > n - 2s.$$

Indeed, one may take $\varphi(y) = n - 2 \operatorname{codim} y$ and apply 4.1, whose condition (i) (resp. (ii)) follows from the corresponding condition (i) (resp. (ii)) of (4.2), taking into account for (ii) the inequality $\dim \mathcal{O}_{Y,y} + \dim \mathcal{O}_{X,x} \leq \dim \mathcal{O}_{X,y}$.

Corollary 4.3. *Let X be of finite type and dimension n over a field k, and let $\ell \neq \operatorname{car} k$. Then $\operatorname{cd}_\ell X \leq 2n + \operatorname{cd}_\ell k$.*

One applies the preceding corollary, replacing n there by $2n + cd_\ell k$, and using 2.1 and 3.2 to verify conditions (i) and (ii) of 4.2, taking into account for (i) the inequality 57

$$\deg . \operatorname{tr} . K(y)/k + \dim \mathcal{O}_{X,y} \leq \dim X.$$

COROLLARY 4.4. *Let X be noetherian, and let $\ell \in \mathbf{P}$ be invertible on X. Suppose that condition* (ii) *of* 4.2 *is satisfied, and moreover that $\ell \neq 2$ or that no residue field of X is orderable. Then one has*

$$\operatorname{cd}_\ell X \leq \operatorname{cd}_\ell R(X) + \dim X \leq 2 \operatorname{cd}_\ell R(X).$$

Put $n = \operatorname{cd}_\ell R(X) + \dim X$ in 4.2, and apply 2.4, which implies condition (i) of 4.2, since

$$\operatorname{cd}_\ell k(y) \leq \operatorname{cd}_\ell R(\mathcal{O}_{X,y}) - \dim \mathcal{O}_{X,y} \leq \operatorname{cd}_\ell R(X) - \dim \mathcal{O}_{X,y} \leq n - 2 \dim \mathcal{O}_{X,y}.$$

The second inequality in 4.4 follows from 2.5.

EXAMPLE 4.5. The inelegant hypothesis on residue fields made in 4.4 is necessary, as is seen by taking $X = \operatorname{Spec} \mathbf{R}[x, y, z]/(x^2 + y^2 + z^2)$, where all the other conditions are satisfied, with $\dim X = \operatorname{cd}_\ell R(X) = \ell = 2$, but the sheaf $\mathbf{Z}/2$ concentrated at the point $(0, 0, 0)$ has nonzero cohomology in every degree, so $\operatorname{cd}_\ell X = +\infty$.

4.6. One is also tempted to try to impose hypotheses in 4.2 only on the closed points of X, but this is not possible. For example, there exist discrete valuation rings with algebraically closed residue field and with function field of arbitrarily large cohomological dimension[34]. The spectrum of such a ring will likewise have very large cohomological dimension.

5. Cohomological dimension: the case $\ell = p$

58

5.0. Using Artin-Schreier theory (IX 3.5), one obtains a better upper bound than 4.3 for the p-cohomological dimension of a scheme X of characteristic p. Let $\operatorname{cdqc} X$ be the largest number n such that $H^n(X, F) \neq 0$ for at least one *quasi-coherent* sheaf of modules F on X (in the Zariski sense). We have the following

THEOREM 5.1. *Let X be a Noetherian scheme of characteristic $p > 0$. Then*

$$\operatorname{cd}_p X \leq \operatorname{cdqc} X + 1.$$

In particular, the p-cohomological dimension of a Noetherian affine scheme X of characteristic $p > 0$ is at most 1.

Proof. Apply IX 5.5:

Taking account of the fact that the $R^q f_* F$, $q > 0$, vanish for a finite morphism f and an abelian sheaf F (VIII 5.5) (resp. a quasi-coherent sheaf F), we are reduced to proving that $H^q(X, i_!(\mathbf{Z}/p)_U) = 0$ for $i : U \to X$ an open immersion and for $q > \operatorname{cdqc} X$.

Let Y be a closed subscheme of X with support $X - U$, and let J be the coherent sheaf of ideals defining Y. The morphism $p : f \mapsto f^p - f$ (IX 3.5) induces a morphism $J \to J$. Examining the exact sequence IX 3.5 and the exact sequence

$$0 \longrightarrow i_!(\mathbf{Z}/p)_U \longrightarrow (\mathbf{Z}/p)_X \longrightarrow (\mathbf{Z}/p)_Y \longrightarrow 0,$$

[34] To see this, it is enough to note that if X is a smooth algebraic scheme over a field k, admitting a point $x \in X(k)$, then one can find a "formal curve arc" passing through x whose "algebraic envelope" is X, and whose function field therefore contains that K of X and induces on it a discrete valuation whose residue field is k.

one sees that there is in fact an exact sequence

$$(*) \qquad 0 \longrightarrow i_!(\mathbf{Z}/p)_U \longrightarrow J \longrightarrow J \longrightarrow 0.$$

Since J is coherent, one has $H^q(X, J) = 0$ for $q > \operatorname{cdqc} X$, whence the result by the long exact cohomology sequence. 59

For a scheme quasi-projective over a separably closed field k, this upper bound can in certain cases be reduced by 1. In fact, we have

COROLLARY 5.2. *Let X be a scheme of finite type over a field k of characteristic $p > 0$. One has*

$$\operatorname{cd}_p X \leq \dim X + 1,$$

and if k is separably closed and X is quasi-projective [34]*, one has*

$$\operatorname{cd}_p X \leq \dim X.$$

Indeed, the first assertion is clear, since $\operatorname{cdqc} X \leq \dim X$ ([6] 4.15.2). Let us treat the second. For such an X, the cohomology groups $H^n(X, F)$ ($n = \dim X$), for a coherent sheaf F, are finite-dimensional vector spaces over k. Let $V = H^n(X, J)$, where J is the sheaf of ideals occurring in the exact sequence $(*)$. The morphism $\varphi : V \to V$ deduced from $(*)$ is evidently of the form $\varphi = \epsilon - \mathrm{id}$ where ϵ satisfies $\epsilon(rv) = r^p\epsilon(V)$ for $r \in k$ and $v \in V$. The Jacobian of the map Φ of the affine scheme deduced from Φ is therefore $-I$, where I is the identity matrix. Consequently Φ is étale, hence surjective because it is additive, and it follows that Φ is surjective since k is separably closed, whence the conclusion immediately.

6. Cohomological dimension for a prescheme of finite type over Spec $\mathbb{Z}$

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PROPOSITION 6.1. *Let X be quasi-finite over* Spec $\mathbf{Z}$. *Suppose that $\ell \neq 2$ or that every residue field of X of characteristic zero is totally imaginary. Then* $\operatorname{cd}_\ell X \leq 3$.

This is a consequence of Theorem 2.2 (ii), of Corollary 4.2, and of the fact that a number field (totally imaginary if $\ell = 2$) has ℓ-cohomological dimension ≤ 2 (CG II 4.4).

A consequence of class field theory is that equality holds if the morphism $X \to$ Spec $\mathbf{Z}$ is proper and surjective.

Unfortunately, the results of n° 4 do not apply as they stand if X is only of finite type over Spec $\mathbf{Z}$, because hypothesis (ii) of (4.2) is not in general satisfied for local rings of *residual characteristic* ℓ. One must therefore use result (5.1) to obtain the correct upper bound; we shall now reap the benefit of Theorem (5.1).

THEOREM 6.2. *Let X be of finite type over* Spec $\mathbf{Z}$, *and let $\ell \in \mathbf{P}$. Suppose, if $\ell = 2$, that no residue field of X is orderable. Then*

$$\operatorname{cd}_\ell X \leq 2 \dim X + 1.$$

More precisely, let F be a sheaf of ℓ-torsion such that for every point x of X such that $F_{\overline{x}} \neq 0$, one has

$$\dim\{\overline{x}\} \leq \begin{cases} (N-1)/2 & \text{if} \quad \ell \neq \operatorname{car} k(x) \\ N - 1 & \text{if} \quad \ell = \operatorname{car} k(x) \end{cases}$$

($\{\overline{x}\}$ is the closure of $\{x\}$); then

$$H^p(X, F) = 0 \text{ if } p > N.$$

Proof. By induction on X, we may suppose the theorem true for every closed 61
subscheme distinct from X. If F is as in the statement and we write $F = \varinjlim F_i$, where the F_i are the constructible subsheaves of F (IX 2.9 (iii)), then every F_i also satisfies the same hypotheses. It therefore suffices to prove the theorem when F is constructible.

In this case, the set E of $x \in X$ such that $F_{\overline{x}} \neq 0$ is a constructible subset, and the induction hypothesis implies that the theorem is true for every F such that E is not dense. Moreover, the theorem is a consequence of 5.1 for a sheaf concentrated on the fiber of X over the point (ℓ) of $\operatorname{Spec} \mathbf{Z}$, and of 4.3 for a sheaf concentrated on the fiber over (p), where $p \neq \ell$, taking account of the fact that $\operatorname{cd}_\ell \mathbf{Z}/p\mathbf{Z} = 1$ (CG II 2.2).

Take X reduced, and let $X = X_0 \cup X_1$, where X_1 is the closed subset which is the union of the irreducible components of X of characteristic ℓ, and where X_0 is the union of the other irreducible components, X_0 and X_1 being endowed with their induced reduced structures. Let $j_\nu : X_\nu \to X$ be the inclusions, let $d_\nu = \dim X_\nu$, and let $N = \sup\{2d_0 + 1, d_1 + 1\}$. For every constructible F on X, there is an exact sequence

$$(*) \qquad 0 \longrightarrow F \longrightarrow j_{0*}j_0{}^*F \oplus j_{1*}j_1{}^*F \longrightarrow C \longrightarrow 0,$$

where C is concentrated on $X_0 \cap X_1$, which has characteristic ℓ and dimension $\leq d_1 - 1$. It follows that

$$H^q(X, C) = 0 \text{ if } q > N - 1,$$

because by 5.1 one has $\operatorname{cd}_\ell(X_0 \cap X_1) \leq \dim(X_0 \cap X_1) + 1 \leq d_1 \leq N - 1$. Examining the relative long exact cohomology sequence of $(*)$, we are reduced to the case $X = X_0$ or $X = X_1$, and the latter is a consequence of 5.1, as already noted.

62 Suppose therefore that every maximal point of X has characteristic different from ℓ, and let $i_\nu : x_\nu \to X$ be the inclusions of the maximal points. Examining the exact sequence

$$0 \longrightarrow K \longrightarrow F \longrightarrow \bigoplus_\nu i_{\nu*}i_\nu{}^*F \longrightarrow C \longrightarrow 0$$

we are reduced by induction to the case $F = i_{\nu*}i_\nu{}^*F$, that is, where F is of the form i_*G for a suitable sheaf G on $\operatorname{Spec} k(x)$, with x a point of X. We may therefore suppose that X is irreducible and reduced, with generic point x, and that the characteristic of $k(x)$ is zero by 4.3.

Consider the spectral sequence

$$(**) \qquad H^p(X, R^q i_* G) \Longrightarrow H^{p+q}(\operatorname{Spec} k(x), G).$$

Let $s = \dim X$. We must prove that

$$(+) \qquad E_2^{p0} = 0 \text{ for } p > 2s + 1.$$

Now $k(x)[i]$ has transcendence degree $s - 1$ over $\mathbf{Q}[i]$ which is a field of cohomological dimension 2. Since $k(x)$ is not orderable if $\ell = 2$, it follows from 2.1 that $\operatorname{cd}_\ell k(x) = s + 1$. This implies that the target of the spectral sequence vanishes for $n > s + 1$. It therefore suffices, for $(+)$, to prove that in $(**)$ one has

$$(++) \qquad E_2^{pq} = 0 \text{ if } p > 2s - q \text{ and } q > 0.$$

Let y be a point of X, and let K be the ring of fractions of a strict localization at a geometric point $\tilde{y}$ of X above y. Then K is a direct product of fields, each of transcendence

[34] Using Chow's lemma, it is easy to replace the condition "X quasi-projective" by "X separated."

degree $s-1$ over a strict localization of $\mathbf{Z}$, and hence, by 2.1 and 2.2, one has $\mathrm{cd}_\ell K \leq s$. Moreover, one may apply 3.2 if the residual characteristic of y is not ℓ. Thus one has

$$(''') \qquad (R^q i_* G)_{\bar{y}} = 0 \begin{cases} \text{if } q > \operatorname{codim} y, \text{ i.e. } \dim \overline{y} > s-q, \text{ and } \ell \neq \operatorname{car} k(y), \\ \text{or if } q > s \text{ in all cases.} \end{cases}$$

In particular, one has $E_2^{pq} = 0$ if $q > s$. To verify (++) in the case $0 < q \leq s$, apply the induction hypothesis to the sheaves $R^q i_*(G)$. 63

According to the statement and (‴), one must set $N' = 2s - q$ and verify the two inequalities

$$s - q \leq (N' - 1)_{/2}$$
$$s - 1 = \dim X_\ell \leq N' - 1,$$

where X_ℓ is the fiber of X over the point (ℓ) of Spec $\mathbf{Z}$. They do indeed hold if $0 < q \leq s$. C.Q.F.D.

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EXPOSÉ XI

Comparison with classical cohomology: the case of a smooth scheme

M. Artin

1. Introduction

In Section 4 we shall prove that étale cohomology with values in a locally constant torsion sheaf agrees with classical cohomology for a scheme X smooth over Spec $\mathbf{C}$. The proof, which is essentially the one given in [1], is elementary once one has the theorem of Grauert recalled below (4.3 (iii)), which says that every finite covering of $X(\mathbf{C})$ comes from an étale covering of X; this is why we give it here, although we shall prove a more complete result later, using Hironaka's resolution of singularities theorem [3][34]. Furthermore, the good neighborhoods constructed in Section 3 could be useful in other situations. 64 65

Note that étale cohomology and classical cohomology do not agree for sheaves that are not torsion. For example, one has $H^1(X_{\text{et}}, \mathbf{Z}) = 0$ for every normal scheme X (IX 3.6 (ii)), but classical cohomology does not necessarily vanish.

2. Existence of sufficiently general hyperplane sections

The following result is classical, but a reference is lacking (pending EGA V):

THEOREM 2.1. *Let k be a field and let $V \subset \mathbf{P}^n_k$ be a projective algebraic scheme purely of dimension r. Let V' be the open subscheme of points of V smooth over* Spec k, *and let P be a rational point of $\mathbf{P}^n$. Then:*

(i) *A "sufficiently general" linear subspace L of a given dimension d in $\mathbf{P}^n$ meets V transversely at every point of $L \cap V'$.*

(ii) *Let $\{H_1, \dots, H_s\}$ $(s \leq r)$ be a system of hypersurfaces of degree $d \geq 2$ passing through P, and let $Y = H_1 \cap \cdots \cap H_s$. If $\{H_1, \dots, H_s\}$ is sufficiently general, then Y meets V transversely at every point of $Y \cap V'$.*

The phrase "for a sufficiently general linear subspace" means that there exists a dense open subset U of the Grassmannian parametrizing the linear subspaces of dimension d, such that the assertion under consideration about L is true whenever the parameter of L lies in U. Thus this does not imply that such an L (defined over k) exists if k is not infinite. Likewise, the hypersurfaces of degree d passing through P are parametrized by a projective space $\mathbf{P}^N$, and the phrase "for $\{H_1, \dots, H_s\}$ sufficiently general" means that there exists a dense open subset U of $(\mathbf{P}^N)^s$ such that the assertion under consideration is true whenever the parameter of $(H_1, \dots, H_s)$ lies in U. 66

Proof of (i): By induction one is reduced immediately to the case where L is a hyperplane. Let (V'_ν) be a finite open covering of V'. If assertion (i) is true when the

[34] Note, moreover, that Hironaka's resolution of singularities also gives a very simple proof of Grauert's result, and in this capacity may in any case be regarded as underlying the comparison theorems. See in this connection the exposé by Mme M. Raynaud in SGA 1 XII (North Holland Pub. Cie).

symbol V' is replaced by V'_ν for every ν, then it is also true for V', because the intersection of finitely many dense open subsets is again open and dense. We may therefore replace the situation by an affine algebraic scheme, say $V \subset \mathbf{E}^n_k = \operatorname{Spec} k[x_1, \dots, x_n]$. The hyperplane L will be the zero set of a linear equation

$$\ell(x) = a_0 + \sum_{i=1}^{n} a_i x_i.$$

Recall that a smooth scheme is locally a complete intersection. We may therefore replace V' by an open subset of points of V such that there exist polynomials $f_1(x), \dots, f_{n-r}(x)$ vanishing on V and indices $1 \leq j_\nu \leq n$ ($\nu = 1, \dots, n-r$) such that the determinant

$$\left| \frac{\partial f_i}{\partial x_{j_\nu}} \right| \quad (i, \nu = 1, \dots, n-r)$$

is invertible. Say $j_\nu = \nu$ ($\nu = 1, \dots, n-r$). We may then solve the equations

$$y_j = \sum_{i=1}^{n-r} c_i \frac{\partial f_i}{\partial x_j}$$

67 for c_i as rational functions of the variables x_i, y_j ($i = 1, \dots, n-r$; $j = 1, \dots, n$), and these functions $c_i(x, y)$ are defined at every point of V'.

Now to say that L meets V transversely at a point Q of V' is equivalent to saying that the value of the matrix

$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_{n-r}}{\partial x_1} & & \frac{\partial f_{n-r}}{\partial x_n} \\ a_1 & \cdots & a_n \end{bmatrix}$$

at Q has rank $n - r + 1$. Thus, for L to meet V transversely at every point of V', it is necessary and sufficient that the system of $r + 1$ equations

$$a_j = \sum_{i=1}^{n-r} c_i(x, a) \frac{\partial f_i}{\partial x_j} \qquad (j = n - r + 1, \dots, n)$$

$$a_0 + \sum_{i=1}^{n} a_i x_i = 0$$

have no solution on V'. Such a condition is plainly constructible in (a): Let $X = V' \times A$, where A is the affine space with coordinates $a_0, \dots, a_n$, and let $Y \subset X$ be the closed subset of solutions of these equations. The condition is that the fiber of Y/A be empty,
68 which is indeed a constructible condition (EGA IV 9.5.1). It is therefore enough to verify that the fiber above the generic point of A is empty, which follows immediately from the form of the equations. This completes the proof of (i). (ii): Consider the rational map $\mathbf{P}^n \to \mathbf{P}^N$ given by the linear system of hypersurfaces of degree d ($d \geq 2$) passing through P. It is well known (and easily verified) that this is a "blowing up" of P, and therefore gives a locally closed immersion

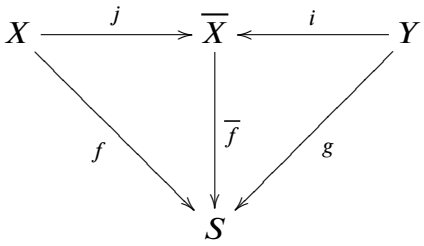
$$f : \mathbf{P}^n - \{P\} \longrightarrow \mathbf{P}^N.$$

Let $V'' = V' - (\{P\} \cap V')$, let $\overline{V}'' = F(V'')$, and let L_i be the hyperplane of $\mathbf{P}^N$ corresponding to the hypersurface H_i ($i = 1, \dots, s$). Then V'' is isomorphic to $\overline{V}''$, and

by the definition of f, to say that Y meets V'' transversely is equivalent to saying that $L_1 \cap \cdots \cap L_s$ meets $\overline{V}''$ transversely. Putting $\overline{V}$ = the closure of $\overline{V}''$, we are reduced to assertion (i) for $\overline{V}$, which implies (ii) with V'' in place of V'. It remains to consider the point P, if it is a smooth point of V. But it is evident that if $\{H_1, \dots, H_s\}$ is sufficiently general and P is smooth on V, then Y meets V transversely at P, whence the result.

3. Construction of good neighborhoods

DEFINITION 3.1. *An elementary fibration is a morphism of schemes $f : X \to S$ which can be embedded in a commutative diagram*



satisfying the following conditions: 69

(i) *j is an open immersion, dense in every fiber, and $X = \overline{X} - Y$.*
(ii) *$\overline{f}$ is smooth and projective, with geometrically irreducible fibers of dimension* 1.
(iii) *g is an étale covering, and every fiber of g is nonempty.*

One easily checks that such an embedding of X, if it exists, is unique up to canonical isomorphism.

DEFINITION 3.2. *A good neighborhood relative to S is an S-scheme X for which there exist S-schemes*

$$X = X_n, \dots, X_0 = S$$

and elementary fibrations $f_i : X_i \to X_{i-1}$, $i = 1, \dots, n$.

PROPOSITION 3.3. [34] *Let k be an algebraically closed field, let $X/\operatorname{Spec} k$ be a smooth scheme, and let $x \in X$ be a rational point. There exists an open subset of X containing x which is a good neighborhood (relative to* Spec k*).*

Proof. Take X irreducible. By induction on $\dim X = n$, it suffices to find a neighborhood U of x in X and an elementary fibration $f : U \to V$, with V smooth and of dimension $n-1$. Indeed, there will exist a neighborhood V' of $v = f(x)$ which is a good neighborhood, and we may take $U' = U \cap V'$ as a good neighborhood of x.

We may suppose that $X \subset \mathbf{E}^r$ is affine. Let X_0 be the closure of X in $\mathbf{P}^r$. Let $\overline{X}$ be the normalization of X_0 and let $Y = \overline{X} - X$, endowed with the induced reduced structure. In addition, let $S \subset \overline{X}$ be the closed subset of singular points. One has $S \subset Y$ and 70

$$\dim \overline{X} = \dim X = n,$$
$$\dim Y = n - 1,$$
$$\dim S \leq n - 2.$$

Embed $\overline{X}$ in a projective space $\mathbf{P}^N$ by means of an invertible sheaf L of the form $M^{\otimes r}$, with M very ample and $r \geq 2$. There then exist hyperplanes $H_1, \dots, H_{n-1}$ of $\mathbf{P}^N$,

[34] This result extends without difficulty to a semilocal assertion.

where H_i is the zero set of

$$\sum_{\nu=0}^{N} a_{i\nu}x_\nu = 0,$$

which contain x and such that the intersection $L = H_1 \cap \cdots \cap H_{n-1}$ has dimension $N-n+1$ and meets $\overline{X}$ and Y transversely (2.1). The intersection $\overline{X} \cap L$ is a smooth connected curve (this follows from Bertini's theorem), and $Y \cap L$ has dimension 0. Choose in addition another hyperplane

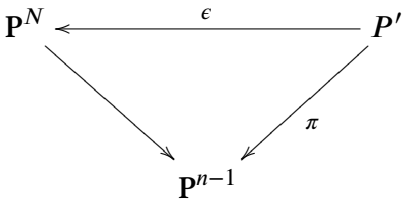
$$H_0 : \sum_{\nu=0}^{N} a_{0\nu}x_\nu = 0$$

such that H_0 meets $X \cap L$ transversely and $H_0 \cap Y \cap L = \emptyset$.

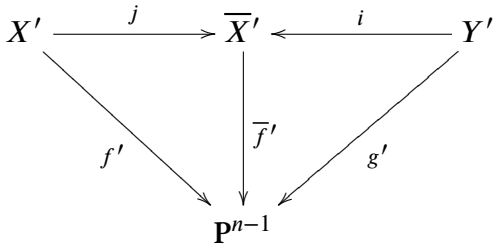
Consider the projection $\mathbf{P}^N \to \mathbf{P}^{n-1}$ obtained by "introducing the projective coordinates"

$$y_i = \sum_{\nu=0}^{N} a_{i_\nu}x_\nu.$$

It is a rational map defined outside the center of projection $C = H_0 \cap \cdots \cap H_{n-1}$. Let $\epsilon : P' \to \mathbf{P}^N$ be the blowup of C, so that there is a diagram



71 where π is a morphism. Let $\overline{X}' \subset P'$ be the "proper" inverse image of $\overline{X}$, that is, the closure of $\epsilon^{-1}(\overline{X} - (\overline{X} \cap C))$. Since, by hypothesis, C meets $\overline{X}$ transversely, the morphism $\overline{X}' \to \overline{X}$ is a blowup of the finite set $\overline{X} \cap C$. Let $X' = X - \overline{X} \cap C$, which is also identified with an open subscheme of $\overline{X}'$, and let $Y' = \overline{X}' - X'$ be the closed subscheme with the induced reduced structure. There is a diagram $\mathbf{D}$ of morphisms



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and I claim that there exists a neighborhood V of $v = f'(x)$ such that the restriction of $\mathbf{D}$ to V satisfies conditions (i), (ii), (iii) of (3.1), hence that $f'_{|V}$ is an elementary fibration. This will complete the proof.

Condition (i) is trivial. For (ii), note that $\overline{X} \cap L$ is a smooth curve by hypothesis, and one sees immediately that there is a one-to-one morphism $\overline{f}'^{-1}(v) \to \overline{X} \cap L$ induced by ϵ. Hence $\left(\overline{f}'^{-1}(v)\right)_{\text{red}} \to \overline{X} \cap L$ is bijective. To check that $\overline{f}'$ is smooth over a neighborhood of v, it suffices, by Hironaka's lemma (SGA 1 II 2.6 [34]), to check that it is

[34] and EGA IV 5.12.10.

smooth at the generic point of $\overline{f}'^{-1}(v)$. At this point, $\overline{X}'$ is isomorphic to $\overline{X}$, and the morphism is smooth because L meets $\overline{X}$ transversely.

It remains to prove that g' is étale in a neighborhood of v (since Y has dimension $n-1$, it is evident that every fiber of g' is nonempty). One has $Y' = \epsilon^{-1}(Y) \amalg D_1 \amalg \cdots \amalg D_r$, where $\overline{X} \cap C = \{P_1, \dots, P_r\}$ and D_i is the blowup of P_i in $\overline{X}$. Every D_i is identified 72
with $\epsilon^{-1}(P_i)$, and maps isomorphically onto $\mathbf{P}^{n-1}$ is defined at every point of Y, and the induced morphism on Y is none other than $g'_{|\epsilon^{-1}(Y)}$. It is étale over v, because L meets Y transversely, hence g' is étale over v.

4. The comparison theorem

4.0. Let X be a scheme locally of finite type over $\operatorname{Spec} \mathbf{C}$, and consider the usual topology on the space $X(\mathbf{C})$ of rational points of X. We shall denote by X_{cl} the site of *local isomorphisms* $f : U \to X(\mathbf{C})$, that is, of morphisms f of topological spaces having the following property:

For every $x \in U$ there exists an open neighborhood U_x such that $f_{|U_x}$ is a homeomorphism of U_x onto an open neighborhood of $f(x)$.

A family $\{U_\nu \to U\}$ of morphisms of X_{cl} is said to be *covering* if U is the union of the images of the U_ν.

Since an open immersion is a local isomorphism, we have a morphism (inclusion) of sites (IV) [34]

$$\delta : X_{\mathrm{cl}} \longrightarrow X(\mathbf{C}) \qquad (*). \tag{4.1}$$

Now for every local isomorphism $U \to X(\mathbf{C})$ there exists, by definition, a "covering" of U by open subsets of $X(\mathbf{C})$, and one deduces from ( ) that the topoi associated with the two sites are equivalent (by δ_*). In particular, one may replace $X(\mathbf{C})$ by X_{cl} when computing ordinary cohomology.

Now let $f : X' \to X$ be an étale morphism of schemes. Then $f(\mathbf{C}) : X'(\mathbf{C}) \to X(\mathbf{C})$ is a local isomorphism. This follows immediately from the Jacobian criterion (SGA 1 II 73
4.10) and the implicit function theorem. The functor $X' \mapsto X'(\mathbf{C})$ therefore gives a morphism of sites

$$\epsilon : X_{\mathrm{cl}} \longrightarrow X_{\mathrm{et}}, \tag{4.2}$$

where X_{et} is the étale site.

Recall the following results[34]:

THEOREM 4.3. (i) *X is connected and nonempty if and only if $X(\mathbf{C})$ is connected and nonempty.*

(ii) *The functor ϵ is fully faithful.*

(iii) *("Grauert–Remmert theorem"). The functor ϵ induces an equivalence between the category of finite coverings of $X(\mathbf{C})$, endowed with its usual topology as a locally compact space, and the category of étale coverings of X.*

We shall accept (i) as known. It follows, for example, from (GAGA) [4]. Assertion (ii) follows easily from (i). Indeed, to verify the surjectivity assertion contained in (ii), let $\varphi : X'(\mathbf{C}) \to X''(\mathbf{C})$ be an $X(\mathbf{C})$-morphism. We may suppose that all the schemes are connected and affine, and hence separated. Then the graph Γ of φ is a connected

[34] More precisely, the inclusion of categories $\mathrm{Ouv}(X(\mathbf{C})) \to X_{\mathrm{cl}}$ defines a morphism of sites (4.1) in the opposite direction.

[34] Which appear in the Exposé cited in the footnote on p. 2.

component of $X' \times_X X''(\mathbf{C})$, whence $\Gamma = Y(\mathbf{C})$ for some connected component Y of $X' \times_X X''$, by (i). Then Y is the graph of a morphism $f : X' \to X''$ inducing φ, i.e. the projection $Y \to X'$ is an isomorphism: indeed, $Y(\mathbf{C}) \to X(\mathbf{C})$ is an isomorphism, from which it follows easily that the same is true of $Y \to X'$ (which is étale, radicial, and surjective!). For (iii), in view of (ii), everything amounts to proving that every finite
74 covering of $X(\mathbf{C})$ is of the form $X'(\mathbf{C})$, where $X' \to X$ is étale.

For this, descent (SGA 1 IX 4.7) immediately reduces us to the case in which X is connected and normal. The problem is local on X (for the Zariski topology), and we may therefore suppose X affine. Let $f : X \to \mathbf{E}^n$ be a finite morphism, and let $T' \to X(\mathbf{C})$ be a connected finite covering. One checks immediately that the composite morphism $\eta' : T' \to \mathbf{E}^n(\mathbf{C})$ is an "analytic covering" in the sense of ([2] § 2, Def. 3). By ([2] § 2, Satz 8), extends to an "analytic covering" $\eta : T \to \mathbf{P}^n(\mathbf{C})$, and it is enough to prove that η is induced by a suitable morphism of schemes $f : Y \to \mathbf{P}^n$, with Y normal. Now the fundamental theorem ([2] § 13 Satz 42) asserts that there is a canonical normal analytic structure on T such that η induces a morphism of analytic spaces. Since η is then a finite morphism, the direct image of the structure sheaf on T is a sheaf $\mathfrak{a}$ of coherent analytic algebras on $\mathbf{P}^n$. It follows from GAGA [4] that $\mathfrak{a}$ is the analytic sheaf associated with an algebraic sheaf A on $\mathbf{P}^n$, and we take $Y = \operatorname{Spec} A$.

We can now prove the following result:

THEOREM 4.4. *Let X be a smooth scheme over* Spec $\mathbf{C}$.

(i) *There is an equivalence between the category of constructible locally constant torsion sheaves on X_{et} and the category of locally constant torsion sheaves with finite fibers on X_{cl}, the equivalence being given by the quasi-inverse functors $_*$ and *.*

(ii) *Let F be a locally constant torsion sheaf with finite fibers on X_{cl}, and also denote by F the induced sheaf $\epsilon_* F$ on X_{et}. Then*

$$R^q \epsilon_* F = 0 \quad \text{if } q > 0.$$

75 (iii) *With the notation of* (i), *the canonical morphism*

$$H^q(X_{\text{et}}, F) \longrightarrow H^q(X_{\text{cl}}, F)$$

is bijective for every q. In particular,

$$H^q(X_{\text{et}}, \mathbf{Z}/n) \xrightarrow{\sim} H^q(X_{\text{cl}}, \mathbf{Z}/n)$$

for every q.

Proof. For (i), note that the sheaves in question are those that are representable by étale coverings of X (resp. by finite coverings of $X(\mathbf{C})$). Since there is an equivalence between the categories of these coverings (4.3), assertion (i) follows immediately. The isomorphism of (iii) follows from (ii) and from the Leray spectral sequence V 5.2 for ϵ. It remains to prove (ii).

Now $R^q \epsilon_* F$ is the sheaf associated with the presheaf $\mathscr{R}^q F$,, where $(\mathscr{R}^q F)(X') = H^q(X'_{\text{cl}}, F)$ for $X' \to X$ étale. To prove that $R^q \epsilon_* F = 0$ if $q > 0$, we must therefore prove the following :

LEMMA 4.5. *Let $\xi \in H^q(X_{\text{cl}}, F)$ and $x \in X_{\text{cl}} = X(\mathbf{C})$. There exists an étale morphism $X' \to X$ whose image contains x and such that the image of ξ in $H^q(X'_{\text{cl}}, F)$ is zero.*

Proof of the lemma. Induction on $n = \dim X$; the problem is local on X for the étale topology, and we may therefore suppose F constant, and by dévissage we may even suppose that $F = \mathbf{Z}/n$. Moreover, replacing X by a Zariski open neighborhood of x, we

may suppose (3.3) that X admits an elementary fibration (3.1) $f : X \to S$. Retain the notation of (3.1): by direct calculation, one sees that $R^q j_* F = 0$ if $q > 1$, that $R^1 j_{\mathrm{cl}*} F$ is 76
the extension by 0 of a constant sheaf on Y_{cl}, and finally that $j_{\mathrm{cl}*} F$ is the constant sheaf $\mathbf{Z}/n$ on $\overline{X}$. Moreover, one may compute $R^q \overline{f}_{\mathrm{cl}*}(j_{\mathrm{cl}*} F)$ fiber by fiber. From the Leray spectral sequence

$$E_2^{pq} = R^p \overline{f}_{\mathrm{cl}*}(R^q j_{\mathrm{cl}*} F) \Longrightarrow R^{p+q} f_{\mathrm{cl}*} F$$

one deduces that $R^{p+q} f_{\mathrm{cl}*} F$ may likewise be computed fiber by fiber, and hence that

$R^0 f_{\mathrm{cl}*} \mathbf{Z}/n = \mathbf{Z}/n$,
$R^1 f_{\mathrm{cl}*} \mathbf{Z}/n$ is a locally constant torsion sheaf on S_{cl},
$R^q f_{\mathrm{cl}*} \mathbf{Z}/n = 0$ if $q > 1$.
The Leray spectral sequence
$E_2^{pq} = H^p(S_{\mathrm{cl}}, R^q f_{\mathrm{cl}*} F) \Longrightarrow H^{p+q}(X_{\mathrm{cl}}, F)$
thus reduces to the exact sequence

$$(*) \qquad \cdots \longrightarrow H^q(S_{\mathrm{cl}}, f_{\mathrm{cl}*} F) \longrightarrow H^q(X_{\mathrm{cl}}, F) \longrightarrow H^{q-1}(S_{\mathrm{cl}}, R^1 f_{\mathrm{cl}*} F) \longrightarrow$$

Let s be the image of x in S. By the induction hypothesis and by (i), for every class $\eta \in H^q(S_{\mathrm{cl}}, G)$ (G locally constant torsion, with finite fibers), there exists an "étale neighborhood" $S' \to S$ of s such that the image of η in $H^q(S'_{\mathrm{cl}}, G)$ is zero. Now the functors $R^q f_{\mathrm{cl}*}$ plainly commute with étale base changes, and if $S' \to S$ is an étale neighborhood of s, then $X' = X \times_S S' \to X$ is an étale neighborhood of x. When $q \geq 1$, we therefore conclude by means of the exact sequence $(*)$, first taking $G = R^1 f_{\mathrm{cl}*} F$ and then $G = f_{\mathrm{cl}*} F$. When $q = 1$, note that $H^1(X_{\mathrm{cl}}, \mathbf{Z}/n\mathbf{Z})$ (resp. $H^1(X, \mathbf{Z}/n\mathbf{Z})$) classifies 77
the principal étale coverings of X_{cl} (resp. X), with group $\mathbf{Z}/n\mathbf{Z}$; hence, by 4.3 (iii), the map $H^1(X, \mathbf{Z}/n\mathbf{Z}) \to H^1(X_{\mathrm{cl}}, \mathbf{Z}/n\mathbf{Z})$ is bijective, from which the effacement result 4.5 for $q = 1$ follows at once.

VARIANT 4.6. One may also prove (4.5) in the following elegant way, due to Serre: one notes that a connected good neighborhood (3.2) X satisfies the following two conditions:

(i) $\pi_n(X(\mathbf{C})) = 0$ if $n > 1$.
(ii) $\pi_1(X(\mathbf{C}))$ is an iterated extension of free groups of finite type.

Indeed, if $X \to S$ is an elementary fibration where S is a good neighborhood, we may suppose (i) and (ii) true for S by induction. Now $X(\mathbf{C}) \to S(\mathbf{C})$ is a locally trivial fiber space, as is readily seen, with fibers isomorphic to $X_0(\mathbf{C})$, where X_0 is an arbitrary fiber of X/S. Now X_0 is a connected nonsingular *noncomplete* curve, and it is well known that this implies that $\pi_n(X_0(\mathbf{C})) = 0$ if $n > 1$, and that $\pi_1(X_0(\mathbf{C}))$ is a free group. Then (i) and (ii) follow from the homotopy exact sequence

$$\longrightarrow \pi_0(X_0(\mathbf{C})) \longrightarrow \pi_n(X(\mathbf{C})) \longrightarrow \pi_n(S(\mathbf{C})) \longrightarrow \ldots$$

Thus $X(\mathbf{C})$ is a $K(\pi, 1)$ space, and this implies that every cohomology class $\xi \in H^n(X(\mathbf{C}), F)$ becomes zero on the universal covering of $X(\mathbf{C})$. It is enough to prove that ξ already becomes zero on a finite covering X' of X, which amounts to proving that $\pi_1 = \pi_1(X(\mathbf{C}))$ is a *good group* (cf. CG 1 16), that is, that the morphism from π_1 to 78
its completion $\hat{\pi}_1 = \varprojlim_N \pi_1/N$ (where N runs through the set of invariant subgroups of π_1 of finite index) induces a bijection

$$H^q(\hat{\pi}_1, M) \xrightarrow{\sim} H^q(\pi_1, M)$$

for every continuous finite π_1-module M (where the symbol on the left denotes the cohomology of $\widehat{\pi}_1$ as a profinite group). Now an iterated extension of free groups of finite type is a good group (cf. CG I p. 15-16, exc. 1,2), whence the result.

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EXPOSÉ XII

The base-change theorem for a proper morphism

M. Artin

1. Introduction

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In this exposé we prove the fundamental theorem which says, in a slightly more precise form, that for a *proper* morphism $f : X \to Y$ of schemes and a torsion abelian sheaf F on X, the fiber of $R^q f_* F$ at a geometric point ξ of Y is isomorphic to the cohomology $H^q(X_\xi, F|X_\xi)$ of the fiber X_ξ.

Recall that the analogous result for a proper morphism of paracompact spaces is well known (Godement II 4.11) and easy. It is considerably more delicate for schemes. For example, in contrast with the case of paracompact spaces, the hypothesis that F be torsion is essential.

The proof proceeds in several stages. By more or less formal dévissage arguments, one reduces to the case where Y is Noetherian, the relative dimension of f is at most 1, and the sheaf F is constant. To handle this special case, one uses the specialization theorem for the fundamental group (5.9), which is a nonabelian variant of the theorem stated above and can be proved directly in the case under consideration (XIII 2).

2. An example

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We show that the hypothesis that F be torsion is necessary. Let X be a nonsingular surface over an algebraically closed field k, let Y be a nonsingular curve over k, and let $f : X \to Y$ be a proper morphism whose generic fiber is smooth and whose geometric fiber at a geometric point $\overline{y}$ of Y is irreducible and has an ordinary double point. Let F be the constant sheaf $\mathbf{Z}_X$.

Recall the following fact. For any point z of a scheme Z, let $i : z \to Z$ be the inclusion. Then $H^1(Z, i_* \mathbf{Z}_z) = 0$ (IX 3.6 (i)). Since X is normal, hence unibranch, it follows that $\mathbf{Z}_X \simeq i_* \mathbf{Z}_x$, where $i : x \to X$ is the inclusion of the generic point. Thus $H^1(X, \mathbf{Z}_X) = 0$ and likewise $R^1 f_* \mathbf{Z}_X = 0$. But for the geometric fiber $X_\circ$ of X/Y having the ordinary double point, I claim that $H^1(X_\circ, \mathbf{Z}_{X_\circ}) \simeq \mathbf{Z}$[34]; this gives the counterexample, since $\mathbf{Z}_{X_\circ} \simeq \mathbf{Z}_X|X_\circ$.

Let $i_\circ : x_\circ \to X_\circ$ be the inclusion of the generic point of $X_\circ$, and let Q be the singular point of $X_\circ$. The curve $X_\circ$ has two "branches" at Q, and it follows that the fiber of $i_{\circ *}(\mathbf{Z}_{x_\circ})$ at Q is isomorphic to $\mathbf{Z} \times \mathbf{Z}$. Away from Q, the curve $X_\circ$ is normal, hence $i_{\circ *}(\mathbf{Z}_{x_\circ}) \simeq \mathbf{Z}_{X_\circ}$ on $X_\circ - Q$. Therefore there is an exact sequence

$$0 \longrightarrow \mathbf{Z}_{X_\circ} \longrightarrow i_{\circ *}(\mathbf{Z}_{x_\circ}) \longrightarrow (\mathbf{Z})_Q \longrightarrow 0,$$

where the last term is extension by zero of the sheaf $\mathbf{Z}$ at the point Q. All three terms have H^0 isomorphic to $\mathbf{Z}$, and $H^1(X_\circ, i_{\circ *}(\mathbf{Z}_{x_\circ})) = 0$. The long exact cohomology sequence therefore gives $H^1(X_\circ, \mathbf{Z}_{X_\circ}) = \mathbf{Z}$. 81

[34] Compare, from the point of view of Π_1, with SGA 3 X (6, Examples) and (1.6).

3. Review of nonabelian H^1

We briefly recall some results on nonabelian cohomology that will be needed below. We omit most proofs, which are well known in various special cases and may be found in Giraud's thesis [1].

Let X be a scheme and F a sheaf of groups on X. For every étale X'/X, one defines a pointed set $H^1(X', F)$, covariantly functorial in F and contravariantly functorial in X'. The functor H^1 commutes with finite products of sheaves. If $f : X \to Y$ is a morphism, one defines a sheaf of pointed sets (that is, with a distinguished section) $R^1 f_* F$. It is the sheaf associated with the presheaf of pointed sets $\mathscr{R}^1 F$ which assigns to every étale Y'/Y the set $H^1(X \times_Y Y', F)$.

There are the usual exact cohomology sequences; we record one of them:

Proposition 3.1. *Let $0 \to F \xrightarrow{u} G$ be an injection of sheaves of groups, and let $C = G/F$ be the sheaf of homogeneous spaces under G. There is an exact sequence of sheaves of pointed sets*

$$0 \longrightarrow f_*F \longrightarrow f_*G \longrightarrow f_*C \xrightarrow{\partial} R^1 f_* F \xrightarrow{u^1} R^1 f_* G.$$

This exact sequence is obtained, of course, by passing to the associated sheaves in the exact sequence of presheaves whose value on an étale Y'/Y is

$$0 \longrightarrow F(X') \longrightarrow G(X') \longrightarrow C(X') \xrightarrow{\partial} H^1(X', F) \longrightarrow H^1(X', G),$$

82 where $X' = X \times_Y Y'$. The equivalence relation induced by ∂ is the one given by the action of f_*G on f_*C. As usual, the equivalence relation induced by u^1 may be made explicit using the notion of "twisting" [1].

Proposition 3.2. *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes. For every étale Z'/Z, putting $Y' = Y \times_Z Z'$ and $X' = X \times_Z Z'$, there is an exact sequence*

$$0 \longrightarrow H^1(Y', f_*F) \longrightarrow H^1(X', F) \longrightarrow H^0(Y', R^1 f_* F).$$

Passing to the associated sheaves gives an exact sequence of pointed sheaves on Z:

$$0 \longrightarrow (R^1 g_*) f_* F \longrightarrow R^1 (gf)_* F \longrightarrow g_* (R^1 f_*) F.$$

Proposition 3.3. *The family of functors of the form $H^1(X', F)$, for X' étale over X, and of the form $R^1 f_* F$, for an arbitrary morphism $f : X \to Y$, is effaceable: there exists an injection $F \xrightarrow{u} G$ such that all the induced morphisms $H^1(X', F) \to H^1(X', G)$ and $R^1 f_* F \to R^1 f_* G$ are zero. More precisely, there exists such an injection for which every $H^1(X', G)$, with X' étale over X, and every pointed sheaf $R^1 f_* G$ vanish. If X is Noetherian and F is a constructible sheaf of ind-**L**-groups, then such an injection exists with G also a sheaf of ind-**L**-groups.*

Proof. The first assertion is presumably true in an arbitrary topos, but for the étale topology it is convenient to perform the effacement using the "Godement resolution"

$$F \longrightarrow G = \prod_{x \in X} i_{x*} i_x^* F. \tag{3.4}$$

83 Here x runs through the points of X and $i_x : \overline{x} \to X$ is a geometric point over x. For

X'/X étale, one has

$$G(X') = \prod_{x\in X} (i_x^* F)(X' \times_X \overline{x}) = \prod_{x\in X} \left(\prod_{\overline{x}'/\overline{x}} F_{\overline{x}'} \right),$$

where the $\overline{x}'$ are geometric points of X'. For each point of X' there is plainly at least one $\overline{x}'$ above it; hence $F(X') \subset G(X')$ (VIII 3), and therefore $F \subset G$. Moreover, every étale $\overline{u}/\overline{x}$ is a disjoint union of spectra of separably closed fields, so $H^1(\overline{u}, \overline{G}) = 0$ for every sheaf $\overline{G}$ on $\overline{x}$. It follows from 3.2 that $H^1(X', i_{x*} i_x^* F) = 0$ for every étale X'/X. Since H^1 commutes with products, $H^1(X', G) = 0$, and consequently $R^1 f_* G = 0$ for every $f : X \to Y$, by the description of $R^1 f_* G$ as the sheaf associated with the presheaf $Y' \mapsto H^1(X \times_Y Y', G)$.

Suppose in addition that F is constructible and X is Noetherian. It is immediate that there is a finite set of points $\{x_1, \dots, x_n\}$ of X such that

$$F \longrightarrow \prod_{1\le i\le n} i_{x_i *} i_{x_i}^* F$$

is already injective. The sheaf $i_{x_i *} i_{x_i}^* F$ is a sheaf of ind-finite groups (IX 1.5), and hence so is the finite product (IX 1.4 and IX 1.6 (iii)). Replacing G by this sheaf gives the last assertion.

3.5. As in the case of abelian cohomology (treated in V 2.4), H^1 may be computed by the Čech procedure [1].

4. The base-change morphism

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Let

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{g} & S' \end{array}$$

be a Cartesian square of morphisms of schemes. From it one obtains a morphism of functors $\varphi = \varphi_g$

$$g^* f_* \xrightarrow{\varphi} f'_* g'^* \tag{4.1}$$

as follows. Giving such a morphism is equivalent, by the adjunction between g^* and g_*, to giving a morphism of functors

$$f_* \longrightarrow g_* f'_* g'^*. \tag{4.1 adj}$$

We take this to be the morphism

$$f_* \xrightarrow{(\mathrm{id}\to g'_* g'^*)} f_* g'_* g'^* = g_* f'_* g'^*,$$

using

$$g_* f'_* = (g f')_* = (f g')_* = f_* g'_*.$$

There is another candidate for (4.1). Indeed, giving such a morphism is also equivalent to giving

$$f'^* g^* f_* \longrightarrow g'^*,$$

and one may take

$$f'^* g^* f_* \simeq g'^* f^* f_* \xrightarrow{(f^* f_* \to \mathrm{id})} g'^*.$$

85 I do not know whether these two choices always coincide [34]; in any event, we shall use only the first. We call it the *base-change morphism.*

Restricting (4.1) to abelian sheaves (respectively, to sheaves of groups) gives a morphism $\varphi^q = \varphi_g^q$ for every $q \geq 0$ (respectively, for $q = 0, 1$):

$$g^*(R^q f_*) \xrightarrow{\varphi^q} (R^q f'_*) g'^*. \tag{4.2}$$

Indeed, this is equivalent to giving

$$(R^q f_*) \longrightarrow g_*(R^q f'_*) g'^*, \tag{4.2 adj}$$

and there are the following "evident" morphisms:

$$(R^q f_*) \xrightarrow{a} (R^q f_*) g'_* g'^* \xrightarrow{b} R^q(f g')_* g'^*$$
$$= R^q(g f')_* g'^* \xrightarrow{c} g_*(R^q f'_*) g'^*.$$

Remark 4.3. Suppose that g (and hence g') is a closed immersion. Then the morphisms b and c above are isomorphisms (VIII 5.7). It follows that (4.2) is essentially just the morphism a above, obtained by applying $R^q f_*$ to the canonical morphism $F \to g'_* g'^* F$.

Proposition 4.4. *(Composition of base-change morphisms).*

(i) *Let*

$$\begin{array}{ccccc} X & \xleftarrow{g'} & X' & \xleftarrow{h'} & X'' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f''} \\ S & \xleftarrow{g} & S' & \xleftarrow{h} & S'' \end{array}$$

86 *be a Cartesian diagram of morphisms of schemes. Then the following diagram is commutative:*

$$\begin{array}{ccccc} (gh)^* f_* & & \xrightarrow{\varphi_{gh}} & & f''_*(g'h')^* \\ \downarrow{\scriptstyle \wr} & & & & \downarrow{\scriptstyle \wr} \\ h^* g^* f_* & \xrightarrow{h^* \varphi_g} & h^* f'_* g'^* & \xrightarrow{\varphi_h g'^*} & f''_* h'^* g'^*. \end{array}$$

The same is true of the diagram obtained by replacing f_*, f'_*, and f''_* by $R^q f_*$, $R^q f'_*$, and $R^q f''_*$, in the natural domains of definition of the functors in question (cf. n° 3).

[34] Mr. Deligne resolved this uncertainty in the affirmative, in a considerably more general context; cf. Exposé XVII.

(ii) *Let*

$$\begin{array}{ccc} Y & \xleftarrow{g''} & Y' \\ {\scriptstyle e}\downarrow & & \downarrow{\scriptstyle e'} \\ X & \xleftarrow{g'} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{g} & S' \end{array}$$

be a Cartesian diagram of morphisms of schemes. Then the following diagram is commutative:

$$\begin{array}{ccccc} g^*(fe)_* & & \xrightarrow{\varphi_g^{fe}} & & (f'e')_* g''^* \\ \| & & & & \| \\ g^* f_* e_* & \xrightarrow{\varphi_g^f e_*} & f'_* g'^* e_* & \xrightarrow{f'_* \varphi_{g'}^e} & f'_* e'_* g''^*, \end{array}$$

where the three horizontal arrows arise respectively from the base-change homomorphisms for fe, f, and e.

In the abelian case, the base-change morphisms for fe are induced by a morphism of spectral sequences

$$\begin{array}{ccc} E_2^{p,q} = g^*(R^p f_*)(R^q e_*)F & \Longrightarrow & g^* R^{p+q}(fe)_* F \\ \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi} \\ (R^p f'_*)(R^q e'_*) g''^* F & \Longrightarrow & R^{p+q}(f'e')_* g''^* F, \end{array}$$

which, on the initial terms, is induced by the base-change morphisms for e and f. For a sheaf of groups, there is a morphism of exact sequences

$$\begin{array}{ccccccc} 0 \longrightarrow & g^*(R^1 f_*)e_* F & \longrightarrow & g^* R^1(fe)_* F & \longrightarrow & g^* f_*(R^1 e_*)F \\ & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi} \\ 0 \longrightarrow & (R^1 f'_*)e'_* g''^* F & \longrightarrow & R^1(f'e')_* g''^* F & \longrightarrow & f'_*(R^1 e'_*) g''^* F. \end{array}$$

We leave the proof to the reader.

5. Statement of the main theorem and some variants

The theorem is as follows:

THEOREM 5.1. *(Proper base-change theorem). Let*

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{g} & S' \end{array}$$

be a Cartesian diagram of schemes, with f proper.

(i) *Let F be a sheaf of sets on X. Then the morphism (4.1)*

$$\varphi : g^* f_* F \longrightarrow f'_* g'^* F$$

is bijective.

87 (ii) *Let F be a sheaf of groups (resp. of ind-finite groups) on X. Then the morphism (4.2)*

$$\varphi^1 : g^*(R^1 f_*)F \longrightarrow (R^1 f'_*)g'^* F$$

is injective (resp. bijective).

(iii) *Let F be a torsion abelian sheaf on X. Then the morphism (4.2)*

$$\varphi^q : g^*(R^q f_*)F \longrightarrow (R^q f'_*)g'^* F$$

is bijective for every $q \geq 0$.

We first state some immediate consequences of the theorem. If $S' = \xi$ is a geometric point of S, so that a sheaf on ξ is determined by its global sections, then $H^0(\xi, R^q f'_* G) = H^q(X', G)$ for every sheaf G on X'. This gives the following corollary (which is, moreover, essentially equivalent to 5.1):

COROLLARY 5.2. *Let $f : X \longrightarrow S$ be proper, let $\xi \longrightarrow S$ be a geometric point, and let X_ξ be the fiber of f at ξ.*

(i) *If F is a sheaf of sets on X, the canonical base-change morphism*

$$(f_* F)_\xi \longrightarrow H^0(X_\xi, F|X_\xi)$$

is bijective.

(ii) *If F is a sheaf of groups (resp. of ind-finite groups), the base-change morphism*

$$(R^1 f_* F)_\xi \longrightarrow H^1(X_\xi, F|X_\xi)$$

is injective (resp. bijective).

88 (iii) *If F is a torsion abelian sheaf, then for every $q \geq 0$ the base-change morphism*

$$(R^q f_* F)_\xi \longrightarrow H^q(X_\xi, F|X_\xi)$$

is bijective.

Taking X 4.3 and X 5.2 into account, we obtain:

COROLLARY 5.3. *Under the hypotheses of X 5.2, let n be the dimension of the fiber X_ξ. Then, for every torsion abelian sheaf F on X, the fiber $(R^q f_* F)_\xi$ vanishes for $q > 2n$. If F is a sheaf of p-torsion groups, where p is the characteristic of ξ, then the fiber of $R^q f_* F$ at ξ vanishes for $q > n$.*

Recall that the *relative dimension of X/S is $\leq n$* if every fiber has dimension $\leq n$. We therefore have:

COROLLARY 5.3 BIS. *Let $f : X \longrightarrow S$ be proper, with X/S of relative dimension $\leq n$. Then $R^q f_* F = 0$ for $q > 2n$ and every torsion abelian sheaf F on X. If S has characteristic p and F is a sheaf of p-torsion abelian groups, then $R^q f_* F = 0$ for $q > n$.*

Applying 5.1 to a morphism between spectra of separably closed fields gives:

COROLLARY 5.4. *(Invariance of cohomology under extension of the base field in the proper case). Let $k \subset K$ be separably closed fields and let X be a proper scheme over* Spec k. *Put $X' = X_K$. Let F be a sheaf on X and let F' be its inverse-image sheaf on X'. Then:*

(i) $H^0(X, F) \xrightarrow{\sim} H^0(X', F')$.

89 (ii) *If F is a sheaf of groups (resp. of ind-finite groups), the morphism*

$$H^1(X,F) \longrightarrow H^1(X',F')$$

is injective (resp. bijective).

(iii) *If F is a torsion abelian sheaf, then*

$$H^q(X,F) \xrightarrow{\sim} H^q(X',F')$$

for every q.

A slightly different variant of 5.2 is the following:

COROLLARY 5.5. *Let S be the spectrum of a henselian ring, and let $S' = s_\circ$ be its closed point. Let $f : X \longrightarrow S$ be proper. Put $X_\circ = X'$ and $F_\circ = g'^* F$. Then:*

(i) $H^0(X,F) \xrightarrow{\sim} H^0(X_\circ, F_\circ)$
for every sheaf of sets F.

(ii) *The morphism $H^1(X,F) \longrightarrow H^1(X_\circ, F_\circ)$*
is injective (resp. bijective) for every sheaf of groups (resp. of ind-finite groups) F.

(iii) $H^q(X,F) \xrightarrow{\sim} H^q(X_\circ, F_\circ)$
for every torsion abelian sheaf F and every q.

Indeed, this statement is just a special case of 5.2 when S is strictly local. Suppose that 5.2 is known. Let $\overline{S}$ be the strict localization of S, let $\overline{s}$ be the closed point of $\overline{S}$, and let G be the Galois group of $\overline{S}/S$ (which is also the Galois group of $\overline{s}/s_\circ$). Then G acts on the fibers of the R^q, and, with the evident notation, one has

$$H^0(X,F) \simeq H^0(\overline{X},\overline{F})^G,$$
$$H^0(X_\circ,F_\circ) \simeq H^0(\overline{X}_\circ,\overline{F}_\circ)^G$$

for a sheaf of sets F. Assertion (i) follows, since $H^0(\overline{X},\overline{F}) \xrightarrow{\sim} H^0(\overline{X}_\circ,\overline{F}_\circ)$ and this 90
isomorphism plainly commutes with the action of G.

If F is a sheaf of groups, one readily defines a morphism of exact sequences of pointed sets

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^1(G, H^0(\overline{X},\overline{F})) & \longrightarrow & H^1(X,F) & \longrightarrow & H^1(\overline{X},\overline{F})^G \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & H^1(G, H^0(\overline{X}_\circ,\overline{F}_\circ)) & \longrightarrow & H^1(X_\circ,F_\circ) & \longrightarrow & H^1(\overline{X}_\circ,\overline{F}_\circ)^G, \end{array} \tag{5.5.1}$$

in which the left-hand vertical arrow is an isomorphism and the right-hand one is injective by 5.2. Let $\alpha, \beta \in H^1(X,F)$ be classes having the same image in $H^1(X_\circ,F_\circ)$, and let F^α be the sheaf obtained by twisting F by α [1]. Let β' be the class in $H^1(X,F^\alpha)$ corresponding to β. Since the class α' corresponding to α is neutral, the image of β' in $H^1(X_\circ,(F^\alpha)_\circ)$ is neutral.

Consider the diagram deduced from (5.5.1) by replacing F with F^α. It shows that the image of β' in $H^1(\overline{X},\overline{F}^\alpha)^G$ is neutral, and hence that β' comes from an element of $H^1(G, H^0(\overline{X},\overline{F}^\alpha))$; this element must itself be neutral. Thus β' is neutral, that is, $\beta' = \alpha'$, whence $\beta = \alpha$. This proves the injectivity of the morphism in (ii).

Now suppose that F is a sheaf of ind-finite groups, and let us prove that the arrow in (ii) is surjective. It is enough to show that the functor $F \mapsto H^1(X_\circ,F_\circ)$ is effaceable.

Indeed, let $0 \longrightarrow F \xrightarrow{u} G$ be an injection which effaces this functor, and put $C = G/F$. There is a morphism of exact sequences (4.4)

$$\begin{array}{ccccc} H^0(X,C) & \longrightarrow & H^1(X,F) & \longrightarrow & H^1(X,G) \\ \downarrow & & \downarrow{\scriptstyle \epsilon} & & \downarrow \\ H^0(X_\circ,C_\circ) & \longrightarrow & H^1(X_\circ,F_\circ) & \xrightarrow{u^1_\circ} & H^1(X_\circ,G_\circ) \end{array}$$

in which $u^1_\circ$ is zero; this indeed implies the surjectivity of ϵ.

91 Let us prove that the functor $F \mapsto H^1(X_\circ, F_\circ)$ is effaceable for sheaves F of ind-finite groups. Let $F \xrightarrow{u} C$ be the "Godement resolution" (3.4). We have

$$H^1(G, H^0(\overline{X}_\circ, \overline{C}_\circ)) \simeq H^1(G, H^0(\overline{X}, \overline{C})) \subset H^1(X, C) = 0.$$

Moreover, there is a commutative square

$$\begin{array}{ccc} H^1(\overline{X},\overline{F}) & \xrightarrow{\sim} & H^1(\overline{X}_\circ,\overline{F}_\circ) \\ \downarrow & & \downarrow \\ 0 = H^1(\overline{X},\overline{C}) & \longrightarrow & H^1(\overline{X}_\circ,\overline{C}_\circ) \end{array}$$

(indeed, $H^1(\overline{X},\overline{C}) = \varinjlim H^1(X',C) = 0$, where X'/X runs through the étale morphisms), so its right-hand vertical arrow is zero. We therefore have a morphism of exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^1(G, H^0(\overline{X}_\circ,\overline{F}_\circ)) & \longrightarrow & H^1(X_\circ,F_\circ) & \longrightarrow & H^1(\overline{X}_\circ,\overline{F}_\circ)^G \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \alpha^G} \\ 0 & \longrightarrow & 0 & \longrightarrow & H^1(X_\circ,C_\circ) & \longrightarrow & H^1(\overline{X}_\circ,\overline{C}_\circ)^G \end{array}$$

in which α^G is zero. It follows that β is also zero, and hence that the functor is effaceable.

Finally, let us prove assertion (iii) of 5.5. There is a morphism of spectral sequences

$$\begin{array}{ccc} E_2^{p,q} = H^p(G, H^q(\overline{X},\overline{F})) & \Longrightarrow & H^n(X,F) \\ \downarrow & & \downarrow{\scriptstyle \varphi} \\ H^p(G, H^q(\overline{X}_\circ,\overline{F}_\circ)) & \Longrightarrow & H^n(X_\circ,F_\circ). \end{array}$$

By 5.2, the morphisms $H^q(\overline{X},\overline{F}) \longrightarrow H^q(\overline{X}_\circ,\overline{F}_\circ)$ are isomorphisms. Consequently the morphism of $E_2^{p,q}$ terms is an isomorphism, and therefore so is the morphism of abutments. This proves the result, C.Q.F.D.

92 **5.6.** Let us examine 5.5 (i) and (ii) when F is constant. Let $F = T_X$ be the constant sheaf of sets on X with value T. Then $F(X)$ is simply the set of locally constant maps from X to T. If $X_\circ$ is a subscheme of X and T has at least two elements, it follows that $F(X) \longrightarrow F(X_\circ)$ is bijective if and only if the map

$$U \longmapsto U_\circ = U \cap X_\circ$$

from the set $\Theta(X)$ of subsets of X that are both open and closed to the analogous set $\Theta(X_\circ)$ is bijective. If the set $\Pi_\circ(X_\circ)$ of connected components of $X_\circ$ is finite, this is equivalent to saying that the map $\Pi_\circ(X_\circ) \longrightarrow \Pi_\circ(X)$ is bijective. Thus 5.5 (i) may also

be restated as follows (bearing in mind that in this case X_0 is of finite type over a field, hence Noetherian, so that $\Pi_0(X_0)$ is finite):

COROLLARY 5.7. *Let S be the spectrum of a henselian ring, let s_0 be its closed point, let $f : X \longrightarrow S$ be a proper morphism, and let X_0 be the closed fiber of X/S. Then the map*

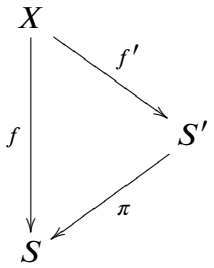
$$\Pi_0(X_0) \longrightarrow \Pi_0(X)$$

induced by $X_0 \longrightarrow X$ is bijective.

(In particular, this implies that $\Pi_0(X)$ is finite.) We shall prove 5.7 directly in the Noetherian case:

LEMMA 5.8. *Statement 5.7 is true if S is Noetherian* [34].

Proof. In the Noetherian case, $\Pi_0(X)$ is certainly finite. Replacing X by a connected component, we may suppose that $\Pi_0(X)$ has one element, that is, that X is connected and nonempty. We must show that X_0 is also connected and nonempty. The image of X in S is closed because f is proper, and is nonempty because X is nonempty; it therefore contains s_0, which shows that X_0 is nonempty. Let



be the Stein factorization of f (EGA III 4.3.3), so that S'/S is finite and S' is connected. 93
Every fiber of f' is known to be connected and nonempty, and the underlying space of X_0 is the union of the underlying spaces of the fibers over the closed points of S'. It remains only to show that S' is the spectrum of a *local* ring. Since S is henselian, the ring A' of S' is a product of local rings (VII 4.1 (i)); because S' is connected, A' is local. This proves 5.8.

5.9.0. Suppose now that $F = G_X$ is a constant sheaf of groups with value a finite group G. Then $H^1(X, G_X)$ classifies the principal Galois coverings of X with group G (VII 2.1). Corollary 5.5 (ii) says that the morphism $X_0 \longrightarrow X$ induces a bijection between the sets of isomorphism classes of these coverings. By the definition of the Π_1, it follows that 5.5 (ii) for constant sheaves of finite groups (in the presence of (i)) is equivalent to:

THEOREM 5.9. *(Specialization theorem for the fundamental group). With the notation of 5.5, suppose that X is connected and nonempty. Then X_0 is connected and nonempty; and, if $\overline{x}_0$ is a geometric point of X_0, the morphism of fundamental groups*

$$\Pi_1(X_0, \overline{x}_0) \longrightarrow \Pi_1(X, \overline{x}_0)$$

is an isomorphism.

On the other hand, by Galois theory (SGA V), statement 5.9 (in the case where X_0 94
is connected, as we may suppose by 5.7) is equivalent to:

[34] Cf. EGA IV 18.5.19.

THEOREM 5.9 BIS. *Let S be the spectrum of a henselian ring, let s_0 be the closed point of S, let $f : X \longrightarrow S$ be a proper morphism, and let X_0 be the closed fiber of X/S. For a scheme Z, denote by* $\mathrm{Et}(Z)$ *the category of étale coverings of Z. Then the restriction functor*

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_0)$$

is an equivalence of categories.

REMARK 5.10. Recall that 5.9 was already proved (SGA 1 X 2.1) when S is the spectrum of a complete Noetherian local ring. Unfortunately, that result does not seem to yield a proof of 5.9 in general [34]. The proof of 5.9 given in this exposé and the next is very different from the one given in SGA 1 X.

5.11. One can give another "noncommutative" variant of 5.5 (ii), using J. GIRAUD's theory of 2-cohomology (cf. J. GIRAUD's thesis, in preparation). Suppose that S is Noetherian. Let L be a *band* on X, and suppose that L is represented locally (for the étale topology) by a sheaf of ind-finite groups. Consider the map

$$\varphi^{(2)} : H^2(X, L) \longrightarrow H^2(X_0, L_0),$$

where L_0 is the restriction of L to X_0. Then

$$(\varphi^{(2)})^{-1}(H^2(X_0, L_0)') = H^2(X, L)', \tag{5.11.1}$$

95 where the "primes" denote the subsets formed by the neutral elements.

In view of 5.5 (iii), this statement is also equivalent to the following:

For every gerbe $\mathcal{G}$ on X, with band L as above, let $\mathcal{G}_0$ denote its restriction to X_0. Then the restriction functor on the categories of sections

$$\mathcal{G}(X) \longrightarrow \mathcal{G}_0(X_0)$$

is an equivalence of categories.

Indeed, the fact that this functor is fully faithful follows from 5.5 (i), and the fact that it is essentially surjective when $\mathcal{G}(X) \neq \emptyset$, that is, when the class $\xi \in H^2(X, L)$ of $\mathcal{G}$ is neutral, follows from 5.5 (ii). It remains to express the fact that $\mathcal{G}_0(X_0) \neq \emptyset$ implies $\mathcal{G}(X) \neq \emptyset$, and this is precisely (5.11.1).

We shall not give the proof of (5.11.1) here. We merely point out that, for every Noetherian scheme X and every closed subscheme of X, one can in fact show that the validity of statements 5.5 (i), (ii) implies that of (5.11.1). The proof is fairly elementary and uses the descent result VIII 9.4 a).

6. First reductions

We use the abbreviation (f, F, g) for the data of 5.1 in any one of cases (i), (ii), or (iii) of that theorem.

LEMMA 6.1. *For 5.1 to hold for the data (f, F, g), where f and F are fixed and g is arbitrary, it is enough that the theorem hold for $(\overline{f}, \overline{F}, \overline{g})$ for every S-scheme $\overline{S}$ that is the*
96 *strict localization of a scheme S' locally of finite type over S at a point closed in its fiber. Here $\overline{f}$ and $\overline{F}$ are obtained from f and F by the base change $\overline{S} \to S$, and $\overline{g} : \overline{s} \to \overline{S}$ is the inclusion of the closed point of $\overline{S}$.*

[34] The situation has changed since these lines were written; cf. Th. 3.1 in M. Artin, Algebraic approximation of structures over complete local rings (to appear).

Proof. We may plainly assume that S and S' are affine, say the spectra of rings A and A'. Then A' is the filtered direct limit of its finite-type A-subalgebras A'_i. Using the theory of passage to the limit (VII 5.11 and 5.14), we are reduced to proving 5.1 for the triples (f, F, g_i), where $g_i : \mathrm{Spec}(A'_i) = S'_i \to S$; this reduces us to the case in which g is of finite type. To prove that the base-change homomorphisms are isomorphisms, it is enough to prove that they induce isomorphisms on the fibers of the sheaves in question, and it is enough to consider geometric fibers at points s' of S' that are closed in their fibers over S (VIII 3.13). Let $\overline{s}'$ be a geometric point of S' corresponding to a separable closure of $k(s')$. Since s' is algebraic over the corresponding point s of S, $k(\overline{s}')$ is also a separable closure of $k(s)$. Denote by $\overline{s}$ the prescheme $\overline{s}'$, regarded as lying over S, that is, as a geometric point of S, and consider the strict localizations $\overline{S}$ and $\overline{S}'$ of S and S' at $\overline{s}$ and $\overline{s}'$, respectively. We then have a commutative square

$$\begin{array}{ccc} \overline{s}' & \longrightarrow & \overline{s} \\ \downarrow & & \downarrow \\ \overline{S}' & \longrightarrow & \overline{S}, \end{array}$$

in which the first horizontal arrow is an isomorphism. In view of VIII 5.2 and 5.3, the homomorphism induced on the fibers at $\overline{s}'$ by the base-change homomorphism associated with (f, F, g) identifies with the homomorphism

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{X}', \overline{F}')$$

induced by $\overline{X}' \to \overline{X}$, where $\overline{X}$ and $\overline{X}'$ denote the schemes obtained from X by the base changes $\overline{S} \to S$ and $\overline{S}' \to S$, respectively. The commutative square above plainly gives a commutative square

$$\begin{array}{ccc} H^i(\overline{X}_{\overline{s}}, \overline{F}) & \longrightarrow & H^i(\overline{X}'_{\overline{s}'}, \overline{F}') \\ \uparrow & & \uparrow \\ H^i(\overline{X}, \overline{F}) & \longrightarrow & H^i(\overline{X}', \overline{F}') \end{array}$$

in which the first horizontal arrow is again an isomorphism. Consequently the second horizontal arrow is an isomorphism as soon as the two vertical arrows are. This is precisely what the hypothesis of 6.1 guarantees, C.Q.F.D.

6.1.1. Thus, to prove 5.1 in a situation $(F, f, -)$, we are reduced to studying the special case 5.5, with S strictly local. This leads us to study in general a morphism $h : Y \to X$ of schemes and to seek general conditions under which the corresponding homomorphisms

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

are bijective. This subsection gives some easy auxiliary results on this situation. For convenience in later references, we state them in greater generality and precision than is needed for the proof of 5.1.

LEMMA 6.2. *Let C and C' be two abelian categories, let $T^\cdot = (T^i)$ and $T'^\cdot = (T'^i)$ be two cohomological functors from C to C', and let $\varphi = (\varphi^i)$ be a morphism of cohomological functors from $T^\cdot$ to $T'^\cdot$. Assume that $T^\cdot$ and $T'^\cdot$ vanish in degrees < 0, and that T^i is effaceable for $i > 0$. Let n be an integer. The following conditions are equivalent:*

a) *φ^i is an isomorphism for $i \leq n$ and a monomorphism for $i = n+1$.*
b) *(If $n \geq -1$.) The homomorphism φ^i is a monomorphism for $i = 0$ and an epimorphism for $i \leq n$.*
c) *(If $n \geq 0$.) The homomorphism φ° is an isomorphism, and the T'^i are effaceable for $0 < i \leq n$.*

If these conditions hold and $A \in \mathrm{Ob}(C)$, then $\varphi^{n+1}(A) : T^{n+1}(A) \to T'^{n+1}(A)$ is an epimorphism (hence an isomorphism) if and only if $T'^{n+1}(A)$ is effaceable, that is, if there exists a monomorphism $A \to B$ such that the corresponding homomorphism $T'^{n+1}(A) \to T'^{n+1}(B)$ is zero. If $C' = (\mathrm{Ab})$, the category of abelian groups, and $\xi \in T'^{n+1}(A)$, then ξ lies in the image of $T^{n+1}(A)$ if and only if ξ is effaceable, that is, if there exists a monomorphism $A \to B$ such that the image of ξ in $T'^{n+1}(B)$ is zero.

Proof. It is clear that (a) implies (b) and (c). Since the hypotheses on $T^\cdot$ imply that the T^i for $i > 0$ are the right satellites of T°, the implication (c) $\Rightarrow$ (a) follows from the axiomatic characterization of satellite functors. It also follows, by induction on n, from the last assertion of 6.2, proved by the familiar argument that we do not repeat here (compare the proof of 6.6 below). The implication (b) $\Rightarrow$ (a) is immediate if $n = -1$. If $n \geq 0$, we proceed by induction on n: the induction hypothesis allows us to suppose that
99 φ^i is an isomorphism for $i \leq n-1$ and a monomorphism for $i = n$. The hypothesis then makes it an isomorphism also for $i = n$, and it remains to show, using the fact that φ^n is an isomorphism, that it is a monomorphism for $i = n+1$. Let $A \to B$ be a monomorphism that effaces T^{n+1}, giving an exact sequence $0 \to A \to B \to C \to 0$ in C and a morphism of exact sequences in C':

$$\begin{array}{ccccccc} T^n(B) & \longrightarrow & T^n(C) & \longrightarrow & T^{n+1}(A) & \longrightarrow & T^{n+1}(B) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ T'^n(B) & \longrightarrow & T'^n(C) & \longrightarrow & T'^{n+1}(A) & \longrightarrow & T'^{n+1}(B), \end{array}$$

where the first two vertical arrows are isomorphisms by what we have just proved, and $T^{n+1}(A) \to T^{n+1}(B)$ is zero by construction. The five lemma then implies that $T^{n+1}(A) \to T'^{n+1}(A)$ is a monomorphism, C.Q.F.D.

Corollary 6.3. *Suppose that $T^\cdot$ and $T'^\cdot$ arise from cohomological functors (denoted by the same symbols) on the right derived category $D^+(C)$ [3], with values in C'; the T^i for $i > 0$ are not necessarily assumed effaceable. Suppose that, for $K^\cdot \in D^+(C)$, the condition $H^i(K^\cdot) = 0$ for $i < 0$ implies $T^i(K^\cdot) = T'^i(K^\cdot) = 0$ for $i < 0$. Let $K^\cdot \in \mathrm{Ob}\, D^+(C)$ be a bounded-below complex in C such that $H^i(K^\cdot) = 0$ for $i < 0$.*

(i) *If condition* (a) *of 6.2 is satisfied, then the homomorphism $\varphi^i(K^\cdot) = T^i(K^\cdot) \to T'^i(K^\cdot)$ is an isomorphism for $i \leq n$ and a monomorphism for $i = n+1$.*
(ii) *Let, in addition, $L^\cdot \in \mathrm{Ob}\, D^+(C)$ satisfy $H^i(L^\cdot) = 0$ for $i < 0$, and let $u : L^\cdot \to K^\cdot$ be a morphism in $D^+(C)$ that induces an injective homomorphism $H^\circ(u) : H^\circ(L^\cdot) \to H^\circ(K^\cdot)$. Then the homomorphism $T^{n+1}(L^\cdot) \to T'^{n+1}(L^\cdot)$*
100 *is surjective if and only if the image of $T'^{n+1}(L^\cdot)$ in $T'^{n+1}(K^\cdot)$ is contained in that of $T^{n+1}(K^\cdot)$. If $C' = (\mathrm{Ab})$ and $\xi \in T'^{n+1}(L^\cdot)$, then ξ lies in the image of $T^{n+1}(L^\cdot)$ if and only if its image in $T'^{n+1}(K^\cdot)$ lies in that of $T^{n+1}(K^\cdot)$.*

Recall that the hypothesis that $T^\cdot$ and $T'^\cdot$ arise from cohomological functors on $D^+(C)$ may be spelled out as follows: they arise from cohomological functors on the

abelian category of bounded-below complexes of C that commute with degree shifts, send morphisms of complexes homotopic to zero to zero morphisms in C', and send every morphism $u : K^\cdot \to L^\cdot$ of complexes that is a quasi-isomorphism (that is, induces isomorphisms $H^i(K^\cdot) \to H^i(L^\cdot)$) to an isomorphism in C'. In practice, all the cohomological functors from C to C' that one encounters arise in this way.

Proof of 6.3. We first prove (i). The assertion is clear if $H^i(K^\cdot) = 0$ for $i \leq n+1$, since the degree properties imposed on $T^\cdot$ and $T'^\cdot$ then give $T^i(K^\cdot) = T'^i(K^\cdot) = 0$ for $i \leq n+1$. On the other hand, let $m \geq 0$ be an integer such that $H^i(K^\cdot) = 0$ for $i < m$; such an integer exists, and for example $m = 0$ will do. We proceed by descending induction on m, the assertion having been proved when $m = n+2$. Thus assume the assertion proved for $m' > m$, and prove it for m. By the hypothesis on the degrees of $K^\cdot$, the complex $K^\cdot$ is quasi-isomorphic to a complex concentrated in degrees $\geq m$, so we may suppose that it is itself concentrated in those degrees. There follows an exact sequence of complexes

$$0 \longrightarrow H^m[-m] \longrightarrow K^\cdot \longrightarrow K'^\cdot \longrightarrow 0,$$

where $H^m = H^m(K^\cdot)$, the notation $[-m]$ denotes a shift of $-m$ in the degrees of a complex, and $K'^\cdot$ is defined as the cokernel. The cohomology objects of $K'^\cdot$ are therefore 101
those of $K^\cdot$ in degrees $\geq m+1$ and are zero in degrees $\leq m$. In particular, the induction hypothesis applies to $K'^\cdot$. The preceding exact sequence gives rise to a morphism of exact sequences

$$\begin{array}{ccccccccc}
T^{i-1}(K'^\cdot) & \longrightarrow & T^{i-m}(H^m) & \longrightarrow & T^i(K^\cdot) & \longrightarrow & T^i(K'^\cdot) & \longrightarrow & T^{i+1-m}(H^m) \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
T'^{i-1}(K'^\cdot) & \longrightarrow & T'^{i-m}(H^m) & \longrightarrow & T'^i(K^\cdot) & \longrightarrow & T'^i(K'^\cdot) & \longrightarrow & T'^{i+1-m}(H^m).
\end{array}$$

Taking $i \leq n$ in this diagram and using the induction hypothesis and $m \geq 0$, we find that the vertical arrows surrounding the middle vertical arrow are isomorphisms, hence in particular epimorphisms, and that the rightmost vertical arrow is a monomorphism. The five lemma therefore shows that the middle vertical arrow is an epimorphism. Taking $i \leq n+1$, we similarly find that the vertical arrows surrounding the middle one are monomorphisms, while the leftmost vertical arrow is an isomorphism, hence in particular an epimorphism. The five lemma then shows that the middle vertical arrow is a monomorphism. This proves 6.3 (i).

To prove the second assertion, 6.3 (ii), we similarly use the exact sequence

$$0 \longrightarrow L^\cdot \longrightarrow K^\cdot \longrightarrow K'^\cdot \longrightarrow 0,$$

where $K'^\cdot$ is defined as the mapping cylinder of $L^\cdot \to K^\cdot$ and again has cohomology in nonnegative degrees. The usual diagram chase in

$$\begin{array}{ccccccccc}
T^n(K^\cdot) & \longrightarrow & T^n(K'^\cdot) & \longrightarrow & T^{n+1}(L^\cdot) & \longrightarrow & T^{n+1}(K^\cdot) & \longrightarrow & T^{n+1}(K'^\cdot) \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
T'^n(K^\cdot) & \longrightarrow & T'^n(K'^\cdot) & \longrightarrow & T'^{n+1}(L^\cdot) & \longrightarrow & T'^{n+1}(K^\cdot) & \longrightarrow & T'^{n+1}(K'^\cdot)
\end{array}$$

gives the last assertion of 6.3, since, by the first part already proved, the first two vertical 102
arrows are isomorphisms and the last two are monomorphisms. This completes the proof of 6.3.

LEMMA 6.4. *Theorem 5.1 holds if f is finite.*

Indeed, this follows at once from VIII 5.5 and 5.8.

PROPOSITION 6.5. *Let $h : Y \to X$ be a quasi-compact and quasi-separated morphism of schemes. In* (i) *below, I denotes a fixed set having at least two elements; in* (ii) *and* (iii), $\mathbf{L}$ *denotes a nonempty subset of the set* $\mathbf{P}$ *of prime numbers; and in* (iii), *n denotes an integer.*

(i) *The following conditions are equivalent:*

a) *For every sheaf of sets F on X, the canonical map*

$$H^\circ(X, F) \longrightarrow H^\circ(Y, h^*(F))$$

is injective (respectively, bijective).

b) *For every finite morphism $X' \to X$, let $h' : Y' \to X'$ be the morphism obtained from h by the base change $X' \to X$. The canonical map*

$$H^\circ(X', I_{X'}) \longrightarrow H^\circ(Y', I_{Y'})$$

induced by h' is injective (respectively, bijective).

b′) *With the notation of* (b), *the map $U \mapsto h'^{-1}(U)$ from the set of subsets of X' that are both open and closed to the set of subsets of Y' that are both open and closed is injective (respectively, bijective).*

103 b″) *(If X is locally Noetherian.) With the notation of* (b), *the induced map on sets of connected components*

$$\Pi_\circ(h') : \Pi_\circ(Y') \longrightarrow \Pi_\circ(X')$$

is surjective (respectively, bijective). Equivalently, if X' is nonempty (respectively, connected and nonempty), then the same is true of Y'.

Moreover, if these conditions hold, then for every sheaf of groups F on X, the functor $P \mapsto h^*(P)$ from the category of torsors (homogeneous principal bundles) under F to the category of torsors under $h^*(F)$ is faithful (respectively, fully faithful). In particular, in the latter case, the canonical map

$$H^1(X, F) \longrightarrow H^1(Y, h^*(F))$$

is injective. Finally, under the same hypotheses, the functor $X' \mapsto Y' = X' \times_X Y$ from the category of étale coverings of X to the category of étale coverings of Y is faithful (respectively, fully faithful).

(ii) *The following conditions are equivalent:*

a) *For every sheaf F of ind-$\mathbf{L}$-finite groups on X, the canonical map*

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

is bijective for $i = 0, 1$.

b) *For every finite morphism $X' \to X$, let $h' : Y' \to X'$ be the morphism obtained from h by base change, and let G be any ordinary finite $\mathbf{L}$-group. The canonical map*

$$H^i(X', G_{X'}) \longrightarrow H^i(Y', G_{Y'})$$

is bijective for $i = 0, 1$.

104 b′) *(If $\mathbf{L} = \mathbf{P}$.) With the notation of* (b), *the inverse-image functor under h' induces an equivalence from the category of étale coverings of X' to the category of étale coverings of Y'.*

b″) *(If X is Noetherian and $\mathbf{L} = \mathbf{P}$.) With the notation of* (b), *if X' is nonempty then so is Y'. If y' is a geometric point of Y' and x' is its image in X', the canonical map*

$$\Pi_i(Y', y') \longrightarrow \Pi_i(X', x') \quad , \quad \text{for } i = 0, 1,$$

is bijective.

(iii) *The following conditions are equivalent:*

a) *For every sheaf F of $\mathbf{L}$-torsion groups on X, the homomorphism*

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n+1$.

a′) *(If $n \geq -1$.) For every F as above, the homomorphism in question is injective for $i = 0$ and surjective for $i \leq n$.*

b) *(If $n \geq -1$.) For every finite morphism $X' \to X$, let $h' : Y' \to X'$ be the morphism obtained from h by the base change $X' \to X$. Then for every $\ell \in \mathbf{L}$ and $\nu > 0$, the canonical homomorphism*

$$H^i(X', (\mathbf{Z}/\ell^\nu\mathbf{Z})_{X'}) \longrightarrow H^i(Y', (\mathbf{Z}/\ell^\nu\mathbf{Z})_{Y'})$$

is injective for $i = 0$ and surjective for $i \leq n$.

In addition, we have the following supplement to the preceding statement:

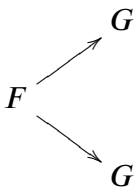
COROLLARY 6.6. (i) *Suppose that the non-parenthesized condition in 6.5 (i) (a) holds. Let F be a sheaf of sets on X, and let $\xi \in H^\circ(Y, h^*(F))$. Suppose that there exists a monomorphism of sheaves of sets $F \to G$ such that the image of ξ in $H^\circ(Y, h^*(G))$ lies in the image of $H^\circ(X, G)$. Then ξ lies in the image of $H^\circ(X, F)$.* 105

(ii) *Suppose that the parenthesized condition in 6.5 (i) (a) holds. Let F be a sheaf of groups on X, and let $\xi \in H^1(Y, h^*(F))$. Suppose that there exists a monomorphism of sheaves of groups $F \to G$ such that the image of ξ in $H^1(Y, h^*(G))$ lies in the image of $H^1(X, G)$. Then ξ lies in the image of $H^1(X, F)$.*

(iii) *Suppose that condition 6.5 (iii) (a) holds for a fixed value of n. Let F be a sheaf of $\mathbf{L}$-torsion groups on X, and let $\xi \in H^{n+1}(Y, h^*(F))$. Suppose that there exists a monomorphism $F \to G$ of sheaves of $\mathbf{L}$-torsion groups such that the image of ξ in $H^{n+1}(Y, h^*(G))$ lies in the image of $H^{n+1}(X, G)$. Then ξ lies in the image of $H^{n+1}(X, F)$.*

Proof of 6.5 and 6.6. We first prove 6.6. The case of 6.6 (iii) is a special case of the last assertion of 6.3, applied when C is the category of abelian $\mathbf{L}$-torsion sheaves on X, C' is the category (Ab), and $T^i(F) = H^i(X, F)$, $T'^i(F) = H^i(Y, h^*(F))$ (note that $T'^\cdot$ is indeed a cohomological functor in F, since the functor h^* is exact). Of course, the use of complexes as in 6.3 is unnecessary here (it will be convenient later, in 6.8); we use only the last assertion of 6.3.

To prove (i), consider the amalgamated sum $H = G \amalg_F G$, the colimit in the category of sheaves of sets on X of the diagram



The usual exactness properties of the topos of sheaves of sets show, since $F \to G$ is a monomorphism, that there is an *exact* diagram of sheaves of sets on X 106

$$F \longrightarrow G \rightrightarrows H$$

(this may be expressed by saying that every monomorphism in a topos is effective, and may be proved by reducing, as usual, to the category of sets). Since the functors H^0 and

h^* are left exact, we obtain a morphism of exact diagrams of sets

$$\begin{array}{ccccc} H^0(X,F) & \longrightarrow & H^0(X,G) & \rightrightarrows & H^0(X,H) \\ \downarrow & & \downarrow & & \downarrow \\ H^0(Y,h^*(F)) & \longrightarrow & H^0(Y,h^*(G)) & \rightrightarrows & H^0(Y,h^*(H)), \end{array}$$

in which the vertical arrows are injective. An immediate diagram chase then proves assertion 6.6 (i).

To prove 6.6 (ii), we first prove the second assertion of 6.5 (i), namely that, under condition a) of 6.5 (i), for every sheaf of groups F on X, the functor $P \mapsto h^*(P)$ from the category of torsors under F to the category of torsors under $h^*(F)$ is faithful (resp. fully faithful), and hence, in the respective case, induces an injection on H^1. For this, if P and P' are two torsors under F, denote by $\mathrm{Isom}_F(P,P')$ the sheaf of F-isomorphisms from P to P'. It is immediate that the formation of this sheaf commutes with every base extension, and in particular with base extension by h. (This assertion is valid for every morphism of sites.) Applying the hypothesis on the effect of h^* on H^0 to the sheaf $\mathrm{Isom}_F(P,P')$, we find that the map

$$\mathrm{Isom}_F(P,P') \longrightarrow \mathrm{Isom}_F(h^*(P),h^*(P'))$$

107 is injective (resp. bijective), which says that the functor $P \mapsto h^*(P)$ is faithful (resp. fully faithful). (Note that every morphism in the category of torsors under a sheaf of groups is an isomorphism.)

On the other hand, note that if F is a subsheaf of groups of a sheaf of groups G, then giving a torsor P under F amounts to giving a torsor Q under G (namely the one obtained from P by extending the structure group along $F \to G$), *together with* a section of Q/F (which is indeed interpreted as a reduction of the structure group from G to F). With the notation of 6.6 (ii), the image of ξ in $H^1(Y,h^*(G))$ comes from an element of $H^1(X,G)$; the latter is represented by a torsor Q under G, so that the image of ξ is the class of the torsor $h^*(Q)$ under $h^*(G)$. The fact that this class comes from $\xi \in H^1(Y,h^*(F))$ means explicitly that ξ is the class of the $h^*(F)$-torsor defined by $h^*(Q)$ and a suitable section of $h^*(Q)/h^*(F)$. Since the functor h^* is right exact, this last sheaf is none other than $h^*(Q/F)$; and since h^* induces a bijection on the H^0 of sheaves of sets, the section in question of $h^*(Q)/h^*(F)$ comes from a section of Q/F. This last section defines an F-torsor by reduction of the structure group, and the class of this torsor in $H^1(X,F)$ plainly has ξ as its image in $H^1(Y,h^*(F))$. This proves 6.6 (ii) and completes the proof of 6.6.

We now prove 6.5, beginning with 6.5 (iii). Clearly a) implies a′), and the converse implication is a special case of 6.2 (again applied when C is the category of $\mathbf{L}$-torsion sheaves on X and C' the category of abelian groups), provided that one proves that the functors $H^i(X,F)$, for $i>0$, are effaceable in C. This presents no difficulty; alter-
108 natively, we can avoid proving that result by noting that F is the filtered colimit of its subsheaves F_i for which there exists an integer $n>0$ annihilating F_i (IX 1.1). Then $h^*(F)$ is the colimit of the $h^*(F_i)$, and, in view of the commutation of cohomology with filtered colimits (VII 3.3), which follows from the quasi-compactness and quasi-separatedness hypotheses on X and Y (hypotheses not used for 6.6), we are reduced to proving condition a) for the F_i. Thus we are reduced to the case in which there exists an integer $m>0$ such that $mF=0$. We may plainly suppose that every prime divisor of m belongs to $\mathbf{L}$, and can then take for C the category of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X. In this

category it is known that the functors $H^i(X, -)$ are effaceable for $i \geq 1$. This proves the equivalence of conditions a) and a′) in 6.5 (iii).

On the other hand, apply a) to the direct image of a **L**-torsion sheaf F' on a scheme X' finite over X; this direct image is a **L**-torsion sheaf by IX 1.2 (v) and IX 1.6 (iii). Using 6.4 and VIII 5.5, we find that the homomorphism

$$H^i(X', F') \longrightarrow H^i(Y', h'^*(F'))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$, whence, a fortiori, condition b). It remains to prove that this condition implies a′). For this, use the fact (IX 2.7.2) that every **L**-torsion sheaf F on X is the filtered colimit of constructible **L**-torsion sheaves. The preceding colimit argument then shows that, in a′), it suffices to consider constructible F. By IX 2.14, there then exist finitely many finite morphisms $p_i : X'_i \to X$, constant **L**-torsion sheaves $G_{X'_i}$ on the X'_i, and an injective homomorphism

$$F \longrightarrow \prod_i p_{i*}(G_{X'_i}).$$

Now if $n = -1$, that is, if the desired conclusion reduces to the assertion that 109

$$H^0(X, F) \longrightarrow H^0(Y, h^*(F))$$

is injective, then in this question we may at once replace F by a sheaf G into which F embeds, and hence may replace F by the product of the $p_{i*}(G_{X'_i})$. In view of 6.4, we are thus reduced to proving the injectivity assertion after replacing $h : Y \to X$ by $h' : Y' \to X'$ and F by a constant sheaf on X' whose value is an ordinary finite **L**-torsion group G. But G is isomorphic to a direct sum of groups of the form $\mathbf{Z}/\ell^\nu\mathbf{Z}$, with $\ell \in \mathbf{L}$. We are therefore indeed reduced to verifying b).

When $n \geq 0$, we proceed similarly. By induction on n, we may suppose that the desired conclusion (and hence a)) has been proved for $n - 1$ in place of n. Applying 6.6 (iii), we see that we may again replace F by a sheaf G into which F embeds, and we again take $G = p_*(G_{X'})$. In view of 6.4 and the vanishing of $R^i p_*$ for $i \geq 1$ when p is finite (VIII 5.5), we are again reduced to proving surjectivity for H^n after replacing $h : Y \to X$ by $h' : Y' \to X'$ and F by a constant sheaf on X' whose value is an ordinary finite **L**-torsion group. Injectivity is already known by the induction hypothesis and the implication a′) ⇒ a); by devissage on G, we are thus again reduced to the case in which G is of the form $\mathbf{Z}/\ell^\nu\mathbf{Z}$, namely the case considered in condition b) of 6.5 (iii). This proves case (iii) of 6.5.

Cases (i) and (ii) are proved in exactly the same way as regards the equivalence of conditions a) and b) in 6.5 (i) resp. 6.5 (ii): one uses IX 2.7.2 and the commutation of 110
H^0 and H^1 with filtered colimits of sheaves (VII 5.14) to reduce to the case in which F is constructible, and then uses IX 2.14, 6.6 (i), (ii), and VIII 5.8 to reduce to the special case considered in b). Moreover, in (i), the equivalence of conditions b), b′), and b″) is trivial and is recorded only for reference, as is the fact that they imply the faithfulness (resp. full faithfulness) of the functor $X' \mapsto Y'$ from the category of etale coverings of X to the category of etale coverings of Y. The same is true of the equivalence of conditions b), b′), and b″) in 6.5 (ii), which were added for good measure. The proof of 6.5 is now complete.

Remark 6.7. [34]

[34] The editor recommends omitting these remarks, as well as 6.13; they were surreptitiously introduced (together with various other more or less felicitous changes to the original text) by an irreverent collaborator.

1) The arguments proving 6.5 and 6.6 are plainly very general and essentially trivial, and it would be useful to extract them as abstract lemmas of the same kind as 6.2 (with the sheaves $p_*(G_{X'})$ playing the role of cogenerators of the category C in which one works). We leave this pleasant exercise to the reader, merely pointing out that essentially the same statement could be given for functors such as $R^i f_*$ in place of the functors H^i. The same remarks apply to 6.8 and 6.11 below.
2) When X is noetherian, the arguments given show that in conditions b) and their variants in 6.5 (i), (ii), and (iii), one may restrict to *integral* X'; when X is universally Japanese, one may even take them to be integral and normal. This
111 is also possible without the Japanese hypothesis on X, provided that *integral* morphisms $p : X' \to X$ are used in place of finite morphisms. None of these variants will be used later in the seminar.
3) The proof of 6.6 does not use the quasi-compactness and quasi-separatedness hypothesis on X. The same is true of the fact that in (i), (ii), and (iii), condition a) implies the other conditions, and of the mutual equivalence of conditions b), b′), and b″). Note also that the form b) of the conditions in 6.5 (i) shows that, for $\mathbf{L} \neq \phi$, they are special cases (when X is quasi-compact and quasi-separated) of 6.5 (iii), for $n = -1$ resp. $n = 0$. This remains true, moreover (if one restricts to the conditions in form a)), without the stated restriction on X. Indeed, for every sheaf of sets F and every prime number ℓ, one can find two abelian ℓ-torsion sheaves G and H, and a homomorphism $u : G \to H$ of the underlying sheaves of sets, such that F is isomorphic to the inverse image of the zero section (one takes for G the free $\mathbf{Z}/\ell\mathbf{Z}$-module generated by F, and for H the constant sheaf $\mathbf{Z}/\ell\mathbf{Z}$ on X). The same argument shows that (without any quasi-compactness or quasi-separatedness restriction) 6.5 (ii) a) (where it even suffices to take $i = 0$) implies the respective condition 6.5 (i) a). Since the latter already implies the injectivity of $H^1(X, F) \to H^1(Y, h^*(F))$ for every sheaf of groups F on X, in the statement of 6.5 (ii) a) it therefore suffices, for $i = 1$, to require the *surjectivity* of $H^1(X, F) \to H^1(Y, h^*(F))$. Unfortunately, 6.5 (ii) cannot be regarded as a special case of 6.5 (iii), which often forces us to repeat in the noncommutative case an argument already carried out in essence in the commutative case. Note also that, apart from 6.10, the results that follow are stated so as to remain valid
112 without any quasi-compactness or quasi-separatedness hypothesis.

PROPOSITION 6.8. *(Descent lemma). With the general notation of 6.5, suppose in addition that a morphism $p : \overline{X} \to X$ is given. Let $\overline{h} : \overline{Y} \to \overline{X}$ denote the morphism obtained from h by the base change p, and let $p_Y : \overline{Y} \to Y$ denote the canonical morphism. We suppose either that p is surjective, or that h is a closed immersion and $X = p(\overline{X}) \cup h(Y)$.*

(i) *Suppose that $\overline{h} : \overline{Y} \to \overline{X}$ satisfies the injectivity (resp. bijectivity) condition of 6.5* (i) a). *In the respective case, suppose in addition that the base-change morphism*

$$(*) \qquad h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F}))$$

is injective for every sheaf of sets $\overline{F}$ on $\overline{X}$. Then the morphism $h : Y \to X$ also satisfies the injectivity (resp. bijectivity) condition of 6.5 (i) a).

(ii) *Suppose that $\overline{h} : \overline{Y} \to \overline{X}$ satisfies condition 6.5* (ii) a), *and that for every sheaf of ind-$\mathbf{L}$-groups $\overline{F}$ on $\overline{X}$ the base-change morphism* (∗) *is bijective. Then $h : Y \to X$ also satisfies condition 6.5* (ii) a).

(iii) *Suppose that* $\overline{h} : \overline{Y} \to \overline{X}$ *satisfies condition* 6.5 (iii) a)*, and that for every abelian* $\mathbf{L}$*-torsion sheaf* $\overline{F}$ *on* $\overline{X}$ *the base-change homomorphism*

$$h^*(R^i p_*(\overline{F})) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is bijective for $i \leq n-1$ *and injective for* $i = n$*, where* $n \geq 1$ *is a given integer. Then* $h : Y \to X$ *also satisfies condition* 6.5 (iii) a).

Proof of 6.8. In the case where p is not assumed surjective, but h is a closed immersion and $X = p(\overline{X}) \cup h(Y)$, consider the disjoint-union scheme $\overline{X}'$ of $\overline{X}$ and Y, and the morphism $p' : \overline{X}' \to X$ induced by p and h. Then p' is surjective; moreover, in each of the three cases (i), (ii), and (iii), it is immediate that the hypotheses on the pair (h, p) remain satisfied for the pair (h, p'). We are therefore reduced to the case in which p is surjective, which we assume from now on. In this case, for every sheaf F on X, the canonical homomorphism

$$F \longrightarrow p_*(p^*(F))$$

is injective. In case (i), to verify the injectivity of $H^0(X, F) \to H^0(Y, h^*(F))$, we are immediately reduced to the case in which F is replaced by $p_*(\overline{F})$, where $\overline{F} = p^*(F)$. Consider the commutative diagram

$$\begin{array}{ccc} H^0(X, p_*(\overline{F})) & \longrightarrow & H^0(Y, h^*(p_*(\overline{F}))) \\ \downarrow \wr & & \downarrow \\ & & H^0(Y, p_{Y*}(\overline{h}^*(\overline{F}))) \\ & & \downarrow \wr \\ H^0(\overline{X}, \overline{F}) & \longrightarrow & H^0(\overline{Y}, \overline{h}^*(\overline{F})), \end{array}$$

where the left vertical arrow and the second right vertical arrow are the canonical isomorphisms, and the first right vertical arrow is induced by the base-change homomorphism for h. By hypothesis the second horizontal arrow is injective, and it follows immediately that the first is as well; this proves the non-respective assertion of (i). For the respective assertion, suppose that the second horizontal arrow is surjective. Since the base-change homomorphism is a monomorphism by hypothesis, the first right vertical arrow is a monomorphism, and hence so is the composite of the two right vertical arrows. It follows immediately that the first horizontal arrow is also surjective, completing the proof of (i).

To prove (ii), what precedes and 6.5 (i) reduce us to proving the surjectivity of $H^1(X, F) \to H^1(Y, h^*(F))$. Applying 6.6 (ii), we are further reduced to proving it for F of the form $p_*(\overline{F})$, where $\overline{F}$ is a sheaf of ind-$\mathbf{L}$-groups on $\overline{X}$ (IX 1.6 (ii)). Writing the exact sequences for H^1 (variants of 3.2) for the morphisms p and p_Y, we obtain a morphism of noncommutative exact sequences:

$$\begin{array}{ccccc} H^1(X, p_*(\overline{F})) & \longrightarrow & H^1(\overline{X}, \overline{F}) & \longrightarrow & H^0(X, R^1 p_*(\overline{F})) \\ \downarrow & & \downarrow & & \downarrow \\ H^1(Y, p_{Y*}(\overline{h}^*(\overline{F}))) & \longrightarrow & H^1(\overline{Y}, \overline{h}^*(\overline{F})) & \longrightarrow & H^0(Y, R^1 p_{Y*}(\overline{h}^*(\overline{F}))), \end{array}$$

where in each row the first arrow is injective and identifies the first term with the inverse image, under the second arrow, of the distinguished point of the third term. By the hypothesis on $\overline{h}$, the middle vertical arrow is bijective. The two outer vertical arrows factor, respectively, through

$$H^1(Y, h^*(p_*(\overline{F}))) \quad \text{and} \quad H^0(Y, h^*(R^1p_*(\overline{F}))),$$

by composing the homomorphisms $H^i(X,-) \to H^i(Y, h^*(-))$ (for $i = 1$ and $i = 0$, respectively) with the homomorphisms obtained by applying $H^i(Y,-)$ (for those same values of i) to the base-change homomorphisms

$$h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F}))$$

and, respectively,

$$h^*(R^1p_*(\overline{F})) \longrightarrow R^1p_{Y*}(\overline{h}^*(\overline{F})).$$

Since the first of these arrows is an isomorphism by hypothesis, the first vertical arrow in the diagram above identifies with the homomorphism $H^1(X,-) \to H^1(Y, h^*(-))$ that we wish to study. To prove that it is bijective, it therefore remains to prove that the last vertical arrow in the diagram is injective. By what we have just said, it is the composite of
115 two maps, the first of which is a canonical homomorphism $H^0(X,-) \to H^0(Y, h^*(-))$, and hence is bijective, while the second is injective because it is obtained by applying the left exact functor $H^0(Y,-)$ to the monomorphism

$$h^*(R^1p_*(\overline{F})) \longrightarrow R^1p_{Y*}(\overline{h}^*(\overline{F})).$$

The latter fact follows readily from the bijectivity hypothesis for the base-change homomorphism in the preceding dimension 0, for example by using the second assertion of 6.5 (i) and the usual calculation of the fibers of the higher direct images (VIII 5.3). This proves 6.8 (ii).

It remains to prove 6.8 (iii). For this, rather than using the injection $F \to p_*(p^*(F))$ directly as above, it is more convenient to use the homomorphism

$$F \longrightarrow Rp_*(\overline{F})$$

in the right derived category $D^+(C)$, where C denotes the category of abelian sheaves on X. Recall that, by definition, $Rp_*(\overline{F})$ is the complex $p_*(C(\overline{F}))$, where $C(\overline{F})$ is an injective resolution of $\overline{F}$ in the category of abelian sheaves on $\overline{X}$. If $K^\cdot$ is a complex on X, we denote by $H^i(X, K^\cdot)$ the hypercohomology groups of X with coefficients in $K^\cdot$, and use analogous notation on Y. By 6.3, we are reduced to proving the surjectivity of

$$H^n(X, Rp_*(\overline{F})) \longrightarrow H^n(Y, h^*(Rp_*(\overline{F}))). \tag{1}$$

By hypothesis, the base-change homomorphism

$$h^*(Rp_*(\overline{F})) \longrightarrow Rp_{Y*}(\overline{h}^*(\overline{F})) \tag{2}$$

induces an isomorphism on the cohomology sheaves in degrees $\leq n-1$ and a monomor-
116 phism in degree n. Equivalently, the "mapping cylinder" $K^\cdot$ of the preceding homomorphism has nonzero cohomology sheaves only in degrees $\geq n$. Hence $H^i(Y, K^\cdot) = 0$ for $i < n$, and the cohomology exact sequence shows that the homomorphisms

$$H^i(Y, h^*(Rp_*(\overline{F}))) \longrightarrow H^i(Y, Rp_{Y*}(\overline{h}^*(\overline{F}))) \tag{3}$$

induced by the homomorphism above are isomorphisms for $i \leq n-1$ and a monomorphism for $i = n$. Consider the composite of (1) and (3):

$$H^n(X, Rp_*(\overline{F})) \longrightarrow H^n(Y, h^*(Rp_*(\overline{F}))) \longrightarrow H^n(Y, Rp_{Y*}(\overline{h}^*(\overline{F}))) \tag{4}$$

The two outer terms are respectively isomorphic, by definition, to $H^n(\overline{X}, \overline{F})$ and $H^n(\overline{Y}, \overline{h}^*(\overline{F}))$, and the composite in (4) is precisely the homomorphism induced by $\overline{h}$, which is bijective by the hypothesis on $\overline{h}$. Since the second arrow in (4) is injective by what we have just seen, the first is also bijective. This completes the proof of 6.8.

Moreover, the preceding proof immediately gives the following slightly more precise and more general result:

COROLLARY 6.9. *The notation is that of 6.8. Suppose in addition that h satisfies the unparenthesized condition in 6.5* (i) *(respectively, the parenthesized condition in 6.5* (i)*, respectively, the condition in 6.5* (iii)*, with $n \geq -1$), and that, for every sheaf of sets $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F}))$$

is injective (respectively, bijective; respectively, that for every sheaf of **L***-torsion groups $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^* R^i p_*(\overline{F}) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

*is bijective for $i \leq n$ and injective for $i = n+1$). Let F be a sheaf of sets (respectively, of ind-***L***-groups, respectively, a sheaf of* **L***-torsion groups) on X, and let $\xi \in H^{n+1}(Y, h^*(F))$ (where $n = -1$ in case* (i) *and $n = 0$ in case* (ii)*). Put $\overline{F} = p^*(F)$. Then ξ lies in the image* 117
of $H^{n+1}(X, F)$ if and only if its inverse image in

$$H^{n+1}(\overline{Y}, p_Y^*(h^*(F))) = H^{n+1}(\overline{Y}, \overline{h}^*(\overline{F}))$$

lies in the image of $H^{n+1}(\overline{X}, \overline{F})$.

We shall use this sharpened form of 6.8 to prove the following result.

COROLLARY 6.10. *The notation is that of 6.5. For simplicity, suppose that the unparenthesized condition in 6.5* (i) *holds.*

(i) *The parenthesized condition in 6.5* (i) *holds (respectively, condition 6.5* (ii) a) *holds, respectively, condition 6.5* (iii) a) *holds) if and only if the following condition is satisfied. For every sheaf F of sets (respectively, of ind-***L***-groups, respectively, of abelian* **L***-torsion groups) on X, every $\xi \in H^\circ(Y, h^*(F))$ (respectively, every $\xi \in H^i(Y, h^*(F))$ with $i = 0, 1$, respectively, every $\xi \in H^i(Y, h^*(F))$ with $i \leq n$), and every nonempty closed subset X' of X, if Y' denotes its inverse image in Y, there exists a morphism $p : \overline{X} \to X$ satisfying the following conditions:*

  1°) *The image $p(\overline{X})$ is contained in X' and contains a nonempty open subset of X'.*

  2°) *For every sheaf of sets $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F}))$$

  *is an isomorphism (respectively, for every sheaf of ind-***L***-groups $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^*(R^i p_*(\overline{F})) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is bijective for $i = 0, 1$; respectively, for every sheaf of abelian **L***-torsion groups $\overline{F}$ on $\overline{X}$, the base-change homomorphism is bijective for $i \leq n$).*

3°) *The inverse image of ξ in*

$$H^i(\overline{Y}, p_Y^*(h^*(F))) = H^i(\overline{Y}, \overline{h}^*(p^*(F)))$$

lies in the image of $H^i(\overline{X}, p^*(F))$.

118 (ii) *Let $n \geq -1$, and suppose that, for every sheaf of sets F (respectively, every sheaf of ind-***L***-groups, respectively, every sheaf of abelian* **L***-torsion groups) on X, the homomorphism*

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

is bijective for $i \leq n$ and injective for $i = n + 1$ (in the first case we take $n = -1$, and in the second case $n = -1$ or 0). Let $\xi \in H^{n+1}(Y, h^*(F))$. Then ξ lies in the image of $H^{n+1}(X, F)$ if and only if it satisfies the condition stated in (i) *above, with $i = n + 1$.*

Proof of 6.10. Using 6.5 and an immediate induction on n, one sees that 6.10 (i) follows from 6.10 (ii), which we now prove.

Suppose that ξ does not lie in the image of $H^{n+1}(X, F)$. Let $\boldsymbol{\Phi}$ be the set of closed subsets X' of X such that the image of ξ in $H^{n+1}(Y', h^*(F)|Y')$ does not lie in the image of $H^{n+1}(X', F|X')$. By hypothesis, $X \in \boldsymbol{\Phi}$, so $\boldsymbol{\Phi}$ is nonempty. Order $\boldsymbol{\Phi}$ by the relation $\supset$, and let us show that $\boldsymbol{\Phi}$ is inductive. It is enough to prove that, if (X'_λ) is a totally ordered family of elements of $\boldsymbol{\Phi}$ and X' is their intersection, then $X' \in \boldsymbol{\Phi}$. Give the X'_λ and X' their induced reduced structures. The morphisms $X'_\lambda \to X$ are affine (being closed immersions), and hence the inverse system of the X'_λ satisfies the hypotheses considered in VII 5. Moreover, one immediately sees that X' is the inverse limit of the X'_λ. Likewise, Y' is the inverse limit of the inverse system of the Y'_λ, which are affine over Y. By VII 5.8, therefore,

$$\varinjlim H^{n+1}(X'_\lambda, F|X'_\lambda) \xrightarrow{\sim} H^{n+1}(X', F|X') \quad , \quad \varinjlim H^{n+1}(Y'_\lambda, h^*(F)|Y'_\lambda) \xrightarrow{\sim} H^{n+1}(Y', h^*(F)|Y').$$

119 It follows at once that, if $(\xi|Y') \in \operatorname{Im} H^{n+1}(X', F|X')$, then there would be an index λ such that $(\xi|Y'_\lambda) \in \operatorname{Im} H^{n+1}(X'_\lambda, F|X'_\lambda)$, which is absurd. Thus $X' \in \boldsymbol{\Phi}$, as asserted. Consequently $\boldsymbol{\Phi}$ is inductive and therefore contains an inclusion-minimal element, say X'. Since $\varnothing \notin \boldsymbol{\Phi}$, the set X' is nonempty. Apply the hypothesis of 6.10 to ξ and X'. This gives a morphism $p : \overline{X} \to X$ satisfying conditions 1°)– 3°) in 6.10. Observe that we may even suppose that p is *surjective onto X'*. Indeed, let U be a nonempty open subset of X' contained in $p(\overline{X})$, and let X'_1 be its complement in X'. By construction, $X'_1 \notin \boldsymbol{\Phi}$. Put $\overline{X}_1 = \overline{X} \amalg X'_1$, and let $p_1 : \overline{X}_1 \to X$ be the morphism defined by p and the inclusion $X'_1 \hookrightarrow X$. Using the relation $X'_1 \notin \boldsymbol{\Phi}$, one sees immediately that p_1 still satisfies the same hypotheses as p; moreover, it is surjective onto X'. We may therefore suppose that p is surjective onto X'. Now apply 6.9, replacing $h : Y \to X$ there by $h' : Y' \to X'$ (with X' endowed, say, with its induced reduced structure), and replacing ξ by $\xi|Y'$. It follows from 6.9 that $(\xi|Y') \in \operatorname{Im} H^{n+1}(X', F|X')$, contrary to $X' \in \boldsymbol{\Phi}$. This completes the proof of 6.10.

We shall use 6.10 to reduce 5.5 to the case in which f is projective. In the non-Noetherian case, however, this requires a non-Noetherian variant of Chow's lemma, which will be given in the next section.

PROPOSITION 6.11. *(Transitivity lemma). The notation is that of 6.8, but no surjectivity hypothesis is imposed on p. Suppose that h satisfies condition* (i) a) *(respectively,* (ii) a), *respectively,* (iii) a)*) of 6.5. In case* (i), *suppose in addition that, for every sheaf of sets* $\overline{F}$ *on* $\overline{X}$, *the base-change homomorphism of 6.8 is a monomorphism (respectively, an isomorphism); in case* (ii), *suppose that* $\overline{X}$ *is Noetherian and that, for every sheaf of ind-***L***-groups* $\overline{F}$, *the base-change homomorphism*

$$h^*(R^i p_*(\overline{F})) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is bijective for $i = 0, 1$*; and finally, in case* (iii), *suppose that the base-change homomorphism is bijective for* $i \leq n$ *and injective for* $i = n+1$. *Then* $\overline{h} : \overline{Y} \to \overline{X}$ *also satisfies condition* (i) *(respectively,* (ii), *respectively,* (iii)*) of 6.5.* 120

Proof of 6.11. In case (i), consider the commutative diagram in the proof of 6.8 (i). Here the hypotheses imply that the first horizontal arrow and the upper right vertical arrow are injective (respectively, bijective), and hence so is the second horizontal arrow, as required. In case (ii), let $\overline{F}$ be a sheaf of ind-**L**-groups on $\overline{X}$. We must show that every element of $H^1(\overline{Y}, \overline{h}^*(\overline{F}))$ comes from an element of $H^1(\overline{X}, \overline{F})$. By 6.6 (ii), applied to $\overline{h}$, it is enough to find a monomorphism $\overline{F} \to \overline{G}$ of sheaves of groups on $\overline{X}$ that *effaces* $H^1(\overline{Y}, \overline{h}^*(\overline{F}))$. Consider the second row of the diagram used in the proof of 6.8 (ii). It is enough, successively, to find a monomorphism $\overline{F} \to \overline{G}$, with $\overline{G}$ a sheaf of ind-**L**-finite groups, that effaces the last term $H^\circ(Y, R^1 p_{Y*}(\overline{h}^*(\overline{F})))$, and then a monomorphism of sheaves of groups $\overline{G} \to \overline{H}$ that effaces $H^1(Y, p_{Y*}(\overline{h}^*(\overline{G})))$. As observed in the proof of 6.8 (i), the two terms to be effaced are respectively isomorphic, by the outer vertical arrows of the diagram in question, to $H^\circ(X, R^1 p_*(\overline{F}))$ and $H^1(X, p_*(\overline{F}))$, in view of the hypotheses made here on the base-change homomorphisms for p and h. These isomorphisms are plainly functorial in $\overline{F}$, so it is enough to efface the latter two terms. By 3.3, one can efface $R^1 p_*(\overline{F})$ (and, a fortiori, $H^\circ(X, R^1 p_*(\overline{F}))$) by embedding $\overline{F}$ into an ind-**L**-group **G**. It remains to efface $H^1(X, p_*(\overline{F}))$. This is possible because, by the diagram in question, the latter term maps into $H^1(\overline{X}, \overline{F})$ by a monomorphism functorial in $\overline{F}$, while $H^1(\overline{X}, \overline{F})$ is again effaceable by 3.3. 121

It remains to treat case (iii), and here again we read the proof of 6.8 backwards[34]. We must prove that the homomorphism

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{Y}, \overline{h}^*(\overline{F}))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n+1$. These homomorphisms are induced by a homomorphism of complexes

$$\mathbf{R}\Gamma_{\overline{X}}(\overline{F}) \longrightarrow \mathbf{R}\Gamma_{\overline{Y}}(\overline{h}^*(\overline{F})),$$

obtained from an injective resolution $C(\overline{F})$ and a homomorphism from $\overline{h}^*(C(\overline{F}))$ (which is a resolution of $\overline{h}^*(\overline{F})$) to an injective resolution of $\overline{h}^*(\overline{F})$. The homomorphism above also identifies with the composite

$$\mathbf{R}\Gamma_X(Rp_*(\overline{F})) \longrightarrow \mathbf{R}\Gamma_Y(h^*(Rp_*(\overline{F}))) \longrightarrow \mathbf{R}\Gamma_Y(Rp_{Y*}(\overline{h}^*(\overline{F}))),$$

[34] Instead of 6.3, one could also use the Leray spectral sequence for p and p_Y.

where the first homomorphism is the homomorphism induced by h^* for $Rp_*(\overline{F})$, and the second is obtained by applying $\mathbf{R}\Gamma_Y$ to the base-change homomorphism

$$h^*(Rp_*(\overline{F})) \longrightarrow Rp_{Y*}(\overline{h}^*(\overline{F})).$$

By the hypothesis on h and 6.3 (i), the first homomorphism therefore induces isomorphisms on the cohomology groups of the complexes in question in the required range. The same is true of the second by our hypothesis on the base-change homomorphisms, using the argument given in the proof of 6.8 (iii). The same conclusion thus holds for their composite, completing the proof of 6.11.

122 Remark 6.12. The argument given for 6.11 in cases (i) and (iii) gives the following more precise result. In order to conclude, for a *given* sheaf $\overline{F}$ on $\overline{X}$, that

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{Y}, \overline{h}^*(\overline{F}))$$

is bijective for $i \leq n$ and injective for $i = n+1$, it is enough to assume that

1°) the base-change homomorphism

$$h^*(R^ip_*(\overline{F})) \longrightarrow R^ip_{Y*}(\overline{h}^*(\overline{F}))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n+1$, and

2°) for every pair of integers $i, j \geq 0$, the homomorphism

$$H^i(X, R^jp_*(\overline{F})) \longrightarrow H^i(Y, h^*(R^jp_*(\overline{F})))$$

induced by h is an isomorphism if $i+j \leq n$ and a monomorphism if $i+j = n+1$.

There is plausibly an analogous result in case (ii), but its statement would necessarily involve noncommutative 2-cohomology. To avoid having to invoke that theory, in 6.11 (ii) we used a different argument, based on the effaceability properties of noncommutative H^1; unlike the argument above, however, it yields no result for a fixed $\overline{F}$.

Remarks 6.13. The most important case in which the conditions of 6.5 (i), (ii), and (iii) hold is that considered in 5.5; it will be proved in the next exposé using the reductions made in the present one. Another interesting case (in which h is no longer a closed immersion as in 5.5) will be studied in XV. Let us also mention the closely related case in which X is the spectrum of a Noetherian ring A, separated and complete for the topology defined by an ideal J, and $Y = \mathrm{Spec}(A/J)$ is the closed subscheme of X defined by J.
123 One then verifies condition 6.5 (ii) directly, in the form b") (with $\mathbf{L} = \mathbf{P}$), by EGA IV 18.3.2. It is very plausible that the conditions of 6.5 (iii) also hold for every n; that is, for every torsion sheaf F on X, the homomorphisms

$$H^n(X, F) \longrightarrow H^n(Y, F|Y)$$

are bijective. One can formulate considerably more general conjectures, related to a generalization of the construction of henselization to nonlocal schemes. First, no example is known in which h is a closed immersion and the conditions of 6.5 (ii) and (iii) (with $\mathbf{L} = \mathbf{P}$) fail when the parenthesized condition in 6.5 (i) holds, that is, when (X, Y) is a *henselian pair* in the terminology of EGA IV 18.5.5. One may therefore conjecture that every such pair satisfies the conditions of 6.5 (ii) and (iii) (with $\mathbf{L} = \mathbf{P}$, for every n).

On the other hand, one may ask under what conditions the following property, for a pair (X, Y) consisting of a quasi-compact and quasi-separated scheme X and a closed subscheme Y, implies that the pair is henselian:

(∗) For every scheme X' étale over X, put $Y' = X' \times_X Y$. The canonical map

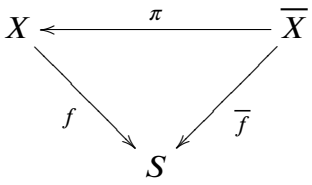
$$\Gamma(X'/X) \longrightarrow \Gamma(Y'/Y)$$

is bijective (cf. EGA IV 18.5.4 b)). This condition is plainly weaker *a priori* than the henselian condition 6.5 (i), and is in fact strictly weaker: it holds, for example, when X is a normal irreducible projective scheme of dimension ≥ 2 over a field k and Y is a hyperplane section of X (as follows readily from SGA 2 XII 2.1, taking $S = \mathrm{Spec}(k)$), whereas the pair (X, Y) is plainly not henselian. It is possible, however, that when X is affine, condition $(*)$ implies that (X, Y) is a henselian pair. If so, for every affine scheme X and every closed subset Y of X, a process of étale localization along Y, analogous to 124
the henselization of local rings, ought to associate to it a *henselian* pair (X', Y), where X' is "pro-étale" over X. Another hypothesis on (X, Y) (suggested by results of Hironaka and Rossi on the contractibility of certain subvarieties of algebraic varieties) under which the henselian property might perhaps be deduced from the weaker property $(*)$ is the following: the immersion $Y \to X$ is regular of codimension 1, and the conormal sheaf $\mathbf{N}_{Y/X} = \mathbf{J}/\mathbf{J}^2$ (where $\mathbf{J}$ is the ideal defining Y in X) is an *ample* invertible sheaf on Y. (In the counterexample considered above, by contrast, this conormal sheaf was anti-ample, that is, its inverse was ample.)

7. A variant of Chow's lemma (*).

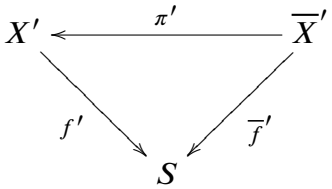
We shall need the following variant of (EGA II 5.6.1), stripped of all Noetherian and irreducibility hypotheses.

LEMMA 7.1. *Let S be a quasi-compact and quasi-separated scheme, and let $f : X \to S$ be a separated morphism of finite type, with X nonempty. There exists a commutative diagram*



with π projective and $\overline{f}$ quasi-projective, and a nonempty open subset U of X, such that 125
π induces an isomorphism from $\overline{U} = U \times_X \overline{X} = \pi^{-1}(U)$ onto U.

Proof. We first make the following observation. Let $i : X' \to X$ be a closed S-subscheme of X, and let $V \subset X'$ be a subscheme that is open in X and dense in X'. If the lemma holds for X', then it holds for X. Indeed, let



be a diagram as in the statement, and let $U' \neq \emptyset$ be an open subset of X' such that π' induces an isomorphism $\overline{U}' = \pi'^{-1}(U') \to U'$. Since V is dense in X', we have $U' \cap V \neq \emptyset$, and we may therefore replace U' by this open subset. Then U' is open in X, and we set $\pi = i\pi'$, $\overline{f} = \overline{f}'$, and $U = U'$.

Now X admits a finite covering by nonempty affine open subsets U_k, $k = 1, \ldots, n$. The hypotheses on S imply that every $U_k \to S$ is quasi-affine, and hence, a fortiori,

[34] A reader interested in the statements of Section 5 only in the Noetherian case may omit the present section.

quasi-compact. We prove the lemma for every X by induction on n. If the intersection U of the U_k is dense in X, we may copy the proof of (EGA II 5.6, B, C, D), with the necessary changes in notation. If the intersection is nonempty, we reduce to the case in which it is dense by replacing X by the closure X' of U (as in the observation above), and U_k by $U'_k = U_k \cap X'$. Finally, suppose that U is empty, and choose r, $1 \leq r < n$, such that

$$V = \bigcap_{k=1}^{r} U_k \neq \emptyset \qquad \text{but} \qquad V \cap U_{r+1} = \emptyset;$$

this is plainly possible since $U_1 \neq \emptyset$. We may now replace X by the closure X' of V and each U_k by $U_k \cap X'$, discarding the empty members. Since $U_{r+1} \cap X' = \emptyset$ (it is open in X' and disjoint from the dense open V), the result follows from the induction hypothesis.

8. Final reductions

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LEMMA 8.1. *To prove any one of the assertions* 5.1 (i), (ii), (iii)*, it is enough to do so under the assumptions that f is* projective *and S is Noetherian.*

Suppose first that 5.1 has been proved when f is projective; we show how the same assertion follows for arbitrary f. Using 6.1, we are reduced to the situation of 5.5, with S strictly local. We now apply 6.10, taking $Y = X_\circ$. Given (ξ, X') as in 6.10, we apply Chow's lemma 7.1 to find an S-morphism $p : \overline{X} \to X'$, with $\overline{X}$ projective over S, such that $p(\overline{X})$ contains a nonempty open subset of X' (condition 1° of 6.10). Since p is projective, the base-change homomorphisms for (p, h) are isomorphisms by hypothesis (condition 2° of 6.10). Finally, since $\overline{X}$ is projective over S, the inclusion $\overline{Y} = \overline{X}_\circ \to \overline{X}$ induces, by hypothesis, an isomorphism on cohomology groups

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{X}_\circ, \overline{h}^*(\overline{F})),$$

which is condition 3° of 6.10. Thus all the conditions of 6.10 are satisfied, and the conclusion follows.

We next show that, in order to verify any one of the three assertions 5.1 for every situation (f, F, g) with f projective, we may further assume that S is Noetherian. By 6.1, we may suppose that S is strictly local and that $S' \to S$ is the inclusion of the closed point; we are thus reduced to verifying 5.5 with S strictly local. Moreover, since X embeds into the projective scheme $\mathbf{P}^r_S$, compatibility 4.4 (ii) allows us to replace X by $\mathbf{P}^r_S$ and F by its direct image on $\mathbf{P}^r_S$. We are therefore reduced to the case $X = \mathbf{P}^r_S$.

Write $S = \mathrm{Spec}(A)$. The ring A is the filtered direct limit of Noetherian Henselian local subrings A_i for which the inclusions $A_i \to A$ are local: one may take the strict henselizations, at the prime ideals induced by the maximal ideal of A, of the finitely generated $\mathbf{Z}$-subalgebras of A. Under these conditions, the residue field k of A is likewise the filtered direct limit of the residue fields k_i of the A_i (EGA IV 5.13.1). Set $S_i = \mathrm{Spec}(A_i)$ and $X_i = \mathbf{P}^r_{S_i}$. Suppose that a direct system (F_i) of sheaves of the type occurring in the assertion under consideration is given on the X_i, as in VII 5.7, and let F be the sheaf it defines on X. Let $F_{i\circ}$ (respectively, $F_\circ$) be the sheaf induced by F_i on the fiber $X_{i\circ}$ over the closed point s_i of S_i (respectively, by F on the fiber $X_\circ$ over the closed point s of S).

We then have a commutative diagram

$$\begin{array}{ccc} \varinjlim_i H^n(X_i, F_i) & \longrightarrow & H^n(X, F) \\ \downarrow & & \downarrow \\ \varinjlim_i H^n(X_{i\circ}, F_{i\circ}) & \longrightarrow & H^n(X_\circ, F_\circ), \end{array}$$

whose two horizontal arrows are isomorphisms by VII 5.7. Consequently, to prove that the second vertical arrow is an isomorphism, it is enough to prove this for the first one; hence it is enough to prove that the homomorphisms

$$H^n(X_i, F_i) \longrightarrow H^n(X_{i\circ}, F_{i\circ})$$

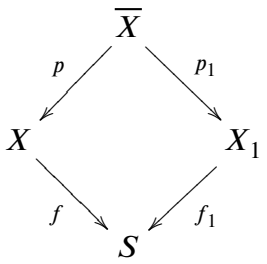
are isomorphisms. On the other hand, every sheaf F of the type under consideration on X is isomorphic to the limit of a direct system of the preceding kind; for example, one may take $F_i = u_{i*}(F)$, where $u_i : X \to X_i$ is the canonical morphism, by 5.2 and IX 1.6 (iii). This completes the proof of 8.1.

Corollary 8.2. *Theorem* 5.1 (i) *is true.*

Indeed, it is enough to combine 8.1, criterion 6.5 (i) b″), and 5.8.

Lemma 8.3. *To prove any one of the assertions* 5.1 (i), (ii), (iii), *it is enough to do so when f is projective of relative dimension ≤ 1 and S is Noetherian.* 127

Indeed, 8.1 already shows that we may restrict ourselves to the case in which f is projective and S is Noetherian, and an argument used above further allows us to suppose that $X = \mathbf{P}^r_S$. We proceed by induction on r, noting that the theorem holds by hypothesis for $r = 1$. We may therefore suppose that $r \geq 2$ and that the theorem has been proved for $\mathbf{P}^{r-1}_S \to S$ (over every base S). Embed $Z = \mathbf{P}^{r-2}_S$ in $X = \mathbf{P}^r_S$ in the usual way, and let $p : \overline{X} \to X$ be the blowup of X along Z. A direct calculation (cf. EGA V) gives a natural morphism $p_1 : \overline{X} \to X_1 = \mathbf{P}^1_S$ that makes $\overline{X}$ an X_1-scheme locally isomorphic to $\mathbf{P}^{r-1}_{X_1}$ (it is the projective bundle over X_1 associated with a suitable locally free sheaf of rank r on X_1). On the other hand, the fibers of p have dimension ≤ 1, as do those of $f_1 : X_1 \to S$. We may therefore apply the relative-dimension-one hypothesis to p and f_1, and the induction hypothesis to p_1, in the commutative diagram



Since p is surjective, 6.8 reduces us to proving 5.1 for the composite $\overline{f} = fp = f_1p_1$. But we may already apply 5.1 to both f_1 and p_1. Taking 6.11 into account, we conclude that 5.1 also holds for their composite f_1p_1. This completes the proof of 8.3.

Lemma 8.4. *To prove* 5.1 (ii), *it is enough to prove the "specialization theorem for the fundamental group"* 5.9 bis *in the case where f is projective of relative dimension ≤ 1 and S is Noetherian and strictly local.* 128

This follows at once from 8.3, 6.1, and 6.5 (ii) (criterion b)).

The specialization theorem in the case of 8.4 will be proved directly in the next exposé (XIII 2).

LEMMA 8.5. *To prove* 5.1 (iii), *it is enough to prove* 5.1 (ii) *and, in addition, the following proposition (which will be proved in the next exposé* (XIII 3), *or in* (EGA IV 21.9.12)*):*

PROPOSITION 8.6. *Let S be Henselian and Noetherian, let $f : X \to S$ be projective of relative dimension ≤ 1, and let $X_\circ$ be the closed fiber of X/S. For every scheme Z, let* $\operatorname{Pic} Z = H^1(Z, (G_m)_Z)$ *denote the Picard group of Z. The restriction homomorphism*

$$\operatorname{Pic} X \longrightarrow \operatorname{Pic} X_\circ$$

is surjective.

Proof of 8.5. By 6.5 (iii) and 8.2, it is enough to prove that

$$\varphi^q : H^q(X, \mathbf{Z}/\ell^\nu\mathbf{Z}) \longrightarrow H^q(X_\circ, \mathbf{Z}/\ell^\nu\mathbf{Z})$$

is surjective for every $q \geq 1$ and every prime ℓ. For $q = 1$, surjectivity is already known from 5.1 (ii). Moreover, since $X_\circ$ is a projective scheme of dimension ≤ 1 over a separably closed field k, we have

$$H^q(X_\circ, \mathbf{Z}/\ell^\nu\mathbf{Z}) = 0$$

for $q > 2$ when $\ell \neq \operatorname{char} k$ (X 4.3), and for $q > 1$ when $\ell = \operatorname{char} k$ (X 5.2). Thus surjectivity is automatic for these values of q.

129 It remains to consider $q = 2$ for a prime $\ell \neq \operatorname{char} k$. Put $n = \ell^\nu$. The sheaf $(\mu_n)_S$ of n-th roots of unity is locally isomorphic to $(\mathbf{Z}/n\mathbf{Z})_S$, and, since S is strictly local, there is a noncanonical isomorphism

$$(\mu_n)_S \simeq (\mathbf{Z}/n\mathbf{Z})_S,$$

and hence $(\mu_n)_X \simeq (\mathbf{Z}/n\mathbf{Z})_X$.

Kummer theory IX 3.2, applied to X and $X_\circ$, gives a morphism of exact sequences

$$\begin{array}{ccccc}
\operatorname{Pic} X & \longrightarrow & H^2(X, (\mu_n)_X) & \longrightarrow & H^2(X, (G_m)_X) \\
\downarrow{\scriptstyle \epsilon} & & \downarrow{\scriptstyle \varphi^2} & & \downarrow \\
\operatorname{Pic} X_\circ & \longrightarrow & H^2(X_\circ, (\mu_n)_{X_\circ}) & \longrightarrow & H^2(X_\circ, (G_m)_{X_\circ}).
\end{array}$$

Now $H^2(X_\circ, (G_m)_{X_\circ}) = 0$ by IX 4.6. Consequently, the surjectivity of ϵ implies that of φ^2, which proves the lemma.

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EXPOSÉ XIII

Base-change theorem for a proper morphism: conclusion of the proof

M. Artin

1. The projective and flat case

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Recall the statement of XII 5.9 bis in the case under consideration here:

Proposition 1.1. *Let S be the spectrum of a Noetherian henselian ring, and let $f : X \to S$ be a projective and flat morphism. Let $X_\circ$ be the closed fiber of X/S. The restriction functor*

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_\circ)$$

is an equivalence of categories.

As we have already observed, it follows from XII 6.5 (i) that the functor is fully faithful. It remains to prove that every étale covering $Y_\circ$ of $X_\circ$ is induced by an étale covering Y of X.

Lemma 1.2. *Let S be locally Noetherian, let $f : X \to S$ be projective and flat, and let $Y \subset X$ be a subscheme. The subfunctor $\epsilon = \prod_{X/S} Y/X$ of the terminal functor, "which expresses the condition $Y = X$", that is, such that for $S' \to S$ one has*

$$\epsilon(S') = \begin{cases} \phi & \text{if} \quad Y' \neq X', \\ \{\phi\} & \text{if} \quad Y' = X', \end{cases}$$

(where a prime denotes the effect of the base change $S' \to S$), is representable by a subscheme Z of S. If Y is open (resp. closed), then so is Z. 133

Proof. Let Y be closed in an open subset U of X, and let C be the closed complement of U. Since f is proper, the image of C in S is a closed subset D, and it is clear that the condition $U = X$ is represented by the open subset $S - D$ of S. We may therefore suppose that $U = X$ already, that is, that Y is a closed subscheme. Consider the surjective morphism $\mathcal{O}_X \xrightarrow{\alpha} \mathcal{O}_Y$ of $\mathcal{O}_X$-modules; saying that $Y = X$ amounts to saying that there exists a morphism $\mathcal{O}_Y \xrightarrow{\beta} \mathcal{O}_X$ such that the composite $\beta\alpha$ is the identity, that is, that there exists a section of $\mathbf{Hom}(\mathcal{O}_Y, \mathcal{O}_X)$ lifting the identity section of $\mathbf{Hom}(\mathcal{O}_X, \mathcal{O}_X)$. Now it follows from EGA III 7.7.8 that the functor which assigns to S'/S the set of sections of $\mathbf{Hom}(\mathcal{O}_Y, \mathcal{O}_X)$ (resp. $\mathbf{Hom}(\mathcal{O}_X, \mathcal{O}_X)$) is represented by a vector bundle V_1 (resp. V_2) over S. Let $\varphi : V_1 \to V_2$ be the morphism $u \mapsto u\alpha$ induced by $\mathcal{O}_X \to \mathcal{O}_Y$, and let $s : S \to V_2$ be the section corresponding to the identity $\mathcal{O}_X \to \mathcal{O}_X$. The inverse image of the section s under φ is identified with a closed subscheme Z of S, and it is clear that Z represents the functor ϵ, which proves the lemma.

Lemma 1.3. *Let S be a locally Noetherian scheme, let $f : X \to S$ be a projective and flat morphism, and let E be a locally free sheaf on X. Let*

$$QE : (\mathrm{Sch})/S \longrightarrow (\mathrm{Ens})$$

be the functor which assigns to S'/S the set $QE(S')$ of quotient sheaves A' of $E' = E \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, endowed with the structure of an étale $\mathcal{O}_{X'}$-algebra. This functor is representable by a scheme locally of finite type over S.

134 *Proof.* The functor Quot(E), with Quot(E)(S') equal to the set of quotients of E' that are flat over S', is representable by a scheme locally of finite type (TDTE IV, exp. 221), say by Z/S. Let M_u be the "universal"quotient of $E \otimes_{\mathcal{O}_S} \mathcal{O}_Z$ over X_Z. The subfunctor of Quot(E) consisting of the quotients which are flat *over* X' is represented by an open subset U of Z. Indeed, let M' be a quotient of E', given by an S-morphism $S' \to Z$. Since M' is flat over S', saying that it is flat over X' amounts to saying that, for every point $s' \in S'$, the module $M'_{s'}$ is flat over the fiber $X'_{s'}$ (EGA IV 5.9), which is equivalent to $M_u|_{X_z}$ being flat over the fiber X_z at the point z which is the image of s'. But the set U of $z \in Z$ such that $M_u|_{X_z}$ is flat over X_z, that is, such that M_u is flat over X_Z at every point of X_z, is open in Z by EGA IV 11.1.5, proving the assertion.

It now suffices to represent the relative functor $QE/U[f]$. We are thus reduced to the following lemma, after replacing S by U and X by $X \times_S U$:

LEMMA 1.4. *Let S be a locally Noetherian scheme, let $f : X \to S$ be a projective and flat morphism, and let M be a locally free $\mathcal{O}_X$-module. The functor* Alget : (Sch)/$S \to$ (Ens)*, which assigns to S'/S the set* Alget(S') *of algebra structures on M' that are étale over $\mathcal{O}_{X'}$, is representable by a scheme of finite type over S.*

Proof. For brevity, we shall say "representable", understanding "by a relative scheme of *finite type*". All the vector bundles encountered below will be defined by *coherent* sheaves, and hence will be of finite type. A bilinear composition law on M' is given by a section of the locally free sheaf $\mathbf{Hom}(M' \otimes M', M')$. Since $\mathbf{Hom}(M' \otimes M', M')$ is
135 flat over S, it follows from EGA III 7.7.6 that the functor "bilinear composition law"is represented by a vector bundle Z over S. We must check that the additional conditions that the law be associative, commutative, and possess a unit section are represented by a subscheme of Z. By considering the relative functor, we may replace S by Z, and may therefore suppose that a composition law on M has been given and prove that the subfunctor of S expressing the preceding conditions is representable by a subscheme of S.

For associativity, the condition is that two sections of $\mathbf{Hom}(M \otimes M \otimes M, M)$ become equal after the base change $S' \to S$. Let Y be the fiber product of these sections over the vector bundle corresponding to $\mathbf{Hom}(M \otimes M \otimes M, M)$. The scheme Y is identified with a closed subscheme of X, and associativity after $S' \to S$ is expressed by the condition $Y' = X'$. It therefore defines a subfunctor of S represented by a closed subscheme of S, by 1.2. For commutativity, the condition is that two sections of $\mathbf{Hom}(M \otimes M, M)$ become equal, which again defines a subfunctor of S represented by a closed subscheme, by 1.2. Finally, suppose, as we now may, that the law is already associative and commutative, and consider the condition that a unit section exist; such a section will then be unique. The functor whose value on S'/S is the set of sections of M' is represented by a vector bundle, by EGA III 7.7.6, and it remains only to represent the subfunctor of this bundle corresponding to the *unit sections* (taking account of the uniqueness just recalled). Passing to the relative situation as usual, we may suppose that a section ϵ of M has already been given, and prove that the subfunctor of S expressing the condition
136 that it become a unit section is representable. But the condition that ϵ be a unit is that the composite of the evident morphisms $M' \to M' \otimes M' \to M'$ be the identity, that is, that two specified sections of $\mathrm{Hom}(M, M)$ become equal. We conclude once again by 1.2.

Finally, suppose that a commutative unital algebra structure has been given on the locally free module M. The locus where it is étale is open in X (SGA I 4.5), and is therefore represented over S by an (open) subscheme. This completes the proof of 1.4, and hence of 1.3.

1.5. Let Z/S now be the object representing the functor considered in 1.3, and consider the condition that Z be smooth over S at a point z. Translating SGA I III 3.1 into the language of the functor QE, one obtains the following condition:

Let B be a local Artinian ring, let $\mathscr{M} = \operatorname{rad} B$, let J be an ideal of B such that $J\mathscr{M} = 0$, put $S' = \operatorname{Spec} B$ and $S'' = \operatorname{Spec} B/J$, and let $S' \to S$ be a morphism. We use the evident notation for the effects of the base changes $S' \to S$ and $S'' \to S$.

(∗) For every quotient A'' of E'', endowed with the structure of an étale $\mathcal{O}_{X''}$-algebra and corresponding to an S-morphism $S'' \to Z$ whose image is z, there exists a quotient A' of E', endowed with the structure of an étale $\mathcal{O}_{X'}$-algebra, which induces it.

Now, since X' and X'' have the same underlying space, an algebra A' inducing A'' is already determined up to unique isomorphism (SGA I 8.3). We may therefore express 137
condition (∗∗) as follows:

(∗∗) Let A' be an étale algebra over $\mathcal{O}_{X'}$, and let A'' be the algebra induced by A' on X''. Every surjective homomorphism

$$E'' \xrightarrow{\varphi''} A''$$

corresponding to a morphism $S'' \to Z$ whose image is z lifts to a homomorphism (necessarily surjective by Nakayama's lemma)

$$E' \xrightarrow{\varphi'} A'.$$

From the exact sequence

$$0 \longrightarrow J \longrightarrow B \longrightarrow B/J \longrightarrow 0$$

we obtain an exact sequence of sheaves on X':

$$0 \longrightarrow J \otimes \mathbf{Hom}(E', A') \longrightarrow \mathbf{Hom}(E', A') \longrightarrow \mathbf{Hom}(E'', A'') \longrightarrow 0.$$

The obstruction to lifting φ'' lies in $H^1(X', J \otimes \mathbf{Hom}(E', A'))$. Let $S_\circ = \operatorname{Spec} B/\mathscr{M}$. Since $J\mathscr{M} = 0$, one has

$$J \otimes \mathbf{Hom}(E', A') \simeq J \otimes \mathbf{Hom}(E_\circ, A_\circ),$$

and it is clear that

$$H^1(X', J \otimes \mathbf{Hom}(E_\circ, A_\circ)) \simeq J \otimes H^1(X_\circ, \mathbf{Hom}(E_\circ, A_\circ)).$$

Taking account of the fact that cohomology commutes with extension of the base fields, we obtain:

COROLLARY 1.6. *Let Z/S be the scheme representing the functor of 1.3, let $X_Z = X \times_S Z$, and let A_Z be the ("general") quotient étale algebra of E_Z. Then Z/S is smooth at every point $z_\circ$ such that*

$$H^1(X_\circ, \mathbf{Hom}(E_\circ, A_\circ)) = 0,$$

where the subscript zero denotes restriction to the fiber $X_\circ$ of X_Z/Z at the point $z_\circ$. 138

We can now complete the proof of 1.1. With the notation of the proposition, let $Y_\circ/X_\circ$ be an étale covering, given by a sheaf of algebras $A_\circ$ that is étale over $\mathcal{O}_{X_\circ}$. For n sufficiently large, $A_\circ(n)$ is generated by its sections (EGA III 2.2.1). We thus have a surjection

$$\mathcal{O}_{X_\circ}^N \longrightarrow A_\circ(n) \longrightarrow 0,$$

and hence a surjection

$$\mathcal{O}_{X_\circ}^N(-n) = E_\circ \longrightarrow A_\circ \longrightarrow 0.$$

Choose n sufficiently large that we also have

$$(+) \qquad H^1(X_\circ, \mathbf{Hom}(E_\circ, A_\circ)) = 0,$$

which is possible by EGA III 2.2.1, since

$$\mathbf{Hom}(E_\circ, A_\circ) = [\mathbf{Hom}(\mathcal{O}_{X_\circ}^N, A_\circ)](n) = [A_\circ(n)]^N.$$

Put $E = \mathcal{O}_X^N(-n)$, and let Z/S be the scheme representing the functor considered in 1.3. The quotient $A_\circ$ of $E_\circ$ corresponds to a point $z_\circ \in Z$ in the closed fiber of Z/S, rational over $k(s)$; by 1.5 and (+), Z/S is smooth at $z_\circ$. Since S is the spectrum of a *henselian* ring, there is a section of Z/S through $z_\circ$; this follows, for example, immediately from the original definition (SGA II 1.1) of smoothness. It follows that $A_\circ$ is induced by an algebra A étale over X (a quotient of E), which completes the proof of 1.1.

2. The case of relative dimension at most one

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The assertion is the following.

Proposition 2.1. *With the notation of* XII 5.9 bis, *the functor* $\mathrm{Et}(X) \to \mathrm{Et}(X_\circ)$ *is an equivalence of categories if* S *is Noetherian and strictly local and if* f *is projective of relative dimension* ≤ 1.

By XII 8.5, *this will complete the proof of* XII 5.1 (ii), *which generalizes both 1.1 and 2.1.*

Lemma 2.2. *Let* S *be the spectrum of a Noetherian local ring* A, *and let* X/S *be projective of relative dimension* $\leq n$. *There exist a projective and flat* S*-scheme* Z/S *of relative dimension* $\leq n$ *and a closed immersion* $X \hookrightarrow Z$ *over* S.

Proof. Let $X \to \mathbf{P}_S^N$ be a projective embedding over S, and let $J \subset A[x_\circ, \dots, x_N]$ be the homogeneous ideal defining X. Put $k = A/\mathfrak{m}_A$, $S_\circ = \operatorname{Spec} k$, and $X_\circ = X \otimes_S k$. Then $X_\circ$ is defined by the homogeneous ideal $J_\circ = \mathrm{Im}(J \to k[x_\circ, \dots, x_N])$. Since $\dim X_\circ \leq n$, it is well known that there exist elements $f_1^\circ, \dots, f_{N-n}^\circ$ of $J_\circ$ defining a subscheme $Z_\circ$ of $\mathbf{P}_k^N$ of dimension n. Let f_i $(1 \leq i \leq N-n)$ be elements of J lifting the f_i°, and let Z be the subscheme of $\mathbf{P}_S^N$ defined by $f_1, \dots, f_{N-n}$. By EGA IV 11.3, Z is flat over S; and by EGA IV 13.1.5 it has relative dimension $\dim Z_\circ = n$. This proves 2.2.

Proof of 2.1. Let X/S be projective of relative dimension ≤ 1, with S Noetherian and strictly local. We may assume that X (and hence $X_\circ$) is connected. Let $X \to Z$ be a closed immersion over S, where Z is projective, flat, and of relative dimension ≤ 1; see
140 2.2. Let $s_\circ$ be the closed point of S. We have a corresponding diagram of categories

$$\begin{array}{ccc} \mathrm{Et}(Z) & \longrightarrow & \mathrm{Et}(X) \\ \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle b} \\ \mathrm{Et}(Z_\circ) & \xrightarrow{\;c\;} & \mathrm{Et}(X_\circ). \end{array}$$

We wish to prove that b is an equivalence. By XII 5.8, b is fully faithful, so it remains to prove that it is essentially surjective. The functor a is an equivalence by 1.1; it therefore suffices to prove that c is essentially surjective. Dropping the subscript $_\circ$, we are reduced to the following proposition.

PROPOSITION 2.3. *Let k be a separably closed field, let Z be a k-scheme locally of finite type and of dimension ≤ 1, and let $X \hookrightarrow Z$ be a connected closed subscheme. Then every étale covering of X is induced by an étale covering of Z.*

Proof. We may suppose that Z and X have dimension 1, the case in which X is zero-dimensional being immediate from the hypothesis on k. Then $Z = X \cup Y$, where Y is the closed subscheme obtained as the closure of $Z - X$. Put $V = X \cap Y$; this is a discrete, that is, locally Artinian, scheme. The morphism $X \amalg Y \to Z$ is finite and surjective, hence is a strict descent morphism for the category of étale coverings (SGA IX 4.7). Clearly

$$(*) \qquad (X \amalg Y) \times_Z (X \amalg Y) = X \amalg Y \amalg V^{(1)} \amalg V^{(2)} \quad (V^{(i)} = V),$$

where pr_i sends $V^{(i)}$ to the component X of $X \amalg Y$ and sends $V^{(j)}$ to the component Y when $j \neq i$.

Let X'/X be an étale covering of degree n, and let Y''/Y be a covering of degree n (for example, the trivial covering of degree n), so that $X' \amalg Y''$ is a covering of $X \amalg Y$ 141
of degree n. It is enough to find descent data for this covering relative to $X \amalg Y \to Z$. Inspection of $(*)$ shows that such descent data amount simply to an isomorphism

$$X' \times_X V = V' \xrightarrow{\sim} V'' = Y'' \times_Y V.$$

Since k is separably closed, so are the residue fields of the points of V; consequently every étale covering of V is completely split. Thus $V' \simeq V''$, which proves the result.

REMARK 2.4. One may avoid the delicate result from SGA IX and descent theory by a direct argument showing that the category of étale coverings of Z is equivalent to the category of triples (X', Y', θ), where X' (respectively Y') is an étale covering of X (respectively Y), and θ is a V-isomorphism $X' \times_X V \simeq Y' \times_Y V$. This result, valid for any scheme Z that is the union of two closed subschemes X and Y, is proved directly from the analogous description of quasi-coherent modules on Z (when $X = V(\mathfrak{J})$, $Y = V(\mathfrak{K})$, and $\mathfrak{J} \cap \mathfrak{K} = 0$).

3. An auxiliary result on the Picard group (*)

PROPOSITION 3.1. *Let X be a Noetherian scheme and let $X_\circ$ be a closed subscheme of X such that* (a) $\dim X_\circ \leq 1$, *and* (b) *for every nonempty connected closed subset Y of X, $Y_\circ = Y \cap X_\circ$ is nonempty and connected.*

(i) *Let $Z_\circ$ be the union of the sets $\overline{\{x\}} \cap X_\circ$, for $x \in \mathrm{Ass}\, \mathcal{O}_X$, that consist of a single point. For every Cartier divisor $D_\circ$ on $X_\circ$ such that $\mathrm{supp}(D_\circ)$ does not meet $Z_\circ$, there exists a Cartier divisor D on X inducing $D_\circ$ (and if $D_\circ \geq 0$, one may take $D \geq 0$).*

(ii) *Suppose there is an affine open subset $U_\circ$ of $X_\circ$ containing the finite set $\mathrm{Ass}\, \mathcal{O}_{X_\circ} \cup Z_\circ$ (this condition is automatically satisfied if $X_\circ$ admits an ample invertible sheaf). Then $\mathrm{Pic}(X) \to \mathrm{Pic}(X_\circ)$ is surjective.*

Proof.

[34] This also appears in EGA IV 21.9.11 and 21.9.12 in a slightly more general form; in particular, in 3.2 it is enough for f to be *separated* rather than proper.

(i) Since $\dim X_\circ \leq 1$, every Cartier divisor $D_\circ$ on $X_\circ$ plainly has discrete support, and is therefore the difference of two effective Cartier divisors whose supports are contained in that of $D_\circ$. We are thus reduced in (i) to the case where $D_\circ$ is effective. For every $x \in \operatorname{Supp} D_\circ$ it is then defined at x by a non-zero-divisor $f_x \in \operatorname{rad}(\mathcal{O}_{X_\circ,x})$. The element f_x is induced by a section g_x of $\mathcal{O}_X$ on an open neighborhood U_x of x. I claim that, after shrinking U_x, the section g_x is a non-zero-divisor in $\mathcal{O}_{U_x}$, that is, $V(g_x) \cap \operatorname{Ass} \mathcal{O}_{U_x} = \emptyset$.

Indeed, conditions (a) and (b) imply that, for every $z \in \operatorname{Ass} \mathcal{O}_X$, the set $\overline{\{z\}} \cap X_\circ$ is a connected closed subset of $X_\circ$ and hence is either a single closed point or a connected union of one-dimensional irreducible components of $X_\circ$. Since $x \notin Z_\circ$, in the first case $x \notin \overline{\{z\}}$. In the second case, one cannot have both $x \in \overline{\{z\}}$ and $z \in V(g_x)$, for then g_x, and therefore f_x, would vanish on an irreducible component of $\overline{\{z\}} \cap X_\circ$, which is also an irreducible component of $X_\circ$; this is incompatible with f_x being a non-zero-divisor in $\mathcal{O}_{X_\circ,x}$. After
142 shrinking U_x, we may therefore suppose that $V(g_x) \cap \operatorname{Ass} \mathcal{O}_{U_x} = \emptyset$, so g_x is a non-zero-divisor.

Let Y_x be the closure of $V(g_x)$ in X. The point x is isolated in $Y_x \cap X_\circ \cap U_x$, hence also in $Y_x \cap X_\circ$. By condition (b), Y_x has a connected component Y'_x such that $Y'_x \cap X_\circ = \{x\}$. Moreover, $Y'_x \subset U_x$: indeed, $T_x = Y'_x - (Y'_x \cap U_x)$ is closed in X and satisfies $T_x \cap X_\circ = \emptyset$, so condition (b) gives $T_x = \emptyset$. Let D'_x be the Cartier divisor on X induced by Y'_x on U_x and by zero on $X - Y'_x$. Then $D = \sum_x D'_x$ is an effective Cartier divisor on X inducing $D_\circ$ on $X_\circ$.

(ii) The conclusion follows from (i) and the fact that every invertible module $\mathbf{L}_\circ$ on $X_\circ$ is isomorphic to the module defined by a Cartier divisor $D_\circ$ whose support does not meet $Z_\circ$. This last assertion means that $\mathbf{L}_\circ$ admits a section on an open set $V_\circ$ containing $\operatorname{Ass} \mathcal{O}_{X_\circ} \cup Z_\circ$ which vanishes at no point of that set. It is enough to prove this after replacing $X_\circ$ by the affine open $U_\circ$; there it follows from the familiar fact that an invertible module on a semilocal scheme is isomorphic to the structure sheaf. Finally, if $X_\circ$ admits an ample invertible sheaf, every finite subset of $X_\circ$ is contained in an affine open subset. This proves (ii).

We have the following corollary, which generalizes XII 8.6 and completes the proof of XII 5.1 (iii).

Corollary 3.2. *Let S be the spectrum of a Noetherian Henselian ring, and let $f : X \to S$ be a proper morphism of relative dimension ≤ 1. For every closed subscheme $X'_\circ$*
143 *of X having the same underlying space as the closed fiber $X_\circ$ of X/S, the canonical map* $\operatorname{Pic}(X) \to \operatorname{Pic}(X'_\circ)$ *is surjective.*

Indeed, apply the proposition to $(X, X'_\circ)$. Condition (a) holds by hypothesis, and condition (b) follows from XII 5.7. It remains to verify the existence of $U_\circ$ required in (ii); this follows because $X'_\circ$ is necessarily projective, as is every proper curve over a field.

Remark. A reader who does not wish to assume or reconstruct the fact that proper curves over a field are projective may use the corollary only under the additional hypothesis that f is projective; this suffices for the application intended here. Conversely, admitting the projectivity of proper algebraic curves over a field, one concludes that under the hypotheses of 3.2, X is necessarily projective over S. Indeed, choose an ample $\xi_\circ \in \operatorname{Pic}(X_\circ)$, lift it to some $\xi \in \operatorname{Pic}(X)$, and apply EGA III 4.7.1.

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EXPOSÉ XIV

Finiteness theorem for a proper morphism; cohomological dimension of affine algebraic schemes

M. Artin

This exposé contains two important theorems whose proofs use the base-change theorem for a proper morphism. 145

1. Finiteness theorem for a proper morphism

THEOREM 1.1. *Let $f : X \to S$ be a proper morphism of finite presentation. Let A be a left Noetherian ring which is a $\mathbf{Z}/n$-algebra for some suitable $n \in \mathbf{N}$. Let F be a* constructible *sheaf of sets (resp. groups, resp. A-modules) on X. Then the sheaf f_*F (resp. $R^1 f_* F$, resp. $R^q f_* F$ for every $q \geq 0$) is also constructible.*

In particular, we have:

COROLLARY 1.2. *Let X be a proper scheme over the spectrum of a separably closed field k, and let F be a constructible torsion abelian sheaf on X. Then the groups $H^q(X, F)$ are finite for every $q \geq 0$.*

REMARK 1.3. The theorem is false for abelian sheaves if the condition that f be proper is omitted, as is seen by taking $F = \mathbf{Z}/p$, $p = \operatorname{car} k$, $X = \mathbf{E}^1_k$, the affine line, and f either the inclusion of X in $\mathbf{P}^1_k$ or the structural morphism $X \to \operatorname{Spec} k$. Nevertheless, it is conjectured[35] that it is true if F has order prime to the residue characteristics of S and 146
S is "excellent" (EGA IV 7.8). When resolution of singularities is available (for example, in characteristic zero), this can be proved for an excellent equicharacteristic scheme S (XIX 5.1).

Proof of 1.1. The assertion is local on S, so we may take S affine. Then S is a limit of spectra of rings of finite type over $\mathbf{Z}$, and the data of the theorem can be obtained, by base change $S \to S_0$, from a proper morphism $f_0 : X_0 \to S_0$, with S_0 of finite type over $\mathbf{Z}$, and a constructible sheaf F_0 on X_0; see EGA IV 8 and IX 2.7.4. Since the inverse image of a constructible sheaf is again constructible (IX 2.4 (iii)), application of XII 5.1 to the base-change morphism $S_0 \leftarrow S$ reduces us to proving the constructibility of $f_{0*}F_0$ (resp. $R^q f_{0*} F_0$), whence the following lemma.

LEMMA 1.4. *It is enough to verify 1.1 when S is of finite type over* $\operatorname{Spec} \mathbf{Z}$.

[35] N.D.E.: Ofer Gabber presented the following generalization of 1.1 at the conferences held in honor of Luc Illusie and Pierre Deligne in 2005. Let $f : X \to Y$ be a morphism of finite type between quasi-excellent Noetherian schemes (EGA IV 7.8), and let F be a constructible sheaf of $\mathbf{Z}/n$-modules on $X_{\text{ét}}$, with n invertible on Y. Then the sheaves $R^i f_* F$ on $Y_{\text{ét}}$ are constructible and vanish for all but finitely many i. There are analogous assertions for constructible sheaves of sets resp. groups of order invertible on Y.
Before Gabber's result, Theorem 1.1 was known in the following case. Let S be a regular scheme of dimension ≤ 1, let $f : X \to Y$ be a morphism of finite type between S-schemes, and let F be a sheaf of left A-modules on $X_{\text{ét}}$. Then the sheaves $R^i f_* F$ are constructible (SGA 4 1/2, Finitude 1.1).

LEMMA 1.5. *The set-theoretic assertion of 1.1 is true: that is, f_*F is a constructible sheaf of sets if F is one.*

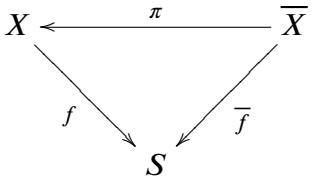
Proof. We may suppose that S is Noetherian (1.4). Let $F \to G = \prod_i \pi_{i*}(C_i)$ be an injection, with $\pi_i : X_i \to X$ finite and C_i constant and constructible on X_i (IX 2.14). We have $f_*F \subset f_*G$; hence, by IX 2.9 (ii), it is enough to treat the case $F = G$. Now $f_*G = \prod_i (f\pi_i)_* C_i$, and a finite product of constructible sheaves on S is constructible (IX 2.6 (ibis)). Replacing X by X_i, we are reduced to the case where $F = D_X$ is a constructible constant sheaf with value D. By XII 5.1 (i), the fiber $(f_*F)_{\bar{s}}$ is isomorphic
147 to $D^{C(s)}$, where $C(s) = \pi_0(X_{\bar{s}})$; it is therefore finite. By IX 2.13 (iii), it is enough to prove that the function "number of elements in the fiber of f_*F" is constructible on S. This function is $\mathrm{card}(D^C)$, where C is the function "number of connected components in the geometric fiber", and C is constructible (EGA IV 9.7.9), which proves the result.

LEMMA 1.6. *Let S be a Noetherian scheme, let $f : X \to S$ be proper, let F be a constructible sheaf of groups (resp. A-modules) on X, and let q be an integer equal to 0 or 1 (resp. an integer ≥ 0). Let S_i $(1 \leq i \leq n)$ be subschemes of S such that $S = \bigcup_i S_i$, and let F_i and $f_i : X_i \to S_i$ be the "restrictions" of the data to S_i. If $R^q f_{i*} F_i$ is constructible for every i, then $R^q f_* F$ is constructible as well.*

Proof. The restriction of $R^q f_* F$ to S_i is isomorphic to $R^q f_{i*} F_i$ by XII 5.1, and the lemma follows from IX 2.8.

We now proceed by arguments analogous to those of XII 8:

LEMMA 1.7. *Let S be Noetherian, and let*



be a commutative diagram of proper morphisms, and let $j : U \to X$ be an open subscheme such that the open subscheme $\overline{U} = U \times_X \overline{X}$ maps isomorphically onto U. Let $Y = X - U$ be the reduced closed subscheme, and let $f_0 : Y \to S$ and $i : Y \to X$ be the canonical morphisms. Suppose that Theorem 1.1 holds[36] for $\overline{f}$, f_0, and π. Then it also holds for f.

148 *Proof.* First consider a sheaf of A-modules F on X. We have the exact sequence

$$0 \longrightarrow j_! j^* F \longrightarrow F \longrightarrow i_* i^* F \longrightarrow 0.$$

In view of the corresponding cohomology exact sequence for the $R^q f_*$ and IX 2.6, it is enough to verify constructibility for the two outer terms. Notice that $i_* i^* F$ is constructible by IX 2.4 (iii) and IX 2.14 (i), and hence so is $j_! j^* F$ (IX 2.6). Since i is a closed immersion,

$$(R^q f_*) i_* (i^* F) \simeq (R^q f_{0*})(i^* F),$$

by VIII 5.6; constructibility in this case therefore follows from the hypothesis on f_0. Moreover, if $\overline{j} : \overline{U} \to \overline{X}$ denotes the inclusion morphism and $\overline{F} = \pi^* F$, then

$$(*) \qquad j_! j^* F \simeq \pi_* \overline{j}_! \overline{j}^* \overline{F},$$

[36] N.D.E.: The hypothesis concerning $R^q \pi_*$ occurs in the proof only when $q = 0$. In that case constructibility is already known from 1.5.

as is seen by first defining a morphism from left to right and then proving it to be invertible fiber by fiber using XII 5.1. At every geometric point $\overline{y}$ of Y, the restriction of $\overline{j}_!\overline{j}^*\overline{F}$ to the fiber $\overline{X}_{\overline{y}}$ of $\overline{X}$ over X is zero; hence, by XII 5.1 (iii), the fiber of $(R^q\pi_*)(\overline{j}_!\overline{j}^*\overline{F})$ at $\overline{y}$ is zero for every $q \geq 0$. On the other hand, π induces an isomorphism over $X - Y$, so

$$(R^q\pi_*)(\overline{j}_!\overline{j}^*\overline{F}) = 0$$

for $q > 0$. The Leray spectral sequence

$$E_2^{p,q} = (R^p f_*)(R^q\pi_*)(\overline{j}_!\overline{j}^*\overline{F}) \Longrightarrow R^n\overline{f}_*(\overline{j}_!\overline{j}^*\overline{F})$$

together with $(*)$ therefore gives isomorphisms

$$(R^q\overline{f}_*)(\overline{j}_!\overline{j}^*\overline{F}) \simeq (R^q f_*)(j_!j^*F)$$

for $q \geq 0$. Since $\overline{j}_!\overline{j}^*\overline{F}$ is constructible (by the same argument as for $j_!j^*F$), the left-hand side is constructible by the hypothesis on $\overline{f}$. Thus $(R^q f_*)(j_!j^*F)$ is constructible, and consequently so is $R^q f_*F$. 149

Now suppose that F is a constructible sheaf of groups on X. Replace $\pi : \overline{X} \to X$ by $\pi \amalg i : (\overline{X} \amalg Y) \to X$. It is then enough to prove that if $\pi : \overline{X} \to X$ is *surjective* and 1.1 holds for π and $\overline{f}$, then it also holds for f. First let $\overline{F}$ be a constructible sheaf of groups on $\overline{X}$. Then $\pi_*\overline{F}$ is constructible on X (1.5), and there is an injection

$$(R^1 f_*)\pi_*\overline{F} \hookrightarrow (R^1\overline{f}_*)\overline{F}.$$

The right-hand side is constructible by hypothesis, and hence so is the left-hand side (IX 2.9 (ii)). Since π is surjective, there is an injection $F \to \pi_*\pi^*F = G$. The sheaf $\overline{F} = \pi^*F$ is constructible, so G and $R^1 f_*G$ are constructible as well, by what we have just proved. We are thus reduced to the following lemma.

LEMMA 1.8. *Let S be Noetherian, let $f : X \to S$ be a proper morphism, and let $F \overset{u}{\hookrightarrow} G$ be an injection of constructible sheaves of groups on X. If $R^1 f_*G$ is constructible, then $R^1 f_*F$ is constructible as well.*

Proof. Let $C = G/F$, which is a constructible sheaf of homogeneous spaces under G (apply IX 2.6). Consider the exact sequence

$$(*) \qquad f_*C \overset{\delta}{\to} R^1 f_*F \overset{u^1}{\longrightarrow} R^1 f_*G.$$

To prove that $R^1 f_*F$ is constructible, it is enough, by Noetherian induction and 1.6, to prove that if $S \neq \emptyset$, there exists a nonempty open subset of S on which $R^1 f_*F$ is constructible. We may therefore suppose that $R^1 f_*G$ is locally constant. Since constructibility is local for the étale topology, we may further suppose that S is irreducible and that $R^1 f_*G$ is constant (and constructible), with value $E = \{e_1, \ldots, e_n\}$. For $i = 1, \ldots, n$, let D_i be the subsheaf of $R^1 f_*F$ which is the inverse image of the section e_i under u^1. Since 150
"coproducts are universal" in a topos, we have $R^1 f_*F = \amalg_i D_i$; it is therefore enough to prove that each D_i is constructible over a suitable nonempty scheme S' étale over S.

Recall that $R^1 f_*F$ is the sheaf associated with the presheaf $\mathscr{R}^1 F$, where $\mathscr{R}^1 F(S') = H^1(X \times_S S', F)$. Consequently, we may suppose (after replacing S by a nonempty scheme S' étale over

S) that D_i is the "empty sheaf" (hence constructible), *or* that there exists an element $\alpha \in H^1(X, F)$ inducing a section $\overline{\alpha}$ of D_i. In the latter case, let

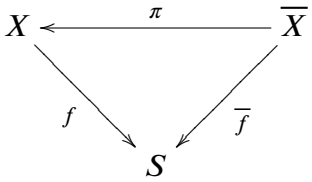
$$f_* C^\alpha \xrightarrow{\delta} R^1 f_* F^\alpha \longrightarrow R^1 f_* G^\alpha$$

be the exact sequence deduced from $(*)$ by twisting with α. We have

$$R^1 f_* F^\alpha \simeq R^1 f_* F \quad , \quad R^1 f_* G^\alpha \simeq R^1 f_* G,$$

and the section of $R^1 f_* F^\alpha$ corresponding to $\overline{\alpha}$ is the unit section. Thus the subsheaf D_i^α of $R^1 f_* F^\alpha$ corresponding to D_i is the inverse image of the unit section of $R^1 f_* G^\alpha$; that is, D_i^α is the image of $f_* C^\alpha$ under δ. Since $f_* C^\alpha$ is constructible (1.5), so is D_i^α, and hence so is D_i (apply IX 2.6). This proves the lemma.

LEMMA 1.9. *Let S be noetherian, and let*



151 *be a commutative diagram of proper morphisms. If Theorem 1.1 holds for f and π, then it also holds for $\overline{f}$.*

Proof. In the case of a constructible sheaf $\overline{F}$ of A-modules, we use the Leray spectral sequence

$$E_2^{p,q} = (R^p f_*)(R^q \pi_*)\overline{F} \Longrightarrow (R^n \overline{f}_*)\overline{F}.$$

By hypothesis, $E_2^{p,q}$ is constructible for every p, q; hence the abutment is constructible as well, as follows from IX 2.6.

Let $\overline{F}$ be a constructible sheaf of groups. We have the exact sequence (XII 3.2)

$$(*) \qquad 0 \longrightarrow (R^1 f_*)\pi_* \overline{F} \longrightarrow (R^1 \overline{f}_*)\overline{F} \longrightarrow f_*(R^1 \pi_*)\overline{F},$$

whose outer terms are constructible by the hypothesis and 1.5. We proceed as in the proof of 1.8: by noetherian induction and 1.6, it suffices to assume $S \neq \emptyset$ and prove constructibility over a suitable nonempty open subset. We may thus assume that $f_*(R^1 \pi_*)\overline{F}$ is locally constant and, after an étale localization, even constant and constructible (IX 2.4 (i)), with value $E = \{e_1, \ldots, e_n\}$. Let D_i be the subsheaf of $(R^1 \overline{f}_*)\overline{F}$ that is the inverse image of e_i, so that $(R^1 \overline{f}_*)\overline{F} = \coprod_i D_i$. It is enough to show that each sheaf D_i induces a constructible sheaf over some suitable nonempty connected étale S'/S. If D_i is not the "empty sheaf", we reduce to the case in which there exists an element $\alpha \in H^1(\overline{X}, \overline{F})$ inducing a section $\overline{\alpha}$ of D_i. Let

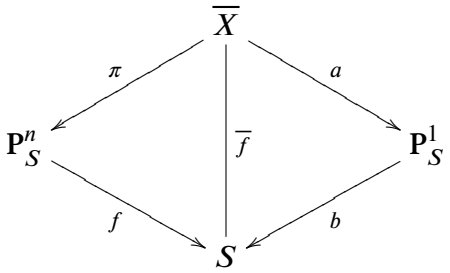
$$0 \longrightarrow (R^1 f_*)\pi_* \overline{F}^\alpha \longrightarrow (R^1 \overline{f}_*)\overline{F}^\alpha \longrightarrow f_*(R^1 \pi_*)\overline{F}^\alpha$$

be the exact sequence deduced from $(*)$ by twisting $\overline{F}$ with α. The section of $(R^1 \overline{f}_*)\overline{F}^\alpha$ corresponding to $\overline{\alpha}$ is the unit section. Consequently the subsheaf D_i^α of $(R^1 \overline{f}_*)\overline{F}^\alpha$

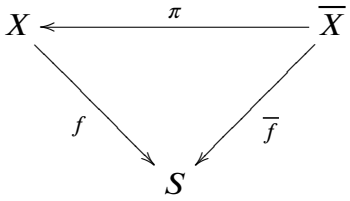
152 corresponding to D_i is the inverse image of the unit section of $f_*(R^1 \pi_*)\overline{F}^\alpha$, whence $D_i^\alpha \simeq (R^1 f_*)\pi_* \overline{F}^\alpha$, which is indeed a constructible sheaf, C.Q.F.D.

LEMMA 1.10. *It suffices to verify 1.1 when S is of finite type over* Spec $\mathbf{Z}$, *and f is projective and of relative dimension* ≤ 1.

Proof. Suppose first that f is projective, and argue by induction on the relative dimension of f, which may be assumed finite after replacing S by an affine open subset. If f has relative dimension $\leq n$, with $n \geq 2$, then locally on S one can find a finite morphism $\pi : X \to \mathbf{P}^n_S$. By 1.9 and IX 2.14 (i), we reduce to the case in which f is the structural morphism of $\mathbf{P}^n_S$. Consider the commutative diagram of morphisms



already considered in the proof of XII 8.3, where $\overline{X}$ is obtained from $\mathbf{P}^n_S$ by blowing up the closed subscheme $Y \simeq \mathbf{P}^{n-2}_S$. The relative dimension of b is ≤ 1, and that of a is $\leq n-1$; therefore the theorem holds for a and b by the induction hypothesis, and hence for $\overline{f}$ by 1.9. Moreover, the relative dimension of π is ≤ 1, while that of $f_0 : Y \to S$ is $\leq n-1$. It therefore follows from 1.7 that 1.1 holds for f, and thus for every projective morphism. Now let f be an arbitrary proper morphism, and argue by noetherian induction on X: we may assume the theorem holds for every proper closed subscheme of X and that $X \neq \emptyset$. Let



be a diagram of the type supplied by Chow's lemma XII 7.1. Applying 1.7 to this diagram 153
yields 1.1 for f, and proves the lemma.

1.11. We can now complete the proof of 1.1. If F is a constructible sheaf of groups, one can find an injection $F \hookrightarrow G = \prod_i \pi_{i,*} C_i$, where each π_i is finite and each C_i is constant and constructible (IX 2.14); by 1.8, we may therefore assume that F is constant. Likewise, if F is a constructible sheaf of A-modules, then by applying IX 2.14 one can find a resolution $F \to G^\cdot$, $G^\cdot = \{0 \to G^0 \to G^1 \to G^2 \to \dots\}$, of F such that every G^ν is of the form $G^\nu = \prod_i \pi_{i,*} C_i$. The spectral sequence of the resolution $G^\cdot$,

$$E_2^{p,q} = H^p((R^q f_*)G^\cdot) \Longrightarrow (R^n f_*)F,$$

then reduces us to the case in which F is constant and constructible, with value M.

Recall that A is a $\mathbf{Z}/n$-algebra. Let ℓ be a prime number dividing n. We have the exact sequence

$$0 \longrightarrow \ell M \longrightarrow M \longrightarrow M/\ell M \longrightarrow 0,$$

and it is again enough to prove 1.1 for ℓM and $M/\ell M$.

An easy induction then reduces us to the case in which A is a $\mathbf{Z}/\ell$-algebra, $\ell \in \mathbf{P}$, and M is a finite-type A-module. Then M is a vector space over $\mathbf{Z}/\ell$, and the canonical morphism

$$H^q(Y, \mathbf{Z}/\ell) \otimes_{\mathbf{Z}/\ell} M \longrightarrow H^q(Y, M)$$

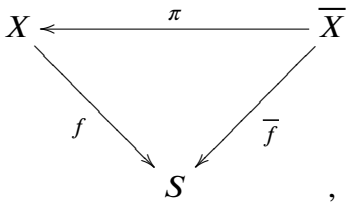
is bijective for every quasi-compact and quasi-separated Y (indeed, this is trivial if M
154 has finite rank, and both sides commute with filtered direct limits (VI 5.1 and VII 3.3)). Hence

$$[R^q f_*(\mathbf{Z}/\ell)] \otimes_{\mathbf{Z}/\ell} M \xrightarrow{\sim} R^q f_*(M).$$

It plainly suffices, therefore, to show that $R^q f_*(\mathbf{Z}/\ell)$ is constructible as a $\mathbf{Z}/\ell$-module; in other words, we are reduced to the case $A = \mathbf{Z}/\ell$, which we assume from now on.

Observe that the case in which f has relative dimension ≤ 0 (that is, f is finite), with F arbitrary, now follows from VIII 5.6 and 1.5.

We proceed by noetherian induction on S. By 1.6, we may assume $S \neq \emptyset$ and may replace S by any nonempty open subset. Moreover, we may assume that S is integral, with generic point s. The fiber X_s is an algebraic scheme of dimension ≤ 1; consequently there is a purely inseparable extension K' of $k(s) = K$ such that the normalization Y of $(X_s \otimes_K K')_{\mathrm{red}}$ is smooth over Spec K' [36]. After replacing S by a nonempty open subset, we may assume that there exists a finite, radicial, surjective morphism $S' \to S$ such that S'_s is K-isomorphic to Spec K', a smooth projective morphism $\overline{X}' \to S'$, and a finite surjective S'-morphism $\pi : \overline{X}' \to X' = X \times_S S'$ inducing $Y \to X_s \otimes_K K'$ on the fibers over the generic point s of S (EGA IV 9.6.1). Now, by VIII 1.1, making a purely inseparable extension is harmless. We may therefore replace S by S' and X by X'; that is, we may assume that f fits into a diagram



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155 in which π is finite and surjective and $\overline{f}$ is smooth. Moreover, by construction, we may assume that there is a closed subscheme Y of X whose generic fiber is finite and such that π induces an isomorphism over $X - Y$. Replacing S once more by a nonempty open subset, we reduce to the case in which $f_0 : Y \to S$ is also finite.

Since F is constant, $\pi^* F$ is constant as well, and $F \to \pi_* \pi^* F$ is injective because π is surjective. If F is a sheaf of groups, it suffices to prove that $(R^1 f_*)(\pi_* \pi^* F) \simeq (R^1 \overline{f}_*)(\pi^* F)$ is constructible (1.8). If F is a sheaf of $\mathbf{Z}/\ell$-modules, we have an exact sequence

$$0 \longrightarrow F \longrightarrow \pi_* \pi^* F \longrightarrow C \longrightarrow 0,$$

where C is supported on Y, because π induces an isomorphism over $X - Y$. It follows that $R^q f_* C = R^q f_{0,*} C$ is constructible (and vanishes if $q > 0$), so it is once again enough to show that $(R^q f_*)(\pi_* \pi^* F) = (R^q \overline{f}_*)(\pi^* F)$ is constructible. Replacing f by $\overline{f}$ and F by $\pi^* F$, we are reduced to the case in which F is constant and constructible and f is smooth.

Let us first treat a constant sheaf of groups, $F = G_X$. All the fibers of $(R^1 f_*) G_X$ are finite. Indeed, by applying XII 5.2 (ii), we reduce, in order to prove this, to the case in which S is the spectrum of a separably closed field and X is smooth of dimension ≤ 1 over S. The result is known in this case (SGA 1 X 2.6). It therefore remains only to prove that the specialization morphisms are injective (IX 2.13 (ii)). For this we reduce to the case in which S is strictly local; by the base-change theorem XII 5.1 (ii), we may

[36] Cf. EGA IV 17.15.14.

even replace S by the spectrum of a strictly local discrete valuation ring. Moreover, we may assume X connected. Since $H^1(X, G)$ classifies the principal Galois coverings of 156
X with group G, it is enough to check that if $X' \to X$ is a connected étale covering, then its geometric fiber is connected as well. This follows readily from the fact that f is smooth and proper with geometrically connected fibers.

We now treat the case $F = (\mathbf{Z}/\ell)_X$. For $q = 0$ and $q = 1$, the result is already known (1.5 and the preceding argument). By XII 5.3 bis, only the case $q = 2$ remains; furthermore, if S has characteristic ℓ, then $R^2 f_*(\mathbf{Z}/\ell) = 0$. Replacing S by a suitable nonempty open subset, we therefore reduce to the case in which ℓ is invertible on S. Then, by XII 5.2 (iii) and X 5.2, the fiber of $R^2 f_*(\mathbf{Z}/\ell)$ at a geometric point $\overline{s}$ above $s \in S$ is isomorphic to $(\mathbf{Z}/\ell)^{c(s)}$, where $c(s)$ is the number of one-dimensional irreducible components of the geometric fiber $X_{\overline{s}}$. The function c is constructible on S (EGA IV 9.7.9), and the result follows (IX 2.13 (iii)).

Remark 1.11. The proof of 1.1 becomes much simpler if one uses the formalism of cohomology with proper supports, which will be developed in XVII as a direct consequence of the base-change theorem for a proper morphism XII 5.1. Moreover, the statement of 1.1 generalizes to the case in which one is given an arbitrary sheaf $\mathscr{A}$ of torsion rings on S and a complex $K^\cdot$ of $f^*(\mathscr{A})$-modules on X satisfying a "constructibility" condition which, in the special case of 1.1, would simply say that the cohomology sheaves $\mathscr{H}^i(K^\cdot)$ of $K^\cdot$ are constructible A-modules and vanish for sufficiently large i.

2. A variant of dimension 157

2.1. Let X be a scheme. We shall write

$$d(x) = \dim \overline{\{x\}} \tag{2.1.1}$$

for $x \in X$, where $\overline{\{x\}}$ is the closure of $\{x\}$. If F is an abelian sheaf on X, we put

$$d(F) = \sup\{d(x) \mid x \in X \quad \text{et} \quad F_{\overline{x}} \neq 0\}. \tag{2.1.2}$$

We shall need a corrected notion in the local case:

Definition 2.2. *Let Y be the spectrum of a universally catenary Noetherian local ring (EGA IV 5.6.2), and let $f : X \to Y$ be a morphism of finite type. Let $x \in X$ and $y = f(x)$. We put*

$$\delta(x) = \dim \overline{\{y\}} + \text{deg.tr. } k(x)/k(y).$$

If F is an abelian sheaf on X, we put

$$\delta(F) = \sup\{\delta(x) \mid x \in X \quad \text{et} \quad F_{\overline{x}} \neq 0\}.$$

Proposition 2.3. (i) *Let $i : X' \to X$ be an immersion and let $x' \in X'$. Then $\delta(i(x')) = \delta(x')$.*

(ii) *Let X_0 be the closed fiber of X/Y. We have*

$$\delta(x) = \begin{cases} d(x) & \text{if } \overline{\{x\}} \cap X_0 \neq \varnothing \\ d(x) + 1 & \text{if } \overline{\{x\}} \cap X_0 = \varnothing; \end{cases}$$

(iii) *$\delta(x) = d(x)$ if f is proper.*

Proof. Assertion (i) follows immediately from the definition. Since $\overline{\{x\}} \cap X_0$ is nonempty 158
when f is proper, (iii) follows from (ii). For (ii), we may replace X by $\overline{\{x\}}$ and Y by $\overline{\{y\}}$

(with their induced reduced structures), as is permitted by (i). Let $x' \in X$ and $y' = f(x')$. We have the formula (EGA IV 5.6.1.1)

$$\dim \mathcal{O}_{X,x'} = \dim \mathcal{O}_{Y,y'} + \text{deg.tr. } k(x)/k(y) - \text{deg.tr. } k(x')/k(y').$$

Suppose that $X_0 \neq \emptyset$. It is then clear that $\sup_{x'} \{\dim \mathcal{O}_{X,x'}\} = \dim X$ is attained by taking x' to be a closed point of X_0, whence

$$\dim X = \dim \mathcal{O}_{X,x'} = \dim Y + \text{deg.tr. } k(x)/k(y) = \delta(x).$$

If $X_0 = \emptyset$, the supremum is plainly attained by taking x' to be a closed point in its fiber $f^{-1}(y')$, with $\dim \overline{\{y'\}} = 1$. Such points exist because the $y' \in Y$ with $\dim \overline{\{y'\}} = 1$ form a very dense set (EGA IV 10.1.3, 10.3.1, 10.4.1, 10.5.9). Thus

$$\begin{aligned} \dim X = \dim \mathcal{O}_{X,x'} &= \dim \mathcal{O}_{Y,y'} + \text{deg.tr. } k(x)/k(y) \\ &= \dim Y - 1 + \text{deg.tr. } k(x)/k(y) = \delta(x) - 1, \end{aligned}$$

which proves the result.

Proposition 2.4. *Let $f : X \to Y$ be a morphism of schemes of finite type over the spectrum of a field k. Let $x \in X$, let $y = f(x)$, and let y_0 be a specialization of y. Let $f' : X' \to Y'$ be the strict localization of f at a geometric point $\overline{y}_0$ above y_0, and let $x' \in X'$ be a point above x. Then*

$$\delta(x') = d(x) - d(y_0).$$

159 *Proof.* We have

$$\begin{aligned} d(x) &= d(y) + \text{deg.tr. } k(x)/k(y) \\ \delta(x') &= d(y') + \text{deg.tr. } k(x')/k(y'), \end{aligned}$$

where $y' = f'(x')$. Moreover,

$$\text{deg.tr. } k(x)/k(y) = \text{deg.tr. } k(x')/k(y'),$$

because the extensions $k(x')/k(x)$ and $k(y')/k(y)$ are algebraic. It therefore remains to prove that

$$d(y') = d(y) - d(y_0).$$

We may assume that y is maximal in Y, and hence that y' is maximal in Y'. Then

$$\begin{aligned} d(y') &= \dim \mathcal{O}_{Y',\overline{y}_0} \\ &= \dim \mathcal{O}_{Y,y_0} \quad \text{(EGA IV 6.1.3)} \\ &= \dim_{y_0} Y - d(y_0) \quad \text{(EGA IV 5.2.3.1)} \\ &= d(y) - d(y_0), \end{aligned}$$

which proves the result.

3. Cohomological dimension of affine algebraic schemes

Theorem 3.1 (D). [37] *Let $f : X \to Y$ be an affine morphism of schemes of finite type over a field k. Let F be a torsion sheaf on X such that $d(F) \leq d$* (cf. 2.1). *Then $d(R^q f_* F) \leq d - q$.*

In particular, taking $Y = \text{Spec}(k)$, we obtain:

160 COROLLARY 3.2. *Let X be an affine scheme of finite type over a separably closed field k. Then*[38]

$$\operatorname{cd} X \leq \dim X.$$

COROLLARY 3.3. *Let $X \subset \mathbf{P}_k^N$ be a projective scheme of dimension n over a separably closed field k, and let $Y = H \cap X$ be a "hyperplane section" of X. Then, for every torsion sheaf F on X, the canonical homomorphism*

$$H_Y^q(X, F) \longrightarrow H^q(X, F)$$

is bijective for $q > n+1$ and surjective for $q = n+1$.

The corollary follows immediately from 3.2 and the exact sequence for a closed subset (V 6.5.4), since $X - Y = U$ is affine of dimension $\leq n$. Notice that 3.3 is a generalization of one of Lefschetz's theorems on hyperplane sections. Indeed, if X is smooth over k and F is locally constant of order prime to the characteristic, the cohomology with support $H_Y^q(X, F)$ can be computed explicitly using the relative cohomological purity theorem (XVI 3) and the spectral sequence (V 6.4.3), and one finds that

$$H_Y^q(X, \mathbf{Z}/n) \simeq H^{q-2}(Y, \mu_n^{-1}) \quad \text{for all } q;$$

the homomorphism $H^{q-2}(Y, \mu_n^{-1}) \to H^q(X, \mathbf{Z}/n)$ furnished by 3.3 is precisely the "Gysin homomorphism", which will be studied later [38].

Theorem 3.1 (d) is equivalent[39] to the following statement:

COROLLARY 3.4 (D). *Let Y' be a strictly local scheme that is a strict localization of an algebraic scheme over a field. Let $f' : X' \to Y'$ be an affine morphism of finite type, and* 161
let F' be a torsion sheaf on X' such that $\delta(F') \leq d$ (cf. (2.2)). *Then $H^q(X', F') = 0$ for $q > d$.*

Indeed, suppose that 3.4 (d) is true. Let $f : X \to Y$ be an affine morphism of algebraic schemes over k, and let F be a torsion sheaf on X such that $d(F) \leq d$. Let y be a point of Y, and let ξ be a geometric point over y. To prove 3.1 (d), it is enough to prove that the stalk $(R^q f_* F)_\xi$ vanishes when $d(y) > d - q$.

Let $f' : X' \to Y'$ be the strict localization of f at the point ξ; that is, let Y' be the strict localization of Y at ξ, and let $X' = X \times_Y Y'$.

Let F' be the sheaf induced by F on X', so that $(R^q f_* F)_\xi = H^q(X', F')$ (VIII 5.2). By (2.4), we have

$$\delta(F') \leq d(F) - d(y) \leq d - d(y),$$

and applying 3.4 (d), we obtain $H^q(X', F') = 0$ if $q > d - d(y)$, which proves the result.

Conversely, suppose that 3.1 (d) is true, and let F' and $f' : X' \to Y'$ be as in the statement of 3.4 (d). Since cohomology commutes with filtered inductive limits (VII 3.3) and F' is the inductive limit of its constructible subsheaves (IX 2.9), we may suppose that F' is constructible. Now Y' is the strict localization of an algebraic scheme Y at

[37] N.D.E.: Theorem 1.1 is now known as the *affine Lefschetz theorem.*

[38] N.D.E.: Consider the analogous assertion in complex analytic geometry: let X be an affine variety over $\mathbf{C}$ and let F be a constructible sheaf on X. Then $H^m(X^{\mathrm{an}}, F) = 0$ for every $m > \dim X$, where X^{an} is the analytic space associated with X. Although this assertion follows from 3.2 by the comparison theorem XVI 4.1, there are also two entirely analytic proofs, due to Pierre Deligne and Hélène Esnault; see Esnault, H., Variation on Artin's vanishing theorem, *Advances in Math.* **198** (2005), 435–438.

[38] Cf. SGA 2 XIV for a systematic study of Lefschetz-type theorems.

[39] N.D.E.: More precisely, the argument in the text gives the following result. Assertion 3.4 (d') for $d' \leq d$ implies 3.1 (d), and assertion 3.1 (d) implies 3.4 (d). This suffices for the conclusion below.

a geometric point ξ over a *closed* point y_0 (X 3.3). Write $Y' = \varprojlim_\alpha Y_\alpha$, where (Y_α) is a filtered inverse system of affine schemes étale over Y (VIII 4.5). By IX 2.7.4, F' is induced by a constructible torsion sheaf F_α on $X_\alpha = X \times_Y Y_\alpha$ for sufficiently large α. Using 2.4 and
162 EGA IV 9.5.5, we see immediately that we may further suppose that $d(F_\alpha) = \delta(F') \leq d$. Replacing Y by Y_α, we are reduced to the case in which F' is induced by a sheaf F on X, with $d(F) \leq d$. Then $R^q f_* F = 0$ for $q > d$ by 3.1 (d), and hence in particular

$$(R^q f_* F)_{\bar{\xi}} = H^q(X', F') = 0 \quad \text{if} \quad q > d,$$

which proves 3.4 (d).

As a special case of 3.4 (d), we have:

Corollary 3.5. *Let Y' be a strictly local scheme of dimension $\leq d$, which is a strict localization of an algebraic scheme over a field k. Let $U \subset Y'$ be an affine open subset. Then* $\operatorname{cd} U \leq d$.

Remark 3.6. It seems very plausible[40] that 3.4, and hence 3.5, remains valid for *every* local Noetherian scheme Y' (not necessarily a strict localization of an algebraic scheme), at least when Y' is *excellent* (EGA IV 7.8). This will be proved in (XIX 6) under the further assumption that Y has characteristic zero.

4. Proof of Theorem 3.1

The proof proceeds by induction on d.

Lemma 4.1. *3.1 (0) and 3.1 (1) hold.*

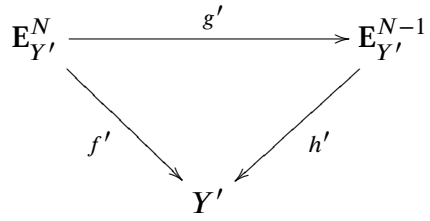
Proof. This is trivial for $d = 0$, in which case supp F is finite over k, and we apply (VIII 5.5). Let F and $f : X \to Y$ be as in the statement of 3.1, and suppose that $d(F) \leq 1$. Replacing F by a constructible subsheaf (IX 2.9 and VII 3.3), we reduce to the case in
163 which F is constructible. The support of F is then contained in a closed subscheme of X of dimension ≤ 1 (IX 2.3), and we may replace X by this subscheme; that is, we may assume that $\dim X \leq 1$. By IX 5.7, the cohomological dimension of an affine algebraic scheme of dimension 1 over a separably closed field is ≤ 1. It is enough, moreover, to treat the case in which k is separably closed. Indeed, let $\overline{f} : \overline{X} \to \overline{Y}$ be the morphism obtained from f by the base change $\operatorname{Spec} \overline{k} \to \operatorname{Spec} k$, where $\overline{k}$ is the separable closure of k. For a geometric point ξ of Y, the strict localization $f' : X' \to Y'$ of f at ξ identifies with a strict localization of $\overline{f}$, and consequently the fiber $(R^q f_* F)_{\bar{\xi}}$ identifies with a fiber of $R^q \overline{f}_* \overline{F}$. Since f is affine, it follows that $R^q f_* F = 0$ for $q > 1$. It is also clear that $d(f_* F) \leq 1$. For $q = 1$, note that the hypothesis $\dim X \leq 1$ implies that there is a finite closed subset Z of Y such that f is finite over $Y - Z$. It follows that $R^1 f_* F$ vanishes on $Y - Z$ (VIII 5.6), and hence $d(R^1 f_* F) \leq 1$. This proves the lemma.

Let $d \geq 2$, and now suppose that 3.1 (d') (or, equivalently, 3.4 (d')) holds for $d' < d$.

Lemma 4.2. *To prove 3.4 (d) in general, it is enough to treat the case in which $X' = \mathbf{E}^1_{Y'}$, is affine one-space over Y' and f is the structural morphism.*

[40] N.D.E.: Ofer Gabber has succeeded in proving the following analogue of 3.1: let $f : X \to Y$ be an affine morphism of finite type, with Y quasi-excellent (EGA IV 7.8) and endowed with a dimension function δ_Y. Let δ_X be the associated rectified dimension function on X. If n is invertible on Y, then for every sheaf F of $\mathbf{Z}/n$-modules on X one has $\delta_Y(R^q f_* F) \leq \delta_X(F) - q$ for every $q \geq 0$. Here, as above, for a sheaf G we define $\delta(G) = \sup\{\delta(x) \mid x \in X \quad \text{et} \quad G_{\bar{x}} \neq 0\}$.

Proof. We may plainly assume that $X' = \mathbf{E}^N_{Y'}$ for a suitable N, since X' can be embedded in some $\mathbf{E}^N_{Y'}$. Suppose that $N > 1$ and that the result is known, for every Y', for $\mathbf{E}^r_{Y'}$, with $r < N$. Let F' be a torsion sheaf on $\mathbf{E}^N_{Y'}$ with $d(F') \leq d$. Consider the diagram 164



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where g' and h' are the canonical projections, so that $\mathbf{E}^N_{Y'}$ is isomorphic to affine one-space over $\mathbf{E}^{N-1}_{Y'}$. Since every strict localization of $\mathbf{E}^{N-1}_{Y'}$ is plainly also the strict localization of an algebraic scheme, we may apply the induction hypothesis to the fibers of $R^q g'_* F'$, and obtain

$$(*) \qquad \delta(R^q g'_* F') \leq d - q.$$

Moreover, 3.4 (d') holds for h' and $d' \leq d$, by the induction hypothesis on N.

Consider the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbf{E}^{N-1}_{Y'}, R^q g'_*(F')) \Longrightarrow H^n(\mathbf{E}^N_{Y'}, F').$$

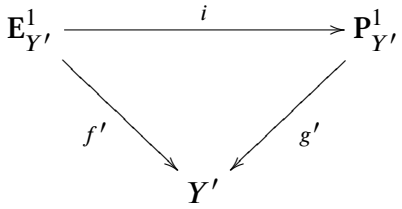
We have $E_2^{p,q} = 0$ if $p > d - q$, as follows by applying $(*)$ and 3.4 (d') to the morphism h'. Consequently the abutment vanishes for $n > d$, proving the lemma.

Consider the following additional statement, which is a special case of 3.5:

Statement 4.3 (d). *Let X' be strictly local of dimension $\leq d$, and the strict localization of an algebraic scheme over a field. Let $f \in \Gamma(X', \mathcal{O}_{X'})$, and let $U \subset X'$ be the open subset $U = X' - V(f)$. Then* $\operatorname{cd} U \leq d$.

Lemma 4.4. 4.3 (d) *implies* 3.4 (d).

Proof. Consider the diagram 165



where Y' is as in the statement of 3.4, and where $\mathbf{P}^1_{Y'}$ is the projective line obtained from $\mathbf{E}^1$ "by adjoining the section Y'_∞ at infinity." Let F' be a torsion sheaf on $\mathbf{E}^1$ with $d(\operatorname{supp} F') \leq d$. We have the spectral sequence

$$H^p(\mathbf{P}^1, R^q i_* F') \Longrightarrow H^n(\mathbf{E}^1, F').$$

Now the $R^q i_* F'$, for $q > 0$, are supported on Y'_∞, which is Y'-isomorphic to Y'. Since Y' is strictly local, it follows that $H^p(\mathbf{P}^1, R^q i_* F') = 0$ if $p, q > 0$. Moreover, by the *proper base-change theorem* (XII 5.1), the cohomology of a torsion sheaf on $\mathbf{P}^1$ can be computed on the closed fiber, which is a scheme of cohomological dimension ≤ 2 (X 4.3). Thus $H^p(\mathbf{P}^1, i_* F') = 0$ if $p > 2$. Since we wish to prove that $H^n(\mathbf{E}^1, F') = 0$ for $n > d$, and since $d \geq 2$, we are reduced to proving that $H^0(\mathbf{P}^1, R^q i_* F') = 0$ for $q > d$. This group, which is also isomorphic to $H^0(Y'_\infty, R^q i_* F')$, is, by (VIII 4.6), isomorphic to

the fiber of $R^q i_* F'$ at the point Q, where Q is the point at infinity of $\mathbf{P}^1$ in the closed fiber.

It is enough (VII 3.3 and IX 2.9) to treat the case in which F' is constructible. Since $\delta(\operatorname{supp} F') \leq d$, there is then a closed subscheme X' of $\mathbf{P}^1$, with $\dim X' \leq d$, which contains the support of F'. Let $\widetilde{X}'$ be the localization of X' at the geometric point Q, put

$$\widetilde{U} = \widetilde{X}' - \widetilde{X}' \times_{\mathbf{P}^1} Y'_\infty,$$

166 and let $\widetilde{F}$ be the sheaf on $\widetilde{U}$ induced by F' via the morphism $\widetilde{U} \to \mathbf{P}^1$. The fiber of $R^q i_* F'$ at Q is then precisely $H^q(\widetilde{U}, \widetilde{F})$ (VIII 5.2), and we apply 4.3 (d) to the strictly local scheme $\widetilde{X}'$ and the open subset $\widetilde{U}$.

The following lemma, together with 4.4, will complete the proof.

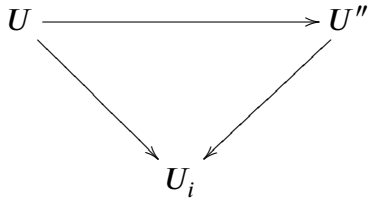
LEMMA 4.5. *3.4 (d') for $d' < d$ implies 4.3 (d') for $d' \leq d$.*

Proof. Suppose that 3.4 (d') holds for $d' < d$. By induction, we may suppose that 4.3 (d') holds for $d' < d$. Let X', f, and U be as in the statement 4.3 (d). Then X' is the strict localization, at a *closed* geometric point, of an algebraic scheme X_0 over a field k. After replacing X_0 by a neighborhood of x_0, we may take X_0 to be affine and of dimension $\leq d$. Then X' is the inverse limit of affine schemes X_i which are étale over X_0 (VIII 4.5).

Let F be a torsion sheaf on U. We must prove that $H^q(U, F) = 0$ for $q > d$, and once again it is enough to treat the case in which F is constructible. The data f, F, and U then arise from analogous data on one of the X_i (IX 2.7.4). Thus there is a sheaf F_i on $U_i = X_i - V(f_i)$ which induces F on U. Since $U = \varprojlim U_i$, we have (VII 5.8)

$$H^q(U, F) = \varinjlim H^q(U_i, F_i).$$

It is therefore enough to prove the following fact: for every i, there is a commutative diagram



167 in which $U \to U_i$ is the given arrow and $\operatorname{cd}(U'') \leq d$. Indeed, this implies that every element of $H^q(U_i, F_i)$, for $q > d$, has zero image in $H^q(U, F)$.

Suppose that $i = 0$, and consider the morphism $g : X_0 \to \mathbf{E}^1_k = Y_0$ defined by $f_0 \in \Gamma(X_0, \mathcal{O}_{X_0})$. Let Y'' be the strict localization of $\mathbf{E}^1_k$ at a geometric point above the origin. Thus Y'' is the spectrum of a discrete valuation ring. Put $X'' = X_0 \times_{Y_0} Y''$.

The morphism $X' \to X_0$ factors through X'', since X' is strictly local. Moreover, the open subset $U'' = X'' \times_{X_0} U_0$ is precisely the generic fiber of X''/Y''. It is therefore an affine algebraic scheme over the generic point of Y'', and plainly $\dim U'' \leq d - 1$.

Let us verify that $\operatorname{cd} U'' \leq d$, which will complete the proof. Let K be the residue field of Y'' at its generic point. This field has cohomological dimension 1; that is, $\operatorname{cd}(G) = 1$, where G is the Galois group of the separable closure $\overline{K}$ of K (X 2.2). Put $\overline{U}'' = U'' \times_{\operatorname{Spec} K} (\operatorname{Spec} \overline{K})$. By the induction hypothesis on d, we have $\operatorname{cd} \overline{U}'' \leq d - 1$ (this is 3.2 for a scheme of dimension $< d$). It then follows from the Hochschild–Serre spectral sequence (VIII 8.4) that $\operatorname{cd} U'' \leq d$:

$$H^p(G, H^q(\overline{U}'', \overline{F})) \Longrightarrow H^n(U'', F),$$

which proves Lemma 4.5.

EXPOSÉ XV

Acyclic morphisms

M. Artin

Introduction

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Let $g : X \to Y$ be a morphism of schemes. In the first section we study conditions under which g is acyclic, that is, such that for every torsion sheaf F on Y one has $H^q(Y, F) \simeq H^q(X, g^*F)$, and such that this remains true when Y is replaced by a Y' étale over Y. A natural necessary condition for acyclicity is that the geometric fibers be acyclic, that is, have trivial cohomology for constant torsion sheaves. We shall see that this condition, together with a local condition called local acyclicity of g (1.11), implies that g is acyclic.

The fundamental theorem is that a smooth morphism is locally acyclic for torsion sheaves of order prime to the residue characteristics 2.1. This will imply the base-change theorem for a smooth morphism XVI 1.1, which will be developed, together with its first consequences, in the next exposé.

The present exposé is independent of Exposés XI to XIV, and in particular of the "base-change theorem for a proper morphism." The latter will again enter in an essential way, but in conjunction with the fundamental theorem of the present exposé, beginning with the next exposé.

1. Generalities on globally and locally acyclic morphisms

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Proposition 1.1. *Let $g : X \to Y$ be a morphism of schemes. The following assertions* (i) *to* (iii) *are equivalent:*

(i) *For every étale morphism $Y' \to Y$ (where Y' may be taken affine over an affine open subset of Y) and every sheaf of sets F on Y', the canonical morphism $F \to g'_* g'^* F$ is injective (resp. bijective).*

(ii) *For every étale morphism $Y' \to Y$ (where Y' may be taken affine over an affine open subset of Y) and every sheaf of sets F on Y', the canonical morphism $H^0(Y', F) \to H^0(X', g'^* F)$ is injective (resp. bijective).*

(iii) *For every locally quasi-finite morphism $Y' \to Y$ and every sheaf of sets F on Y', the canonical morphism $H^0(Y', F) \to H^0(X', g'^* F)$ is injective (resp. bijective).*

(ii bis) *(If g is quasi-compact and quasi-separated.) As in* (ii), *but with F a constant sheaf of the form $I_{Y'}$, where I is a prescribed set with* card $I \geq 2$.

Proof. The equivalence of (i) and (ii) follows immediately from the definitions, while (iii) $\Rightarrow$ (ii) $\Rightarrow$ (ii bis) are trivial. We have (ii bis) $\Rightarrow$ (ii), since we may suppose Y affine, so that X and Y are quasi-compact and quasi-separated, and may then apply XII 6.5 (i). For the implication (i) $\Rightarrow$ (iii), we first record the following lemma:

Lemma 1.2. *Let $f : Y' \to Y$ be a locally quasi-finite morphism. Then f is "locally* 170

finite on Y' for the étale topology," that is, there are commutative diagrams

$$(*) \qquad \begin{array}{ccc} Y_i' & \xrightarrow{a_i} & Y' \\ \downarrow{\scriptstyle b_i} & & \downarrow{\scriptstyle f} \\ Y_i & \xrightarrow{c_i} & Y \end{array}$$

in which the a_i and c_i are étale, the b_i are finite, and the a_i form a covering of Y' [40].

Indeed, let y' be a point of Y' and let $y = f(y')$. It is enough to find a diagram $(*)$ such that y' lies in the image of Y_i'. Let $\widetilde{Y}$ be the spectrum of the henselization of the ring $\mathcal{O}_{Y,y}$ VIII 4.1, and let Z be the (ordinary) localization of $\widetilde{Y}' = Y' \times_Y \widetilde{Y}$ at the unique point lying over y' and the closed point $\tilde{y}$ of $\widetilde{Y}$. By VIII 4.1 (ii), $Z \to \widetilde{Y}$ is finite, and Z is an induced open subscheme of $\widetilde{Y}'$. Since $\widetilde{Y}$ is a limit of schemes Y_i étale over Y, one sees immediately from EGA IV 9 that, for a suitable Y_i, there is an open subscheme Z_i of $Y' \times_Y Y_i$ such that $Z = Z_i \times_{Y_i} \widetilde{Y}$; in $(*)$ it is then enough to take $Y_i' = Z_i$. C.Q.F.D.

Now suppose that (i) of 1.1 holds. It is clear that assertion (iii) of 1.1 is equivalent to the assertion obtained from (i) by replacing the étale morphism $Y' \to Y$ by a locally quasi-finite morphism. In this form, the assertion (for given Y and Y') is local on Y and Y' for the étale topology; hence the lemma reduces us to the case of a *finite* morphism $f : Y' \to Y$. Let $f' : X' \to X$ be the morphism obtained from f by base change along g. Then the base-change morphism $g^* f_* F \to f'_* g'^* F$ is bijective VIII 5.6. By (i), we have $f_*(F) \to g_* g^* f_* F$. Consequently, the morphism $H^0(Y', F) \xrightarrow{\epsilon} H^0(Y', g'_* g'^* F)$ in (iii) is the composite $H^0(Y', F) = H^0(Y, f_* F) \xrightarrow{\epsilon'} H^0(Y, g_* g^* f_* F) = H^0(X, g^* f_* F) \simeq H^0(X, f'_* g'^* F) = H^0(X', g'^* F)$. Since ϵ' is injective (resp. bijective), the same is true of ϵ. This completes the proof of 1.1.

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Definition 1.3. *A morphism $g : X \to Y$ having one of the equivalent properties* (i) *to* (iii) *(resp. the corresponding parenthesized properties) of 1.1 is called a* (-1)-acyclic morphism *(resp. a* 0-acyclic morphism*). The morphism is said to be* universally (-1)-*acyclic (resp.* 0*-acyclic) if, for every base-change morphism $Y' \to Y$, the morphism $g' = g \times_Y Y' : X \times_Y Y' \to Y'$ is (-1)-acyclic (resp.* 0*-acyclic). Finally, a scheme X is said to be (-1)-acyclic (resp.* 0*-acyclic) if it is nonempty (resp. connected and nonempty).*

Corollary 1.4. *Let $g : X \to Y$ be a* 0*-acyclic morphism. Then for every locally quasi-finite morphism $Y' \to Y$ and every sheaf of groups F on Y', the canonical morphism*

$$H^1(Y', F) \longrightarrow H^1(X', g'^* F)$$

is injective.

This follows immediately from XII 6.5 (i).

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Corollary 1.5. *Let $g : X \to Y$ be a morphism. If g is surjective (i.e. its fibers are nonempty, i.e. its fibers are (-1)-acyclic), then g is (-1)-acyclic. The converse holds if g is quasi-compact and quasi-separated.*

The first assertion is clear in the form 1.1 (ii), by considering the fibers of the sheaves in question. The second follows from the definition in the form 1.1 (iii) and the limit

[40] Cf. EGA IV 18.12.1.

argument of VII 5, taking into account that, for every $y \in Y$, $y = \operatorname{Spec} k(y)$ is the inverse limit of affine schemes locally quasi-finite over Y, namely the affine neighborhoods of y in $\overline{\{y\}}$ (endowed with the induced reduced structure). – Note that 1.5 implies that, when g is quasi-compact and quasi-separated, the (-1)-acyclicity of g already implies that g is *universally* (-1)-acyclic.

PROPOSITION 1.6. *Let $g : X \to Y$ be a morphism of schemes, let $\mathbf{L} \subset \mathbf{P}$ be a set of prime numbers, and let $n \in \mathbf{N}$ be a natural number (resp. let $n = 1$). The following assertions* (i) *to* (iii) *(resp. the corresponding parenthesized assertions) are equivalent. If, moreover, g is quasi-compact and quasi-separated, then* (i) *to* (iii) *(resp. the corresponding parenthesized assertions) are also equivalent to* (iv) *(resp. the corresponding parenthesized assertion).*

(i) *For every étale morphism $Y' \to Y$ and every abelian $\mathbf{L}$-torsion sheaf (resp. sheaf of* ind-$\mathbf{L}$*-groups)* IX 1.5 *F on Y', one has*

$$\begin{cases} F \xrightarrow{\sim} g'_* g'^* F, \\ (R^q g'_*) g'^* F = 0 \quad \text{if} \quad 1 \le q \le n \quad (\text{resp. if } q = 1). \end{cases}$$

(ii) *For every étale morphism $Y' \to Y$ and every abelian $\mathbf{L}$-torsion sheaf (resp. sheaf of ind-$\mathbf{L}$-groups) F on Y', the canonical morphism* 173

$$H^q(Y', F) \longrightarrow H^q(X', g'^* F)$$

is bijective if $q \le n$ and injective if $q = n + 1$ (resp. is bijective if $q \le 1$).

(iii) *For every locally quasi-finite morphism $Y' \to Y$ and every abelian $\mathbf{L}$-torsion sheaf (resp. sheaf of ind-$\mathbf{L}$-groups) F on Y', the morphism*

$$H^q(Y', F) \longrightarrow H^q(X', g'^* F)$$

is bijective if $q \le n$ and injective if $q = n + 1$ (resp. is bijective if $q \le 1$).

(iv) *The morphism g is surjective, and for every locally quasi-finite morphism $Y' \to Y$, every $\nu > 0$, and every $\ell \in \mathbf{L}$, the canonical morphism*

$$H^q(Y', \mathbf{Z}/\ell^\nu) \longrightarrow H^q(X', \mathbf{Z}/\ell^\nu)$$

is surjective *for $0 \le q \le n$ (resp. for every locally quasi-finite morphism $Y' \to Y$ and every ordinary finite $\mathbf{L}$-group G, the canonical morphism*

$$H^i(Y', G) \longrightarrow H^i(X', G)$$

is surjective *for $i = 0, 1$).*

Proof of 1.6. The implications (ii) $\Leftarrow$ (iii) $\Rightarrow$ (iv) are trivial. Now $(R^q g'_*) g'^* F$ is the sheaf 174
associated with the presheaf $R^q(Y'') = H^q(X'', g''^* F)$ ($Y'' \to Y'$ étale). If (ii) holds, we have $H^q(Y'', F) \xrightarrow{\sim} H^q(X'', g''^* F)$, so the associated sheaf is zero, i.e. (ii) $\Rightarrow$ (i). The implication (i) $\Rightarrow$ (ii) follows from 1.1 (i) (which corresponds to the case $n = 0$), together with the Leray spectral sequence

$$H^p(Y', (R^q g'_*) g'^* F) \Longrightarrow H^{p+q}(X', g'^* F)$$

in the abelian case, and the exact sequence

$$0 \longrightarrow H^1(Y', g'_* g'^* F) \longrightarrow H^1(X', g'^* F) \longrightarrow H^0(Y', (R^1 g'_*) g'^* F)$$

in the case of sheaves of groups.

(i) $\Rightarrow$ (iii): Since (i) and (ii) are equivalent, it is clear that (iii) is equivalent to the statement obtained from (i) by replacing the word "étale" by the words "locally quasi-finite." In this form, for given Y and Y', it is an assertion local on Y' and on Y for the étale topology, so 1.2 reduces us to the case in which $f : Y' \to Y$ is a *finite* morphism.

Let $f' : X' \to X$ be the morphism . By VIII 5.5 (resp. VIII 5.8) (applied three times), we have

$$f_*(R^q g'_*)g'^* F \simeq R^q(fg')_* g'^* F \simeq R^q(gf')_* g'^* F \simeq (R^q g_*) f'_* g'^* F \simeq (R^q g_*) g^*(f_* F)$$

and the last of these is zero, for the values of q in question, if (i) holds. Thus (i) ⇒ (iii).

175 It remains to prove (iv) ⇒ (iii), and this follows immediately from XII 6.5.

Definition 1.7. *Let $n \in \mathbf{N}$ and $\mathbf{L} \subset \mathbf{P}$. A morphism $g : X \to Y$ satisfying one of the equivalent conditions* (i) *to* (iii) *(resp. the corresponding parenthesized conditions) of 1.6 is called an* n-acyclic morphism *(resp. a* 1-aspherical morphism*) for* $\mathbf{L}$. *We say that g is* acyclic *for* $\mathbf{L}$ *if it is n-acyclic for* $\mathbf{L}$ *for every n. We say that g is* universally n-acyclic *(resp.* universally 1-aspherical*) for* $\mathbf{L}$ *if, for every $Y' \to Y$, the morphism $g' = g \times_Y Y'$ is n-acyclic (resp. 1-aspherical) for* $\mathbf{L}$. *Finally, a scheme X is said to be* n-acyclic *(resp.* 1-aspherical*) for* $\mathbf{L}$ *if it is 0-acyclic in the sense of 1.3 and if, for every $\ell \in \mathbf{L}$ and every $1 \leq q \leq n$, one has $H^q(X, \mathbf{Z}/\ell) = 0$ (resp. and if, for every ordinary finite $\mathbf{L}$-group G, one has $H^1(X, G) = 0$).*

Remarks 1.8.

a) If g is quasi-compact and quasi-separated, then, in order for g to be universally n-acyclic, it is enough that g' be n-acyclic whenever $Y' \to Y$ is locally of finite presentation. We do not need this fact and leave its proof, which is an exercise in passage to the limit, to the reader.

b) We do not know whether a morphism that is n-acyclic for $\mathbf{L}$ is necessarily universally n-acyclic for $\mathbf{L}$. The same question arises for the local variants below 1.11.

c) When $n = 0$, we recover the notion of 1.3. Of course, 1.6 and 1.7 could have been formulated so as to include the case $n = -1$. The following proposition is

176 also valid in that case.

Proposition 1.9.

(i) *Let $X \xrightarrow{g} Y \xrightarrow{h} Z$ be morphisms of schemes, let $\mathbf{L} \subset \mathbf{P}$, and let $n \in \mathbf{N}$. Suppose that g is 0-acyclic (resp. 1-aspherical for $\mathbf{L}$, resp. n-acyclic for $\mathbf{L}$). Put $f = hg$. Then, for every sheaf of sets (resp. of ind $\mathbf{L}$-groups, resp. abelian $\mathbf{L}$-torsion sheaf) F on Z, there is an isomorphism $f_* f^* F \xrightarrow{\sim} h_* h^* F$ (resp. isomorphisms $(R^p f_*) f^* F \xrightarrow{\sim} (R^p h_*) h^* F$ for $p \leq 1$, resp. for $p \leq n$). In particular, h is 0-acyclic (resp. 1-aspherical for $\mathbf{L}$, resp. n-acyclic for $\mathbf{L}$) if and only if f is.*

(ii) *Let $\{g_\alpha : X_\alpha \to Y_\alpha\}$ be an inverse system of morphisms of schemes such that the morphisms $X_\alpha \to X_\beta$ and $Y_\alpha \to Y_\beta$ are affine and the g_α are quasi-compact and quasi-separated. Let $g : X \to Y$ be the inverse limit of the g_α VII 5.1. Then, if all the g_α are 0-acyclic (resp. 1-aspherical for $\mathbf{L}$, resp. n-acyclic for $\mathbf{L}$), the same is true of g.*

Proof. For (i), first observe that

$$h_* h^* F \xrightarrow{\sim} h_*(g_* g^*) h^* F \simeq f_* f^* F.$$

The second assertion of (i) is an immediate consequence of the Leray spectral sequence

$$(R^p h_*)(R^q g_*) f^* F \Longrightarrow (R^{p+q} f_*) f^* F.$$

Finally, the nonabelian assertion follows from the exact sequence XII 3.2.

177 Assertion (ii) follows readily from the theory of passage to the limit VII 5 and VI.

Corollary 1.9.1. *Let $g : X \to Y$ be a quasi-compact and quasi-separated morphism, and let $n \in \mathbf{N}$, $\mathbf{L} \subset \mathbf{P}$. If g is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$), then, for every geometric point $\overline{y}$ of Y algebraic over a point y of Y, the geometric fiber $X_{\overline{y}}$ is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$).*

Indeed, $\overline{y}$ is the inverse limit of affine schemes Y_i that are locally quasi-finite over Y, and since the $X_i = X \times_Y Y_i \to Y_i$ are n-acyclic for $\mathbf{L}$ (resp. ...), the same holds for $X_{\overline{y}} \to \overline{y}$ by 1.9 (ii), which gives the asserted conclusion.

Proposition 1.10. *Let $g : X \to Y$ be a morphism of schemes, let $\mathbf{L} \subset \mathbf{P}$, and let $n \in \mathbf{N}$. Then conditions* (i) *through* (iv) *below are equivalent.*

(i) *Let $\overline{y}$ be a geometric point of Y and $\overline{x}$ a geometric point of X lying over $\overline{y}$. Let $\widetilde{X}$ and $\widetilde{Y}$ be the strict localizations of X and Y at $\overline{x}$ and $\overline{y}$, respectively, and let $\tilde{g} : \widetilde{X} \to \widetilde{Y}$ be the morphism induced by g. Then $\tilde{g}$ is 0-acyclic (resp. 1-aspherical for $\mathbf{L}$, resp. n-acyclic for $\mathbf{L}$).*

(ii) *For every diagram with Cartesian squares*

(∗)
$$\begin{array}{ccccc} X & \xleftarrow{j} & X' & \xleftarrow{j'} & X'' \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g''} \\ Y & \xleftarrow{i} & Y' & \xleftarrow{i'} & Y'' \end{array}$$

in which i is etale and i' is etale and of finite presentation, and for every sheaf 178
of sets (resp. of ind-$\mathbf{L}$-groups, resp. an abelian $\mathbf{L}$-torsion sheaf) F on Y'', the base-change morphism XII 4.1, XII 4.2

$$g'^* i'_* F \longrightarrow j'_* g''^* F$$

is bijective (resp. the morphisms

$$g'^* (R^q i'_*) F \longrightarrow (R^q j'_*) g''^* F$$

are bijective for $q = 0, 1$, resp. are bijective for $0 \leq q \leq n$ and injective for $q = n + 1$).

(iii) *The same statement as* (ii), *except that the morphism i' is assumed only quasi-finite and quasi-separated.*

(iv) *The same statement as* (ii), *but with i assumed only locally quasi-finite and, on the other hand, i' a quasi-compact open immersion.*

Proof. The implication (iii) ⇒ (ii) is trivial. Let us verify that (i) ⇒ (iii). It is enough to check the morphism in (iii) fiber by fiber. Let $\overline{y}$ be a geometric point of Y' and $\overline{x}$ a geometric point of X' lying over $\overline{y}$. Let

$$\begin{array}{ccc} \widetilde{X}' & \xleftarrow{\tilde{j}'} & \widetilde{X}'' \\ \downarrow{\scriptstyle \tilde{g}'} & & \downarrow{\scriptstyle \tilde{g}''} \\ \widetilde{Y}' & \xleftarrow{\tilde{i}'} & \widetilde{Y}'' \end{array}$$

be the Cartesian diagram deduced from (∗), where $\widetilde{X}'$ and $\widetilde{Y}'$ are the strict localizations of X' and Y' at $\overline{x}$ and $\overline{y}$, respectively, and where $\widetilde{Y}'' = Y'' \times_{Y'} \widetilde{Y}'$, $\widetilde{X}'' = X'' \times_{X'} \widetilde{X}'$. By VIII 5.2 and VIII 5.3, we have 179

$$H^q(\widetilde{Y}'', \widetilde{F}) \simeq (R^q {i'}_* F)_{\overline{y}} \simeq (g'^*(R^q i'_*)F)_{\overline{x}},$$

where $\widetilde{F}$ is the inverse image of F on $\widetilde{Y}''$, and

$$H^q(\widetilde{X}'', \tilde{g}''^* \widetilde{F}) \simeq ((R^q j'_*) g''^* F)_{\overline{x}}.$$

Since $\tilde{i}'$ is quasi-finite, we may apply 1.1 (iii) (resp. 1.5 (iii)) to the morphism

$$H^q(\widetilde{Y}', \widetilde{F}) \longrightarrow H^q(\widetilde{X}', \tilde{g}'^* \widetilde{F})$$

for the values of q under consideration, which proves (i) ⇒ (iii).

(ii) ⇒ (i). Assume (ii) and let us verify the acyclicity criterion 1.6 (ii) for a morphism $\tilde{g} : \widetilde{X} \to \widetilde{Y}$. Let $\widetilde{Y}' \to \widetilde{Y}$ be etale, with $\widetilde{Y}'$ affine, and let $\widetilde{F}$ be a sheaf of sets (resp. an abelian **L**-torsion sheaf, resp. a sheaf of ind-**L**-groups) on $\widetilde{Y}'$. We must prove that

$$H^q(\widetilde{Y}', \widetilde{F}) \xrightarrow{\sim} H^q(\widetilde{X}', \tilde{g}'^* \widetilde{F}),$$

where $\tilde{g}' : \widetilde{X}' \to \widetilde{Y}'$ is the morphism obtained from $\tilde{g}$ by base change. Now write $\widetilde{Y} = \varprojlim Y_\alpha$, where the Y_α are affine and etale over Y (assuming Y affine, as we may). We can "descend" the etale morphism $\widetilde{Y}' \to \widetilde{Y}$ to one of the Y_α, say to $Y'_0 \to Y_0$, and we have $\widetilde{Y}' = \varprojlim Y'_\alpha = Y \times_{Y_0} Y'_0$. Every sheaf $\widetilde{F}$ on $\widetilde{Y}'$ is the direct limit of the sheaves ${f'_\alpha}^* f'_{\alpha *} \widetilde{F}$, where $f'_\alpha : \widetilde{Y}' \to Y'_\alpha$ is the canonical morphism. By IX 1.2 and IX 1.6, the sheaves $f'_{\alpha *} \widetilde{F}$ are sheaves of sets (resp. abelian **L**-torsion sheaves, resp. sheaves of ind-**L**-groups). In
180 view of VII 3.3, it is therefore enough to treat the case where $\widetilde{F}$ descends to one of the Y'_α, say to a sheaf F_0 on Y'_0. Applying (ii) to the diagram

$$\begin{array}{ccccc} X & \longleftarrow & X_0 & \xleftarrow{j'_0} & X'_0 \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g_0} & & \downarrow{\scriptstyle g'_0} \\ Y & \longleftarrow & Y_0 & \xleftarrow{i'_0} & Y'_0, \end{array}$$

we deduce that ${g_0}^*(R^q i'_{0*}) F_0 \xrightarrow{\sim} (R^q j'_{0*}) {g'_0}^* F_0$ for the applicable values of q. The geometric points $\overline{y}$ and $\overline{x}$ lift canonically to geometric points of Y_0 and X_0 (which we denote by the same symbols), and by VIII 5.2 and VIII 5.3 we have

$$H^q(\widetilde{Y}', \widetilde{F}) = ((R^q i'_{0*}) F_0)_{\overline{y}} = ({g_0}^*(R^q i'_{0*}) F_0)_{\overline{x}}$$

and

$$H^q(\widetilde{X}', \tilde{g}'^* \widetilde{F}) = ((R^q j'_{0*}) {g'_0}^* F_0)_{\overline{x}}$$

for the values of q under consideration, which proves the result.

It remains to show that conditions (i) through (iii) are equivalent to (iv). Using form (i), we see that these conditions are stable under a locally quasi-finite base change $Y' \to Y$; in form (ii), this immediately shows that they imply (iv). Conversely, (iv) implies (iii), since, in order to verify (iii), it follows at once—using, for example, the Leray spectral sequence or the descent result XII 6.8—that we may assume Y' and Y'' affine. But then, by the "Main Theorem" EGA IV 8.12.8, $Y'' \to Y'$ factors as $Y'' \xrightarrow{i'_1} Y'_1 \xrightarrow{i'_2} Y'$, with i'_1 an
181 open immersion and i'_2 a *finite* morphism. Using VIII 5.5, VIII 5.8 for i'_2, we are reduced to verifying (ii) for i'_1, which follows from (iv). This completes the proof of 1.10.

DEFINITION 1.11. *A morphism $g : X \to Y$ of preschemes satisfying any one of the equivalent conditions* (i) *through* (iv) *of* 1.10 *is called a* locally 0-acyclic *morphism (resp.* locally 1-aspherical *for* **L**, *resp.* locally n-acyclic *for* **L**). *In the obvious way (compare 1.7), one defines the notions of a* locally acyclic *morphism and of a* universally locally 0-acyclic *morphism (resp. ...) for* **L**.

The statements (i) through (iii) of 1.10 can of course be varied. Thus, in (iii) and (iv), it is enough to restrict to a constant sheaf F of the form G_I, with G a finite set (resp. a finite **L**-group, resp. G of the form $\mathbf{Z}/\ell^\nu\mathbf{Z}$, with $\ell \in \mathbf{L}$, $\nu \in \mathbf{N}$); this follows readily from VIII 5.2, VIII 5.3, and XII 6.5. We also record the following variant:

COROLLARY 1.12. *Suppose that $g : X \to Y$ is locally of finite type. To verify local 0-acyclicity (resp. ...), it is enough to verify* 1.10 (i) *in the case where the geometric point $\overline{x}$ is* closed *in the geometric fiber $X_{\overline{y}}$.*

Indeed, in the proof of (i) $\Rightarrow$ (iii) of 1.10, we verified (iii) fiber by fiber, and it was enough to do so for the fibers at geometric points $\overline{x}$ that are closed in their geometric fiber, since g is locally of finite type VIII 3.13 b). 182

PROPOSITION 1.13. (i) *Let $X \xrightarrow{g} Y \xrightarrow{h} Z$ be morphisms of schemes, let $\mathbf{L} \subset \mathbf{P}$, and let $n \in \mathbf{N}$. Suppose that g is surjective and locally 0-acyclic (resp. locally 1-aspherical for* **L**, *resp. locally n-acyclic for* **L**). *Then h is locally 0-acyclic (resp. ...) if and only if $f = hg$ is.*

(ii) *Let $\{g_\alpha : X_\alpha \to Y_\alpha\}$ be an inverse system of morphisms of preschemes such that the transition morphisms $X_\alpha \to X_\beta$ and $Y_\alpha \to Y_\beta$ are affine. Let g be the inverse limit of the $\{g_\alpha\}$ VII 5.1. If every g_α is locally 0-acyclic (resp. ...), then so is g.*

This follows by a chain of reasoning analogous to 1.9.

REMARKS 1.14. To simplify his task, the author omitted from 1.10 through 1.13 a treatment of the case $n = -1$, and leaves it to the reader to verify that the statements, definitions, and proofs apply to this case essentially without change. Thus, with the notation of 1.10, condition (i) says that the induced local morphisms $\widetilde{X} \to \widetilde{Y}$ are (-1)-acyclic, i.e., by 1.5, *surjective*, and conditions (ii) through (iv) say that the base-change morphism ${g'}^*(i'_*(F)) \to j'_*({g''}^*(F))$ is a *monomorphism*. Under these conditions, we shall therefore say that g is *locally* (-1)*-acyclic*. This condition is thus stable under lo- 183
cally quasi-finite base change. As an example, a flat morphism is locally (-1)-acyclic, since for such a morphism the surjectivity of the maps $\widetilde{X} \to \widetilde{Y}$ is a well-known consequence of their flatness. We also note that, when f is locally of finite presentation, f is locally (-1)-acyclic if and only if it is *universally open*, at least when Y is locally Noetherian; this follows readily from EGA IV 14.4.9, 1.10.3. For such a morphism, local (-1)-acyclicity consequently implies *universal* local (-1)-acyclicity. (It is moreover very likely[40] that, in these last statements, the hypothesis that Y be locally Noetherian is unnecessary; this is at any rate so if the set of irreducible components of Y is locally finite EGA IV 14.4.10.)

THEOREM 1.15. *Let $g : X \to Y$ be a quasi-compact and quasi-separated morphism, let $\mathbf{L} \subset \mathbf{P}$ be a set of prime numbers, and let n be a natural number (resp. $n = 1$). Suppose that g is $(n-1)$-acyclic for* **L** *and locally $(n-1)$-acyclic for* **L** 1.3, 1.7, 1.11, 1.14. *Let F be a sheaf on Y, which, for $n \geq 1$, is assumed to have the structure of an abelian* **L***-torsion sheaf (resp. the structure of a sheaf of groups), and let ξ be an element of $H^n(X, g^*(F))$. In order*

[40] This was in fact verified by M. Raynaud. Cf. the new edition of EGA IV 14!

that ξ come from an element of $H^n(Y, F)$ (which is then uniquely determined, thanks to the $(n-1)$-acyclicity hypothesis on g), it is necessary and sufficient that, for every geometric point $\overline{y}$ of Y algebraic over a point y of Y, the image $\xi_{\overline{y}}$ of ξ in $H^n(X_{\overline{y}}, F|X_{\overline{y}})$ lie in the
184 *image of $H^n(\overline{y}, F_{\overline{y}})$ (which, for $n \geq 1$, means that $\xi_{\overline{y}} = 0$).*

We note at once that 1.15 shows that, under the preliminary conditions imposed on g, the morphism g is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$) if and only if, for every $\overline{y}$ as above, $X_{\overline{y}}$ is n-acyclic for $\mathbf{L}$ (resp. is 1-aspherical for $\mathbf{L}$): the "if" direction follows directly from 1.15 and the definitions, while the "only if" direction is in any case immediate from 1.9.1. With this said, an immediate induction on n gives the following result.

COROLLARY 1.16. *Let g be a quasi-compact and quasi-separated morphism that is locally $(n-1)$-acyclic for $\mathbf{L}$, where n is a given natural number (resp. where $n = 1$). Then g is n-acyclic for $\mathbf{L}$ (resp. is 1-aspherical for $\mathbf{L}$) if and only if, for every geometric point $\overline{y}$ of Y algebraic over a point y of Y, the fiber $X_{\overline{y}}$ is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$).*

COROLLARY 1.17. *Let $g : X \to Y$ be a morphism of preschemes, let $\mathbf{L} \subset \mathbf{P}$ be a set of prime numbers, and let n be a natural number (resp. $n = 1$). In order that g be locally n-acyclic for $\mathbf{L}$ (resp. locally 1-aspherical for $\mathbf{L}$), it is necessary and sufficient that, for every geometric point $\overline{x}$ of X, if $\tilde{g} : \widetilde{X} \to \widetilde{Y}$ is the morphism between the strict localizations of X and Y (at $\overline{x}$ and the corresponding geometric point $\overline{y}$ of Y) induced by g, every geometric fiber $X_{\overline{z}}$ of g, with respect to a geometric point $\overline{z}$ of Y algebraic over a point z of Y, is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$).*

185 Necessity follows from criterion 1.10 (i) and 1.9.1. For sufficiency, we proceed by induction on n, which allows us, when $n \geq 1$, to suppose that g is already locally $(n-1)$-acyclic for $\mathbf{L}$. The same then holds for the morphisms $\tilde{g}$, and 1.16 implies that $\tilde{g}$ is n-acyclic (resp. 1-aspherical for $\mathbf{L}$); by criterion 1.10 (i), this implies that g is locally n-acyclic (resp. locally 1-aspherical) for $\mathbf{L}$.

COROLLARY 1.18. *With the notation of 1.17, if g is locally of finite type, then in the stated criterion it is enough to consider geometric points $\overline{x}$ whose underlying point x is closed in its fiber.*

This follows from the proof just given and from 1.12.

Proof of 1.15[40]. We first show that we may suppose Y strictly local. In the abelian case, with $n \geq 1$, the Leray spectral sequence for g and $g^*(F)$, together with $R^i g_*(g^*(F)) = 0$ for $0 < i \leq n-1$ and $F \xrightarrow{\sim} g_* g^*(F)$, which express the $(n-1)$-acyclicity hypothesis on g, yields a canonical exact sequence

$$0 \longrightarrow H^n(Y, F) \longrightarrow H^n(X, g^*(F)) \longrightarrow H^0(Y, R^n g_*(g^*(F))),$$

which shows that it is enough to prove that the image of ξ in the third term is zero. This can be checked fiber by fiber, and in view of VIII 5.2 we are reduced to the case where Y is strictly local. The case $n = 0$, with F a sheaf of sets, is handled in the same way: the hypothesis on g implies that $F \to g_* g^*(F)$ is injective, and ξ is identified with a section of the second sheaf that must be shown to come from a section of the first. This too can
186 be checked fiber by fiber, and one applies VIII 5.3. Essentially the same argument works in the noncommutative case, with $n = 1$, using the exact sequence XII 3.2

$$H^1(Y, F) \longrightarrow H^1(X, g^*(F)) \longrightarrow H^0(Y, R^1 g_*(g^*(F))),$$

and using VIII 5.3.

[40] The reader interested only in the Noetherian case is invited to consult 1.18.

We therefore suppose that Y is strictly local, and hence that X is quasi-compact and quasi-separated. The morphism $g : X \to Y$ satisfies the conditions of XII 6.10, which reduces us to the case in which, for every closed subset Y_1 of Y distinct from Y, the restriction of ξ to $X_1 = X \times_Y Y_1$ lies in the image of $H^n(Y_1, F|Y_1) \to H^n(X_1, g^*(F)|X_1)$, and to checking in this case that one can find a finite morphism $f : Y' \to Y$ whose image in Y contains a nonempty open subset of Y, and such that the inverse image ξ' of ξ on $X' = X \times_Y Y'$ lies in the image of $H^n(Y', F') \to H^n(X', g'^*(F'))$. We note that, in view of VIII 5.5 and VIII 5.8, condition 2°) of *loc. cit.* is automatically satisfied for this morphism $f : Y' \to Y$.

To find such an f, we may suppose that Y is reduced, and we consider a maximal point y of Y. Let $\overline{y}$ be the geometric point corresponding to a separable closure of $k(y)$, so that $\overline{y}$ is the filtered inverse limit of open subsets U_i of finite integral Y-schemes Y_i, each of which has an image in Y containing a neighborhood of y. Using the hypothesis on ξ, we see that there is an index i such that the inverse image of ξ on $X \times_Y U_i$ is zero if $n \geq 1$, resp. lies in the image of $H^0(U_i, F_{U_i})$ if $n = 0$.

We take Y' to be Y_i, and note that we may suppose that $U' = U_i$ is the inverse image of an affine open neighborhood U of y in Y. We then see that the inverse image of ξ on $X \times_Y Z'$ (where $Z' = Y' - U'$ is the inverse image in Y' of $Z = Y - U$) lies in the image of $H^n(Z', F_{Z'})$. After replacing Y by Y', we are therefore reduced to the following case: there is an affine open subset U of Y, with complement Z, such that the restrictions of ξ to X_U and X_Z lie respectively in the image of $H^n(U, F_U)$ and in the image of $H^n(Z, F_Z)$; moreover, if $n \geq 1$, these restrictions may be assumed to be *zero*. 187

When $n \geq 1$, we shall show that under these conditions $\xi = 0$. To simplify notation, we write Y' in place of X, denote by U' and Z' the inverse images of U and Z in Y', and denote the inverse image $g^*(F)$ by F'. Let $i : U \to Y$, $j : Z \to Y$, $i' : U' \to Y'$, and $j' : Z' \to Y'$ be the inclusions,

$$\begin{array}{ccccc} U' & \xrightarrow{i'} & Y' = X & \xrightarrow{j'} & Z' \\ \downarrow{\scriptstyle g_U} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g_Z} \\ U & \xrightarrow{i} & Y & \xrightarrow{j} & Z, \end{array}$$

and consider the exact sequence of sheaves

$$0 \longrightarrow F'_{U',Y'} \longrightarrow F' \longrightarrow F'_{Z',Y'} \longrightarrow 0.$$

Since the image of ξ in $H^n(Y', F'_{Z',Y'}) = H^n(Z', F'_{Z'})$ is zero, ξ comes from an element of $H^n(Y', F'_{U',Y'})$, and the restriction of the latter to U' is equal to that of ξ, and is therefore zero. Since $F'_{U',Y'}$ is isomorphic to the inverse image of $F_{U,Y}$, we are thus reduced to the case of a sheaf of the form $F_{U,Y}$; hence we may suppose that $F_Z = 0$ (which takes account of the hypothesis $\xi|Z' = 0$). Note that if $F \to G$ is a monomorphism of abelian **L**-torsion sheaves (resp. of sheaves of groups), it is enough to prove that the image η of ξ in $H^n(Y', G')$ lies in the image of $H^n(Y, G)$, i.e. is zero XII 6.6. This allows us to replace F by $i_*(F_U)$, since the hypothesis $F_Z = 0$ implies that the canonical homomorphism $F \to i_*(F_U)$ is injective. Now, since g is locally $(n-1)$-acyclic for **L**, and a fortiori 0-acyclic for **L** (because $n \geq 1$), the inverse image of $i_*(F_U)$ on Y' is identified with $i'_*(F'_{U'})$. We are therefore reduced to showing that the image η of ξ in $H^n(Y', i'_*(F'_{U'}))$ is zero. 188

The restriction of η to U' is zero, so it is enough to check that the homomorphism

$$(*) \qquad H^n(Y', i'_*(F'_{U'})) \longrightarrow H^n(U', F'_{U'})$$

is a monomorphism. In the abelian case, we use the Leray spectral sequence for i' and $F'_{U'}$:

$$E_2^{p,q} = H^p(Y', R^q i'_*(F'_{U'})) \Longrightarrow H^*(U', F'_{U'}),$$

and the relations

$$(**) \qquad E_2^{p,q} = 0 \text{ for } 0 < p \leq n-1 \quad , \quad q \leq n-1.$$

To verify the latter, note that the local $(n-1)$-acyclicity hypothesis on g for $\mathbf{L}$ implies that there are isomorphisms

$$R^q i'_*(F'_{U'}) \simeq g^*(R^q i_*(F_U)) \text{ for } q \leq n-1,$$

189 and, in view of the fact that g is $(n-1)$-acyclic for $\mathbf{L}$, so that $H^p(Y, G) \xrightarrow{\sim} H^p(Y', G')$ for $p \leq n-1$ and every abelian $\mathbf{L}$-torsion sheaf G on Y, we obtain $(**)$, since $H^p(Y, G) = 0$ for $p > 0$. In the same way we obtain

$$(***) \qquad E_2^{0,q} \simeq H^0(Y, R^q i_*(F_U)) \text{ for } \quad q \leq n-1.$$

The Leray spectral sequence under consideration then gives rise to an exact sequence that forms the second row of the following commutative diagram:

$$\begin{array}{ccccc} H^0(Y, R^{n-1} i_*(F_U)) & \longrightarrow & H^n(Y, i_*(F_U)) = 0 & & \\ \downarrow \wr & & \downarrow & & \\ H^0(Y', R^{n-1} i'_*(F'_{U'})) & \longrightarrow & H^n(Y', i'_*(F'_{U'})) & \longrightarrow & H^n(U', F'_{U'}), \end{array}$$

where the first vertical arrow is bijective by $(***)$. The diagram shows that $(*)$ is injective, which completes the proof in this case. In the nonabelian case, $n = 1$, we use the same diagram with $n = 1$:

$$\begin{array}{ccccc} H^0(Y, i_*(F_U)) & \longrightarrow & H^1(Y, i_*(F_U)) = 0 & & \\ \downarrow \wr & & \downarrow & & \\ H^0(Y', i'_*(F'_{U'})) & \longrightarrow & H^1(Y', i'_*(F'_{U'})) & \longrightarrow & H^1(U', F'_{U'}), \end{array}$$

XII 3.2, where the first vertical arrow is again bijective by the hypothesis of local and global 0-acyclicity on g; once again we conclude that $(*)$ is injective in this case.

Finally, consider the case $n = 0$. By virtue of , specifying a section ξ of F' amounts to specifying a section $\xi_{U'}$ of $F'_{U'}$ and a section $\xi_{Z'}$ of $F'_{Z'}$ such that the image of $\xi_{Z'}$ under the natural homomorphism

$$F'_{Z'} \longrightarrow j'^*(i'_*(F'_{U'}))$$

190 is the section of the right-hand side induced by $\xi_{U'}$. By hypothesis, there are sections η_Z and η_U of F_Z and F_U, respectively, that induce $\xi_{Z'}$ and $\xi_{U'}$. It remains to see that they satisfy the analogous compatibility condition, which says that they are the restrictions of a single section η of F. We must verify the equality of two sections of $j^*(i_*(F_U))$. Since $Z' \to Z$ is surjective (g being surjective because it is (-1)-acyclic by hypothesis), it is enough to check that the two corresponding sections of

$$g_Z{}^*(j^*(i_*(F_U))) = j'^*(g^*(i_*(F_U)))$$

are equal. Through the base-change homomorphism

$$(*) \qquad g^*(i_*(F_U)) \longrightarrow i'_*({g_U}^*(F_U)) = i'_*(F'_{U'}),$$

the sheaf under consideration maps to ${j'}^*(i'_*(F'_{U'}))$, and this latter homomorphism is injective because $(*)$ is injective, by the hypothesis that g is locally (-1)-acyclic. It therefore suffices to verify that the two sections of ${j'}^*(i'_*(F'_{U'}))$ that are the images of the preceding two sections are equal. But these are immediately seen to be precisely the sections considered above, which are equal by hypothesis. This completes the proof.

Remark 1.18. *There is a refinement of 1.15 that gives a much simpler proof of 1.15 when Y is assumed to be locally Noetherian. In addition to the preliminary hypotheses on g and F, suppose that there is a finite set of geometric points $f_i : \overline{y}_i \to Y$ of Y such that the morphism*

$$F \longrightarrow \prod_i f_{i*}({f_i}^*(F))$$

is injective (a hypothesis automatically satisfied if Y is Noetherian and F is constructible IX 2.14). I claim that under these conditions criterion 1.15 remains valid when one restricts to the geometric points $\overline{y}_i$ alone. Indeed, by XII 6.6, we may suppose that F itself is of the form $f_*(M_{\overline{y}})$, where M is a set resp. a group resp. an abelian $\mathbf{L}$-torsion group, and $f : \overline{y} \to Y$ is a geometric point. Let $f' : X' = X_{\overline{y}} \to X$ be the canonical morphism. Assuming, for definiteness, that $n \geq 1$, the local hypothesis on g implies that

$$g^*(f_*(M_{\overline{y}})) \simeq f'_*(M_{X'}) \quad , \quad R^i f'_*(M_{X'}) = 0 \quad \text{for} \quad 1 \leq i \leq n-1,$$

the second relation following from the fact that its left-hand side is isomorphic to $g^*(R^i f_*(M_{\overline{y}}))$, which is zero since manifestly $R^i f_*(M_{\overline{y}}) = 0$ for $i \geq 1$. From these relations we conclude that

$$H^n(X, g^*(F)) \simeq H^n(X, f'_*(M_{X'})) \longrightarrow H^n(X', M_{X'})$$

is injective, using the Leray spectral sequence for f'. This proves that an element ξ of $H^n(X, g^*(F))$ whose induced element in $H^n(X_{\overline{y}}, M_{X_{\overline{y}}})$ is zero is itself zero. To deduce from this a simplified proof of 1.15 in the case where Y is locally Noetherian, note that we may restrict to the case where Y is Noetherian; an easy passage to the limit, using IX 2.7.2 and VII 5, then reduces us to the case where F is also constructible, to which the statement just proved applies.

2. Local acyclicity of a smooth morphism

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The theorem is the following:

Theorem 2.1. *Let $g : X \to Y$ be a smooth morphism, and let $\mathbf{L} \subset \mathbf{P}$ be the complement of the set of residual characteristics of X. Then g is locally acyclic and locally 1-aspherical for $\mathbf{L}$.*

Let us first record the corollary:

Corollary 2.2. *Let $g : \mathbf{E}^m_Y \to Y$ be the structural morphism of affine m-space. Then g is acyclic and 1-aspherical for the set $\mathbf{L}$ complementary to the set of residual characteristics of Y.*

Indeed, induction on m and 1.9 (i) reduce us to the case $m = 1$. By 2.1, g is locally n-acyclic and locally 1-aspherical for $\mathbf{L}$, so the *local* condition of 1.16 is satisfied. It therefore remains only to prove that the geometric fibers of $\mathbf{E}^1_Y$ are acyclic and 1-aspherical for $\mathbf{L}$; that is, that Spec $k[t]$ is acyclic and 1-aspherical for $\mathbf{L}$ when k is a separably closed field of

characteristic $p \notin \mathbf{L}$. The acyclicity assertion follows from IX 4.6, and the 1-asphericity assertion is well known. More generally, let X' be a normal irreducible Galois covering of $X = \mathbf{P}^1_k$, unramified away from the point ∞ and tamely ramified at that point. We prove that the degree n of the covering is 1. The Hurwitz formula gives

$$2(g' - 1) = 2n(g - 1) + \deg \mathfrak{d}_{X'/X},$$

where g and g' are the genera of X and X', and $\mathfrak{d}_{X'/X}$ is the different divisor. We have $g = 0$, while tame ramification implies $\deg \mathfrak{d}_{X'/X} \leq n - 1$; hence $2(g' - 1) \leq -(n + 1)$, so $n \leq 2(1 - g') - 1$. Since $g' \geq 0$, it follows that $n \leq 1$, and therefore $n = 1$, C.Q.F.D.

Proof of 2.1. The assertion is local on X and Y for the étale topology, so we may assume SGA 1 II 1.1 that g is the structural morphism of affine m-space $\mathbf{E}^m_Y$ and that Y is affine. By the transitivity of local acyclicity, 1.13 (i), and the identity $\mathbf{E}^m_Y = \mathbf{E}^{m-1}_{\mathbf{E}^1_Y}$, we are immediately reduced to $m = 1$, that is, to $X = \operatorname{Spec} \mathcal{O}_Y[t]$.

Apply 1.18. It remains to prove that the geometric fibers of the morphism of strictly local schemes $\tilde{g} : \widetilde{X} \to \widetilde{Y}$ associated with a pair of geometric points $\overline{x}, \overline{y}$, where $\overline{x}$ is closed in the fiber $X_{\overline{y}}$, are acyclic for $\mathbf{L}$ and 1-aspherical for $\mathbf{L}$. We may plainly assume that $Y = \widetilde{Y}$. Let $Y' \to Y$ be a finite, local, surjective morphism, so that Y' is strictly local, and let $\overline{y}'$ be the point of Y' above $\overline{y}$ and $\overline{x}'$ the point of X' above both $\overline{x}$ and $\overline{y}'$. Since $X' \to X$ is finite, the strict localization of X' at $\overline{x}'$ is precisely $\widetilde{X}' = \widetilde{X} \times_Y Y'$. Consequently every geometric fiber of $\widetilde{X}'/Y'$ is obtained from a geometric fiber of $\widetilde{X}/Y$ by an algebraic (necessarily radicial) extension of the base field; moreover, at least one geometric fiber of $\widetilde{X}'/Y'$ lies above each geometric fiber of $\widetilde{X}/Y$. Thus, by VIII 1.1, we may replace Y by Y'. Choosing Y' suitably reduces us to the case $k(\overline{x}) = k(\overline{y})$. Then $\overline{x}$ is a rational point of the fiber $X_{\overline{y}} = \operatorname{Spec} k(\overline{y})[t]$, say the point $t = a^0$. Lift a^0 to a section a of $\mathcal{O}_{\widetilde{Y}}$ and make the "translation" $t_1 = t - a$; we are reduced to the case in which $\overline{x}$ is the point $\{t = 0\}$ of $\operatorname{Spec} k(\overline{y})[t]$. Put $A = \Gamma(\widetilde{Y}, \mathcal{O}_{\widetilde{Y}})$, so that A is a strictly local ring.

NOTATION 2.3. Let A be a Henselian local ring. We denote by $A\{t\}$ the Henselization of $A[t]$ at the prime ideal generated by $\operatorname{rad} A$ and t.

COROLLARY 2.4. *Let A be Henselian and let A' be a finite A-algebra. Then*

$$A'\{t\} \simeq A' \otimes_A A\{t\}.$$

This follows immediately from VIII 4.1. Note that if A is strictly local, then so is $A\{t\}$.

The proof of the theorem is reduced to the following lemma:

LEMMA 2.5. *Let A be a Henselian ring of residual characteristic p. Then every geometric fiber of* $\operatorname{Spec} A\{t\}$ *over* $\operatorname{Spec} A$ *is acyclic and* 1*-aspherical for* $\mathbf{L} = \mathbf{P} - \{p\}$.

Since $A\{t\}$ is the Henselization of $A[t]$, a fiber $\operatorname{Spec}(A\{t\} \otimes_A K)$, where K is a residue
193 field of A, is an inverse limit of affine schemes X_i of finite type over the "line" $\operatorname{Spec} K[t]$, that is, a limit of affine algebraic curves. Hence every geometric fiber of $\operatorname{Spec} A\{t\}$ is a limit of affine curves $(X_i)_{\overline{K}}$ over the separably closed field $\overline{K}$. It follows from IX 5.7 and VII 5.7 that the cohomological dimension of the geometric fibers is at most 1. It is therefore enough to verify that the geometric fibers are 1-*aspherical* for $\mathbf{L}$; that is, that such a geometric fiber $\overline{Z}$ is connected and nonempty and that $H^1(\overline{Z}, G) = 0$ for every finite $\mathbf{L}$-group G.

Write $A = \varinjlim B_\alpha$, where the B_α are rings of finite type over $\mathbf{Z}$. The morphism $B_\alpha \to A$ factors through the Henselization A_α of B_α at the prime ideal induced by $\operatorname{rad} A$,

whence $A = \varinjlim A_\alpha$. One readily checks that $A\{t\} = \varinjlim A_\alpha\{t\}$ as well. Let $\overline{y}$ be a geometric point of Spec A, let $X_{\overline{y}}$ be the geometric fiber of $S = \operatorname{Spec} A\{t\}$ over Spec A at $\overline{y}$, let $\overline{y}_\alpha$ be the point of Spec A_α induced by $\overline{y}$, and let $X_{\alpha,\overline{y}_\alpha}$ be the geometric fiber of Spec $A_\alpha\{t\}$ over Spec A_α at $\overline{y}_\alpha$. We have $X_{\overline{y}} = \varprojlim X_{\alpha,\overline{y}_\alpha}$. By 1.7 (ii), applied to the morphisms $X_{\alpha,\overline{y}_\alpha} \to \overline{y}_\alpha$, the proof of 2.5 is therefore reduced to the case $A = A_\alpha$, that is, to the case in which A is the Henselization of an algebra of finite type over $\mathbf{Z}$ at a given prime ideal. We assume this from now on. Recall that such an algebra is *excellent* EGA IV 7.8.3 (iii), 7.8.6 (i), and hence that A_α is excellent as well (as follows readily from the definition; cf. EGA IV 18.7.6).

Let us verify 0-acyclicity. (For a more general result, see the Appendix.) Since $A\{t\}/A$ has the section "$t = 0$", a geometric fiber $\overline{Z}$ is nonempty. To see that it is connected, 194
we reduce, by replacing A by a quotient 2.4, to the case in which A is integral and $\overline{Z}$ is the geometric fiber over the generic point of Spec A. It is enough to prove that for every finite separable extension K' of the fraction field K of A, the scheme $Z_{K'}$, where $Z = \operatorname{Spec}(A\{t\} \otimes_A K)$, is connected. Let A' be the normalization of A in K' (a finite A-algebra, EGA IV 7.8.2) and let Z' be the generic fiber of Spec $A'\{t\}$ over A'. By 2.4, $A'\{t\} \simeq A' \otimes_A A\{t\}$, and hence $Z_{K'} \simeq Z'$. Replace A by A'. It is then enough to prove that Z is integral, and for this it is enough to prove that $A\{t\}$ is integral. Since A is normal, $A[t]$ is normal, and so is $A\{t\}$ SGA 1 I 9.5 (i). This proves the assertion.

The proof of 2.5, and hence of 2.1, is thus reduced to the following lemma, which will be proved in the next section:

LEMMA 2.6. *Let A be a Henselian excellent ring, and let $\overline{Z}$ be a geometric fiber of* Spec $A\{t\}/$Spec A. *Then every principal Galois covering $\overline{Z}'$ of $\overline{Z}$ with group G of order prime to the residual characteristic p of A is trivial.*

3. Proof of the main lemma

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Replacing A by a finite A-algebra A' (cf. 2.4), assertion 2.6 is reduced to the case where A is integral and $\overline{Z}$ is the geometric generic fiber of Spec $A\{t\}$ over Spec A. Let K be the fraction field of A, and let $Z = \operatorname{Spec} A\{t\} \otimes_A K$. If $\overline{Z}'/\overline{Z}$ is a principal Galois covering, it can be descended to some $Z_{K'}$, where K' is a suitable finite separable extension of K. Replacing A (2.4) by its normalization in K', which is a finite A-algebra EGA IV 7.8.3 (vi), we are reduced to proving the lemma in the following form:

LEMMA 3.1. *With the notation of 2.6, suppose that A is normal, and let Z'/Z be a principal étale Galois covering, with group G of order prime to the residual characteristic p of A. Then Z' is induced by a Galois extension K' of K; that is,* (SGA 1 IX 6.2), *the covering $\overline{Z}'$ of $\overline{Z}$ deduced from Z' is trivial.*

Proof. Let B' be the normalization of $A\{t\}$ in the ring of rational functions of Z', and consider the height-1 prime ideals P of $A\{t\}$ at which the $A\{t\}$-algebra B' is ramified. Since Z'/Z is étale, P does not lie over the generic point of Spec A; that is, $\mathfrak{p} = P \cap A \neq 0$. Thus $A\{t\}/\mathfrak{p}A\{t\} = (A/\mathfrak{p})\{t\}$ is integral and has dimension at most $\dim A$. Since $A\{t\}$ is excellent EGA IV 7.8.6 (i) and 18.7.6, hence catenary, we have $\dim \operatorname{Spec} A\{t\}/P = (\dim A + 1) - 1 = \dim A \geq \dim A\{t\}/\mathfrak{p}A\{t\}$, whence $P = \mathfrak{p}A\{t\}$, since $P \supset \mathfrak{p}A\{t\}$. 196

Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ therefore be the height-1 prime ideals of A such that $B'/A\{t\}$ is ramified at the ideals $\mathfrak{p}_i A\{t\}$, and let s be the least common multiple of the orders of the inertia groups SGA IV 2 $G_i \subset G$ for B' at the prime ideals lying above the $\mathfrak{p}_i A\{t\}$. This is an integer dividing the order of G, and is therefore prime to p. Let $f \in A$ be an

element that vanishes to order 1 at every $\mathfrak{p}_i$ (such an f exists, as follows, for example, from Bourbaki, Alg. Comm., Chap. II, § 3, cor. to prop. 17). Replacing A 2.4 by its normalization A_1 in the finite extension $K_1 = K[x]/(x^s - f)$ of K, and replacing B' by the normalization of $A_1 \otimes_A B'$, Abhyankar's lemma SGA 1 X 3.6 reduces us to the case where B' is ramified at no height-1 prime ideal of $A\{t\}$.

The algebra $B'/A\{t\}$ is then also étale over the fiber of Spec $A\{t\}$ at every codimension-1 point of Spec A. Indeed, let $\mathfrak{p}$ be a height-1 prime ideal of A. Since A is normal, $A_\mathfrak{p}$ is a discrete valuation ring, hence regular, and so $A\{t\} \otimes_A A_\mathfrak{p}$ is regular as well—indeed, it is a limit of schemes étale over $A_\mathfrak{p}[t]$. By hypothesis, the algebra $B'/A\{t\}$ is étale at every point of codimension ≤ 1 of Spec $A\{t\} \otimes_A A_\mathfrak{p}$, hence it is étale everywhere by the Zariski–Nagata purity theorem SGA 1 X 3.1; for a proof, see for example SGA 2, X 3.4.

Replacing A 2.4 by a suitable finite A-algebra, we are reduced to the case where, in addition, the algebra B'/tB' over $A\{t\}/tA\{t\} \simeq A$ is completely split over the generic point of Spec A. It therefore suffices to prove that, under these conditions, the algebra
197 B' over $A\{t\}$ is "trivial," and, since $A\{t\}$ is Henselian, it suffices to prove that B' is étale over $A\{t\}$. Indeed, the fact that B'/tB' splits over the generic point of Spec A implies that it splits over Spec(A) SGA 1 I 10.1; thus the residual extensions of B' over the maximal ideal of $A\{t\}$ are trivial, and we apply EGA IV 18.5.14. We are therefore reduced to the following lemma:

LEMMA 3.2. *Let B be an excellent normal local ring, and let $0 \neq t \in \operatorname{rad} B$ be such that B/tB is normal. Let B' be a finite integral normal B-algebra containing B, such that:*

(i) *The algebra B'/tB' over B/tB is completely split at the maximal points of* Spec(B/tB).
(ii) *The algebra B' over B is étale at the codimension-1 points of* Spec(B/tB).

Then B' is étale over B.

Proof. By (ii), the assertion is trivial for $\dim B \leq 2$, that is, for $\dim B/tB \leq 1$. Suppose that B'/B is not étale; let x be a maximal point of the set of points of Spec B/tB over which B'/B is not étale, and let C be the local ring of Spec B at x. Then C is again an excellent normal ring (EGA IV 7.8.2), we have $0 \neq t \in \operatorname{rad} C$, and (i) and (ii) hold for the C-algebra $C' = C \otimes_B B'$. We may therefore suppose that $B = C$; that is, B' is étale over B above every point of Spec B/tB, except possibly at the origin. Since B/tB is normal
198 (in fact, unibranchness would suffice), condition (i) implies that the algebra B'/tB' is completely split away from the origin. Moreover, we may suppose that $\dim B \geq 3$. We are thus reduced to the following lemma:

LEMMA 3.3. *Let B be an excellent normal local ring, let $0 \neq t \in \operatorname{rad} B$, and suppose that $\dim B \geq 3$. Let B' be a finite integral normal B-algebra containing B, such that the B/tB-algebra B'/tB' is completely split away from the origin. Then B' is étale over B.*

Proof. Let $X = \operatorname{Spec} B$, let $X_1 = X - \{x_0\}$, where x_0 is the closed point, and let $i : U \to X$ be the open immersion whose image consists of the points $x \in X$ at which the algebra B'/B is étale. Then $(X - U) \cap V(t) = \{x_0\}$ has dimension 0. Since $V(t)$ is defined by *one* equation, it follows that $\dim(X - U) \leq 1$ (EGA 0_{IV} 16.2.5); consequently (since X is catenary), $x \in X - U \Rightarrow \dim \mathcal{O}_{X,x} \geq \dim B - 1 \geq 2$; that is, $\operatorname{codim}(X - U, X) \geq 2$.

Let $X' = \operatorname{Spec}(B')$, and let U' be the inverse image of U in X'. Since X is universally catenary, the relation $\operatorname{codim}(X - U, X) \geq 2$ implies $\operatorname{codim}(X' - U', X') \geq 2$ (EGA IV 5.6.10). Since X' is normal, we therefore have

$$B' = \Gamma(X', \mathcal{O}_{X'}) \xrightarrow{\sim} \Gamma(U', \mathcal{O}_{U'}).$$

Let $\mathscr{B}' = \widetilde{B}'$ be the sheaf of $\mathscr{O}_X$-algebras that defines X' over X, and let $\mathscr{C}$ be its restriction to U. The preceding formula is then equivalent to

$$\mathscr{B}' \xrightarrow{\sim} i_*(\mathscr{C}),$$

and 3.3 is consequently reduced to the following statement (in which B' no longer occurs): Let B, t be as in 3.3, let $X = \mathrm{Spec}(B)$, let $Y = \mathrm{Spec}(B/tB)$, let U be an open neighborhood of $Y_1 = Y - \{x_0\}$ in $X_1 = X - \{x_0\}$, and let U' be an étale covering of U whose restriction to Y_1 is completely split. If $\mathscr{C}$ denotes the sheaf of $\mathscr{O}_U$-algebras defining U', then (with $i : U \to X$ denoting the canonical immersion) $i_*(\mathscr{C})$ is a coherent *étale* algebra on X (or, equivalently, in view of the relation $\mathrm{prof}\,\mathscr{O}_{X,x} \geq 2$ for $x \in X - U$ noted above and EGA IV 5.10.5, U' extends to an *étale* covering X' of X). 199

Let $\widehat{X} = \mathrm{Spec}\,\widehat{B}$, where $\widehat{B}$ is the completion of B. Then $\widehat{B}$ is normal (EGA IV 7.8.3 (v)) and $\widehat{B}/B$ is faithfully flat. Let

$$\begin{array}{ccc} \widehat{U} & \xrightarrow{i} & \widehat{X} \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f} \\ U & \xrightarrow{i} & X \end{array}$$

be the Cartesian diagram deduced from $\widehat{X} \to X$. Since f is faithfully flat, the functor i_* commutes with base change by f EGA IV 2.3.1, and we are reduced EGA IV 17.7.1 to proving that $\hat{i}_*(\widehat{\mathscr{C}})$ is an étale algebra on $\widehat{X}$. We may therefore replace B by $\widehat{B}$; that is, we may suppose that B is *complete*. In this case we shall prove that U' is a completely split covering of U (which will plainly imply that it extends to an étale covering of X, and will complete the proof). This assertion is now an immediate consequence of "local Lefschetz theory" SGA 2 X 2.1 (i) and 2.5, as is seen at once, taking account once again of the relation $\mathrm{prof}\,\mathscr{O}_{X,x} \geq 2$ for $\dim\{x\} = 1$. 200

Remark 3.4. One can prove 3.2 and 3.3 under more general conditions, by following essentially the same proof: instead of assuming that B is excellent and normal, it is enough to suppose that the completion $\widehat{B}$ is normal (a condition stable under localization, by the results of EGA IV 7), or merely that the completion of B for the (tB)-adic topology is normal and is a quotient of a regular ring.

4. Appendix: A criterion for local 0-acyclicity

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The results of this section will not be used later in the seminar.

Theorem 4.1. *Let $g : X \to Y$ be a morphism of schemes. Assume that g is flat with separable fibers, and that X and Y are locally noetherian respectively, that g is locally of finite presentation. Then g is universally locally 0-acyclic.*

We may clearly assume that X and Y are affine; in the second case, the standard method of EGA IV 8, using EGA IV 9.7.7, then reduces us to the case in which Y, and hence X, is noetherian. We may therefore confine ourselves to proving the first assertion. Since the hypothesis is stable under base change of finite type, the passage to the limit in 1.13 (ii) reduces us to proving that g is locally 0-acyclic. For this, 1.17 reduces us to proving that, if g is moreover assumed to be strictly local, then the geometric fibers of g are connected. This is precisely EGA IV 18.9.8.

Corollary 4.2. *Let $g : X \to Y$ be a morphism of schemes, with Y discrete. Then g is universally locally 0-acyclic.*

By VIII 1.1, we may assume that Y is the spectrum of a perfect field k. We may also assume that X is affine, and hence is a projective limit of affine schemes of finite type over k; in view of 1.13 (ii), this reduces us to the case in which X is of finite type over k.
202 Finally, we may clearly assume that X is reduced, and it is then separable over k because k is perfect. The conclusion now follows from 4.1.

COROLLARY 4.3. *Assume that g is surjective and satisfies the hypotheses of 4.1 or 4.2; in the non-parenthetical case of 4.1, assume in addition that g is either quasi-compact and quasi-separated or universally open. Then g is a morphism of universal effective descent for the fibered category of étale sheaves on varying preschemes.*

In the cases under consideration, g is universally submersive, and it follows from VIII 9.1 that g is a morphism of universal descent for the fibered category in question. Thus, for the question of effectivity, we may confine ourselves to the case in which Y is affine. Moreover, in each of the cases under consideration, X can be covered by a *finite* number of affine open subsets X_i whose images cover Y. Replacing X by the disjoint union of the X_i, we may therefore assume that X is affine, and hence a fortiori quasi-compact and quasi-separated over Y. We may then conclude by the argument of VIII 9.4.1, using the fact XVI 1.1 that, for a quasi-compact and quasi-separated morphism g, the formation of $g_*(F)$ (where F is an étale sheaf on X) commutes with every base change $Y' \to Y$ that is locally 0-acyclic.

REMARKS 4.4.

a) This gives the proof, which had been left pending, of VIII 9.4 d), as a special case of 4.3. Another, easier proof, not using the rather deli-
203 cate result of EGA IV 18.9.8, is obtained by observing that, by VIII 9.1, we may confine ourselves to the case of the morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$ induced by a field extension K/k. But $\mathrm{Spec}(K)$ is universally locally 0-acyclic over $\mathrm{Spec}(k)$, since it is even universally acyclic for $\mathbf{L} = \mathbf{P} - \{p\}$, $p = \mathrm{car}\, k$, as we shall see in XVI 1.5.

b) It is not known whether every quasi-compact and quasi-separated, surjective, locally (-1)-acyclic (or universally locally (-1)-acyclic) morphism is a morphism of effective descent for the fibered category of étale sheaves on varying preschemes. This statement seems quite plausible and would strengthen 4.3 and VIII 9.4 c), d). Compare the effectivity statement in MURRE, Sém. Bourbaki n° 293, p. 17.

c) It is plausible that, under the hypotheses of 4.2, g is even locally acyclic for L and locally 1-aspherical for L, where L is the set of prime numbers distinct from the residue characteristics of Y. For this question we may clearly assume that Y is the spectrum of an algebraically closed field k, and it can be shown (SGA 1964/65) that the answer is affirmative whenever resolution of singularities is available for schemes of finite type over k. Thus the answer is affirmative when Y has characteristic zero, as follows by using the results of HIRONAKA [40].

[40] Cf. SGA 1 XIII.

EXPOSÉ XVI

Smooth base change theorem and applications

M. Artin

1. The smooth base change theorem

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THEOREM 1.1. *Let* $\mathbf{L} \subset \mathbf{P}$, *let* $n \in \mathbf{N}$, *and let*

(∗)
$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{g} & Y' \end{array}$$

be a cartesian diagram, with f *quasi-compact and quasi-separated*[41], *and* g *universally locally* 0*-acyclic (resp. universally locally* 1*-aspherical for* $\mathbf{L}$, *resp. universally locally* n*-acyclic for* $\mathbf{L}$*)* (XV 1.11). *Then, for every sheaf* F *of sets (resp. of* ind-$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) on* X, *the base-change morphism* (XII 4.1.2)

$$\varphi^q : g^*(R^q f_*)F \longrightarrow (R^q f'_*)g'^* F$$

is bijective for $q = 0$ *(resp.* $q \leq 1$, *resp.* $q \leq n$*).*

In particular, applying XV 2.1, we obtain the following.

COROLLARY 1.2. *(Smooth base change theorem). Suppose that the morphism* g *in* (∗) *is smooth, that* f *is quasi-compact and quasi-separated, and that* $\mathbf{L} \subset \mathbf{P}$ *is the complement of the set of residue characteristics of* Y. *Then, for every sheaf* F *of sets (resp. of* ind-$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) on* X, *the base- change morphism* φ^q *above is bijective* 208
for $q = 0$ *(resp. for* $q = 0, 1$, *resp. for every* q*).*

Proof of 1.1. We first treat the case in which $f : X \to Y$ is quasi-projective: this is simply a combination of the proper base change theorem XII 5.1 and Definition XV 1.9. Indeed, we have a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i} & \overline{X} \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle \overline{f}} & \\ & Y & \end{array}$$

where $\overline{f}$ is projective, hence proper, and i is an open immersion. We know from XII 5.1 that the base change theorem holds for $\overline{f}$, whatever the morphism $g : Y' \to Y$. Since $g \times_Y \overline{X} : Y' \times_Y \overline{X} \to \overline{X}$ is again universally 0-acyclic (resp. ...), it follows immediately from (XII 4.4 (ii)) that it suffices to prove the theorem for the morphism i; that is, we are reduced to the case in which f is an open immersion. The theorem then follows from Definition XV 1.10 (ii).

[41] N.D.E.: That is, coherent in the terminology of the source.

We now treat the general case. The assertion is local on Y, and we may therefore suppose that Y is affine.

LEMMA 1.3. *We may moreover suppose that X is affine.*

Proof. Arguing as in XII 6.1, in order to prove 1.1 for a given f, we are reduced to proving
209 it in the situation deduced from the given situation by the base changes $\widetilde{Y} \to Y, \widetilde{Y}' \to Y'$ for a local morphism $\tilde{g} : \widetilde{Y}' \to \widetilde{Y}$ between strict localizations of Y' and Y induced by g, and to proving that the homomorphisms

$$(*) \qquad H^q(\widetilde{X}, F) \longrightarrow H^q(\widetilde{X}', F') \quad (\text{where } F' = g'^*(F))$$

are bijective for $q \leq n$, where $\widetilde{X} = X \times_Y \widetilde{Y}$, $\widetilde{X}' = X' \times_{Y'} \widetilde{Y}'$. We are thus reduced to the case in which Y, Y', and g are strictly local, and to proving in this case the bijectivity of the preceding maps, assuming that 1.1 has been proved for affine f. Let (X_i) be a finite covering of X by affine open subschemes X_i, and let Z be the disjoint union of the X_i, which is therefore affine and is equipped with a surjective morphism $h : Z \to X$, whence a morphism $h' : Z' \to X'$. Note that h is separated, and that it is affine when f is separated. Applying the descent lemma XII 6.8, we see that, in order to prove the bijectivity of the maps $(*)$ for $q \leq n$ and every F, it suffices to prove the bijectivity of the corresponding maps $H^q(Z, h^*(F)) \to H^q(Z', h'^*(F'))$, provided that the base-change homomorphisms for $Z \to X$ and the base change $X' \to X$ are isomorphisms in dimensions $\leq n$. If f is separated, then h is affine, and this is so by hypothesis; this proves 1.1 when f is assumed separated. In the general case, we may then apply the preceding result to h, which is always separated, and again conclude that 1.1 holds for f, proving 1.3.

REMARK. One could also invoke the Leray spectral sequence for the open covering
210 (X_i) of X, and its noncommutative variants, but this method, which does not differ essentially from using XII 6.8, has the disadvantage of forcing us once again to distinguish the usual three cases, work that has already been done in *loc. cit.*

To complete the proof of the theorem in the general case, it suffices to reduce the proof to the case in which $f : X \to Y$ is affine and of finite type, hence quasi-projective. By 1.3, we may suppose that X is affine. Then let $X = \varprojlim X_\alpha$, where the morphisms $f_\alpha : X_\alpha \to Y$ are affine and of finite type. The usual limit argument (VII 5) completes the proof.

COMPLEMENTS 1.4. a) We leave it to the reader to check that the same proof still yields a conclusion when g is assumed universally locally (-1)-acyclic (XV 1.14): in this case, the base-change homomorphism

$$\phi^0 : g^*(f_*(F)) \longrightarrow f'_*(g'^*(F))$$

is *injective* for every sheaf of sets F on X. On the other hand, suppose again that $n \geq 0$. One may supplement 1.1 with the following injectivity statement, which is also contained in the preceding proof: if g is universally locally 0-acyclic, then for every sheaf of groups F on X, the base-change homomorphism ϕ^1 of 1.1 is a monomorphism; if g is universally locally n-acyclic for $\mathbf{L}$, then the homomorphism ϕ^{n+1} of 1.1 is a monomorphism for every abelian sheaf of $\mathbf{L}$-torsion F on X.

211 b) It is probably also possible to obtain an analogous injectivity conclusion from the hypothesis that g is locally 1-aspherical for $\mathbf{L}$, by introducing the nonabelian

2-cohomology invariants from GIRAUD's thesis; compare XII 5.11 [41]; the same question also arises for (1.6) and (2.3) below.

COROLLARY 1.5. *Let K/k be a field extension, and let $\mathbf{L} = \mathbf{P} - \{p\}$, where $p = \operatorname{car} k$. Then $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is universally locally acyclic for $\mathbf{L}$ and universally locally 1-aspherical for $\mathbf{L}$; if k and K are separably closed, then the preceding morphism is also universally acyclic for $\mathbf{L}$ and universally 1-aspherical for $\mathbf{L}$.*

We immediately note the following consequence of (1.5).

COROLLARY 1.6. *Let K/k be an extension of separably closed fields, let X be a scheme, and let F be a sheaf of sets (resp. of ind-$\mathbf{L}$-groups, resp. of abelian groups of $\mathbf{L}$-torsion) on X, where $\mathbf{L} = \mathbf{P} - \{p\}$, $p = \operatorname{car} k$. Then the map*

$$H^q(X, F) \longrightarrow H^q(X_K, F_K)$$

is bijective for $q = 0$ (resp. for $q \leq 1$, resp. for every q).

Indeed, this follows at once from the fact that $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is universally acyclic for $\mathbf{L}$ and universally 1-aspherical for $\mathbf{L}$, and from the definitions. Note that, by VIII 2.4, the preceding homomorphism on the H^q may also be interpreted as being identified, with the base-change homomorphism

$$g^*(R^q f_*(F)) \longrightarrow R^q f'_*(g'^*(F)),$$

where $f : X \to \operatorname{Spec}(k)$, $f' : X_K \to \operatorname{Spec}(K)$, $g : \operatorname{Spec}(K) \to \operatorname{Spec}(k)$, and $g' : X_K \to X$ are the evident morphisms. Thus, when X is quasi-compact and quasi-separated, the bijectivity of these homomorphisms may also be regarded, by 1.1, as a consequence of the fact that $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is universally *locally* acyclic for $\mathbf{L}$ and universally locally 1-aspherical for $\mathbf{L}$. This latter fact therefore already implies 1.6 for X quasi-compact and quasi-separated, which plainly suffices to imply that $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is universally acyclic for $\mathbf{L}$ and universally 1-aspherical for $\mathbf{L}$ (XV 1.7, XV 1.6 (i)). 212

This shows that, to prove 1.5, it suffices to prove the first assertion of 1.5. After passing to the perfect closure of k, which is permissible by VIII 1.1, we may then suppose that k is *perfect*. Every extension K of k is the filtered direct limit of its finitely generated k-subalgebras A, and since k is perfect, after localizing such an A, we may suppose it smooth, so that K appears as a filtered direct limit of smooth subalgebras. Since the latter are universally locally acyclic for $\mathbf{L}$ and universally locally 1-aspherical for $\mathbf{L}$ by XV 2.1, the conclusion follows from XV 1.13 (ii), C.Q.F.D.

REMARK 1.7. The invariance theorem 1.5 should be compared with XII 5.4, where it was not necessary to suppose that the torsion sheaf under consideration is prime to the residue characteristics, but where, on the other hand, X must be assumed *proper* over k. Already in the case of $H^1(X, \mathbf{Z}/p\mathbf{Z})$, where X is an affine algebraic curve (for example, the affine line) over an algebraically closed field k of characteristic $p > 0$, the analogue of the preceding statements becomes false, because of the phenomena of "wild ramification[42]" at infinity, which imply that, in such a case, the classification of principal étale covers with group $\mathbf{Z}/p\mathbf{Z}$ is essentially "continuous." 213

[41] This is indeed what is established in J. Giraud's book.

[42] N.D.E.: That is, "wild."

2. Specialization theorem for cohomology groups

THEOREM 2.1. *Let* $\mathbf{L} \subset \mathbf{P}$, *let* $n \in \mathbf{N}$, *and let* $f : X \to Y$ *be a proper morphism of finite presentation* [42] *that is locally* 0*-acyclic (resp. locally* 1*-aspherical for* $\mathbf{L}$, *resp. locally* n*-acyclic for* $\mathbf{L}$*). Let* F *be a sheaf of sets (resp. of* ind-$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) that is constructible and locally constant on* X*. Then the* $R^q f_* F$ *are constructible and* locally constant *for* $q = 0$ *(resp. for* $q = 0, 1$*, resp. for* $q \leq n$*), and for every geometric point* $\overline{y}$ *of* Y *one has*

$$(R^q f_* F)_{\overline{y}} \simeq H^q(X_{\overline{y}}, F_{\overline{y}}),$$

for the same values of q.

REMARK. Note that the last assertion is simply the proper base-change theorem XII 5.2.

214 In view of XV 2.1, we immediately obtain the following corollary:

COROLLARY 2.2. *(Specialization theorem for cohomology groups). Let* $f : X \to Y$ *be a proper smooth morphism, and let* $\mathbf{L} \subset \mathbf{P}$ *be the complement of the set of residual characteristics of* Y*. Let* F *be a sheaf of sets (resp. of ind-*$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) that is constructible on* X*. Then the* $R^q f_*(F)$ *are constructible and* locally constant *for* $q = 0$ *(resp. for* $q = 0, 1$*, resp. for every* q*), and for every geometric point* $\overline{y}$ *of* Y *one has*

$$R^q f_*(F)_{\overline{y}} \simeq H^q(X_{\overline{y}}, F_{\overline{y}})$$

for the same values of q.

REMARK. Taking $n = 1$ and translating into the language of fundamental groups, we recover (SGA 1 X 3.8), which compares the fundamental groups of the fibers of a proper smooth morphism.

Proof of 2.1. Since we already know that the $R^q f_*(F)$ are constructible XIV 1.1, it suffices, by IX 2.11, to prove that for every specialization morphism $\overline{y}_1 \to \overline{y}_0$ of geometric points of Y, the corresponding specialization morphism (VIII 7.7)

$$(*) \qquad R^q f_*(F)_{\overline{y}_0} \longrightarrow R^q f_*(F)_{\overline{y}_1}$$

is bijective for the values of q under consideration. For this we have the following slightly
215 more general result (in which it is unnecessary to assume that f is of finite presentation or that F is constructible):

COROLLARY 2.3. *Let* $\mathbf{L} \subset \mathbf{P}$, *let* $n \in \mathbf{N}$, *let* $f : X \to Y$ *be a proper morphism that is locally* 0*-acyclic (resp. locally* 1*-aspherical for* $\mathbf{L}$*, resp. locally* n*-acyclic for* $\mathbf{L}$*), and let* F *be a sheaf of sets (resp. of ind-*$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) on* X *that is locally constant. Then for every specialization morphism* $\overline{y}_1 \to \overline{y}_0$ *of geometric points of* Y*, the corresponding specialization homomorphism* $(*)$ *above is bijective for* $q = 0$ *(resp. for* $q = 0, 1$*, resp. for* $q \leq n$*). Moreover, when* $n = 0$*, i.e. when* f *is locally* 0*-acyclic, for every locally constant sheaf of groups* F *on* X*, the homomorphism* $R^1 f_*(F)_{\overline{y}_0} \to R^1 f_*(F)_{\overline{y}_1}$ *is injective; for arbitrary* n*, and for every locally constant abelian sheaf of* $\mathbf{L}$*-torsion* F *on* X*, the specialization homomorphism* $(*)$ *is injective for* $q = n + 1$.

There exists an integral strictly local scheme Y' and a morphism $g : Y' \to Y$ that maps the closed point y'_0 to y_0 and the generic point y'_1 to y_1: for example, it is enough

[42] For a less restrictive hypothesis than properness that yields the same conclusions, cf. SGA 5 II 3 and SGA 1 XIII.

first to take the strict localization of Y at $\overline{y}_0$, and then its integral closed subscheme defined by a point lying above y_1. Moreover, after replacing Y' by its normalization in an algebraic closure of its function field, we may assume that Y' is normal and that $k(y_1')$ is algebraically closed, so that y_0' and y_1' are geometric points of Y'. Using the base-change theorem (XII 5.1) for (f, F, g), the proof of 2.3 is reduced, after replacing Y, $\overline{y}_0$, 216
$\overline{y}_1$ by Y', y_0', y_1', to proving 2.3 when Y is normal and strictly local and when $\overline{y}_0$ and $\overline{y}_1$ are, respectively, its closed point y_0 and its generic point y_1. Moreover, using XII 4, we see that the specialization homomorphism $(*)$ is then identified with the homomorphism

$$H^q(X, F) \longrightarrow H^q(X_1, g_1^*(F)),$$

where $X_1 = X_{y_1}$ and where $g : y_1 \to Y$ and $g_1 : X_1 \to X$ are the canonical morphisms. Our assertion now follows from the following lemma (in which f is no longer assumed proper):

Lemma 2.4. *With the preliminary notation of 2.3, drop the properness assumption on f, but assume instead that Y is normal and integral, with generic point y_1 such that $k(y_1)$ is separably closed, and let $X_1 = X_{y_1}$ and $g_1 : X_1 \to X$ be the canonical morphism. Then the homomorphism*

$$(**) \qquad H^q(X, F) \longrightarrow H^q(X_1, g_1^*(F))$$

is bijective for $q = 0$ (resp. bijective for $q = 0, 1$, resp. bijective for $q \leq n$ and injective for $q = n+1$). Moreover, when $n = 0$, i.e. when f is locally 0-acyclic, for every sheaf of groups F on X, the homomorphism $(**)$ is injective for $q = 1$.

An immediate argument (XV 1.6 (i) $\Rightarrow$ (ii)) shows that it is enough to prove the relation

$$F \xrightarrow{\sim} {g_1}_* {g_1}^*(F),$$

and, for $n \geq 1$, the additional relations 217

$$R^q {g_1}_* {g_1}^*(F) = 0 \quad \text{for} \quad 1 \leq q \leq n.$$

These relations are local on X for the etale topology, and F is locally constant for the etale topology, so we may assume that F is constant and hence of the form G_X, where G is a set (resp. an ind-**L**-group, resp. an abelian group of **L**-torsion). Note also that the assumption that Y is normal implies $g_*(G_{y_1}) \xleftarrow{\sim} G_Y$ (IX 2.14.1), and the assumption that $k(y_1)$ is separably closed implies that the $R^q g_*(G_{y_1})$ vanish for $q \geq 1$. On the other hand, the assumption that f is locally 0-acyclic (resp. ...) implies that the formation of the $R^q g_*(G_{y_1})$ commutes with the base change $f : X \to Y$ for $q \leq n$. Here we use the fact that we may assume Y affine and that y_1 then appears as the inverse limit of its affine open neighborhoods, which are quasi-compact in Y, so that Definition XV 1.11 and the theory of passage to the limit VII 5 apply. We conclude that indeed

$$G_X \xrightarrow{\sim} {g_1}_*(G_{X_1}) \quad , \quad R^q {g_1}_*(G_{X_1}) = 0 \quad \text{for} \quad 1 \leq q \leq n,$$

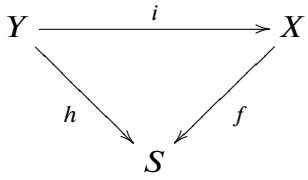
which completes the proof of 2.4, and hence that of 2.1.

Remark. The preceding proof, via 2.4, which is simpler than our original proof, is due to M. Lubkin.

Corollary 2.5. *Under the hypotheses of 2.1, assume that Y is connected, and let $q \leq n$. Then the groups $H^q(X_{\overline{y}}, F)$, for geometric points $\overline{y}$ of Y, are all isomorphic to one another.* 218

3. The relative cohomological purity theorem

DEFINITION 3.1. *Let S be a scheme. A* smooth S-pair (Y, X) *is a closed S-immersion $i : Y \to X$ of smooth S-preschemes, that is, a commutative diagram of preschemes*



such that f and h are smooth and i is a closed immersion. We write $U = X - Y$, denote by $j : U \to X$ the open immersion, and by $g : U \to S$ the structural morphism. The codimension *of (Y, X) at a point $y \in Y$ is the codimension at y of the fiber Y_s in X_s, where $s = h(y)$.*

Thus the codimension is a locally constant function on Y. We shall express the fact that this function is in fact constant, of value c, by saying that "(Y, X) has codimension c."

Let (Y, X) be a smooth S-pair and let $y \in Y$. Recall (SGA II 4.10) that there exist integers m and n, a neighborhood X' of y in X, and an étale morphism

$$\phi : X' \longrightarrow \mathbf{E}^n_S = \operatorname{Spec} \mathcal{O}_S[t_1, \dots, t_n]$$

219 such that $Y' = Y \cap X'$ is the inverse image of the closed subscheme $\mathbf{E}^m_S$ defined by the equations

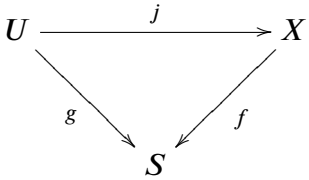
$$t_{m+1} = \cdots = t_n = 0.$$

This result can be expressed by saying that *locally for the étale topology, every smooth S-pair (Y, X) is isomorphic to the standard pair $(\mathbf{E}^m_S, \mathbf{E}^n_S)$ for suitable m and n,* where

$$\mathbf{E}^r_S = \operatorname{Spec} \mathcal{O}_S[t_1, \dots, t_r].$$

In the present section, we propose to calculate the local cohomology sheaves $\mathbf{H}^q_Y(X, F) = (R^q i^!)F$ (V 6 and VIII 6.6) for a smooth S-pair (Y, X), with coefficients in a locally constant abelian torsion sheaf F prime to the residue characteristics. We begin with some preliminary considerations:

PROPOSITION 3.2. *Let*



be a commutative diagram in which f is smooth and j is an open immersion such that the fiber U_s is dense in X_s for every $s \in S$. Let F be a sheaf of sets on S. Then the canonical morphism

$$f^*F \longrightarrow j_*g^*F = j_*j^*f^*F$$

is bijective.

220 *Proof.* We readily reduce to the case where U is quasi-compact in X. Let $\overline{s}$ be a geometric point of S, let $\overline{x}$ be a geometric point of X lying over $\overline{s}$, let $\widetilde{f} : \widetilde{X} \to \widetilde{S}$ be the morphism of the strict localizations of X and S at the corresponding geometric points, and let $\widetilde{U} = U \times_X \widetilde{X}$. With the evident notation, we have

$$(f^*F)_{\overline{x}} = H^0(\widetilde{X}, \widetilde{f}^*\widetilde{F}) \simeq H^0(\widetilde{S}, \widetilde{F}).$$

Consequently, it suffices to prove that $\tilde{g} : \widetilde{U} \to \widetilde{S}$ is 0-acyclic, which will imply

$$(j_* g^* F)_{\overline{x}} \simeq H^0(\widetilde{U}, \tilde{g}^* \widetilde{F}) \simeq H^0(\widetilde{S}, \widetilde{F}).$$

Since $\tilde{g}$ is a limit of smooth morphisms, it is locally 0-acyclic (XV 2.1, XV 1.11), and by XV 1.12 it suffices to prove that the geometric fibers of $\widetilde{U}/\widetilde{S}$ are 0-acyclic, that is, connected and nonempty. Let $\overline{s}'$ be a geometric point of $\widetilde{S}$. Then the fiber $\widetilde{X}_{\overline{s}'}$ is regular, connected, and nonempty, and the hypothesis implies that $\widetilde{U}_{\overline{s}'}$ is a dense open subset of $\widetilde{X}_{\overline{s}'}$, hence connected and nonempty, which proves the result.

COROLLARY 3.2.1. *Under the hypotheses of 3.2, the functor $X' \mapsto X'|U$ from the category of étale coverings of X to the category of étale coverings of U is fully faithful. In particular, for every finite group G, the restriction map*

$$H^1(X, G) \longrightarrow H^1(U, G)$$

is injective.

This follows from 3.2, by a standard argument which, for ease of reference, we shall 221
spell out in a

LEMMA 3.2.2. *Let $f : X \to Y$ be a morphism of schemes, and let $f^* : \mathrm{Et}(Y) \to \mathrm{Et}(X)$ be the inverse-image functor $Y' \mapsto Y' \times_Y X$ from the category $\mathrm{Et}(Y)$ of étale coverings of Y to $\mathrm{Et}(X)$. The following conditions are equivalent:*

(i) *f^* is faithful (resp. fully faithful).*

(ii) *For every étale covering Y' of Y, denoting by $f' : X' \to Y'$ the morphism deduced from f by the base change $Y' \to Y$, the map $U \mapsto f'^{-1}(U)$ from the set of clopen subsets of Y' to the set of clopen subsets of X' is injective (resp. bijective).*

(ii bis) *With the notation of* (ii), *for every constant sheaf of sets C on Y', the canonical map*

$$H^0(Y', C) \longrightarrow H^0(X', f^* C)$$

is injective (resp. bijective).

(iii) *With the notation of* (ii), *the canonical map*

$$\Gamma(Y'/Y) \longrightarrow \Gamma(X'/X)$$

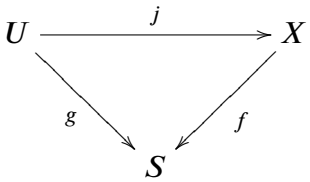
is injective (resp. bijective).

The proof is left to the reader (cf. SGA 2, IX 3.1 and 3.2).

REMARKS 3.2.3. One can generalize Proposition 3.2 considerably, both for sheaves 222
of sets, by weakening the hypotheses on f, and for sheaves of groups, with conclusions concerning the $R^q g_*(f^* F)$ (cf. SGA 2 XIV 1 and SGA 1 XIII). The analogous remark applies to 3.4 through 3.6.

Let us also "recall" the Zariski–Nagata purity theorem in its relative form:

THEOREM 3.3. *(Relative purity theorem). Let*



be a commutative diagram in which f is smooth and j is an open immersion such that, for every $s \in S$, $X_s - U_s$ has codimension ≥ 2 everywhere in X_s. Then the functor $X' \mapsto X'|U$

from the category of étale coverings of X to the category of étale coverings of U is an equivalence of categories.

Proof. We saw in 3.2.1 that the functor is fully faithful. The fact that it is essentially surjective will follow from the classical theorem. Given an étale covering U' of U, we must show that it extends to an étale covering X' of X. There is an immediate and easy reduction first to the case where U is quasi-compact in X, and then to the case where S is affine and of finite type over $\operatorname{Spec}\mathbf{Z}$, X is connected, and f is of finite type.

223 We proceed by induction on $n = \dim S$, the assertion being a consequence of the classical theorem (SGA 2 X 3.4) when $n = 0$. Since uniqueness is available, we can apply the descent technique of SGA 1 IX 4.7 to the normalization $S \leftarrow \overline{S}$ and thereby reduce to the case where S (and hence X) is normal ($\overline{S} \to S$ being finite (EGA IV 7.8.3 (iii) (vi)). Moreover, after enlarging U, we may suppose that U is maximal among the open subsets over which U' can be extended, and we must then prove that $U = X$. Otherwise, let x be a maximal point of $X - U$, and put $s = f(x)$. It is enough to prove that U' can be extended over a neighborhood of x. We may suppose that $\dim \mathcal{O}_{S,s} = n \geq 1$ and that the theorem has already been proved in dimension $< n$. Put $B = \mathcal{O}_{X,x}$, and let B' be the normalization of B in the ring of rational functions on the covering U' of U; thus B' is a finite B-algebra (*loc. cit.*). It plainly suffices to prove that B' is an étale B-algebra.

Let $A = \mathcal{O}_{S,s}$, $0 \neq t \in \max A$, $A_0 = A/tA$, and $B_0 = B/tB$. By the induction hypothesis, the restriction of the covering to $U_0 = U \otimes_A A_0$ extends to an étale covering of $\operatorname{Spec} B_0$, which we denote by $\operatorname{Spec} B_0^*$. After replacing X by a suitable étale covering of a neighborhood of x (which is permissible), we may suppose that $\operatorname{Spec} B_0^*$ is completely split over $\operatorname{Spec} B_0$. We are thereby reduced to the case where the B_0-algebra B_0' is completely split away from the closed point of $\operatorname{Spec} B_0$. The hypotheses of XV 3.3 are then satisfied, which proves the result.

Corollary 3.4. *Under the same hypotheses as in 3.3, we have $(R^1 j_*)g^*F = 0$ for every sheaf F of ind-finite groups on S.*

224 We reduce as usual to the case where S is noetherian, f is of finite type, and F is constructible. Let $0 \to F \to G$ be an injection of sheaves of ind-finite groups on S, and put $C = G/F$. It follows from 3.2 that the functor j_*g^* is exact, so that $j_*g^*G \to j_*g^*C$ is surjective, and hence (XII 3.1) that $(R^1 j_*)g^*F \to (R^1 j_*)g^*G$ is injective. We can therefore replace F by G, and thus reduce, by IX 2.14 and VIII 5.5, to the case where $F = G$ is constant, with value an ordinary finite group G. Let $\overline{x}$ be a geometric point of X, let $\widetilde{X}$ be the strict localization of X at $\overline{x}$, and let $\widetilde{U} = \widetilde{X} \times_X U$. It then follows from 3.3 that

$$(R^q j_* g^* F)_{\overline{x}} = H^q(\widetilde{U}, G_{\widetilde{U}}) \simeq H^q(\widetilde{X}, G_{\widetilde{X}}),$$

and the latter is zero, which proves the result.

Lemma 3.5. *(Relative Abhyankar's lemma). Let (Y, X) be a smooth S-pair of codimension 1, such that Y is defined in X by an equation $t = 0$. Let $U = X - Y$, and let V/U be a principal Galois covering of order n prime to the residue characteristics of Y. Put $X' = \operatorname{Spec} \mathcal{O}_X[z]/(z^n - t)$, $U' = U \times_X X'$, and $V' = V \times_X X'$. Then the covering V' extends uniquely to an étale covering of X'.*

Proof. We immediately reduce to the case where S, X, and Y are affine and of finite type over $\operatorname{Spec}\mathbf{Z}$. Uniqueness is a special case of 3.2.1. We shall deduce existence from XV 1.12: let $\overline{s}$ be a geometric point of S and let $\overline{x}$ be a geometric point of Y lying over $\overline{s}$. Let
225 $\widetilde{f} : \widetilde{X} \to \widetilde{S}$ be the morphism of the corresponding strict localizations induced by f, let

$\widetilde{U} = U \times_X \widetilde{X}$, etc...By descent, in view of uniqueness, it plainly suffices to prove that $\widetilde{V}'$ extends to an étale covering of $\widetilde{X}'$, that is (since $\widetilde{X}'$ admits no nontrivial covering), that $\widetilde{V}'$ is a trivial covering of $\widetilde{U}'$.

Let
$\overline{U} = \varprojlim_m \widetilde{U}_m = \operatorname{Spec} \mathcal{O}_{\widetilde{U}}[t^{1/m}]$, where m runs through the set of integers prime to the residue characteristic of $\widetilde{X}$. Then $\overline{U}/\widetilde{U}$ is an "infinite Galois covering" with abelian group $G = \varprojlim_m \mathbf{Z}/m\mathbf{Z}$, and one sees immediately that it is enough to prove that $\overline{U}$ is 1-aspherical for the set $\mathbf{L} \subset \mathbf{P}$ complementary to the residue characteristic of $\widetilde{X}$.

Consider the morphism $\overline{f} : \overline{U} \to \widetilde{S}$. It is a limit of smooth morphisms, and is therefore locally 1-aspherical for $\mathbf{L}$ (XV 2.1, XV 1.11 (ii)). By XV 1.12, it is enough to prove that the geometric fibres of $\overline{U}/\widetilde{S}$ are 1-aspherical for $\mathbf{L}$. Let $\overline{s}$ be a geometric point of $\widetilde{S}$. The geometric fibre $\overline{U}_{\overline{s}}$ is the limit of the fibres $(\widetilde{U}_m)_{\overline{s}}$. Consequently, it is enough to prove that every principal Galois covering of some $(\widetilde{U}_m)_{\overline{s}}$, of order prime to the residue characteristics, induces a trivial covering of $\overline{U}_{\overline{s}}$. Replacing $\widetilde{X}$ by $\widetilde{X}_m = \operatorname{Spec} \mathcal{O}_{\widetilde{X}}[t^{1/m}]$ (which is still smooth over $\widetilde{S}$), we reduce to the case $m = 1$, that is, $\overline{U}_m = \widetilde{U}$. Let V be such a covering of $\widetilde{U}_{\overline{s}}$. Now $\widetilde{X}_{\overline{s}} = Z$ is a regular strictly local scheme, and $W = \widetilde{U}_{\overline{s}}$ is the open complement of a regular subset of codimension 1. We may therefore apply the usual form of Abhyankar's lemma (SGA 1 X 3.6) to conclude that V induces a covering V_m of W_m which extends to an étale covering of Z_m, for a suitable m. But since Z, and 226
hence Z_m, is strictly local, this covering is trivial; therefore V_m is a trivial covering of W_m, C.Q.F.D.

Remark 3.5.1. When the order n of G is not assumed prime to the residue characteristics of X, one can still prove the following: let m be the largest integer prime to the residue characteristics of Y which divides the order of G, and suppose that for every point $s \in S$, the covering V_s of $U_s = X_s - Y_s$ has, at the maximal points of Y_s, inertia groups whose orders are prime to the residue characteristics of Y. Then the conclusion of 3.5 remains valid on taking $X' = \operatorname{Spec} \mathcal{O}_X[z]/(z^m - t)$. More generally, one may also take for m any common multiple of the orders of the inertia groups just considered, provided these orders are prime to the corresponding residue characteristics.

Corollary 3.6. *(Cohomological purity in dimension 1). Let (Y, X) be a smooth S-pair of codimension 1, and let F be a constructible locally constant sheaf of groups on X, whose orders are prime to the residue characteristics of X. Then, with the notation of 3.1, $(R^1 j_*)j^*F$ is locally isomorphic, as a sheaf of pointed sets, to $i_*(F_Y/\operatorname{int}(F_Y))$, where $F_Y = F|Y$ and where $F_Y/\operatorname{int}(F_Y)$ denotes the quotient of F_Y by its action on itself through inner automorphisms.*

Remark. The isomorphism of Corollary 3.6 is not canonical.

Proof. We immediately reduce to the case where X is noetherian and connected. 227
Since the assertion is local on X for the étale topology, we may suppose that F is a constant sheaf, with value an ordinary finite group G of order m prime to the residue characteristics of X, and moreover that $\boldsymbol{\mu}_{mX} \simeq (\mathbf{Z}/m\mathbf{Z})_X$. Let $U = X - Y$ and let $U' = \operatorname{Spec} \mathcal{O}_U[t^{1/m}]$, which is an "essential" principal Galois covering of U with group $\mathbf{Z}/m\mathbf{Z}$, since $\boldsymbol{\mu}_{mX} \simeq (\mathbf{Z}/m\mathbf{Z})_X$. Principal coverings of U with group G which are trivialized over U' are classified locally on Y by $H = \operatorname{Hom}(\mathbf{Z}/m\mathbf{Z}, G)$ modulo $\operatorname{int}(G)$, which is a pointed set isomorphic to the underlying set of $G/\operatorname{int}(G)$. This gives a morphism from

the constant sheaf H_Y on Y to $(R^1 j_*)G_U$, and it is enough to prove that this morphism is bijective, which follows immediately from 3.5.

THEOREM 3.7. (Cohomological purity). *Let (Y, X) be a smooth S-pair of codimension $c > 0$. Let F be an abelian sheaf on X locally isomorphic (for the étale topology) to a sheaf of the form $f^*(G)$, where G is a torsion sheaf on S, prime to the residue characteristics (for example, F may be a locally constant sheaf of finite groups whose orders are prime to the residue characteristics of X). Then, with the notation of 3.1, one has*

$$\mathrm{H}^q_Y(X, F) = (R^q i^!)F = 0 \qquad \text{if} \quad q \neq 2c,$$

(i.e.)
$$\begin{cases} F \xrightarrow{\sim} j_* j^* F \quad \text{and} \\ (R^q j_*) j^* F = 0 \qquad\qquad \text{if} \quad q \neq 0, \quad 2c-1, \end{cases}$$

228 *and*

$$\mathrm{H}^{2c}_Y(X, F) = i^*[(R^{2c-1} j_*) j^* F]$$

is a sheaf locally isomorphic to $i^*(F)$.

Proof. The assertion is local on X for the étale topology; we may therefore [42] suppose that F is constant, and hence, if desired, that $F \simeq f^*(\mathbf{Z}/n\mathbf{Z})_S$ for the constant sheaf $(\mathbf{Z}/n\mathbf{Z})_S$ on S.

First treat the case $c = 1$. (The assertion concerning $(R^1 j_*) j^* F$, at least in the case where F is locally constant, would follow immediately from 3.6, but we shall derive it again.) Recall that (Y, X) is locally isomorphic to the pair $(\mathbf{E}^{n-1}_S, \mathbf{E}^n_S)$, and that, upon replacing S by $\mathbf{E}^{n-1}_S$, we reduce to the case where $X = \mathbf{E}^1_S = \operatorname{Spec} \mathcal{O}_S[t]$ and Y is the section $t = 0$ of X/S. This situation is locally isomorphic to the inclusion of the section at infinity in the projective line $\mathbf{P}^1_S$. We may therefore take $X = \mathbf{P}^1_S$, take Y to be the section at infinity, and put $U = \mathbf{E}^1_S = X - Y$.

Consider the Leray spectral sequence

$$(R^p f_*)(R^q j_*)(\mathbf{Z}/n) \Longrightarrow (R^{p+q} g_*)(\mathbf{Z}/n).$$

Its abutment is zero for $p+q > 0$, by XV 2.2. Moreover, the $R^q j_*$ are supported on Y when $q > 0$, and Y maps isomorphically onto S under f; on the other hand, $j_*(F|U) \xleftarrow{\sim} F$ by 3.2. It follows that $(R^p f_*)(R^q j_*) = 0$ if $p, q > 0$, that $f_*(R^q j_*)$ determines the sheaf $(R^q j_*)$ when $q > 0$, and finally that $R^p f_* j_*(F|U) = R^p f_*(F)$. The spectral sequence
229 therefore reduces to isomorphisms

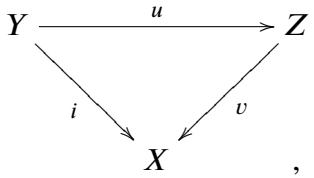
$$f_*(R^q j_*)(F|U) \xrightarrow{\sim} (R^{q+1} f_*)(F) \quad \text{if} \quad q > 0.$$

But $(R^q f_*)(F)$ can be computed fibre by fibre (XII 5.2); it is therefore zero unless $q = 0, 2$, and is a sheaf locally isomorphic to G if $q = 2$, as follows easily from XII 5.2 and IX 4.7. Thus $(R^q j_*)(F|U) = 0$ if $q \neq 1$, and it is a sheaf locally isomorphic to $i^*(F)$ if $q = 1$.

The case $c > 1$ is now handled easily by induction: let (Y, X) be a smooth S-pair of codimension $c > 1$. It is clear that, locally on X, one can find a subscheme Z of X of codimension 1 which contains Y and is smooth over S, so that there is a commutative

[42] At least if F were assumed locally constant. The reader will convince himself that the proof which follows also applies, essentially, to the general case.

diagram



,

in which each pair is a smooth S-pair (one ought to call (Y, Z, X) a smooth S-triple), the codimension of (Z, X) is 1, and that of (Y, Z) is $c - 1$.

Now there is a spectral sequence of composite functors

$$(R^p u^!)(R^q v^!)F \Longrightarrow (R^{p+q} i^!)F.$$

By the case already proved, $(R^q v^!)F = 0$ if $q \neq 2$ and is locally isomorphic to $v^*(F)$ if $q = 2$. The induction hypothesis, applied to $(R^q v^!)F$, then shows that $(R^p u^!)(R^q v^!)F$ is zero if $(p, q) \neq (2c-2, 2)$ and is locally isomorphic to $u^* v^*(F)$ if $(p, q) = (2c-2, 2)$. The result for $(R^{p+q} i^!)F$ follows immediately, C.Q.F.D. 230

Corollary 3.8. *Let (Y, X) be a smooth S-pair of codimension c, let n be a natural number prime to the residue characteristics of X, and put*

$$T_{Y/X} = \mathbf{H}^{2c}_Y(\mathbf{Z}/n\mathbf{Z}),$$

which, by (3.7), is a sheaf on Y locally isomorphic (for the étale topology) to the sheaf $(\mathbf{Z}/n\mathbf{Z})_Y$. Let F be an abelian sheaf on X, annihilated by n, satisfying the condition stated in 3.7. Then (for the record) $\mathbf{H}^i_Y(F) = 0$ for $i \neq 2c$, and moreover there is a canonical isomorphism

$$\mathbf{H}^{2c}_Y(F) \simeq i^*(F) \otimes T_{Y/X}.$$

Proof. One easily defines a canonical homomorphism

$$\varphi : i^*(F) \otimes T_{Y/X} \longrightarrow \mathbf{H}^{2c}_Y(F)$$

that is, a homomorphism

$$i^*(F) \longrightarrow \mathrm{Hom}(\mathbf{H}^{2c}_Y(\mathbf{Z}/n\mathbf{Z}), \mathbf{H}^{2c}_Y(F)),$$

using the canonical homomorphism $F \to \mathrm{Hom}(\mathbf{Z}/n\mathbf{Z}, F)$. It remains to prove that φ is an isomorphism, which is easily checked by following the proof given for 3.7.

Corollary 3.8.1. *Under the conditions of 3.8, the formation of the $\mathbf{H}^i_Y(F)$ commutes with every base change $S' \to S$.* 231

This is clear.

Corollary 3.9. *(Absolute cohomological purity theorem). Let X be a regular scheme, and let Y be a regular closed subprescheme of codimension c at every point. Suppose moreover that X is locally of finite type over a perfect field k. Then, if n is prime to the residue characteristics of X, one has*

$$\mathbf{H}^i_Y(\mathbf{Z}/n\mathbf{Z}) = 0 \text{ if } i \neq 2c,$$

and $\mathbf{H}^{2c}_Y(\mathbf{Z}/n\mathbf{Z})$ is a sheaf on Y locally isomorphic (for the étale topology) to the constant sheaf $\mathbf{Z}/n\mathbf{Z}$.

This is a special case of 3.7, since the hypothesis implies that X and Y are smooth over k.

REMARKS 3.10. a) In 3.8 one can make the structure of the sheaf $T_{Y/X}$ explicit: one finds that it is canonically isomorphic to the sheaf $(\boldsymbol{\mu}_n)_Y^{\otimes -c}$, where $\boldsymbol{\mu}_n$ denotes the sheaf of nth roots of unity (locally isomorphic to the constant sheaf $\mathbf{Z}/n\mathbf{Z}$). The isomorphism in question is studied, under the name of the local *fundamental class* of Y in X, in the slightly more general setting of relative complete intersections, in Exposé XVIII, which is devoted to duality.

232 b) It is very plausible that the conclusions of 3.9 and 3.10 a) remain valid without assuming the existence of a base field k, at least when X is an "excellent" prescheme. This will in any case be established in Exposé XIX when X is of characteristic zero, using Hironaka's resolution of singularities. As we shall see in SGA 5, the "absolute purity theorem" (as well as resolution of singularities) is needed in particular to establish the "local duality formula", which itself is one of the principal ingredients of the LEFSCHETZ–VERDIER formula.

4. Comparison theorem for the cohomology of algebraic schemes over C.

Let X be an algebraic scheme over Spec C. We shall compare the étale and classical sites of X, and we retain the notation of XI 4. The comparison theorem, whose general form is XI 4.4, is as follows:

THEOREM 4.1. *Let $f : X \to S$ be a morphism of finite type between schemes locally of finite type over* Spec C*, so that there is a commutative diagram of morphisms of sites*

$$\begin{array}{ccc} X_{\text{et}} & \xleftarrow{\varepsilon} & X_{\text{cl}} \\ {\scriptstyle f_{\text{et}}}\downarrow & & \downarrow{\scriptstyle f_{\text{cl}}} \\ S_{\text{et}} & \xleftarrow{\varepsilon} & S_{\text{cl}} \end{array}$$

233 *Let F be a sheaf of sets (resp. of ind-finite groups, resp. a torsion abelian sheaf), and suppose that one of the following cases holds:*

(i) *f is proper,*
(ii) *F is constructible.*

Then the base-change morphisms XII 4.2 [42]

$$\varphi : \epsilon^*(R^q f_{\text{et}*})F \longrightarrow (R^q f_{\text{cl}\,*})\epsilon^* F$$

are bijective for $q = 0$ (resp. for $q = 0, 1$, resp. for $q \geq 0$).

Observe that the Grauert–Remmert theorem (XI 4.3 (iii)), which we used in the proof of XI 4.4, is a special case of 4.1.

234 For the proof, we first treat the case in which f is proper. In this case, the result will follow from GAGA and the proper base-change theorem, by an elementary argument that could have appeared in Exposé XIV. To treat the general case, we shall also use resolution of singularities [1] and 3.7.

We may clearly assume that S, and hence also X, is of finite type over C.

The case in which f is proper. We recall the following elementary facts:

(a) The topological space X_{cl} is locally compact.

[42] In fact, we use the base-change homomorphism under more general conditions than those considered in Exposé XII, where we restricted ourselves to a *Cartesian* square of morphisms of *preschemes*. The reader will readily verify that the definition extends verbatim to an *essentially commutative* square of morphisms of *sites*. See the following exposé for further developments on this point.

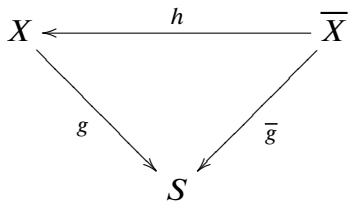
(b) If $f : X \to S$ is proper, then $f_{\mathrm{cl}} : X_{\mathrm{cl}} \to S_{\mathrm{cl}}$ is a *proper* map of topological spaces.

Indeed, (a) is trivial, since locally X_{cl} is a closed subset of some $\mathbf{C}^n$. For (b), Chow's lemma reduces us to the case in which f is projective, and this case is trivial.

Let F be a sheaf of sets (resp. ...) on X_{et}. For both morphisms f_{et} and f_{cl}, formation of the R^q commutes with formation of the fibers (XII 5.2 and [2], 4.11.1). It is therefore enough to verify the theorem fiber by fiber, at the closed points of S_{et}.

In the following lemma, we do not assume that g is proper:

LEMMA 4.2. *Let* 235



be a commutative diagram with h proper, and let U be an open subset of X such that h induces an isomorphism from the open subset $\overline{U} = U \times_X \overline{X}$ of $\overline{X}$ onto U. Let $j : U \to X$ and $\overline{j} : \overline{U} \to \overline{X}$ be the inclusions, let F be a sheaf of sets (resp. ...) on U, and let $\overline{F}$ be the induced sheaf on $\overline{U}$. If Theorem 4.1 is true for $(\overline{g}, \overline{j}_!\overline{F})$, then it is also true for $(g, j_!F)$.

Proof. The theorem is true for $(h, \overline{j}_!\overline{F})$, because we are reduced to checking it fiber by fiber, and there are then two cases: either the restriction of the sheaf to the fiber is zero, or the morphism h is an isomorphism. In both cases the result is trivial. We find that

$$\epsilon^* j_! F \simeq \epsilon^* h_{\mathrm{et}\,*}(\overline{j}_!\overline{F}) \simeq h_{\mathrm{cl}\,*}\epsilon^*(\overline{j}_!\overline{F})$$

and

$$(R^q h_{\mathrm{et}\,*})\overline{j}_!\overline{F} = (R^q h_{\mathrm{cl}\,*})\epsilon^*(\overline{j}_!\overline{F}) = 0 \text{ if } q > 0$$

for the values of q under consideration, and hence that

$$(R^q g_{\mathrm{et}\,*}) j_! F \simeq (R^q \overline{g}_{\mathrm{et}\,*})\overline{j}_!\overline{F}$$

and

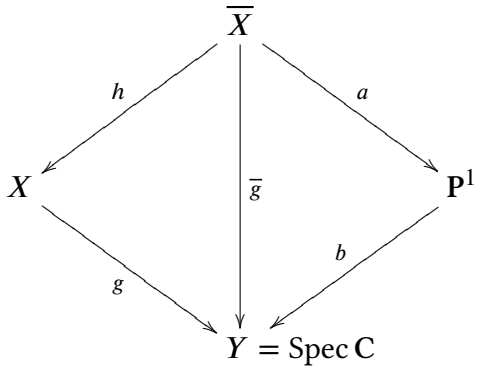
$$(R^q g_{\mathrm{cl}\,*}) j_! F \simeq (R^q \overline{g}_{\mathrm{cl}\,*})\overline{j}_!\overline{F},$$

which proves Lemma 4.2. 236

LEMMA 4.3. *To verify 4.1 for all data (f, F) with f proper, it is enough to do so when, in addition, f has relative dimension ≤ 1 and $S = \operatorname{Spec} \mathbf{C}$ is a point.*

Proof. We proceed by induction on the relative dimension. Suppose the result known in relative dimension $< n$, where $n > 1$. As we have observed, we may check it fiber by fiber; that is, we may suppose that $S = \operatorname{Spec} \mathbf{C}$ is a point, and hence that X has dimension $\leq n$. Moreover, we may assume that X is reduced. Let φ be a rational function on X that is nonconstant on every irreducible component of X, let U be a dense open subset of X on which φ is defined, and let $\overline{X} \subset X \times \mathbf{P}^1$ be the closure of the graph of the morphism

$\varphi : U \to \mathbf{E}^1$. We have a commutative diagram



in which the relative dimensions of a and b are $< n$. By the induction hypothesis, the theorem is true for $(a, .)$ and $(b, .)$, and it follows that it is also true for $(\overline{g}, .)$. Indeed, this
237 is trivial for $q = 0$; if F is a torsion abelian sheaf on $\overline{X}$, it follows from the morphism of Leray spectral sequences

$$\begin{array}{ccc} \varepsilon^*(R^p b_{\mathrm{et}*})(R^q a_{\mathrm{et}*})F & \Longrightarrow & \varepsilon^*(R^{p+q}\overline{g}_{\mathrm{et}*})F \\ \downarrow & & \downarrow \\ (R^p b_{\mathrm{cl}*}(R^q a_{\mathrm{cl}*}))\varepsilon^* F & \Longrightarrow & (R^{p+q}\overline{g}_{\mathrm{cl}*})\varepsilon^* F; \end{array}$$

finally, if F is a sheaf of ind-finite groups on $\overline{X}$, note that since $\epsilon^*(R^1\overline{g}_{\mathrm{et}\,*})$ is an effaceable functor, it is enough to prove (in view of the result for $q = 0$) that $(R^1\overline{g}_{\mathrm{cl}\,*})\epsilon^*$ is also effaceable (cf. XII 8.2). This follows from the exact sequence

$$0 \longrightarrow (R^1 b_{\mathrm{cl}\,*})a_{\mathrm{cl}\,*}\epsilon^* F \longrightarrow (R^1\overline{g}_{\mathrm{cl}\,*})\epsilon^* F \longrightarrow b_{\mathrm{cl}\,*}(R^1 a_{\mathrm{cl}\,*})\epsilon^* F,$$

and from the fact that $(R^1 b_{\mathrm{cl}\,*})\epsilon^*$ and $(R^1 a_{\mathrm{cl}\,*})\epsilon^*$ are effaceable by the induction hypothesis.

Now let F be a sheaf of sets on X. We have $F \hookrightarrow h_{\mathrm{et}\,*}h_{\mathrm{et}}{}^* F = G$. The exact sequence

$$F \to G \rightrightarrows G \amalg_F G, \tag{4.3.1}$$

reduces us (for $q = 0$) to proving 4.1 for G, and hence for a sheaf of the form $h_{\mathrm{et}\,*}\overline{F}$. For such a sheaf, however, the result follows from the theorem for $(\overline{g}, .)$ and for $(h, .)$; and h has relative dimension $< n$, as is immediate.

Let F be a torsion abelian sheaf on X. We have the exact sequence

$$0 \longrightarrow j_! j^* F \longrightarrow F \longrightarrow i_* i^* F \longrightarrow 0. \tag{$*$}$$

238 Since U is dense in X, we have $\dim Y < n$, so 4.1 is true for $(g|Y, i^*F)$, and hence for $(g, i_* i^* F)$. It is therefore enough, by the five lemma, to verify 4.1 for $(g, j_! j^* F)$, and this follows from the theorem for $\overline{g}$ (4.2).

Let F be a sheaf of ind-finite groups on X. It is enough to prove that the functor $(R^1 g_{\mathrm{cl}\,*})\epsilon^*$ is effaceable for the sheaf F. This follows from $(*)$ and from effaceability for $j_! j^* F$ and for $i_* i^* F$, that is, from the theorem for $(g, j_! j^* F)$ (4.2) and for $(g, i_* i^* F)$ (the induction hypothesis), C.Q.F.D.

Lemma 4.4. *To prove (4.1) when f is proper, it is enough to prove the following assertion: let X be a complete regular curve over* Spec C*, and let F be a constant constructible sheaf of sets (resp. ...) on X. Then the morphisms*

$$H^q(X_{\mathrm{et}}, F) \longrightarrow H^q(X_{\mathrm{cl}}, F)$$

are bijective for the values of q under consideration.

Proof. Since it is enough to verify the theorem fiber by fiber, 4.3 reduces us to the case in which X is proper and has dimension ≤ 1. The case of dimension 0 is trivial, and it follows that the theorem is true for a finite morphism. Since f is proper, the cohomology of X_{et} and of X_{cl} commutes with filtered direct limits (VII 3.3 and [2], 4.12.1), and we are therefore reduced to the case in which F is constructible (IX 2.9 (iii)).

We apply IX 2.14. For a sheaf of sets, the exact sequence (4.3.1) and IX 2.14 show that we may take $F = \pi_* C$, where $\pi : X' \to X$ is finite, X' is normal and hence regular, 239
and C is a constant sheaf on X'. For a torsion abelian sheaf, we take a resolution of F by finite products of sheaves of the form $\pi_* C$ and apply the spectral sequence of a resolution, again reducing to the case $F = \pi_* C$. For a sheaf of ind-finite groups, recall that it is enough to prove effaceability of $(R^1 g_{\text{cl}\,*})\epsilon^*$ for the sheaf F, hence for a sheaf G into which F can be embedded; thus we are again reduced to the case $F = \pi_* C$. It is now clear from (VIII 5.5 and VIII 5.8) that

we may now replace X by X', which proves the lemma.

To treat this last case, it is plainly enough (in view of the cohomological-dimension results (IX 5.7) and [2], 4.14.1) to prove that for a complete regular curve one has

($q = 0$): X is connected and nonempty if and only if X_{cl} is connected and nonempty.
($q = 1$): The base-change morphism induces an equivalence between the category of étale coverings of X and the category of finite analytic étale coverings of X_{cl}.
($q = 2$): There is an isomorphism $\epsilon : H^2(X_{\text{et}}, \mu_n) \xrightarrow{\sim} H^2(X_{\text{cl}}, \mu_n)$.

The first two assertions follow immediately from GAGA. For the last, note that $H^2(X_{\text{et}}, \mathbf{G}_m) = 0$ (IX 4.5). Likewise, if $\mathcal{O}^*$ denotes the sheaf of invertible holomorphic functions on X_{cl}, then $H^2(X_{\text{cl}}, \mathcal{O}^*) = 0$. This follows from the exact sequence

$$0 \longrightarrow \mathbf{Z} \longrightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \longrightarrow 0$$

and from the fact that $H^2(X_{\text{cl}}, \mathcal{O}) = H^2(X_{\text{et}}, \mathcal{O}_X) = 0$ by GAGA. Consequently the 240
Kummer exact sequence IX 3.2 (respectively, the exact sequence on X_{cl}

$$0 \to \mu_n \to \mathcal{O}^* \xrightarrow{n} \mathcal{O}^* \to 0)$$

gives

$$H^1(X_{\text{et}}, \mathbf{G}_m)/n \xrightarrow{\sim} H^2(X_{\text{et}}, \mu_n)$$
$$(resp. \quad H^1(X_{\text{cl}}, \mathcal{O}^*)/n \xrightarrow{\sim} H^2(X_{\text{cl}}, \mu_n)).$$

By GAGA, the canonical morphism

$$H^1(X_{\text{et}}, \mathbf{G}_m) = \operatorname{Pic} X \longrightarrow \operatorname{Pic} X_{\text{cl}} = H^1(X_{\text{cl}}, \mathcal{O}^*)$$

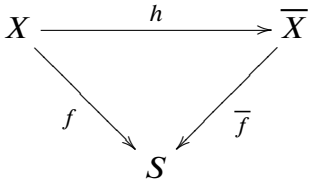
is bijective. This proves the result, and hence proves 4.1 when f is proper.

The case where F is constructible. The proof proceeds by induction on $\dim X$. Suppose the result known in dimensions $< n$, and prove it when $\dim X \leq n$.

Lemma 4.5. *We may suppose that $f : X \to S$ is a dense open immersion and that F is constant.*

Proof. As in the proof of 1.3, we may suppose that X and S are affine, and hence that f is quasi-projective. We then apply the argument of 4.4 to reduce to $F = \pi_* C$, where $\pi : X' \to X$ is finite and C is constant. Replacing X by X' therefore reduces us to the 241

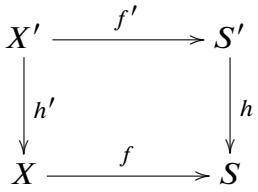
case where F is constant. We may replace X by X_{red}. Let



be a commutative diagram in which $\overline{X}$ is reduced, $\overline{f}$ is projective, and h is a dense open immersion. It is enough to verify the theorem for the two morphisms h and $\overline{f}$ (compare the proof of 4.3); hence it is enough to verify it for h, since $\overline{f}$ is proper. We are thus reduced to a dense open immersion.

LEMMA 4.6. *We may further suppose that S is regular.*

Proof. Let $S' \xrightarrow{h} S$ be a "resolution of the singularities of S", that is, a surjective, proper, birational morphism with S' regular, and let



be the Cartesian diagram obtained from h.

Let F be a constant sheaf of sets on X. We have $F \hookrightarrow G = h'_* h'^* F$, and it follows that, for $q = 0$, it is enough to verify 4.1 for G (compare (4.3.1)), that is, for a sheaf of the
242 form ${h'}_* F'$, where F' is constant on X'. We have a commutative diagram (in which the symbol "ét" is suppressed)

$$\begin{array}{ccc} \varepsilon^* f_*(h'_* F') = \varepsilon^* h_* f'_* F' & \xrightarrow{\quad a \quad} & h_{\mathrm{cl}\,*} \varepsilon^* f'_* F' \\ \downarrow{\scriptstyle d} & & \downarrow{\scriptstyle b} \\ f_{\mathrm{cl}\,*} \varepsilon^* (h'_* F') & \xrightarrow{\quad c \quad} & f_{\mathrm{cl}\,*} h'_{\mathrm{cl}\,*} \varepsilon^* F' = h_{\mathrm{cl}\,*} f'_{\mathrm{cl}\,*} \varepsilon^* F', \end{array}$$

and a and c are bijective by the result for a proper morphism. To prove that d is bijective, it is enough to prove that b is, that is, that

$$\varepsilon^* f'_* F' \xrightarrow{\sim} f'_{\mathrm{cl}\,*} \varepsilon^* F'.$$

We are therefore reduced to $S = S'$, with S regular.
(Remark: we have used, without mentioning it explicitly, the finiteness theorem for a proper morphism XIV 1.1.)

Let F be a constant torsion abelian sheaf on X. It is enough to treat $F = (\mathbf{Z}/n)_X$. The morphism h' is an isomorphism above a dense open subset $j : U \to X$; let $j' : U' \to X'$ be the open subset ${h'}^{-1}(U) \xrightarrow{\sim} U$, and put $Y = X - U$. Since U is dense, $\dim Y < n$, so the theorem holds for $(f, (\mathbf{Z}/n)_Y)$. The exact sequence

$$0 \longrightarrow j_!(\mathbf{Z}/n)_U \longrightarrow (\mathbf{Z}/n)_X \longrightarrow (\mathbf{Z}/n)_Y \longrightarrow 0$$

and the five lemma reduce us to proving the theorem for $j_!(\mathbf{Z}/n)_U$ and f, and hence, by 4.2, to proving it for $(hf', j'_!(\mathbf{Z}/n)_{U'})$. We have an exact sequence

$$0 \longrightarrow j'_!(\mathbf{Z}/n)_{U'} \longrightarrow (\mathbf{Z}/n)_{X'} \longrightarrow (\mathbf{Z}/n)_{Y'} \longrightarrow 0,$$

and the theorem holds for $(hf', (\mathbf{Z}/n)_{Y'})$ by the induction hypothesis. It is therefore 243
enough to prove it for $(hf', (\mathbf{Z}/n)_{X'})$. But the proper case gives the theorem for $(h, \cdot)$, and the Leray spectral sequence for the pair of morphisms h, f' shows that it is enough to prove it for $(f', \mathbf{Z}/n)$. This proves the lemma in the abelian case.

Finally, suppose that F is a constant sheaf of finite groups. Recall that it is enough to prove that the functor $(R^1 f_{\mathrm{cl}\,*})\epsilon^*(\cdot)$ is effaceable for F, hence for a sheaf G into which F embeds, and therefore for $G = h'_* h'^* F$, that is, for a sheaf of the form $h'_* F'$, where F' is constant on X'. We have

$$\epsilon^*(h'_* F') \xrightarrow{\sim} h'_{\mathrm{cl}\,*}(\epsilon^* F')$$

and

$$0 \longrightarrow (R^1 f_{\mathrm{cl}\,*}) h'_{\mathrm{cl}\,*} \epsilon^*(\cdot) \longrightarrow (R^1 (f_{\mathrm{cl}} h'_{\mathrm{cl}})_*)\epsilon^*(\cdot) = (R^1 (h_{\mathrm{cl}} f'_{\mathrm{cl}})_*)\epsilon^*(\cdot).$$

Thus it is enough to prove that the last functor is effaceable for a constant sheaf. But the exact sequence

$$0 \longrightarrow (R^1 h_{\mathrm{cl}\,*}) f'_{\mathrm{cl}\,*} \epsilon^*(\cdot) \longrightarrow (R^1 (h_{\mathrm{cl}} f'_{\mathrm{cl}})_*)\epsilon^*(\cdot) \longrightarrow h_{\mathrm{cl}\,*}(R^1 f'_{\mathrm{cl}\,*})\epsilon^*(\cdot)$$

shows that the first term is effaceable, for example by the Godement resolution (XII 3.3), because h is proper. It is therefore enough to prove that the right-hand term is effaced by the Godement resolution, hence that $(R^1 f'_{\mathrm{cl}\,*})\epsilon^*(\cdot)$ is so effaced, and thus that the theorem holds for (f', F'), where F' is constant. This proves the lemma.

End of the proof. We now suppose that F is a constant sheaf and that $f : X \to S$ is 244
a dense open immersion, with S regular. The assertion for sheaves of sets now follows immediately from 3.2 and the analogous trivial lemma for the classical topology. Indeed, these show that $\epsilon^* f_* F$ and $f_{\mathrm{cl}\,*}\epsilon^* F$ are constant with the same value and hence are isomorphic under φ.

For $q > 0$, we first reduce further to the case $X = S - Y$ with Y regular, that is, to the case in which (Y, S) is a smooth Spec C-pair. Indeed, write $X = S - C_1$, where C_1 is given its induced reduced structure, and define recursively

Y_ν = the dense open subset of regular points of C_ν,

$C_{\nu+1} = C_\nu - Y_\nu$, with its induced reduced structure.

Since Y_ν is dense in C_ν, we have $\dim C_{\nu+1} < \dim C_\nu$, and hence a sequence

$$X = X_1 \subset X_2 = S - C_2 \subset X_3 = S - C_3 \subset \cdots \subset X_r = S,$$

where the inclusions $i_\nu : X_\nu \to X_{\nu+1}$ are open immersions and

$$X_\nu = X_{\nu+1} - Y_\nu, \qquad Y_\nu \text{ closed in } X_{\nu+1} \text{ and regular.}$$

The direct image $i_{\nu *} F$ of a constant sheaf on X is constant (3.2), and the $R^q i_{\nu *} F$ are constructible and concentrated on the subvarieties of Y of dimension at most $n-1$ (3.4, 3.6, 3.7) for the values of q under consideration. The induction hypothesis and the Leray spectral sequence (respectively, the exact sequence XII 3.2) therefore reduce us to the 245
case of i_ν, which proves the asserted reduction.

For a smooth pair (Y, S), however, the values of $(R^q f_*)F$ have been computed explicitly for the étale topology (3.4, 3.6, 3.7); an analogous and easy computation, which we leave to the reader, gives the corresponding results for the classical topology. Direct comparison proves the theorem for f.

5. The finiteness theorem for algebraic schemes in characteristic zero

THEOREM 5.1. *Let k be a field of characteristic zero, and let $f : X \to S$ be a morphism of finite type between schemes locally of finite type over k. Let F be a constructible sheaf of sets (resp. of finite groups, resp. a torsion abelian sheaf) on X. Then the $R^q f_* F$ are constructible for $q = 0$ (resp. for $q = 0, 1$, resp. for every q).*

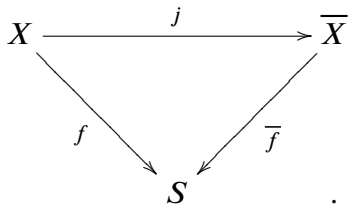
The hypothesis that k have characteristic zero is imposed because we shall use the resolution-of-singularities theorem [1]. The proof is also valid in characteristic $p > 0$ if resolution of singularities is assumed, *provided that the orders of F are prime to p.* Thus in nonzero characteristic one obtains a result for $\dim X \leq 2$ by applying Abhyankar's
246 results [3]. In addition to the ever-present proper base-change theorem, another ingredient of the proof is the "absolute cohomological purity theorem" 3.9.

Recall that the theorem has already been proved for a proper morphism and arbitrary S (XIV 1.1). Moreover, in Exposé XIX we shall obtain (again in characteristic zero) the finiteness theorem under the much weaker hypotheses that S be an *excellent* scheme and f be of finite type, by proving for such schemes the absolute-purity theorem in the form indicated in 3.10.

In nonzero characteristic and dimension at least 3, for the time being we have only the following easy consequence of XI 3.3:

THEOREM 5.2. *Let X be a smooth algebraic scheme over* Spec k, *where k is a separably closed field, and let F be a locally constant sheaf of finite abelian groups (resp. finite groups), of orders prime to the characteristic of k. Then $H^q(X, F)$ is a finite group (resp. a finite pointed set) for every q (resp. for $q = 0, 1$).*

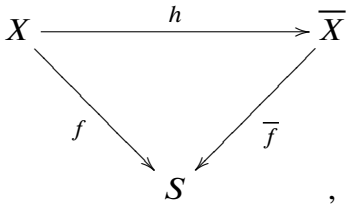
Proof of 5.2. We may suppose that k is algebraically closed (VIII 1.1). We argue by induction on $n = \dim X$. By XI 3.3, there is a hypercovering of X by open subsets that are "good neighborhoods". The spectral sequence of a hypercovering immediately reduces us to the case in which X is a good neighborhood and hence admits an elementary fibration (XI 3.1)



247 By 3.2, 3.6, and 3.7, the sheaf $j_* F$ is locally constant on $\overline{X}$, $(R^1 j_*)F$ is locally constant on $Y = \overline{X} - X$, and $(R^q j_*)F = 0$ for $q > 1$. It follows from 2.1 that $(R^p \overline{f}_*)(R^q j_*)F$ is a locally constant sheaf on S for the values of q under consideration. The result follows from the induction hypothesis and the Leray spectral sequence (resp. the exact sequence XII 3.2).

The proof of 5.1 is very close to the proof of 4.1 in the constructible case. We merely indicate the main steps. Argue by induction on $\dim X = n$. First reduce to X and S affine, and hence f quasi-projective, by applying the method used in the proof of 1.3. Next apply IX 2.14 and VIII 5.5, VIII 5.8 to reduce to the case in which F is constant.

There is a commutative diagram



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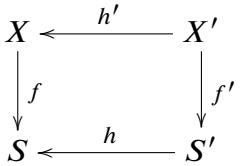
in which h is a dense open immersion and $\overline{f}$ is proper, so the theorem holds for $\overline{f}$ (XIV 1.1). It is therefore enough to prove the theorem for (h, F), that is, for a dense open immersion and a constant sheaf. Let $h : S' \to S$ be a resolution of the singularities of 248
S, and let



be the Cartesian diagram obtained from h. We reduce to proving the theorem for f', and hence to the case in which $S = S'$ is regular. The inductive method used at the end of the proof of 4.1 reduces us to the case in which $Y = S - X$ is nonsingular, that is, (Y, S) is a smooth pair. The proof is completed by applying 3.2, 3.4, 3.6, and 3.7.

Remark 5.3. The proof of 5.1 just outlined is also valid when one is given a left Noetherian ring A annihilated by an integer $n > 0$ (which must be assumed prime to the characteristic when the latter is not assumed to be zero; compare the remarks following 5.1) and considers constructible sheaves of left A-modules.

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EXPOSÉ XVII

Cohomology with proper supports

P. Deligne

Introduction

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In this exposé we develop the formalism of cohomology with proper support. The questions of variance have been treated with considerable care, whereas they were only made plausible in the oral seminar. This explains the length of the exposé, whose Sections 1 to 4 are devoted to categories and general topology. The reader is very strongly advised to read only Sections 5 and 6, where the geometric substance of the exposé is concentrated (§ 5: construction and variance of cohomology with proper supports, base-change, finiteness, finite tor-dimension, and comparison theorems; § 6: theory of the trace morphism for a quasi-finite flat morphism).

In Sections 1 and 2, we "recall" some results on derived categories. The § 1 n° 1 is devoted to the sign formalism. In the § 3, we treat the problem of gluing two variance formalisms. In the § 4, we introduce flat resolutions and study the special properties of GODEMENT's flasque resolutions.

In this exposé, the higher direct-image functors with proper supports $R^q f_!$, and 253
the functor $Rf_!$ from which they arise, are defined only when f is a compactifiable morphism (3.2.1). In the appendix, written by B. SAINT-DONAT, it is shown how to extend the definition to separated morphisms of finite type with quasi-compact quasi-separated target.[43]

The § 5 n° 5 (cohomology of a symmetric product) will not be used again in this seminar. It will be used in SGA 5 to refine the rationality theorem for L-functions.[44]

The § 6 n° 3 (the theory of the trace for *continuous* coefficients) will be used in Exposé XVIII only in the relatively easy case of smooth groups; a reader interested in the Poincaré duality theorem (global duality), and willing to accept a transcendental argument, may even dispense entirely with reading this § 6 n° 3, as well as most of the § 1 of XVIII.

0. Terminological preliminaries

0.1. The sign = placed between two groups of symbols denoting objects of a category will sometimes mean (by abuse of notation) that these objects are canonically

[42]The present exposé and the following one, written in 1968 and 1969, reproduce and supplement the oral lectures of A. GROTHENDIECK (spring 1964). The author of this written version, who did not attend the oral seminar, drew in part on notes by A. GROTHENDIECK.

[43]N.D.E.: The Nagata compactification theorem ensures that every separated morphism of finite type with quasi-compact quasi-separated target is compactifiable. See B. Conrad, "Deligne's notes on Nagata compactifications", *J. Ramanujan Math. Soc.* **22** (2007), 205–257, cited as [C] below, 4.1, as well as "Erratum for 'Deligne's notes on Nagata compactifications"', *J. Ramanujan Math. Soc.* **24** (2009), 427–428.

[44]N.D.E.: See SGA $4\frac{1}{2}$, L-functions modulo ℓ^n and modulo p.

isomorphic. The category and the canonical isomorphism should in principle have been defined beforehand. In a diagram, the sign = will then denote the isomorphism itself.

0.2.0. Let $f : S \to S'$ be a morphism of sites (IV 4.9.3). If U and U' are objects of S and S', an f-morphism from U to U' will by definition be a morphism from U to f^*U'. For the étale sites of schemes, this recovers the usual notion.

254 0.2.1. If $\mathscr{F}$ and $\mathscr{F}'$ are sheaves on S and S', an f-morphism from $\mathscr{F}$ to $\mathscr{F}'$ will mean, equivalently,

(i) a morphism from $f^*\mathscr{F}'$ to $\mathscr{F}$;
(ii) a morphism from $\mathscr{F}'$ to $f_*\mathscr{F}$;
(iii) a function which, to every f-morphism φ from an object U of S to an object U' of S', assigns a function from $\mathscr{F}'(U')$ to $\mathscr{F}(U)$, compatibly with the composition of φ with an arrow of S or of S' [44].

It follows from (i), (ii), and (iii) that sheaves of sets form a fibered and cofibered category over the category of sites.

Likewise for sheaves of left modules on ringed sites. Likewise for étale sheaves on schemes; this is not immediately a special case of the preceding statement, since sites in fact form a 2-category and "the étale site of X" is only a pseudofunctor in X (VII 1.4).

The preceding definition makes sheaves of sets (resp. ...) on varying sites into a fibered category over the category of sites, whose fiber categories are the *opposite categories* of the usual categories of sheaves of sets (resp. ...).

0.3. Let I be a finite set, let ϵ be a function on I with values in $\{+1, -1\}$, let $(\mathscr{A}_i)_{i\in I}$ be a family of additive categories graded by translation functors T_i ([11] I 1.1.0), and let F be a multiadditive multifunctor from the categories $\mathscr{A}_i$ to an additive category $\mathscr{A}$ graded by the translation functor T. We suppose that F is covariant (resp. contravariant) in the
255 variables i for which $\epsilon_i = 1$ (resp. $\epsilon_i = -1$). Modifying and supplementing ([11] II 2.1.1), we shall say that F is a *graded functor* if we are given a family $(\varphi_i)_{i\in I}$ of isomorphisms of functors between $T\circ F$ and $F\circ T_i^{\epsilon_i}$, such that the following diagrams are anticommutative for $i \neq j$ in I:

$$\begin{array}{ccc} F\circ(T_i^{\epsilon_i}, T_j^{\epsilon_j}) & \xrightarrow{\varphi_j * T_i^{\epsilon_i}} & T\circ F\circ T_i^{\epsilon_i} \\ \Big\downarrow{\scriptstyle \varphi_i * T_j^{\epsilon_j}} & - & \Big\downarrow{\scriptstyle T*\varphi_i} \\ T\circ F\circ T_j^{\epsilon_j} & \xrightarrow{T*\varphi_j} & T\circ T\circ F. \end{array}$$

We leave it to the reader to define the composite of two graded functors and verify that it is again a graded functor.

0.4. Let $\mathscr{A}$ be an additive category and let I be a finite set. For $i \in I$, denote by 1_i the i^{th} basis vector of $\mathbf{Z}^I$. An *I-fold complex* (resp. a *naive I-fold complex*) of $\mathscr{A}$ consists of

(a) a family $(K^k)_{k\in\mathbf{Z}^I}$ of objects of $\mathscr{A}$;

[44] These terminologies are not compatible with identifying the objects of S with the associated sheaves.

(b) for every $i \in I$ and every $k \in \mathbf{Z}^I$, an arrow $d_i^k : K^k \to K^{k+1_i}$, these arrows satisfying $d_i^{k+1_i} d_i^k = 0$ and, for $i \neq j$, $d_i^{k+1_j} d_j^k + d_j^{k+1_i} d_i^k = 0$ (resp. $d_i^{k+1_j} d_j^k = d_j^{k+1_i} d_i^k$).

0.5. A *pro-object* of a category $\mathscr{C}$ is a functor from $\mathscr{C}$ to (Ens) which is a filtered direct limit (over a small filtered category) of representable functors (cf. I 8.10). Every pro-object is a direct limit, over a small filtered ordered set, of representable functors. If X_i is an inverse system of objects of $\mathscr{C}$, indexed by a small filtered category, we denote by "$\varprojlim$" X_i the pro-object $\varinjlim h^{X_i}$. 256

0.6. The reader may dualize 0.5 to the case of ind-objects (Cf. I 8.2).

0.7. In accordance with the new terminology, what was formerly called a prescheme is now called a *scheme*, and what was formerly called a scheme is now called a *separated scheme*.

0.8. The category of *ringed schemes* is the category whose objects are schemes whose étale site is endowed with a sheaf of rings, an arrow from $(S, \mathscr{A})$ to $(T, \mathscr{B})$ being a pair consisting of a morphism of schemes f from S to T and a homomorphism φ of sheaves of rings from $f^*\mathscr{B}$ to $\mathscr{A}$. The morphism of schemes f is said to be *induced by* (f, φ).

0.9. If X is a scheme over Y, we denote by $(X/Y)^n$ the n-fold fiber product of X over Y.

0.10. Let S be a scheme. The *big étale site* (resp. the *big fppf site*, resp. the *big fpqc site*) of S is the site Sch/S, endowed with the étale topology (resp. the fppf topology, resp. the fpqc topology) (SGA 3 IV 6.3). Note carefully that this is not a $\mathscr{U}$-site ($\mathscr{U}$ being the fixed universe, and Sch/S consisting of the schemes $\in \mathscr{U}$).

The *small fpqc site* (resp. the *small fppf site*) of S is the site of schemes flat over S (resp. flat and of finite presentation), endowed with the fpqc topology (resp. the fppf topology). When it is necessary to avoid ambiguity, the *small étale site* of S will mean the site S_{et} (VII 1.2). 257

0.11. In many cases, the author has allowed himself to speak of commutative diagrams of morphisms of sites where one should have spoken of essentially commutative diagrams, i.e. diagrams commutative up to isomorphism (cf. IV 3.2.2). The reader may verify that the arguments we give also apply to the general situation.

0.12. The terminology "*coherent scheme*" for "quasi-compact quasi-separated scheme" has been surreptitiously introduced here and there by A. Grothendieck.

0.13. A torsion sheaf F on a scheme S will be said to be *prime to the residual characteristics of* S if it is the direct limit of its subsheaves annihilated by integers n invertible on S.

1. Derived categories

1.1. Exact functors (the sign rules).

1.1.1. For the fundamental theorems concerning derived categories, I refer to VERDIER [11] and [12]. In this subsection, particularly careful attention has been paid to questions of signs.

Recall that if $f : X \to Y$ is a morphism of complexes (in an additive category, which
258 will henceforth be understood), its *cone* $C(f)$ is defined by

$$C(f)^n = X^{n+1} \oplus Y^n \quad d^n = -d_X^{n+1} + f^{n+1} + d_Y^n. \tag{1.1.1.1}$$

When $X = 0$ (resp. $Y = 0$), we have $C(f) = Y$ (resp. $C(f) = X[1]$), so that the commutative diagram

$$\begin{array}{ccccccc} 0 & = & 0 & \longrightarrow & X & = & X \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle f} & & \downarrow \\ X & \xrightarrow{f} & Y & = & Y & \longrightarrow & 0 \end{array}$$

defines a "triangle"

$$X \xrightarrow{f} Y \to C(f) \xrightarrow{j} X[1].$$

In the category of complexes up to homotopy, a triangle is called *distinguished* if it is isomorphic to a triangle of this type, and *antidistinguished* if it becomes distinguished when the signs of its arrows are changed. One checks that the triangle defined by an exact sequence of complexes which is split in each degree ([11] p. 10, cf. the proof of 1.1.5.4) is antidistinguished and that, up to isomorphism, every antidistinguished triangle is obtained in this way.

* *1.1.2.* Sign calculations are most easily carried out using the following "formal rules". Write an element of $C(f)^n$ in the form $1 \otimes x + y$ ($x \in X^{n+1}$, $y \in Y^n$), where 1 is a "cell" of dimension 1, and set $d(1) \otimes x = f(x)$. Thus

$$d(1 \otimes x + y) = d1 \otimes x - 1 \otimes dx + dy = -1 \otimes dx + f(x) + dy.\text{*}$$

259 DEFINITION 1.1.3. (i) *A graded functor (0.3) from a triangulated category $\mathscr{A}$ to a triangulated category $\mathscr{B}$ is said to be* exact *if it sends distinguished triangles to distinguished triangles.*

(ii) *A contravariant graded functor from a triangulated category $\mathscr{A}$ to a triangulated category $\mathscr{B}$ is said to be* exact *if, for every distinguished triangle (X, Y, Z, u, v, w) of $\mathscr{A}$, the triangle $(F(Z), F(Y), F(X), F(v), F(u), TF(w))$ is distinguished (in this definition, FX is identified with $TFTX$ by means of the grading of F).*

(iii) *Let $(\mathscr{A}_i)_{i\in I}$ be a finite family of triangulated categories, let ϵ be a function from I to $\{+1, -1\}$, and let $\mathscr{A}$ be a triangulated category. A graded functor F from the product of the $\mathscr{A}_i$ to $\mathscr{A}$, covariant in those i for which $\epsilon(i) = 1$ and contravariant in those for which $\epsilon(i) = -1$, is said to be* exact *if, for every $i \in I$, the functors from $\mathscr{A}_i$ to $\mathscr{A}$ obtained from F by fixing all variables except the i-th are exact functors.*

A composite of exact functors is again an exact functor.

1.1.4. If $(K^{n_1 \cdots n_p}, d_1, \ldots, d_p)$ is a multicomplex (0.4), its associated simple complex is defined by

$$K^n = \prod_{\sum n_i = n} K^{(n_i)} \quad , \quad d = \sum d_i. \tag{1.1.4.0}$$

When, exceptionally, it is defined using a direct sum rather than a product, this fact will be stated explicitly.

Let I be a finite set. Following CARTAN–EILENBERG and J. P. SERRE, we shall explain
260 how every naive I-fold complex K has canonically associated with it an I-fold complex

(0.4).

For every total order $<$ on I, let $K(<)$ be the following I-fold complex:

$$\begin{cases} K(<)^{\underline{k}} = K^{\underline{k}} \\ d(<)_i^{\underline{k}} = (-1)^{\sum_{j<i} k_j} d_i. \end{cases} \tag{1.1.4.1}$$

If $<_1$ and $<_2$ are two total orders on I, the *canonical isomorphism τ between $K(<_1)$ and $K(<_2)$* is, by definition, the one given by

$$\tau^{\underline{k}} = (-1)^{\epsilon(<_1,<_2)}, \quad \text{where} \quad \epsilon(<_1,<_2) = \sum_{\substack{i<_1 j \\ j<_2 i}} k_i k_j. \tag{1.1.4.2}$$

1.1.4.2.1. One checks that these canonical isomorphisms form a transitive system of isomorphisms among the $K(<)$, as $<$ ranges over the total orders on I; these $K(<)$ therefore determine an I-fold complex, unique up to a unique isomorphism, which in this subsection will be denoted by $K(*)$ and is called the *I-fold complex associated with the naive I-fold complex K*. Note that $K(*)^{\underline{k}}$ is *not canonically isomorphic to $K^{\underline{k}}$: only the choice of a total order on I makes it possible to identify these two objects.*

If K is a naive I-fold complex and $i \in I$, let $K[1_i]$ denote the following naive complex, whose differentials are denoted by $d_j[1_i]$:

$$\begin{cases} K[1_i]^{\underline{k}} = K^{\underline{k}+1_i} \\ d_j[1_i]^{\underline{k}} = d_j^{\underline{k}+1_i} \quad \text{if } i \neq j \\ d_i[1_i]^{\underline{k}} = -d_i^{\underline{k}+1_i}. \end{cases} \tag{1.1.4.3}$$

If K is an I-fold complex, set 261

$$\begin{cases} K[1_i]^{\underline{k}} = K^{\underline{k}+1_i} \\ d_j[1_i]^{\underline{k}} = -d_j[1_i]^{\underline{k}+1_i}. \end{cases} \tag{1.1.4.4}$$

If K is a naive I-fold complex and $<$ is a total order on I, so that $K(*)$ is given by (1.1.4.1), define an isomorphism σ between $K[1_i](*)$ and $K(*)[1_i]$ by the formula

$$\sigma^{\underline{k}} = (-1)^{\sum_{j<i} k_j} \mathrm{id}_{K^{\underline{k}+1_i}}. \tag{1.1.4.5}$$

. One checks that this isomorphism is independent of the choice of the total order $<$; it therefore defines an *isomorphism, called the canonical isomorphism, between $K[1_i](*)$ and $K(*)[1_i]$*.

. For any $i, j \in I$, if K is an I-fold complex or a naive I-fold complex, we identify $K[1_i][1_j]$ and $K[1_j][1_i]$ in the evident way.

For $i \neq j$ in I, the following diagram of canonical isomorphisms of type 1.1.4.6 or 1.1.4.7 is then *anticommutative*:

$$\begin{array}{ccccc} K[1_i][1_j](*) & = & K[1_j][1_i](*) & \longleftrightarrow & K[1_j](*)[1_i] \\ \updownarrow & & & & \updownarrow \\ K[1_i](*)[1_j] & \longleftrightarrow & K(*)[1_i][1_j] & = & K(*)[1_j][1_i]. \end{array} \tag{1.1.4.7.1}$$

1.1.5. Let $F : \mathscr{A} \to \mathscr{B}$ be a contravariant additive functor between additive categories. Its *extension to complexes* $F : C(\mathscr{A}) \to C(\mathscr{B})$ is defined as follows. If K is a complex of $\mathscr{A}$, set 262

$$\text{(1.1.5.1)} \qquad \begin{cases} F(K)^k = F(K^{-k}) \\ d^k_{F(K)} = (-1)^{k+1} F(d^{-k-1}). \end{cases}$$

The *canonical isomorphism* σ *between* $F(K)[1]$ *and* $F(K[-1])$ is defined by

$$\text{(1.1.5.2)} \qquad \sigma^k = (-1)^{k+1} \operatorname{id}_{F(K^{-k-1})}.$$

If $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{B} \to \mathscr{C}$ are two contravariant additive functors, define the canonical isomorphism ρ between $G \circ F(K)$ *and* $G(F(K))$ by

$$\text{(1.1.5.3)} \qquad \rho^k = (-1)^k \operatorname{id}_{G \circ F(K^k)}.$$

Lemma 1.1.5.4. (i) *For every contravariant additive functor* $F : \mathscr{A} \to \mathscr{B}$*, formulas (1.1.5.1) and (1.1.5.2) define an exact contravariant functor from* $K(\mathscr{A})$ *to* $K(\mathscr{B})$.

(ii) *For every pair* (G, F) *of composable contravariant additive functors, formula (1.1.5.3) defines an isomorphism of graded functors between the extension of* GF *to complexes and the composite of the extensions of* G *and* F *to complexes.*

263 Proof. A morphism of degree i, $f : K \to L$, may be regarded as any one of the following:

(a) a morphism from K to $L[i]$;
(b) a morphism from $K[-i]$ to L;
(c) a family $f^n : K^n \to L^{n+i}$ which commutes or anticommutes with the differentials according to the parity of i.

If $f : K \to L[i]$ is a morphism of degree i, then $F(f) : F(L)[-i] \to F(K)$ is again a morphism of degree i; for $i = 1$, we have

$$\text{(1.1.5.5)} \qquad F(f)^n = (-1)^{n+1} F(f^{-n-1}).$$

If $0 \to K \overset{u}{\to} L \overset{v}{\to} M \to 0$ is an exact sequence of complexes split degree by degree, the choice of a splitting allows the differential of L to be written in the form

$$d_L = d_K + d_M + f,$$

where f is a morphism of degree one from M to K. Likewise, the dual exact sequence

$$0 \longrightarrow F(M) \longrightarrow F(L) \longrightarrow F(K) \longrightarrow 0$$

defines a morphism of degree one f^* from $F(K)$ to $F(M)$. We have $f^* = F(f)$. It follows that the image under F of a triangle (K, L, M, u, v, f) defined by a short exact sequence of complexes is again a triangle of the same type and, by 1.1.1, this proves (i).

Under the hypotheses of (ii), for f of degree one, we have $G(F(f))^n = (-1)^{n+1}(-1)^{-n} G \circ F(f)^n = -G \circ F(f)^n$; in particular, the grading isomorphism $G(F(K[1])) \simeq G(F(K))[1]$ is the isomorphism whose components are -1.

264 The diagram

$$\begin{array}{ccc} G \circ F(K)[1] & \overset{\rho[1]}{\longleftrightarrow} & G(F(K))[1] \\ \| & & \updownarrow \\ G \circ F(K[1]) & \overset{\rho}{\longleftrightarrow} & G(F(K[1])) \end{array}$$

is therefore commutative, which proves (ii).

1.1.6. Let $s = +$ or $s = -$; let I be a finite set, let $\epsilon : I \to \{+,-\}$ be a function, let $(\mathscr{A}_i)_{i\in I}$ and $\mathscr{A}$ be additive categories, and let F be a multiadditive multifunctor from the categories $\mathscr{A}_i$ to $\mathscr{A}$, covariant in the arguments indexed by those i for which $\epsilon(i) = +$ and contravariant in the others. We shall define the extension $K^s(F)$ (or simply F) of F to complexes as an exact multifunctor from the categories $K^{s\epsilon(i)}(\mathscr{A}_i)$ ($s\epsilon(i) = \pm$, according to the sign rule) to $K^s(\mathscr{A})$, [44] having the same variance as F. We shall then explain in what sense this construction is compatible with the composition of functors. Other conventions for *degrees* are possible and will be used; that issue is independent of the questions of *sign* considered here.

. If F is contravariant in one variable, $K^s(F)$ is defined by the formulas of 1.1.5.

. If F is covariant, the functor $K^s(F)$ is defined, by passage to the quotient, from the composite of the evident functor from the product of the categories $C^s(\mathscr{A}_i)$ to the category of naive I-fold complexes, the functor (1.1.4) from the category of naive I-fold complexes to that of I-fold complexes (1.1.4.2.1), and the associated simple complex 265
functor (1.1.4.0):

$$K^s(F) = C^s(F)(*)_{s.a}.$$

Let $K_j \in K^s(\mathscr{A}_j)$ be complexes, and let $i \in I$; set $K_j[1_i] = K_j$ for $j \neq i$, and $K_i[1_i] = K_i[1]$. We then have the evident isomorphisms

$$C^s(F)(K_j[1_i]) \simeq C^s(F)(K_j)[1_i],$$

and

$$C^s(F)(K_j)(*)[1_i]_{s.a} \simeq C^s(F)(K_j)(*)_{s.a}[1],$$

whence, via (1.1.4.5), a *canonical isomorphism* σ between $K^s(F)(K_j[1_i])$ and $K^s(F)(K_j)[1]$. It follows from 1.1.4.7 that these isomorphisms define a grading of $K^s(F)$.

. In the general case, F is the composite of the canonical contravariant functors $\mathscr{A}_i \to \mathscr{A}_i^\circ$ (for $\epsilon(i) = -$) with a covariant functor F^+; define the functor $K^s(F)$ by composition using 1.1.6.1 and 1.1.6.2.

Proposition 1.1.7. *Under the preceding hypotheses, the functor $K^s(F)$ is an exact multiadditive functor.*

It suffices to verify 1.1.7 under the hypotheses of 1.1.6.1 (where the proposition is 1.1.5.4 (i)) or under those of 1.1.6.2. To verify that the functors deduced from $K^s(F)$ by fixing all variables except the i-th are exact, one may describe $K^s(F)$ explicitly in terms of an order on I for which i is the least element; a distinguished triangle of the standard form (1.1.1.1) is then sent to a distinguished triangle of the same form. 266

1.1.8. Let I and J be two finite sets, let $\Psi : I \to J$ be a surjective map, and let $\epsilon_I : I \to \{+,-\}$ and $\epsilon_J : J \to \{+,-\}$ be functions, and let us be given multiadditive multifunctors, covariant in the arguments for which $\epsilon = +$ and contravariant in the others:

$$G_j : \prod_{\Psi(i)=j} \mathscr{A}_i \longrightarrow \mathscr{B}_j \qquad (j \in J)$$

$$F : \prod_{j\in J} \mathscr{B}_j \longrightarrow \mathscr{C}.$$

[44] The construction also provides an extension of F to a multifunctor from the categories $C^{s\epsilon(i)}(\mathscr{A}_i)$ to $C^s(\mathscr{A})$.

Choose total orders on I and J such that Ψ is an increasing map. We then define the *canonical isomorphism ρ between the extension to complexes of $F \circ (G_j)$ and the composite of the extensions to complexes of F and of the G_j* by the formula

$$\rho^{\underline{k}} = (-1)^{A(\underline{k})} \ : \ \text{automorphism of } F \circ (G_j)(K_i^{k_i \epsilon_I(i) \epsilon_J(\Psi(i))}). \tag{1.1.8.1}$$

where

$$A(\underline{k}) = \sum_{\epsilon_I(i)=\epsilon_J(\Psi(i))=-} k_i + \sum_{\epsilon_J(j)=-} \sum_{\substack{\Psi(a)=\Psi(b)=j \\ a<b}} k_a k_b.$$

This formula generalizes (1.1.5.3); one verifies that ρ does not depend on the orders chosen on I and J, and is an isomorphism of graded functors (cf. 1.1.5.4 (ii)).

These isomorphisms ρ satisfy the evident cocycle condition for a triple composite.

267 1.1.9. Let I be a finite set, let i and j be two distinct elements of I, let $I' = I\backslash\{i, j\}$, let $(\mathscr{A}_i)_{i\in I}$ and $\mathscr{A}$ be additive categories, and let $F : (\mathscr{A}_i)_{i\in I} \to \mathscr{A}$ and $G : (\mathscr{A}_i)_{i\in I'} \to \mathscr{A}$ be multiadditive multifunctors. Suppose that $\mathscr{A}_i = \mathscr{A}_j$ and that F is contravariant in the i-th variable and covariant in the j-th. For $X, Y \in \mathrm{Ob}\,\mathscr{A}_i$, denote by $F_{X,Y}$ the multifunctor from $(\mathscr{A}_i)_{i\in I'}$ to $\mathscr{A}$ obtained by fixing the i-th and j-th arguments:

$$F_{X,Y}(\dots) = F(\dots, X, \dots, Y, \dots).$$

A *contraction morphism* $c : F \to G$ consists of, for every $X \in \mathscr{A}_i$, a morphism of functors

$$c_X : F_{X,X} \longrightarrow G,$$

such that, for every arrow $f : X \to Y$ of $\mathscr{A}_i$, the diagram

$$\begin{array}{ccc} F_{Y,X} & \longrightarrow & F_{X,X} \\ \downarrow & & \downarrow{\scriptstyle c_X} \\ F_{Y,Y} & \xrightarrow[c_Y]{} & G \end{array}$$

is commutative.

The conventions of 1.1.6 are motivated by the

Lemma 1.1.9.1. *Let $c : F \to G$ be a contraction morphism, and let $(K_\ell)_{\ell\in I}$ be complexes in the $\mathscr{A}_\ell$, with $K_i = K_j$. Let $<$ be a total order on I for which i is the predecessor of j. With the notation of 1.1.6, for $k \in \mathbf{Z}$, let*

$$c^k : K^s(F)(K_\ell)^k \longrightarrow K^s(G)(K_\ell)^k$$

268 *be the morphism whose restriction to $F(\dots K_i^{k_i} K_j^{k_j} \dots)$ is 0 if $k_i \neq k_j$, and is $c_{K_i^{k_i}}$ if $k_i = k_j$. Then $(c^k)_{k\in\mathbf{Z}}$ is a morphism of complexes.*

We leave to the reader the verification of this lemma, as well as the fact that the *contraction morphism* (c^k) does not depend on the order chosen on I.

The standard example of a contraction morphism is the composition law

$$\mathrm{Hom}(Y, Z) \otimes \mathrm{Hom}(X, Y) \longrightarrow \mathrm{Hom}(X, Z).$$

Examples 1.1.10. (i) Applying the definitions of 1.1.6 to the *tensor-product* functor gives the usual notion of the tensor product of two complexes, with differential

$$d(x \otimes y) = dx \otimes y + (-1)^{\deg x} x \otimes dy.$$

(ii) Apply the definitions of 1.1.6 to the functor $\mathrm{Hom}(X,Y)$, *regarding Y as the first variable and X as the second variable*, a convention that we shall always follow. This convention identifies, for two complexes X, Y, the k-th component of the complex $\mathrm{Hom}(X,Y)$ defined in 1.1.6 with $\prod_n \mathrm{Hom}(X^n, Y^{n+k})$, and gives the usual differential: for $f \in \mathrm{Hom}(X^n, Y^{n+k})$, one has

$$df = d \circ f - (-1)^k f \circ d.$$

(iii) The preceding conventions have the virtue that the following systems of arrows are morphisms of complexes, without any signs having to be inserted (1.1.9): 269

$$\mathrm{Hom}^\cdot(L,M) \otimes \mathrm{Hom}^\cdot(K,L) \longrightarrow \mathrm{Hom}^\cdot(K,M) : f \otimes g \mapsto f \circ g$$
$$\mathrm{Hom}^\cdot(L,M) \otimes L \longrightarrow M : f \otimes x \mapsto f(x).$$

* *1.1.10.1.* The convention (1.1.4.5)–(1.1.6.2) is justified as follows for $F = \otimes$: an element of $((K \otimes L)[1])^n$ is written in the form (*1.1.2) (recalling that $(K \otimes L)[1] = C(K \otimes L \to 0)$ (1.1.1.1)) $\sum 1 \otimes (x^p \otimes y^q)$, an element of $(K[1] \otimes L)^n$ in the form $\sum (1 \otimes x^p) \otimes y^q$, and an element of $(K \otimes L[1])^n$ in the form $\sum x^p \otimes (1 \otimes y^q)$; the sign appears when x^p and 1 are interchanged. For $F =$ dual, the convention (1.1.5.2) is justified as follows: an element of $K^*[1]$ is written $1 \otimes \omega$; its value on an element $1' \otimes x$ of $K[-1]$ ($1'$ a cell of dimension -1) is $< 1 \otimes \omega, 1' \otimes x >= (-1)^{\deg \omega} < 1, 1' >< \omega, x >= (-1)^{\deg \omega} < \omega, x >$; this is the convention adopted.*

Remark 1.1.11. "Recall" that a *preadditive category* is a category whose sets of arrows $\mathrm{Hom}(X,Y)$ are endowed with abelian-group structures, the composition of morphisms being biadditive. An *additive functor* F between preadditive categories is a functor satisfying the formula

$$F(f+g) = F(f) + F(g)$$

for $f, g \in \mathrm{Hom}(X,Y)$. The additive functors from a preadditive category $\mathscr{A}$ to another $\mathscr{B}$ form a category denoted by $\mathbf{Hom}_{\mathrm{add}}(\mathscr{A}, \mathscr{B})$.

Let (D) be the unique preadditive category whose set of objects is indexed by $\mathbf{Z}$, $(X^n)_{n \in \mathbf{Z}}$, and for which 270

$$\text{(1.1.11.1)} \qquad \begin{cases} \mathrm{Hom}(X^n, X^n) = \mathbf{Z}, \text{ generated by the identity} \\ \mathrm{Hom}(X^n, X^{n+1}) = \mathbf{Z}, \text{ generated by an arrow denoted } d_n \\ \mathrm{Hom}(X^i, X^j) = 0 \text{ if } j - i \neq 0 \text{ or } 1. \end{cases}$$

Let D be the complex of (D) defined by $D^n = X^n$ and $d^n = d_n$.

For every preadditive category $\mathscr{A}$, the functor $G \mapsto G(D)$ from $\mathbf{Hom}_{\mathrm{add}}((D), \mathscr{A})$ to $C(\mathscr{A})$ is an isomorphism.

The construction of 1.1.6 extends to the case of a multiadditive functor from preadditive categories $\mathscr{A}_i$ to an additive category $\mathscr{A}$.

Let

$$F^\cdot = (F^n, d^n)_{n \in \mathbf{Z}}$$

be a complex of multiadditive functors from additive categories $\mathscr{A}_i$ to an additive category $\mathscr{A}$. One may regard $F^\cdot$ as a functor from (D) to a category of multifunctors, or equivalently as a multiadditive functor

$$F_1 : \prod \mathscr{A}_i \times (D) \longrightarrow \mathscr{A}.$$

The *extension of $F^\cdot$ to complexes* is defined by the formula

$$\text{(1.1.11.2)} \qquad F^\cdot(K_1 \dots K_n) = F_1(K_1 \dots K_n, D).$$

1.1.12. Let $\mathscr{A}$ and $\mathscr{B}$ be two additive categories and let

$$F^{\cdot} = (F^n, d^n)_{n\in\mathbf{Z}}$$

271 be a complex of additive functors from $\mathscr{A}$ to $\mathscr{B}$. Suppose that the functors F^i admit right adjoints F_i; let ${}^t d^i : F_{i+1} \to F_i$ be the transpose of d^i, and let $F_{\cdot}$ be the complex of additive functors from $\mathscr{B}$ to $\mathscr{A}$ whose components are the F_i and whose differential operator is

$$(d_{\cdot})^i = (-1)^{i+1}\, {}^t d^{-i-1} : F_{-i} \longrightarrow F_{-i-1}.$$

Let ^{t} be the *contravariant* functor from (D) to (D) given by

$${}^t X^n = X^{-n} \quad , \quad {}^t d^n = d^{-n-1}.$$

The following diagram is commutative up to canonical isomorphism:

$$\begin{array}{ccccc} \mathscr{A}\times(D)\times\mathscr{B} & \xrightarrow{(F^n(A),\mathrm{Id})} & \mathscr{B}\times\mathscr{B} & \xrightarrow{\mathrm{Hom}_{\mathscr{B}}} & (\mathrm{Ab}) \\ \downarrow{\scriptstyle (\mathrm{Id},{}^t,\mathrm{Id})} & & & & \| \\ \mathscr{A}\times(D)\times\mathscr{B} & \xrightarrow{(\mathrm{Id},(F_{\cdot})^n(B))} & \mathscr{A}\times\mathscr{A} & \xrightarrow{\mathrm{Hom}_{\mathscr{A}}} & (\mathrm{Ab}). \end{array} \tag{1.1.12.1}$$

This expresses the adjunction formula

$$\mathrm{Hom}(F^n(A), B) \xleftrightarrow{\sim} \mathrm{Hom}(A, F_n(B)).$$

The general conventions of 1.1.3 therefore give an isomorphism

$$\mathrm{Hom}^{\cdot}(F^{\cdot}(K), L) \xleftrightarrow{\sim} \mathrm{Hom}^{\cdot}(K, F_{\cdot}(L)).$$

Definition 1.1.13. *If K is a complex in an abelian category $\mathscr{A}$, the* right truncation (resp. left truncation) *of K in dimension n, denoted by $\tau_{\leq n}(K)$* (resp. $\tau_{\geq n}(K)$)*, means the following complexes:*

$$\tau_{\leq n}(K) : \ldots K^{n-2} \longrightarrow K^{n-1} \longrightarrow \mathrm{Ker}(d^n) \longrightarrow 0$$
$$\tau_{\geq n}(K) : 0 \longrightarrow \mathrm{coker}(d^{n-1}) \longrightarrow K^{n+1} \longrightarrow K^{n+2} \ldots .$$

272 Proposition 1.1.14. (i) *$H^i(\tau_{\leq n}(K)) = 0$ for $i > n$, and $H^i(\tau_{\geq n}(K)) = 0$ for $i < n$.*

(ii) *The canonical morphism $\tau_{\leq n}(K) \to K$ (resp. $K \to \tau_{\geq n}K$) induces isomorphisms on the H^i for $i \leq n$ (resp. $i \geq n$).*

The functor $\tau_{\leq n}$ (resp. $\tau_{\geq n}$) sends homotopic morphisms to homotopic morphisms. It therefore follows from 1.1.14 that the functor of 1.1.13 remains meaningful in the derived category.

1.1.15. If K is a double complex, with first (resp. second) differential d' (resp. d''), denote by $\tau''_{\leq n}(K)$ the double subcomplex of K such that

$$\tau''_{\leq n}(K)^{pq} = \begin{cases} K^{pq} & \text{if } q < n \\ \mathrm{Ker}(d'') & \text{if } q = n \\ 0 & \text{if } q > n \end{cases}$$

and define $\tau''_{\geq n}(K)$, $\tau'_{\leq n}(K)$, and $\tau'_{\geq n}(K)$ similarly.

1.1.16. We shall also have occasion to consider the "stupid truncations" of a complex K, defined by

$$\sigma_{\leq n}(K)^i = \begin{cases} K^i & \text{if } i \leq n \\ 0 & \text{if } i > n \end{cases} \qquad \sigma_{\geq n}(K)^i = \begin{cases} K^i & \text{if } i \geq n \\ 0 & \text{if } i < n. \end{cases}$$

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1.2. Derived functors. Let $\mathscr{A}$ be a triangulated category and S a saturated multiplicative system ([11] I 2.1.2). If $A \in \mathrm{Ob}\,\mathscr{A}$, denote by A^+ and A^- the ind- and pro-objects ((0.5) and (0.6))

$$A^+ = \text{“}\varinjlim_{A \xrightarrow{s} A'}\text{”}\ A' \qquad A^- = \text{“}\varprojlim_{A' \xrightarrow{s} A}\text{”}\ A',$$

where s ranges over the filtered category of arrows in S with source or target A, respectively.

The functors $A \mapsto A^+$ and $A \mapsto A^-$ make the elements of S invertible and define fully faithful functors from $\mathscr{A}(S^{-1})$ to the categories of Ind- and pro-objects of $\mathscr{A}$, respectively.

Definition 1.2.1. *Let F be an exact covariant multifunctor from triangulated categories $(\mathscr{A}_i)_{0<i\leq n}$ to the triangulated category $\mathscr{A}$, and let S_i and S be saturated multiplicative systems in the categories $\mathscr{A}_i$ and $\mathscr{A}$.*

(i) *The right derived functor RF of F (relative to the S_i and S) is the functor from the categories $\mathscr{A}_i(S_i^{-1})$ to the category of ind-objects of $\mathscr{A}(S^{-1})$ that makes the following diagram commutative.*

$$\begin{array}{ccc} (\mathscr{A}_i) & \xrightarrow{(A_i)\mapsto F(A_i^+)} & \mathrm{Ind}\,\mathscr{A} \\ \downarrow & & \downarrow \\ \mathscr{A}_i(S_i^{-1}) & \xrightarrow{RF} & \mathrm{Ind}\,\mathscr{A}(S^{-1}). \end{array}$$

(ii) *We say that RF is defined at (A_i) if the ind-object $RF(A_i)$ of $\mathscr{A}(S^{-1})$ is "essentially constant," i.e. comes from an object of $\mathscr{A}(S^{-1})$; this object is called* the value of RF *at (A_i).* 274

(iii) *We say that F is* right derivable *if RF is defined everywhere; in that case we shall again denote by RF the resulting functor from the $\mathscr{A}_i(S_i^{-1})$ to $\mathscr{A}(S^{-1})$.*

(iv) *A family (A_i) of objects of the $\mathscr{A}_i$ is said to be* adapted to RF *if the canonical morphism from $F(A_i)$ to $RF(A_i)$ is an isomorphism.*

We leave to the reader the task of defining by duality the left derived functor LF of F (with values in $\mathrm{Pro}\,\mathscr{A}(S^{-1})$) and of extending the preceding definitions to cases in which F is covariant in some arguments and contravariant in others.

When RF is defined everywhere, the definition given here agrees with Verdier's ([11] p. 39), as is readily verified.

Proposition 1.2.2. *Let (A_1', A_1, A_1'') be a distinguished triangle in $\mathscr{A}_1$, and let $A_i \in \mathrm{Ob}\,\mathscr{A}_i$ $(i \neq 1)$.*

(i) *If $(A_1', (A_i))$ and $(A_1'', (A_i))$ are adapted to RF, then the family $(A_1, (A_i))$ is also adapted to RF.*

(ii) *If RF is defined at $(A_1', (A_i))$ and at $(A_1'', (A_i))$, it is also defined at $(A_1, (A_i))$, and the triangle $(RF(A_1', (A_i)), RF(A_1, (A_i)), RF(A_1'', (A_i)))$ is distinguished in $\mathscr{A}(S^{-1})$.*

275 Let us prove (ii) ⇒ (i). Since F is exact, the arrow

$$(F(A'_1,(A_i)),F(A_1,(A_i)),F(A''_1,(A_i))) \to (RF(A'_1,(A_i)),RF(A_1,(A_i)),RF(A''_1,(A_i)))$$

is a morphism of distinguished triangles. By hypothesis this arrow is an isomorphism at two vertices; it is therefore an isomorphism at the third, as desired.

LEMMA 1.2.2.1. *If $\Delta = (X,Y,Z)$ is a distinguished triangle in a triangulated category $\mathscr{A}$ and S is a saturated multiplicative system in $\mathscr{A}$, then the triangle $\Delta^+ = (X^+,Y^+,Z^+)$ in* Ind $\mathscr{A}$ *is a limit of distinguished triangles in $\mathscr{A}$.*

Let $\mathscr{L}$ denote the category of morphisms of distinguished triangles with source Δ whose arrows belong to S.

(a) If $\Delta = (X,Y,Z)$ and $\Delta' = (X',Y',Z')$ are two distinguished triangles in $\mathscr{A}$, and if $f : \Delta \to \Delta'$ is an $\mathscr{A}(S^{-1})$-morphism of triangles, there exist in $\mathscr{A}$ morphisms of triangles $s : \Delta' \to \Delta''$ and $f_1 : \Delta \to \Delta''$ such that the components of s belong to S and the diagram

(1)
$$\begin{array}{ccc} \Delta & & \\ {\scriptstyle f}\downarrow & \searrow^{f_1} & \\ \Delta' & \xrightarrow{s} & \Delta'' \end{array}$$

is commutative in $\mathscr{A}(S^{-1})$.

There is a commutative diagram

(2)
$$\begin{array}{ccc} X & & \\ {\scriptstyle f}\downarrow & \searrow^{f_1} & \\ X' & \xrightarrow{s_X} & X''_0 \end{array} \qquad (s_X \in S, f_1 \in \mathrm{Fl}\,\mathscr{A})$$

276 and a morphism $\Delta' \overset{s}{\to} \Delta''_0$ in S, having s_X as a component. Argue similarly at the other vertices: one obtains a diagram of type (1), in which, however, f_1 is not yet a morphism of triangles *in* $\mathscr{A}$, since the diagrams

(3)
$$\begin{array}{ccc} Z & \xrightarrow{f_1} & Z''_0 \\ {\scriptstyle w}\downarrow & & \downarrow{\scriptstyle w''} \\ X & \xrightarrow{f_1} & X''_0 \end{array}$$

may, for example, fail to be commutative *in* $\mathscr{A}$, being commutative only in $\mathscr{A}(S^{-1})$. There exists $t_X : X''_0 \to X''_1$ in S such that $t_X w'' f_1 = t_X f_1 w$ and $t : \Delta''_0 \to \Delta''_1$ having t_X as a component. Replace Δ''_0 by Δ''_1, s by ts, and f_1 by tf_1; in the resulting diagram of type (1), (3) is now commutative; proceeding similarly at the other vertices, one obtains (1). We leave it to the reader to verify similarly:

(b) If $f, g : \Delta \rightrightarrows \Delta'$ are two morphisms of distinguished triangles in $\mathscr{A}$, equal in $\mathscr{A}(S^{-1})$, then there exists a morphism of distinguished triangles whose arrows are in S, $s : \Delta' \to \Delta''$, such that $sf = sg$.

(c) The category $\mathscr{L}$ is filtering on the right.

(d) The proof of the lemma is completed by noting that

$$\Delta^+ = \text{“}\varinjlim_{s\in\mathscr{L}}\text{”}\,(\text{target of } s).$$

Let us prove 1.2.2 (ii). Let X' and X'' be objects of $\mathscr{A}(S^{-1})$ representing $RF(A'_1,(A_i))$ 277
and $RF(A''_1,(A_i))$, and complete the morphism of degree 1 from X'' to X' to a distinguished triangle (X',X,X''). Replacing the A_i, A'_1, A_1, and A''_1 by the target of an arrow in S_i or S_1 with source A_i, A'_1, A_1, or A''_1, we reduce to the case where there is a commutative diagram

$$\begin{array}{ccc} F(A''_1,(A_i)) & \xleftarrow{a''} & X'' \\ \downarrow & & \downarrow \\ F(A'_1,(A_i)) & \xleftarrow{a'} & X', \end{array}$$

where a'' and a' are sections of arrows from $F(A''_1,(A_i))$ to X'' and from $F(A'_1,(A_i))$ to X'. By axiom TR 3 of [11], this diagram can be completed to a morphism of triangles

$$\begin{aligned}(X'XX'') &\longrightarrow (F(A'_1,(A_i)),F(A_1,(A_i)),F(A''_1,(A_i)))\\ &\longrightarrow (RF(A'_1,(A_i)),RF(A_1,(A_i)),RF(A''_1,(A_i))).\end{aligned}$$

The composite arrow is a morphism of triangles whose source is a distinguished triangle and whose target is a limit (in $\operatorname{Ind}\mathscr{A}(S^{-1})$) of distinguished triangles (1.2.2.1). For any Y in $\mathscr{A}(S^{-1})$, the following diagram is commutative and its rows are exact:

$$\begin{array}{ccccccccc} \operatorname{Hom}(Y,X''[-1]) & \longrightarrow & \operatorname{Hom}(Y,X') & \longrightarrow & \operatorname{Hom}(Y,X) & \longrightarrow & \operatorname{Hom}(Y,X'') & \longrightarrow & \operatorname{Hom}(Y,X'[1]) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \operatorname{Hom}(Y,RF(A''_1,A_i)[-1]) & \to & \operatorname{Hom}(Y,RF(A'_1,A_i)) & \to & \operatorname{Hom}(Y,RF(A_1,A_i)) & \to & \operatorname{Hom}(Y,RF(A''_1,A_i)) & \to & \operatorname{Hom}(Y,RF(A'_1,A_i)[1]) \end{array}$$

By assumption, all arrows other than the middle arrow are isomorphisms. The 5- 278
lemma shows that the middle arrow is also an isomorphism, for every Y, and hence that the distinguished triangle (X',X,X'') is isomorphic to the triangle $(RF(A'_1,(A_i)),RF(A_1,(A_i)),RF(A''_1,(A_i)))$. This completes the proof.

If $\mathscr{A}$ and $\mathscr{B}$ are two abelian categories and F is an additive covariant functor from $\mathscr{A}$ to $\mathscr{B}$, F defines exact functors from $K(\mathscr{A})$ to $K(\mathscr{B})$, from $K^+(\mathscr{A})$ to $K^+(\mathscr{B})$, and from $K^-(\mathscr{A})$ to $K^-(\mathscr{B})$ (1.1.7).

DEFINITION 1.2.3. (i) *The* right derived functors of F *are the functors from* $D^+(\mathscr{A})$ *to* $\operatorname{Ind}D^+(\mathscr{B})$ *and from* $D(\mathscr{A})$ *to* $\operatorname{Ind}D(\mathscr{B})$ *right derived from the extensions of* F *to complexes* $K^+(\mathscr{A})\to K^+(\mathscr{B})$ *and* $K(\mathscr{A})\to K(\mathscr{B})$.

(ii) *An object* A *of* $\mathscr{A}$ *is said to be* right acyclic *for* F*, or, by abuse of language,* acyclic for RF*, if the complex concentrated at* A *in degree* 0 *is adapted to* RF.

We shall be interested mainly in the case where RF is defined everywhere; it may then be regarded as a functor from $D^+(\mathscr{A})$ to $D^+(\mathscr{B})$, or from $D(\mathscr{A})$ to $D(\mathscr{B})$, as the case may be.

PROPOSITION 1.2.4. *The following diagram is commutative.*

$$\begin{array}{ccc} D^+(\mathscr{A}) & \longrightarrow & D(\mathscr{A}) \\ \downarrow{\scriptstyle RF} & & \downarrow{\scriptstyle RF} \\ \operatorname{Ind}\ D^+(\mathscr{B}) & \longrightarrow & \operatorname{Ind}\ D(\mathscr{B}). \end{array}$$

If K is a complex that is zero in degrees $< n$ and $s : K \to L$ is a quasi-isomorphism, 279
the composite morphism $K \xrightarrow{s} L \to \sigma_{\geq n}(L)$ (1.1.6) is again a quasi-isomorphism. The quasi-isomorphisms from K to a complex bounded below therefore form a cofinal (full)

subcategory of the category of all quasi-isomorphisms with source K. The proposition follows formally.

In particular, if the functor RF is defined everywhere on $D(\mathscr{A})$, it is defined everywhere on $D^+(\mathscr{A})$, and the derived functor $RF : D^+(\mathscr{A}) \to D^+(\mathscr{B})$ is the restriction of the derived functor $RF : D(\mathscr{A}) \to D(\mathscr{B})$.

1.2.5. We leave it to the reader to:

(i) Generalize Definition 1.2.3 to the case of a complex of multifunctors, covariant in some variables and contravariant in others;
(ii) Dualize Definition 1.2.5 (i) to define the left derived functors;
(iii) Extend Proposition 1.2.4 to the case of a bounded complex of multifunctors;
(iv) Dualize Proposition 1.2.5 (iii) by considering left derived functors and the inclusion of $D^-(\mathscr{B})$ into $D(\mathscr{B})$.

PROPOSITION 1.2.6. *Let $\mathscr{A}$ and $\mathscr{B}$ be two abelian categories, let $\mathscr{P}$ be a subset of* $\mathrm{Ob}(\mathscr{A})$, *and let F be an additive functor from $\mathscr{A}$ to $\mathscr{B}$. Assume that*

(a) *every object of $\mathscr{A}$ is a quotient of an element of $\mathscr{P}$;*
(b) *for every short exact sequence*

$$0 \longrightarrow P \longrightarrow Q \longrightarrow R \longrightarrow 0,$$

where Q and R belong to $\mathscr{P}$, P also belongs to $\mathscr{P}$, and the 0*-sequence* $0 \to FP \to$
280 $FQ \to FR \to 0$ *is exact. Then every object of $\mathscr{P}$ is acyclic for LF.*

If $A \in \mathrm{Ob}\,\mathscr{A}$, the left resolutions of A by objects of $\mathscr{P}$ form a cofinal system in the filtering category of left resolutions of A. It therefore suffices to prove that if a complex

$$\cdots \longrightarrow A_2 \longrightarrow A_1 \longrightarrow A_0 \longrightarrow A \longrightarrow 0$$

is acyclic and has its objects in $\mathscr{P}$, its image is acyclic. Splitting it into short exact sequences

$$0 \longrightarrow \mathrm{Im}(d_{i-1}) \longrightarrow A_i \longrightarrow \mathrm{Im}(d_i) \longrightarrow 0 \qquad (i \geq 0) \tag{1.2.6.i}$$

one sees by induction on i that $\mathrm{Im}(d_i) \in \mathbf{P}$ and that the image of (1.2.6.i) under F is a short exact sequence. The image of the complex is therefore acyclic.

PROPOSITION 1.2.7. *Let F be an additive functor from an abelian category to an abelian category $\mathscr{B}$. If every object of $\mathscr{A}$ is a quotient (resp. subobject) of an object acyclic for LF (resp. RF) (1.2.3), the derived functor $LF : D^-(\mathscr{A}) \to D^-(\mathscr{B})$ (resp. $RF : D^+(\mathscr{A}) \to D^+(\mathscr{B})$) is defined everywhere, and every complex bounded above (resp. below) of objects acyclic for LF (resp. RF) is adapted.*

It suffices to consider the left derived functor LF. If K is a complex bounded above, the quasi-isomorphisms $s : L \to K$ with L bounded above and consisting of acyclic objects form a cofinal system in the category of quasi-isomorphisms with target K. It suffices to show that, when K consists of acyclic objects, $F(s) : F(L) \to F(K)$ is a
281 quasi-isomorphism; then, in the general case, one will have $F(L) = LF(K)$. Let M be the cone of s. The complex consists of acyclic objects and $H^*(M) = 0$; one must prove that $H^*(F(M)) = 0$.

Let i be an integer. The adapted complexes form a triangulated subcategory, so that every bounded complex of acyclic objects, in particular $\tau_{\geq i}(M)$, is adapted. Since $H^j(\tau_{\geq i}(M)) = 0$ for $j > i$, one has, for $j > i$, $H^j(F(M)) = H^j(F(\tau_{\geq i}(M))) = H^j LF(\tau_{\geq i}(M)) = 0$.

PROPOSITION 1.2.8. *Let $F^\cdot$ be a complex bounded below of functors from $\mathscr{A}$ to $\mathscr{B}$. If for every $K \in K^+(\mathscr{A})$ there exists a quasi-isomorphism $L \xrightarrow{s} K$, where $L \in K^+(\mathscr{A})$ is adapted to all the RF^i, then the derived functor $RF^\cdot : D^+(\mathscr{A}) \to D^+(\mathscr{B})$ exists, and every complex* bounded below *that is adapted to all the RF^i is adapted to $RF^\cdot$.*

The proof is left to the reader, as is that of the dual statement, or of the variants obtained by taking F contravariant, or by taking a bounded complex of functors and working in $D(\mathscr{A})$ and $D(\mathscr{B})$.

DEFINITION 1.2.9. *Let F be a functor from $D^b(\mathscr{A})$ to $D(\mathscr{B})$, and let d be an interval in* $\mathbf{Z}$. *We say that F has* cohomological amplitude $\subset d$ *if, for every $A \in \mathrm{Ob}\,\mathscr{A}$, $H^i F(A) = 0$ for $i \notin d$.*

Very often, d will be an interval of the form $[0, x]$ or $[-x, 0]$, and we then say that F has cohomological resp. homological dimension $\leq x$. We say that F has finite dimension if there exists a finite interval d in $\mathbf{Z}$ such that F has amplitude $\subset d$.

PROPOSITION 1.2.10. *Assume that the hypotheses of 1.2.7 hold and that the functor* 282
$RF : D^+(\mathscr{A}) \to D^+(\mathscr{B})$ (resp. $LF : D^-(\mathscr{A}) \to D^-(\mathscr{B})$) has finite dimension. Then every complex of objects acyclic for RF (resp. LF) is adapted, and the functor $RF : D(\mathscr{A}) \to D(\mathscr{B})$ (resp. $LF : D(\mathscr{A}) \to D(\mathscr{B})$) is defined everywhere.

For the proof, we refer to Verdier [11]. There is an analogous statement for bounded complexes of functors. The superscript + (resp. −) may also be replaced by b.

2. Categories fibered in derived categories

2.1. Introduction. The editor insists that the reader refrain from reading this §. Here we give the formalism of the trivial duality theorem and resolve the perplexity raised by Artin in XII 4.

For the notions of a category over another category, a fibered category, a cofibered category, and a cleavage, we refer to SGA 1 VI. Among other things, that exposé establishes an equivalence between the notions of normalized split fibered categories over a category $\mathscr{C}$, and pseudofunctors from $\mathscr{C}^\circ$ to (Cat).

DEFINITION 2.1.1. *If X is a ringed site, $D(X)$ denotes the derived category of the abelian category of sheaves of left modules on X.*

When confusion must be avoided, we instead write $D(X, \mathscr{A})$, where $\mathscr{A}$ denotes the sheaf of rings.

If $f : X \to Y$ is a morphism of ringed sites, f defines derived functors 283

(1) $Rf_* : D^+(X) \to D^+(Y)$ and
(2) $Lf^* : D^-(Y) \to D^-(X)$,

and, on passing to the opposite categories, the functors

(1°) $Rf_* : D^+(X)^\circ \to D^+(Y)^\circ$
(2°) $Lf^* : D^-(Y)^\circ \to D^-(X)^\circ$,

"functorial" in f. More precisely, (1°) and (2°) define two pseudofunctors from the category of sites to that of triangulated categories, one covariant and the other contravariant. One thereby obtains two categories over the category of sites, one cofibered, with fibers $D^+(X)^\circ$ and direct-image functors Rf_*, and the other fibered, with fibers $D^-(X)^\circ$ and inverse-image functors Lf^*. If the functor f^* has finite homological dimension (i.e. has finite "tor-dimension"), then in (2) the superscript may be replaced by a +. Thus one

obtains two categories over the category $\mathcal{S}_{\text{tdf}}$ of ringed sites and morphisms of ringed sites of finite tor-dimension; the fibers of these categories are the categories $D^+(X)^\circ$. The "trivial duality theorem" of VERDIER ([11] p. 48) states that these two categories are canonically isomorphic; they are therefore fibered and cofibered; the "direct-image" functors are the functors Rf_* (sic), and the "inverse-image" functors are the functors Lf^* (sic).

2.1.2. The language of fibered categories readily resolves the perplexity raised by ARTIN in XII 4: it suffices to apply the following proposition to the category fibered and
284 cofibered over the category of sites whose fibers are the opposites of the categories of sheaves on the corresponding sites (cf. 0.2).

Let $\mathscr{E}$ be a category over $\mathscr{B}$, let $f : y \to x$ be an arrow of $\mathscr{B}$, and let $F \in \mathrm{Ob}(\mathscr{E}_x)$. Recall that a pair (G, φ) ($G \in \mathrm{Ob}\,\mathscr{E}_y, \varphi \in \mathrm{Hom}_f(G, F)$) is a *strict inverse image* of F under f if, for all $g : z \to y$ and $H \in \mathrm{Ob}\,\mathscr{E}_z$, one has

$$\mathrm{Hom}_g(H, G) \xrightarrow[\sim]{\varphi} \mathrm{Hom}_{fg}(H, F).$$

Similarly for direct images. If $\mathscr{E}$ is fibered or cofibered over $\mathscr{B}$, this notion reduces to that of inverse image.

PROPOSITION 2.1.3. *Let ϵ be a category over $\mathscr{B}$, let $y \in \mathrm{Ob}\,\mathscr{B}$, let $F \in \mathrm{Ob}\,\mathscr{E}_y$, and consider a commutative diagram in $\mathscr{B}$:*

(2.1.3.1)
$$\begin{array}{ccc} y' & \xrightarrow{g'} & y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ x' & \longrightarrow & x. \end{array}$$

Assume that the direct and inverse images f_*F, g^*f_*F, g'^*F, and $f'_*g'^*F$ exist in the strict sense. Then there exists a unique arrow $\varphi \in \mathrm{Fl}\,\mathscr{E}_{x'}$, called the base-change arrow, *making the diagram*

(2.1.3.2)
$$\begin{array}{ccc} & g'^*F & \longrightarrow F \\ \swarrow & & \downarrow \\ f'_*g'^*F & & f_*F \\ & \searrow^{\varphi} \quad \nearrow & \\ & g^*f_*F & . \end{array}$$

285 *commutative. If, in addition, the direct images $g'_*g'^*F$ and $g'_*f'_*g'^*F$ exist in the strict sense, then the arrow φ coincides with the one defined in XII 4. If $f'^*f'_*g'^*F$ and $f'^*f'_*F$ exist in the strict sense, it also coincides with the 2^{nd} arrow defined in XII 4.*

In particular, the two base-change arrows of XII 4 coincide.

By hypothesis, the arrows in diagram (2.1.3.2) induce bijections

$$\mathrm{Hom}_{x'}(f'_*g'^*F, g^*f_*F) = \mathrm{Hom}_{f'}(g'^*F, g^*f_*F) = \mathrm{Hom}_{gf'}(g'^*F, f_*F).$$

Since $fg' = gf'$, $\mathrm{Hom}_{fg'}(g'^*F, f_*F)$ is identical to $\mathrm{Hom}_{gf'}(g'^*F, f_*F)$, and the arrows of diagram (2.1.3.2) induce maps

$$\mathrm{Hom}_y(F, F) \longrightarrow \mathrm{Hom}_{fg'}(g'^*F, f_*F) = \mathrm{Hom}_{x'}(f'_*g'^*F, g^*f_*F),$$

and the image of 1_F is the unique arrow of $\mathscr{E}_{x'}$ making (2.1.3.2) commutative.

By definition, in XII 4, the base change arrow ψ was the one making the following diagram commutative

$$\begin{array}{ccccccc} f'_*g'^*F & \longleftarrow & g'^*F & = & g'^*F & \longrightarrow & F & = & F \\ \downarrow{\scriptstyle\psi} & & \downarrow{\scriptstyle u_4} & & \downarrow{\scriptstyle u_3} & & \downarrow{\scriptstyle u_2} & & \downarrow{\scriptstyle u_1} \\ g'_*f_*F & \longleftarrow & f'^*g^*f_*F & = & g'f^*f_*F & \longrightarrow & f^*f_*F & \longrightarrow & f_*F. \end{array} \tag{2.1.3.3}$$

The horizontal arrows and u_1 are the canonical arrows; u_2 is therefore the adjunction arrow, $u_3 = g'^*(u_2)$, u_4 is deduced from the isomorphism between f'^*g^* and g'^*f^*, and the arrow ψ is deduced from u_4 by adjunction.

The composite map with source g'^*F and target f_*F is the one that appears in (2.1.3.2), and the same therefore holds in the following commutative diagram, whose left-hand square is taken from (2.1.3.3) 286

$$\begin{array}{ccccc} g'^*F & \xrightarrow{u_4} & f'^*g^*f_*F & \longrightarrow & f_*F \\ \downarrow & & \downarrow & & \| \\ f'_*g'^*F & \xrightarrow{\psi} & g^*f_*F & \longrightarrow & f_*F. \end{array}$$

This means that ψ satisfies the defining property of φ.

The definition of φ is self-dual, so there is no need to prove the second part of 2.1.3.

2.2. Categories fibered in triangulated categories.

Definition 2.2.1. *An "additive category over a category $\mathscr{B}$" is a category $\mathscr{A}$ over $\mathscr{B}$ whose sets of arrows $\mathrm{Hom}_f(X,Y)$ (for $f \in \mathrm{Fl}(\mathscr{B})$, with X (resp. Y) over the source (resp. the target) of f) have been endowed with abelian group structures, in such a way that:*

(i) *For any $x \xrightarrow{f} y \xrightarrow{g} z$ in $\mathscr{B}$ and X, Y, Z in $\mathscr{A}$ lying over x, y, and z, respectively, the composition law*

$$\mathrm{Hom}_g(Y,Z) \times \mathrm{Hom}_f(X,Y) \overset{\circ}{\to} \mathrm{Hom}_{gf}(X,Z)$$

is biadditive.

(ii) *The fiber categories are additive.*

We shall not need here the fact that these abelian group structures, when they exist, are uniquely determined by $\mathscr{A}$, $\mathscr{B}$, and ρ.[45] A $\mathscr{B}$-functor $F : \mathscr{A}_1 \to \mathscr{A}_2$ between additive categories over $\mathscr{B}$ is said to be *additive* if it induces homomorphisms

$$F : \mathrm{Hom}_{\mathscr{A}_1,f}(X,Y) \longrightarrow \mathrm{Hom}_{\mathscr{A}_2,f}(X,Y).$$

Definition 2.2.2. *An "abelian category over a category $\mathscr{B}$" is an additive category $\mathscr{A}$ over $\mathscr{B}$ whose fibers are abelian, and such that for every arrow $f : x \to y$ of $\mathscr{B}$ the bifunctor* 287

$$\mathrm{Hom}_f(X,Y) : \mathscr{A}_x^\circ \times \mathscr{A}_y \longrightarrow (\mathrm{Ab})$$

is left exact in X and Y.

We shall not need here the fact that if $\mathscr{A}$ is a category fibered and cofibered over $\mathscr{B}$ (SGA 1 VI 6), it is abelian over $\mathscr{B}$ as soon as its fibers are abelian categories.

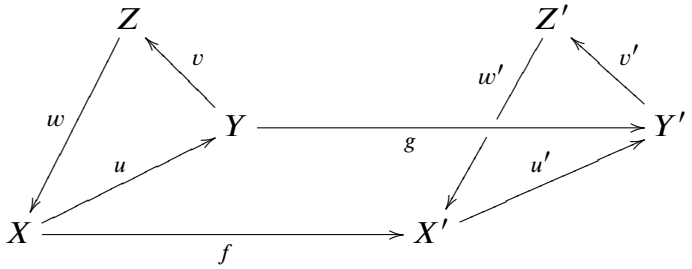
[45] N.D.E.: here ρ denotes the structural functor $\mathscr{A} \to \mathscr{B}$.

DEFINITION 2.2.3. *A "triangulated category over a category $\mathscr{B}$" is an additive category $\mathscr{A}$ over $\mathscr{B}$, endowed with a functor $T : \mathscr{A} \to \mathscr{A}$, and whose fibers $\mathscr{A}_x$, for $x \in \operatorname{Ob}\mathscr{B}$, are endowed with sets Δ_x of triangles, in such a way that:*

(i) *T is an additive $\mathscr{B}$-automorphism of $\mathscr{A}$.*

(ii) *Each fiber category $\mathscr{A}_x$, endowed with the functor T_x induced by T and with the set Δ_x, is a triangulated category* ([11] I 1.1.1).

(iii) *If the triangles (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') belong to Δ_x and $\Delta_{x'}$, respectively, every diagram*



in which $u'f = gu$ can be completed to a morphism of triangles $(f, g, h) : (X, Y, Z) \to (X', Y', Z')$.

288 T will be called the translation functor; we shall often write $A[1]$ rather than $T(A)$, with $A[n]$ then denoting $T^n(A)$ ($n \in \mathbf{Z}$).

The triangles belonging to the sets Δ_x will be called *distinguished* or *exact.*

PROPOSITION 2.2.4. *Let $\mathscr{A}$ be a triangulated category over $\mathscr{B}$, let x be an object of $\mathscr{B}$, and let (X, Y, Z, u, v, w) be a distinguished triangle of $\mathscr{A}_x$. For every $f : y \to x$ and every object $A \in \operatorname{Ob}\mathscr{A}_y$, the following infinite sequence is exact:*

$$\cdots \longrightarrow \operatorname{Hom}_f(A, X) \longrightarrow \operatorname{Hom}_f(A, Y) \longrightarrow \operatorname{Hom}_f(A, Z) \longrightarrow \operatorname{Hom}_f(A, X[1]) \longrightarrow \operatorname{Hom}_f(A, Y[1]) \longrightarrow \dots$$

There is an analogous statement for $f : x \to y$ and $A \in \operatorname{Ob}\mathscr{A}_y$.

We leave it to the reader to prove this proposition by paraphrasing VERDIER [11] p. 4. We also leave it to the reader to give a meaning to the following proposition and to verify it:

PROPOSITION 2.2.5. *Let $\mathscr{A}$ be an additive (resp. abelian, resp. triangulated) category over $\mathscr{B}$. For every functor $\mathscr{B}' \to \mathscr{B}$ with target $\mathscr{B}$, the category $\mathscr{A}' = \mathscr{A} \times_{\mathscr{B}} \mathscr{B}'$ is additive (resp. abelian, resp. triangulated) over $\mathscr{B}$.*

EXAMPLE 2.2.6. The category of sheaves of modules on varying ringed sites is abelian over the category of ringed sites.

EXAMPLE 2.2.7. Let $\mathscr{A}$ be an additive category over $\mathscr{B}$. The category $C(\mathscr{A}/\mathscr{B})$ of complexes of $\mathscr{A}$ over $\mathscr{B}$ has as objects the complexes of the fiber categories. An arrow
289 f from (X^n, d) to (Y^n, d) is a system of arrows $f^n : X^n \to Y^n$ such that $d f^n = f^{n+1} d$. We set $p(f) = p(f^n)$ (this arrow does not depend on n, since the $p(d)$ are identities). $C(\mathscr{A}/\mathscr{B})$ is an additive category over $\mathscr{B}$. A "translation" functor is defined by setting

$$(K[1])^n = K^{n+1} \quad , \quad (d[1])^n = -d^{n+1}.$$

EXAMPLE 2.2.8. Let $\mathscr{A}$ be an additive category over $\mathscr{B}$. *The category $K(\mathscr{A}/\mathscr{B})$ of complexes of $\mathscr{A}$ over $\mathscr{B}$ modulo homotopy* has the same objects as $C(\mathscr{A}/\mathscr{B})$. It is obtained

by declaring the arrows homotopic to zero to be zero in $\boldsymbol{C}(\mathscr{A}/\mathscr{B})$; in other words, for $f : x \to y$ in $\mathscr{B}$, $X^{\cdot} \in \operatorname{Ob} \boldsymbol{C}(\mathscr{A}/\mathscr{B})_x$ and $Y^{\cdot} \in \operatorname{Ob} \boldsymbol{C}(\mathscr{A}/\mathscr{B})_y$, we set

$$\operatorname{Hom}_f(X^{\cdot}, Y^{\cdot}) = H^0 \operatorname{Hom}^{\cdot}_f(X^{\cdot}, Y^{\cdot})$$

The translation functor descends to the quotient.

If $F : \mathscr{B}' \to \mathscr{B}$ is a functor with target $\mathscr{B}$, and if $\mathscr{A}' = \mathscr{A} \times_{\mathscr{B}} \mathscr{B}'$, we leave it to the reader to identify $\boldsymbol{C}(\mathscr{A}'/\mathscr{B}')$ with $\boldsymbol{C}(\mathscr{A}/\mathscr{B}) \times_{\mathscr{B}} \mathscr{B}'$ and $\boldsymbol{K}(\mathscr{A}'/\mathscr{B}')$ with $\boldsymbol{K}(\mathscr{A}/\mathscr{B}) \times_{\mathscr{B}} \mathscr{B}'$. In particular, the fibers of the category $\boldsymbol{K}(\mathscr{A}/\mathscr{B})$ are the categories $\boldsymbol{K}(\mathscr{A}_x)$ and, as such, are triangulated.

PROPOSITION 2.2.9. (i) *With the additional structures defined in 2.2.8, the category $\boldsymbol{K}(\mathscr{A}/\mathscr{B})$ is triangulated over $\mathscr{B}$.*

(ii) *The formation of the additive (resp. triangulated) category $\boldsymbol{C}(\mathscr{A}/\mathscr{B})$ (resp. $\boldsymbol{K}(\mathscr{A}/\mathscr{B})$) is compatible with every change of base category $\mathscr{B}' \to \mathscr{B}$.*

Conditions (i) and (ii) of Definition 2.2.3 are verified fiber by fiber. Condition (iii) is 290
verified by paraphrasing VERDIER [11] p. 4.

REMARK 2.2.10. The preceding discussion also applies to the categories $\boldsymbol{K}^+(\mathscr{A}/\mathscr{B})$, $\boldsymbol{K}^-(\mathscr{A}/\mathscr{B})$, and $\boldsymbol{K}^b(\mathscr{A}/\mathscr{B})$ obtained by restricting respectively to complexes bounded below, bounded above, or bounded.

REMARK 2.2.11. Taking $\mathscr{B}$ to be the final category, one recovers the usual categories of complexes.

2.2.12. Let F be a pseudofunctor from a category $\mathscr{B}^\circ$ to (Cat). It is known (SGA 1 VI) that F defines a category fibered over $\mathscr{B}$, whose fibers are the categories $F(x)$. For any $f : x \to y$ in $\mathscr{B}$ and $X \in \operatorname{Ob} F(x)$, $Y \in \operatorname{Ob} F(y)$, we have by definition

$$\operatorname{Hom}_f(X, Y) = \operatorname{Hom}_{F(x)}(X, F(f)(Y)). \tag{2.2.12.1}$$

It is known (ibid.) that the preceding construction defines an equivalence between the 2-category of pseudofunctors from $\mathscr{B}^\circ$ to (Cat) and the 2-category of categories fibered over $\mathscr{B}$.

Let us start with a pseudofunctor from $\mathscr{B}^\circ$ to the 2-category whose objects are additive (resp. abelian, resp. triangulated) categories and whose 1-arrows are additive (resp. left exact, resp. triangulated) functors. One immediately verifies that the category defined by (2.2.12.1) is additive (resp. abelian, resp. triangulated). Conversely:

PROPOSITION 2.2.13. *The preceding construction establishes an equivalence between* 291

(a) *The 2-category of pseudofunctors from $\mathscr{B}^\circ$ to the 2-category whose objects are additive (resp. abelian, resp. triangulated) categories and whose 1-arrows are additive (resp. left exact, resp. triangulated) functors.*

(b) *The 2-category of additive (resp. abelian, resp. triangulated) categories over $\mathscr{B}$ that are fibered over $\mathscr{B}$.*

The theory developed in SGA 1 VI shows that it suffices to verify the following point:

. Let $\mathscr{A}$ be an additive (resp. abelian, resp. triangulated) category over $\mathscr{B}$, let $f : x \to y$ be an arrow of $\mathscr{B}$, and suppose that an inverse image functor $f^* : \mathscr{A}_y \to \mathscr{A}_x$ exists; then f^* is additive (resp. left exact, resp. triangulated).

The functor f^* gives rise to isomorphisms

$$\operatorname{Hom}_f(X, Y) \simeq \operatorname{Hom}_x(X, f^*(Y)).$$

In the triangulated case, it remains to prove that f^* is triangulated for the isomorphism $Tf^* = f^*T$ (where T is the translation functor) that makes the following diagrams commutative:

(2.2.13.2)
$$\begin{array}{ccccc} \mathrm{Hom}_f(X,Y) & = & \mathrm{Hom}_x(X,f^*Y) & & \\ \| & & & \searrow & \\ \mathrm{Hom}_f(TX,TY) & & & & \mathrm{Hom}_x(TX,Tf^*Y) \\ & \searrow & & \nearrow & \\ & & \mathrm{Hom}_x(TX,f^*TY) & & \end{array}$$

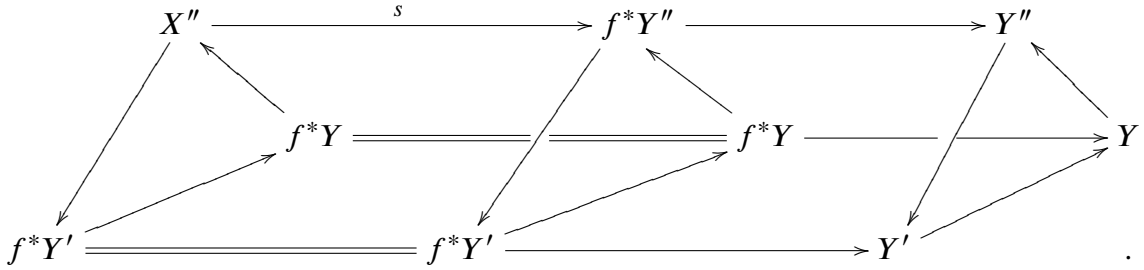
292 *Additive case*: Consider a diagram

(2.2.13.3)
$$\begin{array}{ccc} f^*X & \xrightarrow{p} & X \\ {\scriptstyle f^*(u)}\downarrow\downarrow{\scriptstyle f^*(v)} & & {\scriptstyle u}\downarrow\downarrow{\scriptstyle v} \\ f^*X' & \xrightarrow{p'} & X'. \end{array}$$

We have $p'f^*(u+v) = (u+v)p = up+vp = p'f^*(u)+p'f^*(v) = p'(f^*(u)+f^*(v))$ and hence $f^*(u+v) = f^*(u) + f^*(v)$.

Abelian case: by definition (2.2), the functor $\mathrm{Hom}(X, f^*Y) = \mathrm{Hom}_f(X,Y)$ is left exact in Y, for every X, whence f^* is left exact.

Triangulated case: let (Y', Y, Y'') be a distinguished triangle in $\mathscr{A}_y$ and let (f^*Y', f^*Y, X'') be a distinguished triangle in $\mathscr{A}_x$. By 2.2.3, the canonical morphism between the bases of these triangles can be completed to a morphism of triangles, giving rise to the following diagram:



Let us prove that s is an isomorphism, which will complete the proof. For every X in $\mathrm{Ob}\,\mathscr{A}_x$, we have the commutative diagram:

$$\begin{array}{ccccccccc} \mathrm{Hom}_x(X,f^*Y') & \to & \mathrm{Hom}_x(X,f^*Y) & \to & \mathrm{Hom}_x(X,X'') & \to & \mathrm{Hom}_x(X,f^*Y'[1]) & \to & \mathrm{Hom}_x(X,f^*Y[1]) \\ \| & & \| & & \downarrow & & \| & & \| \\ \mathrm{Hom}_x(X,f^*Y') & \to & \mathrm{Hom}_x(X,f^*Y) & \to & \mathrm{Hom}_x(X,f^*Y'') & \to & \mathrm{Hom}_x(X,f^*Y'[1]) & \to & \mathrm{Hom}_x(X,f^*Y[1]) \\ \| & & \| & & \| & & \| & & \| \\ \mathrm{Hom}_f(X,Y) & \to & \mathrm{Hom}_f(X,Y) & \to & \mathrm{Hom}_f(X,Y'') & \to & \mathrm{Hom}_f(X,Y'[1]) & \to & \mathrm{Hom}_x(X,Y[1]). \end{array}$$

293 The first and last rows are exact by 2.2.4. By the five lemma, the map induced by s from $\mathrm{Hom}(X,X'')$ to $\mathrm{Hom}(X,f^*Y'')$ is always bijective, and s is therefore an isomorphism.

There are, of course, dual results for cofibered categories.

2.3. Trivial duality formula. Let $\mathscr{A}$ be a category over a base category $\mathscr{B}$, and let S be a set of arrows of $\mathscr{A}$, each of which projects onto an identity of $\mathscr{B}$. The category $\mathscr{A}(S^{-1})$ obtained from $\mathscr{A}$ by the calculus of fractions is a category over $\mathscr{B}$ (by the universal property).

Let $\mathscr{B}' \to \mathscr{B}$ be a functor; denote by $\mathscr{A}'$ the fiber product $\mathscr{A} \times_{\mathscr{B}} \mathscr{B}'$ and by S' the inverse image of S in $\mathscr{A}'$. The composite functor $\mathscr{A}' \to \mathscr{A} \to \mathscr{A}(S^{-1})$ makes the arrows of S' invertible and hence factors through $\mathscr{A}'(S'^{-1})$. We thus obtain a canonical $\mathscr{B}'$-functor as follows:

$$\mathscr{A}'(S^{-1}) \longrightarrow \mathscr{B}' \times_{\mathscr{B}} \mathscr{A}(S'^{-1}). \tag{2.3.1}$$

LEMMA 2.3.2. *If $\mathscr{B}'$ is a subcategory of $\mathscr{B}$, and if whenever a composite arrow $h = f \circ g$ of $\mathscr{B}$ belongs to $\mathscr{B}'$, both f and g belong to $\mathscr{B}'$, then the functor (2.3.1) is an isomorphism.*

Note that (2.3.1) always induces a bijection on objects, and that, in the present case, 294
$\mathscr{B}' \times_{\mathscr{B}} \mathscr{A}(S^{-1})$ is a subcategory of $\mathscr{A}(S^{-1})$.

Recall the "explicit" construction of $\mathscr{A}(S^{-1})$ [2, Chap. I]. Given two objects X and Y of $\mathscr{A}$, one considers the "formal composites" $f_1 \dots f_n$, where each f_i is either an arrow of $\mathscr{A}$ or the inverse of an arrow of S. The set $\mathrm{Hom}_{\mathscr{A}(S^{-1})}(X, Y)$ is the quotient of the set of these "formal composites" by the equivalence relation generated by the relations

$$\begin{aligned} f_1 \dots uv \dots f_n &\equiv f_1 \dots (u \circ v) \dots f_n && \text{for } u, v \text{ in } \mathrm{Fl}(\mathscr{A}) \\ f_1 \dots ss^{-1} \dots f_n &\equiv f_1 \dots f_n && \text{for } s \in S \\ f_1 \dots s^{-1}s \dots f_n &\equiv f_1 \dots f_n && \text{for } s \in S. \end{aligned}$$

If these constructions are applied to $\mathscr{A}$ and to the subcategory $\mathscr{A}'$, one checks that $\mathscr{A}'(S^{-1})$ is identical to the inverse-image subcategory of $\mathscr{A}(S^{-1})$ of $\mathscr{B}'$, since if a "formal composite" $f_1 \dots f_n$ has as its composite in $\mathscr{B}$ an arrow of $\mathscr{B}'$, the image of each f_i will also be an arrow of $\mathscr{B}'$.

Suppose now that $\mathscr{A}$ is a triangulated category over $\mathscr{B}$, and that the sets $S_x = S \cap \mathrm{Fl}(\mathscr{A}_x)$, for $x \in \mathrm{Ob}(\mathscr{B})$, are saturated multiplicative systems ([11] Chap. I § 2 n° 1).

DEFINITION 2.3.3. *Under the preceding hypotheses, the category $\mathscr{A}/\mathscr{B}$ is said to be almost derivable (resp. derivable, resp. coderivable) relative to S if, for every functor $F : \mathscr{B}' \to \mathscr{B}$, the functor (2.3.1) is an isomorphism (resp. and if $\mathscr{A}(S^{-1})$ is fibered, resp. cofibered over $\mathscr{B}$).*

We denote by I the category of the following diagram: 295

$$I : \underset{0}{\times} \xrightarrow{F} \underset{1}{\times}. \tag{2.3.4}$$

By 2.2.13, to give a triangulated category $\mathscr{A}$ over I, fibered (resp. cofibered) over I, is equivalently to give its fibers $\mathscr{A}_0$ and $\mathscr{A}_1$ and the triangulated inverse-image functor $F^* : \mathscr{A}_1 \to \mathscr{A}_0$ (resp. and the triangulated direct-image functor $F_* : \mathscr{A}_0 \to \mathscr{A}_1$). If $A_0 \in \mathrm{Ob}\,\mathscr{A}_0$ and $A_1 \in \mathrm{Ob}\,\mathscr{A}_1$, then by (2.2.12.1),

$$\mathrm{Hom}_F(A_0, A_1) = \mathrm{Hom}(A_0, F^*A_1)$$

(resp. $\mathrm{Hom}_F(A_0, A_1) = \mathrm{Hom}(F_*A_0, A_1)$).

PROPOSITION 2.3.5. *Let $\mathscr{A}$ be a triangulated category over I and let S be as in (2.3.3).*

(i) *$\mathscr{A}$ is almost derivable relative to S, and $\mathscr{A}(S^{-1})$ is triangulated over I.*

(ii) *If $\mathscr{A}$ is fibered (resp. cofibered), the functor F^* (resp. F_*) is right derivable (resp. left derivable) relative to S_0 and S_1 (1.2.1) if and only if $\mathscr{A}$ is derivable (resp. coderivable) over I.*

By Lemma 2.3.2, the fiber categories of $\mathscr{A}(S^{-1})$ are $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$, where $S_i = S \cap \mathrm{Fl}(\mathscr{A}_i)$ $(i = 0, 1)$. With the notation of § 1 n° 2, the functors $A_0 \to A_0^-$ and $A_1 \to A_1^+$ identify $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$ with categories of pro- and ind-objects of $\mathscr{A}_0$ and $\mathscr{A}_1$. This makes it possible to define a category over I, with fibers $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$, by setting, for $A_0 \in \mathscr{A}_0$ and $A_1 \in \mathscr{A}_1$,

$$\mathrm{Hom}_F(A_0, A_1) = \mathrm{Hom}_F(A_0^-, A_1^+) = \varinjlim_{A_0' \xrightarrow[s]{} A_0} \varinjlim_{A_1 \xrightarrow[t]{} A_1'} \mathrm{Hom}(A_0', A_1'),$$

296 where s and t range over the elements of S_0 and S_1 with target (respectively source) A_0 (respectively A_1). The category $\mathscr{A}$ maps to this category, and one checks immediately that it satisfies the universal property of $\mathscr{A}(S^{-1})$, over I and after every base change.

To verify (i), it suffices to prove that this category, namely $\mathscr{A}(S^{-1})$, is triangulated, and it suffices to verify condition 2.2.3 (iii). Let (X_0, Y_0, Z_0) and (X_1, Y_1, Z_1) therefore be distinguished triangles of $\mathscr{A}_0$ and $\mathscr{A}_1$, giving rise to the distinguished triangles (X_0^-, Y_0^-, Z_0^-) and (X_1^+, Y_1^+, Z_1^+) of $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$.

(2.3.5.1)
$$\begin{array}{ccc} X_0^- & \xrightarrow{u} & X_1^+ \\ \downarrow & & \downarrow \\ Y_0^- & \xrightarrow{v} & Y_1^+. \end{array}$$

By Lemma 1.2.2.1, the triangles (X_0^-, Y_0^-, Z_0^-) and (X_1^+, Y_1^+, Z_1^+) are limits of distinguished triangles; there therefore exist morphisms of distinguished triangles

(2.3.5.2)
$$\begin{aligned} (X_0^-, Y_0^-, Z_0^-) &\longrightarrow (X_0', Y_0', Z_0') \\ (X_1', Y_1', Z_1') &\longrightarrow (X_1^+, Y_1^+, Z_1^+), \end{aligned}$$

297 and a diagram

$$\begin{array}{ccc} X_0' & \xrightarrow{u'} & X_1' \\ \downarrow & & \downarrow \\ Y_0' & \xrightarrow{v'} & Y_1', \end{array}$$

compatible with (2.3.5.1). Since $\mathscr{A}$ is triangulated over I, this diagram can be extended to a morphism of triangles

$$(X_0', Y_0', Z_0') \longrightarrow (X_1', Y_1', Z_1').$$

Composing the latter with the morphisms (2.3.5.2), we see that diagram (2.3.5.1) can be extended to a morphism of triangles, which proves 2.2.3 (iii).

Let us verify (ii), assuming for example that $\mathscr{A}$ is fibered over I. If $A_1 \in \mathrm{Ob}(\mathscr{A}_1(S_1^{-1}))$, an inverse image of A_1 in $\mathscr{A}(S^{-1})$ is an object B of $\mathscr{A}_0(S_0^{-1})$ such that, in $\mathscr{A}_0(S_0^{-1})$,

$$\begin{aligned} \mathrm{Hom}_{\mathscr{A}_0(S_0^{-1})}(A_0, B) &= \mathrm{Hom}_F(A_0^-, A_1^+) = \mathrm{Hom}(A_0^-, F^*(A_1^+)) \\ &= \mathrm{Hom}_{\mathscr{A}_0(S_0^{-1})}(A_0, F^*(A_1^+)). \end{aligned}$$

Such an object B exists if and only if the ind-object $F^*(A_1^+)$ of $\mathscr{A}_0(S_0^{-1})$ is essentially constant (i.e. representable), and B then represents this ind-object, which is precisely $RF(A_1)$ (1.2.1); this proves the assertion.

COROLLARY 2.3.6. *With the notation of 2.3.3, if $\mathscr{A}$ is almost derivable over $\mathscr{B}$, then $\mathscr{A}(S^{-1})$ is a triangulated category over $\mathscr{B}$.*

First, the fiber category $\mathscr{A}(S^{-1})_x$ can be identified with $\mathscr{A}_x(S_x^{-1})$: apply the hypothesis to the base change $\{x\} \to \mathscr{B}$. This category is triangulated, equipped with a set 298
Δ_x of triangles, which gives a meaning to the preceding statement. To verify condition 2.2.3 (iii), it suffices to do so after every base change $I \to \mathscr{B}$, and the assertion follows from 2.3.5 (i).

It follows immediately from the proof of 2.3.5 (ii) that:

THEOREM 2.3.7. (trivial duality formula).

Let $\mathscr{A}$ be a triangulated category fibered and cofibered over I, and denote by F_* and F^* the direct- and inverse-image functors. Let $S = S_0 \cup S_1$ be a multiplicative system as in 2.3.3. Suppose that LF_* is defined at $A_0 \in \mathrm{Ob}\,\mathscr{A}(S_0^{-1})$ and that RF^* is defined at $A_1 \in \mathrm{Ob}\,\mathscr{A}_1(S_1^{-1})$. Then

$$\mathrm{Hom}_{\mathscr{A}_0(S_0^{-1})}(A_0, RF^*A_1) = \mathrm{Hom}_{\mathscr{A}(S^{-1})}(A_0, A_1) = \mathrm{Hom}_{\mathscr{A}_1(S_1^{-1})}(LF_*A_0, A_1).$$

The positions of the symbols R, L, *, and $_*$ in 2.3.7 are only apparently anomalous: in fact, in the category 0.2 of sheaves on variable sites, fibered and cofibered, the fiber categories are the opposites of the usual categories of sheaves.

2.4. Categories fibered in derived categories.

2.4.0. In this n°, $\mathscr{A}$ denotes a triangulated category cofibered over $\mathscr{B}$, and S a multiplicative system of morphisms of $\mathscr{A}$, all lying over an identity, such that $S_x = S \cap \mathrm{Fl}(\mathscr{A}_x)$ is a saturated multiplicative system of $\mathscr{A}_x$ [11, I 1.2.1] for every $x \in \mathrm{Ob}(\mathscr{B})$.

We seek a criterion for $\mathscr{A}$ to be coderivable relative to S (2.3.3); this implies that 299
$\mathscr{A}(S^{-1})$ is triangulated over $\mathscr{B}$ when it is equipped with the set of triangles isomorphic to the image of a distinguished triangle of $\mathscr{A}$ (2.3.5).

PROPOSITION 2.4.1. *Let $\mathscr{A}$, $\mathscr{B}$, and S be as above. For $\mathscr{A}$ to be coderivable relative to S, it suffices that the following condition hold:*

(∗) *$\forall x \in \mathrm{Ob}\,\mathscr{B}$, $\forall A \in \mathrm{Ob}(\mathscr{A}_x)$, $\exists s : A' \to A$ in S, $\forall g, f : x \xrightarrow{f} y \xrightarrow{g} z$, f_*A' is left adapted to the functor g_*.*

Let $\mathscr{A}^1$ denote the full subcategory of $\mathscr{A}$ consisting of the objects all of whose direct images are adapted to all direct-image functors. Then 1.2.2 shows that $\mathscr{A}^1$ is a triangulated subcategory of $\mathscr{A}$, evidently cofibered. Set $S^1 = S \cap \mathrm{Fl}(\mathscr{A}^1)$.

LEMMA 2.4.1.1. *Suppose that S is stable under direct images. Then $\mathscr{A}/\mathscr{B}$ is coderivable and $\mathscr{A}(S^{-1})$ can be computed by a calculus of right fractions.*

Let us verify condition c) of [2, I 2.2], possibly the least trivial of the conditions. We are given f and s ($s \in S$) and must find

(2.4.1.2)
$$\begin{array}{ccc} A' & \dashrightarrow & B' \\ {\scriptstyle s}\uparrow & & \uparrow{\scriptstyle t} \\ A & \xrightarrow[f]{} & B, \end{array}$$

a commutative diagram (2.4.1.2) with $t \in S$. Let F be the image of f in $\mathscr{B}$. By hypothesis, $F_*(s)$ still belongs to S, so by VERDIER there is a commutative diagram 300

$$\begin{array}{ccc} F_*(A') & \longrightarrow & B' \\ {\scriptstyle F_*(s)}\uparrow & & \uparrow{\scriptstyle t} \\ F_*(A) & \xrightarrow[f]{} & B \end{array}$$

and hence also a diagram (2.4.1.2).

Since $\mathscr{A}(S^{-1})$ is computed by a calculus of right fractions, it is immediate that its formation commutes with base change $c : \mathscr{B}' \to \mathscr{B}$.

This lemma applies in particular to $\mathscr{A}^1$ and S^1; the following lemma therefore implies 2.4.1:

LEMMA 2.4.1.3. *Over $\mathscr{B}$, and after every base change $c : \mathscr{B}' \to \mathscr{B}$, the canonical functor $\Phi : \mathscr{A}^1((S^1)^{-1}) \to \mathscr{A}(S^{-1})$ is an equivalence of categories.*

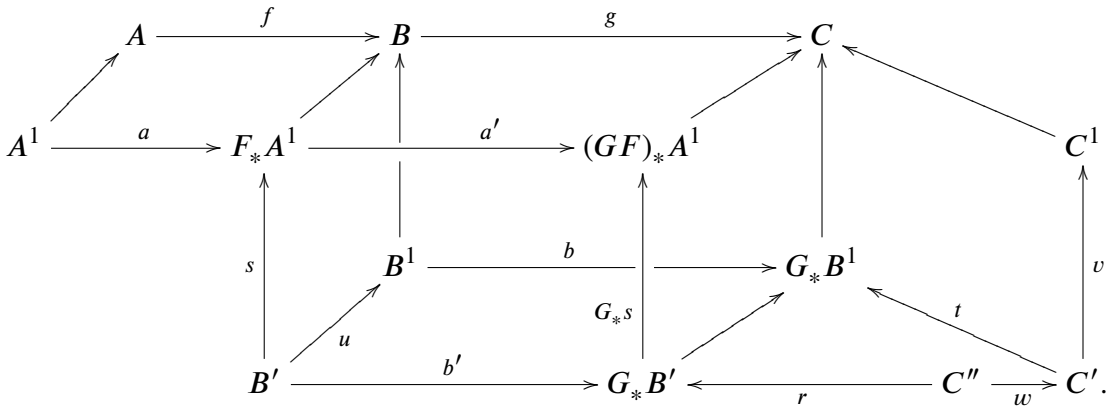
Let us construct a quasi-inverse to Φ. For each $A \in \mathrm{Ob}\,\mathscr{A}$, choose $s_A \in S$ with target A and source $A^1 \in \mathrm{Ob}\,\mathscr{A}^1$, with $s_A = \mathrm{Id}_A$ when $A \in \mathrm{Ob}\,\mathscr{A}^1$. Let f be an arrow of $\mathscr{A}$ with target A, source B, and image F in $\mathscr{B}$. Since $s_B \in S$ and $F_*A^1 \in \mathrm{Ob}\,\mathscr{A}^1$, one can find B', $s \in S^1$ and u making the following diagram commutative:

(2.4.1.4)

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow[s_B]{} & B^1 \\ {\scriptstyle s_A}\uparrow & & \uparrow & & \uparrow{\scriptstyle u} \\ A^1 & \xrightarrow{a} & F_*A^1 & \xleftarrow{s} & B'. \end{array}$$

301 The arrow $f^1 = us^{-1}a$ of $\mathscr{A}^1((S^1)^{-1})$ then depends only on f.

Let f and g be two composable arrows, with images F and G in $\mathscr{B}$. To prove that $(g \circ f)^1 = g^1 \circ f^1$, consider the following diagram, which contains diagrams (2.4.1.4) for f and g:



Since $t \in S$, there exist C'', $r \in S^1$, and w making the diagram commutative. By definition,
$g^1 \circ f^1 = vt^{-1}bus^{-1}a$ and $(g \circ f)^1 = vw(G_*(s)r)^{-1}a'a$, and the conclusion follows from the commutativity of the diagram.

We have thus defined a functor from $\mathscr{A}$ to $\mathscr{A}^1((S^1)^{-1})$; by the universal property of $\mathscr{A}(S^{-1})$, it factors through $\Psi : \mathscr{A}(S^{-1}) \to \mathscr{A}^1((S^1)^{-1})$. The composite $\Phi\Psi$ is isomorphic to the identity because its composite with the functor $J : \mathscr{A} \to \mathscr{A}(S^{-1})$ is isomorphic to J. One checks similarly that $\Psi\Phi$ is isomorphic to the identity.

COROLLARY 2.4.2. *With the notation of 2.4.1, $\mathscr{A}$ is coderivable relative to S provided that, for every finite set B of arrows of $\mathscr{B}$, condition $(*)$ holds when only direct images under the arrows of B are considered.*

For every category $\mathscr{B}'$ with a finite set of arrows, Proposition 2.4.1 is applicable after every base change $\mathscr{B}' \to \mathscr{B}$. We shall take for $\mathscr{B}'$ the categories of the following diagrams: 302

$$\begin{array}{cccc} * & * \longrightarrow * & * \longrightarrow * \longrightarrow * & * \longrightarrow * \longrightarrow * \longrightarrow * \\ (0) & (1) & (2) & (3) \end{array}$$

Consider the categories $\mathscr{A}_x(S_x^{-1})$ ($x \in \mathrm{Ob}\,\mathscr{B}$). Given $f \in \mathrm{Fl}(\mathscr{B})$ and A_1, A_2 in these categories over the source and target of f, define $\mathrm{Hom}_f(A_1, A_2)$ to be the analogous set defined after the base change of type (1) determined by f. Composition of morphisms is defined using base changes of type (3). In this way one constructs a category $\mathscr{A}(S^{-1})^*$ over $\mathscr{B}$, whose formation is compatible with every base change; using base changes of types (0), (1), and (2), one checks that it solves the universal problem defining $\mathscr{A}(S^{-1})$.

The following result is dual to 2.4.2.

PROPOSITION 2.4.3. *Let $\mathscr{A}$ be a triangulated category fibered over $\mathscr{B}$, and let S be as in 2.4.1. For $\mathscr{A}$ to be derivable relative to S, it suffices that, for every finite set B of arrows of $\mathscr{B}$,*

$(*')$ $\forall x \in \mathrm{Ob}(\mathscr{B}), \forall A \in \mathrm{Ob}(\mathscr{A}_x), \exists s : A \to A'$ *in* $S, \forall g, f : Z \xrightarrow{g} Y \xrightarrow{f} X$ *in* B, *f^*A' is right adapted to the functor g^*.*

3. Gluing fibered or cofibered categories

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3.1. Introduction. Let $f : Y \to X$ be a separated morphism of finite type from a scheme Y to a scheme X, or a continuous map from a locally compact space to a locally compact space. One can then define a "direct image with proper supports" functor, denoted $f_!$, from the category of abelian sheaves on Y to the category of abelian sheaves on X (cf. 6.1 in the case of schemes, and VERDIER [13] for separated locally compact spaces). When Y and X are schemes of finite type over $\mathbf{C}$, these two definitions are compatible in the evident sense. Unfortunately, for schemes, unlike for locally compact spaces, the derived functors of $f_!$ are pathological; one can nevertheless define a reasonable "direct image with proper supports" functor directly from $D^+(Y)$ to $D^+(X)$. To define it, factor f as an open immersion followed by a proper morphism: $f = \overline{f}j$. The desired functor $Rf_!$ is then defined as the composite of "extension by zero" $j_!$ and the functor $R\overline{f}_*$, derived from direct image under $\overline{f}$.

These "direct image with proper supports" functors $Rf_!$ are usable only after one verifies for them a "variance formalism" that includes, for a composite $f = gh$, a formula

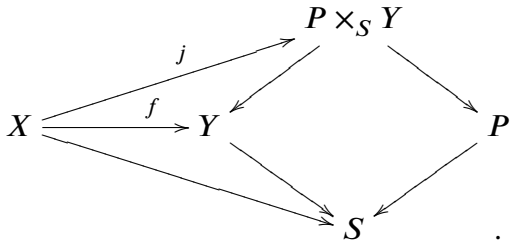
$$Rf_! = Rg_!Rh_!$$

which is best expressed in terms of cofibered categories. Such a formalism is available separately for "extensions by zero" and for direct images under proper morphisms, and 304
the problem is to glue these formalisms. To avoid later being bogged down by pointless computations, we propose in this § to list the elementary verifications required to carry out such a gluing. We do so in a slightly more abstract setting than is strictly necessary, in the hope that the result may be useful again.

The construction of $Rf_!$ sketched above requires that the morphism f be factorable as an open immersion followed by a proper morphism. Following HARTSHORNE [6, III 8 p. 189], in n° 2 we introduce the appropriate notion of a compactifiable morphism.

3.2. Compactifiable morphisms.

DEFINITION 3.2.1. *A morphism f from an S-scheme X to a quasi-compact, quasi-separated S-scheme Y is said to be* S-compactifiable *if there exists an S-scheme P proper over S and a factorization of f as a separated quasi-finite morphism j from X to $P \times_S Y$ followed by the projection from $P \times_S Y$ to Y*



When $S = Y$, we shall simply speak of a compactifiable morphism.

305 PROPOSITION 3.2.2. (i) *An S-compactifiable morphism is separated and of finite type.*

(ii) *If Y is quasi-compact and quasi-separated, a separated quasi-finite morphism with target Y is S-compactifiable.*

(iii) *The composite of two S-compactifiable morphisms is again S-compactifiable.*

(iv) *Let there be a diagram of quasi-compact, quasi-separated S-schemes*

$$\begin{array}{ccc} X \times_Y Y' & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \longrightarrow & Y. \end{array}$$

If f is S-compactifiable, then f' is S-compactifiable.

(v) *Let g be an S-compactifiable morphism;*[46] *for a composite morphism $g \circ f$ to be S-compactifiable, it is necessary that f be so.*

Assertions (i), (ii), and (iv) are immediate. To prove (iii) and (v), consider a composite $f = gh$, with g S-compactifiable.

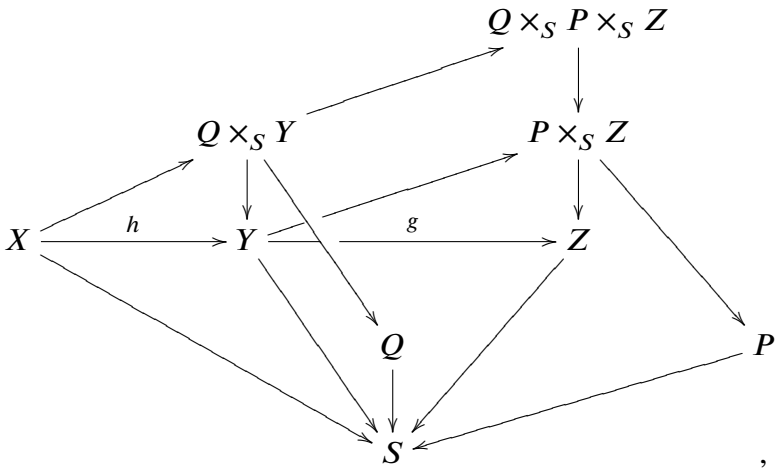
Suppose that gh is S-compactifiable, whence a diagram

$$\begin{array}{ccccc} & & P \times_S Y & \xrightarrow{v} & P \times_S Z \\ & {\scriptstyle u}\nearrow & \downarrow & & \downarrow \\ X & \xrightarrow{h} & Y & \xrightarrow{g} & Z. \end{array}$$

The composite vu is quasi-finite; hence u is quasi-finite, and (v) is proved.

[46] N.D.E.: it suffices to assume that g is a quasi-compact, quasi-separated morphism of S-schemes.

If h is compactifiable, we have a diagram



,

and $f = fh$ factors through a separated quasi-finite morphism to $(Q \times_S P) \times_S Z$. 306

Nagata asserts in [9] that every morphism of finite type between separated integral noetherian schemes is compactifiable.[47] The editor admits to not having understood the proof.

Now fix a scheme S and denote by (S) the subcategory of the category of S-schemes whose objects are the quasi-compact, quasi-separated S-schemes and whose arrows are the S-compactifiable S-morphisms.[48]

Proposition 3.2.3. *In the category (S):*

(i) *The open immersions (resp. the proper morphisms) define a subcategory (S, i) (resp. (S, p)) of (S) having the same objects as (S).*
(ii) *Fiber products exist in (S, p) and are fiber products in (S).*
(iii) *Every morphism f is the composite $f = pj$ of a proper morphism p and an open immersion.*

By 3.2.2 (iii), (iv), and (v), fiber products exist in (S) and are fiber products in the 307
category of all schemes. Since the fiber product $X = X_1 \times_Y X_2$ of two schemes proper over Y is again proper over Y, and a scheme is proper over X if and only if it is proper over X_1 and X_2, condition (ii) is satisfied.

If f is a morphism of (S), by hypothesis it admits, in (S), a factorization $f = pj'$, where p is proper and j' is separated and quasi-finite. By EGA IV 18.12.13, j' admits a factorization $j' = qj$, where q is finite and j is an open immersion; moreover, by 3.2.2 (ii), q and j belong to (S). We then have $f = (pq)j$, which proves (iii). Assertion (i) follows from 3.2.2 (iii).

3.2.4. In the rest of this §, suppose given a category (S), equipped with two subcategories (S, i) and (S, p), such that

(i) $\mathrm{Ob}(S) = \mathrm{Ob}(S, i) = \mathrm{Ob}(S, p)$;
(ii) fiber products exist in (S, p) and are fiber products in (S);
(iii) $\forall f \in \mathrm{Fl}(S), \exists p \in \mathrm{Fl}(S, p), \exists j \in \mathrm{Fl}(S, i), f = pj$.

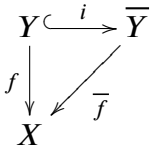
[47] N.D.E.: more generally, every separated morphism of finite type whose target is quasi-compact and quasi-separated is compactifiable. See la N.D.E. (43) page 639.

[48] N.D.E.: Thanks to the compactification theorem, one may also work in the subcategory of the category of (absolute) schemes whose objects are the quasi-compact, quasi-separated schemes and whose arrows are the separated morphisms of finite type (i.e. compactifiable morphisms). Note that the stability of compactifiable morphisms under composition is equivalent to the general form of the compactification theorem.

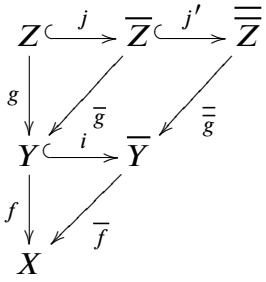
The arrows of (S, i) (resp. of (S, p)) will be called open immersions (resp. proper morphisms).

DEFINITION 3.2.5. (i) *A* compactification *of a morphism f of (S) (resp. of a pair (f, g), resp. of a triple (f, g, h) of composable morphisms) is a commutative diagram (3.2.5.1) (resp. (3.2.5.2), resp. (3.2.5.3)) whose horizontal arrows are open immersions and whose oblique arrows are proper.*

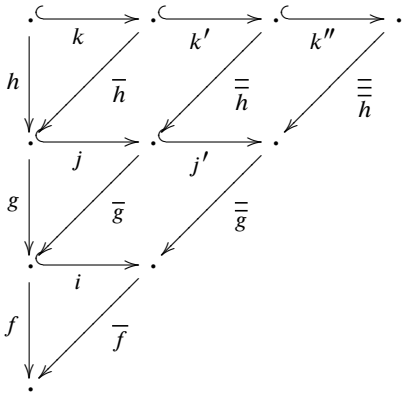
308 (ii) A morphism of compactifications *is a morphism of diagrams (3.2.5.1) (resp. ...) that is the identity on the 1^{st} vertical column and is proper at every vertex*



(3.2.5.1)



(3.2.5.2)



(3.2.5.3)

PROPOSITION 3.2.6. (i) *Every morphism (resp. every pair (f, g), resp. every triple (f, g, h) of composable morphisms) admits a compactification.*

(ii) *The category of compactifications of f is left filtering.*[49]

(iii) *If diagrams $(3.2.5.2)_i$ $(i = 1, 2)$ are compactifications of (f, g), there exists a compactification $(3.2.5.2)_3$ of (f, g) and morphisms of compactifications from $(3.2.5.2)_3$ to the compactifications $(3.2.5.2)_i$.*

(i) The case of a morphism is precisely 3.2.4 (iii). Let us treat the case of a triple; with the notation of (3.2.5.3), it suffices to construct successively compactifications of f, g, and h, of $i\overline{g}$ and $j\overline{h}$, and finally of $j'\overline{\overline{h}}$.

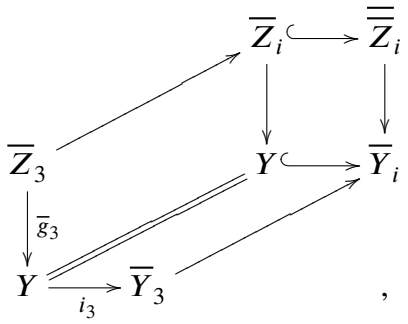
309 (ii) Let $(3.2.5.1)_1$ and $(3.2.5.1)_2$ be two compactifications of f. The projection p from $\overline{Y}_1 \times_S \overline{Y}_2$ to X is proper, and Y maps to it by (i_1, i_2), which (3.2.4 (iii)) can be compactified as $(i_1, i_2) = qj$. The compactification $f = (p \circ q)j$ dominates $(3.2.5.1)_1$ and $(3.2.5.1)_2$.

If r and s are two morphisms from $(3.2.5.1)_1$ to $(3.2.5.1)_2$, the kernel K of the pair of arrows $\overline{Y}_1 \overset{r}{\underset{s}{\rightrightarrows}} \overline{Y}_2$ maps by a proper morphism to $\overline{Y}_1$, and i_1 factors through K: write $i_1 = ki$. If $i = pi_3$ is a compactification of i, the existence of the compactification $f = (\overline{f}_1 kp)i_3$ shows that axiom L_2 is satisfied.

(iii) Apply axiom L_1 for filtering categories to the compactifications of f and g deduced from the $(3.2.5.2)_i$ $(i = 1, 2)$. We thereby obtain $\overline{Y}_3$ and $\overline{Z}_3$ giving rise to

[49] N.D.E.: "left filtering" means "cofiltered" (I 2.7). For axioms L_1 and L_2 in the proof of 3.2.6, cf. V 8.1.1.1.

commutative diagrams $(3.2.5.1)_i$ $(i = 1, 2)$



and it remains to find a compactification of $i_3\overline{g}_3$ by $\overline{\overline{Z}}_3$, such that $\overline{\overline{Z}}_3$ can map by a proper morphism to $\overline{\overline{Z}}_i$, so as to give rise to a commutative cube $(i = 1, 2)$. If the target of a morphism between compactifications of $i_3\overline{g}_3$ has this property, its source has it as well, and by (ii), it will suffice to consider separately the cases $i = 1$ and $i = 2$. Let T be the fiber product of $\overline{Y}_3$ and $\overline{\overline{Z}}_i$ over $\overline{Y}_i$: then T occurs 310
in a commutative cube analogous to the desired one, except that the arrow k from $\overline{Z}_3$ to T might not be an open immersion. Writing $k = pj$ (p proper, j an open immersion) and replacing T by the source $\overline{\overline{Z}}_3$ of p, we obtain the desired cube.

REMARK 3.2.7. *Proposition 3.2.3 would still have been valid if, in Definition 3.2.1 of compactifiable morphisms, we had omitted the requirement that Y be quasi-compact and quasi-separated and required instead that j be a quasi-compact open immersion and not merely a quasi-finite morphism.*

3.3. Gluing. Recall that the hypotheses of 3.2.4 are satisfied.

3.3.1. Suppose given:

α) for each $X \in \mathrm{Ob}(S)$, a category $F(X)$,
β) an (S, i)-cofibered category F_i and an (S, p)- cofibered category F_p,
γ) for each $X \in \mathrm{Ob}(S)$, isomorphisms between $F(X)$, $(F_i)_X$, and $(F_p)_X$,
δ) a normalized cleavage of F_i and a normalized cleavage of F_p.

By (SGA 1 VI), it "amounts to the same thing" to give instead α) and:

(i) For every proper morphism $p : X \to Y$, a functor $p_* : F(X) \to F(Y)$,
(ii) for every composable pair (p, q) of proper morphisms, an isomorphism of functors $c_{p,q} : p_*q_* \leftrightarrow (pq)_*$, 311
(i') and (ii'), analogous data for (S, i),

these data satisfying the following conditions:

(a) if p is an identity, then p_*, $c_{p,q}$, and $c_{q,p}$ are identities,
(b) for every composable triple of proper morphisms, we have

$$c_{p,qr} \circ (p_* * c_{q,r}) = c_{pq,r} \circ (c_{p,q} * r_*),$$

(a') and (b'), analogous conditions for (S, p).

Suppose *given* in addition,

(iii) for every commutative diagram D

$$\begin{array}{ccc} X & \xrightarrow{j'} & Y \\ \downarrow{\scriptstyle p'} & & \downarrow{\scriptstyle p} \\ Z & \xrightarrow[j]{} & T \end{array} \qquad j, j' \in \mathrm{Fl}(S, i);\ p, p' \in \mathrm{Fl}(S, p),$$

an isomorphism of functors $d(D) : p_* j'_* \leftrightarrow j_* p'_*$.

Proposition 3.3.2. *If data* (i) (ii) (i′) (ii′) (iii) *satisfy conditions* (a) (b) (a′) (b′) *as well as* (c) *and* (c′) *stated below, then there exists one and "essentially" only one category F cofibered over S, equipped with isomorphisms $F \times_S (S, i) = F_i$ and $F \times_S (S, p) = F_p$, compatible with datum 3.3.1 γ) and such that the isomorphisms $d(D)$ of* (iii) *are the composite isomorphisms $j_* p'_* = (jp')_* = (pj')_* = p_* j'_*$.*

(c) for every commutative diagram of the following type

$$\begin{array}{ccccc} X_1 & \xhookrightarrow{i} & X_2 & \xhookrightarrow{i'} & X_3 \\ \downarrow{\scriptstyle p_1} & & \downarrow{\scriptstyle p_2} & & \downarrow{\scriptstyle p_3} \\ Y_1 & \xhookrightarrow{j} & Y_2 & \xhookrightarrow{j'} & Y_3, \end{array}$$

312 whose horizontal arrows are open immersions and whose vertical arrows are proper, the following triangle of isomorphisms d from (iii) is commutative:

$$\begin{array}{ccc} p_{3*} i'_* i_* & \longrightarrow & j'_* j_* p_{1*} \\ & \searrow \quad \swarrow & \\ & j'_* p_{2*} i_* & \end{array} .$$

(c') For every commutative diagram of the following type

$$\begin{array}{ccccc} X_1 & \xrightarrow{p} & X_2 & \xrightarrow{p'} & X_3 \\ \downarrow{\scriptstyle i_1} & & \downarrow{\scriptstyle i_2} & & \downarrow{\scriptstyle i_3} \\ Y_1 & \xrightarrow{q} & Y_2 & \xrightarrow{q'} & Y_3, \end{array}$$

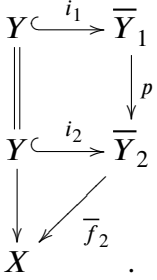
whose horizontal arrows are proper and whose vertical arrows are open immersions, the following triangle of isomorphisms d from (iii) is commutative:

$$\begin{array}{ccc} i_{3*} p'_* p_* & \longrightarrow & q'_* q_* i_{1*} \\ & \searrow \quad \swarrow & \\ & q'_* i_{2*} p_* & \end{array} .$$

313 Definition of f_*. Let f be a morphism from Y to X. Each compactification of f, with diagram (3.2.5.1), defines a composite functor $\overline{f}_* i_*$ from $F(Y)$ to $F(X)$. A morphism of

compactifications defines a commutative diagram

(3.3.2.2)



Datum (iii) therefore provides us with an isomorphism

$$\overline{f}_{2*}i_{2*} = \overline{f}_{2*} \circ (i_{2*} \circ 1_{Y*}) = \overline{f}_{2*} \circ (p_* \circ i_{1*}) = \overline{f}_{1*} \circ i_{1*}.$$

Axiom (c′) also ensures that the isomorphism associated with a composite of morphisms of compactifications is the composite of the isomorphisms associated with each of them. Since the category of morphisms of compactifications is cofiltered (3.2.6 (ii)), we thereby obtain a transitive system of isomorphisms between the functors associated with the various compactifications of f. Choosing one of them, we define the functor f_*.

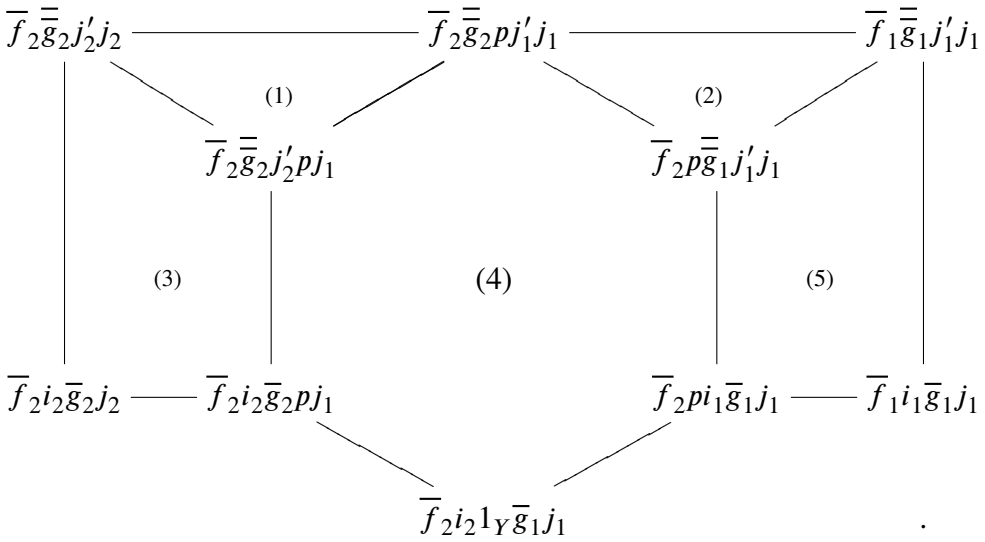
Transitivity homomorphisms. Every compactification of a pair (f, g) of composable morphisms (diagram (3.2.5.2)) defines a compactification of f, one of g, one of $f \circ g = (\overline{f}\overline{\overline{g}})(j'j)$, and an isomorphism $c_{f,g} : (fg)_* \xrightarrow{\sim} f_*g_*$, the composite of the isomorphisms

$$(fg)_* = (\overline{g}\,\overline{\overline{g}})_*(j'j)_* = \overline{f}_*(\overline{\overline{g}}_*j'_*)j_* = \overline{f}_*(i_*\overline{g}_*)j_* = (\overline{f}_*i_*)(\overline{g}_*j_*) = f_*g_*$$

(where the middle isomorphism comes from (iii)). 314

Let us prove that this isomorphism does not depend on the compactification chosen for (f, g). By 3.2.6 (iii), it will suffice to compare the isomorphisms $c_{f,g}$ obtained from two compactifications $(3.2.5.2)_i$ $(i = 1, 2)$ such that there exists a morphism of compactifications from the first to the second. The arrows in these morphisms will be denoted by p.

Consider the following diagram of isomorphisms of functors. For readability, the $*$ have been omitted.



a) The composite isomorphism along the 1st row is the isomorphism 3.3.2.1 between two definitions of $(fg)_*$. 315

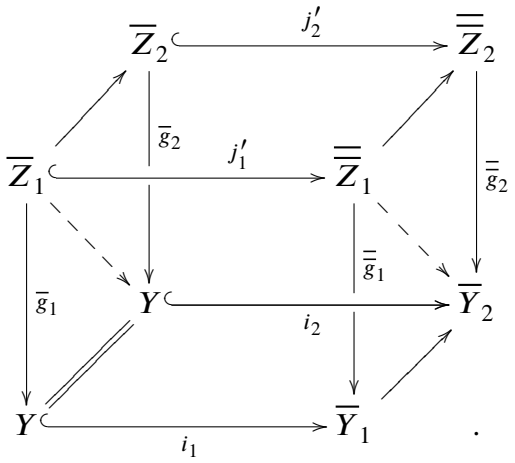
b) The composite isomorphism along the last row is the isomorphism 3.3.2.1 between two definitions of f_*g_*.
c) The two outer vertical isomorphisms are the isomorphisms $c_{f,g}$ between $(fg)_*$ and f_*g_* deduced from one or the other compactification.

The problem is therefore to prove that the outer boundary of the diagram is commutative. A commutative diagram of functors remains commutative when each of them is composed with the same functor. Bearing this in mind, the commutativity of (1) follows from (c) applied to the diagram

$$\begin{array}{ccccc} Z & \xrightarrow{j_1} & \overline{Z}_1 & \xrightarrow{j_1'} & \overline{\overline{Z}}_1 \\ \| & & \downarrow & & \downarrow \\ Z & \xrightarrow{j_2} & \overline{Z}_2 & \xrightarrow{j_2'} & \overline{\overline{Z}}_2. \end{array}$$

The commutativity of (3) and (5) is trivial; that of (2), which essentially concerns proper morphisms, is also trivial. It remains to consider the hexagon (4).

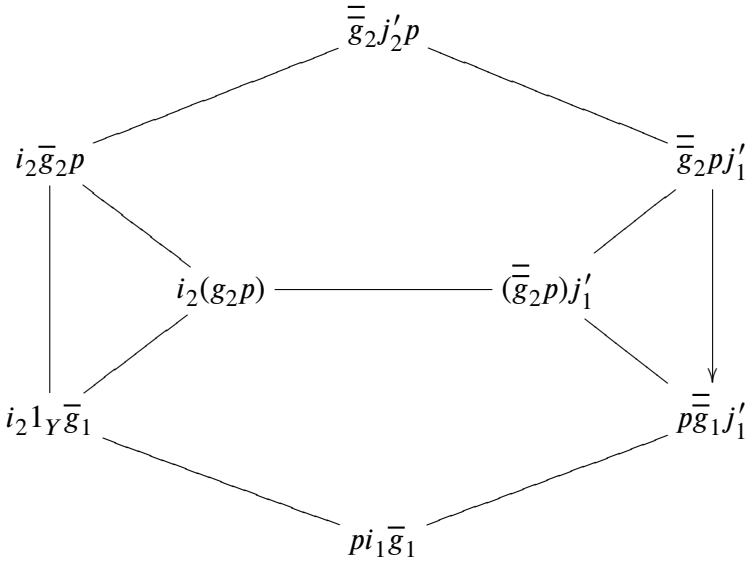
Consider the following commutative cube



316 There are six ways to follow the edges of this cube from $\overline{Z}_1$ to $\overline{Y}_2$, each "adjacent" to two others; there is an isomorphism between the composite functors associated with two adjacent paths (i.e. paths separated by a single face), and the commutativity of (4) will follow from the commutativity of the resulting hexagon.

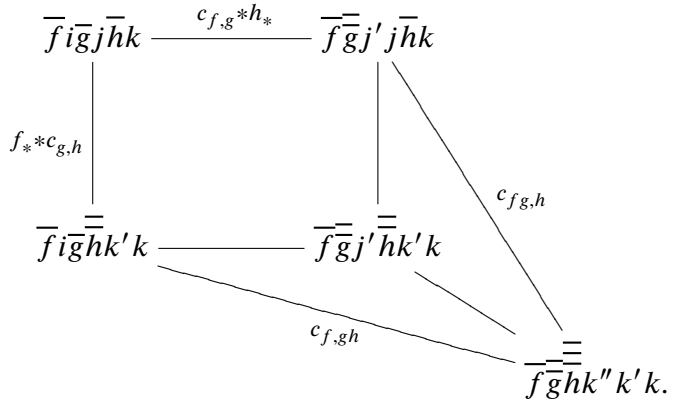
The moral of this kind of diagram is that an edge represents a functor, a face an isomorphism between composite functors, and a volume a compatibility condition (between the faces forming its boundary). We decompose the cube into two triangular prisms (along the "plane" $\overline{Z}_1\overline{\overline{Z}}_1Y\overline{Y}_2$) in order to reduce to the hypotheses (c′).

More precisely, we complete the hexagon to form the following diagram of isomorphisms (for readability, the $*$ have been omitted):



The triangles are trivially commutative, and the pentagons are commutative by (c′). 317

Cocycle condition. Let us verify that the isomorphisms $c_{f,g}$ constructed in 3.3.2.3 satisfy a cocycle condition for a composable triple (f, g, h) of morphisms. Let us introduce a compactification of (f, g, h), with diagram (3.2.5.3), and consider the following diagram of isomorphisms of functors:



The commutativity of the square is trivial, and that of the triangles follows from (c) and (c′). Since the commutativity of the outer boundary is what had to be proved, this completes the proof of 3.3.2.

3.3.3. Suppose that over (S, i) and (S, p) there are given, respectively, fibered categories equipped with normalized cleavages F_i and F_p, and suppose that F_i and F_p have the same fiber $F(X)$ for every $X \in \mathrm{Ob}\, S$.

According to SGA 1 VI, it "amounts to the same thing" to give instead:

(i bis) for every proper morphism $p : X \to Y$, a functor $p^* : F(Y) \to F(X)$,

(ii bis) for every composable pair (p, q) of proper morphisms, an isomorphism of functors $c_{p,q} : q^* p^* \leftrightarrow (pq)^*$,

(i′ bis) and (ii′ bis), analogous data for (S, i), these data satisfying the conditions (a bis) (b bis) (a′ bis) (b′ bis) dual to 3.3.1 (a) (b) (a′) (b′). 318

Suppose *given* in addition,

(iii bis) for every commutative diagram D

$$\begin{array}{ccc} X & \xrightarrow{j'} & Y \\ \downarrow{\scriptstyle p'} & & \downarrow{\scriptstyle p} \\ Z & \xrightarrow{j} & T \end{array} \qquad j, j' \in \mathrm{Fl}(S, i);\ p, p' \in \mathrm{Fl}(S, p)$$

an isomorphism of functors $d(D) : j'^* p^* \leftrightarrow p'^* j^*$.

We leave to the reader the formulation of the conditions (c bis) and (c′ bis) dual to 3.3.2 (c) and (c′).

Arguments parallel to the preceding ones then prove the following proposition.

Proposition 3.3.4. *If data* (i bis) (ii bis) (i′ bis) (ii′ bis) (iii bis) *satisfy the conditions* (a bis) (b bis) (a′ bis) (b′ bis) (c bis) (c′ bis), *there exists one and essentially only one category* F *cofibered over* S *and equipped with isomorphisms* $F \times_S (S, i) = F_i$, $F \times_S (S, p) = F_p$, *these isomorphisms being compatible with the identifications* $F_{iX} = F_{pX}$ $(X \in \mathrm{Ob}\, S)$ *and such that the isomorphisms* $d(D)$ *of* (iii bis) *become the composite isomorphisms* $p'^* j^* = (jp')^* = (pj')^* = j'^* p^*$.

It should be noted, however, that 3.3.4 is not dual to 3.3.2, since the hypotheses imposed on (S, i) and (S, p) are not self-dual.

4. Resolutions. Application to the base change morphism

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4.1. Flat resolutions. Recall that if $f : (X, \mathscr{A}) \to (Y, \mathscr{B})$ is a morphism of ringed sites (or ringed topoi), the inverse image under f of a sheaf of left $\mathscr{B}$-modules $\mathscr{M}$ is the sheaf of left $\mathscr{A}$-modules $\mathscr{A} \otimes_{f^*\mathscr{B}} f^*\mathscr{M}$.

Proposition 4.1.1. *Let* $f : (X, \mathscr{A}) \to (Y, \mathscr{B})$ *be a morphism of ringed sites, let* K *be a sheaf of right* $\mathscr{A}$*-modules, and let* $0 \to L \to M \to N \to 0$ *be an exact sequence of sheaves of left* $\mathscr{B}$*-modules. If* M *and* N *are flat, then*

(i) L *is flat,*
(ii) *the following sequence is exact:*

$$0 \longrightarrow K \otimes_{\mathscr{A}} f^*L \longrightarrow K \otimes_{\mathscr{A}} f^*M \longrightarrow K \otimes_{\mathscr{A}} f^*N \longrightarrow 0.$$

Factor f into the arrows $(X, \mathscr{A}) \xrightarrow{u} (X, f^{-1}\mathscr{B}) \xrightarrow{v} (Y, \mathscr{B})$. The functor v^* is exact and takes flat modules to flat modules (V 1), so it suffices to prove 4.1.1 for u. The isomorphism

$$K \otimes_{\mathscr{A}} (\mathscr{A} \otimes_{f^{-1}\mathscr{B}} P) = K \otimes_{f^{-1}\mathscr{B}} P \tag{4.1.1.1}$$

then reduces us to the case where f is the identity.

Take an exact sequence $0 \to R \to P \to K \to 0$ with P flat. The rows and columns of the following diagram are exact:

$$\begin{array}{ccccccccc}
 & & & & 0 & & & & \\
 & & & & \downarrow & & & & \\
 & & R \otimes L & \longrightarrow & P \otimes L & \longrightarrow & K \otimes L & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & R \otimes M & \longrightarrow & P \otimes M & \longrightarrow & K \otimes M & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & R \otimes N & \longrightarrow & P \otimes N & \longrightarrow & K \otimes M & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array} .$$

Apply the snake lemma to the first two columns: we obtain 320

$$0 \xrightarrow{j} K \otimes L \to K \otimes M,$$

which proves (ii). Now drop the assumption that P is flat. In the preceding diagram the three columns and two rows are short exact sequences; the first row is therefore exact, i.e. L is flat.

4.1.2. Let $\mathcal{S}$ be a subcategory of the category of ringed sites.[50] The sheaves of modules on the ringed sites in $\mathcal{S}$ form a category $\mathscr{M}$, abelian over $\mathcal{S}$ (2.2.2), fibered and cofibered, whose fibers are the opposites of the usual categories of sheaves of modules on a site. We denote by $K(\mathcal{S})$ the category $K(\mathscr{M})$ (2.2.8) and by $K^+(\mathcal{S})$ (resp. $K^-(\mathcal{S})$) the full subcategory of $K(\mathcal{S})$ whose fiber categories are the opposites of the categories $K^+(S)$ (resp. $K^-(S)$) for $S \in \mathrm{Ob}\,\mathcal{S}$. The categories $K(\mathcal{S})$, $K^+(\mathcal{S})$, and $K^-(\mathcal{S})$ are triangulated over $\mathcal{S}$, fibered and cofibered.

Let $f : S \to S'$ be an arrow of $\mathcal{S}$. Every injective object is acyclic (1.2.3) for the 321
functor Rf_*, and every flat sheaf of modules is acyclic for the functor Lf^* (4.1.1 and 1.2.6), whence the derived functors (1.2.7)

$$Rf_* : D^+(S) \longrightarrow D^+(S') \tag{4.1.2.1}$$

$$Lf^* : D^-(S') \longrightarrow D^-(S). \tag{4.1.2.2}$$

If Rf_* (resp. Lf^*) has finite dimension, then in (4.1.2.1) (resp. (4.1.2.2)) one may replace the exponent + (resp. −) by b, by − (resp. +), or omit it (1.2.10).[51]

We denote by $D(\mathcal{S})$ (resp. $D^+(\mathcal{S})$, resp. $D^-(\mathcal{S})$) the category obtained from $K(\mathcal{S})$ (resp. $K^+(\mathcal{S})$, resp. $K^-(\mathcal{S})$) by inverting the quasi-isomorphisms.

Scholium 4.1.3. *("Trivial duality theorem").*

(a) *Assume that all the functors Rf_* (resp. Lf^*) have finite dimension. Then:*
   (i) *The formation of $D(\mathcal{S})$ commutes with every change of base category $\mathcal{S}' \to \mathcal{S}$. The category $D(\mathcal{S})$ is triangulated over $\mathcal{S}$ (2.2.3), and its fibers identify with the opposites of the categories $D(S)$ for $S \in \mathrm{Ob}\,\mathcal{S}$.*

[50] N.D.E.: the theory is valid more generally for categories $\mathcal{S}$ consisting of ringed sites.

[51] N.D.E.: using Spaltenstein resolutions, the exponents may be omitted unconditionally. In the commutative case, see M. Kashiwara and P. Schapira, *Categories and Sheaves*, Springer, 2010, cited as [KS] below, 18.6.

(ii) *The category $D(\mathcal{S})$ is cofibered (resp. fibered) over $\mathcal{S}$. Let $f : S \to S'$; if Lf^* (resp. Rf_*) has finite dimension, then every object of $D(S')$ (resp. $D(S)$) admits an inverse image (resp. direct image) under f in the sense of the category $D(\mathcal{S})$ over $\mathcal{S}$. The inverse image and direct image functors identify with Lf^* and Rf_*.*

322 (iii) *Every object of $D^-(S')$ (resp. $D^+(S)$) admits an inverse image (resp. direct image) in the sense of $D(\mathcal{S})$, and the inverse image (resp. direct image) functor identifies with Lf^* (resp. Rf_*).*[52]

(b) (i) *The formation of $D^-(\mathcal{S})$ (resp. $D^+(\mathcal{S})$) commutes with every change of base category $\mathcal{S}' \to \mathcal{S}$. The category $D^-(\mathcal{S})$ (resp. $D^+(\mathcal{S})$) is triangulated over $\mathcal{S}$, and its fibers identify with the opposites of the categories $D^-(S)$ (resp. $D^+(S)$).*

(ii) *The category $D^-(\mathcal{S})$ (resp. $D^+(\mathcal{S})$) is fibered (resp. cofibered) over $\mathcal{S}$; the "inverse image" (resp. "direct image") functors identify with the derived functors Lf^* (resp. Rf_*).*

(iii) *If $f : S \to S'$ is such that Rf_* (resp. Lf^*) has finite dimension, every object of $D^-(S)$ (resp. $D^+(S')$) admits a direct image in the sense of $D^-(\mathcal{S})$ (resp. an inverse image in the sense of $D^+(\mathcal{S})$), and the "direct image" (resp. "inverse image") functor identifies with Rf_* (resp. Lf^*).*

It suffices to verify that in each case the hypotheses of 2.4.1 or 2.4.3 are satisfied, the other assertions following, after a base change $\{S\} \to \mathcal{S}$, or $I \to \mathcal{S}$, from 2.3.5 and 1.2.10. In case (b), complexes of flat (resp. flasque) sheaves satisfy the hypotheses of 2.4.3 (resp. 2.4.1). In case (a), we use 2.4.1 (resp. 2.4.3) so that we need only consider those f for which the dimension of Rf_* (resp. Lf^*) is bounded by a fixed N. Let us prove that there are "enough" complexes K whose components have their direct (resp. inverse) images all acyclic for the direct image (respectively, inverse image) functors under con-
323 sideration. Let K be a complex and let K^* (resp. K_*) be a Cartan-Eilenberg resolution of K by complexes with flasque components (resp. flat components).[53] If $n > N$, we see that $f_*\tau''_{<n}(K^*) = \tau''_{<n}f_*(K^*)$ (resp. $f^*\tau''_{>-n}(K_*) = \tau''_{>-n}f^*(K_*)$), where τ'' denotes a truncation with respect to the second differential. For $n > 2N$, it follows that $\tau''_{<n}(K^*)$ (resp. $\tau''_{>-n}(K_*)$) satisfies the hypothesis of 2.4.1 (resp. 2.4.3), and every complex K is the source (resp. target) of a quasi-isomorphism whose target (resp. source) is such a complex.

4.1.4. Consider a commutative square of ringed sites (4.1.4.1) (cf. 0.11)

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S. \end{array} \tag{4.1.4.1}$$

We may consider the subcategory of the category of sites consisting of this diagram;[54] we then use 2.1.3 to define a "base change morphism" from Lg^*Rf_* to $Rf'_*Lg'^*$ whenever 4.1.3 is applicable:

Proposition 4.1.5. *The base change morphism of 2.1.3, from Lg^*Rf_* to $Rf'_*Lg'^*$, is defined in each of the following cases:*

[52] N.D.E.: the finite-dimension hypotheses in (a) are superfluous. Cf. the preceding N.D.E.

[53] N.D.E.: it is required only that the augmentation $K^p \to (K^*)^{p\bullet}$ (resp. $(K_*)^{p\bullet} \to K^p$) be a flasque (resp. flat) resolution for every p.

[54] N.D.E.: one must instead consider the category of ringed sites defined by this diagram, which is not strictly speaking a subcategory of the category of ringed sites in general. Cf. la N.D.E. (50) page 673.

(i) *We work in the categories D^+, and Lg^* and Lg'^* have finite dimension.*
(ii) *We work in the categories D^-, and Rf_* and Rf'_* have finite dimension.*
(iii) *We work in the unbounded derived categories, and Rf_*, Rf'_*, Lg^*, and Lg'^* have finite dimension, as do Lf^* and Lf'^* (or: as do Rg_* and Rg'_*).*[55]

It should be noted that the last hypothesis in (iii) looks rather like a practical joke. 324
The editor does not know how to dispense with it except when the sites under consideration have enough points (4.2.12). The functorial properties of this base change morphism follow at once from the description in 2.1.3.

4.1.6. Let $(S, \mathscr{A})$ be a ringed site, and consider the "tensor product" functor

$$\otimes : \mathrm{Mod}(S, \mathscr{A}^\circ) \times \mathrm{Mod}(S, \mathscr{A}) \longrightarrow \mathrm{Mod}(S, C)$$

where C is the center of $\mathscr{A}$.

The recipe of 1.1.6, in applying which the simple complex associated with a double complex is defined using a sum, extends this functor to a functor still denoted $\otimes$:

$$\otimes : K(S, \mathscr{A}^\circ) \times K(S, \mathscr{A}) \longrightarrow K(S, C).$$

Proposition 4.1.7. *With the preceding notation:*

(i) *The tensor-product functor is left derivable (1.2.1) to a "total tensor product" functor*

$$\overset{L}{\otimes} : D^-(S, \mathscr{A}^\circ) \times D^-(S, \mathscr{A}) \longrightarrow D^-(S, C);$$

(ii) *a pair (K, L) of bounded-above complexes is adapted to $\overset{L}{\otimes}$ whenever either K or L has flat components.*

The proof is standard.

Notation 4.1.8. *If $K \in D^-(S, \mathscr{A}^\circ)$ and $L \in D^-(S, \mathscr{A})$, denote by $K \overset{L}{\otimes} L$ the total* 325
tensor product of K and L. The sheaf $H_k(K \overset{L}{\otimes} L) = H^{-k}(K \overset{L}{\otimes} L)$ is denoted by $\mathrm{Tor}_k(K, L)$ and is called the k^{th} local hypertor of K and L.

Let $K \in D^-(S, \mathscr{A}^\circ)$ and $L \in D^-(S, \mathscr{A})$. If $H^i(K) = 0$ for $i > k$ and $H^i(L) = 0$ for $i > \ell$, one checks that $H^i(K \overset{L}{\otimes} L) = 0$ for $i > k + \ell$ and that $H^{k+\ell}(K \overset{L}{\otimes} L) = H^k(K) \otimes H^\ell(L)$. Fix K and regard $K \overset{L}{\otimes} L$ as a functor of L. If Definition 1.2.9 of cohomological amplitude as an interval in $\mathbf{Z}$ is applied, the upper endpoint of this interval is trivial in nature: it is the largest integer i such that $H^i(K) \neq 0$. The lower endpoint is more interesting:

Definition 4.1.9. *A complex $K \in \mathrm{Ob}\, D^b(S, \mathscr{A}^\circ)$ is said to have* tor-dimension $\leq n$ *if the following equivalent conditions hold:*

(i) *For every $\mathscr{A}$-module L, $\mathrm{Tor}_k(K, L) = 0$ for $k > n$.*
(ii) *For every complex of $\mathscr{A}$-modules L such that $H^i(L) = 0$ for $i < \ell$, one has $H^i(K \overset{L}{\otimes} L) = 0$ for $i < \ell - n$.*
(iii) *There is a quasi-isomorphism with target K and source a bounded complex K' whose components are flat and vanish for $i < -n$.*

[55] N.D.E.: the finite-dimension hypotheses in (iii) are superfluous. Cf. la N.D.E. (51) page 673.

4.1.10. Here are some possible variants:

(i) If $K \in \mathrm{Ob}\, D^b(S, \mathscr{A}^\circ)$ has finite tor-dimension and $L \in \mathrm{Ob}\, D(S, \mathscr{A})$, then the functor $\overset{L}{\otimes}$ is defined (1.2.1) at (K, L).
(ii) If $K \in \mathrm{Ob}\, K^b(S, \mathscr{A}^\circ)$ has components of finite tor-dimension, the functor $L \mapsto K \otimes L$ $(L \in K(S, \mathscr{A}))$ is left derivable.
(iii) If every module has finite tor-dimension,[56] the functor $\otimes$ derives to a functor

$$\overset{L}{\otimes} : D(S, \mathscr{A}^\circ) \times D(S, \mathscr{A}) \longrightarrow D(S, \mathscr{C}).$$

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4.2. Godement flasque resolutions.

4.2.1. Let S be a topos and X a set. To give a family of points of S indexed by X is equivalently to give a morphism $k : \mathrm{Top}(X) \to S$ from the topos defined by the discrete topological space X to S. For this family to be conservative (IV 6.4.1), it is necessary and sufficient that, for every sheaf F on S, the adjunction arrow from F to $k_* k^* F$ be a monomorphism (cf. IV 6.4.0).

Definition 4.2.2. *Let $k : \mathrm{Top}(X) \to S$ be a conservative family of points of a topos S. The* Godement resolution, *or* canonical flasque resolution *(relative to X), of an abelian sheaf F on S, denoted $\mathscr{C}^*_X(F)$ (or simply $\mathscr{C}^*(F)$), is the following right resolution of F:*

1) *$\mathscr{C}^0(F) = k_* k^* F$ and $\epsilon : F \to \mathscr{C}^0(F)$ is the adjunction arrow.*
2) *Set $d^{-1} = \epsilon$; recursively, for $n \geq 0$, define*
$\mathscr{C}^{n+1}(F) = \mathscr{C}^0(\mathrm{coker}\, d^{n-1})$, and define d^n as the composite arrow

$$d^n : \mathscr{C}^n(F) \to \mathrm{coker}(d^{n-1}) \to \mathscr{C}^0(\mathrm{coker}(d^{n-1})).$$

Proposition 4.2.3. *Under the hypotheses of 4.2.2,*

(i) *$\mathscr{C}^n(F)$ is a flasque sheaf* V 4.1,
(ii) *$\mathscr{C}^n(F)$ is an exact functor of F,*
(iii) *the stalk at a point $x \in X$ of the complex $\mathscr{C}^*(F)$ is a canonically split resolution of F_x.*

327 The sheaves $\mathscr{C}^n(F)$ are flasque because they are direct images of sheaves (automatically flasque) on the discrete topological space X.

The functors k^* and k_* are exact. Hence the functor $\mathscr{C}^0$ is exact. We prove by induction on $n \geq 0$ that the functor $\mathscr{Z}^n(F) = \mathrm{coker}(d^{n-1})$ is exact, and therefore so is $\mathscr{C}^{n+1} = \mathscr{C}^0 \mathscr{Z}^n$. Set $\mathscr{Z}^{-1}(F) = F$. An exact sequence of sheaves

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

[56] N.D.E.: this condition is superfluous. In the case where $\mathscr{A}$ is commutative, see [KS], 18.6, cited in la N.D.E. (51) page 673.

defines, for $n \geq 0$, a diagram

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathscr{Z}^{n-1}(F') & \longrightarrow & \mathscr{Z}^{n-1}(F) & \longrightarrow & \mathscr{Z}^{n-1}(F'') & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathscr{C}^{n}(F') & \longrightarrow & \mathscr{C}^{n}(F) & \longrightarrow & \mathscr{C}^{n}(F'') & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathscr{Z}^{n}(F') & \longrightarrow & \mathscr{Z}^{n}(F) & \longrightarrow & \mathscr{Z}^{n}(F'') & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array},$$

whose columns are exact, as are, by the induction hypothesis, the first two rows. The 3rd row is therefore exact, proving (ii).

The short exact sequences

$$0 \longrightarrow \mathscr{Z}^{n+1}(F) \longrightarrow \mathscr{C}^{n}(F) \longrightarrow \mathscr{Z}^{n}(F) \longrightarrow 0$$

are special cases of the short exact sequence

$$0 \to F \xrightarrow{\epsilon} \mathscr{C}^{0}(F) \to \mathscr{Z}^{0}(F) \to 0 \tag{4.2.3.1}$$

(make the substitution $F \mapsto \mathscr{Z}^{n-1}(F)$). To prove (iii), it remains to show that the 328
inverse-image sequence of (4.2.3.1) under k is canonically split. Indeed, the composite arrow

$$k^*F \xrightarrow{k^*(\epsilon)} k^*\mathscr{C}^0F = k^*(k_*k^*)F = (k^*k_*)k^*F \longrightarrow k^*F$$

is the identity.

Remark 4.2.4. If $(x_i)_{i\in X}$ is a conservative family of points of a site, this family corresponds to a conservative family of points of the associated topos, which again makes it possible to define canonical flasque resolutions.

Remark 4.2.5. If S is a ringed topos, the functors $\mathscr{C}^n$ carry modules to modules.

Remark 4.2.6. Let $\mathscr{S}$ be a category consisting of sites. Suppose that, for each $S \in \mathrm{Ob}\,\mathscr{S}$, a set S^d and a conservative family $k_S : \mathrm{Top}(S^d) \to S$ of points of S indexed by S^d are given. Suppose that S^d is functorial in S and that the diagrams

$$\begin{array}{ccc}
\mathrm{Top}(S^d) & \xrightarrow{f^d} & \mathrm{Top}(T^d) \\
\downarrow{\scriptstyle k_S} & & \downarrow{\scriptstyle k_T} \\
S & \xrightarrow{f} & T
\end{array}$$

are commutative. The corresponding canonical flasque resolutions are then "functorial in S."

If $\mathscr{S}$ is the category of topological spaces, one may take S^d to be the underlying set. 329

If $\mathscr{S}$ is the category of the étale sites of schemes S locally of finite type over a scheme S_0, and if, for each $s \in S_0$, $\overline{k(s)}$ is an algebraic closure of $k(s)$, one may take for S^d the set of points of S with values in one of the $\overline{k(s)}$.

The condition "of finite type" may be replaced by a cardinality condition if the algebraic closures $\overline{k(s)}$ are replaced by sufficiently large algebraically closed extensions.

4.2.7. One advantage of canonical flasque resolutions is that they make it possible to construct complexes adapted simultaneously to functors of "tensor-product" type and to functors of "direct-image" type. They thereby give a perhaps more comprehensible construction of the base-change morphisms (4.1.5).

LEMMA 4.2.8. *Let $f : (S, \mathscr{A}) \to (S', \mathscr{B})$ be a morphism of ringed sites, let $K \in \mathrm{Ob}\, K(S, \mathscr{A}^\circ)$ and $L \in \mathrm{Ob}\, K(S', \mathscr{B})$ be complexes, and let $k : \mathrm{Top}(X) \to S$ be a conservative set of points of S. Suppose that one of the following conditions holds:*

(a) *The cohomologies of K and L are bounded above;*

(b) *$\mathscr{A}$ has finite tor-dimension over $f^{-1}\mathscr{B}$, and $K \in \mathrm{Ob}\, D^b(S, \mathscr{A}^\circ)$ has finite tor-dimension;*

(c) *$L \in \mathrm{Ob}\, D^b(S', \mathscr{B})$ has finite tor-dimension.*

330 *Then the functor $(K, L) \mapsto K \otimes f^*L$ is derivable on the left (1.2.1) at (K, L); denote its derived functor by $\overset{\mathbf{L}}{\otimes}\mathbf{L}f^*$.*

(i) *If x is a point of S, then $(K \overset{\mathbf{L}}{\otimes} \mathbf{L}f^*L)_x \simeq K_x \overset{\mathbf{L}}{\otimes}_{\mathscr{B}_{f(x)}} L_{f(x)}$.*

(ii) *The pair (K, L) is left adapted to $\overset{\mathbf{L}}{\otimes} f^*$ (1.2.1) if and only if, for every $x \in X$, the pair $(K_x, L_{f(x)})$ is left adapted to the functor $\otimes_{\mathscr{B}_{f(x)}}$.*

(iii) *Under hypotheses (a) or (b), the functor $K \overset{\mathbf{L}}{\otimes} f^*L$ in L has cohomological amplitude $\subset d$ (1.2.9) if and only if, for every $x \in X$, the complex K_x of $\mathscr{B}_{f(x)}$-modules has tor-amplitude $\subset d$.*[57]

This lemma, whose proof is standard, expresses the "pointwise nature" of the functor under consideration.

4.2.9. Let L be a complex. The *Godement resolution* (relative to X) of L is the simple complex (*defined using sums*) associated with the double complex $\mathscr{C}^*L$ (1.1.4). The *Godement resolution truncated at order n*, $\tau''_{\leq n}\mathscr{C}^*L$, is the simple complex associated with the double complex obtained from $\mathscr{C}^*L$ by truncation (1.1.15) in the direction of the Godement differential. By 4.2.3 (iii), L is, point by point on X, homotopy equivalent to the complexes $\tau''_{\leq n}\mathscr{C}^*L$ and $\mathscr{C}^*L$; these complexes are therefore "as good as L" with respect to functors of "tensor-product" type (4.2.8).

PROPOSITION 4.2.10. *Let $(S, \mathscr{A})$ be a ringed site, let X be a conservative set of points of S, and let $N, M \in \mathbf{N} \cup \{\infty\}$, with N or M finite. Let A be the subcategory of $K(S)$ consisting of complexes L such that:*

(a) *If $N = \infty$, then $H^i(L) = 0$ for i sufficiently large; if $M = \infty$, then $H^i(L) = 0$ for i sufficiently small.*

331 (b) *For every $x \in X$ and every $\mathscr{A}_x$-module Q of tor-dimension $\leq N$, the complex L_x is left adapted to the functor $Q \otimes_{\mathscr{A}_x} *$.*

(c) *For every morphism of topoi $f : S \to E$ of cohomological dimension $\leq M$, the complex L is right adapted to the functor f_*.*

Then the inclusion of A into $K(S)$ induces an equivalence between the category obtained from A by inverting quasi-isomorphisms and the category $D^*(S)$, where $* = +$ if $N < M = \infty$, $* = -$ if $M < N = \infty$, and $*$ is blank if $N, M < \infty$.

[57] N.D.E.: hypotheses (a), (b), and (c) are all superfluous. Cf. la N.D.E. (51) page 673.

Let K be a complex satisfying (a). If $N = \infty$, then the canonical arrow $a : K_1 = \tau_{\leq i}K \to K$ is a quasi-isomorphism for i sufficiently large. If $N < \infty$, set $K_1 = K$. Let K_1^* be a flat Cartan–Eilenberg resolution and set $K_2 = \tau''_{\geq -N}K_1$ (truncation in the direction of the new differential). The composite arrow from K_2 to K is a quasi-isomorphism, and K_2 satisfies (b).

Let K be a complex satisfying (a) and (b). If $M = \infty$, then $N < \infty$; for i sufficiently large, $b : K \to \tau_{\geq i}K_1 = K_3$ is a quasi-isomorphism, and K_3 still satisfies (a) and (b). If $M < \infty$, set $K = K_3$. Let $K_4 = \tau''_{\leq M}\mathscr{C}^*K_3$. Then, by 4.2.3 (iii), K_4 satisfies (a), (b), and (c).

The proposition therefore follows from two successive applications of [11, I 2.4.2].

VARIANTS 4.2.11. If $N < \infty$ (resp. if $M < \infty$, resp. if $N, M < \infty$), one could equally have worked with bounded-below (resp. bounded-above, resp. bounded) complexes.

4.2.12. Here is how these constructions may be used to define the *base-change morphism* of 4.1.4 under hypotheses more general than those of *loc. cit.*, when the sites under consideration have enough points. 332

Let there be a commutative diagram (cf. (0.11)) of sites

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S, \end{array} \tag{4.2.12.1}$$

let $\mathscr{A}$ be a sheaf of rings on S, $\mathscr{A}'$ a sheaf of rings on S', and L a complex of $(\mathscr{A}', g^*\mathscr{A})$-bimodules on S'. Suppose that:

(i) the sites X and X' admit conservative families of points, $k : Q \to X$ and $k' : Q' \to X'$, fitting into a commutative diagram

$$\begin{array}{ccc} Q' & \xrightarrow{g'_0} & Q \\ \downarrow{\scriptstyle k'} & & \downarrow{\scriptstyle k} \\ X' & \xrightarrow{g'} & X. \end{array} \tag{4.2.12.2}$$

(ii) One of the following hypotheses holds:
  (a) The functors Rf_* and Rf'_* have finite cohomological dimension, and L is bounded above. Set $s = -$.
  (b) L is bounded below, with components of uniformly bounded tor-dimension (as $g^*\mathscr{A}$-modules). Set $s = +$.
  (c) L is bounded, and the hypotheses of (a) and (b) hold. Set $s =$ blank.

We denote by $L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^* : D^s(S, \mathscr{A}) \to D^s(S', \mathscr{A}')$ the left derived functor (1.2.1) of the functor $K \mapsto L \otimes_{g^*\mathscr{A}} g^*K$ from $K^s(S, \mathscr{A})$ to $K^s(S', \mathscr{A}')$. 333

We shall define a base-change morphism

$$\varphi : L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*Rf_*K \longrightarrow Rf'_*(f'^*L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*K) \tag{4.2.12.3}$$

for K in $D^s(X, \mathscr{A})$.[58]

[58] N.D.E.: Adjunctions in unbounded derived categories make it possible to define such a morphism without hypotheses (i) and (ii). Cf. la N.D.E. (51) page 673.

By 4.2.10, it suffices to define it, functorially in K, for the complexes belonging to the category A of 4.2.10, with

$$N \geq \sup_i \text{ (tor-dimension of } L^i \text{ over } g^*)$$
$$M \geq \text{cohomological dimension of } Rf_*.$$

Such a complex is adapted both to the functor Rf_* and to the functor $f'^*L \overset{\mathbf{L}}{\otimes}_{\mathscr{A}'} \mathbf{L}g'^*$ (4.2.8). The canonical morphisms

$$\alpha : f_*K \longrightarrow Rf_*K$$
$$\beta : f'^*L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*K \longrightarrow f'^*L \otimes g'^*K$$

are therefore quasi-isomorphisms. To compute the left-hand side I of (4.2.12.3), it therefore suffices to take a left resolution (i.e. a quasi-isomorphism)

$$\gamma : K_1 \longrightarrow f_*K,$$

with K_1 adapted to $L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*$; we have

$$I \simeq L \otimes g^*K_1.$$

334 Similarly, to compute the right-hand side II of (4.2.12.3), it suffices to take a right resolution

$$\delta : f'^*L \otimes g'^*K \longrightarrow K_2,$$

with K_2 adapted to Rf'_*; then

$$II \simeq f'_*K_2.$$

We then define the base-change morphism as the composite

$$I \simeq L \otimes g^*K_1 \xrightarrow{\gamma} L \otimes g^*f_*K \xrightarrow{c} f'_*(f'^*L \otimes g'^*K) \xrightarrow{\delta} f'_*K_2 \simeq II,$$

where c is the usual base-change morphism at the level of actual complexes.

This base-change morphism has the following characteristic property, which in principle makes it easy to verify its compatibility properties.

Proposition 4.2.13. *Under the preceding hypotheses, let $K \in \mathrm{Ob}\, C^s(X, \mathscr{A})$, $K_1 \in \mathrm{Ob}\, C^s(S, \mathscr{A})$ and $K_2 \in \mathrm{Ob}\, C^s(X', \mathscr{A})$. Moreover, let u be an f-morphism from K to K_1, i.e. $u \in \mathrm{Hom}(K_1, f_*K) \simeq \mathrm{Hom}(f^*K_1, K)$, and let v be a morphism from $f'^*L \otimes g'^*K$ to K_2. From u, v, and the usual base-change morphism, one obtains a morphism of complexes $c_{u,v} : L \otimes g^*K_1 \to f'_*K_2$. Let u', v', and c'_{uv} denote the composite morphisms*

$$u' : K_1 \xrightarrow{u} f_*K \to Rf_*K$$
$$v' : L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*K \to L \otimes g'^*K \xrightarrow{v} K_2$$
$$c'_{u,v} : L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*K_1 \to L \otimes g'^*K_1 \xrightarrow{c_{uv}} f'_*K_2 \to Rf'_*K_2.$$

335 *The following diagram is commutative:*

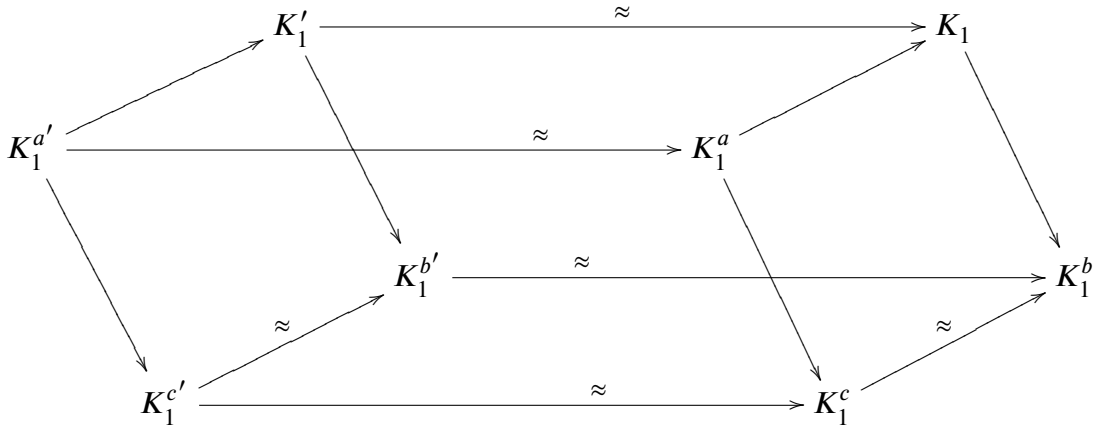
$$\begin{array}{ccc} L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^* Rf_* K & \xrightarrow{\varphi} & Rf'_*(f'^* L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^* K) \\ \uparrow{\scriptstyle L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*(u')} & & \downarrow{\scriptstyle Rf'_*(v')} \\ L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^* K_1 & \xrightarrow{c'_{uv}} & Rf'_* K_2. \end{array} \tag{4.2.13.1}$$

Suppose first that $K_1 = f_* K$ and that $K_2 = f'^* L \otimes g'^* K$, and consider the following commutative diagram of complexes:

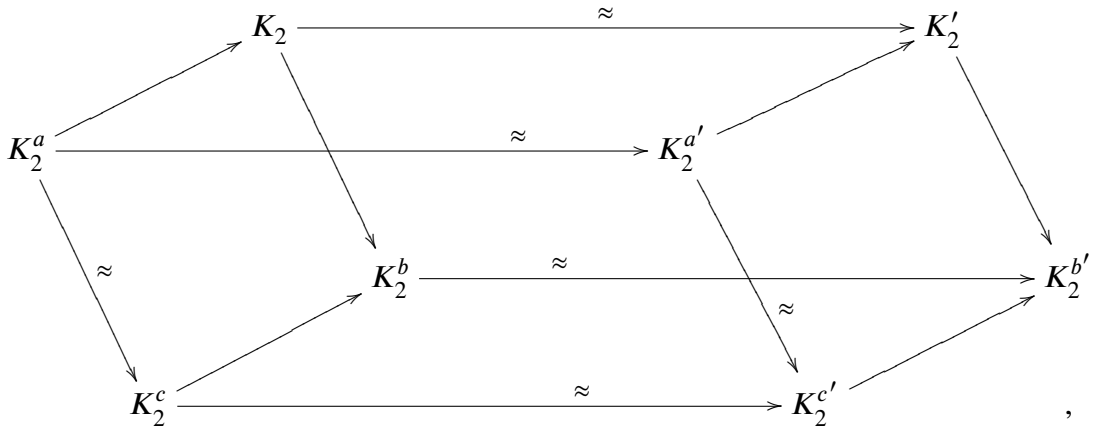
$$\begin{array}{ccc} K^a & \longrightarrow & K \\ \downarrow & & \downarrow \\ K^c & \longrightarrow & K^b \end{array}$$

such that K^a is adapted to $f'^* L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*$, K^b is adapted to Rf_*, and K^c is adapted to both functors. To construct this diagram, we first construct K^a, then apply a canonical flasque resolution (truncated if necessary) to K^a and K.

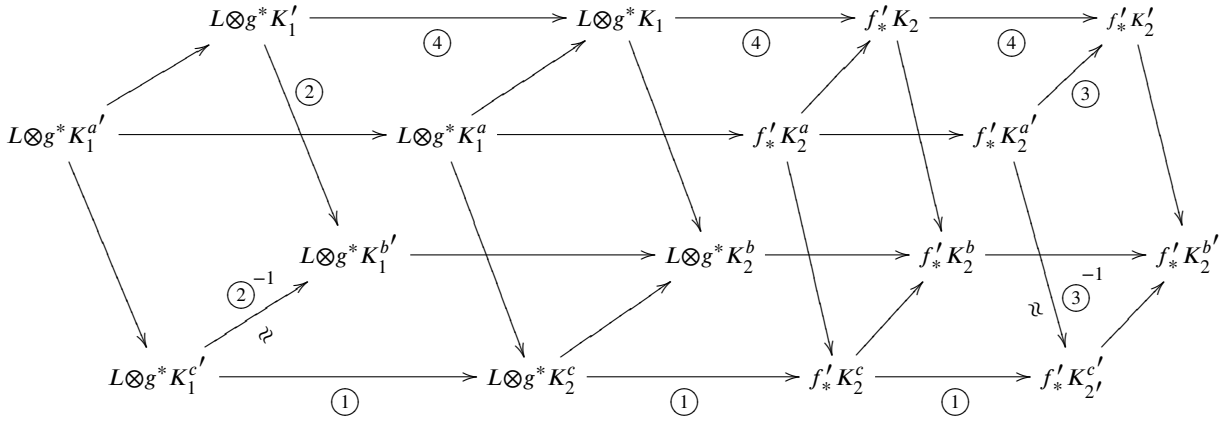
Apply the functor f_* (resp. $L \otimes g'^*$) to this diagram, and resolve the complexes in the diagram $(K_1, K_1^a, K_1^b, K_1^c)$ (resp. $(K_2, K_2^a, K_2^b, K_2^c)$) so as to obtain a commutative cube, in which $\approx$ denotes a quasi-isomorphism:



where K_1', $K_1^{a'}$, $K_1^{b'}$, and $K_1^{c'}$ are adapted to the functor $L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*$ (resp. 336



where K_2', $K_2^{a'}$, $K_2^{b'}$, and $K_2^{c'}$ are adapted to the functor Rf'_*). Using the usual base-change morphisms, one obtains a commutative prism of complexes over S.



337 and one concludes by noting that the morphisms in diagram (4.2.13.1) are identified with the following composites:

for φ: the composite of the morphisms ①;

for $L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*(u')$: the composite of the morphisms ②, one of which is the inverse of a quasi-isomorphism;

for $Rf'_*(v')$: the composite of the morphisms ③, one of which is the inverse of a quasi-isomorphism;

for c'_{uv}: the composite of the morphisms ④.

In the general case, set $K'_1 = f_*K$ and $K'_2 = f'^*L \otimes g^*K$. We then have a diagram

$$\begin{array}{ccccccc} L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*Rf_*K & \xrightarrow{\varphi} & Rf'_*(f'^*L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*K) & & & & \\ \uparrow & \nearrow & \uparrow & \searrow & & & \\ L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*K_1 & \longrightarrow & L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*K'_1 & \longrightarrow & Rf'_*K'_2 & \longrightarrow & Rf'_*K_2, \end{array}$$

whose triangles are commutative, as is the inner rectangle by the preceding argument. The boundary is therefore commutative, and this proves 4.2.13.

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4.3. The base-change theorem. We shall reformulate in terms of derived categories the base-change theorem for proper morphisms XII 5.1 (iii).

THEOREM 4.3.1. *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a Cartesian diagram of schemes, with f proper. Let $\mathscr{A}$ and $\mathscr{A}'$ be sheaves of torsion rings on S and S', and let L be a bounded-above complex of $(\mathscr{A}', g^*\mathscr{A})$-bimodules. Assume that the dimensions of the fibers of f are bounded. Hypothesis 4.2.12 (a) is then satisfied, and the base-change morphism (4.2.12.3), between functors from $D^-(X, f^*\mathscr{A})$ to $D^-(S', \mathscr{A}')$,

$$\varphi_K : L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*Rf_*K \longrightarrow Rf'_*(f'^*L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*K),$$

is an isomorphism.

The functors under consideration are triangulated and "way-out" [6, I 7], so, to verify that φ_K is an isomorphism, it suffices to verify that the $\varphi_{\mathscr{H}^i(K)}$ are isomorphisms. We may therefore assume that K is a sheaf of torsion $f^*\mathscr{A}$-modules concentrated in degree 0.

To verify that φ_K is an isomorphism, it suffices to do so after applying the forgetful 339
functor from the category of $\mathscr{A}'$-modules to that of $\mathbf{Z}/n\mathbf{Z}$-modules, where $n\mathscr{A} = n\mathscr{A}' = 0$.

This reduces us to the case $\mathscr{A}' = \mathbf{Z}/n\mathbf{Z}$, with $n\mathscr{A} = 0$. We may then replace L by an isomorphic complex in the derived category, bounded above and with components flat over $g^*\mathscr{A}$. Since the functors under consideration are triangulated and "way- out" in L, it suffices to verify that φ_K is an isomorphism after replacing L by one of its components. We may therefore assume that L is a sheaf of flat $g^*\mathscr{A}$-modules placed in degree 0. We must prove that the morphisms

$$\varphi^q \,:\, L \otimes g^* R^q f_* K \longrightarrow R^q f'_*(f'^* L \otimes g^* K)$$

are isomorphisms. Let s be a geometric point of S', let X_s be the fiber of X' at s, and let K_s be the inverse image of K on X_s. By XII 5.1 (iii), the fiber at s of φ^q is the morphism

$$\varphi^q_s \,:\, L_s \otimes H^q(X_s, K) \longrightarrow H^q(X_s, L_s \otimes K).$$

By *D.* Lazard's theorem [7], the flat $\mathscr{A}_s$-module L_s is the filtered direct limit of free $\mathscr{A}_s$-modules $L_{s,i}$; the functor $H^q(X_s, *)$ commutes with filtered direct limits (VII 3.3). The morphism φ^q is therefore the direct limit of the morphisms

$$\varphi^q_i \,:\, L_{s,i} \otimes H^q(X_s, K) \longrightarrow H^q(X_s, L_{s,i} \otimes K).$$

The latter are plainly isomorphisms; hence so is φ^q_s, and φ^q is an isomorphism.

VARIANTS 4.3.2. (i) When L is a bounded complex whose components have 340
finite tor-dimension over $g^*\mathscr{A}$, hypotheses (c) of 4.2.12 are satisfied, and one may work in the unbounded derived category.[59]

(ii) If L is bounded below, with components whose tor-dimension is uniformly bounded over $g^*\mathscr{A}$, hypotheses (b) of 4.2.12 are satisfied, even if $\mathscr{A}$ and $\mathscr{A}'$ are not torsion; the base-change morphism φ_K ($K \in D^+(X, f^*\mathscr{A})$) will be an isomorphism if the sheaves $\mathscr{H}^i(K)$ are torsion.

5. The direct-image functors with proper supports

5.1. The fundamental construction. We shall follow the program set out in 3.1 and define the functor $Rf_!$ for a compactifiable morphism f (3.2). In the appendix we shall show how to treat the case of an arbitrary separated morphism of finite type.[60]

5.1.1. Let X be a topos, U an open subtopos of X (IV 8.3), and j the inclusion morphism from U to X. In IV 5.2 we defined an "*extension by the empty set*" functor from the category of sheaves of sets on U to that of sheaves of sets on X. This functor $j_!$ is left adjoint to the restriction functor j^*.

We also denote by $j_!$ the functor from the category of sheaves of pointed sets on U (resp. sheaves of groups, resp. sheaves of $j^*\mathscr{A}$-modules, for $\mathscr{A}$ a sheaf of rings on X) to the analogous category on X, left adjoint to the restriction functor j^* (IV 11.3.1). 341
These three functors $j_!$ are compatible with the forgetful functors from the category of

[59] N.D.E.: One may work in the unbounded derived category without these hypotheses. Cf. la N.D.E. (51) page 673.

[60] N.D.E.: Cf. la N.D.E. (43) page 639.

modules to that of groups, and from the category of groups to that of pointed sets [60]. If $i : F \to X$ is the inclusion of the closed complement of U, then $j^* j_!$ is (canonically isomorphic to) the identity and $i^* j_!$ is the constant functor with value the initial object (or the final object, which here amounts to the same thing). This also characterizes $j_!$. It follows that the three functors $j_!$ under consideration are exact and embed into the corresponding functors j_*. They are called the *extension by zero* functors. The formula $(j_1 j_2)^* = j_2^* j_1^*$ transposes into a transitivity isomorphism

$$(j_1 j_2)_! = j_{1!} j_{2!}.$$

The functors $j_!$ are compatible with base change:

LEMMA 5.1.2. *Let $f : X \to Y$ be a morphism of topoi, let V be an open subtopos of Y, and let U be its inverse image in X:*

$$\begin{array}{ccc} U & \xrightarrow{f'} & V \\ \downarrow{\scriptstyle j'} & & \downarrow{\scriptstyle j} \\ X & \xrightarrow{f} & Y \end{array}$$

There is then a unique base-change isomorphism $f^* j_! \xrightarrow{\sim} j'_! f'^*$ making the diagram

$$\begin{array}{ccc} f^* j_! & \xrightarrow{\sim} & j'_! f'^* \\ \downarrow & & \downarrow \\ f^* j_* & \xrightarrow{\alpha} & j'_* f'^*, \end{array} \tag{5.1.2.1}$$

commutative, where α is the base-change morphism.

342 PROOF. The base-change morphism $\varphi : j^* f_* \to f'_* j'^*$ is trivially an isomorphism. It transposes into a base-change isomorphism $\varphi' : j'_! f'^* \xrightarrow{\sim} f^* j_!$. To verify that diagram (5.1.2.1) is commutative, it suffices to see this after restriction to U, where it is trivial.

5.1.3. Under the hypotheses of 5.1.1, with X ringed by a sheaf of rings $\mathscr{A}$, the functor $j_!$ on modules is exact, hence passes trivially to the derived categories and defines a functor, still denoted $j_!$, from $D(U, j^*\mathscr{A})$ to $D(X, \mathscr{A})$. We still have transitivity isomorphisms

$$(j_1 j_2)! = j_{1!} j_{2!}. \tag{5.1.3.1}$$

At the level of derived categories, we still have an adjunction formula

$$\mathrm{Hom}_{D(X)}(j_! K, L) \xrightarrow{\sim} \mathrm{Hom}_{D(U)}(K, j^* L). \tag{5.1.3.2}$$

5.1.4. Let $f : X \to Y$ be a proper morphism, let $\mathscr{A}$ be a sheaf of *torsion* rings on Y, and endow X with the inverse-image sheaf of rings of $\mathscr{A}$. Assume that the dimensions of the fibers of f are bounded above by an integer n (as is the case if Y is quasi-compact). We then know that, for every sheaf of $f^*\mathscr{A}$-modules F on X, the sheaves $R^q f_* F$ vanish for $q > 2n$ (XII 5.3 bis). The functor f_* therefore admits a derived functor (1.2.10)

$$Rf_* : D(X, f^*\mathscr{A}) \longrightarrow D(Y, \mathscr{A}).$$

343 If f is the composite gh of two proper morphisms, there is a transitivity formula $Rf_* = Rg_* Rh_*$, which, as usual, can be expressed in terms of cofibered categories (Scholium 4.1.3).

[60] But not to that of sets!

5.1.5. Consider a commutative diagram of schemes

$$\begin{array}{ccc} U & \xrightarrow[j_1]{} & X \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ V & \xrightarrow[j_2]{} & Y, \end{array} \tag{5.1.5.1}$$

in which j_1 and j_2 are open immersions and f, g are proper morphisms whose fiber dimensions are bounded above by an integer n. Let $\mathscr{A}$ be a sheaf of torsion rings on Y, and equip U, V, and X with the appropriate inverse images of $\mathscr{A}$. By (5.1.3.2), if $K \in D(U)$, defining a morphism

$$d : j_{2!} R g_* K \longrightarrow R f_* j_{1!} K \tag{5.1.5.2}$$

amounts to defining a morphism

$$d' : R g_* K \longrightarrow j_2^* R f_* j_{1!} K.$$

The fiber product $X' = V \times_Y X$ fits into a commutative diagram

$$\begin{array}{ccccc} U & \xrightarrow{k} & X' & \xrightarrow[j_2']{} & X \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ V & = & V & \xrightarrow[j_2]{} & Y, \end{array} \tag{5.1.5.3}$$

and, trivially, we have $j_2^* R f_* j_{1!} K \simeq R f'_* (j_2'^* j_{1!} K) \simeq R f'_* k_! K$. 344

The morphism k is a proper open immersion; hence $k_! = k_*$, giving an isomorphism $R f'_* k_! K \simeq R f'_* k_* K \simeq R g_* K$, which provides the desired isomorphism d' and the morphism d. Moreover, the restriction of d to V is an isomorphism.

Lemma 5.1.6. *The morphism (5.1.5.2) is an isomorphism.*

It remains to verify that the restriction of d to $Y - V = F$ is an isomorphism. Let $X'' = X \times_Y F$ and apply the proper base-change theorem (4.3.1) to the Cartesian square

$$\begin{array}{ccc} X'' & \xrightarrow[i']{} & X \\ \downarrow{\scriptstyle f''} & & \downarrow{\scriptstyle f} \\ F & \xrightarrow[i]{} & Y, \end{array}$$

with $\mathscr{A}' = i^* \mathscr{A}$ and $L = \mathscr{A}'[0]$. We obtain: $i^* R f_* j_{1!} K \simeq R f''_* i'^* j_{1!} K \simeq R f''_* 0 \simeq 0$, so that both sides of (5.1.5.2) are zero outside V and d is an isomorphism.

5.1.7. Let S be a scheme ringed by a sheaf of torsion rings $\mathscr{A}$. For every S-scheme $\varphi : X \to S$, set $\mathscr{A}_X = \varphi^* \mathscr{A}$. Work in the category (S) of quasi-compact, quasi-separated S-schemes and S-compactifiable morphisms (3.2). By 3.2.3, this category, the subcategory (S, i) of open immersions, and the subcategory (S, p) of proper morphisms satisfy the hypotheses of 3.2.4.

We propose to apply the theory of 3.3.2 to the following data (notation of 3.3.1) 345

α) $F(X) = D(X, \mathscr{A}_X)$ for $\varphi : X \to S$ in S:

(i) For $p : X \to Y$ proper, take the functor p_* to be the functor $R p_*$.

(ii) Use the usual transitivity isomorphisms $R(pq)_* = R p_* R q_*$.

(i′) For an open immersion $i : X \to Y$, take the functor i_* to be the functor $i_!$.

(ii′) Use the transitivity isomorphisms (5.1.3.1).

(iii) For every commutative diagram (5.1.5.1), take the isomorphism (5.1.5.2) for d.

The conditions (a) (b) (c) (a′) (b′) (c′) of 3.3.2 are verified, proving part (a) of:

THEOREM 5.1.8. (a) *In the category (S) defined in 5.1.7, there exists one, and essentially only one, way to define, for every arrow $f : X \to Y$, a functor*

$$Rf_! : D(X, \mathscr{A}_X) \longrightarrow D(Y, \mathscr{A}_Y),$$

and, for every composite morphism $f = gh : X \xrightarrow{h} Y \xrightarrow{g} Z$, transitivity isomorphisms $c_{g,h}$ between $Rf_!$ and $Rg_!\, Rh_!$ such that

(i) *The $c_{f,g}$ satisfy the "cocycle condition"*

$$(c_{f,g} * Rh_!) \circ c_{fg,h} = (Rf_! * c_{g,h}) \circ c_{f,gh}.$$

(ii) *When restricted to proper morphisms (resp. to open immersions), one recovers the variance theory of 5.1.4 (resp. 5.1.3).*

346 (iii) *For every diagram (5.1.5.1), the isomorphism d of (5.1.5.2) is the composite isomorphism $c_{f,j_1} \circ c_{j_2,g}^{-1}$.*

(b) *The functors $Rf_!$ are "way out" [6, I 7] and triangulated.*

Every functor $Rf_!$ is the composite of a functor Rg_*, for g proper with quasi-compact target, hence with fibers of bounded dimension, and of a functor $j_!$ for j an open immersion. These functors are way out and triangulated, and hence so is $Rf_!$.

. The transitivity isomorphisms of 5.1.8 (a) are often used via the "spectral sequence of composite functors" deduced from them, valid for $K \in \mathrm{Ob}\, D(X, \mathscr{A}_X)$:

$$E_2^{pq} = R^p g_! R^q h_!(K) \Longrightarrow R^{p+q} f_!(K). \tag{5.1.8.2}$$

To obtain it, note that by 5.1.8 b) and VERDIER [12], for every $L \in \mathrm{Ob}\, D(Y, \mathscr{A}_Y)$, there is a spectral sequence

$$E_2^{pq} = R^p g_!(\mathscr{H}^q(L)) \Longrightarrow R^{p+q} g_!(L), \tag{5.1.8.3}$$

and (5.1.8.2) is the special case of (5.1.8.3) with $L = Rh_! K$.

DEFINITION 5.1.9. (i) *We call the "* total direct-image functors with proper supports*" the functors constructed in 5.1.8. We shall omit the adjective "total" when there is no risk of confusion.*

(ii) *The q^{th}* direct-image functor with proper support, *denoted $R^q f_!$, is the composite functor $\mathscr{H}^q \circ Rf_!$.*

(iii) *If X is a scheme over a separably closed field k that is embeddable in a proper*
347 *scheme over k, identify sheaves of modules on* $\mathrm{Spec}(k)$ *with ordinary modules, and denote the functor $R^q f_!$ by one of the notations $H_c^q(X, \quad)$, $H_!^q(X, \quad)$, or simply $H_c^q(\quad)$ if there is no risk of confusion. It is then called the "q^{th}* hypercohomology group with proper supports*," or the "q^{th}* cohomology group with proper supports*" if it is applied to a complex concentrated in degree* 0.

. The notation $Rf_!$ is abusive in that *the functor $Rf_!$ is not the right derived functor of the functor $f_!$* that will be defined in 6.1.2. Some prefer the notation $R_! f$.

5.1.10. The ingredients in the proof of 5.1.8 are the proper base-change theorem and the identity between j_* and $j_!$ when j is a proper open immersion. This makes it possible to state various variants of 5.1.8; it is left to the reader to give them a precise meaning and to verify them.

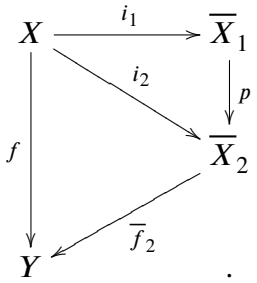
VARIANT 5.1.11. *In the category (S), there exists one, and essentially only one, way to define, for every arrow $f : X \to Y$, a functor $f_!$ from the category of sheaves of pointed sets on X to the category of sheaves of pointed sets on Y, and, for every composite morphism $f = gh$, transitivity isomorphisms satisfying conditions* (i) (ii) (iii) *analogous to those of 5.1.8.*

See 6.1.2 for a direct description of this functor.

VARIANT 5.1.12. *For every compactifiable morphism $f : X \to Y$, one can define* 348
a functor $R^1 f_!$ from the category of sheaves of ind-finite groups on X to the category of sheaves of pointed sets on Y, together with, for every compactification $f = \overline{f} j$ of f, an isomorphism $R^1 f_! \simeq R^1 \overline{f}_* \circ j_!$.

No transitivity formula is postulated here; the only problem is to show that the functor $R^1 f_!$ is independent of the chosen compactification of f. Let $p : (\overline{f}_1, i_1) \to (\overline{f}_2, i_2)$ be a morphism of compactifications, i.e. a diagram

(5.1.12.1)



where $\overline{f}_1 = \overline{f}_2 \circ p$, with p proper. If G is a sheaf of groups on X, there is an exact sequence of pointed sets (with α injective)

$$0 \to R^1 \overline{f}_{2*}(p_* i_{1!} G) \xrightarrow{\alpha} R^1 \overline{f}_{1*}(i_{1!} G) \to \overline{f}_{2*} R^1 p_*(i_{1!} G),$$

and a morphism from $i_{2!}G$ to $p_* i_{1!} G$, whence, by composition, a morphism β from $R^1 \overline{f}_{2*} i_{2!} G$ to $R^1 \overline{f}_{1*} i_{1!} G$. Reasoning as in 3.3.2.1, one sees that the problem is to prove that this morphism is always an isomorphism when G is ind-finite.

Applying the proper base-change theorem to p and the sheaf of sets $i_{1!}G$, one checks 349
that $p_* i_{1!} G = i_{2!} G$, yielding the exact sequence

$$0 \to R^1 \overline{f}_{2*}(i_{2!} G) \xrightarrow{\beta} R^1 \overline{f}_{1*}(i_{1!} G) \to \overline{f}_{2*} R^1 p_* i_{1!} G.$$

The morphism β is injective (since α is). Applying the proper base-change theorem to p and the sheaf of groups G, assumed ind-finite, one finds that $R^1 p_* i_{1!} G = 0$. Thus β is an epimorphism, which completes the proof of 5.1.12.

DEFINITION 5.1.13. *For every site X, denote by $D_{\mathrm{tors}}(X)$ the full subcategory of the derived category of the category of abelian sheaves on X consisting of the complexes whose cohomology sheaves are torsion.*

VARIANT 5.1.14. *In the category (S), there exists one, and essentially only one, way to define, for every arrow $f : X \to Y$, a triangulated and "way out" functor* [6, I 7]

$$Rf_! : D^+_{\mathrm{tors}}(X) \longrightarrow D^+_{\mathrm{tors}}(Y)$$

and, for every composite morphism $f = gh$, transitivity isomorphisms satisfying conditions (i) (ii) (iii) *analogous to those of 5.1.4.*

VARIANT 5.1.15. *Work in the category of S-schemes ringed by sheaves of torsion rings, the arrows being morphisms of ringed schemes inducing (0.8) an S-compactifiable morphism of schemes. For every arrow*

$$f : (X, \mathscr{A}) \longrightarrow (Y, \mathscr{B}),$$

350 *let $Rf_!$ be the composite of the forgetful functor from the category of $\mathscr{A}$-modules to that of $f^{-1}\mathscr{B}$-modules, followed by the usual $Rf_!$. These functors have the variance properties of the usual functors $Rf_!$.*

Note, however, that the notation $Rf_!$ is reasonable here only if $\mathscr{A}$ is finite over $f^{-1}\mathscr{B}$, as one sees by taking $X = Y = \mathrm{Spec}(k)$, with k algebraically closed.

5.1.16. Let $f : X \to S$ be a compactifiable morphism, let $j : U \to X$ be an open immersion into X, let $i : F \to X$ be the closed complement, let $\mathscr{A}_S$ be a sheaf of torsion rings on S, let $\mathscr{A}_X$ be its inverse image on X, and let $K \in \mathrm{Ob}\, D(X, \mathscr{A}_X)$.

The exact sequence of complexes

$$0 \longrightarrow j_!j^*K \longrightarrow K \longrightarrow i_!i^*K \longrightarrow 0$$

gives rise to a distinguished triangle in $D(X, \mathscr{A}_X)$ by VERDIER [11, II 1.1.5]. Its image under $Rf_!$ is a distinguished triangle (5.1.8 (b))

$$R(fj)_!j^*K \longrightarrow Rf_!K \longrightarrow R(fi)_!i^*K \longrightarrow R(fj)_!j^*K[1] \ldots \tag{5.1.16.1}$$

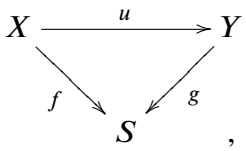
which defines a long exact cohomology sequence

$$\ldots \longrightarrow R^q(fj)_!j^*K \to R^qf_!K \to R^q(fi)_!i^*K \xrightarrow{j} R^{q+1}(fj)_!j^*K \to \ldots. \tag{5.1.16.2}$$

In the special case where S is the spectrum of a separably closed field and K is a sheaf $\mathscr{G}$ concentrated in degree 0, the sequence (5.1.16.2) becomes

$$\ldots H_c^q(U, \mathscr{G}) \to H_c^q(X, \mathscr{G}) \to H_c^q(F, \mathscr{G}) \xrightarrow{j} H_c^{q+1}(U, \mathscr{G}) \ldots \tag{5.1.16.3}$$

351 5.1.17. Consider a diagram of S-compactifiable schemes



with *u proper*. Equip X and Y with the inverse images of a sheaf of torsion rings $\mathscr{A}_S$ on S. For $K \in D(Y, \mathscr{A}_Y)$, the evident (adjunction) morphism $K \to Ru_*u^*K$ defines, upon applying the functor $Rg_!$, a morphism

$$u^* : Rg_!K \longrightarrow Rg_!Ru_*u^*K = Rg_!Ru_!u^*K \simeq Rf_!u^*K \tag{5.1.17.1}$$

(*contravariance of cohomology with proper supports with respect to proper morphisms*).

When S is the spectrum of an algebraically closed field and K is the sheaf $\mathscr{G}$ placed in degree 0, this morphism becomes

$$u^* : H_c^q(Y, \mathscr{G}) \longrightarrow H_c^q(X, \mathscr{G}) \tag{5.1.17.2}$$

5.2. The base-change theorem.

5.2.1. Let $f : X \to Y$ be a morphism of topoi, let V be an open of Y, let $U = f^{-1}(V)$ be its inverse image in X, let $\mathscr{A}$ be a sheaf of rings on Y, let $\mathscr{A}'$ be a sheaf of rings on X, and let L be a bounded-above complex of $(\mathscr{A}', f^*\mathscr{A})$-bimodules:

$$\begin{array}{ccc} U & \xrightarrow{f'} & V \\ {\scriptstyle j'}\downarrow & & \downarrow{\scriptstyle j} \\ X & \xrightarrow[f]{} & Y. \end{array} \tag{5.2.1.1}$$

There is then a base-change isomorphism (5.1.2.1) between functors from $D(V, \mathscr{A})$ to $D(X, f^*\mathscr{A})$: $f^*j_! \xrightarrow{\sim} j'_! f'^*$. Moreover, there is an evident isomorphism between functors from $D^-(U, f'^*\mathscr{A})$ to $D^-(X, \mathscr{A}')$: $L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} j'_! K \simeq j'_!(j'^* L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} K)$. To construct it, it suffices to note that if a complex K is adapted to the functor $j'^* L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}}$, then the complex $j'_! K$ is adapted to the functor $L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}}$, and that there is an isomorphism of the preceding type at the level of complexes. Composing these two isomorphisms gives a base-change isomorphism between functors from $D^-(V, \mathscr{A})$ to $D^-(X, \mathscr{A}')$: 352

$$L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} f^* j_! K \xrightarrow{\sim} j'_!(j'^* L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} f'^* K). \tag{5.2.1.2}$$

If L is bounded with components of finite tor-dimension over $f^*\mathscr{A}$,[61] this isomorphism is still defined for $K \in \mathrm{Ob}\, D(V, \mathscr{A})$.

5.2.2. Let $f : X \to S$ and $g : S' \to S$ be morphisms of schemes, and suppose that f is the composite of an open immersion j and a proper morphism p whose fibers have dimension bounded by an integer n (this is automatic when S is quasi-compact). These morphisms fit into a diagram with cartesian squares

$$\begin{array}{ccccc} X' & & \xrightarrow{g'} & & X \\ & {\scriptstyle j'}\searrow & & & \downarrow{\scriptstyle f} \quad {\scriptstyle j}\searrow \\ {\scriptstyle f'}\downarrow & & \overline{X}' & \xrightarrow[\overline{g}]{} & \overline{X} \\ & {\scriptstyle p'}\swarrow & & & \swarrow{\scriptstyle p} \\ S' & & \xrightarrow[g]{} & & S \end{array} .$$

Let $\mathscr{A}$ be a sheaf of torsion rings on S, $\mathscr{A}'$ a sheaf of torsion rings on S', and L a bounded-above complex of $(\mathscr{A}', g^*\mathscr{A})$-bimodules. We ring X and $\overline{X}$ (resp. X' and $\overline{X}'$) by the inverse images $\mathscr{A}_X$ and $\mathscr{A}_{\overline{X}}$ (resp. $\mathscr{A}'_{X'}$ and $\mathscr{A}'_{\overline{X}'}$) of $\mathscr{A}$ (resp. $\mathscr{A}'$) on X and $\overline{X}$ (resp. X' and $\overline{X}'$). We then have base-change isomorphisms (5.2.1.2) between the functors $p'^* L \overset{\mathbf{L}}{\otimes} \overline{g}^* j_! K$ and $j'_!(f'^* L \overset{\mathbf{L}}{\otimes} g'^* K)$ from $D^-(X, \mathscr{A}_X)$ to $D^-(\overline{X}', \mathscr{A}'_{\overline{X}'})$, and base-change isomorphisms (4.3.1) between the functors $L \overset{\mathbf{L}}{\otimes} g^* R p_* K$ and $R p'_*(p'^* L \overset{\mathbf{L}}{\otimes} \overline{g}^* K)$ 353

[61] N.D.E.: this condition is superfluous. Cf. la N.D.E. (51) page 673.

from $D^-(\overline{X}, \mathscr{A}_{\overline{X}})$ to $D^-(S', \mathscr{A}')$. By composition, one obtains a base-change isomorphism between functors from $D^-(X, \mathscr{A}_X)$ to $D^-(S', \mathscr{A}')$:

$$L \overset{\mathbf{L}}{\otimes} g^* Rp_* j_! K \xrightarrow{\sim} Rp'_* j'_! (f'^* L \overset{\mathbf{L}}{\otimes} g'^* K). \tag{5.2.2.1}$$

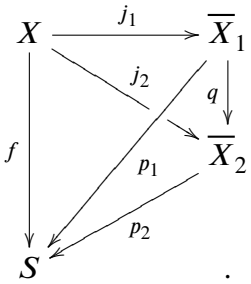
When L is bounded with components of finite tor-dimension, the preceding arguments and 4.3.2 (i) define the base-change isomorphism for $K \in \mathrm{Ob}\, D(X, \mathscr{A}_X)$.[62]

In order to refer to 5.1.8, suppose that S is quasi-compact and quasi-separated. The functors $Rp_* j_! = Rf_!$ and $Rp'_* j'_! = Rf'_!$ then depend only on f and f', giving a meaning to the following statement.

Lemma 5.2.3. *The base-change isomorphism (5.2.2.1) does not depend on the chosen compactification $f = pj$ of X.*

Let $q : \overline{X}_1 \to \overline{X}_2$ be a morphism of compactifications

(5.2.3.1)



354 The base-change isomorphism c_1 relative to $\overline{X}_1$ is obtained by "composing" those relative to j_1 and p_1, or, equivalently, by composing those relative to j_1, q, and p_2. The isomorphism c_2 relative to $\overline{X}_2$ is obtained by composing those relative to j_2 and p_2. To verify that $c_1 = c_2$, it suffices to verify that the isomorphism c'_1 relative to j_2 is the composite c'_2 of those relative to j_1 and q. Use a prime to denote the arrows obtained from the arrows (5.2.3.1) by the base change g; c'_1 and c'_2 are isomorphisms

$$c'_1, c'_2 : L \overset{\mathbf{L}}{\otimes} \overline{g}_2^* * j_{2!} K \xrightarrow{\sim} j'_{2!} (j'^*_2 * L \overset{\mathbf{L}}{\otimes} g'^* K).$$

By (5.1.3.2), to verify that $c'_1 = c'_2$ it suffices to verify it after restricting to $\overline{X}'_2$, which is immediate.

The base-change isomorphisms relative to two compactifications between which a morphism exists are therefore equal, and the conclusion follows from 3.2.6 (ii).

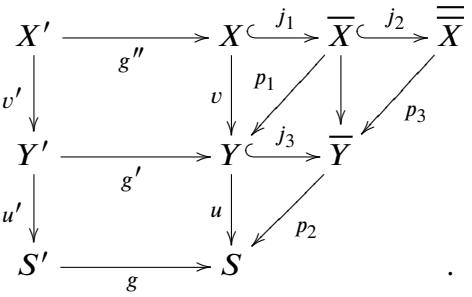
Lemma 5.2.4. *Let $f = uv : X \to Y \to S$ be a composite of S-compactifiable morphisms. The base-change isomorphisms (5.2.2.1) relative to u, v, and f give rise to a commutative diagram (notation as in (5.2.4.1) below):*

$$\begin{array}{ccc}
L \overset{\mathbf{L}}{\otimes} g^* R(uv)_! K & \xrightarrow{\quad\sim\quad} & R(u'v')_! ((u'v')^* L \overset{\mathbf{L}}{\otimes} g''^* K) \\
\| & & \| \\
L \overset{\mathbf{L}}{\otimes} g^* Ru_! Rv_! K & \xrightarrow{\sim} Ru'_! (u'^* L \overset{\mathbf{L}}{\otimes} g'^* Rv_! K) \xrightarrow{\sim} & Ru'_! Rv'_! (v'^* u'^* L \overset{\mathbf{L}}{\otimes} g''^* K).
\end{array}$$

355 Choose a compactification of (u, v) (3.2.5):

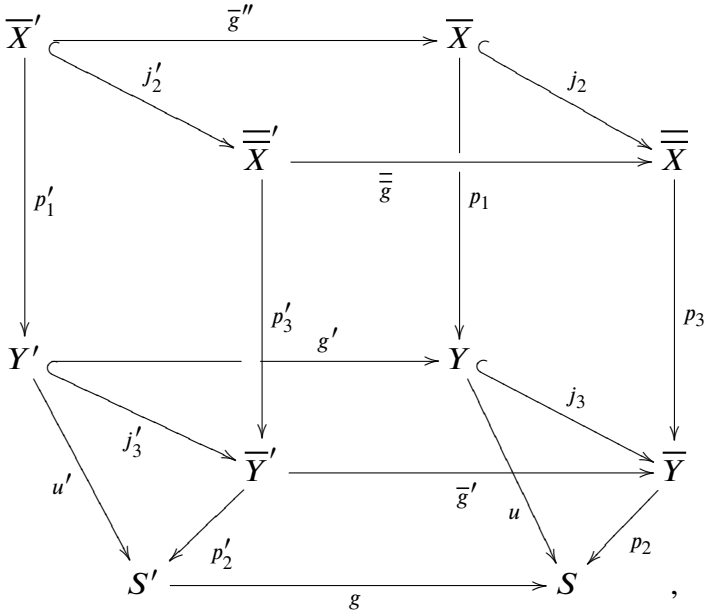
[62] N.D.E.: by combining the preceding N.D.E. and la N.D.E. (58) page 679, this isomorphism can be defined without any condition on L or the hypothesis that the dimension of the fibers of p is bounded.

(5.2.4.1)



Lemma 5.2.4 says that the base-change isomorphism c_1 relative to f is the composite c_2 of those relative to u and v. The isomorphism c_1 is obtained by composing the base-change isomorphisms relative to j_1, j_2, p_3, and p_2. For c_2, one composes those relative to j_1, p_1, j_3, and p_2. It therefore suffices to compare the isomorphisms c'_1, the composite of those relative to p_3 and j_2, and c'_2, the composite of those relative to j_3 and p_1.

Consider the cartesian diagram



$L' = p_2'^{*}L$, $K' = j_{1!}K$ and 356

$$c'_1, c'_2 : L' \overset{\mathbf{L}}{\otimes} \overline{g}'^{*} R(j_3 p_1)_! K' \xrightarrow{\sim} R(j'_3 p'_1)_! ((j'_3 p'_1)^* L' \otimes \overline{g}''^{*} K').$$

By (5.1.3.2), to verify that $c'_1 = c'_2$ it suffices to verify that $j_3'^{*}(c'_1) = j_3'^{*}(c'_2)$, which is immediate.

Lemma 5.2.5. *Let there be morphisms of schemes ringed by sheaves of torsion rings*

$$(S'', \mathscr{A}'') \xrightarrow{g_2} (S', \mathscr{A}') \xrightarrow{g_1} (S, \mathscr{A}),$$

and let $f : X \to S$ be a compactifiable morphism, giving a diagram:

$$\begin{array}{ccccc} X'' & \xrightarrow{g'_2} & X' & \xrightarrow{g'_1} & X \\ \downarrow{\scriptstyle f''} & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S'' & \xrightarrow[g_2]{} & S' & \xrightarrow[g_1]{} & S. \end{array}$$

Ring X, X', and X'' by $f^*\mathscr{A}$, $f'^*\mathscr{A}'$, and $f''^*\mathscr{A}''$, respectively. Then the diagram 357

$$\begin{array}{ccc} L(g_1g_2)^*Rf_! & \xrightarrow{\sim} & Rf''_!L(g'_1g'_2)^* \\ \| & & \| \\ Lg_2^*Lg_1^*Rf_! & \xrightarrow{\sim} Lg_2^*Rf'_!Lg_1'^* \xrightarrow{\sim} & Rf''_!Lg_2'^*Lg_1'^* \end{array}$$

is commutative.

This lemma follows immediately from the analogous results for the functors $j_!$ and Rp_* with p proper. We leave it to the reader to formulate the precise version with modules.

The preceding statements are summarized in the following theorem.

THEOREM 5.2.6. *The functor $Rf_!$ commutes with base change.*

VARIANTS 5.2.7. (i) The functors $f_!$ (for sheaves of pointed sets: 5.1.11) and the functors $R^1f_!$ (for sheaves of groups: 5.1.12) commute with base change.
(ii) The functors $Rf_!$ (5.1.14) commute with base change.
(iii) Under the hypotheses of (5.2.2), if L is bounded and has components of finite tor-dimension,[63] the functors $Rf_! : D(X) \to D(S)$ commute with base change.

358 The conjunction of the following two Propositions 5.2.8 and 5.2.9 is essentially equivalent to Theorem 5.2.6.

PROPOSITION 5.2.8. *Let $f : X \to S$ be a compactifiable morphism and $\mathscr{F}$ a torsion sheaf on X. For every geometric point s of S, let X_s be the geometric fiber of X at s and let $\mathscr{F}_s$ be the induced sheaf. Then canonically*

$$(R^qf_!(\mathscr{F}))_s = H^q_c(X_s, \mathscr{F}_s).$$

COROLLARY 5.2.8.1. *If the fibers of f have dimension $\leq d$, the functor $Rf_!$ has cohomological dimension $\leq 2d$ (1.2.9): for every torsion sheaf $\mathscr{F}$, one has $R^if_!\mathscr{F} = 0$ for $i > 2d$. The functor $R^{2d}f_!$ is therefore right exact on the category of torsion sheaves.*

By base change (5.2.8), we reduce to the case where S is the spectrum of an algebraically closed field. If $\overline{X}$ is then a proper scheme over S containing X as a dense Zariski open subscheme, with $j : X \to \overline{X}$, then $\dim(\overline{X}) = \dim(X) \leq d$. Hence, for $\mathscr{F}$ a torsion sheaf on X and $i > 2d$, by X 4.3, 5.2

$$H^i_c(X, \mathscr{F}) = H^i(\overline{X}, j_!\mathscr{F}) = 0.$$

PROPOSITION 5.2.9. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ be a sheaf of torsion rings on S, let K be a bounded-above complex of left $f^*\mathscr{A}$-modules, and let L be a bounded-above complex of right $\mathscr{A}$-modules. Then canonically*

$$L \overset{\mathbf{L}}{\underset{\mathscr{A}}{\otimes}} Rf_!(K) = Rf_!(f^*L \overset{\mathbf{L}}{\underset{f^*\mathscr{A}}{\otimes}} K).$$

359 This proposition formally implies the following result:

THEOREM 5.2.10. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ be a sheaf of torsion rings on S, let $K \in D^-(X, f^*\mathscr{A})$, and let $d \in \mathbf{Z}$. If K has finite tor-dimension $\leq d$ (4.1.9), then $Rf_!K$ has finite tor-dimension $\leq d$.*

[63] N.D.E.: as observed in 5.2.2, these hypotheses are superfluous.

It suffices to verify that, for every sheaf L of right $\mathscr{A}$-modules on S,

$$\mathscr{H}^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_!K) = 0 \quad \text{for} \quad i > d.$$

By 5.2.9, we know that

$$\mathscr{H}^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_!K) = R^{-i}f_!(f^*L \overset{\mathbf{L}}{\otimes} K),$$

and, by hypothesis,

$$\mathscr{H}^{-i}(f^*L \overset{\mathbf{L}}{\otimes} K) = 0 \quad \text{for} \quad i > d.$$

The conclusion follows by observing that if a complex M of sheaves of modules on X satisfies $\mathscr{H}^{-i}(M) = 0$ for $i > d$, then one still has $\mathscr{H}^{-i}(Rf_!M) = 0$ for $i > d$.

For constant noetherian sheaves of rings, one has the following more general result:

THEOREM 5.2.11. *Let $f : X \to S$ be a morphism of sites, let A be a ring, let C be its center, and let $d \in \mathbf{Z}$. Suppose that*

(i) *the functor $Rf_* : D^+(X, C) \to D^+(S, C)$ has finite cohomological dimension;*
(ii) *A is right noetherian;*
(iii) *S has enough points.*

For every complex $K \in \mathrm{Ob}\, D^-(X, A)$ of sheaves of left A-modules, if K has finite tor-dimension $\leq d$, then the complex $Rf_*K \in \mathrm{Ob}\, D^-(S, A)$ has tor-dimension $\leq d$. 360

PROOF. By (i), the functor f_* is derivable to functors

$$Rf_* : D(X, A) \longrightarrow D(S, A)$$

and

$$Rf_* : D(X, C) \longrightarrow D(S, C).$$

By 4.2.8, it suffices to verify that, for every point s of S, the complex $(Rf_*K)_s$ of A-modules has tor-dimension $\leq d$; for this it suffices that, for every finitely generated right A-module L,

$$H^{-i}(L \overset{L}{\otimes} Rf_*K) = 0 \quad \text{for} \quad i > d.$$

Reasoning as in 5.2.10, we see that it suffices for the "base-change arrow"

$$L \overset{L}{\otimes} Rf_*K \longrightarrow Rf_*(L \overset{L}{\otimes} K) \tag{5.2.11.1}$$

to be an isomorphism.

Let L^* be a resolution of L by finitely generated free A-modules. Since the functors under consideration are way-out to the right, it suffices to prove that (5.2.11.1) is an isomorphism after replacing L by any component of L^*, which reduces us to the immediate case where L is finitely generated and free.

REMARKS 5.2.12. (i) The hypothesis (iii) of 5.2.11 is probably unnecessary. 361

(ii) If the noetherian hypothesis on A is omitted, 5.2.11 remains valid when the functors $R^i f_*$ commute with filtered direct limits.

(iii) The hypothesis that S is ringed by a *constant* sheaf of rings is essential. A counterexample is obtained by taking for S the spectrum of a strictly local discrete valuation ring, for f the inclusion of the generic point η, for the sheaf of rings $\mathscr{A}$ the sheaf equal to $\mathbf{Z}/n\mathbf{Z}$ at the generic point and to $\mathbf{Z}/n^2\mathbf{Z}$ at the closed point s ($n > 1$ prime to the residue characteristics), and for the sheaf the module $f^*\mathscr{A}$.

5.3. Comparison theorem and finiteness theorem.

5.3.1. To a scheme X of finite type over $\mathbf{C}$ are associated its étale site X_{et}, its transcendental site X_{cl}, and a morphism of sites (XI 4)

$$\epsilon : X_{\text{cl}} \longrightarrow X_{\text{et}}.$$

If $j : U \to X$ is an open immersion, the commutative diagram

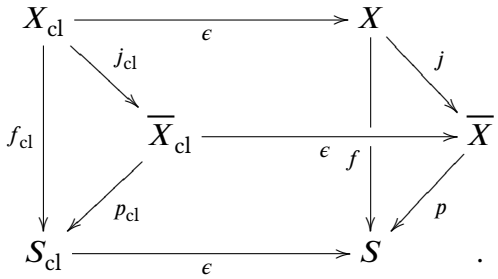
$$\begin{array}{ccc} U_{\text{cl}} & \longrightarrow & U_{\text{et}} \\ \downarrow & & \downarrow \\ X_{\text{cl}} & \longrightarrow & X_{\text{et}} \end{array}$$

identifies $U_{\text{cl}}^{\sim}$ with the open subtopos of $X_{\text{cl}}^{\sim}$ that is the inverse image of the open subtopos $U_{\widetilde{\text{et}}}$ of $X_{\widetilde{\text{et}}}$.

If $p : Y \to X$ is a proper morphism, the continuous map $p^{\text{an}} : Y^{\text{an}} \to X^{\text{an}}$ is proper and separated. To see this, Chow's lemma reduces us to the case where p is projective.

362 Let $f : X \to S$ be a compactifiable morphism of schemes of finite type over $\mathbf{C}$, and let $f = pj$ be a compactification of f.

(5.3.1.1)



The functor $p_{\text{cl}*}j_{\text{cl}!}$ is precisely the direct-image functor with proper supports $f_{\text{cl}!}$ (see Verdier [13]). If S_{cl}, and hence X_{cl}, is separated, the functor $j_{\text{cl}!}$ carries soft sheaves on X_{cl} to soft sheaves on $\overline{X}_{\text{cl}}$ [4, II 3.5.5]. Since the question is local on S, it in any case carries flasque sheaves on X_{cl} to sheaves on $\overline{X}_{\text{cl}}$ that are acyclic for the functor $p_{\text{cl}*}$. Hence (cf. Verdier [13])

$$Rf_{\text{cl}!} = Rp_{\text{cl}*}j_{\text{cl}!}.$$

5.3.2. By 5.1.2, there is a base-change isomorphism between functors from $D(X)$ to $D(\overline{X}_{\text{cl}})$:

$$\epsilon^* j_! \xrightarrow{\sim} j_{\text{cl}!}\epsilon^*.$$

363 By 4.2.12 or 4.1.5, there is a base-change morphism between functors from $D^+(\overline{X})$ to $D^+(S_{\text{cl}})$:

$$\epsilon^* Rp_* \longrightarrow Rp_{\text{cl}*}\epsilon^*.$$

By composition one obtains a *base-change morphism between functors from* $D^+_{\text{tors}}(X)$ to $D^+_{\text{tors}}(S_{\text{cl}})$ (5.1.13):

(5.3.2.1) $$\epsilon^* Rf_! \longrightarrow Rf_{\text{cl}!}\epsilon^*.$$

As in 5.2.3, one checks that it does not depend on the chosen compactification of f; it satisfies a transitivity formula of the type in 5.2.4.

Theorem 5.3.3. (Comparison theorem). *The morphism (5.3.2.1) is an isomorphism.*

By construction, it suffices to treat the case where f is proper. The functors under consideration are triangulated and "way-out" [6, I 7]. It therefore suffices to treat the case where K is a torsion sheaf placed in degree 0, and the result follows from XVI 4.1 (i).

VARIANTS 5.3.4. (i) If $\mathscr{A}$ is a sheaf of torsion rings on S_{et}, one similarly obtains an isomorphism (5.3.2.1) between functors from $D(X, f^*\mathscr{A})$ to $D(S, \epsilon^*\mathscr{A})$.
(ii) In low degrees one obtains the usual variants for sheaves of pointed sets and sheaves of ind-finite groups.

COROLLARY 5.3.5. *Let X be a separated scheme of finite type over* C *which admits a compactification in a proper scheme over* C*, and let F be a torsion sheaf on X. Then* 364

$$H^q_!(X, F) \xrightarrow{\sim} H^q_c(X_{\text{cl}}, \epsilon^* F).$$

THEOREM 5.3.6. *Let $f : X \to S$ be a compactifiable morphism of finite presentation and let A be a left noetherian torsion ring. The functor $Rf_! : D(X, A) \to D(S, A)$ carries complexes with constructible cohomology to complexes with constructible cohomology.*

The assertion is local on S, which may be assumed affine. By passage to the limit, we reduce to the case where S is noetherian. Complexes with constructible cohomology form a triangulated subcategory of $D(S, A)$. Since the functor $Rf_!$ is "way-out," we reduce to verifying that if $\mathscr{F}$ is a constructible sheaf of A-modules, then the sheaves $R^q f_! \mathscr{F}$ are constructible. Write $f = pj$, with p proper and j an open immersion. Then $R^q f_! \mathscr{F} = R^q p_*(j_! \mathscr{F})$. The sheaf $j_! \mathscr{F}$ is constructible, and the conclusion follows from XIV 1.1.

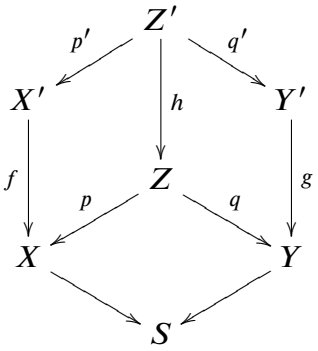
5.3.7. In low degrees, there are analogous statements for direct images with proper supports of a constructible sheaf of ind-finite sets or groups.

COROLLARY 5.3.8. *Let X be a separated scheme of finite type over a separably closed field k, and let $\mathscr{F}$ be a constructible torsion sheaf on X. Suppose that X can be compactified in a proper scheme over k.*[64] *Then the groups $H^q_c(X, \mathscr{F})$ are finite.* 365

5.4. Künneth formula.

5.4.1. Let there be a commutative diagram (0.11) of sites

(5.4.1.1)



.

Suppose S is ringed by a sheaf of rings $\mathscr{A}$ with center $\mathscr{C}$; we again denote by $\mathscr{A}$ and $\mathscr{C}$ the inverse images of these sheaves on the sites of (5.4.1.1).

If K is a sheaf of right $\mathscr{A}$-modules on X and L a sheaf of left $\mathscr{A}$-modules on Y, set

$$K \underset{\mathscr{A}}{\boxtimes} L = p^*K \underset{\mathscr{A}}{\otimes} q^*L$$

(external tensor product). Similarly for K (resp. L) on X' (resp. Y'). Now let K be a sheaf of right $\mathscr{A}$-modules on X' and L a sheaf of left $\mathscr{A}$-modules on Y'. The base-change arrows (2.1.3, XII 4.1) define

(5.4.1.2)

$$\begin{cases} p^* f_* K \longrightarrow h_* p'^* K \\ q^* g_* L \longrightarrow h_* q'^* L, \end{cases}$$

[64] N.D.E.: this hypothesis is automatically satisfied. See la N.D.E. (43) page 639.

366 and hence, by taking their tensor product, an arrow

$$f_*K \boxtimes_{\mathscr{A}} g_*L \longrightarrow h_*p'^*K \otimes_{\mathscr{A}} h_*q'^*L \longrightarrow h_*(K \boxtimes_{\mathscr{A}} L). \tag{5.4.1.3}$$

Suppose that the functors

$$\begin{aligned} Rf_* &: D^+(X', \mathscr{A}^\circ) \longrightarrow D^+(X, \mathscr{A}^\circ) \\ Rg_* &: D^+(Y', \mathscr{A}) \longrightarrow D^+(Y, \mathscr{A}) \\ Rh_* &: D^+(Z', \mathscr{C}) \longrightarrow D^+(Z, \mathscr{C}) \end{aligned}$$

have finite cohomological dimension and therefore extend to the unbounded derived categories. For $K \in \operatorname{Ob} D^-(X', \mathscr{A}^\circ)$ and $L \in \operatorname{Ob} D^-(Y', \mathscr{A})$, one is then led to define a *Künneth arrow*

$$Rf_*K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} Rg_*L \longrightarrow Rh_*(K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L). \tag{5.4.1.4}$$

We define it only when the sites under consideration have enough points, so that 4.2.10 and 4.2.8 may be used. By 4.2.10, it suffices to define it for a pair (K, L) left adapted to the functor $(K, L) \mapsto K \boxtimes_{\mathscr{A}} L$, where K and L are bounded above and right adapted to the functors f_* and g_*.

Let K_1 be a flat resolution of f_*K, let L_1 be a flat resolution of g_*L, and let $u : K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L \to M$ be a resolution of $K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L$ adapted to the functor Rh_*. We have morphisms

$$\begin{aligned} Rf_*K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} Rg_*L \simeq K_1 \boxtimes_{\mathscr{A}} L_1 \longrightarrow f_*K \boxtimes_{\mathscr{A}} g_*L \longrightarrow \\ \longrightarrow h_*(K \boxtimes_{\mathscr{A}} L) \longrightarrow h_*M \simeq Rh_*(K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L) \end{aligned}$$

and define (5.4.1.4) to be their composite.

367 This arrow has the following characteristic property, whose proof is entirely analogous to that of 4.2.13:

. Let $K \in \operatorname{Ob} C^-(X', \mathscr{A}^\circ)$ and $L \in \operatorname{Ob} C^-(Y', \mathscr{A})$. Set $K_1 = f_*K$, $L_1 = g_*L$, and $M_2 = K \boxtimes_{\mathscr{A}} L$. The following diagram is then commutative:

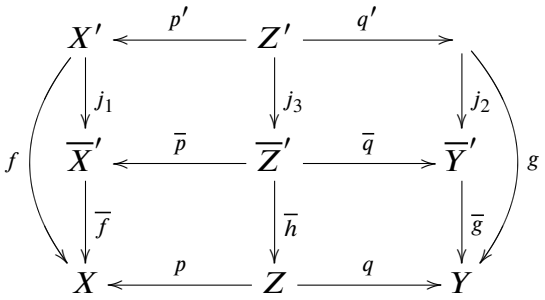
$$\begin{array}{ccccccc} Rf_*K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} Rg_*L & & & & & & Rh_*(K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L) \\ \uparrow & & & & & & \downarrow \\ K_1 \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L_1 & \longrightarrow & K_1 \boxtimes_{\mathscr{A}} L_1 & \longrightarrow & h_*M_2 & \longrightarrow & Rh_*M_2. \end{array}$$

There is an analogous commutativity (cf. 4.2.13) starting instead from morphisms $K_1 \to f_*K$, $L_1 \to g_*L$, and $K \boxtimes_{\mathscr{A}} L \to M_2$.

The commutativity in 5.4.1.5 makes it, in principle, easy to verify the compatibility properties of the Künneth morphism.

5.4.2. Let (5.4.1.1) be a commutative diagram of schemes, and let $\mathscr{A}$ be a torsion sheaf on S. Suppose that f and g are compactifiable (3.2.1), that $Z = X \times_S Y$, and that $Z' = X' \times_S Y'$. Factor f and g as a proper morphism and an open immersion; this

gives a diagram



in which $\overline{Z}' = \overline{X}' \times_S \overline{Y}'$. For complexes K and L on X' and Y', there is an isomorphism 368
of an immediate nature

$$j_{3!}(K \boxtimes_{\mathscr{A}} L) \xrightarrow{\sim} j_{1!}K \boxtimes_{\mathscr{A}} j_{2!}L. \tag{5.4.2.1}$$

By "composing" the inverse of this isomorphism with the Künneth arrow (5.4.1.4) relative to $(\overline{f}, \overline{g}, \overline{h}, j_{1!}K, j_{2!}L)$, one obtains, *for* $K \in \mathrm{Ob}\, D^-(X', \mathscr{A})$ *and* $L \in \mathrm{Ob}\, D^-(Y', \mathscr{A})$, *a Künneth arrow*

$$Rf_!K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} Rg_!L \longrightarrow Rh_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L), \tag{5.4.2.2}$$

which one checks is independent of the chosen compactifications.

THEOREM 5.4.3. *Under the hypotheses of 5.4.2, the Künneth arrow for cohomology with proper supports (5.4.2.2) is an isomorphism.*

PROOF. (a) By construction, it suffices to verify the result when f and g (and hence also h) are proper.

(b) Let there be a commutative diagram

$$\begin{array}{ccccc} X_1 & \xrightarrow{p_1} & S_1 & \xleftarrow{q_1} & Y_1 \\ \downarrow & & \downarrow & & \downarrow \\ X & \xrightarrow{p} & S & \xleftarrow{q} & Y. \end{array} \tag{5.4.3.1}$$

and let $Z_1 = X_1 \times_{S_1} Y_1$, $X_1' = X' \times_X X_1$, $Y_1' = Y' \times_Y Y_1$, and $Z_1' = X_1' \times_{S_1} Y_1' = Z' \times_{Z_1} Z$, so that there is a morphism of diagrams (5.4.1.1),

$$(S_1, X_1, Y_1, Z_1, X_1', Y_1', Z_1') \longrightarrow (S, X, Y, Z, X', Y', Z')$$

all of whose arrows will be denoted uniformly by C.

We shall admit that: 369

LEMMA 5.4.3.2. *The following diagram is commutative:*

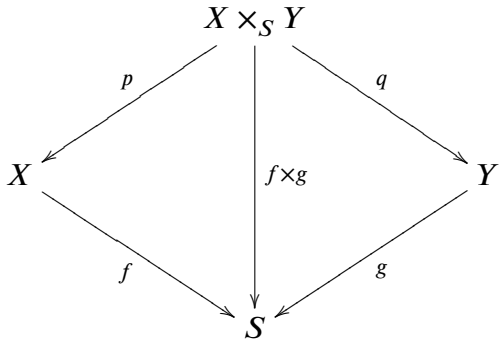
$$\begin{array}{ccccc} C^*(Rf_!K \overset{L}{\boxtimes}_{\mathscr{A}} Rg_!L) & = & C^*Rf_!K \overset{L}{\boxtimes}_{\mathscr{A}} C^*Rg_!L & \xrightarrow{\sim} & Rf_{1!}C^*K \overset{L}{\boxtimes}_{\mathscr{A}} Rg_{1!}C^*L \\ \downarrow & & & & \downarrow \\ C^*(Rh_!(K \overset{L}{\boxtimes}_{\mathscr{A}} L)) & \xrightarrow{\sim} & Rh_{1!}(C^*(K \overset{L}{\boxtimes}_{\mathscr{A}} L)) & = & Rh_{1!}(C^*K \overset{L}{\boxtimes}_{\mathscr{A}} C^*L). \end{array}$$

In this diagram, the horizontal arrows are base-change arrows and the vertical arrows are the Künneth arrows for the two diagrams (5.4.1.1) under consideration.

To prove 5.4.3, it suffices to prove that the Künneth arrow is an isomorphism at every geometric point of Z. By 5.4.3.2, it suffices to verify this after any base change (5.4.3.1) such that S_1 is the spectrum of an algebraically closed field and p_1, q_1 are isomorphisms.

After such a base change, diagram (5.4.1.1) reduces to a diagram

(5.4.3.3)



and the Künneth arrow reduces to an arrow (5.4.4.1)

$$k : Rf_!K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} Rg_!L \longrightarrow R(f \times g)_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L). \tag{5.4.3.4}$$

370 (c) Let (5.4.3.3) be a diagram in which f and g are proper. We shall admit that:

Lemma 5.4.3.5. *The following diagram is commutative:*

$$\begin{array}{ccc} Rf_!K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} Rg_!L & \longrightarrow & R(f \times g)_!(K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} L) \\ \downarrow & & \| \\ Rf_!(K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} f^*Rg_!L) & \longrightarrow & Rf_!Rp_!(p^*K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} q^*L). \end{array}$$

In this diagram, the horizontal arrows are the arrow of Künneth and the base-change arrow. The vertical arrows are a base-change arrow by f and a transitivity isomorphism. The theorem 5.4.3 follows from this lemma and 4.3.1.

The diligent reader may verify that this lemma remains valid if f and g are not necessarily proper.

5.4.4. Let S be a scheme, let $f_i : X'_i \to X_i$ $(i \in I)$ be a finite family of compactifiable morphisms of S-schemes, let X be the product over S of the X_i, let X' be the product over S of the X'_i, and let $f : X' \to X$ be the product of the f_i. If S is ringed by $\mathscr{A}$ as in 5.4.1, and the X_i and X'_i are ringed by the inverse image of $\mathscr{A}$, there is more generally an isomorphism

$$\overset{\mathbf{L}}{\underset{i\in I}{\boxtimes}} Rf_{i!}(K_i) \xrightarrow{\sim} Rf_!(\overset{\mathbf{L}}{\underset{i\in I}{\boxtimes}} K_i), \tag{5.4.4.1}$$

for $K_i \in \mathrm{Ob}\, D^-(X'_i, \mathscr{A})$. Neither the definition of this isomorphism nor that of the two sides of (5.4.4.1) *requires* the choice of a total order on I.

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5.5. Cohomology of symmetric powers. The main result of this n° is Theorem 5.5.21, which in particular gives the cohomology with proper supports of a symmetric power of a scheme X over an algebraically closed field k, for suitable coefficients (for example $\mathbf{Z}/n\mathbf{Z}$), in terms of the cohomology with proper supports of X. The analogue of this

theorem in semisimplicial topology is well known, at least for cohomology with coefficients in a commutative ring A; it can be verified at the level of semisimplicial cochains. Indeed, if X_* is a semisimplicial set, there is a canonical continuous bijection from the symmetric $n-th$ power $(X_*)^n/\mathfrak{S}_n$ of X_* to the n^{th} symmetric power of the geometric realization of X_*, and this bijection induces an isomorphism on cohomology.

5.5.1. Recall some properties (due to D. Lazard) of flat modules, some properties (due to Roby) of the functors TS^n and T^n, and some properties (due to Dold and Puppe) of the derived functors of nonadditive functors.

. *Every flat module over a ring A is a filtered direct limit of finitely generated free modules* (Lazard [7, I 1.2]). This result implies the following statements:

. The functor $\varinjlim$ is an equivalence from the category of ind-objects (I 8.1) of the category of finitely generated free (resp. finitely generated projective) A-modules to the category of flat A-modules.

. If A and B are two rings, every functor $T : \mathrm{Mod}(A) \to \mathrm{Mod}(B)$ that carries finitely generated free modules to flat modules and commutes with filtered direct limits carries flat modules to flat modules (Lazard [7, I 1.5]). 372

. Every exact sequence of A-modules, with L'' flat,

$$0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0$$

is a filtered inductive limit of split exact sequences (Lazard [7, I 2.3]).

5.5.2. Let A be a commutative ring and M an A-module.

. For every integer n, denote by $TS^n(M)$ (or $TS^n_A(M)$) the module of symmetric tensors of degree n of M. Denoting by $\mathfrak{S}_n$ the symmetric group of degree n, by definition one has

$$TS^n(M) = (\bigotimes^n M)^{\mathfrak{S}_n}.$$

If B is an A-algebra, then $TS^n(B)$ is a subalgebra of $\bigotimes^n B$. If, moreover, B is commutative, then $\mathrm{Spec}(TS^n(B))$ is by definition the n^{th} symmetric power of $\mathrm{Spec}(B)$ over $\mathrm{Spec}(A)$:

$$\mathrm{Spec}(TS^n_A(B)) = \mathrm{Sym}^n_{\mathrm{Spec}(A)}(\mathrm{Spec}(B)) \underset{\text{dfn}}{=} (\mathrm{Spec}(B)/\mathrm{Spec}(A))^n/\mathfrak{S}_n.$$

. Recall (Roby [10, I 2 p. 219]) that a *polynomial law* f from an A-module M to an A-module N is the datum, for every commutative A-algebra A', of a map $f_{A'} : M \otimes_A A' \to N \otimes_A A'$, these maps being such that, for every homomorphism of A-algebras $u : A' \to A''$, the diagram

$$\begin{array}{ccc} M \otimes_A A' & \xrightarrow{f_{A'}} & N \otimes_A A' \\ \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle u} \\ M \otimes_A A'' & \xrightarrow{f_{A''}} & N \otimes_A A'' \end{array}$$

is commutative. 373

A polynomial law f is said to be *homogeneous of degree n* (Roby [10, I 8 p. 226]) if, for every $a' \in A'$, one has identically

$$f_{A'}(a'x) = a'^n f_{A'}(x).$$

Such a polynomial law will also be called an n^{ic} *map* (quadratic, cubic, ...); their set will be denoted by

$$\mathrm{Hom}\, n^{\text{ique}}(M, N).$$

More generally, if $(M_i)_{i\in I}$ is a finite family of A-modules and $\underline{n} = (n_i)_{i\in I}$ a family of integers, a multihomogeneous polynomial law of multidegree $\underline{n}$ from the modules M_i to N is a polynomial law from $\oplus M_i$ to N satisfying identically, for $a_i \in A'$,

$$f_{A'}((a_i x_i)_{i\in I}) = \prod_{i\in I} a_i^{n_i} \cdot f_{A'}((x_i)_{i\in I}).$$

We shall denote by $\operatorname{Hom}\underline{n}^{\text{ique}}((M_i), N)$ the set of these polynomial laws.

. For every module M, the covariant functor from A-modules to sets

$$N \longmapsto \operatorname{Hom} n^{\text{ique}}(M, N)$$

374 is represented by an A-module $\Gamma^n(M)$ (or $\Gamma^n_A(M)$), equipped with a *universal n^{ic} map* $\gamma^n : M \to \Gamma^n(M)$ (Roby [10, IV 1 p. 265]). Given $(M_i)_{i\in I}$ and $\underline{n}$ as in 5.5.2.2, the covariant functor

$$N \longmapsto \operatorname{Hom}\underline{n}^{\text{ique}}((M_i), N)$$

is represented by the module $\bigotimes_{i\in I} \Gamma^{n_i}(M_i)$, equipped with the *universal multihomogeneous polynomial law (of multidegree $\underline{n}$)* $(m_i) \mapsto \bigotimes_{i\in I} \gamma^{n_i}(m_i)$.

. The functors TS^n and Γ^n commute with filtered inductive limits (for the latter, Roby [10, IV 7 p. 277]). They take free modules to free modules, and hence flat modules to flat modules (5.5.1.2).

. The n^{ique} map from M to $TS^n(M)$ given by $m \mapsto \overset{n}{\otimes} m$ determines a map φ, called canonical, from $\Gamma^n(M)$ to $TS^n(M)$, satisfying

$$\varphi(\gamma^n(m)) = \overset{n}{\otimes} m.$$

This map is an isomorphism when M is finite free (Roby [10, IV 5 p. 272]), and hence also when M is flat (5.5.2.4 and 5.5.1.1).

Let M be an A-module and P an exact sequence

$$P : P_1 \overset{d}{\to} P_0 \overset{\epsilon}{\to} M \longrightarrow 0.$$

Let $P_1' = P_1 \times P_0$, and let j_0 and j_1 be the morphisms from P_1' to P_0 with coordinates (d, Id) and $(0, \mathrm{Id})$. The sequence

$$\Gamma^n(P_1') \underset{\Gamma^n(j_1)}{\overset{\Gamma^n(j_0)}{\rightrightarrows}} \Gamma^n(P_0) \longrightarrow \Gamma^n(M) \longrightarrow 0$$

375 is then exact (Roby [10, IV 10 p. 284]). *When P is a flat presentation of M*, this yields a description of the functor Γ^n in terms of the functor TS^n

$$TS^n(P_1') \rightrightarrows TS^n(P_0) \to \Gamma^n(M) \to 0. \tag{5.5.2.5.1}$$

. If $M = M' \oplus M''$, the n^{ique} map

$$m \longmapsto (\gamma^{n'} m \otimes \gamma^{n''} m)_{n'+n''=n}$$

defines an isomorphism, called canonical (Roby [10, III 9 p. 262])

$$\Gamma^n(M) \xrightarrow{\sim} \bigoplus_{n'+n''=n} \Gamma^{n'}(M) \otimes \Gamma^{n''}(M). \tag{5.5.2.6.1}$$

If Σ is an exact sequence

$$\Sigma : 0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0,$$

the module $\Gamma^n(M)$ is equipped with a canonical (decreasing) filtration and morphisms

$$\Gamma^i(M'') \otimes \Gamma^{n-i}(M') \longrightarrow \mathrm{Gr}^i(\Gamma^n(M)). \tag{5.5.2.6.2}$$

These data are characterized by being functorial in Σ and satisfying the evident compatibility with (5.5.2.6.1) when Σ is split. The arrows (5.5.2.6.2) are isomorphisms when Σ is split, and hence also when M'' is flat (5.5.1.4).

. The functor Γ^n commutes with extension of scalars (Roby [10, III 8 p. 262]) (the analogous property for TS^n is false in general).

. If $u : M_1 \times M_2 \to N$ is a bilinear map, then $\gamma^n \circ u$ is a bihomogeneous polynomial law of bidegree (n, n) and therefore determines (5.5.2.3) $u^n : \Gamma^n(M_1) \otimes \Gamma^n(M_2) \to \Gamma^n(N)$ satisfying $u^n(\gamma^n x \otimes \gamma^n y) = \gamma^n u(x, y)$. In particular, if B is an A-algebra, multiplication 376
$B \times B \to B$ induces a product on $\Gamma^n(B)$. This product makes $\Gamma^n(B)$ an A-algebra, and the canonical map from $\Gamma^n(B)$ to $TS^n(B)$ is an algebra homomorphism.

5.5.3. Let $\mathscr{A}$ be an abelian category. We shall denote by $\mathrm{ssimpl}^+(\mathscr{A})$ the category of co-semisimplicial objects[65] in $\mathscr{A}$, by $C^{\geq 0}(\mathscr{A})$ the category of differential complexes K with $K^i = 0$ for $i < 0$, and by $K^{\geq 0}(\mathscr{A})$ the image of $C^{\geq 0}(\mathscr{A})$ in $K(\mathscr{A})$. If $K \in \mathrm{Ob}\, \mathrm{ssimpl}^+(\mathscr{A})$, the associated normalized complex $N(K) \in \mathrm{Ob}\, C^{\geq 0}(\mathscr{A})$ is defined by

$$\begin{cases} N(K)^n = \text{the intersection of the kernels of the degeneracy operators} \\ \qquad\qquad \text{from } K^n \text{ to } K^{n-1} \\ d = \sum(-1)^i \partial_i. \end{cases} \tag{5.5.3.1}$$

If K also denotes the complex with components K^n and differential $d = \sum(-1)^i \partial_i$, then $N(K)$ is a direct-summand subcomplex of K, and $K/N(K)$ is homotopic to zero.

By Dold-Puppe [1], the functor N is an equivalence of categories

$$N : \mathrm{ssimpl}^+(\mathscr{A}) \longrightarrow C^{\geq 0}(\mathscr{A}).$$

This equivalence and its inverse take homotopic morphisms to homotopic morphisms. Moreover, for every $K \in \mathrm{Ob}\, \mathrm{ssimpl}^+(\mathscr{A})$, K^n is a direct sum of copies of $N(K)^i$ $(i \leq n)$; if $\mathbf{P}$ is a property of objects of $\mathscr{A}$, stable under finite direct sums and direct summands, it follows that the K^i have property $\mathbf{P}$ for $i \leq n$ if and only if the $N(K)^i$ have this property for $i \leq n$.

The equivalence N is used by Dold and Puppe to extend every functor (not necessarily additive) $T : \mathscr{A} \to \mathscr{B}$ from $\mathscr{A}$ to an arbitrary category $\mathscr{B}$ to a functor $C^{\geq 0}(T)$ (or simply T) from $C^{\geq 0}(\mathscr{A})$ to $C^{\geq 0}(\mathscr{B})$ 377

$$C^{\geq 0}(T) = NTN^{-1}. \tag{5.5.3.2}$$

The functor $C^{\geq 0}(T)$ passes to the quotient to define

$$T \quad \text{or} \quad K^{\geq 0}(T) : K^{\geq 0}(\mathscr{A}) \longrightarrow K^{\geq 0}(\mathscr{B}). \tag{5.5.3.3}$$

. If T is *of degree* $\leq n$, i.e. if the $(n+1)^{\text{th}}$ "cross-effect" vanishes (Dold-Puppe [1, 4.18 p. 232]), then, for every $K \in \mathrm{Ob}\, C^{\geq 0}(\mathscr{A})$ such that $K^i = 0$ for $i > m$, one has $[T(K)]^i = 0$ for $i > nm$ (Dold-Puppe [1, 4.23 p. 233]).

. In particular, if T is a finite-degree functor from the category of A-modules to that of B-modules, which takes flat objects to flat objects, then the image of a bounded flat complex is a bounded flat complex.

Lemma 5.5.4. *Let $\mathscr{A}$ be an abelian category.*

(i) *The quasi-isomorphisms in $K^{\geq 0}(\mathscr{A})$ admit a calculus of left and right fractions.*

[65] N.D.E.: In modern terminology they are called *cosimplicial* objects.

(ii) *The evident functor is an equivalence between the category obtained from* $K^{\geq 0}(\mathscr{A})$ *by inverting the quasi-isomorphisms and the category* $D^{\geq 0}(\mathscr{A})$.

378 The functor $\tau_{\geq 0} : C(\mathscr{A}) \to C^{\geq 0}(\mathscr{A})$ (1.1.13) passes to the quotient to define a functor $\tau_{\geq 0} : K(\mathscr{A}) \to K^{\geq 0}(\mathscr{A})$, left adjoint to the inclusion of $K^{\geq 0}(\mathscr{A})$ in $K(\mathscr{A})$. The functor $\tau_{\geq 0}$ takes quasi-isomorphisms to quasi-isomorphisms, and the adjunction arrow $X \to \tau_{\geq 0}(X)$ is a quasi-isomorphism if and only if $H^i(X) = 0$ for $i < 0$. Lemma 5.5.4 therefore follows from Verdier [11] and parts (i) and (ii) of the following lemma, whose proof is left to the reader:

Lemma 5.5.4.1. *Let* K *be a category,* K' *a full subcategory,* S *a set of morphisms of* K *equal to the inverse image of the set of isomorphisms of* $K(S^{-1})$ *(cf.* [2, Chap. 1]*), and* S' *its trace in* K'. *Suppose that the inclusion* i *of* K' *in* K *has a left adjoint* τ, *and that* $\tau(S) \subset S'$.

(i) *The evident functor from* $K'(S'^{-1})$ *to* $K(S^{-1})$ *is fully faithful, and has* τ *as a left adjoint.*

(ii) *If* $K(S^{-1})$ *admits a calculus of left fractions (resp. right fractions), then* $K'(S'^{-1})$ *has the same property.*

If the inclusion i *does not necessarily have a left adjoint, and if* τ *denotes the not everywhere defined functor left adjoint to* i, *one has:*

(iii) *If* $K(S^{-1})$ *admits a calculus of left fractions (resp. right fractions) and if, for every* $u \in S$ *with target (resp. source) in* K', $\tau(u)$ *is defined and belongs to* S', *then* $K'(S'^{-1})$ *admits a calculus of left fractions (resp. right fractions), and the functor* $K'(S'^{-1}) \to K(S^{-1})$ *is fully faithful.*

379 5.5.5. Let $T : \mathscr{A} \to \mathscr{B}$ be a functor between abelian categories. As in 1.2.1, one defines the derived functors of T

$$LT : D^{\geq 0}(\mathscr{A}) \longrightarrow \mathrm{Pro}(D^{\geq 0}(\mathscr{B}))$$
$$RT : D^{\geq 0}(\mathscr{A}) \longrightarrow \mathrm{Ind}(D^{\geq 0}(\mathscr{B}))$$

by the formulas

$$LT(K) = \text{“}\varprojlim_{s : K' \to K}\text{”}\ T(K')$$
$$RT(K) = \text{“}\varinjlim_{s : K \to K'}\text{”}\ T(K'),$$

where s ranges over the filtered category of quasi-isomorphisms with target (resp. source) K.

We say that T is *left (resp. right) derivable at* K *if* $LT(K)$ ($RT(K)$) is representable. We say that K is *left (resp. right) adapted* to T (or, by abuse of language, adapted to LT (resp. RT)) if the following canonical arrow is an isomorphism:

$$LT(K) \xrightarrow{\sim} TK \text{ (resp. } TK \xrightarrow{\sim} RT(K)).$$

Proposition 5.5.6. *Let* F *be a functor from the category of modules over a ring* A *to an abelian category* $\mathscr{B}$. *Suppose that filtered inductive limits in* $\mathscr{B}$ *are representable and exact, and that* F *commutes with filtered inductive limits. Then every bounded complex* $M^{\cdot} \in \mathrm{Ob}\, K^{\geq 0}(A)$ *of flat modules is adapted to* LF.

380 The proposition follows immediately from the definitions, the following lemma, which will be proved in 5.5.6.4, and the fact that the quasi-isomorphisms $u : N^{\cdot} \to M^{\cdot}$,

with $N^\cdot \in C^{b,\geq 0}(A)$ having flat components, form a cofinal set in the category of quasi-isomorphisms with target $M^\cdot$.

LEMMA 5.5.6.1. *Let $u : N^\cdot \to M^\cdot$ be a quasi-isomorphism between flat complexes that are objects of $C^{b,\geq 0}(A)$. Then $F(u)$ is a quasi-isomorphism.*

LEMMA 5.5.6.2. *Let $\mathscr{A}$ be an additive category and d an interval in $\mathbf{Z}$ that is bounded above. Denote by C^d (resp. $C^{d,b}$, resp. K^d, resp. $K^{d,b}$) the category of complexes that vanish in degrees $i \notin d$ (resp. and are bounded, resp. up to homotopy). There is an equivalence*

$$\ell : \mathrm{Ind}(C^{d,b}(\mathscr{A})) \xrightarrow{\sim} C^d(\mathrm{Ind}\,\mathscr{A});$$

the inverse of this equivalence induces a functor

$$\ell' : K^d(\mathrm{Ind}\,\mathscr{A}) \longrightarrow \mathrm{Ind}(K^{d,b}(\mathscr{A})).$$

A) Let us prove that ℓ is fully faithful. Thus, let K be a bounded complex of objects of $\mathscr{A}$ and $L = \text{“}\varinjlim\text{”}\, L_\alpha$ an ind-complex. One has

$$\mathrm{Hom}(K,L) \overset{\mathrm{dfn}}{=} \varinjlim \mathrm{Hom}(K,L_\alpha) = \varinjlim Z^0(\mathrm{Hom}^\cdot(K,L_\alpha)) \simeq Z^0 \varinjlim_\alpha \mathrm{Hom}^\cdot(K,L_\alpha)$$
$$\xrightarrow{\sim} Z^0\,\mathrm{Hom}^\cdot(K,\ell L) = \mathrm{Hom}(K,\ell L),$$

the penultimate isomorphism following from the fact that K is finite, so that $\varinjlim$ commutes with passage from the double complex to the total complex. This proves the assertion.

B) We leave it to the reader to verify (rather laboriously) that ℓ is essentially surjective by constructing, from left to right and starting in an arbitrarily chosen degree, “enough” bounded complexes mapping to a bounded-above complex of ind-objects K; and likewise to develop the following point: 381

C) If $\ell f : \ell \varinjlim K_\alpha \to \ell \varinjlim L_\beta$ is null-homotopic, $f = dH + Hd$, then, for every α, H lifts to $H^i_\alpha : K^i_\alpha \to L^{i-1}_{\beta(\alpha)}$ which, for β sufficiently large, defines a null-homotopy of $f_\alpha : K_\alpha \to L_{\beta(\alpha)}$.

. Let A be a ring, d an interval in $\mathbf{Z}$ bounded above, $P(A)$ the category of finite projective A-modules, and $\mathrm{Pl}(A)$ the category of flat A-modules. Denote by $D^{\mathrm{Tor}\subset d}(A)$ the image of $C^d(\mathrm{Pl}(A))$ in $D(A)$ and by $D^{\mathrm{parf}\subset d}(A)$ the image of $C^{d,b}(P(A))$ in $D(A)$. By 5.5.1.2, one has $\mathrm{Pl}(A) \simeq \mathrm{Ind}\,P(A)$. On the other hand, $K^{d,b}(P(A)) \simeq D^{\mathrm{parf}\subset d}(A)$. It then follows from 5.5.6.2, and from the fact that H^i commutes with filtered inductive limits, that there exists a commutative diagram of functors

$$\begin{array}{ccc}
C^d(\mathrm{Pl}(A)) & \xleftarrow{\varinjlim\ \sim} & \mathrm{Ind}\,C^{d,b}(P(A)) \\
\downarrow & & \downarrow \\
K^d(\mathrm{Pl}(A)) & \xrightarrow{\ell'} & \mathrm{Ind}\,K^{d,b}(P(A)) \\
\downarrow & & \downarrow \simeq \\
D^{\mathrm{Tor}\subset d}(A) & \xrightarrow{\ell''} & \mathrm{Ind}\,D^{\mathrm{parf}\subset d}(A).
\end{array}$$

. Let us prove 5.5.6.1. With the notation of 5.5.6.3, if d is an interval $[0,k]$, then (because F commutes with $\varinjlim$) there are commutative diagrams 382

$$\begin{array}{ccc}
K^d(\mathrm{Pl}(A)) & \xrightarrow{\ell'} & \mathrm{Ind}\, K^d(P(A)) \\
\downarrow{\scriptstyle K^{\geq 0}(F)} & & \downarrow{\scriptstyle \mathrm{Ind}(K^{\geq 0}(F))} \\
K^+(\mathscr{B}) & & \mathrm{Ind}\, K^+(\mathscr{B}) \\
\downarrow{\scriptstyle H^i} & & \downarrow{\scriptstyle \mathrm{Ind}(H^i)} \\
\mathscr{B} & \xleftarrow{\varinjlim} & \mathrm{Ind}\, \mathscr{B}.
\end{array}$$

If u is as in 5.5.6.1, then by 5.5.6.3, $\ell'(u)$ is an isomorphism, and $H^i K^{\geq 0}(F)(u)$ is therefore an isomorphism, as required.

5.5.7. Let X be a scheme over a scheme S, and let k be an integer. Suppose that the symmetric power $\mathrm{Sym}^k_S(X)$ is defined. A reader who wishes not to make this hypothesis may work in the category of algebraic spaces (Artin[66]). Ring S by a sheaf $\mathscr{A}$ of *commutative* rings. Beginning with 5.5.19, $\mathscr{A}$ will be assumed to be *torsion*. By abuse of notation, we shall *still denote by* $\mathscr{A}$ the inverse image of $\mathscr{A}$ on any S-scheme.

Let $\mathscr{F}$ be a sheaf of modules on X; the sheaf $\overset{k}{\boxtimes}\mathscr{F}$, the external tensor product of k copies of $\mathscr{F}$, is a sheaf on X^k/S, equivariant under the action of the symmetric group 383 $\mathfrak{S}_k$ on X^k/S. Passing to invariants, it defines a sheaf $TS^k_{\mathrm{ext}}(\mathscr{F})$ on $\mathrm{Sym}^k_S(X)$, called the *external symmetric tensor power of* $\mathscr{F}$.

To reason "point by point," it is useful to work with the points of $\mathrm{Sym}^k_S(X)$ described as follows.

. Let s be a geometric point of S. A point x of $\mathrm{Sym}^k_S(X)$ with values in $k(s)$ defines a geometric point of $\mathrm{Sym}^k_S(X)$. Let $(x_1 \dots x_k)$ denote a point of $(X/S)^k$ with values in $k(s)$ whose image is x, let $y_1 \dots y_\ell$ be the distinct elements of the sequence $x_1 \dots x_k$, and let n_i $(1 \leq i \leq \ell)$ be the multiplicity of y_i in this sequence $(\sum n_i = k)$.

With the notation of 5.5.7.1, and writing $\mathfrak{S}$ for a symmetric group, one has

$$TS^k_{\mathrm{ext}}(F)_x \simeq (\bigotimes_1^\ell (\bigotimes^{n_i} F_{y_i}))^{\prod_1^\ell \mathfrak{S}_{n_i}}, \tag{5.5.7.2}$$

where the tensor products are taken over $\mathscr{A}_S$. In particular, *for F flat*, by 5.5.2.5 one has

$$TS^k_{\mathrm{ext}}(F)_x \xleftarrow{\sim} \bigotimes_1^\ell \Gamma^{n_i} F_{y_i}, \tag{5.5.7.3}$$

using induction on ℓ, the flatness of the $\Gamma^{n_i} F_{y_i}$ (5.5.2.4), and the following fact: if G (resp. H) is a finite group acting on an A-module M (resp. N), and if M and N^H are flat, then $M^G \otimes_A N^H \xrightarrow{\sim} (M \otimes_A N)^{G\times H}$.

5.5.8. Let F be a module on X, and let P be a flat presentation of F:

$$P : P_1 \longrightarrow P_0 \longrightarrow F \longrightarrow 0.$$

384 Define P_1', j_0, and j_1 as in 5.5.2.5, and define $\Gamma^k_{\mathrm{ext}}(F)$ (the *external Γ functor*) to be the

[66] N.D.E.: the original text cites an unspecified work here, probably one of the two works of Artin cited in the notes on pages 730 and 743.

cokernel of the pair of arrows $(TS^k_{\mathrm{ext}}(j_0), TS^k_{\mathrm{ext}}(j_1))$. With the notation of 5.5.7.1, one deduces from (5.5.7.3) and 5.5.2.5 isomorphisms

$$\Gamma^k_{\mathrm{ext}}(F)_x \simeq \bigotimes_1^{\ell} \Gamma^{n_i}(F_{y_i}). \tag{5.5.8.1}$$

This formula shows that $\Gamma^k_{\mathrm{ext}}(F)$, which a priori depends functorially on P, depends, up to a unique isomorphism, only on F.

5.5.9. There is a functorial morphism

$$\alpha : \Gamma^k_{\mathrm{ext}}(F) \longrightarrow TS^k_{\mathrm{ext}}(F).$$

The pair (Γ, α) is characterized by the following two properties:

(i) For F flat, α is an isomorphism.

(ii) With the notation of 5.5.2.5, for every presentation

$$P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} F \longrightarrow 0$$

of F, the sequence

$$\Gamma^k_{\mathrm{ext}}(P'_1) \rightrightarrows \Gamma^k_{\mathrm{ext}}(P_0) \to \Gamma^k_{\mathrm{ext}}(F) \to 0$$

is exact.

Moreover:

(iii) The functors TS^k and Γ^k commute with filtered direct limits (by (5.5.7.2), (5.5.8.1), 5.5.2.4).

(iv) The functor Γ^k commutes with every extension $\mathscr{A} \to \mathscr{A}'$ of the base ring.

(v) The functor Γ^k commutes with every base change $S' \to S$: for every cartesian square 385

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S, \end{array}$$

one has

$$\mathrm{Sym}^k_g(g')^* \Gamma^k_{\mathrm{ext}} \simeq \Gamma^k_{\mathrm{ext}} g'^*.$$

((5.5.8.1) and 5.5.2.7).

(vi) The functor Γ^k commutes with inverse images under every *radicial* morphism $i : X' \to X$ (for example a closed immersion): one has

$$\mathrm{Sym}^k_S(i)^* \Gamma^k_{\mathrm{ext}} \simeq \Gamma^k_{\mathrm{ext}} i^*$$

(5.5.8.1).

(vii) For an *immersion* $j : X' \to X$, one has

$$\mathrm{Sym}^k_S(j)_! \Gamma^k_{\mathrm{ext}} \simeq \Gamma^k_{\mathrm{ext}} j_!$$

(5.5.8.1).

5.5.10. Let k_1, k_2, and $k = k_1 + k_2$ be three integers ≥ 0. There is then a finite morphism ( ):[67]

$$\vee_{k_1 k_2} : \mathrm{Sym}_S^{k_1}(X) \times_S \mathrm{Sym}_S^{k_2}(X) \longrightarrow \mathrm{Sym}_S^{k}(X).$$

If F_i $(i = 1, 2)$ is a module on $\mathrm{Sym}_S^{k_i}(X)$, denote by $F_1 \vee F_2$ the sheaf on $\mathrm{Sym}_S^{k}(X)$

$$F_1 \vee F_2 = \vee_{k_1 k_2 *}(F_1 \boxtimes F_2).$$

386 5.5.11. Let there be an exact sequence of modules on X

$$\Sigma : 0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0.$$

With the notation of 5.5.7.1, by (5.5.8.1) and 5.5.2.6 there is a filtration on $\Gamma^k_{\mathrm{ext}}(F)_x$, and there are morphisms

$$\sum_{\sum_1^\ell i_\alpha = i} \bigotimes_1^\ell \Gamma^{i_\alpha}(F'_{y_\alpha}) \otimes \bigotimes_1^\ell \Gamma^{n_i - i_\alpha}(F''_{y_\alpha}) \longrightarrow \mathrm{Gr}^i(\Gamma^k_{\mathrm{ext}}(F)_x).$$

The first member is precisely $(\Gamma^i_{\mathrm{ext}}(F') \vee \Gamma^{k-i}_{\mathrm{ext}}(F''))_x$. One readily checks that these data, constructed point by point, arise from a (unique) filtration with $k+1$ steps on $\Gamma^k_{\mathrm{ext}}(F)$ and from (unique) morphisms

$$\Gamma^i_{\mathrm{ext}}(F') \vee \Gamma^{k-i}_{\mathrm{ext}}(F'') \longrightarrow \mathrm{Gr}^i(\Gamma^k_{\mathrm{ext}}(F)). \tag{5.5.11.1}$$

If the exact sequence Σ is split, one has more precisely (verification point by point from (5.5.2.6.1)):

$$\Gamma^k_{\mathrm{ext}}(F' \oplus F'') \simeq \bigoplus_{k_1 + k_2 = k} \Gamma^{k_1}_{\mathrm{ext}}(F') \vee \Gamma^{k_2}_{\mathrm{ext}}(F''). \tag{5.5.11.2}$$

If the exact sequence Σ is split point by point, or if F'' is flat, the morphisms (5.5.11.1) are isomorphisms (cf. 5.5.1.4).

5.5.12. Let $u : X \to Y$ be a morphism of S-schemes. Suppose that the symmetric powers $\mathrm{Sym}_S^k(X)$ and $\mathrm{Sym}_S^k(Y)$ exist. Let F be a module on X, let G be a module on Y, and let $v \in \mathrm{Hom}(u^*G, F) \simeq \mathrm{Hom}(G, u_* F)$ be a u-homomorphism from F to G (0.2). If $\prod^n u$ is the n^{th} cartesian power of u, the homomorphism v induces a $\prod^n u$-
387 homomorphism $\overset{n}{\boxtimes} v$ from $\overset{n}{\boxtimes} F$ to $\overset{n}{\boxtimes} G$. Passing to invariants, the latter defines

$$TS^n(v) : TS^n_{\mathrm{ext}}(F) \xrightarrow[\mathrm{Sym}^n_S(u)]{} TS^n_{\mathrm{ext}}(G).$$

It follows from Definition 5.5.8 that there exists a unique $\mathrm{Sym}^n(u)$-homomorphism $\Gamma^n_{\mathrm{ext}}(v)$, functorial in (F, G, v),

$$\Gamma^n_{\mathrm{ext}}(v) : \Gamma^n_{\mathrm{ext}}(F) \xrightarrow[\mathrm{Sym}^n_S(u)]{} \Gamma^n_{\mathrm{ext}}(G) \tag{5.5.12.0}$$

[67] N.D.E.: I have not found references for the finiteness of $\vee_{k_1 k_2}$ in general.

making the following diagram commutative (cf. (0.2) for the unusual direction of the arrows):

$$\begin{array}{ccc} \Gamma^n_{\mathrm{ext}}(F) & \xrightarrow[\text{over } \mathrm{Sym}^n_S(u)]{\Gamma^n_{\mathrm{ext}}(v)} & \Gamma^n_{\mathrm{ext}}(G) \\ \uparrow{\scriptstyle \alpha}\ \text{over Id} & & \uparrow{\scriptstyle \alpha}\ \text{over Id} \\ TS^n_{\mathrm{ext}}(F) & \xrightarrow[\text{over } \mathrm{Sym}^n_S(u)]{TS^n_{\mathrm{ext}}(v)} & TS^n_{\mathrm{ext}}(G). \end{array}$$

We shall use this arrow mainly as a "symmetric Künneth arrow": for $G = u_*F$ and v the canonical arrow, the arrow $\Gamma^n_{\mathrm{ext}}(v)$ identifies with a homomorphism of modules

$$k^n : \Gamma^n_{\mathrm{ext}}(u_*F) \longrightarrow \mathrm{Sym}^n(u)_*\Gamma^n_{\mathrm{ext}}(F). \tag{5.5.12.1}$$

5.5.13. Consider the condition:

. $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ and, for every geometric point x of X, the complex K_x is homotopy equivalent to a complex of flat modules $L_x \in \mathrm{Ob}\, C^{b,\geq 0}(\mathscr{A}_x)$.

Lemma 5.5.13.2. *The subcategory $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ of $D^b(X, \mathscr{A})$ consisting of complexes of tor-dimension ≤ 0 4.1.9 is equivalent to the category obtained from the category of complexes $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ satisfying (5.5.13.1) by inverting quasi-isomorphisms. This construction may be performed indifferently by a calculus of left or right fractions.* 388

Let P be the subcategory of $K^-(X, \mathscr{A})$ consisting of complexes that are pointwise homotopy equivalent to a complex of flat modules. It is known (Verdier [11, I 2.4.2]) that $D^-(X, \mathscr{A})$ is obtained from P by inverting quasi-isomorphisms, using a calculus of right or left fractions. Apply 5.5.4.1 to the inclusion of $K^{b,\geq 0}(X, \mathscr{A}) \cap P$ into P. Indeed, if K has tor-dimension ≤ 0 and is pointwise homotopy equivalent to complexes of flat modules L_x, then $\tau_{\geq 0}(K)$ is pointwise homotopy equivalent to the $\tau_{\geq 0}(L_x)$, and the latter are still complexes of flat modules, so $\tau_{\geq 0}(K)$ satisfies 5.5.13.1.

5.5.14. By 5.5.6.1, (5.5.7.2) or (5.5.8.1) and 5.5.2.4, every quasi-isomorphism $v : K' \to K$ between complexes satisfying (5.5.13.1) induces quasi-isomorphisms

$$TS^k(v) : TS^k_{\mathrm{ext}}(K') \xrightarrow{\approx} TS^k_{\mathrm{ext}}(K)$$
$$\Gamma^k(v) : \Gamma^k_{\mathrm{ext}}(K') \xrightarrow{\approx} \Gamma^k_{\mathrm{ext}}(K).$$

It follows that the functors Γ^k and TS^k are derivable (5.5.5) on the subcategory $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ of $D^b(X, \mathscr{A})$, and that complexes satisfying 5.5.13.1 are left adapted to these functors. By 5.5.2.4, Γ^k and TS^k induce functors 389

$$L TS^k : D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A}) \longrightarrow D^{b,\mathrm{tor}\leq 0}(\mathrm{Sym}^k_S(X), \mathscr{A}) \tag{5.5.14.1}$$
$$L\Gamma^k : D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A}) \longrightarrow D^{b,\mathrm{tor}\leq 0}(\mathrm{Sym}^k_S(X), \mathscr{A}). \tag{5.5.14.2}$$

Moreover, by 5.5.9 (i), the canonical morphism α (5.5.9) induces an isomorphism between the functors (5.5.14.1) and (5.5.14.2):

$$\alpha : L\Gamma^k \xrightarrow{\sim} L TS^k. \tag{5.5.14.3}$$

. The isomorphisms of 5.5.9 (iv), (v), (vi), and (vii) pass to the derived category and define isomorphisms:

(iv′) For an $\mathscr{A}$-algebra $\mathscr{A}'$, one has

$$L\Gamma^k_{\mathrm{ext}_{\mathscr{A}}}(K) \overset{\mathbf{L}}{\otimes} \mathscr{A}' \simeq L\Gamma^k_{\mathrm{ext}_{\mathscr{A}'}}(K \overset{\mathbf{L}}{\underset{\mathscr{A}}{\otimes}} \mathscr{A}').$$

(v′) For every base change $g : S' \to S$, one has

$$\mathrm{Sym}^k_g(g')^* L\Gamma^k_{\mathrm{ext}} K \simeq L\Gamma^k_{\mathrm{ext}}(g'^* K).$$

(vi′) For every *radicial* morphism $i : X' \to X$, one has

$$\mathrm{Sym}^k_S(i)^* L\Gamma^k_{\mathrm{ext}} K \simeq L\Gamma^k_{\mathrm{ext}} i^* K.$$

(vii′) For an *immersion* $j : X' \to X$, one has

$$\mathrm{Sym}^k_S(j)_! L\Gamma^k_{\mathrm{ext}} \simeq L\Gamma^k_{\mathrm{ext}} j_!.$$

5.5.15. Let there be an exact sequence of flat complexes in $C^{\geq 0}(X, \mathscr{A})$,

$$0 \longrightarrow K' \longrightarrow K \longrightarrow K'' \longrightarrow 0,$$

and apply the functor Γ^k_{ext} (5.5.3). One finds (as in 5.5.11) that the semisimplicial module
390 $\Gamma^k_{\mathrm{ext}}(N^{-1}K)$ on $\mathrm{Sym}^k_S(X)$ has a filtration whose successive quotients are the semisimplicial modules $\Gamma^{k'}_{\mathrm{ext}}(N^{-1}K') \vee \Gamma^{k''}_{\mathrm{ext}}(N^{-1}K'')$ $(k' + k'' = k)$.

Passing to the derived category, one finds that $L\Gamma^k_{\mathrm{ext}}(K)$ is represented by a filtered complex whose successive quotients represent the objects $L\Gamma^i_{\mathrm{ext}}(K') \overset{\mathbf{L}}{\vee} L\Gamma^{k-i}_{\mathrm{ext}}(K'')$ of the derived category (apply 5.5.2.4 and the Eilenberg–Zilber–Cartier theorem (isomorphism in $\mathscr{K}$ between a bisemisimplicial object and its diagonal: Dold-Puppe [1, 2.9 p. 213]).

In the derived category, this construction can already be made from an exact sequence of complexes satisfying 5.5.13.1.

* In more intrinsic terms, let Δ be a "*true triangle*" of objects of $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$, i.e. an object of the "filtered derived category" with $\mathrm{Gr}^i(\Delta) = 0$ for $i \neq 0, 1$ and $\mathrm{Gr}^i(\Delta) \in \mathrm{Ob}\, D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ (Deligne, in the Appendix to Verdier [12], or 2.5.1, 2.5.2).

The preceding construction then gives an object $L\Gamma^k_{\mathrm{ext}}(\Delta)$ of the filtered derived category of $\mathrm{Sym}^k_S(X)$, isomorphisms

$$\mathrm{Gr}^i(L\Gamma^k_{\mathrm{ext}}(\Delta)) \simeq L\Gamma^i_{\mathrm{ext}}(\mathrm{Gr}^1(\Delta)) \overset{\mathbf{L}}{\vee} L\Gamma^{k-i}_{\mathrm{ext}}(\mathrm{Gr}^0(\Delta))$$

and, denoting by $|\quad|$ the underlying complex of a filtered complex, an isomorphism

$$|L\Gamma^k_{\mathrm{ext}}(\Delta)| \simeq L\Gamma^k_{\mathrm{ext}}(|\Delta|)._*$$

Lemma 5.5.16. *Let N be an integer. The category $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ is obtained from the category of complexes $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ satisfying conditions* (i) *and* (ii) *below by inverting quasi-isomorphisms:*

391 (i) *condition (5.5.13.1).*

(ii) *The complex K is adapted to every direct-image functor of cohomological dimension $\leq N$, relative to an isomorphism of topoi $f : \widetilde{X}_{\mathrm{et}} \to T$.*

The existence of a calculus of right fractions follows from 5.5.13 and from the fact that, by truncated canonical flasque resolutions (4.2.9), every complex K satisfying (i) maps to a complex satisfying (i) and (ii).

5.5.17. Let $u : X \to Y$ be a morphism of S-schemes. Suppose that $\mathrm{Sym}^n_S(X)$ and $\mathrm{Sym}^n_S(Y)$ exist, that there is an integer N such that the functor u_* has dimension $\leq N$ on the category of $\mathscr{A}$-modules, and that the functor Ru_* carries bounded complexes of tor-dimension ≤ 0 to complexes of tor-dimension ≤ 0. The latter two conditions hold automatically if u is proper, $\mathscr{A}$ is torsion, and Y is coherent (i.e. quasi-compact and quasi-separated) (5.2.8.1, 5.2.10).[68]

Let $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ satisfy conditions (i) and (ii) of 5.5.16, with N as above. By 5.5.14, the arrow

$$\beta : L\Gamma^n_{\mathrm{ext}}(K) \xrightarrow{\sim} \Gamma^n_{\mathrm{ext}}(K)$$

is a quasi-isomorphism. Similarly, the arrow

$$\gamma : u_*K \xrightarrow{\sim} Ru_*K$$

is a quasi-isomorphism.

Define the *symmetric Künneth arrow* from $L\Gamma^n_{\mathrm{ext}}Ru_*(K)$ to $R\,\mathrm{Sym}^n(u)_*L\Gamma^n_{\mathrm{ext}}(K)$ to be the composite 392

$$\begin{aligned}(5.5.17.1)\quad k^n : L\Gamma^n_{\mathrm{ext}}Ru_*K \xleftarrow[\gamma]{\sim} L\Gamma^n_{\mathrm{ext}}u_*K \to \Gamma^n_{\mathrm{ext}}u_*K \xrightarrow[5.5.12.1]{} \mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K)\\ \longrightarrow R\,\mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K) \xleftarrow[\beta]{\sim} R\,\mathrm{Sym}^n_S(u)_*L\Gamma^n_{\mathrm{ext}}(K).\end{aligned}$$

It follows from 5.5.16 that the morphisms (5.5.17.1) define a morphism between functors from $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ to $D^+(\mathrm{Sym}^n_S(Y), \mathscr{A})$:

$$(5.5.17.2)\qquad k^n : L\Gamma^n_{\mathrm{ext}}Ru_* \longrightarrow R\,\mathrm{Sym}^n_S(u)_*L\Gamma^n_{\mathrm{ext}},$$

also called the *symmetric Künneth morphism.*

This morphism is characterized by Proposition 5.5.18 below.

. Let S, X, Y, and u be as above, let $K \in \mathrm{Ob}\, C^{b,\geq 0}(X, \mathscr{A})$ be a complex of tor-dimension ≤ 0, let $K_1 \in \mathrm{Ob}\, C^{b,\geq 0}(Y, \mathscr{A})$, and let $K_2 \in \mathrm{Ob}\, C^b(\mathrm{Sym}^n_S(X), \mathscr{A})$. Let v be a u-homomorphism (0.2) from K to K_1, and let t be a homomorphism (in the usual sense, as are all those that follow) from $\Gamma^n_{\mathrm{ext}}(K)$ to K_2. There are morphisms

a) $\Gamma^n_{\mathrm{ext}}(v) : \Gamma^n_{\mathrm{ext}}(K_1) \to \mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K)$ (5.5.12.0)

b) $\mathrm{Sym}^n_S(u)_*(t) : \mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K) \to \mathrm{Sym}^n_S(u)_*K_2$,

whose composite

$$\varphi' : \Gamma^n_{\mathrm{ext}}(K_1) \longrightarrow \mathrm{Sym}^n_S(u)_*K_2,$$

defines, by composition,

$$\varphi : L\Gamma^n_{\mathrm{ext}}(K_1) \longrightarrow R\,\mathrm{Sym}^n(u)_*K_2.$$

Moreover, v and t induce 393

$$\begin{cases} v' : K_1 \longrightarrow Ru_*K \\ t' : L\Gamma^n_{\mathrm{ext}}K \longrightarrow K_2. \end{cases}$$

Then:

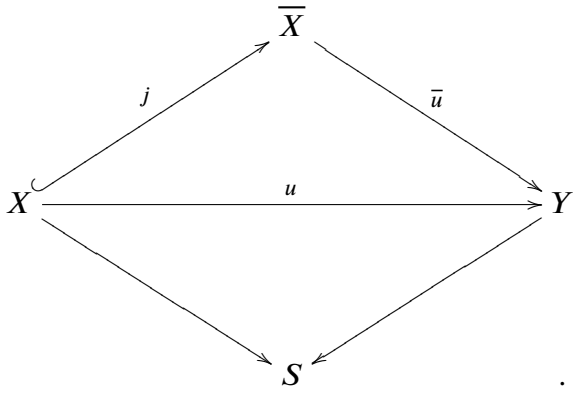
[68] N.D.E.: the coherence of Y may be replaced by the weaker condition that the dimensions of the fibers of u are bounded.

PROPOSITION 5.5.18. *Under the hypotheses of 5.5.17.3, the following diagram is commutative in* $D^+(\mathrm{Sym}^n_S(Y), \mathscr{A})$:

$$\begin{array}{ccc} L\Gamma^n_{\mathrm{ext}} Ru_* K & \xrightarrow[\text{(symmetric Künneth)}]{k^n} & R\,\mathrm{Sym}^n_S(u)_* L\Gamma^n_{\mathrm{ext}} K \\ \uparrow{\scriptstyle L\Gamma^n_{\mathrm{ext}}(v')} & & \downarrow{\scriptstyle R\,\mathrm{Sym}^n_S(u)_*(t')} \\ L\Gamma^n_{\mathrm{ext}} K_1 & \xrightarrow{\varphi} & R\,\mathrm{Sym}^n_S(u)_* K_2. \end{array}$$

The proof is entirely similar to that of 5.4.1.5 and is left to the reader.

5.5.19. *Henceforth* $\mathscr{A}$ *is assumed to be torsion.* Let $u : X \to Y$ be a compactifiable morphism of S-schemes:



394 Suppose that $\mathrm{Sym}^n_S(X)$, $\mathrm{Sym}^n_S(\overline{X})$, and $\mathrm{Sym}^n_S(Y)$ exist. For every module F on X, one then has

$$\Gamma^n_{\mathrm{ext}}(j_! F) \simeq \mathrm{Sym}^n_S(j)_! \Gamma^n_{\mathrm{ext}}(F), \tag{5.5.19.1}$$

and this isomorphism passes immediately to the derived categories (5.5.14.4 (vii')). Define the *symmetric Künneth morphism* between functors from $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ to $D^{b,\mathrm{tor}\leq 0}(\mathrm{Sym}^n_S(Y), \mathscr{A})$[69]

$$k^n_u \quad \text{or} \quad k^n : L\Gamma^n_{\mathrm{ext}} Ru_! \longrightarrow R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} \tag{5.5.19.2}$$

to be the composite

$$\begin{aligned} L\Gamma^n_{\mathrm{ext}} Ru_! \overset{\mathrm{dfn}}{\simeq} L\Gamma^n_{\mathrm{ext}} R\bar{u}_* j_! &\xrightarrow{(5.5.17.2)} R\,\mathrm{Sym}^n_S(\bar{u})_* L\Gamma^n_{\mathrm{ext}} j_! \\ &\xrightarrow[(5.5.14.4\ \text{(vii')})]{\sim} R\,\mathrm{Sym}^n_S(\bar{u})_* \mathrm{Sym}^n_S(j)_! L\Gamma^n_{\mathrm{ext}} \\ &\to R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}. \end{aligned}$$

5.5.20. We leave to the reader the following verifications, which are easy in principle using 5.5.18.

. (Intrinsic character). The morphism (5.5.19.2) is independent of the choice of the compactification $\bar{u}$ of u.

. (Transitivity). Let there be a diagram of S-schemes

$$X \overset{u}{\to} Y \overset{v}{\to} Z,$$

395 with u and v quasi-projective; suppose that $\mathrm{Sym}^n_S(Z)$ exists (and hence also $\mathrm{Sym}^n_S(Y)$

[69] N.D.E.: How can one show that $\mathrm{Sym}^n_S(u)$ is of finite type in general? This condition is necessary for $R\,\mathrm{Sym}^n_S(u)_!$ to be defined.

and $\mathrm{Sym}^n_S(X)$ (SGA 1 V 1.7, EGA II 4.5.4)). The following diagram of functors is then commutative:

$$\begin{array}{ccccc}
L\Gamma^n_{\mathrm{ext}} R(vu)_! & & \xrightarrow{\quad k^n_{vu} \quad} & & R\,\mathrm{Sym}^n_S(vu)_! L\Gamma^n_{\mathrm{ext}} \\
\| & & & & \| \\
L\Gamma^n_{\mathrm{ext}} Rv_! Ru_! & \xrightarrow{k^n_v} & R\,\mathrm{Sym}^n_S(v)_! L\Gamma^n_{\mathrm{ext}} Ru_! & \xrightarrow{k^n_u} & R\,\mathrm{Sym}^n_S(v)_! R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}.
\end{array}$$

. The symmetric Künneth morphism is compatible with changes of base or space of the types 5.5.9 (iv), (v), and (vi). More precisely, one has the following compatibilities with the morphisms 5.5.14.4 (iv′), (v′), and (vi′).

(iv″) For every $\mathscr{A}$-algebra $\mathscr{A}'$ and every quasi-projective S-morphism $u : X \to Y$, the diagram

$$\begin{array}{ccccc}
L\Gamma^n_{\mathrm{ext}} Ru_!(K) \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} \mathscr{A}' & \xrightarrow{\sim} & L\Gamma^n_{\mathrm{ext}}(Ru_!(K) \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} \mathscr{A}') & \xrightarrow{\sim} & L\Gamma^n_{\mathrm{ext}} Ru_!(K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} \mathscr{A}') \\
\downarrow{\scriptstyle k^n_u} & & & & \downarrow{\scriptstyle k^n_u} \\
R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}(K) \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} \mathscr{A}' & \xrightarrow{\sim} & R\,\mathrm{Sym}^n_S(u)_!(L\Gamma^n_{\mathrm{ext}}(K) \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} \mathscr{A}') & \xrightarrow{\sim} & R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}(K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} \mathscr{A}')
\end{array}$$

is commutative.

(v″) For every base change $g : S' \to S$, giving rise to a diagram with cartesian squares 396

$$\begin{array}{ccccc}
X' & & \xrightarrow{\quad g' \quad} & & X \\
& \searrow^{u'} & & & \downarrow{\scriptstyle u} \\
\downarrow & & Y' & \xrightarrow{\quad g'' \quad} & Y \\
& \swarrow & \downarrow & \swarrow & \\
S' & & \xrightarrow[\quad g \quad]{} & & S
\end{array} ,$$

the diagram of functors, entirely analogous to (iv″), is commutative:

$$\begin{array}{ccccc}
\mathrm{Sym}^n_g(g'')^* L\Gamma^n_{\mathrm{ext}} Ru_! & \xrightarrow{\sim} & L\Gamma^n_{\mathrm{ext}} g''^* Ru_! & \xrightarrow{\sim} & L\Gamma^n_{\mathrm{ext}} Ru'_! g'^* \\
\downarrow{\scriptstyle k^n_u} & & & & \downarrow{\scriptstyle k^n_{u'}} \\
\mathrm{Sym}^n_g(g'')^* R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} & \xrightarrow{\sim} & R\,\mathrm{Sym}^n_S(u')_! \,\mathrm{Sym}^n_g(g')^* L\Gamma^n_{\mathrm{ext}} & \xrightarrow{\sim} & R\,\mathrm{Sym}^n_S(u')_! L\Gamma^n_{\mathrm{ext}} g'^*.
\end{array}$$

(vi″) For every *radicial* base change $g : Y' \to Y$, giving rise to a cartesian diagram of S-schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle u'} & & \downarrow{\scriptstyle u} \\ Y' & \xrightarrow{g} & Y, \end{array}$$

the following diagram of functors is commutative:

$$\begin{array}{ccccc} \mathrm{Sym}_S^n(g)^* L\Gamma^n_{\mathrm{ext}} Ru_! & \xrightarrow{\sim} & L\Gamma^n_{\mathrm{ext}} g^* Ru_! & \xrightarrow{\sim} & L\Gamma^n_{\mathrm{ext}} Ru'_! g'^* \\ \downarrow{\scriptstyle k^n_u} & & & & \downarrow{\scriptstyle k^n_{u'}} \\ \mathrm{Sym}_S^n(g)^* R\,\mathrm{Sym}_S^n(u)_! L\Gamma^n_{\mathrm{ext}} & \xrightarrow{\sim} & R\,\mathrm{Sym}_S^n(u')_! g'^* L\Gamma^n_{\mathrm{ext}} & \xrightarrow{\sim} & R\,\mathrm{Sym}_S^n(u')_! L\Gamma^n_{\mathrm{ext}} g'^* \end{array}$$

397 THEOREM 5.5.21. (Symmetric Künneth formula). *Let S be a coherent scheme (= quasi-compact and quasi-separated) ringed by a sheaf $\mathscr{A}$ of commutative torsion rings whose stalks are noetherian rings. Ring each S-scheme by the inverse image of $\mathscr{A}$. Let $u : X \to Y$ be a morphism of quasi-projective S-schemes. For every integer $n \geq 0$ and every $K \in \mathrm{Ob}\, D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$, the symmetric Künneth arrow (5.5.17.2)*

$$k^n : L\Gamma^n_{\mathrm{ext}} Ru_! K \longrightarrow R\,\mathrm{Sym}_S^n(u)_! L\Gamma^n_{\mathrm{ext}} K$$

is an isomorphism in $D^{b,\mathrm{tor}\leq 0}(\mathrm{Sym}_S^n(Y), \mathscr{A})$.

The proof of this theorem will occupy the rest of 5.5.

REMARK 5.5.21.1. The quasi-projectivity hypothesis serves only to ensure that u admits a compactification $\overline{u}$

$$X \xrightarrow{j} \overline{X} \xrightarrow{\overline{u}} Y$$

such that the schemes $\mathrm{Sym}_S^n(X)$, $\mathrm{Sym}_S^n(\overline{X})$, and $\mathrm{Sym}_S^n(Y)$ exist for every n. The language of algebraic spaces should make it possible to replace this hypothesis by the sole hypothesis that u is compactifiable.

398 The first part of the proof consists in reducing 5.5.21, by dévissage, to an "irreducible" case:

LEMMA 5.5.22. *Theorem 5.5.21 follows from the following special case:*

(∗) *For every algebraically closed field k, every smooth projective curve X over k, and every prime number ℓ, the symmetric Künneth arrows (obtained for $Y = S = \mathrm{Spec}(k)$, $\mathscr{A} = \mathbf{Z}/\ell$, $K = \mathbf{Z}/\ell$; see the explanations below)*

$$L\Gamma^n(R\Gamma(X, \mathbf{Z}/\ell)) \longrightarrow R\Gamma(\mathrm{Sym}_k^n(X), \mathbf{Z}/\ell) \tag{5.5.22.1}$$

are isomorphisms ($n \geq 0$).

In (5.5.22.1), the symbol Γ has an unfortunate double use: in $R\Gamma$ it denotes the "global sections" functor, whereas in $L\Gamma^n$ it denotes the functor of 5.5.2.3. The isomorphism $\Gamma^n_{\mathrm{ext}}(\mathbf{Z}/\ell) \simeq \mathbf{Z}/\ell$ is used in this formula.

Let S, X, Y, u, and $\mathscr{A}$ be as in 5.5.21.

LEMMA 5.5.22.2. *Let there be an exact sequence of complexes in $C^{b,\geq 0}(X, \mathscr{A})$, of tor-dimension ≤ 0:*

$$0 \longrightarrow K' \longrightarrow K \longrightarrow K'' \longrightarrow 0.$$

If all the Künneth arrows k^n ($n \geq 0$) relative to two of the complexes K', K, and K'' are isomorphisms, then the Künneth arrows relative to the third are also isomorphisms.

We reduce to the case where u is projective and the complexes K', K, and K'' satisfy conditions (i) and (ii) of 5.5.16, for N at least twice the fiber dimension of u. The symmetric Künneth arrows are then given by (5.5.17.1). 399

We prove by induction on n that if, for $i \leq n$ and two of the complexes K', K, and K'', the arrows k^i are isomorphisms, then the arrows k^i ($i \leq n$) are isomorphisms for the third.

To see this, note that by 5.5.15, the morphism

$$k^n : L\Gamma^n_{\mathrm{ext}} Ru_* K \longrightarrow R\,\mathrm{Sym}^n(u)_* L\Gamma^n_{\mathrm{ext}} K \tag{a}$$

can be represented by a morphism of filtered complexes

$$k'^n : K_0 \longrightarrow K_1$$

with finite filtration; on the successive quotients this morphism induces arrows isomorphic to the composite arrows

$$L\Gamma^i_{\mathrm{ext}} Ru_* K' \overset{\mathbf{L}}{\vee} L\Gamma^{n-i}_{\mathrm{ext}} Ru_* K'' \longrightarrow R\,\mathrm{Sym}^n(u)_* (L\Gamma^i_{\mathrm{ext}} K' \overset{\mathbf{L}}{\vee} L\Gamma^{n-i}_{\mathrm{ext}} K'') \tag{b$_i$}$$

as follows:

$$\begin{aligned} L\Gamma^i_{\mathrm{ext}} Ru_* K' \overset{\mathbf{L}}{\vee} L\Gamma^{n-i}_{\mathrm{ext}} Ru_* K'' &\xrightarrow{k^i \overset{\mathbf{L}}{\vee} k^{n-i}} R\,\mathrm{Sym}^i(u)_* L\Gamma^i_{\mathrm{ext}} K' \overset{\mathbf{L}}{\vee} R\,\mathrm{Sym}^{n-i}(u)_* L\Gamma^{n-i}_{\mathrm{ext}} (K'') \\ &\underset{\text{defn}}{=} \vee_{i,n-i*} (R\,\mathrm{Sym}^i(u)_* L\Gamma^i_{\mathrm{ext}} K' \overset{\mathbf{L}}{\boxtimes} R\,\mathrm{Sym}^{n-i}(u)_* L\Gamma^{n-i}_{\mathrm{ext}} (K'')) \\ &\overset{5.4.3}{\xrightarrow{\sim}} \vee_{i,n-i*} (R\,\mathrm{Sym}^i(u) \times R\,\mathrm{Sym}^{n-i}(u))_* (L\Gamma^i_{\mathrm{ext}} K' \overset{\mathbf{L}}{\boxtimes} L\Gamma^{n-i}_{\mathrm{ext}} K'') \\ = R\,\mathrm{Sym}^n(u)_* \vee_{i,n-i*} (L\Gamma^i_{\mathrm{ext}} K' \overset{\mathbf{L}}{\boxtimes} L\Gamma^{n-i}_{\mathrm{ext}} K'') &\overset{\text{dfn}}{=} R\,\mathrm{Sym}^n(u)_* (L\Gamma^i_{\mathrm{ext}} K' \overset{\mathbf{L}}{\vee} L\Gamma^{n-i}_{\mathrm{ext}} K''). \end{aligned}$$

By the induction hypothesis (and the Künneth formula), these arrows are quasi-isomorphisms for $i \neq 0, n$. Any two of the following assertions therefore imply the third: (a) is a quasi-isomorphism; (b$_0$) is a quasi-isomorphism; (b$_n$) is a quasi-isomorphism. These are precisely the three arrows in question, C.Q.F.D. 400

. Assume assertion (∗) of 5.5.22 and prove 5.5.21. Let S, X, Y, $\mathscr{A}$, K, and u therefore be as in 5.5.21. The problem is local on S, which may be assumed affine. Every complex K is a successive extension of complexes of the form $i_! i^* K$ for i the inclusion in X of a locally closed affine subset. By dévissage (5.5.22.2), we may therefore suppose that K has the form $i_! K_1$, for an immersion $i : X_1 \to X$ with affine source. Compatibility 5.5.20.2 permits us to replace X by X_1 and K by K_1, and hence to suppose that u is affine. Factor u as a composite $u = u_1 \dots u_k$ of morphisms with fibers of dimension ≤ 1. Compatibility 5.5.20.2 shows that it suffices to prove that the arrows k^n relative to the u_i and to any complex K are isomorphisms. We are thus reduced to the case where S is *affine and u has relative dimension* ≤ 1.

. For k^n to be an isomorphism, it suffices that its restrictions to the geometric fibers of $\mathrm{Sym}^n_S(Y)$ be isomorphisms; compatibility 5.5.20.3 (v″) therefore reduces us to the case where S is the *spectrum of an algebraically closed field k*.

. For every rational point y of $\mathrm{Sym}^n_k(Y)$, there is a finite reduced subscheme Y' of Y such that y belongs to the image of $\mathrm{Sym}^n_k(Y')$. Compatibility 5.5.20.3 (vi″) therefore permits us to suppose that Y *is a disjoint union of finitely many copies of* $\mathrm{Spec}(k)$. 401

. If $Y = \amalg Y_i$, then every complex K on X is a sum of complexes supported on one of the schemes $X_i = u^{-1}(Y_i)$. By dévissage (5.5.22.2) and compatibility 5.5.20.3 (vi″), it suffices to prove 5.5.21 for each morphism $u_i : X_i \to Y_i$. By 5.5.22.5, this reduces us to the case where

$$Y = S = \mathrm{Spec}(k),\ k \text{ algebraically closed, } \dim X \leq 1.$$

The sheaf $\mathscr{A}$ now reduces to a noetherian ring A. The following proposition will allow us to reduce to the case where A is a field.

Proposition 5.5.23. *Let A be a (commutative) noetherian ring, let $K \in \mathrm{Ob}\, D^-(A)$ be a bounded-above complex of A-modules, and let n be an integer. The following conditions are equivalent:*

(i) *K has tor-dimension $\leq n$*

(ii) *For every $x \in \mathrm{Spec}(A)$ and every $i < -n$, $H^i(K \overset{\mathbf{L}}{\otimes}_A k(x)) = 0$.*

One has (i) ⇒ (ii) trivially. Suppose (ii). To prove (i), we prove by noetherian induction on the closed subset F of $\mathrm{Spec}(A)$ that if a module N satisfies $N_x = 0$ for $x \notin F$, then $H^i(K \overset{\mathbf{L}}{\otimes}_A N) = 0$ for $i < -n$.

402 Let I be the unique ideal defining F such that A/I is reduced, and let N_k be the submodule of N annihilated by I^k. One has $N = \varinjlim N_k$. Since the Tor functors commute with filtered direct limits, we reduce to the case where there exists k such that $N = N_k$, so that N is a successive extension of modules annihilated by I; the long exact cohomology sequence reduces us to the case where N is already an A/I-module.

Let η be a maximal point of F, and define A/I-modules M_1 and M_2 by the exact sequence

$$0 \longrightarrow M_1 \longrightarrow N \xrightarrow{i} N \otimes k(\eta) \longrightarrow M_2 \longrightarrow 0. \tag{5.5.23.1}$$

One has $(M_j)_\eta = 0$ $(j = 1, 2)$. If, for every closed subset $G \subset F$ with $\eta \notin G$, we denote by $M_{j,G}$ the submodule of M_j consisting of the sections of M_j supported in G, then

$$M_j = \varinjlim_{\eta \notin G} M_{j,G}.$$

By the induction hypothesis and passage to the limit, one therefore has

$$H^i(K \overset{\mathbf{L}}{\otimes}_A M_j) = 0 \quad \text{for} \quad i < -n.$$

The module $N \otimes k(\eta)$ is a vector space over $k(\eta)$, hence a direct sum of copies of $k(\eta)$. By hypothesis, one has

$$H^i(K \overset{\mathbf{L}}{\otimes} (N \otimes k(\eta))) = 0 \quad \text{for} \quad i < -n.$$

If I is the image of the arrow i in (5.5.23.1), there are exact sequences

$$H^{i-1}(K \overset{\mathbf{L}}{\otimes} M_2) \longrightarrow H^i(K \overset{\mathbf{L}}{\otimes} I) \longrightarrow H^i(K \overset{\mathbf{L}}{\otimes} (N \otimes k(\eta)))$$

$$H^i(K \overset{\mathbf{L}}{\otimes} M_1) \longrightarrow H^i(K \overset{\mathbf{L}}{\otimes} N) \longrightarrow H^i(K \overset{\mathbf{L}}{\otimes} I),$$

403 from which it follows that $H^i(K \overset{\mathbf{L}}{\otimes} N) = 0$ for $i < -n$, C.Q.F.D.

Corollary 5.5.24. *For a morphism $u : K \to L$ in $D^-(A)$ to be an isomorphism, it suffices that for every $x \in \mathrm{Spec}(A)$, the morphism*

$$u_x : K \overset{\mathbf{L}}{\otimes}_A k(x) \longrightarrow L \overset{\mathbf{L}}{\otimes}_A k(x)$$

be an isomorphism in $D^-(k(x))$.

Let M be the cone (mapping cylinder) of u. By hypothesis, one has $M \overset{\mathbf{L}}{\otimes}_A k(x) = 0$ for every $x \in \mathrm{Spec}(A)$. By 5.5.23, one therefore has $M = 0$, and u is an isomorphism.

. Let us continue the proof of 5.5.22, interrupted at 5.5.23. By the compatibility 5.5.20.3 (iv″) and 5.5.24, the morphism k^n associated with $K \in \mathrm{Ob}\, D^{b,\mathrm{tor}\geq 0}(X, A)$ is an isomorphism if and only if the morphisms k^n associated with $K \overset{\mathbf{L}}{\otimes}_A k(x) \in \mathrm{Ob}\, D^{b,\mathrm{tor}\geq 0}(X, k(x))$ are isomorphisms, for $x \in \mathrm{Spec}(A)$. This reduces us to the case where *A is a field of characteristic $\ell > 0$*.

. The exact sequence (where C denotes a cone)

$$0 \longrightarrow K[-1] \longrightarrow CK[-1] \longrightarrow K \longrightarrow 0$$

shows that the morphisms k^n are isomorphisms for K if and only if they are so for $K[-1]$ (5.5.22.2).

. The exact sequences 404

$$0 \longrightarrow \sigma_{\geq n}K \longrightarrow K \longrightarrow \sigma_{\leq n}K \longrightarrow 0$$

show that, for the Künneth morphisms to be isomorphisms when K is a flat complex, it suffices that they be so for the components of K, placed in degree 0 (5.5.22.2 and 5.5.25.2).

. Thus let K be a sheaf of A-vector spaces. This sheaf is a filtered direct limit of constructible sheaves (IX 2.7.2), automatically flat since A is a field, and, by passage to the limit, we reduce to the case where K is *constructible* and concentrated in degree zero.

. If a complex K is a direct summand of a complex L, the morphism k^n associated with K is a direct summand of the morphism k^n associated with L, hence is an isomorphism if the latter is one. The trace method (IX 5.8) therefore applies, and it suffices to treat the case where F is of the form $j_! A_{X'}$, for $j : X' \to X$ quasi-finite. The compatibility 5.5.20.2 allows us to replace X by X', whence the reduction

X has dimension ≤ 1 and $K = A_X$.

. We may assume that X is reduced. If $j : U \to X$ is the smooth locus of X, with complement $i : F \to X$, the exact sequence

$$0 \longrightarrow j_! j^* A_X \longrightarrow A_X \longrightarrow i_! i^* A_X \longrightarrow 0$$

reduces us by dévissage to the cases where *X is either a smooth curve or a finite set.* Moreover, the formula

$$A_X = A \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}/\ell} (\mathbf{Z}/\ell)_X$$

reduces us by 5.5.20.3 (iv″) to the case where $A_X = \mathbf{Z}/\ell$.

If X is a curve, we embed X in the corresponding smooth projective curve, and a further dévissage reduces us to one of the following two cases: 405

(i) X is a smooth projective curve, $Y = S = \mathrm{Spec}(k)$ with k algebraically closed, $\mathscr{A} = \mathbf{Z}/\ell$, $K = \mathbf{Z}/\ell$.

(ii) X is a finite disjoint union of copies of $\mathrm{Spec}(k)$, $Y = S = \mathrm{Spec}(k)$ (k algebraically closed), $\mathscr{A} = \mathbf{Z}/\ell$, $K = \mathbf{Z}/\ell$.

A final dévissage of $(\mathbf{Z}/\ell)_X$ into the sheaves $(\mathbf{Z}/\ell)_{X'}$, for X' a connected component of X, reduces us in (i) and (ii) to assuming that X is connected. Case (ii) is then trivial. This proves 5.5.22.

LEMMA 5.5.26. *Theorem 5.5.21 follows from the following two special cases of assertion (*) of 5.5.22:*

Case 1. *Assume that* $k = \mathbf{C}$.

Case 2. *Assume that* k *has characteristic* $p > 0$ *and that* $\ell = p$.

With the notation of 5.5.22, denote by p the characteristic of k. If $p = 0$, the Lefschetz principle (relying here on compatibility 5.5.20.3 (v″) and on XII 5.4) allows us to reduce to the case where $k = \mathbf{C}$.

It remains to treat the case where $p > 0$ and $\ell \neq p$. The curve X lifts to a curve X' over the ring $W(k)$ of Witt vectors over k (SGA 1 III 7.4). Put $S = \mathrm{Spec}(W(k))$, and consider the diagram

$$\begin{array}{ccccc} X'_\eta & \longrightarrow & X' & \longleftarrow & X \\ \downarrow{\scriptstyle f_\eta} & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ \eta & \longrightarrow & S & \longleftarrow & \mathrm{Spec}(k), \end{array}$$

406 and let f_n, f'_n, and $f_{\eta n}$ be the respective projections of $\mathrm{Sym}^n_k(X)$, $\mathrm{Sym}^n_S(X')$, and $\mathrm{Sym}^n_\eta(X'_\eta)$ onto $\mathrm{Spec}(k)$, S, and η. It is known that f'_n is a smooth projective morphism (projectivity is clear; for smoothness, the simplest approach is to reduce to the case where $X' = \mathbf{P}^1_S$, in which case $\mathrm{Sym}^n_S(X') \simeq \mathbf{P}^n$, as follows from the theorem on elementary symmetric functions). By the specialization theorem XVI 2.2, the sheaves $R^i f'_{n*}(\mathbf{Z}/\ell)$ are locally constant on S. The same holds for the sheaves $L^i\Gamma^n(Rf'_*\mathbf{Z}/\ell)$. To verify that the morphism of (5.5.22.1) associated with X over k is an isomorphism, it therefore suffices, by 5.5.20.3 (v″), to prove the analogous statement for the geometric generic fiber $X_{\overline{\eta}}$ of X over S. The field $k(\overline{\eta})$ has characteristic 0, and, in view of the first reduction, this proves 5.5.26.

To prove 5.5.21 in the two special cases of 5.5.26, we compare the symmetric Künneth morphism on the one hand with its transcendental analogue (5.5.27 and 5.5.28) and on the other hand with its coherent analogue (5.5.29 through 5.5.37).

5.5.27. If $p : X \to S$ is a continuous map of topological spaces, with S ringed by a sheaf of commutative rings $\mathscr{A}$, the constructions of 5.5.7 and 5.5.8 apply unchanged to define a functor Γ^n_{ext} from modules on X to modules on $\mathrm{Sym}^n_S(X) = X^n/\mathfrak{S}_n$. The formal arguments of 5.5.7 through 5.5.18 carry over. Likewise, if S, X, and Y are locally compact Hausdorff topological spaces

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array},$$

407 and if the fibers of u have bounded cohomological dimension, the formalism of 5.5.19 and 5.5.20 for the symmetric Künneth morphism in cohomology with proper supports carries over.

If $p : X \to S$ is a morphism of schemes of finite type over $\mathbf{C}$, the functors Γ^n and $L\Gamma^n$ commute with the functors of "passage to the transcendental setting". Finally, if

X, Y, S, $\mathscr{A}$, and K are as in 5.5.21, with S separated and of finite type over $\mathbf{C}$, and if we denote generically by ϵ the morphisms of sites of type $\epsilon : X_{\mathrm{cl}} \to X_{\mathrm{et}}$, the following diagram is commutative:

$$\begin{array}{ccc} \epsilon^* L\Gamma^n_{\mathrm{ext}} Ru_! K & \xrightarrow{k^n} & \epsilon^* R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} K \\ \downarrow & & \downarrow \\ L\Gamma^n_{\mathrm{ext}} Ru^{\mathrm{an}}_! \epsilon^* K & \xrightarrow{k^{n\,\mathrm{an}}} & R\,\mathrm{Sym}^n_S(u^{\mathrm{an}})_! L\Gamma^n_{\mathrm{ext}} \epsilon^* K. \end{array} \tag{5.5.27.1}$$

5.5.28. By the compatibility (5.5.27.1) and an elementary case of the comparison theorem (XI 4.4 or XVI 4.1), to treat Case 1 of 5.5.26, it suffices to verify that if X is a smooth complex projective curve, then the topological analogue of (5.5.22.1) is an isomorphism.

The topological space X^{an} is triangulable (this fact is trivial here, since X is a branched covering of the Riemann sphere). The topological analogue of the dévissage lemma 5.5.22.2 then reduces us to proving

(∗) If D is a closed simplex (of dimension 0, 1, or 2), the topological symmetric Künneth morphism 408

$$k^n : L\Gamma^n R\Gamma(D, \mathbf{Z}/\ell) \longrightarrow R\Gamma(\mathrm{Sym}^n(D), \mathbf{Z}/\ell) \tag{5.5.28.1}$$

is an isomorphism.

One has

$$R\Gamma(D, \mathbf{Z}/\ell) = \mathbf{Z}/\ell \text{ placed in degree } 0,$$

and hence

$$L\Gamma^n R\Gamma(D, \mathbf{Z}/\ell) = \mathbf{Z}/\ell \text{ placed in degree } 0.$$

On the other hand, $\mathrm{Sym}^n(D)$ is contractible, and again one has

$$R\Gamma(\mathrm{Sym}^n(D), \mathbf{Z}/\ell) = \mathbf{Z}/\ell \text{ placed in degree } 0.$$

The H^i of the two sides of (5.5.28.1) therefore vanish in nonzero degrees, and the morphism induced on H^0 identifies with the identity of $\mathbf{Z}/\ell$.

5.5.29. For the application we have in mind, it is useful to regard a quasi-coherent sheaf on a scheme X as a quasi-coherent sheaf of $\mathcal{O}$-modules on the étale site of X.

Let $p : X \to S$ be a quasi-projective morphism of schemes (cf. 5.5.21.1 [69]). If $\mathscr{F}$ is a sheaf of $\mathcal{O}_X$-modules on X_{et}, then

a) We denote by $\overset{n}{\bigotimes} \mathscr{F}$ the sheaf of modules on $(X/S)^n_{\mathrm{et}}$ given by the tensor product of the inverse-image $\mathcal{O}_{(X/S)^n}$-modules of $\mathscr{F}$ on $(X/S)^n_{\mathrm{et}}$;

b) the sheaf $\overset{n}{\bigotimes} \mathscr{F}$ is a $\mathscr{G}_n$-equivariant sheaf. Passing to invariants, it defines a sheaf $TS^n_{\mathrm{ext}}(\mathscr{F})$ on $\mathrm{Sym}^n_S(X)$: if σ is the projection of $(X/S)^n$ onto $\mathrm{Sym}^n_S(X)$, one has

$$TS^n_{\mathrm{ext}}(\mathscr{F}) = (\sigma_* \overset{n}{\bigotimes} \mathscr{F})^{\mathfrak{S}^n}. \tag{5.5.29.1}$$

Assume that p is flat; as in 5.5.8, one then defines the functor Γ^m_{ext} to be the 0^{th} left 409

[69] Everything that follows, including 5.5.34, remains valid under the sole assumption that X and Y admit invertible sheaves relatively ample over S (without assuming that X and Y are of finite type over S).

derived functor of TS^n_{ext}. With the notation of 5.5.2.5 and 5.5.8, if

$$P_1 \longrightarrow P_0 \longrightarrow \mathscr{F} \longrightarrow 0$$

is a flat presentation of $\mathscr{F}$, then the sequence

$$TS^n_{\mathrm{ext}}(P_1') \underset{j_1}{\overset{j_0}{\rightrightarrows}} TS^n_{\mathrm{ext}}(P_0) \longrightarrow \Gamma^n_{\mathrm{ext}}(\mathscr{F})$$

is exact. If x is a geometric point of $\mathrm{Sym}^n_S(X)$ as in 5.5.7.1, one has (cf. (5.5.8.1))

$$\Gamma^n_{\mathrm{ext}}(\mathscr{F})_x = \mathscr{O}_x \otimes_C \Big(\bigotimes_1^\ell \Gamma^{n_i}_{\mathscr{O}_S}(\mathscr{F}_{y_i})\Big), \tag{5.5.29.2}$$

where

$$C = \bigotimes_1^\ell \mathrm{Sym}^{n_i}_{\mathscr{O}_S}(\mathscr{O}_{y_i}).$$

Indeed, it suffices to verify this formula in the trivial case where the fibers of $\mathscr{F}$ are free. Note that $\mathscr{O}_x$ is flat over C, being the strict henselization of C at the closed point that is the image of x.

5.5.30. Assume that X and S are affine, p is flat, and $\mathscr{F}$ is quasi-coherent. Let $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and let $F = H^0(X, \mathscr{F})$, so that $\mathscr{F}$ is the sheaf $F^\sim$ defined by the B-module F. It follows from (5.5.29.1) that

$$TS^n_{\mathrm{ext}}(\mathscr{F}) = TS^n_A(F)^\sim,$$

and, on deriving, one therefore has

$$\Gamma^n_{\mathrm{ext}}(\mathscr{F}) = \Gamma^n_A(F)^\sim. \tag{5.5.30.1}$$

410 The functor Γ^n_{ext} thus appears as a functor Γ^n "relative to A", as (5.5.29.2) already expressed locally.

5.5.31. We denote by $D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X)$ the subcategory of $D^b(X, \mathscr{O}_X)$ formed by the complexes of $\mathscr{O}_X$-modules which, as complexes of $p^{-1}(\mathscr{O}_S)$-modules, have tor-dimension ≤ 0.

Consider the condition

. $K \in \mathrm{Ob}\, C^{b,\geq 0}(X, \mathscr{O}_X)$, and, at every geometric point x of X, K_x is homotopic to a complex of $\mathscr{O}_{X,x}$-modules, in nonnegative degrees, flat over $\mathscr{O}_{S,p(x)}$.

Since the morphism p is flat, so that there are "enough" $\mathscr{O}_X$-modules flat over S, one checks as in 5.5.13.2 that $D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X)$ is obtained from the category of complexes $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{O}_X)$ satisfying (5.5.31.1) by inverting quasi-isomorphisms.

It follows from (5.5.29.2) and 5.5.6.1 that if $u : K \to L$ is a quasi-isomorphism between complexes satisfying (5.5.31.1), then $\Gamma^n_{\mathrm{ext}}(u)$ (in the sense of Dold-Puppe) is again a quasi-isomorphism. The functor Γ^n_{ext} can therefore be derived to give

$$L\Gamma^n_{\mathrm{ext}} : D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X) \longrightarrow D^{b,\mathrm{tor}_S\leq 0}(\mathrm{Sym}^n_S(X), \mathscr{O}_{\mathrm{Sym}^n_S(X)}) \tag{5.5.31.2}$$

5.5.32. We denote by $D^{b,\mathrm{tor}_S\leq 0}_{\mathrm{qcoh}}(X, \mathscr{O}_X)$ the subcategory of $D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X)$ formed by complexes with quasi-coherent cohomology.

411 Standard arguments make it possible, as in 5.5.17, to define a symmetric Künneth arrow of the following type:

Let $u : X \to Y$ be a morphism of quasi-projective S-schemes flat over S; for $K \in \mathrm{Ob}\, D^{b,\mathrm{tor}_S\leq 0}_{\mathrm{qcoh}}(X, \mathscr{O}_X)$, the symmetric Künneth arrow k^n in $D^{b,\mathrm{tor}_S\leq 0}_{\mathrm{qcoh}}(\mathrm{Sym}^n_S(Y), \mathscr{O})$ is an

arrow

$$k^n : L\Gamma^n_{\mathrm{ext}} Ru_* K \longrightarrow R\,\mathrm{Sym}^n_S\, u_* L\Gamma^n_{\mathrm{ext}} K. \tag{5.5.32.1}$$

We leave it to the reader to verify the formal properties of (5.5.32.1) that will be used, notably the following one:

Lemma 5.5.33. *If S is of characteristic $p > 0$, and if $u : X \to S$ is proper and flat, the diagram of symmetric Künneth arrows (5.5.17.2) and (5.5.32.1) is commutative ($Y = S$, $K = \mathbf{Z}/p$ or $\mathcal{O}_X$):*

$$\begin{array}{ccc} L\Gamma^n_{\mathbf{Z}/p} Ru_*(\mathbf{Z}/p) & \xrightarrow{k^n} & R\,\mathrm{Sym}^n_S(u)_*(\mathbf{Z}/p) \\ \downarrow & & \downarrow \\ L\Gamma^n_{\mathcal{O}} Ru_*(\mathcal{O}) & \xrightarrow{k^n} & R\,\mathrm{Sym}^n_S(u)_*\mathcal{O} \end{array} .$$

Proposition 5.5.34. *Under the hypotheses of 5.5.32, the symmetric Künneth arrow (5.5.32.1) is an isomorphism.*

a) The question is local on S, which may be assumed affine.
b) For every point x of $\mathrm{Sym}^n_S(Y)$, there exists an affine open Y_1 of Y such that x lies in $\mathrm{Sym}^n_S(Y_1)$. By localization, we may therefore assume that Y is affine.
c) Every complex K of the type under consideration can be represented by a complex K_0 of quasi-coherent modules on S, zero in degrees < 0. If $\mathscr{U}$ is an affine covering of X, then K admits a finite resolution by sums of complexes of the form $j_* j^* K$, where j is the inclusion of an affine open (alternating Čech complex). Using the quasi-coherent analogue of the dévissage lemma 5.5.22.2, this reduces us to the case where there exist an affine open $j : X_1 \hookrightarrow X$ of X and $K_1 \in D^{b,\mathrm{tor}_S \leq 0}_{\mathrm{qcoh}}(X_1, \mathcal{O})$ such that $K = Rj_* K_1$. 412
d) In view of the quasi-coherent analogue of transitivity 5.5.20.2, it suffices to prove 5.5.34 for uj and j. Reduction b), applied to j, finally reduces us to the case where X, Y, and S are affine. We may then represent K by a complex $K_2 \in C^{b,\geq 0}(X, \mathcal{O}_X)$ with S-flat quasi-coherent components (4.1.9). The functors Ru_* and $L\Gamma^n$ can then be computed without having to resolve K_2 further, and it remains to note that, by (5.5.30.1), for $\mathscr{F}$ quasi-coherent on X, we have

$$\Gamma^n_{\mathrm{ext}} u_* \mathscr{F} \xrightarrow{\sim} \mathrm{Sym}^n_S(u)_* \Gamma^n_{\mathrm{ext}} \mathscr{F},$$

the two sides being identified with

$$\Gamma^n_A(F)^\sim \quad \text{for} \quad \mathscr{F} = F^\sim,$$

C.Q.F.D.

5.5.35. Let us treat case 2 of 5.5.26. With the notation of 5.5.22, the following diagram is commutative (5.5.33):

$$\begin{array}{ccc} L\Gamma^n(R\Gamma(X, \mathbf{Z}/p)) & \xrightarrow[①]{k^n} & R\Gamma(\mathrm{Sym}^n_k(X), \mathbf{Z}/p) \\ \downarrow & & \downarrow \\ L\Gamma^n(R\Gamma(X, \mathcal{O})) & \xrightarrow[②]{k^n} & R\Gamma(\mathrm{Sym}^n_k(X), \mathcal{O}), \end{array} \tag{5.5.35.1}$$

and by 5.5.34, arrow ② is an isomorphism. The Frobenius endomorphism $F : x \to x^p$ of 413

$\mathcal{O}$ acts p-linearly on $R\Gamma(X,\mathcal{O})$ and on $R\Gamma(\mathrm{Sym}^n_k(X),\mathcal{O})$. It therefore also acts p-linearly on $L\Gamma^n(R\Gamma(X,\mathcal{O}))$. One checks that k^n and F commute.

Artin–Schreier theory provides exact sequences

$$0 \longrightarrow H^i(X,\mathbf{Z}/p) \longrightarrow H^i(X,\mathcal{O}) \xrightarrow{F-I} H^i(X,\mathcal{O}) \longrightarrow 0,$$

and similarly for $\mathrm{Sym}^n_S(X)$. To deduce that ① is an isomorphism from the fact that ② is one, it remains to verify the following

LEMMA 5.5.36. *Let k be a perfect field of characteristic $p>0$, let $K \in C^{\geq 0}(\mathbf{Z}/p)$ be a complex of finite-dimensional $\mathbf{Z}/p$-vector spaces, let $L \in C^{\geq 0}(k)$ be a complex of finite-dimensional k-vector spaces, let F be a p-linear endomorphism of L, and let $u : K \to L$ be a morphism. Suppose that $H^*(Fu) = H^*(u)$, and that the sequences*

$$0 \longrightarrow H^i(K) \longrightarrow H^i(L) \xrightarrow{F-I} H^i(L) \longrightarrow 0$$

414 *are exact. Then the sequences*

$$0 \longrightarrow H^i(\Gamma^n K) \longrightarrow H^i(\Gamma^n L) \xrightarrow{F-I} H^i(\Gamma^n L) \longrightarrow 0$$

are exact.

Since $H^*(Fu) = H^*(u)$ and $\mathbf{Z}/p$ is a field, there exists a homotopy

$$Fu - u = dH + Hd.$$

Since the field k is perfect, we can write

$$FH_0 - H_0 = H.$$

Replacing u by $u - (dH_0 + H_0 d)$, we may assume that $Fu = u$.

The functor S that assigns to every k-vector space V equipped with a p-linear endomorphism F the $\mathbf{Z}/p$-vector space V^S of those $v \in V$ such that $Fv = v$ is an exact functor. The hypothesis therefore means that morphism

$$K \longrightarrow L^S$$

is a quasi-isomorphism. It remains to prove that the functor S commutes with the functor Γ^n. This follows readily from the following well-known lemma, whose proof is left to the reader:

LEMMA 5.5.37. *A finite-dimensional vector space V over a perfect field k, equipped with a p-linear endomorphism F, admits a unique decomposition*

$$V = V_0 + V_n$$

with $V^S \otimes k \xrightarrow{\sim} V_0$, $F(V_n) \subset V_n$, and $F|V_n$ nilpotent.

This completes the proof of 5.5.21.

6. The functor $f_!$

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We propose to give a direct construction of the functor $f_!$ of 5.1.11. We will then use this construction to define the "trace morphism" (6.2.3) for quasi-finite flat morphisms of finite presentation.

6.1. New construction of the functor $f_!$. Let $f : X \to S$ be a morphism locally of finite type. A subset P of X is said to be *proper* over S if it is the image of a subscheme of X proper over S.

PROPOSITION 6.1.1. *Let $f : X \to S$ be a morphism locally of finite type and quasi-separated.*

(i) *A subset of X proper over S is locally closed, and is closed if f is separated.*
(ii) *If f is separated, the union of two subsets of X proper over S is proper over S.*
(iii) *For every S-scheme U, let $\Phi_X(U)$ denote the set of subsets of $U \times_S X$ proper over U. The functor Φ_X is a sheaf for the* fpqc *topology. If f is separated, Φ_X is a sheaf of filtered ordered sets for the inclusion relation.*

Assertions (i) and (ii) are trivial, and the second assertion of (iii) follows from (ii). The 416
functor $U \mapsto \mathscr{P}(U \times_S X)$ is a sheaf for every topology whose coverings are surjective. It follows from EGA IV 2.3.14 and EGA IV 2.7.1 (vii) that Φ_X is a subsheaf thereof.

If $\mathscr{F}$ is an étale sheaf of pointed sets on a scheme X, recall (IV 8.5.2) that the support of a section s of $\mathscr{F}$ is the complement of the largest open set on which $s = 0$, where 0 denotes the distinguished section.

DEFINITION 6.1.2. *Let $f : X \to S$ be a separated morphism locally of finite type, and let $\mathscr{F}$ be a sheaf of pointed sets on X. We denote by $f_!\mathscr{F}$ the subsheaf of $f_*\mathscr{F}$ whose sections over U (étale over S) are the sections of $\mathscr{F}$ over $U \times_S X$ with support proper over U.*

It follows from 6.1.1 that $f_!\mathscr{F}$ is indeed a sheaf.

One readily verifies that:

LEMMA 6.1.3. *If f is the composite gh of two separated morphisms locally of finite type, there exists a unique isomorphism of functors $f_! \simeq g_!h_!$ making the following diagram commutative*

$$\begin{array}{ccc} f_! & \xrightarrow{\sim} & g_!h_! \\ \downarrow & & \downarrow \\ f_* & = & g_*h_*. \end{array}$$

If f is an open immersion (resp. a proper morphism), the functor $f_!$ is the extension- 417
by-zero functor (resp. the functor f_*). By 6.1.3, definitions 6.1.2 and 5.1.11 therefore agree on their common domain of validity.

The properties of the functor $f_!$ verified in § 5 remain valid in the more general case 6.1.2.

PROPOSITION 6.1.4. *Let $f : X \to S$ be a separated morphism locally of finite type.*

(i) *The functor $f_!$ is left exact (i.e. commutes with finite projective limits) and commutes with filtered inductive limits.*
(ii) *If f is quasi-finite, the functor $f_!$ is an exact functor from the category of abelian sheaves on X to that of abelian sheaves on S.*

(iii) *For every Cartesian diagram*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

there exists a base-change isomorphism making the diagram

$$\begin{array}{ccc} g^*f_! & \xrightarrow{\sim} & f'_!g'^* \\ \downarrow & & \downarrow \\ g^*f_* & \longrightarrow & f'_*g'^*, \end{array}$$

commute, in which the second horizontal arrow is the base-change arrow XII 4.1.

418 PROOF. For every $U \in \mathrm{Ob}\, S_{\mathrm{et}}$, we have

$$f_!\mathscr{F}(U) = \varinjlim_{P \in \Phi(U)} \Gamma_P(\mathscr{F}),$$

and for U quasi-compact and quasi-separated, the functor $\mathscr{F} \mapsto f_!\mathscr{F}(U)$ commutes with finite projective limits and filtered inductive limits, as a filtered inductive limit (6.1.1 (iii)) of functors having these properties.

This proves (i). Assertion (ii) follows at once from (iii) (take the geometric points of S as base changes).

Assertion (iii) is local on S, which may be assumed affine. Let I be the filtered ordered set of open subsets of X of finite type over S. If $U \overset{j}{\hookrightarrow} X$ is an open subset of X, $(fj)_!(\mathscr{F}|U) = f_!(j_!(\mathscr{F}|U))$ maps to $f_!(\mathscr{F})$. Moreover

$$f_!(\mathscr{F}) = \varinjlim_{U \in I} (fj)_!(\mathscr{F}|U).$$

This allows us to assume that X is of finite type over S and to apply Chow's lemma (EGA II 2nd ed.)[70] to find a projective and surjective morphism $p^0 : X^0 \to X$ such that fp^0 is compactifiable. Let $X^1 = X^0 \times_X X^0$ and let p^1 be the natural arrow from X^1 to X. The sequence

$$0 \to \mathscr{F} \to p^0_* p^{0*}\mathscr{F} \rightrightarrows p^1_* p^{1*}\mathscr{F}$$

is exact (VIII 9.2).

Taking into account the functoriality of the base-change arrow and (i), it suffices to verify (iii) for the sheaves $p^0_* p^{0^*}\mathscr{F}$ and $p^1_* p^{1^*}\mathscr{F}$. By the change of notation ($p^1 \mapsto p^0$),
419 we are thus reduced to assuming that $\mathscr{F}$ is of the form $p^0_*\mathscr{G}$. Assertion (iii) for the pair $(f, p^0_*\mathscr{G})$ is equivalent to assertion (iii) for the pairs $(p^0_*, \mathscr{G})$ and $(fp^0_*, \mathscr{G})$. We conclude by 5.2.7 (i), which applies because p^0 and fp^0 are compactifiable.

COUNTEREXAMPLE 6.1.6. The functor $Rf_!$ does not in general agree with the derived functor of the functor $f_!$. Indeed, if X is affine over an algebraically closed field k, we have

$$f_!\mathscr{F} = \bigoplus_{x \in X(k)} H^0_x(\mathscr{F}),$$

[70] N.D.E.: see [C, 2.6], cited in la N.D.E. (43) page 639.

and the i^{th} derived functor of $f_!$ therefore takes the value at $\mathscr{F}$

$$\bigoplus_{x\in X(k)} H^i_x(\mathscr{F}).$$

On the other hand, if X is smooth, irreducible, and of dimension 1,

$$H^2_c(X, \mu_{\ell^n}) \simeq \mathbf{Z}/\ell^n$$

for ℓ prime to the characteristic of k.

6.2. The trace morphism in relative dimension 0. The main result of this n° is Theorem 6.2.3. The reader is advised not to read the technical preliminaries 6.2.1 and 6.2.2. In the proof of 6.2.3, these allow one to consider only finite morphisms f, a reduction that, via EGA IV 18.12, is in any event fairly evident.

6.2.1. Let $f : X \to S$ be a *quasi-finite separated* morphism. For every $U \in$ Ob S^{et}, let $\Psi_f(U)$ denote the category of pairs consisting of a finite pointed set I, with distinguished point denoted 0, and a decomposition of f^*U into open and closed subsets $f^*U = \prod_{i\in I} V_i$, with V_i finite over U for $i \neq 0$. A morphism from $\varphi = (V_i)_{i\in I}$ to $\varphi' = (V'_i)_{i\in I'}$ is a map σ from I' to I such that, for $i \in I$, one has $V_i = \bigcup_{j\in\sigma^{-1}(i)} V'_j$. We say that φ' *refines* φ if there exists a morphism from φ to φ'. 420

Let $\mathscr{F}$ be a sheaf of abelian groups on S, let $U \in$ Ob S_{et}, let $\varphi = (V_i)_{i\in I} \in$ Ob $\Psi_f(U)$, let $I^* = I\backslash\{0\}$, and let f_i be the projection of V_i onto U. We have

$$f_!f^*\mathscr{F}|U = f_{0!}f_0^*(\mathscr{F}|U) \oplus \bigoplus_{i\neq 0} f_{i*}f_i^*(\mathscr{F}|U),$$

hence a composite arrow

$$\tau_\varphi : \mathscr{F}^{I^*}|U \longrightarrow \bigoplus_{i\neq 0} f_{i*}f_i^*(\mathscr{F}|U) \longrightarrow f_!f^*\mathscr{F}|U. \tag{6.2.1.1}$$

If $\sigma : I \to J$ is a morphism of pointed sets, we denote by ${}^t\sigma : \mathscr{F}^{J^*} \to \mathscr{F}^{I^*}$ the arrow with coordinates ${}^t\sigma^i{}_j = \mathrm{Id}$ if $\sigma(i) = j$, and ${}^t\sigma^i{}_j = 0$ if $\sigma(i) \neq j$. If $\sigma : I' \to I$ is a morphism from φ to φ', the diagram

$$\begin{array}{ccc} \mathscr{F}^{I^*}|U & \xrightarrow{\ {}^t\sigma\ } & \mathscr{F}^{I'^*}|U \\ & {\scriptstyle\tau_\varphi}\searrow \quad \swarrow{\scriptstyle\tau_{\varphi'}} & \\ & f_!f^*\mathscr{T}|U & \end{array} \tag{6.2.1.2}$$

is commutative. Moreover, ${}^t(\sigma\rho) = {}^t\rho{}^t\sigma$. The $\Psi_f(U)$ form a fibered category Ψ_f over S_{et}; the preceding constructions commute with localization and therefore define an S_{et}-functor $\varphi = (V_i)_{i\in I} \in \Psi_f(U) \mapsto \mathscr{F}^{I^*}|U$ from Ψ to the S_{et}-category of sheaves on étale open subschemes of S, and a morphism of functors $\tau_\varphi : \mathscr{F}^{I^*}|U \to f_!f^*\mathscr{F}|U$. 421

Proposition 6.2.2. (i) *The category Ψ_f is locally filtering on the right* (V 8.1.1) *over S_{et}.*

(ii) *The arrows (6.2.1.1) define an isomorphism*

$$\tau : \varinjlim_{\varphi=(V_i)_{i\in I}\in\Psi_f(U)} \mathscr{F}^{I^*}|U \xrightarrow{\sim} f_!f^*\mathscr{F}|U.$$

If S is strictly local with closed point s, the S-scheme X admits a decomposition $\varphi \in \Psi_f(S)$ into open and closed parts $X = \coprod_{i\in I} X_i$ indexed by a pointed set I such that the fiber of X_0 at s is empty, the X_i ($i \neq 0$) are finite over S, and the fiber of X_i ($i \neq 0$) at s consists of a single point (EGA IV 18.12.1). Such a decomposition is a final object of $\Psi_f(S)$. Moreover, by 6.1.4 (iii) (or because this is evident), the fiber at s of the arrow τ_φ (6.2.1.1) is an isomorphism.

In the general case, let x be a geometric point of S and let Ψ_x be the fiber of Ψ_f at x,

$$\Psi_x = \varinjlim_{U(x)} \Psi_f(U),$$

where U ranges over the étale neighborhoods of x. Let S_x be the strict localization of
422 S at x, $X_x = X \times_S S_x$, and $f_x : X_x \to S_x$. The standard limit arguments show that the category Ψ_x is equivalent to the category $\Phi_{f_x}(S_x)$, hence is filtering on the right. It follows that Ψ_f is locally filtering on the right. Moreover, the fiber at x of τ identifies with the arrow

$$\tau_x : \varinjlim_{\Psi_x} \mathscr{F}_x^{I_x} \longrightarrow (f_! f^* \mathscr{F})_x$$

and, by the preceding discussion, this arrow is an isomorphism.

THEOREM 6.2.3. *There is a unique way to define, for every morphism $f : X \to S$ that is separated, flat, of finite presentation, and quasi-finite, and every abelian sheaf $\mathscr{F}$ on S, a trace morphism*

$$\mathrm{Tr}_f : f_! f^* \mathscr{F} \longrightarrow \mathscr{F}$$

such that the following conditions hold:

(Var 1) Tr_f *is functorial in $\mathscr{F}$.*

(Var 2) Tr_f *is compatible with every base change, i.e. for every cartesian diagram*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

the following diagram of functors is commutative:

$$\begin{array}{ccccc} f'_! f'^* g^* & = & f'_! g'^* f^* & = & g^* f_! f^* \\ \downarrow{\scriptstyle \mathrm{Tr}_{f'} * g^*} & & & & \downarrow{\scriptstyle g^* * \mathrm{Tr}_f} \\ g^* & = & g^* & = & g^* \end{array} .$$

423 (Var 3) Tr_f *is compatible with composition, i.e. if f is the composite $f = gh$ of two morphisms that are separated, flat, of finite presentation, and quasi-finite, the following diagram is commutative:*

$$\begin{array}{ccccc} f_! f^* & = & g_! h_! h^* g_* & & \\ & \searrow^{\mathrm{Tr}_f} & \downarrow{\scriptstyle g_! * \mathrm{Tr}_h * g^*} & & \\ & & g_! g^* & & \\ & & \downarrow{\scriptstyle \mathrm{Tr}_g} & & \\ & & \mathrm{Id} . & & \end{array}$$

(Var 4) *(Normalization). If f is finite of constant rank n, the composite homomorphism*

$$\mathscr{F} \to f_* f^* \mathscr{F} = f_! f^* \mathscr{F} \xrightarrow{\mathrm{Tr}_f} \mathscr{F}$$

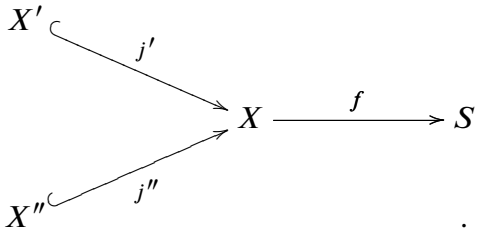
is multiplication by n.

First note that conditions (Var 2) through (Var 4) imply that

LEMMA 6.2.3.1. *If $X = X' \amalg X''$, then, setting $f' = f|X'$ and $f'' = f|X''$, the morphism Tr_f:*

$$f_! f^* \mathscr{F} = f'_! f'^* \mathscr{F} \oplus f''_! f''^* \mathscr{F} \longrightarrow \mathscr{F}$$

is the sum of $\mathrm{Tr}_{f'!}$ and $\mathrm{Tr}_{f''}$.



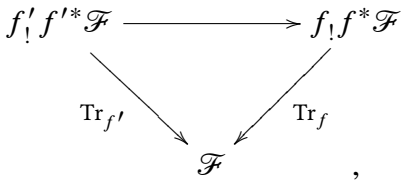
If one applies (Var 2) to j', for base change by j', and (Var 4) to the identity map, 424
one sees that

$$\mathrm{Tr}_{j'} : j'_! j'^* f^* \mathscr{F} \longrightarrow f^* \mathscr{F}$$

is the canonical map defined by the decomposition of $f^*\mathscr{F}$ as the direct sum

$$f^* \mathscr{F} = j'_! j'^* f^* \mathscr{F} \oplus j''_! j''^* f^* \mathscr{F}.$$

If one applies (Var 3) to $f' = f \circ j'$, one sees that the following diagram is commutative



which, together with the analogous statement for f'', is precisely 6.2.3.1.

Let $f : X \to S$ be a separated, flat, quasi-finite morphism of finite presentation. For every $U \in \mathrm{Ob}\, S_{\mathrm{et}}$, let $\Psi'_f(U)$ be the full subcategory of $\Psi_f(U)$ (6.2.1) consisting of the decompositions $(V_i)_{i\in I}$ such that, for $i \neq 0$, V_i is locally free of constant rank over U. The categories $\Psi'_f(U)$ form a fibered subcategory of Ψ_f, cofinal in Ψ_f. 425

Let $\mathscr{F}$ and $\mathscr{G}$ be abelian sheaves on S. By 6.2.2 (ii), giving a morphism $g : f_! f^* \mathscr{F} \to \mathscr{G}$ is equivalent to giving, for each decomposition $\varphi = (V_i)_{i\in I} \in \Psi'_f(U)$, a morphism $g_\varphi : \mathscr{F}^{I^*}|U \to \mathscr{G}|U$, compatible with the morphisms ${}^t\sigma$ of (6.2.1.2). If f_i is the projection from V_i to U, g_φ is the arrow whose coordinate arrows are the composites:

$$g^i_\varphi : \mathscr{F}|U \to f_{i*} f_i^* \mathscr{F}|U \to f_! f^* \mathscr{F}|U \xrightarrow{g} \mathscr{G}|U.$$

It follows that there exists a unique arrow Tr_f, compatible with étale localization and satisfying (Var 4) and 6.2.3.1. It is clear that this arrow satisfies (Var 1) and (Var 2).

To verify (Var 3), by base change one reduces to the easy case in which S is the spectrum of an algebraically closed field.

6.2.4. A *weighting* of a separated quasi-finite morphism $f : X \to S$ is the datum, for each geometric point x of X, of an integer $n(x)$ depending only on the point of X underlying x, such that:

(∗) *For every $U \in \mathrm{Ob}\, S_{\mathrm{et}}$ and every open and closed subspace X_1 of $f^{-1}(U)$ that is finite over U, the function assigning to each geometrical point s of U the number $\sum_{x \in X_1(s)} n(x)$ is locally constant.*

If $f = gh$ is the composite of two weighted morphisms, we weight f by setting $n_f(x) = n_h(x) \cdot n_g(h(x))$.

426 If the morphism f' is obtained by a base change g from a weighted morphism f, we weight f' by setting $n_{f'}(x) = n_f(g'(x))$.

The preceding arguments in fact prove:

Proposition 6.2.5. *There is a unique way to define, for every weighted, separated, quasi-finite morphism $f : X \to S$ and every abelian sheaf $\mathscr{F}$ on S, a trace morphism $\mathrm{Tr}_f : f_! f^* \mathscr{F} \to \mathscr{F}$ that is functorial in $\mathscr{F}$, compatible with base change and with composition, and satisfies:*

(Var 4′) *If f is finite and there exists an integer n such that, for every geometric point s of S, one has $\sum_{x \in X(s)} n(x) = n$, then the composite arrow*

$$\mathscr{F} \longrightarrow f_* f^* \mathscr{F} = f_! f^* \mathscr{F} \xrightarrow{\mathrm{Tr}_f} \mathscr{F}$$

is multiplication by n.

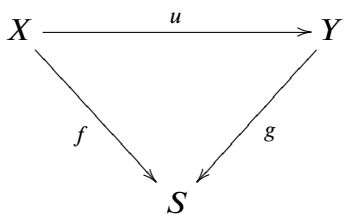
Example 6.2.6. Let S be a reduced noetherian geometrically unibranch scheme (for example, a normal scheme), and let $f : X \to S$ be a separated quasi-finite morphism. For every geometric point x of X with image s in S, let $\mathscr{O}_x$ (resp. $\mathscr{O}_s$) be the strictly local ring of X at x (resp. of S at s). The scheme $\mathrm{Spec}(\mathscr{O}_s)$ is reduced and has a unique maximal point η. Moreover, $\mathscr{O}_x$ is finite over $\mathscr{O}_s$. Set $n(x) = \mathrm{rg}_\eta(\mathscr{O}_x)$. One verifies that $n(x)$ is a weighting of f, whence a trace morphism Tr_f.

427 6.2.7. Let $u : X \to Y$ be a separated, quasi-finite, flat morphism of finite presentation. Since the trace morphism (6.2.3) is functorial, it defines, for every complex of abelian sheaves K on Y, a morphism of complexes

$$\mathrm{Tr}_u : u_! u^* K \longrightarrow K. \tag{6.2.7.1}$$

We shall use the terms *trace morphism, Gysin morphism* and *morphism u_** interchangeably for any morphism obtained from (6.2.7.1) or 6.2.3 by applying a functor; for example, the following ones.

Consider a commutative diagram



with g separated and locally of finite type. For every abelian sheaf $\mathscr{F}$ on Y, the trace morphism defines a morphism

$$\mathrm{Tr}_u \quad \text{or} \quad u_* : f_! u^* \mathscr{F} \simeq g_! u_! u^* \mathscr{F} \xrightarrow{g_!(\mathrm{Tr}_u)} g_! \mathscr{F}. \tag{6.2.7.2}$$

If g is compactifiable, and if K is a complex bounded below with torsion cohomology, or a complex of $g^*\mathscr{A}$-modules for $\mathscr{A}$ a sheaf of torsion rings on S, the trace morphism (6.2.7.1) likewise induces

$$\text{(6.2.7.3)} \qquad \mathrm{Tr}_u \quad \text{or} \quad u_* : Rf_!u^*K \simeq Rg_!u_!u^*K \xrightarrow{Rg_!(\mathrm{Tr}_u)} Rg_!K,$$

$$\text{(6.2.7.4)} \qquad \mathrm{Tr}_u \quad \text{or} \quad u_* : R^qf_!u^*K \simeq R^qg_!u_!u^*K \xrightarrow{R^qg_!(\mathrm{Tr}_u)} R^qg_!K.$$

For S the spectrum of an algebraically closed field, (6.2.7.4) is a morphism 428

$$\text{(6.2.7.5)} \qquad u_* : \mathrm{H}^q_c(X, u^*K) \longrightarrow \mathrm{H}^q_c(Y, K).$$

Thus one sees that *cohomology with proper supports is covariant* with respect to quasi-finite morphisms.

6.2.8. Let $u : X \to Y$ be as in 6.2.7, let $\Delta_n = [0, n]$ be the standard set with $(n+1)$ elements, and let X_n be the $(n+1)^{\text{fold}}$ iterated fiber product of X over Y. The X_n form a simplicial scheme augmented to Y (cf. VBIS). We denote by $u_n : X_n \to Y$ the augmentation morphism. If $\mathscr{F}$ is an abelian sheaf on Y, the sheaves $\mathscr{F}_n = u_{n!}u_n^*\mathscr{F}$ form, by (6.2.7.2) (applied after the change of notation $S \mapsto Y$, $X \mapsto X_n$, $Y \mapsto Y$ or X_m), a *simplicial* sheaf augmented to $\mathscr{F}$. By definition, the differential complex of "*nondegenerate chains*" of this simplicial sheaf has as its n^{th} component the quotient of $\mathscr{F}_n$, namely the cokernel of the direct sum of the degeneracy operators with target $\mathscr{F}_n$ and of the morphisms $\sigma - \epsilon(\sigma)$ for σ a symmetry operator.

Proposition 6.2.9. *Under the preceding hypotheses, if u is separated, étale, of finite type, and surjective, then for every abelian sheaf $\mathscr{F}$ on Y, one has*

(i) *The sheaves $\mathscr{F}_n$ form a (left) resolution of $\mathscr{F}$.*
(ii) *The differential complex of nondegenerate chains of the simplicial sheaf $(\mathscr{F}_n)$ is a (left) resolution of $\mathscr{F}$. If all the geometric fibers of u have at most d points, this resolution has length $\leq d$.*

By base change, it suffices to verify this statement when Y is the spectrum of an alge- 429
braically closed field, in which case 6.2.8 reduces to computing the simplicial homology of a simplex.

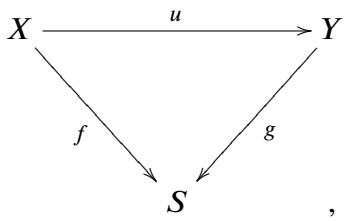
Let $(U_i)_{i\in I}$ be a finite open covering of the scheme Y, let $\mathscr{F}$ be an abelian sheaf on Y, and, for each subset P of I, let j_P be the inclusion of $U_P = \cap_{i\in P}U_i$ into Y. Suppose that a total order on I has been chosen. If $X = \coprod_{i\in I} U_i$ and if u is the projection from X to Y, the resolution 6.2.9 (ii) of $\mathscr{F}$ is given by

$$\text{(6.2.9.1)} \qquad (\mathscr{F}_n)_{n\,\deg} = \bigoplus_{|P|=n+1} j_{P!}j_P^*\mathscr{F}.$$

This resolution still makes sense when I is infinite.

6.2.10. Under the hypotheses of 6.2.9, let $\mathscr{A}$ be a sheaf of torsion rings on Y and let $K \in \mathrm{Ob}^-(Y, \mathscr{A})$. We regard X as ringed by $u^*\mathscr{A}$. Since resolution 6.2.9 (i) (resp. 6.2.9 (ii)) is functorial in $\mathscr{F}$, it defines a resolution of K by a double complex K_1. This resolution is called the extraordinary *Čech resolution* (resp. the *extraordinary alternating* Čech resolution) of K.

Consider a commutative diagram



with g compactifiable. With the notation of 6.2.8, if f_n is the projection from X_n to S,
430 one has $Rf_{n!} = Rg_! u_{n!}$. The complex K_1, filtered by the semisimplicial degree, therefore defines a spectral sequence

$$E_1^{-p,q} = R^q f_{p!}(u_p^* K) \Longrightarrow R^{-p+q} g_! K. \tag{6.2.10.1}$$

The differentials d_1 arise from the simplicial arrows, so that

$$E_2^{-p,q} = \check{\mathscr{H}}^p(R^q f_{p!}(u_p^* K)) \Longrightarrow R^{-p+q} g_! K. \tag{6.2.10.2}$$

Under the hypotheses of (6.2.9.1), one similarly obtains a spectral sequence

$$E_1^{-p,q} = \bigoplus_{|P|=p+1} R^q f_{p!}(j_p^* K) \Longrightarrow R^{-p+q} g_! K. \tag{6.2.10.3}$$

The spectral sequences (6.2.10.1) and (6.2.10.3) are called the *upper localization spectral sequences.*

Proposition 6.2.11. *If f is a separated étale morphism of finite type, the functor $f_!$ is left adjoint to the functor f^*, and its adjunction arrow is the trace morphism.*

It is known that if X is an object of a site S, the functor f^* restricting abelian sheaves to X has a left adjoint $f_!$. If F is an abelian sheaf on X, $f_! F$ is the sheaf generated by the presheaf

$$V \longrightarrow \bigoplus_{\xi \in \mathrm{Hom}(V,X)} F(\xi).$$

In the étale case, provisionally denote this functor by $f_!^0$. Its adjunction property immediately implies that its formation commutes with every base change. The trace morphism
431 defines a morphism t from $f_! \mathscr{F}$ to $f_!^0 \mathscr{F}$, compatible with every base change. It suffices to see that t is an isomorphism fiber by fiber, i.e. when S is the spectrum of a separably closed field, where this is trivial.

. One should note that this interpretation of $f_!^0$ is specific to the case of the étale site and does not extend to the fpqf site of a prescheme, for example.

6.3. Variant of the trace for continuous coefficients. In this no. we prove the assertions of Grothendieck [5, no. 6], and apply these results to the definition of a "trace morphism" for suitable fppf sheaves.

6.3.1. Let A be a commutative ring. If L and M are two finitely generated projective A-modules, the map $u \mapsto \bigwedge^m u$ from $\mathrm{Hom}(L, M)$ to $\mathrm{Hom}(\bigwedge^m L, \bigwedge^m M)$ is compatible with every base change and is $m^{\text{homogeneous}}$ (5.5.2.2). By 5.5.2.3, it defines a morphism

$$\bigwedge^m : \Gamma^m \mathrm{Hom}(L, M) \longrightarrow \mathrm{Hom}(\bigwedge^m L, \bigwedge^m M) \tag{6.3.1.1}$$

such that, after every base change, for $u \in \mathrm{Hom}(L, M)$ one has $\bigwedge^m(\gamma^m(u)) = \bigwedge^m u$. Since $\mathrm{Hom}(L, M)$ is flat, this morphism is identified (5.5.2.5) with a morphism

$$\bigwedge^m : TS^m \mathrm{Hom}(L, M) \longrightarrow \mathrm{Hom}(\bigwedge^m L, \bigwedge^m M). \tag{6.3.1.2}$$

N.B. Such a homomorphism can also be defined whenever 2 is invertible in A: a symmetric element of $\bigotimes^m \mathrm{Hom}(L, M)$ defines a symmetric homomorphism from $\bigotimes^m L$ to $\bigotimes^m M$, and this descends to the quotient to define a homomorphism from $\bigwedge^m L$ to $\bigwedge^m M$.

Given three finitely generated projective A-modules, the diagram 432

$$\begin{array}{ccc} \Gamma^m \mathrm{Hom}(L, M) \otimes \Gamma^m \mathrm{Hom}(M, N) & \longrightarrow & \Gamma^m \mathrm{Hom}(L, N) \\ \downarrow \bigwedge^m \otimes \bigwedge^m & & \downarrow \bigwedge^m \\ \mathrm{Hom}(\bigwedge^m L, \bigwedge^m M) \otimes \mathrm{Hom}(\bigwedge^m M, \bigwedge^m N) & \longrightarrow & \mathrm{Hom}(\bigwedge^m L, \bigwedge^m N), \end{array} \tag{6.3.1.3}$$

in which the horizontal arrows are deduced from the composition maps (and, for the first, from (5.5.2.8)), is commutative. This expresses the formula $\bigwedge^m(vu) = \bigwedge^m v \circ \bigwedge^m u$.

If L is a free A-module of rank n, (6.3.1.2) defines a canonical morphism

$$\det : \Gamma^n \mathrm{End}_A(L) = TS^n \mathrm{End}_A(L) \longrightarrow A, \tag{6.3.1.4}$$

which is an algebra homomorphism by (6.3.1.3). By definition, one has

$$\det(u^{\otimes n}) = \det(u; L). \tag{6.3.1.5}$$

If, in addition, L is a module over an A-algebra B, (6.3.1.4) defines by composition an algebra homomorphism

$$\det_L : TS^n_A(B) \longrightarrow A. \tag{6.3.1.6}$$

Lemma 6.3.2. *Suppose that L occurs in an exact sequence*

$$E : 0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0$$

with L' (resp. L'') locally free of rank n_1 (resp. n_2). Let $n = n_1 + n_2$, let α be the "associated graded" morphism from the group $\mathrm{End}(E)$ of endomorphisms of the extension E to 433 $\mathrm{End}(L') \oplus \mathrm{End}(L'')$*, and, for two flat A-modules X and Y, let β be the morphism from $TS^n(X \oplus Y)$ to $TS^{n_1}(X) \otimes TS^{n_2}(Y)$ deduced from the isomorphism (5.5.2.6)*

$$TS^n(X \oplus Y) \simeq \bigoplus_{n'+n''=n} TS^{n'}(X) \otimes TS^{n''}(Y).$$

The following diagram is commutative:

$$\begin{array}{ccccc} TS^n(\mathrm{End}(E)) & \xrightarrow{TS^n(\alpha)} & TS^n(\mathrm{End}(L') \oplus \mathrm{Ens}(L'')) & \xrightarrow{\beta} & TS^{n_1}(\mathrm{End}(L')) \otimes TS^{n_2}(\mathrm{End}(L'')) \\ \downarrow & & & & \downarrow \det \otimes \det \\ TS^n(\mathrm{End}(L)) & & \xrightarrow{\qquad\qquad \det \qquad\qquad} & & A. \end{array}$$

By the universal property 5.5.2 of TS^n, it suffices to check that the two images of $u^{\otimes n}$ coincide for $u \in \mathrm{End}(E)$. If $\alpha u = (u', u'')$, these images are respectively $\det(u)$ and $\det(u').\det(u'')$, which proves the assertion.

LEMMA 6.3.3. *Let A be a commutative ring, B a commutative A-algebra, C a not necessarily commutative B-algebra, E a B-module that is free of rank n over A, and F a C-module that is free of rank m over B. Suppose that C is flat over A. By the universal property 5.5.2.2 of Γ^{nm} and 5.5.2.3, there is a unique arrow $\sigma : TS_A^{nm}(C) \to TS_A^n TS_B^m(C)$ such that, after every ring extension $A \to A'$, one has $\sigma(c^{\otimes nm}) = (c^{\otimes m})^{\otimes n}$. The following diagram is then commutative:*

$$\begin{array}{ccc} TS_A^n(B) & \xleftarrow{TS_A^n(\det_F)} & TS_A^n TS_B^m(C) \\ \downarrow{\scriptstyle \det_E} & & \uparrow{\scriptstyle \sigma} \\ A & \xleftarrow{\det_{E\otimes_B F}} & TS_A^{nm}(C). \end{array}$$

434 PROOF. By 5.5.2.2 and 5.5.2.3, it suffices to show that the two images of an element $u^{\otimes nm}$ of $TS_A^{nm}(C)$ coincide. These images are $\det_E(\det_F(u))$ and $\det_{E\otimes_B F}(u)$; they are expressed in terms of A, B, E, F, and $u \in \mathrm{End}_B(F)$, and, to check that they coincide, one may first replace B by its (commutative) image B_1 in $\mathrm{End}(E)$ and F by $F \otimes_B B_1$. The formula to be proved is then the block-determinant formula (BOURBAKI, *Alg.* Chap. 8 § 12, Lemma 2, p. 140).

When C is no longer necessarily flat over A, the same proof gives, more generally, a commutative diagram of algebras

$$\begin{array}{ccc} \Gamma_A^n(B) & \xleftarrow{\Gamma_A^n(\det_F)} & \Gamma_A^n \Gamma_B^n(C) \\ \downarrow{\scriptstyle \det_E} & & \uparrow{\scriptstyle \sigma} \\ A & \xleftarrow{\det_{E\otimes_B F}} & \Gamma_A^{nm}(C). \end{array} \tag{6.3.3.1}$$

6.3.4. If B is a commutative A-algebra, $\mathrm{Spec}(TS^n(B))$ is precisely the n^{th} symmetric power of $\mathrm{Spec}(B)$ over $\mathrm{Spec}(A)$, i.e. the quotient of $(\mathrm{Spec}(B)/\mathrm{Spec}(A))^n$ by the symmetric group (SGA 1 V 1).

435 If X is a scheme separated over S, a quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ on X will be called *finite locally free* (resp. of rank n) *over* S if its support is finite over S and its direct image is locally free of finite rank (resp. of rank n) over S. By the *support* of $\mathcal{F}$ we shall again mean the *subscheme* of X defined by the annihilator of $\mathcal{F}$ (which is a quasi-coherent ideal, since $\mathcal{F}$ is quasi-coherent of finite type).

If $\mathcal{F}$ is finite locally free over S, of rank n and with support E, then E is affine over S; by globalizing over S, $\mathcal{F}$ defines, by means of (6.3.1.6), a section $\sigma_E(\mathcal{F})$ of $\mathrm{Sym}_S^n(E)$, hence, by functoriality, a section $\sigma(\mathcal{F})$ of the algebraic space [70] $\mathrm{Sym}_S^n(X)$,

$$\sigma(\mathcal{F}) \in \Gamma(\mathrm{Sym}_S^n(X)/S). \tag{6.3.4.1}$$

In particular, if $f : S' \to S$ is finite locally free of rank n, then, for $\mathcal{F} = \mathcal{O}_{S'}$, one obtains a canonical section

$$t \in \Gamma(\mathrm{Sym}_S^n(S')/S). \tag{6.3.4.2}$$

[70] M. ARTIN, Algebraization of formal moduli I, in *Global Analysis: Papers in Honor of K. Kodaira*, Princeton University Press, 1969, 21–71. We shall use the preceding construction only when $\mathrm{Sym}_S^n(X)$ exists as a scheme, which is the case when X is quasi-projective over S.

PROPOSITION 6.3.5. *Let X be a scheme separated over S, and let $\mathscr{F}_1$ and $\mathscr{F}_2$ be two $\mathscr{O}_X$-modules finite locally free of ranks n_1 and n_2 over S. Put $n = n_1 + n_2$. Let d be the canonical arrow $d : \mathrm{Sym}^{n_1}_S(X) \times \mathrm{Sym}^{n_2}_S(X) \to \mathrm{Sym}^{n}_S(X)$. If $\mathscr{F}$ is an extension of $\mathscr{F}_1$ by $\mathscr{F}_2$, then*

$$\sigma(\mathscr{F}) = d(\sigma(\mathscr{F}_1), \sigma(\mathscr{F}_2)).$$

One may replace X by the support (finite over S) of $\mathscr{F}_1 \oplus \mathscr{F}_2$ and suppose that S is affine. Translated into ring-theoretic terms, the proposition then follows from 6.3.2. 436

LEMMA 6.3.6. *If X is a scheme separated over S, $\mathscr{F}$ an $\mathscr{O}_X$-module finite locally free of rank n over S, and $\mathscr{L}$ an invertible sheaf on X, then*

$$\sigma(\mathscr{F}) = \sigma(\mathscr{F} \otimes \mathscr{L}).$$

Replace X by the support (6.3.4) of $\mathscr{F}$. Locally over S, $\mathscr{L}$ is then isomorphic to $\mathscr{O}_X$, and the assertion is trivial.

The reader may verify:

PROPOSITION 6.3.7. *For every diagram of S-schemes separated over S*

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array},$$

and every $\mathscr{O}_X$-module $\mathscr{F}$ finite locally free of rank n over S, the $\mathscr{O}_Y$-module $u_*\mathscr{F}$ is finite locally free of rank n over S, and

$$(\mathrm{Sym}^n_S u)(\sigma(\mathscr{F})) = \sigma(u_*\mathscr{F}).$$

. Finally, note that the construction of the morphisms σ (6.3.4.1) and t (6.3.4.2) is *compatible with every base change*, which makes sense by 5.5.2.7. 437

Application 1. The case of smooth curves.

6.3.8. Let $f : X \to S$ be a smooth quasi-projective curve over S, i.e. a smooth quasi-projective morphism purely of relative dimension 1. The theory of Hilbert schemes GROTHENDIECK [5]) shows that the fpqc sheaf $\mathrm{Div}^n_{X/S}$ which assigns to each S-scheme S' the set of relative divisors on $X' = X \times_S S'$ that are *finite* over S and of degree n is representable.

For each relative divisor E on X, finite of degree n over S, there is a canonical element (6.3.4.2) $t(E) \in \mathrm{Sym}^n_S(E)$, with image $\sigma(E)$ in $\mathrm{Sym}^n_S(X)$. Since this construction is compatible with every base change, it defines

$$\sigma : \mathrm{Div}^n_{X/S} \longrightarrow \mathrm{Sym}^n_S(X) : E \longrightarrow \sigma(\mathscr{O}_E). \tag{6.3.8.1}$$

Each section s of f defines a relative divisor finite over S of degree 1, namely the divisor $s(S)$. The morphism

$$(X/S)^n \longrightarrow \mathrm{Div}^n_{X/S} : (s_i)_{0<i\leq n} \longmapsto \sum s_i(S)$$

is symmetric, and hence induces a morphism

$$\alpha : \mathrm{Sym}^n_S(X) \longrightarrow \mathrm{Div}^n_{X/S}\,. \tag{6.3.8.2}$$

PROPOSITION 6.3.9. *The morphisms σ and α of 6.3.8 are inverse isomorphisms.*

PROOF. a) $\sigma\alpha = \mathrm{Id}$. It suffices to prove that the composite arrow

$$(X/S)^n \xrightarrow{\pi} \mathrm{Sym}^n_S(X) \xrightarrow{\alpha} \mathrm{Div}^n_{X/S} \xrightarrow{\sigma} \mathrm{Sym}^n_S(X)$$

is π. For $n = 1$, the assertion is clear; the general case follows from the 438

LEMMA 6.3.9.1. *With the notation of 6.3.5 and (6.3.8.1), one has*

$$\sigma(E + F) = d(\sigma(E), \sigma(F)).$$

This lemma follows from 6.3.5, 6.3.6, and the exact sequence

$$0 \longrightarrow \mathcal{O}(-F) \otimes \mathcal{O}_E \longrightarrow \mathcal{O}_{E+F} \longrightarrow \mathcal{O}_F \longrightarrow 0.$$

b) It is known that $\mathrm{Div}^n_{X/S}$ is smooth over S (there is no obstruction to infinitesimally lifting divisors on a curve (cf. SGA 3 XI 1)). The morphism $\alpha\pi : (X/S)^n \to \mathrm{Div}^n_{X/S}$ is a quasi-finite surjective morphism between smooth S-schemes whose source has equidimensional fibers. It is therefore flat (EGA 0_{IV} 17.3.5), so $\alpha\pi$ is an epimorphism of fppf sheaves, and hence so is α. In view of a), this completes the proof of 6.3.9.

LEMMA 6.3.10. *Let k be a commutative ring, M a free k-module, and u an endomorphism of M. Denote by $N_{\det(u-T)=0}$ the norm from $k[T]/\det(u - T)$ to k. Then, for $P \in k[T]$, one has*

$$N_{\det(u-T)=0}(P(T)) = \det(P(u), M).$$

This amounts to checking an algebraic identity between the entries of the matrix of u and the coefficients of P. To prove 6.3.10, it therefore suffices (by familiar arguments) to treat the case in which k is algebraically closed and u is semisimple, with eigenvalues a_i.

439 One then has

$$\det(u - T) = \Pi(a_i - T)$$
$$N_{\det(u-T)=0}(P(T)) = \Pi P(a_i) \quad \text{and}$$
$$\det(P(u), M) = \Pi P(a_i).$$

. Let $f : X \to S$ be a quasi-projective morphism of finite presentation, flat with purely one-dimensional Cohen–Macaulay fibers; these conditions are thus satisfied if X is a smooth quasi-projective curve over S. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module finite locally free *over* S. Since $\mathcal{F}$ vanishes on a fiberwise dense open subset of X, there is a canonical arrow $\psi : \mathcal{O}_X \to \det^*(\mathcal{F})$ (SGA 6 XI or MUMFORD [8, Chap. 5 § 3]). We denote by $\delta(\mathcal{F})$ the divisor defined by the annihilator ideal of $\mathrm{coker}(\psi)$ (ibid.); this divisor is finite over S.

PROPOSITION 6.3.11.1. *Under the hypotheses of 6.3.11.0 and with the notation of 6.3.8 and (6.3.4.1), if $\mathcal{F}$ has rank n, then $\delta(\mathcal{F})$ has degree n and*

$$\sigma(\mathcal{F}) = \sigma(\delta(\mathcal{F})).$$

PROOF. a) *The case where f is smooth.* By a standard reduction, we may assume that S is an Artinian local scheme with algebraically closed residue field. The additivity (6.3.5) of σ allows us to suppose that the support of $\mathcal{F}$ is concentrated at a rational point x of X. One checks at once that if T is a uniformizer at x, defining $p : X \to \mathbf{A}^1_S$, then $p_*(\mathcal{O}_{\delta(\mathcal{F})}) \simeq \mathcal{O}_{\delta(p_*\mathcal{F})}$ (because the constructions in question commute with passage to the completion). By this and 6.3.7, we reduce to the case $X = \mathbf{A}^1_S$.

440 If $X = \mathbf{A}^1_S$ and S is affine, $S = \mathrm{Spec}(k)$, then $\mathscr{F}$ is identified with a locally free k-module M equipped with an endomorphism u (the action of T). As a $k[T]$-module, M has the resolution

$$0 \to k[T] \otimes_k M \xrightarrow{\varphi} k[T] \otimes_k M \xrightarrow{\epsilon} M \to 0,$$

where $\epsilon(P \otimes m) = P(u)(m)$ and $\varphi(1 \otimes m) = 1 \otimes um - T \otimes m$. From the formula $\varphi = u - T$, it follows that the divisor $\delta(\mathscr{F})$ has equation $\det(u - T, M) = 0$.

To check 6.3.11.1, it suffices to show that, after every base change and for every section φ of $\mathscr{O}_X$, one has

$$\det_k(\varphi, M) = \det_k(\varphi, \mathscr{O}_{\delta(\mathscr{F})})(= N_{\delta(\mathscr{F})/S}(\varphi)).$$

For $\varphi = P(T)$, this is precisely the assertion of 6.3.10.

b) The general case. We first state more explicitly the assertion to be proved, in the following more concrete form:

Corollary 6.3.11.2. *Under the hypotheses of 6.3.11.0 on $f : X \to S$, let M be a finite locally free module on X, and let u be an endomorphism of M such that* $\det u$ *is a section of $\mathscr{O}_X$ that is regular on every fiber X_s (whence* $\mathrm{Coker}\, u = M/u(M)$ *and* $\mathrm{Coker} \det u = \mathscr{O}_X/\det u$ *are both flat and of finite presentation over S, with quasi-finite supports). Suppose that the common support*[71] *of these sheaves is finite over S, so that they define finite locally free sheaves over S. Then, for every section φ of $\mathscr{O}_X$, the determinants over $\mathscr{O}_S$ of φ acting on* $\mathrm{Coker}\, u$ *and on* $\mathrm{Coker} \det u$ *are equal:*

$$\det_S(\varphi, M/u(M)) = \det_S(\varphi, \mathscr{O}_X/\det u). \tag{6.3.11.3}$$

We note that this statement implies the equality of the ranks of the two 441
locally free modules under consideration over S, by homogeneity (when φ is replaced by $T\varphi$, with T an "indeterminate" section of $\mathscr{O}_S$). Moreover, replacing φ by $1 + T\varphi$, where T is such an indeterminate (more precisely, one makes a base change $S' = S[T] \to S$), one deduces from (6.3.11.3) the apparently more general formula

$$\mathrm{Pol}_S(\varphi, M/uM)(T) = \mathrm{Pol}_S(\varphi, \mathscr{O}_X/\det u)(T) \tag{6.3.11.4}$$

for the characteristic polynomials.

To prove (6.3.11.3), we shall reduce to the already proved smooth case. By étale localization on S, we reduce to the case where S is strictly local with closed point s; then, by additivity, we reduce to the case where the support of $M/u(M)$ has a single point on X_s, say x. After making a flat extension of the base, we may assume the residue field extension $k(x)/k(s)$ is trivial; and after replacing X by a sufficiently small neighborhood of x, and adding to φ a function that induces the zero operation on $M/u(M)$ and on $\mathscr{O}_X/\det u$ (which does not change the formula to be proved), one readily sees that we may assume that φ induces a section of $\mathscr{O}_{X_s}$ regular at x, and hence (after shrinking X) defines a quasi-finite flat morphism

$$p : X \longrightarrow X' = S[T].$$

[71] N.D.E.: the support of M/uM in the sense of 6.3.4 is a subscheme of the support of $\mathscr{O}/\det u$ defined by a nilideal.

By étale localization on X and on X', we may therefore assume that we have a *finite* and *flat* morphism p as above, with X' smooth over S, and $\varphi = p^*(\varphi')$, where φ' is a section of $\mathcal{O}_{X'}$. Set

$$M' = p_*(M) \quad , \quad u' = p_*(u),$$
$$P = \operatorname{Coker} u \quad , \quad Q = \operatorname{Coker} \det u,$$
$$P' = \operatorname{Coker} u' \quad , \quad Q' = \operatorname{Coker} \det u'.$$

442 We know that

$$\det u' = N_{X/X'} \det u, \quad \text{hence} \quad Q' = \operatorname{Coker} N_{X/X'}(\det u).$$

We trivially have

$$\det_S(\varphi, P) = \det_S(\varphi', P'),$$

and by formula (6.3.11.3), proved in the smooth case,

$$\det_S(\varphi', P') = \det_S(\varphi', Q'),$$

so it remains to prove the formula

$$\det_S(\varphi', Q') = \det_S(\varphi, Q),$$

which is *equivalent* to formula (6.3.11.3) to be established. We thus see that the validity of the latter depends only on the section $\det u$ of $\mathcal{O}_X$, or equivalently that it is equivalent to the same formula in the case where M is replaced by $\mathcal{O}_X$, and φ by $\det u$. But in the case $M = \mathcal{O}_X$, formula (6.3.11.3) is tautological! This completes the proof.

443 Remark 6.3.11.5. In the case where S is Artinian, the equality over S of the ranks of $\operatorname{Coker} u$ and $\operatorname{Coker} \det u$ (notation of 6.3.11.2) is a special case of EGA IV 21.10.17.3, of which (6.3.11.3) may be regarded as a "relative" version.

Proposition 6.3.12. *Let X and Y be two quasi-projective curves of finite presentation, flat with CohenMacaulay fibers over S, and let $u : X \to Y$ be a quasi-finite flat morphism. The following diagram*

$$\begin{array}{ccc} \operatorname{Div}^n_{X/S} & \xrightarrow{u_*} & \operatorname{Div}^n_{Y/S} \\ \downarrow{\scriptstyle \sigma_X} & & \downarrow{\scriptstyle \sigma_Y} \\ \operatorname{Sym}^n_S(X) & \xrightarrow{\operatorname{Sym}^n(u)} & \operatorname{Sym}^n_S(Y) \end{array}$$

is then commutative [71].

Since $u_*(D) = \delta(u_* \mathcal{O}_D)$, 6.3.12 follows from 6.3.7 and 6.3.11.1.

Application 2. The trace morphism.

6.3.13. Let G be an abelian sheaf on the large fppf site of a scheme S (0.10). Suppose that G satisfies the following condition.

. *For every affine flat morphism of finite presentation $f : X \to Y$ of S-schemes, and for every integer $n \geq 1$, the morphism*

$$\mathrm{Hom}_S(\mathrm{Sym}^n_Y(X), G) \longrightarrow \mathrm{Hom}_S((X/Y)^n, G)$$

444 *is injective and has as its image the set of symmetric morphisms from $(X/Y)^n$ to G.*

This condition is stable under base change.

Let $f : T \to S$ be a finite locally free morphism of rank n. Let pr_i be the i^{th} projection from $(T/S)^n$ to T, and let t be the section (6.3.4.2) of $\mathrm{Sym}^n_S(T)$. If $u \in G(T)$, then $\sum \mathrm{pr}_i^*(u)$ is a symmetric element of $G((T/S)^n)$ and therefore defines

$$u_n \in G(\mathrm{Sym}^n_S(T)).$$

We set

$$\mathrm{Tr}_f(u) = t^*(u_n).$$

When f is finite locally free, we define $\mathrm{Tr}_f(u)$, locally on S, by the preceding formula. This construction is compatible with every base change and therefore defines a *trace morphism*

$$\mathrm{Tr}_f : f_* f^* G \longrightarrow G. \tag{6.3.13.2}$$

6.3.14. Condition (6.3.13.1) is satisfied in the following cases:

a) If G is representable, by the definition of symmetric powers.
b) For G the inverse image on S_{fppf} of a sheaf on the small étale site of S (0.10).
c) For G defined by a quasi-coherent $\mathcal{O}_S$-module G_0.

Assertion c) reduces to (5.5.2.7).

Proposition 6.3.15. *For G satisfying (6.3.13.1), we have:* 445

(i) *The morphism Tr_f is functorial in G, and hence additive.*
(ii) *Its formation is compatible with every base change $S' \to S$.*
(iii) *If $T = \coprod_i T_i$, so that $f_* G \simeq \sum f_{i*} G$, then $\mathrm{Tr}_f = \sum \mathrm{Tr}_{f_i}$.*
(iv) *If T has constant rank n over S, the composite morphism*

$$G \longrightarrow f_* f^* G \xrightarrow{\mathrm{Tr}_f} G$$

is multiplication by n.
(v) *The trace morphism is compatible with composition* (cf. 6.2.3 Var 3).

Assertions (i) and (ii) are clear, and (iii) is an easy consequence of 6.3.5. With the notation of 6.3.13, if $u \in G(S)$, then u_n is n times the pullback of u, so that $\mathrm{Tr}_f(f^* u) = t^*(u_n) = nu$.

To prove (v), we reduce, by étale localization on S and using (iii), to proving that for

$$f = gh : S'' \xrightarrow{h} S' \xrightarrow{g} S,$$

with g and h finite locally free of *constant ranks* n and m, the diagram

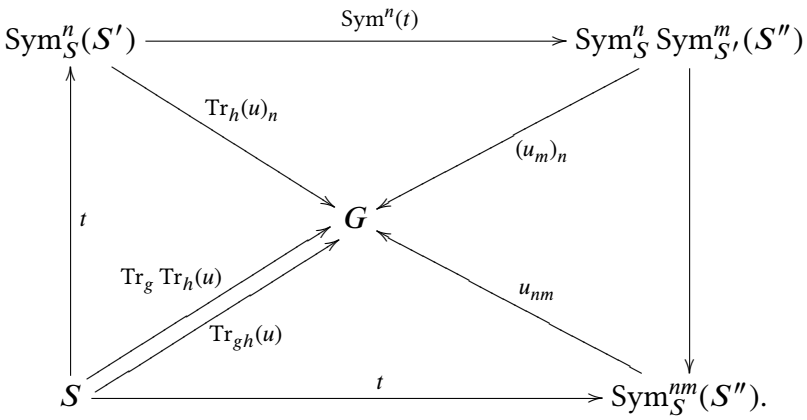
(6.3.15.1)

$$\begin{array}{ccc} G(S'') & \xrightarrow{\mathrm{Tr}_h} & G(S') \\ & {\scriptstyle \mathrm{Tr}_f} \searrow \quad \swarrow {\scriptstyle \mathrm{Tr}_g} & \\ & G(S) & \end{array}$$

[71] The morphism u_* is defined because we restrict ourselves to divisors *finite* over S. We take, for example, $u_*(D) = \delta(u_* \mathcal{O}_D)$ as the definition, a formula that agrees with EGA IV 21.5.5 when u is finite.

is commutative. For $u \in G(S'')$, consider the diagram

(6.3.15.2)



446 The right vertical arrow is defined because S'' is flat over S; every triangle is commutative, and (v) will therefore be a consequence of the commutativity of the outer boundary.

Translating into ring-theoretic terms, this follows from the special case $B = E$ and $C = F$ of 6.3.3.

6.3.16. Now let $f : T \to S$ be a quasi-finite flat morphism of finite presentation. If $\mathscr{F}$ is an fppf sheaf on T, we define its direct image with proper supports $f_!\mathscr{F}$ on S as the presheaf that assigns to S'/S the set of sections of $\mathscr{F}$ on T' whose support is proper over S'. This presheaf is a sheaf by 6.1.1. With the notation of 6.1.1, it may also be expressed as a local inductive limit (V 8.2):

$$f_!\mathscr{F}(U) = \varinjlim_{V \in \Phi(U)} f_*(\mathscr{F}|V),$$

447 a formula that accounts for the fairly evident fact that the properties of $f_!$ reduce to those of f'_* when f' is finite. We thus obtain:

PROPOSITION 6.3.17. *There is a unique way to define, for every $f : T \to S$ separated quasi-finite flat morphism of finite presentation and every fppf sheaf G on S satisfying (6.3.13.1), a trace morphism*

$$\mathrm{Tr}_f : f_!f^*G \longrightarrow G$$

satisfying:

(i) *For finite f, one recovers the morphism (6.3.13.2).*
(ii) *Conditions* (Var 1) *through* (Var 4) *of 6.2.3 are satisfied, mutatis mutandis.*

The quasi-finite morphism f induces a commutative diagram (cf. (0.11)) of sites (in which α_* is the "restriction" functor)

$$\begin{array}{ccc} T_{\mathrm{fppf}} & \xrightarrow{\alpha} & T_{\mathrm{et}} \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_{\mathrm{et}}} \\ S_{\mathrm{fppf}} & \xrightarrow{\alpha} & S_{\mathrm{et}}. \end{array}$$

For G on T_{fppf}, we have

$$f_!\alpha_* \simeq \alpha_* f_!.$$

448 We also have a "base-change" morphism

$$c : f^*_{\mathrm{et}}\alpha_* \longrightarrow \alpha_* f^*.$$

We leave it to the reader to verify the following compatibility between the trace morphisms 6.2.3 and 6.3.17. For G on S_{fppf} satisfying (6.3.13.1), the following diagram is commutative:

$$\begin{array}{ccccc} f_{\mathrm{et}!}f_{\mathrm{et}}^{*}\alpha_{*}G & \xrightarrow{f_!c} & f_{\mathrm{et}!}\alpha_{*}f^{*}G & \xleftarrow{\sim} & \alpha_{*}f_{!}f^{*}G \\ {\scriptstyle \mathrm{Tr}_f}\downarrow{\scriptstyle (6.2.3)} & & & & {\scriptstyle \alpha_*\,\mathrm{Tr}_f}\downarrow{\scriptstyle (6.3.17)} \\ \alpha_{*}G & & = & & \alpha_{*}G. \end{array} \tag{6.3.17.1}$$

In more concrete terms, this means that if G is an fppf sheaf on S whose restriction to the étale site defines an étale sheaf α_*G, and if u is a properly supported section of the inverse image $f_{\mathrm{et}}^{*}(\alpha_{*}G)$, with image u_1 in f^*G, then the elements $\mathrm{Tr}_f(u)$ (in the sense of 6.2.3) and $\mathrm{Tr}_f(u_1)$ (in the sense of 6.3.17) agree in $G(S)$.

Example 6.3.18. Let $f : S' \to S$ be a finite locally free morphism of rank n and let $u \in \Gamma(S', \mathcal{O}_{S'}) = \mathrm{Hom}_S(S', \mathbf{E}^1_S)$.[72] The morphism u defines a morphism

$$\mathrm{Sym}(u) : \mathrm{Sym}^n_S(S') \longrightarrow \mathrm{Sym}^n_S(\mathbf{E}^1_S) = \mathrm{Spec}(S[\sigma_1 \dots \sigma_n],$$

where the σ_i are the elementary symmetric functions (cf. Bourbaki, *Alg.* 5, App. I no. 2[73]). We then have (by an easy calculation left to the reader):

$$[\mathrm{Sym}(u) \circ t]^*(\sigma_i) = \sigma_i(u), \tag{6.3.18.1}$$

where $t \in \Gamma\,(\mathrm{Sym}^n_S(S')/S)$ is the canonical element (6.3.4.2), and where $\sigma_i(u)$ denotes the i^{th} coefficient of the characteristic polynomial of multiplication by u on $f_*\mathcal{O}_{S'}$. It follows that the "trace" maps 449

$$\begin{cases} \mathbf{G}_a(S') \longrightarrow \mathbf{G}_a(S) \\ \mathbf{G}_m(S') \longrightarrow \mathbf{G}_m(S) \end{cases}$$

are respectively the usual *trace* and *norm*; indeed, the morphism $\mathrm{Sym}^n_S(\mathbf{E}^1_S) \to \mathbf{E}^1_S$ induced by the additive law (resp. multiplicative law) on $\mathbf{E}^1_S$ is given by the function σ_1 (resp. σ_n) on $\mathrm{Sym}^n_S(\mathbf{E}^1_S) \simeq \mathrm{Spec}(S[\sigma_1, \dots, \sigma_n])$.

It then follows from (6.3.17.1) that, when n is invertible on S, the diagram of étale sheaves

$$\begin{array}{ccccccccc} 0 & \longrightarrow & f_*\mu_n & \longrightarrow & f_*\mathbf{G}_m & \longrightarrow & f_*\mathbf{G}_m & \longrightarrow & 0 \\ & & {\scriptstyle \mathrm{Tr}_f}\downarrow{\scriptstyle (6.2.3)} & & {\scriptstyle N_f}\downarrow & & {\scriptstyle N_f}\downarrow & & \\ 0 & \longrightarrow & \mu_n & \longrightarrow & \mathbf{G}_m & \longrightarrow & \mathbf{G}_m & \longrightarrow & 0 \end{array} \tag{6.3.18.2}$$

is commutative.

Proposition 6.3.19. *Consider a commutative diagram*

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$$

in which X and Y are smooth quasi-projective curves over S and u is a quasi-finite morphism. 450

[72] N.D.E.: Here $\mathbf{E}^1_S$ denotes the affine line over S.

[73] N.D.E.: reference to be checked. Probably IV § 6 no. 1 in the new version.

(i) *If D is a divisor on X finite over S (see note* p. 735*) and g is a section of G_Y on Y, then*

$$\mathrm{Tr}_{D/S}(u^*g) = \mathrm{Tr}_{(u_*D)/S}(g).$$

(ii) *If E is a divisor on Y finite over S, if g is a section of G_X on X, and if u is finite, then*

$$\mathrm{Tr}_{E/S}(\mathrm{Tr}_{X/Y}(g)) = \mathrm{Tr}_{(u^*E)/S}(g).$$

Proof. (i) Let g_n be the section of G on $\mathrm{Sym}^n_S(Y)$ obtained from g (6.3.13). By 6.3.12, we have

$$\mathrm{Tr}_{D/S}(u^*g) \overset{\text{dfn}}{=} \sigma(D)^*(u^*g)_n = \sigma(D)^*(\mathrm{Sym}^n_S(u)^*g_n) = (\mathrm{Sym}^n_S(u)\circ\sigma(D))^*(g_n)$$
$$= \sigma(u_*D)^*(g_n) = \mathrm{Tr}_{u_*D}(g_n).$$

(ii) Formula (ii) follows from 6.3.15 (ii) and (v):

$$\mathrm{Tr}_{E/S}(\mathrm{Tr}_{X/Y}(g)) = \mathrm{Tr}_{E/S}(\mathrm{Tr}_{u^*E/E}(g)) = \mathrm{Tr}_{u^*E/S}(g).$$

Application 3. Trace of a torsor.

6.3.20. Let G be a sheaf on the large fppf site of S that satisfies (6.3.13.1). Let $u : X \to Y$ be a finite locally free morphism of S-schemes, and let K be a G_X-torsor on X. Suppose that, locally for the fppf topology on Y, the torsor K on X is trivial. This is the case if K is locally trivial for the étale topology.

451 Proposition 6.3.21. *Under the hypotheses of 6.3.20, there exists, up to unique isomorphism, exactly one torsor $\mathrm{Tr}_u(K)$ under G_Y on Y, equipped with a morphism Tr_u or $\mathrm{Tr}_u^K : u_*K \to \mathrm{Tr}_u(K)$, which, for g a local section (in the fppf sense) of G on Y and k a local section of u_*K, satisfies*

$$\mathrm{Tr}_u^K(gk) = \mathrm{Tr}_u(g)\cdot\mathrm{Tr}_u^K(k).$$

The problem is local on Y; we may therefore assume that K is trivial: $K = G_X$. One solution to the problem is to take $K = G_Y$ and as trace morphism the morphism (6.3.13.2). One immediately verifies that it is the only one.

The trace functor 6.3.21 is defined in particular when G satisfies the following condition

For every S-scheme X, every torsor under G_X on X is locally trivial for the étale topology. This condition is satisfied when G is a smooth group, when G is the inverse image of a sheaf on the small étale site of S, or when G is defined by a quasi-coherent sheaf on S.

By construction one has

(6.3.21.2) $$\mathrm{Tr}_u(K) = u_*(K)\times_{u_*G_X} G_Y.$$

6.3.22. To define a trace functor under more general hypotheses, the key point is, for $f : X \to T$ a quasi-projective, flat morphism of finite presentation of S-schemes, to define a reasonable functor $K \mapsto K_n$ from the G_X-torsors on X to the G_Z-torsors on
452 $Z = \mathrm{Sym}^n_T(X)$. We first have the following lemma, where f_n denotes the projection from $\mathrm{Sym}^n_T(X)$ to T; it is proved in the same way as 6.3.21:

Lemma 6.3.22.1. *With the preceding notation, and if, locally on T, the torsor K is trivial, then, up to unique isomorphism, there exists a unique torsor K_n on $\mathrm{Sym}^n_T(X)$, equipped with a morphism $\alpha : f_*K \to f_{n*}(K_n)$, such that for g a local* (fppf) *section of f_*G_X and k a local section of f_*K, one has $\alpha(gk) = g_n\cdot\alpha(k)$ (notation of 6.3.13 for g_n; the definition of g_n here uses the flatness hypothesis etc.).*

The general case will be handled by a somewhat contrived descent argument. First of all:

PROPOSITION 6.3.23. *Consider a commutative diagram*

(6.3.23.1)
$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & \searrow \quad \swarrow & \\ & T & \end{array}$$

of flat, finitely presented, quasi-projective T-schemes, with u étale and surjective. Then the semisimplicial system of schemes $\mathrm{Sym}^n_T((X/Y)^{k+1})$ is a hypercover for the étale topology of the scheme $\mathrm{Sym}^n_T(Y)$.

PROOF. a) We shall use the following auxiliary result.

. *Let (X, Y, T) be as above, let $t \in T$, and let Z be a T-scheme that is the disjoint union of finitely many T-schemes $\mathrm{Spec}(k_i)$, where k_i is a finite extension of k_t,*[74] *and consider a commutative diagram of T-schemes* 453

(6.3.23.3)
$$\begin{array}{ccc} & Z & \\ & x \swarrow \quad \searrow y & \\ X_t & \xrightarrow{u} & Y_t, \end{array}$$

such that x and y are closed immersions. Let $z \in \mathrm{Sym}^n_T(Z)$, and let $x(z)$ and $y(z)$ be its images in $\mathrm{Sym}^n_T(X)$ and $\mathrm{Sym}^n_T(Y)$. Then $\mathrm{Sym}^n_T(u)$ is étale at $x(z)$.

The formation of the scheme $\mathrm{Sym}^n_T(X)$ is compatible with every base change $T' \to T$ (5.5.2.7). By reduction to the noetherian case, it follows that $\mathrm{Sym}^n_T(X)$ is of finite presentation over T; moreover, it is flat over T (5.5.2.4). By the fiberwise flatness criterion, to prove (6.3.23.2) we reduce to the case where T is the spectrum of a field. The completion of $\mathrm{Sym}^n_T(X)$ at z then has as its affine algebra the inverse limit of the affine algebras localized at $x(z)$ of the schemes $\mathrm{Sym}^n_T(Z_k)$, where Z_k ranges over the infinitesimal neighborhoods of Z in X. By hypothesis, u induces isomorphisms between the infinitesimal neighborhoods of Z in X and those in Y. The completion of $\mathrm{Sym}^n_T(u)$ at $x(z)$ and $y(z)$ is therefore an isomorphism, and this proves (6.3.23.2).

It follows from 6.3.23.2 that the scheme given by the disjoint union, over $z \in \mathrm{Sym}^n_T(Z)$, of the henselizations at $y(z)$ of $\mathrm{Sym}^n_T(Y)$ depends only on Z and on the T-scheme X^h_Z that is the disjoint union of the henselizations of X at the various points of Z. We shall denote this scheme by $\mathrm{Sym}^{h\,n}_T(X^h_Z)$; it is functorial in X^h_Z.

b) By passing to the limit, to prove 6.3.23, it suffices to show that for $y \in \mathrm{Sym}^n_T(Y)$, 454 closed in its fiber, the pullback to the henselization of $\mathrm{Sym}^n_T(Y)$ at that point of the simplicial scheme $\mathrm{Sym}^n_T((X/Y)^{k+1})$ is a hypercover for the étale topology. This problem is local on T for the étale topology; we thereby reduce to the case where there exists a diagram (6.3.23.3) with $y = x(z)$ (after killing the residue-field extensions by a separable extension of k_t). The semisimplicial system under consideration is then obtained from the system

$$\mathrm{Sym}^{h\,n}_T((X^h_{u^{-1}(Z)}/Y^h_Z)^{k+1}), \quad \text{augmented to} \quad \mathrm{Sym}^{h\,n}_T(Y^h_Z)$$

[74] N.D.E.: Here k_t denotes the residue field of T at t.

by base change, so it suffices to consider the latter. The semisimplicial system of $(X^h_{u^{-1}(Z)}/Y^h_Z)^{k+1}$ is semisimplicially homotopic to the constant system Y^h_Z, because $X^h_{u^{-1}(Z)}/Y^h_Z$ admits a section. The system $\mathrm{Sym}^{h\,n}_T((X^h_{u^{-1}(Z)}/Y^h_Z)^{h+1})$ is therefore likewise homotopic to $\mathrm{Sym}^{h\,n}_T(Y^h_Z)$, and in particular is a hypercover of it. This proves 6.3.23.

6.3.24. Let $f : Y \to T$ be a flat, quasi-projective morphism of finite presentation between S-schemes, and let K be a G_Y-torsor on Y. Consider the following problem:

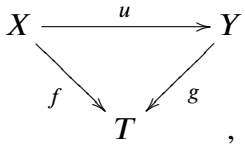
. Define, for every morphism $u : X \to Y$, with X flat, quasi-projective, and of finite presentation over T, a torsor $(u^*K)_n$ on $\mathrm{Sym}^n_T(X)$, and for every $v : X' \to X$ an isomorphism $c_v : (v^*u^*K)_n \simeq \mathrm{Sym}^n_T(v)^*(u^K)_n$, such that

455 (i) the isomorphisms c_v are compatible with composition;
(ii) when u^*K is locally trivial on T, one recovers the torsor of 6.3.22.1 and its functoriality.

By 6.3.23, the existence and uniqueness of a solution to 6.3.24.0, for every torsor obtained from K by an inverse image of the type considered in 6.3.24.0, is a local question for the étale topology on Y. In particular, if G satisfies (6.3.21.1), the problem 6.3.24.0 has a unique solution.

Now suppose that G satisfies:

(6.3.24.1). *For every integer n and every diagram of affine S-schemes*



with f and g flat and of finite presentation and u finite, locally free, and surjective, the truncated semisimplicial scheme $(\mathrm{Sym}^n_T((X/Y)^{k+1}))_{0\le k\le 2}$, *augmented to* $\mathrm{Sym}^n_T(Y)$, *is a descent diagram* (Giraud [3]) *for the category of G-torsors.*

6.3.24.1 bis. It follows in the standard way from 6.3.23 that condition (6.3.24.0) implies the analogous condition in which one assumes only that f and g are flat, quasi-projective, and of finite presentation, and that u is flat and surjective. Indeed, locally for the étale topology, u admits finite locally free quasi-sections, and one concludes by 6.3.23 and Giraud [3], applied to a diagram forming a square of descent data.

456 Given a G_Y-torsor K on a T-scheme Y that is flat, quasi-projective, and of finite presentation, there exists by definition a faithfully flat morphism of finite presentation $u : X \to Y$ such that u^*K is trivial.[75] If u_i denotes the projection from $(X/Y)^{i+1}$ to Y $(i \ge 0)$, the torsors $(u_i^*K)_n$ (6.3.22.1) on the schemes $\mathrm{Sym}^n_T((X/Y)^{i+1})$ form a descent datum $(i = 0, 1, 2)$ on this semisimplicial scheme augmented to $\mathrm{Sym}^n_T(Y)$. Suppose that G satisfies the condition:

(6.3.24.2). *The descent data constructed above are all effective* (cf. 6.3.24.0).

Then we denote by K_n the torsor on $\mathrm{Sym}^n_T(Y)$ (determined up to unique isomorphism) defined by this descent datum.

6.3.25. Suppose that G satisfies (6.3.21.1) or conditions (6.3.24.0) and (456). We leave it to the reader to make sense of the following formalism and to verify it:

[75] N.D.E.: One may assume that T is affine. Then there exists u as stated, which may even be assumed affine so that the preceding paragraph can subsequently be applied.

. *Base change.* Let

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ T' & \xrightarrow[g]{} & T. \end{array}$$

be a commutative square. Denote by g'_n the arrow from $\mathrm{Sym}^n_{T'}(X')$ to $\mathrm{Sym}^n_T(X)$. For X a torsor under G on X, one has

$$g'^*_n K_n \simeq (g'^* K)_n.$$

Now let $f : X \to T$ be a morphism, and let K be a G_X-torsor on X. 457

. *Change of group.* Let $r : G_1 \to G_2$ be a morphism between groups both satisfying (6.3.21.1) or (6.3.24.0), (456). One has

$$r(K_n) \simeq r(K)_n.$$

. *Additivity in K.* One has

$$(K' + K'')_n \simeq K'_n + K''_n.$$

. *Additivity in n.* Let n_1 and n_2 be two integers, $n = n_1 + n_2$, and let

$$\alpha : \mathrm{Sym}_T{}^{n_1}(X) \times_T \mathrm{Sym}_T{}^{n_2}(X) \longrightarrow \mathrm{Sym}^n_T(X)$$

be the canonical arrow; then

$$\alpha^* K_n \simeq \mathrm{pr}_1^* K_{n_1} + \mathrm{pr}_2^* K_{n_2}.$$

. *Multiplicativity in n.* Let n and m be two integers, and let α be the canonical arrow from $\mathrm{Sym}^n_T(\mathrm{Sym}^m_T(X))$ to $\mathrm{Sym}^{nm}_T(X)$ (cf. 6.3.3). One has

$$\alpha^*(K_{nm}) \simeq (K_m)_n.$$

6.3.26. Suppose that G satisfies (6.3.21.1) or (6.3.24.0), (456). If $f : X \to T$ is a finite locally free morphism of rank n between S-schemes, if t is the section (6.3.4.3) of $\mathrm{Sym}^n_T(X)$, and if K is a G_X-torsor on X, we define a G_T-torsor $\mathrm{Tr}_f(K)$ on T by the formula

$$\mathrm{Tr}_f(K) = t^*(K_n). \tag{6.3.26.1}$$

If f is locally free, we define $\mathrm{Tr}_f(K)$ locally on T by the same formula. This defi- 458
nition, via a canonical and functorial isomorphism, is equivalent to the definition 6.3.21 on their common domain of validity (the definition 6.3.21 providing a solution to the descent problem posed). It follows immediately from (6.3.25.1), (6.3.25.2), and (6.3.25.3) that the functor Tr_f is *compatible with base changes* $T' \to T$, *compatible with changes of group* $r : G_1 \to G_2$, *and compatible with addition of torsors.*

The commutativity of the outer boundary of diagram (6.3.15.2), together with (6.3.25.1) and (6.3.25.4), yields *compatibility with composition of morphisms*:

$$\mathrm{Tr}_{fg}(K) \simeq \mathrm{Tr}_f(\mathrm{Tr}_g(K)).$$

6.3.27. In the particular case of relative curves, one also has:

CONSTRUCTION 6.3.27.1. *If $f : X \to S$ is a smooth quasi-projective curve over S, D_1 and D_2 are two relative divisors finite over S, and K is a G-torsor on X, then, setting $\mathrm{Tr}_{D/S}(K) = \mathrm{Tr}_{D/S}(K|D)$, there is a canonical isomorphism*

$$\mathrm{Tr}_{D_1+D_2}(K) \simeq \mathrm{Tr}_{D_1}(K) + \mathrm{Tr}_{D_2}(K).$$

This isomorphism follows immediately from the isomorphism (6.3.25.4) and 6.3.9.1. Finally, the arguments of 6.3.19 also give:

CONSTRUCTION 6.3.27.2. *Consider a commutative diagram*

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array},$$

459 *in which X and Y are smooth quasi-projective curves over S and u is a quasi-finite morphism.*

(i) *If D is a divisor on X finite over S and K is a torsor under G on Y, one has* (cf. note p. 735)*:*

$$\mathrm{Tr}_{D/S}(u^*K) \simeq \mathrm{Tr}_{(u_*D)/S}(K).$$

(ii) *If E is a divisor on Y finite over S, K is a torsor under G on X, and u is finite, one has*

$$\mathrm{Tr}_{E/S}(\mathrm{Tr}_{X/Y}(K)) \simeq \mathrm{Tr}_{u^*E/S}(K).$$

PROPOSITION 6.3.28. *Let A and B be two (commutative) R-algebras and let $u : A \to B$ be a homomorphism making B a faithful, finite-type, locally free A-module. Then the co-semisimplicial complex of algebras $(TS^n_R(\bigotimes_A^{k+1} B))_{k\geq 0}$ is, as a differential co-semisimplicial $TS^n_R(A)$-module, a split resolution of $TS^n_R(A)$.*

PROOF. The complex of modules associated with the co-semisimplicial module $TS^n_R(\bigotimes_A^{k+1} B)$ depends only on the A-module B, equipped with a module homomorphism $u : A \to B$. As an A-module, B can be written $B = A \oplus L$, and this makes it possible to write down an explicit homotopy operator.

Recall that:

PROPOSITION 6.3.29. *Let A be a ring, let B^* be a co-semisimplicial A-algebra, and let $c : A \to B^*$ be the structure morphism. For the inverse-image functor on modules (= extension of scalars) ϵ^* to be faithful (resp. fully faithful), it suffices that the sequence of A-modules*

$$0 \longrightarrow A \longrightarrow B^0$$

460 *(resp. $0 \to A \to B^0 \xrightarrow{j_0 - j_1} B^1$) be exact, and remain so after tensoring with any A-module.*

The proof is immediate and is left to the reader.
It follows in the standard way from (6.3.28) and (6.3.29) that

COROLLARY 6.3.30. *Condition (6.3.24.0) is satisfied by every affine group G.*

(Use the fact that a torsor under G is representable by a scheme affine over T.)

It should not be difficult to extend 6.3.30 to every group G representable by an algebraic space [75]. I conjecture that condition (456) is satisfied by every group representable by a flat algebraic space. We have the tongue-in-cheek statement:

PROPOSITION 6.3.31. *Let the following sequence be exact on S:*

$$0 \to G \xrightarrow{r} G' \xrightarrow{u} H \to 0.$$

Suppose that G' and H, and hence G, satisfy (6.3.13.1), that G' satisfies (6.3.21.1), and that G satisfies (6.3.24.0). Then G satisfies (456).

This applies, for example, to an affine group that is the kernel of an epimorphism of representable smooth groups.

PROOF OF 6.3.31. Let K be a G-torsor on X. By 6.3.23, the question is local on X 461
for the étale topology; it therefore suffices to treat the case where $r(K)$ is trivial. The G-torsors K such that $r(K)$ is trivial are identified with trivial G'-torsors K' equipped with an isomorphism $k : H \xrightarrow{\sim} u(K')$. Put $Z = \mathrm{Sym}^n_T(X)$. The morphism of trivial torsors k induces $k_n : H_Z \xrightarrow{\sim} u(K'_n)$, and the corresponding G-torsor on $\mathrm{Sym}^n_T(X)$ represents the descent datum under consideration.

7. Appendix

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7.0. Preliminaries. In this appendix we shall use the results of VBIS, whose notation we retain.

7.0.1. Let $\mathrm{Sch}_!$ be the category whose objects are quasi-compact, quasi-separated schemes and whose morphisms are separated morphisms of finite type. We denote by $\mathscr{E}$ the bifibred category over $\mathrm{Sch}_!$, with fibre categories opposite to topoi, defined by étale sheaves on schemes (Vbis 4.3.0). Fix once and for all an object R of $\mathrm{Sch}_!$, and denote by $\mathscr{E}_R$ the bifibred category over $\mathrm{Sch}_{!R}$ deduced from $\mathscr{E}$. Finally, let $\mathscr{A}$ be a ring in the topos $\underline{\Gamma}(\mathscr{E}^\circ_R)$ satisfying the following conditions:

(i) $\mathscr{A}$ is a *cocartesian* section of $\mathscr{E}^\circ_R$ over $\mathrm{Sch}^\circ_{!R}$.
(ii) For every scheme X in $\mathrm{Sch}_{!R}$, $\mathscr{A}_X$ is a *torsion* sheaf.

With this understood, we shall work with the category $\mathrm{Mod}(\underline{\Gamma}(\mathscr{E}^\circ_R), \mathscr{A})$ (1.3.5).

Our aim is to define, for every morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$, a functor $Rf_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$, agreeing with the previously defined functor when f is compactifiable, and making the categories $D(X, \mathscr{A}_X)$, for varying X, into a cofibred category over $\mathrm{Sch}_{!R}$.[76]

7.0.2. Although all the results of this appendix are stated in the setting specified in 7.0.1, they could be transcribed, with obvious modifications, into the following setting: take as a section of $\underline{\Gamma}(\mathscr{E}^\circ_R)$ the constant section whose value is the constant sheaf $\mathbf{Z}$, and, for every scheme X, restrict to the derived category $D^*_{\mathrm{tors}}(X)$ defined by complexes of sheaves of abelian groups whose cohomology sheaves are torsion sheaves (4.3.1).

LEMMA 7.0.3. *Let there be given a commutative diagram of schemes* 463

[75] Algebraic Spaces by M. ARTIN. Yale University, Whittemore Lectures, May 1969.

[76] N.D.E.: Every morphism in $\mathrm{Sch}_!$ is compactifiable by Nagata's compactification theorem (see la N.D.E. (43) page 639). This appendix shows how to avoid using the compactification theorem in the definition of $Rf_!$ by relying on Chow's lemma (7.0.6).

(7.0.3.1)
$$\begin{array}{ccc} X' & \xrightarrow{j'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[j]{} & Y. \end{array}$$

where f and f' are proper and j and j' are immersions. Then, for every complex $K^\cdot$ in $D^+_{\mathrm{tors}}(X')$, the canonical arrow[77]

$$j_! \circ R^+ f'_*(K^\cdot) \longrightarrow R^+ f_* \circ j_!(K^\cdot)$$

is an isomorphism.

By the "lemma on way-out functors" ([6] I 7.1), it suffices to show that, for every integer i and every torsion sheaf F, the arrow

(7.0.3.2)
$$j_! \circ R^i f'_*(F) \longrightarrow R^i f_* \circ j_!(F)$$

is an isomorphism.

By replacing j and j' with their closed images, we are reduced to the case where j and j' are open immersions with everywhere dense images. The properness of f and f' then implies that diagram (7.0.3.1) is cartesian. Under these conditions, the proper base-change theorem, in the form XII 5.2, shows that, for every geometric point ξ of Y, the arrow $(j_! \circ R^i f'_*(F))_\xi \to (R^i f_* \circ j_!(F))_\xi$ is an isomorphism.

Construction 7.0.4. Let D be a category, and let X and Y be two D-objects in $\mathrm{Sch}_{!R}$ such that, for every arrow $\alpha \to \beta$ in D, the morphisms $X_\beta \to X_\alpha$ and $Y_\beta \to Y_\alpha$ are proper. Let $j : X \to Y$ be a morphism such that, for every object α of D, $j_\alpha : X_\alpha \to Y_\alpha$ is an immersion. The isomorphism (7.0.3.2) for $i = 0$ satisfies the usual compatibility conditions and allows one to construct an *exact* functor:

$$j_! : \mathrm{Mod}(\underline{\Gamma}(\overline{X}), \mathscr{A}) \longrightarrow \mathrm{Mod}(\underline{\Gamma}(\overline{Y}), \mathscr{A})$$

464 (cf. 1.2.5 for the notation) which induces a triangulated functor, again denoted $j_!$:

$$j_! : D(\underline{\Gamma}(\overline{X}), \mathscr{A}) \longrightarrow D(\underline{\Gamma}(\overline{Y}, \mathscr{A}))$$

Then, for every object α of D, there is a commutative diagram, where $j_{\alpha!}$ denotes the usual "extension by 0" functor:

$$\begin{array}{ccc} D(\underline{\Gamma}(\overline{X}), \mathscr{A}) & \xrightarrow{j_!} & D(\underline{\Gamma}(\overline{Y}), \mathscr{A}) \\ {\scriptstyle Re^*_\alpha}\downarrow & & \downarrow{\scriptstyle Re^*_\alpha} \\ D(X_\alpha, \mathscr{A}_{X_\alpha}) & \xrightarrow[j_{\alpha!}]{} & D(Y_\alpha, \mathscr{A}_{Y_\alpha}). \end{array}$$

Construction 7.0.5. *Let there be given a commutative diagram in* $\mathrm{Sch}_{!R}$

(7.0.5.1)
$$\begin{array}{ccc} X' & \xrightarrow{i'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{i} & Y \end{array}$$

[77] N.D.E.: The definition of this arrow is similar to that of (5.1.5.2), up to an easy dévissage.

where i and i' are immersions, f is a proper morphism, and f' is a proper surjective *morphism. It defines a commutative diagram of simplicial objects in* $\mathrm{Sch}_{!R}$

$$\begin{array}{ccc} [X'|Y'] & \xrightarrow{j} & [X|Y] \\ {\scriptstyle u_{f'}}\downarrow & & \downarrow{\scriptstyle u_f} \\ C^{\Delta}_{Y'} & \xrightarrow[i]{} & C^{\Delta}_{Y} \end{array} \tag{7.0.5.2}$$

where $j_n : [X'|Y']_n \to [X|Y]_n$ is an immersion for every n. With this notation, there is a canonical isomorphism

$$i_! \xrightarrow{\sim} R^+\overline{u}_{f*} \circ j_! \circ L^+\overline{u}^*_{f'}.$$

By replacing i and i' with their closed images, we may suppose that i and i' are open immersions with everywhere dense images. Diagram (7.0.5.1) is then cartesian and f is *surjective*. Moreover, for every integer n, j_n is an open immersion and the diagram 465

$$\begin{array}{ccc} [X'|Y']_n & \xrightarrow{j_n} & [X|Y]_n \\ \downarrow & & \downarrow \\ Y' & \xrightarrow[i]{} & Y \end{array}$$

is cartesian.

Consequently, $L^+\overline{u}^*_f \circ i_!$ is isomorphic to $j_! \circ L^+\overline{u}^*_{f'}$. Using the fact that u_f is a cohomological-descent augmentation (4.3.2), we obtain a sequence of isomorphisms

$$i_! \xrightarrow{\sim} R^+\overline{u}_{f*} \circ L^+(\overline{u}^*_f) \circ i_! \xrightarrow{\sim} R^+\overline{u}_{f*} \circ j_! \circ L^+\overline{u}^*_{f'},$$

which gives the desired isomorphism.[78]

Finally, we shall use the following variant of Chow's lemma, whose proof will appear in EGA II, 2nd edition,[79] and which is an improvement of (XII 7.1):

Lemma 7.0.6. *Let S be a quasi-compact and quasi-separated scheme and let $f : X \to S$ be a separated morphism of finite type. Then there exists a commutative diagram*

$$\begin{array}{ccc} X' & \xrightarrow{p} & X \\ {\scriptstyle f'}\downarrow & \swarrow{\scriptstyle f} & \\ S & & \end{array}$$

satisfying the following conditions:

(i) *p is proper and* surjective*;*
(ii) *there is a dense open subset U of X such that p induces an isomorphism from $p^{-1}(U)$ onto U;*
(iii) *f' is quasi-projective.*

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7.1. Definition of $R^+f_!$.

[78] N.D.E.: More directly, without assuming f' surjective, there is still a sequence of canonical arrows

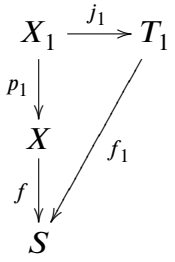
$$i_! \longrightarrow i_!R^+\overline{u}_{f'*}L^+\overline{u}^*_{f'} \xrightarrow{\sim} R^+\overline{u}_{f*}j_!L^+u^*_{f'},$$

of which the second is an isomorphism (7.0.3). When f' is surjective, the first is also an isomorphism by cohomological descent, and the composite agrees with the isomorphism constructed in the text.

[79] N.D.E.: See [C, 2.6], cited in la N.D.E. (43) page 639.

DEFINITION 7.1.1. *Let $f : X \to S$ be a morphism of schemes. A* pseudo-compactification *of f consists of a commutative diagram*

(7.1.1.0)



where p_1 is proper and surjective, j_1 is an immersion, and f_1 is proper.

7.1.2. Let $f : X \to S$ be a morphism in $\mathrm{Sch}_{!R}$. By Chow's lemma (7.0.6), f has a pseudo-compactification (p_1, j_1, f_1). Diagram (7.1.1.0) defines a morphism

$$[X_1|X] \xrightarrow{j_1} [T_1|S]$$

again denoted by j_1, such that j_{1n} is an immersion for every n. We then put $R^{+(1)}f_! = R^+\overline{u}_{f_1*} \circ j_{1!} \circ L^+\overline{u}^*_{p_1}$, thereby obtaining a *triangulated* functor from $D^+(X, \mathscr{A}_X)$ to $D^+(S, \mathscr{A}_S)$.

CONSTRUCTION 7.1.3. *For every pair $((p_1, j_1, f_1), (p_2, j_2, f_2))$ of pseudo-compactifications of f, there is an isomorphism $\alpha_{12} : R^{+(1)}f_! \to R^{+(2)}f_!$ such that, for every other pseudo-compactification (p_3, j_3, f_3), one has $\alpha_{23} \circ \alpha_{12} = \alpha_{13}$.*

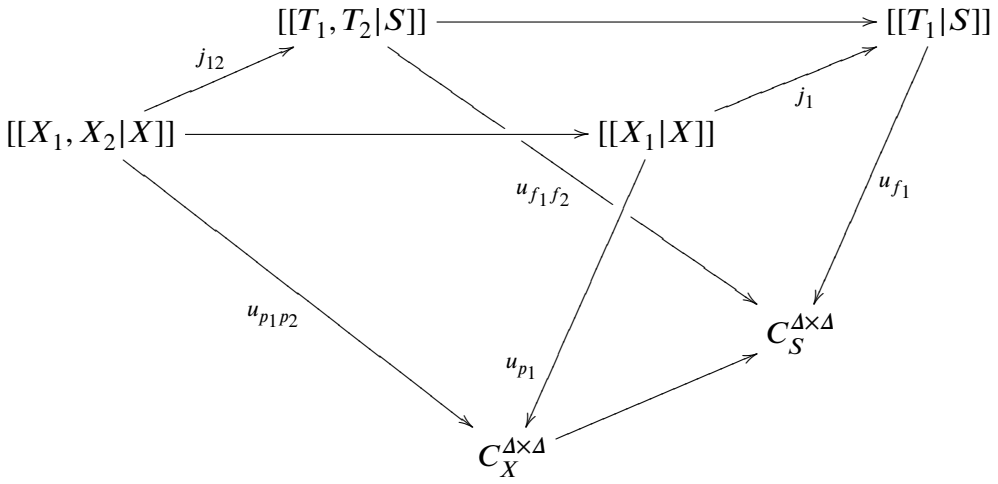
a) Define a bisimplicial system $[[X_1, X_2|X]]$ by setting

$$[[X_1, X_2|X]]_{[n]\times[m]} = [X_1|X]_n \times [X_2|X]_n$$

Similarly, set

$$[[X_1|X]]_{[n]\times[m]} = [X_1|S]_n,$$

all the morphisms $[[X_1|X]]_{[n]\times[m]} \to [[X_1|X]]_{[n]\times[m']}$ induced by morphisms $[m'] \to [m]$ being equal to the identity. Define the bisimplicial systems $[[T_1, T_2|S]]$ and $[[T_1|S]]$ in the same way, so that there is a commutative diagram



467 where j_{12} is an immersion degree by degree. The computational procedure described in Vbis 2.3 makes it possible to construct a morphism

$$R^{+(1)}f_! \xrightarrow{\alpha_1} R^+\overline{u_{f_1f_2*}} \circ j_{12!} \circ L^+\overline{u_{p_1p_2}}^* = R^{+(1,2)}f_!$$

which is obtained from a morphism of triple complexes of sheaves on S.

b) α_1 *is an isomorphism*:

To prove this, one may fix the first index n coming from $\big[[T_1, T_2]\big]_{[n]\times[m]}$ in the preceding morphism of triple complexes and consider the resulting morphism of double complexes. One checks that this morphism is precisely the isomorphism defined in Lemma 7.0.5 for the diagram

$$\begin{array}{ccc} [X_1|X]_n \times X_2 & \longrightarrow & [T_1|S]_n \times T_2 \\ \downarrow & & \downarrow \\ [X_1|X]_n & \longrightarrow & [T_1|S]_n \end{array}$$

c) In the same way, construct an isomorphism

$$R^{+(2)} f_! \xrightarrow{\alpha_2} R^+\overline{u_{f_1 f_{2*}}} \circ j_{12!} \circ L^+\overline{u_{p_1 p_2}}^*$$

and set $\alpha_{12} = \alpha_2^{-1} \circ \alpha_1$.

To verify the relation $\alpha_{23} \circ \alpha_{12} = \alpha_{13}$, one uses a trisimplicial object. We leave it to the reader to check that the resulting diagram is commutative: **468**

$$\begin{array}{ccccc} & & R^{+(2,3)} f_! & \longleftarrow & R^{+(2)} f_! \longrightarrow R^{+(1,2)} f_! \\ & & & \searrow \quad \swarrow & \\ & & & R^{+(1,2,3)} f_! & \\ & & \uparrow & \uparrow & \uparrow \\ & R^{+(3)} f_! & & & R^{+(1)} f_! \\ & & \searrow & \quad \swarrow & \\ & & & R^{+(1,3)} f_! & \end{array}$$

Construction 7.1.4. *Let $Z \xrightarrow{g} Y$ and $Y \xrightarrow{f} X$ be two morphisms in $\mathrm{Sch}_{!R}$. For every triple $((p_1, j_1, f_1), (p_2, j_2 g_2), (p_3, j_3, (fg)_3))$, where (p_1, j_1, f_1) is a pseudo-compactification of f, (p_2, j_2, g_2) a pseudo-compactification of g, and $(p_3, j_3, (fg)_3)$ a pseudo-compactification of fg, there is an isomorphism $R^{+(1)} f_! \circ R^{+(2)} g_! \xrightarrow{\sim} R^{+(3)} (fg)_!$ compatible with the previously defined isomorphisms α_{ij}.*

We proceed in several steps.

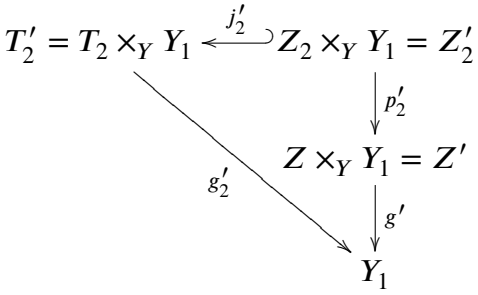
. Let (p_1, j_1, f_1) (resp. (p_2, j_2, f_2)) be a pseudo-compactification of $f : Y \to X$ (resp. of $g : Z \to Y$):

(7.1.4.1.1)

$$\begin{array}{ccccccc} & & & & Z & \xleftarrow{p_2} & Z_2 \xhookrightarrow{j_2} T_2 \\ & & & & \downarrow{\scriptstyle g} & & \swarrow{\scriptstyle g_2} \\ T_1 & \xleftarrow{j_1} & Y_1 & \xrightarrow{p_1} & Y & & \\ & \searrow^{f_1} & & & \downarrow{\scriptstyle f} & & \\ & & & & X & & \end{array}$$

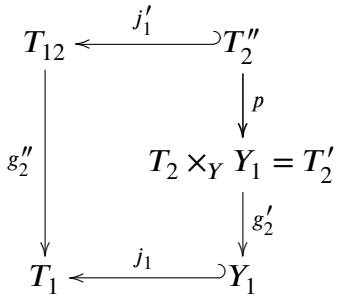
The morphism $Z' = Z \times_Y Y_1 \to Y_1$ obtained from g by base change is equipped with a pseudo-compactification:

(7.1.4.1.2)



469 Now choose a pseudo-compactification of $j_1 \circ g_2'$[80]

(7.1.4.1.3)

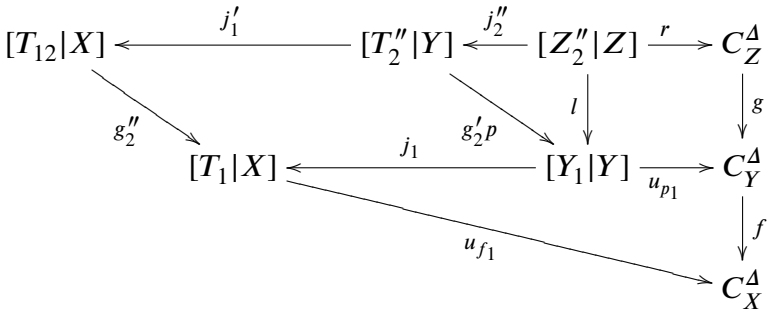


which defines a pseudo-compactification of the composite fg; denote the corresponding functor by $R^{+(12)}(fg)_!$.

. *The preceding data make it possible to define an isomorphism* $R^{+(12)}(fg)_! \xleftarrow{\sim} R^{+(1)}f_! \circ R^{+(2)}g_!$:

There is a commutative diagram, where $Z_2'' = T_2'' \times_{T_2'} Z_2'$ and all the arrows are naturally deduced from (7.1.4.1.1), (7.1.4.1.2), and (7.1.4.1.3).

(7.1.4.2.1)



470 We shall define a natural arrow:

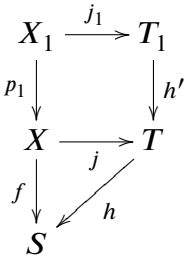
$$\beta : R^{+(1)}f_! \circ R^{+(2)}g_! \longrightarrow R^{+(12)}(fg)_!$$

which we shall show to be an isomorphism.

For this we shall use two lemmas:

[80] N.D.E.: In the original text a compactification was chosen, which is possible by the compactification theorem. Since the purpose of the appendix is to avoid that theorem (cf. la N.D.E. (76) page 743), the editor has taken the liberty of replacing it with a pseudo-compactification and slightly modifying 7.1.4.2. Another solution would be to restrict to pseudo-compactifications (p_1, j_1, f_1) for which f_1 is *projective*.

LEMMA 7.1.4.2.2. *Let there be given a commutative diagram:*



where h and h' are proper, p_1 is proper and surjective, and j and j_1 are immersions. Then the canonical arrow

$$R^+h_* \circ j_! \longrightarrow R^{+(1)}f_!$$

is an isomorphism.

Indeed, by (7.0.5), the arrow

$$j_! \longrightarrow R^+\overline{u}_{h'*} \circ j_{1!} \circ L^+\overline{u}^*_{p_1}$$

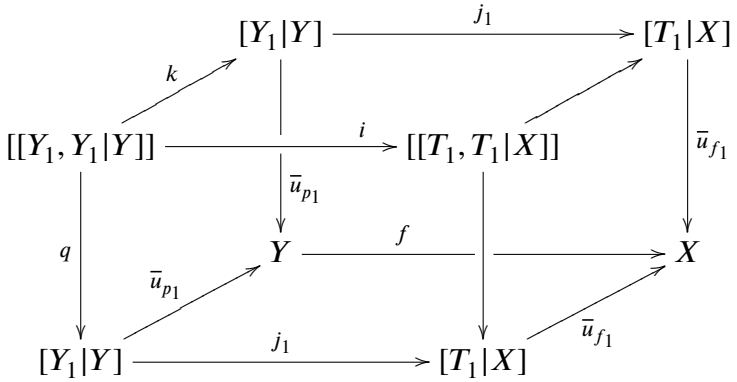
is an isomorphism. The result follows by applying R^+h_*.

LEMMA 7.1.4.2.3. *With the notation of diagram (7.1.4.2.1), the canonical arrows*

$$R^{+(1)}f_! \circ R^+\overline{u}_{p_1*} \longrightarrow R^+\overline{u}_{f_1*} \circ j_{1!}$$
$$R^{+(2)}g_! \circ R^+\overline{r}_* \longrightarrow R^+\overline{u}_{p_1*} \circ R^+(g_2'p)_* \circ j''_{2!}$$

are isomorphisms.

We prove only the first isomorphism, leaving it to the reader to deduce the second. 471
With the evident notation and the notation of 7.1.3 a), there is a commutative diagram



where q and k are the two projections from $[[Y_1,Y_1|Y]]$ to $[Y_1|Y]$. By cohomological descent, $R^+\overline{u}_{f_1*} \circ j_{1!}$ is identified with $R^+\overline{u}_{f_1*} \circ j_{1!} \circ R^+q_*L^+q^*$, which, by 7.0.3 applied to the front and top faces of the cube above, is identified with $R^+\overline{u}_{f_1*} \circ j_{1!} \circ R^+k_* \circ L^+q^*$. By the base-change theorem, this last functor is identified with $R^+\overline{u}_{f_1*} \circ j_{1!} \circ L^+\overline{u}_{p_1*} \circ R^+\overline{u}_{p_1*}$, which by definition is $R^{+(1)}f_! \circ R^+\overline{u}_{p_1*}$.[81]

[81] N.D.E.: the proof in the original text is slightly different and uses 7.1.4.2.2.

Lemma (7.1.4.2.3) having been proved, there is a sequence of isomorphisms

$$\begin{aligned} R^{+(12)}(fg)_! &= R^+\overline{u}_{f_1*} \circ R^+ g_2'' \circ j_{1!}' \circ j_{2!}'' \circ L^+\overline{r}^* \\ &\xleftarrow{\sim} R^+\overline{u}_{f_1*} \circ j_{1!} \circ R^+(g_2' p)_* \circ j_{2!}' \circ L^+\overline{r}^* \\ &\xleftarrow{\sim} R^{+(1)} f_! \circ R^+\overline{u}_{p_1*} \circ R^+(g_2' p)_* \circ j_{2!}' \circ L^+\overline{r}^* \\ &\xleftarrow{\sim} R^{+(1)} f_! \circ R^{+(2)} g_! \circ R^+\overline{r}_* \circ L^+\overline{r}^* \\ &\xleftarrow{\sim} R^{+(1)} f_! \circ R^{+(2)} g_! \end{aligned}$$

472 whose composite gives β.

. If $(p_3, j_3, (fg)_3)$ is a pseudo-compactification of fg, an isomorphism

$$R^{+(1)} f_! \circ R^{+(2)} g_! \xrightarrow{\gamma_{123}} R^{+(3)} fg_!$$

is defined by composing $\alpha_{(12)3}$ with β.

. Now let $(p_{1'}, j_{1'}, f_{1'})$ and $(p_{2'}, j_{2'}, g_{2'})$ be two other pseudo-compactifications of f and g, together with a diagram (7.1.4.1.3); one checks that the following diagram is commutative:

$$\begin{array}{ccc} R^{+(12)}(fg)_1 & \xleftarrow{\beta} & R^{+(1)} f_1 \circ R^{+(2)} g_! \\ {\scriptstyle \alpha_{(12)(1'2')}}\downarrow & & \downarrow{\scriptstyle \alpha_{11'} * \alpha_{22'}} \\ R^{+(1'2')}(fg)_! & \xleftarrow[\beta']{} & R^{+(1')} f_! \circ R^{+(2')} g_! \end{array}$$

by considering a diagram of the type (7.1.4.2.1) in which the simplicial objects are replaced by bisimplicial objects.

It follows that the following diagram is commutative:

$$\begin{array}{ccc} R^{+(3)}(fg)_! & \xleftarrow{\gamma_{123}} & R^{+(1)} f_! \circ R^{+(2)} g_! \\ {\scriptstyle \alpha_{33'}}\downarrow & & \downarrow{\scriptstyle \alpha_{11'} * \alpha_{22'}} \\ R^{+(3')}(fg)_! & \xleftarrow{\gamma_{1'2'3'}} & R^{+(1')} f_! \circ R^{+(2')} g_! \end{array}$$

for every pseudo-compactification $(p_{3'}, j_{3'}, (fg)_{3'})$ of fg, which completes the proof of 7.1.4.

7.1.5. By choosing a pseudo-compactification for every morphism f in $\mathrm{Sch}_{!R}$, in such a way that the trivial pseudo-compactification is associated with an identity morphism, one defines, for every morphism $f : X \to Y$, a triangulated functor

$$R^+ f_! : D^+(X, \mathscr{A}_X) \longrightarrow D^+(Y, \mathscr{A}_Y)$$

473 and, for every pair (f, g) of morphisms, an isomorphism

$$R^+ fg_! \xrightarrow{c_{fg}}{}^{\sim} R^+ f_! \circ R^+ g_!$$

Moreover, if f is an identity, $R^+ f_!$ is the identity functor and c_{fg} is the identity.

Proposition 7.1.6. *The correspondence $f \to R^+ f_!$ defines a normalized pseudofunctor from $\mathrm{Sch}_{!R}$ to the 2-category of categories. In other words, the categories $D^+(X, \mathscr{A}_X)$, for X an object of $\mathrm{Sch}_{!R}$, can be organized into a cofibered category over $\mathrm{Sch}_{!R}$ by setting, for every morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$,*

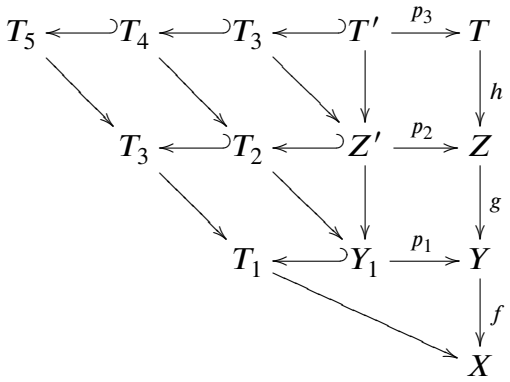
$$\mathrm{Hom}_f(K, L) = \mathrm{Hom}(R^+ f_! K, L)$$

with $K \in D^+(X, \mathscr{A}_X)$ *and* $L \in D^+(Y, \mathscr{A}_Y)$.

It remains to see that the identification $R^+fg_! \overset{c_{fg}}{\xrightarrow{\sim}} R^+f_! \circ R^+g_!$ is compatible with triple composites. Let $h : T \to Z$, $g : Z \to Y$, and $f : Y \to X$ be three morphisms in $\mathrm{Sch}_{!R}$; by what precedes, to verify this compatibility it suffices to verify it after an arbitrary choice of pseudo-compactifications for h, g, and f.

There is a diagram:

(7.1.6.1)

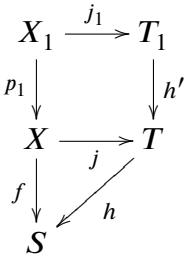


where all the oblique arrows are proper and p_1, p_2, and p_3 are proper and surjective.

A diagram of the type (7.1.4.4) constructed from (7.1.6.1) then gives the conclusion.

PROPOSITION 7.1.7. *If* $f : X \to S$ *is a compactifiable morphism, the functor* $Rf_!$ *coincides with the one defined in Exposé* XVII. 474

Indeed, there is a commutative diagram:



to which one applies (7.1.4.2.2).

DEFINITION 7.1.8. *If* $f : X \to Y$ *is a morphism in* $\mathrm{Sch}_!$, *the functor* $R^+f_!$ *just defined is called the* direct-image functor with proper supports *(in the sense of derived categories). The functor* $H^q \circ R^+f_!$, *denoted* $R^qf_!$, *is called the* q^{th} direct-image functor with proper supports.

With this notation, we leave it to the reader to verify the

CONSTRUCTION 7.1.9. *Let* (π_1, j_1, f_1) *be a pseudo-compactification of* $f : X \to Y$, *and denote by* π_1^n *the canonical morphism* $[X_1|X]_n \to X$. *Then, for every complex* $K^\cdot$ *in* $D^+(X, \mathscr{A}_X)$, *there are two biregular spectral sequences:*

$$E_1^{pq} = R^qf_!(K^p) \Longrightarrow R^{p+q}f_!(K^\cdot) \tag{7.1.9.1}$$

$$E_1^{pq} = R^q(f \circ \pi_1^p)_!(\pi_1^{p*}(K^\cdot)) \Longrightarrow R^{p+q}f_!(K^\cdot) \tag{7.1.9.2}$$

(Take, for example, the canonical flabby resolution of $j_{1!} \circ L^+\overline{u}_{p_1}^*(K^\cdot)$ (cf. XVII (4.2)), and then its direct image under u_{f_1}.)

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7.2. Some properties of $R^+f_!$.

PROPOSITION 7.2.1. *Let $f : X \to Y$ be a morphism in $\mathrm{Sch}_{!R}$ of relative dimension $\leq d$. Then the functor $R^+f_!$ has cohomological dimension $\leq 2d$.*

It is enough to show that, for every sheaf F of $\mathscr{A}_X$-modules on X, one has $R^if_!(F) = 0$ for $i > 2d$. (It then follows by a standard argument that, for every complex $K^\cdot$ in $D^+(X, \mathscr{A}_X)$ such that $H^i(K^\cdot) = 0$ for $i > k$, one has $R^if_!(K^\cdot) = 0$ for $i > k + 2d$.)

Let $\mathscr{U} = (U_i)_{i\in I}$ be a finite covering of X by affine open subsets. One defines (6.2.9) a left resolution $K^\cdot_{\mathscr{U}} \to F$ of F by putting

$$K^n_{\mathscr{U}} = \bigoplus_{|P|=-n+1} j_{P!}j_P^*(F)$$

where, for every subset P of I, j_P is the inclusion of $\cup_{i\in P}U_i$ into X.

From (7.1.9.1) one obtains a spectral sequence

$$E_1^{pq} = \bigoplus_{\substack{|P|=-p+1\\ P\subset I \text{ nonempty}}} R^p(fj_P)_!j_P^*(F) \Longrightarrow R^{p+q}f_!(F)$$

and the fact that fj_P is compactifiable, a case covered by Exposé XVII, shows that $E_1^{pq} = 0$ for $p > 0$ and $q > 2d$; it follows that $R^nf_!(F) = 0$ for $n > 2d$, as required.

PROPOSITION 7.2.2. *Let $f : X \to Y$ be a morphism in $\mathrm{Sch}_{!R}$ and let $K^\cdot$ be an object of $D^b(X, \mathscr{A}_X)$ of finite tor-dimension. Then $R^+f_!(K^\cdot)$ has finite tor-dimension.*

As above, let $\mathscr{U} = (U_i)_{i\in I}$ be a finite covering of X by affine open subsets. The functoriality of $K^n_{\mathscr{U}}$ makes it possible to construct a bounded left resolution of $K^\cdot$ by complexes of finite tor-dimension, since the latter property can be checked pointwise (4.1.9). We denote the double complex so obtained by $K^{\cdot\cdot}_{\mathscr{U}}$, the *second* index being the one corresponding to $K^\cdot$. With the conventions of (Vbis 2.3.2), for every integer n there is an exact sequence, where the operation $\sigma_{\leq n}$ is taken in $C^+(C^+(\mathrm{Mod}(X, \mathscr{A}_X))$:

$$0 \longrightarrow (-1)^{n+1}K^{n+1,\cdot}_{\mathscr{U}} \longrightarrow (\sigma_{\leq n+1}(K^{\cdot\cdot}_{\mathscr{U}}))_s \longrightarrow (\sigma_{\leq n}(K^{\cdot\cdot}_{\mathscr{U}}))_s \longrightarrow 0$$

476 hence a distinguished triangle in $D^+(X, \mathscr{A}_X)$:

$$\begin{array}{ccc} & K^{n+1,\cdot}_{\mathscr{U}} & \\ \nearrow & & \searrow \\ (\sigma_{\leq n}(K^{\cdot\cdot}_{\mathscr{U}}))_s & \longleftarrow & (\sigma_{\leq n+1}(K^{\cdot\cdot}_{\mathscr{U}}))_s \end{array}$$

Using the fact that $R^+f_!(K^{n+1,\cdot}_{\mathscr{U}})$ has finite tor-dimension for every n (5.2.10), it follows that $R^+f_!((\sigma_{\leq 0}(K^{\cdot\cdot}_{\mathscr{U}}))_s) = R^+f_!(K^\cdot)$ has finite tor-dimension.

REMARK 7.2.3. The preceding argument does not show that $R^+f_!(K^\cdot)$ has tor-dimension $\leq k$ whenever $K^\cdot$ has tor-dimension $\leq k$ (cf. 7.3.7).

PROPOSITION 7.2.4. *Let there be given a Cartesian diagram in $\mathrm{Sch}_{!R}$:*

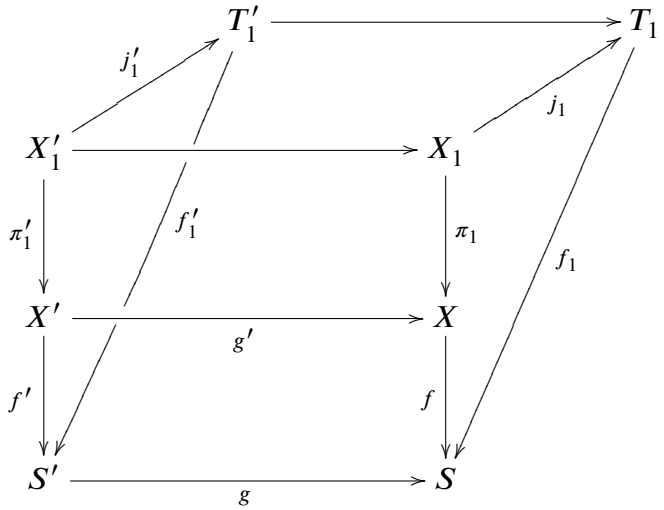
$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow[g]{} & S \end{array}$$

Then, for every complex $K^\cdot$ in $D^+(X, \mathscr{A}_X)$, the canonical morphism

$$g^* \circ R^+ f_!(K^\cdot) \xrightarrow{\mathrm{ch}_{K^\cdot}} R^+ f'_! \circ g'^*(K^\cdot)$$

is an isomorphism.

Let (π_1, j_1, f_1) be a pseudo-compactification of f; by base change it gives a pseudo-compactification (π'_1, j'_1, f'_1) of f' and a commutative diagram:



which makes it possible to define the arrow ch_K. 477

By (7.1.9.2), there is then a morphism of biregular spectral sequences:

$$\begin{array}{ccc} g^* \circ R^q(f \circ \pi_1^p)_!(\pi_1^{p*}(K^\cdot)) & \longrightarrow & g^* \circ R^{p+q} f_!(K^\cdot) \\ \downarrow{\scriptstyle u_1^{pq}} & & \downarrow{\scriptstyle \mathrm{ch}_K} \\ R^q(f' \circ \pi_1'^p)_!(\pi_1'^{p*}(g'^*(K^\cdot))) & \longrightarrow & R^+ f'_! \circ g'^*(K^\cdot) \end{array}$$

and $\mathrm{ch}_{K^\cdot}$ is an isomorphism because the same is true of the u_1^{pq} by (5.2.6).

7.3. **Definition of $Rf_!$.** We now propose to extend the functor $R^+ f_!$ "in a reasonable way" to a functor $Rf_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$, for every morphism f in $\mathrm{Sch}_{!R}$.

7.3.1. Let (p_1, j_1, f_1) be a pseudo-compactification of f, and let F be an $\mathscr{A}_X$-module. Taking the canonical flabby resolution of $j_{1!} \circ L\bar{u}^*_{p_1}(F)$ (cf. 4.2), then its direct image under f_1, and then the simple complex associated with the double complex so obtained gives a complex of $\mathscr{A}_Y$-modules:

$$0 \longrightarrow \mathscr{X}^0(F) \xrightarrow{S^1(F)} \mathscr{X}^1(F) \dots \mathscr{X}^n(F) \xrightarrow{S^n(F)} \mathscr{X}^{n+1}(F) \dots . \tag{7.3.1.1}$$

functorial in F and isomorphic in $D^+(Y, \mathscr{A}_Y)$ to $R^+ f_!(F)$. 478

By 7.2.1, $H^i(\mathscr{X}^\cdot(F)) = 0$ for $i > 2d$, where d is an integer such that the relative dimension of f is $\leq d$; hence the complex

$$0 \longrightarrow \mathscr{X}^0(F) \longrightarrow \mathscr{X}^1(F) \longrightarrow \dots \longrightarrow \mathscr{X}^{2d-1}(F) \longrightarrow \mathrm{Ker}\, S^{2d}(F) \longrightarrow 0 \tag{7.3.1.2}$$

is canonically isomorphic in $D^+(Y, \mathscr{A}_Y)$ to the complex (7.3.1.1). Moreover:

Lemma 7.3.2. *The functors $\mathscr{X}^0, \mathscr{X}^1, \dots, \mathscr{X}^{2d-1}$, and* $\mathrm{Ker}\, S^{2d}$ *are exact in F.*

The exactness of the functors $\mathscr{X}^n$ follows from the exactness of the canonical flabby resolution and from the fact that the direct image under a morphism of a short exact sequence of flabby sheaves is again a short exact sequence.

To prove the exactness of $\operatorname{Ker} S^{2d}$, for a short exact sequence $0 \to F' \to F \to F'' \to 0$ of sheaves on X, consider the diagram

$$\begin{array}{ccccc}
\operatorname{Ker} S^{2d}(F') & \longrightarrow & \mathscr{X}^{2d+1}(F') & \longrightarrow & \mathscr{X}^{2d+2}(F') \dots \\
\downarrow & & \downarrow & & \downarrow \\
\ker S^{2d}(F) & \longrightarrow & \mathscr{X}^{2d+1}(F) & \longrightarrow & \mathscr{X}^{2d+2}(F) \dots \\
\downarrow & & \downarrow & & \downarrow \\
\ker S^{2d}(F'') & \longrightarrow & \mathscr{X}^{2d+1}(F'') & \longrightarrow & \mathscr{X}^{2d+2}(F'') \dots
\end{array}$$

and conclude by using the homology exact sequence associated with the exact sequence

$$0 \longrightarrow \sigma_{\geq 2d+1}(\mathscr{X}^{\cdot}(F')) \longrightarrow \sigma_{\geq 2d+1}(\mathscr{X}^{\cdot}(F)) \longrightarrow \sigma_{\geq 2d+1}(\mathscr{X}^{\cdot}(F'')) \longrightarrow 0.$$

We now have the following general lemma:

LEMMA 7.3.3. *Let A and B be two abelian categories, and let $\mathrm{Je}^{\cdot}$ be an object of $\mathbf{C}^b(\mathscr{F}ex(A, B))$, where $\mathscr{F}ex(A, B)$ denotes the category of exact functors from A to B. Associating with every complex $K^{\cdot}$ in $\mathbf{C}(A)$ the simple complex associated with the double complex $\mathrm{Je}^{\cdot}(K^{\cdot})$ gives a triangulated functor $K(A) \to K(B)$ that preserves quasi-isomorphisms.*

479 The functor in question may be written as a composite

$$C(A) \longrightarrow C^b(C(B)) \xrightarrow{(\quad)_s} C(B),$$

where $(\quad)_s$ is the associated-simple-complex functor, and one uses (Vbis 2.3.2.2. Remark); the details are left to the reader.

EXERCISE 7.3.4. Show that, for every complex $K^{\cdot}$ in $D^+(X, \mathscr{A}_X)$, the canonical arrow

$$\tau_{\leq 2d}\mathscr{X}^{\cdot}(K^{\cdot}) \longrightarrow \mathscr{X}^{\cdot}(K^{\cdot})$$

induces a quasi-isomorphism on the associated simple complexes.

For this, observe first that the canonical arrow

$$\varinjlim_{k \geq 2d} (\tau_{\leq k}\mathscr{X}^{\cdot}(K^{\cdot}))_s \longrightarrow (\mathscr{X}^{\cdot}(K^{\cdot}))_s$$

is an isomorphism, and then that the cohomology sheaves of a complex commute with filtered direct limits.

7.3.5. Let $R^{(1)}f_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$ be the functor defined by (7.3.1.2) using 7.3.3. In view of 7.3.4, it can be computed directly from (7.3.1.1), and one can check that it "does not depend" on the chosen pseudo-compactification. One would then have to verify statements 7.1.3, 7.1.4, and 7.1.6 after deleting the superscripts +, which the editor did not have the courage to do.

DEFINITION 7.3.6. *The functor $Rf_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$ constructed in 7.3.5 is again called the* direct-image functor with proper supports.

7.3.7. One can then show that the functor $Rf_!$ commutes with base change and that statements (5.2.9) and (5.2.10) remain valid for every morphism in $\mathrm{Sch}_{!R}$, which justifies the constructions just made.

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EXPOSÉ XVIII

The global duality formula

P. Deligne

0. Introduction

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0.1. The present exposé is devoted to Poincaré duality. This duality is constructed on the model of VERDIER [1], who treats the case of locally compact Hausdorff topological spaces. We show a priori that the functor $Rf_!$ of Exposé XVII admits a right adjoint $Rf^!$ and establish many of its properties, before computing it more explicitly in the case of smooth morphisms.

The possibility of such a construction seems to rest on the following properties of the functor $Rf_!$:

a) For the existence of an adjoint $Rf^!$ and its formal properties:
   (I) the functor $Rf_!$ commutes with base change;
   (II) the functors $R^i f_!$ commute with filtered inductive limits; they vanish for sufficiently large i;
   (III) the functor $Rf_!$ can be computed "at the level of complexes", i.e. starting from a functor between categories of complexes.

b) For the computation of this adjoint when f is smooth:
   (IV) the computation of $Rf_!((\mathbf{Z}/n\mathbf{Z})_U)$ for "small" U.

0.2. In § 3 n° 1, which is independent of §§ 1, 2, we construct the functor $Rf^!$; constructions specific to the situation are used there, rather than only (I), (II), and (III). This makes this n° rather unpleasant; the author confesses that he did not always fully understand what was happening.

§ 2 is devoted to point (IV); it is purely formal from § 1, where the cohomology of 482
curves is studied.

In § 1, we prove substantially more than is needed for § 2, also studying the case of continuous coefficients. This section was profoundly influenced by SERRE [14]. The points useful in what follows are:

a) n° 1.1, devoted to the trace morphism in the case of curves, and generalized in 2.9;

b) the vanishing theorem 1.6.9. A reader willing to accept a transcendental argument may read n° 1.6, beginning with 1.6.6 (2nd proof), independently of n° 1.2 to 1.5[82]. The essential ingredient in 1.6.9 is either form of Poincaré duality for smooth curves over an algebraically closed field, together with the acyclicity lemma XV 2.6 (a preliminary to the smooth base-change theorem XVI 1.1). Its conclusion is generalized in 2.14.

Number 3.2 collects the consequences of § 2.

[82] N.D.E.: On a first reading, one cannot too strongly recommend following this route: after § 1.1, proceed directly to § 2 and § 3.

0.3. Verdier's method in Poincaré duality makes it possible to define the functor $Rf^!$ for compactifiable f. The price of this generality is that the compatibilities to be verified are unfamiliar and sometimes, even for the most trivial ones, such as 3.2.3, require outlandish proofs. The author admits that he has not proved as many of them as he should have.

An interested reader will have no difficulty in extending, using the appendix to Exposé XVII, the definition of $Rf^!$ to separated morphisms of finite type whose target is a coherent scheme (i.e. a quasi-compact quasi-separated scheme).

0.4. The main result 3.2.5 is stated in concrete terms that would make it possible to return formally to the viewpoint taken in the oral seminar, where the functor $Rf^!$ was
483 defined only for smoothable morphisms, by means of a factorization through a smooth morphism and an immersion. An outline of the proof from the oral seminar can be found in Verdier [16].

1. Cohomology of curves

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1.1. The trace morphism.

1.1.1. Let n be an integer ≥ 1 and let X be a scheme on which n is invertible. The sheaf $\boldsymbol{\mu}_n$ (IX 3.1) on X is then a sheaf of modules locally free of rank 1 over the constant sheaf of rings $\mathbf{Z}/n$. For $i \in \mathbf{Z}$, we denote by $\mathbf{Z}/n(i)$ its i^{th} tensor power:

$$\mathbf{Z}/n(i) = \mathbf{Hom}(\mathbf{Z}/n, \mathbf{G}_m)^{\otimes i}. \tag{1.1.1.1}$$

If $n = dn'$, exponentiation by d gives an isomorphism

$$\boldsymbol{\mu}_n \otimes_{\mathbf{Z}/n} \mathbf{Z}/n' \xrightarrow{\sim} \boldsymbol{\mu}_{n'},$$

and hence isomorphisms, called canonical:

$$\mathbf{Z}/n(i) \otimes_{\mathbf{Z}/n} \mathbf{Z}/n' \xrightarrow{\sim} \mathbf{Z}/n'(i). \tag{1.1.1.2}$$

If F is an abelian sheaf such that $nF = 0$, set

$$F(i) = F \otimes_{\mathbf{Z}/n} \mathbf{Z}/n(i). \tag{1.1.1.3}$$

If already $n'F = 0$, the isomorphism (1.1.1.2) allows us to identify $F \otimes_{\mathbf{Z}/n} \mathbf{Z}/n(i)$ with $F \otimes_{\mathbf{Z}/n'} \mathbf{Z}/n'(i)$, so that $F(i)$ depends only on F and not on the choice of n. More generally, if F is a torsion sheaf prime to the residual characteristics of X (XVII 0.13), and if ${}_mF$ is the kernel of multiplication by m on F, we define $F(i)$ to be the inductive limit, as m tends multiplicatively to infinity through the integers invertible on X:

$$F(i) = \varinjlim_m {}_mF(i). \tag{1.1.1.4}$$

The functor $F \mapsto F(i)$ is exact. If $f : Y \to X$ is a morphism of schemes (resp. a quasi-compact quasi-separated morphism of schemes), and if F is a torsion sheaf on
485 X (resp. on Y), prime to the residual characteristics of X, then $f^*(F(i)) \simeq (f^*F)(i)$ (resp. $(f_*F)(i) \simeq f_*(F(i))$). If $\mathscr{A}$ is a sheaf of rings on X, annihilated by an integer n invertible on X, this extends to an exact functor $K \mapsto K(i)$ from $D(X, \mathscr{A})$ to $D(X, \mathscr{A})$. This functor commutes with inverse-image functors and direct-image functors.

The functors $F \mapsto F(i)$ are sometimes called "*Tate twist*" functors.

DEFINITION 1.1.2. *A* flat curve *over a scheme S is a flat, separated morphism $f : X \to S$ of finite presentation whose fibers are of pure dimension one.*

The expression "of pure dimension one" does not exclude the empty scheme. Contrary to EGA II 7.4.2, a curve over a field is allowed here to be empty, but is required to be separated.

When S is the spectrum of a field, we shall simply speak of a curve over S; a curve over a field is quasi-projective[83].

1.1.3. Let X be a curve over an algebraically closed field k of characteristic exponent p prime to an integer $n \geq 1$. First suppose that X is reduced, and let $\overline{X}$ be a complete curve over k in which X is a dense *open subscheme*.

The complement $Y = \overline{X} - X$ is of dimension 0. The exact cohomology sequence (XVII 5.1.16.3) therefore induces an isomorphism

$$H^2_c(X, \mathbf{Z}/n(1)) \xrightarrow{\sim} H^2(\overline{X}, \mathbf{Z}/n(1)). \tag{1.1.3.1}$$

We also have an isomorphism IX 4.7 (given by Kummer theory)

$$H^2(\overline{X}, \mathbf{Z}/n(1)) = \mathrm{Pic}(\overline{X})/n = (\mathbf{Z}/n)^c \tag{1.1.3.2}$$

where c denotes the set of irreducible components of X, or of $\overline{X}$, which amounts to the same thing.

If X is no longer necessarily reduced, and if c is the set of its irreducible components, then for $i \in c$ let n_i denote the multiplicity of X at the generic point η_i of the i^{th} irreducible component of X: 486

$$n_i = \lg(O_{X,\eta_i}).$$

We know that the restriction map from $H^2_c(X, \mathbf{Z}/n(1))$ to $H^2_c(X_{\mathrm{red}}, \mathbf{Z}/n(1))$ is an isomorphism. Thus, via (1.1.3.1) and (1.1.3.2) applied to X_{red}, there is a canonical isomorphism between $H^2_c(X, \mathbf{Z}/n(1))$ and $(\mathbf{Z}/n)^c$. We denote by Tr_X, or simply by Tr, the composite map

$$\mathrm{Tr} : H^2_c(X, \mathbf{Z}/n(1)) \xrightarrow{\sim} (\mathbf{Z}/n)^c \xrightarrow{t} \mathbf{Z}/n, \tag{1.1.3.3}$$

where

$$t((a_i)_{i\in c}) = \sum n_i a_i.$$

If F is an abelian group annihilated by n, there is an isomorphism

$$H^2_c(X, F(1)) \xleftrightarrow{\sim} F \otimes H^2_c(X, \mathbf{Z}/n(1))$$

and hence, via (1.1.3.3), a *trace morphism*

$$\mathrm{Tr} : H^2_c(X, F(1)) \longrightarrow F. \tag{1.1.3.4}$$

[83] N.D.E.: In EGA II 7.4.10, this statement is proved only under the additional assumption that X is normal, and it is announced (EGA II 7.4.12) that a proof in the general case will appear in EGA V, which has not been published. The reader may fill this gap as an exercise. In any case, this has no bearing on the construction of the trace morphism, since it can be carried out first locally (on quasi-projective open subschemes) and the trace morphisms can then be glued as in the proof of 1.1.6.

If $n = n'd$, the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \boldsymbol{\mu}_n & \longrightarrow & \mathbf{G}_m & \xrightarrow{x^n} & \mathbf{G}_m & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle x^d} & & \downarrow{\scriptstyle x^d} & & \| & & \\ 0 & \longrightarrow & \boldsymbol{\mu}_n & \longrightarrow & \mathbf{G}_m & \xrightarrow{x^{n'}} & \mathbf{G}_m & \longrightarrow & 0 \end{array} \tag{1.1.3.5}$$

is commutative, and therefore so is the diagram

$$\begin{array}{ccc} H^2_c(X, \mathbf{Z}/n(1)) & \xrightarrow{\sim} & (\mathbf{Z}/n)^c \\ \downarrow & & \downarrow \\ H^2_c(X, \mathbf{Z}/n'(1)) & \xrightarrow{\sim} & (\mathbf{Z}/n')^c \end{array} \tag{1.1.3.6}$$

487 by the definition in IX 4.7 of the isomorphism (1.1.3.2).

The map (1.1.3.4) thus depends only on F, and not on the choice of the integer n prime to p such that $nF = 0$. By passage to the limit, we define it for every torsion abelian group F prime to p.

The following lemma follows immediately from the definitions.

LEMMA 1.1.4. *Under the preceding hypotheses, if, for $i \in c$, U_i is a nonempty open subscheme of X contained in its i^{th} irreducible component, the diagram*

$$\begin{array}{ccc} \bigoplus_{i\in c} H^2_c(U_i, F(1)) & \xrightarrow{\sim} & H^2_c(X, F(1)) \\ & {\scriptstyle \bigoplus_{i\in c} Tr_{U_i}} \searrow \quad \swarrow {\scriptstyle Tr_X} & \\ & F & \end{array}$$

is commutative.

LEMMA 1.1.5. *Let X and Y be two curves over an algebraically closed field k, let $u : X \to Y$ be a quasi-finite flat morphism, and let n be an integer prime to the characteristic exponent p of k.*

The following diagram, in which Tr_u *is the map* (XVII 6.2.3), *is then commutative:*

$$\begin{array}{ccc} H^2_c(X, \mathbf{Z}/n(1)) & \xrightarrow{Tr_u} & H^2_c(Y, \mathbf{Z}/n(1)) \\ & {\scriptstyle Tr_X} \searrow \quad \swarrow {\scriptstyle Tr_Y} & \\ & \mathbf{Z}/n & \end{array} .$$

Lemma 1.1.4 allows us to reduce to the case where X and Y are irreducible, with X_{red} and Y_{red} smooth as well. If e (resp. f) is the multiplicity of X (resp. Y) at the generic

point, and if $e = df$, the triangles marked + in the following diagram are commutative:

$$\begin{array}{ccc} H^2_c(X, \mathbf{Z}/n(1)) & \xrightarrow{\sim} & H^2_c(X_{\mathrm{red}}, \mathbf{Z}/n(1)) \\ \Big\downarrow{\scriptstyle \mathrm{Tr}_u} & \overset{\mathrm{Tr}_X \searrow \quad \swarrow e.\mathrm{Tr}_{X_{\mathrm{red}}}}{+} & \Big\downarrow{\scriptstyle d.\mathrm{Tr}_{u_{\mathrm{red}}}} \\ & \mathbf{Z}/n & \\ & \underset{\mathrm{Tr}_Y \nearrow \quad \nwarrow f.\mathrm{Tr}_{Y_{\mathrm{red}}}}{+} & \\ H^2_c(Y, \mathbf{Z}/n(1)) & \xrightarrow{\sim} & H^2_c(Y_{\mathrm{red}}, \mathbf{Z}/n(1)), \end{array} \tag{1.1.5.1}$$

Its outer boundary is commutative, as follows from the commutativity of the dia- 488
gram

$$\begin{array}{ccc} u_!\mathbf{Z}/n_X(1) & \xrightarrow{\sim} & u_{\mathrm{red}!}\mathbf{Z}/n_{X_{\mathrm{red}}}(1) \\ \Big\downarrow{\scriptstyle \mathrm{Tr}_u} & & \Big\downarrow{\scriptstyle d.\mathrm{Tr}_{u_{\mathrm{red}}}} \\ \mathbf{Z}/n_Y(1) & \xrightarrow{\sim} & \mathbf{Z}/n_{Y_{\mathrm{red}}}(1). \end{array}$$

It suffices to verify this last compatibility at the generic point of Y, which is trivial.

To establish the commutativity of the left-hand triangle in (1.1.5.1), it therefore remains only to establish that of the right-hand triangle. Hence we may now assume in addition that X and Y are reduced[84].

Let $\overline{X}$ and $\overline{Y}$ be the complete smooth curves containing X and Y as dense open subschemes (EGA II 7.4.11). The morphism u extends to a flat morphism $\overline{u} : \overline{X} \to \overline{Y}$ (EGA II 7.4.9), and by 1.1.4 it suffices to verify 1.1.5 for $\overline{u}$.

By (XVII 6.3.18.2), the diagram of sheaves on $\overline{Y}$ below, in which N denotes the norm,

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \overline{u}_*\boldsymbol{\mu}_n & \longrightarrow & \overline{u}_*\mathbf{G}_m & \xrightarrow{x^n} & \overline{u}_*G_m & \longrightarrow & 0 \\ & & \Big\downarrow{\scriptstyle \mathrm{Tr}_{\overline{u}}} & & \Big\downarrow{\scriptstyle N} & & \Big\downarrow{\scriptstyle N} & & \\ 0 & \longrightarrow & \boldsymbol{\mu}_n & \longrightarrow & \mathbf{G}_m & \xrightarrow{x^n} & \mathbf{G}_m & \longrightarrow & 0, \end{array}$$

is commutative, and hence so is the diagram

$$\begin{array}{ccc} \mathrm{Pic}(\overline{X})/n & \xrightarrow{\sim} & H^2_c(\overline{X}, \mathbf{Z}/n(1)) \\ \Big\downarrow{\scriptstyle N} & & \Big\downarrow{\scriptstyle \mathrm{Tr}_{\overline{u}}} \\ \mathrm{Pic}(\overline{Y})/n & \xrightarrow{\sim} & H^2_c(\overline{Y}, \mathbf{Z}/n(1)), \end{array}$$

and it remains to observe that if $\mathscr{L}$ is an invertible sheaf of degree 1 on $\overline{X}$, then its norm 489
again has degree 1 (this fact following from the compatibility EGA IV 21.10.17, taking into account the definition EGA IV 21.5.5 of the norm of a divisor).

[84] N.D.E.: This paragraph was added in this edition, and the next two were modified.

PROPOSITION 1.1.6. *There is a unique way to define, for every compactifiable flat curve $f : X \to S$ and every torsion sheaf F on S prime to the residual characteristics of S, a trace morphism*

$$\mathrm{Tr}_f : R^2 f_!(f^* F(1)) \longrightarrow F,$$

functorial in F, whose formation is compatible with every base change (cf. XVII 6.2.3, (VAR 2)), *and which, when S is the spectrum of an algebraically closed field, coincides with the trace morphism (1.1.3.4).*

The morphism Tr_f is prescribed fiber by fiber; its uniqueness is therefore clear.

Let $p : \mathbf{P}^1_S \to S$ be the projective line over S, and let n be an integer invertible on S. We have $\mathbf{Pic}_S(\mathbf{P}^1_S) = \mathbf{Z}_S$, whence Kummer theory (IX 3.2) gives a morphism $\mathbf{Z}/n \to R^1 p_* \mathbf{Z}/n(1)$ which one checks fiber by fiber (XII 5.2) to be an isomorphism. Its inverse Tr_p induces the trace morphism (1.1.3.4) fiber by fiber.

Let $f : X \to S$ be a flat curve over S such that there exists a quasi-finite flat S-morphism u from X to $\mathbf{P}^1_S$. By 1.1.5, the composite morphism $\mathrm{Tr}_p \circ \mathrm{Tr}_u$ (XVII 6.2.3) has, at every geometric point of S, the trace morphism (1.1.3.4) as its fiber. In particular, it does not depend on the choice of u.

490 Suppose that X is a union of open subschemes U_i such that there exists a quasi-finite flat morphism from U_i to $\mathbf{P}^1_S$. If α_i (resp. α_{ij}) is the inclusion of U_i (resp. $U_{ij} = U_i \cap U_j$) in X, the sequence

$$\sum R^2(f\alpha_{ij})_! \mathbf{Z}/n(1) \rightrightarrows \sum R^2(f\alpha_i)_! \mathbf{Z}/n(1) \to R^2 f_! \mathbf{Z}/n(1) \to 0.$$

is exact, by (XVII 6.2.9) and the right exactness of $R^2 f_!$ (since $R^3 f_! = 0$). One checks fiber by fiber, using 1.1.5, that the sum of the trace morphisms $\sum_i R^2(f\alpha_i)_! \mathbf{Z}/n(1) \to \mathbf{Z}/n$ factors through a trace morphism Tr_f from $R^2 f_! \mathbf{Z}/n(1)$ which induces (1.1.3.4) fiber by fiber.

For a general X, let $j : X' \to X$ be the largest open subset of X such that $f|X'$ is a Cohen–Macaulay morphism (cf. EGA IV 12.1.1 (vi)). Every point x of X' has a neighborhood U_x in X' for which there exists a quasi-finite S-morphism u_x from U_x to $\mathbf{P}^1_S$. This morphism is automatically flat (EGA IV 11.3.10 and 15.4.2), and X' therefore satisfies the preceding hypotheses. Moreover, X' is fiberwise dense in X; the morphism from $R^2(fj_!)\mathbf{Z}/n(1)$ to $R^2 f_! \mathbf{Z}/n(1)$ is consequently an isomorphism, as can be checked fiberwise, and the composite arrow

$$\mathrm{Tr}_f : R^2 f_! \mathbf{Z}/n(1) \xleftarrow{\sim} R^2(fj)_! \mathbf{Z}/n(1) \xrightarrow{\mathrm{Tr}_{fj}} \mathbf{Z}/n$$

induces (1.1.3.4) on every fiber.

If F satisfies $nF = 0$, one has (XVII 5.2.6)

$$R^2 f_! \mathbf{Z}/n(1) \otimes F \xrightarrow{\sim} R^2 f_!(f^* F(1)) \tag{1.1.6.1}$$

and Tr_f is defined to be the composite arrow

$$\mathrm{Tr}_f : R^2 f_!(f^* F(1)) \xleftarrow{\sim} R^2 f_! \mathbf{Z}/n(1) \otimes F \to F.$$

In the general case, Tr_f is defined as the filtered colimit of the trace morphisms relative to the sheaves ${}_n F$, for n invertible on S. These arrows induce (1.1.3.4) on every fiber and therefore commute with every base change, C.Q.F.D.

491 One checks fiberwise, using 1.1.4, 1.1.5, and (1.1.6.1), the following results:

LEMMA 1.1.7. *Let $f : Y \to S$ be a compactifiable flat curve over S, let $u : X \to Y$ be a quasi-finite flat morphism of finite presentation, and let F be a torsion sheaf on S, of torsion prime to the residue characteristics of S. The following diagram is commutative:*

$$\begin{array}{ccccc} R^2(fu)_!(fu)^*F(1) & = & R^2f_!(u_!u^*)f^*F(1) & \xrightarrow{\mathrm{Tr}_u} & R^2f_!f^*F(1) \\ \downarrow{\scriptstyle \mathrm{Tr}_{fu}} & & & & \downarrow{\scriptstyle \mathrm{Tr}_f} \\ F & = & F & = & F, \end{array}$$

LEMMA 1.1.8. *Let $u : Y \to S$ be a quasi-finite flat morphism of finite presentation, let $f : X \to Y$ be an S-compactifiable flat curve* (XVII 3.2.1)*, and let F be as in 1.1.7. The following diagram is commutative*[85]*:*

$$\begin{array}{ccccc} R^2(uf)_!(uf)^*F(1) & = & u_!R^2f_!f^*u^*F(1) & \xrightarrow{\mathrm{Tr}_f} & u_!u^*F \\ \downarrow{\scriptstyle \mathrm{Tr}_{uf}} & & & & \downarrow{\scriptstyle \mathrm{Tr}_u} \\ F & = & F & = & F \end{array}$$

One similarly verifies from (1.1.3.2):

LEMMA 1.1.9. *If $f : X \to S$ is a compactifiable smooth curve and the geometric fibers of f are irreducible, then the trace morphism (1.1.6) is an isomorphism.*

1.2. 1-acyclicity of projective space. In this section, the fppf topology is in principle always used (XVII 0.10).

1.2.1. Let ϵ be a locally free module on a scheme S, everywhere of rank ≥ 2, let 492
$p : \mathbf{P}(\epsilon) \to S$ be the corresponding projective bundle, and let G be an abelian sheaf on the big fppf site of S (XVII 0.10). Each pair (K_0, φ) consisting of a torsor K_0 under G and a homomorphism $\varphi : \mathbf{G}_{mS} \to G$ defines a p^*G-torsor $K(K_0, \varphi)$ on $\mathbf{P}(\epsilon)$, the sum of K_0 and the image under φ of the $\mathbf{G}_m$-torsor $\mathcal{O}(1)$[86]:

$$K(K_0, \varphi) = K_0 + \varphi\mathcal{O}(1). \tag{1.2.1.1}$$

The pairs (K_0, φ) are made into a category by setting

$$\mathrm{Hom}((K_0, \varphi), (K_0', \varphi')) = \begin{cases} \varnothing & \text{if} \quad \varphi \neq \varphi' \\ \mathrm{Hom}(K_0, K_0') & \text{if} \quad \varphi = \varphi'. \end{cases}$$

The torsor $K(K_0, \varphi)$ then depends functorially on (K_0, φ). It is compatible with localization and therefore defines a functor from the stack over S of pairs (K_0, φ) to the stack over S of p^*G-torsors on $\mathbf{P}(\epsilon)$ (= the stack whose sections over U/S are the p^*G-torsors

[85] N.D.E.: This compatibility merits a few arguments. To verify it, one may first reduce by base change to the case where S is the spectrum of an algebraically closed field k. One may then suppose that Y is connected and nonempty, in which case $Y \to S$ has a (unique) section $S \to Y$ that is a closed immersion defined by a nilpotent ideal. The arguments in the proof of 1.1.6 allow us to suppose that $X = \mathbf{P}^1_Y$. It remains only to observe that the multiplicity of Y equals the multiplicity of X at its generic point. See also the proof of the commutativity of (2.9.3).

[86] N.D.E.: In this section, we implicitly identify line bundles and $\mathbf{G}_m$-torsors by the correspondence that associates with a line bundle the sheaf of its invertible sections, on which $\mathbf{G}_m$ acts by multiplication. A reader concerned with details may wish to know that, following Grothendieck's convention, the projective scheme $\mathbf{P}(\epsilon)$ is the bundle of hyperplanes in ϵ, and likewise the punctured vector bundle $\mathbf{V}(\epsilon)^* \to \mathbf{P}(\epsilon)$ is an open subset of $\mathbf{V}(\epsilon)$, which is the spectrum of the symmetric algebra of ϵ.

on $U \times_S \mathbf{P}(\epsilon)$). These stacks are stacks in groupoids (= every morphism over a U/S is an isomorphism).

THEOREM 1.2.2. *Under the hypotheses of 1.2.1, if G is a flat commutative group scheme of finite presentation over S, then the functor K of 1.2.1 is an equivalence.*

. This assertion is equivalent to the conjunction of

$$G \xrightarrow{\sim} p_* p^* G \quad \text{and}$$

$$\mathbf{Hom}(\mathrm{G}_m, G) \xrightarrow{\sim} R^1 p_* p^* G.$$

Indeed, a morphism of stacks in groupoids is an equivalence if and only if it induces an isomorphism on the sheaf generated by the presheaf of isomorphism classes of objects (the 2nd formula) and an isomorphism on the sheaf of automorphisms of any local object (the 1st formula).

We shall give two proofs (1.2.4 and 1.2.5) of full faithfulness[87].

493 LEMMA 1.2.3. *Let $f : X \to S$ be a proper flat morphism. Suppose that $f_* \mathcal{O}_X = \mathcal{O}_S$ universally and $R^1 f_* \mathcal{O}_X = 0$ universally. Every S-morphism from X to a finite-type S-group scheme G then factors through f and a section of G; more precisely, $G \xrightarrow{\sim} f_* f^* G$.*

It suffices to prove the 1st assertion.

By the rigidity lemma (MUMFORD [10, 6.1. p. 115]; note that MUMFORD does not use the Noetherian hypothesis imposed on S), it suffices to treat the case where S is the spectrum of an algebraically closed field. Suppose first that G is an abelian scheme, and let G^* be the dual abelian scheme. Since $G = G^{**}$, giving $\varphi : X \to G$ is equivalent to giving an invertible sheaf $\mathscr{L}$ on $X \times G^*$, trivialized along $X \times \{e\}$ and algebraically equivalent to zero on the fibers of the projection pr_1 from $X \times G^*$ to X. If $x : S \to X$ is a rational point of X, then $\mathscr{L} \sim \mathrm{pr}_2^*(x \times G^*)^* \mathscr{L} \otimes \mathscr{M}$, with $\mathscr{M}$ trivialized along $x \times G^*$ and $X \times \{e\}$. Giving $\mathscr{M}$ is equivalent to giving a morphism of schemes from G^* to $\mathbf{Pic}_{X/S}$ that takes e to e. Since $H^1(X, \mathcal{O}_X) = 0$, one has $\mathbf{Pic}^0_{X/S} = 0$[88], $\mathscr{M} \sim \mathcal{O}$, and $\mathscr{L} \sim \mathrm{pr}_2^* \mathscr{L}'$, with $\mathscr{L}'$ algebraically equivalent to zero on G^*, hence defining $g \in G(S)$ such that $\varphi = g \circ f$.

In the general case, G is an extension of an abelian scheme A by an affine group G_0. The preceding result reduces us to the case $A = 0$, and then

$$\mathrm{Hom}_S(\mathrm{Spec}(H^0(X, \mathcal{O}_X), G) \xrightarrow{\sim} \mathrm{Hom}_S(X, G).$$

LEMMA 1.2.4. *Under the hypotheses of 1.2.1, if G is a group scheme of finite presentation over S or is defined by a quasi-coherent sheaf on S, then the functor (1.2.1.1) is fully faithful.*

A. We prove that, if $K(K_0, \varphi)$ is isomorphic to $K(K_0', \varphi')$, then $\varphi = \varphi'$. For quasi-coherent G, one has $\varphi = \varphi' = 0$, and there is nothing to prove. For G of finite
494 presentation, by (SGA 3 IX 5.1), it suffices to prove the assertion when S is the spectrum of an algebraically closed field k. Let T be the largest subtorus of G, and put $H = G/T$. The cohomology exact sequence gives

$$0 \to H^0(\mathbf{P}(\epsilon), T) \to H^0(\mathbf{P}(\epsilon), G) \xrightarrow{\alpha} H^0(\mathbf{P}(\epsilon), H) \xrightarrow{\delta} H^1(\mathbf{P}(\epsilon), T) \xrightarrow{\beta} H^1(\mathbf{P}(\epsilon), G).$$

[87] N.D.E.: With regard to essential surjectivity, the reader should be warned that the proof will use the general proof technique explained in the introduction to the EGA .

[88] N.D.E.: This follows from the canonical isomorphism $\mathrm{Lie}(\mathbf{Pic}_{X/S}) = H^1(X, \mathcal{O}_X) = 0$, obtained by comparing both groups with the kernel of $\mathrm{Pic}(X \times_k \mathrm{Spec}(k[\epsilon])) \to \mathrm{Pic}(X)$ (where $S = \mathrm{Spec}(k)$). For further details on the properties of Picard schemes, see [6].

By 1.2.3, $H^0(\mathbf{P}(\epsilon), G) \sim G(k)$ and $H^0(\mathbf{P}(\epsilon), H) \sim H(k)$, so α is surjective and δ is zero. If $Y(T) = \mathrm{Hom}(\mathbf{G}_m, T)$, then $T \sim G_m \otimes Y(T)$ and $H^1(\mathbf{P}(\epsilon), T) \sim Y(T)$. Thus the injectivity of β expresses the injectivity of the canonical arrow defined by $\mathcal{O}(1)$, $\mathrm{Hom}(\mathbf{G}_m, G) \simeq \mathrm{Hom}(\mathbf{G}_m, T) \to H^1(\mathbf{P}(\epsilon), G)$, proving the assertion.

B. Proving that the functor (1.2.1.1) is fully faithful is local on S. We may therefore suppose that $\mathbf{P}(\epsilon)/S$ has a section s and choose an isomorphism $s^*\mathcal{O}(1) \sim \mathcal{O}$[89]. If $K(K_0, \varphi)$ is isomorphic to $K(K_0', \varphi')$, then $\varphi = \varphi'$ by A, and $K_0 \sim s^*K(K_0, \varphi) \sim s^*K(K_0', \varphi') \sim K_0'$. By the formula

$$\mathrm{Hom}(K(K_0, \varphi), K(K_0', \varphi')) \sim \mathrm{Hom}(K(K_0 - K_0', \varphi - \varphi'), G_{\mathbf{P}(\epsilon)}), \tag{1.2.4.1}$$

it remains only to show that

$$\mathrm{Hom}_{\mathrm{tors}.}(G, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{tors}.}(p^*G, p^*G).$$

This follows from 1.2.3.

1.2.5. Here is another, more Künneth-like, proof of full faithfulness. The question is local on S; by (1.2.4.1), it therefore suffices to check that

$$\mathrm{Hom}_{\mathbf{P}(\epsilon)}(G, K(G, \varphi)) \begin{cases} = \varnothing & \text{if} \quad \varphi \neq 0 \\ \xleftarrow{\sim} G(S) & \text{if} \quad \varphi = 0. \end{cases}$$

This is precisely what the following lemma says[90]:

LEMMA 1.2.6. *Let ϵ be a locally free module everywhere of rank ≥ 2 on S, let G be a group scheme of finite presentation over S, let $\varphi : \mathbf{G}_m \to G$ be a group homomorphism, let $\mathbf{V}(\epsilon)$ be the vector bundle defined by ϵ, let $\mathbf{V}(\epsilon)^*$ be the corresponding punctured vector* 495
bundle, and let $\Psi : \mathbf{V}(\epsilon)^* \to G$ be an S-morphism satisfying the following identity between S-morphisms from $\mathbf{G}_m \times \mathbf{V}(\epsilon)^*$ to G:

$$\Psi(\lambda v) = \Psi(v)\varphi(\lambda).$$

Then $\varphi = e$, and Ψ factors through the projection p from $\mathbf{P}(\epsilon)$ to S.

It suffices to prove the second assertion. We reduce at once to the case where S is Noetherian and then, by (SGA 3 IX 5.1) and the rigidity lemma (MUMFORD [14, 6.1 p. 115]) applied to $\mathbf{P}(\epsilon)$, to the case where S is the spectrum of a field. Let n be an integer, let W_n and G_n be the n^{th} infinitesimal neighborhoods of the zero section in $\mathbf{V}(\epsilon)$ and G, let $\mathfrak{g}_n$ be the affine algebra of G_n, and let $P_n(\Psi) : \mathbf{V}(\epsilon)^* \to \mathbf{Hom}_S(W_n, G)$ be the principal part of n^{th} order of Ψ[91]. Every section of $\mathbf{Hom}_S(W_n, G)$ defines, by right

[89] N.D.E.: This additional necessary datum, which was missing from the original edition, has been added in this edition.

[90] N.D.E.: This translation merits some explanation. The line bundle $\mathcal{O}(1)$ on $\mathbf{P}(\epsilon)$ becomes canonically trivialized after pullback by the canonical projection $\mathbf{V}(\epsilon)^* \to \mathbf{P}(\epsilon)$. After pullback to $\mathbf{V}(\epsilon)^*$, the G-torsor $K(G, \varphi)$ is therefore canonically trivialized as well. *Via* this identification, an isomorphism $G \to K(G, \varphi)$ thus induces an automorphism of the trivial G-torsor over $\mathbf{V}(\epsilon)^*$, that is, an S-morphism $\mathbf{V}(\epsilon)^* \to G$. To descend from $\mathbf{P}(\epsilon)$, this morphism must satisfy a compatibility like the one stated, and the converse follows by faithfully flat descent. Strictly speaking, it seems to the reviewer that the correct condition should be $\Psi(\lambda v) = \varphi(\lambda)^{-1}\Psi(v)$, but this has no effect, since one can apply the lemma to Ψ^{-1} instead of Ψ. The condition $\Psi(\lambda v) = \varphi(\lambda)\Psi(v)$ in the original edition also gives a correct statement (since φ can be replaced by φ^{-1}), but it has been changed so that the proof need not be modified.

[91] N.D.E.: More concretely, $P_n(\Psi)$ is the morphism corresponding by adjunction to the morphism $\mathbf{V}(\epsilon)^* \times_S W_n \to G$ (which we also denote by P_n) obtained by composing Ψ with the morphism $\mathbf{V}(\epsilon)^* \times_S W_n \to \mathbf{V}(\epsilon)^*$ induced on these subschemes by addition on $\mathbf{V}(\epsilon)$. In terms of the functors of points of these

translation by the inverse of the image of the zero section, a morphism from W_n to G_n that takes 0 to e; this defines

$$\Psi_1 = P_n(\Psi).\Psi^{-1} : \mathbf{V}(\epsilon)^* \longrightarrow \mathbf{Hom}_S(W_n, G_n).$$

The hypothesis implies that Ψ_1 commutes with homotheties, which act on the right-hand side by transport of structure from their action on $W_n \subset \mathbf{V}(\epsilon)$[92]. The affine algebra of W_n is $\sum_{i=0}^{n} \mathrm{Sym}^i(\epsilon)$. By homogeneity, Ψ_1 is therefore defined by a section, on $\mathbf{P}(\epsilon)$, of the $\mathcal{O}_{\mathbf{P}(\epsilon)}$-module

$$\sum_{i=0}^{n} p^* \mathbf{Hom}(\mathfrak{g}_n, \mathrm{Sym}^i(\epsilon))(-i).$$[93]

The components of index $i > 0$ of this module have no section other than the zero section, and it follows that

$$\Psi_1 = 0.$$

Since this holds for every n, it implies that Ψ factors through a section of G.

This second proof extends to prove the following analytic variant of 1.2.4. When G is affine, the proof of 1.2.2 can be given independently of this variant.

496 1.2.7. Let L be the fraction field of a complete discrete valuation ring $\mathcal{O}$, let U_m be the rigid analytic "generic fiber" group of the formal completion of $\mathbf{G}_m$, and let $\mathcal{O}(1)^\wedge$ be the U_m-torsor on $\mathbf{P}^r_L$ obtained by formally completing $\mathcal{O}(1)$. If G is a rigid analytic group over L, the functor $(K_0, \varphi) \mapsto p^*K_0 + \varphi\mathcal{O}(1)^\wedge$ (cf. 1.2.1) is a fully faithful functor from the category of pairs (K_0, φ) consisting of a G-torsor on $\mathrm{Spec}(L)$ and $\varphi \in \mathrm{Hom}(U_m, G)$ to the category of p^*G-torsors on $\mathbf{P}^r_L$[94].

Lemma 1.2.8. *Let k be a field of characteristic exponent p, and let $f : X \to \mathrm{Spec}(k)$ be a proper scheme over k satisfying $H^0(X, \mathcal{O}_X) = k$ and $\mathbf{Pic}^\tau_X = 0$[95]. For every finite-type commutative affine group scheme G over k containing no tori, one has $R^1 f_*(f^*G) = 0$.*

It suffices to show that after every base change $c : S \to \mathrm{Spec}(k)$, the functor f^* from the category of G-torsors on S to that of G-torsors on X_S is an equivalence. This problem is local on $\mathrm{Spec}(k)$, so that, in order to solve it, one may reduce to the case where X has a rational point, where G is a successive extension of the groups $\mathbf{G}_a$, α_p, $\boldsymbol{\mu}_p$, $\mathbf{Z}/p$, and $\mathbf{Z}/\ell_i$, with ℓ_i prime to p, and where $\mathbf{Z}/\ell_i \sim \boldsymbol{\mu}_{\ell i}$.

S-schemes, this can be interpreted more concretely by saying that, if v is a section of $\mathbf{V}(\epsilon)^*$ and t a section of W_n, then $P_n(\Psi)(v, t) = \Psi(v + t)$.

[92] N.D.E.: With the notation above, Ψ_1 can be characterized by the relation $\Psi_1(v, t).\Psi(v) = \Psi(v+t)$. It follows at once that, if λ is a section of $\mathbf{G}_m$, then $\Psi_1(\lambda v, \lambda t) = \Psi_1(v, t)$, which is the asserted commutation with homotheties provided that $\mathbf{G}_m$ acts correctly on $\mathbf{Hom}_S(W_n, G_n)$, namely, that a section λ of $\mathbf{G}_m$ acts by right composition with multiplication by the *inverse* of λ on W_n.

[93] N.D.E.: To verify this assertion in detail, one may begin by defining the pullback of this section to $\mathbf{V}(\epsilon)^*$; one then obtains tautologically a section $s = (s_0, \dots, s_n)$ of the trivial bundle obtained by tensoring the structure sheaf with the vector space $\sum_{i=0}^{n} \mathbf{Hom}(\mathfrak{g}_n, \mathrm{Sym}^i(\epsilon))$. It satisfies the relation $s_i(v) = \lambda^i s_i(\lambda v)$. This is indeed descent data giving the desired section on $\mathbf{P}(\epsilon)$. One can also argue that the sections s_i extend to all of $\mathbf{V}(\epsilon)$ (since the origin is a closed subset of codimension ≥ 2 in $\mathbf{V}(\epsilon)$). By the density of $\mathbf{G}_m$ in $\mathbf{A}^1$, the extended sections s_i satisfy the relation $s_i(v) = \lambda^i s_i(\lambda v)$ for $\lambda \in \mathbf{A}^1$. Taking $\lambda = 0$ gives $s_i = 0$ for $i > 0$.

[94] N.D.E.: The reader may find an introduction to the language of rigid analytic geometry in [13]. The nonexistence of nontrivial sections of the sheaves $\mathcal{O}(n)$ for $n < 0$ in this context follows, for example, from a rigid version of GAGA; see [9].

[95] N.D.E.: A definition of $\mathbf{Pic}^\tau$ in a more general setting will be found in 1.6.4.

We know that $\mathrm{Lie}(\mathbf{Pic}_X) = H^1(X, \mathcal{O}_X) = 0$. By 1.2.3 and the existence of a section, the functors f^* under consideration are fully faithful; it therefore suffices to prove that $R^1 f_*(f^*G) = 0$, or, by dévissage, that

$$R^1 f_* \mathbf{G}_a = R^1 f_* \alpha_p = R^1 f_* \boldsymbol{\mu}_p = R^1 f_* \mathbf{Z}/p = R^1 f_* \mathbf{Z}/\ell_i = 0.$$

For $\mathbf{G}_a$, this follows from $H^1(X, \mathcal{O}_X) = 0$. The case of α_p (resp. $\mathbf{Z}/p$) follows from it via 1.2.3, by the long exact cohomology sequence defined by the exact sequence

$$0 \longrightarrow \alpha_p \longrightarrow \mathbf{G}_a \xrightarrow{x^p} \mathbf{G}_a \longrightarrow 0$$

(resp. $$0 \longrightarrow \mathbf{Z}/p \longrightarrow \mathbf{G}_a \xrightarrow{x - x^p} \mathbf{G}_a \longrightarrow 0 \qquad (p \neq 1)).$$

The case of $\mathbf{Z}/\ell_i$ (resp. of $\boldsymbol{\mu}_p$) follows from the isomorphism $\mathbf{Z}/\ell_i \sim \boldsymbol{\mu}_{\ell_i}$, from 1.2.3, and 497
from the long exact sequence deduced from the Kummer exact sequence

$$0 \longrightarrow \boldsymbol{\mu}_n \longrightarrow \mathbf{G}_m \longrightarrow \mathbf{G}_m \longrightarrow 0.$$

Lemma 1.2.9. *Let S be a local Artinian scheme, let $\bar{s}$ be a geometric point of S, let $f : X \to S$ be a proper flat morphism, and let G be a flat commutative group scheme of finite type over S. Assume that f is smooth or that G is affine. If $H^0(X_{\bar{s}}, \mathcal{O}) = k(\bar{s})$, if $\mathbf{Pic}^{\tau}_{X_{\bar{s}}} = 0$, and if $G_{\bar{s}}$ contains no subgroup $\mathbf{G}_m$, then the functor f^* from the category of G-torsors on S to that of f^*G-torsors on X is an equivalence.*

a. The problem in question is local on S when the finite flat morphisms $u : S' \to S$ are taken as localization morphisms. This makes it possible to assume that f admits a section x[96]. The functor f^* is then fully faithful, by 1.2.3 and the existence of x, and it remains to show that every torsor trivialized along x is trivial.
b. Suppose that S is the spectrum of a field. If G is affine, the assertion follows from 1.2.8. If X is smooth and G is an extension of an abelian variety A by an affine group G_0, denote by G^n the inverse image of ${}_nA$ in G. The group $H^1(X, A)$ is torsion (Raynaud [12, XIII 2.6]), and is therefore the union of the images of the $H^1(X, {}_nA)$. The group $H^1(X, G)$ is consequently the union of the images of the $H^1(X, G^n)$. Since the groups G^n are affine, the conclusion again follows from 1.2.8.
c. Let us prove the general case (assuming the existence of a section x) by induction on the length of S. Let $i : S_0 \hookrightarrow S$ be a closed subscheme defined by an ideal I of square zero and length 1.

By the induction hypothesis, the G-torsors P on X for which x^*P is trivial are deformations of the trivial G-torsor P_0 on $X \times S_0$. If G is smooth, and if L is the Lie algebra of G_s, these deformations are classified by

$$H^1(X_s, f^*(L \otimes I)) = 0,$$

which proves 1.2.9 in this case. 498

For G flat over S, this argument generalizes in terms of the *Lie complex* L of G_s. If $L \cap (G_s/k(s))$ denotes the relative cotangent complex of G_s, this complex $L \in D^b(k(s))$ is the dual of $Le^*(L \cap (G_s/k(s)))$. We have $H^i(L) = 0$ for $i \neq 0, 1$(Illusie [7, VII 2.4.1]).

The deformations over S (resp. over X) of the trivial G-torsor over S_0 (resp. over X_0) are classified by $H^1(L \otimes I)$ (resp. by $H^1(X_s, f^*(L \otimes I))$ (Illusie [7, VII 2.4.4 (ii)]).

[96] N.D.E.: The existence of a section after such a base change may be deduced from 2.3 and 2.4.

By hypothesis, we have $H^1(X_s, \mathscr{H}^0(f^*(L \otimes I))) = 0$ (because $H^1(X_s, \mathcal{O}) = 0$) and $H^1(L \otimes I) \sim H^0(X_s, \mathscr{H}^1(f^*L \otimes I)))$. We conclude that

$$H^1(L \otimes I) \xrightarrow{\sim} H^1(X_s, f^*(L \otimes I)),$$

which means that every deformation over X of the trivial G-torsor over X_0 is the inverse image of one and only one deformation over S of the trivial G-torsor over S_0. If this deformation is trivial along a section, it is therefore trivial, which proves 1.2.9.

1.2.10. Let us prove 1.2.2 when S is Artinian. Let T be the largest subtorus of G; since full faithfulness is already known (1.2.4), the problem is local on S for the étale topology: we may assume that T is diagonalizable, $T \sim \mathbf{G}_m^n$. By 1.2.3 and 1.2.9, the exact cohomology sequence gives

$$R^1 f_* T \xrightarrow{\sim} R^1 f_* G,$$

and the assertion follows, since it is trivial for $G = T$ and $\mathrm{Hom}(\mathbf{G}_m, T) \xrightarrow{\sim} \mathrm{Hom}(\mathbf{G}_m, G)$ (cf. 1.2.2.1).

1.2.11. Let us prove 1.2.2 when S is complete Noetherian local. Let s be a section of p[97].

If K is a G-torsor on $\mathbf{P}(\epsilon)$, by 1.2.10 there exists a unique pair $(\widehat{\varphi}, \widehat{\Psi})$ consisting of a formal
499 homomorphism $\widehat{\varphi} : \mathbf{G}_m \to G$ and a formal isomorphism $\widehat{\Psi} : \widehat{p^* s^* K} + \widehat{\varphi}\widehat{\mathcal{O}(1)} \xrightarrow{\sim} \widehat{K}$[98][99]. If G is affine, then $\widehat{\varphi}$ is algebraizable (SGA 3 IX 7.1). Let us temporarily admit that, as will be proved in (1.2.13), $\widehat{\varphi}$ is always algebraizable as $\varphi : \mathbf{G}_m \to G$.

If $K_0 = s^* K$, and if $K_1 = p^* K_0 + \varphi\mathcal{O}(1)$, we have a formal isomorphism $\widehat{\Psi} : \widehat{K_1} = \widehat{K(K_0, \varphi)} \xrightarrow{\sim} \widehat{K}$, i.e. a formal trivialization of $K - K_1$. By formal GAGA, this section is algebraic, which proves 1.2.11. More precisely, if G is quasi-affine, then $K - K_1$ is representable and one uses EGA III 5.1.4. In the general case, one proceeds in the same way with the algebraic space $K - K_1$.

1.2.12. *Algebraicity of $\widehat{\varphi}$: preliminary verification over a trait.* Suppose that S is a complete trait, $S = \mathrm{Spec}(V)$, and let L be the fraction field of V.

a) Over L, there exists a unique triple $(K_{0L}, \varphi_L, \Psi_L)$:

$$\Psi_L : K(K_{0L}, \varphi_L) \xrightarrow{\sim} K|\,\mathrm{Spec}(L).$$

b) By (1.2.10) (1.2.11), and with the notation of 1.2.7, we have a rigid analytic map $\widehat{\varphi_L} : U_m \to \widehat{G}$, a $\widehat{G}$-torsor $\widehat{K_0}$, and a rigid analytic isomorphism

$$\widehat{\Psi} : p^* \widehat{K_0} + \widehat{\varphi_L}\widehat{\mathcal{O}(1)} \xrightarrow{\sim} \widehat{K}.$$

By the uniqueness assertion 1.2.7, $\widehat{\varphi_L}$ is identified with the map deduced from φ_L. Thus the formal morphism $\widehat{\Psi}$ of 1.2.11 induces on $\boldsymbol{\mu}_n$ a map that coincides with the map deduced from the algebraic map $\varphi_L : \mathbf{G}_m \to G$.

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[97] N.D.E.: As above, one must also choose an isomorphism $s^*\mathcal{O}(1) \sim \mathcal{O}$.

[98] N.D.E.: One must require this isomorphism to become the obvious isomorphism after applying s^*.

[99] N.D.E.: What is meant here is that, if A is the ring of S, I its maximal ideal, and $S_n = \mathrm{Spec}(A/I^n)$, a formal morphism consists of a compatible system of morphisms between objects over S_n for every n. The question of algebraicity is whether this compatible system is obtained by inverse image from similar data over S.

1.2.13. *Algebraicity of $\widehat{\varphi}$.* The formal map $\widehat{\varphi} : \mathbf{G}_m \to G$ of 1.2.11 induces, for ℓ invertible on S, maps

$$\varphi_\ell : T_\ell(\widehat{\mathbf{G}_m}) = T_\ell(\mathbf{G}_m) \longrightarrow T_\ell(G).$$

By (1.2.12) and (EGA II 7.1.9), for every point t of S, there exists an algebraic morphism 500
$\varphi_t : \mathbf{G}_{m_t} \to G_t$ such that $T_\ell(\varphi_t) = T_\ell(\varphi)_t$. By (SGA 3 XV 4.1 bis), $\widehat{\varphi}$ is therefore algebraizable.

1.2.14. Let us prove 1.2.2. By passage to the limit, we reduce to the case where S is Noetherian, and then where S is Noetherian local with affine coordinate ring A. Let s be a section of p. Let K be a G-torsor on $\mathbf{P}(\epsilon)$. Over $S' = \mathrm{Spec}(\widehat{A})$, there exists one and only one homomorphism $\varphi' : \mathbf{G}_{mS'} \to G_{S'}$ such that the inverse image K' of K on $\mathbf{P}(\epsilon) \times_S S'$ is of the form $K(K'_0, \varphi')$. This homomorphism φ' descends to $\varphi : \mathbf{G}_m \to G$. To prove the essential surjectivity of the functor 1.2.1, we may replace K by $K - \varphi\mathcal{O}(1)$, i.e. assume that $\varphi = 0$. Over S', there then exists one and only one isomorphism $\Psi : p'^* s^* K' \xrightarrow{\sim} K'$, inducing the identity on the section s. This morphism descends to S (fpqc descent of morphisms of fppf torsors), and this proves 1.2.2.

Variants 1.2.15. *The conclusion of 1.2.2 still holds in the following cases.*

(i) *G is defined by a quasi-coherent module on S.*
(ii) *G is the inverse image of a torsion sheaf on the small étale site of S.*

In case (i), it is standard that the arrows (1.2.2.1) are isomorphisms, taking into account the fact that the fppf cohomology of G coincides with its cohomology for the Zariski topology.

In case (ii), the arrows (1.2.2.1) are isomorphisms by the base change theorem for a proper morphism and the simple connectedness of $\mathbf{P}^r_k$ for k algebraically closed.

1.3. Integration of torsors.

1.3.1. Let S be a scheme and G an abelian sheaf on the big fppf site of S, satisfying the following three conditions.

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(i) *For every projective space $p : \mathbf{P}^r_S \to S$, with $r \geq 1$, one has, on the big* fppf *site of S* (XVII 0.10) 501

$$G \xrightarrow{\sim} p_* p^* G \quad \text{and}$$

$$\mathbf{Hom}(\mathbf{G}_m, G) \xrightarrow{\sim} R^1 p_* p^* G$$

(ii) *For every n and every smooth affine S-morphism $f : X \to T$, if K_1 and K_2 are two G-torsors on $\mathrm{Sym}^n_T(X)$, the set $\mathrm{Hom}(K_1, K_2)$ is identified with the set of symmetric morphisms from the inverse image of K_1 on $(X/T)^n$ to the inverse image of K_2 on $(X/T)^n$.*

This condition is equivalent to the conjunction of:

(ii′) $\mathrm{Hom}_S(\mathrm{Sym}^n_T(X), G) \xrightarrow{\sim} \mathrm{Hom}_S((X/T)^n, G)^{\sigma_n}$, and
(ii″) If the inverse image on $(X/T)^n$ of a torsor K on $\mathrm{Sym}^n_T(X)$ admits a symmetric section, then K is trivial.
(iii) *G satisfies either condition* (XVII 6.3.21.1), *or conditions* (6.3.24.0) *and* (456).

The conclusion of 1.2.2 therefore holds (cf. 1.2.2.1), as does the formula of XVII 6.3.26.

Condition (i) is discussed in 1.2.2 and 1.2.15. Condition (ii) holds whenever G is representable and of finite presentation, or is the inverse image of a sheaf on the small

étale site of S, or is flat quasi-coherent. Condition 1.3.1.1 therefore holds in each of the following cases.

.

(a) G is smooth and of finite presentation (1.2.2 and XVII 6.3.21.1).
(b) G is the inverse image of a torsion sheaf on the small étale site of S (1.2.15 (ii) and XVII 6.3.21.1).
(c) G is affine, and the kernel of an epimorphism of smooth groups (1.2.2 and XVII 6.3.3.1).
(d) G is defined by a flat quasi-coherent sheaf on S (1.2.15 (i) and XVII 6.3.21.1).

502 1.3.2. Let $f : X \to S$ be a smooth projective curve over a scheme S, with nonempty geometrically connected fibers. Let $\mathscr{L}$ be an invertible sheaf on X that is "sufficiently" relatively ample in the following sense.

. For every geometric point s of S, if X_s has genus g, then

$$\deg_{X_s}(\mathscr{L}) > 2g-2,$$
$$\deg_{X_s}(\mathscr{L}) \geq 1 \quad \text{if} \quad g=0, \quad \text{et} \quad \deg_{X_s}(\mathscr{L}) \geq 2 \quad \text{if} \quad g=1$$

We then have $R^1 f_* \mathscr{L} = 0$, and $f_* \mathscr{L}$ is locally free, its formation commutes with every base change, and it has rank ≥ 2 everywhere. If a section s of $f_* \mathscr{L}$ vanishes at no point, the scheme D_s of zeros of s, regarded as a section of $\mathscr{L}$ on X, is a relative divisor (EGA IV 21.15.2), finite locally free over S since X/S has relative dimension 1. We have canonically

$$s : \mathscr{O}(D_s) \xrightarrow{\sim} \mathscr{L}.$$

The divisor D_s depends on s only up to multiplication by a section of $\mathscr{O}_S^*$. The preceding construction therefore gives us a relative divisor $Z_{\mathscr{L}}$ on the scheme obtained from X by the base change

$$p : \mathbf{P}(f_*(\mathscr{L})^\vee) \longrightarrow S.$$

Let K be a torsor under G_X, and let $K|Z_{\mathscr{L}}$ be its inverse image on $Z_{\mathscr{L}}$. The trace of this torsor (XVII 6.3.26), from $Z_{\mathscr{L}}$ to $\mathbf{P}(f_*(\mathscr{L})^\vee)$, is a torsor $K_{\mathscr{L}}$ on $\mathbf{P}(f_*(\mathscr{L})^\vee)$:

$$K_{\mathscr{L}} = \mathrm{Tr}_{Z_{\mathscr{L}}/\mathbf{P}}(K|Z_{\mathscr{L}}).$$

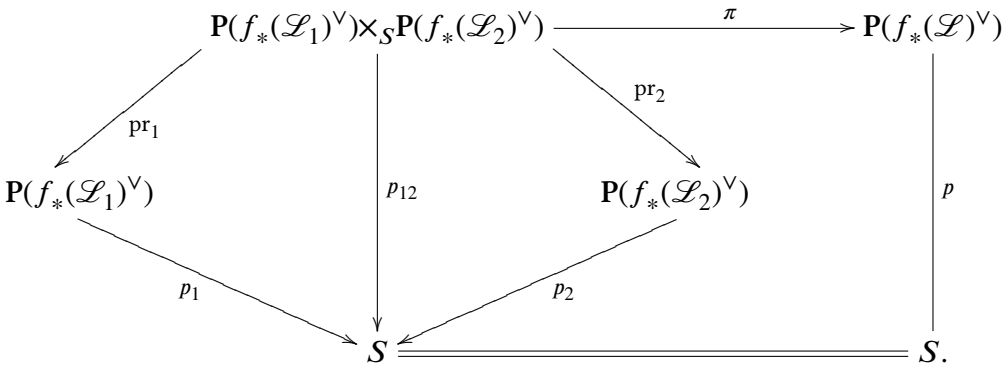
By 1.2.2, there exists a triple, unique up to unique isomorphism, consisting of a G-torsor $\langle \mathscr{L}, K]$ on S, a morphism $\varphi_{\mathscr{L},K} : \mathbf{G}_m \to G$, and an isomorphism

$$K_{\mathscr{L}} \simeq p^* \langle \mathscr{L}, K] + \varphi_{\mathscr{L},K} \mathscr{O}(1).$$

503 1.3.3. If the composition of torsors is written additively, we have canonically

$$\langle \mathscr{L}, K_1 + K_2] \simeq \langle \mathscr{L}, K_1] + \langle \mathscr{L}, K_2] \tag{1.3.3.1}$$
$$\varphi_{\mathscr{L},K_1+K_2} = \varphi_{\mathscr{L},K_1} + \varphi_{\mathscr{L},K_2}. \tag{1.3.3.2}$$

Let $\mathscr{L}_1$ and $\mathscr{L}_2$ be two invertible sheaves on X satisfying (1.3.2.1), and let $\mathscr{L} = \mathscr{L}_1 \otimes \mathscr{L}_2$. The tensor product of sections defines a morphism



We have $\pi^* Z_{\mathscr{L}} \simeq Z_{\mathscr{L}_1} + Z_{\mathscr{L}_2}$, whence a isomorphism (XVII 6.3.27.1)

$$\pi^* K_{\mathscr{L}} = \mathrm{pr}_1^* K_{\mathscr{L}_1} + \mathrm{pr}_2^* K_{\mathscr{L}_2}. \tag{1.3.3.3}$$

There is a canonical isomorphism

$$\pi^* \mathscr{O}(1) \simeq \mathrm{pr}_1^* \mathscr{O}(1) \otimes \mathrm{pr}_2^* \mathscr{O}(1),$$

so that (1.3.3.3) may be rewritten as

$$\begin{aligned}\pi^*(p^*\langle \mathscr{L}, K] + \varphi_{\mathscr{L},K}\mathscr{O}(1)) &= p_{12}^*\langle \mathscr{L}, K] + \mathrm{pr}_1^*\, \varphi_{\mathscr{L},K}\mathscr{O}(1) + \mathrm{pr}_2^*\, \varphi_{\mathscr{L},K}\mathscr{O}(1)\\ &= p_{12}^*\langle \mathscr{L}_1, K] + p_{12}^*\langle \mathscr{L}_2, K] + \mathrm{pr}_1^*\, \varphi_{\mathscr{L}_1,K}\mathscr{O}(1) + \mathrm{pr}_2^*\, \varphi_{\mathscr{L}_2.K}\mathscr{O}(1).\end{aligned}$$

By applying 1.2.2 twice, one obtains a canonical isomorphism

$$\langle \mathscr{L}_1 \otimes \mathscr{L}_2, K] = \langle \mathscr{L}_1, K] + \langle \mathscr{L}_2, K] \tag{1.3.3.4}$$

and an identity 504

$$\varphi_{\mathscr{L}_1} \otimes_{\mathscr{L}_2,K} = \varphi_{\mathscr{L}_1,K} = \varphi_{\mathscr{L}_2,K}.$$

Thus one sees that $\varphi_{\mathscr{L},K}$ depends only on K, and one sets

$$\deg(K) = \varphi_{\mathscr{L},K}. \tag{1.3.3.5}$$

Consequently,

$$K_{\mathscr{L}} \simeq p^*\langle \mathscr{L}, K] + \deg(K)\mathscr{O}(1). \tag{1.3.3.6}$$

$$\deg(K_1 + K_2) = \deg(K_1) + \deg(K_2). \tag{1.3.3.7}$$

Let $\mathscr{U}$ be an invertible sheaf that is locally a direct summand of $f_*\mathscr{L}$, defining a section u of $\mathbf{P}((f_*\mathscr{L})^\vee)$. Let D_u be the relative divisor defined by the local invertible sections of $\mathscr{U}$.

$$u^*\mathscr{O}(1) \simeq \mathscr{U}^\vee$$

It therefore follows from (1.3.3.6) that

$$u^* K_{\mathscr{L}} = \mathrm{Tr}_{D_u/S}(K) \simeq \langle \mathscr{L}, K] - \deg(K)\mathscr{U}^{100}. \tag{1.3.3.8}$$

We leave it to the reader to verify the following compatibility:

[100] N.D.E.: The notation $\mathrm{Tr}_{D_u/S}(K)$ is introduced below in (1.3.5.0).

PROPOSITION 1.3.4. *The following diagram of the isomorphisms (1.3.3.1) and (1.3.3.4) is commutative:*

$$\begin{array}{ccc}
\langle \mathscr{L}_1 \otimes \mathscr{L}_2, K_1 + K_2] & \simeq & \langle \mathscr{L}_1 \otimes \mathscr{L}_2, K_1] + \langle \mathscr{L}_1 \otimes \mathscr{L}_2, K_2] \\
 & & \wr \\
\wr & & \langle \mathscr{L}_1, K_1] + \langle \mathscr{L}_2, K_1] + \langle \mathscr{L}_1, K_2] + \langle \mathscr{L}_2, K_2] \\
 & & \parallel \\
\langle \mathscr{L}_1, K_1 + K_2] + \langle \mathscr{L}_2, K_1 + K_2] & \simeq & \langle \mathscr{L}_1, K_1] + \langle \mathscr{L}_1, K_2] + \langle \mathscr{L}_2, K_1] + \langle \mathscr{L}_2, K_2].
\end{array}$$

The isomorphisms (1.3.3.1) and (1.3.3.4) are moreover compatible with the associativity and commutativity isomorphisms in a sense whose precise formulation we leave to the reader. Formula (1.3.3.4) then makes it possible to extend by "linearity" the definition
505 of $\langle \mathscr{L}, K]$ when $\mathscr{L}$ no longer necessarily satisfies condition (1.3.2.1).

1.3.5. For every effective relative divisor D on X, set

$$\mathrm{Tr}_{D/S}(K) = \mathrm{Tr}_{D/S}(K|D). \tag{1.3.5.0}$$

If $\mathscr{O}(D)$ satisfies (1.3.2.1), the section 1 of $\mathscr{O}(D)$ is a section of $f_*\mathscr{O}(D)$, and as such generates an invertible sheaf that is locally a direct summand of $f_*\mathscr{O}(D)$, of which it defines a trivialization. The isomorphism (1.3.3.8) therefore gives us an isomorphism

$$\langle \mathscr{O}(D), K] \simeq \mathrm{Tr}_{D/S}(K). \tag{1.3.5.1}$$

One immediately verifies that the diagrams

$$\begin{array}{ccc}
\langle \mathscr{O}(D), K_1 + K_2] & \xrightarrow{\sim} & \mathrm{Tr}_{D/S}(K_1 + K_2) \\
\wr \downarrow & & \downarrow \wr \\
\langle \mathscr{O}(D), K_1] + \langle \mathscr{O}(D), K_2] & \xrightarrow{\sim} & \mathrm{Tr}_{D/S}(K_1) + \mathrm{Tr}_{D/S}(K_2)
\end{array} \tag{1.3.5.2}$$

and

$$\begin{array}{ccc}
\langle \mathscr{O}(D_1 + D_2), K] & \xrightarrow{\sim} & \mathrm{Tr}_{D_1+D_2/S}(K) \\
\wr \downarrow & & \downarrow \wr \quad (XVII\ 6.3.27.1) \\
\langle \mathscr{O}(D_1), K] + \langle \mathscr{O}(D_2), K] & \xrightarrow{\sim} & \mathrm{Tr}_{D_1/S}(K) + \mathrm{Tr}_{D_2/S}(K)
\end{array} \tag{1.3.5.3}$$

are commutative. By linearity, this makes it possible to define the isomorphism (1.3.5.1) for every relative divisor.

1.3.6. Let D and E be two effective relative Cartier divisors, and let f be a rational function such that

$$\mathrm{div}(f) = D - E,$$

i.e. an isomorphism between $\mathscr{O}(D)$ and $\mathscr{O}(E)$. The isomorphism f between $\mathscr{O}(D)$ and $\mathscr{O}(E)$ defines, by (1.3.5.1), an isomorphism

$$\langle f, K] : \mathrm{Tr}_{D/S}(K) \xrightarrow{\sim} \mathrm{Tr}_{E/S}(K). \tag{1.3.6.1}$$

506 By (1.3.5.2) for the second formula, one has

$$\langle fg, K] = \langle f, K].\langle g, K]$$
$$\langle f, K_1 + K_2] = \langle f, K_1] + \langle f, K_2].$$

If λ is a section of $\mathcal{O}_S^*$, then, for every invertible sheaf $\mathscr{L}$ on X, multiplication by λ is an automorphism of $\mathscr{L}$. By transport of structure, this automorphism induces on the two sides of (1.3.3.8) automorphisms which correspond to one another under the isomorphism (1.3.3.8). It follows that

$$\langle \lambda, K] = \deg(K)(\lambda) ;$$

if $\deg(K) \neq 0$, the isomorphism $\langle f, K]$ between $\mathrm{Tr}_{D/S}(K)$ and $\mathrm{Tr}_{E/S}(K)$ thus depends on the choice of f.

Let $\lambda \in \Gamma(S, \mathcal{O}_S^*)$, $g \in \Gamma(S, G)$, let $\mathscr{L}$ be an invertible sheaf on X, and let K be a G-torsor on X. If $\langle \lambda, g]$ is the automorphism of $\langle \mathscr{L}, K]$ obtained by transport of structure from the automorphisms λ of $\mathscr{L}$ and g of K, then $\langle \lambda, g]$, identified with a section of G, is given by $\langle \lambda, g] = \langle \lambda, K] + \langle \mathscr{L}, g]$; since $\langle \mathcal{O}(D), g] = \mathrm{Tr}_{D/S}(g)$, the preceding formula gives

$$\langle \lambda, g] = \deg(K)(\lambda) + \deg(\mathscr{L}).g. \tag{1.3.6.2}$$

1.3.7. Let $f : X \to S$ be a smooth projective curve over S, and let G be a group over S satisfying 1.3.1.1. If $X \xrightarrow{f_0} T \xrightarrow{\pi} S$ is the Stein factorization of f, then π is an étale covering of S, and the sheaf $f_*\mathbf{G}_m$ is the torus $\prod_{T/S} \mathbf{G}_m$. Locally on S (for the étale topology), X is the disjoint union of a family of curves $(X_i)_{i\in I}$ over S with nonempty geometrically connected fibers. If K is a torsor under G and $\mathscr{L}$ an invertible sheaf on X, one then sets

$$\langle \mathscr{L}, K] = \sum_i \langle \mathscr{L}|X_i, K|X_i]$$

and defines the degree of $K : f_*\mathbf{G}_m = \mathbf{G}_m^I \to G$ to have coordinates $\deg(K|X_i) : \mathbf{G}_m \to G$. The local constructions globalize and give functors and morphisms 507

$$\langle \mathscr{L}, K]_{X/S} = \mathrm{Tr}_{T/S}(\langle \mathscr{L}, K]_{X/T}) \tag{1.3.7.1}$$

$$\deg(K) : f_*\mathbf{G}_m = \prod_{T/S} \mathbf{G}_m \longrightarrow G : \deg_{X/S}(K) = \prod_{T/S} \deg_{X/T}(K). \tag{1.3.7.2}$$

For an automorphism λ of $\mathscr{L}$, defined by an element λ_0 of $\Gamma(T, \mathcal{O}_T^*)$, one has

$$\langle \lambda, K] = \deg(K)(\lambda). \tag{1.3.7.3}$$

The total degree of K is defined as the composite

$$\deg \mathrm{tot}(K) : \mathbf{G}_m \longrightarrow f_*\mathbf{G}_m \xrightarrow{\deg(K)} G.$$

The preceding results give:

Formulary 1.3.8. For every smooth projective curve $f : X \to S$, there is a functor $\langle\ ,\ ]$, whose formation is compatible with every base change, which associates to an invertible sheaf $\mathscr{L}$ on X and a torsor K under f^*G a torsor $\langle \mathscr{L}, K]$ under G on S. This functor is endowed with the following additional structures and satisfies the following conditions:

. $\langle \mathscr{L}, K]$ is biadditive in $\mathscr{L}$ and K; the biadditivity isomorphisms are compatible with the associativity and commutativity isomorphisms and with base changes, satisfy the compatibility (1.3.4), and are compatible with the isomorphisms (1.3.8.2) below (via XVII 6.3.25.3 and XVII 6.3.27.1).

. For every effective relative divisor E on X/S, there is a canonical isomorphism

$$\langle \mathcal{O}(E), K] \simeq \mathrm{Tr}_{E/S}(K|E).$$

. The degree (1.3.7.2) satisfies, for $\lambda \in f_*\mathbf{G}_m$ an automorphism of $\mathscr{L}$,

$$\langle \lambda, K] = \deg(K)(\lambda).$$

For $\lambda \in \mathbf{G}_m(S)$, the total degree satisfies

$$\langle \lambda, K] = \deg \mathrm{tot}(K)(\lambda).$$

508 Likewise, for $g \in G(S)$ one has $\langle \mathscr{L}, g] = \deg(\mathscr{L}).g$.

One deduces a canonical isomorphism:

(1.3.8.4) $$\langle f^*\mathscr{L}, K] \simeq \deg \mathrm{tot}(K)(\mathscr{L}).$$

It follows immediately from the definitions that, for every homomorphism $\varphi : G \to G'$, there is a canonical isomorphism, compatible with the preceding data:

(1.3.8.5) $$\langle \mathscr{L}, \varphi(K)] \simeq \varphi(\langle \mathscr{L}, K]).$$

. Let $u : X \to Y$ be a flat morphism between smooth projective curves over S. There are then isomorphisms compatible with biadditivity and base changes:

– For $\mathscr{L}$ an invertible sheaf on X and K a G_Y-torsor on Y, one has

$$\langle \mathscr{L}, u^*K] \xrightarrow{\sim} \langle N_{X/Y}(\mathscr{L}), K].$$

– For $\mathscr{L}$ an invertible sheaf on X and K a G_X-torsor on X, one has

$$\langle u^*\mathscr{L}, K] \xrightarrow{\sim} \langle \mathscr{L}, \mathrm{Tr}_{X/Y}\, K].$$

Moreover, these isomorphisms satisfy the evident compatibility for a composite uv of morphisms of curves; for $\mathscr{L} = \mathcal{O}(E)$, they identify with the isomorphisms (XVII 6.3.27.2)

$$\mathrm{Tr}_{E/S}(u^*K) \xrightarrow{\sim} \mathrm{Tr}_{u_*E/S}(K)$$

$$\mathrm{Tr}_{u^*E/S}(K) \xrightarrow{\sim} \mathrm{Tr}_{E/S}\,\mathrm{Tr}_{X/Y}\, K.$$

It remains to prove 1.3.8.6. One reduces to the case in which X and Y have nonempty geometrically connected fibers, and notes that the preceding explicit formulas give, for every section s of $\mathscr{L}$ that does not vanish as a section of the direct image of $\mathscr{L}$, an isomorphism Ψ_s of the desired kind, homogeneous in s, and therefore independent of s when $\mathscr{L}$ satisfies (1.3.2.1). The general case follows.

1.3.9. Under the hypotheses of 1.3.7, we saw in (1.3.8) that every G-torsor K on X
509 defines a functor $\langle *, K]$, denoted below by $F(*)$, endowed with the following additional data and satisfying the following conditions:

(i) F is a morphism of fppf stacks over S (XVII 0.10 and Giraud [4])
  – whose source is the stack whose T-objects, for an S-scheme T, are the invertible sheaves on X_T;
  – whose target is the stack of G-torsors.

(ii) The functor F is endowed with additivity isomorphisms $F(\mathscr{L}_1 \otimes \mathscr{L}_2) \xrightarrow{\sim} F(\mathscr{L}_1) + F(\mathscr{L}_2)$, which are functorial, compatible with the associativity and commutativity isomorphisms, and compatible with base changes.

THEOREM 1.3.10. *Under the hypotheses of 1.3.7, the functor* $\Phi : K \mapsto \langle *, K]$ *is an equivalence between the category of* G*-torsors on* X *and the category of morphisms of stacks considered in (1.3.9).*

One reduces easily to the case in which X has nonempty connected fibers.

Let F satisfy 1.3.9 (i) and (ii). For every S-scheme T and every section t of X_T/T, denote also by t the relative divisor $t(T)$. Then $F(\mathcal{O}(t))$ is a G_T-torsor on T. We denote by $\Psi(F)$ the torsor thus obtained on X, for t the "universal point" of X with values in X, defined by the identity map from X to X. For every section t as above, one has

$$F(\mathcal{O}(t)) \simeq t^*\Psi(F). \tag{1.3.10.1}$$

The isomorphism 1.3.8.2 gives an isomorphism of functors $\Psi \circ \Phi \simeq \mathrm{Id}$. We construct an isomorphism $\Phi \circ \Psi \simeq \mathrm{Id}$.

For every section t of X_T over T, the isomorphism (1.3.10.1) is an isomorphism

$$F(\mathcal{O}(t)) \xrightarrow[\alpha]{\sim} \langle \mathcal{O}(t), \Psi(F)].$$

By additivity, every family $t_1, \dots, t_n$ of sections of X_T over T defines an isomorphism

$$F(\mathcal{O}(\sum t_i)) \xrightarrow[\alpha]{\sim} \langle \mathcal{O}(\sum t_i), \Psi(F)],$$

and by 1.3.9 (ii), this isomorphism is independent of the order of the t_i.

Take $T = (X/S)^n$ and for $t_1 \dots t_n$ the universal n^{tuple} of sections. If $T' = \mathrm{Sym}^n_S(X)$, there is then on $X_{T'}$ a divisor $\sum t_i$ whose inverse image is the divisor $\sum t_i$ on X_T. On T, α is a symmetric isomorphism 510

$$\alpha : F(\mathcal{O}(\sum t_i)) \xrightarrow{\sim} \langle \mathcal{O}(\sum t_i), \Psi(F)]$$

between torsors that are inverse images of torsors on T'. By hypothesis (1.3.1.1) (ii), this isomorphism comes from a unique isomorphism on T':

$$\alpha : F(\mathcal{O}(\sum t_i)) \xrightarrow{\sim} \langle \mathcal{O}(\sum t_i), \Psi(F)] \qquad (\text{on } T').$$

By XVII 6.3.9, one thus has, for every relative divisor D on X, an isomorphism

$$\alpha_D : F(\mathcal{O}(D)) \xrightarrow{\sim} \langle \mathcal{O}(D), \Psi(F)], \tag{1.3.10.2}$$

additive in D, whose formation is compatible with every base change.

Let $\mathscr{L}$ be a sufficiently ample invertible sheaf on X in the sense of 1.3.2.1. Every nowhere-zero section s of $f_*\mathscr{L}$ defines, with the notation of 1.3.2, an isomorphism

$$s : \mathcal{O}(D_s) \xrightarrow{\sim} \mathscr{L},$$

and hence, by (1.3.10.2), an isomorphism

$$\alpha_s : F(\mathscr{L}) \xrightarrow{\sim} \langle \mathscr{L}, \Psi(F)]. \tag{1.3.10.3}$$

Let us prove that this isomorphism is independent of s. Indeed:

(i) The construction of α_s commutes with every base change.

(ii) There exists a homomorphism $\varphi : \mathbf{G}_m \to G$ such that

$$\alpha_{\lambda s} = \varphi(\lambda).\alpha_s$$

(by the functoriality of both sides of (1.3.10.3)).

One necessarily has $\varphi = 0$, since otherwise the torsor on $\mathbf{P}((f_*\mathscr{L})^\vee)$ obtained by pulling back $F(\mathscr{L}) - \langle \mathscr{L}, \Psi(F)]$ would have degree $\neq 0$. The α_s therefore define a
511 morphism from $\mathbf{P}((f_*\mathscr{L})^\vee)$ to G, and this morphism is constant by hypothesis (1.3.1.1) (i). We may therefore set $\alpha_{\mathscr{L}} = \alpha_s$.

The isomorphisms $\alpha_{\mathscr{L}}$ thus form an isomorphism of functors between the restrictions of the functors F and $\langle x, \Psi(F)]$ to sufficiently ample sheaves. This isomorphism is compatible with the additivity isomorphisms, which allows it to be extended by linearity to all invertible sheaves and completes the construction of

$$\alpha : \Phi \circ \Psi \simeq \mathrm{Id}\,.$$

Example 1.3.11. $G = \mathbf{G}_m$. Under the hypotheses of 1.3.7, and in the special case where $G = \mathbf{G}_m$, we shall write $\langle\ ,\ \rangle$ rather than $\langle\ ,\ ]$; if $\mathscr{L}$ and $\mathscr{M}$ are two invertible sheaves on X, then $\langle \mathscr{L}, \mathscr{M} \rangle$ is an invertible sheaf on S. Moreover, there are isomorphisms (1.3.8.2)

$$\langle \mathscr{O}(D), \mathscr{M} \rangle \simeq N_{D/S}(\mathscr{M}).$$

If D and E are disjoint effective relative divisors, then $\mathscr{O}(E)|D$ is canonically isomorphic to $\mathscr{O}$, whence an isomorphism

$$\sigma_{D,E} : \langle \mathscr{O}(D), \mathscr{O}(E) \rangle \simeq N_{D/S}(\mathscr{O}(E)) \simeq N_{D/S}(\mathscr{O}) \simeq \mathscr{O}. \tag{1.3.11.1}$$

Proposition 1.3.12. *Let $(D_i)_{i=1,2}$ and E be effective relative divisors, with D_i disjoint from E $(i = 1, 2)$. Let f be a rational function satisfying* $\mathrm{div}(f) = D_1 - D_2$, *i.e. an isomorphism $f : \mathscr{O}(D_1) \xrightarrow{\sim} \mathscr{O}(D_2)$. The following diagrams of isomorphisms are commutative:*

(1.3.12.1)
$$\begin{array}{ccc} \langle \mathscr{O}(D_1), \mathscr{O}(E) \rangle & \xrightarrow{\sigma_{D_1,E}} & \mathscr{O} \\ \downarrow{\scriptstyle \langle f, \mathscr{O}(E) \rangle} & & \downarrow{\scriptstyle N_{E/S}(f)} \\ \langle \mathscr{O}(D_2), \mathscr{O}(E) \rangle & \xrightarrow{\sigma_{D_2,E}} & \mathscr{O} \end{array}$$

(1.3.12.2)
$$\begin{array}{ccc} \langle \mathscr{O}(E), \mathscr{O}(D_1) \rangle & \xrightarrow{\sigma_{E,D_1}} & \mathscr{O} \\ \downarrow{\scriptstyle \langle \mathscr{O}(E), f \rangle} & & \downarrow{\scriptstyle N_{E/S}(f)} \\ \langle \mathscr{O}(E), \mathscr{O}(D_2) \rangle & \xrightarrow{\sigma_{E,D_2}} & \mathscr{O} \end{array}$$

512 The formula (1.3.12.2) simply expresses the commutativity of the diagram

$$\begin{array}{ccc} N_{E/S}(\mathscr{O}(D_1)) & \xrightarrow{\sim} & \mathscr{O} \\ \downarrow{\scriptstyle N_{E/S}(f)} & & \downarrow{\scriptstyle N_{E/S}(f)} \\ N_{E/S}(\mathscr{O}(D_2)) & \xrightarrow{\sim} & \mathscr{O}. \end{array}$$

For every S-scheme T and every invertible sheaf $\mathscr{L}$ on X_T, set $F(\mathscr{L}) = N_{E_T/T}(\mathscr{L})$. The hypotheses of 1.3.10 are satisfied by this functor, so there exists a unique invertible sheaf $\mathscr{M}$ equipped with an isomorphism of functors compatible with biadditivity:

$$N_{E_T/T}(\mathscr{L}) \xrightarrow{\sim} \langle \mathscr{L}, \mathscr{M} \rangle.$$

Moreover, if $d : T \to X_T$ is a section of X over T, there is a canonical isomorphism

$$d^*\mathscr{M} \xrightarrow{\sim} N_{E_T/T}(\mathscr{O}(d)).$$

Taking for d the universal section of X, $\Delta : X \to X_X$, we obtain

$$\mathscr{M} \xrightarrow{\sim} N_{E_X/X}(\mathscr{O}(\Delta)).$$

Locally on S, E is a sum of sections $E = \sum t_i$ (XVII 6.3); for such an E, one has

$$N_{E_X/X}(\mathscr{O}(\Delta)) = \otimes t_i^*\mathscr{O}(\Delta) = \sum \mathscr{O}(t_i) = \mathscr{O}(E),$$

and this isomorphism, being independent of the order of the t_i, defines a canonical isomorphism[101]

$$N_{E_X/X}(\mathscr{O}(\Delta)) \sim \mathscr{O}(E), \tag{1.3.12.3}$$

so that

$$N_{E_T/T}(\mathscr{L}) \xrightarrow{\sim} \langle \mathscr{L}, \mathscr{O}(E)\rangle. \tag{1.3.12.4}$$

The isomorphism (1.3.12.3) is compatible with biadditivity and base changes and is characterized, in addition, by the fact that for every section d of X_T over T, the isomor- 513
phism

$$N_{E/T}(\mathscr{O}(d)) \xrightarrow{\sim} \langle \mathscr{O}(d), \mathscr{O}(E)\rangle \xrightarrow{\sim} d^*\mathscr{O}(E)$$

is the pullback of the universal isomorphism (1.3.12.3). In particular, for d disjoint from E, the diagram

$$\begin{array}{ccccc} N_{E/T}(\mathscr{O}(d)) & \xrightarrow{\sim} & \langle \mathscr{O}(d), \mathscr{O}(E)\rangle & \xrightarrow{\sim} & d^*\mathscr{O}(E) \\ \| & & & & \| \\ \mathscr{O} & & = & & \mathscr{O} \end{array}$$

is commutative. By additivity[102], it follows that, for D disjoint from E, the diagram

$$\begin{array}{ccccc} N_{E/T}(\mathscr{O}(D)) & \xrightarrow{\sim} & \langle \mathscr{O}(D), \mathscr{O}(E)\rangle & \xrightarrow{\sim} & N_{D/T}(\mathscr{O}(E)) \\ \| & & & & \| \\ \mathscr{O} & & = & & \mathscr{O} \end{array} \tag{1.3.12.5}$$

is commutative. The arrow $\sigma_{E,D}$ is therefore the composite of the inverse of 1.3.12.4 and an evident arrow

$$\langle \mathscr{O}(D), \mathscr{O}(E)\rangle \xleftarrow{\sim} N_{E/S}(\mathscr{O}(D)) \longrightarrow \mathscr{O}$$

and (1.3.12.1) follows at once, since 1.3.12.4 is functorial.

We deduce from 1.3.12 (cf. Serre [14, Ch. III, prop. 7, p. 46])

[101] N.D.E.: The argument seems to assume that, locally, the t_i are well defined up to permutation, which is not true, as can be seen in the case where $S = \operatorname{Spec} k[\epsilon]$ is the spectrum of the algebra of dual numbers, $X = \mathbf{P}^1_S$, and E is the closed subscheme of $\mathbf{A}^1_S = \operatorname{Spec} \mathscr{O}_S[T] \subset X$ defined by the equation $(T-\varepsilon)(T+\varepsilon) = T^2$. It seems instead necessary to apply the "cancelled descent" condition 1.3.1.1 (ii) to obtain the result in the situation of the universal divisor on X after base change by $\operatorname{Sym}^n_S X \to S$, before deducing it for the divisor E.

[102] N.D.E.: The same objection raised above applies here as well. One can argue in the same way by considering "the universal divisor disjoint from E," defined after base change by $\operatorname{Sym}^n_S(X - E) \to S$.

COROLLARY 1.3.13. *Let $(D_i)_{i=1,2}$ and $(E_i)_{i=1,2}$ be effective relative divisors, with D_i disjoint from E_j, let f be a rational function satisfying* $\operatorname{div}(f) = D_1 - D_2$, *and let g be a rational function satisfying* $\operatorname{div}(g) = E_1 - E_2$. *Then*

$$N_{E_1/S}(f).N_{E_2/S}(f)^{-1} = N_{D_1/S}(g).N_{D_2/S}(g)^{-1}$$

or, briefly,

$$f(\operatorname{div}(g)) = g(\operatorname{div}(f)). \tag{1.3.13.1}$$

This formula, in the form

$$N_{D_2/S}(g)N_{E_1/S}(f) = N_{E_2/S}(f)N_{D_1/S}(g),$$

expresses, via (1.3.12.1) and (1.3.12.2), the commutativity of the diagram

$$\begin{array}{ccc} \langle \mathcal{O}(D_1), \mathcal{O}(E_1)\rangle & \xrightarrow{\langle \mathcal{O}(D_1), g\rangle} & \langle \mathcal{O}(D_1), \mathcal{O}(E_2)\rangle \\ \downarrow{\scriptstyle \langle f, \mathcal{O}(E_1)\rangle} & & \downarrow{\scriptstyle \langle f, \mathcal{O}(E_2)\rangle} \\ \langle \mathcal{O}(D_2), \mathcal{O}(E_1)\rangle & \xrightarrow{\langle \mathcal{O}(D_2), g\rangle} & \langle \mathcal{O}(D_2), \mathcal{O}(E_2)\rangle. \end{array}$$

514 COROLLARY 1.3.14. *For $G = \mathbf{G}_m$, the degree (1.3.7) coincides with the usual degree of invertible sheaves.*

It suffices to compare formulas (1.3.7.3) and (1.3.12.1) for f the pullback of $f_0 \in \Gamma(S, \mathcal{O}_S^*)$.

1.3.15. For any sufficiently ample $\mathscr{L}$ and $\mathscr{M}$, there exist locally on S disjoint divisors D and E and isomorphisms

$$\alpha : \mathscr{L} \longrightarrow \mathcal{O}(D) \quad \text{et} \quad \beta : \mathscr{M} \longrightarrow \mathcal{O}(E).$$

Denote by τ the isomorphism making the diagram

$$\begin{array}{ccccc} \langle \mathscr{L}, \mathscr{M}\rangle & \xrightarrow{\langle \alpha, \beta\rangle} & \langle \mathcal{O}(D), \mathcal{O}(E)\rangle & \xrightarrow{\sigma_{D,E}} & \mathcal{O} \\ \wr\downarrow{\scriptstyle \tau} & & & & \| \\ \langle \mathscr{M}, \mathscr{L}\rangle & \xrightarrow{\langle \beta, \alpha\rangle} & \langle \mathcal{O}(E), \mathcal{O}(D)\rangle & \xrightarrow{\sigma_{E,D}} & \mathcal{O}. \end{array}$$

It follows from 1.3.12 that this isomorphism is independent of the particular choice of D, E, α, and β; it therefore globalizes to a canonical isomorphism

$$\tau : \langle \mathscr{L}, \mathscr{M}\rangle \xrightarrow{\sim} \langle \mathscr{M}, \mathscr{L}\rangle. \tag{1.3.15.1}$$

This isomorphism satisfies $\tau^2 = 1$; it is compatible with biadditivity, since (1.3.11.1) is, and extends by linearity to arbitrary invertible sheaves.

It follows from (1.3.12.5) that the morphism (1.3.12.4) identifies with the composite

$$N_{E_T/T}(\mathscr{L}) \xleftarrow{\sim} \langle \mathcal{O}(E), \mathscr{L}\rangle \xrightarrow{\sim\tau} \langle \mathscr{L}, \mathcal{O}(E)\rangle$$

(verify this for $\mathscr{L} = \mathcal{O}(D)$ with D disjoint from E).

515 FORMULARY 1.3.16. *For every smooth projective curve $f : X \to S$, there is a functor $\langle\ ,\ \rangle$, whose construction commutes with every base change, that associates with two invertible sheaves $\mathscr{L}$ and $\mathscr{M}$ on X an invertible sheaf $\langle \mathscr{L}, \mathscr{M} \rangle$ on S. This functor is endowed with the following additional structures and satisfies the following conditions:*

. With the notation of 1.3.8, for $G = \mathbf{G}_m$, one has

$$\langle \mathscr{L}, \mathscr{M} \rangle = \langle \mathscr{L}, \mathscr{M}],$$

in particular, $\langle \mathscr{L}, \mathscr{M} \rangle$ is biadditive in $\mathscr{L}$, $\mathscr{M}$ in the sense specified in 1.3.8.1, and this biadditivity is compatible with that of the right-hand side of a canonical isomorphism (functorial in $\mathscr{M}$)

$$\langle \mathscr{O}(E), \mathscr{M} \rangle \xrightarrow{\sim} N_{E/S}(\mathscr{M}).$$

. There is a symmetry isomorphism

$$\tau : \langle \mathscr{L}, \mathscr{M} \rangle \xrightarrow{\sim} \langle \mathscr{M}, \mathscr{L} \rangle,$$

functorial in $\mathscr{L}$ and $\mathscr{M}$, compatible with base changes and the biadditivity isomorphisms, and satisfying $\tau^2 = 1$. It makes the following diagram commutative, for disjoint divisors D and E:

$$\begin{array}{ccccccc} N_{D/S}(\mathscr{O}(E)) & \longleftarrow & \langle \mathscr{O}(D), \mathscr{O}(E) \rangle & \xrightarrow{\tau} & \langle \mathscr{O}(E), \mathscr{O}(D) \rangle & \longrightarrow & N_{E/S}(\mathscr{O}(D)) \\ \downarrow s & & & & & & \downarrow s \\ \mathscr{O} & = & = & = & = & = & \mathscr{O}. \end{array}$$

These formulas imply formulas (1.3.12.1), (1.3.12.2), (1.3.13.1), and

. For $a, b \in \Gamma(S, \mathscr{O}_S^*)$, denoting by $\langle a, b \rangle$ the automorphism of $\langle \mathscr{L}, \mathscr{M} \rangle$, identified with a section of $\mathbf{G}_m$, obtained by functoriality from the automorphisms a and b of $\mathscr{L}$ and $\mathscr{M}$, one has

$$\langle a, b \rangle = a^{\deg(\mathscr{M})} b^{\deg(\mathscr{L})}.$$

. If $\mathscr{L}_0$ (resp. $\mathscr{M}_0$) is an invertible module on S, there is a canonical isomorphism

$$\langle f^* \mathscr{L}_0, \mathscr{M} \rangle \xrightarrow{\sim} \mathscr{L}^{\otimes \deg(\mathscr{M})}$$

(resp. $\langle \mathscr{L}, f^* \mathscr{M}_0 \rangle \xrightarrow{\sim} \mathscr{M}^{\otimes \deg(\mathscr{L})}$).

. If $u : X \to Y$ is a finite flat morphism of smooth projective curves over S, then, for 516
$\mathscr{L}$ an invertible sheaf on X and $\mathscr{M}$ an invertible sheaf on Y, the isomorphisms 1.3.8.6

$$\langle \mathscr{L}, u^* \mathscr{M} \rangle \sim \langle N_{X/Y} \mathscr{L}, \mathscr{M} \rangle$$
$$\langle u^* \mathscr{M}, \mathscr{L} \rangle \sim \langle \mathscr{M}, N_{X/Y} \mathscr{L} \rangle$$

are deduced from one another by the symmetry (1.3.16.2) (verify this for $\mathscr{L} = \mathscr{O}(E)$ and $\mathscr{M} = \mathscr{O}(F)$ with E and F disjoint).

. For an invertible sheaf $\mathscr{L}$ on X, the symmetry arrow

$$\tau : \langle \mathscr{L}, \mathscr{L} \rangle \longrightarrow \langle \mathscr{L}, \mathscr{L} \rangle$$

is multiplication by $(-1)^{\deg(\mathscr{L})}$.

1.3.17. It remains to prove 1.3.16.6. We shall give two proofs, the first using the theory of the determinant (SGA 6 XI)[103].

Before proceeding to each of the proofs, one may observe that if $\mathscr{L}_1$ and $\mathscr{L}_2$ are two invertible sheaves on X and $\mathscr{L} = \mathscr{L}_1 \otimes \mathscr{L}_2$, then the automorphism $\tau_{\mathscr{L}} : \langle \mathscr{L}, \mathscr{L} \rangle \longrightarrow \langle \mathscr{L}, \mathscr{L} \rangle$ (identified with an invertible function on S) is the product of the automorphisms $\tau_{\mathscr{L}_1}$ and $\tau_{\mathscr{L}_2}$. In the situation where S is the spectrum of a field (and X is geometrically connected), one may therefore suppose that $\mathscr{L} = \mathscr{O}(D)$ with D a nonzero effective divisor. Since $\tau^2 = \mathrm{Id}$, after replacing $\mathscr{L}$ by a sufficiently large odd power (or writing it as a quotient of two ample bundles), one may also make $\mathscr{L}$ as ample as desired, if necessary.

The arguments below will also require reducing to the case where S is the spectrum of a field. For this, it suffices to reduce to the case where S is reduced, and this can be done by showing that fppf-locally on S, the datum $(X, \mathscr{L})$ comes from a similar datum over a reduced base. Before making this reduction, one may reduce to the case where $X \to S$ is equipped with a section $s : S \to X$ and where the geometric fibers of $X \to S$ are connected and have some genus g. Using [10, § 5.2], one can obtain a smooth projective curve $X_0 \to S_0$ of genus g, with S_0 smooth of finite type over $\mathrm{Spec}(\mathbf{Z})$, such that *locally*, every relative curve as above is obtained from it by base change. Using the section $s_0 : S_0 \to X_0$, one can rigidify the Picard scheme $S_1 = \mathbf{Pic}_{X_0/S_0} \to S_0$ and thus obtain a "universal" line bundle $\mathscr{L}_1$ on $X_1 = X_0 \times_{S_0} S_1$. Locally, every relative curve as above equipped with a line bundle comes from $(X_1, \mathscr{L}_1)$ over S_1, which is indeed reduced[104].

1st proof: Let $u : T \to S$ be a finite locally free morphism. If $\mathscr{L}$ is an invertible sheaf on T, there is an isomorphism

$$N_{T/S}(\mathscr{L}) \simeq \det(u_*\mathscr{L}) \det(u_*\mathscr{O}_T)^{-1} \tag{1.3.17.1}$$

uniquely determined by the following three conditions:

a) its construction commutes with base changes;
b) for $\mathscr{L} = \mathscr{O}_T$, it is the identity morphism of $\mathscr{O}_S$;
c) it is functorial in $\mathscr{L}$.

Let D and E be two effective relative divisors on X, let i be the inclusion of D in X, and let u be the projection of D onto S. The short exact sequence

$$0 \longrightarrow \mathscr{O}(-E) \longrightarrow \mathscr{O} \longrightarrow \mathscr{O}_E \longrightarrow 0$$

allows the isomorphism

$$\langle \mathscr{O}(D), \mathscr{O}(-E) \rangle \simeq N_{D/S}(\mathscr{O}(-E)) \simeq \det(u_* i^* \mathscr{O}(-E)) \det(u_* i^* \mathscr{O})^{-1}$$

517 as an isomorphism

$$\langle \mathscr{O}(D), \mathscr{O}(-E) \rangle = \det Ru_*(Li^*\mathscr{O}_E)^{-1} \simeq \det Rf_*(\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_E)^{-1}.$$

or

$$\langle \mathscr{O}(D), \mathscr{O}(E) \rangle \simeq \det Rf_*(\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_E). \tag{1.3.17.2}$$

[103] N.D.E.: That exposé was not written, as explained in [3], which also contains a warning concerning the multiplicativity property of the determinant in distinguished triangles: it requires care, without which counterexamples arise. A reference on this subject is [8].

[104] N.D.E.: The preceding two paragraphs were added in this edition.

For disjoint D and E, one has $\mathcal{O}_D \overset{\mathbf{L}}{\otimes} \mathcal{O}_E = 0$, and the resulting isomorphism of $\langle \mathcal{O}(D), \mathcal{O}(E)\rangle$ with $\mathcal{O}_S$ is the one already considered. It follows that, for disjoint D and E, the symmetry τ and the symmetry isomorphism $\mathcal{O}_D \overset{\mathbf{L}}{\otimes} \mathcal{O}_E \xrightarrow{\sim} \mathcal{O}_E \overset{\mathbf{L}}{\otimes} \mathcal{O}_D$ are compatible via (1.3.17.2). By specialization[105], this remains true for arbitrary D and E.

When $D = E$ is defined by an ideal I, one has

$$\begin{aligned} &\mathscr{H}^0(\mathcal{O}_D \overset{\mathbf{L}}{\otimes} \mathcal{O}_D) = \mathcal{O}_D && \text{symmetry } = +\mathrm{Id} \\ &\mathscr{H}^{-1}(\mathcal{O}_D \overset{\mathbf{L}}{\otimes} \mathcal{O}_D) = I/I^2 && \text{symmetry } = -\mathrm{Id} \\ &\mathscr{H}^{i}(\mathcal{O}_D \overset{\mathbf{L}}{\otimes} \mathcal{O}_D) = 0 && \text{for } i \neq 0, -1. \end{aligned}$$

Consequently, the map

$$\tau : \langle \mathcal{O}(D), \mathcal{O}(D)\rangle \xrightarrow{\sim} \langle \mathcal{O}(D), \mathcal{O}(D)\rangle$$

is $(-1)^{\mathrm{rk}\, I/I^2} = (-1)^{\deg(\mathcal{O}(D))}$, which proves 1.3.16.6.

2nd proof: By reduction to a "universal" case, we may suppose that S is reduced. This case reduces to that in which S is the spectrum of an algebraically closed field k. In this case, let X_1, X_2, Y_1, and Y_2 be four divisors on X, let f and g be two rational functions, and suppose that

$$\begin{aligned} &\mathrm{div}\,(f) = X_2 - X_1 \\ &\mathrm{div}\,(g) = Y_2 - Y_1 \\ &(X_1 \cup Y_2) \cap (X_2 \cup Y_1) = \varnothing. \end{aligned}$$

The maps f and g then define[106]

$$\langle f, g\rangle : \langle \mathcal{O}(X_2), \mathcal{O}(Y_2)\rangle \longrightarrow \langle \mathcal{O}(X_1), \mathcal{O}(Y_1)\rangle.$$

The two sides are canonically isomorphic to k (1.3.11.1), so that $\langle f, g\rangle$ is identified with an element of k^*. 518

Lemma 1.3.18. *One has*

$$\langle f, g\rangle = \prod_{x \in X_1 \cup Y_2} (-1)^{v_x(f)v_x(g)} (f^{v_x(g)}/g^{v_x(f)})(x).$$

Let us show that (1.3.18) implies (1.3.16.6). We may suppose that $\mathscr{L} \sim \mathcal{O}(X) \sim \mathcal{O}(Y)$ with X and Y effective, and that $X \cap Y = \varnothing$[107]. Let f be a rational function such that

[105] N.D.E.: It seems that here one must first reduce to the case where S is the spectrum of a field, as explained above. It is then possible to make D and E occur in two families of divisors parametrized by an integral scheme in such a way that the divisors in the two families are disjoint over the generic point. This may be obtained, for example, by considering the universal situation over which the condition that D and E be disjoint holds on a dense open subset of $\mathrm{Sym}^n_S X \times \mathrm{Sym}^m_S X$, which is integral.

[106] N.D.E.: Compared with the original edition, numerous "signs" have been modified here and there through the end of this paragraph.

[107] N.D.E.: We saw above that we could suppose that $\mathscr{L}$ was very ample.

$\operatorname{div}(f) = Y - X$[108]. The commutative diagram

$$\begin{array}{ccc} \langle \mathscr{L}, \mathscr{L} \rangle & \xrightarrow{\quad\tau\quad} & \langle \mathscr{L}, \mathscr{L} \rangle \\ \| & & \| \\ \langle \mathscr{O}(X), \mathscr{O}(Y) \rangle & \xrightarrow{\langle f^{-1}, f \rangle} & \langle \mathscr{O}(Y), \mathscr{O}(X) \rangle \end{array}$$

shows that

$$\tau = \langle f^{-1}, f \rangle = \prod_{x \in Y} (-1)^{-v_x(f)^2} = (-1)^{\deg(\mathscr{L})}.$$

Let us prove 1.3.18. We reduce to supposing that X_i and Y_i are very ample. There are then factorizations $f = f_1 f_2$; $g = g_1 g_2$ with

$$\begin{aligned} \operatorname{div}(f_1) &= X' - X_1 \qquad & \operatorname{div}(g_1) &= Y' - Y_1 \\ \operatorname{div}(f_2) &= X_2 - X' \qquad & \operatorname{div}(g_2) &= Y_2 - Y', \end{aligned}$$

the divisors X', Y', and $X_1 \cup X_2 \cup Y_1 \cup Y_2$ being pairwise disjoint.

Denote by $\langle\ ,\ \rangle^*$ the right-hand side of (1.3.18), and let us prove that

$$\langle f, g \rangle^* = \langle f_1, g_1 \rangle^* \cdot \langle f_2, g_2 \rangle^*.$$

By Serre [14, Ch. III, prop. 6, p. 44], one has

$$\prod_x (-1)^{v_x(f_1)v_x(g_2)} (f_1^{v_x(g_2)} / g_2^{v_x(f_1)})(x) = 1. \tag{1.3.18.1}$$

It follows that

$$\begin{aligned} \langle f_1, g_1 \rangle^* \cdot \langle f_2, g_2 \rangle^* &= g_1(X_1) f_1(Y') g_2(X') f_2(Y_2) \cdot \\ &\quad \prod_{X_1 \cup X' \cup Y' \cup Y_2} (-1)^{v_x(f_1)v_x(g_2)} (f_1^{v_x(g_2)} \cdot g_2^{-v_x(f_1)})(x) \\ &= \prod_{x \in X'} g_2(x)^{-v_x(f_2)} \cdot g_2(x)^{-v_x(f_1)} \cdot \prod_{x \in Y'} f_1(x)^{v_x(g_1)} \cdot f_1(x)^{v_x(g_2)} \cdot \\ &\quad \prod_{X_1 \cup Y_2} (-1)^{v_x(f)v_x(g)} ((f_1 f_2)^{v_x(g_2)} \cdot (g_1 g_2)^{-v_x(f_1)} = \langle f_1 f_2, g_1 g_2 \rangle^*. \end{aligned}$$

519 One has, by functoriality of $\langle\ ,\ \rangle$, $\langle f, g \rangle = \langle f_1, g_1 \rangle \langle f_2, g_2 \rangle$, and it is readily checked that $\langle f_i, g_i \rangle = \langle f_i, g_i \rangle^*$[109]. The lemma follows.

1.4. Strictly commutative Picard categories. Some results from "general topology" will be needed to express the results of 1.3 by a "universal coefficient formula."

The results of this section are suggested in a letter from A. Grothendieck to J.L. Verdier (1966).

The notions of associativity and commutativity for bifunctors (see below) were introduced by Mac Lane.

[108] N.D.E.: Care should be taken not to confuse X with the ambient curve, which is also denoted by X.

[109] N.D.E.: This is the special case in which one assumes in addition that $X_1 \cap Y_2 = \emptyset$; it follows from 1.3.12.1 and 1.3.12.2.

1.4.1. Let C be a category, let $F : C \times C \to C$ be a functor, and let

$$\sigma : F(F(X,Y),Z) \xrightarrow{\sim} F(X,F(Y,Z))$$

be an isomorphism of trifunctors. The pair (F,σ) is said to be an *associative functor*, or σ is said to be an *associativity datum* on F, if the following condition holds:

(Ass) For every family $(X_i)_{i\in I}$ of objects of C, denoting by $e : I \to M(I)$ the canonical map from I into the free monoid (without identity) generated by I, there exist a map $\mathscr{F} : M(I) \to \operatorname{Ob} C$, isomorphisms $a_i : \mathscr{F}(e_i) \xrightarrow{\sim} X_i$, and isomorphisms $a_{g,h} : \mathscr{F}(gh) \xrightarrow{\sim} F(\mathscr{F}(g),\mathscr{F}(h))$ such that the diagrams

$$\begin{array}{ccccc}
\mathscr{F}(f(gh)) & \xrightarrow[a_{f,gh}]{\sim} & F(\mathscr{F}(f),\mathscr{F}(gh)) & \xrightarrow[a_{g,h}]{\sim} & F(\mathscr{F}(f),F(\mathscr{F}(g),\mathscr{F}(h))) \\
\| & & & & \uparrow{\scriptstyle s}\ {\scriptstyle \sigma} \\
\mathscr{F}((fg)h) & \xrightarrow[a_{fg,h}]{\sim} & F(\mathscr{F}(fg),\mathscr{F}(h)) & \xrightarrow[a_{f,g}]{\sim} & F(F(\mathscr{F}(f),\mathscr{F}(g)),\mathscr{F}(h))
\end{array} \tag{1.4.1.1}$$

are commutative.

We shall not need the fact that axiom (Ass) is equivalent to the "pentagon axiom," namely that the following diagram is commutative:

$$\begin{array}{ccc}
 & F(F(X,Y),F(Z,T)) & \\
\swarrow & & \searrow \\
F(X,F(Y,F(Z,T))) & & F(F(F(X,Y),Z),T) \\
\downarrow & & \downarrow \\
F(X,F(F(Y,Z),T)) & \longleftrightarrow & F(F(X,F(Y,Z)),T).
\end{array}$$

Now suppose given functorial isomorphisms 520

$$\sigma_{X,Y,Z} : F(F(X,Y),Z) \xrightarrow{\sim} F(X,F(Y,Z))$$
$$\tau_{X,Y} : F(X,Y) \xrightarrow{\sim} F(Y,X).$$

We shall say that σ and τ make F an *associative* and *strictly commutative* functor if the following condition holds.

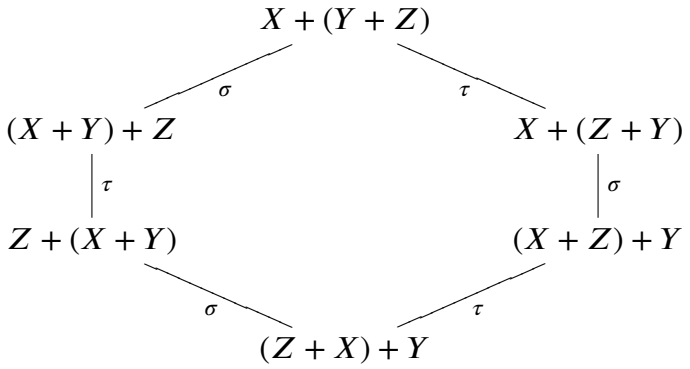
(Ass.s.Com) For every family $(X_i)_{i\in I}$ of objects of C, denoting by $e : I \to N(I)$ the canonical map from I into the free abelian monoid (without identity) generated by I, there exist a map $\mathscr{F} : N(I) \to \operatorname{Ob} C$, isomorphisms $a_i : \mathscr{F}(e_i) \xrightarrow{\sim} X_i$, and isomorphisms $a_{g,h} : \mathscr{F}(gh) \xrightarrow{\sim} F(\mathscr{F}(g),\mathscr{F}(h))$ such that diagram (1.4.1.1) is commutative, as is the diagram

$$\begin{array}{ccc}
\mathscr{F}(gh) & \xrightarrow{a_{g,h}} & F(\mathscr{F}(g),\mathscr{F}(h)) \\
\| & & \downarrow{\scriptstyle \tau} \\
\mathscr{F}(hg) & \xrightarrow{a_{h,g}} & F(\mathscr{F}(h),\mathscr{F}(g)).
\end{array} \tag{1.4.1.2}$$

We shall not need the fact that this axiom is equivalent to the conjunction of:

1) the pentagon axiom;
2) $\tau_{X,X} : X + X \to X + X$ is the identity;

3) $\tau_{Y,X} \circ \tau_{X,Y} : X + Y \to Y + X \to X + Y$ is the identity;
4) the hexagon axiom: the diagram



521 is commutative.

Note that condition 2) is not very often satisfied in practice (hence the terminology *strictly* commutative).

Definition 1.4.2. *A* strictly commutative Picard category $\mathscr{P}$ *is a nonempty category all of whose morphisms are isomorphisms, equipped with a functor* $+ : \mathscr{P} \times \mathscr{P} \to \mathscr{P} : (X, Y) \mapsto X + Y$ *and functorial isomorphisms*

$$\sigma : (X + Y) + Z \xrightarrow{\sim} X + (Y + Z)$$
$$\tau : X + Y \longrightarrow Y + X$$

making $+$ *an associative and strictly commutative functor, and such that, moreover, for every* $X \in \mathrm{Ob}\,\mathscr{P}$*, the functor* $Y \mapsto X + Y$ *is an equivalence of categories.*

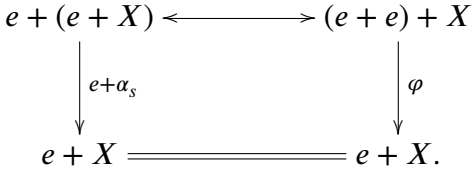
The reader will verify:

Lemma 1.4.3. *If* $(X_i)_{i\in I}$ *is a family of objects of a strictly commutative Picard category* $\mathscr{P}$*, there exist a map* $\Sigma : \mathbf{Z}^{(I)} \to \mathrm{Ob}\,\mathscr{P}$*, isomorphisms* $a_i : \Sigma(e_i) \xrightarrow{\sim} X_i$*, and* $a_{\underline{n},\underline{m}} : \Sigma(\underline{n}+\underline{m}) \xrightarrow{\sim} \Sigma(\underline{n}) + \Sigma(\underline{m})$ *such that the diagrams of types (1.4.1.1) and (1.4.1.2) are commutative. The system* $(\Sigma, (a_i), (a_{\underline{n},\underline{m}}))$ *is unique up to unique isomorphism, and is functorial in* $(X_i)_{i\in I}$*.*

1.4.4. A strictly commutative Picard category admits, up to unique isomorphism, exactly one *neutral object*, which may here be defined as a pair (e, φ) consisting of an object e and an isomorphism $\varphi : e + e \xrightarrow{\sim} e$. If (e, φ) is a neutral object, there is a unique isomorphism of functors

$$\alpha_s : e + X \xrightarrow{\sim} X$$

making the diagram



commutative.

522 One similarly defines $\alpha_d : X + e \xrightarrow{\sim} X$, and φ is a special case of both α_s and α_d; τ interchanges α_s and α_d. The group $\mathrm{Aut}(e)$ is abelian, and for every $X \in \mathrm{Ob}\,\mathscr{P}$, the functors $X+$ and $+X$ induce the same isomorphism between $\mathrm{Aut}(e)$ and $\mathrm{Aut}(X)$.

DEFINITION 1.4.5. *A* strictly commutative Picard stack $\mathscr{P}$ on a *site* $\mathscr{S}$ *is a stack in groupoids* $\mathscr{P}$ *on* $\mathscr{S}$ *(*GIRAUD [4]*), equipped with a functor* $+ : \mathscr{P} \times_{\mathscr{S}} \mathscr{P} \to \mathscr{P}$ *and isomorphisms of functors*

$$\tau_{x,y} : x + y \longrightarrow y + x$$
$$\sigma_{x,y,z} : (x + y) + z \longrightarrow x + (y + z)$$

which, for every $U \in \mathrm{Ob}\,\mathscr{S}$*, make* $\mathscr{P}(U)$ *a strictly commutative Picard category.*

In what follows, we shall simply speak of a *Picard stack*, without specifying "strictly commutative." By 1.4.4, every Picard stack $\mathscr{P}$ admits a global "neutral object" e, and $\mathbf{Aut}(e)$ is an abelian sheaf.

1.4.6. An *additive functor* $F : \mathscr{P}_1 \to \mathscr{P}_2$ between Picard stacks on $\mathscr{S}$ is an $\mathscr{S}$-functor (necessarily Cartesian) equipped with an isomorphism of functors

$$F(x + y) \xrightarrow{\sim} F(x) + F(y)$$

making the diagrams

$$\begin{array}{ccc} F(x+y) & \longrightarrow & F(x) + F(y) \\ \downarrow{\scriptstyle F(\tau)} & & \downarrow{\scriptstyle \tau} \\ F(y+x) & \longrightarrow & F(y) + F(x) \end{array}$$

and

$$\begin{array}{ccccc} F((x+y)+z) & \longrightarrow & F(x+y) + F(z) & \longrightarrow & (F(x) + F(y)) + F(z) \\ \downarrow{\scriptstyle F(\sigma)} & & & & \downarrow{\scriptstyle \sigma} \\ F(x + (y+z)) & \longrightarrow & F(x) + F(y+z) & \longrightarrow & F(x) + (F(y) + F(z)). \end{array}$$

commutative.

An *morphism of additive functors* $u : F \to G$ is a morphism of $\mathscr{S}$-functors (automatically an isomorphism) making the diagram 523

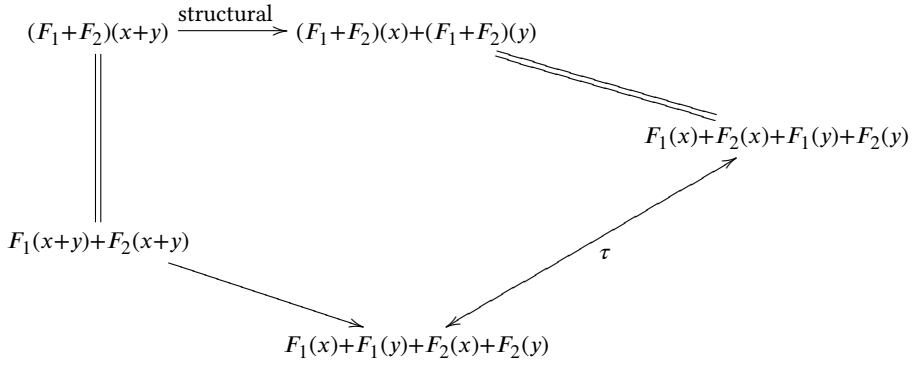
$$\begin{array}{ccc} F(x+y) & \xrightarrow{u_{x+y}} & G(x+y) \\ \downarrow & & \downarrow \\ F(x) + F(y) & \xrightarrow{u_x + u_y} & G(x) + G(y). \end{array}$$

commutative.

1.4.7. If $\mathscr{P}_1$ and $\mathscr{P}_2$ are two Picard stacks on $\mathscr{S}$, the Picard stack $\mathrm{HOM}(\mathscr{P}_1, \mathscr{P}_2)$ is the following Picard stack:

a) if $U \in \mathrm{Ob}\,\mathscr{S}$, its objects over U are the additive functors from $\mathscr{P}_1|U$ to $\mathscr{P}_2|U$; its morphisms are morphisms of additive functors;

b) the sum of two additive functors F_1 and F_2 is defined by the formula $(F_1 + F_2)(x) = F_1(x) + F_2(x)$; the structural isomorphism is the one making the diagram



commutative;

c) the associativity and commutativity isomorphisms are defined by means of the analogous isomorphisms in $\mathscr{P}_2$.

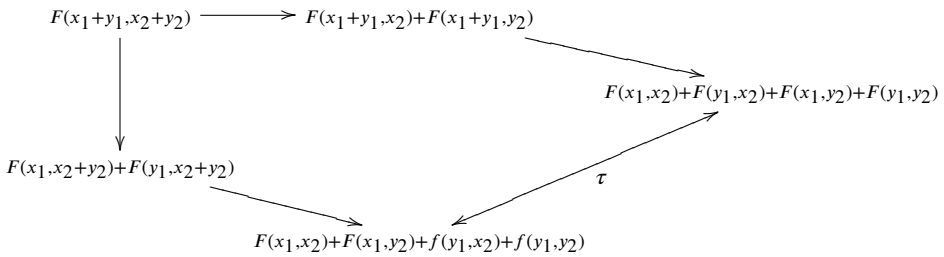
1.4.8. If $\mathscr{P}_1$, $\mathscr{P}_2$, and $\mathscr{P}$ are three Picard stacks, a biadditive functor from $\mathscr{P}_1 \times \mathscr{P}_2$ to $\mathscr{P}$ is an $\mathscr{S}$-functor F from $\mathscr{P}_1 \times_S \mathscr{P}_2$ to $\mathscr{P}$, equipped with isomorphisms of functors

$$F(x_1 + y_1, x_2) \xrightarrow{\sim} F(x_1, x_2) + F(y_1, x_2)$$
$$F(x_1, x_2 + y_2) \xrightarrow{\sim} F(x_1, x_2) + F(x_1, y_2),$$

such that

a) For fixed x_1 (resp. x_2), $F(x_1, *)$ (resp. $F(*, x_2)$) satisfies the compatibility conditions of 1.4.6.

524 b) For $U \in \mathrm{Ob}\,\mathscr{S}$, x_1, $y_1 \in \mathrm{Ob}\,\mathscr{P}_1(U)$ and x_2, $y_2 \in \mathrm{Ob}\,\mathscr{P}_2(U)$, the diagram



is commutative.

For example, the canonical functor

$$\mathrm{HOM}(\mathscr{P}_1, \mathscr{P}_2) \times \mathscr{P}_1 \longrightarrow \mathscr{P}_2 : (F, x) \longmapsto F(x)$$

is biadditive.

As in 1.4.7, one proves that the biadditive functors from $\mathscr{P}_1 \times \mathscr{P}_2$ to $\mathscr{P}$ form a Picard stack $\mathrm{HOM}(\mathscr{P}_1, \mathscr{P}_2; \mathscr{P})$.

We shall see in (1.4.20) that there exists a "2-universal" biadditive functor: $\mathscr{P}_1 \times \mathscr{P}_2 \to \mathscr{P}_1 \otimes \mathscr{P}_2$; more precisely, there exist a Picard stack $\mathscr{P}_1 \otimes \mathscr{P}_2$ and a biadditive functor $\otimes : \mathscr{P}_1 \times \mathscr{P}_2 \to \mathscr{P}_1 \otimes \mathscr{P}_2$ such that, for every Picard stack $\mathscr{P}$, the functor defined by $\otimes$:

$$\mathrm{HOM}(\mathscr{P}_1 \otimes \mathscr{P}_2, \mathscr{P}) \longrightarrow \mathrm{HOM}(\mathscr{P}_1, \mathscr{P}_2; \mathscr{P}) \tag{1.4.8.1}$$

is an equivalence. This stack is unique up to a unique equivalence (itself unique up to a unique isomorphism).

1.4.9. Let $u : \mathcal{S}_1 \to \mathcal{S}_2$ be a morphism of sites. The direct image $u_*\mathscr{P}$ of a Picard stack $\mathscr{P}$ on $\mathcal{S}_1$ is the Picard stack on $\mathcal{S}_2$ defined by

$$u_*\mathscr{P}(V) = \mathscr{P}(u^*V).$$

Its objects over $V \in \mathrm{Ob}\,\mathcal{S}_2$ are the objects of $\mathscr{P}$ over u^*V; its morphisms, its addition law, and its compatibility isomorphisms are those of $\mathscr{P}$. The direct image u_* is a 2-functor.

1.4.10. A *Picard prestack* $\mathscr{P}$ on a site $\mathcal{S}$ is a prestack $\mathscr{P}$ on $\mathcal{S}$ (Giraud [4]), equipped 525
with a functor $+ : \mathscr{P} \times_{\mathcal{S}} \mathscr{P} \to \mathscr{P}$ and associativity and commutativity isomorphisms σ and τ, such that, for every $U \in \mathrm{Ob}\,\mathcal{S}$, $\mathscr{P}(U)$ is a (strictly commutative) Picard category. If $\mathscr{P}_1$ and $\mathscr{P}_2$ are two Picard prestacks, one defines in the evident way (cf. 1.4.6, 1.4.7) the *additive functors* from $\mathscr{P}_1$ to $\mathscr{P}_2$ and the Picard prestack $\mathrm{HOM}(\mathscr{P}_1, \mathscr{P}_2)$ that they form. If $\mathscr{P}$ is a Picard prestack, and $j : \mathscr{P} \to a\mathscr{P}$ is the stack that it generates, there exists, up to an essentially unique equivalence, one and only one pair consisting of a Picard-stack structure on $a\mathscr{P}$ and an additive-functor structure on j; equipped with these data, the pair $(j, a\mathscr{P})$ is called *the Picard stack generated by* $\mathscr{P}$. For every Picard stack $\mathscr{P}_1$, one has an equivalence

$$\mathrm{HOM}(a\mathscr{P}, \mathscr{P}_1) \xrightarrow{\sim} \mathrm{HOM}(\mathscr{P}, \mathscr{P}_1). \tag{1.4.10.1}$$

1.4.11. We shall denote by $C^{[-1,0]}(\mathcal{S})$ the category of complexes of abelian sheaves K on $\mathcal{S}$ such that $K^i = 0$ for $i \notin [-1, 0]$. To every complex $K \in \mathrm{Ob}\,C^{[-1,0]}(\mathcal{S})$

$$K : d : K^{-1} \longrightarrow K^0$$

is associated the following Picard prestack $\mathrm{pch}(K)$:

(I) For $U \in \mathrm{Ob}\,\mathcal{S}$, one has $\mathrm{Ob}\,\mathrm{pch}(K)(U) = K^0(U)$.
(II) If $x, y \in K^0(U)$, an arrow from x to y is an element f of $K^{-1}(U)$ such that $df = y - x$.
(III) The composition law for arrows is the addition law in $K^{-1}(U)$.
(IV) The functor $+$ is given by the addition law in $K^0(U)$ and $K^{-1}(U)$.
(V) The associativity and commutativity isomorphisms are provided by the zero element of $K^{-1}(U)$.

We shall denote by $\mathrm{ch}(K)$ the Picard stack generated (1.4.10) by the Picard prestack $\mathrm{pch}(K)$. The sheaf $\mathscr{H}^0(K)$ is interpreted as the following associated sheaf

$$\mathscr{H}^0(K) \simeq a\,(U \longmapsto \text{group of isomorphism classes of objects of } \mathrm{ch}(K)(U)). \tag{1.4.11.1}$$

On the other hand, 526

$$\mathscr{H}^{-1}(K) \simeq \mathbf{Aut}(e) \tag{1.4.11.2}$$

1.4.12. Every morphism of complexes $f : K \to L$ induces an additive functor $\mathrm{pch}(f) : \mathrm{pch}(K) \to \mathrm{pch}(L)$, hence an additive functor

$$\mathrm{ch}(f) : \mathrm{ch}(K) \longrightarrow \mathrm{ch}(L).$$

It follows from (1.4.11.1) and (1.4.11.2) that $\mathrm{ch}(f)$ is an equivalence if and only if f is a quasi-isomorphism. From (1.4.10.1) one deduces that, for two morphisms $f, g : K \to L$, one has

$$\mathrm{Hom}(\mathrm{ch}(f), \mathrm{ch}(g)) \xrightarrow{\sim} \mathrm{Hom}(\mathrm{pch}(f), \mathrm{pch}(g)).$$

A morphism of *functors* $h : \mathrm{pch}(f) \to \mathrm{pch}(g)$ is a morphism of sheaves $h : K^0 \to L^{-1}$ such that

$$g(x) - f(x) = dh(x) \tag{1.4.12.1}$$

and such that, for every triple (x, y, u) with $y - x = du$, one has

$$h(y) + f(u) = g(u) + h(x) \tag{1.4.12.2}$$

That h is a morphism of additive functors means that

$$h(x + y) = h(x) + h(y). \tag{1.4.12.3}$$

Condition (1.4.12.3) makes it possible to rewrite (1.4.12.2) as

$$g(u) - f(u) = h(du); \tag{1.4.12.2'}$$

conditions (1.4.12.1) through (1.4.12.3) therefore mean that h is a homotopy from f to g, and

$$\mathrm{Hom}(\mathrm{ch}(f), \mathrm{ch}(g)) \simeq \{H : \text{ homotopy } K \longrightarrow L | g - f = dH + Hd\} \tag{1.4.12.4}$$

LEMMA 1.4.13. (I) *For every Picard stack $\mathscr{P}$, there exists a complex K such that $\mathscr{P} = \mathrm{ch}(K)$.*

(II) *For every additive functor $F : \mathrm{ch}(K) \to \mathrm{ch}(L)$, there exist a quasi-isomorphism $k : K' \to K$ and a morphism $\ell : K' \to L$ such that $F \simeq \mathrm{ch}(\ell)\,\mathrm{ch}(k)^{-1}$.*

527 Let us prove (I). Let $(k_i, U_i)_{i \in I}$ be a family such that

(a) $k_i \in \mathrm{Ob}\,\mathscr{P}(U_i)$;

(b) every object of $\mathscr{P}$ is locally isomorphic to an inverse image of one of the k_i.

Set $K^0 = \bigoplus_{i \in I} \mathbf{Z}_{U_i}$. It follows from 1.4.3 that there exists an additive functor F from $\mathrm{ch}(0 \to K^0)$ to $\mathscr{P}$ taking the basis e_i of $\mathbf{Z}_{U_i}$ to k_i. Let K^{-1} be the sheaf of pairs consisting of a local section x of K^0 and an isomorphism $t : F(0) \xrightarrow{\sim} F(x)$. Addition on K^{-1} is defined by $(x_1, t_1) + (x_2, t_2) = (x_1 + x_2, t)$, where t makes the diagram

$$\begin{array}{ccc} F(0) + F(0) & \xrightarrow{t_1 + t_2} & F(x_1) + F(x_2) \\ \updownarrow & & \updownarrow \\ F(0 + 0) & \xrightarrow{t} & F(x_1 + x_2). \end{array}$$

commutative. Define $d : K^{-1} \to K^0$ by $d(x, t) = x$; d is additive.

If $y - x = dt$, there exists a unique morphism $F(t)$ making the diagram

$$\begin{array}{ccc} F(x) & \xrightarrow{F(t)} & F(y) \\ \updownarrow & & \updownarrow \\ F(0) + F(x) & \xrightarrow{t + F(x)} & F(y - x) + F(x) \end{array}$$

commutative, and one verifies that this construction defines an equivalence

$$F : \mathrm{ch}(K) \longrightarrow \mathscr{P}.$$

Let us prove (II). Thus let $F : \mathrm{ch}(K) \to \mathrm{ch}(L)$ be an additive functor. There then exists a family $(k_i, \ell_i, \alpha_i, U_i)_{i\in I}$ such that

(c) $k_i \in K^0(U_i)$, $\ell_i \in L^0(U_i)$, and α_i is a morphism from $F(k_i)$ to ℓ_i;

(d) the homomorphism $k^0 : \bigoplus_{i\in I} \mathbf{Z}_{U_i} \to K^0$ with coordinates K_i is an epimorphism.

Set $K'^0 = \bigoplus_{i\in I} \mathbf{Z}_{U_i}$, and let $\ell^0 : K'^0 \to L^0$ be the homomorphism with coordinates ℓ_i. Since the functor F is additive, there exists a unique family of isomorphisms $\alpha_x : Fk^0(x) \xrightarrow{\sim} \ell^0(x)$ (x a local section of K'^0) such that

(e) for e_i the section 1 of $\mathbf{Z}_{U_i}$, one has $\alpha_{e_i} = \alpha_i$;

(f) The diagrams 528

$$\begin{array}{ccc} Fk^0(x+y) & \xrightarrow{\alpha_{x+y}} & \ell^0(x+y) \\ \updownarrow & & \| \\ Fk^0(x) + Fk^0(y) & \xrightarrow{\alpha_x+\alpha_y} & \ell^0(x) + \ell^0(y) \end{array}$$

are commutative.

Let $K'^{-1} = K^{-1} \times_{K^0} K'^0$; by (d), the evident arrow is a quasi-isomorphism of complexes $k : K' \to K$. If $x, y \in K'^0(U)$, and if $t \in K'^{-1}(U)$ satisfies $y - x = dt$, then there exists a unique section $\ell^{-1}(t; x, y)$ of $L^{-1}(U)$ defining the morphism that makes the diagram

(g)
$$\begin{array}{ccc} F(k^0(x)) & \xrightarrow{F(k^{-1}(t))} & F(k^0(y)) \\ \downarrow{\scriptstyle \alpha_x} & & \downarrow{\scriptstyle \alpha_y} \\ \ell^0(x) & \xrightarrow{\ell^{-1}(t;x,y)} & \ell^0(y). \end{array}$$

commutative. The fact that F is a functor gives

(h) $\ell^{-1}(t; x, y) + \ell^{-1}(t'; y, z) = \ell^{-1}(t + t'; x, z)$.

The functoriality of the additivity isomorphism of F gives

(i) $\ell^{-1}(t_1 + t_2; x_1 + x_2, y_1 + y_2) = \ell^{-1}(t_1; x_1, y_1) + \ell^{-1}(t_2; x_2, y_2)$.

From (h), one obtains

$$\ell^{-1}(0; x, x) + \ell^{-1}(0; x, x) = \ell^{-1}(0, x, x) \qquad \text{i.e.}$$
$$\ell^{-1}(0; x, x) = 0.$$

It then follows from (i) that $\ell^{-1}(t; x, y)$ depends only on t:

$$\ell^{-1}(t; x, y) = \ell^{-1}(t)$$

and that $\ell^{-1}(t)$ is additive in t. One concludes that $\ell = (\ell^{-1}, \ell^0)$ is a morphism of complexes and, by (f) and (g), the arrows α form an isomorphism of additive functors $F \circ \mathrm{ch}(k) = \mathrm{ch}(\ell)$.

1.4.14. Let $\mathrm{Ch}^\flat(\mathcal{S})$ be the category whose objects are the small Picard stacks on $\mathcal{S}$, and whose arrows are the isomorphism classes of additive functors. The construction ch 529
defines a functor from $C^{[-1,0]}(\mathcal{S})$ (1.4.11) to $\mathrm{Ch}^\flat(\mathcal{S})$. Let $D^{[-1,0]}(\mathcal{S})$ be the subcategory of the derived category of $\mathcal{S}$ formed by the complexes K such that $H^i(K) = 0$ for $i \neq 0$ or -1. It follows from 1.4.12 and 1.4.13 that

PROPOSITION 1.4.15. *The functor* ch *induces an equivalence of categories*

$$\mathrm{ch} : D^{[-1,0]}(\mathcal{S}) \xrightarrow{\sim} \mathrm{Ch}^\flat(\mathcal{S}).$$

We shall denote by $\flat$ the inverse equivalence of ch. For every Picard stack $\mathscr{P}$ on $\mathcal{S}$, $\mathscr{P}^\flat \in \mathrm{Ob}\, D^{[-1,0]}(\mathcal{S})$ determines $\mathscr{P}$ up to the isomorphism class of an equivalence.

LEMMA 1.4.16. *Let $L \in \mathrm{Ob}\, C^{[-1,0]}(\mathcal{S})$ be such that L^{-1} is injective.*

(I) $\mathrm{pch}(L)$ *is already a stack.*

(II) *For $K \in \mathrm{Ob}\, C^{[-1,0]}(\mathcal{S})$, every additive functor $F : \mathrm{ch}(K) \to \mathrm{ch}(L)$ is isomorphic to a functor $\mathrm{ch}(f)$ for f a morphism from K to L.*

(I) For any $U \in \mathrm{Ob}\, \mathcal{S}$ and $x \in \mathrm{Ob}\, \mathrm{ch}(L)(U)$, the sheaf of pairs consisting of a local section ℓ of $L^0|U$ and an isomorphism between x and ℓ is a principal homogeneous space under L^{-1}; it therefore has a section, and $j : \mathrm{pch}(L) \to \mathrm{ch}(L)$ is surjective on the isomorphism classes of objects, hence is an equivalence.

(II) Let L' be a complex of injectives, equipped with a quasi-isomorphism $\varphi : L \underset{\approx}{\longrightarrow} L'$, such that φ^i is an isomorphism for $i \leq -1$. By 1.4.13 and [17], there exists a diagram commutative up to homotopy

$$\begin{array}{ccc} L & \xrightarrow[\approx]{\varphi} & L' \\ {\scriptstyle f_0}\uparrow & & \uparrow{\scriptstyle f_1} \\ K' & \xrightarrow[\approx]{\Psi} & K \end{array}$$

such that $F \simeq \mathrm{ch}(f_0)\,\mathrm{ch}(\Psi)^{-1}$. Moreover, $K = \tau_{\leq 0}K$ and $L = \tau_{\leq 0}L'$, so that f_1 factors through $f : K \to L$, and $F \simeq \mathrm{ch}(f)$.

The lemma refines 1.4.15 by the following consequence of 1.4.12.

530 COROLLARY 1.4.17. *The construction* ch *defines an equivalence of* 2*-categories between*

a) *the* 2*-category of Picard stacks on $\mathcal{S}$.*

b) *the* 2*-category whose objects and* 1*-arrows are the objects and arrows of the full subcategory of $C^{[-1,0]}(\mathcal{S})$ formed by the complexes L with L^{-1} injective, and whose* 2*-arrows are homotopies between arrows:* $\mathrm{Hom}(f,g) = \{H | g - f = \mathrm{dH} + \mathrm{Hd}\}$.

CONSTRUCTION 1.4.18. One has

$$\mathrm{HOM}(\mathscr{P}_1, \mathscr{P}_2)^\flat \simeq \tau_{\leq 0} R\,\mathbf{Hom}(\mathscr{P}_1^\flat, \mathscr{P}_2^\flat).$$

Let K, $L \in \mathrm{Ob}\, C^{[-1,0]}(S)$, with L^{-1} injective. By 1.4.16 (II) and 1.4.12, one has

$$\mathrm{ch}(\tau_{\leq 0}\,\mathbf{Hom}(K,L)) \xrightarrow{\sim} \mathrm{HOM}(\mathrm{ch}(K), \mathrm{ch}(L)) \tag{1.4.18.1}$$

and one deduces Construction 1.4.18 from this functorially.

CONSTRUCTION 1.4.19. For $f : \mathcal{S}_1 \to \mathcal{S}_2$ a morphism of sites, one has

$$(f_*\mathscr{P})^\flat \simeq \tau_{\leq 0}\, R f_*(\mathscr{P}^\flat).$$

Let $L \in \mathrm{Ob}\, C^{[-1,0]}$ with L^{-1} injective. By 1.4.16 (I), one has

$$\mathrm{ch}(f_* L) \xrightarrow{\sim} f_* \,\mathrm{ch}(L) \tag{1.4.19.1}$$

and from this one deduces a functorial construction 1.4.19.

We shall not need to use the following

Construction 1.4.20. The tensor product promised in 1.4.8 exists, and

$$(\mathscr{P}_1 \otimes \mathscr{P}_2)^\flat \simeq \tau_{\geq -1}\mathscr{P}_1^\flat \overset{\mathbf{L}}{\otimes} \mathscr{P}_2^\flat.$$

One checks, as in 1.4.12 and 1.4.13 (II), that for all K_1, K_2, and L in $C^{[-1,0]}(\mathscr{S})$, one has

(a) Every morphism $f \in \mathrm{Hom}(K_1 \otimes K_2, L) \simeq \mathrm{Hom}(\tau_{\geq -1}(K_1 \otimes K_2), L)$ defines a biadditive functor $\mathrm{ch}(f) \in \mathrm{Hom}(\mathrm{ch}(K_1), \mathrm{ch}(K_2); \mathrm{ch}(L))$; moreover, morphisms of functors identify with homotopies;

(b) For every biadditive functor $F : \mathrm{ch}(K_1), \mathrm{ch}(K_2) \to \mathrm{ch}(L)$, there exist quasi-isomorphisms $k_i : \mathrm{ch}(K_i') \to \mathrm{ch}(K_i)$ and $f : K_1' \otimes K_2' \to L$ such that $F \simeq \mathrm{ch}(f) \circ (\mathrm{ch}(k_1)^{-1}, \mathrm{ch}(k_2)^{-1})$.

If now K_1^0 or K_2^0 is flat, as is ${K'}_1^0$ or ${K'}_2^0$, then $k_1 \otimes k_2 : \tau_{\geq -1}{K'}_1 \otimes {K'}_2 \to$ 531
$\tau_{\geq -1}K_1 \otimes K_2$ is a quasi-isomorphism, and it follows that $\mathrm{ch}(\tau_{\geq -1}(K_1 \otimes K_2))$ satisfies the universal property (1.4.8.1), whence the existence of $\otimes$ and of a functorial construction 1.4.20.

1.4.21. Let G be an abelian sheaf, and let $G[1]$ be the complex with value G placed in degree -1. The prestack $\mathrm{pch}(G[1])$ identifies with the prestack of trivial torsors under G, and hence $\mathrm{ch}(G[1])$ "is none other than" the stack of torsors under G, with its usual addition law.

1.4.22. Let $K \in C^{[-1,0]}(\mathscr{S})$, and let G be an abelian sheaf. The extensions E of K by G form a Picard stack, $\mathrm{EXT}(K, G)$, under Baer addition.

$$E : 0 \to G[0] \xrightarrow{\alpha} E \xrightarrow{\beta} K \to 0.$$

To every extension E one associates an additive functor from $\mathrm{ch}(K)$ to $\mathrm{ch}(G[1])$ which, when applied to a local section x of K^0, gives the torsor $\beta^{-1}(x)$. The reader will verify the following.

Proposition 1.4.23. *The construction outlined above is an equivalence of Picard stacks*

$$\mathrm{EXT}(K, G) \xrightarrow{\sim} \mathrm{HOM}(\mathrm{ch}(K), \mathrm{ch}(G[1])).$$

1.4.24. Let F be an object of the site $\mathscr{S}$. We shall also write F for the sheaf represented by F, and write $\mathbf{Z}^{(F)}$ for the abelian sheaf it generates. Let C be a Picard stack on $\mathscr{S}$. One checks easily (cf. 1.4.3) that giving any one of the following amounts to the same thing:

a) an additive functor $H : \mathrm{ch}(\mathbf{Z}^{(F)}) \to C$;

b) a morphism of *stacks*:
(stack of local sections of F, with only identity morphisms) $\to C$;

c) an object of $C(F)$.

Suppose (in order to be able to apply Definition 1.4.9) that fiber products exist in $\mathscr{S}$, and let f be the canonical morphism of sites $f : \mathscr{S}/F \to \mathscr{S}$. The preceding construction gives an equivalence

$$\mathrm{HOM}(\mathrm{ch}(\mathbf{Z}^{(F)}), C) \xrightarrow{\sim} f_* f^* C. \tag{1.4.24.1}$$

The isomorphism obtained from (1.4.24.1) by the construction 532

$$\tau_{\leq 0} R\,\mathbf{Hom}(\mathbf{Z}^{(F)}, C^\flat) \xrightarrow{\sim} \tau_{\leq 0} R f_* f^* C^\flat,$$

may also be deduced from the more general isomorphism

$$R\,\mathbf{Hom}(\mathbf{Z}^{(F)}, K) \xrightarrow{\sim} Rf_*f^*K \quad (K \in \mathrm{Ob}\, D^+(\mathcal{S})). \tag{1.4.24.2}$$

1.5. The universal coefficient formula.

1.5.1. Let $f : X \to S$ be a smooth projective curve over S. We denote by $\mathrm{PIC}(X/S)$ the Picard stack on the big fppf site of S (XVII 0.10), namely the direct image (1.4.9) of the stack of invertible sheaves on X. We have

$$\mathrm{PIC}(X/S) = f_*\,\mathrm{ch}\,\mathbf{G}_m[1] \tag{1.5.1.1}$$

$$\mathrm{PIC}(X/S)^\flat = \tau_{\leq 0}Rf_*(\mathbf{G}_m[1]). \tag{1.5.1.2}$$

Each section t of X defines an invertible sheaf $\mathcal{O}(t)$ on X. With the notation of 1.4.24 (cf. 1.4.24 $a \leftrightarrow b$), this construction, being compatible with every base change, defines a morphism of Picard stacks on S

$$\mathrm{ch}\,\mathbf{Z}^{(X)} \longrightarrow \mathrm{PIC}(X/S). \tag{1.5.1.3}$$

If C is any Picard stack on S, the morphisms (1.5.1.3) and (1.4.24.1) define an additive functor

$$\mathrm{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow \mathrm{HOM}(\mathrm{ch}(\mathbf{Z}^{(X)}), C) \xrightarrow{\sim} f_*f^*C. \tag{1.5.1.4}$$

Upon applying the construction $\flat$, the functor (1.5.1.3) defines a morphism[110]

$$\mathbf{Z}^{(X)} \longrightarrow \tau_{\leq 0}Rf_*\mathbf{G}_m[1] \longrightarrow Rf_*\mathbf{G}_m[1], \tag{1.5.1.5}$$

and the analogue of the functor (1.5.1.4) is, by (1.4.24.2), a morphism

$$R\,\mathbf{Hom}(\tau_{\leq 0}Rf_*(\mathbf{G}_m[1]), K) \longrightarrow Rf_*f^*K. \tag{1.5.1.6}$$

If C is the stack $\mathrm{ch}(G[1])$ of torsors under a group G satisfying (1.3.1.1), Theorem
533 1.3.10 asserts that the functor (1.5.1.4) is an equivalence, the inverse equivalence being the one that associates with a torsor K under G_X the additive functor $\langle *, K]$. The assertion that (1.5.1.4) is an equivalence amounts to saying that the morphism deduced from (1.5.1.4) or (1.5.1.6)

$$\tau_{\leq 1}R\,\mathbf{Hom}(\tau_{\leq 0}Rf_*(\mathbf{G}_m[1]), G) \longrightarrow \tau_{\leq 1}Rf_*f^*G \tag{1.5.1.7}$$

is an isomorphism.

Theorem 1.5.2. (universal coefficient formula).

Let $f : X \to S$ be a smooth projective curve over S, let C be a Picard stack on S_{fppf}, and let $K = C^\flat$ (1.4.15). Suppose that K is locally isomorphic (in $D^+(S_{\mathrm{fppf}})$) to complexes of the form $G_{-1} \to G_0$, where the sheaves G_i satisfy (1.3.1.1). Then:

(I) *The functor (1.5.1.4)*

$$\mathrm{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow f_*f^*C \tag{1.5.2.1}$$

is an equivalence of Picard stacks.

[110] N.D.E.: Here the shift functor [1] is to be understood to have higher precedence than the truncation $\tau_{\leq 0}$.

(II) *The morphism deduced from (1.5.1.6) or (1.5.1.4)*

$$\tau_{\leq 0} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_*(\mathrm{G}_m[1]), K) \longrightarrow \tau_{\leq 0} Rf_* f^* K \tag{1.5.2.2}$$

is an isomorphism.

This theorem makes $\tau_{\leq 0} Rf_*(\mathrm{G}_m[1])$ play the role played by homology in the classical universal coefficient formula.

It is clear that, for every stack C, one has $(I) \leftrightarrow (II)$, and both (I) and (II) are local in nature on S. If C is of the form $\mathrm{ch}(G[1])$, then the theorem follows from 1.3.10, as noted above. It remains to show that, if K is of the form

$$K : d : G_{-1} \longrightarrow G_0,$$

and if (1.5.2.2) is an isomorphism for $G_{-1}[1]$ and $G_0[1]$, then (1.5.2.2) is an isomorphism for K. To verify this, it suffices to compare the cohomology long exact sequence

$$0 \longrightarrow R^{-1} f_* K \longrightarrow f_* G_{-1} \longrightarrow f_* G_0 \longrightarrow R^0 f_* K \longrightarrow R^1 f_* G_{-1} \longrightarrow R^1 f_* G_0$$

with the analogous long exact sequence for the left-hand side of (1.5.2.2), and to apply the five lemma. 534

1.5.3. Let u be a flat morphism of smooth projective curves over S

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & {\scriptstyle f_1}\searrow \quad \swarrow{\scriptstyle f_2} & \\ & S & \end{array} .$$

The diagram of additive functors

$$\begin{array}{ccc} \mathbf{Z}^{(X)} & \longrightarrow & \mathrm{PIC}(X/S) \\ \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle N_{X/Y}} \\ \mathbf{Z}^{(Y)} & \longrightarrow & \mathrm{PIC}(Y/S) \end{array} \tag{1.5.3.1}$$

is then essentially commutative.

For every Picard stack C on S, the diagram

$$\begin{array}{ccc} \mathrm{HOM}(\mathrm{PIC}(Y/S), C) & \longrightarrow & f_{2*} f_2^* C \\ \downarrow{\scriptstyle \mathrm{HOM}(N_{X/Y}, C)} & & \downarrow{\scriptstyle u^*} \\ \mathrm{HOM}(\mathrm{PIC}(X/S), C) & \longrightarrow & f_{1*} f_1^* C \end{array} \tag{1.5.3.2}$$

is therefore essentially commutative, and for $K \in \mathrm{Ob}\, D^+(S)$, the diagram

$$\begin{array}{ccc} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{2*} \mathrm{G}_m[1], K) & \longrightarrow & Rf_{2*} f_2^* K \\ \downarrow{\scriptstyle R\,\mathbf{Hom}(N_{X/Y}, K)} & & \downarrow{\scriptstyle u^*} \\ R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{1*} \mathrm{G}_m[1], K) & \longrightarrow & Rf_{1*} f_1^* K \end{array} \tag{1.5.3.3}$$

is commutative.

Now let G be an abelian sheaf on S satisfying (1.3.1.1). The trace XVII 6.3 gives us an additive functor

$$\mathrm{Tr}_u : f_{1*}f_1^* \operatorname{ch}(G[1]) \longrightarrow f_{2*}f_2^* \operatorname{ch}(G[1]) \tag{1.5.3.4}$$

535 and 1.3.8.6 gives us an isomorphism making the diagram essentially commutative

$$\begin{array}{ccc} \mathrm{HOM}(\mathrm{PIC}(X/S), \operatorname{ch}(G[1])) & \longleftarrow & f_{1*}f_1^* \operatorname{ch}(G[1]) \\ \Big\downarrow{\scriptstyle \mathrm{HOM}(u^*, \operatorname{ch}(G[1]))} & & \Big\downarrow{\scriptstyle \mathrm{Tr}_u} \\ \mathrm{HOM}(\mathrm{PIC}(Y/S), \operatorname{ch}(G[1])) & \longleftarrow & f_{2*}f_2^* \operatorname{ch}(G[1]). \end{array} \tag{1.5.3.5}$$

The diagram

$$\begin{array}{ccc} \tau_{\leq 1} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{1*}\mathbf{G}_m[1], G) & \xleftarrow{\sim} & \tau_{\leq 1} Rf_{1*}f_1^* G \\ \Big\downarrow{\scriptstyle R\,\mathbf{Hom}(u^*, G)} & & \Big\downarrow{\scriptstyle Tr_u} \\ \tau_{\leq 1} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{2*}\mathbf{G}_m[1], G) & \xleftarrow{\sim} & \tau_{\leq 1} Rf_{2*}f_2^* G \end{array} \tag{1.5.3.6}$$

is therefore commutative.

1.5.4. With the notation of 1.5.2, let G be an abelian sheaf on S satisfying 1.3.1.1. Locally on S for the étale topology, one has (noncanonically)

$$\tau_{\leq 0} Rf_*\mathbf{G}_m[1] \simeq f_*\mathbf{G}_m[1] + R^1 f_*\mathbf{G}_m.$$

At the level of Picard stacks, one indeed defines a section $\operatorname{ch}(R^1 f_*\mathbf{G}_m) \to f_* \operatorname{ch}(\mathbf{G}_m[1])$[111] by associating with each isomorphism class of invertible sheaves the invertible sheaf belonging to that class, rigidified along suitable sections of X/S.

The isomorphism (1.5.2.2) therefore gives an isomorphism

$$\mathbf{Hom}(R^1 f_*\mathbf{G}_m, G) \xrightarrow{\sim} \mathbf{Hom}_S(X, G) \tag{1.5.4.1}$$

and an exact sequence

$$0 \longrightarrow \mathbf{Ext}^1(R^1 f_*\mathbf{G}_m, G) \longrightarrow R^1 f_* G \longrightarrow \mathbf{Hom}(f_*\mathbf{G}_m, G) \longrightarrow 0. \tag{1.5.4.2}$$

The morphisms in (1.5.4.1) and (1.5.4.2) are interpreted as follows:

536 a) The morphism (1.5.4.1) and the first morphism in (1.5.4.2) are defined by the canonical map from X to $\mathbf{Pic}_{X/S}$.

b) The second morphism in (1.5.4.2) is the degree (1.3.7.2).

Locally on S for the étale topology, one has (noncanonically) $\mathbf{Pic}_{X/S} \sim \mathbf{Pic}^0_{X/S} \times \mathbf{Z}^k$; hence

$$\mathbf{Ext}^1(R^1 f_*\mathbf{G}_m, G) \xrightarrow{\sim} \mathbf{Ext}^1(\mathbf{Pic}^0_{X/S}, G)$$

and (1.5.4.2) may also be written

$$0 \longrightarrow \mathbf{Ext}^1(\mathbf{Pic}^0_{X/S}, G) \longrightarrow R^1 f_* G \longrightarrow \mathbf{Hom}(f_*\mathbf{G}_m, G) \longrightarrow 0. \tag{1.5.4.3}$$

[111] N.D.E.: Recall that this stack identifies with $\mathrm{PIC}(X/S)$, cf. (1.5.1.1).

1.5.5. When, in (1.5.2.2), one takes $K = \mathbf{G}_m[1]$, Theorem 1.5.2 appears as a self-duality theorem for $\tau_{\leq 0}(Rf_*\mathbf{G}_m[1])$. Any complex representing $\tau_{\leq 0}Rf_*\mathbf{G}_m[1]$ admits a canonical three-step filtration, whose successive quotients are, up to unique quasi-isomorphism,

$$\mathbf{Pic}_{X/S}/\mathbf{Pic}^0_{X/S} \quad , \quad \mathbf{Pic}^0_{X/S} \quad , \quad (f_*\mathbf{G}_m)[1].$$

The preceding self-duality corresponds to a duality (with values in $\mathbf{G}_m$) between $\mathbf{Pic}_{X/S}/\mathbf{Pic}^0_{X/S}$ and $f_*\mathbf{G}_m$, and to a self-duality (with values in $\mathbf{G}_m[1]$) on $\mathbf{Pic}^0_{X/S}$ (self-duality of the Jacobian).

1.5.6. The situation is distinctly worse for smooth curves over S : $f : X \to S$ that are not assumed proper over S. The additive group seems to behave uncontrollably. The preceding arguments nevertheless extend to the case of smooth curves obtained from a proper curve over S by removing a part finite over S, when G is an étale torsion sheaf whose torsion is prime to the residue characteristics of S. One may hope that étale p-torsion sheaves also behave reasonably; see Serre [14, Ch. VI n° 11 p. 126].

The role of Theorem 1.2.2 will here be played by the acyclicity theorem XV 2.2, which 537
has the following as a corollary.

Lemma 1.5.7. *Let $p : X \to S$ be the projection of a standard affine space over S, and let G be an abelian torsion sheaf on S_{et}, of torsion prime to the residue characteristics of S. We also write G for the inverse image of G on the site S_{fppf}. Then the functor p^* is an equivalence of categories between the category of torsors under G on S_{et} (or on S_{fppf}, which amounts to the same thing, by Grothendieck [5, Corollaire 11.9]), and the category of torsors under G_X on X.*

1.5.8. Let $\overline{f} : \overline{X} \to S$ be a proper morphism of finite presentation whose fibers are purely one-dimensional, let Y be a closed subscheme of $\overline{X}$ finite over S, defined by an ideal $\mathfrak{m}$, and suppose that $X = \overline{X} - Y$ is a smooth curve over S. Let furthermore G be a torsion sheaf on S_{et}, of torsion prime to the residue characteristics of S. We shall also write G for the sheaf on the big fppf site of S (XVII 0.10) obtained as the inverse image of G.

Let $\mathbf{G}_m(\mathfrak{m})$ be the subsheaf of $\mathbf{G}_{m\overline{X}}$ consisting of sections equal to 1 on Y. A torsor under $\mathbf{G}_m(\mathfrak{m})$ identifies with an invertible sheaf on $\overline{X}$ trivialized on Y. Every relative divisor on X, finite over S, defines such an invertible sheaf. Locally on S for the étale topology, there exist such divisors giving rise to relatively ample invertible sheaves.

Lemma 1.5.9. *Under the hypotheses of 1.5.8, with S affine[112], let $\mathcal{O}_{\overline{X}}(1)$ be a relatively ample invertible sheaf on $\overline{X}$, trivialized along Y, and let $\mathscr{M}$ be an invertible sheaf on $\overline{X}$, trivialized along Y. For every point $s \in S$, there exists an integer n_0 such that, for every integer $n \geq n_0$ and every section m_s of $\mathscr{M}(n) \otimes k(s)$ on the fiber $\overline{X}_s$ which is equal to 1 on Y_s, there exists a section m of $\mathscr{M}(n)$ which is equal to 1 on Y and lifts m_s.*

We reduce to the case where S is Noetherian (since a family of surjective morphisms remains a family of surjective morphisms after base change[113]). Let $\mathscr{N}$ be the coherent

[112] N.D.E.: It is in fact necessary to suppose that S is local with closed point s. If one wishes to retain the original hypothesis, then in the conclusion one should require that m be defined only over the inverse image of an open neighborhood of s in S, as occurs below in 1.5.10.1.

[113] N.D.E.: Eliminating this Noetherian hypothesis is somewhat more subtle than might appear at first sight, because the question is affine rather than linear. If $\overline{X}$, Y, and $\mathscr{M}$ are obtained by a base change $S \to S_0$ (which is assumed to be a local morphism between local schemes) from corresponding data $\overline{X}_0$, Y_0, and $\mathscr{M}_0$ over a Noetherian S_0 (cf. EGA IV 8), and if the desired assertion is known in this Noetherian

538 sheaf of local sections of $\mathscr{M}$ vanishing on $\overline{X}_s$ and Y. For each n, the local solutions to the problem form a principal homogeneous space under $\mathscr{N}(n)$, so that the obstruction to lifting m_s lies in $H^1(X, \mathscr{N}(n)) = H^0(S, R^1\overline{f}_*\mathscr{N}(n))$. But, for n sufficiently large, $R^1\overline{f}_*\mathscr{N}(n) = 0$ (EGA III 2.2.1).

1.5.10. Let $\mathscr{L}$ be an invertible sheaf on $\overline{X}$ trivialized along Y, and let $s \in S$. Consider the conditions

. Every section of $\mathscr{L} \otimes k(s)$ on X_s which is equal to 1 on Y_s lifts to a section of $\mathscr{L}$ on the inverse image U of a neighborhood of s, equal to 1 on $Y \times_S U$.

. There exist four sections m_i of $\mathscr{L} \otimes k(s)$ on X_s, equal to 1 on Y, and linearly independent at the generic point of every irreducible component of X_s.

Consider the following conditions, concerning respectively a section m of $\mathscr{L}$, a pair (m_1, m_2), and a triple (m_1, m_2, m_3) of sections of $\mathscr{L}$.

. m is equal to 1 on Y, and the subscheme $Z(m)$ of X defined by $m = 0$ is a relative divisor over S (which means that it is quasi-finite over S, and hence finite over S, since $X = \overline{X} - Y$ is smooth over S).

. Let λ be the canonical coordinate on the affine line $g : \mathbf{E}^1_s \to S$. The section $\lambda m_1 + (1-\lambda)m_2$ of $g'^*\mathscr{L}$ satisfies (1.5.10.3).

. Similarly, $\lambda m_1 + \mu m_2 + (1-\lambda-\mu)m_3$ satisfies (1.5.10.3).

We leave it to the reader to verify, by a general-position argument[114], the following

Lemma 1.5.11. *Let $\mathscr{L}$ be an invertible sheaf on X, trivialized along Y, and let $\overline{s}$ be a geometric point of S with image s in S. If $\mathscr{L}$ satisfies (1.5.10.1) and (1.5.10.2) at s, then:*

(I) *If n sections m_i of $\mathscr{L}$ satisfy (1.5.10.3), there exist an étale neighborhood U of $\overline{s}$ and a section k of $\mathscr{L}|\overline{f}^{-1}(U)$ such that the pairs (m_i, k) satisfy (1.5.10.4).*

539 (II) *Let m_1, m_2, k_1, k_2 be four sections of $\mathscr{L}$ such that the pairs (m_i, k_j) satisfy (1.5.10.4). There then exist an étale neighborhood U of $\overline{s}$ and a section k of $\mathscr{L}|\overline{f}^{-1}(U)$ such that the triples (m_i, k_j, k) satisfy (1.5.10.5).*

When $k(s)$ is infinite, one could simply take U to be a Zariski neighborhood.

1.5.12. Now let $\mathscr{L}$ be an invertible sheaf on $\overline{X}$ rigidified along Y, and let K be a G-torsor on X, where G is as in 1.5.7. We propose to define a G-torsor $\langle \mathscr{L}, K]$ on S, generalizing the one studied in 1.3, and additive in $\mathscr{L}$.

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situation, then one can obtain the desired result over S at the cost of an unpleasant diagram chase, during which one may have to consider a variant of the desired assertion in which the sections are required to be equal to 0 instead of 1 on Y_s or Y.

[114] N.D.E.: It should be observed here that, in view of (1.5.10.1) and the fact that the second part of condition (1.5.10.3) can be checked in the local situation by considering only the fiber over s, the lemma now involves only sections of the restriction of $\mathscr{L}$ to the fiber X_s. For example, for (I), condition (1.5.10.2) provides an *affine* space V of dimension 3 of sections of $\mathscr{L} \otimes k(s)$ on X_s which are equal to 1 on Y_s, and no section belonging to this space V vanishes at any generic point of X_s. We seek to determine $k \in V$ such that, for every i, (m_i, k) satisfies (1.5.10.2), that is, such that the affine line joining the points m_i and k does not pass through the origin (in any of its incarnations associated with the different irreducible components of X_s); in other words, it suffices that k not belong to the line joining m_i and the origin. One thus obtains a conjunction of conditions satisfied on nonempty open subsets of V. To be completely rigorous and to account for the fact that the irreducible components need not be geometrically irreducible, when one must take a point of V in an extension (if $k(s)$ is finite), it is in fact necessary to introduce a finite Galois extension $s' \to s$ such that the irreducible components of $X_{s'}$ are geometrically irreducible. The dense open subset of $V_{s'}$ obtained by arguing on $X_{s'}$ is Galois-invariant and hence defines the desired dense open subset of V.

Let (m_1, m_2) be a pair of sections of $\mathscr{L}$ that satisfies (1.5.10.4). With the notation of 1.5.10.3 and 1.5.10.4, the trace, from $\lambda m_1 + (1-\lambda)m_2 = 0$ to $\mathbf{E}^1_S$, of the torsor g'^*K is a torsor $K_{1,2}$ on $\mathbf{E}^1_S$. Its pullback by the section $\lambda = 0$ (resp. $\lambda = 1$) is the torsor $K_2 = \mathrm{Tr}_{Z(m_2)/S}(K)$ (resp. $K_1 = \mathrm{Tr}_{Z(m_1)/S}(K)$). By 1.5.7, the torsor $K_{1,2}$ is canonically the pullback of a torsor under G on S; hence there is a canonical isomorphism between K_1 and K_2.

If m_1, m_2, and k are three sections of $\mathscr{L}$ such that the pairs (m_i, k) satisfy (1.5.10.4), composition again gives an isomorphism

$$\Psi_k : K_1 = \mathrm{Tr}_{Z(m_1)/S}(K) \xrightarrow{\sim} K_2 = \mathrm{Tr}_{Z(m_2)/S}(K). \tag{1.5.12.1}$$

If E is a relative divisor on X, finite over S, the sections $m'_i = m_i \otimes 1$ of $\mathscr{L} \otimes \mathscr{O}(E)$ still satisfy (1.5.10.3), and we have (for $i = 0, 1$)

$$K'_i = \mathrm{Tr}_{Z(m'_i)/S}(K) \simeq K_i + \mathrm{Tr}_{E/S}(K) \tag{1.5.12.2}$$

and the isomorphism $\Psi_{k\otimes 1} : K'_1 \to K'_2$ is deduced from Ψ_k:

$$\Psi_{k\otimes 1} = \Psi_k + \mathrm{id}_{\mathrm{Tr}_{E/S}}(K). \tag{1.5.12.3}$$

To verify that Ψ_k does not depend on k, it suffices to do so in an étale neighborhood of every point s of S, and, by 1.5.12.3 and 1.5.9, we reduce to the case in which conditions (1.5.10.1) and (1.5.10.2) hold. With the notation of 1.5.11 (II), $\Psi_{\lambda k_i + (1-\lambda)k} - \Psi_{k_i}$ is a morphism from $\mathbf{E}^1_S$ to G, zero at $\lambda = 1$, and hence zero; thus $\Psi_{k_i} = \Psi_k$, and therefore $\Psi_{k_1} = \Psi_{k_2}$. 540

Let m_1 and m_2 be two sections of $\mathscr{L}$ satisfying 1.5.10.3, and let $K_i = \mathrm{Tr}_{Z(m_i)/S}(K)$. To define an isomorphism $\Psi_{m_2,m_1} : K_1 \to K_2$, it suffices to do so in an étale neighborhood of every point s of S, compatibly with base change. If E is a relative divisor on X, finite over S, such that $\mathscr{L} \otimes \mathscr{O}(E)$ satisfies (1.5.10.1) and (1.5.10.2) (see 1.5.9), and if the pairs $(m_i \otimes 1, k)$ satisfy (1.5.10.4), we set (cf. (1.5.12.3))

$$\Psi_{m_2,m_1} + \mathrm{Tr}_{E/S}(K) = \Psi_k.$$

This defines Ψ_{m_2,m_1}, and one verifies using (1.5.11. (I)) that $\Psi_{m_3,m_2}\Psi_{m_2,m_1} = \Psi_{m_3,m_1}$. These constructions are compatible with base changes. If there exists a section m of $\mathscr{L}$ satisfying (1.5.10.3), we set

$$\langle \mathscr{L}, K] = \mathrm{Tr}_{Z(m)/S}(K)$$

and, by what precedes, this torsor is independent, up to canonical isomorphism, of the choice of m.

In the general case, note that to define $\langle \mathscr{L}, K]$ it suffices to define it in an étale neighborhood of every point s of S, compatibly with base change. If, as above, $\mathscr{L}\otimes\mathscr{O}(E)$ satisfies (1.5.10.1) and (1.5.10.2), then $\mathscr{L} \otimes \mathscr{O}(E)$ has sections satisfying (1.5.10.3), and we set

$$\langle \mathscr{L}, K] = \langle \mathscr{L} \otimes \mathscr{O}(E), K] - \mathrm{Tr}_{E/S}(K).$$

1.5.13. Under the hypotheses of 1.5.8, if x and y are two sections of $\overline{f}_*\mathbf{G}_m(\mathfrak{m})$, then $\lambda x + (1-\lambda)y$ is again a section for λ in an open subset with nonempty fibers of the affine line over S.

This can be expressed by saying that

. "$\overline{f}_*\mathrm{G}_m(\mathfrak{m})$ is path-connected."

Every morphism from an open subset with nonempty fibers of the affine line to G is constant. There is therefore no nontrivial homomorphism from $\overline{f}_*\mathrm{G}_m(\mathfrak{m})$ to G, and the automorphisms of $\mathscr{L}$ act trivially on $\langle\mathscr{L}, K]$. The torsor $\langle\mathscr{L}, K]$ thus depends only on K and on the section of $R^1\overline{f}_*\mathrm{G}_m(\mathfrak{m})$ defined by $\mathscr{L}$. By globalization, this makes it possible to define a symbol $\langle\lambda, K]$ for λ a section of $R^1\overline{f}_*\mathrm{G}_m(\mathfrak{m})$. The torsor $\langle\lambda, K]$ is additive in λ, and therefore corresponds to an extension $e(K)$ of $R^1\overline{f}_*\mathrm{G}_m(\mathfrak{m})$ by G (see 1.4.23 or SGA 7 VII 1.1.6 and 1.2). 541

The formation of $e(K)$ is compatible with every base change and with addition of torsors, and hence gives an additive functor

$$e : \overline{f}_* \operatorname{ch}(G[1]) \longrightarrow \mathrm{EXT}(R^1\overline{f}_*\mathrm{G}_m(\mathfrak{m}), G).$$

Let j be the canonical map from X into $R^1\overline{f}_*\mathrm{G}_m(\mathfrak{m})$, $j : t \to$ (class of $\mathscr{O}(t)$). Pullback by j is an additive functor

$$j^* : \mathrm{EXT}(R^1\overline{f}_*\mathrm{G}_m(\mathfrak{m}), G) \longrightarrow \overline{f}_* \operatorname{ch}(G[1])$$

and trivially $j^*e \sim \mathrm{Id}$. One verifies as in 1.3.10 and 1.5.2 that, for the fppf topology,

Proposition 1.5.14. *Under the hypotheses of 1.5.8, the functors e and j of 1.5.13 are quasi-inverse equivalences. It follows from 1.4.19 and 1.4.23 that there is an isomorphism*

$$\mathrm{Ext}^1(R^1\overline{f}_*\mathrm{G}_m(\mathfrak{m}), G) \xrightarrow{\sim} H^1(X, G).$$

1.5.15. Let $f_i : X_i \to S$ $(i = 1, 2)$ be two compactified curves as in 1.5.8.

Let $\overline{u} : \overline{X}_1 \to \overline{X}_2$ be a morphism such that Y_1 is a subscheme of the inverse image of Y_2 and which induces a finite flat morphism from X_1 to X_2. Then, for $\mathscr{L}$ an invertible sheaf on $\overline{X}_2$ rigidified along Y_2 and K a torsor on X_1, we have

$$\langle\mathscr{L}, \mathrm{tr}_{X_1/X_2}(K)] \xrightarrow{\sim} \langle\overline{u}^*\mathscr{L}, K],$$

542 and we deduce the commutativity of the diagram

$$\begin{array}{ccc} \mathrm{Ext}^1(R^1\overline{f}_{1*}\mathrm{G}_m(\mathfrak{m}_1), G) & \xrightarrow{\sim} & H^1(X_1, G) \\ \downarrow{\scriptstyle \mathrm{Ext}^1(\overline{u}^*,\mathrm{Id})} & & \downarrow{\scriptstyle \mathrm{Tr}_{X_1/X_2}} \\ \mathrm{Ext}^1(R^1\overline{f}_{2*}\mathrm{G}_m(\mathfrak{m}_2), G) & \xrightarrow{\sim} & H^1(X_2, G). \end{array} \tag{1.5.15.1}$$

If $\overline{u} : \overline{X}_1 \to \overline{X}_2$ is finite and flat, and satisfies $Y_1 = \overline{u}^{-1}Y_2$ (as schemes), then, for $\mathscr{L}$ an invertible sheaf on X_1 rigidified along Y_1 and K a torsor on X_2, we have

$$\langle\mathscr{L}, u^*K] \xrightarrow{\sim} \langle N_{\overline{X}_1/\overline{X}_2}\mathscr{L}, K]$$

and we deduce the commutativity of the diagram

$$\begin{array}{ccc} \mathrm{Ext}^1(R^1\overline{f}_{2*}\mathrm{G}_m(\mathfrak{m}_2), G) & \xrightarrow{\sim} & H^1(X_2, G) \\ \downarrow{\scriptstyle \mathrm{Ext}^1(N_{\overline{X}_1/\overline{X}_2},\mathrm{Id})} & & \downarrow{\scriptstyle u^*} \\ \mathrm{Ext}^1(R^1\overline{f}_{1*}\mathrm{G}_m(\mathfrak{m}_1), G) & \xrightarrow{\sim} & H^1(X_1, G). \end{array} \tag{1.5.15.2}$$

One could of course state this result more precisely in terms of Picard stacks.

1.6. A vanishing theorem. The present section, from 1.6.6 (2nd proof) to the end, can be read independently of sections 2 to 5.

LEMMA 1.6.1. *Let X be a scheme and Y a closed subscheme of X with complement U, defined by an ideal $\mathfrak{m}$.*

$$U \overset{j}{\hookrightarrow} X \overset{i}{\hookleftarrow} Y.$$

Let $\mathbf{G}_m(\mathfrak{m})$ be the fpqc *sheaf whose sections on an X-scheme X_1 are the sections of $\mathcal{O}_{X_1}^*$ equal to 1 on the inverse image of Y. For $n \geq 1$ invertible on X, the sequence*

$$0 \to j_!\boldsymbol{\mu}_n \to \mathbf{G}_m(\mathfrak{m}) \xrightarrow{n} \mathbf{G}_m(\mathfrak{m}) \to 0$$

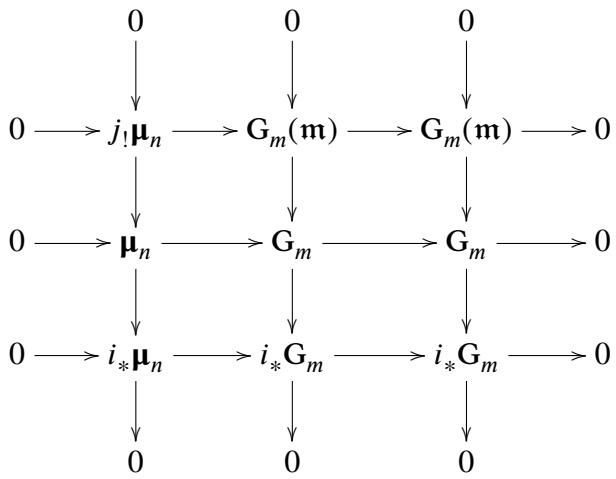
is an exact sequence of sheaves on the big étale site (XVII 0.10) *of X.*

Let i be the morphism of big étale sites $i : Y \to X$ induced by i: 543

$$i^*(X_1/X) = X_1 \times_X Y.$$

The functor i_* is exact. In the diagram

(1.6.1.1)



the columns are exact, as are the last two rows (IX 3.2). The first row is therefore exact.

1.6.2. Let k be a field and

$$G_1' \longrightarrow G_2' \longrightarrow G \longrightarrow G_2'' \longrightarrow G_1''$$

an exact sequence of abelian sheaves on the big fppf site of $\mathrm{Spec}(k)$. Recall that if G_1' and G_2' are representable and of finite type, and G_2'' and G_1'' are representable and locally of finite type, then G is representable and locally of finite type[115].

[115] N.D.E.: In the case where $G_1' = G_2' = 0$, this is clear. In the case where $G_1'' = G_2'' = 0$, using SGA 3 V 10.1.1 (new edition), one can reduce to the case where $G_1' \to G_2'$ is a monomorphism, which is treated in (SGA 3 VI_A 3.2). We are thus reduced to the case where $G_1' = G_1'' = 0$. One can then regard the sheaf G as a torsor under G_2' over the scheme G_2''. If G_2' is assumed affine (the situation to which one is reduced, by (SGA 3 VI_B 11.17), if before making the reductions G_1' and G_2' were assumed affine), then the representability of G follows immediately from faithfully flat descent. This special case is sufficient for the applications made in this paragraph. If G_2' is quasi-projective, the representability of G can be obtained by using (SGA 1 VIII 7.7) and arguments similar to those of [11, Lemma 3.3.1]. In fact, every group scheme G of finite type over a field is quasi-projective: one may suppose that G is connected; this is then [2, Corollary 1.2] if G is smooth, and (M. Raynaud) the general case reduces to it because, by (SGA 3 VII_A 8.3), there exists a finite morphism from G to a smooth group scheme.

If P is a connected commutative algebraic group of finite type over k, and n is an integer invertible in k, the sequence

$$0 \to {}_nP \to P \xrightarrow{n} P \to 0 \tag{1.6.2.1}$$

is exact. If G is a finite étale commutative group over k, killed by n, the first group and the last arrow of the exact sequence

$$\mathrm{Hom}(P,G) \to \mathrm{Hom}({}_nP,G) \xrightarrow{j} \mathrm{Ext}^1(P,G) \xrightarrow{n} \mathrm{Ext}^1(P,G)$$

are zero, whence an isomorphism

$$j : \mathrm{Hom}({}_nP,G) \xrightarrow{\sim} \mathrm{Ext}^1(P,G). \tag{1.6.2.2}$$

544 1.6.3. Under the hypotheses of 1.6.1, suppose that X is proper over the spectrum of a field k

$$p : X \longrightarrow k.$$

The exact sequence

$$0 \longrightarrow \mathbf{G}_m(\mathfrak{m}) \longrightarrow \mathbf{G}_m \longrightarrow i_*\mathbf{G}_m \longrightarrow 0$$

defines an exact sequence of fppf sheaf cohomology on $\mathrm{Spec}(k)$, which is easily interpreted directly as

$$0 \to p_*\mathbf{G}_m(\mathfrak{m}) \to p_*\mathbf{G}_{m_X} \to (pi)_*\mathbf{G}_{m_Y} \xrightarrow{j} R^1p_*\mathbf{G}_m(\mathfrak{m}) \to \mathbf{Pic}_{X/k} \to \mathbf{Pic}_{Y/k}\,. \tag{1.6.3.1}$$

The sheaves $p_*\mathbf{G}_{m_X}$ and $(pi)_*\mathbf{G}_{m_Y}$ are representable by group schemes of finite type and affine. The sheaves $\mathbf{Pic}_{X/k}$ and $\mathbf{Pic}_{Y/k}$ are representable by group schemes locally of finite type. By 1.6.2, the sheaves $p_*\mathbf{G}_m(\mathfrak{m})$ and $R^1p_*\mathbf{G}_m(\mathfrak{m})$ are representable. By (1.5.13.1), the former is represented by a connected group scheme (of finite type); the latter is locally of finite type.

Definition 1.6.4. *Under the preceding hypotheses, we denote by* $\mathbf{Pic}_{\mathfrak{m},X/k}$ *the group scheme representing* $R^1p_*\mathbf{G}_m(\mathfrak{m})$. *We denote by* $\mathbf{Pic}^0_{\mathfrak{m},X/k}$ *its identity component and by* $\mathbf{Pic}^\tau_{\mathfrak{m},X/k}$ *the inverse image in* $\mathbf{Pic}_{\mathfrak{m},X/k}$ *of the torsion subgroup of* $\mathbf{Pic}_{\mathfrak{m},X/k}/\mathbf{Pic}^0_{\mathfrak{m},X/k}$.

The scheme $\mathbf{Pic}_{\mathfrak{m},X/k}$ represents the fppf sheaf generated by the presheaf which associates with every scheme S over k the set of isomorphism classes of invertible modules on $S \times_k X$, trivialized along $S \times_k Y$ (1.5.8).

The Kummer exact sequence 1.6.1 gives, for k algebraically closed and n invertible in k, an isomorphism

$$H^1_c(U,\boldsymbol{\mu}_n) \xrightarrow{\sim} {}_n\,\mathbf{Pic}^\tau_{\mathfrak{m},X/k}(k). \tag{1.6.4.1}$$

545 1.6.5. Let X be a smooth curve over an algebraically closed field k, let $\overline{X}$ be the smooth projective curve containing X as a dense open subscheme, let Y be a closed subscheme of $\overline{X}$ such that $X = \overline{X} - Y$, let $\mathfrak{m}$ be the ideal defining Y, let n be an integer invertible in k, and let G be a finite abelian group killed by n.

The group $\mathbf{Pic}_{\mathfrak{m},X/k}$ is noncanonically the product of the subgroup $\mathbf{Pic}^0_{\mathfrak{m},\overline{X}/k}$ and a free abelian group. By (1.6.4.1), (1.6.2.2), and 1.5.14, we therefore have a canonical composite

isomorphism

$$\mathrm{Hom}(H^1_c(X,\boldsymbol{\mu}_n),G) \xleftarrow{\sim} \mathrm{Hom}({}_n\,\mathbf{Pic}^{\tau}_{\mathfrak{m},\overline{X}/k}(k),G) = \mathrm{Hom}({}_n\,\mathbf{Pic}^{0}_{\mathfrak{m},\overline{X}/k},G)$$
$$\xrightarrow{\sim} \mathrm{Ext}^1(\mathbf{Pic}^{0}_{\mathfrak{m},\overline{X}/k},G) \xleftarrow{\sim} \mathrm{Ext}^1(\mathbf{Pic}_{\mathfrak{m},\overline{X}/k},G) \xrightarrow{\sim} H^1(X,G)$$

If $u : X \to Y$ is a finite morphism between smooth curves over k, we deduce from XVII 6.3. and (1.5.15.2) the commutativity of the diagram

$$\begin{array}{ccc} \mathrm{Hom}(H^1_c(Y,\boldsymbol{\mu}_n),G) & \xrightarrow{\sim} & H^1(Y,G) \\ \downarrow{\scriptstyle \mathrm{Hom}(\mathrm{Tr}_u,G)} & & \downarrow{\scriptstyle u^*} \\ \mathrm{Hom}(H^1_c(X,\boldsymbol{\mu}_n),G) & \xrightarrow{\sim} & H^1(X,G). \end{array} \tag{1.6.5.1}$$

Lemma 1.6.6. *Let X be a smooth curve over an algebraically closed field k, let n be an integer invertible in k, and let $u : X' \to X$ be a tame étale covering. The morphism*

$$\mathrm{Tr}_u : H^1_c(X',\mathbf{Z}/n) \longrightarrow H^1_c(X,\mathbf{Z}/n)$$

is then (noncanonically[116]) isomorphic to the transpose of the morphism

$$u^* : H^1(X,\mathbf{Z}/n) \longrightarrow H^1(X',\mathbf{Z}/n).$$

1st proof. *Since $\boldsymbol{\mu}_n$ is isomorphic to $\mathbf{Z}/n$, 1.6.6 is the special case of (1.6.5.1) with $G = \mathbf{Z}/n$.*

2nd proof. If k is the field $\mathbf{C}$ of complex numbers, then, by the comparison theorems, the morphisms Tr_u and u^* are identified with the analogous morphisms defined by the continuous map of topological spaces $u : X'(\mathbf{C}) \to X(\mathbf{C})$:

$$\mathrm{Tr}_u : H^1_c(X'(\mathbf{C}),\mathbf{Z}/n) \longrightarrow H^1_c(X(\mathbf{C}),\mathbf{Z}/n) \quad \text{and}$$
$$u^* : H^1(X(\mathbf{C}),\mathbf{Z}/n) \longrightarrow H^1(X'(\mathbf{C}),\mathbf{Z}/n).$$

These two homomorphisms are transposes of one another under Poincaré duality. 546

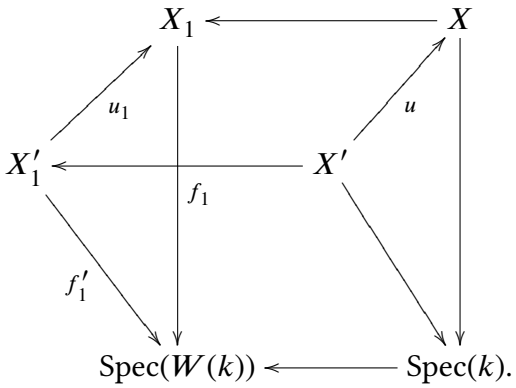
By the Lefschetz principle, the assertion 1.6.6 remains true for k of characteristic 0 (the cohomology groups under consideration are indeed invariant under extension of the algebraically closed base field).

If k has characteristic $p > 0$, let $W(k)$ be the ring of Witt vectors over k. Let $\overline{X}$ be the smooth projective completion of X. By (SGA 1 III 7.4), the curve $\overline{X}$ lifts to a smooth projective curve $\overline{X}_1$ over $W(k)$. Since $\overline{X}$ is smooth, every point $s_i \in S = \overline{X} - X$ lifts to a section s'_i of $\overline{X}$ over $W(k)$; set $S_1 = \bigcup_i s'_i(\mathrm{Spec}(W(k)))$ and $X_1 = \overline{X}_1 - S_1$.

Recall (SGA 1 XIII 2.4) that the covering X' of X lifts to an étale covering X'_1 of X_1, and that X'_1 is obtained from a smooth projective curve $\overline{X}'_1$ over $W(k)$ by removing the

[116] N.D.E.: The isomorphism is canonical if an isomorphism $\boldsymbol{\mu}_n \simeq \mathbf{Z}/n$ has been chosen.

union S_1' of finitely many disjoint sections.



The étale sheaves $R^1 f_{1*}(\mathbf{Z}/n)$, $R^1 f'_{1*}(\mathbf{Z}/n)$, $R^1 f_{1!}(\mathbf{Z}/n)$, and $R^1 f'_{1!}(\mathbf{Z}/n)$ are locally
547 constant and their formation is compatible with base change, as follows from the exact sequences relating the cohomology of $\overline{X}_1$, X_1, and S_1, or of $\overline{X}'_1$, X'_1, and S'_1.

The special or generic geometric fibers of the morphisms

$$\mathrm{Tr}_{u_1} : R^1 f'_{1!}\mathbf{Z}/n \longrightarrow R^1 f_{1!}\mathbf{Z}/n$$

or

$$u_1^* : R^1 f_{1*}\mathbf{Z}/n \longrightarrow R^1 f'_{1*}\mathbf{Z}/n$$

are therefore isomorphic, and this reduces us to the case already treated in which k has characteristic zero.

Lemma 1.6.7. *Let X be a connected smooth curve over an algebraically closed field k of characteristic exponent p, and let n be an integer prime to p. There exists a principal covering* (= finite étale Galois (surjective)) *$u : X' \to X$, of degree dividing a power of n, such that the trace morphism*

$$\mathrm{Tr}_u : H^1_c(X', \mathbf{Z}/n) \longrightarrow H^1_c(X, \mathbf{Z}/n)$$

is zero.

Recall that the group $H^1(X, \mathbf{Z}/n)$ is finite (IX 4.6 or XVI 5.2). Let $u : X' \to X$ be the largest connected abelian principal covering of X whose Galois group is killed by n (its Galois group is denoted by $H_1(X, \mathbf{Z}/n)$). By construction, the homomorphism

$$u^* : H^1(X, \mathbf{Z}/n) \longrightarrow H^1(X', \mathbf{Z}/n)$$

is zero. By 1.6.6, the homomorphism

$$\mathrm{Tr}_u : H^1_c(X', \mathbf{Z}/n) \longrightarrow H^1_c(X, \mathbf{Z}/n)$$

is therefore zero.

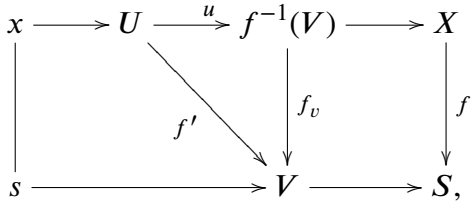
Recall the following lemma (EGA IV 15.6.5):

Lemma 1.6.8. *Let $f : X \to S$ be a smooth morphism and let x be a section of f. Then there exists a Zariski open subset U of X, containing x, such that the geometric fibers of $f|U$ are connected.*

548 Fundamental Lemma 1.6.9. *Let $f : X \to S$ be a compactifiable smooth curve (1.1.2)* (XVII 3.2)*, let x be a geometric point of X, let s be its image in S, and let $n \geq 1$ be an integer*

invertible on S. There exist an étale neighborhood V of s in S and an étale neighborhood U of x in $f^{-1}(V) = V \times_S X$

(1.6.9.1)



such that $R^0 f'_! \mathbf{Z}/n = 0$, the arrow XVII 6.2

$$\mathrm{Tr}_u : R^1 f'_! \mathbf{Z}/n \longrightarrow R^1 f_{v!} \mathbf{Z}/n$$

is zero, and the trace morphism

$$\mathrm{Tr}_{f'} : R^2 f'_! \mathbf{Z}/n(1) \longrightarrow \mathbf{Z}/n$$

is an isomorphism.

For $R^0 f'_! \mathbf{Z}/n = 0$, it suffices that f' be quasi-affine; for $\mathrm{Tr}_{f'}$ to be an isomorphism, it suffices that the geometric fibers of f' be connected (1.1.9); by 1.6.8, these properties of f' are easy to obtain locally on S for the étale topology. It remains to prove, for X with geometrically connected fibers, the existence of a diagram (1.6.9.1) for which the arrow Tr_u under consideration is zero: it will suffice to shrink U and V so that the other two required properties of f' hold.

The problem is local on S in a neighborhood of s; by XVII 5.2.6 (base change), we immediately reduce to the case where S is Noetherian. By passage to the limit, and using XVII 5.3.6 (constructibility), we reduce to the case where S is strictly local Noetherian, with closed point the image of s.

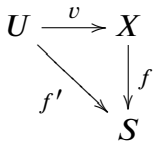
Let t be a geometric point of S. By 1.6.7, there exists a principal covering of degree prime to the characteristic exponent p of $k(s)$, $u : X^1_t \to X_t$, such that the arrow

$$\mathrm{Tr}_u : H^1_c(X^1_t, \mathbf{Z}/n) \longrightarrow H^1_c(X_t, \mathbf{Z}/n)$$

is zero. By the acyclicity theorem XV 2.1 (or XV 2.6), there exists an étale neighborhood $v : U \to X$ of x in X such that X^1_t splits over U_t, so that $v_t : U_t \to X_t$ factors through X^1_t. We therefore still have 549

$$\mathrm{Tr}_{v_t} = 0 : H^1_c(U_t, \mathbf{Z}/n) \longrightarrow H^1_c(X_t, \mathbf{Z}/n).$$

For every geometric point t of S, there therefore exists an étale neighborhood of x in X



such that the fiber at t of

(1.6.9.2)
$$\mathrm{Tr}_v : R^1 f'_! \mathbf{Z}/n \longrightarrow R^1 f_! \mathbf{Z}/n$$

is zero (XVII 5.2.6).

As U varies, the images of the arrows (1.6.9.2) form a filtered decreasing system of subsheaves of the constructible sheaf $R^1 f_! \mathbf{Z}/n$, which becomes zero at any given geometric point of S. By Noetherian induction, one verifies that such a system is always

stationary, with stationary value zero. Hence there exists U such that (1.6.9.2) is zero, and this completes the proof of 1.6.9.

2. The trace morphism

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This § is devoted to the proof of Theorems 2.9 and 2.12. The proof of 2.9 uses only no. 1 of § 1.

LEMMA 2.1. *Let $f : X \to S$ be a compactifiable morphism* (XVII 3.2.1) *of relative dimension $\leq d$, and let $j : U \hookrightarrow X$ be an open subscheme of finite type of X, whose complement has fiberwise dimension $< d$ (for example, U is fiberwise dense in X). If F is a torsion sheaf on X, the canonical arrow*

$$R^{2d}(fj)_!j^*F \xrightarrow{\sim} R^{2d}f_!F$$

is an isomorphism.

Let i be the inclusion of $Y = X - U$ into X. The exact sequence (XVII 5.1.16.2) gives us an exact sequence

$$R^{2d-1}(fi)_!i^*F \longrightarrow R^{2d}(fj)_!j^*F \longrightarrow R^{2d}f_!F \longrightarrow R^{2d}(fi)_!i^*F$$

whose outer terms are zero by (XVII 5.2.8.1).

LEMMA 2.2. *Let $f : X \to S$ be a compactifiable morphism of relative dimension $\leq d$, and let $u_i : X_i \to X$ be a family of separated étale morphisms of finite type; let $u_{ij} : X_{ij} = X_i \times_X X_j \to X$. For every torsion sheaf F on X, the sequence*

$$\bigoplus_{i,j} R^{2d}(fu_{ij})_!u_{ij}^*F \rightrightarrows \bigoplus_i R^{2d}(fu_i)_!u_i^*F \to R^{2d}f_!F \to 0$$

of arrows (XVII 6.2.7.2) *is exact.*

This follows immediately from the right exactness of the functor $R^{2d}f_!$ and from XVII 6.2.9.

LEMMA 2.3. *Let $f : X \to S$ be a flat morphism of finite presentation. The open subset of X on which f is Cohen–Macaulay* (EGA IV 6.8.1 and 12.2.1 (vii) *is relatively of finite presentation and fiberwise dense.*

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The question is local on S, which we reduce to assuming Noetherian, whence the 1st assertion. The second is checked fiberwise, hence with S the spectrum of a field. The local rings of X at the maximal points of X are Artinian, hence Cohen–Macaulay, which proves the assertion since "Cohen–Macaulay" is an "open" property.

LEMMA 2.4. *Let $f : X \to S$ be a flat morphism of finite presentation and Cohen–Macaulay, with S quasi-separated. Every point of X has a (Zariski) open neighborhood U such that there exists a quasi-finite flat S-morphism of finite presentation from U to the standard vector bundle $\mathbf{E}_S^d$.*

This is a special case of EGA IV 15.4.3.

LEMMA 2.5. *Let $\underline{u} = (u_1, \dots, u_d)$ and $\underline{v} = (v_1, \dots, v_d)$ be two systems of parameters of a Cohen–Macaulay local ring A of dimension d and maximal ideal $\mathfrak{m}$. There exists a sequence $\underline{w}_0, \dots, w_n$ of systems of parameters of A such that $\underline{w}_0 = \underline{u}$, $\underline{w}_n = \underline{v}$, and each $\underline{w}_{i+1}$ is obtained from $\underline{w}_i$ by changing only one of the parameters.*

We argue by induction on d, the assertion being vacuous for $d < 2$; we therefore suppose $d \geq 2$. For an element $x \in \mathfrak{m}$ to be A/u_1-regular (resp. A/v_1-regular), it is necessary and sufficient that x belong to no member of $\mathrm{Ass}(A/u_1)$ (resp. $\mathrm{Ass}(A/v_1)$); since no finite union of prime ideals $\neq \mathfrak{m}$ is equal to $\mathfrak{m}$, there exists $x \in \mathfrak{m}$ that is both A/u_1- and A/v_1-regular, and hence there also exist systems of parameters $\underline{a} = (u_1, x, \dots)$ and $\underline{b} = (v_1, x, \dots)$. We conclude by applying the induction hypothesis to $\underline{u} - \{u_1\}$ and $\underline{a} - \{u_1\}$ in A/u_1, to $\underline{a} - \{x\}$ and $\underline{b} - \{x\}$ in A/x, and to $\underline{b} - \{v_1\}$ and $\underline{v} - \{v_1\}$ in A/v_1.

LEMMA 2.6. *Let X be a Cohen–Macaulay scheme of finite type over an algebraically closed field k, and let $u, v : X \to \mathbf{E}^d_k$ be two quasi-finite flat morphisms from X to a standard vector bundle. Every closed point x of X has a (Zariski) open neighborhood U such that there exists a sequence of morphisms $w_0, \dots, w_n : U \to \mathbf{E}^d_k$ such that $w_0 = u|U$,* 552
$w_n = v|U$, and w_{i+1} differs from w_i in only one coordinate.

Suppose first that $u(x) = v(x) = 0$. The germs at x of quasi-finite flat morphisms from X to $\mathbf{A}^d_k$, sending x to 0, then identify with the systems of parameters of the local ring $\mathcal{O}_{X,x}$, and 2.6 follows from 2.5. In the general case, one first joins u to $u - u(x)$ (resp. v to $v - v(x)$) by the chain of morphisms

$$u - (u_1(x), \dots, u_i(x), 0, \dots, 0) (\text{resp. } v - (v_1(x), \dots, v_i(x), 0 \dots 0)).$$

2.7. Let there be a pair of composable S-compactifiable morphisms $X \xrightarrow{g} Y \xrightarrow{f} S$. Suppose that f has relative dimension $\leq d$ and g has relative dimension $\leq e$. For every torsion sheaf F on X, the composition spectral sequence

$$R^p f_! R^q g_! F \Longrightarrow R^{p+q}(fg)!F \tag{2.7.1}$$

then gives us (by the maximum-cycle argument) an isomorphism

$$R^{2d} f_! R^{2e} g_! F \simeq R^{2(d+e)}(fg)_! F. \tag{2.7.2}$$

2.8. Let

$$a^d : \mathbf{E}^d_S \longrightarrow S$$

be the standard vector bundle over a coherent (i.e. quasi-compact, quasi-separated) scheme S. If F is a torsion sheaf on S, prime to the residual characteristics of S, we define as follows, by induction on d, a trace morphism

$$\mathrm{Tr}_{a^d} : R^{2d} a^d_! F(d) \xrightarrow{\sim} F. \tag{2.8.1}$$

For $d = 0$, Tr_{a^0} is the identity. For $d = 1$, Tr_{a^1} is the isomorphism (1.1.6) (cf. 1.1.9). Finally, there is a canonical isomorphism

$$\mathbf{E}^{d+1}_S = \mathbf{E}^1_S \times_S \mathbf{E}^d_S = \mathbf{E}^1_{\mathbf{E}^d_S}$$

and we define $\mathrm{Tr}_{a^{d+1}}$ as the composite of the isomorphisms $\mathrm{Tr}_{a^d_S}$ and $Ra^d_{S!}(\mathrm{Tr}_{a^1_{\mathbf{E}^d_S}})$ (cf. diagram 553
Var 3 of 2.9).

The formation of the source (and the target) of (2.8.1) is compatible with every base change. Every algebraic group over S acting on $\mathbf{E}^d_S$ therefore acts on these sheaves; a connected algebraic group necessarily acts trivially. The group of coordinate permutations, contained in the affine group, therefore acts trivially on the source (and the target) of (2.8.1).

2.8.2. The isomorphism (2.8.1) is invariant under permutation of the coordinates.

This point could also be verified by interpreting (2.8.1) in terms of cup products (XVII 5.4.2.2 and XVII 5.4.3.5).

THEOREM 2.9. *Consider triples (f, d, F) consisting of a compactifiable morphism (XVII 6.3) $f : X \to Y$, an integer d, and a torsion sheaf F on Y prime to the residual characteristics of Y, where the morphism f satisfies the condition $(*)_d$: there exists an open subset U of X such that the restriction of f to U is a flat morphism of finite presentation with fibers of dimension $\leq d$, and such that the fibers of $X - U$ have dimension $< d$.*

There is one and only one way to associate with every triple (f, d, F) as above a trace morphism

$$\mathrm{Tr}_f : R^{2d} f_! f^* F(d) \longrightarrow F$$

such that the following conditions are satisfied:

(Var 1) *(Functoriality). The trace morphism is functorial in F.*

(Var 2) *(Compatibility with base change). For every Cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y \end{array}$$

554 *with Y' quasi-compact and quasi-separated, the diagram*

$$\begin{array}{ccccc} R^{2d} f'_! f'^* g^* F(d) & = & R^{2d} f'_! g'^* f^* F(d) & \xleftarrow{\sim} & g^* R^{2d} f_! f^* F(d) \\ \downarrow{\scriptstyle Tr_{f'}} & & & & \downarrow{\scriptstyle g^*(Tr_f)} \\ g^* F & = & & = & g^{*F} \end{array}$$

in which the horizontal isomorphism is the base-change arrow XVII 5.2.2.1, is commutative.

(Var 3) *(Compatibility with composition). For every pair of composable Z-compactifiable morphisms*

$$X \xrightarrow{g} Y \xrightarrow{f} Z,$$

integers d and e such that f satisfies $(*)_d$ and g satisfies $(*)_e$, and torsion sheaf F on Z, the composite morphism fg satisfies $(*)_{d+e}$ and the diagram

$$\begin{array}{ccc} R^{2d} f_! R^{2e} g_! g^* f^* F(d)(e) & \xrightarrow{R^{2d} f_!(\mathrm{Tr}_g)} & R^{2d} f_! f^* F(d) \\ \updownarrow{\scriptstyle s} & & \downarrow{\scriptstyle \mathrm{Tr}_f} \\ R^{2(d+e)} (fg)_! (fg)^* F(d+e) & \xrightarrow{\mathrm{Tr}_{fg}} & F \end{array}$$

in which the vertical isomorphism is the arrow (2.7.2), is commutative.

(Var 4) *(Normalizations).*

(I) *If f is finite locally free of rank n and $d = 0$, the composite morphism*

$$F \longrightarrow f_* f^* F = f_! f^* F \xrightarrow{\mathrm{tr}_f} F$$

is multiplication by n.

555 (II) *If f is the standard affine line $f : \mathbf{E}^1_Y \to Y$, and if $d = 1$, the trace morphism*

$$\mathrm{Tr}_f : R^2 f_! f^* F(1) \longrightarrow F$$

is the isomorphism (2.8.1).

a) For $d = 0$, condition $(*)_d$ says that f is quasi-finite and flat of finite presentation. By (XVII 6.2.3), for $d = 0$, the trace morphism must coincide with the morphism constructed in *loc. cit.*

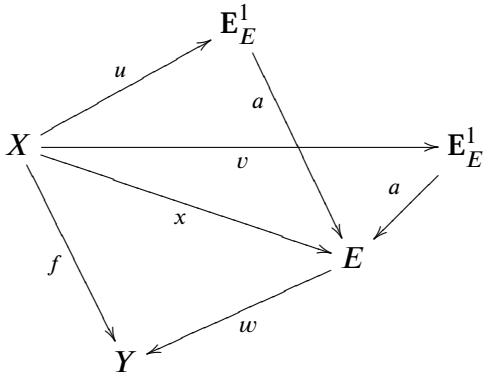
When f is the projection of standard affine space $\mathbf{A}^d_S$ onto S, it follows from (Var 4) (II) and (Var 3) that Tr_f must be the isomorphism (2.8.1).

b) Let $f : X \to Y$ be a compactifiable morphism satisfying the condition $(*)_d$: there exists a quasi-finite flat Y-morphism of finite presentation u from X to the standard vector bundle $a^d : \mathbf{E}^d_Y \to Y$.

Let $t(f,u)$ be the "composite" morphism of the trace morphisms (2.8.1) and (SGA XVII 6.2.3) (cf. (Var 3))

$$t(f,u) = \mathrm{Tr}_{a^d} \circ R^{2d} a^d_! (\mathrm{Tr}_u).$$

This morphism $t(f,u)$ satisfies (Var 1) and (Var 2). By base change, to verify that $t(f,u)$ depends only on f and d, and not on u, it suffices to verify this when Y is the spectrum of an algebraically closed field k. By 2.2, the question is local on X, so by 2.6, it suffices to show that $t(f,u) = t(f,v)$ when u and v differ in only one coordinate, which, by 2.8.2, may be assumed to be the first. Setting $E = \mathbf{E}^{d-1}_Y$, we thus have a commutative diagram



By construction (2.8), 556

$$\mathrm{Tr}_{wa} = \mathrm{Tr}_w \circ R^{2(d-1)} w_! (\mathrm{Tr}_a) ;$$

we deduce that $t(f,u)$ is the "composite" of Tr_w and $t(x,u)$:

$$\begin{aligned} t(f,u) &= \mathrm{Tr}_{wa} \circ R^{2d} (wa)_! (\mathrm{Tr}_u) \\ &= \mathrm{Tr}_w \circ R^{2(d-1)} w_! (\mathrm{Tr}_a) \circ R^{2d} (wa)_! (\mathrm{Tr}_u) \\ &= \mathrm{Tr}_w \circ R^{2(d-1)} w_! (\mathrm{Tr}_a \circ R^2 a_! (\mathrm{Tr}_u)). \end{aligned}$$

Moreover (1.1.7), $t(x,u)$ is precisely the morphism Tr_x of 1.1.6; hence $t(x,u) = t(x,v) = \mathrm{Tr}_x$ and $t(f,u) = t(f,v)$.

We denote by Tr_f the morphism $t(f,u)$ constructed above. By XVII 6.2.3, this trace morphism satisfies (Var 1), (Var 2), (Var 3) for $e = 0$, and (Var 4). Whenever it is defined, it necessarily coincides with the trace morphism sought, if the latter exists.

c) Let $f : X \to Y$ be a compactifiable Cohen–Macaulay morphism, flat of finite presentation and pure of relative dimension d. Let $(U_i)_{i\in I}$ be a cover of X by quasi-compact open subsets such that there exists a quasi-finite flat Y-morphism from U_i to $\mathbf{E}_Y^d$ (2.4). By 2.2 and b), there is then a unique morphism Tr_f making the following diagram commutative, where the notation of 2.2 is used:

$$\begin{array}{ccccc} \bigoplus_{i,j} R^{2d}(fu_{ij})_!(fu_{ij})^*F(d) & \rightrightarrows & \bigoplus_i R^{2d}(fu_i)_!(fu_i)^*F(d) & \longrightarrow & R^{2d}f_!f^*F(d) \\ & & & \searrow^{\bigoplus_i \mathrm{Tr}_{fu_i}} & \downarrow{\scriptstyle \mathrm{Tr}_f} \\ & & & & F \end{array}$$

This trace morphism does not depend on the chosen open cover (compare the two with the sum cover); it satisfies the conditions (Var 1), (Var 2), (Var 3) for $e = 0$, and (Var 4), as follows immediately from b).

d) Let $f : X \to Y$ be a compactifiable morphism satisfying $(*)_d$, and let $j : U \hookrightarrow X$ be the open subset promised by $(*)_d$. By 2.3, after replacing U by a smaller open subset, we may suppose that fj is Cohen–Macaulay and pure of dimension d.

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By 2.1, the canonical morphism

$$R^{2d}(fj)_!(fj)^*F(d) \xrightarrow{\sim} R^{2d}f_!f^*F(d)$$

is an isomorphism, and we define Tr_f to be the composite

$$\mathrm{Tr}_f : R^{2d}f_!f^*F(d) \xleftarrow{\sim} R^{2d}(fj)_!(fj)^*F(d) \xrightarrow{\mathrm{Tr}_{fj}} F.$$

It is immediate, by (Var 3) for $e = 0$ established in c), that Tr_f does not depend on the choice of U. The trace morphism satisfies (Var 1), (Var 2), (Var 3) for $e = 0$, and (Var 4), and is the only one that can satisfy the imposed conditions. It remains to prove that it satisfies (Var 3), and for this purpose the preceding arguments reduce us to considering only pairs (f, d), (g, e) satisfying $(*)_d$ and $(*)_e$.

The reader who wishes to polish the arguments that follow will first verify the

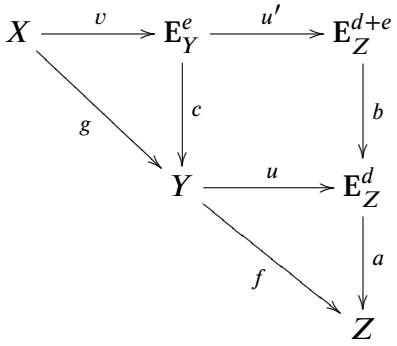
LEMMA 2.9.1. *Let (f, g, h) be three composable T-compactifiable morphisms, satisfying $(*)_k$, $(*)_\ell$, and $(*)_m$.*

$$X \xrightarrow{h} Y \xrightarrow{g} Z \xrightarrow{f} T.$$

If the pairs (f, g) and (g, h) satisfy (Var 3)*, then (f, gh) satisfies* (Var 3) *if and only if (fg, h) satisfies* (Var 3).

The condition in the conclusion also says that Tr_{fgh} is the "composite" of Tr_f, Tr_g, and Tr_h.

e) Let there be a commutative diagram:



The trace morphism relative to fg is the "composite" of the trace morphisms relative to ab and $u'v$; by construction for ab, and by XVII 6.2.3 for $u'v$, this is also the "composite" of the trace morphisms relative to a, b, u', and v. On the other hand, the "composite" of the trace morphisms relative to f and g is the "composite" of those relative to a, u, c, and v. We must therefore verify that, for every Cartesian diagram 558

(2.9.2)
$$\begin{array}{ccc} \mathbf{E}_S^e & \xrightarrow{u'} & \mathbf{E}_T^e \\ \downarrow{\scriptstyle b'} & & \downarrow{\scriptstyle b} \\ S & \xrightarrow{u} & T \end{array}$$

with u quasi-finite and flat of finite presentation, the diagram of trace morphisms (2.8.1) and XVII 6.2.)

(2.9.3)
$$\begin{array}{ccccc} R^{2e}b_!u'_!u'^*b^*F(e) & \xrightarrow{R^{2e}b_!(\mathrm{Tr}_{u'})} & R^{2e}b_!b^*F(e) & \xrightarrow{\mathrm{Tr}_b} & F \\ \| & & & & \| \\ u_!R^{2e}b'_!b'^*u^*F(e) & \xrightarrow{u_!(\mathrm{Tr}_{b'})} & u_!u^*F & \xrightarrow{\mathrm{Tr}_u} & F \end{array}$$

is commutative.

By base change, it suffices to verify this when T is the spectrum of an algebraically closed field k. If S is a sum of schemes S_i, we are reduced to verifying the commutativity of (2.9.3) only for the diagrams (2.9.2) with base $u_i : S_i \to T$. This makes it possible to suppose that S is the spectrum of a local Artinian k-algebra A, of finite degree $n = [A : k]$. If we then identify sheaves on T (resp. on $\mathbf{E}_T^e$) with sheaves on S (resp. on $\mathbf{E}_S^e$) by the functor u^* (resp. u'^*) (VIII 1.1), the morphism Tr_b identifies with the morphism $\mathrm{Tr}_{b'}$ (compatibility of (2.8.1) with base change), while Tr_u and $\mathrm{Tr}_{u'}$ identify with multiplication by n, whence the assertion[117]. This completes the proof of 2.9.

We have seen in the course of the proof that: 559

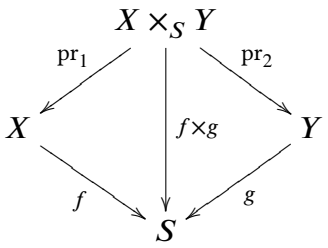
Proposition 2.10. *The trace morphisms* XVII 6.2.3, XVIII 1.1.6, *and (2.8.1) are special cases of the trace morphism 2.9.*

Remark 2.10.1. It can be shown without difficulty that the trace morphism of 2.9 is an isomorphism if and only if the geometric fibers of f have exactly one irreducible

[117] N.D.E.: It was also possible to reduce this compatibility to the case $e = 1$, which is obtained by combining 1.1.7 and 1.1.8. The arguments given here are moreover similar to those that were passed over in silence in 1.1.8.

component of dimension d, and its "multiplicity" is prime to n. We shall not give here the direct proof of this fact, which will follow immediately from the "global duality theorem" of § 3.

2.11. Let $f : X \to S$ and $g : Y \to S$ be two S-compactifiable morphisms satisfying $(*)_d$ and $(*)_e$, respectively.



The Künneth isomorphism (XVII 5.4.3)

$$Rf_!(Z/n)\overset{\mathbf{L}}{\otimes}_{\mathbf{Z}/n}Rg_!(Z/n) \xrightarrow{\sim} R(f\times g)_!(\mathbf{Z}/n)$$

induces an isomorphism

$$R^{2d}f_!\mathbf{Z}/n \otimes R^{2e}g_!\mathbf{Z}/n \xrightarrow{\sim} R^{2(d+e)}(f\times g)_!(\mathbf{Z}/n) \tag{2.11.1}$$

It is clear that $f \times g$ satisfies $(*)_{d+e}$; we have the compatibility:

PROPOSITION 2.12. *The following diagram*

$$\begin{array}{ccc} R^{2d}f_!\mathbf{Z}/n(d) \otimes R^{2e}g_!\mathbf{Z}/n(e) & \xrightarrow{\sim} & R^{2(d+e)}(f\times g)_!\mathbf{Z}/n(d+e) \\ \downarrow{\scriptstyle \mathrm{Tr}_f \otimes \mathrm{Tr}_g} & & \downarrow{\scriptstyle \mathrm{Tr}_{f\times g}} \\ \mathbf{Z}/n & = & \mathbf{Z}/n, \end{array} \tag{2.12.1}$$

560 *in which the first horizontal arrow is a twisted form of (2.11.1), is commutative.*

To verify this, it suffices to use (Var 3) and the asymmetric interpretation of the Künneth arrow used in the proof of XVII 5.4.3 (XVII 5.4.3.5).

2.13. Let $f : X \to S$ be an S-compactifiable morphism satisfying $(*)_d$, let $\mathscr{A}$ be a sheaf of rings on S, and let n be an integer invertible on S such that $n\mathscr{A} = 0$.

For every $K \in D(S, \mathscr{A})$, we have (XVII 5.2.6)

$$K\overset{\mathbf{L}}{\otimes}_{\mathbf{Z}/n}Rf_!\mathbf{Z}/n(d) \xrightarrow{\sim} Rf_!(f^*K(d)). \tag{2.13.1}$$

The trace morphism

$$\mathrm{Tr}_f : R^{2d}f_!\mathbf{Z}/n(d) \longrightarrow \mathbf{Z}/n$$

may also be interpreted as a morphism

$$\mathrm{Tr}_f : Rf_!\mathbf{Z}/n(d)[2d] \longrightarrow \mathbf{Z}/n.$$

Tensoring this morphism with K defines, via (2.13.1), a trace morphism

$$\mathrm{Tr}_f : Rf_!(f^*K(d)[2d]) \longrightarrow K. \tag{2.13.2}$$

As we shall see in 3.2.5, the following theorem is essentially equivalent to the global duality theorem in étale cohomology.

THEOREM 2.14. *Let $f : X \to S$ be a smooth compactifiable morphism purely of relative dimension d, and let n be an integer ≥ 1 invertible on S. For every geometric point x of X with image s in S, there exist an étale neighborhood V of s in S and an étale neighborhood U of x in $f^{-1}(V)$*

(2.14.1)
$$\begin{array}{ccccc} U & \xrightarrow{j} & f^{-1}(V) & \longrightarrow & X \\ & \searrow^{f'_V} & \downarrow{\scriptstyle f_V} & & \downarrow{\scriptstyle f} \\ & & V & \longrightarrow & S \end{array}$$

such that the morphism XVII 6.2.7.4 **561**
$$R^i f'_{V!}\mathbf{Z}/n \longrightarrow R^i f_{V!}\mathbf{Z}/n$$
is zero for $i < 2d$, and the morphism
$$\mathrm{Tr}_{f'_V} : R^{2d} f'_{V!}\mathbf{Z}/n(d) \longrightarrow \mathbf{Z}/n$$
is an isomorphism.

We first prove the following.

LEMMA 2.14.2. *Let $\mathscr{A}$ be an abelian category, let k be an integer ≥ 0, let $(K_i)_{0\leq i\leq 2k}$ be objects of $D^b(\mathscr{A})$ such that $H^p(K_i) = 0$ for $p \notin [0,k]$, let $f_i : K_i \to K_{i+i}$ $(0 \leq i \leq 2k-1)$ be morphisms, and let f be their composite. If $H^p(f_i) = 0$ for $p < k$, then there exists in $D^b(\mathscr{A})$ a morphism φ from the complex $H^k(K_0)[-k]$, consisting of $H^k(K_0)$ placed in degree k, to K_{2k}, making the diagram*
$$\begin{array}{ccc} K_0 & \xrightarrow{f} & K_{2k} \\ \downarrow & & \uparrow{\scriptstyle \varphi} \\ \sigma_{\geq k}(K_0) & = & H^k(K_0)[-k]. \end{array}$$
commute.

The lemma is trivial for $k = 0$. We prove it by induction on k. Applying the induction hypothesis to the complexes $\sigma_{\geq 1}(K_i)[1]$ gives a morphism $\varphi' : H^k(K_0)[-k] \to \sigma_{\geq 1}K_{2k-2}$ making the following diagram commute, where $f' = f_{2k-3} \ldots f_0$:
$$\begin{array}{ccc} K_0 & \xrightarrow{f'} & \sigma_{\geq 1}K_{2k-2} \\ \downarrow & & \uparrow{\scriptstyle \varphi'} \\ \sigma_{\geq k}(K_0) & = & H^k(K_0)[-k]. \end{array}$$

For every complex L, the distinguished triangle
$$\begin{array}{ccccc} & & \sigma_{\geq 1}(L) & & \\ & \swarrow^{+1} & & \nwarrow & \\ \sigma_{\leq 0}(L) & & \longrightarrow & & L \end{array}$$
gives, for every complex M, an exact sequence **562**
$$\cdots \to \mathrm{Hom}(M, \sigma_{\leq 0}(L)) \to \mathrm{Hom}(M, L) \to \mathrm{Hom}(M, \sigma_{\geq 1}L) \xrightarrow{j} \mathrm{Ext}^1(M, \sigma_{\leq 0}L).$$

In particular, we have a diagram

(2.14.3)
$$\begin{array}{ccccccc}
\mathrm{Hom}(\sigma_{\geq k}K_0,\sigma_{\leq 0}K_{2k-2}) & \to & \overset{⑤}{\mathrm{Hom}}(\sigma_{\geq k}K_0,K_{2k-2}) & \to & \overset{③}{\mathrm{Hom}}(\sigma_{\geq k}K_0,\sigma_{\geq 1}K_{2k-2}) & \to & \overset{④}{\mathrm{Ext}^1}(\sigma_{\geq k}K_0,\sigma_{\leq 0}K_{2k-2}) \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
\overset{⑦}{\mathrm{Hom}}(K_0,\sigma_{\leq 0}K_{2k-2}) & \to & \overset{①}{\mathrm{Hom}}(K_0,K_{2k-2}) & \to & \overset{②⑥}{\mathrm{Hom}}(K_0,\sigma_{\geq 1}K_{2k-2}) & \to & \mathrm{Ext}^1(K_0,\sigma_{\leq 0}K_{2k-2})
\end{array}$$

which maps to the analogous diagrams for K_{2k-1} and K_{2k}. By hypothesis, the morphism f' at ① has an image at ② that lifts to ③. The obstruction ④ to the element ③ lifting to ⑤ has zero image in diagram (2.14.3) for K_{2k-1}. In this new diagram, ③ therefore lifts to ⑤, and the difference between ① and the image of ⑤ has zero image at ⑥, and hence is the image of an element ⑦. This element ⑦ has zero image in diagram 2.12.3 for K_{2k}; in this new diagram, f is therefore the image of ⑤, which solves the problem.

We prove that, for a given value of d, assertion 2.14 implies the following.

Corollary 2.14.4. *Under the hypotheses of 2.14, there exists a diagram (2.14.1) such that the morphism* XVII 6.2.4, *in* $D^b(V,\mathbf{Z}/n)$, *admits a factorization*

$$\begin{array}{ccc}
Rf'_{V!}\mathbf{Z}/n(d) & \longrightarrow & Rf_{V!}\mathbf{Z}/n(d) \\
& {\scriptstyle t}\searrow \quad \nearrow{\scriptstyle \varphi} & \\
& \mathbf{Z}/n[-2d], &
\end{array}$$

where t is the arrow

$$t : Rf'_{V!}\mathbf{Z}/n(d) \to \sigma_{\geq 2d}Rf'_{V}\mathbf{Z}/n(d) = H^{2d}(Rf'_{V}\mathbf{Z}/n(d))[-2d] \xrightarrow{\mathrm{Tr}} \mathbf{Z}/n[-2d].$$

563 Set $f'_0 = f$ and construct, inductively, $4d$ diagrams (2.14.1)

$$\begin{array}{ccccc}
U'_{i+1} & \longrightarrow & f_i'^{-1}(V'_{i+1}) & \longrightarrow & U'_i \\
& {\scriptstyle f'_{i+1}}\searrow & \downarrow & & \downarrow{\scriptstyle f'_i} \\
& & V'_{i+1} & \longrightarrow & V'_i
\end{array}$$

satisfying 2.14. Set $V = V'_{4d}$ and $U_i = U'_i \times_{V'_i} V$. We obtain a diagram

$$\begin{array}{ccccccccc}
U_{4d} & \longrightarrow & \cdots & \longrightarrow & U_1 & \longrightarrow & U_0 & \longrightarrow & X \\
& {\scriptstyle f_{4d}}\searrow & & & {\scriptstyle f_1}\searrow & & \downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f} \\
& & & & & & V & \longrightarrow & S,
\end{array}$$

with $U_0 = f^{-1}(V)$, such that the arrows

$$R^i f_{k+1!}\mathbf{Z}/n(d) \longrightarrow R^i f_{k!}\mathbf{Z}/n(d)$$

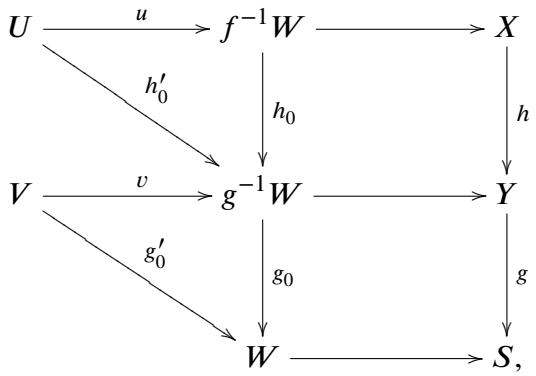
are zero (for $i < 2d$) and the arrow

$$\mathrm{Tr} : R^{2d} f_{4d!}\mathbf{Z}/n(d) \longrightarrow \mathbf{Z}/n$$

is an isomorphism. By 2.14.2, the corollary holds for $U = U_{4d}$.

We are now ready to prove 2.14 by induction on d. The case $d = 0$ is trivial, and the case $d = 1$ is precisely 1.6.9. We may therefore suppose that $d \geq 2$.

Since the problem is local on X and S, we may suppose that f admits a factorization $f = gh$, with g and h smooth and compactifiable, purely of relative dimensions d' and d'', and that $d', d'' < d = d' + d''$. Apply 2.14.4 to g and to the morphisms deduced from h by base change, so as to obtain a diagram



such that the morphisms (XVII 6.2.7.3) admit factorizations 564

$$Rh'_{0!}\mathbf{Z}/n(d'') \xrightarrow{\mathrm{Tr}} \mathbf{Z}/n[-2d''] \xrightarrow{\varphi} Rh_{0!}\mathbf{Z}/n(d'')$$
$$Rg'_{0!}\mathbf{Z}/n(d') \xrightarrow{\mathrm{Tr}} \mathbf{Z}/n[-2d'] \xrightarrow{\Psi} Rg_{0!}\mathbf{Z}/n(d')$$

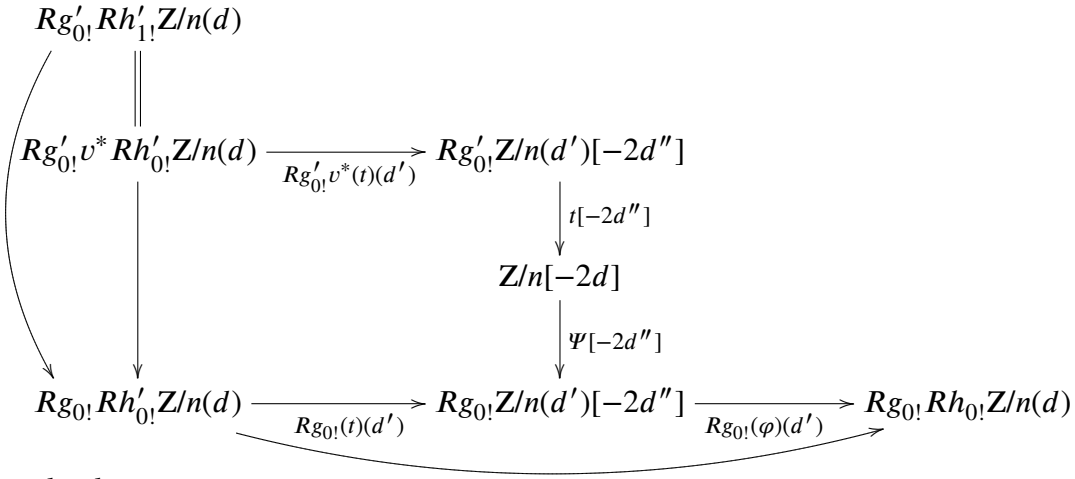
and such that $\mathrm{Tr}_{h'_0}$ and $\mathrm{Tr}_{g'_0}$ are isomorphisms.

Let h_1 and h'_1 be the morphisms deduced from h_0 and h'_0 by base change to V, and let $f' = g'_0 h'_1$, $f_W = g_0 h_0$.

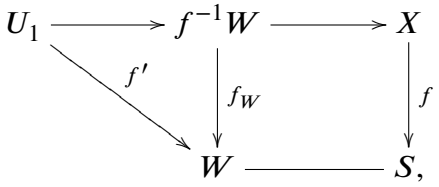
The following diagram is commutative:

$$\begin{array}{ccccc} Rf'\mathbf{Z}/n(d) & & \longrightarrow & & Rf_{W!}\mathbf{Z}/n(d) \\ \| & & & & \| \\ Rg'_{0!}Rh'_{1}\mathbf{Z}/n(d) & \longrightarrow & Rg_{0!}Rh'_{0!}\mathbf{Z}/n(d) & \longrightarrow & Rg_{0!}Rh_{0!}\mathbf{Z}/n(d), \end{array}$$

as is the diagram



For the diagram 565



the canonical arrow from $Rf'_!\mathbf{Z}/n(d)$ to $Rf_W\mathbf{Z}/n(d)$ therefore factors through $\mathbf{Z}/n[-2d]$; hence the arrows from $R^if'_!\mathbf{Z}/n(d)$ to $R^if_W\mathbf{Z}/n(d)$ are zero for $i < 2d$. Finally, $\mathrm{Tr}_{f'}$ is an isomorphism, being the composite of the isomorphisms $\mathrm{Tr}_{g'_0}$ and $R^{2d'}g'_{0!}\,\mathrm{Tr}_{h'_1}$.

This completes the proof of 2.14.

3. The global duality theorem

566

3.1. The functor $Rf^!$. We shall need a variant of the canonical flasque resolutions.

LEMMA 3.1.1. *Let X be a coherent scheme (i.e. quasi-compact and quasi-separated). The functor $\varinjlim$ induces an equivalence between the category of Ind-objects of the category of constructible sheaves of sets (resp. of groups, resp. of abelian groups) and the category of sheaves of sets (resp. of Ind-finite groups, resp. of torsion abelian groups).*

This follows from IX 2.7.2 and IX 2.7.3.

3.1.2. Let $\mathscr{F}$ be a torsion abelian sheaf on a scheme X, the filtered direct limit of constructible sheaves $(\mathscr{F}_i)_{i\in I}$, and let X_0 be a conservative set of points of X. The *modified canonical flasque resolution* of $\mathscr{F}$ is defined by the formula

$$C^*_\ell(\mathscr{F}) = \varinjlim_i C^*(\mathscr{F}_i),$$

where the $C^*(\mathscr{F}_i)$ are the canonical flasque resolutions of XVII 4.2.2. By 3.1.1, for X coherent, this definition is independent of the choice of the $\mathscr{F}_i$. It also commutes with localization; this fact allows us to define $C^*_\ell(\mathscr{F})$ by globalization when $\mathscr{F}$ is a torsion sheaf on an arbitrary scheme X, not necessarily coherent.

The complex $C^*_\ell(\mathscr{F})$ is a resolution of $\mathscr{F}$, functorial in $\mathscr{F}$, whose formation commutes with localization and filtered direct limits; the functors C^n_ℓ are exact.

It follows that, for a sheaf of rings $\mathscr{A}$ on X, the functors C^n_ℓ carry $\mathscr{A}$-Modules to $\mathscr{A}$-Modules; they likewise carry torsion sheaves to torsion sheaves. It is well known that

567 LEMMA 3.1.3. *Let $\mathscr{A}$ be an abelian $\mathscr{U}$-category having a small generating set and such that direct limits indexed by a small filtered ordered set are representable in $\mathscr{A}$, and are exact. A functor F from $\mathscr{A}^\circ$ to* (Ens) *is representable if and only if it carries small direct limits (arbitrary ones) to inverse limits.*

Cf. for example SGA 4 I 8.12.5, where $\mathscr{A}$ is not assumed abelian, nor are filtered $\varinjlim$ assumed exact.

We shall apply this lemma to the category $\mathrm{Mod}(X,\mathscr{A})$ of sheaves of modules on a ringed site $(X,\mathscr{A})$. In this special case, if the functor F is represented by the sheaf $\mathscr{F}$, then

$$\mathscr{F}(U) = F(\mathscr{A}_U). \tag{3.1.3.1}$$

This description of $\mathscr{F}$ gives a direct verification of 3.1.3 in the case at hand.

Recall that if $f : X \to S$ is a compactifiable morphism (XVII 3.2.), then S is quasi-compact and quasi-separated (sic), and the dimensions of the fibers of f are bounded.

THEOREM 3.1.4. *Let $f : X \to S$ be a compactifiable morphism and let $\mathscr{A}$ be a sheaf of torsion rings on S. Then the functor*

$$Rf_! : D(X, f^*\mathscr{A}) \longrightarrow D(S, \mathscr{A}) \tag{3.1.4.1}$$

admits a partial right adjoint

$$Rf^! : D^+(S, \mathscr{A}) \longrightarrow D^+(X, f^*\mathscr{A}) : \tag{3.1.4.2}$$

for $K \in \mathrm{Ob}\, D(X, f^*\mathscr{A})$ and $L \in \mathrm{Ob}\, D^+(S, \mathscr{A})$, there is a functorial isomorphism

$$\mathrm{Hom}(Rf_!K, L) \simeq \mathrm{Hom}(K, Rf^!L). \tag{3.1.4.3}$$

Endowed with the translation morphism making the diagram

$$\begin{array}{ccccc}
\mathrm{Hom}(Rf_!(K[1]), L) & \xleftarrow{\sim} & \mathrm{Hom}(K[1], Rf^!L) & \xrightarrow{\sim} & \mathrm{Hom}(K, (Rf^!L)[-1]) \\
\wr\uparrow & & & & \uparrow\wr \\
\mathrm{Hom}((Rf_!K)[1], L) & \xrightarrow{\sim} & \mathrm{Hom}(Rf_!K, L[-1]) & \xleftarrow[\text{adj.}]{\sim} & \mathrm{Hom}(K, Rf^!(L[-1])),
\end{array} \tag{3.1.4.4}$$

commute, this functor $Rf^!$ is a triangulated functor. 568

N.B. The notation $Rf^!$ is abusive in that $Rf^!$ is not in general the derived functor of a functor $f^!$.

Proof Let d be an integer such that every fiber of f has dimension $< d$, and choose a compactification

$$\begin{array}{ccc}
X & \xrightarrow{\ j\ } & \overline{X} \\
{\scriptstyle f}\downarrow & \swarrow{\scriptstyle \overline{f}} & \\
S. & &
\end{array}$$

If K is a complex of $f^*\mathscr{A}$-Modules on X, then

$$Rf_!K \simeq R\overline{f}_*(j_!K).$$

For every sheaf F on X, the components F^i of the complex $\tau_{\leq 2d}C^*_\ell j_!F$ satisfy

$$R^k\overline{f}_*F^i = 0 \quad \text{for} \quad k > 0. \tag{3.1.4.5}$$

Indeed, for $i \neq 2d$, F^i is a direct limit of flasque sheaves, and for $i = 2d$ and $k > 0$, we have (XVII 5.2.8.1)

$$R^k\overline{f}_*F^{2d} = R^{k+2d}f_!F = 0.$$

The simple complex associated with the truncated modified canonical flasque resolution double complex of K is a resolution of K. By (3.1.4.5), we therefore have (notation of XVII 1.1.15)

$$R\overline{f}_*(j_!K) \xleftarrow{\sim} \overline{f}_*\tau''_{\leq 2d}C^*_\ell j_!K. \tag{3.1.4.6}$$

Denote by $f^\cdot_!$ the functor from sheaves of modules on X to complexes of sheaves on S defined by 569

$$f^\cdot_!(F) = \overline{f}_*\tau_{\leq 2d}C^*_\ell j_!F. \tag{3.1.4.7}$$

LEMMA 3.1.4.8. (i) *The functors $f_!^i$ are exact and commute with filtered direct limits; we have $f_!^i = 0$ for $i \notin [0, 2d]$.*

(ii) *Since the functor $f_!^\cdot$ is bounded with exact components, its derived functor (XVII 1.2.10) is obtained trivially and is denoted $Rf_!^\cdot$, or simply $f_!^\cdot : D(X, f^*\mathscr{A}) \to D(S, \mathscr{A})$. The arrow (3.1.4.6) is an isomorphism of functors*

$$Rf_!^\cdot \xrightarrow{\sim} Rf_!.$$

The functors C_ℓ^i are exact and commute with filtered direct limits (3.1.2). They send every sheaf to an acyclic sheaf. The functors $\overline{f}_*$ and $j_!$ commute with filtered direct limits. It follows that

a) the functors $\overline{f}_* C_\ell^i j_!$, and hence also the functors $f_!^i$, commute with filtered direct limits;
b) for $i < 2d$, the functors $f_!^i = \overline{f}_* C_\ell^i j_!$ are exact. For $i = 2d$, one verifies that $f_!^i$ is exact by (3.1.4.5).

The remaining assertions of 3.1.4.8 are trivial.

REMARK 3.1.4.9. The definition of $f_!^\cdot$ is independent of the sheaf of rings $\mathscr{A}$, and still makes sense for any abelian sheaf. Assertion 3.1.4.8 (i) remains valid in the categories of torsion abelian sheaves on X and S.

By 3.1.4.8 and 3.1.3, the functor

$$f_!^i : \mathrm{Mod}(X, f^*\mathscr{A}) \longrightarrow \mathrm{Mod}(S, \mathscr{A})$$

has a right adjoint $f_i^!$; since $f_!^i$ is exact, $f_i^!$ sends injectives to injectives.

With the sign convention of (XVII 1.1.12), the functors $f_i^!$ form a complex of left exact functors, zero in cohomological degree $n \notin [-2d, 0]$:

$$f_\cdot^! : \mathrm{Mod}(S, \mathscr{A}) \longrightarrow C^b(X, f^*\mathscr{A}). \tag{3.1.4.10}$$

570 The functor $f_\cdot^!$ derives (XVII 1.2.10) to a triangulated functor

$$Rf^! : D^+(S, \mathscr{A}) \longrightarrow D^+(X, f^*\mathscr{A}), \tag{3.1.4.11}$$

right adjoint to the functor $Rf_! \ (= Rf_!^\cdot = Lf_!^\cdot)$ ("trivial duality formula"). More precisely, if $I \in \mathrm{Ob}\, C^+(S, \mathscr{A})$ is a complex of sheaves of injective $\mathscr{A}$-modules, the $\mathscr{A}$-modules $f_i^! I^n$ are injective, and there is an isomorphism of triple complexes (XVII 1.1.12)

$$\mathrm{Hom}^\cdot(f_!^\cdot K, I) \xrightarrow{\sim} \mathrm{Hom}^\cdot(K, f_\cdot^! I). \tag{3.1.4.12}$$

Passing to H^0 of the associated simple complex (computed using products) gives the adjunction statement. We leave it to the reader to verify the compatibility (3.1.4.4).

REMARK 3.1.5. Suppose that the functor $Rf^!$ has finite cohomological amplitude. By XVII 1.2.10, the functor $f_\cdot^!$ then admits a derived functor

$$Rf^! : D(S, \mathscr{A}) \longrightarrow D(X, \mathscr{A}). \tag{3.1.5.1}$$

This derived functor is again right adjoint to the functor $Rf_!$ (3.1.4.1). Indeed, we have

$$\mathrm{Hom}_{D(S)}(Rf_!K, L) \simeq \varinjlim_{L \xrightarrow{\sim} L'} \mathrm{Hom}_{K(S)}(f_!^{\cdot}K, L') \simeq \varinjlim_{\substack{K' \xrightarrow{\sim} K \\ L \xrightarrow{\sim} L'}} \mathrm{Hom}_{K(S)}(f_!^{\cdot}K', L')$$

$$\mathrm{Hom}_{D(X)}(K, Rf^!L) \simeq \varinjlim_{K' \xrightarrow{\sim} K} \mathrm{Hom}_{K(X)}(K', Rf^!L) \simeq \varinjlim_{\substack{K' \xrightarrow{\sim} K \\ L \xrightarrow{\sim} L'}} \mathrm{Hom}_{K(X)}(K', f_!^{\cdot}L')$$

and, by adjunction,

$$\mathrm{Hom}_{K(S)}(f_!^{\cdot}K', L') = H^0 \mathrm{Hom}(f_!^{\cdot}K', L') \simeq H^0 \mathrm{Hom}(K', f_!^{\cdot}L') = \mathrm{Hom}_{K(X)}(K', L').$$

Definition 3.1.6. *The triangulated functor*

$$Rf^! : D^+(S, \mathscr{A}) \longrightarrow D^+(X, f^*\mathscr{A})$$

(or, under the hypotheses of 3.1.5, 571

$$Rf^! : D(S, \mathscr{A}) \longrightarrow D(X, f^*\mathscr{A})),$$

right adjoint to the functor $Rf_!$ (3.1.4.1), is called the exceptional inverse-image functor.

Proposition 3.1.7. *Suppose that the compactifiable morphism $f : X \to S$ has relative dimension $\leq d$.*

(i) *If $L \in \mathrm{Ob}\, D^+(S, \mathscr{A})$ satisfies $H^i(L) = 0$ for $i \leq k$, then $H^i Rf^!L = 0$ for $i \leq k - 2d$.*
(ii) *For $L \in \mathrm{Ob}\, D^b(S, \mathscr{A})$, we have*

$$\dim \mathrm{inj}(Rf^!L) \leq \dim \mathrm{inj}(L).$$

The functor $Rf^!$ is the right derived functor of $f_!^{\cdot} : \mathrm{Mod}(S, \mathscr{A}) \to C^b \mathrm{Mod}(X, f^*\mathscr{A})$. Assertion (i) follows from the fact that $(f_!^{\cdot})^i = 0$ for $i < -2d$, and assertion (ii) follows from the fact that $(f_!^{\cdot})^i = 0$ for $i > 0$, and that the functors $f_i^!$ send injectives to injectives.

Proposition 3.1.8. *Let $f : X \to S$ be a quasi-finite compactifiable morphism.*

(i) *The functor $f_! : \mathrm{Mod}(X, f^*\mathscr{A}) \to \mathrm{Mod}(S, \mathscr{A})$ admits a right adjoint $f^! : \mathrm{Mod}(S, \mathscr{A}) \to \mathrm{Mod}(X, f^*\mathscr{A})$, and $Rf^!$ is the derived functor of $f^!$.*
(ii) *If f is a closed immersion, then $f^!$ is the functor of "sections with support in X."*
(iii) *If f is étale, then $f^!$ identifies with the functor f^*, with*

$$\mathrm{Tr}_f : f_!f^*F \longrightarrow F$$

as adjunction morphism.

Assertion (i) follows immediately from the special case $d = 0$ of the proof of 3.1.4: we have $f_!^{\cdot} = f_!$ and $f_!^{\cdot} = f^!$. Equivalently, the existence of an adjoint functor $f^!$ follows from 3.1.3 and from the fact that $f_!$ is exact and commutes with filtered direct limits; 572
the adjunction $(f_!, Rf^!)$ is a "trivial duality formula."

Assertion (ii) is clear, and (iii) is XVII 6.2.11.

3.1.9. Let $f : X \to S$ be a compactifiable morphism and $\mathscr{A}$ a sheaf of torsion rings on S, and equip X with the inverse-image sheaf of rings of $\mathscr{A}$. Let $K \in \mathrm{Ob}\, D^-(X, f^*\mathscr{A})$, $L \in \mathrm{Ob}\, D^+(X, f^*\mathscr{A})$, and represent L by a bounded-below complex of injective sheaves, so that

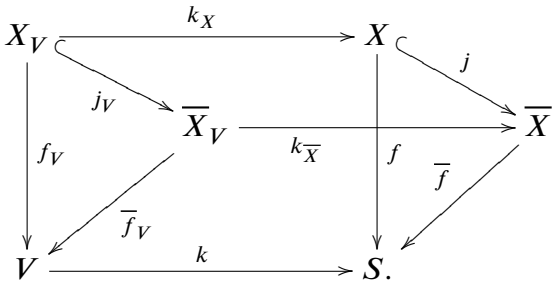
$$R\,\mathbf{Hom}(K, L) \simeq \mathbf{Hom}(K, L).$$

The complex $\mathbf{Hom}(K, L)$ has flasque components, and therefore

$$Rf_*R\,\mathbf{Hom}(K, L) \simeq f_*\,\mathbf{Hom}(K, L). \tag{3.1.9.1}$$

Choose a compactification of f and an integer d as in the proof of 3.1.4, giving a functor $f_!^{\cdot}$ (3.1.4.7).

(3.1.9.2)



For every $V \in \mathrm{Ob}\, S_{\mathrm{et}}$, there is an isomorphism[118]

$$k^* f_!^{\cdot} \xrightarrow{\sim} f_{V!}^{\cdot} k_X^*, \tag{3.1.9.3}$$

so that

$$\mathbf{Hom}^{\cdot}(f_!^{\cdot}K, f_!^{\cdot}L)(V) \simeq \mathbf{Hom}^{\cdot}(f_{V!}^{\cdot}(k_X^*K), f_{V!}^{\cdot}(k_X^*L)).$$

The functoriality of $f_{V!}^{\cdot}$ gives us

$$\mathbf{Hom}^{\cdot}(K, L)(X_V) = \mathrm{Hom}^{\cdot}(k_X^*K, k_X^*L) \longrightarrow \mathrm{Hom}^{\cdot}(f_{V!}^{\cdot}(k_X^*K), f_{V!}^{\cdot}(k_X^*L)),$$

hence an arrow, at the level of complexes,

$$f_*\,\mathbf{Hom}^{\cdot}(K, L) \longrightarrow \mathbf{Hom}^{\cdot}(f_!^{\cdot}K, f_!^{\cdot}L). \tag{3.1.9.4}$$

573 By (3.1.9.1), this arrow derives to

$$Rf_*R\,\mathbf{Hom}(K, L) \longrightarrow R\,\mathbf{Hom}(Rf_!K, Rf_!L). \tag{3.1.9.5}$$

Now let $K \in \mathrm{Ob}\, D^-(X, \mathscr{A})$ and $L \in \mathrm{Ob}\, D^+(S, \mathscr{A})$. The adjunction arrow from $Rf_!Rf^!L$ to L, together with (3.1.9.4), gives a composite arrow

$$Rf_*R\,\mathbf{Hom}(K, Rf^!L) \longrightarrow R\,\mathbf{Hom}(Rf_!K, Rf_!Rf^!L) \longrightarrow R\,\mathbf{Hom}(Rf_!K, L),$$

the local form on S of the adjunction isomorphism:

$$Rf_*R\,\mathbf{Hom}(K, Rf^!L) \longrightarrow R\,\mathbf{Hom}(Rf_!K, L). \tag{3.1.9.6}$$

Proposition 3.1.10. *The arrow 3.1.9.6 is an isomorphism.*

[118] N.D.E.: This morphism is given by the construction of XVII 4.2.6, but there is no reason for it to be an isomorphism unless coherent choices of points have first been made on the topoi under consideration. For this to be the case, the square occurring in XVII 4.2.6 must in some sense be Cartesian. More precisely, one should fix here a family of points $(p)_{p\in P}$ of the topos S_{et}, and for every $V \in \mathrm{Ob}\, S_{\mathrm{et}}$ one considers the modified canonical flasque resolution using the family of points associated by the correspondence of IV 6.7 with the pairs (p, ξ) where $p \in P$ and $\xi \in V_p$.

Let $V \in \operatorname{Ob} S_{\text{et}}$ and retain the notation of (3.1.9.2). The ("transitivity") isomorphism

$$k_! R f_{V!} \simeq R f_! k_{X!},$$

whose existence is the basis of the theory of the functor $Rf_!$, transposes to a *localization isomorphism* (3.1.8. (iii))

$$k_X^* R f^! \simeq R f_V^! k^*, \tag{3.1.10.1}$$

represented at the level of complexes by a morphism

$$k_X^* f^! \longrightarrow f_V^! . k^*.$$

To verify that (3.1.9.6) is an isomorphism, it suffices to verify that for every $V \in \operatorname{Ob} S_{\text{et}}$, (3.1.9.6) induces an isomorphism

$$\begin{aligned} \operatorname{Hom}(k_X^* K, k_X^* R f^! L) &\simeq H^0(V, R f_* R\,\mathbf{Hom}(K, R f^! L)) \\ &\longrightarrow H^0(V, R\operatorname{Hom}(R f_! K, L)) \simeq \mathbf{Hom}(k^* R f_! K, k^* L) \end{aligned} \tag{3.1.10.2}$$

For $V = X$, this is precisely (3.1.4.3). The general case follows from the following compatibility, *which the editor has not verified*:

Lemma 3.1.10.3. *The diagram* 574

$$\begin{array}{ccc} \operatorname{Hom}(k_X^* K, k_X^* R f^! L) & \xrightarrow{(3.1.10.2)} & \operatorname{Hom}(k^* R f_! K, k^* L) \\ \wr \downarrow (3.1.10.1) & & \wr \downarrow (\text{localization}) \\ \operatorname{Hom}(k_X^*, K, R f_V^! k_X^* L) & \xleftarrow[\sim]{(3.1.4.3 \text{ on } V)} & \operatorname{Hom}(R f_! k^* K, k^* L) \end{array}$$

is commutative.

3.1.11. Consider a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S, \end{array} \tag{3.1.11.1}$$

with f compactifiable, $\mathscr{A}$ a sheaf of torsion rings on S, $\mathscr{B}$ a sheaf of torsion rings on S', and M a bounded-above complex of $(f^*\mathscr{A}, \mathscr{B})$-bimodules. At the level of complexes, there is then a morphism, functorial in K ($K \in \operatorname{Ob} C(X, f^*\mathscr{A})$)

$$M \otimes_{\mathscr{A}} g^* f_!^{\cdot} K \longrightarrow f_!'^{\cdot}(f'^* M \otimes_{\mathscr{A}} g'^* K), \tag{3.1.11.2}$$

which transposes to a morphism, functorial in L ($L \in \operatorname{Ob} C(S', \mathscr{B})$)

$$g'_* \mathbf{Hom}_{\mathscr{B}}(f'^* M, f'^{!}_{\cdot} L) \longrightarrow f^{!}_{\cdot} g_* \mathbf{Hom}_{\mathscr{B}}(M, L). \tag{3.1.11.3}$$

Passing to derived functors gives a morphism, functorial in L, from $D^+(S', \mathscr{B})$ to $D^+(X, f^*\mathscr{A})$:

$$R g'_* R\,\mathbf{Hom}_{\mathscr{B}}(f'^* M, R f'^! L) \longrightarrow R f^! R g_* R\operatorname{Hom}_{\mathscr{B}}(M, L), \tag{3.1.11.4}$$

which makes the following diagram commutative, where $K \in \mathrm{Ob}\, D^-(X, f^*\mathscr{A})$ and $L \in \mathrm{Ob}\, D^+(S', \mathscr{B})$:

$$\begin{array}{ccccc}
\mathrm{Hom}(K, Rg'_*\, R\mathbf{Hom}_{\mathscr{B}}(f'^*M, Rf'^!L)) & \xrightarrow{①} & \mathrm{Hom}(K, Rf^! Rg_*\, R\mathbf{Hom}_{\mathscr{B}}(M,L)) & \underset{\text{adj}}{\longleftrightarrow} & \mathrm{Hom}(Rf_!K, Rg_*\, R\mathbf{Hom}_{\mathscr{B}}(M,L)) \\
\updownarrow & & & & \updownarrow \\
\mathrm{Hom}(g'^*K, R\mathbf{Hom}_{\mathscr{B}}(f'^*M, Rf'^!L)) & & & & \mathrm{Hom}(g^*Rf_!K, R\mathbf{Hom}_{\mathscr{B}}(M,L)) \\
\updownarrow & & & & \updownarrow \\
\mathrm{Hom}(f'^*M \otimes_{\mathscr{A}} g'^*K, Rf'^!L) & \underset{\text{adj}}{\longleftrightarrow} & \mathrm{Hom}(Rf'_!(f'^*M \otimes^L_{\mathscr{A}} g'^*K), L) & \xrightarrow{②} & \mathrm{Hom}_{\mathscr{B}}(M \otimes^{\mathbf{L}}_{\mathscr{A}} g^*Rf_!K, L).
\end{array}$$

575 In this diagram, ② (deduced from the arrow XVII 5.2.2.1) is an isomorphism (XVII 5.2.6). The arrow ① is therefore an isomorphism for every K, and this implies that (3.1.11.4) is an isomorphism.

The proper base change theorem thus transposes to the following

PROPOSITION 3.1.12. *The arrow (3.1.11.4) is an isomorphism.*

This isomorphism is called the *induction isomorphism.*

Here are three special cases.

COROLLARY 3.1.12.1. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ and $\mathscr{B}$ be two sheaves of torsion rings on S, and let $\varphi : \mathscr{A} \to \mathscr{B}$ be a homomorphism. Denote by ρ the restriction-of-scalars functor from $\mathscr{B}$ to $\mathscr{A}$. Then, for $K \in \mathrm{Ob}\, D^+(S, \mathscr{B})$, we have*

$$\rho Rf^! K \xrightarrow{\sim} Rf^!(\rho K).$$

It suffices in (3.1.11.3) to take $S = S'$, $X = X'$, $\mathscr{A} = \mathscr{A}$, $\mathscr{B} = \mathscr{B}$, $M = \mathscr{B}$.

Thus the sheaf of rings plays no essential role in the construction of $Rf^!$.

COROLLARY 3.1.12.2. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ be a sheaf of torsion rings on S, and let $K \in \mathrm{Ob}\, D^-(S, \mathscr{A})$, $L \in \mathrm{Ob}\, D^+(S, \mathscr{A})$. We have the* induction formula

$$R\mathbf{Hom}(f^*K, Rf^!L) \xrightarrow{\sim} Rf^! R\mathbf{Hom}(K, L)$$
$$(\text{take } S = S', X = X').$$

COROLLARY 3.1.12.3. *For every square 3.1.11 and every $L \in \mathrm{Ob}\, D^+(S', \mathscr{A})$, we have*

$$Rg'_* Rf'^! L \xrightarrow{\sim} Rf^! Rg_* L$$
$$(\text{take } \mathscr{B} = f^*\mathscr{A} = M).$$

576 3.1.13. For a composite gh of S-compactifiable morphisms

$$X \xrightarrow{h} Y \xrightarrow{g} Z$$

the composition isomorphism

$$c_{g,h} : Rg_! Rh_! \xrightarrow{\sim} R(gh)_!$$

transposes to a composition morphism[119]

(3.1.13.1) $$c^!_{g,h} : Rh^! Rg^! \xrightarrow{\sim} R(gh)^!$$

[119] N.D.E.: The transposed morphism actually goes in the other direction, but this causes no problem since it is an isomorphism.

satisfying the usual cocycle condition for a triple composite. In other words, the *categories* $D^+(X, f^*\mathscr{A})$ *form the fibers of a fibered and cofibered category over the category whose objects are the* S*-schemes* $f : X \to S$, and whose arrows are the S-compactifiable morphisms, *the direct-image* (resp. *inverse-image*) *functors being the functors* $Rf_!$ (resp. $Rf^!$). For every commutative square of compactifiable morphisms,

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow[g']{} & S, \end{array}$$

we therefore have a cobase-change morphism (XVII 2.1.3)

$$ch^! : Rf'_! Rg^! \longrightarrow Rg^! Rf_!. \tag{3.1.13.2}$$

3.1.14. Consider a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S, \end{array}$$

with f compactifiable, and let $\mathscr{A}$ be a sheaf of torsion rings on S. Endow S', X, and X' with the inverse images of $\mathscr{A}$ as their sheaves of rings, again denoted by $\mathscr{A}$ by abuse of notation. The base-change isomorphism 577

$$g^* Rf_! \xrightarrow{\sim} Rf'_! g'^* \tag{3.1.14.1}$$

allows us to regard the categories $D(S, \mathscr{A})$, $D(X, \mathscr{A})$, $D(S', \mathscr{A})$, and $D(X', \mathscr{A})$ as the fibers of a cofibered category over the category defined by the commutative diagram

$$\begin{array}{ccc} X & \longleftarrow & X \\ \downarrow & & \downarrow \\ X & \longleftarrow & X \end{array}$$

the direct-image functors being the functors $Rf_!$, $Rf'_!$, g^*, and g'^* (sic). When we restrict to the categories D^+, we even obtain a fibered and cofibered category over this diagram, the direct-image functors being the functors $Rf^!$, $Rf'^!$, Rg_*, and Rg'_* (sic). We deduce from this

a) a "transitivity isomorphism" (sic) for the "inverse-image" functors

$$Rg_* Rf^! \xrightarrow{\sim} Rf^! Rg_*$$

already obtained in 3.1.12.3 (with the same definition by transposition of (3.1.14.1)).

b) a "base-change morphism" (XVII 2.1.3)

$$g'^* Rf^! \longrightarrow Rf'^! g^* \tag{3.1.14.2}$$

which, by *loc. cit.*, can be described in the following two ways:

a) by adjunction, giving a morphism (3.1.14.2) amounts to giving a morphism

$$Rf'_!g'^*Rf^! \longrightarrow g^*.$$

The base-change theorem provides such a morphism as the composite

$$Rf'_!g'^*Rf^! \xleftarrow{\sim} g^*Rf_!Rf^! \xrightarrow{g^**\text{adj.}} g^*.$$

578 b) by adjunction, giving a morphism (3.1.14.2) amounts to giving a morphism

$$Rf^! \longrightarrow Rg'_*Rf'^!g^*.$$

The isomorphism (3.1.12.3) provides such a morphism as the composite

$$Rf^! \xrightarrow[Rf^!*\text{adj.}]{} Rf^!Rg_*g^* \xleftarrow{\sim} Rg'_*Rf'^!g^*.$$

From their interpretations in terms of a fibered and cofibered category, it is clear that *the morphisms* (3.1.12.3), (3.1.13.2), *and* (3.1.14.2) *each satisfy two compatibilities of the type* XII 4.4 *when* f *or* g *is a composite of two morphisms*[120].

The following results from general topology will be used in the following subsection.

3.1.15. Let $f : X \to S$ be a morphism of sites. Denote by $\Gamma(f)$ the following site:

a) The category $\Gamma(f)$ is the category of triples (U, V, φ) with

$$U \in \mathrm{Ob}(X), V \in \mathrm{Ob}\, S, \quad \varphi \in \mathrm{Hom}(U, f^*V).$$

b) A family of morphisms $(u_i, v_i) : (U_i, V_i, \varphi_i) \to (U, V, \varphi)$ is covering if, in X, the morphisms $u_i : U_i \to U$ cover U.

We then have:

. The functor from $\Gamma(f)$ to X given by

$$(U, V, \varphi) \longmapsto U$$

is an equivalence of sites $\gamma : X \to \Gamma(f)$.

. The functor f^*_Γ from S to $\Gamma(f)$

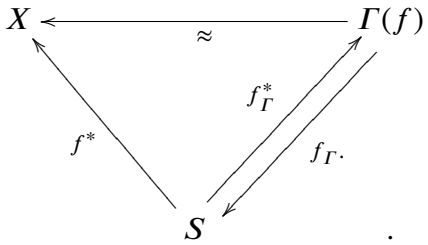
$$V \longmapsto (f^*V, V, \mathrm{Id}_{f^*V})$$

is a morphism of sites, denoted $f_\Gamma : \Gamma(f) \to S$, or simply f by abuse of notation. It satisfies $f_\Gamma\gamma \simeq f$. The functor f^*_Γ has a left adjoint $f_{\Gamma.}$:

$$f_{\Gamma.} : (U, V, \varphi) \longmapsto V.$$

579 The functor $f_{\Gamma.}$ is therefore a comorphism of sites from $\Gamma(f)$ to S[121], and defines the same morphism of topoi as the morphism of sites f_Γ.

(3.1.15.3)



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[120] N.D.E.: These constructions can also be interpreted in the context of exchange structures and cross functors, cf. [1].

[121] N.D.E.: Since this terminology is not familiar, it should be specified here that these are the notions appearing in III 2 and IV 4.7 (one should bear in mind that in the original edition, at the location of this second reference, misprints made it impossible to understand in precisely which direction the arrows went).

We deduce the following rule for computing the inverse image of a sheaf[122]
. Let $\mathscr{F}$ be a sheaf on S, let $f_\Gamma^\cdot\mathscr{F}$ be the presheaf on $\Gamma(f)$ given by

$$f_\Gamma^\cdot\mathscr{F}(U,V,\varphi)=\mathscr{F}(V),$$

and let $af_\Gamma^\cdot\mathscr{F}$ be the associated sheaf. Then

$$f^*\mathscr{F}(U)\simeq af_\Gamma^\cdot\mathscr{F}(U,V,\varphi).$$

3.1.16. Let $f:X\to S$ be a morphism of sites, $\mathscr{A}$ a sheaf of rings on S, $\mathscr{B}$ a sheaf of rings on X, and d an integer. Let K be an object of the following type:
. K associates to every object U of X a complex $K(U)$ of $(\mathscr{A}, f_{|U*}(\mathscr{B}|U))$-modules on S, zero in degrees $>d$, and $K(U)$ is a *covariant* functor of U.

For every sheaf of $\mathscr{A}$-modules $\mathscr{F}$ on S, denote by $K^!\mathscr{F}$ the complex of sheaves on X generated by the complex of presheaves of $\mathscr{B}$-modules

$$U\longmapsto \mathrm{Hom}_{\mathscr{A}}(K(U),\mathscr{F}).$$

Denote by $RK^!$ the derived functor of $K^!$

$$RK^!:D^+(S,\mathscr{A})\longrightarrow D^+(X,\mathscr{B}).$$

If $U:K_1\to K_2$ is a morphism of objects (3.1.11.1), then U induces morphisms of functors 580

$$^tu:K_2^!\longrightarrow K_1^!\quad\text{et}\quad RK_2^!\longrightarrow RK_1^!.$$

Under the preceding hypotheses, we have:

Proposition 3.1.17. (i) *For every point x of X and every $L\in\mathrm{Ob}\,D^+(S,\mathscr{A})$, denoting by $V(x)$ the category of neighborhoods of x, we have*

$$\mathscr{H}^n(RK^!L)_x\simeq\varinjlim_{U\in\mathrm{Ob}\,V(x)}\mathrm{Ext}^n(K(U),L). \tag{3.1.17.1}$$

(ii) *We therefore have a spectral sequence*

$$E_2^{pq}=\varinjlim_{U\in\mathrm{Ob}\,V(x)}\mathrm{Ext}^p(\mathscr{H}^{-q}(K(U)),L)\Longrightarrow\mathscr{H}^n(K^!L)_x. \tag{3.1.17.2}$$

(iii) *If X^0 is a conservative set of points of X, and $u:K_1\to K_2$ is a morphism of objects (3.1.11.1), then, in order that*

$$^tu:RK_2^!\longrightarrow RK_1^!$$

be an isomorphism, it is necessary and sufficient that, for every n and every $x\in X^0$, the morphism of pro-objects of $\mathrm{Mod}(S,\mathscr{A})$

$$u_x:\text{“}\varprojlim_{U\in\mathrm{Ob}\,V(x)}\text{”}\ \mathscr{H}^nK_1(U)\longrightarrow\text{“}\varprojlim_{U\in\mathrm{Ob}\,V(x)}\text{”}\ \mathscr{H}^nK_2(U) \tag{3.1.17.3}$$

be an isomorphism.

To prove (i), take L to be a complex of injectives, so that (3.1.17.1) is immediate; it is clear that (i) ⇒ (ii).

[122] N.D.E.: Here III 2.3 or IV 4.7.3 is being used.

The sufficiency of condition (3.1.17.3) in (iii) follows immediately from (3.1.17.2). We shall not need the fact that it is necessary. If ${}^t u$ is an isomorphism, then, by (3.1.16.1), for every injective sheaf I on $(S, \mathscr{A})$, we have

$$\mathrm{Hom}(\text{“}\varinjlim_{V(x)}\text{”}\, \mathscr{H}^n K_2(U), I) \xrightarrow{\sim} \mathrm{Hom}(\text{“}\varinjlim_{V(x)}\text{”}\, \mathscr{H}^n K_1(U), I),$$

and we conclude by noting that in every abelian category with enough injectives (here, $\mathrm{Mod}(S, \mathscr{A})$), the functors $\mathrm{Hom}(*, I)$, for I an injective $\mathscr{A}$-module, form a conservative family of functors on $\mathrm{Pro}(\mathscr{A})$.

581 Example 3.1.18. Let $f : X \to S$ be a compactifiable morphism of schemes, and let $\mathscr{A}$ be a sheaf of torsion rings on S. Endow X with $f^*\mathscr{A}$ as its sheaf of rings. Choose an integer d that bounds the dimension of the fibers of f, as well as a compactification of f, and set ((3.1.4.7))

$$K(U) = f_!^{\cdot}((f^*\mathscr{A})_U). \tag{3.1.18.1}$$

By formula (3.1.3.1) for the adjoint of $f_!^{\cdot}$, we have an isomorphism

$$f_{\cdot}^! \simeq K^! : \mathrm{Mod}(S, \mathscr{A}) \longrightarrow C^b \mathrm{Mod}(X, f^*\mathscr{A}). \tag{3.1.18.2}$$

and hence

$$Rf^! = RK^! : D^+(S, \mathscr{A}) \longrightarrow D^+(X, f^*\mathscr{A}). \tag{3.1.18.3}$$

Example 3.1.19. Let $f : X \to S$, $\mathscr{A}$, $\mathscr{B}$, and d be as in 3.1.16. Let K be an object of the following type.

. For every $W = (U, V, \varphi) \in \mathrm{Ob}\, \Gamma(f)$ (3.1.15)[123], $K(W)$ is a complex of $(\mathscr{A}_V, \varphi_* B_{|U})$-modules on V, functorial in W: for every morphism $(u, v) : W_1 = (U_1, V_1, \varphi_1) \to W_2 = (U_2, V_2, \varphi_2)$ it is given by

$$K(u, v) : K(W_1) \longrightarrow v^* K(W_2), \quad \text{i.e.}$$
$$K(u, v) : v_! K(W_1) \longrightarrow K(W_2).$$

With every object K of type (3.1.19.1) is associated an object K' of type (3.1.16.1) relative to $f_\Gamma : \Gamma(f) \to S$, given by

$$K'(U, V, \varphi) = j_! K(U, V, \varphi) \tag{3.1.19.2}$$

where j is the “inclusion morphism” of V into S. We set

$$K^! \underset{\text{defn}}{=} K'^! \quad \text{et} \quad RK^! \underset{\text{defn}}{=} RK'^!.$$

Example 3.1.20. With the notation of 3.1.19, take $\mathscr{B} = f^*\mathscr{A}$ and set

$$K(U, V, \varphi) = \mathscr{A}_V \quad \text{(in degree 0)} \tag{3.1.20.1}$$

582 By 3.1.15.4, we then have isomorphisms of functors

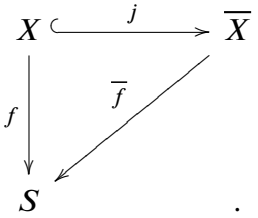
[123] N.D.E.: By definition, φ is an X-morphism $U \to V \times_S X$. To understand this example properly, one must adopt the equivalent viewpoint that interprets φ as an S-morphism $\varphi : U \to V$.

(3.1.20.2) $$K^! \simeq f^*$$

(3.1.20.3) $$RK^! \simeq f^*.$$

3.2. Poincaré duality.

3.2.1. Let f be a compactifiable flat morphism, purely of relative dimension d (or, more generally, satisfying condition $(*)_d$ of 2.9):



Let n be an integer ≥ 1 invertible on S, and suppose that S is ringed by a sheaf of rings $\mathscr{A}$ such that $n\mathscr{A} = 0$. By 3.1.18, the functor $Rf^! : D^+(S,\mathscr{A}) \to D^+(X,\mathscr{A})$ can be computed by the procedure of 3.1.16, for K given by (3.1.18.1).

Let K' be the functor of type (3.1.19.1) that associates to $(U,V,\varphi) \in \mathrm{Ob}\,\Gamma(f)$ the complex of sheaves $\mathbf{Z}/n(-d)[-2d]$ on V, consisting of $\mathbf{Z}/n(-d)$ placed in degree $2d$, and let K'' be the corresponding object of type (3.1.16.1) under (3.1.19.2), relative to $f : \Gamma(f) \to S$.

We may regard K and K'' as objects of 3.1.16.1 relative to $\Gamma(f) \to S$. Moreover, for every $(U,V,\varphi) \in \mathrm{Ob}\,\Gamma(f)$, the trace morphism 2.9 defines a morphism of sheaves on V

$$\mathrm{Tr} : R^{2d}\varphi_!\mathbf{Z}/n \longrightarrow \mathbf{Z}/n(-d),$$

whence, denoting by j the morphism from V to S, a morphism of functors

(3.2.1.1) $$K(U) = \overline{f}_*\tau_{\leq 2d}C^*_\ell\mathbf{Z}/n_U \longrightarrow \mathscr{H}^{2d}(K(U))[-2d] \simeq$$

$$R^{2d}f_!(\mathbf{Z}/n)_U \simeq j_!R^{2d}\varphi!\mathbf{Z}/n \xrightarrow{\mathrm{Tr}_\varphi} j_!\mathbf{Z}/N(-d) = K''(U,V,\varphi)).$$

A twisted form of 3.1.20 shows that 583

$$RK''^! = Rf^*(d)[2d].$$

By 3.1.16, we therefore have a composite morphism

(3.2.1.2) $$t_f : Rf^*(d)[2d] \simeq RK''^! \longrightarrow Rk^! \simeq Rf^!.$$

Remark 3.2.2. If the functor $Rf^!$ has finite cohomological dimension, then t_f is still defined as a morphism between functors from $D(S,\mathscr{A})$ to $D(X,f^*\mathscr{A})$ (definition in 3.1.5).

There is another method for constructing an arrow (3.2.1.2), by adjunction from (2.13.2)

$$\mathrm{Tr} : Rf_!(f^*L(d)[2d]) \longrightarrow L.$$

These two arrows coincide:

LEMMA 3.2.3. *For $L \in \mathrm{Ob}\, D^+(S, \mathscr{A})$ (resp. $D(S, \mathscr{A})$ if the functor $Rf^!$ has finite cohomological dimension), the diagram*

$$\begin{array}{ccc} Rf_!(f^*L(d)[2d]) & \xrightarrow{Rf_!(t_f)} & Rf_!Rf^!L \\ \downarrow{\scriptstyle \mathrm{Tr}_f} & & \downarrow{\scriptstyle \text{adjunction}} \\ L & = & L \end{array}$$

is commutative.

We first give a variant of the definition of t_f, in terms of the functor $f_\cdot^!$ (3.1.4.10). Let $U \in \mathrm{Ob}\, X_{\mathrm{et}}$, $V \in \mathrm{Ob}\, S_{\mathrm{et}}$, and $\varphi \in \mathrm{Hom}_f(U, V)$. By definition,

$$f_\cdot^!(L)(U) = \mathrm{Hom}(f_!^\cdot \mathscr{A}_U, L)\ ;$$

the trace morphism defines a morphism

$$\mathrm{Tr}_f\ :\ f_!^\cdot \mathscr{A}_U \longrightarrow \mathscr{A}_V(-d)[-2d]$$

whose transpose is

$$t_\varphi\ :\ f_\cdot^!(L)(U) \longleftarrow L(d)[2d](V).$$

584 The morphisms t_φ define a morphism

$$t_f\ :\ f^*L(d)[2d] \longrightarrow f_\cdot^!L$$

and we leave it to the reader to verify that, in the derived category, it coincides with (3.2.1.2). This description shows that the following diagrams are commutative at the level of complexes, for L a complex of modules:

$$\begin{array}{ccc} \mathrm{Hom}^\cdot(\mathscr{A}_V, L) & \longrightarrow & \mathrm{Hom}^\cdot(f^*\mathscr{A}_V(d)[2d], f^*L(d)[2d]) \\ \downarrow{\scriptstyle \mathrm{Tr}_f} & & \downarrow{\scriptstyle t_f} \\ \mathrm{Hom}(f_!^\cdot f^*\mathscr{A}_V(d)[2d], L) & \xrightarrow{\sim} & \mathrm{Hom}^\cdot(f^*\mathscr{A}_V(d)[2d], f_\cdot^!L). \end{array}$$

By functoriality of the arrows in this diagram in $\mathscr{A}_V$, it follows that for every sheaf, or every complex of sheaves K, the following diagram is commutative:

$$\begin{array}{ccc} \mathrm{Hom}^\cdot(K, L) & \longrightarrow & \mathrm{Hom}^\cdot(f^*K(d)[2d], f^*L(d)[2d]) \\ \downarrow{\scriptstyle \mathrm{Tr}_f} & & \downarrow \\ \mathrm{Hom}^\cdot(f_!^\cdot f^*K(d)[2d], L) & \xleftrightarrow{\sim} & \mathrm{Hom}^\cdot(f^*K(d)[2d], f_\cdot^!L). \end{array}$$

Taking $K = L$ and $\mathrm{Id} : L \to L$, we obtain 3.2.3 at the level of complexes.

I would be grateful to anyone who has understood this proof if they would explain it to me.

3.2.4. We deduce at once from this lemma and from the definition (3.1.13) of the composition isomorphisms by adjunction that if f is the composite of two flat, finitely presented, compactifiable morphisms that are purely of relative dimensions d_1 and d_2 : $f = gh$, and if $d = d_1 + d_2$, then the diagram

$$\begin{array}{ccc} h^*(g^*L(d_2)[2d_2])(d_1)[2d_1]) & \xrightarrow{\sim} & (gh)^*L(d)[2d] \\ \downarrow{\scriptstyle t_h * t_g} & & \downarrow{\scriptstyle t_{gh}} \\ Rh^!\ Rg^!\ L & \xrightarrow{\sim} & R(gh)^!L \end{array}$$

is commutative. 585

THEOREM 3.2.5 (Poincaré duality). *Let $f : X \to S$ be a compactifiable smooth morphism* (thus S is quasi-compact and quasi-separated (sic)), *let d be the locally constant function on X "relative dimension of f", let $n \geq 1$ be an integer invertible on S, and let $\mathscr{A}$ be a sheaf of rings on S satisfying $n\mathscr{A} = 0$. Then the functor*

$$Rf_! : D(X, f^*\mathscr{A}) \longrightarrow D(S, \mathscr{A})$$

admits as right adjoint the following functor from $D(S, \mathscr{A})$ to $D(X, f^*\mathscr{A})$:

$$K \longmapsto f^*K(d)[2d]$$

with the morphism (2.13.2)

$$\mathrm{Tr}_f : Rf_!f^*K(d)[2d] \longrightarrow K$$

as adjunction arrow:

$$\mathrm{Hom}(K, f^*L(d)[2d]) \xrightarrow{\sim} \mathrm{Hom}(Rf_!K, L).$$

Proof. We reduce to the case where d is constant. By the description 3.2.1 of the arrow t_f (3.2.1.2), Theorem 2.14, together with criterion 3.1.17 (iii), implies that the arrow t_f is an isomorphism (for $L \in D^+(S, \mathscr{A})$)

$$t_f : f^*L(d)[2d] \xrightarrow{\sim} Rf^!L.$$

In particular, the functor $Rf^!$ has finite cohomological dimension and is therefore defined on the unbounded derived category. The morphism t_f remains an isomorphism for $L \in D(S, \mathscr{A})$[124].

By definition, $Rf^!$ is the right adjoint of the functor $Rf_!$; by 3.2.3, the adjunction 586
arrow from $Rf_!Rf^!K$ to K is identified, via t_f, with Tr_f, and this completes the proof of 3.2.5.

Henceforth we identify, by means of t_f, the functors $f^*L(d)[2d]$ and $Rf^!L$.

[124] N.D.E.: More precisely, the fact that $Rf^!$ has finite cohomological dimension makes it possible to show that one can define the functor $Rf^!$ on the unbounded derived category in such a way that, for a complex $I^\cdot$ of injective sheaves, $Rf^!I^\cdot$ is the total complex associated with the double complex $f^!I^\cdot$. Then, for every $n \in \mathbf{Z}$, the morphism induced on the n-th cohomology sheaf by t_f evaluated at $I^\cdot$ is identified with the same morphism for a stupid truncation $I^{\geq k}$ for k sufficiently small, which proves the assertion.

3.2.6. Let S be the spectrum of an algebraically closed field k, let X be a compactifiable scheme, smooth over k and purely of relative dimension d, let n be an integer invertible in k, and let F be a locally constant constructible sheaf of $\mathbf{Z}/n$-modules, i.e. a "coefficient system" killed by n. Denoting by $^\vee$ a $\mathbf{Z}/n$-dual, we have

$$R\,\mathbf{Hom}(F, \mathbf{Z}/n) = F^\vee \quad \text{(in degree 0)}.$$

This formula follows from the fact that $\mathbf{Z}/n$ is an injective $\mathbf{Z}/n$-module. Isomorphism 3.2.5.1, or rather the isomorphism deduced from it,

$$R\Gamma\ R\,\mathbf{Hom}(F, \mathbf{Z}/n(d)[2d]) \xrightarrow{\sim} R\,\mathrm{Hom}(R\Gamma_c(F), \mathbf{Z}/n),$$

is therefore written here as

$$R\Gamma(F^\vee(d)[2d]) \xrightarrow{\sim} R\,\mathrm{Hom}(R\Gamma_c(F), \mathbf{Z}/n). \tag{3.2.6.1}$$

Passing to cohomology groups, and using again the fact that $\mathbf{Z}/n$ is an injective $\mathbf{Z}/n$-module, this gives

$$H^{2d-i}(X, F^\vee(d)) \xrightarrow{\sim} H^i_c(X, F)^\vee, \tag{3.2.6.2}$$

which is the usual form of Poincaré duality.

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EXPOSÉ XIX

Cohomology of excellent preschemes of equal characteristic

M. Artin

In this exposé we develop the theory of étale cohomology for excellent preschemes (EGA IV 7.8.1) of equal characteristic, *assuming the theorem on resolution of singularities for such preschemes*, in the following form: 588 589

Let $X = \operatorname{Spec} A$ be an excellent local scheme of equal characteristic, and let $U \subset X$ be a regular open subset. There exist a proper birational morphism $f : X' \to X$ and an open immersion $U \to X'$ over X such that X' is regular and $Y = X - U$ is everywhere of codimension 1, with normal crossings.

The proofs are therefore valid, "for the time being," only in characteristic zero [4]. One could also deduce from them results for preschemes of characteristic $p > 0$ in low dimensions [1].

Under the resolution hypothesis, we shall obtain more or less satisfactory results for preschemes of equal characteristic and for coefficients prime to the characteristic (cf. 3.2, 4.1, 5.1, 6.1). On the other hand, almost nothing is known about the cohomology of general preschemes of unequal characteristic, even in dimension 2, where resolution is available [2]. For example, the purity theorem 2.1 is not known for any complete ring of unequal characteristic of dimension > 1. This result, for the formal power series rings $k[[x_1, \dots, x_n]]$ (1.2), is one of the main tools in the theory. Let us also note that the resolution theorem is not used in the proof of 1.2. *This theorem is therefore proved in arbitrary characteristic.*

We shall often use without explicit mention the properties of excellent schemes developed in EGA IV 7.8, such as stability under extensions of finite type (EGA IV 7.8.3(ii)) and under strict localizations. The latter fact follows from resolution and EGA IV 7.9.5. 590

1. Purity for the ring $k[[x_1, \dots, x_n]]$

Recall the following corollary of XVI 3.7:

Corollary 1.1. *Let A be a strictly local ring, and denote by $A\{x\}$ the strict localization of $A[x]$ at the prime ideal generated by* rad A *and x. Let $U = \operatorname{Spec} A\{x\}[1/x]$. Then, for n prime to the residual characteristic of A, we have*

$$H^q(U, \boldsymbol{\mu}_n) \simeq \begin{cases} \boldsymbol{\mu}_n(A) & \text{if} \quad q = 0 \\ \mathbf{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1. \end{cases}$$

Indeed, let $X = \operatorname{Spec} A[x]$, and let $Y = V(x)$ be the zero locus of x in X. Then (Y, X) is a smooth pair (XVI 3.1) over Spec A. By (V 4, VIII 5), $H^q(U, \boldsymbol{\mu}_n)$ is identified with the fiber of the sheaf $\mathscr{H}^{q+1}_Y(X, \boldsymbol{\mu}_n)$ at a geometric point above the point (rad A, x) of X, and the corollary therefore follows immediately from (XVI 3.7). Note that $\boldsymbol{\mu}_n$ and $\mathbf{Z}/n$ are both cyclic groups of order n. We have used them here to give the canonical formulas (cf. 3.4). 591

THEOREM 1.2. *Let k be a separably closed field, and let $U = \operatorname{Spec} k[[x_1, \dots, x_r]]\,[1/x_r]$. Then, for n prime to the characteristic of k, we have*

$$H^q(U, \boldsymbol{\mu}_n) = \begin{cases} \boldsymbol{\mu}_n(A) & \text{if} \quad q = 0 \\ \mathbf{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1. \end{cases}$$

REMARK. The proof applies equally to the ring of convergent power series when $k = \mathbf{C}$. Moreover, one easily avoids the hypothesis that k be separably closed by stating the result with a little more care. But to establish the general purity theorem 3.2 (under the resolution-of-singularities hypothesis, of course), we need the result only when k is separably closed, and these other assertions are corollaries of that theorem 3.2.

In dimensions 0 and 1, the result can be proved for every regular strictly local ring:

LEMMA 1.3. *Let A be a regular strictly local ring and let $x \in \operatorname{rad} A$ be a local parameter. Let $U = \operatorname{Spec} A[1/x]$. Then $H^0(U, \boldsymbol{\mu}_n) = \boldsymbol{\mu}_n(A)$, and there are canonical isomorphisms*

$$\mathbf{Z}/n \simeq H^0(U, \mathbf{G}_m)/(H^0(U, \mathbf{G}_m))^n \simeq H^1(U, \boldsymbol{\mu}_n),$$

592 *where the generator of $\mathbf{Z}/n$ is the "residue" of $x \in H^0(U, \mathbf{G}_m)$.*

Proof. For $q = 0$, the assertion is that U is connected and nonempty. This is true. For $q = 1$, apply the Kummer exact sequence (IX 3.7). We have $\operatorname{Pic} U = 0$ because A is regular and factorial, and U is an open subset of $X = \operatorname{Spec} A$. It follows that we have the exact sequence

$$H^0(U, \mathbf{G}_m) \xrightarrow{n} H^0(U, \mathbf{G}_m) \to H^1(U, \boldsymbol{\mu}_n) \to 0,$$

and it remains to prove that $H^0(U, \mathbf{G}_m)/H^0(U, \mathbf{G}_m))^n$ is isomorphic to $\mathbf{Z}/n$ and generated by the residue of x. But there is an obvious exact sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^0(X, \mathbf{G}_m) & \longrightarrow & H^0(U, \mathbf{G}_m) & \longrightarrow & \mathbf{Z} & \longrightarrow & 0 \\ & & \wr\uparrow & & \wr\uparrow & & & & \\ & & A^* & & A[1/x]^* & & & & \end{array}$$

where the image of a section $a \in H^0(U, \mathbf{G}_m)$ in $\mathbf{Z}$ is the order of vanishing of a along $\{x = 0\}$. The group A^* is divisible by n, because A is strictly local and n is prime to the residual characteristic. Indeed, if $u \in H^0(X, \mathbf{G}_m)$, the equation $Y^n - u = 0$ defines a finite étale extension of A, which is therefore completely split. It follows that $H^1(U, \boldsymbol{\mu}_n) \simeq \mathbf{Z}/n$, proving the lemma.

Proof of 1.2. It remains to treat the case $q > 1$. Recall the terminology of (V, Appendix). We have

$$H^q(U, \boldsymbol{\mu}_n) = \varinjlim_{\mathbf{U}} H^q(\mathbf{U}, \boldsymbol{\mu}_n)$$

593 where $\mathbf{U} = \left\{ U_0 \leftleftarrows U_1 \;\substack{\leftarrow\\ \leftarrow\\ \leftarrow}\; \dots \right\}$ ranges over the category (filtered up to homotopy) of *hypercoverings* of U. Let $\mathbf{U}$ be such a hypercovering, and take the U_i to be separated and of finite type. Since we wish to prove that $H^q(U, \boldsymbol{\mu}_n) = 0$, it suffices to find a hypercovering $\mathbf{V}$ and a morphism $\mathbf{V} \to \mathbf{U}$ such that the induced morphism

$$H^q(\mathbf{U}, \boldsymbol{\mu}_n) \longrightarrow H^q(\mathbf{V}, \boldsymbol{\mu}_n)$$

is the zero morphism.

Let $\{W_1, \dots, W_m\}$ be the set of connected components of all the U_i for $i \leq q+1$, and let B_j be the normalization of $A = k[[x_1, \dots, x_r]]$ in the field $R(W_j)$ of rational functions on W_j. Then, since $W_j \to U$ is étale, the scheme W_j is an open subset of Spec B_j, say the open complement of Spec C_j, where C_j is the appropriate reduced quotient ring of B_j.

Choose in addition an element b_j of B_j such that b_j generates the separable extension $R(W_j)$ of $k((x_1, \dots, x_r))$. Let $f_j(Y) \in A[Y]$ be the monic irreducible polynomial of b_j over A, so that B_j is isomorphic to the normalization of the ring $A[Y]/(f_j)$. Finally, let $d_j \in A$ be the discriminant of $f_j(Y)$, which is a nonzero element of A.

Now we may replace the x_i for $i < r$ by other elements of A, the only condition being that $\{x_1, \dots, x_r\}$ form a system of parameters of A.

It is easy to see that, after changing these elements if necessary, we may suppose 594
that the following hypotheses are satisfied:

1.4.

(i) For every j such that d_j is not a unit, $x_1, \dots, x_{r-1}$ and d_j generate an ideal primary for rad A.

(ii) For every j, the images of $x_1, \dots, x_{r-1}$ in C_j generate an ideal primary for rad C_j.

We then have the following lemma:

Lemma 1.5. *Let $A^0 = k[[x_1, \dots, x_{r-1}]]\ \{x_r\}$ be the henselization of $k[[x_1, \dots, x_{r-1}]]\ [x_r]$ at the point $(x_1, \dots, x_r)$, and let $X^0 = \operatorname{Spec} A^0$ and $U^0 = \operatorname{Spec} A^0\ [1/x_r]$. Under the conditions of 1.4 above, there exists a hypercovering $\mathbf{U}^0$ of U^0 (of order $q+1$) and an isomorphism $\mathbf{U} \times_{X^0} X \simeq \mathbf{U}$.*

Assume the lemma. The ring A^0 is excellent (EGA IV 7.8.3 (ii), 7.9.5), and consequently the connected components of the U_i^0 and of the U_i correspond bijectively (EGA IV 7.8.3 (vii)). The isomorphism $\mathbf{U}^0 \times_{X^0} X \simeq \mathbf{U}$ therefore gives an isomorphism

$$H^q(\mathbf{U}^0, \boldsymbol{\mu}_n) \simeq H^q(\mathbf{U}, \boldsymbol{\mu}_n).$$

Since k is separably closed, A^0 is a strictly local ring, and we may apply 1.1 to the ring A^0. It follows that $H^q\ (U^0, \boldsymbol{\mu}_n) = 0$. But

$$H^q(U^0, \boldsymbol{\mu}_n) \simeq \varinjlim_{\mathbf{V}^0} H^q(\mathbf{V}^0, \boldsymbol{\mu}_n),$$

where $\mathbf{V}^0$ ranges over the category of hypercoverings of U^0 of order $q+1$, and H^q 595
$(\mathbf{U}^0, \boldsymbol{\mu}_n)$ is a finite group. Consequently, there exists a morphism of hypercoverings $\mathbf{V}^0 \to \mathbf{U}^0$ such that the induced morphism

$$H^q(\mathbf{U}^0, \boldsymbol{\mu}_n) \longrightarrow H^q(\mathbf{V}^0, \boldsymbol{\mu}_n)$$

is the zero morphism. Put $\mathbf{V} = \mathbf{V}^0 \times_{X^0} X$. Then $H^0\ (\mathbf{V}^0, \boldsymbol{\mu}_n) \simeq H^q(\mathbf{V}, \boldsymbol{\mu}_n)$ (by the same argument as for $\mathbf{U}^0$), so

$$H^q(\mathbf{U}, \boldsymbol{\mu}_n) \longrightarrow H^q(\mathbf{V}, \boldsymbol{\mu}_n)$$

is also the zero morphism, C.Q.F.D.

Proof of 1.5. It suffices to find finite A^0-algebras B_j^0 and C_j^0 and A-isomorphisms

$$B_j^0 \otimes_{A^0} A \xrightarrow{\sim} B_j$$
$$C_j^0 \otimes_{A^0} A \xrightarrow{\sim} C_j.$$

Indeed, by ([3] 1.4) the morphism $B_j \to C_j$ is induced by a morphism $B_j^0 \to C_j^0$, so the open subset $W_j \subset \operatorname{Spec} B_j$ is induced by an open subset $W_j^0 \subset \operatorname{Spec} B_j^0$. Moreover, the U-morphisms among the W_j are induced by morphisms among the A-algebras B_j, hence ([3] 1.4) by morphisms among the B_j^0, and therefore by U^0-morphisms among the W_j^0. It follows that the W_j^0 form the set of connected components of a simplicial object $\mathbf{U}^0$ over U^0 that induces $\mathbf{U}$, and one sees immediately that it is a hypercovering of U^0.

596 Now 1.4 (ii) does imply that C_j is induced by an A^0-algebra C_j^0, as is easily verified [124]. For B_j, it suffices to find a monic polynomial $f_j^0(Y) \in A^0[Y]$ such that the two A-algebras

$$A[Y]/(f_j) \quad , \quad A[Y]/(f_j^0) = A \otimes_{A^0} A^0[Y]/(f_j^0)$$

are isomorphic. Indeed, $A^0[Y]/(f_j^0)$ is then reduced, and we take for B_j^0 the normalization (EGA IV 7.8.3 (vi)) of $A[Y]/(f_j^0)$. The ring $B_j^0 \otimes_{A^0} A$ is still normal (EGA IV 7.8.3 (vii)) and birational to B_j, hence equal to that ring.

By 1.4 (i), the ideal $I_j \subset A$ generated by d_j is induced by an ideal of $k[[x_1, \dots, x_{r-1}]][x_r]$ (Weierstrass preparation theorem [5][6]), hence by an ideal I_j^0 of A^0. Moreover, A/I_j is a finite $k[[x_1, \dots, x_{r-1}]]$-algebra, and is therefore isomorphic to A^0/I_j^0. The latter ring is thus complete, and consequently the I_j^0-adic completion of A^0 is complete and isomorphic to A. We can therefore find a monic polynomial $f_j^0 \in A^0[Y]$ such that in $A[Y]$ we have

$$f_j \equiv f_j^0 (\text{modulo } {d_j}^2 \operatorname{rad} A).$$

By ([3], 1.3), $A[Y]/(f_j) \simeq A[Y]/(f_j^0)$, which proves the lemma.

2. The case of a strictly local ring

The theorem is as follows:

597 THEOREM 2.1. [124]. *Let A be an excellent, regular, strictly local ring of equal characteristic, and let x be a "local parameter"*[124] *of A. Let $U = \operatorname{Spec} A[1/x]$. Then, for n prime to the characteristic of A, we have*

$$H^q(U, \boldsymbol{\mu}_n) = \begin{cases} \boldsymbol{\mu}_n(A) & \text{if} \quad q = 0 \\ \mathbf{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1. \end{cases}$$

The proof proceeds by induction on q. We denote by $P(N)$ the assertion of 2.1 for the values $q \leq N$. By 1.3, the assertion $P(1)$ is true.

[124] Indeed, 1.4 (ii) implies that C_j is finite over $k[[X_1, \dots, X_{r-1}]]$, and a fortiori over A^0, and it suffices to take $C_j^0 = C_j$ with the A^0-module structure obtained by restriction of scalars.

[124] Depends on resolution of singularities (cf. Introduction).

[124] By which we mean here: an element belonging to a *regular* system of parameters.

LEMMA 2.2. *Suppose that $P(N)$ is true. Let $A \to A'$ be a local morphism of excellent, regular, strictly local rings of equal characteristic. Let $\{x_1, \dots, x_r\}$ be part of a regular system of parameters of A whose set of images in A is again part of a regular system of parameters. Let*

$$U = \operatorname{Spec} A[1/x_1, \dots, 1/x_r]$$
$$U' = \operatorname{Spec} A'[1/x'_1, \dots, 1/x'_r].$$

Then the canonical morphism

$$H^q(U, \boldsymbol{\mu}_n) \longrightarrow H^q(U', \boldsymbol{\mu}_n)$$

is bijective for every $q \leq N$. 598

Proof. [124] We argue by induction on r: the assertion is trivial for $r = 0$. Let $r > 0$ and suppose the theorem has been proved for $r-1$. Let $B = A/(x)$, $V = \operatorname{Spec} A[1/x_1, \dots, 1/x_{r-1}]$, and $Y = \operatorname{Spec} B[1/x_1, \dots, 1/x_{r-1}]$ (where the image of x_i in B is denoted by the same symbol). There is an open immersion $i : U \to V$, and Y is the complementary closed subset, defined by the element x_r.

Applying $P(N)$ to the strict localizations of V at the geometric points of Y, we find that

$$R^q i_*(\boldsymbol{\mu}_n)_U = \begin{cases} (\boldsymbol{\mu}_n)_V & \text{if} \quad q = 0 \\ (\mathbf{Z}/n)_Y & \text{if} \quad q = 1 \\ 0 & \text{if} \quad 1 < q \leq N, \end{cases} \tag{2.3}$$

because these strict localizations are excellent regular rings of equal characteristic, and the element x_r is a local parameter at every point of Y.

We use a "prime" to denote the analogous objects deduced from A'. We obtain a morphism of spectral sequences

$$\begin{array}{ccc} E_2^{pq} & = & H^p(V, R^q i_* \boldsymbol{\mu}_n) \Longrightarrow H^{p+q}(U, \boldsymbol{\mu}_n) \\ \downarrow{\scriptstyle \varphi_2^{pq}} & & \downarrow{\scriptstyle \varphi} \qquad\qquad \downarrow{\scriptstyle \varphi} \\ E_2'^{pq} & = & H^p(V', R^q i'_* \boldsymbol{\mu}_n) \Rightarrow H^{p+q}(U', \boldsymbol{\mu}_n). \end{array}$$

Now it is clear from 1.3 and (2.3) that the base change $V \xleftarrow{g} V'$ induces an isomorphism 599

$$g^* R^q i_* \boldsymbol{\mu}_n \xrightarrow{\sim} R^q i'_* \boldsymbol{\mu}_n$$

for $q \leq N$ (only for $q = 0, 1$ is there anything at all to check). Consequently, the induction hypothesis, applied to the morphism

$$H^q(V, \boldsymbol{\mu}_n) \longrightarrow H^q(V', \boldsymbol{\mu}_n),$$

implies that φ_2^{pq} is an isomorphism for $q = 0$ and $p \leq N$.

[124] A shorter proof than the one that follows, but using the cup-product formalism, would consist in showing that, under the hypothesis $P(N)$, the cohomology $H^*(U, \mathbf{Z}/n\mathbf{Z})$ is, in dimensions $\leq N$, the exterior algebra (over $\mathbf{Z}/n\mathbf{Z}$) on $H^1(U, \mathbf{Z}/n\mathbf{Z}) \simeq (\mathbf{Z}/n\mathbf{Z})^r$, which reduces matters to establishing 2.2 for $q = 1$, a well-known case. The calculation indicated under the hypothesis $P(N)$ is made explicit, for example, in SGA 7.

Similarly, we can apply the induction hypothesis to the morphism of rings $B \to B'$; these are regular, excellent, strictly local rings of equal characteristic, and the elements $x_1, \dots, x_{r-1}$ form part of a system of parameters. Therefore

$$H^q(Y, \boldsymbol{\mu}_n) \xrightarrow{\sim} H^q(Y', \boldsymbol{\mu}_n) \quad \text{if} \quad q \leq N.$$

Since $(\boldsymbol{\mu}_n) \simeq (\mathbf{Z}/n)_Y$, it follows that φ_2^{pq} is also bijective for $p \leq N$ and for $q = 1$.

Since $R^q i_* \boldsymbol{\mu}_n = R^q i'_* \boldsymbol{\mu}_n = 0$ if $1 < q \leq N$, we find $E_2^{pq} = {E'}_2^{pq} = 0$ if $1 < q \leq N$, and for $1 \leq m \leq N$ the morphism of spectral sequences therefore reduces to a morphism of exact sequences

$$\begin{array}{ccccccccc}
H^{m-2}(Y, \mathbf{Z}/n) & \longrightarrow & H^m(V, \boldsymbol{\mu}_n) & \longrightarrow & H^m(U, \boldsymbol{\mu}_n) & \xrightarrow{e} & H^{m-1}(Y, \mathbf{Z}/n) & \longrightarrow & H^{m+1}(V, \boldsymbol{\mu}_n) \\
\downarrow a & & \downarrow b & & \downarrow c & & \downarrow d & & \downarrow \\
H^{m-2}(Y', \mathbf{Z}/n) & \to & H^m(V', \boldsymbol{\mu}_n) & \to & H^m(U', \boldsymbol{\mu}_n) & \xrightarrow{e} & H^{m-1}(Y', \mathbf{Z}/n) & \to & H^{m+1}(V', \boldsymbol{\mu}_n).
\end{array}$$

600 Now we have already seen that a, b, and d are bijective, and it follows that c is *injective*. To prove that c' is *surjective*, it suffices to prove that the morphism

$$H^m(U, \boldsymbol{\mu}_n) \xrightarrow{e} H^{m-1}(Y, \mathbf{Z}/n)$$

is surjective for $m \leq N$.

Let $K \subset A$ be an arbitrary separably closed field, and let $A^0 = k\{x_1, \dots, x_r\}$ be the henselization of $k[x_1, \dots, x_r]$ at the origin. Then the hypotheses of 2.1 are satisfied for the morphism $A^0 \to A$ and the elements x_i. With the evident notation, we have a commutative square

$$\begin{array}{ccc}
H^m(U^0, \boldsymbol{\mu}_n) & \xrightarrow{e^0} & H^{m-1}(Y^0, \mathbf{Z}/n) \\
\downarrow c^0 & & \downarrow d^0 \\
H^m(U, \boldsymbol{\mu}_n) & \xrightarrow{e} & H^{m-1}(Y^0, \mathbf{Z}/n)
\end{array}$$

and we already know that the arrow d^0 is bijective. It therefore suffices to prove that the morphism e^0 is surjective. We are thus reduced (dropping the superscript ^{0}) to verifying surjectivity in the case $A = k\{x_1, \dots, x_r\}$.

For this ring, which is a limit of schemes smooth over k, we proved the purity the-
601 orem in XVI 3.7. Thus $R^q i_* \boldsymbol{\mu}_n = 0$ for every $q > 1$, and consequently there is an exact sequence

$$H^m(U, \boldsymbol{\mu}_n) \xrightarrow{e} H^{m-1}(Y, \mathbf{Z}/n) \to H^{m+1}(V, \boldsymbol{\mu}_n) \xrightarrow{f} H^{m+1}(U, \boldsymbol{\mu}_n)$$

for every $m \geq 1$. The surjectivity of e is therefore equivalent to the injectivity of f.

Let $Z = \operatorname{Spec} k\{x_1, \dots, x_{r-1}\}[1/x_1, \dots, 1/x_{r-1}]$. The morphism $\operatorname{Spec} k\{x_1, \dots, x_r\} \to \operatorname{Spec} k\{x_1, \dots, x_{r-1}\}$ given by the inclusion of rings is acyclic. This is the local acyclicity of a smooth morphism (XVI 1.1). Since the open subset V of $\operatorname{Spec} k\{x_1, \dots, x_r\}$ is the inverse image of Z, the morphism

$$H^q(Z, \boldsymbol{\mu}_n) \longrightarrow H^q(V, \boldsymbol{\mu}_n)$$

is bijective for every q (XV 1.3). It therefore suffices to prove that the morphism

$$H^q(Z, \boldsymbol{\mu}_n) \longrightarrow H^q(U, \boldsymbol{\mu}_n)$$

is injective for every q.

Now $k\{x_1, \dots, x_r\}$ is the inductive limit of rings A_α étale over $k\{x_1, \dots, x_{r-1}\}[x_r]$, each equipped with a lifting to that ring of the point $x_1 = x_2 = \cdots = x_r = 0$. Consequently, U is the projective limit of the open subsets $U_\alpha = \operatorname{Spec} A_\alpha[1/x_1, \dots, 1/x_r]$, and we have

$$H^q(U, \boldsymbol{\mu}_n) = \varinjlim_\alpha H^q(U_\alpha, \boldsymbol{\mu}_n).$$

But since $A_\alpha[1/x_r]$ is smooth over the henselian ring $k\{x_1, \dots, x_{r-1}\}$, and since the closed fiber of $A[1/x_r]$ is nonempty, we see immediately that the $k\{x_1, \dots, x_{r-1}\}$-algebra $A_\alpha[1/x_r]$ admits sections. Hence U_α admits sections over Z, and it follows that 602

$$H^q(Z, \boldsymbol{\mu}_n) \longrightarrow H^q(U_\alpha, \boldsymbol{\mu}_n)$$

is injective for every q, and therefore that

$$H^q(Z, \boldsymbol{\mu}_n) \longrightarrow H^q(U, \boldsymbol{\mu}_n)$$

is also injective, which proves the lemma.

LEMMA 2.4. *Suppose that $P(N)$ is true. Let $A \to A'$ be a regular morphism* (EGA IV 6.8.1) *of excellent, strictly local rings of equal characteristic (not necessarily regular), and let $U \subset \operatorname{Spec} A$ be a regular open subset. Let $U' = U \times_{\operatorname{Spec} A} \operatorname{Spec} A'$. Then, for n prime to the residue characteristic of A, the morphism*

$$H^q(U, \boldsymbol{\mu}_n) \longrightarrow H^q(U', \boldsymbol{\mu}_n)$$

is bijective for $q \leq N$.

Proof. Let $f : X \to \operatorname{Spec} A$ be a resolution of singularities of $\operatorname{Spec} A$ such that the open subset U of $\operatorname{Spec} A$ lifts to an open immersion $i : U \to X$, and such that the closed subset $Y = X - U$ is pure of codimension 1 and has normal crossings. We denote by a prime the base change induced by $A \to A'$. The morphism $g : X' \to X$ is regular (EGA IV 6.8.3), so $f' : X' \to \operatorname{Spec} A'$ is a resolution of $\operatorname{Spec} A$, $i' : U' \to X'$ is an open immersion, and $Y' = X' - U'$ is pure of codimension 1 and has normal crossings. All these schemes are excellent. 603

Let p' be a point of X' and let p be its image in X. Since Y has normal crossings in X, there exist elements $x_1, \dots, x_r \in O_{X,p}$ such that $\{x_1, \dots, x_r\}$ is part of a system of parameters of $O_{X,p}$ and such that $\operatorname{Spec} O_{X,p} \cap Y = V(x_1, \dots, x_r)$. Moreover, it follows immediately from the fact that $g : X' \to X$ is regular that the images x_i' of the x_i in $O_{X',p'}$ form part of a system of parameters of this ring, and that $\operatorname{Spec} O_{X',p'} \cap Y = V(x_1', \dots, x_r')$.

Let B' be the strict localization of $O_{X',p'}$ at a geometric point $\overline{p}$ above p', and let B be the strict localization of $O_{X,p}$ at the corresponding geometric point of X (these are excellent rings (EGA IV 7.8.3 (ii), 7.9.5)). Let

$$U_{\overline{p}} = \operatorname{Spec} B[1/x_1, \dots, 1/x_r] \quad , \quad U'_{\overline{p}} = \operatorname{Spec} B'[1/x_1', \dots, 1/x_r'].$$

Lemma 2.2 applies, and we deduce that the morphisms

$$(*) \qquad H^q(U_{\overline{p}}, \boldsymbol{\mu}_n) \longrightarrow H^q(U'_{\overline{p}}, \boldsymbol{\mu}_n)$$

are bijective for $q \leq N$.

Let us compute the sheaves $R^q i_* \boldsymbol{\mu}_n$ and $R^q i'_* \boldsymbol{\mu}_n$ for the open immersions $i : U \to X$ and $i' : U' \to X'$: the stalk $(R^q i'_* \boldsymbol{\mu}_n)_{\overline{p}}$ at the geometric point $\overline{p}$ is precisely $H^q(U'_{\overline{p}}, \boldsymbol{\mu}_n)$

(VIII 5). Likewise, $(R^q i_* \boldsymbol{\mu}_n)_{\overline{p}} = H^q(U_{\overline{p}}, \boldsymbol{\mu}_n)$. Thus $(*)$ implies that the base-change morphisms

$$(**) \qquad g^* R^q i_* \boldsymbol{\mu}_n \longrightarrow R^q i'_* \boldsymbol{\mu}_n$$

604 are bijective for $q \leq N$.

By the base-change theorem for a proper morphism (XII 5.5), we have (since X and X' are proper over the spectra of strictly local rings, and the closed fibers of X and X' are equal)

$$H^q(X, F) \xrightarrow{\sim} H^q(X', g^* F)$$

for every torsion sheaf F and every q. Consequently, $(**)$ implies that in the morphism of spectral sequences

$$\begin{array}{ccc} E_2^{pq} & = & H^p(X, R^q i_* \boldsymbol{\mu}_n) \Longrightarrow H^{p+q}(U, \boldsymbol{\mu}_n) \\ \downarrow & & \downarrow \varphi_2^{pq} \qquad\qquad \downarrow \varphi \\ E_2'^{pq} & = & H^p(X', R^q i'_* \boldsymbol{\mu}_n) \Longrightarrow H^{p+q}(U', \boldsymbol{\mu}_n), \end{array}$$

the arrows φ_2^{pq} are bijective for every $q \leq N$. It is easily seen that this implies that φ_∞^{pq} is bijective if $q \leq N$ and $p + q \leq N + 1$. Consequently, the morphism of the abutments

$$H^m(U, \boldsymbol{\mu}_n) \longrightarrow H^m(U', \boldsymbol{\mu}_n)$$

is bijective for $m \leq N$, which proves the lemma.

The following lemma will complete the proof of 2.1:

LEMMA 2.5. $P(N) \Rightarrow P(N+1)$.

605 *Proof.* Let A be strictly local, regular, excellent, and of equal characteristic, let $x \in \operatorname{rad} A$ be a local parameter, and let $U = \operatorname{Spec} A[1/x]$. Let $\widehat{A}$ be the completion of A, let $\hat{x}$ be the image of x, let $\widehat{U} = \operatorname{Spec} \widehat{A}[1/x]$, and let $g : \widehat{U} \to U$ be the canonical morphism. It suffices to prove that

$$(2.6) \qquad \begin{array}{c} g_*(\boldsymbol{\mu}_n)_{\widehat{U}} \simeq (\boldsymbol{\mu}_n)_U \quad \text{and} \\ R^q g_* \boldsymbol{\mu}_n = 0 \quad \text{if} \quad 1 \leq q \leq N. \end{array}$$

Indeed, the Leray spectral sequence

$$E_2^{pq} = H^p(U, R^q g_* \boldsymbol{\mu}_n) \Longrightarrow H^{p+q}(U', \boldsymbol{\mu}_n)$$

will give isomorphisms

$$H^q(U, \boldsymbol{\mu}_n) \xrightarrow{\sim} H^q(U', \boldsymbol{\mu}_n)$$

for $q \leq N$, and an exact sequence

$$0 \longrightarrow H^{N+1}(U, \boldsymbol{\mu}_n) \longrightarrow H^{N+1}(U', \boldsymbol{\mu}_n) \longrightarrow H^0(U, R^{N+1} g_* \boldsymbol{\mu}_n).$$

But we already know that $P(1)$ is true (1.3), and it is therefore enough to prove that $H^q(U, /u_n) = 0$ if $1 < q \leq N + 1$. We may thus replace A by $\widehat{A}$, which is isomorphic to a ring of formal power series $\widehat{A} \simeq k[[x_1, \ldots, x_r]]$, with k separably closed and $\overline{x} = x_r$. This case was treated in 1.2.

Let us therefore verify (2.6): the fact that $g_*(\boldsymbol{\mu}_n)_{\widehat{U}} \simeq (\boldsymbol{\mu}_n)_U$ is simply the assertion of (2.3) for $q = 0$. Recall that $R^q g_* \boldsymbol{\mu}_n$ is the sheaf associated with the presheaf $\mathscr{R}^q$, where

$$\mathscr{R}^q(V) = H^q(\widehat{V}, \boldsymbol{\mu}_n)$$

606 for $V \to U$ étale. It suffices to take V separated and of finite type; then V is a disjoint

union of integral schemes, and it suffices to consider such V. Let B be the normalization of A in the field of fractions of $\Gamma(V, O_V)$. Then B is a finite A-algebra (EGA IV 7.8.3 (vi)), and we see immediately that V is a regular open subset of Spec B and that $\widehat{V}$ is the corresponding open subset of Spec $\widehat{B}$, where $\widehat{B} = B \otimes_A \widehat{A}$. It follows that the hypotheses of 2.4 are satisfied for the open subset $V \subset \operatorname{Spec} B$ and for the morphism $B \to \widehat{B}$. Hence

$$\mathscr{R}^q(V) \simeq H^q(V, \boldsymbol{\mu}_n) \simeq H^q(\widehat{V}, \boldsymbol{\mu}_n)$$

for $q \leq N$. But the sheaf associated with the presheaf $V \mapsto H^q(V, \boldsymbol{\mu}_n)$ is plainly zero, which proves the lemma.

3. Purity

We shall adopt terminology analogous to that of XVI 3:

Definition 3.1. *A* regular pair (Y, X) of codimension c *is a closed immersion* $i : Y \to X$ *of locally Noetherian regular schemes such that for every* $y \in Y$ *one has* $\operatorname{codim}_y(Y, X) = c$ *(then there exists an open neighborhood* X' *of* y *in* X *and sections* $x_1, \dots, x_c \in \Gamma(X', O_X)$ *that generate the ideal of* Y*, and which form part of a regular system of parameters of* X *at every point of* $Y \cap X'$*). We write* $U = X - Y$*, and* $j : U \to X$ *for the corresponding open immersion.*

A *morphism* $(Y', X') \to (Y, X)$ of regular pairs of codimension c is a morphism $X' \to X$ such that $Y' = Y \times_X X'$. The notion of a regular triple, of a morphism of regular triples, etc., is defined in the analogous evident way. 607

Theorem 3.2. [124]. (Cohomological purity) *Let* (Y, X) *be a regular pair of codimension* $c > 0$*, where* X *is an excellent prescheme of equal characteristic. Let* F *be a locally constant sheaf on* X *of cyclic groups of order* n *prime to the characteristic of* X*. Then, in the notation of* (V 4, VIII 5),

$$\begin{aligned} &\mathscr{H}^q_Y(X, F) = (R^q i^!)F = 0 \quad \text{if} \quad q \neq 2c, \\ &(R^q j_*)j^*F = 0 \quad \text{if} \quad q \neq 0, 2c-1, \quad \textit{and} \\ &\mathscr{H}^{2c}_Y(X, F) = i^*(R^{2c-1} j_*)j^*F \end{aligned}$$

is a locally constant sheaf of cyclic groups of order n on Y.

Proof. (cf. XVI 3.7). We see immediately that

$$\mathscr{H}^q_Y(X, F) = 0 \quad \text{if} \quad q = 0, 1.$$

By (V 4.5),

$$\mathscr{H}^q_Y(X, F) = (R^q i^!)F \simeq i^*(R^{q-1} j_*)j^*F \quad \text{if} \quad q > 1,$$

so the assertions for the three sheaves are equivalent. Since the assertions are local on X for the étale topology, we may assume that $F = \boldsymbol{\mu}_n$. 608

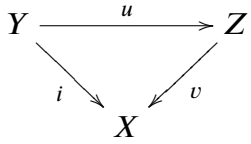
For the case $c = 1$, the calculation of the stalks (VIII 5) of $(R^q j_*)j^*\boldsymbol{\mu}_n$ reduces us to the situation considered in 2.1, whence

$$((R^q j_*)j^*\boldsymbol{\mu}_n)_{\overline{y}} = \begin{cases} \mathbf{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1 \end{cases}$$

for every geometric point $\overline{y}$ of Y. Since these sheaves vanish outside Y in any case, this proves the assertion for $q > 1$. For $q = 1$ $(= 2c - 1)$, we apply the criterion of (IX 2.13 (i)) and the explicit calculation of 1.3.

[124] Depends on resolution of singularities (cf. Introduction).

The case $c > i$ is treated by induction (cf. XVI 3.7): let (Y, X) be a regular pair of codimension $c > i$. By definition, it is clear that, locally on X, we can find a regular triple (Y, Z, X) in which (Z, X) has codimension 1 and (Y, Z) has codimension $c - 1$. Let



be the closed immersions. We have the spectral sequence

$$E_2^{pq} = (R^p u^!)(R^q v^!)F \Longrightarrow (R^{p+q} i^!)F.$$

By the induction hypothesis, $(R^q v^!)F = 0$ if $q \neq 2$, and is locally constant etc. ...if $q = 2$. Thus, again by the induction hypothesis, applied to the morphism u, $(R^p u^!)(R^1 v^!)F = 0$
609 if $(p, q) \neq (2c - 2, 2)$ and is locally constant etc. ...if $(p, q) = (2c - 2, 2)$, whence the result for the abutment $(R^{p+q} i^!)F$, C.Q.F.D.

3.3. It remains to determine the locally constant sheaves of (3.2) up to *canonical* isomorphism. For each prescheme S we introduce the locally constant sheaves $(\boldsymbol{\mu}_n^{\otimes r})_S =$ the r-fold tensor product of the sheaf $(\boldsymbol{\mu}_n)_S$ over the constant sheaf of rings $\mathbf{Z}_S$. We also set $(\boldsymbol{\mu}_n^{\otimes 0})_S = (\mathbf{Z}/n)_S$. It is immediately seen that if n is invertible on S, then $(\boldsymbol{\mu}_n^{\otimes r})_S$ is a locally constant sheaf of cyclic groups of order n. We have formulas of the form $(\boldsymbol{\mu}_n^{\otimes r}) \otimes (\boldsymbol{\mu}_n^{\otimes s}) = (\boldsymbol{\mu}_n^{\otimes r+s})$, and if $S' \to S$, then $(\boldsymbol{\mu}_n^{\otimes r})_{S'}$ is canonically isomorphic to the inverse image on S' of $(\boldsymbol{\mu}_n^{\otimes r})_S$. We shall often omit the symbol S when no confusion can arise.

Let (Y, X) be a regular pair of codimension 1, and suppose that Y is defined by an equation $x = 0$. We have the exact sequence of sheaves

$$0 \longrightarrow (\mathbf{G}_m)_X \longrightarrow j_*(\mathbf{G}_m)_U \longrightarrow \mathbf{Z}_Y \longrightarrow 0,$$

whence, by applying (1.3), canonical isomorphisms

$$(R^1 j_*)\boldsymbol{\mu}_n) \simeq j_*(\boldsymbol{\mu}\mathbf{G}_m)_U/(j_*(\mathbf{G}_m)_U)^n \simeq (\mathbf{Z}/n)_Y, \tag{3.3.1}$$

which plainly do not depend on the choice of x. Consequently, we have a canonical isomorphism

$$\varphi : \mathbf{Z}/n \xrightarrow{\sim} (R_i^{*!})\boldsymbol{\mu}_n \simeq i^*(R^1 j_*)\boldsymbol{\mu}_n. \tag{3.3.2}$$

610 THEOREM 3.4. [124]. *Let C be the family of regular pairs of excellent schemes of equal characteristic. There exists a unique function φ assigning to every regular pair (Y, X) of codimension c an isomorphism (called canonical) of sheaves*

$$\varphi(Y, X) = \varphi : \mathbf{Z}/n \xrightarrow{\sim} \mathscr{H}_Y^{2c}(X, \boldsymbol{\mu}_n^{\otimes c}) = (R^{2c} i^!)\boldsymbol{\mu}_n^{\otimes c}$$

such that φ satisfies the following conditions:

(i) *If $c = 1$, and if Y is defined in X by an equation $x = 0$, then φ is the morphism (3.3′).*

[124] Depends on resolution of singularities (cf. Introduction). This theorem may also be regarded as the conjunction of 3.2 and the theory of variance of Exp. XVIII by means of "local fundamental classes," where homomorphisms $\varphi(Y, X)$ (not necessarily bijective) are studied for Y locally a complete intersection in X.

(ii) *φ is compatible with smooth morphisms of pairs; that is, if $(Y', X') \to (Y, X)$ is a morphism, let $g : Y' \to Y$ be the induced morphism. The diagram of Y'-sheaves*

$$\begin{array}{ccc} \mathbf{Z}/n & \xrightarrow{\varphi(Y',X')} & (R^{2c}i^!)\boldsymbol{\mu}_n^{\boxtimes c} \\ \uparrow & & \uparrow \\ g^*(\mathbf{Z}/n) & \xrightarrow{g^*\varphi(Y,X)} & g^*(R^{2c}i^!)\boldsymbol{\mu}_n^{\boxtimes c} \end{array}$$

is commutative (which implies in particular that the right-hand vertical arrow is an isomorphism).

(iii) Transitivity: *let (Z, Y, X) be a regular triple, that is, a commutative diagram of closed immersions of regular preschemes etc.,*

$$\begin{array}{ccc} Z & \xrightarrow{v} & Y \\ & {\scriptstyle w}\searrow \quad \swarrow{\scriptstyle u} & \\ & X & \end{array} .$$

Let a, b, c be the codimensions of u, v, w, respectively (thus $c = a + b$). *We have* 611
an isomorphism

$$\varphi(Y, X) \otimes {\boldsymbol{\mu}_n}^{\otimes b} : {\boldsymbol{\mu}_n}^{\otimes b} \longrightarrow (R^{2a}u^!)_{\boldsymbol{\mu}_n}{}^{\otimes a} \otimes {\boldsymbol{\mu}_n}^{\otimes b} \simeq \simeq (R^{2a}u^!){\boldsymbol{\mu}_n}^{\otimes c}.$$

The spectral sequence $(R^pv^!)(R^qu^!)F \Rightarrow (R^{p+q}w^!)F$ gives an isomorphism

$$\xi : (R^{2b}v^!)(R^{2a}u^!){\boldsymbol{\mu}_n}^{\otimes c} \xrightarrow{\sim} (R^{2c}w^!){\boldsymbol{\mu}_n}^{\otimes c}.$$

We thus obtain a diagram of isomorphisms

$$\begin{array}{ccc} \mathbf{Z}/n & \xrightarrow{\varphi(Z,Y)} & (R^{2b}v^!)\boldsymbol{\mu}_n \boxtimes b \\ \downarrow{\scriptstyle \varphi(Z,X)} & & \downarrow{\scriptstyle \eta=(R^{2b}v^!)((Y,X)\boxtimes\boldsymbol{\mu}_n^{\boxtimes b})} \\ (R^{2c}w^!)\boldsymbol{\mu}_n & \longleftarrow & (R^{2b}v^!(R^{2a}u^!)\boldsymbol{\mu}_n^{\boxtimes c}) \end{array} \tag{3.4.1}$$

and the transitivity assertion is that this diagram is commutative.

Proof. The proof of existence and uniqueness proceeds by induction on the codimension c. For $c = 1$, we define the morphism φ locally by (3.3.1). Since this morphism does not depend on the choice of the parameter x, gluing gives an isomorphism φ for every (Y, X) of codimension 1; and it is clear from the definition of the morphism and from 1.3 that the φ so constructed satisfies (ii). Transitivity does not enter when $c = 1$.

Now suppose that existence and uniqueness have already been proved when the 612
codimension is $< c$, and that $c > 1$. Applying the induction hypothesis, we immediately find that in the situation of 3.4 (iii), the three arrows of (3.4.1) that are already defined, i.e. $\varphi(Z, Y)$, ξ, η, are compatible with morphisms of regular triples. In this situation set

$$\Psi(Z, Y, X) = \xi\eta\varphi(Z, Y),$$

so Ψ too is compatible with morphisms of triples. Now, if (Z, X) is a regular pair of codimension $c > 1$, locally on X one can find a triple (Z, Y, X) with $0 < \operatorname{codim}(Y, X) < c$. It therefore remains only to prove that the arrow $\Psi(Z, Y, X)$ is independent of Y. Indeed, this will prove the uniqueness of φ in codimension c, and existence will then follow by gluing.

If (Z, Y, X) is arbitrary with $\operatorname{codim}(Y, X) > 1$, locally on X one can find a regular triple (Y, W, X) in which (W, X) has codimension 1. A transitivity diagram chase, which we leave to the reader, therefore reduces us to checking independence of Y when (Y, X) has codimension 1.

Let (Z, Y_0, X) and (Z, Y_1, X) be two such triples. It suffices to check the assertion fiber by fiber, and we are thus reduced to the case in which X is strictly local and Y_i is defined by an equation, say $f_i = 0$ $(i = 0, 1)$. Let $\overline{X} = \operatorname{Spec} O_X[t]$, $\overline{Z} = Z \times_X \overline{X}$, and let $\overline{Y}$ be the closed subscheme of $\overline{X}$ defined by the equation

$$(f_1 - f_0)t + f_0 = 0.$$

613 Then $\overline{Y}$ is plainly regular at the points lying over the sections $\epsilon_0 : \{t = 0\}$ and $\epsilon_1 : \{t = 1\}$ of $\overline{X}/X$. Let C be the closed subset (EGA IV 7.8.3 (iv)) of $\overline{Y}$ consisting of the points at which $\overline{Y}$ is not regular, and replace $\overline{X}, \overline{Y}, \overline{Z}$ by $\overline{X} - C, \overline{Y} - C, Z - C \cap \overline{Z}$, respectively. Then $(\overline{Z}, \overline{Y}, \overline{X})$ is a regular triple, and the sections ϵ_i give morphisms of triples

$$(Z, Y_i, X) \longrightarrow (\overline{Z}, \overline{Y}, \overline{X}) \quad (i = 0, 1).$$

Let $F = (R^{2c}w^!)\boldsymbol{\mu}_n{}^{\otimes c}$ be the sheaf on Z. It is a locally constant sheaf isomorphic to $\mathbf{Z}/n$ (by either of the isomorphisms $\Psi(Z, Y_i, X)$), and one sees immediately that the sheaf $\overline{F} = (R^{2c}\overline{w}^!)\boldsymbol{\mu}_n{}^{\otimes c}$ on $\overline{Z}$ is the inverse image on $\overline{Z}$ of F. Since the fibers of $\overline{Z} \to Z$ are connected and nonempty and $\overline{Z} \to Z$ is smooth, hence open, it follows that the induced morphism

$$H^0(Z, F) \longrightarrow H^0(\overline{Z}, \overline{F})$$

is *bijective*; hence each of the morphisms

$$(*) \qquad H^0(Z, F) \longrightarrow H^0(\overline{Z}, \overline{F}) \begin{cases} \xrightarrow{\epsilon_0^*} H^0(Z, F) \\ \xrightarrow{\epsilon_1^*} H^0(Z, F) \end{cases}$$

is bijective, and consequently $\epsilon_0^* = \epsilon_1^*$.

To give an isomorphism of a sheaf F with $\mathbf{Z}/n$ is equivalent to giving a global section (the image of 1) that is a generator. Let $\overline{\alpha}, \alpha_i$ be the global sections of $\overline{F}, F$ given by the
614 isomorphisms $\Psi(\overline{Z}, \overline{Y}, \overline{X}), \Psi(Z, Y_i, X)$, respectively. Since $\Psi(.,.,.)$ is compatible with morphisms, the diagram $(*)$ implies that

$$\alpha_0 = \epsilon_0^*(\overline{\alpha}) = \epsilon_1^*(\overline{\alpha}) = \alpha_1,$$

whence $\Psi(Z, Y_0, X) = \Psi(Z, Y_1, X)$, C.Q.F.D.

4. Local acyclicity of a regular morphism

THEOREM 4.1. [124] (local acyclicity). *Let $g : X' \to X$ be a regular morphism* (EGA IV 6.8.1) *of excellent schemes of equal characteristic. Then g is universally locally acyclic for n prime to the characteristic.*

[124] Depends on resolution of singularities (cf. Introduction).

Proof. Since the hypothesis is stable under base change of finite type, it follows at once from XV 1.13 ii) that it suffices to prove that g is locally acyclic for n. Apply the criterion of XV 1.17: let $\overline{x}'$ be a geometric point of X' and $\overline{x}$ the corresponding geometric point of X. It follows immediately from the definition of a regular morphism that the morphism of strict localizations at these points induced by g is again regular. We are thus reduced to the case in which $X = \operatorname{Spec} A$ and $X' = \operatorname{Spec} A'$ are strictly local, g is a local morphism, and by XV 1.17 it suffices to prove that the geometric fibers of g are acyclic 615
for n. Replacing X by an integral closed subscheme and X' by the corresponding closed subscheme reduces us to the case of the geometric generic fiber $X'_{\overline{x}}$. Take $\overline{x} = \operatorname{Spec} \overline{K}$, where $\overline{K}$ is the separable closure of the field of rational functions on X. Then

$$\overline{x} = \varprojlim_{\alpha} U_\alpha$$

where U_α ranges over a filtered category of connected étale X-schemes, which may be assumed *regular* (EGA IV 7.8.3 (v)). Let B_α be the normalization of A in the field of rational functions $R(U_\alpha)$. Set $U'_\alpha = X' \times_X U_\alpha$ and $B'_\alpha = A' \otimes_A B_\alpha$. Then U_α is a regular open subscheme of $\operatorname{Spec} B_\alpha$ and $B_\alpha \to B'_\alpha$ is regular, so by (2.3)

$$H^q(U_\alpha, \boldsymbol{\mu}_n) \xrightarrow{\sim} H^q(U'_\alpha, \boldsymbol{\mu}_n)$$

for every q. Now the fiber $X'_{\overline{x}}$ is the limit of the U'_α, hence

$$\begin{aligned} H^q(X'_{\overline{x}}, \boldsymbol{\mu}_n) &= \varinjlim_{\alpha} H^q(U'_\alpha, \boldsymbol{\mu}_n) \\ &= \varinjlim_{\alpha} H^q(U_\alpha, \boldsymbol{\mu}_n) \\ &= H^q(\overline{x}, \boldsymbol{\mu}_n) = 0 \quad \text{if} \quad a > 0, \quad \text{C.Q.F.D.}. \end{aligned}$$

Corollary 4.2. *(base change by a regular morphism). Let*

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{g} & Y' \end{array}$$

be a Cartesian diagram, where $g : Y' \to Y$ is a regular morphism of excellent preschemes of equal characteristic, and f is quasi-compact and quasi-separated. Then for every abelian 616
torsion sheaf F whose torsion is prime to the characteristic, the base-change morphisms (XII 4.2)

$$g^*(R^q f_*)F \longrightarrow (R^q f'_*)g'^* F$$

are bijective for every $q \geq 0$.

This follows from 4.1 and XVI 1.1.

From 4.1 and XV 1.17 we immediately deduce

Corollary 4.3. *Let A be a strictly local excellent ring of equal characteristic, and let $f : X \to \operatorname{Spec} A$ be a morphism. Let $A \to A'$ be a regular morphism, where A' too is strictly local, excellent, and of equal characteristic. Denote by a' the base change induced by $A \to A'$. For every torsion sheaf F on X, of torsion prime to the characteristic of A, we have*

$$H^q(X, F) \xrightarrow{\sim} H^q(X', F')$$

for every $q \geq 0$.

This corollary applies in particular when A' is the completion $\widehat{A}$ of A. (In this case, the hypothesis "A strictly local" can moreover be replaced by "A henselian local," as is seen immediately.)

5. Finiteness theorem

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THEOREM 5.1. [124] (finiteness). *Let $f : X \to S$ be a morphism of finite type between excellent schemes of equal characteristic, and let F be a constructible abelian torsion sheaf on X, of torsion prime to the characteristic. Then $(R^q f_*)F$ is likewise constructible for every q.*

This result generalizes the theorem of XVI 5.1. The proof is nearly the same: one applies the purity result 3.2 instead of XVI 3.7.

COROLLARY 5.2. *Let A be an excellent strictly local ring of equal characteristic and let $f : X \to \operatorname{Spec} A$ be a morphism of finite type. Let F be a constructible abelian sheaf on X, of torsion prime to the characteristic of A. Then the groups $H^q(X, F)$ are finite for every q.*

6. Cohomological dimension of affine morphisms

Let $f : X \to Y$ be a morphism of finite type between excellent preschemes, and let F be a sheaf on X. Recall that we defined (XIV 2.2) an integer $\delta(F)$ when Y is local. We
618 may therefore regard δ as a function $\delta(F, f)$ on an arbitrary Y, by setting

$$\delta(F, f, y) = \delta(F'), \tag{6.0}$$

where $Y' = \operatorname{Spec} O_{Y,y}$, $X' \to Y'$ is the morphism obtained from f by the base change $Y' \to Y$, and F' is the inverse-image sheaf of F on X'.

We have the following result, which generalizes XVI 3.1:

THEOREM 6.1. [124]. *Let $f : X \to Y$ be an affine morphism of finite type, where Y is an excellent scheme of equal characteristic, and let F be an abelian torsion sheaf on X. Then $R^q f_* F$ vanishes at every point $y \in Y$ such that $\delta(F, f, y) < q$.*

The local variant, plainly equivalent to 6.1, is the following:

THEOREM 6.1 BIS. [124]. *Let A be a strictly local excellent ring of equal characteristic, and let $f : X \to Y = \operatorname{Spec} A$ be an affine morphism of finite type. Let F be a torsion sheaf on X. Then*

$$H^q(X, F) = 0 \quad \text{if} \quad q > \delta(F).$$

Recall (X 5.1) that assertions of this kind are of no interest for F of p-torsion, where
619 p is the characteristic. We may therefore suppose that F is of torsion prime to the characteristic.

We have the following corollary:

COROLLARY 6.2. *Let A be a henselian, excellent ring of equal characteristic, with residue field k, and let $U \subset \operatorname{Spec} A$ be an affine open subscheme. Let $\ell \in \mathbf{P}$. We have*

$$\operatorname{cd}_\ell U \leq \operatorname{cd}_\ell k + \dim A.$$

Indeed, let $\widetilde{A}$ be the strict localization of A, which is an infinite Galois étale covering with group $G = G(\overline{k}/k)$, and let $\widetilde{U}$ be the affine open subscheme of $\operatorname{Spec} \widetilde{A}$ induced by U. Applying the Hochschild–Serre spectral sequence (VIII 8.4)

$$H^p(G, H^q(U, .)) \Longrightarrow H^{p+q}(U, .)$$

[124] Depends on resolution of singularities (cf. Introduction).

reduces us to treating the case $A = \widetilde{A}$, and this is then a special case of 6.1 bis.

With the notation above, let $R(A)$ be the ring of rational functions on A. We have $\operatorname{Spec} R(A) = \varprojlim_\alpha U_\alpha$, where U_α is an affine open subscheme of $\operatorname{Spec} A$. The theory of passage to the limit therefore gives

COROLLARY 6.3. *Let A be henselian, excellent, and of equal characteristic, with residue field k. Let $R(A)$ be the field of rational functions on* $\operatorname{Spec} A$. *Then, for $\ell \in \mathbf{P}$, we have*

$$\operatorname{cd}_\ell R(A) \leq \operatorname{cd}_\ell k + \dim A$$

with equality if ℓ is invertible in k, and if moreover $\ell \neq 2$ or k is not orderable. 620

Equality in the latter case follows from X 2.4.

Finally, applying X 4.2, X 4.4, we find

COROLLARY 6.4. *Let X be a noetherian excellent scheme of equal characteristic, and let $R(X)$ be the ring of rational functions on X. Let $\ell \in \mathbf{P}$ be invertible on X, and suppose that $\ell \neq 2$ or that no residue field of X is orderable. Then*

$$\operatorname{cd}_\ell(X) \leq \operatorname{cd}_\ell R(X) + \dim X \leq 2 \operatorname{cd}_\ell R(X).$$

Proof of 6.1. We begin with a reduction analogous to that of XIV 3. Induction on δ: Clearly, 6.1 for $\delta \leq d$ is equivalent to (6.1 bis) for $\delta \leq d$. For $\delta = 0$ this is trivial. We treat the case $\delta \leq 1$ in the form 6.1 bis. Since F is the filtered direct limit of its constructible subsheaves (IX 2.9 (iii)), we may suppose that F is constructible; then the support of F is contained in a closed subset $Z \subset X$ with $\delta(z) \leq 1$ for every $z \in Z$ (XIV 2.2). Each reduced irreducible component Z_i of Z is then either an affine scheme of finite type and dimension ≤ 1 over the closed point y of Y, or a scheme quasi-finite over a reduced closed subscheme C of Y of dimension 1. In the latter case, $C = \operatorname{Spec} B$, where B is a strictly local integral domain of dimension 1. Such a scheme Z_i is the spectrum of a strictly local integral domain of dimension ≤ 1, or of a field finite over the fraction field of B. Thus in all cases the cohomological dimension of Z_i is at most 1, by X 2.3 621 and IX 5.7. We have an exact sequence

$$0 \longrightarrow F \longrightarrow \prod_i F_i \longrightarrow \epsilon \longrightarrow 0$$

where F_i is the restriction of F to Z_i, from which the result for Z follows.

Suppose now that the theorem has been proved for the values $\delta < d$, and let us prove it for $\delta(F) \leq d$, assuming $d \geq 2$.

LEMMA 6.5. *It suffices to treat the case where $X = \mathbf{E}^1_Y = \operatorname{Spec} O_Y[t]$ is affine space of dimension* 1, *and where Y is strictly local.*

The proof is that of XIV 4.2: It is clear that one may suppose Y strictly local, and that $\delta(F) \leq d$. We must prove that $H^q(X, F) = 0$ if $q > d$. We may embed X in $\mathbf{E}^N_Y$, so it suffices to treat the case $X = \mathbf{E}^N_Y$. Induction on N: Consider the diagram

$$\begin{array}{ccc} \mathbf{E}^N_Y & \xrightarrow{\ g\ } & \mathbf{E}^{N-1}_Y \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle h} & \\ & Y & \end{array}$$

so that $\mathbf{E}_Y^N$ is affine space of dimension 1 over $\mathbf{E}_Y^{N-1}$. Applying the induction hypothesis and the Leray spectral sequence

$$H^p(\mathbf{E}^{N-1}, R^q g_* F) \Longrightarrow H^{p+q}(\mathbf{E}^N, F)$$

shows that it suffices to prove that

$$\delta(R^q g_* F) \leq d - q.$$

622 This means that if $z \in \mathbf{E}^{N-1}$ is a point such that $\delta(z) \geq d - q$, then $R^q g_* F = 0$ at the point z. But the induction hypothesis on N applies to the morphism g, and hence $R^q g_* F = 0$ at the point z if $\delta(F, g, z) < q$. Thus it suffices to prove the inequality

$$\delta(F, g, z) + \delta(z) \leq \delta(F).$$

We leave this pleasure to the reader.

LEMMA 6.6. *To prove 6.1 for the values $\delta \leq d$, it suffices to prove the following: Let B be a complete normal local ring of equal characteristic and dimension d. Let $U \subset \operatorname{Spec} B$ be an affine open subset. Suppose that there exists an element $x \in \operatorname{rad} B$ which is invertible on U. Then*

$$H^q(U, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d.$$

This lemma is moreover a special case of Corollary 6.2.

Proof (cf. XIV 4.4). By 6.5, it suffices to treat the case where Y is strictly local and $X = \mathbf{E}_Y^1$. Consider the diagram

$$\begin{array}{ccc} \mathbf{E}_Y^1 & \xrightarrow{\ i\ } & \mathbf{P}_Y^1 \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & Y & \end{array},$$

where $\mathbf{P}_Y^1$ is the projective line and $\mathbf{E}_Y^1$ is obtained from $\mathbf{P}^1$ by removing the section Y_∞ at infinity. Let F be a torsion sheaf on $\mathbf{E}^1$, with $\delta(F) \leq d$. We have the spectral sequence

$$H^p(\mathbf{P}^1, R^q i_* F) \Longrightarrow H^{p+q}(\mathbf{E}^1, F).$$

623 Now the $R^q i_* F$ for $q > 0$ are concentrated on the section Y_∞, which is isomorphic to Y. Since Y is strictly local, it follows that $H^p(\mathbf{P}^1, R^q i_* F) = 0$ if $p > 0$ and $q > 0$. Furthermore, by XII 5.3, we have $H^p(\mathbf{P}^1, i_* F) = 0$ if $p > 2$. Since we wish to prove that $H^n(\mathbf{E}^1, F) = 0$ if $n > d$, and $d \geq 2$, it suffices to prove that

$$H^0(\mathbf{P}^1, R^q i_* F) = H^0(Y_\infty, R^q i_* F) = 0 \quad \text{if} \quad q > d.$$

This group is isomorphic to the stalk of $R^q i_* F$ at the point y_∞ at infinity of $\mathbf{P}^1$ in the closed fiber.

Now it suffices (VII 3.3 and IX 2.9 (iii)) to treat the case where F is constructible. Then, since $\delta(F) \leq d$, there exists a closed subscheme X of $\mathbf{P}^1$ with $\dim X \leq d$ such that X contains the support of F. Let $\operatorname{Spec} A = \widetilde{X}$ be the strict localization of X at the point y_∞, $\widetilde{U} = \widetilde{X} - \widetilde{X} \times_{\mathbf{P}^1} Y_\infty$, and let $\widetilde{F}$ be the sheaf induced by F on $\widetilde{U}$. Note that U is obtained from $\widetilde{X}$ by localizing with respect to an element $x \in \Gamma(X, 0_X)$. We have

$$(R^q i_* F)_{y_\infty} \simeq H^q(\widetilde{U}, \widetilde{F}).$$

Thus it suffices to prove that $H^q(\widetilde{U}, F) = 0$, if $q > d$, for every torsion sheaf $\widetilde{F}$ on $\widetilde{U}$.

We may now apply IX 5.6, with $\varphi = \delta$, and we find that it suffices to prove that

$$H^q(V, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > \delta(V)$$

for every V finite and integral over $\widetilde{U}$ and for invertible ℓ. Let $x : \overline{V} \to V$ be the normalization of V. We have the exact sequence 624

$$0 \longrightarrow (\mathbf{Z}/\ell)_V \longrightarrow x_*(\mathbf{Z}/\ell)_{\overline{V}} \longrightarrow C \longrightarrow 0$$

with $\delta(C) < \delta(V)$, and it follows from this (by induction on δ) that it suffices to take V *normal*.

Let B be the normalization of A in the field of rational functions on V. Then B is an excellent strictly local ring of equal characteristic, and V is an affine open subset of Spec B, obtained by localizing with respect to x. Let $\widehat{B}$ be the completion of B, and $\widehat{V}$ the induced open subset (which is obtained from Spec $\widehat{B}$ by localizing with respect to the image of x in $\widehat{B}$). By 4.3, we have

$$H^q(V, \mathbf{Z}/\ell) \simeq H^q(\widehat{V}, \mathbf{Z}/\ell).$$

We are therefore reduced to proving that

$$H^q(\widehat{V}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > \delta(V) = \dim B = \dim \widehat{B}.$$

By the induction hypothesis, the theorem is true if $\delta(V) < d$, so Theorem 6.1 bis applies in this situation. Consequently, it suffices to treat the case $\dim \widehat{B} = d$, and since $\widehat{B}$ is normal, we are in the situation of the lemma, whence the result.

6.7.0. We shall first prove 6.6 under additional hypotheses, when B has characteristic $p > 0$. Let A be an equidimensional, complete ring of equal characteristic, and let $k \subset A$ be a field such that $A/\operatorname{rad} A$ is a finite extension of k. We say that the k-algebra 625
A is *formally separable* if there exists a k-subalgebra of A of the form $k[[x_1, \dots, x_n]]$ such that the morphism Spec $A \to$ Spec $k[[x]]$ is finite and étale at the generic point of Spec $k[[x]]$ ($x = (x_1, \dots, x_n)$).

Lemma 6.7. *Let $k \subset B$ be a field over which $B/\operatorname{rad} B$ is finite. Under the induction hypothesis on d above, the conclusion of 6.6 holds if the hypotheses of 6.6 are satisfied and if, in addition, either of the following conditions is satisfied:*

(i) *k has characteristic 0.*
(ii) *There exists an element $x \in \operatorname{rad} B$ which is invertible on U and such that $\overline{B} = B/(x)$ is a formally separable k-algebra.*

Proof. Choose elements $x_1 = x, x_2, \dots, x_d$ of rad B such that $\{x_1, \dots, x_d\}$ generates a primary ideal for rad B. Then B is a finite A -algebra, where $A = k[[x_1, \dots, x_d]]$, and $\overline{B}$ is the $[[x_2, \dots, x_d]] = \overline{A}$-algebra $B/x_1 B$. Since in case (ii) $\overline{B}$ is analytically separable, we may in that case choose the elements $x_2, \dots, x_d$ in such a way that $\overline{B}$ becomes a generically étale $k[[x_2, \dots, x_d]]$-algebra. It then follows immediately that B is also a generically étale A-algebra, which is of course also true in case (i).

Let $b \in B$ be an element which generates the field extension $R(A)$ over $k((x))$. Let $\epsilon \in k[[x]]$ be the discriminant of the monic irreducible equation of b over $k[[x]]$. In case (ii), we may choose b so that x_1 does not divide ϵ. Write moreover Spec $B - U =$ 626
$V(x_1) \cup Y$, where Y contains no irreducible component of $V(x_1)$. Let C be the ring $\Gamma(Y, O_Y)$, where Y is given its induced reduced structure. Changing the $x_2, \dots, x_d$ if necessary, we may suppose that $V(x_d)$ is not contained in the image of Y in Spec A, and that, moreover, in case (ii), $V(x_d)$ is not contained in $V(\epsilon)$.

LEMMA 6.8. *Let* $A_0 = k[[x_1, \dots, x_{d-1}]]\,\{x_d\}$ *be the strict localization of* $k[[x_1, \dots, x_{d-1}]]\,[x_d]$ *at the point* $x_1 = x_2 = \cdots = x_d = 0$. *Under the conditions above, there exist an* A^0*-algebra* B^0, *an affine open subset* $U^0 \subset \operatorname{Spec} B^0$, *and an isomorphism* $A \otimes_{A^0} B^0 \simeq B$ *such that the open subset of* $\operatorname{Spec} B$ *induced by* U^0 *is* U.

The proof of this lemma is analogous to that of 1.5: it suffices to find A^0-algebras B^0, C^0 and isomorphisms $A \otimes_{A^0} B^0 \simeq B$, $A \otimes_{A^0} C^0 \simeq C$, because then there will exist ([3] 1.4) a unique (surjective) morphism $B^0 \to C^0$ inducing the morphism $B \to C$. We therefore take $U^0 = \operatorname{Spec} B^0 - \{(\operatorname{Spec} C^0) \cup V(x_1)\}$, and immediately verify that U is induced by U^0. The existence of C^0, and of B^0 in case (ii), is proved as in 1.5. The existence of B^0 in case (i) is a consequence of ([3] 5.1). (It may also be proved directly by applying Abhyankar's lemma to the ramification along $\{x_1 = 0\}$, and descent.)

We can now complete the proof of 6.7. Applying 4.3 once more, we find that

$$H^q(U, \mathbf{Z}/\ell) \simeq H^q(U^0, \mathbf{Z}/\ell),$$

627 from which it follows that it suffices to prove that the latter group is zero for $q > d$. Now $\operatorname{Spec} A^0$ is a limit of schemes affine and étale over $k[[x_1, \dots, x_{d-1}]]\,[x_d]$, so $\operatorname{Spec} B^0$ is a limit of schemes of finite type over $\operatorname{Spec} k[[x_1, \dots, x_{d-1}]]$, each fiber of which has dimension ≤ 1. Since $U^0 \subset \operatorname{Spec} B^0[1/x_1]$, we have

$$U^0 = \varprojlim U^0_\alpha$$

where U^0_α is affine of finite type over the scheme $V^0 = \operatorname{Spec} k[[x_1, \dots, x_{d-1}]]\,[1/x_1]$ and where each fiber of U^0_α / V^0 has dimension ≤ 1.

Let $f^0 : U^0 \to V^0$ be the structural morphism. Since the fibers of f^0_α all have dimension ≤ 1, for $v \in V^0$ we have

$$\delta(\mathbf{Z}/\ell, f^0_\alpha, v) \leq \dim O_{V^0, v} + 1 \leq d - 1$$

(because $\dim O_{V^0,v} \leq d - 2$). The induction hypothesis on d is therefore applicable to the morphism f^0, and, by applying 6.1, we find

$$R^q f^0_{\alpha *}(\mathbf{Z}/\ell) = 0 \quad \text{at the point} \quad v \quad \text{if} \quad q > \dim O_{V^0, v} + 1,$$

i.e. if $q > d - \dim \overline{v}$. Thus $R^q f^0_{\alpha *}(\mathbf{Z}/\ell) \neq 0$ at the point v implies that $\dim \overline{v} \geq d - q$. Therefore, for the inclusion $i : V^0 \to \operatorname{Spec} k[[x_1, \dots, x_{d-1}]]$ (cf. (6.0)), we have

$$\delta(R^q f^0_{\alpha *}(\mathbf{Z}/\ell)) \leq d - q,$$

628 and the induction hypothesis, in the form of 6.1 bis, applied to the morphism i, implies that

$$H^p(V^0, R^q f^0_{\alpha *}(\mathbf{Z}/\ell)) = 0 \quad \text{if} \quad p + q > d.$$

The Leray spectral sequence

$$E_2^{pq} = H^p(V^0, R^q f^0_*(\mathbf{Z}/\ell)) \Longrightarrow H^{p+q}(U^0, \mathbf{Z}/\ell)$$

therefore implies that $H^n(U^0_\alpha, \mathbf{Z}/\ell) = 0$ if $n > d$, and since U^0 is the limit of the U^0_α, we likewise find

$$H^n(U^0, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad n > d.$$

This completes the proof of 6.7, and hence of Theorem 6.1 in characteristic 0.

7. Affine morphisms — end of the proof

Recall that to complete the proof of 6.1 in characteristic $p > 0$ (still under the resolution hypothesis), we had reduced by 6.6 to proving the following assertion for $d \geq 2$:

LEMMA 7.1. *Let B be a complete normal local ring of equal characteristic, with separably closed residue field, and of dimension d. Let $U \subset \operatorname{Spec} B$ be an affine open subset such that there exists $x \in \operatorname{rad} B$ which is invertible on U. Then*

$$H^q(U, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d.$$

Moreover, by the induction hypothesis, we may suppose that 6.1 is true for the values 629
$\delta < d$, and that 7.1 has already been proved in the particular case 6.7 (ii).

First note that the condition that B be normal is not important. Indeed, let B be arbitrary, let $\pi : \overline{B} \to B$ be the normalization of B, and let $\overline{U} \subset \operatorname{Spec} \overline{B}$ be the inverse image of U. We have an exact sequence

$$0 \longrightarrow (\mathbf{Z}/\ell)_U \longrightarrow \pi_*(\mathbf{Z}/\ell)_{\overline{U}} \longrightarrow C \longrightarrow 0,$$

where the dimension of the support of C is $< d$, so $H^q(U, C) = 0$ if $q > d - 1$. Since $H^q(U, pi_* \cdot) \simeq H^q(\overline{U}, \cdot)$ (VIII 5.6), we see that

$$\begin{aligned} &H^q(U, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d \quad \text{is equivalent to} \\ &H^q(\overline{U}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d. \end{aligned} \tag{7.2}$$

7.3.0. We shall briefly recall, without proof, the properties of the notion of formal separability that we need. The reader may consult (EGA IV 18.11.10, 18.11.11): Let A be a complete noetherian local k-algebra whose residue field is a finite extension of k, and let A' denote the completion of the local ring $A \otimes_k k'$ (where $k' = k^{1/p}$). Then the extension $A \to A'$ is *radicial* (possibly infinite), as is easily seen — this is the well-known example of Nagata. It is therefore harmless, for the calculation of étale cohomology, to replace A by A' (VIII 1).

Now the condition that an equidimensional algebra A of dimension d be formally 630
separable, that is, be a finite generically étale extension of $k[[x_1, \dots, x_d]]$, can be expressed in terms of the module $\widehat{\Omega}^1_{A/k}$ (EGA IV 18.11.10), or of the Jacobian criterion (EGA 0_{IV} 22.6, IV 7.1)[124], and one sees that if A is not analytically separable, then the k'-algebra A' is not reduced. For the proof, one may for example reduce to the case where A is given by an equation $f = 0$, $f \in k[[x_1, \dots, x_{d+1}]]$, and apply the Jacobian criterion and the following lemma, whose pleasant proof we leave to the reader:

LEMMA 7.3. *Let $f(x_1, \dots, x_n) \in k[[x_1, \dots, x_n]]$ be a formal power series. Suppose that for each $i = 1, \dots, n$ there exists an invertible element $u_i \in k[[x_1, \dots, x_n]]$ such that $u_i f$ is a series in x_i^p and in the $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n$. Then there exists an invertible element $v \in k[[x_1, \dots, x_n]]$ such that*

$$vf \in k[[x_1^p, \dots, x_n^p]].$$

It follows easily that, by applying a finite number of times the procedure of replacing (A, k) by $(\overline{A}, k')$, where $\overline{A}'$ is the normalization of A', one is reduced to the situation where A is indeed formally separable. Similarly, one can prove that for a given $x \in \operatorname{rad} A$, the ring $(A/(x))_{\text{réd}}$ becomes a formally separable algebra by the same procedure, applied to the algebra A.

[124] The co-editor of this seminar has not succeeded in elucidating the meaning of this reference to a situation which seems quite different to him, nor the justification of the assertion that follows. A. G.

Applying VIII 1.1 and 7.2, we find

COROLLARY 7.4. *It suffices to prove (7.1) under the additional condition that B contain a field k over which $B/\operatorname{rad} B$ is finite, and such that the k-algebra $(B/(x))_{\text{réd}}$ is formally*
631 *separable.*

The proof of 7.1 in this case consists in a reduction to the particular case treated in 6.7 (ii):

With the notation of 7.1, 7.4, let

$$D(x) = \sum_i r_i X_i$$

be the divisor of x, so that each X_i is a closed irreducible subscheme of Spec B of codimension 1 which is formally separable over k (that is, such that $B/\mathfrak{P}(X_i)$ is a formally separable k-algebra, ∞ which is moreover equivalent to the assertion that $(B/(x))_{\text{réd}}$ is formally separable). Choose $y \in \operatorname{rad} B$ such that we have

$$D(y) = \sum_i (r_i - 1)X_i + \Gamma$$

where Γ contains none of the X_i, and consider the blow-up

$$f : Z \longrightarrow \operatorname{Spec} B$$

of the ideal (x, y). The spectra of the rings $B[y/x]$ and $B[x/y]$ form an affine open covering of Z.

Let B_1 be the strict localization of $B[x/y]$ at the maximal ideal generated by $(\operatorname{rad} B, x/y)$. For each closed point Q of the closed fiber of Spec $B[y/x] \to$ Spec B, let A_Q denote the strict localization of $B[y/x]$ at Q, and let U_Q be the affine open subset of Spec A_Q that is the inverse image of U.

632 LEMMA 7.5. *To prove 7.1 for the ring B, it suffices to prove that for $q > d (\geq 2)$ we have*

(i) $$H^q(\operatorname{Spec} B_1[y/x], \mathbf{Z}/\ell) = 0$$

and

(ii) $$H^q(U_Q, \mathbf{Z}/\ell) = 0 \quad \textit{for each } Q.$$

Proof. We have open immersions

$$U \hookrightarrow \operatorname{Spec} B[1/x] \hookrightarrow \operatorname{Spec} B[y/x],$$

and hence a diagram

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & \operatorname{Spec} B[y/x] \\ & {\scriptstyle k}\searrow \quad \swarrow{\scriptstyle j} & \\ & Z & \end{array}.$$

If Q is a closed point of the closed fiber of Spec $B[y/x] \to$ Spec B, we have $(R^q i_* \mathbf{Z}/\ell)_Q = 0$ if $q > d$, by hypothesis. For each point P of Spec $B[y/x]$ which is not closed in the closed fiber of Spec $B[y/x] \to$ Spec B, we have $\delta(F, i, P) < d$, where $F = (\mathbf{Z}/\ell)_U$ (notation of (6.0)). Thus, applying the induction hypothesis and 6.1 to the morphism i, we find that for the morphism Spec $B[y/x] \to$ Spec B we have

$$(R^q i_* \mathbf{Z}/\ell) \leq d - q.$$

Consequently, the induction hypothesis in the form 6.1 bis, applied to the latter morphism, implies that

$$H^p(\text{Spec } B[y/x], R^q i_* \mathbf{Z}/\ell) = 0$$

if $q > 0$ and $p > d - q$. With the Leray spectral sequence 633

$$E_2^{pq} = H^p(\text{Spec } B[y/x], R^q i_* \mathbf{Z}/\ell) \Longrightarrow H^{p+q}(U, \mathbf{Z}/\ell).$$

we are reduced, in proving 7.1, to proving that

$$H^p(\text{Spec } B[y/x], i_* \mathbf{Z}/\ell) = 0 \quad \text{if} \quad p > d.$$

We have an exact sequence

$$0 \longrightarrow \mathbf{Z}/\ell \longrightarrow i_*(\mathbf{Z}/\ell)_U \longrightarrow C \longrightarrow 0 \tag{7.6}$$

where $\delta(C) < d$. It therefore suffices (again by induction) to prove that

$$H^p(\text{Spec } B[y/x], \mathbf{Z}/\ell) = 0 \quad \text{if} \quad p > d. \tag{7.7}$$

Now the open immersion Spec $B[y/x] \xrightarrow{j} Z$ is obtained by removing from Z the closed subset C "at infinity," and in the other open subset Spec $B[x/y]$ of Z it is identified with the open subset $C = V(x/y)$. The $R^q j_* \mathbf{Z}/\ell$ are concentrated on C, and the morphism $C \to$ Spec B is a closed immersion. Consequently, we have $H^p(Z, R^q j_* \mathbf{Z}/\ell) = 0$ if p and $q > 0$. Since the closed fiber of $Z \to$ Spec B has dimension 1 and this morphism is proper, XII 5.3 bis implies that $H^p(Z, j_* \mathbf{Z}/\ell) = 0$ if $p > 2$. The spectral sequence

$$E_2^{pq} = H^p(Z, R^q j_* \mathbf{Z}/\ell) \Longrightarrow H^{p+q}(\text{Spec } B[y/x], \mathbf{Z}\ell)$$

therefore gives isomorphisms

$$H^q(\text{Spec } B[y/x], \mathbf{Z}/\ell) \simeq H^0(Z, R^q j_* \mathbf{Z}/\ell) \simeq (R^q j_* \mathbf{Z}/\ell)_P$$

if $q > 2$, where P is the point "at infinity" of Z. These stalks are none other (VIII 5) than 634

$$(R^q j_* \mathbf{Z}/\ell)_P \simeq H^q(\text{Spec } B_1[y/x], j_* \mathbf{Z}/\ell).$$

By (7.7), it therefore suffices to prove that these latter groups vanish if $q > d$ (≥ 2). We have an exact sequence analogous to (7.6) which, together with the induction hypothesis, shows that it suffices that

$$H^q(\text{Spec } B_1[y/x], \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d,$$

whence the lemma.

Proof of 7.5 (i).

Consider the closed subset $C = V(x/y)$ of Spec $B[x/y]$, with its induced reduced structure. It is immediate that the closed immersion of C into Spec B induced by the morphism $Z \to$ Spec B identifies C with the closed subscheme $X = \bigcup X_i$ of Spec B. Since

$$D(x) = \sum r_i X_i$$
$$D(y) = \sum (r_i - 1) X_i + \Gamma,$$

the element x/y generates the ideal $\mathfrak{J}(X_i)$ locally at the generic point of X_i, where the local ring is a discrete valuation ring because B is normal. Consequently, the morphism Spec $B[x/y] \to$ Spec B is an isomorphism in a neighborhood of such a point, and the element x/y vanishes to order 1 on each of the irreducible components C_i (corresponding to X_i) of C. 635

Since the closed subset C lifts to Spec B_1, this remains true if the ring $B[x/y]$ is replaced by B_1, and hence $B_1/(x/y)$ is a *complete* ring which is a formally separable k-algebra.

To prove 7.5 (i), we may first apply 4.3, which allows us to replace B_1 by its completion $\widehat{B}_1$, without affecting the ring $B_1/(x/y)$. By (7.2), we may moreover replace $\widehat{B}_1$ by its normalization $\overline{B}_1$. Since B_1 (and hence $\widehat{B}_1$) is already regular at the generic points of the C_i, the morphism Spec $\overline{B}_1 \to$ Spec $\widehat{B}_1$ is an isomorphism at these points, and so $\overline{B}_1/(x/y)$ is still a formally separable k-algebra. We are thus reduced to the special case 6.7 (ii), which proves the result.

Proof of 7.5 (ii).

Choose the point Q of the closed fiber of Spec $B[y/x] \to$ Spec B. The strict localization A_Q may be written as a limit

$$A_Q = \varinjlim A$$

where A ranges over a filtered category of rings that are étale and of finite type over $B[y/x]$. Let U_A be the affine open subset of Spec A that is the inverse image of U. Since x is invertible on U_A, the morphism $U_A \to$ Spec B is étale, and U_A is normal. Consequently, if C denotes the normalization of B in the field $R(A)$ of rational functions on A, we have an open immersion

$$U_A \longrightarrow \text{Spec } C.$$

636 It suffices (VII 5.7) to prove that for each A, the image of $H^q(U_A, \mathbf{Z}/\ell)$ in $H^q(U_Q, \mathbf{Z}/\ell)$ is zero if $q > d$. Since x is invertible on U_Q, the morphism $U_Q \to$ Spec A factors through Spec $A[1/f]$, if $f \in B$ is any element such that f is divisible by x in $B[y/x]$ and f/x does not vanish at the point Q. It thus suffices to prove that for a suitable such f, we have

$$H^q(U_{A,f}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad a > d,$$

where $U_{A,f} = U_A - V(f)$ is the affine open subset of Spec $A[1/f]$, which is also an affine open subset of Spec C.

Let $c \in k$ be an element such that $y/x + c$ does not vanish at Q, and let J be the ideal in B of the closed subset $\Gamma \cap X$. Then if

$$f \equiv y + cx (\text{mod } J^N) \quad N > 0, \tag{7.8}$$

the element f satisfies the condition above: Indeed, it is immediate that in Spec $B[y/x]$ we have $V(J) = V(x)$. Thus (7.8) does imply that f is divisible by x, whence

$$f/x \equiv y/x + c (\text{mod } J).$$

Since J vanishes at Q, and $y/x + c$ is nonzero at Q, this proves our assertion.

Now $\Gamma \cap X$ is a closed subset of Spec B of codimension 2. Consequently, we can
637 choose an f satisfying (7.8), call it z, such that $V(z) \cap V(y)$ is likewise of codimension 2. Also choose elements $x_3, \ldots, x_d$ such that $y, z, x_3, \ldots, x_d$ form a system of parameters of the ring B. Then B, and hence C as well, is a finite $k[[y, z, x_3, \ldots, x_d]]$-algebra.

Let C^0 be the normalization of $k[[y, z, x_3, \ldots, x_d]]$ in the separable closure of $k((y, z, x))$ in the field of rational functions $R(C)$, so that C is a radicial extension of C^0, and C^0 is a generically étale extension of $k((y, z, x))$ (where $x = x_3, \ldots, x_d$).

Choose a linear combination

$$f = az + by \quad (a, b \in k),$$

such that the morphism

$$\operatorname{Spec} C^0 \longrightarrow \operatorname{Spec} k[[z, y, x]]$$

is étale over the generic point of $V(f)$, and such that f/x does not vanish at Q in $\operatorname{Spec} B[y/x]$. It suffices to prove that

$$H^q(U_{A,f}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d,$$

and hence (VIII 1.1) to prove that

$$H^q(U^0_{A,f}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d,$$

where $U^0_{A,f}$ is the open subset of $\operatorname{Spec} C^0$ corresponding to the open subset $U_{A,f}$ of $\operatorname{Spec} C$, which is affine, as follows by descent (EGA IV 2.7.1 (xiii)). But by construction, the element f of C^0 is invertible on $U^0_{A,f}$, and $C^0/(f)$ is a formally separable k-algebra. We are therefore in the situation of 6.7 (ii), which completes the proof of 7.5 (ii), and hence that of 6.1.

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Séminaire de Géométrie Algébrique du Bois-Marie

Étale Cohomology

(SGA $4\frac{1}{2}$)

by P. Deligne

with the collaboration of J. F. Boutot,
A. Grothendieck, L. Illusie, and J. L. Verdier

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Chapter 1

Étale cohomology: the starting points

1 Grothendieck topologies

Grothendieck topologies first appeared as the foundation of Grothendieck's theory of descent (cf. SGA 1 VI, VIII); their use in the corresponding cohomology theories came later. We follow the same path here: by formalizing the classical notions of localization, local property, and patching (1.1, 1.2, 1.3), we obtain the general concept of a Grothendieck topology (1.6); to justify its introduction into algebraic geometry, we prove a faithfully flat descent theorem (1.4), generalizing the classical theorem 90 of Hilbert (1.5).

The reader will find a more complete, but concise, exposition of the formalism in Giraud [19]. M. Artin's notes, "Grothendieck topologies" [1] (Chapters I to III), also remain useful. The 866 pages of Exposés I to V of SGA 4 are valuable when one considers exotic topologies, such as the one giving rise to crystalline cohomology; in order to use the étale topology, which is so close to classical intuition, it is not essential to read them.

1.1 Sieves

Let X be a topological space and let $f : X \to \mathbb{R}$ be a real-valued function on X. The continuity of f is a local property; in other words, if f is continuous on every sufficiently small open subset of X, then f is continuous on all of X. To formalize the notion of a "local property," we introduce some definitions.

A set $\mathcal{U}$ of open subsets of X is called a *sieve* if, for every $U \in \mathcal{U}$ and $V \subset U$, one has $V \in \mathcal{U}$. A sieve is called *covering* if the union of all the open subsets belonging to it is equal to X.

Given a family $\{U_i\}$ of open subsets of X, the sieve generated by $\{U_i\}$ is, by definition, the set of open subsets U of X such that U is contained in one of the U_i.

A property $P(U)$, defined for every open subset U of X, is called *local* if, for every covering sieve $\mathcal{U}$ of every open subset U of X, $P(U)$ is true if and only if $P(V)$ is true for every $V \in \mathcal{U}$. For example, given $f : X \to \mathbb{R}$, the property "f is continuous on U" is local.

1.2 Sheaves

Let us make precise the notion of a function given locally on X.

by P. Deligne, written up by J.F. Boutot

1.2.1 The sieve point of view

Let $\mathcal{U}$ be a sieve of open subsets of X. A function given $\mathcal{U}$-locally on X consists of a function f_U on U for every $U \in \mathcal{U}$, such that if $V \subset U$ then $f_V = f_U|V$.

1.2.2 The Čech point of view

If the sieve $\mathcal{U}$ is generated by a family of open subsets U_i of X, then giving a function $\mathcal{U}$-locally amounts to giving a function f_i on each U_i such that $f_i|U_i \cap U_j = f_j|U_i \cap U_j$.

In other words, if $Z = \coprod U_i$, then giving a function $\mathcal{U}$-locally amounts to giving a function on Z that is constant on the fibres of the natural projection $Z \to X$.

1.2.3

The continuous functions form a sheaf. This means that for every covering sieve $\mathcal{U}$ of an open subset V of X, and every function $\{f_U\}$ given $\mathcal{U}$-locally such that each f_U is continuous on U, there exists a unique continuous function f on V such that $f|U = f_U$ for every $U \in \mathcal{U}$.

1.3 Stacks

Let us now make precise the notion of a vector bundle given locally on X.

1.3.1 The sieve point of view

Let $\mathcal{U}$ be a sieve of open subsets of X. A vector bundle given $\mathcal{U}$-locally on X consists of

a) a vector bundle E_U on each $U \in \mathcal{U}$,

b) if $V \subset U$, an isomorphism $\rho_{U,V} : E_V \xrightarrow{\sim} E_U|V$, satisfying

c) if $W \subset V \subset U$, the diagram

$$\begin{array}{ccc} E_W & \xrightarrow{\rho_{U,W}} & E_U|W \\ & \searrow^{\rho_{V,W}} & \uparrow{\scriptstyle \rho_{U,V}|W} \\ & & E_V|W \end{array}$$

commutes, that is, $\rho_{U,V} = (\rho_{U,V}$ restricted to $W) \circ \rho_{V,W}$.

1.3.2 The Čech point of view

If the sieve $\mathcal{U}$ is generated by a family of open subsets U_i of X, then giving a vector bundle $\mathcal{U}$-locally amounts to giving:

a) a vector bundle E_i on each U_i,

b) if $U_{ij} = U_i \cap U_j = U_i \times_X U_j$, an isomorphism $\rho_{ji} : E_i|U_{ij} \xrightarrow{\sim} E_j|U_{ij}$, such that

c) if $U_{ijk} = U_i \times_X U_j \times_X U_k$, the diagram

$$\begin{array}{ccc} E_i|U_{ijk} & \xrightarrow{\rho_{ki}|U_{ijk}} & E_k|U_{ijk} \\ & \searrow^{\rho_{ji}|U_{ijk}} & \uparrow{\scriptstyle \rho_{kj}|U_{ijk}} \\ & & E_j|U_{ijk} \end{array}$$

commutes, that is, $\rho_{ki} = \rho_{kj} \circ \rho_{ji}$ on U_{ijk}.

In other words, if $Z = \coprod U_i$ and $\pi : Z \to X$ is the natural projection, then giving a vector bundle $\mathcal{U}$-locally amounts to giving:

a) a vector bundle E on Z,

b) if x and y are two points of Z such that $\pi(x) = \pi(y)$, an isomorphism $\rho_{yx} : E_x \xrightarrow{\sim} E_y$ between the fibres of E at x and y, depending continuously on (x, y), and such that

c) if x, y, and z are three points of Z such that $\pi(x) = \pi(y) = \pi(z)$, then $\rho_{zx} = \rho_{zy} \circ \rho_{yx}$.

1.3.3

A vector bundle E on X defines a vector bundle $E_{\mathcal{U}}$ given $\mathcal{U}$-locally, namely the system of restrictions E_U of E to the objects of $\mathcal{U}$. The fact that the notion of a vector bundle is local can be expressed as follows: for every covering sieve $\mathcal{U}$ of X, the functor $E \mapsto E_{\mathcal{U}}$ from vector bundles on X to vector bundles given $\mathcal{U}$-locally is an equivalence of categories.

1.3.4

If in 1 one replaces "open subset of X" by "subset of X," one obtains the notion of a sieve of subspaces of X. In this setting there are also patching theorems. For example, let X be a space and let $\mathcal{C}$ be a sieve of subspaces of X generated by a locally finite closed covering of X. Then the functor $E \mapsto E_{\mathcal{C}}$ from vector bundles on X to vector bundles given $\mathcal{C}$-locally is an equivalence of categories.

In algebraic geometry, it is also useful to consider "sieves of spaces over X"; this is what we shall see in the next paragraph.

1.4 Faithfully flat descent

1.4.1

In the setting of schemes, the Zariski topology is not fine enough for the study of nonlinear problems, and one is led to replace the open immersions in the preceding definitions by more general morphisms. From this point of view, descent techniques appear as localization techniques. Thus the following descent statement can be expressed by saying that the properties under consideration are local for the faithfully flat topology (a morphism of schemes is called faithfully flat if it is flat and surjective).

Proposition 1.4.2. *Let A be a ring and B a faithfully flat A-algebra. Then:*

(i) A sequence $\Sigma = (M' \to M \to M'')$ of A-modules is exact if the sequence $\Sigma_{(B)}$ deduced from it by extension of scalars to B is exact.

(ii) An A-module M is of finite type (resp. of finite presentation, flat, locally free of finite rank, invertible (i.e. locally free of rank one)) if the B-module $M_{(B)}$ has the corresponding property.

Proof. (i) Since the functor $M \mapsto M_{(B)}$ is exact (by flatness of B), it suffices to show that if an A-module N is nonzero, then $N_{(B)}$ is nonzero. If N is nonzero, N contains a nonzero cyclic submodule $A/\mathfrak{a}$; hence $N_{(B)}$ contains the cyclic submodule $(A/\mathfrak{a})_{(B)} = B/\mathfrak{a}B$, which is nonzero by surjectivity of the structural morphism $\varphi : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ (if $V(\mathfrak{a})$ is nonempty, then $\varphi^{-1}(V(\mathfrak{a})) = V(\mathfrak{a}B)$ is nonempty).

(ii) For every family (x_i) of elements of $M_{(B)}$, there exists a finitely generated submodule M' of M such that $M'_{(B)}$ contains the x_i. If $M_{(B)}$ is of finite type and the x_i generate $M_{(B)}$, then $M'_{(B)} = M_{(B)}$, and hence $M' = M$ and M is of finite type. □

If $M_{(B)}$ is of finite presentation, the preceding argument yields a surjection $A^n \to M$. If N is the kernel of this surjection, then the B-module $N_{(B)}$ is of finite type, so N is of finite type and M is of finite presentation. The assertion for "flat" follows at once from (i); "locally free of finite rank" means "flat and of finite presentation," and the rank can be tested after extension of scalars to fields.

1.4.3

Let X be a scheme and let $\mathcal{S}$ be a class of X-schemes stable under fibre products over X. A class $\mathcal{U} \subset \mathcal{S}$ is a *sieve* on X (relative to $\mathcal{S}$) if, for every morphism $\varphi : V \to U$ of X-schemes with $U, V \in \mathcal{S}$ and $U \in \mathcal{U}$, one has $V \in \mathcal{U}$. The sieve *generated* by a family $\{U_i\}$ of X-schemes in $\mathcal{S}$ is the class of $V \in \mathcal{S}$ for which there exists a morphism of X-schemes from V to one of the U_i.

1.4.4

Let $\mathcal{U}$ be a sieve on X. A quasi-coherent module given $\mathcal{U}$-locally on X consists of

a) a quasi-coherent module E_U on every $U \in \mathcal{U}$,

b) for every $U \in \mathcal{U}$ and every morphism $\varphi : V \to U$ of X-schemes in $\mathcal{S}$, an isomorphism $\rho_\varphi : E_V \xrightarrow{\sim} \varphi^* E_U$, such that

c) if $\psi : W \to V$ is a morphism of X-schemes in $\mathcal{S}$, the diagram

$$\begin{array}{ccc} E_W & \xrightarrow{\rho_{\varphi\circ\psi}} & \psi^*\varphi^* E_U \\ & \searrow^{\rho_\psi} & \uparrow{\scriptstyle \psi^*\rho_\varphi} \\ & & \psi^* E_V \end{array}$$

commutes, that is, $\rho_{\varphi\circ\psi} = (\psi^*\rho_\varphi) \circ \rho_\psi$.

If E is a quasi-coherent module on X, we write $E_{\mathcal{U}}$ for the module given $\mathcal{U}$-locally whose value is $\varphi_U^* E$ on $\varphi : U \to X$ and such that, for every morphism $\psi : V \to U$, the restriction isomorphism ρ_ψ is the canonical isomorphism $E_V = (\varphi_U \circ \psi)^* E \xrightarrow{\sim} \psi^* \varphi_U^* E = \psi^* E_U$.

Theorem 1.4.5. *Let $\{U_i\} \in \mathcal{S}$ be a finite family of X-schemes flat over X such that X is the union of the images of the U_i, and let $\mathcal{U}$ be the sieve generated by $\{U_i\}$. Then the functor $E \mapsto E_{\mathcal{U}}$ is an equivalence from the category of quasi-coherent modules on X to the category of quasi-coherent modules given $\mathcal{U}$-locally.*

Proof. We treat only the case in which x is affine and $\mathcal{U}$ is generated by an affine X-scheme U faithfully flat over X. The reduction to this case is formal. Put $X = \mathrm{Spec}(A)$ and $U = \mathrm{Spec}(B)$.

If $U \to X$ admits a section, then X belongs to the sieve $\mathcal{U}$ and the assertion is clear. We reduce to this case.

A quasi-coherent module given $\mathcal{U}$-locally defines modules M', M'', and M''' on U, $U \times_X U$, and $U \times_X U \times_X U$, and isomorphisms $\rho : p^*M^\bullet \simeq M^\bullet$ for every projection morphism p between these spaces; this is a *cartesian diagram*

$$M^* : M' \rightrightarrows M'' \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M'''$$

over

$$U_* : U \;\substack{\leftarrow\\ \leftarrow}\; U \times_X U \;\substack{\leftarrow\\ \leftarrow\\ \leftarrow}\; U \times_X U \times_X U.$$

Conversely, M^* determines the module given $\mathcal{U}$-locally: for $V \in \mathcal{U}$ there exists $\varphi : V \to U$, and we put $M_U = \varphi^* M'$; for $\varphi_1, \varphi_2 : V \to U$ there is a natural identification $\varphi_1^* M' \simeq (\varphi_1 \times \varphi_2)^* M'' \simeq \varphi_2^* M'$, and by using M''' one sees that these identifications are compatible, so the definition is legitimate. In short, giving a module $\mathcal{U}$-locally is equivalent to giving a cartesian diagram M^* over U_*.

In algebraic terms, giving M^* amounts to giving a cartesian diagram of modules

$$M' \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}}\; M'' \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}\\ \xrightarrow{\partial_2}}\; M'''$$

over the diagram of rings

$$B \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}}\; B \otimes_A B \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}\\ \xrightarrow{\partial_2}}\; B \otimes_A B \otimes_A B$$

(more precisely, $\partial_i(bm) = \partial_i(b) \cdot \partial_i(m)$, the usual identities such as $\partial_0\partial_1 = \partial_0\partial_0$ hold, and "cartesian" means that the morphisms $\partial_i : M' \otimes_{B,\partial_i} (B \otimes_A B) \to M''$ and $M'' \otimes_{B\otimes_A B,\partial_i} (B \otimes_A B \otimes_A B) \to M'''$ are isomorphisms).

The functor $E \mapsto E_{\mathcal{U}}$ becomes the functor that associates to an A-module M the diagram

$$M^* = (M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M \otimes_A B \otimes_A B \otimes_A B).$$

Its right adjoint is the functor

$$(M' \rightrightarrows M'' \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M''') \longmapsto \ker(M' \rightrightarrows M''').$$

We must prove that the adjunction maps

$$M \to \ker(M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B)$$

and

$$\ker(M' \rightrightarrows M'') \otimes_A B \to M'$$

are isomorphisms. By (1.4.2.i), it suffices to prove this after a faithfully flat base change $A \to A'$ (with B becoming $B' = B \otimes_A A'$). Taking $A' = B$ reduces us to the case in which $U \to X$ admits a section. □

1.5 A special case: Hilbert's theorem 90

1.5.1

Let k be a field, let k' be a Galois extension of k, and let $G = \mathrm{Gal}(k'/k)$. Then the homomorphism

$$\begin{aligned} k' \otimes_k k' &\to \oplus_{\sigma \in G} k' \\ x \otimes y &\mapsto \{x \cdot \sigma(y)\}_{\sigma \in G} \end{aligned}$$

is bijective.

It follows that giving a module locally for the sieve on $\mathrm{Spec}(k)$ generated by $\mathrm{Spec}(k')$ is equivalent to giving a k'-vector space equipped with a semilinear action of G, that is:

a) a k'-vector space V',

b) for every $\sigma \in G$, an endomorphism φ_σ of the underlying group of V' such that $\varphi_\sigma(\lambda v) = \sigma(\lambda)\varphi_\sigma(v)$ for all $\lambda \in k'$ and $v \in V'$, satisfying the condition

c) for all $\sigma, \tau \in G$, $\varphi_{\tau\sigma} = \varphi_\tau \circ \varphi_\sigma$.

Let $V = V'^G$ be the group of invariants under this action of G. It is a k-vector space, and by Theorem 1.4.5 one has:

Proposition 1.5.2. *The inclusion of V in V' defines an isomorphism $V \otimes_k k' \xrightarrow{\sim} V'$.*

In particular, if V' has dimension 1 and $v' \in V$ is nonzero, then φ_σ is determined by the constant $c(\sigma) \in k'^\times$ such that $\varphi_\sigma(v') = c(\sigma)v'$, and condition c) becomes

$$c(\tau\sigma) = c(\tau) \cdot \tau(c(\sigma)).$$

By the proposition, there exists a nonzero invariant vector $v = \mu v'$, $\mu \in k'^\times$. Hence, for every $\sigma \in G$,

$$c(\sigma) = \mu \cdot \sigma(\mu^{-1}).$$

In other words, every 1-cocycle of G with values in $k'^\times$ is a coboundary:

Corollary 1.5.3. *One has* $\mathrm{H}^1(G, k'^\times) = 0$.

1.6 Grothendieck topologies

We now transpose the definitions of the preceding paragraphs to an abstract setting that encompasses both topological spaces and schemes.

1.6.1

Let $\mathcal{S}$ be a category and let U be an object of $\mathcal{S}$. A *sieve* on U is a subset $\mathcal{U}$ of $\mathrm{Ob}(\mathcal{S}/U)$ such that if $\varphi : V \to U$ belongs to $\mathcal{U}$ and $\psi : W \to V$ is a morphism in $\mathcal{S}$, then $\varphi \circ \psi : W \to U$ belongs to $\mathcal{U}$.

If $\{\varphi_i : U_i \to U\}$ is a family of morphisms, the sieve generated by the U_i is, by definition, the set of morphisms $\varphi : V \to U$ that factor through one of the φ_i.

If $\mathcal{U}$ is a sieve on U and $\varphi : V \to U$ is a morphism, the restriction $\mathcal{U}_V$ of $\mathcal{U}$ to V is, by definition, the sieve on V consisting of the morphisms $\psi : w \to V$ such that $\varphi \circ \psi : W \to U$ belongs to $\mathcal{U}$.

1.6.2

A *Grothendieck topology* on $\mathcal{S}$ consists of the choice, for every object U of $\mathcal{S}$, of a set $C(U)$ of sieves on U, called covering sieves, such that the following axioms hold:

a) The sieve generated by the identity of U is covering.

b) If $\mathcal{U}$ is a covering sieve on U and $V \to U$ is a morphism, then the sieve $\mathcal{U}_V$ is covering.

c) A locally covering sieve is covering. In other words, if $\mathcal{U}$ is a covering sieve on U and $\mathcal{U}'$ is a sieve on U such that, for every $V \to U$ belonging to $\mathcal{U}$, the sieve $\mathcal{U}'_V$ is covering, then $\mathcal{U}'$ is covering.

A category equipped with a Grothendieck topology is called a *site*.

1.6.3

Given a site $\mathcal{S}$, a presheaf on $\mathcal{S}$ is a contravariant functor $\mathscr{F}$ from $\mathcal{S}$ to the category of sets. For every object U of $\mathcal{S}$, the elements of $\mathscr{F}(U)$ are called sections of $\mathscr{F}$ over U. For every morphism $V \to U$ and every $s \in \mathscr{F}(U)$, we denote by $s|V$ (s restricted to V) the image of s in $\mathscr{F}(V)$.

If $\mathcal{U}$ is a sieve on U, a section given $\mathcal{U}$-locally consists, for every $V \to U$ belonging to $\mathcal{U}$, of a section $s_V \in \mathscr{F}(V)$ such that, for every morphism $W \to V$, $s_V|W = s_W$. The presheaf $\mathscr{F}$ is called a *sheaf* if, for every object U of $\mathcal{S}$, every covering sieve $\mathcal{U}$ on U, and every section $\{s_V\}$ given $\mathcal{U}$-locally, there exists a unique section $s \in \mathscr{F}(U)$ such that $s|V = s_V$ for every $V \to U$ belonging to $\mathcal{U}$.

Abelian sheaves are defined similarly by replacing the category of sets with the category of abelian groups. One shows that the category of abelian sheaves on $\mathcal{S}$ is an abelian category with enough injectives. A sequence $\mathscr{F} \xrightarrow{f} \mathscr{G} \xrightarrow{g} \mathscr{H}$ of sheaves is exact if, for every object U of $\mathcal{S}$ and every $s \in \mathscr{G}(U)$ such that $g(s) = 0$, there exists locally a t such that $f(t) = s$; i.e. there exists a covering sieve $\mathcal{U}$ on U and, for every $V \in \mathcal{U}$, a section t_V of $\mathscr{F}$ on V such that $f(t_V) = s|V$.

1.6.4 Examples

We have seen two examples above.

a) Let X be a topological space and let $\mathcal{S}$ be the category whose objects are the open subsets of X and whose morphisms are the natural inclusions. The Grothendieck topology on $\mathcal{S}$ corresponding to the usual topology of X is the one for which a sieve $\mathcal{U}$ on an open subset U of X is covering if the union of the open subsets belonging to the sieve equals U. It is clear that the category of sheaves on $\mathcal{S}$ is equivalent to the category of sheaves on X in the usual sense.

b) Let X be a scheme and let $\mathcal{S}$ be the category of schemes over X. The fpqc (faithfully flat quasi-compact) topology on $\mathcal{S}$ is the Grothendieck topology for which a sieve on an X-scheme U is covering if it is generated by a finite family of flat morphisms whose images cover U.

1.6.5 Cohomology

We shall always assume that the category $\mathcal{S}$ has a final object X. The global sections of an abelian sheaf $\mathscr{F}$, denoted by $\Gamma\mathscr{F}$ or $\mathrm{H}^0(X, \mathscr{F})$, form the group $\mathscr{F}(X)$. The functor $\mathscr{F} \mapsto \Gamma\mathscr{F}$ is a left exact functor from the category of abelian sheaves on $\mathcal{S}$ to the category of abelian groups; its derived functors (or satellites) are denoted by $\mathrm{H}^i(X, -)$. These cohomology groups represent the obstructions to passing from the local to the global. By definition, if $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ is an exact sequence of abelian sheaves, then there is a long exact cohomology sequence:

$$0 \longrightarrow \mathrm{H}^0(X, \mathscr{F}) \longrightarrow \mathrm{H}^0(X, \mathscr{G}) \longrightarrow \mathrm{H}^0(X, \mathscr{H}) \longrightarrow \mathrm{H}^1(X, \mathscr{F}) \longrightarrow \cdots$$

$$\ldots \longrightarrow \mathrm{H}^n(X, \mathscr{F}) \longrightarrow \mathrm{H}^n(X, \mathscr{G}) \longrightarrow \mathrm{H}^n(X, \mathscr{H}) \longrightarrow \mathrm{H}^{n+1}(X, \mathscr{F}) \longrightarrow \cdots$$

1.6.6

Given an abelian sheaf $\mathscr{F}$ on $\mathcal{S}$, an *$\mathscr{F}$-torsor* is a sheaf $\mathscr{G}$ equipped with an action $\mathscr{F} \times \mathscr{G} \to \mathscr{G}$ of $\mathscr{F}$ such that locally (after restriction to all the objects of a covering sieve of the final object X), $\mathscr{G}$ with its $\mathscr{F}$-action is isomorphic to $\mathscr{F}$ with the canonical translation action $\mathscr{F} \times \mathscr{F} \to \mathscr{F}$.

One can show that $\mathrm{H}^1(X, \mathscr{F})$ may be interpreted as the set of isomorphism classes of $\mathscr{F}$-torsors.

2 The étale topology

We specialize the definitions of the preceding chapter to the étale topology of a scheme X (2.1, 2.2, 2.3). When X is the spectrum of a field K, the corresponding cohomology agrees with the Galois cohomology of K (2.4).

2.1 The étale topology

We begin by recalling some facts about étale morphisms.

Definition 2.1.1. *Let A be a (commutative) ring. An A-algebra B is called étale if B is an A-algebra of finite presentation and the following equivalent conditions hold:*

a) For every A-algebra C and every square-zero ideal J of C, the canonical map

$$\hom_{A\text{-alg}}(B, C) \to \hom_{A\text{-alg}}(B, C/J)$$

is a bijection.

b) B is a flat A-module and $\Omega_{B/A} = 0$ (where $\Omega_{B/A}$ denotes the module of relative differentials).

c) Let $B = A[X_1, \dots, X_n]/I$ be a presentation of B. Then, for every prime ideal $\mathfrak{p}$ of $A[X_1, \dots, X_n]$ containing I, there exist polynomials $P_1, \dots, P_n \in I$ such that $I_{\mathfrak{p}}$ is generated by the images of $P_1, \dots, P_n$ and $\det(\partial P_i/\partial X_j) \notin \mathfrak{p}$.

(cf. [23, I] or [30, V])

A morphism of schemes $f : X \to S$ is called *étale* if, for every $x \in X$, there exist an affine open neighbourhood $U = \mathrm{Spec}(A)$ of $f(x)$ and an affine open neighbourhood $V = \mathrm{Spec}(B)$ of x in $X \times_S U$ such that B is an étale A-algebra.

2.1.2 Examples

a) If A is a field, an A-algebra B is étale if and only if it is a finite product of separable extensions of A.

b) If X and S are schemes of finite type over $\mathbb{C}$, a morphism $f : X \to S$ is étale if and only if its analytification $f^{\mathrm{an}} : X^{\mathrm{an}} \to Y^{\mathrm{an}}$ is a local isomorphism.

2.1.3 Consequences

a) (base change) If $f : X \to S$ is an étale morphism, then so is $f_{S'} : X \times_S S' \to S'$ for every morphism $S' \to S$.

b) If $f : X \to S$ and $g : Y \to S$ are two étale morphisms, then every S-morphism from X to Y is étale.

c) (descent) Let $f : X \to S$ be a morphism. If there exists a faithfully flat morphism $S' \to S$ such that $f_{S'} : X \times_S S' \to S'$ is étale, then f is étale.

2.1.4

Let X be a scheme, and let $\mathcal{S}$ be the category of étale X-schemes. By (2.1.3.c), every morphism in $\mathcal{S}$ is étale. The *étale topology* on $\mathcal{S}$ is the topology for which a sieve on U is covering if it is generated by a finite family of morphisms $\varphi_i : U_i \to U$ such that the union of the images of the φ_i covers U. The *étale site* of X, denoted by X_{et}, is the site defined by $\mathcal{S}$ with the étale topology.

2.2 Examples of sheaves

2.2.1 Constant sheaf

Let C be an abelian group and, for simplicity, suppose that X is noetherian. We denote by $\underline{C}_X$ (or simply C if there is no ambiguity) the sheaf defined by $U \mapsto C^{\pi_0(U)}$, where $\pi_0(U)$ is the (finite) set of connected components of U. The most important case will be $C = \mathbb{Z}/n$. Thus, by definition,

$$\mathrm{H}^0(X, \mathbb{Z}/n) = (\mathbb{Z}/n)^{\pi_0(X)}.$$

Moreover, $\mathrm{H}^1(X, \mathbb{Z}/n)$ is the set of isomorphism classes of $\mathbb{Z}/n$-torsors (1.6.6), in other words, of finite Galois étale coverings of X with group $\mathbb{Z}/n$. In particular, if X is connected and $\pi_1(X)$ is its fundamental group for a chosen base point, then

$$\mathrm{H}^1(X, \mathbb{Z}/n) = \hom(\pi_1(X), \mathbb{Z}/n).$$

2.2.2 Multiplicative group

We denote by $\mathbb{G}_{m,X}$ (or $\mathbb{G}_m$ if there is no ambiguity) the sheaf defined by $U \mapsto \Gamma(U, \mathscr{O}_U^\times)$; it is indeed a sheaf by the faithfully flat descent theorem (1.4.5). By definition,

$$\mathrm{H}^0(X, \mathbb{G}_m) = \mathrm{H}^0(X, \mathscr{O}_X)^\times;$$

in particular, if X is reduced, connected, and proper over an algebraically closed field k, then

$$\mathrm{H}^0(X, \mathbb{G}_m) = k^\times.$$

Proposition 2.2.3. *There is an isomorphism*

$$\mathrm{H}^1(X, \mathbb{G}_m) = \mathrm{Pic}(X),$$

where $\mathrm{Pic}(X)$ *is the group of isomorphism classes of invertible sheaves on* X*.*

Proof. Let $*$ be the functor that associates to an invertible sheaf $\mathscr{L}$ on X the following presheaf $\mathscr{L}^*$ on X_{et}: for $\varphi : U \to X$ étale,

$$\mathscr{L}^*(U) = \mathrm{Isom}_U(\mathscr{O}_U, \varphi^*\mathscr{L}).$$

By (1.4.2.i) and (1.4.5) (full faithfulness), this presheaf is a sheaf; indeed, it is a $\mathbb{G}_m$-torsor. One checks immediately that

a) the functor $*$ is compatible with (étale) localization;

b) it induces an equivalence from the category of trivial invertible sheaves (i.e. isomorphic to $\mathscr{O}_X$) to the category of trivial $\mathbb{G}_m$-torsors: $\mathscr{L}$ is trivial if and only if $\mathscr{L}^*$ is trivial.

Moreover, by (1.4.2.ii) and (1.4.5),

c) the notion of an invertible sheaf is local for the étale topology.

It follows formally from a), b), and c) that $*$ is an equivalence between the category of invertible sheaves on X and that of $\mathbb{G}_m$-torsors on X_{et}; it induces the desired isomorphism. The inverse equivalence is constructed as follows. If $\mathscr{T}$ is a $\mathbb{G}_m$-torsor, there exists a finite étale covering $\{U_i\}$ of X such that the torsors $\mathscr{T}/U_i$ are trivial; $\mathscr{T}$ is then trivial on every V étale over X belonging to the sieve $\mathcal{U} \subset X_{\mathrm{et}}$ generated by $\{U_i\}$. On every $V \in \mathcal{U}$, $\mathscr{T}|V$ corresponds to an invertible sheaf $\mathscr{L}_V$ (by b)), and the $\mathscr{L}_V$ form an invertible sheaf given $\mathcal{U}$-locally, $\mathscr{L}_{\mathcal{U}}$ (by a)). By c), the latter comes from an invertible sheaf $\mathscr{L}(\mathscr{T})$ on X, and $\mathscr{T} \mapsto \mathscr{L}(\mathscr{T})$ is the desired inverse of $*$. □

2.2.4 Roots of unity

For every integer $n > 0$, the sheaf of nth roots of unity, denoted by $\boldsymbol{\mu}_n$, is the kernel of the nth-power map on $\mathbb{G}_m$. If X is a scheme over a separably closed field k and n is invertible in k, the choice of a primitive nth root of unity $\zeta \in k$ defines an isomorphism $i \mapsto \zeta^i$ from $\mathbb{Z}/n$ to $\boldsymbol{\mu}_n$.

The relation between cohomology with coefficients in $\boldsymbol{\mu}_m$ and cohomology with coefficients in $\mathbb{G}_m$ is given by the cohomology exact sequence deduced from the following theorem.

Theorem 2.2.5 (Kummer theory)**.** *If n is invertible on X, the nth-power map on $\mathbb{G}_m$ is an epimorphism of sheaves. Hence there is an exact sequence*

$$0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0.$$

Proof. Let $U \to X$ be an étale morphism and let $a \in \mathbb{G}_m(U) = \Gamma(U, \mathscr{O}_U^\times)$. Since n is invertible on U, the equation $T^n - a = 0$ is separable; in other words, $U' = \operatorname{Spec}(\mathscr{O}_U[T]/(T^n - a))$ is étale over U. Moreover, $U' \to U$ is surjective and this element has an nth root on U', which proves the result. □

2.3 Fibres, direct images

2.3.1

A *geometric point* of X is a morphism $\bar{x} \to X$, where $\bar{x}$ is the spectrum of a separably closed field $k(\bar{x})$. By abuse of notation, we denote it by $\bar{x}$, leaving the morphism $\bar{x} \to X$ implicit. If x is the image of $\bar{x}$ in X, we say that $\bar{x}$ is centred at x. If $k(\bar{x})$ is an algebraic extension of the residue field $k(x)$, then $\bar{x}$ is called an *algebraic* geometric point of X.

An *étale neighbourhood* of $\bar{x}$ is a commutative diagram

$$\begin{array}{ccc} & & U \\ & \nearrow & \downarrow \\ \bar{x} & \longrightarrow & X, \end{array}$$

where $U \to X$ is étale.

The *strict localization* of X at $\bar{x}$ is the ring $\mathscr{O}_{X,\bar{x}} = \varinjlim \Gamma(U, \mathscr{O}_U)$, where the direct limit is taken over the étale neighbourhoods of $\bar{x}$. It is a strictly henselian local ring whose residue field is the separable closure in $k(\bar{x})$ of the residue field $k(x)$ of X at x. It plays the role of the local ring for the étale topology.

2.3.2

Given a sheaf $\mathscr{F}$ on X_{et}, the *fibre* of $\mathscr{F}$ at $\bar{x}$ is the set (resp. group, ...) $\mathscr{F}_{\bar{x}} = \varinjlim \mathscr{F}(U)$, where the direct limit is again taken over the étale neighbourhoods of X.

A homomorphism of sheaves $\mathscr{F} \to \mathscr{G}$ is a mono-/epi-/isomorphism if and only if the induced morphisms $\mathscr{F}_{\bar{x}} \to \mathscr{G}_{\bar{x}}$ on fibres have the corresponding property at every geometric point of X. If X is of finite type over an algebraically closed field, it suffices that this hold at the rational points of X.

2.3.3

If $f : X \to Y$ is a morphism of schemes and $\mathscr{F}$ is a sheaf on X_{et}, the *direct image* $f_*\mathscr{F}$ of $\mathscr{F}$ under f is the sheaf on Y_{et} defined by $f_*\mathscr{F}(V) = \mathscr{F}(X \times_Y V)$ for every V étale over Y. The functor $f_* : \operatorname{Sh}(X_{\text{et}}) \to \operatorname{Sh}(Y_{\text{et}})$

is left exact. Its right derived functors $\mathsf{R}^q f_*$ are called the higher direct images. If $\bar{y}$ is a geometric point of Y, then

$$(\mathsf{R}^q f_* \mathscr{F})_{\bar{y}} = \varinjlim \mathrm{H}^q(V \times_Y X, \mathscr{F}),$$

where the direct limit is taken over the étale neighbourhoods V of $\bar{y}$.

Let $\mathscr{O}_{Y,\bar{y}}$ be the strict localization of Y at $\bar{y}$, let $\widetilde{Y} = \mathrm{Spec}(\mathscr{O}_{Y,\bar{y}})$, and let $\widetilde{X} = X \times_Y \widetilde{Y}$. The sheaf $\mathscr{F}$ may be extended to $\widetilde{X}_{\mathrm{et}}$ (this is a special case of the general notion of inverse image) as follows. Let $\widetilde{U}$ be a scheme étale over $\widetilde{X}$. Then there exist an étale neighbourhood V of $\bar{y}$ and a scheme U étale over $X \times_Y V$ such that $\widetilde{U} = U \times_V \widetilde{Y}$; set

$$\mathscr{F}(\widetilde{U}) = \varinjlim \mathscr{F}\left(U \times_V V'\right),$$

where the direct limit is taken over the étale neighbourhoods V' of $\bar{y}$ that dominate V. With this definition,

$$(\mathsf{R}^q f_* \mathscr{F})_{\bar{y}} = \mathrm{H}^q(\widetilde{X}, \mathscr{F}).$$

The functor f_* has a left adjoint f^*, the "inverse image" functor. If $\bar{x}$ is a geometric point of X and $f(\bar{x})$ its image in Y, then $(f^*\mathscr{F})_{\bar{x}} = \mathscr{F}_{f(\bar{x})}$. This formula shows that f^* is exact. Consequently, f_* sends injective sheaves to injective sheaves, and the spectral sequence for the composite functor $\Gamma \circ f_*$ (resp. $g_* f_*$) gives the following theorem.

Theorem 2.3.4 (Leray spectral sequence)**.** *Let $\mathscr{F}$ be an abelian sheaf on X_{et} and let $f : X \to Y$ be a morphism of schemes (resp. let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes). There is a spectral sequence*

$$E_2^{pq} = \mathrm{H}^p(Y, \mathsf{R}^q f_* \mathscr{F}) \Rightarrow \mathrm{H}^{p+q}(X, \mathscr{F})$$

(resp. $E_2^{pq} = \mathsf{R}^p g_* \mathsf{R}^q f_* \mathscr{F} \Rightarrow \mathsf{R}^{p+q}(gf)_* \mathscr{F}$*).*

Corollary 2.3.5. *If $\mathsf{R}^q f_* \mathscr{F} = 0$ for every $q > 0$, then $\mathrm{H}^p(Y, f_* \mathscr{F}) = \mathrm{H}^p(X, \mathscr{F})$ (resp. $\mathsf{R}^p g_*(\mathscr{F}_* \mathscr{F}) = \mathsf{R}^p(gf)_* \mathscr{F}$) for every $p \geqslant 0$.*

This applies in particular in the following case:

Proposition 2.3.6. *Let $f : X \to Y$ be a finite morphism (or even, by passage to the limit, an integral morphism), and let $\mathscr{F}$ be an abelian sheaf on X. Then $\mathsf{R}^q f_* \mathscr{F} = 0$ for every $q > 0$.*

Indeed, let $\bar{y}$ be a geometric point of Y, let $\widetilde{Y}$ be the spectrum of the strict localization of Y at y, and let $\widetilde{X} = X \times_Y \widetilde{Y}$. By the preceding discussion, it suffices to prove that $\mathrm{H}^q(\widetilde{X}, \mathscr{F}) = 0$ for every $q > 0$. But $\widetilde{X}$ is the spectrum of a product of strictly henselian local rings (cf. [30, I]), and the functor $\Gamma(\widetilde{X}, -)$ is exact because every surjective étale $\widetilde{X}$-scheme admits a section. This proves the assertion.

2.4 Galois cohomology

For $X = \mathrm{Spec}(K)$, the spectrum of a field, we shall see that étale cohomology can be identified with Galois cohomology.

2.4.1

We begin with a topological analogy. If K is the function field of an integral affine algebraic variety $Y = \mathrm{Spec}(A)$ over $\mathbb{C}$, then $K = \varinjlim_{f \in A} A[1/f]$.

In other words, $X = \varprojlim U$, where U runs over the open subsets of Y. It is known that there exist arbitrarily small Zariski open subsets that are $K(\pi, 1)$ spaces for the classical topology. It is therefore natural to regard $\mathrm{Spec}(K)$ itself as a $K(\pi, 1)$, where π is the fundamental group (in the algebraic sense) of X, namely the Galois group of $\bar{K}/K$, where $\bar{K}$ is a separable closure of K.

2.4.2

More precisely, let K be a field, let $\bar{K}$ be a separable closure of K, and let $G = \mathrm{Gal}(\bar{K}/K)$ be the topological Galois group. To every finite étale K-algebra A (a finite product of separable extensions of K), associate the finite set $\hom_K(A, \bar{K})$. The Galois group G acts on this set through a discrete (hence finite) quotient. If $A = K[T]/(F)$, this set is identified with the set of roots of F in $\bar{K}$. Galois theory, in the form given to it by Grothendieck, says:

Proposition 2.4.3. *The functor*

$$\left\{ \begin{array}{c} K\text{-algebras} \\ \text{finite étale} \end{array} \right\} \to \left\{ \begin{array}{c} \text{finite sets on which} \\ G \text{ acts continuously} \end{array} \right\}$$

that associates $\hom_K(A, \bar{K})$ *to an étale algebra A is an anti-equivalence of categories.*

This yields an analogous description of sheaves for the étale topology on $\mathrm{Spec}(K)$:

Proposition 2.4.4. *The functor*

$$\left\{ \begin{array}{c} \text{étale sheaves} \\ \text{on } \mathrm{Spec}(K) \end{array} \right\} \to \left\{ \begin{array}{c} \text{sets on which} \\ G \text{ acts continuously} \end{array} \right\}$$

that associates to a sheaf $\mathscr{F}$ its fibre $\mathscr{F}_{\bar{K}}$ at the geometric point $\mathrm{Spec}(\bar{K})$ *is an equivalence of categories.*

The group G is said to act continuously on a set E if the stabilizer of every element of E is an open subgroup of G. The inverse functor is described in the evident way: let A be a finite étale K-algebra, let $U = \mathrm{Spec}(A)$, and let $U(\bar{K}) = \hom_K(A, \bar{K})$ be the corresponding G-set; then $\mathscr{F}(U) = \hom_{G\text{-set}}(U(\bar{K}), \mathscr{F}_{\bar{K}})$.

In particular, if $X = \mathrm{Spec}(K)$, then $\mathscr{F}(X) = \mathscr{F}_{\bar{K}}^G$. Restricting to abelian sheaves and passing to derived functors gives canonical isomorphisms

$$\mathrm{H}^q(X_{\mathrm{et}}, \mathscr{F}) = \mathrm{H}^q(G, \mathscr{F}_{\bar{K}})$$

2.4.5 Examples

a) The constant sheaf $\mathbb{Z}/n$ corresponds to $\mathbb{Z}/n$ with the trivial action of G.

b) The sheaf $\boldsymbol{\mu}_n$ of nth roots of unity corresponds to the group $\boldsymbol{\mu}_n(\bar{K})$ of nth roots of unity in $\bar{K}$, with the natural action of G.

c) The sheaf $\mathbb{G}_m$ corresponds to the group $\bar{K}^\times$ with the natural action of G.

3 The cohomology of curves

For topological spaces, unwindings using the Künneth formula and simplicial decompositions reduce the calculation of cohomology to the interval $I = [0, 1]$, for which $\mathrm{H}^0(I, \mathbb{Z}) = \mathbb{Z}$ and $\mathrm{H}^q(I, \mathbb{Z}) = 0$ for $q > 0$.

In our case, the unwindings lead to more complicated objects, namely curves over an algebraically closed field; in this chapter we calculate their cohomology. The situation is more complicated than in the topological case, because the cohomology groups vanish only for $q > 2$. The essential ingredient in the calculations is the vanishing of the Brauer group of the function field of such a curve (Tsen's theorem, 3.2).

3.1 The Brauer group

We first recall its classical definition.

Definition 3.1.1. *Let K be a field and let A be a finite-dimensional K-algebra. The algebra A is called a central simple algebra over K if the following equivalent conditions hold:*

a) A has no nontrivial two-sided ideal, and its centre is K.

b) There exists a finite Galois extension K'/K such that $A_{K'} = A \otimes_K K'$ is isomorphic to an algebra of square matrices over K'.

c) A is K-isomorphic to an algebra of square matrices over a division algebra with centre K.

Two such algebras are called equivalent if the division algebras associated with them by c) are K-isomorphic. If the algebras have the same dimension, this is equivalent to their being K-isomorphic. Tensor product induces, on passing to the quotient, an abelian group structure on the set of equivalence classes. This group is classically called the *Brauer group* of K and is denoted by $\mathrm{Br}(K)$.

3.1.2

We denote by $\mathrm{Br}(n, K)$ the set of K-isomorphism classes of K-algebras A such that there exists a finite Galois extension K' of K for which $A_{K'}$ is isomorphic to the algebra $M_n(K')$ of $n \times n$ square matrices over K'. By definition, $\mathrm{Br}(K)$ is the union of the subsets $\mathrm{Br}(n, K)$ for $n \in \mathbb{N}$. Let $\bar{K}$ be an algebraic closure of K and let $G = \mathrm{Gal}(\bar{K}/K)$. The set $\mathrm{Br}(n, K)$ is the set of "forms" of $M_n(\bar{K})$, and hence is canonically isomorphic to $\mathrm{H}^1\left(G, \mathrm{Aut}(M_n(\bar{K}))\right)$.

Every automorphism of $M_n(\bar{K})$ is inner. Consequently, $\mathrm{Aut}(M_n(\bar{K}))$ is identified with the projective linear group $\mathrm{PGL}(n, \bar{K})$, and there is a canonical bijection

$$\theta_n : \mathrm{Br}(n, K) \xrightarrow{\sim} \mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right).$$

On the other hand, the exact sequence

$$1 \to \bar{K}^\times \to \mathrm{GL}(n, \bar{K}) \to \mathrm{PGL}(n, \bar{K}) \to 1, \tag{3.1.2.1}$$

defines a coboundary operator

$$\Delta_n : \mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right) \to \mathrm{H}^2(G, \bar{K}^\times).$$

Composing θ_n and Δ_n gives a map

$$\delta_n : \mathrm{Br}(n, K) \to \mathrm{H}^2(G, \bar{K}^\times).$$

It is readily checked that the maps δ_n are compatible and define a group homomorphism

$$\delta : \mathrm{Br}(K) \to \mathrm{H}^2(G, \bar{K}^\times).$$

Proposition 3.1.3. *The homomorphism $\delta : \mathrm{Br}(K) \to \mathrm{H}^2(G, \bar{K}^\times)$ is bijective.*

This follows from the next two lemmas.

Lemma 3.1.4. *The map $\Delta_n : \mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right) \to \mathrm{H}^2(G, \bar{K}^\times)$ is injective.*

By [36], Corollary to Proposition I-44, it suffices to check that whenever the exact sequence (3.1.2.1) is twisted by an element of $\mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right)$, the H^1 of the middle group is trivial. This middle group is the group of $\bar{K}$-points of the multiplicative group of a central simple algebra A of rank n^2 over K. To prove that $\mathrm{H}^1(G, A_{\bar{K}}^{\times}) = 0$, interpret $A^{\times}$ as the automorphism group of the free A-module L of rank 1, and H^1 as the set of "forms" of L—A-modules of rank n^2 over K, which are automatically free.

Lemma 3.1.5. *Let $\alpha \in \mathrm{H}^2(G, \bar{K}^{\times})$, let K' be a finite extension of K contained in $\bar{K}$, let $n = [K' : K]$, and let $G' = \mathrm{Gal}(\bar{K}/K')$. If the image of α in $\mathrm{H}^2(G', \bar{K}^{\times})$ is zero, then α belongs to the image of Δ_n.*

First note that

$$\mathrm{H}^2(G', \bar{K}^{\times}) \simeq \mathrm{H}^2\left(G, (\bar{K} \otimes_K K')^{\times}\right).$$

(Geometrically, if $x = \mathrm{Spec}(K)$, $x' = \mathrm{Spec}(K')$, and $\pi : x' \to x$ is the canonical morphism, then $\mathsf{R}^q\pi_*(\mathbb{G}_{m,x'}) = 0$ for $q > 0$ and consequently $\mathrm{H}^q(x', \mathbb{G}_{m,X'}) \simeq \mathrm{H}^q(x, \pi_*\mathbb{G}_{m,X'})$ for $q \geqslant 0$.)

Moreover, the choice of a basis of K' as a vector space over K defines a homomorphism

$$(\bar{K} \otimes_K K')^{\times} \to \mathrm{GL}(n, \bar{K})$$

that sends an element x to the endomorphism of $\bar{K} \otimes_K K'$ given by multiplication by x. There is then a commutative diagram with exact rows:

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \bar{K}^{\times} & \longrightarrow & (\bar{K} \otimes_K K')^{\times} & \longrightarrow & (\bar{K} \otimes_K K')^{\times}/\bar{K}^{\times} & \longrightarrow & 1 \\
 & & \| & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \bar{K}^{\times} & \longrightarrow & \mathrm{GL}(n, \bar{K}) & \longrightarrow & \mathrm{PGL}(n, \bar{K}) & \longrightarrow & 1
\end{array}$$

The lemma follows from the commutative diagram obtained by passing to cohomology:

$$\begin{array}{ccccc}
\mathrm{H}^1\left(G, (\bar{K} \otimes_K K')^{\times}/\bar{K}^{\times}\right) & \longrightarrow & \mathrm{H}^2(G, \bar{K}^{\times}) & \longrightarrow & \mathrm{H}^2\left(G, (\bar{K} \otimes_K K')^{\times}\right) \\
\downarrow & & \| & & \\
\mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right) & \xrightarrow{\Delta_n} & \mathrm{H}^2(G, \bar{K}^{\times}). & &
\end{array}$$

Proposition 3.1.6. *Let K be a field, let $\bar{K}$ be an algebraic closure of K, and let $G = \mathrm{Gal}(\bar{K}/K)$. Suppose that $\mathrm{Br}(K') = 0$ for every finite extension K' of K. Then:*

i) $\mathrm{H}^q(G, \bar{K}^{\times}) = 0$ *for every* $q > 0$.

ii) $\mathrm{H}^q(G, \mathscr{F}) = 0$ *for every torsion G-module $\mathscr{F}$ and every $q \geqslant 2$.*

(For the proof, cf. [36].)

3.2 Tsen's theorem

Definition 3.2.1. *A field K is called C_1 if every nonconstant homogeneous polynomial $f(x_1, \ldots, x_n)$ of degree $d < n$ has a nontrivial zero.*

Proposition 3.2.2. *If a field K is C_1, then $\mathrm{Br}(K) = 0$.*

We must show that every division algebra D with centre K and finite over K is equal to K. Let r^2 be the degree of D over K and let $\mathrm{Nrd} : D \to K$ be the reduced norm.

(Locally for the étale topology on K, D is isomorphic—not canonically—to a matrix algebra M_r, and the reduced norm agrees with the determinant map. This is well defined independently of the chosen isomorphism between D and M_r, because every automorphism of M_r is inner and similar matrices have the same determinant. This map, defined locally for the étale topology, descends, by its local uniqueness, to a map $\mathrm{Nrd} : D \to K$.)

The only zero of Nrd is the zero element of D, since for $x \neq 0$ one has $\mathrm{Nrd}(x) \cdot \mathrm{Nrd}(x^{-1}) = 1$. On the other hand, if $\{e_1, \ldots, e_{r^2}\}$ is a basis of D over K and $x = \sum x_i e_i$, then $\mathrm{Nrd}(x)$ is a homogeneous polynomial $\mathrm{Nrd}(x_1, \ldots, x_{r^2})$ of degree r (this is clear locally for the étale topology). Since K is C_1, one has $r^2 \leqslant r$, that is, $r = 1$ and $D = K$.

Theorem 3.2.3 (Tsen)**.** *Let k be an algebraically closed field and let K be an extension of k of transcendence degree 1. Then K is C_1.*

First suppose that $K = k(X)$. Let

$$f(T) = \sum a_{i_1,\ldots i_n} T_1^{i_1} \cdots T_n^{i_n}$$

be a homogeneous polynomial of degree $d < n$ with coefficients in $k(X)$. After multiplying the coefficients by a common denominator, we may suppose that they lie in $k[X]$. Let $\delta = \sup \deg(a_{i_1 \ldots i_n})$. We seek a nontrivial zero in $k[X]$ by the method of undetermined coefficients, writing each T_i $(i = 1, \ldots, n)$ as a polynomial of degree N in X. The equation $f(T) = 0$ then becomes a system of homogeneous equations in the $n \times (N+1)$ coefficients of the polynomials $T_i(X)$, expressing the vanishing of the coefficients of the polynomial in X obtained by replacing T_i with $T_i(X)$. This polynomial has degree at most $\delta + ND$, so there are $\delta + Nd + 1$ equations in $n \times (N+1)$ variables. Since k is algebraically closed, this system has a nontrivial solution if $n(N+1) > Nd + \delta + 1$, which holds for sufficiently large N when $d < n$.

To prove the theorem in general, it plainly suffices to prove it when K is a finite extension of the purely transcendental extension $k(X)$ of k. Let $f(T) = f(T_1, \ldots, T_n)$ be a homogeneous polynomial of degree $d < n$ with coefficients in K. Let $a = [K : k(X)]$ and let $e_1, \ldots, e_s$ be a basis of K over $k(X)$. Introduce sn new variables U_{ij} such that $T_i = \sum U_{ij} e_j$. For $f(T)$ to have a nontrivial zero in K, it suffices that the polynomial $g(X_{ij}) = N_{K/k}(f(T))$ have a nontrivial zero in $k(X)$. But g is a polynomial of degree sd in sn variables, which proves the result.

Corollary 3.2.4. *Let k be an algebraically closed field and let K be an extension of k of transcendence degree 1. Then the étale cohomology groups $\mathrm{H}^q(\mathrm{Spec}(K), \mathbb{G}_m)$ vanish for every $q > 0$.*

3.3 Cohomology of smooth curves

From now on, unless expressly stated otherwise, all cohomology groups are étale cohomology groups.

Proposition 3.3.1. *Let k be an algebraically closed field and let X be a connected nonsingular projective curve over k. Then*

$$\begin{aligned} \mathrm{H}^0(X, \mathbb{G}_m) &= k^\times, \\ \mathrm{H}^1(X, \mathbb{G}_m) &= \mathrm{Pic}(X), \\ \mathrm{H}^q(X, \mathbb{G}_m) &= 0 \text{ for } q \geqslant 2. \end{aligned}$$

Let η be the generic point of X, let $j : \eta \to X$ be the canonical morphism, and let $\mathbb{G}_{m,X}$ be the multiplicative group of the function field $K(X)$. For every closed point x of X, let $i_x : x \to X$ be the

canonical immersion and let $\mathbb{Z}_x$ be the constant sheaf with value $\mathbb{Z}$ on x. Thus $j_*\mathbb{G}_{m,\eta}$ is the sheaf of nonzero meromorphic functions on X and $\oplus_{x\in X} i_{x_I}\mathbb{Z}_x$ is the sheaf of divisors; hence there is an exact sequence of sheaves:

$$0 \longrightarrow \mathbb{G}_m \longrightarrow j_*\mathbb{G}_{m,\eta} \xrightarrow{\text{div}} \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x \longrightarrow 0. \tag{3.3.1.1}$$

Lemma 3.3.2. *One has* $\mathsf{R}^q j_*\mathbb{G}_{m,\eta} = 0$ *for every* $q > 0$.

It suffices to show that the fibre of this sheaf at every closed point x of X is zero. If $\tilde{\mathscr{O}}_{X,x}$ is the henselization of X at x and K is the fraction field of $\tilde{\mathscr{O}}_{X,x}$, then

$$\mathrm{Spec}(K) = \eta \times_X \mathrm{Spec}(\tilde{\mathscr{O}}_{X,x}),$$

and hence $(\mathsf{R}^q j_*\mathbb{G}_{m,\eta})_x = \mathrm{H}^q(\mathrm{Spec}(K), \mathbb{G}_m)$.

Now K is an algebraic extension of $k(X)$, hence an extension of k of transcendence degree 1; the lemma follows from (3.2.4).

Lemma 3.3.3. *One has* $\mathrm{H}^q(X, j_*\mathbb{G}_{m,\eta}) = 0$ *for every* $q > 0$.

Indeed, (3.3.2) and the Leray spectral sequence for j give

$$\mathrm{H}^q(X, j_*\mathbb{G}_{m,\eta}) = \mathrm{H}^q(\eta, \mathbb{G}_{m,\eta})$$

for every $q \geqslant 0$, and the right-hand side vanishes for $q > 0$ by (3.2.4).

Lemma 3.3.4. *One has* $\mathrm{H}^q\left(X, \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x\right) = 0$ *for every* $q > 0$.

Indeed, for every closed point of X one has $\mathsf{R}^q i_{x*}\mathbb{Z}_x 0$ for $q > 0$, because i_x is a finite morphism (2.3.6), and

$$\mathrm{H}^q(X, i_{x*}\mathbb{Z}_x) = \mathrm{H}^q(x, \mathbb{Z}_x).$$

The right-hand side vanishes for every $q > 0$ because x is the spectrum of an algebraically closed field. (This also shows that the lemma holds more generally for every skyscraper sheaf on X.)

The preceding lemmas and the exact sequence (3.3.1.1) give

$$\mathrm{H}^q(X, \mathbb{G}_m) = 0 \text{ for } q \geqslant 2,$$

and the low-degree exact cohomology sequence

$$1 \to \mathrm{H}^0(X, \mathbb{G}_m) \to \mathrm{H}^0(X, j_*\mathbb{G}_{m,\eta}) \to \mathrm{H}^0\left(X, \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x\right) \to \mathrm{H}^1(X, \mathbb{G}_m) \to 1$$

which is precisely the exact sequence

$$1 \to k^\times \to k(X)^\times \to \mathrm{Div}(X) \to \mathrm{Pic}(X) \to 1.$$

Proposition 3.3.1 implies that the cohomology groups of X with values in $\mathbb{Z}/n$, for n prime to the characteristic of k, have reasonable values:

Corollary 3.3.5. *If X has genus g and n is invertible in k, then $\mathrm{H}^q(X, \mathbb{Z}/n)$ vanishes for $q > 2$ and is free over $\mathbb{Z}/n$ of ranks $1, 2g, 1$ for $q = 0, 1, 2$. Replacing $\mathbb{Z}/n$ by the isomorphic group $\boldsymbol{\mu}_n$, there are canonical isomorphisms*

$$\mathrm{H}^0(X, \boldsymbol{\mu}_n) = \boldsymbol{\mu}_n$$
$$\mathrm{H}^1(X, \boldsymbol{\mu}_n) = \mathrm{Pic}^0(X)_n$$
$$\mathrm{H}^2(X, \boldsymbol{\mu}_n) = \mathbb{Z}/n.$$

Since k is algebraically closed, $\mathbb{Z}/n$ is isomorphic (noncanonically) to μ_n. The Kummer exact sequence

$$0 \to \mu_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0,$$

and Proposition 3.3.1 yield the equalities

$$\mathrm{H}^q(X, \mathbb{Z}/n) = 0 \text{ for } q > 2,$$

and, in low degrees, the exact sequences

$$0 \longrightarrow \mathrm{H}^0(X, \boldsymbol{\mu}_n) \longrightarrow k^\times \xrightarrow{n} k^\times \longrightarrow 0$$

$$0 \longrightarrow \mathrm{H}^1(X, \boldsymbol{\mu}_n) \longrightarrow \mathrm{Pic}(X) \xrightarrow{n} \mathrm{H}^2(X, \boldsymbol{\mu}_n) \longrightarrow 0.$$

Moreover, there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}(X) \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0,$$

and $\mathrm{Pic}^0(X)$ is identified with the group of k-rational points of an abelian variety of dimension g, the Jacobian of X. In such a group, multiplication by n is surjective and its kernel is a free $\mathbb{Z}/n\mathbb{Z}$-module of rank $2g$ (because n is invertible in k); this proves the corollary.

A clever unwinding using the "trace method" yields the following proposition as a corollary.

Proposition 3.3.6 ([2, IX 5.7])**.** *Let k be an algebraically closed field, let X be an algebraic curve over k, and let $\mathscr{F}$ be a torsion sheaf on X. Then:*

i) $\mathrm{H}^q(X, \mathscr{F}) = 0$ *for* $q > 2$.

ii) If X is affine, then even $\mathrm{H}^q(X, \mathscr{F}) = 0$ *for* $q > 1$.

For the proof, as well as for an exposition of the "trace method," we refer to [2, IX 5].

3.4 Unwindings

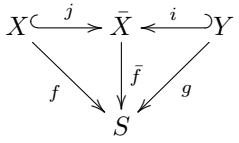
To calculate the cohomology of varieties of dimension > 1, one uses fibrations by curves, thereby reducing to the study of morphisms whose fibres have dimension $\leqslant 1$. This principle has several variants; we mention some of them.

3.4.1

Let A be a k-algebra of finite type, and let $a_1, \ldots, a_n$ be generators of A. If we put $X_0 = \mathrm{Spec}(k)$, $X_i = \mathrm{Spec}(k[a_1, \ldots, a_i])$, and $X_n = \mathrm{Spec}(A)$, the canonical inclusions $k[a_1, \ldots, a_i] \to k[a_1, \ldots, a_i, a_{i+1}]$ define morphisms $X_n \to X_{n-1} \to \cdots \to X_1 \to X_0$ whose fibres have dimension $\leqslant 1$.

3.4.2

For a smooth morphism, one can be more precise. An *elementary fibration* is a morphism of schemes $f : X \to S$ that can be embedded in a commutative diagram



satisfying the following conditions:

i) j is an open immersion, dense in every fibre, and $X = \bar{X} \smallsetminus Y$.

ii) $\bar{f}$ is smooth and projective, with irreducible geometric fibres of dimension 1.

iii) g is an étale covering and no fibre of g is empty.

A *good neighbourhood* relative to S is an S-scheme X for which there exist S-schemes $X = X_n, \ldots, X_0 = S$ and elementary fibrations $f_i : X_i \to X_{i-1}$, $i = 1, \ldots, n$. One can show [2, XI 3.3] that if X is a smooth scheme over an algebraically closed field k, then every rational point of X has an open neighbourhood that is a good neighbourhood (relative to $\mathrm{Spec}(K)$).

3.4.3

A proper morphism $f : X \to S$ can be unwound as follows. By Chow's lemma, there is a commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{\pi} & \bar{X} \\ & \searrow^{f} & \downarrow{\scriptstyle \bar{f}} \\ & & S \end{array}$$

in which π and $\bar{f}$ are projective morphisms, with π in addition an isomorphism over a dense open subset of X. Locally on S, $\bar{X}$ is a closed subscheme of a standard projective space $\mathbb{P}^n_S$.

The latter is unwound by considering the projection $\varphi : \mathbb{P}^n_S \to \mathbb{P}^1_S$ that sends the point with homogeneous coordinates $(x_0, x_1, \ldots, x_n)$ to (x_0, x_1). This is a rational map defined outside the closed subset $Y \simeq \mathbb{P}^{n-2}_S$ of $\mathbb{P}^n_S$ given by the homogeneous equations $x_0 = x_1 = 0$. Let $u : P \to \mathbb{P}^n_S$ be the blow-up with centre Y; the fibres of u have dimension $\leqslant 1$. Moreover, there is a natural morphism $v : P \to \mathbb{P}^1_S$ extending the rational map φ, and v makes P a $\mathbb{P}^1_S$-scheme locally isomorphic to the standard projective space $\mathbb{P}^{n-1}$, which may in turn be projected onto a $\mathbb{P}^1$, and so on.

3.4.4

A smooth projective variety X can be swept out by a *Lefschetz pencil*. The blow-up $\widetilde{X}$ of the intersection of the axis of the pencil with X maps to $\mathbb{P}^1$, and the fibres of this map are the hyperplane sections of X by the hyperplanes of the pencil.

4 The proper base change theorem

4.1 Introduction

This chapter is devoted to the proof and applications of the following theorem.

Theorem 4.1.1. *Let $f : X \to S$ be a proper morphism of schemes and let $\mathscr{F}$ be a torsion abelian sheaf on X. Then, for every $q \geqslant 0$, the fibre of $\mathrm{R}^q f_* \mathscr{F}$ at a geometric point s of S is isomorphic to the cohomology $\mathrm{H}^q(X_s, \mathscr{F})$ of the fibre $X_s = X \otimes_S \mathrm{Spec}\, k(s)$ of f at s.*

For a proper separated continuous map $f : X \to S$ between topological spaces (separated means that the diagonal of $X \times_S X$ is closed) and an abelian sheaf $\mathscr{F}$ on X, the analogous result is well known and elementary. Since f is closed, the $f^{-1}(V)$, with V ranging over the neighbourhoods of s, form a fundamental system of neighbourhoods of X_s, and one checks that $\mathrm{H}^\bullet(X_s, \mathscr{F}) = \varinjlim_U \mathrm{H}^\bullet(U, \mathscr{F})$, where U ranges over the neighbourhoods of X_s. In practice, X_s even has a fundamental system $\mathcal{U}$ of neighbourhoods

U of which it is a deformation retract, and for $\mathscr{F}$ constant one therefore has $\mathrm{H}^\bullet(X_s, \mathscr{F}) = \mathrm{H}^\bullet(U, \mathscr{F})$. Figuratively speaking, the special fibre swallows the general fibre.

For schemes the proof is more delicate, and it is essential to suppose that $\mathscr{F}$ is torsion ([2, XII.2]). In view of the description of the fibres of $\mathsf{R}^q f_* \mathscr{F}$ (2.3.3), Theorem (4.1.1) is essentially equivalent to the following theorem.

Theorem 4.1.2. *Let A be a strictly henselian local ring and let $S = \mathrm{Spec}(A)$. Let $f : X \to S$ be proper, and let X_0 be the closed fibre of f. Then, for every torsion abelian sheaf $\mathscr{F}$ on X and every $q \geqslant 0$, one has $\mathrm{H}^q(X, \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^q(X_0, \mathscr{F})$.*

By passage to the limit, it suffices to prove the theorem when A is the strict henselization of a finitely generated $\mathbb{Z}$-algebra at a prime ideal. We first treat the case $q = 0$ or 1 and $\mathscr{F} = \mathbb{Z}/n$ (4.2). An argument based on constructible sheaves (4.3) shows moreover that it suffices to consider constant $\mathscr{F}$. On the other hand, the unwinding in (3.4.3) allows us to suppose that X_0 is a curve; in that case it remains only to prove the theorem for $q = 2$ (4.4).

Among other applications (4.6), the theorem allows one to define cohomology with proper support (4.5).

4.2 Proof for $q = 0$ or 1 and $\mathscr{F} = \mathbb{Z}/n$

The result for $q = 0$ and $\mathscr{F}$ constant is equivalent to the following proposition (Zariski's connectedness theorem).

Proposition 4.2.1. *Let A be a noetherian henselian local ring and let $S = \mathrm{Spec}(A)$. Let $f : X \to S$ be a proper morphism, and let X_0 be the closed fibre of f. Then the sets of connected components $\pi_0(X)$ and $\pi_0(X_0)$ are in bijection.*

Equivalently, we must show that the sets $\mathrm{Of}(X)$ and $\mathrm{Of}(X_0)$ of open-and-closed subsets are in bijection. The set $\mathrm{Of}(X)$ is known to correspond bijectively to the set of idempotents of $\Gamma(X, \mathscr{O}_X)$, and similarly $\mathrm{Of}(X_0)$ corresponds bijectively to the set of idempotents of $\Gamma(X_0, \mathscr{O}_{X_0})$. Thus we must show that the canonical map

$$\mathrm{Idem}\,\Gamma(X, \mathscr{O}_X) \to \mathrm{Idem}\,\Gamma(X_0, \mathscr{O}_{X_0})$$

is bijective.

Let $\mathfrak{m}$ denote the maximal ideal of A, let $\Gamma(X, \mathscr{O}_X)^\wedge$ denote the completion of $\Gamma(X, \mathscr{O}_X)$ for the $\mathfrak{m}$-adic topology, and, for every integer $n \geqslant 0$, let $X_n = X \otimes_A A/\mathfrak{m}^{n+1}$. By the finiteness theorem for proper morphisms [20, III.3.2], $\Gamma(X, \mathscr{O}_X)$ is a finite A-algebra; since A is henselian, the canonical map

$$\mathrm{Idem}\,\Gamma(X, \mathscr{O}_X) \to \mathrm{Idem}\,\Gamma(X, \mathscr{O}_X)^\wedge$$

is bijective. In particular, the canonical map

$$\mathrm{Idem}\,\Gamma(X, \mathscr{O}_X)^\wedge \to \varprojlim \mathrm{Idem}\,\Gamma(X_n, \mathscr{O}_{X_n})$$

is bijective. But since X_n and X_0 have the same underlying topological space, the canonical map

$$\mathrm{Idem}\,\Gamma(X_n, \mathscr{O}_{X_n}) \to \mathrm{Idem}\,\Gamma(X_0, \mathscr{O}_{X_0})$$

is bijective for every n, which completes the proof.

Since $\mathrm{H}^1(X, \mathbb{Z}/n)$ is in bijection with the set of isomorphism classes of Galois étale coverings of X with group $\mathbb{Z}/n$, the theorem for $q = 1$ and $\mathscr{F} = \mathbb{Z}/n$ follows from the next proposition.

Proposition 4.2.2. *Let A be a noetherian henselian local ring and let $S = \mathrm{Spec}(A)$. Let $f : X \to S$ be a proper morphism, and let X_0 be the closed fibre of f. Then the restriction functor*

$$\mathrm{Rev.\,et}(X) \to \mathrm{Rev.\,et}(X_0)$$

is an equivalence of categories.

(If X_0 is connected and a geometric point of X_0 has been chosen as base point, this says that the canonical map $\pi_1(X_0) \to \pi_1(X)$ on (profinite) fundamental groups is bijective.)

Proposition (4.2.1) shows that this functor is fully faithful. Indeed, if X' and X'' are two étale coverings of X, an X-morphism from X' to X'' is determined by its graph, which is an open-and-closed subset of $X' \times_X X''$.

It remains to show that every étale covering X'_0 of X_0 extends to an étale covering of X. Étale coverings are insensitive to nilpotent elements [23, ch.1]; consequently, X'_0 lifts uniquely to an étale covering X'_n of X_n for every $n \geqslant 0$, in other words, to an étale covering $\mathfrak{X}'$ of the formal scheme $\mathfrak{X}$ obtained by completing X along X_0. By Grothendieck's algebraization theorem for formal coherent sheaves (the existence theorem, [20, III.5]), $\mathfrak{X}'$ is the formal completion of an étale covering $\bar{X}'$ of $\bar{X} = X \otimes_A \hat{A}$.

By passage to the limit, it suffices to prove the proposition when A is the henselization of a finitely generated $\mathbb{Z}$-algebra. We can then apply Artin's approximation theorem to the functor $\mathscr{F} : (A\text{-algebras}) \to (\text{sets})$ that associates to an A-algebra B the set of isomorphism classes of étale coverings of $X \otimes_A B$. Indeed, this functor is locally of finite presentation: if B_i is a filtered direct system of A-algebras and $B = \varinjlim B_i$, then $\mathscr{F}(B) = \varinjlim \mathscr{F}(B_i)$. By Artin's theorem, given an element $\xi \in \mathscr{F}(\hat{A})$, here the isomorphism class of $\bar{X}'$, there exists $\bar{\xi} \in \mathscr{F}(A)$ having the same image as $\bar{\xi}$ in $\mathscr{F}(A/\mathfrak{m})$. In other words, there exists an étale covering X' of X whose restriction to X_0 is isomorphic to X'_0.

4.3 Constructible sheaves

In this paragraph, X is a *noetherian* scheme, and by a sheaf on X we mean an *abelian* sheaf on X_{et}.

Definition 4.3.1. *A sheaf $\mathscr{F}$ on X is called* locally constant constructible *(abbreviated l.c.c.) if it is represented by an étale covering of X.*

Definition 4.3.2. *A sheaf $\mathscr{F}$ on X is called* constructible *if it satisfies the following equivalent conditions:*

(i) There exists a finite surjective family of subschemes X_i of X such that the restriction of $\mathscr{F}$ to X_i is l.c.c.

(ii) There exist a finite family of finite morphisms $p_i : X'_i \to X$, for each i a constant constructible sheaf C_i (i.e. one defined by a finite abelian group) on X'_i, and a monomorphism $\mathscr{F} \to \prod p_{i*}C_i$.

It is readily checked that the category of constructible sheaves on X is an abelian category. Moreover, if $u : \mathscr{F} \to \mathscr{G}$ is a homomorphism of sheaves and $\mathscr{F}$ is constructible, then $\mathrm{im}(u)$ is constructible.

Lemma 4.3.3. *Every torsion sheaf $\mathscr{F}$ is a filtered direct limit of constructible sheaves.*

Indeed, if $j : U \to X$ is an étale scheme of finite type over X, an element $\xi \in \mathscr{F}(U)$ such that $n\xi = 0$ defines a homomorphism of sheaves $j_!\underline{\mathbb{Z}/n} \to \mathscr{F}$ whose image (the smallest subsheaf of $\mathscr{F}$ for which ξ is a local section) is a constructible subsheaf of $\mathscr{F}$. Clearly $\mathscr{F}$ is the direct limit of such subsheaves.

Definition 4.3.4. *Let $\mathcal{C}$ be an abelian category and let T be a functor from $\mathcal{C}$ to the category of abelian groups. The functor T is called* effaceable *in $\mathcal{C}$ if, for every object A of $\mathcal{C}$ and every $\alpha \in T(A)$, there exists a monomorphism $u : A \to M$ in $\mathcal{C}$ such that $T(u)\alpha = 0$.*

Lemma 4.3.5. *The functors* $\mathrm{H}^q(X,-)$ *for* $q>0$ *are effaceable in the category of constructible sheaves on* X.

It suffices to observe that, if $\mathscr{F}$ is a constructible sheaf, there is necessarily an integer $n>0$ such that $\mathscr{F}$ is a sheaf of $\mathbb{Z}/n$-modules. There is then a monomorphism $\mathscr{F}\hookrightarrow\mathscr{G}$, where $\mathscr{G}$ is a sheaf of $\mathbb{Z}/n$-modules and $\mathrm{H}^q(X,\mathscr{G})=0$ for every $q>0$. For example, one can take for $\mathscr{G}$ the Godement resolution $\prod_{x\in X} i_{x*}\mathscr{F}_{\bar{x}}$, where x runs over the points of X and $i_x:\bar{x}\to X$ is a geometric point centred at X. By (4.3.3), $\mathscr{G}$ is a direct limit of constructible sheaves, whence the lemma, because the functors $\mathrm{H}^q(X,-)$ commute with direct limits.

Lemma 4.3.6. *Let* $\varphi^\bullet: T^\bullet\to T'^\bullet$ *be a morphism of cohomological functors defined on an abelian category* $\mathcal{C}$ *with values in the category of abelian groups. Suppose that* T^q *is effaceable for* $q>0$, *and let* $\mathcal{E}$ *be a subset of the objects of* $\mathcal{C}$ *such that every object of* $\mathcal{C}$ *is contained in an object belonging to* $\mathcal{E}$. *Then the following conditions are equivalent:*

(i) $\varphi^q(A)$ *is bijective for every* $q\geqslant 0$ *and every* $A\in\mathrm{Ob}\,\mathcal{C}$.

(ii) $\varphi^0(M)$ *is bijective and* $\varphi^q(M)$ *is surjective for every* $q>0$ *and every* $M\in\mathcal{E}$.

(iii) $\varphi^0(A)$ *is bijective for every* $A\in\mathrm{Ob}\,\mathcal{C}$, *and* T'^q *is effaceable for every* $q>0$.

The proof is by induction on q and presents no difficulty.

Proposition 4.3.7. *Let* X_0 *be a subscheme of* X. *Suppose that, for every* $n\geqslant 0$ *and every scheme* X' *finite over* X, *the canonical map*

$$\mathrm{H}^q(X',\mathbb{Z}/n)\to\mathrm{H}^q(X'_0,\mathbb{Z}/n),$$

where $X'_0=X'\times_X X_0$, *is bijective for* $q=0$ *and surjective for* $q>0$. *Then, for every torsion sheaf* $\mathscr{F}$ *on* X *and every* $q\geqslant 0$, *the canonical map*

$$\mathrm{H}^q(X,\mathscr{F})\to\mathrm{H}^q(X_0,\mathscr{F})$$

is bijective.

By passage to the limit, it suffices to prove the assertion for constructible $\mathscr{F}$. Apply Lemma (4.3.6) with $\mathcal{C}$ the category of constructible sheaves on X, $T^q=\mathrm{H}^q(X,-)$, $T'^q=\mathrm{H}^q(X_0,-)$, and $\mathcal{E}$ the set of constructible sheaves of the form $\prod p_{i*}C_i$, where $p_i:X'_i\to X$ is finite and C_i is a finite constant sheaf on X'_i.

4.4 End of the proof

By the method of fibration by curves (3.4.3), we reduce to proving the theorem in relative dimension $\leqslant 1$. By the preceding paragraph, it suffices to show the following. If S is the spectrum of a strictly henselian noetherian local ring, $f:X\to S$ is a proper morphism whose closed fibre X_0 has dimension $\leqslant 1$, and n is an integer $\geqslant 0$, then the canonical homomorphism

$$\mathrm{H}^q(X,\mathbb{Z}/n)\to\mathrm{H}^q(X_0,\mathbb{Z}/n)$$

is bijective for $q=0$ and surjective for $q>0$.

The cases $q=0$ and 1 were treated above, and $\mathrm{H}^q(X_0,\mathbb{Z}/n)=0$ for $q\geqslant 3$; it therefore suffices to treat $q=2$. We may clearly suppose that n is a power of a prime. If $n=p^r$, where p is the characteristic of the

residue field of S, Artin–Schreier theory shows that $\mathrm{H}^2(X_0, \mathbb{Z}/p^r) = 0$. If $n = \ell^r$, $\ell \neq p$, Kummer theory gives a commutative diagram

$$\begin{array}{ccc} \mathrm{Pic}(X) & \xrightarrow{\alpha} & \mathrm{H}^2(X, \mathbb{Z}/\ell^r) \\ \downarrow & & \downarrow \\ \mathrm{Pic}(X_0) & \xrightarrow{\beta} & \mathrm{H}^2(X_0, \mathbb{Z}/\ell^r) \end{array}$$

in which β is surjective (this was seen in Chapter 3 for a smooth curve over an algebraically closed field, but similar arguments apply to any curve over a separably closed field).

Thus it suffices to prove:

Proposition 4.4.1. *Let S be the spectrum of a henselian noetherian local ring, and let $f : X \to S$ be a proper morphism whose closed fibre X_0 has dimension $\leqslant 1$. Then the canonical restriction map*

$$\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$$

is surjective. (In fact, it is enough that f be a separated *morphism of finite type.)*

To simplify the proof, we suppose that X is integral, although this is not necessary. Every invertible sheaf on X_0 is associated with a Cartier divisor (because X_0 is a curve, hence quasi-projective), so it suffices to show that the canonical map $\mathrm{Div}(X) \to \mathrm{Div}(X_0)$ is surjective.

Every divisor on X_0 is a linear combination of divisors whose support is concentrated at a single nonisolated closed point of X_0. Let x be such a point, let $t_0 \in \mathscr{O}_{X_0,x}$ be a regular nonunit in $\mathscr{O}_{X_0,x}$, and let D_0 be the divisor concentrated at x with local equation t_0. Let U be an open neighbourhood of x in X on which there is a section $t \in \Gamma(U, \mathscr{O}_U)$ lifting t_0. Let Y be the closed subset of U defined by $t = 0$; after making U sufficiently small, we may suppose that x is the only point of $Y \cap X_0$. Then Y is quasi-finite over S at x. Since S is the spectrum of a henselian local ring, it follows that $Y = Y_1 \amalg Y_2$, where Y_1 is finite over S and Y_2 does not meet X_0. Moreover, since X is separated over S, Y_1 is closed in X.

After replacing U by a smaller open neighbourhood of x, we may suppose that $Y = Y_1$, in other words, that Y is closed in X. We then define a divisor D on X lifting D_0 by setting $D|X \smallsetminus Y = 0$ and $D|U = \mathrm{div}(t)$; this makes sense because t is invertible on $U \smallsetminus Y$.

4.4.2 Remark

When f is proper, one could also give a proof similar to that of Proposition (4.2.2). Indeed, since X_0 is a curve, there is no obstruction to lifting an invertible sheaf on X_0 to the infinitesimal neighbourhoods X_n of X_0, and hence to the formal completion $\mathfrak{X}$ of X along X_0. One then concludes by applying successively Grothendieck's existence theorem and Artin's approximation theorem.

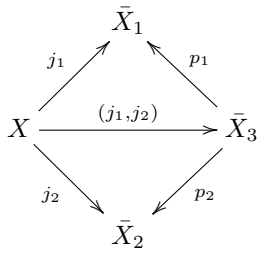
4.5 Cohomology with proper support

Definition 4.5.1. *Let X be a separated scheme of finite type over a field k. By a theorem of Nagata, there exist a scheme $\bar{X}$ proper over k and an open immersion $j : X \to \bar{X}$. For every torsion sheaf $\mathscr{F}$ on X, let $j_!\mathscr{F}$ denote the extension by 0 of $\mathscr{F}$ to $\bar{X}$, and define the groups of* cohomology with proper support *by*

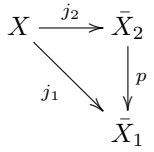
$$\mathrm{H}^q_c(X, \mathscr{F}) = \mathrm{H}^q(\bar{X}, j_!\mathscr{F}).$$

Let us show that this definition is independent of the chosen compactification $j : X \to \bar{X}$. Let $j_1 : X \to \bar{X}_1$ and $j_2 : X \to \bar{X}_2$ be two compactifications. Then X maps into $\bar{X}_1 \times \bar{X}_2$ by $x \mapsto (j_1(x), j_2(x))$,

and the closed image $\bar{X}_3$ of X under this map is a compactification of X. We thus have a commutative diagram



where p_1 and p_2, the restrictions to $\bar{X}_3$ of the natural projections, are proper morphisms. It therefore suffices to treat the case of a commutative diagram



with p proper.

Lemma 4.5.2. *One has $p_*(j_{2!}\mathscr{F}) = j_{1!}\mathscr{F}$ and $\mathsf{R}^q p_*(j_{2!}\mathscr{F}) = 0$ for $q > 0$.*

Let us note at once that the lemma suffices. Using the Leray spectral sequence for p, it gives, for every $q \geqslant 0$,

$$\mathrm{H}^q(\bar{X}_2, j_{2!}\mathscr{F}) = \mathrm{H}^q(\bar{X}_1, j_{1!}\mathscr{F}).$$

To prove the lemma, argue fibre by fibre using the base change theorem (4.1.1) for p. The result is immediate because above a point of X, p is an isomorphism, whereas above a point of $\bar{X}_1 \smallsetminus X$, $j_{2!}\mathscr{F}$ vanishes on the fibre of p.

4.5.3

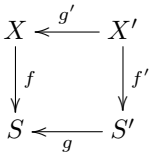
Similarly, if $f : X \to S$ is a separated morphism of finite type between noetherian schemes, there exist a proper morphism $\bar{f} : \bar{X} \to S$ and an open immersion $j : X \to \bar{X}$. The higher direct images with proper support $\mathsf{R}^q f_!$ are defined, for every torsion sheaf $\mathscr{F}$ on X, by

$$\mathsf{R}^q f_!\mathscr{F} = \mathsf{R}^q f_*(j_!\mathscr{F}).$$

As before, one checks that this definition is independent of the chosen compactification.

Theorem 4.5.4. *Let $f : X \to S$ be a separated morphism of finite type between noetherian schemes, and let $\mathscr{F}$ be a torsion sheaf on X. Then the fibre of $\mathsf{R}^q f_!\mathscr{F}$ at a geometric point s of S is isomorphic to the cohomology with proper support $\mathrm{H}^q_c(X_s, \mathscr{F})$ of the fibre X_s of f at s.*

This is a simple variant of the proper base change theorem (4.1.1). More generally, if



is a cartesian diagram, there is a canonical isomorphism

$$g^*(\mathsf{R}^q f_!\mathscr{F}) \simeq \mathsf{R}^q f'_!({g'}^*\mathscr{F}). \tag{4.5.4.1}$$

4.6 Applications

Theorem 4.6.1 (Vanishing theorem)**.** *Let $f : X \to S$ be a separated morphism of finite type whose fibres have dimension $\leqslant n$, and let $\mathscr{F}$ be a torsion sheaf on X. Then $\mathsf{R}^q f_!\mathscr{F} = 0$ for $q > 2n$.*

By the base change theorem, we may suppose that S is the spectrum of a separably closed field. If $\dim X = n$, there exists an affine open U of X such that $\dim(X \smallsetminus U) < n$. We then have an exact sequence $0 \to \mathscr{F}_U \to \mathscr{F} \to \mathscr{F}_{X\smallsetminus U} \to 0$, and induction on n reduces us to proving the theorem for $X = U$ affine. The method of fibration by curves (3.4.1) and the base change theorem then reduce us to a curve over a separably closed field, for which the desired result follows from Tsen's theorem (3.3.6).

Theorem 4.6.2 (Finiteness theorem)**.** *Let $f : X \to S$ be a separated morphism of finite type and let $\mathscr{F}$ be a constructible sheaf on X. Then the sheaves $\mathsf{R}^q f_!\mathscr{F}$ are constructible.*

We consider only the case in which $\mathscr{F}$ is annihilated by an integer invertible on X.

To prove the theorem, one reduces to the case in which $\mathscr{F}$ is the constant sheaf $\underline{\mathbb{Z}/n}$ and $f : X \to S$ is proper and smooth with geometrically connected curve fibres of genus g. If n is invertible on X, the sheaves $\mathsf{R}^q f_*\mathscr{F}$ are then locally free of finite rank and vanish for $q > 2$ (4.6.1). Replacing $\mathbb{Z}/n$ by the sheaf $\boldsymbol{\mu}_n$, which is locally isomorphic to it (over S), there are canonical isomorphisms

$$\begin{aligned} &\mathsf{R}^0 f_*\boldsymbol{\mu}_n = \boldsymbol{\mu}_n \\ &\mathsf{R}^1 f_*\boldsymbol{\mu}_n = \underline{\mathrm{Pic}}(X/S)_n \\ &\mathsf{R}^2 f_*\boldsymbol{\mu}_n = \mathbb{Z}/n. \end{aligned} \tag{4.6.2.1}$$

Theorem 4.6.3 (Comparison with classical cohomology)**.** *Let $f : X \to S$ be a separated morphism of schemes of finite type over $\mathbb{C}$, and let $\mathscr{F}$ be a torsion sheaf on X. Denote by a superscript $(-)^{\mathrm{an}}$ the functor passing to the usual topological spaces, and by $\mathsf{R}^q f_!^{\mathrm{an}}$ the derived functors of direct image with proper support under f^{an}. Then*

$$(\mathsf{R}^q f_!\mathscr{F})^{\mathrm{an}} \simeq \mathsf{R}^q f_!^{\mathrm{an}}\mathscr{F}^{\mathrm{an}}.$$

In particular, when $S =$ a point and $\mathscr{F}$ is the constant sheaf $\mathbb{Z}/n$,

$$\mathrm{H}^q_c(X, \mathbb{Z}/n) \simeq \mathrm{H}^q_c(X^{\mathrm{an}}, \mathbb{Z}/n).$$

Unwindings using the base change theorem reduce us to the case in which X is a smooth proper curve, $S =$ a point, and $\mathscr{F} = \mathbb{Z}/n$. The cohomology groups in question then vanish for $q \neq 0, 1, 2$, and we invoke [32]: indeed, if X is proper over $\mathbb{C}$, then $\pi_0(X) = \pi_0(X^{\mathrm{an}})$ and $\pi_1(X) =$ the profinite completion of $\pi_1(X^{\mathrm{an}})$, which proves the assertion for $q = 0, 1$. For $q = 2$, one uses the Kummer exact sequence and the fact that, again by [32], $\mathrm{Pic}(X) = \mathrm{Pic}(X^{\mathrm{an}})$.

Theorem 4.6.4 (Cohomological dimension of affine schemes)**.** *Let X be an affine scheme of finite type over a separably closed field and let $\mathscr{F}$ be a torsion sheaf on X. Then $\mathrm{H}^1(X, \mathscr{F}) = 0$ for $q > \dim(X)$.*

For the very elegant proof, we refer to [2], XIV §2 and 3.

4.6.5 Remark

This theorem is in a sense a substitute for Morse theory. Indeed, consider the classical case in which X is smooth and affine over $\mathbb{C}$, embedded in a standard affine space $\mathbb{C}^N$. Then, for almost every point $p \in \mathbb{C}^N$, the function "distance from p" on X is a Morse function, and the indices of its critical points are smaller than $\dim(X)$. Thus X is obtained by attaching handles of index less than $\dim(X)$, which gives the classical analogue of (4.6.4).

5 Local acyclicity of smooth morphisms

Let X be a complex analytic variety and let $f : X \to D$ be a morphism from X to the disk. Write $[0,t]$ for the closed line segment in D with endpoints 0 and t, and $(0,t]$ for the half-open segment. If f is *smooth*, the inclusion

$$j : f^{-1}((0,t]) \hookrightarrow f^{-1}([0,t])$$

is a homotopy equivalence; the special fibre $X_0 = f^{-1}(0)$ can be pushed into $f^{-1}([0,t])$.

In practice, for sufficiently small t, $f^{-1}((0,t])$ will be a bundle over $(0,t]$, so that the inclusion

$$X_t = f^{-1}(t) \hookrightarrow f^{-1}((0,t])$$

will also be a homotopy equivalence. The *cospecialization morphism* is then the homotopy class of maps

$$\mathrm{cosp} : X_0 \hookrightarrow f^{-1}([0,t]) \xleftarrow{\sim} f^{-1}((0,t]) \xleftarrow{\sim} X_t.$$

Figuratively speaking, for a smooth morphism the general fibre swallows the special fibre.

No longer suppose that f is necessarily smooth (but suppose that $f^{-1}((0,t])$ is a bundle over $(0,t]$). A morphism $\mathrm{cosp}^\bullet$ can still be defined in cohomology whenever $j_*\mathbb{Z} = \mathbb{Z}$ and $\mathsf{R}^q j_*\mathbb{Z} = 0$ for $q > 0$. Under these hypotheses, the Leray spectral sequence for j shows that

$$\mathrm{H}^\bullet\left(f^{-1}([0,t]), \mathbb{Z}\right) \xrightarrow{\sim} \mathrm{H}^\bullet\left(f^{-1}((0,t]), \mathbb{Z}\right)$$

and $\mathrm{cosp}^\bullet$ is the composite morphism

$$\mathrm{cosp}^\bullet : \mathrm{H}^\bullet(X_t, \mathbb{Z}) \xleftarrow{\sim} \mathrm{H}^\bullet\left(f^{-1}((0,t]), \mathbb{Z}\right) \xleftarrow{\sim} \mathrm{H}^\bullet\left(f^{-1}([0,t]), \mathbb{Z}\right) \to \mathrm{H}^\bullet(X_0, \mathbb{Z})$$

The fibre of $\mathsf{R}^q j_*\mathbb{Z}$ at a point $x \in X_0$ is calculated as follows. In an ambient space, take a ball B_ε centred at x with sufficiently small radius ε, and for sufficiently small η put $E = X \cap B_\varepsilon \cap f^{-1}(\eta t)$; this is the *vanishing-cycle variety* at x. One has

$$(\mathsf{R}^q j_*\mathbb{Z})_x \xleftarrow{\sim} \mathrm{H}^q\left(X \cap B_\varepsilon \cap f^{-1}((0,\eta t]), \mathbb{Z}\right) \xrightarrow{\sim} \mathrm{H}^q(E, \mathbb{Z})$$

and the cospecialization morphism is defined in cohomology whenever the vanishing-cycle varieties are acyclic ($\mathrm{H}^0(E,\mathbb{Z}) = \mathbb{Z}$ and $\mathrm{H}^q(E,\mathbb{Z}) = 0$ for $q > 0$), which is expressed by saying that f is *locally acyclic*.

This chapter is devoted to the analogue of this situation for a smooth morphism of schemes in étale cohomology. In this setting, however, it is essential to restrict to torsion coefficients of order *prime to the residue characteristics*. Paragraph 5.1 gives generalities on locally acyclic morphisms and cospecialization maps. In Paragraph 5.2, we prove that a smooth morphism is locally acyclic. In Paragraph 5.3, this result is combined with those of the preceding chapter to obtain two applications: a specialization theorem for cohomology groups (the cohomology of the geometric fibres of a smooth proper morphism is locally constant), and a base change theorem for a smooth morphism.

Throughout what follows, we fix an integer n, and "scheme" means "scheme on which n is invertible." A "geometric point" will always mean an "algebraic geometric point" (2.3.1) $x : \mathrm{Spec}(k) \to X$, with k algebraically closed.

5.1 Locally acyclic morphisms

5.1.1 Notation

Given a scheme S and a geometric point s of S, we denote by $\widetilde{S}^s$ the spectrum of the strict localization of S at s.

Definition 5.1.2. *A geometric point t of S is called a* generization *of s if it is defined by an algebraic closure of the residue field of a point of $\widetilde{S}^s$. We also say that s is a* specialization *of t, and call the* specialization map *the S-morphism $t \to \widetilde{S}^s$.*

Definition 5.1.3. *Let $f : X \to S$ be a morphism of schemes. Let s be a geometric point of S, t a generization of s, x a geometric point of X over s, and $\widetilde{X}_t^x = \widetilde{X}^x \times_{\widetilde{S}^s} t$. Then $\widetilde{X}_t^x$ is called a* vanishing-cycle variety *of f at x.*

The morphism f is called locally acyclic *if the reduced cohomology of every vanishing-cycle variety $\widetilde{X}_t^x$ vanishes:*

$$\hat{\mathrm{H}}^{\bullet}(\widetilde{X}_t^x, \mathbb{Z}/n) = 0, \tag{5.1.3.1}$$

i.e. $\mathrm{H}^0(X_t^x, \mathbb{Z}/n) = \mathbb{Z}/n$ *and* $\mathrm{H}^q(\widetilde{X}_t^x, \mathbb{Z}/n) = 0$ *for* $q > 0$.

Lemma 5.1.4. *Let $f : X \to S$ be a locally acyclic morphism and let $g : S' \to S$ be a quasi-finite morphism (or a projective limit of quasi-finite morphisms). Then the morphism $f' : X' \to S'$ obtained from f by base change is locally acyclic.*

Indeed, every vanishing-cycle variety of f' is a vanishing-cycle variety of f.

Lemma 5.1.5. *Let $f : X \to S$ be a locally acyclic morphism. For every geometric point t of S giving rise to a cartesian diagram*

$$\begin{array}{ccc} X_t & \xrightarrow{\varepsilon'} & X \\ \downarrow & & \downarrow f \\ t & \xrightarrow{\varepsilon} & S \end{array}$$

one has $\varepsilon'_*\mathbb{Z}/n = f^*\varepsilon_*\mathbb{Z}/n$ and $\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n = 0$ for $q > 0$.

Let $\bar{S}$ be the closure of $\varepsilon(t)$, let S' be the normalization of $\bar{S}$ in $k(t)$, and consider the cartesian diagram

$$\begin{array}{ccccc} X_t & \xrightarrow{f'} & X' & \xrightarrow{\alpha'} & X \\ \downarrow & & \downarrow f' & & \downarrow f \\ t & \xrightarrow{i} & S' & \xrightarrow{\alpha} & S \end{array}$$

The local rings of S' are normal with separably closed fraction fields. They are therefore strictly henselian, and the local acyclicity of f' (5.1.4) gives $f'_*\mathbb{Z}/n = \mathbb{Z}/n$ and $\mathsf{R}^q i'_*\mathbb{Z}/n = 0$ for $q > 0$. Since α is integral, it follows that

$$\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n = \alpha'_*\mathsf{R}^q i'_*\mathbb{Z}/n = \alpha'_* f'^*\mathsf{R}^q i_*\mathbb{Z}/n = f^*\alpha_*\mathsf{R}^q i_*\mathbb{Z}/n = f^*\mathsf{R}^q\varepsilon_*\mathbb{Z}/n,$$

which proves the lemma.

5.1.6

Given a locally acyclic morphism $f : X \to S$ and a specialization map $t \to \widetilde{S}^s$, we shall define canonical homomorphisms, called cospecialization maps,

$$\mathrm{cosp}^\bullet : \mathrm{H}^\bullet(X_t, \mathbb{Z}/n) \to \mathrm{H}^\bullet(X_s, \mathbb{Z}/n),$$

relating the cohomology of the general fibre $X_t = X \times_S t$ to that of the special fibre $X_s = X \times_X s$.

Consider the cartesian diagram

$$\begin{array}{ccccc} X_t & \xrightarrow{\varepsilon'} & \widetilde{X} & \longleftarrow & X_s \\ \downarrow & & \downarrow{\scriptstyle f'} & & \downarrow \\ t & \xrightarrow{\varepsilon} & \widetilde{S}^s & \longleftarrow & s \end{array}$$

obtained from f by base change. By (5.1.4), f' is again locally acyclic. The definition of local acyclicity immediately shows that the restrictions to X_s of the sheaves $\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n$ are $\mathbb{Z}/n$ for $q = 0$ and 0 for $q > 0$. By (5.1.5), we even know that $\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n = 0$ for $q > 0$. We define $\mathrm{cosp}^\bullet$ as the composite map

$$\mathrm{H}^\bullet(X_t, \mathbb{Z}/n) \simeq \mathrm{H}^\bullet(X, \varepsilon'_*\mathbb{Z}/n) \to \mathrm{H}^\bullet(X_s, \mathbb{Z}/n). \tag{5.1.6.1}$$

Variant Let $\bar{S}$ be the closure of $\varepsilon(t)$ in $\widetilde{S}^s$, let S' be the normalization of $\bar{S}$ in $k(t)$, and let X'/S' be obtained from X/S by base change. Diagram (5.1.6.1) may also be written

$$\mathrm{H}^\bullet(X_t, \mathbb{Z}/n) \simeq \mathrm{H}^\bullet(X', \mathbb{Z}/n) \to \mathrm{H}^\bullet(X_s, \mathbb{Z}/n).$$

Theorem 5.1.7. *Let S be a locally noetherian scheme, s a geometric point of S, and $f : X \to S$ a morphism. Suppose that*

a) f is locally acyclic,

b) for every specialization map $t \to \widetilde{S}^s$ and every $q \geqslant 0$, the cospecialization maps $\mathrm{H}^q(X_t, \mathbb{Z}/n) \to \mathrm{H}^q(X_s, \mathbb{Z}/n)$ are bijective.

Then the canonical homomorphism $(\mathsf{R}^q f_*\mathbb{Z}/n)_s \to \mathrm{H}^q(X_s, \mathbb{Z}/n)$ is bijective for every $q \geqslant 0$.

To prove the theorem, we may clearly suppose that $S = \widetilde{S}^s$. We shall in fact show that, for every sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules $\mathscr{F}$ on S, the canonical homomorphism $\varphi^q(\mathscr{F}) : (\mathsf{R}^q f_* f^*\mathscr{F})_s \to \mathrm{H}^q(X_s, f^*\mathscr{F})$ is bijective.

Every sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules is a filtered direct limit of constructible sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules (4.3.3). Moreover, every constructible sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules embeds in a sheaf of the form $\prod i_{\lambda *}C_\lambda$, where $i_\lambda : t_\lambda \to S$ is a finite family of generizations of s and C_λ is a free $\mathbb{Z}/n\mathbb{Z}$-module of finite rank on t_λ. By the definition of the cospecialization maps, condition b) says that the homomorphisms $\varphi^q(\mathscr{F})$ are bijective when $\mathscr{F}$ has this form.

We conclude by a variant of Lemma (4.3.6):

Lemma 5.1.8. *Let $\mathcal{C}$ be an abelian category in which filtered direct limits exist. Let $\varphi^\bullet : T^\bullet \to T'^\bullet$ be a morphism of cohomological functors commuting with filtered direct limits, defined on $\mathcal{C}$ with values in the category of abelian groups. Suppose that there are two subsets $\mathcal{D}$ and $\mathcal{E}$ of the objects of $\mathcal{C}$ such that:*

a) every object of $\mathcal{C}$ is a filtered direct limit of objects belonging to $\mathcal{D}$,

b) every object belonging to $\mathcal{D}$ is contained in an object belonging to $\mathcal{E}$.

Then the following conditions are equivalent:

(i) $\varphi^q(A)$ *is bijective for every* $q \geqslant 0$ *and every* $A \in \mathrm{Ob}\,\mathcal{C}$.

(ii) $\varphi^q(M)$ *is bijective for every* $q \geqslant 0$ *and every* $M \in \mathcal{E}$.

The proof of the lemma proceeds by passage to the direct limit, induction on q, and repeated application of the five lemma to the diagram of exact cohomology sequences deduced from an exact sequence $0 \to A \to M \to A' \to 0$, with $A \in \mathcal{D}$, $M \in \mathcal{E}$, and $A' \in \mathrm{Ob}\,\mathcal{C}$.

Corollary 5.1.9. *Let* S *be the spectrum of a strictly henselian noetherian local ring and let* $f : X \to S$ *be a locally acyclic morphism. Suppose that for every geometric point* t *of* S *one has* $\mathrm{H}^0(X_t, \mathbb{Z}/n) = \mathbb{Z}/n$ *and* $\mathrm{H}^q(X_t, \mathbb{Z}/n) = 0$ *for* $q > 0$ *(in other words, the geometric fibres of* f *are acyclic). Then* $f_*\mathbb{Z}/n\mathbb{Z}/n$ *and* $\mathsf{R}^q f_*\mathbb{Z}/n = 0$ *for* $q > 0$.

Corollary 5.1.10. *Let* $f : X \to Y$ *and* $g : Y \to X$ *be morphisms of locally noetherian schemes. If* f *and* g *are locally acyclic, then so is* $g \circ f$.

We may suppose that X, Y, and Z are strictly local and that f and g are local morphisms. It is then enough to show that, if z is an algebraic geometric point of Z, then $\mathrm{H}^0(X_z, \mathbb{Z}/n) = \mathbb{Z}/n$ and $\mathrm{H}^q(X_z, \mathbb{Z}/n) = 0$ for $q > 0$.

Since g is locally acyclic, one has $\mathrm{H}^0(Y_z, \mathbb{Z}/n) = \mathbb{Z}/n$ and $\mathrm{H}^q(Y_z, \mathbb{Z}/n) = 0$ for $q > 0$. Moreover, the morphism $f_z : X_z \to Y_z$ is locally acyclic (5.1.4), and its geometric fibres are acyclic because they are vanishing-cycle varieties of f. By (5.1.9), $\mathsf{R}^q f_{z*}\mathbb{Z}/n = 0$ for $q > 0$. In addition, $f_{z*}\mathbb{Z}/n$ is constant with fibre $\mathbb{Z}/n$ on Y_z. The result follows from the Leray spectral sequence for f_z.

5.2 Local acyclicity of a smooth morphism

Theorem 5.2.1. *A smooth morphism is locally acyclic.*

Let $f : X \to S$ be a smooth morphism. The assertion is local for the étale topology on X and S, so we may suppose that X is standard affine space of dimension d over S. By passage to the limit, we may suppose that S is noetherian, and the transitivity of local acyclicity (5.1.10) shows that it suffices to treat $d = 1$.

Let s be a geometric point of S and x a geometric point of X centred at a closed point of X_s. We must show that the geometric fibres of $\widetilde{X}^x \to \widetilde{S}^s$ are acyclic. Henceforth put $S = \widetilde{S}^s = \mathrm{Spec}(A)$ and $X = \widetilde{X}^x$. Then $X \simeq \mathrm{Spec}\, A\{T\}$, where $A\{T\}$ is the henselization of $A[T]$ at the point $T = 0$ above s.

If t is a geometric point of S, the fibre X_t is a projective limit of smooth affine curves over t. Thus $\mathrm{H}^q(X_t, \mathbb{Z}/n) = 0$ for $q \geqslant 0$, and it suffices to show that $\mathrm{H}^0(X_t, \mathbb{Z}/n) = \mathbb{Z}/n$ and $\mathrm{H}^1(X_t, \mathbb{Z}/n) = 0$ for n prime to the residue characteristic of S. This follows from the next two propositions.

Proposition 5.2.2. *Let* A *be a strictly henselian local ring,* $S = \mathrm{Spec}(A)$*, and* $X = \mathrm{Spec}\, A\{T\}$*. Then the geometric fibres of* $X \to S$ *are connected.*

By passage to the limit, we may reduce to the case in which A is a strict henselization of a finitely generated $\mathbb{Z}$-algebra.

Let $\bar{t}$ be a geometric point of S lying over t, and let k' be a finite separable extension of $k(t)$ in $k(\bar{t})$. Put $t' = \mathrm{Spec}(k')$ and $X_{t'} = X \times_S \mathrm{Spec}(k')$. We must check that, for all $\bar{t}$ and t', $X_{t'}$ is connected (by which we mean connected and nonempty). Let A' be the normalization of A in k', i.e. the ring of elements of k' integral over the image of A in $k(t)$. One has $A\{t\} \otimes_A A' \xrightarrow{\sim} A'\{T\}$: indeed, the left-hand side is henselian local (because A' is finite over A and local) and is a limit of local étale algebras over $A'\{T\} = A\{T\} \otimes_A A'$. Thus $X_{t'}$ is still the fibre at t' of $X = \mathrm{Spec}\,(A'\{T\})$ over $S' = \mathrm{Spec}(A')$. The local scheme X' is normal, hence integral; its localization $X_{t'}$ is still integral, and a fortiori connected.

Proposition 5.2.3. *Let A be a strictly henselian local ring, $S = \mathrm{Spec}(A)$, and $X = \mathrm{Spec}\,(A\{T\})$. Let $\bar{t}$ be a geometric point of S and $X_{\bar{t}}$ the corresponding geometric fibre. Then every Galois étale covering of $X_{\bar{t}}$ of order prime to the characteristic of the residue field of A is trivial.*

Lemma 5.2.4 (Zariski–Nagata purity theorem in dimension 2)**.** *Let C be a regular local ring of dimension 2 and let C' be a finite normal C-algebra étale over the open complement of the closed point of* $\mathrm{Spec}(C)$*. Then C' is étale over C.*

Indeed, C' is normal of dimension 2, and hence $\mathrm{prof}(C') = 2$. Since $\mathrm{prof}(C') + \dim \mathrm{proj}(C') = \dim(C) = 2$, it follows that C' is free over C. The set of points of C where C' is ramified is then defined by an equation, the discriminant; since it contains no point of height 1, it is empty.

Lemma 5.2.5 (Special case of Abhyankar's lemma)**.** *Let $S = \mathrm{Spec}(V)$ be a trait, let π be a uniformizer, let η be the generic point of S, let X be smooth over S, irreducible, and of relative dimension 1, let $\widetilde{X}_\eta$ be a Galois étale covering of X_η of degree n invertible on S, and let $S_1 = \mathrm{Spec}\left(V[\pi^{1/n}]\right)$. Use a subscript $(-)_1$ to denote base change from S to S_1. Then $\widetilde{X}_{1\eta}$ extends to an étale covering of X_1.*

Let $\widetilde{X}_1$ be the normalization of X_1 in $\widetilde{X}_{1\eta}$. By the structure of the tame inertia groups of the discrete valuation rings obtained by localizing X at the generic points of the special fibre X_s, $\widetilde{X}_1$ is étale over X_1 on the general fibre and at the generic points of the special fibre. By (5.2.4), it is étale everywhere.

5.2.6

Let t denote the point over which $\bar{t}$ lies. We are free to replace A by the normalization of A in a finite separable extension $k(t')$ of $k(t)$ in $k(\bar{t})$ (cf. 5.2.2). This and a preliminary passage to the limit allow us to suppose that

a) A is normal and noetherian, and t is the generic point of S.

b) The étale covering under consideration of $X_{\bar{t}}$ comes from an étale covering of X_t.

c) It comes from an étale covering $\widetilde{X}_U$ of the inverse image X_U of a nonempty open subset U of S (this follows from b): t is the limit of the U).

d) The complement of U has codimension $\geqslant 2$ (at the price of enlarging $k(t)$, by applying (5.2.5) to the discrete valuation rings obtained by localizing S at the points of $S \smallsetminus U$ of codimension 1 in S; there are only finitely many such points).

e) The covering $\widetilde{X}_U$ is trivial over the subscheme $T = 0$. (This may require enlarging $k(t)$ once more.)

Under these conditions, we shall show that the covering $\widetilde{X}_U$ is trivial.

Lemma 5.2.7. *Let A be a normal noetherian strictly henselian local ring, let U be an open subset of* $\mathrm{Spec}(A)$ *whose complement has codimension $\geqslant 2$, let V be its inverse image in $X = \mathrm{Spec}\,(A\{T\})$, and let V' be an étale covering of V. If V' is trivial over $T = 0$, then V' is trivial.*

Let $B = \Gamma(V', \mathscr{O})$. Since V' is the inverse image of V in $\mathrm{Spec}(B)$, it suffices to show that B is finite étale over $A\{T\}$ (and hence split, since $A\{T\}$ is strictly henselian). Let $\widehat{X} = A[\![T]\!]$, and denote by $(-)^\wedge$ base change from X to $\widehat{X}$. The scheme $\widehat{X}$ is faithfully flat over X. Hence $\Gamma(\widehat{V}', \mathscr{O}) = B \otimes_{A\{T\}} A[\![T]\!]$, and it suffices to show that this ring $\widehat{B}$ is finite étale over $A[\![T]\!]$.

Let V_m (resp. V'_m) be the subscheme of $\widehat{V}$ (resp. $\widehat{V}'$) defined by $T^{m+1} = 0$. By hypothesis, V'_0 is a trivial covering of V_0: a sum of n copies of V_0. The same is true of V'_m/V_m, because étale coverings are insensitive to nilpotents. We obtain

$$\varphi : \Gamma(\widehat{V}', \mathscr{O}) \to \varprojlim_m \Gamma(V'_m, \mathscr{O}) = \left(\varprojlim_m \Gamma(V, \mathscr{O})\right)^n .$$

By hypothesis, the complement of U has depth $\geqslant 2$: one has $\Gamma(V_m, \mathscr{O}) = A[T]/(T^{m+1})$, and φ is a homomorphism from $\widehat{B}$ to $A[\![T]\!]^n$. Over U, it provides n distinct sections of $\widehat{V}'/\widehat{V}$; hence $\widehat{V}'$ is trivial, a sum of n copies of $\widehat{V}$. Since the complement of $\widehat{V}$ in $\widehat{X}$ still has codimension $\geqslant 2$ (and therefore depth $\geqslant 2$), it follows that $\widehat{B} = A[\![T]\!]^n$, proving the lemma.

5.3 Applications

Theorem 5.3.1 (Specialization of cohomology groups)**.** *Let $f : X \to S$ be a proper locally acyclic morphism, for example a smooth proper morphism. Then the sheaves $\mathsf{R}^q f_* \mathbb{Z}/n$ are locally constant constructible, and for every specialization map $t \to \widetilde{S}^s$, the cospecialization maps $\mathrm{H}^q(X_t, \mathbb{Z}/n) \to \mathrm{H}^q(X_s, \mathbb{Z}/n)$ are bijective.*

This follows immediately from the definition of the cospecialization maps and the finiteness and proper base change theorems.

Theorem 5.3.2 (Base change by a smooth morphism)**.** *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram with g smooth. For every sheaf $\mathscr{F}$ on X that is torsion of order prime to the residue characteristics of S, one has

$$g^* \mathsf{R}^q f_* \mathscr{F} \xrightarrow{\sim} \mathsf{R}^q f'_* (g'^* \mathscr{F}).$$

After passing to an open covering of X, we reduce to the case where X is affine, and then, by passage to the limit, to the case where X is of finite type over S. The morphism f then factors as an open immersion $j : X \to \bar{X}$ followed by a proper morphism $\bar{f} : \bar{X} \to S$. The Leray spectral sequence for $\bar{f} \circ j$ and the proper base change theorem show that it suffices to prove the theorem when $X \to S$ is an open immersion.

In this case, if $\mathscr{F}$ has the form $\varepsilon_* C$, where $\varepsilon : t \to X$ is a geometric point of X, the theorem is a corollary of (5.1.5). The general case follows from Lemma (5.1.8).

Corollary 5.3.3. *Let K/k be an extension of separably closed fields, let X be a k-scheme, and let n be an integer prime to the characteristic of k. Then the canonical map $\mathrm{H}^q(X, \mathbb{Z}/n) \to \mathrm{H}^q(X_K, \mathbb{Z}/n)$ is bijective for every $q \geqslant 0$.*

It suffices to observe that $\bar{K}$ is a direct limit of smooth $\bar{k}$-algebras.

Theorem 5.3.4 (Relative purity)**.** *Let*

$$\begin{array}{ccccc} U & \xrightarrow{j} & X & \xleftarrow{i} & Y \\ & \searrow & \downarrow{\scriptstyle f} & \swarrow{\scriptstyle h} & \\ & & S & & \end{array} \tag{5.3.4.1}$$

be a commutative diagram with f smooth of pure relative dimension N, h smooth of pure relative dimension $N-1$, i a closed immersion, and $U = X \smallsetminus Y$. If n is prime to the residue characteristics of S, then

$$\begin{aligned} j_*\mathbb{Z}/n &= \mathbb{Z}/n \\ \mathsf{R}^1 j_*\mathbb{Z}/n &= \mathbb{Z}/n(-1)_Y \\ \mathsf{R}^q j_*\mathbb{Z}/n &= 0 \qquad \textit{for } q \geqslant 2 \end{aligned}$$

In these formulas, $\mathbb{Z}/n(-1)$ denotes the $\mathbb{Z}/n$-dual of $\boldsymbol{\mu}_n$. If t is a local equation for Y, the isomorphism $\mathsf{R}^1 j_*\mathbb{Z}/n \simeq \mathbb{Z}/n(-1)_Y$ is defined by the map $a : \mathbb{Z}/n \to \mathsf{R}^1 j_*\boldsymbol{\mu}_n$ sending 1 to the class of the $\boldsymbol{\mu}_n$-torsor of nth roots of t.

The question is local. This allows us to replace (X, Y) by a locally isomorphic pair, for example

$$\begin{array}{ccccc} \mathbb{A}^1_T & \xrightarrow{j} & \mathbb{P}^1_T & \xleftarrow{i} & T \\ & \searrow_{g} & \downarrow{\scriptstyle f} & \swarrow & \\ & & T & & \end{array}$$

with $T = \mathbb{A}^{n-1}_S$ and $i =$ the section at infinity. Corollary (5.1.9) applies to g and gives $\mathsf{R}^q g_{\mathbb{Z}}/n = \mathbb{Z}/n$ for $q = 0$ and 0 for $q > 0$. For f, on the other hand, (4.6.2.1) gives

$$\mathsf{R}^q f_*\mathbb{Z}/n = \mathbb{Z}/n, 0, \mathbb{Z}/n(-1), 0 \text{ for } q > 2.$$

It is readily checked that $j_*\mathbb{Z}/n = \mathbb{Z}/n$, and that the $\mathsf{R}^q j_*\mathbb{Z}/n = 0$ are concentrated on $i(T)$ for $q > 0$. The Leray spectral sequence $E_2^{pq} = \mathsf{R}^p f_* \mathsf{R}^q j_*\mathbb{Z}/n \Rightarrow \mathsf{R}^{p+q} g_*\mathbb{Z}/n$ therefore reduces to

$$\begin{array}{ccccccc} i^*\mathsf{R}^q j_*\mathbb{Z}/n & & 0 & & \cdots & & \\ \\ i^*\mathsf{R}^1 j_*\mathbb{Z}/n & & 0 & & \cdots & & \\ & \searrow^{d_2} & & & & & \\ \mathbb{Z}/n & & 0 & & \mathbb{Z}/n(-1) & 0 & \cdots \end{array}$$

$\mathsf{R}^q j_*\mathbb{Z}/n = 0$ for $q \geqslant 0$, and $\mathsf{R}^q j_*\mathbb{Z}/n$ is the extension by zero of a locally free sheaf of rank one on T (isomorphic, via d_2, to $\mathbb{Z}/n(-1)$). Since the map a defined above is injective (as may be checked fibre by fibre), it is an isomorphism, which proves (5.3.4).

5.3.5

We refer to [2, XVI.4,5] for proofs of the following applications of the acyclicity theorem (5.2.1).

Theorem 5.3.6. *Let $f : X \to S$ be a morphism of schemes of finite type over $\mathbb{C}$ and let $\mathscr{F}$ be a constructible sheaf on X. Then*

$$(\mathsf{R}^q f_* \mathscr{F})^{\mathrm{an}} \sim \mathsf{R}^q f_*^{\mathrm{an}}(\mathscr{F}^{\mathrm{an}})$$

(cf. 4.6.3: for ordinary cohomology it is necessary to assume that $\mathscr{F}$ is constructible, not merely torsion).

Theorem 5.3.7. *Let $f : X \to S$ be a morphism of schemes of finite type over a field k of characteristic 0, and let $\mathscr{F}$ be a constructible sheaf on X. Then the $\mathsf{R}^q f_* \mathscr{F}$ are also constructible.*

The proof uses resolution of singularities and (5.3.4). It is generalized to a morphism of finite type between excellent schemes of characteristic 0 in [2, XIX.5]. Another proof, independent of resolution, is given in this volume (Finiteness theorems in ℓ-adic cohomology, 1.1) and applies to a morphism of schemes of finite type over a field or over a Dedekind ring.

6 Poincaré duality

6.1 Introduction

Let X be an oriented topological manifold of pure dimension N, and suppose that X admits a finite open covering $\mathcal{U} = (U_i)_{1 \leqslant i \leqslant K}$ such that every nonempty intersection of the open subsets U_i is homeomorphic to a ball. For such a manifold, the Poincaré duality theorem can be presented as follows:

A The cohomology of X is the Čech cohomology associated with the covering $\mathcal{U}$. It is the cohomology of the complex

$$0 \to \mathbb{Z}^{A_0} \to \mathbb{Z}^{A_1} \to \cdots \tag{1}$$

where $A_k = \{(i_0, \ldots, i_k) : i_0 < \cdots < i_k \text{ and } U_{i_0} \cap \cdots \cap U_{i_k} \neq \varnothing\}$.

B For $a \in A_k$, $a = (i_0, \ldots, i_k)$, let $U_a = U_{i_1} \cap \cdots \cap U_{i_k}$ and let j_a be the inclusion of U_a into X. The constant sheaf $\mathbb{Z}$ on X admits the (left) resolution

$$\cdots \longrightarrow \bigoplus_{a \in A_1} j_{a!}\mathbb{Z} \longrightarrow \bigoplus_{a \in A_0} j_{a!}\mathbb{Z} \longrightarrow 0 \tag{2}$$

$$\downarrow$$

$$\mathbb{Z}.$$

The compactly supported cohomology $\mathrm{H}_c^\bullet(X, j_{a!}\mathbb{Z})$ is precisely the cohomology with proper support of the (oriented) ball U_a:

$$\mathrm{H}_c^i(X, j_{a!}\mathbb{Z}) = \begin{cases} 0 & \text{if } i \neq N \\ \mathbb{Z} & \text{if } i = N \end{cases}$$

The hypercohomology spectral sequence for the complex (2) and cohomology with proper support therefore says that $\mathrm{H}_c^i(X)$ is the $(i - N)$th cohomology group of the complex

$$\cdots \to \mathbb{Z}^{A_1} \to \mathbb{Z}^{A_0} \to 0. \tag{3}$$

This complex is dual to the complex (1), whence Poincaré duality.

The essential points of this construction are

a) the existence of a theory of cohomology with proper support;

b) the fact that every point x of a manifold X of pure dimension N has a fundamental system of open neighbourhoods U for which

$$\mathrm{H}^i_c(U) = \begin{cases} 0 & \text{for } i \neq N, \\ \mathbb{Z} & \text{for } i = N. \end{cases} \tag{4}$$

Poincaré duality in étale cohomology can be constructed on this model. For X smooth of pure dimension N over an algebraically closed field k, n invertible on X, and x a closed point of X, the key point is to calculate the projective limit over the étale neighbourhoods U of x:

$$\varprojlim \mathrm{H}^i_c(U, \mathbb{Z}/n) = \begin{cases} 0 & \text{if } i \neq 2N \\ \mathbb{Z}/n & \text{if } i = 2N. \end{cases} \tag{5}$$

Just as for topological manifolds one must first treat directly the case of an open ball (or even simply the interval $(0,1)$), here one must first treat directly the case of curves (6.2). The local acyclicity theorem for smooth morphisms then reduces (5) to this special case (6.3).

The isomorphisms (4) and (5) are not canonical: they depend on the choice of an *orientation* of X. For n invertible on a scheme X, $\boldsymbol{\mu}_n$ is a sheaf of free $\mathbb{Z}/n$-modules of rank one. We denote its Nth tensor power by $\mathbb{Z}/n(N)$ ($N \in \mathbb{Z}$). The intrinsic form of the second line of (5) is

$$\varprojlim \mathrm{H}^{2N}_c(U, \mathbb{Z}/n(N)) = \mathbb{Z}/n, \tag{6}$$

and $\mathbb{Z}/n(N)$ is called the *orientation sheaf* of X. Since the sheaf $\mathbb{Z}/n(N)$ is constant and isomorphic to $\mathbb{Z}/n$, it may be taken outside the limit sign (with N fixed), and one may instead write

$$\varprojlim \mathrm{H}^{2N}_c(U, \mathbb{Z}/n) = \mathbb{Z}/n(-N), \tag{7}$$

and Poincaré duality will take the form of a perfect duality, with values in $\mathbb{Z}/n(-N)$, between $\mathrm{H}^i(X, \mathbb{Z}/n)$ and $\mathrm{H}^{2N-i}_c(X, \mathbb{Z}/n)$.

6.2 The case of curves

6.2.1

Let $\bar{X}$ be a smooth projective curve over an algebraically closed field k, and let n be invertible on $\bar{X}$. The proof of (3.3.5) gives, for connected $\bar{X}$, a canonical isomorphism

$$\mathrm{H}^2(\bar{X}, \boldsymbol{\mu}_n) = \mathrm{Pic}(\bar{X})/n\,\mathrm{Pic}(\bar{X}) \xrightarrow[\sim]{\deg} \mathbb{Z}/n$$

Let D be a reduced divisor of $\bar{X}$ and let $X = \bar{X} \smallsetminus D$:

$$X \xhookrightarrow{\ j\ } \bar{X} \xhookleftarrow{\ i\ } D.$$

The exact sequence $0 \to j_!\boldsymbol{\mu}_n \to \boldsymbol{\mu}_n \to i_*\boldsymbol{\mu}_n \to 0$ and the fact that $\mathrm{H}^i(\bar{X}, i_*\boldsymbol{\mu}_n) = \mathrm{H}^i(D, \boldsymbol{\mu}_n) = 0$ for $i > 0$ give an isomorphism

$$\mathrm{H}^2_c(X, \boldsymbol{\mu}_n) = \mathrm{H}^2(\bar{X}, j_!\boldsymbol{\mu}_n) \xrightarrow{\sim} \mathrm{H}^2(\bar{X}, \boldsymbol{\mu}_n) \xrightarrow{\sim} \mathbb{Z}/n.$$

For disconnected $\bar{X}$, similarly

$$\mathrm{H}^2_c(X, \boldsymbol{\mu}_n) = (\mathbb{Z}/n)^{\pi_0(X)}$$

and the *trace morphism* is defined to be the sum

$$\mathrm{tr} : \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \simeq (\mathbb{Z}/n)^{\pi_0(X)} \xrightarrow{\ \Sigma\ } \mathbb{Z}/n.$$

Theorem 6.2.2. *The pairing* $\mathrm{tr}(a \smile b)$ *identifies each of the two groups* $\mathrm{H}^i(X, \mathbb{Z}/n)$ *and* $\mathrm{H}_c^{2-i}(X, \boldsymbol{\mu}_n)$ *with the dual (with values in* $\mathbb{Z}/n$*) of the other.*

Transcendental proof. Suppose that $\bar{X}$ is a smooth projective curve over a trait S and that $j : X \hookrightarrow \bar{X}$ is the inclusion of the complement of a divisor D étale over S. The cohomology (resp. cohomology with proper support) of the special and generic geometric fibres of X/S are "the same," i.e. the fibres of locally constant sheaves on S. This follows from the analogous facts for $\bar{X}$ and D, by means of the exact sequence $0 \to j_!\mathbb{Z}/n \to \mathbb{Z}/n \to \mathbb{Z}/n_D \to 0$ (for cohomology with proper support) and the formulas $j_*\mathbb{Z}/n = \mathbb{Z}/n$, $\mathsf{R}^1 j_*\mathbb{Z}/n \simeq \mathbb{Z}/n_D(-1)$, and $\mathsf{R}^i j_*\mathbb{Z}/n = 0$ $(i \geqslant 2)$ (for ordinary cohomology) (5.3.4).

This specialization principle reduces (6.2.2) to the case in which k has characteristic 0. By (5.3.3), this case reduces to $k = \mathbb{C}$. Finally, when $k = \mathbb{C}$, the groups $\mathrm{H}^\bullet(X, \mathbb{Z}/n)$ and $\mathrm{H}_c^\bullet(X, \boldsymbol{\mu}_n)$ agree with the groups of the same name calculated for the classical topological space X_{cl}, and, through the isomorphism $\mathbb{Z}/n \to \boldsymbol{\mu}_n : x \mapsto \exp\left(\frac{2\pi i x}{n}\right)$, the trace morphism is identified with "integration over the fundamental class." Thus (6.2.2) follows from Poincaré duality for X_{cl}.

6.2.3 Algebraic proof

For a very economical proof, see (Duality, §2). Here is another, related to the autoduality of the Jacobian.

Return to the notation of (6.2.1). We may suppose—and do suppose—that X is connected. The cases $i = 0$ and $i = 2$ being trivial, we also suppose that $i = 1$. Define ${}_D\mathbb{G}_m$ by the exact sequence $0 \to {}_D\,\mathbb{G}_m \to \mathbb{G}_m \to i_*\mathbb{G}_m \to 0$ (sections of $\mathbb{G}_m$ congruent to 1 mod D). The group $\mathrm{H}^1(\bar{X}, {}_D\,\mathbb{G}_m)$ classifies invertible sheaves on $\bar{X}$ trivialized on D. It is the group of points of $\mathrm{Pic}_D(\bar{X})$, an extension of $\mathbb{Z}$ (the degree) by the group of points of Rosenlicht's generalized Jacobian $\mathrm{Pic}_D^0(\bar{X})$ (corresponding to conductor 1 at each point of D), itself an extension of the abelian variety $\mathrm{Pic}^0(\bar{X})$ by the torus $\mathbb{G}_m^D/(\text{diagonal } \mathbb{G}_m)$.

a) The exact sequence $0 \to j_!\boldsymbol{\mu}_n \to {}_D\,\mathbb{G}_m \xrightarrow{n} {}_D\,\mathbb{G}_m \to 0$ gives an isomorphism

$$\mathrm{H}_c^1(X, \boldsymbol{\mu}_n) = \mathrm{Pic}_D^0(\bar{X})_n. \tag{6.2.3.1}$$

b) The map sending $x \in X(k)$ to the class of the invertible sheaf $\mathscr{O}(x)$ on $\bar{X}$, trivialized by 1 on D, comes from a morphism

$$f : X \to \mathrm{Pic}_D(\bar{X}).$$

For the remainder, fix a base point 0 and put $f_0(x) = f(x) - f(0)$. For every homomorphism $v : \mathrm{Pic}_D^0(\bar{X})_n \to \mathbb{Z}/n$, let $\bar{v} \in \mathrm{H}^1\left(\mathrm{Pic}_D^0(\bar{X}), \mathbb{Z}/n\right)$ be the image under v of the class in $\mathrm{H}^1\left(\mathrm{Pic}_D^0(\bar{X}), \mathrm{Pic}_D^0(\bar{X})_n\right)$ of the torsor defined by the extension

$$0 \longrightarrow \mathrm{Pic}_D^0(\bar{X})_n \longrightarrow \mathrm{Pic}_D^0(\bar{X}) \xrightarrow{n} \mathrm{Pic}_D^0(\bar{X}) \longrightarrow 0$$

Geometric class field theory (as presented in Serre [33]) shows that the map $v \mapsto f_0^*(\bar{v})$

$$\hom\left(\mathrm{Pic}_D^0(\bar{X})_n, \mathbb{Z}/n\right) \to \mathrm{H}^1(X, \mathbb{Z}/n) \tag{6.2.3.2}$$

is an isomorphism. To deduce (6.2.2) from (6.2.3.1) and (6.2.3.2), it remains to know that

$$\mathrm{tr}\left(u \smile f_0^*(\bar{v})\right) = -v(u). \tag{6.2.2.3}$$

6.3 The general case

Let X be a smooth algebraic variety of pure dimension N over an algebraically closed field k. To state the Poincaré duality theorem, one must first define the trace morphism

$$\mathrm{tr} : \mathrm{H}_c^{2N}(X, \mathbb{Z}/n(N)) \to \mathbb{Z}/n.$$

The definition is a painful unwinding starting from the case of curves [2, XVIII §2]. One then has:

Theorem 6.3.1. *The pairing* $\mathrm{tr}(a \smile b)$ *identifies each of the groups* $\mathrm{H}_c^i(X, \mathbb{Z}/n(N))$ *and* $\mathrm{H}^{2N-i}(X, \mathbb{Z}/n)$ *with the dual of the other.*

Let $x \in X$ be a closed point and let X_x be the strict localization of X at x. For U ranging over the étale neighbourhoods of x, put

$$\mathrm{H}_c^\bullet(X_x, \mathbb{Z}/n) = \varprojlim \mathrm{H}_c^\bullet(U, \mathbb{Z}/n). \tag{6.3.1.1}$$

It would be preferable to consider instead the pro-object "$\varprojlim$" $\mathrm{H}_c^\bullet(U, \mathbb{Z}/n)$, but since the groups involved are finite, the difference is inessential. As we tried to explain in the introduction, (6.3.1) follows from the facts that

$$\begin{gathered} \mathrm{H}_c^i(X_x, \mathbb{Z}/n) = 0 \text{ for } i \neq 2N \text{ and that} \\ \mathrm{tr} : \mathrm{H}_c^{2N}(X_x, \mathbb{Z}/n(N)) \xrightarrow{\sim} \mathbb{Z}/n \text{ is an isomorphism.} \end{gathered} \tag{6.3.1.2}$$

The case $N = 0$ is trivial. If $N > 0$, let Y_y be the strict localization at a closed point of a smooth scheme Y of pure dimension $N - 1$, and let $f : X_x \to Y_y$ be an essentially smooth morphism (of relative dimension one). The proof uses the Leray spectral sequence for cohomology with proper support for f to reduce to the case of curves. Because the "cohomology with proper support" under consideration is defined as limits (6.3.1.1), the existence of such a spectral sequence raises various passage-to-the-limit problems, treated in too much detail in [2, XVIII]. For every geometric point z of Y_y, one has

$$(\mathsf{R}^i f_! \mathbb{Z}/n)_z = \mathrm{H}_c^i(f^{-1}(z), \mathbb{Z}/n).$$

The geometric fibre $f^{-1}(z)$ is a projective limit of smooth curves over an algebraically closed field. It satisfies Poincaré duality. Its ordinary cohomology is given by the local acyclicity theorem for smooth morphisms:

$$\mathrm{H}^i(f^{-1}(z), \mathbb{Z}/n) = \begin{cases} \mathbb{Z}/n & \text{for } i = 0 \\ 0 & \text{for } i > 0. \end{cases}$$

By duality,

$$\mathrm{H}_c^i(f^{-1}(z), \mathbb{Z}/n) = \begin{cases} \mathbb{Z}/n(-1) & \text{for } i = 2 \\ 0 & \text{for } i \neq 2. \end{cases}$$

and the Leray spectral sequence becomes

$$\mathrm{H}_c^i(X_x, \mathbb{Z}/n(N)) = \mathrm{H}_c^{i-2}(Y_y, \mathbb{Z}/n(N-1)).$$

The result follows by induction on N.

6.4 Variants and applications

One can construct in étale cohomology a "duality formalism" (= functors $\mathsf{R}f_!$, $\mathsf{R}f_!$, and $\mathsf{R}f^!$ satisfying various compatibilities and adjunction formulas) parallel to the one that exists in coherent cohomology. In

this language, the results of the preceding paragraph can be rewritten as follows: if $f : X \to S$ is smooth of pure relative dimension N and $S = \mathrm{Spec}(k)$ with k algebraically closed, then

$$\mathsf{R}f^!\mathbb{Z}/n = \mathbb{Z}/n[2N](N).$$

This result holds without any hypothesis on S. It has the following corollary.

Corollary 6.4.1. *If f is smooth of pure relative dimension N and the sheaves $\mathsf{R}^i f_!\mathbb{Z}/n$ are locally constant, then the sheaves $\mathsf{R}^i f_*\mathbb{Z}/n$ are also locally constant, and*

$$\mathsf{R}^i f_*\mathbb{Z}/n = \underline{\hom}\left(\mathsf{R}^{2N-i} f_!\mathbb{Z}/n(N), \mathbb{Z}/n\right).$$

In particular, under the hypotheses of the corollary, the sheaves $\mathsf{R}^i f_*\mathbb{Z}/n$ are constructible. From this one can show that, if S is of finite type over the spectrum of a field or of a Dedekind ring, then, for every morphism of finite type $f : X \to S$ and every constructible sheaf $\mathscr{F}$ on X, the sheaves $\mathsf{R}^i f_*\mathscr{F}$ are constructible (Finiteness theorems in ℓ-adic cohomology, 1.1).

Chapter 2

Report on the trace formula

In this text, I have attempted to present Grothendieck's cohomological theory of L-functions as directly as possible. I follow very closely some of the lectures given by Grothendieck at the IHES in the spring of 1966. In the spirit of this volume, no appeal will be made to SGA 5—apart from two references to passages in Exposé XI, which is otherwise independent of that seminar.

In §10 (pp. 81–90) of [13], the reader will find a summary of the theory in the crucial case of curves.

1 Some reminders

1.1 Notation

$p, q, \mathbb{F}, \mathbb{F}_q$: p is a prime number, $q = p^f$ a power of p, and $\mathbb{F}$ an algebraic closure of the prime field $\mathbb{F}_p$. For every power p^n of p, denote by $\mathbb{F}_{p^n}$ the subfield with p^n elements of $\mathbb{F}$.

X_0, X: An object over $\mathbb{F}_q$ will often be denoted by a symbol carrying a subscript 0. The same symbol without the 0 then denotes the object obtained from it by extension of scalars to $\mathbb{F}$. For example, if $\mathscr{F}_0$ is a sheaf on a scheme X_0 over $\mathbb{F}_q$, denote by $\mathscr{F}$ its inverse image on $X = X_0 \otimes_{\mathbb{F}_q} \mathbb{F}$.

F: If X_0 is a scheme over $\mathbb{F}_q$, the endomorphism of X_0 that "sends the point with coordinates x to the point with coordinates x^q" is called the *Frobenius morphism* and denoted by F: it is the identity on the underlying topological space and, for x a local section of the structure sheaf, $F^*x = x^q$.

The endomorphism of X obtained from it by extension of scalars is also denoted by F. On the set $X(\mathbb{F}) = X_0(\mathbb{F})$ of rational points of X, F acts as the Frobenius substitution $\varphi \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ defined by $\varphi(x) = x^q$. In particular, $X^F = X_0(\mathbb{F}_q)$.

In case of ambiguity, we shall write $F_{(q)}$ rather than F.

1.2 Frobenius correspondence

Let X_0 be a scheme over $\mathbb{F}_q$, and let $\mathscr{F}_0$ be a sheaf on X_0. I like to view the *Frobenius correspondence* (an F-endomorphism of $\mathscr{F}_0$) as follows:

$$F^* : F^*\mathscr{F}_0 \xrightarrow{\sim} \mathscr{F}_0 \tag{1.2.1}$$

Regard $\mathscr{F}_0$ as the sheaf of local sections of a space $[\mathscr{F}_0]$ étale over X_0 ($[\mathscr{F}_0]$ is an algebraic space in

the sense of M. Artin). The functoriality of F gives a commutative diagram

$$\begin{array}{ccc} [\mathscr{F}_0] & \xrightarrow{F} & [\mathscr{F}_0] \\ \downarrow & & \downarrow \\ X_0 & \xrightarrow{F} & X_0 \end{array}$$

Because $[\mathscr{F}_0]$ is étale over X_0, this diagram is Cartesian; it gives an isomorphism $[\mathscr{F}_0] \xrightarrow{\sim} F^*[\mathscr{F}_0]$ whose inverse is (1.2.1). We also denote by F^* the correspondences or morphisms deduced from F by extension of scalars or by functoriality, such as $F^* : \mathrm{H}_c^\bullet(X, \mathscr{F}) \to \mathrm{H}_c^\bullet(X, \mathscr{F})$.

For a formal definition, I refer to [21, XI.1,2]. Only the case $q = p$ is considered there, but the general case is treated in the same way.

1.3

The Frobenius correspondence is functorial in $(X_0, \mathscr{F}_0)$. If $f_0 : X_0 \to Y_0$ is a morphism of $\mathbb{F}_q$-schemes, $\mathscr{F}_0$ a sheaf on X_0, $\mathscr{G}_0$ a sheaf on Y_0, and $u \in \hom(f_0^*\mathscr{G}_0, \mathscr{F}_0)$ an f_0-morphism from $\mathscr{G}_0$ to $\mathscr{F}_0$, there is a commutative diagram of spaces $f_0F = Ff_0$, and above it a commutative diagram of sheaves

$$\begin{array}{ccc} X_0 & \xrightarrow{f_0} & Y_0 \\ \downarrow{\scriptstyle F} & & \downarrow{\scriptstyle F} \\ X_0 & \xrightarrow{f_0} & Y_0 \end{array} \qquad \begin{array}{ccc} \mathscr{F}_0 & \xleftarrow{u} & \mathscr{G}_0 \\ \uparrow{\scriptstyle F^*} & & \uparrow{\scriptstyle F^*} \\ \mathscr{F}_0 & \xleftarrow{} & \mathscr{G}_0 \end{array}$$

In particular, the correspondence on $f_{0*}\mathscr{F}_0$ deduced from the Frobenius correspondence of $(X_0, \mathscr{F}_0)$ is the Frobenius correspondence of $(U_0, f_{0*}\mathscr{F}_0)$.

1.4

The natural morphism $Y \to Y_0$ is a limit of étale morphisms. It follows that for every abelian sheaf $\mathscr{F}_0$ on X_0, the inverse images on Y of the $\mathsf{R}^i f_0\mathscr{F}_0$ are the $\mathsf{R}^i f_*\mathscr{F}$.

On $\mathsf{R}^i f_*\mathscr{F}$, one has

a) the correspondence $F^*\mathsf{R}^i f_*\mathscr{F} \xrightarrow{\sim} \mathsf{R}^i f_*\mathscr{F} \xrightarrow{F^*} \mathsf{R}^i f_*\mathscr{F}$ deduced by functoriality from the Frobenius correspondence;

b) the correspondence $F^*\mathsf{R}^i f_*\mathscr{F} \to \mathsf{R}^i f_*\mathscr{F}$ that is the inverse image on Y of the Frobenius correspondence on $\mathsf{R}^i i f_{0*}\mathscr{F}_0$.

It follows from (1.3) that these correspondences coincide. They induce the initial term of a Frobenius action on the entire spectral sequence

$$\mathrm{H}^p(Y, \mathsf{R}^q f_*\mathscr{F}) \to \mathrm{H}^{p+q}(X, \mathscr{F}).$$

Likewise for cohomology with proper support, and for the spectral sequence

$$\mathrm{H}_c^p(Y, \mathsf{R}^q f_*\mathscr{F}) \to \mathrm{H}_c^{p+q}(X, \mathscr{F})$$

1.5 Change of finite field

Let $(X_0, \mathscr{F}_0)$ be over $\mathbb{F}_q$, let n be an integer, and let $(X_1, \mathscr{F}_1)$ over $\mathbb{F}_{q^n}$ be obtained from $(X_0, \mathscr{F}_0)$ by extension of scalars. On the one hand there is the Frobenius correspondence $F_{(q)} : X_0 \to X_0$), $F_{(q)}^* : F_{(q)}^* \mathscr{F}_0 \to \mathscr{F}_0$, and on the other there is $F_{(q^n)} : X_1 \to X_1$, $F_{(q^n)}^* : F_{(q^n)}^* \mathscr{F}_1 \to \mathscr{F}_1$. It is easily verified that $F_{(q^n)}$ is obtained from the n-th power of $F_{(q)}$ by extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$; after extension of scalars to $\mathbb{F}$, one still has the same identity $F_{(q^n)} = F_{(q)}^n$ between endomorphisms of X and correspondences on X.

If x is a closed point of X_0, with $[k(x), \mathbb{F}_q] = n$, and $\bar{x} \in X_0(\mathbb{F}) = X(\mathbb{F})$ lies over x, then $\bar{x}$ is a fixed point of $F_{(q^n)} = F_{(q)}^n$. Denote by F_x^* the endomorphism of $\mathscr{F}_{\bar{x}}$ induced by $F_{(q^n)}^*$. Up to isomorphism, $(F_{\bar{x}}^*, \mathscr{F}_{\bar{x}})$ does not depend on the choice of $\bar{x}$. The trace, etc. of F_x^* is denoted by notation such as $\mathrm{tr}(F_x^*, \mathscr{F})$.

1.6 L-function

Let X_0 be a scheme of finite type over $\mathbb{F}_q$. Denote by $|X_0|$ the set of its closed points, and for $x \in |X_0|$ set $\deg(x) = [k(x), \mathbb{F}_p]$. If $\mathscr{F}_0$ is a $\mathbb{Q}_\ell$-sheaf on X_0 (for this notion, see below)—or a constructible flat sheaf of modules over a commutative Noetherian ring A—denote by $L(X_0, \mathscr{F}_0)$ the formal series with constant term 1, in $\mathbb{Q}_\ell[\![t]\!]$ or in $A[\![t]\!]$,

$$L(X_0, \mathscr{F}_0) = \prod_{x \in |X_0|} \det\left(1 - F_x^* t^{\deg(x)}, \mathscr{F}\right)^{-1}.$$

1.7 Remark

In definition (1.6), one has $\deg(x) = [k(x), \mathbb{F}_p]$ (absolute degree), and not $\deg(x) = [k(x), \mathbb{F}_q]$ (degree relative to $\mathbb{F}_q$). The effect is that $L(X_0, \mathscr{F}_0)$ depends only on the scheme X_0 and the sheaf $\mathscr{F}_0$, not on q and the structure morphism $X_0 \to \mathrm{Spec}(\mathbb{F}_q)$. The indeterminate t is an avatar of p^{-s}.

1.8 The Galois point of view

This subsection (n°) will not be used in the remainder of the report. For $\mathscr{F}_0$ an abelian sheaf on X_0, the Galois group $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ acts on $\mathbb{F}$, and by transport of structure on $\mathrm{H}_c^i(X, \mathscr{F})$. In particular, the Frobenius substitution φ (1.1) induces an automorphism, also denoted by φ, of $\mathrm{H}_c^i(X, \mathscr{F})$. The Frobenius correspondence, on the other hand, defines an endomorphism F^* of $\mathrm{H}_c^i(X, \mathscr{F})$. One has

$$F^{*-1} = \varphi \qquad (\text{on } \mathrm{H}_c^i(X, \mathscr{F})). \tag{1.8.1}$$

To see this, apply (1.4) to the case where Y_0 is a point: $Y_0 = \mathrm{Spec}(\mathbb{F}_q)$. One finds that $\mathrm{H}_c^i(X, \mathscr{F})$ is the fiber, at the geometric point $\mathrm{Spec}(\mathbb{F})$ of Y_0, of $\mathsf{R}^i f_! \mathscr{F}_0$, and that F^* is the fiber of the Frobenius correspondence of $\mathsf{R}^i f_! \mathscr{F}_0$. Let $[\mathsf{R}^i f_! \mathscr{F}_0]$ be as in (1.2). One has $\mathrm{H}_c^i(X, \mathscr{F}) = [\mathsf{R}^i f_! \mathscr{F}_0](\mathbb{F})$ and

a) φ acts by transport of structure, via its action on $\mathbb{F}$;

b) F^* is the inverse of $F : [\mathsf{R}^i f_! \mathscr{F}_0](\mathbb{F}) \to [\mathsf{R}^i f_! \mathscr{F}_0](\mathbb{F})$.

The conclusion follows from the identity $F = \varphi$ of (1.1).

One advantage of the Galois point of view is that it permits reasoning by transport of structure.

2 $\mathbb{Q}_\ell$-sheaves

The notion of $\mathbb{Q}_\ell$-sheaves is developed in detail in [21, V,VI]. We shall use only the definitions and results below.

Definition 2.1. *Let X be a Noetherian scheme. A $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ on X is an inverse system of sheaves $\mathscr{F}_n$ ($n \geqslant 0$), with $\mathscr{F}_n$ a constructible sheaf of $\mathbb{Z}/\ell^{n+1}$-modules [2, IX.2], such that the transition morphism $\mathscr{F}_n \to \mathscr{F}_{n-1}$ factors through an isomorphism*

$$\mathscr{F}_n \otimes_{\mathbb{Z}/\ell^{n+1}} \mathbb{Z}/\ell^n \xrightarrow{\sim} \mathscr{F}_{n-1}.$$

The sheaf $\mathscr{F}$ is said to be lisse *if the $\mathscr{F}_n$ are locally constant.*

The terminology of [21] is different: "constructible $\mathbb{Z}_\ell$-sheaf" is used there for "$\mathbb{Z}_\ell$-sheaf," and "constructibly twisted constant" for "lisse."

Likewise below, for $\mathbb{Q}_\ell$-sheaves.

2.2

If k is a separably closed field, a sheaf of $\mathbb{Z}/\ell^n$-modules on $\mathrm{Spec}(k)$ is identified with a $\mathbb{Z}/\ell^n$-module, and the functor $\mathscr{F} \mapsto \varprojlim \mathscr{F}_n$ identifies the $\mathbb{Z}_\ell$-sheaves on $\mathrm{Spec}(k)$ with the $\mathbb{Z}_\ell$-modules of finite type. This justifies the following definition: the *fiber* of the $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ at the geometric point x of X is the $\mathbb{Z}_\ell$-module $\varprojlim(\mathscr{F}_n)_x$.

2.3

Suppose X is connected, and let x be a geometric point of X. It is known that the "fiber at x" functor is an equivalence between the category of constructible locally constant sheaves of $\mathbb{Z}/\ell^n$-modules and the category of finite-type $\mathbb{Z}/\ell^n$-modules equipped with a continuous action of $\pi_1(X,x)$. Passing to the limit gives the following proposition.

Proposition 2.4. *Under the hypotheses above, the "fiber at x" functor is an equivalence between the category of lisse $\mathbb{Z}_\ell$-sheaves on X and that of finite-type $\mathbb{Z}_\ell$-modules equipped with a continuous action of $\pi_1(X,x)$.*

Proposition 2.5. *Let $\mathscr{F}$ be a $\mathbb{Z}_\ell$-sheaf on a Noetherian scheme X. There is a finite partition of X into locally closed parts X_i such that the $\mathscr{F}|X_i$ are lisse.*

Set $\mathrm{gr}_\ell^n \mathscr{F} = \ell^n \mathscr{F}_m / \ell^{m+1}\mathscr{F}_m$ for $m \geqslant n$, and $\mathrm{gr}_\ell \mathscr{F} = \bigoplus \mathrm{gr}_\ell^n \mathscr{F}$. It is a sheaf of modules over the Noetherian ring $\mathrm{gr}_\ell \mathbb{Z} = \mathbb{Z}/\ell[t]$, the graded ring of $\mathbb{Z}$ for the ℓ-adic filtration. It is a quotient of $\mathrm{gr}_\ell^0 \mathscr{F} \otimes_{\mathbb{Z}/\ell} \mathbb{Z}/\ell[t]$, hence a constructible sheaf of $\mathrm{gr}_\ell \mathbb{Z}$-modules [2, IX.2]; and there is a finite partition of X into locally closed parts X_i such that $\mathrm{gr}_\ell \mathscr{F}|X_i$ is locally constant. Each of the sheaves $\mathrm{gr}_\ell^n \mathscr{F}|X_i$ is then locally constant, as are the $\mathscr{F}_m|X_i$, which are successive extensions of them.

2.6

If $\mathscr{F}$ and $\mathscr{G}$ are two $\mathbb{Z}_\ell$-sheaves on X, set $\hom(\mathscr{F},\mathscr{G}) = \varprojlim(\mathscr{F}_n, \mathscr{G}_n)$; this is a $\mathbb{Z}_\ell$-module. Since $\hom(\mathscr{F}_n,\mathscr{G}_n) = \hom(\mathscr{F}_m,\mathscr{G}_n)$ for $m \geqslant n$, one also has

$$\hom(\mathscr{F},\mathscr{G}) = \varprojlim_n \varinjlim_m \hom(\mathscr{F}_m,\mathscr{G}_n),$$

so that the functor $\mathscr{F} \mapsto \text{“}\varprojlim\text{”}\mathscr{F}_n$ is fully faithful from the category of $\mathbb{Z}_\ell$-sheaves to that of pro-sheaves on X (= pro-objects of the category of sheaves on X). We shall use it to identify $\mathbb{Z}_\ell$-sheaves with certain pro-sheaves (those such that every $\mathscr{F}_n = \mathscr{F}/\ell^{n+1}\mathscr{F}$ is a sheaf, and $\mathscr{F} \xrightarrow{\sim} \text{“}\varprojlim\text{”}\mathscr{F}_n$). It is easily verified that if $\mathscr{F} + \text{“}\varprojlim\text{”}\mathscr{F}_n$, with $\mathscr{F}_n$ a constructible sheaf of $\mathbb{Z}/\ell^n$-modules, then $\mathscr{F}$ is a $\mathbb{Z}_\ell$-sheaf if and only if the inverse systems (in n) $\mathscr{F}_n/\ell^{k+1}\mathscr{F}_n$ are essentially constant: the inverse system of $\mathscr{F}'_k = \varprojlim_n \mathscr{F}_n/\ell^{k+1}\mathscr{F}_n$ then satisfies the conditions of (2.1), and $\mathscr{F} = \text{“}\varprojlim\text{”}\mathscr{F}'_k$.

In what follows, we abandon the notation $\mathscr{F}_n$ of (2.1); if $\mathscr{F}$ is a $\mathbb{Z}_\ell$-sheaf, the sheaf $\mathscr{F}_n$ of (2.1) will be denoted by $\mathscr{F} \otimes \mathbb{Z}/\ell^{n+1}$.

2.7

In the abelian category of pro-sheaves, the subcategory of $\mathbb{Z}_\ell$-sheaves is stable under kernels, cokernels, and extensions. This is clear for cokernels. For kernels, if $f : \mathscr{F} \to \mathscr{G}$ is a morphism of $\mathbb{Z}_\ell$-sheaves, it follows readily from (2.4, 2.5) and Artin–Rees that the inverse system of $\ker(f : \mathscr{F} \otimes \mathbb{Z}/\ell^n \to \mathscr{G} \otimes \mathbb{Z}/\ell^n)$ satisfies the criterion given in (2.6) above. There is even an integer r such that for every n

$$\ker(f) \otimes \mathbb{Z}/\ell^n = \ker(f : \mathscr{F} \otimes \mathbb{Z}/\ell^{n+r} \to \mathscr{G} \otimes \mathbb{Z}/\ell^{n+r}) \otimes \mathbb{Z}/\ell^n.$$

The case of extensions is left to the reader.

2.8

A $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ is *torsion-free* if $\ker(\ell : \mathscr{F} \to \mathscr{F}) = 0$. Each $\mathscr{F} \otimes \mathbb{Z}/\ell^n$ is then flat over $\mathbb{Z}/\ell^n$ (i.e. its fibers are finite-rank free $\mathbb{Z}/\ell^n$-modules). It follows from (2.4, 2.5) that for every $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ there is an integer n such that $\mathscr{F}/\ker(\ell^n)$ is torsion-free.

2.9

The abelian category of $\mathbb{Q}_\ell$-*sheaves* on X is the quotient of that of $\mathbb{Z}_\ell$-sheaves by the subcategory of torsion $\mathbb{Z}_\ell$-sheaves. Equivalently, its objects are the $\mathbb{Z}_\ell$-sheaves, and, denoting by $\mathscr{F} \otimes \mathbb{Z}_\ell$ the $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$, viewed as an object of the category of $\mathbb{Q}_\ell$-sheaves, one has

$$\hom(\mathscr{F} \otimes \mathbb{Q}_\ell, \mathscr{G} \otimes \mathbb{Q}_\ell) = \hom(\mathscr{F}, \mathscr{G}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The *fiber* of a $\mathbb{Q}_\ell$-sheaf $\mathscr{F} \otimes \mathbb{Q}_\ell$ at a geometric point x of X is the finite-dimensional $\mathbb{Q}_\ell$-vector space $\mathscr{F}_x \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

2.10

Let X be a separated scheme of finite type over an algebraically closed field k, and let $\mathscr{F}$ be a constructible $\mathbb{Q}_\ell$-sheaf on X. Let $\mathscr{F}'$ be a constructible $\mathbb{Z}_\ell$-sheaf such that $\mathscr{F} \sim \mathscr{F}' \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. We shall see in (4.11) that

$$\left(\varprojlim \mathrm{H}^q_c(X, \mathscr{F}' \otimes \mathbb{Z}/\ell^n)\right) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

is a finite-dimensional $\mathbb{Q}_\ell$-vector space. It depends only on $\mathscr{F}'$, and is denoted by $\mathrm{H}^q_c(X, \mathscr{F})$. The proof does not use the fact that ℓ is prime to the characteristic, but only that case concerns us.

2.11

In this exposé, all important calculations will be carried out at the level of $\mathbb{Z}/\ell^n$-sheaves, with passage to $\mathbb{Q}_\ell$-sheaves taking place only at the very end. In practice, however, it is often convenient to work systematically with $\mathbb{Q}_\ell$-sheaves. In Exposés V and VI of [21], Jouanolou gives the formalism that makes this possible. One of the essential results (VI 2.2) is that if $f : X \to Y$ is a separated morphism of finite type of Noetherian schemes, and $\mathscr{F}$ a $\mathbb{Z}_\ell$-sheaf on Y, then, for every q, the pro-sheaf "$\varprojlim$"$\mathsf{R}^q f_!(\mathscr{F} \otimes \mathbb{Z}/\ell^n)$ is a $\mathbb{Z}_\ell$-sheaf on Y. He even proves a stronger regularity property (of Artin–Rees type) for the inverse system of the $\mathsf{R}^q f_!(\mathscr{F} \otimes \mathbb{Z}/\ell^n)$.

Jouanolou even works in a more general setting, encompassing the very useful case of E_λ-sheaves, for E_λ a finite extension of $\mathbb{Q}_\ell$. They are defined as the $\mathbb{Q}_\ell$-sheaves were, with $\mathbb{Q}_\ell$ replaced by E_λ, $\mathbb{Z}_\ell$ by the ring $\mathscr{O}_\lambda$ of the valuation λ, and $\mathbb{Z}/\ell^n$ by $\mathscr{O}_\lambda/\pi^n$, where π is a uniformizer. It would amount to the same thing to define an E_λ-sheaf as a $\mathbb{Q}_\ell$-sheaf $\mathscr{F}$ equipped with a homomorphism $E_\lambda \to \mathrm{End}(\mathscr{F})$.

3 Trace formula and L-functions

We use the notation of §1, and ℓ is a prime number $\neq p$.

Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$ ($q = p^f$), and let $\mathscr{F}_0$ be a constructible $\mathbb{Q}_\ell$-sheaf on X_0. Grothendieck's cohomological interpretation of L-functions is the following theorem:

Theorem 3.1. $L(X_0, \mathscr{F}_0) = \prod_i \det\left(1 - F^* t^f, \mathrm{H}^i_c(X, \mathscr{F})\right)^{(-1)^{i+1}}$.

It is obtained as a consequence of the trace formula:

Theorem 3.2. *For every integer n, $\sum_{x \in X^{F^n}} \mathrm{tr}\,(F^{n*}, \mathscr{F}_x) = \sum_i (-1)^i \,\mathrm{tr}\left(F^{n*}, \mathrm{H}^i_c(X, \mathscr{F})\right)$.*

Let us briefly recall the classical method for deducing (3.1) from (3.2). Set $T = t^f$; both sides are formal series in T with constant term 1. Since $\mathbb{Q}_\ell$ has characteristic 0, it suffices to prove that the logarithmic derivatives $T\frac{d}{dT}\log$ of the two sides are equal.

To compute these logarithmic derivatives, we shall use the formula

$$T\frac{d}{dT}\log\det(1 - fT)^{-1} = \sum_{n>0} \mathrm{tr}(f^n)T^n,$$

whose proof (in a slightly more general setting) is recalled below. On the left-hand side, one finds

$$\begin{aligned} T\frac{d}{dT}\log L(X_0, \mathscr{F}_0) &= \sum_{x \in |X_0|} \sum_n [k(x) : \mathbb{F}_q]\,\mathrm{tr}\,(F^n_x, \mathscr{F})\, T^{n \cdot [k(x):\mathbb{F}_q]} \\ &= \sum_{n>0} T^n \sum_{x \in X^{F^n}} \mathrm{tr}\,(F^{n*}, \mathscr{F}_x) \end{aligned}$$

and on the right-hand side

$$\sum_{n>0} T^n \sum (-1)^i \,\mathrm{tr}\left(F^{n*}, \mathrm{H}^i_c(X, \mathscr{F})\right)\ ;$$

the terms are compared one by one.

Proposition 3.3. *Let A be a commutative ring, M a projective A-module of finite type, and f an endomorphism of M. For every invertible formal series Q, set $\frac{d}{dT}\log Q = Q^{-1}\frac{d}{dT}Q$. One has*

$$T\frac{d}{dT}\log\det(1 - fT)^{-1} = \sum_{n>0} \mathrm{tr}(f^n)T^n. \tag{3.3.1}$$

Let M' be an A-module such that $M \oplus M'$ is free of finite rank. Replacing (M, f) by $(M \oplus M', f \oplus 0)$ changes neither side of (3.3), and reduces us to the case where M is free. For $M = A^d$, (3.3.1) is an algebraic identity in the coefficients f_i^j of f. By the principle of extension of algebraic identities, it suffices to verify it for A an algebraically closed field of characteristic 0. Since both sides are additive in M in an extension $0 \to M' \to M \to M'' \to 0$, it even suffices to treat only the case where M has rank 1. If f is multiplication by a, (3.3.1) then reduces to the formula

$$T\frac{d}{dT}\log\left((1-aT)^{-1}\right) = \sum_{n>0} a^n T^n.$$

Corollary 3.4. *Let n be an integer, and let $f^{(n)}$ be the endomorphism $(x_1, \ldots, x_n) \mapsto (f(x_n), x_1, \ldots, x_{n-1})$ of M^n. One has*

$$\det(1 - fT^n) = \det(1 - f^{(n)}T). \tag{3.4.1}$$

Proceeding as in (3.3), we reduce to assuming that A has characteristic 0. It then suffices to verify that the logarithmic derivatives $-T\frac{d}{dT}\log$ of the two sides are equal.

$$-T\frac{d}{dT}\log\det(1-fT^n) = -nT^n\frac{d}{dT^n}\log\det(1-fT^n) = n\sum_{m>0}\operatorname{tr}(f^m)T^{nm},$$

$$-T\frac{d}{dT}\log\det\left(1-f^{(n)}T\right) = \sum_{m>0}\operatorname{tr}\left(f^{(n)m}\right)T^m,$$

and one observes that

$$\operatorname{tr}(f^{(n)m}) = 0 \qquad \text{if } n \nmid m$$
$$\operatorname{tr}(f^{(n)nm}) = n\operatorname{tr}(f^m).$$

3.5 Remark

For $A = \mathbb{Q}_\ell$, (3.4.1) is the special case $X_0 = \operatorname{Spec}(\mathbb{F}_{q^n})$ of (3.1). Using this identity, it is easy to verify directly that the right-hand side of (3.1) is independent of q and of the structure morphism $X_0 \to \operatorname{Spec}(\mathbb{F}_q)$.

3.6 Remark

If $j : X_0 \hookrightarrow \bar{X}_0$ is a compactification of X_0, one has $L(X_0, \mathscr{F}_0) = L(\bar{X}_0, j_!\mathscr{F}_0)$ and $\mathrm{H}^i_c(X, \mathscr{F}) = \mathrm{H}^i(\bar{X}, j_!\mathscr{F})$. This reduces (3.1) to the proper case, and explains the use of cohomology with proper support. For X proper, (3.2) has the form of a Lefschetz trace formula. Note that the local terms $\operatorname{tr}(F^{n*}, \mathscr{F}_x)$ $(x \in X^{F^n})$ depend only on the fiber of $\mathscr{F}$ at x and carry no multiplicity. For X and $\mathscr{F}$ smooth, this corresponds to the fact that the graph of F is transverse to the diagonal (since $dF = 0$).

3.7 Remark

If the formalism of $\mathbb{Q}_\ell$-sheaves is admitted (cf. 2.11; one needs the Leray spectral sequence and the base-change theorem for the higher direct images $\mathsf{R}^q f_!$), it is easy to reduce the proof of (3.1, 3.2) to the case where X_0 is a smooth curve and $\mathscr{F}_0$ is lisse. This is clearly explained in [22, §5] (for 3.1; 3.2 is treated in the same way).

3.8

There are two methods for proving (3.2).

Lefschetz–Verdier If X_0 is proper (cf. 3.6), the general Lefschetz–Verdier trace formula expresses the right-hand side of (3.2) as a sum of local terms, one for each point of X^{F^n}. In the original version of [21], this formula was proved only modulo resolution of singularities. The reader will find an unconditional proof in the definitive version. In the case of curves, to which one can reduce (3.7), all the ingredients of the proof were in fact already available.

To deduce (3.2) from the Lefschetz–Verdier formula, one must be able to compute its local terms. For a curve and the Frobenius endomorphism, this had been done by Artin and Verdier [40].

Nielson–Wecken A method inspired by the work of Nielsen and Wecken makes it possible to reduce (3.2) to a special case proved by Weil; this is what will be explained in the following sections.

4 Reduction to theorems in $\mathbb{Z}/\ell^n$-cohomology

We shall prove (3.2) by passage to the limit from an analogous theorem in $\mathbb{Z}/\ell^n$-cohomology. Its statement presents the following difficulty. Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, and let $\mathscr{F}_0$ be a flat constructible sheaf of $\mathbb{Z}/\ell^n$-modules. Its fibers are finite-rank free $\mathbb{Z}/\ell^n$-modules, and the left-hand side of (3.2) therefore makes sense (it is an element of $\mathbb{Z}/\ell^n$). The $\mathrm{H}^i_c(X, \mathscr{F})$, on the other hand, will not in general be free, so the right-hand side is not defined. Overcoming this difficulty is one of Grothendieck's essential uses of the theory of derived categories.

4.1

For the definition of the derived category $\mathsf{D}(\mathcal{A})$ of an abelian category $\mathcal{A}$, and that of the subcategories $\mathsf{D}^+(\mathcal{A})$, $\mathsf{D}^-(\mathcal{A})$, $\mathsf{D}^b(\mathcal{A})$, I refer to Verdier's text in this volume. For Λ a ring, write $\mathsf{D}(\Lambda)$ instead of D(category of left Λ-modules); for X a site, write $\mathsf{D}(X, \Lambda)$ instead of D(category of sheaves of left Λ-modules). Likewise for D^-, etc...

The sign R (resp. L) indicates a right-derived (resp. left-derived) functor. For example, $\overset{\mathsf{L}}{\otimes}$ denotes the derived functor of the tensor product: for K in $\mathsf{D}^-(\Lambda^\circ)$ and L in $\mathsf{D}^-(\Lambda)$ (Λ° the opposite ring of Λ), $K \overset{\mathsf{L}}{\otimes} L$ is computed as follows: take quasi-isomorphisms $K' \to K$, $L' \to L$, with K' and L bounded above and one of them having flat components, and take the simple complex $s(K' \otimes_\Lambda L)$ associated with the double complex $K' \otimes_\Lambda L'$.

4.2

Even to treat only the case of $\mathbb{Z}/\ell^n$-cohomology, we shall need the theory of traces of endomorphisms of modules over rings that are not necessarily commutative (finite-group algebras $\mathbb{Z}/\ell^n[G]$). These traces were introduced by Stallings and Hattori [3].

Let Λ be a ring (with identity, but not necessarily commutative). Denote by $\Lambda^\natural$ the quotient of the additive group of Λ by the subgroup generated by the commutators $ab - ba$. If f is an endomorphism of the free left Λ-module Λ^n, with matrix f_i^j, denote by $\mathrm{tr}(f)$ the image of $\sum f_i^f$ in $\Lambda^\natural$. If f and g are two homomorphisms $f : \Lambda^n \leftrightarrows \Lambda^m : g$, one immediately verifies that $\mathrm{tr}(fg) = \mathrm{tr}(gf)$.

Let f be an endomorphism of a projective Λ-module of finite type P. Write P as a direct factor of a free module Λ^n: this means choosing a module P' and an isomorphism $\alpha : P \oplus P' \xrightarrow{\sim} \Lambda^n$, or equivalently choosing a homomorphism $a : P \to \Lambda^n$ and a retraction $b : \Lambda^n \to P$ (take $P = \ker(b)$). Let $f' = \alpha(f \oplus 0)\alpha^{-1} = afb$. Set $\mathrm{tr}(f) = \mathrm{tr}(f')$. This element of $\Lambda^\natural$ depends only on f: if another choice

$c : P \leftrightarrows \Lambda^m : d$ is made, one has $a = a(dc)(ba)$, whence

$$\mathrm{tr}(afb) = \mathrm{tr}(adcbafb) = \mathrm{tr}(cbafbad) = \mathrm{tr}(cfd).$$

If f is an endomorphism of a $\mathbb{Z}/2$-graded projective Λ-module of finite type, with components $f_i^j : P^j \to P^i$, set

$$\mathrm{tr}(f) = \mathrm{tr}(f_0^0) - \mathrm{tr}(f_1^1).$$

If there is a risk of ambiguity, we shall instead write $\mathrm{tr}(f; P)$. If the homomorphisms $f : P \leftrightarrows Q : g$ are homogeneous, reduction to the free case immediately verifies that

$$\mathrm{tr}(fg) = \mathrm{tr}\left((-1)^{\deg f \deg g} gf\right). \tag{4.2.1}$$

If f is an endomorphism of a filtered $\mathbb{Z}/2$-graded Λ-module (P, F), with finite filtration compatible with the grading, and the $\mathrm{gr}_F^i(P)$ are projective of finite type, then P is projective of finite type, and

$$\mathrm{tr}(f; P) = \sum \mathrm{tr}\left(f; \mathrm{gr}_F^i(P)\right) \tag{4.2.2}$$

(split the filtration to reduce to the case of a direct sum).

4.3

A $\mathbb{Z}$-graded module will be regarded as $\mathbb{Z}/2$-graded by the parity of the degree. In particular, if f is an endomorphism of a bounded complex of projective Λ-modules of finite type, set

$$\mathrm{tr}(f) = \sum (-1)^i \mathrm{tr}(f^i). \tag{4.3.1}$$

By (4.2.1), if f is homotopic to zero, $\mathrm{tr}(f) = 0$:

$$f = dH + hD \to \mathrm{tr}(f) = 0. \tag{4.3.2}$$

Let $\mathsf{K}_{\mathrm{parf}}(\Lambda)$ be the category whose objects are bounded complexes of projective Λ-modules of finite type and whose arrows are morphisms of complexes taken up to homotopy. The functor $\mathsf{K}_{\mathrm{parf}}(\Lambda) \to \mathsf{D}(\Lambda)$ is fully faithful. Denote its essential image by $\mathsf{D}_{\mathrm{parf}}(\Lambda)$. Formulas (4.3.1) and (4.3.2) make it possible to define $\mathrm{tr}(f)$ for f an endomorphism of $K \in \mathrm{Ob}\, \mathsf{D}_{\mathrm{parf}}(\Lambda)$ (and this trace depends only on the isomorphism class of (K, f)).

4.4

Recall that the filtered derived category $\mathsf{DF}(\mathcal{A})$ of an abelian category $\mathcal{A}$ is the category obtained from that of complexes of objects of $\mathcal{A}$ with finite filtration by inverting the filtered quasi-isomorphisms (= morphisms of filtered complexes inducing a quasi-isomorphism on the associated graded). There are functors "underlying complex": $\mathsf{DF}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}) : (K, F) \mapsto K$, and "$p$-th component of the associated graded": $\mathsf{DF}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}) : (K, F) \mapsto \mathrm{gr}_F^p(K)$.

Let $\mathsf{KF}_{\mathrm{parf}}(\Lambda)$ be the category whose objects are bounded filtered complexes (K, F) with finite filtration, with the $\mathrm{gr}_F^p(K^q)$ projective of finite type, and whose arrows are filtration-preserving morphisms of complexes taken up to a filtration-preserving homotopy. The functor $\mathsf{KF}_{\mathrm{parf}}(\Lambda) \to \mathsf{DF}(\Lambda)$ is fully faithful; its essential image $\mathsf{DF}_{\mathrm{parf}}(\Lambda)$ is almost everywhere zero; if (K, F) is in $\mathsf{D}_{\mathrm{parf}}(\Lambda)$, then K is in $\mathsf{D}_{\mathrm{parf}}(\Lambda)$.

By (4.2.2), if f is an endomorphism of $(K, F) \in \mathrm{Ob}\, \mathsf{DF}_{\mathrm{parf}}(\Lambda)$, one has

$$\mathrm{tr}(f, K) = \sum_p \mathrm{tr}(f, \mathrm{gr}_F^p(K)). \tag{4.4.1}$$

4.5

Let Λ be a ring, and let $K \in \mathrm{Ob}\,\mathsf{D}^-(\Lambda)$. We say that K has tor-dimension $\leqslant r$ if, for every right Λ-module N, the hypertors $\mathrm{H}^i(N \overset{\mathsf{L}}{\otimes}_\Lambda K)$ vanish for $i < -r$. It is said to have finite tor-dimension if this condition holds for some r.

The same terminology is used for a complex of sheaves $K \in \mathrm{Ob}\,\mathsf{D}^-(X, \Lambda)$.

Lemma 4.5.1. *Let Λ be a left Noetherian ring, and let $K \in \mathrm{Ob}\,\mathsf{D}^-(\Lambda)$. Then K belongs to $\mathsf{DF}_{parf}(\Lambda)$ if and only if it has finite tor-dimension and the $\mathrm{H}^i(K)$ are of finite type.*

After replacing K by an isomorphic complex (in $\mathsf{D}^-(\Lambda)$), we may suppose that K is bounded above and has finite-rank free components (cf. 4.7 below). If K has tor-dimension $\leqslant r$, the $\mathrm{H}^i(K)$ vanish for $i < -r$, $K^{-r}/\operatorname{im}(d)$ admits the free resolution

$$(\cdots \to K^{-r-1} \to K^{-r}) \to K^{-r}/\operatorname{im}(d),$$

and for every right Λ-module N, $\mathrm{H}^{-r-k}(N \overset{\mathsf{L}}{\otimes}_\Lambda K) = \mathrm{H}^{-r-k}(N \otimes_\Lambda K) = \mathrm{Tor}_k(N, K^r/\operatorname{im} d)$ for $k \geqslant 1$.

By hypothesis, these groups vanish; the module $K^r/\operatorname{im} d$ is therefore flat and finitely presented, i.e. projective of finite type. The quotient

$$0 \to K^r/\operatorname{im}(d) \to K^{r+1} \to K^{r+2} \to \cdots$$

of K is quasi-isomorphic to K, and this shows that $K \in \mathrm{Ob}\,\mathsf{DF}_{\mathrm{parf}}(\Lambda)$.

Proposition–Definition 4.6. *Let X be a Noetherian scheme, and let Λ be a left Noetherian ring. Denote by $\mathsf{D}^b_{\mathrm{ctf}}(X, \Lambda)$ the subcategory of $\mathsf{D}^-(X, \Lambda)$ formed by the complexes $\mathscr{K}$ satisfying the following equivalent conditions:*

i) $\mathscr{K}$ is isomorphic (in $\mathsf{D}^-(X, \Lambda)$) to a bounded complex of flat constructible sheaves of Λ-modules.

ii) $\mathscr{K}$ has finite tor-dimension and the sheaves $\mathrm{H}^i(K)$ are constructible.

The same argument as in (4.5), starting from the following lemma.

Lemma 4.7. *If a complex $\mathscr{K}$ of sheaves of Λ-modules is such that the $\mathrm{H}^i(\mathscr{K})$ are constructible and vanish for large i, there is a quasi-isomorphism $\mathscr{K}' \to \mathscr{K}$, with $\mathscr{K}'$ bounded above and having flat constructible components.*

Construct $\mathscr{K}'$ by proceeding from right to left. For an m such that the $\mathrm{H}^i(\mathscr{K})$ $(i \geqslant m)$ vanish, take $\mathscr{K}'^i = 0$ for $i \geqslant m$. Next suppose that the $\mathscr{K}'^i$ have already been constructed for $i > n$:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & \mathscr{K}^{n-1} & \longrightarrow & \mathscr{K}^n & \longrightarrow & \mathscr{K}^{n+1} & \longrightarrow & \cdots \\ & & & & & & \uparrow & & \\ & & & & & & \mathscr{K}'^{n+1} & \longrightarrow & \cdots, \end{array}$$

with $\mathrm{H}^i(\mathscr{K}') \xrightarrow{\sim} \mathrm{H}^i(\mathscr{K})$ for $i > n+1$, and $\ker(d) \to \mathrm{H}^{n+1}(\mathscr{K})$ surjective in degree $n+1$. Define sheaves $\mathscr{A}$ and $\mathscr{B}$ by the Cartesian diagram

$$\begin{array}{ccccc} \mathscr{K}^n & \longrightarrow & \mathscr{K}^n/\operatorname{im}(d) & \longrightarrow & \ker(d) \\ \uparrow & & \uparrow & & \uparrow \\ \mathscr{A} & \xrightarrow{u} & \mathscr{B} & \longrightarrow & \ker(d). \end{array}$$

Then $\mathscr{B}$ is constructible, u is surjective, and to advance one step it suffices to construct K^n flat and constructible and $v : \mathscr{K}^n \to \mathscr{A}$ such that uv is surjective. The possibility of doing so is guaranteed by the following lemma.

Lemma 4.8. *Let $v : \mathscr{A} \to \mathscr{B}$ be an epimorphism of sheaves of Λ-modules, with $\mathscr{B}$ constructible. There is then $u : \mathscr{K} \to \mathscr{A}$, with $\mathscr{K}$ flat and constructible, such that vu is an epimorphism.*

The sheaf $\mathscr{A}$ is a quotient of a sum of sheaves of the form $\varphi_!\Lambda$, for $\varphi : U \to X$ étale. Since $\mathscr{B}$ is Noetherian ([2, IX.2.10]), a finite sum of these already maps onto $\mathscr{B}$, which proves the lemma.

Theorem 4.9. *Let $f : X \to Y$ be a separated morphism of finite type of Noetherian schemes. If $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X, \Lambda)$, then $\mathsf{R}f_!\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(Y, \Lambda)$.*

Recall that if $j : X \hookrightarrow \bar{X} \xrightarrow{\bar{f}} Y$ is a relative compactification of X/Y, then by definition $\mathsf{R}f_! = \mathsf{R}\bar{f}_* \circ j_!$, where $\mathsf{R}\bar{f}_*$ is the derived functor of $\bar{f}_*$. This formula reduces the proof of (4.9) to the case where f is proper. For f proper, f_* has finite cohomological dimension; this makes it possible to define $\mathsf{R}f_*$ on the entire derived category, and also as a functor

$$\mathsf{R}f_* : \mathsf{D}^-(X, \Lambda) \to \mathsf{D}^-(Y, \Lambda).$$

By composition, one likewise defines in general

$$\mathsf{R}f_! : \mathsf{D}^-(X, \Lambda) \to \mathsf{D}^-(Y, \Lambda).$$

For every right Λ-module N, one has (proof given below)

$$N \overset{\mathsf{L}}{\otimes} \mathsf{R}f_!\mathscr{K} \xrightarrow{\sim} \mathsf{R}f_!(N \overset{\mathsf{L}}{\otimes} \mathscr{K}). \tag{4.9.1}$$

Let us deduce (4.9) from (4.9.1).

a) The spectral sequence
$$E_2^{pq} = \mathsf{R}^p f_!\, \mathrm{H}^q(\mathscr{K}) \to \mathrm{H}^{p+q}\, \mathsf{R}f_!\mathscr{K}\,.$$
and the finiteness theorem for $\mathsf{R}f_!$ show that $\mathsf{R}f_!\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^b(X, \Lambda)$ and has constructible cohomology. It remains to prove finite tor-dimension (4.6).

b) If $\mathscr{K}$ has tor-dimension $\leqslant -r$, the same is true of $\mathsf{R}f_!\mathscr{K}$: one has $\mathrm{H}^i(N \overset{\mathsf{L}}{\otimes} \mathsf{R}f_!\mathscr{K}) = \mathrm{H}^i(\mathsf{R}f_!(N \overset{\mathsf{L}}{\otimes} \mathscr{K}))$, and in the spectral sequence above for $N \overset{\mathsf{L}}{\otimes} \mathscr{K}$, the E_2^{pq} vanish for $q < r - s$, hence a fortiori for $p + q < r$.

Let us prove (4.9.1), remaining at the level of complexes.

a) Using the defining formula $\mathsf{R}f_! = \mathsf{R}f_* \circ j_!$, we reduce to assuming that f is proper.

b) Represent $\mathscr{K}$ by a complex bounded above whose components are acyclic for $f_* : \mathsf{R}^p f_* \mathscr{K}^q = 0$ for $p > 0$. One then has $\mathsf{R}f_*K \sim f_*\mathscr{K}$. It is possible to work in this way with complexes bounded above only because f_* has finite cohomological dimension.

c) If L is a free right Λ-module, one has $\mathsf{R}^p f_*(L \otimes \mathscr{K}^q) = L \otimes \mathsf{R}^p f_* \mathscr{K}^q$: this is trivial when Λ is free of finite type, and the general case follows by passage to the limit. The sheaf $L \otimes \mathscr{K}^q$ is acyclic for f_*, and $f_*(L \otimes \mathscr{K}^q) = L \otimes f_*\mathscr{K}^q$.

d) Let $N_\bullet$ be a free resolution of N. One has $N \overset{\mathsf{L}}{\otimes} \mathscr{K} \sim s(N_\bullet \otimes \mathscr{K})$ (the simple complex associated with the double complex $N_\bullet \otimes \mathscr{K}$). By c),

$$\mathsf{R}f_*(N \overset{\mathsf{L}}{\otimes} \mathscr{K}) \sim \mathsf{R}f_*(s(N_\bullet \otimes \mathscr{K})) \sim f_*(s(N_\bullet \otimes \mathscr{K})) \sim s(N_\bullet \otimes f_*\mathscr{K}) \sim N \overset{\mathsf{L}}{\otimes} \mathsf{R}f_*\mathscr{K},$$

which is the asserted formula.

When Y is the spectrum of a separably closed field, the functor $\mathsf{R}f_!$ is identified with a functor denoted by $\mathsf{R}\Gamma_c : \mathsf{D}^b_{\mathrm{ctf}}(X, \Lambda) \to \mathsf{D}_{\mathrm{parf}}(\Lambda)$.

Theorem 4.10. *With the notation of §1, let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, and let Λ be a Noetherian torsion ring annihilated by an integer prime to p. Let $\mathscr{K}_0 \in \mathrm{Ob}\, \mathsf{D}^b_{\mathrm{ctf}}(X_0, \Lambda)$. With the definition of traces in (4.3), one has for every N*

$$\sum_{x \in X^{F^n}} \mathrm{tr}\,(F^{n*}, \mathscr{K}_x) = \mathrm{tr}\,(F^{n*}, \mathsf{R}\Gamma_c(X, \mathscr{K}))\,.$$

In the remainder of this §, we shall deduce 3.2 from 4.10, and prove 2.10.

4.11 Proof of 2.10

Let X be a separated scheme of finite type over an algebraically closed field k, and let $\mathscr{G}$ be a $\mathbb{Q}_\ell$-sheaf on X. By 2.8, it can be written in the form $\mathscr{G} = \mathscr{F} \otimes \mathbb{Q}_\ell$, with $\mathscr{F}$ a torsion-free $\mathbb{Z}_\ell$-sheaf. Set

$$K_n = \mathsf{R}\Gamma_c(X, \mathscr{F} \otimes \mathbb{Z}/\ell^{n+1}) \in \mathsf{D}(\mathbb{Z}/\ell^{n+1}).$$

By 4.9, one has $K_n \in \mathrm{Ob}\, \mathsf{D}_{\mathrm{parf}}(\mathbb{Z}/\ell^{n+1})$, and it follows from (4.9.1) that the natural arrow

$$K_n \overset{\mathsf{L}}{\otimes}_{\mathbb{Z}/\ell^{n+1}} \mathbb{Z}/\ell^n \longrightarrow K_{n+1}, \tag{4.11.1}$$

is an isomorphism. The problem of defining this arrow is discussed in 4.12.

We may now invoke [21, XI.3.3] (p. 31, lines 1–2 from the end; this passage is independent of the rest of [21]) to realize each K_n by a bounded complex of finite-rank free $\mathbb{Z}/\ell^{n+1}$-modules, and the arrows (4.11.1) by *isomorphisms of complexes*

$$K_n \otimes \mathbb{Z}/\ell^n \xrightarrow{\sim} K_{n-1}.$$

Let K be the complex $\varprojlim K_n$. It is a bounded complex of free $\mathbb{Z}_\ell$-modules, and $K_n \simeq K \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell^{n+1}$. One has

$$\mathrm{H}^i(K) = \varprojlim \mathrm{H}^i(K_n),$$

(indeed, every inverse system of finite groups satisfies the Mittag–Leffler condition), and $\mathrm{H}^i(K)$ is a $\mathbb{Z}_\ell$-module of finite type. One has

$$\mathrm{H}^i_c(X, \mathscr{G}) = \left(\varprojlim \mathrm{H}^i(K_n)\right) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell:$$

this is a finite-dimensional $\mathbb{Q}_\ell$-vector space.

4.12 The arrow (4.11.1)

To *define* the arrow (4.11.1), one cannot proceed as in 4.9, i.e. resolve $\mathbb{Z}/\ell^n$ by a complex of free $\mathbb{Z}/\ell^{n+1}$-modules, since one wishes to construct an isomorphism in $\mathsf{D}_{\mathrm{parf}}(\mathbb{Z}/\ell^n)$. The desired arrow is a special case of the following scalar-extension arrow.

(*) Let $\Lambda \to \Lambda'$ be a homomorphism of Noetherian torsion rings. For X a separated scheme of finite type over an algebraically closed k, and $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}_{\mathrm{ctf}}(X, \Lambda)$, one has an isomorphism

$$\mathsf{R}\Gamma_c(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda' \xrightarrow{\sim} \mathsf{R}\Gamma_c(X, \mathscr{K} \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda').$$

More general arrows are constructed in [2, XVII 4.2.12]. The method is as follows:

a) The definition of $\mathsf{R}\Gamma_c$ and the formula $j_!(\mathscr{K} \otimes_\Lambda \Lambda') = (j_!\mathscr{K}) \otimes_\Lambda \Lambda'$ for an open immersion reduce us to the case where X is proper.

b) For X proper, one must replace $\mathscr{K}$ by a bounded complex whose components are acyclic for the functor Γ, and whose fiber at every geometric point (or merely every closed point) is homotopic to a complex of flat Λ-modules. For such a complex, one has $\Gamma(X, \mathscr{K}) \sim \mathsf{R}\Gamma(X, \mathscr{K})$ and $K \otimes_\Lambda \Lambda' \sim \mathscr{K} \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda'$; the arrow (4.11.1) is defined as the composite

$$\mathsf{R}\Gamma(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda' \to \Gamma(X, \mathscr{K}) \otimes_\Lambda \Lambda' \to \Gamma(X, \mathscr{K} \otimes_\Lambda \Lambda') \xleftarrow{\sim} \mathsf{R}\Gamma(X, \mathscr{K} \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda').$$

c) It remains to show that $\mathscr{K}$ can be replaced by a complex of the desired kind [2, XVII.4.2.10]. First replace $\mathscr{K}$ by a bounded complex with flat components (4.6), then take a truncated canonical flasque resolution of it (this does not modify the homotopy type of the fiber complexes, and one truncates far enough for the acyclicity condition for Γ to hold (i.e. beyond the cohomological dimension of Γ)).

4.13 Proof of (3.2)

To simplify notation, we shall treat only the case $n = 1$ of (3.2). The general case follows by replacing F below by F^n.

Thus let X_0 be separated and of finite type over $\mathbb{F}_q$, and let $\mathscr{G}_0$ be a $\mathbb{Q}_\ell$-sheaf on X_0. One has $\mathscr{G}_0 = \mathscr{F}_0 \otimes \mathbb{Q}_\ell$, for a suitable torsion-free $\mathbb{Z}_\ell$-sheaf $\mathscr{F}_0$. Let $K_n = \mathsf{R}\Gamma_c(X, \mathscr{F} \otimes \mathbb{Z}/\ell^n)$. The Frobenius morphisms induce morphisms

$$F^* : K_n \to K_n \qquad (\text{in } \mathsf{D}_{\mathrm{parf}}(\mathbb{Z}/\ell^{n+1})),$$

which are deduced from one another via the isomorphisms (4.11.1).

As in (4.11), realize the K_n and the isomorphisms (4.11.1) at the level of complexes. The already cited passage [21, XV.3.3] makes it possible to realize the morphisms F^* by endomorphisms of complexes, again denoted by F^*, which reduce to one another. Again denote by F^* their inverse limit and the endomorphisms deduced from it. Since $\mathrm{H}^i_c(X, \mathscr{G}) = \mathrm{H}^i(K) \otimes \mathbb{Q}_\ell = \mathrm{H}^i(K \otimes \mathbb{Q}_\ell)$, one has (with the sign convention of (?.1)

$$\mathrm{tr}\,(F^*, \mathrm{H}^\bullet_c(X, \mathscr{G})) = \mathrm{tr}\,(F^*, K^\bullet \otimes \mathbb{Q}_\ell) = \varprojlim_n \mathrm{tr}\,(F^*, K_n^\bullet) = \varprojlim_n \mathrm{tr}\left(F^*, \mathsf{R}\Gamma_c(X, \mathscr{F} \otimes \mathbb{Z}/\ell^{n+1})\right).$$

Apply (4.10) to $\mathscr{F}_0 \otimes \mathbb{Z}/\ell^{n+1}$ (viewed as a complex concentrated in degree 0), for $\Lambda = \mathbb{Z}/\ell^{n+1}$. One finds that

$$\begin{aligned}\operatorname{tr}\left(F^*, \mathsf{R}\Gamma_c(X, \mathscr{F} \otimes \mathbb{Z}/\ell^{n+1})\right) &= \sum_{x \in X^F} \operatorname{tr}\left(F^*, \mathscr{F}_x \otimes \mathbb{Z}/\ell^{n+1}\right) \\ &= \sum_{x \in X^F} \operatorname{tr}\left(F^*, \mathscr{G}_x\right) \mod \ell^{n+1},\end{aligned}$$

and (3.2) follows by passage to the limit.

5 The Nielsen–Wecken method

In this chapter, we prove two special cases of (4.10). The general case will be considered in the next chapter.

Lemma 5.1. *Theorem (4.10) is true for X_0 of dimension 0.*

In dimension 0, formula (4.10) holds for every correspondence, and not only for the iterates of a Frobenius. It is a special case of the following statement (applied to $X(\mathbb{F})$, K, and the F^{*n}).

(*) Let X be a finite set, $F : X \to X$, $\mathscr{K} \in \operatorname{Ob} \mathsf{D}_{\mathrm{parf}}(X, \Lambda)$, and $F^* : F^*\mathscr{K} \to \mathscr{K}$. One has

$$\sum_{X^F} \operatorname{tr}\left(F^*, \mathscr{K}_x\right) = \operatorname{tr}\left(F^*, \Gamma(X, K)\right).$$

One has $\Gamma(X, \mathscr{K}) = \bigoplus_{x \in X} \mathscr{K}_x$, and the formula follows from the fact that for every morphism $u : \bigoplus_{x \in X} \mathscr{K}_x \to \bigoplus_{x \in X} \mathscr{K}_x$, with matrix u^y_x, one has $\operatorname{tr}(u) = \sum_{x \in X} \operatorname{tr}\left(u^x_x\right)$.

Lemma 5.2. *Let Λ be a Noetherian torsion ring annihilated by a power of the prime number $\ell \neq p$, let X_0 be a smooth projective connected curve over $\mathbb{F}_q$, let $j : U_0 \hookrightarrow X_0$ be a dense open subscheme of X_0, and let $\mathscr{G}_0$ be a locally constant sheaf of projective Λ-modules on U_0. One has*

$$\sum_{x \in U^F} \operatorname{tr}\left(F^*, \mathscr{G}_x\right) = \operatorname{tr}\left(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})\right).$$

More precisely, we shall deduce (5.2) from the following theorem.

Theorem 5.3. *Let X be a nonsingular projective curve over an algebraically closed field k, let f be an endomorphism of X with isolated fixed points, and let $v(f)$ be the number of fixed points of f, each counted with its multiplicity: $v(f)$ is the intersection number of the graph of f with the diagonal of $X \times X$. Then, for ℓ prime to the characteristic exponent of k, one has*

$$v(f) = \sum (-1)^i \operatorname{tr}\left(f^*, \mathrm{H}^i(X, \mathbb{Q}_\ell)\right).$$

This theorem is due to Weil ([41], n° 43, 66, and 68), and Weil's proof is reproduced in Lang [26] (VI §3 combined with VII §2, Th. 3). It can also be deduced from the formalism of the cohomology class associated with a cycle (The cohomology class associated with a cycle, 3.7).

Corollary 5.4. *Let $j : U \hookrightarrow X$ be a dense open subscheme of X, such that $f^{-1}(U) = U$. If the fixed points of f in $X \smallsetminus U$ have multiplicity one, then the number $v_U(f)$ of fixed points of f in U (each counted with its multiplicity) is*

$$v_U(f) = \sum (-1)^i \operatorname{tr}\left(f^*, \mathrm{H}^i_c(U, \mathbb{Q}_\ell)\right).$$

Let F be the finite set $X \smallsetminus U$. The cohomology exact sequence defined by the short exact sequence $0 \to j_!\mathbb{Q}_\ell \to \mathbb{Q}_\ell \to \mathbb{Q}_{\ell F} \to 0$,

$$\xrightarrow{\partial} \mathrm{H}^i_c(U, \mathbb{Q}_\ell) \longrightarrow \mathrm{H}^i(X, \mathbb{Q}_\ell) \longrightarrow \mathrm{H}^i(F, \mathbb{Q}_\ell) \xrightarrow{\partial}$$

gives, with the sign convention of (3.1),

$$\mathrm{tr}\,(f^*, \mathrm{H}^\bullet_c(U, \mathbb{Q}_\ell)) = \mathrm{tr}\,(f^*, \mathrm{H}^\bullet(X, \mathbb{Q}_\ell)) - \mathrm{tr}\,(f^*, \mathrm{H}^\bullet(F, \mathbb{Q}_\ell)) = v(f) - \mathrm{tr}(f^*, \mathbb{Q}_\ell^F).$$

The trace of f^* on $\mathbb{Q}_\ell^F$ is the number $v_F(f)$ of fixed points of f in F; the multiplicity-one hypothesis therefore ensures that

$$\mathrm{tr}\,(f^*, \mathrm{H}^\bullet_c(U, \mathbb{Q}_\ell)) = v_F(f) - v_U(f),$$

as promised.

The proof of (5.2) uses some lemmas concerning a special case of noncommutative traces (4.2), which we now develop. *For brevity, we shall say projective for projective of finite type.*

5.5

Let Λ be a ring, and let H be a finite group. The map $\sum a_h \cdot a \mapsto$ class of a_e, from the group algebra $\Lambda[H]$ to $\Lambda^\natural$, vanishes on the commutators $ab - ba$ of $\Lambda[H]$. It therefore factors through

$$\varepsilon : \Lambda[H]^\natural \to \Lambda^\natural.$$

If F is an endomorphism of a projective $\Lambda[H]$-module P, set

$$\mathrm{tr}^H_\Lambda(F) = \varepsilon\, \mathrm{tr}_{\Lambda[H]}(F)$$

where the trace of (4.2) appears on the right-hand side. To avoid ambiguity, we shall sometimes write this trace as $\mathrm{tr}^H_\Lambda(F, P)$.

Proposition 5.6. $\mathrm{tr}_\Lambda(F) = |H|\,\mathrm{tr}^H_\Lambda(F)$.

The formal properties of traces reduce us to treating only the case $P = \Lambda[H]$. The endomorphism F is then right multiplication by an element $\sum a_h \cdot h$ of $\Lambda[H]$, and $\mathrm{tr}_\Lambda(F) = |H| a_e$, $\mathrm{tr}^H_\Lambda(F) = a_e$.

5.7

Let A be a commutative ring, and let Λ be an A-algebra. Multiplication by an element of A passes to the quotient to define an A-module structure on $A^\natural$. If H is a finite group, P a projective $A[H]$-module, and M a $\Lambda[H]$-module projective as a Λ-module, it is known that the Λ-module $P \otimes_A H$, equipped with the diagonal action of H, is a projective $\Lambda[H]$-module. Indeed:

a) It suffices to verify this for $P = A[H]$.

b) The Λ-module $P \otimes_A M$, equipped with the action of H defined by its action on the first factor, is clearly a projective $\Lambda[H]$-module; denote it by $(P \otimes_A M)'$.

c) The map $h \otimes m \mapsto h \otimes h^{-1}m$ induces an isomorphism

$$A[H] \otimes_A M \xrightarrow{\sim} (A[H] \otimes_A M)'. \tag{5.7.1}$$

Proposition 5.8. *With the preceding notation, if u is an endomorphism of F and v an endomorphism of M, then*

$$\mathrm{tr}^H_\Lambda(u \otimes v) = \mathrm{tr}^H_A(u)\,\mathrm{tr}_\Lambda(v).$$

We reduce to assuming that $P = A[H]$; the endomorphism u is then right multiplication m_x by an element $x = \sum a_h h$ of $A[H]$. Isomorphism (5.7.1) takes $u \otimes v$ to $\sum a_h m_h \otimes h^{-1}v$, whose trace is $a_e \cdot \mathrm{tr}_\Lambda(v)$.

5.9

Let G be a group that is an extension of $\mathbb{Z}$ by a finite group H

$$1 \to H \to G \to \mathbb{Z} \to 1,$$

and let G^+ be the inverse image of $\mathbb{N}$. Let Λ be a ring, and let P be a $\Lambda[G^+]$-module projective as a $\Lambda[H]$-module. If $g \in G^+$, denote by Z_g the centralizer of g in H; multiplication by g is an endomorphism of P, viewed as a $\Lambda[Z_g]$-module. Since P is $\Lambda[Z_g]$-projective, a trace $\mathrm{tr}_\Lambda^{Z_g}(g)$ is defined. By 5.6, one has

$$\mathrm{tr}_\Lambda(g) = |Z_g| \cdot \mathrm{tr}_\Lambda^{Z_g}(g). \tag{5.9.1}$$

Suppose that Λ is an algebra over a commutative ring A. For P an $A[G^+]$-module projective as an $A[H]$-module, and M a $\Lambda[G^+]$-module projective as a Λ-module, the $\Lambda[G^+]$-module $P \otimes_A M$ (diagonal action of G^+) is projective as a $\Lambda[H]$-module and, by 5.8, for every $g \in G^+$ one has

$$\mathrm{tr}_\Lambda^{Z_g}(g, P \otimes_A M) = \mathrm{tr}_A^{Z_g}(g, P) \cdot \mathrm{tr}_\Lambda(g, M). \tag{5.9.2}$$

5.10

Let P be a $\Lambda[G^+]$-module projective as a $\Lambda[H]$-module, and let P_H be the coinvariants of H in P. This is a projective Λ-module. The action of $g \in G^+$ passes to the quotient and defines an endomorphism of P_H that depends only on the image of g in $\mathbb{N}$. We denote by F the action of the elements $g \in G^+$ with image 1.

Proposition 5.11. $\mathrm{tr}_\Lambda(F, P_H) = \sum'_{g \mapsto 1} \mathrm{tr}_\Lambda^{Z_g}(g, P)$, *where* $\sum'_{g\mapsto 1}$ *denotes a sum over the* H*-conjugacy classes of elements of* G^+ *with image* 1 *in* $\mathbb{N}$.

After multiplication by $|H|$, this formula can be verified as follows:

$$|H| \cdot \mathrm{tr}_\Lambda(F, P_H) = \mathrm{tr}_\Lambda\left(\sum_{g\mapsto 1} g, P\right) = \sum_{g\mapsto 1}{}' \frac{|H|}{|Z_g|} \mathrm{tr}_\Lambda(g, P) = \sum_{g\mapsto 1}{}' |H| \cdot \mathrm{tr}_\Lambda^{Z_g}(g, P).$$

Choose an element $g \in G^+$ with image 1 in $\mathbb{N}$, and let σ be the automorphism of $\Lambda[H]$ induced by the automorphism $\mathrm{ad}\, g$ of H. Giving a $\Lambda[G^+]$-module amounts to giving a $\Lambda[H]$–module equipped with a σ-linear endomorphism γ: let $g^n h$ act by $\gamma^n h$. In this language, formula 5.11 takes the form: for P a projective $\Lambda[H]$-module and γ a σ-linear endomorphism of P, one has

$$\mathrm{tr}_\Lambda(\gamma, P_H) = \sum_h{}' \mathrm{tr}_\Lambda^{Z_g}(h\gamma, P),$$

where $\sum'$ is a sum over the $\mathrm{ad}\, g$-conjugacy classes in H (orbits of H acting on itself by the maps $x \mapsto hx\, \mathrm{ad}\, g(h)^{-1}$).

In this form, the properties of traces immediately reduce us to the case $P = \Lambda[H]$. The endomorphism Y then has the form $x \mapsto \sigma(x)a$, with $a = \sum a_h h$ in $\Lambda[H]$. It suffices to treat the universal case where the a_h are indeterminates. In this case, $\Lambda = \mathbb{Z}\langle a_h\rangle_{h\in H}$ (noncommutative polynomials), and $\Lambda^\natural$ is torsion-free: one obtains 5.11 by the argument at the beginning and division by $|H|$. One can also calculate directly.

5.11.1 Remark

Everything said above applies equally well to $\mathbb{Z}/2$-graded modules, or to complexes of modules (cf. 4.2, 4.3).

5.12

Retain the notation of 5.2, and let $A = \mathbb{Z}/\ell^n$, with n large enough that Λ is an A-algebra. Let $f : V_0 \to U_0$ be a finite étale connected covering of U_0, Galois with Galois group H, and such that the sheaf $f^*\mathscr{G}_0$ is constant. Denote its constant value $\mathrm{H}^0(V_0, f^*\mathscr{G}_0)$ by M; it is a $\Lambda[H]$-module projective as a Λ-module.

The sheaf f_*A on U_0, equipped with the natural action of H, is a locally free sheaf (of rank one) of $A[H]$-modules. The complex $\mathsf{R}\Gamma_c(U, f_*A)$ is therefore an object of $\mathsf{D}_{\mathrm{parf}}(A[H])$. It is equipped with a Frobenius endomorphism F^*.

For the natural action of H on $f_*f^*\mathscr{G}_0$, the trace morphism $f_*f^*\mathscr{G}_0 \to \mathscr{G}_0$ factors through an isomorphism

$$(f_*f^*\mathscr{G}_0)_H \to \mathscr{G}_0.$$

Moreover, one has $f_*f^*\mathscr{G}_0 = f_*A \otimes_A M$ (with diagonal action of H). The isomorphism

$$\mathscr{G}_0 \xleftarrow{\sim} (f_*A \otimes_A M)_H = (f_*A \otimes_A M) \otimes_{\Lambda[H]} \Lambda$$

derives (4.12) to an isomorphism

$$\mathsf{R}\Gamma_c(U, \mathscr{G}) \sim (\mathsf{R}\Gamma_c(U, f_*A) \otimes_A M) \otimes_{\Lambda[H]} \Lambda.$$

The endomorphism F^* of the left-hand side is deduced from the endomorphism F^* of $\mathsf{R}\Gamma_c(U, f_*A)$. The complex of $\Lambda[H]$-modules $\mathsf{R}\Gamma_c(U, f_*A)$, equipped with F^*, can be viewed as a complex of $\Lambda[G^+]$-modules, for $G^+ = H \times \mathbb{N}$. Formulas 5.10 and (5.9.2), amplified by 5.11.1, therefore give

$$\mathrm{tr}\,(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})) = \sum_h{}' \mathrm{tr}_A^{Z_g}\,(hF^*, \mathsf{R}\Gamma_c(U, f_*A)) \cdot \mathrm{tr}_\Lambda(h, M)$$

where $\sum'$ is a sum over the conjugacy classes in H. Moreover,

$$|Z_h| \cdot \mathrm{tr}_A^{Z_h}\,(hF^*, \mathsf{R}\Gamma_c(U, f_A)) = \mathrm{tr}_A\left((Fh^{-1})^*, \mathsf{R}\Gamma_c(V, A)\right).$$

This formula remains true for $A = \mathbb{Z}/\ell^m$, with m increasingly large. Passing to the limit, one finds that $\mathrm{tr}_A^{Z_g}\,(hF^*, \mathsf{R}\Gamma_c(U, f^*A))$ is equal to

$$\frac{1}{|Z_h|}\sum(-1)^i \,\mathrm{tr}\left((Fh^{-1})^*, \mathrm{H}_c^i(V, \mathbb{Q}_\ell)\right)$$

(an element of $\mathbb{Z}_\ell$), and

$$\mathrm{tr}\,(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})) = \sum_h{}' \frac{1}{|Z_h|}\,\mathrm{tr}\left((Fh^{-1})^*, \mathrm{H}_c^i(V, \mathbb{Q}_\ell)\right) \cdot \mathrm{tr}_\Lambda(h, M). \tag{5.12.1}$$

5.13

It remains to apply 5.4 (note that if $\bar{V}$ is the nonsingular projective completion of V, all the fixed points of Fh^{-1} have multiplicity one). This can be done without any calculation:

A. If $U^F = \varnothing$, $\mathrm{tr}\,(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})) = 0$.

Indeed, Fh^{-1} has no fixed point on V; 5.4 shows that $\sum(-1)^i \,\mathrm{tr}\left((Fh^{-1})^*, \mathrm{H}_c^i(V, \mathbb{Q}_\ell)\right) = 0$, and one applies (5.12.1).

B. In the general case, let $j : U' = U \smallsetminus U^F \hookrightarrow U$. The filtration $j_!j^*\mathscr{G}_0 \subset \mathscr{G}_0$ of $\mathscr{G}_0$ induces a filtration of $\mathsf{R}\Gamma_c(U, \mathscr{G})$ with successive quotients $\mathsf{R}\Gamma_c(U', \mathscr{G})$ and $\mathsf{R}\Gamma_c(U^F, \mathscr{G})$. By (4.4.1) and 5.1, one has

$$\operatorname{tr}(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})) = \operatorname{tr}(F^*, \mathsf{R}\Gamma_c(U', \mathscr{G})) + \sum_{x \in U^F} \operatorname{tr}(F^*, \mathscr{G}_x),$$

and, since ${U'}^F = 0$, the first term on the right-hand side is 0.

6 End of the proof of 4.10

Let us prove 4.10 under the additional hypotheses that the ring Λ is annihilated by a power of a prime number $\ell \neq p$, and that $n = 1$. The general case will follow: decompose Λ into its ℓ-primary components (with ℓ running through the prime numbers), and replace $\mathbb{F}_q$ by $\mathbb{F}_{q^n}$ and X_0 by $X_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^n}$.

For X_0 separated and of finite type over $\mathbb{F}_q$, and $\mathscr{K}_0 \in \operatorname{Ob} \mathsf{D}_{\mathrm{ctf}}(X_0, \Lambda)$, set

$$T'(X_0, \mathscr{K}_0) = \sum_{x \in X^F} \operatorname{tr}(F^*, \mathscr{K}_x)$$

and

$$T''(X_0, \mathscr{K}_0) = \operatorname{tr}(F^*, \mathsf{R}\Gamma_c(X, \mathscr{K})).$$

We must prove the identity

$$(1; X_0, \mathscr{K}_0) \qquad T'(X_0, \mathscr{K}_0) = T''(X_0, \mathscr{K}_0).$$

We begin by observing that T' and T'' (denoted generically by T) obey the same formalism:

(A) If $j : U_0 \hookrightarrow X_0$ is an open subscheme of X_0, with closed complement Y_0, one has

$$T(X_0, \mathscr{K}_0) = T(U_0, \mathscr{K}_0|U_0) + T(Y_0, \mathscr{K}_0|Y_0).$$

This is clear for T'. For T'', observe that the filtration $0 \subset j_!j^*\mathscr{K}_0 \subset \mathscr{K}_0$ of $\mathscr{K}_0$ induces a filtration of $\mathsf{R}\Gamma_c(X, \mathscr{K})$ with successive quotients $\mathsf{R}\Gamma_c(U, \mathscr{K})$ and $\mathsf{R}\Gamma_c(Y, \mathscr{K})$, and invoke (4.4.1) (more precisely stated: $\mathscr{K}_0$, equipped with the said filtration, defines an object of $\mathsf{DF}(X, \Lambda)$; applying $\mathsf{R}\Gamma_c$ gives an object of $\mathsf{DF}_{\mathrm{parf}}(\Lambda)$ whose underlying object is $\mathsf{R}\Gamma_c(X, \mathscr{K})$ and whose associated graded is as stated).

(B) If $\mathscr{K}_0$ is a bounded complex with constructible flat components,

$$T(X_0, \mathscr{K}_0) = \sum (-1)^i T(X_0, \mathscr{K}^i).$$

This is clear for T'; for T'', apply the argument above to the stupid filtration of $\mathscr{K}_0$ to obtain

$$T''(X_0, \mathscr{K}_0) = \sum T''(X_0, \mathscr{K}_0^i[-i]) = \sum (-1)^i T''(X_0, \mathscr{K}_0^i).$$

(C) For a morphism $f : X_0 \to Y_0$, one has

$$T''(X_0, \mathscr{K}_0) = T''(Y_0, \mathsf{R}f_!\mathscr{K}_0)$$

and the same identity holds for T' if the fibers of f at the rational points of Y_0 satisfy $\big(1; f^{-1}(y), \mathscr{K}_0|f^{-1}(y)\big)$: use the fact that

$$T'(X_0, \mathscr{K}_0) = \sum_{y \in Y^F} T'\left(f^{-1}(y), \mathscr{K}_0|f^{-1}(y)\right)$$

and

$$\mathsf{R}\Gamma_c(Y, \mathsf{R}f_!\mathscr{K}) = \mathsf{R}\Gamma_c(X, \mathscr{K}).$$

Using this formalism and 4.7, the proof of 4.10 is reduced to the two special cases 5.1 and 5.2.

Chapter 3

L-functions modulo ℓ^n and modulo p

In this exposé, I prove a formula analogous to the main theorem of the report on the trace formula (this volume; cited as "Report" below) for a coefficient ring A that is torsion prime to p, or a field of characteristic p. Essential use will be made of [2, XVII 5.5].

1 Symmetric tensors

1.1

Let A be a commutative ring. For a list of properties of the functors $\Gamma^n : (A\text{-modules}) \to (A\text{-modules})$, I refer to [2, XVII 5.5.1 and 2]. We recall only that for M flat (in fact, only the case where M is projective of finite type concerns us), one has

$$\Gamma^n M = \left(M^{\otimes n}\right)^{S_n} \qquad \text{(symmetric tensors of degree } n\text{)}.$$

1.2

Let X be an S-scheme, let X^n/S be the n-th fiber power of X, and let π be the projection from X^n/S onto $(X^n/S)/S_n = \mathrm{Sym}^n_S(X)$ (assumed to exist). If $\mathscr{G}$ is a sheaf of A-modules on X, the external tensor product of n copies of $\mathscr{G}$, namely $\mathscr{G}^{\boxtimes n}$, is an S_n-equivariant sheaf on X^n/S. We shall have to use the *external* functor Γ^n of [2, XVII 5.5.7 to 9]. We recall only that for $\mathscr{G}$ a sheaf of flat A-modules (in fact, only the constructible flat case concerns us), one has

$$\Gamma^n_{\mathrm{ext}}(\mathscr{G}) = \left(\pi_* \mathscr{G}^{\boxtimes n}\right)^{S_n}.$$

Lemma 1.3. *For $S = \mathrm{Spec}(k)$, with k algebraically closed, X a finite disjoint union of copies of S, and G a flat sheaf of A-modules on X, one has*

$$\Gamma\left(\mathrm{Sym}^n_S(X), \Gamma_{ext}(\mathscr{G})\right) = \Gamma^n \Gamma(X, \mathscr{G})$$

Indeed,

$$\begin{aligned}\Gamma\left(\mathrm{Sym}^n_S(X), \Gamma^n_{\mathrm{ext}}(\mathscr{G})\right) &= \Gamma\left(\mathrm{Sym}^n_S(X), \pi_* \mathscr{G}^{\boxtimes n}\right)^{S_n} \\ &= \Gamma\left(X^n/S, \mathscr{G}^{\boxtimes n}\right)^{S_n} = \left(\Gamma(X, \mathscr{G})^{\otimes n}\right)^{S_n} = \Gamma^n \Gamma(X, \mathscr{G}).\end{aligned}$$

1.4

We shall have to use the derived functors of the nonadditive functors Γ^n and Γ^n_{ext}.

The theory of such derived functors is due to Dold and Puppe [17]. For more complete reminders than those given below, I refer to [2, XVII 5.5.3].

Let $\mathcal{A}$ be an abelian category (or even an additive category in which every idempotent endomorphism is the projection onto a direct factor). Denote by N the normalization functor: N : (cosimplicial complexes of objects of $\mathcal{A}$) $\to$ (differential complexes of objects of $\mathcal{A}$, with K^i for $i < 0$). It is an equivalence of categories. Like its inverse, it takes homotopic morphisms to homotopic morphisms. If Γ is a functor $\mathcal{A} \to \mathcal{B}$, and if K is a complex of objects of $\mathcal{A}$, in nonnegative degrees ($K^i = 0$ for $i < 0$), we set $TK = NTN^{-1}K$.

1.5 Example

If K is concentrated in M in degree 0, TK is concentrated in TM in degree 0.

1.6

If K is a bounded complex, in nonnegative degrees, of projective A-modules of finite type, the complex of A-modules $\Gamma^n K$ has the same properties.

Denote by $\mathsf{D}^{\geqslant 0}_{\text{parf}}(A)$ the subcategory of $\mathsf{D}_{\text{parf}}(A)$ (Report, 4.3) consisting of complexes of tor-dimension $\leqslant 0$. One sees that Γ^n induces a functor

$$\mathsf{L}\Gamma^n : \mathsf{D}^{\geqslant 0}_{\text{parf}}(A) \to \mathsf{D}^{\geqslant 0}_{\text{parf}}(A).$$

For the analogous, but more complicated, definition of $\mathsf{L}\Gamma^n_{\text{ext}}$, I refer to [2, XVII 5.5.14].

1.7

Let K be a bounded complex of projective A-modules of finite type, and let u be an endomorphism of K. We denote by $\det(1 - ut, K)$ the formal series with constant term 1

$$\det(1 - ut, K) = \prod_i \det(1 - ut, K^i)^{(-1)^i}.$$

The verification of the following properties is left to the reader.

1.7.1

Let F be a finite filtration stable under u, such that the $\mathrm{gr}^p_F(K^q)$ are projective of finite type. Then

$$\det(1 - ut, K) = \prod_p \det(1 - ut, \mathrm{gr}^p_F(K)).$$

1.7.2

If K is shifted q places to the left, one has

$$\det(1 - ut, K[q]) = \det(1 - ut, K)^{(-1)^q}.$$

Proposition 1.8. *Let u be an endomorphism of a bounded complex, in nonnegative degrees, of projective A-modules of finite type. One has*

$$\det(1-ut,K)^{-1}=\sum_n \mathrm{tr}(u,\Gamma^n K)t^n. \tag{1.8.1}$$

Denote the right-hand side of (1.8.1) by $D(u,K)$.

Lemma 1.9. *Let u be an endomorphism of a short exact sequence of bounded complexes in degrees $\geqslant 0$ of projective A-modules of finite type*

$$0\to K'\to K\to K''\to 0.$$

One has $D(u,K)=D(u,K')\cdot D(u,K'')$.

If $0\to M'\to M\to M''\to 0$ is a split short exact sequence of A-modules, $\Gamma^n M$ has a natural filtration whose successive quotients are the $\Gamma^i M'\otimes\Gamma^j M''$ $(i+j=n)$. Consequently, $\Gamma^n K$ has a natural filtration whose successive quotients are the complexes $N(N^{-1}\Gamma^i K'\otimes N^{-1}\Gamma^j K'')$ $(i+j=n)$, canonically homotopic to the simple complexes $s(\Gamma^i K'\otimes\Gamma^i K'')$ obtained from the double complexes $\Gamma^i K'\otimes\Gamma^i K''$, and

$$\mathrm{tr}(u,\Gamma^n K)=\sum_{i+j=n}\mathrm{tr}(u,\Gamma^i K')\cdot\mathrm{tr}(u,\Gamma^j K''),$$

whence the lemma. From 1.9, one obtains by induction.

Lemma 1.10. *Let F be a finite filtration of K stable under u, such that the $\mathrm{gr}_F^p(K^q)$ are projective of finite type. Then*

$$D(u,K)=\prod_p D\left(u,\mathrm{gr}_F^p(K)\right).$$

Lemma 1.11. *Let u be an endomorphism of a projective A-module of finite type M, and let $M[-q]$ be the complex concentrated in M in degree q. One has*

$$D\left(u,M[-q]\right)=\Big(\sum\mathrm{tr}(u,\Gamma^n M)t^n\Big)^{(-1)^q}\qquad (q\geqslant 0.)$$

By 1.5, this is true for $q=0$. Proceeding by induction, it remains to show that

$$D\left(u,M[-q]\right)\cdot D\left(u,M[-(q+1)]\right)=1.$$

This formula follows from 1.9, applied to the complex K concentrated in M in degrees q and $q+1$ (with $d=\mathrm{id}$), and to its stupid filtration:

$$(\cdots 0\to M\cdots)\to(\cdots M\to M)\to(\cdots M\to 0\to 0\cdots).$$

Indeed, K is homotopic to 0. In the derived category, one therefore has $\Gamma^n(K)=\Gamma^n(0)$: 0 for $n>0$ and A for $n=0$, and $D(u,K)=1$.

1.12

Let us prove 1.8. By 1.10, applied to the "stupid" filtration of K by the subcomplexes

$$0\to\cdots\to 0\to K^q\to K^{q+1}\to\cdots,$$

one has

$$D(u,K)=\prod_q D\left(u,K^q[-q]\right) =_{(1.10)} \prod_q \left(\sum \operatorname{tr}\left(u,\Gamma^n(K^q)\right)t^n\right)^{(-1)^q}.$$

It remains to prove that for u an endomorphism of a projective module of finite type M, one has

$$\det\left(1-ut,M\right)^{-1}=\sum \operatorname{tr}(u,\Gamma^n M)t^n. \tag{1.12.1}$$

If $u=0$, both sides are equal to 1. By adjoining to M a module N on which $u=0$, and applying 1.9, we reduce to the assumption that M is free. Choose a basis for it. It is then a matter of proving algebraic identities among the coordinates of u. The principle of extension of algebraic identities reduces us to assuming that A is an algebraically closed field. The module M then admits a filtration stable under u with successive quotients of rank 1, and 1.10 reduces us to the case where M has rank 1. Formula (1.12.1) then reduces to the identity

$$\frac{1}{1-at}=\sum a^n t^n.$$

Corollary 1.13. *Let u and u' be endomorphisms of bounded complexes K and K' of projective A-modules of finite type. If, in the derived category, K, equipped with u, is isomorphic to K', equipped with u', then*

$$\det(1-ut,K)=\det(1-u't,K').$$

Applying 1.7.2, we reduce to assuming that K and K' are in nonnegative degrees. One then observes that the right-hand side of 1.8 has the required invariance property.

1.14

This corollary makes it possible to define $\det(1-ut,K)$ for u an endomorphism of $K\in \operatorname{Ob}\mathsf{D}_{\mathrm{parf}}(A)$. This construction is ad hoc; in fact, for v an automorphism of $L\in\operatorname{Ob}\mathsf{D}_{\mathrm{parf}}(\Lambda)$, one can define $\det(v)\in\Lambda^\times$, and $\det(1-ut,K)$ is obtained by taking $v=1-ut$, $\Lambda=A[\![t]\!]$, and $L=K\otimes_A\Lambda$.

2 The theorem

2.1

We retain the notation of (Report, §1). For A a commutative Noetherian torsion ring, X_0 a separated scheme of finite type over $\mathbb{F}_q$, and $\mathscr{K}_0\in\operatorname{Ob}\mathsf{D}_{\mathrm{ctf}}(X_0;A)$, set

$$L(X_0,\mathscr{K}_0)=\prod_{x\in|X_0|}\det\left(1-F_x^* t^{\deg(x)},\mathscr{K}\right)^{-1}$$

(1.7, 1.13, and Report, 1.5, 1.6).

Theorem 2.2. *In each of the following two cases*

a) A is torsion prime to p;

b) A is reduced and of characteristic p

one has

$$L(X_0,\mathscr{K}_0)=\det\left(1-F^* t^f,\mathsf{R}\Gamma_c(X,\mathscr{K})\right)^{-1}. \tag{2.2.1}$$

2.3 Remark

In case a), the method of Report, §3 makes it possible to deduce from the trace formula (Report 4.10) that the logarithmic derivatives of the two sides of (2.2.1) are equal. Since the ring A is a torsion ring, this does not suffice to prove 2.2.

2.4 Remark

If A is a field, one has

$$L(X_0, \mathscr{K}_0) = \prod_i L\left(X_0, \underline{\mathrm{H}}^i(\mathscr{K}_0)\right)^{(-1)^i}$$

and the spectral sequence $E_2^{pq} = \mathrm{H}_c^p(X, \underline{\mathrm{H}}^q(\mathscr{K})) \Rightarrow \mathrm{H}_c^{p+q}(X, \mathscr{K})$ shows that

$$\begin{aligned}\det\left(1 - F^*t, \mathsf{R}\Gamma_c(X, \mathscr{K})\right) &= \prod_n \det\left(1 - F^*t^f, \mathrm{H}_c^n(X, \mathscr{K})\right)^{(-1)^n} \\ &= \prod_{p,q} \det\left(1 - F^*t, \mathrm{H}_c^p(X, \underline{\mathrm{H}}^q(\mathscr{K}))\right)^{(-1)^{p+1}}.\end{aligned}$$

This reduces 2.2 (for A a field) to the following special case.

2.4.1

If $\mathscr{G}_0$ is a constructible sheaf of A-vector spaces, one has

$$L(X_0, \mathscr{G}_0) = \prod_n \det\left(1 - F^*t^f, \mathrm{H}_c^n(X, \mathscr{G})\right)^{(-1)^{n+1}}.$$

2.5 Remark

It is easy to reduce case b) to the case where A is a field (or even a finite field) of characteristic p; cf. 4.2.

2.6 Remark

The special case of b) where $A = \mathbb{Z}/p$, and where X is reduced to the constant sheaf $\mathbb{Z}/p$ in degree 0,

$$Z(X_0, t) = \prod_{x \in |X_0|} \left(1 - t^{\deg(x)}\right)^{-1} \equiv \prod_n \det\left(1 - F^*t^f, \mathrm{H}_c^n(X, \mathbb{Z}/p)\right)^{(-1)^{n+1}} \pmod{p}$$

was proved by N. Katz [16, XXII 3.1]. His method, entirely different from the one followed here, starts from Dwork's p-adic theory of $Z(X_0, t)$.

2.7 Remark

Contrary to what I had rashly asserted to friends, in b) one cannot omit the hypothesis "A reduced." For a counterexample, cf. 4.5.

2.8 Outline of the proof of 2.2

A shift in degrees 1.7.2 reduces us to assuming that $\mathscr{K}_0$ is a bounded complex, in nonnegative degrees, of constructible flat sheaves of A-modules. One then has $\mathsf{R}\Gamma_c(X,\mathscr{K}) \in \mathrm{Ob}\,\mathsf{D}^{\geqslant 0}_{\mathrm{parf}}(A)$ (Report, proof of 4.9, b). We shall use 1.8 to expand both sides of (2.2.1) as power series in $T = t^f$. The equality of the coefficients of T^n will follow from [2, XVII 5.5.21] and a trace formula on $\mathrm{Sym}^n(X)$: in case a), the formula (Report, 4.10), and in case b), a formula to be proved in the next chapter. It is first necessary to reduce to the case where the symmetric powers of X_0 exist (for example, the case where X is affine), using the multiplicativity in X_0 of both sides of (2.2.1): for U_0 an open subscheme of X_0, with closed complement Y_0, one has $L(X_0,\mathscr{K}_0) = L(U_0,\mathscr{K}_0)\cdot L(Y_0,\mathscr{K}_0)$, and the analogous formula for the right-hand side follows from 1.7.1 by the method of (Report §6A).

2.9 The case where X_0 is finite

The multiplicativity in X_0 of both sides of (2.2.1) reduces this case to the case where $X_0 = \mathrm{Spec}(\mathbb{F}_{q^k})$. Returning to the definitions, one sees that (2.2.1) is then equivalent to Report 3.4.

2.10 The left-hand side

By 1.8, one has

$$L(X_0,\mathscr{K}_0) = \prod_{x\in|X_0|}\sum_n \mathrm{tr}\,(F_x^*,\Gamma^n\mathscr{K})\,t^{m\deg(x)}.$$

The coefficient of T^n is therefore the sum, over those formal linear combinations $\sum_{x\in|X_0|} n_x\cdot x$ ($n_x\geqslant 0$, with almost all the n_x zero) of elements of $|X_0|$ such that $\sum n_x[k(x):\mathbb{F}_q] = n$, of the products

$$\prod_x \mathrm{tr}\,(F_x^*,\Gamma^{n_x}\mathscr{K})\,. \tag{2.10.1}$$

For $x\in|X_0|$, denote by (x) the 0-cycle consisting of the $[k(x):\mathbb{F}_q]$ points of X above x (an orbit of F). The map $\sum n_x \mapsto \sum n_x\cdot(x)$ identifies the formal linear combinations above, with $\sum n_x[k(x):\mathbb{F}_q] = n$, with the effective 0-cycles of degree n on X fixed under F, or equivalently with the fixed points of F in $\mathrm{Sym}^n(X)$. The coefficient of T^n in $L(X_0,\mathscr{K}_0)$ thus appears as a sum of products (2.10.1), indexed by $\mathrm{Sym}^n(X)^F$. More precisely,

Lemma 2.11. $L(X_0,\mathscr{K}_0) = \sum_n \sum_{x\in\mathrm{Sym}^n(X)^F} \mathrm{tr}\,(F_x^*,\mathsf{L}\Gamma^n_{\mathrm{ext}}\mathscr{K})\,T^n$.

For Y_0 an increasingly large finite subscheme of X_0, one has

$$L(X_0,\mathscr{K}_0) = \lim L(Y_0,\mathscr{K}_0),$$

i.e., for every m, there is a finite subscheme $Z_0\subset X_0$ such that if $Y_0\supset Z_0$, then $L(X_0,\mathscr{K}_0)$ and $L(Y_0,\mathscr{K}_0|Y_0)$ are congruent $\mod T^m$. It suffices for Z_0 to contain the closed points of degree $\leqslant m$. Since the same fact holds for the right-hand side, it suffices to prove 2.11 for X_0 finite. One then has

$$\begin{aligned}
L(X_0,\mathscr{K}_0) &=_{2.9} \det\,(1-F^*T,\Gamma(X,\mathscr{K}))^{-1}\\
&=_{1.8} \sum_n \mathrm{tr}\,(F^*,\Gamma^n\Gamma(X,\mathscr{K}))\,T^n\\
&=_{1.3} \sum_n \mathrm{tr}\,(F^*,\Gamma(\mathrm{Sym}^n(X),\Gamma^n_{\mathrm{ext}}(\mathscr{K}))\,T^n\\
&= \sum_n \sum_{x\in\mathrm{Sym}^n(X)^F} \mathrm{tr}\,(F_x^*,\Gamma^n_{\mathrm{ext}}(\mathscr{K}))\,T^n
\end{aligned}$$

(by the trace formula in the trivial case of a finite set).

2.12 The right-hand side

By 1.8 and [2, XVII 5.5.12], one has

$$\begin{aligned}\det\left(1-F^*T,\mathsf{R}\Gamma_c(X,\mathscr{K})\right)^{-1} &= \sum_n \operatorname{tr}\left(F^*,\mathsf{L}\Gamma^n\mathsf{R}\Gamma_c(X,\mathscr{K})\right)T^n\\ &= \sum_n \operatorname{tr}\left(F^*,\mathsf{R}\Gamma_c(\operatorname{Sym}^n(X),\Gamma^n_{\mathrm{ext}}\mathscr{K})\right)T^n.\end{aligned}$$

2.13

The theorem is therefore equivalent to the trace formulas

$$\operatorname{tr}\left(F^*,\mathsf{R}\Gamma_c(\operatorname{Sym}^n(X,\Gamma^n_{\mathrm{ext}}\mathscr{K})\right) = \sum_{x\in\operatorname{Sym}^n(X)^F}\operatorname{tr}\left(F^*_x,\Gamma^n_{\mathrm{ext}}\mathscr{K}\right).$$

In case a), they follow from Report 4.10. The analogous formula in case b) will be proved in §4.

3 Artin–Schreier theory

3.1

In what follows, a coherent sheaf on a scheme S will always be regarded as a sheaf on $S_{\text{ét}}$. Let S be a scheme of characteristic $p>0$. If G is a locally constant sheaf of finite-rank $\mathbb{Z}/p$-vector spaces, $\mathscr{G}=G\otimes_{\mathbb{Z}/p}\mathscr{O}$ is a locally free coherent sheaf on S. We shall denote by Φ both the endomorphism $f\mapsto f^p$ of $\mathscr{O}$ and its tensor product with the identity of G: $\phi:\mathscr{G}\to\mathscr{G}$. Artin–Schreier theory [2, IX 3.5] asserts that the sequence

$$0\longrightarrow\mathbb{Z}/p\longrightarrow\mathscr{O}\xrightarrow{\Phi-1}\mathscr{O}\longrightarrow 0$$

is exact. Tensoring with G gives an exact sequence

$$0\longrightarrow G\longrightarrow\mathscr{G}\xrightarrow{\Phi-1}\mathscr{G}\longrightarrow 0$$

3.2

Suppose that S is an open subscheme of a Noetherian scheme $\bar S$. Let $j:S\hookrightarrow\bar S$ be the inclusion morphism, and let I be a sheaf of ideals defining the closed complement $\bar S\smallsetminus S$. We again denote by Φ the extension of Φ to $j_*\mathscr{G}$ by functoriality.

Lemma 3.3. *There exist coherent extensions $\bar{\mathscr{G}}\subset j_*\mathscr{G}$ of $\mathscr{G}$ such that*

$$\Phi(\bar{\mathscr{G}})\subset I\cdot\bar{\mathscr{G}}. \tag{3.3.1}$$

If $\bar{\mathscr{G}}$ is such an extension, the sequence

$$0\longrightarrow j_!G\longrightarrow\bar{\mathscr{G}}\xrightarrow{\Phi-1}\bar{\mathscr{G}}\longrightarrow 0 \tag{3.3.2}$$

is exact.

a) *Existence*: Let $\mathscr{G}' \subset j_*\mathscr{G}$ be a coherent extension of $\mathscr{G}$ to $\bar{S}$. For every n, one has $\Phi(I^n\mathscr{G}') \subset I^{p^n}\Phi(\mathscr{G}')$. Let n be large enough that $I^{n(p-1)-1}\Phi(\mathscr{G}') \subset \mathscr{G}'$, and set $\bar{\mathscr{G}} = I^n\mathscr{G}'$. One has $\Phi(\bar{\mathscr{G}}) \subset I^{pn}\Phi(\mathscr{G}') = I \cdot I^n \cdot I^{n(p-1)-1}\Phi(\mathscr{G}') \subset I \cdot I^n\mathscr{G}' = I\bar{\mathscr{G}}$.

b) *Kernel of* $\Phi - 1$: On S, the assertion follows from 3.1; it remains to verify that a section x of $\ker(\Phi - 1)$, on U étale over $\bar{S}$, vanishes in a neighborhood of the inverse image of $\bar{S} \smallsetminus S$. Indeed, since $x = \Phi(x)$, one has by (3.3.1) that $x \in I\bar{\mathscr{G}}$; moreover, $x \in I^n\bar{\mathscr{G}} \Rightarrow x = \Phi(x) \in \Phi(I^n\bar{\mathscr{G}}) \subset I^{np}\bar{\mathscr{G}}$, and x lies in every $I^n\bar{\mathscr{G}}$, hence is zero near $\bar{S} \smallsetminus S$.

c) *Surjectivity of* $\Phi - 1$: We may suppose $\bar{S}$ affine, and write $\mathscr{G}$ as a quotient of $\mathscr{O}^n$; Φ then lifts to a p-linear map $\widetilde{\Phi} : \mathscr{O}^n \to \mathscr{O}^n$, and it suffices to verify the surjectivity of $\widetilde{\Phi} - 1$:

$$\begin{array}{ccc} \mathscr{O}^n & \xrightarrow{\widetilde{\Phi}-1} & \mathscr{O}^n \\ \downarrow & & \downarrow \\ \mathscr{G} & \xrightarrow{\Phi-1} & \mathscr{G} \end{array}$$

The sheaf $\mathscr{O}^n$ is the sheaf of local sections of the group scheme $\mathbb{G}_a^n$, and $\widetilde{\Phi} - 1$ is induced by a homomorphism from $\mathbb{G}_a^n$ to $\mathbb{G}_a^n$. This homomorphism is étale. To verify that it is surjective, it suffices to check this after every base change $\mathrm{Spec}(k) \to \bar{S}$ (k algebraically closed), and over an algebraically closed k, every étale homomorphism between connected groups of the same dimension is surjective. Finally, an étale surjective morphism of $\bar{S}$-schemes induces an epimorphism between the corresponding sheaves on $\bar{S}_{\mathrm{et}}$.

3.4 Frobenius and Frobenius

The absolute Frobenius morphism $F_{\mathrm{abs}} : S \to S$ is the identity on the underlying topological space of S, and $f \mapsto f^p$ on the structure sheaf. For U/S étale, one has a canonical isomorphism $F_{\mathrm{abs}}^*U = U$: the diagram

$$\begin{array}{ccc} U & \xrightarrow{F_{\mathrm{abs}}} & S \\ \downarrow & & \downarrow \\ S & \xrightarrow{F_{\mathrm{abs}}} & S \end{array}$$

is Cartesian. The endomorphism of the ringed site $(\bar{S}_{\mathrm{et}}, \mathscr{O}) \to (\bar{S}_{\mathrm{et}}, \mathscr{O})$ defined by F_{abs} can therefore be described as follows:

$$F_{\mathrm{abs}}^* \text{ is the identity on } S_{\mathrm{et}}\text{; the homomorphism } F_{\mathrm{abs}}^*\mathscr{O} = \mathscr{O} \to \mathscr{O} \text{ is } \Phi.$$

With this description, for every étale sheaf $\mathscr{G}$ on S, the Frobenius correspondence $F_{\mathrm{abs}}^*\mathscr{G} \to \mathscr{G}$ is the identity.

Giving a p-linear morphism of quasi-coherent sheaves $u : \mathscr{M} \to \mathscr{N}$ on S is equivalent to giving a linear morphism $u' : F_{\mathrm{abs}}^*\mathscr{M} \to \mathscr{N}$. In particular, with the notation of 3.5, the homomorphism $\Phi : \mathscr{G} \to \mathscr{G}$ corresponds to $\Phi' : F_{\mathrm{abs}}^*\mathscr{G} \to \mathscr{G}$.

Lemma 3.5. *With the notation of 3.4, the diagram*

$$\begin{array}{ccc} F_{\mathrm{abs}}^*G & \longrightarrow & G \\ \downarrow & & \downarrow \\ F_{\mathrm{abs}}^*\mathscr{G} & \xrightarrow{\Phi'} & \mathscr{G} \end{array}$$

is commutative.

It suffices to verify this locally. We may therefore suppose G constant, in which case the lemma is evident.

The same theory applies to the powers of F_{abs}.

3.6

Suppose now that $\bar{S}$ is a scheme over $\mathbb{F}_q$ ($q = p^f$), and in accordance with the conventions of Report §1, write $S_0, \bar{S}_0, G_0, \mathscr{G}_0, \bar{\mathscr{G}}_0$ in place of $S, \ldots$. Lemma 3.5, for $F^F_{\mathrm{abs}} : S_0 \to S_0$, gives a commutative diagram

$$\begin{array}{ccc} F^*G_0 & \longrightarrow & G_0 \\ \downarrow & & \downarrow \\ F^*\mathscr{G}_0 & \xrightarrow{(\Phi^f)'} & \mathscr{G}_0 \end{array} \qquad \text{, and then} \qquad \begin{array}{ccc} F^*(j_!G_0) & \longrightarrow & j_!G_0 \\ \downarrow & & \downarrow \\ F^*\bar{\mathscr{G}}_0 & \xrightarrow{(\Phi^f)'} & \mathscr{G}_0 \end{array},$$

and then, by extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}$ and passage to cohomology,

$$\begin{array}{ccc} F^*(j_!G) & \xrightarrow{F_*} & j_!G \\ \downarrow & & \downarrow \\ F^*\bar{\mathscr{G}} & \longrightarrow & \bar{\mathscr{G}} \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathrm{H}^\bullet(\bar{S}, j_!G) & \xrightarrow{F^*} & \mathrm{H}^\bullet(\bar{S}, j_!G) \\ \downarrow & & \downarrow \\ \mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) & \longrightarrow & \mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) \end{array},$$

where the second row is obtained by extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}$ of the $\mathbb{F}_q$-linear endomorphism Φ^f of $\mathrm{H}^\bullet(\bar{S}_0, \bar{G}_0)$.

Artin–Schreier theory provides a long exact sequence

$$\cdots \longrightarrow \mathrm{H}^i(\bar{S}, j_!G) \longrightarrow \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}}) \xrightarrow{\Phi - 1} \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}}) \longrightarrow \cdots$$

where Φ is the p-linear extension to $\mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) = \mathrm{H}^\bullet(\bar{S}_0, \bar{\mathscr{G}}_0) \otimes_{\mathbb{F}_q} \mathbb{F}$ of the p-linear endomorphism Φ of $\mathrm{H}^\bullet(\bar{S}_0, \bar{\mathscr{G}}_0)$.

3.7

If $\bar{S}_0$ is proper over $\mathbb{F}_q$, the $\mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})$ are finite-dimensional and $\Phi - 1$ is surjective (cf. 3.6b). Thus

$$\mathrm{H}^i_c(S, G) = \ker\left(\Phi - 1 : \mathrm{H}^i(\bar{S}, \bar{G}) \to \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})\right),$$

and F^* is induced by the $\mathbb{F}$-linear extension to $\mathrm{H}^i(\bar{S}, \bar{\mathscr{G}}) = \mathrm{H}^i(\bar{S}_0, \bar{\mathscr{G}}_0) \otimes_{\mathbb{F}_q} \mathbb{F}$ of the $\mathbb{F}_q$-linear endomorphism Φ^f of $\mathrm{H}^i(\bar{S}_0 m \bar{\mathscr{G}}_0)$.

Applying Lemma 3.8 below, one deduces the identity

$$\mathrm{tr}\left(F^*, \mathrm{H}^i_c(S, G)\right) = \mathrm{tr}\left(\Phi^f, \mathrm{H}^i(\bar{S}_0, \bar{\mathscr{G}}_0)\right). \tag{3.7.1}$$

Lemma 3.8. *Let Φ be a p-linear endomorphism of a vector space V over $\mathbb{F}_q$: $\phi(\lambda x) = \lambda^p \phi(x)$. Set $F = \Phi^f$ (F is linear), and again denote by Φ and F the respective p-linear extensions of Φ and F to $V \otimes_{\mathbb{F}_q} \mathbb{F}$. Then F stabilizes* $\ker(\Phi - 1) \subset V \otimes_{\mathbb{F}_q} \mathbb{F}$ *and*

$$\mathrm{tr}\left(F, \ker(\Phi - 1)\right) = \mathrm{tr}(F, V) = \mathrm{tr}\left(F, V \otimes_{\mathbb{F}_q} \mathbb{F}\right).$$

On $V \otimes_{\mathbb{F}_q} \mathbb{F}$, one has $F\Phi = \Phi F$: the p-linear endomorphism $F\Phi - \Phi F$ vanishes on V, hence everywhere. Write $V = V' + V''$, with F invertible on V' and nilpotent on V''. The theory of p-linear endomorphisms ensures that

$$\ker(\Phi - 1) \otimes_{\mathbb{F}_q} \mathbb{F} \xrightarrow{\sim} V' \otimes_{\mathbb{F}_q} \mathbb{F}.$$

Since $\mathrm{tr}(F, V'') = 0$, one therefore has

$$\begin{aligned} \mathrm{tr}\left(F, \ker(\Phi - 1)\right) &= \mathrm{tr}\left(F, \ker(\Phi - 1) \otimes_{\mathbb{F}_q} \mathbb{F}\right) = \mathrm{tr}\left(F, V' \otimes_{\mathbb{F}_q} \mathbb{F}\right) \\ &= \mathrm{tr}\left(F, V \otimes_{\mathbb{F}_q} \mathbb{F}\right) = \mathrm{tr}(F, V). \end{aligned}$$

4 Trace formula modulo p

Theorem 4.1. *Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, let A be a commutative reduced Noetherian ring of characteristic p, and let $\mathscr{K}_0 \in \mathrm{Ob}\,\mathsf{D}_{\mathrm{ctf}}(X_0, A)$. One has*

$$\sum_{x \in X^F} \mathrm{tr}\left(F_x^*, \mathscr{K}_x\right) = \mathrm{tr}\left(F^*, \mathsf{R}\Gamma_c(X, \mathscr{K})\right). \tag{4.1.1}$$

4.2 Reduction to the case where A is a finite field

A standard passage-to-the-limit argument reduces us to the case where A is of finite type over $\mathbb{Z}/p$. For every maximal ideal $\mathfrak{m}$ of A, the image of each side of (4.1.1) in $A/\mathfrak{m}$ is given by the same formula, with $\mathscr{K}_0$ replaced by $\mathscr{K}_0 \overset{\mathsf{L}}{\otimes}_A A/\mathfrak{m} \in \mathrm{Ob}\,\mathsf{D}_{\mathrm{ctf}}(X_0, A/\mathfrak{m})$ (for the right-hand side, cf. Report 4.12). Since the $A/\mathfrak{m}$ are finite, and the intersection of the maximal ideals of A is reduced to 0, one obtains the stated reduction.

4.3 Reduction to the case where $A = \mathbb{Z}/p$

Suppose that A is a finite field, and denote by $T'_A(\mathscr{K}_0)$ and $T''_A(\mathscr{K}_0)$ the two sides of (4.1.1). If, by restriction of scalars, $\mathscr{K}_0$ is regarded as a complex of sheaves of $\mathbb{Z}/p$-modules, one has

$$T'_{\mathbb{Z}/p}(\mathscr{K}_0) = \mathrm{tr}_{A/(\mathbb{Z}/p)}\, T'_A(\mathscr{K}_0)$$

and likewise for T''. If (4.1.1) holds for $A = \mathbb{Z}/p$, one therefore has

$$\mathrm{tr}_{A/(\mathbb{Z}/p)}\, T'_A(\mathscr{K}_0) = \mathrm{tr}_{A/(\mathbb{Z}/p)}\, T''_A(\mathscr{K}_0). \tag{4.3.1}$$

I claim that, for every $\lambda \in A$, one even has

$$\mathrm{tr}_{A/(\mathbb{Z}/p)}\left(\lambda \cdot T'_A(\mathscr{K}_0)\right) = \mathrm{tr}_{A/(\mathbb{Z}/p)}\left(\lambda \cdot T''_A(\mathscr{K}_0)\right), \tag{4.3.2}$$

and hence $T'_A(\mathscr{K}_0) = T''_A(\mathscr{K}_0)$. This is clear for $\lambda = 0$. For λ invertible, denote by $A(\lambda)$ the inverse image on X of the free rank-one sheaf of A-modules on $\mathrm{Spec}(\mathbb{F}_q)$ for which $F^* = \lambda$. Formula (4.3.2) is precisely (4.3.1) for $A(\lambda) \otimes_A \mathscr{K}_0$.

Theorem 4.4. *Let $\bar{S}_0$ be a smooth projective curve, let $j : S_0 \hookrightarrow \bar{S}_0$ be a dense open subscheme of $\bar{S}_0$, and let G_0 be a locally constant sheaf of finite-rank $\mathbb{Z}/p$-vector spaces on S_0. One has*

$$\sum_{x \in S^F} \mathrm{tr}\left(F_x^*, G_x\right) = \sum_i (-1)^i \,\mathrm{tr}\left(F^*, \mathrm{H}^i_c(S, G)\right). \tag{4.4.1}$$

Proof. Apply 3.3 to $(S_0, \bar{S}_0, G_0)$. The coherent extension $\bar{\mathscr{G}_0}$ of $\mathscr{G}_0$ whose existence is guaranteed by 3.3 is locally free because it is torsion-free (we have $\bar{\mathscr{G}_0} \subset j_*\mathscr{G}_0$) and $\bar{S}_0$ is regular of dimension one. The q-linear endomorphism Φ^f of $\bar{\mathscr{G}_0}$ can be interpreted as a morphism

$$u : F^*\bar{\mathscr{G}_0} \to \bar{\mathscr{G}_0}.$$

It factors through $I\bar{G}$, where I defines the closed subset $\bar{S} \smallsetminus S$. Let $v : j_!G_0 \to \bar{\mathscr{G}_0}$ be the unique extension of the natural map $G_0 \to \mathscr{G}_0 = G_0 \otimes_{\mathbb{Z}/p} \mathscr{O}$. The diagram

$$\begin{array}{ccc} F^*j_!G_0 & \xrightarrow{F^*} & j_!G_0 \\ \downarrow{\scriptstyle F^*v} & & \downarrow{\scriptstyle v} \\ F^*\bar{\mathscr{G}_0} & \xrightarrow{u} & \bar{\mathscr{G}_0} \end{array} \tag{4.4.2}$$

Finally, denote by uF^* the composite map

$$\mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) \xrightarrow{F^*} \mathrm{H}^\bullet(\bar{S}, F^*\bar{\mathscr{G}}) \xrightarrow{u} \mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}),$$

one has by (3.7.1)

$$\mathrm{tr}\left(F^*, \mathrm{H}^i_c(S, G)\right) = \mathrm{tr}\left(uF^*, \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})\right). \tag{4.4.3}$$

For every closed point $i_x : x \hookrightarrow \bar{S}$ of $\bar{S}$, set $\bar{\mathscr{G}_x} = i_x^*\bar{\mathscr{G}}$. The Lefschetz trace formula in coherent cohomology ("Woodshole fixed point formula"), applied to F and u, says that

$$\sum_i (-1)^i \,\mathrm{tr}\left(u^*F^*, \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})\right) = \sum_{x \in \bar{S}^F} \frac{\mathrm{tr}(u_x, \mathscr{G}_x)}{\det(1 - dF_x)}. \tag{4.4.4}$$

On the right-hand side,

a) since $dF = 0$, the denominators are equal to 1;

b) for $x \in \bar{S}^F \smallsetminus S^F$, the endomorphism u_x of $\bar{\mathscr{G}_x}$ is zero, and therefore so is $\mathrm{tr}(u_x, \bar{\mathscr{G}_x})$;

c) for $x \in S^F$, (4.4.4) gives a contribution

$$\mathrm{tr}(u_x, \bar{\mathscr{G}_x}) = \mathrm{tr}(F_x^*, G_x).$$

Identity (4.4.1) therefore follows from (4.4.3) and (4.4.4).

□

4.5 A counterexample

An example shows that the hypothesis that A be reduced cannot be omitted from 4.1.

Let Y_0 be the elliptic curve over $\mathbb{F}_2$ for which the eigenvalues of Frobenius are $\pm\sqrt{-2}$. It has 3 rational points and is supersingular. The involution $\sigma : x \mapsto -x$ therefore has only one fixed point, the point 0. The quotient of Y_0 by σ is a projective line; the quotient of $Y_0 \smallsetminus 0$ by σ is therefore an affine line $\mathbb{F}_0^1$, and $Y_0 \smallsetminus 0$ is an unramified double covering of $\mathbb{F}_0^1$:

$$\pi : Y_0 \smallsetminus 0 \to \mathbb{A}_0^1 \qquad , \pi = \pi\sigma.$$

The sheaf $\pi_*\mathbb{Z}/2$ is a free module of rank one over the algebra A of the two-element group $\{1, \sigma\}$ over $\mathbb{Z}/2$. This algebra is isomorphic to the algebra of numbers over $\mathbb{Z}/2$.

The sheaf of A-modules $\pi_*\mathbb{Z}/2$ is our example:

a) One has $\mathrm{H}^\bullet_c(\mathbb{A}^1, \pi_*\mathbb{Z}/2) = \mathrm{H}^\bullet_c(Y \smallsetminus 0, \mathbb{Z}/2)$. Since Y_0 is supersingular, one has $\mathrm{H}^i(Y, \mathbb{Z}/2) = 0$ for $i > 0$, and the long exact sequence

$$\cdots \to \mathrm{H}^i_c(Y \smallsetminus 0, \mathbb{Z}/2) \to \mathrm{H}^i(Y, \mathbb{Z}/2) \to \mathrm{H}^i(\{0\}, \mathbb{Z}/2) \to \cdots$$

shows that $\mathrm{H}^\bullet_c(Y \smallsetminus 0, \mathbb{Z}/2) = 0$. Thus $\mathrm{H}^\bullet_c(\mathbb{A}^1, \pi_*\mathbb{Z}/2) = 0$ and

$$\mathrm{tr}_A\left(F^*, \mathsf{R}\Gamma_c(\mathbb{A}^1, \pi_*\mathbb{Z}/2)\right) = 0.$$

b) Since $Y_0 \smallsetminus 0$ has two rational points, at one of the rational points of $\mathbb{A}^0_1$, the Frobenius substitution is the identity, and at the other, it is σ. Thus

$$\sum_{x \in \mathbb{A}^1{}^F} \mathrm{tr}\left(F^*, \pi_*\mathbb{Z}/2\right) = 1 + \sigma,$$

and $1 + \sigma \neq 0$.

The equation of the covering π is $y^2 - y = x^3$.

Chapter 4

The cohomology class associated with a cycle

This exposé is inspired by notes of Grothendieck that formed a preliminary version of [21, IV]. It defines the cohomology class of a cycle X in a separated smooth scheme of finite type over a field, and proves that intersection corresponds to the cup product.

Chapter 1 contains some general formalities. In Chapter 2, the class of a cycle is defined in several situations more general than the one just mentioned. The principal compatibility not considered is that between the direct image of a cycle and the trace morphism in cohomology. In Chapter 3, this formalism is used to deduce the Lefschetz trace formula for an endomorphism with isolated fixed points of a proper smooth scheme over an algebraically closed k—and for a Frobenius endomorphism of a curve.

We adopt the following conventions:

1) "scheme" means a separated Noetherian scheme (this is largely a hypothesis of convenience).

2) In Chapter 2, an integer n is fixed, and n is invertible on all schemes under consideration.

3) In Chapter 3, a prime number ℓ is fixed, and ℓ is invertible on all schemes under consideration. The cohomology used is always ℓ-adic cohomology:

$$\mathrm{H}^\bullet(X) = \mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \varprojlim \mathrm{H}^\bullet(X, \mathbb{Z}/\ell^n).$$

Cohomology classes of cycles, trace morphisms, ... are defined by passage to the limit from the case of finite coefficients $\mathbb{Z}/\ell^n$ (cf. [21, VI]).

1 Cohomology with support and cup products

This section contains reminders from general topology, which the reader is invited to consult only as the need arises.

by A. Grothendieck, written up by P. Deligne

1.1 H^1 and torsors

1.1.1

Let $\mathscr{F}$ be an abelian sheaf on a site X. It is known that $\mathrm{H}^1(X,\mathscr{F})$ classifies $\mathscr{F}$-torsors on X. We normalize (= choose the sign of) the isomorphism (set of isomorphism classes of F-torsors) $\to \mathrm{H}^1(X,\mathscr{F})$ in such a way that for every exact sequence $0 \to \mathscr{F} \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{H} \to 0$ and every $h \in \mathrm{H}^0(X,\mathscr{H})$, the $\mathscr{F}$-torsor $\beta^{-1}(h) \subset \mathscr{G}$, on which F acts by $(f,x) \mapsto \alpha(f)+x$, has class ∂h.

1.1.2

Let $\mathscr{P}$ be an $\mathscr{F}$-torsor. If (U_i) is an open covering of X, and p_i a section of $\mathscr{P}$ on U_i, one associates to $\mathscr{P}$ the Čech cocycle

$$p_{ij} = p_j - p_i \qquad (p_{ij} \in \mathrm{H}^0(U_i \times U_j, \mathscr{F})).$$

If, according to the usual rule, the map $\check{\mathrm{H}}^\bullet(X,\mathscr{F}) \to \mathrm{H}^\bullet(X,\mathscr{F})$ is defined so as to be a morphism of δ-functors, the image of $(p_{ij}) \in \check{\mathrm{H}}^1(X,\mathscr{F})$ in $\mathrm{H}^1(X,\mathscr{F})$ is the class of $\mathscr{P}$, normalized as in 1.1.1.

1.1.3

The definition of $\mathrm{H}^\bullet(X,\mathscr{F})$ is as follows: for $\mathscr{F}^\bullet$ a resolution of $\mathscr{F}$ with acyclic components, $\mathrm{H}^i(X,\mathscr{F}) = \mathrm{H}^i\,\Gamma(X,\mathscr{F}^\bullet)$. The δ-functor structure is obtained by associating to a short exact sequence of sheaves a short exact sequence of resolutions that remains exact after application of the functor Γ. If $\mathscr{F}^\bullet$ is a resolution of $\mathscr{F}$, the connecting homomorphism ∂ associated with

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}^0 \xrightarrow{d} \ker(d) \longrightarrow 0$$

induces the negative of the defining isomorphism $\mathrm{H}^1(X,\mathscr{F}) = \Gamma\left(X,\ker(d)\right)/d\Gamma(X,\mathscr{F}^0)$.

1.1.4

Let U be an open part of X (a subsheaf of the final sheaf), and let D be the "closed complement." It is known that $\mathrm{H}^1_D(X,\mathscr{F})$ classifies the $\mathscr{F}$-torsors on X trivialized on U. For every short exact sequence $0 \to \mathscr{F} \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{H} \to 0$, and every section with support in D, $h \in \mathrm{H}^0_D(X,\mathscr{H})$, the torsor $\beta^{-1}(h)$, trivialized on U by the section 0, has class ∂h.

The long exact sequence of cohomology with support

$$\cdots \xrightarrow{\partial} \mathrm{H}^i_D(X,\mathscr{F}) \longrightarrow \mathrm{H}^i(X,\mathscr{F}) \longrightarrow \mathrm{H}^i(U,\mathscr{F}) \xrightarrow{\partial} \cdots$$

is defined from the sequence of functors

$$0 \longrightarrow \Gamma_D \longrightarrow \Gamma \longrightarrow \Gamma(U,-) \longrightarrow 0$$

(exact on injective sheaves). For every section $f \in \mathrm{H}^0(U,\mathscr{F})$, $\partial f \in \mathrm{H}^1_D(X,\mathscr{F})$ is the class of the trivial torsor $\mathscr{F}$, trivialized on U by the section f.

1.1.5

Let $j : U \hookrightarrow X$. If $\mathscr{F}$ injects into $j_*j^*\mathscr{F}$, the exact sequence

$$0 \longrightarrow \mathscr{F} \longrightarrow j_*j^*\mathscr{F} \longrightarrow j_*j^*\mathscr{F}/\mathscr{F} \longrightarrow 0$$

gives $\partial : \mathrm{H}^0(j_*j^*\mathscr{F}/\mathscr{F}) = \mathrm{H}^0_D(j_*j^*\mathscr{F}/\mathscr{F}) \to \mathrm{H}^1_D(X, \mathscr{F})$. The composite map

$$\mathrm{H}^0(U, \mathscr{F}) = \mathrm{H}^0(X, j_*j^*\mathscr{F}) \longrightarrow \mathrm{H}^0(X, j_*j^*\mathscr{F}/\mathscr{F}) \xrightarrow{\partial} \mathrm{H}^1_D(X, \mathscr{F})$$

is the negative of the map considered in 1.1.4.

1.1.6

Recall that if $\mathscr{L}$ is an invertible sheaf on a scheme X, the $\mathbb{G}_m$-torsor corresponding to it is the sheaf $\mathrm{Isom}(\mathscr{O}, \mathscr{L})$, on which $\mathbb{G}_m$ acts by $(\lambda, f) \mapsto f \circ (\lambda\cdot) = \lambda f$. Recall also that if D is a Cartier divisor on X, and $j : U \hookrightarrow X$ is the inclusion of the complementary open subscheme, the invertible sheaf $\mathscr{O}(D)$ is the subsheaf of $j_*\mathscr{O}_U$ consisting of local sections s such that sf lies in $\mathscr{O}_X$, where f is a local equation for D.

1.2 Cup products

In this subsection, we develop some observations about cup products in cohomology with support that will be used again in [Duality] to relate Poincaré duality for curves and self-duality of the Jacobian.

1.2.1

Let X be a site, Y a closed part of X, and $\mathscr{F}, \mathscr{G}$ two abelian sheaves on X. For example: X a scheme, Y a closed subscheme, and $\mathscr{F}, \mathscr{G}$ sheaves on X_{et}. Define a product

$$\Gamma_Y(X, \mathscr{F}) \otimes \Gamma(Y, \mathscr{G}) \to \Gamma_Y(X, \mathscr{F} \otimes \mathscr{G}). \tag{1.2.1.1}$$

The product of a section s of $\mathscr{F}$ with support in Y by a section t of $\mathscr{G}$ on Y is obtained as follows: locally on X, t is the restriction to Y of a section t' of $\mathscr{G}$ on X, and one forms the product $s \otimes t'$. It has support in Y and does not depend on the choice of t', which justifies the definition and permits it to be globalized.

Let $i : Y \hookrightarrow X$ be the inclusion morphism. The local analogue of (1.2.1.1) is the product

$$i^!\mathscr{F} \otimes i^*\mathscr{G} \to i^!(\mathscr{F} \otimes \mathscr{G}), \tag{1.2.1.2}$$

from which (1.2.1.1) follows by applying $\Gamma(Y, -)$.

1.2.2

Let us derive these arrows. Let $\mathscr{K}$, $\mathscr{L}$, and $\mathscr{M}$ be in the derived category, with a bilinear map $\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{L} \to \mathscr{M}$. Deriving (1.2.1.1) gives

$$\mathsf{R}\Gamma_Y(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(Y, \mathscr{L}) \to \mathsf{R}\Gamma_Y(X, \mathscr{M}) \tag{1.2.2.1}$$

inducing

$$\mathrm{H}^i_Y(X, \mathscr{K}) \otimes \mathrm{H}^j(Y, \mathscr{L}) \to \mathrm{H}^{i+j}_Y(X, \mathscr{M}) \tag{1.2.2.2}$$

(the cup product). The local arrow (1.2.1.2) gives

$$\mathsf{R}i^!\mathscr{K} \overset{\mathsf{L}}{\otimes} i^*\mathscr{L} \to \mathsf{R}i^!\mathscr{M}, \tag{1.2.2.3}$$

from which (1.2.2.1) follows by applying $\mathsf{R}\Gamma(Y, -)$.

1.2.3

Above, I passed over in silence the conflicts caused by using right-derived functors ($\mathsf{R}\Gamma$) and left-derived functors ($\overset{\mathsf{L}}{\otimes}$) in the same formula.

a) In the next subsection, all the right-derived functors under consideration will have finite cohomological dimension. This makes it possible to work systematically in the derived categories D^-. To have complexes that are both flat and flasque, one uses canonical flasque resolutions as in [2, XVII].

b) For a more general theory, one can no longer interpret bilinear maps from $\mathscr{K}$ and $\mathscr{L}$ to $\mathscr{M}$ as morphisms from $\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L}$ to $\mathscr{M}$. For example, $\mathsf{R}\Gamma_Y(X,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma(Y,\mathscr{L})$ is not defined if both factors lie in D^+. One solution is to work in D^+, define

$$\mathrm{Bil}(\mathscr{K},\mathscr{L};\mathscr{M}) = \varinjlim \hom(\mathscr{K}'\otimes\mathscr{L}',\mathscr{M}'),$$

where the limit is taken over the quasi-isomorphisms $\mathscr{K}'\xrightarrow{\sim}\mathscr{K}$, $\mathscr{L}'\xrightarrow{\sim}\mathscr{L}$, $\mathscr{M}\xrightarrow{\sim}\mathscr{M}'$ ($\mathscr{K}'$, $\mathscr{L}'$, $\mathscr{M}'$ bounded below), where hom means "morphism of complexes, up to homotopy," and systematically use such bilinear maps without ever mentioning $\otimes$.

1.2.4 Second theme

Let U be an open subscheme of X, and let $j: U\hookrightarrow X$ be the inclusion morphism. For $\mathscr{K}$ in the derived category on U, set $\mathsf{R}\Gamma_!(U,\mathscr{K}) = \mathsf{R}\Gamma(X, j_!\mathscr{K})$. For $\mathscr{K}$, $\mathscr{L}$, and $\mathscr{M}$ on U, with a bilinear map $\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L}\to\mathscr{M}$, we wish to define

$$\mathsf{R}\Gamma(U,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma_!(U,\mathscr{L})\to\mathsf{R}\Gamma_!(U,\mathscr{M}). \tag{1.2.4.1}$$

At the level of sheaves and their global sections, such a product follows from the isomorphism $j_*\mathscr{F}\otimes j_!\mathscr{G}\xleftarrow{\sim} j_!(\mathscr{F}\otimes\mathscr{G})$, but care must be taken because $\mathsf{R}\Gamma_!$ is not in general the derived functor of $\Gamma(X, j_!-)$.

We begin by defining

$$\mathsf{R}j_*\mathscr{K}\overset{\mathsf{L}}{\otimes}j_!\mathscr{L}\xleftarrow{\sim} j_!(\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L})\longrightarrow j_!\mathscr{M}.$$

Applying $\mathsf{R}\Gamma(X,-)$ gives

$$\mathsf{R}\Gamma(U,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma_!(U,\mathscr{L}) = \mathsf{R}\Gamma(X,\mathsf{R}j_*\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma(X,j_!\mathscr{L})\longrightarrow\mathsf{R}\Gamma(X,j_!\mathscr{M}).$$

Here again, the conflict between left and right will be resolved in the next subsection by working in D^-.

1.2.5 Coda

Let $j: U\hookrightarrow X$ be an open part, and let $i: Y\hookrightarrow U$ be a closed part of U. Let $\bar{Y}$ be a closed subset of X such that $\bar{Y}\cap U = Y$ (for example, the complement of $U\smallsetminus Y$). Let $\mathscr{K}$ be on U. Set $\mathsf{R}\Gamma_{Y!}(U,\mathscr{K}) = \mathsf{R}\Gamma_!(Y,\mathsf{R}i^!\mathscr{K})$, where $\mathsf{R}\Gamma_!$ is relative to the inclusion of Y into $\bar{Y}$. For every open subset V of U containing Y, $\mathsf{R}\Gamma_{Y!}(U,\mathscr{K})$ maps to $\mathsf{R}\Gamma_!(V,\mathscr{K})$: if i also denotes the inclusion of $\bar{Y}$ into X, j that of Y into $\bar{Y}$, and k that of V into X, one has $\mathsf{R}i^!\mathscr{K} = j^*\mathsf{R}i^!k_!(k^*\mathscr{K})$, whence a morphism $j_!\mathsf{R}i^!\mathscr{K}\to\mathsf{R}i^!k_!(k^*\mathscr{K})$. Applying $\mathsf{R}\Gamma(\bar{Y},-)$ to it gives $\mathsf{R}\Gamma_{Y!}(U,\mathscr{K})\to\mathsf{R}\Gamma_{\bar{Y}}(k_!k^*\mathscr{K})\to\mathsf{R}\Gamma(k_!k^*\mathscr{K}) = \mathsf{R}\Gamma_!(V,\mathscr{K})$.

For $\mathscr{K},\mathscr{L},\mathscr{M}$ on U, with a bilinear map $\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L}\to\mathscr{M}$, we wish to define

$$\mathrm{H}^n(U,\mathscr{K})\otimes\mathrm{H}^m_!(Y,\mathscr{L})\to\mathrm{H}^{n+m}_{Y!}(U,\mathscr{M}) \tag{1.2.5.1}$$

(the last group itself mapping to $\mathrm{H}^{n+m}_!(V,\mathscr{M})$). In the derived category, this means defining

$$\mathsf{R}\Gamma_Y(U,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma_!(R,\mathscr{L})\to\mathsf{R}\Gamma_{Y!}(U,\mathscr{M}). \tag{1.2.5.2}$$

We identify $\mathsf{R}\Gamma_Y$ with $\mathsf{R}\Gamma(Y,\mathsf{R}i^!\mathscr{K})$. The desired product is then of type (1.2.4.1), relative to the local product (1.2.2.3) on Y: $\mathsf{R}i^!\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L}\to\mathsf{R}i^!\mathscr{M}$.

1.2.6

Above, we developed the absolute formalism. There is a parallel relative formalism, with Γ replaced by f_* for f a morphism $X\to S$.

1.3 The Koszul rule

Let A be a commutative ring, and let $(V_i)_{i\in I}$ be a finite family of graded (or $\mathbb{Z}/2$-graded) A-modules. Recall the definition of the graded tensor product $\bigotimes_{i\in I}V_i$, in the sense of the Koszul rule (cf. [2, XVII 1.1]). For each total ordering a on I, we shall define a module $V(a)$. We shall also define a transitive system of isomorphisms $\varphi_{ab}:V(b)\xrightarrow{\sim}V(a)$, and the graded tensor product of the V_i will be the "common value" $\varprojlim V(a)$ of this system of modules. We take

a) $V(a)=\bigotimes_{i\in I}|V_i|$ (the ordinary tensor product of the underlying ungraded modules of the V_i).

b) if the $x_i\in V_i$ are homogeneous, we take $\varphi_{ab}(\otimes x_i)=(-1)^N\otimes x_i$, where N is the sum of the $\deg(x_i)\deg(x_j)$ over the pairs (i,j) such that $i<_a j$ and $i>_b j$.

Example 1

Take $I=\{1,2\}$, and let a be the ordering in which $1<2$, and b the ordering in which $1>2$. Let $v_i\in V_i$ be homogeneous, and denote by $v_1\otimes v_2$ (resp. $v_2\otimes v_1$) the image in the graded tensor product of the product of the v_i in $V(a)$ (resp. $V(b)$). One has

$$v_1\otimes v_2=(-1)^{\deg(v_1)\deg(v_2)}v_2\otimes v_1$$

(the Koszul rule).

Example 2

If all the V_i have degree 1, there is a canonical isomorphism relating the module underlying the graded tensor product and the ordinary tensor product of the modules $|V_i|$ underlying the V_i:

$$\Big|\bigotimes V_i\Big|\simeq\bigotimes|V_i|\otimes_{\mathbb{Z}}\bigwedge^{|I|}\mathbb{Z}^I.$$

Example 3

For A a field, and X_i a finite family of spaces, the Künneth formula is written $\mathrm{H}^\bullet(\prod X_i,A)=\bigotimes\mathrm{H}^\bullet(X_i,A)$.

Let $V^\vee$ be the graded dual of V. The canonical map

$$V^\vee\otimes V=V\otimes V^\vee\to A \tag{1.3.1}$$

is $v'\otimes v\to v'(v)$.

Suppose now that A is a field, and consider only finite-dimensional vector spaces. The canonical isomorphism

$$W \otimes V^\vee \to \hom(V, W) \tag{1.3.2}$$

is $w \otimes v' \mapsto (v \mapsto w \cdot v'(v))$. Via this isomorphism, the composition $\hom(Y,Z) \otimes \hom(X,Y) \to \hom(X,Z)$ is identified with the morphism induced by (1.3.1): $Z \otimes Y^\vee \otimes Y \otimes X^\vee \to Z \otimes X^\vee$.

The trace of $f : V \to V^\vee$ (zero for homogeneous f of degree $\neq 0$) is the image of f under

$$\mathrm{tr} : \hom(V,V) \xleftarrow[(1.3.2)]{\sim} V \otimes V^\vee = V^\vee \otimes V \to_{(1.3.1)} A. \tag{1.3.4}$$

It is easily verified that, for f of degree 0,

$$\mathrm{tr}(f, V) = \sum (-1)^i \,\mathrm{tr}(f, V^i). \tag{1.3.5}$$

Expressing the commutativity of the two morphisms obtained by composing the morphisms (1.3.1) $V^\vee \otimes V \otimes W^\vee \otimes W \to k$, one finds that for homogeneous $f : V \to W$ and $g : W \to V$, one has

$$\mathrm{tr}(fg) = (-1)^{\deg(f)\deg(g)} \,\mathrm{tr}(gf). \tag{1.3.6}$$

2 The cohomology class associated with a cycle

2.1 The class of a divisor

2.1.1

Let D be a Cartier divisor in a scheme X. Away from D, the invertible sheaf $\mathscr{O}(D)$ is trivialized by the section 1. The class $\mathrm{cl}(D)$ of D in $\mathrm{H}^1_D(X, \mathbb{G}_m)$ is the class of the corresponding $\mathbb{G}_m$-torsor trivialized on $X \smallsetminus D$ (1.1.6 and 1.1.4).

Let $\partial : \mathrm{H}^i(X \smallsetminus D, \mathbb{G}_m) \to \mathrm{H}^i_D(X, \mathbb{G}_m)$ be the morphism of 1.1.4. If D admits a global equation f, multiplication by f is an isomorphism from $\mathscr{O}(D)$, trivialized by 1 on $X \smallsetminus D$, to $\mathscr{O}$, trivialized by f on $X \smallsetminus D$. By 1.1.4, one therefore has

$$\mathrm{cl}(D) = \partial f. \tag{2.1.1}$$

For every morphism $u : X' \to X$ such that u^*D is again a Cartier divisor (i.e., $u^{-1}D$ is disjoint from $\mathrm{Ass}(X')$), one has $\mathrm{cl}(u^*D) = u^* \,\mathrm{cl}(D)$. If such functoriality were desired for every morphism u, one would have to consider not Cartier divisors, but more generally invertible sheaves equipped with a section.

Recall that the integer n is henceforth assumed invertible on the schemes under consideration. Let $\partial : \mathrm{H}^i_D(X, \mathbb{G}_m) \to \mathrm{H}^{i+1}_D(X, \boldsymbol{\mu}_n)$ be the boundary map for the Kummer exact sequence $0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0$.

Definition 2.1.2. *The* class $\mathrm{cl}_n(D)$ *of* D *in* $\mathrm{H}^2_D(X, \boldsymbol{\mu}_n)$ *is* $\partial \,\mathrm{cl}(D)$.

When there is no risk of confusion, the subscript n will be omitted.

2.1.3

The diagram

$$\begin{array}{ccc} \mathrm{H}^0(X \smallsetminus D, \mathbb{G}_m) & \xrightarrow{\partial} & \mathrm{H}^1_D(X, \mathbb{G}_m) \\ \downarrow \partial & & \downarrow \partial \\ \mathrm{H}^1(X \smallsetminus D, \boldsymbol{\mu}_n) & \xrightarrow{\partial} & \mathrm{H}^2_D(X, \boldsymbol{\mu}_n) \end{array}$$

is anticommutative. If D admits a global equation f, $\mathrm{cl}_n(D)$ is therefore the negative of the image under ∂ of the class in $\mathrm{H}^1(X \smallsetminus D, \boldsymbol{\mu}_n)$ of the $\boldsymbol{\mu}_n$-torsor of n-th roots of f.

Proposition 2.1.4. *Let i be the inclusion of D into X. If D and X are regular, the cohomology sheaves with support $\mathsf{R}^p i^! \boldsymbol{\mu}_n$ are zero for $p = 0, 2$, and $\mathsf{R}^2 i^! \boldsymbol{\mu}_n = \underline{\mathbb{Z}/n}$, generated by $\mathrm{cl}_n(D)$.*

It suffices to prove that, for X strictly local and D defined by a regular parameter, one has $\mathrm{H}^p_D(X, d\mu_n) = 0$ for $p = 0, 1$ and $\mathrm{H}^2(X, \boldsymbol{\mu}_n) = \mathbb{Z}/n$, generated by $\mathrm{cl}(D)$. Denoting reduced cohomology by $\sim$, one has $\widetilde{\mathrm{H}}^{p-1}(X \smallsetminus D, \boldsymbol{\mu}_n) \xrightarrow{\sim} \mathrm{H}^p_D(X, \boldsymbol{\mu}_n)$. The assertion for $p = 0, 1$ says that D does not disconnect X, and for $p = 2$ it follows, via 2.1.3, from Abhyankar's lemma.

This is a partial analogue of the relative theorem (Étale cohomology, 5.3.4). Grothendieck conjectures that the $\mathsf{R}^p i^! \boldsymbol{\mu}_n$ vanish for $p \neq 2$, at least for X excellent (purity conjecture), but this is known only in characteristic 0 ([2, XIX]).

Theorem 2.1.5 (Fundamental compatibility)**.** *Let X be a smooth curve over an algebraically closed field k, let P be a closed point of X, and let* tr *be the composite $\mathrm{H}^2_P(X, \boldsymbol{\mu}_n) \to \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \xrightarrow{\mathrm{tr}} \mathbb{Z}/n$. One has*

$$\mathrm{tr}\, \mathrm{cl}(P) = 1.$$

Let $\bar{X}$ be the smooth projective curve completing X. The formula says that the invertible sheaf $\mathscr{O}(P)$ on $\bar{X}$ has degree 1.

2.2 Cohomological method

2.2.1

Let X be a (Noetherian) scheme. Recall that a subscheme Y of X is said to be a local complete intersection of codimension c if, locally (on Y), it is defined by a regular sequence of c equations in X. For X the spectrum of a local ring A with maximal ideal $\mathfrak{m}$, and Y with ideal $\mathfrak{a}$, this means that $\exp^i(A/\mathfrak{a}, A) = 0$ for $i < \dim(\mathfrak{a}/\mathfrak{m}\mathfrak{a}) = c$, and every sequence of elements of $\mathfrak{a}$ whose image in $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$ is a basis of $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$ is a regular sequence of equations for Y.

2.2.2

Let $i : Y \hookrightarrow X$ be a local complete intersection of codimension c. We propose to define a local fundamental class $\mathrm{cl}(Y)$ that is a global section of the cohomology sheaf with support $\mathsf{R}^{2c} i^! \mathbb{Z}/n(c)$ (recall that $\mathbb{Z}/n(c) = \boldsymbol{\mu}_n^{\otimes c}$).

Locally, Y is the intersection of a sequence of c divisors D_i, and we define $\mathrm{cl}(Y)$ as the cup product of the $\mathrm{cl}(D_i)$. Each $\mathrm{cl}(D_i)$ has support in D_i; their product has support in Y. The fact that this product is locally independent of the choice of the D_i follows from 2.2.3 below and the following invariance properties:

a) compatibility with localization;

b) independence of the ordering of the D_i (the $\mathrm{cl}(D_i)$ have degree 2, which is even, so the cup product is commutative);

c) the product depends only on the "flag" $D_1 \supset D_1 \cap D_2 \supset \cdots \supset Y$.

To prove c), note the existence of a product (a variant of 1.2.1)

$$\mathrm{H}^\bullet_{D_1}(X) \otimes \mathrm{H}^\bullet_{D_1 \cap D_2}(D_1) \otimes \cdots \otimes \mathrm{H}^\bullet_Y(D_1 \cap \cdots \cap D_{c-1}) \to \mathrm{H}^\bullet_Y(X);$$

the product of the $\mathrm{cl}(D_i)$ is also the product of the ($\mathrm{cl}(D_i)$ restricted to $D_1 \cap \cdots \cap D_{i-1}$) = ($\mathrm{cl}(D_1 \cap \cdots \cap D_i)$ in $D_1 \cap \cdots \cap D_{i-1}$).

Lemma 2.2.3. *Let A be a local ring with maximal ideal $\mathfrak{m}$, and let $u = (u_1, \ldots, u_c)$, $v = (v_1, \ldots, v_c)$ be two regular sequences generating the same ideal $\mathfrak{a}$. There is then a sequence w_i $(1 \leqslant i \leqslant N)$ of such sequences joining them, such that w_{i+1} is obtained from w_i by a permutation or by changing only the last element.*

Since permutations are available, it would amount to the same thing to allow one element, rather than the last element, to be changed. By 2.2.1, we are thus reduced to verifying that, in the vector space $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$, one can pass from one basis to another by a sequence of permutations and changes of basis modifying only one vector. Indeed, the linear group is generated by diagonal, elementary, and permutation matrices.

2.2.4

The same methods make it possible to define a local fundamental class for every $Y \subset X$ locally definable by c equations (where fewer equations suffice, the class is zero).

2.2.5

One can pass from such a local fundamental class in $\mathrm{H}^0\left(Y, \mathsf{R}^{2c} i^! \mathbb{Z}/n(c)\right)$ to a global fundamental class in $\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c))$ when semipurity results are available:

Proposition 2.2.6. *(i) If $\mathsf{R}^p i^! \mathbb{Z}/n = 0$ for $p < 2c$, then*

$$\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c)) \xrightarrow{\sim} \mathrm{H}^0\left(Y, \mathsf{R}^{2c} i^! \mathbb{Z}/n(c)\right).$$

(ii) Let Z be a closed part of Y, with complement V in Y, and let k be the inclusion of Z into X. If $\mathsf{R}^p i^! \mathbb{Z}/n = 0$ for $p \leqslant 2c$, then

$$\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c)) \hookrightarrow \mathrm{H}^{2c}_V(X, \mathbb{Z}/n(c)).$$

If $\mathsf{R}^p i^! \mathbb{Z}/n = 0$ for $p \leqslant 2c+1$, this arrow is an isomorphism.

The hypothesis $\mathsf{R}^p i^! \mathbb{Z}/n = 0$ is equivalent to $\mathsf{R}^p i^! \mathbb{Z}/n(c) = 0$. With this understood, (i) follows from the spectral sequence $\mathrm{H}^p(Y, \mathsf{R}^q i^!) \Rightarrow \mathrm{H}^{p+q}_Y(X, -)$. If k_1 is the inclusion of Z into Y, one has $k = ik_1$, whence $\mathsf{R}k^! \mathbb{Z}/n = \mathsf{R}k_1^! \mathsf{R}i^!$, and the long exact cohomology sequence for $Z \subset Y$ gives a long exact sequence

$$\cdots \longrightarrow \mathrm{H}^i\left(Z, \mathsf{R}k^! \mathbb{Z}/n(c)\right) \longrightarrow \mathrm{H}^i_Y(X, \mathbb{Z}/n(c)) \longrightarrow \mathrm{H}^i_V(X, \mathbb{Z}/n(c)) \longrightarrow \cdots$$

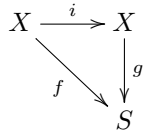
from which (ii) follows.

2.2.7 Amplification

Proposition 2.2.6 remains valid for i any separated morphism of finite type, and $2c$ any integer (positive or negative), provided that $\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c))$ is replaced there by $\mathrm{H}^{2c}\left(Y, \mathsf{R}i^! \mathbb{Z}/n(c)\right)$, and likewise for H_V.

The following semipurity results follow from [24, 1.8, 1.10, 1.15]. We recall their proof.

Theorem 2.2.8. *Let there be a morphism of S-schemes of finite type*



Suppose that X is smooth, purely of relative dimension N, and that Y has fiberwise dimension $\leqslant d$. Let $c = N - d$.

(i) One has $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ *for* $p < 2c$*; likewise with* $\mathbb{Z}/n$ *replaced by a sheaf* $g^*\mathscr{F}$*.*

(ii) If, over a dense open subset U *of* S*,* Y *has fiberwise dimension* $< d$*, one has* $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ *for* $p \leqslant 2c$*. If in addition the complement* Z *of* U *does not locally disconnect* S*, one has* $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ *for* $p \leqslant 2c+1$*.*

Since $\mathsf{R}g^!\mathscr{F} = g^*\mathscr{F}(N)[2N]$, the transitivity formula $\mathsf{R}i^!\mathsf{R}g^! = \mathsf{R}f^!$ shows that $\mathsf{R}^{2c+q}i^!(g^*\mathscr{F}) = 0 \Leftrightarrow \mathsf{R}^{-2d+q}f^!(\mathscr{F})$. This reduces us to studying f, i.e. to assuming that $X = S$. Assertion (i) is then [2, XVIII 3.17].

Let Y' be the inverse image of Z in Y:

$$\begin{array}{ccc} Y' & \xrightarrow{v} & Y \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow{u} & S \end{array}$$

By (i), the $\mathsf{R}^p f^!\mathbb{Z}/n$ have support in Y' for $p \leqslant -2d+1$; and the spectral sequence $\mathsf{R}^a v^!\mathsf{R}^b f^! \Rightarrow \mathsf{R}^{a+b}(fv)^!$ shows that they coincide with the $\mathsf{R}^p(fv)^!\mathbb{Z}/n = \mathsf{R}^p(uf)^!\mathbb{Z}/n$. Applying (i) to Y'/Z and to the spectral sequence $\mathsf{R}^a f^!\mathsf{R}^b u^! \Rightarrow \mathsf{R}^{a+b}(uf)^!$, one finds that (ii) follows from the vanishing of $\mathsf{R}^b u^!\mathbb{Z}/n$ for $b = 0$, or for $b = 0$ and 1, as the case may be.

2.2.9

Grothendieck conjectures the following absolute analogue of 2.2.8 (semipurity conjecture—a consequence of the purity conjecture): for Y of codimension $\geqslant c$ in a regular X, one has $\mathrm{H}^i_Y(X, \mathbb{Z}/n) = 0$ for $i < 2c$, at least if X is excellent.

2.2.10

We are now in a position to define the class of a cycle Y of codimension c in X smooth over a field k. Write $Y = \sum d_i Y_i$, where the Y_i are reduced and irreducible. An open subset U_i of Y_i, whose complement has codimension $> c$, is then a local complete intersection in X. This makes it possible to define the local fundamental class of U_i. By 2.2.6 and 2.2.8, it comes from a unique fundamental class $\mathrm{cl}(Y_i) \in \mathrm{H}^{2c}_{Y_i}(X, \mathbb{Z}/n(c))$, and we set

$$\mathrm{cl}(Y) = \sum d_i \,\mathrm{cl}(Y_i) \in \mathrm{H}^{2c}_{|Y|}(X, \mathbb{Z}/n(c))\,.$$

2.2.11

In complex analytic geometry, a construction of Baum, Fulton, and Mac Pherson [4] makes it possible to define without restriction the cohomology class of a local complete intersection $Y \subset X$. Suppose Y is purely of codimension c, and let $\mathscr{N}$ be the vector bundle on Y that is the normal bundle of Y in X. Its local sections are those of $(\mathscr{I}/\mathscr{I}^2)^\vee$, where $\mathscr{I}$ is the sheaf of ideals of Y. Recall that the sheaf of C^∞ functions on X is defined locally, in terms of local embeddings of X into $\mathbb{C}^n$, as the restriction to X of the quotient of the sheaf of C^∞ functions on $\mathbb{C}^n$ by the ideal generated by the real and imaginary parts of the equations defining X. The bundle $\mathscr{N}$ extends to a complex $C^\infty N$ vector bundle on a neighborhood U of Y in X, and for U sufficiently small, there exist sections f of N whose zero locus is Y and such that, on Y, $df : \mathscr{N} \to N$ is the identity. Two choices of N and f are homotopic on a sufficiently small neighborhood of Y.

The cohomology class $\mathrm{cl}(U)$ of the section 0 of N (denoted $z : U \to N$) is defined: U is a local complete intersection in N, and $\mathsf{R}^i z^! \mathbb{Z} = 0$ for $i \neq 2c$, $\mathsf{R}^{2c} z^! \mathbb{Z} = \mathbb{Z}$. We have $f^* : \mathrm{H}^\bullet_U(N, \mathbb{Z}) \to \mathrm{H}^\bullet_Y(X, \mathbb{Z})$ and set

$$\mathrm{cl}(Y) = f^* \,\mathrm{cl}(U).$$

The same construction works whenever an analytic subspace Y of X has the following normal structure: a locally free sheaf $\mathscr{C}$ of rank c on Y_{red}, and an epimorphism $\mathscr{C} \to \mathscr{I}/\mathscr{I} \cdot \mathscr{I}_{\mathrm{red}}$.

2.3 Homological method

2.3.1

Let $f : Y \to X$ be a flat morphism of finite type with fibers of dimension $\leqslant d$. In [2, XVIII 2.9], we defined a trace morphism

$$\mathrm{tr}_f : \mathsf{R}^{2d} f_! \mathbb{Z}/n(d) \to \mathbb{Z}/n. \tag{2.3.1.1}$$

One has $\mathsf{R}^i f_! \mathbb{Z}/n(d) = 0$ for $i > 2d$, and $\mathsf{R}^i f^! \mathbb{Z}/n = 0$ for $i < -2d$. This, together with the adjunction between $\mathsf{R}f_!$ and $\mathsf{R}f^!$, gives isomorphisms

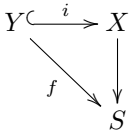
$$\begin{aligned} \hom\left(\mathsf{R}^{2d} f_! \mathbb{Z}/n(d), \mathbb{Z}/n\right) &= \hom\left(\mathsf{R}f_! \mathbb{Z}/n(d), \mathbb{Z}/n[-2d]\right) \\ &= \hom\left(\mathbb{Z}/n, \mathsf{R}f^! \mathbb{Z}/n(-d)[-2d]\right) \\ &= \mathrm{H}^0\left(Y, \mathsf{R}^{-2d} f^! \mathbb{Z}/n\right). \end{aligned} \tag{2.1.3.2}$$

At least when S is a point, the image of tr_f in the last two groups deserves the name fundamental class of Y (in homology).

We again denote by tr_f the image of tr_f in the second group, and the morphisms deduced from it by functoriality. For example, for Y/S proper, the morphism

$$\mathrm{H}^i\left(Y, \mathbb{Z}/n(d)\right) = \mathrm{H}^i\left(S, \mathsf{R}f_! \mathbb{Z}/n(d)\right) \longrightarrow \mathrm{H}^{i-2d}(S, \mathbb{Z}/n). \tag{2.3.1.3}$$

Suppose Y is contained in X, smooth over S and purely of relative dimension N, and set $c = N - d$.



One has $\mathsf{R}f^! = \mathsf{R}i^! \mathsf{R}g^!$, and $\mathsf{R}g^! \mathbb{Z}/n = \mathbb{Z}/n(-N)[-2N]$. The fundamental class of Y is identified with an element of $\mathrm{H}^0\left(Y, \mathsf{R}i^! \mathbb{Z}/n(c)[2c]\right) = \mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right)$, the class $\mathrm{cl}(Y)$ of Y in X. We shall see below that it depends only on $Y \subset X$, not on the projection of X onto S, and that for Y a local complete intersection, it induces the local class of the preceding subsection.

Let us make explicit the isomorphism

$$\mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right) \xrightarrow{\sim} \hom\left(\mathsf{R}^{2d} f_! \mathbb{Z}/n(d), \mathbb{Z}/n\right)$$

that takes $\mathrm{cl}(Y)$ to tr_f: via the isomorphisms

$$\begin{aligned} \mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right) &= \mathrm{H}^{2c}_Y\left(\mathsf{R}i^! \mathbb{Z}/n(c)\right) \\ &= \hom_Y\left(\mathbb{Z}/n(d)[2d], \mathsf{R}i^! \mathbb{Z}/n(N)[2N]\right) \\ &= \hom_X\left(\mathbb{Z}/n(d)[2d]_Y, \mathbb{Z}/n(N)[2n]\right), \end{aligned}$$

it is the top row of

$$\begin{array}{ccccc} \mathrm{H}^{2c}_Y & \longrightarrow & \hom\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathsf{R}f_!\mathsf{R}i^!\mathbb{Z}/n(N)[2n]\right) & \xrightarrow{(1)} & \hom\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathbb{Z}/n\right) \\ & \searrow^{(3)} & \downarrow{\scriptstyle \mathrm{tr}_i} & & \| \\ & & \hom\left(\mathsf{R}g_!\mathbb{Z}/n(d)[2d]_Y, \mathsf{R}g_!\mathbb{Z}/n(N)[2N]\right) & \xrightarrow{\mathrm{tr}_g} & \hom\left(\mathsf{R}g_!\mathbb{Z}/n(d)[2d]_Y, \mathbb{Z}/n\right), \end{array}$$

where (1) is the adjunction arrow $\mathsf{R}f_!\mathsf{R}f^! \to \mathrm{id}$. The commutativity of (2) expresses the fact that the isomorphism $\mathsf{R}f^! = \mathsf{R}i^!\mathsf{R}g^!$ is defined by adjunction. Finally, the arrow deduced from (3)

$$\mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right) \otimes \mathsf{R}^{2d}g_!\mathbb{Z}/n(d) \to \mathsf{R}^{2N}g_!\mathbb{Z}/n(N)$$

is interpreted as a cup product (cf. 1.2).

Definition 2.3.2. *The class* $\mathrm{cl}(Y) \in \mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right)$ *is characterized by the property that, for every local section* u *of* $\mathsf{R}^{2d}f_!\mathbb{Z}/n(d)$*, one has*

$$\mathrm{tr}_f(u) = \mathrm{tr}_g\left(\mathrm{cl}(Y) \smile u\right).$$

Because $\mathrm{cl}(Y)$ already has an image tr_f in $\hom\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathbb{Z}/n\right)$, formula 2.3.2 holds for the arrows deduced from tr_f by functoriality. For example, for X proper over S and $u \in \mathrm{H}^\bullet(Y, \mathbb{Z}/n)$, the formula holds in $\mathrm{H}^\bullet\left(S, \mathbb{Z}/n(-d)\right)$. For $v \in \mathrm{H}^\bullet(X, \mathbb{Z}/n)$, it gives

$$\mathrm{tr}_f(i^*v) = \mathrm{tr}_g\left(\mathrm{cl}(Y) \smile v\right),$$

where the cup product can be computed in $\mathrm{H}^\bullet(X, \mathbb{Z}/n)$.

2.3.3

We shall define the trace morphisms, and hence the class $\mathrm{cl}(Y)$, under more general hypotheses. Consider a diagram

$$\begin{array}{ccc} Y & \hookrightarrow & X \\ & {\scriptstyle f}\searrow & \downarrow{\scriptstyle g} \\ & & S \end{array}$$

with g smooth and purely of relative dimension N, and Y closed in X with fiberwise dimension $\leqslant d$. Equip Y in addition with a "weight" $\mathscr{K}$ of the following kind: $\mathscr{K} \in \mathsf{D}^b_{\mathrm{parf},X}$ is a bounded complex of sheaves of $\mathscr{O}$-modules on X, of finite Tor-dimension over S (or over X, which amounts to the same thing), whose cohomology sheaves are coherent and have support in Y. We propose to define a morphism $\mathrm{tr}_{f,\mathscr{K}} : \mathsf{R}^{2d}f_!\mathbb{Z}/n(d) \to \mathbb{Z}/n$. For Y flat over S and $\mathscr{K} = \mathscr{O}_Y$, this will be the preceding trace morphism. In general, it depends only on the lengths of $\mathscr{K}$ at the generic points y of the irreducible components of Y ($\lg_y(\mathscr{K}) = \sum(-1)^i \lg \mathrm{H}^i(\mathscr{K})_y$). Finally, $\mathrm{cl}(Y, \mathscr{K})$ will denote the class with support in Y such that

$$\mathrm{tr}_g\left(\mathrm{cl}(Y, \mathscr{K}) \smile u\right) = \mathrm{tr}_{f,\mathscr{K}}(u). \tag{2.3.3.1}$$

2.3.4

The construction of $\mathrm{tr}_{f,\mathscr{K}}$ is parallel to that of [2, XVIII.2]; we shall indicate only its main outlines.

A $d = 0$ (f quasi-finite). Let x be a geometric point of Y, s its image in S, $\mathscr{K}_{(x)}$ the inverse image of $\mathscr{K}$ on the strict localization of X at x, $\mathscr{K}_{(x)s} = \mathscr{K}_{(x)} \overset{\mathsf{L}}{\otimes}_{\mathscr{O}^h_{S,s}} k(s)$ its inverse image on the geometric fiber at s, and $n(x) = \sum(-1)^i \dim_{k(s)} \mathrm{H}^i(\mathscr{K}_{(x)s})$. The function $x \mapsto n(x)$ is a weighting of f, and we take the corresponding trace morphism [2, XVII.6.2.5]. Formation of the weighting $n(x)$ and of $\mathrm{tr}_{f,\mathscr{K}}$ is compatible with every base change.

B General case. If $u : Z \to \mathbb{A}^d_S$ is such that ui is quasi-finite, set $\mathrm{tr}_{f,\mathscr{K}} = \mathrm{tr}_{\mathbb{A}^d_S} \circ \mathrm{tr}_{ui,\mathscr{K}}$. This trace morphism is clearly compatible with every base change S'/S. To prove that it does not depend on u, we may therefore suppose that S is the spectrum of an algebraically closed field k. In that case, let Y_i be the irreducible components of Y, and let lg_i be the length of $\mathscr{K}$ at the generic point of Y_i. If α_i is the inclusion of $(Y_i)_{\mathrm{red}}$ into Y, one has

$$\mathrm{tr}_{f,\mathscr{K}} = \sum \mathrm{lg}_i \, \mathrm{tr}_{Y_{i,\mathrm{red}/S}} \, \alpha_i^* \tag{2.3.4}$$

(this formula follows from the analogous and easy formula for tr_{ui}).

This defines $\mathrm{tr}_{f,\mathscr{K}}$ locally on Y; one then proceeds as in [2, XVIII.2.9].

Lemma 2.3.5. *If $g = g'g'' : X \xrightarrow{g''} S' \xrightarrow{g'} S$, with g' and g'' smooth and purely of relative dimensions N' and N'', and if Y still has fiberwise codimension $\geqslant c$ over S', then the class $\mathrm{cl}(Y, \mathscr{K})$ is the same whether computed in terms of g or of g''.*

Let $f' = g''i$ and $d' = d - N'$. The formula $\mathrm{tr}_g = \mathrm{tr}_{g'} \mathrm{tr}_{g''}$ ([2, XVIII.2.9] Var 3) ensures the commutativity of

$$\begin{array}{ccc}
 & \mathrm{H}^{2c}_X(Z, \mathbb{Z}/n(c)) & \\
 & \downarrow & \\
\hom_{S'}(\mathsf{R}f_!\mathbb{Z}/n(d')[2d'], \mathsf{R}g''_!\mathbb{Z}/n(N'')[2N'']) & \longrightarrow & \hom_{S'}\left(\mathsf{R}^{2d'} f'_!\mathbb{Z}/n, \mathbb{Z}/n\right) \\
\downarrow{\scriptstyle \mathsf{R}g'_!(N')} & & \downarrow{\scriptstyle \mathrm{tr}_{g'} \circ \mathsf{R}^{2N'} g'_!} \\
\hom_S(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathsf{R}g'_!\mathbb{Z}/n(N)[2N]) & \longrightarrow & \hom_S\left(\mathsf{R}^{2d} f_!\mathbb{Z}/n, \mathbb{Z}/n\right)
\end{array}$$

while the analogous and easily verified formula $\mathrm{tr}_{f,\mathscr{K}} = \mathrm{tr}_{g'} \mathrm{tr}_{f',\mathscr{K}}$ ensures that the image of $\mathrm{tr}_{f',\mathscr{K}}$ is $\mathrm{tr}_{f,\mathscr{K}}$. The assertion follows.

Lemma 2.3.6. *If Y is a divisor in X, the class of 2.3.2 coincides with the class of 2.1.2.*

By semipurity (2.2.6(i) and 2.2.8(i)), the problem is local on Y; using 2.3.5 to replace S by a suitable S', this allows us to suppose that $N = 1$. Using semipurity once more (2.2.6(ii) and 2.2.8(ii)), it suffices to prove 2.3.6 over the generic points of S. An étale localization then reduces us to assuming that S is the spectrum of an algebraically closed field k. Since the classes of 2.3.2 and 2.1.2 are each additive in Y, this reduces us to the case where Y is a closed point on a smooth curve over k. The fact that the characteristic property 2.3.2 is satisfied is then the fundamental compatibility 2.1.5.

Lemma 2.3.7. *The formation of $\mathrm{cl}(Y, \mathscr{K})$ is compatible with every base change S'/S.*

This follows from the same assertion for the trace morphisms.

Theorem 2.3.8. *(i)* $\mathrm{cl}(Y, \mathscr{K})$ *depends only on* $Y \subset X$ *and on the lengths of* $\mathscr{K}$ *at the generic points of* Y*. This class is additive in* $\mathscr{K}$*. For* $\mathscr{K} = \mathscr{O}_Y$ *and* Y *a local complete intersection, it induces the local class of 2.2.2.*

(ii) Let there be a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{u} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

with X' *smooth over* S'*. If* $u^{-1}(Y)$ *still has fiberwise codimension* $\geqslant c$*, then*

$$\mathrm{cl}\left(u^{-1}(Y), \mathsf{L}u^*\mathscr{K}\right) = u^* \mathrm{cl}(Y, \mathscr{K}).$$

(iii) Let $Y' \subset X$ *have fiberwise codimension* $\geqslant c'$*, let* $\mathscr{K}'$ *be a weight on* Y'*, and suppose that* $Y \cap Y'$ *has fiberwise codimension* $\geqslant c + c'$*. Then*

$$\mathrm{cl}\left(Y \cap Y', \mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{K}'\right) = \mathrm{cl}(Y, \mathscr{K}) \smile \mathrm{cl}(Y', \mathscr{K}').$$

(A) Given $Y \subset X/S$ as above, it follows from semipurity (2.2.6, 2.2.8) that to verify that two classes in $\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c))$ coincide, it suffices to verify this locally at the generic points of Y, or even after every base change $s \to S$ (s a geometric generic point of S), at the generic points of Y.

(B) Proof of (ii) for $S = S'$ and u a closed immersion By localization (A), we may suppose that X' is the inverse image of the section 0 under a smooth morphism $v : X \to \mathbb{A}^{N'}_S$

$$\begin{array}{ccccc} X' & \xrightarrow{u} & X & & \\ \downarrow & & \downarrow{\scriptstyle v} & & \\ S & \xrightarrow{0} & \mathbb{A}^{N'}_S & \xrightarrow{\pi} & S \end{array}$$

Apply 2.3.5 to $g = \pi v$ and 2.3.7 to the base change $S \to \mathbb{A}^{N'}_S$.

(C) In a product situation: $X = X' \times_S X''$, $Y = Y' \times_S Y''$, $\mathscr{K} = \mathscr{K} \overset{\mathsf{L}}{\otimes}_S \mathscr{K}''$, one has $\mathrm{cl}(Y, \mathscr{K}) = \mathrm{cl}(Y, \mathscr{K}') \smile \mathrm{cl}(Y'', \mathscr{K}'')$ (external cup product, i.e. $\mathrm{pr}_1^* \mathrm{cl}(Y', \mathscr{K}') \smile \mathrm{pr}_2^* \mathrm{cl}(Y'', \mathscr{K}'')$).

The functorial properties of tr_g [2, XVIII.2.12] imply the commutativity of

$$\begin{array}{ccccc} \mathrm{H}^{2c'}_{X'} \otimes \mathrm{H}^{2c''}_{X''} & \longrightarrow & \hom_S\left(\mathsf{R}f'_!\mathbb{Z}/n(d')[2d'], \mathsf{R}g'_!\mathbb{Z}/n(N')[2N']\right) \otimes \cdots & \longrightarrow & \hom_S\left(\mathsf{R}^{2d'} f'_!\mathbb{Z}/n, \mathbb{Z}/n\right) \otimes \cdots \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{H}^{2c}_X & \longrightarrow & \hom_S\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathsf{R}g_!\mathbb{Z}/n(N)[2N]\right) & \longrightarrow & \hom_S\left(\mathsf{R}^{2d} f_!\mathbb{Z}/n, \mathbb{Z}/n\right) \end{array}$$

and one verifies that tr_f is the image of $\mathrm{tr}_{f'} \otimes \mathrm{tr}_{f''}$, whence the assertion.

Proof of (ii) (2.3.7) and a preliminary base change reduce us to assuming that $S = S'$. Factor u through its graph:

$$X' \xrightarrow{(\mathrm{id}, u)} X' \times X \xrightarrow{\mathrm{pr}_2} X.$$

This leads us to treat pr_2 and (id, u) separately: the former is covered by (C) with $X' = X'$, $\mathscr{K}' = \mathscr{O}_{x'}$ (with $\mathrm{cl}(Y') = 1 \in \mathrm{H}^0(X, \mathbb{Z}/n)$), and the latter by (B).

Proof of (iii) Let $\delta : X \to X \times_S X$ be the diagonal inclusion. We use the formula $\delta^*(Y_1 \times_S Y_2) = Y_1 \cap Y_2$ to reduce to (B) and (C).

Proof of (i) The base change $S_{\mathrm{red}} \to S$ replaces X by X_{red}; this reduces us to assuming X and S reduced. Localization near the generic points of Y then reduces us to the case where Y_{red} is irreducible and the complete intersection of c divisors D_i. The same localization, together with the definition of tr in the quasi-finite case, then shows that $\mathrm{cl}(Y, \mathscr{K}) = \lg \mathrm{cl}(Y_{\mathrm{red}}, \mathscr{O}_{Y_{\mathrm{red}}})$, where lg is the length of $\mathscr{K}$ at the generic point of Y. By (iv), one therefore has

$$\mathrm{cl}(Y, \mathscr{K}) = \lg \prod_i \mathrm{cl}(D_i, \mathscr{O}_{D_i})$$

and the conclusion follows from 2.3.6.

2.3.9 Remark

For S the spectrum of a field, with the class of a cycle defined as in 2.2.10, assertion (iii) says that if two cycles Y' and Y'' intersect with the correct dimension, then

$$\mathrm{cl}(Y' \cap Y'') = \mathrm{cl}(Y') \smile \mathrm{cl}(Y''),$$

provided that the multiplicities of the components of $Y' \cap Y''$ are computed as alternating sums of Tor.

2.3.10 Remark

Let $X/\operatorname{Spec}(k)$, with k algebraically closed. If two cycles of codimension c in X are algebraically equivalent, the existence of the relative theory above implies that the images in $\mathrm{H}^{2c}(X, \mathbb{Z}/n(c))$ of their classes are the fibers, at two points of a suitable connected S, of a section over S of the constant sheaf $\mathrm{H}^{2c}(X, \mathbb{Z}/n(c))$ (the sheaf $\mathsf{R}^{2c} f_* \mathbb{Z}/n(c)$ for $f = \mathrm{pr}_2 : X \times S \to S$): two algebraically equivalent cycles have the same class in $\mathrm{H}^{2c}(X, \mathbb{Z}/n(c))$.

3 Application: the Lefschetz trace formula in the proper smooth case

3.1

Let X and Y be proper smooth algebraic varieties over an algebraically closed field k, purely of dimensions N and M. Via the Künneth isomorphism $\mathrm{H}^\bullet(X \times Y) = \mathrm{H}^\bullet(X) \otimes \mathrm{H}^\bullet(Y)$, the trace morphism $\mathrm{H}^\bullet(X \times Y)(M) \to \mathrm{H}^\bullet(X)$ is simply $\mathrm{H}^\bullet(X) \otimes$ (the trace morphism $\mathrm{H}^\bullet(Y)(M) \to \mathbb{Q}_\ell$). Since the morphism is homogeneous of even degree, there is no sign problem. We denote it by $\int_Y$.

3.2

A class $\eta \in \mathrm{H}^{2q}(X \times Y)(q)$ defines a morphism of degree $2(q - N)$, $\eta_{XY} : \mathrm{H}^\bullet(Y)(N - q) \to \mathrm{H}^\bullet(X)$, by the formula

$$\beta \mapsto \int_Y \eta \cdot \mathrm{pr}_2^* \beta.$$

Proposition 3.3. *Let p and q be two integers, with $p+q = N+M$, and let $\varepsilon \in \mathrm{H}^{2p}(X\times Y)(p)$, $\eta \in \mathrm{H}^{2q}(X\times Y)(p)$. With the trace of $\eta_{XY}\varepsilon_{YX} : \mathrm{H}^\bullet(X) \to \mathrm{H}^\bullet(X)$ understood in the sense of 1.3, one has*

$$\mathrm{tr}_{X\times Y}(\eta\cdot\varepsilon) = \mathrm{tr}\left(\eta_{XY}\varepsilon_{YX}, \mathrm{H}^\bullet(X)\right).$$

To exorcise the signs, it is helpful to retain from the grading of $\mathrm{H}^\bullet$ only the underlying $\mathbb{Z}/2$-grading. To avoid carrying Tate twists along, also fix an isomorphism $\mathbb{Q}_\ell(1) = \mathbb{Q}_\ell$. Let α_Y be the homomorphism $\mathrm{H}^\bullet(Y) \to \mathrm{H}^\bullet(Y)^\vee$ (an isomorphism) that makes the diagram

$$\begin{array}{ccccc} \mathrm{H}^\bullet(Y)\otimes\mathrm{H}^\bullet(Y) & \longrightarrow & \mathrm{H}^\bullet(Y) & \xrightarrow{\mathrm{tr}} & \mathbb{Q}_\ell \\ \downarrow{\scriptstyle \alpha_Y\otimes 1} & & & & \| \\ \mathrm{H}^\bullet(Y)^\vee\otimes\mathrm{H}^\bullet(Y) & & \xrightarrow{(1.3.1)} & & \mathbb{Q}_\ell. \end{array}$$

commute. The map η_{XY} is the image of η under

$$\mathrm{H}^\bullet(X\times Y) = \mathrm{H}^\bullet(X)\otimes\mathrm{H}^\bullet(Y) \xrightarrow{\alpha_Y} \mathrm{H}^\bullet(X)\otimes\mathrm{H}^\bullet(Y)^\vee \overset{(1.3.2)}{=\!=} \hom\left(\mathrm{H}^\bullet(X), \mathrm{H}^\bullet(Y)\right),$$

and likewise for ε. Definition 1.3.4 of the trace then reduces 3.3 to the formula $\mathrm{tr}_{X\times Y} = \mathrm{tr}_X \otimes \mathrm{tr}_Y$.

3.4 Remark

The proof of 3.4 does not use the fact that α_X and α_Y are isomorphisms. It is valid without assuming X and Y smooth. Outside the smooth case, however, it is difficult to find *cohomology* classes to which one would want to apply 3.3.

3.5 Remark

If ε, η are the cohomology classes of algebraic cycles A and B on $X\times Y$, of codimensions p and q, and $A\cap B$ has dimension 0, then $\mathrm{tr}_{X\times Y}(\eta\varepsilon)$ is the intersection multiplicity $A\cdot B$, computed by an alternating sum of Tor (2.3.9).

Proposition 3.6. *For $q = M$ and $\eta = \mathrm{cl}(B)$ the class of a subscheme $B \subset X\times Y$, finite over X, the map η_{XY} is the composite*

$$\eta_{XY} : \mathrm{H}^\bullet(Y) \xrightarrow{\mathrm{pr}_2^*} \mathrm{H}^\bullet(B) \xrightarrow{\mathrm{tr}_{B/X}} \mathrm{H}^\bullet(X).$$

Indeed, by 3.2 and (2.3.3.1) for $\mathscr{K} = \mathscr{O}_B$, one has

$$\eta_{XY}(\beta) = \mathrm{tr}_{X\times Y/X}\left(\mathrm{cl}(B)\cdot \mathrm{pr}_2^*\,\beta\right) = \mathrm{tr}_{B/X}\left(\mathrm{pr}_2^*\,\beta\right).$$

The most important case is that in which B is the graph of a map $f : X \to Y$. In this case, 3.6 shows that $\eta_{XY} = f^*$.

Corollary 3.7. *Let f be an endomorphism of X proper and smooth over an algebraically closed field k. Suppose that the fixed points of f are isolated. Then the trace $\sum(-1)^i \mathrm{tr}(f^*, \mathrm{H}^i(X))$ is the number of fixed points of f, each counted with its multiplicity.*

In 3.5, take $A =$ the graph of the identity and $B =$ the graph of f, and apply 3.3, 3.6.

Corollary 3.8. *Let X be a curve over an algebraically closed k, obtained by extension of scalars from X_0 over $\mathbb{F}_q$, and let f be the Frobenius morphism. Then $\sum(-1)^i \operatorname{tr}(f^*, \mathrm{H}^\bullet_c(X))$ is the number of fixed points of f.*

One can replace X by X_{red}, and hence suppose that X is reduced. Let U be the open subset of X on which X is smooth of dimension 1, and let $\bar{U}$ be the smooth projective completion of U. The long exact cohomology sequences for (X, U) and $(\bar{U}, U)$ give

$$\operatorname{tr}(f^*, \mathrm{H}^\bullet_c(X)) = \operatorname{tr}(f^*, \mathrm{H}^\bullet_c(U)) + \operatorname{tr}(f^*, \mathrm{H}^\bullet_c(X \smallsetminus U))$$

and

$$\operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(\bar{U})\right) = \operatorname{tr}(f^*, \mathrm{H}^\bullet_c(U)) + \operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(\bar{U} \smallsetminus U)\right).$$

The same formulas hold for the numbers of fixed points. Since formula 3.8 is clear for a scheme of dimension 0, this reduces us to proving 3.8 for $\bar{U}$. This case follows from 3.7. The fact that $df = 0$ guarantees that all fixed points have multiplicity one.

3.9 Remark

For X of dimension $\leqslant 1$ over an algebraically closed k, and $f : X \to X$ such that $df = 0$, it is not difficult to verify that 3.8 is still valid.

Chapter 5

Duality

This exposé contains several theorems and compatibility results, all related to Poincaré duality. In Section 1, we give the local biduality theorem in dimension 1 [21, I.5.1], together with some computations of duals. In Section 2, we give a very economical proof of Poincaré duality for curves, taught to me by M. Artin. In Section 3, we prove a compatibility linking two definitions of the pairing giving rise to Poincaré duality for curves: one by cup product, the other by autoduality of the Jacobian. Finally, in Section 4, we prove the compatibility announced in the title.

Throughout the exposé, all schemes are noetherian and separated, and n is an integer invertible on every scheme under consideration.

1 Local biduality in dimension 1

1.1

Let S be a regular scheme, purely of dimension 1. We shall show that the complex with value $\mathbb{Z}/n$ concentrated in degree 0 is dualizing; that is, for $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}_c^b(S, \mathbb{Z}/n)$ (where $(-)_c$ means constructible), if we put $D\mathscr{K} = \mathsf{R}\mathscr{H}om(\mathscr{K}, \mathbb{Z}/n)$, then $D\mathscr{K}$ is again constructible with bounded cohomology, and the canonical morphism α from $\mathscr{K}$ to $DD\mathscr{K}$ is an isomorphism.

For $\mathscr{K}$ in D^- and arbitrary $\mathscr{L}$, we have

$$\hom(\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{L}, \mathbb{Z}/n) = \hom(\mathscr{L}, \mathscr{H}om(\mathscr{K}, \mathbb{Z}/n)).$$

The morphism from $\mathscr{K}$ to $DD\mathscr{K}$ is defined when $D\mathscr{K}$ lies in D^-: if $\beta : D\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{K} \to \mathbb{Z}/n$ is the canonical pairing, it is the morphism defined by the pairing $\mathscr{K} \overset{\mathsf{L}}{\otimes} D\mathscr{K} = D\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{K} \xrightarrow{\beta} \mathbb{Z}/n$.

1.2

Let $f : X \to S$ be a separated morphism of finite type. Put $\mathscr{K}_X = \mathsf{R}f^!\mathbb{Z}/n$. For $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^-(X, \mathbb{Z}/n)$, put $D\mathscr{K} = \mathsf{R}\mathscr{H}om(\mathscr{K}, \mathscr{K}_X)$. The adjunction between $\mathsf{R}f_!$ and $\mathsf{R}f^!$ gives

$$\mathsf{R}f_*\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathsf{R}f_*D\mathscr{K} \longrightarrow \mathsf{R}f_!(\mathscr{K} \overset{\mathsf{L}}{\otimes} D\mathscr{K}) \longrightarrow \mathsf{R}f_!\mathsf{R}f^!\mathbb{Z}/n \longrightarrow \mathbb{Z}/n.$$

The symmetry of this description shows that, when f is proper, the diagram

$$\begin{array}{ccc} \mathsf{R}f_*\mathscr{K} & \longrightarrow & DD\mathsf{R}f_*\mathscr{K} \\ \downarrow & & \downarrow\wr \\ \mathsf{R}f_*DD\mathscr{K} & \xrightarrow{\sim} & D\mathsf{R}f_*D\mathscr{K} \end{array} \tag{1.2.1}$$

is commutative.

If f is the inclusion of a closed point, we know that $\mathsf{R}f^!\mathbb{Z}/n = \mathbb{Z}/n(-1)[-2]$—that is, up to a twist and a shift, $\mathbb{Z}/n$. Over the spectrum of a field this complex is dualizing (Pontryagin duality for $\mathbb{Z}/n$-modules). By (1.2.1), we therefore have $\mathscr{K} \xrightarrow{\sim} DD\mathscr{K}$ for $\mathscr{K}$ of the form $\mathsf{R}f_!\mathscr{L}$, and hence whenever the support of $\underline{\mathrm{H}}^\bullet(\mathscr{K})$ is finite.

Theorem 1.3. *Let $j : U \hookrightarrow S$ be a dense open subscheme of S, and let $\mathscr{F}$ be a locally constant constructible sheaf of $\mathbb{Z}/n$-modules on U. Then $\mathsf{D}j_*\mathscr{F} = j_*D\mathscr{F}$; that is, $\mathscr{H}om(j_*\mathscr{F}, \mathbb{Z}/n) = j_*\mathscr{H}om(\mathscr{F}, \mathbb{Z}/n)$ and $\mathscr{E}xt^i(j_*\mathscr{F}, \mathbb{Z}/n) = 0$ for $i > 0$.*

On U, $\mathscr{E}xt^i(j_*\mathscr{F}, \mathbb{Z}/n) = 0$ for $i > 0$, since $\mathscr{F}$ is locally constant (and $\mathbb{Z}/n$ is an injective $\mathbb{Z}/n$-module), while for $i = 0$ it is the dual $\mathscr{F}^\vee$ of $\mathscr{F}$. One checks that $\hom(j_*\mathscr{F}, \mathbb{Z}/n) = j_*\mathscr{F}^\vee$, and it remains to verify the vanishing of the $\mathscr{E}xt^i$ for $i > 0$ at the points of $S \smallsetminus U$. The problem is local at these points. We are thus reduced to assuming that S is a strictly local trait and that U consists of its generic point η. Let $I = \mathrm{Gal}(\bar\eta/\eta)$. The sheaf $\mathscr{F}$ is identified with the Galois module $\mathscr{F}_{\bar\eta}$, and the special fiber of $j_*\mathscr{F}$ with $\mathscr{F}_{\bar\eta}^I$.

Let i be the inclusion of the closed point s, and apply D to the exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & j_!\mathscr{F} & \longrightarrow & j_*\mathscr{F} & \longrightarrow & i_*\mathscr{F}_{\bar\eta}^I & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & j_!\mathscr{F}_{\bar\eta}^I & \longrightarrow & \mathscr{F}_{\bar\eta}^I & \longrightarrow & i_*\mathscr{F}_{\bar\eta}^I & \longrightarrow & 0. \end{array}$$

We obtain a morphism of triangles

$$\begin{array}{ccccc} i_*\left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1)[-2] & \longrightarrow & Dj_*\mathscr{F} & \longrightarrow & \mathsf{R}j_*\mathscr{F}^\vee \\ \| & & \downarrow & & \downarrow \\ i_*\left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1)[-2] & \longrightarrow & \left(\mathscr{F}_{\bar\eta}^I\right)^\vee & \longrightarrow & \mathsf{R}j_*\mathscr{F}_{\bar\eta}^I. \end{array}$$

The long exact sequence arising from the first row gives the vanishing of the $\mathscr{E}xt^i$ for $i > 2$, and, on taking the fiber at s, we find

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \left(\underline{\mathrm{H}}^1(Dj_*\mathscr{F}\right)_s & \longrightarrow & \mathrm{H}^1\left(I, \mathscr{F}_{\bar\eta}^\vee\right) & \xrightarrow{\partial} & \left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1) & \longrightarrow & \left(\underline{\mathrm{H}}^2 Dj_*\mathscr{F}\right)_s \longrightarrow 0 \\ & & & & \downarrow & & \| & & \\ & & 0 & \longrightarrow & \mathrm{H}^1\left(I, (\mathscr{F}_{\bar\eta}^I)^\vee\right) & \xrightarrow{\partial} & \left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1) & \longrightarrow & 0. \end{array}$$

Since $\left(\mathscr{F}_{\bar\eta}^I\right)^\vee = \left(\mathscr{F}_{\bar\eta}^\vee\right)_I$, putting $M = \mathscr{F}_{\bar\eta}^\vee$, it remains only to check that

$$\mathrm{H}^1(I, M) \xrightarrow{\sim} \mathrm{H}^1(I, M_I).$$

If p is the residual characteristic exponent, then I is an extension of a group isomorphic to $\widehat{\mathbb{Z}}_{p'} = \varprojlim_{(m,p)=1} \mathbb{Z}/m$ by a p-group P. Since the order of P is prime to the order of M, we have $\mathrm{H}^1(P, M) = 0$ for $i > 0$ and $(M_I)^P = (M^P)_I$. This allows us to replace M by M^P and I by I/P. Finally, there is a functorial isomorphism $\mathrm{H}^1(\widehat{\mathbb{Z}}_{p'}, M) \sim$ (coinvariants of $\widehat{\mathbb{Z}}_{p'}$ in M), which proves the theorem.

Theorem 1.4. *For $\mathscr{K} \in \mathrm{Ob}\, \mathsf{D}_c^b(S, \mathbb{Z}/n)$, one has $\mathscr{K} \xrightarrow{\sim} DD\mathscr{K}$.*

By dévissage, we reduce to the case in which $\mathscr{K}$ is a constructible sheaf $\mathscr{F}$ concentrated in degree 0, and $\mathscr{F}$ either has finite support (1.2) or is of the form $j_*\mathscr{F}_1$ as in 1.3. In the second case, 1.3 reduces us to local biduality for $\mathscr{F}$ locally constant on U.

2 Poincaré duality for curves, after M. Artin

2.1

Let X be a smooth projective curve over an algebraically closed field k. Put $\mathscr{K}_X = \mathbb{Z}/n(1)[2]$ and, for $\mathscr{K} \in \mathrm{Ob}\, \mathsf{D}_c^b(X, \mathbb{Z}/n)$, $D\mathscr{K} = \mathsf{R}\mathscr{H}om(\mathscr{K}, \mathscr{K}_X)$. For M a $\mathbb{Z}/n$-module, also put $DM = \hom(M, \mathbb{Z}/n)$, and similarly for complexes of modules. The trace morphism $\mathrm{H}^0(X, \mathscr{K}_X) \to \mathbb{Z}/n$, or $\mathsf{R}\Gamma(X, \mathscr{K}_X) \to \mathbb{Z}/n$, defines a pairing $\mathsf{R}\Gamma(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(X, D\mathscr{K}) \to \mathbb{Z}/n$. Poincaré duality between the cohomology and compactly supported cohomology of an open subscheme $j : U \hookrightarrow X$ of the curve X says that, for $\mathscr{K} = j_!\mathbb{Z}/n$, this pairing identifies each factor with the dual of the other.

I give below a proof, communicated to me by M. Artin, that for $\mathscr{K} \in \mathrm{Ob}\, \mathsf{D}_c^b(X, \mathbb{Z}/n)$ this pairing is always perfect; that is, it defines an isomorphism

$$\mathsf{R}\Gamma(X, \mathscr{K}) \xrightarrow{\sim} D\mathsf{R}\Gamma(X, D\mathscr{K}). \tag{2.1.1}$$

For every constructible sheaf $\mathscr{F}$, put

$$'\mathrm{H}^i(X, \mathscr{F}) = \text{the } \mathbb{Z}/n\text{-dual of } \mathrm{H}^{-i}(X, D\mathscr{F}).$$

The assertion that (2.1.1) is an isomorphism is equivalent to the following theorem.

Theorem 2.2. *For a constructible sheaf $\mathscr{F}$ of $\mathbb{Z}/n$-modules, one has*

$$\mathrm{H}^i(X, \mathscr{F}) \xrightarrow{\sim} {}'\mathrm{H}^i(X, \mathscr{F}). \tag{2.2.1}$$

Lemma 2.3. *Let $f : X \to Y$ be a generically étale morphism between smooth curves over k. Then $\mathscr{K}_X = f^*\mathscr{K}_Y = \mathsf{R}f^!\mathscr{K}_Y$, with tr_f as the adjunction morphism $\mathsf{R}f_!\mathsf{R}f^!\mathscr{K}_Y \to \mathscr{K}_Y$.*

We reduce to checking that, for f finite, $f_*\mathsf{R}f^!\mathscr{K}_Y = \mathsf{R}\mathscr{H}om(f_*\mathbb{Z}/n, \mathscr{K}_Y)$ is $f_*\mathbb{Z}/n$, with tr_f as the adjunction morphism. The vanishing of the $\mathscr{E}xt^i$ for $i > 0$ follows from 1.3, and the assertion follows.

More explicitly, $\mathsf{R}f^!$ is the derived functor of the functor $f^! : f^!\mathscr{F}(U) = \hom(f_!\mathbb{Z}_U, \mathscr{F})$, and 1.3 implies the vanishing of $\mathsf{R}^i f^!\mathbb{Z}/n$ for $i > 0$.

Corollary 2.4. *Let $f : X' \to X$ be a generically étale morphism of smooth projective curves over k. Then $'\mathrm{H}^i(X, f_*\mathscr{F}) =' \mathrm{H}^i(X', \mathscr{F})$; this isomorphism is functorial, and the diagram*

$$\begin{array}{ccc} \mathrm{H}^i(X, f_*\mathscr{F}) & \longrightarrow & {}'\mathrm{H}^i(X, f_*\mathscr{F}) \\ \| & & \| \\ \mathrm{H}^i(X', \mathscr{F}) & \longrightarrow & {}'\mathrm{H}^i(X', \mathscr{F}) \end{array}$$

is commutative.

The isomorphism is given by 1.2: $Df_*\mathscr{F} = f_*D\mathscr{F}$.

2.5

We prove 2.2. If $\mathscr{F}$ is $\mathbb{Z}/n$ at one point, extended by 0, then $\mathrm{H}^\bullet(D\mathscr{F}) = \mathrm{Ext}^\bullet(\mathscr{F}, \mathscr{K}_X) = \mathbb{Z}/n$ in degree 0, and in the pairing $\mathrm{H}^\bullet(\mathscr{F}) \otimes \mathrm{H}^\bullet(D\mathscr{F}) \to \mathbb{Z}/n$, one has $1 \otimes 1 \mapsto 1$ (Cycle, 2.1.5): the duality is perfect. The case in which $\mathscr{F}$ has finite support is treated in the same way.

For arbitrary constructible $\mathscr{F}$, let $\mathscr{A}$ be the subsheaf of its sections with finite support, and let $\mathscr{G} = \mathscr{F}/\mathscr{A}$. The exact sequence

$$0 \longrightarrow \mathscr{A} \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow 0 \tag{2.5.1}$$

gives a diagram

$$\begin{array}{ccccccccc} & & & & 0 & \longrightarrow & \mathrm{H}^0(X,\mathscr{A}) & \longrightarrow & \mathrm{H}^0(X,\mathscr{F}) \longrightarrow \mathrm{H}^0(X,\mathscr{G}) \longrightarrow 0 \\ & & & & & & \downarrow\wr & & \downarrow \qquad \downarrow \\ 0 \to {}' \mathrm{H}^{-1}(X,\mathscr{F}) & \to & {}' \mathrm{H}^{-1}(X,\mathscr{F}) & \to & {}' \mathrm{H}^0(X,\mathscr{A}) & \to & {}' \mathrm{H}^0(X,\mathscr{F}) & \to & {}' \mathrm{H}^0(X,\mathscr{F}) \to 0 \end{array} \tag{2.5.2}$$

and compatible isomorphisms $\mathrm{H}^i(X,\mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X,\mathscr{G})$, ${}' \mathrm{H}^i(X,\mathscr{F}) \xrightarrow{\sim}{}' \mathrm{H}^i(X,\mathscr{G})$ for $i \neq 0, -1$. If $j : U \to X$ is an open subscheme on which $\mathscr{F}$ is locally constant, there is an exact sequence

$$0 \longrightarrow \mathscr{G} \longrightarrow j_*j^*\mathscr{F} \longrightarrow \mathscr{B} \longrightarrow 0 \tag{2.5.3}$$

with $\mathscr{B}$ of finite support. Since ${}' \mathrm{H}^i(X, j_*j^*\mathscr{F}) = D\,\mathrm{H}^{2-i}(X, j_*(j^*\mathscr{F})^\vee(1))$ by 1.3, one has ${}' \mathrm{H}^i = 0$ for $i < 0$; then (2.5.2) and the long exact sequence defined by (2.5.3) show that ${}' \mathrm{H}^i$ always vanishes for $i < 0$.

If $\mathscr{F} = \mathbb{Z}/n$, the theorem is true for $i = 0$ and 2. This expresses the fact that, for connected X, the trace morphism $\mathrm{H}^2(X, \boldsymbol{\mu}_n) \xrightarrow{\sim} \mathbb{Z}/n$ is an isomorphism. By 2.4, the theorem remains true for $i = 0$ and $\mathscr{F} = f_*\mathbb{Z}/n$.

For arbitrary $\mathscr{F}$ again, with $\mathscr{G}$ as above, there exists $f : X' \to X$ as in 2.4 such that $\mathscr{G}$ embeds into $f_*\mathbb{Z}/n$:

$$0 \longrightarrow \mathscr{G} \longrightarrow f_*\mathbb{Z}/n \longrightarrow \mathscr{Q} \longrightarrow 0.$$

After replacing X' by an X''/X', we may assume that $\mathrm{H}^i(X,\mathscr{F}) \to \mathrm{H}^i(X, f_*\mathbb{Z}/n) = \mathrm{H}^i(X', \mathbb{Z}/n)$ is zero for $i > 0$. Consider then

$$\begin{array}{ccccccccccc} 0 \to \mathrm{H}^0(X,\mathscr{G}) & \to & \mathrm{H}^0(S',\mathbb{Z}/n) & \to & \mathrm{H}^0(X,\mathscr{Q}) & \to & \mathrm{H}^1(X,\mathscr{G}) & \xrightarrow{0} & \mathrm{H}^1(X',\mathbb{Z}/n) & \to & \mathrm{H}^1(X,\mathscr{Q}) \to \cdots \\ \downarrow & & \downarrow\wr & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 \to {}' \mathrm{H}^0(X,\mathscr{G}) & \to & {}' \mathrm{H}^0(X',\mathbb{Z}/n) & \to & {}' \mathrm{H}^0(X,\mathscr{Q}) & \to & {}' \mathrm{H}^1(X,\mathscr{G}) & \to & {}' \mathrm{H}^1(X',\mathbb{Z}/n) & \to & {}' \mathrm{H}^1(X,\mathscr{Q}) \to \cdots \end{array}$$

The isomorphism at $\mathrm{H}^0(X')$ gives $\mathrm{H}^0(X,\mathscr{G}) \hookrightarrow {}' \mathrm{H}^0(X,\mathscr{G})$, and the same injectivity for $\mathscr{F}$ by (2.5.2). Applied to $\mathscr{Q}$, this gives $\mathrm{H}^1(X,\mathscr{G}) \hookrightarrow {}' \mathrm{H}^1(X,\mathscr{G})$. We apply this to $\mathrm{H}^1(X', \mathbb{Z}/n)$. This group has the same order as ${}' \mathrm{H}^1(X', \mathbb{Z}/n) = D(\mathrm{H}^1(X', \mathbb{Z}/n)(1))$, so we obtain an isomorphism, and $\mathrm{H}^1(X,\mathscr{G}) \xrightarrow{\sim}{}' \mathrm{H}^1(X,\mathscr{G})$. The same holds for $\mathscr{F}$. For $i = 2$, we already know that $\mathrm{H}^2(X', \mathbb{Z}/n) \xrightarrow{\sim}{}' \mathrm{H}^2(X', \mathbb{Z}/n)$, and proceed in the same way, then stop for lack of combatants.

3 Poincaré duality for curves

In this section we prove the compatibility (Arcata 6.2.2.3).

3.1

We use the notation of (Arcata 6.2.3): $\bar{X}$ is a smooth, connected, projective curve over an algebraically closed field k, D is a reduced divisor on $\bar{X}$, and 0 is a point of $X = \bar{X} \smallsetminus D$,

$$X \overset{j}{\hookrightarrow} \bar{X} \overset{i}{\hookleftarrow} D,$$

and $\mathrm{Pic}_D(\bar{X}) = \mathrm{H}^1(\bar{X}, {}_D\mathbb{G}_m)$, where ${}_D\mathbb{G}_m$ is the sheaf of sections of $\mathbb{G}_m$ congruent to 1 mod D. Recall that $\mathrm{H}^1(X, \boldsymbol{\mu}_n) = \mathrm{Pic}^0_D(\bar{X})_n$, and that $x \to \mathscr{O}(x)$ defines a canonical map $f : X \to \mathrm{Pic}_D(\bar{X})$. Put $f_0(x) = f(x) - f(0) : X \to \mathrm{Pic}^0_D(\bar{X})$.

Again write j for the inclusion of $X \times X$ in $X \times \bar{X}$. The diagonal Δ of $X \times X$ is closed in $X \times \bar{X}$; it defines a class in $\mathrm{H}^2_\Delta(X \times X, \boldsymbol{\mu}_n) = \mathrm{H}^2_\Delta(X \times \bar{X}, j_!\boldsymbol{\mu}_n)$. Let c denote its image in $\mathrm{H}^2(X \times \bar{X}, j_!\boldsymbol{\mu}_n)$.

The Künneth formula gives

$$\mathrm{H}^\bullet(X \times \bar{X}, j_!\boldsymbol{\mu}_n) = \mathrm{H}^\bullet(X, \mathbb{Z}/n) \otimes \mathrm{H}^\bullet(\bar{X}, j_!\boldsymbol{\mu}_n) = \mathrm{H}^\bullet(X, \mathbb{Z}/n) \otimes \mathrm{H}^\bullet_c(X, \boldsymbol{\mu}_n).$$

We shall compute the $(1,1)$-component of c,

$$c^{1,1} \in \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) = \mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) = \mathrm{H}^1\left(X, \mathrm{Pic}^0_D(\bar{X})_n\right).$$

Consider the exact sequence

$$0 \longrightarrow \mathrm{Pic}^0_D(\bar{X})_n \longrightarrow \mathrm{Pic}^0_D(\bar{X}) \overset{n}{\longrightarrow} \mathrm{Pic}^0_D(\bar{X}) \longrightarrow 0$$

This makes $\mathrm{Pic}^0_D(\bar{X})$ a torsor under $\mathrm{Pic}^0_D(\bar{X})_n$ over $\mathrm{Pic}^0_D(\bar{X})$. Let u be the class, in $\mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) = \mathrm{H}^1(X, \mathrm{Pic}^0_D(\bar{X})_n)$, of its inverse image under f_0.

Proposition 3.2. *With the preceding notation, $c^{1,1} = -u$.*

Let e be the difference between the images in $\mathrm{H}^2(X \times \bar{X}, j_!\boldsymbol{\mu}_n)$ of the cohomology classes of Δ and $X \times \{0\}$. Since the latter is of Künneth type $(0,2)$, one has $c^{1,1} = e^{1,1}$. Consider the Leray spectral sequence for $\mathrm{pr}_1 : X \times \bar{X} \to X$ and the sheaf $j_!\boldsymbol{\mu}_n$:

$$E_2^{pq} = \mathrm{H}^p(X, \mathrm{R}^q\mathrm{pr}_{1*} j_!\boldsymbol{\mu}_n) = \mathrm{H}^p(X, \mathrm{H}^q_c(X, \boldsymbol{\mu}_n)) \Rightarrow \mathrm{H}^{p+q}(X \times \bar{X}, j_!\boldsymbol{\mu}_n).$$

It degenerates: it is the tensor product of $\mathrm{H}^\bullet(X)$ with the (trivial) Leray spectral sequence for $\bar{X} \to \mathrm{Spec}(k)$ and the sheaf $j_!\boldsymbol{\mu}_n$. Fiber by fiber, the divisor $\Delta - X \times \{0\}$ has degree 0, so the image of e in E^{02} is zero. We may therefore speak of its image $\bar{e}$ in $E^{1,1}$, and we must show that $\bar{e} = -u$.

Let $\mathscr{G}$ on $X \times \bar{X}$ be the subsheaf of $\mathbb{G}_m$ consisting of local sections whose restriction to the closed subscheme $X \times D$ is 1. There is again an exact sequence

$$0 \longrightarrow j_!\boldsymbol{\mu}_n \longrightarrow \mathscr{G} \longrightarrow \mathscr{G} \longrightarrow 0 \tag{3.2.1}$$

and e is the image under ∂ of the class $e_1 \in \mathrm{H}^1(X \times \bar{X}, \mathscr{G})$ of the invertible sheaf $\mathscr{O}(\Delta - X \times \{0\})$, trivialized by 1 on $X \times D$.

Applying $\mathsf{R}\mathrm{pr}_{1*}$ to the distinguished triangle defined by (2.2.1) gives a distinguished triangle

$$\xrightarrow{\partial} \mathsf{R}\mathrm{pr}_{1*}j_!\boldsymbol{\mu}_n \longrightarrow \mathsf{R}\mathrm{pr}_{2*}\mathscr{G} \longrightarrow \mathsf{R}\mathrm{pr}_{2*}\mathscr{G} \xrightarrow{\partial} \tag{3.2.2}$$

and the diagram

$$\begin{array}{ccc} \mathrm{H}^1(X\times\bar{X},\mathscr{G}) & \xrightarrow{\partial} & \mathrm{H}^2(X\times\bar{X},j_!\boldsymbol{\mu}_n) \\ \| (1) & & \| \\ \mathrm{H}^1(X,\mathsf{R}\mathrm{pr}_{1*}\mathscr{G}) & \xrightarrow{\partial} & \mathrm{H}^2(X,\mathsf{R}\mathrm{pr}_{1*}j_!\boldsymbol{\mu}_n) \end{array}$$

is commutative. Our problem is thus to identify the image in $E^{11}=\mathrm{H}^1(X,\mathrm{H}^1_c(X,\boldsymbol{\mu}_n))$ (the image in E^{02} being zero) of the image under $\partial_{(3.2.2)}:\mathrm{H}^1(X,\mathsf{R}\mathrm{pr}_{1*}\mathscr{G})\to\mathrm{H}^2(X,\mathsf{R}\mathrm{pr}_{1*}j_!\boldsymbol{\mu}_n)$ of the class, still denoted by e_1, corresponding to e_1 under (1).

The sheaf $\mathsf{R}^1\mathrm{pr}_{1*}\mathscr{G}$ is defined by the group scheme over X obtained by pulling back the algebraic group $\mathrm{Pic}_D(\bar{X})$ over $\mathrm{Spec}(k)$, and the image of e_1 in $\mathrm{H}^0(X,\mathsf{R}^1\mathrm{pr}_*\mathscr{G})=\hom(X,\mathrm{Pic}_D(\bar{X}))$ is f_0.

Represent the triangle (3.2.2) by a short exact sequence of complexes of sheaves:

$$0\longrightarrow\mathscr{A}\longrightarrow\mathscr{B}\longrightarrow\mathscr{C}\longrightarrow 0.$$

Let $\mathscr{B}'$ be the subsheaf of

$$\tau_{\leqslant 1}\mathscr{B}:\cdots\mathscr{B}^0\to\ker(d)\to 0\cdots$$

obtained by replacing $\mathscr{B}'^1=\ker(d)$ by the inverse image, under $\ker(d)\to\underline{\mathrm{H}}^1(\mathscr{B})=\mathrm{Pic}_D(\bar{X})$ over X, of $\mathrm{Pic}^0_D(\bar{X})$ over X. Let $\mathscr{A}'=\mathscr{A}\cap\mathscr{B}'$ and let $\mathscr{C}'$ be the image of $\mathscr{B}'$. We have $\mathscr{A}'=\tau_{\leqslant 1}(\mathscr{A})$, and $\mathscr{C}'$ is obtained from $\mathscr{C}$ in the same way as $\mathscr{B}'$ from $\mathscr{B}$. Hence there is a commutative diagram of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{A} & \longrightarrow & \mathscr{B} & \longrightarrow & \mathscr{C} & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & \tau_{\leqslant 1}\mathscr{A} & \longrightarrow & \mathscr{B}' & \longrightarrow & \mathscr{C}' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^1_c(X,\boldsymbol{\mu}_n)[1] & \longrightarrow & \mathrm{Pic}^0_D(\bar{X})[1] & \longrightarrow & \mathrm{Pic}^0_D(\bar{X})[1] & \longrightarrow & 0 \end{array} \tag{3.2.3}$$

where the last row is to be viewed as an exact sequence of complexes of sheaves concentrated in degree 1 on X.

a) $e\in\mathsf{H}^2(X,\mathscr{A})$ comes from $\widetilde{e}\in\mathsf{H}^2(X,\tau_{\leqslant 1}\mathscr{A})$ (because its image in E^{02} is zero). We seek the image $\bar{e}$ of $\widetilde{e}$ in $\mathsf{H}^2(X,\mathrm{H}^1_c(X,\boldsymbol{\mu}_n)[-1])=\mathrm{H}^1(X,\mathrm{H}^1_c(X,\boldsymbol{\mu}_n))$.

b) $e_1\in\mathsf{H}^1(X,\mathscr{C})$ comes from $\widetilde{e}_1\in\mathsf{H}^1(X,\mathscr{C}')$, because its image f_0 in $\mathrm{H}^0(X,\mathrm{Pic}_D(\bar{X}))$ lies in $\mathrm{H}^0(X,\mathrm{Pic}^0_D(\bar{X}))$. We may take $\widetilde{e}=\partial\widetilde{e}_1$.

c) Thus $\bar{e}=\partial f_0$, where ∂ is defined by (3.2.2), and where we use the usual isomorphisms $\mathsf{H}^2(X,\mathscr{F}[-1])=\mathrm{H}^1(X,\mathscr{F})$. These isomorphisms anticommute with ∂, so that $\bar{e}=-\partial f_0$, for ∂ defined by the exact sequence

$$0\longrightarrow\mathrm{H}^1_c(X,\boldsymbol{\mu}_n)\longrightarrow\mathrm{Pic}^0_D(\bar{X})\longrightarrow\mathrm{Pic}^0_D(\bar{X})\longrightarrow 0$$

The conclusion follows from (Cycles, 1.1.1).

3.3

For every homomorphism $\varphi : \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \to \mathbb{Z}/n$, let $\varphi(u) \in \mathrm{H}^1(X, \mathbb{Z}/n)$ be the image of u under $\varphi : \mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) \to \mathrm{H}^1(X, \mathbb{Z}/n)$. If $x \in \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)$, the cup product $\varphi(u) \smile x$ lies in $\mathrm{H}^2_c(X, \boldsymbol{\mu}_n)$. We shall prove the following.

Proposition 3.4. $\mathrm{tr}(\varphi(u) \smile x) = \varphi(x)$.

This identity follows, after applying φ, from

$$\mathrm{tr}(u \smile x) = x \tag{3.4.1}$$

where tr now denotes the map

$$\mathrm{tr} : \mathrm{H}^2_c(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \otimes \boldsymbol{\mu}_n) \longrightarrow \mathrm{H}^1_c(X, \boldsymbol{\mu}_n),$$

so that $x \mapsto \mathrm{tr}(u \smile x)$ is the composite

$$\begin{aligned} &\mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \xrightarrow{u \otimes x} \mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \xleftarrow{\sim} \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \\ &\xrightarrow{(1)} \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \xrightarrow{\mathrm{tr} \otimes \mathrm{id}} \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \end{aligned} \tag{3.4.2}$$

where (1) is given by $a \otimes b \otimes c \mapsto (a \smile c) \otimes b$.

We express this morphism in terms of the trace morphism $\mathrm{tr} : \mathrm{H}^3_c(X \times X, \boldsymbol{\mu}_n^{\otimes 2}) \to \mathrm{H}^2_c(X, \boldsymbol{\mu}_n)$ relative to the second projection. For $x \in \mathrm{H}^3_c$, $\mathrm{tr}(x)$ is computed as follows: among its Künneth components, retain only the component of type $(2,1)$, and apply $\mathrm{tr} : \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \to \mathbb{Z}/n$.

Let u' be the image of u under the map

$$\begin{aligned} &\mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) \xleftarrow{\sim} \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) = \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1(\bar{X}, j_!\boldsymbol{\mu}_n) \\ &\longrightarrow \mathrm{H}^2(X \times \bar{X}, \mathbb{Z}/n \boxtimes j_!\boldsymbol{\mu}_n). \end{aligned}$$

In view of the relation between the Künneth formula and cup product, one has

$$\mathrm{tr}(u \smile x) = -\,\mathrm{tr}(u' \smile \mathrm{pr}_1^* x).$$

The second cup product is the one in (Cycle 1.2.4):

$$\mathrm{H}^2(X \times \bar{X}, \mathbb{Z}/n \boxtimes j_!\boldsymbol{\mu}_n) \otimes \mathrm{H}^1(\bar{X} \times \bar{X}, j_!\boldsymbol{\mu}_n \boxtimes \mathbb{Z}/n) \longrightarrow \mathrm{H}^3(\bar{X} \times \bar{X}, j_!\boldsymbol{\mu}_n \boxtimes j_!\boldsymbol{\mu}_n).$$

The minus sign comes from permuting two symbols of degree 1 in (1). Since $c \smile \mathrm{pr}_1^* x$ and $c^{1,1} \smile \mathrm{pr}_1^*$ have the same $(2,1)$-component in the Künneth decomposition, and since $u' = c^{1,1}$, one has

$$\mathrm{tr}(u \smile x) = \mathrm{tr}(c \smile \mathrm{pr}_1^* x).$$

The cup product on the right can now be interpreted as the cup product (Cycle 1.2.5) of $\mathrm{cl}(\Delta) \in \mathrm{H}^2_\Delta(X \times X, \boldsymbol{\mu}_n)$ with $\mathrm{pr}_1^* x \in \mathrm{H}^1_c(\Delta, \boldsymbol{\mu}_n)$. On Δ, one has $\mathrm{pr}_1 = \mathrm{pr}_2$, whence

$$\mathrm{tr}(c \smile \mathrm{pr}_1^* x) = \mathrm{tr}(\mathrm{cl}(\Delta) \smile \mathrm{pr}_1^* x) = \mathrm{tr}(\mathrm{cl}(\Delta) \smile \mathrm{pr}_2^* x) = \mathrm{tr}(\mathrm{cl}(\Delta)) \smile x = x.$$

4 Compatibility with SGA 4 XVIII 3.1.10.3

In [2, XVIII], overcome by discouragement, I did not verify the compatibility announced in the title. Although it turns out to be no more than smoke, I feel bound to remedy that omission here. For $f : X \to S$ separated and of finite type, the point was to verify that the adjunction morphism $\mathsf{R}f_!\mathsf{R}f^!\mathscr{L} \to \mathscr{L}$ is compatible with every étale localization $k : V \to S$. Using k and f for several arrows, as below, we must verify commutativity:

$$\begin{array}{ccc} X_V & \xrightarrow{k} & X \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} \\ V & \xrightarrow{k} & S \end{array} \qquad \begin{array}{ccc} k^*\mathsf{R}f_!\mathsf{R}f^!\mathscr{L} & \xrightarrow{k^*(\mathrm{adj})} & k^*\mathscr{L} \\ \downarrow{\scriptstyle \sim} & & \| \\ \mathsf{R}f_!\mathsf{R}f^!k^*\mathscr{L} & \xrightarrow{\mathrm{adj}} & k^*\mathscr{L}. \end{array}$$

At the level of complexes, the required commutativity is written (loc. cit.)

$$\begin{array}{ccc} k^*f_!^\bullet f_\bullet^!\mathscr{L} & \longrightarrow & k^*\mathscr{L} \\ \downarrow{\scriptstyle (1)} & & \| \\ f_!^\bullet f_\bullet^! k^*\mathscr{L} & \longrightarrow & k^*\mathscr{L}. \end{array}$$

It is checked component by component, reducing the problem to a compatibility for $\mathscr{L}$ a sheaf and for each pair of adjoint functors $(f_i^!, f_!^i)$, denoted simply by $f^!$ and $f_!$. The arrow (1) is defined using the isomorphism (2) : $k^*f_! = f_!k^*$, and a morphism (3) : $k^*f^! \to f^!k^*$ deduced from it. In loc. cit., one first introduces a morphism (4) : $k_!f_! \to f_!k_!$, deduced from (2) by the adjunction of $k^* = k^!$ and $k_! : k_!f_! \to k_!f_!k^*k_! \to k_!k^*f_!k_! \to f_!k_!$, and then deduces (3) from (4) by adjunction. Equivalently, one may deduce (3) from (2) by the adjunction of $f_!$ and $f^! : k^* \to f^!f_!k^*f^! \to f^!k^*f_!f^! \to f^!k^*$.

Writing k for k^* and omitting $\mathscr{L}$, the required commutativity is that of the outer boundary of

$$\begin{array}{ccccccc} kf_!f^! & \longrightarrow & f_!kf^! & \longrightarrow & f_!f^!f_!kf^! & \longrightarrow & f_!f^!kf_!f^! \\ \| & & & & & & \| \\ kf_!f^! & & \longleftarrow & & \longleftarrow & & f_!f^!kf_!f^! \\ \downarrow & & & & & & \downarrow \\ k & & \longleftarrow & & \longleftarrow & & f_!f^!k. \end{array}$$

In the first row, we find a homomorphism $k_!f^! \to f_!f^!kf_!f^!$, defined because $kf_!f^!$ is of the form $f_!X$ (with $X = kf^!$). It admits the adjunction morphism in the second row as a retraction, and the commutativity of the lower square is the functoriality of this adjunction morphism.

Chapter 6

Applications of the trace formula to trigonometric sums

In this exposé, I explain how the trace formula can be used to calculate or study various trigonometric sums and how, together with the Weil conjecture, it can be used to bound them.

The first two sections provide a "user's guide" to these tools. Section 3 presents, in cohomological language, Weil's results on sums in 1 variable. Sections 4 through 6 give a detailed study of Gauss and Jacobi sums – including Weil's old and recent results on Hecke characters defined by Jacobi sums. In Section 7, we study a generalization of Kloosterman sums to several variables. Finally, Section 8 gives some indications of other uses that have been, or can be, made of these methods.

Notations

0.1

We use the notation of Report, Section 1. We will often have to consider a finite extension $\mathbb{F}_{q^n} \subset \mathbb{F}$ of $\mathbb{F}_q$. A subscript 0 will denote an object over $\mathbb{F}_q$, and a subscript 1 an object over $\mathbb{F}_{q^n}$. Replacing a subscript 0 by a subscript 1 (resp. omitting the subscript) means extending scalars to $\mathbb{F}_{q^n}$ (resp. to $\mathbb{F}$).

0.2

We denote by ℓ a prime number $\neq p$. We freely use the language of $\mathbb{Q}_\ell$-sheaves, as well as that of E_λ-sheaves, where E_λ is a finite extension of $\mathbb{Q}_\ell$ (cf. Report, Section 2, especially 2.11).

0.3

Let $H^\bullet$ be a graded vector space. If T is an endomorphism of $H^\bullet$, we put (cf. Cycle, 1.3.5)

$$\mathrm{tr}(T, H^\bullet) = \sum (-1)^i \,\mathrm{tr}(T, H^i).$$

1 Principles

1.1

Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, let E_λ be a finite extension of $\mathbb{Q}_\ell$, and let $\mathscr{F}_0$ be an E_λ-sheaf on $X-0$. The trace formula says that

$$\sum_{x\in X^F} \mathrm{tr}(F_x^*,\mathscr{F}) = \sum (-1)^i \,\mathrm{tr}\left(F^*, \mathrm{H}_c^i(X,\mathscr{F})\right). \tag{1.1.1}$$

With the notation 0.3, the right-hand side is simply written $\mathrm{tr}(F^*, \mathrm{H}_c^\bullet(X,\mathscr{F}))$.

This trace formula for E_λ-sheaves can either be deduced from Report 4.10 by passage to the limit (cf. Report 4.11 through 4.13), or be deduced from the trace formula for $\mathbb{Q}_\ell$-sheaves (Report 3.2) by the method of (Functions L mod. ℓ^n, 4.3).

We shall interpret various trigonometric sums as the left-hand side of (1.1.1), for suitable X_0 and $\mathscr{F}_0$.

1.2

Let A be a finite group. An *A-torsor* on a scheme X is a sheaf T on X, equipped with a right action of A, which, locally (for the étale topology) on X, is isomorphic to the constant sheaf A on which A acts by right translation.

If $\tau : A \to B$ is a homomorphism and T is an A-torsor, there exists, up to unique isomorphism, a unique B-torsor $\tau(T)$, equipped with a morphism of sheaves $\tau : T \to \tau(T)$ such that

$$\tau(ta) = \tau(t)\tau(a). \tag{1.2.1}$$

If T is an A-torsor on X and ρ is a linear representation of A, $\rho : A \to \mathrm{GL}(V)$ (where V is a finite-dimensional vector space over E_λ), there exists, up to unique isomorphism, a unique E_λ-sheaf $\mathscr{F}$, smooth of rank $\dim(V)$, equipped with a morphism of sheaves

$$\rho : T \to \underline{\mathrm{Isom}}(V,\mathscr{F}). \tag{1.2.2}$$

such that $\rho(ta) = \rho(t)\rho(a)$. We denote it by $\rho(T)$. For $\tau : A \to B$ and ρ a representation of B, we have a canonical isomorphism $\rho(\tau(T)) = (\rho\tau)(T)$.

In this paragraph (except the appendix), we only consider the case where A is commutative and where $V = E_\lambda$: ρ is a character $A \to E_\lambda^\times$, and $\rho(T)$ is a smooth E_λ-sheaf of rank one. The morphism (1.2.2) is interpreted as an everywhere nonzero morphism

$$\rho : T \to \rho(T) \tag{1.2.3}$$

such that $\rho(ta) = \rho(a)\rho(t)$.

1.3

Let S be a scheme, let G be a commutative group scheme over S, and let G' be an extension of commutative group schemes over S

$$0 \longrightarrow A \longrightarrow G' \xrightarrow{\pi} G \longrightarrow 0.$$

The sheaf T of local sections of π is then an A-torsor on G.

If X is a scheme over S and f, g are two S-morphisms from X to G, the fact that T is defined by an extension gives a canonical isomorphism

$$(f+g)^*T = f^*T + g^*T. \tag{1.3.1}$$

On the left, $f+g$ is the sum of f and g with respect to the group law on G; on the right, $+$ denotes a sum of torsors.

If $f : X \to G$ factors through G', f^*T is trivial (and a factorization gives a trivialization); up to (non-unique) isomorphism, the A-torsor f^*T depends only on the image of f in $\hom_S(X,G)/\pi\hom_S(X,G')$: it is given by the morphism ∂ in the exact sequence

$$\hom(X,G') \longrightarrow \hom(X,G) \xrightarrow{\partial} \mathrm{H}^1(X,A).$$

We will apply these constructions to the exact Kummer sequences $0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0$, and to Lang torsors.

1.4 The Lang torsor

Let G_0 be a connected commutative algebraic group over $\mathbb{F}_q$ (for the noncommutative case, see the appendix). The group law is written *multiplicatively*. Lang's isogeny

$$\mathfrak{L} : G_0 \to G_0 : x \mapsto Fx \cdot x^{-1}$$

is étale; its image, an open subgroup of G_0, can only be G_0 itself; its kernel is the finite group $G_0(\mathbb{F}_q)$. The *Lang torsor* L is the $G_0(\mathbb{F}_q)$-torsor on G_0 defined by the exact sequence

$$0 \longrightarrow G_0(\mathbb{F}_q) \longrightarrow G_0 \xrightarrow{\mathfrak{L}} G_0 \longrightarrow 0. \tag{1.4.1}$$

Notation We also write L and $\mathfrak{L}$ (or $L_{(q)}$ and $\mathfrak{L}_{(q)}$) for the objects obtained from L and $\mathfrak{L}$ by extending the base field. When necessary, we specify them with a subscript 0 or 1.

Examples For $G_0 = \mathbb{G}_a$ or $\mathbb{G}_m$, the sequences (1.4.1) are written

$$0 \longrightarrow \mathbb{F}_q \longrightarrow \mathbb{G}_a \xrightarrow{x^q - x} \mathbb{G}_a \longrightarrow 0 \tag{1.4.2}$$

$$0 \longrightarrow \boldsymbol{\mu}_{q-1} \longrightarrow \mathbb{G}_m \xrightarrow{x^{q-1}} \mathbb{G}_m \longrightarrow 0 \tag{1.4.3}$$

1.5

Let us compute the endomorphism F^* of the fiber of the Lang torsor at $\gamma \in G^F = G_0(\mathbb{F}_q)$. If $g \in \mathfrak{L}^{-1}(\gamma) \subset G$, we have $Fg = Fg \cdot g^{-1} \cdot g = \gamma g$. Therefore (Report, 1.2)

$$\text{on } L_0(G_0)_\gamma \simeq \mathfrak{L}^{-1}(\gamma),\ F^* \text{ is } g \mapsto g\gamma^{-1}. \tag{1.5.1}$$

1.6

On $\mathbb{F}_{q^n}$, the identity $F_{(q^n)} = F_{(q)}^n$ between endomorphisms of G_1 implies that $\mathfrak{L}_{(q^n)} = \mathfrak{L}_{(q)} \circ \prod_{i=0}^{n-1} F_{(q)}^i$. On $G_0(\mathbb{F}_{q^n})$, $F_{(q)}^i$ acts as the element $x \mapsto x^{q^i}$ of $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$. On $G_0(\mathbb{F}_{q^n})$, $\prod F_{(q)}^i$ is therefore the composite of the norm $G_0(\mathbb{F}_{q^n}) \to G_0(\mathbb{F}_q)$ and the inclusion $G_0(\mathbb{F}_q) \subset G_0(\mathbb{F}_{q^n})$: the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & G_0(\mathbb{F}_{q^n}) & \longrightarrow & G_1 & \xrightarrow{\mathfrak{L}_{(q^n)}} & G_1 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle N} & & \downarrow{\scriptstyle \prod_{i=0}^{n-1} F_{(q)}^i} & & \| & & \\ 0 & \longrightarrow & G_0(\mathbb{F}_q) & \longrightarrow & G_q & \xrightarrow{\mathfrak{L}_{(q)}} & G_1 & \longrightarrow & 0 \end{array} \tag{1.6.1}$$

is commutative. In the language of torsors, this gives a canonical isomorphism between $G_0(\mathbb{F}_q)$-torsors on G_1

$$NL_{(q^n)} = L_{(q)}. \tag{1.6.2}$$

Definition 1.7. *Let $f_0 : X_0 \to G_0$ be a morphism and $\chi : G_0(\mathbb{F}_q) \to E_\lambda^\times$ a character. We set $\mathscr{F}(\chi, f_0) = \chi^{-1}(f_0^*(L_0(G_0))) = f_0^*\chi^{-1}(L_0(G_0))$.*

We have the following functoriality properties:

a) $\mathscr{F}(\chi, f_0)$ is bimultiplicative in χ and f_0:

$$\mathscr{F}(\chi, f_0' \cdot f_0'') = \mathscr{F}(\chi, f_0') \otimes \mathscr{F}(\chi, f_0''), \tag{1.7.1}$$
$$\mathscr{F}(\chi'\chi'', \mathscr{F}_0) = \mathscr{F}(\chi', f_0) \otimes \mathscr{F}(\chi'', f_0). \tag{1.7.2}$$

This follows from the definition of $\chi(T)$, for T a torsor, combined with (1.3.1) for (1.7.1).

b) For $g_0 : Y_0 \to X_0$ a morphism, we have

$$\mathscr{F}(\chi, f_0 \circ g_0) = g_0^*\mathscr{F}(\chi, f_0). \tag{1.7.3}$$

In particular, $\mathscr{F}(\chi, f_0) = f_0^*\mathscr{F}(\chi)$, where $\mathscr{F}(\chi)$ denotes $\mathscr{F}(\chi, \mathrm{id}_{G_0})$.

c) For $u_0 : G_0 \to H_0$ a morphism, and $\chi : H_0(\mathbb{F}_q) \to E_\lambda^\times$, we have

$$\mathscr{F}(\chi, u_0 f_0) = \mathscr{F}(\chi u_0, f_0). \tag{1.7.4}$$

d) For $G_0 = \prod_{i\in I} G_0^i$, with χ having coordinates χ_i ($i \in I$) and f_0 having coordinates f_0^i, it follows from (1.7.1) through (1.7.4) that

$$\mathscr{F}(\chi, f_0) = \bigotimes_{i\in I} \mathscr{F}(\chi_i, f_0^i). \tag{1.7.5}$$

Note the χ^{-1} in Definition 1.7. It ensures that, for $x \in X^F$, on $\mathscr{F}(\chi, f_0)_x$ one has

$$F_x^* = \chi f_0(x) \tag{1.7.6}$$

(use the fact that the fiber at x of the morphism (1.2.2), for $\rho = \chi^{-1}$, commutes with F_x^*, together with (1.5.1)).

If $\mathscr{F}(\chi, f_0)_1$ is the sheaf obtained from $\mathscr{F}(\chi, f_0)$ by extension from the base field to $\mathbb{F}_{q^n}$, we deduce from (1.6.2) that

$$\mathscr{F}(\chi, f_0)_1 = \mathscr{F}(\chi \circ N, f_1). \tag{1.7.7}$$

1.8 Abuse of notations

(i) If Ξ denotes the composite map $\chi f_0 : X_0(\mathbb{F}_q) \to E_\lambda^\times$, we will sometimes write $\mathscr{F}(\Xi)$ instead of $\mathscr{F}(\chi, f_0)$. By (1.7.1) through (1.7.5), there is little risk of ambiguity. For example:

a) we write $\mathscr{F}(\chi)$ for $\mathscr{F}(\chi, \mathrm{id}_{G_0})$ (notation already used in 1.7b));

b) we write $\mathscr{F}(\chi f_0)$ for $\mathscr{F}(\chi, f_0)$;

c) with the notation of 1.7d), we write $\mathscr{F}(\prod \chi_i f_0^i)$ for $\mathscr{F}(\chi, f)$).

With this notation, (1.7.4) shows that the notation $\mathscr{F}(\chi u_0 f_0)$ is unambiguous, (1.7.1) (1.7.2) (1.7.5) express the multiplicativity of $\mathscr{F}(\Xi)$ with respect to Ξ, and (1.7.6) becomes $F_x^* = \Xi(x)$ on $\mathscr{F}(\Xi)_x$.

(ii) If X is a scheme over an extension k of $\mathbb{F}_q$, and f is a morphism of X into $G_0 \otimes_{\mathbb{F}_q} k$, we shall again denote by $\mathscr{F}(\chi, f)$, $\mathscr{F}(\chi f)$, or $\mathscr{F}(\chi)$ (for f an inclusion) the inverse image under f of the E_λ-sheaf obtained from $\chi^{-1}(L_0(G))$ by extension of scalars from $\mathbb{F}_q$ to k. If f_0 is the composite $X \to G_0 \otimes_{\mathbb{F}_q} k \to G_0$, this is again the $\mathscr{F}(\chi, f_0)$ of 1.7. If k is a finite extension of $\mathbb{F}_q$, we have $\mathscr{F}(\chi f) = \chi(\chi N_{k/\mathbb{F}_q} f)$.

Applying (1.1.1) to (1.7.6) and (1.7.7), we find:

Theorem 1.9. *Let S_0 be a separated scheme of finite type over $\mathbb{F}_q$, let G_0 be a connected commutative algebraic group over $\mathbb{F}_q$, let $f_0 : S_0 \to G_0$ be a morphism, and let $\chi : G_0(\mathbb{F}_q) \to E_\lambda^\times$. We have*

$$\sum_{s \in S_0(\mathbb{F}_q)} \chi f_0(s) = \mathrm{tr}(F^*, \mathrm{H}_c^\bullet(S, \mathscr{F}(\chi, f_0))) \tag{1.9.1}$$

and, for any integer $n \geqslant 1$,

$$\sum_{s \in S_0(\mathbb{F}_{q^n})} \chi \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q} f_0(s) = \mathrm{tr}(F^{*n}, \mathrm{H}_c^\bullet(S, \mathscr{F}(\chi, f_0))) \,. \tag{1.9.2}$$

1.9.3 Remark

Take G_0 to be a product. With the notation of 1.7d) and 1.8c), the formula (1.9.2) becomes

$$\sum_{s \in S_0(\mathbb{F}_{q^n})} \prod_i \chi_i \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q}(f_0^i(s)) = \mathrm{tr}\left(F^{*n}, \mathrm{H}_c^\bullet\left(S, \mathscr{F}\Big(\prod_i \chi_i f_0^i\Big)\right)\right).$$

1.9.4 Remark

Take $G_0 = \{e\}$. Formula (1.9.1) becomes

$$|S_0(\mathbb{F}_q)| = \sum_{s \in S_0(\mathbb{F}_q)} 1 = \mathrm{tr}(F^*, \mathrm{H}_c^\bullet(S, \mathbb{Q}_\ell)).$$

It is rare to be able to compute the right-hand side of (1.9.1) explicitly. Here is an amusing example, with $G_0 = \{e\}$.

1.10 Example

A nonsingular projective quadric of odd dimension over $\mathbb{F}_q$ has the same number of rational points as projective space of the same dimension over $\mathbb{F}_q$.

If, over $\mathbb{C}$, X is a nonsingular quadric hypersurface in the projective space $\mathbb{P}^{2N}$, and Y a hyperplane, we know that for $i \leqslant 2\dim(X) = 2\dim(Y)$, the inclusions $X \hookrightarrow \mathbb{P}^{2N} \hookleftarrow Y$ induce isomorphisms (in ordinary cohomology)

$$\mathrm{H}^i(X, \mathbb{Q}) \xleftarrow{\sim} \mathrm{H}^i(\mathbb{P}^{2N}, \mathbb{Q}) \xrightarrow{\sim} \mathrm{H}^i(Y, \mathbb{Q}).$$

By specialization, it follows that if X_0' is a nonsingular quadric hypersurface in projective space $\mathbb{P}_0^{2N}$ over $\mathbb{F}_q$, and Y_0' a hyperplane, the inclusions $X_0' \hookrightarrow \mathbb{P}_0^{2N} \hookleftarrow Y_0'$ induce isomorphisms

$$\mathrm{H}^i(X', \mathbb{Q}_\ell) \xleftarrow{\sim} \mathrm{H}^i(\mathbb{P}^{2N}, \mathbb{Q}_\ell) \xrightarrow{\sim} \mathrm{H}^i(Y', \mathbb{Q}_\ell).$$

These isomorphisms commute with F^*, and the trace formula applies.

1.11

The trace formula can also be used to understand the classical formulas giving the number of rational points of linear groups over finite fields, and of certain homogeneous spaces (see Section 8).

1.12

There is a dictionary that translates various kinds of classical manipulations of trigonometric sums into cohomological terms. This dictionary will be given in Section 2. Since the cohomological statement is more "geometric," it can sometimes be applied to situations where the classical argument only applies after a extension of the finite base field. Here is an example (a special case of a theorem of Kazhdan).

Theorem 1.13 (Kazhdan)**.** *Let X_0 be a scheme over $\mathbb{F}_q$. Suppose there is an action ρ of $\mathbb{G}_a$ on X, and a morphism $f : X \to Y$ of schemes over $\mathbb{F}$, making X a $\mathbb{G}_a$-torsor over Y. Then the number of rational points of X_0 is divisible by q.*

First suppose that ρ, Y, and f are defined over $\mathbb{F}_q$.

Classical argument The fibers of $f : X_0(\mathbb{F}_q) \to Y_0(\mathbb{F}_q)$ all have q elements: $|X_0(\mathbb{F}_q)| = q \cdot |Y_0(\mathbb{F}_q)|$, hence the divisibility.

Cohomological translation The higher direct image sheaves $\mathsf{R}^i f_! \mathbb{Q}_\ell$ are

$$\mathsf{R}^i f_! \mathbb{Q}_\ell = \begin{cases} \mathbb{Q}_\ell(-1) & \text{if } i = 2 \\ 0 & \text{for } i \neq 2 \end{cases}$$

The Leray spectral sequence for f therefore degenerates, giving an isomorphism

$$\mathrm{H}_c^i(X, \mathbb{Q}_\ell) = \mathrm{H}_c^{i-2}(Y, \mathbb{Q}_\ell)(-1).$$

To understand the effect of a Tate twist on the eigenvalues of Frobenius, it is most convenient to use the Galois point of view (Report 1.8) and note that the Frobenius substitution φ acts on $\mathbb{Q}_\ell(1)$ by multiplication by q. The eigenvalues of F^* on $\mathrm{H}_c^p(X, \mathbb{Q}_\ell)$ are therefore obtained by multiplying by q the

eigenvalues of F^* acting on $\mathrm{H}_c^{i-2}(Y, \mathbb{Q}_\ell)$. We know that the latter are algebraic integers [16, XXI 5.2.2]; hence

$$\begin{gathered}\text{the eigenvalues of } F^* \text{ acting on } \mathrm{H}_c^i(X, \mathbb{Q}_\ell) \text{ are} \\ \text{algebraic integers divisible by } q.\end{gathered} \tag{$*$}$$

Descent Return to the hypotheses of the theorem. For a suitable n, we may suppose that ρ, Y, and f are defined over $\mathbb{F}_{q^n}$. Since the Frobenius morphism relative to $\mathbb{F}_{q^n}$ is the n-th power of the one relative to $\mathbb{F}_q$, statement $(*)$ for the scheme over $\mathbb{F}_{q^n}$ obtained from X_0 by extension of scalars gives:

$$\begin{gathered}\text{the } n\text{-th powers of the eigenvalues of } F^* \text{ acting on} \\ \mathrm{H}_c^i(X, \mathbb{Q}_\ell) \text{ are algebraic integers divisible by } q^n.\end{gathered} \tag{$**$}$$

This assertion implies $(*)$. The number of rational points of X_0, namely $\sum(-1)^i \operatorname{tr}(F^*, \mathrm{H}^i(X, \mathbb{Q}_\ell))$, is therefore an algebraic integer divisible by q. It follows that, as a rational integer, it is divisible by q.

1.14

Formula 1.9.3 controls the dependence on n of the trigonometric sum on the left-hand side. It implies identities between a trigonometric sum and those which are deduced from it by "extension of scalars." The most famous of these identities is that of Hasse–Davenport. Let $\chi : \mathbb{F}_q^\times \to \mathbb{C}^\times$ and $\psi : \mathbb{F}_q \to \mathbb{C}^\times$ be characters, with ψ nontrivial, and define the Gauss sum $\tau(\chi, \psi)$ by

$$\tau(\chi, \psi) = -\sum_{x \in \mathbb{F}_q^\times} \psi(x)\chi^{-1}(x). \tag{1.14.1}$$

Theorem 1.15 (Hasse-Davenport)**.** *We have*

$$\tau\left(\chi \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q}, \psi \circ \operatorname{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}\right) = \tau(\chi, \psi)^n. \tag{1.15.1}$$

Let $E \subset \mathbb{C}$ be a number field containing the values of χ and ψ, and let E_λ be the completion of E at a place of residue characteristic $\ell \neq p$. Identity (1.15.1) is an identity in E. Proving it in $\mathbb{C}$ is equivalent to proving it in E_λ. We shall prove it in E_λ, regarding χ and ψ as taking values in $E_\lambda^\times$.

Apply 1.9.3 with $X_0 = \mathbb{G}_m$ over $\mathbb{F}_q$ and $G_0 = \mathbb{G}_m \times \mathbb{G}_a$. The groups $\mathrm{H}_c^i(\mathbb{G}_m, \mathscr{F}(\chi^{-1}\psi))$ vanish for $i \neq 1$, and H_c^1 has dimension 1 (4.2). By 1.9.3, $\tau(\chi \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q}, \psi \circ \operatorname{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q})$ is the unique eigenvalue of $(F^*)^n$ on H_c^1, and (1.15.1) follows.

Similar examples are given in Weil [43, App V].

1.16

An E_λ-sheaf $\mathscr{F}_0$ on $X - 0$ is said to be *pointwise pure of weight* n if for any closed point $x \in |X_0|$, the eigenvalues of F_x^* (Report 1.2) are algebraic numbers all of whose complex conjugates have absolute value $q_x^{n/2}$, where q_x is the number of elements of $k(x)$. The following theorem will be proved in [15].

Theorem 1.17. *If $\mathscr{F}_0$ is pointwise pure of weight n, then for any eigenvalue α of the endomorphism F^* of $\mathrm{H}_c^i(X, \mathscr{F})$, there exists an integer $m \leqslant n + i$ such that the complex conjugates of α are all of absolute value $q^{m/2}$.*

The sheaves considered in 1.9 have weight 0. The theorem therefore gives, for the sum 1.9.1 (or rather, for all its complex conjugates), the bound

$$\left|\sum_{s\in S_0(\mathbb{F}_q)}\prod_i \chi_i(f^i(s))\right| \leqslant \sum_i \dim \mathrm{H}^i_c\left(S, \mathscr{F}\left(\prod \chi_i f^i\right)\right)\cdot q^{i/2}. \tag{1.17.1}$$

1.18 Remarks

a) If $\dim S = n$, for any E_λ-sheaf $\mathscr{F}$ on S, we have

$$\mathrm{H}^i_c(S,\mathscr{F}) = 0 \qquad \text{for } i \notin [0,2n].$$

b) The group $\mathrm{H}^0_c(X,\mathscr{F})$ is the group of global sections of $\mathscr{F}$ on S whose support is proper. If S is connected, not proper and $\mathscr{F}$ is smooth, this group is zero; the same holds if S is connected, $\mathscr{F}$ is smooth of rank one, and is nonconstant.

c) If S is smooth and purely of dimension n, and $\mathscr{F}$ is smooth, Poincaré duality says that the vector spaces $\mathrm{H}^i_c(X,\mathscr{F})$ and $\mathrm{H}^{2n-i}(X,\mathscr{F}^\vee)(2n)$ are dual to each other (for the effect of a Tate twist on the eigenvalues of Frobenius, cf. the 2nd part of the proof of 1.13).

d) If $\dim S = n$, Poincaré duality allows us to compute $\mathrm{H}^{2n}_c(S,\mathscr{F})$ as follows: take an open subset U of S_{red} whose complement has dimension $< n$; require this open subset to be smooth and purely of dimension n, with $\mathscr{F}$ smooth on it. We then have $\mathrm{H}^{2n}_c(U,\mathscr{F}) \xrightarrow{\sim} \mathrm{H}^{2n}_c(S,\mathscr{F})$, because in the long exact sequence of cohomology, $\mathrm{H}^i_c(S\smallsetminus U,\mathscr{F}) = 0$ for $i = 2n, 2n-1$. Poincaré duality therefore gives

$$\mathrm{H}^{2n}_c(S,\mathscr{F}) \xrightarrow{\sim} \left(\mathrm{H}^0(U,\mathscr{F}^\vee)(2n)\right)^\vee.$$

Let U be connected, and let u be a geometric point of U. The "local system" $\mathscr{F}$ corresponds to a representation of $\pi_1(U,u)$ on $\mathscr{F}_u$, and $\mathrm{H}^0(U,\mathscr{F}^\vee)$ is $(\mathscr{F}_u^\vee)^{\pi_1(U,u)} = ((\mathscr{F}_u)_{\pi_1(U,u)})^\vee$ (dual of coinvariants). We therefore have

$$\mathrm{H}^{2n}_c(S,\mathscr{F}) \simeq (\mathscr{F}_u)_{\pi_1(U,u)}(-2n).$$

e) These remarks often allow the calculation of b_0 and b_{2n}. Computing the other b_i can be difficult. If $b_{2n} = 0$ and $b_{2n-1} = 0$, the bound (1.17.1) is a Lang–Weil bound, with, for large q, a gain of $\sqrt{q}$ over the trivial $O(q^n)$ bound. One should bear in mind, however, that (1.17.1) does not formally imply the trivial bound

$$\left|\sum_{s\in S_0(\mathbb{F}_q)}\prod_i \chi_i(f^i(s))\right| \leqslant \left|S_0(\mathbb{F}_q)\right|,$$

so it may be useful to take this into account (cf. the proof of 3.8).

Below I explain a method for proving that $b_i = 0$ for $i \neq n$. When it applies, it gives an estimate of order $O(q^{n/2})$. The constant implicit in 0 is $b_n = (-1)^n\chi(S,\mathscr{F}(\prod \chi_i f^i))$, which may be difficult to compute.

Proposition 1.19. *Let $\bar{X}$ be a proper scheme over an algebraically closed field k, let $j : X \hookrightarrow barX$ be an open immersion with domain X, and let $\mathscr{F}$ be an E_λ-sheaf on X. Assume that*

a) $(j_*\mathscr{F})_x = 0$ for $x \in \bar{X}\smallsetminus X$, i.e. $j_!\mathscr{F} \xrightarrow{\sim} j_*\mathscr{F}$;

b) $\mathsf{R}^i j_* \mathscr{F} = 0$ *for* $i > 0$.

Then the maps

$$\mathrm{H}^i_c(X, \mathscr{F}) \to \mathrm{H}^i(X, \mathscr{F})$$

are isomorphisms.

Assumptions a) and b) mean that $j_! \mathscr{F} \xrightarrow{\sim} \mathsf{R} j_* \mathscr{F}$, hence

$$\mathrm{H}^i_c(X, \mathscr{F}) = \mathrm{H}^i(\bar{X}, j_! \mathscr{F}) \xrightarrow{\sim} \mathsf{H}^i(\bar{X}, \mathsf{R} j_* \mathscr{F}) = \mathrm{H}^i(X, \mathscr{F}).$$

In other words, the Leray spectral sequence for j

$$E_2^{pq} = \mathrm{H}^p(\bar{X}, \mathsf{R}^q j_* \mathscr{F}) \Rightarrow \mathrm{H}^{p+q}(X, \mathscr{F})$$

degenerates, by b) into isomorphisms

$$\mathrm{H}^i(\bar{X}, j_* \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X, \mathscr{F}),$$

while, by a)

$$\mathrm{H}^i_c(X, \mathscr{F}) = \mathrm{H}^i(\bar{X}, j_! \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(\bar{X}, j_* \mathscr{F}).$$

1.19.1 Example

If X is a curve, or if $\bar{X}$ is smooth, X is the complement of a normal-crossings divisor, and $\mathscr{F}$ is tamely ramified, hypothesis a) of 1.19 (no invariants under local monodromy) implies hypothesis b).

Proposition 1.20. *Let X_0 be a smooth affine scheme, purely of dimension n, over $\mathbb{F}_q$, and let $\mathscr{F}_0$ be a smooth E_λ-sheaf, pointwise pure of weight m, on X_0. We assume that the maps $\mathrm{H}^i_c(X, \mathscr{F}) \to \mathrm{H}^i(X, \mathscr{F})$ are isomorphisms (cf. 1.17). Then, $\mathrm{H}^i_c(X, \mathscr{F}) = 0$ for $i \neq n$, and the complex conjugates of the eigenvalues of F^* on $\mathrm{H}^n_c(X, \mathscr{F})$ are of absolute value $q^{(m+n)/2}$.*

Poincaré duality pairs $\mathrm{H}^i_c(X, \mathscr{F})$ (resp. $\mathrm{H}^i_c(X, \mathscr{F}^\vee)$) with $\mathrm{H}^{2n-i}(X, \mathscr{F}^\vee(n))$ (resp. $\mathrm{H}^{2n-i}(X, \mathscr{F}(n))$). According to [2, XIV 3.2], the ordinary H^i groups vanish for $i > n$. By duality, the H^i_c are zero for $i < n$; since $\mathrm{H}^i_c(X, \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X, \mathscr{F})$, these groups are zero for $i \neq n$. Apply 1.15 to $\mathscr{F}_0$ and to $\mathscr{F}_0^\vee(n)$ (of weight $-m-2n$). We find that the complex conjugates α of the eigenvalues of F^* on $\mathrm{H}^n_c(X, \mathscr{F}) = \mathrm{H}^n_c(X, \mathscr{F}^\vee(n))^\vee$ satisfy

$$|\alpha| \leqslant q^{(m+n)/2} \qquad \text{and}$$
$$|\alpha^{-1}| \leqslant q^{((-m-2n)+n)/2} = q^{-(m+n)/2},$$

hence the assertion.

1.21 Remark

The last argument shows that if X_0 is a smooth separated scheme over $\mathbb{F}_q$, if $\mathscr{F}_0$ is a smooth E_λ-sheaf pointwise pure of weight m on X_0, and if $\mathrm{H}^i_c(X, \mathscr{F}) \hookrightarrow \mathrm{H}^i(X, \mathscr{F})$, then the complex conjugates of the eigenvalues of F^* on $\mathrm{H}^i_c(X, \mathscr{F})$ are of absolute value $q^{(m+i)/2}$.

Appendix. The Lang isogeny in the noncommutative case

1.22

Let $G-0$ be a connected algebraic group over $\mathbb{F}_q$ and let $\mathfrak{L}$ be the morphism $x \mapsto Fx \cdot x^{-1}$ from $G-0$ to itself. It is a morphism of G_0-homogeneous spaces if, on the source, $G-0$ acts by left translation $x \mapsto (y \mapsto xy)$, and on the target by $x \mapsto (y \mapsto Fx \cdot y \cdot x^{-1})$. We have $\mathfrak{L}(xy) = \mathfrak{L}(x)$ for $y \in G_0(\mathbb{F}_q)$, and $\mathfrak{L}$ induces an isomorphism $G_0/G_0(\mathbb{F}_q) \xrightarrow{\sim} G_0$.

As in the commutative case, $\mathfrak{L}$ therefore makes G_0 a $G_0(\mathbb{F}_q)$-torsor on G_0, the *Lang torsor* L. If $\rho : G_0(\mathbb{F}_q) \to \mathrm{GL}(n, E_\lambda)$ is a linear representation of $G_0(\mathbb{F}_q)$, we denote by $\mathscr{F}(\rho)$ the E_λ-sheaf $\rho(L)$ (1.2).

1.23

Let $\gamma \in G^F = G_0(\mathbb{F}_q)$, and compute the endomorphism F^* of $\mathscr{F}(\rho)_\gamma$. Let $g \in G$ be such that $\mathfrak{L}(g) = \gamma$, and choose the frame $\rho(g) \in \mathrm{Isom}(E_\lambda^n, \mathscr{F}(\rho)_\gamma)$. For $e \in E_\lambda^n$, we have

$$F(\rho(g)(e)) = \rho(Fg)(e) = \rho(\gamma g)(e) = \rho(g \cdot g^{-1}\gamma g)(e).$$

We will see that $g^{-1}\gamma g \in G_0(\mathbb{F}_q)$, hence

$$F(\rho(g)(e)) = \rho(g)\rho(g^{-1}\gamma g)(e)$$

and

$$\rho(g)^{-1}(F_\gamma^*)^{-1}\rho(g) = \rho(g^{-1}\gamma g): \tag{1.23.1}$$

$\rho(g)$ identifies the inverse of F^* at γ with the automorphism $\rho(g^{-1}\gamma g)$ of E_λ^n. It remains to understand $g^{-1}\gamma g$.

Lemma 1.24. *(i) If $Fg \cdot g^{-1} \in G_0(\mathbb{F}_q)$, we have $g^{-1}(Fg \cdot g^{-1})g = g^{-1}Fg \in G_0(\mathbb{F}_q)$.*

(ii) Let $\gamma \in G_0(\mathbb{F}_q)$, and g such that $Fg \cdot g^{-1} = \gamma$. The conjugation class of $\gamma' = g^{-1}Fg$ in $G_0(\mathbb{F}_q)$ only depends on that of γ.

(iii) The map induced by $\gamma \mapsto \gamma'$ on the set $G_0(\mathbb{F}_q)^\natural$ of conjugacy classes of $G_0(\mathbb{F}_q)$ is bijective. Its inverse is the corresponding map for the opposite group.

(i) If $\gamma = Fg \cdot g^{-1} \in G_0(\mathbb{F}_q)$, the identities $Fg = \gamma g$ and $F\gamma = \gamma$ give $F(g^{-1}Fg) = (Fg)^{-1}FFg = g^{-1}\gamma^{-1}F(\gamma g) = g^{-1}\gamma^{-1}F\gamma Fg = g^{-1}Fg$, so that $g^{-1}Fg \in G_0(\mathbb{F}_q)$.

(ii) For $\alpha, \gamma \in G_0(\mathbb{F}_q)$, the g such that $Fg \cdot g^{-1} = \gamma$ form a right coset under $G_0(\mathbb{F}_q)$, and if $Fg \cdot g^{-1} = \gamma$, we have $F(\alpha g \alpha^{-1})(\alpha g \alpha^{-1})^{-1} = \alpha\gamma\alpha^{-1}$. As γ ranges over a conjugacy class in $G_0(\mathbb{F}_q)$, the g such that $Fg \cdot g^{-1} = \gamma$ therefore range over a double coset under $G_0(\mathbb{F}_q)$, and $\gamma' = g^{-1}Fg$ ranges over a conjugacy class.

(iii) Finally, the inverse of $Fg \cdot g^{-1} \mapsto g^{-1}Fg$ is $g^{-1}Fg \mapsto Fg \cdot g^{-1}$.

1.25

Formula (1.23.1) can be read succinctly: F^* at γ is $\rho(\gamma')^{-1}$. If γ belongs to the identity component of its centralizer, we can take g in $Z(\gamma)$. We then have $\gamma = \gamma'$. If this condition is not met, γ and γ' need not be conjugate.

1.26 An example

For $G_0 = \mathrm{SL}(2)$, q odd, $\alpha \in F_q^\times \smallsetminus \mathbb{F}_q^{\times 2}$, and $\bar\gamma$ is the conjugacy class of $-1(1+N)$, with $N^2 = 0$, then $\bar\gamma'$ is the conjugacy class of $-1 \cdot (1+\alpha N)$, so that $\bar\gamma' \neq \bar\gamma$ if $N \neq 0$.

2 Dictionary

Elementary manipulations of trigonometric sums can often be seen, via the trace formula, as counterparts of statements in cohomology. In this section, I list such statements and their counterparts; they carry the same number, with a $*$ appended to the cohomological statement.

2.1

"A sum of algebraic integers is an algebraic integer" has as its cohomological analogue the following theorem, proved in [16, XXI 5.2.2] (see the use of 2.1* in 1.13).

Theorem (2.1*)**.** *Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$ and let $\mathscr{F}_0$ be an E_λ-sheaf on X_0. If, for all $x \in |X_0|$, the eigenvalues of F_x^* on $\mathscr{F}_0$ are algebraic integers, then the eigenvalues of F^* on $\mathrm{H}_c^i(X, \mathscr{F})$ are algebraic integers.*

2.2

Other integrality results are proved in [16, XXI, 5.2.2 et 5.4]: if $\dim(x_0) \leqslant n$, and α is an eigenvalue of F^* on $\mathrm{H}_c^i(X, \mathscr{F})$, we have:

a) under the assumptions of 2.1*, and if $i > n$, then q^{i-n} divides α;

b) if the inverses of the eigenvalues of F_x^* are algebraic integers, then α^{-1} is integral away from p. More precisely, $q^{\inf(n,i)}\alpha^{-1}$ is an algebraic integer.

For the sheaves considered in 1.9, the eigenvalues of F_x^* are roots of unity, so the assumptions of the theorem and of b) above are satisfied.

2.3

Let $f : A \to B$ be a map from one finite set to another, and ε a function on A. We have

$$\sum_{a \in A} \varepsilon(a) = \sum_{b \in B} \sum_{f(a)=b} \varepsilon(a). \tag{2.3.1}$$

2.3*

Let $f : X \to Y$ be a morphism of separated schemes of finite type over an algebraically closed field k, and let $\mathscr{F}$ be an E_λ-sheaf on X. The Leray spectral sequence for f in cohomology with compact support is

$$E_2^{pq} = \mathrm{H}_c^p(Y, \mathsf{R}^q f_! \mathscr{F}) \Rightarrow \mathrm{H}_c^{p+q}(X, \mathscr{F}). \tag{2.3.1*}$$

Let $f_0 : X_0 \to Y_0$ be a morphism of separated schemes of finite type over $\mathbb{F}_q$ and let $\mathscr{F}_0$ be an E_λ-sheaf on X_0. Between (2.3.1) and (2.3.1*) there is the following compatibility.

a) The sheaf $\mathsf{R}^q f_!\mathscr{F}$ is obtained by extending scalars from $\mathbb{F}_q$ to $\mathbb{F}$ in the sheaf $\mathsf{R}^q f_{0!}\mathscr{F}_0$ on Y_0. As such, it is equipped with a Frobenius correspondence $F^* : F^*\mathsf{R}^q f_!\mathscr{F} \to \mathsf{R}^q f_!\mathscr{F}$. It is obtained by applying the functoriality of $\mathsf{R}f_!$ to the Frobenius correspondence of $\mathscr{F}$: namely the composite

$$F^*\mathsf{R}^q f_!\mathscr{F} \xrightarrow{\sim} \mathsf{R}^q f_! F^*\mathscr{F} \xrightarrow{\sim} \mathsf{R}^q f_!\mathscr{F}.$$

b) The trace formula for $\mathscr{F}_q$ reads

$$\sum_{x\in X_0(\mathbb{F}_q)} \operatorname{tr}(F_x^*, \mathscr{F}_0) = \operatorname{tr}(F^*, \mathrm{H}^\bullet(X, \mathscr{F})); \tag{1}$$

while that for $\mathsf{R}^q f_!\mathscr{F}$ reads

$$\sum_{y\in Y_0(\mathbb{F}_q)} \operatorname{tr}(F_y^*, \mathsf{R}^q f_!\mathscr{F}) = \operatorname{tr}(F^*, \mathrm{H}^\bullet(Y, \mathsf{R}^q f_!\mathscr{F})). \tag{2_q}$$

We have, for $y \in Y(\mathbb{F})$, $(\mathsf{R}^q f_!\mathscr{F})_y = \mathrm{H}_c^q(f^{-1}(y), \mathscr{F})$, and if $y \in Y_0(\mathbb{F}_q)$,

$$\operatorname{tr}(F_y^*, \mathsf{R}^q f_!\mathscr{F}) = \operatorname{tr}(F^*, \mathrm{H}_c^q(f^{-1}(y), \mathscr{F})),$$

hence by the trace formula on the fiber

$$\sum_{\substack{x\in X_0(\mathbb{F}_q)\\ f(x)=y}} \operatorname{tr}(F_x^*, \mathscr{F}_0) = \sum (-1)^q \operatorname{tr}(F_y^*, \mathsf{R}^q f_!\mathscr{F}).$$

Taking the alternating sum of the identities (2_q), we therefore find

$$\sum_{y\in Y_0(\mathbb{F}_q)} \sum_{\substack{x\in X_0(\mathbb{F}_q)\\ f(x)=y}} \operatorname{tr}(F_x^*, \mathscr{F}_0) = \operatorname{tr}(F^*, \mathrm{H}^\bullet(Y, \mathsf{R}^\bullet f_!\mathscr{F})). \tag{2}$$

Equality of the left-hand sides of (1) and (2) follows from (2.3.1); equality of the right-hand sides follows from (2.3.1*) (taking into account that F^* is an endomorphism of the entire spectral sequence).

2.4

Let $(A_i)_{i\in I}$ be a finite family of finite sets, $A = \prod_{i\in I} A_i$, ε_i a function on A_i, and ε the function $a \mapsto \prod \varepsilon_i(a_i)$ on A. We have

$$\sum_{a\in A} \varepsilon(a) = \prod_{i\in I} \sum_{a_i\in A_i} \varepsilon_i(a_i). \tag{2.4.1}$$

2.4*

Let $(X_i)_{i\in I}$ be a finite family of separated schemes of finite type over an algebraically closed field k, let $X = \prod_{i\in I} X_i$, let $\mathscr{F}_i$ be an E_λ-sheaf on X_i and $\mathscr{F}$ the external tensor product $\boxtimes_{i\in I} \mathscr{F}_i = \bigotimes_{i\in I} \mathrm{pr}_i^*(\mathscr{F}_i)$ of the $\mathscr{F}_i$. The Künneth formula gives

$$\mathrm{H}_c^\bullet(X, \mathscr{F}) = \bigotimes_{i\in I} \mathrm{H}_c^\bullet(X_i, \mathscr{F}_i). \tag{2.4.1*}$$

For the isomorphism (2.4.1*) to be canonical and independent of the choice of a total ordering of I, the tensor product on the right must be understood as a graded tensor product, with the Koszul sign rule (Cycle, 1.3).

2.5

Let A be a finite set, let B be a subset of A, and let ε be a function on A. We have

$$\sum_{a\in A}\varepsilon(a)=\sum_{a\in A\smallsetminus B}\varepsilon(a)+\sum_{a\in B}\varepsilon(a). \tag{2.5.1}$$

2.5*

Let x be a separated scheme of finite type over an algebraically closed field k, let Y be a closed subscheme, and let $\mathscr{F}$ be an E_λ-sheaf on X. We have a long exact cohomology sequence

$$\cdots\xrightarrow{\partial}\mathrm{H}^i_c(X\smallsetminus Y,\mathscr{F})\longrightarrow\mathrm{H}^i_c(X,\mathscr{F})\longrightarrow\mathrm{H}^i_c(Y,\mathscr{F})\xrightarrow{\partial}\cdots \tag{2.5.1*}$$

More generally, given a finite filtration of X by closed subsets X_p ($p\in\mathbb{Z}$; assume that $X_p\supset X_{p+1}$, that $X_p=X$ for p sufficiently small, and that $X_p=\varnothing$ for p sufficiently large), there is a spectral sequence

$$E_1^{pq}=\mathrm{H}^{p+1}_c(X_p\smallsetminus X_{p+1},\mathscr{F})\Rightarrow\mathrm{H}^{p+q}_c(X,\mathscr{F}). \tag{2.5.2*}$$

2.6

Let A be a finite set, $(A_i)_{i\in I}$ a finite covering of A and ε a function on A. For $J\subset I$, denote by A_J the intersection of A_j ($j\in J$). We have

$$\sum_{J\subset I}(-1)^{|J|}\sum_{a\in A_J}\varepsilon(a)=0. \tag{2.6.1}$$

2.6*

Let X be a separated scheme of finite type over an algebraically closed field k, $(X_i)_{i\in I}$ a closed (resp. open) finite covering of X by subschemes, and $\mathscr{F}$ an E_λ-sheaf on X. For $J\subset I$, let X_J be the intersection of the X_j ($j\in J$). We have the respective Leray spectral sequences

$$E_1^{pq}=\bigoplus_{|J|=p+1>0}\mathrm{H}^q_c(X_J,\mathscr{F})\otimes_{\mathbb{Z}}\bigwedge^{|J|}\mathbb{Z}^J\Rightarrow\mathrm{H}^{p+q}_c(X,\mathscr{F}) \tag{2.6.1*}$$

$$E_1^{pq}=\bigoplus_{|J|=1-p>0}\mathrm{H}^q_c(X_J,\mathscr{F})\otimes_{\mathbb{Z}}\bigwedge^{|J|}\mathbb{Z}^J\Rightarrow\mathrm{H}^{p+q}_c(X,\mathscr{F}) \tag{2.6.2*}$$

If $J=\varnothing$ were not excluded, there would also be spectral sequences converging to 0. The factors $\bigwedge^{|J|}\mathbb{Z}^J$ are included to avoid choosing a total order on I.

2.7

Let A be a finite commutative group, and $\chi:A\to E_\lambda^\times$ a nontrivial character. We have

$$\sum_{a\in A}\chi(a)=0. \tag{2.7.1}$$

We will prove the cohomological analogue of (2.7.1) by transposing the following proof: if $x \in A$, $a \mapsto xa$ is a permutation of A, and

$$\sum_{a\in A} \chi(a) = \sum_{a\in A} \chi(xa) = \chi(x) \sum_{a\in A} \chi(a) \qquad \text{, whence}$$
$$(\chi(x) - 1) \sum_{a\in A} = 0,$$

and by hypothesis there exists x such that $\chi(x) - 1 \neq 0$.

Theorem (2.7*)**.** *Let G_0 be a connected commutative algebraic group over $\mathbb{F}_q$ and $\chi : G_0(\mathbb{F}_q) \to E_\lambda^\times$ a nontrivial character. We have*

$$\mathrm{H}_c^\bullet(G, \mathscr{F}(\chi)) = 0. \tag{2.7.1*}$$

For x a rational point of G, let t_x be the translation $t_x(g) = xg$ of G. The formula $\mathfrak{L}t_x = t_{\mathfrak{L}(x)}\mathfrak{L}$ expresses that $(t_x, t_{\mathfrak{L}(x)})$ is an automorphism of the diagram $G \xrightarrow{\mathfrak{L}} G$ (G equipped with the torsor of Lang). Let $\rho(g)$ be the automorphism of $(G, \mathscr{F}(\chi))$ induced by it. For $g \in G_0(\mathbb{F}_q)$, i.e. for $\mathfrak{L}(g) = e$, it is the identity on G, and multiplication by $\chi(g)^{-1}$ on $\mathscr{F}(\chi)$.

Let $\rho_H(g)$ denote the automorphism of $\mathrm{H}_c^\bullet(G, \mathscr{F}(\chi))$ induced by $\rho(g)$. A homotopy argument (Lemma 2.8 below) shows that $\rho_H(g) = \rho_H(e)$, and hence is the identity. On the other hand, for $g \in G_0(\mathbb{F}_q)$, $\rho_H(g)$ is multiplication by $(\chi^{-1}(g) :$ on $\mathrm{H}^\bullet(G, \mathscr{F}(\chi))$, multiplication by $(\chi^{-1}(g) - 1)$ is zero. Taking g such that $\chi(g) \neq 1$, we obtain 2.7.

Lemma 2.8. *Let X and Y be two schemes over an algebraically closed field k, with X separated and of finite type and Y connected. Let $\mathscr{F}$ be a sheaf on X and let (ρ, ε) be a family of endomorphisms of $(X, \mathscr{F})$ parametrized by Y:*

$$\rho : Y \times_k X \to Y \times_k X \qquad \textit{is a } Y\textit{-morphism, and}$$
$$\varepsilon : \rho^* \mathrm{pr}_2^* \mathscr{F} \to \mathrm{pr}_2^* \mathscr{F} \qquad \textit{a morphism of sheaves.}$$

Assume that ρ is proper. For $y \in Y(k)$, let $\rho_H(y)^*$ denote the endomorphism of $\mathrm{H}_c^\bullet(X, \mathscr{F})$ induced by $\rho_y : X \to X$ and $\varepsilon_y : \rho_y^* \mathscr{F} \to \mathscr{F}$. Then, $\rho_H(y)^*$ is independent of y.

Indeed, $\mathsf{R}^p \mathrm{pr}_{1!} \mathrm{pr}_2^* \mathscr{F}$ is the constant sheaf on Y with fiber $\mathrm{H}^p(X, \mathscr{F})$, and $\rho_H(y)^*$ is the fiber at y of the endomorphism

$$\mathsf{R}^p \mathrm{pr}_{1!} \mathrm{pr}_2^* \mathscr{F} \xrightarrow{\rho^*} \mathsf{R}^p \mathrm{pr}_{1!} \rho^* \mathrm{pr}_2^* \mathscr{F} \xrightarrow{\varepsilon} \mathsf{R}^p \mathrm{pr}_{1!} \mathrm{pr}_2^* \mathscr{F}$$

of this sheaf.

2.9 Remark

Let G_0 be a connected algebraic group over $\mathbb{F}_q$ and $\rho : G_0(\mathbb{F}_q) \to \mathrm{GL}(V)$ a linear representation such that $V^{G_0(\mathbb{F}_q)} = 0$. Then again $\mathrm{H}_c^\bullet(G, \rho(L)) = 0$.

3 Sums in one variable

3.1

Weil was the first to apply "cohomological" methods to the study of trigonometric sums in one variable; since he had proved the analogue of the Riemann hypothesis for Artin's L functions over function fields,

these methods gave him excellent estimates (of order O(square root of the number of terms)) and an understanding of how these sums behave under "extension of the base field."

The substance of this section is a presentation, in cohomological language, of his results.

3.2

Cohomological methods here lead us to compute the groups $\mathrm{H}^i_c(X, \mathscr{F})$, where $\mathscr{F}_0$ is an E_λ-sheaf on a curve $X - 0$ over $\mathbb{F}_q$. These groups vanish for $i \neq 0, 1, 2$, and for $i = 0, 2$, they have a simple interpretation (1.8 a,b,c). Furthermore, the Euler-Poincaré characteristic $\chi_c(\mathscr{F}) = \sum(-1)^i \dim \mathrm{H}^i_c(X, \mathscr{F})$ (which is also equal to $\chi(\mathscr{F}) = \sum(-1)^i \dim \mathrm{H}^i(X, \mathscr{F})$) can be computed in terms of x, the rank of $\mathscr{F}$ at the generic points of X, and the ramification of $\mathscr{F}$. The essential result is the following.

Let $\bar{X}$ be a smooth connected projective curve of genus g over an algebraically closed field k, let X be the dense open subset of $\bar{X}$ complementary to a finite set S of points, and let $\mathscr{F}$ be a smooth E_λ-sheaf on X. Let $\chi(X) = 2 - 2g - \#S$ be the Euler–Poincaré characteristic of X, and let $\mathrm{rg}(\mathscr{F})$ be the rank of $\mathscr{F}$. For each point $s \in S$, we define an integer $\mathrm{Sw}_s(\mathscr{F})$, the Swan conductor, measuring the wild ramification of $\mathscr{F}$, and

$$\chi_c(\mathscr{F}) = \mathrm{rg}(\mathscr{F}) \cdot \chi(X) - \sum_{s \in S} \mathrm{Sw}_s(\mathscr{F}) \tag{3.2.1}$$

[29].

3.3

Let $\bar{X}$, $j : X \hookrightarrow \bar{X}$ and $\mathscr{F}$ as in 3.2. The Poincaré duality theorem takes the particularly convenient form: $\mathrm{H}^i(\bar{X}, j_*\mathscr{F})$ and $\mathrm{H}^{2-i}(\bar{X}, j_*(\mathscr{F}^\vee(1)))$ are dual to each other (Duality, 1.3).

3.4

Take $k = \mathbb{F}$, and suppose that X, $\bar{X}$, and $\mathscr{F}$ arise from X_0, $\bar{X}_0$, and $\mathscr{F}_0$ over $\mathbb{F}_q$. If $\mathscr{F}_0$ becomes trivial on a finite covering of X_0, Weil proved that the complex conjugates of the eigenvalues of F^* on $\mathrm{H}^i(\bar{X}, j_*\mathscr{F})$ have absolute value $q^{i/2}$. Such sheaves $\mathscr{F}_0$ correspond to Artin L-functions over the function field of $\bar{X}_0$.

3.5

Under the same hypotheses, or more generally if E_λ is the completion at a place λ of a number field E and if $\mathscr{F}_0$ belongs to an infinite compatible system of v-adic representations (v a place of E), we can compute

$$\prod_i \det(-F^*, \mathrm{H}^i(X, \mathscr{F}))^{(-1)^i}$$

from *local* information on $\mathscr{F}_0$. This is explained in [13]. For trivial $\mathscr{F}_0$ on a finite covering of X_0, it is the expression of the constant in the functional equation of an Artin L-function as a product of local constants.

3.5 Example

Let ψ be the character $\exp\left(\frac{2\pi i}{p} \mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}\right)$ of $\mathbb{F}_q$; we have $\psi(a^p - a) = 0$. Let X_0 be a smooth projective geometrically irreducible curve of genus g over $\mathbb{F}_q$, and let f be a rational function on X_0, i.e. a morphism

$f : X_0 \to \mathbb{P}^1$, not identically equal to ∞. We are interested in the sum

$$S'_f = \sum_{\substack{x \in X_0(\mathbb{F}_q) \\ f(x) \neq \infty}} \psi(f(x)).$$

This sum is zero for f of the form $g^p - g$. This suggests modifying the sum S'_f as follows:

a) for any closed point x of X_0, set $v_x(f) =$ the order of the pole of f at x if $f(x) = \infty$, and $v_x(f) = 0$ otherwise; also set $v_x^*(f) = \inf v_x(f + g^p - g)$ (the infimum being over g).

b) If $v_x^*(f) = 0$, $x \in X_0(\mathbb{F}_q)$, and $f + g^p - g$ is regular at x, put $\psi(f(x)) = \psi((f + g^p - g)(x))$. Finally set

$$S_f = \sum_{x \in X_0(\mathbb{F}_q)}{}' \psi(f(x)), \tag{3.5.1}$$

where $(-)'$ indicates that the sum runs over those x for which $v_x^*(f) = 0$.

We have $S_f = S_{f+g^p-g}$. We shall verify that if f is not of the form $g^p - g + c^{te}$, then

$$|S_f| \leqslant \left(2g - 2 + \sum_{v_x^*(f) \neq 0} [k(x) : \mathbb{F}_q](1 + v_x^*(f)) \right) q^{1/2}. \tag{3.5.2}$$

We begin by passing from complex coefficients to ℓ-adic coefficients, replacing ψ by a character of the form $\psi_0 \circ \mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}$, with $\psi_0 : \mathbb{F}_p \hookrightarrow E_\lambda^\times$. For $j : U_0 \hookrightarrow X_0$ the inclusion of the open where $f \neq \infty$, we then prove that

$$S_f = \sum (-1)^i \,\mathrm{tr}(F^*, \mathrm{H}^i(X, j_* \mathscr{F}(\psi f))). \tag{3.5.3}$$

The trace formula reduces this statement to the following (for $x \in X_0(\mathbb{F}_q)$):

a) If $v_x^*(f) = 0$, then F_x^* on $j_* \mathscr{F}(\psi f)$ is $\psi(f(x))$;

b) If $v_x^*(f) \neq 0$, then $j_* \mathscr{F}(\psi f)$ is ramified at x: $(j_* \mathscr{F}(\psi f))_{\bar{x}} = 0$.

Formula a) follows from (1.7.6) if $v_x(f) = 0$, and the general case reduces to this one by replacing f by $f + g^p - g$: up to isomorphism, this does not change $\mathscr{F}(\psi f)_0$ on the open where f and g are regular (1.3).

Formula b) follows from

$$\mathrm{Sw}_x \mathscr{F}(f, \psi) = v_x^*(f). \tag{3.5.4}$$

After reduction to the case $v_x(f) = v_x^*(f)$ (hence $p \nmid v_x(f)$) and extension of scalars to $\mathbb{F}$, this formula can be found in [34, 4.4].

Finally, we check that the sheaf $\mathscr{F}(\psi f)$ on U is nonconstant if f is not of the form $g^p - g + c^{te}$: since ψ_0 is injective, the triviality of $\mathscr{F}(\psi f) = \mathscr{F}(\psi_0 f)$ (1.8.ii) is equivalent to that of the $\mathbb{F}_q$-torsor defined by the equation; it would be the inverse image of a $\mathbb{F}_q$-torsor on $\mathrm{Spec}(\mathbb{F}_q)$, with equation $T^p - T - \lambda = 0$, and the $\mathbb{F}_p$-torsor equation $T^p - T - (f - \lambda) = 0$ on U_0 would be trivial and f would therefore be of the form $g^p - g + \lambda$. The $\mathrm{H}^i(\bar{X}, j_* \mathscr{F}(\psi f))$ therefore vanish for $i \neq 1$, and formula (3.5.3) reduces to

$$S_f = -\,\mathrm{tr}(F^*, \mathrm{H}^1(\bar{X}, j_* \mathscr{F}(\psi f))) \tag{3.5.5}$$

and, by (3.2.1) and (3.5.4), the dimension of H^1 is $2g - 2 + \sum_{v_x^*(f) > 0} [k(x) : \mathbb{F}_q](1 + v_x^*(f))$; moreover, $-S_f$ is the sum of this many eigenvalues of F^*, each having complex absolute value $q^{1/2}$.

3.6

Suppose that, for an automorphism σ of X_0, we have

$$f(\sigma x) = -f(x).$$

The sum S_f is then real. If σ is an involution, Poincaré duality allows us to say a little more: set $\mathscr{F}_0 = j_*\mathscr{F}(\psi f)$, $\mathscr{G}_0 = j_*\mathscr{F}(\psi(-f))$, we have

a) $\mathscr{F}_0$ and $\mathscr{G}_0$ are in duality,

b) $\sigma^*\mathscr{F}_0 \xrightarrow{\sim} \mathscr{G}_0$ and $\sigma^*\mathscr{G}_0 \xrightarrow{\sim} \mathscr{F}_0$, for natural isomorphisms such that the composite $\mathscr{F}_0 = (\sigma^2)^*\mathscr{F}_0 \xrightarrow{\sim} \sigma^*\mathscr{G}_0 \xrightarrow{\sim} \mathscr{F}_0$ is the identity.

c) the pairing $\mathscr{F}_0 \otimes \mathscr{G}_0 \to E_\lambda$ satisfies $\sigma^*(f \cdot g) = \sigma^*(f) \cdot \sigma^*(g)$.

Passing to cohomology, we find that $\mathrm{H}^1(X, \mathscr{F})$ and $\mathrm{H}^1(X, \mathscr{F})$ are in perfect duality with values in $E_\lambda(1)$, and that the bilinear form $\alpha \cdot \sigma^*\beta$ on $\mathrm{H}^1(X, \mathscr{F})$ is *alternating*: σ^* acts trivially on $E_\lambda(1)$, and

$$\alpha \cdot \sigma^*\beta = \sigma^*(\alpha \cdot \sigma^*\beta) = \sigma^*\alpha \cdot \beta = -\beta \cdot \sigma^*\alpha.$$

We conclude that $\mathrm{H}^1(X, \mathscr{F})$ is of even dimension (we also check easily that each nonzero $v_x^*(f)$ is odd) and that the eigenvalues of F^* on $\mathrm{H}^1(X, \mathscr{F})$ occur in pairs α and q/α.

3.7 Example

This applies to the Kloosterman sums $\sum_{x \in \mathbb{F}_q^\times} \psi\left(x + \frac{a}{x}\right)$ (take $X_0 = \mathbb{P}^1$, $f = x + \frac{a}{x}$, $\sigma x = -x$; the poles of f are 0 and ∞, and at each of them $v_x^*(f) = 1$; the dimension of H^1 is $-2 + 2 + 2 = 2$). We therefore have (for $a \neq 0$)

$$\sum_{\substack{xy=a \\ x,y \in \mathbb{F}_{q^n}}} \exp\left(\frac{2\pi i}{p} \operatorname{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_p}(x+y)\right) = (-\alpha^n + \alpha^{-n}) \qquad , \alpha\bar{\alpha} = q.$$

This result is due to L. Carlitz [9].

We now give an application of a Lang–Weil method.

Proposition 3.8. *Let $P \in \mathbb{F}_q[X_1, \ldots, X_n]$ be a polynomial in n variables, of degree d, which is not of the form $Q^p - Q + C^{te}$. Again setting $\psi(x) = \exp\left(\frac{2\pi i}{p} \operatorname{tr}_{\mathbb{F}_q/\mathbb{F}_p}(x)\right)$, we have*

$$\left| \sum_{x_i \in \mathbb{F}_q} \psi P(x_1, \ldots, x_n) \right| \leqslant (d+1) q^{n - \frac{1}{2}}.$$

For the left-hand side we have the trivial estimate q^n. It is therefore enough to prove 3.8 when $d - 1 < q^{1/2}$; assume therefore that $d < q + 1$, and let P_d be the homogeneous part of degree d of P. Recall:

Lemma 3.9. *A hypersurface of degree d in $\mathbb{P}^r$ ($r \geqslant i$) cannot contain all rational points over $\mathbb{F}_q$ unless $d \geqslant q + 1$.*

We argue by induction: if there exists a rational hyperplane not entirely contained in the hypersurface (this is not the case for $r = 1$), we apply the induction hypothesis to the intersection of the hypersurface with this hyperplane. Otherwise, the degree d is $\geqslant$ the number of rational hyperplanes, $\geqslant q + 1$.

Let us now distinguish two cases.

Case 1: $p \nmid d$**.** Applying the lemma to P_d, we see that, after making a linear change of variables if necessary, we may assume that $P_d(1,0,\dots,0) \neq 0$, i.e. that the coefficient of X_1^d in P is nonzero. A polynomial in 1 variable

$$S(X) = \sum_{i=0}^{d} a_i x^i,$$

with $a_d \neq 0$ ($p \nmid d$) is never of the form $Q^p - Q + C^{te}$, and the (3.5.2) estimate is reduced to

$$\left|\sum \psi(S(x))\right| \leqslant (d-1)q^{1/2}.$$

Applying this estimate to the partial sums obtained by varying only x_1, we find

$$\left|\sum \psi P(x_1,\dots,x_n)\right| = \left|\sum_{x_2,\dots,x_n}\sum_{x_1} \psi P(x_1,\dots,x_n)\right| \leqslant q^{n-1}(d-1)q^{1/2}$$

as promised.

Case 2: $p \mid d$ **($d > 0$).** If P_d is a p-th power, replacing P by $P - (P_d - P_d^{1/p})$ does not change the sum under consideration and lowers the degree: this case is disposed of by induction on d. Otherwise, the differential of P_d is not identically zero; applying 3.9, one can assume, after a linear change of variables if necessary, that it is nonzero at the point $(1,0,\dots,0)$. If we write

$$P_d = \sum_{i=0}^{d} X_1^{d-i} S_i(X_2,\dots,X_n),$$

this means that the linear form S_1 is not identically zero.

The form S_0 is constant, and by replacing P with $P - (S_0 X_1^d - S_0^{1/p} X_1^{d/p})$, we can assume that it is zero. Finally, denote by $-\lambda$ the coefficient of X_1^{d-1} in P. For fixed $x_2,\dots,x_n$, we have

$$P(X_1,x_2,\dots,x_n) = (S_1(x_2,\dots,x_n) - \lambda)X_1^{d-1} + \text{terms of lower degree in } X_1$$

and if $S_1(x_2,\dots,x_n) \neq \lambda$, we therefore have

$$\left|\sum_{x_1} \psi P(x_1,\dots,x_n)\right| \leqslant (d-2)q^{1/2}.$$

In total,

$$\begin{aligned}\left|\sum \psi P(x_1,\dots,x_n)\right| &\leqslant \left|\sum_{S=\lambda} \psi P(x_1,\dots,x_n)\right| + \sum_{S(X_2,\dots,X_n)\neq\lambda} \left|\sum_{x_1} \psi P(x_1,\dots,x_n)\right| \\ &\leqslant q^{n-1} + (q^{n-1} - q^{n-2})(d-2)q^{1/2} \\ &< (d-2)q^{n-1/2} + q^{n-1} \\ &< (d-1)q^{n-1/2}.\end{aligned}$$

Another result of this nature is given by R.A. Smith [37].

4 Gauss and Jacobi sums

4.1

Let k be a finite field of characteristic p, let χ be a character of $k^\times$, and let ψ be a nontrivial character of the additive group of k. We take the following as the definition of Gauss sums:

$$\tau(\chi,\psi) = -\sum_{x\in k^\times} \psi(x)\chi^{-1}(x) \tag{4.1.1}$$

(note the sign). Classically, χ and ψ are complex-valued. We take them to have values in $E_\lambda^\times$ (see the proof of 1.15). Regard $-\tau(\chi,\psi)$ as a sum over the rational points of the scheme $\mathbb{G}_m$ over k. By Section 1, we have $-\tau(\chi,\psi) = \mathrm{tr}(F^*, \mathrm{H}_c^*(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1}))$. Moreover,

Proposition 4.2. *The cohomology of $\mathbb{G}_m$ with coefficients in $\mathscr{F}(\psi\chi^{-1})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq 1$, *and* $\dim \mathrm{H}_c^1 = 1$.

(ii) F^*, *acting on* H_c^1, *is multiplication by* $\tau(\chi,\psi)$.

(iii) If χ *is nontrivial, we have* $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

Proof. For any n prime to p, denote by K_n the $\boldsymbol{\mu}_n$-torsor on $\mathbb{G}_m$ defined by the exact Kümmer sequence $0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \xrightarrow{x^n} \mathbb{G}_m \to 0$. If k has q elements, then over k one has $\boldsymbol{\mu}_{q-1} = k^\times$, and the Lang torsor on $\mathbb{G}_m/k$ is K_{q-1}. Hence $\mathscr{F}(\chi^{-1}) = \chi(K_{q-1})$. Assertion 4.2(ii) follows from (i,iii), which in turn follow from the following geometric statement, where $\mathscr{F}(\psi)$ denotes the sheaf on $\mathbb{G}_m/\mathbb{F}$ obtained by extending scalars from k to $\mathbb{F}$ in $\mathscr{F}(\psi)$ on $\mathbb{G}_m/k$ (1.8.b). □

Proposition 4.3. *Let* $\chi : \boldsymbol{\mu}_n \to E_\lambda^\times$. *The cohomology of* $\mathbb{G}_m$ *with coefficients in* $\mathscr{F}(\psi)\otimes\chi(K_n)$ *satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq 1$, *and* $\dim \mathrm{H}_c^1 = 1$.

(ii) If χ *is nontrivial, we have* $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

Proof. The sheaf $\mathscr{F}(\psi)$ is the restriction to $\mathbb{G}_m$ of a sheaf locally constant on $\mathbb{G}_a$, wildly ramified at infinity, with Swan conductor 1. The sheaf $\chi(K_n)$ is constant if $\chi = 1$; if $\chi \neq 1$, it is tamely ramified at 0 and ∞.

The sheaf $\mathscr{F}(\psi)\otimes\chi(K_n)$ is therefore ramified at ∞; applying 1.18(b,c), we find that $\mathrm{H}_c^i(\mathbb{G}_m, \mathscr{F}(\psi)\otimes\chi(K_n)) = 0$ for $i \neq 1$. If $\chi \neq 1$, it is ramified at 0 and ∞, and (ii) follows from 1.19.a.

The Swan conductors are 0 at 0 and 1 at ∞. By (3.2.1), the Euler–Poincaré characteristic is therefore -1, which completes the proof. □

4.4 Remark

If χ is nontrivial, 4.2(iii) and Poincaré duality show that $\mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1}))$ and $\mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi^{-1}\chi))$ are in duality (duality with values in $E_\lambda(-1)$). We therefore have $\tau(\chi,\psi)\cdot\tau(\chi^{-1},\psi^{-1}) = q$, i.e. $|\tau(\chi,\psi)| = 1$.

4.5

If k is an extension of degree N of $\mathbb{F}_q$, we can also regard (4.1.1) as a sum in N variables over $\mathbb{F}_q$. More generally, let k be an étale algebra over $\mathbb{F}_q$, of degree N over $\mathbb{F}_q$. It is a product of fields k_i of degrees N_i, and we set

$$\varepsilon(k) = (-1)^{\sum(N_i+1)}. \tag{4.5.1}$$

This is the sign of the permutation of $S = \hom_{\mathbb{F}_q}(k, \mathbb{F})$ induced by the Frobenius substitution $\varphi \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$.

Let $\psi : \mathbb{F}_q \to E_\lambda^\times$ be a nontrivial character, and let $\chi : k^\times \to E_\lambda^\times$. We set

$$\tau_{\mathbb{F}_q}(\chi, \psi) = (-1)^N \sum_{x \in k^\times} \psi \operatorname{tr}_{k/\mathbb{F}_q}(x) \cdot \chi^{-1}(x). \tag{4.5.2}$$

If χ has coordinates $\chi_i : k_i^\times \to E_\lambda^\times$, we have the elementary identity

$$\tau_{\mathbb{F}_q}(\chi, \psi) = \varepsilon(k) \prod \tau(\chi_i, \psi \circ \operatorname{tr}_{k_i/\mathbb{F}_q}). \tag{4.5.3}$$

After some preliminaries, we give in 4.10 a cohomological interpretation of the sums (4.5.2). In 4.12, we will interpret the Hasse–Davenport identity as a twisted form (cf. 1.12) of the following special case of (4.5.3): for $k = \mathbb{F}_q^N$, and χ a character of $\mathbb{F}_q^\times$, we have $\tau_{\mathbb{F}_q}(\chi \circ N_{k/\mathbb{F}_q}, \psi) = \tau(\chi, \psi)^N$.

4.6

Recall that for M a finitely generated projective module over a ring A, the functor $\mathrm{Spec}(B) \mapsto M \otimes_A B$ from affine schemes over $\mathrm{Spec}(A)$ to Ens is represented by the affine scheme $\mathbb{V}(M^\vee)$ (EGA notation), the spectrum of $\mathrm{Sym}_A^\bullet(M^\vee)$. If $M = A^n$, this is the standard affine space of dimension n.

Suppose that M is a unital A-algebra, and let $V = \mathbb{V}(M^\vee)$. By definition, for any B equipped with an A-algebra structure, the set $V(B)$ of points of V with coordinates in B is $M \otimes_A B$. Since the functor $B \mapsto M \otimes_A B$ takes values in unital rings, V is a unital ring scheme over $\mathrm{Spec}(A)$. The norm and trace morphisms $M \otimes_A B \to B$ are functorial in B; they therefore correspond to morphisms of schemes N and T from V to $\mathbb{G}_a$. We shall consider the following schemes obtained from V:

a) V^* is the open subscheme of invertible elements of V (a group scheme under $\cdot$);

b) W is the hyperplane of equation $T = 0$ and $W^* = W \cap V^*$;

c) P is the hyperplane at infinity $V \smallsetminus \{0\}/\mathbb{G}_m$ of the affine space V over $\mathrm{Spec}(A)$. If Q is the hyperplane at infinity of W, $W^*/\mathbb{G}_m$ and $V^*/\mathbb{G}_m$ are open subsets of the projective spaces Q and P over $\mathrm{Spec}(A)$.

$$\begin{array}{ccc} W^* & \hookrightarrow & V^* \\ \downarrow{\scriptstyle\pi} & & \downarrow{\scriptstyle\pi} \\ W^*/\mathbb{G}_m & \hookrightarrow & V^*/\mathbb{G}_m \end{array} \qquad \subset \qquad \begin{array}{ccc} W \smallsetminus \{0\} & \hookrightarrow & V \smallsetminus \{0\} \\ \downarrow{\scriptstyle\pi} & & \downarrow{\scriptstyle\pi} \\ Q & \hookrightarrow & P. \end{array} \tag{4.6.1}$$

If $M = A^I$, I a finite set, then $V \simeq \mathbb{G}_a^I$, $V^* = \mathbb{G}_m^I$, $V^*/\mathbb{G}_m$ is the torus $\mathbb{G}_m^I/(\mathbb{G}_m \text{ diagonal})$, N and T are given by $\prod x_i$ and $\sum x_i$, and Q is therefore the projective hyperplane with equation $\sum x_i = 0$ in P.

The case of interest is that in which M is finite étale over A. The previous description then applies locally to $\mathrm{Spec}(A)$ (for the étale topology) and we can write $V^* = \mathbb{G}_m^I$, for I a locally constant sheaf of finite sets on $\mathrm{Spec}(A)$. If, for example, A is a field with algebraic closure $\bar{A}$, and M is a separable A-algebra, we put $I = \hom_A(M, \bar{A})$, and over $\bar{A}$ we have $V \sim \mathbb{G}_a^I$. Via this isomorphism, N and T are given by $\prod x_i$ and $\sum x_i$.

4.7 The Kummer torsor

Let S be a scheme, let n be an integer invertible on S, and let G be a commutative group scheme with connected fibers (written multiplicatively) over S. The sequence

$$0 \longrightarrow G_n \longrightarrow G \xrightarrow{x^n} G \longrightarrow 0$$

is exact. It defines a G_n-torsor $K_n(G)$ (or simply K_n) on G. For $\chi : G_n \to E_\lambda^\times$ a homomorphism from the étale S-sheaf G_n to the constant sheaf $E_\lambda^\times$, we denote by $\mathscr{K}_n(\chi)$ the E_λ-sheaf $\chi^{-1}(K_n)$.

The K_n torsors form an inverse system of torsors under the projective system of groups G_n (transition morphisms $x \mapsto x^d : G_{nd} \to G_n$). This expresses the commutativity of the diagrams

$$\begin{array}{ccccccccc} 0 & \longrightarrow & G_{nd} & \longrightarrow & G & \xrightarrow{x^{nd}} & G & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle x^d} & & \downarrow{\scriptstyle x^d} & & \| & & \\ 0 & \longrightarrow & G_n & \longrightarrow & G & \xrightarrow{x^n} & G & \longrightarrow & 0. \end{array}$$

We therefore have $\mathscr{K}_n(\chi) = \mathscr{K}_{nd}(\chi \circ x^d)$.

4.8

Apply the construction of 4.6 with $A = \mathbb{F}_q$ and with $M = k$ an étale algebra over $\mathbb{F}_q$. In accordance with our general conventions, we denote by a subscript 0 the $\mathbb{F}_q$-schemes denoted $V, V^*, \ldots$ in 4.6. The same letters without a subscript denote the schemes over $\mathbb{F}$ obtained by extension of scalars.

We put $I = \hom(k, \mathbb{F})$; then $V \sim \mathbb{G}_a^I$ and T is given by $(x_i) \mapsto \sum x_i$.

$$\mathscr{F}(\psi \circ \mathrm{tr}_{k/\mathbb{F}_q}) = T^* \mathscr{F}(\psi) = \bigotimes_i \mathrm{pr}_i^* \mathscr{F}(\psi). \tag{4.8.1}$$

Since $V^* \sim \mathbb{G} + m^I$, a character χ of $V_n^* \sim \boldsymbol{\mu}_n^I$, with values in $E_\lambda^\times$, amounts to a family $(\chi_i)_{i \in I}$ of characters of $\boldsymbol{\mu}_n$ indexed by I. It is defined over $\mathbb{F}_q$ if, for all $\sigma \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$, we have $\chi_{\sigma_i} = \chi_i \circ \sigma^{-1}$. It then defines a E_λ-sheaf $\mathscr{K}_n((\chi_i)_{i \in I})$ on V_0^*. If the product of the χ_i is trivial, χ factors through a character of $(V_0^*/\mathbb{G}_m)_n$ and $\mathscr{K}_n((\chi_i)_{i \in I})$ is the inverse image of a sheaf, denoted in the same way, on $V - 0^*/\mathbb{G}_m$. This construction is related to the Lang torsor as follows.

Lemma 4.9. *(i) For n sufficiently divisible, we have a commutative diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & (V_0^*)_n & \longrightarrow & V_0^* & \xrightarrow{x^n} & V_0^* & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \tau} & & \downarrow{\scriptstyle \tau} & & \| & & \\ 0 & \longrightarrow & k^\times & \longrightarrow & V_0^* & \xrightarrow{\mathfrak{L}} & V_0^* & \longrightarrow & 0; \end{array}$$

if χ is a character $\chi : k^\times \to E_\lambda^\times$, we have $\mathscr{F}(\chi) = \mathscr{K}_n(\chi \circ \tau)$.

(ii) For sufficiently divisible n, τ identifies $k^\times$ with the coinvariants of $\mathrm{Gal}(\bar{\mathbb{F}}/F)$ acting on $V_n^* \sim \boldsymbol{\mu}_n^I$.

(iii) For k a field, $N = [k : \mathbb{F}_q]$, $\omega \in I$ an embedding of k in $\mathbb{F}$, and $n = q^N - 1$, the index component ω of $\chi \circ \tau$ is $\chi \circ \omega^{-1}$.

(iv) If χ is nontrivial on each factor of k, the $(\chi \circ \tau)_i$ are all nontrivial.

It is enough to prove the lemma when k is a field of degree N over $\mathbb{F}_q$. In this case, n is "sufficiently divisible" if $q^N - 1 \mid n$. Choose an embedding ω of k into $\mathbb{F}_q$; then I is identified with $\mathbb{Z}/N$: the element $i \in \mathbb{Z}/N$ corresponds to $\omega_i = \omega^{q^i}$. Via the isomorphism $V_0^*(\mathbb{F}) = F^{\times I}$, we have

a) $x \in k^\times$ corresponds to $(\omega_i(x)) \in \mathbb{F}^{\times I}$;

b) $F((x_i)_{i\in\mathbb{Z}/N}) = (x_{i-1}^q)_{i\in\mathbb{Z}/N}$.

A character $\chi = (\chi_i)$ of $V_n^* \sim \boldsymbol{\mu}_n^I$ is defined over $\mathbb{F}_q$ if $\chi_{-i} = \chi_0(x^{q^i})$ $(i \in \mathbb{Z})$. If $q^N - 1 \mid n$, there are $q^N - 1$ such characters: χ_0 factors through $\boldsymbol{\mu}_{q^N-1}$ and determines the χ_i. To prove (ii), it is therefore sufficient to verify (i) and (iii) (or its corollary (iv)) which ensures that $\chi \mapsto \chi \circ \tau$ is injective.

Let us prove (i) and (iii) for $n = q^N - 1$. Write the group of endomorphisms of $V^* \sim \mathbb{G}_m^I$ additively; in particular, denote by n the $x \mapsto x^n$ operator. If α is the cyclic permutation operator $(x_i) \mapsto (x_{i-1})$, we have $q^N - 1 = (q\alpha)^N - 1 = (q\alpha - 1)((q\alpha)^{N-1} + \cdots + 1)$. This determines τ: we have $\tau((x_i)) = \prod_{0\leqslant j<N} x_{i-j}^{q^j}$, and the induced map from $(V_0^*)_n$ to $k^\times$ is [as in a) above] $(x_i) \mapsto \prod x_{-i}^{q^i}$; hence (iii) follows.

Proposition 4.10. *The cohomology of V^* with coefficients in $\mathscr{F}(\chi^{-1} \cdot \psi \operatorname{tr}_{k/\mathbb{F}_q})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq N$, *and* $\dim \mathrm{H}_c^N = 1$.

(ii) On H_c^N, F^* *is multiplication by* $\tau_{\mathbb{F}_q}(\chi, \psi)$.

(iii) If χ *is nontrivial on each factor of* k, *we have* $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

Applying 4.9(i), over $\mathbb{F}$, if $\chi \circ \tau = (\chi_i)_{i\in I}$, we have $\mathscr{F}(\chi) = \mathscr{K}_n((\chi_i)_{i\in I})$. Points (i) and (iii) therefore follow from the following more geometric statement (for (iii), apply 4.9(iv)), and (ii) follows from the trace formula:

Proposition 4.11. *Let $(\chi_i)_{i\in I}$ be a family of characters of $\boldsymbol{\mu}_n(\mathbb{F})$. The cohomology of $V^* \simeq \mathbb{G}_m^I$ with coefficients in $\mathscr{K}_n((\chi_i)_{i\in I}) \otimes \mathscr{F}(\psi \operatorname{tr}_{k/\mathbb{F}_q})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq N$, *and* $\dim \mathrm{H}_c^N = 1$.

(ii) If the χ_i are all nontrivial, we have $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

We have $V^* \sim \mathbb{G}_m^I$, $\mathscr{K}((\chi_i)) = \bigotimes_i \mathrm{pr}_i^* \chi_i(K_n(\mathbb{G}_m))$ and $\mathscr{F}(\psi \circ \operatorname{tr}_{\mathbb{F}_q/k}) = T^*\mathscr{F}(\psi) = \bigotimes_i \mathrm{pr}_i^* \mathscr{F}(\psi)$. The Künneth formula then applies, and 4.11 follows from 4.3.

4.12

From 2.4* and (Cycle, 1.3 Example 2), it also follows that the group of permutations σ of I such that $\chi_i = \chi_{\sigma i}$ $(i \in I)$ acts on H_c^N by multiplication by the sign $\varepsilon(\sigma)$. We deduce a second proof of the identity of Hasse-Davenport.

If χ is a character of $\mathbb{F}_q^\times$, the fact that, over $\mathbb{F}$, N is given by $(x_i) \mapsto \prod x_i$ gives $\mathscr{F}(\chi \circ N) = N^*\mathscr{F}(\chi) = \bigotimes_i \mathrm{pr}_i^* \mathscr{F}(\psi)$ and the Künneth formula gives

$$\mathrm{H}_c^\bullet(V^*, \mathscr{F}(\chi^{-1} \circ N, \psi \circ \mathrm{tr})) \sim \mathrm{H}_c^\bullet(\mathbb{G}_m, \mathscr{F}(\chi^{-1}\psi))^{\otimes I},$$

where the tensor product on the right-hand side is taken in the sense of 2.4*. Since $\mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1}))$ is of dimension 1, its tensor power $\otimes I$, in the ordinary sense, depends only on the cardinality n of I. Applying (Cycle, 1.3 Example 2), we obtain a canonical isomorphism

$$\mathrm{H}_c^N\left(V, \mathscr{F}(\chi^{-1} \circ N) \otimes T^*\mathscr{F}(\psi)\right) \sim \mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1})) \otimes_{\mathbb{Z}} \bigwedge^N \mathbb{Z}^I. \tag{4.12.1}$$

To compute the action of F^*, it is most convenient to adopt the Galois point of view and observe that, since the isomorphism (4.12.1) is canonical, it is compatible with the transport-of-structure action of $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$. If $\varepsilon(k)$ is the sign of the permutation φ of I, we find that

$$\begin{aligned}\tau_{\mathbb{F}_q}(\chi\circ N_{k/\mathbb{F}_q},\psi) &= \mathrm{tr}\left(\varphi^{-1}, \mathrm{H}_c^N\left(V^*, N^*\mathscr{F}(\chi^{-1})\otimes T^*\mathscr{F}(\psi)\right)\right)\\ &= \varepsilon(k)\,\mathrm{tr}\left(\varphi^{-1}, \mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\chi^{-1}\psi))\right)^N\\ &= \varepsilon(k)\tau(\chi,\psi)^N,\end{aligned}$$

that is,

$$\tau_{\mathbb{F}_q}\left(\chi\circ N_{k/\mathbb{F}_q},\psi\right) = \varepsilon(k)\tau(\chi,\psi)^N. \tag{4.12.2}$$

If k is a field, φ is a cyclic permutation of I, $\varepsilon(k) = (-1)^{N+1}$, and we obtain the Hasse–Davenport identity:

$$\tau\left(\chi\circ N_{k/\mathbb{F}_q}, \psi\circ \mathrm{tr}_{k/\mathbb{F}_q}\right) = (-1)^{N+1}\tau_{\mathbb{F}_q}\left(\chi\circ N_{k/\mathbb{F}_q},\psi\right) = \tau(\chi,\psi)^N.$$

Lemma 4.13. *For $\chi\circ\tau = (\chi_i)_{i\in I}$ as in 4.9(i), the following conditions are equivalent*

(i) $\chi|\mathbb{F}_q^\times$ is trivial.

(ii) The product of χ_i is trivial.

(iii) $\mathscr{F}(\chi) = \mathscr{K}_n((\chi_i))$ is the inverse image of a unique sheaf on $V^*/\mathbb{G}_m$.

(iv) The inverse image of $\mathscr{F}(\chi)$ on $\mathbb{G}_m$ (embedded in V^* by the structural morphism $\mathbb{F}_q\to k$) is trivial.

(i)⇒(ii). We regard χ as a character of $V^*/\mathbb{G}_m(\mathbb{F}_q) = k^\times/\mathbb{F}_q^\times$, and we take $\mathscr{F}(\chi)$ on $V^*/\mathbb{G}_m$.
(ii)⇒(iii). Likewise, we regard (χ_i) as a character of $(V^*/\mathbb{G}_m)_n$. The uniqueness in (iii) follows from the fact that V^* is a fibration with connected fibers over $V^*/\mathbb{G}_m$, and (iii)⇒(iv) is trivial.
(iv)⇒(i), (ii). This inverse image is $\mathscr{F}(\chi|\mathbb{F}_q^\times)$ and $\mathscr{K}_n(\prod\chi_i)$.

4.14

Let k be an étale algebra over $\mathbb{F}_q$, of dimension $N+1$, and let χ be a nontrivial character of $k^\times$, trivial on $\mathbb{F}_q^\times$. The Jacobi sum $J(\chi)$ is defined by

$$J(\chi) = (-1)^{N-1}\sum_{\substack{x\in k^\times/\mathbb{F}_q^\times\\ \mathrm{tr}(x)=0}}\chi^{-1}(x). \tag{4.14.1}$$

We have the following identity between Gauss sums and Jacobi sums.

Proposition 4.15. *For χ as above and ψ a nontrivial additive character of $\mathbb{F}_q$, we have*

$$qJ(\chi) = \tau_{\mathbb{F}_q}(\chi,\psi).$$

In the special case $k=\mathbb{F}_q^n$, formula (4.5.3) rewrites this as

$$qJ(\chi) = \prod_i \tau(\chi_i,\psi), \tag{4.15.1}$$

or equivalently as

$$\chi_0(-1)J(\chi) = (-1)^{N-1} \sum_{\substack{x_1,\dots,x_N \in \mathbb{F}_q^\times \\ \sum x_i = 1}} \prod_{i=1}^{N} \chi_i^{-1}(x_i) = \tau(\chi_0^{-1},\psi)^{-1} \prod_{i=1}^{N} \tau(\chi_i,\psi)$$

$$\chi_0^{-1} = \prod_{i=1}^{N} \chi_i$$

Proof of 4.15.

$$\tau_{\mathbb{F}_q}(\chi,\psi) = (-1)^{N+1} \sum_{x\in k^\times} \chi(x)^{-1}\psi\,\mathrm{tr}(x) = (-1)^{N+1} \sum_{x\in k^\times/\mathbb{F}_q^\times} \chi(x)^{-1} \sum_{\lambda\in\mathbb{F}_q^\times} \psi(\lambda\,\mathrm{tr}\,x).$$

The sum $\sum_{\lambda\in\mathbb{F}_q^\times} \psi(\lambda\,\mathrm{tr}\,x)$ equals $q-1$ if $\mathrm{tr}(x)=0$, and -1 if $\mathrm{tr}(x)\neq 0$. Therefore,

$$\tau_{\mathbb{F}_q}(\chi,\psi) = (-1)^{N+1} \left(q \sum_{x\in k^\times/\mathbb{F}_q^\times} \chi(x)^{-1} - \sum_{x\in k^\times/\mathbb{F}_q^\times} \chi(x)^{-1} \right).$$

The second term on the right-hand side is zero, since it is the sum of the values of a character, and 4.15 follows. □

Proposition 4.15 therefore has the following cohomological counterpart:

Proposition 4.16. *The cohomology of $W^*/\mathbb{G}_m$ (4.6, 4.8) with coefficients in $\mathscr{F}(\chi^{-1})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq N-1$, *and* $\dim \mathrm{H}_c^{N-1} = 1$.

(ii) On H_c^{N-1}, F^* *is multiplication by* $J(\chi)$.

(iii) H_c^{N-1} *is canonically isomorphic to* $\mathrm{H}_c^{N+1}\left(V^*, \mathscr{F}(\chi^{-1}, \psi\circ \mathrm{tr}_{k/\mathbb{F}_q})\right)(1)$.

Assertion (ii) follows from (i) and the trace formula; (i) and (iii) follow from 4.11 and the following more geometric statement.

Proposition 4.17. *Let $(\chi_i)_{i\in I}$ be a family of nontrivial characters of $\boldsymbol{\mu}_n(\mathbb{F})$ whose product is 1. Then* $\mathrm{H}_c^{i-1}\left(W^*/\mathbb{G}_m, \mathscr{K}_m((\chi_i)_{i\in I})\right)$ *is canonically isomorphic to* $\mathrm{H}_c^{i+1}\left(V^*, \mathscr{K}_m((\chi_i)_{i\in I}) \otimes \mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q})\right)(1)$.

The first line of 4.15, namely the projection formula, gives (π as in (4.6.1)).

$$\mathsf{R}^i\pi_!\left(\mathscr{K}_n((\chi_i)_{i\in I}) \otimes \mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q})\right) = \mathscr{K}_n((\chi_i)_{i\in I}) \otimes \mathsf{R}^i\pi_!\mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q}) \tag{4.17.1}$$

(on $V^*/\mathbb{G}_m$). Let us compute the sheaves $\mathsf{R}^i\pi_!\mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q}) = \mathsf{R}^i\pi_!T^*\mathscr{F}(\psi)$. Let $v_0 : \widetilde{V}_0 \to V_0$ be the blow-up of V_0 at $\{0\}$. In the diagram

$$\begin{array}{ccccc} V_0 \smallsetminus \{0\} & \hookrightarrow & \widetilde{V}_0 & \xrightarrow{v_0} & V_0 \\ & \searrow^{\pi} & \downarrow{\scriptstyle \bar{\pi}} & & \\ & & P_0 & & \end{array} \tag{4.17.2}$$

π is a fibration whose fibers are punctured affine lines, $\widetilde{V}_0$ is the corresponding line bundle, and $Z_0 = v_0^{-1}(0)$ is its section 0. The $T^*\mathscr{F}(\psi)$ sheaf on $V - 0 \smallsetminus \{0\}$ extends to $(Tv_0)^*\mathscr{F}(\psi)$ on $\widetilde{V}_0$, and the restriction of this sheaf to Z_0 is the constant sheaf E_λ, because $T_0 v_0$ vanishes on Z_0. The long exact sequence for cohomology with compact supports is therefore

$$\longrightarrow \mathsf{R}^i\pi_!T^*\mathscr{F}(\psi) \longrightarrow \mathsf{R}^i\bar{\pi}_!(Tv)^*\mathscr{F}(\psi) \longrightarrow (E_\lambda \text{ for } i = 0,\ 0 \text{ otherwise}) \longrightarrow \tag{4.17.3}$$

The inverse image $\bar{\pi}^{-1}(Q_0)$ is the blow-up $\widetilde{W_0}$ of W_0 at 0; T_0V_0 vanishes on $\widetilde{W_0}$; hence $(Tv)^*\mathscr{F}(\psi) = E_\lambda$ on $\pi^{-1}(Q_0)$ and, since $\bar{\pi}$ is an affine-line fibration,

$$\mathsf{R}^i\bar{\pi}_!(Tv)^*\mathscr{F}(\psi)|Q = \begin{cases} 0 & \text{for } i \neq 2 \\ E_\lambda(-1) & \text{for } i = 2. \end{cases} \tag{4.17.4}$$

If $x \in P$, $x \notin Q$, the line $D = \pi^{-1}(x)$ is mapped isomorphically onto $\mathbb{G}_a$ by Tv; therefore (by 2.7*)

$$\mathrm{H}_c^\bullet(D, (Tv)^*\mathscr{F}(\psi)) = \mathrm{H}^\bullet(\mathbb{G}_a, \mathscr{F}(\psi)) = 0 \qquad \text{and}$$

$$\mathsf{R}^i\bar{\pi}_!(Tv)^*\mathscr{F}(\psi) = \begin{cases} 0 & \text{for } i \neq 2 \\ E_\lambda(-1)_Q & \text{for } i = 2 \end{cases} \tag{4.17.5}$$

Combining (4.17.3) and (4.17.5), we finally find

$$\mathsf{R}^i\pi_!T^*\mathscr{F}(\psi) = \begin{cases} 0 & \text{for } i \neq 1, 2 \\ E_\lambda & \text{for } i = 1 \\ E_\lambda(-1)_Q & \text{for } i = 2. \end{cases} \tag{4.17.6}$$

4.18

Let us compute the compactly supported cohomology of the sheaf $\mathscr{K}_n((\chi_i)) \otimes T^*\mathscr{F}(\psi)$ on V^* using the Leray spectral sequence of $\pi : V^* \to V^*/\mathbb{G}_m$. Applying (4.17.1) and (4.17.6), we obtain as initial terms $E_2^{p,1} = \mathrm{H}_c^p(V^*/\mathbb{G}_m, \mathscr{K}_n(\chi_i)) = 0$ (because $(\chi_i) \neq 0$) and $E_2^{p,2} = \mathrm{H}_c^p(W^*/\mathbb{G}_m, \mathscr{K}(\chi_i))(-1)$.

The spectral sequence reduces to an isomorphism, and 4.17 follows.

4.19

The sheaves $\mathscr{K}_n((\chi_i)_{i\in I})$ make sense over any field, or even over any base scheme. This allows us to generalize 4.16(i). With the notation of 4.6, suppose that M is finite étale of constant rank N over A, and let n be invertible in A. Let a be the projection of $W^*/\mathbb{G}_m$ to $\mathrm{Spec}(A)$. Let also χ be a character (morphism of sheaves) $\chi : (V^*/\mathbb{G}_m)_n \to E_\lambda^\times$, and $\mathscr{K}_n(\chi)$ the sheaf corresponding. Locally for the étale topology, we can regard χ as a family of characters $(\chi_i)_{i\in I}$ of $\boldsymbol{\mu}_n$, whose product is trivial.

Proposition 4.20. *(i) If χ is everywhere nontrivial, we have $\mathsf{R}^i a_!(\mathscr{K}_n(\chi)) = 0$ for $i \neq N-1$, and $\mathsf{R}^{N-1}a_!(\mathscr{K}_n(\chi))$ is smooth of rank 1.*

(ii) An automorphism σ of M which preserves χ acts on this $\mathsf{R}^{N-1}a_!$ by multiplication by the sign $\varepsilon(\sigma)$ of σ, seen as a permutation of I.

(iii) If the χ_i are all nontrivial, we have $\mathsf{R}a_! \xrightarrow{\sim} \mathsf{R}a_*$.

Proof. The scheme $W^*/\mathbb{G}_m$ is the complement of a relative normal-crossings divisor in the scheme Q, which is proper and smooth over $\mathrm{Spec}(A)$, and $\mathscr{K}_n(\chi)$ is locally constant on $W^*/\mathbb{G}_m$ and is tamely ramified along the boundary. Consequently, the $\mathsf{R}^i a_!$ and $\mathsf{R}^i a_*$ are smooth sheaves, compatible with every change of base. A standard argument then reduces us to the case in which A is a finite field, and (i) follows from 4.17 and 4.11(i). To prove (ii), use the compatibility of the action of σ with the isomorphism 4.17, together with 4.12. For (iii), note that if the χ_i are all nontrivial, then $\mathscr{K}_n(\chi)$ on $W^*/\mathbb{G}_m \subset Q$ is ramified along every divisor at infinity, and apply 1.19.1. □

5 Hecke characters

5.1

Let F be a number field (of finite degree over $\mathbb{Q}$) and let k be a field of characteristic 0. Here are several equivalent descriptions of an *algebraic* homomorphism $\alpha : F^\times \to k^\times$.

(i) Let $\{e_i\}$ be a basis of F over $\mathbb{Q}$. The homomorphism α is algebraic if it is given by a formula $\alpha(\sum x^i e_i) = A(x^i)$, $A \in k(X^i)$.

This means that α agrees on $F^\times$ with a rational map defined over k from $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ to $\mathbb{G}_m$. By the Zariski density of $F^\times$, and because a rational homomorphism is defined everywhere, such a map is a homomorphism of group schemes:

(ii) α is induced by a homomorphism of group schemes over k

$$\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m) \otimes_{\mathbb{Q}} k \to \mathbb{G}_m.$$

If k is a number field, the adjunction property of restriction of scalars R shows that this is equivalent to

(iii) (k a number field) α is induced by a homomorphism of group schemes over $\mathbb{Q}$: $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m) \to \mathrm{R}_{k/\mathbb{Q}}(\mathbb{G}_m)$.

Let $\bar{k}$ be an algebraic closure of k, and let $I = \hom(F, \bar{k})$. Over $\bar{k}$, the character group of $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ is $\mathbb{Z}^I$, with basis the embeddings of F into $\bar{k}$. The characters defined over k are those invariant under $\mathrm{Gal}(\bar{k}/k)$; they can be described either as those sending $F^\times$ into $k^\times \subset \bar{k}$, or in terms of orbits of $\mathrm{Gal}(\bar{k}/k)$ on I, which in turn correspond to the factors of $F \otimes k$.

(iv) α is of the form $\alpha = \prod_{\omega \in I} \omega^{n_\omega}$. The allowed families of exponents (n_ω) are those such that $n_\omega = n_{\omega'}$ for ω and ω' in the same orbit of $\mathrm{Gal}(\bar{k}/k)$. Equivalently, these are those such that $\prod_\omega (x)^{n_\omega} \in k$ for every $x \in F$.

(v) Write $F \otimes k = \prod_{j \in J} F_j$, where the F_j are fields. Then α can be written $\alpha = \prod N_{F_j/k}^{m_j}$.

In the special case where k contains a normal closure of F, these expressions become: $\alpha = \prod \omega^{n_\omega}$, where ω runs over the $[F : \mathbb{Q}]$ embeddings of F into k.

5.2

Let k be a number field, and let α be an algebraic homomorphism from $F^\times$ to $k^\times$. There is then a unique homomorphism, also denoted α, of the group $I(F)$ of fractional ideals of F to that of k, such that $\alpha((x)) = (\alpha(x))$. Uniqueness follows from the fact that every ideal a has a power that is principal, and that $I(k)$ is torsion-free. To prove existence, we use for example 5.1(v): we have $\alpha(a) = \prod N_{F_j/k}^{m_j}((a))$.

5.3

Let us recall the definition of algebraic Hecke characters, also called Weil Hecke characters (or grössencharaktere) of type A_0. Let F be a number field, let $\mathfrak{m}$ be an ideal of F (i.e., of the ring of integers of F), let $I_{\mathfrak{m}}$ be the group of fractional ideals of F prime to $\mathfrak{m}$, and let k be a field of characteristic 0. A homomorphism $\chi : I_{\mathfrak{m}} \to k^\times$ is an *algebraic Hecke character* (of conductor $\leqslant \mathfrak{m}$) if there exists an algebraic homomorphism $\chi_{\mathrm{alg}} : F^\times \to k^\times$ satisfying

$$\text{For } x \in F^\times, \text{ prime to } \mathfrak{m}, \text{ totally positive, and } \equiv 1 \pmod{\mathfrak{m}}, \text{ one has } \chi((x)) = \chi_{\mathrm{alg}}(x). \qquad (*)$$

By the density of the set of x in $(*)$, χ_{alg} is entirely determined by χ. This is the *algebraic part* of χ. If $\chi((x)) = \chi_{\mathrm{alg}}(x)$ for x totally positive and $\equiv 1 \pmod{\mathfrak{m}'}$, χ extends to a Hecke character of conductor $\leqslant \inf(\mathfrak{m}, \mathfrak{m}')$: $I_{\mathfrak{m}+\mathfrak{m}'} \to k^\times$. We identify Hecke characters that coincide on their common domain of definition, and we call *conductor* of χ the smallest $\mathfrak{m}'$ such that χ is of conductor $\leqslant \mathfrak{m}'$.

5.4 Remark

If an algebraic Hecke character χ takes its values in a subfield k' of k, it is already a Hecke character with values in k': it suffices to observe that $\chi_{\mathrm{alg}} : \mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m) \to \mathbb{G}_m$ is already defined over k', and this follows from the Zariski density in $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ of the set of totally positive $x \equiv 1 \pmod{\mathfrak{m}}$. We can always take for k' a subfield of k of finite degree over $\mathbb{Q}$.

5.5

If ε is a totally positive unit $\equiv 1 \pmod{\mathfrak{m}}$, we have $\chi_{\mathrm{alg}}(\varepsilon) = \chi((\varepsilon)) = 1$. The homomorphism χ_{alg} therefore factors through the quotient $T_{\mathfrak{m}}$ of $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ by the Zariski closure of the group $E_{\mathfrak{m}} \subset F^\times$ of totally positive units $\equiv 1 \pmod{\mathfrak{m}}$.

According to Serre [35, II.3], if k is an algebraic closure of $\mathbb{Q}$, and $\mathfrak{m}$ is large enough, the characters $\prod \omega^{n_\omega}$ of $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ of the form χ_{alg} are characterized as follows: there must exist an integer N, the *weight* of χ (or χ_{alg}), such that for any element σ of $\mathrm{Gal}(k/\mathbb{Q})$ conjugate to complex conjugation, we have $n_\omega + n_{\sigma\omega} = N$. If F_1 is the largest subfield of F which is a quadratic extension that is totally imaginary over a totally real field F_1', this amounts to saying that χ_{alg} is of the form $\chi_1 \circ N_{F/F_1}$, and that $\chi_1 | F_1 = (N_{F_1/\mathbb{Q}})^N$.

If χ is of weight N, for any ideal $\mathfrak{a}$ of F, $\chi(\mathfrak{a})$ is a algebraic number all of whose conjugates have absolute value $N(\mathfrak{a})^{N/2}$ [35, II.3 prop.2].

5.6

For a fixed number field k, additional conditions are imposed on χ_{alg}. For any ideal $\mathfrak{a}$ of F prime to the conductor, we have in fact

$$\chi_{\mathrm{alg}}(\mathfrak{a}) = (\chi(\mathfrak{a})), \qquad (5.6.1)$$

so that $\chi_{\mathrm{alg}}(\mathfrak{a})$ is principal. Since the group of fractional ideals of k is torsion-free, it is enough to prove the n-th power of (5.6.1), for a suitable $n \neq 0$; this allows us to replace $\mathfrak{a}$ by $\mathfrak{a}^n$, therefore to assume $\mathfrak{a} = (x)$, with x totally positive $\equiv 1 \pmod{\mathfrak{m}}$. All that remains is to use the definitions.

Together with 5.5, this shows that χ_{alg} determines the valuations of $\chi(\mathfrak{a})$ at all places of k.

5.7

Serre's group $S_{\mathfrak{m}}$ can be characterized as the group of multiplicative type whose group of characters (on any field) is the group of algebraic Hecke characters of conductor $\leqslant \mathfrak{m}$ (see [35, II 2.1 et 2.2]). Its relation with ℓ-adic representations is explained in [35, II 2.3].

Theorem 5.8 ([35])**.** *Let χ be an algebraic Hecke character of F with values in E_λ, for E_λ a finite extension of $\mathbb{Q}_\ell$. There is then a unique homomorphism $\chi_\lambda : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to E_\lambda$, such that*

(i) χ_λ is unramified outside the conductor $\mathfrak{f}$ of χ and outside ℓ.

(ii) For $\mathfrak{p}$ a prime ideal of F prime to $\mathfrak{f}$ and to ℓ, and $F_\mathfrak{p} \in \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$ the geometric Frobenius at $\mathfrak{p}$, we have

$$\chi_\lambda(F_\mathfrak{p}) = \chi(\mathfrak{p}).$$

5.9

At the end of this section, we give a criterion for a λ-adic representation to arise from a Hecke character, and in the next section we apply it to Jacobi sums. We thereby recover, in a somewhat generalized form, Weil's results [42, 44], with the second part of Weil's proof replaced by a ℓ-adic cohomology argument.

Theorem 5.10. *Let F and k be two number fields, let λ be a place of k of residue characteristic ℓ, and let $\chi_\lambda : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to k_\lambda^\times$ be a homomorphism, unramified outside ℓ and a finite set S of places. We assume that there exists a set T of places (containing S and the places above ℓ), of density 0, and an algebraic homomorphism $\chi_0 : F^\times \to k^\times$ such that*

(i) For $\bar{k}$ an algebraic closure of k, $\chi_0 : F^\times \to k^\times \hookrightarrow \bar{k}^\times$ is of the type considered in 5.5, of weight N;

(ii) For $\mathfrak{p}$ prime $\notin T$, $\chi_\lambda(F_\mathfrak{p})$ lies in $k^\times$, $(\chi_\lambda(F_\mathfrak{p})) = \chi_0(\mathfrak{p})$, and all complex conjugates of $\chi_\lambda(F_\mathfrak{p})$ have absolute value $(N\mathfrak{p})^{N/2}$.

Then χ_λ is defined (5.8) by an algebraic Hecke character of F with values in k, whose algebraic part is χ_0.

Let χ' be an algebraic Hecke character with values in a finite Galois extension k' of k, of algebraic part χ_0 (5.5). After enlarging T if necessary, we may suppose that the conductor of χ' is supported in T. Let λ' be a place of k' above λ, and let $\chi_{\lambda'}$ be the composite $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to k_\lambda^\times \to {k'}_{\lambda'}^\times$, and $\chi'_{\lambda'} : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to {k'}_{\lambda'}^\times$ defined by χ' (5.8). Set $\varepsilon = \chi_{\lambda'} \cdot {\chi'}_{\lambda'}^{-1}$.

Hypothesis (ii), together with 5.5, 5.6, and 5.8, ensures that for $\mathfrak{p} \notin T$, we have $\varepsilon(F_\mathfrak{p}) \in {k'}^\times$, and that in every completion of k' this number has absolute value 1. It follows that $\varepsilon(F_\mathfrak{p})$ belongs to the finite group $\boldsymbol{\mu}$ of roots of unity in k'. By the Chebotarev density theorem and the density-zero hypothesis on T, the $F_\mathfrak{p}$ ($\mathfrak{p} \notin T$) are dense in $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$, so that $\varepsilon(\sigma) \in \mu$ for all $\sigma \in \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$: the character ε is a finite order character $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to \mu$. By class field theory, it corresponds to a finite-order character of the idèle class group of F, i.e. to an algebraic Hecke character with trivial algebraic part. After correcting χ' by this character, we are reduced to the case in which $\chi_{\lambda'} = \chi'_{\lambda'}$, and it remains to show that χ', a priori taking values in k', in fact takes values in k. If $\sigma \in \mathrm{Gal}(k'/k)$, χ' and ${\chi'}^\sigma$ coincide on every prime ideal $\mathfrak{p} \notin T$, since $\chi'(\mathfrak{p}) = \chi'_{\lambda'}(F_\mathfrak{p}) = \chi_{\lambda'}(F_\mathfrak{p}) \in k^\times$. The characters $\chi'_{\lambda'}$, and ${\chi'_{\lambda'}}^\sigma$, coincide on a dense subset of $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$ and hence are equal. Thus $\chi' = {\chi'}^\sigma$, so χ' takes values in k.

6 Hecke characters defined by Jacobi sums

The idea of this section is as follows. Consider a Gauss sum $\tau_{\mathbb{F}_q}(\chi,\psi)$. Both this sum and the corresponding sheaf $\mathscr{F}(\chi^{-1}\psi)$ are tied in two ways to the characteristic p and to the base field $\mathbb{F}_q$:

a) because the additive character ψ and the sheaf (ψ) occur in it;

b) because $\mathscr{F}(\chi)$ is defined from the exact Lang sequence

$$0 \longrightarrow k^\times \longrightarrow V^* \longrightarrow V^* \longrightarrow 0.$$

These present two difficulties if we wish to lift the corresponding cohomological constructions to a number field F and find ℓ-adic representations of $\mathrm{Gal}(\bar{F}/F)$ whose Frobenius eigenvalues are Gauss sums.

If χ is trivial on $\mathbb{F}_q^\times$, the sum $\tau_{\mathbb{F}_q}(\chi,\psi)$ is independent of ψ and is q times a Jacobi sum. The latter is related to the cohomology of $W^*/\mathbb{G}_m$, with coefficients in a sheaf obtained from χ: thus ψ, and with it difficulty a), disappear. To resolve b), it suffices to describe the sheaves in question by means of exact Kümmer sequences (which make sense in every characteristic). This is possible by 4.9.

Once the ℓ-adic representations have been found, we use 5.10 and Stickelberger's theorem to show that they come from algebraic Hecke characters.

6.1

If E is a field with algebraic closure $\bar{E}$, and λ is a character, of order dividing n, of $\mu_d(\bar{E})$, we denote by ${}_n\lambda$ the character of $\mu_n(\bar{E})$ such that λ and ${}_n\lambda$ induce the same character of $\widehat{\mathbb{Z}}(1)_{\bar{E}} = \varinjlim \mu_n(\bar{E})$. The same notation is used when λ is a character of $\widehat{\mathbb{Z}}(1)_{\bar{E}}$.

Let F and k be two number fields, $\bar{F}$ an algebraic closure of F, I a finite set equipped with an action of $\mathrm{Gal}(\bar{F}/F)$ and $\lambda = (\lambda_i)_{i\in I}$ a family of characters $\lambda_i : \widehat{\mathbb{Z}}(1)_{\bar{F}} \to k^\times$. It is assumed that $\lambda_i \neq 1$ $(i \in I)$, that the product of λ_i is trivial and that the family λ is defined over F, i.e. that for every $\sigma \in \mathrm{Gal}(\bar{F}/F)$, we have $\lambda_{\sigma(i)} = \lambda_i \circ \sigma^{-1}$.

The Galois set I is the set of homomorphisms into $\bar{F}$ of an algebra E separable over F. If C is the set of orbits of $\mathrm{Gal}(\bar{F}/F)$ in I, we have a decomposition of E as a product of fields $E = \prod_{\alpha\in C} E_\alpha$, where α is identified with the set of embeddings of E_α in $\bar{F}$. If the characters λ_i, for $i \in \alpha$, are of order d_α, the field E_α contains the d_α-th roots of 1 and there exists $\lambda_\alpha : \mu_{d_\alpha}(E_\alpha) \hookrightarrow k^\times$ such that, if $i \in \alpha$ corresponds to the embedding σ_i of E_α in $\bar{F}$, then ${}_{d_\alpha}\lambda_i = \lambda_\alpha \sigma_i^{-1}$.

6.2

Let $n > 0$ be a multiple of the d_α, let Σ be the set of places of F at which E/F ramifies or which divide n, let μ be a place of k of residue characteristic ℓ, and let A be the ring of elements of F integral outside Σ and ℓ. The integral closure M of A is finite étale of rank $N = |I|$ over A. With the notation of 4.6, 4.8, ${}_n\lambda_i$ define

$${}_n\lambda : (V^*/\mathbb{G}_m)_n \to k^\times \subset k_\mu^\times.$$

Let a be the projection of $W^*/\mathbb{G}_m$ onto $\mathrm{Spec}(A)$. By 4.20, the sheaf $\mathsf{R}^{N-2}a_!\mathscr{K}_n({}_n\lambda)$ is smooth of rank one; it defines an ℓ-adic representation

$$j_\mu[\lambda] : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to k_\mu^\times$$

(or simply j_μ) unramified outside σ and ℓ.

6.3

Let us compute $j_\mu(F_\mathfrak{p})$. By 4.20, it suffices to do so after reduction modulo $\mathfrak{p}$. Put $f = A/\mathfrak{p}$, $e = M/\mathfrak{p}M$, and let $\bar{f}$ be the algebraic closure of f determined by a place of $\bar{F}$ above $\mathfrak{p}$. We still have $I = \hom_f(e, \bar{f})$. After reduction, ${}_n\lambda$ still has a description as in 6.1:

a) Let D be the set of orbits of $\mathrm{Gal}(\bar{f}/f)$ in I. The decomposition of e as a product of fields has the form $e = \prod_{\beta\in D} e_\beta$, where σ_i $(i \in \beta)$ are identified with the embeddings of e_β into F. For $\beta \in D$, denote by $\alpha(\beta)$ the element of C containing β, and put $d_\beta = d_{\alpha(\beta)}$. Let

$$\lambda_\beta : \mu_{d_\beta}(e_\beta) \xleftarrow{\sim} \mu_{d_\beta}(E_{\alpha(\beta)}) \xrightarrow{\lambda_{d(\beta)}} k^\times.$$

b) If $i \in I$ corresponds to the embedding σ_i of e_β into $\bar{f}$, the reduction ${}_{d\beta}\bar{\lambda}_i : \mu_{d\beta}(\bar{f}) \to k^\times$ of ${}_{d\beta}\lambda_i$, itself the reduction of ${}_{d\beta}\lambda_i$, is $\lambda_\beta \circ \sigma_i^{-1}$.

Let q_β be the number of elements of e_β, $\chi_\beta =_{(q_{\beta-1})} \lambda_\beta$, and $\chi : e^\times \to k^\times$ the character with coordinates χ_β. On the affine space over $A/\mathfrak{p}$ defined by e (4.6), or rather on the torus $e_\beta^\times$, we have isomorphisms $\mathscr{K}_n((_n\bar{\lambda}_i)_{i\in\beta}) = \mathscr{K}_{d_\beta}((\lambda_\beta \circ \sigma_i^{-1})_{i\in\beta}) = \mathscr{K}_{q_{\beta-1}}((\chi_\beta \circ \sigma_i^{-1})_{i\in I}) = \mathscr{F}(\chi_\beta)$ (4.9). On $e^\times$, we therefore have $\mathscr{K}_n((_n\bar{\lambda}_i)_{i\in\beta}) = \mathscr{F}(\chi)$, and this isomorphism descends to the reduction modulo $\mathfrak{p}$ of $V^*/\mathbb{G}_m$. It follows that

Proposition 6.4. *With the notations of 6.2 and 6.3, $j_\mu(F_\mathfrak{p})$ is the Jacobi sum $J(\chi)$.*

Let us rewrite the formulas linking $(\lambda_i)_{i\in I}$ to $(\chi_\beta)_{\beta\in B}$: for $i \in \beta \subset \alpha$,

$$\begin{aligned} {}_{d_\alpha}\lambda_i &= \lambda_\alpha \circ \sigma_i^{-1} && \lambda_\alpha : \mu_{d_\alpha}(E_\alpha) \to k^\times \\ \lambda_\beta &= \text{reduction of } \lambda_\alpha && : \mu_{d_\alpha}(e_\beta) \to k^\times \\ \chi_\beta &=_{q_{\beta-1}} \lambda_\beta : x \mapsto \lambda_\beta(x^{(q_\beta-1)/d_\alpha}) && : e_\beta^\times \to k^\times \end{aligned}$$

Theorem 6.5. *The representation j_μ is defined by an algebraic Hecke character of F with values in $k^\times$.*

We shall verify this using 5.10. Here we follow Weil's proof [42, 44], using Stickelberger's theorem, and I shall give only a few indications.

6.6

Given $j : \mathrm{Gal}(\bar{F}/F) \to k_\lambda^\times$ and a finite extension F'/F, one verifies using 5.10 that j is defined by a Hecke character if and only if $j|\,\mathrm{Gal}(\bar{F}/F')$ is. This remark allows us, in 6.5, to reduce to the case where Galois acts trivially on I and where F contains the n-th roots of unity. We may then reduce to the case $F = \mathbb{Q}(\sqrt[n]{1})$ and take $k = F$. In this case, each ${}_n\lambda_i$ is defined by $a_i \in \mathbb{Z}/n$: it is $x \mapsto x^{a_i} : \mu_n \to k^\times$. We have $\sum a_i = 0$, and $a_i \neq 0$.

6.7

In a number field, the places of degree greater than 1 always form a set of density 0. By applying 5.10, we may therefore disregard them and use Stickelberger only for a prime field. Let p be prime, let $0 < a < p-1$, and let $\widetilde{x} : \mathbb{F}_p^\times \to \mathbb{Q}_p^\times$ be the "multiplicative lift" character; put $g = -\sum_{x\in\mathbb{F}_p^\times} \widetilde{x}^{-a}\zeta^z \in \mathbb{Q}_p(\zeta)$, for ζ a p-th root of 1. Again denote by x the integer in $[0, p-1]$ lifting x, and expand $\zeta^x = (1+\pi)x$ by the binomial

formula. We find $g = -\sum_{0=0}^{p-1} A_i \pi^i$ with $A_i \in \mathbb{Z}_p$, whose reduction $\bmod p$ is the sum $\sum_{x\in\mathbb{F}_p^\times} x^{-a}\binom{x}{i}$. For b not divisible by $p-1$, we have $\sum_{x\in\mathbb{F}_p^\times} x^b = 0$. If $i < a$, $\binom{x}{i}$ is a polynomial in x of degree $i < a$, whence $A_i \equiv 0 \pmod p$. Likewise, $A_a \equiv \frac{p-1}{a!} \pmod p$, and $g \sim \frac{\pi^a}{a!}$: the valuation of g is given by $v(g)/v(p) = \frac{a}{p-1}$.

6.8

Let σ_i ($i \in (\mathbb{Z}/n)^\times$) be the element of $\mathrm{Gal}(\mathbb{Q}(\sqrt[n]{1})/\mathbb{Q})$ such that $\sigma_i(\zeta) = \zeta^i$ for $\zeta^n = 1$, and write the group of algebraic homomorphisms $\mathbb{Q}(\sqrt[n]{1})^\times \to \mathbb{Q}(\sqrt[n]{1})^\times$ additively. For λ as in 6.6, defined by a family $(a_i)_{i\in I}$ of nonzero integers $\bmod n$ whose sum is zero, 5.10, (4.15.1) and 6.7 show that $j_\mu[\lambda]$ is defined by an algebraic Hecke character $j[a]$ and that, if N denotes the norm, its algebraic part is given by

$$(Nj[a])_{\mathrm{alg}} = \sum_{i\in(\mathbb{Z}/n)^\times} \sum_{j\in I} \left\lfloor \frac{ia_j}{n} \right\rfloor \sigma_i^{-1} \tag{6.8.1}$$

where $\lfloor\cdot\rfloor$ is the fractional part.

6.9

The most interesting case is that in which $F \subset \mathbb{Q}(\sqrt[n]{1})$ and the family of ${}_n\lambda_i$ is constructed as follows:

a) Take a finite family $(Aj)_{j\in J}$ of orbits of $\mathrm{Gal}(\sqrt[n]{1}/F) = H \subset (\mathbb{Z}/n)^\times$ in $\mathbb{Z}/n$. Assume that $A_j \neq \{0\}$ and that $\sum_{j\in J}\sum_{a\in A_j} a = 0$.

b) Take $I = \sqcup_{j\in J} A_j$; Galois acts on each A_j.

c) Take $\bar{F} \supset \mathbb{Q}(\sqrt[n]{1})$, $k = \mathbb{Q}(\sqrt[n]{1})$ and, if $i \in I$ corresponds to $a \in A_j \subset (\mathbb{Z}/n)^\times$, ${}_n\lambda_i(x) = x^a$.

Using 4.20, one verifies that a general j_μ (relative to F_1) is obtained from a j_μ as above (for $F \subset \mathbb{Q}(\sqrt[n]{1})$, with F_1 an extension of F) by restriction to $\mathrm{Gal}(\bar{F}/F_1)$ and multiplication by a character of order two.

For $a \in \mathrm{Gal}(k/\mathbb{Q}) = (\mathbb{Z}/n)^\times$, a transforms the family of A_j into the family of aA_j. If $a \in H$, the family of A_j is therefore preserved, and the $j_\mu(F_\mathfrak{p})$ are invariant under H:

Proposition 6.10. *Under the assumptions of 6.9, j_μ is defined by a character that is an algebraic Hecke character of F with values in F.*

Let us again take $F = \mathbb{Q}(\sqrt[n]{1})$ and consider the Hecke characters $j[a]$ defined by a family $(a_i)_{i\in I}$, $i \in \mathbb{Z}/n$, $a_i \neq 0$, $\sum a_i = 0$. The following theorem was obtained independently by Vishik.

Theorem 6.11. *Every algebraic Hecke character of $\mathbb{Q}(\sqrt[n]{1})$ has a power, with exponent a, in the group generated by the $j[a]$ and the norm.*

Since we work "up to torsion," only the algebraic part of the Hecke characters under consideration matters. Applying (6.8.1) to $j[a]$, with $a =$ an element of $\mathbb{Z}/n$ repeated n times, we are reduced to the following lemma.

Lemma 6.12. *Write $\sum x_j\sigma_j$ for the elements of the group algebra $\mathbb{Q}[(\mathbb{Z}/n)^\times]$, and let x be the subgroup consisting of those for which $x_j + x_{-j} = C\frac{te}{}$. Then X is generated over $\mathbb{Q}$ by $N = \sum\sigma_j$ and by the elements $g_a = \sum \left\lfloor \frac{ia}{n} \right\rfloor \sigma) i^{-1}$ ($a \in \mathbb{Z}/n$, $a \neq 0$).*

Both X and $Y = \langle N, (g_a)\rangle$ are ideals of the group algebra, and it suffices to check that they are annihilated by the same complex characters χ of $(\mathbb{Z}/n)^\times$. This amounts to saying that a odd character ($\chi(j) = -\chi(-j)$) never annihilates all the g_a. If χ is primitive, we have

$$\chi(g_1) = \sum \frac{i}{n}\chi^{-1}(i) = -L(\chi, 0) \neq 0.$$

In the general case, if χ comes from a primitive character of $(\mathbb{Z}/n)^\times$, $\chi(g_{n/d})$ reduces to an analogous sum over $(\mathbb{Z}/n)^\times$, hence the lemma.

7 Generalized Kloosterman sums

7.1

Let $\mathbb{F}_q$ be a finite field with q elements, and let ψ be a nontrivial additive character. In this section, we undertake a cohomological study of generalized Kloosterman sums

$$\mathrm{K}_{n,a} = \sum_{x_1\cdots x_n = a} \psi(x_1 + \cdots + x_n) \qquad (n \geqslant 1). \tag{7.1.1}$$

For $a = 0$, this sum is elementary (see 7.7):

$$\mathrm{K}_{n,0} = (-1)^{n-1}. \tag{7.1.2}$$

The interesting case is $a \neq 0$. The x_i then belong to $\mathbb{F}_q^\times$. In this case we shall prove the bound

$$|K_{n,a}| \leqslant nq^{\frac{n-1}{2}}. \tag{7.1.3}$$

Our essential tools will be the cohomological counterparts of the following identities:

$$\begin{aligned} \mathrm{K}_{n,a} &= \sum_{x_1} \psi(x_i) \sum_{x_2\cdots x_n = a/x_1} \psi(x_2 + \cdots + x_n) \qquad (7.1.4)\\ &= \sum_{x\in\mathbb{F}_q^\times} \psi(x)\mathrm{K}_{n-1,a/x} \qquad (a \neq 0, n \geqslant 2) \\ \sum_a \mathrm{K}_{n,a} &= \sum_a \sum_{x_1\cdots x_n = a} \psi\left(\sum x_i\right) = \sum_{x_1,\ldots,x_n} \psi\left(\sum x_i\right) = 0 \qquad (7.1.5) \end{aligned}$$

For χ a nontrivial character of $\mathbb{F}_q^\times$,

$$\sum \chi(a)\mathrm{K}_{n,a} = \sum_{x_1,\ldots,x_n} \chi\left(\prod x_i\right)\cdot\psi\left(\sum x_i\right) = (-\tau(\chi^{-1}, \psi))^n \tag{7.1.6}$$

(Gauss sum).

7.2

Let k be an étale algebra of degree n over $\mathbb{F}_q$, and let $\varepsilon(k)$ be as in (4.5.1). Put

$$\mathrm{K}_{k,a} = \sum_{N_{k/\mathbb{F}_q}(x) = a} \psi\,\mathrm{tr}(x).$$

We likewise have

$$\mathrm{K}_{k,0} = (-1)^{n-1} \cdot \varepsilon(k) \tag{7.2.1}$$

$$\sum_a \mathrm{K}_{k,a} = 0 \tag{7.2.2}$$

$$\sum_a \chi(a)\mathrm{K}_{k,a} = (-1)^n \tau_{\mathbb{F}_q}\left(\chi^{-1} \circ N_{k/\mathbb{F}_q}, \psi\right). \tag{7.2.3}$$

By Fourier inversion on $\mathbb{F}_q^\times$, we deduce an expression of $\mathrm{K}_{k,a}$ in terms of Gauss sums:

$$\mathrm{K}_{k,a} = \frac{1}{q-1}\sum_\chi \chi(a) \sum_{b\in\mathbb{F}_q^\times} \chi^{-1}(b)\mathrm{K}_{k,b}$$

$$(-1)^n\mathrm{K}_{k,a} = \frac{1}{q-1}\left(\varepsilon(k) + \sum_{\chi\neq 1} \chi(a)\tau_{\mathbb{F}_q}\left(\chi \circ N_{k/\mathbb{F}_q}, \psi\right)\right). \tag{7.2.4}$$

The Hasse–Davenport identity now makes it possible to compare $\mathrm{K}_{k,a}$ with $\mathrm{K}_{n,a} = \mathrm{K}_{\mathbb{F}_{q^n},a}$:

$$\mathrm{K}_{k,a} = \varepsilon(k)\mathrm{K}_{n,a}. \tag{7.2.5}$$

I know of no cohomological interpretation of (7.2.4), but I shall give one for (7.2.5). Let us return to the sums $\mathrm{K}_{n,a}$.

7.3

Let $\mathbb{F}$ be an algebraic closure of $\mathbb{F}_q$ and, for $a \in \mathbb{F}$, let V_a^{n-1}, or simply V_a, be the hypersurface in $\mathbb{A}^n$ with equation $x_1 \cdots x_n = a$. As usual, we shall regard ψ as taking ℓ-adic rather than complex values. With the notation of 1.8(ii), (7.1.3) follows from

Theorem 7.4. *The cohomology of V_a^{n-1} with coefficients in $\mathscr{F}(\psi(\sum x_i))$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq n-1$;

(ii) $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$;

(iii) for $a \neq 0$, $\dim \mathrm{H}_c^{n-1} = n$;

(iv) for $a = 0$, H_c^{n-1} *is canonically isomorphic to* E_λ.

7.5

As usual, this theorem also controls the dependence of $\mathrm{K}_{n,a}$ on q: for each $a \in \mathbb{F}_q^\times$, there are n Frobenius eigenvalues $\alpha_1, \ldots, \alpha_n$, of complex absolute value $q^{\frac{n-1}{2}}$, such that

$$\sum_{\substack{x_1\cdots x_n = a \\ x_i \in \mathbb{F}_{q^m}}} \psi \circ \mathrm{tr}\left(\sum x_i\right) = (-1)^{n-1}\left(\alpha_1^m + \cdots + \alpha_n^m\right).$$

For even n, $x \mapsto -x$ is an involution of V_a, which transforms $\mathscr{F}(\psi)$ into its dual. Arguing as in 3.6, we deduce a canonical nondegenerate alternating form on $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi))$, with values in $\mathbb{Q}_\ell(-(n-1))$, and hence the following.

Corollary 7.6. *With the notations of 7.5, if n is even, the α_i group together into $\frac{n}{2}$ pairs of roots $\alpha, \bar{\alpha}$ whose product is q^{n-1}.*

7.7

Let us verify the case $a = 0$ of 7.5. The hypersurface V_0 is the union of the coordinate hyperplanes $x_i = 0$. Let us compute the Leray spectral sequence (2.6.1*) for this covering of V_0, for cohomology with or without supports and with coefficients in $\mathscr{F}(\psi)$. By 2.7*, all the initial terms vanish except the cohomology of the intersection $\{0\}$ of all the coordinate hyperplanes. What remains is an isomorphism $\mathrm{H}^0_c \xrightarrow{\sim} \mathrm{H}^0 = E_\lambda$ which justifies 7.4.

We shall prove 7.4 and 7.8 below by simultaneous induction on n, using the spectral sequence of which (7.1.4) is the counterpart. This requires control, as a varies, of $\mathrm{K}_{n,a}$.

Let $\pi : \mathbb{A}^n \to \mathbb{A}$ be the map "product of the coordinates," and let σ be the sum of the coordinates. The hypersurfaces V_a are the fibers of π.

Theorem 7.8. *(i) The sheaf $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ is smooth of rank n on $\mathbb{A}^1 \smallsetminus \{0\}$.*

(ii) Its extension by 0 to $\mathbb{P}^1$ is the direct image of its restriction to $\mathbb{A}^1 \smallsetminus \{0\}$.

(iii) At 0, the monodromy is unipotent, with a single Jordan block.

(iv) At ∞, wild inertia acts without fixed points $\neq 0$, and the Swan conductor is 1.

(v) We have $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}^i\pi_*\mathscr{F}(\psi\sigma)$ (zero for $i \neq n-1$).

Proof. that 7.4 (for a given value of n) implies 7.8 (for the same value of n).

The fiber of $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma)$ at a is $\mathrm{H}^i_c(V_a, \mathscr{F}(\psi\sigma))$, and on the dense open subset of $\mathbb{A}^1$ where the $\mathsf{R}^i\pi_!\mathscr{F}(\psi^{-1}\sigma)$ are locally constant, the fiber of $\mathsf{R}^i\pi_*\mathscr{F}(\psi\sigma)$ is $\mathrm{H}^i(V_a, \mathscr{F}(\psi\sigma))$ (Finiteness Theorem 2.1). The hypothesis 7.4(n) therefore gives us the following.

Lemma 7.9. *(i) $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma) = 0$ for $i \neq n-1$.*

(ii) For $i = n-1$, the fiber of this sheaf at any point $a \neq 0$ has constant rank n. At 0, it has rank 1.

(iii) On a dense open U, we have $\mathsf{R}^i\pi_!\mathscr{F}(\psi) \xrightarrow{\sim} \mathsf{R}^i\pi_*\mathscr{F}(\psi)$.

7.10

We shall now use the Leray spectral sequence of π for cohomology with or without compact supports, with coefficients in $\mathscr{F}(\psi\sigma)$. The abutment is known: $\mathrm{H}^\bullet_c(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = \mathrm{H}^\bullet(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = 0$ (2.7*). For cohomology with compact supports, 7.9(i) ensures that $E_2^{pq} = 0$ for $q \neq n-1$. The terms E_2 are therefore all zero:

$$\mathrm{H}^p_c(\mathbb{A}^1, \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) = 0. \tag{7.10.1}$$

For $p = 0$, this means that $\mathsf{R}^{n-1}\pi+!\mathscr{F}(\psi\sigma)$ has no sections with point support. In view of the constancy of rank in 7.9(ii), we obtain

$$\begin{array}{l}\text{The sheaf } \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma) \text{ on } \mathbb{A}^1 \text{ is smooth on } \mathbb{A}^1 \smallsetminus \{0\}, \text{ and is} \\ \text{a subsheaf of the direct image } \mathscr{G} \text{ of its restriction to } \mathbb{A}^1 \smallsetminus \{0\}.\end{array} \tag{7.10.2}$$

The long exact cohomology sequence of

$$0 \longrightarrow \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma) \longrightarrow \mathscr{G} \longrightarrow \mathscr{Q} \longrightarrow 0.$$

where the support of $\mathscr{Q}$ is concentrated at 0, gives

$$\mathrm{H}^0_c(\mathbb{A}^1, \mathscr{G}) \longrightarrow \mathscr{Q}_0 \longrightarrow \mathrm{H}^1_c(\mathbb{A}^1, \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)).$$

Since the outer terms vanish, $\mathscr{Q} = 0$ and

$$\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma) \text{ is the direct image of its restriction to } \mathbb{A}^1 \smallsetminus \{0\}. \tag{7.10.3}$$

Let j be the inclusion of $\mathbb{A}^1$ into $\mathbb{P}^1$. The "derived-category version" of the Leray spectral sequences is

$$\mathsf{H}^\bullet(\mathbb{P}^1, j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma)) = \mathsf{H}^\bullet_c(\mathbb{A}^1, \mathsf{R}\pi_!\mathscr{F}(\psi\sigma)) = \mathrm{H}^\bullet_c(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = 0$$
$$\mathsf{H}^\bullet(\mathbb{P}^1, \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma)) = \mathsf{H}^\bullet(\mathbb{A}^1, \mathsf{R}\pi_*\mathscr{F}(\psi\sigma)) = \mathrm{H}^\bullet(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = 0.$$

Let Δ be the mapping cylinder of $j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \to \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma)$. According to 7.9(iii), the cohomology sheaves of this complex of sheaves have finite support. On the other hand, the long exact sequence in cohomology of the triangle $(j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma), \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma), \Delta)$ shows that $\mathsf{H}^\bullet(\mathbb{P}^1, \Delta) = 0$. We have $\mathsf{H}^\bullet(\mathbb{P}^1, \Delta) = \mathrm{H}^0(\mathbb{P}^1, \mathscr{H}^\bullet(\Delta))$, and finally $\Delta = 0$:

$$j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma).$$

In particular, $\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}\pi_*\mathscr{F}(\psi\sigma)$ and, since these complexes of sheaves have only one nonzero cohomology sheaf,

$$j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} j_*\mathsf{R}\pi_!\mathscr{F}(\psi\sigma). \tag{7.10.4}$$

This completes the proof of (i)(ii)(v).

7.11

Let Sw_0 and Sw_∞ be the Swan conductors at 0 and at ∞ of $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$. We have $\chi_c(\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) = 0$ (7.10.1) and the formula of Euler–Poincaré (3.2.1) reduces to

$$\mathrm{Sw}_0 + \mathrm{Sw}_\infty = 1: \tag{7.11.1}$$

one of these conductors is 0 (tame ramification), and the other is 1.

7.12

Let $\mathscr{G}$ be a sheaf of rank 1 on $\mathbb{G}_m$, of Kummer type 4.7 ($\mathscr{G} = \mathscr{K}_n(\chi)$) and nonconstant ($\chi \neq 1$). The sheaf $\mathscr{F}(\psi\sigma) \otimes \pi^*\mathscr{G}$ on $V^* = \pi^{-1}(\mathbb{G}_m) = \mathbb{A}^n \smallsetminus \pi^{-1}(0)$ is then of the type encountered in the study of Gauss sums, and 4.11(ii) gives us

$$\mathrm{H}^\bullet_c(V^*, \pi^*\mathscr{G} \otimes \mathscr{F}(\psi\sigma)) \xrightarrow{\sim} \mathrm{H}^\bullet(V^*, \pi^*\mathscr{G} \otimes \mathscr{F}(\psi\sigma)). \tag{7.12.1}$$

The Leray spectral sequence of π transforms this statement into

$$\mathrm{H}^\bullet_c(\mathbb{G}_m, \mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) \xrightarrow{\sim} \mathrm{H}^\bullet(\mathbb{G}_m, \mathscr{G} \otimes \mathsf{R}^{n-1}\pi_*\mathscr{F}(\psi\sigma)). \tag{7.12.2}$$

Let i be the inclusion of $\mathbb{G}_m$ into $\mathbb{P}^1$, and let Δ be the mapping cylinder of $\delta : i_!(\mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi)) \to \mathsf{R}i_*(\mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma))$ (where $\mathsf{R}^{n-1}\pi_! = \mathsf{R}^{n-1}\pi_*$). Reasoning as in 7.10, we deduce from (7.12.2) that $\Delta = 0$. In particular

$$\text{At 0 and at } \infty, \text{ inertia acts without fixed points} \neq 0 \text{ on } \mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma). \tag{7.12.3}$$

At ∞, this holds even if $\mathscr{G}$ is trivial (7.10.4), so wild inertia acts without a fixed point. In particular, the ramification is wild: by 7.11, we have $\mathrm{Sw}_\infty = 1$, and the ramification at 0 is tame.

At 0, (7.12.3) forces the ramification – which is tame – of $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ to be unipotent. By (7.10.3) and 7.9(ii), there is a single Jordan block. This completes the proof of 7.8(n).

7.13

For $n = 1$, 7.4 is trivial. To complete the proof of 7.4 and 7.8, we still have to prove that 7.8, for a given value of n, implies 7.4 for $n + 1$. The case $a = 0$ having been treated (7.7), we assume that $a = 0$. We shall write $x_0, \dots, x_n$ for the coordinates of $\mathbb{A}^{n+1}$. Let $g = x_0 : V_a \to \mathbb{G}_m$, τ the involution $x \mapsto ax^{-1}$ of $\mathbb{G}_m$, and let π, as above, be the product of the coordinates: $\mathbb{A}^{n+1} \to \mathbb{A}^1$, with σ their sum. We shall denote by $\mathscr{F}(\psi\sigma)$ the sheaf $\mathscr{F}(\psi(\sum x_i))$ both on $\mathbb{A}^{n+1}$ and on $\mathbb{A}^n$.

7.14

Let us write down the Leray spectral sequence for g. Since the sheaf $\mathscr{F}(\psi\sigma)$ on $\mathbb{A}^{n+1}$ is the external tensor product of the sheaves $\mathscr{F}(\psi)$ on $\mathbb{A}^1$ and $\mathscr{F}(\psi\sigma)$ on $\mathbb{A}^n$, we find (cf. (7.1.4)).

$$'E_2^{pq} = \mathrm{H}_c^p(\mathbb{G}_m, \mathscr{F}(\psi) \otimes \tau^* \mathsf{R}^q\pi_! \mathscr{F}(\psi\sigma)) \Rightarrow \mathrm{H}_c^{p+q}(V_a, \mathscr{F}(\psi\sigma))$$
$$''E_2^{pq} = \mathrm{H}^p(\mathbb{G}_m, \mathscr{F}(\psi) \otimes \tau^* \mathsf{R}^q\pi_* \mathscr{F}(\psi\sigma)) \Rightarrow \mathrm{H}^{p+q}(V_a, \mathscr{F}(\psi\sigma)).$$

By hypothesis, $\mathsf{R}^q\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}^q\pi_*\mathscr{F}(\psi\sigma)$, and this sheaf is zero for $q \neq n-1$. The sheaf $\mathscr{F}(\psi)$ on $\mathbb{G}_m$ is wildly ramified at ∞ and unramified at 0, while $\tau^*\mathsf{R}^q\pi_!\mathscr{F}(\psi\sigma)$ is wildly ramified at 0 (without invariants under wild inertia, and with Swan conductor 1) and tamely ramified at ∞. Their tensor product is therefore

a) wildly ramified at 0 and ∞, without invariants under wild inertia;

b) of Swan conductor 1 at 0 and n (the rank of $\mathsf{R}^q\pi_!\mathscr{F}(\psi)$) at ∞.

We have ${}'E_2^{pq} \xrightarrow{\sim} {}''E_2^{pq}$, and ${}'E_2^{pq} = 0$ except for $p = 1$ and $q = n-1$. The Euler-Poincaré formula finally gives

$$\dim {}'E_2^{pq} = 1 + n;$$

this completes the proof. □

7.15 Remark

The isomorphism obtained in 7.14

$$\mathrm{H}_c^n(V_a^n, \mathscr{F}(\psi\sigma)) = \mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi) \otimes \tau^* \mathsf{R}^{n-1}\pi_! \mathscr{F}(\psi\sigma)) \tag{7.15.1}$$

allows us, with the notation of 7.5, to calculate the product $\det(F)$ of the Frobenius eigenvalues α_i. The formalism of [13] applies, since $\mathscr{F}(\psi)$ and $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ belong to infinite compatible systems of ℓ-adic representations. This reduces us to local problems, which can be simplified by noting that (cf. [13, 9.5])

a) the global constants for $\mathscr{F}(\psi)$ and $\tau^*\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi)$ are easy to calculated easily, since the cohomology of these sheaves is elementary.

b) at every point, for either of these sheaves, the corresponding ℓ-adic representation of the decomposition group has unramified semisimplification.

The result, for the sum with n variables, is that

$$\det\left(F, \mathrm{H}_c^{n-1}(V_a^{n-1}, \mathscr{F}(\psi))\right) = q^{\frac{n(n-1)}{2}}. \tag{7.15.2}$$

7.16 Remark

For $p = 2$, the sheaf (ψ) is orthogonal. For $a \neq 0$ and n odd, we deduce from 7.4(ii) and Poincaré duality a symmetric bilinear form on $\mathrm{H}_c^{n-1}(V_a^{n-1}, \mathscr{F}(\psi))$ with values in $\bar{\mathbb{Q}}_\ell(1-n)$. With respect to this form, F is an orthogonal similitude of multiplier q^{n-1}, and $q^{\frac{1-n}{2}}F$ belongs to the special orthogonal group, by (7.15.2). One of the eigenvalues of F is therefore $q^{\frac{1-n}{2}}$, and the others occur in pairs of roots α, $\bar{\alpha}$ whose product is q^{n-1}.

For $n = 3$, L. Carlitz [8] obtained a more precise result, equivalent to the following proposition.

Proposition 7.17. *For $p = 2$, and $\psi(x) = (-1)^{\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_q}(x)}$, we have an isomorphism*

$$\mathrm{H}_c^2(V_a^2, \mathscr{F}(\psi\sigma)) = \mathrm{Sym}^2 \mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma)).$$

The second tensor power of $\mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma))$ is $\mathrm{H}_c^1(V_a^1 \times V_a^1, \mathscr{F}(\psi\sigma) \boxtimes \mathscr{F}(\psi\sigma))$ (Künneth), with the symmetry $x \otimes y \mapsto y \otimes x$ represented geometrically by $-\tau^*$, where τ is the automorphism $(x, y) \mapsto (y, x)$ of $V_a \times V_a$. If X' is the quotient of $V_a^1 \times V_a^1$ by $\{\mathrm{id}, \tau\}$ and π is the projection of $V_a^1 \times V_a^1$ onto X', the group $\mathrm{Sym}^2 \mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma))$ is therefore identified with the cohomology of X' with coefficients in the part anti-invariant under τ of $\pi_*(\mathscr{F}(\psi\sigma) \boxtimes \mathscr{F}(\psi\sigma))$. Denote by s the function on X' such that $s\pi = \sigma\mathrm{pr}_1 + \sigma\mathrm{pr}_2$. We have $\mathscr{F}(\psi\sigma) \boxtimes \mathscr{F}(\psi\sigma) = \mathscr{F}(\psi(\sigma\mathrm{pr}_1 + \sigma\mathrm{pr}_2)) = \mathscr{F}(\psi s\pi) = \pi^*\mathscr{F}(\psi s)$ (an isomorphism compatible with τ). On the image $\pi(\Delta)$ of the diagonal, the desired sheaf of anti-invariants is therefore zero, while on the complement $X = X' \smallsetminus \pi(\Delta)$ it is the tensor product of $\mathscr{F}(\psi s)$ by $\varepsilon(P)$, where ε is the unique character of order 2 of $\{1, \tau\}$, and P the $\{1, \tau\}$-torsor on X given by the double covering induced by $V_a^1 \times V_a^1$. We have

$$\mathrm{Sym}^2 \mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma)) = \mathrm{H}_c^2(X, \mathscr{F}(\psi\sigma) \otimes \varepsilon(P)). \tag{7.17.1}$$

Let us compute X, s and $\varepsilon(P)$. On $V_a^1 \times V_a^1$, put $x_1 = x_1 \circ \mathrm{pr}_1$, $x_2 = x_2 \circ \mathrm{pr}_2$, $x_1' = x_1 \circ \mathrm{pr}_2$, $x_2' = x_2 \circ \mathrm{pr}_2$. We have $x_1x_2 = x_1'x_2' = a$. Identify functions on X' with functions on X invariant under τ. Let $s_1 = x_1 + x_1'$, $s_2 = x_2 + x_2'$, $p_1 = x_1x_1'$, $p_2 = x_2x_2'$. We have $s_1p_2 = as_2$, $s_2p_1 = as_1$ and X is defined in X' by $s_1 \neq 0$, $s_2 \neq 0$. On X, we therefore have $p_1 = as_1s_2^{-1}$, and (s_1, s_2) identifies x with $\mathbb{G}_m \times \mathbb{G}_m$. On X, the covering P has equation $T^2 - s_1T + as_1s_2^{-1}$ (take T to be the coordinate x_1). If $T = s_1t$, this is rewritten $t^2 - t + as_1^{-1}s_2^{-1}$. We therefore have $\varepsilon(P) = \mathscr{F}(\psi(as_1^{-1}s_2^{-1}))$. Since $s = s_1 + s - 2$, we have $\mathscr{F}(\psi s) \otimes \varepsilon(P) = \mathscr{F}(\psi(s_1 + s_2 + as_1^{-1}s_2^{-1}))$ and the right-hand side of (7.17.1) is identified with the left-hand side of 7.17.

Remark $\mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma))$ is also the first cohomology group of the elliptic curve obtained by completing the affine curve with equations

$$y^2 - y = x_1 + x_2$$
$$x_1x_2 = a;$$

its modular invariant is $j = a^{-2}$.

7.18 Remark

For χ a multiplicative character of $\mathbb{F}_q^\times$, and $a \neq 0$, the method of 7.14 applies to the study of the sum

$$s = \sum_{x_0 \cdots x_n = a} \chi(x_0)\psi(x_0 + \cdots + x_n)$$

(replace the sheaf $\mathscr{F}(\psi)$ on $\mathbb{G}_m$ by $\mathscr{F}(\chi \cdot \psi)$). We again find a cohomology group of dimension $n + 1$, and $|S| \leqslant (n + 1)q^{n/2}$. Such sums were considered by Salie and Mordell [28].

One may hope that our methods make it possible to study the sums

$$S_{k,\chi,a} = \sum_{N(x)=a} \chi(x)\psi\,\mathrm{tr}(x)$$

(k an étale algebra over $\mathbb{F}_q$, $a \in \mathbb{F}_q^\times$, χ a character of $k^\times$). The analog of 7.4 (for $a \neq 0$) should be true.

7.19

The symmetric group S_n acts on $\mathbb{A}^n$, preserving V_a and the map σ, and hence also the sheaf $(\psi\sigma)$, the inverse image under σ of the sheaf (ψ) on $\mathbb{G}_a$. It therefore acts on $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi\sigma))$. Formula (7.2.5) admits the following cohomological interpretation.

Proposition 7.20. *The action of S_n on $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi\sigma))$ is multiplication by the sign character.*

It suffices to show that a transposition τ acts by multiplication by -1. Since $\tau^2 = 1$, the only possible eigenvalues of τ are ± 1. Let V_a/τ be the quotient of V_a for $\{\mathrm{id}, \tau\}$. The map σ, and hence the sheaf $(\psi\sigma)$, descend to the quotient, and the subspace of $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi))$ fixed by τ is $\mathrm{H}_c^{n-1}(V_a/\tau, \mathscr{F}(\psi\sigma))$. We must prove that this cohomology is zero.

Suppose $a \neq 0$. For $\tau = (1,2)$ the quotient V_a/τ is identified with $\mathbb{G}_a \times \mathbb{G}_m^{n-2}$, the quotient map being $(x_1, x_2, x_3, \ldots, x_n) \mapsto (x_1 + x_2, x_3, \ldots, x_n)$. Indeed, x_1 and x_2 are determined up to order by $x_1 + x_2$ and $x_1 x_2 = a/x_3 \cdots x_n$. In the coordinates $(t, x_3, \ldots, x_n)$, the sheaf takes the form $\mathscr{F}(\psi(t + x_3 + \cdots + x_n)) = \mathscr{F}(\psi(t)) \otimes \mathscr{F}(\psi(x_3 + \cdots + x_n))$. The Künneth formula and $\mathrm{H}_0^\bullet(\mathbb{G}_a, \mathscr{F}(\psi)) = 0$ therefore give the required vanishing.

For $a = 0$, we can deduce 7.20 from calculation 7.7.

7.21

The sums $\mathrm{K}_{n,a}$ in this section were studied by S. Sperber, in his thesis, and in [38] by methods p-adic methods inspired by Dwork and by Bombieri's article [6]. For $p \neq 2$, these methods determine the dependence on q in 7.5, the functional equation $\alpha_i = q^{n-1}\beta_i^{-1}$ between the Frobenius eigenvalues for the sums $\mathrm{K}_{n,a}$ and $\mathrm{K}_{n,b}$ if $b = (-1)^n a$ (see 7.6), and the value 7.15 of the product of the roots. For $p \geqslant n+3$, they also determine the p-adic valuations of the roots. The result is very beautiful: if we arrange the roots α_i in a suitable order, with $0 \leqslant i \leqslant n$, $\alpha_i q^{-i}$ are units. The restrictions on p are probably not essential.

8 Other applications

8.1

The identity 1.9.4 can sometimes be used to calculate the number of rational points of an algebraic variety over $\mathbb{F}_q$. In most cases where an exact result has been obtained, cohomology is expressed in terms of algebraic cycles; this is the case for rational surfaces, quadrics, smooth intersections of two quadrics of odd dimension, flag varieties.... This is not, however, always the case; in Lusztig's beautiful article [27], algebraic cycles do not appear.

8.2

For reductive groups, cohomology can be calculated by lifting to characteristic 0: if B is a Borel subgroup of G, G is a B-torsor over G/B; the cohomology of the smooth projective variety G/B is invariant under

specialization, and we use the Leray spectral sequence of $G \to G/B$ to prove the same result for G. It can be expressed conveniently in terms of *the* maximal torus T of G and the action *of the* Weyl group W on T (for the definition *of the* maximal torus, cf. [7, VIII §2 Rem.2] or [10, p.105]). It is the exterior algebra on its primitive part, which agrees up to a shift with the quotient $I(\mathrm{H}^\bullet(BG, \mathbb{Q}_\ell))$ of the cohomology of BG formed by the "indecomposable elements," and $\mathrm{H}^\bullet(BG, \mathbb{Q}_\ell) = \mathrm{H}^\bullet(BT, \mathbb{Q}_\ell)^W$. If X is the group of characters of T, we have altogether

$$\mathrm{H}^\bullet(G, \mathbb{Q}_\ell) = \bigwedge \left(I \left(\mathrm{Sym}^\bullet(X \otimes \mathbb{Q}_\ell(-1))^W \right) [-1] \right).$$

If G is defined over a finite field $\mathbb{F}_q$, $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ acts on X and this isomorphism is Galois-compatible. Passing to cohomology with compact supports, we recover the classical formulas for the number of rational points of G. If $F^* : X \to X$ is defined by the Frobenius morphism $F : T \to T$, and $\deg_G(F)$ is the degree $q^{\dim G}$ of $F : G \to G$, we have

$$|G_0(\mathbb{F}_q)| = \deg_G(F) \cdot \det \left(1 - {F^*}^{-1}, I(\mathrm{Sym}^\bullet(X \otimes \mathbb{Q})^W) \right). \tag{8.2.1}$$

The same formula applies to the Ree and Suzuki groups; these groups are of the form G^F (reductive G, F^2 a Frobenius), and it suffices to verify that the Lefschetz trace formula holds for F acting on G. If B is a Borel subgroup stable by F, we verify it directly for F acting on B; for the action on G/B, which is proper and smooth, we can invoke the general theorems.

8.3

The trace formula was used by Deligne–Lusztig [10], Kazhdan [25] and Springer [39] in the study of complex representations of finite groups $G_0(\mathbb{F}_q)$, for G_0 reductive over $\mathbb{F}_q$. The work of Kazhdan and Springer contains admirable examples of how the trace formula permits "analytic continuation" from a split situation to a nonsplit situation (see 1.13).

Chapter 7

Finiteness theorems in ℓ-adic cohomology

1 Statement of the theorems

Throughout this exposé, S will be a Noetherian scheme and A a left Noetherian torsion ring annihilated by an integer invertible on S. Our main result is the following.

Theorem 1.1. *Suppose that S is regular of dimension 0 or 1. Let $f : X \to Y$ be a morphism of S-schemes of finite type, and let $\mathscr{F}$ be a constructible sheaf of left A-modules on X. Then the sheaves $\mathsf{R}^i f_* \mathscr{F}$ are constructible.*

The proof will be given in Section 2 and in 3.10.

1.2 Remark

In characteristic 0, this result is less general than [2, XIX paragraphe 5], which proves the conclusion of the theorem for every morphism of finite type of excellent schemes of characteristic 0. The proof of [2, XIX] uses, on the one hand, resolution of singularities and, on the other, the assumption that the schemes have equal characteristic (in order to deduce the "purity theorem" from resolution).

1.3 Remark

We shall say that a complex $K \in \mathrm{Ob}\,\mathsf{D}(X, A)$ is *constructible* if its cohomology sheaves are constructible, and shall use a subscript c for the subcategory of $\mathsf{D}(X, A)$ (or D^+, D^-, D^b) consisting of constructible complexes. In the language of derived categories, which we shall use freely, 1.1 says that $\mathsf{R}f_* : \mathsf{D}^+(X, A) \to \mathsf{D}^+(Y, A)$ sends D^+_c into D^+_c. More explicitly:

a) For K concentrated in a sheaf $\mathscr{F}$ in degree 0, one has $\mathscr{H}^i \mathsf{R}f_* K = \mathsf{R}^i f_* \mathscr{F}$; the derived statement therefore implies the theorem.

b) Conversely, one invokes the spectral sequence

$$E_1^{pq} = \mathsf{R}^q f_* \mathscr{H}^p(K) \Rightarrow \mathscr{H}^{p+q} \mathsf{R}f_* K.$$

Speaking the language of derived categories has the advantage of replacing, by the simple transitivity formula $\mathsf{R}(fg)_* = \mathsf{R}f_*\mathsf{R}g_*$, the Leray spectral sequence $E_2^{pq} = \mathsf{R}^p f_* \mathsf{R}^q g_* \Rightarrow \mathsf{R}^{p+q}(fg)_*$.

Under the hypotheses of 1.1, it follows from [2] that $\mathsf{R}f_*$ has finite cohomological dimension. This makes it possible above to replace D^+ by D, D^-, or D^b.

1.4 Ideas of the proof

a) Poincaré duality treats the case where X is smooth, $\mathscr{F}$ is locally constant, and $Y = S$. The dimension hypothesis is seen naïvely when calculating the $\mathsf{R}\mathscr{H}om$ on S.

b) Factor f as gj: g proper and j an open immersion. One has $\mathsf{R}f_* = \mathsf{R}g_*\mathsf{R}j_*$, and the finiteness theorem for proper morphisms controls $\mathsf{R}g_*$. By dévissage, one reduces to assuming that x is smooth, $\mathscr{F}$ is locally constant, f is an open immersion, and Y is proper over S.

c) Naively, an induction on $\dim X$ (with a change of S) makes it possible to suppose that the theorem is true outside a part of Y finite over S. Denoting the structure morphisms by $a : X \to S$ and $b : Y \to S$, one then shows that if the theorem were false for $\mathscr{F}$ and $\mathsf{R}f_*$, it would also be false for $\mathscr{F}$ and $\mathsf{R}b_*\mathsf{R}f_* = \mathsf{R}a_*$: a contradiction.

Corollary 1.5. *Under the hypotheses of the theorem, the categories* $\mathsf{D}_c(X, A)$ *and* $\mathsf{D}_c(Y, A)$ *are taken into one another by the* 4 *operations* $\mathsf{R}f_*$, $\mathsf{R}f_!$, f^*, $\mathsf{R}f^!$.

The case of $\mathsf{R}f_*$ is treated above, that of $\mathsf{R}f_!$ in [2, XVII 5.3], and that of f^* is clear. It remains to consider $\mathsf{R}f^!$. The problem is local, which makes it possible to suppose that f factors as $X \xrightarrow{i} Z \xrightarrow{g} Y$ (i a closed immersion and g smooth, purely of relative dimension n). Poincaré duality $\mathsf{R}g_!K(n)[2n]$ and the transitivity $\mathsf{R}f^! = \mathsf{R}i^!\mathsf{R}g^!$ reduce us to proving 1.5 for i. For every $L \in \mathsf{D}(Z, A)$, if j is the inclusion into Z of $U = Z \smallsetminus X$, then $i_*\mathsf{R}i^!L$ is the mapping cylinder of $K \to \mathsf{R}j_*j^*K$, and 1.5 for i follows from 1.1 for j.

Corollary 1.6. *Let* X *be as in the theorem, and suppose that* A *is commutative. Then, if* $\mathscr{F}$ *and* $\mathscr{G}$ *are constructible sheaves of* A*-modules on* X*, the* $\mathscr{E}xt^i_A(\mathscr{F}, \mathscr{G})$ *are constructible:* $\mathsf{R}\mathscr{H}om$ *sends* $\mathsf{D}^-_c(X, A) \times \mathsf{D}^+_c(X, A)$ *into* $\mathsf{D}^+_c(X, A)$.

By dévissage of $\mathscr{F}$, we reduce to assuming that $\mathscr{F}$ has the form $j_!\mathscr{F}_1$ ($j : Y \hookrightarrow X$ a locally closed immersion and $\mathscr{F}_1$ locally constant on Y). One then has

$$\mathsf{R}\mathscr{H}om(j_!\mathscr{F}_1, \mathscr{G}) = \mathsf{R}j_*\,\mathscr{H}om(\mathscr{F}_1, \mathsf{R}j^!\mathscr{G}):$$

by 1.3, we are reduced to the case where $\mathscr{F}$ is locally constant—or even constant, since the problem is local. For $\mathscr{F}$ locally constant and constructible, one has $\mathscr{H}om(\mathscr{F}, \mathscr{G})_x = \hom(\mathscr{F}_x, \mathscr{G}_x)$, and likewise for $\mathsf{R}\mathscr{H}om$. For $\mathscr{F}$ constant, the $\mathscr{E}xt^i$ can therefore be calculated using a projective resolution of finite type of its constant value, and the $\mathscr{E}xt^i$ are constructible if $\mathscr{G}$ is—whence the corollary.

1.7 Remark

Since $\mathsf{R}f_*$ has finite cohomological dimension, it takes complexes of finite tor-dimension (resp. $\leqslant d$) to complexes of finite tor-dimension (resp. $\leqslant d$) [2, XVII 5.2.11]. [2, XVII 5.2.10] and the proof of 1.5 then show that finite tor-dimension is stable under the 4 operations $\mathsf{R}f_*$, $\mathsf{R}f_!$, f^*, $\mathsf{R}f^!$. A variant of the argument of 1.6 (apply dévissage to K along a partition of x so that the cohomology sheaves become locally constant, then localize so as to replace K by a finite complex of locally free sheaves of finite type) then shows that, for A commutative, $\mathsf{R}\mathscr{H}om$ induces

$$\mathsf{R}\mathscr{H}om : \mathsf{D}^b_{\mathrm{ctf}}(X, A) \times \mathsf{D}^b_{\mathrm{tf}}(X, A) \to \mathsf{D}^+_{\mathrm{tf}}(X, A)$$

(t.f.: finite tor-dimension).

1.8

Under the hypotheses of the theorem, the same method allows us to prove a local biduality theorem $K \xrightarrow{\sim} DDK$ (Section 4). We also prove a finiteness theorem for vanishing-cycle sheaves. For arbitrary S, one still obtains a "generic" theorem:

Theorem 1.9. *Let $f : X \to Y$ be a morphism of S-schemes of finite type, and let $\mathscr{F}$ be a constructible sheaf of A-modules on X. There is a dense open subset U of S such that*

(i) Above U, the $\mathsf{R}^i f_* \mathscr{F}$ are constructible, and all but finitely many of them vanish.

(ii) Formation of the $\mathsf{R}^i f_* \mathscr{F}$ is compatible with every base change $S' \to U \subset S$.

For S the spectrum of a field, one has $U = S$.

If S is the spectrum of a field, a passage-to-the-limit argument gives compatibility with S-base changes for the $\mathsf{R}^i f_*$ for every quasi-compact quasi-separated morphism of S-schemes and every sheaf $\mathscr{F}$.

Corollary 1.10. *Let x be a scheme of finite type over a separably closed k, and let $\mathscr{F}$ be a constructible sheaf of A-modules. Then the $\mathrm{H}^i(X, \mathscr{F})$ are of finite type.*

This is the special case $S = \mathrm{Spec}(k)$, $Y = S$.

Corollary 1.11. *Let X and Y be two schemes of finite type over a separably closed k, and let $K \in \mathrm{Ob}\, \mathsf{D}^-(X, A^\circ)$, $L \in \mathrm{Ob}\, \mathsf{D}^-(Y, A)$ be complexes of sheaves of right and left modules. The Künneth arrow*

$$\mathsf{R}\Gamma(X, K) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(Y, L) \to \mathsf{R}\Gamma(X \times Y, \mathrm{pr}_1^* K \overset{\mathsf{L}}{\otimes} \mathrm{pr}_2^* L)$$

is an isomorphism.

Proceed as in [2, XVII 5.4.3]:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\mathrm{pr}_1} & X \\ \downarrow{\scriptstyle \mathrm{pr}_2} & & \downarrow{\scriptstyle a} \\ Y & \xrightarrow{b} & \mathrm{Spec}(k) \end{array}$$

base change by b shows that $\mathsf{R}\mathrm{pr}_{2,*}\mathrm{pr}_1^* K$ is $b^* \mathsf{R}\Gamma(X, K)$. One then has (cf. [2, XVII 5.2.11] and the proof)

$$\begin{aligned} \mathsf{R}\Gamma(X \times Y, \mathrm{pr}_1^* K \overset{\mathsf{L}}{\otimes} \mathrm{pr}_2^* L) &= \mathsf{R}\Gamma(Y, \mathsf{R}\mathrm{pr}_{2,*}(\mathrm{pr}_1^* K \overset{\mathsf{L}}{\otimes} \mathrm{pr}_2^* L)) \\ &= \mathsf{R}\Gamma(Y, (\mathsf{R}\mathrm{pr}_{2,*}\mathrm{pr}_1^* K) \overset{\mathsf{L}}{\otimes} L) \\ &= \mathsf{R}\Gamma(Y, b^* \mathsf{R}\Gamma(X, K) \overset{\mathsf{L}}{\otimes} L) \\ &= \mathsf{R}\Gamma(X, K) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(Y, L) \end{aligned}$$

1.12

Finally, as a local counterpart of 1.9(ii), we shall prove a generic local acyclicity theorem (2.13).

2 Generic theorems

In this section, we prove 1.9 and the generic local acyclicity theorem 2.13. To prove 1.9, we immediately reduce to the case where S is integral. Let η be its generic point.

2.1

We begin by proving 1.9 under the following additional assumptions: X is smooth over S, purely of relative dimension n, $A = \mathbb{Z}/m$, $\mathscr{F}$ is locally constant, and $Y = S$. Let $\mathscr{F}' = \mathscr{H}om(\mathscr{F}, \mathbb{Z}/m)$. We shall show that if the $\mathsf{R}^i f_! \mathscr{F}'$ are locally constant, the conclusions of 1.9 hold with $U = S$. Recall that if $\mathscr{L}$ is a constructible locally constant sheaf, the $\mathscr{E}xt^i(\mathscr{L}, \mathscr{G})$ may be computed fiber by fiber, and are constructible if $\mathscr{G}$ is. Recall also that $\mathbb{Z}/m$ is an injective $\mathbb{Z}/m$-module and is the dualizing module, so that $\mathscr{F} = \mathsf{R}\mathscr{H}om(\mathscr{F}', \mathbb{Z}/m)$. Poincaré duality

$$\mathsf{R}f_* \, \mathsf{R}\mathscr{H}om(K, \mathsf{R}f^! L) = \mathsf{R}\mathscr{H}om(\mathsf{R}f_! K, L),$$

for $K = \mathscr{F}'$, $L = \mathbb{Z}/m$, $\mathsf{R}f^! L = \mathbb{Z}/m[2n](n)$, therefore gives

$$\mathsf{R}^{2n-i} f_* \mathscr{F} = \mathscr{H}om(\mathsf{R}^i f_! \mathscr{F}', \mathbb{Z}/m)(-n).$$

The sheaf on the right is locally constant and its formation commutes with every base change, which proves the assertion.

2.2

Let us prove 1.9 under the additional assumptions that X is smooth over S, $\mathscr{F}$ is locally constant, and $Y = S$.

a) By decomposing X into connected components, we reduce to the case where it is purely of some relative dimension n over S.

b) By decomposing A as a product, we reduce to the case where $\ell^m A = 0$, with ℓ a prime invertible on S; we then filter $\mathscr{F}$ by the $\ell^k \mathscr{F}$ and use the corresponding spectral sequence to reduce to the case $\ell \mathscr{F} = 0$. We may then replace A by $A/\ell A$ and assume that $\ell A = 0$.

c) After replacing S by a suitable U, we may assume that there exists a finite étale Galois covering X_1/X, with Galois group G, such that the inverse image $\mathscr{F}_1$ of $\mathscr{F}$ on X_1 is a constant sheaf, with constant value F. Writing f_1 for the projection of X_1 onto S, we then have

$$\mathsf{R}^i f_{1*} \mathscr{F}_1 = (\mathsf{R}^i f_{1*} \mathbb{Z}/\ell) \otimes_{\mathbb{Z}/\ell} F.$$

We also have the Hochschild–Serre spectral sequence (or the Leray spectral sequence for the covering X_1/X)

$$\underline{\mathrm{H}}^p(G, \mathsf{R}^q f_{1*} \mathscr{F}_1) \Rightarrow \mathsf{R}^{p+q} f_* \mathscr{F}.$$

By 2.1, after shrinking U if necessary, we may assume that the $\mathsf{R}^i f_{1*} \mathbb{Z}/\ell$ are locally constant and that their formation commutes with every base change. The same property then holds for the $\mathsf{R}^i f_* \mathscr{F}$. Finally, if $\bar{\eta}$ is a geometric point localized at the generic point η of S, almost all the $\mathsf{R}^i f_* \mathscr{F}$ vanish, because the $(\mathsf{R}^i f_* \mathscr{F})_{\bar{\eta}} = \mathrm{H}^i(X_{\bar{\eta}}, \mathscr{F})$ do.

2.3

We prove by induction on n that

$$\begin{array}{c} \text{The conclusions of 1.9 hold when } \dim X_\eta \leqslant n \\ \text{and } f \text{ is an open immersion with dense image.} \end{array} \qquad (*_n)$$

For $n = 0$, after shrinking S if necessary, we have $X = Y$: $(*_0)$ is clear. Assume $(*_{n-1})$, and prove $(*_n)$. In $(*_{n-1})$, "open immersion" may be replaced by "immersion," as is seen by factoring it as an open immersion followed by a closed immersion.

Lemma 2.4. *After shrinking S if necessary, the conclusions of 1.9 hold over $Y' \subset Y$, where the complement Y_1 of Y' is finite over S.*

The assertion is local on Y, which we may assume affine: $Y \subset \mathbb{A}^n_S$. The induction hypothesis $(*_{n-1})$ applies to

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow & \downarrow \mathrm{pr}_i \\ & & \mathbb{A}^1_S \end{array}$$

Thus, for each i, there exists a dense open subset U_i of $\mathbb{A}^1_S$ such that the conclusions of 1.9 hold over $\mathrm{pr}_i^{-1}(U_i)$; they hold over the union of the $\mathrm{pr}_i^{-1}(U_i)$, and 2.4 follows.

2.6

Let us prove $(*_n)$ when X is smooth over S and $\mathscr{F}$ is locally constant on X. Since the problem is local on Y, we may first assume Y affine and then projective (replacing Y by its closure in a projective space). Let $i : Y_1 \hookrightarrow Y$ and $j : Y' \to Y$ be as in 2.4.

$$\begin{array}{ccccc} X & \hookrightarrow & Y & \xleftarrow{i} & Y_1 \\ & \searrow^{a} & \downarrow{\scriptstyle b} & \swarrow_{b_1} & \\ & & S & & \end{array}$$

After shrinking S if necessary, we know that $j^*\mathsf{R}f_*\mathscr{F}$ is constructible and that its formation commutes with every base change in S; we also know that $\mathsf{R}a_*\mathscr{F} = \mathsf{R}b_*\mathsf{R}f_*\mathscr{F}$ is constructible and that its formation commutes with every base change.

Apply $\mathsf{R}b_*$ to the triangle defined by the exact sequence

$$0 \longrightarrow j_!j^*\mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}f_*\mathscr{F} \longrightarrow i_!i^*\mathsf{R}f_*\mathscr{F} \longrightarrow 0: \tag{1}$$

we obtain a triangle

$$\longrightarrow \mathsf{R}b_*j_!j^*\mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}a_*\mathscr{F} \longrightarrow b_{1*}i^*\mathsf{R}f_*\mathscr{F} \longrightarrow \tag{2}$$

in which the first two terms are constructible and their formation commutes with every base change in S (for the first, by the finiteness theorem for the proper morphism b). The same therefore holds for the third. Since b_1 is finite, it follows that $i^*\mathsf{R}f_*\mathscr{F}$ is constructible and its formation commutes with every base change in S; the same holds for $\mathsf{R}f_*\mathscr{F}$ by (1).

2.7

Let us prove $(*_n)$ in general. We first reduce to the case where X contains a dense open subset V smooth over S. When S is the spectrum of a perfect field, it is enough to replace X by X_{red} (and Y by Y_{red}). In general, we must shrink S, make a finite, radicial, surjective base change $S' \to S$, and replace X and Y by X_{red} and Y_{red}. Since the étale topology is insensitive to finite, radicial, surjective morphisms, this is harmless. After shrinking V, we may assume that $\mathscr{F}$ is locally constant on V.

$$V \xhookrightarrow{i} X \xhookrightarrow{f} Y.$$

Define Δ by the triangle

$$\longrightarrow \mathscr{F} \longrightarrow \mathsf{R}j_*j^*\mathscr{F} \longrightarrow \Delta \longrightarrow . \tag{1}$$

The cohomology sheaves of Δ are supported on $X \smallsetminus V$, and $\dim(X \smallsetminus V)_\eta < n$. The induction hypothesis therefore allows us to assume that $\mathsf{R}f_*\Delta$ is constructible. Applying $\mathsf{R}f_*$ to the triangle (1), we obtain a triangle

$$\longrightarrow \mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}(fj)_*j^*\mathscr{F} \longrightarrow \mathsf{R}f_*\Delta \longrightarrow$$

in which two of the vertices are constructible and their formation commutes with every base change in S. The same therefore holds for the third.

2.8

Let us prove 1.9. The problem is local on Y, which we may assume affine. Taking an affine covering of X and invoking the Leray spectral sequence, we reduce to the case where X is affine as well. This ensures that we can factor f as an open immersion followed by a proper morphism: $f = gj$, hence $\mathsf{R}f_* = \mathsf{R}g_*\mathsf{R}j_*$. The open immersion is covered by $(*_n)$, and the proper morphism by the finiteness theorem.

The following corollaries are proved as in Section 1.

Corollary 2.9. *Under the hypotheses of the theorem, for K in $\mathsf{D}^b_c(X, A)$ or $\mathsf{D}^b_c(Y, A)$ respectively, there exists a nonempty open subset U of S over which $\mathsf{R}f_*K$, $\mathsf{R}f_!K$, f^*K, and $\mathsf{R}f^!K$ belong to $\mathsf{D}^b_c(Y, A)$ or $\mathsf{D}^b_c(X, A)$, and their formation commutes with every base change $S' \to U \subset S$.*

Corollary 2.10. *Let X be as in the theorem, and suppose that A is commutative. Then, if $\mathscr{F}$ and $\mathscr{G}$ are constructible sheaves of A-modules on X, over a dense open subset U of S, the $\mathscr{E}x\ell^i_A(\mathscr{F}, \mathscr{G})$ are again constructible and their formation commutes with every base change $S' \to U \subset S$.*

2.11

If x is a geometric point of a scheme X, we denote by X_x the strict henselization of X at x. For $f : X \to S$ and t a geometric point of S, we denote by X_t the geometric fiber of X at t. Finally, for x a geometric point of X and t a geometric point of $S_{f(x)}$, $(X_x)_t$ is the fiber at t of $X_x \to S_{f(x)}$.

Definition 2.12. *Let $f : X \to S$ and $K \in \mathrm{Ob}\,\mathsf{D}^+(X, A)$. We say that f is* locally acyclic at x relative to *K if, for every geometric point t of $S_{f(x)}$, we have $K_x = \mathsf{R}\Gamma(X_x, K) \xrightarrow{\sim} \mathsf{R}\Gamma(X_{x,t}, K)$. We say that f is* locally acyclic relative to *K if this holds for every x, and* universally locally acyclic relative to *K if it remains true after every base change $S' \to S$.*

Theorem 2.13. *Let $f : X \to S$ be a morphism of finite type and let $\mathscr{F}$ be constructible on X. There exists a dense open subset U of S over which f is universally locally acyclic relative to $\mathscr{F}$.*

We shall use the following result, proved in the appendix.

Lemma 2.14. *Let $X \xrightarrow{f} S_1 \xrightarrow{g} S_2$. If g is smooth and f is universally locally acyclic relative to K, then so is gf.*

We reduce to the case where S is integral, with generic point η, and argue by induction on $\dim X_\eta$. We first deduce from the induction hypothesis that

$$\text{After shrinking } S, \text{ there exists } T \subset X, \text{ finite over } S, \text{ such that } f \text{ is universally locally acyclic outside } T. \tag{A}$$

Since this question is local, we localize on X and factor f as $X \xrightarrow{u} \mathbb{A}^1_S \xrightarrow{S}$. The induction hypothesis applies to u, and we conclude by 2.14 and the usual arguments.

To prove the theorem, we may assume that X is proper, since the problem is local. We shrink S so that the $\mathsf{R}^i f_* \mathscr{F}$ are locally constant and (A) applies, and use the following lemma.

Lemma 2.15. *Let $f : X \to S$ be proper and let $\mathscr{F}$ be constructible. Suppose that the $\mathsf{R}^i f_* \mathscr{F}$ are locally constant and that, outside $T \subset X$ finite over S, f is locally acyclic relative to $\mathscr{F}$. Then f is locally acyclic relative to $\mathscr{F}$.*

Let s be a geometric point of S, let S_s be the strict localization of S at s, and let t be a geometric point of S_s. Let $X_{(s)}$ be the inverse image of X on S_s, let $\bar{j} : X_t \to X_{(s)}$, and let $i : X_s \hookrightarrow X_{(s)}$. Again denoting by $\mathscr{F}$ the inverse image of $\mathscr{F}$ on $X_{(s)}$ or X_t, we must prove that $i^*\mathscr{F} \xrightarrow{\sim} i^*\mathsf{R}\bar{j}_*\mathscr{F}$. Let Δ be the mapping cylinder of this map. Its cohomology sheaves are supported on T_s. To prove that $\Delta = 0$, it therefore suffices to prove that $\mathsf{R}\Gamma(X_s, \Delta) = 0$. This is the mapping cylinder of $(\mathsf{R}f_*\mathscr{F})_s = \mathsf{R}\Gamma(X_s, \mathscr{F}) \to \mathsf{R}\Gamma(X_s, i^*\mathsf{R}\bar{j}_*\mathscr{F}) = \mathsf{R}\Gamma(X_{(s)}, \mathsf{R}\bar{j}_*\mathscr{F}) = \mathsf{R}\Gamma(X_t, \mathscr{F}) = (\mathsf{R}f_*\mathscr{F})_t$. By hypothesis this specialization morphism is an isomorphism, and $\Delta = 0$.

Corollary 2.16. *When S is the spectrum of a field, every S-scheme X is universally locally acyclic relative to K, for every $K \in \operatorname{Ob} \mathsf{D}^+(X, A)$.*

This follows from 2.13 by passage to the limit (which is possible because the U of 2.13 is always equal to S.

3 Proof of 1.1 and constructibility of the vanishing-cycle sheaves

3.1

Let S be a strictly local trait (the spectrum of a henselian discrete valuation ring with separably closed residue field). Denote its closed and generic points by s and η, and let $\bar{\eta}$ be a geometric point localized at η (a separable closure of $k(\eta)$). Let X be over S and let $\mathscr{F}$ be a sheaf on X_η. Recall the definition of the vanishing-cycle sheaves $\mathsf{R}^i\Psi_\eta(\mathscr{F})$ (sheaves on X_s, endowed with an action of $\mathrm{Gal}(\bar{\eta}/\eta)$). Let $i : X_s \hookrightarrow X$, $j : X_\eta \hookrightarrow X$, and let $\bar{j}$ be the composite $X_{\bar{\eta}} \to X_\eta \hookrightarrow X$; then

$$\mathsf{R}^i\Psi_\eta(\mathscr{F}) = i^*\mathsf{R}^i\bar{j}_*(\bar{j}^*\mathscr{F}).$$

Here is the main theorem of this subsection. It improves [16, XIII 2.3.1, 2.4.2].

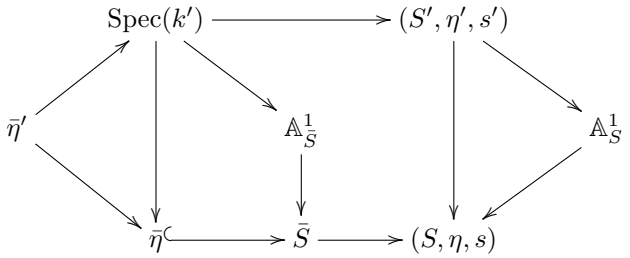
Theorem 3.2. *Suppose that X is of finite type over S and that $\mathscr{F}$ is constructible. Then the vanishing-cycle sheaves $\mathsf{R}^i\Psi_\eta(\mathscr{F})$ are constructible.*

We argue by induction on the dimension n of X_η. We may moreover assume—and do assume—that X_η is dense in X (replacing X by $\bar{X}_\eta$ does not change the $\mathsf{R}^i\Psi$, which are supported on $\bar{X}_\eta$).

Lemma 3.3. *If 3.2 holds when $X_\eta < n$, and if $\dim X_\eta = n$, then there exist constructible subsheaves $\mathscr{G}_i$ of the $\mathsf{R}^i\Psi_\eta(\mathscr{F})$ such that the supports of the local sections of $\mathsf{R}^i\Psi_\eta(\mathscr{F})/\mathscr{G}$ are finite.*

Let s' be a geometric generic point of the affine line over s, and let S' be the strict localization at s' of

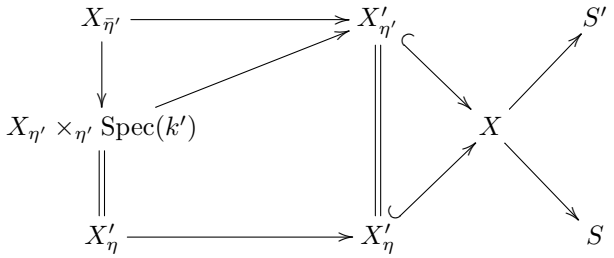
the affine line $\mathbb{A}^1_S$ over S. It is again a strictly local trait, and uniformizers for S are uniformizers on S'.



Let $k' = k(\bar\eta) \otimes_{k(\eta)} k(\eta')$. If $\bar S$ is the normalization of S in $\bar\eta$, then k' is the fraction field of the strict localization of $\mathbb{A}^1_{\bar S}$ at s'; it is therefore a field, and $\mathrm{Gal}(\bar\eta/\eta) = \mathrm{Gal}(k'/\eta')$. Let $\bar\eta'$ be the spectrum of an algebraic closure of k'. The Galois group $P = \mathrm{Gal}(\bar\eta'/k')$ is a pro-p group, where p is the characteristic exponent of $k(s)$.

Lemma 3.4. *For every scheme X' over S', and every sheaf $\mathscr{F}$ on $X'_\eta = X'_{\eta'}$, the vanishing-cycle sheaves for X'/S' and for X'/S are related by $\mathsf{R}^i\Psi_\eta(\mathscr{F}) = \mathsf{R}^i\Psi_{\eta'}(\mathscr{F})^P$.*

The following diagram compares the morphisms used in the definition of $\mathsf{R}\Psi$ for X'/S' and X'/S:



and the lemma follows from the Hochschild–Serre spectral sequence, since P is a pro-p group and p is invertible in A.

Let us prove 3.3. The question is local; this allows us to assume that X is affine, $X \subset \mathbb{A}^n_S$. Let f be one of the projections $X \subset \mathbb{A}^n_S \to \mathbb{A}^1_S$. Let X' be the "localization" $X_{\mathbb{A}^1_S} S'$ of x, and let $\mathscr{F}'$ be the inverse image of $\mathscr{F}$ on x'.

$$\begin{array}{ccccc} \mathscr{F}' \text{ on } X' & \longrightarrow & S' & & \\ \downarrow{\scriptstyle\lambda} & & \downarrow{\scriptstyle\lambda} & & \\ \mathscr{F} \text{ on } X & \xrightarrow{f} & \mathbb{A}^1_S & \longrightarrow & S. \end{array}$$

On $X'_s = X'_{s'}$, we have

$$\lambda^* \mathsf{R}^i\Psi_\eta(\mathscr{F}) = \mathsf{R}^i\Psi_\eta(\mathscr{F}') = \mathsf{R}^i\Psi_{\eta'}(\mathscr{F}')^P:$$

this sheaf is constructible because the induction hypothesis applies to X'/S', and we use the following lemma, applied to $X_s \subset \mathbb{A}^n_S$ and to the vanishing-cycle sheaves.

Lemma 3.5. *Let $\mathbb{A}^n$ be affine n-space over a field k, let $X \subset \mathbb{A}^n$, let $]sF$ be a sheaf of A-modules on X, and let $\bar\eta$ be a geometric generic point of $\mathbb{A}^1$. Let $X_{\bar\eta,i}$ be the geometric generic fiber of $X \subset \mathbb{A}^n \xrightarrow{\mathrm{pr}_i} \mathbb{A}^1$, and let $\mathscr{F}_{\bar\eta,i}$ be the restriction of $\mathscr{F}$ to $X_{\bar\eta,i}$. If the $\mathscr{F}_{\bar\eta,i}$ are constructible, there exists a constructible $\mathscr{F}' \subset \mathscr{F}$ such that the local sections of $\mathscr{F}/\mathscr{F}'$ have finite support.*

By passage to the limit, for each i

a) There exists an étale neighborhood U of $\bar{\eta}$ and a constructible sheaf $\mathscr{H}$ on $X_{U,i} = X \times_{\mathbb{A}^1,\mathrm{pr}_i} U$ whose restriction $\mathscr{F}_{\bar{\eta},i}$ to $X_{\bar{\eta},i}$ is the given one.

b) For a suitable U, the isomorphism $\mathscr{H}_{\bar{\eta},i} \xrightarrow{\sim} \mathscr{F}_{\bar{\eta},i}$ comes from $u : \mathscr{H} \to \mathscr{F}_{U,i}$, where $\mathscr{F}_{U,i}$ is the inverse image of $\mathscr{F}$ on $X_{U,i}$. Let $\varphi : X_{U,i} \to X$. The morphism u defines $v_i : \varphi_!\mathscr{H} \to \mathscr{F}$, with image $\mathscr{F}'_i$. We have $(\mathscr{F}/\mathscr{F}'_i)_{\bar{\eta},i} = 0$, and for $\mathscr{F}'$ we take the sum of the $\mathscr{F}'_i$.

3.6

Let us prove 3.2 in dimension n. We may assume that X is affine (because the problem is local on X_s), and then projective over S (by taking the closure of X in a projective embedding). We then have a spectral sequence [16, I 2.2.3] (the Leray spectral sequence for $\bar{j}$, taking into account the base-change theorem for the proper morphism $X \to S$).

$$E_2^{pq} = \mathrm{H}^p(X_s, \mathsf{R}^i\Psi_\eta(\mathscr{F})) \Rightarrow \mathrm{H}^{p+q}(X_{\bar{\eta}}, \mathscr{F}). \tag{1}$$

Let $\mathscr{G}^q \subset \mathsf{R}^q\Psi_\eta(\mathscr{F})$ be as in 3.3, and let $\mathscr{H}^q$ be the quotient sheaf. We have $\mathrm{H}^i(X_s, \mathscr{H}^q) = 0$ for $i \neq 0$, and for $\mathscr{H}^q$, hence $\mathsf{R}^q\Psi_\eta(\mathscr{F})$, to be constructible, it suffices that $\mathrm{H}^0(X_s, \mathscr{H}^q)$ be of finite type.

Compute (1) modulo modules of finite type (i.e. in the quotient of the category of A-modules by the thick subcategory of A-modules of finite type). The long cohomology exact sequence deduced from $0 \to \mathscr{G}^q \to \mathsf{R}^q\Psi_\eta(\mathscr{F}) \to \mathscr{H}^q \to 0$, together with 1.10 applied to $\mathscr{G}^q$, gives $E_2^{pq} \sim \mathrm{H}^q(X_s, \mathscr{H}^q)$. In particular, $E_2^{pq} \sim 0$ for $p \neq 0$, $E_2^{0q} \sim \mathrm{H}^q(X_{\bar{\eta}}, \mathscr{F}) \sim 0$, and this completes the proof.

Parallel arguments give the following result, which improves [16, XIII 2.1.12, 2.4.2].

Proposition 3.7. *The formation of the vanishing-cycle sheaves commutes with changes of traits.*

Let $g : (S', \eta', s') \to (S, \eta, s)$ be a (surjective) morphism of strictly local traits, let X/S, let $\mathscr{F}$ be on X_η, of torsion prime to the residual characteristic, and let $(X', \mathscr{F}')$ be their inverse images on S'. Again denoting by g the morphisms parallel to g, such as $X'_{s'} \to X_s$, we must prove that

$$g^*\mathsf{R}\Psi_\eta(\mathscr{F}) \xrightarrow{\sim} \mathsf{R}\Psi_{\eta'}(\mathscr{F}'). \tag{3.7.1}$$

This morphism is defined by the commutative diagram

$$\begin{array}{ccccc} X'_{s'} & \xrightarrow{i'} & X' & \xleftarrow{\bar{j}'} & X'_{\eta'} \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g} \\ X_s & \xrightarrow{i} & X & \xleftarrow{\bar{j}} & X_{\bar{\eta}} \end{array}$$

A passage to the limit reduces us to the case where X is of finite type over S, and we argue by induction on $\dim X_\eta$. Since the question is local, we may assume that X is affine, and then projective. Assume (3.7.1) when $\dim X_\eta < n$. When $\dim X_\eta = n$, it then follows from the following two assertions, where Δ is the mapping cylinder of (3.7.1).

(A) The support of the local sections of the cohomology sheaves of Δ is finite.

(B) $\mathsf{R}\Gamma(X_s, \Delta) = 0$, hence $\Delta = 0$, by (A).

3.8

The proof of (A) parallels that of 3.3. We localize on X and consider maps $X \to \mathbb{A}^1_S$; denoting by $S(x)$ the strict henselization of $\mathbb{A}^1_S$ at the generic point of the special fiber, we apply the induction hypothesis to $X_1/S(x)$ ($X_1 = X \times_{\mathbb{A}^1} S(x)$) and to $S'(x)$. Taking invariants under a pro-p group, we find that Δ vanishes on the geometric generic fiber of $X_{s'} \to \mathbb{A}^1_s$. Since this holds locally on X and for every projection, we obtain (A).

3.9

For (B), the properness of X over S ensures that

$$\mathsf{R}\Gamma(X'_{s'}, \mathsf{R}\Psi_{\eta'}(\mathscr{F}')) = \mathsf{R}\Gamma(X'_{\bar{\eta}}, \mathscr{F}')$$

and

$$\mathsf{R}\Gamma(X_s, \mathsf{R}\Psi_{\eta}(\mathscr{F})) = \mathsf{R}\Gamma(X_{\bar{\eta}}, \mathscr{F}).$$

We have $\mathsf{R}\Gamma(X_s, \mathsf{R}\Psi_\eta(\mathscr{F})) \xrightarrow{\sim} \mathsf{R}\Gamma(X'_{s'}, g^*\mathsf{R}\Psi_\eta(\mathscr{F}))$ (invariance under change of the separably closed base field); thus $\mathsf{R}\Gamma(X'_{s'}, \Delta)$ is the mapping cylinder of $\mathsf{R}\Gamma(X_{\bar{\eta}}, \mathscr{F}) \to \mathsf{R}\Gamma(X'_{\bar{\eta}}, \mathscr{F}')$, an isomorphism by the same invariance theorem.

3.10

Let us prove 1.1. We reduce to the case where S is connected, hence integral. If $\dim S = 0$ (S is the spectrum of a field), apply 1.9, 1.10. If S has dimension 1, Theorem 1.9 ensures that the $\mathsf{R}^i f_* \mathscr{F}$ are constructible over the complement of a finite set T of points of S. It remains to see that their restrictions to the Y_t ($t \in T$) are constructible. It suffices to see this after localization at t: we may assume that S is a strictly local trait. The essential case is the following.

Lemma 3.11. *Let X be of finite type over S, let $j : X_\eta \hookrightarrow X$, $i : X_s \hookrightarrow X$, and let $\mathscr{F}$ be constructible on X_η. Then $i^*\mathsf{R}j_*\mathscr{F}$ is constructible.*

Let $I = \mathrm{Gal}(\bar{\eta}/\eta)$ be the inertia group. It is known to be an extension of $\widehat{\mathbb{Z}}_{p'}(1)$ by a pro-p group P (p being the characteristic exponent of the residue field). The Hochschild–Serre spectral sequence gives

$$E_2^{pq} = \mathscr{H}^p(I, \mathsf{R}^q\Psi_\eta(\mathscr{F})) \Rightarrow i^*\mathsf{R}^{p+q}j_*\mathscr{F}.$$

Let $R^q_t = \mathsf{R}^q\Psi_\eta(\mathscr{F})^P$ (t for tame). If σ is a generator of $\widehat{\mathbb{Z}}_{p'}(1)$, then $E_2^{0q} = \ker(\sigma - 1, R^q_t)$, $E_2^{1q} = \mathrm{coker}(\sigma - 1, R^q_t)$, and $E_2^{pq} = 0$ for $p \neq 0, 1$. These sheaves are constructible because the vanishing-cycle sheaves are, and 3.11 follows.

3.12

Let us treat the general case, with S still a strictly henselian trait. If $X = X_\eta$, then f is the composite $X \to Y_\eta \to Y$, and we apply 1.9 and 3.11. In the general case, let $j : X_\eta \hookrightarrow X$ and let Δ be the mapping cylinder of $\mathscr{F} \to \mathsf{R}j_*j^*\mathscr{F}$. Its cohomology sheaves are supported on X_s; by 1.9 applied to $X_s \to Y_{s'}$, $\mathsf{R}f_*\Delta$ is therefore constructible. The complex $\mathsf{R}f_*\mathsf{R}j_*j^*\mathscr{F} = \mathsf{R}(fj)_*j^*\mathscr{F}$ is also constructible, and we conclude from the triangle

$$\longrightarrow \mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}f_*\mathsf{R}j_*j^*\mathscr{F} \longrightarrow \mathsf{R}f_*\Delta \longrightarrow .$$

4 Local biduality

4.1

Let S be regular of dimension 0 or 1, and put $K_S = A$ (the constant sheaf with value A). For $K \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(S, A)$, put $DK = \mathsf{R}\mathscr{H}om(K, K_S) \in \mathrm{Ob}\,\mathsf{D}^+(S, A^\circ)$ (a complex of sheaves of right A-modules). We still have $DK \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(S, A^0)$, and an explicit calculation, possible because S has dimension 1, shows that $K \xrightarrow{\sim} DDK$ (for $A = \mathbb{Z}/n$, Duality 1.4).

4.2

We assume that A is commutative, an assumption that is probably unnecessary. For $a : X \to S$ of finite type over S, put $K_X = \mathsf{R}a^! K_S$. For $K \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(S, A)$, put $DK = \mathsf{R}\mathscr{H}om(K, K_X) \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X, A)$. Our main result is the following.

Theorem 4.3. *We have $K \xrightarrow{\sim} DDK$ ($K \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X, A)$).*

Lemma 4.4. *If a is proper, then $\mathsf{R}a_* K \xrightarrow{\sim} \mathsf{R}a_* DDK$.*

Poincaré duality

$$\mathsf{R}a_! \, \mathsf{R}\mathscr{H}om(K, \mathsf{R}a^! K_S) = \mathsf{R}\mathscr{H}om(\mathsf{R}a_! K, K_S)$$

gives us an isomorphism $\mathsf{R}a_* D = D\mathsf{R}a_! = D\mathsf{R}a_*$. The lemma follows from the commutativity of the diagram (Duality, 1.2)

$$\begin{array}{ccc} \mathsf{R}a_* K & \longrightarrow & \mathsf{R}a_* DDK \\ \downarrow \wr & & \downarrow \wr \\ DD\mathsf{R}a_* K & \xrightarrow{\sim} & D\mathsf{R}a_* DK. \end{array}$$

4.5

By localization, it is enough to treat the cases where S is the spectrum of a field or a trait. When S is the spectrum of a field, we argue by induction on $\dim X$. Since the problem is local, we may assume that X is proper. Moreover, the induction hypothesis, by the usual arguments, ensures that the mapping cylinder Δ of $K \to DDK$ has skyscraper cohomology sheaves. The lemma then ensures that $\mathsf{R}a_* \Delta = 0$, hence that $\Delta = 0$.

4.6

When S is a trait, we already know that 4.3 holds on the generic fiber (4.5). We may also assume that X is proper over S, and we argue by induction on $\dim X_s$, with X proper over S. The mapping cylinder δ again has skyscraper cohomology sheaves on X_s, and we conclude as above.

4.7

Suppose that $A = \mathbb{Z}/m$. We then have a variant of the preceding theory in which K is arbitrary in $\mathsf{D}^b_c(X, A)$. The complex $\mathsf{R}a^! \mathbb{Z}/m$ has finite injective dimension [2, XVIII 3.1.7]. This ensures that DK still belongs to $\mathsf{D}^b_c(X, A)$. To prove local biduality over S, we use the fact that $\mathbb{Z}/m$ is the dualizing $\mathbb{Z}/m$-module. We then proceed as above.

5 Appendix, by L. Illusie

In this appendix, which takes up certain parts of the late [21, II], and a letter from P. Deligne to the author, we prove 2.14, as well as various generalizations and variants of the specialization theorems [2, XVI 2.1] and (Étale cohomology 5.1.7).

The schemes and morphisms considered will be assumed quasi-compact and quasi-separated. If X is a scheme and x a geometric point of X, we denote by $X(x)$ the strict localization of X at x.

Fix a base scheme S and a sheaf of rings A on S (not necessarily commutative or noetherian). If X is an S-scheme, we write $\mathsf{D}(X)$ for $\mathsf{D}(X, A_X)$.

5.1 Cohomological properness

Proposition 5.1.1. *Let $f : X \to Y$ be an S-morphism and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. The following conditions are equivalent:*

(i) The formation of $\mathsf{R}f_*E$ commutes with every finite base change $S' \to S$.

(ii) The formation of $\mathsf{R}f_*E$ commutes with every quasi-finite base change (or inverse limit of quasi-finite morphisms) $S' \to S$.

(iii) For every specialization arrow $i : t \to S(s)$ (Étale cohomology, 5.1.2), if $\widetilde{S}$ denotes the normalization in $k(\bar{t})$ of the integral scheme that is the closure of $i(t)$ in $S(s)$ (by [20, IV, 6.15.5, 18.6.11, 18.8.16], $\bar{S}$ is therefore local and integral, with closed point radicial over s), if $\bar{f} : \bar{X} \to \bar{Y}$ is the morphism deduced from f by the base change $\bar{S} \to S$, and if $f_s : X_s \to Y_s$ is the fiber of f at s, then the base-change arrow

$$(\mathsf{R}\bar{f}_*(E|\bar{X}))|Y_s \to \mathsf{R}f_{s*}(E|X_s)$$

is an isomorphism.

The equivalence of (i) and (ii) follows from Zariski's Main Theorem, and it is clear that (ii) implies (iii). Let us prove that (iii) implies (ii). For every $S' \to S$ that is an inverse limit of quasi-finite morphisms, and every geometric point s of S', denote by $C(S', s)$ the cone of the base-change arrow

$$(\mathsf{R}f'_*(E|X'))|Y_s \to \mathsf{R}f_{s*}(E|X_s),$$

where $f' : X' \to Y'$ is deduced from f by the base change $S' \to S$, and $f_s : X_s \to Y_s$ is the fiber of f at s. We must prove that, for every (S', s) as above, $C(S', s)$ is acyclic. We shall prove by induction on N that $\mathrm{H}^i\, C(S', s) = 0$ for $i \leqslant N$. By the Main Theorem, passage to the limit, and elimination of radicial morphisms, we may assume that S and S' are strictly local, that S' is integral with closed point s, and that $S' \to S$ is finite. Let t be a geometric generic point of S', and let $\bar{S}$ be the normalization of S' in $k(t)$. Let $\bar{S}_\bullet$ be the simplicial scheme that is the coskeleton of $S' \to S'$ ($\bar{S}_0 = \bar{S}, \ldots \bar{S}_n = (\bar{S}/S')^{n+1}, \ldots$), and let $X_\bullet = X \times_S \bar{S}_\bullet$. Let $g_\bullet : \bar{X}_\bullet \to X'$ be the canonical projection. Put $G|X' = G'$. The bicomplex $g_{\bullet *} g_\bullet^* G'$, whose columns are the $g_{n*} g_n^* G'$, is a resolution of G' [2, VIII 8]. We may therefore represent $C(S', s)$ by a bicomplex whose columns identify with the $C(\bar{S}_n, s)$. By (iii), $C(\bar{S}_0, s) = C(\bar{S}, s)$ is acyclic. On the other hand, for $n > 0$, we have $\mathrm{H}^i(\bar{S}_n, s) = 0$ for $i \leqslant N$ by the induction hypothesis. It follows that $\mathrm{H}^i\, C(S', s) = 0$ for $i \leqslant N + 1$, which completes the proof.

5.1.2 Remark

Let (f, E) satisfy condition (i) of 5.1.1.

a) If $E \in \mathrm{Ob}\,\mathsf{D}^b(X)$ and f_* has finite cohomological dimension, then for every $F \in \mathrm{Ob}\,\mathsf{D}^-(S, A^\circ)$, the canonical arrow

$$\mathsf{R}f_*(E) \overset{\mathsf{L}}{\otimes}_A q^*F \to \mathsf{R}f_*(E \overset{\mathsf{L}}{\otimes}_A p^*F), \qquad (*)$$

where $p : X \to S$ and $q : Y \to S$ are the projections, is an isomorphism. The same conclusion holds with $E \in \mathrm{Ob}\,\mathsf{D}^-(X)$.

Indeed, by dévissage, it is enough to verify the assertion for F of the form $i_!A_{S'}$, with $i : S' \to S$ quasi-finite. By the Main Theorem, we reduce to the case where i is finite, and the conclusion is then equivalent to the commutation of $\mathsf{R}f_*(E)$ with base change by i.

b) Suppose that A is noetherian and constructible. Then, for every constructible $F \in \mathrm{Ob}\,\mathsf{D}^b(S, A^\circ)$ of finite tor-dimension, the arrow $(*)$ above is an isomorphism.

Indeed, by dévissage, we reduce to the case where F is of the form $i_!M$, with $i : S' \to S$ locally closed, A locally constant on S', and M constructible, of finite tor-dimension, and with locally constant cohomology on S'. After base change from S to S', we are thus reduced to the case where F is perfect [5, I], hence finally, by localization and dévissage, to the case $F = A$, for which the conclusion is immediate.

5.1.3

Let $f : X \to Y$ be an S-morphism and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. We say that (f, E) is *cohomologically proper* rel. to S if the formation of $\mathsf{R}f_*E$ commutes with every base change $S' \to S$. Equivalently, after every base change $S' \to S$, the formation of $\mathsf{R}f'_*E'$ (where $f' = f \times_S S'$ and $E' =$ the inverse image of E on $X \times_S S'$) commutes with every finite base change $S'' \to S'$ (cf. [2, XII 6.1]). Here are some examples of cohomological properness (for a study of the analogous notion for sheaves of sets or of noncommutative groups, the reader may consult [23, XIII]).

1. Suppose that A is torsion and f is proper. Then, for every $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$, (f, E) is cohomologically proper rel. to S (proper base-change theorem, [2, XII 5.1]).

2. Suppose that S is finite, that X and Y are of finite type over S, and that A is annihilated by an integer invertible on S. Then, for every $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$, (f, E) is cohomologically proper rel. to S (1.9).

3. Suppose that A is constant, noetherian, and annihilated by an integer n invertible on S, and that $f : X = Y \smallsetminus D \to Y$ is the inclusion of the complement of a divisor D on Y with normal crossings relative to S [23, XIII 2.1]. Let E be a sheaf of A-modules on X satisfying one of the following conditions:

   (i) E is constructible locally constant and tamely ramified along D;

   (ii) E is, locally for the étale topology on Y and on S, the inverse image of a sheaf of A-modules on S.

   Then (f, E) is cohomologically proper rel. to S, and $\mathsf{R}f_*E$ is constructible (compare [23, XIII 2.4]).

Let us sketch the proof in case (i). The question is local near a point y of Y. We may assume that D is a sum of smooth divisors, $D = \sum_{i=1}^m D_i$. By the relative Abhyankar lemma [23, XIII 5.5], near y we may trivialize E by a finite covering $\widetilde{Y} \to Y$ of the form $\widetilde{Y} = Y[T_1, \ldots, T_r]/(T_1^{n_1} - t_1, \ldots, T_r^{n_r} - t_r)$, where the t_i are local equations of the smooth divisors passing through y, and the n_i are integers prime to the char. of $k(y)$. The inverse image $\widetilde{D}$ of D in Y still has normal crossings rel. to S, and if $g : \widetilde{X} = \widetilde{Y} \smallsetminus \widetilde{D} \to X$ is

the projection, g^*E is constant. Since E injects into g_*g^*E and the quotient is tamely ramified, an easy dévissage reduces us to the case where E is constant. Denote the projections by $p : X \to S$, $q : Y \to S$, and, for $1 \leqslant i \leqslant m$, let $f_i : Y \smallsetminus D_i \to D$ be the canonical inclusion. From the relative purity theorem [2, XVI 3.7], one easily deduces by induction on m that, for every locally constant A-module M on S, the canonical arrow

$$\mathsf{R}f_*(\mathbb{Z}/n)\overset{\mathsf{L}}{\otimes}q^*M \to \mathsf{R}f_*(p^*M) \tag{5.1.3.1}$$

is an isomorphism, and that on the other hand we have:

$$\begin{aligned} f_{i*}(\mathbb{Z}/n) &= \mathbb{Z}/n, \\ \mathsf{R}^1 f_{i*}(\mathbb{Z}/n) &= (\mathbb{Z}/n)(-1)_{D_i}, \end{aligned} \tag{5.1.3.2}$$

where a canonical isomorphism is given by the fundamental class $\mathrm{cl}(D_i)$ (Cycle 2.1.4),

$$\begin{aligned} \mathsf{R}^q f_{i*}(\mathbb{Z}/n) &= 0 \text{ for } q > 1, \\ \mathsf{R}f_*(\mathbb{Z}/n) &= \bigotimes_i^{\mathsf{L}} \mathsf{R}f_{i*}(\mathbb{Z}/n) \text{ , i.e.:} \\ \mathsf{R}^1 f_*(\mathbb{Z}/n) &= \bigoplus_{i=1}^{m} (\mathbb{Z}/n)(-1)_{D_i}, \\ \textstyle\bigwedge^{\bullet} \mathsf{R}^1 f_*(\mathbb{Z}/n) &\xrightarrow{\sim} \mathsf{R}^{\bullet} f_*(\mathbb{Z}/n). \end{aligned}$$

The conclusion of 3, in case (i), therefore follows from (5.1.3.1) and (5.1.3.2). In case (ii), by passage to the limit we reduce to the case where $E = p^*M$, with M constructible, and then, by dévissage and base change (using (i)), to the already treated case where M is constant.

It follows from the preceding proof that (5.1.3.1) is an isomorphism for every $M \in \mathrm{Ob}\,\mathsf{D}^+(S)$.

One may compare the formulas (5.1.3.2) with the analogous formulas in integral cohomology when $S = \mathrm{Spec}(\mathbb{C})$ (see, for example, [12, 3.1]), which follow from the fact that, locally for the classical topology, the complement of D has the homotopy type of a torus.

By passage to the limit, one deduces from (5.1.3.2) similar formulas in ℓ-adic cohomology, i.e. with $\mathbb{Z}/n$ replaced by $\mathbb{Z}_\ell$ (ℓ a prime invertible on S). Suppose that $q : Y \to S$ is proper and smooth. It then follows from the Weil conjectures [14, 15] that the Leray spectral sequence

$$E_2^{ij} = \mathsf{R}^i q_* \mathsf{R}^j f_* \mathbb{Z}_\ell \Rightarrow \mathsf{R}^{\bullet} p_* \mathbb{Z}_\ell$$

degenerates at E_3 modulo torsion [11, §6]. The same phenomenon occurs in integral cohomology when $S = \mathrm{Spec}(\mathbb{C})$; see (loc. cit.) and [12, 3.2.13].

5.2 Specialization and cospecialization

Let $f : X \to S$ and $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$.

5.2.1

Suppose that the formation of $\mathsf{R}f_*E$ commutes with every finite base change. For every specialization $u : t \to S(s)$ of geometric points of S, we then define an arrow, called the *specialization arrow*,

$$\mathrm{sp}(u) : \mathsf{R}\Gamma(X_s, E|X_s) \to \mathsf{R}\Gamma(X_t, E|X_t), \tag{5.2.1.1}$$

as follows. Let, as in 5.1.1(iii), $\bar{S}$ be the normalization in $k(t)$ of the closure of $u(t)$ in $S(s)$, and let $\bar{f} : \bar{X} \to \bar{S}$ be the morphism deduced from f by the base change $\bar{S} \to S$. By the immediate implication (i)⇒(iii) of 5.1.1, the base-change arrow

$$\mathsf{R}\Gamma(\bar{X}, E|\bar{X}) = \mathsf{R}\bar{f}_*(E|\bar{X})_s \to \mathsf{R}\Gamma(X_s, E|X_s) \tag{5.2.1.2}$$

is an isomorphism. We define (5.2.1.1) as the composite of the inverse of (5.2.1.2) with the restriction arrow

$$\mathsf{R}\Gamma(\bar{X}, E|\bar{X}) \to \mathsf{R}\Gamma(X_t, E|X_t). \tag{5.2.1.3}$$

5.2.2

Suppose that f is locally acyclic rel. to E (2.12). For every specialization $u : t \to S(s)$ of geometric points of S, we then define an arrow, called the *cospecialization arrow*,

$$\mathrm{cosp}(u) : \mathsf{R}\Gamma(X_t, E|X_t) \to \mathsf{R}\Gamma(X_s, E|X_s), \tag{5.2.2.1}$$

as follows. Let $g : X_t \to X$ and $g' : \bar{X} \to X$ be the canonical arrows. Viewed vis-à-vis these two maps, the canonical arrow

$$\mathsf{R}g'_*(E|\bar{X}) \to \mathsf{R}g_*(E|X_t) \tag{5.2.2.2}$$

is an isomorphism by the local-acyclicity hypothesis. Indeed, we may assume that S is strictly local and integral, with closed point s, and that t is a geometric generic point. Compute the fiber of (5.2.2.2) at a geometric point x of X, with image y in S. After replacing S by the strict localization of S at y, we may assume that $y = s$. Since $\bar{S} \to S$ is an inverse limit of finite morphisms and is radicial at s, we find that $\mathsf{R}g'_*(E|\bar{X})_x = E_x$ and that the fiber of (5.2.2.2) at x identifies with the restriction $E_x = \mathsf{R}\Gamma(X(x), E) \to \mathsf{R}\Gamma(X(x)_t, E|X(x)_t)$, which is an isomorphism by hypothesis. Applying $\mathsf{R}\Gamma(X, -)$ to (5.2.2.2), we deduce that the restriction arrow (5.2.1.3) is an isomorphism. We define (5.2.2.1) as the composite of the inverse of (5.2.1.3) with the base-change arrow (5.2.1.2).

5.2.3

If $s'' \xrightarrow{u'} s' \xrightarrow{u} s$ are specialization arrows of geometric points of S, one verifies that

$$\mathrm{sp}(u')\,\mathrm{sp}(u'') = \mathrm{sp}(uu'),$$
$$\mathrm{cosp}(u)\,\mathrm{cosp}(u') = \mathrm{cosp}(uu'),$$

whenever both sides are defined.

5.2.4

Suppose that the formation of $\mathsf{R}f_*E$ commutes with every finite base change and that f is locally acyclic rel. to E. It follows from the definitions that, for every specialization $u : t \to S(s)$ of geometric points of S, the arrows $\mathrm{sp}(u)$ and $\mathrm{cosp}(u)$ are isomorphisms, inverse to one another.

The preceding hypotheses hold in particular when A is noetherian and annihilated by an integer n invertible on S, when f is proper and smooth, and when E has locally constant cohomology (cohomological properness of proper morphisms (5.1.3), and local acyclicity of smooth morphisms [2, XV]). This recovers the "specialization theorem" [2, XVI 2.1].

On the other hand, 5.1.1 has the following consequence:

Proposition 5.2.5. *Suppose that f is locally acyclic relative to E and that, for every specialization $u : t \to S(s)$, the cospecialization arrow $\mathrm{cosp}(u)$ is an isomorphism. Then the formation of $\mathsf{R}f_*E$ commutes with every finite base change.*

In view of 5.2.4, 5.2.5 generalizes (Étale cohomology 5.1.7).

Corollary 5.2.6. *Suppose that f is locally acyclic rel. to E. Let x be a geometric point of X, with image s in S, and denote by $f_{(x)} : X(x) \to S(s)$ the morphism induced by f. Then the formation of $\mathsf{R}f_{(x)_*}(E|X(x))$ commutes with every finite base change $S' \to S(s)$.*

By 5.2.5, it is enough to verify that the cospecialization arrows for $(f_{(x)}, E|X(x))$ are isomorphisms. By the transitivity of the cospecialization arrows (5.2.3), it is enough to show that if $u : t \to S(s)$ is a specialization, then $\mathrm{cosp}(u) : \mathsf{R}\Gamma(X(x)_t, E|X(x)_t) \to \mathsf{R}\Gamma(X(x)_s, E|X(x)_s)$ is an isomorphism. But since s is closed in $S(s)$, $X(x)_s$ is strictly local with closed point x, $\mathsf{R}\Gamma(X(x)_s, E|X(x)_s) = E_x$, and $\mathrm{cosp}(u)$ is an isomorphism, inverse to the restriction isomorphism $E_x \to \mathsf{R}\Gamma(X(x)_t, E|X(x)_t)$.

Corollary 5.2.7. *Let $f : X \to Y$ be an S-morphism, let $p : X \to S$, $q : Y \to S$ be the projections, and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. Suppose that A is locally constant, that f is locally acyclic rel. to E, and that q is locally acyclic at every geometric point y of Y rel. to every sheaf locally constant in a neighborhood of y. Then p is locally acyclic relative to E.*

Let x be a geometric point of x, with images y and s in Y and S, let $t \to S(s)$ be a specialization, and let $f_{(x)} : X(x) \to Y(y)$ be the morphism induced by f. We must show that the restriction

$$E_x \to \mathsf{R}\Gamma(X(x)_t, E|X(x)_t) \qquad (*)$$

is an isomorphism. We may assume that A is constant. By 5.2.4 and 5.2.5, for every specialization $z \to Y(y)$, the specialization arrow

$$E_x = \mathsf{R}f_{(x)_*}(E|X(x))_y \to \mathsf{R}f_{(x)_*}(E|X(x)|_z = \mathsf{R}\Gamma(X(x)_z, E|X(x)_z)$$

is an isomorphism; in other words, $\mathsf{R}f_{(x)_*}(E|X(x))$ is constant with value E_x. The acyclicity hypothesis on q therefore plays the decisive rôle and ensures that the restriction

$$E_x \to \mathsf{R}\Gamma(Y(y)_y, (E_x)|Y(y)_t) = \mathsf{R}\Gamma\left(Y(y)_t, \mathsf{R}f_{(x)_*}(E|X(x))|Y(y)_t\right)$$

is an isomorphism. But this arrow identifies with $(*)$, which proves the assertion.

The hypothesis of 5.2.7 on q holds in particular when q is smooth and A is annihilated by an integer n invertible on S [2, XV]. On the other hand, if the hypotheses of 5.2.7 hold after every base change $S' \to S$, then it follows that p is universally locally acyclic rel. to E. These remarks imply, in particular, the statement of 2.14.

Corollary 5.2.8. *Let $f : X \to S$ and $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. Suppose that f is universally locally acyclic rel. to E, that the formation of $\mathsf{R}f_*E$ commutes with every finite base change, and that, for every morphism $s' \to s$, where s is a geometric point of S and s' is the spectrum of a separably closed field, the base-change arrow $\mathsf{R}\Gamma(X_s, E|X_s) \to \mathsf{R}\Gamma(X_{s'}, E|X_{s'})$ is an isomorphism. Then (f, E) is cohomologically proper (5.1.3).*

Let $f' : X' \to S'$ be the morphism deduced from f by a base change $S' \to S$; we must show that the formation of $\mathsf{R}f'_*(E|X')$ commutes with every finite base change. By 5.2.5, it is enough to show that, for every specialization $u' : t' \to S'(s')$, the cospecialization arrow $\mathrm{cosp}(u')$: $\mathsf{R}\Gamma(X'_{t'}, E|X'_{t'}) \to \mathsf{R}\Gamma(X'_{s'}, E|X'_{s'})$

is an isomorphism. Let s (resp. t) be the geometric point that is the image of s (resp. t) in S, and let $u : t \to S(s)$ be the specialization defined by u'. One readily sees that there is a commutative square

$$\begin{array}{ccc} \mathsf{R}\Gamma(X_t, E|X_t) & \xrightarrow{\mathrm{cosp}(u)} & \mathsf{R}\Gamma(X_s, E|X_s) \\ \downarrow & & \downarrow \\ \mathsf{R}\Gamma(X'_{t'}, E|X'_{t'}) & \xrightarrow{\mathrm{cosp}(u')} & \mathsf{R}\Gamma(X'_{s'}, E|X'_{s'}) \end{array}$$

in which the vertical arrows are the base-change arrows. By hypothesis, these are isomorphisms. On the other hand, (5.2.4) $\mathrm{cosp}(u)$ is an isomorphism, hence $\mathrm{cosp}(u')$ is an isomorphism, which completes the proof.

Here is an example of an application of 5.2.8. Suppose that A is annihilated by an integer n invertible on S, and that f is locally of finite type and universally locally acyclic. Let x be a geometric point of X, with image s in S, and let $f_{(x)} : X(x) \to S(s)$ be the morphism induced by f. It follows from 5.2.6 and 1.9 (by passage to the limit) that the hypotheses of 5.2.8 are satisfied by the pair $(f_{(x)}, E|X(x))$, which is therefore cohomologically proper.

5.2.9

Suppose that A is constant. Let $f : X \to S$, and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$ be a complex of finite tor-dimension. We say that f is *strongly locally acyclic* rel. to E if, for every geometric point x of X, with image s in S, every specialization $t \to S(s)$, and every A°-module M, the restriction arrow

$$E_x \overset{\mathsf{L}}{\otimes}_A M \to \mathsf{R}\Gamma(X(x)_t, (E|X(x)_t) \overset{\mathsf{L}}{\otimes}_A M)$$

is an isomorphism. I do not know whether "locally acyclic" implies "strongly locally acyclic."

Proposition 5.2.10. *Under the hypotheses of 5.2.9, suppose that A is noetherian and torsion, that S is noetherian, and that f is strongly locally acyclic rel. to E. Then, for every diagram with cartesian squares*

$$\begin{array}{ccccc} X & \xleftarrow{j} & X' & \xleftarrow{j'} & X'' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f''} \\ S & \xleftarrow{i} & S' & \xleftarrow{i'} & S'', \end{array}$$

with i quasi-finite and i' an open immersion, and every $F \in \mathrm{Ob}\,\mathsf{D}^+(S'', A^\circ)$, the canonical arrow

$$j^* E \overset{\mathsf{L}}{\otimes} f'^* \mathsf{R}i'_* F \to \mathsf{R}j'_*((jj')^* E \overset{\mathsf{L}}{\otimes} f''^* F) \qquad (*)$$

is an isomorphism.

Compare this statement with [2, XV 1.17].

Let us prove 5.2.10. We readily reduce to the case where $S = S'$ is strictly local, and to F constructible and concentrated in degree 0. We then resolve F on the right by finite sums of sheaves of the form $u_* M$, where $u : t \to S$ is a geometric point and M is an A°-module. The hypothesis, vis-à-vis this resolution, implies that $(*)$ is an isomorphism for $F = u_* M$, and the result follows.

Chapter 8

Derived categories

1 Triangulated categories: definitions and examples

1.1 Definition of triangulated categories

1.1.0

Let $\mathcal{A}$ be an additive category and let T be an additive automorphism of $\mathcal{A}$. For two objects X and Y of $\mathcal{A}$, we set

$$\hom^i(X,Y) = \hom_{\mathcal{A}}(X, T^i(Y)) \qquad i \in \mathbb{Z}$$

Let X, Y, and Z be three objects of $\mathcal{A}$, let $\alpha \in \hom^i(X,Y)$, and let $\beta \in \hom^j(X,Y)$. We define the composite $\beta \circ \alpha$ belonging to $\hom^{i+j}(X,Z)$ by

$$\beta \circ \alpha = \beta \circ T^i(\alpha).$$

One checks that this gives a new category whose groups of morphisms between objects are graded. By an abuse of language, this new category will be called the category $\mathcal{A}$ graded by the translation functor T.

Let $\mathcal{A}$ be a category graded by a translation functor T. Whenever a morphism is named or denoted without its degree being specified, it will always mean a morphism of degree zero.

Let $\mathcal{A}$ and $\mathcal{A}'$ be two additive categories graded by the functors T and T', and let F be an additive functor from $\mathcal{A}$ to $\mathcal{A}'$. We say that F *is graded* if an isomorphism of functors* has been specified:

$$F \circ T = T' \circ F$$

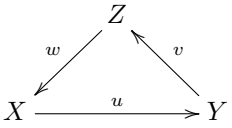
The definition of morphisms of graded functors is given in 4.1.1.

Let $\mathcal{A}$ be an additive category graded by the functor T. A triangle is a collection of three objects X, Y, Z and three morphisms $u : X \to Y$, $v : Y \to Z$, $w : Z \to T(X)$ ($\deg w = 1$). We use either the

by Jean-Louis Verdier

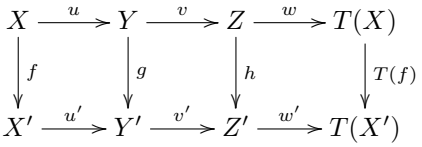
*Text amended in 1976: the original text required an equality $F \circ T = T' \circ F$. The case of functors in several variables is discussed in [2, XVIII 0.2]. It raises a problem of signs.

notation (X, Y, Z, u, v, w) or the following diagram for a triangle:



$\deg(w) = 1$

Let (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') be two triangles of $\mathcal{A}$. A morphism from the first triangle to the second is a collection of three morphisms $f : X \to X'$, $g : Y \to Y'$, $h : Z \to Z'$ such that the following diagram is commutative:



1.1.1

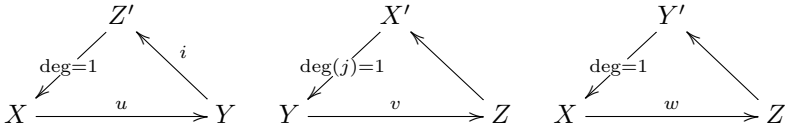
A *triangulated category* is an additive category graded by a translation functor and equipped with a family of triangles, called the family of distinguished triangles. This family must also satisfy the following axioms.

TR1 Every triangle isomorphic to a distinguished triangle is distinguished. Every morphism $u : X \to Y$ is contained in a distinguished triangle (X, Y, Z, u, v, w). The triangle $(X, X, 0, \mathrm{id}_{X'}, 0, 0)$ is distinguished.

TR2 A triangle (X, Y, Z, u, v, w) is distinguished if and only if $(Y, Z, T(X), v, w, -T(u))$ is distinguished.

TR3 If (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') are distinguished, then for every morphism $(f, g) : u \to u'$, there exists a morphism $h : Z \to Z'$ such that (f, g, h) is a morphism of triangles.

TR4



If these are three distinguished triangles such that $w = v \circ u$, then there exist two morphisms $f : Z' \to Y'$, $g : Y' \to X'$ such that

1. (id_X, v, f) is a morphism of triangles;
2. (u, id_Z, g) is a morphism of triangles;
3. $(Z', Y', X', f, g, T(i)j)$ is a distinguished triangle.

If $\mathcal{C}$ and $\mathcal{C}'$ are two triangulated categories, an *exact functor* from $\mathcal{C}$ to $\mathcal{C}'$ is an additive graded functor that takes distinguished triangles to distinguished triangles. Morphisms of exact functors are morphisms of graded functors.

The following property follows immediately from axioms TR1, TR2, and TR3.

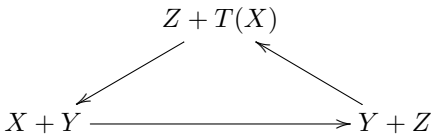
Proposition 1.1.2. *Let (X, Y, Z, u, v, w) be a distinguished triangle and let M be an object of a triangulated category. Then the following sequences are exact:*

$$\cdots \longrightarrow \hom^i(M, X) \xrightarrow{\hom(M,u)} \hom^i(M, Y) \xrightarrow{\hom(M,v)} \hom^i(M, Z) \xrightarrow{\hom(M,w)} \hom^{i+1}(M, X) \longrightarrow \cdots$$

$$\cdots \longrightarrow \hom^i(Z, M) \xrightarrow{\hom(v,M)} \hom^i(Y, M) \xrightarrow{\hom(u,M)} \hom^i(X, M) \xrightarrow{\hom(w,M)} \hom^{i+1}(Z, M) \longrightarrow \cdots$$

From this proposition one obtains statements of the following kind:

- If (f, g, h) is a morphism of distinguished triangles and f, g are isomorphisms.
- The distinguished triangles constructed on a morphism (TR1) are all isomorphic (*but the isomorphisms are not, in general, uniquely determined*).
- When one of the morphisms in a distinguished triangle has a kernel or a cokernel, the latter splits; that is, the triangle is of the form



- The sum of two distinguished triangles is distinguished. (One uses axiom TR4.)

1.2 Examples

1.2.1

We first fix some notation. Let $\mathcal{A}$ be an additive category. The following category will be denoted by $\mathsf{C}(\mathcal{A})$:

- The objects of $\mathsf{C}(\mathcal{A})$ are the complexes of $\mathcal{A}$, with no restriction on degree and with differential of degree $+1$.
- The morphisms of $\mathsf{C}(\mathcal{A})$ are the degree-preserving morphisms of complexes (those commuting with the differential).

Let T be the following functor. For every object $X^\bullet$ of $\mathsf{C}(\mathcal{A})$,

$$T(X^\bullet)_i = (X^\bullet)_{i+1}$$
$$d_i(T(X^\bullet)) = -d_{i+1}(X^\bullet).$$

On morphisms the functor T acts as follows: for every morphism $f : X^\bullet \to Y^\bullet$, the morphism $T(f) : T(X^\bullet) \to T(Y^\bullet)$ is defined by

$$T(f)_i = f_{i+1}.$$

It is immediate that T is an additive automorphism of $\mathsf{C}(\mathcal{A})$. The morphisms of degree n defined using this translation functor are then the morphisms that raise degree by n and commute or anticommute with the differential according to the parity of n. Thus one recovers the usual definition.

Let $\mathscr{C}$ denote the following family of triangles. A triangle $(X^\bullet, Y^\bullet, Z^\bullet, u, v, w)$ belongs to the family if[†]

[†]Text amended in 1976 to ensure the validity of 1.2.4.

- $Z^\bullet$ is the simple complex associated with the double complex

$$\cdots \longrightarrow 0 \longrightarrow X^\bullet \xrightarrow{u} Y^\bullet \longrightarrow 0 \longrightarrow \cdots$$

in which the objects of $Y^\bullet$ have first degree zero and the objects of $X^\bullet$ have first degree -1.

- v is the image, under the functor: double complex $\mapsto$ simple complex, of the following morphism of double complexes:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & 0^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & X^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

- w is the negative of the image, under the same functor, of the following morphism of double complexes:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & X^\bullet & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & X^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

1.2.2

The set of null-homotopic morphisms is a two-sided ideal (if f and g are composable and either f or g is null-homotopic, then the composite is null-homotopic; if f and g are null-homotopic, then $f \oplus g$ is null-homotopic). For every pair of objects $X^\bullet$, $Y^\bullet$, let $\mathrm{Htp}(X^\bullet, Y^\bullet)$ denote the subgroup of $\hom_{\mathsf{C}(\mathcal{A})}(X^\bullet, Y^\bullet)$ consisting of the null-homotopic morphisms. The objects of $\mathsf{K}(\mathcal{A})$ are then the objects of $\mathsf{C}(\mathcal{A})$, and its group of morphisms with source $X^\bullet$ and target $Y^\bullet$ is

$$\hom_{\mathsf{C}(\mathcal{A})}(X^\bullet, Y^\bullet) / \mathrm{Htp}(X^\bullet, Y^\bullet).$$

The composition law on the morphisms of $\mathsf{K}(\mathcal{A})$ is defined by passing the composition law on the morphisms of $\mathsf{C}(\mathcal{A})$ to the quotient. The category $\mathsf{K}(\mathcal{A})$ is additive. The translation functor T on $\mathsf{C}(\mathcal{A})$ passes to the quotient (the translate of a null-homotopic morphism is null-homotopic). The resulting functor is an additive automorphism of $\mathsf{K}(\mathcal{A})$, again denoted by T. By an abuse of notation, $\mathsf{K}(\mathcal{A})$ will henceforth denote the category $\mathsf{K}(\mathcal{A})$ graded by the functor T.

A distinguished triangle in $\mathsf{K}(\mathcal{A})$ is any triangle *isomorphic* to a triangle arising from $\mathscr{C}$. The family of distinguished triangles satisfies axioms TR1, TR2, TR3, and TR4. By an abuse of notation, $\mathsf{K}(\mathcal{A})$ will denote the category $\mathsf{K}(\mathcal{A})$ triangulated by the family of triangles defined above.

1.2.3

There is also a whole arsenal of subcategories:

- A complex is said to be bounded below if all but finitely many of its objects of negative degree are zero. Bounded-above and bounded complexes are defined similarly. Let $\mathsf{K}^+(\mathcal{A})$, $\mathsf{K}^-(\mathcal{A})$, and $\mathsf{K}^b(\mathcal{A})$ denote the full subcategories of $\mathsf{K}(\mathcal{A})$ generated by these families of objects. These subcategories are "*triangulated subcategories*" of $\mathsf{K}(\mathcal{A})$: every distinguished triangle two of whose objects belong to the subcategory is isomorphic to a triangle all three of whose objects belong to the subcategory.

- Suppose that $\mathcal{A}$ is an abelian category. A complex is said to be cohomologically bounded below if all but finitely many of its cohomology objects of negative degree are zero. Cohomologically bounded-above complexes, cohomologically bounded complexes, and acyclic complexes are defined similarly. Let $\mathsf{K}^{\infty,+}(\mathcal{A})$, $\mathsf{K}^{\infty,-}(\mathcal{A})$, $\mathsf{K}^{\infty,b}(\mathcal{A})$, and $\mathsf{K}^{\infty,\varnothing}(\mathcal{A})$ denote the full subcategories generated by these families of objects. They are triangulated subcategories.

  We shall also use the triangulated subcategories $\mathsf{K}^{+,b}(\mathcal{A})$, $\mathsf{K}^{+,\varnothing}(\mathcal{A})$, ... etc ... (the first superscript gives information about the objects of the complexes, and the second about the cohomology objects).

- Without assuming that $\mathcal{A}$ is abelian, let O be a set of objects of $\mathcal{A}$ stable under isomorphisms and direct sums. Let $O(\mathcal{A})$ denote the full subcategory of $\mathsf{C}(\mathcal{A})$ generated by the complexes whose objects in every degree belong to O. Let $\underline{O}(\mathcal{A})$ denote the corresponding triangulated subcategory of $\mathsf{K}(\mathcal{A})$. When $\mathcal{A}$ is abelian, one may take for O either the set I of injectives or the set P of projectives.

1.2.4

The category $\mathsf{K}(\mathcal{A})$ is uniquely determined by the data of $\mathsf{C}(\mathcal{A})$ and the family $\mathscr{C}$. More precisely, every graded additive functor from $\mathsf{C}(\mathcal{A})$ to a triangulated category that takes every triangle of $\mathscr{C}$ to a distinguished triangle factors uniquely through $\mathsf{K}(\mathcal{A})$, and the resulting functor is exact.

For every complex $X^\bullet$ of $\mathsf{C}(\mathcal{A})$, let $\bar{X}^\bullet$ denote the complex obtained from $X^\bullet$ by setting the differential equal to zero. A three-term sequence

$$0 \longrightarrow X^\bullet \longrightarrow Y^\bullet \longrightarrow Z^\bullet \longrightarrow 0$$

is called a semi-split sequence if the sequence

$$0 \longrightarrow \bar{X}^\bullet \longrightarrow \bar{Y}^\bullet \longrightarrow \bar{Z}^\bullet \longrightarrow 0$$

is split; that is, if it is isomorphic to the sequence

$$0 \longrightarrow \bar{X}^\bullet \longrightarrow \bar{X}^\bullet + \bar{Z}^\bullet \longrightarrow \bar{Z}^\bullet \longrightarrow 0.$$

Let $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ be a semi-split sequence. Choose a splitting, that is, for every n an isomorphism

$$(Y^\bullet)^n \simeq (X^\bullet)^n + (Z^\bullet)^n.$$

The matrix of the differential of $Y^\bullet$ in the above direct-sum decomposition has the form

$$\begin{pmatrix} d_{X^\bullet} & \varphi_n \\ 0 & d_{Y^\bullet} \end{pmatrix}$$

where φ_n is a morphism from $(Z^\bullet)^n$ to $(X^\bullet)^{n+1}$. The φ_n determine a morphism of complexes

$$\varphi : Z^\bullet \to T(X^\bullet).$$

When the splitting is changed, the homotopy class of φ does not change. Moreover, if $\underline{u}$, $\underline{v}$, and $\underline{\varphi}$ denote the images in $\mathsf{K}(\mathcal{A})$ of the morphisms u, v, and φ, then the triangle $(X^\bullet, Y^\bullet, Z^\bullet, \underline{u}, \underline{v}, \underline{\varphi})$ is distinguished. Let $\mathsf{SSS}(\mathcal{A})$ denote the category of semi-split sequences of $\mathsf{C}(\mathcal{A})$ and let $\mathsf{TrK}(\mathcal{A})$ denote the category of distinguished triangles of $\mathsf{K}(\mathcal{A})$. The preceding construction defines a functor

$$\rho : \mathsf{SSS}(\mathcal{A}) \to \mathsf{TrK}(\mathcal{A}).$$

This functor is essentially surjective.

1.3 Examples of cohomological and exact functors

Definition 1.3.1. *Let $\mathcal{A}$ be a triangulated category and let $\mathcal{B}$ be an abelian category. An additive functor $F : \mathcal{A} \to \mathcal{B}$ is called a* cohomological functor *if, for every distinguished triangle (X, Y, Z, u, v, w), the sequence*

$$F(X) \xrightarrow{F(u)} F(Y) \xrightarrow{F(v)} F(Z)$$

is exact.

The functor $F \circ T^i$ will often be denoted by F^i. By axiom TR2 for triangulated categories, there is an unbounded exact sequence

$$\cdots \longrightarrow F^i(X) \longrightarrow F^i(Y) \longrightarrow F^i(Z) \longrightarrow F^{i+-}(X) \longrightarrow \cdots$$

1.3.2 Examples

1. Let $\mathcal{A}$ be a triangulated category. For every object X of $\mathcal{A}$, the functor $\hom_{\mathcal{A}}(X, -)$ is cohomological. The functor $\hom_{\mathcal{A}}(-, X)$ is a cohomological functor $\mathcal{A}^\circ \to \mathsf{Ab}$.

2. Let $\mathcal{A}$ be an abelian category and let $\mathsf{C}(\mathcal{A})$ be the category of complexes of $\mathcal{A}$. Define the functor $\mathrm{H}^0 : \mathsf{C}(\mathcal{A}) \to \mathcal{A}$ as follows. For an object $X^\bullet$ of $\mathsf{C}(\mathcal{A})$, set

   $$\mathrm{H}^0(X^\bullet) = \ker d^0 / \operatorname{im} d^{-1}.$$

   The functor H^0 annihilates null-homotopic morphisms and therefore factors uniquely through $\mathsf{K}(\mathcal{A})$. The resulting functor will again be denoted by $\mathrm{H}^0 : \mathsf{K}(\mathcal{A}) \to \mathcal{A}$.

1.3.3 Example of an exact bifunctor

Let $\mathcal{A}$, $\mathcal{A}'$, and $\mathcal{A}''$ be three additive categories, and let

$$P : \mathcal{A} \times \mathcal{A}' \to \mathcal{A}''$$

be a bilinear functor (that is, additive in each argument separately). It induces the bilinear functor

$$P^* : \mathsf{C}(\mathcal{A}) \times \mathsf{C}(\mathcal{A}') \to \mathsf{C}(\mathcal{A}'')$$

as follows.

Let $X^\bullet$ be an object of $\mathsf{C}(\mathcal{A})$ and let $Y^\bullet$ be an object of $\mathsf{C}(\mathcal{A}')$. Then $P(X^\bullet, Y^\bullet)$ is a double complex of $\mathcal{A}''$. Set $P^*(X^\bullet, Y^\bullet) =$ the simple complex associated with $P(X^\bullet, Y^\bullet)$.

Let f be a null-homotopic morphism of $\mathsf{C}(\mathcal{A})$ (respectively $\mathsf{C}(\mathcal{A}')$), and let $Z^\bullet$ be an object of $\mathsf{C}(\mathcal{A}')$ (respectively $\mathsf{C}(\mathcal{A})$). The morphism $P^*(f, Z^\bullet)$ (respectively $P^\bullet(Z^\bullet, f)$) is then null-homotopic. It follows that P^* uniquely defines a functor

$$P^\bullet : \mathsf{K}(\mathcal{A}) \times \mathsf{K}(\mathcal{A}') \to \mathsf{K}(\mathcal{A}'').$$

The bifunctor $P^\bullet$ is exact.

In particular, let $\mathcal{A}$ be an additive category. For P one may take the functor

$$\begin{aligned} \mathcal{A}^\circ \times \mathcal{A} &\to \mathsf{Ab} \\ (X, Y) &\mapsto \hom(X, Y) \end{aligned}$$

The preceding construction then gives a functor

$$\hom^\bullet : \mathsf{K}(\mathcal{A})^\circ \times \mathsf{K}(\mathcal{A}) \to \mathsf{K}(\mathsf{Ab})$$

which, when composed with the functor $\mathrm{H}^0 : \mathsf{K}(\mathsf{Ab}) \to \mathsf{Ab}$, plainly gives back the functor $\hom_{\mathsf{K}(\mathcal{A})}$.

2 Quotient categories

2.1 Thick subcategories. Multiplicative systems of morphisms

We begin with two definitions.

Definition 2.1.1. *A subcategory $\mathcal{B}$ of a triangulated category $\mathcal{A}$ is said to be* thick *if $\mathcal{B}$ is a full triangulated subcategory of $\mathcal{A}$ and, in addition, $\mathcal{B}$ has the following property:*

For every morphism $f : X \to Y$ that factors through an object of $\mathcal{B}$ and is contained in a distinguished triangle (X, Y, Z, f, g, h) in which Z is an object of $\mathcal{B}$, the source of f and the target of f are objects of $\mathcal{B}$.

The set of thick subcategories of $\mathcal{A}$ will be denoted by $\mathscr{N}$.

The intersection of an arbitrary family of thick subcategories is a thick subcategory.

Definition 2.1.2. *Let $\mathcal{A}$ be a triangulated category. A set S of morphisms of $\mathcal{A}$ is called a* multiplicative system of morphisms *if it has the following five properties.*

FR1 *If $f, g \in S$ and f and g are composable, then $f \circ g \in S$. For every object X of $\mathcal{A}$, the identity morphism of X belongs to S.*

FR2 *In the category $\mathcal{A}$, every diagram*

$$\begin{array}{ccc} & & Y \\ & & \downarrow{\scriptstyle s\in S} \\ Z & \xrightarrow{f} & X \end{array}$$

can be completed to a commutative diagram

$$\begin{array}{ccc} P & \xrightarrow{g} & Y \\ \downarrow{\scriptstyle t\in S} & & \downarrow{\scriptstyle s\in S} \\ Z & \xrightarrow{f} & X \end{array}$$

Moreover, the symmetric property holds.

FR3 *If f and g are morphisms, the following properties are equivalent:*

i) there exists $s \in S$ such that $s \circ f = s \circ g$;

ii) there exists $t \in S$ such that $f \circ t = g \circ t$.

FR4 *Let T be the translation functor of $\mathcal{A}$. For every element s of S, one has $T(s) \in S$.*

FR5 *Let (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') be two distinguished triangles, and let f and g be two elements of S such that the pair (f, g) is a morphism from u to u'. There exists a morphism $h \in S$ such that (f, g, h) is a morphism of triangles.*

A multiplicative system of morphisms is said to be *saturated* if it has the following property:

A morphism f belongs to S if and only if there exist two morphisms g and g' such that $g \circ f \in S$ and $f \circ g' \in S$.

The intersection of an arbitrary family of saturated multiplicative systems is a saturated multiplicative system.

The set of saturated multiplicative systems will be denoted by $\mathscr{S}$.

Let $\mathcal{B}$ be a thick subcategory. Denote by $\varphi(\mathcal{B})$ the set of morphisms f contained in a distinguished triangle (X, Y, Z, f, g, h) in which Z is an object of $\mathcal{B}$. Then $\varphi(\mathcal{B})$ is a saturated multiplicative system.

Let S be a saturated multiplicative system. Denote by $\psi(S)$ the full subcategory generated by the objects Z contained in a distinguished triangle (X, Y, Z, f, g, h) in which f is an element of S. The subcategory $\psi(S)$ is thick.

The main result of this subsection is the following: φ is an order-preserving isomorphism from $\mathscr{N}$ onto $\mathscr{S}$, where the order relation is defined by inclusion. Its inverse is precisely ψ.

2.2 Universal problems

Let $\mathcal{A}$ and $\mathcal{A}'$ be two triangulated categories and let F be an exact functor from $\mathcal{A}$ to $\mathcal{A}'$. Let $S(F)$ be the set of morphisms of $\mathcal{A}$ that F takes to isomorphisms, and let $\mathcal{B}(F)$ be the full subcategory generated by the objects of $\mathcal{A}$ that F takes to zero objects of $\mathcal{A}'$. Then $S(F)$ is a saturated multiplicative system and $\mathcal{B}(F)$ is a thick subcategory. Moreover, $S(F) = \varphi(\mathcal{B}(F))$. Now let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory, and let $S = \varphi(\mathcal{B})$ be the corresponding saturated multiplicative system. The following two universal problems are equivalent.

Problem 1 Find a triangulated category $\mathcal{A}/\mathcal{B}$ and an exact functor $Q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ such that every exact functor from $\mathcal{A}$ to a triangulated category $\mathcal{A}'$ that takes the objects of $\mathcal{B}$ to zero objects of $\mathcal{A}'$ admits a unique factorization of the form $G \circ Q$, where G is exact.

Problem 2 Find a triangulated category $\mathcal{A}_S$ and an exact functor $Q : \mathcal{A} \to \mathcal{A}_S$ such that every exact functor from $\mathcal{A}$ to a triangulated category $\mathcal{A}'$ that takes the morphisms of S to isomorphisms of $\mathcal{A}'$ admits a unique factorization of the form $G \circ Q$, where G is exact.

(The author asks the reader to excuse this use of obsolete and reactionary language.) In the next subsection we shall show that Problem 2 has a solution. Consequently, the corresponding Problem 1 has the same solution.

2.3 Calculus of fractions

2.3.1

Let $\mathcal{A}$ be a triangulated category and let S be a saturated multiplicative system. Denote by $\mathcal{A}[S^{-1}]$ the category of fractions of $\mathcal{A}$ with respect to the set of morphisms S, and by Q the canonical functor from $\mathcal{A}$ to $\mathcal{A}[S^{-1}]$. The pair $(\mathcal{A}[S^{-1}], Q)$ solves the following problem: every functor from $\mathcal{A}$ to an arbitrary category (not necessarily additive) that takes morphisms of S to isomorphisms factors uniquely through $(\mathcal{A}[S^{-1}], Q)$. Such a pair always exists, without any hypothesis on the set of morphisms S [18, 1.1][‡]. However, because the set of morphisms S is a saturated multiplicative system, it has properties FR1, FR2, and FR3. Consequently, by [18], the category $\mathcal{A}[S^{-1}]$ can be obtained by a right or left calculus of

[‡]The original text references [C.G.G] here. Most likely, this is the lost work *Catégories et foncteurs* by Chevalley, Gabriel and Grothendieck, referenced in[31].

fractions. We recall how $\mathcal{A}[S^{-1}]$ is obtained by a right calculus of fractions. (The left calculus of fractions is obtained by reversing the arrows.)

2.3.2

The objects of $\mathcal{A}[S^{-1}]$ are the objects of $\mathcal{A}$.

- For every object X of $\mathcal{A}$, let S_X denote the category of arrows belonging to S and having target X. To every object Y of $\mathcal{A}$ one associates, functorially in Y, the following set-valued functor on S_X° (S_X° denotes the opposite category):

$$H_Y : s \mapsto \hom(\text{source}(s), Y).$$

  Set

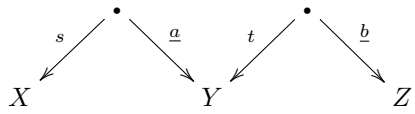
$$\hom_{\mathcal{A}[S^{-1}]}(X, Y) = \varinjlim_{S_X^\circ} H_Y.$$

  For all matters concerning direct limits, we refer to [1].

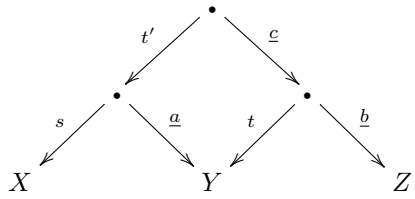
  One then observes with pleasure, using FR1, FR2, and FR3, that the category S_X° has properties L1, L2, and L3. Direct limits therefore have all the good properties of direct limits over filtered ordered sets.

- Let X, Y, and Z be three objects of $\mathcal{A}$, let a be an element of $\hom_{\mathcal{A}[S^{-1}]}(X, Y)$, and let b be an element of $\hom_{\mathcal{A}[S^{-1}]}(Y, Z)$. Let s be an object of S_X, let $\underline{a}$ be an element of $\hom(\text{source}(s), Y)$ whose image is a, and let an object of S_y, $\underline{b}$ be an element of $\hom(\text{source}(t), Z)$ whose image is b. There is then a diagram

(1)



  which, by FR2, can be completed to a diagram



  Let $b \circ a$ denote the image in $\hom_{\mathcal{A}[S^{-1}]}(X, Z)$ of the morphism $\underline{b} \circ \underline{c}$. Using properties FR1, FR2, and FR3, one checks that $b \circ a$ is independent of the chosen representatives $\underline{a}$ and $\underline{b}$, and also of the way in which diagram (1) is completed. One further checks that this defines a category $\mathcal{A}[S^{-1}]$.

We denote by Q the evident functor $\mathcal{A} \to \mathcal{A}[S^{-1}]$.

Since $\mathcal{A}$ is additive, $\mathcal{A}[S^{-1}]$ is an additive category. Moreover, using FR4, one proves that there exists a unique translation functor on $\mathcal{A}[S^{-1}]$, denoted by T, satisfying

$$Q \circ T = T \circ Q.$$

Finally, using FR5, one proves that there exists a unique triangulated structure on $\mathcal{A}[S^{-1}]$ for which Q is exact. The distinguished triangles of $\mathcal{A}[S^{-1}]$ are precisely the triangles isomorphic to the images under Q of the distinguished triangles of $\mathcal{A}$. Denoting by $\mathcal{A}_S$ the category $\mathcal{A}[S^{-1}]$ equipped with this triangulated structure, one proves without difficulty that the pair $(\mathcal{A}_S, Q)$ is a solution to Problem 2.

Definition 2.3.3. *Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory of $\mathcal{A}$, and let $(Q, \mathcal{A}/\mathcal{B})$ be the solution to Problem 1 (§2.2). The category $\mathcal{A}/\mathcal{B}$ is called the* quotient category *of $\mathcal{A}$ by $\mathcal{B}$, and Q is called the canonical quotient functor. More generally, let $\mathcal{N}$ be a triangulated subcategory of $\mathcal{A}$, and let $\underline{\mathcal{N}}$ be the smallest thick subcategory containing $\mathcal{N}$. By definition, the quotient category of $\mathcal{A}$ by $\mathcal{N}$, denoted $\mathcal{A}/\mathcal{N}$, is $\mathcal{A}/\underline{\mathcal{N}}$.*

2.4 Properties of quotient categories

2.4.1

Let $\mathcal{A}$ be a triangulated category and let H be a cohomological functor with values in an abelian category $\mathcal{G}$. Let $\mathcal{B}_H$ be the full subcategory generated by the objects all of whose translates are taken by H to zero objects of $\mathcal{G}$, and let S_H be the set of morphisms all of whose translates are taken by H to isomorphisms of $\mathcal{G}$. Then $\mathcal{B}_H$ is a thick subcategory of $\mathcal{A}$, and S_H is a saturated multiplicative system. Moreover, $S_H = \varphi(\mathcal{B}_H)$ (§2.1). The functor H factors uniquely through $(Q, \mathcal{A}/\mathcal{B}_H)$.

Theorem 2.4.2. *Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a triangulated subcategory, let $\mathcal{N}$ be a thick subcategory of $\mathcal{A}$, and let $S = \varphi(\mathcal{N})$ be the corresponding saturated multiplicative system.*

(a) The category $\mathcal{N} \cap \mathcal{B}$ is a thick subcategory of $\mathcal{B}$. The corresponding multiplicative system is $S \cap \mathcal{B}$.

(b) The following two properties are equivalent:

(i) For every object X of $\mathcal{B}$ and every morphism $s : R \to X$, where s belongs to S, there exists a morphism $s' : R' \to R$ such that $s \circ s'$ belongs to $S \cap \mathcal{B}$; and $R' \in \operatorname{Ob} \mathcal{B}$.

(ii) Every morphism from an object X of $\mathcal{B}$ to an object Y of $\mathcal{N}$ factors through an object of $\mathcal{N} \cap \mathcal{B}$.

(c) The properties (i)' and (ii)' obtained from (i) and (ii) by reversing the arrows are equivalent.

(d) If properties (i) and (ii) hold, or if properties (i)' and (ii)' hold, the canonical functor

$$\mathcal{B}/\mathcal{N} \cap \mathcal{B} \longrightarrow \mathcal{A}/\mathcal{N}$$

is faithful.

Since this functor is injective on objects, it realizes $\mathcal{B}/\mathcal{N} \cap \mathcal{B}$ as a subcategory of $\mathcal{A}/\mathcal{N}$.

(e) If, in addition, $\mathcal{B}$ is a full subcategory of $\mathcal{A}$, then $\mathcal{B}/\mathcal{N} \cap \mathcal{B}$ is a full subcategory of $\mathcal{A}/\mathcal{N}$.

Corollary 2.4.3. *Let $\mathcal{A} \subset \mathcal{B} \subset \mathcal{C}$ be three triangulated categories, with $\mathcal{A}$ a thick subcategory of $\mathcal{B}$ and $\mathcal{B}$ a thick subcategory of $\mathcal{C}$. Then $\mathcal{A}$ is a thick subcategory of $\mathcal{C}$, and $\mathcal{B}/\mathcal{A}$ is a thick subcategory of $\mathcal{C}/\mathcal{A}$. The canonical functor $\mathcal{C}/\mathcal{B} \to (\mathcal{C}/\mathcal{A})/(\mathcal{B}/\mathcal{A})$ is an isomorphism.*

2.5 Properties of the quotient functor

Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory, let S be the corresponding multiplicative system, and let $Q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ be the canonical quotient functor.

Proposition 2.5.1. *For objects X and Y of $\mathcal{A}$, the following assertions are equivalent:*

(a) Every morphism $f : X \to Y$ for which there exists a morphism $s : Z \to X$ in S satisfying $f \circ s = 0$ is zero.

(b) Every morphism $f : X \to Y$ for which there exists a morphism $s : Y \to Z$, $s \in S$, satisfying $s \circ f = 0$ is zero.

(c) Every morphism $f : X \to Y$ that factors through an object of $\mathcal{B}$ is zero.

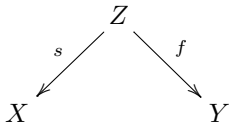
(d) The canonical map

$$\hom_{\mathcal{A}}(X, Y) \to \hom_{\mathcal{A}/\mathcal{B}}(Q(X), Q(Y))$$

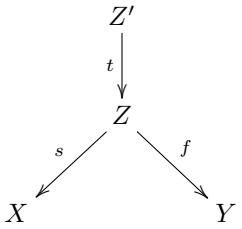
is injective.

Proposition 2.5.2. *Let X and Y be two objects of $\mathcal{A}$. The following assertions are equivalent:*

(a) Every diagram ($s \in S$)



can be completed to a diagram: ($s, t \in S$):



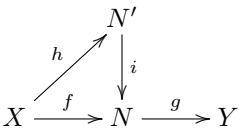
in which $f \circ t$ factors through $(X, s \circ t)$.

(b) The assertion obtained by reversing the direction of the arrows in (a) and interchanging X and Y.

(c) Every diagram

$$X \xrightarrow{f} N \xrightarrow{g} Y$$

in which $g \circ f = 0$ and N is an object of $\mathcal{B}$ can be completed to a diagram



in which $f = i \circ h$ and $g \circ i = 0$.

(d) The assertion obtained by reversing the arrows in (c) and interchanging X and Y.

(e) The canonical map

$$\hom_{\mathcal{A}}(X, Y) \xrightarrow{Q} \hom_{\mathcal{A}/\mathcal{B}}(Q(X), Q(Y))$$

is surjective.

Proposition 2.5.3. *Let X be an object of $\mathcal{A}$. The following assertions are equivalent:*

(a) For every object Y of $\mathcal{A}$, the canonical map

$$\hom_{\mathcal{A}}(X,Y) \xrightarrow{Q} \hom_{\mathcal{A}/\mathcal{B}}(Q(X),Q(Y))$$

is an isomorphism.

(b) Every morphism from X to an object of $\mathcal{B}$ is zero.

Similarly, the following two assertions are equivalent:

(a)' For every object Y of $\mathcal{A}$, the canonical map

$$\hom_{\mathcal{A}}(X,Y) \xrightarrow{Q} \hom_{\mathcal{A}/\mathcal{B}}(Q(X),Q(Y))$$

is an isomorphism.

(b)' Every morphism from an object of $\mathcal{B}$ to X is zero.

Definition 2.5.4. *Every object having the equivalent properties (a) and (b) of the preceding proposition is called an* object free on the left over $\mathcal{A}/\mathcal{B}$, *or a* left Q-free object. *Similarly, every object having the equivalent properties (a)' and (b)' is called an* object free on the right over $\mathcal{A}/\mathcal{B}$, *or a* right Q-free object. *It is determined up to isomorphism by the knowledge of $Q(X) \in \operatorname{Ob} \mathcal{A}/\mathcal{B}$.*

2.6 Functors adjoint to the quotient functor

Definition 2.6.1. *Two triangulated subcategories $\mathcal{N}$ and $\mathcal{N}'$ of a triangulated category $\mathcal{A}$ are said to be* orthogonal *if, for every object X of $\mathcal{N}$ and every object Y of $\mathcal{N}'$, one has*

$$\hom_{\mathcal{A}}(X,Y) = 0$$

The category $\mathcal{N}'$ is then said to be right orthogonal to $\mathcal{N}$, and $\mathcal{N}$ left orthogonal to $\mathcal{N}'$.

Proposition 2.6.2. *Let $\mathcal{N}$ be a triangulated subcategory of a triangulated category $\mathcal{A}$. The full category $\mathcal{N}^{\perp}$ (respectively ${}^{\perp}\mathcal{N}$) generated by the objects X of $\mathcal{A}$ such that, for every object Y of $\mathcal{N}$, $\hom_{\mathcal{A}}(Y,X) = 0$ (respectively $\hom_{\mathcal{A}}(X,Y) = 0$), is a thick subcategory of $\mathcal{A}$. By an abuse of language, the category $\mathcal{N}^{\perp}$ is called the right orthogonal of $\mathcal{N}$.*

This proposition allows us to restate Proposition 2.5.3.

Proposition 2.6.3. *Let $\mathcal{B} \subset \mathcal{A}$ be a thick subcategory of a triangulated category $\mathcal{A}$. The full category generated by the objects free on the right over $\mathcal{A}/\mathcal{B}$ (respectively on the left) is precisely the right (respectively left) orthogonal of $\mathcal{B}$.*

Proposition 2.6.4. *Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory of $\mathcal{A}$, let $\mathcal{B}^{\perp}$ be the right orthogonal of $\mathcal{B}$, let $S(\mathcal{B})$ and $s(\mathcal{B}^{\perp})$ be the corresponding multiplicative systems, and let Q and $Q^{\perp}$ be the canonical quotient functors. For an object x of $\mathcal{A}$, consider the following properties:*

(i) X maps by a morphism of $S(\mathcal{B})$ to an object of $\mathcal{B}^{\perp}$.

(ii) X receives, by a morphism of $S(\mathcal{B}^{\perp})$, an object of $\mathcal{B}$.

(iii) The category $S_X(\mathcal{B})$ of arrows of $S(\mathcal{B})$ with source X has a final object.

(iv) The category $S^X(\mathcal{B}^\perp)$ of arrows of $S(\mathcal{B}^\perp)$ with target X has an initial object.

(v) The category $\mathcal{B}/X$ of objects of $\mathcal{B}$ over X has a final object.

(vi) The category $\mathcal{B}^\perp/X$ of objects of $\mathcal{B}^\perp$ under X has an initial object.

Then

$$(i) \Longleftrightarrow (ii) \Longleftrightarrow (iii) \Longleftrightarrow (v) \Rightarrow (iv), \quad (v) \Rightarrow (vi)$$

If, in addition, ${}^\perp(\mathcal{B}^\perp) = \mathcal{B}$, then all the properties are equivalent.

Let $\mathcal{B}$ be a thick subcategory of a triangulated category $\mathcal{A}$, let i be the inclusion functor from $\mathcal{B}$ into $\mathcal{A}$, let Q be the quotient functor from $\mathcal{A}$ to $\mathcal{A}/\mathcal{B}$, let $\mathcal{B}^\perp$ be the right orthogonal of $\mathcal{B}$, and let $i^\perp$ and $Q^\perp$ be the corresponding functors. Proposition 2.6.4 immediately gives the following propositions.

Proposition 2.6.5. *The following two properties are equivalent:*

(i) The functor i has a right adjoint.

(ii) The functor Q has a right adjoint.

The proposition remains true when "right" is replaced by "left."

Proposition 2.6.6. *The following two properties are equivalent:*

(i) The functor i has a right adjoint.

(ii) The functor $i^\perp$ has a left adjoint. The left orthogonal of the category $\mathcal{B}^\perp$ is equal to $\mathcal{B}$.

Proposition 2.6.7. *Suppose the properties of Propositions 2.6.5 and 2.6.6 hold. Let i^* and Q^* be the right adjoints of i and Q. Similarly, let ${}^*i^\perp$ and ${}^*Q^\perp$ be the left adjoints of $i^\perp$ and $Q^\perp$. All these functors are exact. Moreover:*

There is a functorial isomorphism $Q^* \circ Q \xrightarrow{\sim} i^\perp \circ^* i^\perp$ such that the following diagram is commutative:

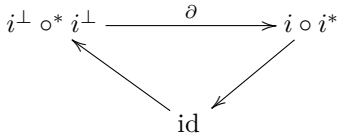
$$\begin{array}{ccc} & \mathrm{id} & \\ \swarrow & & \searrow \\ Q^* \circ Q & \xrightarrow{\sim} & i^\perp \circ^* i^\perp \end{array}$$

(the oblique arrows are defined by the adjunction properties).

Similarly, there is an isomorphism ${}^*Q^\perp \circ Q^\perp \xrightarrow{\sim} i \circ i^*$ such that the following diagram is commutative:

$$\begin{array}{ccc} {}^*Q^\perp \circ Q^\perp & \xrightarrow{\sim} & i \circ i^* \\ & \searrow \quad \swarrow & \\ & \mathrm{id} & \end{array}$$

Finally, there is a functorial morphism of degree 1, $\partial : i^{\perp} \circ^* i^{\perp} \to i \circ i^*$, *such that the following triangle is distinguished* ($\deg \partial = 1$)*:*



Such a morphism δ *is unique and satisfies* $\delta(TX_0) = -T\delta(X)$ *(*X *an object* $q \circ q$, T *the translation).*

3 Derived categories of an abelian category

3.1 Derived categories

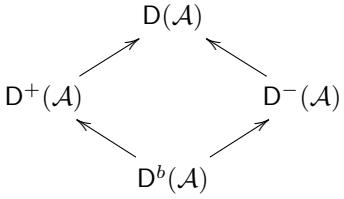
We use the notation of 1.2.3. We shall also use the following notation.

3.1.1 Notation

Let $\mathcal{A}$ be an abelian category. Set

$$\begin{aligned}
\mathsf{D}(\mathcal{A}) &= \mathsf{K}^{\infty,\infty}(\mathcal{A})/\mathsf{K}^{\infty,\varnothing}(\mathcal{A}) \\
\mathsf{D}^b(\mathcal{A}) &= \mathsf{K}^{b,b}(\mathcal{A})/\mathsf{K}^{b,\varnothing}(\mathcal{A}) \\
\mathsf{D}^+(\mathcal{A}) &= \mathsf{K}^{+,+}(\mathcal{A})/\mathsf{K}^{+,\varnothing}(\mathcal{A}) \\
\mathsf{D}^-(\mathcal{A}) &= \mathsf{K}^{-,-}(\mathcal{A})/\mathsf{K}^{-,\varnothing}(\mathcal{A}).
\end{aligned}$$

Proposition 3.1.2. *The natural functors appearing in the following diagram*



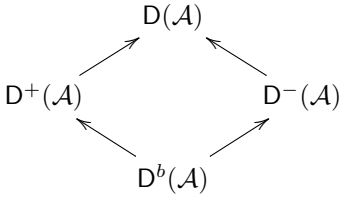
are fully faithful.

The natural functors

$$\begin{aligned}
\mathsf{D}^b(\mathcal{A}) = \mathsf{K}^{b,b}(\mathcal{A})/\mathsf{K}^{b,\varnothing}(\mathcal{A}) &\to \mathsf{K}^{+,b}(\mathcal{A})/\mathsf{K}^{+,\varnothing}(\mathcal{A}) \\
\mathsf{D}^b(\mathcal{A}) = \mathsf{K}^{b,b}(\mathcal{A})/\mathsf{K}^{b,\varnothing}(\mathcal{A}) &\to \mathsf{K}^{-,b}(\mathcal{A})/\mathsf{K}^{-,\varnothing}(\mathcal{A})
\end{aligned}$$

are equivalences of categories.

Since the functors in the diagram



are injective on objects, they realize the categories $\mathsf{D}^b(\mathcal{A})$, $\mathsf{D}^+(\mathcal{A})$, *and* $\mathsf{D}^-(\mathcal{A})$ *as full subcategories of* $\mathsf{D}(\mathcal{A})$.

The proof of this proposition is contained in Theorem 2.4.2.

Definition 3.1.3. *The category* $\mathsf{D}(\mathcal{A})$ *is called the* derived category *of* $\mathcal{A}$. *The objects of* $\mathsf{D}(\mathcal{A})$ *isomorphic to objects of* $\mathsf{D}^b(\mathcal{A})$ *are called the* bounded *objects of* $\mathsf{D}(\mathcal{A})$. *The objects of* $\mathsf{D}(\mathcal{A})$ *isomorphic to objects of* $\mathsf{D}^+(\mathcal{A})$ *are called the* left-bounded *objects, and the objects of* $\mathsf{D}(\mathcal{A})$ *isomorphic to objects of* $\mathsf{D}^-(\mathcal{A})$ *are called the* right-bounded *objects.*

Let D denote the canonical functor

$$D : \mathsf{C}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}).$$

The "restriction" of D to the full subcategory $\mathcal{A}$ of $\mathsf{C}(\mathcal{A})$ will again be denoted by D. The functor D restricted to $\mathcal{A}$ is fully faithful. The functor D factors uniquely through $\mathsf{K}(\mathcal{A})$. Let X and Y be two objects of $\mathcal{A}$.

One readily checks that, for every $m \geqslant 0$, the groups

$$\hom_{\mathsf{D}(\mathcal{A})}(D(X), T^{-m}D(Y)) \qquad (T \text{ is the translation functor})$$

are zero. The cohomological dimension of $\mathcal{A}$ is the least integer n such that, for every $m > n$,

$$\hom_{\mathsf{D}(\mathcal{A})}(D(X), T^{m}D(Y)) = 0 \qquad \text{for every pair } X, Y \text{ of objects of } \mathcal{A}.$$

If there is no such integer, the cohomological dimension of $\mathcal{A}$ is said to be infinite.

Let $\mathsf{I}^+(\mathcal{A})$ (respectively $\mathsf{I}(\mathcal{A})$) denote the full triangulated subcategory of $\mathsf{K}^+(\mathcal{A})$ (respectively of $\mathsf{K}(\mathcal{A})$) defined by the complexes whose objects are injective in every degree.

Proposition 3.1.4. *Suppose that the category* $\mathcal{A}$ *has enough injectives. The natural functor*

$$Q^+ : \mathsf{K}^+(\mathcal{A}) \to \mathsf{D}^+(\mathcal{A})$$

induces an equivalence of categories

$$\mathsf{I}^+(\mathcal{A}) \to \mathsf{D}^+(\mathcal{A}).$$

The quasi-inverse functor is a right adjoint of Q^+.

If, in addition, $\mathcal{A}$ has finite cohomological dimension, the preceding assertion remains true after the superscripts + are omitted.

There is an analogous statement concerning projectives.

The proof of Proposition 3.1.4 relies on 2.6.4.

The triangulated structure of $\mathsf{D}(\mathcal{A})$ is made more explicit by the following proposition.

Let $\mathsf{E}(\mathcal{A})$ be the category of three-term exact sequences in $\mathsf{C}(\mathcal{A})$. The category $\mathsf{SSS}(\mathcal{A})$ of semi-split exact sequences (§1.2.4) is a full subcategory of $\mathsf{E}(\mathcal{A})$. We defined (loc. cit.) a functor

$$\rho : \mathsf{SSS}(\mathcal{A}) \to \mathsf{TrK}(\mathcal{A})$$

where $\mathsf{TrK}(\mathcal{A})$ denotes the category of distinguished triangles of $\mathsf{K}(\mathcal{A})$. This gives a functor

$$\sigma : \mathsf{SSS}(\mathcal{A}) \to \mathsf{TrD}(\mathcal{A})$$

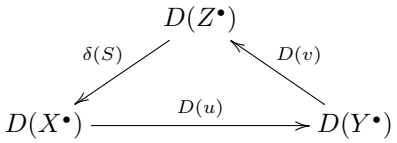
where $\mathsf{TrD}(\mathcal{A})$ denotes the category of distinguished triangles of $\mathsf{D}(\mathcal{A})$.

Proposition 3.1.5. *There exists a unique functor*

$$\Pi : \mathsf{E}(\mathcal{A}) \to \mathsf{TrD}(\mathcal{A})$$

of the following form ($\deg \delta(S) = 1$*):*

$$(S = 0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0) \mapsto$$



whose restriction to $\mathsf{SSS}(\mathcal{A})$ *is* σ*.* This functor is essentially surjective.

Proposition 3.1.6. *The canonical cohomological functor*

$$\mathrm{H}^0 : \mathsf{K}(\mathcal{A}) \to \mathcal{A}$$

factors uniquely through a functor that we again denote by H^0*,*

$$\mathrm{H}^0 : \mathsf{D}(\mathcal{A}) \to \mathcal{A}.$$

On $\mathsf{D}(\mathcal{A})$*, the functor* H^0 *and its translates* H^i *form a conservative system; that is, a morphism* $f : X^\bullet \to Y^\bullet$ *is an isomorphism if and only if, for every integer* i*,*

$$\mathrm{H}^i(f) : \mathrm{H}^i(X) \to \mathrm{H}^i(Y)$$

is an isomorphism.

Definition 3.1.7. *A morphism* $f : X^\bullet \to Y^\bullet$ *of* $\mathsf{C}(\mathcal{A})$ *(respectively of* $\mathsf{K}(\mathcal{A})$*) is called a* quasi-isomorphism *if, for every integer* i*,*

$$\mathrm{H}^i(f) : \mathrm{H}^i(X) \to \mathrm{H}^i(Y)$$

is an isomorphism.

By the preceding proposition, the quasi-isomorphisms are the morphisms that become isomorphisms in $\mathsf{D}(\mathcal{A})$.

Definition 3.1.8. *Let* σ *be a set of objects of* $\mathcal{A}$ *and let* $X^\bullet$ *be an object of* $\mathsf{C}(\mathcal{A})$ *(respectively* $\mathsf{K}(\mathcal{A})$*). A right (respectively left) resolution of type* σ *of* $X^\bullet$ *is a quasi-isomorphism* $X^\bullet \to V^\bullet$ *(respectively* $V^\bullet \to X^\bullet$*), where* $V^\bullet$ *is a complex all of whose objects belong to* σ*. We say* injective resolution *(respectively projective resolution) instead of right (respectively left) resolution of injective (respectively projective) type.*

In general, we shall omit the words "right" or "left" when no ambiguity can result.

Proposition 3.1.9. *1. Suppose that every object of* $\mathcal{A}$ *embeds in an object of* σ *(respectively is a quotient of an object of* σ*). Then every object of* $\mathsf{C}^+(\mathcal{A})$ *(respectively* $\mathsf{C}^-(\mathcal{A})$*) has a right (respectively left) resolution of type* σ *in* $\mathsf{C}^+(\mathcal{A})$ *(respectively in* $\mathsf{C}^-(\mathcal{A})$*).*

2. Suppose, in addition, that there exists an integer n *such that for every exact sequence*

$$Y^0 \to Y^1 \to \cdots \to Y^{n-1} \to Y^n \to 0$$

$$\text{(respectively } 0 \to Y^n \to Y^{n-1} \to \cdots \to Y^1 \to Y^0\text{)}$$

of objects of $\mathcal{A}$ *in which, for every* i*,* $0 \leqslant i \leqslant n-1$*,* Y^i *is an object of* σ*, the object* Y^n *is also an object of* σ*. Then every object of* $\mathsf{C}(\mathcal{A})$ *has a right (respectively left) resolution of type* σ*.*

3.2 Study of Ext

Let $X^\bullet$ and $Y^\bullet$ be two objects of $\mathsf{C}(\mathcal{A})$ (respectively $\mathsf{K}(\mathcal{A})$), and let $D : \mathsf{C}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ (respectively $Q : \mathsf{K}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$) be the canonical functor.

Definition 3.2.1. *We call the* ith hyper-Ext*, and denote by*

$$(X^\bullet, Y^\bullet) \mapsto \mathrm{Ext}^i(X^\bullet, Y^\bullet)$$

the functor $\hom_{\mathsf{D}(\mathcal{A})}(D(X^\bullet), T^i D(Y^\bullet))$ *(respectively* $\hom_{\mathsf{D}(\mathcal{A})}(Q(X^\bullet), T^i Q(Y^\bullet))$*).*

Proposition 3.2.2. *Let* $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ *be an exact sequence in* $\mathsf{C}(\mathcal{A})$*, and let* $V^\bullet$ *be an object of* $\mathsf{C}(\mathcal{A})$*. There are unbounded exact sequences*

$$\cdots \longrightarrow \mathrm{Ext}^i(V^\bullet, X^\bullet) \longrightarrow \mathrm{Ext}^i(V^\bullet, Y^\bullet) \longrightarrow \mathrm{Ext}^i(V^\bullet, Z^\bullet) \xrightarrow{\delta} \mathrm{Ext}^{i+1}(V^\bullet, X^\bullet) \longrightarrow \cdots$$

$$\cdots \longrightarrow \mathrm{Ext}^i(Z^\bullet, V^\bullet) \longrightarrow \mathrm{Ext}^i(Y^\bullet, V^\bullet) \longrightarrow \mathrm{Ext}^i(X^\bullet, V^\bullet) \xrightarrow{\delta} \mathrm{Ext}^{i+1}(Z^\bullet, V^\bullet) \longrightarrow \cdots$$

Let $X^\bullet$ and $Y^\bullet$ be two objects of $\mathsf{K}(\mathcal{A})$. Denote by $\mathsf{Qis}/X^\bullet$ (respectively $\mathsf{Qis}^+/X^\bullet$, $\mathsf{Qis}^-/X^\bullet$, $\mathsf{Qis}^b/X^\bullet$) the category of quasi-isomorphisms with target $X^\bullet$ and source in $\mathsf{K}(\mathcal{A})$ (respectively $\mathsf{K}^+(\mathcal{A})$, $\mathsf{K}^-(\mathcal{A})$, $\mathsf{K}^b(\mathcal{A})$). Similarly, define the categories $Y^\bullet/\mathsf{Qis}$, $Y^\bullet/\mathsf{Qis}^+$,... (quasi-isomorphisms with source $Y^\bullet$).

The following proposition summarizes the various equivalent definitions of Ext^i.

Proposition 3.2.3. *1. There are isomorphisms of functors*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) = \hom_{\mathsf{D}(\mathcal{A})}(Q(X^\bullet), Q(Y^\bullet)) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}/X^\bullet)^\circ} \hom_{\mathsf{K}(\mathcal{A})}(-, Y^\bullet) \xrightarrow{\sim}$$

$$\varinjlim_{Y^\bullet/\mathsf{Qis}} \hom_{\mathsf{K}(\mathcal{A})}(X^\bullet, -) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}/X^\bullet)^\circ \times Y^\bullet/\mathsf{Qis}} \hom_{\mathsf{K}(\mathcal{A})}(-, -).$$

If $X^\bullet$ *is an object of* $\mathsf{K}^-(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}^-/X^\bullet)^\circ} \hom_{\mathsf{K}(\mathcal{A})}(-, Y^\bullet)$$

If $Y^\bullet$ *is an object of* $\mathsf{K}^+(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{Y^\bullet/\mathsf{Qis}} \hom_{\mathsf{K}(\mathcal{A})}(X^\bullet, -)$$

If $X^\bullet$ *is an object of* $\mathsf{K}^-(\mathcal{A})$ *and* $Y^\bullet$ *is an object of* $\mathsf{K}^b(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{Y^\bullet/\mathsf{Qis}^b} \hom_{\mathsf{K}(\mathcal{A})}(X^\bullet, -)$$

If $X^\bullet$ *is an object of* $\mathsf{K}^b(\mathcal{A})$ *and* $Y^\bullet$ *is an object of* $\mathsf{K}^+(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}^b/X^\bullet)^\circ} \hom_{\mathsf{K}(\mathcal{A})}(-, Y^\bullet)$$

If the category $\mathcal{A}$ has enough injectives and $Y^\bullet$ is an object of $\mathsf{K}^+(\mathcal{A})$, then $Y^\bullet$ has an injective resolution, unique up to isomorphism in $\mathsf{K}^+(\mathcal{A})$,

$$Y^\bullet \to I(Y^\bullet)$$

and

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \mathrm{hom}_{\mathsf{K}(\mathcal{A})}(X^\bullet, I(Y^\bullet)).$$

There is an analogous statement for projectives. If the category $\mathcal{A}$ has enough injectives and if $\mathcal{A}$ has finite cohomological dimension*, then every object $Y^\bullet$ of $\mathsf{K}(\mathcal{A})$ has a unique injective resolution*

$$Y^\bullet \to I(Y^\bullet)$$

and

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \mathrm{hom}_{\mathsf{K}(\mathcal{A})}(X^\bullet, I(Y^\bullet))$$

There is an analogous statement for projectives.

Remark: On making explicit the isomorphism of Proposition 3.2.3(3), one recovers the Yoneda definition.

4 Derived functors

4.1 Definition of derived functors

Definition 4.1.1. *Let $\mathcal{C}$ and $\mathcal{C}'$ be two graded categories (we denote by T the translation functor of $\mathcal{C}$ and $\mathcal{C}'$), and let F and G be two graded functors from $\mathcal{C}$ to $\mathcal{C}'$. A* morphism of graded functors *is a morphism of functors*

$$u : F \to G$$

having the following property:

For every object X of $\mathcal{C}$, the following diagram is commutative:

$$\begin{array}{ccc} F(TX) & \xrightarrow{u(TX)} & G(TX) \\ \uparrow \wr & & \uparrow \wr \\ TF(X) & \xrightarrow{Tu(X)} & TG(X) \end{array}$$

Let $\mathcal{C}$ and $\mathcal{C}'$ be two triangulated categories. Denote by $\mathsf{Fex}(\mathcal{C}, \mathcal{C}')$ the category of exact functors from $\mathcal{C}$ to $\mathcal{C}'$, the morphisms between two functors being morphisms of graded functors.

Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $\Phi : \mathsf{K}^*(\mathcal{A}) \to \mathsf{K}^{*'}(\mathcal{B})$ be an exact functor ($*$ and $*'$ denote one of the signs $+$, $-$, b, or v, "empty"). By composition, the canonical functor $Q : \mathsf{K}^*(\mathcal{A}) \to \mathsf{D}^*(\mathcal{A})$ gives a functor

$$\mathsf{Fex}(\mathsf{D}^*(\mathcal{A}), \mathsf{D}^{*'}(\mathcal{B})) \longrightarrow \mathsf{Fex}(\mathsf{K}^*(\mathcal{A}), \mathsf{D}^{*'}(\mathcal{B}))$$

and hence (also denoting by Q' the canonical functor $\mathsf{K}^{*'}(\mathcal{B}) \to \mathsf{D}^{*'}(\mathcal{B})$) a functor $\%$ (respectively $\%'$) $\mathsf{Fex}(\mathsf{D}^*(\mathcal{A}), \mathsf{D}^{*'}(\mathcal{B})) \to \mathsf{Ab}$:

$$\Psi \mapsto \mathrm{hom}(Q' \circ \Phi, \Psi \circ Q)$$
$$(\text{respectively } \Psi \mapsto \mathrm{hom}(\Psi \circ Q, Q' \circ \Phi) \ ).$$

Definition 4.1.2. *We say that* Φ *admits a* total right derived functor *(respectively total left derived functor) if the functor* $\%$ *(respectively* $\%'$*) is representable. An object representing* $\%$ *(respectively* $\%'$*) is called the total right (respectively left) derived functor of* Φ *and is denoted by* $\mathsf{R}\Phi$ *(respectively* $\mathsf{L}\Phi$*).*[§]

4.1.3 Notation

Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor. The *total right derived functor* of $\mathsf{K}^+(f) : \mathsf{K}^+(\mathcal{A}) \to \mathsf{K}^+(\mathcal{B})$ is denoted by R^+f. The functors R^-f, $\mathsf{R}f$, L^+f, L^-f, and $\mathsf{L}f$ are defined similarly.

Likewise, let $F : \mathsf{K}^*(\mathcal{A}) \to \mathcal{B}$ be a cohomological functor. By composition, the canonical quotient functor $Q : \mathsf{K}^*(\mathcal{A}) \to \mathsf{D}^*(\mathcal{A})$ defines a functor

$$\mathsf{Foco}(\mathsf{D}^*(\mathcal{A}), \mathcal{B}) \longrightarrow \mathsf{Foco}(\mathsf{K}^*(\mathcal{A}), \mathcal{B})$$

(where $\mathsf{Foco}(-,-)$ denotes the category of cohomological functors), and hence a functor $\%$ (respectively $\%'$) $\mathsf{Foco}(\mathsf{D}^*(\mathcal{A}), \mathcal{B}) \to \mathsf{Ab}$ given by $G \mapsto \hom(F, G \circ Q)$ (respectively $G \mapsto \hom(G \circ Q, F)$).

Definition 4.1.4. *We say that the functor* F *admits a* right cohomological derived functor *(respectively left cohomological derived functor) if the functor* $\%$ *(respectively* $\%'$*) is representable. An object representing* $\%$ *(respectively* $\%'$*) is called the right (respectively left) cohomological derived functor and is denoted by* $\mathsf{R}F$ *(respectively* $\mathsf{L}F$*).*

4.1.5 Notation

Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor. The right cohomological derived functor of

$$\mathrm{H}^0 \circ \mathsf{K}^+f : \mathsf{K}^+(\mathcal{A}) \to \mathcal{B}$$

is denoted by R^+f. When no confusion can result, the functor $\mathsf{R}^+f \circ T^i$ (T is the translation functor) is denoted by R^if. The right cohomological derived functor of $\mathrm{H} \circ \mathsf{K}f$ is denoted by $\mathsf{R}f$. When no confusion can result, the functor $\mathsf{R}f \circ T^i$ is denoted by R^if.

The functors L^-f, $\mathsf{L}f$, and L^if are defined similarly.

4.1.6 Remarks

As we shall see in the next subsection, these definitions include the classical definition of derived functors when the categories have enough injectives or projectives and the complexes are suitably bounded. They are nevertheless more general, but without further hypotheses they are not very manageable. For example, let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor between two abelian categories. Suppose that the functors R^+f and $\mathsf{R}f$ exist. I do not know how to prove that $\mathsf{R}f$ induces on $\mathsf{D}^+(\mathcal{A})$ the functor R^+f. Suppose, in addition, that the functor R^+f (cohomological derived) exists. I do not know how to prove that the functor $\mathrm{H}^0 \circ \mathsf{R}^+f$ is isomorphic to R^+f. In practice, however, these properties will hold.

4.2 Existence of derived functors

Proposition 4.2.1. *The hypotheses are those of Definition 4.1.4. Suppose, in addition, that* $\mathcal{A}$ *is a* $\mathscr{U}$*-category, where* $\mathscr{U}$ *is a universe such that the set* $\mathbb{Z}$ *of rational integers belongs to* $\mathscr{U}$*, and such that the set of objects of* $\mathcal{A}$ *belongs to* $\mathscr{U}$*. Suppose also that* $\mathcal{B}$ *is the category of abelian* $\mathscr{U}$*-groups, or more*

[§]A more manageable definition (and one equivalent in practice) is given in [2, XVII 1.2].

generally the category of sheaves of abelian $\mathscr{U}$-groups on an arbitrary $\mathscr{U}$-site. (A $\mathscr{U}$-site is a $\mathscr{U}$-category whose set of objects belongs to $\mathscr{U}$, equipped with a topology.) The functor F admits a right cohomological derived functor. This functor is calculated as follows:

Let X be an object of $\mathsf{D}^*(\mathcal{A})$, and let $\underline{X}$ be the object of $\mathsf{K}^*(\mathcal{A})$ above X. There is a functorial isomorphism

$$\mathsf{R}F(X) = \varinjlim_{\underline{X}/\mathsf{Qis}^*} F(-).$$

In particular, let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor. The functors $\mathsf{R}f$ and R^+f exist and are calculated as above. The functor $\mathsf{R}f$ induces on $\mathsf{D}^+(\mathcal{A})$ the functor R^+f. Since no confusion can then result, we shall use the notation $\mathsf{R}^i f$.

4.2.2 Remarks

Proposition 4.2.1 can be expressed in a more general form in the setting of triangulated categories. The hypotheses on the category $\mathcal{B}$ needed to ensure the validity of Proposition 4.2.1 are hypotheses asserting the existence and exactness of direct limits over pseudo-filtered $\mathscr{U}$-categories whose sets of objects belong to $\mathscr{U}$. Proposition 4.2.1 introduces an asymmetry between right and left derived functors, at least in practice.

Theorem 4.2.3. *The data are those of Definition 4.1.2. Suppose, in addition, that a full triangulated subcategory of $\mathsf{K}^*(\mathcal{A})$, denoted $\mathcal{C}$, is given, and denote by $\mathcal{N}$ the full triangulated subcategory of the acyclic objects of $\mathcal{C}$. Suppose that*

(1) Every object of $\mathcal{N}$ is taken by Φ to an acyclic object of $\mathsf{K}^{*'}(\mathcal{B})$.

(2) Every object X of $\mathsf{K}^*(\mathcal{A})$ is the source (respectively target) of a quasi-isomorphism whose target (respectively source) is an object of $\mathcal{C}$.

a) Then Φ admits a total right (respectively left) derived functor.

b) The functor $\mathsf{R}\Phi$ (respectively $\mathsf{L}\Phi$) is obtained as follows:

The restriction of the functor $Q' \circ \Phi : \mathsf{K}^*(\mathcal{A}) \to \mathsf{D}^{*'}(\mathcal{B})$ to $\mathcal{C}$ vanishes on the objects of $\mathcal{N}$ and therefore factors through $\mathcal{C}/\mathcal{N}$. This gives a functor $\Phi' : \mathcal{C}/\mathcal{N} \to \mathsf{D}^{*'}(\mathcal{B})$. But the natural functor $\mathcal{C}/\mathcal{N} \to \mathsf{D}^*(\mathcal{A})$ is an equivalence of categories (Theorem 2.4.2). Composing with its quasi-inverse gives the functor $\mathsf{R}\Phi$ (respectively $\mathsf{L}\Phi$).

c) Let X be an object of $\mathsf{K}^*(\mathcal{A})$, let Y be an object of $\mathcal{C}$, and let $X \to Y$ (respectively $Y \to X$) be a quasi-isomorphism. The object $\mathsf{R}\Phi \circ Q(X)$ (respectively $\mathsf{L}\Phi \circ Q(X)$) is isomorphic to $Q \circ \Phi(Y)$.

d) The functor $X \mapsto \mathsf{R}\Phi \circ Q(X)$ (respectively $X \mapsto \mathsf{L}\Phi \circ Q(X)$) is isomorphic to the functor $X \mapsto \varinjlim_{X/\mathsf{Qis}^*} Q \circ \Phi(-)$ (respectively $X \mapsto \varinjlim_{\mathsf{Qis}^*/X} Q \circ \Phi(-)$).

e) The functor $\mathrm{H}^0 \circ \Phi : \mathsf{K}^*(\mathcal{A}) \to \mathcal{B}$ admits a right (respectively left) cohomological derived functor, which is precisely $\mathrm{H}^0 \circ \mathsf{R}\Phi$ (respectively $\mathrm{H}^0 \circ \mathsf{L}\Phi$).

Corollary 4.2.4. *Suppose that $* = +$ (respectively $* = -$) in the hypotheses of the preceding theorem and that $\mathcal{A}$ has enough injectives (respectively projectives). Then Theorem 4.2.3 applies with $\mathcal{C}$ the category of complexes of injectives (respectively projectives). Suppose in addition that $\mathcal{A}$ has finite cohomological dimension. Then the theorem applies without restriction on $*$, with $\mathcal{C}$ the category of complexes of injectives (respectively projectives).*

Definition 4.2.5. *Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor.*

We say that f has finite cohomological dimension *on the right (respectively left) if R^+f (respectively L^-f) exists and there is an integer $m \geqslant 0$ such that, for every integer $n > m$, R^nf (respectively $\mathsf{L}^{-n}f$) is zero on the objects arising from $\mathcal{A}$ under the functor D. The* cohomological dimension *on the right (respectively left) of f is then the least integer m having this property. If R^+f (respectively L^-f) exists and f does not have finite cohomological dimension, we say that f has infinite cohomological dimension.*

An object X of $\mathcal{A}$ is said to be f-acyclic *on the right (respectively on the left) if, for every integer $n > 0$, $\mathsf{R}^nf(D(X)) = 0$ (respectively $\mathsf{L}^{-n}f(D(X)) = 0$).*

Corollary 4.2.6. *The data are those of Definition 4.2.5. Suppose there exists a set σ of objects of $\mathcal{A}$, stable under direct sums, having the following properties:*

1. *Every acyclic complex $\in \mathrm{Ob}\,\mathsf{K}^+(\mathcal{A})$ (respectively $\in \mathrm{Ob}\,\mathsf{K}^-(\mathcal{A})$) of objects of σ is taken by f to an acyclic complex.*
2. *Every object X of $\mathcal{A}$ embeds in (respectively is a quotient of) an object of σ. The hypotheses of Theorem 2.4.3 hold with $* = +$ (respectively $* = -$) and with $\mathcal{C}$ the category of complexes whose objects in every degree belong to σ. The elements of σ are right (respectively left) f-acyclic objects.*

If, in addition, f has finite right (respectively left) cohomological dimension, the set of right (respectively left) f-acyclic objects has, in addition to properties (1) and (2) above, property (2) of Proposition 3.1.9. Theorem 4.2.3 then applies to the functor $\mathsf{K}f$, with $\mathcal{C}$ the category of complexes whose objects are right (respectively left) f-acyclic in every degree. The functor $\mathsf{R}f$ (respectively $\mathsf{L}f$) exists and induces on $\mathsf{D}^+(\mathcal{A})$ the functor R^+f (respectively L^+f), and on $\mathsf{D}^-(\mathcal{A})$ the functor R^-f (respectively L^-f).

4.3 Product. Composition

Let $\mathcal{A}$ be an abelian category, let $F : \mathsf{D}(\mathcal{A}) \to \mathsf{Ab}$ be a cohomological functor, and let $X^\bullet$ and $Y^\bullet$ be two objects of $\mathsf{C}(\mathcal{A})$. The functor F defines a map

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \to \mathrm{hom}_{\mathsf{Ab}}(F(D(X^\bullet)), F(D(Y^\bullet)))$$

and hence a pairing $F(D(X^\bullet)) \times \mathrm{Ext}^0(X^\bullet, Y^\bullet) \to F(D(Y^\bullet))$.

Applying translations to this pairing gives pairings

$$F^i(D(X^\bullet)) \times \mathrm{Ext}^j(X^\bullet, Y^\bullet) \to F^{i+j}(D(Y^\bullet)).$$

By definition, when these pairings are applied to cohomological derived functors, they are precisely the Yoneda products. The properties of these products follow immediately from this definition. We shall not develop them here.

Let $\mathcal{A}$, $\mathcal{A}'$, and $\mathcal{A}''$ be three abelian categories, and let $f : \mathcal{A} \to \mathcal{A}'$ and $g : \mathcal{A}' \to \mathcal{A}''$ be two additive functors. Suppose that the functors R^+f and R^+g exist. I do not know how to deduce that the functor $\mathsf{R}^+g \circ f$ exists, and even if the latter functor exists, the formula

$$\mathsf{R}^+g \circ f \xrightarrow{\sim} \mathsf{R}^+g \circ \mathsf{R}^+f$$

is probably not true in general. Nevertheless, one has the following proposition.

Proposition 4.3.1. *Suppose that the category $\mathcal{A}'$ has enough right g-acyclic objects and that $\mathcal{A}$ has enough right f-acyclic objects that f takes to right g-acyclic objects. Then the functor $\mathsf{R}^+g \circ f$ exists and there is an isomorphism*

$$\mathsf{R}^+g \circ f \xrightarrow{\sim} \mathsf{R}^+g \circ \mathsf{R}^+f.$$

There are analogous statements for left derived functors.

5 Examples

5.1 The functor $\mathsf{Rhom}^\bullet$

Let $\mathcal{A}$ be an abelian category with enough injectives.

The functor $Y \mapsto \hom^\bullet(X^\bullet, Y^\bullet)$ (1.3), where $X^\bullet$ is an object of $\mathsf{K}(\mathcal{A})$ and $Y^\bullet$ an object of $\mathsf{K}^+(\mathcal{A})$, has a total right derived functor (Corollary 4.2.4), denoted by $\mathsf{R}\hom^\bullet(X^\bullet, -)$. Let Y be an object of $\mathsf{D}^+(\mathcal{A})$. The functor

$$\begin{aligned} \mathsf{K}(\mathcal{A}) &\to \mathsf{D}(\mathsf{Ab}) \\ X^\bullet &\mapsto \mathsf{R}\hom^\bullet(X^\bullet, Y) \end{aligned}$$

is exact and vanishes on acyclic complexes, hence gives an exact bifunctor

$$\mathsf{D}(\mathcal{A})^\circ \times \mathsf{D}^+(\mathcal{A}) \longrightarrow \mathsf{D}(\mathsf{Ab})$$

which we denote by $\mathsf{Rhom}^\bullet$.

When $\mathcal{A}$ has finite cohomological dimension, the functor $\mathsf{Rhom}^\bullet$ extends to $\mathsf{D}(\mathcal{A})^\circ \times \mathsf{D}(\mathcal{A})$ (loc. cit.).

If, in addition, $\mathcal{A}$ has enough projectives, the functor hom can be derived with respect to the first argument. The two definitions coincide on their common domain of validity.

5.2 $\mathscr{T}or$ of sheaves

Let E be a ringed site (in particular, a ringed topological space), let $\mathscr{A}$ be its sheaf of rings, and let $\mathsf{Mod}(E)$ be the category of $\mathscr{A}$-modules. An object X of $\mathsf{Mod}(E)$ is said to be *$\mathscr{A}$-flat* if, for every exact sequence

$$0 \longrightarrow Y' \longrightarrow Y \longrightarrow Y'' \longrightarrow 0$$

the sequence $0 \to Y' \otimes_{\mathscr{A}} X \to Y \otimes_{\mathscr{A}} X \to Y'' \otimes_{\mathscr{A}} X \to 0$ is exact. There are enough $\mathscr{A}$-flat objects: the direct sums of objects $\mathscr{A}_U$, where U is an object of E and $\mathscr{A}_U$ is the sheaf $\mathscr{A}$ "extended by zero outside U." Thus Theorem 4.2.3 applies, and one can define the functor

$$\begin{aligned} \mathsf{D}^-(E, \mathscr{A}) \times \mathsf{D}^-(E, \mathscr{A}) &\to \mathsf{D}^-(E, \mathscr{A}) \\ (X, Y) &\mapsto X \overset{\mathsf{L}}{\otimes} Y \end{aligned}$$

the total tensor product, which is the left derived functor of the tensor product. Passing to cohomology defines the $\mathscr{T}or_i^{\mathscr{A}}(X, Y)$, the local hyper-Tor groups.

Since the category $\mathsf{Mod}(E)$ has enough injectives, one can derive the functor $\hom^\bullet(X, Y)$, the complex of local homomorphisms, and thus obtain $\mathsf{Rhom}(X, Y)$ ($X \in \mathsf{D}$, $Y \in \mathsf{D}^+$).

Let X and Y be two objects of $\mathsf{D}^-(E, \mathscr{A})$ and let Z be an object of $\mathsf{D}^+(E, \mathscr{A})$. There is a functorial isomorphism

$$\hom_{\mathsf{D}(E,\mathscr{A})}(X \overset{\mathsf{L}}{\otimes} Y, Z) \xrightarrow{\sim} \hom_{\mathsf{D}(E,\mathscr{A})}(X, \mathsf{Rhom}(Y, Z)).$$

5.3 Left derived functor of the inverse image

Let $f : E \to E'$ be a map of ringed topological spaces. The inverse-image functor

$$f^* : \mathsf{Mod}(E') \to \mathsf{Mod}(E)$$

is not necessarily exact. But one knows how to define f^*-flat objects, and there are enough such objects. Thus one can define the functor $\mathsf{L}^- f^*$. There is likewise a duality formula:

Let X be an object of $\mathsf{D}^-(E, \mathscr{A})$ and let Y be an object of $\mathsf{D}^+(E', \mathscr{A}')$. There is a functorial isomorphism

$$\hom_{\mathsf{D}(E',\mathscr{A}')}(\mathsf{L}^- f^*(X), Y) \xrightarrow{\sim} \hom_{\mathsf{D}(E,\mathscr{A})}(X, \mathsf{R}^+ f_*(Y)).$$

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SGA 5 — Exposé I

Dualizing Complexes

by A. Grothendieck
(written up by L. Illusie)

Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A XVIII (SGA A = SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very analogous to the one developed in HARTSHORNE [H] for the Zariski topology and coherent sheaves: the latter served, to a very large extent, as its model. But, whereas in the case of "continuous" coefficients the existence of dualizing complexes presents no difficulties, in the case of "discrete" coefficients, on the other hand, it is much more delicate to prove, and it seems to require resolution of singularities in an essential way. See however Deligne's biduality theorem (SGA $4^{1/2}$, Finiteness Theorem 4.3.).

Paragraph 1 contains the definition of dualizing complexes and the formal properties that follow from it, notably the interpretation of dualizing functors as "exchangers of functors."

In paragraph 2, one shows that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to a translation of degrees and tensoring by an invertible sheaf.

In paragraph 3, one comes to the central theorem of the exposé (3.4.1.), which asserts that, on a "good" regular prescheme X, for example on an excellent noetherian regular prescheme of characteristic zero[1], the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing.

Paragraph 4 gives the applications of the theory to the local duality theorem. The latter expresses that, on a "good" prescheme X endowed with a closed point x whose residue field is separably closed, and for a constructible sheaf F, one has a perfect pairing between $\underline{\mathrm{H}}^i(F')_x$ and $\underline{\mathrm{H}}^{-i}_x(F)$, where F' is the dual of F, i.e. $F' = \mathrm{R}\,\underline{\underline{\mathrm{Hom}}}(F, K_X)$, where K_X is a dualizing complex.

Finally, in paragraph 5, one proves directly that on a regular prescheme of dimension 1 the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing (n prime to the residual characteristics).

1 Definition and formal properties of dualizing complexes

We fix a coefficient ring $A = \mathbb{Z}/n\mathbb{Z}$. If X is a prescheme, we denote by A_X the constant sheaf on $X_{\text{ét}}$ with value A, and by $\mathrm{D}(X)$ the derived category of the category of A_X-modules. We denote, on the other hand, by $\mathrm{D}_c(X)$ the full triangulated subcategory of $\mathrm{D}(X)$ formed by the complexes F such that $\mathrm{H}^i(F)$ is a constructible A_X-module for every i. We put $\mathrm{D}^*_c(X) = \mathrm{D}^*(X) \cap \mathrm{D}_c(X)$, where $* = \varnothing$, $+$, $-$, or b.

[1] This last restriction will become unnecessary once one has resolution of singularities à la Hironaka for these preschemes, and the "purity theorem."

Definition 1.1

An A_X-module I is said to be *quasi-injective* if one has

$$\underline{\mathrm{Ext}}^i(F, I) = 0$$

for every *constructible* A_X-module F and every $i > 0$.

Remarks 1.2

a) Contrary to what happens in the case of "continuous" coefficients (cf. [H] II.7.17), a quasi-injective sheaf is not injective in general. For example, let $X = \mathrm{Spec(R)}$ and $A = \mathbb{Z}/2\mathbb{Z}$: the sheaf A_X is quasi-injective, but is not injective, nor even of finite injective dimension, since $\mathrm{H}^*(X; A_X) = \mathrm{H}^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$ is a polynomial algebra on one generator of dimension 1.

b) Let I be a quasi-injective A_X-module. It is false in general that the relation $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > 0$, valid for constructible F, is valid for every F. Indeed, take $A = \mathbb{F}_\ell$. Let k be a field, $\overline{k}$ a separable closure of k, and $f : \overline{x} = \mathrm{Spec}(\overline{k}) \to X = \mathrm{Spec}(k)$ the canonical morphism, and suppose that, for every connected étale covering $X' \to X$, there exists a connected étale covering $X'' \to X'$ such that $[X'' : X']$ is divisible by ℓ (take, for example, $k = \mathbb{Q}$). Consider the exact sequence

$$(*) \qquad 0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0$$

defined by the adjunction arrow ε. In view of the hypothesis, the section of $\underline{\mathrm{Ext}}^1(F, A_X)$ defined by $(*)$ does not vanish above any U étale over X; hence $\underline{\mathrm{Ext}}^1(F, A_X)_{\overline{x}} \neq 0$. But one checks trivially that A_X is quasi-injective.

Proposition 1.3 (cf. [H] 1.7.6)

Let X be a prescheme and $N \in \mathbb{Z}$. The following conditions on $K \in \mathrm{ob\,D}^+(X)$ are equivalent:

(i) K is isomorphic, in $\mathrm{D}(X)$, to a bounded complex I such that I^i is quasi-injective for every i and zero for $i > N$;

(ii) for every $F \in \mathrm{ob\,D}^b_c(X)$ such that $\mathrm{H}^i(F) = 0$ for $i < 0$, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$;

(iii) for every constructible A_X-module F, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$.

Proof. $(i) \Rightarrow (ii)$. Let $F \in \mathrm{ob\,D}^b_c(X)$ be such that $\mathrm{H}^i(F) = 0$ for $i < 0$, and let $I \in \mathrm{ob\,D}^+(X)$ be bounded and such that I^i is quasi-injective for every i and zero for $i > N$; we show that $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > N$. Arguing by induction on the number of indices i such that $I^i \neq 0$, one is reduced, by dévissage, to the case where $N = 0$ and I is concentrated in degree 0. One then has a biregular spectral sequence

$$\mathrm{E}_2^{pq} = \underline{\mathrm{Ext}}^p(\mathrm{H}^{-q}(F), I) \Rightarrow \underline{\mathrm{Ext}}^*(F, I).$$

Since I is quasi-injective, $\mathrm{E}_2^{pq} = 0$ for $p > 0$ and every q, and hence one has $\mathrm{E}_2^{0i} = \underline{\mathrm{Hom}}(\mathrm{H}^{-i}(F), I) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(F, I)$, which gives the conclusion.

$(ii) \Rightarrow (iii)$ is trivial.

$(iii) \Rightarrow (i)$ We may suppose that K is bounded below and has injective components. The hypothesis, applied to $F = A_X$, implies that $\mathrm{H}^i(K) = 0$ for $i > N$. We may therefore replace K by an isomorphic complex K', bounded, and such that K'^i is injective for $i < N$ and zero for $i > N$. It is then observed that K'^N is quasi-injective, which completes the proof.

Remark 1.3.1

The writer does not know whether (1.3) is still true when, in condition (ii), the hypothesis "$F \in \mathrm{ob}\,\mathrm{D}_c^b(X)$" is replaced by "$F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$".

Definition 1.4

The *quasi-injective dimension* of K, denoted $\dim.q.inj(K)$, is the lower bound (in $\overline{\mathrm{R}}$) of the integers N satisfying the equivalent conditions of (1.3).

If K is of finite quasi-injective dimension, the functor $\mathrm{R}\,\underline{\mathrm{Hom}}(\quad, K) : \mathrm{D}(X) \to \mathrm{D}(X)$ induces a functor $\mathrm{D}_c^b(X) \to \mathrm{D}^b(X)$.

We shall see below that if X is smooth of dimension d over a field k and n is prime to the characteristic of k, then $\dim.q.inj(A_X) = 2d$.

The example of (1.2 a)) shows that a complex of finite quasi-injective dimension is not necessarily of finite injective dimension. Nevertheless one has:

Proposition 1.5

Let X be a locally noetherian prescheme and $K \in \mathrm{ob}\,\mathrm{D}^+(X)$. Suppose that there exists an integer N such that $\mathrm{cd}_{\mathbf{n}}(X) \leqslant N$, where $\mathbf{n}$ is the set of prime divisors of n (notation of SGAA X 1). Then, for $N' \in \mathbb{Z}$, one has

$$\dim.q.inj(K) \leqslant N' \quad \Longrightarrow \quad \dim.inj(K) \leqslant N + N'.$$

We shall use, in the proof, the following

Lemma 1.5.1

Let C be an abelian category satisfying (AB5) and possessing a family of generators $(U_i)_{i \in I}$. Let $K \in \mathrm{ob}\,\mathrm{D}^+(C)$ and $N \in \mathbb{Z}$. The following conditions are equivalent:

(i) $\dim.inj(K) \leqslant N$;

(ii) for every $i \in I$ and every quotient F of U_i, one has

$$\mathrm{Ext}^n(F, K) = 0 \qquad \text{for } n > N.$$

Proof. It suffices to prove $(ii) \Rightarrow (i)$. By a well-known argument (cf. for example [H] 1.7.6), one is reduced to showing that if $K' \in \mathrm{ob}(C)$ is such that $\mathrm{Ext}^1(F, K') = 0$ every time F is a quotient of a U_i, then K' is injective. In other words, one has to prove that, if every morphism from a subobject

V of a U_i into K' extends to U_i, then K' is injective. But this follows immediately from the proof of Lemma 1, p. 136 of (Tohoku).

Proof of (1.5). Suppose $\dim.q.inj(K) \leqslant N'$ and prove $\dim.inj(K) \leqslant N + N'$. Since the category of A_X-modules admits as generators the $A_{U,X}$ where U is étale of finite type over X, it suffices, by (1.5.1), to prove that $\mathrm{Ext}^i(X; F, K) = 0$ for $i > N + N'$ when F is a quotient of such an $A_{U,X}$, hence constructible (SGAA IX 2.9). Consider the spectral sequence

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(X; \underline{\mathrm{Ext}}^q(F, K)) \Rightarrow \mathrm{Ext}^*(X; F, K).$$

The hypothesis $\mathrm{cd}_{\mathbf{n}}(X) \leqslant N$ (resp. $\dim.q.inj(K) \leqslant N'$) implies $\mathrm{E}_2^{pq} = 0$ for $p > N$ (resp. $q > N'$). Hence $\mathrm{E}_2^{pq} = 0$ for $p + q > N + N'$, whence the conclusion.

Remark 1.5.2

The hypothesis of (1.5) is verified (with $N = 2d$) if X is a prescheme of finite type and dimension d over a separably closed field k, with n prime to $\mathrm{char}(k)$ (SGAA X 4).

Proposition 1.6

Let $f : X \to Y$ be a separated quasi-finite morphism of noetherian preschemes, and let $K \in \mathrm{ob}\,\mathrm{D}^+(Y)$. Then, for $N \in \mathbb{Z}$, one has

$$\dim.q.inj(K) \leqslant N \quad \Longrightarrow \quad \dim.q.inj(\mathrm{R}^!f(K)) \leqslant N.$$

Proof. According to the "Main theorem" (EGA IV 8.12.6), there exists a factorization of f as $f = uf'$, where f' is an open immersion and u is a finite morphism. We have $\mathrm{R}^!f = \mathrm{R}^!f'\,\mathrm{R}^!u$, which reduces us to examining separately the case of an open immersion and that of a finite morphism. If f is an open immersion, the assertion is trivial (since every constructible sheaf on X is then induced by a constructible sheaf on Y). Suppose therefore that f is a finite morphism, and let F be a constructible A_X-module. We then have the duality isomorphisms (SGAA XVIII 3.1.9.6)

$$f_*\underline{\mathrm{Ext}}^i(F, \mathrm{R}^!f(K)) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(f_*F, K).$$

Since $f_*(F)$ is constructible (SGAA IX 2.14), it follows that, if $\dim.q.inj(K) \leqslant N$, then $f_*\underline{\mathrm{Ext}}^i(F, \mathrm{R}^!f(K)) = 0$ for $i > N$, hence $\underline{\mathrm{Ext}}^i(F, \mathrm{R}^!f(K)) = 0$ for $i > N$ (SGAA VIII 5.5), as required.

Let now X be a prescheme, and let $K \in \mathrm{ob}\,\mathrm{D}^+(X)$. Denote by $\underline{D}_K$ the functor $\mathrm{R}\,\underline{\mathrm{Hom}}(\ \ , K)$. We shall define, for $F \in \mathrm{ob}\,\mathrm{D}(X)$, a functorial homomorphism $F \to \underline{D}_K\underline{D}_K F$ (cf. [H] V.1.2). The construction which follows would be valid more generally on a ringed topos. We may first suppose that K is formed of injectives, which allows us to "remove the R". The identity morphism $\underline{\mathrm{Hom}}(F, K) \to \underline{\mathrm{Hom}}(F, K)$ defines, by the formula dear to Cartan, a homomorphism $F \otimes \underline{\mathrm{Hom}}(F, K) \to K$, whence, by a second application of the said formula, a homomorphism $F \to \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}(F, K), K)$, which is the announced homomorphism, and which we baptize the canonical homomorphism. We say that the pair (F, K) is *bidualizing*, or *satisfies biduality*, or again that F is *reflexive* relative to K, if the canonical homomorphism $F \to \underline{D}_K\underline{D}_K F$ is an isomorphism.

From now on, unless expressly mentioned otherwise, all preschemes considered will be assumed locally noetherian.

Definition 1.7

Let X be a prescheme. A complex $K \in \mathrm{ob}\,\mathrm{D}^+(X)$ is said to be *dualizing* if the following conditions are satisfied:

(i) K is of finite quasi-injective dimension;

(ii) for every $F \in \mathrm{ob}\,\mathrm{D}_c^-(X)$, one has $\underline{D}_K(F) \in \mathrm{ob}\,\mathrm{D}_c^+(X)$;

(iii) every $F \in \mathrm{ob}\,\mathrm{D}_c^b(X)$ is reflexive relative to K.

Remarks 1.8

a) By a standard dévissage (truncating F and using the fact that $\mathrm{D}_c(X)$ is triangulated), one sees that condition (ii) is equivalent to

(ii bis): for every constructible A_X-module F and every $i \in \mathbb{Z}$, $\underline{\mathrm{Ext}}^i(F, K)$ is constructible.

b) Condition (ii bis) implies $K \in \mathrm{ob}\,\mathrm{D}_c^+(X)$. Contrary to the example of coherent sheaves (EGA 0_{III} 12.3.3), the converse is perhaps not automatically verified. It is however verified (see no. 3) in the "good" cases, for example when one has on X resolution of singularities and purity (in particular if X is an excellent prescheme of characteristic zero).

c) By dévissage on F, one sees that condition (iii) is equivalent to (iii bis): every constructible A_X-module F is reflexive relative to K.

d) If $K \in \mathrm{ob}\,\mathrm{D}^+(X)$ satisfies (i) and (ii) (resp. is dualizing), the functor $\underline{D}_K$ induces a functor (resp. an equivalence of categories) $(\mathrm{D}_c^b(X))^\circ \to \mathrm{D}_c^b(X)$.

e) If $K \in \mathrm{ob}\,\mathrm{D}^+(X)$ is of finite *injective* dimension (cf. 1.5.2) and satisfies (ii), then, for every $F \in \mathrm{ob}\,\mathrm{D}_c(X)$, one has $\underline{D}_K(F) \in \mathrm{D}_c(X)$, and $\underline{D}_K$ induces a functor $(\mathrm{D}_c^+(X))^\circ \to \mathrm{D}_c^-(X)$.

If moreover K is dualizing, every $F \in \mathrm{ob}\,\mathrm{D}_c(X)$ is reflexive relative to K, and $\underline{D}_K$ induces equivalences of categories

$$(\mathrm{D}_c(X))^\circ \xrightarrow{\approx} \mathrm{D}_c(X), \quad (\mathrm{D}_c^-(X))^\circ \xrightarrow{\approx} \mathrm{D}_c^+(X), \quad (\mathrm{D}_c^+(X))^\circ \xrightarrow{\approx} \mathrm{D}_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that K be of finite injective dimension.

We now turn to the "exchange formulas." We shall not return to the notion of the tor-dimension of a complex, see SGA A, Exposé XVII, 4.1.9.[2] Retain only the following point: if X is a prescheme and $G \in \mathrm{ob}\,\mathrm{D}^-(X)$, the condition $\underline{\mathrm{Tor}}_i(F, G) = 0$ for $i \geqslant N$ and all A_X-modules F is equivalent to the same condition with F constructible, by SGA A, Exposé IX, 2.9 and the commutation of $\otimes^{\mathrm{L}}$ with filtered inductive limits. Thus there is no need to introduce a notion of "quasi-tor-dimension."

Lemma 1.9, cf. [H] II.4.3.

a) Let X be a prescheme, let $F \in \mathrm{ob}\,\mathrm{D}_c(X)$, and let $G \in \mathrm{ob}\,\mathrm{D}_c^-(X)$. One has $F \otimes^{\mathrm{L}} G \in \mathrm{ob}\,\mathrm{D}_c(X)$ if $F \in \mathrm{ob}\,\mathrm{D}_c^-(X)$, or if G has finite tor-dimension.

[2] See also SGA 6, I, 5.

b) Let $f : X \to Y$ be a morphism of preschemes and let $G \in \operatorname{ob} \mathrm{D}_c(Y)$. Then $f^*(G) \in \operatorname{ob} \mathrm{D}_c(X)$.

Proof. Part (b) follows trivially from SGA A, Exposé IX, 2.4. We prove (a). Either hypothesis on (F, G) implies that the spectral sequence

$$E_2^{pq} = \sum_{q_1+q_2=q} \underline{\mathrm{Tor}}_{-p}(\mathrm{H}^{q_1}(F), \mathrm{H}^{q_2}(G)) \Rightarrow \mathrm{H}^*(F \otimes^{\mathrm{L}} G)$$

is biregular. Hence one is reduced to the case where F and G are concentrated in degree 0, that is, are constructible A_X-modules. The question is local, so one may suppose X noetherian. Then F admits, by SGA A, Exposé IX, 2.7, a left resolution F' in which the F'^i are of the form $A_{U,X}$ with U étale and of finite type over X. We have

$$F \otimes^{\mathrm{L}} G \xleftarrow{\sim} F' \otimes G,$$

and the result follows since the $A_{U,X} \otimes G$ are constructible. □

Notation 1.10. If X is a prescheme, we denote by $\mathrm{D}^b(X)_{\mathrm{torf}}$, respectively $\mathrm{D}^b(X)_{\mathrm{qinjf}}$, the full subcategory of $\mathrm{D}(X)$ consisting of objects of finite tor-dimension, respectively of finite quasi-injective dimension.

Proposition 1.11, cf. [H] V.2.6. Let X be a prescheme and $K \in \operatorname{ob} \mathrm{D}^+(X)$.

a) For $F, G \in \operatorname{ob} \mathrm{D}^-(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}^b(X)_{\mathrm{torf}}$, there is a functorial isomorphism
$$\underline{D}_K(F \otimes^{\mathrm{L}} G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, \underline{D}_K(G)).$$

b) If K satisfies conditions (i) and (ii) of Definition 1.7, then
$$G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \Longrightarrow \underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}}$$
and $\underline{D}_K(G)$ satisfies condition 1.7(ii).

c) If K is dualizing, then for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^b(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^b(X)$ with $\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$, there is a functorial isomorphism
$$\underline{D}_K(F \otimes^{\mathrm{L}} \underline{D}_K(G)) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G).$$
Moreover,
$$\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \Longrightarrow G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}},$$
and G satisfies condition 1.7(ii).

d) If K is dualizing and has finite injective dimension, then the implications in (b) and (c) are equivalences.

Proof. Part (a) is precisely Cartan's adjunction formula, SGA 6, I, 7.4.

For (b), let $G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$. If F is a variable constructible A_X-module, then the cohomology objects of $F \otimes^{\mathrm{L}} G$ are constructible, by Lemma 1.9, and vanish outside two fixed bounds independent

of F. The same therefore holds for the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$, and (a) gives the assertion.

For (c), the first assertion follows from (a) applied to $\underline{D}_K(G)$ in place of G, together with the reflexivity of G. The second follows by a similar argument to (b).

For (d), prove for instance the converse implication of (b). Suppose that $\underline{D}_K(G) \in \mathrm{ob}\,\mathrm{D}^b_c(X)_{\mathrm{qinjf}}$ and satisfies condition 1.7(ii). Then, as F varies through constructible modules, the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$ are constructible and vanish outside two fixed bounds. The same is consequently true for the cohomology objects of $\underline{D}_K\underline{D}_K(F \otimes^{\mathrm{L}} G)$; but

$$F \otimes^{\mathrm{L}} G \xrightarrow{\sim} \underline{D}_K\underline{D}_K(F \otimes^{\mathrm{L}} G)$$

because K is dualizing and of finite injective dimension, cf. 1.8(d). Hence G has finite tor-dimension. The converse implication in (c) is proved analogously. □

Remarks 1.11.1.

a) The editor does not know whether assertion (d) of Proposition 1.11 is valid under the sole hypothesis that K be dualizing.

b) One expresses the preceding proposition pictorially by saying that the dualizing functor $\underline{D}_K$ exchanges the functors $\otimes^{\mathrm{L}}$ and $\mathrm{R}\,\underline{\mathrm{Hom}}$, as well as the notions of finite tor-dimension and finite quasi-injective dimension. The exchange is, however, fully satisfactory only when K has finite injective dimension.

Proposition 1.12. Let Y be a noetherian prescheme, let $f : X \to Y$ be a morphism of finite type, and let $K \in \mathrm{ob}\,\mathrm{D}^+(Y)$. Suppose that n is prime to the residual characteristics of Y, and put

$$K_X = \mathrm{R}^! f(K_Y), \qquad \underline{D}_X = \underline{D}_{K_X}, \qquad \underline{D}_Y = \underline{D}_{K_Y},$$

in the notation of SGA A, Exposé XVIII, 3.1.

a) For $F \in \mathrm{ob}\,\mathrm{D}^-(X)$ there is a functorial isomorphism

$$\mathrm{R}f_*\underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y\mathrm{R}_!f(F). \tag{i}$$

If, moreover, K_X and K_Y are dualizing,[3] then for $F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$ there is a functorial isomorphism

$$\mathrm{R}_!f\,\underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y\mathrm{R}f_*(F). \tag{ii}$$

b) For $F \in \mathrm{D}^-_c(Y)$ there is a functorial isomorphism[4]

$$\mathrm{R}^!f\,\underline{D}_Y(F) \xrightarrow{\sim} \underline{D}_Xf^*(F). \tag{i}$$

If, moreover, K_X and K_Y are dualizing, then for $F \in \mathrm{D}^b_c(Y)$ there is a functorial isomorphism

$$f^*\underline{D}_YF \xrightarrow{\sim} \underline{D}_X\mathrm{R}^!f\,F. \tag{ii}$$

[3] We shall see below that under fairly general conditions K_Y dualizing implies K_X dualizing.

[4] If the functor $\mathrm{R}^!f$ has finite cohomological dimension, as for example when Y is of finite type over a field (SGA A, Exposé X, 4.3 and Exposé XVIII, 3.1.7), the isomorphism (b)(i) is valid for $F \in \mathrm{ob}\,\mathrm{D}_c(Y)$.

Proof. The isomorphism (a)(i) is just the duality isomorphism of SGA A, Exposé XVIII, 3.1.9.6.

The isomorphism (a)(ii) is obtained by applying $\underline{D}_Y$ to both sides of (a)(i), written with argument $\underline{D}_X F$, and using the finiteness theorem SGA A, Exposé XVII, 5.3.6, which asserts that $\mathrm{R}_! f(\underline{D}_X(F)) \in \mathrm{ob}\,\mathrm{D}_c(Y)$.

The isomorphism (b)(i) is the induction formula of SGA A, Exposé XVIII, 3.1.12.2.

The isomorphism (b)(ii) is obtained by applying $\underline{D}_X$ to both sides of (b)(i), written with argument $\underline{D}_Y F$, and using 1.9(b). □

Scholium 1.12.1. In figurative language, the dualizing functors $\underline{D}_X$ and $\underline{D}_Y$ exchange the usual and unusual direct images, $\mathrm{R}f_*$ and $\mathrm{R}_! f$, and also the usual and unusual inverse images, f^* and $\mathrm{R}^! f$.

Corollary 1.13. Under the hypotheses of Proposition 1.12, suppose in addition that f is proper, and let $F \in \mathrm{ob}\,\mathrm{D}^-(X)$. If (F, K_X) is bidualizing, then $(\mathrm{R}f_* F, K_Y)$ is bidualizing. Conversely, if f is finite and $(\mathrm{R}f_* F, K_Y)$ is bidualizing, then (F, K_X) is bidualizing.

Proof. Suppose (F, K_X) is bidualizing. Applying $\mathrm{R}f_*$ to the canonical isomorphism $F \xrightarrow{\sim} \underline{D}_X^2 F$, one obtains[5]

$$\begin{aligned} \mathrm{R}f_* F &\xrightarrow{\sim} \mathrm{R}f_* \underline{D}_X \underline{D}_X F \\ &\xrightarrow{\sim} \underline{D}_Y \mathrm{R}f_* \underline{D}_X F \qquad \text{by (a)(i)} \\ &\xrightarrow{\sim} \underline{D}_Y \underline{D}_Y \mathrm{R}f_* F \qquad \text{by (a)(i).} \end{aligned}$$

The courageous reader will convince himself that the isomorphism obtained is the canonical one; this will prove the first assertion.

Now suppose f finite and $(f_* F, K_Y)$ bidualizing, since $\mathrm{R}f_* = f_*$ for f finite. We prove that the canonical morphism $F \to \underline{D}_X^2 F$ is an isomorphism. By Lemma 1.14(a) below, it suffices to show that the induced morphism

$$f_* F \longrightarrow f_* \underline{D}_X^2 F$$

is an isomorphism. But by applying (a)(i) twice, one convinces oneself that this morphism is precisely the canonical isomorphism

$$f_* F \longrightarrow \underline{D}_Y^2 f_* F,$$

which completes the proof. □

Lemma 1.14. Let $f : X \to Y$ be a finite morphism of preschemes.

a) For a morphism $u : F \to G$ in $\mathrm{D}(X)$, u is an isomorphism if and only if $f_* u$ is an isomorphism.

b) For $F \in \mathrm{ob}\,\mathrm{D}(X)$, one has $F \in \mathrm{ob}\,\mathrm{D}_c(X)$ if and only if $f_* F \in \mathrm{ob}\,\mathrm{D}_c(Y)$.

Proof. For (a), since f_* is exact, one may restrict to the case where F and G are concentrated in degree zero. The assertion then follows immediately from the computation of the fibres of $f_* F$ and $f_* G$, SGA A, Exposé VIII, 5.5.

[5] *Editorial note.* The printed source gives the intermediate term $\underline{D}_Y \mathrm{R}f_* \underline{D}_X F$ and cites Proposition 1.12(a)(i) for both successive arrows. This edition reproduces that chain. The mathematically coherent emendation is $\underline{D}_Y \mathrm{R}_! f\, \underline{D}_X F$ in the intermediate line, with Proposition 1.12(a)(ii) cited for the final arrow.

For (b), again one may restrict to the case where F is concentrated in degree zero. Necessity is known, SGA A, Exposé IX, 2.14. We prove sufficiency: f_*F constructible implies F constructible. The question is local upstairs, hence a fortiori downstairs, so we may suppose Y and hence X affine. By EGA IV, 17.16.6, there exists a finite family (Y_i) of affine subpreschemes of Y, pairwise disjoint and with union Y, such that, if $X_i = X \times_Y Y_i$, the induced morphism $f_i : X_i \to Y_i$ factors as

$$X_i \xrightarrow{u_i} X_i' \xrightarrow{v_i} Y_i,$$

where u_i is finite radicial and surjective and v_i is finite étale. Put $F|_{X_i} = F_i$. By the base-change theorem for finite morphisms, SGA A, Exposé VIII, 5.5, one has

$$f_{i*}(F_i) \xrightarrow{\sim} (f_*F)|_{Y_i},$$

and by SGA A, Exposé IX, 2.8, one is reduced to proving the assertion for (f_i, F_i). Put $F_i' = u_{i*}(F_i)$, so that $f_{i*}(F_i) = v_{i*}(F_i')$. It is clear from the definition of constructibility, respectively follows from SGA A, Exposé VIII, 1.1, that F_i, respectively F_i', is constructible if and only if F_i', respectively $v_{i*}(F_i')$, is constructible. This proves the claim. □

Corollary 1.15. Let $f : X \to Y$ be a separated quasi-finite morphism of noetherian preschemes. Let $K_Y \in \operatorname{ob} \mathrm{D}^+(Y)$, and put, as in Proposition 1.12,

$$K_X = \mathrm{R}^! f(K_Y), \qquad \underline{D}_X = \underline{D}_{K_X}, \qquad \underline{D}_Y = \underline{D}_{K_Y}.$$

If K_Y is dualizing, then K_X is dualizing.

Proof. We already know, by Proposition 1.6, that if K_Y has finite quasi-injective dimension, then the same is true of K_X. We show that if K_Y satisfies condition (ii), respectively condition (iii), of Definition 1.7, then K_X satisfies the same condition.

By the Main Theorem, EGA IV, 8.12.6, f factors as $f = f'i$, where i is an open immersion and f' is finite. Since

$$\mathrm{R}^! f \simeq \mathrm{R}^! i \, \mathrm{R}^! f',$$

one must treat separately the case of an open immersion and the case of a finite morphism. If f is an open immersion, the assertion is trivial, since a constructible sheaf on X is induced by a constructible sheaf on Y, and formation of $\mathrm{R}\,\underline{\mathrm{Hom}}$ commutes with restriction to an open.

Suppose then that f is finite. It follows from Corollary 1.13 that if K_Y satisfies (iii), then K_X satisfies (iii), taking account of the finiteness theorem SGA A, Exposé IX, 2.14. It remains to prove that if K_Y satisfies (ii), then K_X satisfies (ii). Let $F \in \operatorname{ob} \mathrm{D}_c^-(X)$. We must show that $\underline{D}_X(F) \in \operatorname{ob} \mathrm{D}_c^+(X)$. By the finiteness theorem, $f_*F \in \operatorname{ob} \mathrm{D}_c^-(Y)$. Hence, by the hypothesis on K_Y,

$$\underline{D}_Y f_*F \in \operatorname{ob} \mathrm{D}_c^+(Y).$$

But

$$\underline{D}_Y f_*F \xrightarrow{\sim} f_* \underline{D}_X(F)$$

by Proposition 1.12(a)(i), and Lemma 1.14(b) completes the proof. □

2. Uniqueness of the dualizing complex

In this section we suppose that A is local, that is,

$$A = \mathbb{Z}/\ell^{\nu}\mathbb{Z},$$

where ℓ is prime.

Theorem 2.1, cf. [H] V.3.1. Let X be a connected locally noetherian prescheme, and let K be a dualizing complex on X in the sense of Definition 1.7. Let also $K' \in \operatorname{ob} \mathrm{D}_c^+(X)$. Then K' is dualizing if and only if there exist an invertible A_X-module L and an integer r, necessarily determined up to unique isomorphism, such that

$$K' \xrightarrow{\sim} K \otimes^{\mathrm{L}} L[r].$$

Proof. Let L be an invertible A_X-module and r an integer, and put $L^{\vee} = \underline{\mathrm{Hom}}(L, A_X)$. For $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}^+(X)$ there are canonical isomorphisms

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F \otimes^{\mathrm{L}} L^{\vee}[-r], G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G \otimes^{\mathrm{L}} L[r]) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \otimes^{\mathrm{L}} L[r]. \tag{$*$}$$

It follows immediately that, if $K' \simeq K \otimes^{\mathrm{L}} L[r]$, then K' is dualizing.

Moreover, if we write $\underline{D}_X = \underline{D}$ and $\underline{D}_{K'} = \underline{D}'$, then for $F \in \operatorname{ob} \mathrm{D}(X)$ the isomorphism $(*)$ gives

$$\underline{D}'F \xrightarrow{\sim} \underline{D}F \otimes^{\mathrm{L}} L[r].$$

Since $\underline{D}K \simeq A_X$, one obtains

$$\underline{D}'K \xrightarrow{\sim} L[r],$$

which shows that $L[r]$, hence L and r, are determined essentially uniquely by K'.

Conversely, suppose that K' is dualizing, and put

$$P = \underline{D}'K = \underline{D}'\underline{D}A_X = \mathrm{R}\,\underline{\mathrm{Hom}}(K, K').$$

We shall prove:

(a) $K' \xrightarrow{\sim} K \otimes^{\mathrm{L}} P$,

(b) $P \xrightarrow{\sim} L[r]$ for an invertible A_X-module L.

This will complete the proof of the theorem.

First observe that, by Proposition 1.9(b)(ii), for $F \in \operatorname{ob} \mathrm{D}_c(X)$ there is an isomorphism

$$F \otimes^{\mathrm{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}F. \tag{$**$}$$

Applying $(**)$ to $F = K$ gives

$$K \otimes^{\mathrm{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}K \xrightarrow{\sim} \underline{D}'A_X = K',$$

which proves (a).

Applying $(**)$ to

$$F = P' = \underline{D}\underline{D}'A_X$$

gives

$$P' \otimes^{\mathrm{L}} P = \underline{D'}\underline{DD}D'A_X \xrightarrow{\sim} \underline{D'}\underline{D'}A_X \xrightarrow{\sim} A_X.$$

Thus (b) follows from the following lemma. □

Lemma 2.2, cf. [H] V.3.3. Let X be a connected locally noetherian prescheme, and let $P, P' \in \operatorname{ob} \mathrm{D}_c^-(X)$ be such that

$$P \otimes^{\mathrm{L}} P' \xrightarrow{\sim} A_X.$$

Then there exist an invertible A_X-module L and an integer r such that

$$P \xrightarrow{\sim} L[r], \qquad P' \xrightarrow{\sim} L^\vee[-r].$$

Proof. The proof has two steps.

First suppose that $X = \operatorname{Spec}(k)$, with k separably closed. One merely repeats the easy proof given in [H]. The essential point is this: by EGA 0_{I}, 5.4.3, if M and M' are finite-type A-modules such that

$$M \otimes M' \xrightarrow{\sim} A,$$

then M and M' are invertible and $M' \simeq M^\vee$. This is where the hypothesis that A is local is used.

We now pass to the general case. By the special case just treated, for every $x \in X$ there exists an integer $r(x)$ such that

$$P_{\overline{x}} \simeq A_{\overline{x}}[r(x)], \qquad P'_{\overline{x}} \simeq A_{\overline{x}}[-r(x)].$$

Equivalently, $\underline{\mathrm{H}}^i(P_{\overline{x}}) = 0$, respectively $\underline{\mathrm{H}}^i(P'_{\overline{x}}) = 0$, except for $i = -r(x)$, respectively $i = r(x)$, and the corresponding nonzero cohomology modules are isomorphic to $A_{\overline{x}}$. It remains to show that the function $x \mapsto r(x)$ is locally constant.

Assume this for the moment. Since X is connected, r is constant, say equal to r_0. Then

$$\underline{\mathrm{H}}^i(P) = 0 \ (i \neq -r_0), \qquad \underline{\mathrm{H}}^i(P') = 0 \ (i \neq r_0),$$

and

$$\underline{\mathrm{H}}^{-r_0}(P) \otimes \underline{\mathrm{H}}^{r_0}(P') \xrightarrow{\sim} A.$$

Put $L = \underline{\mathrm{H}}^{-r_0}(P)$ and $L' = \underline{\mathrm{H}}^{r_0}(P')$. Let $u : \overline{x} \to \overline{y}$ be a specialization morphism, and let

$$u^* : L_{\overline{y}} \to L_{\overline{x}}, \qquad u'^* : L'_{\overline{y}} \to L'_{\overline{x}}$$

be the corresponding specialization morphisms. There is a commutative square

$$\begin{array}{ccc} L_{\overline{y}} \otimes L'_{\overline{y}} & \xrightarrow{\sim} & A_{\overline{y}} \\ {\scriptstyle u^* \otimes u'^*}\downarrow & & \downarrow \\ L_{\overline{x}} \otimes L'_{\overline{x}} & \xrightarrow{\sim} & A_{\overline{x}} \end{array} \qquad (*)$$

Choose a basis e, respectively f, of $L_{\overline{x}}$, respectively $L_{\overline{y}}$, and let e', respectively f', be the dual basis of $L'_{\overline{x}}$, respectively $L'_{\overline{y}}$, meaning that $e \otimes e'$ and $f \otimes f'$ map to 1. If $u^*(f) = \lambda e$ and $u'^*(f') = \lambda' e'$, the commutativity of $(*)$ gives $\lambda\lambda' = 1$. Hence u^* is an isomorphism, and u'^* is the contragredient

isomorphism. By SGA A, Exposé IX, 2.13, it follows that L and L' are locally constant, hence of finite presentation as A_X-modules. Consequently, for every $x \in X$ the canonical map

$$\underline{\mathrm{Hom}}(L, F)_{\overline{x}} \longrightarrow \mathrm{Hom}(L_{\overline{x}}, F_{\overline{x}})$$

is an isomorphism for every A_X-module F. Since the geometric fibres of L are invertible, L itself is invertible. Similarly L' is invertible, and $L' \simeq L^\vee$. Hence

$$P \simeq L[-r_0], \qquad P' \simeq L^\vee[r_0].$$

It remains only to prove that r is locally constant. For $m \in \mathbb{Z}$, the condition $r(x) = m$ is equivalent to

$$\underline{\mathrm{H}}^m(P')_{\overline{x}} \neq 0.$$

Thus, by SGA A, Exposé IX, 2.4, the function r is locally constructible. Therefore, to verify that it is locally constant, EGA IV, 1.10.1, allows one to suppose that X is local. Let x be its closed point and U the open complement.

We prove that r is constant. By induction on the dimension of X, we may already suppose that $r(y)$ is constant and equal to r_0 for $y \in U$; the point is to prove $r(x) = r_0$.

Assume the contrary, and derive a contradiction. By hypothesis

$$P|_U \xrightarrow{\sim} L[r_0]$$

with L an invertible A_U-module. After translating degrees we may suppose $r_0 = 0$. On the other hand $r(x) \neq 0$; write $r(x) = a > 0$. Let

$$i : U \to X, \qquad j : x \to X$$

be the canonical inclusions. Consider the exact sequence

$$0 \longrightarrow i_!(P|_U) \longrightarrow P \longrightarrow j_*j^*P \longrightarrow 0. \tag{$**$}$$

By hypothesis the only nonzero cohomology objects of P are $\underline{\mathrm{H}}^0(P)$ and $\underline{\mathrm{H}}^{-a}(P)$, and

$$\underline{\mathrm{H}}^0(P) \xrightarrow{\sim} i_!L.$$

It follows that $(**)$ splits naturally, via the map $P \to \underline{\mathrm{H}}^0(P)$, hence

$$P \xrightarrow{\sim} i_!(P|_U) \oplus j_*j^*P.$$

Therefore

$$\begin{aligned} P \otimes^{\mathrm{L}} P' &\xrightarrow{\sim} i_!(P|_U) \otimes^{\mathrm{L}} P' \oplus j_*j^*P \otimes^{\mathrm{L}} P' \\ &\xrightarrow{\sim} i_!(P|_U \otimes^{\mathrm{L}} P'|_U) \oplus j_*(j^*P \otimes^{\mathrm{L}} j^*P') \\ &\xrightarrow{\sim} i_!A_U \oplus j_*A_x, \end{aligned}$$

which is impossible, since $i_!A_U \oplus j_*A_x$ is plainly not isomorphic to A_X; already the sheaves $\underline{\mathrm{H}}^0$ are not isomorphic. This proves Lemma 2.2, and hence Theorem 2.1. □

3. Existence of dualizing complexes

3.1. Preliminaries

As was said in the introduction, the object of this section is to prove that, on a "good" regular prescheme X, the sheaf A_X is dualizing, where n is assumed prime to the residual characteristics.

In fact, one would like this to hold on excellent regular preschemes of equal characteristic; but there is no doubt that the proof given below will extend to them once resolution of singularities and the purity theorem are available in that case.

We first present the various ingredients which will be used.

3.1.1. Singular locus. Let X be a prescheme, locally noetherian. We shall say that X satisfies condition *(reg)* if the following equivalent properties hold (EGA IV, 6.12.3):

(i) for every sub-prescheme Y of X, the set $\mathrm{Reg}(Y)$ of regular points of Y is open in Y;

(ii) for every closed integral sub-prescheme Y of X, the set $\mathrm{Reg}(Y)$ contains a non-empty open subset of Y.

The condition (reg) is satisfied if X is locally of finite type over a complete noetherian local ring, for example over a field (EGA IV, 6.12.8), or over a Dedekind ring whose field of fractions has characteristic zero, for example over $\mathbb{Z}$ (EGA IV, 6.12.6); more generally, it is satisfied if X is excellent (EGA IV, 7.8.6).

3.1.2. Dévissage of constructible sheaves. Let X be a noetherian prescheme, and let C be a strictly full subcategory of the category $\mathrm{Cons}(X)$ of constructible A_X-modules. Suppose that C is stable under direct factors and extensions. The following conditions are equivalent:

(i) $C = \mathrm{Cons}(X)$;

(ii) C contains sheaves of the form $i_!(F)$, where $i : Y \to X$ is the immersion of an integral sub-prescheme and F is locally constant on Y;

(iii) C contains sheaves of the form $f_* i_!(A'_U)$, where $i : U \to Y$ is an open subset of an integral prescheme, $f : Y \to X$ is a finite morphism such that the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$, for ℓ a prime divisor of n.

If X satisfies (reg) (3.1.1), these conditions are again equivalent to the following:

(i bis) the analogue of (ii), with Y moreover regular;

(ii bis) the analogue of (iii), with U regular.

If, in addition, X is affine, then in (ii), respectively in (ii bis), one may suppose Y closed in an open subset of the form $D(f)$, with $f \in \Gamma(X, \mathcal{O}_X)$.

Proof. The equivalence of (i)–(iii) was proved in SGA A, Exposé IX, 2.5 and 5.8. The remaining assertions follow by noetherian induction. □

3.1.3. Cohomological dimension. Let X be a locally noetherian prescheme. We denote by $(\mathrm{cdloc})_n$ the following condition. For every sub-prescheme Y of X, every geometric point $\overline{y}$ of Y, and every

$$f \in \Gamma(\widetilde{Y}(\overline{y}), \mathcal{O}_{Y,\overline{y}}),$$

where $\widetilde{Y}(\overline{y})$ denotes the strict localization of Y at $\overline{y}$ (SGA A, Exposé VIII, 7.1), one has

$$\mathrm{cd}_{\mathbf{n}}(\widetilde{Y}(\overline{y}) - V(f)) \leq \dim \mathcal{O}_{Y,\overline{y}} \quad (= \dim \mathcal{O}_{Y,y} \text{ by EGA IV, 6.1.3}),$$

where $\mathbf{n}$ is the set of prime divisors of n.

The condition $(\mathrm{cdloc})_n$ is satisfied if X is locally of finite type over a field (SGA A, Exposé XIV, 3.5), or is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 6.2).[6]

3.1.4. Absolute cohomological purity. Let X be a regular prescheme, and let Y be a regular closed sub-prescheme of X, of codimension d at every point. One says that the pair (X, Y) satisfies the absolute cohomological purity theorem, relative to n, if

$$\underline{\mathrm{H}}^i_Y(A_X) = 0 \qquad (i \neq 2d),$$

and if the "local fundamental class" homomorphism (SGA $4^{1/2}$, Cycle 2.2)

$$A_Y \longrightarrow \underline{\mathrm{H}}^{2d}_Y((\mu_n)^{\otimes d}_X)$$

is an isomorphism.

One says that the absolute cohomological purity theorem is true on X if every pair (U, Y) as above satisfies it, where U is an open subset of X.

Absolute purity is true on X if X is smooth over a perfect field of characteristic prime to n (SGA A, Exposé XVI, 3.9), or if X is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 3.2), or if X is regular of dimension 1 (cf. 5.1).[7]

3.1.5. Hironaka-style resolution of singularities.

a) Let X be a regular prescheme, and let $D \geq 0$ be a divisor on X. One says that D has *strict normal crossings* if there exists a finite family $(f_i)_{i \in I}$ of elements of $\Gamma(X, \mathcal{O}_X)$ such that

$$D = \sum_{i \in I} \mathrm{Div}(f_i)$$

(EGA IV, 21), and such that, for every $x \in \mathrm{Supp}(D)$, the germs $(f_i)_x$ which lie in $\mathfrak{m}_x$ form part of a regular system of parameters. One says that D has *normal crossings* if, locally for the étale topology, D has strict normal crossings.

The standard example of a normal crossings divisor is furnished by a locally finite union of coordinate hyperplanes in affine space.

b) Let X be a locally noetherian prescheme. We shall say that X is *strongly desingularizable* if the following condition is satisfied: for every finite morphism $f : Y \to X$, where Y is integral and its generic point is separable over its image, and every regular open subset U of Y, there exists a proper birational morphism $h : Y' \to Y$, with Y' regular, such that h induces an isomorphism from $U' = h^{-1}(U)$ onto U, and $Y' - U'$ is a normal crossings divisor.

[6] Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.
[7] Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

Here are cases where X is strongly desingularizable:

a) X is excellent of characteristic zero;

b) X is locally noetherian, regular, and of dimension ≤ 1;

c) X is of finite type over $\mathbb{Z}$ or over a perfect field, and of dimension ≤ 2.[8]

3.1.6. Cohomological finiteness. Let $f : X \to Y$ be a morphism of preschemes, and let $F \in \operatorname{ob} \mathrm{D}_c^+(X)$. One has $\mathrm{R}f_*(F) \in \operatorname{ob} \mathrm{D}_c^+(Y)$ in the following cases:

a) f is proper and of finite presentation (SGA A, Exposé XIV, 1.1);

b) "when resolution of singularities and the purity theorem are available", in particular:

(i) if X and Y are locally of finite type over a field k of characteristic zero and f is a k-morphism of finite type (SGA A, Exposé XVI, 5.1);

(ii) if Y is excellent of characteristic zero and f is of finite type (SGA A, Exposé XIX, 5.1);

(iii) if X is locally of finite type over a field and $\dim(X) \leq 2$ (SGA A, Exposé XIX, 5.1).

3.2. The quasi-injective dimension of A_X

Proposition 3.2.1. Let X be a regular prescheme whose residual characteristics are prime to n. Suppose that X satisfies (reg) (3.1.1) and $(\mathrm{cdloc})_n$ (3.1.3), and that absolute purity is true on X (3.1.4). Then, for every constructible A_X-module F and every $x \in X$, one has

$$\underline{\mathrm{Ext}}^i(F, A_X)_x = 0 \qquad \text{for } i > 2 \dim \mathcal{O}_{X,x}. \tag{*}$$

Proof. The question being local, we may suppose X affine. By dévissage (3.1.2), it is enough to verify $(*)$ for $F = i_!(G)$, where $i : Y \to X$ is the immersion of a regular integral sub-prescheme, closed in an open subset $D(f)$, $f \in A(X)$, and G is a locally constant constructible sheaf on Y. We must compute

$$\underline{\mathrm{Ext}}^q(F, A_X) = \mathrm{H}^q(\mathrm{R}\,\underline{\mathrm{Hom}}(F, A_X)).$$

The projection formula for i (SGA A, Exposé XVIII) gives

$$\mathrm{R}\,\underline{\mathrm{Hom}}(i_!G, A_X) \xrightarrow{\sim} \mathrm{R}i_*\mathrm{R}\,\underline{\mathrm{Hom}}(G, \mathrm{R}^!i(A_X)).$$

By purity,

$$\mathrm{R}^!i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes -d}[-2d],$$

where $d = \operatorname{codim}(Y, X)$. Put $T = (\mu_n)_Y^{\otimes -d}$. Since G is locally constant constructible, for every geometric point $\overline{y}$ of Y and every p one has

$$\underline{\mathrm{Ext}}^p(G, T)_{\overline{y}} \xrightarrow{\sim} \mathrm{Ext}^p(G_{\overline{y}}, T_{\overline{y}}).$$

[8] Added in 1977: more generally, if X is excellent noetherian of dimension ≤ 2, by H. Hironaka, *Desingularization of excellent surfaces*, notes by B. Bennett, Bowdoin College, 1967, to be supplemented if necessary by H. Hironaka, *Certain numerical characters of singularities*, J. Math. Kyoto Univ. 10 (1970), 151–187.

But $T_{\overline{y}}$ is injective, being isomorphic to $A = \mathbb{Z}/n\mathbb{Z}$. Hence $\underline{\mathrm{Ext}}^p(G,T) = 0$ for $p \neq 0$, and

$$\mathrm{R}\,\underline{\mathrm{Hom}}(G,T) \xrightarrow{\sim} \underline{\mathrm{Hom}}(G,T).$$

Thus

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F,A_X) \xrightarrow{\sim} \mathrm{R}i_*(\mathcal{H})[-2d], \qquad \mathcal{H} = \underline{\mathrm{Hom}}(G,T).$$

Let Z be the schematic closure of Y in X (EGA I, 9.5.10). Then $Y = Z \cap D(f)$ (EGA 0_{I}, 2.1.6). Denote by $j : Y \to Z$ and $k : Z \to X$ the canonical immersions. We have $\mathrm{R}i_* = k_*\mathrm{R}j_*$, so we are reduced to computing $\mathrm{R}j_*(\mathcal{H})$. If $\overline{x}$ is a geometric point of Z, let $\widetilde{Z} = \mathrm{Spec}(\mathcal{O}_{Z,\overline{x}})$ be the strict localization, let $\widetilde{f}$ be the image of f in $\mathcal{O}_{Z,\overline{x}}$, let $\widetilde{Y} = \widetilde{Z} \times_X Y = \widetilde{Z} - V(\widetilde{f})$, and let $\widetilde{\mathcal{H}}$ be the inverse image of $\mathcal{H}$ on $\widetilde{Y}$. By SGA A, Exposé VIII, 5.2,

$$\mathrm{R}^q j_*(\mathcal{H})_{\overline{x}} \xrightarrow{\sim} \mathrm{H}^q(\widetilde{Y},\widetilde{\mathcal{H}}),$$

hence

$$\underline{\mathrm{Ext}}^q(F,A_X)_{\overline{x}} \xrightarrow{\sim} \mathrm{H}^{q-2d}(\widetilde{Y},\widetilde{\mathcal{H}}).$$

The condition $(\mathrm{cdloc})_n$ implies that this group is zero for

$$q - 2d > \dim \mathcal{O}_{Z,x}.$$

Using

$$d = \mathrm{codim}_x(Z,X) \leq \dim \mathcal{O}_{X,x}$$

(EGA IV, 5.1.3) and

$$\dim \mathcal{O}_{Z,x} = \dim \mathcal{O}_{X,x} - \mathrm{codim}_x(Z,X)$$

(EGA 0_{IV}, 16.5.12 and EGA IV, 5.1.9), one obtains

$$\underline{\mathrm{Ext}}^q(F,A_X)_x = 0 \qquad (q > 2\dim \mathcal{O}_{X,x}),$$

as required. □

Remark 3.2.2. The bound obtained in 3.2.1 is the best possible. Indeed, let x be a point of X, and let Y be the closed integral sub-prescheme of X whose generic point is x. Since X satisfies (reg), there exists a regular open subset U of Y containing x. By purity,

$$\underline{\mathrm{Ext}}^{2d}(A_U,A_X)|U = (\mu_n)_U^{\otimes d},$$

where $d = \dim \mathcal{O}_{X,x} = \mathrm{codim}(Y,X)$.

Corollary 3.2.3. Let X be a regular prescheme of dimension d. If X is smooth over a field k of characteristic prime to n, or if X is excellent of characteristic zero, then X satisfies the conclusion of 3.2.1; consequently

$$\dim . q. inj(A_X) \leq 2d.$$

Proof. If X is excellent of characteristic zero, or if k is perfect of characteristic prime to n and X is smooth over k, then the hypotheses of 3.2.1 are satisfied by 3.1.1, 3.1.3, and 3.1.4. The conclusion is still valid when X is smooth over an arbitrary field k, as follows immediately after the base change $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$, where k' is a perfect closure of k (SGA A, Exposé VIII, 1.1). □

Lemma 3.2.4. Let X be a prescheme satisfying the conclusion of 3.2.1 and of dimension $\leq d$. If there exists an integer N such that X has cohomological dimension $\leq N$, then

$$\dim . inj(A_X) \leq 2d + N.$$

If, moreover, for every closed subset Y of X, the implication

$$\operatorname{codim}(Y, X) \geq p \quad \Rightarrow \quad \operatorname{cd}_{\mathbf{n}}(Y) \leq N - 2p$$

holds, then

$$\dim . inj(A_X) \leq N.$$

Proof. The first assertion follows immediately from 1.5. For the second, one argues as in 1.5. We are reduced to proving, for F constructible on X, the relation

$$\operatorname{Ext}^i(X; F, A_X) = 0 \qquad (i > N).$$

For this, inspect the spectral sequence

$$E_2^{pq} = \mathrm{H}^p(X; \underline{\operatorname{Ext}}^q(F, A_X)) \Rightarrow \operatorname{Ext}^*(X; F, A_X).$$

By 3.2.1, the sheaves $\underline{\operatorname{Ext}}^{2i}(F, A_X)$ and $\underline{\operatorname{Ext}}^{2i-1}(F, A_X)$ are concentrated in codimension $\geq i$. Therefore $E_2^{p,2i} = E_2^{p,2i-1} = 0$ for $p > N - 2i$, by the hypothesis. Hence $E_2^{pq} = 0$ for $p + q > N$, and the assertion follows. □

Corollary 3.2.5. If X is a smooth prescheme of dimension d over a separably closed field k of characteristic prime to n, then

$$\dim . inj(A_X) = 2d.$$

Proof. By 3.2.3 and SGA A, Exposé X, 4.2, we are in the situation of 3.2.4. The hypotheses of SGA A, Exposé X, 4.2 are satisfied by SGA A, Exposé X, 2.1 and 3.2; hence $\dim . inj(A_X) \leq 2d$. The equality follows from purity: if x is a closed point of codimension d in X, then

$$\operatorname{Ext}^{2d}(X, A_x, A_X) = \mathrm{H}^{2d}_x(A_X) \xrightarrow{\sim} (\mu_n)^{\otimes d}(k) \neq 0.$$

□

3.3. Constructibility of the sheaves $\underline{\operatorname{Ext}}^i(F, A_X)$

Proposition 3.3.1. Let X be a prescheme.

a) The following conditions are equivalent:

(i) for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^+(X)$, one has

$$\mathrm{R}\,\underline{\operatorname{Hom}}(F, G) \in \operatorname{ob} \mathrm{D}_c^+(X);$$

(i bis) for every pair (F, G) of constructible A_X-modules, one has $\mathrm{R}\,\underline{\operatorname{Hom}}(F, G) \in \operatorname{ob} \mathrm{D}_c^+(X)$, i.e. the sheaves $\underline{\operatorname{Ext}}^i(F, G)$ are constructible for all i;

(ii) for every open immersion $i : U \to X$ and every constructible A_U-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(ii bis) for every immersion $i : Y \to X$ and every constructible A_Y-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(ii ter) for every immersion $i : Y \to X$ and every locally constant constructible A_Y-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(iii) for every finite morphism, respectively every closed immersion, $f : Y \to X$, and every $F \in \mathrm{ob}\, \mathrm{D}_c^+(X)$, one has $\mathrm{R}^! f(F) \in \mathrm{ob}\, \mathrm{D}_c^+(Y)$.

b) These conditions are satisfied in each of the following two cases:

(i) $\dim(X) \leq 1$;

(ii) resolution of singularities and purity are available on X in the sense of 3.1.6(b).

Proof. The equivalence of (i) and (i bis) follows from the standard spectral sequence passing from sheaf-Ext to hyper-Ext.

(i bis) $\Rightarrow$ (ii). Let $\overline{F}$ be a constructible A_X-module extending F, for example $\overline{F} = i_! F$. By duality (SGA A, Exposé XVIII, 3.1.10),

$$\mathrm{R}i_* F = \mathrm{R}i_* \mathrm{R}\,\underline{\mathrm{Hom}}(A_U, i^! \overline{F}) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(i_! A_U, \overline{F}),$$

and this proves the claim.

(ii) $\Rightarrow$ (ii bis). Factor i as a closed immersion followed by an open immersion.

(ii bis) $\Rightarrow$ (i bis). The question is local, so we may suppose X affine. By the usual dévissage (SGA A, Exposé IX, 2.5), it is enough to verify (ii) for $F = i_! M$, where $i : Y \to X$ is the immersion of a sub-prescheme and M is a locally constant constructible A_Y-module. By the projection formula (SGA A, Exposé XVIII),

$$\mathrm{R}\,\underline{\mathrm{Hom}}(i_! M, G) \xrightarrow{\sim} \mathrm{R}i_* \mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G),$$

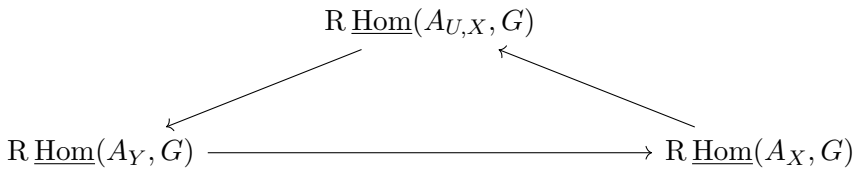
and by a spectral sequence we are reduced to proving that

$$\mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G) \in \mathrm{ob}\, \mathrm{D}_c^+(Y).$$

First prove $\mathrm{R}i^! G \in \mathrm{ob}\, \mathrm{D}_c^+(Y)$. After factoring i, we may suppose that Y is closed. Let $j : U = X - Y$ be the complementary open immersion. Applying $\mathrm{R}\,\underline{\mathrm{Hom}}(\ , G)$ to

$$0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0$$

gives a distinguished triangle.



Since $\mathrm{R}\,\underline{\mathrm{Hom}}(A_X, G) = G$, and

$$\mathrm{R}\,\underline{\mathrm{Hom}}(A_{U,X}, G) \xrightarrow{\sim} \mathrm{R}j_* j^! G$$

belongs to $\mathrm{D}_c^+(X)$ by hypothesis, it follows that

$$\mathrm{R}\,\underline{\mathrm{Hom}}(A_Y, G) \xrightarrow{\sim} i_*\mathrm{R}i^!G$$

belongs to $\mathrm{D}_c^+(X)$. Resolving M locally on Y, for the étale topology, by free A_Y-modules, another dévissage gives

$$\mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^!G) \in \mathrm{ob}\,\mathrm{D}_c^+(Y).$$

This proves the implication.

(ii bis) $\Rightarrow$ (ii ter) is trivial.

(ii ter) $\Rightarrow$ (ii). Let $i : U \to X$ be an open immersion and let F be a constructible A_U-module. We must prove $\mathrm{R}i_*F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$. The question being local, suppose X noetherian and argue by noetherian induction on X. We may suppose $U \neq \varnothing$. Then there exists a non-empty open subset $j : V \to U$ such that j^*F is locally constant (SGA A, Exposé IX, 2.4). Let M be the mapping cylinder of

$$F \longrightarrow \mathrm{R}j_*j^*F.$$

Applying $\mathrm{R}i_*$ gives a distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}i_*(M) & \\ {\scriptstyle (+1)}\swarrow & & \nwarrow \\ \mathrm{R}i_*(F) & \longrightarrow & \mathrm{R}i_*\mathrm{R}j_*j^*(F) \end{array}$$

Condition (ii ter) implies

$$\mathrm{R}i_*\mathrm{R}j_*j^*F \xrightarrow{\sim} \mathrm{R}(ij)_*j^*F \in \mathrm{ob}\,\mathrm{D}_c^+(X).$$

Consequently,

$$\mathrm{R}j_*j^*F \xrightarrow{\sim} i^*\mathrm{R}(ij)_*j^*F \in \mathrm{ob}\,\mathrm{D}_c^+(U).$$

Thus $M \in \mathrm{ob}\,\mathrm{D}_c^+(U)$, since $F \in \mathrm{ob}\,\mathrm{D}_c^+(U)$. Moreover, the cohomology of M is supported on a closed subset Z of U distinct from U. Applying the induction hypothesis to $U \cap \overline{Z} \to \overline{Z}$, we deduce $\mathrm{R}i_*M \in \mathrm{ob}\,\mathrm{D}_c^+(X)$, and hence $\mathrm{R}i_*F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$.

The equivalence of (i)–(ii ter) with condition (iii) is left to the reader; use 1.14.

For b)(i), normalization reduces one to the case where X is regular, and one uses purity, which will be proved in 5.1. For b)(ii), see 3.1.6(b). □

3.4. The biduality theorem on regular preschemes

Theorem 3.4.1, cf. [H], V, 2.2. Let X be a regular locally noetherian prescheme of finite dimension. Suppose that X is strongly desingularizable (3.1.5), satisfies (reg) (3.1.1) and $(\mathrm{cdloc})_n$ (3.1.3), and that absolute purity (3.1.4) is true on every prescheme smooth over X. Then A_X is dualizing.

Proof. We must prove that A_X satisfies conditions (i)–(iii) of 1.6. Condition (i) follows from 3.2.1. Condition (ii) follows from 3.3.1(b). It remains to prove condition (iii). Thus, by 1.7, for every constructible A_X-module F we must prove that the canonical homomorphism

$$F \longrightarrow \underline{D}_X\underline{D}_X F, \qquad \underline{D}_X = \mathrm{R}\,\underline{\mathrm{Hom}}(\ , A_X),$$

is an isomorphism. The question being local, suppose X noetherian. By dévissage (3.1.2, iii bis), it is enough to treat sheaves

$$F = f_* i_!(A'_U),$$

where $i : U \to Y$ is the inclusion of a regular open subset of an integral prescheme, $f : Y \to X$ is finite and the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$ for a prime divisor ℓ of n.

Because X is strongly desingularizable, there is a proper birational morphism $h : Y' \to Y$, with Y' regular, inducing an isomorphism from $U' = h^{-1}U$ to U, and such that $Y' - U'$ is a normal crossings divisor. By base change (SGA A, Exposé XVII),

$$i_!(A'_U) \xrightarrow{\sim} \mathrm{R}h_* i'_!(A'_{U'}),$$

where $i' : U' \to Y'$ is the inclusion. Hence, with $g = fh$,

$$f_* i_!(A'_U) \xrightarrow{\sim} f_* \mathrm{R}h_* i'_!(A'_{U'}) \xrightarrow{\sim} \mathrm{R}g_* i'_!(A'_{U'}).$$

By 1.11, it suffices to prove that the pair

$$(i'_!(A'_{U'}), \mathrm{R}^! g(A_X))$$

is bidualizing. Locally immersing Y' in a prescheme X' smooth over X, and applying purity on X', one sees that $\mathrm{R}^! g(A_X)$ is locally isomorphic to $A_{Y'}$, up to a translation of degrees. We are therefore reduced to the following special case.

Lemma 3.4.2. Let X be a locally noetherian regular prescheme on which absolute purity is true. Let $Y = Y_1 \cup \cdots \cup Y_p$ be a strict normal crossings divisor on X, and put $U = X - Y$. Then the pair $(A'_{U,X}, A_X)$ is bidualizing.

Proof. By the exact sequence

$$0 \longrightarrow A'_{U,X} \longrightarrow A'_X \longrightarrow A'_Y \longrightarrow 0,$$

it is equivalent to verify that (A'_Y, A_X) is bidualizing. We argue by induction on p.

If $p = 1$, let $i : Y \to X$ be the immersion. By 1.11, it is enough to verify that

$$(A'_Y, \mathrm{R}^! i(A_X))$$

is bidualizing. By purity,

$$\mathrm{R}^! i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes -1}[-2].$$

Thus the canonical homomorphism $A'_Y \to \underline{D}_Y \underline{D}_Y(A'_Y)$ is an isomorphism, as is checked trivially fibre by fibre, using Pontryagin duality for A-modules.

Suppose $p \geq 2$, and put $Y' = Y_1 \cup \cdots \cup Y_{p-1}$. We have an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_Y \longrightarrow A'_{Y'} \longrightarrow 0.$$

By the induction hypothesis, $(A'_{Y'}, A_X)$ is bidualizing. By the five lemma, we are reduced to biduality for $(A'_{(Y-Y'),X}, A_X)$. But again there is an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_{Y_p} \longrightarrow A'_{Y_p \cap Y'} \longrightarrow 0.$$

Since (A'_{Y_p}, A_X) is bidualizing, it remains to prove the same for $(A'_{Y_p\cap Y'}, A_X)$. Applying 1.11 again and using purity, we are finally reduced to biduality for $(A'_{Y_p\cap Y'}, A_{Y_p})$ on Y_p, where the induction hypothesis applies. □

This completes the proof of Theorem 3.4.1.

Corollary 3.4.3. Let S be an excellent noetherian regular prescheme of characteristic zero. Then A_S is dualizing. Moreover, if $f : X \to S$ is a morphism of finite type, then $\mathrm{R}^!f(A_S)$ is dualizing.

Proof. The first assertion follows because S satisfies the hypotheses of 3.4.1, as recalled in 3.1. For the second, one must first verify that $\mathrm{R}^!f(A_S)$ has finite quasi-injective dimension. Cover S by finitely many affine open subsets S_i such that $X_i = f^{-1}(S_i)$ is a finite union of affine open subsets X_{ij} immersed as closed subschemes in preschemes X'_{ij} smooth of finite dimension over S_i. From 3.2.3 and 1.6 it follows that

$$\dim . q. inj(\mathrm{R}^!f(A_S)) \leq 2 \sup \dim X'_{ij}.$$

It remains to verify conditions (ii) and (iii) of 1.7. Since these are local on X for the étale topology, we may suppose that f factors as $f = f'i$, where $f' : X' \to S$ is smooth of relative dimension d, and $i : X \to X'$ is a closed immersion. Then

$$\mathrm{R}^!f(A_S) \simeq \mathrm{R}^!i\,\mathrm{R}^!f'(A_S).$$

But $\mathrm{R}^!f'(A_S)$ is locally isomorphic to $A_{X'}$, up to the translation by $2d$, hence is dualizing by the first assertion. By 1.15, $\mathrm{R}^!f(A_S)$ is dualizing, and in particular satisfies conditions (ii) and (iii) of 1.7. □

Corollary 3.4.4. Let k be a field, and let $f : X \to S = \mathrm{Spec}(k)$ be a morphism locally of finite type, with $\dim(X) \leq 2$. Then $\mathrm{R}^!f(A_S)$ is dualizing.

Proof. Replacing k by its perfect closure, which is harmless for the étale topology (SGA A, Exposé VIII, 1.1), we may suppose k perfect. Cover X by affine open subsets X_i of finite type over k. Since $\dim(X_i) \leq 2$, the normalization lemma makes X_i finite over a scheme X'_i smooth over S and of dimension ≤ 2. By 1.15 this reduces us to the case where f is smooth, hence locally

$$f^!(A_S) \simeq A_X[2d].$$

Thus it remains to show that A_X is dualizing in this case; this is exactly 3.4.1, since the hypotheses are satisfied (see 3.1). □

Remark 3.4.5. We shall see below that if X is regular of dimension ≤ 1, then A_X is dualizing.

4. Local duality

The object of this section is to present the notion of "biduality" developed in the preceding sections in a form closer to geometric intuition.

4.1

Let X be a prescheme, let $i : x \to X$ be a closed point whose residue field is separably closed, and let K_X be a dualizing complex on X (1.7). We know (1.15) that $K_{\{x\}} = \mathrm{R}^! i(K_X)$ is a dualizing complex on x, hence, by (2.1) and (3.4.4), isomorphic to $A[r]$ for some $r \in \mathbb{Z}$ (if A is local). We shall say that K_X is *normalized at* x if $r = 0$ and if an isomorphism $K_{\{x\}} \xrightarrow{\sim} A$ has been given.

For example, if X is smooth of relative dimension d over the spectrum of a separably closed field, the complex $(\mu_n)_X^{\otimes d}[2d]$ is, by the purity theorem (SGA XVI 3, SGA $4^{1/2}$ Cycle 2.3), canonically normalized at every closed point of X.

4.2

Let X be a prescheme, $i : Y \to X$ a closed sub-prescheme, and K_X a dualizing complex on X. Then (1.15) $K_Y = \mathrm{R}^! i(K_X)$ is a dualizing complex on Y, and one has the *complementary induction formula* (1.12 b) (ii)), for $F \in \mathrm{ob}\, \mathrm{D}_c^b(X)$,

$$(4.2.1) \qquad i^* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathrm{R}^! i(F),$$

where $\underline{D}_X = \mathrm{R}\,\underline{\mathrm{Hom}}(\ , K_X)$ and $\underline{D}_Y = \mathrm{R}\,\underline{\mathrm{Hom}}(\ , K_Y)$.

Apply (4.2.1) in particular in the case where $Y = \{x\}$ is a closed point of X with separably closed residue field, K_X being normalized at x (4.1). Taking into account that A_x is injective, (4.2.1) is then written

$$(4.2.2) \qquad \underline{D}_X(F)_x \xrightarrow{\sim} \mathrm{Hom}^*(\mathrm{R}^! i(F), A).$$

Noting that $\mathrm{R}^! i(F)$ is also written, by definition, $\mathrm{R}\Gamma_x(F)$, one therefore has a perfect pairing in $\mathrm{D}(A)$:

$$(4.2.2)' \qquad \underline{D}_X(F)_x \times \mathrm{R}\Gamma_x(F) \longrightarrow A,$$

giving rise to perfect pairings between finite groups:

$$(4.2.2)'' \qquad \underline{\mathrm{H}}^i(\underline{D}_X(F))_x \times \underline{\mathrm{H}}_x^{-i}(F) \longrightarrow A.$$

For example, let X be a prescheme smooth of dimension d over a separably closed field, and let $x \in X$ be a closed point. Then, for $F \in \mathrm{ob}\, \mathrm{D}_c^b(X)$, one has a canonical perfect duality between finite groups:

$$(4.2.2)''' \qquad \mathrm{Ext}^{2d-i}(F, (\mu_n)_X^{\otimes d})_x \times \underline{\mathrm{H}}_x^i(F) \longrightarrow A.$$

Exercise 4.2.3

Let X be a strictly local scheme (SGAA VIII 4.2), with closed point x, and let $i : x \to X$ be the canonical morphism. Let K be a dualizing complex on X, normalized at x. Denote by

$$\mathbf{L}\, i_? : \mathrm{D}^b(A) \longrightarrow \mathrm{D}^-(X)$$

the functor defined by

$$\mathbf{L}\, i_?(M) = M \overset{\mathbf{L}}{\otimes}_A K.$$

(If K has finite tor-dimension over A, $\mathbf{L}\, i_?$ extends to a functor from $\mathrm{D}^+(A)$ to $\mathrm{D}(X)$.)

Show that there exists, for $L \in \mathrm{D}^-(X)$ and $M \in \mathrm{D}^b(A)$, a canonical functorial isomorphism

$$(*) \qquad \mathrm{R}\,\mathrm{Hom}(L, \mathbf{L}\, i_?(M)) \xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(\mathrm{R}^!i(L), M),$$

(assuming of course that $\mathrm{R}^!i : \mathrm{D}^-(X) \to \mathrm{D}^-(A)$ is defined ...).

Hint: First define a "trace" morphism $\mathrm{R}^!i\,\mathbf{L}\,i_?(M) \to M$; then, to check that $(*)$ is an isomorphism, reduce by dévissage to the case where $L \in \mathrm{ob}\,\mathrm{D}^b_c(X)$ and $M = A$, and apply (4.2.2). Note that, for $M = A$, $(*)$ reduces to the perfect pairing ("local duality", cf. [H] V 6.2)

$$\mathrm{R}\Gamma_x(L) \times \mathrm{R}\,\mathrm{Hom}(L, K) \longrightarrow A,$$

or

$$\mathrm{H}^i_x(L) \times \mathrm{Ext}^{-i}(L, K) \longrightarrow A.$$

4.3

We shall now generalize (4.2.2) to the case of a geometric point $\overline{x}$ of X localized at a point x which is not necessarily closed. For this we need a few preliminaries.

Lemma 4.3. Let X be a prescheme, X' the strict localization of X at a geometric point $\overline{x}$, and $f : X' \to X$ the canonical map. Then, for $F \in \mathrm{ob}\,\mathrm{D}^-_c(X)$ and $G \in \mathrm{ob}\,\mathrm{D}^+(X)$, the canonical functorial homomorphism (SGAA VI I 3.7)

$$f^*\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G)$$

is an isomorphism.

Proof. As usual, one reduces to F and G concentrated in degree 0. Since the question is local in a neighborhood of x, one may suppose X affine. Then F admits (SGAA IX 2.7) a left resolution by sheaves of the form $i_!(A_U)$, where $i : U \to X$ is étale of finite presentation; hence, by dévissage, one is reduced to $F = i_!(A_U)$. Form the cartesian square

$$\begin{array}{ccc} U & \xleftarrow{g} & U' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i'} \\ X & \xleftarrow[f]{} & X' \end{array}$$

One has canonical isomorphisms, trivial special cases of duality (SGAA XVIII 3.1.10):

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \xleftarrow{\sim} \mathrm{R}i_*i^*G, \qquad \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G) \xleftarrow{\sim} \mathrm{R}i'_*g^*i^*G.$$

Under these isomorphisms, the canonical arrow

$$f^*\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G)$$

is identified with the base-change arrow (SGAA XVII)

$$f^*\mathrm{R}i_*i^*(G) \longrightarrow \mathrm{R}i'_*g^*i^*(G).$$

But this is an isomorphism, as one sees at once by passing to the limit (SGAA VII 5.11).

Lemma 4.4. Let X be a prescheme, and let $K \in \mathrm{ob}\,\mathrm{D}_c^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K be dualizing, it is necessary and sufficient that, for every strict localization $f : \overline{X}(\overline{x}) \to X$, f^*K be dualizing.

Proof. Use the fact that every constructible sheaf on a strict localization $\overline{X}(\overline{x})$ is induced by a constructible sheaf on an étale $\overline{x}$-pointed scheme over X (SGAA IX 2.7.4), and apply (4.3).

4.5

Let X be a prescheme, $\overline{x}$ a geometric point of X, and $f : X' = \overline{X}(\overline{x}) \to X$ the corresponding strict localization.

Let Y be the integral closed sub-prescheme which is the closure of x in X; denote by $g : \overline{x} \to Y$ and $i : Y \to X$ the canonical maps, and form the cartesian square

$$\begin{array}{ccc} Y & \xrightarrow{i} & X \\ {\scriptstyle g}\uparrow & & \uparrow{\scriptstyle f} \\ \overline{x} & \xrightarrow[i']{} & X' \end{array}$$

($\overline{x}$ is both the closed point of X' and the strict localization of Y at $\overline{x}$).

We shall denote by

$$\mathrm{R}\Gamma_{\overline{x}} : \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(\overline{x})$$

the functor defined by

$$\mathrm{R}\Gamma_{\overline{x}} = g^*\mathrm{R}^!i. \tag{4.5.1}$$

Using the well-known relation between the functors $\mathrm{R}^!i$ and $\mathrm{R}j_*j^*$, where $j : U \to X$ is the open complement of Y, and applying passage to the limit (SGAA VII 5.11), one obtains a canonical isomorphism

$$\mathrm{R}\Gamma_{\overline{x}} \xrightarrow{\sim} \mathrm{R}^!i'f^*. \tag{4.5.2}$$

When x is a closed point with separably closed residue field, the functor $\mathrm{R}\Gamma_{\overline{x}}$ coincides with the usual functor $\mathrm{R}\Gamma_x$ (SGAA V 4.3), which justifies the notation.

Let $K_X \in \mathrm{ob}\,\mathrm{D}^+(X)$, and put $K_Y = \mathrm{R}^!i(K_X)$, $K_{X'} = f^*K_X$, and $K_{\overline{x}} = \mathrm{R}\Gamma_{\overline{x}}(K_X)$ (warning: $K_{\overline{x}}$ is not the fiber of K_X at $\overline{x}$!). Suppose that K_X is dualizing. Then, by (1.15) and (4.4), the same is true of $K_{X'}$, K_Y, and $K_{\overline{x}}$. Hence (2.1) one has $K_{\overline{x}} \simeq A[d]$ for some $d \in \mathbb{Z}$.

We shall say that K_X is *normalized at* $\overline{x}$ if $d = 0$ and if an isomorphism $K_{\overline{x}} \simeq A$ has been chosen. In the case where x is closed with separably closed residue field, this recovers the notion introduced in (4.1).

Let K_X be a dualizing complex, and put $\underline{D}_X = \mathrm{R}\underline{\mathrm{Hom}}(\ , K_X)$, $\underline{D}_Y = \mathrm{R}\underline{\mathrm{Hom}}(\ , K_Y)$, etc. Let $F \in \mathrm{ob}\,\mathrm{D}_c^b(X)$; let us compute $(\underline{D}_X F)_{\overline{x}}$. One has:

$$\begin{aligned} (\underline{D}_X F)_{\overline{x}} &\xrightarrow{\sim} g^*i^*\underline{D}_X F \\ &\xrightarrow{\sim} g^*\underline{D}_Y \mathrm{R}^!i F \qquad \text{(by the complementary induction formula (4.2.1))} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}} g^*\mathrm{R}^!i(F) \qquad \text{(by (4.3)), that is,} \end{aligned}$$

$$(\underline{D}_X F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}}\mathrm{R}\Gamma_{\overline{x}}(F). \tag{4.5.3}$$

Of course, one could have made an analogous computation by passing through X' instead of Y; one would then have obtained a canonical isomorphism

$$(\underline{D}_X F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}^! i'^* f^*(F),$$

compatible with (4.5.3) by means of the identification (4.5.2).

If moreover K_X is normalized at $\overline{x}$, one obtains perfect pairings generalizing (4.2.2)' and (4.2.2)":

$$(4.5.3)' \qquad \underline{D}_X(F)_{\overline{x}} \times \mathrm{R}\Gamma_{\overline{x}}(F) \longrightarrow A,$$

$$(4.5.3)'' \qquad \underline{\mathrm{H}}^i(\underline{D}_X(F)_{\overline{x}}) \times \underline{\mathrm{H}}_{\overline{x}}^{-i}(F) \longrightarrow A,$$

where $\underline{\mathrm{H}}_{\overline{x}}^i = \underline{\mathrm{H}}^i \mathrm{R}\Gamma_{\overline{x}}$.

Conversely, let $K_X \in \mathrm{ob}\,\mathrm{D}^+(X)$ satisfy conditions (i) and (ii) of (1.7), and suppose that $K_{\overline{x}} = \mathrm{R}\Gamma_{\overline{x}}(K_X)$ is dualizing for every geometric point $\overline{x}$ of X. Then, for every $\overline{x}$, one can define (cf. 1.12 b) (ii)) a canonical arrow

$$(4.5.3)^{\mathrm{bis}} \qquad \underline{D}_X(F)_{\overline{x}} \longrightarrow \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}}(F) \qquad (F \in \mathrm{ob}\,\mathrm{D}_c^b(X)).$$

If this is an isomorphism for every $\overline{x}$ and every F, then K_X is dualizing. Indeed, it remains to verify that the canonical arrow

$$F \longrightarrow \underline{D}_X \underline{D}_X(F)$$

is an isomorphism for every $F \in \mathrm{ob}\,\mathrm{D}_c^b(X)$. At an arbitrary geometric point $\overline{x}$, one has

$$\begin{aligned} F_{\overline{x}} &\xrightarrow{\sim} \underline{D}_{\overline{x}} \underline{D}_{\overline{x}}(F_{\overline{x}}) \qquad \text{(since } K_{\overline{x}} \text{ is dualizing)} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}} \underline{D}_X(F) \qquad \text{(by (4.3) and the induction formula 1.12 b) (i))} \\ &\xrightarrow{\sim} \underline{D}_X \underline{D}_X(F)_{\overline{x}} \qquad \text{(by hypothesis),} \end{aligned}$$

and therefore, modulo a compatibility check, (F, K_X) is bidualizing, which proves the assertion. In summary:

Proposition 4.5.4. Let X be a prescheme, and let $K_X \in \mathrm{ob}\,\mathrm{D}^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K_X be dualizing, it is necessary and sufficient that, for every geometric point $\overline{x}$ of X, the complex $K_{\overline{x}}$ be dualizing and the canonical arrow $(4.5.3)^{\mathrm{bis}}$ be an isomorphism for every $F \in \mathrm{ob}\,\mathrm{D}_c^b(X)$.

Remark 4.5.5. The existence of dualizing complexes should be regarded, in a sense, as an important complement to the global duality theorem. Indeed, the latter expresses that, for a separated morphism of finite type $f : X \to Y$ (Y quasi-compact), one has a canonical isomorphism

$$\underline{D}_Y \mathrm{R}_! f(F) \xrightarrow{\sim} \mathrm{R}f_* \underline{D}_X(F)$$

(with the notation of 1.12). Its use for computing $\mathrm{R}_! f(F)$ (when $F \in \mathrm{ob}\,\mathrm{D}_c^b(X)$) requires information about $\underline{D}_X(F)$. When K_X is dualizing, this information is supplied by the isomorphism (4.5.3), which makes possible, at least in theory, a computation of the fibers of $\underline{D}_X(F)$ as invariants of "purely spatial" cohomology.

Exercise 4.5.6. Let X be a prescheme endowed with a dualizing complex (1.7). The family of functors (cf. (4.5.3)")

$$\underline{\mathrm{H}}^i_{\overline{x}} : \mathrm{D}^b_c(X) \longrightarrow (A\text{-modules})$$

(i running through $\mathbb{Z}$ and $\overline{x}$ through the set of geometric points of X) is "conservative", i.e. a morphism $F \to G$ in $\mathrm{D}^b_c(X)$ such that the maps $\mathrm{H}^i_{\overline{x}}(F) \to \mathrm{H}^i_{\overline{x}}(G)$ are isomorphisms, is an isomorphism.

4.6

Let X be a prescheme, $j : Y \to X$ a closed sub-prescheme, and $i : \overline{x} \to Y$ a geometric point of Y. Let, on the other hand, K_X be a dualizing complex on X, normalized at $\overline{x}$ (cf. (4.5)). We saw in (4.5) that, if Y is the closure of x, one has, for $F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$, a perfect pairing

$$\mathrm{R}^!j(F)_{\overline{x}} \times \underline{D}_X(F)_{\overline{x}} \longrightarrow A.$$

It is not difficult to define an analogous pairing in the general case. Introduce the factorization of i as

$$\overline{x} \xrightarrow{i'} \overline{\{x\}} \xrightarrow{i''} Y.$$

For $F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$, one has canonical isomorphisms:[9]

$$\begin{aligned}
\mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F) &\xrightarrow{\sim} \mathrm{R}\Gamma_{\overline{x}} \underline{D}_Y \mathrm{R}^! j(F) \qquad \text{(complementary induction formula (4.2.1))} \\
&\xrightarrow{\sim} i'^* \mathrm{R}^! i'' \underline{D}_Y \mathrm{R}^! j(F) \qquad \text{(definition of } \mathrm{R}\Gamma_{\overline{x}} \text{ (4.5.1))} \\
&\xrightarrow{\sim} i'^* \underline{D}_{\overline{x}} i''^* \mathrm{R}^! j(F) \qquad \text{(ordinary induction (1.12 b) (i))} \\
&\xrightarrow{\sim} \underline{D}_{\overline{x}} i^* \mathrm{R}^! j(F) \qquad \text{(by 4.3)), i.e. finally a}
\end{aligned}$$

canonical isomorphism

$$\text{(4.6.1)} \qquad \mathrm{R}^!j(F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F),$$

and hence perfect pairings

$$\text{(4.6.1)}' \qquad \mathrm{R}^!j(F)_{\overline{x}} \times \mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F) \longrightarrow A,$$

$$\text{(4.6.1)}'' \qquad \underline{\mathrm{H}}^i_Y(F)_{\overline{x}} \times \underline{\mathrm{H}}^{-i}_{\overline{x}}(\underline{D}_X(F)|Y) \longrightarrow A.$$

Example 4.6.2. Let X be a regular prescheme, let $\overline{x}$ be a geometric point of X, and put $d = \dim(\mathcal{O}_{X,\overline{x}})$. Set $K_X = (\mu_n)^{\otimes d}_X[2d]$. If X satisfies condition (reg) (3.1.1) and the purity theorem (3.1.4), then K_X is canonically normalized at $\overline{x}$. Thus, if K_X is dualizing, one has perfect pairings

$$\underline{\mathrm{H}}^i_Y(F)_{\overline{x}} \times \underline{\mathrm{H}}^{2d-i}_{\overline{x}}(F'|Y) \longrightarrow A,$$

for every closed subset Y containing x and every $F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$, with $F' = \mathrm{R}\,\mathrm{Hom}(F, (\mu_n)^{\otimes d}_X)$.

To finish, let us give a strictly local variant of (4.6.1).

[9] *Editorial note.* The ordinary-induction line is not unambiguously legible in the printed source. The current source-checked French TeX reads $i'^* \underline{D}_{\overline{x}} i'''^* \mathrm{R}^! j(F)$; type checking through $i'' : \overline{\{x\}} \to Y$ instead suggests the subscript $\overline{\{x\}}$, while the source glyph can resemble X and its first pullback mark is also unclear. This edition retains the French-control reading without claiming that the scan adjudicates it.

4.7

Let X be a noetherian strictly local scheme (SGAA VIII 4.2), with closed point x. Let Y be a non-empty closed subset of X, and consider the canonical immersions

$$U = X - Y \xrightarrow{i} V = X - \{x\} \xrightarrow{j} X.$$

Finally, let K_X be a dualizing complex normalized at x (4.1), giving dualizing complexes "corresponding" on U, V, Y, x (and by hypothesis $\underline{D}_x = \mathrm{R}\,\mathrm{Hom}(\ , A)$):

$$K_V = j^* K_X, \qquad K_U = i^* K_V, \qquad K_Y = \mathrm{R}\Gamma_Y(K_X).$$

Let $G \in \mathrm{ob}\,\mathrm{D}^b_c(U)$, and put $G' = \underline{D}_U(G)$. We propose to define a natural pairing

$$(4.7.1) \qquad \mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A,$$

i.e. a functorial homomorphism

$$(4.7.1)' \qquad \mathrm{R}\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A.$$

(Since the ring $A = \mathbb{Z}/\ell\mathbb{Z}$ has infinite cohomological dimension for $\ell \geqslant 2$, one knows how to define only the tensor products

$$\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}^-(A) \longrightarrow \mathrm{D}^-(A)$$

and

$$\overset{\mathbf{L}}{\otimes} : \mathrm{D}(A) \times \mathrm{D}^b(A)_{\mathrm{torf}} \longrightarrow \mathrm{D}(A) \quad (*).$$

We therefore leave it to the reader to make suitable hypotheses ensuring that the two factors of $\overset{\mathbf{L}}{\otimes}$ appearing in (4.7.1)' are in $\mathrm{D}^b(A)$, or that one of them has finite tor-dimension.)

First note that, by the complementary duality formula (1.12 a) (ii), one has a canonical isomorphism

$$(4.7.2) \qquad i_!(G') \xrightarrow{\sim} \underline{D}_V(\mathrm{R}i_* G).$$

On the other hand, one has the ordinary duality isomorphism

$$(4.7.3) \qquad \mathrm{R}\,\mathrm{Hom}(A_U, G) \xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(A_V, \mathrm{R}i_* G).$$

By the definition of $\underline{D}_V$ (and independently of the fact that K_V is dualizing), one has a canonical arrow

$$(4.7.4) \qquad \mathrm{R}i_* G \overset{\mathbf{L}}{\otimes} \underline{D}_V(\mathrm{R}i_* G) \longrightarrow K_V.$$

From (4.7.2), (4.7.3), and (4.7.4) one obtains a (canonical) arrow

$$(4.7.5) \qquad \mathrm{R}\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A \mathrm{R}\Gamma(V, i_! G') \longrightarrow \mathrm{R}\,\mathrm{Hom}(A_V, K_V).$$

But $K_V = j^! K_X$, and one has the duality isomorphism (SGAA 3.1.10):

$$(4.7.6) \qquad \mathrm{R}\,\mathrm{Hom}(A_V, K_V) \xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(A_{V,X}, K_X).$$

Finally, the exact sequence

$$(4.7.7) \qquad 0 \longrightarrow A_{V,X} \longrightarrow A_X \longrightarrow A_x \longrightarrow 0$$

gives a homomorphism of degree +1:

$$\text{(4.7.8)} \qquad \mathrm{R}\,\mathrm{Hom}(A_{V,X}, K_X) \longrightarrow \mathrm{R}\,\mathrm{Hom}(A_x, K_X) \xrightarrow{\sim} A.$$

Composing (4.7.5), (4.7.6), and (4.7.8), one obtains the announced arrow (4.7.1)'.

(*) (added in 1977) Inexact: in fact it is easy to extend $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}^-(A) \to \mathrm{D}^-(A)$ *to* $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}(A) \to \mathrm{D}(A)$.

We shall now see that the pairing (4.7.1) is "perfect", i.e. that the arrow (4.7.1)' identifies $\mathrm{R}\Gamma(V, i_!G')[-1]$ with $\underline{D}_x\mathrm{R}\Gamma(U, G)$. Indeed suppose, as is allowed, that

$$G = (ji)^*F,$$

with $F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$. Put $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!G'$. One has the exact sequence

$$\text{(4.7.9)} \qquad 0 \longrightarrow F'_{U,X} \longrightarrow F' \longrightarrow F'|Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}\Gamma_x(F'_{U,X}) & \\ {\scriptstyle +1}\nearrow & & \searrow \\ \mathrm{R}\Gamma_x(F'|Y) & \longleftarrow & \mathrm{R}\Gamma_x(F') \end{array} \qquad (b)$$

By definition of $\mathrm{R}\Gamma_x$, one has

$$\mathrm{R}\Gamma_x(F'_{U,X}) = \mathrm{R}\,\mathrm{Hom}(A_x, F'_{U,X});$$

hence, using (4.7.7) and the fact that $(F'_{U,X})_x = \mathrm{R}\Gamma(X, F'_{U,X}) = 0$, one has a canonical isomorphism

$$\mathrm{R}\,\mathrm{Hom}(A_{V,X}, F'_{U,X})[1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

But one has

$$\begin{aligned} \mathrm{R}\,\mathrm{Hom}(A_{V,X}, F'_{U,X}) &= \mathrm{R}\,\mathrm{Hom}(j_!A_V, j_!i_!G') \\ &\xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(A_V, i_!G') \qquad \text{by duality (SGAA 3.1.10),} \end{aligned}$$

and hence finally a canonical isomorphism

$$\text{(4.7.10)} \qquad \mathrm{R}\Gamma(V, i_!G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

On the other hand one has the exact sequence

$$\text{(4.7.11)} \qquad 0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}\Gamma(U, F|U) & \\ {\scriptstyle +1}\swarrow & & \nwarrow \\ \mathrm{R}\Gamma_Y(F) & \longrightarrow & \mathrm{R}\Gamma_X(F) \end{array} \qquad (a)$$

By virtue of (4.7.1) and (4.7.10), one has a natural pairing

$$(4.7.12) \qquad \mathrm{R}\Gamma(U, F|U) \times \mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow A.$$

By virtue of (4.6.1), one has a perfect pairing

$$(4.7.13) \qquad \mathrm{R}\Gamma_Y(F) \times \mathrm{R}\Gamma_x(F'|Y) \longrightarrow A.$$

Finally, by the ordinary induction formula (1.12 b) (i)), one has a perfect pairing

$$(4.7.14) \qquad \mathrm{R}\Gamma(X, F) \times \mathrm{R}\Gamma_x(F') \longrightarrow A.$$

The reader who has the patience to orient himself in this maze of "canonical arrows" will check that the pairings (4.7.12), (4.7.13), and (4.7.14) are "compatible" with the arrows of the triangles (a) and (b), and consequently that the pairings (4.7.12) and (4.7.1) are perfect.

The pairing of the triangles (a) and (b) gives rise to two exact sequences "transpose to one another":

$$(4.7.15) \qquad \cdots \longrightarrow \mathrm{H}^i_Y(X, F) \longrightarrow \mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F) \longrightarrow \cdots$$
$$\cdots \longleftarrow \mathrm{H}^{-i}_x(Y, F'|Y) \longleftarrow \mathrm{H}^{-i}_x(X, F') \longleftarrow \mathrm{H}^{-i-1}(V, F'_{U,V}) \longleftarrow \cdots .$$

In summary, we may state:

Proposition 4.7.16. Let X be a noetherian strictly local scheme, with closed point x. Let Y be a non-empty closed subset of X, let $U = X - Y$, $V = X - x$, and let $i : U \to V$ and $j : V \to X$ be the canonical inclusions. Let K_X be a dualizing complex on X normalized at x (hence giving associated dualizing complexes on V, U, Y). For $G \in \mathrm{ob}\, \mathrm{D}^b_c(U)$, one has a perfect pairing:

$$(*) \qquad \mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_!G')[-1] \longrightarrow A$$

(where $G' = \underline{D}_U(G)$). If $G = F|U$, where $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$, one has a canonical isomorphism

$$(**) \qquad \mathrm{R}\Gamma(V, i_!G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X})$$

(where $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!(F'|U) = (ji)_!(G')$), and a perfect pairing between the distinguished triangles

$$\begin{array}{ccc} & \mathrm{R}\Gamma(U, F|U) & \\ {\scriptstyle +1}\swarrow & & \searrow\nwarrow \\ \mathrm{R}\Gamma_Y(F) & \longrightarrow & \mathrm{R}\Gamma(X, F) \end{array} \qquad \text{and} \qquad \begin{array}{ccc} & \mathrm{R}\Gamma_x(F'_{U,X}) & \\ {\scriptstyle +1}\nearrow & & \searrow \\ \mathrm{R}\Gamma_x(F'|Y) & \longleftarrow & \mathrm{R}\Gamma_x(F') \end{array}$$

compatible with the pairing $(*)$ and the isomorphism $(**)$, and giving rise to two exact sequences transpose to one another (4.7.15).

Remark 4.7.17. In the particular case where $Y = \{x\}$, the pairing (4.7.1) is defined under the sole hypothesis that K_X is normalized at x (and not necessarily dualizing), i.e. such that $\mathrm{R}\Gamma_x(K_X) \simeq A$ (the isomorphism being given): the isomorphism (4.7.2) (which used biduality) then reduces to the identity. Since the pairing (4.7.14) is in any case perfect, one sees therefore that local duality on X (in the form (4.2.2)') is equivalent to the "global" duality on $U = X - x$, i.e. to the assertion that the pairing (4.7.1)

$$\mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(U, G')[-1] \longrightarrow A$$

is perfect, or equivalently that the pairings

$$\mathrm{H}^i(U,G)\times\mathrm{H}^{-i-1}(U,G')\longrightarrow A$$

are perfect.

Suppose that one has on X a dualizing complex K_X normalized at x. If X is excellent and U is regular of dimension d (x may therefore be an isolated singularity), one should be able to check that $K_U = K_X|U$ is canonically isomorphic to $(\mu_n)_U^{\otimes d}[2d]$ (this is easily checked, at any rate, if X is the strict localization of a prescheme locally of finite type over a field of characteristic 0). Then local duality on X implies that, for every locally constant constructible A_U-module G, one has perfect pairings:

$$\mathrm{H}^i(U,G)\times\mathrm{H}^{2d-1-i}(U,G^\circ)\longrightarrow A,$$

where $G^\circ = \underline{\mathrm{Hom}}(G,(\mu_n)_U^{\otimes d})$, corresponding formally to Poincaré duality on a scheme of dimension $2d-1$.

5. Local duality on curves

Theorem 5.1. Let X be a noetherian regular prescheme of dimension 1. Suppose that n is prime to the residual characteristics of X. Then the purity theorem is true on X (3.1.4), and the complex A_X is dualizing (1.7).

Proof. Suppose purity has been proved. Since X plainly satisfies condition (reg) (3.1.1), and since condition $(\mathrm{cdloc})_n$ is verified by (SGAA X 2.2), it follows from (3.2.1) that $\dim.q.inj(A_X)=2$. On the other hand, condition (ii) of (1.7) is verified by (3.3.1 b) (i)) (which was proved assuming purity). Hence, by (4.4), we are reduced to proving that A_X is dualizing when X is strictly local. But, to prove purity, which is a local question in a neighborhood of a closed point, one may equally well suppose X strictly local. Finally, we are reduced to proving the theorem when X is strictly local.

Thus suppose that X is strictly local. Let x be its closed point, let $(\pi = \mathrm{car}(k(x)))$, let η be its generic point, and let $U = X - x = \mathrm{Spec}(k(\eta))$. Let $G = \mathrm{Gal}(k(\overline{\eta})/k(\eta))$ be the fundamental group of U. It is known [CL, Chap. IV § 2 Exer. 2] that one has a canonical exact sequence

$$1\longrightarrow P\longrightarrow G\longrightarrow H\longrightarrow 1, \tag{5.1.1}$$

where $H=\prod_{\ell\neq\pi}\mathbb{Z}_\ell(1)$, $\mathbb{Z}_\ell(1)=\varprojlim\mu_{\ell^\nu}$, and P is a pro-π-group. (Of course, H is isomorphic, but not canonically, to $\prod_{\ell\neq\pi}\mathbb{Z}_\ell$.)

It is also known (SGAA VIII 2) that the étale cohomology of U is interpreted as the Galois cohomology of G, sheaves (respectively constructible sheaves) of A_U-modules corresponding to A-modules (respectively finite type A-modules) on which G acts continuously.

This being so, the $\underline{\mathrm{H}}^i_x(A_X)$ are determined by the exact sequence

$$0\longrightarrow\underline{\mathrm{H}}^0_x(A_X)\longrightarrow\mathrm{H}^0(X,A_X)\longrightarrow\mathrm{H}^0(U,A_U)\longrightarrow\underline{\mathrm{H}}^1_x(A_X)\longrightarrow 0, \tag{5.1.2}$$

and the isomorphisms

$$\underline{\mathrm{H}}^i_x(A_X)\xrightarrow{\sim}\mathrm{H}^{i-1}(U,A_U)\qquad\text{for } i\geqslant 2. \tag{5.1.3}$$

But trivially $\mathrm{H}^0(X, A_X) = \mathrm{H}^0(U, A_U) = A$, so $\underline{\mathrm{H}}^0_x(A_X) = \underline{\mathrm{H}}^1_x(A_X) = 0$.

On the other hand, for a Galois G-A-module M, one has the Hochschild–Serre spectral sequence

$$E_2^{pq} = \mathrm{H}^p(H, \mathrm{H}^q(P, M)) \Longrightarrow \mathrm{H}^*(G, M).$$

Since n is prime to π and P is a pro-π-group, one has

$$\mathrm{H}^q(P, M) = 0 \quad \text{for } q \neq 0,$$

hence $E_2^{pq} = 0$ for $q \neq 0$, giving a canonical isomorphism

$$\mathrm{H}^p(G, M) \xrightarrow{\sim} \mathrm{H}^p(H, M^P). \tag{5.1.4}$$

By an analogous argument, one sees moreover that $\mathrm{H}^p(H, N)$ is identified with $\mathrm{H}^p(\widehat{\mathbb{Z}}(1), N)$ when the factor $\widehat{\mathbb{Z}}(1)$ is made to act trivially on N. Thus, finally, with the preceding convention, one has a canonical isomorphism

$$\mathrm{H}^p(G, M) \xrightarrow{\sim} \mathrm{H}^p(\widehat{\mathbb{Z}}(1), M^P). \tag{5.1.5}$$

Now the Galois cohomology of $\widehat{\mathbb{Z}}$ is well known (CG chap. I p. 31): for a G-A-module N, one has

$$\mathrm{H}^i(\widehat{\mathbb{Z}}, N) = 0 \quad \text{for } i \geqslant 2, \qquad \mathrm{H}^0(\widehat{\mathbb{Z}}, N) = N^{\widehat{\mathbb{Z}}}, \qquad \mathrm{H}^1(\widehat{\mathbb{Z}}, N) = N_{\widehat{\mathbb{Z}}}. \tag{5.1.6}$$

From (5.1.3), (5.1.5), and (5.1.6) one immediately deduces that

$$\mathrm{H}^i(U, A_U) = 0 \quad \text{for } i \geqslant 2,$$

and

$$\mathrm{H}^1(U, A_U) \xrightarrow{\sim} A.$$

This last isomorphism is not canonical, since it depends on the choice of an identification of $\widehat{\mathbb{Z}}(1)$ with $\widehat{\mathbb{Z}}$. But, using the Kummer exact sequence, one finds (SGAA XIX 1.3) a canonical isomorphism (given by the "local fundamental class")

$$\mathrm{H}^1(U, \mu_n) \xrightarrow{\sim} A. \tag{5.1.7}$$

Purity is therefore proved.

Let us now prove that A_X is dualizing. By (4.7.17), it is a matter of showing that, for every locally constant constructible A_U-module M, the pairing

$$\mathrm{H}^i(U, M) \times \mathrm{H}^{1-i}(U, \underline{\mathrm{Hom}}(M, A_U)) \longrightarrow \mathrm{H}^1(U, A_U) \xrightarrow{\sim} (\mu_n)_X^{\otimes -1} \tag{5.1.8}$$

is a perfect duality. We shall interpret this pairing in terms of Galois cohomology. We shall use the following elementary lemma.

Lemma 5.1.9. Let $D = \mathrm{Hom}(\ , A)$ denote the Pontrjagin dualizing functor on the category of finite type A-modules. Let M be a finite type A-module on which a group G acts A-linearly.

(i) One has a canonical isomorphism

$$D(M^G) \xrightarrow{\sim} (DM)_G.$$

(ii) If G is a profinite group acting continuously on M, and if G has pro-order prime to n, then the canonical arrow

$$M^G \longrightarrow M_G$$

is an isomorphism.

Proof. By definition one has

$$M^G = \varprojlim_{s\in G} \mathrm{Ker}(M \xrightarrow{s-1} M),$$

hence

$$D(M^G) \xrightarrow{\sim} \varinjlim_{s\in G} \mathrm{Coker}(DM \xrightarrow{s-1} DM) \xrightarrow{\sim} (DM)_G,$$

which proves (i).

Let us prove (ii). Let M' be the A-module defined by the exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M_G \longrightarrow 0.$$

One has $\mathrm{H}^1(G, M') = 0$ because G has pro-order prime to n. The exact cohomology sequence therefore implies that the arrow $M^G \to M_G$ is an epimorphism. But, applying the functor D and taking (i) into account, one deduces that it is a monomorphism, which completes the proof.

Applying (5.1.9) to M and P, one finds that $\underline{\mathrm{Hom}}(M, A)^P$ is canonically identified with $\underline{\mathrm{Hom}}(M^P, A)$. If one chooses an isomorphism from $\widehat{\mathbb{Z}}(1)$ to $\widehat{\mathbb{Z}}$, the pairing (5.1.8) is written, taking (5.1.5) into account, as

$$(5.1.8)' \qquad \mathrm{H}^i(\widehat{\mathbb{Z}}, M^P) \times \mathrm{H}^{1-i}(\widehat{\mathbb{Z}}, D(M^P)) \longrightarrow \mathrm{H}^1(\widehat{\mathbb{Z}}, A) \xrightarrow{\sim} A.$$

It follows immediately from (5.1.6) and (5.1.9(i)) that (5.1.8)' is a perfect pairing. This completes the proof of (5.1).

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Appendix

by L. Illusie

The preceding theory extends easily to base Rings more general than the constant Ring $\mathbb{Z}/n\mathbb{Z}$. We shall briefly indicate how the theory is written for locally constant base Rings, killed by $n > 0$, and whose geometric fibres are noetherian rings. This is, moreover, only a provisional framework[10]: sooner or later one will have to globalize, by taking as "coefficients" on a prescheme X relative preschemes over the site $(X_{\text{ét}}, \mathbb{Z}/n\mathbb{Z})$.

1. Constructible sheaves

1.1. Let X be a prescheme, and let A_X be a sheaf of rings on X (for the étale topology), locally constant, whose geometric fibres are left noetherian rings. We denote by ${}_AX$ the category of left A_X-Modules, and by $\mathrm{D}({}_AX)$ (or $\mathrm{D}(X)$ if there is no risk of confusion) the corresponding derived category. The notion of a constructible left A_X-Module is defined as in (SGAA IX 2.3), by replacing the expression "locally constant with value of finite presentation" by "of finite presentation as an A_{U_i}-Module".

We denote by $\mathrm{D}_c(X)$ the full subcategory of $\mathrm{D}(X)$ consisting of complexes E such that $\underline{\mathrm{H}}^i(E)$ is a constructible A_X-Module for every i. It is a triangulated subcategory of $\mathrm{D}(X)$.

1.2. Let $E \in \mathrm{ob}\,\mathrm{D}(X)$. We say (cf. Exp. II 3.11) that E is pseudo-coherent if, for every i, $\underline{\mathrm{H}}^i(E)$ is an A_X-Module of finite presentation (or locally constant constructible, if one prefers). We denote by $\mathrm{D}(X)_{\mathrm{coh}}$ (or $\mathrm{D}_{\mathrm{coh}}(X)$) the full subcategory of $\mathrm{D}(X)$ consisting of pseudo-coherent objects. It follows from (SGAA IX 2.1) that this is a triangulated subcategory of $\mathrm{D}_c(X)$.

Proposition 1.3 (dévissage lemma). Let X be a quasi-compact and quasi-separated prescheme (respectively a noetherian prescheme), and let C be a strictly full subcategory of the category $\mathrm{Cons}(X)$ of constructible left A_X-Modules. Suppose that C is stable under extensions. The following conditions are equivalent:

(i) $C = \mathrm{Cons}(X)$;

(ii) C contains the sheaves of the form $i_!(F)$, where $i : Y \to X$ is a sub-prescheme of finite presentation (respectively an integral sub-prescheme), and F is locally constant constructible on Y.

If A_X is constant with value A, a left noetherian ring killed by an integer $n > 0$, the preceding conditions are equivalent to:

(iii) C is stable under direct factors and contains the sheaves of the form $f_*i_!(M_U)$, where $i : U \to Y$ is an open immersion (respectively an open subset of an integral prescheme), $f : Y \to X$ is a finite morphism (respectively finite and such that the generic point of Y is separable over its image), and M_U is a left A_U-Module constant with value a monogenic A-module killed by a prime divisor of n.

[10] According to Grothendieck!

Finally, if X is noetherian and satisfies condition (reg) (I 3.1.1), condition (ii) (respectively (iii)) is equivalent to the analogous condition in which one further assumes Y (respectively U) regular.

Proof. The equivalence of conditions (i) and (ii) is proved as in the case where A_X is constant; see (SGAA IX 2.4 and 2.5).

Suppose that A_X is constant with value A and killed by n, and prove that (iii) implies (ii). Let $E \in \operatorname{ob} \operatorname{Cons}(X)$; we show that $E \in \operatorname{ob} C$. By the equivalence (i)$\Leftrightarrow$(ii), we may suppose that $E = j_! F$, where $j : Z \to X$ is the immersion of a sub-prescheme of finite presentation (respectively an integral one), and F is a constructible locally constant A_Z-Module, which we may even suppose, after a little more dévissage, to be trivialized by an étale covering $g : Z' \to Z$ with group G. The sheaf F is therefore defined by a finite type A-module M on which G acts. If ℓ is a prime divisor of n, the subgroup of ℓ-torsion in M is a sub-G-A-module M_ℓ of M, and

$$M = \bigoplus_\ell M_\ell,$$

where ℓ runs over the set of prime divisors of n. Since C is stable under direct factors, we are reduced to the case where M is ℓ-torsion. Let H be an ℓ-Sylow subgroup of G, and let $h : U = Z'/H \to Z$ be the corresponding intermediate covering. Since the degree of h is prime to ℓ, the "trace method" (SGAA IX 5.1) shows that the adjunction map

$$F \longrightarrow h_* h^*(F)$$

is an isomorphism onto a direct factor; it is therefore enough to prove that $j_! h_* h^*(F) \in \operatorname{ob} C$. By the "Main theorem" (EGA IV 8.12.6), there exists a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{i} & Y \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow[j]{} & X, \end{array}$$

where i is an open immersion and f is finite. We have

$$j_! h_* h^*(F) \simeq f_* i_! h^*(F).$$

But $h^*(F)$ is defined by M considered as an H-A-module. Thus (iii)$\Rightarrow$(ii) follows from the next lemma.

Lemma 1.3.1. Let A be a left noetherian ring, ℓ a prime number, H a finite ℓ-group, and M a left H-A-module of finite type killed by ℓ^ν ($\nu \geq 1$). There exists a finite filtration of M by sub-H-A-modules such that the successive quotients are monogenic A-modules killed by ℓ, and on which H acts trivially.

Proof. The exact sequence of H-A-modules

$$0 \longrightarrow \ell M \longrightarrow M \longrightarrow M/\ell M \longrightarrow 0$$

reduces, by induction on ν, to the case where M is killed by ℓ. We show that then M has a finite filtration by sub-H-A-modules on whose successive quotients H acts trivially. Arguing by noetherian induction on the set of sub-H-A-modules of M having the stated property, we are reduced to proving that $M \neq 0$ implies $M^H \neq 0$. But the kernel of the augmentation $\mathbb{F}_\ell[H] \to \mathbb{F}_\ell$ is plainly nilpotent, and Nakayama's lemma implies that every nonzero H-$\mathbb{F}_\ell$-module has a nonzero invariant vector.

This proves the assertion. Thus M has a finite filtration whose successive quotients are finite type A-modules killed by ℓ and on which H acts trivially. By noetherian induction one may even arrange for the successive quotients to be monogenic A-modules, completing the proof of (1.3.1).

The last assertion of (1.3) is immediate and left to the reader.

2. Quasi-injective dimension

With the notation of (1.1), one defines, as in (I 1.1 and 1.4), the notions of a quasi-injective A_X-Module and of quasi-injective dimension of an object of $\mathrm{D}(X)$. Propositions (I 1.3, 1.5 and 1.6) extend without change.

It is sometimes convenient to introduce a notion of pointwise quasi-injective dimension. Let $x \in X$ and let $K \in \mathrm{ob}\,\mathrm{D}^+(X)$. We say that K has quasi-injective dimension $\leq N$ at $\bar{x}$ if, for every constructible A_X-Module F, one has

$$\underline{\mathrm{Ext}}^i(F, K)_{\bar{x}} = 0 \qquad \text{for } i > N.$$

The quasi-injective dimension of K at $\bar{x}$, denoted $\dim.\mathrm{q.inj}_{\bar{x}}(K)$, is the greatest lower bound of the N satisfying the preceding condition. Of course,

$$\dim.\mathrm{q.inj}(K) = \sup_{\bar{x}} \dim.\mathrm{q.inj}_{\bar{x}}(K),$$

where $\bar{x}$ runs through the set of geometric points of X.

It is interesting to compare this with the notion of strict topological dimension introduced by Verdier in [1]. We say that $K \in \mathrm{ob}\,\mathrm{D}^+(X)$ has strict topological dimension $\leq N$ at a geometric point $\bar{x}$ of X if, for every sub-prescheme Y of X,

$$\underline{\mathrm{Ext}}^i(A_Y, K)_{\bar{x}} = \underline{\mathrm{H}}^i_Y(K)_{\bar{x}} = 0 \qquad \text{for } i > N.$$

The strict topological dimension of K at $\bar{x}$, denoted $\dim.\mathrm{top}_{\bar{x}}(K)$, is the greatest lower bound of the N satisfying the preceding condition. Whereas quasi-injective dimension depends a priori on the base Ring, strict topological dimension depends only on the underlying complex of abelian sheaves.

One plainly has

$$\dim.\mathrm{top}_{\bar{x}}(K) \leq \dim.\mathrm{q.inj}_{\bar{x}}(K).$$

We shall see below examples where equality holds.

3. Dualizing complexes: definition and formal properties

Keep the preceding notation, but suppose A_X is commutative. The notions of bidualizing pair and dualizing complex are defined as in (I 1.7). The statements (I 1.9, 1.11, 1.14) remain valid without modification. The statements (I 1.12, 1.13, 1.15) are also valid, but on condition that A be killed by an integer $n > 0$ prime to the residual characteristics of Y. Finally, the uniqueness theorem (I 2.1) is valid provided one assumes $\mathrm{Spec}(A_{\bar{x}})$ connected for every geometric point $\bar{x}$ of X.

4. Absolute purity

Return to the situation of absolute cohomological purity (I 3.1.4). Let X be a locally noetherian regular prescheme, let $n > 0$ be an integer prime to the residual characteristics of X, and let $i : Y \to X$ be a regular sub-prescheme of codimension $d > 0$ at every point. To say that the pair (Y, X) satisfies absolute cohomological purity means, recall, that the "local fundamental class" homomorphism (SGA $4^{1/2}$ Cycle 2.2)

$$(\mu_n^{\otimes -d})_Y[-2d] \longrightarrow \mathrm{R}i^!(\mathbb{Z}/n\mathbb{Z})_X \tag{4.1}$$

is an isomorphism.

Now put on X a sheaf of rings A_X killed by n. For the moment set $G = (\mathbb{Z}/n\mathbb{Z})_X$. Then, for every $F \in \mathrm{ob}\,\mathrm{D}_A^+(X)$, one can define a canonical arrow in $\mathrm{D}_A^+(X)$:

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G) \longrightarrow \mathrm{R}i^!(F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G). \tag{4.2}$$

Indeed, by duality (SGAA XVIII 3.1.6), this is equivalent to defining an arrow

$$i_!(i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G)) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G;$$

we take this to be the composite of the Künneth arrow (SGAA XVII 5.4.2.2)

$$i_!(i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G)) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} i_!\mathrm{R}i^!(G),$$

and the arrow deduced from the trace homomorphism (SGAA XVIII 3.1.6)

$$F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} i_!\mathrm{R}i^!(G) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G.$$

Notice that in the definition of (4.2) only the fact that G and $\mathrm{R}i^!(G)$ have finite Tor-dimension over $\mathbb{Z}/n\mathbb{Z}$ was used. Suppose on the other hand that the functor $i^!$ has finite cohomological dimension on the category of $(\mathbb{Z}/n\mathbb{Z})_X$-Modules: this is the case, for example, if X is of finite type over a field (SGAA XIV 3.1), or more generally if X is excellent of equal characteristic (SGAA XIX 6.1) when resolution of singularities is available, or again if X is of finite type over $\mathrm{Spec}(\mathbb{Z})$ (SGAA X 6). Then the arrow (4.2) is defined for all $F \in \mathrm{ob}\,\mathrm{D}({}_AX)$.

Composing (4.1) and (4.2), one obtains, for $F \in \mathrm{ob}\,\mathrm{D}_A^+(X)$, a canonical arrow

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n^{\otimes -d})_Y[-2d] \longrightarrow \mathrm{R}i^!(F). \tag{4.3}$$

Assume that A_X is locally constant, and let $F \in \mathrm{ob}\,\mathrm{D}_A^+(X)$ be such that every $\underline{\mathrm{H}}^i(F)$ is a locally constant A_X-Module. Then, if $i^!$ has finite cohomological dimension, the arrow (4.2), and hence the arrow (4.3) if (Y, X) satisfies purity, is an isomorphism.

Indeed, by dévissage, one first reduces to the case where F is concentrated in degree zero. Since the question is local, one may suppose X affine and A_X and F constant. Resolving F on the left by free A_X-Modules and applying the hypothesis that $i^!$ has finite cohomological dimension, one reduces to the case where F is free. By an immediate passage to the limit, one reduces to F free of finite type, and finally to $F = A_X$. It is therefore enough to prove the assertion when $A_X = (\mathbb{Z}/n\mathbb{Z})_X$. But the preceding dévissage reduces to $F = (\mathbb{Z}/n\mathbb{Z})_X$, and in that case (4.2) is the identity.

Definition 4.4. Given the Ring A_X (locally constant and killed by n), we shall say that the pair (Y, X) satisfies cohomological purity in the strong sense if, for every $F \in \mathrm{ob}\,\mathrm{D}_A^+(X)$ such that the $\underline{\mathrm{H}}^i(F)$ are locally constant, the canonical arrow (4.3) is an isomorphism.

We shall say that the purity theorem is true in the strong sense on X if, locally for the étale topology, every pair (Y, X) as above satisfies purity in the strong sense. From what has been shown, for (Y, X) to satisfy purity in the strong sense it is enough that (Y, X) satisfy ordinary purity (I 3.1.) and that the functor $i^!$ have finite cohomological dimension.

5. Bound for quasi-injective dimension

Proposition 5.1. Let X be a locally noetherian regular prescheme, equipped with a locally constant sheaf of rings A_X, killed by an integer $n > 0$ prime to the residual characteristics of X, whose geometric fibres are left noetherian rings. Suppose that X satisfies conditions (reg) (I 3.1.1) and $(\mathrm{cdloc})_n$ (I 3.1.3), and that the purity theorem in the strong sense is true on X (4.4) (conditions all satisfied if X is smooth over a perfect field, or if X is excellent of characteristic zero).

Let $F \in \mathrm{ob}\, \mathrm{D}_A^+(X)$ be such that the $\underline{\mathrm{H}}^i(F)$ are locally constant, and let x be a point of X such that $F_{\bar{x}} \neq 0$ and $\dim .\mathrm{inj}_{A_{\bar{x}}}(F_{\bar{x}}) < +\infty$. Then

$$\dim .\mathrm{q.inj}_{\bar{x}}(F) = 2\dim(\mathcal{O}_{X,x}) + \dim .\mathrm{inj}(F_{\bar{x}}),$$

with the notation of no. 2.

Proof. One merely repeats step by step the proof of (I 3.2.1). First show the inequality

$$\dim .\mathrm{q.inj}_{\bar{x}}(F) \leq 2\dim(\mathcal{O}_{X,x}) + \dim .\mathrm{inj}(F_{\bar{x}}) = N,$$

that is, the relation

$$\underline{\mathrm{Ext}}^i(E, F)_{\bar{x}} = 0 \quad \text{for } i > N \text{ and every constructible } A_X\text{-Module } E. \tag{*}$$

Since the question is local, by dévissage (1.3) one may restrict to checking $(*)$ for $E = i_!(M)$, where $i : Y \to X$ is an integral regular sub-prescheme closed in an open $D(f)$ (with $f \in \Gamma(X, \mathcal{O}_X)$), of codimension d at every point of $D(f)$, and M is a locally constant constructible A_Y-Module ($A_Y = i^* A_X$).

Now there is the duality isomorphism (SGAA XVIII)

$$\mathrm{R}\underline{\mathrm{Hom}}(i_!(M), F) \simeq \mathrm{R}i_* \mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F).$$

By the purity theorem, there is also the isomorphism (4.3)

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n^{\otimes -d})_Y[-2d] \xrightarrow{\sim} \mathrm{R}i^!(F).$$

Let $\bar{y}$ be a geometric point of Y. Since M is locally constant constructible, one has canonically

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)_{\bar{y}} \simeq \mathrm{R}\underline{\mathrm{Hom}}(M_{\bar{y}}, (\mathrm{R}i^! F)_{\bar{y}}),$$

hence, by (4.3),

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)_{\bar{y}} \simeq \mathrm{R}\underline{\mathrm{Hom}}(M_{\bar{y}}, F_{\bar{y}})[-2d].$$

After replacing X by a connected neighbourhood of x, one has $F_{\bar{x}} \simeq F_{\bar{y}}$ and $A_{\bar{x}} \simeq A_{\bar{y}}$, since A_X and the $\underline{\mathrm{H}}^i(F)$ are locally constant. Thus

$$\underline{\mathrm{Ext}}^i(M_{\bar{y}}, F_{\bar{y}}) = 0 \quad \text{for } i > \dim .\mathrm{inj}(F_{\bar{x}}),$$

and hence

$$\underline{\mathrm{Ext}}^q(M, \mathrm{R}i^!F) = 0 \quad \text{for } q > 2d + \dim.\mathrm{inj}(F_{\bar{x}}).$$

Set $G = \mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^!F)$ and compute $\mathrm{R}i_*(G)_{\bar{x}}$. Introduce, as in (I 3.2.1), the schematic closure Z of Y in X, giving a cartesian diagram of immersions

$$\begin{array}{ccc} Y & \longrightarrow & D(f) \\ {\scriptstyle j}\downarrow & & \downarrow \\ Z & \xrightarrow[k]{} & X. \end{array}$$

Then $\mathrm{R}i_*(G) \simeq k_*\mathrm{R}j_*(G)$. Assume of course that $x \in Z$, since otherwise $\mathrm{R}i_*(G) = 0$. Let $\bar{Z} = \mathrm{Spec}(\mathcal{O}_{Z,\bar{x}})$ be the strict localization of Z at $\bar{x}$, let $\bar{f}$ be the image of f in $\mathcal{O}_{Z,\bar{x}}$, let $\bar{Y} = \bar{Z} \times_Z Y = \bar{Z} \cap D(\bar{f})$, and let $\bar{G}$ be the inverse image of G on $\bar{Y}$. By (SGAA VIII 5.2),

$$\mathrm{R}i_*(G)_{\bar{x}} = \mathrm{R}j_*(G)_{\bar{x}} \simeq \mathrm{R}\Gamma(\bar{Y}, \bar{G}).$$

Condition $(\mathrm{cdloc})_n$ therefore implies, by a spectral sequence argument, that

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > 2d + \dim.\mathrm{inj}(F_{\bar{x}}) + \dim(\mathcal{O}_{Z,x}).$$

But (cf. the proof of I 3.2.1)

$$\dim(\mathcal{O}_{Z,x}) = \dim(\mathcal{O}_{X,x}) - d \geq 0,$$

so

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > d + \dim(\mathcal{O}_{X,x}) + \dim.\mathrm{inj}(F_{\bar{x}}),$$

and a fortiori for $p > 2\dim(\mathcal{O}_{X,x}) + \dim.\mathrm{inj}(F_{\bar{x}})$. This proves $(*)$.

It remains to prove that the bound $(*)$ is the best possible. Put $f = \dim.\mathrm{inj}(F_{\bar{x}})$, and choose a finite type $A_{\bar{x}}$-Module M' such that

$$\underline{\mathrm{Ext}}^f(M', F_{\bar{x}}) \neq 0.$$

Let Z be the integral closed sub-prescheme of X which is the closure of x, and let $Y = U \cap Z$ be a regular open subset of Z containing x and of codimension $d = \dim(\mathcal{O}_{X,x})$ at every point. After possibly shrinking Y, there exists a locally constant constructible A_Y-Module M such that $M_{\bar{x}} = M'$. Applying the purity theorem, one finds

$$\underline{\mathrm{Ext}}^{2d+f}(M, F|U)_{\bar{x}} \simeq \underline{\mathrm{Ext}}^f(M', F_{\bar{x}}) \neq 0,$$

which completes the proof.

If F is an A_X-Module such that $F_{\bar{x}}$ is injective, then in the last part of the proof one may take $M = A_Y$; hence:

Corollary 5.1.1. Under the hypotheses of (5.1), suppose that F is a locally constant A_X-Module such that $F_{\bar{x}}$ is injective. Then

$$\dim.\mathrm{q.inj}_{\bar{x}}(F) = \dim.\mathrm{top}_{\bar{x}}(F) = 2\dim(\mathcal{O}_{X,x}),$$

with the notation of no. 2.

Remarks 5.1.2. a) With the notation of (5.1), if $F_{\bar{x}} = 0$, then $F = 0$ in a neighbourhood of x, hence $\mathrm{R}\underline{\mathrm{Hom}}(E, F)_{\bar{x}} = 0$ for all $E \in \mathrm{ob}\,\mathrm{D}^-(X)$.

b) If $\dim.\mathrm{inj}(F_{\bar{x}}) = +\infty$, then $\dim.\mathrm{q.inj}_{\bar{x}}(F) = +\infty$; indeed, for every p there exists a finite type $A_{\bar{x}}$-module M' such that $\underline{\mathrm{Ext}}^p(M', F_{\bar{x}}) \neq 0$, hence a constructible A_X-Module M, locally constant in a neighbourhood of x, such that $\underline{\mathrm{Ext}}^p(M, F)_{\bar{x}} \neq 0$.

6. Constructibility of $\mathrm{R}\underline{\mathrm{Hom}}$

The equivalence of conditions a) of (I 3.3.1) remains valid when the Ring $(\mathbb{Z}/n\mathbb{Z})_X$ is replaced by a commutative Ring, locally constant, whose geometric fibres are noetherian rings: nothing has to be changed in the proof.

The preceding conditions are verified when one has resolution of singularities and purity in the strong sense (4.4), in particular if X has dimension ≤ 1, or if X is excellent of characteristic zero, or locally of finite type over a field and of dimension ≤ 2.

7. Existence of dualizing complexes

In this number, X denotes a prescheme equipped with a commutative sheaf of rings A_X, locally constant, whose geometric fibres are noetherian rings.

Lemma 7.1. If $F \in \mathrm{ob}\,\mathrm{D}^-_{\mathrm{coh}}(X)$ and $G \in \mathrm{ob}\,\mathrm{D}^+_{\mathrm{coh}}(X)$, then $\mathrm{R}\underline{\mathrm{Hom}}(F,G) \in \mathrm{ob}\,\mathrm{D}^+_{\mathrm{coh}}(X)$ (notation of (1.2)).

Proof. This is standard. One reduces, by a spectral sequence, to F and G concentrated in degree zero, and then argues as in (EGA 0_{III} 12.3.3).

Lemma 7.2. Let $g : X' \to X$ be a morphism of preschemes, and endow X' with the sheaf of rings $A_{X'} = g^*(A_X)$. Then, for $F \in \mathrm{ob}\,\mathrm{D}^-_{\mathrm{coh}}(X)$ and $G \in \mathrm{ob}\,\mathrm{D}^+(X)$, the canonical map

$$g^*\mathrm{R}\underline{\mathrm{Hom}}(F,G) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(g^*F, g^*G)$$

is an isomorphism.

Proof. This too is standard: one reduces by dévissage to $F = A_X$ (cf. EGA 0_{III} 12.3.5).

Remark 7.2.1. The preceding lemmas are special cases of more general statements appearing in (SGA 6 I).

Lemma 7.3. Suppose X is quasi-compact, and let $K \in \mathrm{ob}\,\mathrm{D}^+_{\mathrm{coh}}(X)$ be such that, for every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ is a dualizing complex in the sense of [H] V 2.1. Then, for every $F \in \mathrm{ob}\,\mathrm{D}^b_{\mathrm{coh}}(X)$, one has $\mathrm{R}\underline{\mathrm{Hom}}(F,K) \in \mathrm{ob}\,\mathrm{D}^b_{\mathrm{coh}}(X)$, and F is reflexive relative to K.

Proof. It suffices, of course, to check this for F concentrated in degree zero. By (7.1), $\mathrm{R}\underline{\mathrm{Hom}}(F,K) \in \mathrm{ob}\,\mathrm{D}^+_{\mathrm{coh}}(X)$. To prove boundedness, let $\bar{x}$ be a geometric point of X. By (7.2),

$$\mathrm{R}\underline{\mathrm{Hom}}(F,K)_{\bar{x}} \simeq \mathrm{R}\underline{\mathrm{Hom}}(F_{\bar{x}}, K_{\bar{x}}).$$

Putting $\dim.\mathrm{inj}(K_{\bar{x}}) = N(\bar{x})$, one obtains

$$\underline{\mathrm{Ext}}^i(F,K)_{\bar{x}} = 0 \quad \text{for } i > N(\bar{x}).$$

Since the sheaves $\underline{\mathrm{Ext}}^i(F,K)$ are locally constant, there is an open neighbourhood U of x such that $\underline{\mathrm{Ext}}^i(F,K)|U = 0$ for $i > N(\bar{x})$, and quasi-compactness of X gives the assertion. It remains to check that the canonical arrow

$$F \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(F,K),K)$$

is an isomorphism. It is enough to evaluate at a geometric point $\bar{x}$ of X. In the commutative

diagram

$$\begin{array}{ccc} F_{\bar{x}} & \longrightarrow & \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(F,K),K)_{\bar{x}} \\ {\scriptstyle \mathrm{Id}}\downarrow & & \downarrow \\ F_{\bar{x}} & \longrightarrow & \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(F_{\bar{x}},K_{\bar{x}}),K_{\bar{x}}), \end{array}$$

the right vertical arrow is an isomorphism by (7.2), and the bottom horizontal arrow is an isomorphism because $K_{\bar{x}}$ is dualizing; hence the top horizontal arrow is an isomorphism, q.e.d.

Lemma 7.4. Suppose X is a noetherian regular prescheme, A_X is killed by an integer $n > 0$ prime to the residual characteristics of X, and the purity theorem in the strong sense (4.4) is true on X.

Let D be a divisor with normal crossings on X (I 3.1.5 a)), and denote by $j : Y = V(D) \to X$ and $i : X - Y = U \to X$ the canonical inclusions. Let $K \in \mathrm{ob}\,\mathrm{D}^{+}_{\mathrm{coh}}(X)$ be such that, for every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ is a dualizing complex for the ring $A_{\bar{x}}$ (in the sense of [H] V 2.1). Then, for every $E \in \mathrm{ob}\,\mathrm{D}^{b}_{\mathrm{coh}}(X)$, one has $\mathrm{R}\underline{\mathrm{Hom}}(i_!i^*(E),K) \in \mathrm{ob}\,\mathrm{D}^{b}_{c}(X)$, and the pair $(i_!i^*(E),K)$ is bidualizing.

Proof. We may suppose E is concentrated in degree zero, i.e. a locally constant constructible A_X-Module. Since the question is local on X for the étale topology, we may suppose that (a) A_X is constant with value A, and E is constant with value E; and (b) D has strict normal crossings, say $D = \sum_{i\in I} \mathrm{div}(s_i)$ with $s_i \in \Gamma(X,\mathcal{O}_X)$.

Argue by induction on $p = \mathrm{card}(I)$. For $p = 0$, the assertions follow from (7.3). Suppose $I = \{1,\dots,p\}$, put $Y_i = V(s_i)$, $Y' = V(\sum_{1\le i\le p-1} \mathrm{div}(s_i))$, and $U' = X - Y'$. We get a cartesian diagram of immersions

$$\begin{array}{ccccc} \varnothing & \longrightarrow & U'\cap Y_p & \xrightarrow{i'_p} & Y_p \\ \downarrow & & {\scriptstyle j'_p}\downarrow & & \downarrow{\scriptstyle j_p} \\ U & \longrightarrow & U' & \xrightarrow[i']{} & X. \end{array}$$

Consider the exact sequence

$$0 \longrightarrow E_{U,X} \longrightarrow E_{U',X} \longrightarrow E_{U'\cap Y_p,X} \longrightarrow 0.$$

By the induction hypothesis the theorem is true for $(E_{U',X},K)$. For fixed K, the F such that $\mathrm{R}\underline{\mathrm{Hom}}(F,K) \in \mathrm{ob}\,\mathrm{D}^{b}_{c}(X)$ and (F,K) is bidualizing form a triangulated subcategory of $\mathrm{D}_c(X)$; hence we are reduced to the case $F = E_{U'\cap Y_p,X}$. But

$$E_{U'\cap Y_p,X} = j_{p*}i'_{p!}(E_{U'\cap Y_p}),$$

and, by (I 1.13), it is equivalent to prove the theorem for $(i'_{p!}(E_{U'\cap Y_p}),\mathrm{R}j^{!}_{p}(K))$ on Y_p. By purity, however,

$$\mathrm{R}j^{!}_{p}(K) \simeq j^{*}_{p}(K) \otimes^{L}_{\mathbb{Z}/n\mathbb{Z}} (\mu_n)^{\otimes -1}_{Y_p}[-2].$$

Thus $\mathrm{R}j^{!}_{p}(K) \in \mathrm{ob}\,\mathrm{D}^{+}_{\mathrm{coh}}(Y_p)$ and, for every geometric point $\bar{y}$ of Y_p, $(\mathrm{R}j^{!}_{p}(K))_{\bar{y}}$ is dualizing. The induction hypothesis then applies.

Theorem 7.5. Assume the following: X is a noetherian regular prescheme; A_X is killed by an integer $n > 0$ prime to the residual characteristics of X; X is strongly desingularizable (I 3.1.5), satisfies conditions (reg) (I 3.1.1) and $(\mathrm{cdloc})_n$ (I 3.1.3); and the purity theorem in the strong sense (4.4) is true on every prescheme smooth over X (conditions satisfied if X is excellent of characteristic zero, or of dimension ≤ 2 and smooth over a perfect field).

Then, if $K \in \mathrm{ob}\, \mathrm{D}^+_{\mathrm{coh}}(X)$ is such that $K_{\bar{x}}$ is dualizing (in the sense of [H] V 2.1) for every geometric point $\bar{x}$ of X, the complex K is dualizing.

Proof. We must show that K satisfies conditions (i) to (iii) of (I 1.7). For every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ has finite injective dimension, and since $K \in \mathrm{ob}\, \mathrm{D}^+_{\mathrm{coh}}(X)$, the function $x \mapsto \dim .\mathrm{inj}(K_{\bar{x}})$ is locally constant; it is therefore bounded because X is noetherian. It follows from (5.1) that K has finite quasi-injective dimension.

By what was seen in no. 6, condition (ii) of loc. cit. is also satisfied.

It remains to prove condition (iii), i.e. that every $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ is reflexive relative to K. By dévissage we may suppose F is concentrated in degree zero. Since the question is local on X for the étale topology, we may suppose A_X constant with value A. Dévissage (1.3) then reduces us to $F = f_* i_!(E_U)$, where $f : Y \to X$ is finite, Y is integral with generic point separable over its image, $i : U \to Y$ is a nonempty regular open, and E_U is an A_U-Module constant with value a finite type A-module E. Since X is strongly desingularizable, there exists (I 3.1.5) a projective birational morphism $h : Y' \to Y$ such that Y' is regular, h induces an isomorphism $h^{-1}(U) \to U$, and $h^{-1}(U)$ is the complement of a normal crossings divisor in Y'. Identifying $h^{-1}(U)$ with U via h, we have a commutative diagram

$$\begin{array}{ccccc} & & Y' & & \\ & \overset{i'}{\nearrow} & \downarrow{\scriptstyle h} & \overset{g}{\searrow} & \\ U & \xrightarrow[i]{} & Y & \xrightarrow[f]{} & X. \end{array}$$

Consequently,

$$f_* i_!(E_U) \simeq f_* \mathrm{R}h_* i'_!(E_U) \simeq \mathrm{R}g_* i'_!(E_U) \simeq \mathrm{R}g_* i'_! i'^*(E_{Y'}).$$

By (I 1.13), it is enough to prove biduality for the pair $(i'_! i'^*(E_{Y'}), \mathrm{R}g^!(K))$. But, locally embedding Y' into a prescheme X' smooth over X and applying strong purity on X' (4.4), one finds that $\mathrm{R}g^!(K)$ is locally isomorphic, up to degree shift, to $g^*(K)$. Hence $\mathrm{R}g^!(K) \in \mathrm{ob}\, \mathrm{D}^+_{\mathrm{coh}}(Y')$ and $\mathrm{R}g^!(K)_{\bar{x}}$ is dualizing for every geometric point $\bar{x}$ of Y'. It remains only to apply (7.4).

The preceding proof also shows that $\mathrm{R}\underline{\mathrm{Hom}}(F, K) \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ (using the finiteness theorem for a proper morphism (SGAA XIV)), which will reassure the reader who did not trust the spiel of no. 6.

This completes the proof.

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Bibliography

[1] Verdier, J.-L., *Dimension des espaces localement compacts*, Note aux C.R. Ac. Sc. Paris, 261, pp. 5293–5296 (1965). See also: Verdier, J.-L., *Dualité dans la cohomologie des espaces localement compacts*, Séminaire Bourbaki, no. 300, 1965–66.

SGA A = SGA 4.

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SGA 5 — Exposé III

LEFSCHETZ FORMULA

by A. Grothendieck, written up by L. Illusie

In this exposé we prove the Lefschetz formula in étale cohomology, for correspondences between complexes of sheaves, in Verdier's formulation (cf. [18]). In an earlier version, distributed by the IHES about ten years ago, the formula was stated, under resolution hypotheses, but the required compatibilities were not proved. Deligne's finiteness theorems (SGA $4^{1/2}$, Finiteness) made it possible to dispense with these hypotheses, provided one works with schemes separated and of finite type over a field. As for the compatibilities, the editor has verified them (painfully).

Let us say at once that the Lefschetz formula in question here, just like the trace formula for constant coefficients (SGA $4^{1/2}$, Cycle), which it generalizes, is essentially trivial: it only translates certain compatibilities in the formalism of duality. Roughly speaking, it says that, under certain conditions, the trace of an endomorphism of a hypercohomology complex $R\Gamma(X, L)$, where L is a complex of sheaves on X, can be calculated as an "integral" of certain cohomology classes on $X \times X$. In practice, the endomorphisms one considers are associated with correspondences $C \subset X \times X$, and the classes recovered on the product have support in $C \cap \Delta$, where Δ is the diagonal; these classes are "local in nature" in a neighbourhood of the fixed points, which in principle facilitates their calculation. See 4.2 and 4.7 for precise statements. In fact, the calculation of these "local" classes in terms of more comprehensible invariants, without which the present formula would have no interest, has so far been undertaken only in very special situations (see 4.10, 4.11, and the following exposé, where in particular the case of the Frobenius correspondence on a curve is treated).

For lack of a good derived category of ℓ-adic sheaves (Jouanolou's thesis unfortunately not having been published), we work systematically with torsion coefficients (prime to the residual characteristics). Despite minor technical difficulties in the definition of traces, this point of view in fact leads to finer formulae, also more manageable insofar as, for the calculation of local terms, one wants to trivialize the sheaves involved by coverings (see notably (SGA $4^{1/2}$, Report), III B, and XII).

In no. 1, we fix the notation and recall various Künneth formulae from (SGA $4^{1/2}$, Finiteness), which play an essential role in what follows. We also introduce a certain number of fibred and cofibred categories, whose consideration later facilitates certain verifications of compatibilities. No. 2, which is very technical, may be skipped on first reading. Its main result (2.5) expresses a compatibility between external tensor product of RHom, direct image, and base change. In no. 3, we define cohomological correspondences between complexes, and show how they act on hypercohomology; the result quoted above makes it possible to describe this action in several equivalent ways. The core of the exposé is no. 4, where Verdier's pairing between cohomological correspondences is defined and the Lefschetz formula is proved, in a relative situation appreciably more general than the one usually considered, with a view to later applications to computations of local terms. We give the most evident corollaries for correspondences with constant coefficients. In no. 5, we indicate rapidly how to generalize to correspondences between complexes certain usual constructions on correspondences with constant coefficients (cf. [12]), such as passage to the dual, composition, and external tensor product. Most of the verifications, analogous to those in the preceding numbers, have been omitted. Finally, in the appendix, we paraphrase the preceding theory in the context of coherent sheaves, using the formalism of duality of [8]. Here one may work over any noetherian base S, provided that, when necessary, one imposes hypotheses of properness, finite tor-dimension, and tor-independence on the S-schemes considered. We show that the very general Lefschetz formula thereby obtained contains as a special case the Woods Hole formula [1], and we point out some applications.

Contents

1. Notation and recalls of Künneth formulae.
2. Functoriality of RHom and external tensor products.
3. Cohomological correspondences.
4. Pairings of correspondences. Lefschetz formula.
5. Complements.
6. Appendix. Lefschetz formula for coherent sheaves.

2 Notation and recalls of Künneth formulae

1.1

We denote by S the spectrum of a field. Unless otherwise stated, the S-schemes considered in what follows will be S-schemes separated and of finite type.

We denote by A a commutative noetherian ring, annihilated by an integer invertible on S. If X is an S-scheme, we denote by $D(X) = D(X, A)$ the derived category of complexes of A-modules on X (for the étale topology), and by $D_{\mathrm{ctf}}(X)$ the full subcategory consisting of complexes of finite tor-dimension, with constructible cohomology. The category $D_{\mathrm{ctf}}(X)$ is a triangulated subcategory of $D(X)$. If $E, F \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, it follows readily from the definitions that $E \otimes^L F \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$. According to (SGA $4^{1/2}$, Finiteness Theorem 1.6, 1.7), one also has $\mathrm{RHom}(E, F) \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$.

1.2

Let $f : X \to Y$ be an S-morphism. One has trivially

$$f^* D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

and, by (loc. cit.),

$$Rf_* D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \qquad Rf_! D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \qquad Rf^! D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

(the inclusion relative to $Rf_!$ uses only the finiteness theorem for proper morphisms (SGA 4 XIV) and the sorites (SGA 4 XVII 5.2.11), and hence holds more generally for every compactifiable morphism whose fibres have bounded dimension). In what follows, we shall sometimes write f_*, $f_!$, $f^!$ instead of Rf_*, $Rf_!$, $Rf^!$; the hypotheses of 1.1 imply that these functors are defined on the whole derived category.

1.3

For $f : X \to S$, we shall put

$$K_X = f^! A,$$

and shall denote by D_X (or D) the functor $\mathrm{RHom}(-, K_X)$. According to (SGA $4^{1/2}$, Finiteness Theorem 4.3), the complex K_X is dualizing: for $E \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, one has canonically

$$E \xrightarrow{\sim} DDE.$$

1.4

Let $f : X \to Y$ be an S-morphism, $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)$. We put

$$\operatorname{Hom}_f^{(1)}(L, M) = \operatorname{Hom}(f^*M, L) \quad (\simeq \operatorname{Hom}(M, f_*L)),$$

$$\operatorname{Hom}_f^{(2)}(L, M) = \operatorname{Hom}(L, f^*M),$$

$$\operatorname{Hom}_f^{(3)}(L, M) = \operatorname{Hom}(f_!L, M) \quad (\overset{\text{(SGA 4 XVIII 3.1.5)}}{\simeq} \operatorname{Hom}(L, f^!M)),$$

$$\operatorname{Hom}_f^{(4)}(L, M) = \operatorname{Hom}(f^!M, L).$$

The elements of $\operatorname{Hom}_f^{(i)}(L, M)$ will be called *arrows of type i from L to M over f* (or f-arrows of type i from L to M). The transitivity isomorphisms for the functors f^*, f_*, $f_!$, $f^!$ allow one to compose arrows of type i for fixed i, and one thus obtains certain fibred or cofibred categories over the category of S-schemes: for example, for $i = 1$ (resp. $i = 3$) one obtains the fibred and cofibred category considered in (SGA 4 XVII 4.1.3) (resp. (SGA 4 XVIII 3.1.13)), whose fibre over X is the category opposite to $D(X)$ (resp. $D(X)$).

1.5

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, and $L_i \in \operatorname{ob} D^-(X_i)$. We put

$$L_1 \otimes_S^L L_2 = \operatorname{pr}_1^* L_1 \otimes^L \operatorname{pr}_2^* L_2,$$

where $\operatorname{pr}_i : X \to X_i$ is the canonical projection.

1.6

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \to Y = Y_1 \times_S Y_2,$$

$L_i \in \operatorname{ob} D^-(X_i)$, $M_i \in \operatorname{ob} D^-(Y_i)$. The evident canonical isomorphism

$$f_1^*M_1 \otimes_S^L f_2^*M_2 \xrightarrow{\sim} f^*(M_1 \otimes_S^L M_2) \tag{1.6.1}$$

allows one to define, for $u_i \in \operatorname{Hom}_{f_i}^{(1)}(L_i, M_i)$ (resp. $\operatorname{Hom}_{f_i}^{(2)}(L_i, M_i)$),

$$u_1 \otimes_S^L u_2 \in \operatorname{Hom}_f^{(1)}(L, M) \quad (\text{resp. } \operatorname{Hom}_f^{(2)}(L, M)), \tag{1.6.2}$$

where $L = L_1 \otimes_S^L L_2$, $M = M_1 \otimes_S^L M_2$. If u_i, v_i are composable, one has

$$(v_1u_1) \otimes_S^L (v_2u_2) = (v_1 \otimes_S^L v_2)(u_1 \otimes_S^L u_2). \tag{1.6.3}$$

Taking $M_i = f_{i*}L_i$, and u_i of type 1, given by the identity of $f_{i*}L_i$, 1.6.2 gives a *Künneth arrow* (cf. SGA 4 XVII 5.4.1.4)

$$f_{1*}L_1 \otimes_S^L f_{2*}L_2 \longrightarrow f_*(L_1 \otimes_S^L L_2). \tag{1.6.4}$$

According to (SGA $4^{1/2}$, Finiteness Theorem 1.9), 1.6.4 is an isomorphism: by 1.6.3 one is indeed reduced to the case where $X_2 = Y_2$, $f_2 = 1$, and one applies (loc. cit. App.). When f_1 and f_2 are proper, it is not necessary to invoke (loc. cit.): 1.6.4 is a special case of the Künneth isomorphism (SGA 4 XVII 5.4.3).

1.7

Choose a compactification $f_i = p_i j_i$, where j_i is an open immersion and p_i is a proper morphism. Composing the evident canonical isomorphism (where $j = j_1 \times_S j_2$)

$$j_{1!}L_1 \otimes_S^L j_{2!}L_2 \xrightarrow{\sim} j_!(L_1 \otimes_S^L L_2)$$

with the isomorphism 1.6.4 relative to $(p_i, j_{i!}L_i)$, one obtains a Künneth isomorphism (SGA 4 XVII 5.4.3)

$$f_{1!}L_1 \otimes_S^L f_{2!}L_2 \xrightarrow{\sim} f_!(L_1 \otimes_S^L L_2). \tag{1.7.1}$$

One checks easily (loc. cit.) that 1.7.1 does not depend on the compactifications chosen, and is compatible with the transitivity isomorphisms for composites $g_i f_i$ (cf. (loc. cit. 5.2.4)). Thanks to 1.7.1, one can define, for $u_i \in \mathrm{Hom}_{f_i}^{(3)}(L_i, M_i)$,

$$u_1 \otimes_S^L u_2 \in \mathrm{Hom}_f^{(3)}(L, M) \tag{1.7.2}$$

with L and M as in 1.6.2, and one has a transitivity formula analogous to 1.6.3.

For $L_i = f_i^! M_i$, and u_i given by the identity of L_i, 1.7.2 gives a *Künneth arrow*

$$f_1^! M_1 \otimes_S^L f_2^! M_2 \longrightarrow f^!(M_1 \otimes_S^L M_2). \tag{1.7.3}$$

Proposition 1.7.4

The arrow 1.7.3 is an isomorphism.

By the transitivity formula, it suffices to treat the case where $X_2 = Y_2$, $f_2 = 1$. By dévissage, one reduces to $M_2 = g_* j_! A_V$, where $j : V \to U$ is an open immersion and $g : U \to Y_2$ a finite morphism, and then, by (SGA 4 XVIII 3.1.12.3), to the case where $M_2 = A_{Y_2}$. By localization on X_2, one may suppose moreover that $f_1 = g_1 j_1$, where j_1 is a closed immersion and g_1 a smooth morphism purely of relative dimension n_1. By transitivity, it suffices to treat the cases

$$\text{a) } f_1 = g_1, \qquad \text{b) } f_1 = j_1.$$

In case a), the trace morphism supplies by adjunction (SGA 4 XVIII 3.2.5) an isomorphism

$$f_1^* M_1(n_1)[2n_1] \xrightarrow{\sim} f_1^! M_1;$$

this is compatible with the base change $Y_2 \to S$, because the trace morphism is so (SGA 4 XVIII 2.9), whence 1.7.4 in this case. In case b), if i_1 denotes the inclusion of the complementary open, one has an exact sequence

$$0 \longrightarrow j_{1*} R j_1^! M_1 \longrightarrow M_1 \longrightarrow R i_{1*} i_1^* M_1 \longrightarrow 0,$$

and the compatibility of Ri_{1*} with base change, a special case of 1.6.4, gives the conclusion.

Thanks to 1.7.3, one can define, for $u_i \in \mathrm{Hom}_f^{(4)}(L_i, M_i)$,

$$u_1 \otimes_S^L u_2 \in \mathrm{Hom}_f^{(4)}(L, M) \tag{1.7.5}$$

(L and M as in 1.6.2), with a transitivity formula analogous to 1.6.3.

On the other hand, with the notation of 1.3, 1.7.3 gives an isomorphism

$$K_{X_1} \otimes_S^L K_{X_2} \xrightarrow{\sim} K_X. \tag{1.7.6}$$

1.8

Let $f : X \to Y$ be an S-morphism, $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)$, S' an S-scheme, $P \in \operatorname{ob} D(S')$, and $f' : X' \to Y'$ the morphism deduced from f by the base change $S' \to S$. The identity arrow of P may be regarded indifferently as an arrow of type i $(1 \leq i \leq 4)$ over the identity arrow of S'; for $u \in \operatorname{Hom}_f^{(i)}(L, M)$ $(1 \leq i \leq 4)$, one may therefore, by 1.6.2, 1.7.2, 1.7.5, consider

$$u \otimes_S^L \operatorname{Id}_P \in \operatorname{Hom}_{f'}^{(i)}(L \otimes_S^L P, M \otimes_S^L P). \tag{1.8.1}$$

We shall sometimes write $u \otimes_S^L P$ instead of $u \otimes_S^L \operatorname{Id}_P$.

3 Functoriality of RHom and external tensor products

2.1

Let $f : X \to Y$ be an S-morphism, $P \in \operatorname{ob} D(Y)$, $Q \in \operatorname{ob} D^+(Y)$. The evident canonical arrow at the level of complexes

$$f^*\mathcal{H}om^\bullet(P, Q) \longrightarrow \mathcal{H}om^\bullet(f^*P, f^*Q)$$

gives by derivation a canonical arrow

$$f^*\mathrm{RHom}(P, Q) \longrightarrow \mathrm{RHom}(f^*P, f^*Q). \tag{2.1.1}$$

Let $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D^+(X)$. Put

$$\operatorname{Hom}_f^{(i,j)}((L, M), (P, Q)) = \operatorname{Hom}_f^{(i)}(L, P) \times \operatorname{Hom}_f^{(j)}(M, Q), \tag{2.1.2}$$

with the notation of 1.4. For $(u, v) \in \operatorname{Hom}_f^{(2,1)}((L, M), (P, Q))$ one defines

$$\mathrm{RHom}(u, v) \in \operatorname{Hom}_f^{(1)}(\mathrm{RHom}(L, M), \mathrm{RHom}(P, Q)) \tag{2.1.3}$$

by composing 2.1.1 with the arrow $\mathrm{RHom}(f^*P, f^*Q) \to \mathrm{RHom}(L, M)$ defined by u and v. For $L = f^*P$, $Q = f_*M$, u (resp. v) given by the identity of L (resp. Q), $\mathrm{RHom}(u, v)$ is the "trivial duality" isomorphism

$$\mathrm{RHom}(P, f_*M) \xrightarrow{\sim} f_*\mathrm{RHom}(f^*P, M). \tag{2.1.4}$$

For $(u, v) \in \operatorname{Hom}_f^{(3,4)}((L, M), (P, Q))$, one defines

$$\mathrm{RHom}(u, v) \in \operatorname{Hom}_f^{(1)}(\mathrm{RHom}(L, M), \mathrm{RHom}(P, Q)) \tag{2.1.5}$$

as the composite of the arrows

$$\mathrm{RHom}(P, Q) \xrightarrow{a} \mathrm{RHom}(f_!L, Q) \xrightarrow{b} f_*\mathrm{RHom}(L, f^!Q) \xrightarrow{c} f_*\mathrm{RHom}(L, M),$$

where a (resp. c) is defined by u (resp. v), and b is the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6).

It follows easily from the definitions that the formation of $\mathrm{RHom}(u, v)$ is compatible with composition: for composable (u, v), (u', v') of type $(2, 1)$ (resp. $(3, 4)$), one has

$$\mathrm{RHom}(u'u, v'v) = \mathrm{RHom}(u', v')\,\mathrm{RHom}(u, v). \tag{2.1.6}$$

2.2

Let X be an S-scheme. Recall that for $L \in \operatorname{ob} D^-(X)$, $M \in \operatorname{ob} D(X)$, $N \in \operatorname{ob} D^+(X)$, one has the adjunction isomorphism ("dear to Cartan," as Grothendieck says)

$$\operatorname{Hom}(L \otimes^L M, N) = \operatorname{Hom}(L, \operatorname{RHom}(M, N)). \tag{2.2.1}$$

For $E \in \operatorname{ob} D^-(X)$, $F \in \operatorname{ob} D^+(X)$, the identity arrow of $\operatorname{RHom}(E, F)$ defines via 2.2.1 an arrow

$$E \otimes^L \operatorname{RHom}(E, F) \longrightarrow F, \tag{2.2.2}$$

called the *evaluation arrow*: it is derived from the arrow at the level of complexes

$$\mathcal{H}om^\bullet(E, F) \otimes E \longrightarrow F, \qquad f \otimes x \longmapsto f(x).$$

Let, for $i = 1, 2$, $E_i, F_i \in \operatorname{ob} D_{\mathrm{ctf}}(X)$. By adjunction (2.2.1), the arrow

$$E_1 \otimes^L E_2 \otimes^L \operatorname{RHom}(E_1, F_1) \otimes^L \operatorname{RHom}(E_2, F_2) \longrightarrow F_1 \otimes^L F_2,$$

composed of a symmetry isomorphism and of the tensor product of the evaluation arrows relative to (E_1, F_1) and (E_2, F_2), supplies an arrow

$$\operatorname{RHom}(E_1, F_1) \otimes^L \operatorname{RHom}(E_2, F_2) \longrightarrow \operatorname{RHom}(E_1 \otimes^L E_2, F_1 \otimes^L F_2). \tag{2.2.3}$$

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the canonical projection, $E_i, F_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, $E = E_1 \otimes_S^L E_2$, $F = F_1 \otimes_S^L F_2$. One has a canonical arrow

$$\operatorname{RHom}(E_1, F_1) \otimes_S^L \operatorname{RHom}(E_2, F_2) \longrightarrow \operatorname{RHom}(E, F), \tag{2.2.4}$$

obtained by composing the tensor product of the arrows $p_i^* \operatorname{RHom}(E_i, F_i) \to \operatorname{RHom}(p_i^* E_i, p_i^* F_i)$ and 2.2.3 for $(p_i^* E_i, p_i^* F_i)$.

Proposition 2.3

The arrow 2.2.4 is an isomorphism.

By localization and dévissage, one reduces to $E_i = f_{i!} A_{U_i}$ for $f_i : U_i \to X_i$ étale. One has canonical isomorphisms (duality)

$$\operatorname{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} f_{i*} f_i^* F_i, \qquad \operatorname{RHom}(E, F) \xrightarrow{\sim} f_* f^* F \quad (f = f_1 \times_S f_2). \tag{1}$$

Taking into account the Künneth formula 1.6.4, it suffices, to conclude, to verify that these isomorphisms make the square

$$\begin{array}{ccc} \operatorname{RHom}(f_{1!} A_{U_1}, F_1) \otimes_S^L \operatorname{RHom}(f_{2!} A_{U_2}, F_2) & \xrightarrow{2.2.3} & \operatorname{RHom}(f_! A_U, F) \\ \downarrow \sim & & \downarrow \sim \\ f_{1*} f_1^* F_1 \otimes_S^L f_{2*} f_2^* F_2 & \xrightarrow[1.6.4]{} & f_* f^* F \end{array} \tag{2}$$

commute, where $U = U_1 \times_S U_2$. Since f_i is étale, the isomorphism (1) relative to f_i (resp. f) is adjoint to the composite arrow

$$f_i^* \operatorname{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} \operatorname{RHom}(f_i^* f_{i!} A_{U_i}, f_i^* F_i) \longrightarrow f_i^* F_i, \tag{3}$$

where the second arrow is deduced from the adjunction $A_{U_i} \to f_i^* f_{i!} A_{U_i}$ (resp. the analogous composite arrow with f). The commutativity of (2) is equivalent to that of the square

$$\begin{array}{ccc} f_1^*\mathrm{RHom}(f_{1!}A_{U_1}, F_1) \otimes_S^L f_2^*\mathrm{RHom}(f_{2!}A_{U_2}, F_2) & \longrightarrow & f^*\mathrm{RHom}(f_!A_U, F) \\ \downarrow & & \downarrow \\ f_1^*F_1 \otimes_S^L f_2^*F_2 & \xrightarrow[\sim]{} & f^*F, \end{array} \tag{4}$$

where the vertical arrows are given by (3). But the commutativity of (4) follows from the fact that 2.2.3 is functorial in E_i and reduces to the identity for $E_i = A$, whence the proposition.

2.4

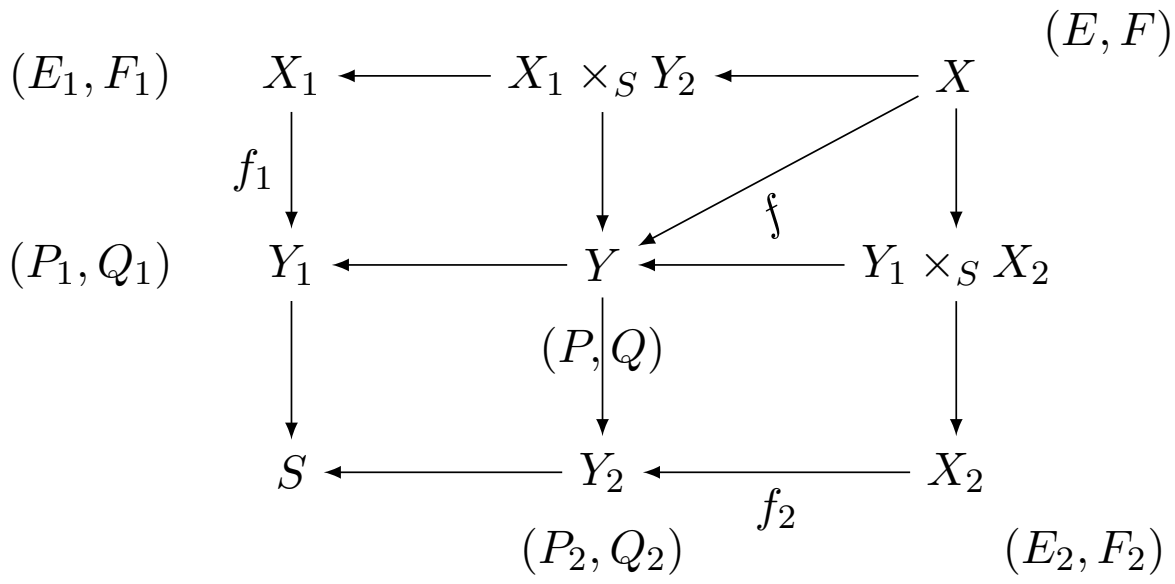
Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X \to Y,$$

$E_i, F_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$, $P_i, Q_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y_i)$,

$$E = E_1 \otimes_S^L E_2, \quad F = F_1 \otimes_S^L F_2, \quad P = P_1 \otimes_S^L P_2, \quad Q = Q_1 \otimes_S^L Q_2.$$

Consider the diagram



(2.4.0)

Give ourselves $(u_1, v_1) \in \mathrm{Hom}_{f_1}^{(3,4)}((E_1, F_1), (P_1, Q_1))$, $(u_2, v_2) \in \mathrm{Hom}_{f_2}^{(2,1)}((E_2, F_2), (P_2, Q_2))$ (notation of 2.1.2 and 1.4). By 1.8, one deduces from (u_i, v_i) a square

$$\begin{array}{ccc} (E, F) & \xleftarrow{(E_1 \otimes_S^L u_2,\, F_1 \otimes_S^L v_2)} & (E_1 \otimes_S^L P_2, F_1 \otimes_S^L Q_2) \\ \downarrow{\scriptstyle (u_1 \otimes_S^L E_2,\, v_1 \otimes_S^L F_2)} & & \downarrow{\scriptstyle (u_1 \otimes_S^L P_2,\, v_1 \otimes_S^L Q_2)} \\ (P_1 \otimes_S^L E_2, Q_1 \otimes_S^L F_2) & \xrightarrow[(P_1 \otimes_S^L u_2,\, Q_1 \otimes_S^L v_2)]{} & (P, Q), \end{array} \tag{2.4.1}$$

where the vertical arrows (resp. horizontal arrows) are of type $(3,4)$ (resp. $(2,1)$).

We may now state the main compatibility of this number, which will play an essential role in the verification of the Lefschetz–Verdier formula.

Proposition 2.5

The diagram 2.5.1 below, where the square on the right is deduced from 2.4.1 by applying RHom in the sense of 2.1.3 and 2.1.5, is commutative:

$$\begin{array}{ccccc}
\mathrm{RHom}(E_1,F_1)\otimes_S^L \mathrm{RHom}(E_2,F_2) & & \xrightarrow{2.2.4} & & \mathrm{RHom}(E,F) \\
\downarrow & & & \swarrow \quad \searrow & \\
\mathrm{RHom}(u_1,v_1)\otimes_S^L \mathrm{RHom}(u_2,v_2) & & \mathrm{RHom}(E_1\otimes_S^L P_2, F_1\otimes_S^L Q_2) & & \mathrm{RHom}(P_1\otimes_S^L E_2, Q_1\otimes_S^L F_2) \\
\downarrow & & & \searrow \quad \swarrow & \\
\mathrm{RHom}(P_1,Q_1)\otimes_S^L \mathrm{RHom}(P_2,Q_2) & & \xrightarrow[2.2.4]{} & & \mathrm{RHom}(P,Q)
\end{array} \tag{2.5.1}$$

2.6

The proof of 2.5 will occupy the rest of no. 2. It follows from 1.6.3 that, if $w_i : L_i \to M_i$ is an arrow of type 1 over f_i, the arrow $w = w_1 \otimes_S^L w_2$ makes the diagram

$$\begin{array}{ccccc}
 & & L_1\otimes_S^L L_2 & & \\
 & \swarrow^{L_1\otimes_S^L w_2} & & \searrow^{w_1\otimes_S^L L_2} & \\
L_1\otimes_S^L M_2 & & & & M_1\otimes_S^L L_2 \\
 & \searrow_{w_1\otimes_S^L M_2} & & \swarrow_{M_1\otimes_S^L w_2} & \\
 & & M_1\otimes_S^L M_2 & &
\end{array}$$

commute. Applying this remark to $w_i = \mathrm{RHom}(u_i, v_i)$, one sees that it suffices to prove the following special cases of 2.5.

Corollary 2.6.1

Let $f : X \to Y$ be an S-morphism, S' an S-scheme, $f' : X' \to Y'$ the morphism deduced from f by the base change $S' \to S$, $E, F \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, $P, Q \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$, $P', Q' \in \mathrm{ob}\, D_{\mathrm{ctf}}(S')$, and $(u,v) \in \mathrm{Hom}_f^{(2,1)}((E,F),(P,Q))$ (resp. $\mathrm{Hom}_f^{(3,4)}((E,F),(P,Q))$). Then one has a commutative square, whose vertical arrows are of type 1 over f':

$$\begin{array}{ccc}
\mathrm{RHom}(E,F)\otimes_S^L \mathrm{RHom}(P',Q') & \xrightarrow{2.2.4} & \mathrm{RHom}(E\otimes_S^L P', F\otimes_S^L Q') \\
\downarrow{\scriptstyle \mathrm{RHom}(u,v)\otimes_S^L \mathrm{RHom}(P',Q')} & & \downarrow{\scriptstyle \mathrm{RHom}(u\otimes_S^L P', v\otimes_S^L Q')} \\
\mathrm{RHom}(P,Q)\otimes_S^L \mathrm{RHom}(P',Q') & \xrightarrow[2.2.4]{} & \mathrm{RHom}(P\otimes_S^L P', Q\otimes_S^L Q').
\end{array} \tag{2.6.1.1}$$

First examine the case where (u, v) is of type $(2, 1)$. By functoriality, one may suppose that $E = f^*P$, $F = f^*Q$, (u, v) being given by the identity of (f^*P, f^*Q). The commutativity of 2.6.1.1 in the case $P' = Q' = A_{S'}$ is evident; to establish it in the general case, it therefore suffices to verify that, for $L, M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$, the square

$$\begin{array}{ccc} f^*\mathrm{RHom}(E,F) \otimes^L f^*\mathrm{RHom}(L,M) & \xrightarrow{f^*(2.2.3)} & f^*\mathrm{RHom}(E \otimes^L L, F \otimes^L M) \\ \downarrow{\scriptstyle 2.1.1 \otimes 2.1.1} & & \downarrow{\scriptstyle 2.1.1} \\ \mathrm{RHom}(f^*E, f^*F) \otimes^L \mathrm{RHom}(f^*L, f^*M) & \xrightarrow[2.2.3]{} & \mathrm{RHom}(f^*(E \otimes^L L), f^*(F \otimes^L M)) \end{array}$$

is commutative. But one sees immediately that the two composites of this square are adjoint to the tensor product of the evaluation arrows

$$f^*E \otimes^L f^*\mathrm{RHom}(E,F) \longrightarrow f^*F \quad \text{and} \quad f^*L \otimes^L f^*\mathrm{RHom}(L,M) \longrightarrow f^*M,$$

which proves the assertion.

Now pass to the case where (u, v) is of type $(3, 4)$. By functoriality, one may suppose that $P = f_!E$, $F = f^!Q$, (u, v) being defined by $(\mathrm{Id}_P, \mathrm{Id}_F)$. One must prove the commutativity of the square

$$\begin{array}{ccc} f_*\mathrm{RHom}(E, f^!Q) \otimes^L_S \mathrm{RHom}(P', Q') & \xrightarrow{b} & f'_*\mathrm{RHom}(E \otimes^L_S P', f^!Q \otimes^L_S Q') \\ \downarrow{\scriptstyle a \otimes^L_S \mathrm{RHom}(P',Q')} & & \downarrow{\scriptstyle a'} \\ \mathrm{RHom}(f_!E, Q) \otimes^L_S \mathrm{RHom}(P', Q') & \xrightarrow[2.2.4]{} & \mathrm{RHom}(f_!E \otimes^L_S P', Q \otimes^L_S Q'), \end{array}$$

where b is the composite of $f'_*(2.2.4)$ and a base-change isomorphism, a is the duality isomorphism (SGA 4 XVIII 3.1.9.6), and a' is the composite of the duality isomorphism (relative to $E \otimes^L_S P'$ and $Q \otimes^L_S Q'$) and the base-change isomorphisms

$$f_!E \otimes^L_S P' \xrightarrow[\sim]{c} f'_!(E \otimes^L_S P'), \qquad f^!Q \otimes^L_S Q' \xrightarrow[\sim]{d} f'^!(Q \otimes^L_S Q').$$

Recall that a is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_*\mathrm{RHom}(E, f^!Q) \longrightarrow \mathrm{RHom}(f_!E, f_!f^!Q)$$

and of the arrow defined by the adjunction arrow $f_!f^!Q \to Q$. There is an analogous description of a', as the composite of the quoted arrow

$$f'_*\mathrm{RHom}(E \otimes^L_S P', f^!Q \otimes^L_S Q') \longrightarrow \mathrm{RHom}(f'_!(E \otimes^L_S P'), f'_!(f^!Q \otimes^L_S Q')), \qquad (*)$$

of the arrows defined by c and d, and of the arrow defined by the adjunction arrow

$$f'_!f'^!(Q \otimes^L_S Q') \longrightarrow Q \otimes^L_S Q'.$$

By definition, d is adjoint to the arrow

$$f'_!(f^!Q \otimes^L_S Q') \longrightarrow Q \otimes^L_S Q'$$

defined by the base-change isomorphism $f'_!(f^!Q \otimes^L_S Q') \simeq f_! f^! Q \otimes^L_S Q'$ and the adjunction arrow $f_! f^! Q \to Q$; in other words, the square

$$\begin{array}{ccc} f'_!(f^!Q \otimes^L_S Q') & \xrightarrow{f'_! d} & f'_! f'^!(Q \otimes^L_S Q') \\ {\scriptstyle \text{base ch.}\ \sim}\downarrow & & \downarrow{\scriptstyle \text{adj}} \\ f_! f^! Q \otimes^L_S Q' & \xrightarrow[\text{adj}\otimes 1]{} & Q \otimes^L_S Q' \end{array}$$

is commutative. It follows that one may also describe a' as the composite of (*), of the arrow defined by c, and of those defined by the base-change isomorphism

$$f'_!(f^!Q \otimes^L_S Q') \xrightarrow{\sim} f_! f^! Q \otimes^L_S Q'$$

and the adjunction arrow $f_! f^! Q \to Q$. This decomposition of a' and that of a given above, together with the commutativity of the square

$$\begin{array}{ccc} \mathrm{RHom}(f_!E, f_! f^! Q) \otimes^L_S \mathrm{RHom}(P', Q') & \longrightarrow & \mathrm{RHom}(f_!E \otimes^L_S P', f_! f^! Q \otimes^L_S Q') \\ \downarrow & & \downarrow \\ \mathrm{RHom}(f_!E, Q) \otimes^L_S \mathrm{RHom}(P', Q') & \longrightarrow & \mathrm{RHom}(f_!E \otimes^L_S P', Q \otimes^L_S Q'), \end{array}$$

defined by 2.2.4 and the adjunction arrow $f_! f^! Q \to Q$, show that, to establish the commutativity of 2.6.2, it suffices to prove the following compatibility.

Lemma 2.6.3

Let f, f', E, F, P', Q' be as in 2.6.1. Then one has a commutative square

$$\begin{array}{ccc} f_*\mathrm{RHom}(E, F) \otimes^L_S \mathrm{RHom}(P', Q') & \xrightarrow{(1)} & f'_*\mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \\ {\scriptstyle (2)}\downarrow & & \downarrow{\scriptstyle (4)} \\ \mathrm{RHom}(f_!E, f_!F) \otimes^L_S \mathrm{RHom}(P', Q') & \xrightarrow[(3)]{} & \mathrm{RHom}(f_!E \otimes^L_S P', f_!F \otimes^L_S Q'), \end{array}$$

where (1) is the composite of the base-change isomorphism

$$f_*\mathrm{RHom}(E, F) \otimes^L_S - \xrightarrow{\sim} f'_*(\mathrm{RHom}(E, F) \otimes^L_S -)$$

and f'_*(2.2.4), (2) is deduced from the canonical arrow

$$f_*\mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_!E, f_!F)$$

(SGA 4 XVIII 3.1.9.5) by applying $\otimes^L_S \mathrm{RHom}(P', Q')$, (3) is given by 2.2.4, and finally (4) is the composite of the canonical arrow

$$f'_*\mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \longrightarrow \mathrm{RHom}(f'_!(E \otimes^L_S P'), f'_!(F \otimes^L_S Q'))$$

(loc. cit.) and the arrows defined by the base-change isomorphisms

$$f'_!(E \otimes_S^L P') \xrightarrow{\sim} f_!E \otimes_S^L P', \qquad f'_!(F \otimes_S^L Q') \xrightarrow{\sim} f_!F \otimes_S^L Q'.$$

Returning to the definition of the arrow (SGA 4 XVIII 3.1.9.5), choose a compactification of f[11]; one is then reduced to proving the lemma separately in the case where f is an open immersion and in the case where f is a proper morphism.

a) Case where f is an open immersion. By adjunction (2.2.1), it suffices to see that, if one tensors the square 2.6.3 by $f_!E \otimes_S^L P'$ and composes with the evaluation arrow 2.2.2

$$(f_!E \otimes_S^L P') \otimes^L \mathrm{RHom}(f_!E \otimes_S^L P', f_!F \otimes_S^L Q') \longrightarrow f_!F \otimes_S^L Q',$$

the square obtained is commutative. But the sums of this square are zero outside X', so that it suffices to verify that the square induced on X' commutes, which is trivial.

b) Case where f is a proper morphism. Then $f_! = f_*$, $f'_! = f'_*$. One sees easily that the composite of

$$f_*\mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_*E, f_*F)$$

with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E, f_*F) \xrightarrow{\sim} f_*\mathrm{RHom}(f^*f_*E, F)$$

(2.1.4) is none other than the arrow deduced from the adjunction arrow $f^*f_*E \to E$ by applying $f_*\mathrm{RHom}(-, F)$. Similarly, the composite of (4) with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E \otimes_S^L P', f_*F \otimes_S^L Q') \xrightarrow{\sim} f'_*\mathrm{RHom}(f^*f_*E \otimes_S^L P', F \otimes_S^L Q')$$

($f_*F \otimes_S^L Q'$ being identified with $f'_*(F \otimes_S^L Q')$ by the canonical isomorphism) is none other than the arrow deduced from the adjunction arrow $f^*f_*E \to E$ by applying $f'_*\mathrm{RHom}(- \otimes_S^L P', F \otimes_S^L Q')$. Thus one is reduced to verifying the commutativity of the square

$$\begin{array}{ccc} \mathrm{RHom}(f_*E, f_*F) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow{(3)} & \mathrm{RHom}(f_*E \otimes_S^L P', f_*F \otimes_S^L Q') \\ \downarrow & & \downarrow \\ f_*\mathrm{RHom}(f^*f_*E, F) \otimes_S^L \mathrm{RHom}(P', Q') & \xrightarrow[(1')]{} & f'_*\mathrm{RHom}(f^*f_*E \otimes_S^L P', F \otimes_S^L Q'), \end{array}$$

where the vertical arrows are the trivial-duality isomorphisms and $(1')$ is (1) with E replaced by f^*f_*E. But this is a special case of 2.6.1.1 in the case, already treated, where (u, v) is of type $(2, 1)$ (take $u : f^*f_*E \to f_*E$ given by the identity of f^*f_*E, and $v : F \to f_*F$ given by the identity of f_*F). We have therefore proved 2.6.3. As we have seen, the commutativity of 2.6.2 follows, which completes the proof of 2.6.1 and, finally, of 2.5.

[11] The arrow quoted is defined with the aid of such a compactification, but it does not depend on it: this point, tacitly admitted in (loc. cit.), is easily verified.

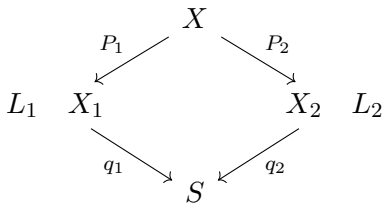
4 Cohomological Correspondences

3.1

Let, for $i = 1, 2$, X_i be an S-scheme and $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$, and let

$$X = X_1 \times_S X_2, \qquad p_i : X \longrightarrow X_i, \qquad q_i : X_i \longrightarrow S$$

be the canonical projections:



With the notation of 1.3, one has a canonical isomorphism (2.2.4)

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2).$$

Composing with the Künneth isomorphism

$$q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} p_2^! L_2 \qquad (1.7.3),$$

one obtains an isomorphism

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \mathrm{RHom}(p_1^* L_1, p_2^! L_2). \tag{3.1.1}$$

This isomorphism will play a fundamental role throughout. For $X_i = S$, 3.1.1 reduces to

$$L_1^\vee \overset{\mathbf{L}}{\otimes} L_2 \xrightarrow{\sim} \mathrm{RHom}(L_1, L_2),$$

a special case of an isomorphism valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.7). We set

$$\mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}_S(L_1, L_2), \qquad \mathrm{Hom}(p_1^* L_1, p_2^! L_2) = \mathrm{Hom}_S(L_1, L_2). \tag{3.1.2}$$

The elements of $\mathrm{Hom}_S(L_1, L_2)$ will be called *cohomological correspondences from L_1 to L_2*. Instead of "cohomological correspondence from A_{X_1} to A_{X_2}", we shall simply say *cohomological correspondence from X_1 to X_2* (understood: with coefficients in A).

If X_1 is *smooth*, purely of relative dimension r_1, then the same is true of p_2, and by the duality theorem (SGA 4 XVIII 3.2.5) one has

$$p_2^! L_2 \xrightarrow{\sim} p_2^* L_2(r_1)[2r_1],$$

therefore

$$\mathrm{RHom}_S(L_1, L_2) = \mathrm{RHom}(p_1^* L_1, p_2^* L_2)(r_1)[2r_1], \qquad \mathrm{Hom}_S(L_1, L_2) = \mathrm{H}^{2r_1}(X, \mathrm{RHom}(p_1^* L_1, p_2^* L_2)(r_1)). \tag{3.1.3}$$

In particular, the group of cohomological correspondences from X_1 to X_2 is identified in this case with $\mathrm{H}^{2r_1}(X, A(r_1))$.

3.2

The cohomological correspondences met in practice are associated with S-morphisms $X_2 \to X_1$, or more generally with cycles on X, or with schemes over X. Let

$$c : C \longrightarrow X$$

be a scheme over X (one sometimes says that c is a correspondence between X_1 and X_2), and put $c_i = p_i c : C \to X_i$ for the projections. We shall call *cohomological correspondences from L_1 to L_2 with support in c* (or in C, if there is no risk of confusion) the elements of the group

$$\mathrm{H}^0(X, c_!\mathrm{RHom}(c_1^*L_1, c_2^!L_2)) \quad (= \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \text{ if } c \text{ is proper}).$$

By the induction formula (SGA 4 XVIII 3.1.12.2), one has a canonical isomorphism

$$c^!\mathrm{RHom}(p_1^*L_1, p_2^!L_2) \xrightarrow{\sim} \mathrm{RHom}(c_1^*L_1, c_2^!L_2), \tag{3.2.1}$$

whence, applying $c_!$ and composing with the adjunction arrow $c_!c^! \to \mathrm{Id}$, canonical arrows

$$c_!\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \longrightarrow \mathrm{RHom}(p_1^*L_1, p_2^!L_2), \tag{3.2.2}$$

$$\mathrm{H}^0(X, c_!\mathrm{RHom}(c_1^*L_1, c_2^!L_2)) \longrightarrow \mathrm{Hom}_S(L_1, L_2). \tag{3.2.3}$$

Thus every correspondence from L_1 to L_2 with support in C canonically defines, by 3.2.3, a correspondence (without support) from L_1 to L_2.

Examples. a) Suppose that c is proper and that c_2 has finite tor-dimension and relative dimension $\leq N$. The trace morphism (SGA $4^{1/2}$ Cycle 2.3.3)

$$\mathrm{Tr}_{c_2} : c_{2!}A_C \longrightarrow A_{X_2}(-N)[-2N]$$

defines by adjunction a cohomological correspondence from A_{X_1} to $A_{X_2}(-N)[2N]$ with support in C, which will be called the *class of* (c, N), and will be denoted

$$\mathrm{cl}(c, N) \in \mathrm{H}^{-2N}(C, c_2^!A_{X_2}(-N)). \tag{3.2.4}$$

The most important case is that in which $N = 0$, i.e. c_2 is finite (and of finite tor-dimension); in that case $\mathrm{cl}(c)$ is a cohomological correspondence from X_1 to X_2 with support in C. If X_1 is smooth, purely of relative dimension r_1, then, by 3.1.3 and 3.2.1,

$$\mathrm{H}^{-2N}(C, c_2^!A_{X_2}(-N)) = \mathrm{H}^{2(r_1-N)}(C, c^!A_X(r_1 - N)). \tag{3.2.5}$$

If moreover c is a closed immersion, the second member rewrites as

$$\mathrm{H}_C^{2(r_1-N)}(X, A_X(r_1 - N)),$$

and the class of (c, N), considered as an element of this group, is precisely the cohomology class associated with c in the sense of (SGA $4^{1/2}$ Cycle 2.3.1), as follows immediately from the definitions.

b) Let $F : X_2 \to X_1$ be an S-morphism, and denote by $c(F) : C(F) \to X$ its graph, i.e. the image of the section of p_2 defined by F ("the set of points (Fx, x)"): c_2 is an isomorphism and $c_1c_2^{-1} = F$. One therefore has canonical isomorphisms

$$\mathrm{RHom}(c_1(F)^*L_1, c_2(F)^!L_2) = \mathrm{RHom}(F^*L_1, L_2), \qquad \mathrm{Hom}(c_1(F)^*L_1, c_2(F)^!L_2) = \mathrm{Hom}(F^*L_1, L_2). \tag{3.2.6}$$

In other words, the cohomological correspondences from L_1 to L_2 with support in the graph of F are the "lifts" of F to a homomorphism $F^*L_1 \to L_2$. Here are two important examples of such correspondences:

(i) $X_1 = X_2$, $L_1 = L_2$, $F = 1$ is the identity of X_1, whose graph Δ is the diagonal. The correspondence with support in Δ defined by the identity of L_1 is called the *diagonal correspondence*. It is the element

$$1 \in \mathrm{Hom}(L_1, L_1) = \mathrm{H}^0(X, \Delta_*\Delta^!\mathrm{RHom}_S(L_1, L_1))$$

(a group that is identified with $\mathrm{H}^{2r}_\Delta(X, \mathrm{RHom}_S(L_1, L_1)(r))$ when X_1 is smooth, purely of relative dimension r).

(ii) Suppose that S is the spectrum of a finite field $\mathbb{F}_q$, that $X_1 = X_2$, $L_1 = L_2$, and take for $F : X_1 \to X_1$ the relative Frobenius endomorphism over $\mathbb{F}_q$, the identity on the underlying space and raising to the q-th power on $\mathcal{O}_{X_1}$ (cf. XIV), and take as lift of F to L_1 the canonical isomorphism

$$F^*L_1 \xrightarrow{\sim} L_1$$

(loc. cit. §2, Prop. 1) (more precisely, the a-th iterate of the isomorphism of (loc. cit.) if $q = p^a$). The cohomological correspondence from L_1 to L_1 with support in the graph of F so defined is called the *Frobenius correspondence* (relative to $\mathbb{F}_q$).

c) Certain cohomological correspondences between sheaves on curves, related to Hecke operators, have been studied by Langlands in ([13] §7).

d) Cohomological correspondences between constant sheaves (the most frequent ones in nature) furnish in a natural way, by "integration", cohomological correspondences between complexes, as we shall now see.

3.3

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism, and put

$$f = f_1 \times_S f_2 : X \to Y, \qquad f_1' = f_1 \times_S Y_2, \qquad f_2' = Y_1 \times_S f_2,$$
$$p_1' = X_1 \times_S f_2, \qquad p_2' = f_1 \times_S X_2.$$

$$\begin{array}{ccc} X_1 \times_S Y_2 & \xleftarrow{p_1'} & X \\ {\scriptstyle f_1'}\downarrow & \swarrow_{f} & \downarrow{\scriptstyle p_2'} \\ Y & \xleftarrow[f_2']{} & Y_1 \times_S X_2 \end{array} \tag{3.3.0}$$

Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. There exists a unique isomorphism

$$f_*\mathrm{RHom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2) \tag{3.3.1}$$

whose inverse makes the square

$$\begin{array}{ccc} D(f_{1!}L_1) \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2 & \xrightarrow{3.1.1} & \mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2) \\ {\scriptstyle (1)}\downarrow & & \downarrow \\ f_{1*}DL_1 \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2 & \xrightarrow[(2)]{} & f_*\mathrm{RHom}_S(L_1, L_2), \end{array}$$

commutative, where (1) is deduced from the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6)

$$Df_{1!}L_1 \xrightarrow{\sim} f_{1*}DL_1$$

by applying $- \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2$, and (2) is the composite of the Künneth isomorphism

$$f_{1*}DL_1 \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2 \xrightarrow{\sim} f_*(DL_1 \overset{\mathbf{L}}{\otimes}_S L_2)$$

and $f_*(3.1.1)$. Applying $R\Gamma(Y, -)$ to 3.3.1, one obtains an isomorphism

$$f_* : \mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_S(f_{1!}L_1, f_{2*}L_2). \tag{3.3.2}$$

For $u \in \mathrm{Hom}_S(L_1, L_2)$, the correspondence f_*u will be called the *direct image* of u by f. When $Y_1 = Y_2 = S$, we shall write u_* rather than f_*u. If $Y_1 = Y_2 = S$ is the spectrum of a separably closed field, 3.3.2 reads

$$\mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2)). \tag{3.3.3}$$

Thus cohomological correspondences "operate" on cohomology. The following statement will allow us to interpret this operation in several ways, and in particular to relate it to the usual definitions in situations such as a) or b) above.

Proposition 3.4. With the notation of 3.3.0, one has a commutative diagram of isomorphisms in $D_{\mathrm{ctf}}(Y)$:

$$\begin{array}{ccccc} & & \mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2) & & \\ & \overset{(1)}{\swarrow} & \downarrow (5) & \overset{(2)}{\searrow} & \\ f'_{1*}\mathrm{RHom}_S(L_1, f_{2*}L_2) & & & & f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2) \\ & {\scriptstyle f'_{1*}(3)}\searrow & \downarrow & \swarrow{\scriptstyle f'_{2*}(4)} & \\ & & f_*\mathrm{RHom}_S(L_1, L_2) & & \end{array} \tag{3.4.1}$$

where (5) is the inverse of 3.3.1 and the arrows (1) to (4) are defined by the commutative diagrams below; in these diagrams "base ch." denotes an arrow deduced from a base-change or Künneth isomorphism of one of the types considered in no. 1, "dual." an arrow deduced from a duality isomorphism of type 2.1.4 or (SGA 4 XVIII 3.1.9.6), and "triv." a trivial isomorphism:

$$\begin{array}{ccc} \mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) & \xrightarrow{\text{base ch.}} & \mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2) \\ \downarrow{\scriptstyle \text{base ch.}} & & \\ \mathrm{RHom}(f'_{1!}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) & & \downarrow{\scriptstyle (1)} \\ \downarrow{\scriptstyle \text{dual.}} & & \\ f'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, f_1^{\prime !}(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2)) & & \\ \downarrow{\scriptstyle \text{base ch.}} & & \\ f'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) & \xrightarrow[\text{base ch.}]{} & f'_{1*}\mathrm{RHom}_S(L_1, f_{2*}L_2), \end{array}$$

$$
\begin{array}{ccc}
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) & \xrightarrow{\text{base ch.}} & \mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2) \\
\downarrow{\scriptstyle \text{base ch.}} & & \\
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, f'_{2*}(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2)) & & \\
\downarrow{\scriptstyle \text{dual.}} & & \\
f'_{2*}\mathrm{RHom}(f_2'^{*}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & & \\
\downarrow{\scriptstyle \text{triv.}} & & \\
f'_{2*}\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow[\text{base ch.}]{} & f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2),
\end{array}
\qquad (2)
$$

$$
\begin{array}{ccc}
\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) & \xrightarrow{\text{base ch.}} & \mathrm{RHom}_S(L_1, f_{2*}L_2) \\
\downarrow{\scriptstyle \text{base ch.}} & & \\
\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, p'_{1*}(K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2)) & & \\
\downarrow{\scriptstyle \text{dual.}} & & \\
p'_{1*}\mathrm{RHom}(p_1'^{*}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}), K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) & & \\
\downarrow{\scriptstyle \text{triv.}} & & \\
p'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow[\text{base ch.}]{} & p'_{1*}\mathrm{RHom}_S(L_1, L_2),
\end{array}
\qquad (3)
$$

$$
\begin{array}{ccc}
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow{\text{base ch.}} & \mathrm{RHom}_S(f_{1!}L_1, L_2) \\
\downarrow{\scriptstyle \text{base ch.}} & & \\
\mathrm{RHom}(p'_{2!}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & & \\
\downarrow{\scriptstyle \text{dual.}} & & \\
p'_{2*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, p_2'^{!}(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2)) & & \\
\downarrow{\scriptstyle \text{base ch.}} & & \\
p'_{2*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow[\text{base ch.}]{} & p'_{2*}\mathrm{RHom}_S(L_1, L_2).
\end{array}
\qquad (4)
$$

This is only an explication of 2.5 in the particular case where $(u_1, v_1) : (L_1, K_{X_1}) \to (f_{1!}L_1, K_{Y_1})$ is given by the identity of $f_{1!}L_1$ and the canonical isomorphism $f_1^! K_{Y_1} \xrightarrow{\sim} K_{X_1}$, and where $(u_2, v_2) : (A_{X_2}, L_2) \to (A_{Y_2}, f_{2*}L_2)$ is given by the canonical isomorphism $A_{X_2} \xrightarrow{\sim} f_2^! A_{Y_2}$ and the identity of $f_{2*}L_2$.

Corollary 3.5. Assume that $Y_1 = Y_2 = S$. Let $u \in \mathrm{Hom}_S(L_1, L_2)$. The direct image $f_*u \in \mathrm{Hom}(f_{1!}L_1, f_{2*}L_2)$ (3.3.2) can be obtained in the following two equivalent ways:

a) Identify u, by trivial duality, with an arrow

$$L_1 \longrightarrow p_{1*}p_2^! L_2;$$

compose with the base-change isomorphism

$$p_{1*}p_2^!L_2 \xrightarrow{\sim} f_1^! f_{2*}L_2;$$

f_*u is the adjoint of this composite arrow.

b) Identify u, by adjunction, with an arrow

$$p_{2!}p_1^*L_1 \longrightarrow L_2;$$

compose with the base-change isomorphism

$$f_2^* f_{1!}L_1 \xrightarrow{\sim} p_{2!}p_1^*L_1;$$

f_*u is the adjoint of this composite arrow.

Description a) (respectively b)) indeed corresponds to the path $(f_{1*}(3) \circ (1))^{-1}$ (respectively $f_{2*}(4) \circ (2)$) in 3.4.1.

3.6

In practice, it is mainly description 3.5 b) that is used. Let us spell it out at the level of cohomology classes, assuming, in order to lighten the notation, that S is the spectrum of a separably closed field. Let

$$u \in \mathrm{Hom}_S(L_1, L_2) = \mathrm{Hom}(p_1^*L_1, p_2^!L_2).$$

For $x \in \mathrm{H}_c^i(X_1, L_1)$, $u_*(x) \in \mathrm{H}^i(X_2, L_2)$ is defined as follows: consider the element $p_1^*x \in \mathrm{H}^i(X_2, p_{2!}p_1^*L_1)$ deduced from x by inverse image, apply u to it, i.e. form the product

$$u \cdot p_1^*x \in \mathrm{H}^i(X_2, p_{2!}p_2^!L_2),$$

then apply the arrow

$$a : \mathrm{H}^i(X_2, p_{2!}p_2^!L_2) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

defined by the adjunction arrow $p_{2!}p_2^! \to \mathrm{Id}$; in short:

$$u_*(x) = a(u \cdot p_1^*x). \tag{3.6.1}$$

Let $c : C \to X$ be a proper morphism, and let $u \in \mathrm{Hom}(c_1^*L_1, c_2^!L_2)$. Denote by

$$u_* \in \mathrm{Hom}\big(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2)\big)$$

the image of u under the composite of 3.2.3 and 3.3.3. It follows from 3.6.1 that, for $x \in \mathrm{H}_c^i(X_1, L_1)$, one has

$$u_*(x) = a(u \cdot c_1^*x), \tag{3.6.2}$$

where c_1^*x is the inverse image of x in $\mathrm{H}^i(X_2, c_{2!}c_1^*L_1)$ and

$$a : \mathrm{H}^i(X_2, c_{2!}c_2^!L_2) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

is the arrow deduced from the adjunction $c_{2!}c_2^!L_2 \to \mathrm{Id}$. In the situation of Example 3.2 a), one thus finds

$$\mathrm{cl}(c, N)_*(x) = \mathrm{Tr}_{c_2}(c_1^*x), \tag{3.6.3}$$

where

$$\mathrm{Tr}_{c_2} : \mathrm{H}^i(X_2, Rc_{2!}A_C) \longrightarrow \mathrm{H}^{i-2N}(X_2, A(-N))$$

is the arrow induced by the trace morphism relative to c_2. For $N = 0$, this formula agrees with (SGA $4^{1/2}$ Cycle 3.6): $\mathrm{cl}(c)_*$ is the homomorphism denoted η_{X_2,X_1} there. If X_1 is smooth, purely of relative dimension r_1, and c is a closed immersion, it follows from 3.6.1 and from the remark made at the end of 3.2 a) that one also has

$$\mathrm{cl}(c, N)_*(x) = \mathrm{Tr}_{p_2}(\mathrm{cl}(c)\, p_1^* x), \tag{3.6.4}$$

where $\mathrm{cl}(C) \in \mathrm{H}_C^{2(r_1-N)}(X, A(r_1 - N))$ is the cohomology class of C and

$$\mathrm{Tr}_{p_2} : \mathrm{H}^*(X_2, Rp_{2!}A_X) \longrightarrow \mathrm{H}^{*-2r_1}(X_2, A(-r_1))$$

is the arrow defined by the trace morphism relative to p_2. This recovers the usual definition of the homomorphism induced on cohomology by a correspondence ([12] 1.3), (SGA $4^{1/2}$ Cycle 3.2).

3.6.5

In the situation of 3.2 b), it follows from 3.6.2 that, for $u \in \mathrm{Hom}(F^*L_1, L_2)$,

$$u_* : \mathrm{H}_c^i(X_1, L_1) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

is the composite of the "inverse image" homomorphism

$$\mathrm{H}_c^i(X_1, L_1) \longrightarrow \mathrm{H}^i(X_2, F^*L_1)$$

and the homomorphism induced by u (more generally, $f_*u : Rf_{1!}L_1 \to Rf_{2*}L_2$ is the composite of the canonical homomorphism

$$Rf_{1!}L_1 \longrightarrow Rf_{2*}F^*L_1$$

and of $Rf_{2*}u$). In particular, the diagonal correspondence induces the canonical arrow

$$Rf_{1!}L_1 \longrightarrow Rf_{1*}L_1$$

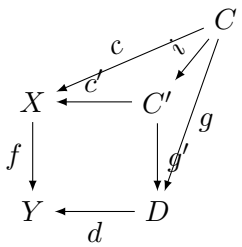
(the identity if X_1 is proper over S).

3.7

Return to the situation of the beginning of 3.3, assuming in addition that a commutative square

$$\begin{array}{ccc} X & \xleftarrow{c} & C \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ Y & \xleftarrow[d]{} & D \end{array} \tag{3.7.1}$$

is given, with c proper. Put $c_i = p_i c$, $d_i = q_i d$, where $p_i : X \to X_i$, $q_i : Y \to Y_i$ are the projections. From 3.7.1 one deduces a commutative diagram whose inner square is cartesian:



(3.7.2)

For $K \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, there is a canonical arrow

$$f_* c_! c^! K \longrightarrow d_* d^! f_* K, \tag{3.7.3}$$

which is the composite of the arrow

$$f_* c_! c^! K = f_* c'_* (i_* i^!) c'^! K \longrightarrow f_* c'_* c'^! K = d_* g'_* c'^! K$$

defined by the adjunction arrow $i_* i^! \to \mathrm{Id}$, and the isomorphism

$$d_* g'_* c'^! K \xrightarrow{\sim} d_* d^! f_* K$$

defined by the canonical isomorphism $g'_* d'^! \xrightarrow{\sim} d^! f_*$ (SGA 4 XVIII 3.1.12.3). Taking $K = \mathrm{RHom}_S(L_1, L_2)$, and composing 3.7.3 with the isomorphism deduced from 3.3.1, one obtains a canonical arrow

$$f_* c_* c^! \mathrm{RHom}_S(L_1, L_2) \longrightarrow d_* d^! \mathrm{RHom}_S(f_{1!} L_1, f_{2*} L_2), \tag{3.7.4}$$

which, by 3.2.1, may be rewritten as

$$f_* c_! \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow d_* \mathrm{RHom}(d_1^* (f_{1!} L_1), d_2^! (f_{2*} L_2)). \tag{3.7.5}$$

One deduces

$$f_* : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \longrightarrow \mathrm{Hom}(d_1^* (f_{1!} L_1), d_2^! (f_{2*} L_2)). \tag{3.7.6}$$

The arrow 3.7.5 (respectively 3.7.6) generalizes 3.3.1 (respectively 3.3.2), with which it coincides when c and d are the identity. Also note that, for c proper, 3.2.2 (respectively 3.2.3) is nothing other than 3.7.5 (respectively 3.7.6) when f_1, f_2, and d are the identity arrows. When $Y_1 = Y_2 = S = D$, we shall abbreviate $f_* u$ as u_*, as above.

5 Pairings of Correspondences. Lefschetz Formula

4.1

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the projections, $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, and K_{X_i} the dualizing complex (1.3). The canonical arrows

$$p_i^* \mathrm{RHom}(L_i, K_{X_i}) \overset{\mathbf{L}}{\otimes} p_i^* L_i \longrightarrow p_i^* K_{X_i}, \tag{4.1.1}$$

composed from 2.1.1 and the evaluation arrow 2.2.2, give, by tensor product on X and composition with the Künneth isomorphism 1.7.6, a pairing

$$(DL_1 \overset{\mathbf{L}}{\otimes}_S L_2) \overset{\mathbf{L}}{\otimes} (L_1 \overset{\mathbf{L}}{\otimes}_S DL_2) \longrightarrow K_X, \tag{4.1.2}$$

which may be rewritten, by 3.1.1 and 3.1.2, as

$$\mathrm{RHom}_S(L_1, L_2) \overset{\mathbf{L}}{\otimes} \mathrm{RHom}_S(L_2, L_1) \longrightarrow K_X. \tag{4.1.3}$$

One deduces a pairing, called cup product,

$$\langle\ ,\ \rangle : \mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_2, L_1) \longrightarrow \mathrm{H}^0(X, K_X). \tag{4.1.4}$$

One may also regard 4.1.2 as the external tensor product of the evaluation arrows

$$DL_i \overset{\mathbf{L}}{\otimes} L_i \longrightarrow K_{X_i}.$$

In view of the Künneth isomorphism 2.2.4, one thus sees that 4.1.2 is a "perfect duality", i.e. identifies by the adjunction isomorphism 2.2.1 each of the complexes $DL_1 \overset{\mathbf{L}}{\otimes}_S L_2$ and $L_1 \overset{\mathbf{L}}{\otimes}_S DL_2$ with the dual of the other (with values in K_X).

When $X_1 = X_2 = S$, 4.1.3 is a special case of a pairing valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.8), and, with the notation of (SGA 6 I 8.3), one has

$$\langle u, v\rangle = \mathrm{Tr}(vu) = \mathrm{Tr}(uv). \tag{4.1.5}$$

4.2

Let $c : C \to X$, $d : D \to X$, and $e = c \times_X d : E = C \times_X D \to X$. For $P, Q \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, one has a canonical arrow (which is not in general an isomorphism)

$$c^!P \overset{\mathbf{L}}{\otimes}_X d^!Q \longrightarrow e^!(P \overset{\mathbf{L}}{\otimes} Q), \tag{4.2.1}$$

defined, in the same way as 1.7.3, as the adjoint of the composite arrow

$$e_!(c^!P \overset{\mathbf{L}}{\otimes}_X d^!Q) \underset{(1)}{\xrightarrow{\sim}} c_!c^!P \overset{\mathbf{L}}{\otimes} d_!d^!Q \xrightarrow{(2)} P \overset{\mathbf{L}}{\otimes} Q,$$

where (1) is the Künneth isomorphism (SGA 4 XVII 5.4.2.2) and (2) is the tensor product of the adjunction arrows. Applying $e_!$ to 4.2.1 and composing with the Künneth isomorphism just cited, one obtains a canonical arrow

$$c_!c^!P \overset{\mathbf{L}}{\otimes} d_!d^!Q \longrightarrow e_!e^!(P \overset{\mathbf{L}}{\otimes} Q). \tag{4.2.2}$$

Similarly, composing the Künneth arrow

$$c_*c^!P \overset{\mathbf{L}}{\otimes} d_*d^!Q \longrightarrow e_*(c^!P \overset{\mathbf{L}}{\otimes}_X d^!Q)$$

with the arrow deduced from 4.2.1 by applying e_*, one obtains a canonical arrow

$$c_*c^!P \overset{\mathbf{L}}{\otimes} d_*d^!Q \longrightarrow e_*e^!(P \overset{\mathbf{L}}{\otimes} Q). \tag{4.2.3}$$

For $P = DL_1 \overset{\mathbf{L}}{\otimes}_S L_2$, $Q = L_1 \overset{\mathbf{L}}{\otimes}_S DL_2$, one deduces from 4.2.2 and 4.2.3, by composition with 4.1.2 and the isomorphism $e^!K_X = K_E$, and with the identifications 3.1.1 and 3.2.1, pairings

$$c_!\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \overset{\mathbf{L}}{\otimes} d_!\mathrm{RHom}(d_2^*L_2, d_1^!L_1) \longrightarrow e_!K_E, \tag{4.2.4}$$

$$c_*\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \overset{\mathbf{L}}{\otimes} d_*\mathrm{RHom}(d_2^*L_2, d_1^!L_1) \longrightarrow e_*K_E, \tag{4.2.4'}$$

$$\langle\ ,\ \rangle : \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \otimes \mathrm{Hom}(d_2^*L_2, d_1^!L_1) \longrightarrow \mathrm{H}^0(E, K_E). \tag{4.2.5}$$

These pairings are compatible with étale localization: given a commutative square (not necessarily cartesian)

if c', d', e' denote the composites $C' \to C \to X$, $D' \to D \to X$, $E' \to E \to X$, one has a commutative square.

$$\begin{array}{ccc} \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(d_2^* L_2, d_1^! L_1) & \xrightarrow{4.2.5} & \mathrm{H}^0(E, K_E) \\ \downarrow & & \downarrow \\ \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(d_2'^* L_2, d_1'^! L_1) & \longrightarrow & \mathrm{H}^0(E', K_{E'}) \end{array} \tag{4.2.6}$$

where the vertical arrows are the restrictions, and the lower horizontal arrow is the composite of the arrow analogous to 4.2.5 and the restriction

$$\mathrm{H}^0(E'', K_{E''}) \longrightarrow \mathrm{H}^0(E', K_{E'}),$$

with $E'' = C' \times_{X'} D'$ (there are analogous commutative squares with arrows of type 4.2.4, 4.2.4', which we leave to the reader to write down). If Z is étale over E, we shall denote, for

$$u \in \mathrm{Hom}(c_1^* L_1, c_2^! L_2), \qquad v \in \mathrm{Hom}(d_2^* L_2, d_1^! L_1),$$
$$\langle u, v \rangle_Z \in \mathrm{H}^0(Z, K_Z) \tag{4.2.7}$$

the image of $\langle u, v \rangle$ in $\mathrm{H}^0(Z, K_Z)$ by restriction (in practice, Z will be a connected component of E). If z is an isolated closed point of E, and $\bar{z}$ a geometric point above z, it follows from 4.2.6 that

$$\langle u, v \rangle_z \in \mathrm{H}^0(z, K_z) = A$$

depends only on the situation deduced by strict localization at the images of $\bar{z}$ in C, D, X.

Example 4.3. Assume that X_i is smooth, purely of relative dimension r_i, and that c and d are closed immersions, with c_2 (respectively d_1) finite. For $L_1 = A_{X_1}$, $L_2 = A_{X_2}$, we have, by 3.1.3,

$$DL_1 \otimes_S^{\mathbf{L}} L_2 = A_X(r_1)[2r_1], \qquad L_1 \otimes_S^{\mathbf{L}} DL_2 = A_X(r_2)[2r_2],$$

and 4.1.2 is identified with the product

$$A_X(r_1)[2r_1] \otimes^{\mathbf{L}} A_X(r_2)[2r_2] \longrightarrow A_X(r)[2r],$$

where $r = r_1 + r_2$, and hence, taking 3.2.1 into account, 4.2.5 is identified with the usual cup product

$$\mathrm{H}^{2r_1}_C(X, A(r_1)) \otimes \mathrm{H}^{2r_2}_D(X, A(r_2)) \longrightarrow \mathrm{H}^{2r}_{C \cap D}(X, A(r)).$$

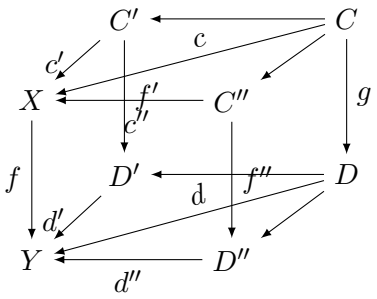
Take as cohomological correspondence from X_1 to X_2 (respectively from X_2 to X_1) the class of c (respectively d) (3.2 a)). It follows from (SGA $4^{1/2}$ Cycle 2.3.8) that, at an isolated point z of $C \cap D$, $\langle \mathrm{cl}(c), \mathrm{cl}(d) \rangle_z$ is nothing but the image in A of the intersection multiplicity of C and D at z.

We now give the main result of the exposé.

Theorem 4.4. Let, for $i = 1, 2$, $f_i : X_i \longrightarrow Y_i$ be a proper S-morphism,

$$f = f_1 \times_S f_2 : X \longrightarrow Y, \qquad p_i : X \longrightarrow X_i, \qquad q_i : Y \longrightarrow Y_i$$

the projections. Let moreover



(4.4.0)

be a commutative cube with cartesian horizontal faces, with c', c'', d', d'' proper. Put

$$c'_i = p_i c', \qquad c''_i = p_i c'', \qquad d'_i = q_i d', \qquad d''_i = q_i d''.$$

Finally, let $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$. Then there is a commutative square

$$\begin{array}{ccc} f_* c'_* \mathrm{RHom}({c'_1}^* L_1, {c'_2}^! L_2) \otimes^{\mathbf{L}} f_* c''_* \mathrm{RHom}({c''_2}^* L_2, {c''_1}^! L_1) & \xrightarrow{(1)} & f_* c_* K_C \\ \downarrow{\scriptstyle (2)} & & \downarrow{\scriptstyle (4)} \\ {d'_*}\mathrm{RHom}({d'_1}^* M_1, {d'_2}^! M_2) \otimes^{\mathbf{L}} {d''_*}\mathrm{RHom}({d''_2}^* M_2, {d''_1}^! M_1) & \xrightarrow[(3)]{} & d_* K_D, \end{array} \tag{4.4.1}$$

where $M_i = f_{i*} L_i$, (3) is given by 4.2.4, (1) is deduced from 4.2.4 by applying f_* and composing with the canonical arrow

$$f_* c'_*(-) \otimes^{\mathbf{L}} f_* c''_*(-) \longrightarrow f_*(c'_*(-) \otimes^{\mathbf{L}} c''_*(-)),$$

(2) is the tensor product of the arrows 3.7.5 relative to the faces containing f, and (4) is defined by the adjunction arrow $g_* K_C = g_! g^! K_D \longrightarrow K_D$.

Proof. Put

$$DL_1 \otimes^{\mathbf{L}}_S L_2 = P, \qquad L_1 \otimes^{\mathbf{L}}_S DL_2 = Q, \qquad DM_1 \otimes^{\mathbf{L}}_S M_2 = E, \qquad M_1 \otimes^{\mathbf{L}}_S DM_2 = F.$$

Consider the diagram

$$\begin{array}{ccccccc} f_* c'_* {c'}^! P \otimes^{\mathbf{L}} f_* c''_* {c''}^! Q & & \xrightarrow{(1)} & & f_* c_* c^! \big(P \otimes^{\mathbf{L}} Q\big) & \xrightarrow{(2)} & f_* c_* c^! K_X \\ \downarrow{\scriptstyle (8)} & & & & \downarrow{\scriptstyle (9)} & & \downarrow{\scriptstyle (10)} \\ {d'_*}{d'}^! f_* P \otimes^{\mathbf{L}} {d''_*}{d''}^! f_* Q & \xrightarrow[(3)]{} & d_* d^! \big(f_* P \otimes^{\mathbf{L}} f_* Q\big) & \xrightarrow[(4)]{} & d_* d^! f_* \big(P \otimes^{\mathbf{L}} Q\big) & \xrightarrow[(5)]{} & d_* d^! f_* K_X \\ \downarrow{\scriptstyle (11)} & & \downarrow{\scriptstyle (12)} & & & & \downarrow{\scriptstyle (13)} \\ {d'_*}{d'}^! E \otimes^{\mathbf{L}} {d''_*}{d''}^! F & \xrightarrow[(6)]{} & d_* d^! \big(E \otimes^{\mathbf{L}} F\big) & & \xrightarrow[(7)]{} & & d_* d^! K_Y, \end{array} \tag{4.4.2}$$

The arrows are as follows: (1) is deduced from 4.2.2 by applying f_*; (2) is defined by 4.1.2; (3) is given by 4.2.2; (4) is given by the canonical arrow $f_*P \otimes^{\mathbf{L}} f_*Q \longrightarrow f_*(P \otimes^{\mathbf{L}} Q)$; (5) by 4.1.2; (6) by 4.2.2; (7) by 4.1.2; (8) is the tensor product of arrows 3.7.3; (9) and (10) are defined by 3.7.3; (11) and (12) by the isomorphisms $f_*P \longrightarrow E$, $f_*Q \longrightarrow F$ (3.3.1); and (13) by the adjunction arrow $f_*K_X \longrightarrow K_Y$. By definition, 4.4.1 is identified, via 3.1.1 and 3.2.1, with the outer boundary of 4.4.2. It remains only to prove that the squares (A), (B), (C), (D) are commutative. The commutativity of (B) and (C) is clear by functoriality. It remains to prove that of (A) and (D).

a) Commutativity of (D). It is enough to prove that the square

$$\begin{array}{ccc} f_*P \otimes^{\mathbf{L}} f_*Q & \longrightarrow & f_*K_X \\ \downarrow & & \downarrow \\ E \otimes^{\mathbf{L}} F & \longrightarrow & K_Y \end{array} \tag{4.4.3}$$

(corresponding to (D) for $d = \mathrm{Id}$) commutes. Recall that the pairing

$$P \otimes^{\mathbf{L}} Q \longrightarrow K_X \quad (\text{respectively } E \otimes^{\mathbf{L}} F \longrightarrow K_Y)$$

is the external tensor product of the evaluation arrows

$$DL_i \otimes^{\mathbf{L}} L_i \longrightarrow K_{X_i} \quad (\text{respectively } DM_i \otimes^{\mathbf{L}} M_i \longrightarrow K_{Y_i}).$$

By the Künneth isomorphisms

$$f_{1*}DL_1 \otimes^{\mathbf{L}}_S f_{2*}L_2 \xrightarrow{\sim} f_*P, \qquad f_{1*}L_1 \otimes^{\mathbf{L}}_S f_{2*}DL_2 \xrightarrow{\sim} f_*Q,$$

4.4.3 is therefore identified with the external tensor product of the squares

$$\begin{array}{ccc} f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i & \xrightarrow{(1)} & f_{i*}K_{X_i} \\ {\scriptstyle (3)}\downarrow & & \downarrow{\scriptstyle (4)} \\ DM_i \otimes^{\mathbf{L}} M_i & \xrightarrow[(2)]{} & K_{Y_i}, \end{array} \tag{4.4.4}$$

where (2) is the evaluation arrow, (1) the composite of the canonical arrow

$$f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i \longrightarrow f_{i*}(DL_i \otimes^{\mathbf{L}} L_i)$$

and the arrow defined by the evaluation arrow, (3) is the tensor product of the duality isomorphism $f_{i*}DL_i \longrightarrow DM_i$ and the identity arrow of M_i, and (4) is the adjunction arrow. It is therefore enough to verify the commutativity of the squares 4.4.4. Since the above duality isomorphism is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i})$$

and the arrow deduced from the adjunction arrow

$$f_{i*}K_{X_i} \longrightarrow K_{Y_i},$$

it is enough to see that the following square commutes:

$$\begin{array}{ccc} f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i & \xrightarrow{(1)} & f_{i*}K_{X_i} \\ \downarrow{\scriptstyle (3)} & & \downarrow{\scriptstyle \mathrm{Id}} \\ \mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i & \xrightarrow[(2)]{} & f_{i*}K_{X_i}, \end{array} \tag{4.4.5}$$

where (1) is the arrow (1) of 4.4.4, (2) the evaluation arrow, and (3) the tensor product by $f_{i*}L_i$ of the canonical arrow just mentioned. More generally, for any arrow $N_i \longrightarrow f_{i*}L_i$ in $D_{\mathrm{ctf}}(X_i)$, i.e. any arrow $L_i \xrightarrow{u} N_i$ of type 1 over f_i, the triangle

$$\begin{array}{ccc} f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} N_i & & \\ \downarrow & \searrow & \\ \mathrm{RHom}(N_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} N_i & \longrightarrow & f_{i*}K_{X_i} \end{array}$$

deduced from 4.4.5 commutes: indeed, by functoriality, one may assume that $L_i = f_i^* N_i$, with u given by the identity of L_i; the arrow

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(N_i, f_{i*}K_{X_i})$$

is then the inverse of the trivial duality isomorphism, and the commutativity of the triangle is verified immediately at the level of complexes, resolving N_i flatly and K_{X_i} injectively.

b) Commutativity of (A). Referring to the definition of 3.7.3, one sees that it is enough to verify this commutativity in each of the following cases:

(i) $X_i = Y_i$, $f_i =$ the identity arrow of X_i; (ii) the lateral faces of 4.4.0 are cartesian.

Place ourselves in case (i). The square (A) is the outer boundary of the diagram

$$\begin{array}{ccccc} c'_*c'^!P \otimes^{\mathbf{L}} c''_*c''^!Q & \xrightarrow{\sim} & c_*(c'^!P \otimes^{\mathbf{L}}_X c''^!Q) & \longrightarrow & c_*c^!(P \otimes^{\mathbf{L}} Q) \\ \downarrow{\scriptstyle (8)} & A_1 & \downarrow & A_2 & \downarrow{\scriptstyle (9)} \\ d'_*d'^!P \otimes^{\mathbf{L}} d''_*d''^!Q & \xrightarrow[\sim]{} & d_*(d'^!P \otimes^{\mathbf{L}}_X d''^!Q) & \longrightarrow & d_*d^!(P \otimes Q), \end{array}$$

in which the horizontal isomorphisms of A_1 are Künneth isomorphisms, the horizontal arrows of A_2 are given by 4.2.1, and the middle vertical arrow is deduced by applying d_* to the arrow

$$g_!(c'^!P \otimes^{\mathbf{L}}_X c''^!Q) \longrightarrow d'^!P \otimes^{\mathbf{L}}_X d''^!Q$$

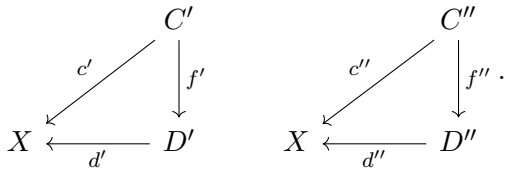
defined, via Künneth, by the external tensor product of the adjunction arrows

$$f'_!c'^!P \longrightarrow d'^!P, \qquad f''_!c''^!Q \longrightarrow d''^!Q.$$

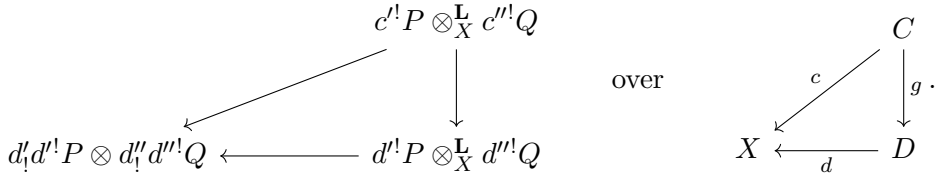
These adjunction arrows and the identity arrows of $d'_!d'^!P$, $d''_!d''^!Q$ define commutative triangles of arrows of type 3 (1.4)

$$\begin{array}{ccc} & c'^!P & \\ \swarrow & \downarrow & \\ d'_!d'^!P & \longleftarrow & d'^!P \end{array} \qquad \begin{array}{ccc} & c''^!Q & \\ \swarrow & \downarrow & \\ d''_!d''^!Q & \longleftarrow & d''^!Q \end{array}$$

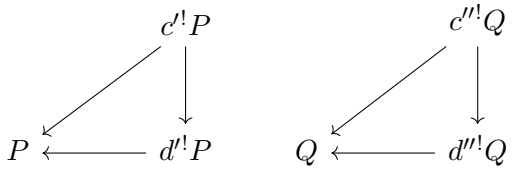
over



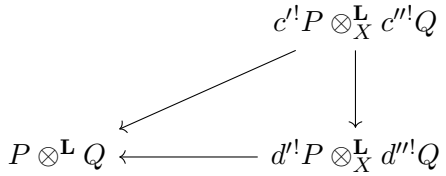
By external tensor product (over X), these triangles give a commutative triangle



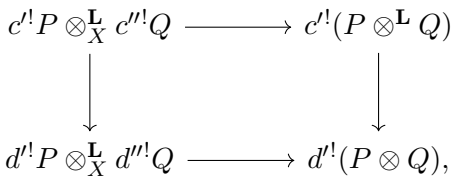
The commutativity of this triangle is equivalent to that of the square deduced from A_1 by inverting the horizontal arrows. On the other hand, there are commutative triangles of arrows of type 3



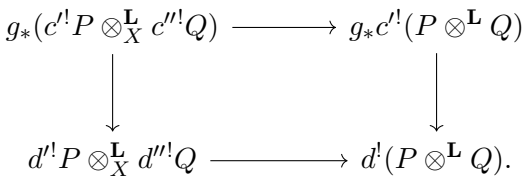
defined by the identity arrows of $d'^!P$, $c'^!P$, $d''^!Q$, $c''^!Q$. By external tensor product, one obtains a commutative triangle



which can be interpreted as a morphism of arrows of type 3 over $g : C \longrightarrow D$:



or again as a commutative square over D:



But the square deduced from the latter by applying d_* is none other than A_2, whence the commutativity of (A) in case (i).

It remains to treat case (ii). Since the faces of 4.4.0 containing f are cartesian, the arrows

$$f'_*c'^!P \longrightarrow d'^!f_*P, \qquad f''_*c''^!Q \longrightarrow d''^!f_*Q$$

(3.7.3) are isomorphisms. Their inverses define arrows of type 1

$$c'^!P \longrightarrow d'^!f_*P, \qquad c''^!Q \longrightarrow d''^!f_*Q$$

over f', f'', whence, by direct image, commutative squares of arrows of type 1

$$\begin{array}{ccc} c'_*c'^!P & \longleftarrow & c'^!P \\ \downarrow & & \downarrow \\ d'_*d'^!P & \longleftarrow & d'^!f_*P \end{array} \qquad \begin{array}{ccc} c''_*c''^!Q & \longleftarrow & c''^!Q \\ \downarrow & & \downarrow \\ d''_*d''^!Q & \longleftarrow & d''^!f_*Q \end{array}$$

over the faces of 4.4.0 containing f. Taking the tensor product, one obtains a commutative square in $D_{\mathrm{ctf}}(Y)$

$$\begin{array}{ccc} d'_*d'^!f_*P \otimes^{\mathbf{L}} d''_*d''^!f_*Q & \longrightarrow & d_*\big(d'^!f_*P \otimes^{\mathbf{L}}_Y d''^!f_*Q\big) \\ \downarrow & & \downarrow \\ f_*c'_*c'^!P \otimes^{\mathbf{L}} f_*c''_*c''^!Q & \longrightarrow & d_*g_*\big(c'^!P \otimes^{\mathbf{L}}_X c''^!Q\big), \end{array} \tag{4.4.6}$$

where the left vertical arrow is the inverse of (8), the horizontal arrows are given by Künneth, and the right vertical arrow by the external tensor product over Y of the arrows

$$f'_*c'^!P \longrightarrow d'^!f_*P, \qquad f''_*c''^!Q \longrightarrow d''^!f_*Q.$$

The same tensor product defines the left vertical arrow of the square

$$\begin{array}{ccc} d'^!f_*P \otimes^{\mathbf{L}}_Y d''^!f_*Q & \longrightarrow & d^!f_*(P \otimes^{\mathbf{L}} Q) \\ \downarrow & & \downarrow \\ g_*(c'^!P \otimes^{\mathbf{L}}_X c''^!Q) & \longrightarrow & g_*c'^!(P \otimes^{\mathbf{L}} Q), \end{array} \tag{4.4.7}$$

where the right vertical arrow is 3.7.3, and the horizontal arrows are defined by 4.2.1. The outer boundary of the diagram composed of 4.4.6 and d_*(4.4.7) is the square deduced from (A) by inverting the vertical arrows. It is therefore enough to prove that 4.4.7 commutes, in other words that the arrows 4.2.1

$$c'^!P \otimes^{\mathbf{L}}_X c''^!Q \longrightarrow c'^!(P \otimes^{\mathbf{L}} Q), \qquad d'^!f_*P \otimes^{\mathbf{L}}_Y d''^!f_*Q \longrightarrow d^!f_*(P \otimes^{\mathbf{L}} Q)$$

define a morphism from

$$c'^!P \otimes^{\mathbf{L}}_X c''^!Q \longrightarrow d'^!f_*P \otimes^{\mathbf{L}}_Y d''^!f_*Q$$

to

$$c'^!(P \otimes^{\mathbf{L}} Q) \longrightarrow d^!f_*(P \otimes^{\mathbf{L}} Q),$$

viewed as arrows of type 1 over g. By adjunction, we are reduced to verifying that the tensor products of adjunction arrows

$$c'_*c'^!P \otimes^{\mathbf{L}}_X c''_*c''^!Q \longrightarrow P \otimes^{\mathbf{L}} Q, \qquad d'_*d'^!f_*P \otimes^{\mathbf{L}}_Y d''_*d''^!f_*Q \longrightarrow f_*P \otimes^{\mathbf{L}} f_*Q$$

define a morphism, between arrows of type 1 over f, from the tensor product of the arrows

$$c'_*c'^!P \longrightarrow d'_*d'^!f_*P, \qquad c''_*c''^!Q \longrightarrow d''_*d''^!f_*Q$$

to the tensor product of the arrows

$$P \longrightarrow f_*P, \qquad Q \longrightarrow f_*Q.$$

For this it is enough to verify that the adjunction arrows

$$c'_*c'^!P \longrightarrow P, \qquad d'_*d'^!f_*P \longrightarrow f_*P$$

define a morphism from

$$c'_*c'^!P \longrightarrow d'_*d'^!f_*P$$

to

$$P \longrightarrow f_*P,$$

and likewise for the pair (c'', d''). It is enough to consider the case of (c', d'), the statement being the same in the two cases. We have to see that the triangle

$$\begin{array}{ccc} & & f_*c'_*c'^!P \\ & \swarrow & \downarrow \sim \\ f_*P & \longleftarrow & d'_*d'^!f_*P \end{array}$$

where the horizontal and oblique arrows are given by adjunction and the vertical arrow by the isomorphism 3.7.3, is commutative. Taking into account the definition of 3.7.3 (SGA 4 XVIII 3.1.12.3) as the transpose of a base-change isomorphism, we are reduced, by applying the functor $\mathrm{Hom}(R, -)$ $(R \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y))$, to verifying the commutativity of the square

$$\begin{array}{ccc} c'_*c'^*f^*R & \xrightarrow{\mathrm{Id}} & c'_*f'^*d'^*R \\ \uparrow & & \uparrow \sim \\ f^*R & \longrightarrow & f^*d'_*d'^*R \end{array}$$

where the left vertical arrow (respectively the lower horizontal arrow) is defined by the adjunction arrow $1 \longrightarrow c'_*c'^*$ (respectively $1 \longrightarrow d'_*d'^*$), and the right vertical arrow by the base-change isomorphism

$$f^*d'_* \xrightarrow{\sim} c'_*f'^*.$$

But this square commutes by the very definition of the base-change arrow. The commutativity of (A) is therefore established, and this completes the proof of the theorem.

Corollary 4.5. Under the hypotheses of 4.4, there is a commutative square

$$\begin{array}{ccc} \mathrm{Hom}(c'_1{}^*L_1, c'_2{}^!L_2) \otimes \mathrm{Hom}(c''_2{}^*L_2, c''_1{}^!L_1) & \xrightarrow{4.2.5} & \mathrm{H}^0(C, K_C) \\ \downarrow{\scriptstyle 3.7.6 \otimes 3.7.6} & & \downarrow{\scriptstyle g_*} \\ \mathrm{Hom}(d'_1{}^*f_{1*}L_1, d'_2{}^!f_{2*}L_2) \otimes \mathrm{Hom}(d''_2{}^*f_{2*}L_2, d''_1{}^!f_{1*}L_1) & \xrightarrow[4.2.5]{} & \mathrm{H}^0(D, K_D), \end{array}$$

where g_* is defined by the adjunction arrow

$$g_*K_C = g_*g^!K_D \longrightarrow K_D.$$

In other words, for

$$u \in \mathrm{Hom}({c'_1}^*L_1, {c'_2}^!L_2), \qquad v \in \mathrm{Hom}({c''_2}^*L_2, {c''_1}^!L_1),$$

one has

$$\langle f_*u, f_*v\rangle = g_*\langle u, v\rangle. \tag{4.5.1}$$

For every proper S-scheme $t : T \longrightarrow S$, denote by

$$\int_T : \mathrm{H}^0(T, K_T) \longrightarrow \mathrm{H}^0(S, K_S) = \mathrm{H}^0(S, A) \tag{4.6}$$

the arrow defined by the adjunction arrow

$$t_!K_T \longrightarrow A.$$

Corollary 4.7 – Lefschetz–Verdier formula. Under the hypotheses of 4.4, suppose that $Y_1 = Y_2 = S = D' = D''$. For

$$u \in \mathrm{Hom}({c'_1}^*L_1, {c'_2}^!L_2), \qquad v \in \mathrm{Hom}({c''_2}^*L_2, {c''_1}^!L_1),$$

one has

$$\langle u_*, v_*\rangle = \int_C \langle u, v\rangle, \tag{4.7.1}$$

where

$$u_* \in \mathrm{Hom}(f_{1*}L_1, f_{2*}L_2), \qquad v_* \in \mathrm{Hom}(f_{2*}L_2, f_{1*}L_1)$$

are the homomorphisms deduced from u, v by direct image over S (3.7.6), and

$$\langle u_*, v_*\rangle = \mathrm{Tr}(u_*v_*) = \mathrm{Tr}(v_*u_*)$$

is the cup product, in the sense of (SGA 6 I 8.3), of the homomorphisms of perfect complexes u_*, v_* (4.1.5).

Taking 3.6.5 into account, in particular one has:

Corollary 4.8. Let $t : T \longrightarrow S$ be a proper S-scheme, $\Delta \subset T \times_S T$ the diagonal, $F : T \longrightarrow T$ an S-endomorphism of T, $T^F = (\text{graph of } F) \times_{(T\times_S T)} \Delta$ the scheme of fixed points of F, $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(T)$, $u \in \mathrm{Hom}(F^*L, L)$ a correspondence from L to L with support in the graph of F, and $1 \in \mathrm{Hom}(L, L)$ the identity correspondence, with support in Δ (3.2 b)). Then

$$\mathrm{Tr}(u_*) = \int_{T^F} \langle u, 1\rangle,$$

where u_* is the endomorphism of t_*L obtained by composing $t_*L \longrightarrow t_*F^*L$ (inverse image) with $t_*u : t_*F^*L \longrightarrow t_*L$.

From the remarks of 3.2 a) and 3.6 one also deduces:

Corollary 4.9. Under the hypotheses of 4.7, assume that X_i is smooth over S, purely of relative dimension r_i, and that C' (respectively C'') is a closed subscheme of X finite and of finite tor-dimension over X_2 (respectively X_1). Let

$$\mathrm{cl}(C') \in \mathrm{H}^{2r_1}_{C'}(X, A(r_1)), \qquad \mathrm{cl}(C'') \in \mathrm{H}^{2r_2}_{C''}(X, A(r_2))$$

be the cohomology classes of C' and C'', and let

$$\mathrm{cl}(C')_* \in \mathrm{Hom}(\mathrm{R}f_{1*}A, \mathrm{R}f_{2*}A), \qquad \mathrm{cl}(C'')_* \in \mathrm{Hom}(\mathrm{R}f_{2*}A, \mathrm{R}f_{1*}A)$$

be the corresponding homomorphisms (3.2 a), 3.7.6). Then

$$\langle \mathrm{cl}(C')_*, \mathrm{cl}(C'')_* \rangle = \int_{C' \cap C''} \mathrm{cl}(C)\,\mathrm{cl}(D),$$

where

$$\mathrm{cl}(C)\,\mathrm{cl}(D) \in \mathrm{H}^{2(r_1+r_2)}_{C' \cap C''}(X, A(r_1+r_2)) = \mathrm{H}^0(C' \cap C'', K_{C' \cap C''})$$

is the usual cup product.

Taking 4.3 into account, in particular one has:

Corollary 4.10. Under the hypotheses of 4.9, suppose that $C' \cap C''$ is finite. For $z \in C' \cap C''$, denote by $(C' \cdot C'')_z$ the image in A of the intersection multiplicity of C' and C'' at z. Then

$$\langle \mathrm{cl}(C')_*, \mathrm{cl}(C'')_* \rangle = \sum_{z \in C' \cap C''} (C' \cdot C'')_z.$$

4.11. Under the hypotheses of 4.9, one has in general

$$\langle \mathrm{cl}(C')_*, \mathrm{cl}(C'')_* \rangle = \sum_{Z \in \pi_0(C' \cap C'')} (\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z, \tag{4.11.1}$$

where $\pi_0(C' \cap C'')$ denotes the set of connected components of $C' \cap C''$, and $(\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z$ the restriction of $\mathrm{cl}(C')\,\mathrm{cl}(C'')$ to $\mathrm{H}^0(Z, K_Z)$. When Z is a component of dimension ≥ 1, the calculation of $(\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z$ presents problems, because one is in an "excess" situation. If Z is smooth, purely of dimension d, if C' and C'' are smooth in a neighbourhood of Z, and if X is projective, one can show that

$$\int_Z (\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z = \int_Z c_d(N_Z), \tag{4.11.2}$$

where N_Z is the locally free sheaf of rank d defined by the exact sequence of normal bundles

$$0 \longrightarrow N_{Z/C'} \oplus N_{Z/C''} \longrightarrow N_{Z/X} \longrightarrow N_Z \longrightarrow 0,$$

and $c_d(N_Z) \in \mathrm{H}^{2d}(Z, A(d))$ denotes its d-th Chern class (SGA 5 VII) (use (SGA 6 XIV 6.3) to reduce to a calculation of intersection number in K-theory, and apply (SGA 6 VII 2.5), cf. ([10] 4.1); the projectivity hypothesis is no doubt unnecessary).

Let T be a smooth projective S-scheme; take as correspondence C' (respectively C'') the graph of an S-endomorphism F (respectively the diagonal), and suppose that the scheme of fixed points $T^F = C' \cap C''$ is smooth. The sheaf $N_{(T^F)}$ above is nothing other than the tangent sheaf to T^F, so

that 4.11.2 implies, by the Gauss–Bonnet formula (SGA 5 VII 4.9), the analogue of a formula well known to topologists:

$$\mathrm{Tr}(F, \mathrm{R}\Gamma(T, A)) = \mathrm{EP}(T^F), \tag{4.11.3}$$

where $\mathrm{EP}(T^F)$ denotes the Euler–Poincaré characteristic of T_s^F, for s a geometric point above S (loc. cit.).

The calculation of $(\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z$ for a non-smooth component Z of dimension ≥ 1 has not been treated in general, at least to the knowledge of the editor.

4.12. In the situation of 4.7, let z be an isolated point of C, with projection z_i on X_i, a' (respectively a'') on C' (respectively C''), and let $\bar{z}$ be a geometric point above z, with image $\bar{z}_i$ above z_i. Assume that c_2' (respectively c_1'') is quasi-finite and flat at a' (respectively a''), that L_i is perfect in a neighbourhood of $\bar{z}_i$, and that u (respectively v) is given in a neighbourhood of a' (respectively a'') by

$$\mathrm{Tr}_{c_2'} : c_{2!}' {c_2'}^* L_2 \longrightarrow L_2 \quad \text{and} \quad s \in \mathrm{Hom}({c_1'}^* L_1, {c_2'}^* L_2)$$

(respectively

$$\mathrm{Tr}_{c_1''} : c_{1!}'' {c_1''}^* L_1 \longrightarrow L_1 \quad \text{and} \quad t \in \mathrm{Hom}({c_2''}^* L_2, {c_1''}^* L_1)).$$

Let $s_{\bar{z}} \in \mathrm{Hom}(L_{1\bar{z}_1}, L_{2\bar{z}_2})$ (respectively $t_{\bar{z}} \in \mathrm{Hom}(L_{2\bar{z}_2}, L_{1\bar{z}_1})$) be the homomorphism deduced from s (respectively t) by passage to the fibres. Let again, as in 3.2 a), $\mathrm{cl}(c')$ (respectively $\mathrm{cl}(c'')$) denote the correspondence from X_1 to X_2 (respectively from X_2 to X_1) defined, in a neighbourhood of a' (respectively a''), by c_2' (respectively c_1''), and let $(C' \cdot C'')_z$ be the corresponding local term $\langle \mathrm{cl}(c'), \mathrm{cl}(c'') \rangle_z$ (equal to the image in A of the intersection multiplicity of C' and C'' at z when C' and C'' are closed subschemes of X, with X_1 and X_2 smooth). Then:

$$\langle u, v \rangle_z = (C' \cdot C'')_z \langle s_{\bar{z}}, t_{\bar{z}} \rangle. \tag{4.12.1}$$

This follows easily from 4.2.6.

4.13. If T is a scheme over S, denote by $DF(T)$ the filtered derived category of complexes of A_T-modules with finite filtration ([9] V 1), and by $DF_{\mathrm{ctf}}(T)$ the full subcategory formed by the L such that $\mathrm{gr}(L) \in \mathrm{ob}\, D_{\mathrm{ctf}}(T)$. The kind of compatibility of filtered derived functors with passage to associated graded objects implies that DF_{ctf} is stable under the six operations f^*, f_*, $f_!$, $f^!$, $\otimes^{\mathbf{L}}$, RHom. The definitions and results of the preceding numbers are valid, mutatis mutandis, in DF_{ctf}. The pairing 4.1.2 is compatible with passage to associated graded objects (K_X being regarded as endowed with the trivial filtration), and it follows that, with the notation of 4.2.5, for $L_i \in \mathrm{ob}\, DF_{\mathrm{ctf}}(X_i)$,

$$u \in \mathrm{Hom}_{DF}(c_1^* L_1, c_2^! L_2), \qquad v \in \mathrm{Hom}_{DF}(d_2^* L_2, d_1^! L_1),$$

one has the following equality in $\mathrm{H}^0(E, K_E)$:

$$\langle u, v \rangle = \sum_i \langle \mathrm{gr}^i u, \mathrm{gr}^i v \rangle. \tag{4.13.1}$$

As Deligne observes, this remark applies in particular to the situation considered by Langlands in ([13] §7). The correspondences that he considers in (loc. cit. 7.11) indeed preserve, near the fixed points, finite filtrations on the sheaves with associated graded objects inducing, on the strict localization at the image of a fixed point, either a constant sheaf or a constant sheaf on

the complement of the closed point extended by zero. The calculation of the local terms is easily reduced, by 4.13, to 4.12.1 and to the case of sheaves concentrated on the closed points that are the images of the fixed points, a case treated directly. Applying 4.7, one obtains the Lefschetz formula required in ([13] 7.11). We point out that the general statement (loc. cit. 7.12) is true; the proof, more delicate, will be given in III B.

5. Complements

5.1. Dual of a correspondence

Let, for $i = 1, 2$, X_i be an S-scheme, let $X = X_1 \times_S X_2$, and let $p_i : X \to X_i$ be the projections. Let $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$. Every cohomological correspondence u from L_1 to L_2 gives, by applying D_X and identifying $Dp_2^!$ with $p_2^* D$ and Dp_1^* with $p_1^! D$, a correspondence Du from DL_2 to DL_1, called the *dual of* u. In other words, D defines a homomorphism

$$D : \operatorname{Hom}(p_1^* L_1, p_2^! L_2) \longrightarrow \operatorname{Hom}(p_2^* DL_2, p_1^! DL_1). \tag{5.1.1}$$

One checks that 5.1.1 is obtained, by applying $H^0(X, -)$, from the isomorphism

$$D : \operatorname{RHom}(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \operatorname{RHom}(p_2^* DL_2, p_1^! DL_1) \tag{5.1.2}$$

which makes the square

$$\begin{array}{ccc} p_1^* DL_1 \otimes^{\mathbf{L}} p_2^* L_2 & \xrightarrow{3.1.1} & \operatorname{RHom}(p_1^* L_1, p_2^! L_2) \\ \downarrow & & \downarrow \\ p_2^* DDL_2 \otimes^{\mathbf{L}} p_1^* DL_1 & \xrightarrow{3.1.1} & \operatorname{RHom}(p_2^* DL_2, p_1^! DL_1), \end{array} \tag{5.1.3}$$

commutative. Here the left vertical arrow is the composite of the symmetry isomorphism and the one defined by the biduality isomorphism $L_2 \to DDL_2$. It follows easily from the definitions that the following square is commutative:

$$\begin{array}{ccc} \operatorname{RHom}_S(L_1, L_2) \otimes^{\mathbf{L}} \operatorname{RHom}_S(L_2, L_1) & \xrightarrow{4.1.3} & K_X \\ \downarrow{\scriptstyle D \otimes^{\mathbf{L}} D} & & \downarrow{\scriptstyle \mathrm{Id}} \\ \operatorname{RHom}_S(DL_2, DL_1) \otimes^{\mathbf{L}} \operatorname{RHom}_S(DL_1, DL_2) & \xrightarrow{4.1.3} & K_X. \end{array} \tag{5.1.4}$$

Let $c : C \to X$ be an S-morphism and put $c_i = p_i c$. Applying $c_* c^!$ to 5.1.2, and using (3.2.1), one deduces an isomorphism

$$D : \operatorname{Hom}(c_1^* L_1, c_2^! L_2) \xrightarrow{\sim} \operatorname{Hom}(c_2^* DL_2, c_1^! DL_1), \tag{5.1.5}$$

which is checked to identify with the one obtained by applying the functor D_X and the identifications of Dc_1^* with $c_1^! D$ and $Dc_2^!$ with $c_2^* D$. If $c' : C' \to X$ and $c'' : C'' \to X$ are S-morphisms, it follows from 5.1.4 that, for

$$u \in \operatorname{Hom}({c'_1}^* L_1, {c'_2}^! L_2), \qquad v \in \operatorname{Hom}({c''_2}^* L_2, {c''_1}^! L_1),$$

one has

$$\langle u, v\rangle = \langle Du, Dv\rangle, \tag{5.1.6}$$

with the notation of 4.2.5.

On the other hand, one checks that, in the situation of 3.3, the following square is commutative:

$$\begin{array}{ccc} f_*\mathrm{RHom}_S(L_1, L_2) & \xrightarrow{3.3.1} & \mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2) \\ {\scriptstyle f_*D}\downarrow & & \downarrow{\scriptstyle D} \\ f_*\mathrm{RHom}_S(DL_2, DL_1) & \xrightarrow{3.3.1} & \mathrm{RHom}_S(f_{2!}DL_2, f_{1*}DL_1), \end{array} \tag{5.1.7}$$

and consequently, in the situation of 3.7.6, one has

$$f_*Du = Df_*u. \tag{5.1.8}$$

5.2. Composition of correspondences

Let, for $i = 1, 2, 3$, X_i be an S-scheme, let $X_{ij} = X_i \times_S X_j$, $X = X_1 \times_S X_2 \times_S X_3$, and let $p_i : X \to X_i$, $p_{ij} : X \to X_{ij}$ be the projections. Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. One has $X = X_{12} \times_{X_2} X_{23}$, and

$$(DL_1 \otimes^{\mathbf{L}}_S L_2) \otimes^{\mathbf{L}}_{X_2} (DL_2 \otimes^{\mathbf{L}}_S L_3) = p_1^*DL_1 \otimes^{\mathbf{L}} p_2^*L_2 \otimes^{\mathbf{L}} p_2^*DL_2 \otimes^{\mathbf{L}} p_3^*L_3.$$

The evaluation arrow $L_2 \otimes^{\mathbf{L}} DL_2 \to K_{X_2}$ therefore defines a pairing

$$(DL_1 \otimes^{\mathbf{L}}_S L_2) \otimes^{\mathbf{L}}_{X_2} (DL_2 \otimes^{\mathbf{L}}_S L_3) \longrightarrow p_{13}^!(DL_1 \otimes^{\mathbf{L}}_S L_3), \tag{5.2.1}$$

the second member being identified with $(DL_1 \otimes^{\mathbf{L}}_S L_3) \otimes^{\mathbf{L}}_S K_{X_2}$ by Künneth (1.7.3). Taking 3.1.1 and 3.1.2 into account, one may rewrite 5.2.1 in the form

$$\mathrm{RHom}_S(L_1, L_2) \otimes^{\mathbf{L}}_{X_2} \mathrm{RHom}_S(L_2, L_3) \longrightarrow p_{13}^!\mathrm{RHom}_S(L_1, L_3) = \mathrm{RHom}(p_1^*L_1, p_3^!L_3), \tag{5.2.2}$$

the identification with the second member being given by the induction isomorphism (SGA 4 XVIII 3.1.12.2). Hence one obtains a pairing, called *composition of correspondences*,

$$\mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_2, L_3) \longrightarrow \mathrm{Hom}(p_1^*L_1, p_3^!L_3), \tag{5.2.3}$$

which will be denoted $(u, v) \mapsto vu$. One checks that, if $p_{ij,i} : X_{ij} \to X_i$ and $p_{ij,j} : X_{ij} \to X_j$ denote the projections, 5.2.2 can also be obtained by composing the arrow

$$\mathrm{RHom}_S(L_1, L_2) \otimes^{\mathbf{L}}_{X_2} \mathrm{RHom}_S(L_2, L_3) \longrightarrow \mathrm{RHom}(p_1^*L_1, p_{12}^*p_{12,2}^!L_2) \otimes \mathrm{RHom}(p_{23}^!p_{23,2}^*L_2, p_3^!L_3)$$

(the tensor product of the canonical arrows $p_{12}^*\mathrm{RHom}_S(L_1, L_2) \to \mathrm{RHom}(p_1^*L_1, p_{12}^*p_{12,2}^!L_2)$ and $p_{23}^*\mathrm{RHom}_S(L_2, L_3) \to \mathrm{RHom}(p_{23}^!p_{23,2}^*L_2, p_3^!L_3)$), the arrow defined by the base-change arrow $p_{12}^!p_{12,2}^*L_2 \to p_{23}^*p_{23,2}^*L_2$, and a usual composition arrow on the RHom's.

Let $c' : C' \to X_{12}$ and $c'' : C'' \to X_{23}$ be correspondences, and let $c : C = C' \times_{X_2} C'' \to X$ be the "composite" correspondence. From 5.2.2 one deduces, or else defines directly by a composition analogous to the one just described, a pairing

$$\mathrm{RHom}({c'_1}^*L_1, {c'_2}^!L_2) \otimes^{\mathbf{L}}_{X_2} \mathrm{RHom}({c''_2}^*L_2, {c''_3}^!L_3) \longrightarrow \mathrm{RHom}(c_1^*L_1, c_3^!L_3), \tag{5.2.4}$$

hence a "composition" pairing $(u, v) \mapsto vu$:

$$\mathrm{Hom}({c'_1}^* L_1, {c'_2}^! L_2) \otimes \mathrm{Hom}({c''_2}^* L_2, {c''_3}^! L_3) \longrightarrow \mathrm{Hom}(c_1^* L_1, c_3^! L_3). \tag{5.2.5}$$

One checks that composition of correspondences is compatible with direct images in the following sense. Let, for $1 \leq i \leq 3$, $f_i : X_i \to Y_i$ be an S-morphism, with f_2 proper, and let c', c'', c be morphisms as above, with c' and c'' proper. Let $d' : D' \to Y_{12}$ and $d'' : D'' \to Y_{23}$ be proper morphisms, $d : D = D' \times_{Y_2} D'' \to Y$, and suppose given the commutative squares

$$\begin{array}{ccc} X_{12} & \longleftarrow & C' \\ {\scriptstyle f_{12}}\downarrow & & \downarrow \\ Y_{12} & \longleftarrow & D' \end{array} \qquad \begin{array}{ccc} X_{23} & \longleftarrow & C'' \\ {\scriptstyle f_{23}}\downarrow & & \downarrow \\ Y_{23} & \longleftarrow & D'' \end{array}$$

Then one has a commutative square

$$\begin{array}{ccc} \mathrm{Hom}({c'_1}^* L_1, {c'_2}^! L_2) \otimes \mathrm{Hom}({c''_2}^* L_2, {c''_3}^! L_3) & \xrightarrow{5.2.5} & \mathrm{Hom}(c_1^* L_1, c_3^! L_3) \\ \downarrow{\scriptstyle f_{12*} \otimes f_{23*}} & & \downarrow{\scriptstyle f_{13*}} \\ \mathrm{Hom}({d'_1}^* M_1, {d'_2}^! M_2) \otimes \mathrm{Hom}({d''_2}^* M_2, {d''_3}^! M_3) & \xrightarrow{5.2.5} & \mathrm{Hom}(d_1^* M_1, d_3^! M_3), \end{array} \tag{5.2.6}$$

where $M_1 = f_{1!} L_1$, $M_2 = f_{2*} L_2$, and $M_3 = f_{3*} L_3$.

Finally, suppose that $X_1 = X_3$ and $L_1 = L_3$. The fundamental pairing 4.1.2 may be interpreted as the composite of the pairing

$$q_1^* D L_1 \otimes^{\mathbf{L}} q_2^* L_2 \otimes^{\mathbf{L}} q_2^* D L_2 \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow q_1^* D L_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \tag{5.2.7}$$

defined by $L_2 \otimes^{\mathbf{L}} D L_2 \to K_{X_2}$, or, equivalently, induced by 5.2.1 on X_{12} via the diagonal $X_{12} \to X_1 \times_S X_2 \times_S X_1$ ($q_i : X_{12} \to X_i$ denoting the projection), and the pairing

$$q_1^* D L_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow K_{X_{12}} \tag{5.2.8}$$

defined by $D L_1 \otimes^{\mathbf{L}} L_1 \to K_{X_1}$. It follows that, for $u \in \mathrm{Hom}_S(L_1, L_2)$ and $v \in \mathrm{Hom}_S(L_2, L_1)$, one has, in $H^0(X_1 \times_S X_2, K_{X_1 \times_S X_2})$,

$$\langle u, v \rangle = \langle vu, 1 \rangle, \tag{5.2.9}$$

where 1 denotes the identity correspondence from L_1 to L_1 on $X_1 \times_S X_1$, supported on the diagonal (3.2 b)), and the cup product in the second member is taken in the sense of 4.2.5, taking account of the fact that the fibred product $X \times_{(X_1 \times_S X_1)} X_1$ (via $p_{13} : X \to X_{13} = X_1 \times_S X_1$ and the diagonal $X_1 \to X_1 \times_S X_1$) identifies with $X_1 \times_S X_2$. Likewise, more generally, if $c' : C' \to X_{12}$ and $c'' : C'' \to X_{21}$ are correspondences, if $c : C = C' \times_{X_2} C'' \to X_1 \times_S X_2 \times_S X_1$ is the composite correspondence, and if $d : C' \times_{X_{12}} C'' = D \to X_{12}$ is the intersection correspondence, then, for $u \in \mathrm{Hom}({c'_1}^* L_1, {c'_2}^! L_2)$ and $v \in \mathrm{Hom}({c''_2}^* L_2, {c''_1}^! L_1)$, one has the following equality in $H^0(D, K_D)$:

$$\langle u, v \rangle = \langle vu, 1 \rangle, \tag{5.2.10}$$

where 1 denotes, as above, the diagonal correspondence, and D is identified with the fibred product of $C \to X_1 \times_S X_1$ and the diagonal $X_1 \to X_1 \times_S X_1$. Formulas 5.2.9 and 5.2.10 generalize a well-known formula in the case of constant coefficients ([12] 1.3.6 (ii) c)).

5.3. External tensor product of correspondences

Let, for $1 \leq i \leq 4$, X_i be an S-scheme, let $X = X_1 \times_S X_2 \times_S X_3 \times_S X_4$, $X_{ij} = X_i \times_S X_j$, and let $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$. The Künneth isomorphism (1.7.6, 2.2.4)

$$DL_1 \otimes_S^{\mathbf{L}} DL_3 \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3)$$

defines an isomorphism on X

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_S^{\mathbf{L}} (DL_3 \otimes_S^{\mathbf{L}} L_4) \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3) \otimes_S^{\mathbf{L}} (L_2 \otimes_S^{\mathbf{L}} L_4), \tag{5.3.1}$$

which, by 3.1.1, may be rewritten in the form

$$\mathrm{RHom}_S(L_1, L_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}_S(L_3, L_4) \xrightarrow{\sim} \mathrm{RHom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4). \tag{5.3.2}$$

Hence a pairing, called the *external tensor product*,

$$\mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_3, L_4) \longrightarrow \mathrm{Hom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4), \tag{5.3.3}$$

which will be denoted $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$. The pairing 5.3.2 may also be interpreted as the composite of the external tensor product

$$\mathrm{RHom}(E_1, E_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}(E_3, E_4) \xrightarrow{\sim} \mathrm{RHom}(E_1 \otimes_S^{\mathbf{L}} E_3, E_2 \otimes_S^{\mathbf{L}} E_4)$$

(where $E_1 = p_{12,1}^* L_1$, etc., with the evident notation) and of the isomorphism defined by Künneth

$$E_2 \otimes_S^{\mathbf{L}} E_4 \xrightarrow{\sim} p_{24}^!(L_2 \otimes_S^{\mathbf{L}} L_4) \quad (1.7.3).$$

Let $c' : C' \to X_{12}$ and $c'' : C'' \to X_{34}$ be correspondences, and let $c : C = C' \times_S C'' \to X = X_{13} \times_S X_{24}$ be the product correspondence. From 5.3.2 one deduces, or else defines analogously, a pairing

$$\mathrm{RHom}({c'_1}^* L_1, {c'_2}^! L_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}({c''_3}^* L_3, {c''_4}^! L_4) \longrightarrow \mathrm{RHom}(c_{13}^*(L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^!(L_2 \otimes_S^{\mathbf{L}} L_4)), \tag{5.3.4}$$

hence an external tensor product pairing, $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$,

$$\mathrm{Hom}({c'_1}^* L_1, {c'_2}^! L_2) \otimes \mathrm{Hom}({c''_3}^* L_3, {c''_4}^! L_4) \longrightarrow \mathrm{Hom}(c_{13}^*(L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^!(L_2 \otimes_S^{\mathbf{L}} L_4)). \tag{5.3.5}$$

One checks that these pairings are compatible with direct images, in the following sense. Let, for $1 \leq i \leq 4$, $f_i : X_i \to Y_i$ be an S-morphism, let c', c'', c be as above, with c' and c'' proper, let $d' : D' \to Y_{12}$ and $d'' : D'' \to Y_{34}$ be proper morphisms, let $d : D = D' \times_S D'' \to Y = Y_{12} \times_S Y_{34}$, and suppose given the commutative squares

$$\begin{array}{ccc} X_{12} & \longleftarrow & C' \\ {\scriptstyle f_{12}}\downarrow & & \downarrow \\ Y_{12} & \longleftarrow & D' \end{array} \qquad \begin{array}{ccc} X_{34} & \longleftarrow & C'' \\ {\scriptstyle f_{34}}\downarrow & & \downarrow \\ Y_{34} & \longleftarrow & D'' \end{array}$$

Then one has a commutative square

$$\begin{array}{ccc} \mathrm{Hom}({c'_1}^* L_1, {c'_2}^! L_2) \otimes \mathrm{Hom}({c''_3}^* L_3, {c''_4}^! L_4) & \xrightarrow{5.3.5} & \mathrm{Hom}(c_{13}^* L_{13}, c_{24}^! L_{24}) \\ \downarrow{\scriptstyle f_{12} \otimes f_{34}} & & \downarrow{\scriptstyle f_{13,24*}} \\ \mathrm{Hom}({d'_1}^* M_1, {d'_2}^! M_2) \otimes \mathrm{Hom}({d''_3}^* M_3, {d''_4}^! M_4) & \xrightarrow{5.3.5} & \mathrm{Hom}(d_{13}^* M_{13}, d_{24}^! M_{24}), \end{array} \tag{5.3.6}$$

where $M_i = f_{i!}L_i$ if $i = 1, 3$ and $f_{i*}L_i$ otherwise, $L_{ij} = L_i \otimes_S^{\mathbf{L}} L_j$, and $M_{ij} = M_i \otimes_S^{\mathbf{L}} M_j$ (there is also an analogous square with the RHom's).

Suppose that $X_3 = X_2$, $X_4 = X_1$, $L_3 = L_2$, and $L_4 = L_1$. A reasoning analogous to that of the last paragraph of 5.2 shows that, for $u \in \mathrm{Hom}_S(L_1, L_2)$ and $v \in \mathrm{Hom}_S(L_2, L_1)$, one has

$$\langle u, v\rangle = \langle u \otimes_S^{\mathbf{L}} v, 1\rangle, \tag{5.3.7}$$

where 1 denotes the diagonal correspondence from L_{12} to L_{12} on X_{12}, and the cup product in the second member is taken in the sense of 4.2.5. More generally, if $c' : C' \to X_{12}$ and $c'' : C'' \to X_{21}$ are correspondences, their "intersection" $C = C' \times_{X_{12}} C''$ identifies with the fibred product of $C' \times_S C'' \to X = X_{12} \times_S X_{21}$ and the diagonal $X_{12} \to X$, and, for $u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2)$ and $v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1)$, one has the following equality in $H^0(C, K_C)$:

$$\langle u, v\rangle = \langle u \otimes_S^{\mathbf{L}} v, 1\rangle, \tag{5.3.8}$$

where 1 denotes, as above, the diagonal correspondence. Thus there are two "diagonal" procedures (5.2.10, 5.3.8) for the calculation of the "Verdier local term" $\langle u, v\rangle$.

6. Appendix. Lefschetz formula for coherent sheaves

In this number, S denotes a noetherian scheme. The S-schemes considered will be assumed separated and of finite type. Recall that, by Nagata, every S-morphism between such schemes is compactifiable, i.e. factors as an open immersion followed by a proper morphism. For every S-scheme X, we shall write $D(X)$ for the derived category of the category of $\mathcal{O}_X$-modules.

6.1

Recall ([8] App.) that for every S-morphism $f : X \to Y$ one has a functor

$$f^! : D^+(Y) \longrightarrow D^+(X)$$

(and transitivity isomorphisms $(gf)^! = f^! g^!$ for a composite gf, with a cocycle condition). If f is proper, $f^!$ is right "adjoint" to the functor $f_* : D^+(X)_{\mathrm{coh}} \to D^+(Y)_{\mathrm{coh}}$ (where the subscript "coh" denotes the full subcategory formed by complexes with coherent cohomology), this adjunction being made precise by an isomorphism

$$f_* \mathrm{RHom}(E, f^! F) \xrightarrow{\sim} \mathrm{RHom}(f_* E, F) \tag{6.1.1}$$

for $E \in \mathrm{ob}\, D(X)_{\mathrm{coh}}$, $F \in \mathrm{ob}\, D^+(Y)$, defined analogously to the isomorphism (SGA 4 XVIII 3.1.9.6). The considerations of ([8] App.5), together with the duality theorem for projective space, show that, if f is smooth, purely of dimension d, there is a canonical isomorphism

$$f^! L \xleftarrow{\sim} f^* L \otimes \Omega^d_{X/Y}[d]; \tag{6.1.2}$$

and, if moreover f is proper, the arrow

$$\mathrm{Tr}_f : f_*(f^* L \otimes \Omega^d_{X/Y}[d]) \longrightarrow L \tag{6.1.3}$$

deduced, by 6.1.2, from the adjunction arrow $f_* f^! \to I$, coincides with the trace morphism ([8] VI).

6.2

Contrary to what happens in the case of discrete coefficients, there is no reasonable subcategory of D here that is stable under the "six operations" $(f^*, f_*, f^!, f_!, \otimes^{\mathbf{L}}, \mathrm{RHom})$. The most natural finiteness notion seems to be relative perfection (SGA 6 III 4). If X is an S-scheme and $L \in \mathrm{ob}\, D(X)$, recall that L is said to be perfect relative to S if L has finite tor-dimension relative to S and coherent cohomology. We shall denote by $D(X)_{\mathrm{parf}/S}$ the full subcategory of $D(X)$ formed by the complexes perfect relative to S. It is not stable in general either under $\otimes^{\mathbf{L}}$ or under RHom. However, if E is perfect in the absolute sense and F is perfect relative to S, then $\mathrm{RHom}(E, F)$ is perfect relative to S (SGA 6 III 4.9). On the other hand, let, for $i = 1, 2$, X_i be an S-scheme and $L_i \in \mathrm{ob}\, D^-(X_i)$. We shall put, as in 1.5,

$$L_1 \otimes^{\mathbf{L}}_S L_2 = Lp_1^* L_1 \otimes^{\mathbf{L}} Lp_2^* L_2,$$

where $p_i : X_1 \times_S X_2 \to X_i$ is the projection. Considering this object is reasonable only if X_1 and X_2 are tor-independent over S, i.e. (SGA 6 III 1.5) if $\mathrm{Tor}_i^S(\mathcal{O}_X, \mathcal{O}_Y) = 0$ for $i \neq 0$. Suppose this is the case: then, if L_1 and L_2 are perfect relative to S, so is $L_1 \otimes^{\mathbf{L}}_S L_2$ (SGA 6 III 4.7).

Let $f : X \to Y$ be an S-morphism. If f is proper, then $f_* D(X)_{\mathrm{parf}/S} \subset D(Y)_{\mathrm{parf}/S}$ (SGA 6 III 4.8). If f has finite tor-dimension, then $f^* D(Y)_{\mathrm{parf}/S} \subset D(X)_{\mathrm{parf}/S}$ (SGA 6 III 4.5), and $f^! D(Y)_{\mathrm{parf}/S} \subset D(X)_{\mathrm{parf}/S}$ (SGA 6 III 4.9).

For every S-scheme $a : X \to S$, we put

$$K_X = a^! \mathcal{O}_S, \tag{6.2.1}$$

and write D_X (or simply D) for the functor $\mathrm{RHom}(-, K_X)$. If X has finite tor-dimension over S, then K_X is perfect relative to S and, for every $L \in \mathrm{ob}\, D(X)_{\mathrm{parf}/S}$, one has $DL \in \mathrm{ob}\, D(X)_{\mathrm{parf}/S}$, and the canonical arrow $L \to DDL$ is an isomorphism (SGA 6 III 4.9).

Let $f : X \to Y$ be an S-morphism, $L \in \mathrm{ob}\, D(X)$, and $M \in \mathrm{ob}\, D(Y)$. If $f^* M$ is defined (for example if $M \in \mathrm{ob}\, D^-(Y)$, or if f has finite tor-dimension), put

$$\mathrm{Hom}_f^{(1)}(L, M) = \mathrm{Hom}(f^* M, L), \qquad \mathrm{Hom}_f^{(2)}(L, M) = \mathrm{Hom}(L, f^* M).$$

If $f^! M$ is defined (for example if $M \in \mathrm{ob}\, D^+(Y)$, or if f is smooth, or finite of finite tor-dimension, or a composite of such morphisms, since in these cases $f^!$ extends, by 6.1.1 and 6.1.2, to a functor defined on the whole category $D(Y)$), put

$$\mathrm{Hom}_f^{(3)}(L, M) = \mathrm{Hom}(L, f^! M), \qquad \mathrm{Hom}_f^{(4)}(L, M) = \mathrm{Hom}(f^! M, L).$$

The elements of $\mathrm{Hom}_f^{(i)}(L, M)$ will be called, as in 1.4, *arrows of type i from L to M over f* (or f-arrows of type i from L to M). We leave to the reader the task of making explicit the composition of arrows of type i, for fixed i, and the corresponding fibred and cofibred categories. We recall only the partial adjunction formulas

$$\mathrm{Hom}_f^{(1)}(L, M) = \mathrm{Hom}(M, f_* L) \quad (M \in \mathrm{ob}\, D^-(Y),\ L \in \mathrm{ob}\, D^+(X)),$$

$$\mathrm{Hom}_f^{(3)}(L, M) = \mathrm{Hom}(f_* L, M) \quad (f \text{ proper},\ L \in \mathrm{ob}\, D(X)_{\mathrm{coh}},\ M \in \mathrm{ob}\, D^+(Y)).$$

6.3

In the situation of 1.6, suppose that X_1 and X_2 are tor-independent over S, and likewise Y_1 and Y_2. Then, for $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$, the Künneth arrow 1.6.4 is defined (use the fact that finite relative tor-dimension is preserved by direct image (SGA 6 III 3.7.1)) and is an isomorphism (to verify this, one may assume S and the Y_i affine, and, for a Čech calculation, reduce to also assuming X_1 and X_2 affine; the conclusion is then a trivial consequence of the tor-independence hypotheses). On the other hand, for $M_i \in \mathrm{ob}\, D(Y_i)_{\mathrm{parf}/S}$, with $f_1^! M_1$ or $f_2^! M_2$ in D^b, the Künneth arrow 1.7.3 is defined[12], and is an isomorphism when f_1 and f_2 have finite tor-dimension, as is easily checked with the help of the Künneth isomorphism of type 1.6.4 just indicated, conjugated with 6.1.2, and with 6.1.1 in the case where f_i is a closed immersion. In particular, if X_1 and X_2 have finite tor-dimension over S and are tor-independent, 1.7.6 is an isomorphism.

One defines $\mathrm{RHom}(u, v)$ as in 2.1 for (u, v) of type $(2, 1)$ (respectively $(3, 4)$, with f proper in order to avoid the pro-objects caused by an $f_!$). Under suitable degree hypotheses (see for example ([9] V 2.3)), the evaluation arrows (2.2.2) and tensor-product arrows (2.2.3, 2.2.4) are defined, and the arrows 2.2.3 and 2.2.4 (provided they are defined) are isomorphisms as soon as E_1 and E_2 are perfect in the absolute sense.

In the situation of 2.4, suppose that the pairs (X_1, X_2), (X_1, Y_2), (Y_1, X_2), (Y_1, Y_2) are tor-independent over S, that f_1 is proper, that E_i, F_i, P_i, Q_i, $\mathrm{RHom}(E_i, F_i)$, $\mathrm{RHom}(P_i, Q_i)$ are perfect relative to S, and that the Künneth arrows

$$f_1^! Q_1 \otimes_S^{\mathbf{L}} Q_2 \to (f_1 \times_S Y_2)^!(Q_1 \otimes_S^{\mathbf{L}} Q_2), \qquad f_1^! Q_1 \otimes_S^{\mathbf{L}} F_2 \to (f_1 \times_S Y_2)^!(Q_1 \otimes_S^{\mathbf{L}} F_2)$$

are isomorphisms. Then 2.4.1 and the diagram of 2.5 are defined, and the latter is commutative (however, the horizontal arrows are not necessarily isomorphisms). The verification is the same as for 2.5.

6.4

Let, for $i = 1, 2$, X_i be an S-scheme, let $X = X_1 \times_S X_2$, and let $p_i : X \to X_i$, $q_i : X_i \to S$ be the canonical projections. Suppose that X_1 and X_2 are tor-independent and that q_1 has finite tor-dimension. Let $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$. By the analogue of 2.2.4, one has a canonical arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \mathrm{RHom}(L_1 \otimes_S^{\mathbf{L}} \mathcal{O}_{X_2}, q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2), \tag{6.4.1}$$

and hence, by composing with the Künneth isomorphism (6.3) $q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2 \xrightarrow{\sim} p_2^! L_2$, an arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \mathrm{RHom}(p_1^* L_1, p_2^! L_2). \tag{6.4.2}$$

The arrow 6.4.1, and therefore also 6.4.2, is an isomorphism, as one sees by reducing to the case where q_1 is a closed immersion.

As in 3.1.2, we shall put

$$\mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}_S(L_1, L_2), \qquad \mathrm{Hom}(p_1^* L_1, p_2^! L_2) = \mathrm{Hom}_S(L_1, L_2), \tag{6.4.3}$$

[12] In the proper or lissifiable case, the definition of the arrow is immediate; the general case can be reduced to the lissifiable case by cohomological descent.

and call the elements of $\mathrm{Hom}_S(L_1, L_2)$ *cohomological correspondences from* L_1 *to* L_2. If $c : C \to X$ is a correspondence (proper in practice), then by the induction formula ([8] III 8.8)[13]

$$c^!\mathrm{RHom}(p_1^*L_1, p_2^!L_2) = \mathrm{RHom}(c_1^*L_1, c_2^!L_2), \tag{6.4.4}$$

and the elements of

$$\mathrm{Hom}(c_1^*L_1, c_2^!L_2) = H^0(X, c_*c^!\mathrm{RHom}(p_1^*L_1, p_2^!L_2))$$

will be called *cohomological correspondences from* L_1 *to* L_2 *with support in* C. We shall see below examples analogous to those of 3.2.

6.5

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be a proper S-morphism, and let $f = f_1 \times_S f_2 : X \to Y$. Suppose that the X_i and Y_i have finite tor-dimension over S, and that the squares of 2.4.0 are tor-independent. Let $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$. As in 3.3.1, one defines an isomorphism

$$f_*\mathrm{RHom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{RHom}_S(f_{1*}L_1, f_{2*}L_2), \tag{6.5.1}$$

hence, by applying $R\Gamma(Y, -)$, an isomorphism

$$f_* : \mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_S(f_{1*}L_1, f_{2*}L_2), \tag{6.5.2}$$

which we shall simply denote by $u \mapsto u_*$ when $Y_1 = Y_2 = S$. There is a commutative diagram analogous to 3.4.1, as follows from the analogue of 2.5 indicated in 6.3. One deduces from it a corollary analogous to 6.5 and, in the case where $Y_1 = Y_2 = S$ is the spectrum of a field, a description, analogous to that of 3.6, of the operations u_* on cohomology.

Given a commutative square 3.7.1, with c proper, one has, for $K \in \mathrm{ob}\, D^+(C)$, a canonical arrow

$$f_*c_*c^!K \longrightarrow d_*d^!f_*K, \tag{6.5.3}$$

defined in the same way as 3.7.3 (the isomorphism $g'_*c'^! \xrightarrow{\sim} d^!f_*$ is also valid in the present context: definition of the arrow by transposition of a base-change arrow, and verification that it is an isomorphism by reduction to the cases where d is smooth and where d is a closed immersion). Taking $K = \mathrm{RHom}_S(L_1, L_2)$, one deduces from 6.5.3 and 6.5.1 canonical arrows

$$f_*c_*\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \longrightarrow d_*\mathrm{RHom}(d_1^*(f_{1*}L_1), d_2^!(f_{2*}L_2)), \tag{6.5.4}$$

$$f_* : \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \longrightarrow \mathrm{Hom}(d_1^*(f_{1*}L_1), d_2^!(f_{2*}L_2)), \tag{6.5.5}$$

analogous to 3.7.5 and 3.7.6.

6.6

Let, for $i = 1, 2$, X_i be S-schemes of finite tor-dimension and tor-independent over S, and let $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$. The external tensor product of the evaluation arrows $DL_i \otimes^{\mathbf{L}} L_i \to K_{X_i}$ defines, as in 4.1, "cup-product" pairings analogous to 4.1.2, 4.1.3, and 4.1.4.

[13] The "lissifiability" hypothesis of (loc. cit.) is unnecessary.

Let $c : C \to X$ and $d : D \to X$ be proper morphisms, let $e = c \times_X d : E \to X$, and let $P, Q \in \operatorname{ob} D(X)_{\mathrm{parf}/S}$. By 6.1.1, one has

$$c_* c^! P = \mathrm{RHom}(Rc_* \mathcal{O}_C, P)$$

(and an analogous isomorphism with d). Assuming c and d of finite tor-dimension at the points above $e(E)$, one deduces, by tensor product, a canonical arrow

$$c_* c^! P \otimes^{\mathbf{L}} d_* d^! Q \longrightarrow \mathrm{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, P \otimes^{\mathbf{L}} Q). \tag{6.6.1}$$

When c and d are tor-independent, the second member is rewritten, by Künneth and 6.1.1, as $e_* e^!(P \otimes^{\mathbf{L}} Q)$, and 6.6.1 takes the form 4.2.3. For $P = DL_1 \otimes^{\mathbf{L}}_S L_2$, $Q = L_1 \otimes^{\mathbf{L}}_S DL_2$, one deduces from 6.6.1 and the pairings indicated above pairings analogous to 4.2.4 and 4.2.5:

$$c_* \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \otimes^{\mathbf{L}} d_* \mathrm{RHom}(d_2^* L_2, d_1^! L_1) \longrightarrow \mathrm{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X), \tag{6.6.2}$$

$$\langle\ ,\ \rangle : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(d_2^* L_2, d_1^! L_1) \longrightarrow \mathrm{Hom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X). \tag{6.6.3}$$

Just like the pairings of 4.2, these are compatible with étale localization. When c and d are tor-independent, the second member of 6.6.2 (respectively 6.6.3) identifies canonically with $e_* K_E$ (respectively $H^0(E, K_E)$).

Theorem 4.4 has the following analogue:

Theorem 6.7. Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be a proper S-morphism, let $f = f_1 \times_S f_2 : X \to Y$, and let $p_i : X \to X_i$, $q_i : Y \to Y_i$ be the projections. Suppose given, on the other hand, a cube 4.4.0 with cartesian horizontal faces, with c', c'', d', d'' proper, and with c', c'' (respectively d', d'') of finite tor-dimension at the points above $c(C)$ (respectively $d(D)$). Put $c'_i = p_i c'$, $c''_i = p_i c''$, $d'_i = q_i d'$, and $d''_i = q_i d''$. Finally, let $L_i \in \operatorname{ob} D(X_i)_{\mathrm{parf}/S}$. Assume that the pairs (X_1, X_2), (X_1, Y_2), (Y_1, X_2), (Y_1, Y_2) are tor-independent over S, and that X_i and Y_i have finite tor-dimension over S. Then one has a commutative square

$$\begin{array}{ccc} f_* c'_* \mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes^{\mathbf{L}} f_* c''_* \mathrm{RHom}(c_2''^* L_2, c_1''^! L_1) & \xrightarrow{(1)} & f_* \mathrm{RHom}(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''}, K_X) \\ \downarrow{\scriptstyle (2)} & & \downarrow{\scriptstyle (4)} \\ d'_* \mathrm{RHom}(d_1'^* M_1, d_2'^! M_2) \otimes^{\mathbf{L}} d''_* \mathrm{RHom}(d_2''^* M_2, d_1''^! M_1) & \xrightarrow{(3)} & \mathrm{RHom}(Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''}, K_Y), \end{array} \tag{6.7.1}$$

where $M_i = f_{i*} L_i$, (3) is given by 6.6.2, (1) is deduced from 6.6.2 by application of f_* and composition with the canonical arrow $f_* c'_* \otimes^{\mathbf{L}} f_* c''_* \to f_*(c'_* \otimes^{\mathbf{L}} c''_*)$, (2) is the tensor product of the arrows 6.5.4 relative to the faces containing f, and (4) is the composite of the duality isomorphism 6.1.1 and of the arrow defined by the canonical arrow

$$Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''} \longrightarrow f_*(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''})$$

which is adjoint to the tensor product of the base-change arrows $f^* Rd'_* \mathcal{O}_{D'} \to Rc'_* \mathcal{O}_{C'}$, $f^* Rd''_* \mathcal{O}_{D''} \to Rc''_* \mathcal{O}_{C''}$.

The proof is essentially the same as that of 4.4; we leave to the reader the small modifications to be made.

When c' and c'' are tor-independent, and likewise d' and d'', the arrow (4) of 6.7.1 is rewritten in the form $f_*c_*K_C \to d_*K_D$; it is simply defined by the adjunction arrow $g_*K_C = g_*g^!K_D \to K_D$ (as is arrow (4) of 4.4.1). However, it would not be natural to consider only this case, since that would exclude correspondences with excess intersection.

From 6.7 one deduces formulas analogous to 4.5.1 and 4.7.1: under the hypotheses of 6.7, one has, for $u \in \mathrm{Hom}({c'_1}^*L_1, {c'_2}^!L_2)$ and $v \in \mathrm{Hom}({c''_2}^*L_2, {c''_1}^!L_1)$,

$$\langle f_*u, f_*v\rangle = f_*\langle u, v\rangle, \tag{6.7.2}$$

where f_*u (respectively f_*v) denotes the direct image correspondence of u (respectively v) in the sense of 6.5.5, and f_* in the second member is the homomorphism

$$\mathrm{Hom}(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''}, K_X) \longrightarrow \mathrm{Hom}(Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''}, K_Y) \tag{6.7.3}$$

which is the composite of the duality isomorphism 6.1.1 and the arrow induced by the canonical arrow

$$Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''} \longrightarrow f_*(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''})$$

cited in 6.7. When $Y_1 = Y_2 = S = D' = D''$, 6.7.2 is the *Lefschetz-Verdier formula.*

As an illustration, we shall show that this formula implies the Woods Hole formula [1].

6.8

For this purpose we shall need a few reminders on the "Hodge" cohomology class associated with a regularly embedded subscheme of a smooth scheme and on Grothendieck residues ([8] III 9). Let $f : X \to S$ be a smooth morphism purely of relative dimension N, and let $i : Y \hookrightarrow X$ be a regular immersion of codimension d, i.e. ((SGA IV 16.9.2), (SGA 6 VII 1)) a closed subscheme defined locally by a regular sequence of d equations, or, in equivalent terminology ((8 III 1), (SGA 6 VIII 1)), a closed immersion, locally a complete intersection, of codimension d. Put $g = fi$. Then one associates with Y an element

$$\mathrm{cl}^d_{X/S}(Y) \in \mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}), \tag{6.8.1}$$

called the *cohomology class associated with* Y, defined as follows. By the "purity" theorem ([8] III 7.2), one has

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}) = \begin{cases} 0, & i \neq d, \\ \mathrm{Hom}(\bigwedge^d N_{Y/X}, \Omega^d_{X/S} \otimes \mathcal{O}_Y), & i = d, \end{cases} \tag{6.8.2}$$

where $N_{Y/X} = I/I^2$ is the conormal sheaf ($I = \mathrm{Ker}\ \mathcal{O}_X \to \mathcal{O}_Y$). Therefore

$$\mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}) = H^0(Y, \underline{\mathrm{Hom}}(\bigwedge^d N_{Y/X}, \Omega^d_{X/S} \otimes \mathcal{O}_Y)). \tag{6.8.3}$$

The differential $d_{X/S} : \mathcal{O}_X \to \Omega^1_{X/S}$ induces an $\mathcal{O}_Y$-linear map

$$d_{X/S} \otimes 1 : N_{Y/X} \longrightarrow \Omega^1_{X/S} \otimes \mathcal{O}_Y.$$

The d-th exterior power of this map is a global section, on Y, of $\Lambda^d N^\vee_{Y/X} \otimes \Omega^d_{X/S}$; its image in $\mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S})$ under the isomorphism 6.8.3 is, by definition, the class 6.8.1.

Assume g finite and flat (hence $d = N$). Let $\omega \in H^0(Y, \Lambda^d N_{Y/X}^\vee \otimes \Omega^d_{X/S})$. By 6.8.2 and 6.1, one may view ω as a homomorphism

$$\mathcal{O}_Y \longrightarrow g^!\mathcal{O}_S (= \Lambda^d N_{Y/X}^\vee \otimes \Omega^d_{X/S}),$$

and therefore, by adjunction, as a homomorphism $g_*\mathcal{O}_Y \to \mathcal{O}_S$. The residue (relative to g) of ω is defined as the section of $\mathcal{O}_S$ which is the image of 1 under this homomorphism, and is denoted $\mathrm{Res}_{Y/S}(\omega)$ (cf. ([8] III 9)). Let, on the other hand, $i_*\omega \in H^N(X, \Omega^N_{X/S})$ denote the image of ω by the composite of the isomorphism 6.8.3 and the canonical arrow

$$\mathrm{Ext}^N_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^N_{X/S}) \longrightarrow H^N(X, \Omega^N_{X/S}).$$

When f is proper, it follows from transitivity for trace morphisms (or adjunction arrows) $i_*i^! \to 1$, $f_*f^! \to 1$, that

$$\mathrm{Res}_{Y/S}(\omega) = \mathrm{Tr}_f(i_*\omega), \tag{6.8.4}$$

where $\mathrm{Tr}_f : H^N(X, \Omega^N_{X/S}) \to H^0(S, \mathcal{O}_S)$ is defined by the adjunction arrow $f_*f^!\mathcal{O}_S = Rf_*\Omega^N_{X/S}[N] \to \mathcal{O}_S$ (6.1). On the other hand, without any properness hypothesis, one has the important compatibility ([8] III 9 R6)

$$\mathrm{Res}_{Y/S}(y\, \mathrm{cl}_{X/S}(Y)) = \mathrm{Tr}_{Y/S}(y), \tag{6.8.5}$$

for $y \in H^0(Y, \mathcal{O}_Y)$, where $\mathrm{Tr}_{Y/S} : g_*\mathcal{O}_Y \to \mathcal{O}_S$ denotes the trace in the classical sense, i.e. $\mathrm{Tr}_{Y/S}(y)$ is the trace of the endomorphism of $g_*\mathcal{O}_Y$ defined by multiplication by y (this compatibility is not proved in loc. cit.; the case $N = 1$ is treated in [Raynaud, Anneaux locaux henséliens, VII 1, Lecture Notes in Math. n° 169, Springer Verlag, 1970]). In other words, $\mathrm{Tr}_{Y/S}$ corresponds, by adjunction, to the homomorphism $\mathcal{O}_Y \to g^!\mathcal{O}_S$ defined by multiplication by $\mathrm{cl}_{X/S}(Y)$. It follows that, for $F \in \mathrm{ob}\, D(S)_{\mathrm{coh}}$, the arrow

$$g^*F \longrightarrow g^!F = g^*F \otimes^{\mathbf{L}} g^!\mathcal{O}_S \tag{6.8.6}$$

given by multiplication by $\mathrm{cl}_{X/S}(Y)$ corresponds by adjunction to the arrow

$$g_*g^*F = F \otimes^{\mathbf{L}} g_*\mathcal{O}_Y \longrightarrow F \tag{6.8.7}$$

defined by $\mathrm{Tr}_{Y/S} : g_*\mathcal{O}_Y \to \mathcal{O}_S$.

6.9

Let X (respectively Y) be a proper smooth S-scheme of relative dimension m (respectively n), $Z = X \times_S Y$, and let $a : A \hookrightarrow Z$, $b : B \hookrightarrow Z$ be regular immersions such that $a_2 : A \to Y$, $b_1 : B \to X$ are finite and flat (write $p_1 : Z \to X$, $p_2 : Z \to Y$ for the projections, and $a_i = p_i a$, $b_i = p_i b$). Let $c : C = A \times_Z B \to Z$, $L \in \mathrm{ob}\, D(X)_{\mathrm{parf}}$, $M \in \mathrm{ob}\, D(Y)_{\mathrm{parf}}$ (the subscript parf means "perfect in the absolute sense", which is also equivalent to being perfect rel. to S, since X and Y are smooth). Assume that C is finite and flat over S. The hypotheses made therefore imply that c is a regular immersion and that a and b are tor-independent.

One has $p_1^*\Omega^1_{X/S} = \Omega^1_{Z/Y}$, $p_2^*\Omega^1_{Y/S} = \Omega^1_{Z/X}$, and a decomposition

$$\Omega^1_{Z/S} = \Omega^1_{Z/Y} \oplus \Omega^1_{Z/X},$$

whence, for every integer r, a decomposition

$$\Omega^r_{Z/S} = \bigoplus_{i+j=r} \Omega^i_{Z/Y} \otimes \Omega^j_{Z/X}. \tag{6.9.0}$$

On the other hand, by 6.1.2, one has $p_2^! M = p_2^* M \otimes \Omega^m_{Z/Y}[m]$, hence

$$H^0(Z, a_* a^! \mathrm{RHom}(p_1^* L, p_2^! M)) = \mathrm{Ext}^m_{\mathcal{O}_Z}(a_1^* L, a_2^* M \otimes \Omega^m_{Z/Y}),$$

whence, by 6.4.4 and 6.8.2, an isomorphism

$$\mathrm{Hom}(a_1^* L, a_2^! M) = H^0(A, \mathrm{RHom}(a_1^* L, a_2^* M) \otimes \Lambda^m N^\vee_{A/Z} \otimes \Omega^m_{Z/Y}). \tag{6.9.1}$$

Similarly one has an isomorphism

$$\mathrm{Hom}(b_2^* M, b_1^! L) = H^0(B, \mathrm{RHom}(b_2^* M, b_1^* L) \otimes \Lambda^n N^\vee_{B/Z} \otimes \Omega^n_{Z/X}). \tag{6.9.2}$$

Let $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$, and write

$$\langle u, v \rangle \in H^0(C, \mathcal{O}_C) \tag{6.9.3}$$

for the section $\mathrm{Tr}(u_C v_C) = \mathrm{Tr}(v_C u_C)$, where $u_C \in \mathrm{Hom}(c_1^* L, c_2^* M)$ and $v_C \in \mathrm{Hom}(c_2^* M, c_1^* L)$ are induced by u and v. Consider the cohomological correspondence $u^! \in \mathrm{Hom}(a_1^* L, a_2^! M)$ (respectively $v^! \in \mathrm{Hom}(b_2^* M, b_1^! L)$) deduced from u (respectively v) by composition with the trace homomorphism 6.8.7 $a_{2*} a_2^* M \to M$ (respectively $b_{1*} b_1^* L \to L$), or equivalently (6.8.6) by product with $\mathrm{cl}_{Z/Y}(A)$ (respectively $\mathrm{cl}_{Z/X}(B)$). Note in passing that $\mathrm{cl}_{Z/Y}(A)$ (respectively $\mathrm{cl}_{Z/X}(B)$) is the component in $H^0(A, \Lambda^m N^\vee_{A/Z} \otimes \Omega^m_{Z/Y})$ (respectively $H^0(B, \Lambda^n N^\vee_{B/Z} \otimes \Omega^n_{Z/X})$) of $\mathrm{cl}_{Z/S}(A)$ (respectively $\mathrm{cl}_{Z/S}(B)$) defined by the projection $\Omega^m_{Z/S} \to \Omega^m_{Z/Y}$ (respectively $\Omega^n_{Z/S} \to \Omega^n_{Z/X}$) of 6.9.0. One checks easily that the cup-product

$$\langle u^!, v^! \rangle \in \mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) = H^0(C, \Lambda^{m+n} N^\vee_{C/Z} \otimes \Omega^{m+n}_{Z/S})$$

defined in 6.6.3 is given by

$$\langle u^!, v^! \rangle = \langle u, v \rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B). \tag{6.9.4}$$

On the other hand one checks that the homomorphism 6.7.3

$$\mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) \longrightarrow H^0(S, \mathcal{O}_S)$$

is the composite of the canonical arrow $\mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) \to H^{m+n}(Z, \Omega^{m+n}_{Z/S})$ and the trace morphism $\mathrm{Tr}_{Z/S} : H^{m+n}(Z, \Omega^{m+n}_{Z/S}) \to H^0(S, \mathcal{O}_S)$. The Lefschetz-Verdier formula 6.7.2 therefore gives the relation

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Tr}_{Z/S}(\langle u, v \rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B)). \tag{6.9.5}$$

Thanks to 6.8.4 one may rewrite the second member as a residue; thus, in summary, one has obtained:

Theorem 6.10. Under the hypotheses of the first paragraph of 6.9, let $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$, and consider the cohomological correspondence $u^!$ (respectively $v^!$) from L to M (respectively from M to L), with support in A (respectively B), deduced from u (respectively v) by composition with the trace $a_{2*} a_2^* M \to M$ (respectively $b_{1*} b_1^* L \to L$) (6.8.7), and the homomorphism $(u^!)_* : f_* L \to g_* M$ (respectively $(v^!)_* : g_* M \to f_* L$) deduced from $u^!$ (respectively $v^!$) by direct

image (6.5.5) ($f : X \to S$, $g : Y \to S$ denoting the projections). Then, with the notation of 6.8, 6.9.3, one has

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Res}_{C/S}(\langle u, v \rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B)). \tag{6.10.1}$$

Remarks 6.11. a) As in 3.6, using the description 3.5 b) of the direct image of a correspondence, one checks that, if $h : Z \to S$ is the projection, $(u^!)_*$ (respectively $(v^!)_*$) is the composite of the arrows

$$f_*L \xrightarrow{(1)} (ha)_* a_1^* L \xrightarrow{(ha)_* u} (ha)_* a_2^* M = g_* a_{2*} a_2^* M \xrightarrow{(2)} g_* M,$$

$$(\text{respectively } g_*M \xrightarrow{(1)} (hb)_* b_2^* M \xrightarrow{(hb)_* v} (hb)_* b_1^* L = f_* b_{1*} b_1^* L \xrightarrow{(2)} f_* L),$$

where (1) is the usual functoriality arrow (defined by the adjunction arrow $1 \to a_{1*}a_1^*$ (respectively $1 \to b_{2*}b_2^*$)) and (2) is given by the trace $a_{2*}a_2^*M \to M$ (respectively $b_{1*}b_1^*L \to L$) (6.8.7).

b) Suppose that S is the spectrum of a field. The intersection $C = A \times_{(X \times_S Y)} B$ is then formed of isolated points. At each point z of C, choose, near the images of z, regular sequences $(s_1, \ldots, s_m)$, $(t_1, \ldots, t_n)$ of equations of A, B. Then 6.10.1 is written, with the notation of the residue symbols of ([8] III 9), as

$$\langle (u^!)_*, (v^!)_* \rangle = \sum_{z \in C} \mathrm{Res}_z \left(\langle u, v \rangle \prod_{1 \le i \le m} \frac{d_{Z/Y} s_i}{s_i} \prod_{1 \le j \le n} \frac{d_{Z/X} t_j}{t_j} \right), \tag{6.11.1}$$

where $d_{Z/Y}s_i$ (respectively $d_{Z/X}t_j$) denotes the component of type $(1,0)$ (respectively $(0,1)$) of $d_{Z/S}s_i$ (respectively $d_{Z/S}t_j$) in the decomposition 6.9.0.

Corollary 6.12. (Woods-Hole formula [1]). Assume that S is the spectrum of a field. Let X be a proper smooth S-scheme, F an S-endomorphism of X, L an object of $D(X)_{\mathrm{parf}}$ (i.e. a perfect complex on X), and $u \in \mathrm{Hom}(F^*L, L)$. Assume that the scheme of fixed points X^F is finite and consists of transverse points (i.e. points where the graph of F meets the diagonal transversely). Then

$$\sum_i (-1)^i \, \mathrm{Tr}((F,u), H^i(X,L)) = \sum_{x \in X^F} \frac{\mathrm{Tr}(u_x)}{\det(1 - dF_x)},$$

where (F,u) is the endomorphism of $H^i(X,L)$ which is the composite of $H^i(X,L) \to H^i(X, F^*L)$ (inverse image) and $H^i(X,u)$, $\mathrm{Tr}(u_x)$ denotes the value at x of the endomorphism induced by u on the fibre of L at x, and $\det(1 - dF_x)$ is the value at x of the determinant of the map (1 − map induced by F on $\Omega^1_{X,x}$).

Proof. Apply 6.10, taking account of 6.11 a), b). Put $Z = X \times_S X$. Let $x \in X^F$. Near x, choose a system of local coordinates $(x_1, \ldots, x_n)$ (defining an étale morphism into $\mathbb{A}^n_S$), put $F_i = x_i F$, and write $(x_1, \ldots, x_n, y_1, \ldots, y_n)$ for the corresponding local coordinates on Z near $z = (x,x)$ ($x_i = x_i \otimes 1$, $y_i = 1 \otimes x_i$). Then, near z, $(x_1 - y_1, \ldots, x_n - y_n)$ (respectively $(y_1 - F_1, \ldots, y_n - F_n)$) is a regular system of equations for the diagonal A (respectively for the graph B of F). The contribution at z of the second member of 6.11.1 is

$$\langle u, 1 \rangle_z = \mathrm{Res}_z \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n \, dy_1 \cdots dy_n}{(x_1 - y_1) \cdots (x_n - y_n)(y_1 - F_1) \cdots (y_n - F_n)} \right).$$

Observe that in the numerator one may replace dy_i by

$$dy_i - dF_i = dy_i - \sum_{1 \le j \le n} (dF_i / dx_j) dx_j.$$

Applying then formula ([8] III 9 R3) to the situation $(B \hookrightarrow Z \to S)$, $(s_1, \dots, s_p) = (y_1 - F_1, \dots, y_n - F_n)$, $(t_1, \dots, t_n) = (x_1 - y_1, \dots, x_n - y_n)$, $\omega = (\mathrm{Tr}(u) dx_1 \cdots dx_n)$, one obtains

$$\langle u, 1 \rangle_z = \mathrm{Res}_x \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right). \tag{*}$$

Since x is transverse, the $x_i - F_i$ form a regular system of parameters of $\mathcal{O}_{X,x}$, and hence one may write, near x,

$$x_i = \sum_j (x_j - F_j) c_{ij},$$

where (c_{ij}) is a matrix of sections of $\mathcal{O}_X$ invertible at x, whose determinant has value at x equal to $\det(1 - dF_x)^{-1}$. Thanks to ([8] III 9 R1), one therefore deduces from (*) that

$$\langle u, 1 \rangle_z = \mathrm{Res}_x \left(\mathrm{Tr}(u) \det(c_{ij}) \frac{dx_1 \cdots dx_n}{x_1 \cdots x_n} \right),$$

whence, by ([8] III 9 R6),

$$\langle u, 1 \rangle_z = \mathrm{Tr}(u(x)) \det(c_{ij}(x)) = \mathrm{Tr}(u(x)) \det(1 - dF_x)^{-1}.$$

The corollary therefore follows from 6.11.1.

Remark 6.12.1. If X^F is assumed only finite, the preceding calculation shows that one has

$$\sum_i (-1)^i \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \mathrm{Res}_x \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right),$$

where (x_i) is a system of local coordinates at x, and the F_i are the coordinates of F (near x) in this system.

The reader will convince himself, moreover, that the preceding formula is still valid if X is proper and is assumed only to be smooth at the points of X^F.

6.13

Assume that S is the spectrum of an algebraic closure of the finite field $\mathbb{F}_q$ and that X/S is obtained by extension of scalars from a proper smooth scheme $X_0/\mathbb{F}_q$. Take for F the Frobenius endomorphism of X/S "defined by raising coordinates to the q-th power". One has $X^F = X_0(\mathbb{F}_q)$, and X^F consists of transverse points since $dF = 0$. If L is a locally free sheaf of finite type on X and $u \in \mathrm{Hom}(F^*L, L)$, the Woods-Hole formula is therefore written as

$$\sum_i (-1)^i \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X_0(\mathbb{F}_q)} \mathrm{Tr}(u_x, L_x). \tag{6.13.1}$$

In particular, for $L = \mathcal{O}_X$, and u the canonical isomorphism $F^*\mathcal{O}_X \xrightarrow{\sim} \mathcal{O}_X$, one finds:

$$\sum_i (-1)^i \mathrm{Tr}(F, H^i(X, \mathcal{O}_X)) = \mathrm{Card}(X_0(\mathbb{F}_q)) \pmod p. \tag{6.13.2}$$

Artin-Schreier theory makes it possible to rewrite the first member in the form $\sum_i (-1)^i \mathrm{Tr}(F, H^i(X, \mathbb{F}_p))$, where $H^*(X, \mathbb{F}_p)$ denotes the étale cohomology of X with values in the constant sheaf $\mathbb{F}_p$. By a

refinement of this argument, Deligne, in (SGA $4^{1/2}$ Fonctions L), deduced from 6.13.1 the following generalization of 6.13.2:

Theorem 6.13.3 (Deligne). Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, let X be the scheme deduced from X_0 by extension of scalars to an algebraic closure k of $\mathbb{F}_q$, and let F be the Frobenius endomorphism of X/k. Then, for every $L_0 \in \operatorname{ob} D_{\mathrm{ctf}}(X_0, \mathbb{F}_p)$, if L denotes the inverse image of L_0 on X, one has

$$\sum_i (-1)^i \operatorname{Tr}(F, H^i_c(X, L)) = \sum_{x \in X^F} \operatorname{Tr}(F, L_x).$$

Let us point out that the particular case of 6.13.3 corresponding to $L_0 = \mathbb{F}_p$ was established by Katz in (SGA 7 XXII) by a completely different method, based on Dwork's calculation of the zeta function of a hypersurface. Moreover, 6.13.2 in the proper smooth case is also an immediate consequence of the Lefschetz formula in crystalline cohomology ([5] VII 3.1.9). In fact, crystalline cohomology gives the more precise information that each factor $\det(1 - Ft, H^i(X_0/W)) \in W[t]$ of the zeta function is congruent mod p to $\det(1 - Ft, H^i(X_0, \mathcal{O}_{X_0}))$ (at least if $H^*(X_0/W)$ is torsion-free). Katz showed [11] that, using estimates of B. Mazur [14], one could obtain higher congruences (involving the $H^i(X_0, \Omega^j_{X_0})$ and the Cartier operator) for smooth hypersurfaces (or complete intersections) for which the Hodge numbers (in the interesting dimension) are assumed to vanish up to a certain level.

On the other hand, if X_0 is a proper scheme over $\mathbb{F}_q$, geometrically integral, and such that $H^i(X_0, \mathcal{O}_{X_0}) = 0$ for $i > 0$, the Lefschetz-Verdier formula 6.7.2, applied to $X = X_0 \otimes k$ and the Frobenius endomorphism of X, immediately implies that X_0 has at least one rational point over $\mathbb{F}_q$ (since $\operatorname{Tr}(F, R\Gamma(X, \mathcal{O}_X)) = 1$). Serre asks whether this conclusion is still true when one replaces $\mathbb{F}_q$ by a field satisfying (C_1) ([17] II 3), for example $\mathbb{C}(t)$ (Tsen), or $\mathbb{C}((t))$ (Lang) (see loc. cit. for other examples); indeed, he observes that if X_0 is a complete intersection of multidegree $(m_1, \ldots, m_r)$ in $\mathbb{P}^n$, the condition of triviality of cohomology is equivalent ([16] no. 78) to the condition $m_1 + \cdots + m_r < n + 1$, which is also the condition that occurs when one writes condition (C_1).

6.14

The Woods Hole formula has many other applications, for which we refer to [2] and to Beauville's exposé [4].

6.15

The editor has not examined the case of components of fixed points of dimension ≥ 1, but it is not impossible that one could deduce from 6.7.2 (probably conjugated with Riemann-Roch) Lefschetz-Riemann-Roch formulae in the style of those of Donovan [7], Iversen [10], Nielsen [15] (at least with values in the base field). Nor is it impossible, on the other hand, that the formula 6.7.2, applied over the dual numbers, might provide residue formulae for vector fields analogous to those of Bott [6], [3].

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Exposé III B
Calculations of Local Terms

by Luc Illusie

Introduction

This exposé, written in January 1977, does not correspond to any oral exposé of the seminar. It comprises two independent parts. In the first, we compute the Lefschetz–Verdier local terms for cohomological correspondences between smooth curves over an algebraically closed field, supported on correspondences satisfying certain general-position hypotheses. The formula that we obtain gives as corollaries Verdier's formula [6] for transversal endomorphisms (which applies in particular to the case of the Frobenius endomorphism), and the formula stated by Langlands in ([4] 7.12). The method of proof is parallel to that followed by Artin–Verdier in [6]. One reduces to proving the vanishing of the local term when the given complexes of sheaves have zero fibre at the images of the fixed point under consideration, and in fact one proves a stronger property, namely that a certain restriction map is zero (2.5). To establish this property, one first reduces to the tamely ramified case by covers of the same degree of the two given curves (so as not to destroy the general-position hypotheses), and then treats this case directly by means of Abhyankar's lemma and the relative purity theorem. The second part of this exposé, of a much more technical nature, is inspired by the method used by Grothendieck to establish the Lefschetz formula for certain cohomological correspondences on curves (see XII and (SGA $4^{1/2}$ Rapport)). In no. 5, for complexes of modules on a topos, we develop a theory of non-commutative traces which generalizes the commutative theory of (SGA 6 I) and, in the case of a punctual topos, recovers the non-commutative theory developed by Grothendieck in the oral seminar (this was written up in Exposé XI, unfortunately lost). We apply the techniques of no. 5 to define, in no. 6, Lefschetz–Verdier local terms for cohomological correspondences between complexes of modules over rings that are not necessarily commutative. These local terms behave like non-commutative traces and make it possible, in particular when one takes as base ring the algebra of a group, to divide canonically the ordinary local terms attached to equivariant correspondences. As an application, we prove Grothendieck's divisibility conjecture of (XII 4.5), in a more general form. We also show that the local terms defined by Grothendieck in the Lefschetz formula of XII are indeed the Lefschetz–Verdier local terms, and we generalize the formula of (loc. cit.).

I thank P. Deligne for the useful suggestions he made to me during many discussions; I am particularly grateful to him for having pointed out, and helped me correct, an error in an early proof of Theorem 1.2.

Summary

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I. Correspondences in General Position between Curves

In this part, S denotes the spectrum of an algebraically closed field k of characteristic p, and Λ a noetherian commutative ring killed by an integer prime to p. For every scheme X over S we denote by $D(X)$ the derived category of sheaves of Λ-modules on X, and by $D_{\mathrm{ctf}}(X)$ the full subcategory of $D(X)$ formed by complexes of finite tor-dimension and constructible cohomology.

1. Statement of the theorem and corollaries

1.1

Let X (resp. Y) be the strict localization, at a closed point, of a smooth separated S-curve, let A (resp. B) be a strictly local S-scheme, and let

$$a : A \longrightarrow X \times_S Y, \qquad b : B \longrightarrow X \times_S Y$$

be S-morphisms. We denote by $p_1 : X \times_S Y \to X$, $p_2 : X \times_S Y \to Y$ the projections, put $a_i = p_i a$, $b_i = p_i b$, and denote by x (resp. y) the closed point of X (resp. Y). We assume that a_2 and b_1 are finite and surjective, and that $A \times_{(X \times_S Y)} B$ is reduced to the closed point z whose image in $X \times_S Y$ is (x, y).

Let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, $M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$, $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, and $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$. Denote by

$$u_z \in \mathrm{Hom}(L_x, M_y) \qquad (\text{resp. } v_z \in \mathrm{Hom}(M_y, L_x))$$

the composite homomorphism

$$L_x = R\Gamma(X, L) \xrightarrow{a_1^*} R\Gamma(A, a_1^* L) \xrightarrow{R\Gamma(A,u)} R\Gamma(A, a_2^! M) \xrightarrow{a_{2*}} R\Gamma(Y, M) = M_y,$$

where a_{2*} is defined by the adjunction map $a_{2*} a_2^! M \to M$ (resp. the analogous composite homomorphism defined by means of (b, v)). Since L_x and M_y are perfect complexes of Λ-modules, one can form the cup-product (SGA 6 I 8.3)

$$\langle u_z, v_z \rangle = \mathrm{Tr}(u_z v_z) = \mathrm{Tr}(v_z u_z) \in \Lambda. \tag{1.1.1}$$

Put $\Lambda_X(1)[2] = K_X$, $\Lambda_Y(1)[2] = K_Y$. By (I 5) (or (SGA $4^{1/2}$ Duality 1, or Finitude 4.1)) the functor $D_X = \mathrm{RHom}(-, K_X)$ (resp. $D_Y = \mathrm{RHom}(-, K_Y)$) on $D_{\mathrm{ctf}}(X)$ (resp. $D_{\mathrm{ctf}}(Y)$) is dualizing, i.e. for

every $E \in \operatorname{ob} D_{\mathrm{ctf}}(X)$ (resp. $E \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$), one has $E \xrightarrow{\sim} D_X D_X E$ (resp. $E \xrightarrow{\sim} D_Y D_Y E$). We shall sometimes write D instead of D_X, D_Y.

On the other hand, denote by $p_i^!$ the functor $p_i^*(-)(1)[2]$. We have trivially

$$K_X \otimes_S^{\mathbf{L}} M \xrightarrow{\sim} p_2^! M, \qquad L \otimes_S^{\mathbf{L}} K_Y \xrightarrow{\sim} p_1^! L,$$

and it follows from (III 2.3), by passage to the limit, that the Künneth maps analogous to (III 3.1.1)

$$DL \otimes_S^{\mathbf{L}} M \longrightarrow \mathrm{RHom}(p_1^* L, p_2^! M), \qquad L \otimes_S^{\mathbf{L}} DM \longrightarrow \mathrm{RHom}(p_2^* M, p_1^! L)$$

are isomorphisms. Since a_2 is finite, a is finite, and hence the functor $a^!$ is defined; one verifies easily that one has the transitivity isomorphism $a^! p_2^! = a_2^!$. By the induction formula for a (SGA 4 XVIII 3.1.12.2), one therefore has a canonical isomorphism analogous to (III 3.2.1)

$$a^! \mathrm{RHom}(p_1^* L, p_2^! M) \xrightarrow{\sim} \mathrm{RHom}(a_1^* L, a_2^! M),$$

whence an isomorphism

$$\mathrm{Hom}(a_1^* L, a_2^! M) \xrightarrow{\sim} H^0(X \times_S Y, a_* a^! (DL \otimes_S^{\mathbf{L}} M)). \tag{*}$$

Likewise one has an isomorphism

$$\mathrm{Hom}(b_2^* M, b_1^! L) \xrightarrow{\sim} H^0(X \times_S Y, b_* b^! (L \otimes_S^{\mathbf{L}} DM)). \tag{**}$$

As in (III 4.1.2), one defines a natural pairing

$$(DL \otimes_S^{\mathbf{L}} M) \otimes^{\mathbf{L}} (L \otimes_S^{\mathbf{L}} DM) \longrightarrow K_X \otimes_S^{\mathbf{L}} K_Y = \Lambda_{X \times_S Y}(2)[4] \overset{\text{dfn}}{=} K_{X \times_S Y}.$$

Composing with the pairing analogous to (III 4.2.3)

$$a_* a^! P \otimes^{\mathbf{L}} b_* b^! Q \longrightarrow c_* c^! (P \otimes^{\mathbf{L}} Q),$$

where $P = DL \otimes_S^{\mathbf{L}} M$, $Q = L \otimes_S^{\mathbf{L}} DM$, and $c : Z = A \times_{(X \times_S Y)} B \to X \times_S Y$ is the canonical projection, and taking into account the isomorphisms (*), (**), one deduces a pairing analogous to (III 4.2.5)

$$\langle\ ,\ \rangle : \mathrm{Hom}(a_1^* L, a_2^! M) \otimes \mathrm{Hom}(b_2^* M, b_1^! L) \longrightarrow H^0(X \times_S Y, c_* c^! K_{X \times_S Y}).$$

One verifies easily, by passage to the limit, that $c^! K_{X \times_S Y} = \Lambda_z$, so that $H^0(X \times_S Y, c_* c^! K_{X \times_S Y}) = \Lambda$. Thus one has a "cup-product"

$$\langle u, v \rangle \in \Lambda. \tag{1.1.2}$$

1.1.3

Let X' (resp. Y') be a smooth S-curve, A' (resp. B') a separated S-curve,

$$a' : A' \longrightarrow X' \times_S Y', \qquad b' : B' \longrightarrow X' \times_S Y'$$

finite S-morphisms, x' (resp. y') a closed point of X' (resp. Y'), z' an isolated closed point of $A' \times_{(X' \times_S Y')} B'$, with image s' (resp. t') in A' (resp. B'), and such that $a'_2 (= p_2 a')$ (resp. $b'_1 (= p_1 b')$) is quasi-finite at s' (resp. t'),

$$L' \in \operatorname{ob} D_{\mathrm{ctf}}(X'), \quad M' \in \operatorname{ob} D_{\mathrm{ctf}}(Y'), \quad u' \in \mathrm{Hom}(a_1'^* L', a_2'^! M'), \quad v' \in \mathrm{Hom}(b_2'^* M', b_1'^! L').$$

One can then consider the local term (III 4.2.7) $\langle u', v' \rangle_{z'} \in \Lambda$. On the other hand, if X, Y, A, B are the strict localizations of X', Y', A', B' at x', y', s', t', if $a : A \to X \times_S Y$, $b : B \to X \times_S Y$ are the morphisms deduced from a', b', if $L = L'|X$, $M = M'|Y$, and if $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$ are the maps deduced from u', v', then the data (X, Y, a, b, u, v) satisfy the hypotheses at the beginning of 1.1, and hence a cup-product 1.1.2 $\langle u, v \rangle \in \Lambda$ is defined. It follows from the compatibility of the formation of local terms (III 4.2.7) with localization (III 4.2.6) that $\langle u', v' \rangle_{z'} = \langle u, v \rangle$. It is also easy to see that any data (X, Y, a, b, u, v) satisfying the hypotheses at the beginning of 1.1 are induced by data $(X', Y', a', b', x', y', z', s', t', u', v')$ as above, where one may also assume, if desired, that $A' \times_{(X' \times_S Y')} B'$ is reduced to z'.

In general the cup-products 1.1.1 and 1.1.2 are unequal. Nevertheless one has the following theorem, which is the main result of this first part:

Theorem 1.2. Under the hypotheses at the beginning of 1.1, assume moreover that A and B are in general position, i.e. that the intersection of the tangent cones at (x, y) to the images of A and B in $X \times_S Y$ is reduced to (x, y), and that likewise A and $X \times \{y\}$ are in general position, as are $\{x\} \times Y$ and B. Then

$$\langle u_z, v_z \rangle = \langle u, v \rangle. \tag{1.2.1}$$

By the Lefschetz–Verdier formula (III 4.7), taking 1.1.3 into account, one deduces:

Corollary 1.3. Let F be an endomorphism of a proper smooth curve X/S such that the fixed-point scheme X^F is finite and formed of points where the graph of F is transverse to the diagonal. Let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, and $u \in \mathrm{Hom}(F^* L, L)$. Then

$$\mathrm{Tr}((F, u), R\Gamma(X, L)) = \sum_{x \in X^F} \mathrm{Tr}(u_x, L_x),$$

where the trace on both sides is taken in the sense of (SGA 6 I 8).

For L concentrated in degree 0, 1.3 is the result of Artin–Verdier ([6] 4.1). By a passage to the limit explained in XV, one deduces from 1.3 an analogous result in ℓ-adic cohomology ([6] 1.1), (XV §2 no. 3 prop. 2):

Corollary 1.3.1. With X and F as in 1.3, let L be a constructible $\mathbb{Q}_\ell$-sheaf on X (ℓ prime to p), and let $u \in \mathrm{Hom}(F^* L, L)$. Then:

$$\sum_i (-1)^i \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \mathrm{Tr}(u_x, L_x).$$

The hypotheses of 1.3 and 1.3.1 apply in particular to the case where (F, u) is a Frobenius correspondence (XV 2), whence Grothendieck's trace formula ([2] 5.1), (SGA $4^{1/2}$ Rapport 3.2).

1.4

With the notation of 1.1, assume that b_2 and a_1 are finite and surjective (instead of a_2 and b_1), and consider the dual correspondences (III 5.1)

$$Du \in \mathrm{Hom}(a_2^* DM, a_1^! DL), \qquad Dv \in \mathrm{Hom}(b_1^* DL, b_2^! DM),$$

and the homomorphisms $(Du)_z \in \mathrm{Hom}((DM)_y, (DM)_x)$, $(Dv)_z \in \mathrm{Hom}((DL)_x, (DM)_y)$. Assume that $(A, \{x\} \times Y)$ is in general position, as are $(X \times \{y\}, B)$ and (A, B). Applying 1.2 to Du, Dv, one obtains

$$\langle Du, Dv \rangle = \langle (Du)_z, (Dv)_z \rangle,$$

whence, by (III 5.1.6) and (III 4.2.6),

$$\langle u, v \rangle = \langle (Du)_z, (Dv)_z \rangle. \tag{1.4.1}$$

From this remark and from 1.2 one deduces, by the Lefschetz–Verdier formula (III 4.7):

Corollary 1.5. Let X, Y be proper smooth curves over S,

$$a : A \longrightarrow X \times_S Y, \qquad b : B \longrightarrow X \times_S Y$$

correspondences where A and B are proper curves over S, let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, $M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$, $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, and $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$. Assume that $C = A \times_{(X \times_S Y)} B$ is finite over S and of the form $C = C_1 \amalg C_2$, with the following properties: for $z \in C_1$, the morphisms deduced from a_2, b_1 by strict localization at the images of z are finite, and, if (x, y) denotes the image of z in $X \times_S Y$, each of the pairs (A, B), $(A, X \times \{y\})$, $(\{y\} \times Y, B)$ is in general position at (x, y); for $z \in C_2$, the morphisms deduced from a_1, b_2 by strict localization at the images of z are finite, and each of the pairs (A, B), $(A, \{x\} \times Y)$, $(B, X \times \{y\})$ is in general position at (x, y) ("general position" meaning that the intersection of the tangent cones at (x, y) to the images of the curves of the pair under consideration is reduced to (x, y)). Then, if $u_* \in \mathrm{Hom}(R\Gamma(X, L), R\Gamma(Y, M))$, $v_* \in \mathrm{Hom}(R\Gamma(Y, M), R\Gamma(X, L))$ are the morphisms defined by u, v (III 3.7.6), one has, with the notation of 1.1.1, (III 5.1.5),

$$\langle u_*, v_* \rangle = \sum_{z \in C_1} \langle u_z, v_z \rangle + \sum_{z \in C_2} \langle (Du)_z, (Dv)_z \rangle. \tag{1.5.1}$$

1.6

Assuming the formalism of ℓ-adic cohomology, one deduces from 1.5, by the passage-to-the-limit technique of XIV, that, under the hypotheses of 1.5 on X, Y, a, b, for $L \in \mathrm{ob}\, D^b(X, \mathbb{Q}_\ell)$ (ℓ prime $\neq p$), $M \in \mathrm{ob}\, D^b(Y, \mathbb{Q}_\ell)$, $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$, one has the analogue of 1.5.1 in $\mathbb{Q}_\ell$. This formula contains, as a special case, Proposition 7.12 of Langlands [4]: apply the formula with $X = Y$, M the sheaf F considered by Langlands; $L = \alpha M$ ($\alpha \in \mathbb{Q}_\ell^*$), A the diagonal of $X \times X$, b the correspondence ϕ of (loc. cit.), u the correspondence supported on A given by $\alpha^{-1} : L \to M$, v the correspondence Φ of (loc. cit.); with the notation of (loc. cit.), C_1 (resp. C_2) is the set of fixed points where $a < d$ (resp. $a > d$).

1.7

Under the hypotheses of 1.2, take for u (resp. v) the correspondence $\mathrm{cl}(A)$ (resp. $\mathrm{cl}(B)$) defined by $\mathrm{Tr}_{a_2} : a_{2*}\Lambda \to \Lambda$ (resp. $\mathrm{Tr}_{b_1} : b_{1*}\Lambda \to \Lambda$). Formula 1.2.1 becomes

$$\deg(a_2) \deg(b_1) = \langle \mathrm{cl}(A), \mathrm{cl}(B) \rangle, \tag{1.7.1}$$

where $\deg(a_2)$ (resp. $\deg(b_1)$) denotes the generic degree of a_2 (resp. b_1). Taking into account the hypotheses of general position, the left hand side of 1.7.1 is none other than the intersection multiplicity at (x, y) of the cycles a_*A, b_*B, the direct images of A and B. It is not difficult to deduce 1.7.1 directly from the compatibility of the formation of the cohomology class associated with a cycle with intersections (SGA $4^{1/2}$ Cycle 2.3.8).

2. Reduction to a vanishing theorem

2.1

Under the hypotheses of 1.1, let s (resp. t) denote the image of z in A (resp. B), and let $i : U = X - \{x\} \hookrightarrow X$ and $j : V = Y - \{y\} \hookrightarrow Y$ be the canonical inclusions. Since a_2 and b_1 are finite, one has $a_2^{-1}(y) = s$, $b_1^{-1}(x) = t$, hence

$$a_1^{-1}(U) \subset a_2^{-1}(V), \qquad b_2^{-1}(V) \subset b_1^{-1}(U),$$

and there are commutative diagrams

$$\begin{array}{ccccc} U & \xleftarrow{a_1'} & a_1^{-1}(U) & \xrightarrow{a_2'} & V \\ \downarrow{\scriptstyle i} & (1) & \downarrow & (2) & \downarrow{\scriptstyle j} \\ X & \xleftarrow[a_1]{} & A & \xrightarrow[a_2]{} & Y \end{array} \qquad \begin{array}{ccccc} U & \xleftarrow{b_1'} & b_2^{-1}(V) & \xrightarrow{b_2'} & V \\ \downarrow{\scriptstyle i} & (3) & \downarrow & (4) & \downarrow{\scriptstyle j} \\ X & \xleftarrow[b_1]{} & B & \xrightarrow[b_2]{} & Y, \end{array}$$

where squares (1) and (4) are cartesian.

Lemma 2.2. To prove 1.2, it is enough to consider the following two cases:

$$\text{a) } L_x = 0, \quad M_y = 0; \qquad \text{b) } L|U = 0, \quad M|V = 0.$$

Consider the decreasing filtrations on L and M defined by the subcomplexes

$$F^nL = \begin{cases} L, & n \leq 0, \\ i_!i^*L, & n = 1, \\ 0, & n > 1, \end{cases} \qquad F^nM = \begin{cases} M, & n \leq 0, \\ j_!j^*M, & n = 1, \\ 0, & n > 1. \end{cases}$$

The complex $a_{2*}a_1^*L$ (resp. $b_{1*}b_2^*M$) is then filtered by

$$F^n a_{2*}a_1^*L = a_{2*}a_1^*F^nL.$$

We have $F^n a_{2*}a_1^*L = a_{2*}a_1^*L$ for $n \leq 0$, and 0 for $n > 1$; moreover, since a_2 is finite and square (1) is cartesian,

$$F^1 a_{2*}a_1^*L = j_!(a_2')_!(a_1')^*i^*L.$$

Similarly,

$$F^n b_{1*}b_2^*M = \begin{cases} b_{1*}b_2^*M, & n \leq 0, \\ i_!(b_1')_!(b_2')^*j^*M, & n = 1, \\ 0, & n > 1. \end{cases}$$

We may suppose that u (resp. v) is defined by a homomorphism of complexes

$$u : a_{2*}a_1^*L \longrightarrow M \quad (\text{resp. } v : b_{1*}b_2^*M \longrightarrow L).$$

The preceding formulas show that the composite

$$F^1 a_{2*}a_1^*L \longrightarrow a_{2*}a_1^*L \xrightarrow{u} M$$

(resp. $F^1 b_{1*}b_2^*M \to b_{1*}b_2^*M \xrightarrow{v} L$) factors uniquely through F^1M (resp. F^1L); in other words, u (resp. v) is a homomorphism of filtered complexes. Thus u (resp. v) underlies a filtered correspondence, in the sense of (III 4.13), which we again denote by

$$u \in \mathrm{Hom}_{DF}(a_1^*L, a_2^!M), \qquad v \in \mathrm{Hom}_{DF}(b_2^*M, b_1^!L).$$

By additivity of cup-products ([3] V 3.8.7), (III 4.13.1), one has

$$\langle u_z, v_z\rangle = \langle \mathrm{gr}^0 u_z, \mathrm{gr}^0 v_z\rangle + \langle \mathrm{gr}^1 u_z, \mathrm{gr}^1 v_z\rangle,$$

and

$$\langle u, v\rangle = \langle \mathrm{gr}^0 u, \mathrm{gr}^0 v\rangle + \langle \mathrm{gr}^1 u, \mathrm{gr}^1 v\rangle.$$

The lemma follows, since $\mathrm{gr}^1 L = i_!i^*L$ (resp. $\mathrm{gr}^1 M = j_!j^*M$) has zero fibre at x (resp. y), whereas $\mathrm{gr}^0 L$ (resp. $\mathrm{gr}^0 M$) is concentrated at x (resp. y).

Lemma 2.3. Under the hypotheses of 1.1, suppose that $L|U = 0$ and $M|V = 0$. Then

$$\langle u_z, v_z\rangle = \langle u, v\rangle.$$

The idea of the proof is that then u and v are direct images of correspondences between complexes concentrated at the closed points, and that the formula to be proved merely expresses the Lefschetz formula for the projections $x \to S$, $y \to S$. In view of 1.1.3, the data of 1.1 may be replaced by the following data: X (resp. Y) is a smooth S-curve endowed with a closed point x (resp. y), A (resp. B) is a separated S-curve, $a : A \to X \times_S Y$, $b : B \to X \times_S Y$ are finite S-morphisms such that $A \times_{X\times_S Y} B$ is reduced to a closed point z above (x, y),

$$u \in \mathrm{Hom}(a_1^*L, a_2^!M), \qquad v \in \mathrm{Hom}(b_2^*M, b_1^!L),$$

with

$$L = h_{1*}L', \quad M = h_{2*}M', \quad L' \in \mathrm{ob}\, D_{\mathrm{ctf}}(\{x\}), \quad M' \in \mathrm{ob}\, D_{\mathrm{ctf}}(\{y\}),$$

where $h_1 : \{x\} \to X$ and $h_2 : \{y\} \to Y$ are the canonical inclusions. We must show that

$$\langle u_z, v_z\rangle = \langle u, v\rangle_z,$$

where

$$u_z \in \mathrm{Hom}(\mathrm{R}\Gamma(X, L), \mathrm{R}\Gamma(Y, M)), \qquad v_z \in \mathrm{Hom}(\mathrm{R}\Gamma(Y, M), \mathrm{R}\Gamma(X, L))$$

are defined as in 1.1, and $\langle u, v\rangle_z$ is the value at z of the cup-product (III 4.2.7). We have $\mathrm{R}\Gamma(X, L) = \mathrm{R}\Gamma_c(X, L) = L_x$, $\mathrm{R}\Gamma(Y, M) = \mathrm{R}\Gamma_c(Y, M) = M_y$, and u_z (resp. v_z) is nothing other than the direct image u_* (resp. v_*) of u (resp. v) on S in the sense of (III 3.7.6), as is shown by the calculation procedure (III 3.5 b)).

On the other hand, if $h = h_1 \times_S h_2 : (x,y) \to X \times_S Y$ is the inclusion, and if $a' : A' \to (x,y)$, $b' : B' \to (x,y)$ denote the fibres of a, b at (x,y), each reduced to a point, then there are canonical isomorphisms

$$h_* a'_* a'^{!}(DL_x \overset{\mathrm{L}}{\otimes}_\Lambda M_y) \xrightarrow{\sim} a_* a^{!}(DL \overset{\mathrm{L}}{\otimes}_\Lambda M),$$

$$h_* b'_* b'^{!}(L_x \overset{\mathrm{L}}{\otimes}_\Lambda DM_y) \xrightarrow{\sim} b_* b^{!}(L \overset{\mathrm{L}}{\otimes}_\Lambda DM),$$

and the arrows

$$h_* : \mathrm{Hom}(a_1'^{*} L_x, a_2'^{!} M_y) \to \mathrm{Hom}(a_1^{*} L, a_2^{!} M),$$

$$h_* : \mathrm{Hom}(b_2'^{*} M_y, b_1'^{!} L_x) \to \mathrm{Hom}(b_2^{*} M, b_1^{!} L)$$

(III 3.7.6) are isomorphisms. The correspondences u and v are therefore written uniquely as

$$u = h_* u', \qquad v = h_* v', \tag{1}$$

with

$$u' \in \mathrm{Hom}(a_1'^{*} L_x, a_2'^{!} M_y), \qquad v' \in \mathrm{Hom}(b_2'^{*} M_y, b_1'^{!} L_x).$$

By the Lefschetz formula (III 4.5.1) for $(h_1, h_2, A' \to A, B' \to B)$, one has

$$\langle u, v \rangle_z = \langle u', v' \rangle_z. \tag{2}$$

On the other hand, by (1), $u_* = u'_*$ and $v_* = v'_*$; hence, by the Lefschetz formula for $\{x\} \to S$, $\{y\} \to S$, one has

$$\langle u_*, v_* \rangle = \langle u', v' \rangle_z. \tag{3}$$

In view of the preceding remark, the conclusion follows by comparing (2) and (3).

Lemma 2.4. To prove 1.2, one may suppose that a and b are closed immersions, and it is enough to prove that, if $L_x = 0$ and $M_y = 0$, the restriction map

$$H^0_A(X \times_S Y, DL \overset{\mathrm{L}}{\otimes}_S M) \longrightarrow H^0_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_S M)) \tag{2.4.1}$$

is zero.

Put $X \times_S Y = Z$. The closed subschemes $a' : A' \to Z$, $b' : B' \to Z$, images of a and b, satisfy the same hypotheses as a and b. Let u' (resp. v') be the correspondence supported on A' (resp. B') which is the direct image of u (resp. v) by the arrow deduced from the adjunction arrow $f_* f^! \to 1$ (resp. $g_* g^! \to 1$), where $f : A \to A'$ (resp. $g : B \to B'$) is the projection. It is immediate that $u_z = u'_z$ and $v_z = v'_z$. On the other hand, it follows easily from the definition of the cup-product 1.1.2 that

$$\langle u, v \rangle = \langle u', v' \rangle,$$

hence the first assertion.

By 2.2 and 2.3, one may suppose that $L_x = 0$ and $M_y = 0$, and the conclusion of 1.2 then becomes $\langle u, v \rangle = 0$. Let $P, Q \in \mathrm{ob}\, D_{\mathrm{ctf}}(Z)$. The cup-product

$$H^0_A(Z, P) \otimes H^0_B(Z, Q) \longrightarrow H^0_z(Z, P \overset{\mathrm{L}}{\otimes} Q)$$

decomposes as

$$H^0_A(Z, P) \otimes H^0_B(Z, Q) \xrightarrow{b^* \otimes 1} H^0_z(B, b^* P) \otimes H^0_B(Z, Q) \longrightarrow H^0_z(Z, P \overset{\mathrm{L}}{\otimes} Q),$$

where b^* is the restriction and the second arrow, a variant of the product considered in (SGA $4^{1/2}$ Cycle 1.2.1), is derived from the evident product at the level of sections of sheaves. This assertion is easily checked, for example by means of the flasque resolutions of (SGA 4 XVII 4.2). If $a' : \{z\} \to B$, $b' : \{z\} \to A$, $c : \{z\} \to Z$ are the inclusions, one may also invoke the standard decomposition of the Künneth arrow (III 4.2.1)

$$a'^! P \overset{\mathrm{L}}{\otimes}_Z b'^! Q \longrightarrow c^!(P \overset{\mathrm{L}}{\otimes} Q)$$

into a "base-change arrow"

$$a'^! P \overset{\mathrm{L}}{\otimes}_Z b'^! Q \longrightarrow a'^! b^* P \overset{\mathrm{L}}{\otimes} a^* b'^! Q,$$

followed by "projection arrows"

$$a'^! b^* P \overset{\mathrm{L}}{\otimes} a^* b'^! Q \longrightarrow a'^!(b^* P \overset{\mathrm{L}}{\otimes} b'^! Q) \longrightarrow a'^! b'^!(P \overset{\mathrm{L}}{\otimes} Q)).$$

Taking $P = DL \overset{\mathrm{L}}{\otimes}_S M$, $Q = L \overset{\mathrm{L}}{\otimes}_S DM$, and returning to the definition of $\langle u, v \rangle$ (III 4.2.5), one sees that it is enough to prove that $b^* u = 0$, whence the conclusion.

In view of 2.4, 1.2 will therefore follow from the following more precise result.

Proposition 2.5. Under the hypotheses of 1.2, suppose that a and b are closed immersions and that $L_x = 0$, $M_y = 0$. Let T be the strict localization of $X \times_S Y$ at $z = (x, y)$, and identify A, B with closed subschemes of T. Then the restriction arrows

$$b^* : \mathrm{R}\Gamma_A(T, (DL \overset{\mathrm{L}}{\otimes}_S M)|T) \longrightarrow \mathrm{R}\Gamma_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_S M)),$$

$$a^* : \mathrm{R}\Gamma_B(T, (L \overset{\mathrm{L}}{\otimes}_S DM)|T) \longrightarrow \mathrm{R}\Gamma_z(A, a^*(L \overset{\mathrm{L}}{\otimes}_S DM))$$

are zero.

2.6

The proof of 2.5 will occupy the rest of number 2 and numbers 3 and 4. By symmetry of the statement, it is enough to prove the assertion concerning b^*. We shall rewrite it in a form which will be more convenient. First note that

$$\mathrm{R}\Gamma(T, (DL \overset{\mathrm{L}}{\otimes}_S M)|T) = (DL \overset{\mathrm{L}}{\otimes}_S M)_z = (DL)_x \overset{\mathrm{L}}{\otimes} M_y = 0,$$

because $M_y = 0$, and similarly

$$\mathrm{R}\Gamma(B, b^*(DL \overset{\mathrm{L}}{\otimes}_S M)) = (DL \overset{\mathrm{L}}{\otimes}_S M)_z = 0.$$

Consequently, the relative cohomology triangles give isomorphisms

$$\mathrm{R}\Gamma(T - A, (DL \overset{\mathrm{L}}{\otimes}_S M)|T - A) \xrightarrow{\sim} \mathrm{R}\Gamma_A(T, (DL \overset{\mathrm{L}}{\otimes}_S M)|T)[1],$$

$$\mathrm{R}\Gamma(B - z, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|B - z) \xrightarrow{\sim} \mathrm{R}\Gamma_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M))[1],$$

under which the arrow b^* of 2.5, shifted by 1, is identified with the restriction

$$b^* : \mathrm{R}\Gamma(T - A, (DL \overset{\mathrm{L}}{\otimes}_S M)|T - A) \longrightarrow \mathrm{R}\Gamma(B - z, (DL \overset{\mathrm{L}}{\otimes}_S M)|B - z). \tag{2.6.1}$$

Since $L_x = 0$, $M_y = 0$, one may write

$$L = i_! DP, \qquad M = j_! Q, \tag{2.6.2}$$

where $i : X - x \to X$, $j : Y - y \to Y$ are the inclusions, and $P \in \mathrm{ob}\, D_{\mathrm{ctf}}(X - x)$, $Q \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y - y)$. By the duality isomorphism (SGA 4 XVIII 3.19.6), together with passage to the limit, one has

$$DL = Ri_* DDP = Ri_* P,$$

hence

$$DL \overset{\mathrm{L}}{\otimes}_S M = Ri_* P \overset{\mathrm{L}}{\otimes}_S j_! Q.$$

Put $X \times_S Y = Z$, and identify $X \times \{y\}$ and $\{x\} \times Y$ with closed subschemes of Z (resp. T), which we shall simply denote by X and Y. Let

$$f' : Z - Y \hookrightarrow Z, \qquad g' : Z - (X \cup Y) \hookrightarrow Z - Y,$$

$$f : T - Z \hookrightarrow T, \qquad g : T - (X \cup Y) \hookrightarrow T - Y$$

be the inclusions. The Künneth formula (III 1.6.4) gives, by passage to the limit,

$$DL \overset{\mathrm{L}}{\otimes}_S M = Rf'_* g'_! (P \overset{\mathrm{L}}{\otimes}_S Q),$$

and hence

$$(DL \overset{\mathrm{L}}{\otimes}_S M)|T = Rf_* g_! ((P \overset{\mathrm{L}}{\otimes}_S Q)|T - (X \cup Y)). \tag{2.6.3}$$

With this identification, 2.6.1 becomes

$$b^* : \mathrm{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) \longrightarrow \mathrm{R}\Gamma(B - z, g_!(E)|B - z), \tag{2.6.4}$$

where

$$E = (P \overset{\mathrm{L}}{\otimes}_S Q)|T - (X \cup Y).$$

In the next two numbers, we shall show that the arrow 2.6.4 is zero: first we shall reduce to the case where the cohomology sheaves of E have tame ramification, more precisely where the inertia action occurs through an ℓ-group, for a prime number $\ell \neq p$, and then we shall treat that case directly by means of a purity theorem.

3. Reduction to the tame case

3.1

Recall that the base ring Λ is annihilated by an integer prime to p, hence is a direct product of noetherian rings Λ_i, each Λ_i being annihilated by a power of a prime number $\ell_i \neq p$. For every scheme S' over S, the derived category $D(S', \Lambda)$ is the product of the categories $D(S', \Lambda_i)$, so that, in order to prove the vanishing of 2.6.4, we may suppose that Λ is annihilated by a power of a prime number $\ell \neq p$. We may also suppose that P (resp. Q) is bounded and has locally free components of finite type.

3.2

Put $X - x = U$, $Y - y = V$. Let U''/U be a finite Galois covering with group G which trivializes the components of P, let H be an ℓ-Sylow subgroup of G, let $U' = U''/H$ be the quotient covering, and let X' be the normalized strictly local trait of X in U'. Then X'/X is a finite connected covering of degree d_1 prime to ℓ, and the representation of $\pi_1(U')$ defined by the components of $P' = P|U'$ factors through a finite ℓ-group. Proceeding similarly with Q, one finds a finite connected covering Y'/Y of degree d_2 prime to ℓ, inducing on V an étale covering V' such that the representation of $\pi_1(V')$ defined by the components of $Q' = Q|V'$ factors through a finite ℓ-group. Let

$$d = d_1 e_1 = d_2 e_2 = \operatorname{lcm}(d_1, d_2).$$

After adjoining an e_1-th root (resp. an e_2-th root) of a uniformizer of X' (resp. Y'), we may suppose that X' (resp. Y') has the same properties as above, and moreover that $d_1 = d_2 = d$.

Let x' (resp. y') be the closed point of X' (resp. Y'), T' the strict localization of $X' \times_S Y'$ at (x', y'), z' the closed point of T', A' (resp. B') the inverse image of A (resp. B) in T', and

$$E' = (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T' - (X' \cup Y') = (P' \overset{\mathrm{L}}{\otimes}_\Lambda Q')|T' - (X' \cup Y').$$

We then have a commutative square of restriction arrows

$$\begin{array}{ccc} \mathrm{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) & \xrightarrow{b^*} & \mathrm{R}\Gamma(B - z, g_!(E)|B - z) \\ \downarrow & & \downarrow (1) \\ \mathrm{R}\Gamma(T' - (Y' \cup A'), g'_!(E')|T' - (Y' \cup A')) & \xrightarrow{b'^*} & \mathrm{R}\Gamma(B' - z', g'_!(E')|B' - z'), \end{array}$$

where $b' : B' \to T'$, $g' : T' - (X' \cup Y') \to T' - Y'$ are the inclusions. The covering T'/T is finite, of degree d^2, and étale outside $X \cup Y$, hence induces an étale covering of degree d^2 from $B' - z'$ onto $B - z$. The composite of (1) and the trace morphism

$$\mathrm{R}\Gamma(B' - z', g'_!(E')|B' - z') \longrightarrow \mathrm{R}\Gamma(B - z, g_!(E)|B - z)$$

is therefore multiplication by d^2, which is an isomorphism, since ℓ does not divide d. To prove that the arrow 2.6.4 is zero, it is therefore enough to prove that $b'^* = 0$. We are thus reduced to the case where the representation of $\pi_1(U)$ (resp. $\pi_1(V)$) defined by the components of P (resp. Q) factors through a finite ℓ-group, provided one ensures that A', B' satisfy the general-position hypotheses of 1.2. This will follow from the following lemma.

Lemma 3.3. Under the hypotheses of 1.2, a and b being closed immersions, let X'/X, Y'/Y be finite coverings of strictly local S-traits, of the same degree d. Let A' (resp. B') denote the inverse image of A (resp. B) in $X' \times_S Y'$. Then the pairs (A', B'), (A', X'), (Y', B') are each in general position in the sense of 1.2.[14]

The proof of 3.3 will be given after a few preliminaries.

Lemma 3.4. Let $X \xleftarrow{s_1} W \xrightarrow{s_2} Y$ be finite morphisms of S-traits, let $s = (s_1, s_2) : W \to X \times_S Y$, and let u, v, w be uniformizers of X, Y, W respectively, so that

$$s_1^* u = a w^m \pmod{w^{m+1}}, \qquad s_2^* v = b w^n \pmod{w^{n+1}},$$

[14] With the usual abuse of notation $X' = X' \times \{y'\}$, etc. (where x' (resp. y') is the closed point of X' (resp. Y').

with $a, b \in k^*$. Then: (i) if $m < n$, $s(W)$ is tangent to X at $z = (x, y)$; (ii) if $m = n$, $s(W)$ is tangent to the subscheme of $X \times_S Y$ with equation

$$ap_2^*(v) - bp_1^*(u) = 0.$$

Let $Z = X \times_S Y$, and let W' be the reduced closed subscheme which is the image of s. In the tangent space to Z at z, the slope, with respect to the tangent to X, of the tangent to W' is the value at the origin of the function induced on W' by $p_2^*(v)/p_1^*(u)$; this is also the value at the origin of the function induced by $p_2^*(v)/p_1^*(u)$ on W, and is therefore 0 (resp. b/a) in case (i) (resp. (ii)).

In the situation of 3.4, we shall define the slope of W with respect to $((X, u), (Y, v))$ as the element of $k \cup \{\infty\}$ equal to 0 if $m < n$, to ∞ if $m > n$, and to b/a if $m = n$. This element is independent of the choice of w.

Lemma 3.5. Let $f : X' \to X$, $g : Y' \to Y$ be finite morphisms of S-traits with the same ramification index d, and let u, v, u', v' be uniformizers of X, Y, X', Y' respectively. Then there exists $C \in k^*$ such that, for every commutative diagram of finite morphisms of S-traits

$$\begin{array}{ccccc} X' & \xleftarrow{s_1'} & W' & \xrightarrow{s_2'} & Y' \\ & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle g} \\ X & \xleftarrow[s_1]{} & W & \xrightarrow[s_2]{} & Y, \end{array}$$

one has $\lambda = C\lambda'^d$, where λ (resp. λ') is the slope of W (resp. W') with respect to $((X, u), (Y, v))$ (resp. $((X', u'), (Y', v'))$).

By hypothesis,

$$f^*u = \alpha u'^d \pmod{u'^{d+1}}, \qquad g^*v = \beta v'^d \pmod{v'^{d+1}},$$

with $\alpha, \beta \in k^*$. We show that $C = \beta/\alpha$ has the required properties. Let w (resp. w') be a uniformizer of W (resp. W'). We have

$$s_1^*u = aw^m \pmod{w^{m+1}}, \qquad s_2^*v = bw^n \pmod{w^{n+1}},$$

$$s_1'^*u' = a'w'^{m'} \pmod{w'^{m'+1}}, \qquad s_2'^*v' = b'w'^{n'} \pmod{w'^{n'+1}},$$

$$h^*w = \gamma w'^r \pmod{w'^{r+1}},$$

with $a, b, a', b', \gamma \in k^*$. Commutativity of the squares in the diagram implies

$$m'd = rm, \qquad n'd = rn,$$

hence $m'/n' = m/n$. Suppose first that $m = n$. Then $\lambda = b/a$, $\lambda' = b'/a'$. If $N = m'd = rm = n'd = rn$, the preceding congruences give

$$(fs_1')^*u = \alpha a'^d w'^N \pmod{w'^{N+1}}, \qquad (s_1h)^*u = a\gamma^m w'^N \pmod{w'^{N+1}},$$

whence $\alpha a'^d = a\gamma^m$. Similarly $\beta b'^d = b\gamma^m$, whence $\lambda = C\lambda'^d$, for $\lambda \neq 0, \infty$. But, since $m'/n' = m/n$, $\lambda = 0$ (resp. ∞) is equivalent to $\lambda' = 0$ (resp. ∞), which completes the proof.

We now prove 3.3. Let $Z = X \times_S Y$, $Z' = X' \times_S Y'$, let $\widetilde{A}$ (resp. $\widetilde{B}$) be the normalization of A (resp. B), and let $\widetilde{A}'$ (resp. $\widetilde{B}'$) be the normalization of A' (resp. B'). Since $\widetilde{A}'$ (resp. $\widetilde{B}'$) is also the normalization of $\widetilde{A} \times_Z Z'$ (resp. $\widetilde{B} \times_Z Z'$), we have commutative squares

$$\begin{array}{ccccc} \widetilde{A}' & \longrightarrow & Z' & \longleftarrow & \widetilde{B}' \\ \downarrow & & \downarrow & & \downarrow \\ \widetilde{A} & \longrightarrow & Z & \longleftarrow & \widetilde{B}. \end{array} \tag{*}$$

Let $(x_i)_{1\le i\le m}$ (resp. $(y_j)_{1\le j\le n}$) be the points of $\widetilde{A}$ (resp. $\widetilde{B}$) above x (resp. y), and let $(x'_i)_{1\le i\le m'}$ (resp. $(y'_j)_{1\le j\le n'}$) be the points of $\widetilde{A}'$ (resp. $\widetilde{B}'$) above x' (resp. y'). Since A and B are strictly local, we have decompositions

$$\widetilde{A} = \coprod_{1\le i\le m} A_i, \qquad \widetilde{B} = \coprod_{1\le j\le n} B_j, \qquad \widetilde{A}' = \coprod_{1\le i\le m'} A'_i, \qquad \widetilde{B}' = \coprod_{1\le j\le n'} B'_j,$$

where A_i, B_j, A'_i, B'_j are strictly local traits with respective closed points x_i, y_j, x'_i, y'_j. Choose uniformizers u, v, u', v' of X, Y, X', Y' respectively; let λ_i (resp. μ_j) be the slope of A_i (resp. B_j) with respect to $((X,u),(Y,v))$, and let λ'_i (resp. μ'_j) be that of A'_i (resp. B'_j) with respect to $((X',u'),(Y',v'))$. The general-position hypotheses on A and B are expressed by

$$\lambda_i \ne 0 \text{ for all } i, \qquad \mu_j \ne \infty \text{ for all } j, \qquad \lambda_i \ne \mu_j \text{ for all } i,j, \tag{1}$$

and we have to show that

$$\lambda'_i \ne 0 \text{ for all } i, \qquad \mu'_j \ne \infty \text{ for all } j, \qquad \lambda'_i \ne \mu'_j \text{ for all } i,j. \tag{2}$$

Let $i' \in [1,m']$, $j' \in [1,n']$, and let x_i (resp. y_j) be the image of $x'_{i'}$ (resp. $y'_{j'}$). The preceding diagram provides commutative squares of finite morphisms of S-traits

$$\begin{array}{ccccc} X' & \longleftarrow & A'_{i'} & \longrightarrow & Y' \\ & & \downarrow & & \downarrow \\ X & \longleftarrow & A_i & \longrightarrow & Y \end{array} \qquad \begin{array}{ccccc} X' & \longleftarrow & B'_{j'} & \longrightarrow & Y' \\ & & \downarrow & & \downarrow \\ X & \longleftarrow & B_j & \longrightarrow & Y. \end{array}$$

By 3.5, there exists $C \in k^*$ such that

$$\lambda_i = C{\lambda'_{i'}}^d, \qquad \mu_j = C{\mu'_{j'}}^d.$$

Thus (1) implies (2), which completes the proof of 3.3.

4. End of the proof of 1.2

4.1

To complete the proof of 2.5, hence of 1.2, it remains to establish the vanishing of 2.6.4 under the following additional hypotheses: Λ is annihilated by a power of a prime number $\ell \ne p$, and

P and Q are bounded, with locally free components of finite type, such that the corresponding representations of the fundamental groups factor through finite ℓ-groups. We have

$$g_!(E)|T-(Y\cup A) = g_{1!}(E_1),$$

where $g_1 : T-(X\cup Y\cup A)\hookrightarrow T-(Y\cup A)$ is the inclusion, and

$$E_1 = (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T-(X\cup Y\cup A).$$

Now E_1 is a complex with bounded, constructible, locally constant cohomology such that the corresponding representation of $\pi_1(T-(X\cup Y\cup A))$ factors through an ℓ-group. The assertion to be proved will therefore follow from the following more general statement.

Proposition 4.2. Let T be the strict localization at a closed point of a smooth S-scheme of dimension 2, let t be the closed point of T, and let A, B, X be closed subschemes of dimension 1 of T. Let $i : T-(A\cup X)\hookrightarrow T-A$ be the inclusion. Suppose that X is regular, and that A is in general position with respect to B and X; that is, the intersection of the tangent cones at t to A and $B\cup X$ is reduced to t. Let $E\in \mathrm{ob}\, D^b(T-(A\cup X))$ be such that $H^*(E)$ is constructible and locally constant, and that the corresponding representation of $\pi_1(T-(A\cup X))$ factors through an ℓ-group. Then the restriction arrow

$$\mathrm{R}\Gamma(T-A, i_!(E)) \longrightarrow \mathrm{R}\Gamma(B-t, i_!(E)|B-t)$$

is zero.

The proof will rely on the following lemma.

Lemma 4.3. Let Z be a connected regular noetherian scheme of dimension 2, with residual characteristics prime to ℓ, and let X, Y be regular divisors meeting transversely at a point z, giving a diagram of inclusions

$$\begin{array}{ccc} Z-(X\cup Y) & \xrightarrow{i'} & Z-Y \\ \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle j} \\ Z-X & \xrightarrow[i]{} & Z \\ \downarrow{\scriptstyle m} & & \downarrow{\scriptstyle m} \\ Y-z & \xrightarrow[i_1]{} & Y. \end{array}$$

Assume the purity hypothesis

$$(P)\qquad R^q j_*(\mathbb{Z}/\ell) = 0 \quad \text{for } q>1.$$

Let $L\in \mathrm{ob}\, D^b(Z-(X\cup Y),\Lambda)$ have constructible locally constant cohomology, with the corresponding representation of $\pi_1(Z-(X\cup Y))$ factoring through an ℓ-group. Then:

a) There is a canonical isomorphism

$$i_{1!}m'^* Rj_* i'^*_! L \xrightarrow{\sim} m^* Rj_*(i'^*_! L),$$

and the complex $m'^* Rj_* i'^*_! L$ has constructible, locally constant cohomology.

b) Suppose that Y is connected, and let $\bar{s}$ be a geometric point of Y localized at a closed point $s \neq z$. Then the restriction arrow

$$\mathrm{R}\Gamma(Y, m^* R j_*(i'_! L)) \longrightarrow (R j_*(i'_! L))_{\bar{s}} \tag{4.3.1}$$

is zero.

c) Under the hypotheses of b), let C be a closed subscheme of dimension 1 of Z such that $Y \cap C = s$, and let $\widetilde{C}$ be the strict localization of C at $\bar{s}$. Then the restriction arrow

$$\mathrm{R}\Gamma(Z - Y, i'_! L) \longrightarrow \mathrm{R}\Gamma(\widetilde{C} - \bar{s}, i'_!(L)|\widetilde{C} - \bar{s}) \tag{4.3.2}$$

is zero.

The trivial isomorphism

$$i^* R j_* i'_! L \xrightarrow{\sim} R j_*(i'^* i'_! L) = R j_* L$$

gives an isomorphism

$$m'^* R j_* i'_! L \xrightarrow{\sim} i_1^* m^* R j_* i'_! L,$$

hence, by adjunction, an arrow

$$i_{1!} m'^* R j_* i'_! L \longrightarrow m^* R j_*(i'_! L),$$

which induces an isomorphism outside z. We show that it is an isomorphism, that is, that if $\bar{z}$ is a geometric point above z, then

$$(R j_*(i'_! L))_{\bar{z}} = 0.$$

For this, one may replace Λ by $\mathbb{Z}/\ell^\nu$, where ℓ^ν annihilates Λ, and by dévissage and passage to the limit reduce to the case where L is a locally constant $\mathbb{Z}/\ell^\nu$-module of finite type corresponding to a representation of π_1 through an ℓ-group. Since the only finite abelian group of ℓ-torsion which is simple under the action of an ℓ-group is $\mathbb{Z}/\ell$ with trivial action, a further dévissage reduces to the case where L is the constant sheaf $\mathbb{Z}/\ell$.

Let $\widetilde{Z}$ be the strict localization of Z at $\bar{z}$, $\widetilde{X} = X \times_Z \widetilde{Z}$, $\widetilde{Y} = Y \times_Z \widetilde{Z}$, and let $i' : \widetilde{Z} - (\widetilde{X} \cup \widetilde{Y}) \hookrightarrow \widetilde{Z} - \widetilde{Y}$ be the inclusion. One has

$$(R j_*(i'_!(\mathbb{Z}/\ell)))_{\bar{z}} = \mathrm{R}\Gamma(\widetilde{Z} - \widetilde{Y}, i'_!(\mathbb{Z}/\ell)),$$

so it remains to prove that the restriction arrow

$$H^n(\widetilde{Z} - \widetilde{Y}, \mathbb{Z}/\ell) \longrightarrow H^n(\widetilde{X} - \bar{z}, \mathbb{Z}/\ell)$$

is an isomorphism for all n. For $n \leq 1$ this is a consequence of Abhyankar's lemma, and for $n > 1$ both sides vanish by the purity hypothesis. The first assertion of a) is therefore proved. We prove the second. By dévissage, we may suppose L concentrated in degree 0. The hypothesis (P) implies, by dévissage, that $R^q j_* L = 0$ for $q > 2$. It remains to prove constructibility and local constancy of $m'^* R^q j_* i'_! L$ for $q \leq 1$, which is standard. For constructibility, resolving L on the right by sheaves of the form $p_* p^* L$, where $p : Z' \to Z$ is a finite covering tamely ramified along Y, reduces to the case where L is constant, of value L_0: then

$$m'^* j_* i'_! L = (L_0)_{Y-z}, \qquad m'^* R^1 j_* i'_! L = (L_0)_{Y-z}(-1)$$

by Abhyankar. For local constancy, one has to see that the specialization arrows are isomorphisms. One may, for this, replace Λ by $\mathbb{Z}/\ell^\nu$ as above, and then reduce by dévissage to the case where L is the constant sheaf $\mathbb{Z}/\ell$; the conclusion follows from Abhyankar's lemma.

We have therefore proved a). We prove b). Let $\bar\eta$ be a geometric generic point of Y. Choose specialization arrows

$$a : \bar\eta \to Y(\bar z), \qquad b : \bar\eta \to Y(\bar s),$$

where $Y(\bar z)$ (resp. $Y(\bar s)$) is the strict localization of Y at $\bar z$ (resp. $\bar s$). Put

$$M = m^* R j_*(i'_! L).$$

The commutative square

$$\begin{array}{ccc} Y & \longrightarrow & Y(\bar s) \\ \uparrow & & \uparrow{\scriptstyle b} \\ Y(\bar z) & \xrightarrow[a]{} & \bar\eta \end{array}$$

induces a commutative square

$$\begin{array}{ccc} \mathrm{R}\Gamma(Y, M) & \xrightarrow{4.3.1} & M_{\bar s} \\ \downarrow & & \downarrow{\scriptstyle b^*} \\ M_{\bar z} & \xrightarrow[a^*]{} & M_{\bar\eta}. \end{array}$$

Since $M|Y - z$ has locally constant cohomology, b^* is an isomorphism. On the other hand, since $M = i_{1!} i_1^* M$, one has $M_{\bar z} = 0$, hence the arrow 4.3.1 is zero. It remains to prove c). We have $\widetilde C = C \times_Z \widetilde Z$, where $\widetilde Z$ denotes, as above, the strict localization of Z at $\bar z$. Put $\widetilde Y = Y \times_Z \widetilde Z$. Then there is a commutative diagram of restriction arrows

$$\begin{array}{ccc} \mathrm{R}\Gamma(Z - Y, i'_! L) = \mathrm{R}\Gamma(Z, R j_*(i'_! L)) & \xrightarrow{4.3.2} & \mathrm{R}\Gamma(\widetilde C - \bar s, i'_!(L)|\widetilde C - \bar s) \\ \downarrow & & \uparrow \\ \mathrm{R}\Gamma(Y, m^* R j_*(i'_! L)) & \xrightarrow[4.3.1]{} & (R j_*(i'_! L))_{\bar s} = \mathrm{R}\Gamma(\widetilde Z - \widetilde Y, (i'_! L)|\widetilde Z - \widetilde Y) \end{array}$$

The vanishing of 4.3.2 therefore follows from b), completing the proof of 4.3.

We prove 4.2. Let $f : T' \to T$ be the blowing-up of the maximal ideal of T, let $D = f^{-1}(z) (\simeq \mathbb{P}^1_S)$ be the exceptional divisor, and let A', B', X' be the respective strict transforms of A, B, X. Then T' is regular, connected, of dimension 2, X' is a regular divisor meeting D transversely at one point z', and the general-position hypotheses on A, B, X mean that A' is disjoint from $B' \cup X'$. Let E' be the inverse image of E on $T' - (A' \cup D \cup X')$, and let

$$i' : T' - (A' \cup D \cup X') \hookrightarrow T' - (A' \cup D)$$

be the inclusion. We must prove that the restriction arrow

$$\mathrm{R}\Gamma(T' - (A' \cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B' \cup D, i'_!(E')|B' - (B' \cup D)) \qquad (*)$$

is zero. But B' is a disjoint union of strictly local schemes B'_j $(1 \le j \le n)$, of dimension 1, such that for every j, $B'_j \cap D$ is reduced to a point b_j. Applying 4.3 c) to $(Z = T' - A', X = X', Y =$

$D-(D\cap A'), L=E', C=B'_j$), the hypothesis (P) being verified by virtue of the relative purity theorem (SGA 4 XVI 3), since (Z,Y) is a filtered projective limit of smooth pairs of codimension 1 over S, with affine étale transition morphisms, one obtains that the restriction arrow

$$\mathrm{R}\Gamma(T'-(A'\cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B'_j-b_j, i'_!(E')|B'_j-b_j)$$

is zero. Therefore (*) is zero, which completes the proof of 4.2, hence of 2.5 and 1.2.

II. Equivariant correspondences

5. Non-commutative traces

Throughout this number, fix a topos T and a commutative unital ring K of T. The rings under consideration will be assumed associative and unital, but not necessarily commutative. By a K-algebra we shall mean any (associative, unital) ring A of T endowed with a (unital) homomorphism $K\to A$ whose image is contained in the center.

5.0. Introduction

Let E be a K-module locally a direct factor of a free module of finite type. There is then a trace homomorphism, local in nature,

$$\mathrm{Tr}_K : \underline{\mathrm{Hom}}_K(E,E)\longrightarrow K, \tag{5.0.1}$$

defined by composing the evaluation arrow

$$\underline{\mathrm{Hom}}_K(E,K)\otimes_K E\longrightarrow K,\qquad f\otimes x\longmapsto f(x), \tag{5.0.2}$$

with the inverse of the canonical tensor-product isomorphism

$$\underline{\mathrm{Hom}}_K(E,K)\otimes_K E\longrightarrow \underline{\mathrm{Hom}}_K(E,E),\qquad f\otimes y\longmapsto (x\longmapsto f(x)y). \tag{5.0.3}$$

The arrows 5.0.2 and 5.0.3 are defined more generally for every K-module E, and from them one obtains, for every complex of K-modules E, homomorphisms of complexes

$$\underline{\mathrm{Hom}}^\bullet_K(E,K)\otimes_K E\longrightarrow K,\qquad f\otimes x\longmapsto f(x), \tag{5.0.4}$$

$$\underline{\mathrm{Hom}}^\bullet_K(E,K)\otimes_K E\longrightarrow \underline{\mathrm{Hom}}^\bullet_K(E,E),\qquad f\otimes y\longmapsto (x\longmapsto f(x)y). \tag{5.0.5}$$

Upon deriving, 5.0.4 and 5.0.5 give, for $E\in \mathrm{ob}\, D^-(K)$, arrows in $D(K)$

$$\underline{\mathrm{RHom}}_K(E,K)\overset{\mathrm{L}}{\otimes}_K E\longrightarrow K, \tag{5.0.6}$$

$$\underline{\mathrm{RHom}}_K(E,K)\overset{\mathrm{L}}{\otimes}_K E\longrightarrow \underline{\mathrm{RHom}}_K(E,E). \tag{5.0.7}$$

If E is perfect (SGA 6 I), 5.0.7 is an isomorphism, as is readily checked by reduction to the case $E=K$, and, by composing the inverse of 5.0.7 with 5.0.6, one obtains a trace homomorphism, global in nature,

$$\mathrm{Tr}_K : \underline{\mathrm{RHom}}_K(E,E)\longrightarrow K, \tag{5.0.8}$$

which generalizes 5.0.1 (for more details, see SGA 6 I 7, 8).

We propose here to examine to what extent, in the preceding constructions, K can be replaced by a K-algebra A (we shall be interested chiefly in the case where A is the K-algebra of an ordinary finite group), and in particular to define a trace arrow analogous to 5.0.8, generalizing the notion of non-commutative trace introduced by Stallings in [5] and developed by Grothendieck in Exposé XI of this seminar. As was said in the introduction, that exposé, written by I. Bucur, was unfortunately lost; the reader may nevertheless refer to Deligne's exposé (SGA $4^{1/2}$, Rapport), where the essential points are indicated. Although replacing K by A in 5.0.3 presents no difficulty, the same is not true in 5.0.2: if E is a left A-module, the evaluation arrow

$$\underline{\mathrm{Hom}}_A(E,A)\otimes_K E \longrightarrow A, \qquad f\otimes x \longmapsto f(x),$$

does not in general factor through $\underline{\mathrm{Hom}}_A(E,A)\otimes_A E$, but, upon passage to the quotient, gives an arrow

$$\underline{\mathrm{Hom}}_A(E,A)\otimes_A E \longrightarrow A_\natural, \tag{5.0.9}$$

where $A_\natural$ is the quotient K-module of A by the sub-K-module locally generated by local sections of the form $ab-ba$. If E is locally a direct factor of a free left A-module of finite type, the tensor-product arrow, analogous to 5.0.3,

$$\underline{\mathrm{Hom}}_A(E,A)\otimes_A E \longrightarrow \underline{\mathrm{Hom}}_A(E,E) \tag{5.0.10}$$

is an isomorphism, and, by composing the inverse of 5.0.10 with 5.0.9, one obtains a trace homomorphism

$$\mathrm{Tr}_A : \underline{\mathrm{Hom}}_A(E,E) \longrightarrow A_\natural, \tag{5.0.11}$$

which reduces to 5.0.1 when $A=K$. One also checks readily (see 5.7) that, when T is punctual, 5.0.11 coincides with the trace homomorphism of Stallings and Grothendieck. In this number we shall examine the extension to complexes and the derivation of the arrows 5.0.9, 5.0.10, 5.0.11, and the behavior of the resulting arrows with respect to extension and restriction of scalars and to certain external tensor products. In fact, we shall derive arrows slightly more general than 5.0.9 and 5.0.10, so that the constructions may be applied, in no. 6, to the definition of "non-commutative" Lefschetz–Verdier local terms.

5.1. Notation

If A is a K-algebra, we denote by A° the algebra opposite to A, and by $A^e = A\otimes_K A^\circ$ the enveloping algebra of A. Giving an A-bimodule, a left A^e-module, or a right A^e-module amounts to the same thing. In particular, A is an A-bimodule, and hence both a left and a right A^e-module. If E is a bimodule, recall ([1] IX 4) that

$$H_0(A,E) = A\otimes_{A^e} E = E\otimes_{A^e} A$$

is the quotient K-module of E by the sub-K-module locally generated by local sections of the form $ax-xa$, for $x\in E$, $a\in A$[15]. We shall sometimes write $E_\natural$ instead of $H_0(A,E)$. In particular, $A_\natural$ is the K-module defined in 5.0.9. If $A=K[G]$, where G is an ordinary finite group, $A_\natural$ is simply the free K-module on the set $G_\natural$ of conjugacy classes of G.

[15] If M is a sheaf, the notation $x\in M$ means that x is a local section of M.

If A, B are K-algebras, we shall use the Cartan–Eilenberg notation ${}_AE$, E_B, ${}_AE_B$ to denote respectively a left A-module, a right B-module, and a module that is left over A and right over B. Finally, we shall denote by $D(A)$ (resp. $D(A,B)$) the derived category of the category of left A-modules (resp. of A-B-bimodules, left over A and right over B).

5.2

Let A, B be K-algebras. Recall that, in the situation $({}_AE, {}_BF_A, {}_BG)$, there is a canonical isomorphism

$$\underline{\mathrm{Hom}}_A(E, \underline{\mathrm{Hom}}_B(F,G)) \xrightarrow{\sim} \underline{\mathrm{Hom}}_B(F \otimes_A E, G), \qquad f \longmapsto (y \otimes x \longmapsto f(x)(y)). \tag{5.2.1}$$

This derives to an isomorphism

$$\underline{\mathrm{RHom}}_A(E, \underline{\mathrm{RHom}}_B(F,G)) \xrightarrow{\sim} \underline{\mathrm{RHom}}_B(F \overset{\mathrm{L}}{\otimes}_A E, G) \tag{5.2.2}$$

for $E \in \mathrm{ob}\, D^-(A)$, $F \in \mathrm{ob}\, D(B,A)$, $G \in \mathrm{ob}\, D^+(B)$. Indeed, one may suppose that E is bounded above with flat terms and that G is bounded below with injective terms; one then has $\underline{\mathrm{RHom}}_B(F \overset{\mathrm{L}}{\otimes}_A E, G) = \mathrm{Hom}^\bullet_B(F \otimes_A E, G)$, $\underline{\mathrm{RHom}}_B(F,G) = \mathrm{Hom}^\bullet_B(F,G)$, and $\underline{\mathrm{RHom}}_A(E, \mathrm{Hom}^\bullet_B(F,G)) = \mathrm{Hom}^\bullet_A(E, \mathrm{Hom}^\bullet_B(F,G))$, since, if E is acyclic, $F \otimes_A E$ is acyclic, and hence $\mathrm{Hom}^\bullet_A(E, \mathrm{Hom}^\bullet_B(F,G)) \simeq \mathrm{Hom}^\bullet_B(F \otimes_A E, G)$ is acyclic. For E and G as above, 5.2.2 is given by the isomorphism of complexes

$$\mathrm{Hom}^\bullet_A(E, \mathrm{Hom}^\bullet_B(F,G)) \xrightarrow{\sim} \mathrm{Hom}^\bullet_B(F \otimes_A E, G)$$

defined by 5.2.1. Note that, if A and B are flat over K, 5.2.2 still holds under the conditions $E \in \mathrm{ob}\, D(A)$, $F \in \mathrm{ob}\, D^-(B,A)$, $G \in \mathrm{ob}\, D^+(B)$, as is seen by resolving F by flat $(B \otimes_K A^\circ)$-modules. From 5.2.2 one deduces an isomorphism

$$\underline{\mathrm{Hom}}_A(E, \underline{\mathrm{RHom}}_B(F,G)) \xrightarrow{\sim} \underline{\mathrm{Hom}}_B(F \overset{\mathrm{L}}{\otimes}_A E, G). \tag{5.2.3}$$

5.3

Let A be a K-algebra. In the situation $(E_A, {}_AF)$, there is a canonical isomorphism

$$E \otimes_A F \xrightarrow{\sim} (E \otimes_K F) \otimes_{A^e} A, \qquad x \otimes y \longmapsto (x \otimes y) \otimes 1. \tag{5.3.1}$$

On the other hand, in the situation $({}_AE, {}_AF_A)$, the evaluation arrow

$$\underline{\mathrm{Hom}}_A(E,F) \otimes_K E \longrightarrow F, \qquad f \otimes x \longmapsto f(x) \tag{5.3.2}$$

is A^e-linear, and therefore induces, via 5.3.1, upon applying $- \otimes_{A^e} A$, an arrow

$$\underline{\mathrm{Hom}}_A(E,F) \otimes_A E \longrightarrow F \otimes_{A^e} A \ (= F_\natural). \tag{5.3.3}$$

In other words, the composite of 5.3.2 and the canonical arrow $F \to F_\natural$, $x \mapsto x \otimes 1$, factors uniquely through 5.3.3.

Suppose that A is flat over K. Then 5.3.1 derives to an isomorphism

$$E \overset{\mathrm{L}}{\otimes}_A F \xrightarrow{\sim} (E \overset{\mathrm{L}}{\otimes}_K F) \overset{\mathrm{L}}{\otimes}_{A^e} A \tag{5.3.4}$$

for $E \in \mathrm{ob}\, D^-(A^\circ)$, $F \in \mathrm{ob}\, D^-(A)$ (resolve E and F by flat modules). On the other hand, 5.3.3 derives to an arrow of $D(K)$, called the evaluation arrow,

$$\underline{\mathrm{RHom}}_A(E,F) \overset{\mathrm{L}}{\otimes}_A E \longrightarrow F \overset{\mathrm{L}}{\otimes}_{A^e} A, \tag{5.3.5}$$

for $E \in \mathrm{ob}\, D^-(A)$, $F \in \mathrm{ob}\, D^b(A^e)$ such that $\underline{\mathrm{RHom}}_A(E,F) \in \mathrm{ob}\, D^b(A^\circ)$. One first derives 5.3.2 to an arrow of $D^-((A^e)^\circ)$,

$$\underline{\mathrm{RHom}}_A(E,F) \overset{\mathrm{L}}{\otimes}_K E \longrightarrow F, \tag{5.3.6}$$

by resolving E by flat modules and F by injective A^e-modules, and then one applies the functor $- \overset{\mathrm{L}}{\otimes}_{A^e} A$ to 5.3.6, taking 5.3.4 into account.

5.4

Let A be a K-algebra. In the situation $({}_AE, {}_AF_A, {}_AG)$, there is a canonical arrow

$$\underline{\mathrm{Hom}}_A(E,F) \otimes_A G \longrightarrow \underline{\mathrm{Hom}}_A(E, F \otimes_A G), \qquad u \otimes y \longmapsto (x \longmapsto u(x) \otimes y), \tag{5.4.1}$$

which corresponds by adjunction (5.2.1) to the A-linear arrow

$$E \otimes_K \underline{\mathrm{Hom}}_A(E,F) \otimes_A G \longrightarrow F \otimes_A G$$

defined by the evaluation arrow $E \otimes_K \underline{\mathrm{Hom}}_A(E,F) \to F$. The arrow 5.4.1 derives to an arrow of $D(K)$, called the tensor-product arrow,

$$\underline{\mathrm{RHom}}_A(E,F) \overset{\mathrm{L}}{\otimes}_A G \longrightarrow \underline{\mathrm{RHom}}_A(E, F \overset{\mathrm{L}}{\otimes}_A G), \tag{5.4.2}$$

for $E \in \mathrm{ob}\, D^-(A)$, $F \in \mathrm{ob}\, D^+(A^e)$, $G \in \mathrm{ob}\, D^-(A)$, with $F \overset{\mathrm{L}}{\otimes}_A G \in \mathrm{ob}\, D^+(A)$. If E and G are bounded above with flat terms, and F is bounded below with injective terms, the first member of 5.4.2 identifies with $\mathrm{Hom}^\bullet_A(E,F) \otimes_A G$, and it is sent to the second member by the composite of 5.4.1 and the natural arrow

$$\mathrm{Hom}^\bullet_A(E, F \otimes_A G) \longrightarrow \underline{\mathrm{RHom}}_A(E, F \otimes_A G) = \underline{\mathrm{RHom}}_A(E, F \overset{\mathrm{L}}{\otimes}_A G).$$

The arrow 5.4.2 is not an isomorphism in general. It is, however, an isomorphism in certain important cases, for example if E is perfect (as is checked by reducing to $E = A$). We shall see another interesting example later (6.2.3).

The arrows 5.3.5 and 5.4.2 will play an essential role throughout the rest of the exposé. The examples of objects F that we have in mind are: a) $F = A$, or $F = A$ regarded as a right A-module in the natural way and as a left A-module through a homomorphism of K-algebras $\varphi : A \to A$ (cf. 5.13); b) $F = M \overset{\mathrm{L}}{\otimes}_K A$, with $M \in \mathrm{ob}\, D^+(K)$; in no. 6, M will be a dualizing complex, see especially 6.3.2, 6.4.1, 6.5.6.

5.5

Denote by $DF(A)$ the derived category of complexes of left A-modules endowed with a finite filtration ([3] V). Then, in the situation of 5.2, there is a canonical isomorphism in $DF(K)$, analogous to 5.2.2, for $E \in \mathrm{ob}\, D^-F(A)$, $F \in \mathrm{ob}\, D^-F(B,A)$ $(= D^-F(B \otimes_K A^\circ))$, $G \in \mathrm{ob}\, D^+F(B)$. Similarly, there are "filtered derived" arrows analogous to 5.3.4, 5.3.5, 5.3.6, 5.4.2. These arrows are compatible with passage to associated graded objects, in the sense of ([3] V 1,2).

5.6

Assume that A is flat over K. Denote by $D(A)_{\mathrm{parf}}$ the full subcategory of $D(A)$ formed by complexes of finite perfect amplitude, that is, perfect complexes of finite tor-dimension (SGA 6 I). We define a canonical arrow, called the trace,

$$\mathrm{Tr}_A = \underline{\mathrm{RHom}}_A(E, F \overset{\mathrm{L}}{\otimes}_A E) \longrightarrow F \overset{\mathrm{L}}{\otimes}_{A^e} A \tag{5.6.1}$$

for $E \in \mathrm{ob}\, D(A)_{\mathrm{parf}}$, $F \in \mathrm{ob}\, D^b(A^e)$, by composing the inverse isomorphism of 5.4.2 for $G = E$ with the evaluation arrow 5.3.5. From this one deduces a trace arrow

$$\mathrm{Tr}_A : \underline{\mathrm{Hom}}_A(E, F \overset{\mathrm{L}}{\otimes}_A E) \longrightarrow H^0(T, F \overset{\mathrm{L}}{\otimes}_{A^e} A). \tag{5.6.2}$$

For $F = A$, by composing with the canonical arrow $A \overset{\mathrm{L}}{\otimes}_{A^e} A \to A \otimes_{A^e} A = A_\natural$ (5.1), one obtains arrows, again called traces,

$$\mathrm{Tr}_A : \underline{\mathrm{RHom}}_A(E, E) \longrightarrow A_\natural, \tag{5.6.3}$$

$$\mathrm{Tr}_A : \underline{\mathrm{Hom}}_A(E, E) \longrightarrow H^0(T, A_\natural). \tag{5.6.4}$$

Notice that 5.6.4 is a rather innocuous arrow, local in nature, whereas 5.6.3 is a "serious" arrow, global in nature.

Denote by $DF(A)_{\mathrm{parf}}$ the full subcategory of $DF(A)$ formed by filtered complexes whose associated graded object has finite perfect amplitude. Then there is an arrow in $DF(K)$ analogous to 5.6.1 for $E \in \mathrm{ob}\, DF(A)_{\mathrm{parf}}$, $F \in \mathrm{ob}\, D^b F(A^e)$, whence arrows analogous to 5.6.2, 5.6.3, 5.6.4. It follows from the remarks in 5.5 that the filtered arrow 5.6.1 is compatible with passage to the associated graded object (cf. [3] V 3.7.2), and one deduces, as in loc. cit., that, if the filtration on F is concentrated in degree 0, then, for $u \in \mathrm{Hom}_{DF(A)}(E, F \overset{\mathrm{L}}{\otimes}_A E)$, one has

$$\mathrm{Tr}_A(u) = \sum_i \mathrm{Tr}_A(\mathrm{gr}^i u). \tag{5.6.5}$$

This formula implies in particular that, for $E \in \mathrm{ob}\, D(A)_{\mathrm{parf}}$, $F \in \mathrm{ob}\, D^b(A^e)$, $u \in \underline{\mathrm{Hom}}_A(E, F \overset{\mathrm{L}}{\otimes}_A E)$, one has

$$\mathrm{Tr}(u[1]) = -\mathrm{Tr}(u), \tag{5.6.6}$$

where $u[1] : E[1] \to (F \overset{\mathrm{L}}{\otimes}_A E)[1]$ is the translate of u.

5.7

The traces defined in 5.6 simultaneously generalize the usual commutative traces ((SGA 6 I), ([3] V 3)) and the non-commutative traces of Stallings [5] and Grothendieck.

(a) Suppose that $A = K$. Then $A^e = A_\natural = K$, the arrow 5.6.1 is written

$$\mathrm{Tr}_K : \underline{\mathrm{RHom}}_K(E, F \overset{\mathrm{L}}{\otimes}_K E) \longrightarrow F$$

and coincides with the trace arrow defined in ([3] V 3.7.1), in the special case where the complex K of loc. cit. is reduced to A. In particular,

$$\mathrm{Tr}_K : \underline{\mathrm{RHom}}_K(E, E) \longrightarrow K$$

coincides with the trace defined in (SGA 6 I 8).

(b) Assume again that A is a flat K-algebra, not necessarily commutative. We propose to make 5.6.4 explicit for E strictly perfect, that is, bounded with terms locally direct factors of free modules of finite type. First, take $E = A$; $\underline{\mathrm{RHom}}_A(A, A) = \underline{\mathrm{Hom}}_A(A, A)$ identifies with A by $f \mapsto f(1)$, $a \mapsto (x \mapsto xa)$, and 5.6.3 identifies with the canonical projection $A \to A_\natural$, which we shall denote by $a \mapsto a_\natural$; thus, if d_a denotes right multiplication by a,

$$\mathrm{Tr}_A(d_a) = a_\natural. \tag{5.7.1}$$

For $1 \leq i \leq n$, let E_i be a copy of A, and let $E = \bigoplus_{1\leq i\leq n} E_i$. The isomorphism 5.4.2, for $F = A$, $G = E$, is written

$$\bigoplus_{1\leq i,j\leq n} \underline{\mathrm{Hom}}_A(E_i, A) \otimes_A E_j \longrightarrow \underline{\mathrm{Hom}}_A(E, E),$$

and, for $u = \sum u_{ij} \otimes 1 \in \underline{\mathrm{Hom}}_A(E, E)$ ($u_{ij} \otimes 1 \in \underline{\mathrm{Hom}}_A(E_i, A) \otimes_A E_j$), one has

$$\mathrm{Tr}_A(u) = \sum_i (u_{ii})_\natural, \tag{5.7.2}$$

by definition of the evaluation arrow 5.3.5. Now take for E a direct factor of A^n, the image of a projector p, and denote by $i : E \to A^n$ the inclusion; then, for $u \in \underline{\mathrm{Hom}}_A(E, E)$, one has

$$\mathrm{Tr}_A(u) = \mathrm{Tr}_A(iup), \tag{5.7.3}$$

as is seen either with the aid of 5.6.5 or directly from the definition. Finally, suppose that E is strictly perfect; if $u \in \mathrm{Hom}_A(E, E)$ is a homomorphism of complexes, one has, by 5.6.5 and 5.6.6,

$$\mathrm{Tr}_A(u) = \sum_i (-1)^i \mathrm{Tr}_A(u^i). \tag{5.7.4}$$

The fact that 5.6.4 is obtained, by applying H^0, from a homomorphism of complexes $\underline{\mathrm{Hom}}^\bullet_A(E, E) \to A$ (5.6.3) implies that, if u, $v \in \mathrm{Hom}_A(E, E)$ are homotopic, then

$$\mathrm{Tr}_A(u) = \mathrm{Tr}_A(v). \tag{5.7.5}$$

Since every endomorphism of E in $D(A)$ is locally defined by an endomorphism of complexes, locally unique up to homotopy (SGA 6 I 4.10), the formulas 5.7.2 to 5.7.4 therefore permit the calculation of $\mathrm{Tr}_A(u)$ for every endomorphism u of E in $D(A)$ and, combined with 5.7.5, provide a direct definition of $\mathrm{Tr}_A(u)$. When the topos T is punctual, this recovers the definition of (SGA $4^{1/2}$, Rapport 4.3).

5.8

Let P, Q be A-bimodules. Then $P \otimes_A Q$ and $Q \otimes_A P$ are A-bimodules, and there are canonical isomorphisms

$$(P \otimes_A Q) \otimes_{A^e} A \xrightarrow{\sim} P \otimes_{A^e} Q \xrightarrow{\sim} (Q \otimes_A P) \otimes_{A^e} A \xrightarrow{\sim} A \otimes_{A^e} (P \otimes_K Q) \otimes_{A^e} A, \tag{5.8.1}$$

each of the objects above identifying canonically with the quotient K-module of $P \otimes_K Q$ by the submodule generated by sections of the form $axb \otimes y - x \otimes bya$, for $x \in P$, $y \in Q$, $a, b \in A$. For A flat over K, these isomorphisms derive to isomorphisms

$$(P \overset{\mathrm{L}}{\otimes}_A Q) \overset{\mathrm{L}}{\otimes}_{A^e} A \simeq P \overset{\mathrm{L}}{\otimes}_{A^e} Q \simeq (Q \overset{\mathrm{L}}{\otimes}_A P) \overset{\mathrm{L}}{\otimes}_{A^e} A \simeq A \overset{\mathrm{L}}{\otimes}_{A^e} (P \overset{\mathrm{L}}{\otimes}_K Q) \overset{\mathrm{L}}{\otimes}_{A^e} A \tag{5.8.2}$$

for $P, Q \in \operatorname{ob} D^-(A^e)$. On the other hand, the composition arrow at the level of modules

$$\underline{\operatorname{Hom}}_A(E, P \otimes_A F) \otimes_K \underline{\operatorname{Hom}}_A(F, Q \otimes_A E) \longrightarrow \underline{\operatorname{Hom}}_A(E, P \otimes_A Q \otimes_A E),$$

$$f \otimes g \longmapsto (P \otimes_A g)f,$$

derives to

$$\underline{\operatorname{RHom}}_A(E, P \overset{\mathrm{L}}{\otimes}_A F) \overset{\mathrm{L}}{\otimes}_K \underline{\operatorname{RHom}}_A(F, Q \overset{\mathrm{L}}{\otimes}_A E) \longrightarrow \underline{\operatorname{RHom}}_A(E, P \overset{\mathrm{L}}{\otimes}_A Q \overset{\mathrm{L}}{\otimes}_A E) \tag{5.8.3}$$

for $P, Q \in \operatorname{ob} D^b(A^e)$, $E, F \in \operatorname{ob} D^b(A)$ such that the tensor products and $\underline{\operatorname{RHom}}$'s appearing above belong to D^b. When E and F are perfect, this arrow identifies, via 5.4.2, with the arrow deduced from the evaluation arrow $F \overset{\mathrm{L}}{\otimes}_K \underline{\operatorname{RHom}}_A(F, Q) \to Q$; from this one readily deduces that there is a commutative square

$$\begin{array}{ccc} \underline{\operatorname{RHom}}_A(E, P \overset{\mathrm{L}}{\otimes}_A F) \overset{\mathrm{L}}{\otimes}_K \underline{\operatorname{RHom}}_A(F, Q \overset{\mathrm{L}}{\otimes}_A E) & \xrightarrow{(1)} & \underline{\operatorname{RHom}}_A(E, P \overset{\mathrm{L}}{\otimes}_A Q \overset{\mathrm{L}}{\otimes}_A E) \\ \downarrow{\scriptstyle (2)} & & \downarrow{\scriptstyle \mathrm{Tr}_A} \\ \underline{\operatorname{RHom}}_A(F, Q \overset{\mathrm{L}}{\otimes}_A P \overset{\mathrm{L}}{\otimes}_A F) & \xrightarrow[\mathrm{Tr}_A]{} & P \overset{\mathrm{L}}{\otimes}_{A^e} Q, \end{array} \tag{5.8.4}$$

where (1) and (2) are given by 5.8.3. The point is that the two composite arrows in 5.8.4 are defined by the tensor product of the evaluation arrows

$$F \overset{\mathrm{L}}{\otimes}_K \underline{\operatorname{RHom}}_A(F, Q) \longrightarrow Q, \qquad E \overset{\mathrm{L}}{\otimes}_K \underline{\operatorname{RHom}}_A(E, P) \longrightarrow P.$$

It follows that, for $u \in \underline{\operatorname{Hom}}_A(E, P \overset{\mathrm{L}}{\otimes}_A F)$, $v \in \underline{\operatorname{Hom}}_A(F, Q \overset{\mathrm{L}}{\otimes}_A E)$, one has

$$\mathrm{Tr}_A(vu) = \mathrm{Tr}_A(uv). \tag{5.8.5}$$

We shall sometimes denote the common value in 5.8.5 by $\langle u, v \rangle_A$.

5.9. Traces and extension of scalars

Let $A \to B$ be a morphism of flat K-algebras. It induces a morphism of K-algebras $A^e \to B^e$, hence an extension-of-scalars functor $- \overset{\mathrm{L}}{\otimes}_{A^e} B^e : D(A^e) \longrightarrow D(B^e)$. For E, F as in 5.3.5, if $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e \in \operatorname{ob} D^b(B^e)$ and $\operatorname{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \in \operatorname{ob} D^b((B^e)^\circ)$, there is a commutative square in $D(K)$

$$\begin{array}{ccc} \operatorname{RHom}_A(E, F) \overset{\mathrm{L}}{\otimes}_{A^e} E & \xrightarrow{(1)} & F \overset{\mathrm{L}}{\otimes}_{A^e} A \\ \downarrow{\scriptstyle (3)} & & \downarrow{\scriptstyle (4)} \\ \operatorname{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \overset{\mathrm{L}}{\otimes}_{B^e} (B \overset{\mathrm{L}}{\otimes}_A E) & \xrightarrow[(2)]{} & (F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \overset{\mathrm{L}}{\otimes}_{B^e} B, \end{array} \tag{5.9.1}$$

where the arrows (1) and (2) are defined by 5.3.6, (4) by $A \to B$, and (3) by the canonical arrow $F \to F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$. The verification is immediate. Similarly, for E, F, G as in 5.4.2, if $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e \in \mathrm{ob}\, D^+(B^e)$ and $(F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \overset{\mathrm{L}}{\otimes}_B (B \overset{\mathrm{L}}{\otimes}_A G) \in \mathrm{ob}\, D^+(B)$, one has a commutative square in $D(K)$

$$\begin{array}{ccc} \mathrm{RHom}_A(E,F) \overset{\mathrm{L}}{\otimes}_A G & \xrightarrow{\quad (1) \quad} & \mathrm{RHom}_A(E, F \overset{\mathrm{L}}{\otimes}_A G) \\ \downarrow{\scriptstyle (3)} & & \downarrow{\scriptstyle (4)} \\ \mathrm{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \overset{\mathrm{L}}{\otimes}_B (B \overset{\mathrm{L}}{\otimes}_A G) & \xrightarrow[(2)]{} & \mathrm{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, (F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \overset{\mathrm{L}}{\otimes}_B (B \overset{\mathrm{L}}{\otimes}_A G)), \end{array} \tag{5.9.2}$$

where (1) and (2) are defined by 5.4.2, and (3) and (4) by $F \to F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$. From this one deduces, for E, F as in 5.6.1, with $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e \in \mathrm{ob}\, D^b(B^e)$, a commutative square in $D(K)$, analogous to the preceding ones, in which the vertical arrows are defined by 5.6.1:

$$\begin{array}{ccc} \mathrm{RHom}_A(E, F \overset{\mathrm{L}}{\otimes}_A E) & \xrightarrow{\mathrm{Tr}_A} & F \overset{\mathrm{L}}{\otimes}_{A^e} A \\ \downarrow & & \downarrow \\ \mathrm{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, (F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \overset{\mathrm{L}}{\otimes}_B (B \overset{\mathrm{L}}{\otimes}_A E)) & \xrightarrow[\mathrm{Tr}_B]{} & (F \overset{\mathrm{L}}{\otimes}_{A^e} B^e) \overset{\mathrm{L}}{\otimes}_{B^e} B. \end{array} \tag{5.9.3}$$

Finally, if one is given $F' \in \mathrm{ob}\, D^b(B^e)$ such that $\mathrm{RHom}_A(E, F') \in \mathrm{ob}\, D^b((B^\circ))$, and $f : F \overset{\mathrm{L}}{\otimes}_{A^e} B^e \to F'$, then there is a compatibility slightly more general than 5.9.1, expressed by an analogous commutative square, with F' in place of $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$ in the lower row and with the vertical arrows defined by f. One has similarly commutative squares analogous to 5.9.2 and 5.9.3 with F' in place of $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$. Taking in particular $F = A$ and, for $f : A \overset{\mathrm{L}}{\otimes}_{A^e} B^e \to B$, the arrow defined by $A \to B$, one obtains a commutative square

$$\begin{array}{ccc} \mathrm{RHom}_A(E,E) & \xrightarrow{\mathrm{Tr}_A} & A \overset{\mathrm{L}}{\otimes}_{A^e} A \\ \downarrow & & \downarrow \\ \mathrm{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, B \overset{\mathrm{L}}{\otimes}_A E) & \xrightarrow[\mathrm{Tr}_B]{} & B \overset{\mathrm{L}}{\otimes}_{B^e} B, \end{array} \tag{5.9.4}$$

where the left vertical arrow, respectively the right vertical arrow, is given by extension of scalars, respectively is the canonical arrow deduced from $A \to B$.

5.10. Traces and restriction of scalars

Let $A \to B$ be a morphism of flat K-algebras such that B, as a left A-module, is flat and of finite presentation, i.e. locally a direct factor of a free A-module of finite type. This is the case for example if $A \to B$ is the morphism $K[H] \to K[G]$ defined by an injective morphism of discrete groups $H \hookrightarrow G$ of finite index. The scalar-restriction homomorphism

$$\mathrm{Hom}_B(B, B) \longrightarrow \mathrm{Hom}_A(B, B),$$

where B is considered as a left module, identifies, by the canonical isomorphisms

$$B \xrightarrow{\sim} \mathrm{Hom}_B(B,B), \qquad \mathrm{Hom}_A(B,A) \otimes_A B \xrightarrow{\sim} \mathrm{Hom}_A(B,B),$$

with an arrow of B-bimodules

$$B \longrightarrow \mathrm{Hom}_A(B,A) \otimes_A B. \tag{5.10.1}$$

The composite of 5.10.1 with the evaluation arrow $\mathrm{Hom}_A(B,A) \otimes_A B \to A_\natural$ (5.3.3) is the homomorphism $f \mapsto \mathrm{Tr}_A(f)$ $(= \mathrm{Tr}_A(d_f)$, where d_f is right multiplication by $f)$; hence, by the formula $\mathrm{Tr}_A(f_g) = \mathrm{Tr}_A(gf)$ (5.8.5), it factors through $B_\natural$ as an arrow again denoted $\mathrm{Tr}_{B/A} : B_\natural \to A_\natural$. Thus there is a commutative square

$$\begin{array}{ccc} B & \xrightarrow{\mathrm{Tr}_B} & B_\natural \\ {\scriptstyle 5.10.1}\downarrow & & \downarrow{\scriptstyle \mathrm{Tr}_{B/A}} \\ \mathrm{Hom}_A(B,A) \otimes_A B & \xrightarrow[\mathrm{ev}]{} & A_\natural. \end{array} \tag{5.10.2}$$

In the situation $({}_BE, {}^BP_A)$, one has a homomorphism of right B-modules

$$\mathrm{Hom}_B(E,P) \otimes_A B \longrightarrow \mathrm{Hom}_B(E, P \otimes_A B), \qquad u \otimes b \longmapsto (x \longmapsto u(x) \otimes b), \tag{5.10.3}$$

which is an isomorphism, as is checked immediately: it is enough to see that the underlying K-linear homomorphism is an isomorphism, and this follows from the fact that B is locally a direct factor of a free A-module of finite type. This isomorphism derives to an isomorphism of $D(B)$

$$\mathrm{RHom}_B(E,P) \overset{\mathrm{L}}{\otimes}_A B \xrightarrow{\sim} \mathrm{RHom}_B(E, P \overset{\mathrm{L}}{\otimes}_A B), \tag{5.10.4}$$

for $E \in \mathrm{ob}\, D^-(B)$, $P \in \mathrm{ob}\, D^+(B \otimes_K A^\circ)$: resolve E by modules of the form $B \otimes_A L$, where L is flat, and P by injective $(B \otimes_K A^\circ)$-modules, which are necessarily injective as B-modules, since A is flat over K; this allows one to remove the R on the left, and the resolution chosen for E also allows one to remove it on the right.

For $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^+(K)$, $F \in \mathrm{ob}\, D^-(B)$, define an arrow in $D(K)$[16]

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F \longrightarrow \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F \tag{5.10.5}$$

as the composite of the arrows

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F \xrightarrow{(1)} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B,A) \overset{\mathrm{L}}{\otimes}_A (B \overset{\mathrm{L}}{\otimes}_K M)) \overset{\mathrm{L}}{\otimes}_B F$$

$$\xrightarrow{(2)} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B,A) \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A B \overset{\mathrm{L}}{\otimes}_B F$$

$$\xrightarrow{(3)} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, A \overset{\mathrm{L}}{\otimes}_K M)) \overset{\mathrm{L}}{\otimes}_A F$$

$$\xrightarrow{(4)} \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F,$$

[16] In no. 6 we shall apply the constructions that follow to the case where M is a dualizing complex (6.7).

where (1) is defined by 5.10.1, (2) by the inverse isomorphism of 5.10.4 with $P = \mathrm{Hom}_A(B, A) \overset{\mathrm{L}}{\otimes}_K M$, (3) by the canonical isomorphism

$$\mathrm{Hom}_A(B, A) \overset{\mathrm{L}}{\otimes}_K M \xrightarrow{\sim} \mathrm{Hom}_A(B, A \overset{\mathrm{L}}{\otimes}_K M)$$

(5.4.2), and (4) by the adjunction isomorphism 5.2.2.

Proposition 5.10.6. Suppose that $M \overset{\mathrm{L}}{\otimes}_K F \in \mathrm{ob}\, D^+(B)$. Then there is a commutative square in $D(K)$

$$\begin{array}{ccc} \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F & \xrightarrow{5.4.2} & \mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K F) \\ \downarrow{\scriptstyle 5.10.5} & & \downarrow{\scriptstyle \mathrm{res}} \\ \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F & \xrightarrow[5.4.2]{} & \mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K F), \end{array} \tag{5.10.7}$$

where the right vertical arrow is restriction of scalars.

One has to verify that there is a commutative square

$$\begin{array}{ccc} E \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F & \longrightarrow & M \overset{\mathrm{L}}{\otimes}_K F \\ \downarrow & & \downarrow{\scriptstyle \mathrm{res}} \\ E \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F & \longrightarrow & M \overset{\mathrm{L}}{\otimes}_K F, \end{array}$$

where the horizontal arrows are given by the evaluations (5.3.6). This square is deduced, by applying $- \overset{\mathrm{L}}{\otimes}_B F$, from the analogous square with $F = B$, so that one may suppose $F = B$. The commutativity of 5.10.7 in this case is equivalent to that of the triangle

$$\begin{array}{ccc} \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) & & \\ \downarrow{\scriptstyle (1)} & \searrow^{(2)} & \\ \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, B \overset{\mathrm{L}}{\otimes}_K M)) & \xrightarrow[(3)]{} & \mathrm{RHom}_A(E, B \overset{\mathrm{L}}{\otimes}_K M), \end{array}$$

where (1) is defined by 5.10.1, (2) is restriction of scalars, and (3) is the adjunction isomorphism. But the arrow $B \overset{\mathrm{L}}{\otimes}_K M \to \mathrm{Hom}_A(B, B \overset{\mathrm{L}}{\otimes}_K M)$ given by 5.10.1 is nothing other than the adjunction arrow corresponding to the functors restriction of scalars and $\mathrm{Hom}_A(B, -)$; the commutativity follows at once.

Let $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^b(K)$ be such that $\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \in \mathrm{ob}\, D^b(B^\circ)$. Then, by 5.3.5, there is an evaluation arrow

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_{B^e} E \longrightarrow M \overset{\mathrm{L}}{\otimes}_K (B \overset{\mathrm{L}}{\otimes}_{B^e} B),$$

which, by composition with the canonical arrow $B \overset{\mathrm{L}}{\otimes}_{B^e} B \to B_\natural$, gives an arrow

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E \longrightarrow M \overset{\mathrm{L}}{\otimes}_K B_\natural. \tag{5.10.8}$$

Proposition 5.10.9. Suppose that A is of finite presentation and flat as a K-module, and that $A_\natural$, $B_\natural$ are flat over K. Let $E \in \operatorname{ob} D^-(B)$, $M \in \operatorname{ob} D^b(K)$ be such that $\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \in \operatorname{ob} D^b(B^\circ)$. Then $\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \in \operatorname{ob} D^b(A^\circ)$, and there is a commutative square

$$\begin{array}{ccc} \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E & \xrightarrow{5.10.8} & M \overset{\mathrm{L}}{\otimes}_K B_\natural \\ \downarrow{\scriptstyle 5.10.5} & & \downarrow{\scriptstyle M\otimes_K \mathrm{Tr}_{B/A}} \\ \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A E & \xrightarrow[5.10.8]{} & M \overset{\mathrm{L}}{\otimes}_K A_\natural. \end{array} \tag{5.10.10}$$

The additional hypotheses on A and B (verified, for example, if A and B are K-algebras of finite groups) are no doubt useless, and one should have a commutative square analogous to 5.10.10 with $A_\natural$ (resp. $B_\natural$) replaced by $A \overset{\mathrm{L}}{\otimes}_{A^e} A$ (resp. $B \overset{\mathrm{L}}{\otimes}_{B^e} B$).

The first assertion of 5.10.9 is trivial. We prove the second. We may suppose that M has injective components and that E has components of the form $B \otimes_K L$, with L flat. Then

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E = \mathrm{Hom}_B^\bullet(E, B \otimes_K M) \otimes_B E,$$

$$\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A E = \mathrm{Hom}_A^\bullet(E, A \otimes_K M) \otimes_A E,$$

and the arrows in 5.10.10 are defined at the level of complexes. Thus we are reduced to verifying that, if L, M are K-modules and E a left B-module, there is a commutative square

$$(*) \qquad \begin{array}{ccc} \mathrm{Hom}_B(B \otimes_K L, B \otimes_K M) \otimes_B (B \otimes_K L) & \xrightarrow{5.3.3} & M \otimes_K B_\natural \\ \downarrow{\scriptstyle (1)} & & \downarrow{\scriptstyle M\otimes_K \mathrm{Tr}_{B/A}} \\ \mathrm{Hom}_A(B \otimes_K L, A \otimes_K M) \otimes_A (B \otimes_K L) & \xrightarrow[5.3.3]{} & M \otimes_K A_\natural, \end{array}$$

where the arrow (1) is defined by sending $B \otimes_K M$ into $\mathrm{Hom}_A(B, A) \otimes_A B \otimes_K M$ by 5.10.1 and composing with an isomorphism of the type 5.10.3 and an adjunction isomorphism. Since A is flat and of finite presentation as a K-module, the "tensor-product" arrows

$$(2) \quad B \otimes_K \mathrm{Hom}_K(L, M) = \mathrm{Hom}_B(B, B) \otimes_K \mathrm{Hom}_K(L, M) \to \mathrm{Hom}_B(B \otimes_K L, B \otimes_K M),$$

$$(3) \quad \mathrm{Hom}_A(B, A) \otimes_K \mathrm{Hom}_K(L, M) \to \mathrm{Hom}_A(B \otimes_K L, A \otimes_K M)$$

are isomorphisms, by which (1) identifies with the tensor product (over K) of 5.10.1 by $\mathrm{Hom}_K(L, M) \otimes_K L$. On the other hand, the upper (resp. lower) horizontal arrow of (*) identifies by (2) (resp. (3)) with the tensor product (over K) of $\mathrm{Tr}_B : B \to B_\natural$ (resp. $\mathrm{ev} : \mathrm{Hom}_A(B, A) \otimes_A B \to A_\natural$) by the evaluation arrow $\mathrm{Hom}_K(L, M) \otimes_K L \to M$. The commutativity therefore follows from that of 5.10.2, completing the proof of 5.10.9.

Corollary 5.10.11. Under the hypotheses of 5.10.9, for $E \in \operatorname{ob} D(B)_{\mathrm{parf}}$, $M \in \operatorname{ob} D^b(K)$, there is

a commutative square in $D(K)$

$$\begin{array}{ccc} \mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow{\mathrm{Tr}_B} & M \overset{\mathrm{L}}{\otimes}_K B_\natural \\ \downarrow & & \downarrow{\scriptstyle M \overset{\mathrm{L}}{\otimes} \mathrm{Tr}_{B/A}} \\ \mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow[\mathrm{Tr}_A]{} & M \overset{\mathrm{L}}{\otimes}_K A_\natural, \end{array} \tag{5.10.12}$$

where the left vertical arrow is restriction of scalars.

It is enough indeed to conjugate 5.10.6 and 5.10.9.

5.11. Application to finite groups: divisibility properties

Let G be a finite group, with neutral element denoted e, and put $K[G] = B$. The K-module $B_\natural$ is free with basis the set $G_\natural$ of conjugacy classes of G; every section s of $B_\natural$ is written uniquely

$$s = \sum_{x \in G_\natural} s(x)x.$$

Proposition 5.11.1. Cf. SGA $4^{1/2}$, Rapport 4.5. Let $E \in \mathrm{ob}\, D(B)_{\mathrm{parf}}$, $M \in \mathrm{ob}\, D^b(K)$. For $u \in \mathrm{Hom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E)$, one has, in $H^0(T, M)$,

$$\mathrm{Tr}_K(u) = |G|^{17}\ \mathrm{Tr}_B(u)(e). \tag{*}$$

It is enough to apply 5.10.11 with $A = K$, noting that $\mathrm{Tr}_{B/K} : B_\natural \to K$ is given by

$$\mathrm{Tr}_{B/K}(g) = \begin{cases} |G|, & g = e, \\ 0, & g \neq e. \end{cases} \tag{5.11.2}$$

Proposition 5.11.3. Under the hypotheses of 5.11.1, let $g \in G$, let Z_g be the centralizer of g in G, and put $A = K[Z_g]$. Let $u \in \mathrm{Hom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E)$, and let $gu \in \mathrm{Hom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E)$ be the composite of the image of u in $\mathrm{Hom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E)$ with left multiplication by g. Then, in $H^0(T, M)$,

$$\mathrm{Tr}_A(gu)(e) = \mathrm{Tr}_B(u)(g^{-1}).$$

We shall show more generally that there is a commutative square

$$\begin{array}{ccccc} \mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow{\mathrm{Tr}_B} & M \overset{\mathrm{L}}{\otimes}_K B_\natural & \xrightarrow{x \mapsto x(g^{-1})} & M \\ \downarrow & & & & \| \\ \mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow[\mathrm{Tr}_A]{} & M \overset{\mathrm{L}}{\otimes}_K A_\natural & \xrightarrow[x \mapsto x(e)]{} & M, \end{array} \tag{5.11.4}$$

[17] If X is a set, $|X|$ denotes the cardinality of X.

where the left vertical arrow is the composite of restriction of scalars and left multiplication by g. This square is the boundary of the diagram

$$\begin{array}{ccccc}
\mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow{\mathrm{Tr}_B} & M \overset{\mathrm{L}}{\otimes}_K B_\natural & \xrightarrow{x\mapsto x(g^{-1})} & M \\
\Big\downarrow{\scriptstyle \mathrm{rest}} & & \Big\downarrow{\scriptstyle \mathrm{Tr}_{B/A}}\ (1) & & \Big\| \\
\mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow[\mathrm{Tr}_A]{} & M \overset{\mathrm{L}}{\otimes}_K A_\natural & & M \\
\Big\downarrow{\scriptstyle \mathrm{RHom}_A(E, s_g)} & & \Big\downarrow{\scriptstyle s_g}\ (2) & (3) & \\
\mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) & \xrightarrow[\mathrm{Tr}_A]{} & M \overset{\mathrm{L}}{\otimes}_K A_\natural & \xrightarrow[x\mapsto x(e)]{} & M,
\end{array}$$

where rest denotes restriction of scalars and s_g left multiplication by g. Square (1) commutes by 5.10.11, and (2) by the definition of Tr_A. It therefore remains to prove the commutativity of (3), for which it is enough to show that, for $h \in G$,

$$\mathrm{Tr}_A(x \mapsto gxh)(e) = \text{the value at } g^{-1} \text{ of the image of } h \text{ in } B,$$

i.e.

$$\mathrm{Tr}_A(x \mapsto gxh)(e) = \begin{cases} 1, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases} \tag{*}$$

For this one may replace T by the punctual topos and K by $\mathbb{Z}$, and it is then enough to prove (*) after multiplication by $|Z_g|$. In view of 5.11.1, we are therefore reduced to showing that

$$\mathrm{Tr}_{\mathbb{Z}}(x \mapsto gxh) = \begin{cases} |Z_g|, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases}$$

But this follows from the fact that the left hand side is the number of fixed points of the map $x \mapsto gxh$ from G to G, whence the commutativity of 5.11.4 and the conclusion of 5.11.3.

Corollary 5.11.5. Under the hypotheses of 5.11.3, one has

$$\mathrm{Tr}_K(gu) = |Z_g|\ \mathrm{Tr}_B(u)(g^{-1}) = |Z_g|\ \mathrm{Tr}_A(gu)(e). \tag{5.11.5}$$

5.12. Traces and external tensor products

Let, for $i = 1, 2$, A_i be a flat K-algebra, and put $A = A_1 \otimes_K A_2$. Let $E_i \in \mathrm{ob}\, D^-(A_i)$, $F_i \in \mathrm{ob}\, D^b(A_i^e)$ be such that $\mathrm{RHom}_{A_i}(E_i, F_i) \in \mathrm{ob}\, D^b(A_i^\circ)$, let $F = F_1 \overset{\mathrm{L}}{\otimes}_K F_2 \in \mathrm{ob}\, D^b(A^e)$, and suppose $\mathrm{RHom}_A(E, F) \in \mathrm{ob}\, D^b(A^\circ)$, where $E = E_1 \overset{\mathrm{L}}{\otimes}_K E_2$. Then there is a commutative square

$$\begin{array}{ccc}
(\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{A_1} E_1) \overset{\mathrm{L}}{\otimes}_K (\mathrm{RHom}_{A_2}(E_2, F_2) \overset{\mathrm{L}}{\otimes}_{A_2} E_2) & \xrightarrow{(1)} & (F_1 \overset{\mathrm{L}}{\otimes}_{A_1^e} A_1) \overset{\mathrm{L}}{\otimes}_K (F_2 \overset{\mathrm{L}}{\otimes}_{A_2^e} A_2) \\
\Big\downarrow{\scriptstyle (2)} & & \Big\downarrow{\scriptstyle (3)} \\
\mathrm{RHom}_A(E, F) \overset{\mathrm{L}}{\otimes}_A E & \longrightarrow & F \overset{\mathrm{L}}{\otimes}_{A^e} A,
\end{array} \tag{5.12.1}$$

where (1) is the tensor product of the arrows 5.3.5, (2) is defined by the tensor-product arrow

$$\mathrm{RHom}_{A_1}(E_1,F_1)\overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2,F_2)\longrightarrow \mathrm{RHom}_A(E,F), \tag{5.12.2}$$

and (3) is the evident canonical isomorphism. The verification is essentially trivial. One checks in the same way that, for $E_i\in \mathrm{ob}\, D^-(A_i)$, $F_i\in \mathrm{ob}\, D^+(A_i^e)$, $G_i\in \mathrm{ob}\, D^-(A_i)$, with $F_i\overset{\mathrm{L}}{\otimes}_{A_i} G_i\in \mathrm{ob}\, D^b(A_i)$, $\mathrm{RHom}_{A_i}(E_i,F_i)\in \mathrm{ob}\, D^b(A_i^\circ)$, $\mathrm{RHom}_{A_i}(E_i,F_i\overset{\mathrm{L}}{\otimes}_{A_i} G_i)\in \mathrm{ob}\, D^b(K)$, $F\overset{\mathrm{L}}{\otimes}_A G\in \mathrm{ob}\, D^b(A)$, where $G=G_1\overset{\mathrm{L}}{\otimes}_K G_2$, there is a commutative square

$$\begin{array}{ccc}
\begin{array}{c}(\mathrm{RHom}_{A_1}(E_1,F_1)\overset{\mathrm{L}}{\otimes}_{A_1} G_1)\overset{\mathrm{L}}{\otimes}_K\\ (\mathrm{RHom}_{A_2}(E_2,F_2)\overset{\mathrm{L}}{\otimes}_{A_2} G_2)\end{array} & \xrightarrow{(1)} & \begin{array}{c}\mathrm{RHom}_{A_1}(E_1,F_1\overset{\mathrm{L}}{\otimes}_{A_1} G_1)\overset{\mathrm{L}}{\otimes}_K\\ \mathrm{RHom}_{A_2}(E_2,F_2\overset{\mathrm{L}}{\otimes}_{A_2} G_2)\end{array}\\
\downarrow{\scriptstyle (2)} & & \downarrow{\scriptstyle (3)}\\
\mathrm{RHom}_A(E,F)\overset{\mathrm{L}}{\otimes}_A G & \xrightarrow[5.4.2]{} & \mathrm{RHom}_A(E,F\overset{\mathrm{L}}{\otimes}_A G),
\end{array} \tag{5.12.3}$$

where (2) and (3) are tensor-product arrows of the type 5.12.2, and (1) is the tensor product of the arrows 5.4.2. From the commutativity of the squares 5.12.1 and 5.12.3 it follows that, for $E_i\in \mathrm{ob}\, D(A_i)_{\mathrm{parf}}$, $F_i\in \mathrm{ob}\, D^b(A_i^e)$ such that $F=F_1\overset{\mathrm{L}}{\otimes}_K F_2\in \mathrm{ob}\, D^b(A^e)$, there is a commutative square

$$\begin{array}{ccc}
\mathrm{RHom}_{A_1}(E_1,F_1\overset{\mathrm{L}}{\otimes}_{A_1} E_1)\overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2,F_2\overset{\mathrm{L}}{\otimes}_{A_2} E_2) & \xrightarrow{(1)} & (F_1\overset{\mathrm{L}}{\otimes}_{A_1^e} A_1)\overset{\mathrm{L}}{\otimes}_K (F_2\overset{\mathrm{L}}{\otimes}_{A_2^e} A_2)\\
\downarrow{\scriptstyle (2)} & & \downarrow{\scriptstyle (3)}\\
\mathrm{RHom}_A(E,F\overset{\mathrm{L}}{\otimes}_A E) & \xrightarrow[5.6.1]{} & F\overset{\mathrm{L}}{\otimes}_{A^e} A,
\end{array} \tag{5.12.4}$$

where (1) is the tensor product of the arrows Tr_{A_i} (5.6.1), (2) is defined by 5.12.2, and (3) is the canonical isomorphism appearing in 5.12.1. Consequently, for $u_i\in \mathrm{Hom}_{A_i}(E_i,F_i\overset{\mathrm{L}}{\otimes}_{A_i} E_i)$, one has

$$\mathrm{Tr}_A(u_1\overset{\mathrm{L}}{\otimes}_K u_2)=\mathrm{Tr}_{A_1}(u_1)\,\mathrm{Tr}_{A_2}(u_2), \tag{5.12.5}$$

the product on the right being defined by the canonical arrow

$$H^0(T,F_1\overset{\mathrm{L}}{\otimes}_{A_1^e} A_1)\otimes H^0(T,F_2\overset{\mathrm{L}}{\otimes}_{A_2^e} A_2)\longrightarrow H^0(T,F\overset{\mathrm{L}}{\otimes}_{A^e} A).$$

5.13. Traces of semilinear endomorphisms

Let A be a K-algebra and φ an endomorphism of A. If E, F are left A-modules, recall that an additive homomorphism $u:E\to F$ is called φ-A-linear if $u(ax)=\varphi(a)u(x)$ for all $a\in A$, $x\in E$, i.e. if u is an A-linear homomorphism from E to the A-module ${}_{(\varphi)}F$ deduced from F by restriction of scalars along φ. If F is an A-bimodule and G a left A-module, there is an evident isomorphism

$${}_{(\varphi)}F\otimes_A G\xrightarrow{\sim}{}_{(\varphi)}(F\otimes_A G),$$

which derives trivially to

$${}_{(\varphi)}F\overset{\mathrm{L}}{\otimes}_A G\xrightarrow{\sim}{}_{(\varphi)}(F\overset{\mathrm{L}}{\otimes}_A G).$$

Assuming A flat over K, one deduces from 5.4.2 the arrow

$$\mathrm{RHom}_A(E, {}_{(\varphi)}F) \overset{\mathrm{L}}{\otimes}_A G \longrightarrow \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A G)) \tag{5.13.1}$$

for $E \in \mathrm{ob}\, D^-(A)$, $F \in \mathrm{ob}\, D^+(A^e)$, $G \in \mathrm{ob}\, D^-(A)$, with $F \overset{\mathrm{L}}{\otimes}_A G \in \mathrm{ob}\, D^+(A)$. For $E \in \mathrm{ob}\, D(A)_{\mathrm{parf}}$, 5.13.1 is an isomorphism, which allows one to define, by 5.3.5, a trace arrow

$$\mathrm{Tr}_{A,\varphi} : \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A E)) \longrightarrow {}_{(\varphi)}F \overset{\mathrm{L}}{\otimes}_{A^e} A. \tag{5.13.2}$$

In particular, for $F = A$, one obtains

$$\mathrm{Tr}_{A,\varphi} : \mathrm{Hom}_A(E, {}_{(\varphi)}E) \longrightarrow H^0(T, A_{\natural,\varphi}), \tag{5.13.3}$$

where $A_{\natural,\varphi} = {}_{(\varphi)}A \otimes_{A^e} A$ is the quotient K-module of A by the submodule locally generated by local sections of the form $ab - \varphi(b)a$. In the case where $A = K[G]$, G is a discrete group, and φ is defined by an endomorphism φ of G, $A_{\natural,\varphi}$ is the free K-module with basis $G_{\natural,\varphi}$, where $G_{\natural,\varphi}$ is the quotient of G by the action $(g, x) \mapsto \varphi(g)xg^{-1}$; the orbits are sometimes called φ-conjugacy classes of G. The traces 5.13.2 coincide, for T punctual, with those considered by Grothendieck and discussed in Exposé XII.

In the situation of 5.10, if φ, ψ are respective endomorphisms of A, B, compatible with $A \to B$, there is a unique arrow

$$\mathrm{Tr}_A : B_{\natural,\psi} \longrightarrow A_{\natural,\varphi}$$

making the square, analogous to 5.10.2,

$$\begin{array}{ccc} {}_{(\psi)}B = \mathrm{Hom}_B(B, {}_{(\psi)}B) & \xrightarrow{\mathrm{Tr}_{B,\psi}} & B_{\natural,\psi} \\ \downarrow & & \downarrow{\scriptstyle \mathrm{Tr}_A} \\ \mathrm{Hom}_A(B, {}_{(\varphi)}A) \otimes_A B & \xrightarrow[\mathrm{ev}]{} & A_{\natural,\varphi}, \end{array} \tag{5.13.4}$$

commute, where ev is the evaluation arrow, and the left vertical arrow is restriction of scalars. One defines an arrow analogous to 5.10.5, with $A \otimes_K M$ (resp. $B \otimes_K M$) replaced by ${}_{(\varphi)}A \otimes_K M$ (resp. ${}_{(\psi)}B \otimes_K M$), and there are compatibilities analogous to 5.10.6, 5.10.9, 5.10.11, which we leave to the reader to state.

Let G be a finite group and φ an endomorphism of G. For $g \in G$, let $Z_{g,\varphi}$ be the φ-centralizer of g in G, i.e. the set of $x \in G$ such that $\varphi(x)gx^{-1} = g$. Put $A = K[G]$, and let $E \in \mathrm{ob}\, D(A)_{\mathrm{parf}}$, $M \in \mathrm{ob}\, D^b(K)$. The preceding compatibilities easily imply that, for $u \in \mathrm{Hom}_A(E, {}_{(\varphi)}(M \overset{\mathrm{L}}{\otimes}_K E))$ and $g \in G$,

$$\mathrm{Tr}_K(gu) = |Z_{g^{-1},\varphi}|\ \mathrm{Tr}_{A,\varphi}(u)(g^{-1}), \tag{5.13.5}$$

the details of the verification being left to the reader.

6. Equivariant Correspondences and Divisibility of Local Terms

6.1. Notation

In this number, S denotes the spectrum of a field of characteristic p, and Λ a noetherian commutative ring annihilated by an integer not divisible by p. The S-schemes considered will be assumed separated

and of finite type. The Λ-algebras considered will be associative, unital, and of finite type, but not necessarily commutative. Most often, and in particular for everything concerning "non-commutative Lefschetz–Verdier local terms," they will be assumed flat; in applications, they will be algebras of finite groups.

If X is an S-scheme and A a Λ-algebra, we write $D({}_AX)$ (resp. $D(X_A)$) for the derived category of the category of sheaves of left (resp. right) A-modules on X. For A commutative, we shall also write $D(X, A)$ for $D({}_AX)$, and $D(X) = D(X, \Lambda)$. We denote by $D_c(-)$ (resp. $D_{\mathrm{ctf}}(-)$) the full subcategory formed by complexes with constructible cohomology (resp. finite tor-dimension and constructible cohomology). If A, B are Λ-algebras, we write $D({}_AX_B)$ for the derived category of sheaves of (A, B)-bimodules, left over A and right over B. Finally, we use the notations A^o, A^e, $A_\natural$ of 5.1.

6.2

The formalism of (III 1, 2, 3) extends without difficulty to the "non-commutative" case. We briefly indicate the main points of this generalization.

Let X be an S-scheme and let A, B be Λ-algebras. If $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_A)$ and $M \in \mathrm{ob}\, D({}_AX_B)$, with M of finite tor-dimension and constructible cohomology as an object of $D(X_B)$, then

$$L \overset{\mathrm{L}}{\otimes}_A M \in \mathrm{ob}\, D(X_B)_{\mathrm{ctf}}$$

as is immediate. On the other hand, it follows from (SGA $4^{1/2}$ Finiteness 1.6) that, for M as above and $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AX)$, one has

$$\mathrm{RHom}_A(L, M) \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_B).$$

It also follows from loc. cit. that, if f is an S-morphism, $D_{\mathrm{ctf}}({}_A-)$ is stable under the operations $(f^*, f_*, f_!, f^!)$, the case of f^* being trivial.

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \longrightarrow Y = Y_1 \times_S Y_2,$$

and let A be a Λ-algebra. For $L_1 \in \mathrm{ob}\, D^-(X_{1A})$, $L_2 \in \mathrm{ob}\, D^-({}_AX_2)$, there is a Künneth arrow in $D^-(Y)$

$$f_{1!}L_1 \overset{\mathrm{L}}{\otimes}_{S,A} f_{2!}L_2 \longrightarrow f_*(L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2), \tag{6.2.1}$$

where

$$L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \overset{\mathrm{dfn}}{=} p_1^*L_1 \overset{\mathrm{L}}{\otimes}_A p_2^*L_2.$$

As in (III 1.6), one deduces that it is an isomorphism. If B, C are Λ-algebras, with B or C assumed flat over Λ in order to define

$$\overset{\mathrm{L}}{\otimes}_A : D^-({}_B-_A) \times D^-({}_A-_C) \longrightarrow D^-({}_B-_C),$$

there is an isomorphism analogous to 6.2.1 in $D^-({}_BY_C)$, for $L_1 \in \mathrm{ob}\, D^-({}_BX_{1A})$, $L_2 \in \mathrm{ob}\, D^-({}_AX_{2B})$. One checks similarly that, for $M_1 \in \mathrm{ob}\, D^-(Y_{1A})$, $M_2 \in \mathrm{ob}\, D^-({}_AY_2)$, there is a Künneth isomorphism in $D^-(X)$

$$f^!M_1 \overset{\mathrm{L}}{\otimes}_{S,A} f^!M_2 \longrightarrow f^!(M_1 \overset{\mathrm{L}}{\otimes}_{S,A} M_2), \tag{6.2.2}$$

and an analogous isomorphism in $D^-({}_BX_C)$ for $M_1 \in \mathrm{ob}\, D^-({}_BY_{1A})$, $M_2 \in \mathrm{ob}\, D^-({}_AY_{2C})$.

Finally, let, for $i = 1, 2$, A_i be a Λ-algebra, A_2 being flat, $E_1 \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{A_1}X_1)$, $F_2 \in \mathrm{ob}\, D^+({}_{A_2}X_2)$, $F_1 \in \mathrm{ob}\, D({}_{A_1}X_{1A_2})$, with F_1 of finite tor-dimension and constructible cohomology as an object of $D(X_{1A_2})$. Then

$$\mathrm{RHom}_{A_1}(E_1, F_1) \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_{1A_2}),$$

as seen above, and there is a canonical arrow in $D^+(X)$ ($D^+(X, A_2)$ when A_2 is commutative)

$$\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{S,A_2} F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^*E_1, F_1 \overset{\mathrm{L}}{\otimes}_{S,A_2} F_2), \tag{6.2.3}$$

defined by composing the "inverse-image" arrow (III 2.1.1)

$$p_1^*\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^*F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^*E_1, p_1^*F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^*F_2$$

with the arrow analogous to 5.4.2

$$\mathrm{RHom}_{A_1}(p_1^*E_1, p_1^*F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^*F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^*E_1, p_1^*F_1 \overset{\mathrm{L}}{\otimes}_{A_2} p_2^*F_2),$$

adjoint, by 5.2.3, to the arrow

$$p_1^*E_1 \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{A_1}(p_1^*E_1, p_1^*F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^*F_2 \longrightarrow p_1^*F_1 \overset{\mathrm{L}}{\otimes}_{A_2} p_2^*F_2$$

defined by the evaluation arrow 5.3.6. One deduces from 6.2.1, as in (III 2.3), that 6.2.3 is an isomorphism. In particular, for $A_2 = \Lambda$, $F_2 = \Lambda_{X_2}$, 6.2.3 gives an isomorphism

$$p_1^*\mathrm{RHom}_{A_1}(E_1, F_1) \xrightarrow{\sim} \mathrm{RHom}_{A_1}(p_1^*E_1, p_1^*F_1). \tag{6.2.4}$$

6.3

If $t : T \to S$ is an S-scheme, and A is a Λ-algebra, we put

$$t^!_A = KA_T \qquad (\in \mathrm{ob}\, D^+({}_AT_A)). \tag{6.3.1}$$

For $A = \Lambda$, we write K_T instead of $K\Lambda_T$. It follows from 6.2.2 that there is a canonical isomorphism in $D^+({}_AT_A)$

$$K_T \overset{\mathrm{L}}{\otimes} A \xrightarrow{\sim} KA_T. \tag{6.3.2}$$

We denote by D_A the functor

$$\mathrm{RHom}_A(-, KA_T).$$

It sends $D_{\mathrm{ctf}}({}_AT)$ into $D_{\mathrm{ctf}}(T_A)$ and conversely, and the proof of (SGA $4^{1/2}$ Finiteness Theorem 4) shows that, for $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AT)$, the canonical arrow

$$L \longrightarrow D_AD_AL$$

is an isomorphism. For $A = \Lambda$, we write D instead of D_A.

6.4

Let A be a flat Λ-algebra, and let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the projections, and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$. By 6.2.3 applied to $A_1 = A_2 = A$, $E_1 = L_1$, $F_2 = L_2$, $E_2 = KA_{X_1}$, there is an isomorphism

$$D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^* L_1, KA_{X_1} \overset{\mathrm{L}}{\otimes}_{S,A} L_2),$$

whence, by composing with the Künneth isomorphism 6.2.2

$$KA_{X_1} \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} p_2^! L_2,$$

an isomorphism

$$D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^* L_1, p_2^! L_2), \tag{6.4.1}$$

which generalizes (III 3.1.1).

Let $c : C \to X$ be an S-morphism, and put $c_i = p_i c$. By the induction formula (SGA 4 XVIII 3.1.12.2), there is a canonical isomorphism, analogous to (III 3.2.1),

$$c^! \mathrm{RHom}_A(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}_A(c_1^* L_1, c_2^! L_2), \tag{6.4.2}$$

and hence, by composing with 6.4.1, an isomorphism

$$H^0(X, c_* c^!(D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2)) \xrightarrow{\sim} \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2). \tag{6.4.3}$$

The elements of $\mathrm{Hom}_A(c_1^* L_1, c_2^! L_2)$ will be called cohomological correspondences from L_1 to L_2 with support in c. The terminology differs slightly from that of loc. cit.; the two terminologies coincide when c is proper, which is the case in practice.

Given S-morphisms $f_i : X_i \to Y_i$ $(i = 1, 2)$ and a commutative square (III 3.7.1) with c proper, one defines, as in loc. cit., direct-image arrows analogous to (III 3.7.5), (III 3.7.6), for $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$.

6.5

Let A be a flat Λ-algebra, X an S-scheme, $X^2 = X \times_S X$, $p_1, p_2 : X^2 \to X$ the canonical projections, $\delta : X \to X^2$ the diagonal, and

$$c : C \longrightarrow X^2$$

an S-morphism. Put $c_i = p_i c$, and let $X^c = C \times_{X^2} X$ be the scheme of fixed points of c, giving a cartesian square

$$\begin{array}{ccc} X^c & \xrightarrow{i^c} & X \\ {\scriptstyle \delta^c}\downarrow & & \downarrow{\scriptstyle \delta} \\ C & \xrightarrow[c]{} & X^2. \end{array} \tag{6.5.1}$$

Put

$$KA_{X,L_\natural} = K_X \overset{\mathrm{L}}{\otimes}_\Lambda (A \overset{\mathrm{L}}{\otimes}_{A^e} A), \qquad KA_{X,\natural} = K_X \overset{\mathrm{L}}{\otimes}_\Lambda A_\natural. \tag{6.5.2}$$

Let $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X)$. We define a "local trace" homomorphism

$$\mathrm{Tr}_A : \delta^* \mathrm{RHom}_A(p_1^* L, p_2^! L) \longrightarrow KA_{X,L_\natural}, \tag{6.5.3}$$

by composing the isomorphism

$$\delta^*\mathrm{RHom}_A(p_1^*L, p_2^!L) \xrightarrow{\sim} \delta^*(D_AL \overset{\mathrm{L}}{\otimes}_{S,A} L) \tag{6.5.4}$$

deduced from the inverse of 6.4.1 by applying δ^*, the trivial isomorphism

$$\delta^*(D_AL \overset{\mathrm{L}}{\otimes}_{S,A} L) = D_AL \overset{\mathrm{L}}{\otimes}_A L, \tag{6.5.5}$$

and the evaluation arrow 5.3.5

$$D_AL \overset{\mathrm{L}}{\otimes}_A L \longrightarrow KA_{X,L_\natural}. \tag{6.5.6}$$

(Here $KA_X \overset{\mathrm{L}}{\otimes}_{A^e} A$ is identified with $KA_{X,L_\natural}$ by means of 6.3.2, $K_X \otimes A = KA_X$.) It follows at once from the definition that, for $X = S$, 6.5.3 coincides with the arrow

$$\mathrm{Tr}_A : \mathrm{RHom}_A(L, L) \longrightarrow A \overset{\mathrm{L}}{\otimes}_{A^e} A$$

of 5.6.1. Composing the arrow deduced from 6.5.3 by applying $i^c c^!$ with the arrow

$$\delta^* c^c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) \longrightarrow i^c_* i^{c!} \delta^* \mathrm{RHom}_A(p_1^*L, p_2^!L) \tag{6.5.7}$$

deduced from 6.4.2 and the base-change arrow

$$\delta^* c_* c^! \longrightarrow i^c_* i^{c!} \delta^*,$$

one obtains an arrow

$$\mathrm{Tr}_A : \delta^* c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) \longrightarrow i^c_* i^{c!} KA_{X,L_\natural} = i^c_* KA_{X^c,L_\natural}, \tag{6.5.8}$$

and hence the arrow

$$\mathrm{Tr}_A : \mathrm{Hom}_A(c_1^*L, c_2^!L) \longrightarrow H^0(X^c, KA_{X^c,L_\natural}). \tag{6.5.9}$$

We again write Tr_A for the composites of 6.5.3, 6.5.8, 6.5.9 with the arrows deduced from the canonical arrow

$$KA_{X,L_\natural} \longrightarrow KA_{X,\natural}.$$

One checks easily that, for $A = \Lambda$, one has, for $u \in \mathrm{Hom}(c_1^*L, c_2^!L)$,

$$\mathrm{Tr}(u) = \langle u, 1 \rangle, \tag{6.5.10}$$

with the notation of (III 4.2.5), where 1 denotes the diagonal correspondence. More precisely, the arrow

$$c_* \mathrm{RHom}(c_1^*L, c_2^!L) \longrightarrow \delta_* i^c_* K_{X^c}$$

adjoint to 6.5.8 is the one defined by (III 4.2.4') with $X_1 = X_2$, $L_1 = L_2$, $d = \delta$, and the diagonal correspondence $1 \in \mathrm{Hom}(L, L)$.

6.6

Let A be a flat Λ-algebra, and let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$. We define a pairing

$$\mathrm{RHom}_A(p_1^* L_1, p_2^! L_2) \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_A(p_2^* L_2, p_1^! L_1) \longrightarrow KA_{X, L_\natural} \tag{6.6.1}$$

as follows. One identifies the left hand side with

$$\mathrm{RHom}_A(p_1^* L_1, p_1^* KA_{X_1}) \overset{\mathrm{L}}{\otimes}_A p_2^* L_2 \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_A(p_2^* L_2, p_2^* KA_{X_2}) \overset{\mathrm{L}}{\otimes}_A p_1^* L_1$$

by the isomorphisms 6.4.1, then with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} \mathrm{RHom}_A(\ ) \overset{\mathrm{L}}{\otimes}_\Lambda p_2^* L_2 \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_A(\ ) \overset{\mathrm{L}}{\otimes}_\Lambda p_1^* L_1 \overset{\mathrm{L}}{\otimes}_{A^e} A$$

by 5.3.1. This object is sent to

$$A \overset{\mathrm{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\mathrm{L}}{\otimes}_\Lambda KA_{X_2}) \overset{\mathrm{L}}{\otimes}_{A^e} A = R$$

by the tensor product of the evaluation arrows 5.3.6. Finally R is identified with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\mathrm{L}}{\otimes}_A KA_{X_2})$$

by 5.8.2, and then with $KA_{X, L_\natural}$ by 6.5.2 and (III 1.7.6).

If $c : C \to X$, $d : D \to X$ are correspondences, and

$$e = c \times_X d : E \to X,$$

one deduces from 6.6.1, as in (III 4.2), a pairing

$$c_* \mathrm{RHom}_A(c_1^* L_1, c_2^! L_2) \overset{\mathrm{L}}{\otimes}_\Lambda d_* \mathrm{RHom}_A(d_2^* L_2, d_1^! L_1) \longrightarrow e_* KA_{E, L_\natural}, \tag{6.6.2}$$

and hence a cup-product

$$\langle\ ,\ \rangle_A : \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}_A(d_2^* L_2, d_1^! L_1) \longrightarrow H^0(E, KA_{E, L_\natural}). \tag{6.6.3}$$

It is not difficult to extend to the non-commutative case the notion of composition of correspondences studied in (III 5.2), and one checks that, for

$$u \in \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2), \qquad v \in \mathrm{Hom}_A(d_2^* L_2, d_1^! L_1),$$

one has the formula analogous to (III 5.2.9)

$$\langle u, v \rangle_A = \mathrm{Tr}_A(\bar{v} u) = \mathrm{Tr}_A(u \bar{v}). \tag{6.6.4}$$

The Lefschetz formula (III 4.4) generalizes without difficulty in this context. Given proper S-morphisms $f_i : X_i \to Y_i$ $(i = 1, 2)$ and a commutative cube (III 4.4.0), for $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$ one has a commutative square analogous to (III 4.4.1), with K_C (resp. K_D) replaced by $KA_{C, L_\natural}$ (resp. $KA_{D, L_\natural}$), and a formula analogous to (III 4.5.1), in $H^0(D, KA_{D, L_\natural})$:

$$\langle f_* u, f_* v \rangle_A = g_* \langle u, v \rangle_A. \tag{6.6.5}$$

The arrow

$$f_* c_* KA_{C, L_\natural} \longrightarrow d_* KA_{D, L_\natural}$$

is deduced from arrow (4) of (III 4.4.1) by tensoring with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} A.$$

6.7

Let X and c be as in 6.5, and let $A \to B$ be a morphism of flat Λ-algebras. With respect to extension and restriction of scalars, the arrows Tr_A (6.5.3, 6.5.8, 6.5.9) behave like the arrows Tr_A of 5.6.1, 5.6.2. The point is that, if f is an S-morphism, the functors $(f^*, f_*, f_!, f^!)$ commute with extension and restriction of scalars; consequently, the study of the behavior of Tr_A (6.5.3) with respect to extension and restriction of scalars reduces to the behavior of the "tensor-product" arrow

$$\mathrm{RHom}_A(p_1^*L, p_1^*KA_X) \overset{\mathrm{L}}{\otimes}_A p_2^*L \longrightarrow \mathrm{RHom}_A(p_1^*L, p_1^*KA_X \overset{\mathrm{L}}{\otimes}_A p_2^*L)$$

and of the "evaluation" arrow

$$\mathrm{RHom}_A(L, KA_X) \overset{\mathrm{L}}{\otimes}_A L \longrightarrow KA_{X,L_\natural}.$$

Thus the commutativity of the squares 5.9.1, 5.9.2 implies the commutativity of the square

$$\begin{array}{ccc} \delta^* c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) & \xrightarrow{\mathrm{Tr}_A} & i_*^c KA_{X^c,L_\natural} \\ \downarrow & & \downarrow \\ \delta^* c_* \mathrm{RHom}_B(c_1^*(B \overset{\mathrm{L}}{\otimes}_A L), c_2^!(B \overset{\mathrm{L}}{\otimes}_A L)) & \xrightarrow[\mathrm{Tr}_B]{} & i_*^c KB_{X^c,L_\natural}, \end{array} \tag{6.7.1}$$

where the left vertical arrow is extension of scalars and the right vertical arrow is defined by the canonical arrow

$$A \overset{\mathrm{L}}{\otimes}_{A^e} A \longrightarrow B \overset{\mathrm{L}}{\otimes}_{B^e} B.$$

It follows that, for $u \in \mathrm{Hom}_A(c_1^*L, c_2^!L)$,

$$\mathrm{Tr}_B(B \overset{\mathrm{L}}{\otimes}_A u) = \varphi\, \mathrm{Tr}_A(u), \tag{6.7.2}$$

where

$$\varphi : H^0(X^c, KA_{X^c,L_\natural}) \longrightarrow H^0(X^c, KB_{X^c,L_\natural})$$

is the canonical map.

Assume A is finite type and flat as a Λ-module, B finite type and flat as a left A-module, and $A_\natural$, $B_\natural$ flat over Λ. There is a map $\mathrm{Tr}_{B/A} : B_\natural \to A_\natural$; we shall again denote by

$$\mathrm{Tr}_{B/A} : KB_{X,\natural} \longrightarrow KA_{X,\natural} \tag{6.7.3}$$

the arrow deduced from it by applying $K_X \overset{\mathrm{L}}{\otimes} -$ (6.5.2). It follows from 5.10.6 and 5.10.9 that, for $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_BX)$, there is a commutative square

$$\begin{array}{ccc} \delta^* c_* \mathrm{RHom}_B(c_1^*L, c_2^!L) & \xrightarrow{\mathrm{Tr}_B} & i_*^c KB_{X^c,\natural} \\ \downarrow & & \downarrow{\scriptstyle \mathrm{Tr}_{B/A}} \\ \delta^* c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) & \xrightarrow[\mathrm{Tr}_A]{} & i_*^c KA_{X^c,\natural}, \end{array} \tag{6.7.4}$$

where the left vertical arrow is restriction of scalars. Consequently, for $u \in \mathrm{Hom}_B(c_1^*L, c_2^!L)$, one has, in $H^0(X^c, KA_{X^c,\natural})$,

$$\mathrm{Tr}_A(u) = \mathrm{Tr}_{B/A}(\mathrm{Tr}_B(u)). \tag{6.7.5}$$

6.8

Let A, B be flat Λ-algebras, X, Y S-schemes, $Z = X \times_S Y$, and let $c : C \to X^2$, $d : D \to Y^2$ be correspondences. Let

$$e = c \times_S d : E = C \times_S D \longrightarrow Z^2$$

be their product. Then

$$Z^e = X^c \times_S Y^d.$$

Put $R = A \otimes_\Lambda B$. As in (III 5.3), one defines the external tensor product

$$\mathrm{Hom}_A(c_1^* L, c_2^! L) \otimes \mathrm{Hom}_B(d_1^* M, d_2^! M) \xrightarrow{\overset{\mathrm{L}}{\otimes}_S} \mathrm{Hom}_R(e_1^*(L \overset{\mathrm{L}}{\otimes}_S M), e_2^!(L \overset{\mathrm{L}}{\otimes}_S M)), \tag{6.8.1}$$

and one deduces easily from the commutativity of the squares 5.12.1, 5.12.3 that, for

$$u \in \mathrm{Hom}_A(c_1^* L, c_2^! L), \qquad v \in \mathrm{Hom}_B(d_1^* M, d_2^! M),$$

one has, in $H^0(Z^e, KR_{Z^e, L_\natural})$,

$$\mathrm{Tr}_R(u \overset{\mathrm{L}}{\otimes}_S v) = \mathrm{Tr}_A(u)\,\mathrm{Tr}_B(v), \tag{6.8.2}$$

the product on the right being defined by the canonical arrow

$$H^0(X^c, KA_{X^c, L_\natural}) \otimes H^0(Y^d, KB_{Y^d, L_\natural}) \longrightarrow H^0(Z^e, KR_{Z^e, L_\natural}), \tag{6.8.3}$$

coming from the isomorphism

$$KA_{X^c, L_\natural} \overset{\mathrm{L}}{\otimes}_S KB_{Y^d, L_\natural} \longrightarrow K(A \otimes B)_{Z^e, L_\natural}$$

given by (III 1.7.6) and arrow (4) of 5.9.1.

6.9

Let G be a finite group. If X is a G-S-scheme, that is, an S-scheme endowed with an action of G by S-automorphisms,[18] and A is a Λ-algebra,[19] we denote by $D(G, {}_A X)$ the derived category of the category of sheaves of left G-A-modules on X. For the notions of G-sheaf of sets, rings, modules, and so on, on X for the étale topology, we refer the reader to (X 1). We denote by the subscript c (resp. ctf) the full subcategory formed by complexes of G-A-modules which, as complexes of A-modules, have constructible cohomology (resp. finite tor-dimension and constructible cohomology).

Similarly, if B is a Λ-algebra, we denote by $D(G, {}_A X_B)$ the derived category of sheaves of G-(A, B)-bimodules on X. If G acts trivially on X, a left G-A-module on X is just a left $A[G]$-module on X without operators, hence

$$D(G, {}_A X) = D({}_{A[G]} X).$$

Since G is finite, one also has

$$D_c({}_A X) = D_c({}_{A[G]} X),$$

but only an inclusion

$$D_{\mathrm{ctf}}({}_{A[G]} X) \subset D_{\mathrm{ctf}}(G, {}_A X).$$

[18] We agree that G acts on spaces on the right.

[19] We are chiefly interested in the case of commutative Λ-algebras.

The duality formalism of (SGA 4 XVII, XVIII) extends without difficulty to G-S-schemes. Since the functor forgetting the action of G is faithful, it is enough to extend naturally the definition of the "six operations" $(\overset{\mathrm{L}}{\otimes}, \mathrm{RHom}, f^*, f_*, f_!, f^!)$ and the canonical arrows considered in loc. cit. (base change, Künneth, duality, etc.). This extension presents no difficulty: for the definition of the functors $f_!$, $f^!$, it is enough to use equivariant compactifications. We leave to the reader the task of paraphrasing the definitions and results 6.2 to 6.8 in the setting of G-S-schemes.

If, for $i = 1, 2$, X_i is a G-S-scheme, $c : C \to X = X_1 \times_S X_2$ is a G-S-morphism, and $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{G,A}X_i)$, the elements of

$$\mathrm{Hom}_A(c_1^* L_1, c_2^! L_2),$$

where Hom_A is taken in the category $D({}_{G,A}-)$, will be called equivariant cohomological correspondences from L_1 to L_2 with support in c. Note that, when G acts trivially on X_i $(i = 1, 2)$ and $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{A[G]}X_i)$, the isomorphism 6.4.1 in $D(G, X)$ refines to an isomorphism

$$D_{A[G]} L_1 \overset{\mathrm{L}}{\otimes}_{S,A[G]} L_2 \xrightarrow{\sim} \mathrm{RHom}_{A[G]}(p_1^* L_1, p_2^! L_2).$$

It follows that, in a G-equivariant situation of type 6.5, with trivial action of G on the spaces, and with $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{A[G]}X) \subset D_{\mathrm{ctf}}({}_{G,A}X)$, one may, for

$$u \in \mathrm{Hom}_{A[G]}(c_1^* L, c_2^! L),$$

define a local term

$$\mathrm{Tr}_{A[G]}(u) \in H^0(X^c, KA[G]_{X^c, L_\natural}),$$

finer, in the evident sense, than the local term

$$\mathrm{Tr}_A(u) \in H^0(G, X^c, KA_{X^c, L_\natural})$$

obtained by using only finite tor-dimension of L with respect to A.

6.10

Let X, c be as in 6.5, and let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[G]}X)$. For

$$u \in \mathrm{Hom}_{\Lambda[G]}(c_1^* L, c_2^! L),$$

6.5.9 gives a local trace

$$\mathrm{Tr}_{\Lambda[G]}(u) \in H^0(X^c, K\Lambda[G]_{X^c,\natural}) = H^0(X^c, K_{X^c} \otimes \Lambda[G_\natural]), \tag{6.10.1}$$

where $G_\natural$ is the set of conjugacy classes of G. For every $x \in G_\natural$, one may consider the "value of $\mathrm{Tr}_{\Lambda[G]}(u)$ at x,"

$$\mathrm{Tr}_{\Lambda[G]}(u)(x) \in H^0(X^c, K_{X^c}),$$

the image of $\mathrm{Tr}_{\Lambda[G]}(u)$ under the projection

$$H^0(X^c, K_{X^c} \otimes \Lambda[G_\natural]) \longrightarrow H^0(X^c, K_{X^c})$$

defined by x. It follows from 6.7.5 and 5.11.2 that

$$\mathrm{Tr}_\Lambda(u) = |G|\ \mathrm{Tr}_{\Lambda[G]}(u)(e), \tag{6.10.2}$$

where e denotes the identity element of G.

More generally, let $g \in G$, let Z_g be the centralizer of g, and let

$$gu \in \mathrm{Hom}_{\Lambda[Z_g]}(c_1^* L, c_2^! L)$$

be the correspondence obtained by composing u with left multiplication by $g : c_1^* L \to c_1^* L$. We have $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[Z_g]}X)$, and hence 6.10.2 implies

$$\mathrm{Tr}_\Lambda(gu) = |Z_g|\ \mathrm{Tr}_{\Lambda[Z_g]}(gu)(e). \tag{6.10.3}$$

By the same argument as in 5.11.3, one deduces from 6.7.4 the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gu)(e) = \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}), \tag{6.10.4}$$

which allows 6.10.3 to be rewritten, if desired, in the form

$$\mathrm{Tr}_\Lambda(gu) = |Z_g|\ \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}). \tag{6.10.5}$$

6.11

Let $f : X \to Y$ be a proper G-S-morphism, and let

$$\begin{array}{ccc} C & \xrightarrow{c} & X \times_S X \\ \downarrow & & \downarrow{\scriptstyle f\times_S f} \\ D & \xrightarrow[d]{} & Y \times_S Y \end{array} \tag{6.11.1}$$

be a commutative G-equivariant square of G-S-schemes, with c and d proper. Assume that G acts trivially on Y and D. We denote by

$$f^{c/d} \quad (\text{or } f^c) : X^c \longrightarrow Y^d \tag{6.11.2}$$

the morphism induced by 6.11.1 on the schemes of fixed points of c and d, with the notation of 6.5.1. Let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(G, {}_A X)$, and let

$$u \in \mathrm{Hom}(c_1^* L, c_2^! L)$$

be a G-equivariant correspondence with support in c. By direct image, it gives a G-equivariant correspondence

$$f_* u \in \mathrm{Hom}_{\Lambda[G]}(d_1^* f_* L, d_2^! f_* L).$$

For $g \in G$, let Z_g denote the centralizer of g, and let

$$gu \in \mathrm{Hom}((gc)_1^* L, c_2^! L) \tag{6.11.3}$$

be the Z_g-equivariant correspondence with support in

$$gc \overset{\mathrm{dfn}}{=} (gc_1 = c_1 g, c_2) : C \to X \times_S X, \tag{6.11.4}$$

defined as the composite of the arrows

$$g^*(c_1^* L) \xrightarrow{\alpha_g} c_1^* L \xrightarrow{u} c_2^! L,$$

where α_g is the arrow giving the action of G on c_1^*L. It is clear that

$$f_*(gu) = g(f_*u). \tag{6.11.5}$$

Proposition 6.12. Under the hypotheses of 6.11, assume that

$$f_*L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[G]}Y).$$

Then, for every $g \in G$,

$$f_*^{gc} \mathrm{Tr}_\Lambda(gu) = |Z_g|\ \mathrm{Tr}_{\Lambda[G]}(f_*u)(g^{-1}), \tag{6.12.1}$$

where

$$f_*^{gc} : H^0(X^{gc}, K_{X^{gc}}) \longrightarrow H^0(Y^d, K_{Y^d})$$

is the "Gysin" morphism defined by the adjunction arrow

$$f_*^{gc} K_{X^{gc}} = f_*^{gc} f^{gc!} K_{Y^d} \longrightarrow K_{Y^d}.$$

Indeed, by 6.10.5 applied to f_*u, and by 6.11.5, one has

$$\mathrm{Tr}_\Lambda(f_*(gu)) = |Z_g|\ \mathrm{Tr}_{\Lambda[G]}(f_*u)(g^{-1}),$$

and 6.12.1 therefore follows from the Lefschetz formula (III 4.5.1).

6.13

Let ℓ be a prime number distinct from p. For every integer $n \geq 1$, put

$$\Lambda_n = \mathbb{Z}/\ell^n.$$

Under the hypotheses of 6.11, let $L = (L_n)$ be a projective system of objects of

$$D_{\mathrm{ctf}}(G, X, \Lambda_n)$$

such that

$$L_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n \xrightarrow{\sim} L_n \quad (m \geq n),$$

and let

$$u \in \mathrm{Hom}(c_1^*L, c_2^!L)$$

be a projective system of equivariant correspondences $u_n \in \mathrm{Hom}(c_1^*L_n, c_2^!L_n)$ such that, for $m \geq n$,

$$u_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n = u_n$$

modulo the isomorphism $L_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n = L_n$. By compatibility of local traces with extension of scalars, the local terms, for $g \in G$,

$$\mathrm{Tr}_{\Lambda_n}(gu_n) \in H^0(X^{gc}, K\Lambda_{n,X^{gc}})$$

define an element

$$\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(X^{gc}, K_{\ell,X^{gc}}) \overset{\mathrm{dfn}}{=} \varprojlim_n H^0(X^{gc}, K\Lambda_{n,X^{gc}}),$$

and the terms

$$f_*^{gc}\,\mathrm{Tr}_{\Lambda_n}(gu_n) = \mathrm{Tr}_{\Lambda_n}(gf_*u_n)$$

an element

$$f_*\,\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(Y^d, K_{\ell,Y^d}) \overset{\mathrm{dfn}}{=} \varprojlim_n H^0(Y^d, K\Lambda_{n,Y^d}).$$

Assume moreover that

$$f_*M_n \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda_n[G]}Y)$$

for all n. Then the local traces

$$\mathrm{Tr}_{\Lambda_n[G]}(f_*u_n)$$

similarly define an element

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u) \in H^0(Y^d, K\mathbb{Z}_\ell[G]_{Y^d,\natural}) \overset{\mathrm{dfn}}{=} \varprojlim_n H^0(Y^d, K\Lambda_n[G]_{Y^d,\natural}).$$

It follows from 6.12 that

$$f_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) = |Z_g|\ \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u)(g^{-1}). \tag{6.13.1}$$

6.14

Let X be a G-S-scheme, let $j : U \to X$ be an equivariant open immersion, and let

$$c : C \to X \times_S X$$

be a morphism of G-S-schemes such that $c_2(= p_2c) : C \to X$ is quasi-finite and of finite tor-dimension. Suppose that

$$c_1^{-1}(U) \subset c_2^{-1}(U).$$

Then there are commutative squares

$$\begin{array}{ccccc}
U & \longleftarrow & c_1^{-1}(U) & & \\
\downarrow{\scriptstyle i} & & \downarrow & & \\
X & \xleftarrow{c_1} & C & \xrightarrow{c_2} & X \\
\uparrow{\scriptstyle j} & & \uparrow & & \uparrow \\
X-U & \longleftarrow & c_2^{-1}(X-U) & \longrightarrow & X-U
\end{array} \tag{6.14.1}$$

The trace morphism

$$\mathrm{Tr}_{c_2} : c_{2!}\Lambda_C \longrightarrow \Lambda_X$$

then defines a filtered G-equivariant correspondence

$$\underline{c} \in \mathrm{Hom}(c_1^*\Lambda_X, c_2^!\Lambda_X), \tag{6.14.2}$$

where Λ_X is endowed with the filtration defined by the submodule $i_!\Lambda_U$, that is,

$$F^n\Lambda_X = \Lambda_X \quad (n \le 0), \qquad F^1\Lambda_X = i_!\Lambda_U, \qquad F^n\Lambda_X = 0 \quad (n \ge 1).$$

The associated graded consists of two G-equivariant correspondences

$$\mathrm{gr}^0\,\underline{c} = \underline{c}_{X-U,X} \in \mathrm{Hom}(c_1^*\Lambda_{X-U,X}, c_2^!\Lambda_{X-U,X}),$$
$$\mathrm{gr}^1\,\underline{c} = \underline{c}_{U,X} \in \mathrm{Hom}(c_1^*\Lambda_{U,X}, c_2^!\Lambda_{X-U,X}), \tag{6.14.3}$$

where we have put

$$\Lambda_{X-U,X} = j_*\Lambda_{X-U}, \qquad \Lambda_{U,X} = i_!\Lambda_U.$$

We have

$$\underline{c}_{X-U,U} = j_*\underline{c}_{X-U}, \tag{6.14.4}$$

where

$$\underline{c}_{X-U} \in \mathrm{Hom}(c_1^*\Lambda_{X-U}, c_2^!\Lambda_{X-U})$$

is the correspondence defined by $\mathrm{Tr}_{c_2,X-U}$. We write $\underline{c}$, $\underline{c}_{U,X}$, and so on, when we want to specify the base ring.

Forgetting the action of G, there are local terms

$$\mathrm{Tr}_\Lambda(\underline{c}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{U,X}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) \in H^0(X^c, K_{X^c}), \tag{6.14.5}$$

which, by additivity of traces (III 4.13.1), satisfy the relation

$$\mathrm{Tr}_\Lambda(\underline{c}) = \mathrm{Tr}_\Lambda(\underline{c}_{U,X}) + \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}). \tag{6.14.6}$$

Moreover, by 6.14.4,

$$\mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) = j_*^c\,\mathrm{Tr}_\Lambda(\underline{c}_{X-U}), \tag{6.14.7}$$

where

$$j_*^c : H^0((X-U)^c, K_{(X-U)^c}) \longrightarrow H^0(X^c, K_{X^c})$$

is the direct-image morphism defined by $j^c : (X-U)^c \to X^c$. Recall (III 4.3) that, if X is smooth, c a closed immersion, and X^c finite, $\mathrm{Tr}_\Lambda(\underline{c})$ is the image in Λ of the intersection multiplicity $(C \cdot X)$ of C with the diagonal. If, moreover, $X-U$ is smooth, $\mathrm{Tr}_\Lambda(\underline{c}_{X-U})$ is the image in Λ of $(c_2^{-1}(X-U)\cdot(X-U))$. Note also that, if $X-U$ is proper over S and $c|X-U$ is the diagonal, then the Lefschetz formula (III 4.7) implies that

$$\int_{X-U} \mathrm{Tr}(\underline{c}_{X-U}) \in \Lambda$$

is the Euler–Poincaré characteristic of $X-U$ with values in Λ.

6.15

Let there be a G-equivariant diagram 6.11.1, with f, c, d proper, and an equivariant open immersion $U \to X$. Suppose that c_2 is quasi-finite and of finite tor-dimension, that

$$c_1^{-1}(U) \subset c_2^{-1}(U),$$

that G acts trivially on C and D, and that

$$f|U : U \longrightarrow f(U) = V$$

is an étale Galois covering with group G. This last hypothesis implies that

$$Rf_*(\Lambda_{U,X}) \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[G]}Y) :$$

indeed $Rf_*(\Lambda_{U,X})$ is concentrated in degree 0 and is extension by zero of a sheaf on V locally isomorphic to $\Lambda[G]$. Hence one may apply 6.12 to $L = \Lambda_{U,X}$, $u = \underline{c}_{U,X}$, and, in view of 6.14.6 and 6.14.7, one obtains:

Proposition 6.16. Under the hypotheses of 6.15, for every $g \in G$ one has

$$f_*^{gc}\,\mathrm{Tr}_\Lambda(g\underline{c}) - (fj)_*^{gc}\,\mathrm{Tr}_\Lambda(g\underline{c}_{X-U}) = |Z_g|\ \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1}), \tag{6.16.1}$$

where $j : X - U \to X$ is the inclusion, and Z_g denotes the centralizer of g.

Applying 6.16 to the projective system of correspondences $\underline{c}_\ell = (\underline{c}_{\Lambda_n})$, where $\Lambda_n = \mathbb{Z}/\ell^n$ with ℓ prime distinct from p, and taking into account that 6.14.6 and 6.14.7 give, after passage to the limit and application of f_*^{gc},

$$f_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) = f_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,U,X}) + (fj)_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}),$$

one deduces, by 6.13:

Corollary 6.17. Under the hypotheses of 6.15, for every $g \in G$ one has

$$f_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) - (fj)_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}) = |Z_g|\ \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*\underline{c}_{\ell,U,X})(g^{-1}), \tag{6.17.1}$$

with the notation of 6.16.

In view of the compatibility of local traces with localization and of the remarks made in 6.14, one sees that 6.17 gives an affirmative solution, in a more general formulation, to Grothendieck's divisibility conjecture (XII 4.5), in the case where the endomorphism φ of G considered in loc. cit. is the identity.

In fact, by following 5.13, it is not difficult to give divisibility statements analogous to 6.12, 6.13, 6.16, 6.17, in the case where c is not G-equivariant, but where c_2 is and c_1 satisfies

$$c_1(xg) = c_1(x)\varphi(g)$$

for every point x of G, φ being a given endomorphism of G. The divisibility formulas are then written as above, with Z_g replaced by the φ-centralizer of g^{-1} (5.13.5).

6.18

Put

$$N_G = \sum_{g\in G} g \in \Lambda[G]. \tag{6.18.1}$$

Let X be an S-scheme. For $L \in \mathrm{ob}\, D({}_{\Lambda[G]}X)$, left multiplication by N_G induces an arrow in $D(X)$

$$N_G : \Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} L \longrightarrow \mathrm{RHom}_{\Lambda[G]}(\Lambda, L), \tag{6.18.2}$$

where Λ is regarded as a $\Lambda[G]$-algebra by the natural augmentation sending every $g \in G$ to 1.

Lemma 6.18.3. If $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}X)$, the arrow 6.18.2 is an isomorphism.

By dévissage, one reduces to the case where L is perfect, then to $L = \Lambda[G]$, and the assertion follows from the fact that N_G (6.18.1) is a basis of the module of G-invariants of the left $\Lambda[G]$-module $\Lambda[G]$.

Proposition 6.19. In the situation of 6.11, let $M \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$,

$$v \in \operatorname{Hom}(d_1^* M, d_2^! M),$$

and let $a : M \to f_* L$ be an arrow in $D({}_{\Lambda[G]}Y)$, where M is regarded as an object of $D({}_{\Lambda[G]}Y)$ by the augmentation $\Lambda[G] \to \Lambda$, such that the square

$$\begin{array}{ccc} d_1^* M & \xrightarrow{v} & d_2^! M \\ \downarrow & & \downarrow \\ d_1^* f_* L & \xrightarrow[f_* u]{} & d_2^! f_* L \end{array}$$

is commutative. Suppose that $f_* M \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}Y)$ and that a induces an isomorphism

$$M \xrightarrow{\sim} \operatorname{RHom}_{\Lambda[G]}(\Lambda, f_* L).$$

Then

$$\operatorname{Tr}_\Lambda(v) = \sum_{g \in G_\natural} \operatorname{Tr}_{\Lambda[G]}(f_* u)(g^{-1}). \tag{6.19.1}$$

By 6.18.3, a, composed with N_G^{-1}, gives an isomorphism

$$M \xrightarrow{\sim} \Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} f_* L,$$

under which v is identified with

$$\Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} f_* u.$$

By 6.7.2, $\operatorname{Tr}_\Lambda(v)$ is therefore the image of $\operatorname{Tr}_{\Lambda[G]}(f_* u)$ under the arrow induced by the augmentation

$$\Lambda[G] \longrightarrow \Lambda.$$

This augmentation, by definition, sends every conjugacy class to 1, whence the conclusion.

6.20

Let, for $i = 1, 2$, X_i be a G-S-scheme, and let

$$X = X_1 \times_S X_2, \qquad c : C \to X$$

be a G-S-morphism such that c_2 is quasi-finite and of finite tor-dimension. Let $L_i \in \operatorname{ob} D^+(G, {}_A X_i)$. The trace morphism

$$\operatorname{Tr}_{c_2} : c_{2!} c_2^* L_2 \longrightarrow L_2$$

defines by adjunction an arrow in $D(G,X)$, again denoted

$$\mathrm{Tr}_{c_2} : c_2^* L_2 \longrightarrow c_2^! L_2, \tag{6.20.1}$$

which associates to every G-equivariant morphism

$$u \in \mathrm{Hom}(c_1^* L_1, c_2^* L_2)$$

a G-equivariant correspondence

$$u^! = \mathrm{Tr}_{c_2} \circ u \in \mathrm{Hom}(c_1^* L_1, c_2^! L_2). \tag{6.20.2}$$

Proposition 6.21. Let $f : X \to Y$ be a proper G-morphism, and let 6.11.1 be an equivariant commutative square, with trivial action of G on Y and D. Suppose c, d are proper, c_2, d_2 quasi-finite and of finite tor-dimension, and the square

$$\begin{array}{ccc} C & \xrightarrow{c_2} & X \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle f} \\ D & \xrightarrow[d_2]{} & Y \end{array}$$

is cartesian and tor-independent (SGA 6 III 1.5). Let $M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$,

$$v \in \mathrm{Hom}(d_1^* M, d_2^* M),$$

hence a G-equivariant homomorphism

$$h^* v \in \mathrm{Hom}(c_1^* f^* M, c_2^* f^* M),$$

and G-equivariant correspondences

$$(h^* v)^! \in \mathrm{Hom}(c_1^* f^* M, c_2^! f^* M), \qquad f_*((h^* v)^!) \in \mathrm{Hom}(d_1^*(f_* f^* M), d_2^!(f_* f^* M)).$$

Then there is a commutative square of arrows in $D({}_{\Lambda[G]}Y)$, where M is regarded as an object of $D({}_{\Lambda[G]}Y)$ by the trivial action,

$$\begin{array}{ccc} d_1^* M & \xrightarrow{v^!} & d_2^! M \\ \downarrow & & \downarrow \\ d_1^* f_* f^* M & \xrightarrow[f_*((h^* v)^!)]{} & d_2^! f_* f^* M, \end{array}$$

where the vertical arrows are given by the adjunction arrow $M \to f_* f^* M$.

Put $h^* v = u$, $f^* M = L$, and consider the diagram

$$\begin{array}{ccccc} d_1^* M & \xrightarrow{v} & d_2^* M & \xrightarrow{\mathrm{Tr}_{d_2}} & d_2^! M \\ \downarrow & & \downarrow & & \downarrow \\ d_1^* f_* L & \longrightarrow & h_* c_1^* L & \xrightarrow{h_* \mathrm{Tr}_{c_2}} & h_* c_2^! L \longrightarrow d_2^! f_* L. \end{array}$$

The arrows (1) and (3) are given by the adjunction arrow $M \to f_*L$, (2) by the adjunction arrow $d_2^*M \to h_*h^*(d_2^*M) = h_*c_2^*L$, (4) by the base-change arrow, and (5) is the canonical isomorphism (SGA 4 XVIII 3.1.12.3). By definition, $v^!$ is the composite of the upper row. On the other hand, it follows from the interpretation of the direct image of a correspondence (III 3.5 b), 3.7) that $f_*u^!$ is the composite of the lower row. It remains only to prove commutativity of the squares (A) and (B). The commutativity of (A) follows immediately from the definition of the base-change arrow. As for (B), it follows from the fact that, in view of the tor-independence hypothesis, Tr_{d_2} is compatible with base change by f (SGA $4^{1/2}$ Cycle 2.3.3).

Applying 6.19, one obtains:

Corollary 6.22. Under the hypotheses of 6.21, assume that

$$f_*f^*M \in \mathrm{ob}\, D_{\mathrm{ctf}}(_{\Lambda[G]}Y),$$

and that the adjunction arrow

$$M \to f_*f^*M$$

induces an isomorphism

$$M \xrightarrow{\sim} \mathrm{RHom}_{\Lambda[G]}(\Lambda, f_*f^*M).$$

Then

$$\mathrm{Tr}_\Lambda(v^!) = \sum_{g \in G_\natural} \mathrm{Tr}_{\Lambda[G]}(f_*((f^*v)^!))(g^{-1}). \tag{6.22.1}$$

Proposition 6.23. Under the hypotheses of 6.21, suppose moreover that there is an open immersion

$$i : V \hookrightarrow Y$$

such that $(fc_1)^{-1}(V)$ is connected, that $d_1^{-1}(V) \subset d_2^{-1}(V)$, and that f induces a principal Galois covering with group G,

$$f_V : U = f^{-1}(V) \to V.$$

Suppose also that M is of the form $M = i_!E$, where $E \in \mathrm{ob}\, D(V)$, and that there is an isomorphism

$$\alpha : f_V^*E \xrightarrow{\sim} \Lambda_X \overset{\mathrm{L}}{\otimes}_\Lambda E_0$$

in $D(G, X)$, with $E_0 \in \mathrm{ob}\, D(\Lambda[G])$, perfect over Λ. Let

$$v \in \mathrm{Hom}(d_1^*M, d_2^*M)$$

be as in 6.21, and let

$$v_0 \in \mathrm{Hom}_{\Lambda[G]}(E_0, E_0)$$

be the homomorphism defined, thanks to α, by $f^*v|c_1^{-1}(U)$. Then, if $\underline{c}_{U,X}$ denotes the correspondence defined in 6.14.3, with the notation 6.20.2 one has

$$\mathrm{Tr}_\Lambda(v^!) = \sum_{g \in G_\natural} \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1})\, \mathrm{Tr}_\Lambda(gv_0). \tag{6.23.1}$$

Note that, by the central character of Tr_Λ, the right hand side does not depend on the choice of α.

By the projection formula

$$f_*f^*\Lambda_Y \overset{\mathrm{L}}{\otimes} M \xrightarrow{\sim} f_*f^*M$$

and 6.15, M satisfies the hypotheses of 6.22. On the other hand, α defines an isomorphism

$$f^*M \xrightarrow{\sim} \Lambda_{U,X} \otimes E_0$$

under which $(f^*v)^!$ is identified with the external tensor product $\underline{c}_{U,X} \otimes v_0$. By the projection formula, f_*f^*M is therefore identified with $f_*(\Lambda_{U,X}) \otimes E_0$, and $f_*((f^*v)^!)$ with the external tensor product $f_*\underline{c}_{U,X} \otimes v_0$. For $g \in G$, the Z_g-equivariant correspondence $gf_*((f^*v)^!)$ is therefore identified with

$$gf_*\underline{c}_{U,X} \otimes gv_0.$$

Since, by 6.10.4,

$$\mathrm{Tr}_{\Lambda[G]}(f_*((f^*v)^!))(g^{-1}) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_*((f^*v)^!))(e),$$

it is enough, in view of 6.22.1, to prove the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1})\,\mathrm{Tr}_\Lambda(gv_0),$$

or, equivalently by 6.10.4,

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X})(e)\,\mathrm{Tr}_\Lambda(gv_0). \tag{6.23.2}$$

Replacing G by Z_g if necessary, we may suppose that $g = e$. Returning to the definition of the traces, one sees that it is enough to establish the following lemma.

Lemma 6.23.3. (cf. SGA $4^{1/2}$ Report 4.7). Let P be a G_V-module locally a direct factor of a free module of finite type, let $P' = i_!P$, and let Q be a complex of $\Lambda[G]$-modules, perfect over Λ. Then $P \overset{\mathrm{L}}{\otimes}_\Lambda Q$, considered as an object of $D({}_{\Lambda[G]}V)$ by the diagonal action of G, is perfect over $\Lambda[G]$, and there is a commutative square

$$\begin{array}{ccc} \delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_{\Lambda[G]}(Q,Q) & \xrightarrow{(1)} & K_Y \\ \downarrow{\scriptstyle (2)} & & \| \\ \delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P' \overset{\mathrm{L}}{\otimes}_\Lambda Q, p_2^!P' \overset{\mathrm{L}}{\otimes}_\Lambda Q) & \xrightarrow[(3)]{} & K_Y, \end{array} \tag{*}$$

where $\delta : Y \to Y \times_S Y$ is the diagonal, the upper horizontal arrow is the tensor product of the arrows

$$\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \xrightarrow{6.5.3} K\Lambda[G]_{Y,\natural} \xrightarrow{x \mapsto x(e)} K_Y$$

and

$$\mathrm{RHom}_{\Lambda[G]}(Q,Q) \xrightarrow{\mathrm{Tr}_\Lambda} \Lambda,$$

the vertical arrow is deduced from the external tensor product by restriction along the diagonal

$$\Lambda[G] \to \Lambda[G] \otimes \Lambda[G],$$

and the lower horizontal arrow is the composite of 6.5.3 (defined because $P' \overset{\mathrm{L}}{\otimes}_\Lambda Q \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[G]}Y)$) and of $x \mapsto x(e)$.[20]

[20] One should be able to replace P by a perfect complex (over $\Lambda[G]$) (or even P' by an object of $D_{\mathrm{ctf}}({}_{\Lambda[G]}Y)$).

The first assertion is evident, for one reduces to $P = \Lambda[G]_V$, and $P \overset{\mathrm{L}}{\otimes}_\Lambda Q$ is then isomorphic to the tensor product of P by Q considered with the trivial action of G. For the second, first note that, by 6.5.4 and 6.5.5, one has

$$\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \xrightarrow{\sim} D_{\Lambda[G]}(i_!P) \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} i_!P \xrightarrow{\sim} i_!(D_{\Lambda[G]}P \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} P),$$

so that we are reduced to checking that the square induced by (*) on V commutes. We may therefore suppose that $V = Y$. The square (*) then reads

$$\begin{array}{ccc} \mathrm{RHom}_{\Lambda[G]}(P, K_Y \overset{\mathrm{L}}{\otimes} P) \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q,Q) & \xrightarrow{(1)} & K_Y \\ \downarrow{\scriptstyle (2)} & & \| \\ \mathrm{RHom}_{\Lambda[G]}(P \overset{\mathrm{L}}{\otimes} Q, K_Y \overset{\mathrm{L}}{\otimes} P \overset{\mathrm{L}}{\otimes} Q) & \xrightarrow[(3)]{} & K_Y, \end{array} \qquad (**)$$

where (2) is the external tensor product, (1) is given by

$$\mathrm{Tr}_{\Lambda[G]}(-)(e) \otimes \mathrm{Tr}_\Lambda,$$

(traces in the sense of no. 5), and (3) by $\mathrm{Tr}_{\Lambda[G]}(-)(e)$. By descent, one reduces to $P = \Lambda[G]_Y$, and, by the initial remark, one may then suppose that G acts trivially on Q. The assertion is then an immediate consequence of 5.12.5.

6.24

If Λ is annihilated by a power of a prime number $\ell \neq p$, the right hand side of 6.23.1 may be rewritten in the form

$$\sum_{g \in G_\natural} \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_* \underline{c}_{\ell,U,X})(g^{-1}) \, \mathrm{Tr}_\Lambda(gv_0),$$

with the notation of 6.17. Formula 6.17.1 shows that the local terms defined by Grothendieck in (XII 6.2) coincide with Verdier's local terms, at least when the endomorphism φ of G considered in loc. cit. is the identity; but, as already observed, this restriction is not serious. If Y is proper over S, then by conjugating 6.23.1 with the Lefschetz formula for $Y \to S$ (III 4.7), one obtains a formula generalizing (XII 6.3), and a fortiori the Lefschetz formula for Frobenius proved in (SGA $4^{1/2}$ Report) and Part I of the present exposé.

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In SGA $4^{1/2}$:

Finiteness Theorem: finiteness theorems in ℓ-adic cohomology;

Cycle: the cohomology class associated with a cycle;

Report: report on the trace formula.

Exposé V. J-adic Projective Systems

by J.-P. Jouanolou

Throughout this exposé, the letter A denotes a ring with unit. The present exposé is intended to develop a formalism which will be used in the following exposé for the study of ℓ-adic cohomology.

1. Generalities on abelian A-categories

Definition 1.1. An A-category (resp. an abelian A-category) is a pair $(\mathcal{C}, \rho)$ formed by an additive (resp. abelian) category $\mathcal{C}$ and a homomorphism of rings

$$\rho : A \longrightarrow \mathrm{End}(\mathrm{id}_{\mathcal{C}})$$

from A to the ring of endomorphisms of the identity functor from $\mathcal{C}$ to $\mathcal{C}$.

Given two A-categories (resp. abelian A-categories) $(\mathcal{C}, \rho)$ and $(\mathcal{C}', \rho')$, a morphism of A-categories (resp. of abelian A-categories) from $(\mathcal{C}, \rho)$ to $(\mathcal{C}', \rho')$ is an additive (resp. exact) functor $F : \mathcal{C} \to \mathcal{C}'$ such that, for every object M of $\mathcal{C}$ and every element a of A, the equality below is satisfied:

$$F(\rho(a)_M) = \rho'(a)_{F(M)}.$$

Example 1.1.1

Let $\mathcal{C}$ be an additive (resp. abelian) category. Recall that the category of left $(A, \mathcal{C})$-modules, denoted $(A, \mathcal{C})$-mod, is the category described as follows:

(i) Its objects are pairs (X, α) consisting of an object X of $\mathcal{C}$ and a homomorphism of rings

$$\alpha : A \longrightarrow \mathrm{Hom}_{\mathcal{C}}(X, X).$$

(ii) A morphism $(X, \alpha) \to (Y, \beta)$ is a $\mathcal{C}$-morphism $u : X \to Y$ such that, for every element a of A, the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ {\scriptstyle \alpha(a)}\downarrow & & \downarrow{\scriptstyle \beta(a)} \\ X & \xrightarrow[u]{} & Y. \end{array}$$

The category $(A, \mathcal{C})$-mod is evidently an $A^\natural$-category (resp. an abelian $A^\natural$-category), where $A^\natural$ is the centre of A. Moreover, it has the following properties:

(a) The functor $(A, \mathcal{C})$-mod $\to \mathcal{C}$ obtained by forgetting the A-module structure commutes with inductive and projective limits.

(b) Let I be a category. In order that the I-indexed inductive (resp. projective) limits of $(A, \mathcal{C})$-mod be representable, it is enough that this be so for the I-indexed inductive (resp. projective) limits of $\mathcal{C}$.

1.2. External tensor product

Let $(\mathcal{C}, \rho)$ be an abelian A-category. If M is a right A-module and X an object of $\mathcal{C}$, the external tensor product of X by M, denoted $M \otimes_A X$, is the covariant functor from $\mathcal{C}$ to the category of sets, defined on objects Y of $\mathcal{C}$ by the equality below and in the evident way on arrows:

$$(M \otimes_A X)(Y) = \mathrm{Hom}_A(M, \mathrm{Hom}_{\mathcal{C}}(X, Y)),$$

where $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is regarded as a right A-module via the action of A on X. Interpreting $M \otimes_A X$ and Y as objects of the category $\widehat{\mathcal{C}} = \mathrm{Hom}(\mathcal{C}, \mathrm{Ens})^\circ$, the preceding equality takes the more familiar form

$$\mathrm{Hom}_{\widehat{\mathcal{C}}}(M \otimes_A X, Y) = \underline{\mathrm{Hom}}_A(M, \mathrm{Hom}_{\mathcal{C}}(X, Y)).$$

If the functor $M \otimes_A X$ is representable, a representative will be denoted and named in the same way.

If $u : M \to M'$ and $v : X \to X'$ are two morphisms, respectively in (A-mod) and in $\mathcal{C}$, one defines in the usual way a $\mathcal{C}$-morphism

$$u \otimes_A v : M \otimes_A X \longrightarrow M' \otimes_A X',$$

which allows us to speak of the external tensor-product bifunctor, covariant in both variables. Its properties are summarized in the following proposition.

Proposition 1.2.1. The tensor-product bifunctor is additive and commutes with inductive limits.

The proof, classical in the case of modules over a ring, is immediate.

Proposition 1.2.2 (Representability criteria).

(a) For every object X of $\mathcal{C}$, $A \otimes_A X$ is represented by X.

(b) For every finitely presented A-module M and every object X of $\mathcal{C}$, the functor $M \otimes_A X$ is representable.

(c) If the category $\mathcal{C}$ has arbitrary direct sums, the functor $M \otimes_A X$ is representable for every pair (M, X).

Proof. Assertion (a) is clear. The other two assertions then follow from the lemma below, whose proof is immediate.

Lemma 1.2.3. Let $M \to N \to P \to 0$ be an exact sequence of A-modules and let X be an object of $\mathcal{C}$. If the functors $M \otimes_A X$ and $N \otimes_A X$ are represented by objects Y and Z of $\mathcal{C}$, then the functor $P \otimes_A X$ is represented by the cokernel of the morphism $Y \to Z$ corresponding to $\alpha \otimes_A \mathrm{id}_X$.

Remark 1.2.4

Let $\mathfrak{G}$ be an ideal of finite type of A. For every object X of $\mathcal{C}$, the functor $(A/\mathfrak{G}) \otimes_A X$ is representable, and the arrow

$$p_X : X \longrightarrow (A/\mathfrak{G}) \otimes_A X$$

coming from the canonical epimorphism from A onto $A/\mathfrak{G}$ and from part (a) of 1.2.2 is an epimorphism. The same statement remains valid when the ideal $\mathfrak{G}$ is no longer assumed to be of finite type, but $\mathcal{C}$ has arbitrary direct sums. Put $\mathfrak{G}X = \mathrm{Ker}(p_X)$. When the category has filtered inductive limits, it is clear that

$$\mathfrak{G}X = \sum_{a \in \mathfrak{G}} aX.$$

If $u : X \to Y$ is an epimorphism, u maps $\mathfrak{G}X$ epimorphically onto $\mathfrak{G}Y$. Finally, for an object X to be annihilated by every element of $\mathfrak{G}$, it is necessary and sufficient that $\mathfrak{G}X = 0$. Then every morphism from an object Z to X factors, by means of p_Z, through a morphism from $(A/\mathfrak{G}) \otimes_A Z$ to X.

2. The Mittag-Leffler-Artin-Rees condition

Throughout this paragraph, the letter $\mathcal{C}$ denotes an abelian category and we put

$$P = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}),$$

the category of projective systems indexed by the ordered set $\mathbb{N}$ of positive integers and with values in $\mathcal{C}$.

2.1. The Mittag-Leffler-Artin-Rees condition

Let $X = (X_n, u_n)_{n \geq 0}$ be an object of the category P. If r denotes an integer ≥ 0, we agree to write $X[r]$ for the projective system $(X_{n+r}, u_{n+r})_{n \geq 0}$. A morphism $h : X \to Y$ of P defines in the evident way a morphism denoted $h[r]$ from $X[r]$ to $Y[r]$, functorially. If r and s are two integers satisfying $s \geq r \geq 0$, the transition morphisms $X_{n+s} \to X_{n+r}$ $(n \geq 0)$ define a morphism

$$w_{sr} : X[s] \longrightarrow X[r],$$

and it is clear that, if t is a third integer with $t \geq s \geq r$, the equality below is satisfied:

$$w_{ts} \circ w_{sr} = w_{tr}.$$

In other words, we have thus associated to X an object $\overline{X}$ of the category $\mathrm{Hom}(\mathbb{N}^\circ, P)$. Moreover, if $h : X \to Y$ is a morphism of P, there corresponds to it in the evident way a morphism $\overline{h}$ from $\overline{X}$ to $\overline{Y}$, functorially. Finally, if

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

is an exact sequence of P, the corresponding sequence below in $\overline{P} = \mathrm{Hom}(\mathbb{N}^\circ, P)$ is exact:

$$0 \longrightarrow \overline{X} \xrightarrow{\overline{u}} \overline{Y} \xrightarrow{\overline{v}} \overline{Z} \longrightarrow 0.$$

Definition 2.1.1. Let an object of the category P be given. It is said to satisfy the Mittag-Leffler-Artin-Rees condition (abbreviated MLAR) if there exists an integer r such that, for every integer $s \geq r$, the equality below is satisfied:

$$\mathrm{Im}(X[s] \xrightarrow{w_{s0}} X) = \mathrm{Im}(X[r] \xrightarrow{w_{r0}} X).$$

Equivalently, for every integer $n \geq 0$, the images of X_{n+r} and X_{n+s} in X_n by the "transition morphisms" are the same. Finally, a third interpretation is that the associated projective system $\overline{X}$ satisfies the Mittag-Leffler condition (abbreviated ML) (EGA 0_{III} 13.1.2). The following proposition is deduced from this by (EGA 0_{III} 13.2.1).

Proposition 2.1.2. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence of P.

(i) If Y satisfies (MLAR), then Z satisfies (MLAR).

(ii) If X and Z satisfy (MLAR), then Y satisfies (MLAR).

2.2. AR-null projective systems

Definition 2.2.1. Let X be an object of the category P. It is called AR-null if there exists an integer $r \geq 0$ such that the canonical morphism $X[r] \to X$ is zero.

It is clear in particular that an AR-null projective system satisfies the condition (MLAR).

Proposition 2.2.2. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence of P. The following assertions are equivalent:

(i) Y is AR-null;

(ii) X and Z are AR-null.

In other words, the full subcategory P_0 of P, whose objects are the AR-null projective systems, is thick in P.

Proof. It is clear that (i) implies (ii). Conversely, let r and t be integers satisfying the condition of Definition 2.2.1 for X and Z respectively. One sees immediately that the integer $r+t$ satisfies the conditions of Definition 2.2.1 for Y.

2.3. Reminders on the calculus of fractions and applications

Let $\mathcal{E}$ be a category and S a subset of $\mathrm{Fl}(\mathcal{E})$. A category of fractions of $\mathcal{E}$ with respect to S is a pair $(p, \mathcal{E}(S^{-1}))$ consisting of a category $\mathcal{E}(S^{-1})$ and a functor $p : \mathcal{E} \to \mathcal{E}(S^{-1})$ satisfying the following universal property: for a functor u from $\mathcal{E}$ to a category $\mathcal{F}$ to factor through p, it is necessary and sufficient that it make all arrows belonging to S invertible; the factorization is then unique. The uniqueness of such a pair up to isomorphism is clear. We refer the reader to Exposé I of the Séminaire Homotopique de Strasbourg for the proof of existence, and also for most of the results stated without proof in what follows in this paragraph. See also P. Gabriel and M. Zisman, *Calculus of fractions and homotopy theory*, Erg. der Math., 35, Springer-Verlag, 1967.

Certain properties of the multiplicative part S ensure a simple description of the category $\mathcal{E}(S^{-1})$.

Definition 2.3.1. A subset S of $\mathrm{Fl}(\mathcal{E})$ is said to admit a right calculus of fractions if the following conditions hold:

(a) The identity arrows of $\mathcal{E}$ belong to S.

(b) If $u : X \to Y$ and $v : Y \to Z$ belong to S, the composite $v \circ u$ belongs to S.

(c) Every diagram of type (D_1) below, in which t belongs to S, can be completed to a diagram of type (D_2), in which s belongs to S:

$$(D_1) \qquad X \xrightarrow{u} Y \xleftarrow{t} Y'$$

$$(D_2) \qquad \begin{array}{ccc} X' & \xrightarrow{u'} & Y' \\ {\scriptstyle s}\downarrow & & \downarrow{\scriptstyle t} \\ X & \xrightarrow[u]{} & Y. \end{array}$$

(d) For every commutative diagram

$$X \underset{g}{\overset{f}{\rightrightarrows}} Y \xrightarrow{s} Z,$$

in which s belongs to S, there exists a commutative diagram

$$T \xrightarrow[t]{} X \underset{g}{\overset{f}{\rightrightarrows}} Y,$$

in which t belongs to S.

Dually one defines the following notion.

Definition 2.3.2. A subset S of $\mathrm{Fl}(\mathcal{E})$ is said to admit a left calculus of fractions if the following conditions hold:

(a') The identity arrows of $\mathcal{E}$ belong to S.

(b') The composite of two arrows belonging to S belongs to S.

(c') Every diagram of type $(D_1)'$ below, in which s belongs to S, can be completed to a diagram of type $(D_2)'$, in which t belongs to S:

$$(D_1)' \qquad \begin{array}{ccc} X' & \xrightarrow{u'} & Y' \\ {\scriptstyle s}\downarrow & & \\ X & & \end{array} \qquad (D_2)' \qquad \begin{array}{ccc} X' & \xrightarrow{u'} & Y' \\ {\scriptstyle s}\downarrow & & \downarrow{\scriptstyle t} \\ X & \xrightarrow[u]{} & Y. \end{array}$$

(d') For every commutative diagram Δ of the form below, in which s belongs to S, there exists a morphism t with source Y and belonging to S such that $t \circ f = t \circ g$:

$$(\Delta) \qquad X' \xrightarrow{s} X \underset{g}{\overset{f}{\rightrightarrows}} Y.$$

2.3.3

Suppose that S is a subset of $\mathrm{Fl}(\mathcal{E})$ admitting a right calculus of fractions. It follows from part (d) of 2.3.1 that, for every object X of $\mathcal{E}$, the category $(S/X)^\circ$, opposite to the category of arrows over X belonging to S, is filtering (SGA 4 I 2.7). Consider then the category $\mathcal{E}_S$ described as follows:

- the objects of $\mathcal{E}_S$ are the objects of $\mathcal{E}$;
- if X and Y are two objects of $\mathcal{E}$,

$$\mathrm{Hom}_{\mathcal{E}_S}(X, Y) = \varinjlim_{(S/X)^\circ} \mathrm{Hom}_{\mathcal{E}}(\mathrm{source}(\cdot), Y).$$

If P_S denotes the evident functor from $\mathcal{E}$ to $\mathcal{E}_S$, the pair $(P_S, \mathcal{E}_S)$ is a solution of the universal problem described at the beginning of the paragraph. In this case we agree to write $\mathcal{E}(S^{-1})$ for this particular solution.

2.3.4

Let now S be a subset of $\mathrm{Fl}(\mathcal{E})$ admitting a left calculus of fractions. Then the functor $P_S : \mathcal{E} \to \mathcal{E}(S^{-1})$ commutes with finite inductive limits. In particular, if $\mathcal{E}$ is additive, $\mathcal{E}(S^{-1})$ is additive and P_S is an additive functor.

2.3.5

If $\mathcal{E}$ is an abelian category and if S is a subset of $\mathrm{Fl}(\mathcal{E})$ admitting both a right calculus of fractions and a left calculus of fractions, the category $\mathcal{E}(S^{-1})$ is abelian and the functor P_S is exact. Moreover, the pair $(P_S, \mathcal{E}(S^{-1}))$ is a quotient of $\mathcal{E}$ by the thick abelian subcategory $P_S^{-1}(0)$.

2.4. The category $P_{AR} = \mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$

Returning to the notation of 2.1, let AR be the set of canonical arrows (cf. 2.1)

$$v_{rX} : X[r] \longrightarrow X,$$

where X runs through the set of objects of P and r through the set of integers ≥ 0.

Proposition 2.4.1. The subset AR of $\mathrm{Fl}(P)$ admits both a right calculus of fractions and a left calculus of fractions.

Proof. The parts (a) $\Rightarrow$ (a') and (b) $\Rightarrow$ (b') of 2.3.1 and 2.3.2 are clear.

We prove (c). Let (D_1) be the commutative diagram below, in which v_{rY} denotes the canonical arrow from $Y[r]$ to Y:

$$\begin{array}{ccc} & & Y[r] \\ & & \downarrow{\scriptstyle v_{rY}} \\ X & \longrightarrow & Y. \end{array}$$

The diagram (D_2) below is commutative, and v_{rX}, the canonical morphism from $X[r]$ to X, belongs to AR, whence the assertion:

$$\begin{array}{ccc} X[r] & \xrightarrow{u[r]} & Y[r] \\ {\scriptstyle v_{rX}}\downarrow & & \downarrow{\scriptstyle v_{rY}} \\ X & \xrightarrow[u]{} & Y. \end{array}$$

We prove (d). Let a commutative diagram be given:

$$X \underset{g}{\overset{f}{\rightrightarrows}} Z[r] \xrightarrow{v_{rZ}} Z.$$

Applying translation by r, one obtains a commutative diagram

$$X[r] \underset{g[r]}{\overset{f[r]}{\rightrightarrows}} Z[2r] \xrightarrow{v_{rZ}[r]} Z[r].$$

It is clear that the following diagram is commutative, so that $f \circ v_{rX} = g \circ v_{rX}$:

$$\begin{array}{ccc} X[r] & \underset{g[r]}{\overset{f[r]}{\rightrightarrows}} & Z[2r] \\ {\scriptstyle v_{rX}}\downarrow & & \downarrow{\scriptstyle v_{rZ}[r]} \\ X & \underset{g}{\overset{f}{\rightrightarrows}} & Z[r]. \end{array}$$

We prove (c'). First observe that, if Y denotes an object of P and r a positive integer, there exists an object T of P such that $Y = T[r]$ and such that, for every object X of P, the homomorphism given by translation by r

$$\mathrm{Hom}_P(X, T) \longrightarrow \mathrm{Hom}_P(X[r], Y)$$

is a bijection. The projective system (T_n, w_n) defined as follows plainly works. If $Y = (Y_n, u_n)_{n\geq 0}$, then T_n is the zero object of $\mathcal{C}$ for $n \geq r-1$ and is Y_{n-r} otherwise, and

$$w_n = 0 \quad \text{for } n \geq r-1, \qquad w_n = u_{n-r}.$$

This said, suppose a diagram of the following type is given, and show that it can be completed as indicated in the statement of (c'):

$$\begin{array}{ccc} X[r] & \xrightarrow{u'} & Y' \\ {\scriptstyle v_{rX}}\downarrow & & \\ X. & & \end{array}$$

For this, let Y be a projective system such that $Y' = Y[r]$ and such that the canonical map

$$\mathrm{Hom}_P(X, Y) \longrightarrow \mathrm{Hom}_P(X[r], Y')$$

is a bijection. Then it is clear that the following diagram is commutative, where u denotes the unique morphism from X to Y such that $u' = u[r]$:

$$\begin{array}{ccc} X[r] & \xrightarrow{u'} & Y' = Y[r] \\ {\scriptstyle v_{rX}}\downarrow & & \downarrow{\scriptstyle v_{rY}} \\ X & \xrightarrow[u]{} & Y. \end{array}$$

We prove (d'). Consider the commutative diagram below, and show that there exists a morphism w from Y such that $w \circ f = w \circ g$:

$$X[r] \xrightarrow{v_{rX}} X \underset{g}{\overset{f}{\rightrightarrows}} Y.$$

It is first of all clear that the following diagram is commutative:

$$\begin{array}{ccc} X[r] & \underset{g[r]}{\overset{f[r]}{\rightrightarrows}} & Y[r] \\ {\scriptstyle v_{rX}}\downarrow & & \downarrow{\scriptstyle v_{rY}} \\ X & \underset{g}{\overset{f}{\rightrightarrows}} & Y. \end{array}$$

In particular $v_{rY} \circ f[r] = v_{rY} \circ g[r]$. Let now T be the system associated to Y and r as at the beginning of the proof of (c'), so that in particular $Y = T[r]$. The translates by r of the morphisms $v_{rT} \circ f$ and $v_{rT} \circ g$ are respectively equal to $v_{rY} \circ f$ and $v_{rY} \circ g[r]$, hence are equal. It follows from the fact that translation by r is injective on morphisms from X to T that the equality below holds, whence the assertion:

$$v_{rT} \circ f = v_{rT} \circ g.$$

Definition 2.4.2. Let $\mathcal{C}$ be an abelian category. The category of projective systems of $\mathcal{C}$ up to translation, denoted $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$, is the category described as follows:

(i) The objects of $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$ are the objects of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$.

(ii) If X and Y are two objects of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, a morphism from X to Y in the category $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$ is an element of the set

$$\mathrm{Hom}_{AR}(X, Y) = \varinjlim_r \mathrm{Hom}(X[r], Y),$$

where, for every pair (r, s) of integers with $s \geq r$, the map

$$\mathrm{Hom}(X[r], Y) \longrightarrow \mathrm{Hom}(X[s], Y)$$

comes from the canonical morphism

$$X[s] \longrightarrow X[r].$$

The composition of morphisms is defined in the evident way.

In what follows, the category $\mathcal{C}$ being fixed, we put, to simplify notation,

$$P = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}), \qquad P_{AR} = \mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C}).$$

2.4.3

We have seen (2.4.1) that the subset AR of $\mathrm{Fl}(P)$ admits a right calculus of fractions. This allows us (2.3.3) to consider the category

$$P'_{AR} = P(AR^{-1}),$$

which, recall, has the same objects as P. Let X and Y be two objects of P, and denote by $\mathrm{Hom}'_{AR}(X, Y)$ the set of isomorphisms from X to Y in P'_{AR}. An element of $\mathrm{Hom}_{AR}(X, Y)$ defines in the evident way an element of $\mathrm{Hom}'_{AR}(X, Y)$. More precisely, one thus defines a functor

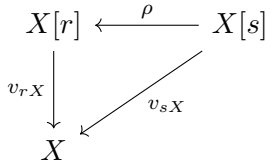
$$m : P_{AR} \longrightarrow P'_{AR}.$$

Proposition 2.4.3. The functor m is an isomorphism of categories.

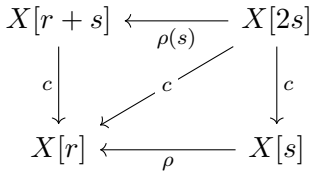
Proof. We must see that, for every pair (X, Y) of objects of P, the corresponding map

$$m_{XY} : \mathrm{Hom}_{AR}(X, Y) \longrightarrow \mathrm{Hom}'_{AR}(X, Y)$$

is a bijection. It is plainly a surjection. For injectivity, one is immediately reduced to seeing that, for every commutative diagram of the type below, there exists an integer t greater than or equal to r and s such that the composite of ρ and the canonical morphism from $X[t]$ to $X[s]$ is the canonical morphism from $X[t]$ to $X[r]$:



But this follows from commutativity of the diagram below, in which the letter c denotes all canonical morphisms involved:



Proposition 2.4.4.

(i) The category P_{AR} is abelian.

(ii) The functor $p_{AR} : P \to P_{AR}$ is exact.

(iii) For an object X of P to be AR-null, it is necessary and sufficient that $p_{AR}(X) = 0$.

(iv) The pair (p_{AR}, P_{AR}) is a quotient of the abelian category P by the thick abelian subcategory of AR-null objects.

Proof. If (iii) is admitted, the other assertions are consequences of (2.3.5). It follows from (2.3.4) that, for an object X of P to be AR-null, it is necessary and sufficient that $p_{AR}(\mathrm{id}_X) = 0$, whence (iii).

Corollary 2.4.5. Let $u : X \to Y$ be a morphism in the category P. The following assertions are equivalent:

(i) $p_{AR}(u)$ is an isomorphism.

(ii) The kernel and cokernel of u are AR-null.

A morphism $u : X \to Y$ in the category P of projective systems, satisfying the equivalent conditions (i) and (ii), will be called an AR-isomorphism.

Definition 2.4.6. Two objects X and Y of the category P are called AR-isomorphic if $p_{AR}(X)$ and $p_{AR}(Y)$ are isomorphic in the category P_{AR}. If X and Y are two AR-isomorphic objects of the category P, there need not exist in general an AR-isomorphism u from X to Y.

2.5. Characterization of projective systems satisfying the Mittag-Leffler-Artin-Rees condition

Proposition 2.5.1. The following assertions for an object X of P are equivalent:

(i) X satisfies (MLAR) (2.1.1).

(ii) There exists a strict projective system Y and an AR-isomorphism $u : Y \to X$ (2.4.5).

(iii) There exists a strict projective system Y such that X and Y are isomorphic in the category P_{AR}.

(iv) X satisfies the Mittag-Leffler condition and, denoting by X' the projective system of universal images of X, the projective system X/X' is AR-null.

Proof. We shall prove the chain of implications

$$(i) \Rightarrow (iv) \Rightarrow (iii) \Rightarrow (ii) \Rightarrow (i).$$

$(i) \Rightarrow (iv)$. There exists an integer m such that the canonical morphism w_{mX} from $X[m]$ to X factors through the canonical inclusion i of X' in X. It follows that $p_{AR}(i)$ is a direct epimorphism and therefore, since p_{AR} is exact, that $p_{AR}(X/X') = 0$. We conclude by (2.4.4) (iii).

$(iv) \Rightarrow (iii)$. Condition (iv) means, with the preceding notation, that $p_{AR}(i)$ is an epimorphism. But because p_{AR} is exact, it is also a monomorphism, hence an isomorphism. For Y one takes the projective system X'.

$(iii) \Rightarrow (ii)$. The P_{AR}-isomorphism from Y to X whose existence is assumed is the image under p_{AR} of a P-morphism $Y[r] \to X$, where r denotes a positive or zero integer. The conclusion follows from the fact that $Y[r]$ is strict.

$(ii) \Rightarrow (i)$. We may suppose Y contained in X and u the canonical injection. Let T be the cokernel of u. The hypothesis means that T is AR-null, hence there exists an integer r such that, for every $s \geq r$, the canonical morphism from $T[s]$ to T is zero. It follows that, for $s \geq r$, the image of $X[s]$ in X is contained in Y; since it also plainly contains the image, equal to Y, of $Y[s]$ in Y, it is equal to Y. Thus Y is the image of $X[s]$ for s sufficiently large, whence X satisfies (MLAR), and Y is the projective system of universal images of X.

Corollary 2.5.2. Let $u : Y \to X$ be an AR-isomorphism, with Y strict. Under these conditions, X satisfies (MLAR) and $u(Y)$ is the projective system of universal images of X.

3. J-adic and AR-J-adic projective systems

Let A be a commutative ring with unit, let J be an ideal of A, and let $(\mathcal{C}, \rho)$ be an abelian A-category. We suppose either that J is of finite type, or that $\mathcal{C}$ has infinite direct sums. The notation is the same as in the preceding paragraph. In particular, the symbols P and P_{AR} denote respectively the categories $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ and $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$.

3.1. J-adic projective systems

Definition 3.1.1. Let X be an object of P. It is called J-adic if it satisfies the following conditions:

(i) for every integer $n \geq 0$, $J^{n+1}X_n = 0$;

(ii) for every pair (m, n) of integers with $m \geq n \geq 0$, the morphism

$$A/J^{n+1} \otimes_A X_m \longrightarrow X_n$$

deduced from the transition morphism $X_m \to X_n$ is an isomorphism.

We shall write J-ad($\mathcal{C}$) for the full subcategory of P generated by the J-adic projective systems.

Remark 3.1.2

Let I and J be two ideals of A and suppose either that I and J are of finite type, or that $\mathcal{C}$ has infinite direct sums. Then one can show that, if I and J define the same topology on A, the categories I-ad($\mathcal{C}$) and J-ad($\mathcal{C}$) are equivalent.

Proposition 3.1.3 (Stability under exact sequences). Let

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

be an exact sequence of P.

(i) If X is strict and Y is J-adic, then Z is J-adic.

(ii) If Y is strict and Z is J-adic, then X is strict.

(iii) If X and Z are J-adic, then Y is J-adic.

Proof. If $m \geq n$, it is clear that the diagram (T) below, in which the vertical arrows denote the transition morphisms, is commutative and exact:

$$\begin{array}{ccccccccc} & & A/J^{n+1} \otimes_A X_m & \xrightarrow{\mathrm{id} \otimes u_m} & A/J^{n+1} \otimes_A Y_m & \xrightarrow{\mathrm{id} \otimes v_m} & A/J^{n+1} \otimes_A Z_m & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f_{nm}} & & \downarrow{\scriptstyle g_{nm}} & & \downarrow{\scriptstyle h_{nm}} & & \\ 0 & \longrightarrow & X_n & \longrightarrow & Y_n & \longrightarrow & Z_n & \longrightarrow & 0. \end{array}$$

In case (i), f_{nm} is an epimorphism and g_{nm} an isomorphism. In case (ii), g_{nm} is an epimorphism and h_{nm} an isomorphism. In case (iii), f_{nm} and h_{nm} are isomorphisms. In all three cases, the snake lemma applied to the diagram (T) gives the conclusion.

3.2. AR-J-adic projective systems

Proposition 3.2.1. The following assertions, for an object X of P such that $J^{n+1}X_n = 0$ for every $n \geq 0$, are equivalent:

(i) There exists a J-adic projective system Y and an AR-isomorphism $u : Y \to X$ (2.4.5).

(ii) There exists a J-adic projective system Y, isomorphic to X in the category P_{AR}.

Proof. We have to show that (ii) implies (i). A P_{AR}-isomorphism from Y to X is represented, for some integer r, by an AR-isomorphism $v : Y[r] \to X$. Since $J^{n+1}X_n = 0$ for every $n \geq 0$ and Y is J-adic, v is the composite of a P-morphism $u : Y \to X$ and the canonical morphism from $Y[r]$ to Y. It is then clear that u is an AR-isomorphism, whence the assertion.

Definition 3.2.2. A projective system indexed by $\mathbb{N}$ with values in $\mathcal{C}$ is called AR-J-adic if:

(a) $J^{n+1}X_n = 0$ for every $n \geq 0$;

(b) it satisfies the equivalent conditions (i) and (ii) of Proposition 3.2.1.

Proposition 3.2.3. Let X be a projective system indexed by $\mathbb{N}$ with values in $\mathcal{C}$. Suppose that $J^{n+1}X_n = 0$ for every $n \geq 0$. Then, in order that X be AR-J-adic, it is necessary and sufficient that it satisfy (MLAR) and that, denoting by X' its projective system of universal images, there exist an integer $r \geq 0$ such that, for every pair (m, n) of integers with $m \geq n + r$, the "transition morphism" below is an isomorphism:

$$X'_m/J^{n+1}X'_m \longrightarrow X'_{n+r}/J^{n+1}X'_{n+r}.$$

Proof.

(a) The conditions of the statement are sufficient. If X satisfies (MLAR), it is AR-isomorphic to its projective system of universal images X'. Moreover, the hypothesis made on X' means that the projective system

$$\left(X'_{n+r}/J^{n+1}X'_{n+r}\right)_{n\geq 0},$$

with the evident transition morphisms, is J-adic. It is clear that this projective system is AR-isomorphic (2.4.6) to X'.

(b) The conditions of the statement are necessary. If X is AR-J-adic, it is AR-isomorphic to a strict projective system, hence satisfies (MLAR). The projective system X' of universal images of X is AR-isomorphic to X, hence is AR-J-adic. To prove the second point, it remains to see that, if X is a strict AR-J-adic projective system, there exists an integer r such that the projective system $\ell_r(X)$ defined by

$$\ell_r(X)_n = X_{n+r}/J^{n+1}X_{n+r}$$

is J-adic. For this, let $u : Y \to X$ be an AR-isomorphism, with Y J-adic. Since the kernel T of u is AR-null, let r be an integer such that the canonical morphism $T[r] \to T$ is zero. It is immediate that the canonical morphism

$$\ell_r(u) : \ell_r(Y) \longrightarrow \ell_r(X)$$

is an isomorphism, whence the assertion.

Proposition 3.2.4 (Stability under exact sequences). Let

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

be an exact sequence of P, the projective systems X, Y, and Z being such that

$$J^{n+1}X_n = J^{n+1}Y_n = J^{n+1}Z_n = 0 \qquad (n \geq 0).$$

(i) If X satisfies (MLAR) and Y is AR-J-adic, then Z is AR-J-adic.

(ii) If Y satisfies (MLAR) and Z is AR-J-adic, then X satisfies (MLAR).

Proof. We use freely, for every projective system X and every integer r, the notation $\ell_r(X)$ already made explicit.

Proof of (i). If X' and Y' denote the projective systems of universal images of X and Y, then u maps X' into Y', whence the evident exact commutative diagram below, in which the arrows are evident:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X' & \longrightarrow & Y' & \longrightarrow & \widetilde{Z'} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & X & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & 0. \end{array}$$

Since the category P_{AR} is abelian and the functor $P \to P_{AR}$ is exact, it follows from this diagram, and from the fact that the inclusions of X' and Y' into X and Y respectively are AR-isomorphisms (2.5.1), that the morphism $\widetilde{Z} \to Z$ is an AR-isomorphism.

This allows us to reduce to the case where X and Y, and therefore Z, are strict. Then there exists an integer r such that the projective system $\ell_r(Y)$ is J-adic. The functor ℓ_r is plainly right exact, whence an exact sequence

$$\ell_r(X) \xrightarrow{\ell_r(u)} \ell_r(Y) \xrightarrow{\ell_r(v)} \ell_r(Z) \longrightarrow 0.$$

After replacing $\ell_r(X)$ by its image under $\ell_r(u)$, we are under the hypotheses for applying (3.1.3 (i)), which shows that $\ell_r(Z)$ is J-adic, whence the assertion.

(ii) Let Y' and Z' be the projective systems of universal images of Y and Z. Consider the evident exact commutative diagram below, in which the two vertical arrows on the right are the canonical inclusions:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \widetilde{X} & \longrightarrow & Y' & \longrightarrow & Z' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & X & \xrightarrow[u]{} & Y & \xrightarrow[v]{} & Z & \longrightarrow & 0. \end{array}$$

It shows that the morphism $\widetilde{X} \to X$ is an AR-isomorphism, which allows us to reduce to the case where Y and Z are strict. In this case, since Z is strict and AR-J-adic, it follows from 3.2.3 that there exists an integer r such that $\ell_r(Z)$ is J-adic. Considering the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \widetilde{\widetilde{X}} & \longrightarrow & \ell_r(Y) & \longrightarrow & \ell_r(Z) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & X & \xrightarrow[u]{} & Y & \xrightarrow[v]{} & Z & \longrightarrow & 0, \end{array}$$

in which, as before, $\widetilde{\widetilde{X}} \to X$ is an AR-isomorphism, allows us to reduce to the case where Y is strict and Z is J-adic. Then (3.1.3 (ii)) gives the conclusion.

4. Filtrations and graduations

4.1. Generalities

4.1.1

Let $\mathcal{C}$ be an abelian category and let $X = (X_n)_{n\in\mathbb{Z}}$ be a projective system indexed by $\mathbb{Z}$ with values in $\mathcal{C}$. Suppose that

$$X_i = 0 \quad (i < 0).$$

For every element p of $\mathbb{Z}$, define a subobject F^pX of X in the category $P_1 = \mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{C})$ by setting

$$(F^pX)_n = F^pX_n = \begin{cases} \mathrm{Ker}(X_n \to X_{p-1}), & n \geq p-1, \\ 0, & n < p-1, \end{cases}$$

with the transition morphisms defined in the evident way. Thus one obtains a decreasing filtration $(F^pX)_{p\in\mathbb{Z}}$ of X satisfying

$$F^pX = X \quad \text{if } p \leq 0, \tag{a}$$

$$(F^pX)_{p-1} = 0 \quad \text{for all } p. \tag{b}$$

Moreover, the snake lemma applied to the exact commutative diagram (D) below proves that, as soon as $n \geq p$, the canonical morphism

$$F^pX_n/F^{p+1}X_n \longrightarrow \mathrm{Im}(X_n \to X_p) \cap \mathrm{Ker}(X_p \to X_{p-1})$$

is an isomorphism:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Ker}(X_n \to X_p) & \longrightarrow & X_n & \longrightarrow & \mathrm{Im}(X_n \to X_p) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & 0 & \longrightarrow & X_{p-1} & \xrightarrow{\sim} & X_{p-1} & \longrightarrow & 0. \end{array}$$

If now X is a projective system indexed by $\mathbb{N}$ with values in $\mathcal{C}$, we extend it to a projective system indexed by $\mathbb{Z}$ by putting $X_n = 0$ for $n < 0$, with the evident transition morphisms, which allows the preceding constructions to be applied to it. This allows us also to consider the category

$$P = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$$

as a full subcategory of P_1.

Definition 4.1.2. Let X be a projective system indexed by $\mathbb{Z}$. We say that X is essentially constant if there exists an integer p such that, for every pair (r, s) of integers greater than or equal to p and satisfying $s \geq r$, the transition morphism

$$X_s \longrightarrow X_r$$

is an isomorphism. It is clear that it is enough to require this when $r = p$.

Proposition 4.1.3 (The property (MLAR) and associated gradeds). The following assertions for an object X of the category P are equivalent:

(i) X satisfies (MLAR).

(ii) The projective system with values in the category of objects of $\mathcal{C}$ graded by $\mathbb{N}$,

$$r \longmapsto (\mathrm{gr}^{\alpha}(X_{\alpha+r}))_{\alpha\in\mathbb{N}},$$

with the evident transition morphisms, is essentially constant.

(iii) The projective system with values in the category of objects of $\mathcal{C}$ graded by $\mathbb{N}$,

$$n \longmapsto (\mathrm{gr}^{p}(X_n))_{p\in\mathbb{N}},$$

with the evident transition morphisms, satisfies (MLAR).

Proof. $(i) \Rightarrow (ii)$ follows from the expression of $F^pX_n/F^{p+1}X_n$ given in 4.1.1.

$(ii) \Rightarrow (iii)$. Let r be an integer such that, for every $s > r$ and every p, the canonical morphism

$$\mathrm{gr}^{p}(X_{p+s}) \longrightarrow \mathrm{gr}^{p}(X_{p+r})$$

is an isomorphism. We shall see that, for every pair (m, n) of integers with $m \geq n + r$, the equality below is satisfied for every integer p:

$$\mathrm{Im}(\mathrm{gr}^{p}(X_m) \to \mathrm{gr}^{p}(X_n)) = \mathrm{Im}(\mathrm{gr}^{p}(X_{n+r}) \to \mathrm{gr}^{p}(X_n)).$$

Indeed:

(a) If $p \geq n + 1$, then $\mathrm{gr}^{p}(X_n) = 0$,

(b) if $p \leq n$, the chain of inequalities $m \geq n + r \geq p + r$ shows that the canonical morphism

$$\mathrm{gr}^{p}(X_m) \longrightarrow \mathrm{gr}^{p}(X_{n+r})$$

is an isomorphism, whence the assertion.

$(iii) \Rightarrow (i)$. Denote by Y the projective system of (iii), and let r be an integer such that

$$\mathrm{Im}(Y[s] \to Y) = \mathrm{Im}(Y[r] \to Y) \qquad (s \geq r).$$

If n and ℓ are two integers, with $\ell \geq n$, the filtration induced on $\mathrm{Im}(X_\ell \to X_n)$ by that of X_n is also the "quotient" filtration of that of X_ℓ, and consequently, endowing $\mathrm{Im}(X_\ell \to X_n)$ with this filtration, the equality below is satisfied:

$$\mathrm{gr}^{p}(\mathrm{Im}(X_\ell \to X_n)) = \mathrm{Im}(\mathrm{gr}^{p}(X_\ell) \to \mathrm{gr}^{p}(X_n)).$$

In particular, condition (iii) is expressed by saying that, as soon as $\ell \geq n + r$, the canonical monomorphism

$$j : \mathrm{Im}(X_\ell \to X_n) \longrightarrow \mathrm{Im}(X_{n+r} \to X_n)$$

induces an isomorphism on associated gradeds. Since the gradings are finite, it follows that j itself is an isomorphism, whence (i).

4.1.4

If X is an object of $\mathcal{P}$ satisfying (ML), it follows from the expression of the associated graded recalled in (4.1.1) that, for every integer p, the projective system

$$\mathrm{gr}^p(X) = \left(\mathrm{gr}^p(X_n)\right)_{n\in\mathbb{N}}$$

is essentially constant.

Its projective limit, which exists without any hypothesis on the abelian category $\mathcal{C}$, will be called the p-th component of the strict associated graded of X, and will be denoted

$$\mathrm{grs}^p(X).$$

4.2. Projective systems adapted to J and the Artin-Rees lemma

We suppose given a commutative ring A, an ideal J of A, and an abelian A-category $\mathcal{C}$.

Definition 4.2.1. An object X of the category $\mathcal{P} = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ is said to be adapted to J if its filtration $(F^pX)_{p\in\mathbb{N}}$ as a projective system (4.1.1) is compatible with the J-adic filtration of X, i.e. if for every pair (p, q) of non-negative integers the inclusion below holds:

$$J^pF^q(X) \subset F^{p+q}(X).$$

Lemma 4.2.2. Let X be an object of the category $\mathcal{P}$. The following assertions are equivalent:

(i) X is adapted to J.

(ii) For every pair (n, p) of integers with $p \leq n + 1$, one has

$$J^pF^{n-p+1}(X_n) = 0.$$

Proof. $(i) \Rightarrow (ii)$: then

$$J^pF^{n-p+1}(X_n) \subset F^{n+1}(X_n) = 0.$$

$(ii) \Rightarrow (i)$: if $n < p + q$, $F^{p+q}(X_n) = 0$ by definition. Moreover

$$J^pF^q(X_n)$$

is contained in $J^{n+1-q}F^q(X_n)$, and the latter is zero by hypothesis. If $n > p + q$, the image of $J^pF^q(X_n)$ in X_{p+q-1} under the transition morphism is contained in $J^pF^q(X_{p+q-1})$, which is zero by hypothesis. The inclusion

$$J^pF^q(X_n) \subset F^{p+q}(X_n)$$

therefore follows by definition of the filtration of a projective system.

In particular, it follows from (4.2.2) that every projective subsystem of a projective system adapted to J is adapted to J.

Examples. a) A J-adic projective system is adapted to J. It follows that every projective subsystem of a J-adic projective system is adapted to J.

b) If a projective system X satisfies condition (ML) and is adapted to J, then its projective system of universal images is also adapted to J.

4.2.3. The associated graded of a projective system adapted to J

Let X be a projective system adapted to J. Since the filtration of X is compatible with J, it is clear that the graded object

$$\left(\mathrm{gr}^p(X)\right)_{p\in\mathbb{N}}$$

is canonically endowed with the structure of a graded $\mathrm{gr}_J(A)$-module. In particular, by passage to the limit, one deduces that when X satisfies condition (ML), the graded object

$$\mathrm{grs}(X) = \left(\mathrm{grs}^p(X)\right)_{p\in\mathbb{N}}$$

is canonically endowed with the structure of a graded $\mathrm{gr}_J(A)$-module.

Let us briefly spell out how, when moreover X is strict, one defines the corresponding morphisms:

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q}).$$

Since the projective system X is strict, we know (4.1.1 and 4.1.2) that the following canonical morphism is an isomorphism:

$$\varphi : F^q(X_{p+q})/F^{q+1}(X_{p+q}) \longrightarrow F^q(X_q).$$

Moreover, the fact that the filtration of X is adapted to J gives a morphism

$$J^p \otimes_A F^q(X_{p+q}) \longrightarrow F^{p+q}(X_{p+q})$$

which factors, since

$$J^{p+1}F^q(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0, \qquad J^pF^{q+1}(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0,$$

through

$$J^p/J^{p+1} \otimes_A \left(F^q(X_{p+q})/F^{q+1}(X_{p+q})\right)$$

as a morphism

$$\beta : (J^p/J^{p+1}) \otimes_A \left(F^q(X_{p+q})/F^{q+1}(X_{p+q})\right) \longrightarrow F^{p+q}(X_{p+q}).$$

The morphism ω_{pq} is then given by the following expression:

$$\omega_{pq} = \beta \circ (\mathrm{id}_{J^p/J^{p+1}} \otimes_A \varphi)^{-1}.$$

Lemma 4.2.4. Let X be a J-adic projective system. The canonical morphism

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X),$$

coming from the graded $\mathrm{gr}_J(A)$-module structure of $\mathrm{grs}(X)$, is an epimorphism.

Proof. This follows immediately from the description of the morphisms ω_{pq} given in (4.2.3).

Proposition 4.2.5. Characterization of strict AR-J-adic projective systems by means of their associated gradeds. Let X be a strict projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. Suppose that for every integer n, one has

$$J^{n+1}X_n = 0.$$

The following assertions are equivalent:

(i) X is AR-J-adic.

(ii) There exists an integer $r > 0$ such that, for every integer n, the following inclusion holds:

$$F^n(X_n) \subset J^{n-r}X_n.$$

(iii) There exists an integer r such that, for every integer m, the following canonical morphism is an isomorphism whenever $n \geq m + r$:

$$\mathrm{gr}^n(J^m X) \longrightarrow \mathrm{gr}^n(X).$$

In the statement of (ii), we agree to set $J^{n-r}X_n = X_n$ for $n < r$.

Proof. The equivalence of (ii) and (iii) is clear, since X and $J^m X$ are strict. Let us prove the equivalence of (i) and (ii). We know (3.2.3) that (i) is equivalent to the existence of an integer r such that, for every pair (m, n) of integers with $m \geq n + r$, the transition morphism below is an isomorphism:

$$X_{m+1}/J^{n+1}X_{m+1} \longrightarrow X_m/J^{n+1}X_m.$$

Consider the following exact commutative diagram:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & J^{n+1}X_{m+1} & \longrightarrow & J^{n+1}X_m & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & & & \\
0 & \longrightarrow & F^{m+1}(X_{m+1}) & \longrightarrow & X_{m+1} & \longrightarrow & X_m & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & & & \\
 & & X_{m+1}/J^{n+1}X_{m+1} & \longrightarrow & X_m/J^{n+1}X_m & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & 0 & & 0. & & & &
\end{array}$$

It shows the existence of a canonical isomorphism between

$$\mathrm{Ker}(X_{m+1}/J^{n+1}X_{m+1} \longrightarrow X_m/J^{n+1}X_m)$$

and

$$F^{m+1}(X_{m+1})/(F^{m+1}(X_{m+1}) \cap J^{n+1}X_{m+1}).$$

In particular, for the transition morphism after reduction modulo J^{n+1} to be an isomorphism, it is necessary and sufficient that $F^{m+1}(X_{m+1})$ be contained in $J^{n+1}X_{m+1}$, which is precisely assertion (ii).

Theorem 4.2.6. Artin-Rees lemma. Let X be a projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. Suppose that:

(i) X satisfies (MLAR);

(ii) X is adapted to J;

(iii) $\mathrm{grs}(X)$ is a graded $(\mathrm{gr}_J(A), \mathcal{C})$-module of finite type, i.e. for every increasing sequence $(M_n)_{n\in\mathbb{N}}$ of graded sub-$\mathrm{gr}_J(A)$-modules of $\mathrm{grs}(X)$ whose union is $\mathrm{grs}(X)$, there exists an integer p such that

$$\mathrm{grs}(X) = M_p.$$

Then X is AR-J-adic.

Proof. Recall that if X' is the projective system of universal images of X, we have set

$$\mathrm{grs}(X) = \mathrm{gr}(X').$$

Moreover, X' is adapted to J, so it remains to show that X' is AR-J-adic. Suppose that the following lemma has been proved.

Lemma 4.2.7. Let X be a strict projective system adapted to J. For every integer s, the graded subobject $\big((M_s)_n\big)_{n\in\mathbb{N}}$ of $\mathrm{grs}(X)$ defined by the equalities

$$(M_s)_n = J^{n-s}X_n \cap F^n(X_n) \qquad (J^i = A \text{ if } i < 0),$$

is a graded $\mathrm{gr}_J(A)$-submodule of $\mathrm{grs}(X)$.

To prove the theorem, we may suppose X strict. Then $\mathrm{grs}(X)$ is the union of the increasing sequence of the M_s, hence is equal to one of them. This means that there exists an integer s_0 such that

$$F^n(X_n) \subset J^{n-s_0}X_n$$

for every integer n. We conclude by (4.2.5).

Proof of 4.2.7. Return to the notation of (4.2.3). We have to see that the morphism

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q})$$

sends

$$J^p/J^{p+1} \otimes_A \big(J^{q-s}X_q \cap F^q(X_q)\big)$$

into

$$J^{p+q-s}X_{p+q} \cap F^{p+q}(X_{p+q}).$$

Now the canonical morphism

$$\rho : X_{p+q}/F^{q+1}(X_{p+q}) \longrightarrow X_q$$

is an isomorphism and sends

$$\big(J^{q-s}X_{p+q} + F^{q+1}(X_{p+q})\big)/F^{q+1}(X_{p+q})$$

onto $J^{q-s}X_q$. It follows that the inverse image under $\mathrm{id}(J^p/J^{p+1}) \otimes_A \varphi$ (4.2.3) of

$$J^p/J^{p+1} \otimes_A \big(J^{q-s}X_q \cap F^q(X_q)\big)$$

is

$$J^p/J^{p+1} \otimes_A \frac{(J^{q-s}X_{p+q} \cap F^q(X_{p+q})) + F^{q+1}(X_{p+q})}{F^{q+1}(X_{p+q})}.$$

It is now clear that the image under β (4.2.3) of this last object is contained in

$$J^{p+q-s}X_{p+q} \cap J^p F^q(X_{p+q}),$$

hence a fortiori in

$$J^{p+q-s}X_{p+q} \cap F^{p+q}(X_{p+q}).$$

This proves the lemma.

5. Noetherian J-adic and AR-J-adic projective systems

Throughout this paragraph, a noetherian commutative ring A, an ideal J of A, and an abelian A-category $\mathcal{C}$ are fixed.

5.1. Generalities

Definition 5.1.1. An object X of the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ is said to be noetherian J-adic (resp. artinian) if it is J-adic and if its components are noetherian (resp. artinian) objects of the category $\mathcal{C}$.

We shall denote by $J\text{-adn}(\mathcal{C})$ (resp. $J\text{-adf}(\mathcal{C})$) the full subcategory of $\mathcal{P} = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ whose objects are noetherian (resp. artinian) J-adic projective systems. It is clear (3.1.3) that an extension of noetherian (resp. artinian) J-adic projective systems is of the same type.

Proposition 5.1.2. Let X be a J-adic object of $\mathcal{P}$. In order for X to be noetherian (resp. artinian) J-adic, it is necessary and sufficient that it be J-adic and that X_0 be noetherian (resp. artinian).

Proof. Indeed, let X be a J-adic projective system such that X_0 is noetherian (resp. artinian). Consider the canonical epimorphism (4.2.4):

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X).$$

Since X_0 is noetherian (resp. artinian) and J^n is of finite type for every n, the existence of this epimorphism shows that for every integer n, the kernel of the canonical morphism

$$X_n \longrightarrow X_{n-1}$$

is noetherian (resp. artinian). It follows at once by induction on n that the X_n are noetherian (resp. artinian).

Definition 5.1.3. Let X be a projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. It is said to be noetherian (resp. artinian) AR-J-adic if it is AR-J-adic and if its components are noetherian (resp. artinian) objects.

We shall denote by AR-J-adn$(\mathcal{C})$ (resp. AR-J-adf$(\mathcal{C})$) the full subcategory of $\mathrm{Hom}_{\mathrm{AR}}(\mathbb{N}^\circ, \mathcal{C})$ (2.4) generated by the images of noetherian (resp. artinian) AR-J-adic projective systems.

It is then clear that the canonical functors

$$P^n_{\mathrm{AR}} : J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{C}),$$

$$P^f_{\mathrm{AR}} : J\text{-adf}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adf}(\mathcal{C}),$$

obtained by restriction of P_{AR} (2.4), are equivalences of categories.

Let us briefly recall the statement and proof of Hilbert's theorem.

Theorem 5.1.4. Hilbert's theorem. Let A be a commutative ring, $\mathcal{C}$ an abelian A-category, and M an object of finite type (resp. noetherian) of $\mathcal{C}$. For every A-algebra of finite type B, graded by $\mathbb{N}$, the graded $(B,\mathcal{C})$-module

$$B \otimes_A M$$

is of finite type (resp. noetherian).

N.B. If $B = \bigoplus_{n\geq 0} B_n$, then $B \otimes_A M$ denotes the graded object $(B_n \otimes_A M)_{n\in\mathbb{N}}$.

Proof. One reduces immediately to the case where B is the polynomial ring $A[T]$, endowed with its usual grading. The $(B,\mathcal{C})$-module $A[T] \otimes_A M = (M_n)_{n\in\mathbb{N}}$ is then obtained by putting $M_n = M$ for every n, with T sending M_n to M_{n+1} by the identity of M, for every integer n.

Let N be a graded sub-$(A[T],\mathcal{C})$-module of $A[T] \otimes_A M$. The components $(N_n)_{n\in\mathbb{N}}$ of N define an increasing sequence

$$N_0 \subset N_1 \subset \cdots \subset N_n \subset \cdots$$

of subobjects of M. Suppose now that M is noetherian (resp. of finite type), and let

$$N^1 \subset N^2 \subset \cdots \subset N^p \subset \cdots$$

be an increasing sequence of graded sub-$(A[T],\mathcal{C})$-modules of $A[T] \otimes_A M$, whose union, in the first hypothesis, when M is of finite type, is $A[T] \otimes_A M$.

Consider the following double-entry table:

$$\begin{array}{ccccccc}
N^1_0 & \subset & N^2_0 & \subset \cdots \subset & N^p_0 & \subset \cdots \\
\cap & & \cap & & \cap & \\
N^1_1 & & N^2_1 & & N^p_1 & \\
\vdots & & \vdots & & \vdots & \\
\cap & & \cap & & \cap & \\
N^1_n & \subset & N^2_n & \subset \cdots \subset & N^p_n & \subset \cdots .
\end{array}$$

It is clear by hypothesis on M that there exists a pair (p_0, n_0) such that $N^{p_0}_{n_0}$ is maximal (resp. equal to M). If p_1 is an integer $\geq p_0$ such that the $N^{p_1}_n$ $(n \leq n_0)$ are the largest elements of the sequences $(N^p_n)_{p\in\mathbb{N}}$, it is clear that $N^p = N^{p_1}$ as soon as $p \geq p_1$, which allows one to conclude in both cases.

Corollary 5.1.5. Suppose moreover that the category $\mathcal{C}$ has denumerable direct sums. Then for every object M of finite type (resp. noetherian) of $\mathcal{C}$, and every A-algebra of finite type B, the $(B,\mathcal{C})$-module $B \otimes_A M$ is of finite type (resp. noetherian).

Proof. One reduces to the case where $B = A[T]$. Let N be a sub-$(A[T],\mathcal{C})$-module of $A[T] \otimes_A M$, considered as an object of $\mathcal{C}$. We shall denote by $\mathrm{gr}(N)$ the associated graded of N for the

filtration induced by the natural filtration of the graded object $A[T] \otimes_A M$: $\mathrm{gr}(N)$ is a graded sub-$(A[T], \mathcal{C})$-module of the graded $(A[T], \mathcal{C})$-module $A[T] \otimes_A M$.

Now let

$$N^0 \subset N^1 \subset \cdots \subset N^p \subset \cdots$$

be an increasing sequence of sub-$(A[T], \mathcal{C})$-modules of $A[T] \otimes_A M$, whose union is $A[T] \otimes_A M$ in the first hypothesis. In the first hypothesis (M of finite type), there exists an integer i such that $\mathrm{gr}(N^i) = A[T] \otimes_A M$, whence

$$N^i = A[T] \otimes_A M.$$

In the second hypothesis (M noetherian), the sequence of the $\mathrm{gr}(N^i)$ is stationary, hence the sequence of the N^i is stationary as well.

In what follows, Hilbert's theorem will be useful to us especially through the following proposition.

Proposition 5.1.6. Let A be a noetherian commutative ring, J an ideal of A, and $\mathcal{C}$ an abelian A-category. For every noetherian J-adic projective system X (5.1.1), the graded object $\mathrm{grs}(X)$ is a noetherian $(\mathrm{gr}_J(A), \mathcal{C})$-module.

Proof. Since $\mathrm{gr}_J(A)$ is an A-algebra of finite type, Hilbert's theorem (5.1.4) implies that

$$\mathrm{gr}_J(A) \otimes_A X_0$$

is a noetherian graded $(\mathrm{gr}_J(A), \mathcal{C})$-module. The assertion then follows from the existence of the epimorphism (4.2.4):

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X).$$

5.2. Structure of J-adn$(\mathcal{C})$ and AR-J-adn$(\mathcal{C})$

In this section, we shall denote by $\mathcal{E}_{\mathcal{C}}$, or by $\mathcal{E}$ if there is no possible confusion, the full subcategory of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ whose objects are noetherian AR-J-adic projective systems.

Proposition 5.2.1. The category $\mathcal{E}_{\mathcal{C}}$ is stable under kernels and cokernels in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$. In other words, $\mathcal{E}_{\mathcal{C}}$ is an abelian category and the inclusion functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$$

is exact.

Proof. Let $u : X \to Y$ be a morphism of $\mathcal{E}_{\mathcal{C}}$. We shall successively see that the cokernel, the image, and the kernel of u are noetherian AR-J-adic. Of course, the proof that the image of u is noetherian AR-J-adic is only a technical intermediate step.

a) $\mathrm{Coker}(u)$ is noetherian AR-J-adic. Since X satisfies (MLAR) and Y is AR-J-adic, $\mathrm{Coker}(u)$ is AR-J-adic (3.2.4 (i)).

b) $\mathrm{Im}(u)$ is noetherian AR-J-adic. From the exact sequence

$$0 \longrightarrow \mathrm{Im}(u) \longrightarrow Y \longrightarrow \mathrm{Coker}(u) \longrightarrow 0,$$

the assertion will follow from Lemma 5.2.1.1 below.

Lemma 5.2.1.1. Let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be an exact sequence in the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$. If M and N are noetherian AR-J-adic, then L is noetherian AR-J-adic.

Proof. It is plainly enough to see that it is AR-J-adic. One reduces, in the same way as in the proof of (3.2.4), to the case where M and N are strict, and then even noetherian J-adic. In this case, it follows from 3.1.3 (ii) that L is strict. The projective system L is strict, adapted to J, and its associated graded, as a graded subobject of $\mathrm{grs}(M)$, is noetherian: $\mathrm{grs}(M)$, a quotient of $\mathrm{gr}_J(A) \otimes_A M_0$, is noetherian (5.1.6). The assertion then follows from the Artin-Rees lemma (4.2.6).

c) $\mathrm{Ker}(u)$ is noetherian AR-J-adic. It is enough to apply (5.2.1.1) to the evident exact sequence below:

$$0 \longrightarrow \mathrm{Ker}(u) \longrightarrow X \longrightarrow \mathrm{Im}(u) \longrightarrow 0.$$

Proposition 5.2.2. The objects of the abelian category AR-J-adn($\mathcal{C}$) are noetherian.

Proof. One reduces immediately to proving the following. Suppose we have a noetherian J-adic object X, a sequence of noetherian J-adic objects $(Y^n)_{n\in\mathbb{N}}$, and morphisms in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$

$$v_n : Y^n \longrightarrow X$$

such that

(i) the v_n become monomorphisms in AR-J-adn($\mathcal{C}$);

(ii) for every integer n, v_n factors through v_{n+1} in AR-J-adn($\mathcal{C}$),

then the v_n remain stationary, up to isomorphism, for sufficiently large n, in AR-J-adn($\mathcal{C}$).

Now, to say that v_n factors through v_{n+1} in AR-J-adn($\mathcal{C}$) means that there exists an integer r and a morphism

$$f : Y^n[r] \longrightarrow Y^{n+1}$$

such that, with a denoting the canonical morphism

$$Y^n[r] \longrightarrow Y^n,$$

the following equality holds:

$$v_n \circ a = v_{n+1} \circ f.$$

Since the morphism a is strict, it follows in particular that the image of v_n is contained in that of v_{n+1}. Replacing each Y^n by $\mathrm{Im}(v_n)$, we are reduced to seeing that every increasing sequence of strict projective subsystems of X is stationary. But if $(Z^n)_{n\in\mathbb{N}}$ is such a sequence, then, since $\mathrm{grs}(X)$ is noetherian, the increasing sequence of the $\mathrm{grs}(Z^n)$ is stationary, hence so is the sequence of the Z^n.

Summarizing:

Theorem 5.2.3. The categories J-adn($\mathcal{C}$) and AR-J-adn($\mathcal{C}$) are abelian and noetherian.

As for AR-J-adn($\mathcal{C}$), this is what we have just proved. As for J-adn($\mathcal{C}$), it is equivalent to it.

Proposition 5.2.4. Stability under extension. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. Suppose that:

(i) Z is AR-J-adic;

(ii) X is artinian AR-J-adic.

Then Y is AR-J-adic.

Proof. As in the proof of (3.2.4), one reduces to the case where Y and Z are strict and Z is J-adic. Then X is strict by (3.1.3 (ii)). There then exists an integer r such that the projective system $(X_{n+r}/J^{n+1}X_{n+r})_{n\in\mathbb{N}}$ is artinian J-adic. Since X is a quotient of it and is AR-isomorphic to it, we are reduced to proving the following lemma.

Lemma 5.2.4.1. Let

$$0 \longrightarrow T \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. If X is artinian J-adic, Z is J-adic, and T is AR-zero, then Y is AR-J-adic.

Proof. To prove the lemma, we first show that, for every integer n, the evident projective system

$$\left(Y_{n+k}/J^{n+1}Y_{n+k}\right)_{k\in\mathbb{N}}$$

is essentially constant. Let S_n^k denote the kernel of the evident morphism

$$X_{n+k}/J^{n+1}X_{n+k} \longrightarrow Y_{n+k}/J^{n+1}Y_{n+k}.$$

Consider the following evident exact commutative diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & S_n^k & \longrightarrow & X_{n+k}/J^{n+1}X_{n+k} & \longrightarrow & Y_{n+k}/J^{n+1}Y_{n+k} & \longrightarrow & Z_{n+k}/J^{n+1}Z_{n+k} & \longrightarrow & 0 \\
 & & \downarrow & & \sim\downarrow & & \downarrow & & \sim\downarrow & & \\
0 & \longrightarrow & T_n & \longrightarrow & X_n & \longrightarrow & Y_n & \longrightarrow & Z_n & \longrightarrow & 0.
\end{array}$$

With n fixed, the objects S_n^k are identified with subobjects of T_n and form a decreasing sequence. Since T_n is artinian, there exists an integer k_0 such that $S_n^k = S_n^{k_0}$ for every integer $k \geq k_0$; it follows at once that the canonical morphisms

$$Y_{n+k}/J^{n+1}Y_{n+k} \longrightarrow Y_{n+k_0}/J^{n+1}Y_{n+k_0}$$

are isomorphisms.

This being said, denote by Y' the projective system defined by

$$Y'_n = \varprojlim_r Y_{n+r}/J^{n+1}Y_{n+r}$$

on objects, and in the evident way on morphisms. The projective system Y' is J-adic. Moreover, by passage to the limit one obtains an exact commutative diagram of the following type:

$$\begin{array}{ccccccccc} 0 & \to & S & \to & X & \to & Y' & \to & Z & \to & 0 \\ & & \downarrow & & \downarrow \mathrm{id} & & \downarrow & & \downarrow \mathrm{id} & & \\ 0 & \to & T & \to & X & \to & Y & \to & Z & \to & 0. \end{array}$$

Since T is AR-zero, the snake lemma applied to this diagram shows that the evident morphism $Y' \to Y$ is an AR-isomorphism, whence Y is AR-J-adic.

Corollary 5.2.5. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. Suppose that:

(i) X is artinian AR-J-adic (resp. artinian and noetherian);

(ii) Z is artinian AR-J-adic (resp. noetherian).

Then Y is artinian AR-J-adic (resp. noetherian).

5.3. Noetherian AR-J-adic projective systems and δ-functors

Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A-categories, and let $F : \mathcal{C} \to \mathcal{D}$ be an additive functor. The functor F extends in the clear way to a functor, again denoted F,

$$F : \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{D}),$$

which is additive and "commutes with translations".

Proposition 5.3.1. Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A-categories and

$$T = (T^i)_{i \in \mathbb{N}}$$

an exact A-linear δ-functor from $\mathcal{C}$ to $\mathcal{D}$. Suppose that:

(i) For every noetherian object M of $\mathcal{C}$, annihilated by a power of J, and every integer n, $T^n(M)$ is noetherian.

(ii) For every graded A-algebra B of finite type, annihilated by J, and every noetherian graded $(B, \mathcal{C})$-module N, the graded $(B, \mathcal{D})$-modules $T^n(N)$, $n \in \mathbb{N}$, are noetherian.

Then, for every noetherian AR-J-adic object X of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, the $T^n(X)$ are noetherian AR-J-adic.

Proof. Since the functors T^n commute with translations, it is enough to prove the assertion when X is noetherian J-adic. In that case, since $\mathrm{grs}(X)$ is a noetherian graded $\mathrm{gr}_J(A)$-module (5.1.6), it follows from hypothesis (ii) and from a variant of a corollary of Shih's theorem (cf. Appendix A.3) that for every integer n the following assertions hold:

(a) $T^n(X)$ satisfies (MLAR),

(b) grs $T^n(X)$ is a noetherian graded $(\mathrm{gr}_J(A), \mathcal{D})$-module.

The Artin-Rees lemma (4.2.6) allows us to conclude.

Corollary 5.3.2. The δ-functor

$$T = (T^i)_{i\in\mathbb{N}} : \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{D})$$

induces an exact A-linear δ-functor

$$\mathcal{E}_\mathcal{C} \longrightarrow \mathcal{E}_\mathcal{D},$$

commuting with translations, and hence, by passage to the quotient, a new exact A-linear δ-functor, again denoted $(T^i)_{i\in\mathbb{N}}$,

$$\text{AR-}J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{D}).$$

Appendix: Shih's theorem

The purpose of the present appendix is to give a proof of Shih's theorem in the form used in the proof of (5.3.1). This form being more general than the one appearing in EGA 0_{III}, a direct proof, adapted from that of loc. cit., has proved necessary.

Terminology. Let r_0 be an integer ≥ 1. The datum, for every integer $r \geq r_0$, of a bigraded object

$$E_r = (E_r^{pq})_{p\in\mathbb{Z},\ q\in\mathbb{Z}},$$

endowed with a differential d_r of bidegree $(r, 1-r)$, and, for every r, of an isomorphism

$$H(E_r) \xrightarrow{\sim} E_{r+1},$$

will be called below a spectral filter. The notion of a morphism of spectral filters is defined in the evident way.

A.1. Recollections on the spectral sequences associated with δ-functors

Let $\mathcal{C}$ be an abelian category and X an object of $\mathcal{C}$, endowed with a decreasing filtration $(F^pX)_{p\in\mathbb{Z}}$. For every exact δ-functor T from $\mathcal{C}$ to an abelian category $\mathcal{D}$, the method of "exact couples" (cf., e.g., Maclane, *Homology*, p. 336) makes it possible to describe a spectral sequence

$$E(X): \quad E_1^{pq} = T^{p+q}(F^pX/F^{p+1}X) \Rightarrow E^n = T^n(X),$$

for which the filtration of E^n is defined as follows:

$$F^p(E^n) = \mathrm{Im}(T^n(F^pX) \longrightarrow T^n(X)) = \mathrm{Ker}(T^n(X) \longrightarrow T^n(X/F^pX)).$$

Moreover, if the filtration of X is finite, the spectral sequence so obtained is weakly convergent (EGA 0_{III} 11.1.3).

Functoriality. If $f : X \to Y$ is a morphism of filtered objects of $\mathcal{C}$, one constructs functorially a morphism of spectral sequences

$$E(f) : E(X) \longrightarrow E(Y),$$

coinciding with $T^{p+q}(\mathrm{gr}^p f)$ (resp. $T^n(f)$) on $E_1^{pq}(X)$ (resp. $E^n(X)$). In particular, if $u : X \to X$ is an endomorphism of X such that

$$u(F^pX) \subset F^{p+h}(X)$$

for an integer h independent of p, then $E(u)$ is an endomorphism of bidegree $(h, -h)$ of the spectral filter associated with $E(X)$, commuting with the differentials.

A.2. The case of projective systems

Let now $X = (X_n)_{n\in\mathbb{Z}}$ be a projective system of objects of an abelian category $\mathcal{C}$, such that

$$X_i = 0 \qquad (i < 0).$$

If T denotes an exact δ-functor from $\mathcal{C}$ to an abelian category $\mathcal{D}$, T extends naturally to an exact δ-functor, again denoted T,

$$\mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{D}).$$

Thus, endowing X with its natural filtration as a projective system (EGA 0_{III} 13.4.1), one obtains a spectral sequence

$$E(X) = E(X_n)_{n\in\mathbb{Z}},$$

where the $E(X_n)$ denote the spectral sequences associated with T and the X_n endowed with the induced filtration.

Properties of $E(X)$. a) For every integer n, the filtration of the abutment $E^n(X)$ in the spectral sequence $E(X)$ is identical with its natural filtration as a projective system when X is strict.

Indeed, denoting by the letter F (as in A.1) the first filtration and by the letter Φ the second, one has for every integer α:

$$F^pE^n(X)_\alpha = \mathrm{Ker}\big(T^n(X_\alpha) \longrightarrow T^n(X_\alpha/F^pX_\alpha)\big)$$

$$= \begin{cases} 0, & \alpha \le p-1, \\ \mathrm{Ker}\big(T^n(X_\alpha) \longrightarrow T^n(X_{p-1})\big), & \alpha \ge p-1. \end{cases}$$

In both cases one recognizes the expression of $\Phi^pE^n(X)$.

This shows in particular that, for $\beta \ge \alpha \ge p$, the transition morphism

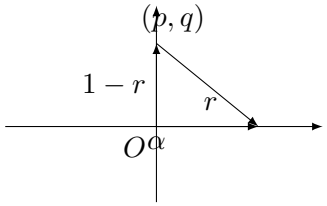
$$E_\infty^{pq}(X_\beta) \longrightarrow E_\infty^{pq}(X_\alpha)$$

which is identified with

$$\mathrm{gr}_T^p\, T^{p+q}(X_\beta) \longrightarrow \mathrm{gr}_T^p\, T^{p+q}(X_\alpha),$$

is a monomorphism, by EGA 0_{III} 13.4.1.3.

b) Fixing $\alpha \in \mathbb{Z}$, it is clear (cf. the adjoining diagram) that, since $F^i(X_\alpha) = 0$ for $i \geq \alpha + 1$, one has:



$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \quad \text{for } r > p,$$
$$Z_r^{pq}(X_\alpha) = Z_{r+1}^{pq}(X_\alpha) \quad \text{for } r > \alpha - p.$$

c) When X satisfies the Mittag-Leffler condition, the various projective systems $\mathrm{gr}^p(X)$ are essentially constant (EGA 0_{III} 13.4.3). It follows at once that, for $r < +\infty$, the projective systems $E_r^{pq}(X)$ are essentially constant. We shall denote by

$$\mathcal{E}_r^{pq}(X)$$

their projective limits. Since the differential operators pass to the limit, one thus obtains a spectral filter $\mathcal{E}_r^{pq}(X)$, $p, q \in \mathbb{Z}$, $r \geq 1$. The objects

$$\mathcal{Z}_r^{pq}(X), \quad \mathcal{B}_r^{pq}(X) \quad (r < +\infty)$$

associated with this spectral filter are precisely the limits of the essentially constant projective systems $Z_r^{pq}(X)$ and $B_r^{pq}(X)$.

It also follows from part b) that

$$B_\infty^{pq}(X) = B_{p+1}^{pq}(X),$$

which allows one to put

$$\mathcal{B}_\infty^{pq}(X) = \mathcal{B}_{p+1}^{pq}(X).$$

Still supposing that X satisfies (ML), since, for every α,

$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \qquad (r > p),$$

one has a sequence of canonical monomorphisms

$$\cdots \subset \mathcal{E}_r^{pq}(X) \subset \cdots \subset \mathcal{E}_{p+2}^{pq}(X) \subset \mathcal{E}_{p+1}^{pq}(X). \tag{S}$$

Suppose from now on that X is strict. The following expressions for $Z_r^{pq}(X_n)$ and $B_r^{pq}(X_n)$:

$$Z_r^{pq}(X_n) = \mathrm{Im}\Big[T^{p+q-1}\big(F^{p-r+1}X_n/F^pX_n\big) \xrightarrow{\partial} T^{p+q}\big(F^pX_n/F^{p+1}X_n\big)\Big],$$

$$B_r^{pq}(X_n) = \mathrm{Im}\Big[T^{p+q}\big(F^pX_n/F^{p+r}X_n\big) \longrightarrow T^{p+q}\big(F^pX_n/F^{p+1}X_n\big)\Big],$$

show that

$$\begin{cases} Z_r^{pq}(X_m) \xrightarrow{\sim} Z_r^{pq}(X_n) & \text{for } m \geq n \geq p, \\ B_r^{pq}(X_m) \xrightarrow{\sim} B_r^{pq}(X_n) & \text{for } m \geq n \geq p + r - 1, \end{cases} \tag{d_1}$$

where the canonical morphisms are isomorphisms. In particular, the canonical morphism

$$\mathcal{E}_r^{pq}(X) \longrightarrow E_r^{pq}(X_{p+r-1})$$

is an isomorphism. Moreover, it follows from b) that

$$\begin{cases} Z_\infty^{pq}(X_{p+r-1}) = Z_r^{pq}(X_{p+r-1}) & (r \geq 1), \\ B_\infty^{pq}(X_{p+r-1}) = B_r^{pq}(X_{p+r-1}) & (r > p). \end{cases} \tag{d_2}$$

This makes it possible to define, for every r, a canonical epimorphism

$$\theta_r^{pq} : \mathcal{E}_r^{pq}(X) \longrightarrow E_\infty^{pq}(X_{p+r-1})$$

which is an isomorphism when $r > p$. In addition, if a is an integer such that

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \qquad (r \geq a),$$

for example $a = p + 1$ suffices, so that

$$\cdots \hookrightarrow \mathcal{E}_r^{pq}(X) \hookrightarrow \cdots \hookrightarrow \mathcal{E}_{a+1}^{pq}(X) \hookrightarrow \mathcal{E}_a^{pq}(X), \tag{S bis}$$

then, for $s \geq r \geq a$, the diagrams

$$\begin{array}{ccc} \mathcal{E}_s^{pq}(X) & \xrightarrow[\text{epi}]{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\ \downarrow & & \downarrow \text{can} \\ \mathcal{E}_r^{pq}(X) & \xrightarrow[\text{epi}]{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1}) \end{array} \tag{d_3}$$

are commutative.

A.3. A variant of Shih's theorem

Suppose given a ring S filtered by ideals $F^i(S)$, $i \geq 0$, acting on X compatibly with its filtration as a projective system.

It follows from the generalities of A.1 that, for every integer i, $\mathrm{gr}^i(S)$ acts on the bigraded objects $E_r(X)$ by morphisms of bidegree $(i, -i)$. In particular, for every integer n,

$$E_r^{(n)}(X) = \left(E_r^{p,q}(X)\right)_{p+q=n}$$

is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module. The same holds by passage to the limit for

$$\mathcal{E}_r^{(n)}(X) = \left(\mathcal{E}_r^{p,q}(X)\right)_{p+q=n}.$$

The ring S, acting on X compatibly with its filtration as a projective system, likewise acts on $T^n(X)$ for every integer n. Consequently, the strict associated graded $\mathrm{grs}(X)$ is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{C})$-module, and similarly, when $T^n(X)$ satisfies (ML), the strict associated graded $\mathrm{grs}\, T^n(X)$ is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module.

Theorem A.3. Suppose that, for an integer n, the graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules

$$\mathcal{E}_1^{(n)}(X) = T^n(\mathrm{grs}^\bullet X) \quad \text{and} \quad \mathcal{E}_1^{(n-1)}(X) = T^{n-1}(\mathrm{grs}^\bullet X)$$

are noetherian. Then:

(i) $T^n(X)$ satisfies (MLAR);

(ii) grs $T^n(X)$ is a noetherian $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module.

Proof. i) For every pair (m, r) of integers,

$$\mathcal{B}_r^{(m)}(X) = \left(\mathcal{B}_r^{p,m-p}(X)\right)_{p\in\mathbb{N}}$$

is a graded sub-$(\mathrm{gr}^\bullet(S), \mathcal{D})$-module of $\mathcal{E}_1^{(m)}(X)$. In particular, for every integer m, the $\mathcal{B}_r^{(m)}(X)$ form an increasing sequence, indexed by r, of graded sub-$(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules of $\mathcal{E}_1^{(m)}(X)$. Using the fact that $\mathcal{E}_1^{(n)}(X)$ and $\mathcal{E}_1^{(n-1)}(X)$ are noetherian graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules, one deduces that the sequences

$$\left[\mathcal{B}_r^{(n)}(X)\right]_{r\in\mathbb{N}} \qquad \text{and} \qquad \left[\mathcal{B}_r^{(n-1)}(X)\right]_{r\in\mathbb{N}}$$

are stationary.

Thus there exists an integer $a \geq 1$ such that, for every (p, q), with $p + q = n$ or $n - 1$,

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \qquad (r \geq a). \tag{A.3.1}$$

Consequently, for $r \geq a + 1$, the arrows d_r with source and target $\mathcal{E}_r^{pq}(X)$, $p + q = n$, are zero. In other words, the monomorphisms

$$\mathcal{E}_s^{pq}(X) \hookrightarrow \mathcal{E}_r^{pq}(X) \qquad (s \geq r \geq a + 1)$$

deduced from (A.3.1) with $p+q = n$ are in fact isomorphisms. For $s \geq r \geq a+1$, the commutativity of diagram (d_3), with $p + q = n$,

$$\begin{array}{ccc} \mathcal{E}_s^{pq}(X) & \xrightarrow[\text{epi}]{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\ \downarrow\sim & & \downarrow \\ \mathcal{E}_r^{pq}(X) & \xrightarrow[\text{epi}]{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1}) \end{array}$$

shows that the transition morphism

$$E_\infty^{pq}(X_{p+s-1}) \longrightarrow E_\infty^{pq}(X_{p+r-1}),$$

which is in any case a monomorphism (cf. a)), is an epimorphism, hence an isomorphism. In other words, the projective system

$$r \longmapsto \left[\bigoplus_p \mathrm{gr}^p\, T^n(X_{p+r})\right]_{r\geq 1}$$

is constant from $r = a$ on, and therefore $T^n(X)$ satisfies (MLAR), by (4.1.3).

ii) Denoting by

$$\mathcal{E}_\infty^{(n)}(X), \quad \mathcal{Z}_\infty^{(n)}(X), \quad \mathcal{B}_\infty^{(n)}(X)$$

the "limits" of the various projective systems indexed by r considered above, we see that

$$\mathcal{E}_\infty^{(n)}(X) = \mathcal{Z}_\infty^{(n)}(X)/\mathcal{B}_\infty^{(n)}(X)$$

is a noetherian $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module, since $\mathcal{Z}_\infty^{(n)}(X)$ is contained in $\mathcal{E}_1^{(n)}(X)$. Moreover, it follows from A.2 a) that

$$\mathcal{E}_\infty^{(n)}(X) = \mathrm{grs}\ T^n(X).$$

Exposé VI

ℓ-ADIC COHOMOLOGY

by J.-P. Jouanolou

Throughout this exposé, a prime number ℓ is fixed.

As usual, $X_{\text{ét}}$ denotes the étale site of a prescheme X.

1. Constructible ℓ-adic sheaves

Throughout this paragraph, fix once and for all a locally noetherian prescheme X.

1.1. Definition and generalities

Definition 1.1.1. A constructible ℓ-adic sheaf on X is a projective $(\ell\mathbb{Z})$-adic system of constructible abelian sheaves on $X_{\text{ét}}$ (V 3.2.1).

For a projective $(\ell\mathbb{Z})$-adic system

$$F = (F_n)_{n\in\mathbb{N}}$$

of abelian sheaves on $X_{\text{ét}}$ to be a constructible ℓ-adic sheaf, it is enough that F_0 be constructible. Indeed, the assertion is local and follows, in the noetherian case, from (V 5.1.2).

Let $\mathrm{Abc}(X)$ denote the category of constructible abelian sheaves on X. We shall denote by

$$\ell\text{-}\,\mathrm{adc}(X)$$

the full subcategory of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Abc}(X))$ generated by the constructible ℓ-adic sheaves. With the notation of Exposé V,

$$\ell\text{-}\,\mathrm{adc}(X) = (\ell\mathbb{Z})\text{-}\,\mathrm{ad}(\mathrm{Abc}(X)).$$

Letting $\mathrm{Ab}(X)$ denote the category of abelian sheaves on $X_{\text{ét}}$, one knows (SGA 4 IX 2.9) that, when X is noetherian, the constructible abelian sheaves on X are the noetherian abelian sheaves. In this case, the category $\ell\text{-}\,\mathrm{adc}(X)$ is clearly identified with the category denoted

$$(\ell\mathbb{Z})\text{-}\,\mathrm{adn}(\mathrm{Ab}(X))$$

in Exposé V.

Proposition 1.1.2. Let $F = (F_n)_{n\in\mathbb{N}}$ be an object of the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$. For F to be constructible ℓ-adic, it is necessary and sufficient that it be so locally for the étale topology of X.

Proof. It is clear that the property of F of being ℓ-adic may be read above a covering of the final object of the site. Moreover, it follows from (SGA 4 IX 2.8) that the F_n are constructible if they are so locally.

Proposition 1.1.3. The category $\ell\text{-}\,\mathrm{adc}(X)$ of constructible ℓ-adic sheaves on X is abelian. It is noetherian when X is noetherian.

Proof. When X is noetherian, the two assertions are consequences of (V 5.2.3). In the general case, one must see that every morphism

$$u : F \longrightarrow G$$

of constructible ℓ-adic sheaves on X has a kernel and a cokernel.

As for the cokernel, this follows from (V 3.1.3). Let us prove the existence of the kernel of u. For every noetherian open V of X, one knows that the restriction of u to V,

$$u_V : F_V \longrightarrow G_V,$$

has a kernel in the category $\ell\text{-}\operatorname{adc}(V)$, admitting the following simple description. If K is the kernel of u_V in the category $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(V))$, K satisfies the Mittag-Leffler-Artin-Rees condition and, denoting by K' its projective system of universal images, there exists an integer r such that the projective system

$$m_r(K') = \left(K'_{n+r}/\ell^{n+1}K'_{n+r}\right)_{n\in\mathbb{N}}$$

(with the evident transition morphisms) is constructible ℓ-adic. The evident composite below is a kernel of u_V in the category $\ell\text{-}\operatorname{adc}(V)$:

$$m_r(K') \longrightarrow K' \longrightarrow K.$$

The preceding construction shows in particular that, if W is an open contained in V, the restriction to W of a kernel for u_V is a kernel for u_W. Let $(V_i)_{i\in I}$ be a covering of X by noetherian opens. If i and j are two elements of I, the restrictions to $V_i \cap V_j$ of the kernels of u_{V_i} and u_{V_j} are canonically isomorphic, which allows them to be glued to construct a kernel of u.

1.1.4

Let $F = (F_n)_{n\in\mathbb{N}}$ be a constructible ℓ-adic sheaf. Since the ring $\mathbb{Z}_\ell$ acts evidently on each sheaf F_n, it also acts on F. Moreover, if F and G are two constructible ℓ-adic sheaves and

$$u : F \longrightarrow G$$

is a morphism, u is compatible with the actions of $\mathbb{Z}_\ell$ on F and G, respectively. This shows that $\ell\text{-}\operatorname{adc}(X)$ is canonically endowed with the structure of an abelian $\mathbb{Z}_\ell$-category.

This justifies the following terminology.

Terminology. In what follows, constructible ℓ-adic sheaves will be called constructible $\mathbb{Z}_\ell$-sheaves.

One should take care not to confuse this notion with that of a $(\mathbb{Z}_\ell)_X$-module, which will not be used below.

The category $\ell\text{-}\operatorname{adc}(X)$ will also be denoted

$$\mathbb{Z}_\ell\text{-}\operatorname{fc}(X).$$

1.2. Constructible twisted constant $\mathbb{Z}_\ell$-sheaves

Definition 1.2.1. A constructible $\mathbb{Z}_\ell$-sheaf

$$F = (F_n)_{n\in\mathbb{N}}$$

is said to be twisted constant if and only if the sheaves F_n are locally constant on $X_{\text{ét}}$.

The category of constructible twisted constant $\mathbb{Z}_\ell$-sheaves is the full subcategory of $\mathbb{Z}_\ell$- $\mathrm{fc}(X)$ generated by the constructible twisted constant $\mathbb{Z}_\ell$-sheaves.

Example 1.2.2. The $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$.

Assume that the integer ℓ is prime to the residual characteristics of X. For every integer n, raising to the ℓ-th power defines a morphism of locally constant sheaves:

$$\mu_{\ell^{n+1}} \longrightarrow \mu_{\ell^n}.$$

This evidently defines a projective system:

$$\mathbb{Z}_\ell(1) = (\mu_{\ell^{n+1}})_{n\in\mathbb{N}}.$$

Reducing, for each integer n, to the case where μ_{ℓ^n} and $\mu_{\ell^{n+1}}$ are constant, one sees that the projective system $\mathbb{Z}_\ell(1)$ is ℓ-adic.

When $X = \mathrm{Spec}(K)$, where K is a field of finite type over the prime field, the $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$ is an example of a constructible twisted constant ℓ-adic sheaf (1.2.1) for which there is no finite étale extension K' of K such that the inverse images of the F_n on $X' = \mathrm{Spec}(K')$ are constant. Indeed, K' would contain, for every integer n, a primitive ℓ^n-th root of unity, which is not possible for a finite-type extension of the prime field.

Reminders 1.2.3. (SGA 1 V 7). One knows that the category of constructible locally constant abelian torsion sheaves on a prescheme T is equivalent to the category of étale coverings of T endowed with the structure of an abelian X-group.

Now let X be a connected locally noetherian prescheme. Let R_X denote the category of étale coverings of X, and let Ensf denote the category of finite sets. If a is a geometric point of X, let F_a be the functor

$$F_a : R_X \longrightarrow \mathrm{Ensf},$$

which associates to every étale covering of X its geometric fibre at a. The fundamental group

$$\pi_1(X) = \pi_1(X, a)$$

of X at a, being by definition the group of automorphisms of F_a, suitably topologized, allows the functor F_a to be interpreted as a functor

$$R_X \longrightarrow \mathrm{Ensf}(\pi_1)$$

from the category R_X to the category of finite sets endowed with a continuous action of $\pi_1 = \pi_1(X, a)$.

Recall the following proposition.

Proposition 1.2.3.1. The functor

$$F_a : R_X \longrightarrow \mathrm{Ensf}(\pi_1)$$

is an equivalence of categories. It commutes with inductive limits and finite projective limits.

Let $Lc(X)$ and $\mathrm{Abf}(\pi_1)$ denote respectively the category of constructible locally constant abelian torsion sheaves and the category of finite abelian groups endowed with a continuous operation of the fundamental group. The functor F_a induces an exact functor, still denoted F_a,

$$F_a : Lc(X) \longrightarrow \mathrm{Abf}(\pi_1),$$

which is an equivalence of categories.

1.2.4. Twisted constant $\mathbb{Z}_\ell$-sheaves on a connected prescheme

The reminders in (1.2.3) make the lemma below immediate.

Lemma 1.2.4.1. The extension $\widehat{F}_a$ of F_a to the category of projective systems indexed by $\mathbb{N}$ induces an equivalence from the category of constructible twisted constant ℓ-adic sheaves to the category

$$(\ell\mathbb{Z})\text{-}\,\mathrm{ad}(\mathrm{Abf}(\pi_1)).$$

Now let $F = (F_n)_{n\in\mathbb{N}}$ be an object of the category

$$(\ell\mathbb{Z})\text{-}\,\mathrm{ad}(\mathrm{Abf}(\pi_1)).$$

Its projective limit

$$\varprojlim_{\mathbb{N}} F_n$$

is naturally endowed with a structure of $\mathbb{Z}_\ell$-module of finite type on which π_1 acts continuously for the ℓ-adic topology. Recall that if G and M denote respectively a profinite group and a $\mathbb{Z}_\ell$-module of finite type, a continuous operation of G on M means a continuous map

$$G \times M \longrightarrow M, \qquad (g, m) \longmapsto g \cdot m,$$

such that

$$\text{(i)} \quad g \cdot (m + n) = g \cdot m + g \cdot n \quad (g \in G,\ m, n \in M),$$

and

$$\text{(ii)} \quad (gh) \cdot m = g \cdot (h \cdot m) \quad (g, h \in G,\ m \in M).$$

It is then clear that, for every ℓ-adic integer a,

$$g \cdot (am) = a(g \cdot m) \qquad (g \in G,\ m \in M).$$

Moreover, the group of invertible $\mathbb{Z}_\ell$-linear maps from M to M, denoted $\mathrm{GL}(M)$, is identified with the projective limit of the groups $\mathrm{GL}(M/\ell^{n+1}M)$. Since these groups are finite, the group $\mathrm{GL}(M)$ is thus naturally endowed with the structure of a profinite group, and in particular with the structure

of a compact topological group. One checks easily that to give a continuous operation of the profinite group G on M is the same as to give a morphism of topological groups

$$G \longrightarrow \mathrm{GL}(M),$$

where the latter group is endowed with the topology just described. This said, one evidently defines a functor

$$\varprojlim_{\mathbb{N}} : (\ell\mathbb{Z})\text{-}\,\mathrm{ad}(\mathrm{Abf}(\pi_1)) \longrightarrow \mathbb{Z}_\ell\text{-}\,\mathrm{mod}(\pi_1),$$

where $\mathbb{Z}_\ell\text{-}\,\mathrm{mod}(\pi_1)$ denotes the category of finite-type $\mathbb{Z}_\ell$-modules endowed with a continuous operation of π_1.

This allows us to state the following lemma.

Lemma 1.2.4.2. The preceding functor

$$\varprojlim_{\mathbb{N}} : (\ell\mathbb{Z})\text{-}\,\mathrm{ad}(\mathrm{Abf}(\pi_1)) \longrightarrow \mathbb{Z}_\ell\text{-}\,\mathrm{mod}(\pi_1)$$

is an equivalence of categories.

Proof. It is clear that it is essentially surjective: a finite-type $\mathbb{Z}_\ell$-module E on which π_1 acts continuously is the image of the projective system

$$(E/\ell^{n+1}E)_{n\in\mathbb{N}},$$

with the evident operations of π_1. Now let

$$F = (F_n)_{n\in\mathbb{N}}, \qquad G = (G_n)_{n\in\mathbb{N}}$$

be two objects of the category $(\ell\mathbb{Z})\text{-}\,\mathrm{ad}(\mathrm{Abf}(\pi_1))$, and, for simplicity, write $\mathrm{Hom}_{\pi_1}(F,G)$ for the set of morphisms from F to G in this category. Forgetting the action of π_1, one may regard F and G as objects of the category of ℓ-adic projective systems of finite abelian groups, and write $\mathrm{Hom}_0(F,G)$ for the set of morphisms from F to G in this category. The group π_1 acts on $\mathrm{Hom}_0(F,G)$ as follows: an element s of π_1 clearly defines two automorphisms, denoted in the same way, of F and G respectively in the category of ℓ-adic projective systems of finite abelian groups, and if u belongs to $\mathrm{Hom}_0(F,G)$, one puts

$$s \cdot u = sus^{-1}.$$

Under these conditions, it is clear that $\mathrm{Hom}_{\pi_1}(F,G)$ is canonically identified with the set of π_1-invariants in $\mathrm{Hom}_0(F,G)$, and similarly that $\mathrm{Hom}_{\pi_1}(F_n,G_n)$ is identified for every n with

$$\mathrm{Hom}_{\mathbb{Z}}(F_n,G_n)^{\pi_1},$$

the operation of π_1 being defined analogously. It follows, for example, from (Serre, Groupes proalgébriques, Prop. 12) that, denoting by M and N the projective limits of F and G respectively, one has

$$\mathrm{Hom}_{\mathbb{Z}_\ell}(M,N) = \varprojlim_{n} \mathrm{Hom}_{\mathbb{Z}}(F_n,G_n).$$

Taking the invariants of both sides under π_1 and using the fact that this operation, being associated with a projective limit, commutes with projective limits, one obtains the assertion.

Lemmas 1.2.4.1 and 1.2.4.2 allow one to state the following less precise proposition.

Proposition 1.2.5. The category of constructible twisted constant $\mathbb{Z}_\ell$-sheaves on a connected locally noetherian prescheme X is equivalent to the category of finite-type $\mathbb{Z}_\ell$-modules on which $\pi_1(X)$ acts continuously for the ℓ-adic topology.

Remark. Let F be a constructible twisted constant $\mathbb{Z}_\ell$-sheaf, and let M be the corresponding finite-type $\mathbb{Z}_\ell$-module endowed with a continuous operation of π_1. For F to be locally constant, it is clear that it is enough for π_1 to act on M through a finite quotient; this is not always the case (1.2.2). It is even necessary when X is normal. After decomposing X into connected components, one may, in order to see this, reduce to the case where X is integral. I claim that then, if K denotes its field of rational functions, and if there exists a finite separable extension K' of K such that the inverse image $F_{K'}$ of F above $\mathrm{Spec}(K')$ is constant, then the group $\pi_1(X)$ acts on M through a finite group. Indeed (SGA 1 V 8.2), $\pi_1(K) \to \pi_1(X)$ is surjective, and the hypothesis means that the kernel of the composite

$$\pi_1(K) \longrightarrow \pi_1(X) \longrightarrow \mathrm{Aut}(M)$$

has finite index. The same is therefore true for the kernel of $\pi_1(X) \to \mathrm{Aut}(M)$.

Proposition 1.2.6. Let X be a locally noetherian prescheme and let F be a projective system, indexed by $\mathbb{N}$, of abelian sheaves on $X_{\text{ét}}$. The following assertions are equivalent:

(i) F is a constructible $\mathbb{Z}_\ell$-sheaf;

(ii) every noetherian open U of X is a finite union of locally closed subsets $(Z_i)_{1\leq i\leq q}$ above which the inverse image of F is a constructible twisted constant $\mathbb{Z}_\ell$-sheaf;

(iii) X admits a covering by opens, each a finite union of locally closed subsets, above which the inverse image of F is a constructible twisted constant $\mathbb{Z}_\ell$-sheaf.

Proof. It is clear that (ii) implies (iii).

(iii) $\Longrightarrow$ (i): If F satisfies (iii), its geometric fibres are ℓ-adic, hence F is ℓ-adic by (SGA 4 VIII 3.6). Moreover, its components are constructible (SGA 4 IX 2.8).

To see that (i) implies (ii), one immediately reduces to the case where X is noetherian. If F is constructible ℓ-adic, its strict graded object $\mathrm{gr}^s(F)$ is a noetherian $(\mathrm{gr}_\ell\,\mathbb{Z}, \mathrm{Ab}(X))$-module (V 5.1.6). It follows (SGA 4 IX 2.9) that X is a finite union of locally closed subsets $(Z_i)_{1\leq i\leq q}$ above which the inverse image of $\mathrm{gr}^s(F)$ is a locally constant sheaf. Since, for a sheaf with finite fibres, the property of being locally constant is stable under extensions, one deduces that, for every integer n, the inverse images on each Z_i of the component F_n of F are locally constant sheaves.

1.3. Operations $\otimes$ and Hom on $\mathbb{Z}_\ell$-sheaves

1.3.1. Tensor product. Let F and G be two objects of the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$. If

$$F = (F_n)_{n\in\mathbb{N}} \quad \text{and} \quad G = (G_n)_{n\in\mathbb{N}},$$

the tensor product $F \otimes G$ of F and G is defined by the following formula on objects, and in the evident way on arrows:

$$(F \otimes G)_n = F_n \otimes G_n.$$

More precisely one clearly defines a right exact additive bifunctor, called the tensor product:

$$\otimes : \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \times \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)).$$

Proposition 1.3.2. If F and G are two constructible $\mathbb{Z}_\ell$-sheaves, then the tensor product $F \otimes G$ is a constructible $\mathbb{Z}_\ell$-sheaf.

Proof. The ℓ-adic property is clear. Moreover, it follows immediately from the definition of constructible sheaves (SGA 4 IX 2.3) that a tensor product of two constructible sheaves is constructible.

By restriction one therefore defines a right exact additive bifunctor, again called tensor product:

$$\otimes : \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X) \times \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X) \longrightarrow \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X).$$

1.3.3. Sheaf of homomorphisms

Let $F = (F_n)_{n\in\mathbb{N}}$ and $G = (G_n)_{n\in\mathbb{N}}$ be two constructible $\mathbb{Z}_\ell$-sheaves. Suppose that F is locally free, meaning that, for every integer n, the sheaf F_n is locally free over the constant sheaf

$$(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X.$$

For every integer n, the structural exact sequence

$$G_{n+1} \xrightarrow{\ell^{n+1}} G_{n+1} \longrightarrow G_n \longrightarrow 0$$

gives, since F_{n+1} is locally free over $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$, an exact sequence

$$\mathcal{H}om(F_{n+1}, G_{n+1}) \xrightarrow{\ell^{n+1}} \mathcal{H}om(F_{n+1}, G_{n+1}) \longrightarrow \mathcal{H}om(F_{n+1}, G_n) \longrightarrow 0. \tag{1}$$

Since $\ell^{n+1}G_n = 0$, the structural exact sequence

$$F_{n+1} \xrightarrow{\ell^{n+1}} F_{n+1} \longrightarrow F_n \longrightarrow 0$$

clearly defines an isomorphism

$$\mathcal{H}om(F_{n+1}, G_n) \xleftarrow{\sim} \mathcal{H}om(F_n, G_n). \tag{2}$$

Combining (1) and (2), one obtains, for every integer n, a morphism

$$\mathcal{H}om(F_{n+1}, G_{n+1}) \longrightarrow \mathcal{H}om(F_n, G_n).$$

Hence a projective system

$$\mathcal{H}om(F, G) = \big(\mathcal{H}om(F_n, G_n)\big)_{n\in\mathbb{N}},$$

which is clearly a $\mathbb{Z}_\ell$-sheaf.

1.3.4. Generalities on invertible ℓ-adic sheaves. Application to the sheaves $\mathbb{Z}_\ell(r)$ ($r \in \mathbb{Z}$)

Let L be a constructible $\mathbb{Z}_\ell$-sheaf. We shall say that it is invertible if, for every integer n, its n-th component L_n is an invertible $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$-module, i.e. locally free of rank one. This is equivalent to saying that L is locally free in the sense of the preceding paragraph and that the geometric fibres of the L_n are nonzero cyclic abelian groups.

If L is an invertible $\mathbb{Z}_\ell$-sheaf, we agree to put

$$L^{-1} = \mathcal{H}om(L, \mathbb{Z}_\ell).$$

It is clear that there is an isomorphism

$$L \otimes L^{-1} \xrightarrow{\sim} \mathbb{Z}_\ell,$$

deduced componentwise from the analogous isomorphism (EGA 0_I 5.4.3.1). Following (EGA 0_I 5.4.4), we agree to put $L^{\otimes 0} = \mathbb{Z}_\ell$ and, for $n \geq 1$,

$$L^{\otimes -n} = (L^{-1})^{\otimes n}.$$

With this notation, there is then a canonical functorial isomorphism

$$L^{\otimes m} \otimes L^{\otimes n} \xrightarrow{\sim} L^{\otimes (m+n)}$$

for all rational integers m and n. Moreover, this isomorphism has the usual associativity properties.

In particular, the graded objects

$$(L^{\otimes i})_{i \in \mathbb{N}} \quad \text{and} \quad (L^{\otimes i})_{i \in \mathbb{Z}}$$

are $\mathbb{Z}_\ell$-algebras, graded respectively by $\mathbb{N}$ and $\mathbb{Z}$.

Example. The $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$ defined in (1.2.2) is invertible, and in what follows we shall agree to put

$$\mathbb{Z}_\ell(r) = \big(\mathbb{Z}_\ell(1)\big)^{\otimes r} \qquad (r \in \mathbb{Z}).$$

1.4. Various categories constructed from the category of constructible $\mathbb{Z}_\ell$-sheaves

1.4.1. Constructible A-sheaves (A a finite $\mathbb{Z}_\ell$-algebra). Let A be a finite $\mathbb{Z}_\ell$-algebra. Define a category denoted

$$A\text{-}\operatorname{fc}(X),$$

whose objects are called constructible A-sheaves, as follows.

(i) An object of $A\text{-}\operatorname{fc}(X)$ is a pair (F, u) consisting of a constructible $\mathbb{Z}_\ell$-sheaf F and a morphism of $\mathbb{Z}_\ell$-algebras

$$u : A \longrightarrow \operatorname{End}(F).$$

(ii) If (F, u) and (F', u') are two constructible A-sheaves, an arrow $(F, u) \to (F', u')$ is a morphism of $\mathbb{Z}_\ell$-sheaves

$$f : F \longrightarrow F'$$

such that, for every element a of A, the diagram below is commutative:

$$\begin{array}{ccc} F & \xrightarrow{f} & F' \\ {\scriptstyle u(a)}\downarrow & & \downarrow{\scriptstyle u'(a)} \\ F & \xrightarrow[f]{} & F'. \end{array}$$

A constructible twisted constant A-sheaf is a constructible A-sheaf whose underlying constructible $\mathbb{Z}_\ell$-sheaf is twisted constant. The category of constructible twisted constant A-sheaves is the full subcategory of the category of constructible A-sheaves generated by the constructible twisted constant A-sheaves.

It follows immediately from (1.2.4) that, if X is connected, the category of constructible twisted constant A-sheaves is equivalent to the category of finite-type A-modules endowed with a continuous A-linear operation of $\pi_1(X)$ on the underlying finite-type $\mathbb{Z}_\ell$-module (1.2.4).

1.4.2. Constructible $\mathbb{Q}_\ell$-sheaves. It is clear that the full subcategory of $\mathbb{Z}_\ell$- fc(X) generated by the constructible $\mathbb{Z}_\ell$-sheaves of torsion, i.e. killed by a power of ℓ, is thick.

Definition 1.4.3. The category of constructible $\mathbb{Q}_\ell$-sheaves on X, denoted

$$\mathbb{Q}_\ell\text{- fc}(X),$$

is the quotient abelian category, in the sense of Gabriel's thesis, of $\mathbb{Z}_\ell$- fc(X) by the thick abelian subcategory of $\mathbb{Z}_\ell$-sheaves of torsion. An object of $\mathbb{Q}_\ell$- fc(X) is called a constructible $\mathbb{Q}_\ell$-sheaf.

The category $\mathbb{Q}_\ell$- fc(X) admits the following simple description. It has the same set of objects as $\mathbb{Z}_\ell$- fc(X), and, if F and G are two such objects,

$$\mathrm{Hom}_{\mathbb{Q}_\ell}(F, G) = \mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \mathrm{Hom}_{\mathbb{Z}_\ell}(F, G),$$

where $\mathrm{Hom}_{\mathbb{Z}_\ell}(F, G)$ and $\mathrm{Hom}_{\mathbb{Q}_\ell}(F, G)$ denote respectively the morphisms in $\mathbb{Z}_\ell$- fc(X) and $\mathbb{Q}_\ell$- fc(X). Indeed, if

$$u : \mathbb{Z}_\ell\text{- fc}(X) \longrightarrow D$$

denotes an exact functor from $\mathbb{Z}_\ell$- fc(X) to an abelian category D, then to say that u makes multiplication by the elements of $\mathbb{Z}_\ell$ invertible is the same as to say that u vanishes on the torsion objects; this immediately shows that the abelian category described above by objects and morphisms solves the universal problem of passage to the quotient.

A constructible $\mathbb{Q}_\ell$-sheaf is said to be twisted constant if it is the image of a constructible twisted constant $\mathbb{Z}_\ell$-sheaf. The category of constructible twisted constant $\mathbb{Q}_\ell$-sheaves is the full subcategory of

$$\mathbb{Q}_\ell\text{- fc}(X)$$

generated by the constructible twisted constant $\mathbb{Q}_\ell$-sheaves.

It follows from (1.2.4) that, if X is connected, the category of constructible twisted constant $\mathbb{Q}_\ell$-sheaves is equivalent to the category of finite-dimensional $\mathbb{Q}_\ell$-vector spaces V endowed with a continuous operation of $\pi_1(X)$, i.e. a continuous map

$$u : \pi_1(X) \times V \longrightarrow V$$

satisfying the properties made explicit in (1.2.4), which in particular implies that $u(g, v)$ is $\mathbb{Q}_\ell$-linear in v for every $g \in \pi_1(X)$.

It is immediate to see that this is equivalent to giving a morphism of topological groups

$$\pi_1(X) \longrightarrow \mathrm{GL}(V),$$

the latter being endowed with its natural topology as an ℓ-adic analytic group.

1.4.3. Constructible L-sheaves (L a finite $\mathbb{Q}_\ell$-algebra). Let L be a finite $\mathbb{Q}_\ell$-algebra. In a manner analogous to what was done in (1.4.1) starting from constructible $\mathbb{Z}_\ell$-sheaves, one defines, starting from the category of constructible $\mathbb{Q}_\ell$-sheaves, the category of constructible L-sheaves, and then the category of constructible twisted constant L-sheaves. If X is connected, the category of constructible twisted constant L-sheaves is equivalent to the category of finite-dimensional L-vector spaces endowed with a continuous operation of $\pi_1(X)$, i.e. a continuous operation of $\pi_1(X)$ on the underlying $\mathbb{Q}_\ell$-vector space, commuting with the operation of L.

1.4.4

If A is a finite commutative $\mathbb{Z}_\ell$-algebra and F, G are two constructible A-sheaves, the projective system

$$F \otimes_A G = (F_n \otimes_{A_X} G_n)_{n\in\mathbb{N}},$$

with the evident transition morphisms, is a $\mathbb{Z}_\ell$-sheaf. Moreover A acts evidently on each of its components, which allows it to be endowed naturally with the structure of an A-sheaf. Thus one defines a right exact bifunctor, called tensor product,

$$A\text{-}\,\mathrm{fc}(X) \times A\text{-}\,\mathrm{fc}(X) \longrightarrow A\text{-}\,\mathrm{fc}(X).$$

Now let F and G be two A-sheaves and suppose that F is locally free, i.e. that, for every integer n, F_n is a locally free $(A/\ell^{n+1}A)_X$-module. Then, as in (1.3.2), one sees that the projective system

$$\mathcal{H}om_A(F, G) = \left(\mathcal{H}om_{A_n}(F_n, G_n)\right)_{n\in\mathbb{N}},$$

with the evident transition morphisms, is a $\mathbb{Z}_\ell$-sheaf. Moreover, since A acts naturally on its components, it is canonically endowed with the structure of an A-sheaf.

As in (1.3.3), one defines the notion of an invertible A-sheaf and, if L is such an A-sheaf, gives a meaning to the expression $L^{\otimes r}$ for every natural integer r. Paragraph (1.3.3) adapts without difficulty to this situation.

1.4.5

Let F be a constructible $\mathbb{Z}_\ell$-sheaf, and let p be the canonical functor

$$\mathbb{Z}_\ell\text{-}\operatorname{fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-}\operatorname{fc}(X).$$

The composite of p with the functor $G \mapsto F \otimes G$ makes multiplication by the elements of $\mathbb{Z}_\ell$ invertible and hence defines a functor

$$\mathbb{Q}_\ell\text{-}\operatorname{fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-}\operatorname{fc}(X),$$

which is right exact. Applying the same argument to the first variable, one thus defines a right exact bifunctor, called tensor product,

$$\mathbb{Q}_\ell\text{-}\operatorname{fc}(X) \times \mathbb{Q}_\ell\text{-}\operatorname{fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-}\operatorname{fc}(X).$$

If X is connected and F and G are two constructible twisted constant $\mathbb{Q}_\ell$-sheaves corresponding to representations of $\pi_1(X)$ on vector spaces V and W, respectively, then the tensor product $F \otimes G$ is again locally constant and corresponds to the diagonal representation of $\pi_1(X)$ on $V \otimes W$.

1.4.6

Let F be a locally free constructible $\mathbb{Z}_\ell$-sheaf (1.3.3). The composite of p with the functor $G \mapsto \mathcal{H}om(F, G)$ (1.3.2) makes multiplication by elements of $\mathbb{Z}_\ell$ invertible and hence defines a functor

$$\mathcal{H}om(F, \cdot) : \mathbb{Q}_\ell\text{-}\operatorname{fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-}\operatorname{fc}(X).$$

If F and F' are two locally free constructible $\mathbb{Z}_\ell$-sheaves defining isomorphic $\mathbb{Q}_\ell$-sheaves, then there exists a morphism of $\mathbb{Z}_\ell$-sheaves

$$u : F \longrightarrow F'$$

whose kernel and cokernel are torsion. This immediately shows that the functors $\mathcal{H}om(F, \cdot)$ and $\mathcal{H}om(F', \cdot)$ from $\mathbb{Q}_\ell\text{-}\operatorname{fc}(X)$ to itself are isomorphic.

Now let F be a constructible twisted constant $\mathbb{Q}_\ell$-sheaf. If we again denote by F the corresponding $\mathbb{Z}_\ell$-sheaf, it follows from the interpretation of the restrictions of F to the connected components Y of X as continuous $\pi_1(Y)$-modules that F may be written uniquely, up to isomorphism, in the form

$$F = P \oplus T,$$

where P is locally free and T is torsion. This allows one to define, for every $\mathbb{Q}_\ell$-sheaf G, a $\mathbb{Q}_\ell$-sheaf

$$\mathcal{H}om(F, G) = \mathcal{H}om(P, G).$$

An isomorphism $u : F \to F'$ of $\mathbb{Q}_\ell$-sheaves defines, for every G, an isomorphism

$$\mathcal{H}om(F', G) \xrightarrow{\sim} \mathcal{H}om(F, G).$$

More precisely, one thereby defines a bifunctor

$$\mathcal{H}om : (\mathbb{Q}_\ell\text{-}\operatorname{fct}(X))^\circ \times \mathbb{Q}_\ell\text{-}\operatorname{fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-}\operatorname{fc}(X),$$

where $\mathbb{Q}_\ell\text{-}\operatorname{fct}(X)$ denotes the category of constructible twisted constant $\mathbb{Q}_\ell$-sheaves. This bifunctor has the usual exactness properties.

1.4.7

Let L now be a finite-type extension of $\mathbb{Q}_\ell$, and let A denote the integral closure of $\mathbb{Z}_\ell$ in L, so that A is a free $\mathbb{Z}_\ell$-algebra of finite type and a Dedekind ring. The category of constructible L-sheaves is isomorphic to the quotient category of the category of constructible A-sheaves by the category of constructible torsion A-sheaves. This allows one, as in (1.4.6), to define from the tensor product of (1.4.4) a tensor product:

$$L\text{-}\operatorname{fc}(X) \times L\text{-}\operatorname{fc}(X) \longrightarrow L\text{-}\operatorname{fc}(X).$$

Since A is a Dedekind ring, the argument used in (1.4.6) shows that every twisted constant A-sheaf is the direct sum of a locally free A-sheaf and a torsion A-sheaf. Denoting by $L\text{-}\operatorname{fct}(X)$ the category of constructible twisted constant L-sheaves, one defines, again as in (1.4.6), a bifunctor

$$\mathcal{H}om : (L\text{-}\operatorname{fct}(X))^\circ \times L\text{-}\operatorname{fc}(X) \longrightarrow L\text{-}\operatorname{fc}(X),$$

with the usual exactness properties.

1.5. Constructible $(AR, \mathbb{Z}_\ell)$-sheaves

Definition 1.5.1. Let X be a locally noetherian prescheme. An object F of $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$ is called a constructible $(AR, \mathbb{Z}_\ell)$-sheaf if X admits a covering

$$(U_i)_{i\in I}$$

by noetherian opens such that the restriction of F above each U_i is a $(AR, \ell\mathbb{Z})$-noetherian object (V 1.3).

It follows immediately from (V 5.2.1) that the full subcategory of $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$ generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves is stable under kernels and cokernels. In particular, it is an abelian category.

1.5.2. Constructible $(AR, \mathbb{Z}_\ell)$-sheaves which are locally AR-null

It is clear that an AR-null object (V 2.2.1) of $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$ is a constructible $(AR, \mathbb{Z}_\ell)$-sheaf if and only if its components are constructible. It follows in particular from (V 2.2.2) and (SGA 4 IX 2.6) that the category of AR-null constructible $(AR, \mathbb{Z}_\ell)$-sheaves is a thick abelian subcategory of the full category generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves in $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$.

Agreeing to say that an object P of $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$ is locally AR-null if there exists an open covering of X for the étale topology (or, equivalently, for the Zariski topology) above each open of which P is AR-null, one sees, by reducing through localization to the preceding case, that the category of locally AR-null constructible $(AR, \mathbb{Z}_\ell)$-sheaves is a thick abelian subcategory of that of constructible $(AR, \mathbb{Z}_\ell)$-sheaves.

Observe that, if X is noetherian, the constructible $(AR, \mathbb{Z}_\ell)$-sheaves are the noetherian $(AR, \ell\mathbb{Z})$-adic objects of $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$, and locally AR-null is equivalent to AR-null.

Definition 1.5.3. Let X be a locally noetherian prescheme. The category of constructible $(AR, \mathbb{Z}_\ell)$-sheaves on X, denoted

$$(AR, \mathbb{Z}_\ell)\text{-}\operatorname{fc}(X),$$

is the quotient abelian category of the full subcategory generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves in

$$\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$$

by the full thick subcategory generated by the locally AR-null constructible $(AR, \mathbb{Z}_\ell)$-sheaves.

Notation 1.5.4. Whenever useful, $U(X)$ will denote the full subcategory of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$ generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves.

Since every constructible $\mathbb{Z}_\ell$-sheaf is in particular a constructible $(AR, \mathbb{Z}_\ell)$-sheaf, one defines in the evident way a functor

$$u : \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X) \longrightarrow (AR, \mathbb{Z}_\ell)\text{-}\,\mathrm{fc}(X).$$

Proposition 1.5.5. The above functor u is an equivalence of categories.

Proof. To prove that u is fully faithful and essentially surjective, one reduces "by gluing" as in the proof of 1.1.3 to the analogous statement (V 5.1.3).

2. Formalism of ℓ-adic cohomology

2.1. Inverse image

Let

$$f : X \longrightarrow Y$$

be a morphism of locally noetherian preschemes. The inverse-image functor

$$f^* : \mathrm{Ab}(Y) \longrightarrow \mathrm{Ab}(X)$$

is exact and therefore extends in the evident way to an exact functor, denoted in the same way,

$$f^* : \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(Y)) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)).$$

This latter functor evidently carries constructible $\mathbb{Z}_\ell$-sheaves to constructible $\mathbb{Z}_\ell$-sheaves, and locally AR-null projective systems to locally AR-null projective systems. With the notation of (1.5.4), it therefore induces two "inverse image" functors

$$f^* : \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(Y) \longrightarrow \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X),$$

$$f^* : U(Y) \longrightarrow U(X),$$

the second of which is exact.

The equivalence of categories (1.5.5), together with the fact that f^* carries locally AR-null projective systems to locally AR-null projective systems, then shows that the inverse-image functor

$$f^* : \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(Y) \longrightarrow \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X)$$

is also exact.

If $g : Y \to Z$ is another morphism of locally noetherian preschemes, the functors f^* and g^* are evidently related by an isomorphism

$$(g \circ f)^* \xrightarrow{\sim} f^* \circ g^*,$$

with the usual associativity property (SGA 1 VI 7 ff.).

2.2. Higher direct images

Let

$$f : X \longrightarrow Y$$

be a separated morphism of finite type between locally noetherian preschemes. Recall the following result of étale cohomology (SGA 4 XVII; for the proper case, cf. SGA 4 XIV 1.1).

Lemma 2.2.1. Let A be a commutative ring annihilated by a nonzero integer n, and let M be a constructible $(A, \mathrm{Ab}(X))$-module. For every integer i, the sheaf

$$R^i_! f(M)$$

is a constructible $(A, \mathrm{Ab}(Y))$-module.

The following lemma follows from this by applying (V 5.3.1) after restriction over the noetherian opens of Y.

Lemma 2.2.2. For every integer i and every constructible $(AR, \mathbb{Z}_\ell)$-sheaf F on X, the projective system

$$R^i_! f(F)$$

is a constructible $(AR, \mathbb{Z}_\ell)$-sheaf.

This said, the exact δ-functor

$$(R^i_! f)_{i\in\mathbb{Z}} : \mathrm{Ab}(X) \longrightarrow \mathrm{Ab}(Y)$$

extends in the evident way to an exact δ-functor, denoted in the same way,

$$\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(Y)),$$

which by restriction gives an exact δ-functor

$$U(X) \longrightarrow U(Y). \tag{1}$$

Moreover, the functors $R^i_! f$ carry locally AR-null projective systems to locally AR-null projective systems; hence, after passage to the quotient, the δ-functor (1) gives an exact δ-functor

$$(R^i_! f)_{i\in\mathbb{Z}} : (AR, \mathbb{Z}_\ell)\text{- fc}(X) \longrightarrow (AR, \mathbb{Z}_\ell)\text{- fc}(Y).$$

Using now the equivalence of categories (1.5.5), this latter δ-functor is interpreted as an exact δ-functor

$$\mathbb{Z}_\ell\text{- fc}(X) \longrightarrow \mathbb{Z}_\ell\text{- fc}(Y).$$

Definition 2.2.3. The functor

$$R^i_! f : \mathbb{Z}_\ell\text{- fc}(X) \longrightarrow \mathbb{Z}_\ell\text{- fc}(Y)$$

is called the i-th direct image functor with proper supports of f.

Remark. Assuming the analogue of Lemma (2.2.1) proved for the ordinary higher direct images when X and Y are excellent, which is done in characteristic zero (SGA 4 XIX 5.1), one defines in

the same way higher direct images in ℓ-adic cohomology in that case. One could also develop the result of the following paragraph in this framework.

Proposition 2.2.4 (Leray spectral sequence). Let $f : X \to Y$ and $g : Y \to Z$ be two separated morphisms of finite type between locally noetherian preschemes, and let F be a constructible $\mathbb{Z}_\ell$-sheaf, respectively a constructible $(AR, \mathbb{Z}_\ell)$-sheaf. There exists in $\mathbb{Z}_\ell$- fc(Z) a biregular spectral sequence (EGA 0_{III} 11.1.3), functorial in F:

$$R^i_!g(R^j_!f(F)) \Longrightarrow R^{i+j}_!(g \circ f)(F).$$

Proof. By the equivalence of categories (1.5.5) and the way in which higher direct images are defined for constructible $\mathbb{Z}_\ell$-sheaves, it is enough to show the analogous statement for constructible $(AR, \mathbb{Z}_\ell)$-sheaves. For any abelian category C, let us agree to denote by $\mathrm{Sp}(C)$ the category of spectral sequences beginning at degree 2 with values in C. Associating to every object F of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$ the ordinary Leray spectral sequence, obtained by extension to projective systems,

$$R^i_!g(R^j_!f(F)) \Longrightarrow R^{i+j}_!(g \circ f)(F), \tag{S}$$

it is clear that one obtains a functor

$$\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \longrightarrow \mathrm{Sp}(\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(Z))),$$

which by restriction gives, by (2.2.1) and because $U(Z)$ is thick in $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(Z))$, a functor

$$U(X) \longrightarrow \mathrm{Sp}(U(Z)). \tag{1}$$

Since the canonical functor

$$U(Z) \longrightarrow (AR, \mathbb{Z}_\ell)\text{- fc}(Z)$$

is exact, (1) yields an additive functor

$$E : U(X) \longrightarrow \mathrm{Sp}((AR, \mathbb{Z}_\ell)\text{- fc}(Z)). \tag{2}$$

For every object F of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$, the associated spectral sequence (S) is biregular: this follows from the analogous assertion for the components of the projective system F and from the fact that the spectral sequence under consideration is supported in the first quadrant. This shows that, for every object of $U(X)$, the corresponding spectral sequence is biregular.

Moreover, if $u : F \to G$ is a morphism of $U(X)$ whose kernel and cokernel are locally AR-null, then, because the higher direct images carry locally AR-null projective systems to locally AR-null projective systems, $E(u)$ is an isomorphism on the terms E_2^{pq}; it is therefore an isomorphism, since the spectral sequences considered are biregular. It follows that (2) defines, by passage to the quotient, a functor

$$(AR, \mathbb{Z}_\ell)\text{- fc}(X) \longrightarrow \mathrm{Sp}((AR, \mathbb{Z}_\ell)\text{- fc}(Z)), \tag{3}$$

which associates in a precise way to every constructible $(AR, \mathbb{Z}_\ell)$-sheaf F a biregular spectral sequence

$$R^i_!g(R^j_!f(F)) \Longrightarrow R^{i+j}_!(g \circ f)(F).$$

2.2.3. Other properties

A) Cohomological dimension. Let $f : X \to Y$ be a separated morphism of finite type between locally noetherian preschemes. Suppose that the relative dimension of f is at most an integer d. Then, for every constructible $(AR, \mathbb{Z}_\ell)$-sheaf F,

$$R^i_! f(F) = 0 \qquad \text{for } i > 2d.$$

This follows from the analogous assertion for the components of F (SGA 4 X 4.1 and 5.2).

B) Proper base change. Consider a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y. \end{array}$$

Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms of locally noetherian preschemes, with f separated of finite type. Let $X' = X \times_Y Y'$, which is locally noetherian since f is of finite type, and let

$$f' : X' \to Y', \qquad g' : X' \to X$$

be the canonical morphisms. For every integer i and every constructible $\mathbb{Z}_\ell$-sheaf F, there exists an isomorphism in $\mathbb{Z}_\ell$- fc(Y'), functorial in F:

$$w_i : g^*(R^i_! f(F)) \xrightarrow{\sim} R^i_! f'(g'^* F).$$

Moreover, if

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

is an exact sequence of $\mathbb{Z}_\ell$- fc(X) (resp. of $(AR, \mathbb{Z}_\ell)$- fc(X)), then the $(w_i)_{i\in\mathbb{N}}$ commute with boundary operators; in other words, the diagrams below are commutative:

$$\begin{array}{ccc} g^*(R^i_! f(F'')) & \xrightarrow{w_i} & R^i_! f'(g'^* F'') \\ {\scriptstyle g^*(\delta)}\downarrow & & \downarrow{\scriptstyle \delta} \\ g^*(R^{i+1}_! f(F')) & \xrightarrow[w_{i+1}]{} & R^{i+1}_! f'(g'^* F'). \end{array}$$

Proof. It is enough to prove the analogous assertion for constructible $(AR, \mathbb{Z}_\ell)$-sheaves. But it is deduced by passage to the quotient from the analogous assertion for projective systems of torsion sheaves, which is an immediate consequence of the proper base-change theorem (cf. SGA 4 XII 5.1 in the proper case and SGA 4 XVII in the general case).

C) Exact sequence relative to an open and its complement. Let $f : X \to Y$ be a compactifiable morphism of locally noetherian preschemes, let U be an open of X, and let Z be the closed complement. For every constructible $\mathbb{Z}_\ell$-sheaf F, there exists an exact sequence in $\mathbb{Z}_\ell$- fc(Y), functorial in F:

$$\cdots \longrightarrow R^i_! f_U(F/U) \longrightarrow R^i_! f(F) \longrightarrow R^i_! f_Z(F/Z) \longrightarrow R^{i+1}_! f_U(F/U) \longrightarrow \cdots .$$

Proof. Analogous to that of B).

2.2.5

The formalism developed in this paragraph for $\mathbb{Z}_\ell$-sheaves extends without difficulty to A-sheaves, where A is a finite $\mathbb{Z}_\ell$-algebra, and to L-sheaves, where L is a finite extension of $\mathbb{Q}_\ell$. For example, in the case of $\mathbb{Q}_\ell$-sheaves, the fundamental point in the proofs is that all the functors considered for $\mathbb{Z}_\ell$-sheaves carry $\mathbb{Z}_\ell$-sheaves of torsion to $\mathbb{Z}_\ell$-sheaves of torsion.

3. ℓ-adic cohomology class associated with a cycle

3.1

If X is a proper scheme, or a smooth finite-type scheme, over an algebraically closed field k, one knows that, for every constructible abelian torsion sheaf M prime to the characteristic of k, the cohomology groups

$$H^i(X, M)$$

are abelian groups of finite length. This permits (2.2.1, Remark) the definition, for every $\mathbb{Z}_\ell$-sheaf F, with ℓ prime to the characteristic of k, of a noetherian (AR, ℓ)-adic projective system of abelian groups

$$H^i(X, F)$$

with the usual properties. In particular, the method used in number 2 makes it possible to define the cup-product, the inverse-image homomorphism for a morphism of schemes over k, and the direct-image homomorphism for a proper morphism of smooth schemes over k (cf. Exposé IV for the definitions in the case of torsion sheaves).

Moreover, one knows (1.2.4) that the category of $\mathbb{Z}_\ell$-sheaves on an algebraically closed field is equivalent to the category of finite-type $\mathbb{Z}_\ell$-modules, the equivalence being supplied by the projective-limit functor. In practice it will be convenient to identify the finite-type $\mathbb{Z}_\ell$-module $H^i(X, F)$ with the corresponding $\mathbb{Z}_\ell$-sheaf.

3.2

Let ℓ be a prime number $\neq \operatorname{car} k$, and let X be a smooth scheme of finite type over k. If Z is an algebraic cycle of codimension d on X, then in Exposé IV[21] one associated to it, for every integer n, a cohomology class

$$\gamma_X^{(n)}(Z) \in H^{2d}(X, (\mu_X^{(n)})^{\otimes d}),$$

where $\mu_X^{(n)}$ denotes the locally free $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$-module of the ℓ^{n+1}-st roots of unity. Moreover, one saw that, if $n' \geq n$, $\gamma_X^{(n)}(Z)$ is the image of $\gamma_X^{(n')}(Z)$ in

$$H^{2d}(X, (\mu_X^{(n)})^{\otimes d})$$

by the morphism

$$H^{2d}(X, (\mu_X^{(n')})^{\otimes d}) \longrightarrow H^{2d}(X, (\mu_X^{(n)})^{\otimes d})$$

deduced from raising to the power $\ell^{n'-n}$:

$$\mu_X^{(n')} \longrightarrow \mu_X^{(n)}.$$

[21] See (SGA $4^{1/2}$ Cycle).

Thus one associates to Z an element, denoted $\gamma_X(Z)$:

$$\gamma_X(Z) \in H^{2d}(X, \mathbb{Z}_\ell(d)),$$

where $\mathbb{Z}_\ell(d)$ is defined in (1.3.4). More generally, if Z is a cycle on X, one associates to it by linearity an ℓ-adic cohomology class in

$$\bigoplus_d H^{2d}(X, \mathbb{Z}_\ell(d)),$$

which is in fact the projective limit of the cohomology classes associated with Z in

$$\bigoplus_d H^{2d}(X, (\mu_X^{(n)})^{\otimes d}).$$

3.3

By passage to the limit, it is evident that the ℓ-adic cohomology classes of the preceding paragraph satisfy the same formal properties as the cohomology classes studied in Exposé IV, namely:

(i) If $f : X \to Y$ is a proper morphism of smooth preschemes over k and Z is a cycle on X, then

$$f_*(\gamma_X(Z)) = \gamma_Y(f_*(Z)). \qquad (*)$$

(ii) If $f : X \to Y$ is a morphism of smooth preschemes over k and Z is a cycle on Y, then

$$\gamma_X(f^*Z) = f^*(\gamma_Y(Z)). \qquad (**)$$

(iii) If X is a smooth prescheme over k and Z_1 and Z_2 are two cycles on X, then

$$\gamma_X(Z_1 \cdot Z_2) = \gamma_X(Z_1) \cdot \gamma_X(Z_2), \qquad (**)$$

where the dot on the left-hand side denotes the intersection product of cycles, and the one on the right-hand side the cup-product in cohomology.

(iv) Two algebraically equivalent cycles on a smooth k-prescheme have the same ℓ-adic cohomology class.

($*$) This compatibility is not proved in (SGA $4^{1/2}$ Cycle), but is not difficult to deduce from the results of loc. cit.

($**$) In the non-excessive case, see (SGA $4^{1/2}$ Cycle 2.3.8) for a precise statement.

Exposé VII

COHOMOLOGY OF SOME CLASSICAL SCHEMES AND THE COHOMOLOGICAL THEORY OF CHERN CLASSES

by

J.-P. Jouanolou

Throughout the text, an integer $\nu > 0$ is fixed and we put

$$A = \mathbb{Z}/\nu\mathbb{Z};$$

the residual characteristics of all schemes considered are assumed to be prime to the integer ν. If X is such a scheme, μ_X denotes the sheaf of ν-th roots of unity and $A^\bullet(X)$ denotes the cohomology ring

$$\bigoplus_i H^{2i}(X, \mu_X^{\otimes i}),$$

with multiplication given by cup-product.

1. Vector bundles

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module, understood to be of finite type. Put $X = \mathrm{V}(E)$ and $U = \mathrm{V}(E)^*$, the complement of the zero section, and denote by

$$p : X \to S, \qquad q : U \to S$$

the canonical projections.

Proposition 1.1. Let L be an A-module.

(i) a) For every integer $j > 0$, one has

$$R^j p_*(p^*L) = 0.$$

b) The adjunction morphism

$$L \longrightarrow p_* p^*(L)$$

is an isomorphism.

(ii) Suppose that E has constant rank r.

a)

$$R^i_! p(p^*L) = 0 \quad \text{if } i \neq 2r.$$

b) The trace morphism (SGA 4 XVIII)

$$R^{2r}_! p(p^*L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

is an isomorphism.

Proof. Part (i) follows immediately by localization for the Zariski topology from (SGA 4 XV 2.2). Assertion (ii) is essentially contained in (SGA 4 XVIII).

Corollary 1.2. Let L be an A_S-module. For every integer i, the inverse-image morphism

$$H^i(S, L) \longrightarrow H^i(X, p^*L)$$

is an isomorphism.

Proof. By (1.1), the adjunction morphism $L \to Rp_*(p^*L)$ is an isomorphism. Since $H^*(X, p^*L) \simeq H^*(S, Rp_* p^*(L))$, the assertion follows.

Proposition 1.3. Suppose that E has constant rank r. For every A_S-module L, the following properties hold.

(i) a)
$$R^i q_*(q^*L) = 0 \quad \text{if } i \neq 0, 2r-1.$$

b) The adjunction morphism
$$L \longrightarrow q_* q^*(L)$$
is an isomorphism.

c) If Y denotes the complement of U in X, then the morphism
$$R^{2r-1} q_*(q^*L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$
obtained as the composite of the boundary morphism
$$R^{2r-1} q_*(q^*L) \longrightarrow R^{2r}_Y p_*(p^*L) \tag{1}$$
and the purity isomorphism (SGA 4 XVIII)
$$R^{2r}_Y p_*(p^*L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$
is an isomorphism.

(ii) a)
$$R^i_! q(q^*L) = 0 \quad \text{if } i \neq 1, 2r.$$

b) The boundary morphism
$$L \longrightarrow R^1_! q(q^*L)$$
coming from the unlimited exact sequence, where $f : Y \to S$ is the canonical projection,
$$\cdots \longrightarrow R^i_! q(q^*L) \longrightarrow R^i_! p(p^*L) \longrightarrow R^i f_*(f^*L) \longrightarrow R^{i+1}_! q(q^*L) \longrightarrow \cdots$$
is an isomorphism.

c) The trace morphism
$$R^{2r}_! q(q^*L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$
is an isomorphism.

Proof. Let $\alpha : U \to X$ and $\beta : Y \to X$ be the canonical immersions. One knows (SGA 4 XVI 3.7) that, putting $F = p^*(L)$,
$$F \xrightarrow{\sim} \alpha_* \alpha^*(F) \quad \text{and} \quad (R^j \alpha_*)\alpha^*(F) = 0 \quad \text{if } j \neq 0, 2r-1.$$
Moreover, the sheaves $(R^j \alpha_*)\alpha^* F$ for $j \geq 1$ have support in Y, and hence are acyclic for the functor p_*. From these remarks one obtains assertions (i) a) and b) by using the Leray spectral sequence
$$R^i p_*(R^j \alpha_*)(q^*L) \Longrightarrow R^n q_*(q^*L).$$
Assertion (i) c) follows from (1.1 (i)) and from the unlimited exact sequence
$$\cdots \longrightarrow R^i_Y p_*(p^*L) \longrightarrow R^i p_*(p^*L) \longrightarrow R^i q_*(q^*L) \longrightarrow R^{i+1}_Y p_*(p^*L) \longrightarrow \cdots .$$
The sheaf $R^{2r}_Y p_*(p^*L)$ is associated with the presheaf
$$U \longmapsto H^{2r}_{Y \times_S U}(X \times_S U, p^*L).$$

We prove (ii). Assertion c) comes from the fact that q is smooth and has connected geometric fibres of dimension r. The exact sequence mentioned in b) is obtained by applying the functor $R_!p$ to the canonical exact sequence

$$0 \longrightarrow \alpha_!(q^*L) \longrightarrow p^*L \longrightarrow \beta_!(\beta^*p^*L) \longrightarrow 0.$$

Using (1.1 (i)), one obtains assertions (ii) a) and b) without difficulty from this unlimited exact sequence.

Remark 1.4. For $S = \mathrm{Spec}(\mathbb{C})$, (1.3) says that the complement of the origin in $\mathbb{C}^r$ has the same cohomology as the real sphere of dimension $2r-1$, a deformation retract of $\mathbb{C}^r - 0$.

Corollary 1.5 (Gysin exact sequence). The Leray spectral sequence

$$H^i(S, R^jq_*(q^*L)) \Rightarrow H^{i+j}(U, q^*L)$$

defines an unlimited exact sequence, called the Gysin exact sequence,

$$\cdots \longrightarrow H^i(S, L) \longrightarrow H^i(U, q^*L) \longrightarrow H^{i-2r+1}(S, L \otimes \mu_S^{\otimes -r}) \longrightarrow H^{i+1}(S, L) \longrightarrow \cdots .$$

Proof. Lemma of two lines.

2. Projective schemes

In what follows, the mode of exposition adopted is based on a suggestion of L. Illusie, and the method followed is close to a method of P. Deligne for proving the degeneration of certain spectral sequences [6].

2.1

Let X and Y be two schemes and let $f : X \to Y$ be a morphism. We shall define an arrow

$$Rf_*(A_X) \otimes^L Rf_*(A_X) \longrightarrow Rf_*(A_X) \tag{2.1.1}$$

in $D(Y)$, having the usual associativity and commutativity properties, and therefore endowing $Rf_*(A_X)$ with a structure of generalized algebra. For this, write

$$C^*(X, A_X)$$

for the Godement resolution of A_X (SGA 4 XVII). Proceeding as in (TF 6.6), one defines a morphism of resolutions

$$C^*(X, A_X) \otimes C^*(X, A_X) \longrightarrow C^*(X, A_X \otimes A_X).$$

Composing it with the morphism

$$C^*(X, A_X \otimes A_X) \longrightarrow C^*(X, A_X)$$

deduced from the multiplication of A_X, one finally obtains a morphism

$$C^*(X, A_X) \otimes C^*(X, A_X) \longrightarrow C^*(X, A_X). \tag{2.1.2}$$

Applying the functor f_*, one obtains without difficulty a morphism of complexes of A_Y-modules

$$f_*C^*(X, A_X) \otimes f_*C^*(X, A_X) \longrightarrow f_*C^*(X, A_X),$$

and hence (2.1.1), using flat resolutions of the components of the expression on the left.

2.2

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of rank $r+1$. We denote by

$$P = P_S(E) \xrightarrow{p} S$$

the corresponding projective bundle. The canonical $\mathcal{O}_P$-module $\mathcal{O}_P(1)$ defines, as is known, by means of the Kummer exact sequence, an element

$$\xi \in H^2(P, \mu),$$

and we shall denote by η the image of ξ under the canonical morphism

$$H^2(P, \mu) \longrightarrow H^0(S, R^2p_*(\mu))$$

coming from the Leray spectral sequence

$$H^i(S, R^jp_*(\mu)) \Rightarrow H^{i+j}(P, \mu).$$

The class ξ is identified with an arrow $A_P \to \mu_P[2]$ in $D(P)$, which by transposition defines $\mu_P^{\otimes -1}[-2] \to A_P$, and hence by the usual adjunction an arrow

$$c : \mu_S^{\otimes -1}[-2] \longrightarrow Rp_*(A_P),$$

coinciding with η on cohomology in dimension 2.

By tensor product, the arrow c defines, by means of (2.2.1), arrows

$$c_i : \mu_S^{\otimes -i}[-2i] \longrightarrow Rp_*(A_P),$$

for every integer $i \geq 1$, and we agree to denote by c_0 the adjunction arrow

$$A_S \longrightarrow Rp_*(A_P).$$

Thus one defines an arrow

$$\gamma = \bigoplus_{i=0}^{r} c_i : \bigoplus_{i=0}^{r} \mu_S^{\otimes -i}[-2i] \longrightarrow Rp_*(A_P)$$

of $D(S)$.

Theorem 2.2.1. The arrow γ is an isomorphism.

Once the arrow γ has been defined, (2.2.1) is equivalent to the following corollary.

Proposition 2.2.2. Under the preceding hypotheses, one has:

a)
$$R^{2i+1}p_*(A_P) = 0 \quad \text{for every } i \in \mathbb{N}.$$

b) When $\bigoplus_i R^{2i}p_*(\mu_P^{\otimes i})$ is endowed with its A_S-algebra structure deduced from cup-product, the morphism of A_S-algebras
$$A_S[T] \longrightarrow \bigoplus_i R^{2i}p_*(\mu_P^{\otimes i})$$
defined by η has kernel the ideal (T^{r+1}) and hence defines an isomorphism of A_S-algebras
$$\theta : A_S[T]/(T^{r+1}) \xrightarrow{\sim} \bigoplus_i R^{2i}p_*(\mu_P^{\otimes i}).$$

c) The morphism

$$A_S \longrightarrow R^{2r}p_*(\mu^{\otimes r})$$

obtained by composing θ with the restriction to A_S of multiplication by T^r on $A_S[T]/(T^{r+1})$ is the inverse of the trace morphism (SGA 4 XVIII)

$$R^{2r}p_*(\mu^{\otimes r}) \longrightarrow A_S;$$

in other words,

$$\mathrm{Tr}(\eta^r) = 1.$$

Proof. Since the data are stable under base change, the proper base-change theorem (SGA 4 XII 5.1) and compatibility of the trace morphism with base change (SGA 4 XVIII) allow one to reduce to checking the analogous assertion on the fibres of P. In other words, one may suppose that S is the spectrum of a separably closed field k and that p is the canonical morphism $\mathbb{P}^r_k \to k$. In this case, the proof uses the following lemma.

Lemma 2.2.3. Let k be a separably closed field and let $p : \mathbb{P}^r_k \to k$ be the canonical projection. Then:

a)

$$R^{2i+1}p_*(A_P) = 0 \quad \text{for every } i.$$

b)

$$R^{2i}p_*(A_P) \simeq A_k \quad \text{for } i \in [0, r], \qquad R^{2i}p_*(A_P) = 0 \quad \text{if } i > r.$$

The lemma, manifestly true for $r = 0$, is proved by induction on r. Suppose therefore that it is true for $r-1$ and prove it for r. The projective space $\mathbb{P}^{r-1}_k$ embeds as a closed subspace in $\mathbb{P}^r_k$ with complement the open $\mathbb{E}^r_k$, whence an exact sequence of cohomology with proper supports

$$\cdots \longrightarrow H^i_!(\mathbb{E}^r_k) \longrightarrow H^i(\mathbb{P}^r_k) \longrightarrow H^i(\mathbb{P}^{r-1}_k) \longrightarrow H^{i+1}_!(\mathbb{E}^r_k) \longrightarrow \cdots .$$

The computation of cohomology with proper supports of $\mathbb{E}^r_k$ (1.1) shows that, if $i \neq (2r, 2r-1)$, the inverse-image map

$$H^i(\mathbb{P}^r_k) \longrightarrow H^i(\mathbb{P}^{r-1}_k)$$

is an isomorphism, whence the assertion in this case. If $i = 2r-1$, one similarly sees that $H^{2r-1}(\mathbb{P}^r_k)$ is contained in $H^{2r-1}(\mathbb{P}^{r-1}_k)$, hence is zero by the induction hypothesis. Finally, if $i = 2r$, the relations

$$H^{2r-1}(\mathbb{P}^{r-1}_k) = H^{2r}(\mathbb{P}^{r-1}_k) = 0$$

show that

$$H^{2r}_!(\mathbb{E}^r_k) \xrightarrow{\sim} H^{2r}(\mathbb{P}^r_k),$$

whence the assertion by (1.1).

The lemma implies part a) of 2.2.1. We now prove that θ is an isomorphism. The morphism θ may be interpreted as a morphism of rings

$$A[T]/(T^{r+1}) \longrightarrow \bigoplus_i H^{2i}(P, \mu^{\otimes i}).$$

Thus, by (2.2.3), the assertion will be proved if one shows that, for every $i \in [0, r]$, ξ^i is a free element of the A-module $H^{2i}(P, \mu^{\otimes i})$. It is enough to check this for ξ^r, which will be an immediate consequence of 2.2.2 c).

We now prove part c). In practice, this is the definition of the trace morphism in this situation; in any case, it expresses the compatibility of the trace morphism and the fundamental class in the diagram

$$\begin{array}{ccc} x_0 & \xrightarrow{\text{inj}} & \mathbb{P}^r_k \\ & \searrow & \downarrow p \\ & & k, \end{array}$$

where $\text{inj} : x_0 \to \mathbb{P}^r_k$ is the injection of a closed point of $\mathbb{P}^r_k$.

Corollary 2.2.4. Let L be a complex of A_S-modules. The arrow

$$\mathrm{R}\Gamma(S, \gamma \overset{\mathrm{L}}{\otimes}_A \mathrm{id}(L)) : \bigoplus_{i=0}^{r} \mathrm{R}\Gamma(S, L \otimes \mu_S^{\otimes -i})[-2i] \longrightarrow \mathrm{R}\Gamma(P, p^*(L))$$

is an isomorphism.

Indeed, it is enough to apply the functor $\mathrm{R}\Gamma_S$ to the isomorphism $\gamma \overset{\mathrm{L}}{\otimes}_A \mathrm{id}_L$.

2.2.5

To state the next corollary, for every scheme X we shall put

$$A^*(X) = \bigoplus_i H^{2i}(X, \mu_X^{\otimes i})$$

and shall use this notation systematically in what follows.

Corollary 2.2.6. The inverse-image homomorphism

$$p^* : A^*(S) = \bigoplus_i H^{2i}(S, \mu_S^{\otimes i}) \longrightarrow \bigoplus_i H^{2i}(P, \mu_P^{\otimes i}) = A^*(P)$$

is an injective morphism of unital rings, and the elements $1, \xi, \ldots, \xi^r$ form a basis of the $A^*(S)$-module $A^*(P)$. In particular, $A^*(P)$ is a free $A^*(S)$-module of the same rank as E.

3. Chern Classes

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of constant rank r. Denoting by $\check{E}$ the dual of E, put $P = P_S(\check{E})$. We have seen (2.2.6) that, if ξ denotes the class of $\mathcal{O}_P(1)$ in $H^2(P, \mu)$, then the $A^*(S)$-module $A^*(P)$ is free with basis $1, \xi, \ldots, \xi^{r-1}$. In particular, it is immediate for reasons of homogeneity that there exists a unique relation of the form

$$\xi^r + c_1 \xi^{r-1} + \cdots + c_i \xi^{r-i} + \cdots + c_r = 0, \tag{3.1}$$

with $c_i \in H^{2i}(S, \mu^{\otimes i})$ for every $i \in [0, r]$.

Now let E be any locally free $\mathcal{O}_S$-module. For every integer r, the set S_r of points of S at which E has rank r is open. Thus one obtains a partition

$$S = \coprod_r S_r$$

of S into disjoint opens; in particular, for every A_S-module M and every integer i, the product of the inverse-image morphisms

$$H^i(S, M) \longrightarrow \prod_r H^i(S_r, M/S_r)$$

is a bijection.

The preceding considerations justify the following definition.

Definition 3.2. Let S be a scheme and E a locally free $\mathcal{O}_S$-module. For every integer $i \geq 0$, the i-th Chern class of E, denoted $c_i(E)$, is the element of $H^{2i}(S, \mu^{\otimes i})$ whose restriction, for every integer $r \geq 0$, to the open subset S_r of S consisting of the points where E has rank r, is the element c_i defined by relation 3.1. If the rank of E is bounded, the total Chern class of E is the element

$$c(E) = \sum_i c_i(E)$$

of $A^*(S)$; in the general case, this denotes the corresponding element of

$$\widehat{A}^*(S) = \prod_i A^i(S).$$

3.3

It is evident that, for all S and E, $c_0(E) = 1$, and that, if the rank of E is bounded by d, then $c_i(E) = 0$ for $i > d$.

Proposition 3.4. The Chern classes just defined satisfy the following properties.

(i) *Normalization.* Let L be an invertible $\mathcal{O}_S$-module. If $[L]$ denotes the image of the class of L under the boundary morphism

$$H^1(S, \mathbb{G}_m) \longrightarrow H^2(S, \mu),$$

then

$$c(L) = 1 + [L].$$

(ii) *Stability under base change.* Let $f : S' \to S$ be a morphism of schemes and let E be a locally free $\mathcal{O}_S$-module. For every integer i,

$$f^*(c_i(E)) = c_i(f^*E).$$

(iii) *Additivity.* Let

$$0 \to E' \to E \to E'' \to 0$$

be an exact sequence of locally free $\mathcal{O}_S$-modules. For every integer n, one has

$$c_n(E) = \sum_{i+j=n} c_i(E')c_j(E'').$$

Before beginning the proof, we extract the following corollary.

Corollary 3.5. For every exact sequence $0 \to E' \to E \to E'' \to 0$ of locally free $\mathcal{O}_S$-modules, one has

$$c(E) = c(E')c(E'')$$

in $\widehat{A}^*(S)$.

Proof of 3.4. Following A. Grothendieck: *La théorie des classes de Chern*, Bull. S.M.F. (1958), pp. 137–154.

(i) Let E be a locally free $\mathcal{O}_S$-module; put $P = P_S(\check{E})$ and let $p : P \to S$ be the canonical projection. Recall that there is a canonical surjective homomorphism

$$p^*(\check{E}) \longrightarrow \mathcal{O}_P(1).$$

In particular, if E is invertible and we put $E = L$, in agreement with the notation of (i), then $P = S$ and there is an isomorphism

$$L \xrightarrow{\sim} \mathcal{O}_P(1).$$

Therefore $\xi + [L] = 0$, and the assertion follows from the definition of Chern classes.

(ii) Immediate verification, left to the reader.

(iii) By using a partition of S into opens on which the modules in question have constant rank, we may suppose that E, E', E'' have constant ranks r, r', r'', respectively. Suppose that the following lemma has been proved; we show how it implies the assertion.

Lemma 3.6. Let E be a locally free $\mathcal{O}_S$-module endowed with a finite filtration

$$0 = E_0 \subset E_1 \subset \cdots \subset E_i \subset \cdots \subset E_r = E$$

whose consecutive quotients $L_i = E_i/E_{i-1}$ $(1 \leq i \leq r)$ are invertible. Then

$$c(E) = (1 + [L_1])(1 + [L_2]) \cdots (1 + [L_r]).$$

Let $h : Z \to S$ be the fibre product of the flag schemes of E' and E''. By the definition of flag schemes, it is clear that $h^*(E')$ and $h^*(E'')$ admit finite filtrations with invertible quotients (L'_i) $(1 \leq i \leq r')$ and (L''_j) $(1 \leq j \leq r'')$, respectively. It follows that $h^*(E)$ admits a filtration with consecutive quotients the L'_i and the L''_j. Applying Lemma 3.6 then shows that

$$c(h^*(E)) = \prod_{1 \leq i \leq r'} c(L'_i) \prod_{1 \leq j \leq r''} c(L''_j) = c(h^*(E'))c(h^*(E'')),$$

hence, by (ii),

$$h^*(c(E)) = h^*(c(E')c(E'')). \tag{3.5.1}$$

But h is a composite of morphisms of the form

$$P_T(F) \longrightarrow T \qquad (F \text{ locally free } \mathcal{O}_T\text{-module}),$$

so by (2.2.5) the inverse-image homomorphism $h^* : A^*(S) \to A^*(Z)$ is injective. This gives the desired assertion by (3.5.1).

We prove Lemma 3.6. By the preceding argument, it is enough to prove the analogous relation

$$c(p^*E) = \prod_i c(p^*(L_i)),$$

where, recall, $p : P = P_S(\check{E}) \to S$ is the canonical projection. For every $i \in [1, r]$, the monomorphism $E_i \to E$ defines by duality an epimorphism $\check{E} \to \check{E}_i$; putting $Y_i = P_S(\check{E}_i)$, one obtains a sequence of closed subschemes of P:

$$S \simeq Y_1 \subset Y_2 \subset \cdots \subset Y_i \subset \cdots \subset Y_r = P.$$

For every pair (i, j) of integers with $1 \leq i \leq j \leq r$, let

$$\alpha_{ji} : Y_i \to Y_j$$

be the canonical closed immersion, and put $\alpha_i = \alpha_{r,i}$. For every integer $i \in [1, r-1]$, it follows from the evident lemma (3.6.1) below that Y_i is defined in Y_{i+1} by an invertible ideal isomorphic to

$$\alpha_{i+1}^*(p^*(\check{L}_{i+1}) \otimes \mathcal{O}_P(-1)).$$

Lemma 3.6.1. Let $0 \to L \to E \to F \to 0$ be an exact sequence of locally free $\mathcal{O}_S$-modules. Then the sequence

$$0 \to L \otimes_S S(E) \xrightarrow{u} S(E) \xrightarrow{v} S(F) \to 0,$$

where v denotes the evident morphism and u is the degree-one morphism corresponding to the inclusion of L in E, is an exact sequence of graded $\mathcal{O}_S$-modules.

Moreover, Y_1 is identified with S and we have an isomorphism

$$\alpha_1^*(p^*(\check{L}_1)) \simeq \mathcal{O}_{Y_1}(1),$$

i.e.

$$\alpha_1^*(p^*(\check{L}_1) \otimes \mathcal{O}_P(-1)) \simeq \mathcal{O}_{Y_1}. \tag{3.6.2}$$

We shall be able to conclude by using the following two properties of the Gysin homomorphism (SGA 4 XVIII) associated with a closed immersion $u : Z \to X$ of smooth S-schemes defined by an ideal J of $\mathcal{O}_X$. It satisfies the projection formula

$$u_*(x\, u^*(y)) = u_*(x)y, \tag{3.6.3}$$

and, if the ideal J is invertible, the relation

$$u_*(1_Z) = -[J] = [\check{J}]. \tag{3.6.4}$$

Indeed, from this we shall prove, by induction increasing on the integer i, that

$$(\alpha_i)_*(\alpha_i)^*\Big(\prod_{j\leq i}[p^*(\check{L}_j)\otimes\mathcal{O}_P(-1)]\Big)=0. \tag{3.6.5}$$

For $i=r$, assertion (3.6.5) says

$$\prod_i[p^*(\check{L}_i)\otimes\mathcal{O}_P(-1)]=0,$$

or equivalently

$$\prod_i(\xi+c_1(p^*(\check{L}_i)))=0,$$

which immediately gives (3.6). Assertion (3.6.5) for $i=1$ follows from (3.6.2). Suppose it true for i and prove it for $i+1$. For simplicity put

$$N=\prod_{j\leq i}[p^*(\check{L}_j)\otimes\mathcal{O}_P(-1)].$$

The equality (3.6.5) for i can also be written

$$(\alpha_{i+1})_*(\alpha_{i+1,i})_*(\alpha_{i+1,i})^*(\alpha_{i+1})^*(N)=0,$$

hence, in view of the projection formula for $\alpha_{i+1,i}$,

$$(\alpha_{i+1})_*\big[(\alpha_{i+1})^*(N)(\alpha_{i+1})^*(p^*(\check{L}_{i+1})\otimes\mathcal{O}_P(-1))\big]=0,$$

which is exactly the desired result.

Corollary 3.7.

(i) For every locally free $\mathcal{O}_S$-module E and every integer i,

$$c_i(\check{E})=(-1)^ic_i(E).$$

(ii) If E is a locally free $\mathcal{O}_S$-module of rank r, then

$$c_1(E)=c_1(\Lambda^rE).$$

(iii) More generally, the Chern classes of a tensor product or of an exterior power are expressed by means of the universal polynomials made explicit in (SGA 6 V 6.1).

Proof. We prove, for example, (ii); the other assertions are seen similarly. Using the flag scheme of E, one reduces to the case where E admits a filtration with invertible quotients L_i. The assertion then follows from the well-known isomorphism

$$\Lambda^r(E)\simeq\bigotimes_i L_i.$$

3.8

Given a scheme X locally of finite type over $\mathrm{Spec}(\mathbb{C})$, the set $X(\mathbb{C})$ of its rational points is canonically endowed with the structure of a locally compact topological space. Thus, by algebraic topology, one has a theory of Chern classes [3], assigning to every continuous complex vector bundle V on $X(\mathbb{C})$ classes $c_i(V) \in H^{2i}(X(\mathbb{C}), \mathbb{Z})$. Now let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of continuous complex-valued functions on $X(\mathbb{C})$, and let $\mathcal{E}$ be a locally free $\mathcal{O}_{\mathbb{C}}$-module of finite type. One defines Chern classes

$$c_i(\mathcal{E}) \in H^{2i}(X(\mathbb{C}), \mathbb{Z}),$$

namely those of the complex vector bundle whose sheaf of continuous sections is $\mathcal{E}$. Finally, using the evident map $\mathbb{Z} \to \mathbb{Z}/\nu\mathbb{Z}$, one obtains Chern classes, denoted by the same symbol when no confusion is possible,

$$c_i(\mathcal{E}) \in H^{2i}(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z}). \tag{3.8.1}$$

On the other hand, when $X_{\text{ét}}$ is equipped with its structural sheaf $\mathcal{O}_a$ and $X(\mathbb{C})$ with the ring $\mathcal{O}_{\mathbb{C}}$, there is a morphism of ringed topoi (SGA 4 XI)

$$\varepsilon : X(\mathbb{C}) \to X_{\text{ét}},$$

the structural morphism of rings

$$\varepsilon^*(\mathcal{O}_a) \to \mathcal{O}_{\mathbb{C}} \tag{3.8.2}$$

coming from the fact that every algebraic function is continuous. The morphism ε induces a morphism of abelian sheaves

$$\varepsilon^*(\mu_{X_{\text{ét}}}) \to \mu_{X(\mathbb{C})}, \tag{3.8.3}$$

where the letter μ denotes, in each of the two contexts, the sheaf of ν-th roots of unity. Finally, we shall canonically identify $(\mathbb{Z}/\nu\mathbb{Z})_{X(\mathbb{C})}$ and $\mu_{X(\mathbb{C})}$ by means of the isomorphism

$$(\mathbb{Z}/\nu\mathbb{Z})_{X(\mathbb{C})} \xrightarrow{\sim} \mu_{X(\mathbb{C})} \tag{3.8.4}$$

sending the section 1 to the section $e^{2i\pi/\nu}$.

Proposition 3.8.5. Let E be a locally free $\mathcal{O}_X$-module. For every integer $i \geq 0$, let

$$\varepsilon^* : H^{2i}(X_{\text{ét}}, \mu_X^{\otimes i}) \to H^{2i}(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z})$$

be the inverse-image morphism deduced from 3.8.3 and 3.8.4. Then

$$c_i(\varepsilon^* E) = \varepsilon^* c_i(E).$$

Proof. It is enough to check this when E is invertible. Then, with the usual identifications, $\varepsilon^*(E)$ is the image of E under the canonical map

$$\varepsilon^* : H^1(X_{\text{ét}}, \mathbb{G}_m) \to H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*), \tag{3.8.6}$$

and one has to see that the diagram

$$\begin{array}{ccc} H^1(X, \mathbb{G}_m) & \xrightarrow{\varepsilon^*} & H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*) \\ \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle c_1} \\ H^2(X, \mu_X) & \xrightarrow[\varepsilon^*]{} & H^2(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z}) \end{array} \tag{3.8.7}$$

where ∂ is the boundary morphism coming from the canonical exact sequence

$$0 \to \mu_X \to \mathbb{G}_m \xrightarrow{\nu} \mathbb{G}_m \to 0 \tag{3.8.8}$$

and c_1 is the first Chern class (3.8.1), is commutative. In fact, using that the morphism "first Chern class"

$$H^1(X(\mathbb{C}), \mathcal{O}^*_{\mathbb{C}}) \to H^2(X, \mathbb{Z})$$

is the boundary morphism of the canonical exact sequence

$$0 \to \mathbb{Z} \to \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}^*_{\mathbb{C}} \to 0, \qquad t \mapsto 2i\pi t,$$

one easily sees that the morphism c_1 of (3.8.7) is nothing other than the boundary morphism associated with the exact sequence

$$0 \to \mathbb{Z}/\nu\mathbb{Z} \to \mathcal{O}^*_{\mathbb{C}} \xrightarrow{\nu} \mathcal{O}^*_{\mathbb{C}} \to 0, \qquad t \mapsto e^{2i\pi t/\nu},$$

which, in view of (3.8.4), is the analogue of (3.8.8). The assertion follows at once.

3.9

Let X be a scheme and E a locally free $\mathcal{O}_X$-module. Immediate passage to the limit allows one to define, for every prime number ℓ prime to the residual characteristics of X, ℓ-adic Chern classes

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_\ell(i)) \overset{\text{def}}{=} \varprojlim_{\nu} H^{2i}(X, \mu_{\ell^\nu}^{\otimes i}),$$

which have the usual properties made explicit in 3.4 and 3.7 and which, in the context of 3.8, are manifestly compatible, in an evident sense, with the Chern classes defined topologically.

Tensoring by $\mathbb{Q}_\ell$, one obtains ℓ-adic Chern classes modulo torsion

$$c_i(E) \in H^{2i}(X, \mathbb{Q}_\ell(i)) = H^{2i}(X, \mathbb{Z}_\ell(i)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell,$$

which manifestly satisfy the same properties.

Now suppose that X is a quasi-projective smooth scheme over a field k, and denote here by $C^*(X)$ its Chow ring (Sém. Chevalley: Anneaux de Chow et applications III 4). Recall that in IV we defined a morphism of graded rings

$$\varphi_X : C^*(X) \to A^*(X)$$

compatible with direct-image and inverse-image morphisms. Moreover, the canonical isomorphism

$$\varepsilon_X : \mathrm{Pic}(X) \xrightarrow{\sim} C^1(X)$$

which sends every invertible $\mathcal{O}_X$-module L to the class of the divisor of a rational section of L not vanishing on any connected component of X, gives, on the model of the aforementioned article of Grothendieck, where k was assumed algebraically closed, a theory of Chern classes with values in $C^*(X)$. The composite

$$\varphi_X \circ \varepsilon_X : \mathrm{Pic}(X) \longrightarrow H^2(X, \mathbb{Z}_\ell(1))$$

is just the first ℓ-adic Chern class. Indeed, it is enough to check this on elements of $\mathrm{Pic}(X)$ of the form $\check{J}$, with J an invertible ideal of $\mathcal{O}_X$, and then this follows from (3.6.4). By the uniqueness properties of theories of Chern classes, it follows that the morphism φ_X sends Chern classes to Chern classes.

4. Self-intersection Formula and Applications

Let S be a scheme, X an S-scheme, and Y a closed subscheme of X, smooth over S and everywhere of codimension d. Finally suppose that X is smooth over S in a neighbourhood of Y. Let $i : Y \to X$ be the canonical closed immersion and J the ideal of $\mathcal{O}_X$ defining Y. Recall (SGA $4^{1/2}$ Cycle) that in this case one has Gysin isomorphisms

$$i_* : H^{2p}(Y, \mu_Y^{\otimes p}) \to H_Y^{2p+2d}(X, \mu_X^{\otimes(p+d)}),$$

namely cup-products with the fundamental class. If no confusion is possible, we shall denote by the same symbol the Gysin morphisms

$$i_* : H^{2p}(Y, \mu_Y^{\otimes p}) \to H^{2p+2d}(X, \mu_X^{\otimes(p+d)})$$

obtained from the preceding ones by forgetting supports.

Theorem 4.1. In the preceding situation, if

$$N = J/J^2$$

denotes the normal sheaf of i, then for every $y \in A^*(Y)$ one has

$$i^* i_*(y) = y\, c_d(\check{N}).$$

Before beginning the proof of this theorem, we indicate a few variants, proved in exactly the same way.

a) If ℓ is a prime number prime to the residual characteristics of S, the analogous formula is true in $H^*(Y, \mathbb{Z}_\ell(*))$.

b) One may put in supports. If, for example, T is a closed subset of Y and $y \in H_T^{2p}(Y, \mu_Y^{\otimes p})$, formula 4.1 is still true in $H_T^{2p+2d}(Y, \mu_Y^{\otimes(p+d)})$. Of course, i_* then denotes the Gysin morphism

$$H_T^{2p}(Y, \mu_Y^{\otimes p}) \to H_T^{2p+2d}(X, \mu_X^{\otimes(p+d)}),$$

which is again an isomorphism (SGA $4^{1/2}$ Cycle).

c) Finally, the two cases may be combined.

Since the fundamental tool in the proof of 4.1 is the theory of blown-up schemes, we now recall a few facts about it. Let

$$f : X' \to X$$

be the X-scheme obtained by blowing up Y, and let

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X \end{array}$$

be the cartesian diagram constructed from f and i. Recall (cf. SGA 6 VII 3.1, whose notation we adopt) that j is a regular immersion of codimension 1 defined by an ideal

$$J' \simeq \mathcal{O}_{X'}(1)$$

and that $Y' \simeq P_Y(N)$. Moreover, one has an exact sequence of $\mathcal{O}_{Y'}$-modules

$$0 \to F \to g^*(N) \to \mathcal{O}_{Y'}(1) \to 0.$$

Lemma 4.2. For every A_X-module M, the homomorphisms

$$f^* : H^*_Y(X, M) \to H^*_{Y'}(X', f^*(M))$$

and

$$f_* : H^*_{Y'}(X', f^*(M)) \to H^*_Y(X, M)$$

are related by $f_* f^* = \mathrm{id}$. Moreover, f_* has degree 0.

Proof. Since f is a complete-intersection morphism of relative dimension zero (SGA 6 VII 1.8), the last assertion follows from the definition of the Gysin homomorphism (SGA $4^{1/2}$ Cycle). For the first, the projection formula for f,

$$f_*(x\, f^*(x')) = f_*(x)x',$$

where $x \in H^*(X, A)$ and $x' \in H^*_Y(X, M)$, reduces the claim to proving that $f_*(1) = 1$. It is enough to check the analogous assertion over $U = X - Y$, which is evident because f induces an isomorphism from $U' = X' - Y'$ onto U.

Remark 4.3. Of course, the analogue of 4.2 obtained by eliminating supports is also true and is proved in the same way.

We now come to the proof of 4.1. First, by a general result of (SGA $4^{1/2}$ Cycle), which is proved in an essentially formal way, one has

$$i^* i_*(y) = y\, i^* i_*(1),$$

so it is enough to prove

$$i^* i_*(1_Y) = c_d(\check{N}),$$

where i_* here denotes the Gysin isomorphism

$$H^0(Y, A_Y) \to H^{2d}_Y(X, \mu_X^{\otimes d}).$$

Since $f^* i_* 1_Y \in H^{2d}_{Y'}(X', \mu_{X'}^{\otimes d})$, and since the Gysin morphism

$$j_* : H^{2d-2}(Y', \mu_{Y'}^{\otimes(d-1)}) \to H^{2d}_{Y'}(X', \mu_{X'}^{\otimes d})$$

is an isomorphism, there exists

$$u \in H^{2d-2}(Y', \mu_{Y'}^{\otimes(d-1)})$$

such that

$$f^* i_*(1_Y) = j_*(u). \tag{4.1.1}$$

Applying j^* to both sides of 4.1.1, one obtains

$$j^* f^* i_*(1_Y) = j^* j_*(u).$$

Putting $\xi = c_1(\mathcal{O}_{Y'}(1))$, it follows from 3.6.4 that

$$g^* i^* i_*(1_Y) = -u\,\xi. \tag{4.1.2}$$

The element u admits a unique decomposition of the form

$$u = a_0 + a_1\xi + \cdots + a_{d-1}\xi^{d-1}, \qquad a_i \in H^{2d-2-2i}(Y, \mu_Y^{\otimes(d-1-i)}). \tag{4.1.3}$$

In view of the defining relation for the Chern classes of $\check{N}$,

$$\xi^d + c_1(\check{N})\xi^{d-1} + \cdots + c_{d-1}(\check{N})\xi + c_d(\check{N}) = 0, \tag{4.1.4}$$

we deduce the equality

$$u\xi = -a_{d-1}c_d(\check{N}) + (a_0 - a_{d-1}c_{d-1}(\check{N}))\xi + \cdots + (a_{d-2} - a_{d-1}c_1(\check{N}))\xi^{d-1}.$$

Comparing this last expression with (4.1.2), and using the injectivity of g^* (2.2.6), one obtains

$$i^* i_*(1_Y) = a_{d-1}c_d(\check{N}). \tag{4.1.5}$$

It remains to determine a_{d-1}. First, the homomorphism g_* lowers degrees by $2d-2$, hence

$$g_*(\xi^i) = 0 \quad (0 \leq i \leq d-2),$$

and moreover (2.2.2)

$$g_*(\xi^{d-1}) = 1.$$

Applying g_* to both sides of 4.1.3, we obtain

$$g_*(u) = a_{d-1}. \tag{4.1.6}$$

On the other hand, applying f_* to both sides of 4.1.1, we obtain, in view of 4.2,

$$i_*(1_Y) = i_* g_*(u). \tag{4.1.7}$$

But the homomorphism i_* in question is injective; hence

$$a_{d-1} = g_*(u) = 1_Y,$$

which completes the proof.

Remarks 4.4. a) The essential ingredient in what precedes is the purity theorem. Once it is available in full generality, statement 4.1 will be valid with Y regular and purely of codimension d in X, and X excellent and regular in a neighbourhood of Y. In this form, the result is in any case established in characteristic 0, by (SGA 4 XIX).

b) When S is the spectrum of a separably closed field and X and Y are quasi-projective and smooth over S, the analogous statement in the Chow ring of Y was established by Mumford (cf. 9.2).

Corollary 4.5. Under the hypotheses of 4.1, let x and y belong to $A^*(Y)$. Then

$$i_*(x)i_*(y) = i_*(xy\, c_d(\check{N})).$$

Proof. The projection formula for the morphism i shows that

$$i_*(x)i_*(y) = i_*(x\, i^*i_*(y)),$$

and the conclusion follows from 4.1.

Corollary 4.6. Let S be a scheme, let E be a locally free $\mathcal{O}_S$-module of constant rank r, and let $g : X = V(\check{E}) \to S$ be the vector bundle whose sheaf of sections is E. For every section f of X over S, one has

$$f_*(1_S) = g^*(c_r(E))$$

in $H^{2r}(X, \mu_X^{\otimes r})$.

Proof. Since g^* is an isomorphism (1.2) and $g \circ f = \mathrm{id}$, f^* is bijective. It is therefore enough to check that

$$f^*f_*(1_S) = f^*g^*(c_r(E)).$$

If N denotes the normal sheaf of f, this last equality is equivalent, by 4.1, to

$$c_r(\check{N}) = c_r(E).$$

In fact one even has an isomorphism $\check{N} \simeq E$, since the exact sequence of (EGA IV 17.2.5), applied to the situation

$$S \xrightarrow{f} X \xrightarrow{g} S,$$

gives an isomorphism $N \simeq f^*\Omega^1_{X/S}$, and moreover $\Omega^1_{X/S} \simeq g^*\check{E}$ (EGA IV 16.4.8).

Corollary 4.7. Under the hypotheses of 4.6, assume in addition that S is smooth and quasi-projective over a separably closed field. Let σ_X denote the zero-section cycle of X, and let $Y = f^{-1}(\sigma_X)$ be the cycle of zeros of f. Then

$$\mathrm{cl}_S(Y) = c_r(E)$$

in $H^{2r}(S, \mu_S^{\otimes r})$.

Proof. By IV,

$$\mathrm{cl}_S(Y) = f^* \mathrm{cl}_X(\sigma_X).$$

Let s_0 be the zero-section morphism of X. Then

$$\mathrm{cl}_X(\sigma_X) \overset{\text{dfn}}{=} (s_0)_*(1_S) \overset{(4.6)}{=} g^*(c_r(E)),$$

and the assertion follows by applying f^* to the extreme terms.

Remark 4.8. Let ℓ be a prime number distinct from the residual characteristics of S. By simple passage to the projective limit, one deduces from 4.6 and 4.7 the corresponding statements in ℓ-adic cohomology.

Now let ℓ be a prime number and let X be a proper scheme over a separably closed field k of characteristic prime to ℓ. Recall (IV) that the ℓ-adic Euler-Poincaré characteristic of X is the integer

$$\mathrm{EP}_\ell(X) = \sum_i (-1)^i \dim H^i(X, \mathbb{Q}_\ell).$$

Corollary 4.9. Let X be a smooth, proper, connected scheme of dimension n over a separably closed field k, with $\mathrm{char}(k) \neq \ell$. If $p : X \to k$ denotes the canonical projection, then

$$\mathrm{EP}_\ell(X) = p_* c_n(\check{\Omega}^1_X)$$

in $\mathbb{Q}_\ell = H^0(k, \mathbb{Q}_\ell)$.

Proof. Let $j : X \to X \times X$ be the diagonal immersion of X, and let $q : X \times X \to k$ be the canonical projection. Applying 4.1 to the morphism j gives

$$c_n(\check{\Omega}^1_X) = j^* j_*(1_X),$$

hence

$$p_*(c_n(\check{\Omega}^1_X)) = p_* j^* j_*(1_X) = q_* j_* j^* j_*(1_X).$$

Using the projection formula for j, one deduces

$$p_*(c_n(\check{\Omega}^1_X)) = q_*(j_*(1_X) \cup j_*(1_X)),$$

where, by IV, the second member is equal to the ℓ-adic Euler-Poincaré characteristic of X.

4.10

When X is projective over k, the result recalled from IV shows that $\mathrm{EP}_\ell(X)$ is, in the Chow ring of $X \times X$, the self-intersection of the diagonal cycle, and consequently is independent of the prime number $\ell \neq \mathrm{char}(k)$. We shall therefore be able to speak of the Euler-Poincaré characteristic $\mathrm{EP}(X)$ of X, without reference to the integer ℓ.

Proposition 4.11. Let X be a smooth, projective, connected scheme of dimension n over a separably closed field k. Then

$$\mathrm{EP}(X) = \sum_{p,q} (-1)^{p+q} \dim H^p(X, \Omega^q_X).$$

Proof. We use the notation of the proof of 4.9, and let J be the ideal of the diagonal immersion of X. As observed in 4.10, $\mathrm{EP}(X)$ is, in the Chow ring of $X \times X$, the intersection product $\Delta \cdot \Delta$ of the diagonal cycle Δ with itself. To compute it, one may neglect torsion and therefore work in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$. This shows that $\mathrm{EP}(X)$ is the image under the direct-image morphism

$$q_* : K(X \times X) \to K(k) = \mathbb{Z}$$

of the square of the class of $\mathcal{O}_{X \times X}/J$. But, by (SGA 6 VII 2.5), this square is equal to

$$\sum_i (-1)^i \lambda^i([\Omega^1_X]),$$

and the desired formula follows immediately, by the definition of q_*.

5. Flag Schemes

Let S be a scheme, let E be a locally free $\mathcal{O}_S$-module of rank r, and let $(p_i)_{1\leq i\leq m}$ be m integers >0 satisfying

$$p_1 + p_2 + \cdots + p_m = r.$$

Recall that a flag of type $(p_1, \ldots, p_m)$ of E is a finite filtration

$$0 = E_m \subset E_{m-1} \subset \cdots \subset E_0 = E$$

of E such that, for every integer $i \in [1, m]$, the quotient E_{i-1}/E_i is locally free of rank p_i.

One defines a functor $(\mathrm{Sch}\,/S)^\circ \to (\mathrm{Ens})$ by assigning to an S-scheme $u : T \to S$ the set of flags of type $(p_1, \ldots, p_m)$ of $u^*(E)$, and to a morphism of S-schemes the inverse image, in the evident sense, on flags. It is proved (EGA I, 2nd edition) that this functor is representable, and any representative is called an S-scheme of flags of type $(p_1, \ldots, p_m)$ of E.

In what follows, $f : X \to S$ will denote the S-scheme of flags of type $(p_1, \ldots, p_m)$ of E. Moreover, the scheme of flags of type $(1, 1, \ldots, 1)$ of E will be denoted D and called the flag scheme of E, without further specification, if no confusion is possible.

It is clear from the definition that the $\mathcal{O}_X$-module $f^*(E)$ admits a canonical finite filtration

$$0 = F_m \subset F_{m-1} \subset \cdots \subset F_0 = f^*(E),$$

for which the quotient $E_i = F_{i-1}/F_i$ $(1 \leq i \leq m)$ is locally free of rank p_i. For every integer $i \in [1, m]$, let

$$g_i : D_i \to X$$

be the X-scheme of flags of E_i, and let $g : D_1 \times_X \cdots \times_X D_m \to X$ be the fibre product of the X-schemes D_i. It is clear that the composite

$$h = f \circ g : \prod_i D_i \to S \tag{5.1}$$

is S-isomorphic to the S-scheme of flags of E; up to S-isomorphism, one may therefore write

$$D = \prod_i D_i.$$

Proposition 5.2. In $D^b(S, A)$ there is an isomorphism

$$\mathrm{R}f_*(A) \simeq \bigoplus_i R^{2i} f_*(A)[-2i].$$

Proof. Since h is a composite of finitely many morphisms of the form

$$P_T(F) \to T \qquad (F \text{ a locally free } \mathcal{O}_T\text{-module}), \tag{Q}$$

it follows from (2.2.1) that

$$\mathrm{R}h_*(A) \simeq \bigoplus_j R^j h_*(A)[-j].$$

For the same reason, one also has

$$\mathrm{R}g_*(A) \simeq \bigoplus_j R^j g_*(A)[-j].$$

Using ([6] 2.16), one obtains an isomorphism

$$\mathrm{R}f_*(g_*A) \simeq \bigoplus_j R^j f_*(g_*A)[-j].$$

Since g is smooth with geometrically connected fibres, $A_X \simeq g_*(A_D)$; hence it remains to show that

$$R^{2i+1} f_*(A) = 0 \qquad (i \in \mathbb{N}). \tag{5.2.1}$$

For this, one immediately reduces, using conservativity of the system of fibre functors (SGA 4 VIII 3.5), to the case where S is the spectrum of a separably closed field k. Then, since g is a composite of projective bundles, (2.2.4) shows that the inverse-image map

$$H^{2i+1}(X, A_X) \to H^{2i+1}(D, A_D)$$

is a monomorphism, so one may suppose that $X = D$. In this case, since the projection $D \to k$ is itself a composite of projective bundles, the assertion follows by induction from (2.2.4).

Corollary 5.3. For every morphism of schemes $u : S \to T$, the Leray spectral sequence

$$E_2^{pq} = R^p u_*\, R^q f_*(A) \Rightarrow R^{p+q}(u \circ f)_*(A)$$

is degenerate.

Proof. Cf. [6] 1.2.

We shall now specify the multiplicative structure of the cohomology of X, first examining a particular case. With the preceding notation, one obtains a morphism of rings

$$\alpha : A^*(S)[T_{ij_i}]_{1 \leq i \leq m,\, 1 \leq j_i \leq p_i} \to A^*(X)$$

by sending $A^*(S)$ into $A^*(X)$ by inverse image f^* and, for every pair (i, j_i), as above, sending the element T_{ij_i} to the j_i-th Chern class $c_{j_i}(E_i)$. For every integer p, one has, by (3.4 (iii)),

$$c_p(f^*E) = \sum_{i_1 + i_2 + \cdots + i_m = p} c_{i_1}(E_1) \cdots c_{i_m}(E_m),$$

so that, by functoriality of Chern classes, the morphism vanishes on the ideal J_X generated by the elements

$$c_p(E) - \sum_{i_1 + \cdots + i_m = p} T_{1,i_1} \cdots T_{m,i_m}$$

for varying p. Hence there is a morphism of rings

$$\varphi_X : \big(A^*(S)[T_{ij_i}]_{(i,j_i)}\big)/J_X \to A^*(X).$$

Proposition 5.4. The above morphism of rings φ_X is an isomorphism. The $A^*(S)$-module $A^*(X)$ is free of rank

$$\frac{r!}{p_1! p_2! \cdots p_m!}.$$

Proof. We follow Grothendieck's exposé (Séminaire Chevalley 1958, Anneaux de Chow et applications, Exp. IV). We shall need the following lemma (cf. Lemma 1, p. 19 of loc. cit.).

Lemma 5.4.1. a) Let R be a commutative ring, and let $(a_i)_{1\leq i\leq r}$ be r elements of R. Let σ_i be the i-th elementary symmetric function of the elements $(U_i)_{1\leq i\leq r}$, and let I be the ideal of the polynomial ring $R[U_1, U_2, \ldots, U_r]$ generated by the elements $\sigma_i - a_i$. Then the R-algebra

$$R[U_1, U_2, \ldots, U_r]/I$$

is a free R-module of rank $r!$.

b) Let $r, p_1, p_2, \ldots, p_m$ be integers > 0 satisfying $p_1 + \cdots + p_m = r$. Then the morphism of polynomial rings

$$V = R[T_{i,j_i}]_{1\leq i\leq m,\, 1\leq j_i\leq p_i} \longrightarrow R[U_i]_{1\leq i\leq r} = W$$

obtained by sending T_{i,j_i} to the j_i-th elementary symmetric function of the elements

$$(U_{p_1+\cdots+p_{i-1}+s})_{1\leq s\leq p_i}$$

is injective and makes W a free V-module of rank $p_1!\cdots p_m!$.

In particular, the morphism of R-algebras

$$R[T_1, \ldots, T_r] \to R[U_1, \ldots, U_r]$$

obtained by sending T_i to the i-th elementary symmetric function of the U_j is injective and makes the second member a free module of rank $r!$ over the first.

We show how (5.4.1) implies (5.4). First suppose that X is the scheme D of flags of E, and for simplicity put $T_{i1} = U_i$. Since the canonical projection $D \to S$ is a composite of $r-1$ morphisms of the form

$$P_{Y_i}(M_i) \to Y_i \quad (1 \leq i \leq r-1),$$

where M_i is a locally free $\mathcal{O}_{Y_i}$-module of rank $r-i+1$, it follows from (2.2.6) that $A^*(D)$ is a free $A^*(S)$-module of rank $r!$, and that the morphism φ_D is surjective. But Lemma 5.4.1 a) shows that the source of φ_D is also a free $A^*(S)$-module of rank $r!$; hence the assertion in the case $X = D$.

In the general case, it is clear that the diagram

$$\begin{array}{ccc} A^*(X) & \xrightarrow{\quad g^*\quad} & A^*(D) \\ {\scriptstyle\varphi_X}\uparrow & & \uparrow{\scriptstyle\varphi_D} \\ C = A^*(S)[T_{i,j_i}]/J_X & \xrightarrow[\gamma]{} & A^*(S)[U_k]/J_D \end{array}$$

commutes; here the map γ is obtained by passage to the quotient from the morphism of $A^*(S)$-algebras

$$A^*(S)[T_{i,j_i}] \to A^*(S)[U_k]$$

defined by sending T_{i,j_i} to the j_i-th elementary symmetric function of the elements

$$(U_{p_1+\cdots+p_{i-1}+s})_{1\leq s\leq p_i}.$$

Since φ_D is an isomorphism, 5.4.1 b) implies that $A^*(D)$ is a free C-module of rank $p_1!\cdots p_m!$. On the other hand, expressing g as a fibre product of X-schemes of flags shows that $A^*(D)$ is also a free $A^*(X)$-module of rank $p_1!\cdots p_m!$. It follows at once that φ_X is an isomorphism.

In the statement of the next proposition, we agree to put, for every scheme Y and every A_Y-module M,

$$Q(Y,M)=\bigoplus_{n,i} H^n(Y,M\otimes\mu_Y^{\otimes i}).$$

The A-module $Q(Y,M)$ is canonically endowed with a structure of $A^*(Y)$-module, coming from cup-product. If $u:Z\to Y$ is a morphism of schemes, the inverse-image morphism

$$u^*:Q(Y,M)\to Q(Z,u^*M)$$

is compatible with the $A^*(Y)$-module structure on $Q(Y,M)$ and the $A^*(Z)$-module structure on $Q(Z,u^*M)$. One deduces a morphism of $A^*(Z)$-modules, called canonical,

$$A^*(Z)\otimes_{A^*(Y)} Q(Y,M)\to Q(Z,u^*M).$$

Proposition 5.5. Let S be a scheme, and let E be a locally free $\mathcal{O}_S$-module of rank r. Let $f:X\to S$ be the S-scheme of flags of type $(p_1,\dots,p_m)$ of E. Then, for every A_S-module L, the canonical morphism

$$\alpha:A^*(X)\otimes_{A^*(S)} Q(S,L)\to Q(X,f^*L)$$

is an isomorphism.

Proof. When $X=P_S(E)$, this is an immediate consequence of (2.2.4) and (2.2.6). Hence it follows that it is true for $X=D$. In the general case, since the canonical morphisms $g:D\to X$ and $h:D\to S$ are composites of projective bundles, the canonical morphisms

$$A^*(D)\otimes_{A^*(S)} Q(S,L)\to Q(D,h^*L)$$

and

$$A^*(D)\otimes_{A^*(X)} Q(X,g^*L)\to Q(D,g^*L)$$

are isomorphisms. Consequently $\mathrm{id}_{A^*(D)}\otimes\alpha$ is an isomorphism, whence the result, since $A^*(D)$ is free over $A^*(X)$.

We shall now give a relative variant of (5.4) and (5.5). With the preceding notation, let $u:S\to T$ be a morphism of schemes. Put

$$A^*(X/T)=\bigoplus_i R^{2i}(u\circ f)_*(A_X)$$

and similarly

$$A^*(S/T)=\bigoplus_i R^{2i}u_*(A_S).$$

Finally, for every A_S-module L, put

$$Q(S/T,L)=\bigoplus_{n,i} R^n u_*(L\otimes\mu_S^{\otimes i})$$

and

$$Q(X/T,f^*L)=\bigoplus_{n,i} R^n(u\circ f)_*(f^*L\otimes\mu_X^{\otimes i}).$$

For every pair (i,j) of integers, with $1 \leq i \leq m$ and $1 \leq j \leq p_i$, the image of the Chern class $c_j(E_i)$ by the canonical morphism

$$H^{2j}(X, \mu_X^{\otimes j}) \to H^0(T, R^{2j}(u \circ f)_*(\mu_X^{\otimes j}))$$

defines an A_T-morphism

$$c_{ij} : A_T \to R^{2j}(u \circ f)_*(\mu_X^{\otimes j}).$$

The data of the c_{ij} and the inverse image f^* allow one to define a morphism of $A^*(S/T)$-algebras

$$\theta_1 : A^*(S/T)[T_{i,j_i}]_{(i,j_i)} \to A^*(X/T).$$

Denoting by c_i the sections of $A^*(S/T)$ defined by the Chern classes $c_i(E)$, one sees as in (5.4) that θ_1 vanishes on the ideal J locally generated by the relations

$$\sum_{i_1+\cdots+i_m=p} T_{1,i_1} \cdots T_{m,i_m} = c_p \qquad (p \in \mathbb{N}).$$

Hence there is a morphism of $A^*(S/T)$-algebras

$$\theta : A^*(S/T)[T_{i,j_i}]_{(i,j_i)}/J \to A^*(X/T).$$

Proposition 5.6. With the preceding notation:

a) The morphism of $A^*(S/T)$-algebras

$$\theta : A^*(S/T)[T_{i,j_i}]_{(i,j_i)}/J \to A^*(X/T)$$

is an isomorphism.

b) For every A_S-module L, the canonical morphism of $A^*(X/T)$-modules

$$A^*(X/T) \otimes_{A^*(S/T)} Q(S/T, L) \to Q(X/T, f^*L)$$

is an isomorphism.

Proof. Simple sheafification of (5.4) and (5.5), respectively.

6. Group Schemes

We begin with a general property of objects twisted by torsors.

Proposition 6.1. Let S be a scheme, G a quasi-compact smooth S-group scheme with connected fibres, and X an S-scheme endowed with a left action of G. If $f : E \to S$ is a right S-torsor under G, denote by X_E the S-scheme obtained from X by twisting by means of E and the action of G on X. Let $p : X \to S$ and $q : X_E \to S$ be the canonical projections. Under these conditions, for every A_S-module M and every integer i, the A_S-module $R^i q_*(q^*M)$ is canonically isomorphic to $R^i p_*(p^*M)$.

Proof.

$$\begin{array}{ccc} E \times_S X & \xrightarrow{\varphi} & X_E \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle q} \\ E & \xrightarrow[f]{} & S \end{array} \qquad \begin{array}{ccc} E \times_S X & \xrightarrow{\mathrm{pr}_2} & X \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle p} \\ E & \xrightarrow[f]{} & S. \end{array}$$

It is known that diagram (D_1), where φ denotes the canonical passage-to-the-quotient map, is cartesian. Since f is smooth, the smooth base-change theorem gives an isomorphism

$$u : f^* R^i q_*(q^* M) \xrightarrow{\sim} R^i(\mathrm{pr}_1)_*(\varphi^* q^* M).$$

Similarly, the cartesian diagram (D_2) gives an isomorphism

$$v : f^* R^i p_*(p^* M) \xrightarrow{\sim} R^i(\mathrm{pr}_1)_*((\mathrm{pr}_2)^* p^* M).$$

But $p \circ \mathrm{pr}_2 = q \circ \varphi$, so after the evident identification of $\varphi^* \circ q^*$ with $(\mathrm{pr}_2)^* \circ p^*$ one obtains an isomorphism

$$v^{-1} \circ u : f^* R^i q_*(q^* M) \xrightarrow{\sim} f^* R^i p_*(p^* M).$$

Since the geometric fibres of f are non-empty and connected, it follows from (SGA 4 XII 6.5 (i)) that the adjunction morphism

$$\mathrm{adj} : \mathrm{id} \to f_* f^*$$

is an isomorphism, and therefore

$$(\mathrm{adj})^{-1} \circ f_*(v^{-1} \circ u) \circ (\mathrm{adj})$$

is the desired isomorphism.

Proposition 6.2. (Mme Raynaud [5].) Let S be a scheme and let $p : G \to S$ be a reductive S-group scheme (SGA 3 XIX). For every locally constant constructible A_S-module M and every integer i, the A_S-module $R^i p_*(p^* M)$ is locally constant and constructible. Moreover, for every morphism $u : S' \to S$ defining a cartesian diagram

$$\begin{array}{ccc} G' & \xrightarrow{u'} & G \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle p} \\ S' & \xrightarrow[u]{} & S, \end{array}$$

the base-change morphism

$$u^* R^i p_*(p^* M) \to R^i p'_*(p \circ u')^*(M)$$

is an isomorphism.

Proof. The two assertions of the statement are local for the étale topology of S. By (SGA 3 XII 2.3), we may therefore suppose that G is split. Let B be a Borel subgroup of G of the type indicated in (SGA 3 XXII 5.1.1), so that the quotient G/B exists by (SGA 3 XXII 5.8.5), and let $q : G \to G/B$ and $r : G/B \to S$ be the canonical morphisms. Since r is proper and smooth, it satisfies the proper base-change theorem and the functors $R^i r_*$ send locally constant constructible abelian sheaves to sheaves of the same type (SGA 4 XVI 2.2). Moreover, by (SGA 4 XXII 5.5.1), the

S-scheme B is isomorphic to a finite product of vector bundles and punctured vector bundles, and consequently it follows immediately from (1.1) and (1.2) that, if $t : B \to S$ denotes the canonical morphism, the sheaves $R^i t_*(t^*M)$, with M a locally constant constructible A_S-module, are locally constant constructible and stable under base change. Now let M be a locally constant constructible A_S-module. The preceding considerations and (6.1) imply that the sheaves $R^i q_*(p^*M)$ are locally constant constructible; using the Leray spectral sequence

$$R^i r_*(R^j q_*)(p^*M) \Rightarrow R^{i+j} p_*(p^*M),$$

one deduces the same for the sheaves $R^i p_*(p^*M)$. Finally, by (6.1), the sheaves $R^i q_*(p^*M)$ are stable under base change; using the proper base-change theorem for r, it follows that the sheaves $R^i p_*(p^*M)$ are also stable under base change.

Remark 6.3. Since every reductive S-group is locally, for the étale topology, an inverse image by the canonical morphism $S \to \mathrm{Spec}(\mathbb{Z})$ of a reductive $\mathrm{Spec}(\mathbb{Z})$-scheme (SGA 3 XXIII 5.7 and XXV 1.3), (6.2) shows that the computation of the cohomology, with coefficients of torsion prime to the characteristic, of reductive group schemes over an algebraically closed field reduces to the transcendental problem of computing the cohomology of reductive group schemes of the same type over the field of complex numbers; this is known by the methods of algebraic topology, in view of the comparison theorem (SGA 4 XI 4.4).

Corollary 6.4. Let S be a scheme and let $p : G \to S$ be an S-group scheme which is an extension of a finite étale S-group scheme F by a reductive S-group scheme H. Then the conclusions of (6.2) are valid for every locally constant constructible A_S-module M.

Proof. After the base change $F \to S$, we may suppose that F is of the form Γ_S, where Γ is an ordinary finite group. Then there is an isomorphism of S-schemes

$$G \simeq \coprod_{\gamma \in \Gamma} H,$$

and the conclusion follows from (6.2).

Corollary 6.5. Let S be a scheme, G a reductive S-group scheme, and H a reductive sub-S-group scheme of G, such that the quotient sheaf of G by H for the faithfully flat quasi-compact topology is representable. Then, if

$$p : G/H \to S$$

is the canonical morphism, the conclusions of (6.2) are valid for every locally constant constructible A_S-module M.

Proof. Let $f : G \to G/H$ be the canonical morphism; it is smooth, hence of descent for étale sheaves (SGA 4 VIII 9.1). For every integer $r \geq 0$, put

$$G_r = G \times_{G/H} G \times_{G/H} \cdots \times_{G/H} G \quad (r+1 \text{ times}),$$

which, by SGA 3 V 1.5, is canonically identified with

$$G \times_S H \times_S \cdots \times_S H,$$

where H occurs r times, and let $f_r : G_r \to G/H$ and $g_r : G_r \to S$ be the canonical morphisms, so that in particular $f_0 = f$. Finally, for every integer $q \geq 0$, let $H^q(M)$ denote the complex

$$R^q(g_0)_*(f_0^*M) \to R^q(g_1)_*(f_1^*M) \rightrightarrows \cdots \to R^q(g_r)_*(f_r^*M) \cdots .$$

By (SGA), one has a spectral sequence of descent, or Čech spectral sequence,

$$E(M) : E_2^{pq} = H^p(H^q(M)) \Rightarrow E^n = R^n p_*(M).$$

Since the schemes G_r are reductive group schemes, it follows from 6.2 that the terms E_2^{pq} are locally constant constructible and compatible with every base change. It follows at once that the same is true for the E^n.

7. Complete Intersections

In this paragraph a separably closed field k is given. If ℓ is a prime number $\neq \mathrm{char}(k)$, we denote, for every proper k-scheme X and every integer $i \geq 0$, by

$$b_\ell^i(X) \quad \text{or} \quad b^i(X)$$

when no confusion is possible, the i-th Betti number of X for coefficients $\mathbb{Z}_\ell$. If X is projective over k, a hyperplane section of X means the zero locus of a section transverse to the zero section of a vector bundle $V(\check{\mathcal{L}})$, with $\mathcal{L}$ an ample invertible $\mathcal{O}_X$-module.

Theorem 7.1. (Lefschetz theorem.) Let X be a smooth, connected, projective k-scheme of dimension n, let Y be a subscheme of X which is the intersection of d hyperplane sections of X, and let L be a locally constant constructible A_X-module. Then:

(i) The inverse-image homomorphism

$$H^i(X, L) \to H^i(Y, L|Y)$$

is an isomorphism for $i \leq n - d - 1$ and a monomorphism for $i = n - d$.

(ii) Suppose Y is smooth over k, of dimension $n - d$. Then the Gysin homomorphism

$$H^i(Y, (L|Y) \otimes \mu_Y^{\otimes -d}) \to H^{i+2d}(X, L)$$

is an isomorphism for $i \geq n - d + 1$ and an epimorphism for $i = n - d$.

Proof. It is well known that the complement of a hyperplane section is an affine open; consequently $U = X - Y$ is a union of d affine opens. Assertion (i) then follows immediately from the exact cohomology sequence

$$\cdots \to H_!^i(U, L|U) \to H^i(X, L) \to H^i(Y, L|Y) \xrightarrow{\partial} H_!^{i+1}(U, L|U) \to \cdots$$

and from the following lemma.

Lemma 7.1.1. Let U be a k-scheme of finite type of dimension n, which is a union of d affine open subsets, and let F be an A_U-module. Then:

a) if F is constructible,

$$H^i(U, F) = 0 \qquad \text{for } i \geq n + d;$$

b) if F is locally constant and U is smooth over k,

$$H^i_c(U, F) = 0 \qquad \text{for } i \leq n - d.$$

For $d = 1$, assertion a) is just (SGA 4 XIV 3.2). In the general case, writing $\mathfrak{U}$ for a covering of U by d affine open subsets, one deduces a) from the Čech spectral sequence

$$H^p(\mathfrak{U}, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(U, F).$$

We prove b). Since the functors $H^i_c(U, \cdot)$ commute with filtered inductive limits (SGA 4 VII 3.3), one may suppose F constructible; the assertion then follows from a), by Poincaré duality (SGA 4 XVIII).

Finally, in view of the definition of the Gysin morphism through the canonical isomorphism

$$H^{i-2d}(Y, (L|Y) \otimes \mu_Y^{\otimes -d}) \simeq H^i_Y(X, L), \tag{7.1.2}$$

assertion (ii) follows from 7.1.1 a) and from the exact cohomology sequence associated with an open subset and its complement

$$\cdots \to H^{i-1}(U, L|U) \xrightarrow{\partial} H^i_Y(X, L) \to H^i(X, L) \to H^i(U, L|U) \to \cdots .$$

Remark 7.1.3. In fact, it follows from excision that the canonical isomorphism (7.1.2) generalizes to the case where the smoothness hypothesis on X, and the hypothesis that L is constructible locally constant, are satisfied only in a neighbourhood of Y; hence 7.1 (ii) generalizes in this setting.

For substantially more general formulations, see (SGA 2 XIV 4.6), which, however, is proved only by means of resolution of singularities, and hence provisionally in characteristic 0.

Corollary 7.2. Let X be a smooth connected projective k-scheme of dimension n, let Y be a subscheme of X which is the intersection of d hyperplane sections of X, and let L be a constructible twisted constant $\mathbb{Z}_\ell$-sheaf ($\ell \neq \mathrm{car}(k)$). Then:

(i) the inverse-image homomorphism

$$\alpha_i : H^i(X, L) \to H^i(Y, L|Y)$$

is an isomorphism for $i \leq n - d - 1$, and a monomorphism for $i = n - d$. If moreover L is constructible locally free, the cokernel of α_{n-d} is torsion-free;

(ii) suppose that Y is smooth over k, of dimension $n - d$. Then the Gysin homomorphism

$$\beta_i : H^i(Y, (L|Y)(-d)) \to H^{i+2d}(X, L)$$

is an isomorphism for $i \geq n - d + 1$ and an epimorphism for $i = n - d$. If moreover L is constructible locally free, the epimorphism β_{n-d} is split and its kernel is torsion-free.

Proof. We prove (i), assertion (ii) following from it by Poincaré duality. In the general case, a simple passage to the limit starting from 7.1 (i) gives the conclusion. When L is a locally free $\mathbb{Z}_\ell$-sheaf, apply the following lemma to the inverse-image morphism

$$R\Gamma(X, L) \to R\Gamma(Y, L|Y)$$

in $D(\mathbb{Z}_\ell)$; hypothesis (iii) is satisfied by 7.1 (i).

Lemma 7.2.1. Let V be a discrete valuation ring, let k_V be its residue field, and let $u : K \to L$ be a morphism in $D(V)$. Suppose that K and L have cohomology of finite type, and denote by M the mapping-cylinder of u. Then for every $m \in \mathbb{Z}$ the following conditions are equivalent:

(i) $H^p(M) = 0$ for $p \leq m-1$, and $H^m(M)$ is torsion-free;

(ii) for every V-module N, the morphism

$$H^p(u \otimes^L \mathrm{id}_N) : H^p(K \otimes^L N) \to H^p(L \otimes^L N)$$

is an isomorphism for $p \leq m-1$ and a monomorphism for $p = m$;

(ii bis) for every V-module N, one has

$$H^p(M \otimes^L N) = 0 \qquad (p \leq m-1);$$

(iii) the morphism

$$H^p(u \otimes^L \mathrm{id}) : H^p(K \otimes^L k_V) \to H^p(L \otimes^L k_V)$$

is an isomorphism for $p \leq m-1$ and a monomorphism for $p = m$;

(iii bis) one has

$$H^p(M \otimes^L k_V) = 0 \qquad (p \leq m-1).$$

Moreover these equivalent conditions imply:

(i bis) the morphism

$$H^p(u) : H^p(K) \to H^p(L)$$

is an isomorphism for $p \leq m-1$, and a monomorphism with free cokernel for $p = m$.

Proof. The following diagram of implications is evident:

$$\begin{array}{ccc} \text{(ii)} & \Leftrightarrow & \text{(ii bis)} \\ \Downarrow & & \\ \text{(iii)} & \Leftrightarrow & \text{(iii bis)} \end{array} \qquad \text{(i)} \Rightarrow \text{(i bis)}.$$

We shall prove separately that (i)⇒(ii bis) and that (iii bis)⇒(i), which will suffice. The implication (i)⇒(ii bis) is immediate from the Künneth exact sequence

$$0 \to H^p(M) \otimes_V N \to H^p(M \otimes^L N) \to \mathrm{Tor}_1^V(H^{p+1}(M), N) \to 0.$$

The same exact sequence, for $N = k_V$, shows that if (iii bis) holds, then

$$H^p(M) \otimes k_V = 0 \quad (p \leq m-1), \qquad \mathrm{Tor}_1^V(H^m(M), k_V) = 0.$$

One deduces (i) by Nakayama's lemma and the local criterion of flatness (EGA 0_{III} 10.1.3).

Under the preceding hypotheses, one sees that knowledge of the cohomology groups of X with coefficients in $\mathbb{Q}_\ell$ implies knowledge of those of Y, except in dimension $n-1$. The calculation of that remaining group is immediately reduced, under these conditions, to the calculation of the Euler-Poincaré characteristic of Y. This is the calculation to which we now turn, in a slightly more general setting.

Proposition 7.3. Let X be a proper smooth connected k-scheme of dimension n, let E be a locally free $\mathcal{O}_X$-module of constant rank r, and let Y be the zero-locus of a section of $V(\check{E})$ transverse to the zero-section. Let $p : X \to k$ be the canonical morphism and p_* the corresponding Gysin morphism. Then

$$\mathrm{EP}(Y) = p_*(c(\check{\Omega}^1_X)c(E)^{-1}c_r(E))$$

in $H^0(k, \mathbb{Q}_\ell) = \mathbb{Q}_\ell$.

Proof. By (EGA IV 17.13.2), Y is smooth and purely of dimension $n-r$. If $i : Y \to X$ denotes the canonical immersion, the conormal sheaf of i is the inverse image by i of the conormal sheaf $\check{E}$ to the zero-section of $V(\check{E})$. Thus one has an exact sequence

$$0 \to i^*(\check{E}) \to i^*(\Omega^1_X) \to \Omega^1_Y \to 0, \tag{7.3.1}$$

whence immediately

$$c(\check{\Omega}^1_Y) = i^*(c(\check{\Omega}^1_X)c(E)^{-1}). \tag{7.3.2}$$

By (3.10) we therefore get

$$\mathrm{EP}(Y) = p_*i_*i^*(c(\check{\Omega}^1_X)c(E)^{-1}),$$

or, using the projection formula for i,

$$\mathrm{EP}(Y) = p_*(c(\check{\Omega}^1_X)c(E)^{-1}i_*(1_Y)).$$

The stated formula follows immediately from (4.7).

Corollary 7.4. Let X be a proper smooth connected k-scheme of dimension n, and let Y be a smooth closed subscheme of dimension $n-1$ of X, hence defined by a section of an invertible $\mathcal{O}_X$-module L. Let p be the canonical morphism from X to k and p_* the corresponding Gysin morphism. Then

$$\mathrm{EP}(X) - \mathrm{EP}(Y) = p_*(c(\check{\Omega}^1_X)c(L)^{-1}) \tag{7.4.1}$$

in $H^0(k, \mathbb{Q}_\ell) = \mathbb{Q}_\ell$. Moreover,

$$b^{n-1}(Y) = 2b^{n-1}(X) - b^n(X) + (-1)^n p_*(c(\check{\Omega}^1_X)c(L)^{-1}). \tag{7.4.2}$$

Proof. Formula (7.4.1) follows from 7.3 and 4.2, taking into account the evident identity

$$1 - c(L)^{-1}c_1(L) = c(L)^{-1}.$$

Using (7.2), one sees that

$$b^{n-1}(Y) = b^{n-1}(X) - b^n(X) + b^{n+1}(X) + (-1)^n(\mathrm{EP}(X) - \mathrm{EP}(Y)).$$

But $b^{n-1}(X) = b^{n+1}(X)$ by Poincaré duality, and (7.4.2) follows.

Corollary 7.5. Let X be a smooth connected subscheme of dimension n of $\mathbb{P}^{n+r}_k$, a global complete intersection of r divisors $X_1, X_2, \ldots, X_r$ of $\mathbb{P}^{n+r}_k$. For each $i \in \{1, \ldots, r\}$, let d_i be the degree of X_i. Then:

(i) the cohomology groups $H^p(X, \mathbb{Z}_\ell)$ are free $\mathbb{Z}_\ell$-modules of finite type;

(ii) if p is an integer $\leq 2n$ and $\neq n$, then

$$H^p(X, \mathbb{Z}_\ell) = \begin{cases} 0, & \text{if } p \text{ is odd,} \\ \mathbb{Z}_\ell, & \text{if } p \text{ is even;} \end{cases}$$

(iii) the Euler-Poincaré characteristic of X is given by

$$\mathrm{EP}(X) = d_1 \cdots d_r \sum_{\substack{0 \leq i \leq n \\ j_1 + \cdots + j_r = n-i}} (-1)^{n-i} \binom{n+r+1}{i} d_1^{j_1} \cdots d_r^{j_r}.$$

Proof. Assertion (i) follows immediately from (7.2), and so does (ii), in view of the cohomology of projective spaces (no. 2). We prove (iii). Let P be the ambient projective space, let $p : P \to k$ be the canonical projection, let p_* be the corresponding Gysin morphism, and let ξ be the class of $\mathcal{O}_P(1)$. For every $i \in \{1, \ldots, r\}$, the scheme X_i is defined by a section of an invertible $\mathcal{O}_P$-module L_i, so that X is the zero-locus of a section of $V(\check{L}_1 \oplus \cdots \oplus \check{L}_r)$ transverse to the zero-section. On the other hand, one has an exact sequence

$$0 \to \Omega^1_P \otimes \mathcal{O}_P(1) \to \mathcal{O}_P^{n+r+1} \to \mathcal{O}_P(1) \to 0,$$

hence

$$c(\check{\Omega}^1_P) = (1+\xi)^{n+r+1}.$$

Formula (7.3) gives

$$\begin{aligned} \mathrm{EP}(X) &= p_*(c(\check{\Omega}^1_P) c(\check{L}_1 \oplus \cdots \oplus \check{L}_r)^{-1} c_1(\check{L}_1) \cdots c_1(\check{L}_r)) \\ &= p_*(d_1 \cdots d_r \, \xi^r (1+\xi)^{n+r+1} (1+d_1\xi)^{-1} \cdots (1+d_r\xi)^{-1}). \end{aligned}$$

Using the fact that $p_*(\xi^{n+r}) = 1$ (2.2.2 c)), the formula of the statement follows without difficulty.

Remark 7.6. One may also obtain the expression for the Euler-Poincaré characteristic by using (4.11) and by transcribing Hirzebruch's calculations ([3], pp. 160–161), which use the Riemann-Roch theorem. For a more complete calculation and the determination of the Hodge cohomology of complete intersections, the reader is referred to (SGA 7 XI).

7.7

By analogy with what happens in the transcendental case, where for constant coefficients one has another well-known theorem of Lefschetz ([4]), one is led to conjecture the following statement. If X is a smooth connected projective k-scheme of dimension n, and if ξ is the class in $H^2(X, \mathbb{Z}_\ell(1))$ of a very ample invertible $\mathcal{O}_X$-module, then for every twisted constant $\mathbb{Q}_\ell$-sheaf L and every integer $p \leq n$, multiplication by ξ^{n-p}:

$$H^p(X, L) \to H^{2n-p}(X, L(n-p))$$

is an isomorphism. As Deligne observed, the statements (7.7) are false without an additional hypothesis on L. However, Deligne proved them in *La conjecture de Weil II* when, in addition, L is pure.

This conjectural statement has the following consequences, likewise not proved in car $\neq 0$.

Let X be a smooth connected projective k-scheme of dimension n, let Y be a smooth hyperplane section of X, and let L be a twisted constant $\mathbb{Q}_\ell$-sheaf on X.

(i) The inverse-image morphism
$$H^p(X, L) \to H^p(Y, L|Y)$$
is an epimorphism for $p \geq n$.

(ii) The Gysin morphism
$$H^p(Y, (L|Y)(-1)) \to H^{p+2}(X, L)$$
is a monomorphism for $p \leq n-2$.

Proof. Let $j : X \hookrightarrow \mathbb{P}^r$ be a closed immersion and let ξ_X (resp. ξ_Y) be the class in $H^2(X, \mathbb{Q}_\ell(1))$ (resp. $H^2(Y, \mathbb{Q}_\ell(1))$) of $\mathcal{O}_X(1)$ (resp. $\mathcal{O}_Y(1)$). Let $\alpha : Y \hookrightarrow X$ denote the canonical injection. For $q \leq n-1$, the diagram
$$\begin{array}{ccc} H^q(X, L) & \xrightarrow{\alpha^*} & H^q(Y, L|Y) \\ \cup \xi_X^{n-1-q} \downarrow & & \downarrow \cup \xi_Y^{n-1-q} \\ H^{2n-2-q}(X, L(n-1-q)) & \xrightarrow{\alpha^*} & H^{2n-2-q}(Y, (L|Y)(n-1-q)) \end{array}$$
is commutative and the right column is conjecturally an isomorphism. Put $p = 2n-2-q$. For $q \leq n-2$, hence $p \geq n$, the top row is an isomorphism by (7.2), giving (i). Assertion (ii) is proved similarly by considering, this time, the following diagram, with $p \leq n-1$:
$$\begin{array}{ccc} H^p(Y, (L|Y)(-1)) & \xrightarrow{\text{Gysin}} & H^{p+2}(X, L) \\ \cup \xi_Y^{n-1-p} \downarrow & & \downarrow \cup \xi_X^{n-1-p} \\ H^{2n-2-p}(Y, (L|Y)(n-2-p)) & \xrightarrow{\text{Gysin}} & H^{2n-p}(X, L(n-1-p)). \end{array}$$
In this diagram, the bottom row is an isomorphism for $2n-2-p \geq n$, i.e. for $p \leq n-2$ (7.2 (ii)).

For a discussion of these conjectures and their consequences, the reader is referred to Kleiman's exposé, *Algebraic cycles and the Weil conjectures*, in *Dix exposés sur la cohomologie des schémas.*

8. Blown-up varieties

In this section we give a cohomological variant of the calculations, for the group K and for the "group of cycle classes" of a blown-up scheme, given in SGA 6 VII 3.7 and 4.8. The hypotheses and notations are those of the beginning of paragraph 4 (4.1 to 4.3). Under these conditions, the "fundamental class" morphism
$$\gamma : j_*(\mu_{Y'}^{\otimes -1})[-2] \to A_{X'}$$
gives, after applying the functor Rf_*, a morphism
$$Rf_*(\gamma) : i_* Rg_*(\mu_{Y'}^{\otimes -1})[-2] \to Rf_*(A_{X'}).$$

Theorem 8.1. The morphism in $D^+(X, A)$
$$A_X \oplus i_* Rg_*(\mu_{Y'}^{\otimes -1})[-2] \to Rf_*(A_{X'}) \oplus i_*(\mu_Y^{\otimes -d})[-2d] \tag{8.1.1}$$

defined by the matrix

$$\begin{pmatrix} \mathrm{adj} & Rf_*(\gamma) \\ 0 & i_*(\mathrm{trace}) \end{pmatrix}$$

is an isomorphism.

Proof. By conservativity of the system of fibre functors, it is enough to verify the assertion above each point x of X. It is evident above $U = X - Y$. For $x \in Y$, proper base change reduces us to proving the following lemma, with $r = d - 1$.

Lemma 8.2. Let k be a separably closed field, let $r \geq 1$ be an integer, and let $P = \mathbb{P}^r_k$. Then:

a) the canonical map $A \to H^0(P, A)$ is a bijection;

b) $H^1(P, A) = H^{2r-1}(P, A) = 0$;

c) for $0 \leq p \leq 2r - 2$, multiplication by $\xi = c_1(\mathcal{O}_P(1)) \in H^2(P, \mu)$

$$H^p(P, \mu_P^{\otimes -1}) \to H^{p+2}(P, A)$$

is an isomorphism;

d) the trace morphism

$$H^{2r}(P, \mu_P^{\otimes -1}) \to H^0(k, \mu_k^{\otimes -(r+1)})$$

is a bijection.

The lemma is an immediate consequence of 2.2.2.

For the statement of the following corollary, we agree that, for every X-scheme $h : T \to X$, every closed subset Z of T, and every complex M of A_X-modules, we write

$$H^*(T, M) = H^*(T, h^*(M)), \qquad H^*_Z(T, M) = H^*_Z(T, h^*(M)).$$

Corollary 8.3. For every complex M of A_X-modules and every $p \in \mathbb{Z}$, the maps

$$H^p(X, M) \oplus H^{p-2}(Y', M \otimes \mu^{\otimes -1}) \to H^p(X', M) \oplus H^{p-2d}(Y, M \otimes \mu^{\otimes -d}) \tag{8.3.1}$$

and

$$H^p_Y(X, M) \oplus H^{p-2}(Y', M \otimes \mu^{\otimes -1}) \to H^p_{Y'}(X', M) \oplus H^{p-2d}(Y, M \otimes \mu^{\otimes -d}) \tag{8.3.2}$$

defined by the matrix

$$\begin{pmatrix} f^* & j_* \\ 0 & g_* \end{pmatrix}$$

are bijections.

Proof. Temporarily denote by u the morphism (8.1.1). By the projection formula, the arrows (8.3.1) and (8.3.2) are those induced on the p-th cohomology objects by the morphisms $R\Gamma(X, u \otimes \mathrm{id}_M)$ and $R\Gamma_Y(X, u \otimes \mathrm{id}_M)$ respectively. The assertion follows.

8.4

We now make explicit the inverse isomorphism of (8.3.2), and for this we need a few lemmas.

Lemma 8.4.1. Let S be a scheme, let E be a locally free $\mathcal{O}_S$-module of constant rank d, and let $p : P = \mathbb{P}_S(E) \to S$ be the associated projective bundle. Let F be the $\mathcal{O}_P$-module defined by the canonical exact sequence

$$0 \to F \to p^*(E) \to \mathcal{O}_P(1) \to 0.$$

Then

$$p_*(c_{d-1}(\check{F})) = 1.$$

Proof. Let c_i denote the Chern classes of E, and put $\xi = c_1(\mathcal{O}_P(1))$. From

$$p^*(c(E)) = c(F)(1 + \xi)$$

it follows that

$$c(F) = p^*(c(E))(1 + \xi)^{-1},$$

hence

$$c_{d-1}(F) = \sum_{0 \le i \le d-1} (-1)^{d-1-i} c_i \, \xi^{d-1-i}. \tag{8.4.2}$$

Now p_* lowers degrees by $2d - 2$, and $p_*(\xi^{d-1}) = 1$ (2.2.2). Applying p_* to both sides of (8.4.2) and using the projection formula gives the lemma.

Lemma 8.4.3. Let F denote the locally free $\mathcal{O}_{Y'}$-module defined by the canonical exact sequence

$$0 \to F \to g^*(N) \to \mathcal{O}_{Y'}(1) \to 0.$$

Let M be an A_X-module which is locally constant constructible in a neighbourhood of Y, and let $p \ge 0$ be an integer. For every $y \in H^p(Y, M)$ one has

$$f^*(i_*(y)) = j_*(g^*(y) c_{d-1}(\check{F}))$$

in $H^{p+2d}_{Y'}(X', M \otimes \mu^{\otimes d})$.

Proof. Put

$$z = f^*(i_*(y)) - j_*(g^*(y) c_{d-1}(\check{F})).$$

Then

$$j^*(z) = g^* i^* i_*(y) - j^* j_*(g^*(y) c_{d-1}(\check{F})),$$

so, by (4.1) and the linearity lemma of (SGA 4 XVIII),

$$j^*(z) = g^*(y)(c_d(g^*\check{N}) - c_{d-1}(\check{F}) c_1(\mathcal{O}_{Y'}(-1))) = 0. \tag{8.4.4}$$

Writing $U = X - Y$ and $U' = X' - Y'$, one has an evident exact commutative diagram

$$\begin{array}{ccccc} H^{p+2d-1}(U') & \xrightarrow{\partial} & H^{p+2d}_{Y'}(X') & \xrightarrow{j^*} & H^{p+2d}(Y') \\ \uparrow \wr & & \uparrow f^* & & \uparrow g^* \\ H^{p+2d-1}(U) & \xrightarrow{\partial} & H^{p+2d}_{Y}(X) & \xrightarrow{i^*} & H^{p+2d}(Y), \end{array} \tag{8.4.5}$$

with evident simplifications of notation. Considering (8.4.5) and (8.4.4), one sees that z is of the form

$$z = f^*(t),$$

with $t \in H_Y^{p+2d}(X, M \otimes \mu^{\otimes d})$. By (4.2), $t = f_*(z)$, so it is enough to see that $f_*(z) = 0$. But

$$f_*(z) = f_* f^* i_*(y) - f_* j_*(g^*(y) c_{d-1}(\check{F})),$$

and using (4.2) gives

$$f_*(z) = i_*(y) - i_* g_*(g^*(y) c_{d-1}(\check{F})).$$

The conclusion follows from the projection formula for g and from (8.4.2).

Proposition 8.5. Let M be an A_X-module which is locally constant constructible in a neighbourhood of Y.

a) Let

$$\gamma : H^*_{Y'}(X', M \otimes \mu^{\otimes d}) \to H^*(Y', M \otimes \mu^{\otimes (d-1)})(-2)$$

be the unique map defined by the relation

$$f^* \circ f_* = j_* \circ \gamma + \mathrm{id}\,. \tag{8.5.1}$$

Then

$$g_* \circ \gamma = 0, \tag{8.5.2}$$

and the inverse matrix of (8.3.2) is

$$\begin{pmatrix} f_* & -i_* \\ -\gamma & g^*(\cdot) c_{d-1}(\check{F}) \end{pmatrix}. \tag{8.5.3}$$

b) For every integer $p \geq 0$, the sequence

$$0 \to H^{p-2d}(Y, M(-d)) \xrightarrow{\lambda} H^{p-2}(Y', M(-1)) \oplus H^p(X, M) \xrightarrow{\mu} H^p(X', M) \to 0,$$

where

$$\lambda(x) = (g^*(x) c_{d-1}(\check{F}), -i_*(x)), \qquad \mu(x, y) = j_*(x) + f^*(y),$$

is exact. Moreover, the map λ has the left inverse

$$\lambda'(x, y) = g_*(x).$$

Proof. We prove a). In view of 4.2 ($f_* f^* = \mathrm{id}$), in order to see that

$$\begin{pmatrix} f_* & -i_* \\ -\gamma & g^*(\cdot) c_{d-1}(\check{F}) \end{pmatrix} \circ \begin{pmatrix} f^* & j_* \\ 0 & g_* \end{pmatrix} = \mathrm{id},$$

it is enough to prove

$$\gamma j_* + \mathrm{id} = g^*(g_*(\cdot) c_{d-1}(\check{F})), \tag{8.5.4}$$

and

$$\gamma f^* = 0. \tag{8.5.5}$$

From (8.5.1) one deduces

$$f^*f_*f^* = j_*\gamma f^* + f^*,$$

and hence, by (4.2), $j_*\gamma f^* = 0$; this gives (8.5.5), since j_* is an isomorphism. We prove (8.5.4). Again (8.5.1) gives

$$f^*f_*j_* = j_*\gamma j_* + j_*,$$

so, since $f_*j_* = i_*g_*$ and using (8.4.3),

$$j_*g^*(g_*(\cdot)c_{d-1}(\check{F})) = j_*\gamma j_* + j_*.$$

Multiplying on the left by the inverse of j_* gives (8.5.4). Finally, (8.5.2) is obtained by writing that the product of the two matrices in the other order is the identity.

We prove b). From 8.4.3 and 8.4.1 respectively one gets $\mu \circ \lambda = 0$ and $\lambda' \circ \lambda = \mathrm{id}$. Moreover, μ is surjective by (8.3). It remains to prove that $\mathrm{Ker}(\mu) \subset \mathrm{Im}(\lambda)$. But $\mu \circ (\lambda \circ \lambda') = 0$ and $\lambda' \circ (\lambda \circ \lambda') = \mathrm{id}$, so $\mu(x) = 0$ implies

$$(\mu \oplus \lambda')(\lambda \circ \lambda'(x)) = (\mu \oplus \lambda')(x),$$

and hence $x = \lambda \circ \lambda'(x)$ by 8.3.

8.6

We shall now try to determine the multiplicative structure of the cohomology of X'. For every $p \in \mathbb{Z}$ and every A_X-module M which is locally constant constructible in a neighbourhood of Y, put

$$\nabla^p(X', M) = H^{p-2}(Y', M(-1)) \oplus H^p(X, M).$$

For $p, q \in \mathbb{Z}$ and two such modules M, N, define a composition law

$$\cup : \nabla^p(X', M) \otimes_A \nabla^q(X', N) \to \nabla^{p+q}(X', M \otimes_A N) \tag{8.6.1}$$

by

$$(y'_1, x_1) \cup (y'_2, x_2) = (-y'_1y'_2\xi + y'_1g^*i^*(x_2) + (-1)^{pq}y'_2g^*i^*(x_1),\ x_1x_2). \tag{8.6.2}$$

This composition law is clearly anticommutative, and an easy but tedious calculation shows that it is associative. In particular,

$$\nabla^*(X', A) = \bigoplus_{p \in \mathbb{Z}} \nabla^p(X', A)$$

is also endowed with a structure of graded-commutative A-algebra, in the sense of algebras graded by $\mathbb{Z}$, and

$$\nabla^*(X', M) = \bigoplus_{p \in \mathbb{Z}} \nabla^p(X', M)$$

is endowed with a graded module structure over the graded ring $\nabla^*(X', A)$.

Proposition 8.6.3.

a) For every pair (M, N) of A_X-modules locally constant constructible in a neighbourhood of Y, the diagram

$$\begin{array}{ccc} \nabla^*(X', M) \otimes \nabla^*(X', N) & \xrightarrow{\mu \otimes \mu} & H^*(X', M) \otimes H^*(X', N) \\ \Big\downarrow{\scriptstyle (8.6.1)} & & \Big\downarrow{\scriptstyle \cup} \\ \nabla^*(X', M \otimes N) & \xrightarrow{\mu} & H^*(X', M \otimes N) \end{array}$$

is commutative. In particular, the surjective map

$$\mu_A = \mu : \nabla^*(X', A) \to H^*(X', A)$$

is a ring morphism, and

$$\mu_M = \mu : \nabla^*(X', M) \to H^*(X', M)$$

is a morphism of modules over variable rings.

b) If $\nabla^*(X', A)$ is made to act on $H^*(Y, M(-d))(-2d)$ by the map

$$\nabla^p(X', A) \otimes H^{q-2d}(Y, M(-d)) \to H^{p+q-2d}(Y, M(-d)), \qquad (y', x) \otimes y \mapsto i^*(x)y,$$

then the map λ of 8.5 b) becomes a morphism of $\nabla^*(X', A)$-modules.

Proof. We prove a). With the notation of 8.6.2, one must prove the equality

$$j_*(-y'_1 y'_2 \xi + y'_1 g^* i^*(x_2) + (-1)^{pq} y'_2 g^* i^*(x_1)) + f^*(x_1 x_2)$$
$$= (j_*(y'_1) + f^*(x_1))(j_*(y'_2) + f^*(x_2)).$$

It follows at once from

$$\begin{cases} f^*(x_1) f^*(x_2) = f^*(x_1 x_2), & \text{multiplicativity of } f^*, \\ j_*(-y'_1 y'_2 \xi) = j_*(y'_1) j_*(y'_2), & (4.5), \\ j_*(y' g^* i^*(x)) = j_*(y' j^* f^*(x)) = j_*(y') f^*(x), & \text{projection formula.} \end{cases}$$

We prove b). The equality to be proved,

$$\lambda(i^*(x) \cdot y) = (y', x) \cdot \lambda(y),$$

breaks up into the two following equalities:

$$g^*(i^*(x) \cdot y) c_{d-1}(\check{F}) = -y' g^*(y) c_{d-1}(\check{F}) \xi + y' g^* i^*(-i_*(y)) + g^*(i^* x) g^*(y) c_{d-1}(\check{F}),$$

and

$$-i_*(i^*(x) y) = -x\, i_*(y).$$

The second is just the projection formula for i. The first follows immediately from

$$-\xi\, c_{d-1}(\check{F}) = g^*[c_d(\check{N})] \quad \text{(exact sequence of 8.4.3)}$$

and

$$i^* i_*(y) = y\, c_d(\check{N}) \qquad (4.1).$$

9. Chow ring of a blown-up variety and the self-intersection formula in the Chow ring

In this paragraph we propose to give Mumford's proof of the analogue of 4.1 in the Chow ring. We shall then show how one can deduce from it the "key formula" (8.4.3), and hence determine the structure of the Chow ring of a blown-up variety.

In accordance with the conventions of 3.9, $C^*(X)$ denotes the Chow ring of a quasi-projective smooth scheme X over a field k. The theory of Chern classes used is that of [1]. We shall use the following lemma several times.

Lemma 9.1. Let S be a quasi-projective smooth scheme over a separably closed field k, let E be a locally free $\mathcal{O}_S$-module of constant rank $r+1$, and let $p : P = \mathbb{P}_S(E) \to S$ be the associated projective bundle. Let F be the $\mathcal{O}_P$-module defined by the canonical exact sequence

$$0 \to F \to p^*(E) \to \mathcal{O}_P(1) \to 0, \tag{9.1.0}$$

and put $\xi = c_1(\mathcal{O}_P(1))$. Then:

(i) for every $y \in C^*(S)$,

$$y = p_*(p^*(y)c_r(\check{F})); \tag{9.1.1}$$

(ii) suppose that $E = N \oplus \mathcal{O}_S$, with N a locally free $\mathcal{O}_S$-module of rank r, so that P identifies with the projective completion $\widehat{N}$ of $V(N)$, and let

$$s : S \to \mathbb{P}_S(E)$$

be the section obtained by composing the zero-section of $V(N)$ with the canonical inclusion $V(N) \hookrightarrow \widehat{N}$. Then:

(a) for every $y \in C^*(S)$ one has

$$s_*(y) = p^*(y)c_r(\check{F}) = p^*(y)\left(\sum_{i=0}^{r} p^*(c_i(\check{N}))\xi^{r-i}\right) \quad \text{in } C^*(\widehat{N}), \tag{9.1.2}$$

$$s^*s_*(y) = y\, c_r(\check{N}) \quad \text{in } C^*(S); \tag{9.1.3}$$

(b) for every $z \in C^*(\widehat{N})$,

$$s^*(z) = p_*(z\, c_r(\check{F})) \quad \text{in } C^*(S). \tag{9.1.4}$$

Proof. We prove (9.1.1). By the projection formula for p, it is enough to show that $p_*(c_r(\check{F})) = 1$. For this it suffices to copy the proof of 8.4.1, using the relations ([CH] 4 (17)). The equality of the first and third members of (9.1.2) is just Lemma 3 of [1]. Paraphrasing the proof of (8.4.1), one sees that

$$c_r(\check{F}) = \sum_{i=0}^{r} p^*(c_i(\check{N}))\xi^{r-i}, \tag{9.1.2 bis}$$

which proves (9.1.2). Applying s^* to the two extreme members of (9.1.2), and using $p \circ s = \mathrm{id}$, one sees that (9.1.3) will follow from

$$s^*\left(\sum_{i=0}^{r-1} p^*(c_i(\check{N}))\xi^{r-i}\right) = 0,$$

and a fortiori from

$$s^*(\xi^i) = 0 \qquad (i > 0). \tag{9.1.5}$$

We prove (9.1.5). If $\alpha : V(N) \hookrightarrow \widehat{N}$ denotes the canonical inclusion, then evidently $\alpha^*(\mathcal{O}_P(1)) \simeq \mathcal{O}_{V(N)}$, whence

$$\alpha^*(\xi) = \alpha^*(c_1(\mathcal{O}_P(1))) = c_1(\mathcal{O}_{V(N)}) = 0$$

by functoriality of Chern classes. The assertion follows immediately from the definition of s. We prove (9.1.4). We shall use the equality

$$\xi\, c_r(\check{F}) = c_{r+1}(p^*(\check{N}) \oplus \mathcal{O}'_P) = 0, \tag{9.1.6}$$

which follows at once from the exact sequence (9.1.0). The element z has a unique decomposition

$$z = \sum_{i=0}^{r} p^*(a_i)\xi^i, \qquad a_i \in C^*(S),$$

([CH] 4 Th. 1). It follows from (9.1.5) that $s^*(z) = a_0$. On the other hand, (9.1.6) shows that $z\, c_r(\check{F}) = p^*(a_0)c_r(\check{F})$, and the conclusion follows from (9.1.1).

We now state the self-intersection formula in the Chow ring.

Theorem 9.2 (Mumford). Let k be a separably closed field and let $i : Y \hookrightarrow X$ be a closed immersion, purely of codimension d, of quasi-projective smooth k-schemes. If N denotes the normal sheaf of i, then for every $y \in C^*(Y)$ one has

$$i^* i_*(y) = y\, c_d(\check{N})$$

in $C^*(Y)$.

For the proof we shall need a number of notations and auxiliary schemes. Embed X in $Z = X \times_k \mathbb{P}^1_k$ by the morphism $u : t \mapsto (t, 0)$, so that there is a cartesian diagram

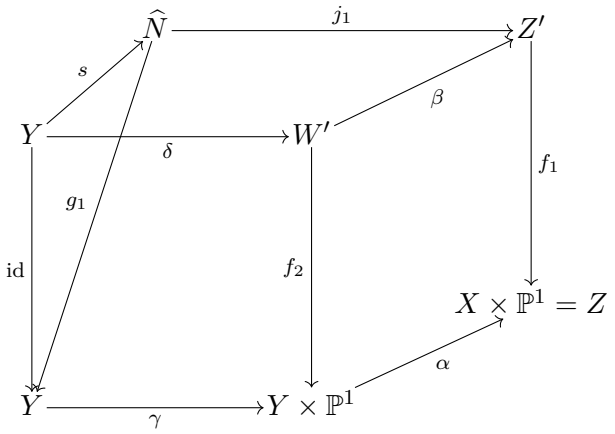
$$\begin{array}{ccc} \widehat{N} & \xrightarrow{j_1} & Z' \\ {\scriptstyle g_1}\downarrow & & \downarrow{\scriptstyle f_1} \\ Y & \xrightarrow[i_1]{} & Z, \end{array}$$

where Z' denotes the blowing-up of Z along Y, and $\widehat{N} = \mathbb{P}_Y(N \oplus \mathcal{O}_Y)$ is the projective completion of the vector bundle $V(N)$. As in the statement of 9.1, let $s : Y \hookrightarrow \widehat{N}$ be the morphism obtained by composing the zero-section of $V(N)$ with the canonical inclusion $V(N) \subset \widehat{N}$.

The inclusion $Y \subset X$ identifies $Y \times \mathbb{P}^1$ with a closed subscheme of $X \times \mathbb{P}^1$. Let W' be the strict transform of $Y \times \mathbb{P}^1$ by f_1, i.e. the subscheme of Z' obtained by blowing up $Y \times \mathbb{P}^1$ along $Y = Y \times 0$. It is clear that $W' \cap \widehat{N}$ identifies with the zero-section of $\widehat{N}$ and that, since Y has codimension 1 in $Y \times \mathbb{P}^1$, the canonical morphism $f_2 : W' \to Y \times \mathbb{P}^1$ is an isomorphism. The remaining notation

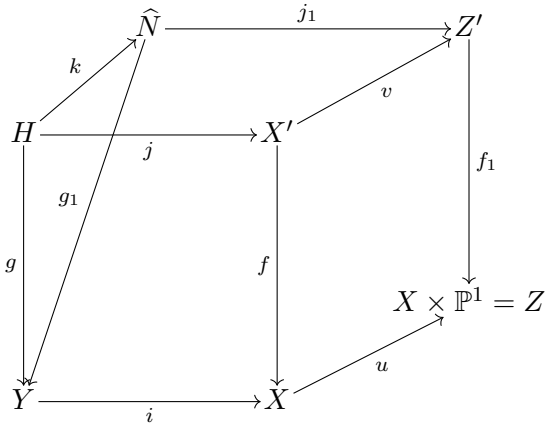
relative to W' is fixed by the canonical diagram (D_1).

(D_1)



Similarly, if X' denotes the closed subscheme of Z' which is the blowing-up of $X = X \times 0$ along Y, it is clear that $\mathbb{P}(N) = X' \cap \widehat{N}$ identifies with the hyperplane at infinity H of $\widehat{N}$. The remaining notation relative to X' is fixed by the canonical diagram (D_2).

(D_2)



We now give, in lemma form, a number of relations useful below. Denote by E the locally free $\mathcal{O}_{\widehat{N}}$-module defined by the canonical exact sequence

$$0 \to E \to g_1^*(N \oplus \mathcal{O}_Y) \to \mathcal{O}_{\widehat{N}}(1) \to 0, \tag{9.3.0}$$

and put

$$\bar{\xi} = c_1(\mathcal{O}_{\widehat{N}}(1)). \tag{9.3.1}$$

Lemma 9.3. With the preceding notation, one has

$$j_1^*(\beta_*(1)) = c_d(\check{E}) \quad \text{in } C^*(\widehat{N}), \tag{9.3.2}$$

$$j_1^*(v_*(1)) = \bar{\xi} \quad \text{in } C^*(\widehat{N}), \tag{9.3.3}$$

$$f_1^* u_*(1) = (j_1)_*(1) + v_*(1) \quad \text{in } C^*(Z'). \tag{9.3.4}$$

Proof. All the k-schemes under consideration are smooth, so it is immediate for dimension reasons that the morphisms j_1 and β (resp. j_1 and v) are transverse. From ([CH] 4 Cor. 1 to Prop. 2) one obtains

$$\begin{cases} j_1^*\beta_*(1) = s_*(1), \\ j_1^* v_*(1) = k_*(1). \end{cases}$$

Equations (9.3.2) and (9.3.3) then come from (9.1.2) and from the normalization of Chern classes ([CH] 4–18), respectively. We prove (9.3.4). Since the cycle X' of Z' is the strict transform of $X \times 0$ by f_1, one has a priori an equality of the form

$$f_1^* u_*(1) = (j_1)_* g_1^*(\mu) + v_*(1), \tag{9.3.5}$$

with $\mu \in \Gamma(Y, \mathbb{N})$; it remains to show that $\mu = 1$. Applying j_1^* to both sides of (9.3.5), one obtains, cf. (D_2),

$$g_1^* i^* u^* u_*(1) = j_1^*(j_1)_* g_1^*(\mu) + j_1^* v_*(1). \tag{9.3.6}$$

A very special case of (9.1.3), or a direct verification, shows that $u^* u_*(1) = 0$. Using (9.3.3), it follows from (9.3.6) that

$$0 = j_1^*(j_1)_* g^*(\mu) + \bar{\xi}. \tag{9.3.7}$$

After restricting to a neighbourhood of a connected component of Y, one may suppose Y connected, so that (9.3.7) takes the form

$$0 = \mu\, j_1^*(j_1)_*(1) + \bar{\xi}, \qquad \mu \in \mathbb{N}. \tag{9.3.7 bis}$$

Since $j_1^*(j_1)_*(1)$ is homogeneous of degree 1, it is of the form

$$j_1^*(j_1)_*(1) = g_1^*(\rho) + \nu\bar{\xi},$$

with $\rho \in C^1(Y)$ and $\nu \in \mathbb{Z}$. Substituting this expression into (9.3.7 bis) gives at once $\rho = 0$ and $\mu\nu = -1$, with $\mu \in \mathbb{N}$; hence $\mu = 1$ and $\nu = -1$, completing the proof of (9.3.4).

The key to the proof of 9.2 lies in the proof of the following lemma, which would be evident if one knew, as in cohomology, that

$$j_1^*(j_1)_*(y) = -y\bar{\xi}$$

for every $y \in C^*(\widehat{N})$.

Lemma 9.4. For every $y \in C^*(\widehat{N})$, one has

$$c_d(\check{E})\, j_1^*(j_1)_*(y) = 0.$$

Proof. The element y has a unique decomposition in $C^*(\widehat{N})$ of the form

$$y = \sum_{0 \le p \le d} g_1^*(a_p)\bar{\xi}^p,$$

or, by (9.3.3),

$$y = \sum_{0 \le p \le d} g_1^*(a_p)\big(j_1^* v_*(1)\big)^p. \tag{9.4.1}$$

Using the projection formula ([CH] 4 I 9) for j_1 and (9.1.6), it follows that

$$c_d(\check{E})\, j_1^*(j_1)_*(y) = c_d(E)\, j_1^*(j_1)_* g_1^*(a_0);$$

in other words, we are reduced to proving 9.4 when y is of the form $g_1^*(a)$, with $a \in C^*(Y)$. Then $j_1^*(j_1)_* g_1^*(a)$ has in turn a unique decomposition in $C^*(\widehat{N})$ of the form

$$j_1^*(j_1)_* g_1^*(a) = \sum_{0 \le p \le d} g_1^*(b_p)\bar{\xi}^p, \tag{9.4.2}$$

and, using (9.1.6) once again, it remains to prove that $g_1^*(b_0)c_d(\check{E}) = 0$. In fact, we shall prove the stronger statement $b_0 = 0$. In view of (9.3.3), we may rewrite (9.4.2) as

$$j_1^*(j_1)_* g_1^*(a) = g_1^*(b_0) + z\, j_1^* v_*(1), \tag{9.4.3}$$

with $z \in C^*(\widehat{N})$. Applying $(j_1)_*$ to both sides of (9.4.3) and using the projection formula for j_1, one gets

$$(j_1)_* g_1^*(b_0) = \big[(j_1)_* g_1^*(a)\big](j_1)_*(1) - v_*(1)(j_1)_*(z). \tag{9.4.4}$$

Substituting in (9.4.4) the expression for $(j_1)_*(1)$ deduced from (9.3.4) gives

$$(j_1)_* g_1^*(b_0) = (j_1)_* g_1^*(a) f_1^* u_*(1) - v_*(1)\big[(j_1)_* g_1^*(a) + (j_1)_*(z)\big]. \tag{9.4.5}$$

Since $X \times 0$ is linearly equivalent to $X \times 1$ in $X \times \mathbb{P}^1$, one has

$$(j_1)_* g_1^*(a) f_1^* u_*(1) = 0. \tag{9.4.6}$$

Comparing with (9.4.5), we get

$$(j_1)_* g_1^*(b_0) = -v_*(1)\big[(j_1)_* g_1^*(a) + (j_1)_*(z)\big]. \tag{9.4.7}$$

Moreover, it is clear that $W' \cap X' = \emptyset$: they could meet at most above Y, where one gives the zero-section and the other the hyperplane at infinity of $\widehat{N}$. Hence $\beta^* v_*(1) = 0$, and from (9.4.7)

$$\beta^*(j_1)_* g_1^*(b_0) = 0. \tag{9.4.8}$$

But, as already observed in the proof of 9.3.2, β and j_1 are transverse; thus (9.4.8) gives

$$0 = \delta_* s^* g_1^*(b_0) = \delta_*(b_0). \tag{9.4.9}$$

Finally, applying $(f_2)_*$ to (9.4.9), one obtains $\gamma_*(b_0) = 0$, whence $b_0 = 0$, since γ_* is injective (γ admits an evident retraction).

9.5

We finally show how the preceding results imply the theorem. Let $p : X \times \mathbb{P}^1_k \to X$ be the first projection, so that $i = p \circ i_1$. If $y \in C^*(Y)$, then evidently

$$i^* i_*(y) = (i_1)^*(\bar{x}), \qquad \bar{x} = p^* i_*(y). \tag{9.5.1}$$

By (9.1.1) it follows that

$$i^* i_*(y) = (g_1)_*\big[g_1^* i_1^*(\bar{x}) c_d(\check{E})\big],$$

hence

$$i^* i_*(y) = (g_1)_*\big[j_1^* f_1^*(\bar{x}) \cdot c_d(\check{E})\big]. \tag{9.5.2}$$

To prove the theorem, by linearity we may suppose that y is the irreducible cycle associated with a subvariety T of Y, so that $\bar{x}$ corresponds to the cycle $T \times \mathbb{P}^1$ of $Z = X \times \mathbb{P}^1$. Let θ denote the strict transform of $T \times \mathbb{P}^1$ by f_1, i.e. the closure of the inverse image of $T \times \mathbb{P}^1$ over $Z \setminus Y$. Then

$$f_1^*(\bar{x}) = \theta + (j_1)_*(w), \qquad w \in C^*(\widehat{N}). \tag{9.5.3}$$

Substituting this expression into (9.5.2), 9.4 gives

$$i^* i_*(y) = (g_1)_*\big[j_1^*(\theta) c_d(\check{E})\big], \tag{9.5.4}$$

and, by (9.1.4),

$$i^* i_*(y) = s^* j_1^*(\theta). \tag{9.5.5}$$

But the reduced scheme underlying θ is the blowing-up of $T \times \mathbb{P}^1$ along $T = (T \times \mathbb{P}^1) \cap Y$; it meets the zero-section of $\widehat{N}$ transversely along the image of T by that zero-section. Hence

$$j_1^*(\theta) = s_*(y),$$

which, together with (9.5.5) and (9.1.3), completes the proof of the theorem.

We now turn to the "key formula" mentioned in the introduction. The adapted notation, recalled for the most part, is that of 4. Observe that it is compatible with the notation introduced in the present paragraph, except that the scheme denoted Y' in 4 is here denoted H (for "hyperplane"). We shall use both notations concurrently.

Proposition 9.6. Let k be a separably closed field and let $i : Y \hookrightarrow X$ be a closed immersion, purely of codimension d, of quasi-projective smooth k-schemes. Let $f : X' \to X$ be the X-scheme obtained by blowing up X along Y, and let

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X \end{array}$$

be the cartesian diagram obtained from f and i. If F denotes the locally free $\mathcal{O}_{Y'}$-module defined by the canonical exact sequence

$$0 \to F \to g^*(N) \to \mathcal{O}_{Y'}(1) \to 0, \tag{9.6.1}$$

then

$$f^* i_*(y) = j_*(g^*(y) c_{d-1}(\check{F})) \tag{9.6.2}$$

in $C^*(X')$, for every $y \in C^*(Y)$.

Before proving 9.6, we isolate in a lemma some properties of the diagrams (D_1) and (D_2) which have not yet been established.

Lemma 9.7. One has

$$c_d(\check{E}) = k_*(c_{d-1}(\check{F})) + (g_1)^*(c_d(\check{N})), \tag{9.7 i}$$

$$\beta_* f_2^* \gamma_*(y) = f_1^* \alpha_* \gamma_*(y) \qquad \text{for every } y \in C^*(Y), \tag{9.7 ii}$$

$$f_1^* u_*(t) = (j_1)_*((i \circ g_1)^*(t)) + v_* f^*(t) \qquad \text{for every } t \in C^*(X). \tag{9.7 iii}$$

Proof. We prove (i). Put $\xi = c_1(\mathcal{O}_{Y'}(1)) = k^*(\bar{\xi})$. Paraphrasing the proof of (8.4.1), one obtains

$$c_d(\check{E}) = \sum_{p=0}^{d} g_1^*(c_p(\check{N}))\bar{\xi}^{d-p}, \tag{9.7.1}$$

$$c_{d-1}(\check{F}) = \sum_{q=0}^{d-1} g^*(c_q(\check{N}))\xi^{d-1-q}. \tag{9.7.2}$$

From (9.7.2) one gets

$$k_*[c_{d-1}(\check{F})] = \sum_{q=0}^{d-1} k_* k^*[g_1^*(c_q(\check{N}))\bar{\xi}^{d-1-q}],$$

and hence, by the projection formula for k and the fact that $k_*(1) = \bar{\xi}$,

$$k_*[c_{d-1}(\check{F})] = \sum_{q=0}^{d-1} g_1^*(c_q(\check{N}))\bar{\xi}^{d-q}. \tag{9.7.3}$$

Assertion (i) follows at once by comparing (9.7.1) and (9.7.3). We prove (ii). We may suppose that y is an irreducible cycle of Y, corresponding to a subvariety denoted by the same letter. Then the cycle $\gamma_*(y) = y \times 0$ is rationally equivalent to $y \times 1$, so it is enough to verify the equality of cycles

$$\beta_* f_2^*(y \times 1) = f_1^* \alpha_*(y \times 1). \tag{9.7.4}$$

Since $(y \times 1) \cap (y \times 0) = \emptyset$, f_1 is an isomorphism above an open neighbourhood of $y \times 1$, which gives (9.7.4). We prove (iii). If $p : X \times \mathbb{P}^1 \to X$ is the first projection, then (9.1.2) gives

$$u_*(t) = p^*(t)u_*(1), \tag{9.7.5}$$

and comparing with (9.3.4) yields

$$f_1^* u_*(t) = (p \circ f_1)^*(t)\big[(j_1)_*(1) + v_*(1)\big]. \tag{9.7.6}$$

The projection formulas for j_1 and v then give

$$f_1^* u_*(t) = (j_1)_*\big[(p \circ f_1 \circ j_1)^*(t)\big] + v_*\big[(p \circ f_1 \circ v)^*(t)\big], \tag{9.7.7}$$

which is (iii), since $p \circ f_1 \circ j_1 = i \circ g_1$ and $p \circ f_1 \circ v = f$.

9.8

We now prove 9.6. We shall first prove

$$v_* f^* i_*(y) = v_* j_*(g^*(y) c_{d-1}(\check{F})), \tag{9.8.1}$$

and then prove that v_* is injective. From (9.7)(iii) one gets

$$v_* f^* i_*(y) = f_1^* u_* i_*(y) - (j_1)_*[g_1^* i^* i_*(y)],$$

and therefore, by (9.2),

$$v_* f^* i_*(y) = f_1^*(i_1)_*(y) - (j_1)_*[g_1^*(y\, c_d(\check{N}))]. \tag{9.8.2}$$

On the other hand,

$$v_*j_*(g^*(y)c_{d-1}(\check{F})) = (j_1)_*[k_*k^*g_1^*(y)c_{d-1}(\check{F})],$$

or, by the projection formula for k,

$$v_*j_*(g^*(y)c_{d-1}(\check{F})) = (j_1)_*[g_1^*(y)k_*(c_{d-1}(\check{F}))].$$

Using (9.7)(i), this gives

$$v_*j_*(g^*(y)c_{d-1}(\check{F})) = (j_1)_*[g_1^*(y)c_d(\check{E})] - (j_1)_*[g_1^*(y\,c_d(\check{N}))]. \tag{9.8.3}$$

Comparing (9.8.2) and (9.8.3), one sees that (9.8.1) is equivalent to

$$f_1^*(i_1)_*(y) = (j_1)_*[g_1^*(y)c_d(\check{E})], \tag{9.8.4}$$

i.e. to the equality corresponding to (9.6.2) when X is replaced by $X \times \mathbb{P}^1$. By (9.1.2),

$$(j_1)_*[g_1^*(y)c_d(\check{E})] = (j_1)_*s_*(y) = \beta_*\delta_*(y).$$

But it is immediate (cf. (D_1)) that $\delta_* = f_2^*\gamma_*$, whence:

$$(j_1)_*[g_1^*(y)c_d(\check{E})] = \beta_*f_2^*\gamma_*(y). \tag{9.8.5}$$

Also, evidently,

$$f_1^*(i_1)_*(y) = f_1^*\alpha_*\gamma_*(y), \tag{9.8.6}$$

and (9.8.4) follows by comparing (9.8.5), (9.8.6), and (9.7)(ii).

Lemma 9.8.7. The map v_* is injective.

Proof. Let Z'' be the blowing-up of Z' along W', or equivalently along $\widehat{N} \cup W' = f_1^{-1}(Y \times \mathbb{P}^1)$. The strict transform X'' of X in Z'' is also the strict transform of X' in Z''. The notation is determined by the canonical diagram

$$\begin{array}{ccc} X'' & \xrightarrow{v''} & Z'' \\ {\scriptstyle f'}\big\downarrow\wr & & \big\downarrow{\scriptstyle f_1'} \\ X' & \xrightarrow[v]{} & Z'. \end{array}$$

Since $X' \cap W' = \emptyset$, the morphism f_1' is an isomorphism above an open neighbourhood of X', and in particular f' is an isomorphism. Moreover, since f_1' is transverse to v, one has

$$(f_1')^* \circ v_* = v_*'' \circ (f')^*, \tag{9.8.8}$$

so it is enough to show that v_*'' is injective. But, by the commutativity property of blow-ups (see, for example, Hironaka: *Smoothing of algebraic cycles*, Amer. Journal of Math. (1968), Lemma 4.1), Z'' can also be obtained by first blowing up $X \times \mathbb{P}^1$ along $Y \times \mathbb{P}^1$, yielding a scheme Z_1', and then blowing up Z_1' along the inverse image of Y. Clearly $Z_1' = X' \times \mathbb{P}^1$, and the projection $Z_1' \to X \times \mathbb{P}^1$ is the product $f \times \mathrm{id}_{\mathbb{P}^1}$; hence the evident cartesian diagram

$$\begin{array}{ccccc} H & \xrightarrow{t\mapsto(t,0)} & H \times \mathbb{P}^1 & \xrightarrow{j\times\mathrm{id}} & X' \times \mathbb{P}^1 = Z_1' \\ {\scriptstyle g}\big\downarrow & & \big\downarrow{\scriptstyle g\times\mathrm{id}} & & \big\downarrow{\scriptstyle f\times\mathrm{id}} \\ Y & \xrightarrow{\gamma} & Y \times \mathbb{P}^1 & \xrightarrow{i\times\mathrm{id}} & X \times \mathbb{P}^1 \end{array}$$

The strict transform of $X\times 0$ by $f\times\mathrm{id}$ identifies with $X'\times 0$, which contains $H = H\times 0 = (f\times\mathrm{id})^{-1}(Y)$. If we now blow up Z'_1 along H, then, since H is purely of codimension 1 in X', the strict transform of X' is canonically isomorphic to X' (as was already known); hence the canonical commutative diagram

$$\begin{array}{ccccc} & & X' & \xrightarrow{v''} & Z'' \\ & & \downarrow{\scriptstyle f''} & & \downarrow{\scriptstyle f''_1} \\ H & \longrightarrow & X' & \xrightarrow{v'} & X'\times\mathbb{P}^1 \end{array}$$

$$t \longmapsto (t,0)$$

Recall that we wish to prove that v''_* is injective. Comparing with (D_2), we see that this is equivalent to proving that v_* is injective when, in addition, Y has codimension 1 in X; this is what we now do. But then the identity $(f_1)_*\circ v_* = u_*\circ f_*$ gives the conclusion, since f_* is an isomorphism and u_* is a direct monomorphism (u admits a retraction).

We now state the theorem corresponding to (8.5 b).

Theorem 9.9. For every integer $p\geq 0$, the sequence

$$0\to C^{p-d}(Y)\xrightarrow{\lambda} C^{p-1}(Y')\oplus C^p(X)\xrightarrow{\mu} C^p(X')\to 0$$

in which λ and μ are the maps

$$\lambda : x\mapsto (g^*(x)c_{d-1}(\check{F}), -i_*(x)), \qquad \mu : (x,y)\mapsto j_*(x)+f^*(y)$$

is exact. Moreover, the map λ has the left inverse

$$\lambda' : (x,y)\mapsto g_*(x).$$

Proof. The last assertion is just (9.1.1). Likewise, (9.6.2) says that $\mu\circ\lambda = 0$. We prove that μ is surjective. We shall need

$$f_*f^*(z) = z \qquad (z\in C^*(X)), \tag{9.9.1}$$

which follows immediately from the projection formula for f and the evident fact that $f_*(1)=1$. Let $t\in C^p(X')$. By (9.9.1),

$$f_*(t - f^*f_*(t)) = 0.$$

Since f induces an isomorphism from $U' = X' - Y'$ to $U = X - Y$, the restriction of $t - f^*f_*(t)$ to U' is zero, and hence ([CH] 4 th. 2) there is an equality

$$t - f^*f_*(t) = j_*(a) \qquad \text{with } a\in C^*(Y').$$

We now prove exactness in the middle. Let $a\in C^{p-1}(Y')$ and $b\in C^p(X)$ satisfy

$$j_*(a) + f^*(b) = 0. \tag{9.9.2}$$

Applying f_* to both sides gives

$$f_*j_*(a) + f_*f^*(b) = 0,$$

and in view of (9.9.1),

$$b = -i_* g_*(a). \tag{9.9.3}$$

Putting $x = g_*(a)$, it remains only to prove that

$$a = g^*(x) c_{d-1}(\check{F}). \tag{9.9.4}$$

Put

$$z = a - g^*(x) c_{d-1}(\check{F}) \qquad (x = g_*(a)),$$

and prove $z = 0$. We shall first show that $j_*(z) = 0$ and $g_*(z) = 0$, and then that an element z satisfying these two relations is necessarily zero. By definition,

$$j_*(z) = j_*(a) - j_*(g^*(x) \cdot c_{d-1}(\check{F})),$$

so, by (9.6.2),

$$j_*(z) = j_*(a) - f^* i_*(x).$$

Replacing $i_*(x)$ by $-b$ using (9.9.3), one deduces from (9.9.2) that $j_*(z) = 0$. By the definition of x,

$$g_*(z) = x - g_*(g^*(x) c_{d-1}(\check{F})) = 0$$

by the last assertion of the theorem. It remains to explain why the vanishing of z follows from $j_*(z) = 0$ and $g_*(z) = 0$. Since $Y' = \mathbb{P}_Y(N)$, z has a unique expression

$$z = g^*(a_0) + g^*(a_1)\xi + \cdots + g^*(a_{d-1})\xi^{d-1},$$

where $a_i \in C^{p-1-i}(Y)$. Since $g_*(z) = 0$, the projection formula for g, together with the fact that g_* lowers degrees by $d-1$, gives $a_{d-1} = 0$. On the other hand, by (9.2),

$$z\,\xi = j^* j_*(z) = 0,$$

hence

$$g^*(a_0)\xi + \cdots + g^*(a_{d-1})\xi^{d-1} = 0.$$

Thus all a_i are zero and $z = 0$.

9.10

With the notation of (9.9), the map

$$\omega = \lambda \circ \lambda'$$

is a projector satisfying $\mathrm{Im}(\omega) = \mathrm{Im}(\lambda)$. Therefore μ induces an isomorphism

$$\bar{\mu} : \mathrm{Ker}(\omega) \xrightarrow{\sim} C^p(X').$$

But

$$\mathrm{Ker}(\omega) = \mathrm{Ker}(g_*) \oplus C^p(X) \hookrightarrow C^{p-1}(Y') \oplus C^p(X). \tag{9.10.1}$$

Indeed, an element (x, y) of $C^{p-1}(Y') \oplus C^p(X)$ belongs to $\mathrm{Ker}(\omega)$ if and only if

$$g^* g_*(x) \cdot c_{d-1}(\check{F}) = -i_* g_*(x) = 0,$$

that is, in view of (9.1.1), if and only if $g_*(x) = 0$. On the other hand, the structure of the Chow ring of a projective bundle, and in particular the relations

$$g_*(\xi^i) = 0 \quad \text{for } i < d-1, \qquad g_*(\xi^{d-1}) = 1,$$

show that

$$\mathrm{Ker}(g_*) = g^*C^{p-1}(Y) \oplus g^*C^{p-2}(Y)\xi \oplus \cdots \oplus g^*C^{p-d+1}(Y)\xi^{d-2}.$$

In other words, $\bar{\mu}$ may be interpreted as an isomorphism

$$C^{p-1}(Y) \oplus C^{p-2}(Y) \oplus \cdots \oplus C^{p-d+1}(Y) \oplus C^p(X) \xrightarrow{\sim} C^p(X'), \tag{9.10.2}$$

whose inverse is easy to exhibit.

As Murre pointed out to me, the statement (9.10.2) was proved for $d = 2$ by Samuel in *Relations d'équivalence*, Proceedings of the International Congress, Edinburgh 1958.

Of course, in the framework of étale or ℓ-adic cohomology one has an analogue of (9.10.2), deduced this time from (8.5).

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Exposé VIII

GROUPS OF CLASSES OF ABELIAN AND TRIANGULATED CATEGORIES. PERFECT COMPLEXES.

by A. Grothendieck, written up by I. Bucur

In the present exposé, we quickly outline the notions and results which will be indispensable to us in the following exposés, concerning in particular pseudo-coherent and perfect complexes. A more complete exposé on the latter, with more detailed proofs, will appear in the first exposés of the seminar on the Riemann-Roch theorem for schemes, which will follow the present seminar.[22]

1. The case of abelian categories

Let $\mathcal{C}$ be an abelian category and let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$. A function f from $\mathcal{D}$ to an abelian group G is said to be additive if, for every exact sequence in $\mathcal{C}$

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

such that E', E'', and E are objects of $\mathcal{D}$, one has

$$f(E) = f(E') + f(E'').$$

Let f be an additive function from $\mathcal{D}$ to an abelian group G. The following properties are easy consequences of the definition:

$\alpha)$ $f(0) = 0$, $\quad$ $\beta)$ $f(X) = f(Y)$ if X and Y are isomorphic,

$\gamma)$ $f(X \oplus Y) = f(X) + f(Y)$.

$\delta)$ Let $E = (E^i, d^i)$ be a finite complex in $\mathcal{D}$

such that the objects $B^{i+1} = \mathrm{Im}(d^i)$, $Z^i = \mathrm{Ker}(d^i)$ and $H^i(E) = Z^i/B^i$ are in $\mathcal{D}$. Then

$$\sum_i (-1)^i f(E^i) = \sum_i (-1)^i f(H^i(E)).$$

$\varepsilon)$ If $F \in \mathrm{ob}\,\mathcal{D}$ has a finite filtration

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = F,$$

and if one assumes that, for $0 < i \leq n$, F_i and F_i/F_{i-1} are objects of $\mathcal{D}$, then

$$f(F) = \sum_{i=1}^{n} f(F_i/F_{i-1}).$$

$\varphi)$ If $\mathcal{D}$ is stable under infinite direct sums, then $f = 0$.

[22] SGA 6 I

To prove δ), one uses the exact sequences

$$0 \longrightarrow Z^i \longrightarrow E^i \longrightarrow B^{i+1} \longrightarrow 0,$$

$$0 \longrightarrow B^i \longrightarrow Z^i \longrightarrow H^i(E) \longrightarrow 0.$$

The proof of ε) is by induction on n, using the exact sequence

$$0 \longrightarrow F_{n-1} \longrightarrow F \longrightarrow F/F_{n-1} \longrightarrow 0.$$

To prove φ), let $E \in \operatorname{ob}\mathcal{D}$ and

$$F = \bigoplus_{n\in\mathbb{Z}^+} E_n, \qquad E_n = E, \quad n \in \mathbb{Z}^+.$$

Then $E \oplus F \simeq F$, and applying γ) gives

$$f(F) = f(F) + f(E),$$

hence $f(E) = 0$.

If, for every abelian group G, one considers the abelian group $A_{\mathcal{D}}(G)$ of additive functions from $\mathcal{D}$ to G, one obtains in an evident way a covariant functor

$$A_{\mathcal{D}} : \mathrm{Ab} \longrightarrow \mathrm{Ab}\,.$$

The functor $A_{\mathcal{D}}$ is representable. A representing pair $(K_{\mathcal{D}}(\mathcal{C}), \varphi)$ may be obtained as follows. Let L be the free abelian group generated by the objects of $\mathcal{D}$, and let R be the subgroup of L generated by the elements of the form $E - E' - E''$ for each exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

of $\mathcal{C}$. Put

$$K_{\mathcal{D}}(\mathcal{C}) = L/R.$$

The map φ is obtained by composing the two canonical maps

$$\operatorname{ob}\mathcal{D} \longrightarrow L \longrightarrow L/R.$$

When there is no danger of confusion, one may abbreviate the notation:

$$K_{\mathcal{D}}(\mathcal{C}) = K(\mathcal{D}).$$

2. The case of triangulated categories

Let $\mathcal{C}$ be a triangulated category and let G be an abelian group. A function f from $\mathcal{C}$ to G is said to be additive if, for every distinguished triangle

$$\begin{array}{ccccc} & & Z & & \\ & {\scriptstyle w}\swarrow & & \nwarrow{\scriptstyle v} & \\ X & & \xrightarrow[u]{} & & Y \end{array}$$

one has

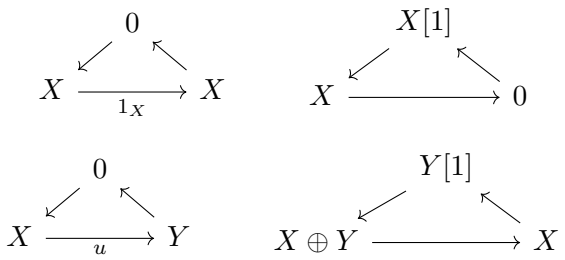
$$f(Y) = f(X) + f(Z).$$

Given an additive function f, the following relations are immediate consequences of the definition:

$$\text{a) } f(0) = 0, \qquad \text{b) } f(X[n]) = (-1)^n f(X),$$

$$\text{c) } f(X) = f(Y) \text{ if } X \text{ and } Y \text{ are isomorphic}, \qquad \text{d) } f(X \oplus Y) = f(X) + f(Y).$$

Indeed the following triangles are distinguished:

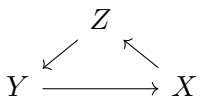


(u an isomorphism). To form the last triangle, we used the fact that the sum of two distinguished triangles is distinguished.

As in the case of abelian categories, one obtains in an evident way a functor

$$A : \mathrm{Ab} \longrightarrow \mathrm{Ab}$$

which associates to each abelian group G the abelian group $A(G)$ of additive functions on $\mathcal{C}$ with values in G. This functor is representable, and it is easy to construct a representing pair $(K(\mathcal{C}), \mathrm{cl}_{\mathcal{C}})$. Namely, one first considers the free abelian group $\mathbb{Z}^{(\mathrm{ob}\,\mathcal{C})}$ generated by the objects of $\mathcal{C}$, and then the subgroup R generated by the elements of the form $X - Y - Z$, where



is a distinguished triangle. The quotient group $\mathbb{Z}^{(\mathrm{ob}\,\mathcal{C})}/R$ is $K(\mathcal{C})$, and $\mathrm{cl}_{\mathcal{C}}$ is obtained by composing the two canonical maps

$$\mathrm{ob}\,\mathcal{C} \longrightarrow \mathbb{Z}^{(\mathrm{ob}\,\mathcal{C})} \longrightarrow K(\mathcal{C}).$$

3. Functorial character

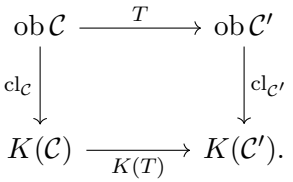
Let $\mathcal{C}$ and $\mathcal{C}'$ be two triangulated categories, and let $T : \mathcal{C} \to \mathcal{C}'$ be an exact functor from $\mathcal{C}$ to $\mathcal{C}'$, that is, an additive "graded" functor carrying distinguished triangles to distinguished triangles. Denote, as above, by

$$A, A' : \mathrm{Ab} \longrightarrow \mathrm{Ab}$$

the functors associated with the categories $\mathcal{C}$ and $\mathcal{C}'$. There results a functorial morphism $A' \to A$ given by composition with T, hence a homomorphism

$$K(T) : K(\mathcal{C}) \longrightarrow K(\mathcal{C}')$$

uniquely determined by the commutativity of the diagram



It follows that, if the functors $T, T_1 : \mathcal{C} \to \mathcal{C}'$ are isomorphic, then $K(T) = K(T_1)$. In particular, if T is an equivalence of categories, then $K(T)$ is an isomorphism.

If $\mathcal{C}''$ is a third triangulated category and

$$F : \mathcal{C} \times \mathcal{C}' \longrightarrow \mathcal{C}''$$

is a bifunctor exact in each variable, then it is clear that there results a bilinear map

$$K(F) : K(\mathcal{C}) \times K(\mathcal{C}') \longrightarrow K(\mathcal{C}'')$$

such that the following relation holds:

$$K(F)(\mathrm{cl}_{\mathcal{C}}(X), \mathrm{cl}_{\mathcal{C}'}(X')) = \mathrm{cl}_{\mathcal{C}''}(F(X, X')).$$

Proposition 3.1. Let $\mathcal{A}$ be a triangulated category, $\mathcal{B}$ a thick triangulated subcategory of $\mathcal{A}$, $\mathcal{A}/\mathcal{B}$ the quotient category of $\mathcal{A}$ by $\mathcal{B}$, $Q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ the canonical quotient functor, and $i : \mathcal{B} \to \mathcal{A}$ the inclusion functor. Then the following sequence is exact:

$$K(\mathcal{B}) \xrightarrow{K(i)} K(\mathcal{A}) \xrightarrow{K(Q)} K(\mathcal{A}/\mathcal{B}) \longrightarrow 0.$$

Proof. It is clear that there is a natural homomorphism

$$\mathrm{Coker}(K(i)) \xrightarrow{\alpha} K(\mathcal{A}/\mathcal{B}).$$

On the other hand, there is an evident map

$$\mathrm{ob}(\mathcal{A}/\mathcal{B}) \xrightarrow{\gamma} \mathrm{Coker}(K(i)),$$

since $\mathrm{ob}(\mathcal{A}/\mathcal{B}) = \mathrm{ob}(\mathcal{A})$. Let $X, Y \in \mathrm{ob}(\mathcal{A}/\mathcal{B})$ and suppose that they are isomorphic. We want to show that $\gamma(X) = \gamma(Y)$. The category $\mathcal{A}/\mathcal{B}$ is obtained by a calculus of fractions on the right or on the left, the multiplicative system S being formed by the morphisms f which are contained in a distinguished triangle (X', Y', Z', f, g, h) in which Z' is an object of $\mathcal{B}$. (S is a saturated multiplicative system.) It follows that there exist X_1 and morphisms

$$X_1 \xrightarrow{s} X, \qquad X_1 \xrightarrow{t} Y$$

such that $s, t \in S$. It is therefore enough to prove that $\gamma(X) = \gamma(X_1)$, and this follows from the fact that there is a distinguished triangle of the form

$$(X_1, X, Z, s, \ldots),$$

with $Z \in \mathrm{ob}\,\mathcal{B}$.

The map γ is an additive function. Indeed, if (X, Y, Z, f, g, h) is a distinguished triangle of $\mathcal{A}/\mathcal{B}$, one must prove that

$$\gamma(X) + \gamma(Z) = \gamma(Y).$$

But the relation now follows because the triangle (X, Y, Z, f, g, h) is isomorphic to a distinguished triangle of $\mathcal{A}$. Consequently one obtains

$$\beta : K(\mathcal{A}/\mathcal{B}) \longrightarrow \mathrm{Coker}(K(i)),$$

and it is immediate to check that β is the inverse of α.

4. Comparison with the case of abelian categories

Let $\mathcal{C}$ be an abelian category and $D(\mathcal{C})$ the derived category of $\mathcal{C}$. As usual, denote by $D^b(\mathcal{C})$ the full subcategory of $D(\mathcal{C})$ generated by the bounded complexes. By associating with every object X of $\mathcal{C}$ the complex whose components are all zero except the component in dimension zero, which is equal to X, one obtains a homomorphism

$$K(\mathcal{C}) \xrightarrow{\mu} K(D^b(\mathcal{C})).$$

This homomorphism is in fact an isomorphism. Indeed, we may use Proposition 3.1 to construct an inverse to μ, taking into account that

$$D^b(\mathcal{C}) \simeq K^{b,b}(\mathcal{C})/K^{b,\phi}(\mathcal{C}),$$

where $K^{b,\phi}(\mathcal{C})$ is the thick subcategory of $K^{b,b}(\mathcal{C})$ consisting of the acyclic complexes. Let $X^\bullet = (X_i) \in \mathrm{ob}(K^{b,b}(\mathcal{C}))$. If one sets

$$f(X^\bullet) = \sum_i (-1)^i \,\mathrm{cl}_{\mathcal{C}}(X_i),$$

one obtains an additive function from $K^{b,b}(\mathcal{C})$ with values in $K(\mathcal{C})$. Indeed, one first observes that, if $X^\bullet$ and $Y^\bullet$ are homotopic, then $f(X^\bullet) = f(Y^\bullet)$, since $f(X^\bullet) = f(H(X^\bullet))$. It follows that there is a homomorphism

$$K(D^b(\mathcal{C})) \xrightarrow{\nu} K(\mathcal{C}),$$

and it is immediate to check that it is the inverse of μ.

Since the invariant $K(\mathcal{C})$ is often of no interest when $\mathcal{C}$ is too large, it is useful to relativize it with respect to a subcategory $\mathcal{C}_o$ of $\mathcal{C}$. Thus let $\mathcal{C}_o$ be a full subcategory of $\mathcal{C}$. We may define, on the one hand, the group

$$K_o(\mathcal{C}) = K_{\mathcal{C}_o}(\mathcal{C}).$$

Suppose, on the other hand, that the following condition is satisfied:

$$(A) \qquad \mathcal{C}_o \text{ is stable under finite direct sums.}$$

It is then easy to define the triangulated category $D^b_o(\mathcal{C})$ by using complexes in $\mathcal{C}$ with bounded cohomology and with components in the subcategory $\mathcal{C}_o$, and passing to the quotient by quasi-isomorphisms of such complexes. As above, there results a homomorphism

$$(*) \qquad K_o(\mathcal{C}) \longrightarrow K(D^b_o(\mathcal{C})).$$

Suppose that the subcategory $\mathcal{C}_o$ also satisfies the following condition (B):

$$(B) \qquad \text{If } X, Y \in \mathrm{ob}\,\mathcal{C}_o \text{ and } X \xrightarrow{u} Y \text{ is an epimorphism (in } \mathcal{C}\text{, of course)},$$

then $\mathrm{Ker}\, u \in \mathrm{ob}\,\mathcal{C}_o$. Reasoning as above, one can show that the homomorphism $(*)$ is an isomorphism.

The canonical inclusion functor

$$D^b_o(\mathcal{C}) \longrightarrow D^b(\mathcal{C})$$

is not fully faithful in general. However:

Proposition 4.1. Suppose that the subcategory $\mathcal{C}_o$ satisfies conditions (A), (B), and the following condition (C):

(C) Every diagram $\begin{array}{ccc} A & \xrightarrow{u} & B \\ & & \uparrow{\scriptstyle v} \\ & & L \end{array}$

where u is an epimorphism and $L \in \mathrm{ob}\,\mathcal{C}_o$, can be completed to a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ \uparrow & & \uparrow{\scriptstyle v} \\ L_1 & \xrightarrow[w]{} & L \end{array}$$

such that $L_1 \in \mathrm{ob}\,\mathcal{C}_o$ and w is an epimorphism. Under these conditions, the canonical functors

$$D^b_o(\mathcal{C}) \longrightarrow D^b(\mathcal{C}), \qquad D^-_o(\mathcal{C}) \longrightarrow D^-(\mathcal{C})$$

are fully faithful. Moreover, the essential image of $D^b_o(\mathcal{C})$ (resp. $D^-_o(\mathcal{C})$) is stable under direct factors.

The proof of this proposition essentially uses a criterion of Verdier ([2], 4–2 Theorem) and will be given in the first exposé of the seminar on Riemann-Roch (SGA 6 I 2.7).

Corollary. Suppose, in addition, that the subcategory $\mathcal{C}_o$ is stable under Ker and Coker, hence is an abelian category, and denote by $D^b(\mathcal{C})_o$ the full triangulated subcategory of $D^b(\mathcal{C})$ generated by the complexes of $D^b(\mathcal{C})$ whose cohomology objects are isomorphic to objects of $\mathcal{C}_o$. Then the inclusion functor

$$D^b(\mathcal{C}_o) \longrightarrow D^b(\mathcal{C})_o$$

is an equivalence between the two categories, and consequently one obtains the sequence of isomorphisms

$$K(\mathcal{C}_o) \simeq K_o(\mathcal{C}) \simeq K(D^b(\mathcal{C}_o)) \simeq K(D^b(\mathcal{C})_o). \qquad (*)$$

Observation. Suppose that the subcategory $\mathcal{C}_o$ satisfies conditions (A) and (B), and that every object of $\mathcal{C}$ admits a finite resolution on the left by elements of $\mathcal{C}_o$. In this case, $\mathcal{C}_o$ also satisfies condition (C), and the canonical functor

$$D^b_o(\mathcal{C}) \longrightarrow D^b(\mathcal{C})$$

is in fact an equivalence of categories. Consequently, one has the isomorphism

$$K_{\mathcal{C}_o}(\mathcal{C}) \simeq K(\mathcal{C}).$$

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[23] cf. (SGA 6 IV 1.6).

5. Pseudo-coherent complexes

Let $\mathcal{C}$ be an abelian category and $\mathcal{C}_o$ a full subcategory of $\mathcal{C}$ satisfying conditions (A), (B), (C). We shall denote by $D_o^-(\mathcal{C})_{\mathrm{coh}}$ the essential image of $D_o^-(\mathcal{C})$ under the canonical functor

$$D_o^-(\mathcal{C}) \longrightarrow D(\mathcal{C}).$$

The objects of $D_o^-(\mathcal{C})_{\mathrm{coh}}$ will be called $\mathcal{C}_o$-pseudo-coherent complexes. Thus a complex of $\mathcal{C}$ is $\mathcal{C}_o$-pseudo-coherent if it is isomorphic in $D(\mathcal{C})$ to a bounded-above complex whose components are objects of $\mathcal{C}_o$.

The category $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is a triangulated subcategory of $D(\mathcal{C})$, in the sense that every distinguished triangle of $D(\mathcal{C})$ with two vertices in the subcategory $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is isomorphic to a triangle whose three objects are in the subcategory $D_o^-(\mathcal{C})_{\mathrm{coh}}$.

First, it is clear that the subcategory $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is stable under the translation automorphism $T : D(\mathcal{C}) \to D(\mathcal{C})$. Therefore, to prove that $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is a triangulated subcategory of $D(\mathcal{C})$, it is enough to check that if $U^\bullet$, $V^\bullet$ are objects of $D_o^-(\mathcal{C})_{\mathrm{coh}}$ and if the triangle

$$\begin{array}{ccc} & W^\bullet & \\ {\scriptstyle w}\swarrow & & \nwarrow{\scriptstyle v} \\ U^\bullet & \xrightarrow[u]{} & V^\bullet \end{array}$$

is distinguished for the triangulated category structure of $D(\mathcal{C})$, then $W^\bullet$ is also an object of $D_o^-(\mathcal{C})_{\mathrm{coh}}$. By hypothesis, there exist complexes $X^\bullet$, $Y^\bullet$, isomorphic in $D(\mathcal{C})$ to $U^\bullet$ and $V^\bullet$ respectively, whose components are objects of $\mathcal{C}_o$. We obtain a commutative square in $D(\mathcal{C})$:

$$\begin{array}{ccc} U^\bullet & \xrightarrow{u} & V^\bullet \\ {\scriptstyle\sim}\downarrow & & \downarrow{\scriptstyle\sim} \\ X^\bullet & \xrightarrow[f]{} & Y^\bullet. \end{array}$$

But, after replacing $X^\bullet$ by another isomorphic complex, we may suppose, using the fact that the inclusion functor

$$D_o^-(\mathcal{C}) \longrightarrow D^-(\mathcal{C})$$

is fully faithful (Prop. 4.1), that f is an actual morphism of complexes. Let $Z^\bullet$ be the mapping cylinder of f, and let $g : Y^\bullet \to Z^\bullet$ be the canonical morphism. By the construction itself, the components of $Z^\bullet$ are objects of $\mathcal{C}_o$ (condition (A)). On the other hand,

$$\begin{array}{ccc} & Z^\bullet & \\ {\scriptstyle h}\swarrow & & \nwarrow{\scriptstyle g} \\ X^\bullet & \xrightarrow[f]{} & Y^\bullet \end{array}$$

being a distinguished triangle in $D(\mathcal{C})$, it follows that $Z^\bullet$ is isomorphic to $W^\bullet$.

The category $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is therefore endowed with a structure of triangulated category: a triangle in $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is distinguished if and only if it is distinguished in the triangulated category $D(\mathcal{C})$.

The intersection of $D_o^-(\mathcal{C})_{\mathrm{coh}}$ with $D^b(\mathcal{C})_{\mathrm{coh}}$ will be denoted by $D_o^b(\mathcal{C})_{\mathrm{coh}}$.

An object M of $\mathcal{C}$ will be called pseudo-coherent if it is so as a complex reduced to degree zero; one proves that it is necessary and sufficient for this that it admit a resolution on the left whose components are objects of $\mathcal{C}_o$.

Proposition 5.1. A complex X is $\mathcal{C}_o$-pseudo-coherent if and only if it is isomorphic to a complex which is bounded on the right and whose components are $\mathcal{C}_o$-pseudo-coherent.

The proof of this proposition will appear in the first exposé of the seminar on Riemann-Roch (SGA 6 I 2.7).

6. Perfect complexes

An object of the category $D(\mathcal{C})$ will be called a $\mathcal{C}_o$-perfect complex if it is isomorphic to an object of $D^b_o(\mathcal{C})$, that is, to a bounded complex all of whose components are in $\mathcal{C}_o$. We shall denote by $D_o(\mathcal{C})_{\mathrm{parf}}$ the full subcategory of $D(\mathcal{C})$ generated by the perfect complexes.

Thus the category $D_o(\mathcal{C})_{\mathrm{parf}}$ is nothing other than the essential image of $D^b_o(\mathcal{C})$ under the inclusion functor

$$D^b_o(\mathcal{C}) \longrightarrow D(\mathcal{C}).$$

It is easy to see that $D_o(\mathcal{C})_{\mathrm{parf}}$ is a triangulated subcategory of $D(\mathcal{C})$, and hence is endowed with a structure of triangulated category.

7. An important special case

Let A be a ring and let $\mathcal{C}$ be the category of A-modules, on the left for example. Denote by $\mathcal{C}_o$ the full subcategory of $\mathcal{C}$ generated by the projective A-modules of finite type. The subcategory $\mathcal{C}_o$ satisfies conditions (A), (B), (C). In this case we use the following notation:

$$D^-(A)_{\mathrm{coh}} = D^-_o(\mathcal{C})_{\mathrm{coh}}, \qquad D^b(A)_{\mathrm{coh}} = D^b_o(\mathcal{C})_{\mathrm{coh}},$$

$$D(A)_{\mathrm{parf}} = D_o(\mathcal{C})_{\mathrm{parf}}.$$

The complexes of $D(\mathcal{C})$ which are $\mathcal{C}_o$-pseudo-coherent (resp. $\mathcal{C}_o$-perfect) will simply be called pseudo-coherent (resp. perfect) complexes.

It will be useful to consider the following groups:

$$K_\bullet(A) = K(D^b(A)_{\mathrm{coh}}),$$

$$K^\bullet(A) = K(D(A)_{\mathrm{parf}}).$$

As for the group $K^\bullet(A)$, we have, by Proposition 4.1 and its corollary,

$$K^\bullet(A) = K(D(A)_{\mathrm{parf}}) \simeq K(D^b_o(\mathcal{C})) \simeq K_{\mathcal{C}_o}(\mathcal{C}),$$

which shows that it is also the classical group of classes of projective modules of finite type over A.

In the special case where A is noetherian, the group $K_\bullet(A)$ also admits a classical description. For this, let $\mathcal{C}_1$ be the abelian category of modules of finite type. Then

$$K_\bullet(A) \simeq K(\mathcal{C}_1).$$

Indeed, using Proposition 4.1, one has

$$K(\mathcal{C}_1) \simeq K(D^b(\mathcal{C}_1)) \simeq K(D^b_{\mathcal{C}_1}(\mathcal{C})),$$

because $D^b(\mathcal{C}_1) \simeq D^b_{\mathcal{C}_1}(\mathcal{C})$ immediately.

In fact, if A is noetherian and $L^\bullet \in D^-(A)$, then it follows from Proposition 5.1 that $L^\bullet$ is pseudo-coherent if and only if the $H^i(L^\bullet)$ are of finite type.

For an arbitrary ring A, one has a homomorphism

$$K^\bullet(A) \xrightarrow{\widetilde{\gamma}_A} K_\bullet(A),$$

which comes from the inclusion functor

$$(*) \qquad D(A)_{\text{parf}} \longrightarrow D^b(A)_{\text{coh}}.$$

In general, the homomorphism $\widetilde{\gamma}_A$ is neither injective nor surjective. In the special case of a regular ring A, every A-module M of finite type has finite projective dimension and it follows, thanks to 5.1, that the inclusion functor $(*)$ is in fact an equivalence between the two categories. In that case, $\widetilde{\gamma}_A$ is therefore an isomorphism.

8. Tor of complexes

Let A be a not necessarily commutative ring, and let ${}_A\mathcal{C}$ (resp. $\mathcal{C}_A$) be the abelian category of left (resp. right) A-modules. By $K^-({}_A\mathcal{C})$ we shall denote the triangulated category of complexes of ${}_A\mathcal{C}$ bounded above, and by ${}_A\mathcal{D}$ the full triangulated subcategory of $K^-({}_A\mathcal{C})$ consisting of complexes whose components are flat left A-modules.

Let $X^\bullet$ be an object of $K^-(\mathcal{C}_A)$, and consider the functor

$$\Phi_{X^\bullet} = X^\bullet \otimes_A \cdot \ : \ K^-({}_A\mathcal{C}) \longrightarrow K^-(\text{Ab}).$$

Under these conditions, the following propositions hold.

1) Every acyclic object of ${}_A\mathcal{D}$ is transformed by $\Phi_{X^\bullet}$ into an acyclic object of $K^-(\text{Ab})$. Indeed, if $N^\bullet$ is an object of ${}_A\mathcal{D}$ such that $Z^i(N^\bullet)$, $B^i(N^\bullet)$, and $H^i(N^\bullet)$ are flat, one has the classical isomorphism

$$\bigoplus_{p+q=n} H_p(X^\bullet) \otimes H_q(N^\bullet) \longrightarrow H_n(X^\bullet \otimes_A N^\bullet).$$

2) Every object $Y^\bullet$ of $K^-({}_A\mathcal{C})$ is the target of a quasi-isomorphism whose source is an object of ${}_A\mathcal{D}$. It is enough for this to take a Cartan-Eilenberg resolution. We can therefore apply Theorem 2 of §2, no. 2 of [2] to conclude that $\Phi_{X^\bullet}$ admits a left derived functor:

$$\mathbf{L}\Phi_{X^\bullet} : D^-({}_A\mathcal{C}) \longrightarrow D^-(\text{Ab}).$$

The object $\mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$ may be obtained, up to isomorphism, as follows. Let $P^\bullet$ be an object of ${}_A\mathcal{D}$ and let $P^\bullet \to Y^\bullet$ be a quasi-isomorphism. Then $\mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$ is isomorphic to

$$Q(X^\bullet \otimes_A P^\bullet),$$

where, as usual, Q denotes the canonical functor

$$K^-(\mathrm{Ab}) \longrightarrow D^-(\mathrm{Ab}).$$

It follows that for each object $Y^\bullet$ of $D^-({}_A\mathcal{C})$, the functor

$$K^-(\mathcal{C}_A) \longrightarrow D^-(\mathrm{Ab}), \qquad X^\bullet \longmapsto \mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$$

is exact and vanishes on acyclic complexes, hence gives an exact bifunctor

$$D^-(\mathcal{C}_A) \times D^-({}_A\mathcal{C}) \longrightarrow D^-(\mathrm{Ab}),$$

$$(X^\bullet, Y^\bullet) \longmapsto X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet.$$

Passing to homology, one defines the $\mathrm{Tor}_i^A(X^\bullet, Y^\bullet)$ (cf. also [1], 6.3). In the particular case of a commutative ring A, one obtains in fact an exact bifunctor

$$D^-(\mathcal{C}) \times D^-(\mathcal{C}) \longrightarrow D^-(\mathcal{C}),$$

the tensor product in the sense of derived categories ($\mathcal{C} = {}_A\mathcal{C} = \mathcal{C}_A$).

Definition. We shall say that the complex X of $D(\mathcal{C}_A)$ has finite Tor-dimension if it lies in D^- and if there exists $n_o \in \mathbb{Z}^+$ such that

$$\mathrm{Tor}_n^A(X, Y) = 0$$

for every $Y \in \mathrm{ob}({}_A\mathcal{C})$ and every $n > n_o$.

We shall use the notion of Tor-dimension to obtain a characterization of perfect complexes.

Proposition 8.1. Let A be a not necessarily commutative ring and let $L^\bullet$ be a complex of A-modules. The following conditions are equivalent:

a) $L^\bullet$ is perfect;

b) $L^\bullet$ is pseudo-coherent, has bounded cohomology, and has finite Tor-dimension.[24]

For the proof, one uses the following lemma.

Lemma. (Bourbaki, Alg. Comm. I, §2, ex. 15). Every A-module E which is flat and of finite presentation is projective.

Proof of the lemma. Let $u : F \to F'$ be an epimorphism. We must prove that

$$\mathrm{Hom}_A(E, F) \longrightarrow \mathrm{Hom}_A(E, F')$$

is an epimorphism. It is enough to prove that, for every divisible group G,

$$\mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E, F'), G) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E, F), G)$$

[24] cf. SGA 6 I 5.8.1.

is a monomorphism. But the diagram

$$\begin{array}{ccc} \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E,F'),G) & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E,F),G) \\ \sim\downarrow & & \downarrow\sim \\ \mathrm{Hom}_{\mathbb{Z}}(F',G)\otimes_A E & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(F,G)\otimes_A E \end{array}$$

is commutative, and the vertical arrows are isomorphisms, by the hypothesis on E.

Proof of Proposition 5.2. The implication a) $\Rightarrow$ b) is immediate. To prove b) $\Rightarrow$ a), let

$$\cdots \longrightarrow P_n \xrightarrow{d_n} \cdots \longrightarrow P_i \longrightarrow 0 \longrightarrow \cdots$$

be a complex isomorphic to $L^\bullet$ in $D(\mathcal{C})$, where the P_j are projective A-modules of finite type. Since $L^\bullet$ has finite Tor-dimension and bounded cohomology, it follows that there exists $n_o \in \mathbb{Z}$ such that:

I) The sequence

$$\cdots \longrightarrow P_{n_o-p} \xrightarrow{d_{n_o-p}} \cdots \longrightarrow P_{n_o} \longrightarrow \mathrm{Ker}\, d_{n_o+1} \longrightarrow 0$$

is exact.

II) $\mathrm{Ker}\, d_{n_o+1}$ is flat.

III) The complex

$$0 \longrightarrow \mathrm{Ker}\, d_{n_o+1} \longrightarrow P_{n_o+1} \longrightarrow \cdots \longrightarrow P_i \longrightarrow 0 \longrightarrow \cdots$$

is isomorphic to $L^\bullet$ in $D(\mathcal{C})$.

But $\mathrm{Ker}\, d_{n_o+1}$ is of finite presentation, hence projective of finite type by the lemma, which proves 8.1.

If A is a commutative ring, and if $X^\bullet$, $Y^\bullet$ are two perfect complexes, then their tensor product

$$X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet$$

in the derived category is again a perfect complex. We obtain a bifunctor

$$D(A)_{\mathrm{parf}} \times D(A)_{\mathrm{parf}} \longrightarrow D(A)_{\mathrm{parf}}$$

which is exact in each variable. Consequently, one defines a product on $K^\bullet(A)$, and it is easy to see that $K^\bullet(A)$ is thereby endowed with the structure of a commutative ring with unit. In fact, $K^\bullet(A)$ can be endowed with the structure of a λ-ring; for details, see the seminar on Riemann-Roch following this one.[25]

The tensor product

$$X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet$$

of a perfect complex $X^\bullet$ and a bounded pseudo-coherent complex $Y^\bullet$ is a bounded pseudo-coherent complex (A still a commutative ring). Consequently, one can introduce on $K_\bullet(A)$ the structure of a $K^\bullet(A)$-module.

[25] SGA 6 V 2.

9. Functorial properties

a) Let A and B be two not necessarily commutative rings, and let $u : A \to B$ be a homomorphism. Every left (resp. right) B-module M can be endowed with the structure of a left (resp. right) A-module, denoted $M_{[u]}$. In particular, $B_{[u]}$ is a right A-module.

The functor

$$E \longmapsto B_{[u]} \otimes_A E$$

allows one to define the exact functor

$$D(A)_{\mathrm{parf}} \longrightarrow D(B)_{\mathrm{parf}}$$

and consequently the homomorphism

$$K^\bullet(A) \xrightarrow{u^*} K^\bullet(B).$$

Thus one obtains a covariant functor from the category of rings (resp. commutative rings) to the category of abelian groups (resp. commutative rings).

b) Suppose now that the left A-module $B_{[u]}$ is pseudo-coherent. It follows that the functor

$$F \longmapsto F_{[u]}$$

transforms a free B-module of finite type into a pseudo-coherent A-module. By using Proposition 5.1, one obtains an exact functor

$$D^b(B)_{\mathrm{coh}} \longrightarrow D^b(A)_{\mathrm{coh}}$$

and consequently a canonical homomorphism

$$K_\bullet(B) \xrightarrow{u_*} K_\bullet(A).$$

The homomorphisms u_* and u^* satisfy the following identity ("the projection formula"):

$$u_*(u^*(\alpha) \cdot \beta) = \alpha\,(u_*(\beta)) \quad \text{for } \alpha \in K^\bullet(A),\ \beta \in K_\bullet(B).$$

This easy formula will be proved in a more general form in the exposés of the seminar on Riemann-Roch (SGA 6 IV 2.11.1.2).

10. Relative construction[26]

Let $u : A \to B$ be a homomorphism of rings, and let

$$D(B) \longrightarrow D(A)$$

be the associated exact functor. In $D(B)$, we can consider the complexes that are pseudo-coherent relative to A, those of finite Tor-dimension relative to A, and also the complexes perfect relative to A. In particular, it is natural to consider the triangulated subcategories

$$D(B)_{A-\mathrm{parf}}, \qquad D^b(B)_{A-\mathrm{coh}}$$

[26] cf. (SGA 6 III, IV).

of $D(B)$ generated by the complexes that are perfect, respectively pseudo-coherent and bounded, relative to A. These two triangulated categories allow one to define the following groups:

$$K^\bullet(B \text{ rel } A) = K(D(B)_{A-\text{parf}}),$$

$$K_\bullet(B \text{ rel } A) = K(D^b(B)_{A-\text{coh}}).$$

Examples. Let G be a group and let Λ be a commutative ring. The homomorphism

$$\Lambda \xrightarrow{\varepsilon} \Lambda[G]$$

therefore allows one to define the groups

$$K^\bullet(\Lambda[G] \text{ rel } \Lambda), \qquad K_\bullet(\Lambda[G] \text{ rel } \Lambda),$$

which will be denoted by

$$R^\bullet_\Lambda(G), \qquad R^\Lambda_\bullet(G).$$

In fact, $R^\bullet_\Lambda(G)$ is a commutative ring and $R^\Lambda_\bullet(G)$ is a module over it.

Remarks.

a) If the ring Λ is not necessarily commutative, one may set, by definition,

$$\Lambda[G] = \Lambda \otimes_{\mathbb{Z}} \mathbb{Z}G$$

and consequently obtains a homomorphism

$$\Lambda \xrightarrow{\varepsilon} \Lambda[G].$$

In this way, one may give a meaning to the groups $R^\bullet_\Lambda(G)$ and $R^\Lambda_\bullet(G)$ even when Λ is non-commutative.

b) The preceding construction can be generalized to the case of a group prescheme, or of a pro-prescheme in groups, G over $\operatorname{Spec}\Lambda$, by starting from the abelian category of Λ-modules on which G operates; this is no longer made explicit in terms of modules over a ring. One then uses the natural inclusion functor into the category of Λ-modules. If Λ is a field, it can be made explicit in terms of modules over a pro-ring, and in the general case, in terms of comodules, at least if G is affine over Λ; cf. SGA 3 I 4.7.2).

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Exposé X — Euler–Poincaré Formula in Étale Cohomology

EULER–POINCARÉ FORMULA IN ÉTALE COHOMOLOGY

by A. Grothendieck

(written up by I. Bucur)

The purpose of the present exposé is to state and prove formula (7.2), of Euler–Poincaré type. This formula generalizes a formula obtained independently by Ogg and Chafarevitch, whose results are presented (from the viewpoint of étale cohomology) in M. Raynaud's exposé [1].

The Euler–Poincaré formula uses in an essential way the formalism of derived categories. As in loc. cit. the proof is made through a variant (5.1) of a well-known formula of Weil.

1. Sheaves on a scheme with operators

Let X be a scheme endowed with a group G acting by automorphisms. Proceeding as in ([T] chap. V), one may define the notion of an étale sheaf on X with group of operators G, or simply a "sheaf on $(X_{\text{ét}}, G)$".

By definition, to give an étale sheaf on X with group of operators G is to give a sheaf $F \in \mathrm{ob}(\widetilde{X}_{\text{ét}})$ and a morphism

$$S_g : F \longrightarrow g^*(F)$$

for each $g \in G$, in such a way that the following conditions are satisfied:

$$\text{a)} \quad S_e = 1_F,$$

and, for $g, h \in G$, the diagram

$$\begin{array}{ccc} F & \xrightarrow{S_{hg}} & (hg)^*(F) \\ {\scriptstyle S_h}\downarrow & & \downarrow{\scriptstyle C_{h,g}(F)} \\ h^*(F) & \xrightarrow[h^*(S_g)]{} & h^*(g^*(F)) \end{array}$$

is commutative, where

$$C_{h,g}(F) : (hg)^*(F) \longrightarrow h^*(g^*(F))$$

is the canonical isomorphism.

Morphisms between two sheaves (F, S_g), (F', S'_g) on X with group of operators G are defined in the evident way, and one obtains a category $\widetilde{X}_{\text{ét},G}$ which is a topos, as follows readily from Giraud's criterion (SGA 4 II 4.12).

Given a ring A of $\widetilde{X}_{\text{ét},G}$, one may consider the abelian category of A-modules, which will be denoted $\mathrm{Mod}(A)$.

For example, given a ring Λ, one may equip the constant sheaf Λ_X associated with Λ with the trivial structure of a sheaf with group of operators G, using the canonical transitivity isomorphisms

$$S_g : \Lambda_X \xrightarrow{\sim} g^*(\Lambda_X).$$

All the results proved in [1], chap. V, can be extended to this setting, and we leave the details to the reader.

2. Where one proves that a certain complex of $\Lambda[G]$-modules is perfect

Recall the following result of SGA 4 XVII.

Lemma 2.1. Let X and Y be two schemes and let $f : X \to Y$ be a proper morphism, with Y quasi-compact. Let A be a torsion ring, and let $F^\bullet$ be a complex of sheaves of A-modules on X of finite Tor-dimension (SGA 4 XVII). Then $Rf_*(F^\bullet)$ is of finite Tor-dimension.

Proposition 2.2. Let X be a scheme proper over a separably closed field k, let G be a finite group acting on X, and let V be an open subset of X stable under G and such that G acts admissibly (SGA 1 V 1.7) and freely on V. Let $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ a prime number different from the characteristic of k, and let Λ_V be the constant sheaf on V associated with Λ, endowed with its trivial structure of a sheaf with group of operators G. Let

$$i : V \longrightarrow X$$

be the immersion of V into X, and put $\Lambda_{V,X} = i_!(\Lambda_V)$. Under these conditions the complex of $\Lambda[G]$-modules

$$R\Gamma_X(\Lambda_{V,X})$$

is a perfect complex (VIII 7).

Proof. If $p : X \to Y = X/G$ is the canonical projection, it is well known that there is a canonical isomorphism

$$R\Gamma_X(\Lambda_{V,X}) \simeq R\Gamma_Y(Rp_*(\Lambda_{V,X}))$$

in the category $D(\Lambda[G])$. On the other hand, the complex $Rp_*(\Lambda_{V,X})$ of $\Lambda[G]$-modules on Y is of finite Tor-dimension. Indeed, since $p : X \to Y$ is a finite morphism of schemes, one has

$$R^q p_*(\Lambda_{V,X}) = 0 \qquad (q \neq 0)$$

(SGA 4 VIII 5.5). Hence

$$Rp_*(\Lambda_{V,X}) = p_*(\Lambda_{V,X}).$$

SGA 4 VIII 5.5 also gives the possibility of determining the fibres of the sheaf $p_*(\Lambda_{V,X})$, and, taking account of the fact that G acts freely on V, one finds

$$\big(p_*(\Lambda_{V,X})\big)_{\bar{y}} \simeq \begin{cases} \prod_{x \in p^{-1}(y)} \Lambda_{V,X,\bar{x}} \simeq \Lambda[G], & y \in p(V), \\ 0, & y \notin p(V), \end{cases}$$

where $\bar{y}$ is the geometric point of Y localized at y.

Consequently $p_*(\Lambda_{V,X})$ is flat over $\Lambda[G]$, fibre by fibre (SGA 4 XVII), and therefore is of finite Tor-dimension. Applying 2.1 to the canonical morphism $Y \to \mathrm{Spec}(k)$, we deduce that

$$R\Gamma_X(\Lambda_{V,X}) = R\Gamma_Y(Rp_*(\Lambda_{V,X}))$$

is of finite Tor-dimension.

To conclude that $R\Gamma_X(\Lambda_{V,X})$ is perfect, it remains to show that it is pseudo-coherent and has bounded cohomology (VIII 8.1). Since $\Lambda[G]$ is noetherian, to verify that $R\Gamma_X(\Lambda_{V,X})$ is pseudo-coherent it is enough to show that the groups $H^q(X, \Lambda_{V,X})$ are of finite type over $\Lambda[G]$ (VIII 7.2), or equivalently over Λ; this follows from (SGA 4 XIV 1.1), applied to the Λ-module $\Lambda_{V,X}$.

The Weil formula 6.1 will give an expression for the class $\mathrm{cl}_{K^\bullet(\Lambda[G])}(R\Gamma_X(\Lambda_{V,X}))$, when X is a curve, in terms of known global invariants and local invariants that we shall define in no. 5.

3. Reminders on linear representations of finite groups

In this number, Λ denotes a commutative ring, and G a group.

3.1. Characters

Consider the full subcategory $\mathcal{E}$ of $\mathrm{Mod}(\Lambda[G])$ consisting of $\Lambda[G]$-modules M whose underlying Λ-module is projective of finite type. For such a module M and for $g \in G$, consider the endomorphism, for the Λ-module structure,

$$g_M : M \longrightarrow M, \qquad g_M(x) = gx.$$

One obtains a function

$$\chi_M : G \longrightarrow \Lambda, \qquad \chi_M(g) = \mathrm{Tr}^*_\Lambda(g),$$

which is a central function on G and is called the character of the $\Lambda[G]$-module M.

Associating to each $M \in \mathrm{ob}(\mathcal{E})$ its character χ_M gives an additive function (VIII 1). Using the formalism of traces for endomorphisms of perfect complexes, one even obtains an additive function Tr on the category of complexes of $\Lambda[G]$-modules which are perfect as complexes of Λ-modules. Hence an homomorphism

$$\mathrm{Tr} : R^\bullet(G) \longrightarrow \mathcal{F}(G, \Lambda),$$

where $R^\bullet_\Lambda(G)$ is the ring of representations of G over Λ (VIII 10), and $\mathcal{F}(G, \Lambda)$ is the module of central functions $G \to \Lambda$.

When G is finite, the composite of the canonical homomorphism

$$K^\bullet(\Lambda[G]) \longrightarrow R^\bullet_\Lambda(G)$$

with Tr will be denoted by tr.

Recall the following result (IX th. 1, cor. 3).

Proposition 3.2. Let $\mathbb{Z}_\ell$ be the ring of ℓ-adic integers and let G be a finite group. Then the homomorphism

$$\mathrm{tr} : K^\bullet(\mathbb{Z}_\ell[G]) \longrightarrow \mathcal{F}(G, \mathbb{Z}_\ell)$$

is a monomorphism. Two projective $\mathbb{Z}_\ell[G]$-modules having the same trace function in $\mathcal{F}(G, \mathbb{Z}_\ell)$ are isomorphic.

3.3. Restriction and induction

Let $\alpha : G \to G'$ be a homomorphism from a group G to a group G'. By restriction of operators one obtains an exact functor

$$J : \mathrm{Mod}(\Lambda[G']) \longrightarrow \mathrm{Mod}(\Lambda[G]).$$

This functor has a left adjoint

$$I : \mathrm{Mod}(\Lambda[G]) \longrightarrow \mathrm{Mod}(\Lambda[G']),$$

namely

$$I(F) = \Lambda[G'] \otimes_{\Lambda[G]} F.$$

If F is projective of finite type, respectively of finite type, then so is $I(F)$.

If $\varphi \in \mathcal{F}(G', \Lambda)$ is a central function on G', the function $\varphi \circ \alpha$ is a central function on G, and therefore one obtains an application

$$\alpha^* : \mathcal{F}(G', \Lambda) \longrightarrow \mathcal{F}(G, \Lambda).$$

If φ is the character of a $\Lambda[G']$-module M in $\mathcal{C}(G')$, then $\alpha^*(\varphi)$ is the character of the $\Lambda[G]$-module $J(M)$.

Now suppose that $\alpha : G \to G'$ is injective and that G has finite index in G'. If F is a $\Lambda[G]$-module in $\mathcal{C}(G)$ whose character is $\psi : G \to \Lambda$, then $I(F)$ lies in $\mathcal{C}(G')$ and the character $\psi^* : G' \to \Lambda$ of $I(F)$ is the character induced by ψ, i.e. the function on G' deduced from ψ by the formula

$$\psi^*(g) = \sum_{t \in G'/G} \psi(t^{-1}gt), \tag{3.3.1}$$

with the convention that $\psi(t^{-1}gt) = 0$ if $t^{-1}gt \notin G$.

3.4. Additional structures on complexes

Let $X^\bullet$ and $Y^\bullet$ be two complexes of A-modules. As in the case of modules, if one of the complexes $X^\bullet$, $Y^\bullet$ has an additional structure of an A-B-bimodule, it is possible to introduce on $\mathrm{Hom}_A^\bullet(X^\bullet, Y^\bullet)$ a structure of complex of B-modules, and consequently a corresponding structure on $\mathrm{RHom}_A(X^\bullet, Y^\bullet)$.

For example, suppose that $X^\bullet$ is an object of $D^-(A)$ and that $Y^\bullet$ is a complex of A-B-bimodules; to be precise, suppose that we are in the case described by the symbol ${}_{A-B}Y^\bullet$. In particular, $Y^\bullet$ is an object of $D(A)$. Let $P^\bullet \to X^\bullet$ be a projective resolution of $X^\bullet$. Then $\mathrm{Hom}_A^\bullet(P^\bullet, Y^\bullet)$ is evidently a complex of left B-modules, i.e. an object of $D(B)$. If another projective resolution $P_1^\bullet \to X^\bullet$ is used, the complex of B-modules $\mathrm{Hom}_A^\bullet(P_1^\bullet, Y^\bullet)$ is canonically isomorphic in $D(B)$ to $\mathrm{Hom}_A^\bullet(P^\bullet, Y^\bullet)$. It follows that one may define $\mathrm{RHom}_A(X^\bullet, Y^\bullet)$ as an object of $D(B)$, determined up to unique isomorphism, and functorial in $X^\bullet$ in $D^-(A)$ and in $Y^\bullet$ in $D(A \otimes_{\mathbb{Z}} B)$.

Analogous considerations apply to the complex $X^\bullet \otimes_A^{\mathbf{L}} Y^\bullet$. For example, if $Y^\bullet$ is a complex of A-B-bimodules as above and $X^\bullet$ is a right-bounded complex of right A-modules, one may define $X^\bullet \otimes_A^{\mathbf{L}} Y^\bullet$ as an object of $D(B)$ determined up to unique isomorphism.

3.5

Let Λ be a commutative ring, let A and B be two algebras over Λ, and put $C = A \otimes_\Lambda B$. Thus C-modules are A-B-bimodules.

If Δ is a full triangulated subcategory of $D(A)$, denote by $D(C)_\Delta$ the full triangulated subcategory of $D(C)$ generated by the complexes which, when regarded as objects of $D(A)$, belong to Δ.

Proposition 3.6. With the notation of 3.4 and 3.5, let Δ be a full triangulated subcategory of $D(A)$ stable under direct factors. If $P^\bullet$ is an object of $D(B)_{\mathrm{parf}}$ (VIII 7) and $M^\bullet$ is an object of $D(C)_\Delta$, then the complex of A-modules $\mathrm{RHom}_B(P^\bullet, M^\bullet) \in D(A)$ defined in 4.1 is an object of Δ. In particular, there is a pairing of triangulated categories

$$\mathrm{RHom} : D(B)_{\mathrm{parf}} \times D(C)_\Delta \longrightarrow \Delta, \tag{3.6.1}$$

that is, a bifunctor exact in each variable, and consequently (VIII 3) a bilinear map

$$K^\bullet(B) \times K^\bullet(D(C)_\Delta) \xrightarrow{\mathrm{hom}} K(\Delta) \tag{3.6.2}$$

such that

$$K(a)(\mathrm{cl}_{K^\bullet(B)}(P^\bullet), \mathrm{cl}_{K(D(C)_\Delta)}(M^\bullet)) = \mathrm{cl}_{K(\Delta)}(\mathrm{RHom}_B(P^\bullet, M^\bullet)). \tag{3.6.3}$$

One obtains an analogous statement on replacing B by its opposite B° and $\mathrm{RHom}_B(P^\bullet, M^\bullet)$ by $P^\bullet \otimes^{\mathbf{L}}_B M^\bullet$.

Proof. First one proves that $\mathrm{RHom}_B(P^\bullet, M^\bullet)$ is an object of Δ in the special case where $P^\bullet$ has a single nonzero component, and then argues by induction on the number of nonzero components of $P^\bullet$.

3.7

Let P be a $\Lambda[G]$-module. It is in particular a Λ-module. The Λ-module $\check{P} = \mathrm{Hom}_\Lambda(P, \Lambda)$ can be endowed with a left $\Lambda[G]$-module structure by putting, for every $u \in \mathrm{Hom}_\Lambda(P, \Lambda)$ and $g \in G$, $(gu)(x) = u(g^{-1}x)$, i.e.

$$g_{\check{P}} = {}^t(g^{-1})_P. \tag{3.7.1}$$

The $\Lambda[G]$-module $\check{P}$ thus obtained is called the dual of P.

One may therefore define the dual $\check{P}^\bullet = \mathrm{Hom}^\bullet_\Lambda(P^\bullet, \Lambda)$ of a complex $P^\bullet$ of $\Lambda[G]$-modules as a complex of $\Lambda[G]$-modules.

If χ_P is the character of P, the character $\chi_{\check{P}}$ of its dual is given by

$$\chi_{\check{P}}(g) = \chi_P(g^{-1}), \tag{3.7.2}$$

which follows immediately from (4.3.1), using the formula

$$\mathrm{Tr}^*_\Lambda(v) = \mathrm{Tr}^*_\Lambda({}^t v).$$

If P is a projective $\Lambda[G]$-module of finite type, it is clear that $\check{P}$ is also a projective $\Lambda[G]$-module of finite type. Moreover, in this case the canonical homomorphism of Λ-modules

$$P \longrightarrow \check{\check{P}}$$

is in fact an isomorphism of $\Lambda[G]$-modules. In this way one defines on $K^\bullet(\Lambda[G])$ an involution

$$x \longmapsto \check{x},$$

which associates with the class of P the class of $\check{P}$. It is immediate that if $\mu : \Lambda \to \Gamma$ is a homomorphism of commutative rings and $u : \Lambda[G] \to \Gamma[G]$ the associated homomorphism, then the homomorphism

$$u^* : K^\bullet(\Lambda[G]) \longrightarrow K^\bullet(\Gamma[G])$$

commutes with the involutions $x \longmapsto \check{x}$ on $K^\bullet(\Lambda[G])$ and $K^\bullet(\Gamma[G])$.

Let A be an algebra over Λ, and put

$$A[G] = \Lambda[G] \otimes_\Lambda A.$$

Given a complex $X^\bullet$ of $A[G]$-modules, denote by $X^{\bullet G}$ the complex of A-modules which is the image of $X^\bullet$ under the functor that associates to each $A[G]$-module X the A-module formed by the elements of X invariant under G.

Proposition 3.8. Let G be a finite group, let $P^\bullet$ be a bounded complex of $\Lambda[G]$-modules all of whose components are projective of finite type, and let $M^\bullet$ be a complex of $A[G]$-modules. Under these conditions there are canonical isomorphisms of complexes of A-modules

$$\mathrm{Hom}^\bullet_{\Lambda[G]}(P^\bullet, M^\bullet) \simeq \mathrm{Hom}^\bullet_\Lambda(P^\bullet, M^\bullet)^G \simeq (\check{P}^\bullet \otimes_\Lambda M^\bullet)^G \simeq \check{P}^\bullet \otimes_{\Lambda[G]} M^\bullet. \tag{3.8.1}$$

The proof uses the well-known isomorphisms in the case of modules and is left to the reader. Note that, in the second and third terms, the operations of G on $\mathrm{Hom}^\bullet_\Lambda(P^\bullet, M^\bullet)$, respectively on $\check{P}^\bullet \otimes_\Lambda M^\bullet$, are defined by transport of structure from its operations on $P^\bullet$ and $M^\bullet$, respectively $\check{P}^\bullet$ and $\check{M}^\bullet$.

Corollary 3.9. If $P^\bullet \in \mathrm{ob}(D(\Lambda[G])_{\mathrm{parf}})$ and $M^\bullet \in D^+(A[G])$, then in the derived category $D(A)$ one has the following isomorphisms:

$$\mathrm{RHom}_{\Lambda[G]}(P^\bullet, M^\bullet) = R\Gamma^G(\check{P}^\bullet \otimes^{\mathbf{L}}_\Lambda M^\bullet) \simeq \check{P}^\bullet \otimes_{\Lambda[G]} M^\bullet. \tag{3.9.1}$$

Indeed, since $P^\bullet \in D(\Lambda[G])_{\mathrm{parf}}$, we may suppose that $P^\bullet$ is bounded and that all its components are projective of finite type. Hence

$$\mathrm{RHom}_{\Lambda[G]}(P^\bullet, M^\bullet) = \mathrm{Hom}^\bullet_{\Lambda[G]}(P^\bullet, M^\bullet)$$

and one can use the isomorphisms (4.3.4). But $\check{P}^\bullet$ is, like $P^\bullet$, bounded with projective finite-type components, hence

$$\check{P}^\bullet \otimes_{\Lambda[G]} M^\bullet \simeq \check{P}^\bullet \otimes^{\mathbf{L}}_{\Lambda[G]} M^\bullet.$$

Similarly, $\check{P}^\bullet \otimes_\Lambda M^\bullet = \check{P}^\bullet \otimes^{\mathbf{L}}_\Lambda M^\bullet$. Moreover the components of the complex $\check{P}^\bullet \otimes_\Lambda M^\bullet$ are cohomologically trivial ([3] IX 3), and therefore

$$R\Gamma^G(\check{P}^\bullet \otimes^{\mathbf{L}}_\Lambda M^\bullet) \simeq (\check{P}^\bullet \otimes_\Lambda M^\bullet)^G.$$

4. The Swan representation

Let E be a smooth connected algebraic curve over the algebraically closed field k. If $x \in E$ is a closed point of E, its local ring $\mathcal{O}_{E,x}$, or simply $\mathcal{O}_x$, is a discrete valuation ring whose field of fractions is isomorphic to the field of rational functions on E and whose residue field is k. The corresponding valuation will be denoted v_x.

Lemma 4.1. Let E be a smooth algebraic curve over the algebraically closed field k, and let $g : E \to E$ be an endomorphism of E different from the identity. Suppose that $x \in E$ is an isolated fixed point of g, and let π be a uniformizer of $\mathcal{O}_{E,x}$. If v_x is the corresponding valuation of $\mathcal{O}_{E,x}$, then the multiplicity $v_x(g)$ of the fixed point x of the map g (IV) is equal to $v_x(g(\pi) - \pi)$.

In the product $E \times E$, the diagonal and the graph of the endomorphism g will be denoted respectively by Δ and Γ_g. Let

$$\gamma : E \longrightarrow E \times E, \qquad \delta : E \longrightarrow E \times E$$

be the corresponding immersions. On the completed local rings at x and at (x, x), they are given by homomorphisms

$$\gamma^*, \delta^* : A \widehat{\otimes} A \longrightarrow A \qquad (A = \widehat{\mathcal{O}}_x),$$

where

$$\delta^*(a \otimes b) = ab, \qquad \gamma^*(a \otimes b) = a\, g^*(b).$$

We have to evaluate the multiplicity $v_x(g)$ of the point $\delta(x)$ in the intersection $\Delta\Gamma_g$ on $E \times E$. The divisor Δ corresponds to the ideal generated by the element $1 \otimes \pi - \pi \otimes 1$ of $A \widehat{\otimes} A$. Hence the multiplicity of $\delta(x)$ in the intersection $\Delta\Gamma_g$ is

$$v_x(g) = v_x(\gamma^*(1 \otimes \pi - \pi \otimes 1)) = v_x(g(\pi) - \pi),$$

as required.

4.2

Let C be a smooth, proper, connected algebraic curve over the algebraically closed field k. If η is its generic point, let $K = k(\eta)$ be its field of rational functions.

Let

$$K' \supset K, \qquad [K' : K] = n$$

be a finite Galois extension of degree n with Galois group G. Let C' be the normalization of C in K', and let

$$p : C' \longrightarrow C$$

be the canonical projection. The Galois group G acts naturally on C'. The stabilizer of $y' \in C'$ in G will be denoted by $G_{y'}$.

If π is a uniformizer of $\mathcal{O}_{C',y'}$ and if $s \in G_{y'}$, $s \neq e$, put

$$v_{y'}(s(\pi) - \pi) = v_{y'}(s) \tag{4.2.1}$$

(the multiplicity of the fixed point y' for the map $s : C' \to C'$ (Lemma 5.1)).

The Swan representation associated with y' is defined by its character $\sigma_{y'}$, given by

$$\sigma_{y'}(s) = \begin{cases} 1 - v_{y'}(s), & s \neq e, \\ \sum_{s \neq e} (v_{y'}(s) - 1), & s = e. \end{cases} \tag{4.2.2}$$

The results of Artin–Serre–Swan (IX Th. 4) show that, for a prime number ℓ distinct from the characteristic of k, there exists a projective $\mathbb{Z}_\ell[G_{y'}]$-module of finite type $Sw_{y'}$, determined up to isomorphism, whose character is $\sigma_{y'}$.

It is also useful to consider the $\mathbb{Z}_\ell[G_{y'}]$-module

$$Sw'_{y'} = Sw_{y'} \otimes \mathbb{Z}_\ell[G_{y'}]. \tag{4.2.3}$$

The character $\sigma'_{y'}$ of $Sw'_{y'}$ is given by

$$\sigma'_{y'}(s) = \begin{cases} 1 - v_{y'}(s), & s \neq e, \\ 1 + \sum_{s \neq e} v_{y'}(s) = 1 + v_{y'}(\mathfrak{D}_{C'/C}), & s = e, \end{cases} \tag{4.2.4}$$

where $\mathfrak{D}_{C'/C}$ is the different of the covering C'/C; the last formula is just [3, Chap. VI 2, prop. 4] with $H = \{1\}$.

Finally, the function

$$a_{y'} = \sigma_{y'} + u_{G_{y'}} = \sigma'_{y'} - 1_{G_{y'}}, \tag{4.2.5}$$

where $u_{G_{y'}}$ denotes the character of the augmentation ideal of $\mathbb{Z}_\ell[G_{y'}]$, is also the character of a $\mathbb{Q}_\ell[G_{y'}]$-module, the Artin representation of $G_{y'}$, which will be denoted by $Art_{y'}$.

Thus

$$a_{y'}(s) = \begin{cases} -v_{y'}(s), & s \neq e, \\ \sum_{s \neq e} v_{y'}(s) = v_{y'}(\mathfrak{D}_{C'/C}), & s = e. \end{cases} \tag{4.2.6}$$

Now consider the inclusion $G_{y'} \subset G$. We put

$$Sw_y = Sw_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \qquad Sw'_y = Sw'_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \tag{4.2.7}$$

$$Art_y = Art_{y'} \otimes_{\mathbb{Q}_\ell[G_{y'}]} \mathbb{Q}_\ell[G]. \tag{4.2.8}$$

The characters of the preceding representations are denoted by σ_y, σ'_y, and a_y.

4.3

The modules so obtained depend, up to isomorphism, only on the point $y \in C$, and not on the choice of the point y' and C' above y. Let y'' be another such point. There exists $g \in G$ such that $y'' = gy'$. The situation (C', y'', G) is deduced from the situation (C, y', G) by the isomorphism g on C' and by $\mathrm{int}(g) : h \mapsto ghg^{-1}$ on G. It follows by transport of structure that the modules Sw_y, etc., associated with y'' are deduced from those associated with y' by extension of the operators by means of $\mathrm{int}(g) : G \to G$. This extension is known to transform a module into an isomorphic module.

Proposition 4.4. With the notation of 3.7, with ($\Lambda = \mathbb{Z}_\ell$) and with $M^\bullet$ an object of $D^+(A[G])$, one has isomorphisms

$$(*) \qquad Sw_y \simeq \check{S}w_y,$$

$$(**) \qquad \mathrm{Hom}^\bullet_{\mathbb{Z}_\ell[G]}(Sw_y, M^\bullet) \simeq Sw_y \otimes_{\mathbb{Z}_\ell[G]} M^\bullet.$$

Proof. For two projective $\mathbb{Z}_\ell[G]$-modules of finite type to be isomorphic, it is enough that they have the same character (3.2). If $\check{\sigma}_y$ denotes the character of $\check{S}w_y$, then by (3.7.2)

$$\check{\sigma}_y(s) = \sigma_y(s^{-1}).$$

On the other hand

$$v_{y'}(s^{-1}(\pi) - \pi) = v_{y'}(s(s^{-1}(\pi) - \pi)) = v_{y'}(\pi - s(\pi)) = v_{y'}(s(\pi) - \pi),$$

so $Sw_{y'} \simeq \check{S}w_{y'}$, whence formula $(*)$ by definition (4.2.7). Formula $(**)$ follows from 3.8.

4.5

We shall need below an explicit formula for the character of the Swan representation. In fact one has the formulas

$$\sigma^*_{y'}(s) = \sigma^*_{y''}(s) = \sigma_y(s) = \begin{cases} \sum_{\substack{p(z')=y \\ s(z')=z'}} (1 - v_{z'}(s)), & s \neq e, \\ \sum_{p(z')=y} (1 + v_{z'}(\mathfrak{D}_{C'/C})) - n, & s = e, \end{cases} \tag{4.5.1}$$

$$\sigma'^*_{y'}(s) = \sigma'^*_{y''}(s) = \sigma'_y(s) = \begin{cases} \sum_{\substack{p(z')=y \\ s(z')=z'}} (1 - v_{z'}(s)), & s \neq e, \\ \sum_{p(z')=y} (1 + v_{z'}(\mathfrak{D}_{C'/C})), & s = e, \end{cases} \tag{4.5.2}$$

$$a^*_{y'}(s) = a^*_{y''}(s) = a_y(s) = \begin{cases} -\sum_{\substack{p(z')=y \\ s(z')=z'}} v_{z'}(s), & s \neq e, \\ \sum_{p(z')=y} v_{z'}(\mathfrak{D}_{C'/C}), & s = e. \end{cases} \tag{4.5.3}$$

To deduce these formulas one uses (3.3.1) and takes into account the following fact: if one fixes a point $y'_0 \in C'$, then for every $y' \in C'$ such that $p(y') = p(y'_0)$ one can find an element $g' \in G$ such that $g'(y') = y'_0$. The elements g' form a system of representatives for the right cosets of $G_{y'_0}$ in G, and moreover

$$G_{y'_0} = g' G_{y'} g'^{-1}.$$

This system of representatives can be used in the second member of formula (3.3.1), and one obtains the formulas displayed above.

4.6

The invariants Sw_y, Sw'_y, Art_y can be defined under more general conditions. Let C' be a smooth algebraic curve over k, not necessarily connected. Suppose that the finite group G acts on C' and that there exists an open subset U of $C = C'/G$, dense in C, such that $C'|U$ is a principal covering of U with group G. For a point $y \in C$, one can define the modules Sw_y, Sw'_y, Art_y by proceeding as above. One can reduce to the case where C' is connected in the following way. Let $y' \in C'$ be such that $p(y') = y$, let C'_1 be the irreducible component of C' containing y', and let C_1 be the irreducible component of C containing y. If $G_{C'_1}$ is the stabilizer of the component C'_1, and if η'_1 (resp. η_1) is the generic point of C'_1 (resp. C_1), then the extension

$$K'_1 = k(\eta'_1) \supset k(\eta_1) = K_1$$

is Galois with Galois group $G_{C'_1}$ and C'_1 is the normalization of C_1 in K'_1 (SGA 1 I 10). One can therefore apply the preceding considerations to the curves C'_1, C_1 and the group $G_{C'_1}$. In this way one obtains projective $\mathbb{Z}_\ell[G_{C'_1}]$-modules of finite type

$$Sw_{G_{C'_1},y}, \qquad Sw'_{G_{C'_1},y}, \qquad Art_{G_{C'_1},y},$$

and Sw_y, Sw'_y, Art_y are the induced modules associated with the inclusion $G_{C'_1} \subset G$.

Remark 4.6. The modules $Sw_{y'}$, $Sw'_{y'}$, $Art_{y'}$ are purely local invariants: they are known from the group $G_{y'}$ acting on the completion of the local ring $\mathcal{O}_{C',y'}$. Moreover $Sw_{y'} = 0$ if and only if $p : C' \to C$ is tamely ramified at y', or equivalently, if $G_{y'}$ has order prime to the characteristic of k [3 VI].

5. The Weil formula (compare [3 VI §4])

5.0

All schemes considered in this number will be defined over an algebraically closed base field k.

We shall use the following notation:

$$\Lambda_\nu = \mathbb{Z}/\ell^\nu\mathbb{Z},$$

where ℓ is a fixed prime number distinct from the characteristic of k.

Let $\mathbb{Z}_\ell \to \Lambda_\nu$ be the canonical homomorphism. It induces a homomorphism

$$\lambda_\nu : \mathbb{Z}_\ell[G] \longrightarrow \Lambda_\nu[G]$$

and consequently [VIII 9] a reduction homomorphism

$$\lambda^*_\nu : K^\bullet(\mathbb{Z}_\ell[G]) \longrightarrow K^\bullet(\Lambda_\nu[G]).$$

The images by λ^*_ν of the elements associated with the modules Sw_y, Sw'_y will be denoted by

$$Sw_y^{\Lambda_\nu}, \qquad Sw_y'^{\,\Lambda_\nu}.$$

The reduction homomorphism λ^*_ν is in fact an isomorphism [IX 1.5]. The inverse of λ^*_ν will be denoted by μ_ν.

The constant sheaf on the scheme X associated with the ring Λ_ν will be denoted by $\Lambda_{\nu,X}$.

Let V be an open subset of X, and let $j : V \hookrightarrow X$ be the canonical immersion. We put

$$j_!(\Lambda_{\nu,V}) = \Lambda_{\nu,V,X}.$$

Proposition 5.1. Let C' be a smooth proper algebraic curve over the algebraically closed field k, let G be a finite group acting on C', let $C = C'/G$, let $p : C' \to C$ be the canonical projection, let U be a dense open subset of C such that $U' = p^{-1}(U)$ is a principal covering of U with group G, let $Y = C - U$, let $Y' = p^{-1}(Y) = C' - U'$, and let $\mathrm{EP}(C)$ be the Euler–Poincaré characteristic of C (equal to $2 - 2g$ if C is connected of genus g). Under these conditions one has the Weil formula:

$$\mathrm{cl}_{K^\bullet(\Lambda_\nu[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{\nu,U',C'})) = \mathrm{EP}(C)\,\mathrm{cl}_{K^\bullet(\Lambda_\nu[G])}(\Lambda_\nu[G]) - \sum_{y\in Y} Sw'^{\Lambda_\nu}_y, \tag{5.2}$$

where the first member of (5.2) is defined using 2.1.

Proof. Denote by S_1 (resp. S_2) the first (resp. second) member of the formula to be proved. Since the reduction homomorphism λ_ν^* is in fact an isomorphism, it is enough to prove that

$$\mu_\nu(S_1) = \mu_\nu(S_2).$$

On the other hand, by (3.3.1), the trace homomorphism

$$K^\bullet(\mathbb{Z}_\ell[G]) \xrightarrow{tr} \mathcal{F}(G, \mathbb{Z}_\ell)$$

is a monomorphism. Consequently, it is enough to prove that

$$\mathrm{tr}(\mu_\nu(S_1)) = \mathrm{tr}(\mu_\nu(S_2)).$$

The character $\sigma_2 = \mathrm{tr}(\mu_\nu(S_2))$ is plainly given by

$$\sigma_2 = \mathrm{EP}(C) r_G - \sum_{y\in Y} \sigma'_y, \tag{5.2.1}$$

where σ'_y is the character of Sw'_y ((4.2.7) and (4.2.3)), and r_G is the regular representation.

Now consider the reduction homomorphisms for traces:

$$\mathcal{F}(G, \mathbb{Z}_\ell) \xrightarrow{\lambda_n^*} \mathcal{F}(G, \mathbb{Z}/\ell^n\mathbb{Z}).$$

One plainly has

$$\mathcal{F}(G, \mathbb{Z}_\ell) = \varprojlim_n \mathcal{F}(G, \mathbb{Z}/\ell^n\mathbb{Z}).$$

It is therefore enough to prove, for every n, that

$$\lambda_n^*(\mathrm{tr}(\mu_\nu(S_1))) = \lambda_n^*(\sigma_2).$$

The first member is equal to $\mathrm{tr}(\lambda_n^*(\mu_\nu(S_1)))$, while

$$\lambda_n^*(\mu_\nu(S_1)) = \mathrm{cl}_{K^\bullet(\Lambda_n[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'})),$$

by the Künneth formula (SGA 4 XVIII)

$$\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}) \otimes^L_{\Lambda_n} \Lambda_\nu \simeq \mathrm{R}\Gamma_{C'}(\Lambda_{\nu,U',C'}) \quad \text{for } n \geq \nu.$$

Thus we are reduced to proving the equality

$$\mathrm{tr}\big(\mathrm{cl}_{K^\bullet(\Lambda_n[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}))\big) = \lambda_n^*(\sigma_2).$$

Let τ denote the first member of this equality, and let $g \in G$. The element g plainly induces an endomorphism of the distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'}) & \\ \swarrow & & \searrow \\ \mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}) & \longrightarrow & \mathrm{R}\Gamma_{C'}(\Lambda_{n,C'}) \end{array}$$

Therefore

$$\tau(g) = \mathrm{Tr}^*\, g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'})} = \mathrm{Tr}^*\, g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} - \mathrm{Tr}^*\, g_{\mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'})}.$$

For brevity, Tr^* has been written instead of $\mathrm{Tr}^*_{\Lambda_n}$.

Applying a trivial Lefschetz formula to the discrete space Y' on which g acts gives

$$\begin{aligned} \mathrm{Tr}^*\, g_{\mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'})} &= \text{the number of fixed points of the restriction } g_{Y'} \text{ of } g \text{ to } Y' \\ &= \sum_{\substack{y' \in Y' \\ gy' = y'}} 1. \end{aligned}$$

On the other hand, the Lefschetz formula (III 4.7) for g acting on C' gives, if $g \neq e$,

$$\mathrm{Tr}^*\, g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} = \sum_{y' \in Y'} v_{y'}(g),$$

the sum of the multiplicities of the fixed points of the map $g_{C'} : C' \to C'$. Also

$$\mathrm{Tr}^*\, g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} = \mathrm{EP}(C') \cdot 1_\Lambda \quad \text{for } g = e,$$

by the explicit computation of $H^1(C', \Lambda_{n,C'})$, via (SGA 4 IX 47) and Künneth (SGA 4 XVII). We therefore obtain the following formula for τ:

$$\tau(g) = \begin{cases} \displaystyle\sum_{\substack{y' \in Y' \\ g(y') = y'}} (v_{y'}(g) - 1), & g \neq e, \\ \mathrm{EP}(C') - \mathrm{card}(Y'), & g = e. \end{cases} \tag{5.2.2}$$

Comparison with (5.2.1) gives easily $\tau = \lambda_n^*(\sigma_2)$. Indeed, if $g \neq e$, the equality between $\tau(g)$ and $\lambda_n^*(\sigma_2)$ is immediate from (4.5.2). If $g = e$, the formula to be checked is

$$\mathrm{EP}(C') - \mathrm{card}(Y') = N\,\mathrm{EP}(C) - \sum_{y \in Y} \sigma'_y(e), \qquad N = \mathrm{card}(G),$$

that is, by (4.5.2),

$$\mathrm{EP}(C') - N\,\mathrm{EP}(C) = -\deg(\mathfrak{D}_{C'/C}),$$

which is nothing other than the classical Hurwitz formula.

6. Definition of the local terms $\varepsilon_x^{\Delta}(F)$

6.1

Let C be a connected smooth algebraic curve over the algebraically closed field k, let A be a Λ-algebra ($\Lambda = \mathbb{Z}/\ell^{\nu}\mathbb{Z}$, with ℓ distinct from the characteristic of k), and let $\Delta \subset D^+(A)$ be a triangulated subcategory of $D^+(A)$ stable under direct factors. Let F be a complex of sheaves of A-modules on C satisfying the following conditions:

i) For every geometric point $\bar{x} \in C(k)$, one has $F_{\bar{x}} \in \mathrm{ob}(\Delta)$.

ii) There exists a non-empty open subset of C on which F is locally constant; equivalently, there exist a non-empty open subset U of C and a connected Galois covering $C' \xrightarrow{p} C$ of C, with group G, such that, if $F' = p^*(F)$ is the inverse image of F by p, then $F'|U'$ is constant ($U' = p^{-1}(U)$).

The sheaf $F'|U'$ is therefore defined by a complex M of A-modules on which G acts. If η (resp. η') denotes the generic point of C (resp. C'), and if $\bar{\eta}$ (resp. $\bar{\eta}'$) is a geometric point above η (resp. η'), then

$$p(\eta') = \eta, \qquad F_{\bar{\eta}} \simeq F'_{\bar{\eta}'} = M. \tag{6.1.1}$$

As a complex of A-modules, M is isomorphic to $F_{\bar{\eta}}$ and, by condition i), is an object of Δ. Hence, by (3.6), for every $x \in C(k)$,

$$\mathrm{Hom}^{\bullet}_{\Lambda[G]}(Sw_x^{\Lambda}, M)$$

is an object of Δ, and so defines an element of $K(\Delta)$.

In fact the object of Δ defined by $\mathrm{Hom}^{\bullet}_{\Lambda[G]}(Sw_x^{\Lambda}, M)$ is independent, up to isomorphism, of the Galois covering $C' \to C$ which trivializes $F|U$. To show this it is enough to compare the elements associated with two coverings

$$C' \xrightarrow{p} C, \qquad C'' \xrightarrow{p'} C' \xrightarrow{p} C.$$

In this case, if G' is the Galois group of the covering $C'' \xrightarrow{p''} C$, then G is a quotient group of G'. Let

$$\varphi : \mathbb{Z}_{\ell}[G'] \longrightarrow \mathbb{Z}_{\ell}[G]$$

be the associated canonical epimorphism. The Swan modules are then related by

$$Sw_{G,x} \simeq Sw_{G',x} \otimes_{\mathbb{Z}_{\ell}[G']} \mathbb{Z}_{\ell}[G],$$

which follows immediately from the fact that the character $a_{G,x}$ is the character induced from G' to G ([3] VI Prop. 3).

On the other hand, there is an isomorphism of complexes of $A[G']$-modules

$$F_{\bar{\eta}''} \simeq [F_{\bar{\eta}'}]_{(\varphi)}.$$

It follows that there is a canonical isomorphism of complexes of A-modules between

$$\mathrm{Hom}^{\bullet}_{\Lambda[G]}(Sw^{\Lambda}_{G,x}, F_{\bar{\eta}}) \quad \text{and} \quad \mathrm{Hom}^{\bullet}_{\Lambda[G']}(Sw^{\Lambda}_{G',x}, F_{\bar{\eta}''})$$

(the adjunction formula).

6.2

We shall use the following notation:

$$\alpha_x^{\Delta}(F) = \mathrm{cl}_{K(\Delta)}(\mathrm{Hom}^{\bullet}_{\Lambda[G]}(Sw_x^{\Lambda}, F_{\bar{\eta}})), \tag{6.2.1}$$

$$\varepsilon_x^{\Delta}(F) = \alpha_x^{\Delta}(F) + (\mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \mathrm{cl}_{K(\Delta)}(F_x)). \tag{6.2.2}$$

Of course, in formula (6.2.1), $F_{\bar{\eta}}$ is endowed with its $A[G]$-module structure by the isomorphism (6.1.1). Note that if $x \in C(k)$ is such that C' is tamely ramified at x, then by (4.6) $Sw_x = 0$, hence $\alpha_x^{\Delta}(F) = 0$, and in this case, in conformity with Ogg-Shafarevich, one has

$$\varepsilon_x^{\Delta}(F) = \mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \mathrm{cl}_{K(\Delta)}(F_x). \tag{6.2.3}$$

It is clear that $\varepsilon_x^{\Delta}(F) = 0$ if $x \notin Y = C - U$, and therefore only finitely many x satisfy $\varepsilon_x^{\Delta}(F) \neq 0$.

We now give examples of categories Δ and complexes of sheaves F for which the conditions above are satisfied.

a) Let $\Delta = D^b(A)_{\mathrm{coh}}$ (VIII 5), and let F be a constructible complex of sheaves of A-modules (SGA 4 XVII) with bounded cohomology. In this case $\varepsilon_x^{\Delta}(F) \in K_{\bullet}(A)$.

b) Take $\Delta = D(A)_{\mathrm{parf}}$ (VIII 6), and let F be a constructible complex of sheaves of A-modules of finite tor-dimension. In this case $\varepsilon_x^{\Delta}(F) \in K^{\bullet}(A)$.

c) Let A, B be two Λ-algebras and let $\varphi : A \to B$ be a homomorphism of Λ-algebras. Take $\Delta = D^b(B \,\mathrm{rel}\, A)_{\mathrm{coh}}$ (VIII 10), and let F be a complex of sheaves of B-modules, constructible as a sheaf of complexes of A-modules and with bounded cohomology. One obtains $\varepsilon_x^{\Delta}(F) \in K_{\bullet}(B \,\mathrm{rel}\, A)$. Of course, one can vary this as above to obtain, for example, elements $\varepsilon_x^{\Delta}(F) \in K^{\bullet}(B \,\mathrm{rel}\, A)$.

7. Euler–Poincaré formula

Theorem 7.1. Let C be a connected, smooth, proper algebraic curve over the algebraically closed field k, let η be the generic point of C, let $\Delta \subset D^+(A)$ be a triangulated subcategory of $D^+(A)$ stable under direct factors, and let F be a complex of sheaves of A-modules satisfying conditions i), ii) of 7.1. Under these conditions one has $\mathrm{R}\Gamma_C(F) \in \mathrm{ob}(\Delta)$, and the following formula (Euler–Poincaré formula) is true:

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) = \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \sum_{x \in C(k)} \varepsilon_x^{\Delta}(F), \tag{7.2}$$

where the $\varepsilon_x^{\Delta}(F)$ are defined in 6.2 (formula 6.2.2).

The proof of this theorem will be made in a few steps.

a) Let Ψ be a bounded below complex of sheaves of $\Lambda[G]$-modules on C. There is an isomorphism

$$R\Gamma^G(\mathrm{R}\Gamma_C(\Psi)) \simeq \mathrm{R}\Gamma_C(R\underline{\Gamma}^G(\Psi)), \tag{7.3}$$

where $\underline{\Gamma}^G$ is the functor "subsheaf of G-invariant sections." Indeed, Γ_C sends an injective sheaf of $\Lambda[G]$-modules to an injective $\Lambda[G]$-module, and $\underline{\Gamma}^G$ sends an injective sheaf of $\Lambda[G]$-modules to an injective sheaf of Λ-modules (using, for example, the evident adjoints of Γ_C and $\underline{\Gamma}^G$). Moreover

$$\Gamma^G \circ \Gamma_C = \Gamma_C \circ \underline{\Gamma}^G.$$

Taking derived functors on both sides and applying [5] prop. 3.1) gives (7.3).

b) Let Φ be a bounded below complex of sheaves of Λ-modules on C, zero outside U. There is an isomorphism

$$\Phi \simeq R\underline{\Gamma}^G(p_*(\Phi')), \qquad \Phi' = p^*(\Phi). \tag{7.4}$$

First note that the functors p^* and $\underline{\Gamma}^G \circ p_*$ establish quasi-inverse equivalences between the category of sheaves of Λ-modules on C' on which G acts and which are zero outside $U' = p^{-1}(U)$, and the category of sheaves of Λ-modules on C which are zero outside U. Hence one has the isomorphism

$$\Phi \simeq R((\underline{\Gamma}^G \circ p_*) \circ p^*)(\Phi).$$

On the other hand, one can again apply [5] Prop. 3.1), which gives, since $Rp^* = p^*$ and $Rp_* = p_*$:

$$\Phi \simeq R(\underline{\Gamma}^G \circ p_*)(p^*(\Phi)) \simeq (R\underline{\Gamma}^G)(p_*p^*(\Phi)),$$

which is the desired isomorphism.

c) Let Φ be a complex of sheaves as in b). One has the isomorphism

$$\mathrm{R}\Gamma_C(\Phi) = (R\Gamma^G)(\mathrm{R}\Gamma_{C'}(\Phi')), \qquad \Phi' = p^*(\Phi). \tag{7.5}$$

Indeed,

$$(R\Gamma^G)(\mathrm{R}\Gamma_{C'}(\Phi')) \simeq (R\Gamma^G)(\mathrm{R}\Gamma_C(p_*(\Phi'))) \simeq \mathrm{R}\Gamma_C(R\underline{\Gamma}^G(p_*(\Phi')))$$

by (7.3), and (7.5) follows from the isomorphism (7.4).

d) Let $i' : U' \to C'$, $i : U \to C$ be the canonical inclusions and put

$$\Phi = F_{U,C} = i_!(F|U).$$

To compute $\mathrm{R}\Gamma_C(\Phi)$, one may apply formula (7.5). On the other hand, with the notation of 7.1,

$$\Phi' \simeq \Lambda_{U',C'} \otimes^L_\Lambda M,$$

and the Künneth formula (SGA 4 XVII) gives

$$\mathrm{R}\Gamma_{C'}(\Phi') \simeq \mathrm{R}\Gamma_{C'}(\Lambda_{U',C'} \otimes^L_\Lambda M) \xleftarrow{\sim} \mathrm{R}\Gamma_{C'}(\Lambda_{U',C'}) \otimes^L_\Lambda M. \tag{7.6}$$

This last isomorphism is in fact an isomorphism between the objects of $D(A[G])$ defined by the two sides. Combining (7.6), (7.5), and formula (3.9.1), which is applicable by 2.1, one obtains

$$\mathrm{R}\Gamma_C(F_{U,C}) \simeq R\Gamma^G(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'}) \otimes^L_\Lambda M) \simeq \mathrm{RHom}_{\Lambda[G]}(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'})^\vee, M). \tag{7.7}$$

Thus, since $M \simeq F_{\bar\eta} \in \mathrm{ob}(\Delta)$ (condition (i)) and by (3.6), we find

$$\mathrm{R}\Gamma_C(F_{U,C}) \in \mathrm{ob}(\Delta).$$

On the other hand, there is the distinguished triangle (where $Y = C - U$)

$$\begin{array}{ccc} & \mathrm{R}\Gamma_Y(F|Y) & \\ \swarrow & & \nwarrow \\ \mathrm{R}\Gamma_C(F_{U,C}) & \longrightarrow & \mathrm{R}\Gamma_C(F) \end{array} \tag{7.8}$$

where obviously

$$\mathrm{R}\Gamma_Y(F) = \coprod_{y\in Y} F_y \in \mathrm{ob}(\Delta),$$

because $F_y \in \mathrm{ob}(\Delta)$ (condition (i)). It follows that $\mathrm{R}\Gamma_C(F) \in \mathrm{ob}(\Delta)$, which is the first assertion of the theorem. The distinguished triangle (7.8) furthermore gives the relation

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F_{U,C})) + \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y),$$

and, using the isomorphism 7.7,

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{RHom}_{\Lambda[G]}(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'})^\vee, F_{\bar\eta})) + \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y).$$

To obtain the Euler–Poincaré formula it remains only to use the Weil formula (5.2). Indeed, first 4.4, (5.2), and 3.6 (in the particular case $B = \Lambda[G]$) give

$$\mathrm{cl}_{K^\bullet(\Lambda[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'})) = \mathrm{cl}_{K^\bullet(\Lambda[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'})^\vee).$$

$$\begin{aligned}
\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) &= \hom\Big(\mathrm{cl}_{K^\bullet(\Lambda[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'})), \mathrm{cl}_{K(D(A[G])_\Delta)}(F_{\bar\eta})\Big) + \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y) \\
&= \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y) \\
&\quad + \hom\Big(\mathrm{EP}(C)\,\mathrm{cl}_{K^\bullet(\Lambda[G])}(\Lambda[G]) - \sum_{y\in Y}(Sw_y^\Lambda + \mathrm{cl}_{K^\bullet(\Lambda[G])}(\Lambda[G])), \\
&\qquad\qquad \mathrm{cl}_{K(D(A[G])_\Delta)}(F_{\bar\eta})\Big) \\
&= \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y) + \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \sum_{y\in Y}(\alpha_y^\Delta(F) + \mathrm{cl}_{K(\Delta)}(F_{\bar\eta})) \\
&= \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \sum_{y\in Y} \varepsilon_y^\Delta(F).
\end{aligned}$$

Since $\varepsilon_x^\Delta(F) = 0$ for $x \notin Y$, this is exactly formula (7.2). For a complex of sheaves F satisfying the conditions of 7.1, we introduce the notation

$$\mathrm{EP}_\Delta(C, F) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)).$$

Corollary 7.9. If F, F' are complexes of sheaves satisfying the conditions of 7.1 and if they are locally isomorphic (for the étale topology) as complexes of A-modules, then

$$\mathrm{EP}_\Delta(C, F) = \mathrm{EP}_\Delta(C, F'). \tag{7.10}$$

In particular, if F is locally constant of value $M^\bullet$, then

$$\mathrm{EP}_\Delta(C, F) = \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(M^\bullet). \tag{7.11}$$

Corollary 7.12. Let C be a connected smooth proper algebraic curve over k, let V be a non-empty open subset of C, let Δ be a subcategory of $D^+(A)$ as in 7.1, and let F be a complex of sheaves of A-modules on V such that its restriction to a suitable non-empty open subset of V is trivialized by

a Galois covering of V. Suppose moreover that $F_{\bar{x}} \in \mathrm{ob}(\Delta)$ for every $x \in V$. Under these conditions one has

$$(*) \qquad \mathrm{R}\Gamma_! V(F),\ \mathrm{R}\Gamma_V(F) \in \mathrm{ob}(\Delta),$$

$$(**) \qquad \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_! V(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_V(F)).$$

Put $\mathrm{EP}_\Delta(V,F) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_V(F))$ and

$$\mathrm{EP}(V) = \mathrm{EP}_{D(F_\ell)_{\mathrm{coh}}}(V, F_\ell) = \mathrm{EP}(C) - \mathrm{card}(Z),$$

where $Z = C - V$ and where F_ℓ is the sheaf on C defined in dimension zero by the constant sheaf associated with $F_\ell = \mathbb{Z}/\ell\mathbb{Z}$. Then one has the formula

$$\mathrm{EP}_\Delta(V,F) = \mathrm{EP}(V)\,\mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \sum_{x \in V(k)} \varepsilon_x^\Delta(F) - \sum_{x \in Z} \alpha_x^\Delta(F), \tag{7.13}$$

where ε_x^Δ and α_x^Δ are defined by (6.2.1), (6.2.2).

We shall use the following lemma.

Lemma 7.14. Let K be the field of fractions of a strictly local discrete valuation ring R, let $W = \mathrm{Spec}\, K$, let A be a noetherian ring annihilated by an integer n prime to the residual characteristic p of R, and let F be a complex of sheaves of A-modules on W trivialized by a finite extension of K. If Δ is a triangulated subcategory of $D^+(A)$ stable under direct factors and if $F_{\bar{\eta}} \in \mathrm{ob}(\Delta)$ (η the generic point of S), then $\mathrm{R}\Gamma_W(F) \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_W(F)) = 0.$$

Proof of 7.14. Let π be the fundamental group of W. Then F corresponds to a complex of A-modules $M^\bullet$ with π as group of operators. The underlying complex of A-modules of $M^\bullet$ is isomorphic to $F_{\bar{\eta}}$ and consequently belongs to $\mathrm{ob}(\Delta)$. Introduce the quotient group $\bar{\pi}$ of π by its unique Sylow p-subgroup P [4, II 33], that is, the projective limit of discrete quotient groups of π which are of order prime to p, where p is the characteristic exponent of the residue field of R (if $p = 1$, put $\bar{\pi} = \pi$). Thus there is an exact sequence

$$1 \longrightarrow P \longrightarrow \pi \longrightarrow \bar{\pi} \longrightarrow 0,$$

and one obtains

$$\mathrm{R}\Gamma_W(F) \simeq R\Gamma^\pi(M^\bullet) \simeq R\Gamma^{\bar{\pi}}(R\Gamma^P(M^\bullet))$$

(Hochschild–Serre isomorphism). Now P acts on $M^\bullet$ through a finite quotient group, which is therefore a p-group, hence of order prime to n. Thus

$$R\Gamma^P(M^\bullet) \simeq \Gamma^P(M^\bullet),$$

and $\Gamma^P(M^\bullet)$ is a direct factor of $M^\bullet$, hence belongs to $\mathrm{ob}(\Delta)$. On the other hand

$$\bar{\pi} = \prod_{\ell \neq p} \mathbb{Z}_\ell$$

([4], II 33). We are therefore reduced to proving that if $M^\bullet \in \mathrm{ob}(\Delta)$, then $R\Gamma^{\bar{\pi}}(M^\bullet) \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}(R\Gamma^{\bar{\pi}}(M^\bullet)) = 0.$$

For this, let f be a topological generator of $\bar{\pi}$. We may suppose that the components M^i of M are $\bar{\pi}$-modules such that

$$H^j(\bar{\pi}, M^i) = 0 \quad \text{if } j \geq 1.$$

There is then a distinguished triangle

$$\begin{array}{ccc} & M^\bullet & \\ {\scriptstyle 1-f}\swarrow & & \nwarrow \\ R\Gamma^{\bar{\pi}}(M^\bullet) & \longrightarrow & M^\bullet \end{array} \tag{7.15}$$

Indeed, on the one hand it is clear that $R\Gamma^{\bar{\pi}} = \mathrm{Ker}(1-f)$. On the other hand $1-f : M^\bullet \to M^\bullet$ is surjective. Indeed

$$H^1(\bar{\pi}, M) = M^\bullet/(1-f)M^\bullet$$

(see for this calculation [4] pp. 197, in a substantially analogous case), and therefore $M^\bullet = (1-f)M^\bullet$. The triangle (7.15) gives at the same time $R\Gamma^{\bar{\pi}}(M^\bullet) \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}(R\Gamma^{\bar{\pi}}(M^\bullet)) = 0,$$

which proves 7.14.

Proof of 7.12. The fact that $\mathrm{R}\Gamma_! V(F) \in \mathrm{ob}(\Delta)$ follows immediately from 7.1 applied to the complex $i_!(F)$, where $i : V \to C$; indeed

$$\mathrm{R}\Gamma_! V(F) = \mathrm{R}\Gamma_C(i_! F).$$

To prove that $\mathrm{R}\Gamma_V(F) \in \mathrm{ob}(\Delta)$, consider the distinguished triangle

$$\begin{array}{ccc} & Q^\bullet & \\ \swarrow & & \nwarrow \\ Ri_!(F) & \longrightarrow & Ri_*(F), \end{array}$$

where $Q^\bullet$ is concentrated on $Z = C - V$. Applying $\mathrm{R}\Gamma_C$ gives the distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}\Gamma_Z(Q^\bullet|Z) & \\ \swarrow & & \nwarrow \\ \mathrm{R}\Gamma_! V(F) & \longrightarrow & \mathrm{R}\Gamma_V(F). \end{array} \tag{7.16}$$

Since $\mathrm{R}\Gamma_! V(F) \in \mathrm{ob}(\Delta)$, it follows that $\mathrm{R}\Gamma_V(F) \in \mathrm{ob}(\Delta)$ if and only if $\mathrm{R}\Gamma_Z(Q^\bullet|Z) \in \mathrm{ob}(\Delta)$, that is, if and only if $(Ri_*(F))_{\bar{x}} \in \mathrm{ob}(\Delta)$ for every $x \in Z$. Recall that to calculate $(Ri_*(F))_{\bar{x}}$, one considers the cartesian square

$$\begin{array}{ccc} \bar{V} & \overset{j}{\hookrightarrow} & C_x \\ {\scriptstyle u}\downarrow & & \downarrow \\ V & \hookrightarrow & C \end{array}$$

where C_x is the strictly localized scheme corresponding to the geometric point x. One has

$$Ri_*(F)_{\bar{x}} \simeq \mathrm{R}\Gamma_{\bar{V}}(u^*(F)),$$

by virtue of 7.14, and therefore $(Ri_*(F))_{\bar{x}} \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}((Ri_*(F))_{\bar{x}}) = 0.$$

It follows that

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_Z(Q^\bullet|Z)) = 0,$$

and the distinguished triangle (7.16) gives

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_! V(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_V(F)).$$

Finally, to establish (7.13), it is enough to use the Euler–Poincaré formula (7.2) for the complex of sheaves $i_!(F)$:

$$\mathrm{EP}_\Delta(V,F) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_! V(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(i_!(F))) = \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \sum_{x\in C(k)} \varepsilon_x^\Delta(i_!(F)).$$

But

$$\sum_{x\in C(k)} \varepsilon_x^\Delta(i_!(F)) = \sum_{x\in V(k)} \varepsilon_x^\Delta(F) + \sum_{x\in Z} \varepsilon_x^\Delta(F).$$

For $x \in Z$ one has $(i_!(F))_{\bar{x}} = 0$, and consequently, for such x,

$$\varepsilon_x^\Delta(i_!(F)) = \alpha_x^\Delta(F) + \mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}),$$

which gives formula (7.13).

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Exposé XII

NIELSEN–WECKEN AND LEFSCHETZ FORMULAS IN ALGEBRAIC GEOMETRY

by A. Grothendieck

written up by I. BUCUR

1.

Let X' be a scheme proper over the algebraically closed field k, and let G be a finite group acting on X'.

For each element $g \in G$, denote by

$$g_{X'} : X' \to X'$$

the automorphism of X' induced by g.

Suppose that there exists an open subset U' of X', stable under G, such that G acts freely on U', and let

$$\Lambda_n = \mathbb{Z}/\ell^n\mathbb{Z},$$

where ℓ is a prime number different from the characteristic of k.

Let $\Lambda_{n,U',X'}$ denote the sheaf $i_!(\Lambda_{n,U'})$ on X', where $\Lambda_{n,U'}$ is the constant sheaf on U' associated with the ring Λ_n and where $i : U' \hookrightarrow X'$ is the canonical immersion of U' into X'. The sheaf $\Lambda_{n,U',X'}$ is endowed in a natural way with the structure of a sheaf with group G of operators. Consequently the complex

$$H_{n,U',X'} = R\Gamma_{X'}(\Lambda_{n,U',X'})$$

is a complex of $\Lambda_n[G]$-modules, and more precisely a perfect complex (VIII, 7) of $\Lambda_n[G]$-modules (IX, 2).

We shall see below that it is more convenient for our purpose to consider, instead of the complex $R\Gamma_{X'}(\Lambda_{n,U',X'})$, its dual

$$H^\vee_{n,U',X'} = R\Gamma_{X'}(\Lambda_{n,U',X'})^\vee$$

with respect to the ring Λ_n, in the sense of derived categories:

$$H^\vee_{n,U',X'} = R\Gamma_{X'}(\Lambda_{n,U',X'})^\vee = R\operatorname{Hom}_{\Lambda_n}(R\Gamma_{X'}(\Lambda_{n,U',X'}), \Lambda_n).$$

The complex $R\Gamma_{X'}(\Lambda_{n,U',X'})^\vee$ is a perfect complex of $\Lambda_n[G]$-modules.

Suppose now that an endomorphism $f' : X' \to X'$ of the scheme X' and an endomorphism $\varphi : G \to G$ of the group G are given, satisfying the following two conditions:

$$f' g_{X'} = \varphi(g)_{X'} f' \qquad \forall g \in G. \tag{1.1}$$

$$\text{The complement } Y' = X' - U' \text{ is stable under } f'. \tag{1.2}$$

Under the hypotheses already made, Y' is stable under G. In general one does not require the open subset U' to be stable under f'.

We therefore obtain homomorphisms of sheaves

$$f'_{\Lambda_{n,X'}} : \Lambda_{n,X'} \longrightarrow \Lambda_{n,X'}, \qquad f'_{\Lambda_{n,Y'}} : \Lambda_{n,Y'} \longrightarrow \Lambda_{n,Y'},$$
$$f'_{\Lambda_{n,U',X'}} : \Lambda_{n,U',X'} \longrightarrow \Lambda_{n,U',X'}, \qquad g_{\Lambda_{n,U',X'}} : \Lambda_{n,U',X'} \longrightarrow \Lambda_{n,U',X'},$$

such that the following diagram, whose rows are exact, is commutative:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \Lambda_{n,U',X'} & \longrightarrow & \Lambda_{n,X'} & \longrightarrow & \Lambda_{n,Y'} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f'_{\Lambda_{n,U',X'}}} & & \downarrow{\scriptstyle f'_{\Lambda_{n,X'}}} & & \downarrow{\scriptstyle f'_{\Lambda_{n,Y'}}} & & \\ 0 & \longrightarrow & \Lambda_{n,U',X'} & \longrightarrow & \Lambda_{n,X'} & \longrightarrow & \Lambda_{n,Y'} & \longrightarrow & 0. \end{array} \tag{1.3}$$

We deduce in particular homomorphisms of complexes of Λ_n-modules

$$\begin{aligned}
f'_{H_{n,U',X'}} &= R\Gamma_{X'}(f'_{\Lambda_{n,U',X'}}) : H_{n,U',X'} \to H_{n,U',X'},\\
f'^{\vee}_{H_{n,U',X'}} &= (f'_{H_{n,U',X'}})^{\vee} : H^{\vee}_{n,U',X'} \to H^{\vee}_{n,U',X'},\\
g_{H_{n,U',X'}} &= R\Gamma_{X'}(g_{\Lambda_{n,U',X'}}) : H_{n,U',X'} \to H_{n,U',X'},\\
g^{\vee}_{H_{n,U',X'}} &= (g_{H_{n,U',X'}})^{\vee} : H^{\vee}_{n,U',X'} \to H^{\vee}_{n,U',X'},
\end{aligned}$$

such that the following relations hold:

$$g_{H_{n,U',X'}} \circ f'_{H_{n,U',X'}} = f'_{H_{n,U',X'}} \circ \varphi(g)_{H_{n,U',X'}}, \tag{1.4}$$

$$f'^{\vee}_{H_{n,U',X'}} \circ g^{\vee}_{H_{n,U',X'}} = \varphi(g)^{\vee}_{H_{n,U',X'}} \circ f'^{\vee}_{H_{n,U',X'}}. \tag{1.5}$$

Relation (1.5) shows (cf. Exposé XI on the general theory of traces, 5.2.1, 7.1) that the trace

$$\mathrm{Tr}^*_{\Lambda_n[G],\varphi}(f'^{\vee}_{H_{n,U',X'}}) \in \Lambda_n[G]_{\natural,\varphi} \simeq \mathcal{F}_!(G_{\natural,\varphi}, \Lambda_n) \qquad (*)^{28}$$

is defined.

Our first aim will be to give, under some additional hypotheses, an explicit formula for $\mathrm{Tr}^*_{\Lambda_n[G],\varphi}(f'^{\vee}_{H_{n,U',X'}})$ in terms of local invariants associated with f', terms which we shall define below.

2.

In fact, one can do better. For this, let n, m be two positive integers such that $m \geq n$. The ring homomorphisms

$$\alpha_{m,n} : \Lambda_m \longrightarrow \Lambda_n$$

give rise to homomorphisms of sheaves of rings on X':

$$\begin{aligned}
\alpha_{m,n,X'} &: \Lambda_{m,X'} \longrightarrow \Lambda_{n,X'},\\
\alpha_{m,n,U',X'} &: \Lambda_{m,U',X'} \longrightarrow \Lambda_{n,U',X'}.
\end{aligned}$$

Application of the Künneth formula gives

$$R\Gamma_{X'}(\Lambda_{m,X'}) \otimes^{\mathbb{L}}_{\Lambda_m} \Lambda_n \simeq R\Gamma_{X'}(\Lambda_{n,X'}), \tag{2.1}$$

$$R\Gamma_{X'}(\Lambda_{m,U',X'}) \otimes^{\mathbb{L}}_{\Lambda_m} \Lambda_n \simeq R\Gamma_{X'}(\Lambda_{n,U',X'}), \tag{2.2}$$

$$R\Gamma_{X'}(f'_{\Lambda_{m,X'}}) \otimes^{\mathbb{L}}_{\Lambda_m} \Lambda_n \simeq R\Gamma_{X'}(f'_{\Lambda_{n,X'}}), \tag{2.3}$$

$$R\Gamma_{X'}(f'_{\Lambda_{m,U',X'}}) \otimes^{\mathbb{L}}_{\Lambda_m} \Lambda_n \simeq R\Gamma_{X'}(f'_{\Lambda_{n,U',X'}}). \tag{2.4}$$

One deduces easily that

$$R\,\mathrm{Hom}_{\Lambda_m}(R\Gamma_{X'}(\Lambda_{m,U',X'}), \Lambda_m) \otimes^{\mathbb{L}}_{\Lambda_m} \Lambda_n \simeq R\,\mathrm{Hom}_{\Lambda_n}(R\Gamma_{X'}(\Lambda_{n,U',X'}), \Lambda_n), \tag{2.5}$$

$$f'^{\vee}_{H_{m,U',X'}} \otimes^{\mathbb{L}}_{\Lambda_m} \Lambda_n \simeq f'^{\vee}_{H_{n,U',X'}}. \tag{2.6}$$

[28] Cf. III B 5.12.

The morphism of perfect complexes $R\Gamma_{X'}(f'_{\Lambda_{n,X'}})$ (resp. $R\Gamma_{X'}(f'_{\Lambda_{n,U',X'}})$, $f'^{\vee}_{H_{n,U',X'}}$) is therefore obtained by extension of scalars (XI, 6.3) from the morphisms $R\Gamma_{X'}(f'_{\Lambda_{m,X'}})$ (resp. $R\Gamma_{X'}(f'_{\Lambda_{m,U',X'}})$, $f'^{\vee}_{H_{m,U',X'}}$) by means of the morphisms $\alpha_{m,n}$ (resp. $\alpha_{m,n}$, $\alpha_{m,n,G} : \Lambda_m[G] \to \Lambda_n[G]$).

Using (XI, 6.3, 6.3.2, 6.3.3), 2.3, and 2.6, one obtains the relations

$$\mathrm{Tr}^*_{\Lambda_n}(R\Gamma_{X'}(f'_{\Lambda_{n,X'}})) = \alpha_{m,n}(\mathrm{Tr}^*_{\Lambda_m}(R\Gamma_{X'}(f'_{\Lambda_{m,X'}}))), \tag{2.7}$$

$$\mathrm{Tr}^*_{\Lambda_n[G],\varphi}(f'^{\vee}_{H_{n,U',X'}}) = \alpha_{m,n,G}(\mathrm{Tr}^*_{\Lambda_m[G],\varphi}(f'^{\vee}_{H_{m,U',X'}})). \tag{2.8}$$

The system of functions

$$(\mathrm{Tr}_{\Lambda_n[G],\varphi}(f'^{\vee}_{H_{n,U',X'}}))_{n>0}$$

defines a function

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}(f'^{\vee}_{H_{U',X'}(\ell)}) \in \mathcal{F}_!(G_{\natural\varphi}, \mathbb{Z}_\ell),$$

which is φ-central on G and has values in the ring $\mathbb{Z}_\ell$ of ℓ-adic integers.

Similarly, the elements $\mathrm{Tr}^*_{\Lambda_n}(R\Gamma_{X'}(f'_{\Lambda_{n,X'}}))$, $n \geq 1$, define an ℓ-adic integer

$$\mathrm{Tr}^*_{\mathbb{Z}_\ell}(f'_{R\Gamma_{X'}(\mathbb{Z}_\ell,X')}).$$

Our aim will be to find a formula for

$$\mathrm{Tr}^*_{\mathbb{Z}_\ell[G],\varphi}(f'^{\vee}_{H_{U',X'}(\ell)}).$$

3. The local Nielsen–Wecken invariant (cf. [1])

Suppose, in addition to the hypotheses described in number 1, that every orbit of G is contained in an affine open subset. Then a quotient $X = X'/G$ exists and, by formula 1.1, the endomorphism f' of X' descends to the quotient:

$$f : X \longrightarrow X.$$

We shall use the following notation:

$$\begin{aligned} &p : X' \longrightarrow X = X'/G \quad \text{for the canonical projection,}\\ &Y = p(Y') = Y'/G,\\ &U = X - Y. \end{aligned}$$

The open U' is a principal covering with structural group G. Let $x \in U$ be a fixed point of f. We shall associate with it a φ-conjugacy class (XI, 7.1), denoted $\mathrm{NW}_f^{G,\varphi}(x)$, as follows.

Let $x' \in U'$ be such that $p(x') = x$. Since U' is a principal covering with structural group G and since x is fixed by f, there exists a unique element $g \in G$ such that $f'(x') = gx'$. The φ-conjugacy class of g depends only on x. Indeed, if $p(x'') = x$, there exists $h \in G$ such that $hx'' = x'$. Consequently, if $f'(x'') = g'x''$, then

$$\begin{aligned} gx' = f'(x') = f'(hx'') = \varphi(h)f'(x'') = (\varphi(h)g'h^{-1})x',\\ g = \varphi(h)g'h^{-1}, \end{aligned}$$

that is, g and g' are φ-conjugate. Thus one obtains a well-defined φ-conjugacy class, the local Nielsen–Wecken invariant associated with the fixed point $x \in U$ of f, which will be denoted by $\mathrm{NW}_f^{G,\varphi}(x)$.

The φ-conjugacy class $\mathrm{NW}_f^{G,\varphi}(x)$ defines a φ-central function (cf. XI, 7.1)

$$\mathrm{NW}_f^{G,\varphi}(x) : G \longrightarrow \mathbb{Z}$$

with integer values, characterized by the following relations:

$$\mathrm{NW}_f^{G,\varphi}(x)(g) = \begin{cases} 1 & \text{if there exists } x' \in X' \text{ such that } (g^{-1}f')(x') = x' \\ & \qquad \text{and } p(x') = x, \\ 0 & \text{otherwise.} \end{cases} \tag{3.1}$$

4. Generalities on local terms

Suppose that X' is smooth over k and, in addition, that the number of fixed points of f is finite. It follows in particular that the number of fixed points of $g^{-1}f'$ is finite for every $g \in G$.

For each fixed point $x \in X$ of f, introduce the φ-central function

$$S_x(f', f, G, \varphi) : G \longrightarrow \mathbb{Q},$$

with rational values, defined by the formula

$$S_x(f', f, G, \varphi)(g) = \frac{1}{c^\varphi(g)} \sum_{x' \in X'^{g^{-1}f'}_x} (\nu_{x'}(g^{-1}f') - 1). \tag{4.1}$$

We have used the following notation:

$$c^\varphi(g) = \operatorname{card} G_g^\varphi, \qquad G_g^\varphi = \{x \in G \mid \varphi(x) g x^{-1} = g\}, \tag{4.2}$$

$$X'^{g^{-1}f'}_x = \{x' \in X' \mid p(x') = x, (g^{-1}f')(x') = x'\}, \tag{4.3}$$

$$\nu_{x'}(g^{-1}f') = \text{the multiplicity of the fixed point } x' \text{ of } g^{-1}f'. \tag{4.4}$$

By composition with the canonical inclusions

$$\mathbb{Q} \hookrightarrow \mathbb{Q}_\ell,$$

the function $S_x(f', f, G, \varphi)$ gives φ-central functions $S_{x,\ell}(f', f, G, \varphi)$ on G with values in the field $\mathbb{Q}_\ell$ of ℓ-adic numbers.

Conjecture 4.5. *For each $\ell \neq \mathrm{car}(k)$, the values of the function $S_{x,\ell}(f', f, G, \varphi)$ are ℓ-adic integers.*

We shall prove below that Conjecture 4.5 is true when the dimension of X is equal to 1.

Proposition 4.6. *Let R be a discrete valuation ring, $\mathfrak{m}$ its maximal ideal, π a uniformizer, $k = R/\mathfrak{m}$ its residue field, assumed algebraically closed, v its valuation, $f'_R : R \to R$ an endomorphism of R, and G a finite group of automorphisms of R. Suppose that the endomorphism f'_R and the*

automorphisms g_R induce the identity on k, and suppose in addition that an endomorphism of G is given,

$$\varphi : G \longrightarrow G,$$

satisfying

$$f'_R \circ g_R = \varphi(g)_R \circ f'_R.$$

If, for every $g \in G$, one writes

$$\nu(g_R^{-1} f'_R) = v(g_R^{-1} f'_R(\pi) - \pi),$$

and if $c^{\varphi}(g)$ (cf. 4.2) is decomposed in the form

$$c^{\varphi}(g) = p^s q, \qquad (p, q) = 1, \qquad p = \mathrm{car}(k),$$

then $q \mid (\nu(g_R^{-1} f'_R) - 1)$.

To prove Proposition 4.6 one reduces to the case $g = e$ and $\varphi = \mathrm{id}_G$. Indeed one first reduces to the case $g = e$ by associating with the situation (f'_R, G, g, φ) the situation

$$(f''_R = g_R^{-1} f'_R,\ G,\ e,\ \varphi' = \mathrm{int}(g) \circ \varphi),$$

where $\varphi'(a) = g^{-1}\varphi(a)g$, and by observing easily that

$$\nu(g_R^{-1} f') - 1 = \nu(f''_R) - 1$$

and

$$c^{\varphi}(g) = c^{\varphi'}(e) \qquad (G_g^{\varphi} = G_e^{\varphi'}) \qquad \text{(cf. 4.2)}.$$

To reduce simultaneously to the case $g = e$, $\varphi = \mathrm{id}_G$, note that one can associate with the situation (f''_R, G, e, φ') the situation

$$(f''_R, G_e^{\varphi'}, e, \mathrm{id}_{G_e^{\varphi'}}),$$

and that

$$G_e^{\varphi'} = (G_e^{\varphi'})_e^{\mathrm{id}_{G_e^{\varphi'}}}.$$

The proof of Proposition 4.6 essentially uses the following lemma.

Lemma 4.7. *With the hypotheses and notation of Proposition 4.6, suppose moreover that $\varphi = \mathrm{id}_G$, that is, that the operations of G on R commute with f'_R, and that $r = \nu(f'_R) \geq 2$. Let*

$$c = \mathrm{card}(\mathrm{Im}(G \longrightarrow \mathrm{Aut}_k(\mathfrak{m}/\mathfrak{m}^2))).$$

Then the following divisibility relation holds:

$$c \mid (r - 1).$$

Proof. It is clear that the integer r satisfies the following conditions:

a) the homomorphism

$$f'_r : A/\mathfrak{m}^r \longrightarrow A/\mathfrak{m}^r$$

induced by f'_R is the identity;

b) the homomorphism

$$f'_{r+1} : A/\mathfrak{m}^{r+1} \longrightarrow A/\mathfrak{m}^{r+1}$$

is different from the identity.

On the other hand, it is evident that G acts on all the vector spaces $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ over k.

We first show that under the hypotheses of the lemma there exists a unique homomorphism of $k[G]$-modules

$$u(f'_R) : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \mathfrak{m}^r/\mathfrak{m}^{r+1}$$

satisfying the two following conditions:

1) $u(f'_R) \neq 0$;

2) $u(f'_R)(x \bmod \mathfrak{m}^2) = (f'_R(x) - x) \bmod \mathfrak{m}^{r+1}$, for every $x \in \mathfrak{m}$.

It is clear that condition a) implies $f'_R(x) - x \in \mathfrak{m}^r$ for every $x \in R$, which gives meaning to condition 2). This observation also gives the possibility of defining a homomorphism of abelian groups

$$\mathfrak{m} \longrightarrow \mathfrak{m}^r/\mathfrak{m}^{r+1}, \qquad x \longmapsto (f'_R(x) - x) \bmod \mathfrak{m}^{r+1}.$$

This homomorphism vanishes on $\mathfrak{m}^2$. Indeed, if $x_1, x_2 \in \mathfrak{m}$, then $f'_R(x_1x_2) - x_1x_2 \in \mathfrak{m}^{r+1}$ because, by condition a),

$$f'_R(x_1) = x_1 + \alpha_1, \quad \alpha_1 \in \mathfrak{m}^r,$$
$$f'_R(x_2) = x_2 + \alpha_2, \quad \alpha_2 \in \mathfrak{m}^r,$$

and hence

$$f'_R(x_1x_2) = f'_R(x_1)f'_R(x_2) = (x_1 + \alpha_1)(x_2 + \alpha_2) = x_1x_2 + x_1\alpha_2 + x_2\alpha_1 + \alpha_1\alpha_2,$$

and, plainly, $x_1\alpha_2 + x_2\alpha_1 + \alpha_1\alpha_2 \in \mathfrak{m}^{r+1}$.

It follows by passage to the quotient that we get a homomorphism of abelian groups

$$u(f'_R) : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \mathfrak{m}^r/\mathfrak{m}^{r+1}.$$

The homomorphism $u(f'_R)$ is in fact a homomorphism of vector spaces over k. To see this, one must show that for all $x \in \mathfrak{m}$, $\alpha \in R$,

$$f'_R(\alpha x) - \alpha x \equiv \alpha(f'_R(x) - x) \pmod{\mathfrak{m}^{r+1}},$$

that is,

$$f'_R(\alpha x) - \alpha f'_R(x) \in \mathfrak{m}^{r+1}. \tag{4.7.1}$$

But

$$f'_R(\alpha x) - \alpha f'_R(x) = (f'_R(\alpha) - \alpha)f'_R(x),$$

which gives (4.7.1), since

$$f'_R(\alpha) - \alpha \in \mathfrak{m}^r \quad \text{(condition a))}, \qquad f'_R(x) \in \mathfrak{m} \quad (x \in \mathfrak{m}).$$

The homomorphism $u(f'_R)$ is a homomorphism of $k[G]$-modules. Indeed, for $x \in \mathfrak{m}$ one has

$$\begin{aligned} u(f'_R)(g(x \bmod \mathfrak{m}^2)) &= (f'_R(gx) - gx) \bmod \mathfrak{m}^{r+1} \\ &= g(f'_R(x) - x) \bmod \mathfrak{m}^{r+1} = g\, u(f'_R)(x \bmod \mathfrak{m}^2). \end{aligned}$$

Finally, $u(f'_R) \neq 0$. Indeed, by condition b) there exists an element $x_0 \in R$ such that $f'_R(x_0) - x_0 \notin \mathfrak{m}^{r+1}$. But we need such an element in $\mathfrak{m}$. To obtain one, write x_0 in the form

$$x_0 = \frac{x}{x'}, \qquad x, x' \in \mathfrak{m}, \quad x' \notin \mathfrak{m}^{r+1}.$$

At least one of the elements x, x' satisfies the required condition.

Suppose, for contradiction, that simultaneously

$$f'_R(x) - x \in \mathfrak{m}^{r+1}, \qquad f'_R(x') - x' \in \mathfrak{m}^{r+1}.$$

Then, in the field of fractions K of A, one has

$$\frac{f'_R(x)}{x} \in 1 + \mathfrak{m}^r, \qquad \frac{f'_R(x')}{x'} \in 1 + \mathfrak{m}^r.$$

We may suppose that $f'_R(x') \neq 0$, since otherwise x' is the desired element. Since $1 + \mathfrak{m}^r$ is a multiplicative group, it follows that

$$\frac{x'}{f'_R(x')} \in 1 + \mathfrak{m}^r,$$

and consequently

$$\frac{f'_R(x)}{x} \cdot \frac{x'}{f'_R(x')} \in 1 + \mathfrak{m}^r.$$

This last relation implies

$$\frac{f'_R(x_0)}{x_0} \in 1 + \mathfrak{m}^r.$$

Indeed,

$$\begin{aligned} \frac{f'_R(x_0)}{x_0} - \frac{f'_R(x)}{x} \cdot \frac{x'}{f'_R(x')} &= \frac{f'_R(\frac{x}{x'})}{\frac{x}{x'}} - \frac{f'_R(x)}{x} \cdot \frac{x'}{f'_R(x')} \\ &= \frac{x' f'_R(\frac{x}{x'})}{x} - \frac{f'_R(x)}{x} \cdot \frac{x'}{f'_R(x')} \\ &= \frac{1}{x}\left(\frac{x' f'_R(x') f'_R(\frac{x}{x'}) - x' f'_R(x)}{f'_R(x')}\right) = 0. \end{aligned}$$

Thus $f'_R(x_0) - x_0 \in \mathfrak{m}^{r+1}$, a contradiction.

If the vector space

$$\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1})$$

is endowed with its usual G-module structure, its element $u(f'_R)$ belongs to the subspace

$$\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1})^G$$

of G-invariants.

It is classical that the vector space $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ identifies canonically with the i-th tensor power $\bigotimes_k^i(\mathfrak{m}/\mathfrak{m}^2)$. Consequently $\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1})$ identifies by the homomorphism of contraction with the $(r-1)$-st tensor power $\bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2)$:

$$\gamma : \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1}) \longrightarrow \bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2).$$

Through γ, the G-module structure on the Hom-space is transported to the G-module structure on $\bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2)$ induced by that on $\mathfrak{m}/\mathfrak{m}^2$:

$$g(\xi_1 \otimes \xi_2 \otimes \cdots \otimes \xi_{r-1}) = g\xi_1 \otimes \cdots \otimes g\xi_{r-1}.$$

Denote by $\mu(f'_R)$ the image of $u(f'_R)$ under γ. Thus:

I) $0 \neq \mu(f'_R) \in \bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2)$;

II) $\mu(f'_R)$ is invariant under G.

We can now conclude the proof of Lemma 4.7. Since the vector space $\mathfrak{m}/\mathfrak{m}^2$ has dimension one,

$$\mathrm{Aut}_k(\mathfrak{m}/\mathfrak{m}^2) = k^*.$$

Therefore the image of $G \to \mathrm{Aut}_k(\mathfrak{m}/\mathfrak{m}^2)$ is a finite cyclic subgroup of k^*, generated by a primitive c-th root of unity ε. It is generated by the automorphism

$$\chi : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \mathfrak{m}/\mathfrak{m}^2, \qquad \xi \longmapsto \varepsilon\xi.$$

Since $\mu(f'_R)$ is invariant under G,

$$\begin{aligned}\mu(f'_R) &= \chi\mu(f'_R)\\ &= \chi \sum_i \xi_{1i} \otimes \xi_{2i} \otimes \cdots \otimes \xi_{r-1,i}\\ &= \sum_i \varepsilon\xi_{1i} \otimes \varepsilon\xi_{2i} \otimes \cdots \otimes \varepsilon\xi_{r-1,i}\\ &= \varepsilon^{r-1}\mu(f'_R),\end{aligned}$$

that is,

$$(1 - \varepsilon^{r-1})\mu(f'_R) = 0.$$

Since $\mu(f'_R) \neq 0$, it follows that $1 - \varepsilon^{r-1} = 0$, that is, $c \mid (r-1)$.

Proof of Proposition 4.6. We have already noted that one may reduce to the case $g = e$, $\varphi = \mathrm{id}_G$. If G_1 denotes the kernel of the homomorphism

$$G \longrightarrow \mathrm{Aut}_k(\mathfrak{m}/\mathfrak{m}^2),$$

then ramification theory gives

$$\mathrm{card}(G_1) = p^t.$$

Consequently, with the notation of Lemma 4.7,

$$\mathrm{card}(G) = p^t c.$$

Since, on the other hand, $c^{\mathrm{id}_G}(e) = \mathrm{card}(G)$, the assertion of Proposition 4.6 in the case $g = e$, $\varphi = \mathrm{id}_G$, and hence in general, follows from Lemma 4.7.

Corollary 4.8. *For each $\ell \neq \mathrm{car}(k)$, the image of the rational number*

$$\frac{\nu(g_R^{-1} f'_R) - 1}{c^{\varphi}(g)}$$

under the canonical inclusion

$$\mathbb{Q} \hookrightarrow \mathbb{Q}_\ell$$

is an ℓ-adic integer.

Proposition 4.9. *Conjecture 4.5 is true if the dimension of the scheme X is equal to* 1.

Proof. We introduce the endomorphism

$$\psi : G \longrightarrow G$$

defined by

$$\psi = \mathrm{int}_g \circ \varphi, \qquad \mathrm{int}_g(\xi) = g^{-1}\xi g.$$

It is easy to see that:

i) $\psi(G_{x'}) \subset G_{g^{-1}f'x'}$, where $G_{x'}$ is the stabilizer of the point x';

ii) $c^{\varphi}(g) = \mathrm{card}(G^{\psi})$, where $G^{\psi} = \{\xi \mid \psi(\xi) = \xi\}$;

iii) G^{ψ} acts on ${X'_x}^{g^{-1}f'}$;

iv) $\nu_{x'}(g^{-1}f') = \nu_{hx'}(g^{-1}f')$ for every $h \in G^{\psi}$;

v) $(g^{-1}f')(g_1 x) = \psi(g_1)(g^{-1}f')(x)$.

Consequently one can decompose ${X'_x}^{g^{-1}f'}$ as the union of its orbits and obtain

$$\sum_{x' \in {X'_x}^{g^{-1}f'}} (\nu_{x'}(g^{-1}f') - 1) = \sum_{x' \in {X'_x}^{g^{-1}f'}/G^{\psi}} \frac{\nu_{x'}(g^{-1}f') - 1}{\mathrm{card}(G^{\psi}_{x'})}\ \mathrm{card}(G^{\psi}), \tag{4.9.1}$$

where

$$\nu_{x'}(g^{-1}f') = \nu((g^{-1}f')_{\mathcal{O}_{x'}}),$$

that is,

$$S_x(f', f, G, \varphi)(g) = \sum_{x' \in {X'}^{g^{-1}f'}/G^{\psi}} \frac{\nu_{x'}(g^{-1}f') - 1}{\mathrm{card}(G^{\psi}_{x'})}. \tag{4.9.2}$$

Proposition 4.9 follows from Proposition 4.6.

5. Generalization of a Nielsen–Wecken type formula (cf. [1])

Proposition 5.1. With the hypotheses and notation of 1, 2, 3, and 4, suppose in addition that Y' is a finite subset of X'.

If ℓ is a prime number different from the characteristic of k, then:

a) the values of the φ-central function

$$\sum_{x\in Y^f} S_{x,\ell}(f,f',G,\varphi)$$

are ℓ-adic integers;

b) the following formula holds:

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}(f'^{\vee}_{H_{U',X'}(\ell)}) = \sum_{x\in U^f}\nu_x(f)\mathrm{NW}_f^{G,\varphi}(x) + \sum_{x\in Y^f} S_{x,\ell}(f,f',G,\varphi). \tag{5.1.1}$$

Proof. It is clear that if equality b) is proved with everything regarded as functions with values in $\mathbb{Q}_\ell$, assertion a) follows at once, since $\mathrm{NW}_f^{G,\varphi}(x)$ and $\mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}(f'^{\vee}_{H_{U',X'}(\ell)})$ are functions with values in $\mathbb{Z}_\ell$.

First, reducing modulo ℓ^n for every $n\geq 1$ and using (XI, 7.2.3.6), one sees that for every $g\in G$,

$$\begin{aligned} c^\varphi(g)\ \mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}(f'^{\vee}_{H_{U',X'}(\ell)})(g) &= \mathrm{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')^{\vee}_{H_{U',X'}(\ell)} \\ &= \mathrm{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{H^{\vee}_{U',X'}(\ell)}. \end{aligned} \tag{5.2}$$

The commutative diagram (1.3), whose rows are exact, gives (XI, §4)

$$\mathrm{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{H^{\vee}_{U',X'}(\ell)} = \mathrm{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{R\Gamma_{X'}(\mathbb{Z}_{\ell,X'})} - \mathrm{Tr}^*_{\mathbb{Z}_\ell}(g_{Y'}^{-1}f'_{Y'})_{R\Gamma_{Y'}(\mathbb{Z}_{\ell,Y'})}. \tag{5.3}$$

The two terms on the right-hand side of (5.3) can be evaluated using the classical Lefschetz formula (cf.). (For the last term this is in fact a trivial Lefschetz formula):

$$\mathrm{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{R\Gamma_{X'}(\mathbb{Z}_{\ell,X'})} = \sum_{x\in X^f}\sum_{x'\in X'^{\,g_{X'}^{-1}f'}_x}\nu_{x'}(g_{X'}^{-1}f'), \tag{5.4}$$

$$\mathrm{Tr}^*_{\mathbb{Z}_\ell}(g_{Y'}^{-1}f'_{Y'})_{R\Gamma_{Y'}(\mathbb{Z}_{\ell,Y'})} = \text{the number of fixed points of } g_{Y'}^{-1}f'_{Y'}. \tag{5.5}$$

Consequently,

$$\mathrm{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{H^{\vee}_{U',X'}(\ell)} = \sum_{x\in U^f}\sum_{x'\in X'^{\,g_{X'}^{-1}f'}_x}\nu_{x'}(g_{X'}^{-1}f') - \sum_{x\in Y^f}\sum_{x'\in X'^{\,g_{X'}^{-1}f'}_x}(1-\nu_{x'}(g_{X'}^{-1}f')). \tag{5.6}$$

But for a given $x\in U^f$, the set $X'^{\,g_{X'}^{-1}f'}_x$ is nonempty if and only if $\mathrm{NW}_f^{G,\varphi}(x)=1$. Moreover,

$$\mathrm{card}\, X'^{\,g_{X'}^{-1}f'}_x = c^\varphi(g), \qquad x\in U, \tag{5.7}$$

$$\nu_{x'}(g_{X'}^{-1}f') = \nu_x(f), \qquad x\in U. \tag{5.8}$$

Relations (5.2) and (5.6) therefore give

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}(f'^{\vee}_{H_{U',X'}(\ell)})(g) = \sum_{x\in U^f}\nu_x(f)\mathrm{NW}_f^{G,\varphi}(x)(g) - \sum_{x\in Y^f}\sum_{x'\in X'^{\,g_{X'}^{-1}f'}_x}\frac{1-\nu_{x'}(g_{X'}^{-1}f')}{c^\varphi(g)}, \tag{5.9}$$

which is the formula to be proved.

5.10. Special cases

a) The unramified case: $Y' = \emptyset$, that is, $Y = \emptyset$.

In this case formula (5.1.1) becomes

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}(f'^{\vee}_{H_{U'},X'}(\ell)) = \sum_{x\in X^f} \nu_x(f)\mathrm{NW}_f^{G,\varphi}(x). \tag{5.11}$$

b) The case of multiplicities 1: $\nu_{x'}(g_{X'}^{-1}f') = 1$ for every $x' \in Y'^{g_{Y'}^{-1}f'_{Y'}}$ and every $g \in G$.

In this case formula 5.1.1 reduces to formula 5.11. This case occurs in particular if $\dim X = 1$ and $\nu_x(f) = 1$ for every $y \in Y$.

6. Application to a Lefschetz formula

6.1

We keep the notation and the conditions of number 1[29], except where the contrary is expressly stated. Let A be a commutative noetherian Λ_n-algebra, let F be a constructible sheaf, for the étale topology, of perfect A-modules on X, and let

$$u : f^*(F) \longrightarrow F$$

be a homomorphism of sheaves of A-modules.

In particular, one obtains an endomorphism of the pair (X, F):

$$(u, f) : (X, F) \longrightarrow (X, F),$$

and consequently an endomorphism of complexes of A-modules:

$$R\Gamma_X(u, f) : R\Gamma_X(F) \longrightarrow R\Gamma_X(F).$$

Since the scheme X is proper over k by hypothesis and the sheaf F is constructible and torsion, the finiteness theorem (SGA 4, XVII 5) ensures that the complex $R\Gamma_X(F)$ is a perfect complex of A-modules. The trace

$$\mathrm{Tr}_A(R\Gamma_X(u, f))$$

is therefore defined, with

$$\mathrm{Tr}_A(R\Gamma_X(u, f)) \in A.$$

Our aim will be to give, under some additional hypotheses, an explicit formula for $\mathrm{Tr}_A(R\Gamma_X(u, f))$ in terms of local invariants associated with the fixed points of f.

[29] X^f denotes the set of closed fixed points of X.

6.2. Definition of the local terms

Suppose that X' is irreducible of dimension one and that the restriction to U' of the sheaf $F' = p^*(F)$ is constant.

If the number of fixed points of f' is finite, as will be assumed from now on, one may always suppose that $X^f \subset Y^f$.

The sheaf $F'|_{U'}$ is therefore defined by an A-module M on which G acts:

$$F'|_{U'} \simeq M_{U'}.$$

Indeed, if η (resp. η') denotes the generic point of X (resp. X'), and $\bar{\eta}$ (resp. $\bar{\eta}'$) a geometric point over η (resp. η'), then

$$p(\eta') = \eta, \qquad F_{\bar{\eta}} \simeq F'_{\bar{\eta}'}.$$

The complex F' is isomorphic to M, and consequently M is a perfect A-module.

On the other hand, if the morphism f' is not constant, it fixes the generic point of X'. Consequently the morphism $(p^*(u), f')$ of the pair (X', F') induces a morphism on the fibre $F'_{\bar{\eta}'}$,

$$u'_{\bar{\eta}'} : F'_{\bar{\eta}'} \longrightarrow F'_{\bar{\eta}'},$$

and hence, evidently, a morphism of A-modules

$$u_M : M \longrightarrow M.$$

One easily sees that the following relation is satisfied:

$$u_M(\varphi(g)m) = g u_M(m) \qquad \forall m \in M, \quad \forall g \in G. \tag{6.2.1}$$

The dual $M^\vee$ of M with respect to A is therefore endowed with an $A[G]$-module structure and an endomorphism

$$u_M^\vee : M^\vee \longrightarrow M^\vee$$

such that

$$u_M^\vee(gx) = \varphi(g) u_M^\vee(x) \qquad \forall g \in G, \quad \forall x \in M^\vee.$$

Moreover, $M^\vee$ is a perfect A-module, and consequently the trace $\mathrm{Tr}_A(g^{-1}u_M^\vee)$ is defined for every $g \in G$.

Let $x \in X^f$ be a closed fixed point of f. The morphism u induces an endomorphism of the module F_x, the fibre of the sheaf F at the point x (the closed points of X are identified with the points of $X(x)$):

$$u_x : F_x \longrightarrow F_x.$$

The trace $\mathrm{Tr}_A(u_x)$ gives a first local invariant associated with the fixed point x.

We introduce another local invariant $\alpha_x(f, u, f', G, \varphi)$, an element of the ring A, by the formula

$$\alpha_x(f, u, f', G, \varphi) = \sum_{g \in G_{\natural, \varphi}} S_{x, \ell^n}(f, f', G, \varphi)(g) \, \mathrm{Tr}_A(g^{-1} u_M^\vee), \tag{6.2.2}$$

where the function

$$S_{x, \ell^n}(f, f', G, \varphi) : G \longrightarrow \Lambda_n$$

is obtained from the function $i_\ell \circ S_x(f, F', G, \varphi)$ (cf. 4) by reduction modulo $\ell^n\mathbb{Z}$, i.e.

$$S_{x, \ell^n}(f, f', G, \varphi)(g) = (i_\ell \circ S_x(f, f', G, \varphi))(g) \bmod \ell^n\mathbb{Z}.$$

6.3. A Lefschetz formula

Under the hypotheses and with the notation of 5.1, 6.1, and 6.2, one has the formula

$$\mathrm{Tr}_A(\mathrm{R}\Gamma_X(u,f)) = \sum_{x\in X^f} \tau_x(f,u,f',G,\varphi), \tag{6.3.1}$$

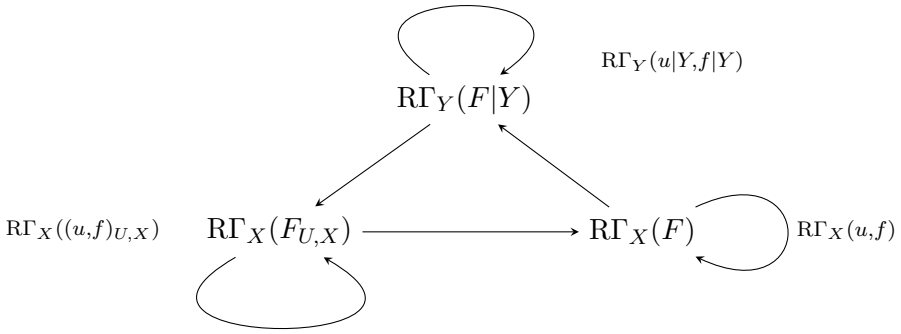
where

$$\tau_x(f,u,f',G,\varphi) = \mathrm{Tr}_A(u_x) - \alpha_x(f,u,f',G,\varphi). \tag{6.3.2}$$

Proof. Since the subscheme Y is stable under f, the morphism

$$(u,f) : (X,F) \longrightarrow (X,F)$$

induces an endomorphism of distinguished triangles



(6.3.3)

and hence the relation

$$\mathrm{Tr}_A(\mathrm{R}\Gamma_X(u,f)) = \mathrm{Tr}_A(\mathrm{R}\Gamma_X((u,f)_{U,X})) + \mathrm{Tr}_A(\mathrm{R}\Gamma_Y(u|Y,f|Y)). \tag{6.3.4}$$

The last term in (6.3.4) can be made explicit by the elementary Lefschetz formula and the fact that one may always suppose $X^f = Y^f$:

$$\mathrm{Tr}_A(\mathrm{R}\Gamma_Y(u|_Y,f|_Y)) = \sum_{x\in X^f} \mathrm{Tr}_A(u_x). \tag{6.3.5}$$

On the other hand,

$$\mathrm{R}\Gamma_X((u,f)_{U,X}) \simeq \mathrm{R}\Gamma^G \mathrm{R}\Gamma_{X'}((u',f')_{U',X'}), \quad (u' = p^*(u)), \tag{6.3.6}$$

$$F_{U',X'} \simeq \Lambda_{n,U',X'} \otimes^{\mathbf{L}}_{\Lambda_n} M, \qquad (u',f')_{U',X'} = f'_{\Lambda_{n,U',X'}} \otimes_{\Lambda_n} M, \tag{6.3.7}$$

whence, by the Künneth formula,

$$\mathrm{R}\Gamma_{X'}(u',f')_{U',X'} \simeq \mathrm{R}\Gamma_{X'}(f'_{\Lambda_{n,U',X'}}) \otimes^{\mathbf{L}}_{\Lambda_n} u_M. \tag{6.3.8}$$

It follows from (6.3.8) that

$$\mathrm{R}\Gamma_{X'}((u',f')_{U',X'})^\vee \simeq (\mathrm{R}\Gamma_{X'}(f'_{\Lambda_{n,U',X'}}))^\vee \otimes^{\mathbf{L}}_{\Lambda_n} A \otimes^{\mathbf{L}}_A u^\vee_M. \tag{6.3.9}$$

Using formula 6.3.6, one obtains

$$\mathrm{R}\Gamma_X((u,f)_{U,X}) = \mathrm{R}\Gamma_{X'}(f'_{\Lambda_{n,U',X'}}) \otimes^{\mathbf{L}}_{A[G]} u^\vee_M, \tag{6.3.10}$$

and hence, taking account of the general formula (cf. Exposé XI, 7.5.9),

$$\mathrm{Tr}_A((u,f)_{U,X}) = \sum_{g\in G_{\natural,\varphi}} \mathrm{Tr}_{\Lambda_n[G],\varphi}({f'}^{\vee}_{H_{n,U',X'}})(g)\,\mathrm{Tr}_A(g^{-1}u_M^{\vee}). \tag{6.3.11}$$

But the trace $\mathrm{Tr}_{\Lambda_n[G],\varphi}({f'}^{\vee}_{H_{n,U',X'}})$ is calculated using the Nielsen–Wecken formula (cf. Proposition 5.1 b)), evidently after reduction modulo $\ell^n\mathbb{Z}$.

Taking account of the relation $X^f = Y^f$, one obtains

$$\mathrm{Tr}_{\Lambda_n[G],\varphi}({f'}^{\vee}_{H_{n,U',X'}})(g) = \sum_{x\in X^f} S_{x,\ell^n}(f,f',G,\varphi)(g). \tag{6.3.12}$$

Combining relations 6.3.12, 6.3.11, 6.3.5, and 6.3.4 gives the formula in the statement.

6.4. Remarks on the local terms

a)

It is easy to see that, if $g \in G$, then one has the relation

$$\alpha_x(f,u,g^{-1}f',G,\varphi_g) = \alpha_x(f,u,f',G,\varphi), \tag{6.4.1}$$

where

$$\varphi_g(\xi) = (\mathrm{int}_g \circ \varphi)(\xi) = g^{-1}\varphi(\xi)g, \qquad \forall \xi \in G. \tag{6.4.2}$$

Relation 6.4.1 allows us always to reduce, after replacing f' by $g^{-1}f'$ and φ by φ_g, to the case where there exists at least one point $x' \in {X'_x}^{f'}$.

b)

Let $x' \in {X'_x}^{f'}$ and let $G_{x'}$ be the stabilizer of the point x'. Evidently $G_{x'}$ is stable under φ, and hence one obtains an endomorphism

$$\varphi_{x'} = \varphi|_{G_{x'}} : G_{x'} \longrightarrow G_{x'}.$$

Let $X'/G_{x'}$ denote the quotient scheme obtained from X' by the equivalence relation defined by $G_{x'}$, let $F'/G_{x'}$ be the endomorphism of $X'/G_{x'}$ obtained from the endomorphism f' of X' by the same equivalence relation, and let x' denote the class of the point x' modulo this equivalence relation.

The canonical projection

$$p : X' \longrightarrow X'/G = X$$

factors as

$$p : X' \xrightarrow{p_1} X'/G_{x'} \xrightarrow{p_2} X = X'/G.$$

Taking the inverse image by p_2 of the homomorphism u, one obtains a homomorphism

$$v = p_2(u) : (f'/G_{x'})^*(p_2^*(F)) \longrightarrow p_2^*(F).$$

It is easy to see that

$$\alpha_{x'}(f'/G_{x'},v,f',G_{x'},\varphi_{x'}) = \alpha_x(f,u,f',G,\varphi) \qquad (x = p(x')). \tag{6.4.3}$$

Thus one may always reduce to the "totally ramified" case, that is, to the case where the fibre of the point x is reduced to one point.

c) The "tamely ramified" case.

Suppose that there exists $x' \in p^{-1}(x)$ such that

$$f'(x') = x', \qquad p(x') = x,$$

a case to which one can always reduce, cf. case a), and suppose that the order of the group $G_{x'}$ is prime to the characteristic of the field k. Then one has

$$\alpha_x(f, u, f', G, \varphi) = S_{x,\ell^n}(f, f', G, \varphi)(e)\ \mathrm{Tr}_A(u_M^\vee). \tag{6.4.4}$$

Taking case b) into account, one may suppose that $G = G_{x'}$. Formula (6.4.4) then follows from the following lemma.

Lemma 6.4.5

If the following relations hold:

$$\nu_{x'}(f') > 1,$$
$$\nu_{x'}(g^{-1}f') > 1,$$

then g belongs to the first ramification group G_1 (cf. 4.7):

$$g \in G_1,$$

that is, g induces the identity on the vector space $\mathfrak{m}_{x'}/\mathfrak{m}_{x'}^2$.

Proof

Let $\widehat{\mathcal{O}}_{x'} = k[[T]]$. The automorphism of $\mathfrak{m}/\mathfrak{m}^2$ induced by g^{-1} is determined by a root of unity $\varepsilon(g^{-1}) \in k$ by the condition

$$g^{-1}T = \varepsilon(g^{-1})T \mod \mathfrak{m}^2,$$

where $\mathfrak{m}$ is the maximal ideal of the ring $k[[T]]$.

On the other hand, the hypothesis $\nu_{x'}(f') > 1$ gives

$$f'(T) = T + a_\nu T^\nu + a_{\nu+1}T^{\nu+1} + \cdots, \qquad \nu \geq 2.$$

Consequently

$$g^{-1}f'(T) = \varepsilon(g^{-1})T + b_\nu T^\nu + b_{\nu+1}T^{\nu+1} + \cdots,$$

and the hypothesis $\nu(g^{-1}f') > 1$ implies $\varepsilon(g^{-1}) = 1$, as required.

d) The case of a sheaf F which is not ramified at the fixed points of the endomorphism f.

In this case the local term $\tau_x(f, u, f', G, \varphi)$ is given by the formula

$$\tau_x(f, u, f', G, \varphi) = \nu_x(f)\ \mathrm{Tr}_A(u_x), \tag{6.4.6}$$

and consequently depends only on the pair (f, u).

Indeed, in this case one has

$$S'_{x,\ell^n}(f, f', G, \varphi)(g) = \begin{cases} 0 & \text{if there exists } x' \in X' \text{ such that } (g^{-1}f')(x') = x' \text{ or} \\ & \text{equivalently if the } \varphi\text{-conjugacy class of } g \text{ coincides with} \\ & \text{the class } \mathrm{NW}_f^{G,\varphi}(x) \text{ (cf. 3);} \\ \nu_x(f) - 1 & \text{otherwise.} \end{cases} \tag{6.4.7}$$

It follows in particular that

$$S_{x,\ell^n}(f, f'; G, \varphi)(g) = 0 \qquad \text{if } g \notin \mathrm{NW}_f^{G,\varphi}(x). \tag{6.4.8}$$

We therefore find

$$\begin{aligned} \tau_x(f, u, f', G, \varphi) &= \nu_x(f)\,\mathrm{Tr}_A(g^{-1}u_M^\vee) = \nu_x(f)\,\mathrm{Tr}_A(g^{-1}u')_{x'} \\ &= \nu_x(f)\,\mathrm{Tr}_A(u_x). \end{aligned} \tag{6.4.9}$$

7. Comments on the conditions of validity of the Lefschetz formula

7.1

Of course, in practice one starts with a scheme X proper over k and an endomorphism

$$(u, f) : (X, F) \longrightarrow (X, F),$$

such that:

- F is a constructible sheaf of A-modules,
- the set X^f of fixed points of f is finite.

The problem is to define, under these conditions, local terms $\tau_x(u, f)$, $x \in X^f$, such that

$$(*) \qquad \mathrm{Tr}_A(R\Gamma_X(u, f)) = \sum_{x \in X^f} \tau_x(u, f).$$

Using the two duality theorems, global and local, Verdier succeeds in defining terms $\tau_x(u, f)$ such that formula (*) is true (cf. III). Unfortunately, the explicit determination of the $\tau_x(u, f)$, as defined by Verdier, in terms of known local invariants seems difficult; hence the interest of the definitions proposed in the preceding chapter.[30]

By contrast, these latter definitions require additional hypotheses on F, f, and u, namely the existence of U, U', G, f', φ having the properties required in 5.1, 6.1, 6.2.

Proposition 7.2. Let X be a smooth scheme of dimension 1, let F be constructible, and let

$$(u, f) : (X, F) \longrightarrow (X, F)$$

[30] See, however, III B for the comparison of the local terms of 6.3.1 with Verdier's local terms.

be an endomorphism such that the morphism of sheaves

$$u : f^*(F) \longrightarrow F$$

is an isomorphism. Under these conditions there exist U, U', G, f', φ satisfying the conditions of 5.1, 6.1, 6.2.

Proof. Let U be the largest open subset of X on which F is locally constant, and let Y denote its complement. The set Y is stable under f:

$$f(Y) \subset Y.$$

Indeed, suppose by contradiction that there is a $y \in Y$ such that $f(y) \in U$. The sheaf $f^*(F)$ is then locally constant in a neighbourhood of y, and using the isomorphism u it follows that F is locally constant in a neighbourhood of y, which is absurd.

If f is constant, the proof of the proposition is trivial. We shall therefore suppose that f is not constant. If η is the generic point of X, then $f(\eta) = \eta$.

Let $i_\eta : \mathrm{Spec}(k(\eta)) \hookrightarrow X$ be the canonical inclusion, put $F_\eta = i_\eta^*(F)$, and let

$$u_\eta : f_\eta^*(F_\eta) \longrightarrow F_\eta$$

be the morphism, in fact the isomorphism, of sheaves induced by u, where f_η is of course the restriction of f to $\mathrm{Spec}(k(\eta))$.

Translating everything into terms of H-modules,

$$H = \mathrm{Gal}(k(\eta)_s/k(\eta)), \qquad k(\eta)_s \text{ a separable closure of } k(\eta),$$

one obtains the isomorphisms

$$H \xrightarrow{\Psi} H \qquad F_{\bar\eta} \xrightarrow{u_{\bar\eta}} F_{\bar\eta},$$

where $F_{\bar\eta}$ has an H-module structure and the following relation holds:

$$u_{\bar\eta}(gx) = \Psi(g)\, u_{\bar\eta} x, \qquad g \in H, \quad x \in F_{\bar\eta}.$$

If H_1 is the largest distinguished subgroup of H of finite index such that $F_{\bar\eta}^{H_1} = F_{\bar\eta}$, then $\Psi(H_1) = H_1$. Such a subgroup exists because the sheaf F_η is constructible.

Let $K \supset k(\eta)$ be the Galois extension of $k(\eta)$ corresponding to the subgroup H_1, and let X' be the normalization of X in K. It is clear that one can define an endomorphism

$$f' : X' \longrightarrow X'$$

and an endomorphism

$$H/H_1 = G \xrightarrow{\varphi} G$$

such that the system U, U', f', G, φ satisfies the required conditions.

Example 7.3. Let X be a prescheme defined over the finite field $\mathbb{F}_p$, where p is a prime number, let $\mathrm{fr}_X : X \to X$ be the Frobenius endomorphism (cf. XIV), and let F be a sheaf of Λ-modules on X.

If Y, S are schemes over $\mathbb{F}_p$, and if Y is a scheme over S, we denote by

$$\mathrm{Fr}_{Y/S} : Y \longrightarrow Y^{(p/S)}$$

the relative Frobenius morphism (cf. XIV 2).

We obtain a morphism of sheaves on $X_{\text{ét}}$

$$(\mathrm{fr}_X)_*(F) \longrightarrow F$$

by setting, for every étale X-prescheme U,

$$(\mathrm{fr}_X)_*(F)(U) = F(\mathrm{fr}_X^{-1}(U)) = F(U^{(p/X)}) \xrightarrow{F(\mathrm{Fr}_{U/X})} F(U).$$

This morphism is in fact an isomorphism. Its inverse

$$F(\mathrm{Fr}_{/X})^{-1} : F \longrightarrow (\mathrm{fr}_X)_*(F)$$

defines by adjunction a morphism

$$\mathrm{Fr}^*_{F/X} : (\mathrm{fr}_X)^*(F) \longrightarrow F$$

which is an isomorphism.

If in particular X is smooth of dimension 1 and F is constructible, the pair

$$(\mathrm{Fr}^*_{F/X}, \mathrm{fr}_X) : (X, F) \longrightarrow (X, F)$$

therefore falls under Proposition 7.2.

[1] Wecken, F., Fixpunktklassen I, Math. Ann. 117 (1941), p. 659–671; Fixpunktklassen II, III, Math. Ann. 118 (1942–1943), p. 216–234 and p. 544–577.

Exposé XV

FROBENIUS MORPHISM AND RATIONALITY OF L-FUNCTIONS

by C. Houzel

§ 1. Frobenius morphism

No. 1. Frobenius endomorphism

Definition 1. Let p be a prime number. A prescheme X is said to be of characteristic p if $p \cdot 1_{\mathcal{O}_X} = 0$.

This condition is equivalent to the following one: the unique morphism $X \to \mathrm{Spec}(\mathbb{Z})$ factors through $\mathrm{Spec}(\mathbb{F}_p) \to \mathrm{Spec}(\mathbb{Z})$; moreover, such a factorization is unique.

We shall denote by $(\mathrm{Sch})_p$ the full subcategory of the category of preschemes whose objects are the preschemes of characteristic p. By the preceding remark, the category $(\mathrm{Sch})_p$ is naturally isomorphic to the category of $\mathbb{F}_p$-preschemes.

If X is a prescheme of characteristic p, the pair

$$\mathrm{fr}_X = (\mathrm{id}_{|X|}, \psi), \qquad \psi : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad \psi(f) = f^p$$

for every section f of $\mathcal{O}_X$ over an open set, is an endomorphism of X, because ψ is visibly an endomorphism of the sheaf of rings $\mathcal{O}_X$.

Definition 2. Let X be a prescheme of characteristic p. The endomorphism fr_X thus defined is called the Frobenius endomorphism of X.

The Frobenius endomorphism fr_X depends functorially on the prescheme X of characteristic p; that is, for every morphism $g : X \to Y$ of preschemes of characteristic p, the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{fr}_X} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g} \\ Y & \xrightarrow[\mathrm{fr}_Y]{} & Y. \end{array}$$

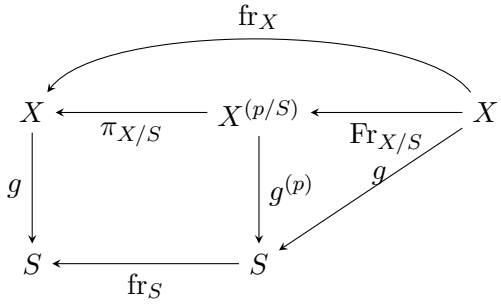
In other words, we have defined an endomorphism fr of the identity functor $\mathrm{id}_{(\mathrm{Sch})_p}$.

No. 2. Relative Frobenius morphism

Let S be a prescheme of characteristic p; all S-preschemes are then also of characteristic p. If X is an S-prescheme, one sees that fr_X factors through the projection

$$\pi_{X/S} : X \times_S (S, \mathrm{fr}_S) \longrightarrow X,$$

where (S, fr_S) denotes S regarded as an S-prescheme by means of the morphism fr_S. We shall denote the fibre product $X \times_S (S, \mathrm{fr}_S)$ by $X^{(p/S)}$, or even by $X^{(p)}$ if no confusion can result. The unique S-morphism from X to $X^{(p/S)}$ whose composite with $\pi_{X/S}$ is fr_X is denoted by $\mathrm{Fr}_{X/S}$; thus one obtains a commutative diagram in which the square is cartesian.



It is clear that $X^{(p/S)}$ depends functorially on the S-prescheme X; the functor

$$p_S : X \longmapsto X^{(p/S)}$$

is simply base change by the morphism $\mathrm{fr}_S : S \to S$. Like every base change, it commutes with projective limits and with sums, and preserves properties such as being of finite type, separated, affine, projective, flat, étale, and so on. Commutation with projective limits shows that if X is endowed with an algebraic structure, for example a group or algebra structure, then $X^{(p/S)}$ is naturally endowed with an algebraic structure of the same kind.

Using the functoriality of fr_X with respect to X, one sees that the morphism $\mathrm{Fr}_{X/S}$ depends functorially on the S-prescheme X, so that a functorial morphism

$$\mathrm{Fr}_{X/S} : \mathrm{id}_{(\mathrm{Sch})/S} \longrightarrow p_S$$

has been defined.

Definition 3. Let S be a prescheme of characteristic p and X an S-prescheme. The S-morphism

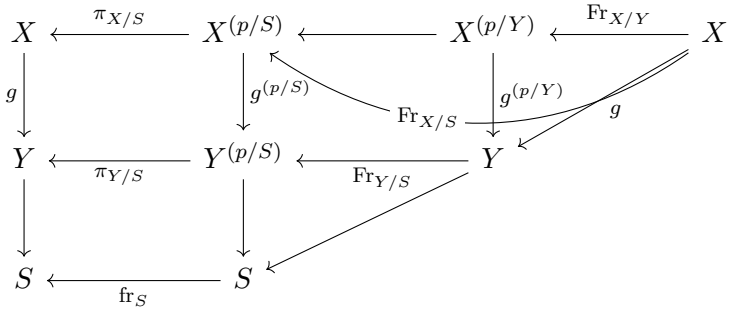
$$\mathrm{Fr}_{X/S} : X \longrightarrow X^{(p/S)}$$

is called the Frobenius morphism of X relative to S.

The first formal properties of the Frobenius morphism are stated in the following proposition.

Proposition 1.

a) *(Transitivity.)* Let S be a prescheme of characteristic p, and let $g : X \to Y$ be an S-morphism of S-preschemes. The natural diagram relating X, $X^{(p/S)}$, $X^{(p/Y)}$, Y, $Y^{(p/S)}$, and S is commutative and its three squares are cartesian.



b) *(Base change.)* Let $g : S' \to S$ be a morphism of preschemes of characteristic p, and let X be an S-prescheme. There is a canonical isomorphism

$$X^{(p/S)} \times_S S' \simeq (X \times_S S')^{(p/S')}$$

functorial in X, that is, $(X^{(p/S)})' \simeq X'^{(p/S')}$ when the result of base change by g is marked with a prime; moreover

$$\mathrm{Fr}_{X'/S'} = (\mathrm{Fr}_{X/S})'$$

modulo the identification allowed by the preceding isomorphism.

The proof is immediate.

Proposition 2. Let S be a prescheme of characteristic p and X an S-prescheme.

a) The relative Frobenius morphism

$$\mathrm{Fr}_{X/S} : X \longrightarrow X^{(p/S)}$$

is integral, surjective, and radicial; hence it is a universal homeomorphism.

b) If $X \to S$ is an open immersion, then $X^{(p)} = X$ and $\mathrm{Fr}_{X/S} = \mathrm{id}_X$. If $X \to S$ is a closed immersion, identifying X with a closed ringed subspace $(|X|, \mathcal{O}_S/\mathcal{J})$ of S, then

$$X^{(p)} = (|X|, \mathcal{O}_S/\mathcal{J}^p)$$

and $\mathrm{Fr}_{X/S} = (\mathrm{id}_{|X|}, \varphi)$, where $\varphi : \mathcal{O}_S/\mathcal{J}^p \to \mathcal{O}_S/\mathcal{J}$ is the evident augmentation.

c) Suppose that X is locally of finite presentation over S.

1) The following three conditions are equivalent:

(i) X is unramified over S,
(ii) $\mathrm{Fr}_{X/S}$ is unramified,
(iii) $\mathrm{Fr}_{X/S}$ is a monomorphism.

2) Likewise, the following three conditions are equivalent:

(i') X is étale over S,
(ii') $\mathrm{Fr}_{X/S}$ is étale,
(iii') $\mathrm{Fr}_{X/S}$ is an isomorphism.

We prove a). The composite $\pi_{X/S} \circ \mathrm{Fr}_{X/S} = \mathrm{fr}_X$ is visibly integral, surjective, and radicial; it follows that $\mathrm{Fr}_{X/S}$ is radicial (EGA I 3.5.6 (ii)). Moreover $\pi_{X/S}$ is separated and radicial, because it is deduced from fr_S by the base change $X \to S$; it follows that $\mathrm{Fr}_{X/S}$ is integral (EGA II 6.1.5 (v)) and surjective.

The proof of b) is evident, observing that

$$\mathcal{J}\mathcal{O}_{(S,\mathrm{fr}_S)} = \mathcal{J}^p$$

for every ideal $\mathcal{J}$ of $\mathcal{O}_S$.

Proof of c). It is known that for a morphism to be a monomorphism, respectively an isomorphism, it is necessary and sufficient that it be radicial and unramified, respectively radicial, étale, and surjective. This gives the equivalence of (ii) and (iii), respectively of (ii') and (iii'). It remains to prove that (i) is equivalent to (ii), and (i') to (ii').

For this, one uses the differential criterion of unramifiedness (SGA I 3) and the isomorphism

$$\Omega^1_{X/S} \simeq \Omega^1_{X/X^{(p)}}$$

given by the canonical exact sequence

$$\mathrm{Fr}^*_{X/S}(\Omega^1_{X^{(p)}/S}) \longrightarrow \Omega^1_{X/S} \longrightarrow \Omega^1_{X/X^{(p)}} \longrightarrow 0$$

(SGA II 4), in which the first arrow is visibly zero, because $\Omega^1_{X^{(p)}/S}$ is generated by the differentials of sections of $\mathcal{O}_{X^{(p)}}$, and the differential $d_{X/S}$ vanishes on $(\mathcal{O}_X)^p$ and on $\mathcal{O}_S$. Thus (i) is equivalent to (ii).

Now suppose X is étale over S; the same is true of $X^{(p)}$ by base change, and consequently the morphism $\mathrm{Fr}_{X/S} : X \to X^{(p)}$ is étale. Conversely, suppose that $\mathrm{Fr}_{X/S}$ is étale; it is unramified, hence X is unramified over S. In other words, the structural morphism $g : X \to S$ is locally on X, for the étale topology, a closed immersion (SGA I, cor. 7.8). We may therefore suppose that g is a closed immersion of finite presentation, and it remains to see that, if $\mathrm{Fr}_{X/S}$ is an isomorphism, then g is an open immersion. Now X is defined by an ideal $\mathcal{J}$ of finite type,

$$X = (|X|, \mathcal{O}_S/\mathcal{J}),$$

and, by b),

$$X^{(p)} = (|X|, \mathcal{O}_S/\mathcal{J}^p), \qquad \mathrm{Fr}_{X/S} = (\mathrm{id}_{|X|}, \mathcal{O}_S/\mathcal{J}^p \to \mathcal{O}_S/\mathcal{J}).$$

To say that $\mathrm{Fr}_{X/S}$ is an isomorphism is therefore to say that $\mathcal{J} = \mathcal{J}^p$; since $\mathcal{J}$ is of finite type, Nakayama's lemma implies that $\mathcal{J}_s = 0$, or $\mathcal{J}_s = \mathcal{O}_{S,s}$, for every point $s \in S$, whence the conclusion.

No. 3. Effect on a module. Examples

Let S be a prescheme of characteristic p and X an S-prescheme. For every $\mathcal{O}_X$-Module $\mathcal{E}$, a sheaf for the Zariski topology, set

$$\mathcal{E}^{(p/S)} = \pi_{X/S}^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{O}_{X^{(p)}} = \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S,\mathrm{fr}_S)}.$$

This is an $\mathcal{O}_{X^{(p)}}$-Module, and one has

$$\mathcal{E}^{(p/X)} \simeq \mathrm{Fr}_{X/S}^*(\mathcal{E}^{(p/S)}).$$

The most interesting case is the case $X = S$; one may then write $\mathcal{E}^{(p)}$ instead of $\mathcal{E}^{(p/S)}$ without risk of confusion, and $\mathcal{E}^{(p)} = \mathrm{fr}_S^*(\mathcal{E})$. The adjunction morphism

$$\mathcal{E} \longrightarrow \mathrm{fr}_{S*}\,\mathrm{fr}_S^*(\mathcal{E})$$

is a morphism of abelian sheaves $\mathcal{E} \to \mathcal{E}^{(p)}$ which is not $\mathcal{O}_S$-linear, but is "p-linear" in the evident sense; it is visibly a universal p-linear morphism from $\mathcal{E}$ to an $\mathcal{O}_S$-Module.

It is clear that $\mathcal{E}^{(p)}$ depends functorially on $\mathcal{E}$, and that the functor $\mathcal{E} \mapsto \mathcal{E}^{(p)}$ is right exact, but not exact in general; it commutes with tensor products:

$$(\mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{F})^{(p)} \simeq \mathcal{E}^{(p)} \otimes_{\mathcal{O}_{X^{(p)}}} \mathcal{F}^{(p)},$$

and preserves properties such as being quasi-coherent, of finite type, of finite presentation, flat, locally free, and so on. In particular, if $\mathcal{L}$ is an invertible sheaf on S, one immediately sees that $\mathcal{L}^{(p)}$ is canonically isomorphic to $\mathcal{L}^{\otimes p}$, because the morphism $\mathcal{L} \to \mathcal{L}^{\otimes p}$ defined by raising sections to their p-th tensor power is a universal p-linear morphism.

If $\mathcal{A}$ is an $\mathcal{O}_S$-Algebra, then $\mathcal{A}^{(p)}$ has an $\mathcal{O}_S$-Algebra structure deduced from that of $\mathcal{A}$. Suppose $\mathcal{A}$ is quasi-coherent, respectively graded and quasi-coherent; the same is true of $\mathcal{A}^{(p)}$, and

$$\mathrm{Spec}(\mathcal{A}^{(p)}) = \mathrm{Spec}(\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S,\mathrm{fr}_S)}) \simeq \mathrm{Spec}(\mathcal{A}) \times_S (S, \mathrm{fr}_S) = \mathrm{Spec}(\mathcal{A})^{(p)},$$

respectively

$$\mathrm{Proj}(\mathcal{A}^{(p)}) = \mathrm{Proj}(\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S,\mathrm{fr}_S)}) \simeq \mathrm{Proj}(\mathcal{A}) \times_S (S, \mathrm{fr}_S) = \mathrm{Proj}(\mathcal{A})^{(p)}.$$

The projection $\pi_{\mathrm{Spec}(\mathcal{A})/S} : \mathrm{Spec}(\mathcal{A})^{(p)} \to \mathrm{Spec}(\mathcal{A})$, respectively $\pi_{\mathrm{Proj}(\mathcal{A})/S}$, is identified with the morphism deduced from $\mathcal{A} \to \mathcal{A}^{(p)}$. On the other hand, $\mathrm{Fr}_{\mathrm{Spec}(\mathcal{A})/S}$, respectively $\mathrm{Fr}_{\mathrm{Proj}(\mathcal{A})/S}$, comes from a homomorphism of $\mathcal{O}_S$-Algebras

$$\mathcal{A}^{(p)} = \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S,\mathrm{fr}_S)} \longrightarrow \mathcal{A}$$

which sends $a \otimes f$ to $a^p f$, for every section a of $\mathcal{A}$ and every section f of $\mathcal{O}_S$. To see this, observe that the composite of this homomorphism with $\mathcal{A} \to \mathcal{A}^{(p)}$, on the right, is raising to the p-th power in $\mathcal{A}$.

Consider the particular case where $\mathcal{A}$ is the symmetric algebra $\mathrm{Sym}(\mathcal{E})$ of a quasi-coherent $\mathcal{O}_S$-Module $\mathcal{E}$; it is clear that $\mathcal{A}^{(p)} \simeq \mathrm{Sym}(\mathcal{E}^{(p)})$, hence

$$\mathbb{V}(\mathcal{E}^{(p)}) \simeq \mathrm{Spec}(\mathcal{A}^{(p)}) \simeq \mathrm{Spec}(\mathcal{A})^{(p)} = \mathbb{V}(\mathcal{E})^{(p)},$$

$$\mathbb{P}(\mathcal{E}^{(p)}) \simeq \mathrm{Proj}(\mathcal{A}^{(p)}) \simeq \mathrm{Proj}(\mathcal{A})^{(p)} = \mathbb{P}(\mathcal{E})^{(p)}.$$

As for $\mathrm{Fr}_{\mathbb{V}(\mathcal{E})/S}$, respectively $\mathrm{Fr}_{\mathbb{P}(\mathcal{E})/S}$, it factors in the following way:

$$\mathbb{V}(\mathcal{E}) \longrightarrow \mathbb{V}(\mathrm{Sym}^p(\mathcal{E})) \longrightarrow \mathbb{V}(\mathcal{E}^{(p)}),$$

respectively

$$\mathbb{P}(\mathcal{E}) \longrightarrow \mathbb{P}(\mathrm{Sym}^p(\mathcal{E})) \longrightarrow \mathbb{P}(\mathcal{E}^{(p)}),$$

where the first arrow is defined by the canonical morphism

$$\mathrm{Sym}(\mathrm{Sym}^p(\mathcal{E})) \longrightarrow \mathrm{Sym}(\mathcal{E})$$

which comes from $\mathrm{Sym}^p(\mathcal{E}) \to \mathrm{Sym}(\mathcal{E})$, while the second is defined by the morphism of $\mathcal{O}_S$-Modules

$$\mathcal{E}^{(p)} \longrightarrow \mathrm{Sym}^p(\mathcal{E})$$

coming from the p-linear morphism $\mathcal{E} \to \mathrm{Sym}^p(\mathcal{E})$ given by raising sections to their p-th symmetric power. In the case $X = \mathbb{P}(\mathcal{E})$, let us compute the fundamental sheaf $\mathcal{O}_{X^{(p)}}(1)$:

$$\mathcal{O}_{X^{(p)}}(1) = (\mathrm{Sym}(\mathcal{E})^{(p)}(1))^{\sim} = (\mathrm{Sym}(\mathcal{E})(1)^{(p)})^{\sim} \simeq \mathcal{O}_X(1)^{(p/S)}.$$

When $\mathcal{E} = \mathcal{O}_S^n$, $\mathrm{Sym}(\mathcal{E}) = \mathcal{O}_S[T_1, \ldots, T_n]$ and $X = \mathbb{V}(\mathcal{E}) = S[T_1, \ldots, T_n]$, respectively $X = \mathbb{P}(\mathcal{E}) = \mathbb{P}_S^{n-1}$, it is clear that $\mathcal{E}^{(p)} \simeq \mathcal{O}_S^n = \mathcal{E}$, so that $X^{(p)} = X$. The Frobenius morphism $\mathrm{Fr}_{X/S}$ comes from the endomorphism of the algebra $\mathcal{O}_S[T_1, \ldots, T_n]$ which sends T_i to T_i^p $(i = 1, \ldots, n)$.

Now consider a prescheme X, affine of finite type, respectively projective, over S, defined by an ideal, respectively a homogeneous ideal,

$$\mathcal{J} \subset \mathcal{O}_S[T_1, \ldots, T_n].$$

Thus $X = \mathrm{Spec}(\mathcal{A})$, respectively $X = \mathrm{Proj}(\mathcal{A})$, where

$$\mathcal{A} = \mathcal{O}_S[T_1, \ldots, T_n]/\mathcal{J}.$$

One sees that

$$\mathcal{A}^{(p)} \simeq \mathcal{O}_S[T_1, \ldots, T_n]/\mathcal{J}',$$

where $\mathcal{J}'$ is the image in $\mathcal{O}_S[T_1, \ldots, T_n]$ of $\mathcal{J}^{(p)} = \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S,\mathrm{fr}_S)}$; the ideal $\mathcal{J}'$ is deduced from $\mathcal{J}$ by raising the coefficients of the polynomials to the p-th power. One has $X^{(p)} = \mathrm{Spec}(\mathcal{A}^{(p)})$, respectively $\mathrm{Proj}(\mathcal{A}^{(p)})$, and the Frobenius morphism

$$\mathrm{Fr}_{X/S} : X \longrightarrow X^{(p)}$$

is defined by raising the coordinates T_i to the p-th power; this determines a morphism $\mathcal{A}^{(p)} \to \mathcal{A}$.

4. Iterated Frobenius morphism; perfect preschemes

For every integer $\nu \geq 0$, the iterated endomorphism fr_X^ν depends functorially on the prescheme X of characteristic p. Thus, if S is a prescheme of characteristic p and X is an S-prescheme, fr_X^ν factors, in a unique way, as an S-morphism

$$\mathrm{Fr}_{X/S}^\nu : X \longrightarrow X^{(p^\nu/S)},$$

where

$$X^{(p^\nu/S)} = X \times_S (S, \mathrm{fr}_S^\nu),$$

followed by the projection $X^{(p^\nu/S)} \to X$. The morphism $\mathrm{Fr}_{X/S}^\nu$ depends functorially on the S-prescheme X and is compatible with change of the base S (cf. no. 2, prop. 1b)). Moreover it is clear that

$$X^{(p^\nu/S)} = X^{(p^\nu)} \simeq \left(X^{(p^{\nu-1})}\right)^{(p)}$$

and that

$$\mathrm{Fr}_{X/S}^\nu = (\mathrm{Fr}_{X/S}^{\nu-1})^{(p)} \circ \mathrm{Fr}_{X/S},$$

modulo the identification allowed by the preceding canonical isomorphism.

Definition 4. A prescheme S of characteristic p is said to be perfect if fr_S is an automorphism of S.

If S is a perfect prescheme of characteristic p and X is an S-prescheme, one sees immediately that $X^{(p/S)}$ is identified with the prescheme X considered as an S-prescheme by means of the morphism $\mathrm{fr}_S^{-1} \circ g : X \to S$, where g is the structural morphism of X into S; modulo this identification, $\mathrm{Fr}_{X/S} = \mathrm{fr}_X$.

As an example of a perfect prescheme, we shall meet $S = \mathrm{Spec}(\mathbb{F}_q)$, where $q = p^\nu$ is a power of the characteristic p; indeed, in this case $\mathrm{fr}_S^\nu = \mathrm{id}_S$. If X is an S-prescheme, one sees that

$$X^{(p^\nu/S)} = X, \qquad \mathrm{Fr}_{X/S}^\nu = \mathrm{fr}_X^\nu .$$

Note that for $\nu = 1$ one obtains $S = \mathrm{Spec}(\mathbb{F}_p)$, for which $\mathrm{fr}_S = \mathrm{id}_S$; the S-preschemes are the preschemes of characteristic p; over this base S, $X^{(p/S)} = X$ and $\mathrm{Fr}_{X/S} = \mathrm{fr}_X$.

§ 2. Frobenius correspondence

1. Definition

Let X be a prescheme of characteristic p, and let F be a sheaf of sets on $X_{\text{ét}}$. For every étale X-prescheme U one has

$$(\mathrm{fr}_X)_*(F)(U) = F(\mathrm{fr}_X^{-1}(U)) = F(U^{(p/X)}),$$

and the Frobenius morphism $\mathrm{Fr}_{U/X} : U \to U^{(p/X)}$ defines a map

$$F(\mathrm{Fr}_{U/X}) : \mathrm{fr}_{X*}(F)(U) \longrightarrow F(U),$$

which is visibly functorial in U (since $\mathrm{Fr}_{U/X}$ is functorial in U). Thus one obtains a morphism of sheaves on $X_{\text{ét}}$:

$$(\mathrm{fr}_X)_*(F) \longrightarrow F.$$

Now, when U is étale over X, $\mathrm{Fr}_{U/X}$ is an isomorphism (§ 1 no. 2, prop. 2c)), hence the same is true of $F(\mathrm{Fr}_{U/X})$; it follows that the preceding morphism is an isomorphism. In the case where F is endowed with an algebraic structure, for example if it is a sheaf of Λ-modules, where Λ is a given base ring, this isomorphism is compatible with the algebraic structure. The inverse

$$F(\mathrm{Fr}_{/X})^{-1} : F \longrightarrow \mathrm{fr}_{X*}(F)$$

of this isomorphism defines, since $(\mathrm{fr}_X)^*$ is left adjoint to the functor $(\mathrm{fr}_X)_*$, a morphism

$$\mathrm{Fr}^*_{F/X} : (\mathrm{fr}_X)^*(F) \longrightarrow F.$$

One can even say more: since fr_X is a universal homeomorphism (§ 1 no. 2, prop. 2a)), the functors $(\mathrm{fr}_X)^*$ and $(\mathrm{fr}_X)_*$ are quasi-inverse to each other; consequently $\mathrm{Fr}^*_{F/X}$ is an isomorphism, just as is the morphism $F(\mathrm{Fr}_{/X})^{-1}$ from which it comes.

Definition 1. Given a prescheme X of characteristic p and a sheaf of sets F on $X_{\text{ét}}$, the Frobenius correspondence on (X, F) is the pair $(\mathrm{fr}_X, \mathrm{Fr}^*_{F/X})$ formed by the Frobenius endomorphism fr_X of X and the isomorphism

$$\mathrm{Fr}^*_{F/X} = (F(\mathrm{Fr}_{/X})^{-1})^\sharp : (\mathrm{fr}_X)^*(F) \longrightarrow F.$$

Of course one defines in the same way a class of iterated Frobenius correspondence $(\mathrm{fr}^\nu_X, (\mathrm{Fr}^*_{F/X})^\nu)$ for every integer $\nu \geq 0$; the isomorphism

$$(\mathrm{Fr}^*_{F/X})^\nu : (\mathrm{fr}^\nu_X)^*(F) \longrightarrow F$$

is equal to

$$(\mathrm{Fr}^*_{F/X})^{\nu-1} \circ (\mathrm{fr}^{\nu-1}_X)^*(\mathrm{Fr}^*_{F/X})$$

or also to

$$\mathrm{Fr}^*_{F/X} \circ \mathrm{fr}^*_X((\mathrm{Fr}^*_{F/X})^{\nu-1})$$

for $\nu \geq 1$.

Let us examine the particular case where F is representable by an étale prescheme V over X; then $(\mathrm{fr}_X)^*(F) = V^{(p/X)}$, $F(V) = \mathrm{Hom}_X(V, V)$,

$$(\mathrm{fr}_X)^*(F)(V) = F(V^{(p/X)}) = \mathrm{Hom}_X(V^{(p/X)}, V),$$

and $F(\mathrm{Fr}_{V/X})$ is the map from $\mathrm{Hom}_X(V^{(p/X)}, V)$ to $\mathrm{Hom}_X(V, V)$ which sends a morphism ψ to $\psi \circ \mathrm{Fr}_{V/X}$. Thus $\mathrm{Fr}^*_{F/X}$ is identified with a morphism from $V^{(p/X)}$ to V, which is the image of id_V under the inverse map $F(\mathrm{Fr}_{V/X})^{-1}$ of the preceding one. This shows that for $F = V$ étale over X one has

$$\mathrm{Fr}^*_{V/X} = (\mathrm{Fr}_{V/X})^{-1}.$$

Note that if the sheaf F is constant, then it is clear that

$$(\mathrm{fr}_X)^*(F) = F, \qquad \mathrm{Fr}^*_{F/X} = \mathrm{id}_F\,.$$

Proposition 1.

a) The isomorphism $\mathrm{Fr}^*_{F/X}$ depends functorially on the sheaf F; in other words there is a functorial isomorphism

$$\mathrm{Fr}^*_{/X} : \mathrm{fr}^*_X \longrightarrow \mathrm{id}_{\widetilde{X}_{\text{ét}}},$$

where $\widetilde{X}_{\text{ét}}$ is the category of sheaves of sets on $X_{\text{ét}}$.

b) The Frobenius correspondence $(\mathrm{fr}_X, \mathrm{Fr}^*_{F/X})$ on (X, F) is compatible with change of the base X. That is, if $\mathcal{F}$ denotes the fibred category over $(\mathrm{Sch})_p$ of sheaves, for the étale topology, on variable preschemes of characteristic p, the functors fr^*_X, $X \in \mathrm{ob}(\mathrm{Sch})_p$, define a cartesian $(\mathrm{Sch})_p$-functor

$$\mathrm{fr}^* : \mathcal{F} \longrightarrow \mathcal{F},$$

and the morphisms $\mathrm{Fr}^*_{F/X}$, $X \in \mathrm{ob}(\mathrm{Sch})_p$, $F \in \mathrm{ob}\,\widetilde{X}_{\text{ét}}$, define a functorial $(\mathrm{Sch})_p$-morphism

$$\mathrm{Fr}^*_{/} : \mathrm{fr}^* \longrightarrow \mathrm{id}_{\mathcal{F}}$$

which is an isomorphism.

c) Conversely, these properties characterize $\mathrm{Fr}^*_{/}$; that is, there exists a unique functorial $(\mathrm{Sch})_p$-morphism from fr^* to $\mathrm{id}_{\mathcal{F}}$.

Notice that statement b) implies statement a); the latter is entirely evident.

Let us prove b). Let $g : X' \to X$ be a morphism of preschemes of characteristic p, and let F be a sheaf on $X_{\text{ét}}$. Put $F' = g^*(F)$ and, more generally, mark by a prime the inverse image by g of any object given over X. Since

$$g \circ \mathrm{fr}_{X'} = \mathrm{fr}_X \circ g,$$

one has

$$\mathrm{fr}^*_{X'}(F') = \mathrm{fr}^*_{X'}(g^*F) = g^*(\mathrm{fr}^*_X F) = (\mathrm{fr}^*_X F)',$$

whence the fact that the fr^*_X define a cartesian functor fr^* over $(\mathrm{Sch})_p$. It remains to prove that

$$\mathrm{Fr}^*_{F'/X'} = (\mathrm{Fr}^*_{F/X})'.$$

Using the properties of adjoint functors, one sees that this amounts to showing that the image of $F'(\mathrm{Fr}_{/X'})^{-1}$ under the bijection

$$\mathrm{Hom}_{X'}(F', \mathrm{fr}_{X'*}(F')) = \mathrm{Hom}_{X'}(g^*F, \mathrm{fr}_{X'*}\, g^*F) \simeq \mathrm{Hom}_X(F, g_*\, \mathrm{fr}_{X'*}\, g^*F)$$

(with g^* left adjoint to g_*) coincides with the image of $F(\mathrm{Fr}_{/X})^{-1}$ under the map

$$\mathrm{Hom}_X(F, \mathrm{fr}_{X*}\, F) \longrightarrow \mathrm{Hom}_X(F, \mathrm{fr}_{X*}\, g_* g^* F) = \mathrm{Hom}_X(F, g_*\, \mathrm{fr}_{X'*}\, g^* F)$$

defined by the adjunction morphism $F \xrightarrow{\alpha} g_* g^* F$; in other words, one must show that

$$\alpha \circ F(\mathrm{Fr}_{/X}) = g_*(F'(\mathrm{Fr}_{/X'})) \circ \mathrm{fr}_{X*}(\alpha),$$

This says that for every prescheme U étale over X there is a commutative diagram

$$\begin{array}{ccc} F(U^{(p/X)}) & \xrightarrow{F(\mathrm{Fr}_{U/X})} & F(U) \\ \downarrow & & \downarrow \\ F'(U'^{(p/X')}) & \xrightarrow[F'(\mathrm{Fr}_{U'/X'})]{} & F'(U'), \end{array}$$

where the vertical arrows are the canonical maps; but this follows immediately from

$$\mathrm{Fr}_{U'/X'} = (\mathrm{Fr}_{U/X})'$$

(§ 1, no. 2, prop. 1b)).

To prove c), after composing with $(\mathrm{Fr}^*_{/})^{-1}$, we are reduced to proving that $\mathrm{id}_{\mathcal{F}}$ has only one $(\mathrm{Sch})_p$-endomorphism, namely the identity. If ψ is a $(\mathrm{Sch})_p$-endomorphism of $\mathrm{id}_{\mathcal{F}}$, let F be a sheaf on a prescheme X of characteristic p, and let U be an étale prescheme over X. We know that $F(U)$ is identified with $\mathrm{Hom}_U(U, F|U)$, and $\psi_F(U) : F(U) \to F(U)$ with the map

$$s \longmapsto \psi_{F|U} \circ s = s \circ \psi_U,$$

where $s \in \mathrm{Hom}_U(U, F|U)$; since the final sheaf U of $\widetilde{U}_{\text{ét}}$ has no endomorphism other than the identity, one has $\psi_U = \mathrm{id}_U$. It follows that $\psi_F(U)$ is also the identity of $F(U)$, and hence that $\psi = \mathrm{id}_{\mathcal{F}}$.

Of course, if the sheaf F is endowed with an algebraic structure, the isomorphism $\mathrm{Fr}^*_{F/X}$ is compatible with this structure; for example, if F is a sheaf of groups on the prescheme X of characteristic p, $\mathrm{Fr}^*_{F/X}$ is an isomorphism of sheaves of groups. Let Λ be a ring and let F be a sheaf of Λ-modules on X; then $\mathrm{Fr}^*_{F/X}$ is an isomorphism of sheaves of Λ-modules. More generally, if $F^\bullet$ is a complex of Λ-modules, one immediately defines an isomorphism

$$\mathrm{Fr}^*_{F^\bullet/X} : \mathrm{fr}^*_X(F^\bullet) \longrightarrow F^\bullet$$

which is equal to $\mathrm{Fr}^*_{F^i/X}$ on the component of degree i, thanks to proposition 1a). In this case the pair $(\mathrm{fr}_X, \mathrm{Fr}^*_{F^\bullet/X})$ is indeed a correspondence, in the usual sense, on $(X, F^\bullet)$; this correspondence is functorial in $F^\bullet$ and compatible with change of the base X (Proposition 1a), b)).

2. Effect on cohomology

Proposition 2. Let X be a prescheme of characteristic p.

a) For every sheaf of sets F on $X_{\text{ét}}$, the endomorphism of $H^0(X, F)$ defined by $(\mathrm{fr}_X, \mathrm{Fr}^*_{F/X})$, that is, the composite

$$H^0(X, F) \longrightarrow H^0(X, \mathrm{fr}^*_X(F)) \longrightarrow H^0(X, F),$$

where the first arrow is the canonical map arising from the contravariant character of $H^0(X, F)$ with respect to X, and the second is $H^0(X, \mathrm{Fr}^*_{F/X})$, is the identity of $H^0(X, F)$.

b) For every sheaf of groups F on $X_{\text{ét}}$, the endomorphism of $H^1(X, F)$ defined by $(\mathrm{fr}_X, \mathrm{Fr}^*_{F/X})$, that is,

$$H^1(X, F) \longrightarrow H^1(X, \mathrm{fr}^*_X F) \longrightarrow H^1(X, F),$$

is the identity.

c) For every left bounded complex F of Λ-modules on $X_{\text{ét}}$, the endomorphism of $\mathrm{R}\Gamma_X(F)$ defined by $(\mathrm{fr}_X, \mathrm{Fr}^*_{F/X})$, that is, the composite

$$\mathrm{R}\Gamma_X(F) \longrightarrow \mathrm{R}\Gamma_X(\mathrm{fr}^*_X F) \longrightarrow \mathrm{R}\Gamma_X(F),$$

is the identity.

We first establish a); b) and c) will be deduced from it. Let ψ_F be the considered endomorphism of $H^0(X,F)$. It is clear that ψ_F depends functorially on F and that $\psi_X = \mathrm{id}_{H^0(X,X)}$. Every element s of $H^0(X,F)$ is identified with a morphism of sheaves $s : X \to F$ such that s is the image under $H^0(X,s)$ of the unique element of $H^0(X,X)$; since

$$\psi_F \circ H^0(X,s) = H^0(X,s) \circ \psi_X = H^0(X,s),$$

by the preceding remarks, one sees that $\psi_F(s) = s$ for every s, hence $\psi_F = \mathrm{id}_{H^0(X,F)}$.

Note that ψ_F is also the composite

$$H^0(X,F) \longrightarrow H^0(X, \mathrm{fr}_{X*}F) \longrightarrow H^0(X,F),$$

where the first arrow is $H^0(X, F(\mathrm{Fr}_{/X})^{-1})$ and the second is the canonical map, which is nothing but the identity. Similarly, the endomorphisms considered in b) and c) factor as

$$H^1(X,F) \longrightarrow H^1(X, \mathrm{fr}_{X*}F) \longrightarrow H^1(X,F),$$

and

$$\mathrm{R}\Gamma_X(F) \longrightarrow \mathrm{R}\Gamma_X(\mathrm{fr}_{X*}F) \longrightarrow \mathrm{R}\Gamma_X(F).$$

To establish b), we shall use a monomorphism $F \to G$ of F into a sheaf of groups G such that $H^1(X,G) = 0$. Then $\mathrm{fr}_{X*}F \to \mathrm{fr}_{X*}G$ is a monomorphism and $H^1(X, \mathrm{fr}_{X*}G) = 0$. Introduce the quotient homogeneous sheaf of sets $C = G/F$. Since fr_X is integral, surjective, and radicial, the functors fr_X^* and fr_{X*} are quasi-inverse equivalences of categories (SGA VIII theorem 1.1). It follows that

$$\mathrm{fr}_{X*}(C) \simeq \mathrm{fr}_{X*}(G)/\,\mathrm{fr}_{X*}(F),$$

and the exact cohomology sequence (SGA XII proposition 3.1) shows that the considered endomorphism of $H^1(X,F)$ arises, by passage to the quotient, from

$$\psi_C : H^0(X,C) \longrightarrow H^0(X,C),$$

which is the identity by a). This endomorphism is therefore $\mathrm{id}_{H^1(X,F)}$.

Assertion c) is proved analogously, using a complex I that is injective in each degree and isomorphic to F in the derived category D^+, built from the Λ-modules. The complex $\mathrm{fr}_{X*}(I)$ is injective in each degree and isomorphic to $\mathrm{fr}_{X*}(F)$ in D^+, since fr_{X*} is an equivalence of categories. One may then identify $\mathrm{R}\Gamma_X(F)$ with $H^0(X,I)$ and $\mathrm{R}\Gamma_X(\mathrm{fr}_{X*}F)$ with $H^0(X, \mathrm{fr}_{X*}I)$; one sees that the considered endomorphism of $\mathrm{R}\Gamma_X(F)$ is identified with ψ_I, which is the identity. This endomorphism is therefore $\mathrm{id}_{\mathrm{R}\Gamma_X(F)}$.

3. Behavior of products

Let X and X' be preschemes of characteristic p. Their product $X \times X'$ is identical with the fibred product $X \times_e X'$, where $e = \mathrm{Spec}(\mathbb{F}_p)$. It is clear that

$$\mathrm{fr}_{X\times X'} = \mathrm{fr}_X \times \mathrm{fr}_{X'} = (\mathrm{fr}_X \times \mathrm{id}_{X'}) \circ (\mathrm{id}_X \times \mathrm{fr}_{X'}) = (\mathrm{id}_X \times \mathrm{fr}_{X'}) \circ (\mathrm{fr}_X \times \mathrm{id}_{X'}).$$

It follows immediately that, for every X-prescheme Y and every X'-prescheme Y', there is a canonical isomorphism

$$(Y \times Y')^{(p/X\times X')} \simeq Y^{(p/X)} \times Y'^{(p/X')};$$

modulo this isomorphism one can write

$$\mathrm{Fr}_{Y\times Y'/X\times X'} = \mathrm{Fr}_{Y/X} \times \mathrm{Fr}_{Y'/X'}\,.$$

Proposition 3. Let X and X' be preschemes of characteristic p, let Λ be a commutative ring, let F be a sheaf of Λ-modules on $X_{\text{ét}}$, and let F' be a sheaf of Λ-modules on $X'_{\text{ét}}$. One has

$$\begin{aligned}\mathrm{Fr}^*_{F\boxtimes_\Lambda F'/X\times X'} &= \mathrm{Fr}^*_{F/X} \otimes \mathrm{Fr}^*_{F'/X'} \\ &= (\mathrm{Fr}^*_{F/X} \otimes \mathrm{id}_{\mathrm{fr}^*_{X'}F'}) \circ (\mathrm{id}_F \otimes \mathrm{Fr}^*_{F'/X'}) \\ &= (\mathrm{id}_{\mathrm{fr}^*_X F} \otimes \mathrm{Fr}^*_{F'/X'}) \circ (\mathrm{Fr}^*_{F/X} \otimes \mathrm{id}_{F'}).\end{aligned}$$

To write these equalities, one has identified

$$\mathrm{fr}^*_{X\times X'}(F \boxtimes_\Lambda F')$$

with the tensor product $\mathrm{fr}^*_X(F) \otimes_\Lambda \mathrm{fr}^*_{X'}(F')$ by means of the canonical isomorphism.

The two sides of the first equality depend functorially on F and F', and are compatible with change of the base X or the base X'. Moreover they are compatible with passage to the inductive limit in F or in F', since the tensor product commutes with inductive limits and it is clear that

$$\mathrm{Fr}^*_{(\varinjlim F_i)/X} = \varinjlim \mathrm{Fr}^*_{F_i/X};$$

this permits reduction to the case where $F = \Lambda_X$ and $F' = \Lambda_{X'}$, in which case it is evident.

Corollary. With the data of proposition 3, the endomorphisms of $\mathrm{R}\Gamma_{X\times X'}(F \boxtimes F')$ defined respectively by

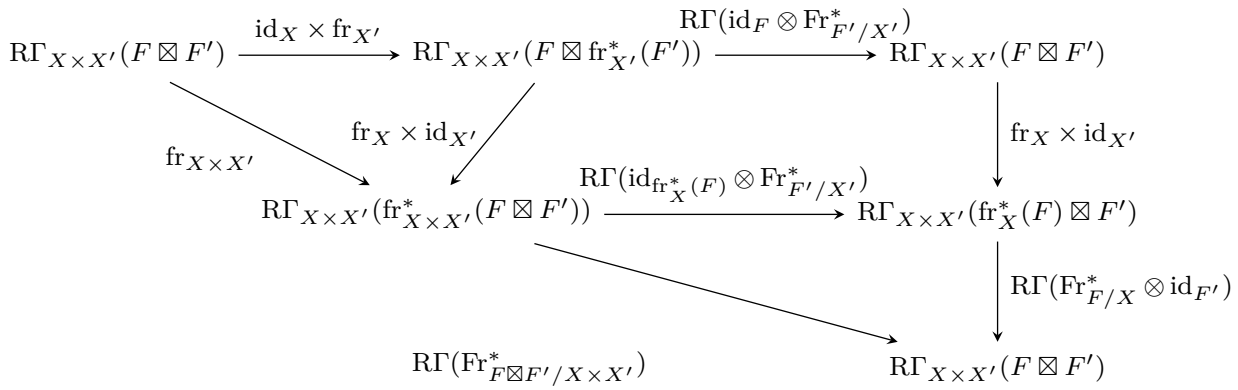
$$(\mathrm{id}_X \times \mathrm{fr}_{X'},\ \mathrm{id}_F \otimes \mathrm{Fr}^*_{F'/X'})$$

and by

$$(\mathrm{fr}_X \times \mathrm{id}_{X'},\ \mathrm{Fr}^*_{F/X} \otimes \mathrm{id}_{F'})$$

are inverse isomorphisms of each other.

Denoting these endomorphisms by ψ' and ψ, let us show for example that $\psi \circ \psi' = \mathrm{id}$. This follows from the commutative diagram



which is commutative because

$$\mathrm{fr}_{X\times X'} = (\mathrm{fr}_X \times \mathrm{id}_{X'}) \circ (\mathrm{id}_X \times \mathrm{fr}_{X'})$$

for the upper triangle and because

$$\mathrm{Fr}^*_{F\boxtimes F'/X\times X'} = (\mathrm{Fr}^*_{F/X} \otimes \mathrm{id}_{F'}) \circ (\mathrm{id}_{\mathrm{fr}^*_X F} \otimes \mathrm{Fr}^*_{F'/X'})$$

for the lower triangle, and in which the composite of the two oblique arrows is the identity by proposition 2c). With an analogous diagram exchanging the roles of (X, F) and (X', F'), one obtains $\psi' \circ \psi = \mathrm{id}$.

We shall apply this result to the particular case where $X' = \bar{e} = \mathrm{Spec}(\overline{\mathbb{F}}_p)$ is the spectrum of an algebraic closure $\overline{\mathbb{F}}_p$ of $\mathbb{F}_p$ and where $F' = \Lambda_{\bar{e}}$ is the constant sheaf with fibre Λ. In this case put

$$\overline{X} = X \times_e \bar{e} = X \otimes_{\mathbb{F}_p} \overline{\mathbb{F}}_p, \qquad \overline{F} = F \otimes_\Lambda \Lambda_{\bar{e}},$$

the inverse image of F by the projection $\overline{X} \to X$. The Frobenius endomorphism $\mathrm{fr}_{\bar{e}}$ is identified with the canonical generator

$$f : \lambda \longmapsto \lambda^p \qquad (\lambda \in \overline{\mathbb{F}}_p)$$

of the Galois group

$$\pi_e = \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p),$$

where f determines an isomorphism $\widehat{\mathbb{Z}} \simeq \pi_e$, and

$$\mathrm{Fr}^*_{\Lambda_{\bar{e}}/\bar{e}} = \mathrm{id}_{\Lambda_{\bar{e}}}$$

since $\Lambda_{\bar{e}}$ is a constant sheaf. Put

$$\overline{\mathrm{fr}}_X = \mathrm{fr}_X \times \mathrm{id}_{\bar{e}},$$

the inverse image of fr_X by $\overline{X} \to X$,

$$f_X = \mathrm{id}_X \times \mathrm{fr}_{\bar{e}},$$

the automorphism of $\overline{X}$ defined by f by transport of structure, and

$$\overline{\mathrm{Fr}}^*_{F/X} = \mathrm{Fr}^*_{F/X} \otimes \mathrm{id}_{\Lambda_{\bar{e}}},$$

the inverse image of $\mathrm{Fr}^*_{F/X}$ by $\overline{X} \to X$. We may write

$$\mathrm{fr}_{\overline{X}} = \overline{\mathrm{fr}}_X \circ f_X = f_X \circ \overline{\mathrm{fr}}_X, \qquad \mathrm{Fr}^*_{\overline{F}/\overline{X}} = \overline{\mathrm{Fr}}^*_{F/X}.$$

The corollary of Proposition 3 shows that the endomorphism of $\mathrm{R}\Gamma_{\overline{X}}(\overline{F})$ defined by the pair $(\overline{\mathrm{fr}}_X, \overline{\mathrm{Fr}}^*_{F/X})$ is the inverse of the automorphism defined by f_X. We shall say that $(\overline{\mathrm{fr}}_X, \overline{\mathrm{Fr}}^*_{F/X})$ is the geometric Frobenius correspondence; it is the inverse image by $\overline{X} \to X$ of the Frobenius correspondence on (X, F). Denote by

$$\mathrm{fr}_{\mathrm{R}\Gamma_{\overline{X}}(\overline{F})}$$

the automorphism of $\mathrm{R}\Gamma_{\overline{X}}(\overline{F})$ which it defines. The arithmetic Frobenius operation, coming by transport of structure from $f \in \pi_e$, is its inverse.

§ 3. The L-function

1. Definition

Let p be a prime number; consider a prime number $\ell \neq p$ and a finite extension Ω of $\mathbb{Q}_\ell$. To every pair (X, F) consisting of a scheme X of finite type over $e = \mathrm{Spec}(\mathbb{F}_p)$ and a constructible Ω-sheaf F on X, we shall associate a formal series

$$L_F \in \Omega[[t]],$$

with constant term 1.

First consider the special case where $X = e$; let $\bar{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$ and let $\bar{e} = \mathrm{Spec}(\bar{\mathbb{F}}_p)$. The sheaf F is entirely determined by its fibre $F_{\bar{e}}$, regarded as a finite-dimensional Ω-vector space on which the Galois group

$$\pi_e = \mathrm{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_p)$$

acts continuously. Since π_e is topologically generated by the Frobenius element

$$f : \lambda \longmapsto \lambda^p \qquad (\lambda \in \bar{\mathbb{F}}_p),$$

the data of the sheaf F is equivalent to that of the vector space $F_{\bar{e}}$ endowed with an automorphism $f_{F_{\bar{e}}}$ which is "topologically unipotent", that is, whose k-th power tends to the identity in the analytic group $\mathrm{Aut}(F_{\bar{e}})$ as k tends "multiplicatively" to infinity. We set

$$L_F(t) = 1/\det(1 - f_{F_{\bar{e}}}^{-1} t) \in \Omega[[t]].$$

Now suppose that $X = \mathrm{Spec}(\mathbb{F}_q)$, with $q = p^d$ (d an integer ≥ 1). Let $\bar{x}$ be a geometric point of X, defined by an embedding of $\mathbb{F}_q$ into $\bar{\mathbb{F}}_p$; the Galois group

$$\pi_x = \mathrm{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_q)$$

is identified with the subgroup of π_e topologically generated by f^d. Thus the data of the sheaf F is equivalent to that of its fibre $F_{\bar{x}}$, a finite-dimensional vector space over Ω, endowed with a topologically unipotent automorphism $(f^d)_{F_{\bar{x}}}$, which we write as $f^d_{F_{\bar{x}}}$ for simplicity. Let g be the unique morphism from X to e; the direct image $g_*(F)$ is likewise defined by its fibre $(g_*F)_{\bar{e}}$ with the operations of π_e, which is computed as an "induced module"

$$(g_*F)_{\bar{e}} \simeq F_{\bar{x}} \otimes_{\pi_x} \pi_e.$$

We set

$$L_F(t) = L_{g_*F}(t) = 1/\det(1 - f^{-1}_{(g_*F)_{\bar{e}}} t);$$

an easy determinant calculation shows that this expression is equal to

$$L_F(t) = 1/\det(1 - f^{-d}_{F_{\bar{x}}} t^d).$$

In the general case of a scheme X of finite type over e, endowed with a constructible Ω-sheaf F, define

$$L_F = \prod_{x \in X^0} L_{F_x},$$

where X^0 denotes the set of closed points of X; we shall see that this product is multipliable in $\Omega[[t]]$. Indeed, let us make it explicit by introducing, for each $x \in X^0$, a geometric point $\bar{x}$ of X localized at x, defined by an embedding of the residue field $k(x)$ into $\bar{\mathbb{F}}_p$, and by setting

$$d(x) = [k(x) : \mathbb{F}_p]$$

(the degree of x). With this notation

$$L_{F_x}(t) = 1/\det(1 - f_{F_{\bar{x}}}^{-d(x)} t^{d(x)}) = 1 + \mathrm{Tr}(f_{F_{\bar{x}}}^{-d(x)}) t^{d(x)} + \cdots .$$

Since for every integer n the set of points $x \in X^0$ such that $d(x) \leq n$ is finite, it follows that the family $(L_{F_x})_{x \in X^0}$ is multipliable in $\Omega[[t]]$. Thus the L-function of (X, F) is

$$L_F(t) = \prod_{x \in X^0} 1/\det(1 - f_{F_{\bar{x}}}^{-d(x)} t^{d(x)}).$$

The first formal properties of the L-function are collected in the following statement.

Proposition 1.

a) Multiplicativity in F. Let

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence of constructible Ω-sheaves on X; then

$$L_F = L_{F'} L_{F''}.$$

b) Let Y be a closed subscheme of X and let $U = X - Y$ be its open complement. Then

$$L_F = L_{F|U}\, L_{F|Y}$$

for every constructible Ω-sheaf F.

c) Let $X \to S$ be a morphism of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω-sheaf on X; then

$$L_F = \prod_{s \in S^0} L_{F|X_s}.$$

Assertion a) is clear; b) follows from $X^0 = U^0 \cup Y^0$ as a disjoint union, and c) follows from

$$X^0 = \coprod_{s \in S^0} X_s^0$$

using associativity of multipliable infinite products. These formulas on closed points are valid because the image of a closed point under a morphism of preschemes of finite type over $\mathbb{F}_p$ is again a closed point.

2. Statement of the rationality theorem and reduction to the case of a curve

Theorem. Let X be a scheme of finite type over $\mathbb{F}_p$, of dimension n, let $g : X \to e$ be its structural morphism, and let F be a constructible Ω-sheaf on X. Then

$$L_F = \prod_{i=0}^{2n} (L_{R^i_! g(F)})^{(-1)^i}. \tag{1}$$

In this statement the $R^i_! g(F)$ are the higher direct images with proper support, in the sense of ℓ-adic cohomology. We put

$$H^i_!(\bar{X}, \bar{F}) = R^i_! \bar{g}(\bar{F}) = R^i_! g(F)_{\bar{e}}$$

(by the base-change theorem), where $\bar{X}$, $\bar{F}$, and $\bar{g}$ are obtained from X, F, and g by the base change $\bar{e} \to e$, with $\bar{e}$ the spectrum of an algebraic closure of $\mathbb{F}_p$. With this notation, by no. 1,

$$L_{R^i_! g(F)}(t) = 1/\det(1 - f^{-1}_{H^i_!(\bar{X},\bar{F})} t) = L_{f^{-1}_{H^i_!(\bar{X},\bar{F})}}(t),$$

where $L_u(t) = 1/\det(1-ut)$ for every endomorphism u of a finite-dimensional vector space, according to the formalism developed in an earlier exposé.

More generally, consider a morphism $u : X \to Y$ of schemes of finite type over $\mathbb{F}_p$ and a constructible Ω-sheaf F on X. Let $(P_{F,u})$ denote the following property:

$$(P_{F,u}): \qquad L_F = \prod_i (L_{R^i_! u(F)})^{(-1)^i}.$$

The theorem for the pair (X, F) is precisely $(P_{F,g})$, where $g : X \to e$ is the structural morphism of X. We shall reduce it to the case where X is the projective line by means of a few lemmas.

Lemma 1. Let $u : X \to Y$ be a morphism of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω-sheaf. If $(P_{F|X_y,u_y})$ is true for every $y \in Y^0$, then $(P_{F,u})$ is true.

Indeed,

$$L_F = \prod_{y \in Y^0} L_{F|X_y}$$

(no. 1, Proposition 1c), and hence

$$= \prod_{y \in Y^0} \prod_i (L_{R^i_! u_y(F|X_y)})^{(-1)^i}$$

using $(P_{F|X_y,u_y})$; but

$$R^i_! u_y(F|X_y) \simeq R^i_! u(F)_y$$

by the base-change theorem, which gives

$$L_F = \prod_i \left(\prod_{y \in Y^0} L_{(R^i_! u(F))_y} \right)^{(-1)^i} = \prod_i (L_{R^i_! u(F)})^{(-1)^i}$$

by the definition of the L-function on Y.

Corollary. Let $u : X \to Y$ be a finite morphism of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω-sheaf on X. The property $(P_{F,u})$ is true.

The lemma allows us to reduce to the case where Y is punctual: $Y = \mathrm{Spec}(\mathbb{F}_q)$. Then X is finite and

$$u_*(F) \simeq \bigoplus_{x \in X} u_*(F_x).$$

To establish the formula $(P_{F,u})$, namely $L_F = L_{u_*F}$, one is therefore reduced to the case where X itself is punctual: $X = \mathrm{Spec}(\mathbb{F}_{q'})$. Let $g : X \to e$ and $h : Y \to e$ be the structural morphisms. By definition (no. 1),

$$L_F = L_{g_*F} = L_{h_*u_*F} = L_{u_*F},$$

since $g = h \circ u$.

Lemma 2. Let $u : X \to Y$ and $v : Y \to Z$ be morphisms of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω-sheaf on X. If the properties $(P_{F,u})$ and $(P_{R^i_!u(F),v})$ for all i are true, then $(P_{F,v\circ u})$ is true.

Indeed,

$$L_F = \prod_i (L_{R^i_!u(F)})^{(-1)^i} = \prod_i \prod_j (L_{R^j_!v(R^i_!u(F))})^{(-1)^i(-1)^j}$$

$$= \prod_{i,j} (L_{R^j_!v(R^i_!u(F))})^{(-1)^{i+j}}$$

by $(P_{F,u})$ and $(P_{R^i_!u(F),v})$. The Leray spectral sequence

$$R^j_!v(R^i_!u(F)) \Longrightarrow R^k_!(v \circ u)(F)$$

and the multiplicativity of L_F with respect to F (no. 1, Proposition 1a) finally give

$$L_F = \prod_k (L_{R^k_!(v\circ u)(F)})^{(-1)^k},$$

as required.

Lemma 3. Let X be a scheme of finite type over $\mathbb{F}_p$, and let $g : X \to e$ be its structural morphism.

a) Let F be a constructible Ω-sheaf on X, let Y be a closed subscheme of X, and let $U = X - Y$ be the open complement. If the theorem is true for two of the three pairs (X, F), $(Y, F|Y)$, $(U, F|U)$, then it is true for the third.

b) Let

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence of constructible Ω-sheaves on X. If the theorem is true for two of the three pairs (X, F'), (X, F), (X, F''), then it is true for the third.

We shall use only a), but we also give b) for symmetry. The proofs of a) and b) are analogous; let us prove a), for example. The exact sequence

$$\cdots \longrightarrow R^i_!g_U(F|U) \longrightarrow R^i_!g(F) \longrightarrow R^i_!g_Y(F|Y) \longrightarrow R^{i+1}_!g_U(F|U) \longrightarrow \cdots$$

and Proposition 1a) (no. 1) give

$$\prod_i (L_{R^i_! g(F)})^{(-1)^i} = \prod_i (L_{R^i_! g_U(F|U)}\, L_{R^i_! g_Y(F|Y)})^{(-1)^i},$$

and Proposition 1b) (no. 1) then allows one to conclude.

The arguments of Lemmas 1, 2, and 3 may be summarized by saying that, by Proposition 1 of no. 1, the function L_F has formal properties analogous to those of cohomology with proper support.

We now use the preceding lemmas to reduce to the case where X is the projective line.

First reduction: by noetherian induction one may construct a partition $(U_j)_{1\leq j\leq r}$ of X by affine schemes such that U_{j+1} is open in $X - \bigcup_{i=1}^{j} U_i$ for $j = 1, \ldots, r-1$. Suppose that the theorem has been established for every pair (U, G), where U is an affine scheme of finite type over $\mathbb{F}_p$ and G is a constructible Ω-sheaf on U. It is then true for $(U_j, F|U_j)$, $j = 1, \ldots, r$; by Lemma 3a), descending induction on j shows that it is true for $(\bigcup_{i\geq j} U_i, F|\bigcup_{i\geq j} U_i)$, and hence for (X, F), since $X = \bigcup_{i\geq 1} U_i$. It is therefore enough to prove the theorem in the case where X is affine.

Second reduction: suppose X affine of dimension n. There exists a finite morphism $u : X \to \mathbb{E}^n$ from X into numerical space of dimension n. We reason by induction on n. For $n = 0$, X is finite over e and the theorem is true by the corollary to Lemma 1. If $n \geq 1$, suppose the theorem established for affine schemes of dimension $n-1$; let $u : X \to \mathbb{E}^n$ be a finite morphism and let $v : \mathbb{E}^n \to \mathbb{E}^1$ be a projection morphism, defined from an isomorphism $\mathbb{E}^n \simeq \mathbb{E}^1 \times \mathbb{E}^{n-1}$. Composing, one obtains a morphism $w = v \circ u : X \to \mathbb{E}^1$ whose fibres are affine schemes of dimension $n-1$. Lemma 1 then shows that the property $(P_{F,w})$ is true by the induction hypothesis. By Lemma 2, it is enough to establish the theorem for the pairs $(\mathbb{E}^1, R^i_! w(F))$. Thus one is reduced to the case where $X = \mathbb{E}^1$ is the affine line.

Third reduction: suppose $X = \mathbb{E}^1$. Let $i : \mathbb{E}^1 \to \mathbb{P}^1$ be the immersion of the affine line into the projective line $\mathbb{P}^1$, and let $Z = \mathbb{P}^1 - i(\mathbb{E}^1)$ be the reduced closed subscheme at the point at infinity of $\mathbb{P}^1$. Since the theorem is true for $(Z, i_!(F)|Z)$, it is equivalent, by Lemma 3a), to establish it for $(\mathbb{E}^1, F)$ or for $(\mathbb{P}^1, i_!(F))$; since i is an open immersion one has $F = i_!(F)|\mathbb{E}^1$.

3. Proof of the theorem

After these reductions, it remains to establish the theorem in the case where X is the projective line. We shall first transform formula (1) of the statement into a Lefschetz-type formula involving traces; this latter formula will be proved in the particular case where X is a projective smooth curve.

Return to formula (1):

$$L_F = \prod_i (L_{R^i_! g(F)})^{(-1)^i} = \frac{P_F^1 \cdots P_F^{2n-1}}{P_F^0 \cdots P_F^{2n}}, \quad \text{where} \quad P_F^i(t) = \det(1 - f^{-1}_{H^i_!(\bar{X},\bar{F})} t).$$

The logarithm of the left-hand side is

$$\sum_{x\in X^0} \sum_{n=1}^{\infty} \operatorname{Tr}(f_{F_{\bar{x}}}^{-nd(x)})\, t^{nd(x)}/n = \sum_{n=1}^{\infty} \left(\sum_{d(x)|n} d(x) \operatorname{Tr}(f_{F_{\bar{x}}}^{-n}) \right) t^n/n,$$

while that of the right-hand side is

$$\sum_{n\geq 1} \left(\sum_i (-1)^i \operatorname{Tr}(f^{-n}_{H^i_!(\bar{X},\bar{F})}) \right) t^n/n.$$

Comparing the coefficients of t^n/n, one sees that it is necessary to establish

$$\sum_{d(x)|n} d(x)\,\mathrm{Tr}(f_{F_{\bar{x}}}^{-n}) = \sum_i (-1)^i\,\mathrm{Tr}(f_{H_!^i(\bar{X},\bar{F})}^{-n}) \tag{2}$$

for all $n \geq 1$; these calculations are legitimate because the field Ω has characteristic zero.

Transform formula (2) further by supposing that X is proper, which allows us to write ordinary cohomology instead of cohomology with proper support. Recall that the category of $\mathbb{Q}_\ell$-sheaves on X is a category of fractions of the category of ℓ-adic sheaves. By abuse of notation, we shall henceforth denote by $F = (F_\nu)_{\nu\in\mathbb{N}}$ the ℓ-adic sheaf[31] on X corresponding to the $\mathbb{Q}_\ell$-sheaf underlying F. Then $\bar{F} = (\bar{F}_\nu)$ and the Frobenius morphisms $\overline{\mathrm{Fr}}^*_{F_\nu/X}$ define a morphism of ℓ-adic sheaves

$$\overline{\mathrm{Fr}}^*_{F/X} : \mathrm{fr}_X^*(\bar{F}) \longrightarrow \bar{F},$$

hence an ℓ-adic correspondence on $(\bar{X}, \bar{F})$. For every integer ν, the endomorphism $\mathrm{fr}_{R\Gamma_{\bar{X}}(\bar{F}_\nu)}$ defined by $(\bar{\mathrm{fr}}_X, \overline{\mathrm{Fr}}^*_{F_\nu/X})$ is the inverse of $f_{R\Gamma_{\bar{X}}(\bar{F})}$ (§2, no. 3, Corollary to Proposition 3). Using the definition of ℓ-adic cohomology, it follows that the endomorphism $\mathrm{fr}_{H^i(\bar{X},\bar{F})}$ of $H^i(\bar{X},\bar{F})$ defined by $(\bar{\mathrm{fr}}_X, \overline{\mathrm{Fr}}^*_{F/X})$ is the inverse of $f_{H^i(\bar{X},\bar{F})}$; thus

$$f_{H^i(\bar{X},\bar{F})}^{-n} = \mathrm{fr}_{H^i(\bar{X},\bar{F})}^n$$

for every $n \geq 1$ and every i.

Apply this result in the particular case where $X = x = \mathrm{Spec}(\mathbb{F}_q)$ $(q = p^d)$ is reduced to a point whose degree d divides n. We obtain

$$f_{F_{\bar{x}}}^{-n} = \mathrm{fr}_{F_{\bar{x}}}^n = (\mathrm{Fr}^*_{F_{\bar{x}}/\bar{x}})^n$$

($\bar{x}$ is the geometric point defined by an embedding of $\mathbb{F}_q$ into $\bar{\mathbb{F}}_p$).

On the other hand, $\bar{\mathrm{fr}}_X$ and f_X act in inverse ways on the points of $\bar{X}$. It follows that the projection $\bar{X} \to X$ establishes a bijective correspondence between the set of closed points of degree d of X and the set of d-element orbits of $\bar{\mathrm{fr}}_X$ acting on $\bar{X}$. Noting that the set $\bar{X}^{\bar{\mathrm{fr}}_X^n}$ of points of $\bar{X}$ fixed by $\bar{\mathrm{fr}}_X^n$ is the union of the orbits of $\bar{\mathrm{fr}}_X$ whose cardinality divides n, one can rewrite (2) in the "geometric" form

$$\sum_{\bar{x}\in\bar{X}^{\bar{\mathrm{fr}}_X^n}} \mathrm{Tr}(\mathrm{fr}_{F_{\bar{x}}}^n) = \sum_i (-1)^i\,\mathrm{Tr}(\mathrm{fr}_{H^i(\bar{X},\bar{F})}^n). \tag{2'}$$

This is a Lefschetz-type formula on the scheme $\bar{X}$ endowed with the sheaf $\bar{F}$ and the correspondence $(\bar{\mathrm{fr}}_X, \overline{\mathrm{Fr}}^*_{F/X})$.

To lighten notation, we shall henceforth write X, F, h, and h_F instead of $\bar{X}$, $\bar{F}$, $\bar{\mathrm{fr}}_X^n$, and $(\overline{\mathrm{Fr}}^*_{F/X})^n$; since X and F will no longer appear in their former sense, no confusion can arise. We shall suppose that X is a smooth projective curve over $k = \bar{\mathbb{F}}_p$. The fixed points of h on X all have multiplicity 1: indeed, h acts by $T \mapsto T^{p^n}$ on the completed local ring of a fixed point, where T is a uniformizer. The theorem will follow from the following result.

Proposition 2. Let X be a smooth projective curve over an algebraically closed field k of characteristic p, let F be a constructible ℓ-adic sheaf on X_{et} (with ℓ a prime number distinct from p),

[31] To simplify the notation, for the remainder of the proof we shall suppose that $\Omega = \mathbb{Q}_\ell$. In the general case, one would have to replace $\mathbb{Z}_\ell$ by the normal closure of $\mathbb{Z}_\ell$ in Ω. (Details left to the reader.)

let h be an endomorphism of X whose fixed points all have multiplicity 1, and let $h_F : h^*(F) \to F$ be a morphism of sheaves. The following formula, where $h_{H^i(X,F)}$ denotes the endomorphism of $H^i(X, F)$ defined by (h, h_F), is true:

$$\sum_{x \in X^h} \mathrm{Tr}((h_F)_x) = \sum_i (-1)^i \,\mathrm{Tr}(h_{H^i(X,F)}). \tag{2"}$$

In this seminar, an analogous Lefschetz formula has been established for a constructible torsion sheaf F_ν, annihilated by $\ell^{\nu+1}$ on the curve X, endowed with a morphism $h_{F_\nu} : h^*(F_\nu) \to F_\nu$. It is written

$$\sum_{x \in X^h} \mathrm{Tr}((h_{F_\nu})_x) = \mathrm{Tr}^*\, h_{R\Gamma_X(F_\nu)}, \tag{3}$$

where both sides belong to $\mathbb{Z}/\ell^{\nu+1}\mathbb{Z}$.

Suppose that the ℓ-adic sheaf in the statement is written as $F = (F_\nu)_{\nu \in \mathbb{N}}$, with F annihilated by $\ell^{\nu+1}$. For each ν we may write formula $(3)_\nu$, and one sees that the left-hand sides of these formulas, as ν varies, are the components of an ℓ-adic integer which is precisely the left-hand side of formula (2") to be proved. It remains to establish that the right-hand sides of the formulas $(3)_\nu$ define, as ν varies, an ℓ-adic integer equal to

$$\sum_i (-1)^i \,\mathrm{Tr}(h_{H^i(X,F)}), \quad \text{where} \quad H^i(X, F) = \varprojlim_\nu H^i(R\Gamma_X(F_\nu)).$$

This follows from purely algebraic lemmas that we shall now prove to finish.

Consider a projective system $(K_\nu)_{\nu \in \mathbb{N}}$ of perfect complexes, with $K_\nu \in D_{\mathrm{parf}}(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z})$, such that for each ν the transition morphism $K_{\nu+1} \to K_\nu$ is isomorphic, in the sense of derived categories, to the canonical morphism

$$K_{\nu+1} \to K_{\nu+1}/\ell^{\nu+1} K_{\nu+1}.$$

Suppose each complex K_ν is endowed with an endomorphism $h_\nu \in \mathrm{Fl}(D_{\mathrm{parf}}(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z}))$, so that $(h_\nu)_{\nu \in \mathbb{N}}$ is compatible with the transition morphisms, thereby defining an endomorphism of the projective system (K_ν). In the application to the proof of Proposition 2 one takes

$$K_\nu = R\Gamma_X(F_\nu), \qquad h_\nu = h_{R\Gamma_X(F_\nu)};$$

the condition on transition morphisms follows by the "Künneth formula" from the analogous condition verified by the transition morphisms of the ℓ-adic projective system (F_ν).

We propose to find a projective system $(K'_\nu)_{\nu \in \mathbb{N}}$, where K'_ν is a complex of free $(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z})$-modules of finite type, zero outside an interval of degrees $[a, b]$ independent of ν, and endowed with an endomorphism of complexes $h'_\nu : K'_\nu \to K'_\nu$, such that the following conditions are satisfied:

1) the transition morphism $K'_{\nu+1} \to K'_\nu$ is isomorphic, as a morphism of complexes, to

$$K'_{\nu+1} \to K'_{\nu+1}/\ell^{\nu+1} K'_{\nu+1}$$

for every $\nu \in \mathbb{N}$;

2) $(h'_\nu)_{\nu \in \mathbb{N}}$ is compatible with the transition morphisms;

3) (K'_ν, h'_ν) is isomorphic to (K_ν, h_ν) in $D_{\mathrm{parf}}(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z})$ for every ν, these isomorphisms being choosable so as to be compatible with the transition morphisms of (K'_ν) and (K_ν).

Assuming the construction of the systems (K'_ν) and (h'_ν), we may write

$$\operatorname{Tr}^*(h_\nu) = \operatorname{Tr}^*(h'_\nu) = \sum_i (-1)^i \operatorname{Tr}(h'^i_\nu)$$

for every $\nu \in \mathbb{N}$. Let

$$K' = \varprojlim_\nu K'_\nu, \qquad h' = \varprojlim_\nu h'_\nu$$

be the induced endomorphism of K'. Then $(\operatorname{Tr}^*(h_\nu))_\nu$ is an ℓ-adic integer equal to $\sum_i (-1)^i \operatorname{Tr}(h'^i)$. Moreover, the projective system (K'_ν) satisfies the Mittag-Leffler condition, and so does the system $(H^{i-1}(K'_\nu))_\nu$ for every i; it follows that

$$H^i(K') \simeq \varprojlim_\nu H^i(K'_\nu)$$

for every i (EGA 0_{III} 13.2.3). In the case of interest to us, where $K_\nu = R\Gamma_X(F_\nu)$ and $h_\nu = h_{R\Gamma_X(F_\nu)}$, $H^i(K')$ is identified with $H^i(X,F)$, and the endomorphism of $H^i(K')$ defined by h' is identified with $h_{H^i(X,F)}$. Consequently

$$\sum_i (-1)^i \operatorname{Tr}(h'^i) = \sum_i (-1)^i \operatorname{Tr}(h_{H^i(X,F)}),$$

and formula (2") in the statement is thereby proved.

The existence of the systems (K'_ν) and (h'_ν) rests on two lemmas, in which we consider a commutative ring A, an ideal I of A, and the functor

$$T : L \longmapsto L \otimes_A A_0$$

where L is a complex of A-modules bounded on the right, with $A_0 = A/I$.

Lemma 1. Suppose that the ideal I is nilpotent. Consider a complex $L \in D^-(A)$ flat in every degree, a complex $M \in D^-(A_0)$ projective in every degree, and an isomorphism

$$u : T(L) \xrightarrow{\sim} M$$

in $D^-(A_0)$. There exist a complex L' of A-modules, bounded on the right and projective in every degree, an isomorphism

$$v : L \xrightarrow{\sim} L'$$

in $D^-(A)$, and an isomorphism of complexes

$$w : T(L') \xrightarrow{\sim} M,$$

such that $u = w \circ T(v)$ in $D^-(A_0)$.

Remark. If M is assumed to be free, respectively of finite type, in every degree, then the complex L' is also free, respectively of finite type, in every degree. Indeed, if a flat A-module N is such that $T(N)$ is free, respectively of finite type, it is easy to show that N is free, respectively of finite type (Bourbaki, Commutative Algebra, Chap. II, §3, Prop. 5). Note the corollary: let $L \in \operatorname{ob} D^-(A)$ be such that $L \overset{\mathrm{L}}{\otimes}_A A_0$ is a pseudo-coherent, respectively perfect, complex of A_0-modules, that is, isomorphic in $D^-(A)$ to a complex bounded above, respectively bounded, and projective of finite type in every degree. Then L is a pseudo-coherent, respectively perfect, complex of A-modules.

Lemma 2. Consider a complex L of A-modules bounded on the right, projective in every degree, endowed with an endomorphism φ, and an endomorphism φ_0 of $M = T(L)$ coinciding with $T(\varphi)$ in the sense of the derived category $D^-(A_0)$. There exists an endomorphism φ' of L which coincides with φ in the sense of $D^-(A)$ and satisfies $T(\varphi') = \varphi_0$.

Before proving these lemmas, let us indicate how they are applied to prove the existence of the systems (K'_ν) and (h'_ν). We argue by induction on ν, supposing that $K'_0, h'_0, \ldots, K'_{\nu-1}, h'_{\nu-1}$ have already been constructed. Apply the lemmas with

$$A = \mathbb{Z}/\ell^{\nu+1}\mathbb{Z}, \qquad I = \ell^\nu\mathbb{Z}/\ell^{\nu+1}\mathbb{Z}$$

so that $I^2 = 0$, and

$$A_0 = \mathbb{Z}/\ell^\nu\mathbb{Z}.$$

First take $L = K_\nu$ and $M = K'_{\nu-1}$ in the first lemma. In view of the remark, this gives the existence of $K'_\nu = L'$ with transition morphism $K'_\nu \to K'_{\nu-1}$ given by w, so that condition 1) above is satisfied. The isomorphism v in $D^-(A)$ allows h_ν to be transported to an endomorphism of K' in the sense of $D^-(A)$. We may suppose that the latter comes from an endomorphism of complexes $h''_\nu : K'_\nu \to K'_\nu$. Applying the second lemma with $L = K'_\nu$, $\varphi = h''_\nu$, and $\varphi_0 = h'_{\nu-1}$, we obtain the existence of $h'_\nu : K'_\nu \to K'_\nu$ satisfying condition 2); condition 3) is also satisfied. Note that, with the notation of Lemma 1, if $M^i = 0$ for an index i, then $L'^i = 0$ follows from Nakayama's lemma; the complexes K'_ν are therefore indeed all zero outside one and the same interval of degrees.

To prove Lemmas 1 and 2 we shall need the following result.

Lemma 3.

(i) Let P and Q be A-modules. If P is projective, then for every A_0-linear map $f_0 : T(P) \to T(Q)$ there exists an A-linear map $f : P \to Q$ such that $T(f) = f_0$.

(ii) Suppose that the ideal I is nilpotent. For every projective A_0-module P_0 there exists a projective A-module P such that $T(P) \simeq P_0$. If moreover P' is a flat A-module such that $T(P') \simeq P_0$, then $P' \simeq P$.

Assertion (i) is clear: if u_P (resp. u_Q) denotes the canonical map from P onto $T(P)$ (resp. from Q onto $T(Q)$), then the composite $f_0 \circ u_P$ is an A-linear map from P to $T(Q)$, and u_Q is surjective. Since P is projective, there is a factorization of $f_0 \circ u_P$ through u_Q, namely $f_0 \circ u_P = u_Q \circ f$, where f is an A-linear map from P to Q; then $T(f) = f_0$.

To prove (ii), first observe that the case where P_0 is free is evident, with no hypothesis on the ideal I. The general case, where P_0 is projective, is reduced to this by introducing a free A_0-module L_0 of which P_0 is a direct factor, and a free A-module L such that $T(L) \simeq L_0$. The module P_0 is identified with the image of a projector $e_0 : L_0 \to L_0$, and it remains to see that there exists a projector $e : L \to L$ lifting e_0. This follows from Bourbaki, Commutative Algebra, Chap. III, §4, no. 6, Lemma 2, applied to the sub-A-algebra B of $\mathrm{End}(L)$ generated by an endomorphism of L lifting e_0 (whose existence is ensured by (i)) and to the nilpotent ideal which is the trace on B of the kernel of the canonical homomorphism $\mathrm{End}(L) \to \mathrm{End}(L_0)$. This proves the existence of P in (ii). If moreover P' is as in (ii), then the composite isomorphism $T(P) \simeq P_0 \simeq T(P')$ comes, by (i), from a homomorphism $f : P \to P'$. Since P and P' are flat over A and $T(f)$ is an isomorphism, it follows that $\mathrm{Gr}_I(f)$ is an isomorphism; hence f is an isomorphism, since I is nilpotent. This proves (ii).

Corollary. Suppose the ideal I is nilpotent. For every acyclic complex N_0 of projective A_0-modules, bounded on the right, there exists an acyclic complex N of projective A-modules, bounded on the right, such that $T(N) \simeq N_0$.

The complex N_0, being acyclic, decomposes into short exact sequences

$$0 \longrightarrow Z_0^i \longrightarrow N_0^i \longrightarrow Z_0^{i+1} \longrightarrow 0,$$

where Z_0^i denotes, for each i, the submodule of cycles of N_0^i; moreover $N_0^i = 0$ for all sufficiently large i. By descending induction on i, one sees that the preceding exact sequences are split and that Z_0^i is projective for all i. One lifts N_0 by lifting all these exact sequences step by step, by descending induction on i. It is therefore enough to show that if

$$S_0 : 0 \longrightarrow P_0 \longrightarrow Q_0 \longrightarrow R_0 \longrightarrow 0$$

is an exact sequence of A_0-modules with Q_0 projective, and if R is an A-module such that $T(R) \simeq R_0$, then there exists an exact sequence

$$S : 0 \longrightarrow P \longrightarrow Q \longrightarrow R \longrightarrow 0$$

of A-modules, with Q projective, such that $T(S) \simeq S_0$. Now Q_0 can be lifted to a projective A-module Q by Lemma 3(ii), and $Q_0 \to R_0$ can be lifted to an A-linear map $Q \to R$ by Lemma 3(i). This map is surjective by Nakayama's lemma; one sets $P = \ker(Q \to R)$ to complete the exact sequence S.

Proof of Lemma 2: since $M = T(L)$ is projective in every degree, the hypothesis of the lemma says that $T(\varphi)$ is homotopic to φ_0. Thus there exists a homotopy operator $k_0 = (k_0^i)_i$, with $k_0^i : M^i \to M^{i-1}$, such that

$$T(\varphi) - \varphi_0 = d_0 k_0 + k_0 d_0,$$

where d_0 is the differential of M. By Lemma 3(i), k_0^i lifts to a linear map $k^i : L^i \to L^{i-1}$. This gives a homotopy operator $k = (k^i)_i$ in L. It is enough to set

$$\varphi' = \varphi - kd - dk$$

where d is the differential of L, to verify Lemma 2.

Proof of Lemma 1: after replacing L by a quasi-isomorphic complex if necessary, we may suppose that the complex L, and hence also $T(L)$, is projective in every degree; we shall assume this henceforth. Thus we may suppose that the isomorphism u of $D^-(A_0)$ comes from a quasi-isomorphism $T(L) \to M$, again denoted by u. We shall first reduce to the case where u is surjective. The complex N_0 defined by

$$N_0^i = M^i \oplus M^{i-1}, \quad d^i(x, y) = (d^i x, x - d^{i-1} y)$$

($x \in M^i$, $y \in M^{i-1}$) is visibly acyclic and projective in every degree; an epimorphism of complexes $N_0 \to M$ is defined by the projection $N_0^i \to M^i$ in degree i. The corollary to Lemma 3 allows us to lift N_0 to an acyclic complex N, bounded on the right and projective in every degree. We may then replace L by $L \oplus N$, which is quasi-isomorphic to it, since N is acyclic, and projective in every degree. At the same time, replace u by the quasi-isomorphism

$$T(L \oplus N) = T(L) \oplus N_0 \longrightarrow M$$

defined by u on $T(L)$ and by the epimorphism $N_0 \to M$ on N_0; it is clear that this last quasi-isomorphism is surjective.

Suppose therefore that $u : T(L) \to M$ is a surjective quasi-isomorphism, and let R_0 be its kernel. The complex R_0 is acyclic because u is a quasi-isomorphism, and it is projective in every degree because the exact sequence

$$0 \longrightarrow R_0 \longrightarrow T(L) \longrightarrow M \longrightarrow 0$$

is split, since M is projective in every degree. Hence, by the corollary to Lemma 3, R_0 can be lifted to a complex R of A-modules, bounded on the right, acyclic, and projective in every degree. We now show that the morphism $f_0 : R_0 \to T(L)$ lifts to a morphism of complexes $f : R \to L$. For this, observe that R_0 is homotopically trivial, so that f_0 is homotopic to 0 and can be written

$$f_0 = h_0 d_{R_0} + d_{T(L)} h_0,$$

where $h_0 = (h_0^i)_i$, with $h_0^i : R_0^i \to T(L)^{i-1}$. Lemma 3(i) allows h_0 to be lifted to $h = (h^i)_i$, with $h^i : R^i \to L^{i-1}$, and one sees that

$$f = h d_R + d_L h$$

is a morphism of complexes lifting f_0. A standard lemma (compare SGA 60/61, IV 5.7) then implies that f is injective and that $\operatorname{coker} f = L'$ is flat in every degree. Since

$$T(L') \simeq \operatorname{coker} T(f) = \operatorname{coker} f_0 \simeq M$$

is projective in every degree, it follows, by Lemma 3(ii), that the same is true of L'. On the other hand, the canonical homomorphism $L \to L'$ is a quasi-isomorphism, being surjective with acyclic kernel R; thus L' satisfies the desired conclusion of Lemma 1, q.e.d.

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Seminar on Algebraic Geometry of Bois Marie

1966–67

Intersection Theory and the Riemann–Roch Theorem

(SGA 6)

a seminar directed by

P. Berthelot, A. Grothendieck, L. Illusie

with the collaboration of

D. Ferrand, J.-P. Jouanolou, O. Jussila, S. Kleiman, M. Raynaud, J.-P. Serre

Preface

The present edition of *Séminaire de Géométrie Algébrique 6* reproduces, up to minor changes, the first edition, distributed by the I.H.E.S. in the form of mimeographed exposés. The corrections made concern essentially printing errors, except in Exposés I and III, where slight modifications were made by L. Illusie to a few statements which were incorrect in the primitive version. Footnotes have also been added, especially in Exposé XIV by A. Grothendieck, in order to point out recent progress on questions that were still open at the time of the seminar.

Finally, note that Exposé XI, by D. Ferrand, on determinant theory for perfect complexes, was not written up and therefore will not be published; to avoid errors in references, we have left unchanged the numbering of the following exposés.

June 1971

Introduction

The purpose of the present seminar is to develop a global theory of intersections, and a Riemann–Roch formula, for arbitrary schemes. The reader will find a description of the programme of the seminar in Exposé 0. The possibility in principle of proving a Riemann–Roch formula for general regular schemes, along the lines of the well-known Borel–Serre report of 1958, was already clear at that time, at least for a projective morphism. The extension to the general case is less evident; the programme in which it belongs, and which is partly carried out in this seminar, goes back to 1960. As was also the case for the duality theory of coherent sheaves, an essential tool for formulating a satisfactory theory is Verdier's theory of derived categories; familiarity with it is indispensable for understanding the seminar.

Apart from this theory, the study of the seminar requires little more than a general knowledge of the foundations of algebraic geometry as set out in EGA, chapters I, II, III. We shall only occasionally refer to EGA IV, for certain facts concerning flat or smooth morphisms, which for the most part also appear in the first exposés of SGA 1. Finally, to develop some results with the full generality desirable for applications, we sometimes use the notion of a ringed site and of a ringed topos, for which we refer to SGA 4, Exposés I to IV. A reader unfamiliar with the language of sites and topoi may everywhere replace these objects by ordinary topological spaces, the objects of the topos then being replaced by open subsets of these spaces; nevertheless,

we recommend that the reader assimilate the language of topoi, which provides an extremely convenient unifying principle.

The fundamental notion for the theory presented here is that of a perfect complex of modules, together with its various variants, developed in Exposés I, II, III. It seems clear that these notions, also important in other contexts in algebraic geometry, notably for formulas of Lefschetz–Verdier type in cohomology with discrete coefficients (SGA 5) or coherent coefficients, will also have a role to play outside algebraic geometry, especially in the formulation of a complex analytic variant of the Riemann–Roch theorem presented here, or of suitable variants of the Atiyah–Singer index theorem. Some indications in this direction will be given in Exposé II. Unfortunately, at present there is still no statement, even heuristic, which would encompass these two theorems, the first of which remains conjectural for the moment. A new idea seems to be missing, just as one is also missing in algebraic geometry for proving Riemann–Roch beyond projective hypotheses (cf. Exposé XIV, no. 2).

We shall make no use in this seminar of the local theory of intersections, set out in J.-P. Serre, *Algèbre locale. Multiplicités* (Lecture Notes in Mathematics no. 11, Springer), and merely point out here that Serre's course had an evident influence on the genesis of the theory in 1957. Nor shall we use the theory of the Chow ring developed in the Chevalley Seminar of 1958, Exposés 2 and 3 (École Normale Supérieure). That theory is essentially tied to nonsingularity hypotheses, whereas the aim of our seminar is, on the contrary, to develop a theory of intersections on general schemes, and even on general locally ringed topoi. On this matter see the comments in Exposé XIV, nos. 4 and 8, which give the relations between our theory and that of the Chow ring and raise the question of a generalization of the latter. Thus one may say that the seminar presents a coherent and self-contained theory of intersections, but one which is by no means exhaustive of the different points of view useful, indeed indispensable, in intersection theory; consequently it should be complemented by the cited exposés of Serre and Chevalley.

We have appended to the seminar, as an appendix to Exposé 0, Grothendieck's 1957 report, "Classes of sheaves and the Riemann–Roch theorem", which served as the basis for the Borel–Serre seminar at Princeton in the same year and for their already cited article. That report sketches two proofs of the Riemann–Roch theorem for nonsingular quasi-projective varieties. The first, valid for the time being only in characteristic zero but giving, in return, a slightly more precise result in the case of an immersion, has not yet appeared in the literature, apart from the work of Borel–Serre and the present seminar. The report presupposes, of course, none of the other exposés of the seminar, and can even serve, like Exposé 0, as an introduction to the study of the latter, especially for a reader not yet familiar with Borel–Serre. The proof just alluded to applies essentially unchanged to the case of a projective morphism of complex analytic spaces, and will doubtless also apply in arbitrary characteristic once the problem of the operations "exterior powers" in the derived category is solved (cf. Exposé XIV, nos. 1, 2, 3). The absence of a study of this question is no doubt the most serious gap in this seminar, from the point of view we have adopted.

One will notice the absence, in the table of contents of the present seminar, of two of the participants mentioned on the cover page. One of them, J.-P. Jouanolou, took an active part in the technical elaboration of the first part of the seminar, but was prevented from taking part in the oral exposés; in particular, he is responsible for the linear-independence assertion contained in the important statement VI 1.1, giving the structure of the ring K° of classes of locally free modules on a projective bundle, which had previously been proved only under the assumption that the base scheme has an ample invertible module. The other, J.-P. Serre, gave two oral exposés which were logically independent of the rest of the seminar, and therefore preferred to publish them in the form of a separate article (cf. J.-P. Serre, "Grothendieck groups of split reductive group schemes", to appear in *Publications Mathématiques*, no. 34).

As in the other parts of the *Séminaire de Géométrie Algébrique du Bois Marie*, the sigla SGA 1 to SGA 6 refer to the different parts of that seminar; the Roman numeral following the siglum indicates the number of the exposé, and the Arabic numerals following it correspond to the internal numbering of the exposé in question. For internal references within the present seminar the siglum SGA 6 is omitted, and references of the form I 4.9 are used, meaning: Exposé I, statement 4.9. The siglum EGA refers to the *Éléments de Géométrie Algébrique* of J. Dieudonné and A. Grothendieck.

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Exposé 0

Outline of a programme for a theory of intersections on general schemes

by A. Grothendieck

The present exposé is introductory in nature, and its reading is not logically indispensable for the study of the seminar. It is addressed more particularly to readers familiar with the provisional version of the Riemann–Roch theorem, as it is presented in the Borel–Serre report or in Grothendieck's report already mentioned in the introduction, cited below as [RRR], and reproduced as an appendix at the end of the present exposé.

1. Reminder on the Riemann–Roch formula

Recall the Riemann–Roch formula for a proper morphism

$$f : X \longrightarrow Y$$

of quasi-projective smooth schemes over a field k, and a coherent sheaf F on X, whose class in the group $K(X)$ of classes of coherent sheaves on X is denoted by $\mathrm{cl}(F)$:

$$\mathrm{Todd}(T_Y)\ \mathrm{ch}_Y\big(f_!(\mathrm{cl}(F))\big) = f_*\big(\mathrm{Todd}(T_X)\ \mathrm{ch}_X(F)\big). \tag{1.1}$$

This formula is valid in $A(Y)\otimes\mathbb{Q}$, where $A(Y)$ is the Chow ring of Y. The f_* on the right hand side is obtained by tensoring with $\mathbb{Q}$ the homomorphism "direct image of cycles"

$$f_* : A(X) \longrightarrow A(Y),$$

while the term on the left is the Euler–Poincaré characteristic of F relative to f:

$$f_!(\mathrm{cl}(F)) = \sum_i (-1)^i\, \mathrm{cl}\big(R^i f_*(F)\big).$$

The symbols ch_X, ch_Y denote the Chern characters on X and on Y, and T_X, T_Y the tangent bundles of X and Y. As is known, $\mathrm{Todd}(-)$ and $\mathrm{ch}(-)$ are universal polynomials with coefficients in $\mathbb{Q}$ in the Chern classes of the argument. Since the constant term of $\mathrm{Todd}(-)$ is 1, it is an invertible element for every value of the argument. Thus, after multiplying by $\mathrm{Todd}(T_Y)^{-1}$, formula (1.1) takes the equivalent form, more convenient for our purposes,

$$\mathrm{ch}_Y\big(f_!(\mathrm{cl}(F))\big) = f_*\big(\mathrm{Todd}(T_f)\ \mathrm{ch}_X(F)\big), \tag{1.2}$$

where

$$T_f = T_X - f^*T_Y \in K(X). \tag{1.3}$$

Thus T_f plays the role of a virtual relative tangent bundle of X over Y. When f is smooth, one simply has $T_f = T_{X/Y}$, whereas when $f : X \to Y$ is an immersion, one obtains

$$T_f = -\check{N}_{X/Y},$$

where $N_{X/Y}$ denotes the normal sheaf of X in Y.

One of the main aims of the seminar is to generalize (1.2) simultaneously in two directions:

a) to get rid of the hypothesis that a base field k exists;

b) to replace the regularity hypotheses on Y, X by a hypothesis of "local regularity" on f.

Finally, along the way, we shall also deal with the following problem:

c) to eliminate the quasi-projectivity hypotheses which, in the absence of a base field, are expressed by the existence of ample invertible modules on X and Y.

2. Removing the base field

Let us first examine generalization a), while retaining the hypotheses of regularity and of the existence of ample invertible modules on X and on Y.

The definitions of $K(X)$, $K(Y)$ and of the homomorphism $f_! : K(X) \to K(Y)$ then present no new problem, thanks to the fact that X and Y are regular. The most natural way to give

a meaning to (1.2) would seem to consist in defining Chow rings $A(X), A(Y)$ and a group homomorphism

$$f_* : A(X) \longrightarrow A(Y),$$

in establishing a theory of Chern classes, giving maps

$$c_i : K(X) \longrightarrow A(X),$$

and similarly for Y, and finally in describing an element, the virtual relative tangent bundle,

$$T_f \in K(X).$$

2.1

For the latter, a definition presents itself rather evidently. One uses the fact that the morphism f, thanks to the hypothesis that ample invertible modules exist on X and on Y, may be factored as

$$X \xrightarrow{i} X' \xrightarrow{f'} Y,$$

where i is a closed immersion and f' is smooth; for example, one may take $X' = \mathbb{P}^r_Y$ for suitable r. One then sets

$$T_f = i^* T_{X'/Y} - \check{N}_{X/X'} \in K(X), \tag{2.1}$$

where $\Omega^1_{X'/Y}$ is the locally free module of relative 1-differentials of X' over Y, or the relative cotangent module of X' over Y, and $N_{X/X'}$ is the conormal module of X in X', by definition equal to J/J^2, where J is the ideal on X' defining X. The regularity hypotheses on X and X' also imply that $N_{X/X'}$ is locally free. Finally, $\check{N}$ denotes the dual of N. One verifies without difficulty, in Exposé VIII, that the element T_f defined by (2.1) is independent of the chosen factorization of f.

2.2

As for a definition, under the conditions considered, of a Chow ring and a corresponding theory of Chern classes, although it is plausible that such a theory should be developable on the model of smooth varieties over a field, it seems that this work has not been done at the present time. We shall therefore adopt another viewpoint, by directly defining a ring which replaces the Chow ring, and whose tensor product with $\mathbb{Q}$ is indeed isomorphic to $A(X) \otimes \mathbb{Q}$ in the classical case.

A first natural filtration of $K(X)$, the one considered in [RRR], is the topological filtration by the codimension of the support of coherent sheaves: $\mathrm{Fil}^i K(X)$ is generated by the $\mathrm{cl}(F)$ with F coherent and $\mathrm{codim}(\mathrm{Supp}\, F, X) \geq i$. A first question is the compatibility of this filtration with the multiplicative structure of $K(X)$. In [RRR], this compatibility is obtained, for X smooth over a field, by means of Chow's moving lemma. Assuming this assertion, one obtains a graded ring $\mathrm{Gr}_{\mathrm{top}}(X)$ and a canonical surjective homomorphism

$$\varphi : A(X) \longrightarrow \mathrm{Gr}_{\mathrm{top}}(X),$$

sending the class of a prime cycle Z to $\mathrm{cl}(\mathcal{O}_Z)$; it then turns out, Exposé XIV 4.2, that the kernel is a torsion subgroup. Moreover, [RRR] shows how the images in $\mathrm{Gr}_{\mathrm{top}}(X)$ of the Chern classes $c_i(F)$ can be computed directly in terms of $\mathrm{cl}(F) \in K(X)$ and the exterior-power operations λ^i in $K(X)$.

This shows that, for a noetherian regular scheme X endowed with an ample invertible module, the ring $\mathrm{Gr}_{\mathrm{top}}(X)$ is a convenient substitute for the hypothetical Chow ring $A(X)$. It also has

the advantage over the latter that certain computations of [RRR], leading to important formulas in this ring, are performed directly in it–formulas which, even for X projective and smooth over a field, have not at present been established in $A(X)$ itself (Exp. XIV, 4.3 and 6.4). To justify this viewpoint completely, however, one would have to solve the problem mentioned above of the compatibility of the topological filtration of $K(X)$ with its ring structure, a problem which seems essentially connected with Chow's moving lemma, itself the essential technical ingredient of the theory of the Chow ring which we have claimed to be able to avoid.

2.3

To avoid this problem, one endows $K(X)$ with a second filtration, finer than the topological filtration, and describable in purely algebraic terms from the structure of $K(X)$ as an augmented λ-ring. The definition of this filtration is suggested by the expression given in [RRR] for the Chern classes of an element $x \in K(X)$ in $\mathrm{Gr}_{\mathrm{top}}(X)$:

$$c_i(x) = \gamma^i(x - \epsilon(x)) \mod \mathrm{Fil}^{i+1} K(X), \tag{2.3}$$

where γ^i denotes a certain linear combination of the λ^j $(j \leq i)$, made explicit in loc. cit., and where ϵ is the augmentation. One shows that $\gamma^i(x - \epsilon(x))$ is indeed of topological filtration at least i, which gives meaning to formula (2.3). One then defines the λ-filtration of $K(X)$ by the ideals $\mathrm{Fil}^i_\lambda K(X)$, where $\mathrm{Fil}^i_\lambda K(X)$ consists of finite sums of expressions of the form

$$\gamma^{i_1}(x_1 - \epsilon(x_1)) \cdots \gamma^{i_r}(x_r - \epsilon(x_r)), \qquad i_1 + \cdots + i_r \geq i.$$

The associated graded ring for this filtration will be denoted $\mathrm{Gr}^\bullet(X)$; it is this ring which will replace the Chow ring of X in the seminar. In the case under consideration here, one can moreover show that the canonical homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}_{\mathrm{top}}(X) \tag{2.4}$$

is an isomorphism modulo torsion, justifying the viewpoint that, after tensoring with $\mathbb{Q}$, this ring can play exactly the same role as the Chow ring tensored with $\mathbb{Q}$. On the other hand, exactly what was needed has been done so that formula (2.3), with $\mathrm{Fil}_{\mathrm{top}}$ replaced by Fil_λ, defines a theory of Chern classes with values in $\mathrm{Gr}^\bullet(X)$; the Chern classes in $\mathrm{Gr}_{\mathrm{top}}(X)$ are then deduced by means of the homomorphism (2.4). It is shown in Exposé V, using the fact proved in Exposé VI that $K(X)$ is a special λ-ring in the sense of [RRR], that is, a λ-ring in the sense of Exposé V, that this notion of Chern class satisfies all the usual formal properties reviewed in [RRR].

2.4

One should note a very appreciable advantage of the definition of $\mathrm{Gr}^\bullet(X)$ over that of $A(X)$ and $\mathrm{Gr}_{\mathrm{top}}(X)$: it remains meaningful for an absolutely arbitrary scheme X, and more generally for an arbitrary locally ringed topos X. One then takes for $K(X)$ the group of classes constructed from locally free modules on X, which is again an augmented λ-ring, hence equipped with a corresponding filtration and an associated graded $\mathrm{Gr}^\bullet(X)$. The price of this generality, however, is that when one is not willing to neglect torsion phenomena, i.e. to tensor with $\mathbb{Q}$, the properties of $\mathrm{Gr}^\bullet(X)$ are in several respects less satisfactory than those of its two competitors (Exp. XIV no. 4). Thus, assuming again that $f : X \to Y$ is a proper morphism of noetherian regular schemes endowed with ample invertible modules, it is not true in general that the homomorphism

$$f_* : K(X) \longrightarrow K(Y)$$

maps $\mathrm{Fil}^i_\lambda K(X)$ into $\mathrm{Fil}^{i-d}_\lambda K(Y)$, where d is the relative dimension of X over Y, and hence defines, by passage to associated gradeds, a degree $-d$ homomorphism from $\mathrm{Gr}^\bullet(X)$ to $\mathrm{Gr}^\bullet(Y)$. However, Exposé VII proves that this is true after tensoring with $\mathbb{Q}$, so that one obtains a degree $-d$ homomorphism

$$f_* : \mathrm{Gr}^\bullet(X) \otimes \mathbb{Q} \longrightarrow \mathrm{Gr}^\bullet(Y) \otimes \mathbb{Q}, \tag{2.5}$$

which plays the role of the homomorphism "direct image of cycles" $A(X) \to A(Y)$.

Having said this, we see that we possess all the structural elements needed to give meaning to formula (1.2) when the existence of a base field is not postulated, with $f : X \to Y$ a proper morphism of noetherian regular schemes admitting ample invertible modules.

3. The case of general schemes

We now show how one can still give meaning to formula (1.2) when the regularity hypothesis on X, Y is abandoned and replaced by a local hypothesis on the morphism f, to be specified below.

3.1

A first difficulty comes from the fact that the observation which served as the starting point of the theory [RRR], namely that the group $K(X)$ of classes of sheaves on X is the same whether it is defined in terms of coherent sheaves or in terms of locally free sheaves, is essentially tied to the regularity hypothesis on X. For a general scheme X, say noetherian for simplicity, one must distinguish between these two groups; we shall denote them respectively by $K_\bullet(X)$ and $K^\bullet(X)$. The first is covariant in X for proper morphisms, the second contravariant in X for arbitrary morphisms. Notice that $K^\bullet(X)$ is naturally endowed with a ring structure, indeed with an augmented λ-ring structure, whereas $K_\bullet(X)$ no longer has, in general, a ring structure, but only a module structure over $K^\bullet(X)$. The homomorphism $x \mapsto x \cdot \mathrm{cl}(\mathcal{O}_X)$ from $K^\bullet(X)$ to $K_\bullet(X)$ is no longer an isomorphism in general. For our formulation of the generalized Riemann–Roch theorem, we shall work exclusively with $K^\bullet(X)$ and $K^\bullet(Y)$, to the exclusion of $K_\bullet(X)$ and $K_\bullet(Y)$. As noted in 2.3, these augmented λ-rings, equipped with their natural λ-filtration, give rise to graded rings $\mathrm{Gr}^\bullet(X)$, $\mathrm{Gr}^\bullet(Y)$, and to Chern-class homomorphisms

$$c_i : K^\bullet(X) \longrightarrow \mathrm{Gr}^\bullet(X),$$

and similarly for Y. The difficulty, however, lies in defining a direct-image homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(Y), \tag{3.1}$$

which can no longer be defined by the naive formula

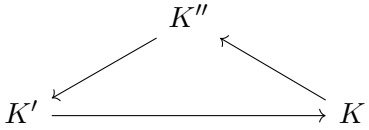
$$f_*(\mathrm{cl}(F)) = \sum_i (-1)^i \,\mathrm{cl}(R^i f_*(F)),$$

because even if F is locally free, the modules $R^i f_*(F)$ are in general no longer locally free, nor even of finite Tor-dimension, and the preceding formula loses its meaning. A solution to this difficulty is furnished by the viewpoint of derived categories. Indeed, whereas the $R^i f_*(F)$ are in general "bad" sheaves, say of infinite Tor-dimension, to which one would not know how to associate a class in $K^\bullet(Y)$, the complex $Rf_*(F)$ on Y, the direct image of F in the sense of derived categories, whose cohomology sheaves are the $R^i f_*(F)$, tends to be excellent. More precisely, suppose that F has finite Tor-dimension over $\mathcal{O}_Y$; this is the case if $\mathcal{O}_X$ has finite Tor-dimension over $\mathcal{O}_Y$ and, in addition, F is locally free. Then one proves (Exp. III) that the complex $Rf_*(F)$ is "perfect", by which we mean here that it has coherent cohomology

sheaves (as was already known in any case) and globally finite Tor-dimension; equivalently, on every affine open U of Y, this complex is isomorphic, in the derived category $D(U)$, to a bounded complex of locally free modules of finite type. From the viewpoint of homological algebra, perfect complexes are "as good as" locally free modules, of which they are the natural generalization. They moreover form a "triangulated subcategory" of the derived category $D(Y)$, i.e. a subcategory stable under the cones (or "mapping cylinders") of morphisms. On the other hand, to every triangulated category C one associates a group $K(C)$ of classes, a natural variant of the construction of the group of classes of an abelian category, by taking the free group generated by the objects of C, modulo the relations

$$\mathrm{cl}(K) - \mathrm{cl}(K') - \mathrm{cl}(K'')$$

for every distinguished triangle



What then replaces the fundamental observation of [RRR] in the present case is the fact that, when the scheme Y admits an ample invertible module, every perfect complex $L_\bullet$ on Y is globally (and not merely locally) isomorphic in $D(Y)$ to a bounded complex L' of locally free modules of finite type (Exp. II). It follows easily (Exp. IV) that the canonical homomorphism

$$K^\bullet(Y) \longrightarrow K(\mathrm{Parf}(Y))$$

is an isomorphism, where $\mathrm{Parf}(Y)$ is the triangulated category of perfect complexes on Y. Thus to every perfect complex $L_\bullet$ on Y one can associate a class

$$\mathrm{cl}(L_\bullet) \in K^\bullet(Y),$$

which is none other than

$$\mathrm{cl}(L_\bullet) = \sum_i (-1)^i \,\mathrm{cl}(L'_i),$$

where L' is as above, so that the L'_i are indeed locally free modules on Y whose classes may be taken in $K^\bullet(Y)$.

3.2

To give a meaning to (1.2), it remains to impose on f a hypothesis making it possible to give a meaning to the homomorphism (2.5), i.e. ensuring that the homomorphism (3.1) just defined is, up to a shift and modulo torsion, compatible with the filtrations of these rings; and finally one must again define an element, the "virtual relative tangent bundle",

$$T_f \in K^\bullet(X).$$

For the latter, returning to the definition proposed in 2.1, one sees that a hypothesis is needed ensuring that, with the notation of loc. cit., the conormal module $N_{X/X'}$ is locally free. The most natural hypothesis to this end is that i be a "regular immersion", i.e. identify X with a subscheme of X' defined by an ideal which locally can be defined by an $\mathcal{O}_{X'}$-regular sequence of sections of $\mathcal{O}_{X'}$. This hypothesis, purely local in nature, in fact does not depend on the choice of the factorization of f into an immersion followed by a smooth morphism (Exp. VIII), and, as above, the same is true for the class T_f defined by formula (2.1). When the hypothesis just

considered is satisfied, the morphism f is called a "locally complete intersection" morphism. Such a morphism is also locally of finite Tor-dimension.

We shall also allow ourselves to omit the word "locally" in this terminology, which cannot cause any confusion.

If f is a locally complete intersection morphism, one introduces a natural notion of "virtual relative dimension" $d(f)$ by the formula

$$d(f)(x) = \dim_x(f^{-1}(f(x))) - \operatorname{codim}_x(X, X').$$

It is an integer-valued function, locally constant at the point $x \in X$, hence constant if X is connected; moreover it does not depend on the choice of the factorization (2.1). When this virtual relative dimension d is constant, when f is proper, and when X and Y admit ample invertible modules, one can prove (Exposé VIII), although this is far from a trivial result, that

$$f_*(\mathrm{Fil}^i K^\bullet(X)_\mathbb{Q}) \subset \mathrm{Fil}^{i-d} K^\bullet(Y)_\mathbb{Q}, \tag{3.4}$$

whence a homomorphism (3.3) of degree $-d$.

3.3

We have therefore assembled all the structural elements needed to give a meaning to formula (1.2) in the case of a proper locally complete intersection morphism of noetherian schemes X, Y, both admitting ample invertible modules. With a little additional care, this formulation in fact retains a meaning independently of any noetherian hypothesis.

In this general form, this Riemann–Roch formula will be proved in Exposé VIII. The reader will observe that the proof given in this seminar combines elements of the two proofs which appear in [RRR] and which were discussed in the Introduction. The need to resort to the methods of the first of these proofs, involving more K-formalism than the second, followed in Borel–Serre, is due to the need for a proof of the inclusion (3.4), a question which did not arise in the more restricted framework of [RRR] and Borel–Serre [2], where one could work with the Chow ring or the ring $\mathrm{Gr}^\bullet_{\mathrm{top}}$. From the second proof, we retain the introduction of the scheme obtained by blowing up Y along X in the case where f is an immersion. This technical artifice will no doubt disappear, and the proof in the case of an immersion will extend to the case where one no longer postulates the existence of an ample invertible module on Y, once the question raised in Exposé XIV, no. 1, concerning the introduction of exterior-power operations in the derived category $D^-(Y)$ has been resolved.

4. Removing the ample invertible module hypothesis

We must finally give a few indications of how one can still give a meaning to (1.2) in the absence of a hypothesis asserting the existence of ample invertible modules on X and Y.

4.1

First note that for an arbitrary scheme X, or more generally an arbitrary locally ringed topos, it seems beyond doubt that the "right" invariant which should replace the ring $K(X)$ of [RRR] is not the ring $K^\bullet(X)$ considered above, defined in terms of locally free modules on X, but rather the group $K(\mathrm{Parf}(X))$ considered in 3.1, defined in terms of perfect complexes on X. We shall denote this group here by $K^\bullet(X)$, and distinguish it from the "naive" group of 3.1, denoted

$K^\bullet_{\text{naive}}(X)$. This point of view is forced by the need to define a direct-image homomorphism (3.1) for a proper morphism

$$f : X \longrightarrow Y$$

of finite Tor-dimension, say between noetherian schemes; the non-noetherian case requires somewhat stronger local finiteness properties on f, cf. Exposé III. With the definition of $K^\bullet(X)$ now adopted, this homomorphism f_* is indeed defined in a trivial way by formula (3.2), where now F is an arbitrary perfect complex on X, which implies that $Rf_*(F)$ is also perfect.

4.2

This new definition of $K^\bullet(X)$ raises, however, a new problem in homological algebra: that of defining a λ-structure on $K^\bullet(X)$ compatible with its natural ring structure coming from tensor product in the derived category. If L and L' are two perfect complexes, then so is their total tensor product $L \otimes^{\mathbf{L}} L'$. More precisely, one would need to define "exterior-power" operations in the derived category $D^-(X)$, transforming perfect complexes into perfect complexes. This question has not yet been clarified at present, and, as was said in the Introduction, it constitutes the most serious gap in the seminar. Once this question is solved, and proceeding as in 2.3, one obtains a natural filtration on $K^\bullet(X)$, allowing one to form the associated graded ring $\mathrm{Gr}^\bullet(X)$, which will play the role of the Chow ring, and to define a theory of Chern classes

$$c_i : K^\bullet(X) \longrightarrow \mathrm{Gr}^i(X).$$

4.3

To give a meaning to formula (1.2) for a proper locally complete intersection morphism

$$f : X \longrightarrow Y$$

of noetherian schemes, one must still include in the statement of this formula the nontrivial inclusions (3.4), where d is the virtual relative dimension, in order to be able to define (3.3); and finally one must define the virtual relative tangent class

$$T_f \in K^\bullet(X).$$

When f admits a factorization (2.1) through an immersion into a relative smooth scheme, the definition of T_f presents no new difficulty. It can moreover be reformulated in that case, in a way more consonant with the spirit of derived categories, by considering the well-known canonical homomorphism

$$N_{X/X'} \longrightarrow \Omega^1_{X'/Y} \otimes_{\mathcal{O}_{X'}} \mathcal{O}_X \tag{4.1}$$

induced by the exterior differential, and the chain complex $L_{X/Y}$ defined from (4.1) by "extending by zeros", with the complex equal in degree 1, respectively 0, to the source, respectively the target, of the homomorphism (4.1). Formula (2.2) then takes the more satisfactory form

$$T_f = \mathrm{cl}(L^\vee_{X/Y}), \tag{4.2}$$

a formula which makes sense in $K^\bullet(X)$ when f is a locally complete intersection morphism, since in that case the complex $L_{X/Y}$ is manifestly perfect. The assertion that T_f is invariant relative to the chosen factorization of f can then be made precise as an invariance assertion for the complex $L_{X/Y}$: up to canonical isomorphism in the derived category $D(X)$, this complex is likewise independent of the chosen factorization (Exposé VIII).

4.4

When $f : X \to Y$ is an arbitrary morphism of schemes, one can still construct a canonical object $L_{X/Y}$ of the derived category $D(X)$, defined up to unique isomorphism, which on each affine open of X lying above an affine open of Y is isomorphic to the complex constructed according to the principle explained in the preceding paragraph. In the general case, however, "smooth" should be replaced by "formally smooth", so that every algebra B over a ring A occurs as a quotient of a formally smooth algebra, for example a polynomial algebra. For such a definition it is not possible to proceed by simple gluing from the affine constructions, since the objects of the derived category $D(X)$ are, essentially, not glueable.

Here is a general construction of $L_{X/Y}$, valid more generally for every morphism

$$f : X \longrightarrow Y$$

of commutatively ringed topoi. Let $B = \mathcal{O}_X$ and $A = f^{-1}\mathcal{O}_Y$, so that A is a ring on the space X and B is an A-algebra. Let $T \to B$ be a homomorphism of sheaves of sets which generates B as an A-algebra, i.e. such that the corresponding homomorphism of A-algebras

$$C = A[T] \longrightarrow B$$

is an epimorphism of sheaves of sets; here $A[T]$ denotes the commutative A-algebra freely generated by T, that is, the sheaf associated with the presheaf

$$U \longmapsto A(U)[T(U)].$$

Let J be the kernel of $C \to B$, and consider, as in 4.3, the chain complex of length 1 defined by the natural homomorphism

$$J/J^2 \longrightarrow \Omega^1_{C/A} \otimes_C B.$$

It is not difficult to see that, up to canonical isomorphism in the category $D(X)$, the complex so constructed is independent of the choice of T and of $T \to B$, so that it deserves the designation of the relative cotangent complex $L_{X/Y}$ of X over Y. One also verifies that, on every affine open of X over an affine open of Y, it is given, up to canonical isomorphism, by the procedure of 4.3.

In the case where f is locally complete intersection, one sees moreover that the result is a perfect complex, which makes it possible again to define an element T_f of $K^\bullet(X)$ by formula (4.2). Here again, in general, it would not have been possible to define T_f as an element of $K^\bullet_{\text{naive}}(X)$, and nothing permits one to suppose that T_f is globally isomorphic in $D(X)$ to a bounded complex with locally free components of finite type. This is therefore an additional justification for the point of view advanced in 4.1 in favour of the "sophisticated" $K^\bullet(X)$, in preference to $K^\bullet_{\text{naive}}(X)$.

4.5

Subject to the question raised in 4.2, we have therefore again assembled all the structural elements needed to give a meaning to (1.2) for a proper locally complete intersection morphism of arbitrary schemes; the noetherian hypothesis implicitly made in the preceding sections can in fact be eliminated without any essential difficulty. Unfortunately, this does not mean that, even subject to this reservation, the formula in question is established, even in the case where Y is the spectrum of an algebraically closed field of characteristic $p > 0$ and X is, say, a proper smooth algebraic scheme of dimension 3. See the comments in Exposé XIV, no. 2.

5. Analytic variant

In the form presented in no. 4, the Riemann–Roch formula (1.2) also has a meaning for a proper locally complete intersection morphism of complex analytic spaces, or of rigid analytic spaces in the sense of Tate. But of course the proof given in [RRR] applies to this case only when one assumes in addition that the morphism f is projective; and, as in the algebraic case, it seems that an essentially new idea is needed to treat the general case. The formula we have in view here is a formula with values in an "analytic Chow ring" $\mathrm{Gr}^\bullet(X)$, or rather $\mathrm{Gr}^\bullet(X)_{\mathbb{Q}}$, constructed according to the principle explained in 4.2, the difficulty encountered there disappearing, moreover, because one is in characteristic zero, as is noted in Exposé XIV, no. 1.

When X and Y are nonsingular, it is possible to give another version of the Riemann–Roch formula by proceeding as follows. First let X be an arbitrary complex analytic space, and let X^{top} be the same space equipped with the sheaf of rings of continuous complex-valued functions. We therefore have a canonical morphism of ringed spaces

$$X^{\mathrm{top}} \longrightarrow X,$$

which, by the evident contravariant nature of the functor $K^\bullet(X)$ in the variable ringed space X, defines a homomorphism of rings

$$K^\bullet(X) \longrightarrow K^\bullet(X^{\mathrm{top}}). \tag{5.1}$$

For simplicity suppose that X^{top} is compact; then, as in the case of a scheme admitting an ample invertible module, one proves (Exposé II) that every perfect complex on X^{top} is globally isomorphic to a bounded complex, locally free in each degree, i.e. to a complex of complex vector bundles on X^{top}, and hence that $K^\bullet(X^{\mathrm{top}})$ is canonically isomorphic to the well-known group $K(X^{\mathrm{top}})$ of topologists [1], defined in terms of complex vector bundles on the compact space X^{top}. Thus one may interpret (5.1) as a homomorphism from the $K^\bullet(X)$ defined in terms of the analytic structure to the well-known ring $K(X^{\mathrm{top}})$ of topologists. If now

$$f : X \longrightarrow Y$$

is a morphism of compact nonsingular analytic spaces, one knows how to associate to it, thanks to Atiyah–Hirzebruch [1], a group homomorphism

$$f_*^{\mathrm{top}} : K^\bullet(X^{\mathrm{top}}) \longrightarrow K^\bullet(Y^{\mathrm{top}}). \tag{5.2}$$

Using likewise the homomorphism f_* of (3.1), defined by formula (3.2), which here implicitly uses Grauert's finiteness theorem [4] for a proper morphism of analytic spaces, one obtains a square of natural homomorphisms involving only groups of type $K^\bullet$, to the exclusion of rings of Chow-ring or cohomology type:

$$\begin{array}{ccc} K^\bullet(X) & \longrightarrow & K^\bullet(X^{\mathrm{top}}) \\ \downarrow f_* & & \downarrow f_*^{\mathrm{top}} \\ K^\bullet(Y) & \longrightarrow & K^\bullet(Y^{\mathrm{top}}). \end{array} \tag{5.3}$$

The Riemann–Roch type formula to which we are alluding consists in asserting the commutativity of this square. Note the difference in nature between this assertion, which does not neglect torsion phenomena and is situated in a group $K^\bullet(Y^{\mathrm{top}})$ of an essentially discrete nature, and the Riemann–Roch formula of the type considered in no. 4, which is situated in a group of "Chow" type no longer of a discrete nature, but which, on the other hand, must be tensored with $\mathbb{Q}$. Let us note that formula (5.3) is proved at any rate when X and Y are nonsingular projective varieties [7], as is the formula of the type considered in no. 4, proved in this case in

[RRR]. What is involved here is a plausible variant of the Atiyah–Singer index theorem, and of its generalizations due to Shih [7] and Atiyah, which obviously calls for a common generalization. Once the commutativity of (5.3) is proved, one would immediately deduce from it, using the "differentiable Riemann–Roch theorem" of Atiyah–Hirzebruch [1], a cohomological variant of the complex analytic Riemann–Roch formula, namely the commutativity of the diagram

$$\begin{array}{ccc} K^\bullet(X) & \xrightarrow{x\mapsto \mathrm{ch}(x)\,\mathrm{Todd}(T_X)} & H^{2\bullet}(X,\mathbb{Q}) \\ \downarrow f_* & & \downarrow f_* \\ K^\bullet(Y) & \xrightarrow{y\mapsto \mathrm{ch}(y)\,\mathrm{Todd}(T_Y)} & H^{2\bullet}(Y,\mathbb{Q}), \end{array} \tag{5.4}$$

where the horizontal arrows are deduced from cohomological Chern classes, themselves deduced from the homomorphism (5.1) and from the ordinary Chern-class homomorphism

$$K^\bullet(X^{\mathrm{top}}) \longrightarrow H^{2\bullet}(X,\mathbb{Z}),$$

the first vertical arrow is the same as in (5.3), and the second is the Gysin homomorphism in rational cohomology.

6. The covariant invariant $K_\bullet(X)$

In all that precedes, scarcely anything has been said except about the contravariant invariant $K^\bullet(X)$, to the exclusion of the covariant invariant $K_\bullet(X)$ whose existence was indicated in 3.1. If one regards $K^\bullet(X)$ as giving an analogue of the integral cohomology ring of a topological space, one should regard $K_\bullet(X)$ as giving an analogue of integral homology. The module structure of $K_\bullet(X)$ over $K^\bullet(X)$ then corresponds to the cap-product operation. The fact that, when X is regular, the natural homomorphism $K^\bullet(X) \to K_\bullet(X)$ is an isomorphism should be regarded as the analogue of the Poincaré duality theorem for oriented topological varieties, asserting that cap product with the fundamental homology class defines an isomorphism from cohomology to homology. These analogies moreover suggest the usefulness of also defining, for a finite polyhedron X, say, a covariant invariant $K_\bullet(X)$ whose relations to integral homology – spectral sequence, Chern homomorphism, and so on – would be analogous to those existing between $K^\bullet(X)$ and integral cohomology. Nor does it seem excluded that there might exist, in terms of these groups $K_\bullet(X)$, variants of the differentiable Riemann–Roch theorem valid for spaces more general than compact differentiable manifolds.

Returning to the case where X is an arbitrary noetherian scheme, one should endow $K_\bullet(X)$ with an increasing filtration, $\mathrm{Fil}_i\, K_\bullet(X)$ being generated by classes $\mathrm{cl}_\bullet(F)$, with F a coherent sheaf on X such that $\dim \mathrm{Supp}\, F \leq i$. One will prove (Exposé X) the compatibility of this filtration with the module structure and with the filtration of $K^\bullet(X)$, i.e. the formula

$$\mathrm{Fil}^i\, K^\bullet(X) \cdot \mathrm{Fil}_j\, K_\bullet(X) \subset \mathrm{Fil}_{j-i}\, K_\bullet(X), \tag{6.1}$$

which allows the graded group associated with the filtered module $K_\bullet(X)$, denoted $\mathrm{Gr}_\bullet(X)$, to be endowed with a module structure over $\mathrm{Gr}^\bullet(X)$. Its elements may in fact be interpreted as classes of algebraic cycles on X, for an equivalence relation less fine than rational equivalence when X is of finite type over a base field k. If

$$f : X \longrightarrow Y$$

is a proper morphism of noetherian schemes, then

$$f_* : K_\bullet(X) \longrightarrow K_\bullet(Y) \tag{6.2}$$

is compatible with the filtrations and defines a homomorphism of graded groups

$$f_* : \mathrm{Gr}_\bullet(X) \longrightarrow \mathrm{Gr}_\bullet(Y), \tag{6.3}$$

giving rise to a "projection formula", i.e. a formula which says that it is $\mathrm{Gr}^\bullet(Y)$-linear; this formula is obtained by passage to associated gradeds from the analogous projection formula for the homomorphism (6.2), which is $K^\bullet(Y)$-linear.

From the point of view proposed in the seminar, a manageable theory of intersections on noetherian schemes X, not necessarily regular, should consist in the combined study of the contravariant invariants $K^\bullet(X), \mathrm{Gr}^\bullet(X)$ and the covariant invariants $K_\bullet(X), \mathrm{Gr}_\bullet(X)$. These play roles of essentially different nature, and their properties complement each other. Thus the covariant groups $K_\bullet(X), \mathrm{Gr}_\bullet(X)$ are easier to compute for certain nonproper fibrations, thanks to the "homotopy exact sequence" which one can establish for them ([RRR] and Exposé IX). For example, one proves (Exposé IX) canonical isomorphisms such as

$$K_\bullet(X[t]) \simeq K_\bullet(X), \qquad \mathrm{Gr}_\bullet(X[t]) \simeq \mathrm{Gr}_\bullet(X),$$

which fail in general for $K^\bullet$ when X is a nonregular scheme. On the other hand, $K^\bullet(X)$ and $\mathrm{Gr}^\bullet(X)$ lend themselves well to computations, thanks to their ring structures.

When one considers schemes X proper over a field, the projection

$$\pi_X : X \longrightarrow (\text{point})$$

defines a homomorphism, the "Euler–Poincaré characteristic",

$$\chi_X = \pi_{X*} : K_\bullet(X) \longrightarrow K_\bullet(\text{point}) = \mathbb{Z}, \tag{6.4}$$

which induces on the subgroup $\mathrm{Gr}_0(X) = \mathrm{Fil}_0 K_\bullet(X)$ the homomorphism "degree of zero-cycles"

$$\deg_X : \mathrm{Gr}_0(X) \longrightarrow \mathbb{Z}. \tag{6.5}$$

The multiplication homomorphism

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathrm{Gr}_0(X),$$

composed with the degree homomorphism (6.5), then furnishes a pairing

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathbb{Z}, \tag{6.6}$$

which explains that, in certain respects, $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}_\bullet(X)$ play dual roles, just as cohomology and homology do. Thus, if $f : X \to Y$ is a morphism of proper algebraic schemes, the two homomorphisms

$$f^* : \mathrm{Gr}^\bullet(Y) \longrightarrow \mathrm{Gr}^\bullet(X), \qquad f_* : \mathrm{Gr}_\bullet(X) \longrightarrow \mathrm{Gr}_\bullet(Y)$$

are formally transposes of one another, in the sense of the pairings (6.6).

The homomorphisms (6.4) and (6.5), and the pairing (6.6) deduced from them, may be regarded as the source of numerical relations in intersection theory on proper algebraic schemes over a given field k. This point of view will be developed in some detail in Exposé X, which is essentially independent of Exposés VI, VII, VIII, centred on the proof of the Riemann–Roch theorem, and of Exposé IX. Some notions concerning the behaviour of $K^\bullet(X)$ and $\mathrm{Gr}^\bullet(X)$ under specialization will also be given there.

7. Picard groups and the determinant functor

As in every intersection theory, it is important to specify the place occupied in the theory by the Picard group, or group of classes of divisors in the classical terminology. Subject to a slight restriction, automatically satisfied when X is for instance quasi-compact, one establishes in Exposé X a canonical isomorphism

$$c_1 : \mathrm{Pic}(X) \xrightarrow{\sim} \mathrm{Gr}^1(X), \tag{7.1}$$

which shows the exact role of the Picard group in the theory we are proposing. This role justifies a more detailed study of the Picard group in Exposés XII and XIII, which are logically independent of the text of the seminar. Kleiman, in Exposé XIII, studies numerical equivalence and certain finiteness theorems for the Picard group, announced without proof in [6], using purely numerical techniques, not relying on the theory of the Picard functor and its representability, due to himself and Mumford, and markedly simpler than Grothendieck's initial proofs. The only result refractory to Kleiman's methods, and which he is obliged to use, is [6, C-07 (i)]; its proof and certain refinements, including certain representability theorems for the Picard functor, occupy Exposé XII. The latter is moreover of a spirit and technique entirely foreign to the rest of the seminar, and is connected with [5]; this exposé is logically independent of all the others.

Finally, in Exposé XI, of a nature appreciably more general than XII and XIII and closer to the general spirit of the seminar, one studies, on an arbitrary locally ringed topos, a remarkable functor called the "determinant functor":

$$\overset{\bullet}{\det} : \mathrm{Parf}_{\mathrm{is}}(X) \longrightarrow \mathrm{Inv}(X) \tag{7.2}$$

from the category of perfect complexes on X, whose morphisms are only the isomorphisms in the sense of the derived category $D(X)$, to the category of invertible modules on X. This functor is described, up to very little, by the requirement that it be of "local nature" and that it be given, for a bounded complex L with locally free components, by the formula

$$\overset{\bullet}{\det}(L) = \bigotimes_i \det(L_i)^{(-1)^i}, \tag{7.3}$$

where $\det(L_i)$ denotes the exterior power of maximal degree of the locally free module L_i. Although this exposé is also logically independent of the rest of the seminar, except for Exposé I, it nevertheless constitutes an important element of the general "yoga" proposed in the seminar and developed essentially in Exposés I to XI.

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Appendix to Exposé 0: RRR

Classes of Sheaves and the Riemann–Roch Theorem

by A. Grothendieck[1]

Demonstration of the Riemann–Roch theorem (1 November 1957).

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Chapter I. λ-rings (formal preliminaries)

1. Definitions

A λ-*ring*[2] means a commutative ring K with unit, provided with a family of maps

$$\lambda^i : K \longrightarrow K \qquad (i \text{ an integer } \geq 0) \tag{1.1}$$

satisfying the following conditions:

$$\lambda_x^0 = 1, \qquad \lambda_x^1 = x, \qquad \lambda^n(x+y) = \sum_{i=0}^{n} \lambda_x^i \lambda_y^{n-i}. \tag{1.2}$$

For every $x \in K$ put

$$\lambda_t(x) = \sum_{n \geq 0} \lambda^n(x) t^n \in K[[t]]. \tag{1.3}$$

[1]This is the literal reproduction of the report cited in the Introduction to the present seminar. The footnotes indicated by the signs (*), (**) were added in October 1967.

[2]In this seminar we rather say *pre-*λ*-ring* (V 2.1), reserving the name λ-ring for those satisfying the supplementary condition of page 3 (cf. V 2.4).

The preceding relations then say that $x \mapsto \lambda_t(x)$ is a homomorphism from the additive group K to the multiplicative group $1 + K[[t]]^+$ of formal series over K with augmentation 1, lifting the canonical homomorphism

$$1 + \sum_{i>0} x_i t^i \longmapsto x_1.$$

Thus on the ring $\mathbb{Z}$ there exists a unique λ-structure compatible with its ring structure and such that

$$\lambda_t(1) = 1 + t. \tag{1.4}$$

One then has

$$\lambda_t(n) = (1 + t)^n, \tag{1.5}$$

whence

$$\lambda^i(n) = \binom{n}{i}.$$

The notion of a λ-homomorphism of λ-rings is clear; an *augmented* λ*-ring* is a λ-ring provided with a λ-homomorphism to $\mathbb{Z}$.

To define the notion of a *special* λ-ring, the following considerations are useful. Let k be a commutative ring with unit and let

$$K = 1 + k[[t]]^+$$

be the group of formal series with coefficients in k and augmentation 1. We shall introduce on it a λ-ring law whose additive structure is the multiplication of formal series. The multiplication of this ring will be denoted $f \circ g$. It is entirely determined by the condition that the coefficients $c_n = c_n(f \circ g)$ of $f \circ g$ are given by universal polynomials with integral coefficients in the coefficients $a_i = c_i(f)$ and $b_i = c_i(g)$ of f and g, respectively:

$$c_n = P_n(a_1, \ldots, a_n; b_1, \ldots, b_n), \tag{1.6}$$

that it be distributive with respect to the "addition" fg, and be given for linear factors by

$$(1 + at) \circ (1 + bt) = 1 + abt. \tag{1.7}$$

The P_n are computed by calculations with symmetric functions; P_n is isobaric of weight n in the a_i and isobaric of weight n in the b_i (where a_i and b_i have weight i), and moreover is symmetric in $a = (a_i)$ and $b = (b_i)$.

Similarly, the $\lambda^i(f)$ are determined without ambiguity by the condition that the coefficients

$$c_n(\lambda^i(f)) = c_n(\lambda^i(f))$$

of $\lambda^i(f)$ be calculated by universal polynomials with integral coefficients in the coefficients $c_n(f) = a_n$ of f:

$$c_n = P_{i,n}(a_1, \ldots, a_{in}), \tag{1.8}$$

that the relations (1.2) be satisfied, and that, for $i > 1$, one have

$$\lambda^i(1 + at) = 1 \tag{1.9}$$

(1 is the zero element of the ring $K = 1 + k[[t]]^+$). The $P_{i,n}$ are again computed by calculations with symmetric functions, and one observes that $P_{i,n}$ is an isobaric polynomial of weight in in the a_i. Provided with the preceding operations, K becomes a λ-ring; its zero element is 1 and its unit element is $1 + t$. Also note the relation

$$(1 + at) \circ f = f(at). \tag{1.7 bis}$$

Let now K be an arbitrary λ-ring. We say that K is *special* if the additive homomorphism λ_t from K to $1 + K[[t]]^+$ is a λ-homomorphism. Thus this means that one has the formulas

$$\lambda_t(1) = 1 + t, \quad \lambda_t(xy) = \lambda_t(x) \circ \lambda_t(y), \quad \lambda_t(\lambda^i(x)) = \lambda^i(\lambda_t(x)), \tag{1.10}$$

or, more explicitly,

$$\begin{cases} \lambda^i(1) = 0 & \text{if } i > 1, \\ \lambda^n(xy) = P_n(\lambda^1 x, \dots, \lambda^n x; \lambda^1 y, \dots, \lambda^n y), & \\ \lambda^n(\lambda^i(x)) = P_{i,n}(\lambda^1 x, \dots, \lambda^{in} x), & \end{cases} \tag{1.11}$$

where the P_n and $P_{i,n}$ are the universal polynomials considered above. In particular the formulas (1.5) follow. One may show that the λ-ring $1 + k[[t]]^+$ constructed from a commutative ring k is special, but we shall not need this fact.

2. Examples

1) Let G be a group and k a commutative ring. Consider the group $K(G)$, the quotient of the free group generated by the classes of representations of G in finite-type k-modules by the subgroup generated by elements of the form

$$(\rho \oplus \rho') - (\rho) - (\rho')$$

(where $\oplus$ denotes the direct sum). Tensor product and exterior powers define on $K(G)$ a λ-ring structure. One may also pass to the quotient by elements of the form

$$(\rho) - (\rho') - (\rho''),$$

where ρ is a representation which is an extension of ρ' by ρ'', trivial as an extension of k-modules. One obtains the group $K_r(G)$ of reduced classes of representations of G, and one checks that the λ-ring law passes to the quotient.

When k is an algebraically closed field of characteristic 0,[3] one can show that $K(G)$, and a fortiori $K_r(G)$, is a special λ-ring. This is no longer true if the characteristic is not 0; however, if k is algebraically closed, $K_r(G)$ is still a special λ-ring. To prove this one is reduced to the case where G is a product $\mathrm{Gl}(m,k) \times \mathrm{Gl}(n,k)$ and where only rational representations of G are considered; our assertion follows from the explicit determination of $K_r(G)$ to be made below (Example 3). Finally, when k has characteristic 0, complete reducibility of rational representations of G shows that $K(G) = K_r(G)$.

Variants of the preceding example: G is an algebraic group defined over k and one considers only rational representations of G defined over k; one may consider the case of a topological group, or of a Lie group, and continuous representations, etc.[4]

2) Let X be an algebraic variety, irreducible or not, defined over the field k. We may consider the quotient group of the free group generated by the classes of vector bundles on X (defined over k) by the subgroup generated by the elements of the form

$$(E \oplus E') - (E) - (E'),$$

[3] It is unnecessary for k to be algebraically closed; cf. VI 3.3 for a still more general result, encompassing all those that will be used in the present report.

[4] One could build other examples of λ-rings; for example, the set of equivalence classes of quadratic forms, the set of representation classes of a Lie algebra, etc. We promise that we shall not need all these examples in order to treat the Riemann–Roch theorem.

on which the λ-ring operations are defined by tensor product and exterior powers. If X is complete, this is a free group generated by the indecomposable vector bundles on X (thanks to Krull–Remak–Schmidt); knowledge of this ring and of its basis is equivalent to the complete classification of vector bundles on X and to knowledge of the elementary tensor operations on these bundles. If k is algebraically closed of characteristic 0, all other tensor operations on classes of vector bundles are then known by means of what follows in 3). One may make a more drastic passage to the quotient by the group generated by the elements

$$(E) - (E') - (E''),$$

where E is a vector bundle extension of E' by E''. One then obtains a λ-ring, denoted $K(\xi(X))$, where $\xi(X)$ denotes the additive category of vector bundles on X defined over k. Using the results of Example 1), one shows that, if k is algebraically closed,[5] the λ-ring $K(\xi(X))$ is special, and that the same is true of the first λ-ring defined here if the characteristic of k is 0, but not in the general case.

3) Determination of $K(G)$ and of $K_r(G)$ in some important cases. Suppose the field k is algebraically closed and that the group G is affine algebraic, connected, and "globally reductive" (the radical of G is isomorphic to $(k^*)^s$), and consider only rational linear representations. In characteristic 0, complete reducibility of rational linear representations of G shows that $K(G) = K_r(G)$ is the free group generated by the classes of irreducible rational representations. If T is a maximal torus of G, then

$$K(T) = K_r(T) = \mathbb{Z}(\widehat{T}),$$

the algebra of the group $\widehat{T}$ of rational characters of T; one has $\widehat{T} \simeq \mathbb{Z}^r$ if r is the rank of G (this is independent of the characteristic). The Weyl group W operates on T, hence on $K(T)$, in a way which depends only on the operations of W on the free group $\widehat{T}$. There is an evident homomorphism of λ-rings, deduced from the injection of T into G:

$$K_r(G) \longrightarrow K_r(T)^W = K(T)^W, \tag{1.13}$$

where $K(T)^W$ denotes the set of fixed points of W in $K(T)$. The theory of linear representations of G[6] then proves the following theorem.

Theorem 1.1. The homomorphism (1.13) is an isomorphism.

Thus one sees that $K_r(G)$, once the "type" of G is known (characterized by W operating on a group $\mathbb{Z}^r$), does not depend on the characteristic, and that it is a special λ-ring. There is a canonical basis of $K_r(G)$ corresponding to the orbits of W in $\widehat{T}$; another basis is obtained as follows. Let $(\rho_i)_{1\leq i\leq r'}$ be the classes of representations of G corresponding to the fundamental representations of $G/\operatorname{rad} G$, and let $(\sigma_j)_{1\leq j\leq r''}$ be a basis of the group of rational homomorphisms from G to k^*. Thus $r' + r'' = r$ (the rank of G).

Corollary. The ring $K_r(G)$ is identified with the subring

$$\mathbb{Z}[\rho_1, \ldots, \rho_{r'}; \sigma_1, \ldots, \sigma_{r''}, \sigma_1^{-1}, \ldots, \sigma_{r''}^{-1}]$$

of the field of rational functions in the ρ_i and the σ_j with coefficients in $\mathbb{Q}$.[7]

In particular, $K_r(G)$ has no zero divisors. If $G = \prod_i \mathrm{Gl}(n_i, k)$, the ring $K_r(G)$ is made explicit as follows. Let ρ_i be the representation of G deduced from the identical representation of $\mathrm{Gl}(n_i, k)$. If one then considers the field of rational functions over $\mathbb{Q}$ in the indeterminates

[5] Unnecessary condition; cf. the preceding footnote.

[6] Cf. Chevalley Seminar 1956/58, *Groupes de Lie algébriques* (École Normale Supérieure).

[7] The derived group G' of G must be supposed simply connected for this statement to be correct.

$\lambda^j(\rho_i)$, $1 \leq j \leq n_i$, then $K_r(G)$ is identified with the subring generated by these indeterminates and the inverses of the $\lambda^{n_i}(\rho_i)$. This justifies our assertion that, at least in characteristic 0, the operations λ^i make it possible to reconstruct all tensor operations on vector bundles; this remains true over an algebraically closed field of arbitrary characteristic, provided one takes reduced classes.

Remarks. a) In Example 1), the ring $K(G)$ is augmented, the augmentation being defined by the degree of representations; by passage to the quotient, this augmentation defines an augmentation on $K_r(G)$. Likewise, the λ-rings considered in Example 2) are augmented when X is k-irreducible; in $K(\xi(X))$, the augmentation is defined by the rank of a vector bundle, that is, the dimension of its fibres.

b) When the field k is not algebraically closed, it is plausible that $K(G)$ is a special λ-ring in characteristic 0, and likewise for $K_r(G)$ in all characteristics.

c) In non-zero characteristic, Theorem 1.1 was graciously supplied by Chevalley.

3. The λ-ring defined by a graded ring

Let A be a graded commutative ring with unit. To simplify, assume that the degrees of A are positive and that $A^0 = \mathbb{Z}$. Denote by $\widehat{A}$ the product of the A^i (this is a commutative ring with unit), and by $\widehat{A}^+$ the kernel of the evident augmentation $\widehat{A} \to A^0 = \mathbb{Z}$. Then $1 + \widehat{A}^+$ is a subgroup of the multiplicative group of invertible elements of $\widehat{A}$. Consider the abelian group

$$\widetilde{A} = \mathbb{Z} \times (1 + \widehat{A}^+), \tag{1.14}$$

whose elements will be denoted $[n, x]$, with $n \in \mathbb{Z}$ and

$$x = 1 + \sum_{i \geq 1} x^i \in 1 + \widehat{A}^+ \qquad (x^i \in A^i).$$

We write this group additively, thus:

$$[n, x] + [n', x'] = [n + n', xx']. \tag{1.15}$$

The zero element of this group is $[0, 1]$. We shall introduce on it an augmented λ-ring structure, compatible with its additive structure, the augmentation being $\varepsilon : [n, x] \mapsto n$. The product $[m, x][n, y]$ is completely characterized by the fact that it is written

$$[mn, x^n y^m (x * y)],$$

where $x * y$ is an element of $1 + \widehat{A}^+$ whose components are expressed as universal polynomials with integral coefficients in the x^i and the y^i:

$$(x * y)^i = Q_i(x^1, \ldots, x^i; y^1, \ldots, y^i) \qquad (i \geq 1), \tag{1.16}$$

the polynomial Q_i being isobaric of weight i. Moreover, $x * y$ must be distributive with respect to the "addition" xy and, for "linear" factors, must reduce to

$$[1, 1 + x^1][1, 1 + y^1] = [1, 1 + x^1 + y^1]. \tag{1.17}$$

The polynomials Q_i are evaluated by calculations with elementary symmetric functions. This is the calculation of the Chern classes of a tensor product of vector bundles by means of the Chern classes x^i and y^i of the factors and of their respective dimensions m and n. From (1.17) one deduces

$$(1 + x^1) * (1 + y^1) = \frac{1 + x^1 + y^1}{(1 + x^1)(1 + y^1)}, \tag{1.17 bis}$$

and an arbitrary product $x * y$ is computed term by term by decomposing formally

$$x = \prod_{i=1}^{n}(1+\alpha_i), \qquad y = \prod_{i=1}^{n}(1+\beta_i),$$

where the α_i and β_i have degree 1.

In this way $\widetilde{A}$ becomes a commutative augmented ring with unit $[1,1]$; it may be interpreted as the ring obtained by adjoining a unit to the ring $1+\widehat{A}^+$. Moreover, on $\widetilde{A}$ one defines a filtration by putting $\widetilde{A}^0 = \widetilde{A}$ and denoting by $\widetilde{A}^n$ the set of the $[0,x]$ with $x^i = 0$ for $1 \leq i < n$. This filtration is compatible with the product, for

$$\left(1+\sum_{i\geq m} x^i\right) * \left(1+\sum_{j\geq n} y^j\right) = 1 - \frac{(m+n-1)!}{(m-1)!(n-1)!}x^m y^n + \cdots, \tag{1.18}$$

the omitted terms having degree $> m+n$. This formula shows that the graded ring $G(\widetilde{A})$ associated with $\widetilde{A}$ is identified with A from the additive point of view, but not from the multiplicative point of view. One can only say that the map η' from A to $G(\widetilde{A})$ defined by

$$\eta'(x^i) = (-1)^{i-1}(i-1)!\,x^i \qquad (x^i \in A^i) \tag{1.19}$$

is a ring homomorphism.

On $\widetilde{A}$ introduce operations λ^i satisfying the axioms of λ-rings (Section 1, formulas (1.2)), well determined by the condition that the components of $\lambda^i[0,x]$ be deduced from the components x^i of x by universal polynomials with integral coefficients:

$$\lambda^i[0,x] = [0,\lambda^i x], \qquad (\lambda^i x)^{(n)} = Q_{i,n}(x^1,\ldots,x^n), \tag{1.20}$$

where $Q_{i,n}$ is isobaric of weight n, and such that

$$\lambda^i[1,1+x^1] = 0 \qquad \text{for } i > 1. \tag{1.21}$$

The polynomials $Q_{i,n}$ are evaluated by a calculation with elementary symmetric functions: it is the calculation of the Chern classes of the exterior powers of a vector bundle by means of the Chern classes and the rank of that bundle. One then observes that $\widetilde{A}$ is a special augmented λ-ring and that the $\widetilde{A}^i$ are stable under the operations λ^j.

The augmented λ-ring $\widetilde{A}$ may be considered as a functor in A. Consequently, the principal automorphism

$$x = \sum_{i\geq 0} x^i \longmapsto \check{x} = \sum_{i\geq 0}(-1)^i x^i$$

of A defines an automorphism of $\widetilde{A}$, denoted by the same symbol $\check{}$.

Proposition 1.2. Let $x \in \widetilde{A}$ be of the form

$$x = [n, 1 + x^1 + \cdots + x^n]$$

(so $x^i = 0$ for $i > n$). Then $\lambda^i(x) = 0$ for $i > n$, one has $\lambda^n(x) = [1, 1+x^1]$, and finally

$$\sum_{i=0}^{n}(-1)^i\lambda^i(x) = [0, 1-(n-1)!x^n + \cdots]. \tag{1.22}$$

From this proposition, and from the fact that the $\widetilde{A}^i$ are stable under the operations λ^j, it follows that formula (1.22) is valid whenever $\varepsilon(x) = n$, whether or not $x^i = 0$ for $i > n$.

Formula (1.22) admits the following generalizations. Let K be any λ-ring, and let n be arbitrary. For $x \in K$, put

$$\gamma^n(x) = (-1)^n \sum_{i=0}^{n} (-1)^i \lambda^i(x+n) = \lambda^n(x+n-1) \tag{1.23}$$

(the identity of the last two members follows from an immediate induction). In particular $\gamma^0(x) = 1$ and $\gamma^1(x) = x$. Next put

$$\gamma_t(x) = \sum_{n \geq 0} \gamma^n(x) t^n \in K[[t]]. \tag{1.24}$$

One deduces the formula

$$\gamma_t(x) = \lambda_{t/(1-t)}(x), \tag{1.25}$$

which is equivalent to

$$\lambda_s(x) = \gamma_{s/(1+s)}(x). \tag{1.25 bis}$$

This proves that the λ^i are expressed in terms of the γ^i, and that

$$\gamma_t(x+y) = \gamma_t(x)\gamma_t(y).$$

Consequently the maps γ^i define a new λ-ring structure on K. Formula (1.22) then shows that, for $K = \widetilde{A}$ and every $x \in \widetilde{A}$, one has

$$\gamma^n(x - \varepsilon(x)) = [0, 1 + (-1)^{n-1}(n-1)! x^n + \cdots], \tag{1.22 bis}$$

and hence the component x^n is obtained, up to the factor $(-1)^{n-1}(n-1)!$, by reducing modulo $\widetilde{A}^{n+1}$ the element $\gamma^n(x - \varepsilon(x))$ of $\widetilde{A}^n$.

The Chern homomorphism. For every graded ring A satisfying the preceding conditions, we shall define a homomorphism of augmented rings

$$\mathrm{ch} : \widetilde{A} \longrightarrow \widehat{A} \otimes \mathbb{Q}. \tag{1.26}$$

This homomorphism is defined without ambiguity by the condition that it be an additive homomorphism, send 1 to 1, and be such that the components of $\mathrm{ch}(x)$ in degree > 0 are given by universal isobaric polynomials with rational coefficients in the components of x, and that

$$\mathrm{ch}([1, 1 + x^1]) = \exp x^1 = \sum_{n \geq 0} (x^1)^n / n!. \tag{1.27}$$

One immediately obtains the explicit expression of ch. Let η be the additive endomorphism of $\widehat{A} \otimes \mathbb{Q}$ defined, for x^i of degree i, by

$$\eta(x^i) = \frac{1}{(-1)^{i-1}(i-1)!} x^i. \tag{1.28}$$

Then

$$\mathrm{ch}\left([n, 1 + \sum_{i \geq 1} x^i]\right) = n + \eta\left(\log\left(1 + \sum_{i \geq 1} x^i\right)\right). \tag{1.29}$$

Since this formula can be inverted, at least if A has only finitely many homogeneous components, one sees that in this case one has an isomorphism ch from the ring $\widetilde{A} \otimes \mathbb{Q}$ onto the ring $\widehat{A} \otimes \mathbb{Q}$, which completely elucidates the structure of $\widetilde{A} \otimes \mathbb{Q}$.

Recall that, according to Hirzebruch [1], a formal series $f \in \mathbb{Q}[[t]]$ defines by universal polynomial formulas an additive homomorphism

$$1 + \widehat{A}^+ \longrightarrow 1 + (\widehat{A} \otimes \mathbb{Q})^+,$$

denoted $\mathfrak{C}_f$. When $f(t) = t/(1-\exp(-t))$, we write simply $\mathfrak{C}_f = \mathfrak{C}$ (the initial of Todd). We extend the definition of $\mathfrak{C}(x)$ to $x \in \widetilde{A}$. This said, we have the following result.

Proposition 1.3. Let $N \in \widetilde{A}$ be of the form

$$N = [q, 1 + N^1 + \cdots + N^q]$$

(so $N^i = 0$ for $i > q$). If one puts

$$\lambda_{-1}(N) = \sum_{i \geq 0} (-1)^i \lambda^i(N) = \sum_{i=0}^{n} (-1)^i \lambda^i(N),$$

then

$$\mathrm{ch}(\lambda_{-1}(N)) = (-1)^q N^q \mathfrak{C}(\check{N})^{-1}, \tag{1.30}$$

which is equivalent to

$$\mathrm{ch}(\lambda_{-1}(\check{N})) = N^q \mathfrak{C}(N)^{-1}. \tag{1.30 bis}$$

4. The operations $\lambda^p(N, x)$

Let A be the ring of formal series, with coefficients in $\mathbb{Z}$, in two infinite sequences of indeterminates denoted $\lambda^i N$ and $\lambda^i x$ $(i \geq 1)$. Put $\lambda^1 N = N$ and $\lambda^1 x = x$, and introduce on A the unique special λ-ring structure for which

$$\lambda^i(N) = \lambda^i N, \qquad \lambda^i(x) = \lambda^i x,$$

and for which the operations λ^i are continuous. If one restricted oneself to polynomials in the indeterminates $\lambda^i N$ and $\lambda^i x$, one would have the λ-ring freely generated by N and x; our ring A is its completion. One may then form

$$\lambda_{-1}(N) = \sum_{i \geq 0} (-1)^i \lambda^i N, \tag{1.31}$$

which is a formal series with constant term 1, hence invertible. Thus an element $\lambda^p(N, x)$ of A can be defined without ambiguity by putting

$$\lambda^p(N, x)\lambda_{-1}(N) = \lambda^p(x\lambda_{-1}(N)). \tag{1.32}$$

Theorem 1.4. Let J_q be the closed ideal of A generated by the $\lambda^i N$ with $i > q$. Identify A/J_q with the ring of formal series in the $\lambda^i x$ and the $\lambda^j N$ $(1 \leq j \leq q)$. Then $\lambda^p(N, x)$ reduced modulo J_q is a polynomial, isobaric of weight p with respect to the variables $\lambda^i x$.

Let N and F be two vector spaces of finite dimensions q, q' over an algebraically closed field k of characteristic 0. Let $N^{(p)}$ be the kernel of the map

$$(a_1, \ldots, a_p) \longmapsto a_1 + \cdots + a_p$$

from N^p to N. We make the symmetric group $\mathfrak{S}_p$ operate on N^p, hence on $N^{(p)}$, hence on $\Lambda(N^{(p)})$, and then on

$$\Lambda(N^{(p)}) \otimes \bigotimes^p F.$$

Consider this vector space as a graded module over $\mathfrak{S}_p \times \mathrm{Gl}(N) \times \mathrm{Gl}(F)$, and take its "alternating part", which is a graded module over $\mathrm{Gl}(N) \times \mathrm{Gl}(F) = G$. Then

$$\lambda^p(N, F) = E_G \left(\left(\Lambda(N^{(p)}) \otimes \bigotimes^p F \right)^{\mathrm{alt}} \right). \tag{1.33}$$

N.B. If M is a complex of G-modules, $E_G(M)$ denotes the alternating sum in $K(G)$ of the classes of the components of M, that is, the Euler–Poincaré characteristic of M. In formula (1.33), the first member $\lambda^p(N,F)$ denotes the element of the ring $K(G)$ obtained by substituting, for the variables λ^i_N and λ^i_F, the classes in $K(G)$ of the G-modules $\lambda^i(N)$ and $\lambda^i(F)$, respectively.

Proof. Expanding the second member of (1.32) by the second formula (1.11), one sees that the first assertion is proved if one proves the analogous assertion obtained by setting $x = 1$, i.e. if one proves that in the λ-ring $\mathbb{Z}[\lambda^1 N, \ldots, \lambda^q N]$, where $\lambda^i N = 0$ for $i > q$, the element $\lambda^p(\lambda_{-1}(N))$ is divisible by $\lambda_{-1}(N)$. A priori, the quotient

$$\lambda^p(N,1) = \lambda^p(\lambda_{-1}(N))/\lambda_{-1}(N)$$

is a formal series contained in the quotient field K of the ring of formal series in the $\lambda^i N$ for $1 \leq i \leq q$. On the other hand, also denote by N a vector space of dimension q over an algebraically closed field k of characteristic 0, and consider the element

$$\mathscr{L}^p(N) = E_G((\Lambda(N^{(p)}))^{\mathrm{alt}}) \in K(G),$$

where now $G = \mathrm{Gl}(N)$. We know (Section 2, Theorem 1.3) that $K(G)$ is identified with the subring of K generated by the λ^i_N for $1 \leq i \leq q$ and by the inverse of λ^q_N. Moreover, in this ring one verifies easily that

$$\mathscr{L}^p(N)\lambda_{-1}(N) = \lambda^p(\lambda_{-1}(N)).$$

Since we are in a large field K and $\lambda_{-1}(N) \neq 0$, it follows that $\mathscr{L}^p(N) = \lambda^p(N,1)$. Both members lie in the intersection of $K(G)$ with the ring of formal series in the λ^i_N, hence are polynomials in the λ^i_N.

To prove (1.33) in the general case, one verifies that the second member satisfies in $K(G)$ the formula analogous to (1.32), that is, that its product by $\lambda_{-1}(N)$ is $\lambda^p(F\lambda_{-1}(N))$. Likewise, the product of $\lambda^p(N,F) \in K(G)$ by $\lambda_{-1}(N)$ is $\lambda^p(F\lambda_{-1}(N))$, as follows from the fact that $K(G)$ is special. Since $K(G)$ is integral (cf. Section 2), and $\lambda_{-1}(N) \neq 0$ (same reference), formula (1.33) is proved. QED.

Remarks. a) When the dimension q' of F is at least p, formula (1.33) characterizes the polynomial $\lambda^p(N,x)$ in the λ^i_N $(1 \leq i \leq q)$ and the λ^j_x $(1 \leq j \leq p)$.

b) It is plausible that (1.33) remains valid if the field k is not algebraically closed.[8] One would have to resolve this question if one wanted to prove the Riemann–Roch theorem over a non-algebraically-closed field of characteristic 0 by the method to be explained. The passage from an $\mathfrak{S}_p$-module to its alternating part has reasonable meaning only when the base field has characteristic 0.

Let now K be any special λ-ring and let N be an element of K such that $\lambda^i N = 0$ for all sufficiently large i. Then for every $x \in K$ and every integer $p \geq 1$ one can consider the elements $\lambda^p(N,x)$, which are expressed by universal polynomials with integral coefficients in the λ^i_N and the λ^j_x. One has $\lambda^1(N,x) = x$ and formula (1.32) in K. To express the properties of the operations $x \mapsto \lambda^p(N,x)$, it is convenient to introduce a new λ-ring, denoted K_N. As a commutative ring, K_N is obtained by adjoining a unit to the ring whose underlying additive group is K and whose multiplication is given by

$$x_N y = xy\mu \qquad (\mu = \lambda_{-1}(N)). \tag{1.34}$$

The map λ_t from K_N to $K_N[[t]]$ is defined by

$$\lambda_t(n,x) = (1+t)^n \sum_{p \geq 0} \lambda^p(N,x)t^p, \tag{1.35}$$

[8] This is in fact proved by the same method.

where $\lambda^0(N,x) = (1,0)$ (the unit element of K_N). I say that, with these operations, K_N is a special λ-ring. Thus one has to verify the formulas

$$\lambda_t(X+Y) = \lambda_t(X)\lambda_t(Y), \qquad \lambda_t(XY) = \lambda_t(X)\lambda_t(Y), \qquad \lambda_t(\lambda^i(X)) = \lambda^i(\lambda_t(X))$$

for $X, Y \in K_N$. One may restrict oneself to verifying these formulas when X and Y lie in K, and one checks that it suffices to verify them when K is the free special λ-ring generated by X, Y and N subject to the relations $\lambda^i N = 0$ for $i > q$. But the group homomorphism

$$K_N \longrightarrow K, \tag{1.36}$$

which sends unit to unit and on K reduces to the homomorphism $x \mapsto x\mu$, is a homomorphism compatible with the ring structures and the maps λ^i, as one verifies immediately. In the present case, its restriction to K is injective. Since K is a special λ-ring, it follows easily that the above relations are indeed verified for $X, Y \in K$, completing the proof of our assertion.

According to formula (1.23), it is natural, for $x \in K$ and $n \geq 1$, to put

$$\gamma^n(N,x) = \sum_{i=0}^{n-1} \binom{n-1}{i} \lambda^i(N,x), \tag{1.37}$$

in other words, $\gamma^n(N,x) = \gamma^n(x_N)$, where x_N denotes x considered as an element of the λ-ring K_N.

Proposition 1.5. Let A be a graded commutative ring with unit, let N be an element of $\widetilde{A}$ of the form

$$N = [q, 1 + N^1 + \cdots + N^q],$$

and let x be an arbitrary element of $\widetilde{A}$:

$$x = [\nu, 1 + \sum_{i \geq 1} x^i].$$

Then, for every integer $j \geq 0$, one has $\gamma^{q+j}(N,x) \in \widetilde{A}^j$, and the component of degree j of $\gamma^{q+j}(N,x)$ is of the form

$$\left(\gamma^{q+j}(N,x)\right)^{(j)} = (-1)^{j-1}(j-1)!\, G_{q,j}(\nu, x^1, \ldots, x^j; N^1, \ldots, N^j), \tag{1.38}$$

where $G_{q,j}$ is a universal polynomial with integral coefficients characterized by the condition

$$G_{q,j}(\nu, x^1, \ldots, N^j)\,(N^q) = (-1)^q (x * \lambda_{-1}(N))^{(q+j)}. \tag{1.39}$$

Proof. We may suppose that A is the ring of polynomials with integral coefficients in indeterminates x^i and N^j $(1 \leq j \leq q)$. Since, combining (1.32) and (1.37), one has

$$\gamma^n(N,x)\lambda_{-1}(N) = \gamma^n(x\lambda_{-1}(N)), \tag{1.40}$$

and, for $n = q + j$, the second member has filtration $\geq q + j$ (use formula (1.22 bis), taking account of the fact that $x\lambda_{-1}(N)$ has zero augmentation), while $\lambda_{-1}(N)$ has filtration q (formula (1.22)), it follows that $\gamma^n(N,x)$ has filtration j, since by (1.18) the associated graded of A is integral here. Using the preceding formulas, one immediately obtains the indicated form for $(\gamma^{q+j}(N,x))^{(j)}$. QED.

Chapter II. Classes of coherent algebraic sheaves and Chern classes

1. The theory of Chow

Throughout this chapter, a base field k is fixed; for simplicity it will be assumed algebraically closed. A large part of what follows, perhaps all of it, is nevertheless valid without this restriction. The algebraic spaces and the regular maps considered will be defined over k.

An algebraic space is called *quasi-projective* if it is isomorphic to a locally closed subset of a projective space.

Let X be a nonsingular connected quasi-projective variety of dimension n. Denote by $A(X)$, and call the *Chow ring* of X, the graded ring of classes of cycles on X modulo rational equivalence. Thus $A^i(X)$ is formed from the classes of cycles of dimension $n-i$. The fact that $A(X)$ is indeed a graded ring follows from the following theorem of Chow.

Theorem 2.1.[9] Every cycle on X is rationally equivalent to a nonsingular cycle. Given elements of $A^i(X)$ and $A^j(X)$, one can find nonsingular representatives of these elements, say Y^{n-i} and Z^{n-j}, which meet everywhere transversely.

A cycle is called nonsingular if its components are nonsingular subvarieties. Two cycles are said to meet everywhere transversely if every component of one cuts every component of the other transversely, i.e. if the two secant varieties are nonsingular at their intersection points and their tangent varieties there are in general position.

Let f be a morphism, i.e. a regular map, from X to Y (where X and Y are quasi-projective and nonsingular). Then f defines, using the preceding theorem, a homomorphism of graded rings

$$f^* : A(Y) \longrightarrow A(X), \tag{2.1}$$

namely inverse image of cycle classes. Thus $A(X)$ becomes a contravariant functor in X.

Let f be a proper morphism from X^n to Y^m (i.e. in Weil's terminology, the graph of f is complete over Y, cf. [3]). Then f also defines an additive homomorphism

$$f_* : A(X^n) \longrightarrow A(Y^m), \tag{2.2}$$

preserving the dimension of cycles, hence increasing the degree by $m-n$.

Thus, relative to proper morphisms, and ignoring multiplicative structures and gradings, $A(X)$ is a covariant functor in X. Let us record the classical formula

$$f_*(x \cdot f^*(y)) = f_*(x) \cdot y. \tag{2.3}$$

We shall also use Chow's theory of Chern classes. It consists in attaching to every vector bundle E on X (still assumed quasi-projective and nonsingular) Chern classes

$$c^i(E) \in A^i(X) \qquad (i \geq 1),$$

in such a way as to satisfy the following conditions. Put $c^0(E) = 1$ and

$$\widetilde{C}(E) = \left[\operatorname{rank} E, \sum_{i \geq 0} c^i(E)\right] \in \widetilde{A}(X) \tag{2.4}$$

[9] As a result of a misunderstanding on the author's part, the stated theorem is more than is presently known. In fact, one uses only the "moving lemma" of Chow, saying that two cycles can be moved in their classes so that their intersection is not excessive; cf. Chevalley Seminar 1958, *Anneau de Chow et applications.*

(see Chapter I, §3 for the definition of $\widetilde{A}(X)$). The conditions to be verified by a theory of Chern classes are then expressed as follows:

$$\widetilde{C}(E) = \widetilde{C}(E') + \widetilde{C}(E'') \tag{2.5}$$

if the vector bundle E is an extension of E' by E''. If one puts

$$C(E) = \sum_i c^i(E),$$

this condition is expressed by $C(E) = C(E')C(E'')$. Formula (2.5) permits one to extend $\widetilde{C}$ to an additive homomorphism from $K(\xi(X))$ (cf. Chapter I, §2, Example 2) to $\widetilde{A}(X)$:

$$\widetilde{C} : K(\xi(X)) \longrightarrow \widetilde{A}(X), \tag{2.6}$$

and one wants $\widetilde{C}$ to be a λ-homomorphism of λ-rings. This last condition may also be written

$$\widetilde{C}(E \otimes F) = \widetilde{C}(E)\widetilde{C}(F), \qquad \widetilde{C}(\lambda^i E) = \lambda^i \widetilde{C}(E), \tag{2.7}$$

for two vector bundles E and F; of course $\widetilde{C}$ is automatically compatible with augmentations. Let D be a divisor on X and let $L(D)$ be the vector bundle it defines. One wants

$$c^1(L(D)) = \ell(D), \qquad c^i(L(D)) = 0 \quad (i > 1), \tag{2.8}$$

where, for any cycle Z, $\ell(Z)$ denotes its class in $A(X)$. Finally, one imposes that the c^i be functorial, i.e. that if f is a morphism from X to Y and E is a vector bundle on Y, then

$$c^i(f^!(E)) = f^*(c^i(E)), \tag{2.9}$$

where $f^!(E)$ is the inverse image of E on X. One may also write (2.9) in the following form: the notion of inverse image of vector bundles defines a λ-homomorphism of augmented λ-rings

$$f^! : K(Y) \longrightarrow K(X),^{10} \tag{2.10}$$

where here one writes $K(X)$ instead of $K(\xi(X))$, and (2.9) means that there is commutativity in the diagram

$$\begin{array}{ccc} K(Y) & \xrightarrow{f^!} & K(X) \\ \Big\downarrow{\scriptstyle \widetilde{C}_Y} & & \Big\downarrow{\scriptstyle \widetilde{C}_X} \\ A(Y) & \xrightarrow[f^*]{} & A(X). \end{array}$$

Here the space on which $\widetilde{C}$ is considered has been put as a subscript.

Theorem 2.2. There exists a theory of Chern classes satisfying conditions (2.5), (2.7), (2.8), and (2.9).

We shall also admit the following proposition.

Proposition 2.3. Let Y be a closed subset of X and let U be its complement. If x is an element of $A(X)$ inducing 0 on U, then there exists a representative of x which is a cycle carried by Y.

Corollary. Let $p = \dim X - \dim Y$. Then, if $i < p$, the homomorphism

$$A^i(X) \longrightarrow A^i(U)$$

is bijective. Every element of the kernel of $A^p(X) \to A^p(U)$ is a linear combination of components of dimension $n - p$ of Y; hence it is proportional to $\ell(Y)$ if Y is irreducible.

[10] The notation $f^!$ used here is in conflict with that used in duality theory (cf., for example, R. Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics, no. 20, Springer, 1966). This is why in this seminar we replace it by the notation f^*.

2. Definition of the Chern classes of coherent algebraic sheaves

Let $\mathcal{C}$ be an abelian category, and let $\mathcal{C}_0$ be a subclass of $\mathcal{C}$. Denote by $K(\mathcal{C}_0)$ the quotient group of the free group generated by the isomorphism classes of elements of $\mathcal{C}_0$ by the subgroup generated by the elements of the form

$$(E) - (E') - (E''),$$

where E is an extension of E' by E'' and $E, E', E'' \in \mathcal{C}_0$. We must suppose that the classes of elements of $\mathcal{C}_0$, for the isomorphism relation, form a set. Denote by $\gamma_{\mathcal{C}_0}(E)$ the class of an object $E \in \mathcal{C}_0$ in $K(\mathcal{C}_0)$, and omit the subscript $\mathcal{C}_0$ when there is no danger of confusion. The group $K(\mathcal{C}_0)$ is the solution of a universal problem relative to functions

$$E \longmapsto f(E)$$

defined on $\mathcal{C}_0$, with values in an arbitrary abelian group A, which are additive in the sense that $f(E) = f(E') + f(E'')$ whenever E is an extension of E' by E''.

Let F be an additive functor from $\mathcal{C}$ to $\mathcal{C}'$ and let $\mathcal{C}'_0$ be a subclass of $\mathcal{C}'$ satisfying the same condition as $\mathcal{C}_0$, such that for every $A \in \mathcal{C}_0$ the $R^q F(A)$ are in $\mathcal{C}'_0$ and are zero for q sufficiently large. We suppose that injective resolutions exist in $\mathcal{C}$, so that the derived functors $R^q F$ of F exist. Then F defines a homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}'_0)$, still denoted F, by the formula

$$F(\gamma_{\mathcal{C}_0}(E)) = \sum_{i \geq 0} (-1)^i \gamma_{\mathcal{C}'_0}(R^i F(E)). \tag{2.11}$$

More generally, if one has a cohomological functor (F^i) from $\mathcal{C}$ to $\mathcal{C}'$ such that for every $E \in \mathcal{C}_0$ the $F^i(E)$ are in $\mathcal{C}'_0$ and are zero except for a finite number of i, one obtains a homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}'_0)$. These remarks extend to multifunctors; moreover, the classical spectral sequences give transitivity properties which we shall not make explicit.

In particular, if F is an exact functor, formula (2.11) simplifies to

$$F(\gamma_{\mathcal{C}_0}(E)) = \gamma_{\mathcal{C}'_0}(F(E)). \tag{2.11 bis}$$

This applies in particular if $\mathcal{C} = \mathcal{C}'$, if F is the identity functor, and if $\mathcal{C}_0 \subset \mathcal{C}'_0$; one has a canonical homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}'_0)$.

Theorem 2.3. Let $\mathcal{C}$ be an abelian category and let $\mathcal{C}_0 \subset \mathcal{C}_1$ be two subclasses such that the isomorphism classes of objects of $\mathcal{C}_1$ form a set. Suppose that the following conditions are satisfied:

(i) For every exact sequence $0 \to A' \to A \to A'' \to 0$ in $\mathcal{C}$ such that A and A'' are in $\mathcal{C}_0$ (respectively $\mathcal{C}_1$), one has $A' \in \mathcal{C}_0$ (respectively $A' \in \mathcal{C}_1$).

(ii) Let $u : A \to B$ and $v : C \to B$, with $A, B, C \in \mathcal{C}_1$ and u surjective. Then there exist $L \in \mathcal{C}_0$ and homomorphisms $u' : L \to C$ and $v' : L \to A$ such that u' is surjective and $vu' = uv'$.

(iii) For every $A \in \mathcal{C}_1$ there exists an integer $d = d(A)$ such that for every left resolution L of A by objects of $\mathcal{C}_0$ one has $Z_d(L) \in \mathcal{C}_0$, where Z_d denotes the cycles in dimension d.

Under these conditions, the canonical homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}_1)$ is an isomorphism.

Principle of the proof. Let $A \in \mathcal{C}_1$. By (ii) and (iii) there exists a finite resolution L of A by objects of $\mathcal{C}_0$; put

$$\ell(A) = \sum_{i \geq 0} (-1)^i \gamma_{\mathcal{C}_0}(L_i).$$

One shows that the element of $K(\mathcal{C}_0)$ thus constructed is independent of the resolution L chosen. If one has a second resolution L', one "caps" L and L' by a third resolution L'', with surjective homomorphisms from L'' to L and L'; this is possible by (i) and (ii). One is then reduced to proving that $E_{\mathcal{C}_0}(L) = E_{\mathcal{C}_0}(L'')$. But the difference of the two is $E_{\mathcal{C}_0}(N)$, where N is the kernel of the surjective homomorphism from L'' to L; this kernel is in $\mathcal{C}_0$ by (i). Since N is an acyclic complex in all dimensions, $E_{\mathcal{C}_0}(N) = 0$, whence the conclusion. Here $E_{\mathcal{C}_0}(L)$, for a finite complex with values in $\mathcal{C}_0$, denotes the Euler–Poincaré characteristic

$$\sum_i (-1)^i \gamma_{\mathcal{C}_0}(L_i).$$

One proves analogously that the function $\ell : \mathcal{C}_1 \to K(\mathcal{C}_0)$ thus constructed is additive; hence it defines a homomorphism from $K(\mathcal{C}_1)$ to $K(\mathcal{C}_0)$, and it is immediate that this homomorphism is inverse to the canonical homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}_1)$. QED.

Remark. To verify (ii), it is enough to verify that $\mathcal{C}_0$ is contained in a subclass $\mathcal{C}_2$ of $\mathcal{C}$ satisfying the two conditions

(ii a) every $A \in \mathcal{C}_1$ is isomorphic to a quotient of an $L \in \mathcal{C}_2$;

(ii b) if $u : A \to B$, with $A, B \in \mathcal{C}_2$, then $\operatorname{Ker} u \in \mathcal{C}_1$.

Corollary. Let X be a quasi-projective algebraic space. Let $\mathscr{F}(X)$ be the category of coherent algebraic sheaves on X, let $\xi(X)$ be the category of coherent locally free algebraic sheaves, equivalently vector bundles on X, and let $\mathscr{F}_0(X)$ be the category of coherent algebraic sheaves F such that, for every $x \in X$, the stalk F_x is an $\mathcal{O}_x$-module of finite cohomological dimension. Thus $\xi(X) \subset \mathscr{F}_0(X) \subset \mathscr{F}(X)$. Then the canonical homomorphism

$$K(\xi(X)) \longrightarrow K(\mathscr{F}_0(X)) \tag{2.12}$$

is an isomorphism.

We verify the conditions of Theorem 2.3. Conditions (i) and (iii) follow from the definition of cohomological dimension and from the fact that a finite-type projective module over a noetherian local ring $\mathcal{O}_x$ is free. To verify (ii), we shall prove that $\mathcal{C} = \mathscr{F}(X)$ satisfies conditions (ii a) and (ii b) of the remark. The second is trivial; therefore it remains to prove the following result.

Proposition 2.4 (Serre). Let X be a quasi-projective algebraic space. Then every coherent algebraic sheaf on X is isomorphic to the quotient of a locally free algebraic sheaf.

This proposition is true if X is a closed subset of projective space by [2], since, for n large enough, $F(n)$ is generated by its sections, which means that F is a quotient of $\mathcal{O}(-n)^k$. In the general case, X is an open subset of a closed subset of a projective space, and it suffices to use the following result, due to Cartier and Serre in particular cases.

Proposition 2.5. Let F be a coherent algebraic sheaf on an open subset U of an algebraic space X. Then F is isomorphic to the restriction of a coherent algebraic sheaf on X.

By applying Zorn's lemma to the opens containing U on which one can extend F, one is reduced to the case where X is affine. Let $\overline{F}$ be the algebraic sheaf on X whose sections on an open V are the sections of F on $U \cap V$. One checks easily that $\overline{F}$ is quasi-coherent, i.e. defined on every affine open V—hence here on all of X—by a module, not necessarily of finite type, over the coordinate ring of that open. This is immediate if U itself is affine, since $\overline{F}$ is then defined on the open V by $\Gamma(U \cap V, F)$ considered as a module over $\Gamma(V, \mathcal{O}_X)$. In the general case, U is a finite union of affine opens U_i, hence $\overline{F}$ is the kernel of the evident homomorphism of quasi-coherent sheaves

$$\prod_i \overline{F}|_{U_i} \longrightarrow \prod_{i,j} \overline{F}|_{U_i \cap U_j},$$

and is therefore itself quasi-coherent. Since X is affine, $\overline{F}$ is generated by its sections. Since $\overline{F}|_U = F$ is coherent, there exists a finite number of sections of $\overline{F}$ which generate F on U. These sections generate a coherent subsheaf of $\overline{F}$ whose restriction to U is isomorphic to F. QED.

Now suppose that X is nonsingular and quasi-projective. By the theorem on syzygies, one has, with the notation of the corollary to Theorem 2.3,

$$\mathscr{F}_0(X) = \mathscr{F}(X).$$

In this case, therefore,

$$K(\xi(X)) = K(\mathscr{F}(X)). \tag{2.13}$$

One may express this isomorphism as follows.

Corollary 2 to Theorem 2.3. Let X be a nonsingular quasi-projective variety. For every abelian group A, there is a one-to-one correspondence between additive functions f of vector bundles on X with values in A, and additive functions g of coherent algebraic sheaves on X with values in A. To every f there corresponds the unique g such that

$$f(E) = g(\mathcal{O}_X(E)) \tag{2.14}$$

for every vector bundle E on X, where $\mathcal{O}_X(E)$ denotes the sheaf of germs of regular sections of E on X.

To compute $g(F)$ for an arbitrary coherent algebraic sheaf, one considers a finite resolution of F by sheaves of the form $\mathcal{O}_X(E_i)$, and one then has

$$g(F) = \sum_i (-1)^i f(E_i). \tag{2.14 bis}$$

In particular, if X is connected, in §1 we considered an additive function $\widetilde{C}(E)$ of the vector bundle E, with values in the group $A = \widetilde{A}(X)$. We deduce from it an additive function of coherent sheaves, say $\widetilde{C}(F)$, and, by passing to components, functions

$$F \longmapsto c^i(F) \in A^i(X).$$

The $c^i(F)$ are called the *Chern classes of the coherent sheaf* F. Theoretically they are computed from the Chern classes of vector bundles by (2.14 bis), with the understanding that in the second member one must pass to multiplicative notation. As for the component of $\widetilde{C}(F)$ in $\mathbb{Z}$, it is the rank of the sheaf F, corresponding to the rank of a module over an integral domain: if X is affine, to which one can always reduce, F is defined by a finite-type module over the coordinate ring of X, and one takes the rank of this module.

3. Functorial generalities on $K(X)$

If X is an algebraic space, put, for brevity,

$$K(X) = K(\mathscr{F}(X)).$$

When X is nonsingular and quasi-projective, identify this group with $K(\xi(X))$ by the corollary 2 to Theorem 2.3. Since $K(\xi(X))$ is not only an abelian group but a λ-ring, it follows by transport of structure that $K(X)$ is also a λ-ring, and is therefore provided with a multiplicative structure and operations λ^i. It is important to define these structures directly, without passing through vector bundles; this will be done below.

If E is a vector bundle on X, denote by $\gamma(E)$ its class in $K(X)$. If F is a coherent algebraic sheaf on X, $\gamma(F)$ denotes in the same way its class in $K(X)$. If Y is an irreducible subvariety of

X, put $\gamma(Y) = \gamma(\mathcal{O}_Y)$, where $\mathcal{O}_Y$ denotes the sheaf of local rings of Y, considered as a coherent algebraic sheaf on X. By linearity, one defines the symbol $\gamma(Y)$ when Y is any cycle on X. When there is risk of confusion, one writes γ_X for γ.

Let X be an arbitrary algebraic space. We introduce on $K(X)$ a filtration, denoting by $K(X)^i$ the subgroup of $K(X)$ generated by the $\gamma(F)$ with

$$\dim \operatorname{Supp} F \leq n - i,$$

where n is the dimension of X. From Serre's dévissage lemma, in the form given in [3], follows the following result.

Proposition 2.6. The group $K(X)^i$ is generated by the $\gamma(Y)$, where Y runs through the subvarieties of dimension $n - i$ of X. If X is irreducible, then $\mathrm{Gr}^0(K(X)) = \mathbb{Z}$.

The latter isomorphism is given by the homomorphism from $K(X)$ to $\mathbb{Z}$ defined with the aid of the notion of rank of a coherent algebraic sheaf.

Now suppose that X is nonsingular. Following the developments at the beginning of §2, define a multiplication in $K(X)$ by putting

$$\gamma(F)\gamma(G) = \sum_{i \geq 0} (-1)^i \gamma(\mathrm{Tor}_i(F, G)), \tag{2.15}$$

the Tor_i being taken, of course, with respect to the local rings of X. The exact sequence of the Tor_i shows that formula (2.15) indeed defines a biadditive map from $K(X) \times K(X)$ to $K(X)$; associativity follows from the spectral sequence of multiple Tors. One also verifies in this way that one can compute the product of several factors by the formula

$$\gamma(F_1) \cdots \gamma(F_n) = \sum_{i \geq 0} (-1)^i \gamma(\mathrm{Tor}_i(F_1, \ldots, F_n)), \tag{2.16}$$

where, in the second member, the Tor_i are simultaneous Tors, computed by simultaneous projective resolution of the arguments F_i.

If in (2.15) one supposes that G is locally free, one finds

$$\gamma(F)\gamma(G) = \gamma(F \otimes G), \tag{2.15 bis}$$

which shows in particular that $\gamma(\mathcal{O}_X)$ is the unit element of $K(X)$, and that the canonical homomorphism from $K(\xi(X))$ to $K(X)$ is a homomorphism of rings.

Formula (2.15) is evidently suggested by Serre's "Tor formula" in intersection theory. This formula proves the second part of the following proposition, the first part following immediately from (2.15) and from a local Tor calculation.

Proposition 2.7. Let X be a nonsingular variety, and let Y and Z be two cycles in X. If Y and Z meet everywhere transversely, then

$$\gamma(Y)\gamma(Z) = \gamma(Y \cdot Z). \tag{2.17}$$

If Y and Z are homogeneous and of respective dimensions $n - i$ and $n - j$, and if they meet without an excessive component, then

$$\gamma_X(Y)\gamma_X(Z) = \gamma_X(Y \cdot Z) \mod K(X)^{i+j+1}. \tag{2.18}$$

One must be careful: equality does not in general hold in (2.18). We shall see in §4 that, if X is nonsingular and quasi-projective, the ring structure of $K(X)$ is compatible with its filtration, and

that the associated graded ring $\mathrm{Gr}(K(X))$ is identified with a quotient of $A(X)$ by a subgroup of torsion, perhaps always zero.[11]

I do not know how to define directly the $\lambda^i(F)$ in terms of sheaves except when the field k has characteristic 0; this is the reason that has prevented us from proving the Riemann–Roch theorem when k is not of characteristic 0. Let F be a coherent algebraic sheaf on X on which the group $\mathfrak{S}_p$ of permutations of p letters operates. Denote by F^{alt} the set of $x \in F$ such that

$$s \cdot x = \varepsilon(s)x$$

for every $s \in \mathfrak{S}_p$, where $\varepsilon(s)$ is the signature of s. It is evidently a coherent algebraic subsheaf of F. This said, one has:

Theorem 2.8. Let X be a nonsingular quasi-projective variety over a field k of characteristic 0. Let F be a coherent algebraic sheaf on X. Then

$$\lambda^p(\gamma(F)) = \sum_{i \geq 0} (-1)^i \gamma\left((\mathrm{Tor}_i(F, F, \ldots, F))^{\mathrm{alt}}\right). \tag{2.19}$$

Here it is clear that the group $\mathfrak{S}_p$ operates on the sheaf $\mathrm{Tor}_i(F, \ldots, F)$.

Principle of the proof. One considers a finite resolution L of F by locally free sheaves L_i, and uses the following general lemma.

Lemma 2.9. Let $L = (L_i)$ be a graded locally free coherent algebraic sheaf with only finitely many nonzero components. In $K(X)$ one has the formula

$$E\left((\otimes^p L)^{\mathrm{alt}}\right) = \lambda^p(E(L)). \tag{2.20}$$

In this formula, $\otimes^p L$ is considered as a sheaf on which the group $\mathfrak{S}_p$ acts, taking the gradings into account; the gradings therefore introduce signs in the usual way, cf. Cartan–Eilenberg. Moreover, if M is a graded coherent algebraic sheaf on X, put

$$E(M) = \sum_i (-1)^i \gamma(M_i).$$

To prove formula (2.20), one is reduced to proving the analogous formula in $K(G)$, where G is an algebraic group, for instance a product of groups $\mathrm{Gl}(n, k)$, and where L is a finite-dimensional G-module over k (cf. Chapter I, §2, Example 2, for the definition of $K(G)$). In that case, this formula has already been used in Theorem 1.4 to show that the second member of (1.33) satisfies the same functional equation as the first member; the verification is elementary. QED.

Let f be a morphism from X to Y, where Y is an algebraic space. The notion of inverse image of vector bundle defines a homomorphism

$$f^! : K(\xi(Y)) \longrightarrow K(\xi(X))$$

of λ-rings. Identifying vector bundles with locally free coherent algebraic sheaves, the notion of inverse image of vector bundle corresponds to the notion of inverse image of algebraic sheaf, already used in [3, no. 8]. When Y is nonsingular, one defines more generally a homomorphism of groups

$$f^! : K(Y) \longrightarrow K(X) \tag{2.21}$$

[11] No: for a counterexample see this seminar, XIV 4.7.

by a formula involving an alternating sum of Tors, inspired by the developments at the beginning of §2, and which the reader will spell out. One then has commutativity in the following diagram:

$$\begin{array}{ccc} K(\xi(Y)) & \xrightarrow{f^!} & K(\xi(X)) \\ \downarrow & & \downarrow \\ K(Y) & \xrightarrow[f^!]{} & K(X), \end{array}$$

which shows in particular, when X and Y are both nonsingular and quasi-projective, that the two definitions of $f^!$ coincide under the identifications of $K(X)$ with $K(\xi(X))$ and of $K(Y)$ with $K(\xi(Y))$.

Another case where one can define a homomorphism (2.21), without Y necessarily nonsingular, is the case where the functor $F \mapsto f^!(F)$ from $\mathscr{F}(Y)$ to $\mathscr{F}(X)$ is exact, i.e. where the local rings of X are flat modules over the local rings of Y. It is enough then to put

$$f^!(\gamma_Y(F)) = \gamma_X(f^!(F)).$$

This case occurs in particular when X is a locally trivial fibre space over Y; to see this, one is reduced to the case of a product $X = Y \times T$. Note that one also verifies that in this case one has $f^!(\mathcal{O}_Z) = \mathcal{O}_{f^{-1}(Z)}$ if Z is an irreducible subvariety of Y, whence

$$f^!(\gamma_Y(Z)) = \gamma_X(f^{-1}(Z)) \tag{2.22}$$

if X is flat over Y. Without this latter condition, formula (2.22) becomes false in general, but one has the following analogue of Proposition 2.7.

Proposition 2.8. Let $f : X \to Y$ be a morphism of nonsingular varieties, and let Z be a cycle on Y. If Z is everywhere transverse to f, formula (2.22) is exact. If Z is homogeneous of codimension i and if $f^{-1}(Z)$ has no excessive component, then

$$f^!(\gamma_Y(Z)) = \gamma_X(f^{-1}(Z)) \mod K(X)^{i+1}. \tag{2.23}$$

The proof is analogous to that of Proposition 2.7.

Finally, let X and Y again be arbitrary algebraic spaces, and let f be a proper morphism from X to Y. For every coherent algebraic sheaf F on X, the direct image sheaf $f_*(F)$ is coherent algebraic, and the same holds more generally for the sheaves $R^i f_*(F)$ (cf. [3]). Recall that, by definition, $R^i f_*(F)$ is the sheaf on Y associated with the presheaf which sends an open U to the group $H^i(f^{-1}(U), F)$; in particular

$$R^0 f_*(F) = f_*(F), \qquad R^q f_*(F) = 0 \quad (q > \dim X), \tag{2.24}$$

and moreover

$$\Gamma(U, R^q f_*(F)) = H^q(f^{-1}(U), F) \tag{2.25}$$

if the open U is affine. According to §2, one defines a homomorphism from $K(X)$ to $K(Y)$, still denoted f_*, by the formula

$$f_*(\gamma(F)) = \sum_{q=0}^{\infty} (-1)^q \gamma(R^q f_*(F)). \tag{2.26}$$

Thus $K(X)$ becomes a covariant functor in X relative to proper morphisms. The relation $(fg)_* = f_* g_*$, just like the relation $(fg)^! = g^! f^!$ which we forgot to mention in its place, is a

particular case of the transitivity relations alluded to in §2 and which follow from the spectral sequences of composed functors.

Let us note the following formula, valid for Y nonsingular and particularly easy to verify when Y is also quasi-projective by taking for y the class of a vector bundle on Y:

$$f_*(x \cdot f^!(y)) = f_*(x) \cdot y \qquad (x \in K(X),\ y \in K(Y)). \tag{2.27}$$

In the particular case where the inverse images of the points of Y by f are finite and contained in an affine open,[12] one checks that the functor $F \mapsto f_*(F)$ is exact and that $R^q f_*(F) = 0$ for $q > 0$. In this case, therefore, $f_*(\gamma_X(F)) = \gamma_Y(f_*(F))$. In particular, if X is an algebraic subspace of Y and if i is the canonical injection of X into Y, then

$$i_*(\gamma_X(F)) = \gamma_Y(F), \tag{2.28}$$

where, in the second member, F is considered as a sheaf on Y which is zero outside X.

4. Some technical results

Let X be an algebraic space. We have seen in §3 that the projection p from $X \times k$ to X defines a homomorphism

$$p^! : K(X) \longrightarrow K(X \times k). \tag{2.29}$$

Proposition 2.9. The homomorphism (2.29) is surjective.

We argue by induction on $n = \dim X$. The assertion is trivial if $n < 0$; suppose $n \geq 0$. By Proposition 2.6, it is enough to verify that for every irreducible closed subset Y of $X \times k$, $\gamma(Y)$ is in the image of $p^!$. It is enough to continue the reasoning by supposing X irreducible. If $p(Y)$ is not dense in X, our assertion follows from the induction hypothesis. Otherwise, either $Y = X \times k$ and the assertion is trivial, or Y is a hypersurface in $X \times k$. Let U be an affine open of X formed of nonsingular points. If D is the divisor on $U \times k$ induced by Y, one knows by commutative algebra that it is linearly equivalent to a divisor of the form $D' \times k$, where D' is a divisor on U. Then

$$\gamma_{U\times k}(D) - \gamma_{U\times k}(D' \times k)$$

belongs to $K(U \times k)$[2] by the corollary to Proposition 2.10, and by what has already been proved this element is of the form $p^!(x')$ with $x' \in K(U)$. Thus

$$\gamma_{U\times k}(D) = p^!(x' + \gamma_U(D')),$$

but x' is the restriction of an element y of $K(X)$ by Proposition 2.10, and consequently the restriction of $\gamma_{X\times k}(D) - p^!(y)$ to $U \times k$ is zero. By Proposition 2.10, $\gamma_{X\times k}(D) - p^!(y)$ comes from an element of

$$K(X \times k - U \times k) = K(X' \times k), \qquad X' = X - U.$$

The proof is finished by applying the induction hypothesis again. QED.

Corollary. Let X be an algebraic space and let p be the projection from $X \times k^n$ to X. Then the homomorphism

$$p^! : K(X) \longrightarrow K(X \times k^n)$$

is surjective; it is even bijective if X is nonsingular.

[12] This latter condition is evidently a consequence of the first, something the author still did not know in 1957!

The first assertion is by induction on n. For the second, put $f(x) = (x, 0)$, whence $pf = 1$ and $f^! p^! = 1$, which proves that $p^!$ is injective.

Proposition 2.10. Let X be an algebraic space, let U be an open subset of X, and let Y be its complement in X. Then the sequence of natural homomorphisms

$$K(Y) \longrightarrow K(X) \longrightarrow K(U) \longrightarrow 0 \tag{2.30}$$

is exact.

N.B. Compare with Proposition 2.3.

The fact that $K(X) \to K(U)$ is surjective follows at once from Proposition 2.5, or also from Proposition 2.6. It remains to prove that if $x \in K(X)$ has zero restriction in $K(U)$, then it is in the image of $K(Y)$. For this, return to the explicit definition of $K(X)$ and $K(U)$. There are two coherent algebraic sheaves F and G on X with

$$x = \gamma_X(F) - \gamma_X(G)$$

and

$$\gamma_U(F) - \gamma_U(G) = 0.$$

This latter relation means that one can find coherent sheaves P' and Q' on U, provided with filtrations whose factors are pairwise isomorphic on U, and such that one has an isomorphism on U

$$P = F + Q' \longrightarrow Q = G + P'.$$

By Proposition 2.5, the sheaves P' and Q' are restrictions to U of coherent sheaves on X, denoted by the same letters; in the same way, their filtrations come from filtrations on the corresponding sheaves on X, by Lemma 2.11 below. One then has

$$\gamma_X(F) - \gamma_X(G) = (\gamma_X(P) - \gamma_X(Q)) + (\gamma_X(P') - \gamma_X(Q')).$$

Introducing the factors P_i and Q_i of the filtrations of P' and Q' on X, the last term is the sum of terms $\gamma_X(P_i) - \gamma_X(Q_i)$, where $P_i|_U \simeq Q_i|_U$. We are thus reduced to proving the following particular case: if R and S are coherent sheaves on X such that $R|_U \simeq S|_U$, then $\gamma_X(R) - \gamma_X(S)$ is in the image of $K(Y)$. Let T be the graph of the given isomorphism from $R|_U$ to $S|_U$; it is a coherent subsheaf of $(R \times S)|_U$, hence by Lemma 2.11 it is the restriction to U of a coherent subsheaf, again denoted T, of $R \times S$. Then

$$\gamma_X(R) - \gamma_X(S) = (\gamma_X(R) - \gamma_X(T)) - (\gamma_X(S) - \gamma_X(T)),$$

and it is enough to prove, for instance, that the first term of the second member is in the image of $K(Y)$. But the projection $u : T \to R$ is an isomorphism over U, hence

$$\gamma_X(R) - \gamma_X(T) = \gamma_X(\operatorname{Coker} u) - \gamma_X(\operatorname{Ker} u),$$

and $\operatorname{Ker} u$ and $\operatorname{Coker} u$ have support in Y.

It remains, then, to prove the following lemma.

Lemma 2.11. Let X be an algebraic space, let F be a coherent algebraic sheaf on X, let U be an open subset of X, and let G be a coherent subsheaf of $F|_U$. Then G is the restriction to U of a coherent subsheaf of F.

Let $\overline{G}$ be the subsheaf of F whose sections on an open V of X are the sections of F on V whose restriction to $U \cap V$ is a section of G. Everything amounts to proving that this sheaf $\overline{G}$ is coherent. For this one may suppose X and U affine, because U is a union of affine opens U_i

and $\overline{G}$ is then the intersection of the analogous sheaves $\overline{G|U_i}$. Then the coordinate ring of U is a ring of fractions A_S of the coordinate ring A of X, and one immediately verifies that if F is defined by the A-module M, then $F|_U$ is defined by the A_S-module

$$M_S = A_S \otimes_A M.$$

The subsheaf G of $F|_U$ is then defined by a submodule N' of M_S, and one sees that $\overline{G}$ is precisely defined by the submodule N of M, inverse image of N' under the canonical homomorphism from M to M_S. QED.

Corollary 1 to Proposition 2.10. Let $x \in K(X)$. Then $x \in K(X)^i$ if and only if there exists a closed subset Y of X, of dimension $\leq n - i$, such that the restriction of x to $X - Y$ is zero.

Corollary 2. Suppose X is connected of dimension n, with no singularities in dimension $n - 1$. If D and D' are two linearly equivalent divisors on X, then

$$\gamma(D) = \gamma(D') \mod K(X)^2.$$

By Corollary 1, one may suppose X nonsingular. The corollary then follows from the general formula

$$\gamma(D) = 1 - \gamma(L(-D)) \mod K(X)^2, \tag{2.31}$$

where $L(-D)$ denotes the rank-one vector bundle defined by the divisor $-D$. To verify (2.31), one writes it in the form

$$\gamma(L(-D)) = 1 - \gamma(D) \mod K(X)^2, \tag{2.31 bis}$$

and notes that the two members are homomorphisms from the group of divisors to the multiplicative group of invertible elements of the ring $K(X)/K(X)^2$. Here $K(X)^1/K(X)^2$ has square zero, which is very easily verified on the explicit formula (2.15), using Corollary 1. Thus one is reduced to the case where D is an irreducible hypersurface. More generally, if D is a hypersurface whose components are disjoint, one has the equality

$$\gamma(D) = 1 - \gamma(L(-D)), \tag{2.32}$$

as follows from the exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X(L(-D)) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

Theorem 2.12. Let X be a nonsingular quasi-projective variety, and let Z and Z' be two homogeneous cycles of codimension p on X which are linearly equivalent. Then

$$\gamma_X(Z) = \gamma_X(Z') \mod K(X)^{p+1}. \tag{2.33}$$

By hypothesis, one can find a cycle Y of dimension $n - p + 1$ on $X \times k$ (n being the dimension of X), and two points $a, a' \in k$, such that

$$Z = Y \cdot (X \times a), \qquad Z' = Y \cdot (X \times a').$$

By formula (2.23), one then has

$$\gamma_X(Z) = i_a^!(\gamma_{X\times k}(Y)) \mod K(X)^{p+1},$$

and an analogous formula for $\gamma_X(Z')$. It is therefore enough to prove that $i_a^! = i_{a'}^!$, which follows at once from Proposition 2.9. QED.

Corollary 1. The filtration of $K(X)$ is compatible with its ring structure, i.e. one has

$$K(X)^i K(X)^j \subset K(X)^{i+j}$$

for all integers $i, j \geq 0$. Moreover there is a natural surjective homomorphism of graded rings

$$\varphi : A(X) \longrightarrow \mathrm{Gr}(K(X)), \tag{2.34}$$

where the second member denotes the graded ring associated with $K(X)$, provided with the filtration $(K(X)^i)_{i\geq 0}$, defined by the condition

$$\varphi(\ell(Z^{n-p})) = \gamma(Z^{n-p}) \mod K(X)^{p+1}.$$

This corollary follows immediately from Theorem 2.12, Propositions 2.6 and 2.7, and Chow's theorem 2.1, by induction on the integer $i + j$.

Corollary 2. Let X and Y be two nonsingular quasi-projective varieties and let f be a morphism from X to Y. Then the homomorphism $f^!$ from $K(Y)$ to $K(X)$ is compatible with the filtrations, and one has a commutative diagram

$$\begin{array}{ccc} A(Y) & \xrightarrow{\varphi_Y} & \mathrm{Gr}(K(Y)) \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle \mathrm{Gr}(f^!)} \\ A(X) & \xrightarrow[\varphi_X]{} & \mathrm{Gr}(K(X)). \end{array}$$

Using the graph of f, decompose f as the product of an immersion of X into $X \times Y$ and of a projection of $X \times Y$ onto Y. For the latter the corollary is trivial, and for an immersion one proceeds as for Corollary 1, using Proposition 2.8 in place of Proposition 2.7.

Proposition 2.13 ("cellular decomposition"). Let P be an algebraic space, irreducible or not, provided with an increasing family

$$P^0 \subset P^1 \subset \cdots \subset P^n = P$$

of closed subsets, such that for every i with $0 \leq i \leq n$, the connected components of $P^i - P^{i-1}$ are algebraic spaces $V_{i,\alpha}$ isomorphic to k^i; put $P^{-1} = \varnothing$. Then:

a) For every algebraic space X, the natural map from $K(X) \otimes K(P)$ to $K(X \times P)$, deduced from the tensor product of sheaves on X and P, is surjective.

b) The $\gamma_P(V_{i,\alpha})$ form a system of generators of the group $K(P)$.

The proof is by induction on n, using the corollary to Proposition 2.9 and Proposition 2.10. N.B. Unfortunately, I do not know whether the $\gamma_P(V_{i,\alpha})$ form a basis of $K(P)$ or whether the homomorphism considered in a) is bijective. We shall not need to know this below.

Theorem 2.14. Let x_i $(1 \leq i \leq q)$ and E_j $(1 \leq j \leq r)$ be letters, and let n_j $(1 \leq j \leq r)$ be positive integers. Consider the symbols $\lambda^k(x_i)$ with $k \geq 1$ and $\lambda^k(E_j)$ with $1 \leq k \leq n_j$ as indeterminates, and let R be a polynomial with integral coefficients in these indeterminates. Suppose that, for every graded ring A, when one substitutes in R elements ξ_i for the x_i and elements

$$\eta_j = [n_j, 1 + \eta_j^{(1)} + \cdots + \eta_j^{(n_j)}]$$

of $\widetilde{A}$ for the E_j, subject to the restrictions $\varepsilon(\xi_i) = 0$ and $\eta_j^{(k)} = 0$ for $k > n_j$, one has

$$R((\lambda^k(\xi_i)), (\lambda^k(\eta_j))) \in \widetilde{A}^p.$$

Then, if X is an irreducible nonsingular quasi-projective variety, for arbitrary elements x'_i of $K(X)$[1] and vector bundles E'_j of dimension n_j on X $(1 \leq i \leq q,\ 1 \leq j \leq r)$, one has

$$R((\lambda^k(x'_i)), (\lambda^k(\gamma(E'_j)))) \in K(X)^p. \qquad (*)$$

Proof. By §2, the x'_i are themselves differences of elements of the form $\gamma(E)$, with E a vector bundle. On the other hand, assuming X locally closed in a projective space P, it follows from §2 and Proposition 2.5 that, for every $F \in \mathscr{F}(X)$, the sheaf $F(n)$ is generated by its sections for n sufficiently large. If F is defined by a vector bundle E, this proves that $E(n)$ is the inverse image of the canonical vector bundle on a Grassmannian variety G by a morphism from X to G. Considering the corresponding morphism from X to $P \times G$, one sees that F is the inverse image of a vector bundle on $P \times G$. Applying this to all vector bundles which occur, and using Corollary 2 to Theorem 2.12, we are reduced to the case where X is a product of Grassmannians. In this case the conclusion will follow from the following lemma.

Lemma 2.15. Suppose that X is a product of Grassmannians. Then one can find a λ-homomorphism from $K(X)$ to a torsion-free λ-ring of the form $\widetilde{A}$, where A is a graded ring, compatible with filtrations, and such that the corresponding homomorphism $\mathrm{Gr}(K(X)) \to \mathrm{Gr}(A)$ is injective.

To prove the lemma, we shall use Chow's theory of Chern classes. If one could prove the lemma, or Theorem 2.14 directly, one would deduce a purely "sheaf-theoretic" theory of Chern classes; cf. §5.

We shall consider the homomorphism $\widetilde{C}$ from $K(X)$ to $\widetilde{A}(X)$ and put $A = A(X)$. It remains to prove that the homomorphism $\psi = \mathrm{Gr}(\widetilde{C})$ from $\mathrm{Gr}(K(X))$ to $\mathrm{Gr}(A(X)) \simeq A(X)$ is injective. The fact that $\widetilde{C}$ is compatible with filtrations is true for every nonsingular quasi-projective variety and follows easily from the corollary to Proposition 2.3. On the other hand, since X is a product of Grassmannians G_j, there are vector bundles E_j on X, inverse images of the canonical bundles on the factors. We shall use the following known facts:

a) On X, linear equivalence of cycles is equivalent to numerical equivalence, and $A(X)$ is a group of finite rank.

b) The $c^i(E_j)$ generate the ring $A(X)$.

From a) it follows that

$$\varphi : A(X) \longrightarrow \mathrm{Gr}(K(X))$$

is injective, and hence bijective, by the following lemma.

Lemma 2.15. Let X^n be a nonsingular projective variety and let Z^{n-p} be a cycle on X such that $\varphi(Z) \in \mathrm{Gr}^p(K(X))$ is zero. Then Z^{n-p} is numerically equivalent to zero.

To prove this lemma, it suffices to use the additive function

$$\chi(F) = \sum_{i \geq 0} (-1)^i \dim H^i(X, F)$$

and the fact that this function is equal to 1 for $F = \mathcal{O}_{(x)}$ for every $x \in X$.

Thus, in the case under consideration, $A(X)$ and $\mathrm{Gr}(K(X))$ have the same finite rank in every degree, and $\mathrm{Gr}(K(X))$ is torsion-free, since this is true of $A(X)$. From assertion b) above it follows, in a formal way, that the map $\widetilde{C}$ from $K(X)$ to $\widetilde{A}(X)$ is surjective modulo finite groups. Since $\mathrm{Gr}(K(X))$ and $A(X)$ have the same finite rank in every degree, it then follows easily that the homomorphism $\mathrm{Gr}(\widetilde{C})$ from $\mathrm{Gr}(K(X))$ to $\mathrm{Gr}(\widetilde{A}(X))$ is bijective modulo finite

groups in every degree; one uses induction on the degree. Since $\mathrm{Gr}(K(X))$ is torsion-free, this homomorphism $\mathrm{Gr}(\widetilde{C})$ is therefore injective. QED.

N.B. The homomorphism $\widetilde{C}$ is not surjective in general, even if X is a projective space of dimension 2.

5. Sheaf-theoretic definition of Chern classes. Application to the study of injection morphisms

Theorem 2.16. Let X be a nonsingular quasi-projective variety. Then, for every $x \in K(X)$ and every integer $i \geq 0$, the element $\gamma^i(x - \varepsilon(x))$ is of filtration $\geq i$. Let $c^i(x) \in \mathrm{Gr}^i(K(X))$ be the class of this element modulo $K(X)^{i+1}$. If one puts

$$\widetilde{c}(x) = \left[\varepsilon(x), 1 + \sum_{i \geq 0} c^i(x)\right] \in \mathrm{Gr}\widetilde{(K(X))},$$

then $\widetilde{c}$ satisfies conditions analogous to those of Theorem 2.2.

The first assertion follows from Theorem 2.14, as does the fact that $x \mapsto \widetilde{c}(x)$ is a homomorphism of λ-rings; formula (1.25) implies $\gamma_t(x + y) = \gamma_t(x)\gamma_t(y)$, which already shows that $\widetilde{c}$ is an additive homomorphism. The functoriality of $\widetilde{c}$ is verified immediately, and (2.8) follows from (2.31). QED.

Corollary 1. One has

$$c^i(x) = \varphi(c^i(x)).$$

This follows from a uniqueness theorem for Chern classes, proved as in Hirzebruch by passage to flag varieties. One only has to show that, if Y is a flag variety associated with a vector bundle on X, then $\mathrm{Gr}(K(X)) \to \mathrm{Gr}(K(Y))$ is injective; this is verified by the explicit determination of $K(Y)$, with all its structures, from $K(X)$. I shall not give the details here.

Corollary 2. Suppose that for every element $Z \in A(X)$ there exists a morphism f from X to a product X' of Grassmannians such that Z is in the image of f^*. Then the canonical homomorphisms

$$\varphi : A(X) \longrightarrow \mathrm{Gr}(K(X)), \qquad \psi : \mathrm{Gr}(K(X)) \longrightarrow A(X)$$

satisfy, in dimension i, the relations

$$\psi\varphi = (-1)^{i-1}(i-1)!\,1, \qquad \varphi\psi = (-1)^{i-1}(i-1)!\,1. \tag{2.35}$$

Since φ is surjective, it is clearly enough to prove the first formula, and for this one may suppose that X is a product of Grassmannians. Since then φ is injective, it is enough to prove the second formula (2.35), which by Corollary 1 is written

$$\gamma^i(x - \varepsilon(x)) = (-1)^{i-1}(i-1)!\,x \mod K(X)^{i+1}$$

for every $x \in K(X)^i$. Since ψ is injective, it is enough to prove the analogous relation for $\widetilde{C}(x) \in A(X)^i$, and this is then a consequence of (1.22 bis).

Remark. Corollary 2 makes plausible the validity of (2.35) in all cases; there should exist a direct proof of it. In every case one proves easily the second formula (2.35) modulo the torsion subgroup of $\mathrm{Gr}(K(X))$, and in characteristic 0 this same formula is a consequence of the Riemann–Roch theorem. The interest of (2.35), when true, lies in the fact that it shows that φ is injective modulo torsion groups, perhaps always zero. Thus one does not lose much by

defining Chern classes as elements of $\mathrm{Gr}(K(X))$. If one tensors with $\mathbb{Q}$, as will be done later for the Riemann–Roch theorem, the two definitions of Chern classes would then be identical.

Let X still be nonsingular and quasi-projective. Let in addition Y be a nonsingular subvariety of X, and let i be the injection of Y into X. For i_* and $i^!$ one then has the following formulas:

(i) $i^!$ is a homomorphism of λ-rings.

(ii) $i_*(x \cdot i^!(y)) = i_*(x) \cdot y$, whence in particular

$$i_* i^!(y) = \xi y, \qquad \text{with } \xi = i_* i^!(1) = \gamma_X(N).$$

(iii) $i^! i_*(x) = x \cdot \lambda_{-1}(N)$, whence in particular

$$i^!(\xi) = i^! i_*(1) = \lambda_{-1}(N).$$

(iv) $i_*(x) i_*(x') = i_*(x x' \lambda_{-1}(N))$.

Here, by definition, N is the class in $K(Y)$ of the conormal bundle of Y in X. Formulas (i) and (ii) are true for arbitrary morphisms, not necessarily injections, and (iii) and (iv) are verified from the definition (2.15) and the analogous formula for inverse images, taking x and x' to be classes of vector bundles and using the fact that

$$\mathrm{Tor}(\mathcal{O}_Y, \mathcal{O}_Y) = \Lambda(N). \tag{2.36}$$

Formula (iv) expresses that i_* is a homomorphism of rings if one introduces on $K(Y)$ a new multiplicative structure, by the definition

$$x \cdot_N x' = x x' \lambda_{-1}(N)$$

(cf. Chapter I, §4). It is completed by the formula

(v) $i_*(\lambda^p(N, x)) = \lambda^p(i_*(x))$ for $p \geq 1$,

which means that i_* defines a λ-homomorphism from $K(Y)_N$ into $K(X)$. This formula (v), certainly true in all cases, is for the moment proved only in characteristic 0, because it follows from the conjunction of Theorems 1.4 and 2.8.

There are analogous formulas for i_* and i^*:

(i bis) i^* is a homomorphism of graded rings.

(ii bis) $i_*(x \cdot i^*(y)) = i_*(x) y$, whence in particular

$$i_* i^*(y) = \xi' y, \qquad \text{with } \xi' = i_*(1).$$

(iii bis) $i^* i_*(x) = x \cdot (-1)^q N^q$, whence in particular

$$i^*(\xi') = (-1)^q N^q.$$

(iv bis) $i_*(x) i_*(x') = i_*(x x' (-1)^q N^q)$.

(v bis) i_* increases degrees by q units.

In these formulas, q is the codimension of Y in X, and one puts $N^i = c^i(N)$. Only formulas (iii bis) and (iv bis) require proof. I do not know whether they are well known, or even whether they are true, and I confine myself to noting that they follow by reduction from formulas (ii) and (iv), provided one works in $\mathrm{Gr}(K(Y))$ and $\mathrm{Gr}(K(X))$ and not in $A(Y)$ and $A(X)$. One then uses the fact that i_* and i^* come by passage to the quotient from i_* and $i^!$, noting that i_* shifts the filtration by q units.

All universal formulas relating i_* and $i^!$, respectively i_* and i^*, are formal consequences of the preceding formulas, as one can convince oneself without great difficulty.

Formula (v) gives

$$i_*(\gamma^p(N,x)) = \gamma^p(i_*(x)).$$

The conjunction of Theorem 2.14 and Proposition 1.5 shows that $\gamma^p(N,x)$, for $p = q + j$, is of filtration $\geq j$. If one denotes by $G_{q,j}(N,x)$ its class in $\mathrm{Gr}^j(K(Y))$, Theorem 2.16 shows that

$$i_*(G_{q,j}(N,x)) = c^p(i_*(x)).$$

Finally, Proposition 1.5 joined with Theorem 2.14 permits one to express $G_{q,j}(N,x)$ as a polynomial with integral coefficients in the $c^i(N)$, the $c^i(x)$, and $\varepsilon(x)$. One deduces the following theorem.

Theorem 2.17. Suppose the field k has characteristic 0, and let $i : Y \to X$ be an injection morphism of nonsingular quasi-projective varieties. Then one has formulas (i) to (v), and moreover, for every integer $j \geq 0$, the formula

$$c^{q+j}(i_*(x)) = i_*\left(\left(\frac{\widetilde{c}(x) * \lambda_{-1}(\widetilde{c}(N))}{(-1)^q N^q}\right)^{(j)}\right). \tag{2.37}$$

The second member of formula (2.37) is an abbreviated way of denoting the polynomial in the components of $\widetilde{c}(x)$ and $\widetilde{c}(N)$ described in Proposition 1.5.

Corollary. Under the preceding conditions one has

$$\mathrm{ch}(\widetilde{c}(i_*(x))) = i_*\left(\mathrm{ch}(\widetilde{c}(x))\,\mathfrak{C}(\check{N})^{-1}\right), \tag{2.38}$$

where ch is the Chern homomorphism defined by formula (1.26).

It is enough to use (2.37) written in the form

$$\widetilde{c}(i_*(x)) = 1 + i_*\left(\frac{\widetilde{c}(x) * \lambda_{-1}(\widetilde{c}(N)) - 1}{(-1)^q N^q}\right), \tag{2.37 bis}$$

the explicit formula (1.29), formula (iv bis), and finally (1.30).

One may write (2.38) in the form

$$\mathrm{ch}_X(F) = i_*\left(\mathrm{ch}_Y(F)\,\mathfrak{C}(T_{X/Y})^{-1}\right), \tag{2.38 bis}$$

where F is a coherent algebraic sheaf on Y (considered in the first member as a sheaf on X which is zero outside Y), where ch_X and ch_Y are the Chern homomorphisms taken respectively on X and Y, and where $T_{X/Y}$ is the normal bundle of Y in X.

Remarks. Contrary to (2.38), formula (2.37) is a "formula without denominators". It has been proved, even in characteristic 0, only by placing oneself in $\mathrm{Gr}(K(X))$ and not in $A(X)$, which may not be the same thing. It is certainly true in every characteristic and in $A(X)$. Let us note that one verifies at once that the images of the two members by i^* are the same; hence, applying i_*, the product of their difference by N^q is zero. This suggests a completely different

method of proof, by trying to obtain Y as a transverse inverse image of a subvariety Y' in a variety X' such that, in dimension $\leq n = \dim X$, the class of Y' in $A(X')$ is not a divisor of 0. Unfortunately one cannot hope for X' to be a Grassmannian or something close to it.

Note that, if torsion elements are neglected, formula (2.38) is equivalent to (2.37).

6. The Riemann–Roch theorem

Its statement is as follows: let f be a proper morphism from a non-singular quasi-projective variety Y into a variety X of the same kind. Then one has

$$f_*\big(\mathrm{ch}_Y(x)\,\mathfrak{C}(Y)\big) = \mathrm{ch}_X\big(f_!(x)\big)\,\mathfrak{C}(X). \tag{2.39}$$

Both sides are in $A(X)\otimes\mathbb{Q}$, and $\mathfrak{C}(X) = \mathfrak{C}(T_X)$.

This formula makes it possible to compute, up to torsion, the Chern classes of $f_!(x)$ from the Chern classes of x, those of T_X and T_Y, and the homomorphism f_*. Or, equivalently, it makes it possible to determine the homomorphism

$$f_! : K(Y)\otimes\mathbb{Q} \longrightarrow K(X)\otimes\mathbb{Q}$$

when these groups are identified, by means of the homomorphisms ch_Y and ch_X, respectively with $A(Y)\otimes\mathbb{Q}$ and $A(X)\otimes\mathbb{Q}$ (provided that formula (2.35) is true), from f_* and the Chern classes of T_X and T_Y.

I can prove formula (2.39) only in characteristic 0, and as an equality in $\mathrm{G}(K(X))\otimes\mathbb{Q}$. For this, since Y is locally closed in a projective space P, f is decomposed as the product of an injection of Y into $P\times X$ and a projection morphism from $P\times X$ to X, and (2.39) is proved for each of these two morphisms.

For an injection morphism, one sees immediately that formulas (2.38) and (2.39) are equivalent. The characteristic 0 hypothesis enters only through Theorem 2.8, which permits one to express the classes $\lambda^i(\gamma(F))$, for a coherent algebraic sheaf F, by local cohomological invariants of F.

For a projection morphism $f : P\times X \to X$, Proposition 2.13 permits us to suppose that x is of the form $y\otimes z$, with $y\in K(P)$ and $z\in K(X)$. Then the Künneth formula gives

$$f_!(x) = \chi(y)\,z,$$

where, for a coherent algebraic sheaf F on P, one puts

$$\chi(F) = \sum_{i\geq 0}(-1)^i \dim H^i(P,F).$$

One is therefore at once reduced to proving the formula

$$\chi(F) = \pi_r\big(\mathrm{ch}_p(F)\,\mathfrak{C}(P)\big), \tag{2.40}$$

where π_r denotes the component of degree $r = \dim P$. By Proposition 2.13, one may suppose that $F = \mathcal{O}_Q$, where Q is a linear subvariety of P. Then $\chi(F) = 1$, since this is the arithmetic genus of Q; it remains only to see that the second member of (2.40) is 1.

Indeed, if V is a hyperplane of P and if one puts

$$\xi = \gamma_P(\mathcal{O}_V), \qquad L = \ell(V),$$

then

$$1-\xi = \gamma_P(E(-L)),$$

where $E(-L)$ is the vector bundle defined by $-L$. Hence

$$C_p(1-\xi) = [1, 1-L],$$

and consequently

$$\mathrm{ch}_p(1-\xi) = e^{-L}, \qquad \mathrm{ch}_p(\xi) = 1 - e^{-L},$$

and therefore

$$\mathrm{ch}_p(\xi^p) = (1-e^{-L})^p.$$

If p is the dimension of Q in P, one has $\xi^p = \gamma_P(Q)$; hence

$$\mathrm{ch}_p(F)\,\mathfrak{C}(P) = \left(L/(1-e^{-L})\right)^{r+1}(1-e^{-L})^p.$$

It remains to verify that the coefficient of L^r in this formal series in L is 1. This follows from the characteristic property of the formal series $L/(1-e^{-L})$, namely that the coefficient of L^q in $(L/(1-e^{-L}))^{q+1}$ is 1 (put $q = r+1-p$).

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Proof of the Riemann–Roch theorem

(Grothendieck, 1 November 1957)

Reminder of notation

These are the notations of the "Report on Riemann–Roch," cited as RRR below.

All varieties considered are non-singular and quasi-projective, that is, isomorphic to a locally closed subvariety of a projective space. The base field is algebraically closed, of arbitrary characteristic.

If X is such a variety, $K(X)$ denotes the abelian group generated by the classes of coherent sheaves on X, modulo the identification coming from an extension into a sum; the alternating sum of Tor makes $K(X)$ a ring. If F is a sheaf, respectively E a vector bundle, respectively Y a subvariety of X, one denotes by $\gamma_X(F)$, respectively $\gamma_X(E)$, respectively $\gamma_X(Y)$, the element of $K(X)$ corresponding to F, respectively to E, respectively to $\mathcal{O}_Y$. In fact, one often writes simply F, E instead of $\gamma_X(F)$.

Let $A(X)$ be the graded ring of classes of cycles on X for rational equivalence (cf. Chow); $A^i(X)$ is formed by the cycles of codimension i. If E is a vector bundle, $C(E) = \sum C^i(E)$, with $C^i(E) \in A^i(X)$, is its Chern class (defined as inverse image of Schubert cycles). By linearity one deduces the classes $C^i(y)$, for all $y \in K(X)$. If

$$C(y) = \prod(1 + a_i),$$

one sets, following Hirzebruch,

$$\mathrm{ch}(y) = \sum e^{a_i}, \qquad T(y) = \prod \frac{a_i}{1-e^{-a_i}}.$$

These are elements of $A(X) \otimes \mathbb{Q}$. When y is the tangent bundle of X, one writes $T(X)$ instead of $T(y)$.

If G is a vector bundle, $\lambda^p G$ denotes its p-th exterior power and

$$\lambda_{-1}(G) = \sum (-1)^p \lambda^p(G);$$

this is an element of $K(X)$.

If f is a proper regular map from a variety Y into a variety X, one defines homomorphisms

$$f^* : A(X) \longrightarrow A(Y), \qquad f_* : A(Y) \longrightarrow A(X).$$

The first is homogeneous of degree 0 and multiplicative. The second is homogeneous of degree $\dim X - \dim Y$. One also defines homomorphisms

$$f^* : K(X) \longrightarrow K(Y), \qquad f_! : K(Y) \longrightarrow K(X).$$

If F is a coherent sheaf on X, $f^*(F)$ is, by definition, the alternating sum of the $\mathrm{Tor}_p^{\mathcal{O}_X}(\mathcal{O}_Y, F)$. For vector bundles, this is simply the inverse-image operation.

If G is a coherent sheaf on Y, $f_*(G)$ is coherent on X. If $R^p f_*$ denotes the p-th derived functor of the functor f_* so defined, one has, by definition,

$$f_!(G) = \sum (-1)^p R^p f_*(G).$$

Riemann–Roch theorem

Let $f : Y \to X$ be a proper morphism (regular map) from a non-singular quasi-projective variety Y into another such variety X, and let $y \in K(Y)$. Then

$$f_*(\mathrm{ch}(y)T(Y)) = \mathrm{ch}(f_!(y))T(X) \quad \text{in } A(X) \otimes \mathbb{Q}. \tag{1}$$

Proof (abridged).

Lemma 1. Consider proper morphisms of varieties

$$Z \xrightarrow{g} Y \xrightarrow{f} X.$$

If the theorem R.R. is true for f and for g, it is true for fg. If it is true for fg and for g, it is true for f. If f_* is injective, and if the theorem R.R. is true for fg and for f, it is true for g.

This is trivial. In fact, when one supposes R.R. true for fg and g, one must still suppose that $g_!$ is surjective in order to be able to assert that R.R. is true for f.

Since a proper morphism is composed of an injection and a projection $X \times P \to X$, where P is a projective space, one is reduced to treating these two cases separately. The second, as was seen above, is reduced to the standard calculation, by means of Proposition 2.13 ("cellular decomposition") of RRR. Thus suppose $Y \subset X$, with injection morphism i. Formula (1) becomes

$$\mathrm{ch}(i_!(y)) = i_*(\mathrm{ch}(y)T(E)^{-1}), \tag{2}$$

where $E = T_{X/Y}$ is the normal bundle of Y in X.

Lemma 2. Suppose that there exists on X a vector bundle G of rank $p = \mathrm{codim}_X Y$, such that

$$\gamma_X(Y) = i_*(1) = \lambda_{-1}(G), \qquad i^*(G) = E, \qquad (-1)^p C^p(G) = Y. \tag{3}$$

If then

$$y = i^*(x), \qquad x \in K(X), \tag{4}$$

formula (2) is valid.

The proof is immediate from the fact that ch is a homomorphism of rings, and that

$$\mathrm{ch}(\lambda_{-1}(G)) = (-1)^p C^p(G)\, T(G)^{-1}. \tag{5}$$

Corollary. R.R. is true if Y is a hypersurface and if y is of the form $i^*(x)$, with $x \in K(X)$.

Lemma 3. Suppose that $X = Y \times P$, where P is a projective space, and that i is the injection $u \mapsto (u, v_0)$, $v_0 \in P$. Then i_* is injective and R.R. is true.

This is pure standard calculation.

Suppose again that X, Y, y are arbitrary, always with $Y \subset X$. Let X' be the variety obtained by blowing up X along Y, and let Y' be the hypersurface inverse image of Y. One has a diagram of maps

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X. \end{array}$$

The inverse image of E in Y' admits a subbundle of rank 1, whose dual is denoted L. Moreover L^{-1} is also the restriction to Y' of the line bundle $L(Y')$ defined on X' by the divisor Y'. Put $F = E/L^{-1}$. Elementary computations give the following lemma.

Lemma 4. One has the formulas, where $p = \mathrm{codim}_X Y$:

$$f^* i_!(y) = j_*(\lambda_{-1}(\check{F})y), \tag{6}$$

$$\lambda_{-1}(\check{F}) \equiv \gamma^{p-1}(E)\lambda^p(\check{E}) \pmod{1 - L}, \tag{7}$$

$$f_*(1) = 1, \quad \text{hence } f_* f^* = 1, \tag{8}$$

$$f^* i_*(\alpha) \equiv j_*(\alpha C^{p-1}(F)) \pmod{\mathrm{Ker}\, f_*} \quad \text{for all } \alpha \in A(Y). \tag{9}$$

To simplify notation, $K(Y)$ has been made to operate on $K(Y')$, and $A(Y)$ on $A(Y')$, by the homomorphisms g^*. In formula (7), it seems that $\gamma^{p-1}(E)$ denotes $\lambda^{p-1}(E-2)$, that is,

$$\sum_{i+j=p-1} (\cdots)\lambda^i(E);$$

the congruence takes place in $K(Y')$.

Formula (6) follows from the equality

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y(v), \mathcal{O}_Y(v')) = \lambda^i \check{F}(v') \quad (v' \in Y',\ v = g(v')). \tag{6 bis}$$

Formula (7) follows from the formulas

$$g_*(L^{-i}) = 0 \quad \text{for } 0 < i \leq p-1, \qquad g_*(1) = 1, \tag{10}$$

themselves consequences of Künneth and of the cohomology of projective spaces.

Formula (8) is trivial. Finally, (9) follows from $g_*(C^{p-1}(F)) = 1$, which is deduced, for example, from the immediate formulas

$$g_*(\xi^i) = 0 \quad \text{if } 0 \leq i < p-1, \qquad g_*(\xi^{p-1}) = 1, \tag{10 bis}$$

with $\xi = C^1(L) \in A^1(Y')$. In fact, (9) is very likely an identity, and not merely a congruence. The unnecessarily complicated formula (7) may be replaced, for example, by VII 4.1.

From formula (7) it follows at once that $\lambda_{-1}(\check{F}) \in j^*(K(X'))$, provided that

$$y\,\gamma^{p-1}(E)\lambda^p(\check{E}) \in i^*(K(X)). \tag{11}$$

Suppose this condition (11) satisfied. By the corollary to Lemma 2, R.R. is true for $(Y', X', y\lambda_{-1}(\check{F}))$. On the other hand, to prove (2), formula (8) reduces the matter to proving that

$$f^*\,\mathrm{ch}(i_!(y)) \equiv f^*\big(i_*(\mathrm{ch}(y)T(E)^{-1})\big) \pmod{\mathrm{Ker}\, f_*}. \tag{12}$$

This last formula is verified immediately, using (6), (9), R.R. for $(Y', X', y\lambda_{-1}(\check{F}))$, and the formula

$$\mathrm{ch}(\lambda_{-1}(\check{F})) = C^{p-1}(F)T(F)^{-1} \tag{13}$$

(cf. formula (5)).

Thus R.R. is proved every time $y\,\gamma^{p-1}(E) = 0$, in particular every time $p - 1 > \dim Y$, since $\gamma^{p-1}(E)$ is of filtration $\geq p-1$ in $K(Y)$, filtered by the codimension of supports; that is, every time

$$\dim Y < \frac{\dim X - 1}{2}.$$

One is reduced to this case by combining Lemma 1 and Lemma 3, taking for P a projective space of sufficiently large dimension, for example $r > \dim X + 1$. This proves the assertion.

Remark. We have used the fact that $\gamma^{p-1}(E)$ was of filtration $\geq p-1$. One can prove this fact directly, without passing through Grassmannians as in RRR, 2.16, by studying the projective bundle associated to E. This method more generally gives Theorem 2.14 of RRR.

Extract from a letter of Grothendieck

"Morally," the new method rests on the determination of $K(X)$ and $A(X)$ when X is either the projective bundle associated to a vector bundle, or a variety obtained by blowing up a non-singular subvariety. In this second case I have not made the complete determination; a small equality is still missing. With the notation of the short note, one must prove formula (9), but as an equality:

$$f^*(i_*(\alpha)) = j_*(\alpha C^{p-1}(F)), \qquad \alpha \in A(Y). \tag{9}$$

One observes that the two members have the same image by f_*, and the same image by j^*, so that the question is equivalent to

$$\mathrm{Ker}\, f_* \cap \mathrm{Ker}\, j^* = 0. \tag{9 bis}$$

In any case, (9) is true after reducing to $G(K(X'))$ (which destroys a little torsion, at most), for it then follows from (6); there is therefore hardly any doubt that it is true.

I also point out that equality (9) follows very easily from (6), without passing through the troublesome (6 bis).

P.C.C.J-P.S.

Exposé I. Generalities on finiteness conditions in derived categories

by L. Illusie

0. Introduction

The object of Exposés I to IV is to develop, with the degree of generality appropriate in this Seminar, the formalism of finiteness conditions in derived categories of ringed topoi.

As was said in Exposé 0, the need to define, on arbitrary schemes, "Grothendieck groups" possessing good variance properties forced one to generalize and to soften the finiteness notions used up to now.

A classical notion such as that of a coherent sheaf on a ringed space $(X, \mathcal{O}_X)$ makes sense only when $\mathcal{O}_X$ is coherent. One might think of replacing it by the notion of finite presentation. But this has the inconvenience that the kernel of an epimorphism of modules of finite presentation is no longer, in general, of finite presentation. A satisfactory notion is reached by observing that, when $\mathcal{O}_X$ is coherent, a coherent sheaf F is not only of finite presentation but of n-finite presentation for every $n \in \mathbb{N}$, where n-finite presentation means that locally there exists an exact sequence

$$L_n \longrightarrow L_{n-1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow F \longrightarrow 0,$$

with the L_i free of finite type. When $\mathcal{O}_X$ is no longer assumed coherent, say that an $\mathcal{O}_X$-module F is *pseudo-coherent* if it satisfies the preceding condition, that is, if it is of n-finite presentation for every n. Then the notion of pseudo-coherence has, with respect to short exact sequences, the same stability property as coherence: if two terms of a short exact sequence of $\mathcal{O}_X$-modules are pseudo-coherent, then so is the third.

The preceding notion is easily generalized to complexes of sheaves. A complex F of $\mathcal{O}_X$-modules is said to be *pseudo-coherent* if it is n-pseudo-coherent for every $n \in \mathbb{Z}$, where n-pseudo-coherent means that there locally exists a quasi-isomorphism $L \to F$, with L a complex bounded above and with components free of finite type in degrees $> n$. For a complex F reduced to degree 0, this is the same as saying that F is pseudo-coherent as a complex or as a module.

The notion of pseudo-coherent complex has excellent stability properties, described in Exposé I, §§1 and 2. First of all it is stable under isomorphism in the derived category $\mathrm{D}(X)$ of the category of $\mathcal{O}_X$-modules; better still, if two objects of a distinguished triangle of $\mathrm{D}(X)$ are pseudo-coherent, then so is the third. In other words, the full subcategory $\mathrm{D}(X)_{\mathrm{coh}}$ of $\mathrm{D}(X)$ formed by pseudo-coherent complexes is a triangulated subcategory. Moreover, pseudo-coherence is preserved under inverse image and derived tensor product. In the case where $\mathcal{O}_X$ is coherent, to say that a complex F is pseudo-coherent simply means that F has locally bounded-above cohomology and that all the $H^i(F)$ are coherent.

If $\mathcal{O}_X$ is a sheaf of regular local rings, every coherent $\mathcal{O}_X$-module F admits locally a finite resolution to the left by finite-type free modules, that is, there exists locally an exact sequence

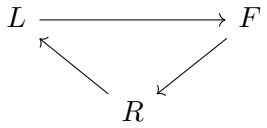
$$0 \longrightarrow L_n \longrightarrow \cdots \longrightarrow L_0 \longrightarrow F \longrightarrow 0,$$

where the L_i are free of finite type. It is therefore natural, when $\mathcal{O}_X$ is an arbitrary sheaf of local rings, to study the modules which have the preceding property. Such modules are called *perfect*. More generally, a complex F of $\mathcal{O}_X$-modules is said to be *perfect* if there exists locally a quasi-isomorphism $L \to F$, where L is a bounded complex with components free of finite type. It is shown in Exposé I, §3, that the full subcategory $\mathrm{D}(X)_{\mathrm{parf}}$ of $\mathrm{D}(X)$ formed by perfect

complexes is, like $\mathrm{D}(X)_{\mathrm{coh}}$, a triangulated subcategory, stable under inverse image and derived tensor product. One obviously has

$$\mathrm{D}(X)_{\mathrm{parf}} \subset \mathrm{D}(X)_{\mathrm{coh}},$$

but the inclusion is strict in general. Pseudo-coherence can be redefined "by passage to the limit" from perfection: a complex F is pseudo-coherent if and only if, for every $n \in \mathbb{Z}$, F can be locally approximated to order n by a perfect complex, meaning that there exists locally a distinguished triangle



with L perfect and R acyclic in degrees $> n$. Conversely, perfection is recovered from pseudo-coherence by means of an additional regularity condition: a complex is perfect if and only if it is pseudo-coherent and locally of finite Tor-dimension (Exposé I, §5). Thus one recovers the fact that, if the local rings are regular, every coherent sheaf is perfect, and more generally every pseudo-coherent complex with locally bounded cohomology is perfect.

Pseudo-coherence and perfection are the two fundamental finiteness notions with which we shall work in this Seminar. Sections 1 to 5 of Exposé I are devoted to their definition and to establishing their elementary stability properties. For this purpose, only two or three local properties of the category of finite-type free modules, relative to the category of all modules, are used. It is therefore convenient and useful, notably in Exposé II, to axiomatize the situation by introducing a notion of pseudo-coherence (respectively of perfection) in a category fibered over a site, relative to a fibered subcategory satisfying suitable conditions.

Sections 6 to 8 of Exposé I generalize to perfect complexes a certain number of notions well known for finite-type locally free sheaves: rank, duality, and the trace of an endomorphism.

In Exposé II, one examines the problem of the existence of "global resolutions": under what conditions, on a ringed topos $(X, \mathcal{O}_X)$, is a pseudo-coherent (respectively perfect) complex globally isomorphic in $\mathrm{D}(X)$ to a complex of locally free modules of finite type? The importance of this question lies essentially in the fact that at present one does not know how to generalize to perfect complexes certain usual constructions on locally free modules, such as exterior and symmetric powers or Chern classes. That is why it is useful to have workable sufficient criteria for the existence of global resolutions, which reduce certain questions about pseudo-coherent (respectively perfect) complexes to analogous questions about locally free sheaves of finite type. Such criteria are given in Exposé II.

In particular, one shows that global resolutions exist in the following cases: X is the Zariski topos of an affine scheme, or more generally of a quasi-compact scheme possessing an ample invertible sheaf, or even an ample family of invertible sheaves in the sense of Kleiman, for example a regular scheme; or again X is the topos of sheaves of sets on a compact topological space and $\mathcal{O}_X$ is the sheaf of continuous functions with complex values. By way of illustration, and to show the flexibility of the notions introduced, the appendix gives a "purely sheaf-theoretic" definition of the index of a family of elliptic operators.

Exposé III studies the stability of finiteness conditions under derived direct image. To obtain reasonable statements, one must relativize the notions of pseudo-coherence and perfection. We place ourselves here in the framework of ordinary schemes, which is sufficient for the Seminar, but there is no doubt that one will sooner or later need to develop an analogous theory for relative analytic schemes or spaces. Let $p : X \to S$ be an S-scheme locally of finite type, and let F be a complex of $\mathcal{O}_X$-modules. One says that F is pseudo-coherent (respectively perfect) relative to p if one can locally extend X, by a closed immersion, into a smooth S-scheme X',

so that F, extended by zero to X', is pseudo-coherent (respectively perfect). The notion of pseudo-coherence relative to p is especially interesting when S is not locally noetherian; in the opposite case it coincides with the ordinary notion of pseudo-coherence. Moreover, as expected, to say that F is perfect relative to p is equivalent to saying that F is pseudo-coherent relative to p and locally of finite Tor-dimension relative to p (i.e. relative to the sheaf of rings $p^{-1}(\mathcal{O}_S)$).

The central theorem of Exposé III is the following finiteness theorem, stated below in a slightly more precise form: if $f : X \to Y$ is a proper morphism of S-schemes locally of finite type, the functor Rf_* transforms complexes pseudo-coherent relative to S into complexes pseudo-coherent relative to S. This theorem is in fact a conjecture, but it is nevertheless proved in the two particular cases S locally noetherian and f projective. It seems likely, unfortunately, that the extension to the general case is of the same order of difficulty as the analogous theorem in analytic geometry, Grauert's theorem. Combining the finiteness theorem with an essentially trivial formula called the projection formula, one obtains workable criteria for the stability of relative perfection under direct image. As a corollary one recovers Grauert's continuity and semicontinuity theorems (EGA III 7.6).

Exposé IV translates the results of Exposés I to III into the language of "Grothendieck groups." On a ringed topos $(X, \mathcal{O}_X)$, write $K^*(X)$ (respectively $K_\bullet(X)$) for the Grothendieck group of the category of perfect complexes and complexes of finite Tor-dimension (respectively pseudo-coherent complexes with bounded homology). The group $K^*(X)$ is a ring, and $K_\bullet(X)$ is a module over $K^*(X)$. The functor K^* is contravariant, as suggested above, while $K_\bullet$ is covariant for proper pseudo-coherent morphisms of schemes, or of analytic spaces, subject to the availability of the finiteness theorem. There are also unusual variants: covariance of K^*, contravariance of $K_\bullet$, for morphisms satisfying suitable regularity hypotheses. Finally, the globalization criteria of Exposé II make it possible to link these groups with the "Grothendieck groups" defined naïvely from coherent sheaves or finite-type locally free sheaves.

1. Preliminary definitions

1.1. Fibered categories with additive fibers (respectively abelian, respectively triangulated fibers)

1.1.1. Let S be a category, and let $\mathcal{C}$ be an S-category fibered in categories with additive (respectively triangulated) fibers (SGA 1 VI 6.1). We shall say that $\mathcal{C}$ is an *additive* (respectively *triangulated*) S-category, or that $\mathcal{C}$ is additive (respectively triangulated) over S, if, for every arrow $f : X \to Y$ of S, the inverse-image functor

$$f^* : \mathcal{C}_Y \longrightarrow \mathcal{C}_X$$

(determined up to unique isomorphism) is additive (respectively exact). We shall say that $\mathcal{C}$ is an *abelian* S-category if $\mathcal{C}$ is an additive S-category with abelian fibers, and that $\mathcal{C}$ is a *flat abelian* S-category if, moreover, the inverse-image functors are exact.

1.1.2. Let $\mathcal{C}$ be a triangulated S-category. One defines, for example with the aid of a cleavage of $\mathcal{C}$ (SGA 1 VI 7.1), a fibered S-category $\mathrm{Tr}(\mathcal{C})$ whose fiber over each $X \in \mathrm{ob}(S)$ is the category $\mathrm{Tr}(\mathcal{C}_X)$ of distinguished triangles of $\mathcal{C}_X$.

1.1.3. Let $\mathcal{C}$ and $\mathcal{C}'$ be additive (respectively triangulated) S-categories, and let $F : \mathcal{C} \to \mathcal{C}'$ be a functor. We say that F is an additive (respectively exact) S-functor if F is a Cartesian S-functor that is additive (respectively exact) on each fiber.

1.1.3 (continued). If $\mathcal{C}$ and $\mathcal{C}'$ are triangulated S-categories and if F is an exact S-functor,

then F defines a Cartesian S-functor

$$\mathrm{Tr}(\mathcal{C}) \longrightarrow \mathrm{Tr}(\mathcal{C}').$$

1.1.4. Let $\mathcal{C}$ be a triangulated S-category, and let $\mathcal{C}'$ be a full fibered sub-S-category of $\mathcal{C}$. We say that $\mathcal{C}'$ is a *triangulated sub-S-category* of $\mathcal{C}$ if, for every $X \in \mathrm{ob}\, S$, $\mathcal{C}'_X$ is a triangulated subcategory of $\mathcal{C}_X$ [V], Chap. I, §1, 2.3.

1.1.5. Let $\mathcal{C}$ be an additive S-category. One defines (SGA 4 XVII 2.2, 2.4) an additive (respectively triangulated) S-category

$$\mathrm{C}(\mathcal{C}) \qquad \text{respectively} \qquad \mathrm{K}(\mathcal{C}),$$

whose fiber over $X \in \mathrm{ob}\, S$ is the category $\mathrm{C}(\mathcal{C}_X)$ of complexes with differential of degree $+1$ (respectively the category $\mathrm{K}(\mathcal{C}_X)$ of complexes of $\mathcal{C}_X$ "up to homotopy"). If $\mathcal{C}$ is a flat abelian S-category, one defines, loc. cit., a triangulated S-category $\mathrm{D}(\mathcal{C})$, whose fiber over $X \in \mathrm{ob}\, S$ is the derived category $\mathrm{D}(\mathcal{C}_X)$; if $f : X \to Y$ is an arrow of S, we shall write

$$f^* : \mathrm{D}(\mathcal{C}_Y) \longrightarrow \mathrm{D}(\mathcal{C}_X)$$

for the inverse-image functor, obtained from

$$f^* : \mathrm{K}(\mathcal{C}_Y) \longrightarrow \mathrm{K}(\mathcal{C}_X)$$

by passage to the quotient. One defines full triangulated sub-S-categories

$$\mathrm{K}^*(\mathcal{C}) \qquad \text{respectively} \qquad \mathrm{D}^*(\mathcal{C}) \quad (* = +, -, b)$$

by the usual degree conditions.

1.2. Terminology

Let S be a site, let $\mathcal{C}$ be a flat abelian S-category (1.1), and let $\mathcal{C}_0$ be a strictly full fibered sub-S-category.

Let $F \in \mathrm{ob}\, \mathcal{C}_X$, for $X \in \mathrm{ob}\, S$. The set of arrows $f : U \to X$ in S/X such that there exists an epimorphism

$$E \longrightarrow f^*F, \qquad E \in \mathrm{ob}\, \mathcal{C}_{0,U},$$

is a sieve on X. If this sieve is covering, we shall say that F is of $\mathcal{C}_0$-*finite type* (or simply of finite type when no confusion is to be feared).

We shall say that $\mathcal{C}_0$ is *locally stable under kernels of epimorphisms* if the following condition is satisfied: for every $X \in \mathrm{ob}\, S$ and every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$, with $E, F \in \mathrm{ob}\, \mathcal{C}_{0,X}$, the kernel of u is locally in $\mathcal{C}_0$; in other words the set of $f : U \to X$ in S/X such that

$$f^* \mathrm{Ker}(u) \in \mathrm{ob}\, \mathcal{C}_{0,U}$$

is a refinement of X.

By the standard process of decomposing a long exact sequence into short exact sequences, one sees that the preceding condition is equivalent to the following apparently stronger one: for every $X \in \mathrm{ob}\, S$ and every exact sequence in $\mathcal{C}_X$

$$0 \longrightarrow E^0 \longrightarrow E^1 \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0 \qquad (*)$$

with $E^i \in \mathrm{ob}\,\mathcal{C}_{0,X}$ for $1 \leq i \leq n$, the object E^0 is locally in $\mathcal{C}_0$.

We shall say that $\mathcal{C}_0$ is *locally quasi-stable under kernels of epimorphisms* if, for every exact sequence such as $(*)$, E^0 is of $\mathcal{C}_0$-finite type.

On the other hand, we shall say that $\mathcal{C}_0$ is *locally liftable in* $\mathcal{C}$ if, for every $X \in \mathrm{ob}\,S$, every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$ with $F \in \mathrm{ob}\,\mathcal{C}_{0,X}$ locally admits a section. Equivalently, for $\mathcal{C}_0$ to be locally liftable in $\mathcal{C}$, it is necessary and sufficient that, for every diagram of $\mathcal{C}_X$, $X \in \mathrm{ob}\,S$,

$$\begin{array}{ccc} & & H \\ & & \downarrow \\ E & \xrightarrow[u]{} & F \end{array}$$

where u is an epimorphism and $H \in \mathrm{ob}\,\mathcal{C}_0$, the set of U over X for which there exists a commutative diagram

$$\begin{array}{ccc} & \nearrow & H_U \\ & & \downarrow \\ E_U & \xrightarrow[u_U]{} & F_U \end{array}$$

is a refinement of X.

We shall say that $\mathcal{C}_0$ is *locally quasi-liftable in* $\mathcal{C}$ if the following weaker condition is satisfied: for every $X \in \mathrm{ob}\,S$, and every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$, with $F \in \mathrm{ob}\,\mathcal{C}_{0,X}$, the set of arrows $f : U \to X$ of S/X for which there exists

$$v : F' \longrightarrow f^*E, \qquad F' \in \mathrm{ob}\,\mathcal{C}_{0,U},$$

such that $f^*(u)v$ is an epimorphism, is a refinement of X. This condition is equivalent to the following: for every diagram in $\mathcal{C}_X$

$$\begin{array}{ccc} & & H \\ & & \downarrow \\ E & \xrightarrow[u]{} & F, \end{array}$$

where u is an epimorphism and $H \in \mathrm{ob}\,\mathcal{C}_0$, the set of U over X such that there exists a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{v} & H_U \\ \downarrow & & \downarrow \\ E_U & \xrightarrow[u_U]{} & F_U \end{array}$$

with $G \in \mathrm{ob}\,\mathcal{C}_0$ and v an epimorphism, is a refinement of X.

If S is a chaotic site, for example the punctual site (one object, one morphism), we shall omit the adverb "locally" in the preceding definitions.

1.3. Examples

a) Let S be the punctual site, let A be a ring, and let $\mathcal{C}$ be the category of left A-modules. The full subcategory $\mathcal{C}_0$ of $\mathcal{C}$ formed by the projective A-modules of finite type is stable under kernels

of epimorphisms and is liftable in $\mathcal{C}$; an A-module is of $\mathcal{C}_0$-finite type if and only if it is of finite type in the ordinary sense. The full subcategory $\mathcal{C}_1$ of $\mathcal{C}$ formed by the A-modules of finite type is quasi-liftable in $\mathcal{C}$, but is not liftable in general; if A is noetherian, $\mathcal{C}_1$ is stable under kernels of epimorphisms. The full subcategory $\mathcal{C}'_0$ of $\mathcal{C}$ formed by the free A-modules of finite type is liftable in $\mathcal{C}$ and quasi-stable, though not stable in general, under kernels of epimorphisms.

b) Let $(S, \mathcal{O}_S)$ be a space ringed by local rings, let S be the site of open subsets of S, and let $\mathcal{C}$ be the fibered S-category, even split (SGA 1 VI 9), defined by the functor

$$U \longmapsto \{\text{category of left } \mathcal{O}_U\text{-modules}\}.$$

Then $\mathcal{C}$ is a flat abelian S-category (1.1). Let $\mathcal{C}_0$ be the strictly full additive sub-S-category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } \mathcal{O}_U\text{-modules}\}.$$

An $\mathcal{O}_S$-module M is locally of $\mathcal{C}_0$-finite type if and only if M is of finite type in the ordinary sense (EGA 0_{I}, 5.2). From EGA 0_{I}, 5.2.7 and 5.4.9, it follows that $\mathcal{C}_0$ is locally stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$.

c) Let G be a group, let S be the topos of G-sets (SGA 4 V 3.11), and let k be a field, regarded as the constant ring on S. Let $\mathcal{C}$ be the split S-category defined by

$$U \longmapsto \{\text{category of left } k_U\text{-modules}\};$$

then $\mathcal{C}$ is a flat abelian S-category. Let $\mathcal{C}_0$ be the strictly full additive sub-S-category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } k_U\text{-modules}\}.$$

Since localization in S is equivalent to forgetting the operations of G, it is clear that a G-k-module M is locally of $\mathcal{C}_0$-finite type if and only if M is finite-dimensional as a vector space over k, and that $\mathcal{C}_0$ is locally stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$.

The reader is left to look at the "Galois" variant of c), where G is a profinite group and S is the topos of sets on which G operates continuously.

In fact, examples b) and c) are particular cases of the following example.

d) Let $(S, \mathcal{O}_S)$ be a ringed topos (cf. SGA 4 IV 2.10), and let $\mathcal{C}$ be the split S-category defined by

$$U \longmapsto \{\text{category of left } \mathcal{O}_U\text{-modules}\};$$

then $\mathcal{C}$ is a flat abelian S-category. Let $\mathcal{C}_0$ be the strictly full additive sub-S-category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } \mathcal{O}_U\text{-modules}\}.$$

An $\mathcal{O}_S$-module locally of $\mathcal{C}_0$-finite type will simply be called *of finite type*. The category $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$, as follows from:

Lemma 1.3.1. Let P be an $\mathcal{O}_S$-module. The following conditions are equivalent:

(i) P is of finite type and the functor $\mathrm{Hom}(P, \cdot)$ is exact;

(ii) P is of finite type and every epimorphism $M \to P$ locally admits a section;

(iii) P is locally a direct factor of a free module of finite type.

The proof is immediate and is left as an exercise to the reader.

1.4. An "extension" lemma (cf. EGA 0_{III}, 11.9.1)

Proposition 1.4.1. Let S be a site, let $\mathcal{C}$ be a flat abelian S-category (1.1), and let $\mathcal{C}_0$ be a strictly full additive sub-S-category of $\mathcal{C}$. Assume that $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$ (1.2). Let $S \in \mathrm{ob}\, S$, let

$$u : E \longrightarrow F$$

be a morphism, of degree 0, of complexes of $\mathcal{C}_S$ (differentials of degree +1), let G be the cone of u (cf. [V], Chap. I, §1, 2.1), let $n \in \mathbb{Z}$, and assume that $\mathrm{H}^n(G)$ is of $\mathcal{C}_0$-finite type (1.2). Then the set of $X \in \mathrm{ob}(S/S)$ satisfying property (P) below is a refinement of S.

(P): there exist an object $M \in \mathcal{C}_{0,X}$ and arrows of $\mathcal{C}_X$

$$r : M \longrightarrow Z^{n+1}(E_X) = \mathrm{Ker}\left(E_X^{n+1} \xrightarrow{d^{n+1}} E_X^{n+2}\right), \qquad s : M \longrightarrow F_X^n$$

such that the following diagram of $\mathcal{C}_X$

$$\begin{array}{ccccccc}
\cdots \to E_X^{n-1} & \xrightarrow{\binom{d_E^{n-1}}{0}} & E_X^n \oplus M & \xrightarrow{(d_E^n, r)} & E_X^{n+1} \to \cdots \\
\Big\downarrow{\scriptstyle u_X^{n-1}} & & \Big\downarrow{\scriptstyle (u_X^n, s)} & & \Big\downarrow{\scriptstyle u_X^{n+1}} \\
\cdots \to F_X^{n-1} & \xrightarrow[d_F^{n-1}]{} & F_X^n & \xrightarrow[d_F^n]{} & F_X^{n+1} \to \cdots
\end{array}$$

is a morphism of complexes $u' : E' \to F_X$, that is,

$$d_F^n s = u_X^{n+1} r,$$

and such that

(i) $\mathrm{H}^{n+1}(u')$ is a monomorphism, with

$$\Im \mathrm{H}^{n+1}(u') = \Im \mathrm{H}^{n+1}(u_X);$$

(ii) $\mathrm{H}^n(u')$ is an epimorphism.

In pictorial terms, one may say that, after modifying E and u locally in degree n only by adding, in degree n, an object of $\mathcal{C}_0$, one can locally make $\mathrm{H}^{n+1}(u)$ injective, without changing its image, and $\mathrm{H}^n(u)$ surjective.

Proof. The procedure is as follows: first one shows that r and s may be chosen locally so as to make $\mathrm{H}^{n+1}(u)$ injective; one verifies that the finiteness hypothesis on $\mathrm{H}^n(G)$ is preserved; a new local correction in degree n then makes it possible to obtain $\mathrm{H}^n(u)$ surjective.

The exact cohomology sequence

$$\cdots \longrightarrow \mathrm{H}^n(G) \longrightarrow \mathrm{H}^{n+1}(E) \longrightarrow \mathrm{H}^{n+1}(F) \longrightarrow \cdots$$

shows that the kernel of $\mathrm{H}^{n+1}(u)$ is of $\mathcal{C}_0$-finite type. In other words, the set of X over S for which there is an exact sequence

$$M \longrightarrow \mathrm{H}^{n+1}(E_X) \xrightarrow{\mathrm{H}^{n+1}(u)} \mathrm{H}^{n+1}(F_X), \qquad M \in \mathrm{ob}\, \mathcal{C}_{0,X},$$

is a refinement of S. Since $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$, one may moreover impose that the arrow

$$M \longrightarrow \mathrm{H}^{n+1}(E_X)$$

factor through $Z^{n+1}(E_X)$:

$$M \xrightarrow{r} Z^{n+1}(E_X) \longrightarrow \mathrm{H}^{n+1}(E_X).$$

The composite

$$M \xrightarrow{u_X^{n+1} r} Z^{n+1}(F_X) \longrightarrow \mathrm{H}^{n+1}(F_X)$$

is zero by construction, so $u_X^{n+1} r$ factors through

$$B^{n+1}(F_X) = \mathfrak{F}\left(F_X^n \xrightarrow{d_F^n} F_X^{n+1}\right).$$

Applying quasi-liftability once more, one may impose that $u_X^{n+1} r$ factor through F_X^n:

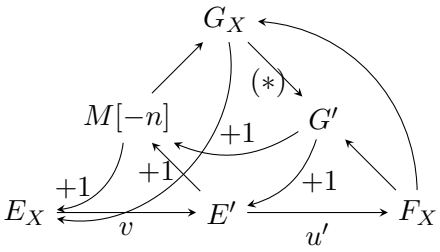
$$M \xrightarrow{s} F_X^n \longrightarrow B^{n+1}(F_X) \longrightarrow Z^{n+1}(F_X).$$

Then u', defined as in (1.4.1), is plainly a morphism of complexes satisfying (i).

Let G' be the cone of $u' : E' \to F_X$. We show that $\mathrm{H}^n(G')$ is of $\mathcal{C}_0$-finite type. By construction there is an exact sequence

$$0 \longrightarrow E_X \xrightarrow{v} E' \longrightarrow M[-n] \longrightarrow 0,$$

with $u'v = u_X$. To compare G_X and G', one may, for instance, form the octahedron in $\mathrm{K}(\mathcal{C}_X)$:



The exact cohomology sequence of the distinguished triangle marked by the octahedron shows that

$$\mathrm{H}^n(G_X) \longrightarrow \mathrm{H}^n(G')$$

is an epimorphism; hence $\mathrm{H}^n(G')$ is indeed of $\mathcal{C}_0$-finite type.

The preceding correction therefore reduces us to the case where $\mathrm{H}^{n+1}(u)$ is a monomorphism. We suppose this from now on. The exact cohomology sequence of $E \to F \to G$ then gives an exact sequence

$$\mathrm{H}^n(E) \xrightarrow{\mathrm{H}^n(u)} \mathrm{H}^n(F) \longrightarrow \mathrm{H}^n(G) \longrightarrow 0.$$

Since $\mathrm{H}^n(G)$ is of $\mathcal{C}_0$-finite type, the set of X over S for which there exists an epimorphism

$$M \longrightarrow \mathrm{H}^n(G_X), \qquad M \in \mathrm{ob}\,\mathcal{C}_{0,X},$$

is, by definition, a refinement of S. Using the quasi-liftability of $\mathcal{C}_0$ in $\mathcal{C}$, one may impose that this epimorphism factor through $Z^n(F_X)$:

$$M \xrightarrow{s} Z^n(F_X) \longrightarrow \mathrm{H}^n(F_X) \longrightarrow \mathrm{H}^n(G_X).$$

It is clear that u', defined as in (1.4.1) with $r = 0$, is a morphism of complexes and that

$$\mathrm{H}^n(u') : \mathrm{H}^n(E_X) \oplus M \longrightarrow \mathrm{H}^n(F_X)$$

is an epimorphism. This completes the proof of (1.4.1).

Lemma 1.4.2. Let $\mathcal{C}$ be an abelian category, let $u : E \to F$ be a morphism of complexes of $\mathcal{C}$ (respectively an arrow of $\mathrm{D}(\mathcal{C})$), and let G be the cone of u. For a given $n \in \mathbb{Z}$, the following conditions are equivalent:

(i) $\mathrm{H}^i(G) = 0$ for $i \geq n$;

(ii) $\mathrm{H}^i(u)$ is an isomorphism for $i > n$ and an epimorphism for $i = n$.

The proof is immediate from the exact cohomology sequence.

Definition 1.4.3. We shall say that u is an *n-quasi-isomorphism* (respectively an *n-isomorphism*) if u satisfies the equivalent conditions of (1.4.2).

That being so, one deduces at once from (1.4.1) the following:

Corollary 1.4.4. Let $S, S, \mathcal{C}, \mathcal{C}_0$ be as in (1.4.1). Let $u : E \to F$ be an $(n+1)$-quasi-isomorphism of complexes of $\mathcal{C}_S$, with $n \in \mathbb{Z}$ fixed, such that $\mathrm{H}^n(G)$ is of $\mathcal{C}_0$-finite type. Then the set of X over S for which there exist (M, r, s) as in (1.4.1) such that u' is an n-quasi-isomorphism is a refinement of S.

In pictorial terms, one may, using the finiteness condition on $\mathrm{H}^n(G)$, locally "gain one notch," that is, locally modify u in degree n by adding to E^n an object of $\mathcal{C}_0$ in such a way as to make u an n-quasi-isomorphism.

Corollary 1.4.5. Let $\mathcal{C}$ be an abelian category, let $n \in \mathbb{Z}$, and let $u : E \to F$ be an n-quasi-isomorphism of complexes of $\mathcal{C}$. There exists a commutative diagram in $\mathrm{C}(\mathcal{C})$

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ {\scriptstyle u}\downarrow & \swarrow_{u'} & \\ F & & \end{array}$$

where u' is a quasi-isomorphism and v is a monomorphism inducing the identity in degrees $\geq n$, that is,

$$E'^i = E^i, \qquad v^i = \mathrm{Id}(E^i) \qquad \text{for } i \geq n.$$

Proof. Applying (1.4.4) in the case where S is the punctual site and $\mathcal{C}_0 = \mathcal{C}$, one obtains a commutative triangle in $\mathrm{C}(\mathcal{C})$

$$\begin{array}{ccc} E & \xrightarrow{v_{n-1}} & E'_{n-1} \\ {\scriptstyle u}\downarrow & \swarrow_{u'_{n-1}} & \\ F & & \end{array}$$

where u'_{n-1} is an $(n-1)$-quasi-isomorphism and v_{n-1} is a monomorphism inducing the identity in degrees $\geq n$. Iterating the process, one finds, for $i \leq n-1$, a sequence of commutative triangles in $\mathrm{C}(\mathcal{C})$

$$\begin{array}{ccc} E'_i & \xrightarrow{v_{i-1}} & E'_{i-1} \\ {\scriptstyle u'_i}\downarrow & \swarrow_{u'_{i-1}} & \\ F & & \end{array}$$

where u'_i is an i-quasi-isomorphism and v_{i-1} is a monomorphism inducing the identity in degrees $\geq i$. The corollary follows by passing to the limit. More precisely, writing d'_i for the differential of E'_i, define E' by

$$E'^i = E^i, \quad d'^i = d^i \quad (i \geq n),$$

and

$$E'^i = (E'_i)^i, \quad d'^i = (d'_i)^i \quad (i < n),$$

and then define v and u' by

$$v^j = (v_i)^j, \qquad u'^j = (u'_i)^j \qquad (i \leq j).$$

One immediately verifies that E', v, and u', so defined, answer the question.

In abbreviated form: one can extend an n-quasi-isomorphism into a quasi-isomorphism "without changing anything given in degree $> n$."

2. Pseudo-coherent complexes

2.0. In this section, S denotes a site, S an object of S, $\mathcal{C}$ a flat abelian S-category (1.1), and $\mathcal{C}_0$ a strictly full additive sub-S-category of $\mathcal{C}$. We assume that $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms (1.2) and locally quasi-liftable in $\mathcal{C}$ (1.2).

Definition 2.1. Let $X \in \operatorname{ob} S$, let E be a complex of $\mathcal{C}_X$, and let $n \in \mathbb{Z}$. We shall say that E is *strictly $\mathcal{C}_0$-n-pseudo-coherent* (respectively $\mathcal{C}_0$-pseudo-coherent, respectively $\mathcal{C}_0$-perfect) if $E^i = 0$ for i sufficiently large and $E^i \in \operatorname{ob} \mathcal{C}_{0,S}$ for $i \geq n$ (respectively for every i, respectively for every i and $E^i = 0$ for almost all i).

When no confusion is to be feared, we shall sometimes say "strictly n-pseudo-coherent" (respectively, and so on) instead of "strictly $\mathcal{C}_0$-n-pseudo-coherent".

Lemma 2.1.1. Let $E \in \operatorname{ob} \mathrm{C}(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. If E is strictly n-pseudo-coherent and if $\mathrm{H}^i(E) = 0$ for $i > n$, then $\mathrm{H}^n(E)$ is of finite type.

Proof. The sequence

$$0 \longrightarrow Z^n(E) \longrightarrow E^n \longrightarrow E^{n+1} \longrightarrow \cdots \longrightarrow 0$$

is exact, and $E^i \in \operatorname{ob} \mathcal{C}_0$ for $i \geq n$. Since $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms, it follows that $Z^n(E)$ is of finite type, hence so is $\mathrm{H}^n(E)$, which is a quotient of it.

Proposition 2.2. Let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. Then:

a) If F is strictly n-pseudo-coherent (relative to $\mathcal{C}_0$) and if

$$u : E \longrightarrow F$$

is a quasi-isomorphism, then the set of X over S such that there exists a quasi-isomorphism

$$v : F' \longrightarrow E_X$$

with F' strictly n-pseudo-coherent is a refinement of S.

b) The following conditions are equivalent:

(i) the set of X over S such that there exists a quasi-isomorphism

$$u : E \longrightarrow F_X$$

of $\mathrm{C}(\mathcal{C}_X)$, with E strictly n-pseudo-coherent, is a refinement of S;

(i′) the same condition as (i), but with u an isomorphism of $\mathrm{D}(\mathcal{C}_X)$;

(ii) the set of X over S such that there exists an n-quasi-isomorphism

$$u : E \longrightarrow F_X$$

of $\mathrm{C}(\mathcal{C}_X)$, with E strictly perfect, is a refinement of S;

(ii′) the same condition as (ii), but with u an n-isomorphism of $\mathrm{D}(\mathcal{C}_X)$.

Proof. For a), there exists, for i sufficiently large, an i-quasi-isomorphism

$$v_i : F'_i \longrightarrow E$$

with F'_i strictly n-pseudo-coherent (take $F'_i = 0$). We show that, if $i > n$, there exists, after refining S, an $(i-1)$-quasi-isomorphism

$$v_{i-1} : F'_{i-1} \longrightarrow E$$

with F'_{i-1} strictly n-pseudo-coherent. By (1.4.4), it is enough to show that $\mathrm{H}^{i-1}(M')$, where M' is the cone of v_i, is of $\mathcal{C}_0$-finite type. Now if M'' denotes the cone of uv_i, the hypothesis that u is a quasi-isomorphism and (1.4.2) imply that

$$\mathrm{H}^j(M') \xrightarrow{\sim} \mathrm{H}^j(M'') \qquad \text{for all } j,$$

and

$$\mathrm{H}^j(M') = 0 \qquad \text{for } j \geq i.$$

The cone formula

$$M''^j = F_i'^{\,j+1} \oplus F^j$$

shows, on the other hand, that M'' is strictly $(i-1)$-pseudo-coherent. Hence, by (2.1.1),

$$\mathrm{H}^{i-1}(M'') = \mathrm{H}^{i-1}(M')$$

is of finite type, whence the assertion. It follows, by iteration and after refining S, that there is an n-quasi-isomorphism

$$v_n : F'_n \longrightarrow E$$

where F'_n is strictly n-pseudo-coherent. But, by (1.4.5), one can extend v_n into a quasi-isomorphism

$$v : F' \longrightarrow E$$

without changing anything in degrees $\geq n$; in particular F' is strictly n-pseudo-coherent. This proves a).

For b), the equivalence of (i) and (i′) follows at once from a) and from the definition of isomorphisms of $\mathrm{D}^-(\mathcal{C}_X)$. If $E \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$ is strictly n-pseudo-coherent, there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

where E' is strictly perfect and $E''^i = 0$ for $i \geq n$. Consequently $E' \to E$ is an n-quasi-isomorphism, which gives the implication (i) ⇒ (ii). The implication (ii) ⇒ (ii′) is trivial, so it remains to prove (ii′) ⇒ (i). Let

$$u : E \longrightarrow F$$

be an n-isomorphism of $\mathrm{D}^-(\mathcal{C}_S)$, with E strictly perfect. Then u is represented by a diagram in $\mathrm{C}^-(\mathcal{C}_S)$

$$\begin{array}{ccc} & E' & \\ {\scriptstyle s}\swarrow & & \searrow{\scriptstyle u'} \\ E & & F, \end{array}$$

where s is a quasi-isomorphism and u' an n-quasi-isomorphism. By a), after refining S, there exists a quasi-isomorphism

$$t : E'' \longrightarrow E',$$

with E'' strictly n-pseudo-coherent. Applying (1.4.5) to $u't : E'' \to F$, we extend it to a quasi-isomorphism

$$E''' \longrightarrow F$$

with E''' strictly n-pseudo-coherent. This proves (ii$'$) $\Rightarrow$ (i) and completes the proof of (2.2).

Definition 2.3. Under the hypotheses of (2.2), we shall say that $F \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$ is n-$\mathcal{C}_0$-pseudo-coherent if F satisfies the equivalent conditions b) of (2.2). We shall say that F is $\mathcal{C}_0$-pseudo-coherent if F is n-$\mathcal{C}_0$-pseudo-coherent for every n.

When this cannot cause confusion, we shall sometimes allow ourselves to say n-pseudo-coherent, pseudo-coherent, instead of n-$\mathcal{C}_0$-pseudo-coherent, $\mathcal{C}_0$-pseudo-coherent.

Remark 2.3.1. Let $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C}_S)$ be a pseudo-coherent complex. The set of X over S such that there exists a quasi-isomorphism

$$E \longrightarrow F_X$$

with E strictly pseudo-coherent (2.1) is not in general a refinement of S. However, we shall see later (2.7 a) and Exposé II) important cases in which this is so.

The proof of (2.2 a) gives the following result.

Proposition 2.4. Let $n, i \in \mathbb{Z}$, and let

$$u : E \longrightarrow F$$

be an i-quasi-isomorphism of $\mathrm{C}(\mathcal{C}_S)$, with F n-pseudo-coherent and E strictly n-pseudo-coherent. Then, after refining S, there exists a commutative triangle in $\mathrm{C}(\mathcal{C}_S)$

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ {\scriptstyle u}\downarrow & \swarrow{\scriptstyle u'} & \\ F & & \end{array}$$

where E' is strictly n-pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degree $\geq i$.

Proposition 2.5. a) If $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C}_S)$ is n-pseudo-coherent (respectively pseudo-coherent), then $F[m]$ is $(n-m)$-pseudo-coherent (respectively pseudo-coherent).

b) Let

$$\begin{array}{ccc} E & \longrightarrow & F \\ & {\scriptstyle +1}\nwarrow \quad \swarrow & \\ & G & \end{array} \qquad (*)$$

be a distinguished triangle of $\mathrm{D}(\mathcal{C}_S)$. If E and G are n-pseudo-coherent (respectively pseudo-coherent), then F is n-pseudo-coherent (respectively pseudo-coherent).

Proof. a) is trivial from the definitions.

b) It is enough, of course, to prove the non-parenthesized assertion. The triangle $(*)$ can also be written

$$\begin{array}{ccc} G[-1] & \longrightarrow & E \\ & {\scriptstyle +1}\nwarrow \quad \swarrow & \\ & F & \end{array}$$

By a), $G[-1]$ is $(n+1)$-pseudo-coherent. Thus we are reduced to proving that, if E is $(n+1)$-pseudo-coherent and F is n-pseudo-coherent, then G is n-pseudo-coherent. After localizing, we

may assume E (respectively F) strictly $(n+1)$-pseudo-coherent (respectively n-pseudo-coherent). The arrow

$$E \xrightarrow{u} F$$

of $\mathrm{D}^-(\mathcal{C}_S)$ is represented by a pair of arrows of $\mathrm{C}^-(\mathcal{C}_S)$

$$\begin{array}{ccc} & E' & \\ {\scriptstyle s}\swarrow & & \searrow{\scriptstyle u'} \\ E & & F, \end{array}$$

where s is a quasi-isomorphism. By (2.2 a)), after localizing, there exists a quasi-isomorphism

$$t : E'' \longrightarrow E',$$

where E'' is strictly $(n+1)$-pseudo-coherent. We are therefore reduced to the case where u is defined by a true arrow of $\mathrm{C}^-(\mathcal{C}_S)$. The assertion then follows from the definitions and from the formula

$$C(u)^i = E^{i+1} \oplus F^i,$$

where $C(u)$ denotes the cone of u.

Scholium 2.6. By rotation of triangles, one sees that pseudo-coherence properties for two of the objects of the triangle $(*)$ imply another such property for the third. The various cases may be summarized in the table below; the lower-case letters denote degrees of pseudo-coherence, and in each row the conclusion is underlined:

$$\begin{array}{ccc} E & F & G \\ n+1 & n & \underline{n} \\ n & \underline{n} & n \\ \underline{n} & n & n-1. \end{array}$$

We shall write

$$\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-}n\mathrm{coh}} \quad \text{respectively} \quad \mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$$

for the strictly full fibered subcategory of $\mathrm{D}(\mathcal{C})$ whose fiber over each $X \in \mathrm{ob}\, S$ is formed by the objects of $\mathrm{D}(\mathcal{C}_X)$ that are n-pseudo-coherent (respectively pseudo-coherent) relative to $\mathcal{C}_0$. By (2.5), the category

$$\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$$

is a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$ (1.1).

Proposition 2.7. Suppose that S is the punctual site, so that

$$\mathcal{C} = \mathcal{C}_S, \qquad \mathcal{C}_0 = \mathcal{C}_{0,S}.$$

Then:

a) If $F \in \mathrm{ob}\, \mathrm{D}(\mathcal{C})$ is pseudo-coherent, there exists a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly pseudo-coherent (2.1).

b) Let $\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0)$ denote the full triangulated subcategory of $\mathrm{K}^-(\mathcal{C}_0)$ formed by acyclic complexes. The canonical functor

$$\mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

is fully faithful and has essential image $\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$.

Proof. a) Let $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$, and suppose constructed an i-quasi-isomorphism

$$u_i : E_i \longrightarrow F$$

of $\mathrm{C}(\mathcal{C})$, with E_i strictly pseudo-coherent; this is possible for i sufficiently large, taking $E_i = 0$. Let G_i be the cone of u_i. It follows from (1.4.2) and (2.1.1) that $\mathrm{H}^{i-1}(G_i)$ is of $\mathcal{C}_0$-finite type. Hence, by (1.4.4), there exists a commutative triangle in $\mathrm{C}(\mathcal{C})$

$$\begin{array}{ccc} E_i & \xrightarrow{v_{i-1}} & E_{i-1} \\ {\scriptstyle u_i}\downarrow & \swarrow_{u_{i-1}} & \\ F & & \end{array}$$

where E_{i-1} is strictly pseudo-coherent, u_{i-1} an $(i-1)$-quasi-isomorphism, and v_{i-1} a monomorphism inducing the identity in degrees $\geq i$. Passing to the limit (cf. the proof of (1.4.5)) gives a quasi-isomorphism

$$u : E \longrightarrow F$$

with E strictly pseudo-coherent.

b) Apply the criterion of [*V*], Chap. I, §2, 4.2 b)(i), together with a).

We note that the proof of a) gives the following more precise result, which generalizes (1.4.5).

Proposition 2.7.1. Under the hypotheses of (2.7), let $u : E \to F$ be an n-quasi-isomorphism of $\mathrm{C}(\mathcal{C})$, with F pseudo-coherent and E strictly pseudo-coherent. Then there exists a commutative triangle in $\mathrm{C}(\mathcal{C})$

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ {\scriptstyle u}\downarrow & \swarrow_{u'} & \\ F & & \end{array}$$

where E' is strictly pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degree $\geq n$.

Definition 2.8. Let $n \in \mathbb{N}$ and $F \in \mathrm{ob}\,\mathcal{C}_S$. We shall say that F is of n-$\mathcal{C}_0$-finite presentation if the set of X over S such that there exists an exact sequence

$$E^{-n} \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F_X \longrightarrow 0, \qquad E^i \in \mathrm{ob}\,\mathcal{C}_{0,X} \quad (-n \leq i \leq 0),$$

is a refinement of S. We shall say that F is of ∞-$\mathcal{C}_0$-finite presentation if F is of n-$\mathcal{C}_0$-finite presentation for every $n \in \mathbb{N}$.

Of course, we shall sometimes simply say n-finite presentation when there is no possible ambiguity about $\mathcal{C}_0$. We shall say "finite presentation" for "1-finite presentation," and "finite type" for "0-finite presentation," the latter notion coinciding with the one already introduced in (1.2).

Proposition 2.9. Let $n \in \mathbb{N}$ and $F \in \mathrm{ob}\,\mathcal{C}_S$. The following conditions are equivalent:

(i) F is of n-finite presentation (respectively of ∞-finite presentation);

(ii) F is $(-n)$-pseudo-coherent (respectively pseudo-coherent).

In the case where S is the punctual site, F is pseudo-coherent if and only if there exists an exact sequence

$$\cdots \longrightarrow E^i \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F \longrightarrow 0, \qquad E^i \in \mathrm{ob}\,\mathcal{C}_0 \quad \text{for all } i.$$

Proof. For the equivalence (i) $\Leftrightarrow$ (ii), it is plainly enough to treat the non-parenthesized case. The implication (i) $\Rightarrow$ (ii) is immediate from the definitions, and (ii) $\Rightarrow$ (i) follows at once from

(2.4): the arrow $0 \to F$ is a 1-quasi-isomorphism, and from this one deduces, after localizing, a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly $(-n)$-pseudo-coherent and $E^i = 0$ for $i \geq 1$, that is, an n-finite presentation of F. The second assertion is proved in the same way, using (2.7.1).

Proposition 2.10. a) Let $m, n \in \mathbb{Z}$, with $n \leq m$. If $F \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$ is n-pseudo-coherent and acyclic in degrees $\geq m$, then there exists locally a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly n-pseudo-coherent and $E^i = 0$ for $i \geq m$. If S is punctual, and if F is pseudo-coherent and acyclic in degrees $\geq m$, there exists a quasi-isomorphism

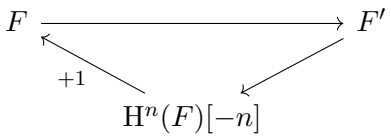
$$E \longrightarrow F$$

with E strictly pseudo-coherent and $E^i = 0$ for $i \geq m$.

b) Let $F \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$ be acyclic in degrees $> n$. For F to be n-pseudo-coherent, it is necessary and sufficient that $\mathrm{H}^n(F)$ be of finite type.

Proof. a) Since F is acyclic in degrees $\geq m$, the arrow $0 \to F$ is an m-quasi-isomorphism. The assertion follows from (2.4) and (2.7.1).

b) There exists a distinguished triangle in $\mathrm{D}^-(\mathcal{C}_S)$:



(to see this, begin by truncating F, replacing F^n by $Z^n(F)$ and F^i by 0 for $i > n$). The exact cohomology sequence implies

$$\mathrm{H}^i(F') = 0 \quad \text{for } i \geq n-1,$$

so F' is $(n-1)$-pseudo-coherent. We conclude by (2.5) and (2.6).

Exposé I, continued: pseudo-coherent complexes and perfect complexes

Proposition 2.11. Let $n \in \mathbb{Z}$.

a) Let $F \in \mathrm{ob}\,\mathrm{C}^-(\mathcal{C}_S)$. If, for every $i \geq n$ (respectively for every i), F^i is of finite $(i-n)$-presentation (respectively of finite ∞-presentation), then F is n-pseudo-coherent (respectively pseudo-coherent).

b) Let $F \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$. If, for every $i \geq n$ (respectively for every i), $\mathrm{H}^i(F)$ is of finite $(i-n)$-presentation (respectively of finite ∞-presentation), then F is pseudo-coherent.

Proof. It is enough to treat the non-parenthesized assertions.

a) We argue by induction on N such that $F^i \neq 0$ only for $i \leq n+N$. If $N = 0$, the assertion follows at once from (2.10 b). Suppose $N \geq 1$. We have an exact sequence

$$0 \longrightarrow F^{n+N}[-n-N] \longrightarrow F \longrightarrow F' \longrightarrow 0.$$

By the induction hypothesis, F' is n-pseudo-coherent; on the other hand $F^{n+N}[-n-N]$ is n-pseudo-coherent by hypothesis, using (2.9 and 2.5 a). Hence, by (2.5 b), F is n-pseudo-coherent.

b) We argue by induction on N such that $\mathrm{H}^i(F) = 0$ for $i > n+N$. If $N = 0$, the assertion follows from (2.10 b). Suppose $N \geq 1$. As in the proof of (2.10 b), we have a distinguished triangle in $\mathrm{D}^-(\mathcal{C}_S)$

$$F' \longrightarrow F \longrightarrow \mathrm{H}^{n+N}(F)[-n-N] \longrightarrow F'[1].$$

more plainly, with cone $\mathrm{H}^{n+N}(F)[-n-N]$. The exact cohomology sequence implies

$$\mathrm{H}^i(F') = 0 \quad \text{for } i > n+N-1,$$

that is, for $i > (n-1)+(N-1)$, and

$$\mathrm{H}^i(F') \xrightarrow{\sim} \mathrm{H}^{i+1}(F) \quad \text{for } i < n+N-1.$$

By the induction hypothesis, applied with $n-1$ in place of n, F' is therefore $(n-1)$-pseudo-coherent. On the other hand, $\mathrm{H}^{n+N}(F)[-n-N]$ is n-pseudo-coherent by (2.9 and 2.5 a). Hence, by (2.6), F is n-pseudo-coherent.

Remark 2.11.1. If F is pseudo-coherent, the F^i (respectively the $\mathrm{H}^i(F)$) are not in general pseudo-coherent, nor even of finite type. Nevertheless, in the following paragraph, in the excursions on the notions of coherence, we shall see important examples where pseudo-coherence of F implies that of the $\mathrm{H}^i(F)$.

Proposition 2.12. Let $F', F'' \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$, and let $F = F' \oplus F''$. For F to be n-pseudo-coherent (respectively pseudo-coherent), it is necessary and sufficient that F' and F'' be n-pseudo-coherent (respectively pseudo-coherent).

Proof. It is enough to prove the non-parenthesized assertion. Sufficiency is obvious; let us prove necessity. By induction we may suppose that it has already been proved that F' and F'' are $(n+1)$-pseudo-coherent. After localizing, we may even suppose that F' and F'' are strictly $(n+1)$-pseudo-coherent and that $F = F' \oplus F''$ in $\mathrm{C}^-(\mathcal{C}_S)$. Let $F'_{>n}$ (respectively $F'_{\leq n}$) denote the complex obtained from F' by replacing F'^i by 0 for $i \leq n$ (respectively for $i > n$), and define $F''_{>n}$ and $F''_{\leq n}$ similarly. We have an exact sequence

$$0 \longrightarrow F'_{>n} \oplus F''_{>n} \longrightarrow F \longrightarrow F'_{\leq n} \oplus F''_{\leq n} \longrightarrow 0.$$

The complex $F'_{>n} \oplus F''_{>n}$ is strictly perfect; since F is n-pseudo-coherent, (2.6) implies that $F'_{\leq n} \oplus F''_{\leq n}$ is n-pseudo-coherent. This means, by (2.10 b), that

$$\mathrm{H}^n(F'_{\leq n} \oplus F''_{\leq n})$$

is of finite type. But

$$\mathrm{H}^n(F'_{\leq n} \oplus F''_{\leq n}) = \mathrm{H}^n(F'_{\leq n}) \oplus \mathrm{H}^n(F''_{\leq n}),$$

so $\mathrm{H}^n(F'_{\leq n})$ (respectively $\mathrm{H}^n(F''_{\leq n})$) is of finite type, and again by (2.10 b) $F'_{\leq n}$ (respectively $F''_{\leq n}$) is n-pseudo-coherent. Applying (2.5) to the exact sequence

$$0 \longrightarrow F'_{>n} \longrightarrow F' \longrightarrow F'_{\leq n} \longrightarrow 0$$

(and similarly for F''), we deduce that F' and F'' are n-pseudo-coherent.

Proposition 2.13. Let $(S', \mathcal{C}', \mathcal{C}'_0)$ be a triple satisfying the same hypotheses (2.0) as $(S, \mathcal{C}, \mathcal{C}_0)$. Let $f : S' \to S$ be a morphism of sites, corresponding to an underlying functor $f^{-1} : S \to S'$, and let

$$u : \mathrm{D}^*(\mathcal{C}) \longrightarrow \mathrm{D}^*(\mathcal{C}') \qquad (* = +, -, b)$$

be a Cartesian functor above f, exact on each fiber. Suppose that there exist $a, b \in \mathbb{Z}$ such that, for every $S \in \mathrm{ob}\,S$, the following conditions hold:

(i) if $E \in \operatorname{ob} \mathcal{C}_{0,S}$, then $u(E)$ is a-$\mathcal{C}'_0$-pseudo-coherent (respectively $\mathcal{C}'_0$-pseudo-coherent);

(ii) if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is acyclic in degrees > 0, then $u(F)$ is acyclic in degrees $> b$.

Then, if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is m-pseudo-coherent relative to $\mathcal{C}_0$ and acyclic in degrees $> n$, the object

$$u(F) \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}'_{S'}), \qquad S' = f^{-1}(S),$$

is $\sup(m+b, n+a)$-pseudo-coherent relative to $\mathcal{C}'_0$ (respectively $(m+b)$-pseudo-coherent), and is acyclic in degrees $> n+b$. In particular, in the parenthesized case, if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is pseudo-coherent relative to $\mathcal{C}_0$, then $u(F)$ is pseudo-coherent relative to $\mathcal{C}'_0$.

Proof. We may restrict to the non-parenthesized case. First suppose that $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is strictly perfect (2.1), with $F^i = 0$ for $i > n$. Then $u(F)$ is $(n+a)$-$\mathcal{C}'_0$-pseudo-coherent. We argue by induction on the number N of indices i such that $F^i \neq 0$. If $N = 1$, then $F = F^p[-p]$ for some $p \leq n$, so $u(F)$ is $(p+a)$-$\mathcal{C}'_0$-pseudo-coherent, hence a fortiori $(n+a)$-pseudo-coherent. If $N > 1$, there is an exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0,$$

where F' and F'' are strictly perfect of length $< N-1$, and where $F'^i = F''^i = 0$ for $i > n$. Applying (2.5 b) to the distinguished triangle

$$u(F') \longrightarrow u(F) \longrightarrow u(F'') \longrightarrow u(F')[1].$$

and using the induction hypothesis, we get that $u(F)$ is $(n+a)$-pseudo-coherent.

Now pass to the general case: F is m-$\mathcal{C}_0$-pseudo-coherent and acyclic in degrees $> n$. After localizing, we may, by (2.10 a), suppose F strictly m-pseudo-coherent and such that $F^i = 0$ for $i > n$. There is then an exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0,$$

where F'' is zero in degrees $\geq m$, and F' is strictly perfect and zero in degrees $> n$. In the distinguished triangle

$$u(F') \longrightarrow u(F) \longrightarrow u(F'') \longrightarrow u(F')[1].$$

the object $u(F'')$ is acyclic in degrees $> m+b$, hence $(m+b)$-pseudo-coherent, while $u(F')$ is $(n+a)$-pseudo-coherent by what was just proved. We conclude by (2.5 b).

Corollary 2.14. Let $\mathcal{C}_1$ be a strictly full additive sub-S-category of $\mathcal{C}$ such that

$$\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}.$$

Assume that every object of $\mathcal{C}_1$ is $\mathcal{C}_0$-pseudo-coherent. Then:

(i) $\mathcal{C}_1$ is locally quasi-liftable in $\mathcal{C}$ (1.2), and locally quasi-stable under kernels of epimorphisms (1.2);

(ii) for $S \in \operatorname{ob} S$ and $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$, in order that F be n-pseudo-coherent (respectively pseudo-coherent) relative to $\mathcal{C}_1$, it is necessary and sufficient that it be so relative to $\mathcal{C}_0$.

Proof. For (i), let $E \to F$ be an epimorphism in $\mathcal{C}_S$, with $F \in \operatorname{ob} \mathcal{C}_{1,S}$. Since F is $\mathcal{C}_0$-pseudo-coherent, hence a fortiori of $\mathcal{C}_0$-finite type, there exists, after localizing, an epimorphism $E' \to F$ with $E' \in \operatorname{ob} \mathcal{C}_{0,S}$. Using the quasi-liftability of $\mathcal{C}_0$ in $\mathcal{C}$, we may suppose, after localizing again, that this epimorphism factors through E; this gives the quasi-liftability of $\mathcal{C}_1$. On the other

hand, if $E \in \mathrm{ob}\,\mathcal{C}_S$ admits a finite right resolution by objects of $\mathcal{C}_1$, then E is $\mathcal{C}_0$-pseudo-coherent by (2.11 a), hence a fortiori of $\mathcal{C}_1$-finite type.

For (ii), it is enough to prove necessity. For this one may apply, at choice, (2.11 a) or (2.13), with $S' = S$, $\mathcal{C}' = \mathcal{C}$, $f = \mathrm{Id}$, $u = \mathrm{Id}$, and with $\mathcal{C}'_0$ and $\mathcal{C}_0$ replaced respectively by $\mathcal{C}_0$ and $\mathcal{C}_1$.

Scholium 2.14.1. There is therefore a largest strictly full additive sub-S-category of $\mathcal{C}$ giving the same notion of n-pseudo-coherence (respectively pseudo-coherence) as $\mathcal{C}_0$, namely the category formed by the $\mathcal{C}_0$-pseudo-coherent objects. This latter category is locally stable under kernels of epimorphisms.

Example 2.15. Let (S, A) be a ringed topos, and let $\mathcal{C}, \mathcal{C}_0$ be the S-categories considered in (1.3 d). The fibered category $\mathrm{D}(\mathcal{C})$ will often be denoted by $\mathrm{D}(S, A)$, or $\mathrm{D}_A(S)$, when one wants to suggest that the modules are left A-modules, or even simply $\mathrm{D}(S)$ if there is no ambiguity about A. We denote by S the final object of S, and put $\mathrm{D}(\mathcal{C}_S) = \mathrm{D}_A(S)$ (or $\mathrm{D}(S)$). An object of $\mathrm{D}(S)$ will be said to be n-pseudo-coherent (respectively pseudo-coherent) if it is n-$\mathcal{C}_0$-pseudo-coherent (respectively $\mathcal{C}_0$-pseudo-coherent), and similarly for the strict notions. The category fibered in $\mathrm{D}(\mathcal{C})$, denoted $\mathrm{D}(\mathcal{C})_{n-\mathcal{C}_0-\mathrm{coh}}$ (respectively $\mathrm{D}(\mathcal{C})_{\mathcal{C}_0-\mathrm{coh}}$) in (2.6), will be denoted $\mathrm{D}_A(S)_{n-\mathrm{coh}}$ (respectively $\mathrm{D}_A(S)_{\mathrm{coh}}$). One observes that, by (2.14), the same notion of pseudo-coherence is obtained if $\mathcal{C}_0$ is replaced by the S-category $\mathcal{C}_1$ defined by

$U \longmapsto$ the category of left A_U-modules locally direct summands of free finite-type modules.

If an A-module M is a direct summand of a free finite-type A-module, it does not follow in general that M is locally free of finite type. One does, however, have the following result.

Lemma 2.15.1. Suppose that S satisfies one of the following two hypotheses:

(i) S has enough points (SGA 4 IV 6.4), and for every point s of S, the left A_s-modules which are projective of finite type are free; this last condition is satisfied, for example, if $A_s/\operatorname{rad}(A_s)$ is a field or if $\operatorname{rad}(A_s)$ is nilpotent;

(ii) (S, A) is a "locally ringed" topos in the sense of [1], which means that A is commutative and that, for every $X \in \mathrm{ob}\,S$ and every $f \in \Gamma(X, A)$, one has

$$X = X_f \cup X_{1-f},$$

where X_f (respectively X_{1-f}) denotes the largest subobject of X on which f (respectively $1 - f$) is invertible.

Then every direct summand of a free finite-type left A-module is locally free of finite type.

Proof. Suppose first that (i) holds. We note that, if E and F are two A-modules and E is of finite presentation, the canonical arrow

$$\mathrm{Hom}(E, F)_s \longrightarrow \mathrm{Hom}(E_s, F_s)$$

is an isomorphism for every point s of S. It follows that, if E and F are both of finite presentation and if $E_s \xrightarrow{\sim} F_s$ at a point s of S, then there is an s-pointed object U of S such that $E|U \xrightarrow{\sim} F|U$. The conclusion follows at once.

Suppose next that (ii) holds. Let $n \in \mathbb{N}$, and let $\mathbb{D}^n$ denote the functor on the category of locally ringed topoi belonging to a fixed universe, defined by

$$T \longmapsto \Gamma(T, \mathcal{O}_T)^n.$$

It is known [1] that the functor $\mathrm{End}_{\mathbb{D}}(\mathbb{D}^n)$ is represented by

$$M_n = \operatorname{Spec} \mathbb{Z}[T_{ij}] \quad (1 \leq i \leq n, 1 \leq j \leq n),$$

endowed with the endomorphism U_n of $(\mathcal{O}_{M_n})^n$ defined by the matrix (T_{ij}). Now let

$$p : A^n \longrightarrow A^n$$

be a projector. We show that $p(A^n)$ is locally free of finite type. From what has just been said, there exists a morphism

$$s : S \longrightarrow M_n$$

of locally ringed topoi such that $s^*(u_n) = p$. We may interpret s as a point of M_n with values in $\Gamma(S, A)$, that is, as a matrix of order n with coefficients in $\Gamma(S, A)$. Since this matrix is a projector, s factors through the closed subscheme

$$i : P_n \hookrightarrow M_n$$

corresponding to projectors; say $S \xrightarrow{s'} P_n \xrightarrow{i} M_n$. Hence

$$p = s'^* i^*(u_n),$$

and we are done, because, by the assertion under hypothesis (i), the image of $i^*(u_n)$ is a locally free finite-type sheaf.

Corollary 2.16. Let

$$f : (S', A') \longrightarrow (S, A)$$

be a morphism of ringed topoi, defined by a functor $f^{-1} : S \to S'$ and a morphism of sheaves of rings $f^{-1}(A) \to A'$. Let $\mathrm{D}({}_{A'}S'_A)$ denote the derived category of A'-A-bimodules, left over A' and right over A, and let

$$\otimes^{\mathrm{L}}_A : \mathrm{D}^-({}_{A'}S'_A) \times_{S'} \mathrm{D}^-({}_AS) \longrightarrow \mathrm{D}^-({}_{A'}S')$$

be the derived functor of the tensor product

$$(E', E) \longmapsto E' \otimes_A f^{-1}(E).$$

Let $X \in \mathrm{ob}\, S$, put $X' = u(X)$, and let $E' \in \mathrm{ob}\, \mathrm{D}^-({}_{A'}X'_A)$, $E \in \mathrm{ob}\, \mathrm{D}^-({}_AX)$. If E' is m'-pseudo-coherent (as an object of $\mathrm{D}^-({}_AS')$) and acyclic in degrees $> n'$, and if E is m-pseudo-coherent and acyclic in degrees $> n$, then

$$E' \otimes^{\mathrm{L}}_A E$$

is $\sup(m + n', m' + n)$-pseudo-coherent and acyclic in degrees $> n + n'$.

Proof. After replacing S by S/X, we may suppose $X = S$, and apply (2.13) with

$$u = E' \otimes^{\mathrm{L}}_A : \mathrm{D}^-({}_AS) \longrightarrow \mathrm{D}^-({}_{A'}S').$$

Let $U \in \mathrm{ob}\, S$, let $U' = f^{-1}(U)$, and let E be a complex of left A_U-modules. If E is acyclic in degrees $> n$, then $E' \otimes^{\mathrm{L}}_A E$ is acyclic in degrees $> n + n'$, by the definition of $\otimes^{\mathrm{L}}_A$. On the other hand, if $E = (A_U)^p$, then

$$E' \otimes^{\mathrm{L}}_A E = (E'|U')^p,$$

which is m'-pseudo-coherent by hypothesis and (2.12). The conditions of (2.13) are therefore satisfied, with $a = m'$ and $b = n'$, whence the conclusion.

In particular:

Corollary 2.16.1. Under the hypotheses of (2.16), one has:

a) The functor

$$\mathrm{L}f^* : \mathrm{D}^-({}_AS) \longrightarrow \mathrm{D}^-({}_{A'}S')$$

(which is none other than the functor $A' \otimes_A^{\mathrm{L}}$) induces a functor

$$\mathrm{L}f^* : \mathrm{D}^-({}_A S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^-({}_{A'} S')_{\mathrm{coh}}$$

(with the notation of (2.6)).

b) If A is commutative, the functor

$$\otimes_A^{\mathrm{L}} : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\otimes_A^{\mathrm{L}} : \mathrm{D}^-(S)_{\mathrm{coh}} \times_S \mathrm{D}^-(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^-(S)_{\mathrm{coh}}.$$

Exercise 2.16.2. Let S be a topos, and let $A \to A'$ be a morphism of rings of S. Assume one of the following hypotheses:

(i) $A \to A'$ is surjective, with nilpotent ideal as kernel;

(ii) the functor $A' \otimes_A -$, on the category of left A-modules, is exact and faithful.

Show that $E \in \mathrm{ob}\, \mathrm{D}^-({}_A S)$ is n-pseudo-coherent (respectively pseudo-coherent) if and only if

$$A' \otimes_A E \in \mathrm{ob}\, \mathrm{D}^-({}_{A'} S)$$

is n-pseudo-coherent (respectively pseudo-coherent).

Hints: first show that a left A-module L is of finite type if and only if $A' \otimes_A L$ is of finite type (cf. Bourbaki, Commutative Algebra, Chap. I, §3, Prop. 11 and Chap. II, §3, Prop. 4), and then reduce to this case by truncation, using (2.5) and (2.10 b).

3. Link with the classical notion of coherence

3.0. The hypotheses and notation are those of (2.0), except that $\mathcal{C}_0$ is no longer assumed locally quasi-stable under kernels of epimorphisms.

Definition 3.1. Let $F \in \mathrm{ob}\, \mathcal{C}_S$. We shall say that F is *precoherent* relative to $\mathcal{C}_0$ if, for every U over S and every morphism

$$u : E \longrightarrow F|U,$$

where $E \in \mathrm{ob}\, \mathcal{C}_{0,U}$, $\mathrm{Ker}(u)$ is of $\mathcal{C}_0$-finite type (1.2). We shall say that F is *coherent* relative to $\mathcal{C}_0$ if F is precoherent and of finite type.

Technical note. We introduce this notion here only as a technical intermediary; the only notion to retain is that of coherence.

Remark 3.2. Let $F \in \mathrm{ob}\, \mathcal{C}_S$. It is clear from the definitions that, if F is precoherent (respectively of finite type), then every subobject (respectively quotient) of F is precoherent (respectively of finite type).

Lemma 3.3.

a) Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H$$

be an exact sequence of $\mathcal{C}_S$. If G is of finite type and H is precoherent, then F is of finite type.

b) Let

$$F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be an exact sequence of $\mathcal{C}_S$. If F is of finite type and G is precoherent, then H is precoherent.

c) Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be an exact sequence of $\mathcal{C}_S$. If F and H are precoherent (respectively of finite type), then so is G.

Proof. a) After localizing, there is an epimorphism $G' \to G$ with $G' \in \mathrm{ob}\,\mathcal{C}_0$. Hence there is a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & F' & \longrightarrow & G' & \longrightarrow & H \\ & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \mathrm{Id}} \\ 0 & \longrightarrow & F & \longrightarrow & G & \longrightarrow & H. \end{array}$$

Since H is precoherent, F' is of finite type; but $F' \to F$ is an epimorphism, so F is of finite type.

b) Let U be over S, and let $H' \xrightarrow{f} H|U$ be a morphism of $\mathcal{C}_U$, with $H' \in \mathrm{ob}\,\mathcal{C}_{0,U}$. After a change of notation, we may suppose $U = S$. By the quasi-liftability of $\mathcal{C}_0$ in $\mathcal{C}$, there exists, after localizing, a commutative square

$$\begin{array}{ccc} G' & \xrightarrow{p} & H' \\ \downarrow & & \downarrow{\scriptstyle f} \\ G & \longrightarrow & H, \end{array}$$

where the horizontal arrows are epimorphisms and $G' \in \mathrm{ob}\,\mathcal{C}_0$. Since $\mathrm{Ker}(f)$ is a quotient of $\mathrm{Ker}(fp)$, it is enough to show that $\mathrm{Ker}(fp)$ is of finite type; hence we may suppose $H' = G'$. The arrows $F \to G$ and $H' \to G$ define

$$g : F \oplus H' \longrightarrow G.$$

Putting $K = \mathrm{Ker}(G \to H)$, we have a commutative diagram, with exact rows and with q an epimorphism,

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F & \longrightarrow & F \oplus H' & \longrightarrow & H' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} & & \\ 0 & \longrightarrow & K & \longrightarrow & G & \longrightarrow & H & \longrightarrow & 0. \end{array}$$

The snake diagram shows that $\mathrm{Ker}(g) \to \mathrm{Ker}(f)$ is an epimorphism. But, since $F \oplus H'$ is of finite type and G is precoherent, $\mathrm{Ker}(g)$ is of finite type by a); hence we are done.

c) First prove the non-parenthesized assertion. Up to a change of notation, it is a matter of showing that, if $g : G' \to G$ is a morphism of $\mathcal{C}_S$, with $G' \in \mathrm{ob}\,\mathcal{C}_0$, then $\mathrm{Ker}(g)$ is of finite type. Let H' be the image of the composed arrow $G' \to H$. We have a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F' & \longrightarrow & G' & \longrightarrow & H' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & F & \longrightarrow & G & \longrightarrow & H & \longrightarrow & 0. \end{array}$$

Since h is a monomorphism, the snake diagram gives an isomorphism $\mathrm{Ker}(f) \xrightarrow{\sim} \mathrm{Ker}(g)$. But F' is of finite type by a), hence $\mathrm{Ker}(f)$ is of finite type, again by a). This proves the non-parenthesized assertion. For the parenthesized assertion, strictly speaking one cannot invoke (2.5 b), since $\mathcal{C}_0$ is not assumed locally quasi-stable under kernels of epimorphisms. After localizing, there are epimorphisms

$$F' \xrightarrow{u} F, \qquad H' \xrightarrow{v} H,$$

with $F', H' \in \mathrm{ob}\,\mathcal{C}_0$. After localizing further, we may suppose that v factors through G, whence a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F' & \longrightarrow & F' \oplus H' & \longrightarrow & H' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle w} & & \downarrow{\scriptstyle v} & & \\ 0 & \longrightarrow & F & \longrightarrow & G & \longrightarrow & H & \longrightarrow & 0. \end{array}$$

Since u and v are epimorphisms, so is w, and this proves the assertion. The proof of (3.3) is complete.

It follows immediately that:

Corollary 3.4. The full (respectively strictly full) subcategory of $\mathcal{C}_S$ formed by coherent objects is stable under finite projective (respectively inductive) limits. In particular, the kernel, cokernel, and image of an arrow between two coherent objects are coherent. Let

$$E^0 \longrightarrow E^1 \longrightarrow E^2 \longrightarrow E^3 \longrightarrow E^4$$

be an exact sequence of $\mathcal{C}_S$. If E^0, E^1, E^3, E^4 are coherent, then E^2 is coherent as well.

Corollary 3.5. Suppose that the objects of $\mathcal{C}_0$ are $\mathcal{C}_0$-coherent (which in particular implies that $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms, that is, that we are under the conditions of (2.0)).

a) For $F \in \mathrm{ob}\,\mathcal{C}_S$, the following conditions are equivalent:

(i) F is of finite presentation;

(ii) F is coherent;

(iii) F is of finite ∞-presentation (2.8).

b) Let $F \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{C}_S)$, and let $n \in \mathbb{Z}$. Then

$$F \text{ is } n\text{-pseudo-coherent} \iff \begin{array}{l} \mathrm{H}^i(F) \text{ is coherent for } i \geq n+1, \\ \text{and } \mathrm{H}^n(F) \text{ is of finite type.} \end{array}$$

In particular, for F to be pseudo-coherent it is necessary and sufficient that $\mathrm{H}^n(F)$ be coherent for every n.

Proof. Assertion a) and the implication $\Rightarrow$ in b) follow trivially from (3.4). The implication $\Leftarrow$ in b) follows at once from a) and (2.11 b).

Examples 3.6. Let (S, A) be a ringed topos, and let $\mathcal{C}, \mathcal{C}_0$ be the S-categories of (1.3 d), also as in (2.15). In order that the objects of $\mathcal{C}_0$ be coherent, it is necessary and sufficient that A be coherent, that is, precoherent. Here are examples where this is the case:

- A is constant with value a left noetherian ring (SGA 4 IX 2.1);
- S is the topos of sheaves of sets, for the Zariski topology, on a locally noetherian scheme S, and $A = \mathcal{O}_S$;
- S is the topos of sheaves of sets on a complex analytic space S, and $A = \mathcal{O}_S$.

On the other hand, the sheaf of continuous complex-valued functions on a topological space is not coherent in general.

Of course, if (S, A) is defined by a ringed space, the notion of coherence considered here coincides with the usual notion of Serre (EGA 0_I, 5.3.1).

Proposition 3.7. a) Let

$$f : (S', A') \longrightarrow (S, A)$$

be a morphism of ringed topoi. Assume that f is flat on the right, that is, that the functor $f^* = A' \otimes_A -$ is exact. There is, for $F \in \operatorname{ob} \mathrm{D}_A^-(S)$ and $G \in \operatorname{ob} \mathrm{D}_A^+(S)$, a functorial canonical arrow

$$f^*\mathrm{RHom}(F, G) \longrightarrow \mathrm{RHom}(f^*(F), f^*(G)).$$

This arrow is an isomorphism when F is pseudo-coherent.

b) Suppose A is commutative and coherent, and let $F \in \operatorname{ob} \mathrm{D}^-(S)$, $G \in \operatorname{ob} \mathrm{D}^+(S)$. If $\mathrm{H}^i(F)$ and $\mathrm{H}^i(G)$ are coherent for all i, then the same is true of $\mathrm{Ext}^i(F, G)$.

Proof. a) To define the arrow (cf. SGA 4 XVII), one may suppose that G has injective components, and one takes the composite arrow

$$f^* \operatorname{Hom}(F, G) \longrightarrow \operatorname{Hom}(f^*F, f^*G) \longrightarrow \mathrm{RHom}(f^*F, f^*G).$$

Let us show that it is an isomorphism for F pseudo-coherent. For brevity write T (respectively T') for the functor $f^*\mathrm{RHom}(\cdot, G)$ (respectively $\mathrm{RHom}(f^*\cdot, f^*G)$), and let $a \in \mathbb{Z}$ be such that $\mathrm{H}^i(G) = 0$ for $i < a$. It is clear that one has $TA \xrightarrow{\sim} T'A$, hence by dévissage $TF \xrightarrow{\sim} T'F$ for F strictly perfect (2.1). On the other hand, if F is acyclic in degrees $> n$, then TF and $T'F$ are acyclic in degrees $< a - n$. To verify that $TF \xrightarrow{\sim} T'F$ for F pseudo-coherent, the question being local, one may suppose F strictly n-pseudo-coherent; that is, there is a distinguished triangle

$$F' \longrightarrow F \longrightarrow F'' \longrightarrow F'[1].$$

with F' strictly perfect and F'' acyclic in degrees $> n - 1$. From this one obtains a morphism of distinguished triangles

$$\begin{array}{ccccc} TF'' & \to & TF & \to & TF' \\ \downarrow & & \downarrow & & \downarrow \\ T'F'' & \to & T'F & \to & T'F'. \end{array}$$

Now $TF' \xrightarrow{\sim} T'F'$, since F' is strictly perfect, and TF'' and $T'F''$ are acyclic in degrees $\leq a - n$. Writing the induced morphism on the exact cohomology sequences of the preceding triangles, one deduces that $TF \to T'F$ induces an isomorphism

$$\mathrm{H}^i(TF) \xrightarrow{\sim} \mathrm{H}^i(T'F)$$

for $i \leq a - n$. Since n may be chosen arbitrarily negative, this proves the assertion.

b) The proof is left as an exercise to the reader; see, for example, [H] II 3.3.

Remark 3.7.1. We shall see below (7.1.2) another case of compatibility of RHom with $\mathrm{L}f^*$.

Corollary 3.7.2. (cf. EGA 0_I, 5.2.7.) Let (S, A) be a ringed topos, let s be a point of S (SGA 4 VIII 7.8), and let

$$E, F \in \operatorname{ob} \mathrm{D}^b_{A,\mathrm{coh}}(S).$$

If $E_s \xrightarrow{\sim} F_s$, then there exists an s-pointed object U of S such that

$$E|U \xrightarrow{\sim} F|U.$$

Proof. Let $\alpha : E_s \to F_s$ and $\beta : F_s \to E_s$ be reciprocal isomorphisms. By (2.22 a), one has

$$\operatorname{Hom}(E, F)_s \xrightarrow{\sim} \operatorname{Hom}(E_s, F_s), \qquad \operatorname{Hom}(F, E)_s \xrightarrow{\sim} \operatorname{Hom}(F_s, E_s).$$

There are therefore a neighbourhood U of s and a section u (respectively v) of $\mathrm{Hom}(E,F)$ (respectively of $\mathrm{Hom}(F,E)$) over U such that $u_s = \alpha$ (respectively $v_s = \beta$). The morphisms $vu : E|U \to E|U$ and $uv : F|U \to F|U$ induce the identity at s; hence on a suitable V over U, as required.

4. Perfect complexes

4.0. In this section, S denotes a site, S an object of S, $\mathcal{C}$ a flat abelian S-category, and $\mathcal{C}_0$ a strictly full additive sub-S-category of $\mathcal{C}$ (1.1). Unless otherwise mentioned, we suppose that $\mathcal{C}_0$ is locally liftable in $\mathcal{C}$ and locally stable under kernels of epimorphisms (1.2).

Lemma 4.1. Let

$$u : E \longrightarrow F$$

be an arrow of $\mathrm{C}(\mathcal{C}_S)$. If E is strictly $\mathcal{C}_0$-perfect (2.1) and F is acyclic, then u is locally homotopic to zero.

Proof. One has to construct locally a homotopy k such that

$$u = dk + kd.$$

Since E is strictly perfect, there are $a, b \in \mathbb{Z}$, with $a \leq b$, such that $E^i = 0$ for $i \notin [a,b]$. Thus one may take $k^i = 0$ for $i \notin [a,b]$. We construct the k^i, for $i > a$, by descending induction on i. Suppose $k^i : E^i \to F^{i-1}$ has been constructed so that

$$u^i = k^{i+1}d_E^i + d_F^{i-1}k^i$$

for $i > n > a$. We show that, after localizing, there exists

$$k^{n-1} : E^{n-1} \longrightarrow F^{n-2}$$

such that

$$u^{n-1} = k^n d_E^{n-1} + d_F^{n-2}k^{n-1}.$$

By the induction hypothesis, the arrow

$$v^{n-1} = u^{n-1} - k^n d_E^{n-1} : E^{n-1} \longrightarrow F^{n-1}$$

factors through $Z^{n-1}(F)$. But $Z^{n-1}(F) = B^{n-1}(F)$, since F is acyclic. Since $\mathcal{C}_0$ is locally liftable in $\mathcal{C}$, it follows, after localizing, that there exists $k^{n-1} : E^{n-1} \to F^{n-2}$ such that $v^{n-1} = d_F^{n-2}k^{n-1}$, as required.

Corollary 4.2. Let

$$\begin{array}{ccc} P & & \\ {\scriptstyle v}\downarrow & & \\ F & \xleftarrow[f]{} & E \end{array}$$

be a diagram of $\mathrm{C}(\mathcal{C}_S)$, where P is strictly $\mathcal{C}_0$-perfect and f is a quasi-isomorphism. The set of X over S for which there exists an arrow

$$u : P_X \longrightarrow E_X$$

of $\mathrm{C}(\mathcal{C}_X)$ such that the diagram

$$\begin{array}{ccc} & & P_X \\ & \overset{u}{\nearrow} & \downarrow{\scriptstyle v_X} \\ E_X & \xrightarrow[f_X]{} & F_X \end{array}$$

is commutative in $\mathrm{K}(\mathcal{C}_X)$, is a refinement of S.

Proof. Let G be the cone of f. There is an exact sequence

$$\longrightarrow \mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,E) \longrightarrow \mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,F) \longrightarrow \mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,G) \longrightarrow .$$

Since G is acyclic, the image of v in $\mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,G)$ is locally zero by (4.1). Therefore, after localizing, there exists $u : P \to E$ such that $v = fu$ in $\mathrm{K}(\mathcal{C}_S)$, as required.

Corollary 4.3. Let $f : E \to F$ be a quasi-isomorphism of $\mathrm{C}(\mathcal{C}_S)$, with F strictly perfect relative to $\mathcal{C}_0$. The set of X over S for which there exists a quasi-isomorphism

$$g : F_X \longrightarrow E_X$$

such that $f_X g$ is homotopic to Id_{F_X}, is a refinement of S.

Proof. This is a special case of (4.2).

Scholium 4.4. Let $F \in \mathrm{ob}\,\mathrm{K}(\mathcal{C}_S)$. One may define, for example with the aid of a cleavage of $\mathcal{C}$ (SGA 1 VI 7.1), a category $(\mathrm{Qis}\,/F)$ fibered over S/S, whose fiber over X over S is the category $(\mathrm{Qis}\,/F_X)^\circ$, opposite to the category of quasi-isomorphisms with target F_X. One may then express (4.3) by saying that, if F is strictly perfect, the arrow Id_F is locally cofinal in $(\mathrm{Qis}\,/F)$.

Notation 4.5. Let $\mathcal{M}$ be a fibered S-category. Let $X \in \mathrm{ob}\,S$, and let $E, F \in \mathrm{ob}\,\mathcal{M}_X$. We shall denote by

$$\underline{\mathrm{Hom}}_{\mathcal{M}}(E,F)$$

the sheaf on S/X associated to the presheaf of S-morphisms from E to F in $\mathcal{M}$. For the definition of this presheaf, see [G] I 2.6. One must be careful that, even if $\mathcal{C}$ is a stack ([G]), the corresponding presheaf for $\mathcal{M} = \mathrm{K}(\mathcal{C})$ or $\mathrm{D}(\mathcal{C})$ is not in general a sheaf.

Corollary 4.6. Let $E, F \in \mathrm{ob}\,\mathrm{C}(\mathcal{C}_S)$, with E strictly perfect. The canonical arrow

$$\underline{\mathrm{Hom}}_{\mathrm{K}(\mathcal{C})}(E,F) \longrightarrow \underline{\mathrm{Hom}}_{\mathrm{D}(\mathcal{C})}(E,F)$$

is an isomorphism.

Proof. This is an immediate consequence of (4.4) and of the formula

$$\underline{\mathrm{Hom}}_{\mathrm{D}(\mathcal{C}_X)}(E_X,F_X) = \varinjlim_{(\mathrm{Qis}\,/E_X)^\circ} \underline{\mathrm{Hom}}_{\mathrm{K}(\mathcal{C}_X)}(\,\cdot\,,F_X)$$

for X over S.

Definition 4.7. Let $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C}_S)$, and let $a, b \in \mathbb{Z}$, with $a \leq b$. We shall say that F has $\mathcal{C}_0$-perfect amplitude contained in the interval $[a,b]$, and we shall write

$$\mathcal{C}_0\text{-}\,\mathrm{Parf}\,\text{-}\,\mathrm{amp}(F) \subset [a,b],$$

if the set of X over S for which there exists a quasi-isomorphism

$$F' \longrightarrow F$$

in $\mathrm{C}(\mathcal{C}_X)$, with F' strictly $\mathcal{C}_0$-perfect (2.1) and such that $F'^i = 0$ for $i \notin [a,b]$, is a refinement of S. By (4.3), this condition is equivalent to the one obtained by replacing "quasi-isomorphism in $\mathrm{C}(\mathcal{C}_X)$" by "isomorphism in $\mathrm{D}(\mathcal{C}_X)$." We shall say that F has finite $\mathcal{C}_0$-perfect amplitude if there are integers a and b such that $\mathcal{C}_0$-Parf-amp$(F) \subset [a,b]$. We shall say that F is $\mathcal{C}_0$-perfect if F is locally of finite $\mathcal{C}_0$-perfect amplitude.

Example 4.8. Let (S,A) be a ringed topos, let $\mathcal{C}$ be the S-category defined by

$$U \longmapsto \text{the category of left } A_U\text{-modules},$$

and let $\mathcal{C}_0$ be the sub-S-category defined by

$U \longmapsto$ the category of left A_U-modules which are direct summands of free finite-type modules.

The pair $(\mathcal{C}, \mathcal{C}_0)$ satisfies the hypotheses (4.0) by (1.3.1). An object F of $\mathrm{D}_A(S)$ (cf. (2.15)) will be said to have perfect amplitude contained in $[a, b]$ (respectively finite perfect amplitude, respectively to be perfect) if F has $\mathcal{C}_0$-perfect amplitude contained in $[a, b]$ (respectively the corresponding property). When the direct summands of free finite-type modules are locally free (cf. (2.15.1)), to say that F is perfect means that F is locally isomorphic, in $\mathrm{D}(S)$, to a bounded complex whose components are free of finite type.

Notation 4.9. We shall denote by $\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{- Parf}}$, or $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$, the strictly full sub-$S$-category of $\mathrm{D}(\mathcal{C})$ formed by the $\mathcal{C}_0$-perfect objects. In the case of Example (4.8), we shall write $\mathrm{D}_A(S)_{\mathrm{Parf}}$, or $\mathrm{Parf}_A(S)$, or simply $\mathrm{Parf}(S)$ if no ambiguity about A is possible.

Proposition 4.10. The category $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$ (1.1.4). The canonical functor

$$\mathrm{K}^b(\mathcal{C}_0) \longrightarrow \mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$$

(respectively

$$\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0)) \longrightarrow \mathrm{Tr}(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0))$$

(1.1.3)) induces an S-equivalence on the associated stacks ([G] II 2.2.4), which means:

(i) for every $X \in \mathrm{ob}\, S$ and all $E, F \in \mathrm{ob}\, \mathrm{K}^b(\mathcal{C}_{0,X})$ (respectively $\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_{0,X}))$), the canonical arrow

$$\mathrm{Hom}_{\mathrm{K}^b(\mathcal{C}_0)}(E, F) \longrightarrow \mathrm{Hom}_{\mathrm{D}(\mathcal{C})}(E, F)$$

(respectively

$$\mathrm{Hom}_{\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))}(E, F) \longrightarrow \mathrm{Hom}_{\mathrm{Tr}(\mathrm{D}(\mathcal{C}))}(E, F)$$

) is an isomorphism; and

(ii) every object of $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ (respectively of $\mathrm{Tr}(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0))$) is locally isomorphic in $\mathrm{D}(\mathcal{C})$ (respectively in $\mathrm{Tr}(\mathrm{D}(\mathcal{C}))$) to an object of $\mathrm{K}^b(\mathcal{C}_0)$ (respectively of $\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))$).

Proof. The non-parenthesized assertion (i) is a special case of (4.6), and the non-parenthesized assertion (ii) is true by the definition (4.7) and (4.9). It is clear, moreover, that $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is stable under translation of degrees. To prove that $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$, it remains to show that, if

$$E \xrightarrow{u} F \longrightarrow G \longrightarrow E[1]. \qquad (*)$$

is a distinguished triangle of $\mathrm{D}(\mathcal{C})$ with E and F perfect, then G is perfect. After localizing, we may suppose that E and F are objects of $\mathrm{K}^b(\mathcal{C}_0)$. After localizing further, we may suppose, by the non-parenthesized assertion (i), that u is an arrow of $\mathrm{K}^b(\mathcal{C}_0)$; then G is perfect, since it is isomorphic to the cone of u, which is strictly perfect. Hence $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is indeed a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$. The preceding argument also proves the parenthesized assertion (ii). It remains to prove the parenthesized assertion (i). Let $X \in \mathrm{ob}\, S$, and let

$$E = (E_1 \xrightarrow{a_1} E_2 \xrightarrow{a_2} E_3 \xrightarrow{a_3} E_1[+1]), \quad F = (F_1 \xrightarrow{b_1} F_2 \xrightarrow{b_2} F_3 \xrightarrow{b_3} F_1[+1])$$

be distinguished triangles of $\mathrm{K}^b(\mathcal{C}_{0,X})$. We must show that

$$\mathrm{Hom}_{\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))}(E, F) \longrightarrow \mathrm{Hom}_{\mathrm{Tr}(\mathrm{D}(\mathcal{C}))}(E, F) \qquad (**)$$

is an isomorphism. By the non-parenthesized assertion (i), we already know that $(**)$ is injective. Let $(u_i) : E \to F$ be a morphism of triangles in $\mathrm{D}(\mathcal{C})$. After localizing, the non-parenthesized assertion (i) permits us to suppose that u_i, for $1 \leq i \leq 3$, is an arrow of $\mathrm{K}^b(\mathcal{C}_0)$. The arrow

$$u_{i+1}a_i - b_i u_i,$$

for $1 \leq i \leq 3$ (with the convention $u_4 = u_1$), is zero in $\mathrm{D}(\mathcal{C})$, hence locally zero in $\mathrm{K}^b(\mathcal{C}_0)$ by the non-parenthesized assertion (i). In other words, (u_i) locally defines a morphism of triangles in $\mathrm{K}^b(\mathcal{C}_0)$. This proves that $(**)$ is surjective and completes the proof of (4.10).

Remark 4.10.1. In (4.10), one may replace $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ by

$$\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{- Parf}} = \mathrm{D}^*(\mathcal{C}) \cap \mathrm{Parf}(\mathcal{C}, \mathcal{C}_0) \qquad (* = - \text{ or } b),$$

and nothing changes in the proof.

Complement 4.11. The triangulated structure of $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ can be made more precise in terms of perfect amplitude (4.7). If $E \in \mathrm{ob}\,\mathrm{D}(\mathcal{C})$ has perfect amplitude contained in $[a, b]$, then $E[n]$ has perfect amplitude contained in $[a - n, b - n]$.

Let

$$E \longrightarrow F \longrightarrow G \longrightarrow E[1].$$

be a distinguished triangle of $\mathrm{D}(\mathcal{C})$. One may draw up a table analogous to (2.6), where the entries indicate perfect amplitudes and each row corresponds to an implication whose target is underlined:

E	F	G
$[a+1, b+1]$	$[a, b]$	$\underline{[a, b]}$
$[a, b]$	$\underline{[a, b]}$	$\underline{[a, b]}$
$\underline{[a, b]}$	$\underline{[a, b]}$	$[a-1, b-1]$.

From (4.10) one immediately deduces:

Corollary 4.12. Suppose that S is a punctual site. For $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C})$ to have perfect amplitude relative to $\mathcal{C}_0$ contained in $[a, b]$, it is necessary and sufficient that there exist a quasi-isomorphism $F' \to F$, with F' strictly perfect and such that $F'^i = 0$ for $i \notin [a, b]$. The canonical functors

$$\mathrm{K}^b(\mathcal{C}_0) \longrightarrow \mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$$

and

$$\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0)) \longrightarrow \mathrm{Tr}(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0))$$

are equivalences of categories.

Remark 4.12.1. When S is a punctual site, the objects of $\mathcal{C}_0$ are projective. When $\mathcal{C}_0$ contains all the projective objects of $\mathcal{C}$, one says "projective amplitude" instead of "$\mathcal{C}_0$-perfect amplitude."

Exposé I, continued: perfect amplitude, finite Tor-dimension, and rank

Remark 4.12.2. Let $\mathcal{C}$ be an abelian category (which may be regarded as fibered over a punctual site), and let $\mathcal{C}_0$ be a strictly full additive subcategory. Suppose that $\mathcal{C}_0$ is stable under kernels of epimorphisms and is quasi-liftable in $\mathcal{C}$. Then an acyclic object of $\mathrm{K}^b(\mathcal{C}_0)$ is not in general zero; hence there is no hope that the functor

$$\mathrm{K}^b(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

should be faithful. However, if $\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0)$ denotes the thick subcategory of $\mathrm{K}^b(\mathcal{C}_0)$ formed by the objects which are acyclic in $\mathcal{C}$, the canonical functor

$$\mathrm{K}^b(\mathcal{C}_0)/\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

is fully faithful.

Indications: one already knows from (2.7) that the functor

$$\mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}^-(\mathcal{C})$$

is fully faithful. Show, for example with the aid of the criterion of [V], 4.2 b)(ii), that the functor

$$\mathrm{K}^b(\mathcal{C}_0)/\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0)$$

is fully faithful.

Lemma 4.13. Let $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C}_S)$, and let $a, b, c \in \mathbb{Z}$ be such that $a \leq b$ and $a \leq c$. If F is of perfect amplitude contained in $[a, c]$ (4.7), and if $\mathrm{H}^i(F) = 0$ for $i > b$, then F is of perfect amplitude contained in $[a, d]$, where $d = \inf(b, c)$.

Proof. We may suppose $b \leq c$, and F strictly perfect with $F^i = 0$ for $i \notin [a, c]$. Then F is isomorphic to its truncation F' defined by

$$F'^i = F^i \quad (i < b), \qquad F'^b = Z^b(F), \qquad F'^i = 0 \quad (i > b).$$

But $Z^b(F)$ admits a finite right resolution by objects of $\mathcal{C}_0$, hence is locally an object of $\mathcal{C}_0$, whence the assertion.

Corollary 4.14. Let $F \in \mathrm{ob}\,\mathcal{C}_S$ and $n \in \mathbb{N}$. The following conditions are equivalent:

(i) $\mathrm{perf.amp}(F) \subset [-n, 0]$;

(ii) F locally admits a left resolution of length n by objects of $\mathcal{C}_0$, that is, the set of objects X over S such that there exists an exact sequence

$$0 \longrightarrow L^{-n} \longrightarrow L^{-n+1} \longrightarrow \cdots \longrightarrow L^0 \longrightarrow F_X \longrightarrow 0,$$

with $L^i \in \mathrm{ob}\,\mathcal{C}_{0,X}$ for all i, is a refinement of S.

Proof. This is an immediate consequence of (4.7) and (4.13).

Definition 4.14.1. We shall say that F is of perfect dimension $\leq n$, and shall write $\mathrm{perf.dim}(F) \leq n$, if F satisfies the equivalent conditions of (4.14). We shall call the perfect dimension of F the smallest integer n such that $\mathrm{perf.dim}(F) \leq n$; if there is no such integer, we shall say that F is of infinite perfect dimension.

When S is the punctual site and $\mathcal{C}_0$ is the full subcategory of $\mathcal{C}$ formed by the projective objects, the perfect dimension is just the usual projective dimension.

Proposition 4.15. Let $a, b, n \in \mathbb{Z}$, with $a \leq b$ and $n \geq 0$, and let $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C})$ be such that $F^i = 0$ (respectively $\mathrm{H}^i(F) = 0$) for $i \notin [a, b]$. If, for every $i \in [a, b]$, F^i (respectively $\mathrm{H}^i(F)$) is of perfect dimension $\leq n + (i - a)$, then F is of perfect amplitude contained in $[a - n, b]$. Consequently, if for every $i \in [a, b]$, F^i (respectively $\mathrm{H}^i(F)$) is perfect, then F is perfect.

Proof. Proceed by induction on b (with a and n fixed), by truncating F and applying (4.11), cf. the proof of (2.11).

Lemma 4.16. Let $a, b \in \mathbb{Z}$, $a \leq b$, and let $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C})$ be such that $F^i = 0$ for $i \notin [a, b]$. Suppose that $\mathrm{perf.amp}(F) \subset [a, b]$, and that $F^i \in \mathrm{ob}\,\mathcal{C}_0$ for $i > a$. Then F^a is locally an object of $\mathcal{C}_0$.

Proof. There is an exact sequence, obtained by truncating F,

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F^a[-a] \longrightarrow 0,$$

with $\mathrm{perf.amp}(F') \subset [a+1, b+1]$. From (4.11) and (4.13) one deduces that $\mathrm{perf.dim}(F^a) = 0$, i.e. that F^a is locally an object of $\mathcal{C}_0$.

Proposition 4.17. Let $F', F'' \in \mathrm{ob}\,\mathrm{D}(\mathcal{C}_S)$, and let $a, b \in \mathbb{Z}$, with $a \leq b$. Consider the conditions:

(i) $\mathrm{perf.amp}(F') \subset [a, b]$ and $\mathrm{perf.amp}(F'') \subset [a, b]$;

(ii) $\mathrm{perf.amp}(F' \oplus F'') \subset [a, b]$.

Then (i) implies (ii). If $\mathcal{C}_0$ is locally stable under direct factors, i.e. if the relation $E' \oplus E'' \in \mathrm{ob}\,\mathcal{C}_0$, for $E', E'' \in \mathrm{ob}\,\mathcal{C}_X$ and $X \in \mathrm{ob}\,S$, implies that locally E' and E'' are objects of $\mathcal{C}_0$, then (i) and (ii) are equivalent.

Proof. The implication (i) $\Rightarrow$ (ii) is a particular case of (4.11). Let us prove the converse. Put $F = F' \oplus F''$. Since F is pseudo-coherent, F' and F'' are also pseudo-coherent (2.12). On the other hand, the relation $\mathrm{H}^i(F) = 0$ for $i \notin [a, b]$ implies $\mathrm{H}^i(F') = \mathrm{H}^i(F'') = 0$ for $i \notin [a, b]$. By (2.10 a), we may therefore suppose, after localizing, that F'^i and F''^i are in $\mathrm{ob}\,\mathcal{C}_0$ for $i \geq a+1$, and that $F'^i = F''^i = 0$ for $i > b$. Then, by truncating, we get $F'^i = F''^i = 0$ for $i < a$. Since $\mathrm{perf.amp}(F) \subset [a, b]$, it follows from (4.16) that F^a is locally an object of $\mathcal{C}_0$. But $F^a = F'^a \oplus F''^a$, hence F'^a and F''^a are locally objects of $\mathcal{C}_0$, since $\mathcal{C}_0$ is locally stable under direct factors.

Proposition 4.18. Let $(S', \mathcal{C}', \mathcal{C}'_0)$ be a triple satisfying the same hypotheses (4.0) as $(S, \mathcal{C}, \mathcal{C}_0)$. Let $f : S' \to S$ be a morphism of sites, defined by $f^{-1} : S \to S'$, and let

$$u : \mathrm{D}^*(\mathcal{C}) \longrightarrow \mathrm{D}^*(\mathcal{C}') \qquad (* = -, +, b)$$

be a functor over f (cf. (2.13)). If u sends objects of $\mathcal{C}_0$ to $\mathcal{C}'_0$-perfect complexes, then u defines a functor

$$\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}} \longrightarrow \mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}.$$

Proof. Let $F \in \mathrm{ob}\,\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}}$. We show that $u(F) \in \mathrm{ob}\,\mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}$. Since the question is local, one may suppose F strictly perfect. It remains only to use dévissage, taking into account the fact that $\mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}$ is a triangulated subcategory of $\mathrm{D}^*(\mathcal{C}')$ (4.10).

Remark 4.18.1. Suppose that there exist $m, n \in \mathbb{Z}$, $m \leq n$, such that $F \in \mathrm{ob}\,\mathcal{C}_0$ implies

$$\mathcal{C}'_0\text{-}\,\mathrm{perf.amp}(u(F)) \subset [m, n].$$

Then, if $F \in \mathrm{ob}\,\mathrm{D}^*(\mathcal{C})$, one has

$$\mathcal{C}_0\text{-}\,\mathrm{perf.amp}(F) \subset [a, b] \quad \Rightarrow \quad \mathcal{C}'_0\text{-}\,\mathrm{perf.amp}(u(F)) \subset [a+m, b+n].$$

The proof is left as an exercise to the reader (use (4.11)).

Corollary 4.19. Under the hypotheses of (2.16), if $E' \in \mathrm{ob}\,\mathrm{D}^-(A, S')$ is perfect as an object of $\mathrm{D}^-(A, S')$, and $E \in \mathrm{ob}\,\mathrm{D}^-(A, S)$ is perfect, then

$$E' \overset{\mathrm{L}}{\otimes}_A E$$

is perfect.

Proof. We may suppose that $E' \in \mathrm{ob}\,\mathrm{D}^-(A, S'_A)$, where S' is the final object of S', and apply (4.18) to

$$u = E' \overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S').$$

In particular:

Corollary 4.19.1. a) The functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S')$$

induces a functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^-(A, S')_{\mathrm{parf}}.$$

b) If A is commutative, the functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S)_{\mathrm{parf}} \times_S \mathrm{D}^-(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^-(S)_{\mathrm{parf}}.$$

Remark 4.19.2. One can refine (4.19) in terms of perfect amplitude:

$$\mathrm{perf.\,amp}(E) \subset [a, b], \qquad \mathrm{perf.\,amp}(E') \subset [a', b'] \quad \Rightarrow \quad \mathrm{perf.\,amp}(E' \overset{\mathrm{L}}{\otimes}_A E) \subset [a + a', b + b'].$$

In particular,

$$\mathrm{perf.\,amp}(E) \subset [a, b] \quad \Rightarrow \quad \mathrm{perf.\,amp}(\mathrm{L}f^*(E)) \subset [a, b].$$

5. Finite Tor-dimension and perfection

5.0. The reader will surely have asked the question: what is missing from a pseudo-coherent complex E in order that it be perfect? Is it enough, for example, that E have bounded cohomology? The answer is no. Indeed, take for $\mathcal{C}$ the category of modules over a commutative noetherian ring A, and for $\mathcal{C}_0$ the full subcategory of projective A-modules of finite type. Any A-module of finite type E is pseudo-coherent and has bounded cohomology, but E is perfect only if, in addition, it is of finite projective dimension (or, equivalently, of finite Tor-dimension). This example suggests that the difference between a pseudo-coherent complex and a perfect complex lies in a question of cohomological dimension. This is what we shall now examine in the case of the category of modules of a ringed topos (cf. (3.3)). We shall need a few preliminary sorties on the notion of finite Tor-dimension. We shall be brief, since this question already appears in the literature: cf. [H] II 4 and SGA 4 XVII 4.1.9.

Proposition 5.1. Let (S, A) be a ringed topos, $X \in \mathrm{ob}\,S$, $F \in \mathrm{ob}\,\mathrm{D}^-(A, X)$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. The following conditions are equivalent:

(i) F is isomorphic, in $\mathrm{D}^-(A, X)$, to a complex F' such that F'^i is flat for every i, and zero for $i \notin [m, n]$;

(ii) for every right A_X-module E, one has

$$\mathrm{H}^i(E \overset{\mathrm{L}}{\otimes}_A F) = 0 \quad \text{for } i \notin [m, n].$$

Proof. The implication (i) ⇒ (ii) is trivial; the implication (ii) ⇒ (i) follows, by truncation, from the following lemma.

Lemma 5.1.1. (cf. (4.16)). Let $F \in \mathrm{ob}\,\mathrm{D}^-(A,X)$. Suppose that F satisfies hypothesis (ii) of (5.1), that $F^i = 0$ for $i \notin [m,n]$, and that F^i is flat for $i \neq m$. Then F^m is flat.

Proof. There is a distinguished triangle

$$F' \longrightarrow F \qquad (+1: F \to F^m[-m],\ F^m[-m] \to F')$$

where F' has flat components and is zero outside the interval $[m+1,n]$ (suppose $m < n$). Apply the functor $E \overset{\mathrm{L}}{\otimes}_A \cdot$, for an arbitrary right A_X-module E, and write the exact cohomology sequence.

Definition 5.2. An object F of $\mathrm{D}^-(A,S)$ satisfying the equivalent conditions of (5.1) will be said to be of flat amplitude contained in the interval $[m,n]$, and we shall write

$$\mathrm{tor.amp}(F) \subset [m,n].$$

We shall say that F is of finite flat amplitude, or of finite Tor-dimension, if there exist integers m and n such that the preceding condition is satisfied.

5.3. Let $X \in \mathrm{ob}\,S$. We shall write $\mathrm{D}^-(A,X)_{\mathrm{torf}}$ for the full subcategory of $\mathrm{D}^-(A,X)$ formed by the objects of finite Tor-dimension. By criterion (ii) of (5.1), this is a triangulated subcategory of $\mathrm{D}^-(A,X)$. As X varies, the categories $\mathrm{D}^-(A,X)_{\mathrm{torf}}$ form a triangulated sub-S-category (1.1.4) of $\mathrm{D}^-(A,S)$, which we shall denote by $\mathrm{D}^-(S)_{A\text{-}\mathrm{torf}}$. The functor $\overset{\mathrm{L}}{\otimes}_A$ induces an exact S-functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^b(S_A) \times_S \mathrm{D}(A/S)_{\mathrm{torf}} \longrightarrow \mathrm{D}^b(S_{\mathbb{Z}}).$$

5.4. In order that a left A-module F be of flat amplitude contained in $[-n,0]$, $n \in \mathbb{N}$, it is necessary and sufficient that F be of Tor-dimension $\leq n$ in the ordinary sense, that is, that F admit a left resolution of length n by flat A-modules.

5.5. Let $X \in \mathrm{ob}\,S$, and let $m,n \in \mathbb{Z}$, with $m \leq n$. In order that $F \in \mathrm{ob}\,\mathrm{D}^-(A,X)$ be of flat amplitude contained in $[m,n]$, it is necessary and sufficient that there exist a covering family $(X_i \to X)$ such that $F|X_i$ is of flat amplitude contained in $[m,n]$. In particular, if S/X is of finite type (SGA 4 VI 1.7), and if F is locally of finite Tor-dimension, then F is of finite Tor-dimension.

Proposition 5.6. Under the hypotheses of (2.16), if E' is of flat amplitude contained in $[m',n']$ as an object of $\mathrm{D}^-(A,S')$, and if E is of flat amplitude contained in $[m,n]$, then

$$E' \overset{\mathrm{L}}{\otimes}_A E$$

is of flat amplitude contained in $[m+m', n+n']$.

Proof. This is immediate: use form (5.1 (i)) of the notion of finite Tor-dimension and the fact that, if E is a flat left A-module, then $f^{-1}(E)$ is a flat left $f^{-1}(A)$-module (SGA 4 IV, App.).

In particular:

Corollary 5.6.1. a) The functor

$$\mathrm{L}f^* : \mathrm{D}^-(A,S) \longrightarrow \mathrm{D}^-(A,S')$$

induces a functor

$$\mathrm{L}f^* : \mathrm{D}(A,S)_{\mathrm{torf}} \longrightarrow \mathrm{D}(A,S')_{\mathrm{torf}}.$$

b) If A is commutative, the functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}(S)_{\mathrm{torf}} \times_S \mathrm{D}(S)_{\mathrm{torf}} \longrightarrow \mathrm{D}(S)_{\mathrm{torf}}.$$

Remark 5.6.2. Let $E \in \mathrm{ob}\,\mathrm{D}^-(A,S)$, where S is the final object of S, and let $m,n \in \mathbb{Z}$, with $m \leq n$. If E is of flat amplitude contained in $[m,n]$, then so is E_s, for every point s of S. The converse is true if S has enough points; use form (5.1(ii)) of the notion of flat amplitude and the fact that $\overset{\mathrm{L}}{\otimes}_A$ commutes with passage to the fibers.

Exercises 5.7. a) Suppose that A is commutative and that (S',A') is faithfully flat over (S,A) (cf. (2.16.2)). In order that $E \in \mathrm{ob}\,\mathrm{D}^-(A,S)$[13] be of flat amplitude contained in $[m,n]$, it is necessary and sufficient that $f^*(E)$ have that property.

b) Let $F \in \mathrm{ob}\,\mathrm{D}^+(A,S)$[13]. Show the equivalence of the following conditions:

(i) F is isomorphic, in $\mathrm{D}^+(A,S)$, to a complex F' such that F'^i is injective for every i and zero for $i \notin [m,n]$;

(ii) for every left A_S-module E, one has $\mathrm{Ext}^i(E,F) = 0$ for $i \notin [m,n]$.

If F satisfies the preceding conditions, one says that F is of injective amplitude contained in $[m,n]$. The full subcategory of $\mathrm{D}^+(A,S)$ formed by the objects of finite injective amplitude is a triangulated subcategory, denoted $\mathrm{D}^+(A,S)_{\mathrm{injf}}$. Show that, if A is commutative, the functor

$$\mathrm{R}\,\mathrm{Hom} : \mathrm{D}(S)^\circ \times \mathrm{D}^+(S) \longrightarrow \mathrm{D}(S)$$

induces a functor

$$\mathrm{R}\,\mathrm{Hom} : \mathrm{D}(S)^\circ_{\mathrm{torf}} \times \mathrm{D}(S)_{\mathrm{injf}} \longrightarrow \mathrm{D}(S)_{\mathrm{injf}}.$$

We can now answer the question of (5.0) and state the main result of this section.

Theorem 5.8. Let (S,A) be a ringed topos, $E \in \mathrm{ob}\,\mathrm{D}(A,S)$, and let $m,n \in \mathbb{Z}$, with $m \leq n$. The following conditions are equivalent:

(i) $\mathrm{perf.amp}(E) \subset [m,n]$ (4.8);

(ii) E is $(m-1)$-pseudo-coherent (2.15) and $\mathrm{tor.amp}(E) \subset [m,n]$ (5.2).

Since "perfect" plainly implies "pseudo-coherent", one deduces:

Corollary 5.8.1. In order that E be perfect (4.8), it is necessary and sufficient that E be pseudo-coherent (2.15) and locally of finite Tor-dimension (5.2).

For the proof, we shall need a few lemmas.

Lemma 5.8.2. Let B be a ring of S. If E is a left A-module, G a right B-module, and F an A-B-bimodule (left over A, right over B), there is a canonical functorial homomorphism

$$\mathrm{Hom}_B(F,G) \otimes_A E \longrightarrow \mathrm{Hom}_B(\mathrm{Hom}_A(E,F),G). \tag{5.8.2.1}$$

[13] Here S denotes the final object of the topos S.

It is an isomorphism for G injective and E of finite presentation.

Proof. First define the arrow. By the adjunction formula dear to Cartan, it amounts to defining a homomorphism of B-modules:

$$\mathrm{Hom}_B(F,G)\otimes_A E\otimes_{\mathbb{Z}_S}\mathrm{Hom}_A(E,F)\longrightarrow G.^{14} \qquad (*)$$

There is a canonical homomorphism of A-B-bimodules

$$E\otimes_{\mathbb{Z}_S}\mathrm{Hom}_A(E,F)\longrightarrow F \qquad (**)$$

coming, by adjunction, from the identity arrow of $\mathrm{Hom}_A(E,F)$. Similarly, there is a canonical homomorphism of B-modules

$$\mathrm{Hom}_B(F,G)\otimes_A F\longrightarrow G. \qquad (***)$$

We then define (*) as the composite of (**) and (***). To see that the arrow (5.8.2.1) is an isomorphism when G is injective and E is of finite presentation, it is enough to note that, if G is injective, the source and the target of the arrow are right-exact functors in E, and that the arrow induces the identity for $E=A$.

Here is now a particular case of (5.8).

Lemma 5.8.3. Let E be a left A-module. The following conditions are equivalent:

(i) E is locally a direct factor of a free module of finite type;

(ii) E is flat and of finite presentation.

Proof (after Bourbaki, Alg. comm., Chap. I, §2, Exer. 15). It is clear that (i) implies (ii). Let us prove (ii) $\Rightarrow$ (i). By (1.3.1), it is enough to show that the functor $\mathrm{Hom}_A(E,\cdot)$ is exact. Let

$$F\longrightarrow F''\longrightarrow 0$$

be an exact sequence of left A-modules. We must show that the sequence

$$\mathrm{Hom}_A(E,F)\longrightarrow\mathrm{Hom}_A(E,F'')\longrightarrow 0$$

is exact. For this it is enough to prove that, for every injective $\mathbb{Z}_S$-module I, the sequence obtained by applying the functor $\mathrm{Hom}_{\mathbb{Z}_S}(\cdot,I)$ is exact. But there is a commutative diagram, where the vertical arrows are the canonical arrows of (3.17.1):

$$\begin{array}{ccccc} 0 & \to & \mathrm{Hom}_{\mathbb{Z}_S}(F'',I)\otimes_A E & \to & \mathrm{Hom}_{\mathbb{Z}_S}(F,I)\otimes_A E \\ & & \downarrow & & \downarrow \\ 0 & \to & \mathrm{Hom}_{\mathbb{Z}_S}(\mathrm{Hom}_A(E,F''),I) & \to & \mathrm{Hom}_{\mathbb{Z}_S}(\mathrm{Hom}_A(E,F),I). \end{array}$$

The vertical arrows are isomorphisms by (5.8.2). The top row is exact because E is flat. Hence the bottom row is exact, q.e.d.

Proof of (5.8). The implication (i) $\Rightarrow$ (ii) follows trivially from the definitions. Let us prove (ii) $\Rightarrow$ (i). Since the question is local, one may suppose E strictly $(m-1)$-pseudo-coherent. Since, on the other hand, E is acyclic in degrees $<m$, the truncation arrow

$$\begin{array}{ccccccc} \cdots & \to & E^{m-1} & \to & E^m & \to & E^{m+1}\to\cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \to & 0 & \to & E^m/B^m & \to & E^{m+1}\to\cdots \end{array}$$

[14] We write S for the final object of the topos S, and $\mathbb{Z}_S$ for the constant sheaf with value $\mathbb{Z}$.

is a quasi-isomorphism. Let E' be the truncated complex. By construction, E' is of flat amplitude contained in $[m,n]$, and E'^i is free of finite type for $i > m$. Therefore, by (5.1.1), E'^m is flat. On the other hand there is a distinguished triangle

$$E'^m[-m] \xrightarrow{+1} E'' \longrightarrow E'$$

where E'' is strictly perfect. By (2.6), $E'^m[-m]$ is $(m-1)$-pseudo-coherent; i.e. by (2.9), E'^m is of finite presentation. We conclude from (5.8.3) that E'^m is locally a direct factor of a free module of finite type, hence that $\operatorname{perf.amp}(E) \subset [m,n]$.

Remarks 5.8.4. a) Using (2.5), one recovers the fact that perfect complexes form a triangulated subcategory of $\mathrm{D}(A,S)$.

b) A perfect complex, even with bounded cohomology, is not necessarily of finite Tor-dimension. It is so, however, if S is of finite type (5.5). Write $\mathrm{D}(A,S)_{\mathrm{parf}}$ for the full sub-S-category of $\mathrm{D}^b(A,S)$ formed by perfect complexes of finite Tor-dimension. It follows at once from (2.16.1 b) and (5.3) that, if A is commutative, the functor $\overset{\mathrm{L}}{\otimes}_A$ induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}(S)_{\mathrm{parf}} \times_S \mathrm{D}^b(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}}.$$

This formula is the starting point for the theory of intersections on schemes. Note, on the other hand, that the derived tensor product of two pseudo-coherent complexes with bounded cohomology is not, in general, of bounded cohomology.

Corollary 5.9. Suppose that $S = (\mathrm{Ens})$ (so that A "is" an ordinary ring) and that every left A-module is of finite Tor-dimension; this is the case, for example, if A is of finite global cohomological dimension, cf. (MULT VI 2.6) and (EGA 0_{IV} 17.2.8). Then

$$\text{(i)}\quad \mathrm{D}^b(A,S) = \mathrm{D}(A,S)_{\mathrm{torf}}, \qquad \text{and} \qquad \text{(ii)}\quad \mathrm{D}^b(A,S)_{\mathrm{coh}} = \mathrm{D}(A,S)_{\mathrm{parf}}.$$

In other words, in order that $E \in \operatorname{ob}\mathrm{D}(A,S)$ be of finite Tor-dimension (respectively finite Tor-dimension and perfect), it is necessary and sufficient that E have bounded cohomology (respectively bounded and pseudo-coherent cohomology).

Proof. By (5.8), it is enough to prove (i). Let $E \in \operatorname{ob}\mathrm{D}^b(A,S)$. We may suppose, after extending E by 0, that $E \in \operatorname{ob}\mathrm{D}^b(A,S)$, where S is the final object of S. By induction on the number of non-zero cohomology objects of E, one is reduced, by dévissage, to the case where E is an ordinary A-module; then one wins.

More generally, one may state:

Corollary 5.10. Let $X \in \operatorname{ob} S$. Suppose that S/X has enough points (SGA 4 IV 6.4.1) and that, for every point s of S/X, every left A_s-module is of finite Tor-dimension. Then

$$\mathrm{D}^b(A,X)_{\mathrm{coh}} = \mathrm{D}^b(A,X)_{\mathrm{parf}}.$$

Equivalently, in order that $E \in \operatorname{ob}\mathrm{D}^b(A,X)$ be pseudo-coherent, it is necessary and sufficient that E be perfect.

Proof. Let $E \in \operatorname{ob}\mathrm{D}^b(A,X)_{\mathrm{coh}}$ and let s be a point of S/X. One must show that E is perfect in a neighbourhood of s, that is, that there exists an s-pointed object U of S over X such that $E|U$ is perfect. Now, since the functor s^* is exact, (2.16.1 a) gives $E_s \in \operatorname{ob}\mathrm{D}^b(A_s)_{\mathrm{coh}}$. Hence, by (5.9), E_s is perfect. The assertion is therefore a consequence of (3.7.2) and of the following lemma.

Lemma 5.10.1. Let s be a point of S, and let $F \in \mathrm{ob}\,\mathrm{D}(A_s)_{\mathrm{parf}}$. There exist an s-pointed object U of S and $F' \in \mathrm{ob}\,\mathrm{D}(A_U)_{\mathrm{parf}}$ such that $F'_s = F$.

Proof. This is evident if F is a free A_s-module of finite type. Suppose now that F is projective of finite type, hence the image of a projector

$$p : L \longrightarrow L,$$

where L is free of finite type. There is then a neighbourhood U of s, a free A_U-module L', and a homomorphism $p' : L' \to L'$ such that $p'_s = p$. Since $p_s^2 = p_s$, we may suppose, after shrinking U, that $p'^2 = p'$. Thus $F' = \mathrm{Im}(p')$ is a direct factor of L' and satisfies $F'_s = F$. The case where F is strictly perfect, that is, a bounded complex made up of projective modules of finite type, follows immediately.

Examples 5.11. Corollary (5.10) applies in particular when (S, A) is defined by a locally ringed space in regular local rings, for example a regular scheme or a smooth complex analytic space over $\mathbb{C}$.

6. Rank of a perfect complex

6.0. Let $(X, \mathcal{O}_X)$ be a ringed space in local rings. It is well known that one can attach, to every locally free sheaf of finite type L on an open set U of X, a function $\mathrm{rg}_U(L)$, locally constant on U and with integral values, called the rank of L. The family of functions $\mathrm{rg}_U(L)$ is characterized by the following properties:

(i) if $V \subset U$ is an inclusion of open sets, then $\mathrm{rg}_V(L|V) = \mathrm{rg}_U(L)|V$;

(ii) for every exact sequence of locally free $\mathcal{O}_U$-modules of finite type

$$0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0,$$

one has $\mathrm{rg}_U(L) = \mathrm{rg}_U(L') + \mathrm{rg}_U(L'')$;

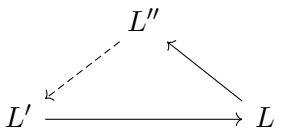
(iii) $\mathrm{rg}_X(\mathcal{O}_X) = 1$.

We propose in this section to extend the notion of rank to perfect complexes.

Definition 6.1 (cf. SGA 5 VIII 2). Let $\mathcal{C}$ be an additive (respectively triangulated) category, and let F be an abelian group. A function

$$f : \mathrm{ob}(\mathcal{C}) \longrightarrow F$$

is said to be an additive function from $\mathcal{C}$ to F if, for all $L, L', L'' \in \mathrm{ob}(\mathcal{C})$ such that $L \simeq L' \oplus L''$ (respectively such that there is a distinguished triangle



), one has

$$f(L) = f(L') + f(L'').$$

The additive functions from $\mathcal{C}$ to F form an abelian group, denoted $\mathrm{Add}(\mathcal{C}, F)$.

6.2. Let $\mathcal{C}$ be a triangulated category, and let f be an additive function from $\mathcal{C}$ to an abelian group F. Then f induces an additive function on the underlying additive category $\mathcal{C}^{\mathrm{ad}}$ of $\mathcal{C}$; hence an inclusion

$$\mathrm{Add}(\mathcal{C}, F) \subset \mathrm{Add}(\mathcal{C}^{\mathrm{ad}}, F).$$

In particular, $f(0) = 0$ and $f(L) = f(M)$ if $L \simeq M$. On the other hand,

$$f(L[n]) = (-1)^n f(L)$$

for all $L \in \mathrm{ob}(\mathcal{C})$ and $n \in \mathbb{Z}$.

6.3. Let $\mathcal{C}$ be an additive (respectively triangulated) category. The group $\mathrm{Add}(\mathcal{C}, F)$ depends functorially on F. The functor $\mathrm{Add}(\mathcal{C}, \cdot)$ is representable (cf. SGA 5 VIII 2) by an abelian group, denoted $k(\mathcal{C})$[15], endowed with an additive application

$$\mathrm{cl}_{\mathcal{C}} : \mathrm{ob}(\mathcal{C}) \longrightarrow k(\mathcal{C}).$$

If $\mathcal{C}$ is additive, one takes for $k(\mathcal{C})$ the symmetrized monoid of isomorphism classes of objects of $\mathcal{C}$. If $\mathcal{C}$ is triangulated, one takes for $k(\mathcal{C})$ the quotient of the free abelian group generated by $\mathrm{ob}(\mathcal{C})$ by the relations identifying L with $L' + L''$ whenever there is a distinguished triangle

$$L' \longrightarrow L \longrightarrow L'' \longrightarrow L'[1].$$

The group $k(\mathcal{C})$ depends functorially on $\mathcal{C}$, with respect to additive (respectively exact) functors. If $u, u' : \mathcal{C} \to \mathcal{D}$ are isomorphic additive (respectively exact) functors, then $k(u) = k(u')$.

If $\mathcal{C}$ is a triangulated category, one has a canonical epimorphism

$$k(\mathcal{C}^{\mathrm{ad}}) \longrightarrow k(\mathcal{C})$$

(cf. (6.2)).

Lemma 6.4. Let $\mathcal{C}$ be an additive category. The canonical functor

$$\mathcal{C} \longrightarrow \mathrm{K}^b(\mathcal{C})$$

induces an isomorphism, functorial in $\mathcal{C}$,

$$k(\mathcal{C}) \xrightarrow{\sim} k(\mathrm{K}^b(\mathcal{C})).$$

Proof. This is an immediate consequence of the following formula, which follows trivially from (6.2):

$$f(L) = \sum_i (-1)^i f(L^i) \tag{6.4.1}$$

for $L \in \mathrm{ob}\,\mathrm{K}^b(\mathcal{C})$ and f an additive function on $\mathrm{K}^b(\mathcal{C})$.

Definition 6.5. Let S be a category, $\mathcal{C}$ an additive (respectively triangulated) S-category (1.1.1), and F an abelian presheaf on S. A family of functions

$$f_X : \mathrm{ob}(\mathcal{C}_X) \longrightarrow F(X), \qquad X \in \mathrm{ob}\, S,$$

is said to be an additive S-function from $\mathcal{C}$ to F if the following conditions are satisfied:

(i) for every $X \in \mathrm{ob}\, S$, f_X is an additive function (6.1);

[15] We shall not use here the usual notation $K(\mathcal{C})$, since $K(\mathcal{C})$ already denotes, when $\mathcal{C}$ is additive, the category of complexes of $\mathcal{C}$ "up to homotopy."

(ii) for every arrow $u : X \to Y$ of S, the following diagram is commutative:

$$\begin{array}{ccc} \mathrm{ob}\,\mathcal{C}_Y & \xrightarrow{f_Y} & F(Y) \\ {\scriptstyle u^*}\downarrow & & \downarrow{\scriptstyle F(u)} \\ \mathrm{ob}\,\mathcal{C}_X & \xrightarrow[f_X]{} & F(X). \end{array}$$

The additive S-functions from $\mathcal{C}$ to F form an abelian group, denoted $\mathrm{Add}_S(\mathcal{C}, F)$.

Remarks 6.6. **a)** Let $\mathcal{C}^{\mathrm{is}}$ denote the fibered S-category whose fiber over $X \in \mathrm{ob}\,S$ is the category with the same objects as $\mathcal{C}_X$ but whose arrows are the isomorphisms of $\mathcal{C}_X$. Let $\underline{F}$ denote the fibered S-category whose fiber over $X \in \mathrm{ob}\,S$ is the discrete category (i.e. the only arrows are identities) $F(X)$, the inverse-image functor by $u \in \mathrm{Fl}(S)$ being $F(u)$. Then an additive S-function from $\mathcal{C}$ to F is simply a Cartesian S-functor from $\mathcal{C}^{\mathrm{is}}$ to $\underline{F}$, inducing on each fiber an additive function in the sense of (6.1). It will be convenient for us to interpret additivity as follows. Suppose, for definiteness, that $\mathcal{C}$ is triangulated. The category

$$\mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$$

(respectively $\mathrm{Cart}_S(\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F})$ (1.1.2)) of Cartesian S-functors from $\mathcal{C}^{\mathrm{is}}$ (respectively $\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}$) to $\underline{F}$ is the discrete category associated to an abelian group, and there is a homomorphism

$$\rho : \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F}) \longrightarrow \mathrm{Cart}_S(\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F}),$$

defined by

$$\rho(f)(T) = f(L) - f(L') - f(L'')$$

for $f \in \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$ and $T = (L' \to L \to L'' \to L'[1]) \in \mathrm{ob}\,\mathrm{Tr}(\mathcal{C})$. To say that $f \in \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$ is additive means that $\rho(f) = 0$, or equivalently that one has a functorial canonical exact sequence

$$0 \longrightarrow \mathrm{Add}_S(\mathcal{C}, F) \longrightarrow \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F}) \xrightarrow{\rho} \mathrm{Cart}_S(\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F}). \tag{6.6.1}$$

b) Let $u : X \to Y$ be an arrow of S. The inverse-image functors by u, being all isomorphic to one another, define by (6.3) a unique homomorphism

$$k(Y) \longrightarrow k(X).$$

Consequently, $X \mapsto k(X)$ defines an abelian presheaf on S, denoted $k(\mathcal{C})$. An additive S-function from $\mathcal{C}$ to F is then simply interpreted as a homomorphism of presheaves

$$k(\mathcal{C}) \longrightarrow F,$$

or in other words

$$\mathrm{Add}_S(\mathcal{C}, F) = \mathrm{Hom}(k(\mathcal{C}), F).$$

The presheaf $k(\mathcal{C})$ of course depends functorially on $\mathcal{C}$ with respect to additive (respectively exact) S-functors.

Proposition 6.7. Let S be a site, $\mathcal{C}$ a flat abelian S-category, and $\mathcal{C}_0$ a strictly full additive sub-S-category of $\mathcal{C}$ satisfying the hypotheses (4.0). Then the morphism of abelian presheaves

$$k(\mathcal{C}_0) \longrightarrow k(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)) \qquad \text{(cf. (4.9))}$$

induced by the inclusion of $\mathcal{C}_0$ in $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$, defines an isomorphism on the associated sheaves. In other words, for every abelian sheaf F on S, the restriction homomorphism

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0), F) \longrightarrow \mathrm{Add}_S(\mathcal{C}_0, F)$$

is an isomorphism.

Proof. Let F be an abelian sheaf on S. We show that the arrow

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C},\mathcal{C}_0),F) \longrightarrow \mathrm{Add}_S(\mathcal{C}_0,F)$$

is an isomorphism. It factors as

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C},\mathcal{C}_0),F) \xrightarrow{a} \mathrm{Add}_S(\mathrm{K}^b(\mathcal{C}_0),F) \xrightarrow{b} \mathrm{Add}_S(\mathcal{C}_0,F).$$

By (6.4), the arrow b is an isomorphism. We are reduced to proving that a is an isomorphism.

Consider the following commutative diagram, whose rows are exact by (6.6.1), and whose vertical arrows are the restriction arrows:

$$\begin{array}{ccccccc} 0 & \to & \mathrm{Add}_S(\mathrm{Parf}(\mathcal{C}),F) & \to & \mathrm{Cart}_S(\mathrm{Parf}(\mathcal{C})^{\mathrm{is}},\underline{F}) & \to & \mathrm{Cart}_S(\mathrm{Tr}(\mathrm{Parf}(\mathcal{C}))^{\mathrm{is}},\underline{F}) \\ & & \downarrow a & & \downarrow & & \downarrow \\ 0 & \to & \mathrm{Add}_S(\mathrm{K}^b(\mathcal{C}_0),F) & \to & \mathrm{Cart}_S(\mathrm{K}^b(\mathcal{C}_0)^{\mathrm{is}},\underline{F}) & \to & \mathrm{Cart}_S(\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))^{\mathrm{is}},\underline{F}). \end{array}$$

Since F is a sheaf, $\underline{F}$ is a stack (G II 2.2.4); hence, by (4.10), the two right vertical arrows are isomorphisms. The same is therefore true of a, q.e.d.

Remark 6.7.1 (cf. 4.10.1). In (6.7), one may replace $\mathrm{Parf}(\mathcal{C},\mathcal{C}_0)$ by $\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}}$ ($* = -$ or b).

6.8. Let (S,A) be a ringed topos, and let $\mathcal{C}$ and $\mathcal{C}_0$ be the fibered S-categories considered in (4.8). There are homomorphisms of abelian presheaves

$$k(\mathcal{C}_0) \xrightarrow{\alpha} k(\mathrm{D}^-(A,S)_{\mathrm{parf}}) \xrightarrow{\beta} k(\mathrm{Parf}(A,S)),$$

which, by (6.7), induce isomorphisms on the associated sheaves. The homomorphism α is compatible with inverse images: that is, if

$$u : (S',A') \longrightarrow (S,A)$$

is a morphism of ringed topoi, there is a commutative diagram, whose vertical arrows are induced by the functor $\mathrm{L}u^*$ (cf. (4.19.1)):

$$\begin{array}{ccc} k(\mathcal{C}_0) & \longrightarrow & k(\mathrm{D}^-(A,S)_{\mathrm{parf}}) \\ \downarrow & & \downarrow \\ k(\mathcal{C}'_0) & \longrightarrow & k(\mathrm{D}^-(A',S')_{\mathrm{parf}}), \end{array}$$

where $\mathcal{C}'_0$ is defined analogously to $\mathcal{C}_0$. If A is commutative, the derived tensor product endows $k(\mathcal{C}_0)$ and $k(\mathrm{D}^-(A,S)_{\mathrm{parf}})$ with structures of presheaves of rings (cf. (4.19.1)), and α is a homomorphism of presheaves of rings.

6.9. In the situation of (6.8), suppose that (S,A) is locally ringed (cf. (2.15.1)). Then, for every $X \in \mathrm{ob}\, S$, $\mathcal{C}_{0,X}$ is the category of locally free A_X-modules of finite type. Let $\mathcal{C}_1$ be the fibered S-category defined by

$$X \longmapsto \text{the category of free } A_X\text{-modules of finite type.}$$

It is clear that $\mathcal{C}_1$ satisfies the hypotheses (4.0). An object of $\mathrm{D}(S)$ is perfect if and only if it is perfect relative to $\mathcal{C}_1$. There are homomorphisms of abelian presheaves

$$k(\mathcal{C}_1) \longrightarrow k(\mathcal{C}_0) \longrightarrow k(\mathrm{D}^-(S)_{\mathrm{parf}}) \longrightarrow k(\mathrm{Parf}(S))$$

(the two on the left being homomorphisms of presheaves of rings), which, by (6.7), induce isomorphisms on the associated sheaves. We shall show that the associated sheaves are in fact canonically isomorphic to the constant sheaf $\mathbb{Z}_S$.

Lemma 6.10. Let (S, A) be a locally ringed topos, let U be an object of S distinct from the empty object, and let $p, q \in \mathbb{N}$. If

$$A_U^p \simeq A_U^q,$$

then $p = q$.

Proof. By an argument similar to that given in the proof of (2.15.1), one is reduced to the case where (S, A) is the locally ringed topos defined by a prescheme; in this case the assertion is trivial: take a point in U.

Definition 6.11. The rank, denoted rg, is the additive S-function from $\mathcal{C}_1$ to $\mathbb{Z}_S$ defined by

$$\operatorname{rg}(L) = p \quad \text{if } L \simeq A_X^p, \qquad X \neq \varnothing,$$

$$\operatorname{rg}(L) = 0 \quad \text{if } L \in \operatorname{ob} \mathcal{C}_{1,X}, \qquad X = \varnothing.$$

By (6.7), the rank function extends uniquely to an additive S-function from $\operatorname{Parf}(S)$ to $\mathbb{Z}_S$, which we shall again call rank and denote by rg.

Proposition 6.12. a) For $L, M \in \operatorname{ob} \mathrm{D}^-(X)_{\mathrm{parf}}$, with $X \in \operatorname{ob} S$, one has

$$\operatorname{rg}(L \overset{\mathrm{L}}{\otimes} M) = \operatorname{rg}(L)\ \operatorname{rg}(M).$$

b) Let

$$u : (S', A') \longrightarrow (S, A)$$

be a morphism of locally ringed topoi. For $M \in \operatorname{ob} \mathrm{D}^-(S)_{\mathrm{parf}}$, one has

$$\operatorname{rg}(\mathrm{L}u^*(M)) = u^*(\operatorname{rg}(M)).$$

In other words, the rank homomorphism

$$\operatorname{rg} : k(\mathrm{D}^-(S)_{\mathrm{parf}}) \longrightarrow \mathbb{Z}_S$$

is a homomorphism of presheaves of rings, compatible with inverse images.

Proof. It is enough to show that the homomorphism induced by rank on the sheaf associated to $k(\mathrm{D}^-(S)_{\mathrm{Parf}})$ is a homomorphism of sheaves of rings compatible with inverse images. By (6.9), one may replace $k(\mathrm{D}^-(S)_{\mathrm{Parf}})$ by $k(\mathcal{C}_1)$. We are thus reduced to verifying a) and b) for free modules of finite type, which is trivial.

6.13. With (S, A) still denoting a locally ringed topos, let $X \in \operatorname{ob} S$ and $n \in H^0(X, \mathbb{Z}_S)$. We define an object of $\operatorname{Parf}(X)$, denoted A_X^n, as follows. If n is constant of value an integer $n \geq 0$ (respectively $n < 0$), put $A_X^n = A_X^{\oplus n}$ (respectively $A_X^n = A_X^{-n}[1]$). In the general case, decompose X into disjoint pieces X_i on which $n|X_i$ is constant, and define A_X^n as the unique object of $\operatorname{Parf}(X)$ inducing $A_{X_i}^{n|X_i}$ on each X_i. More generally, for $L \in \operatorname{ob} \operatorname{Parf}(X)$, one may define L^n by putting $L^n = A_X^n \otimes_A L$.

The map

$$i_X : H^0(X, \mathbb{Z}_S) \longrightarrow k(\operatorname{Parf}(X)), \qquad i_X(n) = \operatorname{cl}(A_X^n),$$

is a homomorphism of rings, with $i_X(1) = 1$, and defines, as X varies, a homomorphism of presheaves of rings

$$i : \mathbb{Z}_S \longrightarrow k(\operatorname{Parf}(S)).$$

Proposition 6.14. Under the hypotheses of (6.13), one has

$$\mathrm{rg} \circ i = \mathrm{Id}(\mathbb{Z}_S).$$

In other words, $k(\mathrm{Parf}(S))$, considered as a presheaf of $\mathbb{Z}_S$-algebras by means of i, is augmented toward $\mathbb{Z}_S$ by the rank. Moreover, the rank

$$\mathrm{rg} : k(\mathrm{Parf}(S)) \longrightarrow \mathbb{Z}_S$$

defines an isomorphism on the associated sheaves.

Proof. The first assertion follows trivially from the relations

$$\mathrm{cl}(L[1]) = -\mathrm{cl}(L), \qquad \mathrm{cl}(L^n) = n\,\mathrm{cl}(L), \qquad L \in \mathrm{ob}\,\mathrm{Parf}(S).$$

Let us prove the second. First observe that i factors in the evident way through $k(\mathcal{C}_1)$. Since $\mathrm{rg} \circ i$ is the identity, we are reduced, by (6.9), to proving that the homomorphism

$$i \circ \mathrm{rg} : k(\mathcal{C}_1) \longrightarrow k(\mathcal{C}_1)$$

induces an isomorphism on the associated sheaves; equivalently, for $L \in \mathrm{ob}\,\mathcal{C}_1$, that $i(\mathrm{rg}(L))$ is locally isomorphic to L. This is immediate.

One deduces, by (6.6 b):

Corollary 6.15. Under the hypotheses of (6.13), for every abelian sheaf F on S, the group of S-additive functions from $\mathrm{Parf}(S)$ to F is canonically identified with $H^0(S, F)$, where S is the final object of S. The rank function corresponds, for $F = \mathbb{Z}_S$, to the unit section of $\mathbb{Z}_S$.

7. Duality of perfect complexes

7.0. In this section, (S, A) denotes a topos ringed in commutative rings, and S also denotes the final object of S. We propose to generalize to perfect complexes the duality notions that are well known for locally free sheaves of finite type.

Proposition 7.1. Let $E \in \mathrm{ob}\,\mathrm{D}^-(S)_{\mathrm{Parf}}$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$.

a) If F has locally bounded cohomology, then so does $\mathrm{RHom}(E, F)$. If F is pseudo-coherent (respectively locally of finite Tor-dimension, respectively perfect), then the same is true of $\mathrm{RHom}(E, F)$.

b) Suppose, moreover, that E is of finite Tor-dimension, a condition satisfied if S is of finite type. Then, if F has bounded cohomology (respectively finite Tor-dimension), the same is true of $\mathrm{RHom}(E, F)$. More precisely, let a, b, m, n be integers such that

$$\mathrm{tor.amp}(E) \subset [a, b]$$

(or, equivalently by (5.8), $\mathrm{parf.amp}(E) \subset [a, b]$), and such that $\mathrm{H}^i(F) = 0$ for $i \notin [m, n]$ (respectively $\mathrm{tor.amp}(F) \subset [m, n]$). Then

$$\mathrm{H}^i(\mathrm{RHom}(E, F)) = 0 \quad (i \notin [m - b, n - a]),$$

respectively

$$\mathrm{tor.amp}(\mathrm{RHom}(E, F)) \subset [m - b, n - a].$$

Proof. Assertion a) is local, so we may suppose E strictly perfect, and then conclude by devissage on E, reducing to $E = A$. For assertion b), we may suppose E strictly perfect with $E^i = 0$ for

$i \notin [a,b]$, and $F^i = 0$ for $i \notin [m,n]$ (respectively F^i flat for every i and $F^i = 0$ for $i \notin [m,n]$). It is then enough to apply the following lemma.

Lemma 7.1.1. Let $E \in \mathrm{ob}\,\mathrm{D}^-(S)$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$. If E is strictly pseudo-coherent, that is, in the sense of (2.1), with components direct factors of free finite-type modules, then the canonical arrow of $\mathrm{D}^+(S)$

$$\mathrm{Hom}^\bullet(E,F) \longrightarrow \mathrm{RHom}(E,F)$$

is an isomorphism.

Proof. The spectral sequence

$$E_2^{p0} = \mathrm{H}^p(\mathrm{Hom}^\bullet(E,F)) \Longrightarrow \mathrm{Ext}^p(E,F)$$

where $\mathrm{Ext}^q(E^i,F^j) = 0$ for every $q > 0$ by (1.3.1), degenerates. One obtains isomorphisms

$$E_2^{p0} = \mathrm{H}^p(\mathrm{Hom}^\bullet(E,F)) \xrightarrow{\sim} \mathrm{Ext}^p(E,F) = \mathrm{H}^p(\mathrm{RHom}(E,F)),$$

which prove the assertion.

Proposition 7.1.2. Let

$$f : (S',A') \longrightarrow (S,A)$$

be a morphism of ringed topoi. Let $E \in \mathrm{ob}\,\mathrm{D}^-(S)$ and $F \in \mathrm{ob}\,\mathrm{D}^b(S)$ be such that $\mathrm{RHom}(E,F) \in \mathrm{ob}\,\mathrm{D}^b(S)$ and $\mathrm{L}f^*(F) \in \mathrm{ob}\,\mathrm{D}^b(S')$. There is a functorial canonical arrow

$$\mathrm{L}f^*\mathrm{RHom}(E,F) \longrightarrow \mathrm{RHom}(\mathrm{L}f^*E, \mathrm{L}f^*F).$$

This is an isomorphism if E is perfect.

Proof. By the adjunction of $\mathrm{L}f^*$ and $\mathrm{R}f_*$, it is enough to define an arrow

$$\mathrm{RHom}(E,F) \longrightarrow \mathrm{R}f_*\mathrm{RHom}(\mathrm{L}f^*E, \mathrm{L}f^*F).$$

For this we use the adjunction arrow $F \to \mathrm{R}f_*\mathrm{L}f^*F$ and the duality isomorphism

$$\mathrm{R}f_*\mathrm{RHom}(\mathrm{L}f^*E, \mathrm{L}f^*F) \xrightarrow{\sim} \mathrm{RHom}(E, \mathrm{R}f_*\mathrm{L}f^*F).$$

To see that the arrow of (7.1.2) is an isomorphism for E perfect, one reduces trivially, by localization and devissage, to the case $E = A$, where it is a tautology.

Proposition 7.2. There exists (cf. SGA 5 I 1.6), for $E \in \mathrm{ob}\,\mathrm{D}(S)$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$, a functorial canonical arrow

$$E \longrightarrow \mathrm{RHom}(\mathrm{RHom}(E,F),F).$$

This is an isomorphism if $E \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{Parf}}$ and $F = A$.

Proof. Define the arrow by resolving F injectively and using the canonical arrow in $\mathrm{K}(S)$

$$E \longrightarrow \mathrm{Hom}^\bullet(\mathrm{Hom}^\bullet(E,F),F).$$

To see that the arrow in question is an isomorphism if E is perfect and $F = A$, one reduces, by localization and devissage, to $E = A$, where it is evident.

Definition 7.3. Let $E \in \mathrm{ob}\,\mathrm{D}(S)$. We shall put

$$\mathrm{RHom}(E,A) = E^\vee,$$

and we shall say that $E^\vee$ is the dual of E.

By (7.1) and (7.2), if $E \in \mathrm{ob}\,\mathrm{D}(S)$ is perfect (respectively perfect and of finite Tor-dimension), then so is $E^\vee$, and the canonical arrow $E \to E^{\vee\vee}$ is an isomorphism. In other words, the functor

$\mathrm{RHom}(\,\cdot\,, A)$ defines an anti-auto-equivalence of the category $\mathrm{D}(S)_{\mathrm{Parf}}$ (respectively $\mathrm{D}(S)_{\mathrm{parf}}$, cf. (5.8.4 b)) with itself.

Lemma 7.4 (Cartan isomorphism) (cf. ([H] II 5.15)). For $E \in \mathrm{ob}\,\mathrm{D}(S)$, $F \in \mathrm{ob}\,\mathrm{D}^-(S)$, and $G \in \mathrm{ob}\,\mathrm{D}^+(S)$, there are functorial canonical isomorphisms

$$\begin{aligned}\mathrm{RHom}(E \overset{\mathrm{L}}{\otimes} F, G) &\xrightarrow{\sim} \mathrm{RHom}(E, \mathrm{RHom}(F, G)),\\ \mathrm{RHom}(E \otimes F, G) &\xrightarrow{\sim} \mathrm{RHom}(E, \mathrm{RHom}(F, G)),\\ \mathrm{Hom}(E \otimes F, G) &\xrightarrow{\sim} \mathrm{Hom}(E, \mathrm{RHom}(F, G)).\end{aligned}$$

Proof. This is standard: resolve G injectively and F flatly.

Lemma 7.5. For $E \in \mathrm{ob}\,\mathrm{D}^-(S)$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$, there is a functorial canonical homomorphism

$$E \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(E, F) \longrightarrow F.$$

Proposition 7.6 (cf. ([H] II 5.13)). Let $E \in \mathrm{ob}\,\mathrm{D}(S)$, $F \in \mathrm{ob}\,\mathrm{D}^+(S)$, and $G \in \mathrm{ob}\,\mathrm{D}^+(S)$. Under either of the following hypotheses:

(i) G is of finite Tor-dimension;

(ii) $E \in \mathrm{ob}\,\mathrm{D}^-(S)$, F is of finite Tor-dimension, and $G \in \mathrm{ob}\,\mathrm{D}^b(S)$,

there is a functorial canonical homomorphism

$$\mathrm{RHom}(E, F) \overset{\mathrm{L}}{\otimes} G \longrightarrow \mathrm{RHom}(E, F \overset{\mathrm{L}}{\otimes} G).$$

This is an isomorphism when E is perfect.

Proof. Let us define the arrow. Under hypothesis (i), it is enough to resolve F injectively and G flatly. Under hypothesis (ii), by (7.4) it is equivalent to define an arrow

$$E \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(E, F) \overset{\mathrm{L}}{\otimes} G \longrightarrow F \overset{\mathrm{L}}{\otimes} G.$$

Apply the functor $- \otimes^{\mathrm{L}} G$ to the arrow of (7.5).

To see that the arrow is an isomorphism when E is perfect, use the standard method (cf. the proof of (7.2)): reduce, by localization and devissage, to the case $E = A$.

Corollary 7.7. There exists, for $E \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{Parf}}$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$, a functorial isomorphism

$$E^\vee \overset{\mathrm{L}}{\otimes} F \xrightarrow{\sim} \mathrm{RHom}(E, F),$$

with the notation of (7.3).

Proof. Apply (7.6), with F and G replaced respectively by A and F, and use the fact (7.1) that $E^\vee \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{Parf}}$.

Corollary 7.8. Let $E, F \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{Parf}}$. There is a natural arrow

$$\mathrm{RHom}(E, F) \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(F, E) \longrightarrow A,$$

which is a "perfect duality," that is, modulo (7.4), it identifies each of the complexes $\mathrm{RHom}(E, F)$ and $\mathrm{RHom}(F, E)$ with the dual of the other, in the sense of (7.3).

Proof. By (7.5), one has natural pairings $E^\vee \otimes^{\mathrm{L}} E \to A$ and $F^\vee \otimes^{\mathrm{L}} F \to A$. Hence, by tensor product, one obtains a natural pairing

$$E^\vee \overset{\mathrm{L}}{\otimes} F \overset{\mathrm{L}}{\otimes} F^\vee \overset{\mathrm{L}}{\otimes} E \longrightarrow A,$$

which, in view of (7.7), gives the announced pairing on the RHom's. Let us show that it is a perfect duality. There are canonical isomorphisms

$$\begin{aligned} \mathrm{RHom}(F^\vee \overset{\mathrm{L}}{\otimes} E, A) &\xrightarrow{\sim} \mathrm{RHom}(F^\vee, E^\vee) && (7.4)\\ &\xrightarrow{\sim} F^{\vee\vee} \overset{\mathrm{L}}{\otimes} E^\vee && (7.6)\\ &\xrightarrow{\sim} F \overset{\mathrm{L}}{\otimes} E^\vee && (7.2). \end{aligned}$$

Thus, up to a compatibility verification, the result follows. One could also localize and devissage to reduce to $E = F = A$.

Exercise 7.9. Let $E, F \in \mathrm{ob}\, \mathrm{D}(S)_{\mathrm{Parf}}$. Show that there is a canonical isomorphism

$$E^\vee \overset{\mathrm{L}}{\otimes} F^\vee \xrightarrow{\sim} (E \overset{\mathrm{L}}{\otimes} F)^\vee.$$

8. Traces and cup-products

8.0. In this section we generalize to perfect complexes the notion of the trace of an endomorphism of a projective module of finite type (see SGA 5 III for applications). The hypotheses and notation are those of (7.0).

8.1. Let $E \in \mathrm{ob}\, \mathrm{D}(S)_{\mathrm{parf}}$ (5.8.4 b). There is a natural arrow

$$\tau : \mathrm{RHom}(E, E) \longrightarrow A,$$

obtained by composing the inverse of the isomorphism (7.7)

$$\mathrm{RHom}(E, E) \xrightarrow{\sim} E^\vee \overset{\mathrm{L}}{\otimes} E$$

with the canonical arrow (7.5) $E^\vee \otimes^{\mathrm{L}} E \to A$.

Let $U \in \mathrm{ob}\, S$. Applying to τ the functor $H^0(U, \cdot)$, and using the fact that $H^0(U, \mathrm{RHom}(E, E))$ identifies with $\mathrm{Hom}_{\mathrm{D}(U)}(E|U, E|U)$, one obtains a homomorphism of $H^0(U, A)$-modules

$$\mathrm{Tr}_{E|U} : \mathrm{Hom}_{\mathrm{D}(U)}(E|U, E|U) \longrightarrow H^0(U, A),$$

which we shall call the trace homomorphism. The finite Tor-dimension hypothesis on E is in fact superfluous for the definition of $\mathrm{Tr}_{E|U}$; see (8.5).

Before going further, we give some properties of the trace that follow immediately from the definition.

8.1.1. If $E = A$, then

$$\mathrm{Tr}_{A|U} : \mathrm{Hom}(A|U, A|U) \longrightarrow H^0(U, A)$$

is just the well-known isomorphism. If E is a direct factor of a free module of finite type, an endomorphism of E over U is locally defined by

$$e \longmapsto \sum_i \langle e, x_i^\vee \rangle x_i,$$

where x_i (respectively $x_i^\vee$) is a local section of E (respectively $E^\vee$); the corresponding trace is $\sum_i \langle x_i, x_i^\vee \rangle$. It follows easily that, if E and F are direct factors of free modules of finite type, and if

$$u : (E+F)|U \longrightarrow (E+F)|U$$

is defined by a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$\mathrm{Tr}_{(E+F)|U}(u) = \mathrm{Tr}_{E|U}(a) + \mathrm{Tr}_{F|U}(d).$$

8.1.2. Let E be a strictly perfect complex, let $U \in \mathrm{ob}\, S$, and let

$$u : E|U \longrightarrow E|U$$

be a homomorphism of complexes, with components $u^i : E^i|U \to E^i|U$. Then

$$\mathrm{Tr}_{E|U}(u) = \sum_i (-1)^i \mathrm{Tr}_{E^i|U}(u^i).$$

Indeed, the question being local, one may suppose that u is defined by a section of $H^0(E^\vee \otimes E)$, and even that this section is of the form

$$\sum_{i,j} x_{ij}^\vee \otimes x_{ij},$$

where x_{ij} (respectively $x_{ij}^\vee$) is a section of E^i (respectively $E^{i\vee}$). Making explicit the arrow (7.5) $E^\vee \otimes^{\mathrm{L}} E \to A$, one finds

$$\mathrm{Tr}_{E|U}(u) = \sum_{i,j} (-1)^i \langle x_{ij}, x_{ij}^\vee \rangle = \sum_i (-1)^i \mathrm{Tr}_{E^i|U}(u^i).$$

8.2. Let $E, F \in \mathrm{ob}\, \mathrm{D}(S)_{\mathrm{Parf}}$. We have seen in (7.8) that there is a canonical homomorphism

$$\chi : \mathrm{RHom}(E,F) \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(F,E) \longrightarrow A.$$

Let $U \in \mathrm{ob}\, S$, and let

$$u \in H^0(U, \mathrm{RHom}(E,F)), \qquad v \in H^0(U, \mathrm{RHom}(F,E)).$$

One may interpret u (respectively v) as a homomorphism, in $\mathrm{D}(U)$, from $A|U$ to $\mathrm{RHom}(E,F)|U$ (respectively $\mathrm{RHom}(F,E)|U$), and form

$$u \overset{\mathrm{L}}{\otimes} v : A|U \longrightarrow (\mathrm{RHom}(E,F) \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(F,E))|U.$$

The map $(u,v) \mapsto \langle u, v \rangle = \chi \circ (u \otimes^{\mathrm{L}} v)$ defines an $H^0(U,A)$-bilinear pairing

$$\mathrm{Hom}_{\mathrm{D}(U)}(E|U, F|U) \otimes_{H^0(U,A)} \mathrm{Hom}_{\mathrm{D}(U)}(F|U, E|U) \longrightarrow H^0(U,A),$$

which we shall call the cup-product. The same comment as for the definition of $\mathrm{Tr}_{E|U}$ applies here.

The relation between trace and cup-product is expressed in the following proposition.

Proposition 8.3. In the situation of (8.2), for $u \in \mathrm{Hom}(E|U, F|U)$ and $v \in \mathrm{Hom}(F|U, E|U)$, one has

$$\mathrm{Tr}_{E|U}(vu) = \mathrm{Tr}_{F|U}(uv) = \langle u, v \rangle.$$

In particular, if $E = F$, then

$$\mathrm{Tr}_{E|U}(u) = \langle u, \mathrm{Id}\rangle = \langle \mathrm{Id}, u\rangle.$$

Proof. For simplicity we suppose $U = S$. We therefore have two arrows

$$u : A \longrightarrow E^\vee \overset{\mathrm{L}}{\otimes} F, \qquad v : A \longrightarrow F^\vee \overset{\mathrm{L}}{\otimes} E,$$

and hence

$$u \overset{\mathrm{L}}{\otimes} v : A \longrightarrow E^\vee \overset{\mathrm{L}}{\otimes} F \overset{\mathrm{L}}{\otimes} F^\vee \overset{\mathrm{L}}{\otimes} E.$$

On the other hand, we have seen that $vu : A \to E^\vee \otimes^{\mathrm{L}} E$ and $uv : A \to F^\vee \otimes^{\mathrm{L}} F$. The proposition follows from the commutativity of the following diagrams, whose verification is purely mechanical:

$$\begin{array}{ccc} A & \xrightarrow{u \otimes^{\mathrm{L}} v} & E^\vee \otimes^{\mathrm{L}} F \otimes^{\mathrm{L}} F^\vee \otimes^{\mathrm{L}} E \\ & \searrow^{vu} & \downarrow{\scriptstyle \mathrm{Id} \otimes \tau_F} \\ & & E^\vee \otimes^{\mathrm{L}} E \end{array} \qquad \begin{array}{ccc} A & \xrightarrow{u \otimes^{\mathrm{L}} v} & E^\vee \otimes^{\mathrm{L}} F \otimes^{\mathrm{L}} F^\vee \otimes^{\mathrm{L}} E \\ & \searrow^{uv} & \downarrow{\scriptstyle \tau_E \otimes \mathrm{Id}} \\ & & F^\vee \otimes^{\mathrm{L}} F \end{array}$$

$$\begin{array}{ccc} E^\vee \otimes^{\mathrm{L}} F \otimes^{\mathrm{L}} F^\vee \otimes^{\mathrm{L}} E & \xrightarrow{\mathrm{Id} \otimes \tau_E} & F^\vee \otimes^{\mathrm{L}} F \\ \downarrow{\scriptstyle \mathrm{Id} \otimes \tau_F} & & \downarrow{\scriptstyle \tau_F} \\ E^\vee \otimes^{\mathrm{L}} E & \xrightarrow[\tau_E]{} & A. \end{array}$$

Corollary 8.4. Let E, F, U be as in (8.2), let $u \in \mathrm{Hom}(E|U, E|U)$, and let $s : E|U \to F|U$ be an isomorphism. Then $sus^{-1} \in \mathrm{Hom}(F|U, F|U)$, and

$$\mathrm{Tr}_{F|U}(sus^{-1}) = \mathrm{Tr}_{E|U}(u).$$

Proof. Immediate.

Remark 8.5. If $\mathcal{C}$ is a fibered S-category, denote by $\mathcal{E}(\mathcal{C})$ the fibered S-category whose fiber over $U \in \mathrm{ob}\, S$ is the category whose objects are pairs (L, u), with $L \in \mathrm{ob}\, \mathcal{C}_U$ and $u \in \mathrm{End}_S(L)$, and whose arrows from (L, u) to (M, v) are the arrows $f : L \to M$ such that $fu = vf$. By (8.4) one may regard the trace homomorphism as an S-functor from $\mathcal{E}(\mathrm{D}(S)_{\mathrm{Parf}})^{\mathrm{is}}$ to $\mathcal{A}$, where the exponent is means that only isomorphisms are considered, and where $\mathcal{A}$ is the S-category with discrete fibers defined by A (cf. (6.6 a)). Since every perfect complex is locally of finite Tor-dimension, the forgetful S-functor

$$\mathcal{E}(\mathrm{D}(S)_{\mathrm{Parf}})^{\mathrm{is}} \longrightarrow \mathcal{E}(\mathrm{D}(S)_{\mathrm{parf}})^{\mathrm{is}}$$

induces an isomorphism on the associated stacks. Consequently, the trace homomorphism extends uniquely to an S-functor from $\mathcal{E}(\mathrm{D}(S)_{\mathrm{Parf}})^{\mathrm{is}}$ to $\mathcal{A}$.

For $E \in \mathrm{ob}\, \mathrm{Parf}(U)$, $U \in \mathrm{ob}\, S$, we again write

$$\mathrm{Tr}_E : \mathrm{Hom}(E, E) \longrightarrow H^0(U, A)$$

for the corresponding homomorphism, which is an $H^0(U, A)$-module homomorphism. One sees in the same way that the definition of the cup-product (8.2), as well as assertions (8.3) and (8.4), generalize to perfect complexes, not necessarily of finite Tor-dimension.

Proposition 8.6. Let $U \in \operatorname{ob} S$, let $E, F \in \operatorname{Parf}(U)$, and let $u \in \operatorname{Hom}(E,E)$, $v \in \operatorname{Hom}(F,F)$; hence $u \otimes^{\mathrm{L}} v \in \operatorname{Hom}(E \otimes^{\mathrm{L}} F, E \otimes^{\mathrm{L}} F)$. Then

$$\operatorname{Tr}_{E \otimes^{\mathrm{L}} F}(u \overset{\mathrm{L}}{\otimes} v) = \operatorname{Tr}_E(u)\ \operatorname{Tr}_F(v).$$

Proof. The question being local, by (8.4) and (4.10) one may suppose that u and v are arrows of $K^b(\mathcal{C}_0)$. Using (8.1.2), one is reduced to the case where u and v are arrows of $\mathcal{C}_0$. One finishes by (8.1.1).

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Exposé II. Existence of global resolutions

by L. Illusie

1. General globalization criteria

1.0

In this section we are given a site S, a flat abelian S-category $\mathcal{C}$ (I 1.1), and a strictly full additive sub-S-category $\mathcal{C}_0$ (I 1.1). We assume that $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$ (I 1.2) and locally quasi-stable under kernels of epimorphisms (I 1.2).

Proposition 1.1. Let $S \in \operatorname{ob}(S)$, and suppose that:

(i) $\mathcal{C}_{0S}$ is quasi-liftable in $\mathcal{C}_S$ (I 1.2), where $\mathcal{C}_S$ is regarded as a category fibered over a punctual site;

(ii) every object of $\mathcal{C}_S$ which is of finite $\mathcal{C}_0$-type (I 1.2), respectively of finite $\mathcal{C}_0$-presentation (I 2.8), is of finite $\mathcal{C}_{0S}$-type.

Then:

a) $\mathcal{C}_{0S}$ is quasi-stable under kernels of epimorphisms (I 1.2), where $\mathcal{C}_{0S}$ is regarded as a category fibered over a punctual site.

b) Let $E \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. If E is n-$\mathcal{C}_0$-pseudo-coherent (I 2.2), then E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. In other words,

$$\mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_0\text{-coh}} = \mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_{0S}\text{-coh}},$$

and, respectively,

$$\mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_0\text{-coh}} \subset \mathrm{D}^-(\mathcal{C}_S)_{(n+1)-\mathcal{C}_{0S}\text{-coh}} \subset \mathrm{D}^-(\mathcal{C}_S)_{(n+1)-\mathcal{C}_0\text{-coh}},$$

inclusions of strictly full subcategories of $\mathrm{D}(\mathcal{C}_S)$, the inclusion on the right being trivial.

c) For an object $E \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ to be pseudo-coherent relative to $\mathcal{C}_0$ (I 2.3), it is necessary and sufficient that it be pseudo-coherent relative to $\mathcal{C}_{0S}$; equivalently,

$$\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}} = \mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_{0S}\text{-coh}}.$$

Moreover, the canonical functor

$$\mathrm{K}^-(\mathcal{C}_{0S})/\mathrm{K}^{-,\delta}(\mathcal{C}_{0S}) \longrightarrow \mathrm{D}(\mathcal{C}_S),$$

where $\mathrm{K}^{-,\delta}(\mathcal{C}_{0S})$ denotes the thick subcategory of $\mathrm{K}^-(\mathcal{C}_{0S})$ formed by acyclic complexes, is fully faithful and has as essential image $\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}}$. If $\mathcal{C}_{0S}$ is liftable in $\mathcal{C}_S$ (I 1.2), that is, if the objects of $\mathcal{C}_{0S}$ are projective, then $\mathrm{K}^{-,\delta}(\mathcal{C}_{0S})$ is reduced to zero objects, and consequently the functor

$$\mathrm{K}^-(\mathcal{C}_{0S}) \longrightarrow \mathrm{D}(\mathcal{C}_S)$$

is fully faithful and has essential image $\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}}$.

Proof. a) Let

$$0 \longrightarrow L \longrightarrow L^0 \longrightarrow \cdots \longrightarrow L^m \longrightarrow 0$$

be an exact sequence in $\mathcal{C}_S$, with $L^i \in \operatorname{ob} \mathcal{C}_{0S}$ for all i. It is enough to prove that L is of finite $\mathcal{C}_{0S}$-type. By (I 2.2), L is $\mathcal{C}_0$-pseudo-coherent, hence, by (I 2.9), of finite $\mathcal{C}_0$-presentation; the assertion follows from (ii).

b) We prove by descending induction on i that E is i-$\mathcal{C}_{0S}$-pseudo-coherent for $i \geq n$, respectively for $i \geq n+1$. One may already suppose that E is strictly $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent, respectively strictly $(n+2)$-$\mathcal{C}_{0S}$-pseudo-coherent (I 2.1); the point is then to prove that E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. Thus there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

where E' is strictly $\mathcal{C}_{0S}$-perfect (I 2.1) and $E''^i = 0$ for $i > n$, respectively for $i > n+1$. Replacing E by E'', one may therefore suppose (I 2.6) that $E^i = 0$ for $i > n$, respectively for $i > n+1$. Then, by (I 2.10 a)), $H^n(E)$, respectively $H^{n+1}(E)$, is of finite $\mathcal{C}_0$-type, respectively of finite $\mathcal{C}_0$-presentation. Hence, by (ii), $H^n(E)$, respectively $H^{n+1}(E)$, is of finite $\mathcal{C}_{0S}$-type. Applying (I 2.10 b)) again, one obtains that E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. The hypothesis (i) and part a) have been used implicitly in the references to (I 2.6 and 2.10).

c) The first assertion follows from b), the second from (I 2.7), and the last is evident.

Proposition 1.2. Suppose that $\mathcal{C}_0$ is locally liftable in $\mathcal{C}$ (I 1.2) and locally stable under kernels of epimorphisms (I 1.2), and make the hypotheses (1.1 (i) and (ii), respectively). Let $\mathcal{C}_{1S}$ denote the strictly full subcategory of $\mathcal{C}_S$ formed by the objects which are locally in $\mathcal{C}_0$. Then:

a) $\mathcal{C}_{1S}$ is stable under kernels of epimorphisms (I 1.2) and is quasi-liftable in $\mathcal{C}_S$ (I 1.2), as a category fibered over a punctual site.

b) Let $E \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ and let $[m,n]$ be an interval of $\mathbb{Z}$. For E to have $\mathcal{C}_0$-perfect amplitude contained in $[m,n]$, it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(\mathcal{C}_S)$, to a complex F such that $F^i = 0$ for $i \notin [m,n]$, $F^i \in \operatorname{ob} \mathcal{C}_{0S}$ for $i > m$, and $F^m \in \operatorname{ob} \mathcal{C}_{1S}$.

c) The canonical functor
$$\mathrm{K}^b(\mathcal{C}_{1S})/\mathrm{K}^{b,\delta}(\mathcal{C}_{1S}) \longrightarrow \mathrm{D}(\mathcal{C}_S)$$
is fully faithful and has as essential image the full subcategory of $\mathrm{D}(\mathcal{C}_S)$ formed by complexes of finite $\mathcal{C}_0$-perfect amplitude.

d) If $\mathcal{C}_{1S}$ is not merely quasi-liftable but liftable in $\mathcal{C}_S$, one may replace, in b), the expression "E is isomorphic to F in $\mathrm{D}(\mathcal{C}_S)$" by "there exists a quasi-isomorphism $F \to E$"; and, in c), $\mathrm{K}^{b,\delta}(\mathcal{C}_{1S})$ is reduced to zero objects.

Proof. a) It is clear that $\mathcal{C}_{1S}$ is stable under kernels of epimorphisms. On the other hand, the objects of $\mathcal{C}_{1S}$ are $\mathcal{C}_0$-pseudo-coherent, hence $\mathcal{C}_{0S}$-pseudo-coherent by (1.1 c)); the second assertion follows from (I 2.14), taking S to be the punctual site.

b) It suffices to prove necessity. By (1.1 c)), E is already $\mathcal{C}_{0S}$-pseudo-coherent. Since E is acyclic in degree $> n$, one may, after replacing E by an isomorphic complex in $\mathrm{D}(\mathcal{C}_S)$, suppose (I 2.10) that E is strictly $(m+1)$-$\mathcal{C}_{0S}$-pseudo-coherent and that $E^i = 0$ for $i > n$. Since E is acyclic in degree $< m$, one may also replace E by its truncation
$$0 \longrightarrow E^m/B^m \longrightarrow E^{m+1} \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0,$$
so that $E^i = 0$ for $i < m$. Then, by (I 4.16), E^m is locally an object of $\mathcal{C}_0$, and therefore an object of $\mathcal{C}_{1S}$.

c) In view of (1.2 a)), the assertion follows from (1.2 b)) and (I 4.12.2). d) The fact that $\mathrm{K}^{b,\delta}(\mathcal{C}_{1S})$ is reduced to zero objects is evident. The other assertion follows from (I 4.6), applied over the punctual site.

2. Application to certain categories of modules

2.0. Notation

Let (S, A) be a ringed topos. For $U \in \operatorname{ob} S$, write $\mathrm{Mod}(U)$ for the category of left A_U-modules, and $\mathrm{Lib}(U)$, respectively $\mathrm{Loclib}(U)$, for the full subcategory of $\mathrm{Mod}(U)$ formed by free A_U-modules, respectively modules locally direct factors of free modules, of finite type.

2.1. Lifting lemmas

Lemma 2.1.1 (cf. EGA 0_{I}, 5.2.3). Let (S, A) be a ringed topos, and let $S \in \operatorname{ob} S$. Suppose that S is quasi-compact, that is, every covering family $(S_i \to S)$ is dominated by a finite covering family.

a) Let F be a left A_S-module which is the filtered inductive limit, over a filtered category Λ, of A_S-modules F_λ, and let
$$u = \varinjlim u_\lambda : F \longrightarrow G$$
be an epimorphism, where G is an A_S-module of finite type. Then there exists $\lambda \in \Lambda$ such that u_λ is an epimorphism.

b) Let F be a left A_S-module generated by its sections, that is, such that there exists an epimorphism $A_S^{(I)} \to F$ for a suitable set of indices I. If F is of finite type, then there exists an epimorphism $A_S^p \to F$ for some $p \in \mathbb{N}$.

The proof of a) is analogous to the proof of the cited result and is left to the reader; b) follows immediately from a).

Lemma 2.1.2. Let (S, A) be a ringed topos, let $\mathcal{C}$ be the S-category fibered in left A-modules, let $\mathcal{C}_0$ be the fibered subcategory of free A-modules of finite type (I 1.3 d)), let $S \in \operatorname{ob} S$, and let $\mathcal{C}'_S$ be a strictly full additive subcategory of $\mathcal{C}_S$ containing $\mathcal{C}_{0S}$ and stable under kernels, cokernels, and extensions. Suppose that

$$H^1(S, F) = 0 \quad \text{for every } F \in \operatorname{ob} \mathcal{C}'_S.$$

Then:

a) Every exact sequence in $\mathcal{C}_S$

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0,$$

such that $\operatorname{Hom}(G, E) \in \operatorname{ob} \mathcal{C}'_S$ and such that G is locally a direct factor of a free module of finite type, is split. The category $\mathcal{C}_{0S}$ is liftable in $\mathcal{C}'_S$ (I 2.2).

b) Let $\operatorname{Mod}(A(S))$ denote the category of $A(S)$-modules. The functor

$$\Gamma_S : \mathcal{C}'_S \longrightarrow \operatorname{Mod}(A(S))$$

is exact. Moreover:

(i) if every object F of $\mathcal{C}'_S$ admits a finite global presentation

$$A_S^q \longrightarrow A_S^p \longrightarrow F \longrightarrow 0,$$

then Γ_S is fully faithful and has as essential image the full subcategory of $\operatorname{Mod}(A(S))$ formed by $A(S)$-modules of finite presentation;

(ii) if $\mathcal{C}'_S$ contains the free A_S-modules, if every object F of $\mathcal{C}'_S$ admits a global presentation

$$A_S^{(J)} \longrightarrow A_S^{(I)} \longrightarrow F \longrightarrow 0,$$

and if Γ_S commutes with filtered inductive limits, then Γ_S is an equivalence of categories.

Proof. a) Since G is locally a direct factor of a free module of finite type, the spectral sequence

$$E_2^{pq} = H^p(S, \mathcal{E}xt^q(G, E)) \Longrightarrow \operatorname{Ext}^{p+q}(G, E)$$

degenerates (I 1.3.1) and gives an isomorphism

$$H^1(S, \operatorname{Hom}(G, E)) \simeq \operatorname{Ext}^1(G, E),$$

whence the first assertion. The second is a particular case, since for $G \in \operatorname{ob} \mathcal{C}_{0S}$ and $E \in \operatorname{ob} \mathcal{C}'_S$ one has $\operatorname{Hom}(G, E) \in \operatorname{ob} \mathcal{C}'_S$.

b) The first assertion is trivial. Let us prove, in case (i) or (ii), the full faithfulness of $\Gamma_S : \mathcal{C}'_S \to \operatorname{Mod}(A(S))$. Let $F, G \in \operatorname{ob} \mathcal{C}'_S$ and let

$$A_S^{(J)} \longrightarrow A_S^{(I)} \longrightarrow F \longrightarrow 0 \qquad (*)$$

be a global presentation of F, with I and J finite in case (i). Applying Γ_S, which is exact on $\mathcal{C}'_S$ and, in case (ii), commutes with filtered inductive limits, one obtains an exact sequence

$$A(S)^{(J)} \longrightarrow A(S)^{(I)} \longrightarrow \Gamma_S(F) \longrightarrow 0. \qquad (**)$$

Applying the left exact functors $\mathrm{Hom}_{\mathcal{C}_S}(-, G)$ and $\mathrm{Hom}_{A(S)}(-, \Gamma_S(G))$ to $(*)$ and $(**)$ gives a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hom}_{\mathcal{C}_S}(F, G) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}_S}(A_S^{(I)}, G) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}_S}(A_S^{(J)}, G) \\ & & \downarrow & & \downarrow\sim & & \downarrow\sim \\ 0 & \longrightarrow & \mathrm{Hom}_{A(S)}(\Gamma_S F, \Gamma_S G) & \longrightarrow & \mathrm{Hom}_{A(S)}(A(S)^{(I)}, \Gamma_S G) & \longrightarrow & \mathrm{Hom}_{A(S)}(A(S)^{(J)}, \Gamma_S G). \end{array}$$

The two vertical arrows on the right are visibly isomorphisms, hence so is the arrow on the left. The description of the essential image in cases (i) and (ii) is immediate and is left to the reader.

Lemma 2.1.3. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, and let $\varphi : \mathcal{A} \to \mathcal{B}$ be an exact functor.

a) For $M \in \mathrm{ob}\,\mathcal{A}$, the following conditions are equivalent:

   (i) for every $L \in \mathrm{ob}\,\mathcal{A}$, the canonical map induced by φ

   $$\mathrm{Ext}^n_{\mathcal{A}}(L, M) \longrightarrow \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$$

   is an isomorphism for all n;

   (ii) for every $L \in \mathrm{ob}\,\mathcal{A}$, the canonical map

   $$\mathrm{Hom}_{\mathcal{A}}(L, M) \longrightarrow \mathrm{Hom}_{\mathcal{B}}(\varphi L, \varphi M)$$

   is an isomorphism, and for every $n \geq 1$ and every $x \in \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$ there exists an epimorphism $L' \to L$ in $\mathcal{A}$ which kills x.

b) The following conditions are equivalent:

   (i) for all $L, M \in \mathrm{ob}\,\mathcal{A}$ and all $n \in \mathbb{Z}$, the canonical map

   $$\mathrm{Ext}^n_{\mathcal{A}}(L, M) \longrightarrow \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$$

   is an isomorphism;

   (ii) the canonical functor induced by φ

   $$\mathrm{D}^b(\mathcal{A}) \longrightarrow \mathrm{D}^b(\mathcal{B})$$

   is fully faithful and has as essential image the full subcategory of $\mathrm{D}^b(\mathcal{B})$ formed by complexes F such that $H^i(F)$ is in the essential image of φ for every i.

Proof. For (i)⇒(ii) in a), it is enough to show that, if $n \geq 1$, $L \in \mathrm{ob}\,\mathcal{A}$, and $y \in \mathrm{Ext}^n_{\mathcal{A}}(L, M)$, then there exists an epimorphism $L' \to L$ killing y. Using Yoneda's interpretation ([18], III 3.2), y is the class of an exact sequence

$$0 \longrightarrow M \longrightarrow E_{n-1} \longrightarrow \cdots \longrightarrow E_0 \longrightarrow L \longrightarrow 0,$$

and the epimorphism $E_0 \to L$ kills y ([18], III 3.2.5 and 3.2.8). Conversely, put

$$F^n(L) = \mathrm{Ext}^n_{\mathcal{A}}(L, M), \qquad G^n(L) = \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M).$$

The effaceability hypothesis says, for $n \geq 1$ and $L \in \mathrm{ob}\,\mathcal{A}$, that

$$G^n(L) = \varinjlim \mathrm{Coker}\, igl(G^{n-1}(L_0) \to G^{n-1}(L_1)),$$

the limit being taken over the exact sequences $L_1 \to L_0 \to L \to 0$. As above, one has the same formula for $F^n(L)$. Since $F^0 \to G^0$ is an isomorphism, induction on n shows that the morphism $F^n \to G^n$ induced by φ is an isomorphism for every n. The equivalence in b) follows by an immediate dévissage, left to the reader.

Remarks 2.1.3.1. There is of course a dual version of a), which we leave to the reader to state. On the other hand, if $\mathcal{A}$ and $\mathcal{B}$ have enough injectives, the conditions (i) and (ii) of b) are equivalent to:

(iii) for $L \in \mathrm{ob}\,\mathrm{D}^-(\mathcal{A})$ and $M \in \mathrm{ob}\,\mathrm{D}^+(\mathcal{A})$, the canonical map

$$\mathrm{RHom}(L, M) \longrightarrow \mathrm{RHom}(\varphi L, \varphi M)$$

is an isomorphism,

as is seen at once by a spectral sequence argument.

2.2. Noetherian schemes and divisorial schemes

Notation 2.2.1. Let S be a scheme. We write $\mathrm{Qcoh}(S)$, respectively $\mathrm{Coh}(S)$, for the abelian category of quasi-coherent, respectively coherent, $\mathcal{O}_S$-modules. Let $[m, n]$ be an interval of $\mathbb{Z}$. We shall say that an object E of $\mathrm{D}(\mathrm{Qcoh}(S))$ is n-pseudo-coherent, respectively pseudo-coherent, respectively of perfect amplitude contained in $[m, n]$, respectively perfect, if it is so relative to the category fibered over the Zariski site of S defined by

$$U \longmapsto \mathrm{Loclib}(U)$$

((I 2.3), (I 4.7), and (2.0)). We denote by $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{coh}}$, respectively $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$, the full triangulated subcategory of $\mathrm{D}(\mathrm{Qcoh}(S))$ formed by pseudo-coherent, respectively perfect, complexes.

Proposition 2.2.2. Let S be a noetherian scheme. The canonical functor

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

is fully faithful and has as essential image $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{coh}}$.

Proof. Apply (1.1), taking for the site the Zariski site of S, and for $\mathcal{C}$, respectively $\mathcal{C}_0$, the fibered categories $U \mapsto \mathrm{Qcoh}(U)$, respectively $U \mapsto \mathrm{Coh}(U)$. Since $\mathcal{O}_S$ is coherent, the objects of $\mathcal{C}_0$ are pseudo-coherent (I 3.5); hence an object of $\mathrm{D}(\mathrm{Qcoh}(U))$ is pseudo-coherent, in the sense of (2.2.1), if and only if it is pseudo-coherent relative to $\mathcal{C}_0$ (I 2.14.1). By (1.1 c)), it remains to verify conditions (i) and (ii) of (1.1). For (ii) this is trivial, since a quasi-coherent module which is a quotient of a coherent module is coherent, and coherence is local. For (i), one uses (2.1.1 a)) and the fact that every quasi-coherent $\mathcal{O}_S$-module is a filtered inductive limit of coherent $\mathcal{O}_S$-modules (EGA I 9.4.9).

Granting (3.5) and (3.7), which are independent of the present section, one easily deduces the following.

Corollary 2.2.2.1. Let S be a noetherian scheme, respectively a noetherian scheme of finite dimension. The canonical functor

$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

respectively

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}^b(S)_{\mathrm{coh}}$, respectively $\mathrm{D}^-(S)_{\mathrm{coh}}$ (I 2.15).

Proposition 2.2.3 (cf. EGA II 4.5.2 and 4.5.5). Let S be a quasi-compact and quasi-separated scheme, and let $(L_i)_{i\in I}$ be a family of invertible sheaves on S. For every $\mathcal{O}_S$-module F and every pair $(i,n) \in I \times \mathbb{Z}$, let $F_{(i,n)}$ denote the sub-$\mathcal{O}_S$-module of $F \otimes L_i^{\otimes n}$ generated by the sections of $F \otimes L_i^{\otimes n}$ over S. The following conditions are equivalent:

(i) the open sets S_f, for $f \in \Gamma(S, L_i^{\otimes n})$, $i \in I$ and $n \in \mathbb{N}$, form a base for the topology of S;

(ii) those of the S_f which are affine cover S;

(iii) for every quasi-coherent $\mathcal{O}_S$-module F of finite type, there exists a finite family (i_α, n_α) with $n_\alpha \geq 0$ and an epimorphism

$$\bigoplus_\alpha L_{i_\alpha}^{\otimes(-n_\alpha)} \longrightarrow F;$$

(iv) for every $F \in \mathrm{ob}\,\mathrm{Qcoh}(S)$, one has

$$F = \sum_{(i,n)\in I\times\mathbb{N}} F_{(i,n)} \otimes L_i^{\otimes(-n)},$$

a sum, not necessarily direct, of sub-sheaves of F;

(v) same condition as (iv), but with F a quasi-coherent ideal of $\mathcal{O}_S$.

Proof. This is a simple paraphrase of the cited passages; we recall the argument briefly for the reader's convenience. For (i)⇒(ii), choose for $x \in S$ an affine open U containing x. An open S_f with $x \in S_f \subset U$, where $f \in \Gamma(S, L_i^{\otimes n})$ and $n \geq 0$, is necessarily affine, by the following lemma.

Lemma 2.2.3.1 (EGA II 5.5.8). Let X be a scheme, L an invertible $\mathcal{O}_X$-module, and $f \in \Gamma(X, L)$. The canonical immersion $X_f \to X$ is affine.

Proof. This is trivial: the question is local on X, so one may suppose X affine and $L \simeq \mathcal{O}_X$.

For (ii)⇒(iii), choose a finite covering of S by affine opens $S_{f_{ij}}$, with $f_{ij} \in \Gamma(S, L_i^{\otimes m_{ij}})$. Replacing the f_{ij} by suitable powers, one may suppose that the m_{ij} with fixed i are all equal to an integer m_i. If F is quasi-coherent of finite type, then, for each i, j, there is an integer $q_i \geq 0$ such that every section of F over $S_{f_{ij}}$ extends to a section of $F \otimes L_i^{\otimes q_i m_i}$ over S. Thus $F \otimes L_i^{\otimes q_i m_i}$ is generated over $S_{f_{ij}}$ by finitely many global sections. Proceeding as in the proof of EGA II 4.5.5, one obtains, for each i, an integer $n_i \geq 0$ such that $F \otimes L_i^{\otimes n_i}$ is generated over the union of the corresponding $S_{f_{ij}}$ by finitely many sections over S. Hence there are integers $r_i \geq 0$ and a morphism

$$L_i^{-n_i r_i} \longrightarrow F$$

which induces an epimorphism over that union; the assertion follows.

For (iii)⇒(iv), since every quasi-coherent sheaf on S is the inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9 and IV 1.7.7), it is enough to treat the case where F is of finite type. There exist morphisms

$$u_i : L_i^{-n_i r_i} \longrightarrow F$$

whose sum is an epimorphism. But

$$\mathrm{Im}(u_i) = \mathrm{Im}(u_i \otimes \mathrm{Id}(L_i^{n_i})) \otimes L_i^{-n_i},$$

and $\mathrm{Im}(u_i \otimes \mathrm{Id}(L_i^{n_i}))$ is a subsheaf of $F_{(i,n_i)}$, which gives the assertion. The implication (iv)⇒(v) is trivial. Finally, for (v)⇒(i), let $x \in S$, let U be an open containing x, and let F be a quasi-coherent ideal defining $S - U$. By (v), there are $(i,n) \in I \times \mathbb{N}$ and a section f of $F \otimes L_i^{\otimes n}$ over S with $f(x) \neq 0$. Then $x \in S_f \subset U$, proving (i).

Definition 2.2.4. Let S be a quasi-compact and quasi-separated scheme, and let $(L_i)_{i\in I}$ be a family of invertible sheaves on S. We say that the L_i form an *ample family* if the equivalent conditions of (2.2.3) hold. In particular, a family consisting of a single sheaf L is ample if and only if L is ample in the usual sense (EGA II 4.5.3).

Definition 2.2.5. A scheme S is called *divisorial* if S is quasi-compact and quasi-separated and if there exists an ample family (2.2.4) of invertible $\mathcal{O}_S$-modules.

Lemma 2.2.6. Let X be a locally noetherian scheme. Let U be an open of X such that the inclusion $U \to X$ is affine, for example U affine and X separated. Then $X - U$ is purely of codimension 0 or 1; that is, at every maximal point x of $X - U$ one has $\dim \mathcal{O}_{X,x} = 0$ or 1. It is purely of codimension 1 if U contains the maximal points of X.

Proof. Put $Y = X - U$. Let x be a maximal point of Y and make the base change $X' = \mathrm{Spec}\,\mathcal{O}_{X,x} \to X$; one obtains a cartesian diagram

$$\begin{array}{ccc} U' & \longrightarrow & U \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X. \end{array}$$

Let $Y' = X' - U'$. Since x is a maximal point of Y, Y' is just the closed point of X'. Since X' is affine, U' is an affine open of X'. We show that the dimension d of X' is 0 or 1. Indeed, one knows (SGA 2 V 5.7) that $H^d_{Y'}(X', \mathcal{O}_{X'}) \neq 0$. But if $d \geq 2$, then

$$H^d_{Y'}(X', \mathcal{O}_{X'}) = H^{d-1}(U', \mathcal{O}_{X'}),$$

and the latter group is zero because U' is affine and $d - 1 \geq 1$, a contradiction.

Remark 2.2.6.1. Without the separation hypothesis on X, (2.2.6) would be false for an affine open U of X, as shown by the example of the affine plane with the origin doubled (EGA I 5.5.11).

Proposition 2.2.7. Let S be a noetherian separated locally factorial scheme, that is, a scheme whose local rings are factorial. Then S is divisorial (2.2.5). In particular (EGA IV 21.11.1):

Corollary 2.2.7.1. Every noetherian regular separated scheme is divisorial.

Proof of 2.2.7. One may suppose that S is integral. Let $(S_i)_{i\in I}$ be a finite covering of S by nonempty affine opens. By (2.2.6), $S - S_i$ is purely of codimension 1; hence, by EGA IV 21.7.4, it is the support of a positive divisor. In other words, for each $i \in I$, there is an invertible sheaf L_i and a section $f_i \in \Gamma(S, L_i)$ such that $S_i = S_{f_i}$. The family (L_i) is then ample by (2.2.4), using (2.2.3 (ii)).

Remark 2.2.7.2. As in (2.2.6), the separation hypothesis on S is essential. Let S again be the affine plane with the origin doubled. Recall that S is obtained by gluing two copies of the affine plane $\mathbb{A}^2_k$ along the identity on the complement U of the origin O; hence there is a canonical projection

$$p : S \longrightarrow \mathbb{A}^2_k$$

which is an isomorphism over U, while $p^{-1}(O)$ consists of two points O_1 and O_2. An invertible sheaf on S is obtained by gluing two invertible sheaves on $\mathbb{A}^2_k$ by an isomorphism over U. Since every invertible sheaf on $\mathbb{A}^2_k$ is trivial, an invertible sheaf on S is defined by an element

$g \in \Gamma(U, \mathcal{O}_U^*)$. But g is necessarily constant, since the restriction $\Gamma(\mathbb{A}_k^2, \mathcal{O}_{\mathbb{A}^2}) \to \Gamma(U, \mathcal{O}_U)$ is an isomorphism, the origin being of depth 2 in the plane. Thus every invertible sheaf on S is trivial. It is immediate, on the other hand, that p induces an isomorphism

$$\Gamma(\mathbb{A}_k^2, \mathcal{O}_{\mathbb{A}^2}) \xrightarrow{\sim} \Gamma(S, \mathcal{O}_S).$$

Consequently every global section of $\mathcal{O}_S$ which vanishes at O_1 also vanishes at O_2, and the opens S_f, for $f \in \Gamma(S, \mathcal{O}_S)$, do not form a base for the topology of S.

Theorem 2.2.8. Let S be a divisorial scheme (2.2.5), and let $(L_i)_{i\in I}$ be an ample family of invertible $\mathcal{O}_S$-modules. Let $L(S)$ be the strictly full additive subcategory of $\mathrm{Qcoh}(S)$ generated by the $L_i^{\otimes(-n)}$, for $(i, n) \in I \times \mathbb{N}$. Let $E \in \mathrm{ob}\,\mathrm{D}(\mathrm{Qcoh}(S))$.

a) The categories $L(S)$ and $\mathrm{Loclib}(S)$ are quasi-liftable in $\mathrm{Qcoh}(S)$ (I 1.2) and quasi-stable under kernels of epimorphisms (I 1.2).

b) Let $n \in \mathbb{Z}$. For E to be n-pseudo-coherent, respectively pseudo-coherent, in the sense of (2.2.1), it is necessary and sufficient that E be so relative to $L(S)$ or to $\mathrm{Loclib}(S)$ (I 2.3). The canonical functors

$$\mathrm{K}^-(L(S))/\mathrm{K}^{-,\delta}(L(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S)),$$

$$\mathrm{K}^-(\mathrm{Loclib}(S))/\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

are fully faithful and have as essential image $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{coh}}$.

c) Let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have perfect amplitude contained in $[m, n]$ (2.2.1), it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(\mathrm{Qcoh}(S))$, to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\,L(S)$ for $i > m$, and F^m is locally free of finite type. The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S))/\mathrm{K}^{b,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

is fully faithful and has as essential image $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$.

Proof. Let $F \to G$ be an epimorphism in $\mathrm{Qcoh}(S)$, with G of finite type. Since F is the inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9), there exists, by (2.1.1 a)), a quasi-coherent subsheaf F' of finite type in F such that $F' \to F \to G$ is an epimorphism. By (2.2.3 (iii)), there is an epimorphism $E \to F'$, with $E \in \mathrm{ob}\,L(S)$; hence $E \to F' \to F \to G$ is an epimorphism. This proves that the category of quasi-coherent $\mathcal{O}_S$-modules of finite type is quasi-liftable in $\mathrm{Qcoh}(S)$, and a fortiori the same is true for $\mathrm{Loclib}(S)$ and for $L(S)$.

For the second assertion of a), apply (1.1) on the Zariski site of S, with $\mathcal{C}$ the fibered category $U \mapsto \mathrm{Qcoh}(U)$ and $\mathcal{C}_0$ the fibered category $U \mapsto L(U)$. Since every object of L is pseudo-coherent, the notions of n-pseudo-coherence and pseudo-coherence in the fibered category D(Qcoh) coincide with the same notions relative to L (I 2.14). Condition (ii) of (1.1) is satisfied by (2.2.3 (iii)), and thus (1.1 a)) gives the part of a) concerning $L(S)$. The assertion for $\mathrm{Loclib}(S)$ is proved similarly. Part b) follows immediately from (1.1 b), c)), taking into account that, if $E \in \mathrm{ob}\,\mathrm{D}(\mathrm{Qcoh}(S))$ is n-pseudo-coherent, then E lies in $\mathrm{D}^-(\mathrm{Qcoh}(S))$, since S is quasi-compact. An object of the fibered category Qcoh is locally in L if and only if it is locally free of finite type; hence L is locally liftable in Qcoh and locally stable under kernels of epimorphisms. Thus c) follows from (1.2), with $\mathcal{C} = \mathrm{Qcoh}$, $\mathcal{C}_0 = L$, and $\mathcal{C}_1 = \mathrm{Loclib}(S)$.

Proposition 2.2.9. The hypotheses and notation are those of (2.2.8), but assume moreover that S is separated or noetherian.

a) The canonical functors

$$\mathrm{K}^-(L(S))/\mathrm{K}^{-,\delta}(L(S)) \longrightarrow \mathrm{D}(S), \qquad \mathrm{K}^-(\mathrm{Loclib}(S))/\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

are fully faithful and have as essential image $\mathrm{D}^-(S)_{\mathrm{coh}}$. Moreover, if S is of finite cohomological dimension in the sense of (3.7), for example if the underlying space of S is noetherian of finite Krull dimension, then the functors

$$\mathrm{K}(L(S))/\mathrm{K}^{\delta}(L(S)) \longrightarrow \mathrm{K}(\mathrm{Loclib}(S))/\mathrm{K}^{\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^-(S)_{\mathrm{coh}}$$

are equivalences of categories.

b) The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S))/\mathrm{K}^{b,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}(S)_{\mathrm{parf}}$ (I 4.9). For $E \in \mathrm{ob}\,\mathrm{D}(S)$ to have perfect amplitude contained in $[m,n]$ (I 4.8), it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(S)$, to a complex F such that $F^i = 0$ for $i \notin [m,n]$, $F^i \in \mathrm{ob}\,L(S)$ for $i > m$, and F^m is locally free of finite type.

Proof. In view of (3.5) and (3.7), this is an immediate corollary of (2.2.8). The only point to check, and it follows trivially from (3.5), respectively (3.7), is that for an object E of $\mathrm{D}^-(\mathrm{Qcoh}(S))$, respectively of $\mathrm{D}(\mathrm{Qcoh}(S))$ when S has finite cohomological dimension in the sense of (3.7), the notions of pseudo-coherence and perfection of (2.2.1) coincide with the usual notions (I 2.15 and 4.9) after regarding E as an object of $\mathrm{D}(S)$.

Remark 2.2.10. a) If $S = \mathrm{Spec}\,A$ is affine, the family consisting of the single sheaf $\mathcal{O}_S$ is ample, and for it $L(S) = \mathrm{Lib}(S)$. It is also clear that the categories $\mathrm{K}^{-,\delta}(\mathrm{Lib}(S))$ and $\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S))$ are reduced to zero objects. It follows from (2.2.9) that, writing $\mathrm{Mod}(A)$ for the category of A-modules and $\mathrm{D}(A) = \mathrm{D}(\mathrm{Mod}(A))$, the functor $R\Gamma_S : \mathrm{D}(S) \to \mathrm{D}(A)$ induces fully faithful functors

$$\mathrm{D}^-(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}(A), \qquad \mathrm{D}(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}(A),$$

whose essential images are respectively the triangulated subcategories $\mathrm{D}^-(A)_{\mathrm{coh}}$ and $\mathrm{D}(A)_{\mathrm{parf}}$, formed by complexes with bounded-above cohomology and pseudo-coherent, respectively perfect, relative to the subcategory $\mathrm{Lib}(A)$, respectively $\mathrm{Loclib}(A)$, of free, respectively projective, modules of finite type. There are commutative diagrams of equivalences of categories

$$\begin{array}{ccccc}
\mathrm{K}^{-,b}(\mathrm{Lib}(S)) & \longrightarrow & \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) & \longrightarrow & \mathrm{D}^b(S)_{\mathrm{coh}} \\
{\scriptstyle \Gamma_S}\downarrow & & {\scriptstyle \Gamma_S}\downarrow & & \downarrow{\scriptstyle R\Gamma_S} \\
\mathrm{K}^{-,b}(\mathrm{Lib}(A)) & \longrightarrow & \mathrm{K}^{-,b}(\mathrm{Loclib}(A)) & \longrightarrow & \mathrm{D}^b(A)_{\mathrm{coh}},
\end{array}$$

$$\begin{array}{ccc}
\mathrm{K}^b(\mathrm{Loclib}(S)) & \longrightarrow & \mathrm{D}(S)_{\mathrm{parf}} \\
{\scriptstyle \Gamma_S}\downarrow & & \downarrow{\scriptstyle R\Gamma_S} \\
\mathrm{K}^b(\mathrm{Loclib}(A)) & \longrightarrow & \mathrm{D}(A)_{\mathrm{parf}}.
\end{array}$$

In the upper diagram, one may replace b by $-$ when S has finite cohomological dimension in the sense of (3.7).

b) For an arbitrary scheme S, it follows from a) that a complex of $\mathcal{O}_S$-modules which is pseudo-coherent and has bounded cohomology is locally isomorphic, in $\mathrm{D}(S)$, to a strictly pseudo-coherent complex, which was by no means evident a priori (cf. I 2.3.1).

2.3. Soft sheaves

2.3.1. Recall a few definitions and results from [T.F.]. Let X be a paracompact space. A sheaf of sets F on X is called *soft* if, for every closed subset Y of X, every section of F over Y extends to a section of F over X. Soft abelian sheaves are acyclic: if F is soft, then $H^i(X, F) = 0$ for $i > 0$ (T.F. II 4.4.3).

Let A be a sheaf of rings on X. If F is a soft A-module, then F is generated by its sections (cf. 2.1.1): indeed, the homomorphism $A_X^{(I)} \to F$ defined by $I = H^0(X, F)$ is an epimorphism. Moreover, if A is soft, every A-module is soft (T.F. II 3.7.1). Examples of soft sheaves of rings include the sheaf of continuous real or complex-valued functions on a paracompact space and the sheaf of C^∞ functions on a C^∞ manifold.

Proposition 2.3.2. Let S be a compact space equipped with a soft sheaf of rings A. Write $\mathrm{Mod}(S)$ for the category of left A-modules and $\mathrm{D}(S) = \mathrm{D}(\mathrm{Mod}(S))$.

a) The category $\mathrm{Lib}(S)$, respectively $\mathrm{Loclib}(S)$ (2.0), is liftable in $\mathrm{Mod}(S)$ and quasi-stable, respectively stable, under kernels of epimorphisms (I 1.2). This assertion holds more generally for S paracompact.

b) Let $E \in \mathrm{ob}\,\mathrm{D}(S)$ and $n \in \mathbb{Z}$. For E to be n-pseudo-coherent, respectively pseudo-coherent, it is necessary and sufficient that it be so relative to $\mathrm{Lib}(S)$ or to $\mathrm{Loclib}(S)$. The canonical functors
$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{D}(S), \qquad \mathrm{K}^-(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$
are fully faithful and have as essential image $\mathrm{D}^-(S)_{\mathrm{coh}}$ (I 2.15).

c) Let $E \in \mathrm{ob}\,\mathrm{D}(S)$ and let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have perfect amplitude contained in $[m, n]$ (I 4.8), it is necessary and sufficient that there exist a quasi-isomorphism $F \to E$, where F is a complex such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$, and $F^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$. The canonical functor
$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$
is fully faithful and has as essential image $\mathrm{D}(S)_{\mathrm{parf}}$ (I 4.9).

d) Let $\mathrm{Mod}(A(S))$ be the category of left $A(S)$-modules and $\mathrm{D}(A(S))$ the corresponding derived category. The functor
$$H^0(S, -) : \mathrm{Mod}(S) \longrightarrow \mathrm{Mod}(A(S))$$
is exact and is an equivalence of categories. Moreover it defines equivalences of categories
$$\mathrm{D}(S) \xrightarrow{\sim} \mathrm{D}(A(S)), \qquad \mathrm{D}^-(S)_{\mathrm{coh}} \xrightarrow{\sim} \mathrm{D}^-(A(S))_{\mathrm{coh}}, \qquad \mathrm{D}(S)_{\mathrm{parf}} \xrightarrow{\sim} \mathrm{D}(A(S))_{\mathrm{parf}},$$
where pseudo-coherence, respectively perfection, in $\mathrm{D}(A(S))$ is understood relative to the category of free, respectively projective, modules of finite type. It also gives an equivalence from the full subcategory of $\mathrm{D}(S)$ formed by n-pseudo-coherent objects, respectively by objects of perfect amplitude contained in $[m, n]$, onto the analogous full subcategory of $\mathrm{D}(A(S))$.

Proof. The first assertion of a) follows from (2.1.2 a)), applied with S the topos of sheaves of sets on S and $\mathcal{C}'_S = \mathrm{Mod}(S)$, using (2.3.1). The second follows from (1.1 a)), respectively (1.2 a)), with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$, respectively Loclib; the condition (ii) of (1.1) is verified by (2.3.1) and (2.1.1 b)). Part b) follows from (1.1 b), c)) with the same $\mathcal{C}$ and $\mathcal{C}_0$. Part c) follows from (1.2 b), c), d)), with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$. The first assertion of d) follows, using (2.3.1), from (2.1.2 b) (ii)) with $\mathcal{C}'_S = \mathcal{C}_S = \mathrm{Mod}(S)$. The other assertions in d) follow from it, together with b) and c), and from the fact that Γ_S carries free modules, respectively direct factors of free modules, of finite type to free, respectively projective, modules of finite type.

2.4. Stein spaces

Definition 2.4.1. Let S be a locally ringed topos (I 2.15.1 (ii)), with final object also denoted S. We say that S is *Stein* if the structural sheaf $\mathcal{O}_S$ is coherent (I 3.1) and if the following properties hold:

Theorem A: every coherent $\mathcal{O}_S$-module (I 3.1) is generated by its sections;

Theorem B: for every coherent $\mathcal{O}_S$-module F, one has $H^i(S, F) = 0$ for $i > 0$.

Remark 2.4.1.1. If S has enough closed points s such that the ideal $\mathfrak{m}_s$ defined by s is coherent, then the relation $H^1(S, F) = 0$ for every coherent sheaf F implies Theorem A. Indeed, let F be a coherent $\mathcal{O}_S$-module and let s be a closed point of S. There is an exact sequence

$$0 \longrightarrow \mathfrak{m}_s F \longrightarrow F \longrightarrow F/\mathfrak{m}_s F \longrightarrow 0,$$

where $F/\mathfrak{m}_s F$ is a coherent sheaf concentrated at s. Applying $H^0(S, -)$ and writing the cohomology exact sequence, one finds that

$$H^0(S, F) \longrightarrow H^0(S, F/\mathfrak{m}_s F)$$

is an epimorphism. Since

$$H^0(S, F/\mathfrak{m}_s F) = F_s/\mathfrak{m}_s F_s,$$

where $\mathfrak{m}_s$ is the maximal ideal of $\mathcal{O}_s$, Nakayama's lemma gives the conclusion. The editor does not know whether the preceding implication is true in general when one assumes only that $\mathcal{O}_S$ is a coherent sheaf of local rings. The editor also does not know whether the relation $H^1(S, F) = 0$ for every coherent sheaf F implies Theorem B.

Examples 2.4.2. A complex analytic Stein space, a compact convex subset of $\mathbb{C}^n$ equipped with the sheaf induced by the sheaf of holomorphic functions, the spectrum of a noetherian ring, and an affine rigid analytic space define Stein topoi.

Lemma 2.4.3. Let S be a Stein topos (2.4.1). Suppose that the final object S is quasi-compact (2.1.1), and let $\mathrm{Coh}(S)$ be the abelian category of coherent $\mathcal{O}_S$-modules, that is, modules of finite presentation. Then:

a) $\mathrm{Lib}(S)$, respectively $\mathrm{Loclib}(S)$, is liftable in $\mathrm{Coh}(S)$ and quasi-stable, respectively stable, under kernels of epimorphisms (I 1.2).

b) The canonical functors

$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^-(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^-(\mathrm{Coh}(S))$$

are equivalences of categories.

c) Put $A = \Gamma(S, \mathcal{O}_S)$, and write $\mathrm{Mod}(A)$ for the category of A-modules and $\mathrm{D}(A)$ for the corresponding derived category. The functor

$$\Gamma_S : \mathrm{Coh}(S) \longrightarrow \mathrm{Mod}(A)$$

is exact, fully faithful, and has as essential image the strictly full additive subcategory of $\mathrm{Mod}(A)$ formed by A-modules of finite presentation. It induces a fully faithful functor

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(A),$$

whose essential image is the full subcategory $\mathrm{D}^-(A)_{\mathrm{coh}}$ formed by complexes pseudo-coherent relative to the subcategory of free modules of finite type.

Here "enough closed points" means enough fiber functors (SGA 4 IV 6.4.1) of the form $i^* : S \to s$, with i_* the inclusion of a closed subtopos of S; and $\mathfrak{m}_s$ denotes the sections of $\mathcal{O}_S$ whose fiber at s belongs to the maximal ideal of $\mathcal{O}_s$.

Proof. For the first assertion of a), apply (2.1.2 a)) with $\mathcal{C}'_S = \mathrm{Coh}(S)$; the hypothesis of (2.1.2) is verified by Theorem B, since if E is coherent and G is locally free of finite type, then $\mathrm{Hom}(G, E)$ is coherent. Thus $\mathrm{Loclib}(S)$, and a fortiori $\mathrm{Lib}(S)$, is liftable in $\mathrm{Coh}(S)$. The second assertion of a) follows from (1.1 a)), respectively (1.2 a)), applied with $\mathcal{C}_0 = \mathrm{Lib}$, respectively Loclib; condition (ii) of (1.1) is verified by Theorem A and (2.1.1 b)). Part b) follows from (1.1 c)), applied successively with $\mathcal{C}_0 = \mathrm{Lib}$ and $\mathcal{C}_0 = \mathrm{Loclib}$. The first assertion of c) follows from (2.1.2 b)(i)), applied with $\mathcal{C}'_S = \mathrm{Coh}(S)$. To prove that $\Gamma_S : \mathrm{D}^-(\mathrm{Coh}(S)) \to \mathrm{D}(A)$ is fully faithful, it is enough, by b), to prove that the composite

$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(A)$$

is so. But this composite factors as

$$\mathrm{K}^-(\mathrm{Lib}(S)) \xrightarrow{\Gamma_S} \mathrm{K}^-(\mathrm{Lib}(A)) \longrightarrow \mathrm{D}(A),$$

where $\mathrm{Lib}(A)$ denotes the category of free A-modules of finite type; the first arrow is an equivalence of categories, and the second is fully faithful by (I 2.7 b)), applied with $\mathcal{C} = \mathrm{Mod}(A)$ and $\mathcal{C}_0 = \mathrm{Lib}(A)$, since $\mathrm{K}^{-,\delta}(\mathrm{Lib}(A)) = 0$. This proves the last assertion of c).

Lemma 2.4.4. Under the hypotheses of (2.4.3), the canonical functor

$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}^b(S)_{\mathrm{coh}}$ (I 2.15).

Proof. Apply (2.1.3) with $\mathcal{A} = \mathrm{Coh}(S)$, $\mathcal{B} = \mathrm{Mod}(S)$, and $\varphi : \mathcal{A} \to \mathcal{B}$ the inclusion functor. Since this functor is fully faithful, it is enough, by (2.1.3 a)(ii)), to check that for fixed $M \in \mathrm{ob}\,\mathrm{Coh}(S)$ and $n > 0$ the functor

$$L \longmapsto \mathrm{Ext}^n_{\mathrm{Mod}(S)}(L, M)$$

is coeffaceable on $L \in \mathrm{ob}\,\mathrm{Coh}(S)$. By Theorem A and (2.1.1 b)), every coherent $\mathcal{O}_S$-module is a quotient of a free module of finite type; and if L is free of finite type, then $\mathrm{Ext}^n_{\mathrm{Mod}(S)}(L, M) = 0$ by Theorem B. The assertion follows. Hence the functor $\mathrm{D}^b(\mathrm{Coh}(S)) \to \mathrm{D}(S)$ is fully faithful and has as essential image the full subcategory of $\mathrm{D}(S)$ formed by complexes with bounded coherent cohomology, which is precisely $\mathrm{D}^b(S)_{\mathrm{coh}}$ since $\mathcal{O}_S$ is coherent (I 3.5).

Proposition 2.4.5. With the hypotheses and notation of (2.4.3), one has:

a) The canonical functors

$$\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}}$$

are equivalences of categories. Moreover, the functor $R\Gamma_S : \mathrm{D}^b(S) \to \mathrm{D}^b(A)$ induces a fully faithful functor

$$R\Gamma_S : \mathrm{D}^b(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^b(A),$$

whose essential image is $\mathrm{D}^b(A)_{\mathrm{coh}}$, the triangulated subcategory of $\mathrm{D}(A)$ formed by complexes with bounded cohomology and pseudo-coherent relative to the category of free modules of finite type.

b) Let $E \in \mathrm{ob}\,\mathrm{D}(S)$, and let $[m, n]$ be an interval of $\mathbb{Z}$. If E has perfect amplitude contained in $[m, n]$ (I 4.5), then E is isomorphic to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$, and $F^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$.

c) The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)_{\mathrm{parf}}$$

is an equivalence of categories (I 4.9). Moreover, $R\Gamma_S : \mathrm{D}^b(S) \to \mathrm{D}^b(A)$ induces a fully faithful functor

$$R\Gamma_S : \mathrm{D}(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^b(A),$$

whose essential image is $\mathrm{D}^b(A)_{\mathrm{parf}}$, the triangulated subcategory of $\mathrm{D}(A)$ formed by complexes perfect relative to the category of projective modules of finite type.

Proof. a) The functors

$$\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^b(A)_{\mathrm{coh}}$$

are equivalences of categories by (2.4.3 b), c). Since the fully faithful functor $\Gamma_S : \mathrm{D}^b(\mathrm{Coh}(S)) \to \mathrm{D}(A)$ factors as

$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}} \xrightarrow{R\Gamma_S} \mathrm{D}(A),$$

the conclusion follows from (2.4.4).

b) Suppose $\mathrm{parf.amp}(E) \subset [m, n]$. In particular E is pseudo-coherent; by a), after replacing E by an isomorphic object, we may assume that $E^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$ and $E^i = 0$ for $i > n$. Truncating, we may also suppose that $E^i = 0$ for $i < m$. By (I 4.16), applied with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$, one then has $E^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$.

c) The hypotheses (1.1 (i)) and, respectively, (1.1 (ii)) are verified for $\mathcal{C} = \mathrm{Coh}(S)$ and $\mathcal{C}_0 = \mathrm{Lib}$, by (2.4.3 a)) and by (2.1.1 b)) together with Theorem A. Hence (1.2 c), d)) shows that the functor

$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(\mathrm{Coh}(S))$$

is fully faithful. Therefore, by (2.4.4), the same is true for

$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S),$$

and its essential image is $\mathrm{D}(S)_{\mathrm{parf}}$ by b). By an argument analogous to that used in the proof of (2.4.3 c)), one sees that the functor $\mathrm{K}^b(\mathrm{Loclib}(S)) \to \mathrm{D}(S)$ induced by Γ_S is fully faithful and has essential image $\mathrm{D}^b(A)_{\mathrm{parf}}$; one finishes the proof as in a).

Remark 2.4.6. The editor does not know whether in (2.4.4) one may replace D^b by D^-, and consequently whether (2.4.5 a)) remains true when one replaces $\mathrm{K}^{-,b}$ by K^- and D^b by D^-.

We finish with an interesting complement to (2.4.5). First we give a definition.

Definition 2.4.7. A *complex pseudo-analytic space* is a ringed space locally isomorphic to a ringed space of the form $(Y, \mathcal{O}_X|Y)$, where X is a complex analytic space and Y is a closed subset of X.

The structural sheaf of a complex pseudo-analytic space is therefore coherent. We say that a complex pseudo-analytic space is Stein if the corresponding topos is Stein (2.4.1).

Exposé II. Global resolutions (continued)

Proposition 2.4.8. Let S be a compact Stein complex pseudo-analytic space (2.4.7). Suppose that, for every open U of S and every reduced closed subspace X of U, the normalization of X is locally connected. This condition is verified, for example, when S is locally defined by real analytic equations or inequalities in a space $\mathbb{R}^n$. Then

$$A = \Gamma(S, \mathcal{O}_S)$$

is a noetherian ring.

Proof. It is enough to show that, if M is an A-module of finite presentation, then the set of submodules of M, or equivalently the set of submodules of finite type of M, satisfies the maximal condition. Indeed, if N is a submodule of finite type of M, then M/N is of finite presentation by I 2.6 and 2.9, so that N is itself of finite presentation by the equivalence (2.4.3 c). Using again the equivalence (2.4.3 c) between the category of coherent $\mathcal{O}_S$-modules and the category of A-modules of finite presentation, together with the compactness of S, one is reduced to proving the following lemma.

Lemma 2.4.8.1. Let S be a complex pseudo-analytic space satisfying the local connectedness condition of (2.4.8). Let F be a coherent $\mathcal{O}_S$-module, let (E_n) be an increasing sequence of coherent sub-$\mathcal{O}_S$-modules of F, and let s be a point of S. Then there exists an open neighbourhood U of s such that the sequence $(E_n|U)$ is stationary.

Proof. Since the ring $\mathcal{O}_{S,s}$ is noetherian, there exists an integer n_0 such that $(E_n)_s = (E_{n_0})_s$ for $n \geq n_0$. Replacing the sequence (E_n) by the sequence (E_n/E_{n_0}), one may therefore suppose that $(E_n)_s = 0$ for every n. On the other hand $F_s = M$ is a finite $\mathcal{O}_{S,s}$-module; hence it has a finite filtration

$$M = M_0 \supset M_1 \supset \cdots \supset M_q = 0$$

such that

$$M_i/M_{i+1} \simeq \mathcal{O}_s/\mathfrak{p}_i,$$

where $\mathfrak{p}_i$ is a prime ideal of $\mathcal{O}_s$. Since the category of finite-type $\mathcal{O}_s$-modules is equivalent to the category of germs at s of coherent $\mathcal{O}_S$-modules, after shrinking around s one may suppose that the M_i, respectively the $\mathfrak{p}_i$, are the fibres at s of coherent sub-$\mathcal{O}_S$-modules F_i, respectively coherent ideals $\mathfrak{p}_i$, of F, respectively of $\mathcal{O}_S$, and that one has

$$F = F_0 \supset F_1 \supset \cdots \supset F_q = 0, \qquad F_i/F_{i+1} \simeq \mathcal{O}_S/\mathfrak{p}_i.$$

To prove that the sequence (E_n) is stationary in a neighbourhood of s is therefore equivalent to proving the same assertion for the induced sequences on the graded pieces of this filtration. Thus one may suppose that $F = \mathcal{O}_S$ and that $\mathcal{O}_{S,s}$ is an integral domain; the E_n then form an increasing sequence of ideals of $\mathcal{O}_S$ with $(E_n)_s = 0$.

Since $\mathcal{O}_{S,s}$ is an integral domain, hence in particular reduced, one may moreover suppose, by Oka's theorem, after localizing at s, that S is reduced. We have to show that there is an open neighbourhood U of s such that $E_n|U = 0$ for n large enough. Consider, to do this, the normalization $\widetilde{S}$ of S. It is the analytic spectrum of a finite $\mathcal{O}_S$-algebra $\widetilde{\mathcal{O}}_S$ whose fibre at each $t \in S$ is the integral closure of $\mathcal{O}_{S,t}$. Since $E_n \subset E_n\widetilde{\mathcal{O}}_S$, it is enough to prove that $E_n\widetilde{\mathcal{O}}_S = 0$ for n large. This reduces us to the case where S is normal.

In that case, since $(E_n)_s = 0$, the closed immersion

$$i_n : S_n \longrightarrow S$$

defined by E_n is étale at s. But, S being normal, the set of points at which i_n is étale is both open and closed by the lemma below. Hence its image in S is simultaneously open and closed. The restriction of E_n to the connected component of s in S is therefore zero; this component is open because S is locally connected. The lemma is proved.

Lemma 2.4.8.2 (cf. EGA IV 18.10.3). Let Y be a normal pseudo-analytic space, and let $f : X \to Y$ be a non-ramified morphism. The set of points of X at which f is étale is both open and closed.

Proof. It is immediate that the set of points at which f is étale is open. It remains to show that the set of points at which f is not étale is open. Since a non-ramified morphism is locally an

immersion, one may suppose that f is a closed immersion defined by a coherent ideal I. Let x be a point of X such that $I_x \neq 0$. Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_{Y,x}$ containing I_x. One has

$$\dim(\mathcal{O}_{Y,x}/\mathfrak{p}) > 0,$$

for otherwise $\mathfrak{p} = 0$, since $\mathcal{O}_{Y,x}$ is integral. Thus Y has positive codimension at x, hence positive codimension in a neighbourhood of x; the assertion follows.

3. Complements on quasi-coherent sheaves on schemes

3.0. In this section we fix a quasi-compact and quasi-separated scheme S. The canonical, fully faithful inclusion functor

$$\iota : \mathrm{Qcoh}(S) \longrightarrow \mathrm{Mod}(S)$$

extends to a functor, again denoted ι, from $\mathrm{D}(\mathrm{Qcoh}(S))$ to $\mathrm{D}(S) = \mathrm{D}(\mathrm{Mod}(S))$, which we shall study.

Lemma 3.1. The category $\mathrm{Qcoh}(S)$ is an abelian category satisfying AB5, and the quasi-coherent sheaves of finite type form a small family of generators. In particular $\mathrm{Qcoh}(S)$ has enough injectives.

Proof. The assertion about generators follows at once from the fact that every quasi-coherent sheaf on S is the filtered inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9 and IV 1.7.7). The other assertions are evident.

Remark 3.1.1. The editor does not know whether $\mathrm{Qcoh}(S)$ still has enough injectives when one assumes only that S is quasi-separated.

Lemma 3.2. The functor

$$\iota : \mathrm{Qcoh}(S) \longrightarrow \mathrm{Mod}(S)$$

has a right adjoint

$$Q : \mathrm{Mod}(S) \longrightarrow \mathrm{Qcoh}(S),$$

called the coherer.

Proof. If S is affine, the existence of Q is immediate: one takes $Q = \widetilde{\Gamma_S}$ (EGA I 1.3.7). Let now $f : U \to S$ be an affine open of S. It is clear that, for $E \in \mathrm{ob}\,\mathrm{Qcoh}(S)$ and $F \in \mathrm{ob}\,\mathrm{Mod}(U)$, one has a canonical isomorphism

$$\mathrm{Hom}(E, f_*F) \simeq \mathrm{Hom}(E, f_*Q(F)). \tag{1}$$

Let $(f_i : U_i \to S)$ be a finite covering of S by affine opens, and, for each pair (i,j), let $(U_{ijk} \to U_i \cap U_j)_k$ be a finite covering of $U_i \cap U_j$ by affine opens, which is possible since S is quasi-separated. For $F \in \mathrm{ob}\,\mathrm{Mod}(S)$ one has an exact sequence

$$0 \longrightarrow F \longrightarrow \prod_i f_{i*}(F|U_i) \longrightarrow \prod_{i,j,k} f_{ijk*}(F|U_{ijk}), \tag{2}$$

where f_{ijk} denotes the inclusion of U_{ijk} into S. If $g : V \to U$ is an inclusion of affine opens of S, there is an evident natural arrow

$$Q(F|U) \longrightarrow g_*Q(F|V).$$

Consequently (2) gives an arrow

$$\prod_i f_{i*}Q(F|U_i) \longrightarrow \prod_{i,j,k} f_{ijk*}Q(F|U_{ijk}), \tag{3}$$

whose kernel we denote by QF. If $E \in \mathrm{ob}\,\mathrm{Qcoh}(S)$, applying the functor $\mathrm{Hom}(E, \cdot)$ to (2), and using (1), one obtains a functorial isomorphism in E,

$$\mathrm{Hom}(E, F) \simeq \mathrm{Hom}(E, QF).$$

This proves the existence of Q; its uniqueness is evident, since it is an adjoint.

Lemma 3.3.

a) The coherer is an additive functor and transforms injectives into injectives.

b) Let

$$RQ : \mathrm{D}^+(S) \longrightarrow \mathrm{D}^+(\mathrm{Qcoh}(S))$$

be the right derived functor of the coherer. For $E \in \mathrm{ob}\,\mathrm{D}(\mathrm{Qcoh}(S))$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$ there are canonical functorial isomorphisms

$$R\,\mathrm{Hom}_{\mathrm{Mod}}(E, F) \simeq R\,\mathrm{Hom}_{\mathrm{Qcoh}}(E, RQ(F)), \tag{i}$$
$$\mathrm{Hom}_{\mathrm{Mod}}(E, F) \simeq \mathrm{Hom}_{\mathrm{Qcoh}}(E, RQ(F)). \tag{ii}$$

Proof. Assertion a) is an immediate consequence of the definition. For b), it is enough to establish the isomorphism (i), since (ii) follows by passing to cohomology. For this purpose one may suppose that F has injective components. Then QF has injective components by a), and the isomorphism (i) follows directly from the definition of Q.

Lemma 3.4. Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes. For $E \in \mathrm{ob}\,\mathrm{Mod}(X)$ there is a canonical functorial isomorphism

$$Q_Y f_*(E) \simeq f_* Q_X(E).$$

Proof. Since f is quasi-compact and quasi-separated (EGA IV 1.2.3 and 1.2.4), f_* transforms quasi-coherent sheaves into quasi-coherent sheaves (EGA III 1.4.10 and IV 1.7.21). The functors $Q_Y f_*$ and $f_* Q_X$ from $\mathrm{Mod}(X)$ to $\mathrm{Qcoh}(Y)$ are therefore both right adjoints to the functor

$$f^* : \mathrm{Qcoh}(Y) \longrightarrow \mathrm{Mod}(X),$$

and hence are canonically isomorphic.

Proposition 3.5. If S is noetherian, or quasi-compact and separated, the functor

$$\iota : \mathrm{D}^+(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}^+(S)$$

is fully faithful and has as essential image the full subcategory $\mathrm{D}^+(S)_{\mathrm{qcoh}}$ of $\mathrm{D}^+(S)$ formed by complexes with quasi-coherent cohomology.

Proof. One may restrict to proving the assertion in the case where S is separated, since the noetherian case is already known. It is plainly equivalent to prove the following two assertions.

(3.5.1) For $E \in \mathrm{ob}\,\mathrm{D}^+(\mathrm{Qcoh}(S))$, the adjunction arrow

$$E \longrightarrow RQ(E)$$

is an isomorphism.

(3.5.2) For $E \in \mathrm{ob}\,\mathrm{D}^+(S)_{\mathrm{qcoh}}$, the adjunction arrow

$$RQ(E) \longrightarrow E$$

is an isomorphism.

Proof of (3.5.1). By an immediate dévissage one is reduced to proving that quasi-coherent sheaves are acyclic for the coherer, that is, for $E \in \mathrm{ob}\,\mathrm{Qcoh}(S)$, one has $R^iQ(E) = 0$ for $i > 0$, and that, moreover, the adjunction arrow $E \to QE$ is an isomorphism. If S is affine, this follows from the formula $Q = \widetilde{\Gamma_S}$ used in the proof of (3.2). In the general case, cover S by a finite number of affine opens U_i. Since S is separated, the inclusions

$$f_{i_0\dots i_p} : U_{i_0} \cap \cdots \cap U_{i_p} \longrightarrow S$$

are affine morphisms. Let $E \in \mathrm{ob}\,\mathrm{Qcoh}(S)$, and consider the resolution of E defined by this covering,

$$0 \longrightarrow E \longrightarrow \prod_i f_{i*}f_i^*E \longrightarrow \cdots \longrightarrow \prod_{i_0\dots i_p} f_{i_0\dots i_p*}f_{i_0\dots i_p}^*E \longrightarrow \cdots . \qquad (*)$$

It gives a spectral sequence

$$E_1^{pq} = \prod_{i_0\dots i_p} R^qQ(f_{i_0\dots i_p*}f_{i_0\dots i_p}^*E) \Longrightarrow R^{p+q}Q(E). \qquad (**)$$

Since $f_{i_0\dots i_p}$ is affine, (3.4) gives, for every q,

$$R^qQ(f_{i_0\dots i_p*}f_{i_0\dots i_p}^*E) \simeq f_{i_0\dots i_p*}R^qQ(f_{i_0\dots i_p}^*E). \qquad (***)$$

In particular $E_1^{pq} = 0$ for $q > 0$, by the affine case already treated. From (**) one deduces isomorphisms

$$R^nQ(E) \simeq H^n(QE') \quad (n \geq 0),$$

where E' is the resolution of E defined by the exact sequence (*). But by (***) one has $QE^n \simeq E^n$ for $n \geq 0$; hence finally

$$R^nQ(E) = 0 \quad (n > 0),$$

and $Q(E) \simeq E$. This proves (3.5.1).

Proof of (3.5.2). The spectral sequence

$$R^pQ(H^q(E)) \Longrightarrow R^{p+q}Q(E)$$

reduces us immediately to the case where $E \in \mathrm{ob}\,\mathrm{Qcoh}(S)$. The composite

$$E \longrightarrow RQE \longrightarrow E$$

of the adjunction arrows is the identity, and the assertion follows from (3.5.1). This completes the proof of (3.5).

Remark 3.6. The conclusion of (3.5) would be false if one omitted the hypothesis "separated or noetherian", as is shown by a neat counterexample of J.-L. Verdier; see Appendix I to the present exposé.

Proposition 3.7. Under the hypothesis of (3.5), suppose moreover that there is an integer N such that, for every $F \in \mathrm{ob}\,\mathrm{Mod}(S)$ and every quasi-compact open U of S, one has

$$H^i(U, F) = 0 \quad \text{for } i > N.$$

Then:

a) The coherer has finite cohomological dimension.

b) The functor

$$\iota : \mathrm{D}(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image the full subcategory $\mathrm{D}(S)_{\mathrm{qcoh}}$ of $\mathrm{D}(S)$ formed by complexes with quasi-coherent cohomology.

Proof of a). If S is affine, the formula $Q = \widetilde{\Gamma_S}$ implies that Q is of cohomological dimension $\leq N$. We pass to the general case. Let us show that there is an integer a such that $R^qQ(E) = 0$ for every $E \in \mathrm{ob}\,\mathrm{Mod}(S)$ and every $q > a$. The hypothesis implies that every $E \in \mathrm{ob}\,\mathrm{Mod}(S)$ admits a finite right resolution E' of length N such that the E^i are acyclic, that is, $H^q(U, E^i) = 0$ for every open U of S and every $q > 0$. Thus it is enough to prove that there is an integer b such that

$$R^qQ(E) = 0$$

for every acyclic $\mathcal{O}_S$-module E and every $q > b$; one may then take $a = b + N$.

Let (U_i) be a finite covering of S by N' affine opens. Let E be an acyclic $\mathcal{O}_S$-module and let $C'(E)$ be the resolution of E defined by the covering (U_i). There are two cases.

If S is separated, the components of $C'(E)$ are sums of terms of the form f_*f^*E, where $f : U \to S$ is the inclusion of an affine open, hence an affine morphism. One has

$$RQ(Q_X f_*)f^*E \simeq RQ\, Rf_*f^*E \simeq RQ\, f_*f^*E,$$

since f^*E is acyclic. On the other hand, by (3.4),

$$R(Q_X f_*)f^*E \simeq R(f_*Q_X)f^*E \simeq f_*RQf^*E,$$

since f is affine. Hence

$$R^nQ(f_*f^*E) \simeq f_*R^nQ(f^*E) \quad (n \geq 0).$$

By the affine case,

$$R^nQ(f_*f^*E) = 0 \quad (n > N).$$

Since $C'(E)$ has length N', it follows that

$$R^nQ(E) = 0 \quad (n > N + N').$$

If S is noetherian, the components of $C'(E)$ are sums of terms of the same form f_*f^*E, where now $f : U \to S$ is the inclusion of a quasi-compact and separated open. As in the first case, one has

$$R(Q_X f_*)f^*E \simeq RQ\, f_*f^*E.$$

Since U is noetherian, $\mathrm{Qcoh}(U)$ has enough injective objects in $\mathrm{Mod}(U)$, and hence the right derived functor of $f_* : \mathrm{Qcoh}(U) \to \mathrm{Qcoh}(S)$ is induced by $Rf_* : \mathrm{D}^+(U) \to \mathrm{D}^+(S)$ when $\mathrm{D}^+(\mathrm{Qcoh}(U))$ and $\mathrm{D}^+(\mathrm{Qcoh}(S))$ are identified with full subcategories of $\mathrm{D}^+(U)$ and $\mathrm{D}^+(S)$ by (3.5). Thus

$$R(f_*Q_X)f^*E \simeq Rf_*RQf^*E.$$

One deduces a spectral sequence

$$E_2^{pq} = R^pf_*R^qQ(f^*E) \Longrightarrow R^{p+q}(f_*Q)f^*E.$$

By the first case there is an integer $n(U)$ depending only on U such that Q is of cohomological dimension $< n(U)$ on $\mathrm{Mod}(U)$; hence $E_2^{pq} = 0$ for $q > n(U)$. Also $E_2^{pq} = 0$ for $p > N$, by the cohomological-dimension hypothesis on S. Thus

$$R^n(f_*Q)f^*E = 0 \quad (n > N + N''),$$

where N'' is an integer greater than all the $n(U)$, with U running over the intersections $U_{i_0} \cap \cdots \cap U_{i_p}$. Taking account of the preceding displayed isomorphism, one obtains

$$R^n Q(f_* f^* E) = 0 \quad (n > N + N''),$$

and hence finally

$$R^n Q(E) = 0 \quad (n > N + N' + N'').$$

This proves a).

Proof of b). The coherer, being of finite cohomological dimension by a), admits a right derived functor

$$RQ : \mathrm{D}(S) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

whose restriction to $\mathrm{D}^+(S)$ coincides with the functor RQ defined above. For $E \in \mathrm{ob}\,\mathrm{D}(\mathrm{Qcoh}(S))$ there is a canonical functorial isomorphism

$$E \xrightarrow{\sim} RQ(E), \tag{1}$$

since, by (3.5.1), the E^i are acyclic for Q and satisfy $E^i \xrightarrow{\sim} Q(E^i)$. On the other hand, for $E \in \mathrm{ob}\,\mathrm{D}(S)$ one has a canonical functorial homomorphism

$$RQ(E) \longrightarrow E, \tag{2}$$

which, in the case where the E^i are Q-acyclic, comes from the ordinary adjunction homomorphism between ι and Q. It is checked directly that these homomorphisms make

$$RQ : \mathrm{D}(S) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

a right adjoint of

$$\iota : \mathrm{D}(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}(S).$$

The fact that (1) is an isomorphism implies that ι is fully faithful. Finally, using the spectral sequence

$$R^p Q(H^q(E)) \Longrightarrow R^{p+q} Q(E),$$

which converges since Q has finite cohomological dimension, one sees that (2) is an isomorphism for $E \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{qcoh}}$; hence ι has image $\mathrm{D}(S)_{\mathrm{qcoh}}$. This proves b) and completes the proof of (3.7).

Appendix I. A counterexample of Verdier

0. Introduction

If S is a scheme, write $\mathrm{Mod}(S)$, respectively $\mathrm{Qcoh}(S)$, for the category of $\mathcal{O}_S$-modules, respectively of quasi-coherent $\mathcal{O}_S$-modules. The purpose of this appendix is to draw attention to certain pathological properties of the category $\mathrm{D}(\mathrm{Qcoh}(S))$, and in particular to prove the following facts.

0.1. Let X be an affine scheme with ring A, and let I be an injective A-module. We know that, if A is noetherian, the sheaf $\widetilde{I}$ is an injective $\mathcal{O}_X$-module. If A is not noetherian, $\widetilde{I}$ is not necessarily injective, nor even flasque, even if the underlying space of X is noetherian of finite Krull dimension. Moreover, the restriction of $\widetilde{I}$ to an open U of X is not, in general, an injective object of $\mathrm{Qcoh}(U)$.

0.2. Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes. The functor

$$Rf_* : \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(Y)$$

does not in general induce, on $\mathrm{D}^+(\mathrm{Qcoh}(X))$ and $\mathrm{D}^+(\mathrm{Qcoh}(Y))$, the right derived functor of

$$f_* : \mathrm{Qcoh}(X) \longrightarrow \mathrm{Mod}(Y).$$

More precisely, an injective object of $\mathrm{Qcoh}(X)$ is not necessarily acyclic for $f_* : \mathrm{Mod}(X) \to \mathrm{Mod}(Y)$.

0.3. Let S be a quasi-compact and quasi-separated scheme. An object of $\mathrm{Qcoh}(S)$, even an injective object, is not necessarily acyclic for the coherer (II 3.2). The functor

$$\iota : \mathrm{D}^b(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}^b(S)$$

is not in general fully faithful.

1. The counterexamples

We first recall the following lemma (SGA 2 II 9).

Lemma 1.1. Let A be a ring, $X = \mathrm{Spec}(A)$, let $\mathbf{f} = (f_\alpha)$ be a finite system of elements of A, and let Y be the closed subset defined by $\mathbf{f}(x) = 0$. For $i > 0$, the following conditions are equivalent:

(i) for every injective A-module I, one has $H^i_Y(X, \widetilde{I}) = 0$;

(ii) the pro-object $n \mapsto H_i(\mathbf{f}^n, A)$ is zero, that is, for every n there exists $n' \geq n$ such that

$$H_i(\mathbf{f}^{n'}, A) \longrightarrow H_i(\mathbf{f}^n, A)$$

is zero.

1.2. We shall use (1.1) to prove (0.1). Let k be a field,

$$B = k[x, y], \qquad \mathbf{f} = (x, y), \qquad M = \bigoplus_{n>0} B/(\mathbf{f}^n),$$

and let A be the B-algebra $D_B(M)$ (EGA 0_{IV} 18.2.3). One has trivially

$$H_2^A(\mathbf{f}^n, A) \simeq H_2^B(\mathbf{f}^n, A).$$

On the other hand,

$$H_2^B(\mathbf{f}^n, A) \simeq H_2^B(\mathbf{f}^n, B) \oplus H_2^B(\mathbf{f}^n, M),$$

and

$$H_2^B(\mathbf{f}^n, B) = 0,$$

since $\mathbf{f}$ is a regular sequence. Thus

$$H_2^B(\mathbf{f}^n, A) \simeq H_2^B(\mathbf{f}^n, M).$$

Finally,

$$H_2^B(\mathbf{f}^n, M) \simeq H_B^0(\mathbf{f}^n, M),$$

where $H_B^0(\mathbf{f}^n, M)$ is canonically identified with the submodule $(\mathbf{f}^n, M)$ of M annihilated by $\mathbf{f}^n$. Hence

$$H_2^A(\mathbf{f}^n, A) \simeq (\mathbf{f}^n, M). \qquad (*)$$

Since the transition morphism

$$H_2^A(\mathbf{f}^{n'}, A) \longrightarrow H_2^A(\mathbf{f}^n, A)$$

is induced by multiplication by $\mathbf{f}^{n'-n}$, one checks immediately from (*) that the pro-object $n \mapsto H_2^A(\mathbf{f}^n, A)$ is not zero. Hence, by (1.1), there is an injective A-module I such that

$$H_Y^2(X, \widetilde{I}) \neq 0,$$

and therefore $\widetilde{I}$ is not flasque. Notice also that the projection $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ is a homeomorphism, so that the underlying space of X is noetherian of dimension 2. If U denotes the complement of Y in X, then by SGA 2 II 4,

$$H_Y^2(X, I) \simeq H^1(U, I) \neq 0.$$

But, since U is separated, the functor $\mathrm{D}^+(\mathrm{Qcoh}(U)) \to \mathrm{D}^+(U)$ is fully faithful by II 3.5, so $\widetilde{I}|U$ is not injective in $\mathrm{Qcoh}(U)$. This proves assertion (0.1).

Let Z be the quasi-compact and quasi-separated scheme obtained by gluing two copies of X along U; let j_1, j_2 be the two canonical open immersions of X into Z. Thus one has a Cartesian square

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j_1} \\ X & \xrightarrow[j_2]{} & Z. \end{array}$$

One has

$$j_2^* R^1 j_{1*}(\widetilde{I}) \simeq R^1 i_*(i^* \widetilde{I}),$$

and hence

$$H^0(X, R^1 j_{1*}(\widetilde{I})) \simeq H^1(U, \widetilde{I}) \neq 0,$$

so that $R^1 j_{1*}(\widetilde{I}) \neq 0$, which proves (0.2).

On the other hand, by II 3.4 and 3.5 one has

$$RQ\, Rj_{1*}(\widetilde{I}) \simeq j_{1*}(\widetilde{I}).$$

It follows from this a spectral sequence whose exact sequence in low degrees gives the isomorphism

$$R^1 j_{1*}(\widetilde{I}) \simeq R^2 Q(j_{1*} \widetilde{I}).$$

Since the sheaf $j_{1*}\widetilde{I}$ is injective in $\mathrm{Qcoh}(Z)$, the first part of (0.3) is proved.

Finally, let $E \in \mathrm{ob}\,\mathrm{Qcoh}(Z)$. We have the duality isomorphism

$$R\,\mathrm{Hom}_{\mathrm{Mod}}(E, Rj_{1*}\widetilde{I}) \simeq R\,\mathrm{Hom}_{\mathrm{Mod}}(j_1^* E, \widetilde{I}),$$

but, since X is affine, II 3.5 gives

$$R\,\mathrm{Hom}_{\mathrm{Mod}}(j_1^* E, \widetilde{I}) \simeq R\,\mathrm{Hom}_{\mathrm{Qcoh}}(j_1^* E, I) \simeq \mathrm{Hom}(j_1^* E, I).$$

Consequently

$$R\,\mathrm{Hom}_{\mathrm{Mod}}(E, Rj_{1*}\widetilde{I}) \simeq \mathrm{Hom}(j_1^* E, I).$$

A spectral sequence then gives, in low degree, the isomorphism

$$\mathrm{Ext}^2_{\mathrm{Mod}}(E, j_{1*}\widetilde{I}) \simeq \mathrm{Hom}(E, R^1 j_{1*}\widetilde{I}).$$

Thus the identity morphism of $E = R^1 j_{1*}\widetilde{I}$ supplies a non-zero element of

$$\mathrm{Ext}^2_{\mathrm{Mod}}(E, j_{1*}\widetilde{I}).$$

It follows that the functor

$$\mathrm{D}^b(\mathrm{Qcoh}(Z)) \longrightarrow \mathrm{D}^b(Z)$$

is not fully faithful. This completes the proof of (0.3).

Appendix II. Definition of the analytic index of a relative elliptic complex

In this appendix we propose to show that the notion of a perfect complex leads to a simple and natural definition of the "analytic index" of a continuous family of elliptic differential operators, or more generally of a relative elliptic complex. For the initiated reader, let us say that the definition given here has the advantage, compared with the classical definitions (see for example [2] and [17]), of not requiring any deformation of the elliptic complex under consideration.

Sections 1 and 2 consist of reminders on the classical notion of a "mixed structure". [16] We give, of course, only the minimum of definitions and results required in order to state and prove, in Section 3, the central theorem we have in view. It says, more precisely, that the direct image, by the projection (assumed proper) onto the base (assumed locally compact), of a relative elliptic complex is a perfect complex. When the base S is compact, the definition of the analytic index as an element of $K(S)$, the Grothendieck group of complex vector bundles of finite rank on S, then follows from the globalization sorite of II 2.3.2.

Throughout what follows, if S is a topological space, $\mathcal{O}_S$ will denote the sheaf of continuous real-valued functions on S.

1. The sorite of mixed structures

In this section a topological space S is fixed.

1.1. Definition of mixed varieties

Let V be an open subset of $\mathbb{R}^n$. For $r \in \mathbb{N}$ one writes $C^r(V)$ for the Fréchet space of real-valued functions of class C^r on V. Put

$$C^\infty(V) = \lim_r C^r(V).$$

Let U be an open subset of S. For $r \in \mathbb{N} \cup \{\infty\}$, denote by ${}^r\mathcal{O}_{U\times V}$ the sheaf of rings on $U \times V$ defined by

$$\Gamma(U' \times V', {}^r\mathcal{O}_{U\times V}) = C(U', C^r(V')),$$

where U' (respectively V') is an open subset of U (respectively of V), and $C(U', C^r(V'))$ denotes the space of continuous maps from U' into $C^r(V')$. There is a canonical injection

$$\mathrm{pr}_1^{-1}(\mathcal{O}_U) \longrightarrow {}^r\mathcal{O}_{U\times V}$$

which makes $(U \times V, {}^r\mathcal{O}_{U\times V})$ into a ringed space over U.

A ringed space X over S is said to be a *mixed variety of class* C^r ($r \in \mathbb{N} \cup \{\infty\}$) if X is locally isomorphic, as a ringed space over S, to a space of the type just described. If S is reduced to one point, a mixed variety of class C^r over S is just a variety of class C^r in the usual sense.

Let $f : X \to S$ be a mixed variety of class C^r. For $s \in S$, the fiber $X_s = f^{-1}(s)$, endowed with the sheaf $i_s^{-1}(\mathcal{O}_X)/I_s$, where $i_s : X_s \to X$ is the injection and I_s is the ideal of sections which vanish on X_s, is a variety of class C^r.

[16] The informed reader is advised to skip them.

1.2. Relative differential operators

It is easy, on the model of algebraic geometry (EGA IV 16, SGA 3 VII), to develop an infinitesimal study of mixed varieties. We shall only review rapidly the notions we shall need.

1.2.1. Let $f : X \to S$ be a mixed variety of class C^∞. For each $n \in \mathbb{N}$ one defines the sheaf $P^n_{X/S}$ of principal parts of order n on X. In two different ways, corresponding to the two projections pr_1 and pr_2 of $X \times_S X$ onto X, it is an $\mathcal{O}_X$-algebra augmented toward $\mathcal{O}_X$, and is locally free of finite type as an $\mathcal{O}_X$-module. When the $\mathcal{O}_X$-module structure on $P^n_{X/S}$ is not specified, it will be understood to be the one coming from pr_1. One has the canonical identifications

$$P^0_{X/S} = \mathcal{O}_X, \qquad P^1_{X/S}/\mathcal{O}_X = \Omega^1_{X/S}.$$

The sheaf $\Omega^1_{X/S}$ is the sheaf of relative differentials of X over S, or relative cotangent sheaf. Its dual is denoted $T_{X/S}$ and called the relative tangent sheaf. The sheaf $T_{X/S}$, respectively $\Omega^1_{X/S}$, is the sheaf of sections of a vector bundle of finite rank on X, called the relative tangent, respectively cotangent, bundle; these bundles are mixed varieties over X.

The $f^{-1}(\mathcal{O}_S)$-linear homomorphism, corresponding to the $\mathcal{O}_X$-algebra structure defined by pr_2,

$$d^n_{X/S} : \mathcal{O}_X \longrightarrow P^n_{X/S},$$

which associates to a section of $\mathcal{O}_X$ its principal part of order n, is called the universal differential operator of order n on X relative to S.

1.2.2. More generally, if E is a locally free $\mathcal{O}_X$-module of finite type, one may consider the sheaf of principal parts of order n of E,

$$P^n_{X/S}(E) = P^n_{X/S} \otimes_{\mathcal{O}_X} E,$$

where the tensor product is defined by means of the second $\mathcal{O}_X$-module structure on $P^n_{X/S}$. One has a universal differential operator

$$d^n_{X/S}(E) : E \longrightarrow P^n_{X/S}(E).$$

Let E, F be locally free $\mathcal{O}_X$-modules of finite type. By definition, a differential operator from E to F of order $\leq n$ is an $f^{-1}(\mathcal{O}_S)$-linear homomorphism from E to F which factors through $d^n_{X/S}(E)$ as an $\mathcal{O}_X$-linear homomorphism $P^n_{X/S}(E) \to F$; the factorization is then unique. In other words, $d^n_{X/S}(E)$ gives a bijection

$$\mathrm{Hom}_{\mathcal{O}_X}(P^n_{X/S}(E), F) \xrightarrow{\sim} \mathrm{Diff}^n(E, F), \tag{1.2.2.1}$$

where $\mathrm{Diff}^n(E, F)$ denotes the set of differential operators from E to F of order $\leq n$.

1.2.3. We shall also have to consider differential operators with complex coefficients. Let $\mathcal{O}_X \otimes \mathbb{C}$ (respectively $\mathcal{O}_S \otimes \mathbb{C}$) be the complexification of $\mathcal{O}_X$ (respectively $\mathcal{O}_S$). Let E, F be locally free $(\mathcal{O}_X \otimes \mathbb{C})$-modules of finite type. By a differential operator from E to F of order $\leq n$ one means an $f^{-1}(\mathcal{O}_S \otimes \mathbb{C})$-linear homomorphism from E to F which factors through $d^n_{X/S}(E)$ as an $(\mathcal{O}_X \otimes \mathbb{C})$-linear homomorphism $P^n_{X/S}(E) \to F$. If $\mathrm{Diff}^n_{\mathbb{C}}(E, F)$ denotes the set of such operators, $d^n_{X/S}(E)$ therefore defines a bijection

$$\mathrm{Hom}_{\mathcal{O}_X \otimes \mathbb{C}}(P^n_{X/S}(E), F) \xrightarrow{\sim} \mathrm{Diff}^n_{\mathbb{C}}(E, F). \tag{1.2.3.1}$$

1.2.4. For $n \geq 1$ there is a canonical exact sequence

$$0 \longrightarrow S^n(\Omega^1_{X/S}) \longrightarrow P^n_{X/S} \longrightarrow P^{n-1}_{X/S} \longrightarrow 0.$$

Hence, tensoring over $\mathcal{O}_X$ with a locally free $\mathcal{O}_X$-module, respectively $(\mathcal{O}_X \otimes \mathbb{C})$-module, E of finite type, one obtains an exact sequence

$$0 \longrightarrow S^n(\Omega^1_{X/S}) \otimes E \longrightarrow P^n_{X/S}(E) \longrightarrow P^{n-1}_{X/S}(E) \longrightarrow 0. \tag{1.2.4.1}$$

Here $S^n(\Omega^1_{X/S})$ denotes the n-th symmetric power.

Let F be a locally free $\mathcal{O}_X$-module, respectively $(\mathcal{O}_X \otimes \mathbb{C})$-module, of finite type. Applying $\mathrm{Hom}(\ ,F)$, respectively $\mathrm{Hom}_{\mathcal{O}_X \otimes \mathbb{C}}(\ ,F)$, to (1.2.4.1), and using (1.2.2.1) and (1.2.3.1), one obtains an exact sequence

$$0 \longrightarrow \mathrm{Diff}^{n-1}(E,F) \longrightarrow \mathrm{Diff}^n(E,F) \xrightarrow{\sigma_n} \mathrm{Hom}_{\mathcal{O}_X}\big(S^n(\Omega^1_{X/S}) \otimes E, F\big), \tag{1.2.4.2}$$

and analogously in the complex case. If X is paracompact and $\mathcal{O}_X$ is soft, the sequence may be completed on the right by adding a zero. The homomorphism σ_n is called the *symbol*. It associates to a differential operator from E to F of order $\leq n$ a homogeneous polynomial map of degree n from $T^*_{X/S}$ into $\mathrm{Hom}(E,F)$, or equivalently a homomorphism $p^*E \to p^*F$, where p is the projection of the relative cotangent bundle onto X, satisfying the usual homogeneity condition.

Let E, F, G be locally free $\mathcal{O}_X$-modules of finite type, $u \in \mathrm{Diff}^n(E,F)$ and $v \in \mathrm{Diff}^m(F,G)$. Then $vu \in \mathrm{Diff}^{m+n}(E,G)$ and one checks at once the relation

$$\sigma_{m+n}(vu) = \sigma_m(v)\sigma_n(u). \tag{1.2.4.3}$$

The same relation holds for differential operators with complex coefficients. In particular, if the composite of two differential operators is zero, then the composite of their symbols is zero.

1.2.5. Let $E^\bullet$ be a bounded complex of differential operators with complex coefficients on X relative to S, and let $d^i : E^i \to E^{i+1}$ be of order n_i. By (1.2.4.3),

$$\sigma_{n_{i+1}}(d^{i+1})\sigma_{n_i}(d^i) = 0.$$

The complex $E^\bullet$ is called an *elliptic complex of order* (n_i) if the symbol complex on the relative cotangent bundle, pulled back to that bundle, is acyclic outside the zero section.

When S is reduced to one point, this notion of elliptic complex on X coincides with the usual one, studied in detail in [4] and [16].

1.2.6. Let $s \in S$, and let i_s be the inclusion of the fiber X_s in X, considered as a monomorphism of ringed spaces (1.1). For $n \in \mathbb{N}$ one has a canonical isomorphism

$$i_s^* P^n_{X/S} \simeq P^n_{X_s},$$

and hence a restriction homomorphism

$$i_s^* : \mathrm{Diff}^n(E,F) \longrightarrow \mathrm{Diff}^n(i_s^*E, i_s^*F),$$

for E, F locally free $\mathcal{O}_X$-modules of finite type, and similarly for operators with complex coefficients. If $E^\bullet$ is a bounded complex of differential operators with complex coefficients on X, then $E^\bullet$ is elliptic if and only if, for every $s \in S$, $i_s^*(E^\bullet)$ is an elliptic complex. The verification is left to the reader.

2. Rigidity lemmas

2.0. It is "well known" that compact differentiable varieties are rigid, that is, they admit no continuous deformation other than the trivial one. Since this result will be used essentially in

Section 3, and references seem rather scarce, it is perhaps useful to give precise statements and a few proofs.

A topological space S is fixed. The mixed varieties under consideration are assumed to be of class C^r, with $r \gg 1$; possibly $r = \infty$.

2.1. Notation. Let X, Y be mixed S-varieties. We denote by $\mathrm{Hom}(X, Y)$ the set of S-morphisms from X to Y, and by $\mathcal{H}om(X, Y)$ the sheaf on X of germs of S-morphisms from X to Y, defined by

$$\mathcal{H}om(X, Y)(U) = \mathrm{Hom}(U, Y)$$

for U open in X.

Let $s \in S$, and write i_s for the inclusion of X_s in X. There is a canonical induction homomorphism

$$i_s^{-1}\mathcal{H}om(X, Y) \longrightarrow \mathcal{H}om(X_s, Y_s) \tag{2.1.1}$$

which is an epimorphism, as is immediate. In the case $Y = S \times \mathbb{R}$, the sheaf $\mathcal{H}om(X, Y)$ is canonically identified with $\mathcal{O}_X$, and the epimorphism (2.1.1) with the epimorphism

$$i_s^{-1}(\mathcal{O}_X) \longrightarrow i_s^*(\mathcal{O}_X)$$

considered in (1.1).

Proposition 2.2. Let X, Y be mixed S-varieties, and let $s \in S$. Suppose that the fiber X_s is paracompact and that s has a fundamental system of paracompact neighborhoods. Let A be a closed subset of X_s, and let

$$f \in H^0(A, \mathcal{H}om(X_s, Y_s)).$$

The subsheaf $\mathcal{F}$ of $i_s^{-1}\mathcal{H}om(X, Y)$ consisting of the germs which extend f, that is, whose image by (2.1.1) coincides with f over A, is soft.

Proof. Let X' be an open neighborhood in X which is isomorphic, as a mixed variety, to $S' \times U$, where S' is a paracompact neighborhood of s and U is a relatively compact open subset of $\mathbb{R}^n$, and such that the image by f of $X'_s \cap A$ is contained in an open subset of Y isomorphic to $S' \times V$, where V is a convex open subset of $\mathbb{R}^q$ containing the origin. By T.F. II 3.4.1 it is enough to show that every section of $\mathcal{F}$ over a closed subset of X_s contained in X'_s extends to X'_s. Since $X'_s = \{s\} \times U$ is relatively compact, a closed subset of X_s contained in X'_s is compact. Identifying X' with $S' \times U$ and Y' with $S' \times V$, putting $A' = A \cap X'_s$, and using T.F. II 3.3.1, the assertion is reduced to the following elementary lemma.[17]

Lemma 2.2.1. Let U be an open subset of $\mathbb{R}^n$, and let V be a convex open subset of $\mathbb{R}^q$ containing 0. Let A_1, A_2 be closed subsets of $\{s\} \times U$. Let U_1, respectively U_2, be an open subset of $S \times U$, respectively of $\{s\} \times U$, containing A_1, respectively A_2. Let $f_1 : U_1 \to V$ be a morphism and let $f_2 : U_2 \to S \times V$ be an S-morphism which, on $U_1 \cap (\{s\} \times U)$, defines the same germ as f_1 near A_1. Then there exists an open subset U' of $S \times U$ containing $\{s\} \times U$ and an S-morphism

$$F : U' \longrightarrow S \times V$$

such that the germ of F along $\{s\} \times U$ extends the germ defined by f_1, and the morphism induced by F on $\{s\} \times U$ defines the same germ as f_2 near A_2.

Proof. Replacing f_1 by af_1, where a is a C^∞ function on U equal to 1 near A_1 and with support in U_1, one may suppose $U_1 = U$. Let b be a C^∞ function on U, equal to 1 on A_2 and with support in $U_2 \cap (\{s\} \times U)$. Then

$$U' = U_1 \cup \big(S \times (U - \mathrm{Supp}\, b)\big)$$

[17] Communicated to the writer by A. Douady. A closed subset of a paracompact space has a fundamental system of paracompact neighborhoods.

is an open subset of $S \times U$ containing $\{s\} \times U$. The morphism $F : U' \to S \times V$ defined by

$$F(t,x) = \big(t,\, b(x)\,\mathrm{pr}_2\, f_2(x) + (1 - b(x)) f_1(t,x)\big)$$

visibly satisfies the required conditions.

Corollary 2.3. Under the hypotheses of (2.2), the sheaves $i_s^{-1}\mathcal{H}om(X,Y)$ and $\mathcal{H}om(X_s,Y_s)$ are soft. In particular, the sheaves $i_s^{-1}(\mathcal{O}_X)$ and $\mathcal{O}_{X_s}$ are soft. For every closed subset A of X_s, the map

$$H^0(X_s, i_s^{-1}\mathcal{H}om(X,Y)) \longrightarrow H^0(A, \mathcal{H}om(X_s,Y_s))$$

deduced from (2.1.1) is surjective.

Proof. The softness of $i_s^{-1}\mathcal{H}om(X,Y)$ follows from (2.2) applied with $A = \varnothing$. Taking $S = \{s\}$, one obtains the softness of $\mathcal{H}om(X_s,Y_s)$. The last assertion follows from (2.2).[18]

Corollary 2.4. Under the hypotheses of (2.2), let E be an $\mathcal{O}_X$-module. For every closed subset A of X_s, the canonical homomorphism

$$H^0(X_s, i_s^{-1}E) \longrightarrow H^0(A, i_s^*E)$$

deduced from the epimorphism

$$i_s^{-1}E \longrightarrow i_s^*E = i_s^{-1}E \otimes_{i_s^{-1}\mathcal{O}_X} \mathcal{O}_{X_s}$$

is an epimorphism.

Proof. Since $i_s^{-1}\mathcal{O}_X$ is soft by (2.3), the functor $H^0(A,\ )$ is exact on the category of $i_s^{-1}\mathcal{O}_X$-modules (T.F. II 3.7.1, 3.4.2, 4.4.3). Hence

$$H^0(A, i_s^{-1}E) \longrightarrow H^0(A, i_s^*E)$$

is an epimorphism. Since $i_s^{-1}E$ is soft, the homomorphism

$$H^0(X_s, i_s^{-1}E) \longrightarrow H^0(A, i_s^{-1}E)$$

is an epimorphism, and this proves the assertion.

Corollary 2.5. Let $p : X \to S$ be a mixed variety and let $s \in S$. Suppose that s has a fundamental system of paracompact neighborhoods S' such that the $p^{-1}(S')$ form a fundamental system of paracompact neighborhoods of X_s. This condition is satisfied, for example, if S is locally compact and p is a proper map of topological spaces. Then, for every $\mathcal{O}_X$-module E,

$$(R^i p_* E)_s = 0 \qquad (i > 0).$$

Proof. By T.F. II 4.17.1,

$$(R^i p_* E)_s = H^i(X_s, i_s^{-1}E).$$

But $i_s^{-1}\mathcal{O}_X$ is soft by (2.3), hence $H^i(X_s, i_s^{-1}E) = 0$ for $i > 0$.

Corollary 2.6. Let X, Y be mixed S-varieties, and let $s \in S$. Suppose that X satisfies the hypotheses of (2.5). For every morphism $f : X_s \to Y_s$, there exists an open neighborhood S' of s and a morphism

$$F : X|S' \longrightarrow Y|S'$$

such that $F_s = f$.

[18] For $\mathcal{O}_{X_s}$ this is very well known. The writer does not know whether $\mathcal{O}_X$ is soft when the hypotheses of (2.2) are satisfied for every $s \in S$ and X is paracompact. This would trivially imply, under those hypotheses, the softness of $i_s^{-1}(\mathcal{O}_X)$ as well as (2.5) below.

Proof. This is an immediate consequence of (2.3), with $A = X_s$, and of the fact that the hypotheses on X give

$$H^0(X_s, i_s^{-1}\mathcal{H}om(X,Y)) = \varinjlim_{S' \ni s} \mathrm{Hom}(X|S', Y|S')$$

for open S' containing s (T.F. II 8.5.5).

Proposition 2.7. Let X, Y be mixed S-varieties, let $f : X \to Y$ be a morphism, let $s \in S$, and let $x \in X_s$. If the tangent linear map of

$$f_s : X_s \longrightarrow Y_s$$

is invertible at x, then f is a local isomorphism at x; that is, there is an open neighborhood U of x in X and an open subset V of Y such that f induces an isomorphism $U \simeq V$.

Proof. This is an easy consequence of the implicit function theorem. The details are left to the reader.

Corollary 2.8. Let X, Y be mixed S-varieties and let $s \in S$. Suppose that s has a fundamental system of neighborhoods S' such that the $X|S'$, respectively $Y|S'$, form a fundamental system of neighborhoods of X_s, respectively Y_s; this is the case, for example, if S is locally compact and X, respectively Y, is proper over S. Let $f : X \to Y$ be an S-morphism such that

$$f_s : X_s \longrightarrow Y_s$$

is an isomorphism. Then there exists a neighborhood S' of s such that

$$f|X|S' : X|S' \longrightarrow Y|S'$$

is an isomorphism.

Proof. By (2.7), there exists an open neighborhood U of X_s in X such that $f|U$ is étale, that is, a local isomorphism. We may suppose that $U = X|S'$ for a neighborhood S' of s; after shrinking S, we may therefore assume that f is étale. Then the fiber product $Z = X \times_Y X$ is naturally a mixed S-variety, étale over X for each projection, and the diagonal morphism $\Delta : X \to Z$ is an open immersion. Moreover the sets $Z|S'$ form a fundamental system of neighborhoods of Z_s. Since

$$\Delta_s : X_s \longrightarrow Z_s$$

is an isomorphism, it follows, after shrinking S, that Δ is an isomorphism. Thus f is an open immersion, and after shrinking S again it is an isomorphism.

Corollary 2.9. Let X be a mixed S-variety and let $s \in S$. Suppose that the hypothesis of (2.5) is satisfied. Then there exists a neighborhood S' of s in S such that $X|S'$ is S'-isomorphic to $S' \times X_s$.

Proof. Apply (2.6) and (2.8).

Corollary 2.10. Let $s \in S$ and let V be a paracompact variety. Put $X = S \times V$, and let i_s be the injection of V into X defined by s. Suppose that the hypothesis of (2.5) is satisfied. Let E be a locally free $\mathcal{O}_X$-module, respectively $\mathcal{O}_X \otimes \mathbb{C}$-module, of finite type. Then there exists a neighborhood S' of s in S such that $E|X|S'$ is isomorphic to $\mathrm{pr}_2^*(i_s^*E)$, where $\mathrm{pr}_2 : S' \times V \to V$ is the second projection.

Proof. Put $F = \mathrm{pr}_2^*(i_s^*E)$. By (2.4), applied to $\mathcal{H}om(E,F)$, the induction homomorphism

$$\mathrm{Hom}(i_s^{-1}E, i_s^{-1}F) \longrightarrow \mathrm{Hom}(i_s^*E, i_s^*F)$$

is an epimorphism. On the other hand, by T.F. II 3.3.1,

$$\mathrm{Hom}(i_s^{-1}E, i_s^{-1}F) = \lim_{S' \ni s} \mathrm{Hom}(E|X|S', F|X|S').$$

Thus the canonical isomorphism $i_s^*E \simeq i_s^*F$ is induced, after shrinking S, by a homomorphism $f : E \to F$. Since f_s is an isomorphism, f is an isomorphism near X_s by Nakayama, hence over $X|S'$ for a suitable neighborhood S' of s.

3. The finiteness theorem

3.0. Conventions and notation

3.0.1. Let X and Y be topological spaces. We write $C(X, Y)$ for the set of continuous maps from X to Y, and $\mathcal{C}(X, Y)$ for the sheaf on X defined by $U \mapsto C(U, Y)$.

3.0.2. Let E, F be Banach spaces, real or complex. We write $L(E, F)$ for the Banach space of continuous linear maps from E to F.

3.0.3. The mixed varieties occurring in this section are assumed to be of class C^∞. If X is a mixed S-variety, we shall often identify a locally free $\mathcal{O}_X$-module, respectively $\mathcal{O}_X \otimes \mathbb{C}$-module, of finite type with the corresponding real, respectively complex, vector bundle.

3.0.4. Let X be a compact ordinary variety, and let E be a real or complex vector bundle of finite rank on X. Write $C^\infty(X, E)$ for the Fréchet space of C^∞ sections of E on X. For $r \in \mathbb{Z}$, write $C_r(X, E)$ for the Hilbert space of sections of E which are r times differentiable in L^2 ([16] IX). For $s \leq r$, $C_r(X, E)$ maps to $C_s(X, E)$ by a continuous linear injective map with dense image, and

$$C^\infty(X, E) = \lim_r C_r(X, E).$$

3.0.5. Let S be a topological space. We shall need the notion of a Hilbert bundle over S, for which we refer the reader to [12] or [14]. There is an additive functor from the category of real, respectively complex, Hilbert bundles over S to the category of $\mathcal{O}_S$-, respectively $\mathcal{O}_S \otimes \mathbb{C}$-, modules: it sends a Hilbert bundle E to the sheaf of continuous sections of E.

3.1. Auxiliary technical results

Lemma 3.1.1. Let X, Y be topological spaces, with X locally compact. Let $\mathcal{C}'(X, Y)$ be the presheaf, on the basis of relatively compact open subsets of X, defined by

$$\mathcal{C}'(X, Y)(U) = C(\overline{U}, Y).$$

The canonical restriction morphism

$$\mathcal{C}'(X, Y) \longrightarrow \mathcal{C}(X, Y)$$

induces an isomorphism on the associated sheaves.

Proof. Immediate.

Lemma 3.1.2. Let E, F be locally convex topological vector spaces over $\mathbb{C}$, and let $u : E \to F$ be a continuous linear map, respectively a continuous linear map with dense image. Let X be compact. Endow $C(X, E)$ and $C(X, F)$ with the topology of uniform convergence. Then the induced linear map

$$u_X : C(X, E) \longrightarrow C(X, F)$$

is continuous, respectively continuous with dense image.

Proof. It is clear that u_X is continuous. On the other hand, by Bourbaki, *Intégration*, chapter II, §2, lemmas 1 and 2, the subspace $C(X,\mathbb{C})\otimes E$ of $C(X,E)$, formed by the continuous maps from X to E whose values lie in a finite-dimensional vector subspace, is dense; and similarly $C(X,\mathbb{C})\otimes F$ is dense in $C(X,F)$. If u has dense image, then $u_X(C(X,E))$ is dense in $C(X,\mathbb{C})\otimes F$, and taking closures gives the assertion.

3.1.3. Let V be a compact variety, let E and F be complex vector bundles of finite rank on V, and let $n\in\mathbb{N}$. The space $\mathrm{Diff}^n_{\mathbb{C}}(E,F)$, identified by (1.2.3.1) with

$$\mathrm{Hom}(P^n(E),F)\simeq C^\infty(V,\mathrm{Hom}(P^n(E),F)),$$

is naturally a Fréchet space. Let $u\in\mathrm{Diff}^n_{\mathbb{C}}(E,F)$. One checks easily that the linear map $C^\infty(V,E)\to C^\infty(V,F)$ defined by u is continuous. If $L(C^\infty(V,E),C^\infty(V,F))$ denotes the space of continuous linear maps, one therefore has a linear map

$$\delta:\mathrm{Diff}^n_{\mathbb{C}}(E,F)\longrightarrow L(C^\infty(V,E),C^\infty(V,F)).$$

Moreover, for $r\in\mathbb{Z}$, u acts by extension of scalars on the C_r sections of E and defines a linear map

$$\delta_r:\mathrm{Diff}^n_{\mathbb{C}}(E,F)\longrightarrow L(C_r(V,E),C_{r-n}(V,F)).$$

The maps δ_r are continuous, and for $u\in\mathrm{Diff}^n_{\mathbb{C}}(E,F)$, $\delta(u)$ is the inverse-limit element of the projective system of the $\delta_r(u)$. These assertions follow easily from [16] IX: one reduces immediately to $n=0$; the statement on the C_r follows directly from the definitions, while the statement for C^∞ follows, after localizing by a partition of unity, from the formula for successive derivatives of a product. The analogous real statements are also valid.

Lemma 3.1.4. Let S be a topological space, let V be a compact variety, let E,F be complex vector bundles of finite rank on V, and let $n\in\mathbb{N}$. Put $X=S\times V$. For

$$u\in\mathrm{Diff}^n_{\mathbb{C}}(\mathrm{pr}_2^*E,\mathrm{pr}_2^*F),$$

the map $s\mapsto i_s^*u$ from S into $\mathrm{Diff}^n_{\mathbb{C}}(E,F)$ is continuous. The linear map

$$\mathrm{Diff}^n_{\mathbb{C}}(\mathrm{pr}_2^*E,\mathrm{pr}_2^*F)\longrightarrow C(S,\mathrm{Diff}^n_{\mathbb{C}}(E,F))$$

so defined is an isomorphism.

Proof. By (1.2.3.1) one may restrict to the case $n=0$. Putting $G=\mathrm{Hom}(E,F)$, one is reduced to showing that for $u\in\Gamma(X,\mathrm{pr}_2^*G)$, the map $s\mapsto u_s=i_s^*u$ from S into $C^\infty(V,G)$ is continuous and that the resulting map

$$\Gamma(X,\mathrm{pr}_2^*G)\longrightarrow C(S,C^\infty(V,G))$$

is an isomorphism. This follows immediately from the definition of mixed varieties.

3.1.5. Under the hypotheses of (3.1.4), let

$$u\in\mathrm{Diff}^n_{\mathbb{C}}(\mathrm{pr}_2^*E,\mathrm{pr}_2^*F).$$

Identifying u, by (3.1.4), with a continuous map from S into $\mathrm{Diff}^n(E,F)$, one sees that, for U open in S, the morphism

$$\mathrm{pr}_{1*}(u)(U):C^\infty(U,E)\longrightarrow C^\infty(U,F)$$

is canonically identified with the linear map

$$C(U,C^\infty(V,E))\longrightarrow C(U,C^\infty(V,F))$$

deduced from

$$\delta u : U \longrightarrow L(C^\infty(V,E), C^\infty(V,F)).$$

Suppose S locally compact. By (3.1.1), the morphism of sheaves

$$\mathrm{pr}_{1*}(u) : \mathrm{pr}_{1*}\,\mathrm{pr}_2^*\,E \longrightarrow \mathrm{pr}_{1*}\,\mathrm{pr}_2^*\,F$$

is the morphism associated to the morphism of presheaves

$$\mathcal{C}'(S, C^\infty(V,E)) \longrightarrow \mathcal{C}'(S, C^\infty(V,F))$$

defined by δu. On the other hand, for $r \in \mathbb{Z}$,

$$\delta_r u : S \longrightarrow L(C_r(V,E), C_{r-n}(V,F))$$

defines a morphism of presheaves

$$u_r : \mathcal{C}'(S, C_r(V,E)) \longrightarrow \mathcal{C}'(S, C_{r-n}(V,F)).$$

By (3.1.3), the u_r form a projective system whose limit is the previous morphism. It will be convenient to interpret $\delta_r u$ as a morphism of trivial Hilbert bundles

$$S \times C_r(V,E) \longrightarrow S \times C_{r-n}(V,F),$$

inducing on sheaves of sections the sheaf morphism associated to u_r.

3.2. Definition of the index

Theorem 3.2.1. Let S be a locally compact space, let $p : X \to S$ be a proper mixed variety, and let $E^\bullet$ be an elliptic complex of order (n_i) of complex vector bundles on X (1.2.5). Then the complex $Rp_*(E^\bullet)$, which by (2.5) is isomorphic to $p_*(E^\bullet)$ in the derived category of $\mathcal{O}_S \otimes \mathbb{C}$-modules, is perfect.

Proof. The question being local on S, one may suppose by (2.9) that $X = S \times V$, where V is a compact variety, and that $p = \mathrm{pr}_1$. Moreover, by (2.10), one may suppose that for all i,

$$E^i = \mathrm{pr}_2^*(F^i),$$

where F^i is a complex vector bundle of finite rank on V. By (3.1.5), $p_*(E^\bullet)$ is associated to the complex of presheaves

$$\mathcal{C}'(S, C^\infty(V, F^\bullet)),$$

where the differentials are induced by the given differential operators. Put, for each $i \in \mathbb{Z}$,

$$m_i = \sum_{j<i} n_j,$$

with the convention $n_i = 0$ if $E^i = 0$. For each $r \in \mathbb{Z}$ one has, again by (3.1.5), a complex of presheaves

$$\mathcal{C}'(S, C_{r-m_i}(V, F^i)),$$

which we denote by $\mathcal{C}'(S, C_r(V, F^\bullet))$. These complexes form a projective system, and

$$\mathcal{C}'(S, C^\infty(V, F^\bullet)) = \lim_r \mathcal{C}'(S, C_r(V, F^\bullet)). \tag{3.2.1.1}$$

Finally, denote by $F_r^\bullet$ the complex of trivial Hilbert bundles over S defined by the corresponding $\delta_r(d^i)$, and by $\mathcal{F}_r^\bullet$ the complex of sheaves of sections; by (3.1.5), it is just the complex of

sheaves associated to $\mathcal{C}'(S, C_r(V, F^\bullet))$. To prove the theorem it is enough to prove the following assertions:

(a) For $r \in \mathbb{Z}$, the canonical morphism

$$\mathcal{C}'(S, C^\infty(V, F^\bullet)) \longrightarrow \mathcal{C}'(S, C_r(V, F^\bullet))$$

is a quasi-isomorphism of complexes of presheaves, that is, it induces isomorphisms on the presheaves of cohomology.

(b) For $r \in \mathbb{Z}$, the complex $\mathcal{F}_r^\bullet$ is perfect.

Proof of (b). The ellipticity of $E^\bullet$ implies ([16] XI, theorem 6) that, for every $s \in S$, the differential of $(F_r^\bullet)_s$ has closed image and the spaces $H^i((F_r^\bullet)_s)$ are finite-dimensional vector spaces. In the terminology of [11], $F_r^\bullet$ is a direct quasi-acyclic complex of Hilbert bundles. Hence, by [12] 2.3, $F_r^\bullet$ is locally homotopically equivalent to a bounded complex of vector bundles of finite rank. Since the sheaf-of-sections functor is additive (3.0.5), $\mathcal{F}_r^\bullet$ is locally homotopically equivalent to a bounded complex of locally free sheaves of finite type; hence it is perfect.

For the proof of (a) we shall need the following intermediate assertion:

(a') For $p \gg q$, the canonical morphism

$$F_p^\bullet \longrightarrow F_q^\bullet$$

is a quasi-isomorphism.

Proof of (a'). Let $M_{pq}^\bullet$ be the complex of Hilbert bundles which is the cone of the morphism $F_p^\bullet \to F_q^\bullet$. The ellipticity of $E^\bullet$ implies that, for every $s \in S$, the morphism $(F_p^\bullet)_s \to (F_q^\bullet)_s$ is a quasi-isomorphism ([16] XI, corollary 2 and theorem 6). In other words, $(M_{pq}^\bullet)_s$ is a direct acyclic complex. By [11] 2.3, $M_{pq}^\bullet$ is locally homotopically trivial; hence its sheaf complex, the cone of $\mathcal{F}_p^\bullet \to \mathcal{F}_q^\bullet$, is acyclic.

Proof of (a). It is enough to show that, for every relatively compact open subset U of S, the morphism

$$C^\infty(U, C^\infty(V, F^\bullet)) \longrightarrow C^\infty(U, C_r(V, F^\bullet))$$

is a quasi-isomorphism. Replacing S by $\overline{U}$ and $E^\bullet$ by the induced complex over $\overline{U}$, one is reduced to proving that

$$C(S, C^\infty(V, F^\bullet)) \longrightarrow C(S, C_r(V, F^\bullet))$$

is a quasi-isomorphism, with S compact. By (3.2.1.1),

$$C(S, C^\infty(V, F^\bullet)) = \lim_i C(S, C_i(V, F^\bullet)).$$

For each i, $C(S, C_i(V, F^j))$ is a Banach space for the topology of uniform convergence, and the transition morphisms

$$C(S, C_i(V, F^j)) \longrightarrow C(S, C_{i-1}(V, F^j))$$

are continuous linear maps with dense image by (3.0.4) and (3.1.2). On the other hand, applying to the quasi-isomorphisms (a') the functor "sections", which is exact since $\mathcal{O}_S$ is soft, one sees that the transition morphisms are quasi-isomorphisms. Assertion (a) follows from EGA 0_{I} 13.2.3, topological variant. This completes the proof of (3.2.1).

3.2.2. If S is a topological space, we denote by $\mathrm{Parf}(S)$ the triangulated category of perfect complexes of $(\mathcal{O}_S \otimes \mathbb{C})$-modules, and by $K^0(S)$ the corresponding Grothendieck group.[19]

[19] See I 6.3. For a general review of K-functors, see Exposé IV.

Let $S, X, E^\bullet$ be as in (3.2.1). The *analytic index* of $E^\bullet$ is the element

$$\mathrm{ind}_{\mathrm{an}}(E^\bullet) = [Rp_*(E^\bullet)] \in K^0(S).$$

If S is compact, $\mathrm{ind}_{\mathrm{an}}(E^\bullet)$ canonically defines an element in the usual group $K(S)$ of topologists. Indeed, in general, for every topological space S, let $\mathrm{Loclib}(S)$ be the additive category of locally free $(\mathcal{O}_S \otimes \mathbb{C})$-modules of finite type, and let $K(S)$ be the Grothendieck group of the triangulated category

$$K^b(\mathrm{Loclib}(S))/K^b_{\mathrm{ac}}(\mathrm{Loclib}(S)).$$

This category is in fact identified with $K^b(\mathrm{Loclib}(S))$ when S is paracompact (II 2.2.2(a)), so in this case, by I 6.4, $K(S)$ is canonically isomorphic to the Grothendieck group of the additive category $\mathrm{Loclib}(S)$. There is a natural homomorphism

$$K(S) \longrightarrow K^0(S).$$

By II 2.3.2 this is an isomorphism for compact S, proving the assertion. More generally, $\mathrm{ind}_{\mathrm{an}}(E^\bullet)$ defines an element in the completed group

$$\widehat{K}(S) = \lim_A K(A),$$

the projective limit being taken over the compact subsets A of S.

3.2.3. The writer does not know whether the finiteness theorem (3.2.1) remains valid if, instead of assuming S locally compact, one merely assumes that each point of S has a fundamental system of paracompact neighborhoods. Moreover, the local compactness hypothesis on S does not seem very natural. The space of "universal local parameters" for elliptic complexes of fixed order is in fact a subspace of infinite dimension in a Fréchet space. More precisely, let V be a compact variety, let $F = (F^i)$ be a sequence of complex vector bundles of finite rank on V, zero except for finitely many indices, and let $n = (n_i)$ be a sequence of integers ≥ 0. Let $\mathrm{Comp.\,diff}_n(F)$ be the space of complexes of differential operators of base F and order n, that is, the closed subspace of

$$\prod_i \mathrm{Diff}^{n_i}(F^i, F^{i+1})$$

defined by the equations $d^{i+1}d^i = 0$. This product is a Fréchet space. The subset $\mathrm{Ell}_n(F)$ formed by the elliptic complexes of order n is open. It is this space, in general not locally compact, which plays the role of a space of universal local parameters.

3.2.4. Nevertheless, even if the generalization of (3.2.1) just mentioned is impossible, one can always associate to $E^\bullet$, under the sole hypothesis that the fibration p is proper and locally trivial, an element of $K^0(S)$ deserving the name analytic index of $E^\bullet$. Indeed, suppose, for simplicity, that we are in the local situation where $X = S \times V$ with V compact, and $E^\bullet$ is given by a continuous map from S into $\mathrm{Ell}_n(F)$ (3.2.3). Assertions (a') and (b) in the proof of (3.2.1) are in fact valid without any hypothesis on S; it is therefore reasonable to put, with the notation of that proof,

$$\mathrm{ind}_{\mathrm{an}}(E^\bullet) = [\mathcal{F}_r^\bullet] \in K^0(S),$$

this element being independent of r.

3.2.5. Suppose S paracompact. Let H be a complex Hilbert space of infinite Hilbert dimension, and let $I(H)$ be the subspace of $L(H, H)$ formed by the Fredholm operators. The complexes $F_r^\bullet$ define [12] a canonical element in the group $[S, I(H)]$ of homotopy classes of continuous maps from S to $I(H)$. But the writer does not know, for non-compact S (even when S is locally compact), what the relations are between $[S, I(H)]$ and $K^0(S)$: is there a natural arrow from

one to the other? are they isomorphic? A fortiori, the relation between the two indices thus defined remains unclear.

For a generalization, under the hypotheses of (3.2.1), of the Atiyah–Singer index theorem, we refer to the recent work of Shih [17], or to work in preparation of M. F. Atiyah and I. M. Singer.[20]

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[20] For the general case, use the notion of "weak Hilbert bundle", or continuous field of Hilbert spaces, of [12].

Exposé III. Relative finiteness conditions

by L. Illusie

1. Relative pseudo-coherence

Theorem 1.1. Let $n \in \mathbb{Z}$. For each locally finite type morphism of schemes

$$f : X \longrightarrow Y$$

there exists one and only one full subcategory $D(f)_{n\text{-coh}}$ of $D(X)$ such that the following conditions are satisfied:

(i) If $E \in \operatorname{ob} D(f)_{n\text{-coh}}$, then $i^*(E) \in \operatorname{ob} D(fi)_{n\text{-coh}}$ for every open immersion $i : U \to X$.

(ii) If $E \in \operatorname{ob} D(X)$, and if there exists an open covering $(i_\alpha : U_\alpha \to X)_{\alpha \in A}$ such that $i_\alpha^*(E) \in \operatorname{ob} D(fi_\alpha)_{n\text{-coh}}$ for every $\alpha \in A$, then $E \in \operatorname{ob} D(f)_{n\text{-coh}}$.

(iii) If

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow^{h} & \downarrow{\scriptstyle g} \\ & & Z \end{array}$$

is a commutative triangle of locally finite type morphisms, where f is a closed immersion and g is smooth, then one has

$$E \in \operatorname{ob} D(h)_{n\text{-coh}} \quad \Longleftrightarrow \quad f_*(E) \in D(Y)_{n\text{-coh}}.$$

Here $D(X)$ denotes the derived category of $\mathcal{O}_X$-modules, and $D(Y)_{n\text{-coh}}$ denotes the full subcategory of $D(Y)$ formed by the n-pseudo-coherent objects.

Proof. Uniqueness is evident. Let indeed $f : X \to Y$ be a locally finite type morphism. Choose an open covering $(i_\alpha : X_\alpha \to X)_{\alpha \in A}$ such that, for every α, fi_α factors as

$$X_\alpha \xrightarrow{i'_\alpha} Y_\alpha \xrightarrow{f'_\alpha} Y,$$

where i'_α is a closed immersion and f'_α is smooth. It follows at once from (i), (ii), (iii) that the objects of $D(f)_{n\text{-coh}}$ are precisely the objects $E \in D(X)$ such that

$$i'_{\alpha *} i_\alpha^*(E) \in \operatorname{ob} D(Y_\alpha)_{n\text{-coh}};$$

therefore $D(f)_{n\text{-coh}}$ is well determined, since it is a full subcategory of $D(X)$. To prove existence we shall need several lemmas.

Lemma 1.1.1. Let $i : X \to Y$ be a closed immersion of schemes such that $i_*(\mathcal{O}_X)$ is pseudo-coherent, and let $E \in \operatorname{ob} D(X)$. In order that E be n-pseudo-coherent, it is necessary and sufficient that $i_*(E)$ be so.

Proof. The necessity follows from the general stability criterion for pseudo-coherence under application of a functor. Conversely, suppose that $i_*(E)$ is n-pseudo-coherent, and prove that the same is true of E. Since $i_*(E)$ has locally bounded-above cohomology, the same is true of E. The question being local, one may suppose that E is m-pseudo-coherent for m sufficiently large. We prove by descending induction on m, with E assumed m-pseudo-coherent for some

$m \geq n$. Suppose already proved that E is $(n+1)$-pseudo-coherent. After localization, there is a distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1]$$

where E' is perfect and $H^i(E'') = 0$ for $i > n$. By Exposé I, Proposition 2.6, it remains to show that E'' is n-pseudo-coherent. Applying i_* to the preceding triangle, we get a distinguished triangle

$$i_*(E') \longrightarrow i_*(E) \longrightarrow i_*(E'') \longrightarrow i_*(E')[1]$$

and, by the first part of the proof, $i_*(E')$ is pseudo-coherent. Hence, again by Exposé I, Proposition 2.6, $i_*(E'')$ is n-pseudo-coherent. Since $i_*(E'')$ is zero in degrees $> n$, this means, by Exposé I, Proposition 2.10 b), that $H^n(i_*(E''))$ is of finite type. But

$$H^n(i_*(E'')) = i_* H^n(E''),$$

so the assertion is reduced to showing, by Exposé I, Proposition 2.10 b), that if M is an $\mathcal{O}_X$-module such that $i_*(M)$ is of finite type, then M is of finite type. This is immediate and independent of the hypothesis on $i_*(\mathcal{O}_X)$.

Remark 1.1.2. The preceding lemma applies in particular to the case where i is a regular immersion. Indeed, by means of the Koszul complex, $i_*(\mathcal{O}_X)$ has locally a finite left resolution by finite-type free $\mathcal{O}_Y$-modules; hence $i_*(\mathcal{O}_X)$ is perfect, and a fortiori pseudo-coherent.

Lemma 1.1.3. Let

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ {\scriptstyle i'}\downarrow & & \downarrow{\scriptstyle i} \\ Y' & \xrightarrow[g]{} & Y \end{array}$$

be a Cartesian square of schemes, where i is a closed immersion, and let E be a complex of $\mathcal{O}_X$-modules. Then the canonical base-change arrow

$$g^* i_*(E) \longrightarrow i'_* f^*(E)$$

is an isomorphism.

If $E' = f^*(E)$, if g is flat and $i_*(E)$ is n-pseudo-coherent, then $i'_*(E')$ is n-pseudo-coherent. Conversely, if g is faithfully flat and $i'_*(E')$ is n-pseudo-coherent, then $i_*(E)$ is n-pseudo-coherent.

Proof. The assertion is verified trivially on the fibers.

Lemma 1.1.4. Let

$$\begin{array}{ccccc} X' & \xleftarrow{i'} & X & \xrightarrow{i''} & X'' \\ & {\scriptstyle f'}\nwarrow & & \nearrow{\scriptstyle f''} & \\ & & Y & & \end{array}$$

be a commutative diagram of schemes, where i' (respectively i'') is a closed immersion and f' (respectively f'') is smooth, and let $E \in \operatorname{ob} D(X)$. Then

$$i'^*(E) \text{ is } n\text{-pseudo-coherent} \quad \Longleftrightarrow \quad i''^*(E) \text{ is } n\text{-pseudo-coherent}.$$

Proof. Following a device due to Lichtenbaum, send X diagonally into $X' \times_Y X''$. One is at once reduced to proving the assertion in the case where one of the morphisms f', f'' is the identity. Changing notation, one has a commutative triangle

$$\begin{array}{ccc} & & Y \\ & {\scriptstyle j}\nearrow & \downarrow{\scriptstyle f} \\ X & \xrightarrow[i]{} & Z \end{array}$$

where i and j are closed immersions and f is smooth, and one must show that $i_*(E)$ is n-pseudo-coherent if and only if $j_*(E)$ is n-pseudo-coherent. The question being local on Y and on Z in a neighborhood of a point of X, one may suppose that X, Y, Z are affine and that there is a factorization

$$\begin{array}{ccc} Y & \xrightarrow{i'} & Z[T_1, \ldots, T_r] \\ {\scriptstyle f}\downarrow & \swarrow & \\ Z & & \end{array}$$

where i' is a closed immersion and the oblique arrow is the canonical projection. Then, by Exposé VII, 1.10, i' is a regular immersion; hence by Lemma 1.1.1 an object F of $D(Y)$ is n-pseudo-coherent if and only if $i'_*(F)$ is so.

Thus one may suppose that $Y = Z[T_1, \ldots, T_r]$, f being the canonical projection. The preceding diagram then corresponds to a diagram of rings

$$A \xleftarrow{v} C \xrightarrow{u} B = C[T_1, \ldots, T_r]$$

where u and v are epimorphisms and $c : C \to B$ is the canonical inclusion. For $1 \leq i \leq r$, put $u(T_i) = x_i$, and let $z_i \in C$ be such that $v(z_i) = x_i$. The retraction

$$\sigma : B \longrightarrow C, \qquad \sigma(T_i) = z_i,$$

is defined by $c(T_i) = z_i$, is such that $v\sigma = u$. Geometrically, it corresponds to a section s of f such that $i = sj$. By Exposé VII, 1.10, s is a regular immersion; the assertion therefore follows from Lemma 1.1.1.

End of the proof of Theorem 1.1. Let $f : X \to Y$ be a locally finite type morphism. There exists an open covering (X_α) of X such that each X_α embeds by a closed immersion into a scheme Y_α smooth over Y. Put $\coprod X_\alpha = X'$, $\coprod Y_\alpha = Y'$. We have a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{i'} & Y' \\ {\scriptstyle p'}\downarrow & \swarrow_{f'} & \\ X & & \\ {\scriptstyle f}\downarrow & & \\ Y & & \end{array}$$

where p' is the morphism which is the sum of a surjective family of open immersions (we shall say *special*, for short), i' is a closed immersion, and f' is smooth. We define $D(f)_{n\text{-coh}}$ as the full subcategory of $D(X)$ formed by the objects E such that $i'_* p'^*(E)$ is n-pseudo-coherent. First we show that this definition makes sense, that is, that $D(f)_{n\text{-coh}}$ is independent of the choice of i', p', f'. Let

$$\begin{array}{ccc} X'' & \xrightarrow{i''} & Y'' \\ {\scriptstyle p''}\downarrow & \swarrow_{f''} & \\ X & & \\ {\scriptstyle f}\downarrow & & \\ Y & & \end{array}$$

be an analogous commutative diagram, with p'' special, i'' a closed immersion, and f'' smooth.

From the two diagrams one deduces a commutative diagram

$$\begin{array}{ccc}
X' \times_X X'' & \longrightarrow & Y' \times_Y Y'' \\
\downarrow & & \downarrow \\
X' & \xrightarrow{i'} & Y' \\
\downarrow & & \downarrow{\scriptstyle f'} \\
X'' & \xrightarrow[i'']{} & Y'' \\
\downarrow & & {\scriptstyle f''}\downarrow \\
X & \xrightarrow[f]{} & Y
\end{array}$$

where the horizontal arrows, except for f, are closed immersions, and the arrows of the left (respectively right) Cartesian square are special morphisms (respectively smooth morphisms). Thus one is reduced to proving the following assertion.

$$\begin{array}{ccc}
X_1 & \xrightarrow{i_1} & Y_1 \\
{\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\
X_2 & \xrightarrow[i_2]{} & Y_2
\end{array}$$

Let this be a commutative square of schemes, where p is special, q is smooth, i_1 and i_2 are closed immersions, and let $E \in \operatorname{ob} D(X_2)$. Then $i_{2*}(E)$ is n-pseudo-coherent if and only if $i_{1*}p^*(E)$ is so.

Proof of the assertion. By definition, p is the sum of a family of opens U_α of X_2 covering X_2. Each U_α is induced by an open V_α of Y_2, and the V_α cover an open neighborhood X_2' of X_2 in Y_2. There is therefore a Cartesian commutative diagram

$$\begin{array}{ccccc}
X_1 & \xrightarrow{i_1'} & X_1' & & \\
{\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p'} & & \\
X_2 & \xrightarrow[i_2']{} & X_2' & \xrightarrow[j]{} & Y_2
\end{array}$$

where i_1' and i_2' are closed immersions, p and p' are special morphisms, and j is an open immersion. The condition that $i_{2*}(E)$ be n-pseudo-coherent is plainly equivalent to the condition that $i'_{2*}(E)$ be n-pseudo-coherent. On the other hand, by Lemma 1.1.3, the condition that $i'_{2*}(E)$ be n-pseudo-coherent is equivalent to the condition that $i'_{1*}p^*(E)$ be n-pseudo-coherent. Applying Lemma 1.1.4 to the factorizations $(jp')i_1'$ and qi_1 of i_2p, and to $p^*(E) \in \operatorname{ob} D(X_1)$, one finally concludes that $i'_{2*}(E)$ is n-pseudo-coherent if and only if $i_{1*}p^*(E)$ is n-pseudo-coherent, as required.

We have thus shown that the category $D(f)_{n\text{-coh}}$ defined above by means of a diagram as above depends in fact only on f. Moreover it is trivial from the definition that it satisfies (i) to (iii). This proves Theorem 1.1.

Definition 1.2. Let $f : X \to Y$ be a locally finite type morphism, let $E \in \operatorname{ob} D(X)$, and let $n \in \mathbb{Z}$. We say that E is n-pseudo-coherent relative to f if $E \in \operatorname{ob} D(f)_{n\text{-coh}}$ (cf. 1.1). We say that E is pseudo-coherent, or $(-\infty)$-pseudo-coherent, relative to f if E is n-pseudo-coherent relative to f for every $n \in \mathbb{Z}$. We write $D(f)_{\text{coh}}$ for the full subcategory of $D(X)$ formed by the objects pseudo-coherent relative to f.

We shall say that f is n-pseudo-coherent (respectively pseudo-coherent) if $\mathcal{O}_X$ is n-pseudo-coherent (respectively pseudo-coherent) relative to f. When this cannot lead to confusion, we

shall sometimes say "n-pseudo-coherent (respectively pseudo-coherent) relative to Y" instead of "relative to f," and we shall also write

$$D(X)_{Y,n\text{-coh}} \quad \text{(respectively } D(X)_{Y,\text{coh}}\text{)}$$

instead of

$$D(f)_{n\text{-coh}} \quad \text{(respectively } D(f)_{\text{coh}}\text{)}.$$

Finally, for $* = +, -, b$, put

$$D^*(f)_{n\text{-coh}} = D^*(X) \cap D(f)_{n\text{-coh}}, \qquad D^*(f)_{\text{coh}} = D^*(X) \cap D(f)_{\text{coh}}.$$

Remarks 1.2.1.

a) The categories $D(f)_{n\text{-coh}}$, $D(f)_{\text{coh}}$ are not in general contained in $D^-(X)$. This is the case, however, if X is quasi-compact, since an object of $D(X)$ with locally bounded-above cohomology then has bounded-above cohomology.

b) The objects of $D(f)_{\text{coh}}$ have quasi-coherent cohomology. Indeed, if $E \in \text{ob}\, D(f)_{\text{coh}}$, the question being local, one may suppose that f is a closed immersion; then $f_*(E)$ is pseudo-coherent, hence has quasi-coherent cohomology, and one uses the fact that an $\mathcal{O}_X$-module M is quasi-coherent if and only if $f_*(M)$ is.

c) When it is a matter of an $\mathcal{O}_X$-module, we shall use the terminology "of finite presentation relative to f" in preference to "(-1)-pseudo-coherent relative to f." For an $\mathcal{O}_X$-module E to be of finite presentation relative to f, it is necessary and sufficient, by Exposé I, Proposition 2.9, that there exist locally a closed immersion $i : X \to X'$, with X' smooth over Y, such that $i_*(E)$ is of finite presentation in the ordinary sense. In particular, to say that f is (-1)-pseudo-coherent is equivalent to saying that f is locally of finite presentation in the sense of EGA IV 1.4.

Examples 1.3. A smooth morphism and a regular immersion are pseudo-coherent morphisms. The same is true of an arbitrary locally finite type morphism $f : X \to Y$ if Y is locally noetherian.

Proposition 1.4. Let $f : X \to Y$ be a locally finite type morphism and let

$$E \longrightarrow F \longrightarrow G \longrightarrow E[1]$$

be a distinguished triangle of $D(X)$. The following table, where $n \in \mathbb{Z} \cup \{-\infty\}$ indicates a degree of pseudo-coherence relative to f, has the property that each line represents an implication whose conclusion is underlined:

E	F	G
$n+1$	n	$\underline{n}$
n	$\underline{n}$	n
$\underline{n}$	n	$n-1$.

In particular, $D(f)_{\text{coh}}$ is a triangulated subcategory of $D(X)$.

Proof. By the properties (i) to (iii) of Theorem 1.1, one may suppose that f is a closed immersion. We then have a distinguished triangle

$$f_*(E) \longrightarrow f_*(F) \longrightarrow f_*(G) \longrightarrow f_*(E)[1]$$

and the assertion follows from Exposé I, Proposition 2.6.

Definition 1.5. Let X and Y be two S-schemes. We shall say that X and Y are *Tor-independent over S* if

$$\text{Tor}_i^S(\mathcal{O}_X, \mathcal{O}_Y) = 0 \quad \text{for } i \neq 0,$$

where the Tor_i^S are the local hyper-Tor sheaves defined in EGA III 6.5. This is a condition of local nature. If X, Y, S are affine, with respective rings B, C, A, it means that the canonical map

$$B \otimes_A^{\mathbf{L}} C \longrightarrow B \otimes_A C$$

is an isomorphism. It is therefore satisfied in particular if X or Y is flat over S.

Lemma 1.5.1. Let

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \uparrow & & \uparrow & & \uparrow \\ Y & \longleftarrow & Y' & \longleftarrow & Y'' \end{array}$$

be Cartesian squares of schemes. If X and Y' are Tor-independent over Y, and if X' and Y'' are Tor-independent over Y', then X and Y'' are Tor-independent over Y.

Proof. This is an easy exercise on associativity of the tensor product.

Notation 1.6. Let X and Y be two S-schemes, $E \in \mathrm{ob}\, D^-(X)$, $F \in \mathrm{ob}\, D^-(Y)$. We put

$$E \otimes_S^{\mathbf{L}} F = L\,\mathrm{pr}_1^*(E) \otimes^{\mathbf{L}} L\,\mathrm{pr}_2^*(F),$$

where $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections from $X \times_S Y$.

Lemma 1.7. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be a closed S-immersion. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Then, for $E_i \in \mathrm{ob}\, D^-(X_i)_{\mathrm{qcoh}}$, there is a canonical isomorphism

$$f_{1*}(E_1) \otimes_S^{\mathbf{L}} f_{2*}(E_2) \xrightarrow{\sim} (f_1 \times_S f_2)_*(E_1 \otimes_S^{\mathbf{L}} E_2).$$

Proof. Put $f_1 \times_S f_2 = f$. One easily defines, by adjunction, a canonical arrow

$$f_{1*}(E_1) \otimes_S^{\mathbf{L}} f_{2*}(E_2) \longrightarrow f_*(E_1 \otimes_S^{\mathbf{L}} E_2).$$

We show that it is an isomorphism. The question being local, one may suppose S, X_i, Y_i affine with respective rings C, A_i, B_i. By dévissage, one is reduced to $E_i = \mathcal{O}_{X_i}$. One must then verify that

$$A_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} A_2 \longrightarrow A$$

is an isomorphism, where $A = A_1 \otimes_C A_2$, $B = B_1 \otimes_C B_2$. In view of the Tor-independence hypotheses, this follows at once from associativity of the tensor product.

Proposition 1.8. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be a locally finite type S-morphism. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Let $[a_i, b_i]$ be intervals of $\mathbb{Z}$, and let

$$E_i \in \mathrm{ob}\, D^-(X_i)_{\mathrm{qcoh}}$$

be a complex which is a_i-pseudo-coherent relative to f_i and acyclic in degree $> b_i$. Then

$$E_1 \otimes_S^{\mathbf{L}} E_2$$

is a-pseudo-coherent relative to $f = f_1 \times_S f_2$ and acyclic in degree $> b$, where

$$a = \sup(a_1 + b_2, a_2 + b_1), \qquad b = b_1 + b_2.$$

Proof. The question being local on $X_1 \times_S X_2$, one may suppose that f_i factors as

$$X_i \xrightarrow{i_i} Y_i' \xrightarrow{p_i} Y_i,$$

where i_i is a closed immersion and p_i a smooth morphism. Put $f' = f_1' \times_S f_2'$. We must show that $f'_*(E_1 \otimes_S^{\mathbf{L}} E_2)$ is a-pseudo-coherent and acyclic in degree $> b$. By Lemma 1.5.1, Y_1' and Y_2' are Tor-independent over S. After changing notation, one may suppose that $f_i = f_i'$, in other words that the f_i are closed immersions. By hypothesis, $f_{i*}(E_i)$ is a_i-pseudo-coherent and acyclic in degree $> b_i$; hence by Exposé I, Proposition 2.16,

$$f_{1*}(E_1) \otimes_S^{\mathbf{L}} f_{2*}(E_2)$$

is a-pseudo-coherent and acyclic in degree $> b$. One concludes by Lemma 1.7.

Corollary 1.9. Under the hypotheses of Proposition 1.8, the functor

$$\otimes_S^{\mathbf{L}} : D^-(X_1) \times D^-(X_2) \longrightarrow D^-(X)$$

induces a functor

$$\otimes_S^{\mathbf{L}} : D^-(f_1)_{\mathrm{coh}} \times D^-(f_2)_{\mathrm{coh}} \longrightarrow D^-(f)_{\mathrm{coh}}.$$

In particular, if f_1 and f_2 are pseudo-coherent, then so is f.

Corollary 1.10. Let

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[h]{} & Y \end{array}$$

be a Cartesian square, with f locally of finite type. Then:

a) Suppose that X and Y' are Tor-independent over Y. For $n \in \mathbb{Z} \cup \{-\infty\}$, the functor $Lg^* : D^-(X) \to D^-(X')$ induces a functor

$$Lg^* : D^-(f)_{n\text{-}\mathrm{coh}} \longrightarrow D^-(f')_{n\text{-}\mathrm{coh}}.$$

In particular, if f is n-pseudo-coherent, then so is f'.

b) Suppose that h is faithfully flat. Let $n \in \mathbb{Z} \cup \{-\infty\}$, and let $E \in \mathrm{ob}\, D^-(X)$ be a complex with quasi-coherent cohomology. In order that E be n-pseudo-coherent relative to f, it is necessary and sufficient that $g^*(E)$ be n-pseudo-coherent relative to f'. In particular, in order that f be n-pseudo-coherent, it is necessary and sufficient that f' be so.

Proof. Taking Remark 1.2.1 b) into account, all assertions of Corollaries 1.9 and 1.10 follow trivially from Proposition 1.8, except the necessity in 1.10 b). For this one may, the question being local on X, suppose that f factors as a closed immersion followed by a smooth morphism. One is then reduced by Lemma 1.1.3 to the case where f (respectively f') is the identity of X (respectively X'). The assertion then follows, by a standard dévissage, from the fact that a quasi-coherent sheaf L on X is of finite type if and only if g^*L is of finite type.

Proposition 1.11. Let

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & {\scriptstyle h}\searrow & \downarrow{\scriptstyle g} \\ & & Z \end{array}$$

be a commutative triangle of locally finite type morphisms. Let $E \in \mathrm{ob}\, D(X)$, $F \in \mathrm{ob}\, D(Y)$, and let $[a, b]$, $[c, d]$ be intervals of $\mathbb{Z}$.

a) If E is a-pseudo-coherent relative to f and acyclic in degree $> b$, and if F is c-pseudo-coherent relative to g and acyclic in degree $> d$, then

$$E \otimes^{\mathbf{L}}_{\mathcal{O}_Y} F$$

is m-pseudo-coherent relative to h and acyclic in degree $> n$, where

$$m = \sup(a+d, b+c), \qquad n = b+d.$$

b) If E is a-pseudo-coherent relative to h and acyclic in degree $> b$, and if g is $(a+1-b)$-pseudo-coherent, then E is a-pseudo-coherent relative to f.

Proof. The question being local on X, one may suppose that there is a Cartesian commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{i} & E^r_Y & \xrightarrow{j} & E^{r+s}_Z \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ Y & \xrightarrow{k} & E^s_Z & \longrightarrow & Z \end{array}$$

where i, j, k are closed immersions and p, q are the canonical projections of affine spaces. If F is c-pseudo-coherent relative to Z, then $k_*(F)$ is c-pseudo-coherent by Theorem 1.1 (iii), and $j_*p^*(F)$ is c-pseudo-coherent by Lemma 1.1.3. Similarly, if g is x-pseudo-coherent, then $j_*(\mathcal{O}_{E^r_Y})$ is x-pseudo-coherent. By the definition of relative pseudo-coherence, one is thus reduced to proving a) and b) when f and g are closed immersions.

Proof of a). By Theorem 1.1 (iii), one has to show the following. Suppose $f_*(E)$ is a-pseudo-coherent and acyclic in degree $> b$, and $g_*(F)$ is c-pseudo-coherent and acyclic in degree $> d$. Then

$$h_*(E \otimes^{\mathbf{L}}_Y F)$$

is m-pseudo-coherent and acyclic in degree $> n$. There is a canonical isomorphism

$$f_*(E) \otimes^{\mathbf{L}}_Y F \xrightarrow{\sim} f_*(E \otimes^{\mathbf{L}}_Y F),$$

a special case of the isomorphism used in the proof of Proposition 1.8. Since the functor $g_*(- \otimes^{\mathbf{L}} F)$ takes finite-type locally free sheaves to c-pseudo-coherent complexes, and complexes acyclic in degree > 0 to complexes acyclic in degree $> d$, the assertion follows from Exposé I, Proposition 2.13.

Proof of b). One must prove the following assertion: suppose $f_*(E)$ is a-pseudo-coherent and acyclic in degree $> b$, and $g_*(\mathcal{O}_Y)$ is $(a+1-b)$-pseudo-coherent. Then $f_*(E)$ is a-pseudo-coherent. Since the assertion concerns $f_*(E)$, one may suppose $X = Y$ and $f = \mathrm{Id}_X$. One then proceeds as in the proof of Lemma 1.1.1, proving by descending induction that E is pseudo-coherent in degree a. Suppose already proved that E is $(a+1)$-pseudo-coherent. By Exposé I, Proposition 2.10, after localizing there is a distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1]$$

where E'^i is locally free for $i \in [a+1, b]$, $E'^i = 0$ for $i \notin [a+1, b]$, and $E''^i = 0$ for $i > a$. The hypothesis that $g_*(\mathcal{O}_Y)$ is $(a+1-b)$-pseudo-coherent implies, by Exposé I, Proposition 2.11 a), that $g_*(E')$ is $(a+1)$-pseudo-coherent. Since $g_*(E)$ is a-pseudo-coherent, it follows from Exposé I, Proposition 2.6, that $g_*(E'')$ is a-pseudo-coherent. We conclude by Exposé I, Proposition 2.10 b), and by the fact that an $\mathcal{O}_Y$-module M is of finite type if and only if $g_*(M)$ is. Corollaries 1.12–1.14 follow immediately.

Corollary 1.12. In the situation of Proposition 1.11, if g is pseudo-coherent, then, for $r \in \mathbb{Z} \cup \{-\infty\}$,

$$D(f)_{r\text{-coh}} = D(h)_{r\text{-coh}}.$$

In particular, taking $X = Y$, the notions of r-pseudo-coherence over Y, in the absolute sense or relative to g, coincide.

Corollary 1.13. In the situation of Proposition 1.11, if f is pseudo-coherent, the functor

$$\mathcal{O}_X \otimes^{\mathbf{L}}_Y - = Lf^* : D^-(Y) \longrightarrow D^-(X)$$

induces, for $r \in \mathbb{Z} \cup \{-\infty\}$, a functor

$$Lf^* : D^-(g)_{r\text{-coh}} \longrightarrow D^-(h)_{r\text{-coh}}.$$

Corollary 1.14. In the situation of Proposition 1.11, if f is r-pseudo-coherent and g is s-pseudo-coherent, with $r, s \in \mathbb{Z} \cup \{-\infty\}$, then h is $\sup(r, s)$-pseudo-coherent. If h is r-pseudo-coherent and g is $(r+1)$-pseudo-coherent, then f is r-pseudo-coherent.

Remark 1.14.1. In particular, for $r = s = -1$, one recovers the assertions (ii) and (v) of EGA IV 1.4.3.

2. The finiteness theorem

Conjecture 2.1. Let S be a scheme, let $f : X \to Y$ be a proper S-morphism of schemes locally of finite type over S, and let $E \in \operatorname{ob} D^+(X)$. If E is pseudo-coherent relative to S, then the same is true of $\mathbf{R}f_*(E)$.

Theorem 2.2. The preceding conjecture is true in the following cases:

(i) S is locally noetherian; (ii) f is projective.

Proof in case (i). Since S is locally noetherian, the projection of X (respectively Y) onto S is a pseudo-coherent morphism, and the notions of pseudo-coherence over X (respectively Y) and relative to S coincide (Corollary 1.12). On the other hand, since X (respectively Y) is locally noetherian, hence $\mathcal{O}_X$ (respectively $\mathcal{O}_Y$) coherent, an object M of $D(X)$ (respectively $D(Y)$) is pseudo-coherent if and only if its $H^i(M)$ are coherent and M is locally in D^- (Exposé I, 3.5). The theorem therefore follows from:

Theorem 2.2.1. Let $f : X \to Y$ be a proper morphism of locally noetherian schemes, and let $E \in \operatorname{ob} D^+(X)$. If the $H^i(E)$ are coherent, then the same is true of $R^i f_*(E)$. If moreover E is locally in D^-, then the same is true of $\mathbf{R}f_*(E)$.

Proof. The first assertion follows from the spectral sequence

$$R^p f_*(H^q(E)) \Longrightarrow R^{p+q} f_*(E)$$

and from the finiteness theorem EGA III 3.2.1. For the second, one may suppose Y affine. Then X is quasi-compact and separated; suppose it covered by r affine opens. One knows (EGA III 1.4.12) that under these conditions, for a quasi-coherent $\mathcal{O}_X$-module M, one has $R^i f_*(M) = 0$ for every $i > r$. If E is locally in D^-, then $E \in D^-$, since X is quasi-compact; say $H^i(E) = 0$ for $i > N$. The preceding spectral sequence then gives $R^i f_*(E) = 0$ for $i > N + r$, which proves the assertion.

Proof in case (ii). The question being local on Y, one may suppose S and Y affine. There is then a Cartesian commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{i} & \mathbb{P}^r_Y & \xrightarrow{j} & \mathbb{P}^r_{Y'} \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p'} \\ Y & \longrightarrow & Y' & \xrightarrow[g]{} & S \end{array}$$

where i, j, j' are closed immersions, g' is smooth and affine, and p, p' are projections of projective spaces. Suppose that E is pseudo-coherent relative to S. By Theorem 1.1 (iii), this means that $(ji)_*(E)$ is pseudo-coherent. We must show that $\mathbf{R}f_*(E)$ is pseudo-coherent relative to S, that is, by Theorem 1.1 (iii), that

$$j'_*\mathbf{R}f_*(E)$$

is pseudo-coherent. But

$$j'_*\mathbf{R}f_*(E) \simeq \mathbf{R}p'_*((ji)_*(E)),$$

so the assertion is a consequence of the following more precise result.

Theorem 2.2.2. Let Y be an affine scheme, let

$$f : X = \mathbb{P}^r_Y \longrightarrow Y$$

be the projection of a projective space of finite type, and let $E \in \operatorname{ob} D^b(X)_{\mathrm{coh}}$. Then $\mathbf{R}f_*(E)$ is pseudo-coherent. Moreover, if E is acyclic in degree $> a$, then $\mathbf{R}f_*(E)$ is acyclic in degree $> a + r$, and there exists an integer N such that, for all $n \geq N$,

$$\mathbf{R}f_*(E(n)), \qquad E(n) = E \otimes \mathcal{O}_X(n),$$

is acyclic in degree $> a$.

Proof. Since Y is affine, X is quasi-compact and separated. By Exposé II, 2.2.9 a), E is isomorphic, in $D(X)$, to a complex F bounded above such that, for every i, $F^i \in \operatorname{ob} L(X)$, that is, F^i is a finite direct sum of sheaves of the type $\mathcal{O}_X(-n)$, $n \geq 0$. If E is acyclic in degree $> a$, F may be chosen so that $F^i = 0$ for $i > a$; this follows formally from the preceding assertion, taking account of Exposé I, 2.10 a), and Exposé II, 2.2.8 a). Replacing E by F, one may suppose that $E^i \in \operatorname{ob} L(X)$ for every i, and that $E^i = 0$ for $i > a$. Since X is covered by $r + 1$ affine opens, EGA III 1.4.10 and the hypercohomology spectral sequence give

$$R^i f_*(E) = 0 \qquad \text{for } i > a + r.$$

Let $t \leq a$ be an integer and put

$$E' = (0 \to E^t \to \cdots \to E^a \to 0).$$

There is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

where E'' satisfies the same conditions as E, but with a replaced by $t - 1$. Applying $\mathbf{R}f_*$, one obtains a distinguished triangle

$$\mathbf{R}f_*(E') \longrightarrow \mathbf{R}f_*(E) \longrightarrow \mathbf{R}f_*(E'') \longrightarrow \mathbf{R}f_*(E')[1]$$

Since t can be chosen arbitrarily negative, it is enough, for the first assertion, to prove that $\mathbf{R}f_*(E')$ is $(t + r)$-pseudo-coherent. But $\mathbf{R}f_*(E'')$ is acyclic in degree $\geq t + r$, hence $(t + r)$-pseudo-coherent. Thus one wins by Exposé I, Proposition 2.6, if one proves that $\mathbf{R}f_*(E')$ is perfect.

For this one is reduced by dévissage to the case where E' is concentrated in degree 0 and equal to $\mathcal{O}_X(-n)$, $n > 0$. Then, by EGA III 2.1.16,

$$\mathbf{R}f_*(\mathcal{O}_X(-n)) \simeq R^r f_*(\mathcal{O}_X(-n))[-r].$$

Therefore $\mathbf{R}f_*(\mathcal{O}_X(-n))$ is perfect, and this ends the proof of the first assertion of Theorem 2.2.2. We have already seen that $\mathbf{R}f_*(E)$ is acyclic in degree $> a + r$. It remains to prove the last assertion. For $t < -r + a$, $\mathbf{R}f_*(E'')$ is acyclic in degree $> a$. Taking account of the preceding triangle, one is reduced to doing the proof for E' instead of for E. Since E' is bounded above and has components in $L(X)$, there is an integer N such that, for all $n \geq N$, the components of $E'(n)$ are finite direct sums of sheaves of type $\mathcal{O}_X(p)$ with $p \geq 0$. Then, by EGA III 2.1.16 and the hypercohomology spectral sequence, $\mathbf{R}f_*(E'(n))$ is acyclic in degree $> a$ for all $n \geq N$, which completes the proof of Theorem 2.2.2 and hence of Theorem 2.2.

We also have the following classical complement to 2.2 (ii), which will be given for later reference.

Corollary 2.3 (EGA III 2.2.1). Let S be a quasi-compact scheme, $f : X \to Y$ a projective S-morphism of finite type schemes over S, and let

$$L = i^*\mathcal{O}_P(1)$$

be the invertible sheaf defined by an extension $i : X \to P = P(F)$, where F is a quasi-coherent $\mathcal{O}_Y$-module of finite type. For $M \in \operatorname{ob} D(X)$ and $n \in \mathbb{Z}$, put

$$M(n) = M \otimes L^{\otimes n}.$$

If $E \in \operatorname{ob} D^+(X)$ is pseudo-coherent relative to S and acyclic in degree $> a$, then there exists an integer N such that, for all $n \geq N$, $\mathbf{R}f_*(E(n))$ has the same properties.

Proof. One reduces trivially, as in the proof of Theorem 2.2 (ii), to the situation of Theorem 2.2.2: $Y = S$, $f : X = \mathbb{P}^r_Y \to Y$ the canonical projection, $L = \mathcal{O}_X(1)$ the canonical sheaf, $E \in \operatorname{ob} D^b(X)$ a pseudo-coherent complex acyclic in degree $> a$. The conclusion then follows from Theorem 2.2.2.

Corollary 2.4. Let S be a quasi-compact scheme and let $f : X \to Y$ be a proper S-morphism of schemes of finite type over S. If S is noetherian, or if f is projective, the functor

$$\mathbf{R}f_* : D^+(X) \longrightarrow D^+(Y)$$

induces a functor

$$\mathbf{R}f_* : D^b(X)_{S\text{-coh}} \longrightarrow D^b(Y)_{S\text{-coh}}.$$

Proof. Consequence of Theorem 2.2 and of the fact that there is then an integer N such that, for every quasi-coherent $\mathcal{O}_X$-module E, $R^i f_*(E) = 0$ for all $i > N$.

Corollary 2.5. Let $f : X \to Y$ be a proper and pseudo-coherent morphism, and let $E \in \operatorname{ob} D^+(X)$. Suppose Y locally noetherian or f projective. If E is pseudo-coherent in the absolute sense, then the same is true of $\mathbf{R}f_*(E)$. If Y is quasi-compact, the functor $\mathbf{R}f_*$ induces a functor

$$\mathbf{R}f_* : D^b(X)_{\text{coh}} \longrightarrow D^b(Y)_{\text{coh}}.$$

Proof. Consequence of Theorem 2.2, Corollary 2.4, and Corollary 1.12.

Scholie 2.6. To prove Conjecture 2.1 in general, new techniques will surely have to be introduced. One may hope that they will one day make it possible to prove the analogous conjecture in analytic geometry, for a proper morphism of analytic spaces over a suitably topologized locally ringed topos in $\mathbb{C}$-algebras. For the present this conjecture has been proved only over a complex analytic space, by Grauert's theorem.[21]

[21] Added in April 1971: see a forthcoming article of R. Kiehl and J.-L. Verdier, where the finiteness theorem is proved by different, simpler techniques and under more general hypotheses.

3. Relative finite Tor-dimension

Definition 3.1. Let

$$f : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

be a morphism of ringed topoi, let $E \in \mathrm{ob}\, D^-(X)$, and let $[m, n]$ be an interval of $\mathbb{Z}$. We say that E is of flat amplitude relative to f (or to Y, if there is no possible confusion) contained in $[m, n]$, and write

$$\mathrm{tor.amp}_f(E) \subset [m, n] \qquad (\text{or } \mathrm{tor.amp}_Y(E) \subset [m, n]),$$

if E, considered as a complex of $f^{-1}(\mathcal{O}_Y)$-modules, is of flat amplitude contained in $[m, n]$ in the sense of Exposé I, 5.2. We say that f is of flat amplitude contained in $[-a, 0]$, or of Tor-dimension $\leq a$, if $\mathcal{O}_X$ is of flat amplitude relative to f contained in $[-a, 0]$. We say that E is of finite flat amplitude relative to f if there exists an interval $[m, n]$ such that $\mathrm{tor.amp}_f(E) \subset [m, n]$.

Notation 3.2. We write

$$D(f)_{\mathrm{torf}}, \qquad D(X)_{f\text{-}\mathrm{torf}}, \qquad D(X)_{Y\text{-}\mathrm{torf}}$$

if no confusion is possible, for the full subcategory of $D^-(X)$ formed by complexes of finite flat amplitude relative to f. By Exposé I, 5.3, this is a triangulated subcategory of $D^b(X)$.

Proposition 3.3. With the notation of Definition 3.1, suppose that X has enough points (SGA 4 XVII IV 6.4). The following conditions are then equivalent:

(i) $\mathrm{tor.amp}_f(E) \subset [m, n]$.

(ii) For every point x of X, one has

$$\mathrm{tor.amp}_y(E_x) \subset [m, n],$$

where $y = f(x)$, ringed by $\mathcal{O}_{Y,y}$.

(iii) For every object V of Y and every $\mathcal{O}_V$-module M, one has

$$H^i\left(E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} M\right) = 0 \quad \text{for } i \notin [m, n].$$

Equivalently, for every $\mathcal{O}_V$-module M,

$$E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} M \simeq E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} L!f^*(M).$$

(iv) The same condition as (iii), but with M the cokernel of a homomorphism

$$\mathcal{O}_V^q \longrightarrow \mathcal{O}_V^p, \qquad p, q \in \mathbb{N}.$$

Proof. The equivalence (i)$\Longleftrightarrow$(ii) follows from Exposé I, 5.6.2, and the implications (ii)$\Rightarrow$(iii)$\Rightarrow$(iv) are trivial. We prove (iv)$\Rightarrow$(ii). Let x be a point of X, $y = f(x)$. One has to show that

$$H^i(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} M) = 0$$

for every $\mathcal{O}_{Y,y}$-module M and every $i \notin [m, n]$. Since every $\mathcal{O}_{Y,y}$-module is an inductive filtered limit of modules of finite presentation, one may restrict to M of finite presentation. If M is a finitely presented $\mathcal{O}_{Y,y}$-module, the same argument as for ordinary ringed spaces (EGA I, 5.3.9) shows that there exists a y-pointed object V of Y and an $\mathcal{O}_V$-module P, cokernel of a homomorphism

$$\mathcal{O}_V^q \longrightarrow \mathcal{O}_V^p, \qquad p, q \in \mathbb{N},$$

such that $P_y \simeq M$. One then has

$$\left(E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} P\right)_x \simeq E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} M,$$

by the commutation of tensor product with passage to fibers, and hypothesis (iv) implies the desired vanishing.

Remark 3.3.1. The editor does not know whether the equivalences (i)$\Longleftrightarrow$(iii)$\Longleftrightarrow$(iv) are true without the hypothesis that X has enough points.

Corollary 3.4. Let

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow^{h} & \downarrow{\scriptstyle g} \\ & & Z \end{array}$$

be a commutative triangle of ringed topoi, where X is assumed to have enough points. Let $E \in \operatorname{ob} D(X)$, $F \in \operatorname{ob} D(Y)$, and let $[a,b]$, $[c,d]$ be intervals of $\mathbb{Z}$. Then

$$\operatorname{tor.amp}_f(E) \subset [a,b] \quad \text{and} \quad \operatorname{tor.amp}_g(F) \subset [c,d]$$

imply

$$\operatorname{tor.amp}_h(E \otimes^{\mathbf{L}}_Y F) \subset [a+c, b+d].$$

Proof. By Proposition 3.3, one must show that the relation on the left implies

$$H^i\left(((E \otimes^{\mathbf{L}}_Y F)|_{h^{-1}(Z')}) \otimes^{\mathbf{L}}_{\mathcal{O}_{Z'}} M\right) = 0$$

for every $i \notin [a+c, b+d]$ and every $\mathcal{O}_{Z'}$-module M, where Z' is an object of Z. Put $Y' = g^{-1}(Z')$, $X' = f^{-1}(Y') = h^{-1}(Z')$. Then

$$((E \otimes^{\mathbf{L}}_Y F)|_{X'}) \otimes^{\mathbf{L}}_{Z'} M \simeq (E|_{X'}) \otimes^{\mathbf{L}}_{Y'} ((F|_{Y'}) \otimes^{\mathbf{L}}_{Z'} M),$$

and the assertion follows at once from the hypotheses.

In particular:

Corollary 3.4.1. Under the hypotheses of Corollary 3.4:

a) If g is of Tor-dimension $\leq n$, then

$$\operatorname{tor.amp}_f(E) \subset [a,b] \Longrightarrow \operatorname{tor.amp}_h(E) \subset [a-n, b].$$

b) If f is of Tor-dimension $\leq m$, then

$$\operatorname{tor.amp}_g(F) \subset [c,d] \Longrightarrow \operatorname{tor.amp}_h(Lf^*(F)) \subset [c-m, d].$$

Remark 3.4.2. Let $f : X \to Y$ be a morphism of finite Tor-dimension and $E \in \operatorname{ob} D(X)$. By Corollary 3.4.1 a), if E is of finite flat amplitude in the absolute sense, then E is of finite flat amplitude relative to f. But the converse is false, even if f is flat, as is shown by the example of a singularity of a scheme over a field.

Proposition 3.5. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be an S-morphism, with $E_i \in \operatorname{ob} D^-(X_i)$, and let $[a_i, b_i]$ be an interval of $\mathbb{Z}$. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Put

$$a = a_1 + a_2, \qquad b = b_1 + b_2, \qquad f = f_1 \times_S f_2 : X \to Y.$$

Then:

a) If $\mathrm{tor.amp}_{f_1}(E_1) \subset [a_1, b_1]$, and if E_2 is acyclic outside $[a_2, b_2]$, then

$$E_1 \otimes_S^{\mathbf{L}} E_2$$

is acyclic outside $[a, b]$.

b) If $\mathrm{tor.amp}_{f_i}(E_i) \subset [a_i, b_i]$ for $i = 1, 2$, then

$$\mathrm{tor.amp}_f(E_1 \otimes_S^{\mathbf{L}} E_2) \subset [a, b].$$

Proof. By Proposition 3.3, it is enough to verify the assertion on the fibers. Let s be a point of S, x a point of X above s, $y = f(x)$, and let x_i (respectively y_i) be the canonical projection of x (respectively y) onto X_i (respectively Y_i). Put

$$(E_i)_{x_i} = M_i, \quad \mathcal{O}_{X_i,x_i} = A_i, \quad \mathcal{O}_{Y_i,y_i} = B_i, \quad \mathcal{O}_{S,s} = C,$$

and

$$\mathcal{O}_{X,x} = A = A_1 \otimes_C A_2, \qquad \mathcal{O}_{Y,y} = B = B_1 \otimes_C B_2.$$

Then

$$(E_1 \otimes_S^{\mathbf{L}} E_2)_x \simeq M_1 \otimes_{A_1}^{\mathbf{L}} A \otimes_{A_2}^{\mathbf{L}} M_2.$$

The Tor-independence hypotheses imply on the other hand that

$$A \simeq A_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} A_2.$$

Consequently, the second member of the displayed isomorphism is canonically isomorphic, in the derived category $D(B)$ of B-modules, to

$$M_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} M_2.$$

The relations $\mathrm{tor.amp}_{f_i}(E_i) \subset [a_i, b_i]$, $i = 1, 2$, imply

$$\mathrm{tor.amp}_{B_i}(M_i) \subset [a_i, b_i],$$

hence, by Exposé I, 5.5,

$$\mathrm{tor.amp}_B(M_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} M_2) \subset [a, b],$$

which proves b). If, on the other hand, $\mathrm{tor.amp}_{f_1}(E_1) \subset [a_1, b_1]$, M_1 is isomorphic to a complex of flat B_1-modules, zero outside $[a_1, b_1]$, and if E_2, hence M_2, is acyclic outside $[a_2, b_2]$, then

$$M_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} M_2$$

is acyclic outside $[a, b]$. This proves a) and completes the proof.

Corollary 3.5.1. Under the hypotheses of Proposition 3.5, if f_1 and f_2 are of finite Tor-dimension, then so is f.

Corollary 3.5.2. Let

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[h]{} & Y \end{array}$$

be a Cartesian square of schemes, where X and Y' are assumed Tor-independent over Y, and let $E \in \mathrm{ob}\, D(X)$. Then one has

$$\mathrm{tor.amp}_f(E) \subset [a, b] \Longrightarrow \mathrm{tor.amp}_{f'}(Lg^*(E)) \subset [a, b],$$

and consequently the functor $Lg^* : D^-(X) \to D^-(X')$ induces a functor

$$Lg^* : D(f)_{\mathrm{torf}} \longrightarrow D(f')_{\mathrm{torf}}.$$

Proposition 3.6. Let

$$\begin{array}{ccc} X & \xrightarrow{\ i\ } & X' \\ {\scriptstyle f}\downarrow & \swarrow_{f'} & \\ Y & & \end{array}$$

be a commutative triangle of schemes, where i is a closed immersion and f' is smooth, and let $E \in \mathrm{ob}\, D(X)$. For E to be locally of finite flat amplitude relative to f, it is necessary and sufficient that $i_*(E)$ be locally of finite flat amplitude in the absolute sense. More precisely, if f' is of relative dimension $\leq d$, then

$$\text{(i)} \quad \mathrm{tor.amp}(i_*E) \subset [a,b] \Longrightarrow \mathrm{tor.amp}_f(E) \subset [a,b],$$
$$\text{(ii)} \quad \mathrm{tor.amp}_f(E) \subset [a,b] \Longrightarrow \mathrm{tor.amp}(i_*E) \subset [a-d,b].$$

Proof. If $[m,n]$ is an interval in $\mathbb{Z}$, one has evidently

$$\mathrm{tor.amp}(E) \subset [m,n] \Longleftrightarrow \mathrm{tor.amp}(i_*E) \subset [m,n]$$

in the reduction in which one tests by modules after the closed immersion. Thus it remains to treat the case $X = X'$, $i = \mathrm{Id}_X$, $f = f'$. Assertion (i) is immediate from (3.4.1 a). For (ii), consider the diagonal section

$$s : X \longrightarrow X \times_Y X$$

of the two projections. If f is smooth of relative dimension d, then, by VII 1.10, s is a regular immersion of codimension $\leq d$, hence of Tor-dimension $\leq d$ (as is seen from the Koszul complex). By (3.5.2),

$$\mathrm{tor.amp}_f(E) \subset [a,b] \Longleftrightarrow \mathrm{tor.amp}_{\mathrm{pr}_2}(\mathrm{pr}_1^* E) \subset [a,b].$$

By (3.4.1 b) it follows that

$$\mathrm{tor.amp}(s^* \mathrm{pr}_1^* E) \subset [a-d,b],$$

and the implication follows because $s^* \mathrm{pr}_1^* \simeq \mathrm{Id}$.

Remark 3.6.1. The statement (3.6) would be false if one supposed only that f' were flat; see (3.4.2).

Proposition 3.7 (projection formula). Let S be a quasi-compact scheme, let $f : X \to Y$ be a quasi-compact and quasi-separated S-morphism of quasi-compact schemes over S, let $E \in \mathrm{ob}\, D^-(X)_{\mathrm{qcoh}}$, and let $M \in \mathrm{ob}\, D^b(S)_{\mathrm{qcoh}}$. If M has finite flat amplitude, or if E has finite flat amplitude relative to S, there is a functorial canonical isomorphism in $D(Y)_{\mathrm{qcoh}}$

$$Rf_*(E) \otimes_S^{\mathbf{L}} M \xrightarrow{\sim} Rf_*(E \otimes_S^{\mathbf{L}} M).$$

Proof. First, by EGA III 1.4.12 and IV 1.7.21, there is an integer N such that $R^i f_*(F) = 0$ for $i > N$ and for every quasi-coherent $\mathcal{O}_X$-module F. Hence Rf_* sends $D^-(X)_{\mathrm{qcoh}}$ into $D^-(Y)_{\mathrm{qcoh}}$. The adjunction arrow

$$Lf^* Rf_*(E) \longrightarrow E$$

yields, after tensoring with M, an arrow

$$Lf^*(Rf_*(E) \otimes_S^{\mathbf{L}} M) \longrightarrow E \otimes_S^{\mathbf{L}} M,$$

and therefore, by adjunction, the canonical arrow

$$Rf_*(E) \otimes_S^{\mathbf{L}} M \longrightarrow Rf_*(E \otimes_S^{\mathbf{L}} M). \qquad (*)$$

Both sides have quasi-coherent cohomology. The assertion is local on Y, so assume S and Y affine. If M has finite flat amplitude, devissage and the way-out property of the two functors in $(*)$ reduce successively to M an ordinary quasi-coherent module, then to M free, then to M free of finite type, and finally to $M = \mathcal{O}_S$, where $(*)$ is tautological. If instead E has finite flat amplitude relative to S, devissage reduces to the case $M = P$, with P a flat A-module and $S = \operatorname{Spec} A$. Put $A' = D_A(P)$ (EGA 0_{IV}, 18.2.3) and $S' = \operatorname{Spec} A'$. Then the assertion is the flat base-change assertion that

$$q^* Rf_*(E) \longrightarrow Rf'_* p^*(E)$$

is an isomorphism for the base change to S', which is EGA III 1.4.15 together with IV 1.7.21.

Corollary 3.7.1. Let S be a scheme, let $f : X \to Y$ be a quasi-compact and quasi-separated S-morphism of quasi-compact schemes over S, and let $E \in \operatorname{ob} D^-(X)_{\mathrm{qcoh}}$. If E is locally of finite flat amplitude relative to S, then the same is true of $Rf_*(E)$. More precisely, if S is quasi-compact and if N is an integer such that $R^i f_*(F) = 0$ for $i > N$ and every quasi-coherent $\mathcal{O}_X$-module F, then

$$\text{tor. amp}_S(E) \subset [a, b] \Longrightarrow \text{tor. amp}_S(Rf_* E) \subset [a, b + N].$$

Proof. By (3.3 (1) (iv)), and using that Rf_* commutes with restriction to an open of S, it suffices to prove that the hypothesis implies

$$H^i(Rf_* E \otimes_S^{\mathbf{L}} M) = 0$$

for $i \notin [a, b + N]$ and every quasi-coherent $\mathcal{O}_S$-module M. But the hypothesis says that $H^i(E \otimes_S^{\mathbf{L}} M) = 0$ for $i \notin [a, b]$, and (3.7) gives the result.

Corollary 3.7.2. Let $f : X \to Y$ be quasi-compact, quasi-separated, and locally of finite Tor-dimension, and let $E \in \operatorname{ob} D^-(X)_{\mathrm{qcoh}}$. If E is locally of finite flat amplitude in the absolute sense, then so is $Rf_*(E)$.

Proof. Combine (3.4.1 a) and (3.7.1).

4. Relative perfection

Definition 4.1. Let $f : X \to Y$ be a morphism locally of finite type of schemes, let $E \in \operatorname{ob} D(X)$, and let $[a, b]$ be an interval of $\mathbb{Z}$. We say that E is of perfect amplitude relative to f (or to Y, if there is no ambiguity), contained in $[a, b]$, and write

$$\text{parf. amp}_f(E) \subset [a, b]$$

if E, relative to f, is pseudo-coherent (1.2) and of flat amplitude contained in $[a, b]$ (3.1). We say that f is of perfect amplitude contained in $[-n, 0]$, or of perfect dimension $\leq n$, if $\mathcal{O}_X$ has perfect amplitude contained in $[-n, 0]$. We say that E is of finite perfect amplitude relative to f if such an interval $[a, b]$ exists. We say that E is perfect relative to f if it is locally of finite perfect amplitude relative to f. We say that f is perfect if $\mathcal{O}_X$ is locally of finite perfect amplitude relative to f.

Example 4.1.1. A regular immersion of codimension d (VII 1.4) is of perfect dimension $\leq d$, as follows from the Koszul complex. A morphism locally a complete intersection, i.e. locally the composite of a regular immersion and a smooth morphism, is a perfect morphism.

Notation 4.2. Let $f : X \to Y$ be locally of finite type. We write

$$D(f)_{\mathrm{parf}}, \qquad D(f)_{\mathrm{parf}}, \qquad D(f)^{-}_{\mathrm{parf}}$$

when no confusion is possible, for the full subcategories of $D(X)$ formed by complexes perfect, respectively of finite perfect amplitude, relative to f. By (I 5.3) and (1.4), these are triangulated subcategories of $D(X)$. If X is quasi-compact, the two finite-amplitude/perfect notations coincide in the evident bounded sense.

Proposition 4.3 (cf. I 5.8). Let $f : X \to Y$ be locally of finite type, let $E \in \mathrm{ob}\, D(X)$, let $[a, b]$ be an interval of $\mathbb{Z}$, and let $x \in X$, $y = f(x)$. The following conditions are equivalent:

i) $\mathrm{parf.\,amp}_f(E) \subset [a, b]$ in a neighbourhood of x;

ii) in a neighbourhood of x, E is $(a-1)$-pseudo-coherent relative to f, is acyclic outside $[a, b]$, and
$$\mathrm{tor.\,amp}_{\mathcal{O}_{Y,y}}(E_x) \subset [a, b].$$

Proof. The implication i) $\Rightarrow$ ii) is immediate. For the converse, the question is local around x, and by the definitions one reduces to the case where f is smooth. Then E is $(a-1)$-pseudo-coherent near x; after localizing, one may assume that E is strictly $(a-1)$-pseudo-coherent and zero in degrees $> b$. After truncating, one may represent it by a complex with E^i free of finite type for $i > a$, E^a of finite presentation, and zero outside $[a, b]$. By (I 5.1.1), the term in degree a modulo boundaries is flat relative to f near x. Since f is smooth, one embeds locally into an affine space over Y and descends finite presentation and flatness from a noetherian approximation; pseudo-coherence follows from (1.10 a), and the desired perfect-amplitude assertion follows.

Corollary 4.3.1. Let $f : X \to Y$ be locally of finite type and let E be an $\mathcal{O}_X$-module. One has $\mathrm{parf.\,amp}_f(E) \subset [0, 0]$ if and only if E is flat and of finite presentation relative to f (1.2.1 c). In particular, f is of perfect dimension zero if and only if f is flat and locally of finite presentation.

Corollary 4.3.2. Let $f : X \to Y$ be locally of finite type, let E be a complex of $\mathcal{O}_X$-modules flat and of finite presentation relative to f, and let $[a, b]$ be an interval. If $E^i = 0$ for $i \notin [a, b]$, then $\mathrm{parf.\,amp}_f(E) \subset [a, b]$. If E is bounded above, then E is pseudo-coherent relative to f.

Proof. This follows from (4.3.1) and from I 4.15 and I 4.8.

Proposition 4.4. Let

$$\begin{array}{ccc} X & \xrightarrow{\ i\ } & X' \\ {\scriptstyle f}\downarrow & \swarrow{\scriptstyle g} & \\ Y & & \end{array}$$

be a commutative triangle of schemes, where i is a closed immersion and g is smooth, and let $E \in \mathrm{ob}\, D(X)$. Then E is perfect relative to f if and only if $i_*(E)$ is perfect in the absolute sense. More precisely, if g has relative dimension $\leq d$, then

$$\mathrm{parf.\,amp}(i_*E) \subset [a, b] \Longrightarrow \mathrm{parf.\,amp}_f(E) \subset [a, b],$$
$$\mathrm{parf.\,amp}_f(E) \subset [a, b] \Longrightarrow \mathrm{parf.\,amp}(i_*E) \subset [a - d, b].$$

Proof. This is an immediate consequence of (3.6).

Proposition 4.5. Let

$$\begin{array}{ccc} X & \xrightarrow{\ f\ } & Y \\ & {\scriptstyle h}\searrow & \downarrow{\scriptstyle g} \\ & & S \end{array}$$

be a commutative triangle of morphisms locally of finite type, let $E \in \operatorname{ob} D(X)$, $F \in \operatorname{ob} D(Y)$, and let $[a,b]$, $[c,d]$ be intervals of $\mathbb{Z}$. Then

$$\operatorname{parf.amp}_f(E) \subset [a,b], \quad \operatorname{parf.amp}_g(F) \subset [c,d] \Longrightarrow \operatorname{parf.amp}_h(E \otimes_Y^{\mathbf{L}} F) \subset [a+c, b+d].$$

In particular, if E is perfect relative to f and F is perfect relative to g, then $E \otimes_Y^{\mathbf{L}} F$ is perfect relative to h.

Proof. Combine (1.1) and (3.4).

Corollary 4.5.1. Under the hypotheses of (4.5):

a) if g is of perfect dimension $\leq n$, then

$$\operatorname{parf.amp}_f(E) \subset [a,b] \Longrightarrow \operatorname{parf.amp}_h(E) \subset [a-n, b];$$

therefore, if g is perfect, E perfect relative to f implies E perfect relative to h;

b) if f is of perfect dimension $\leq m$, then

$$\operatorname{parf.amp}_g(F) \subset [c,d] \Longrightarrow \operatorname{parf.amp}_h(Lf^*F) \subset [c-m, d];$$

therefore, if f is perfect, F perfect relative to g implies Lf^*F perfect relative to h.

Remark 4.5.2. With the preceding notation, if g is perfect, E perfect relative to h does not necessarily imply that E is perfect relative to f; the usual example of a singular point of a scheme over a field already shows this. The following criterion is available.

Proposition 4.6 (criterion of perfection by fibers). Let

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow^{h} & \downarrow{\scriptstyle g} \\ & & S \end{array}$$

be a commutative triangle of morphisms of schemes. Suppose that f and h are locally of finite type, and that g is locally of finite presentation and flat. Let $x \in X$, $y = f(x)$, $s = h(x) = g(y)$, and let $E \in \operatorname{ob} D(X)$. Let $[a,b]$ be an interval of $\mathbb{Z}$, and let $f_s : X_s \to Y_s$ be the fiber of f over s, i.e. the morphism obtained from f by the base change $\operatorname{Spec} k(s) \to S$. Then the following conditions are equivalent:

i) $\operatorname{parf.amp}_h(E) \subset [a,b]$ near x, and

$$\operatorname{tor.amp}_{\mathcal{O}_{Y_s,y}}(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y_s,y}) \subset [a,b];$$

ii) $\operatorname{parf.amp}_f(E) \subset [a,b]$ near x.

Proof. The implication ii) $\Rightarrow$ i) follows because g has perfect dimension zero by (4.3.1), hence (4.5.1 a) gives $\operatorname{parf.amp}_h(E) \subset [a,b]$ near x; the asserted Tor-amplitude condition follows from (4.3). Conversely, the question is local around x. We may factor f as a closed immersion followed by a smooth morphism. Replacing E by its direct image under the immersion and replacing the smooth factor accordingly does not change the conditions; hence one may suppose f smooth. Since g is perfect of dimension zero and h is perfect by hypothesis, E is pseudo-coherent. After localizing, use (II 2.2.10 b) and (I 2.10 a) to suppose that E is strictly pseudo-coherent, zero in degrees $> b$, and quasi-isomorphic to the truncation

$$t : E \longrightarrow (0 \to E^a/B^a \to E^{a+1} \to \cdots \to E^b \to 0).$$

It remains to prove that E^a/B^a is f-flat near x. Let $i : X_s \to X$ be the canonical map. One has

$$E_x \otimes^{\mathbf{L}}_{mathcalO_{Y,y}} \mathcal{O}_{Y_s,y} \simeq i^*(E)_x.$$

Using (3.5.2), the Tor-amplitude hypothesis on the fiber shows that the truncated complex $i^*(t)$ is a quasi-isomorphism, so $i^*(E^a/B^a)_x$ is flat over $\mathcal{O}_{Y_s,y}$. On the other hand, the Tor-amplitude of E relative to h implies that E^a/B^a is h-flat. The usual criterion of flatness by fibers (EGA IV 11.3.10) then gives that E^a/B^a is f-flat near x.

Corollary 4.6.1. Let $f : X \to S$ be locally of finite presentation and flat, let $x \in X$, $s = f(x)$, let $E \in \operatorname{ob} D(X)$, and let $[a, b]$ be an interval. If $i_s : X_s = X \times_S \operatorname{Spec} k(s) \to X$ is the canonical projection and $E_s = Li_s^*E$, then

$$\operatorname{parf.amp}_S(E) \subset [a, b] \text{ near } x, \quad \operatorname{tor.amp}(E_s) \subset [a, b]$$

if and only if

$$\operatorname{parf.amp}(E) \subset [a, b] \text{ near } x.$$

Consequently E is perfect if and only if E is perfect relative to f and E_s is perfect for every $s \in S$.

Proof. Apply (4.6) with $X = Y$ and $f = \mathrm{Id}_X$.

Remark 4.6.2. From (4.6.1) and I 5.10 it follows that, if f is locally of finite presentation, flat, and has regular schemes as fibers, then E perfect relative to f is equivalent to E perfect. This equivalence was already obtained in (4.4) when f is smooth.

Proposition 4.7. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be an S-morphism locally of finite type, with $E_i \in \operatorname{ob} D(X_i)$ and intervals $[a_i, b_i]$. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Put $a = a_1 + a_2$, $b = b_1 + b_2$, and $f = f_1 \times_S f_2$. Then:

a) if $\operatorname{parf.amp}_{f_1}(E_1) \subset [a_1, b_1]$, and if E_2 is pseudo-coherent relative to f_2 and acyclic outside $[a_2, b_2]$, then $E_1 \otimes^{\mathbf{L}}_S E_2$ is pseudo-coherent relative to f and acyclic outside $[a, b]$;

b) if $\operatorname{parf.amp}_{f_i}(E_i) \subset [a_i, b_i]$ for $i = 1, 2$, then

$$\operatorname{parf.amp}_f(E_1 \otimes^{\mathbf{L}}_S E_2) \subset [a, b].$$

Proof. Combine (1.9) and (3.5).

Corollary 4.7.1. Under the hypotheses of (4.7), if f_i has perfect dimension $\leq n_i$ for $i = 1, 2$, then f has perfect dimension $\leq n_1 + n_2$. Consequently, if f_1 and f_2 are perfect, then so is f.

Corollary 4.7.2. In a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[h]{} & Y, \end{array}$$

where f is locally of finite type and X and Y' are Tor-independent over Y, one has

$$\operatorname{parf.amp}_f(E) \subset [a, b] \Longrightarrow \operatorname{parf.amp}_{f'}(Lg^*E) \subset [a, b].$$

Consequently $Lg^* : D^-(X) \to D^-(X')$ sends $D(f)_{\mathrm{parf}}$, respectively $D(f)_{\mathrm{parf}}$, into the corresponding categories for f'.

Proposition 4.8. Let S be a scheme, let $f : X \to Y$ be a proper S-morphism of S-schemes locally of finite type, and let $E \in \mathrm{ob}\, D^-(X)$. Under the proviso that (2.1) is valid for f (hence at least if S is locally noetherian or if f is projective):

a) if E is perfect relative to S, then so is $Rf_*(E)$;

b) if S is quasi-compact, X and Y are of finite type over S, $[a, b]$ is an interval, and N is such that $R^i f_*(M) = 0$ for $i > N$ and every quasi-coherent $\mathcal{O}_X$-module M, then

$$\mathrm{parf.\,amp}_S(E) \subset [a, b] \Longrightarrow \mathrm{parf.\,amp}_S(Rf_*E) \subset [a, b + N].$$

Consequently Rf_* sends perfect complexes relative to S to perfect complexes relative to S.

Proof. Assertion a) follows from b), and b) is obtained by combining (2.1) and (3.7).

Corollary 4.8.1. Let $f : X \to Y$ be proper and perfect, and let $E \in \mathrm{ob}\, D^-(X)$. Under the proviso that (2.1) holds for f, if E is perfect then $Rf_*(E)$ is perfect. If moreover Y is quasi-compact, Rf_* sends $D(X)_{\mathrm{parf}}$ to $D(Y)_{\mathrm{parf}}$.

Proof. If E is perfect, then E is perfect relative to f by (4.5.1 a).

Proposition 4.9. Let S be noetherian, let X and Y be S-schemes of finite type, let $f : X \to Y$ be a compactifiable S-morphism, i.e. the composite of an open immersion and a proper morphism, and let $E \in \mathrm{ob}\, D(X)$, $F \in \mathrm{ob}\, D^-(Y)$. If E is perfect relative to f and F is perfect relative to S, then

$$\mathrm{RHom}(E, f^!F)$$

is perfect relative to S.

Proof. The question is local on X. One may suppose that f factors as

$$X \xrightarrow{i} X' \xrightarrow{f'} Y,$$

where i is a closed immersion and f' is smooth. If E is perfect relative to f, then it is perfect relative to i by (4.4); if F is perfect relative to S, then $f'^!F$ is perfect relative to S by (4.5.1 b), since $f'^!F$ is locally isomorphic to f'^*F up to translation. As $f^! = i^! f'^!$, it remains to prove the assertion when f is a closed immersion. Then f_*E is perfect and, by global duality ([H] III 6.7),

$$f_*\mathrm{RHom}(E, f^!F) \simeq \mathrm{RHom}(f_*E, F).$$

Since f_*E is perfect, (I 7.7) gives

$$\mathrm{RHom}(f_*E, F) \simeq (f_*E)^\vee \otimes^{\mathbf{L}} F,$$

and the result follows from (4.5) with the identity morphism on Y.

Corollary 4.9.1. Under the hypotheses of (4.8), if f is perfect, then $f^!$ sends $D(Y)_{\mathrm{parf}}$ into $D(X)_{\mathrm{parf}}$.

Proof. Take $E = \mathcal{O}_X$ in (4.9).

Corollary 4.9.2. Let S be noetherian and let $f : X \to S$ be compactifiable. Put $K_X = f^!(\mathcal{O}_S)$, and $D_X(\cdot) = \mathrm{RHom}(\cdot, K_X)$. For $E \in \mathrm{ob}\, D(f)_{\mathrm{parf}}$, one has $D_X(E) \in \mathrm{ob}\, D(f)_{\mathrm{parf}}$, and the canonical arrow

$$E \longrightarrow D_X D_X(E)$$

is an isomorphism.

Proof. For the first assertion, apply (4.9) with $Y = S$ and $F = \mathcal{O}_S$. For the second, one reduces easily to the case where f is a closed immersion. It is enough then to apply f_* to the canonical arrow. By global duality one has $f_* D_X = D_S f_*$, and one is reduced to ordinary biduality for perfect complexes (I 7.2).

5. Applications: exchange and semi-continuity theorems

Lemma 5.1. Let $(X, \mathcal{O}_X)$ be a commutatively ringed topos, let $\mathrm{Mod}(X)$ be the category of $\mathcal{O}_X$-modules, and write $D(X) = D(\mathrm{Mod}(X))$. Let $E \in \mathrm{ob}\, D(X)$ and let $p \in \mathbb{Z}$. Consider the following conditions:

a) the functor $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ from $\mathrm{Mod}(X)$ to $\mathrm{Mod}(X)$ is right exact;

a′ the canonical map
$$H^p(E) \otimes M \longrightarrow H^p(E \otimes^{\mathbf{L}} M)$$
is an isomorphism for all $M \in \mathrm{ob}\, \mathrm{Mod}(X)$;

b the functor $M \mapsto H^{p+1}(E \otimes^{\mathbf{L}} M)$ is left exact;

b′ the canonical map defined below
$$H^{p+1}(E \otimes^{\mathbf{L}} M) \longrightarrow \mathrm{Hom}(H^{-p-1}(E^\vee), M)$$
is an isomorphism for all $M \in \mathrm{ob}\, \mathrm{Mod}(X)$.

Then
$$\mathrm{a} \Longleftrightarrow \mathrm{a}' \Longleftrightarrow \mathrm{b} \Longleftarrow \mathrm{b}'.$$
If E is pseudo-coherent, then a), a'), b), and b') are equivalent.

The map in b') is obtained as follows. The canonical evaluation arrow $E^\vee \otimes^{\mathbf{L}} E \to \mathcal{O}_X$, tensorized with M, gives an arrow
$$E \otimes^{\mathbf{L}} M \longrightarrow \mathrm{RHom}(E^\vee, M).$$
On H^{p+1}, and then by the canonical map
$$\mathrm{Ext}^{p+1}(E^\vee, M) o \mathrm{Hom}(H^{-p-1}(E^\vee), M),$$
one obtains the arrow b').

Proof. The implications $\mathrm{a}' \Rightarrow \mathrm{a} \Leftrightarrow \mathrm{b} \Leftarrow \mathrm{b}'$ are formal. To prove $\mathrm{a} \Rightarrow \mathrm{a}'$, present M by a sequence $M^0 \to M^1 \to M \to 0$ in which M^0 and M^1 are direct sums of sheaves of the form $\mathcal{O}_{U,X}$, with U an object of X. Right exactness and compatibility with direct sums reduce the assertion to $M = \mathcal{O}_{U,X}$. Then both sides are identified with extension by zero from U, and exactness of that extension gives the result.

Assume now that E is pseudo-coherent and prove $\mathrm{b} \Rightarrow \mathrm{b}'$. It is enough to show that the preceding comparison arrow
$$E \otimes^{\mathbf{L}} M \longrightarrow \mathrm{RHom}(E^\vee, M)$$
is an isomorphism when M is injective. Let $T(F) = F \otimes^{\mathbf{L}} M$ and $T'(F) = \mathrm{RHom}(F^\vee, M)$. Both functors have the same boundedness behaviour under truncation; the arrow is local and is compatible with restriction to opens, and an injective remains injective after such a restriction. After localizing and truncating, one may suppose E strictly perfect. A devissage then reduces to $E = \mathcal{O}_X$, where the assertion is immediate.

Remarks 5.2.

i) Conditions a), a'), b), and b') are local on the topos: if (X_α) is an open covering of the final object and the condition holds for each X_α and $E|X_\alpha$, then it holds on X.

ii) If x is a point of X, write a_x, a'_x, b_x, etc. for the corresponding conditions for E_x and $\mathrm{Mod}(\mathcal{O}_{X,x})$. If X has enough points, condition a), respectively a'), b), is equivalent to the corresponding stalkwise condition for every x; if E is pseudo-coherent, this is also equivalent to b') stalkwise. Moreover, a') may be tested on quotients of modules of the form $\mathcal{O}_{U,x}^{(I)} \to \mathcal{O}_{U,x}$.

iii) The criterion EGA III 7.4.2 extends word for word: if E is represented by flat terms, condition a) says that $\mathrm{Coker}(E^p \to E^{p+1})$ is flat, hence locally a direct factor of a free module of finite type when E^p and E^{p+1} are of finite presentation. In particular, for E perfect, condition a) for degree p is equivalent to the corresponding condition for $E^\vee$ in degree $-p-1$.

iv) Let X be a quasi-separated scheme and let $E \in \mathrm{ob}\, D(X)$ have locally bounded quasi-coherent cohomology. Let a_{qcoh}, a'_{qcoh}, etc. be the analogous conditions with $\mathrm{Mod}(X)$ replaced by $\mathrm{Qcoh}(X)$. Then

$$a \Rightarrow a_{\mathrm{qcoh}}, \qquad a' \Rightarrow a'_{\mathrm{qcoh}}, \qquad b \Rightarrow b_{\mathrm{qcoh}},$$

and these conditions are all equivalent when E is pseudo-coherent. This follows by reducing to affine opens, using II 3.5 and II 2.2.10 a), and the fact that quasi-coherent sheaves extend from affine opens on a quasi-separated scheme.

Proposition 5.3 (exchange property; EGA III 7.7.5 II). Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes, let $E \in \mathrm{ob}\, D^-(X)$ have quasi-coherent cohomology and finite flat amplitude relative to f, and let $p \in \mathbb{Z}$. Consider:

a) $M \mapsto R^p f_*(E \otimes^{\mathbf{L}}_Y M)$, from $\mathrm{Qcoh}(Y)$ to $\mathrm{Qcoh}(Y)$, is right exact;

a′ the canonical arrow

$$R^p f_*(E) \otimes M \longrightarrow R^p f_*(E \otimes^{\mathbf{L}}_Y M)$$

is an isomorphism for every $M \in \mathrm{ob}\, \mathrm{Qcoh}(Y)$;

b $M \mapsto R^{p+1} f_*(E \otimes^{\mathbf{L}}_Y M)$ is left exact;

b′ the canonical arrow

$$R^{p+1} f_*(E \otimes^{\mathbf{L}}_Y M) \longrightarrow \mathrm{Hom}(H^{-p-1}((Rf_*E)^\vee), M)$$

is an isomorphism for every $M \in \mathrm{ob}\, \mathrm{Qcoh}(Y)$;

c for every base change $Y' \to Y$, the canonical arrow

$$R^p f_*(E) \otimes_Y \mathcal{O}_{Y'} \longrightarrow H^p(Rf_*E \otimes^{\mathbf{L}}_Y \mathcal{O}_{Y'})$$

is an isomorphism.

Then

$$a \iff a' \iff c \iff b \impliedby b'.$$

If, moreover, f is proper, E is pseudo-coherent relative to f (hence perfect relative to f), and (2.1) is valid for f, for instance if f is projective or Y is noetherian, then the preceding conditions are all equivalent.

Proof. The implications among a), a'), b), and b') follow at once from (3.7), (5.1), and (5.2 iv); the implication $b \Rightarrow b'$ in the proper pseudo-coherent case follows from (2.1). It remains to compare a') and c). This is supplied by the following lemma.

Lemma 5.3.1. Let X be a quasi-separated scheme, and let $E \in \mathrm{ob}\, D^-(X)$ have locally bounded quasi-coherent cohomology. Then condition (5.1 a) is equivalent to:

c) for every $X' \to X$, the canonical map

$$H^p(E) \otimes_X \mathcal{O}_{X'} \longrightarrow H^p(E \otimes_X^{\mathbf{L}} \mathcal{O}_{X'})$$

is an isomorphism.

If E is pseudo-coherent, condition (5.1 a) also implies that, for every $X' \to X$, the map

$$H^{-p-1}(E^\vee) \otimes_X \mathcal{O}_{X'} \longrightarrow H^{-p-1}((E \otimes_X^{\mathbf{L}} \mathcal{O}_{X'})^\vee) \tag{5.3.1.1}$$

is an isomorphism.

Proof of the lemma. For $a \Rightarrow c$, localize on X', and assume $X = \operatorname{Spec} A$, $X' = \operatorname{Spec} A'$, and $E = \widetilde{F}$. By the affine case the hypothesis gives

$$H^p(F) \otimes_A A' \longrightarrow H^p(F \otimes_A^{\mathbf{L}} A')$$

an isomorphism, which is exactly c). Conversely, by (5.2 iv) it suffices to prove $c \Rightarrow a'_{\mathrm{qcoh}}$. Again localizing, write $X = \operatorname{Spec} A$, $E = \widetilde{F}$, and let $M = \widetilde{N}$ be quasi-coherent. Taking $A' = D_A(N)$ and applying c) to $\operatorname{Spec} A' \to X$ gives the required map

$$H^p(F) \otimes_A N \longrightarrow H^p(F \otimes_A^{\mathbf{L}} N)$$

as an isomorphism.

For the last assertion, the question is local on X'. By (II 2.2.10 b) and truncation, one may suppose E strictly perfect. Then $(E \otimes_X^{\mathbf{L}} \mathcal{O}_{X'})^\vee$ identifies canonically with $E^\vee \otimes_X^{\mathbf{L}} \mathcal{O}_{X'}$, and (5.3.1.1) is the canonical map

$$H^{-p-1}(E^\vee) \otimes_X \mathcal{O}_{X'} \longrightarrow H^{-p-1}(E^\vee \otimes_X^{\mathbf{L}} \mathcal{O}_{X'}). \tag{$*$}$$

If E satisfies (5.1 a) for p, then by (5.2 iii) $E^\vee$ satisfies the same condition for $-p-1$, and $(*)$ is an isomorphism by the first part of the lemma. The lemma, and therefore (5.3), follow.

Remark 5.3.2. Proposition (5.3) applies in particular when E is a bounded complex of quasi-coherent flat $\mathcal{O}_X$-modules of finite presentation relative to f: by (4.3.2), such a complex has finite perfect amplitude relative to f.

Definition 5.4. Let X be a topos, let $\mathrm{Pt}(X)$ be its set of points (SGA 4 IV 6.1), let M be an ordered set, and let $\varphi : \mathrm{Pt}(X) \to M$ be a function. We say that φ is upper semi-continuous at x (respectively lower semi-continuous, respectively locally constant) if there exists an x-pointed object U of X such that for every point y of U, one has $\varphi(y) \leq \varphi(x)$ (respectively $\varphi(y) \geq \varphi(x)$, respectively $\varphi(y) = \varphi(x)$). We say that φ is upper semi-continuous (respectively lower semi-continuous, respectively locally constant) if it is so at every point.

Examples 5.4.1.

a) If $\underline{M}$ is the constant sheaf with value M and $s \in H^0(X, \underline{M})$, then $x \mapsto x^*s$ is locally constant. In particular, if X is locally ringed and E is a perfect complex on X, the rank of E (I 6.11) defines a locally constant function on $\mathrm{Pt}(X)$ with values in $\mathbb{Z}$.

b) If X is locally ringed and P is an $\mathcal{O}_X$-module of finite type, then for $x \in \mathrm{Pt}(X)$, $P_x \otimes_{\mathcal{O}_{X,x}} k(x)$ is a finite-dimensional vector space over $k(x)$, and

$$x \longmapsto \dim_{k(x)}(P_x \otimes k(x))$$

is upper semi-continuous. This is the usual Nakayama argument: an epimorphism $\mathcal{O}_{X,x}^n \to P_x$ lifts over a pointed neighbourhood.

Lemma 5.5. Let $(X, \mathcal{O}_X)$ be a locally ringed topos, let $E \in \mathrm{ob}\, D(X)$, and let $p \in \mathbb{Z}$. Then:

i) If E is perfect, the vector spaces

$$H^i(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} k(x))$$

are finite-dimensional over $k(x)$, zero except for finitely many i, and the function

$$x \longmapsto \sum_i (-1)^i \dim_{k(x)} H^i(E_x \otimes^{\mathbf{L}} k(x))$$

is locally constant on $\mathrm{Pt}(X)$.

ii) If E is $(p-1)$-pseudo-coherent, then

$$H^p(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} k(x))$$

is finite-dimensional over $k(x)$, and its dimension is upper semi-continuous on $\mathrm{Pt}(X)$. If moreover $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ is exact on $\mathrm{Mod}(X)$, then $H^p(E)$ is locally free of finite type and the preceding function is locally constant.

iii) If E is p-pseudo-coherent (respectively $(p-1)$-pseudo-coherent), and if for a point x the functor

$$M \longmapsto H^p(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} M)$$

is right exact (respectively left exact) on $\mathrm{Mod}(\mathcal{O}_{X,x})$, then there exists an x-pointed object U such that

$$M \longmapsto H^p(E|_U \otimes^{\mathbf{L}} M)$$

is right exact (respectively left exact) on $\mathrm{Mod}(U)$.

Proof. For i), by I 4.19.1 the complex $E_x \otimes^{\mathbf{L}} k(x)$ is a perfect complex of $k(x)$-modules; hence its cohomology spaces have finite rank and vanish except for finitely many degrees. The identity

$$\mathrm{rank}(E_x \otimes^{\mathbf{L}} k(x)) = \sum_i (-1)^i \dim_{k(x)} H^i(E_x \otimes^{\mathbf{L}} k(x))$$

follows by devissage from the additivity of rank. The assertion follows from the compatibility of the rank with inverse images (I 6.12).

For ii), after truncation and translation, one may assume for the first assertion that E is strictly perfect, $p = 0$, and $E^i = 0$ for $i \notin [-1, 1]$. Put $E(x) = E_x \otimes^{\mathbf{L}} k(x)$. Then

$$0 = \mathrm{rank}(E(x)) = \dim H^{-1}(E(x)) - \dim H^0(E(x)) + \dim H^1(E(x))$$

up to the corresponding alternating-rank convention; hence upper semi-continuity of $\dim H^0(E(x))$ follows from that of the neighbouring cohomology dimensions and from local constancy of the rank of the terms. For H^1 this is immediate from (5.4.1 b). For H^{-1}, replacing E by the truncated complex $(0 \to E^{-1} \to E^0 \to 0)$ reduces to the same argument.

For the second assertion in ii), after truncating one may suppose E strictly perfect and $E^i = 0$ for $i < p-1$. Then $Z'^{p+1}(E) = \mathrm{Coker}(E^p \to E^{p+1})$ is of finite presentation and one has an exact sequence

$$0 \to H^p(E) \to Z'^p(E) \to E^{p+1} \to Z'^{p+1}(E) \to 0.$$

Exactness of $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ implies, by (5.2 iii), that $Z'^p(E)$ and $Z'^{p+1}(E)$ are flat, hence locally free of finite type by I 5.8.3. The exact sequence then shows that $H^p(E)$ is locally free of finite type, and the function $x \mapsto \dim_{k(x)} H^p(E)_x$ is locally constant.

For iii), one may assume that E is strictly perfect and that $E^i = 0$ for $i < p$ (respectively for $i < p-1$). Saying that the functor

$$M \longmapsto H^p(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} M)$$

is right exact (respectively left exact) on $\mathrm{Mod}(\mathcal{O}_{X,x})$ is equivalent, by (5.2 iii), to saying that $Z'^{p+1}(E_x)$ (respectively $Z'^p(E_x)$) is flat, hence free of finite type since it is of finite presentation. Since

$$Z'^{p+1}(E_x) \simeq Z'^{p+1}(E)_x \qquad (\text{respectively } Z'^p(E_x) \simeq Z'^p(E)_x),$$

the usual argument (EGA 0_I, 5.2.7) gives an x-pointed object U of X on which $Z'^{p+1}(E|_U)$ (respectively $Z'^p(E|_U)$) is locally free of finite type. This proves the exactness assertion.

Corollary 5.6 (EGA III 7.6.9). Let X be a scheme, let $p \in \mathbb{Z}$, and let $E \in \mathrm{ob}\, D^-(X)$ be $(p-1)$-pseudo-coherent. Then:

i) For $x \in X$, $H^p(E \otimes^{\mathbf{L}}_X k(x))$ has finite rank over $k(x)$, and

$$x \longmapsto \dim_{k(x)} H^p(E \otimes^{\mathbf{L}}_X k(x))$$

is constructible and upper semi-continuous on X.

ii) Suppose X is quasi-separated and E has locally bounded quasi-coherent cohomology. If $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ is exact on the category of quasi-coherent $\mathcal{O}_X$-modules, then the preceding function is locally constant and $H^p(E)$ is locally free of finite type. Conversely, if X is locally noetherian and reduced and that function is locally constant, then the same functor is exact on quasi-coherent modules.

Proof. If X is integral with generic point x, the functor $M \mapsto H^p(E_x \otimes M)$ is exact on $\mathcal{O}_{X,x}$-modules, since $\mathcal{O}_{X,x} = k(x)$. By (5.5 iii) there is a non-empty open neighbourhood U of x on which the corresponding functor is exact, and by (5.5 ii) the rank function is locally constant on U. This gives constructibility. The other assertions follow from (5.5 ii, iii) and (5.2 iv), except for the converse in ii), for which one reduces locally and by truncation to the strictly perfect case and applies EGA III 7.6.2 and 7.6.5 to the functor $T_M = H^p(E \otimes^{\mathbf{L}} M)$.

Proposition 5.7 (EGA III 7.7.5 and 7.9.4). Let $f : X \to Y$ be a proper morphism of quasi-compact schemes, let $E \in \mathrm{ob}\, D(X)_{\mathrm{parf}}$, and let $p \in \mathbb{Z}$. Assuming (2.1) is valid for f, for instance if f is projective or if Y is noetherian, the following hold:

i) For $y \in Y$, the vector spaces $H^i(Rf_*E \otimes^{\mathbf{L}}_Y k(y))$ are finite-dimensional over $k(y)$, zero except for finitely many i, and their alternating rank is locally constant on Y.

ii) The function

$$y \longmapsto \dim_{k(y)} H^p(Rf_*E \otimes^{\mathbf{L}}_Y k(y))$$

is constructible and upper semi-continuous on Y.

iii) Suppose Y is quasi-separated. If $M \mapsto R^p f_*(E \otimes_Y^{\mathbf{L}} M)$ is exact on the category of quasi-coherent $\mathcal{O}_Y$-modules, then $R^p f_*(E)$ is locally free of finite type and the function in ii) is locally constant. Conversely, if Y is noetherian and reduced and the function in ii) is locally constant, then the same functor is exact on quasi-coherent modules.

Proof. By (4.8), $Rf_*(E)$ is a perfect complex. Assertion i), respectively ii), follows from (5.5 i), respectively (5.6 i). If M is quasi-coherent on Y, the canonical arrow

$$Rf_*(E) \otimes_Y^{\mathbf{L}} M \xrightarrow{\sim} Rf_*(E \otimes_Y^{\mathbf{L}} M)$$

is an isomorphism by (3.7), and iii) follows from (5.6 ii).

Remarks 5.8.

a) In the situation of (5.7), let $i_y : X_y \to X$ be the inclusion of the fiber over a point $y \in Y$. One may show, under the usual flatness hypotheses, the Künneth formula

$$H^i(Rf_*E \otimes_Y^{\mathbf{L}} k(y)) \simeq H^i(X_y, E_y),$$

where $E_y = Li_y^*E$, or $E_y = i_y^*E$ in the flat case.

b) One can develop for complex analytic spaces a theory analogous to that of this section: exchange, semi-continuity, and continuity properties. Its essential inputs are Grauert's finiteness theorem for proper morphisms and the analytic analogue of the projection formula. More precisely, for a morphism $f : X \to Y$ of analytic spaces, with X of bounded dimension, and for $E \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)_{\mathrm{coh}}$, the canonical arrow

$$Rf_*(E) \otimes_Y^{\mathbf{L}} M \longrightarrow Rf_*(E \otimes_Y^{\mathbf{L}} M)$$

is an isomorphism. Unlike Grauert's theorem, this assertion is essentially formal; it follows by the same devissage as in the proof of (3.7), and requires no hypothesis on E beyond belonging to the appropriate derived category.

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Exposé IV. Grothendieck groups of ringed topoi

by L. Illusie

1. Reminders and generalities on Grothendieck groups

1.1. Let C be a triangulated category. Recall that a map f from $\operatorname{ob} C$ to an abelian group G is called *additive* if

$$f(E) = f(E') + f(E'')$$

for every distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1].$$

The additive maps from $\operatorname{ob} C$ to G form an abelian group, depending functorially on G. The functor so obtained is represented by an abelian group $k(C)$ and a universal additive map

$$\mathrm{cl}_C : \operatorname{ob} C \longrightarrow k(C),$$

which we shall write simply as cl when no confusion is possible.

The group $k(C)$ depends functorially on C with respect to exact functors. If C, C' are triangulated categories, two isomorphic exact functors from C to C' induce the same homomorphism $k(C) \to k(C')$; in particular, if $u : C \to C'$ is an exact equivalence of categories, then $k(u) : k(C) \to k(C')$ is an isomorphism.

Lemma 1.2. Let C be a triangulated category and let $L \in \operatorname{ob} C$.

a) For every $n \in \mathbb{Z}$,

$$\mathrm{cl}(L[n]) = (-1)^n \,\mathrm{cl}(L).$$

If L', L'' are objects of C such that $L \simeq L' \oplus L''$, then

$$\mathrm{cl}(L) = \mathrm{cl}(L') + \mathrm{cl}(L'').$$

b) Suppose that there exists $n \in \mathbb{Z}$ such that

$$\coprod_{i \geq 0} L[2ni]$$

is representable. Then $\mathrm{cl}(L) = 0$.

Proof. Assertion a) follows directly from the definitions. For b), it is enough to observe that

$$\coprod_{i \geq 0} L[2ni] \simeq L \oplus \left(\coprod_{i \geq 0} L[2ni] \right) [2n],$$

and to apply a).

Lemma 1.3. Let C be a triangulated category and let A be a thick subcategory of C ([V] I 2.1). The inclusion and quotient functors define an exact sequence

$$k(A) \longrightarrow k(C) \longrightarrow k(C/A) \longrightarrow 0.$$

Proof. See SGA 5 VIII 3.1.

Lemma 1.4. Let A be an additive category, respectively an abelian category, and let f be an additive function on $\operatorname{ob} K^b(A)$, respectively on $\operatorname{ob} D^b(A)$. For $E \in \operatorname{ob} K^b(A)$, respectively $E \in \operatorname{ob} D^b(A)$, one has

$$f(E) = \sum_i (-1)^i f(E^i),$$

respectively

$$f(E) = \sum_i (-1)^i f(H^i(E)).$$

Proof. Left as an exercise to the reader.

1.5. Let C be an abelian category and let A be a full subcategory of C, stable under finite sums. A map f from $\operatorname{ob} A$ to an abelian group G is called *additive* if

$$\sum_i (-1)^i f(E^i) = 0$$

for every acyclic bounded complex E of C whose components belong to A. When A is stable under kernels of epimorphisms (I 1.2), this amounts to saying that

$$f(L) = f(L') + f(L'')$$

for every exact sequence

$$0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0$$

in C whose terms belong to A. The additive maps from $\operatorname{ob} A$ to G form an abelian group depending functorially on G. Denote by $K^{b,\emptyset}(A)$ the thick subcategory of $K^b(A)$ consisting of acyclic complexes. It follows easily from (1.3) and (1.4) that the map

$$\mathrm{cl}_A : \operatorname{ob} A \longrightarrow k\big(K^b(A)/K^{b,\emptyset}(A)\big) \tag{1.5.1}$$

induced by $\mathrm{cl}_{K^b(A)/K^{b,\emptyset}(A)}$ is a universal additive map, i.e. that the pair

$$\Big(k\big(K^b(A)/K^{b,\emptyset}(A)\big), \mathrm{cl}_A\Big)$$

represents the functor which associates to each abelian group G the abelian group of additive maps from $\operatorname{ob} A$ to G. We put

$$k\big(K^b(A)/K^{b,\emptyset}(A)\big) = k(A), \tag{1.5.2}$$

and write $k(A, C)$ instead of $k(A)$ when it is necessary to avoid ambiguity about the ambient abelian category.[22] In particular, for $A = C$, one has

$$k(D^b(C)) = k(C). \tag{1.5.3}$$

Lemma 1.6. Let C be an abelian category and let A be a thick abelian subcategory, i.e. a full subcategory stable under kernels, cokernels, and extensions. Let $D_A(C)$ be the full subcategory of $D(C)$ formed by the objects E such that $H^i(E) \in \operatorname{ob} A$ for every i. Then $D_A(C)$ is a triangulated subcategory ([V] I 2.3) of $D(C)$, and the canonical functor

$$D^b(A) \longrightarrow D^b_A(C) = D^b(C) \cap D_A(C)$$

induces an isomorphism

$$k(A) \xrightarrow{\sim} k(D^b_A(C)),$$

where $k(A)$ is defined by (1.5.2).

Proof. The first assertion is an immediate consequence of the long exact cohomology sequence. For the second, consider the map

$$f : \operatorname{ob} D^b_A(C) \longrightarrow k(A)$$

defined by

$$f(E) = \sum_i (-1)^i \,\mathrm{cl}_A(H^i(E)).$$

[22] The notation $k(A)$ had been introduced in I 6.3 for the group $k(K^b(A))$, which is not in general isomorphic to the group defined by (1.5.2). We therefore change convention here: in this exposé the notation $k(A)$ will mean only the group defined by (1.5.2).

By the long exact cohomology sequence, f is additive in the sense of (1.1), hence defines a homomorphism

$$\varphi : k(D^b_A(C)) \longrightarrow k(A).$$

It follows at once from (1.4) that φ is inverse to the canonical homomorphism $k(A) \to k(D^b_A(C))$.

Remark 1.6.1. In the situation of (1.6), the canonical functor

$$D^b(A) \longrightarrow D^b_A(C)$$

is not in general an equivalence of categories. This is shown by the example where S is a scheme, C is the category of $\mathcal{O}_S$-modules, and A is the category of quasi-coherent $\mathcal{O}_S$-modules (II App. I 0.3). See, however, the criterion (II 2.1.3).

Definition 1.7. Let C be a triangulated category, respectively an abelian category, and let C' be a subset of $\operatorname{ob} C$. We shall say that C' is an *exact subset* if, for every distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1],$$

respectively for every exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

such that two of the terms E', E, E'' belong to C', the third also belongs to C'. We shall say that C' is *dense* in $\operatorname{ob} C$ if $\operatorname{ob} C$ is the smallest exact subset of $\operatorname{ob} C$ containing C'.

Lemma 1.8. Let C be a triangulated category, respectively an abelian category, and let C' be a dense subset of $\operatorname{ob} C$. Then the elements $\operatorname{cl}_C(E)$, for $E \in C'$, generate $k(C)$.

Proof. Put $C' = C_0$, and for $i \geq 1$ define C_i recursively as the set of objects of C occurring in a distinguished triangle, respectively a short exact sequence, two of whose terms belong to C_{i-1}. The C_i form an increasing filtration of $\operatorname{ob} C$, which is exhaustive by hypothesis. For $i \geq 0$, let $k(C)_i$ be the subgroup of $k(C)$ generated by the elements $\operatorname{cl}_C(E)$, $E \in C_i$. It is immediate from the definitions that $k(C)_i = k(C)_{i+1}$ for every $i \geq 0$, hence $k(C)_i = k(C)_0$. If $x \in k(C)$, one can write $x = \operatorname{cl}(E') - \operatorname{cl}(E'')$ for $E', E'' \in \operatorname{ob} C$ (in the non-respectable case one may even arrange $E'' = 0$). Since the filtration (C_i) is exhaustive, there exists $i \geq 0$ such that E' and E'' belong to C_i; hence $x \in k(C)_i = k(C)_0$, as required.

Exercise 1.9. Let C be an abelian category and A a full subcategory stable under finite sums, subobjects, and quotients, such that $\operatorname{ob} A$ is dense in $\operatorname{ob} C$. Then the homomorphism

$$k(A, C) \longrightarrow k(C)$$

defined by the additive function $E \mapsto \operatorname{cl}_C(E)$ on $\operatorname{ob} A$ (cf. (1.5)) is an isomorphism.

Hint. Put $A = A_0$, and define A_i, for $i \geq 1$, as the full subcategory of C consisting of objects of C which are extensions of two objects of A_{i-1}. Prove that A_i satisfies the same hypotheses as A, and then prove that the canonical homomorphism

$$\varinjlim_i k(A_i, C) \longrightarrow k(C)$$

is an isomorphism. The desired conclusion follows at once.

2. The functors $K_\bullet$ and K' of a ringed topos

2.1. Let (X, A) be a ringed topos. Recall the following notation:

$$D^b(A_X)_{\mathrm{coh}}$$

is the category of pseudo-coherent complexes with bounded cohomology of left A-modules (I 2.15);

$$D(A_X)_{\mathrm{parf}}$$

is the category of complexes of left A-modules of finite perfect amplitude (I 5.8.4);

$$\mathrm{Loclib}(A_X)$$

is the category of left A-modules locally direct factors of finite free modules (II 2.0);

$$\mathrm{Coh}(A_X)$$

is the category of coherent left A-modules, relative to the category of finite-type free A-modules (I 3.1).

The last two categories are full subcategories of the category $\mathrm{Mod}(A_X)$ of left A-modules, and the first two are full triangulated subcategories of

$$D(A_X) = D(\mathrm{Mod}(A_X)).$$

Definition 2.2. Let (X, A) be a ringed topos. Unless otherwise stated, we shall put

$$K_\bullet(A_X) = k(D^b(A_X)_{\mathrm{coh}}), \qquad K'(A_X) = k(D(A_X)_{\mathrm{parf}}),$$

where $k(\ )$ is defined in (1.1), and

$$K_\bullet(A_X)_{\mathrm{naif}} = k(\mathrm{Coh}(A_X)), \qquad K'(A_X)_{\mathrm{naif}} = k(\mathrm{Loclib}(A_X)),$$

where $k(\ )$ is defined by (1.5.2), the ambient abelian category being $\mathrm{Mod}(A_X)$.[23] We say that $K_\bullet(A_X)$, respectively $K'(A_X)$, is the Grothendieck group of pseudo-coherent, respectively perfect, A-module complexes on X, and that $K_\bullet(A_X)_{\mathrm{naif}}$, respectively $K'(A_X)_{\mathrm{naif}}$, is the Grothendieck group of coherent, respectively locally free, left A-modules.

Remarks 2.3.

a) One can define variants of $K_\bullet(\ )$ and $K'(\ )$ by putting

$$K_\bullet^{\mathrm{loc}}(A_X) = k(D^{\mathrm{loc}b}(A_X)_{\mathrm{coh}}),$$

where "loc b" means "with locally bounded cohomology", and

$$K'_{\mathrm{loc}}(A_X) = k(D(A_X)_{\mathrm{parf}}).$$

These groups coincide with those defined in (2.2) when the final object of X is quasi-compact, but in general they lend themselves less well to computations in intersection theory.

b) The group $K_\bullet(A_X)_{\mathrm{naif}}$ has practically no interest unless A is coherent as a left A-module; otherwise there is a strong risk that very few coherent sheaves exist.

c) Considering the k of triangulated categories which are "too large" is of no interest. Thus, from (1.2 b), one immediately deduces the relations

$$k(D(A_X)) = k(D^-(A_X)) = k(D^+(A_X)) = k(D^b(A_X)) = k(D^-(A_X)_{\mathrm{coh}}) = 0,$$

for the last equality observing that $L \in \mathrm{ob}\, D^-(A_X)_{\mathrm{coh}}$ implies

$$\coprod_{i\geq 0} L[2i] \in \mathrm{ob}\, D^-(A_X)_{\mathrm{coh}}.$$

[23] We shall allow ourselves to omit the index A when no confusion can result. We shall sometimes also drop the index "naive", but will then say so explicitly.

2.4. Case of a coherent sheaf of rings. Let (X, A) be a ringed topos, with A assumed coherent as a left A-module (I 3.1 and 3.6). Every coherent left A-module is a pseudo-coherent complex (I 3.5), and one has a fully faithful functor

$$\mathrm{Coh}(A_X) \longrightarrow D^b(A_X)_{\mathrm{coh}}. \tag{2.4.1}$$

Moreover $\mathrm{Coh}(A_X)$ is a thick subcategory of $\mathrm{Mod}(A_X)$ (I 3.4), and, by (I 3.5), $D^b(A_X)_{\mathrm{coh}}$ is the full subcategory of $D^b(A_X)$ formed by the objects L such that $H^i(L) \in \mathrm{ob}\,\mathrm{Coh}(A_X)$ for all i. Consequently, it follows from (1.6) that the homomorphism

$$K_\bullet(A_X)_{\mathrm{naif}} \longrightarrow K_\bullet(A_X) \tag{2.4.2}$$

defined by (2.4.1) is an isomorphism.

2.5. The homomorphism $K' \to K_\bullet$. Let (X, A) be a ringed topos. The inclusion functor

$$D(A_X)_{\mathrm{parf}} \longrightarrow D^b(A_X)_{\mathrm{coh}}$$

defines a homomorphism

$$K'(A_X) \longrightarrow K_\bullet(A_X). \tag{2.5.1}$$

It is not in general injective or surjective. However, assume the following hypotheses:

(i) the final object of X is quasi-compact;

(ii) X has enough points, and for every point x of X, every left A_x-module has finite Tor-dimension.

Then, by (I 5.10),

$$D(A_X)_{\mathrm{parf}} = D^b(A_X)_{\mathrm{coh}},$$

and (2.5.1) is an isomorphism. The hypotheses (i) and (ii) are satisfied, for example, by a compact space ringed by regular local rings.

2.6. Pairings between K' and K' in the non-commutative case. Let

$$f : (X', A') \longrightarrow (X, A)$$

be a morphism of ringed topoi, and denote by $\mathrm{Mod}(A', X', A)$ the category of bimodules which are left A'-modules and right $f^{-1}(A)$-modules. Let

$$\mathrm{Loclib}^b_A(A', X', A)$$

be the full subcategory formed by the objects which belong, as objects over A', to $\mathrm{Loclib}(A', X')$. By analogy with (2.2), put

$$K'_{\mathrm{naif}}(A', X', A)_{A'} = k(\mathrm{Loclib}^b_A(A', X', A)).$$

Tensor product

$$\otimes_A : \mathrm{Mod}(A', X', A) \times \mathrm{Mod}(A_X) \longrightarrow \mathrm{Mod}(A', X')$$

sends

$$(E', E) \longmapsto E' \otimes_{f^{-1}(A)} f^{-1}(E),$$

and induces a functor

$$\mathrm{Loclib}^b_A(A', X', A) \times \mathrm{Loclib}(A_X) \longrightarrow \mathrm{Loclib}(A', X'). \tag{2.6.1}$$

The map $(E', E) \mapsto \mathrm{cl}(E' \otimes_A E)$ deduced from (2.6.1) is biadditive in the sense of (1.5), and therefore defines a homomorphism

$$K'_{\mathrm{naif}}(A', X', A)_{A'} \otimes K'_{\mathrm{naif}}(A_X) \longrightarrow K'_{\mathrm{naif}}(A', X'). \tag{2.6.2}$$

Let $D^-(A', X', A)$ be the derived category of $\mathrm{Mod}(A', X', A)$, and let

$$D^-(A', X', A)_{A'\text{-}\,\mathrm{parf}}$$

be the full subcategory formed by the complexes of finite perfect amplitude relative to A', i.e. as objects of $D^-(A', X')$. Put, as in (2.2),

$$K'(A', X', A)_{A'} = k(D^-(A', X', A)_{A'\text{-}\,\mathrm{parf}}).$$

The inclusion functors define evident homomorphisms

$$K'_{\mathrm{naif}}(A', X', A)_{A'} \longrightarrow K'(A', X', A)_{A'}, \tag{2.6.3.1}$$
$$K'_{\mathrm{naif}}(A_X) \longrightarrow K'(A_X), \tag{2.6.3.2}$$
$$K'_{\mathrm{naif}}(A', X') \longrightarrow K'(A', X'). \tag{2.6.3.3}$$

On the other hand, one knows (I 4.18, 4.19) that the derived tensor functor

$$\otimes^{\mathbf{L}}_A : D^-(A', X', A) \times D^-(A_X) \longrightarrow D^-(A', X')$$

induces an exact functor

$$D^-(A', X', A)_{A'\text{-}\,\mathrm{parf}} \times D(A_X)_{\mathrm{parf}} \longrightarrow D(A', X')_{\mathrm{parf}}. \tag{2.6.4}$$

The latter defines a homomorphism

$$K'(A', X', A)_{A'} \otimes K'(A_X) \longrightarrow K'(A', X'). \tag{2.6.5}$$

Since (2.6.1) is induced by (2.6.4), the homomorphisms (2.6.2) and (2.6.5) are compatible with the homomorphisms (2.6.3.1), (2.6.3.2), and (2.6.3.3), in other words one has a commutative diagram

$$\begin{array}{ccc} K'_{\mathrm{naif}}(A', X', A)_{A'} \otimes K'_{\mathrm{naif}}(A_X) & \to & K'_{\mathrm{naif}}(A', X') \\ \downarrow & & \downarrow \\ K'(A', X', A)_{A'} \otimes K'(A_X) & \longrightarrow & K'(A', X'), \end{array} \tag{2.6.6}$$

where the upper and lower horizontal arrows are respectively (2.6.2) and (2.6.5), and the vertical arrows are defined by (2.6.3.1), (2.6.3.2), and (2.6.3.3).

2.7. Contravariant law and ring structure on K'. We spell out some important particular cases of the pairings (2.6), although their direct study is immediate.

a) Let

$$f : (X', A') \longrightarrow (X, A)$$

be a morphism of ringed topoi. The functor

$$Lf^* : D^-(A_X) \longrightarrow D^-(A', X')$$

sends $D(A_X)_{\mathrm{parf}}$ to $D(A', X')_{\mathrm{parf}}$ (I 4.19.1), and likewise sends $\mathrm{Loclib}(A_X)$ to $\mathrm{Loclib}(A', X')$. Thus it defines a homomorphism

$$f^* : K'(A_X) \longrightarrow K'(A', X') \tag{2.7.1}$$

and, respectively,

$$f^* : K'(A_X)_{\mathrm{naif}} \longrightarrow K'(A', X')_{\mathrm{naif}}. \tag{2.7.2}$$

This can also be interpreted as multiplication by $\mathrm{cl}(A')$ in (2.6.2), respectively (2.6.5). One has a commutative diagram, a special case of (2.6.6),

$$\begin{array}{ccc} K'(A_X)_{\mathrm{naif}} & \xrightarrow{f^*} & K'(A', X')_{\mathrm{naif}} \\ \downarrow & & \downarrow \\ K'(A_X) & \xrightarrow{f^*} & K'(A', X'), \end{array}$$

where the left and right vertical arrows are respectively (2.6.3.2) and (2.6.3.3).

Let $g : (X, A) \to (X'', A'')$ be a morphism of ringed topoi. The transitivity of inverse-image functors implies that the homomorphisms of type (2.7.1), respectively (2.7.2), satisfy

$$(gf)^* = f^* g^*. \tag{2.7.3}$$

Thus $K'(A_X)$, respectively $K'(A_X)_{\mathrm{naif}}$, is a contravariant functor of (X, A).

b) Let $(X, \mathcal{O}_X)$ be a commutatively ringed topos. Thanks to the associativity and commutativity of tensor product, (2.6.5), respectively (2.6.2), with $A = \mathcal{O}_X$, $(X', A') = (X, A)$, and $f = \mathrm{Id}$, endows $K'(X)$, respectively $K'(X)_{\mathrm{naif}}$, with the structure of a commutative ring with unit

$$1 = \mathrm{cl}(\mathcal{O}_X).$$

The canonical homomorphism

$$K'(X)_{\mathrm{naif}} \longrightarrow K'(X)$$

is a homomorphism of unital rings.

If $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of commutatively ringed topoi, compatibility of tensor product with inverse image shows that

$$f^* : K'(Y) \longrightarrow K'(X),$$

respectively

$$f^* : K'(Y)_{\mathrm{naif}} \longrightarrow K'(X)_{\mathrm{naif}},$$

is a homomorphism of unital rings.

2.8. Augmentation of K' toward $H^0(\ , \mathbb{Z})$. Let $(X, \mathcal{O}_X)$ be a locally ringed topos (I 2.15.1). Recall that one has defined (I 6.11 to 6.14) a homomorphism of unital rings

$$i : H^0(X, \mathbb{Z}) \longrightarrow K'(X)_{\mathrm{naif}},$$

thereby endowing $K'(X)_{\mathrm{naif}}$, and then $K'(X)$ through the canonical homomorphism $K'(X)_{\mathrm{naif}} \to K'(X)$, with the structure of a unital algebra over $H^0(X, \mathbb{Z})$. Moreover, $K'(X)_{\mathrm{naif}}$, respectively $K'(X)$, is augmented toward $H^0(X, \mathbb{Z})$ by the rank, and the canonical homomorphism $K'(X)_{\mathrm{naif}} \to K'(X)$ is a homomorphism of augmented algebras.

If $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of locally ringed topoi, the homomorphisms

$$f^* : K'(Y) \longrightarrow K'(X), \qquad f^* : K'(Y)_{\mathrm{naif}} \longrightarrow K'(X)_{\mathrm{naif}}$$

are compatible (I 6.12, 6.13) with the preceding augmented-algebra structures; this is expressed by the commutativity of the diagram, where rg denotes the rank,

$$\begin{array}{ccccccc} H^0(Y,\mathbb{Z}) & \xrightarrow{i} & K'(Y)_{\text{naif}} & \to & K'(Y) & \xrightarrow{\text{rg}} & H^0(Y,\mathbb{Z}) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H^0(X,\mathbb{Z}) & \xrightarrow{i} & K'(X)_{\text{naif}} & \to & K'(X) & \xrightarrow{\text{rg}} & H^0(X,\mathbb{Z}). \end{array}$$

Later in the Seminar (V 3.9.1 and VI) one will see that $K'(X)_{\text{naif}}$ can be endowed with a structure richer than that of an algebra augmented toward $H^0(X,\mathbb{Z})$, namely with a structure of a λ-algebra augmented toward $H^0(X,\mathbb{Z})$, viewed as a binomial ring (V 2.8), this structure being compatible with inverse-image homomorphisms. Deligne has recently proved that $K'(X)$ can be endowed with an analogous structure, at least when X is of characteristic $p \geq 0$, and that the canonical homomorphism

$$K'(X)_{\text{naif}} \longrightarrow K'(X)$$

is a homomorphism of augmented λ-algebras, thereby answering affirmatively a conjecture of Grothendieck (XIV 1).

2.9. Comparison of K' and K'_{naif}. Let (X,A) be a ringed topos. There is in general no reason for the canonical homomorphism

$$K'(X)_{\text{naif}} \longrightarrow K'(X)$$

to be injective or surjective. However, the "global resolution existence criteria" of Exposé II show that it is an isomorphism in the following cases:

(i) X is the Zariski topos of a separated or noetherian divisorial scheme, and $A = \mathcal{O}_X$ (II 2.2.9);

(ii) X is the topos defined by a compact space and A is a soft sheaf (II 2.3.2);

(iii) X is a Stein topos whose final object is quasi-compact and $A = \mathcal{O}_X$ (II 2.4.5).

2.10. Pairing between K' and $K_\bullet$. Let $(X,\mathcal{O}_X)$ be a commutatively ringed topos. One knows (I 5.8.4 b) that there is a pairing

$$\otimes^{\mathbf{L}}_{\mathcal{O}_X} : D(X)_{\text{parf}} \times D^b(X)_{\text{coh}} \longrightarrow D^b(X)_{\text{coh}}.$$

It defines a homomorphism

$$K'(X) \otimes K_\bullet(X) \longrightarrow K_\bullet(X). \tag{2.10.1}$$

Thanks to associativity of tensor product, (2.10.1) endows $K_\bullet(X)$ with the structure of a $K'(X)$-module. More generally, if

$$f : (X,\mathcal{O}_X) \longrightarrow (Y,\mathcal{O}_Y)$$

is a morphism of commutatively ringed topoi, $K_\bullet(X)$ may be regarded as a $K'(Y)$-module through the homomorphism

$$f^* : K'(Y) \longrightarrow K'(X).$$

2.11. Covariant law on $K_\bullet$ and projection formula. Let (X,A) be a ringed topos. In contrast with $K'(A_X)$, the group $K_\bullet(A_X)$ has no natural variance with respect to (X,A). Nevertheless, in algebraic geometry and analytic geometry, $K_\bullet(X)$ tends to be covariant in X for proper morphisms. We now make this precise.

2.11.1. Let $f : X \to Y$ be a proper and pseudo-coherent morphism (III 1.2) of quasi-compact schemes. Then, subject to the validity of the finiteness conjecture for f (III 2.1), hence in particular if Y is noetherian (in which case the pseudo-coherence hypothesis on f is automatically verified (III 1.3)) or if f is projective, the functor Rf_* induces, by (III 2.6), a functor

$$Rf_* : D^b(X)_{\mathrm{coh}} \longrightarrow D^b(Y)_{\mathrm{coh}}.$$

It therefore defines a homomorphism

$$f_* : K_\bullet(X) \longrightarrow K_\bullet(Y). \tag{2.11.1.1}$$

This homomorphism is $K'(Y)$-linear (cf. (2.10)); in other words, for $y \in K'(Y)$ and $x \in K_\bullet(X)$, one has the *projection formula*

$$f_*(yx) = y f_*(x). \tag{2.11.1.2}$$

Indeed, if $x = \mathrm{cl}(E)$ for $E \in \mathrm{ob}\, D^b(X)_{\mathrm{coh}}$ and $y = \mathrm{cl}(M)$ for $M \in \mathrm{ob}\, D(Y)_{\mathrm{parf}}$, then by (III 3.7) one has

$$Rf_*(E) \otimes_Y^{\mathbf{L}} M \simeq Rf_*(E \otimes_Y^{\mathbf{L}} M).$$

Finally, if $g : Y \to Z$ is a proper morphism satisfying the same hypotheses as f, and for which the finiteness conjecture (III 2.1) is true, the transitivity of direct-image functors gives, for the homomorphisms (2.11.1.1), the formula

$$g_* f_* = (gf)_*. \tag{2.11.1.3}$$

2.11.2. Let $f : X \to Y$ be a proper morphism of finite-dimensional analytic spaces. Then, by Grauert's finiteness theorem [2], Rf_* sends $D^b(X)_{\mathrm{coh}}$ into $D^b(Y)_{\mathrm{coh}}$, and hence defines a homomorphism

$$f_* : K_\bullet(X) \longrightarrow K_\bullet(Y). \tag{2.11.2.1}$$

Like its algebraic analogue (2.11.1.1), this homomorphism is $K'(Y)$-linear; equivalently, formula (2.11.1.2) is valid, as follows from the projection formula at the level of derived categories (III 5.9 b). One also has the transitivity formula (2.11.1.3).

2.11.3. Outside the algebraic or analytic framework, this covariance of $K_\bullet$ for proper morphisms disappears. For instance, let

$$f : X \to Y$$

be a proper morphism of topological spaces ringed by the sheaves of complex continuous functions. Then, if $E \in \mathrm{ob}\, D^b(X)_{\mathrm{coh}}$, in general $Rf_*(E)$ need not belong to $D^b(Y)_{\mathrm{coh}}$, as is already evident for $E = \mathcal{O}_X$ and Y reduced to a point. In fact, the definition (2.2) of $K_\bullet$ is especially adapted to the needs of algebraic and analytic geometry, and a good definition of $K_\bullet$ for a topological space remains to be found. It seems, moreover, that one will have to leave the framework of $\mathcal{O}_X$-linear complexes, and define, for a morphism $f : X \to Y$ of topological spaces, or of mixed varieties (II App. II), a suitable class of complexes of topological vector-sheaves on X, meant to contain, when f is the structural morphism of a mixed variety, the elliptic complexes on X relative to Y. This is suggested by the finiteness theorem (II App. II), as well as by the analogous theorem, where Y is assumed reduced to a point, proved by Grothendieck in [3].

2.12. Unusual variances of K' and $K_\bullet$. Let

$$f : (X, A) \longrightarrow (Y, B)$$

be a morphism of finite Tor-dimension of ringed topoi (III 3.1). The functor

$$Lf^* : D^-(B_Y) \longrightarrow D^-(A_X)$$

sends $D^b(B_Y)_{\mathrm{coh}}$ into $D^b(A_X)_{\mathrm{coh}}$ (I 2.16.1), and therefore defines a homomorphism

$$f^* : K_\bullet(B_Y) \longrightarrow K_\bullet(A_X). \tag{2.12.1}$$

If A and B are commutative, commutation of inverse image with tensor product shows that (2.12.1) is compatible with the module structures of $K_\bullet(Y)$ and $K_\bullet(X)$ over $K'(Y)$ and $K'(X)$ respectively (2.10). In other words, for $a \in K'(Y)$ and $y \in K_\bullet(Y)$, one has

$$f^*(ay) = af^*(y). \tag{2.12.2}$$

If $g : (Y, B) \to (Z, C)$ is a morphism of finite Tor-dimension of ringed topoi, transitivity of inverse-image functors implies that the homomorphisms (2.12.1) satisfy

$$f^*g^* = (gf)^*. \tag{2.12.1.1}$$

Now suppose that we are in the situation of (2.11.1), respectively (2.11.2), with f of finite Tor-dimension. Then by (III 4.8.1 and 5.9 b), the functor Rf_* sends $D(X)_{\mathrm{parf}}$ into $D(Y)_{\mathrm{parf}}$, and hence defines a homomorphism

$$f_* : K'(X) \longrightarrow K'(Y). \tag{2.12.3}$$

This is not generally a ring homomorphism. Nevertheless, for $y \in K_\bullet(Y)$, respectively $K'(Y)$, and $x \in K'(X)$, one has

$$f_*(f^*(y)x) = yf_*(x), \tag{2.12.4}$$

where f^* is the homomorphism (2.12.1), respectively (2.7.1). The proof is analogous to that of (2.11.1.2). If $g : Y \to Z$ is a morphism satisfying the same hypotheses as f, the homomorphisms (2.12.3) satisfy the transitivity formula

$$g_*f_* = (gf)_*. \tag{2.12.3.1}$$

The homomorphism (2.12.3) is one of the essential ingredients of the Riemann–Roch theorem, both in algebraic geometry and in analytic geometry, for the indications in the analytic case see Exposé 0, 5.

Suppose that X and Y are separated and divisorial schemes (II 2.2.5), for example schemes possessing ample invertible sheaves. Then (2.9 (i)) identifies $K'(X)$, respectively $K'(Y)$, with $K'(X)_{\mathrm{naif}}$, respectively $K'(Y)_{\mathrm{naif}}$, and (2.12.3) therefore defines a homomorphism

$$f_* : K'(X)_{\mathrm{naif}} \longrightarrow K'(Y)_{\mathrm{naif}}. \tag{2.12.3'}$$

As was already explained in Exposé 0, 3.1, the definition of (2.12.3') was far from evident a priori. In practice, it is precisely one of the main motivations of this Seminar, from Exposé I through Exposé IV.

3. Complements on Grothendieck groups of schemes

The results of this section will not be used in Exposés V to VIII.

Proposition 3.1.0. Let

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[h]{} & Y \end{array}$$

be a cartesian square of schemes, where Y is quasi-compact and f is quasi-compact and quasi-separated. Let $E \in \mathrm{ob}\, D^b(X)$ be a complex with quasi-coherent cohomology. Suppose that X and Y' are Tor-independent over Y (III 1.5). Suppose also either that h has finite Tor-dimension, or that E has finite flat amplitude relative to f. Under these conditions, the base-change arrow

$$Lh^*\, Rf_*E \longrightarrow Rf'_*\, Lg^*E$$

is defined and is an isomorphism.

Proof. To define the arrow, one uses the trivial duality formula (SGA 4 XVII 2.3) for h, which says that there is a canonical functorial isomorphism

$$\mathrm{Hom}(Lh^*L, M) \xrightarrow{\sim} \mathrm{Hom}(L, Rh_*M)$$

for $L \in \mathrm{ob}\, D^-(Y)$ and $M \in \mathrm{ob}\, D^+(Y')$. Using (III 3.5.2) and (III 3.7.1), one easily verifies that one may take $L = Rf_*E$ and $M = Rf'_*Lg^*E$, so that applying this formula reduces the problem to defining an arrow

$$Rf_*E \longrightarrow Rf_*Rg_*Lg^*E.$$

For this arrow we take the image under Rf_* of the adjunction map

$$E \longrightarrow Rg_*Lg^*E,$$

which comes from the trivial duality formula for g. The base-change arrow is therefore defined.

It remains to show that it is an isomorphism. For this, one may first suppose that Y and Y' are affine. Then, by the Leray spectral sequence of an open covering of X, one is reduced to the case where X and X' are affine. One thus has a cartesian square of rings

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A', \end{array}$$

and one may suppose that E is defined by a bounded-above complex of B-modules. The Tor-independence of A' and B over A, written

$$B \otimes^{\mathbf{L}}_A A' \xrightarrow{\sim} B',$$

then implies, in $D(A')$,

$$E \otimes^{\mathbf{L}}_A A' \xrightarrow{\sim} E \otimes^{\mathbf{L}}_B B',$$

which proves the proposition.

Translating (3.1.0) into terms of Grothendieck groups gives:

Proposition 3.1.1. In the cartesian square of schemes

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[h]{} & Y, \end{array}$$

suppose that Y and Y' are quasi-compact, that X and Y' are Tor-independent over Y (III 1.5), that f is proper and pseudo-coherent, and that the finiteness conjecture (III 2.1) is valid for f and f', for example if Y and Y' are noetherian, or if f is projective. Then, if f is perfect, respectively if h has finite Tor-dimension, for $x \in K'(X)$, respectively $x \in K_\bullet(X)$, one has

$$h^* f_*(x) = f'_* g^*(x),$$

where h^*, g^* are defined by (2.7.1), respectively (2.12.1), and f_*, f'_* by (2.12.3), respectively (2.11.1.1). Notice that by (III 4.7.1) f' is perfect if f is, and by (III 3.5.1) g has finite Tor-dimension if h does.

3.2. Passage to the limit on K'_{naif} and $K_\bullet$. Let $(X_i, f_{ij})_{i\in I}$ be a filtered projective system of quasi-compact and quasi-separated schemes whose transition morphisms are affine (cf. EGA IV 8.2.2). Put

$$X = \varprojlim_i X_i,$$

and let $f_i : X \to X_i$ denote the canonical morphism.

3.2.1. The homomorphisms (2.7.2)

$$f_i^* : K'(X_i)_{\mathrm{naif}} \longrightarrow K'(X)_{\mathrm{naif}}$$

define, by the transitivity formula (2.7.3), a homomorphism

$$\varinjlim_i K'(X_i)_{\mathrm{naif}} \longrightarrow K'(X)_{\mathrm{naif}}. \tag{3.2.1.1}$$

I say that this homomorphism is an isomorphism. Indeed, let E be a locally free $\mathcal{O}_X$-module of finite type. By EGA IV 8.5.2 and 8.5.5, there exists an index i and a locally free $\mathcal{O}_{X_i}$-module of finite type E_i such that

$$E \simeq f_i^*(E_i).$$

By EGA IV 8.5.2.5, the image of $\mathrm{cl}(E_i)$ in $\varinjlim_i K'(X_i)_{\mathrm{naif}}$ does not depend on the choice of E_i; denote it by $u(E)$. By EGA IV 8.5.2 and 8.5.6, the map

$$u : \mathrm{ob}\,\mathrm{Loclib}(X) \longrightarrow \varinjlim_i K'(X_i)_{\mathrm{naif}}$$

so defined is additive in the sense of (1.5). It therefore defines a homomorphism

$$K'(X)_{\mathrm{naif}} \longrightarrow \varinjlim_i K'(X_i)_{\mathrm{naif}},$$

and one verifies immediately, using EGA IV 8.5.2, that it is inverse to (3.2.1.1). This proves the assertion.

3.2.2. Likewise one defines a canonical homomorphism

$$\varinjlim_i K'(X_i) \longrightarrow K'(X). \tag{3.2.2.1}$$

Using Deligne's localization technique (SGA 4 XVII), which implies that the category of perfect complexes on X is equivalent to the category

$$\varinjlim_i \mathrm{Parf}(X_i)$$

of the analogous categories of perfect complexes on the X_i, one proves again that (3.2.2.1) is an isomorphism.

3.2.3. Suppose that the X_i, and X, are noetherian, and that the morphisms f_i are flat (then so are the f_{ij}). Then the homomorphisms (2.12.1)

$$f_i^* : K_\bullet(X_i) \longrightarrow K_\bullet(X)$$

define an isomorphism

$$\varinjlim_i K_\bullet(X_i) \xrightarrow{\sim} K_\bullet(X).$$

The proof is analogous to that of (3.2.1), using the fact (2.4) that $K_\bullet(X) \simeq K_\bullet(X)_{\mathrm{naif}}$ is computed in terms of coherent sheaves on X.

Moreover, the module structures of the $K_\bullet(X_i)$ over the $K'(X_i)_{\mathrm{naif}}$ are compatible with the inverse images f_{ij}^* by (2.7.1) and (2.12.2). Passing to the limit, one finds that $\varinjlim_i K_\bullet(X_i)$ is a module over $\varinjlim_i K'(X_i)_{\mathrm{naif}}$; this is compatible, through the isomorphisms (3.2.1.1) and (3.2.2.1), with the module structure of $K_\bullet(X)$ over $K'(X)_{\mathrm{naif}}$. In other words, one has a commutative diagram

$$\begin{array}{ccc} \varinjlim K'(X_i)_{\mathrm{naif}} \otimes \varinjlim K_\bullet(X_i) & \to & \varinjlim K_\bullet(X_i) \\ \downarrow \sim & & \downarrow \sim \\ K'(X)_{\mathrm{naif}} \otimes K_\bullet(X) & \longrightarrow & K_\bullet(X). \end{array}$$

3.3. Relative Grothendieck groups.

3.3.1. Let

$$f : X \longrightarrow Y$$

be a morphism of schemes locally of finite type. Put

$$K'(f) = k(D(f)_{\mathrm{parf}}), \qquad K_\bullet(f) = k(D^b(f)_{\mathrm{coh}}),$$

where $D^b(f)_{\mathrm{coh}}$ denotes the category of complexes which are pseudo-coherent relative to f and have bounded cohomology (III 1.2), and $D(f)_{\mathrm{parf}}$ the category of complexes of finite perfect amplitude relative to f (III 4.2). When there is no possible ambiguity, we shall sometimes write $K'(X/Y)$, respectively $K_\bullet(X/Y)$, instead of $K'(f)$, respectively $K_\bullet(f)$.

The groups $K'(f)$ and $K_\bullet(f)$ do not possess the rich structure of their absolute analogues $K'(X)$ and $K_\bullet(X)$, for tensor product does not in general define homomorphisms

$$K'(f) \otimes K'(f) \longrightarrow K'(f), \qquad K'(f) \otimes K_\bullet(f) \longrightarrow K_\bullet(f).$$

On the other hand, they satisfy interesting variance properties, which we now review.

3.3.2. Let

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle h}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array}$$

be a commutative triangle of morphisms locally of finite type of schemes.

3.3.2.1. If g is pseudo-coherent, one has

$$K_\bullet(f) = K_\bullet(h)$$

(III 1.12). If g is smooth and X quasi-compact, one has

$$K'(f) = K'(h)$$

(III 4.4.1).

3.3.2.2. By (III 1.11 and 4.5), the functor $\otimes_Y^{\mathbf{L}}$ defines pairings

$$D(f)_{\mathrm{parf}} \times D^b(g)_{\mathrm{coh}} \longrightarrow D^b(h)_{\mathrm{coh}},$$

$$D(f)_{\mathrm{parf}} \times D(g)_{\mathrm{parf}} \longrightarrow D(h)_{\mathrm{parf}},$$

which induce homomorphisms

$$K'(f) \otimes K_\bullet(g) \longrightarrow K_\bullet(h),$$

$$K'(f) \otimes K'(g) \longrightarrow K'(h).$$

In particular, taking $X = Y$ and $f = \mathrm{Id}$, one deduces that $K_\bullet(g)$ is a $K'(Y)$-module. On the other hand, if f is perfect, product by

$$\mathrm{cl}(\mathcal{O}_X) \in K'(f)$$

defines $K'(Y)$-linear homomorphisms

$$f^* : K_\bullet(Y/S) \longrightarrow K_\bullet(X/S),$$

and

$$f^* : K'(Y/S) \longrightarrow K'(X/S).$$

3.3.2.3. Suppose S quasi-compact, X and Y of finite type over S, and f proper. Then, subject to the validity of the finiteness conjecture (III 2.1), in particular if Y is noetherian or if f is projective, the functor Rf_* sends $D^b(h)_{\mathrm{coh}}$ into $D^b(g)_{\mathrm{coh}}$, and therefore defines a homomorphism

$$f_* : K_\bullet(X/S) \longrightarrow K_\bullet(Y/S).$$

If in addition f is perfect, Rf_* sends $D(h)_{\mathrm{parf}}$ into $D(g)_{\mathrm{parf}}$ (III 4.8), and defines a homomorphism

$$f_* : K'(X/S) \longrightarrow K'(Y/S).$$

From the projection formula (III 3.7) one deduces

$$f_*(ax) = af_*(x),$$

for f proper, $a \in K'(S)$, $x \in K_\bullet(X)$, respectively for f proper and perfect, $a \in K_\bullet(S)$, $x \in K'(X/S)$.

3.3.2.4. Suppose S noetherian, X and Y of finite type over S, and f compactifiable and perfect. Then by (III 4.9.1), the functor $f^!$ sends $D(g)_{\mathrm{parf}}$ into $D(h)_{\mathrm{parf}}$, and defines a homomorphism

$$f^! : K'(Y/S) \longrightarrow K'(X/S).$$

3.3.3. Under the hypotheses of (III 4.7), the functor $\otimes^{\mathbf{L}}_S$ defines pairings

$$D(f_1)_{\mathrm{parf}} \times D^b(f_2)_{\mathrm{coh}} \longrightarrow D^b(f)_{\mathrm{coh}},$$

$$D(f_1)_{\mathrm{parf}} \times D(f_2)_{\mathrm{parf}} \longrightarrow D(f)_{\mathrm{parf}},$$

which induce homomorphisms

$$K'(f_1) \otimes K_\bullet(f_2) \longrightarrow K_\bullet(f),$$

$$K'(f_1) \otimes K'(f_2) \longrightarrow K'(f).$$

Of course, for $X_i = Y_i = S$, one recovers the $K'(S)$-module structure on $K_\bullet(S)$.

If X is flat over S, one deduces, for $n \geq 1$, homomorphisms

$$K'(X/S)^{\otimes n} \longrightarrow K'((X/S)^n),$$

which should, in a sense, replace the absent ring structure on $K'(X/S)$.

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Exposé V. Generalities on λ-rings

by P. Berthelot

The theory of λ-rings is useful whenever one uses rings such as the ring of classes of vector bundles on a ringed site, the representation ring of a group, and so on, for which the exterior-power operations define maps λ^i. We shall use it abundantly later in the Seminar. It is therefore necessary first to develop the general formalism. This exposé is accordingly entirely independent of the rest of the Seminar, and may be useful to users of λ-rings outside algebraic geometry.

The first paragraph essentially gives the theory of Hirzebruch's "multiplicative sequences of polynomials" (cf. [2]). Besides its generality, the functorial viewpoint adopted here has the advantage of considerably reducing the calculations. The following paragraphs will show how, to every augmented λ-ring, one can associate a "Chern ring" which permits the development of a theory of Chern classes. This is the theory which will allow us to state the Riemann–Roch theorem in the general form of Exposé VIII.

1. Universal polynomials

1.1. Let K be a commutative ring with unit, fixed throughout this paragraph. Let $\mathcal{C}$ denote the category of filtered commutative K-algebras with unit, whose filtration is decreasing and such that the group of elements of filtration zero is the whole algebra. We define on $\mathcal{C}$ a functor with values in the category of abelian groups as follows. If $A \in \mathrm{Ob}(\mathcal{C})$, associate to A the set of formal series $\psi \in A[[t]]$ satisfying the following conditions: if

$$\psi = a_0 + a_1 t + \cdots + a_n t^n + \cdots ,$$

then

$$a_0 = 1, \qquad \forall i \geq 1, \quad a_i \in \mathrm{Fil}^i A. \tag{1.1.1}$$

This set will be denoted $\widehat{G}_K(A)$, or $\widehat{G}(A)$ when no confusion is possible. It is clearly an abelian group for multiplication of formal series, and this defines a functor

$$\widehat{G}_K : K\text{-Alg.fil.} \longrightarrow \mathrm{Ab}.$$

If A has the gross filtration, then $\widehat{G}(A)$ is the group of formal series with augmentation 1, which will also be denoted

$$1 + A[[t]]^+.$$

Lemma 1.2. The functor $\widehat{G}_K$ is representable.

Let $(X_i)_{i\in\mathbb{N},\ i\geq 1}$ be a family of indeterminates. Assign to X_i the weight i, and filter the ring $K[X_i]_{i\geq 1}$ by weights with respect to the X_i. One then obtains a bijection

$$\widehat{G}_K(A) \xrightarrow{\sim} \mathrm{Hom}_{K\text{-Alg.fil.}}(K[X_i]_{i\geq 1}, A)$$

by associating to every series

$$\psi = 1 + a_1 t + \cdots + a_n t^n + \cdots \in \widehat{G}_K(A)$$

the morphism f_ψ determined by $f_\psi(X_i) = a_i$. This homomorphism is compatible with the filtrations, and one thus obtains an isomorphism of functors.

Put

$$\varphi = 1 + X_1 t + \cdots + X_n t^n + \cdots .$$

For every filtered K-algebra A and every series $\psi \in \widehat{G}(A)$, there exists a unique homomorphism

$$f_\psi \in K[X_i]_{i\geq 1} \longrightarrow A$$

such that

$$\widehat{G}(f_\psi)(\varphi) = \psi.$$

1.3. For every $A \in \mathrm{Ob}(\mathcal{C})$ we define a filtration on $\widehat{G}(A)$ as follows: a series

$$\psi = 1 + a_1 t + \cdots$$

is said to be of filtration $\geq k$ if $a_i = 0$ for $1 \leq i < k$. Let $F^k\widehat{G}(A)$ be the subgroup so obtained. This is a functor on $\mathcal{C}$, and from now on we shall regard $\widehat{G}$ as a functor with values in the category Ab fil of filtered abelian groups by means of this filtration, with $F^0\widehat{G} = \widehat{G}$.

Let H be a functor on $\mathcal{C}$ with values in Ab fil. We shall suppose that H satisfies the following two conditions:

i) $\forall A \in \mathrm{Ob}(\mathcal{C})$, $H(A)$ is separated and complete,

ii) if $A \in \mathrm{Ob}(\mathcal{C})$ is a graded K-algebra, with the filtration associated to its grading, and if a finite group G acts on A by graded morphisms, then $H(A^G) \simeq H(A)^G$.

Here $H(A)^G$ denotes the invariants under G.

In fact we shall need these conditions only when A is a polynomial ring and G is a finite product of symmetric groups acting by permutation of the indeterminates.

Examples.

1) $H(A) = \widehat{G}(A)$,
2) $H(A) = 1 + A[[t]]^+$.

The latter functor will be denoted $\widehat{G}_g$, with g standing for "gross". Finally,

3) $H(A) = \widehat{G}_a(A)$,

where $\widehat{G}_a(A)$ denotes the group, for addition, of formal series

$$a = a_0 + a_1 t + \cdots$$

such that $a_0 = 0$ and $a_i \in \mathrm{Fil}^i A$ for $i \geq 1$, endowed with the filtration analogous to that on $\widehat{G}$.

1.4. For every n, denote by f_n the endomorphism of $K[X_1, \ldots, X_n]$ defined by

$$f_n(X_i) = X_i \quad (i < n), \qquad f_n(X_n) = 0.$$

It factors through the inclusion of $K[X_1, \ldots, X_{n-1}]$ in $K[X_1, \ldots, X_n]$. Hence the homomorphism

$$H(K[X_1, \ldots, X_{n-1}]) \longrightarrow H(K[X_1, \ldots, X_n])$$

is injective. This will allow us to identify the first group with a subgroup of the second; similarly, we shall regard $H(K[X_1, \ldots, X_n])$ as contained in $H(K[X_i]_{i \geq 1})$.

One should note that f_1 is none other than the augmentation of $K[X]$.

1.5. Embed $K[X_1, \ldots, X_n]$ in $K[T_1, \ldots, T_n]$ by sending X_i to the i-th elementary symmetric function in the T_j. Let the symmetric group Sym_n act by permutation of the T_j. Using the theorem on symmetric polynomials and (1.3.ii), one sees that the induced homomorphism

$$H(K[X_1, \ldots, X_n]) \longrightarrow H(K[T_1, \ldots, T_n])$$

is injective, which again permits us to identify the former with a subgroup of the latter:

$$H(K[X_1, \ldots, X_n]) \simeq H(K[T_1, \ldots, T_n])^{\mathrm{Sym}_n} \subset H(K[T_1, \ldots, T_n]).$$

Let

$$u_i^n : K[X] \longrightarrow K[T_1, \ldots, T_n]$$

be the homomorphism defined by $u_i^n(X) = T_i$, and put

$$u_H^n = \sum_{i=1}^{n} H(u_i^n) : H(K[X]) \longrightarrow H(K[T_1, \ldots, T_n]).$$

The symmetric group permutes the morphisms $H(u_i^n)$, and so leaves u_H^n invariant; thus u_H^n takes its values in

$$H(K[T_1, \ldots, T_n])^{\mathrm{Sym}_n} \simeq H(K[X_1, \ldots, X_n])$$

by (1.3.ii). Notice, finally, that if $H = \widehat{G}$, then

$$u_H^n(1 + Xt) = \prod_{i=1}^{n} (1 + T_i t) = 1 + X_1 t + \cdots + X_n t^n.$$

Theorem 1.6. The map from $\mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G}, H)$ to $H(K[X])$ which, to a morphism of functors, associates its value on the element $1 + Xt$ of $\widehat{G}(K[X])$, is injective. Its image is the subgroup of $H(K[X])$ formed by the elements θ such that, with the notation of 1.4 for the f_n,

i) $H(f_1)(\theta) = 0$,

ii) $\forall n > 1, \quad H(f_n)(u_H^n(\theta)) \equiv u_H^n(\theta) \pmod{F^n H(K[X_1, \ldots, X_n])}$.

First prove injectivity. By Lemma 1.2 it is enough to see that a morphism of functors ξ which is zero on $1 + Xt$ is zero on

$$\varphi = 1 + X_1 t + \cdots + X_n t^n + \cdots .$$

The commutative diagram

$$\begin{array}{ccc} \widehat{G}(K[X]) & \xrightarrow{u^n_{\widehat{G}}} & \widehat{G}(K[X_1, \ldots, X_n]) \\ \xi \big\downarrow & & \big\downarrow \xi \\ H(K[X]) & \xrightarrow[u_H^n]{} & H(K[X_1, \ldots, X_n]) \end{array} \tag{1.6.1}$$

shows that, for every n,

$$\xi(1 + X_1 t + \cdots + X_n t^n) = 0.$$

But

$$\varphi \equiv 1 + X_1 t + \cdots + X_n t^n \pmod{F^{n+1}\widehat{G}},$$

hence

$$\xi(\varphi) \equiv \xi(1 + X_1 t + \cdots + X_n t^n) \pmod{F^{n+1}H};$$

therefore $\xi(\varphi) \equiv 0 \pmod{F^{n+1}H}$ for every n. Since $H(K[X_i]_{i\geq 1})$ is separated and complete, $\xi(\varphi) = 0$.

Now let ξ be any element of $\mathrm{Hom}(\widehat{G}, H)$, and set

$$\theta = \xi(1 + Xt).$$

We show that θ satisfies conditions i) and ii). First,

$$H(f_1)(\theta) = H(f_1)\xi(1 + Xt) = \xi\, \widehat{G}(f_1)(1 + Xt) = \xi(1) = 0.$$

Second,

$$u_H^n(\theta) = \xi(1 + \cdots + X_n t^n)$$

by (1.6.1), while

$$1 + \cdots + X_n t^n \equiv 1 + \cdots + X_{n-1} t^{\,n-1} \pmod{F^n \widehat{G}},$$

and

$$1 + \cdots + X_{n-1} t^{\,n-1} = \widehat{G}(f_n)(1 + \cdots + X_n t^n).$$

Since $H \circ \widehat{G}(f_n) = H(f_n) \circ \xi$, and since ξ is a morphism of filtered functors, condition ii) follows.

It remains to show that if $\theta \in H(K[X])$ satisfies i) and ii), then there exists $\xi \in \mathrm{Hom}(\widehat{G}, H)$ with

$$\theta = \xi(1 + Xt).$$

First determine the value of ξ on the element $1 + \cdots + X_n t^n$ of $\widehat{G}(K[X_1, \ldots, X_n])$. In view of (1.6.1), it must be, by definition,

$$\xi(1 + \cdots + X_n t^n) = \sum_{i=1}^{n} \xi(1 + T_i t),$$

where the $\xi(1 + T_i t)$ are known by functoriality from the morphisms u_i^n.

One then glues the $\xi(1 + \cdots + X_n t^n)$, for variable n, by means of the commutative diagram

$$\begin{array}{ccc} K[X_1, \ldots, X_n] & \xrightarrow{f_n} & K[X_1, \ldots, X_n] \\ {\scriptstyle \sigma_n}\downarrow & & \downarrow{\scriptstyle \sigma'_n} \\ K[T_1, \ldots, T_n] & \xrightarrow[f'_n]{} & K[T_1, \ldots, T_n], \end{array} \tag{1.6.2}$$

where f'_n is determined by $f'_n(T_i) = T_i$ if $i < n$, and $f'_n(T_n) = 0$; where σ_n sends X_i to the i-th elementary symmetric function in $T_1, \ldots, T_n$; and where σ'_n sends X_i to the i-th elementary symmetric function in $T_1, \ldots, T_{n-1}$ for $i < n$, and sends X_n to 0.

Indeed one obtains from this

$$\begin{aligned} H(f_n)(\xi(1 + \cdots + X_n t^n)) &= H(f'_n)\left(\sum_{i=1}^{n} \xi(1 + T_i t)\right) \\ &= \sum_{i=1}^{n-1} \xi(1 + T_i t) = \xi(1 + \cdots + X_{n-1} t^{\,n-1}), \end{aligned}$$

inside $H(K[T_1,\dots,T_{n-1}])$ by the identifications of 1.4. By 1.6.ii the elements $\xi(1+\cdots+X_nt^n)$ define an element ψ of $H(K[X_i]_{i\geq 1})$, since this is a separated complete group. Put

$$\xi(\varphi)=\psi.$$

Finally, if $A\in \mathrm{Ob}(\mathcal{C})$ and $x\in \widehat{G}(A)$, define

$$\xi_A(x)=H(f_x)(\psi),$$

where $f_x : K[X_i]_{i\geq 1}\to A$ is the homomorphism defined in 1.2. Since ξ has already been defined on the elements $1+\cdots+X_nt^n=\varphi_n$, one must verify that the two definitions give the same result, i.e.

$$H(f_{\varphi_n})(\psi)=\xi(1+\cdots+X_nt^n).$$

But for every p one has

$$\psi\equiv \xi(1+\cdots+X_pt^p)\pmod{F^{p+1}H(K[X_i]_{i\geq 1})}.$$

Thus, for $p\geq n$,

$$H(f_{\varphi_n})(\psi)\equiv H(f_{\varphi_n})(\xi(1+\cdots+X_pt^p))\pmod{F^{p+1}H(K[X_i]_{i\geq 1})}.$$

Using the identifications of 1.4 and the gluing of the $\xi(\varphi_m)$ for variable m, one obtains

$$H(f_{\varphi_n})(\xi(\varphi_p))=\xi(\varphi_n).$$

Hence finally

$$\forall p\geq n,\quad H(f_{\varphi_n})(\psi)\equiv \xi(1+\cdots+X_nt^n)\pmod{F^pH(K[X_1,\dots,X_n])}.$$

Since H is separated, the required equality follows.

We have thus defined a system of maps

$$\xi_A : \widehat{G}(A)\longrightarrow H(A).$$

To verify that they constitute a morphism of functors with values in groups, one must prove first that, for every morphism $h : A\to A'$, the square

$$\begin{array}{ccc}\widehat{G}(A) & \xrightarrow{\widehat{G}(h)} & \widehat{G}(A')\\ {\scriptstyle\xi_A}\downarrow & & \downarrow{\scriptstyle\xi_{A'}}\\ H(A) & \xrightarrow[H(h)]{} & H(A')\end{array}$$

is commutative. If $x\in\widehat{G}(A)$, then $h\circ f_x=f_{\widehat{G}(h)(x)}$, whence

$$H(h)(\xi_A(x))=H(h)H(f_x)(\psi)=H(f_{\widehat{G}(h)(x)})(\psi)=\xi_{A'}(\widehat{G}(h)(x)).$$

It remains only to show that the morphism of functors so constructed is compatible with the structures of filtered groups.

Suppose $x\in F^n\widehat{G}(A)$. To say that x is of filtration $>n$ is to say that $f_x : K[X_i]_{i\geq 1}\to A$ factors through the endomorphism g_n of $K[X_i]_{i\geq 1}$ defined by

$$g_n(X_i)=0\quad(1\leq i\leq n-1),\qquad g_n(X_i)=X_i\quad(i\geq n).$$

To show that ξ is compatible with filtrations, it is enough to show that

$$H(g_n)(\xi(\varphi)) \in F^n H.$$

But by construction

$$\xi(\varphi) \equiv \xi(1 + \cdots + X_{n-1}t^{n-1}) \pmod{F^n H}.$$

Therefore

$$H(g_n)(\xi(\varphi)) = H(g_n)(\xi(1 + \cdots + X_{n-1}t^{n-1})) \equiv 0 \pmod{F^n H}.$$

To verify the additivity of ξ, it suffices to do this for the elements of finite degree of $\widehat{G}(A)$, and then pass to the limit. Let X_1, X_2 be two elements of degree n of $\widehat{G}(A)$, and let X be their product. Consider the commutative diagram

$$\begin{array}{ccc} K[X_1, \ldots, X_{2n}] & \xrightarrow{f_X} & A \\ \downarrow & \nearrow_{f'_X} & \\ (K[T_1, \ldots, T_{2n}])^{\mathrm{Sym}_n \times \mathrm{Sym}_n} & & \end{array}$$

where the first factor of $\mathrm{Sym}_n \times \mathrm{Sym}_n$ permutes $T_1, \ldots, T_n$ and the second permutes $T_{n+1}, \ldots, T_{2n}$. The ring $K[X_1, \ldots, X_{2n}]$ is embedded into the ring of invariants by sending X_j to the j-th elementary symmetric function of the T_i; f' sends the i-th elementary symmetric function of $T_1, \ldots, T_n$, respectively of $T_{n+1}, \ldots, T_{2n}$, to the i-th coefficient of X_1, respectively X_2. Then

$$\xi(X) = H(f'_X)\left(\xi\left(\prod_{i=1}^{2n}(1 + T_i t)\right)\right),$$
$$\xi(X_1) = H(f'_X)\left(\xi\left(\prod_{i=1}^{n}(1 + T_i t)\right)\right),$$

and similarly for X_2, using $i = n+1, \ldots, 2n$. Since $H(f'_X)$ is a homomorphism of groups, we are reduced to the case

$$X = \prod_{i=1}^{2n}(1 + T_i t),$$

and in this case the additivity of ξ follows from the definition of the $\xi(1 + X_1 t + \cdots + X_k t^k)$.

1.7. One can generalize the preceding considerations to the case of a product of functors $\widehat{G}$. The functor $(\widehat{G})^k$ is representable by the ring

$$K[X_i^{(j)}]_{j=1,\ldots,k;\ i\geq 1},$$

where $X_i^{(j)}$ has weight i. We denote by $\varphi^{(j)}$ the series

$$1 + X_1^{(j)} t + \cdots + X_n^{(j)} t^n + \cdots .$$

The element $(\varphi^{(j)})_j$ of

$$\left(\widehat{G}(K[X_i^{(j)}])\right)^k$$

has the universal property analogous to that of φ (cf. 1.2).

One can introduce the morphisms $f_n^{(j)}$ and $u_H^{n,(j)}$ as above, by repeating the construction for each family $(X_i^{(j)})_{i\geq 1}$ separately. One obtains a result analogous to Theorem 1.6 by considering

morphisms of functors from $(\widehat{G})^k$ to H which are $\mathbb{Z}$-multilinear and compatible with the filtrations in each of the k arguments separately. They are determined by their value on

$$(1 + X^{(j)}t)_{j=1,\dots,k} \in \big(\widehat{G}(K[X^{(j)}]_j)\big)^k.$$

An element

$$\theta \in H(K[X^{(j)}]_j)$$

determines such a morphism of functors if and only if it satisfies

$$\begin{aligned} &\text{i)} \quad \forall j, \quad H(f_1^{(j)})(\theta) = 0, \\ &\text{ii)} \quad \forall n, j, \quad H(f_n^{(j)})(u_H^{n,(j)}(\theta)) \equiv u_H^{n,(j)}(\theta) \pmod{F^n H}. \end{aligned}$$

The reader will easily convince himself that, apart from notational complications, it is enough to copy the proof of 1.6.

1.8. Let $\mathcal{C}_0$ be the category of K-algebras, and let

$$i : \mathcal{C}_0 \longrightarrow \mathcal{C}$$

be the functor which assigns to a K-algebra A the K-algebra A endowed with the gross filtration. Let

$$g : \mathcal{C} \longrightarrow \mathcal{C}_0$$

be the forgetful functor which forgets the filtration. These functors satisfy the following relations:

$$g \circ i = \mathrm{Id}_{\mathcal{C}_0}, \qquad \widehat{G} \circ i \circ g = \widehat{G}_g, \qquad \widehat{G}_g \circ i \circ g = \widehat{G}_g$$

(cf. 1.3, example 2).

Lemma 1.9. Let $H : \mathcal{C}_0 \to \mathrm{Ab\,fil}$ be a functor on $\mathcal{C}_0$. Then the canonical homomorphism

$$\mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G}, H \circ g) \longrightarrow \mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G} \circ i, H)$$

is an isomorphism.

To see that it is injective, one uses the sequence of canonical homomorphisms

$$\mathrm{Hom}(\widehat{G}, H \circ g) \to \mathrm{Hom}(\widehat{G} \circ i, H) \to \mathrm{Hom}(\widehat{G}_g, H \circ g) \to \mathrm{Hom}(\widehat{G}, H \circ g),$$

the last arrow coming from the fact that $\widehat{G}$ is a subfunctor of $\widehat{G}_g$. To verify that the composite is the identity, observe that a morphism of functors

$$\xi : \widehat{G} \longrightarrow H \circ g$$

is determined by its values on objects of $\mathcal{C}$ whose filtration is gross. Indeed, if $A \in \mathrm{Ob}(\mathcal{C})$, and if A_g is A endowed with the gross filtration, the canonical morphism $A \to A_g$ defines a commutative diagram

$$\begin{array}{ccc} \widehat{G}(A) & \xrightarrow{\xi_A} & (H \circ g)(A) \\ \downarrow & & \downarrow \\ \widehat{G}(A_g) & \xrightarrow[\xi_{A_g}]{} & (H \circ g)(A_g), \end{array}$$

which shows that ξ_A is the restriction of ξ_{A_g} to $\widehat{G}(A)$.

Next consider the composite homomorphism

$$\mathrm{Hom}(\widehat{G} \circ i, H) \longrightarrow \mathrm{Hom}(\widehat{G}_g, H \circ g) \longrightarrow \mathrm{Hom}(\widehat{G}, H \circ g) \longrightarrow \mathrm{Hom}(\widehat{G} \circ i, H).$$

It is clear that this is the identity, whence the asserted result.

Theorem 1.10. Let H be a functor on the category $\mathcal{C}_0$ of K-algebras, with values in the category of separated complete filtered abelian groups, and commuting with invariants under a finite group. Then the map from

$$\mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G} \circ i, H)$$

to $H(K[X])$ which, to a morphism of functors, associates its value on the element $1 + Xt$ of $\widehat{G} \circ i(K[X])$, is injective. Its image is the subgroup of $H(K[X])$ formed by the elements θ such that, with notation analogous to that of 1.4 and 1.5,

i) $H(f_1)(\theta) = 0$,

ii) $\forall n > 1, \quad H(f_n)(u_H^n(\theta)) \equiv u_H^n(\theta) \pmod{F^n H(K[X_1, \ldots, X_n])}$.

The theorem follows immediately from Theorem 1.6 and Lemma 1.9, since $H \circ g(K[X])$ does not depend on the chosen filtration on $K[X]$.

The analogous theorem obtained by replacing $\widehat{G}$ by $(\widehat{G})^k$ is also true (cf. 1.7).

When there is no risk of confusion, we write the functor $\widehat{G} \circ i$ simply as $\widehat{G}$. This is legitimate because i identifies $\mathcal{C}_0$ with a subcategory of $\mathcal{C}$.

1.11. Let $\mathcal{C}_g$ denote the category of graded K-algebras, and let

$$j : \mathcal{C}_g \longrightarrow \mathcal{C}$$

be the functor which assigns to a graded K-algebra B the K-algebra B endowed with the filtration canonically defined by its grading. The associated graded functor

$$\mathrm{Gr} : \mathcal{C} \longrightarrow \mathcal{C}_g$$

will be denoted Gr. There is an isomorphism of functors

$$\varepsilon : \mathrm{Id}_{\mathcal{C}_g} \xrightarrow{\sim} \mathrm{Gr} \circ j.$$

If $B \in \mathrm{Ob}(\mathcal{C}_g)$, denote by

$$1 + \widehat{B}^+$$

the multiplicative group of elements of

$$\prod_{i=0}^{\infty} B^i$$

having augmentation 1. With the filtration analogous to that on $\widehat{G}$, one obtains a functor

$$\mathcal{C}_g \longrightarrow \mathrm{Ab\,fil}.$$

Furthermore, for every $A \in \mathrm{Ob}(\mathcal{C})$, one defines a homomorphism of filtered groups

$$\widehat{G}(A) \longrightarrow 1 + \widehat{\mathrm{Gr}\, A}^+$$

by sending the series $1 + \cdots + a_n t^n + \cdots$ to the series $1 + \cdots + \bar{a}_n + \cdots$, where $\bar{a}_n$ is the class of a_n in $\mathrm{Gr}^n A$. Its kernel is the group $\widehat{G}'(A)$ of formal series such that

$$a_0 = 1, \qquad \forall i \geq 1, \quad a_i \in \mathrm{Fil}^{i+1} A.$$

Thus one has a functorial isomorphism in A:

$$\widehat{G}(A)/\widehat{G}'(A) \xrightarrow{\sim} 1 + \widehat{\mathrm{Gr}\, A}^+.$$

Lemma 1.12. Let $H : \mathcal{C}_g \to \mathrm{Ab\,fil}$ be a functor on $\mathcal{C}_g$. Then the canonical homomorphism

$$\rho : \mathrm{Hom}_{\mathrm{Ab\,fil}}(1 + \hat{\cdot}^+, H) \longrightarrow \mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G}, H \circ \mathrm{Gr})$$

is an isomorphism.

The homomorphism ρ is defined as the composite

$$\mathrm{Hom}(1 + \hat{\cdot}^+, H) \longrightarrow \mathrm{Hom}(1 + \widehat{\mathrm{Gr}(\cdot)}^+, H \circ \mathrm{Gr}) \longrightarrow \mathrm{Hom}(\widehat{G}, H \circ \mathrm{Gr}),$$

the last arrow being defined by the homomorphism

$$\widehat{G} \longrightarrow 1 + \widehat{\mathrm{Gr}(\cdot)}^+.$$

We construct an inverse morphism

$$\rho' : \mathrm{Hom}(\widehat{G}, H \circ \mathrm{Gr}) \longrightarrow \mathrm{Hom}(1 + \hat{\cdot}^+, H).$$

For this, we show that every morphism of functors from $\widehat{G}$ to $H \circ \mathrm{Gr}$ factors through the canonical morphism

$$\widehat{G} \longrightarrow 1 + \widehat{\mathrm{Gr}(\cdot)}^+.$$

Equivalently, it suffices to show that it vanishes on $\widehat{G}'$, by 1.11. Let $A \in \mathrm{Ob}(\mathcal{C})$ and let $x \in \widehat{G}'(A)$. Define a homomorphism of filtered K-algebras

$$K[X_i]_{i \geq 1} \longrightarrow A$$

by sending X_1 to 0 and X_i to the $(i-1)$-st coefficient of x for $i > 1$. Using this homomorphism, it is enough to verify that the morphism of functors considered vanishes on the element

$$1 + X_2 t + \cdots + X_{n+1} t^n + \cdots .$$

Use the endomorphism f of $K[X_i]_{i \geq 1}$ defined by

$$f(X_i) = X_{i+1} \quad \text{for all } i,$$

and the commutative diagram

$$\begin{array}{ccc} \widehat{G}(K[X_i]) & \longrightarrow & (H \circ \mathrm{Gr})(K[X_i]) \\ {\scriptstyle \widehat{G}(f)}\downarrow & & \downarrow{\scriptstyle H(\mathrm{Gr}(f))} \\ \widehat{G}(K[X_i]) & \longrightarrow & (H \circ \mathrm{Gr})(K[X_i]). \end{array}$$

Now $\mathrm{Gr}(f) = \mathrm{Gr}(f_1)$, where f_1 is the augmentation homomorphism of $K[X_i]_{i \geq 1}$. Since the latter annihilates φ, the analogous diagram written for f_1 shows that $H(\mathrm{Gr}(f))$ annihilates the image of φ in $H \circ \mathrm{Gr}(K[X_i]_{i \geq 1})$, and the result follows.

Passing to the quotient, one associates to every morphism of functors

$$\widehat{G} \longrightarrow H \circ \mathrm{Gr}$$

a morphism of functors

$$1 + \widehat{\mathrm{Gr}}^+ \longrightarrow H \circ \mathrm{Gr}\,.$$

By composing with j and using the isomorphism ε, one obtains a morphism

$$1 + \hat{\cdot}^+ \longrightarrow H,$$

which defines the desired homomorphism ρ'. It is immediate to verify that ρ and ρ' are inverses of one another.

Theorem 1.13. Let H be a functor on $\mathcal{C}_g$, with values in the category of separated complete filtered abelian groups, and commuting with invariants under a finite group. Then the map from

$$\mathrm{Hom}_{\mathrm{Ab\,fil}}(1+\hat{\cdot}^+, H)$$

to $H(K[X])$, which to a morphism of functors associates its value on the element $1+X$ of $1+\widehat{K[X]}^+$, is injective. Its image is the subgroup of $H(K[X])$ formed by the elements θ such that, with notation as in 1.4 and 1.5,

i) $H(f_1)(\theta) = 0$,

ii) $\forall n > 1, \quad H(f_n)(u_H^n(\theta)) \equiv u_H^n(\theta) \pmod{F^n H(K[X_1, \ldots, X_n])}$.

This is immediate from 1.6 and 1.12. There is an analogous statement replacing $\widehat{G}$ by $(\widehat{G})^k$.

Remark 1.14. If H is a functor on $\mathcal{C}_g$, with values in abelian groups, commuting with invariants under a finite group, and satisfying the condition

$$\forall B \in \mathrm{Ob}(\mathcal{C}_g), \qquad H(B) \simeq \varprojlim_n H(B/B^{\geq n}), \qquad B^{\geq n} = \prod_{i \geq n} B^i,$$

then condition ii) of 1.13 is always satisfied if one takes for the filtration on H the filtration by the kernels of

$$H(B) \longrightarrow H(B/B^{\geq n}),$$

which is separated and complete. Indeed, if p_n is the canonical homomorphism

$$K[X_1, \ldots, X_n] \longrightarrow K[X_1, \ldots, X_n]/K[X_1, \ldots, X_n]^{\geq n},$$

then

$$p_n \circ f_n = p_n,$$

and hence

$$H(p_n)[H(f_n) \circ u_H^n(\theta) - u_H^n(\theta)] = 0.$$

This says precisely that

$$H(f_n) \circ u_H^n(\theta) - u_H^n(\theta) \in F^n H(K[X_1, \ldots, X_n]).$$

Thus, in order to give a morphism of functors from $1+\hat{\cdot}^+$ to H, it is enough to give an element of $H(K[X])$ satisfying only condition i) of 1.13.

1.15. Let $H : \mathcal{C} \to \mathrm{Ab\,fil}$ (respectively $\mathcal{C}_0 \to \mathrm{Ab\,fil}$, $\mathcal{C}_g \to \mathrm{Ab\,fil}$) be a functor satisfying the conditions of 1.3 (respectively 1.10, 1.13), and let

$$\xi : (\widehat{G})^k \to H$$

(respectively $(\widehat{G} \circ i)^k \to H$, $(1+\hat{\cdot}^+)^k \to H$) be a morphism of functors. We have seen in 1.7 that ξ is determined by its value on the element

$$(\varphi^{(j)})_{j=1,\ldots,k}$$

of

$$\left(\widehat{G}(K[X_i^{(j)}]_{j=1,\ldots,k;\ i\geq 1})\right)^k$$

(respectively similarly). Suppose H is one of the functors

$$\widehat{G}, \quad \widehat{G}_g, \quad \widehat{G}_a, \quad 1+\widehat{\mathrm{Gr}}^+, \quad \text{etc.}$$

Then the element $\xi((\varphi^{(j)}))$ is a formal series with coefficients in

$$K[X_i^{(j)}]_{i\geq 1,\ j=1,\ldots,k}.$$

The n-th coefficient of this series is called the n-th universal polynomial of ξ. The knowledge of these polynomials is equivalent to the knowledge of ξ.

According to the proof of 1.6, these polynomials are themselves determined by the element

$$\xi(1+X^{(1)}t,\ldots,1+X^{(k)}t).$$

The calculation is performed by the method of the proof: one computes the elements

$$\xi(1+X_1^{(1)}t+\cdots+X_n^{(1)}t^n,\ldots,1+X_1^{(k)}t+\cdots+X_n^{(k)}t^n),$$

by decomposing

$$1+X_1^{(1)}t+\cdots+X_n^{(1)}t^n=\prod_{j=1}^{n}(1+T_j^{(1)}t)$$

and by using the additivity of ξ. The polynomials thus obtained are symmetric with respect to the $T_j^{(i)}$, for fixed i and variable j, and can be expressed as functions of the $X_j^{(i)}$. Gluing by means of the filtration of H, one thus determines the universal polynomials of ξ. If

$$(\psi^{(1)},\ldots,\psi^{(k)})\in(\widehat{G}(A))^k,$$

the calculation of $\xi(\psi^{(1)},\ldots,\psi^{(k)})$ is made by substituting the coefficients of the $\psi^{(i)}$ for the $X_j^{(i)}$ in the universal polynomials.

Suppose that there exists an integer m such that the coefficient of t^n in

$$\theta=\xi(1+X^{(1)}t,\ldots,1+X^{(k)}t)$$

is a polynomial whose isobaric components have weights between n and m. Then the k-th universal polynomial of ξ has isobaric components whose weights range between k and km. This follows immediately from the calculation just described and from the theorem on symmetric functions.

In particular, if $H=\widehat{G}$, then the n-th universal polynomial of ξ has no component of weight $<n$. Now consider the canonical morphism

$$H\longrightarrow 1+\widehat{\mathrm{Gr}}^+.$$

Composing it with ξ gives a morphism

$$(\widehat{G})^k\longrightarrow 1+\widehat{\mathrm{Gr}}^+,$$

which, by the proof of 1.12, factors through

$$(1+\widehat{\mathrm{Gr}}^+)^k$$

and defines a morphism

$$\bar{\xi}:(1+\hat{\cdot}^+)^k\longrightarrow 1+\hat{\cdot}^+.$$

One sees at once that the n-th universal polynomial of $\bar{\xi}$ is the same as that of

$$(\widehat{G})^k \longrightarrow 1 + \widehat{\mathrm{Gr}}^+,$$

that is, the isobaric component of weight n of the n-th universal polynomial of ξ.

The same remarks apply to the case $H = \widehat{G}_a$. Then one may define a morphism of functors

$$H \longrightarrow \widehat{\mathrm{Gr}}^+$$

analogously to 1.11, by putting

$$\widehat{B}^+ = \prod_{i \geq 1} B^i$$

for a graded algebra B; one repeats the same reasoning and reaches the conclusions of the preceding paragraph.

1.16. We finish with an example. Suppose $K = \mathbb{Q}$. Define an endomorphism of the functor $1 + \hat{\cdot}^+$, denoted Todd and called the Todd character, by putting

$$\mathrm{Todd}(1 + X) = \frac{X}{1 - e^{-X}}.$$

The conditions i) and ii) of Theorem 1.13 are immediately verified, which shows that an endomorphism of $1 + \hat{\cdot}^+$ has indeed been defined.

2. Definition of λ-rings; examples

In this paragraph the base ring of paragraph 1 will be $\mathbb{Z}$, except in 2.9. We place ourselves in the category $\mathcal{C}_0$ of 1.8, and write $\widehat{G}$ for the restriction of $\widehat{G}$ to $\mathcal{C}_0$, as in 1.10.

Definition 2.1. A *pre-λ-ring* is a commutative ring K with unit, endowed with a homomorphism of abelian groups

$$\lambda_t : K \longrightarrow 1 + K[[t]]^+$$

such that, for every $x \in K$,

$$\lambda_t(x) = 1 + xt + \cdots .$$

We shall write

$$\lambda_t(x) = \sum_{i \geq 0} \lambda^i(x) t^i.$$

Thus giving a pre-λ-ring structure on a ring K is equivalent to giving a family of maps

$$\lambda^i : K \longrightarrow K \qquad (i \in \mathbb{N})$$

such that

i) $\forall x \in K, \quad \lambda^0(x) = 1,$

ii) $\forall x \in K, \quad \lambda^1(x) = x,$

iii) $\forall x, y \in K, \ \forall n \in \mathbb{N}, \quad \lambda^n(x+y) = \sum_{p+q=n} \lambda^p(x)\lambda^q(y).$

If K, K' are two pre-λ-rings and $f : K \to K'$ is a ring homomorphism, we say that f is a homomorphism of pre-λ-rings, or a λ-homomorphism, if the diagram

$$\begin{array}{ccc} K & \xrightarrow{\lambda_t} & 1 + K[[t]]^+ \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \widehat{G}(f)} \\ K' & \xrightarrow[\lambda_t]{} & 1 + K'[[t]]^+ \end{array}$$

is commutative. This is equivalent to the relations

$$\forall x \in K,\ \forall i \in \mathbb{N}, \qquad \lambda^i(f(x)) = f(\lambda^i(x)).$$

Examples 2.2.

a) One can give all pre-λ-ring structures on $\mathbb{Z}$. It is enough to give $\lambda_t(1)$, which is then a formal series with integral coefficients subject to the sole condition that it begin with $1 + t$.

b) Let (T, A) be a commutatively ringed topos, and let K be the "Grothendieck group" of A-modules locally isomorphic to a direct factor of a free A-module of finite type. We know that K can be given a ring structure by the tensor product (Exposé IV), and we now endow it with a pre-λ-ring structure as follows. For every A-module E locally a direct factor of a free module of finite type, let $\lambda^i(E)$ denote the class in K of the i-th exterior power of E, which is again locally a direct factor of a free module of finite type. This defines a function from the category of such A-modules to the abelian group $1 + K[[t]]^+$. It remains to show that this function is additive, which will imply that it factors through K and hence gives the desired structure.

Lemma 2.2.1. Let

$$0 \longrightarrow E' \xrightarrow{u} E \xrightarrow{v} E'' \longrightarrow 0$$

be an exact sequence of A-modules which is locally split, for example with E'' locally a direct factor of a free module of finite type. There exists on $\Lambda^n(E)$ a finite filtration and isomorphisms

$$\Lambda^{n-p}(E') \otimes_A \Lambda^p(E'') \xrightarrow{\sim} \mathrm{Gr}^p\, \Lambda^n(E).$$

Take for $F^p\Lambda^n(E)$ the image of the canonical morphism

$$\Lambda^{n-p}(E') \otimes_A \Lambda^p(E) \longrightarrow \Lambda^n(E) \quad (0 \le p \le n).$$

If $p > n$, put $F^p\Lambda^n(E) = \Lambda^n(E)$, and if $p < 0$, put $F^p\Lambda^n(E) = 0$. This does indeed define a finite filtration on $\Lambda^n(E)$.

Define a homomorphism

$$f_p : \Lambda^{n-p}(E') \otimes_A \Lambda^p(E'') \longrightarrow F^p\Lambda^n(E)/F^{p-1}\Lambda^n(E)$$

by

$$f_p(x'_1 \wedge \cdots \wedge x'_{n-p} \otimes x''_1 \wedge \cdots \wedge x''_p) = \mathrm{cl}\big(u(x'_1) \wedge \cdots \wedge u(x'_{n-p}) \wedge x_1 \wedge \cdots \wedge x_p\big),$$

where the x_i are lifts in E of the x''_i in E''. This visibly does not depend on the choice of the x_i. If the exact sequence is locally split, then f_p is an isomorphism, since locally

$$\Lambda^n(E) \simeq \bigoplus_{p+q=n} \Lambda^p(E') \otimes_A \Lambda^q(E'').$$

When E', E, E'' are locally direct factors of free modules of finite type, the $\mathrm{Gr}^p\, \Lambda^n(E)$ are locally direct factors of free modules of finite type, and hence so are the $F^p\Lambda^n(E)$. Thus one may write in K

$$\lambda^n(E) = \sum_{p=0}^{n} \mathrm{cl}(\mathrm{Gr}^p\, \Lambda^n(E)) = \sum_{p=0}^{n} \lambda^{n-p}(E')\lambda^p(E''),$$

which gives the required additivity.

Let (T, A) and (T', A') be two commutatively ringed topoi, and let

$$f : (T, A) \longrightarrow (T', A')$$

be a morphism of ringed topoi. We know (Exposé IV, 2.7) that f defines a morphism of rings on the corresponding K's, namely

$$f^* : K^*_{\text{naif}}(T', A') \longrightarrow K^*_{\text{naif}}(T, A).$$

Since inverse image commutes with exterior powers, the corresponding morphism f^* on K is a λ-homomorphism for the structure just defined.

We point out two special cases:

- If X is a scheme, then $K^*_{\text{naif}}(X)$ is thus endowed with a pre-λ-ring structure. This is the case which will be used later in the Seminar, and it may be regarded as the model for the general theory of λ-rings.
- If A is a ring, $X = \operatorname{Spec} A$, and G is a group acting trivially on A, then, taking for (T, A) the topos of sheaves of sets on X on which G acts, the ring K thus obtained is the representation ring of G in the projective A-modules of finite type.

2.3. Let K be a commutative ring with unit. We propose to endow the abelian group

$$\widehat{G}(K) = 1 + K[[t]]^+$$

with a pre-λ-ring structure, functorially in K. To this end, we apply the results of 1.10 to the functor $\widehat{G}$.

First define multiplication by giving a $\mathbb{Z}$-bilinear morphism of functors

$$\widehat{G} \times \widehat{G} \longrightarrow \widehat{G}$$

through the element $1 + XYt$ of $\widehat{G}(\mathbb{Z}[X, Y])$. This element satisfies 1.7.i; to see that it satisfies 1.7.ii, one must show that in

$$\prod_{i=1}^{n} (1 + T_i Y t),$$

when expressed by means of the elementary symmetric functions X_j in the T_i, the variable X_n does not occur in the coefficients of t^k for $k < n$. This is evident by degree considerations.

The law so defined will be denoted by the sign $\circ$. Thus, by definition,

$$(1 + Xt) \circ (1 + Yt) = 1 + XYt. \tag{2.3.1}$$

It is distributive by construction, and it is associative and commutative by the diagrams

$$\begin{array}{ccc} \widehat{G} \times \widehat{G} \times \widehat{G} & \to & \widehat{G} \times \widehat{G} \\ \downarrow & & \downarrow \\ \widehat{G} \times \widehat{G} & \longrightarrow & \widehat{G} \end{array} \qquad \begin{array}{ccc} \widehat{G} \times \widehat{G} & \to & \widehat{G} \times \widehat{G} \\ \downarrow & & \downarrow \\ \widehat{G} & \xrightarrow[\text{Id}]{} & \widehat{G}, \end{array}$$

where the meaning of the arrows is evident, and where it is enough to verify commutativity on the elements $1 + Xt, 1 + Yt, 1 + Zt$, respectively $1 + Xt, 1 + Xt$, of $\widehat{G}(\mathbb{Z}[X, Y, Z])$, respectively $\widehat{G}(\mathbb{Z}[X, Y])$, by 1.10; this verification is immediate.

By 1.8, the product under this law of two series is calculated by substituting their coefficients into the universal polynomials of the law. The n-th universal polynomial will be denoted P_n. It is a polynomial in

$$\mathbb{Z}[X_i, Y_j]_{1 \leq i,j \leq n}$$

which is isobaric of weight n with respect to each of the two groups of variables (see 1.15), and symmetric with respect to these two groups. For example,

$$P_2(X_1, X_2; Y_1, Y_2) = X_2 Y_1^2 + Y_2 X_1^2 - 2X_2 Y_2.$$

One also has the following relation: if $K \in \mathrm{Ob}(\mathcal{C}_0)$, $\psi \in \widehat{G}(K)$, and $a \in K$, then

$$\psi(t) \circ (1 + at) = \psi(at). \tag{2.3.2}$$

As usual, by the standard functorial argument this reduces to the case $K = \mathbb{Z}[X, Y]$, $\psi = 1 + Xt$, $a = Y$, where it is nothing other than (2.3.1).

Applying (2.3.2) to $\widehat{G}(K) = 1 + K[[t]]^+$ with $a = 1$, one sees that $1 + t$ is the unit for this multiplication. To avoid confusion, we sometimes denote it by 1, or by $1_\circ$.

The functor $\widehat{G}$ can now be considered as a functor from $\mathcal{C}_0$ to the category of commutative rings with unit, and hence can be composed with itself. To avoid confusion, choose different indeterminates for each of the two factors in the product of $\widehat{G}$ with itself; we indicate the indeterminate by an exponent when necessary:

$$\widehat{G}^t, \quad \widehat{G}^u, \quad \text{etc.}$$

For $A \in \mathrm{Ob}(\mathcal{C}_0)$ one has

$$\widehat{G}^u \circ \widehat{G}^t(A) = 1 + \{1 + A[[t]]^+\}[[u]]^+.$$

To define the operations λ^i on $1 + K[[t]]^+$, we are therefore going to give a morphism of functors

$$\widehat{G} \longrightarrow \widehat{G}^u \circ \widehat{G}^t.$$

For this, one must first define a filtration of the functor $\widehat{G}^u \circ \widehat{G}^t$ which permits the gluing of 1.6.

First note that $F^k \widehat{G}$ is an ideal for the law $\circ$. For this follows, for the product

$$\left(1 + \sum_{i \geq k} X_i t^i\right) \circ (1 + Yt),$$

by reducing by functoriality to the case $A = \mathbb{Z}[X_i, Y]_{i \geq 1}$, and using (2.3.2). Therefore $F^n(\widehat{G}^u \circ \widehat{G}^t)$ is defined to be the set of series of the form

$$1 + \psi_1 u + \cdots + \psi_i u^i + \cdots, \qquad \psi_i \in F^{[n/i]} \widehat{G},$$

where $[n/i]$ denotes the integral part of n/i.

The element defining the morphism λ_u is

$$1 + \{1 + Xt\}u.$$

The reader will verify as an exercise that the conditions of 1.6 are indeed fulfilled; in fact the filtration was chosen ad hoc.

Again, the operations λ^i are calculated by substitution from

$$\lambda_u(1 + \cdots + X_n t^n + \cdots),$$

and so are determined by universal polynomials. Denote by

$$Q_{i,j}(X_1, \ldots, X_{ij})$$

the coefficient of t^j in

$$\lambda^i(1 + \cdots + X_n t^n + \cdots).$$

It is an isobaric polynomial of weight ij with respect to the variables $X_1, \ldots, X_{ij}$ (by a verification analogous to that of 1.15). It is calculated by symmetric functions from the defining element

$$\lambda_u(1 + Xt) = 1 + \{1 + Xt\}u. \tag{2.3.3}$$

For example,

$$Q_{2,2}(X_1, X_2, X_3, X_4) = X_1X_3 - X_4.$$

Finally, we indicate the origin of formulas (2.3.1) and (2.3.3). Consider the example 2.2.b, where the operations λ^i are defined by exterior powers of A-modules. We want

$$\lambda_t : K \longrightarrow \widehat{G}(K)$$

to be a λ-homomorphism. Thus on $\widehat{G}$ one must define laws such that, for all A-modules E, F,

$$\lambda_t(EF) = \lambda_t(E) \circ \lambda_t(F), \qquad \lambda_u(\lambda_t(E)) = \widehat{G}^u_{\lambda_t}(\lambda_t(\lambda_u(E))).$$

The relations (2.3.1) and (2.3.3) are precisely the preceding relations applied to the case where E and F are invertible modules.

We shall see in Exposé VI to what extent the desired property is verified.

Definition 2.4. A pre-λ-ring K will be called a *λ-ring* if the homomorphism of abelian groups

$$\lambda_t : K \longrightarrow 1 + K[[t]]^+$$

is a λ-homomorphism for the pre-λ-ring structure on the second member defined in 2.3. Thus the following relations hold for all $x, y \in K$:

$$\lambda_t(1) = 1 + t, \qquad \lambda_t(xy) = \lambda_t(x) \circ \lambda_t(y), \qquad \lambda_u(\lambda_t(x)) = \widehat{G}(\lambda_t)(\lambda_u(x)). \tag{2.4.1}$$

In terms of the maps λ^i, the last two relations are equivalent, for $x, y \in K$ and $i, j \in \mathbb{N}$, to

$$\lambda^i(xy) = P_i(\lambda^1(x), \ldots, \lambda^i(x); \lambda^1(y), \ldots, \lambda^i(y)), \tag{2.4.2}$$

$$\lambda^j(\lambda^i(x)) = Q_{i,j}(\lambda^1(x), \ldots, \lambda^{ij}(x)). \tag{2.4.3}$$

We note that here we depart from the usual terminology (cf. [1]), where what we call a pre-λ-ring was called a λ-ring, and our λ-rings were called "special" λ-rings. The reason is that only "special" λ-rings are useful in practice.

Examples 2.5.

a) We have determined all pre-λ-ring structures on $\mathbb{Z}$. It follows from (2.4.1) that only the structure defined by

$$\lambda_t(1) = 1 + t$$

is capable of being a λ-ring structure. One then has

$$\lambda_t(n) = (1 + t)^n,$$

hence

$$\forall n, n' \in \mathbb{Z}, \quad \lambda_t(n) \circ \lambda_t(n') = (1+t)^n \circ (1+t)^{n'} = (1+t)^{nn'} = \lambda_t(nn'),$$

and

$$\lambda_u((1+t)^n) = (\lambda_u(1+t))^n = (1+\{1+t\}u)^n = \widehat{G}(\lambda_t)((1+u)^n) = \widehat{G}(\lambda_t)(\lambda_u(n)).$$

Thus this is indeed a λ-ring structure. If $\binom{X}{k}$ denotes the polynomial with rational coefficients

$$\frac{1}{k!}X(X-1)\cdots(X-k+1),$$

then

$$\lambda^i(n) = \binom{n}{i}. \tag{2.5.1}$$

b) For every ring K, the pre-λ-ring $1 + K[[t]]^+$ is a λ-ring. First,

$$\lambda_u(1) = \lambda_u(1+t) = 1 + \{1+t\}u = 1 + 1 \cdot u.$$

The other relations are verified by the commutativity of the diagrams

$$\begin{array}{ccc} \widehat{G}^t \times \widehat{G}^t & \xrightarrow{\lambda_u \times \lambda_u} & (\widehat{G}^u \circ \widehat{G}^t) \times (\widehat{G}^u \circ \widehat{G}^t) \\ {\scriptstyle\circ}\downarrow & & \downarrow{\scriptstyle\circ} \\ \widehat{G}^t & \xrightarrow[\lambda_u]{} & \widehat{G}^u \circ \widehat{G}^t, \end{array} \qquad \begin{array}{ccc} \widehat{G}^t & \xrightarrow{\lambda_v} & \widehat{G}^v \circ \widehat{G}^t \\ {\scriptstyle\lambda_u}\downarrow & & \downarrow{\scriptstyle\widehat{G}^v(\lambda_u)} \\ \widehat{G}^u \circ \widehat{G}^t & \underset{\lambda_v}{\to} & \widehat{G}^v \circ \widehat{G}^u \circ \widehat{G}^t, \end{array}$$

which are checked by 1.10 on the elements $1 + Xt$ and $1 + Yt$. For the first,

$$\begin{aligned} \lambda_u(1+Xt) \circ \lambda_u(1+Yt) &= (1+\{Xt\}u) \circ (1+\{1+Yt\}u) \\ &= 1 + \{(1+Xt) \circ (1+Yt)\}u \\ &= 1 + \{1+XYt\}u \\ &= \lambda_u((1+Xt) \circ (1+Yt)). \end{aligned}$$

For the second,

$$\lambda_v(\lambda_u(1+Xt)) = 1 + \{1 + \{1+Xt\}u\}v = \widehat{G}^v(\lambda_u)\lambda_v(1+Xt).$$

c) The geometric examples of 2.2 will be treated later.

2.6. One can generalize Example 2.5.a as follows. Let K be a commutative ring with unit such that:

i) K is torsion-free,
ii) $\forall x \in K,\ \forall n \in \mathbb{N}, \quad \binom{x}{n} \in K.$

Then, for all $x \in K$ and all $n \in \mathbb{N}$, put

$$\lambda^n(x) = \binom{x}{n},$$

that is,

$$\lambda_t(x) = \sum_{n=0}^{\infty} \binom{x}{n} t^n = (1+t)^x = \exp(x \operatorname{Log}(1+t)), \tag{2.6.1}$$

where the last identity is considered in $K \otimes_{\mathbb{Z}} \mathbb{Q}[[t]]$. By multiplicativity of the exponential, this defines a pre-λ-ring structure on K.

To see that it is a λ-ring structure, one must prove the relations (2.4.2), and it is enough to prove them in $K \otimes \mathbb{Q}$, by i). One is then reduced by functoriality to the case $K = \mathbb{Q}[X, Y]$ and to the elements X, Y. In view of the definition of the λ^i, the two sides of the equalities (2.4.2) are polynomials in X, Y, with rational coefficients. It is therefore enough, in order to prove their equality, to show that they take the same value for all integral values of X, Y. But the relations obtained by substituting integral values in X, Y are precisely those expressing that the pre-λ-ring $\mathbb{Z}$ of 2.5.a is a λ-ring. They are therefore satisfied.

Definition 2.7. A *binomial ring* is a ring satisfying the conditions 2.6.i and ii), endowed with the λ-ring structure defined by (2.6.1).

Example 2.8. Let X be a site, K a binomial ring, and $\mathcal{K}$ the sheaf associated to the constant presheaf with value K on X. Then $H^0(X, \mathcal{K})$ is a binomial ring. Indeed, the conditions 2.6 are preserved under passage to an inductive limit. In particular, $H^0(X, \mathbb{Z})$ is a binomial ring.

2.9. Let K be a binomial ring and A a K-algebra. For every series

$$\varphi \in \widehat{G}(A)$$

and every $a \in K$, one can define a series φ^a in the usual way. Writing $\varphi = 1 + \psi$, put

$$\varphi^a = (1+\psi)^a = \sum_{n=0}^{\infty} \binom{a}{n} \psi^n \, (= \exp(a \operatorname{Log} \varphi)), \tag{2.9.1}$$

where the last equality can be written only modulo torsion. These formulas are obtained by substitution from the analogous formulas for $\varphi = 1 + t$. Therefore the following relations, the classical properties of the exponential, remain valid for $a, b \in K$ and $\varphi, \psi \in \widehat{G}(A)$:

$$\varphi^a \varphi^b = \varphi^{a+b}, \qquad \varphi^{ab} = (\varphi^a)^b, \qquad \varphi^a \psi^a = (\varphi\psi)^a. \tag{2.9.2}$$

Continuation of 2.9

We still suppose that K is binomial. Multiplication, for the law $\circ$, by $\lambda_t(a)$, where $a \in K$, defines a morphism of functors

$$\widehat{G}_K \longrightarrow \widehat{G}_K$$

which is characterized by its value on the element $1 + Xt$ of $G(K[X])$. Thus

$$\lambda_t(a) \circ (1 + Xt) = \lambda_{Xt}(a),$$

and hence

$$\lambda_t(a) \circ (1+Xt) = (1+Xt)^a = \exp(a \operatorname{Log}(1+Xt)) = \sum_{n=0}^{\infty} \binom{a}{n} X^n t^n. \tag{2.9.3}$$

It follows that

$$\lambda_t(a) \circ \varphi = \varphi^a \quad (= \exp(a \operatorname{Log} \varphi)). \tag{2.9.4}$$

Indeed, the two members are additive functors which coincide on $1 + Xt$. From (2.9.3) and 1.15 one obtains that the universal polynomial of degree n defining multiplication by $\lambda_t(a)$ is isobaric of weight n. We denote it by $R_{n,a}$.

Finally, if A is a K-λ-algebra and $b \in A$, then

$$\lambda_t(ab) = (\lambda_t(b))^a. \tag{2.9.5}$$

This follows from $\lambda_t(ab) = \lambda_t(a) \circ \lambda_t(b)$ and from (2.9.4).

3. The operations γ

We still work over the category $\mathscr{C}_0$.

3.1. Let K be a λ-ring. We shall define from the operations λ^i new operations denoted γ^i. They are the operations which are adapted to Chern classes. If E is a vector bundle of rank d, the top Chern class of E is represented, in K-theoretic language, by the alternating sum of the exterior powers. To obtain the n-th class, one first replaces E by $E \oplus \mathbf{1}^{n-d}$, so that the same top-class recipe applies. This motivates the formula below.

Proposition 3.2. Let K be a λ-ring.

i) For every $x \in K$ and every $n \in \mathbb{N}$,

$$\lambda^n(x+n-1) = (-1)^n \sum_{i=0}^{n} (-1)^i \lambda^i(x+n).$$

ii) If

$$\gamma^n(x) = \lambda^n(x+n-1), \qquad \gamma_t(x) = \sum_{n\geq 0} \gamma^n(x) t^n,$$

then

$$\gamma_t(x) = \lambda_{t/(1-t)}(x), \qquad \text{hence} \qquad \lambda_t(x) = \gamma_{t/(1+t)}(x).$$

Thus knowledge of the operations γ is equivalent to knowledge of the operations λ.

Proof. For i),

$$\lambda^n(x+n-1) = \sum_{i=0}^{n} \lambda^i(x+n)\lambda^{n-i}(-1).$$

Since $\lambda_t(1) = 1+t$, one has $\lambda_t(-1) = 1/(1+t)$, and hence $\lambda^k(-1) = (-1)^k$, giving the desired relation. For ii),

$$\lambda_{t/(1-t)}(x) = \sum_{n\geq 0} \lambda^n(x) t^n (1-t)^{-n},$$

and

$$t^n(1-t)^{-n} = \sum_{p\geq n} \binom{p-1}{p-n} t^p.$$

The coefficient of t^p is therefore

$$\sum_{n=0}^{p} \lambda^n(x) \binom{p-1}{p-n} = \sum_{n=0}^{p} \lambda^n(x)\lambda^{p-n}(p-1) = \lambda^p(x+p-1).$$

3.3. If $f : K \to K'$ is a homomorphism of λ-rings, then f is a λ-homomorphism if and only if, for every $i \in \mathbb{N}$,

$$f(\gamma^i(x)) = \gamma^i(f(x)),$$

or equivalently

$$\gamma_t(f(x)) = \widehat{G}(f)(\gamma_t(x)).$$

3.4. From 3.2 one obtains

$$\gamma^0(x) = 1, \qquad \gamma^1(x) = x, \qquad \gamma^n(x+y) = \sum_{p+q=n} \gamma^p(x)\gamma^q(y). \tag{3.4.1}$$

Thus γ_t defines a new pre-λ-ring structure. It is not, in general, a λ-ring structure. Let ρ_t denote the automorphism of the functor $\widehat{G}$ induced by the change of variable

$$t \longmapsto \frac{t}{1-t}.$$

Then $\gamma_t = \rho_t \lambda_t$. Transporting the λ-ring functor structure of $\widehat{G}$ by ρ_t gives a second structure on $\widehat{G}$, for which $\gamma_t : K \to \widehat{G}(K)$ is a λ-homomorphism whenever K is a λ-ring.

3.5. The multiplication in this transported structure is the composite

$$\widehat{G} \times \widehat{G} \xrightarrow{\rho_t^{-1} \times \rho_t^{-1}} \widehat{G} \times \widehat{G} \xrightarrow{\circ} \widehat{G} \xrightarrow{\rho_t} \widehat{G}.$$

It will be denoted ∇. It is determined by

$$\begin{aligned}(1+Xt)\nabla(1+Yt) &= \rho_t\Big(\rho_t^{-1}(1+Xt) \circ \rho_t^{-1}(1+Yt)\Big) \\ &= \rho_t\left(\frac{1+(X+1)t}{1+t} \circ \frac{1+(Y+1)t}{1+t}\right) \\ &= \rho_t\left(\frac{(1+((X+1)(Y+1))t)(1+t)}{(1+(X+1)t)(1+(Y+1)t)}\right) \\ &= \frac{1+(1+X+Y+XY)t}{(1+Xt)(1+Yt)}. \end{aligned} \tag{3.5.1}$$

Thus, for $x, y \in K$,

$$\gamma_t(xy) = \gamma_t(x) \nabla \gamma_t(y). \tag{3.5.2}$$

The universal polynomial of degree n for this law will be denoted P'_n. Its isobaric components have weights between n and $2n$ in each of the two groups of variables, and

$$\gamma^n(xy) = P'_n(\gamma^1(x), \dots, \gamma^n(x), \gamma^1(y), \dots, \gamma^n(y)). \tag{3.5.3}$$

The unit for ∇ is $\gamma_t(1) = 1/(1-t)$.

3.6. The operations λ' of the transported structure are defined by

$$\begin{array}{ccc} & \widehat{G} & \\ 1 + K[[t]]^+ & \xrightarrow{\lambda_u} & 1 + \{1 + K[[t]]^+\}[[u]]^+ \\ {\scriptstyle \rho_t}\downarrow & & \downarrow{\scriptstyle \widehat{G}(\rho_t)} \\ 1 + K[[t]]^+ & \xrightarrow[\lambda'_u]{} & 1 + \{1 + K[[t]]^+\}[[u]]^+. \end{array} \tag{3.6.1}$$

If γ' denotes the γ-operations attached to these operations λ', then

$$\gamma'_u(1+Xt) = \frac{1}{1-u} + \{1+Xt\}u. \tag{3.6.2}$$

Let $Q'_{i,j}$ be the coefficient of t^j in $\gamma'^i(1 + X_1 t + X_2 t^2 + \cdots)$. It is a polynomial in $\mathbb{Z}[X_1, \dots, X_{ij}]$, whose isobaric components have weights from j to ij. For a λ-ring K and $x \in K$,

$$\gamma_t(\lambda^i(x)) = \lambda'^i(\gamma_t(x)), \qquad \gamma_t(\gamma^i(x)) = \gamma'^i(\gamma_t(x)), \tag{3.6.3}$$

and therefore

$$\gamma^j(\gamma^i(x)) = Q'_{i,j}(\gamma^1(x), \dots, \gamma^{ij}(x)). \tag{3.6.4}$$

The transported pre-λ-ring structure on $\widehat{G}$ is itself a λ-ring structure.

3.7. We denote by $\widehat{G}_\circ$ the functor $\widehat{G}$ with the structure of §2, and by $\widehat{G}_\nabla$ the same functor with the structure of §3. Giving a λ-ring structure on a ring A is equivalent either to giving a ring homomorphism $A \to \widehat{G}_\circ(A)$ satisfying the usual diagram for λ_t, or to giving a ring homomorphism $A \to \widehat{G}_\nabla(A)$ satisfying the corresponding diagram for γ_t. Explicitly, the two diagrams are

$$\begin{array}{ccc} A & \xrightarrow{\lambda_u} & \widehat{G}^u_\circ(A) \\ {\scriptstyle\lambda_t}\downarrow & & \downarrow{\scriptstyle\widehat{G}^u_\circ(\lambda_t)} \\ \widehat{G}^t_\circ(A) & \underset{\lambda_u}{\to} & \widehat{G}^u_\circ(\widehat{G}^t_\circ(A)) \end{array} \tag{3.7.1}$$

and

$$\begin{array}{ccc} A & \xrightarrow{\gamma_u} & \widehat{G}^u_\nabla(A) \\ {\scriptstyle\gamma_t}\downarrow & & \downarrow{\scriptstyle\widehat{G}^u_\nabla(\gamma_t)} \\ \widehat{G}^t_\nabla(A) & \underset{\gamma_u}{\to} & \widehat{G}^u_\nabla(\widehat{G}^t_\nabla(A)). \end{array} \tag{3.7.2}$$

3.8. Let K be a binomial ring. From (2.6.1), for $x \in K$,

$$\gamma_t(x) = \left(\frac{1}{1-t}\right)^x = \exp(-x \operatorname{Log}(1-t)). \tag{3.8.1}$$

For $a \in K$, multiplication by $\gamma_t(a)$ for the law ∇ is characterized by its value on $1 + Xt$:

$$\gamma_t(a)\nabla(1+Xt) = (1+Xt)^a = \exp(a \operatorname{Log}(1+Xt)) = \sum_{n=0}^{\infty} \binom{a}{n} X^n t^n. \tag{3.8.2}$$

Thus the universal polynomial of degree n defining this multiplication is again $R_{n,a}$.

Definition 3.9. Let K be a λ-ring. An *augmented K-λ-algebra* is a K-λ-algebra A, together with a λ-homomorphism of K-algebras

$$\epsilon : A \longrightarrow K.$$

Morphisms are λ-homomorphisms compatible with the augmentations.

Example 3.9.1. Let (X, A) be a locally ringed topos. The Grothendieck group $K^\circ(X)$ of locally free finite-type A-modules has a natural pre-λ-ring structure by exterior powers. There is a natural λ-homomorphism from $H^0(X, \mathbb{Z})$ to $K^\circ(X)$: a section a of the constant sheaf $\mathbb{Z}$ partitions the final object into opens U_n on which $a = n$, and is sent to the class of the locally free module which is A^n on U_n for $n \geq 0$, minus the analogous class on the negative pieces. The rank map is the augmentation. This is the augmented structure used below.

3.10. Let A be an augmented K-λ-algebra. For $n > 0$, define $\operatorname{Fil}^n A$ to be the ideal generated by products

$$\gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k), \qquad i_1 + \cdots + i_k \geq n, \qquad \epsilon(x_i) = 0. \tag{3.10.1}$$

If A is generated as an abelian group by (x_α), then $\operatorname{Fil}^n A$ is generated over K by

$$\gamma^{i_1}(x_{\alpha_1} - \epsilon(x_{\alpha_1})) \cdots \gamma^{i_k}(x_{\alpha_k} - \epsilon(x_{\alpha_k})), \qquad i_1 + \cdots + i_k \geq n. \tag{3.10.2}$$

The filtration is decreasing and multiplicative, and

$$\operatorname{Fil}^0 A = A, \qquad \operatorname{Fil}^1 A = \operatorname{Ker} \epsilon.$$

It is functorial for homomorphisms of augmented K-λ-algebras.

Proposition 3.11. Let A be a λ-ring and let $y_1, \ldots, y_n$ be elements of A such that $\gamma^j(y_i) = 0$ for $j \geq 2$. Put $x = \sum_i y_i$. Then $\gamma^k(x - n) = 0$ for $k > n$, and, for $1 \leq k \leq n$, $\gamma^k(x - n)$ is the k-th elementary symmetric function of the $y_i - 1$. This follows from

$$\gamma_t(x - n) = \prod_{i=1}^{n} \gamma_t(y_i - 1).$$

4. λ-rings generated by generators and relations

4.1. Let K be a λ-ring and let $(X_i)_{i \in I}$ be indeterminates. Put

$$A = K[X_i, {}^n X_i]_{i \in I,\ n \geq 2}.$$

Define

$$\lambda_t(X_i) = 1 + X_i t + \sum_{n \geq 2} {}^n X_i t^n, \qquad \lambda_t({}^n X_i) = \lambda^n(\lambda_t(X_i)). \tag{4.1.1}$$

This gives a λ-ring structure on A extending that of K.

Proposition 4.2. For every K-λ-algebra B and every family $(b_i)_{i \in I}$ in B, there exists a unique K-λ-homomorphism $A \to B$ carrying X_i to b_i. Necessarily it carries ${}^n X_i$ to $\lambda^n(b_i)$, and this condition is sufficient.

4.3. The same construction may be made using γ's. Let

$$A' = K[X_i, \overset{\gamma}{n} X_i]_{i \in I,\ n \geq 2},$$

and define

$$\gamma_t(X_i) = 1 + X_i t + \sum_{n \geq 2} \overset{\gamma}{n} X_i t^n, \qquad \gamma_t(\overset{\gamma}{n} X_i) = \gamma'^n(\gamma_t(X_i)). \tag{4.3.1}$$

This gives a λ-ring satisfying the same universal property, hence canonically isomorphic to A.

Definition 4.4. The preceding λ-ring is the *free λ-ring* over K generated by the family (X_i).

Definition 4.5. An ideal J of a λ-ring A is a *λ-ideal* if $\lambda^n(x) \in J$ for every $x \in J$ and $n > 1$. Kernels of λ-homomorphisms are λ-ideals; conversely, if J is a λ-ideal, A/J receives a unique λ-ring structure. The intersection of λ-ideals is a λ-ideal. Thus every family of elements has a smallest containing λ-ideal, generated by the elements and all their iterated λ^n's.

Definition 4.6. The quotient of the free λ-ring by the λ-ideal generated by prescribed relations (P_i) is called the λ-ring generated by the generators (X_i) subject to the relations (P_i). Equivalently, one may formulate the same construction in terms of γ-operations.

Proposition 4.7. This quotient has the expected universal property: every family (b_i) in a K-λ-algebra satisfying the relations (P_i) determines a unique K-λ-homomorphism from the quotient to that algebra.

Proposition 4.8. Let $n \geq 0$. In the free λ-ring over K on one generator X, the ideal generated by the $\lambda^k X$, $k > n$, is a λ-ideal. Indeed, after quotienting, one has only to check that if $\varphi \in \widehat{G}(C)$ has no terms of degree $> n$, then $\lambda^k(\varphi) = 0$ for $k > n$. By the splitting principle, it is enough to write

$$\varphi = \prod_{i=1}^{n} (1 + T_i t),$$

and then

$$\lambda_u(\varphi) = \prod_{i=1}^{n} (1 + \{1 + T_i t\} u),$$

which has degree n in u.

Corollary 4.9. The λ-ring generated over K by one element X, subject to $\lambda^k(X) = 0$ for $k > n$, is

$$K[X, \lambda^2 X, \ldots, \lambda^n X].$$

More generally, for a family (X_i) with bounds (n_i), the quotient subject to $\lambda^k(X_i) = 0$ for $k > n_i$ is the polynomial ring generated by $X_i, \lambda^2 X_i,$
$ldots, \lambda^{n_i} X_i$.

4.10. Let K be binomial and let A be an augmented K-λ-algebra. Put

$$B = A[X, \gamma^2 X, \gamma^3 X, \ldots],$$

the free A-λ-algebra on X, written in γ-coordinates. Extend the augmentation by $\epsilon(X) = 0$, hence $\epsilon(\gamma^i X) = 0$ for $i > 0$. Let M_k be the set of monomials in the variables $\gamma^j X$, counted with weight j, of total weight k, and set $M_0 = \{1\}$.

Lemma 4.11. If $x \in \mathrm{Fil}^p A$, then every coefficient of $\gamma^n(x + X)$ which is homogeneous of weight k in the $\gamma^j X$ belongs to $\mathrm{Fil}^{p+n-k} A$.

Proposition 4.12. For every $k \geq 0$,

$$\mathrm{Fil}^k B = \mathrm{Fil}^k A + \mathrm{Fil}^{k-1} A\, M_1 + \cdots + \mathrm{Fil}^1 A\, M_{k-1} + \mathrm{Fil}^0 A\, M_k + B\, M_{k+1}. \tag{4.12.1}$$

The inclusion from right to left follows from Lemma 4.11. The reverse inclusion is obtained by writing the generators of $\mathrm{Fil}^k B$ as products of γ-operations and reducing, term by term, to the displayed weights.

Remarks 4.13. The proof gives the same assertion for a free λ-algebra on any finite or infinite family of generators, with the evident multi-weight. It is also unchanged when one uses the λ-coordinates, since λ and γ coordinates differ by triangular universal formulas.

Corollary 4.14. Passing to the associated graded ring, one has

$$\mathrm{Gr}^k B \simeq \mathrm{Gr}^k A \oplus \mathrm{Gr}^{k-1} A\, M_1 \oplus \cdots \oplus \mathrm{Gr}^0 A\, M_k. \tag{4.14.1}$$

Corollary 4.15. If $B' = K[X, \gamma^2 X, \ldots, \gamma^n X]$ is the K-λ-algebra generated by one element satisfying $\gamma^k(X) = 0$ for $k > n$, then the λ-ring filtration of B' is exactly the weight filtration in the variables $\gamma^j X$.

5. The elements $\lambda^p(N, x)$

This section will be used only in Exposé VII, §4, in the study of regular immersions; it may be omitted on first reading.

5.1. Let $d \geq 1$. Let A be the $\mathbb{Z}$-λ-algebra generated by N and x, subject to the relations $\lambda^i(N) = 0$ for $i > d$. Thus

$$A = \mathbb{Z}[N, \ldots, \lambda^d N; x, \ldots, \lambda^j x, \ldots].$$

Put

$$\lambda_{-1}(N) = \sum_{i=0}^{d} (-1)^i \lambda^i(N). \tag{5.1.1}$$

Lemma 5.2. For every $p \geq 1$, the element $\lambda^p(x\lambda_{-1}(N))$ is divisible by $\lambda_{-1}(N)$.

It is enough, by isobaricity, to prove this for $\lambda^p(\lambda_{-1}(N))$. Let B be the λ-ring generated by $N_1, \dots, N_d$, with $\lambda^i(N_j) = 0$ for $i \geq 2$. The map $N \mapsto \sum_i N_i$ identifies A with the symmetric polynomials in the N_i, and

$$\lambda_{-1}(N) = \prod_{i=1}^{d} (1 - N_i).$$

If some $N_i = 1$, then $\lambda_t(\lambda_{-1}(N)) = \lambda_t(0) = 1$, so every positive coefficient vanishes. Thus each $\lambda^p(\lambda_{-1}(N))$ is divisible by every $1 - N_i$, hence by their product.

Definition 5.3. Since A is integral, define $\lambda^p(N, x)$ by

$$\lambda^p(N, x)\lambda_{-1}(N) = \lambda^p(x\lambda_{-1}(N)), \qquad p \geq 1. \tag{5.3.1}$$

Thus $\lambda^p(N, x)$ is a polynomial in the $\lambda^i(x)$ and $\lambda^j(N)$, isobaric of weight p in the $\lambda^i(x)$. If a λ-ring B contains an element N' with $\lambda^k(N') = 0$ for $k > d$, then $\lambda^p(N', x')$ is defined by substitution; it is independent of the chosen bound d.

5.4. In the λ-ring generated over $\mathbb{Z}$ by N, x, y, subject to $\lambda^k(N) = 0$ for $k > d$, formula (5.3.1) gives

$$\begin{aligned}
\lambda^p(N, x+y)\lambda_{-1}(N) &= \lambda^p\big((x+y)\lambda_{-1}(N)\big) \\
&= \sum_{i+j=p} \lambda^i\big(x\lambda_{-1}(N)\big)\lambda^j\big(y\lambda_{-1}(N)\big) \\
&= \sum_{\substack{i+j=p \\ i,j\neq 0}} \lambda^i(N, x)\lambda^j(N, y)\big(\lambda_{-1}(N)\big)^2 + \big(\lambda^p(N, x) + \lambda^p(N, y)\big)\lambda_{-1}(N).
\end{aligned}$$

Since the ring under consideration is an integral domain, this yields the relation, valid by substitution in every λ-ring,

$$\lambda^p(N, x+y) = \sum_{\substack{i+j=p \\ i,j\neq 0}} \lambda^i(N, x)\lambda^j(N, y)\lambda_{-1}(N) + \lambda^p(N, x) + \lambda^p(N, y). \tag{5.4.1}$$

Now consider the λ-ring generated over $\mathbb{Z}$ by N, N', x, subject to $\lambda^k(N) = 0$ for $k > d$ and $\lambda^k(N') = 0$ for $k > d'$. Then $\lambda^k(N + N') = 0$ for $k > d + d'$, and

$$\begin{aligned}
\lambda^p(N + N', x)\lambda_{-1}(N)\lambda_{-1}(N') &= \lambda^p\big(x\lambda_{-1}(N)\lambda_{-1}(N')\big) \\
&= \lambda^p\big(N', x\lambda_{-1}(N)\big)\lambda_{-1}(N').
\end{aligned}$$

Here we used $\lambda_{-1}(N + N') = \lambda_{-1}(N)\lambda_{-1}(N')$. Cancelling in the integral domain gives the relation, again valid by substitution,

$$\lambda^p(N + N', x)\lambda_{-1}(N) = \lambda^p\big(N', x\lambda_{-1}(N)\big). \tag{5.4.2}$$

5.5. Let A be a λ-ring and let $N \in A$ satisfy $\lambda^k(N) = 0$ for $k > d$. Define on the underlying abelian group of A a new multiplication by

$$x \cdot_N y = xy\lambda_{-1}(N).$$

After adjoining a unit, one obtains a ring $\mathbb{Z} \times A$ with

$$(n, x)(m, y) = (nm, ny + mx + x \cdot_N y).$$

The formulas

$$\lambda_t(n, x) = (1+t)^n \left(1 + \sum_{p\geq 1} \lambda^p(N, x)t^p\right) \tag{5.5.1}$$

define a pre-λ-ring structure. In fact, when A is a λ-ring, this is a λ-ring structure; it is the one transported from the original structure by multiplication by $\lambda_{-1}(N)$. We denote this augmented λ-ring by $\widehat{A}_N$. Its γ-operations satisfy

$$\gamma^p(N,x)\lambda_{-1}(N) = \gamma^p(x\lambda_{-1}(N)). \tag{5.5.2}$$

5.6. Suppose that, in addition, L is an element such that $\lambda^k(L) = 0$ for $k > 1$. Then $\lambda^p(N+L,x)$ is the coefficient of t^p in the series

$$\frac{\sum_{k\geq 1} \lambda^k(N,x)(1 + L + \cdots + L^{k-1})t^k}{1 + \sum_{k\geq 1} \lambda^k(N,x)\lambda_{-1}(N)L^k t^k}. \tag{5.6.1}$$

This is obtained by applying (5.4.2) to $N + L$ and using $\lambda_{-1}(L) = 1 - L$.

Proposition 5.7. Let K be a binomial ring, A an augmented K-λ-algebra, and $N \in A$ such that $\lambda^k(N) = 0$ for $k > d$. Then, for every $x \in A$ and every $p \geq 1$,

$$\gamma^p(N,x) \in \mathrm{Fil}^{p-d} A. \tag{5.7.1}$$

Indeed, let A_0 be the K-λ-algebra generated by N_0', X_0', subject to $\gamma^k(N_0') = 0$ for $k > d$, and augmented by $\epsilon(N_0') = \epsilon(X_0') = 0$. This is a polynomial algebra over K in the variables $\gamma^i(N_0')$, $1 \leq i \leq d$, and $\gamma^j(X_0')$, $j \geq 1$, and its λ-filtration is the weight filtration, with weight i assigned to $\gamma^i(N_0')$ and $\gamma^i(X_0')$ (4.12 and 4.15). Put

$$N_0 = N_0' + d, \qquad X_0 = X_0' + \epsilon(x).$$

For N_0, X_0, the assertion is equivalent to

$$\gamma^p(N_0, X_0)\gamma^d(N_0') \in \mathrm{Fil}^p A_0.$$

Since $\gamma^d(N_0') = (-1)^d\lambda_{-1}(N_0)$, this becomes

$$(-1)^d\gamma^p(X_0\gamma^d(N_0')) \in \mathrm{Fil}^p A_0,$$

by (5.5.4). This holds because $\gamma^d(N_0')$ has zero augmentation. By 4.7 there is a K-λ-homomorphism $A_0 \to A$, compatible with augmentations and sending N_0' to $N - d$ and X_0' to $x - \epsilon(x)$, hence sending N_0 to N and X_0 to x. Compatibility with the filtrations gives the result.

6. Chern ring: opening

Throughout this section the base ring is a fixed binomial ring K. We use the functor G of 1.1, and return to the notation of §1: G denotes the functor there defined and G_i its restriction to $\mathscr{C}_0$.

6.1. In 3.5 and 3.6 we defined morphisms of functors

$$G_i \times G_i \xrightarrow{\star} G_i, \qquad G_i^\circ \xrightarrow{\gamma'} G_i^\circ \circ G_S, \qquad G_i \xrightarrow{a} G_i,$$

where the last map is multiplication, for the law ∇, by $\gamma_t(a)$ with $a \in K$. By 1.9 these give morphisms

$$G \times G \xrightarrow{\star} G, \qquad G \xrightarrow{\gamma'} G \circ G_S, \qquad G \xrightarrow{a} G.$$

The isobaricity of the universal polynomials involved implies that the first and third images lie in G, while the second lies in the subgroup of formal series in u whose non-zero terms have the prescribed degrees.

Exposé V, 6. Chern ring: completion

Definition 6.2. The functor defined in 6.1 on the category of graded K-algebras, with values in the category of K-λ-algebras, is called the *Chern ring*; its value on $A \in \mathrm{Ob}(\mathscr{C})$ is denoted $\mathrm{Ch}(A)$, or, if there is a risk of confusion, $\mathrm{Ch}_K(A)$.

From formulas (6.1.1), (6.1.2), (6.1.4), and (3.6.2), one deduces the relations

$$(1, 1+X)(1, 1+Y) = (1, 1+X+Y), \qquad \lambda^i(1, 1+X) = 0 \quad \text{for } i \geq 2, \tag{6.2.1}$$

the relations which served in [1] to define the Chern ring. Conversely, from these formulas one can recover the relations (6.1.1) and (6.1.4).

Let us indicate why the Chern ring is introduced. We place ourselves in the situation of Example 3.9, and write A for the graded ring associated, by the filtration defined in 3.10, with the $K^\bullet$ of a commutatively locally ringed topos; for simplicity suppose that $K^\bullet$ is a λ-ring. The augmentation ring is K. We shall see below that for every element of an augmented λ-ring one can define Chern classes with values in the associated graded ring (6.7). To every locally free sheaf of finite type one can therefore associate an element of $\mathrm{Ch}(A)$, whose component on the factor K is the rank of the sheaf and whose component on the factor $1 + \widehat{A}^+$ is the sum of the Chern classes. Formulas (6.2.1) are then the usual formulas giving the Chern classes and the rank of the tensor product of two invertible sheaves and of the exterior powers of an invertible sheaf; in other words, the map just defined from $K^\bullet$ to $\mathrm{Ch}(A)$ is a λ-homomorphism.

6.3. For every graded K-algebra A, denote by $A_\mathbb{Q}$ the K-algebra $A \otimes \mathbb{Q}$, graded in the evident way. By 1.13 one defines a functorial homomorphism

$$1 + \widehat{A}^+ \xrightarrow{\mathrm{ch}} \widehat{A}^+_\mathbb{Q}$$

by giving its value on $1 + X$:

$$\mathrm{ch}(1+X) = \exp(X) - 1. \tag{6.3.1}$$

This homomorphism is compatible with the multiplication defined in 6.1 on $1 + \widehat{A}^+$. By 1.13 it is enough to show this for the elements $1 + X$, $1 + Y$ of the universal ring $1 + K[X, Y]^+$. One has

$$\begin{aligned}
\mathrm{ch}\big((1+X) * (1+Y)\big) &= \mathrm{ch}\left(\frac{1+X+Y}{(1+X)(1+Y)}\right) \\
&= \mathrm{ch}(1+X+Y) - \mathrm{ch}(1+X) - \mathrm{ch}(1+Y) \\
&= \exp(X+Y) - \exp(X) - \exp(Y) + 1 \\
&= \mathrm{ch}(1+X)\ \mathrm{ch}(1+Y).
\end{aligned}$$

Let η be the endomorphism of $\widehat{A}^+_\mathbb{Q}$, functorial in A, defined by

$$\eta\left(\sum_{k\geq 1} a_k t^k\right) = \sum_{k\geq 1} \frac{(-1)^{k-1}}{(k-1)!}\, a_k t^k.$$

Then, for every $\varphi \in 1 + \widehat{A}^+$,

$$\mathrm{ch}(\varphi) = \eta(\mathrm{Log}\,\varphi). \tag{6.3.2}$$

Indeed, ch and $\eta \circ \mathrm{Log}$ are two functorial homomorphisms of abelian groups, and it is enough for them to agree on $1 + X$, which is evident.

It follows that ch commutes with the operations of K: again it is enough to show this for $1+X$, and for $a \in K$ one has

$$\begin{aligned}\mathrm{ch}((1+X)^a) &= \mathrm{ch}(\exp(a\,\mathrm{Log}(1+X))) \\ &= \eta(a\,\mathrm{Log}(1+X)) \\ &= a\,\eta(\mathrm{Log}(1+X)) \\ &= a\ \mathrm{ch}(1+X).\end{aligned}$$

Thus one can extend ch in the evident way to a homomorphism of K-algebras

$$K \times (1+\widehat{A}^+) \longrightarrow K \oplus \widehat{A}^+_{\mathbb{Q}},$$

namely

$$\mathrm{ch}(a,\varphi) = a + \eta(\mathrm{Log}\,\varphi). \tag{6.3.3}$$

Definition 6.4. The homomorphism

$$\mathrm{ch} : \mathrm{Ch}\,A \longrightarrow K \oplus \widehat{A}^+_{\mathbb{Q}}$$

just defined is called the *Chern character*.

If A is of characteristic zero, that is, is a $\mathbb{Q}$-algebra, the Chern character is an isomorphism. Indeed, if $\psi \in K \oplus \widehat{A}^+$, one may write $\psi = a + \psi_1$, where ψ_1 is of positive degree, and then an inverse morphism is defined by

$$\psi \longmapsto (a, \exp(\eta^{-1}(\psi_1))).$$

Proposition 6.5. Let A be a graded K-algebra. On $\mathrm{Ch}(A)$ consider the filtration defined as follows, the degree filtration:

$$(\mathrm{Ch}(A))_0 = \mathrm{Ch}(A), \qquad (\mathrm{Ch}(A))_n = 0 \times (1+\widehat{A}^{\geq n}) \quad (n \geq 1), \tag{6.5.1}$$

where $\widehat{A}^{\geq n} = \prod_{i \geq n} A^i$. The following congruences hold.

i) If $n \geq 1$ and $\varphi \in (\mathrm{Ch}\,A)_n$, writing $\varphi = 1 + a_n + \cdots$, then

$$\mathrm{ch}(\varphi) = \frac{(-1)^{n-1}}{(n-1)!}\,a_n + \cdots . \tag{6.5.2}$$

ii) If $\varphi \in (\mathrm{Ch}\,A)_m$, $\psi \in (\mathrm{Ch}\,A)_n$, and if $\varphi = 1 + a_m + \cdots$, $\psi = 1 + b_n + \cdots$, then

$$\varphi * \psi = 1 - \frac{(m+n-1)!}{(m-1)!(n-1)!}\,a_m b_n + \cdots . \tag{6.5.3}$$

Consequently the degree filtration is a ring filtration.

For (i), the Chern character is by construction a morphism of filtered functors; from the congruence $\varphi \equiv 1 + a_n \pmod{(\mathrm{Ch}\,A)_{n+1}}$ one obtains

$$\mathrm{ch}(\varphi) = \mathrm{ch}(1+a_n) \pmod{\widehat{A}^{\geq n+1}},$$

and, since $\mathrm{ch}(1+a_n) = \eta(\mathrm{Log}(1+a_n))$, this is

$$\frac{(-1)^{n-1}}{(n-1)!}a_n + \cdots .$$

For (ii), by substitution it is enough to prove (6.5.3) for the torsion-free universal polynomial algebra

$$A = K[X_i, Y_j]_{i,j\in\mathbb{N}}$$

and for $\varphi = 1+\sum_{k\geq m} X_k$, $\psi = 1+\sum_{k\geq n} Y_k$. Since K is torsion-free in the universal case, one can embed this algebra in its tensor product with $\mathbb{Q}$, so that ch is injective. Put $\varphi * \psi = 1 + Z_p + \cdots$, where Z_p is the first non-zero term different from 1. The multiplicativity of ch, together with (6.5.2), gives

$$1 + \frac{(-1)^{p-1}}{(p-1)!} Z_p + \cdots = \left(1 + \frac{(-1)^{m-1}}{(m-1)!} X_m + \cdots\right)\left(1 + \frac{(-1)^{n-1}}{(n-1)!} Y_n + \cdots\right),$$

which gives the stated value of Z_p by comparing the terms of lowest degree.

6.6. The λ-ring $\mathrm{Ch}\, A$ is augmented over K: one defines a λ-homomorphism $\mathrm{Ch}\, A \to K$ by $(a, \varphi) \mapsto a$. It is therefore equipped with the canonical filtration of an augmented λ-ring. We compare it with the filtration just defined.

Proposition 6.6.1. Let $x \in \mathrm{Ch}\, A$ be an element of the form $x = (n, 1 + \cdots + a_n)$. Then:

i) $\lambda^i(x) = 0 \quad \text{for } i > n,$

ii) $\lambda^n(x) = (1, 1 + a_1),$

iii) $\displaystyle\sum_{i=0}^{n} (-1)^i \lambda^i(x) = (0, 1 - (n-1)! a_n + \cdots),$

where the unwritten terms have degree $> n$.

By functoriality, it is enough to prove this in the case where $A = K[X_1, \ldots, X_n]$, filtered by weight with respect to the X_i, X_i having weight i, and for the element $(n, 1 + X_1 + \cdots + X_n)$. Embed A in $A' = K[T_1, \ldots, T_n]$ by the symmetric functions. Then

$$(n, 1 + X_1 + \cdots + X_n) = \left(n, \prod_{i=1}^{n} (1 + T_i)\right) = \sum_{i=1}^{n} (1, 1 + T_i).$$

Hence, by (6.2.1),

$$\lambda_t(x) = \prod_{i=1}^{n} (1 + (1, 1 + T_i)t),$$

so that

$$\lambda^k(x) = 0 \quad (k > n), \qquad \lambda^n(x) = \prod_{i=1}^{n} (1, 1 + T_i) = \left(1, 1 + \sum_{i=1}^{n} T_i\right) = (1, 1 + X_1).$$

Finally,

$$\sum_{i=1}^{n} (-1)^i \lambda^i(x) = \lambda_{-1}(x) = \prod_{i=1}^{n} (1 - (1, 1 + T_i)) = (-1)^n \prod_{i=1}^{n} (0, 1 + T_i).$$

By (6.5.3),

$$\prod_{i=1}^{n} (0, 1 + T_i) = \left(0, 1 + (-1)^{n-1}(n-1)! \prod_{i=1}^{n} T_i + \cdots\right),$$

hence

$$\lambda_{-1}(x) = (0, 1 - (n-1)! X_n + \cdots).$$

Corollary 6.6.2. Relation 6.6.1(iii) is true for every element X of augmentation n.

For this we shall use the following lemma.

Lemma 6.6.3. For every n, $(\mathrm{Ch}\, A)_n$ is a λ-ideal of $\mathrm{Ch}\, A$.

We may suppose $n > 1$. Let $x = (0, \varphi) \in (\mathrm{Ch}\, A)_n$, and write

$$\lambda^i(0, \varphi) = (0, 1 + Q^i_{i,1} + \cdots + Q^i_{i,j} + \cdots),$$

where the $Q^i_{i,j}$ are the isobaric components of weight j of the Q^i_i of (3.6). For every j, the polynomial $Q^i_{i,j}$ is a polynomial in the first j coefficients of φ. If $j < n$, these coefficients are zero; hence the $Q^i_{i,j}$ are zero, and so the $\lambda^i(x)$ belong to $(\mathrm{Ch}\, A)_n$. Thus $(\mathrm{Ch}\, A)_n$ is a λ-ideal.

Let $x = (n, 1 + a_1 + \cdots + a_n + \cdots)$ be an element of augmentation n in $\mathrm{Ch}\, A$. Then

$$x \equiv (n, 1 + \cdots + a_n) \pmod{(\mathrm{Ch}\, A)_{n+1}},$$

and therefore

$$\lambda_{-1}(x) \equiv \lambda_{-1}(n, 1 + \cdots + a_n) = (0, 1 - (n-1)!a_n + \cdots),$$

which proves the corollary.

Corollary 6.6.4. Let ϵ be the augmentation homomorphism of $\mathrm{Ch}\, A$. Then, for every $x \in \mathrm{Ch}\, A$, one has

$$\gamma^n(x - \epsilon(x)) = (0, 1 + (-1)^{n-1}(n-1)!a_n + \cdots).$$

By definition,

$$\gamma^n(x - \epsilon(x)) = (-1)^n \sum_{i=0}^{n} (-1)^i \lambda^i(x + n - \epsilon(x)).$$

Since $x + n - \epsilon(x)$ has augmentation n, Corollary 6.6.2 applies to it, giving the result.

Corollary 6.6.5. With the notation of 3.10,

$$\mathrm{Fil}^n\, \mathrm{Ch}\, A \subset (\mathrm{Ch}\, A)_n.$$

This follows from the preceding corollary, which gives $\gamma^n(x - \epsilon(x)) \in (\mathrm{Ch}\, A)_n$, and from the fact that the two filtrations are ring filtrations and that the elements $\gamma^n(x - \epsilon(x))$, for n and x variable, generate the λ-ring filtration.

Corollary 6.6.7. Let A be a graded K-algebra of characteristic zero, bounded by degree. Then the two filtrations defined on $\mathrm{Ch}\, A$ are identical.

Suppose that $\mathrm{Fil}^p A = 0$, and let $x = (0, 1 + a_n + \cdots + a_p)$ be an element of $(\mathrm{Ch}\, A)_n$. Put

$$x' = \left(0, 1 + \frac{(-1)^{n-1}}{(n-1)!}a_n + a_{n+1} + \cdots + a_p\right).$$

Then $\gamma^n(x') = (0, 1 + a_n + \cdots + a'_p)$, since x' has zero augmentation. Hence $x - \gamma^n(x') \in (\mathrm{Ch}\, A)_{n+1}$. Iterating the operation $p - n$ times, one reaches $(\mathrm{Ch}\, A)_{p+1} = 0$, and therefore x is written as an element of $\mathrm{Fil}^n\, \mathrm{Ch}\, A$. In view of (6.6.5), the two filtrations are identical.

6.7. Now suppose that A is a λ-ring augmented over K, and let ϵ be the augmentation homomorphism. Then A is equipped with its canonical filtration of augmented λ-ring, and one can associate to $\mathrm{Gr}\, A$ its Chern ring.

If $x \in A$, then $\gamma_t(x - \epsilon(x)) \in \widehat{G}(A)$ by definition of the filtration of an augmented λ-ring. We write $c(x)$ for the image of $\gamma_t(x - \epsilon(x))$ in

$$\widehat{G}(A)/\widehat{G}'(A) = 1 + \widehat{\mathrm{Gr}\, A}^+.$$

It is called the *total Chern class* of x. We write $c^i(x)$, and call it the i-th Chern class of x, for the i-th coefficient of $c(x)$. Thus

$$c^i(x) = \mathrm{cl}_{\mathrm{Gr}^i A}\Big(\gamma^i(x - \epsilon(x))\Big), \qquad c(x) = 1 + c^1(x) + \cdots . \tag{6.7.1}$$

Put

$$\widetilde{c}(x) = (\epsilon(x), c(x)). \tag{6.7.2}$$

This is called the *completed Chern class* of x.

Proposition 6.8. The completed Chern class

$$\widetilde{c} : A \longrightarrow \mathrm{Ch}(\mathrm{Gr}\, A)$$

is a K-λ-homomorphism of augmented K-λ-algebras.

It is clear that the homomorphism so defined is additive. To verify that it is multiplicative, write

$$\begin{aligned}\gamma_t(xy - \epsilon(xy)) &= \gamma_t(\epsilon(y)(x - \epsilon(x)))\gamma_t(\epsilon(x)(y - \epsilon(y)))\gamma_t((x - \epsilon(x))(y - \epsilon(y))) \\ &= (\gamma_t(x - \epsilon(x)))^{\epsilon(y)}(\gamma_t(y - \epsilon(y)))^{\epsilon(x)}(\gamma_t(x - \epsilon(x)) \circ \gamma_t(y - \epsilon(y))),\end{aligned}$$

which gives the result by (6.1.2) after passage to the quotient. If $a \in K$, then $\widetilde{c}(a) = (a, 1)$, so $\widetilde{c}$ is a homomorphism of K-algebras; moreover it is compatible with augmentations.

It remains to see that it is a λ-homomorphism. One must prove the commutativity of the diagram

$$\begin{array}{ccc} A & \xrightarrow{\gamma_t} & \widehat{G}(A) \\ \Big\downarrow{\scriptstyle \widetilde{c}} & & \Big\downarrow{\scriptstyle \widehat{G}(\widetilde{c})} \\ \mathrm{Ch}\, A & \xrightarrow[\gamma_t]{} & \widehat{G}(\mathrm{Ch}\, A). \end{array}$$

By additivity, it is enough to prove it for the elements of K, where it is evident, and for the elements of zero augmentation in A, where it follows, by the definition of c and (6.1.4), from diagram (3.7.2).

Proposition 6.9. Let A be an augmented K-λ-algebra and let $z \in \mathrm{Fil}^m A$. Then

$$\gamma^m(z) = (-1)^{m-1}(m-1)!z \pmod{\mathrm{Fil}^{m+1} A},$$

that is,

$$c^m(z) = \mathrm{cl}_{\mathrm{Gr}^m A}\Big((-1)^{m-1}(m-1)!z\Big).$$

One must see that these two elements give the same image in $\mathrm{Gr}^m A$, that is, that the m-th Chern class of z is the image of $(-1)^{m-1}(m-1)!z$.

One may write z as a sum

$$\sum a\, \gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k),$$

with $a \in A$, $\epsilon(x_i) = 0$ for all i, and $i_1 + \cdots + i_k \geq m$. We then have

$$\widetilde{c}(z) = \sum \widetilde{c}(a)\, \widetilde{c}(\gamma^{i_1}(x_1)) \cdots \widetilde{c}(\gamma^{i_k}(x_k)).$$

Since $\widetilde{c}$ is a λ-homomorphism, one may write

$$\widetilde{c}(\gamma^i(x_j)) = \gamma^i(\widetilde{c}(x_j)).$$

Taking account of the definition of c and of (6.6.4), one obtains

$$\widetilde{c}(\gamma^{i_j}(x_j)) = \left(0, 1 + (-1)^{i_j-1}(i_j-1)!\,\gamma^{i_j}(x_j) + \cdots\right).$$

By (6.5.3), and since $\widetilde{c}(a) = (a, 1)$, one finally gets

$$\begin{aligned}\widetilde{c}(z) &= \sum \left(0, \left(1 + (-1)^{i_1+\cdots+i_k-1}(i_1+\cdots+i_k-1)!\,\gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k) + \cdots\right)^a\right) \\ &= \left(0, 1 + \sum_{\substack{(x_1,\ldots,x_k)\\ i_1+\cdots+i_k=m}} (-1)^{m-1}(m-1)!a\,\gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k) + \cdots\right) \\ &= (0, 1 + (-1)^{m-1}(m-1)!z + \cdots),\end{aligned}$$

which proves the assertion.

Proposition 6.10. Let K be a binomial ring, R an augmented K-λ-algebra, $N \in R$ an element such that $\lambda^k(N) = 0$ for $k > d$, and $x \in \mathrm{Fil}^i R$. Then

$$\gamma^{d+i}(N, x) - (-1)^{d+i-1}(d+i-1)!x \in \mathrm{Fil}^{i+1} R.$$

One can find elements $z_{\alpha,j} \in R$ and $a_\alpha \in K$ such that

$$x = \sum_\alpha a_\alpha\,\gamma^{i_1}(x_{\alpha,1})\cdots\gamma^{i_k}(x_{\alpha,k}),$$

with $i_1 + \cdots + i_k \geq i$ and $\epsilon(x_{\alpha,j}) = 0$ for every α and every j. Let R_0 be the K-λ-algebra generated by N_0' and $X_{\alpha,j}$, subject to the relations

$$\gamma^k(N_0') = 0 \quad \text{for } k > d.$$

Define an augmentation homomorphism $R_0 \to K$ by $\epsilon(N_0') = \epsilon(X_{\alpha,j}) = 0$. By (4.9.1), R_0 is the ring of polynomials with coefficients in K in the indeterminates $\gamma^k(N_0')$, $1 \leq k \leq d$, and $\gamma^p(X_{\alpha,j})$, $p \geq 1$; the λ-filtration of R_0 is its weight filtration, where $\gamma^k(N_0')$ and $\gamma^k(X_{\alpha,j})$ have weight k (4.12 and 4.15).

Put

$$X_0 = \sum_\alpha a_\alpha\,\gamma^{i_1}(X_{\alpha,1})\cdots\gamma^{i_k}(X_{\alpha,k}).$$

We shall prove 6.10 first for X_0 and $N_0' = N_0 + d$, which do satisfy the hypotheses of 6.10. In view of the preceding remark about the filtration of R_0, it is enough to show that

$$\gamma^{d+i}(N_0, X_0)\gamma^d(N_0') - (-1)^{d+i-1}(d+i-1)!X_0\gamma^d(N_0') \in \mathrm{Fil}^{d+i+1} R_0.$$

But $\lambda_{-1}(N_0) = (-1)^d\gamma^d(N_0')$. Thus this relation becomes

$$(-1)^d\gamma^{d+i}(X_0, (-1)^d\gamma^d(N_0')) - (-1)^{d+i-1}(d+i-1)!X_0\gamma^d(N_0') \in \mathrm{Fil}^{d+i+1} R_0.$$

Putting $z = (-1)^d X_0\gamma^d(N_0')$, we have $z \in \mathrm{Fil}^{d+i} R_0$, and the statement is reduced to

$$\gamma^{d+i}(z) - (-1)^{d+i-1}(d+i-1)!z \in \mathrm{Fil}^{d+i+1} R_0,$$

which is just 6.9.

To deduce 6.10 from this particular case, consider the K-λ-homomorphism $R_0 \to R$ which sends N_0' to $N - d$ and $X_{\alpha,j}$ to $x_{\alpha,j}$, for every α and j. This homomorphism is compatible with the λ-filtrations and sends N_0 to N and X_0 to x. Hence the result follows.

Proposition 6.11. Suppose that K is of characteristic zero, and let $n \geq 0$ be an integer. Then the functor Ch is an equivalence from the category of graded K-algebras A such that $A_0 = K$ and $A_k = 0$ for $k > n$, to the category of augmented K-λ-algebras Λ such that $\mathrm{Fil}^{n+1}\,\Lambda = 0$. A quasi-inverse functor is given by $\Lambda \mapsto \mathrm{Gr}\,\Lambda$.

Let A be a graded K-algebra such that $A_k = 0$ for $k > n$, equipped with the filtration associated to its grading. Then $(\mathrm{Ch}\,A)_{n+1} = 0$, and hence $\mathrm{Fil}^{n+1}\,\mathrm{Ch}\,A = 0$ by (6.6.5). Conversely, if Λ is an augmented K-λ-algebra such that $\mathrm{Fil}^{n+1}\,\Lambda = 0$, then $\mathrm{Gr}^0\,\Lambda = K$ and $\mathrm{Gr}^k\,\Lambda = 0$ for $k > n$.

We shall define isomorphisms of functors

$$A \longrightarrow \mathrm{Gr}(\mathrm{Ch}\,A), \qquad \Lambda \longrightarrow \mathrm{Ch}(\mathrm{Gr}\,\Lambda).$$

Since A is of finite grading, the associated gradeds of the λ-ring filtration on $\mathrm{Ch}\,A$ and of the degree filtration are the same by (6.6.7). Define

$$u : A \longrightarrow \mathrm{Gr}_{\mathrm{deg}}(\mathrm{Ch}\,A)$$

to be the identity on A_0, and, for $i \geq 1$ and $a_i \in A_i$,

$$u(a_i) = \mathrm{cl}_{\mathrm{Gr}^i}\left(0, 1 + (-1)^{i-1}(i-1)!a_i\right),$$

then extend by additivity. By relation (6.5.3) this is a homomorphism of graded rings, and it is easy to see that it is compatible with the operations of K. The inverse morphism u^{-1} may be defined as follows: for $i \geq 1$ and $\bar{\varphi} \in \mathrm{Gr}^i(\mathrm{Ch}\,A)$, the i-th coefficient a_i of φ is independent of the lift of $\bar{\varphi}$ in $(\mathrm{Ch}\,A)_i$, and one puts

$$u^{-1}(\bar{\varphi}) = \frac{(-1)^{i-1}}{(i-1)!}a_i.$$

Thus u is an isomorphism.

The isomorphism $\Lambda \to \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$ is defined by the completed Chern class $\widetilde{c}$. To verify that it really defines an isomorphism, it is enough to verify that the homomorphism obtained by passing to associated graded rings is one, the filtrations being discrete. One gets

$$\mathrm{Gr}\,\Lambda \xrightarrow{\mathrm{Gr}(\widetilde{c})} \mathrm{Gr}_\lambda(\mathrm{Ch}(\mathrm{Gr}\,\Lambda)) \simeq \mathrm{Gr}_{\mathrm{deg}}(\mathrm{Ch}(\mathrm{Gr}\,\Lambda)) \xrightarrow{u^{-1}} \mathrm{Gr}\,\Lambda.$$

It remains to show that the composite is the identity. Since the homomorphisms are homomorphisms of graded K-algebras, it is enough, by the definition of the λ-ring filtration of Λ, to show that it is the identity on the $\gamma^i(x)$, where $\epsilon(x) = 0$. One then obtains

$$\mathrm{Gr}(\widetilde{c})(\gamma^i(x)) = \mathrm{cl}_{\mathrm{Gr}^i}(\gamma^i(c(x))).$$

But $\gamma^i(\widetilde{c}(x)) = (0, 1 + (-1)^{i-1}(i-1)!c^i(x) + \cdots)$, by (6.6.4). Therefore

$$\mathrm{cl}_{\mathrm{Gr}^i}(\gamma^i(\widetilde{c}(x))) = u(c^i(x)),$$

which shows that $u^{-1} \circ \mathrm{Gr}(\widetilde{c}) = \mathrm{Id}$.

Corollary 6.12. If K is of characteristic zero, and if Λ is a K-λ-algebra with discrete filtration, then the completed Chern class

$$\widetilde{c} : \Lambda \longrightarrow \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$$

is an isomorphism.

The proof is contained in that of 6.11.

Definition 6.13. Let A be a graded K-algebra and Λ an augmented K-λ-algebra. A *Chern theory for Λ with values in A* is a λ-homomorphism

$$\Lambda \longrightarrow \mathrm{Ch}(A)$$

compatible with augmentations.

Thus one defines a functor on the category of graded K-algebras, with values in the category of sets, by associating with a graded K-algebra A the set of Chern theories of Λ with values in A, namely

$$\mathrm{Hom}_\lambda(\Lambda, \mathrm{Ch}(A)).$$

Proposition 6.14. The functor defined in 6.13 is representable. If K is of characteristic zero and Λ has discrete filtration, it is represented by the graded ring associated with Λ.

First prove the first assertion. Consider the graded K-algebra generated by generators $c^i(x)$, with $i \geq 1$, $x \in \mathrm{Fil}^1 \Lambda$, $c^i(x)$ of degree i, and take its quotient by the following relations:

$$\begin{aligned}
&\text{i)} \quad c^k(x+y) = \sum_{i+j=k} c^i(x)c^j(y), \quad k \geq 1,\ x, y \in \mathrm{Fil}^1 \Lambda,\\
&\text{ii)} \quad c^k(xy) = P''_k(c^i(x), c^j(y)) \quad \text{(cf. 6.1)}, \quad k \geq 1,\ x, y \in \mathrm{Fil}^1 \Lambda,\\
&\text{iii)} \quad c^k(ax) = R_{k,a}(c^i(x)) \quad \text{(cf. 6.1)}, \quad k \geq 1,\ a \in K,\ x \in \mathrm{Fil}^1 \Lambda,\\
&\text{iv)} \quad c^k(\gamma^j(x)) = Q''_{j,k}(c^i(x)) \quad \text{(cf. 6.1)}, \quad j, k \geq 1,\ x \in \mathrm{Fil}^1 \Lambda.
\end{aligned}$$

These relations are isobaric, so the quotient is a graded K-algebra Ω. Define a morphism

$$\widetilde{c}_0 : \Lambda \longrightarrow \mathrm{Ch}(\Omega)$$

by

$$x \longmapsto (\epsilon(x), 1 + c^1(x - \epsilon(x)) + \cdots).$$

The imposed relations show that this is a Chern theory for Λ with values in Ω. Conversely, if f is a Chern theory with values in a graded K-algebra A, then there is a unique homomorphism $\Omega \to A$ through which f factors through $\mathrm{Ch}(\Omega)$: it sends, for every $i \geq 1$ and every $x \in \mathrm{Fil}^1 \Lambda$, the element $c^i(x)$ to the i-th coefficient of $f(x)$. This is well defined because f is a λ-homomorphism.

Now suppose that K is of characteristic zero and Λ has discrete filtration. Let n be such that $\mathrm{Fil}^n \Lambda = 0$. Then the graded algebra Ω is zero in degrees $> n$. To show this, it is enough to prove that, for

$$x_1, \ldots, x_p \in \mathrm{Fil}^1 \Lambda \quad \text{and} \quad k_1 + \cdots + k_p \geq n,$$

one has

$$c^{k_1}(x_1) \cdots c^{k_p}(x_p) = 0.$$

By hypothesis,

$$\gamma^{k_1}(x_1) \cdots \gamma^{k_p}(x_p) = 0.$$

Applying the λ-homomorphism $\widetilde{c}_0$, one gets

$$\gamma^{k_1}(\widetilde{c}_0(x_1)) \cdots \gamma^{k_p}(\widetilde{c}_0(x_p)) = (0, 1).$$

By (6.6.4) and (6.5.3), this implies

$$(k_1 + \cdots + k_p)!\, c^{k_1}(x_1) \cdots c^{k_p}(x_p) = 0.$$

Since K is of characteristic zero, the desired result follows.

The universal property of Ω, applied to the canonical homomorphism $\widetilde{c}$, gives a commutative diagram

$$\begin{array}{ccc} \Lambda & \xrightarrow{\widetilde{c}} & \mathrm{Ch}(\mathrm{Gr}\,\Lambda) \\ {\scriptstyle \widetilde{c}_0}\downarrow & \nearrow & \\ \mathrm{Ch}(\Omega) & & \end{array}$$

By Proposition 6.11, the homomorphism $\Omega \to \mathrm{Gr}\,\Lambda$ is an isomorphism if and only if the homomorphism

$$\mathrm{Ch}(\Omega) \longrightarrow \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$$

is one. Since it is visibly surjective, the latter is also surjective; it remains to show that $\mathrm{Ch}(\Omega) \to \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$ is injective. By functoriality of Ch, one obtains the diagram

$$\begin{array}{ccc} \Lambda & \xrightarrow{\widetilde{c}_0} & \mathrm{Ch}(\Omega) \\ {\scriptstyle \widetilde{c}}\downarrow & & \downarrow{\scriptstyle \widetilde{c}} \\ \mathrm{Ch}(\mathrm{Gr}\,\Lambda) & \xrightarrow[\mathrm{Ch}(\widetilde{c}_0)]{} & \mathrm{Ch}(\mathrm{Gr}_\Lambda(\mathrm{Ch}(\Omega))). \end{array}$$

Since Ω has bounded degrees, the right vertical arrow is an isomorphism by (6.12). Using 6.11, one therefore obtains a morphism

$$\mathrm{Gr}\,\Lambda \longrightarrow \Omega$$

whose two composites with $\Omega \to \mathrm{Gr}\,\Lambda$ are the identity, by the universal property of Ω.

7. Appendix: the Adams operations ψ^k

We follow here a presentation due to J.-P. Serre.

7.1. Let K be a pre-λ-ring. For $x \in K$, write

$$\frac{d}{dt}(\lambda_t(x))$$

for the derivative with respect to t of $\lambda_t(x)$. Define elements of K, denoted $\psi^k(x)$, by putting

$$\frac{d}{dt}(\lambda_t(x))/\lambda_t(x) = \sum_{k=1}^{\infty}(-1)^{k-1}\psi^k(x)t^{k-1}. \tag{7.1.1}$$

The maps $\psi^k : K \to K$ are additive. Indeed,

$$\begin{aligned} \frac{d}{dt}(\lambda_t(x+y))/\lambda_t(x+y) &= \frac{d}{dt}(\lambda_t(x)\lambda_t(y))/\lambda_t(x)\lambda_t(y) \\ &= \frac{d}{dt}(\lambda_t(x))/\lambda_t(x) + \frac{d}{dt}(\lambda_t(y))/\lambda_t(y), \end{aligned}$$

and therefore, for every k,

$$\psi^k(x+y) = \psi^k(x) + \psi^k(y).$$

By (7.1.1), the ψ^k are isobaric polynomials of weight k in the λ^i, $i = 1, \ldots, k$. For example,

$$\psi^1(x) = \lambda^1(x) = x, \qquad \psi^2(x) = x^2 - 2\lambda^2(x).$$

7.2. Let K be a ring of characteristic zero. Suppose given additive maps $\psi^k : K \to K$ such that $\psi^1 = \mathrm{Id}_K$. Define a pre-λ-structure on K by

$$\lambda_t(x) = \exp\left(\sum_{k=1}^{\infty}(-1)^{k-1}\psi^k(x)t^k/k\right). \tag{7.2.1}$$

Multiplicativity follows from the additivity of the ψ^k. Thus there is an equivalence between giving a pre-λ-structure on K and giving the corresponding operations ψ^k.

7.3. It is clear from formula (7.1.1) that the ψ^k commute with λ-homomorphisms. If K, K' are pre-λ-rings of characteristic zero, it is necessary and sufficient for a homomorphism $f : K \to K'$ to be a λ-homomorphism that it commute with the ψ^k. This follows from 7.2, since the λ^i are expressed as polynomials in the ψ^k.

Examples 7.4. i) For $K = \mathbb{Z}$, with its λ-ring structure, one has

$$\frac{d}{dt}(\lambda_t(n))/\lambda_t(n) = n(1+t)^{n-1}/(1+t)^n = n\sum_{k=1}^{\infty}(-1)^{k-1}t^{k-1}.$$

Consequently the ψ^k are all equal to the identity map of $\mathbb{Z}$.

ii) Work in the category $\mathscr{C}_0$ of $\mathbb{Z}$-algebras (1.8). By 7.3, the operations ψ^k define additive endomorphisms of the functor G on λ-rings. By 1.10 they are characterized by their value on $1 + Xt$. One has

$$\frac{d}{dt}(\lambda_u(1+Xt))/\lambda_u(1+Xt) = \frac{1+Xt}{1+\{1+Xt\}u} = \sum_{k=1}^{\infty}(-1)^{k-1}\{1+Xt\}^{\circ k}u^{k-1}.$$

Thus

$$\psi^k(1+Xt) = 1 + X^k t. \tag{7.4.1}$$

Proposition 7.5. i) If K is a λ-ring, the operations ψ^k satisfy the identities

$$\psi^k(1) = 1, \qquad \psi^k(xy) = \psi^k(x)\psi^k(y), \qquad \psi^k \circ \psi^{k'} = \psi^{kk'}. \tag{7.5.1}$$

ii) Conversely, if the ψ^k satisfy these identities and if K is of characteristic zero, then K is a λ-ring.

For (i), if K is a λ-ring, $\lambda_t(1) = 1 + t$, hence

$$\frac{d}{dt}(\lambda_t(1))/\lambda_t(1) = \frac{1}{1+t} = \sum_{k=1}^{\infty}(-1)^{k-1}t^{k-1},$$

and therefore $\psi^k(1) = 1$.

To prove the second relation of (7.5.1), consider the diagram

$$\begin{array}{ccccc} K \times K & \xrightarrow{\lambda_t \times \lambda_t} & \widehat{G}(K) \times \widehat{G}(K) & \xrightarrow{g^k \times g^k} & K \times K \\ \downarrow & & \downarrow \circ & & \downarrow \\ K & \xrightarrow[\lambda_t]{} & \widehat{G}(K) & \xrightarrow[g^k]{} & K, \end{array} \tag{7.5.2}$$

where the vertical arrows are defined by multiplication, and the maps g^k as follows: the logarithmic derivative is a group homomorphism from $G(K)$ to $K[[t]]$, and the map which to

a series assigns its $(k-1)$-st coefficient, multiplied by $(-1)^{k-1}$, is an additive homomorphism from $K[[t]]$ to K; g^k is their composite. Clearly $g^k \circ \lambda_t = \psi^k$ by definition (7.1.1).

The second of formulas (7.5.1) therefore expresses the commutativity of the diagram just written. If K is a λ-ring, the left square is commutative, and it is enough to verify the commutativity of the right square. This is a diagram of functor homomorphisms on $\mathscr{C}_0$, with source $\widehat{G} \times \widehat{G}$. By 1.10 it is enough to verify commutativity on $(1+Xt, 1+Yt)$. Then

$$g^k(1+Xt) = X^k, \qquad g^k(1+Yt) = Y^k,$$

and

$$g^k((1+Xt) \circ (1+Yt)) = g^k(1+XYt) = (XY)^k = g^k(1+Xt)g^k(1+Yt),$$

which gives the result.

To prove the last relation, first observe that the diagram

$$\begin{array}{ccc} \widehat{G}_t(K) & \xrightarrow{\lambda_u} & \widehat{G}_u\widehat{G}_t(K) \\ {\scriptstyle g^k}\downarrow & & \downarrow{\scriptstyle g_u^k} \\ K & \xrightarrow[\lambda_u]{} & \widehat{G}^u(K) \end{array} \tag{7.5.3}$$

is commutative. As above, it is enough to see this for $1+Xt$, and one has

$$g_u^{k'}\lambda_u(1+Xt) = g_u^{k'}(1+\{1+Xt\}u) = (1+Xt)^{\circ k'} = 1+X^{k'}t,$$

and then

$$g^k(1+X^{k'}t) = X^{kk'} = g^k(1+Xt).$$

Consequently

$$\psi^k \circ \psi^{k'} = g^k \circ g_u^{k'} \circ \lambda_u \circ \lambda_t = g^k \circ \lambda_t \circ \psi^{k'}.$$

Moreover

$$\psi^k \circ \psi^{k'} = g^k \circ \lambda_t \circ \psi^{k'} = \psi^k \circ \lambda_t \circ \psi^{k'}.$$

If K is a λ-ring, the diagram

$$\begin{array}{ccc} K & \xrightarrow{\lambda_t} & \widehat{G}(K) \\ {\scriptstyle \psi^{k'}}\downarrow & & \downarrow{\scriptstyle \psi^{k'}} \\ K & \xrightarrow[\lambda_t]{} & \widehat{G}(K) \end{array} \tag{7.5.4}$$

is commutative by 7.3, since λ_t is a λ-homomorphism. The desired relations follow at once.

For (ii), if K is of characteristic zero, the logarithmic derivative is an isomorphism from $\widehat{G}(K)$ onto $K[[t]]$, since one can integrate. Therefore two elements of $\widehat{G}(K)$ are equal if and only if, for every k, they have the same image by g^k.

To show that λ_t is a ring homomorphism, it is enough to show that the left square in (7.5.2) is commutative. By the preceding remark, it is enough to show that, for every k, diagram (7.5.2) is commutative; the right square always is. But this is exactly the relation

$$\psi^k(xy) = \psi^k(x)\psi^k(y).$$

To show that λ_t is a λ-homomorphism, it is enough, by 7.3, to show that diagram (7.5.4) is commutative for every k'. By the preceding remark, this is equivalent to showing, for all k, k', that

$$g^k \circ \psi^{k'} \circ \lambda_t = g^k \circ \lambda_t \circ \psi^{k'},$$

that is, the relations $\psi^{kk'} = \psi^k \circ \psi^{k'}$. Finally, the relation $\lambda_t(1) = 1+t$ is obtained by integration from the relations $\psi^k(1) = 1$.

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Exposé VI. The $K^\bullet$ of a projective bundle: computations and consequences

Pierre Berthelot

1. Computation of $K^\bullet$ of a projective bundle: the case of locally free sheaves of finite type

In this section $K^\bullet(X)$ denotes the Grothendieck group of locally free sheaves of finite type on a scheme X. The aim is to prove the following structure theorem.

Theorem 1.1. Let S be a quasi-compact scheme, let E be a locally free $\mathcal{O}_S$-module of rank $r+1$, and let

$$X = \mathbf{P}(E)$$

be the associated projective bundle. The structural morphism $f : X \to S$ gives $K^\bullet(X)$ the structure of a $K^\bullet(S)$-algebra. If $\xi \in K^\bullet(X)$ denotes the class of $\mathcal{O}_X(1)$, then $K^\bullet(X)$ is a free $K^\bullet(S)$-module with basis

$$1, \xi, \ldots, \xi^r.$$

1.2. Plan of the proof

The proof is divided into three parts.

First, one proves that the powers of ξ generate $K^\bullet(X)$ as a $K^\bullet(S)$-module. For this it is enough to establish the following facts. If F is locally free of finite type on X, one can find a graded $\mathrm{Sym}(E)$-module M, of finite presentation and flat over $\mathcal{O}_S$, such that $F = \mathrm{Proj}_0(M)$. Such an M admits a finite graded resolution

$$0 \longrightarrow L_{r+1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0$$

by graded $\mathrm{Sym}(E)$-modules locally free of finite type. Finally, each L_i has a finite filtration whose quotients are of the form

$$N \otimes_{\mathcal{O}_S} \mathrm{Sym}(E)(n),$$

with N locally free of finite type on S. Applying Proj_0 then gives the required generation by the powers of ξ.

Second, one proves that $1, \xi, \ldots, \xi^{r+1}$ are linearly dependent over $K^\bullet(S)$. This is obtained from the canonical epimorphism $f^*E \to \mathcal{O}_X(1)$.

Third, one proves that $1, \xi, \ldots, \xi^r$ are linearly independent over $K^\bullet(S)$.

Let S be a scheme, let $B = \bigoplus_{n\geq 0} B_n$ be a graded $\mathcal{O}_S$-algebra, and let M be a graded B-module. If N is a graded B-module, $N(n)$ denotes the graded module defined by

$$N(n)_k = N_{n+k}.$$

A graded B-module is said to be free, respectively locally free, respectively locally free of finite type, if locally on S it is a direct sum, respectively locally a direct sum, respectively locally a finite direct sum, of modules $B(n)$. It is said to be of finite presentation if, locally on S, it admits a presentation by free graded modules of finite type.

For every integer n there is a canonical homomorphism of graded B-modules

$$M_n \otimes_{\mathcal{O}_S} B(-n) \longrightarrow M, \tag{1.2.1}$$

defined by $x \otimes a \mapsto ax$. A free graded module of finite type admits a decomposition

$$M \simeq \bigoplus_{n=n_0}^{n_1} B(-n)^{k_n}. \tag{1.2.2}$$

Proposition 1.2. The integers k_n occurring in a decomposition (1.2.2) are independent of the chosen decomposition.

Indeed, after restricting to an open set where $B_0 = \mathcal{O}_S$, one compares the first non-zero homogeneous component. This determines k_{n_0}; passing to the quotient and arguing by induction gives all the k_n.

Proposition 1.3. Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$-module of finite type, let $X = \mathbf{P}(E)$, and let F be an $\mathcal{O}_X$-module of finite presentation which is flat over S. Then there exists an integer n_0 such that, for all $n \geq n_0$,

(i) $R^i f_*(F(n)) = 0 \quad (i \geq 1)$,

(ii) $f_*(F(n))$ is locally free of finite type on S,

(iii) $M = \bigoplus_{n \geq n_0} f_*(F(n))$ is a graded $\mathrm{Sym}(E)$-module of finite presentation.

The statement is local on S. Since S is quasi-compact, one is reduced to the affine case. If S is noetherian, (i) is Serre's theorem, (ii) follows from EGA III, 7.9.10, and (iii) from the usual comparison between a coherent sheaf on a projective bundle and its associated graded module.

In the general affine case one uses passage to the limit. Write $S = \mathrm{Spec}(A)$ as a filtered projective limit of noetherian affine schemes $S_\lambda = \mathrm{Spec}(A_\lambda)$, with affine transition morphisms. The locally free module E, the scheme X, and the sheaf F descend, after increasing λ, to corresponding data $E_\lambda, X_\lambda, F_\lambda$, with F_λ flat over S_λ. The noetherian case applies to F_λ, and the theorem of change of base gives the assertions after pullback to S.

In this reduction one uses the Cartesian square

$$\begin{array}{ccc} X & \xrightarrow{g} & X_0 \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f_0} \\ S & \xrightarrow[g_0]{} & S_0. \end{array}$$

Here S_0 is noetherian, $X_0 = \mathbf{P}(E_0)$, and $F = g^*(F_0)$. The previous paragraph proves the assertion for F_0, and EGA III, 7.7.5 and 7.8.5 give the required base-change identifications for F.

Corollary 1.4. Under the hypotheses of Proposition 1.3 there exists a graded $\mathrm{Sym}(E)$-module M, of finite presentation and flat over $\mathcal{O}_S$, such that

$$F = \mathrm{Proj}_0(M).$$

Choose an integer n_0 for which the properties of Proposition 1.3 hold. For

$$M = \bigoplus_{n \geq n_0} f_*(F(n)),$$

we have seen that M is of finite presentation; it is flat over S, since the $f_*(F(n))$, for $n \geq n_0$, are locally free on S. Moreover, $F = \mathrm{Proj}_0(M)$ by EGA II 3.4. We now turn to part b) of the proof.

Lemma 1.5.1. Let A be a ring, let B be a positively graded A-algebra with $B_0 = A$, and let M be a graded B-module. Suppose that M, regarded as an ungraded B-module, is projective of finite type. Let $\widetilde{B}$ be the sheaf of algebras on $\mathrm{Spec}(A)$ defined by B, and let $\widetilde{M}$ be the associated graded $\widetilde{B}$-module. Then, locally on $\mathrm{Spec}(A)$, $\widetilde{M}$ is free of finite type over $\widetilde{B}$ as a graded module (1.2).

Indeed, since M is of finite type, its degrees are bounded below. Further, $M \otimes_B A$ is projective of finite type over A, and therefore has only finitely many non-zero homogeneous components, each projective of finite type over A. After localizing on $\mathrm{Spec}(A)$, one may thus suppose that $M \otimes_B A$ is free of finite type over A. Finally, projectivity of M gives

$$\mathrm{Tor}_1^B(M, A) = 0.$$

These properties imply that M is free of finite type over B as a graded module (Bourbaki, *Algèbre*, ch. II, § 11, prop. 7), and hence $\widetilde{M}$ is free of finite type over $\widetilde{B}$, as required.

Lemma 1.5.2. Let A be a ring, let B be an A-algebra of finite presentation, and let M be a B-module of finite presentation. Suppose that B and M are flat over A, and that, for every residue field k of A, the $B \otimes_A k$-module $M \otimes_A k$ has projective dimension at most n (as is the case if $B = A[t_0, \ldots, t_r]$). Then M has a resolution of length n by projective B-modules of finite type:

$$0 \longrightarrow L_n \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0.$$

This is the local criterion of EGA IV, 12.3.2.

Proposition 1.6. Let S be quasi-compact, let E be locally free of rank $r+1$, let $B = \mathrm{Sym}(E)$ with its usual grading, and let M be a graded B-module of finite presentation, flat over S. Then M admits a graded resolution

$$0 \longrightarrow L_{r+1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0,$$

where the L_i are graded B-modules locally free of finite type.

First, M admits a graded epimorphism $L \to M$ with L locally free of finite type: choose an integer N such that M is generated in degrees $\leq N$, and take

$$L = \bigoplus_{i \leq N} M_i \otimes_{\mathcal{O}_S} B(-i),$$

with the morphism (1.2.1). Repeating the construction for successive kernels gives a global graded resolution. After $r+1$ steps, Lemma 1.5.2 shows locally on affine opens that the final kernel is projective over B, and Lemma 1.5.1 then shows that it is locally free as a graded B-module; truncation gives the asserted resolution.

Proposition 1.7. Let S be quasi-compact, let E be locally free of finite type, let $B = \mathrm{Sym}(E)$, and let L be a graded B-module locally free of finite type. Then L admits a finite decreasing filtration by graded submodules $L^{(m)}$ such that

(i) $L^{(m)} = L$ for $m \leq m_0$,
(ii) $L^{(m)} = 0$ for $m \geq m_1$,
(iii) $\mathrm{gr}^m L \simeq N_m \otimes_{\mathcal{O}_S} B(m)$,

where N_m is locally free of finite type on S.

The question is local on S. Since S is quasi-compact, the degrees of homogeneous bases that occur locally lie between two fixed integers. One argues by induction on the length of this interval. The first step is the canonical map from the lowest non-zero homogeneous component,

$$L_{n_0} \otimes_{\mathcal{O}_S} B(-n_0) \longrightarrow L,$$

which locally identifies the corresponding direct summand. The quotient has local homogeneous bases in higher degrees, and induction applies.

1.8. Generation by the powers of ξ

Let F be locally free of finite type on X. By Corollary 1.4 and Proposition 1.6, F comes from a graded module M admitting a finite graded resolution by L_i's. Applying Proj_0 gives an exact sequence of locally free sheaves on X, and therefore in $K^\bullet(X)$

$$\mathrm{cl}(F) = \sum_i (-1)^i \,\mathrm{cl}(\mathrm{Proj}_0(L_i)).$$

Using the filtration of Proposition 1.7, each term is a sum of classes

$$\mathrm{cl}\big(f^* N_{ij}(j)\big) = f^* \,\mathrm{cl}(N_{ij})\, \xi^j.$$

Thus $K^\bullet(X)$ is generated as a $K^\bullet(S)$-module by the powers of ξ.

1.9. The relation

For linear dependence, use the canonical epimorphism $f^*E \to \mathcal{O}_X(1)$. Its kernel is locally free of rank r. Hence

$$\lambda^{r+1}(f^*E - \xi) = 0.$$

Since $\lambda_t(-\xi) = 1/(1+\xi t)$, one obtains the relation

$$\sum_{i=0}^{r+1} (-1)^i \,\lambda^i(E)\, \xi^{r+1-i} = 0, \tag{1.9.1}$$

equivalently the characteristic equation of ξ associated with the locally free module E.

1.10–1.11. Linear independence

The proof of the linear-independence assertion is due to J.-P. Jouanolou.

Lemma 1.10. With the hypotheses of Theorem 1.1, let F and G be locally free sheaves of finite type on X, and suppose that they have the same class in $K^\bullet(X)$. Then there is an integer n_0 such that, for all $n \geq n_0$,

(i) $f_*(F(n))$ and $f_*(G(n))$ are locally free of finite type on S,
(ii) $\mathrm{cl}\, f_*(F(n)) = \mathrm{cl}\, f_*(G(n))$ in $K^\bullet(S)$.

An equality of classes in $K^\bullet(X)$ is a finite integral combination of relations coming from short exact sequences of locally free sheaves. Twisting by $\mathcal{O}_X(n)$ and applying f_*, for n large enough all higher direct images vanish and all direct images are locally free of finite type; the same relations then hold in $K^\bullet(S)$.

Suppose that

$$f^*(x_0) + f^*(x_1)\xi + \cdots + f^*(x_r)\xi^r = 0. \tag{1.11.1}$$

Writing each x_i as a difference of classes of locally free sheaves and adding suitable summands, (1.11.1) is converted into an equality $\mathrm{cl}(F) = \mathrm{cl}(G)$ for locally free sheaves of finite type on X. By Lemma 1.10, for all sufficiently large n one has

$$\sum_{i=0}^{r} x_i\, \sigma_{n+i}(E) = 0 \quad \text{in } K^\bullet(S), \tag{1.11.2}$$

where $\sigma_p(E)$ denotes the class of $\mathrm{Sym}^p(E)$, and $\sigma_p(E) = 0$ for $p < 0$.

The Koszul complex of the symmetric algebra gives, for $p \geq 1$,

$$\sum_{i=0}^{r+1} (-1)^i \lambda^i(E)\, \sigma_{p-i}(E) = 0. \tag{1.11.3}$$

Using this relation, one descends by induction from large n in (1.11.2). It follows that (1.11.2) holds for all $n \geq -r$. Taking $n = -r$ gives $x_r = 0$; repeating the argument gives successively all $x_i = 0$. This proves the linear independence and hence Theorem 1.1.

Remark 1.12. Let

$$\sigma_t(E) = \sum_{i \geq 0} \sigma_i(E) t^i.$$

The relation (1.11.3) is equivalent to

$$\lambda_t(E)\, \sigma_{-t}(E) = 1.$$

It follows in particular that σ_t is additive and extends to a homomorphism

$$K^\bullet(S) \longrightarrow 1 + K^\bullet(S)[[t]]_+.$$

Remark 1.13. Any sequence of $r+1$ consecutive powers of ξ is a basis of $K^\bullet(X)$ over $K^\bullet(S)$. If

$$x = 1 - \xi^{-1}$$

is the class of a hyperplane section, then $1, x, \ldots, x^r$ is also a basis. The dependence relation becomes

$$\sum_{k=0}^{r+1} \lambda^{r+1-k}(\check{E} - r - 1)\, x^k = 0. \tag{1.13.1}$$

For $E = \mathcal{O}_S^{r+1}$, so that $X = \mathbf{P}_S^r$, this reduces to

$$x^{r+1} = 0.$$

Using relative schemes in Hakim's sense, Theorem 1.1 also makes sense when S is a locally ringed topos; if the final object of the topos is quasi-compact, the proof transports without essential change.

2. Computation of $K^\bullet$ of a projective bundle: the case of perfect complexes

The present section will not be used in the rest of the Seminar. In this section the notation $K^\bullet(X)$ denotes the Grothendieck group of perfect complexes on X (IV 2.2). We propose to prove the following theorem, analogous to 1.1:

Theorem 2.1. Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$-module of rank $r+1$, let $X = \mathbf{P}(E)$ be the associated projective bundle; let ξ be the class in $K^\bullet(X)$ of the complex equal to $\mathcal{O}_X(1)$ in degree zero, and zero in the other degrees. Then $K^\bullet(X)$, endowed with its natural $K^\bullet(S)$-algebra structure, is a free $K^\bullet(S)$-module with basis

$$1, \xi, \ldots, \xi^r.$$

2.2. Notation

Fix the notation: B will denote the symmetric algebra of E; write $\mathcal{A}$ for the category of $\mathcal{O}_X$-modules, $\mathcal{G}$ for the category of graded B-modules with graded morphisms as morphisms, $\mathcal{G}_0$ for the full subcategory of $\mathcal{G}$ whose objects are the B-modules locally free of finite type as graded B-modules, and $\mathcal{G}'$ for the category of $\mathcal{O}_S$-modules. For every n, and every complex $L^\bullet$ of $D^+(\mathcal{A})$, we shall write $L^\bullet(n)$ for the complex

$$L^\bullet(n) = L^\bullet \otimes_{\mathcal{O}_X} \mathcal{O}_X(n).$$

2.3. The functor Proj_0

We shall need to apply the functor Proj_0 to arbitrary graded B-modules, and not only to quasi-coherent graded B-modules as in the construction of EGA II 3.2. We shall therefore indicate briefly how this construction can be made in the general case.

2.3.1. First let S be a scheme, let A be a quasi-coherent $\mathcal{O}_S$-algebra, let $Y = \mathrm{Spec}(A)$, and let $f : Y \to S$ be the structural morphism. From the isomorphism

$$A \simeq f_*\mathcal{O}_Y$$

one obtains a canonical homomorphism

$$f^{-1}(A) \longrightarrow \mathcal{O}_Y, \tag{2.3.1.1}$$

where f^{-1} denotes inverse image in the sense of sheaves of sets; in other words, one has a morphism of ringed spaces

$$\bar{f} : (Y, \mathcal{O}_Y) \longrightarrow (S, A). \tag{2.3.1.2}$$

It gives an inverse-image functor $\bar{f}^*$ from the category of A-modules to the category of $\mathcal{O}_Y$-modules, left adjoint to the direct-image functor $\bar{f}_*$ from the category of $\mathcal{O}_Y$-modules to the category of $f_*(\mathcal{O}_Y)$-modules. Thus

$$\bar{f}^*(M) = f^{-1}(M) \otimes_{f^{-1}A} \mathcal{O}_Y \tag{2.3.1.3}$$

for every A-module M.

Since the homomorphism (2.3.1.1) is flat (in each fiber it corresponds to a localization), the functor $\bar{f}^*$ just defined is exact (cf. [4], p. 164). If M is a quasi-coherent A-module, this construction gives the module $\widetilde{M}$ defined in EGA II 1.4.3. Indeed, the isomorphism

$$M \simeq f_*(\widetilde{M}) = \bar{f}_*(\widetilde{M})$$

gives, by adjunction, a homomorphism

$$\bar{f}^*(M) \longrightarrow \widetilde{M}$$

and one checks immediately on each fiber that it is an isomorphism.

2.3.2. Now let B be a quasi-coherent graded $\mathcal{O}_S$-algebra, let M be a graded B-module, and let $X = \mathrm{Proj}(B)$. For every affine open U of S, and every section s of B over U, homogeneous of degree $d > 0$, write $M_{(s),U}$ for the homogeneous component of degree 0 of the module of fractions M_s; thus

$$M_{(s),U} = \big((M|_U) \otimes_{B|_U} (B|_U)_s\big)_0.$$

If V is an affine open of U, then $M_{(s),V}$ is the restriction to V of $M_{(s),U}$.

Consider the base of opens of X formed by the affine opens

$$D_+(U,s) \simeq \mathrm{Spec}\big((B|_U)_{(s)}\big)$$

Write f for the structural morphism $X \to S$, and $f_{U,s}$ for the affine morphism

$$f_{U,s} : D_+(U,s) \longrightarrow U.$$

Put

$$\mathrm{Proj}_0(M)_{U,s} = \bar{f}^*_{U,s}(M_{(s),U}).$$

With the notation of 2.3.1, we shall show that the $\mathrm{Proj}_0(M)_{U,s}$ glue as U and s vary, thereby defining an $\mathcal{O}_X$-module denoted $\mathrm{Proj}_0(M)$. The gluing for variable U and fixed s being evident, one is reduced (cf. EGA II 2.5.2) to showing that, if t is a section of B on U, homogeneous of degree $e > 0$, then the restriction of $\mathrm{Proj}_0(M)_{U,s}$ to $D_+(U,st)$ is canonically isomorphic to $\mathrm{Proj}_0(M)_{U,st}$. Consider

$$\begin{array}{ccc} D_+(st) & \xrightarrow{\ i\ } & D_+(s) \\ & \searrow^{f_{U,st}} & \downarrow{\scriptstyle f_{U,s}} \\ & & U \end{array}$$

It is a matter of showing that the modules

$$i^*\bar{f}^*_{U,s}(M_{(s),U}) \quad \text{and} \quad \bar{f}^*_{U,st}(M_{(st),U})$$

are canonically isomorphic. On the one hand, one has the canonical isomorphisms

$$\begin{aligned} i^*\bar{f}^*_{U,s}(M_{(s),U}) &\simeq i^*\Big(f^{-1}_{U,s}(M_{(s),U}) \otimes_{f^{-1}_{U,s}((B|_U)_{(s)})} \mathcal{O}_{D_+(s)}\Big) \\ &\simeq i^{-1}\Big(f^{-1}_{U,s}(M_{(s),U}) \otimes_{f^{-1}_{U,s}((B|_U)_{(s)})} \mathcal{O}_{D_+(s)}\Big) \otimes_{i^{-1}(\mathcal{O}_{D_+(s)})} \mathcal{O}_{D_+(st)} \\ &\simeq f^{-1}_{U,st}(M_{(s),U}) \otimes_{f^{-1}_{U,st}((B|_U)_{(s)})} \mathcal{O}_{D_+(st)}. \end{aligned}$$

On the other hand,

$$\bar{f}^*_{U,st}(M_{(st),U}) \simeq f^{-1}_{U,st}(M_{(st),U}) \otimes_{f^{-1}_{U,st}((B|_U)_{(st)})} \mathcal{O}_{D_+(st)}.$$

Finally, it follows from EGA II 2.2.2, translated into sheaf language, that one has the canonical isomorphisms

$$M_{(st),U} \simeq (M_{(s),U})_{t^d/s^e} \simeq M_{(s),U} \otimes_{(B|_U)_{(s)}} (B|_U)_{(st)},$$

and hence the desired isomorphism.

It is clear that, if M is quasi-coherent, the module $\mathrm{Proj}_0(M)$ thus constructed is indeed the one defined in EGA II.

2.3.3. It follows from 2.3.1 and 2.3.2 that the functor Proj_0 is exact. Returning to the notation of 2.2, by passage to derived categories one therefore obtains a functor, still denoted Proj_0:

$$\mathrm{Proj}_0 : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{A}).$$

It is also clear from the preceding constructions that the functor Proj_0, on the category of graded B-modules, is compatible with tensor product; the same remains true of the functor obtained on the derived categories.

2.4. The functor F_k

For an $\mathcal{O}_X$-module L and an integer k, set

$$F_k(L) = \bigoplus_{n \geq k} f_*(L(n)). \tag{2.4.1}$$

The functor F_k is left exact and therefore has a right derived functor

$$RF_k : D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{G}). \tag{2.4.2}$$

If L is locally free of finite type, then for k sufficiently large it is F_k-acyclic, by the vanishing theorem used above in 1.3.

For every k there is a functorial morphism

$$\beta : \mathrm{Proj}_0(F_k(F)) \longrightarrow F, \tag{2.5.1}$$

constructed on each $D_+(U, s)$ by the usual localization formula: a fraction represented by a homogeneous section of sufficiently high degree is sent to the corresponding local section of F. In the quasi-coherent case this is the canonical morphism of EGA. Passing to derived categories gives

$$\mathrm{Proj}_0 \circ RF_k \longrightarrow \mathrm{id}_{D^+(\mathcal{A})}\,. \tag{2.5.2}$$

Proposition 2.6. Let S be quasi-compact, let E be locally free of finite type, let $X = \mathbf{P}(E)$, and let $L^\bullet$ be a perfect complex on X. Then, for k sufficiently large,

(i) $RF_k(L^\bullet)$ is $\mathcal{G}_0$-perfect,

(ii) $\mathrm{Proj}_0\, RF_k(L^\bullet) \longrightarrow L^\bullet$ is an isomorphism in $D^+(\mathcal{A})$.

The assertion is local on S. One may therefore assume S affine. Since $\mathcal{O}_X(1)$ is ample, the perfect complex $L^\bullet$ is locally, and then globally after refining, represented by a bounded complex whose terms are locally free of finite type. For k sufficiently large all those finitely many terms are F_k-acyclic, and the graded modules $F_k(L^i)$ are $\mathcal{G}_0$-perfect. This gives (i), and the morphism (2.5.2) is then an isomorphism term by term, giving (ii).

2.7. Decomposition of a $\mathcal{G}_0$-perfect complex

On $\mathcal{G}$, the functor which associates to a graded B-module its n-th homogeneous component is exact with values in $\mathcal{G}'$; it defines a functor

$$\mathrm{Comp}_n : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{G}'). \tag{2.7.1}$$

The homomorphism (1.2.1) gives a morphism of functors

$$\mathrm{Comp}_n \otimes_{\mathcal{O}_S} B(-n) \longrightarrow \mathrm{id}_{D^+(\mathcal{G})}\,. \tag{2.7.2}$$

Let $M^\bullet$ be a $\mathcal{G}_0$-perfect complex. We shall show that there exists a finite sequence of distinguished triangles

$$\begin{array}{ccc} & P_1^\bullet & \\ \swarrow & & \nwarrow \\ N_1^\bullet & \longrightarrow & M^\bullet, \end{array} \qquad \begin{array}{ccc} & P_2^\bullet & \\ \swarrow & & \nwarrow \\ N_2^\bullet & \longrightarrow & P_1^\bullet, \end{array} \quad \cdots, \quad \begin{array}{ccc} & P_k^\bullet & \\ \swarrow & & \nwarrow \\ N_k^\bullet & \longrightarrow & P_{k-1}^\bullet, \end{array}$$

where each $P_k^\bullet$ at the last stage is acyclic, the terms $N_i^\bullet$ are of the form

$$N_i^\bullet = K_i^\bullet \otimes_{\mathcal{O}_S} B(d_i),$$

and the $K_i^\bullet$ are perfect complexes on S in the ordinary sense.

Indeed, because $M^\bullet$ is $\mathcal{G}_0$-perfect and S is quasi-compact, there exists a finite covering of S and an integer d_0 such that, on every open of the covering, $M^\bullet$ is represented by a bounded complex of locally free B-modules of finite type, with no homogeneous generators in degrees $< d_0$. Taking d_0 minimal, the morphism (2.7.2) extracts the part of the lowest degree. Its cone has all homogeneous generators in strictly higher degrees. Repeating the process, and using the finiteness of the degree interval from Proposition 1.7, one obtains the desired finite sequence of triangles.

2.8. Proof of Theorem 2.1

Let $L^\bullet$ be a perfect complex on X. For k sufficiently large put

$$M^\bullet = RF_k(L^\bullet).$$

By Proposition 2.6, $M^\bullet$ is $\mathcal{G}_0$-perfect and $\mathrm{Proj}_0(M^\bullet) \simeq L^\bullet$. Applying the decomposition of 2.7 and then Proj_0, one obtains in $K^\bullet(X)$

$$\mathrm{cl}(L^\bullet) = \sum_i \mathrm{cl}(f^* K_i^\bullet(d_i)) = \sum_i f^* \,\mathrm{cl}(K_i^\bullet)\, \xi^{d_i}.$$

Thus the powers of ξ generate $K^\bullet(X)$ as a $K^\bullet(S)$-module.

The dependence relation is the same relation (1.9.1), now interpreted in the Grothendieck group of perfect complexes. For linear independence one uses the $K^\bullet(S)$-linear homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(S).$$

For the structure sheaf and negative twists of the projective bundle one has

$$f_*(1) = 1, \qquad f_*(\xi^n) = 0 \quad (-r \le n < 0).$$

Multiplying a relation among $1, \xi, \ldots, \xi^r$ by suitable negative powers of ξ and applying f_* successively gives all its coefficients equal to zero. This completes the proof of Theorem 2.1.

This completes the induction, for if $0 \le s \le r$ and if $x_j = 0$ for $s < j \le r$, then from a relation

$$\sum_{i=0}^{s} f^*(x_i)\xi^i = 0$$

after multiplication by ξ^{r-s} one gets

$$\sum_{i=0}^{s} f^*(x_i)\xi^{i+r-s} = 0,$$

and hence $x_s = 0$ by what precedes.

3. Consequences of the structure theorem for the $K^\bullet$ of a projective bundle

Proposition 3.1. Let S be a quasi-compact scheme, let $E_1, \ldots, E_n$ be $\mathcal{O}_S$-modules locally free of finite type, and let $D(E_i)$ be the flag scheme of E_i. Put

$$X = D(E_1) \times_S \cdots \times_S D(E_n).$$

Then the canonical homomorphism

$$K^\bullet(S) \longrightarrow K^\bullet(X)$$

is injective, where $K^\bullet$ is defined either from $\mathcal{O}_S$-modules locally free of finite type or from perfect complexes.

One verifies this immediately by using the universal property of flag schemes. The scheme X can be constructed as follows: one considers a sequence of S-schemes $X_1, \ldots, X_n$, defined by induction by taking X_k to be the flag scheme of the inverse image of E_k on X_{k-1}; then $X = X_n$. Thus, to prove the proposition, it is enough to prove it when $n = 1$, the general case following by induction. We may therefore suppose that X is the flag scheme of an $\mathcal{O}_S$-module E, locally free of finite type.

The scheme X may then be constructed as follows. Put $E = F_0$, $S = X_0$, and define successively schemes $X_1, \ldots, X_n$ and, for each i, an $\mathcal{O}_{X_i}$-module locally free of finite type F_i, by

$$X_i = \mathbb{P}(F_{i-1}), \qquad F_i = \mathrm{Ker}\big(f_i^*(F_{i-1}) \longrightarrow \mathcal{O}_{X_i}(1)\big),$$

where $f_i : X_i \to X_{i-1}$ is the structural morphism. It follows that it is enough to prove the proposition for a projective bundle. This is precisely the content of 1.1, respectively of 2.1, according to which definition of $K^\bullet$ is being used.

Theorem 3.2. Let S be a quasi-compact scheme. The pre-λ-ring $K^\bullet(S)$, defined from $\mathcal{O}_S$-modules locally free of finite type, is a λ-ring.

The homomorphism

$$\lambda_t : K^\bullet(S) \longrightarrow 1 + K^\bullet(S)[[t]]^+$$

being additive, it is enough, in order to verify that it is a λ-homomorphism, to check the universal relations on elements of the form $\mathrm{cl}(E)$, where E is an $\mathcal{O}_S$-module locally free of finite type. Thus one has to show that for every pair E, F of such modules, with classes again denoted E, F, the relations V 2.4.2 are satisfied:

$$\lambda^i(EF) = P_i(\lambda^1(E), \ldots, \lambda^i(E); \lambda^1(F), \ldots, \lambda^i(F)),$$

$$\lambda^j(\lambda^i(E)) = Q_{i,j}(\lambda^1(E), \ldots, \lambda^{ij}(E)).$$

Since S is quasi-compact, it admits a finite covering by disjoint opens such that on each of them E and F have constant rank. Since the $K^\bullet$ of a finite disjoint union is the product of the corresponding $K^\bullet$'s, it is enough to check the relations on the members of this covering; hence we may suppose that E and F have constant ranks on S.

Put then

$$X = D(E) \times_S D(F).$$

By Proposition 3.1 the λ-homomorphism $K^\bullet(S) \to K^\bullet(X)$ is injective. Consequently, in order to prove the relations V 2.4.2, we may work in $K^\bullet(X)$. But in $K^\bullet(X)$ the elements E and F split as sums of classes of invertible sheaves. By additivity it is therefore enough to prove the same relations for elements ξ, η which are classes of invertible sheaves. Written in the form V 2.4.1, the relations are

$$1 + \xi\eta t = (1 + \xi t) \circ (1 + \eta t), \qquad \lambda_u(1 + \xi t) = 1 + \{1 + \xi t\}u.$$

These are just the relations V 2.3.1 and V 2.3.3 defining the λ-ring structure on $1 + K^\bullet(S)[[t]]^+$.

Let us also record the following more general result.

Theorem 3.3. Let (T, A) be a commutatively ringed topos, and let K be the Grothendieck group of A-modules locally free of bounded rank on T. Then K, endowed with its canonical pre-λ-structure, is a λ-ring.

To prove that K is a λ-ring, it is necessary to prove the relations V 2.4.1; by additivity, it is enough to prove them for classes of A-modules E, F locally free of bounded rank. Since the ranks of E and F are bounded, one may decompose T as a finite disjoint sum of opens over which E and F have constant rank. Using the compatibility of K with finite disjoint sums, and writing the relations V 2.4.1 in the form V 2.4.2, we are reduced to the case of modules of constant rank.

Let E be an A-module locally free of finite rank n. There is a torsor P under $\mathrm{GL}(n, A)$ such that

$$E = P \times_{\mathrm{GL}(n,A)} A^n,$$

where A^n is considered as a $\mathrm{GL}(n, A)$-module by the standard representation. One can then define a λ-homomorphism from the representation ring $R_{\mathbb{Z}}(\mathrm{GL}(n))$ to K by associating to a representation ρ of $\mathrm{GL}(n)$ in a free $\mathbb{Z}$-module of rank p the class of

$$P \times_{\mathrm{GL}(n,A)} A^p.$$

Under this homomorphism, the class of E is the image of the standard representation. Thus any polynomial relation among the $\lambda^i(E)$'s follows from the corresponding relation for the standard representation in $R_{\mathbb{Z}}(\mathrm{GL}(n))$.

Similarly, for two locally free modules E, F of ranks n, p, one uses a torsor under $\mathrm{GL}(n, A) \times \mathrm{GL}(p, A)$, with $\mathrm{GL}(n, A)$ acting through the standard representation and $\mathrm{GL}(p, A)$ through the trivial one on the first factor, and conversely for F. We are reduced to proving that, for every group scheme G over $\operatorname{Spec}\mathbb{Z}$ which is a finite product of linear group schemes, the pre-λ-ring $R_{\mathbb{Z}}(G)$ is a λ-ring. This may be obtained by the explicit computation of $R_{\mathbb{Z}}(G)$ in [5]; alternatively, one can reason as in 3.2, using 1.14.

4. Computation of the $K^\bullet$ of a flag bundle

4.1

Let S be a scheme and let E be an $\mathcal{O}_S$-module locally free of rank n. Let

$$\pi = (p_1, \ldots, p_k), \qquad p_i \geq 1, \qquad \sum_{i=1}^{k} p_i = n.$$

The flag functor of type π of E is the functor on the category of S-schemes, with values in sets, which to an S-scheme $f : X \to S$ associates the set of chains of quotients of f^*E

$$(L_{k-1}, \ldots, L_1),$$

where, for each i, L_i is a quotient of L_{i+1}, locally free of rank $p_1 + \cdots + p_i$. This functor is representable by an S-scheme $D_\pi(E)$, endowed with a canonical sequence of quotients of the inverse image of E, locally free of ranks $p_1, p_1 + p_2, \ldots$. If $\pi = (1, n-1)$, then $D_\pi(E) = \mathbb{P}(E)$; if $\pi = (r, n-r)$, then $D_\pi(E) = G_{r,n}(E)$, the Grassmannian of type $(r, n-r)$ of E. If all p_i's are equal to 1, one obtains the usual flag bundle of E, denoted $D(E)$.

Lemma 4.2. Let S be a scheme, let E be an $\mathcal{O}_S$-module locally free of rank n, and let

$$\pi = (p_1, \ldots, p_i, \ldots, p_k), \qquad p_1 + \cdots + p_k = n.$$

Let $p', p'' \geq 1$ be such that $p' + p'' = p_i$, and put

$$\pi' = (p_1, \ldots, p_{i-1}, p', p'', p_{i+1}, \ldots, p_k), \qquad \pi'' = (p', p'').$$

Let $L_1, \ldots, L_{k-1}$ be the canonical quotients of the inverse image of E on $D_\pi(E)$, and let F be the kernel of $L_i \to L_{i-1}$. Then there is a canonical S-isomorphism

$$D_{\pi'}(E) \simeq D_{\pi''}(F).$$

The proof, immediate from the functorial definition of flag schemes, is left to the reader.

Lemma 4.3. Let A be a commutative ring with unit, let $n > 0$, and let $(c_i)_{1 \leq i \leq n}$ be elements of A. Let C be the A-algebra generated by $\xi_1, \ldots, \xi_n$, subject to the relations

$$\sigma_i(\xi_1, \ldots, \xi_n) = c_i, \qquad i = 1, \ldots, n,$$

where σ_i denotes the elementary symmetric functions. Let $p_1, \ldots, p_k$ be integers ≥ 1 whose sum is n. For $1 \leq j \leq k$ put

$$\xi^{(j)} = (\xi_1^{(j)}, \ldots, \xi_{p_j}^{(j)}), \qquad \xi_i^{(j)} = \xi_{\sum_{\alpha<j} p_\alpha + i}.$$

Put

$$c_i^{(j)} = \sigma_i(\xi_1^{(j)}, \ldots, \xi_{p_j}^{(j)}), \qquad 1 \leq j \leq k, \quad 1 \leq i \leq p_j,$$

and let B be the subalgebra of C generated by the $c_i^{(j)}$'s. Then:

i) The elements

$$(\xi^{(1)})^{\alpha(1)} \cdots (\xi^{(k)})^{\alpha(k)}, \qquad 0 \leq \alpha_i^{(j)} \leq p_j - i,$$

form a basis of C, considered as a B-module.

ii) The generators $c_i^{(j)}$ of the A-algebra B satisfy the n isobaric relations deduced from

$$\sum_{i=1}^{n} c_i = \prod_{j=1}^{k} \left(\sum_{i=1}^{p_j} c_i^{(j)} \right),$$

where c_i and $c_i^{(j)}$ are assigned weight i; these n relations generate the ideal of relations among the $c_i^{(j)}$'s.

Corollary 4.4. The elements of C of the form

$$\xi_1^{\alpha_1} \cdots \xi_{n-1}^{\alpha_{n-1}}, \qquad 0 \leq \alpha_i \leq n - i,$$

form a basis of C as an A-module.

For the proof of the lemma and corollary we refer to [2], Exposé 4, pages 4.19, from which they are borrowed.

4.5

Let S be quasi-compact, let E be an $\mathcal{O}_S$-module locally free of rank n, let $\pi = (p_1, \ldots, p_k)$ be a sequence of integers ≥ 1 with sum n, and put $X = D_\pi(E)$. The scheme X is equipped with canonical quotients L_i. Put $L_0 = 0$, $L_k = f^*E$, where $f : X \to S$ is the structural morphism. For $1 \leq i \leq k$, let F_i be the kernel of $L_i \to L_{i-1}$. Let $K^\bullet(X)$ be the Grothendieck group of

locally free finite-type sheaves on X, respectively of perfect complexes. Denote by F_j the class of F_j, and by $\lambda_i^{(j)}$ the class of $\lambda^i(F_j)$. In the case where $K^\bullet(X)$ is defined by perfect complexes, one gives the same evident meaning to the series $\lambda_t(E)$ and $\lambda_t(F_j)$, where E is the class of E. Finally,

$$\sum_{j=1}^{k} F_j = E.$$

Theorem 4.6. With the hypotheses and notations of 4.5, the $K^\bullet(S)$-algebra $K^\bullet(X)$ is generated by the $\lambda_i^{(j)}$, and the ideal of relations among the $\lambda_i^{(j)}$'s is generated by the relations deduced from

$$\lambda_t(E) = \prod_{j=1}^{k} \lambda_t(F_j).$$

Equivalently, $K^\bullet(X)$ is the $K^\bullet(S)$-λ-algebra generated by $F_1, \ldots, F_k$, subject to the relations

$$\text{(i)} \quad \lambda^p(F_i) = 0 \quad (p > p_i), \qquad \text{(ii)} \quad \sum_i F_i = E.$$

We first treat the case where all p_i's are equal to 1. Then the theorem reads as follows.

Corollary 4.7. Let S be quasi-compact, let E be an $\mathcal{O}_S$-module locally free of rank n, let $X = D(E)$ be the flag bundle of E, and let $\xi_1, \ldots, \xi_n$ be the classes of the invertible sheaves $F_1, \ldots, F_n$. Then $K^\bullet(X)$ is the $K^\bullet(S)$-algebra generated by the ξ_i's, subject to the relations

$$\sigma_i(\xi_1, \ldots, \xi_n) = \lambda^i(E).$$

We argue by induction on n. For $n = 1$ the assertion is trivial. Suppose it known for $n - 1$. Consider $\mathbb{P}(E)$, and on $\mathbb{P}(E)$ the exact sequence

$$0 \longrightarrow F \longrightarrow g^*E \longrightarrow \mathcal{O}_{\mathbb{P}(E)}(1) \longrightarrow 0,$$

where g is the structural morphism of $\mathbb{P}(E)$. By Lemma 4.2, X identifies with the usual flag bundle $D(F)$ over $\mathbb{P}(E)$, and the canonical sheaves correspond under this identification. The diagram

$$\begin{array}{ccc} D(E) = D(F) & \xrightarrow{\ h\ } & \mathbb{P}(E) \\ & \searrow^{f} & \downarrow{\scriptstyle g} \\ & & S \end{array}$$

shows that $K^\bullet(S) \to K^\bullet(X)$ factors through $K^\bullet(\mathbb{P}(E))$. By the induction hypothesis applied to $D(F)$, and by Corollary 4.4, $K^\bullet(X)$ has over $K^\bullet(\mathbb{P}(E))$ a basis formed by

$$\xi_2^{\alpha_2} \cdots \xi_n^{\alpha_{n-1}}, \qquad 0 \le \alpha_i \le n - i.$$

Moreover $K^\bullet(\mathbb{P}(E))$ has over $K^\bullet(S)$ the basis $1, \xi_1, \ldots, \xi_1^{n-1}$. Hence $K^\bullet(X)$ has over $K^\bullet(S)$ the basis

$$\xi_1^{\alpha_1} \cdots \xi_{n-1}^{\alpha_{n-1}}, \qquad 0 \le \alpha_i \le n - i.$$

In addition, the additivity of λ_t gives $\prod_i (1 + \xi_i t) = \lambda_t(E)$, so the displayed relations are satisfied. If C is the $K^\bullet(S)$-algebra generated by ξ_i's and subjected to these relations, then there is a surjective homomorphism $C \to K^\bullet(X)$. By Corollary 4.4 this homomorphism sends a basis of C to a basis of $K^\bullet(X)$, hence it is an isomorphism.

For a general flag bundle of type π, the usual flag bundle $D(E)$ is obtained from $D_\pi(E)$ by constructing successively the flag bundles of the F_j's. It follows that $K^\bullet(D(E))$, considered as a

module over $K^\bullet(X)$, has a basis formed by the elements appearing in 4.3.i), the $\xi_i^{(j)}$'s being the classes of the sheaves analogous to the F_j's in the splitting of the inverse images of the F_j's on $D(E)$. Thus $K^\bullet(X)$ is a subring of $K^\bullet(D(E))$. Moreover, additivity gives

$$\lambda_i^{(j)} = \sigma_i(\xi_1^{(j)}, \ldots, \xi_{p_j}^{(j)}),$$

and therefore $K^\bullet(X)$ contains the sub-$K^\bullet(S)$-algebra of $K^\bullet(D(E))$ generated by the $\lambda_i^{(j)}$'s. If B is this subalgebra, then by 4.3.i) the same family of elements is a basis both for the B-module structure and for the $K^\bullet(X)$-module structure. Hence $K^\bullet(X) = B$, proving generation. The ideal of relations among the $\lambda_i^{(j)}$'s is then given by 4.3.ii), taking into account the structure of $K^\bullet(D(E))$ over $K^\bullet(S)$ determined by 4.7, and the fact that the $\xi_i^{(j)}$'s are precisely the classes involved in the splitting of the inverse image of E on $D(E)$. This proves the first form of 4.6.

For the second form, one observes that the elements F_i of the $K^\bullet(S)$-λ-algebra $K^\bullet(X)$ satisfy (i) and (ii). If A is the $K^\bullet(S)$-λ-algebra generated by elements F_i' satisfying (i) and (ii), there is therefore a $K^\bullet(S)$-λ-homomorphism $A \to K^\bullet(X)$ sending F_i' to F_i. By 4.6 this homomorphism is surjective. Since the λ-ideal generated by (i) and (ii) in $K^\bullet(S)[F_1, \ldots, F_k]$ contains the usual ideal generated by the first form of the relations, the homomorphism is also injective. Thus it is an isomorphism.

Corollary 4.8. With the hypotheses and notations of 4.5, put

$$c_i^{(j)} = \gamma^i(F_j - p_j), \qquad 1 \leq i \leq p_j.$$

Then the $K^\bullet(S)$-algebra $K^\bullet(X)$ is generated by the $c_i^{(j)}$'s, and the ideal of relations among them is generated by the relations deduced from

$$\gamma_t(E - n) = \prod_{j=1}^{k} \gamma_t(F_j - p_j).$$

This is merely another formulation of 4.6. Indeed, the $c_i^{(j)}$'s may be expressed as functions of the $\lambda_i^{(j)}$'s and conversely. Moreover the relation $\lambda_t(E) = \prod \lambda_t(F_j)$ is equivalent to $\gamma_t(E - n) = \prod \gamma_t(F_j - p_j)$.

4.9

Let S be a scheme. Denote by $G_S^{n,r}$ the Grassmannian of type $(n, r-n)$ of $\mathcal{O}_S^r$, with $r \geq n$. For variable r, the $G_S^{n,r}$ form an inductive system, the morphism

$$G_S^{n,r} \longrightarrow G_S^{n,r+1}$$

being defined by considering $\mathcal{O}^r_{G_S^{n,r}}$ as the quotient of $\mathcal{O}^{r+1}_{G_S^{n,r}}$ by the $(r+1)$-st factor. In this system of morphisms the canonical quotients therefore correspond by inverse image. The inductive system so defined will be denoted $G_S^{n,\infty}$.

We define its $K^\bullet$ by the formula

$$K^\bullet(G_S^{n,\infty}) = \varprojlim_r K^\bullet(G_S^{n,r}). \tag{4.9.1}$$

Let F_1, F_2 be the canonical sheaves on $G_S^{n,r}$ (4.5), and F_1, F_2 their classes in $K^\bullet(G_S^{n,r})$. Put

$$c_i = \gamma^i(F_1 - n).$$

When r varies, the c_i correspond under the transition morphisms and consequently define elements, still denoted c_i, of $K^\bullet(G_S^{n,\infty})$.

Proposition 4.10. Let S be a quasi-compact scheme. Then, with the notation of 4.9, one has an isomorphism

$$K^\bullet(G_S^{n,\infty}) \simeq K^\bullet(S)[[c_1, \ldots, c_n]].$$

By 4.8, $K^\bullet(G_S^{n,r})$ is the $K^\bullet(S)$-algebra generated by

$$c_i = \gamma^i(F_1 - n), \quad 1 \leq i \leq n,$$

and by

$$\gamma^j(F_2 - r + n), \quad 1 \leq j \leq r - n,$$

subject to the relations deduced from

$$\gamma_t(F_1 - n) \cdot \gamma_t(F_2 - r + n) = \gamma_t(r - r) = 1,$$

or equivalently

$$\gamma_t(F_2 - r + n) = 1/\gamma_t(F_1 - n).$$

This relation makes it possible to express the $\gamma^j(F_2 - r + n)$ in terms of the c_i, and gives a series of isobaric relations among the c_i (where c_i is assumed to have weight i) by writing that the coefficients of $1/\gamma_t(F_1 - n)$ vanish in degree $> r - n$; moreover, these relations generate the ideal of relations among the c_i. We may therefore write

$$K^\bullet(G_S^{n,r}) = K^\bullet(S)[c_1, \ldots, c_n]/I_r,$$

where I_r is a homogeneous ideal of $K^\bullet(S)[c_1, \ldots, c_n]$, of degree $> r - n$. It then follows from the definition (4.9.1) that $K^\bullet(G_S^{n,\infty})$ is the separated completion of $K^\bullet(S)[c_1, \ldots, c_n]$ for the topology defined by the I_r. This topology is finer than the topology defined by weight; we shall show that it is equivalent to it, which will imply the proposition.

Put $A = K^\bullet(S)[c_1, \ldots, c_n]/I_r$, and let C be the A-algebra generated by generators $x_1, \ldots, x_n$ subject to the relations

$$\sigma_j(x_1, \ldots, x_n) = c_j,$$

where the σ_j denote the elementary symmetric functions. By 4.4, the canonical homomorphism $A \to C$ is injective. We obtain

$$\gamma_t(F_1 - n) = 1 + c_1 t + \cdots + c_n t^n = \prod_{i=1}^{n} (1 + x_i t).$$

Since $1/\gamma_t(F_1 - n)$ is a polynomial, so is $1/(1 + x_i t)$, for every i. Hence the x_i are nilpotent, and there exists an integer N such that every polynomial of degree $> N$ in the x_i, with coefficients in $K^\bullet(S)$, is zero. Consequently every polynomial of A of weight $> N$ is zero, i.e. every polynomial of $K^\bullet(S)[c_1, \ldots, c_n]$ of weight $> N$ belongs to I_r. It follows that the two topologies under consideration are equivalent, completing the proof.

5. Applications to projective bundles; study of f_*

5.1

Let S be quasi-compact, let E be locally free of rank $r + 1$, and put $X = \mathbb{P}(E)$, with structural morphism $f : X \to S$. On X one has the canonical exact sequence

$$0 \longrightarrow F \longrightarrow f^*E \longrightarrow \mathcal{O}_X(1) \longrightarrow 0.$$

Let E, F, ξ denote the corresponding classes in $K^\bullet$, and put $x = \xi - 1$.

5.2

For a locally free sheaf G on X, the higher direct images $R^i f_* G$ need not be locally free of finite type on S. Thus the direct image $f_* : K^\bullet(X) \to K^\bullet(S)$ is not defined here by alternating sums of higher direct images. Theorem 1.1 gives a substitute: every $y \in K^\bullet(X)$ is uniquely expressible in the form

$$y = a_0 + a_1\xi + \cdots + a_r\xi^r, \qquad a_i \in K^\bullet(S),$$

and we put

$$f_*(y) = a_0.$$

This additive homomorphism satisfies the projection formula. Moreover,

$$f_*(1) = 1, \qquad f_*(\xi^n) = 0 \quad (-r \le n < 0), \qquad f_*(\xi^n) = s_n \quad (n \ge 0),$$

where s_n is the n-th complete symmetric function in the roots of E. The last relations follow by induction from the dependence relation and the Koszul relation.

Proposition 5.3. For every $k \ge 0$,

$$\mathrm{Fil}^k K^\bullet(X) = \mathrm{Fil}^k K^\bullet(S) + \mathrm{Fil}^{k-1} K^\bullet(S)x + \cdots + \mathrm{Fil}^{k-r} K^\bullet(S)x^r,$$

as a direct sum of $K^\bullet(S)$-modules, where the filtration is the canonical filtration of augmented λ-rings.

Here x has augmentation zero and

$$\gamma_t(x) = \lambda_{t/(1-t)}(1 - \xi^{-1}) = \frac{1}{1 - xt},$$

so that $\gamma^i(x) = x^i$. The result follows from the computation of the filtration for free λ-rings in V, 4.12 and V, 4.13, together with the relation (1.9.1). Directness follows from the independence of $1, x, \ldots, x^r$.

Corollary 5.4. The filtration induced on $K^\bullet(S)$ by $K^\bullet(X)$ is the original filtration of $K^\bullet(S)$.

Corollary 5.5. If X is a product over S of flag bundles of locally free sheaves, then

$$\mathrm{Gr}\, K^\bullet(S) \longrightarrow \mathrm{Gr}\, K^\bullet(X)$$

is injective.

Corollary 5.6. With the notation of the projective bundle above, $\mathrm{Gr}\, K^\bullet(X)$ is a free $\mathrm{Gr}\, K^\bullet(S)$-module with basis $1, x, \ldots, x^r$, and

$$\mathrm{Gr}^k K^\bullet(X) = \mathrm{Gr}^k K^\bullet(S) \oplus \mathrm{Gr}^{k-1} K^\bullet(S)x \oplus \cdots \oplus \mathrm{Gr}^{k-r} K^\bullet(S)x^r.$$

The elements $1, x, \ldots, x^{r+1}$ satisfy

$$\sum_{i=0}^{r+1} (-1)^i c_i(E) x^{r+1-i} = 0,$$

where $c_i(E)$ is the i-th Chern class of E.

Corollary 5.7. For a flag bundle as in 4.5, the $\mathrm{Gr}\, K^\bullet(S)$-algebra $\mathrm{Gr}\, K^\bullet(X)$ is generated by the classes $\xi_i^{(\alpha)}$ of $\xi_i^{(\alpha)} - 1$, with relations generated by

$$c_t(E) = \prod_{i,\alpha} c_t(\xi_i^{(\alpha)}),$$

where c_t denotes the total Chern class.

Corollary 5.8. For every k,

$$f_*(\mathrm{Fil}^k K^\bullet(X)) \subset \mathrm{Fil}^k K^\bullet(S).$$

This follows immediately from Proposition 5.3 and the projection formula.

Proposition 5.9. With the notation of 5.1, for every $y \in K^\bullet(X)$, one has

$$f_*(\lambda_{-1}(F)y) = 0, \qquad f_*(\gamma^*(F)y) = 0.$$

In particular,

$$f_*(\lambda_{-1}(F)) = 1.$$

If

$$\xi_t : 1 + K^\bullet(X)[[t]]^+ \longrightarrow 1 + K^\bullet(S)[[t]]^+$$

is defined coefficientwise by

$$\xi_t\left(\sum a_i t^i\right) = \sum f_*(a_i)t^i,$$

then, although it is not in general a homomorphism of groups, it satisfies

$$\xi_t(mf^*y) = \xi_t(m)y.$$

Using $E = F + \xi$ and $\lambda_t(E) = \lambda_t(F)(1 + \xi t)$, one obtains the asserted identities.

Proposition 5.10. Let $z \in K^\bullet(X)$. The condition

$$z(1 - \xi) = 0$$

is equivalent to the existence of an element $y \in K^\bullet(S)$ such that

$$\text{(i)} \quad z = y\lambda_{-1}(F), \qquad \text{(ii)} \quad y\lambda_{-1}(E) = 0.$$

Sufficiency follows from

$$\lambda_{-1}(E) = \lambda_{-1}(F)\lambda_{-1}(\xi) = \lambda_{-1}(F)(1 - \xi).$$

Conversely, write $z = \sum_{i=0}^r a_i\xi^i$. Combining the dependence relation among the powers of ξ with $z(1 - \xi) = 0$, and using the independence of $1, \ldots, \xi^r$, expresses z as a multiple of $\lambda_{-1}(F)$. The top relation gives $y\lambda_{-1}(E) = 0$.

Proposition 5.11. Let $z \in \mathrm{Gr}\, K^\bullet(X)$. The condition $zx = 0$ is equivalent to the existence of $y \in \mathrm{Gr}\, K^\bullet(S)$ such that

$$\text{(i)} \quad z = y\, c_{\mathrm{top}}(F), \qquad \text{(ii)} \quad y\, c_{\mathrm{top}}(E) = 0.$$

The proof is the same as that of 5.10, after passing to associated graded rings and using 5.6.

6. Study of the filtration of $K^\bullet(X)$ when X has an ample sheaf

Let X be a scheme possessing an ample sheaf. Then the two definitions of $K^\bullet(X)$, by locally free finite-type sheaves and by perfect complexes, coincide. Let

$$\varepsilon : K^\bullet(X) \longrightarrow H^0(X, \mathbb{Z})$$

be the augmentation.

Proposition 6.1. If X is quasi-compact and has an ample sheaf, then $\mathrm{Fil}^1 K^\bullet(X)$ is a nilideal.

It is enough to show that $E - \varepsilon(E)$ is nilpotent, where E is the class of a locally free sheaf. By decomposing X into finitely many disjoint opens and then passing to the flag bundle of E, one reduces to the case where E is the class of an invertible sheaf. Let L be the class of an ample invertible sheaf. For n sufficiently large, $E^\vee(n)$ is generated by global sections, giving a surjection $(E(-n))^m \to \mathcal{O}_X$ with locally free kernel. Thus in $K^\bullet(X)$

$$mEL^{-n} = 1 + F.$$

The corresponding γ-relation shows that $1 - EL^{-n}$, and therefore $L^n - E$, is nilpotent. Since $E - 1 = (L^n - 1) - (L^n - E)$, and since $L^n - 1$ is also nilpotent for n large, the assertion follows.

Proposition 6.2. Let R be an augmented sub-λ-algebra of $K^\bullet(X)$, generated by the classes of finitely many sheaves. Then the λ-ring filtration of R is discrete and $\mathrm{Fil}^1 R$ is nilpotent. Moreover, R is a finite module over its image in $H^0(X, \mathbb{Z})$.

After reducing to constant ranks, R is generated by finitely many elements $x_i = E_i - \varepsilon(E_i)$ and their γ-operations. Proposition 6.1 shows that all these elements are nilpotent. Hence the augmentation ideal of R is nilpotent; only finitely many monomials in the x_i's and the $\gamma^j(x_i)$'s remain. This gives both discreteness and finiteness.

Corollary 6.3. Let A be a commutative ring. An element y of $K^\bullet(X) \otimes_{\mathbb{Z}} A$ is invertible if and only if its augmentation is invertible. In particular, the class of every nowhere-zero locally free sheaf of finite type becomes invertible in $K^\bullet(X) \otimes_{\mathbb{Z}} A$.

Indeed, the augmentation ideal is nil.

Corollary 6.4. Let X be quasi-compact with an ample sheaf, and let $f : X' \to X$ be proper and perfect. Assume either that f is projective or that X is noetherian, so that the finiteness theorem applies. If there is an element $x' \in K^\bullet(X') \otimes A$ such that the augmentation of $f_*(x')$ is invertible, then

$$f^* : K^\bullet(X) \otimes A \longrightarrow K^\bullet(X') \otimes A$$

is injective. Indeed, by the projection formula,

$$f_*(f^* y \cdot x') = y \cdot f_*(x'),$$

and $f_*(x')$ is invertible by 6.3.

6.5. The codimension filtration

Assume now that X is noetherian of dimension d, and has an ample sheaf. Define a filtration $K^\bullet(X)^i$ as follows: an element $x \in K^\bullet(X)$ lies in $K^\bullet(X)^i$ if, for every closed subscheme $Y \subset X$, it can be represented by a perfect complex $M^\bullet(Y)$ such that

a) $\mathrm{cl}(M^\bullet) = x$;

b) $\mathrm{codim}(\mathrm{Supp}\, H^j(M^\bullet) \cap Y, Y) \geq i + j$, and $H^j(M^\bullet) = 0$ for $j \gg 0$.

These $K^\bullet(X)^i$'s are additive subgroups.

Proposition 6.6. The filtration of 6.5 is a ring filtration, and $K^\bullet(X)^{d+1} = 0$.

If x and x' are represented by perfect complexes $M^\bullet$ and $M'^\bullet$ satisfying the above codimension estimates, then xx' is represented by $M^\bullet \otimes M'^\bullet$. The support of $H^n(M^\bullet \otimes M'^\bullet)$ is contained in the union of the intersections of supports with total degree n, and the codimension estimate

follows from EGA 0_{IV}, 14.2.2.3. Finally $K^\bullet(X)^{d+1} = 0$, because a support in codimension greater than d is empty.

Lemma 6.7. Let $X = \operatorname{Spec} A$ be affine noetherian, let $x_1, \dots, x_n$ be finitely many points of X, and let E be invertible. There exists a section $s \in \Gamma(X, E)$ which is a basis of E_{x_i} for every i.

The proof uses closed specializations x_i' with distinct maximal ideals $\mathfrak{m}_i$, the isomorphisms $M/\mathfrak{m}_i M \simeq A/\mathfrak{m}_i$, for $M = \Gamma(X, E)$, and Nakayama's lemma.

Lemma 6.8. Let X be noetherian with an ample sheaf L, let E be invertible, and let $x_1, \dots, x_n$ be points of X. There exists an integer N such that, for every $k \geq 1$, $E(kN)$ has a global section which is a basis at every x_i.

Choose a power of the ample sheaf and a section whose non-vanishing locus is affine and contains all the x_i's; apply Lemma 6.7 there, and extend after multiplying by a further power.

Theorem 6.9. Let X be a noetherian scheme of dimension d with an ample sheaf. Then

$$\operatorname{Fil}^{d+1} K^\bullet(X) = 0,$$

and therefore $(\operatorname{Fil}^1 K^\bullet(X))^{d+1} = 0$.

First, if E and E' are invertible, with classes still denoted E, E', then $E - E' \in K^\bullet(X)^1$. For a closed $Y \subset X$, choose one point in each irreducible component. Lemma 6.8 gives morphisms from suitable negative powers of the ample sheaf to E and E' which are isomorphisms at those points. The two-term complex built from these morphisms has class $E - E'$, and is acyclic at the chosen generic points of Y; hence its cohomology supports have codimension at least 1 in Y.

Now take $x \in \operatorname{Fil}^{d+1} K^\bullet(X)$. It is enough to treat products

$$\gamma^{i_1}(E_1 - F_1) \cdots \gamma^{i_k}(E_k - F_k), \qquad \sum i_j \geq d + 1,$$

where the E_i, F_i are represented by locally free sheaves of the same constant rank. Pull back to the product of the flag bundles of these sheaves. The pullback is injective on $K^\bullet$, and each class splits into sums of invertible classes. For invertible classes ξ, η,

$$\gamma_t(\xi - \eta) = \frac{\gamma_t(\xi - 1)}{\gamma_t(\eta - 1)} = \frac{1 + (\xi - 1)t}{1 + (\eta - 1)t},$$

so

$$\gamma^i(\xi - \eta) = (-1)^{i-1}(\xi - \eta)(\eta - 1)^{i-1} \in K^\bullet(X')^i.$$

Thus the pullback of x belongs to $K^\bullet(X')^{d+1}$. Multiplication by a basis element of $K^\bullet(X')$ over $K^\bullet(X)$ of relative codimension r gives an element of $K^\bullet(X')^{d+1+r} = 0$, since $\dim X' = d + r$. The basis property and injectivity then give $x = 0$.

Problem 6.10. Under the hypotheses of Theorem 6.9, is it true that for every k one has

$$K^\bullet(X)^k = \operatorname{Fil}^k K^\bullet(X)?$$

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Exposé VII. Regular immersions and the computation of $K^\bullet$ of a blow-up scheme

by Pierre Berthelot

1. Generalities on regular immersions

The classical notion of regular immersion (EGA IV 16.9.2) proves inadequate in the non-noetherian case. We shall define a weaker notion, one which gives the desired properties, and which is equivalent to the usual definition in the noetherian case.

Definition 1.1. Let A be a commutative ring, let E be a projective A-module of finite type, and let $u : E \to A$ be an A-linear homomorphism. We say that u is *regular* if the Koszul complex $\mathrm{K}(u)$ defined by u is acyclic in dimensions > 0, and hence is a resolution of $B = A/u(E)$.

Recall that the Koszul complex associated with u is the graded A-module $\bigwedge(E)$, with boundary the interior multiplication by u, regarded as an element of the dual $\check{E}$. This boundary is an antiderivation of degree -1, equal to u on E.

Lemma 1.2. Let $u : E \to A$ be regular. Let A' be an A-algebra and let $u' : E' \to A'$ be obtained from u by base change; suppose that u' is regular. Put $J = u(E)$, $J' = JA' = u(E')$. For every $n \geq 0$ one has

$$J^n/J^{n+1} \otimes_A A' \xrightarrow{\sim} J'^n/J'^{n+1}, \tag{1.2.1}$$

$$\mathrm{Tor}_i^A(J^n/J^{n+1}, A') = 0 \quad (i \geq 1), \tag{1.2.2}$$

$$J^n \otimes_A A' \xrightarrow{\sim} J'^n, \tag{1.2.3}$$

$$\mathrm{Tor}_i^A(J^n, A') = 0 \quad (i \geq 1). \tag{1.2.4}$$

First assume that the modules J^n/J^{n+1} are projective over $B = A/J$. Since J is of finite type they are locally free of finite type over B. The assertions are checked pointwise on $\mathrm{Spec}(A)$, so that one may suppose A local and $u(E) \subset \mathrm{rad}(A)$. Then J^n/J^{n+1} is free over B, and (1.2.2) is equivalent to $\mathrm{Tor}_i^A(B, A') = 0$ for $i \geq 1$. This follows by computing with the Koszul resolution $\mathrm{K}(u)$, since after tensoring with A' it becomes $\mathrm{K}(u')$, acyclic in positive degrees.

The exact sequences

$$0 \to J^n/J^{n+1} \to A/J^{n+1} \to A/J^n \to 0, \qquad 0 \to J^n \to A \to A/J^n \to 0$$

then give (1.2.1), (1.2.3), and (1.2.4) by induction on n. It remains to know that J^n/J^{n+1} is projective over B. Locally one reduces to $E = A^d$, with u given by $f_1, \ldots, f_d$, and to the universal homomorphism $\mathbb{Z}[T_1, \ldots, T_d]^d \to \mathbb{Z}[T_1, \ldots, T_d]$. Since the sequence $(T_1, \ldots, T_d)$ is regular, the conclusion follows by base change.

Proposition 1.3. Let A be a ring, E a projective A-module of finite type, $u : E \to A$ a regular homomorphism, and $J = u(E)$. Then J is quasi-regular: J is of finite type, J/J^2 is a projective A/J-module of finite type, and the canonical homomorphism

$$\mathrm{Sym}_{A/J}(J/J^2) \longrightarrow \mathrm{gr}_J(A) = \bigoplus_{n \geq 0} J^n/J^{n+1} \tag{1.3.1}$$

is an isomorphism. This is obtained from the preceding proof, again by reduction to the universal polynomial case.

Definition 1.4. Let (X, A) be a commutatively locally ringed site, let E be an A-module locally free of finite type, and let $u : E \to A$ be A-linear. We say that u is regular if $\mathrm{K}(u)$ is acyclic in degrees > 0. If J is an ideal of A, we say that J is regular if, locally on X, it is the image of a surjective regular homomorphism from a free A-module of finite type.

Let $i : Y \to X$ be an immersion of preschemes. If U is an open subset of X containing $i(Y)$, and if i is a closed immersion into U, then i is said to be regular if the ideal defining $i(Y)$ in $\mathcal{O}_U$ is regular. This condition is independent of U, and regularity may be tested in a neighbourhood of every point.

The definition replaces here that of EGA IV 16.9.2, and in the noetherian case the two definitions are equivalent. A closed immersion locally defined by a regular sequence is regular, and a regular immersion is quasi-regular by Proposition 1.3. A minimal system of generators of an ideal J at a point is a family whose images form a basis of $J_x \otimes k(x)$; regularity is equivalently the local acyclicity of the corresponding Koszul complex. If J is of finite type and J/J^2 is locally free, these systems correspond to local bases of J/J^2.

Proposition 1.5. If $i : Y \to X$ is regular, it remains regular after every flat base change. Conversely, i is regular if and only if it is so locally for the faithfully flat quasi-compact topology.

Definition 1.6. If $i : Y \to X$ is defined by an ideal J, the $\mathcal{O}_Y$-module J/J^2 is the conormal sheaf of Y in X, denoted $N_{X/Y}$. If i is regular it is locally free of finite type, and its rank is the codimension. For closed immersions $Z \xrightarrow{j} Y \xrightarrow{i} X$ one has the exact conormal sequence

$$j^* N_{X/Y} \longrightarrow N_{X/Z} \longrightarrow N_{Y/Z} \longrightarrow 0. \tag{1.6.1}$$

Proposition 1.7. Let $i : Y \to X$ and $j : Z \to Y$ be immersions. If i and j are regular, then $i \circ j$ is regular and the conormal sequence, completed by 0 on the left, is exact. If $i \circ j$ and i are regular and if this sequence is exact and locally split, then j is regular. If X is noetherian and if $i \circ j$ and j are regular, then i is regular at the points of Z.

Locally, write $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $Z = \mathrm{Spec}(C)$, with $B = A/I$ and $C = B/K$. Choosing liftings $f_1, \ldots, f_n$ of a basis of I/I^2 and $g_1, \ldots, g_m$ of a basis of K/K^2, one obtains the diagram

$$\begin{array}{ccc} A_0 = \mathbb{Z}[T_1, \ldots, T_{n+m}] & \longrightarrow & A \\ \downarrow & & \downarrow \\ B_0 = \mathbb{Z}[T_1, \ldots, T_m] & \longrightarrow & B \\ \downarrow & & \downarrow \\ C_0 = \mathbb{Z} & \longrightarrow & C, \end{array} \tag{1.7.1}$$

Therefore:

$$\begin{aligned} i \text{ regular} &\iff \forall i \geq 1, \quad \mathrm{Tor}_i^{A_0}(B_0, A) = 0, \\ i \circ j \text{ regular} &\iff \forall i \geq 1, \quad \mathrm{Tor}_i^{A_0}(C_0, A) = 0, \\ j \text{ regular} &\iff \forall i \geq 1, \quad \mathrm{Tor}_i^{B_0}(C_0, B) = 0. \end{aligned} \tag{1.7.2}$$

One then uses the transitivity spectral sequence

$$E^2_{pq} = \mathrm{Tor}_p^{B_0}(C_0, \mathrm{Tor}_q^{A_0}(B_0, A)) \Rightarrow \mathrm{Tor}_n^{A_0}(C_0, A). \tag{1.7.3}$$

This gives the first two assertions; the noetherian assertion is that of EGA IV 19.1.5, with the definitions identified in the noetherian case.

Proposition 1.8. Let X be a scheme, E a locally free $\mathcal{O}_X$-module of finite type, $u : E \to \mathcal{O}_X$ a regular homomorphism, and $J = u(E)$. Let Y denote the closed subscheme of X defined by J, let X' be the scheme obtained by blowing up Y, let $P = \mathbf{P}(E)$ be the projective bundle

associated with E, and let $p : P \to X$ be the structural morphism. On P consider the canonical exact sequence

$$0 \to F \to p^*E \to \mathcal{O}_P(1) \to 0, \tag{1.8.1}$$

and the composite morphism

$$v : F \longrightarrow p^*E \xrightarrow{p^*(u)} \mathcal{O}_P.$$

Then:

1. the formation of X' commutes with every base change $f : T \to X$ such that $f^*(u)$ is regular;

2. the closed immersion $j : X' \to P$ defined by the surjective homomorphism

$$\mathcal{S}(E) \longrightarrow \bigoplus_{n\geq 0} J^n$$

defined by u in degree 1 is a regular immersion, its ideal being the image of the regular morphism v. In particular, if $\operatorname{codim}(i)$ is constant, then $\operatorname{codim}(j) = \operatorname{codim}(i) - 1$;

3. the $\mathcal{S}(E)$-module $\bigoplus_{n\geq 0} J^n$ is perfect as a graded $\mathcal{S}(E)$-module, i.e. locally there is a finite graded resolution of $\bigoplus_{n\geq 0} J^n$ by graded $\mathcal{S}(E)$-modules locally free of finite type.

For (i), since $X' = \operatorname{Proj}(\bigoplus_{n\geq 0} J^n)$, the result is an immediate consequence of the relations (1.2.3) and of EGA II.

For (ii), the asserted properties being local on X, one may suppose that $X = \operatorname{Spec}(A)$, and moreover that E is free on X. We first prove the assertion in the standard case

$$A = \mathbb{Z}[T_1, \ldots, T_d],$$

where E is given the basis $e_1, \ldots, e_d$, with $u(e_i) = T_i$. Then

$$P = \operatorname{Proj}(\mathbb{Z}[T_1, \ldots, T_d, X_1, \ldots, X_d]),$$

the grading being taken with respect to the X_i. To show that v is a regular morphism whose image is the ideal of j, it suffices to do so locally on P, hence on the open sets $D_+(X_i)$; let us place ourselves, for instance, on $D_+(X_1)$. Then $p^*(E)$ is defined by the module

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1] \otimes E,$$

and $\mathcal{O}_P(1)$ by

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1],$$

the canonical morphism sending e_i to X_i/X_1 for every i. Thus F has as a basis the elements

$$e_i - e_1 X_i/X_1 \qquad (i > 1).$$

One has

$$v(e_i - e_1 X_i/X_1) = T_i - T_1 X_i/X_1; \qquad T'_i = T_i - T_1 X_i/X_1.$$

Since the sequence of the T_i is regular in

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1],$$

the same is true of the sequence $T_1, T'_2, \ldots, T'_d$, which is obtained from it by an invertible matrix, and therefore also of the extracted sequence $T'_2, \ldots, T'_d$.

It remains to show that the T_i' generate the ideal of j. In $\bigoplus_{n\geq 0} J^n$, for $m \leq p$, we shall write $x^{(m)}$ for the element x of J^p, considered as being of degree m. Over $D_+(X_1)$,

$$X' = \operatorname{Spec}\left(\left(\bigoplus_{n\geq 0} J^n\right)_{(T_1^{(1)})}\right),$$

and j is defined by the morphism sending X_i to $T_i^{(1)}$. In this morphism the image of T_i' is

$$T_i^{(0)} - T_1^{(0)}T_i^{(1)}/T_1^{(1)} = \left((T_iT_1)^{(1)} - (T_1T_i)^{(1)}\right)/T_1^{(1)} = 0.$$

Thus the T_i' belong to the ideal of j. Since every element of

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1]$$

can be expressed as a polynomial in $T_1, T_2', \ldots, T_d', X_2/X_1, \ldots, X_d/X_1$, it suffices, in order to show that the T_i' generate the ideal of j, to show that no non-zero element of

$$\mathbb{Z}[T_1, X_2/X_1, \ldots, X_d/X_1]$$

belongs to the ideal of j. This amounts to showing that the homomorphism

$$\mathbb{Z}[T_1, X_1, \ldots, X_d] \longrightarrow \bigoplus_{n\geq 0} J^n$$

is injective, since $T_1^{(1)}$ is not a zero divisor in $\bigoplus_{n\geq 0} J^n$. If one grades $\mathbb{Z}[T_1, X_1, \ldots, X_d]$ by assigning degree 0 to T_1 and degree 1 to the X_i, the kernel of the homomorphism in question is a graded ideal. But it is contained in the kernel of the composite homomorphism

$$\mathbb{Z}[T_1, X_1, \ldots, X_d] \longrightarrow \bigoplus_{n\geq 0} J^n \longrightarrow A,$$

the last homomorphism being defined in degree n by the injection of J^n into A. The kernel of the composite homomorphism is the principal ideal generated by $X_1 - T_1$, which, by integrality, contains no non-zero homogeneous element. Therefore the first homomorphism is injective, which completes the proof in this case.

Return to the general case where A is arbitrary, with E still assumed free on X. Put index 0 on the special case just treated. By part (i), the situation under consideration is obtained from the preceding one by the base change $A_0 \to A$, where the homomorphism $A_0 \to A$ is defined by choosing a basis of E, its image $f_1, \ldots, f_d$ in J, and sending T_i to f_i. Put

$$B_0 = \left(\bigoplus_{n\geq 0} J_0^n\right)_{(T_1^{(1)})}, \qquad B = \left(\bigoplus_{n\geq 0} J^n\right)_{(f_1^{(1)})}.$$

Then $B = B_0 \otimes_{A_0} A$, and the Koszul complex defined by $v : F \to \mathcal{O}_P$ is obtained by base change from the Koszul complex defined by v_0. Since the latter is a projective resolution of B_0 on a suitable open set, the homology groups of $K_\bullet(v)$ are, on this open set, the groups

$$\operatorname{Tor}_i^{A_0}(A, B_0).$$

These are the components of degree 0 of

$$\operatorname{Tor}_i^{A_0}\left(A, \left(\bigoplus_{n\geq 0} J_0^n\right)_{T_1^{(1)}}\right) = \left(\bigoplus_{n\geq 0} \operatorname{Tor}_i^{A_0}(A, J_0^n)\right)_{T_1^{(1)}}.$$

But the $\mathrm{Tor}_i^{A_0}(A, J_0^n)$ vanish for $i \geq 1$ by the relations 1.2; hence $K_\bullet(v)$ is a resolution of B, which shows that j is regular.

For (iii), the property being local on X, one may again suppose X affine and E free of finite type. We then resume the notation of (ii). First consider the standard case of (ii). The ring A_0 is noetherian and of finite cohomological dimension, and so is the algebra $\mathcal{S}(E_0)$. One may then construct, step by step, a partial graded resolution of $\bigoplus_{n\geq 0} J_0^n$ by graded free $\mathcal{S}(E_0)$-modules of finite type, of length greater than the cohomological dimension of A_0. The last kernel is then a graded $\mathcal{S}(E_0)$-module, projective of finite type as an ungraded $\mathcal{S}(E_0)$-module, hence locally free of finite type on $\mathrm{Spec}(A_0)$ as a graded $\mathcal{S}(E_0)$-module by VI 1.5.1. In this way one obtains the announced finite resolution, replacing the last module of the partial resolution by the corresponding kernel.

Tensoring the resolution thus constructed by A, one obtains a graded complex of graded $\mathcal{S}(E)$-modules locally free of finite type, and it remains to see that this is a resolution of $\bigoplus_{n\geq 0} J^n$. But the i-th homology module of the resulting complex is a graded $\mathcal{S}(E)$-module whose n-th component is $\mathrm{Tor}_i^{A_0}(A, J_0^n)$, which is zero for $i \geq 1$ by (1.2.4). Consequently, by (1.2.3), one has thus constructed a resolution of the required type.

Proposition 1.9. If $i : Y \to X$ is a regular closed immersion, if X' is the blow-up of Y, and if $f : X' \to X$ is structural, then i and f are perfect morphisms: K resolves $\mathcal{O}_Y$ by finite locally free $\mathcal{O}_X$-modules, and f factors through a regular immersion into a projective bundle followed by a smooth morphism.

Proposition 1.10. If X and Y are smooth over S, every S-immersion from X into Y is regular. In particular, every section of a smooth morphism is a regular immersion.

2. Computations on regular immersions

2.1–2.3. For two A-algebras B, C, the graded module $\bigoplus_{i\geq 0} \mathrm{Tor}_i^A(B, C)$ is given a natural structure of associative graded $B \otimes_A C$-algebra by choosing a projective resolution $P_\bullet \to B$, using the Kunneth morphisms, and extending the augmentation $B \otimes_A B \to B$. If the resolution is an alternating differential graded algebra, the Tor algebra is alternating. Globally this gives, for two X-schemes X', X'', a graded algebra structure on

$$\bigoplus_i \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''}).$$

The multiplication is represented by the commutative square of complexes

$$\begin{array}{ccc} P_i \otimes P_j & \longrightarrow & (P_{i-1} \otimes P_j) \oplus (P_i \otimes P_{j-1}) \\ \downarrow & & \downarrow \\ P_{i+j} & \longrightarrow & P_{i+j-1}. \end{array} \tag{2.2.1}$$

Lemma 2.4. For two X-schemes X', X'' and a flat $\mathcal{O}_{X'}$-module M, one has canonically

$$\mathrm{Tor}_i^{\mathcal{O}_X}(M, \mathcal{O}_{X''}) \simeq M \otimes_{\mathcal{O}_{X'}} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''}). \tag{2.4.1}$$

This is immediate locally from a projective resolution, and then glues.

Proposition 2.5. Let Y and Z be closed subschemes of X, with $T = Y \times_X Z$. Under the local Koszul hypotheses of this proposition, one has

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z), \tag{2.5.1}$$

and

$$\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Z)\simeq I\cap J/IJ\simeq \mathrm{Ker}(i^*N_{Y/X}\times j^*N_{Z/X}\to N_{T/X}). \tag{2.5.2}$$

In particular, for a regular closed immersion $Y\to X$,

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y)\simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y),\qquad \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y)\simeq N_{Y/X}. \tag{2.5.3}$$

The proof is the standard comparison of Koszul complexes, together with the bicomplex spectral sequence and the preceding multiplicative construction.

2.6–2.7. Assume that X has an ample sheaf, so that the two definitions of $K^\bullet$ coincide. If $i:Y\to X$ is a regular closed immersion, one defines

$$i_*:K^\bullet(Y)\to K^\bullet(X)$$

using the class of the direct image, and one has, for every $x\in K^\bullet(Y)$,

$$i^*i_*(x)=x\lambda_{-1}(N),\qquad \lambda_{-1}(N)=\sum_{r\geq 0}(-1)^r\lambda^r(N), \tag{2.7.1}$$

where N is the conormal sheaf. Indeed, if x is represented by a locally free finite type $\mathcal{O}_Y$-module M, the homology sheaves of the derived inverse image of i_*M are $M\otimes \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y)$, and Proposition 2.5 gives the displayed formula.

Corollary 2.8. Under the hypotheses of 2.7, for $x,y\in K^\bullet(Y)$,

$$i_*(xy\lambda_{-1}(N))=i_*(x)i_*(y). \tag{2.8.1}$$

This follows from (2.7.1) and the projection formula.

Corollary 2.9. Under the same hypotheses, if $i_*(1)$ is invertible in $K^\bullet(X)$, then i_* is an isomorphism of $K^\bullet(Y)$ onto a direct summand.

3. Computation of $K^\bullet$ of a blow-up scheme

3.1. Let $i:Y\to X$ be a regular closed immersion. Let X' be the blow-up of Y, and let $Y'=Y\times_X X'$. By EGA II 8.1.8, Y' is a closed subscheme of X' defined by an ideal canonically isomorphic to $\mathcal{O}_{X'}(1)$. Thus $j:Y'\to X'$ is a regular closed immersion of codimension 1. Let J be the ideal of Y in X, and J' the ideal of Y' in X'; then

$$J'\simeq J\mathcal{O}_{X'}\simeq \mathcal{O}_{X'}(1).$$

Writing $f:X'\to X$ and $g:Y'\to Y$ for the structural morphisms, one has the commutative diagram

$$\begin{array}{ccc} Y' & \xrightarrow{\ j\ } & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[\ i\ ]{} & X. \end{array} \tag{3.1.1}$$

Moreover,

$$X'=\mathrm{Proj}(S),\qquad S=\bigoplus_{n\geq 0}J^n,$$

and

$$Y'=\mathrm{Proj}(i^*S)=\mathrm{Proj}(\bigoplus_{n\geq 0}J^n/J^{n+1}).$$

Since i is regular, $\bigoplus J^n/J^{n+1} \simeq \mathrm{Sym}_{\mathcal{O}_Y}(J/J^2)$. If $N = J/J^2$, then $Y' = \mathbf{P}(N)$. On Y' there is the exact sequence

$$0 \to F \to g^*N \to \mathcal{O}_{Y'}(1) \to 0. \tag{3.1.2}$$

The sheaf $\mathcal{O}_{Y'}(1) = j^*\mathcal{O}_{X'}(1) = J'/J'^2$ is the conormal sheaf of j; denote it by L.

Lemma 3.2. With the hypotheses and notation of 3.1, one has

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y), \tag{3.2.i}$$

$$\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq F. \tag{3.2.ii}$$

For (i), since $\mathcal{O}_Y$ can locally be resolved by a Koszul complex, the algebra $\bigoplus_i \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y)$ is alternating. Hence there is a unique algebra homomorphism

$$\bigwedge \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \longrightarrow \bigoplus_{i\geq 0} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \tag{3.2.1}$$

extending the injection of degree one; it is enough to verify locally that it is an isomorphism. On an affine open, choose a regular homomorphism $E \to A$ defining J, with basis images $f_1, \ldots, f_d$. On the open of X' where $f_1^{(1)}$ is invertible, change the basis by

$$e'_1 = e_1, \qquad e'_i = e_i - (f_i^{(1)}/f_1^{(1)})e_1 \quad (i \neq 1).$$

The Koszul differential then gives local isomorphisms

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq (\mathcal{O}_{X'}/f_1^{(0)}\mathcal{O}_{X'}) \otimes_{\mathcal{O}_{X'}} \bigwedge^i(\mathcal{O}_{X'}^{d-1}) \simeq \bigwedge^i(\mathcal{O}_{Y'}^{d-1}). \tag{3.2.2}$$

In particular, for $i = 1$ one obtains $\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \mathcal{O}_{Y'}^{d-1}$, and hence

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y). \tag{3.2.3}$$

For (ii), tensor the exact sequence $0 \to J \to \mathcal{O}_X \to \mathcal{O}_Y \to 0$ with $\mathcal{O}_{X'}$. Since the image of $J \otimes \mathcal{O}_{X'} \to \mathcal{O}_{X'}$ is $J\mathcal{O}_{X'} = \mathcal{O}_{X'}(1)$, and then tensoring with $\mathcal{O}_{Y'}$, one obtains

$$0 \to \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \to g^*N \to \mathcal{O}_{Y'}(1) \to 0, \tag{3.2.4}$$

which identifies this Tor sheaf with F in (3.1.2).

3.3. Suppose now that X has an ample sheaf. Then the same is true for X', which is projective over X, and for Y and Y'. Moreover i and j are perfect morphisms, g is smooth hence perfect because $Y' = \mathbf{P}(N)$, and f is perfect by 1.9. Since these morphisms are projective, one has additive homomorphisms

$$i_* : K^\bullet(Y) \to K^\bullet(X), \quad j_* : K^\bullet(Y') \to K^\bullet(X'), \quad f_* : K^\bullet(X') \to K^\bullet(X), \quad g_* : K^\bullet(Y') \to K^\bullet(Y).$$

Proposition 3.4. With the hypotheses and notation of 3.1 and 3.3, for every $x \in K^\bullet(Y)$ one has

$$f^*(i_*(x)) = j_*(g^*(x)\lambda_{-1}(F)). \tag{3.4.1}$$

By additivity, one may assume x represented by a locally free finite type $\mathcal{O}_Y$-module M. The class of $f^*i_*(x)$ is represented by the complex obtained from a projective resolution of i_*M and application of f^*. Its homology sheaves are $\mathrm{Tor}_i^{\mathcal{O}_X}(M, \mathcal{O}_{X'})$. By 2.4 and 3.2,

$$\mathrm{Tor}_i^{\mathcal{O}_X}(M, \mathcal{O}_{X'}) \simeq M \otimes_{\mathcal{O}_Y} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq M \otimes_{\mathcal{O}_Y} \bigwedge^i F,$$

and the alternating sum of their classes is exactly the right hand side of (3.4.1).

Lemma 3.5. With the hypotheses and notation of 3.1, one has the following isomorphisms:

i) for all $n \geq 0$ and $i \geq 1$,

$$R^i f_*(\mathcal{O}_{X'}(n)) = 0;$$

ii) for all $n \geq 0$,

$$f_*(\mathcal{O}_{X'}(n)) = J^n.$$

One may suppose X affine, hence quasi-compact. For (i) one uses the following lemma: if $f : X' \to X$ is projective and perfect, with X quasi-compact, and if L' is a perfect complex on X' acyclic in degrees > 0, then there exists n_0 such that, for $n \geq n_0$, $Rf_*(L' \otimes \mathcal{O}_{X'}(n))$ is acyclic in degrees > 0. Applied to $\mathcal{O}_{X'}$, this gives the vanishing for large n. The exact sequence

$$0 \to \mathcal{O}_{X'}(n+1) \to \mathcal{O}_{X'}(n) \to \mathcal{O}_{Y'}(n) \to 0$$

and the fact that $Y' = \mathbf{P}(N)$ then allow descending induction to $n = 0$.

For (ii), the canonical graded homomorphism

$$\bigoplus_{n \geq 0} J^n \longrightarrow \bigoplus_{n \geq 0} f_*(\mathcal{O}_{X'}(n))$$

is shown to be an isomorphism locally. Write $S = \bigoplus J^n$ and use a surjective graded homomorphism $\mathcal{O}_X[T_1, \ldots, T_d] \to S$ given by a set of generators of J. This realizes X' as a regular closed subscheme of $\mathbf{P}_X^{d-1}$. With a finite graded presentation $L'' \to L' \to S \to 0$, one obtains on $\mathbf{P}_X^{d-1}$ a presentation $M'' \to M' \to \mathcal{O}_{X'} \to 0$. For each n there is a commutative diagram

$$\begin{array}{ccccccc} L''_n & \longrightarrow & L'_n & \longrightarrow & J^n & \longrightarrow & 0 \\ \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} & & \downarrow & & \downarrow \\ p_*M''(n) & \longrightarrow & p_*M'(n) & \longrightarrow & p_*\mathcal{O}_{X'}(n) & \longrightarrow & 0, \end{array}$$

where $p : \mathbf{P}_X^{d-1} \to X$. For n sufficiently large the bottom row is exact; hence the middle vertical arrow is an isomorphism for large n. Descending induction, using

$$0 \to J^{n+1} \to J^n \to J^n/J^{n+1} \to 0$$

and the analogous direct-image exact sequence, completes the proof.

Proposition 3.6. With the hypotheses and notation of 3.1 and 3.3,

$$f_* f^* = \mathrm{Id}_{K^\bullet(X)}\,. \tag{3.6.1}$$

Indeed, by the projection formula, $f_* f^*(x) = x f_*(1)$, and Lemma 3.5 says that $Rf_*\mathcal{O}_{X'}$ has $\mathcal{O}_X$ in degree 0 and no higher cohomology.

Theorem 3.7. Let $i : Y \to X$ be a regular closed immersion, let X' be the blow-up of Y, and let $Y' = X' \times_X Y$, so that one has

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X. \end{array}$$

Assume that X is quasi-compact and has an ample sheaf. Then the sequence

$$0 \to K^\bullet(Y) \xrightarrow{u} K^\bullet(Y') \times K^\bullet(X) \xrightarrow{v} K^\bullet(X') \tag{3.7.0}$$

is exact, where

$$u(y) = \big(g^*(y)\lambda_{-1}(F), -i_*(y)\big), \tag{3.7.u}$$

and, for $y' \in K^\bullet(Y')$, $x \in K^\bullet(X)$,

$$v(y', x) = j_*(y') + f^*(x). \tag{3.7.v}$$

Moreover u has a left inverse u', defined by $u'(y', x) = g_*(y')$. If, in addition, X is noetherian and Y regular, then v is surjective and

$$0 \to K^\bullet(Y) \xrightarrow{u} K^\bullet(Y') \times K^\bullet(X) \xrightarrow{v} K^\bullet(X') \to 0 \tag{3.7.2}$$

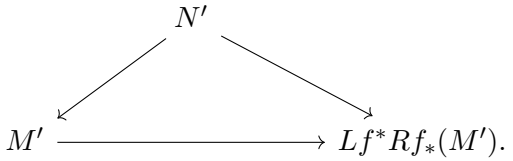
is exact.

First, $u' \circ u = 1$, because $g_*(g^* y \lambda_{-1}(F)) = y\, g_*(\lambda_{-1}(F)) = y$. Also $v \circ u = 0$, by Proposition 3.4. If (y', x) belongs to the kernel of v, then $j_*(y') + f^*(x) = 0$. One must find $y \in K^\bullet(Y)$ such that

$$x = -i_*(y), \qquad y' = g^*(y)\lambda_{-1}(F). \tag{3.7.1}$$

Necessarily $y = u'(y', x) = g_*(y')$. Applying f_* to the relation $f^*(x) = -j_*(y')$, and using Proposition 3.6, gives $x = -i_*(y)$. To prove the second equality in (3.7.1), put $z = y' - g^*(g_* y')\lambda_{-1}(F)$. Then $g_* z = 0$, and the preceding formulas give $j_* z = 0$. Since $\operatorname{Ker} j_* \cap \operatorname{Ker} g_* = 0$, it follows that $z = 0$. This proves exactness at the middle term.

For the final surjectivity assertion, assume X noetherian and Y regular. Then Y' is regular, being a projective bundle over Y, and local rings of X' along Y' are regular because Y' is cut out by one regular element. Therefore coherent modules on X' with support in Y' are perfect. If such a module E has class in $K^\bullet(X')$, filtration by the powers $J'^n E$ shows that its class is in the image of j_*. For arbitrary $x \in K^\bullet(X')$, choose a perfect complex M' representing x, and form the distinguished triangle



The horizontal arrow is the canonical morphism. It is an isomorphism off Y', so the cohomology sheaves of N' are coherent sheaves supported on Y'. Their class lies in $\operatorname{Im}(j_*)$. Hence

$$x - f^* f_*(x) \in \operatorname{Im}(j_*),$$

and v is surjective.

Remark 3.8

In the case where X is noetherian and Y regular, 3.7 determines the abelian group $K^\bullet(X')$ entirely from the λ-rings $K^\bullet(X)$, $K^\bullet(Y)$, the maps i_*, i^* between these rings, and the element $N \in K^\bullet(Y)$. Indeed, $K^\bullet(Y')$ can be computed from $K^\bullet(Y)$ and N by VI 1.1. We shall see in the following paragraph that the multiplicative structure of $K^\bullet(X')$ is determined by the same data, and so is the λ-structure after tensoring with $\mathbb{Q}$ (this restriction is probably unnecessary; see 4.5).

4. Regular immersions and the filtration of $K^\bullet$

Throughout this paragraph we keep the notation and hypotheses of 3.1.

Lemma 4.1. Suppose that there exists on Y a locally free sheaf N' of finite type such that, if N, N' denote the classes in $K^\bullet(Y)$ of N, N', then $N = N' + 2$. Then the following congruence is verified:

$$\lambda_{-1}(F) \equiv 0 \pmod{1 - L}.$$

Let d be the rank of N, which we may suppose constant; the rank of F is then $d - 1$. By definition,

$$\begin{aligned}\lambda_{-1}(F) &= \sum_{i\geq 0}(-1)^i\lambda^i(F)\\ &= (-1)^{d-1}\lambda^{d-1}(F-1)\\ &= (-1)^{d-1}\lambda^{d-1}(F+L-1-L)\\ &= (-1)^{d-1}\lambda^{d-1}(N-2+1-L).\end{aligned}$$

Now

$$\lambda_t(1-L) = \frac{\lambda_t(1)}{\lambda_t(L)} = \frac{1+t}{1+Lt}.$$

Thus, for every $i \geq 1$, $\lambda^i(1 - L)$ is divisible by $1 - L$. Consequently

$$\begin{aligned}\lambda_{-1}(F) &\equiv (-1)^{d-1}\lambda^{d-1}(N-2) \pmod{1-L}\\ &= (-1)^{d-1}\lambda^{d-1}(N') \pmod{1-L}.\end{aligned}$$

Since N' has rank $d - 2$, $\lambda^{d-1}(N') = 0$, proving the result.

4.2

Let $i : Y \to X$ be a regular immersion. Let $r \geq 1$ be an integer and let

$$s : X \longrightarrow \mathbf{P}^r_X$$

be the section defined by a homomorphism $(\mathcal{O}_X)^{r+1} \to \mathcal{O}_X$, after passing to the quotient by the sub-bundle generated by the corresponding sections. If $\mathbf{P}^r_X$ is defined by the graded $\mathcal{O}_X$-algebra $\mathcal{O}_X[T_0, \ldots, T_r]$, the ideal I defining the section s is generated by the sections $T_1, \ldots, T_r$. Therefore I/I^2 is isomorphic to $(\mathcal{O}_X[T_0])^r$, so the ideal corresponding to $\mathcal{O}_{\mathbf{P}^r}$ is $(\mathcal{O}_X)^r$.

Since the section s is evidently a regular immersion, the same is true for $s \circ i$, and one may write the corresponding exact sequence of conormal sheaves (1.7). Thus, if N denotes the conormal sheaf of i, and N_1 the conormal sheaf of $s \circ i$, one obtains the equality

$$N_1 = N + r.$$

It follows that every regular immersion i can be made to satisfy the property of Lemma 4.1 by composing it with a section $X \to \mathbf{P}^2_X$, or with a section $X \to \mathbf{P}^1_X$.

Theorem 4.3. Let $i : Y \to X$ be a regular closed immersion, and let N be the class in $K^\bullet(Y)$ of the conormal sheaf of Y in X. Suppose that X is quasi-compact and has an ample sheaf. Then, with the notation of 3.3, for every $y \in K^\bullet(Y)$ and every $p \geq 1$,

$$i_*(\lambda^p(N, y)) \equiv \lambda^p(i_*(y)) \pmod{\text{torsion}}, \tag{4.3.1}$$

where $\lambda^p(N, y)$ is the element defined in V 5.3.

First suppose that the hypothesis of 4.1 is satisfied; in that case we shall show

$$i_*(\lambda^p(N,y)) = \lambda^p(i_*(y)).$$

Since $f_*f^* = \mathrm{Id}$, this relation is equivalent to

$$f^*(i_*(\lambda^p(N,y))) = f^*(\lambda^p(i_*(y))).$$

Now one has

$$\begin{aligned} f^*(\lambda^p(i_*(y))) &= \lambda^p(f^*(i_*(y))) \\ &= \lambda^p(j_*g^*(y)\lambda_{-1}(F)), \end{aligned} \tag{3.4}$$

whereas

$$f^*(i_*(\lambda^p(N,y))) = j_*(g^*(\lambda^p(N,y))\lambda_{-1}(F)) \tag{3.4}$$

by Proposition 3.4. Since $g^*(N) = F + L$, formula V 5.4.2 gives

$$g^*(\lambda^p(N,y))\lambda_{-1}(F) = \lambda^p(g^*(N), g^*(y))\lambda_{-1}(F) = \lambda^p(L, g^*(y)\lambda_{-1}(F)).$$

Putting $z = g^*(y)\lambda_{-1}(F)$, we are reduced to showing

$$j_*(\lambda^p(L,z)) = \lambda^p(j_*(z)).$$

This is the analogous relation for the immersion j and the element z. By hypothesis, $\lambda_{-1}(F)$ is a multiple of $1 - L$, so one can find z' such that $z = z'(1-L)$. Then

$$z = j_*(u') \quad \text{with } u' = g_*(z')$$

by (2.7), and since $L = j^*(J')$, it remains only to verify

$$j_*(j^*(\lambda^p(J',u'))) = \lambda^p(j_*j^*(u')).$$

The exact sequence

$$0 \longrightarrow J' \longrightarrow \mathcal{O}_{X'} \longrightarrow \mathcal{O}_{Y'} \longrightarrow 0$$

gives

$$j_*j^*(1) = 1 - J'. \tag{4.3.2}$$

The desired relation is therefore, after applying the projection formula,

$$\lambda^p(J',u')(1-J') = \lambda^p(u'(1-J')).$$

Since $1 - J' = \lambda_{-1}(J')$, this is precisely the formula which serves to define $\lambda^p(J',u')$.

We now pass to the general case. Compose i first with a section $X \to \mathbf{P}^1_X$, and then with a section $\mathbf{P}^1_X \to \mathbf{P}^1_{\mathbf{P}^1_X}$, in such a way that the composed immersion satisfies the condition of 4.1 (see 4.2). Proposition 4.3 is then true for the immersion $s \circ i$ which precedes it; it remains to descend to X. We descend first to $\mathbf{P}^1_X$, and then to X, so that the proof is made by applying twice the following lemma, due to J.-P. Jouanolou.

Lemma. Under the conditions of 4.3, let $s : X \to \mathbf{P}^1_X$ be the section defined in 4.2. If Proposition 4.3 is true for $s \circ i$ and for s, then it is true for i.

Suppose 4.3 true for $s \circ i$. Then, for every $p \geq 1$ and every $y \in K^\bullet(Y)$,

$$(s \circ i)_*(\lambda^p(N+1,y)) \equiv \lambda^p((s \circ i)_*(y)) \pmod{\text{torsion}}. \tag{4.3.3}$$

The section s is defined by an ideal isomorphic to $\mathcal{O}_{\mathbf{P}}(-1)$ (4.2). Let ξ be its class in $K^\bullet(\mathbf{P}^1_X)$. Then, for every $x \in K^\bullet(X)$,

$$s_*(x) = p^*(x)(1-\xi),$$

where p is the structural morphism $\mathbf{P}^1_X \to X$. Indeed, since $p \circ s = \mathrm{Id}_X$, one has $x = s^*(p^*(x))$, whence $s_*(x) = s_*(s^*p^*(x)) = p^*(x)s_*(1)$; and $s_*(1) = 1 - \xi$. Thus

$$\begin{aligned}(s \circ i)_*(\lambda^p(N+1, y)) &\equiv \lambda^p(p^*(i_*(y))(1-\xi)) \pmod{\text{torsion}} \\ &= \lambda^p(\xi, p^*(i_*(y)))(1-\xi) \pmod{\text{torsion}}.\end{aligned}$$

Now $\lambda^p(\xi, p^*(i_*(y))) - \lambda^p(1, p^*(i_*(y)))$ is divisible by $1 - \xi$. To see this, one may work in the universal λ-ring used to define $\lambda^p(\xi, x)$ (V 5). Then $\lambda^p(\xi, x) - \lambda^p(1, x)$ is a polynomial in ξ which vanishes for $\xi = 1$, hence is divisible by $\xi - 1$. By the independence equation for ξ on $K^\bullet(X)$, one has $(1-\xi)^2 = 0$ (VI 1.13.1). Hence

$$\begin{aligned}(s \circ i)_*(\lambda^p(N+1, y)) &\equiv \lambda^p(1, p^*(i_*(y)))(1-\xi) \pmod{\text{torsion}} \\ &= p^*(\lambda^p(1, i_*(y)))(1-\xi) \pmod{\text{torsion}}.\end{aligned}$$

Applying p_*, we obtain

$$i_*(\lambda^p(N+1, y)) \equiv \lambda^p(1, i_*(y)) \pmod{\text{torsion}}, \tag{4.3.4}$$

since $p_*(\xi) = 0$.

By formula V 5.6, applied with $L = 1$, $\lambda^p(N+1, y)$ is the coefficient of t^p in the series

$$\frac{\sum_{k\geq 1} \lambda^k(N, y)t^k}{1 + \sum_{k\geq 1} \lambda^k(N, y)\lambda_{-1}(N)t^k}.$$

Similarly, the same formula applied to $N = 0$ and $L = 1$ shows that $\lambda^p(1, y)$ is the coefficient of t^p in

$$\frac{\sum_{k\geq 1} \lambda^k(y)t^k}{\sum_{k\geq 0} \lambda^k(y)t^k}.$$

The congruences (4.3.4) can therefore be written

$$i_*\left(\frac{\sum_{k\geq 1} k\lambda^k(N, y)t^k}{1 + \sum_{k\geq 1} \lambda^k(N, y)\lambda_{-1}(N)t^k}\right) = \frac{\sum_{k\geq 1} k\lambda^k(i_*(y))t^k}{\sum_{k\geq 0} \lambda^k(i_*(y))t^k} \pmod{\text{torsion}}.$$

Applying formula 2.8, one obtains

$$\frac{\sum_{k\geq 0} k\, i_*(\lambda^k(N, y))t^k}{1 + \sum_{k\geq 1} i_*(\lambda^k(N, y))t^k} = \frac{\sum_{k\geq 0} k\lambda^k(i_*(y))t^k}{\sum_{k\geq 0} \lambda^k(i_*(y))t^k} \pmod{\text{torsion}}.$$

If

$$u(t) = 1 + \sum_{k\geq 1} i_*(\lambda^k(N, y))t^k, \qquad v(t) = \sum_{k\geq 0} \lambda^k(i_*(y))t^k,$$

this last relation can also be written

$$\frac{d}{dt}\left(\frac{u(t)}{v(t)}\right) \equiv 0 \pmod{\text{torsion}}.$$

Since the constant term of $u(t)/v(t)$ is 1, and since the congruences are modulo torsion, one may integrate and write the congruence $u(t) = v(t)$. This gives the announced congruences.

Corollary 4.4. With the notation and hypotheses of 4.3, for every $p \geq 1$ and every $y \in K^\bullet(Y)$, one has

$$i_*(\psi^p(N, y)) \equiv \psi^p(i_*(y)) \pmod{\text{torsion}}.$$

This follows immediately from 4.3, using the expression of the ψ^p as integral linear combinations of the λ^p.

Remark 4.5. The proof of 4.3 itself suggests that (4.3.1) should be true not merely modulo torsion, but also as an equality. In RRR, the equality was proved in characteristic 0 by means of a geometric characterization of $\lambda^p(N,x)$. An extension of this method to all characteristics would probably be possible if one knew how to define exterior-power operations on perfect complexes (cf. XIV).

Theorem 4.6. Let X be a quasi-compact scheme having an ample sheaf, let $d>0$ be an integer, and let $i: Y \to X$ be a regular immersion of codimension d. Then, for every k,

$$i_*\big((\mathrm{Fil}^k K^\bullet(Y)) \otimes \mathbb{Q}\big) \subset (\mathrm{Fil}^{k+d} K^\bullet(X)) \otimes \mathbb{Q}.$$

First note that $i_*(K^\bullet(Y)) \subset \mathrm{Fil}^1 K^\bullet(X)$. Indeed, $X-Y$ is dense in X; otherwise Y would contain an affine open, and on this open the ideal J defining Y would be a nilideal, hence a locally nilpotent ideal, since it is of finite type. This is impossible, since the J^n/J^{n+1} are non-zero locally free modules. The augmentation in $K^\bullet(X)$ of an element of $i_*(K^\bullet(Y))$ is a locally constant function on X; since it vanishes on $X-Y$, it vanishes on all of X.

For every abelian group A, write $A_\mathbb{Q} = A \otimes \mathbb{Q}$. Let $x \in \mathrm{Fil}^k K^\bullet(Y)_\mathbb{Q}$. Then x can be written in the form

$$x = \sum a\, \gamma^{i_1}(x_1) \cdots \gamma^{i_n}(x_n),$$

with $\epsilon(x_i)=0$ for all i, $a \in \mathbb{Q}$, and $i_1 + \cdots + i_n \geq k$. The x_i can be expressed as integral linear combinations of the classes E_α of finitely many locally free finite-type $\mathcal{O}_Y$-modules E_α. Let R be the sub-$\mathbb{Q}$-λ-algebra of $K^\bullet(Y)_\mathbb{Q}$ generated by the E_α and by N. Then $x \in \mathrm{Fil}^k R$.

The filtration of the λ-ring R is discrete (VI 6.2); choose m such that $\mathrm{Fil}^m R = 0$. Finally, after decomposing X as a finite sum of disjoint open subsets and proving the assertion on each such open, one may suppose that the E_α have constant rank r_α. The augmentation ring of R is then $\mathbb{Q}$.

We shall show that, for every i,

$$i_*(\mathrm{Fil}^i R) \subset \mathrm{Fil}^{i+d} K^\bullet(X)_\mathbb{Q}.$$

This will imply in particular

$$i_*(x) \in \mathrm{Fil}^{k+d} K^\bullet(X),$$

and so prove the theorem.

The proof is by descending induction on i, the case $i=m$ being trivial. Let $y \in \mathrm{Fil}^i R$, and set, for j,

$$a_j = (-1)^{j-1}(j-1)!.$$

Then, by V 6.10,

$$\gamma^{d+i}(N,y) - a_{d+i} y \in \mathrm{Fil}^{i+1} R.$$

The induction hypothesis can therefore be applied, giving

$$i_*(\mathrm{Fil}^{i+1} R) \subset \mathrm{Fil}^{i+d+1} K^\bullet(X)_\mathbb{Q}.$$

Thus

$$i_*\big(\gamma^{d+i}(N,y) - a_{d+i} y\big) \in \mathrm{Fil}^{i+d+1} K^\bullet(X)_\mathbb{Q}.$$

By 4.4 this implies

$$\gamma^{d+i}\big(i_*(y)\big) - a_{d+i} i_*(y) \in \mathrm{Fil}^{i+d+1} K^\bullet(X)_\mathbb{Q}.$$

Since $i_*(y)$ has zero augmentation,

$$\gamma^{d+i}\big(i_*(y)\big) \in \mathrm{Fil}^{i+d+1} K^\bullet(X).$$

As we are working modulo torsion, this gives

$$i_*(y) \in \mathrm{Fil}^{i+d} K^\bullet(X)_\mathbb{Q}.$$

Remark 4.6.1. One sees that, even if 4.3 were true without tensoring by $\mathbb{Q}$, the preceding proof works only modulo torsion. For a counterexample, see XIV 4.6.

4.7

We return to the situation of 3.7, where one had the commutative diagram

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X, \end{array}$$

where i is a regular immersion, X' is the X-scheme obtained by blowing up Y, and $Y' = X' \times_X Y$. Suppose X noetherian with an ample sheaf, and Y regular. We saw in 3.7 that there is then the exact sequence

$$0 \longrightarrow K^\bullet(Y) \xrightarrow{u} K^\bullet(Y') \times K^\bullet(X) \xrightarrow{v} K^\bullet(X') \longrightarrow 0.$$

We shall show that this exact sequence permits one to describe the whole situation formed by the $K^\bullet$ of X, Y, X', Y', from the data of the λ-rings $K^\bullet(X)$, $K^\bullet(Y)$, the maps i^* and i_*, and the element $N \in K^\bullet(Y)$, at least up to torsion, up to one further point.

Let K, K' be two λ-rings, let $i_* : K \to K'$ be an additive homomorphism, let $i^* : K' \to K$ be a λ-homomorphism, let d be an integer, and let $N \in K$ be such that $\lambda^d(N) \neq 0$ for $i > d$ and $\lambda^d(N)$ is invertible. Suppose, moreover, that the following properties hold:

i) i_* is K'-linear (the "projection formula");

ii) $i^* i_*$ is multiplication by $\lambda_{-1}(N)$;

iii) for all $n \geq 1$ and all $y \in K$,

$$i_*(\lambda^n(N, y)) = \lambda^n(i_*(y))$$

(relation (4.3.1)).

We first use VI 1.1 to construct a K-λ-algebra K_1 which plays the role of $K^\bullet(Y')$. Let L be an indeterminate, and let $K[L]$ be the K-λ-algebra generated by L, subject to the relations $\lambda^i(L) = 0$ for $i \geq 2$ (V 4.9). Put

$$P(L) = \lambda^d(N - L) = \sum_{i=0}^{d} (-1)^i \lambda^{d-i}(N) L^i.$$

The ideal generated by P is a λ-ideal. To see this, first observe that the relations $\lambda^i(N) = 0$ for $i > d$ imply that the $\lambda^j(N - L)$, for $j \geq d$, are multiples of $\lambda^d(N - L)$. If one works in $1 + K[(P)][[t]]$, the image $\lambda_t(N - L)$ is a polynomial of degree $< d$, and a universal argument shows that its λ^i for $i \geq d$ are equal to 1 (cf. V 4.8). Since in

$$1 + K[[t]]^+[L]/(L^N)$$

one has $\lambda_t(\lambda^d(N - L)) = \lambda(\lambda_t(N - L))$, it follows that the $\lambda^i(\lambda^d(N - L))$, $i \geq 1$, are multiples of P. By the relations V 2.4.2, the ideal generated by P is therefore a λ-ideal.

Put $K_1 = K/(P)$. The natural homomorphism $K \to K_1$ will be denoted g^*; it is a λ-homomorphism. Moreover K_1 is a free K-module with basis

$$1, L, \ldots, L^{d-1}.$$

Indeed, from

$$\lambda^d(N) = \lambda^{d-1}(N-L)L$$

and the invertibility hypothesis on $\lambda^d(N)$, it follows that L is invertible. Thus K_1 admits as a basis over K the elements $1, L^{-1}, \ldots, L^{-d+1}$. Define an application $g_* : K_1 \to K$ by assigning to

$$y' = y_0 + y_1 L + \cdots + y_{d-1} L^{d-1}$$

the element y_{d-1} of K. It is clear that the application so defined is an additive, K-linear homomorphism, and that $g_* g^* = \mathrm{Id}$.

We next equip $K_1 \times K'$ with a λ-ring structure by putting

$$(y_1', x_1) * (y_2', x_2) = (y_1' y_2'(1-L) + x_1 y_2' + x_2 y_1',\ x_1 x_2), \tag{4.7.1}$$

$$\lambda_*^n(y', x) = \sum_{p=1}^{n} \lambda^{n-p}(x) \lambda^p(L, y') + \lambda^n(x). \tag{4.7.2}$$

In these formulas K_1 is regarded as a K'-algebra by the homomorphism

$$K' \xrightarrow{i^*} K \xrightarrow{g^*} K_1.$$

Next, for $y_1, y_2 \in K$ and $n \geq 1$, put

$$y_1 *_y y_2 = -\, y_1 y_2 \lambda_{-1}(N), \tag{4.7.3}$$

$$\lambda_*^n(y_1) = -\, \lambda^n(N, -y_1), \tag{4.7.4}$$

and define a homomorphism

$$u : K \longrightarrow K_1 \times K'$$

by

$$u(y) = \big(g^*(y)\lambda_{-1}(F), -i_*(y)\big), \qquad F = N - L.$$

Proposition 4.7.5. The operations defined by (4.7.1) and (4.7.2) furnish $K_1 \times K'$ with a λ-ring structure, and u is an injective homomorphism which identifies K, equipped with the operations defined by (4.7.3) and (4.7.4), with a λ-ideal of $K_1 \times K'$.

It is immediate to verify that $K_1 \times K'$ is a λ-ring; for example, one may refer to V 5.5, where an entirely analogous construction is made. The verification of the second part of the proposition is also straightforward from the preceding relations, the relation

$$i_*(y_1 y_2 \lambda_{-1}(N)) = i_*(y_1)\, i_*(y_2),$$

which follows immediately from the hypotheses, the relation $g_*(\lambda_{-1}(F)) = 1$ (VI 5.9), and the relations of V 5.

Put

$$K_1' = (K_1 \times K')/K,$$

and let $v : K_1 \times K' \to K_1'$ be the canonical homomorphism. Regard K_1' as a λ-ring via the quotient structure. Define additive homomorphisms

$$j_* : K_1 \to K_1', \quad j^* : K_1' \to K_1, \quad f_* : K_1' \to K', \quad f^* : K' \to K_1'.$$

The homomorphism j_* is the composite $K_1 \to K_1 \times K' \to K'_1$. Similarly, f^* is the composite $K' \to K_1 \times K' \to K'_1$. Thus

$$v = j_* + f^*.$$

Define next a homomorphism $K_1 \times K' \to K_1$ by assigning to (y', x) the element

$$y'(1 - L) + g^* i^*(x).$$

This homomorphism vanishes on K, and hence defines a homomorphism

$$j^* : K'_1 \longrightarrow K_1.$$

Finally, define a homomorphism $K_1 \times K' \to K'$ by assigning to (y', x) the element

$$i_* g_*(y') + x.$$

This homomorphism vanishes on K, and hence defines a homomorphism

$$f_* : K'_1 \longrightarrow K'.$$

Proposition 4.7.6.

i) j^* and f^* are λ-homomorphisms;

ii) j_* is K'_1-linear and f_* is K'-linear;

iii) $j^* j_*$ is the homomorphism "multiplication by $1 - L$";

iv) for every $y' \in K_1$ and every $p \geq 1$,

$$j_*(\lambda^p(L, y')) = \lambda^p(j_*(y'));$$

v) $f_* f^* = \mathrm{Id}$.

It suffices to let the machine run.

If one returns to the geometric situation recalled above, one sees that $K^\bullet(X')$ is constructed from $K^\bullet(Y)$, $K^\bullet(X)$, the maps i^* and i_*, and the element $N \in K^\bullet(Y)$, by the preceding method. However, the λ-operations on $K^\bullet(X')$ agree with the λ-operations defined above on K'_1 only up to torsion, so long as relation (4.3.1) has been proved only modulo torsion.

Theorem 4.8. Let $i : Y \to X$ be a regular closed immersion of constant codimension d, let X' be the scheme obtained by blowing up Y, and let $Y' = X' \times_X Y$, so that one has the commutative diagram

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X. \end{array}$$

Assume that X is quasi-compact and has an ample sheaf. Then, for every k, there is an exact sequence

$$0 \longrightarrow \mathrm{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q} \xrightarrow{u^{\mathrm{gr}}} \mathrm{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q} \times \mathrm{Gr}^k K^\bullet(X)_\mathbb{Q} \xrightarrow{v^{\mathrm{gr}}} \mathrm{Gr}^k K^\bullet(X')_\mathbb{Q},$$

where, for $y \in \mathrm{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q}$,

$$u^{\mathrm{gr}}(y) = \big(g^{\mathrm{gr}}(y)(-1)^{d-1} c^{d-1}(F), -i_{\mathrm{gr}}(y)\big),$$

and, for $y' \in \mathrm{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q}$ and $x \in \mathrm{Gr}^k K^\bullet(X)_\mathbb{Q}$,

$$v^{\mathrm{gr}}(y', x) = j_{\mathrm{gr}}(y') + f^{\mathrm{gr}}(x).$$

The notation is that of 3.1 and 3.3, and $c^{d-1}(F)$ denotes the $(d-1)$-st Chern class of F (V 6.7). Moreover, u^{gr} has a left inverse u'^{gr}, defined by

$$u'^{\mathrm{gr}}(y', x) = g_{\mathrm{gr}}(y').$$

If, in addition, X is noetherian and Y regular, one has the exact sequence

$$0 \longrightarrow \mathrm{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q} \xrightarrow{u^{\mathrm{gr}}} \mathrm{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q} \times \mathrm{Gr}^k K^\bullet(X)_\mathbb{Q} \xrightarrow{v^{\mathrm{gr}}} \mathrm{Gr}^k K^\bullet(X')_\mathbb{Q} \longrightarrow 0.$$

Proof. Notice that one could give, in the case where Y is regular, a purely algebraic proof in terms of the theory of λ-rings, using the description 4.7 of the situation. We shall instead compute it from the proof of 3.7.

We know that the inverse-image homomorphisms f^*, g^* are λ-homomorphisms, are compatible with the filtrations of the λ-rings, and therefore define homomorphisms on the associated gradeds:

$$f^{\mathrm{gr}} : \mathrm{Gr}^k K^\bullet(X) \to \mathrm{Gr}^k K^\bullet(X'), \qquad g^{\mathrm{gr}} : \mathrm{Gr}^{k-d} K^\bullet(Y) \to \mathrm{Gr}^{k-d} K^\bullet(Y'),$$

and similarly after tensoring with $\mathbb{Q}$. Moreover, 4.6 shows that, after tensoring with $\mathbb{Q}$, the homomorphisms i_*, j_* pass to the associated gradeds, giving

$$i_{\mathrm{gr}} : \mathrm{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q} \to \mathrm{Gr}^k K^\bullet(X)_\mathbb{Q}, \qquad j_{\mathrm{gr}} : \mathrm{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q} \to \mathrm{Gr}^k K^\bullet(X')_\mathbb{Q}.$$

Finally, since

$$\lambda_{-1}(F) = (-1)^{d-1}\gamma^{d-1}(F),$$

the morphisms u^{gr} and v^{gr} are obtained by passage to the quotient from the morphisms u and v of 3.7. Thus $v^{\mathrm{gr}}u^{\mathrm{gr}} = 0$.

By VI 5.8,

$$g_*(\mathrm{Fil}^{k-1} K^\bullet(Y')) \subset \mathrm{Fil}^{k-d} K^\bullet(Y)$$

for every k, since $Y' = \mathbf{P}(N)$ and N has rank d. We therefore obtain a homomorphism

$$g_{\mathrm{gr}} : \mathrm{Gr}^{k-1} K^\bullet(Y') \to \mathrm{Gr}^{k-d} K^\bullet(Y),$$

and an analogous homomorphism after tensoring with $\mathbb{Q}$. The homomorphism u'^{gr} is obtained from u' (3.7) by passage to the associated graded. Since u' is a left inverse of u, u'^{gr} is a left inverse of u^{gr}, proving at the same time the injectivity of u^{gr}.

To show that $\mathrm{Ker}(v^{\mathrm{gr}}) = \Im(u^{\mathrm{gr}})$, suppose that

$$y' \in \mathrm{Fil}^{k-1} K^\bullet(Y')_\mathbb{Q}, \qquad x \in \mathrm{Fil}^k K^\bullet(X)_\mathbb{Q}$$

are such that

$$j_*(y') + f^*(x) \in \mathrm{Fil}^{k+1} K^\bullet(X')_\mathbb{Q}. \tag{4.6.1}$$

We must show that there exists $y \in \mathrm{Fil}^{k-d} K^\bullet(Y)_\mathbb{Q}$ such that

$$\text{i)} \quad x + i_*(y) \in \mathrm{Fil}^{k+1} K^\bullet(X)_\mathbb{Q},$$
$$\text{ii)} \quad y' - g^*(y)\lambda_{-1}(F) \in \mathrm{Fil}^k K^\bullet(Y')_\mathbb{Q}.$$

Put

$$y = g_*(y').$$

By VI 5.8, $y \in \mathrm{Fil}^{k-d} K^\bullet(Y)$. Since f_* preserves the filtration (using results of Exposé VIII, which is inconvenient here, for 4.8 is not used to prove this fact), it follows from 1.8 ii) that f is a complete-intersection morphism of virtual relative dimension 0 (VIII 4.2). Since f is projective, the result follows from VIII 3.2. Applying f_* to (4.6.1), we get

$$f_*(j_*(y')) + f_*f^*(x) \in \mathrm{Fil}^{k+1} K^\bullet(X)_\mathbb{Q}.$$

Thus, by 3.6,

$$i_*(y) + x \in \mathrm{Fil}^{k+1} K^\bullet(X)_\mathbb{Q}.$$

To prove (ii), one must show that

$$y' - g^*(g_*y')\lambda_{-1}(F) \in \mathrm{Fil}^k K^\bullet(Y')_\mathbb{Q}.$$

Put

$$z = y' - g^*(g_*y')\lambda_{-1}(F).$$

Then

$$\begin{aligned} j_*(z) &= j_*(y') - j_*\big(g^*(g_*y')\lambda_{-1}(F)\big) \\ &= j_*(y') - f^*(i_*g_*y') \qquad \text{by 3.4.} \end{aligned}$$

But relation (i) gives

$$f^*(i_*g_*y') + f^*(x) \in \mathrm{Fil}^{k+1} K^\bullet(X')_\mathbb{Q}.$$

Together with (4.6.1), this gives

$$j_*(z) \in \mathrm{Fil}^{k+1} K^\bullet(X')_\mathbb{Q}.$$

Consequently

$$j^*(j_*(z)) = z(1-L) \in \mathrm{Fil}^{k+1} K^\bullet(Y')_\mathbb{Q}.$$

Let z' be the image of z in $\mathrm{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q}$, and let $(1-L)$ denote the class of $1-L$ in $\mathrm{Gr}^1 K^\bullet(Y')_\mathbb{Q}$. We know that $z'(1-L) = 0$. We may then apply VI 5.11 (which remains valid after tensoring with $\mathbb{Q}$, since its proof uses only the structure of $K^\bullet(Y')$ as a free $K^\bullet(Y)$-algebra) and find $z'' \in K^\bullet(Y)_\mathbb{Q}$ such that

$$z = g^*(z'')\lambda_{-1}(F) \in \mathrm{Fil}^k K^\bullet(Y')_\mathbb{Q}.$$

Applying g_*, one obtains, since $g_*(z) = 0$,

$$z'' \in \mathrm{Fil}^{k+1-d} K^\bullet(Y)_\mathbb{Q},$$

and hence, since $\lambda_{-1}(F) \in \mathrm{Fil}^{d-1} K^\bullet(Y')_\mathbb{Q}$,

$$z \in \mathrm{Fil}^k K^\bullet(Y')_\mathbb{Q},$$

which finishes the proof.

Assume now that X is noetherian and Y regular. Then v is surjective (3.7). We saw in the proof of 3.7 that for every $x' \in K^\bullet(X')$ there exists $y' \in K^\bullet(Y')$ such that

$$x' = f^*(f_*x') + j_*(y').$$

The same property remains true after tensoring with $\mathbb{Q}$. Hence, if $x' \in \mathrm{Fil}^k K^\bullet(X')_\mathbb{Q}$, then there exist $(y', x) \in K^\bullet(Y')_\mathbb{Q} \times K^\bullet(X)_\mathbb{Q}$ such that

$$x' = v(y', x), \qquad x \in \mathrm{Fil}^k K^\bullet(X)_\mathbb{Q}.$$

Indeed it suffices to take $x = f_*(x')$.

Filter $K^\bullet(Y')_\mathbb{Q} \times K^\bullet(X)_\mathbb{Q}$, as in 4.7, by putting

$$\mathrm{Fil}^k(K^\bullet(Y')_\mathbb{Q} \times K^\bullet(X)_\mathbb{Q}) = \mathrm{Fil}^{k-1} K^\bullet(Y')_\mathbb{Q} \times \mathrm{Fil}^k K^\bullet(X)_\mathbb{Q}.$$

It follows immediately from 4.7.1 that this is a ring filtration. To show that for every k the homomorphism

$$\mathrm{Fil}^1(K^\bullet(Y')_\mathbb{Q} \times K^\bullet(X)_\mathbb{Q}) \longrightarrow \mathrm{Fil}^1 K^\bullet(X')_\mathbb{Q}$$

is surjective, it is enough to show that, if $x' = y(x'_1)$ with $\epsilon(x'_1) = 0$, there is $(y', x) \in \mathrm{Fil}^1(K^\bullet(Y')_\mathbb{Q} \times K^\bullet(X)_\mathbb{Q})$ such that $x' = v(y', x)$.

By what precedes, there exists $(y'_1, x_1) \in K^\bullet(Y')_\mathbb{Q} \times K^\bullet(X)_\mathbb{Q}$ such that $x'_1 = v(y'_1, x_1)$ and $x_1 \in \mathrm{Fil}^1 K^\bullet(X)_\mathbb{Q}$. By 4.7,

$$x' = v(\gamma^k_*(y'_1, x_1)).$$

But

$$\gamma^k_*(y'_1, x_1) = \sum_{p=1}^{k} \gamma^{k-p}(x_1)\gamma^p(L, y'_1),$$

which is an immediate consequence of (4.7.2). Since x_1 has zero augmentation, $\gamma^k(x_1) \in \mathrm{Fil}^1 K^\bullet(X)_\mathbb{Q}$, and by V 5.7

$$\sum_{p=1}^{k} \gamma^{k-p}(x_1)\gamma^p(L, y'_1) \in \mathrm{Fil}^{k-1} K^\bullet(Y')_\mathbb{Q}.$$

This proves the assertion.

Remark 4.9. In the case where X is noetherian and Y regular, one can, proceeding as in 4.7, describe the structure of graded ring of $\mathrm{Gr}^* K^\bullet(X')_\mathbb{Q}$ in terms of the graded rings $\mathrm{Gr}^* K^\bullet(X)_\mathbb{Q}$ and $\mathrm{Gr}^* K^\bullet(Y)_\mathbb{Q}$, of the additive homomorphisms i_{gr} and i^{gr} induced by i^* and i_*, and of the element $c(N) \in \mathrm{Gr}^* K^\bullet(Y)$. Similarly, one can describe, from the same data, the homomorphisms $f^{\mathrm{gr}}, f_{\mathrm{gr}}, j^{\mathrm{gr}}, j_{\mathrm{gr}}$ defined from f^*, f_*, j^*, j_* by passage to the associated gradeds.

4.10

Let X be a regular scheme having an ample sheaf. We know (IV) that the Grothendieck groups formed with locally free finite-type $\mathcal{O}_X$-modules, and with coherent $\mathcal{O}_X$-modules, are isomorphic. We shall use this fact to define on $K^\bullet(X)$ a "topological" filtration: take $\mathrm{Fil}^i_{\mathrm{top}} K^\bullet(X)$ to be the subgroup generated by the classes of coherent sheaves whose support has codimension at least i. One shows easily that $\mathrm{Fil}^i_{\mathrm{top}} K^\bullet(X)$ is generated by the classes of the sheaves $\mathcal{O}_Z$, where Z ranges over the closed integral subschemes of X of codimension $\geq i$.

Proposition 4.11. Let X be a regular scheme having an ample sheaf. Then, for every $k \geq 0$,

$$\mathrm{Fil}^k_{\mathrm{top}} K^\bullet(X)_\mathbb{Q} \subset \mathrm{Fil}^k K^\bullet(X)_\mathbb{Q}.$$

The reverse inclusion, valid without tensoring by $\mathbb{Q}$, will be proved in X.

By 4.10, it is enough to show that, if Z is an integral closed subscheme of X, of codimension k, then

$$\mathrm{cl}(\mathcal{O}_Z) \in \mathrm{Fil}^k K^\bullet(X)_\mathbb{Q}.$$

To this end we proceed by noetherian induction on the integral closed subschemes of X. Suppose the property already proved for the proper integral closed subschemes of Z, and let us prove it for Z. Let J be the ideal of $\mathcal{O}_X$ defining Z, and let z be the generic point of Z. Since Z is integral, J_z is the maximal ideal of $\mathcal{O}_{X,z}$; and since X is regular, J_z can be generated by an

$\mathcal{O}_{X,z}$-regular sequence. One can therefore find a sequence of sections of J, in a neighbourhood of z, forming an $\mathcal{O}_X$-regular sequence. Consequently there is an open neighbourhood U of z, such that $Z' = X - U$ is contained in Z, and such that $J\mathcal{O}_U$ is a regular ideal of $\mathcal{O}_U$. The immersion $Z \cap U \to U$ is then a regular immersion of codimension

$$\dim(\mathcal{O}_{X,z}) = \operatorname{codim}(Z, X)$$

(EGA IV 5.1.2). Moreover, one has the exact sequence (IV)

$$K^\bullet(Z') \xrightarrow{j_*} K^\bullet(X) \xrightarrow{u^*} K^\bullet(U) \to 0,$$

where j is the immersion $Z' \to X$ and u is the immersion $U \to X$. After tensoring with $\mathbb{Q}$, one obtains

$$\operatorname{cl}(\mathcal{O}_Z) \equiv i_*(1_{\mathcal{O}_{Z\cap U}}),$$

and by 4.6 the image under u^* of the class of $\mathcal{O}_Z$ belongs to $\operatorname{Fil}^k K^\bullet(U)_\mathbb{Q}$. Since u^* is a surjective λ-homomorphism, it induces a surjection

$$\operatorname{Fil}^k K^\bullet(X) \longrightarrow \operatorname{Fil}^k K^\bullet(U).$$

Thus $u^*(1_Z) = u^*(x)$, with $x \in \operatorname{Fil}^k K^\bullet(X)_\mathbb{Q}$. Therefore there exists

$$y \in j_*(K^\bullet(Z'))$$

such that

$$\operatorname{cl}(\mathcal{O}_Z) - y \in \operatorname{Fil}^k K^\bullet(X)_\mathbb{Q}.$$

But $Z' \subset Z$, and since $z \notin Z'$, $\operatorname{codim}(Z', X) > k$. Thus y can be expressed as a combination of classes of sheaves $\mathcal{O}_{Z''_\alpha}$, where the Z''_α are integral closed subschemes of Z, different from Z. By the induction hypothesis, $y \in \operatorname{Fil}^k K^\bullet(X)_\mathbb{Q}$, and this completes the proof.

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Exposé VIII. The Riemann–Roch theorem

by Pierre Berthelot

1. Complete-intersection morphisms

Definition 1.1. Let $f : X \to Y$ be a morphism of schemes. One says that f is a *complete-intersection morphism* if, for every point x of X, there exist an open neighbourhood U of x and a smooth Y-scheme V such that the restriction of f to U factors as

$$\begin{array}{ccc} U & \xrightarrow{i} & V \\ & \searrow^{f} & \downarrow{\scriptstyle g} \\ & & Y, \end{array}$$

where i is a regular immersion (VII 1.4). After replacing V by an open subset, one may suppose that i is a regular closed immersion.

In this definition one may replace V by a vector bundle $Y[T_1, \ldots, T_n]$. Indeed, since the notion is local, one may suppose that V is affine and that its image in Y is contained in an affine open W of Y. There then exists an integer n and a closed W-immersion

$$V \longrightarrow W[T_1, \ldots, T_n],$$

hence a Y-immersion $V \to Y[T_1, \ldots, T_n]$, which is regular by VII 1.10. Since the composite of two regular immersions is regular, one obtains a regular Y-immersion

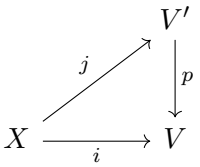
$$U \longrightarrow Y[T_1, \ldots, T_n]$$

factorizing f.

Proposition 1.2. Let $f : X \to Y$ be a complete-intersection morphism. Then every Y-immersion $i : X \to Z$ of X in a smooth Y-scheme is regular.

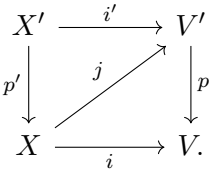
We first prove the following corollary.

Corollary 1.3. Let



be a commutative diagram in which i and j are immersions, and p is a smooth morphism. Then i is a regular immersion if and only if j is a regular immersion.

(i) Suppose i regular. Put $X' = V' \times_V X$. The datum of j defines a section j' of X' over X, such that $j = i'j'$:



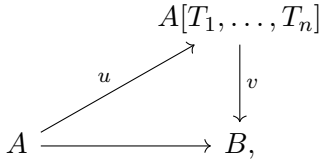
Now j' is a regular immersion, being a section of a smooth morphism (VII 1.10); i' is a regular immersion because it is obtained from i by the smooth, hence flat, base change (VII 1.5). Therefore j is a regular immersion (VII 1.7).

(ii) Suppose j regular and p étale. Then j' is an open immersion (EGA IV 17.9.3), hence an isomorphism of X onto $j'(X)$; put $U = j'(X)$. Let i'' be the restriction of i' to U, and let U' be an open subset of V' such that $U = U' \cap X'$. Then

$$i'' = j \circ (j')^{-1},$$

so that i'' is a regular immersion of U into U'. But the morphism $U' \to V$, being smooth, is flat and locally of finite presentation, hence open. Consequently it is faithfully flat and quasi-compact from U' onto an open neighbourhood of X in V. It follows from VII 1.5 that i is a regular immersion.

(iii) Suppose j regular, and suppose that V' is a vector bundle over V. Since the property to be proved is local on V, one may suppose V affine, i closed, and $V' = V[T_1, \ldots, T_n]$. Put $V = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$. One has the commutative diagram



where the morphisms u and v are surjective. Choose elements $t_i \in A$, $1 \leq i \leq n$, such that $u(t_i) = v(T_i)$ for every i. Define a homomorphism

$$w : A[T_1, \ldots, T_n] \longrightarrow A, \qquad w(T_i) = t_i.$$

This gives a closed immersion $s : V \to V'$ which is a section of p, and such that $j = s \circ i$. Moreover, being a section of a smooth morphism, s is a regular immersion (VII 1.10). Since s and $s \circ i$ are regular, it will follow that i is regular if one proves that the sequence

$$0 \longrightarrow i^*(K/K^2) \longrightarrow J/J^2 \longrightarrow I/I^2 \longrightarrow 0$$

is exact and locally split; here I, J, K denote respectively the ideals defining i, j, s (VII 1.7). It is enough to show that $i^*(K/K^2)$ is a direct factor of J/J^2. But j' is obtained from s by the base change $i' : X' \to V'$. Moreover j' and s are regular immersions (VII 1.10). Thus, by VII 1.2, the ideal defining j' is $i'^*(K)$, and the conormal sheaf of j' is $i'^*(K/K^2)$. The exact sequence of conormal sheaves attached to the decomposition $j = i'j'$ therefore gives in particular a homomorphism

$$J/J^2 \longrightarrow i^*(K/K^2).$$

It is immediate to verify that the composite

$$i^*(K/K^2) \longrightarrow J/J^2 \longrightarrow i^*(K/K^2)$$

is the identity. In affine terms, this corresponds to the decomposition of J as the direct sum of an ideal of A and the ideal generated by the $T_i - t_i$. This gives the required splitting.

(iv) Finally suppose j regular and p smooth. For every $x \in X$, one can find an open neighbourhood W of $i(x)$, an open neighbourhood W' of $j(x)$ lying over W, and a commutative diagram

$$\begin{array}{ccc} W' & \xrightarrow{q} & W[T_1, \ldots, T_n] \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p_0} \\ W & = & W, \end{array}$$

where p_0 is the structural morphism, and q is an étalemorphism (EGA IV 17.11.4). One may then suppose $V = W$ and $V' = W'$. It is easy to see that qj is an immersion, by introducing the fiber product $X[T_1, \ldots, T_n]$. Applying (ii) and (iii), one deduces that qj is regular, and then that i is regular.

We now prove Proposition 1.2. Since regularity is local on Z, we may work in a neighbourhood of a point $x \in X$. There is then an open neighbourhood U of x in X, a smooth Y-scheme V, and a regular Y-immersion $i' : U \to V$. We have a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{i'} & V \\ {\scriptstyle i}\downarrow & & \downarrow \\ Z & \longrightarrow & Y, \end{array}$$

where the unlabeled vertical arrows are the structural morphisms. From it one deduces a closed immersion

$$i'' : U \longrightarrow Z \times_Y V,$$

and a commutative diagram

$$\begin{array}{ccccc} & & U & & \\ & {\scriptstyle i}\swarrow & & \searrow{\scriptstyle i'} & \\ Z & \xleftarrow{p} & Z \times_Y V & \xrightarrow{p'} & V. \end{array}$$

Since i' is regular and p, p' are smooth, the corollary implies that i'' and i are regular.

The following proposition will not be used in the sequel of the Seminar.

Proposition 1.4. Let $f : X \to Y$ be a flat morphism locally of finite presentation. For f to be a complete-intersection morphism in the sense of 1.1, it is necessary and sufficient that it be a complete-intersection morphism in the sense of EGA IV 19.3.6.

Suppose that f is a complete-intersection morphism in the sense of 1.1, and let $x \in X$. Then there are an open neighbourhood U of x and a factorization

$$\begin{array}{ccc} U & \xrightarrow{i} & V \\ & \searrow^{f} & \downarrow{\scriptstyle g} \\ & & Y, \end{array}$$

where i is a regular immersion and g is smooth. Let $y = f(x)$. Then the immersion $i_y : U_y \to V_y$ is regular at x. Indeed, after possibly restricting V, one may suppose that there is a finite-type free $\mathcal{O}_V$-module E and a regular homomorphism $u : E \to \mathcal{O}_V$ such that the Koszul complex associated to u is a resolution of $\mathcal{O}_U$. Let u_y be the homomorphism induced on the fiber over y, and let $K_\bullet(u_y) = K_\bullet(u)_y$ be the associated Koszul complex. Since g is smooth, hence flat, $K_\bullet(u)$ is a Y-flat resolution of $\mathcal{O}_U$. Consequently the homology groups of $K_\bullet(u)_y$ are the groups

$$\operatorname{Tor}_i^{\mathcal{O}_Y}(\mathcal{O}_U, k(y)),$$

where $k(y)$ is the residue field of $\mathcal{O}_{Y,y}$. Since f is flat, these groups are zero; hence i_y is regular at x. Since g is smooth, the local ring of V_y at x is regular. The local ring of U_y at y is therefore the quotient of a regular ring by a regular ideal, hence a complete-intersection ring (EGA IV 19.3.2). It follows that f is a complete intersection at x in the sense of EGA IV 19.3.6.

Conversely, suppose that f is a complete-intersection morphism in the sense of EGA IV 19.3.6, and let $x \in X$. Let U be an affine neighbourhood of x lying over an affine neighbourhood U' of $y = f(x)$. One can then find a smooth U'-scheme V, and a closed immersion $U \to V$ (for instance by taking for V a vector bundle over U'). We shall show that i is regular. By EGA IV 19.3.7, i is transversally regular at x, since V is smooth over Y, hence V_y is regular at x, and U is flat over Y. It follows from EGA IV 19.2.4, a)3c), that i is regular at x. Therefore f is a complete-intersection morphism in the sense of 1.1.

Proposition 1.5. Let $f : X \to Y$ and $g : Y \to Z$ be two locally complete-intersection morphisms. Then gf is locally complete-intersection.

Let $x \in X$. One can find open neighbourhoods U, V of $x, f(x)$ and regular immersions i, j such that one has factorizations

$$\begin{array}{ccc} U & \xrightarrow{i} & V[T_1, \ldots, T_n] \\ & \searrow^{f} & \downarrow \\ & & V \end{array} \qquad \begin{array}{ccc} V & \xrightarrow{j} & Z[T'_1, \ldots, T'_m] \\ & \searrow^{g} & \downarrow \\ & & Z. \end{array}$$

This diagram can be completed by the commutative square

$$\begin{array}{ccc} V[T_1, \ldots, T_n] & \xrightarrow{j'} & Z[T'_1, \ldots, T'_m, T_1, \ldots, T_n] \\ \downarrow & & \downarrow \\ V & \xrightarrow[j]{} & Z[T'_1, \ldots, T'_m], \end{array}$$

where j' is obtained from j by base change. This base change is flat; hence j' is a regular immersion, and so $j'i$ is a regular immersion of U in a scheme smooth over W, where W is the relevant open of Z. This proves the assertion.

Proposition 1.6. Let $f : X \to Y$ be a morphism and let $g : Y' \to Y$ be a flat base change. If f is complete-intersection, then the morphism

$$f' : X \times_Y Y' \longrightarrow Y'$$

deduced from f is complete-intersection. The converse is true if, moreover, g is quasi-compact and surjective.

One can find an open covering of X by open subsets U_i such that, for every i, the restriction of f to U_i factors as

$$U_i \longrightarrow V_i \longrightarrow Y,$$

where $U_i \to V_i$ is a regular immersion and $V_i \to Y$ is smooth. The $U_i \times_Y Y'$ form an open covering of $X \times_Y Y'$, the immersions

$$U_i \times_Y Y' \longrightarrow V_i \times_Y Y'$$

are regular by VII 1.5, and the morphisms $V_i \times_Y Y' \to Y'$ are smooth, giving the first assertion. For the converse, EGA IV 2.7.1 (iv) first implies that f is locally of finite presentation; this reduces us easily, using 1.2, to the case where f is a closed immersion. In that case the assertion is just VII 1.5.

Proposition 1.7. Let $f : X \to Y$ be a complete-intersection morphism. Then f is a perfect morphism.

Since the property is local on X, one may work in a neighbourhood of a point $x \in X$ and suppose that there is an open subset U containing x and a factorization

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & V \\ & \searrow^{f} & \downarrow{\scriptstyle g} \\ & & Y, \end{array}$$

where i is a regular immersion and g is smooth. Since g is smooth, it is a perfect morphism (III 4.1). By VII 1.9, i is also a perfect morphism. Since the composite of two perfect morphisms is perfect (III 4.5), the proposition follows.

Proposition 1.8. Let $f : X \to Y$ be a complete-intersection morphism, let x be a point of X, and let

$$\begin{array}{ccc} U & \xrightarrow{\ i\ } & V \\ & \searrow^{f} & \downarrow{\scriptstyle g} \\ & & Y \end{array}$$

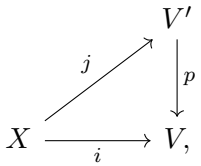
be a factorization in which U is an open neighbourhood of x, i is a regular immersion, and g is smooth. Let J be the ideal of $\mathcal{O}_V$ defining U. Then the integer

$$d_x = \operatorname{rank}\big((\Omega^1_{V/Y})_{i(x)}\big) - \operatorname{rank}\big((J/J^2)_x\big)$$

is independent of the chosen factorization of f.

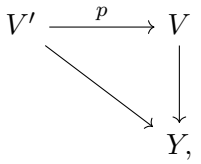
Suppose that the restriction of f to U factors through two smooth Y-schemes V and V', and put $V'' = V \times_Y V'$. Then V'' is smooth over Y, and also over V and V'. Moreover, the immersions

$X \to V$ and $X \to V'$ define an immersion $X \to V''$, which is regular by 1.2. It is therefore enough to compare the integers d_x attached to two factorizations through smooth Y-schemes V and V' such that there exists a commutative diagram over Y



where p is smooth. We may also suppose that $\Omega^1_{V/Y}$, $\Omega^1_{V'/Y}$, and $\Omega^1_{V'/V}$ have constant rank, as do the conormal sheaves of i and j.

From the commutative diagram, all morphisms being smooth,



one deduces

$$\operatorname{rank}(\Omega^1_{V'/Y}) = \operatorname{rank}(\Omega^1_{V/Y}) + \operatorname{rank}(\Omega^1_{V'/V}). \tag{1.8.1}$$

Let $X' = V' \times_V X$; then the rank of $\Omega^1_{V'/V}$ is equal to that of $\Omega^1_{X'/X}$. Further, j defines a section $j' : X \to X'$, which is a regular immersion of codimension equal to the rank of $\Omega^1_{X'/X}$ (EGA IV 17.10.4). If i' is the immersion obtained from i by the base change $V' \to V$, then $j = i'j'$, and consequently (VII 1.7)

$$\operatorname{rank}(N_{V'/X}) = \operatorname{rank}(N_{V/X}) + \operatorname{rank}(\Omega^1_{V'/V}). \tag{1.8.2}$$

Comparing (1.8.1) and (1.8.2) gives the proposition.

Definition 1.9. The integer d_x of Proposition 1.8 is called the *relative virtual dimension* of X over Y (or of f) at the point x.

The relative virtual dimension of X over Y is therefore a locally constant function on X, with integral values, positive or negative. If f is a smooth morphism of relative dimension n, its relative virtual dimension is n. If f is a regular immersion of codimension n, its relative virtual dimension is $-n$.

Proposition 1.10. Let $f : X \to Y$ and $g : Y \to Z$ be two complete-intersection morphisms, and let $x \in X$. Then

$$\text{dim. rel. virt.}_x(gf) = \text{dim. rel. virt.}_x(f) + \text{dim. rel. virt.}_{f(x)}(g).$$

This follows immediately from 1.9 and from the proof of 1.5.

2. The relative cotangent complex

2.1. We give here a definition of the cotangent complex which, without having the greatest possible generality (see Exposé 0, 4.4), will be sufficient for the needs of the Riemann–Roch theorem. Of course the definition given in Exposé 0 coincides, in the case considered there, with the definition given here.

Let $f : X \to Y$ be a smoothable morphism, that is, a morphism admitting a factorization

$$\begin{array}{ccc} X & \xrightarrow{i} & V \\ & \searrow^{f} & \downarrow{\scriptstyle g} \\ & & Y, \end{array}$$

where g is smooth and i is an immersion, which may be supposed closed. Let J be the ideal of X in V. The canonical derivation

$$d : \mathcal{O}_V \longrightarrow \Omega^1_{V/Y}$$

defines a homomorphism

$$J \longrightarrow \mathcal{O}_V \xrightarrow{d} \Omega^1_{V/Y} \longrightarrow \Omega^1_{V/Y} \otimes_{\mathcal{O}_V} \mathcal{O}_X$$

which vanishes on J^2, and hence defines a homomorphism, still denoted d,

$$J/J^2 \xrightarrow{d} i^*(\Omega^1_{V/Y}). \tag{2.1.1}$$

One checks easily that this homomorphism is $\mathcal{O}_X$-linear. One then defines a chain complex $L_\bullet^{X/Y}$ by putting

$$L_0^{X/Y} = i^*(\Omega^1_{V/Y}), \qquad L_1^{X/Y} = J/J^2, \qquad L_i^{X/Y} = 0 \quad (i \neq 0, 1),$$

and taking as homomorphism $L_1^{X/Y} \to L_0^{X/Y}$ the homomorphism (2.1.1).

Proposition 2.2. The complex $L_\bullet^{X/Y}$ is independent, up to canonical isomorphism in the derived category $D(X)$, of the factorization chosen through a smooth Y-scheme.

Let V, V' be two smooth Y-schemes through which f factors. Put $V'' = V \times_Y V'$. Then V'' is smooth over Y, and also over V and V'. Moreover the immersions $X \to V$ and $X \to V'$ define an immersion $X \to V''$ which factors f. It is therefore enough to compare the complexes $L_\bullet^{X/Y}$ attached to two factorizations through smooth Y-schemes V and V' for which there is a commutative diagram over Y

$$\begin{array}{ccc} & & V' \\ & \overset{j}{\nearrow} & \downarrow{\scriptstyle p} \\ X & \xrightarrow[i]{} & V, \end{array}$$

where i and j are closed immersions, and p is smooth. Functoriality then gives homomorphisms

$$I/I^2 \longrightarrow J/J^2, \qquad p^*(\Omega^1_{V/Y}) \longrightarrow \Omega^1_{V'/Y},$$

hence a diagram

$$\begin{array}{ccc} I/I^2 & \longrightarrow & J/J^2 \\ \downarrow{\scriptstyle d} & & \downarrow{\scriptstyle d} \\ i^*(\Omega^1_{V/Y}) & \longrightarrow & j^*(\Omega^1_{V'/Y}). \end{array} \tag{2.2.1}$$

Since V and V' are smooth over Y, one obtains the exact sequence

$$0 \longrightarrow p^*(\Omega^1_{V/Y}) \longrightarrow \Omega^1_{V'/Y} \longrightarrow \Omega^1_{V'/V} \longrightarrow 0.$$

Since the modules of differentials considered are locally free on $\mathcal{O}_V$, applying the functor j^* gives the exact sequence

$$0 \longrightarrow i^*(\Omega^1_{V/Y}) \longrightarrow j^*(\Omega^1_{V'/Y}) \longrightarrow j^*(\Omega^1_{V'/V}) \longrightarrow 0. \tag{2.2.2}$$

Furthermore, j defines a section j' of $X' = V' \times_V X$:

$$\begin{array}{ccc} X' & \xrightarrow{i'} & V' \\ {\scriptstyle p'}\downarrow & \overset{j}{\nearrow} & \downarrow{\scriptstyle p} \\ X & \xrightarrow[i]{} & V. \end{array}$$

Let I, J, K be the ideals defining i, j, j'. Then $j = i'j'$, and the immersions i', j, j' are regular; moreover i' is defined by the ideal $p^*(I)$, because p is flat, and its conormal sheaf is therefore

$$i'^*(p^*(I)) = p'^*(I/I^2).$$

One obtains the exact sequence

$$0 \longrightarrow I/I^2 \longrightarrow J/J^2 \longrightarrow K/K^2 \longrightarrow 0. \tag{2.2.3}$$

We shall show: (a) the diagram (2.2.1) is commutative; this will give a homomorphism of complexes θ from the complex $L^{X/Y}_\bullet$ defined by V to the complex $L^{X/Y}_\bullet$ defined by V'; (b) the homomorphism

$$K/K^2 \longrightarrow j^*(\Omega^1_{V'/V})$$

defined by (2.2.2), (2.2.3), and (2.2.1) is an isomorphism. This will imply that the mapping cylinder of θ is acyclic, hence that θ is a quasi-isomorphism, and therefore defines an isomorphism in the derived category.

The properties (a) and (b) are local on X. One may therefore suppose X, Y, V, V' affine. Put $Y = \mathrm{Spec}(A)$, $V = \mathrm{Spec}(B)$, $V' = \mathrm{Spec}(B')$, $X = \mathrm{Spec}(C)$, and keep the same letters for the corresponding homomorphisms and ideals. One has the diagram

$$\begin{array}{ccccc} B \otimes_A B & \xrightarrow{p\otimes p} & B' \otimes_A B' & \longleftarrow & B' \otimes_B B' \\ {\scriptstyle\Delta}\downarrow & & {\scriptstyle\Delta}\downarrow & & \downarrow \\ B & \xrightarrow{p} & B' & & B' \otimes_B C = B'/IB' \\ & {\scriptstyle i}\searrow & {\scriptstyle j}\downarrow & \swarrow{\scriptstyle p'} & \\ & & C & & \end{array}$$

where the left and right displayed squares are denoted (1) and (2), respectively. The homomorphism $I/I^2 \to J/J^2$ is deduced from the homomorphism $IB' \to J$ and coincides with that deduced from $p : I \to J$; the homomorphism

$$i^*(\Omega^1_{V/Y}) \longrightarrow j^*(\Omega^1_{V'/Y})$$

is deduced from the commutative square (1). Hence property (a) is evident.

The homomorphism $K/K^2 \to j^*(\Omega^1_{V'/V})$ considered in (b) may be defined as follows. Let $\overline{x} \in K/K^2$, and let x be an element of J lifting $\overline{x}$ (such an element exists because $K = J/IB'$). To $\overline{x}$ one associates the image of x by the composite

$$J \xrightarrow{d} \Omega^1_{B'/A} \otimes_{B'} C \longrightarrow \Omega^1_{B'/B} \otimes_{B'} C.$$

On the other hand, let H be the augmentation ideal of

$$B' \otimes_B B' \longrightarrow B'.$$

The commutative square (2) gives a homomorphism $H \to K$, hence a homomorphism

$$H/H^2 \otimes_{B'} C \longrightarrow K/K^2.$$

But $H/H^2 = \Omega^1_{B'/B}$. It remains only to check that the homomorphisms thus constructed are inverse to each other, which is straightforward.

Definition 2.3. Let $f : X \to Y$ be a smoothable morphism. The complex $L_\bullet^{X/Y}$, defined up to canonical isomorphism in the derived category $D(X)$, is called the *relative cotangent complex* of X over Y (or of f).

Proposition 2.4. Let $f : X \to Y$ be a smoothable complete-intersection morphism. Then $L_\bullet^{X/Y}$ is perfect. More precisely, for every factorization of f as the composite of an immersion and a smooth morphism, the complex $L_\bullet^{X/Y}$ defined in 2.1 is strictly perfect.

Let $f = gi$, where g is smooth and i is an immersion. By 1.2, i is a regular immersion. If J is the ideal defining i, then J/J^2 is a locally free finite-type $\mathcal{O}_X$-module. Likewise, since $g : V \to Y$ is smooth, $i^*(\Omega^1_{V/Y})$ is locally free of finite type. This proves 2.4.

Corollary 2.5. Let $f : X \to Y$ be a smoothable complete-intersection morphism, and let $f = gi$ be a factorization of f through a smooth Y-scheme V. Let J be the ideal of $\mathcal{O}_V$ defining i. If $K^\bullet(X)$ denotes the Grothendieck group of locally free finite-type $\mathcal{O}_X$-modules (respectively of perfect complexes on X), the element $T_f \in K^\bullet(X)$ defined by

$$T_f = \mathrm{cl}(i^*\Omega^1_{V/Y}) - \mathrm{cl}(J/J^2)$$

respectively

$$T_f = \mathrm{cl}(L_\bullet^{X/Y})$$

does not depend on the chosen factorization of f. The value at a point x of X of the augmentation of T_f is precisely the relative virtual dimension of X over Y at x (1.9).

The element T_f is well defined in both cases by 2.4; its class in $K^\bullet(X)$ is independent of the factorization, by 2.2 in the second case and by the proof of 2.2 in the first. The last assertion is trivial.

Proposition 2.6. Let $f : X \to Y$ and $g : Y \to Z$ be two smoothable complete-intersection morphisms, and suppose in addition that the following condition is satisfied:

(G) There exists a smooth Y-scheme V factorizing f such that, for every closed immersion $Y \to V'$ into a smooth Z-scheme, one can find a smooth V'-scheme V'' with

$$V \simeq V'' \times_{V'} Y.$$

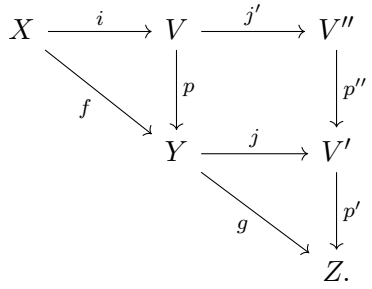
This condition is in particular satisfied if f is quasi-projective and if Y has an ample sheaf, by taking for V some $\mathbb{P}^r_Y$.

Then gf is complete-intersection and smoothable, and there is a distinguished triangle

$$\begin{array}{ccc} & L_\bullet^{X/Y} & \\ \swarrow & & \nwarrow \\ f^*L_\bullet^{Y/Z} & \longrightarrow & L_\bullet^{X/Z} \end{array}$$

in the derived category.

Let V' be a smooth Z-scheme factorizing g. Consider the commutative diagram



Then the immersion j' is regular, since it is obtained from j by a flat base change. Hence $j'i$ is a regular immersion factorizing gf, which is therefore smoothable and complete-intersection. Moreover, one has the exact sequence of conormal sheaves

$$0 \longrightarrow i^*(J'/J'^2) \longrightarrow K/K^2 \longrightarrow I/I^2 \longrightarrow 0,$$

where I, J, J', K denote the sheaves defining respectively the immersions $i, j, j', j'i$. Since p'' is flat, $J' = p''^*(J)$, and the sequence can be written

$$0 \longrightarrow f^*(J/J^2) \longrightarrow K/K^2 \longrightarrow I/I^2 \longrightarrow 0. \tag{2.6.1}$$

Furthermore, since V' and V'' are smooth over Z, there is an exact sequence

$$0 \longrightarrow p''^*(\Omega^1_{V'/Z}) \longrightarrow \Omega^1_{V''/Z} \longrightarrow \Omega^1_{V''/V'} \longrightarrow 0.$$

Applying $(j'i)^*$, one obtains an exact sequence

$$0 \longrightarrow (p''ji)^*(\Omega^1_{V'/Z}) \longrightarrow (j'i)^*(\Omega^1_{V''/Z}) \longrightarrow (j'i)^*(\Omega^1_{V''/V'}) \longrightarrow 0.$$

Since

$$j'^*(\Omega^1_{V''/V'}) = \Omega^1_{V/Y}$$

by hypothesis (G), one obtains

$$0 \longrightarrow f^*(j^*\Omega^1_{V'/Z}) \longrightarrow (j'i)^*(\Omega^1_{V''/Z}) \longrightarrow i^*(\Omega^1_{V/Y}) \longrightarrow 0. \tag{2.6.2}$$

Combining (2.6.1) and (2.6.2), one is reduced to checking that the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & f^*(J/J^2) & \longrightarrow & K/K^2 & \longrightarrow & I/I^2 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f^*(d)} & & \downarrow{\scriptstyle d} & & \downarrow{\scriptstyle d} & & \\
0 & \longrightarrow & f^*(j^*\Omega^1_{V'/Z}) & \longrightarrow & (j'i)^*\Omega^1_{V''/Z} & \longrightarrow & i^*\Omega^1_{V/Y} & \longrightarrow & 0
\end{array}$$

is commutative. This is verified locally without difficulty by returning to the definitions of the homomorphisms involved. The announced distinguished triangle follows from the resulting exact sequence of complexes.

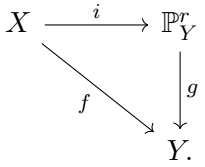
Corollary 2.7. With the notation and hypotheses of 2.6, one has in $K^\bullet(X)$ the equality

$$T_{gf} = T_f + f^*(T_g).$$

3. Riemann–Roch theorem: statement

3.1. In no. 3, Y is a quasi-compact scheme having an ample sheaf.

Let $f : X \to Y$ be a projective complete-intersection morphism (1.1); suppose moreover that the relative virtual dimension of f (1.9) is constant on X. Since f is projective, there is a closed immersion $X \to P$, where P is a projective bundle over Y; since Y has an ample sheaf, one may suppose that $P = \mathbb{P}^r_Y$. Thus there is a factorization



Since f is complete-intersection, the immersion i is regular (1.2).

By 1.7, f is a perfect morphism. Moreover Y has an ample sheaf, and therefore so do X and P. One can therefore define an additive homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(Y)$$

(IV 2.12), the groups $K^\bullet$ defined in terms of locally free sheaves of finite type, or of perfect complexes, being identical. One can also define additive homomorphisms

$$i_* : K^\bullet(X) \longrightarrow K^\bullet(P), \qquad g_* : K^\bullet(P) \longrightarrow K^\bullet(Y),$$

and one has $f_* = g_* i_*$.

Let d be the relative virtual dimension of X over Y. The codimension of i is then $r - d$. By VII 4.6, the homomorphism i_* gives, after tensoring with $\mathbb{Q}$, inclusions

$$i_*\big(\mathrm{Fil}^k K^\bullet(X)_\mathbb{Q}\big) \subset \mathrm{Fil}^{k+r-d} K^\bullet(P)_\mathbb{Q}.$$

By VI 5.8, the homomorphism g_* gives inclusions

$$g_*\big(\mathrm{Fil}^{k'} K^\bullet(P)\big) \subset \mathrm{Fil}^{k'-r} K^\bullet(Y).$$

Composing, one obtains the following.

Proposition 3.2. Let $f : X \to Y$ be a projective complete-intersection morphism whose relative virtual dimension d is constant. Suppose that Y is quasi-compact and has an ample sheaf. Then for every $k \in \mathbb{Z}$,

$$f_*\big(\mathrm{Fil}^k K^\bullet(X)_\mathbb{Q}\big) \subset \mathrm{Fil}^{k-d} K^\bullet(Y)_\mathbb{Q}.$$

Consequently one obtains an additive homomorphism

$$f_*^{\mathrm{gr}} : \mathrm{Gr}\, K^\bullet(X)_\mathbb{Q} \longrightarrow \mathrm{Gr}\, K^\bullet(Y)_\mathbb{Q}$$

sending $\mathrm{Gr}^k K^\bullet(X)_\mathbb{Q}$ into $\mathrm{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q}$.

3.3. Recall that in V 6.1 we defined a functor

$$A \longmapsto \mathrm{Ch}(A)$$

from the category of graded K-algebras to the category of augmented K-λ-algebras, where K is a binomial ring (V 2.7). If A is a graded K-algebra, then

$$\mathrm{Ch}(A) = K \times (1 + \widehat{A}^+).$$

For every augmented K-λ-algebra Λ we defined a K-λ-homomorphism, called the completed Chern class (V 6.7),

$$\widetilde{c} = (\varepsilon, c) : \Lambda \longrightarrow \mathrm{Ch}(\mathrm{Gr}_\Lambda).$$

Finally, we defined a ring homomorphism, functorial in A, called the Chern character (V 6.4),

$$\mathrm{ch} : \mathrm{Ch}(A) \longrightarrow K \oplus \widehat{A}^+_{\mathbb{Q}},$$

and a group endomorphism, called the Todd character (V 1.16),

$$\mathrm{Todd} : 1 + \widehat{A}^+ \longrightarrow 1 + \widehat{A}^+.$$

We shall apply these definitions to the augmented λ-rings $K^\bullet(X)$ and $K^\bullet(Y)$; the augmentation rings $H^0(X, \mathbb{Z})$ and $H^0(Y, \mathbb{Z})$ are binomial (V 2.8). To simplify notation, for every scheme Z and every $z \in K^\bullet(Z)$ we write

$$\mathrm{ch}(z) = \mathrm{ch}(\widetilde{c}(z)), \qquad \mathrm{Todd}(z) = \mathrm{Todd}(c(z)).$$

For every graded algebra A we regard the group $1 + \widehat{A}^+$ as contained in $\widehat{A}$. Finally, if $f : A \to B$ is additive and compatible with the gradings up to a shift, we shall still write f for the homomorphism $\widehat{A} \to \widehat{B}$ deduced from it.

3.4. Let X be a scheme. For every locally free finite-type $\mathcal{O}_X$-module E, denote by $\check{E}$ the class in $K^\bullet(X)$ of the dual $\check{E}$ of E. It is clear that the function $E \mapsto \check{E}$ is additive on the category of locally free finite-type $\mathcal{O}_X$-modules, and therefore defines an endomorphism of $K^\bullet(X)$, denoted $x \mapsto \check{x}$.

Proposition 3.5. Let X be a quasi-compact scheme. Then, for all $x \in K^\bullet(X)$ and all $i \geq 0$, one has

$$c^i(\check{x}) = (-1)^i c^i(x).$$

Equivalently,

$$c_t(\check{x}) = c_{-t}(x), \qquad c_t(x) = \sum_{i \geq 0} c^i(x) t^i.$$

By additivity one may suppose that x is the class of a locally free finite-type $\mathcal{O}_X$-module E. Moreover, after replacing X by the flag scheme of E, which gives an injection on the groups $\mathrm{Gr}\, K^\bullet$ (VI 5.5), one may suppose that E is an invertible sheaf. Then $\check{E} = E^{-1}$, so $\check{x} = x^{-1}$.

The $c^i(\check{x})$ and $c^j(x)$ are zero for $i, j \geq 2$. It is enough to prove

$$c^1(x^{-1}) = -c^1(x).$$

Now $c^1(x)$ is the class in $\mathrm{Gr}^1 K^\bullet(X)$ of $x - 1$, while $c^1(x^{-1})$ is that of $x^{-1} - 1$. The equality

$$x^{-1} - 1 = 1 - x + (x - 1)(1 - x^{-1})$$

gives

$$x^{-1} - 1 = 1 - x \pmod{\mathrm{Fil}^2 K^\bullet(X)},$$

and the result follows.

Theorem 3.6 (Riemann–Roch). Let $f : X \to Y$ be a projective complete-intersection morphism (1.1), whose relative virtual dimension d is constant (1.9). Suppose that Y is quasi-compact and admits an ample sheaf. With the notation of 3.3, for every $x \in K^\bullet(X)$ one has

$$\mathrm{ch}(f_*(x)) = f_*^{\mathrm{gr}}(\mathrm{ch}(x)\, \mathrm{Todd}(\check{T}_f)),$$

where $\check{T}_f$ denotes the dual element of the class of the relative cotangent complex $L_\bullet^{X/Y}$ (2.5 and 2.3).

In other words, the diagram

$$\begin{array}{ccc} K^\bullet(X) & \xrightarrow{\mathrm{ch}\circ\widetilde{c}} & \mathrm{Gr}\,K^\bullet(X)_{\mathbb{Q}} \\ \Big\downarrow{\scriptstyle f_*} & & \Big\downarrow{\scriptstyle x\mapsto f_*^{\mathrm{gr}}(x\,\mathrm{Todd}(\check{T}_f))} \\ K^\bullet(Y) & \xrightarrow[\mathrm{ch}\circ\widetilde{c}]{} & \mathrm{Gr}\,K^\bullet(Y)_{\mathbb{Q}} \end{array}$$

is commutative.

Continuation of the proof of Theorem 3.6

The proof of Theorem 3.6 will occupy the end of this exposé. Let us give some preliminaries.

Proposition 3.7. For arbitrary

$$x \in \mathrm{Gr}\,K^\bullet(X)_{\mathbb{Q}}, \qquad y \in \mathrm{Gr}\,K^\bullet(Y)_{\mathbb{Q}},$$

one has

$$f_*^{\mathrm{gr}}\big(x \cdot f^{\mathrm{gr}}(y)\big) = f_*^{\mathrm{gr}}(x)\, y.$$

By additivity, one may suppose x and y homogeneous; say

$$x \in \mathrm{Gr}^i\,K^\bullet(X)_{\mathbb{Q}}, \qquad y \in \mathrm{Gr}^j\,K^\bullet(Y)_{\mathbb{Q}}.$$

Let

$$x' \in \mathrm{Fil}^i\,K^\bullet(X)_{\mathbb{Q}}, \qquad y' \in \mathrm{Fil}^j\,K^\bullet(Y)_{\mathbb{Q}}$$

be elements lifting x and y. Then the projection formula gives

$$f_*(x' \cdot f^*(y')) = f_*(x')\, y'.$$

By 3.2, this equality is written in $\mathrm{Fil}^{i+j-d}\,K^\bullet(Y)_{\mathbb{Q}}$. One deduces from it an equality in $\mathrm{Gr}^{i+j-d}\,K^\bullet(Y)_{\mathbb{Q}}$, and one verifies immediately that this is the required equality.

Lemma 3.8. Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms satisfying the hypotheses of 3.6. Then $g \circ f$ also satisfies them. If x is an element of $K^\bullet(X)$, and if 3.6 is verified by f for the element x, and by g for the element $f_*(x)$, then it is verified by $g \circ f$ for the element x.

By hypothesis, one has

$$\mathrm{ch}(f_*(x)) = f_*^{\mathrm{gr}}\big(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_f)\big), \tag{3.8.1}$$

and

$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}}\big(\mathrm{ch}(f_*(x))\,\mathrm{Todd}(\check{T}_g)\big). \tag{3.8.2}$$

Combining (3.8.1) and (3.8.2), one obtains

$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}}\Big(f_*^{\mathrm{gr}}\big(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_f)\big) \cdot \mathrm{Todd}(\check{T}_g)\Big).$$

Applying 3.7, one finds

$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}}\Big(f_*^{\mathrm{gr}}\big(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_f) \cdot f^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_g))\big)\Big).$$

The functoriality of c and of Todd implies

$$f^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_g)) = \mathrm{Todd}(f^*(\check{T}_g)).$$

By 2.7, one has

$$\check{T}_{g\circ f} = \check{T}_f + f^*(\check{T}_g).$$

The multiplicativity of the Todd character therefore gives

$$\mathrm{Todd}(\check{T}_f)\, f^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_g)) = \mathrm{Todd}(\check{T}_{g\circ f}).$$

Thus finally

$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}} f_*^{\mathrm{gr}}(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_{g\circ f})),$$

which is the announced result.

3.9. If one applies 3.8 to the factorization of f given in 3.1, one sees that, to prove the Riemann–Roch theorem, it is enough to prove it in the case of a regular immersion of constant codimension, and in the case of the structural morphism

$$\mathbb{P}^r_Y \longrightarrow Y.$$

Moreover, one may suppose that the immersion i satisfies the condition of VII 4.1. Indeed, let s be the section of

$$\mathbb{P}^2_{\mathbb{P}^r_Y}$$

introduced in VII 4.2. Then $s \circ i$ is a regular immersion satisfying the condition of VII 4.1 (VII 4.2), and by 3.8 it is enough to prove 3.6 for

$$s \circ i, \qquad \mathbb{P}^2_{\mathbb{P}^r_Y} \longrightarrow \mathbb{P}^r_Y, \qquad \mathbb{P}^r_Y \longrightarrow Y.$$

We shall therefore prove 3.6 by proving the following two particular cases:

i) f is a regular immersion of constant codimension, satisfying the condition of VII 4.1;

ii) f is the structural morphism $\mathbb{P}^r_Y \to Y$.

4. Riemann–Roch theorem: case of a regular closed immersion

4.1. Let $i : Y \to X$ be a regular closed immersion. The Riemann–Roch theorem for i is essentially relation VII 4.4, which makes it possible to compute the Chern classes of $i_*(x)$ from the $\gamma^p(N, x)$. More precisely, if one introduces the λ-ring $K^\bullet(Y)_{\mathbb{Q}N}$ defined in V 5.5, then the homomorphism

$$K^\bullet(Y)_{\mathbb{Q}N} \longrightarrow K^\bullet(X)_{\mathbb{Q}}$$

deduced from i_* is a λ-homomorphism by VII 2.8 and VII 4.3. The Riemann–Roch theorem is then reduced to an analogous formula concerning the additive homomorphism

$$K^\bullet(Y) \longrightarrow K^\bullet(Y)_{\mathbb{Q}N}$$

defined by $y \mapsto (0, y)$, that is, to a formal relation valid for every λ-ring. We shall not develop this proof here, in order to avoid introducing still more formalism on λ-rings (cf. RRR, I 1.5), and we shall resume the method of [1], which used blow-up schemes. This is the method already employed for the proof of VII 4.3, and also the method at the end of RRR.

As indicated in 3.9, one may restrict to the case of a regular immersion of constant codimension d satisfying the condition of VII 4.1; the case of a general regular immersion is then deduced

from 3.8. In this paragraph we resume the notation of VII 3.1. Thus one has the Cartesian square

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X, \end{array}$$

where X' is the X-scheme obtained by blowing up Y. We suppose moreover that X is quasi-compact and has an ample sheaf.

Proposition 4.2. The morphism f is projective, a complete-intersection morphism, and of virtual relative dimension zero. In addition one has the relation

$$f_*^{\mathrm{gr}} f^{\mathrm{gr}} = \mathrm{Id}_{\mathrm{Gr}\, K^\bullet(X)_{\mathbb{Q}}}.$$

The first assertion follows immediately from VII 1.8. The second relation is a consequence of the relation

$$f_* f^* = \mathrm{Id}_{K^\bullet(X)}$$

proved in VII 3.6.

Proposition 4.3. For every $y \in K^\bullet(Y)_{\mathbb{Q}}$, one has the relation

$$f_*^{\mathrm{gr}} i_*^{\mathrm{gr}}(y) = j_*^{\mathrm{gr}}\big(g^{\mathrm{gr}}(y)\, c^{d-1}(\check{F})\big),$$

where c^{d-1} denotes the $(d-1)$-st Chern class.

This relation is contained in VII 4.8, for it is nothing other than

$$v^{\mathrm{gr}} \circ u^{\mathrm{gr}} = 0.$$

Proposition 4.4. Let X be a quasi-compact scheme. If E is a locally free sheaf of finite type on X, of class E in $K^\bullet(X)$, and of rank n, then

$$\mathrm{ch}(\lambda_{-1}(E)) = c^n(\check{E})\, \mathrm{Todd}(\check{E})^{-1}.$$

The two sides of the relation to be proved are multiplicative. This is clear for $\mathrm{ch} \circ \lambda_{-1}$ and for Todd; it is also true for c^n, which is the class of $(-1)^n E\lambda_{-1}(E)$. Consequently, by passing to the flag bundle of E, one may suppose that E is a sum of classes of invertible sheaves, and reduce to proving the formula for an invertible sheaf. Let ξ be the class of such a sheaf. Then

$$\mathrm{ch}(\lambda_{-1}(\xi)) = \mathrm{ch}(1-\xi) = 1 - \mathrm{ch}(\xi).$$

Since $\widetilde{c}(\xi) = (1, 1 + c^1(\xi))$, the definition of ch gives

$$\mathrm{ch}(\lambda_{-1}(\xi)) = 1 - \exp(c^1(\xi)).$$

Moreover, ξ has augmentation 1; the Chern class which intervenes in formula 4.4 relative to ξ is therefore c^1. Taking into account the definition of Todd, and 3.5, the relation to be proved is written

$$1 - \exp(c^1(\xi)) = -c^1(\xi) \left(\frac{-c^1(\xi)}{1 - \exp(c^1(\xi))} \right)^{-1},$$

and is therefore verified.

Remark 4.4.1. It is also possible to deduce 4.4 from a corresponding purely algebraic statement; cf. RRR, I 1.3.

4.5. We shall now show that, in order to prove the Riemann–Roch theorem in the case considered in this paragraph, one may reduce to the case where Y is a divisor of X, and where the element to which the formula is applied is the inverse image of an element of $K^\bullet(X)$.

By 3.2, the formula to be proved,

$$\mathrm{ch}(i_*(x)) = i_*^{\mathrm{gr}}(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_i)),$$

is equivalent to the formula

$$f^{\mathrm{gr}}(\mathrm{ch}(i_*(x))) = f^{\mathrm{gr}} i_*^{\mathrm{gr}}(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_i)).$$

Taking account of the functoriality of ch, and of 4.3, it may also be written

$$\mathrm{ch}\big(f^*(i_*(x))\big) = j_*^{\mathrm{gr}}\big(\mathrm{ch}(g^*(x))\,\mathrm{Todd}(g^*(\check{T}_i))\,c^{d-1}(\check{F})\big),$$

or, by VII 3.4,

$$\mathrm{ch}\big(j_*(g^*(x)\lambda_{-1}(F))\big) = j_*^{\mathrm{gr}}\big(\mathrm{ch}(g^*(y))\,\mathrm{Todd}(g^*(\check{T}_i))\,c^{d-1}(\check{F})\big).$$

By definition,

$$T_i = -c^1(J/J^2) = -N, \qquad T_j = -c^1(J'/J'^2) = -L.$$

Moreover, the exact sequence

$$0 \longrightarrow F \longrightarrow g^*(N) \longrightarrow \mathcal{O}_{Y'}(1) \longrightarrow 0$$

gives the relation

$$g^*(T_i) = -(F + L),$$

hence

$$\mathrm{Todd}(g^*(\check{T}_i)) = \mathrm{Todd}(\check{T}_j)\,\mathrm{Todd}(\check{F})^{-1}.$$

Taking account of 4.4, the relation to be proved is therefore written

$$\mathrm{ch}\big(j_*(g^*(x)\lambda_{-1}(F))\big) = j_*^{\mathrm{gr}}\big(\mathrm{ch}(g^*(x)\lambda_{-1}(F))\,\mathrm{Todd}(\check{T}_j)\big).$$

Putting

$$z = g^*(x)\lambda_{-1}(F),$$

this relation is nothing but the Riemann–Roch theorem applied to the regular immersion of codimension 1, namely j, and to the element $z \in K^\bullet(Y')$.

Furthermore, since i was supposed to satisfy the hypothesis of VII 4.1, $\lambda_{-1}(F)$ is divisible by $1 - L$. Thus one can find z' such that $z = z'(1 - L)$, or still

$$z = j^*(j_*(z'))$$

(Exp. VII 2.7). We are therefore reduced to the case where Y is a divisor, and to an element x for which there exists y, with $x = i^*(y)$.

4.6. It remains to show the relation

$$\mathrm{ch}\big(i_*(i^*(y))\big) = i_*^{\mathrm{gr}}\big(\mathrm{ch}(i^*(y))\,\mathrm{Todd}(\check{T}_i)\big).$$

The projection formula gives

$$i_*(i^*(y)) = y\, i_*(1),$$

and from the exact sequence

$$0 \longrightarrow J \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0,$$

where J is here invertible, one obtains

$$i_*(1) = 1 - J.$$

Thus

$$\mathrm{ch}(i_*(i^*(y))) = \mathrm{ch}(y)\ \mathrm{ch}(1 - J).$$

Since J is invertible, one has

$$\widetilde{c}(J) = (1, 1 + c^1(J)), \qquad \mathrm{ch}(J) = \exp(c^1(J)).$$

Consequently

$$\mathrm{ch}(i_*(i^*(y))) = \mathrm{ch}(y)(1 - \exp(c^1(J))).$$

On the other hand, the functoriality of ch and the projection formula 3.7 imply

$$i_*^{\mathrm{gr}}\big(\mathrm{ch}(i^*(y))\,\mathrm{Todd}(\check{T}_i)\big) = \mathrm{ch}(y)\, i_*^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_i)).$$

Now

$$T_i = -c^1(J/J^2) = -i^*(J).$$

Hence

$$i_*^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_i)) = i_*^{\mathrm{gr}}\, i^{\mathrm{gr}}(\mathrm{Todd}(-\check{J})) = i_*^{\mathrm{gr}}(1)\,\mathrm{Todd}(-\check{J}).$$

Since $i_*(1) = 1 - J$, $i_*^{\mathrm{gr}}(1)$ is the image in $\mathrm{Gr}^1\, K^\bullet(X)$ of $1 - J$, hence it is $-c^1(J)$. Moreover

$$\mathrm{Todd}(-\check{J}) = \mathrm{Todd}(J)^{-1} = \frac{1 - \exp(c^1(J))}{-c^1(J)}.$$

Thus finally

$$i_*^{\mathrm{gr}}\big(\mathrm{ch}(i^*(y))\,\mathrm{Todd}(\check{T}_i)\big) = \mathrm{ch}(y)(1 - \exp(c^1(J))),$$

which proves the relation to be shown.

5. Riemann–Roch theorem: case of the structural morphism of a projective bundle

5.1. By 3.9, in order to prove the Riemann–Roch theorem, it remains to prove it for the structural morphism

$$f : \mathbb{P}_Y^r \longrightarrow Y,$$

where Y is a quasi-compact scheme admitting an ample sheaf. Put

$$P = \mathbb{P}_Y^r, \qquad \xi = c^1(\mathcal{O}_P(1)), \qquad E = (\mathcal{O}_Y)^{r+1}.$$

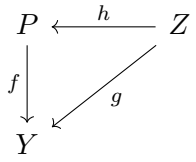
Proposition 5.2. On P one has the exact sequence

$$0 \longrightarrow \Omega^1_{P/Y} \longrightarrow f^*(E)(-1) \longrightarrow \mathcal{O}_P \longrightarrow 0,$$

i.e. one has an isomorphism

$$\Omega^1_{P/Y} \simeq F(-1), \qquad F = \mathrm{Ker}\big(f^*(E) \longrightarrow \mathcal{O}_P(1)\big).$$

Consider the vector bundle over Y defined by E, and denote by Z the complement of the zero section of this bundle. One has the commutative diagram



The morphism h identifies Z with the affine P-scheme defined by the sheaf of algebras

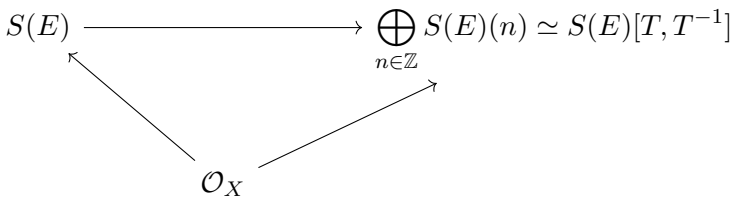
$$B = \bigoplus_{n\in\mathbb{Z}} \mathcal{O}_P(n);$$

B is a graded $\mathcal{O}_P$-algebra.

Since f, g, and h are smooth morphisms, one may write the exact sequence of sheaves of relative differentials. Since Z is affine over P, one may write this sequence in terms of B-modules; thus one obtains

$$0 \longrightarrow \Omega^1_{P/Y} \otimes_{\mathcal{O}_P} B \longrightarrow \Omega^1_{B/Y} \longrightarrow \Omega^1_{B/\mathcal{O}_P} \longrightarrow 0. \tag{5.2.1}$$

This exact sequence is an exact sequence of graded B-modules. Indeed, it follows from the commutative diagram



where the arrows are algebra homomorphisms, compatible with the grading of $S(E)[T, T^{-1}]$ (grading defined by the degree in the indeterminate T), the two other algebras being supposed to be reduced to degree 0.

The module of differentials $\Omega^1_{B/\mathcal{O}_P}$ is isomorphic to

$$B(-1) \otimes_{\mathcal{O}_P} \mathcal{O}_P(1),$$

for one may define an $S(E)$-derivation

$$S(E)[T, T^{-1}] \longrightarrow S(E)[T, T^{-1}] \otimes_{S(E)} S(E)\, dT$$

satisfying the universal property which defines $\Omega^1_{B/\mathcal{O}_P}$, by setting

$$d(aT^n) = naT^{n-1} \otimes dT;$$

moreover, d is compatible with the gradings if the degree 1 is attributed to the factor $S(E)\, dT$.

Let V be the vector bundle defined by E over Y. Then $\Omega^1_{Z/Y}$ is the restriction to Z of $\Omega^1_{V/Y}$. But V is the affine Y-scheme defined by $S(E)$, so $\Omega^1_{V/Y}$ is defined by

$$S(E) \otimes_{\mathcal{O}_Y} E$$

(since E is locally free of finite type), and therefore $\Omega^1_{V/Y}$ is equal to $g^*(E)$. Consequently

$$\Omega^1_{B/Y} = f^*(E) \otimes_{\mathcal{O}_P} B.$$

To see its grading, one looks locally, and one sees at once that $f^*(E)$ must be considered as being of degree 1. Taking the terms of degree 0 in the exact sequence (5.2.1), one obtains the announced exact sequence.

Remark 5.2.1. Using the general results on the differential study of Hilbert functors ([2], 5), one directly obtains an isomorphism

$$\Omega^1_{P/Y} \xrightarrow{\sim} F(-1)$$

from the functorial characterization given of $g^*(\Omega^1_{P/Y})$ for every Y-morphism $g : Y' \to P$.

5.3. By 5.2, one has

$$T_f = c_1(\Omega^1_{P/Y}) = f^*(E)\xi^{-1} - 1.$$

Consequently

$$\widetilde{c}(T_f) = \widetilde{c}(E)\,\widetilde{c}(\xi^{-1}) - \widetilde{c}(1).$$

Since $E = (\mathcal{O}_Y)^{r+1}$, one has $E = r + 1$, and hence

$$\widetilde{c}(E) = (r + 1, 1).$$

Put

$$x = c^1_{\mathrm{Gr}^1 K^\bullet(P)}(\xi - 1) = c^1_{\mathrm{Gr}^1 K^\bullet(P)}(1 - \xi^{-1}) = c^1(\xi).$$

Since $\xi^{-1} = \check{\xi}$, one has

$$\widetilde{c}(\xi^{-1}) = (1, 1 - x).$$

By V (6.2.1), one deduces

$$\widetilde{c}(T_f) = (r, (1 - x)^{r+1}).$$

Taking 3.5 into account, one finally obtains

$$\widetilde{c}(\check{T}_f) = (r, (1 + x)^{r+1}).$$

By the definition of Todd, one therefore has

$$\mathrm{Todd}(\check{T}_f) = \left(\frac{x}{1 - \exp(-x)}\right)^{r+1}. \tag{5.3.1}$$

5.4. Using the basis of $K^\bullet(P)$ over $K^\bullet(Y)$ formed by the elements ξ^k, with $-r \leq k \leq 0$, one sees by additivity that it is enough to prove the theorem for elements of the form

$$f^*(a)\xi^k.$$

Then

$$\mathrm{ch}(f_*(f^*(a)\xi^k)) = \mathrm{ch}(a f_*(\xi^k)) = \mathrm{ch}(a)\ \mathrm{ch}(f_*(\xi^k)).$$

On the other hand,

$$f^{\mathrm{gr}}_*(\mathrm{ch}(f^*(a)\xi^k)\,\mathrm{Todd}(\check{T}_f)) = f^{\mathrm{gr}}_*(f^{\mathrm{gr}}(\mathrm{ch}(a))\,\mathrm{ch}(\xi^k)\,\mathrm{Todd}(\check{T}_f)) = \mathrm{ch}(a)\, f^{\mathrm{gr}}_*(\mathrm{ch}(\xi^k)\,\mathrm{Todd}(\check{T}_f)).$$

Thus it is enough to prove the theorem for the ξ^k, with $-r \leq k \leq 0$. By VI (5.2.3), one has

$$f_*(\xi^k) = 0 \quad \text{for } -r \leq k < 0, \qquad f_*(1) = 1.$$

Consequently

$$\mathrm{ch}(f_*(\xi^k)) = 0 \quad \text{if } -r \leq k < 0, \qquad \mathrm{ch}(f_*(1)) = 1. \tag{5.4.1}$$

We now calculate the second member of the Riemann–Roch formula. For every k, one has

$$\widetilde{c}(\xi^k) = (\widetilde{c}(\xi))^k = (1, 1+x)^k = (1, 1+kx)$$

by V (6.2.1). Hence

$$\mathrm{ch}(\xi^k) = \exp(kx). \tag{5.4.2}$$

The second member of the Riemann–Roch formula is therefore

$$f_*^{\mathrm{gr}}\left(\exp(kx)\,\frac{x^{r+1}}{(1-\exp(-x))^{r+1}}\right). \tag{5.4.3}$$

Since x is the class in $\mathrm{Gr}^1 K^\bullet(P)$ of $1-\xi^{-1}$, $f_*^{\mathrm{gr}}(x^n)$ is, for $0 \leq n \leq r$, the class in $\mathrm{Gr}^{n-r} K^\bullet(Y)$ of $f_*((1-\xi^{-1})^n) = 1$. Thus

$$f_*^{\mathrm{gr}}(x^n) = 0 \quad \text{if } 0 \leq n < r, \qquad f_*^{\mathrm{gr}}(x^r) = 1. \tag{5.4.4}$$

Finally recall that the elements

$$1, x, \ldots, x^{r+1}$$

are linearly dependent over $\mathrm{Gr}\, K^\bullet(Y)$ (VI 5.6), and satisfy the equation

$$x^{r+1} = 0, \tag{5.4.5}$$

and that $1, \ldots, x^r$ are linearly independent over $\mathrm{Gr}\, K^\bullet(Y)$.

One can therefore write the series

$$\varphi_k = \exp(kx)\,\frac{x^{r+1}}{(1-\exp(-x))^{r+1}}$$

in a unique way as a linear combination, with coefficients in $\mathrm{Gr}\, K^\bullet(Y)_{\mathbb{Q}}$, of the elements $1, \ldots, x^r$. The image under f_*^{gr} of φ_k is then, by (5.4.4) and 3.7, its coefficient of x^r. Taking account of (5.4.1), the Riemann–Roch theorem for f is therefore equivalent to

$$\begin{cases} \text{for } -r \leq k < 0, \text{ the coefficient of } x^r \text{ in } \varphi_k \text{ is } 0, \\ \text{for } k = 0, \text{ the coefficient of } x^r \text{ in } \varphi_k \text{ is } 1. \end{cases} \tag{RR}$$

This coefficient is the residue of the differential form

$$\frac{\exp(kx)\,dx}{(1-\exp(-x))^{r+1}}.$$

Putting $y = 1-\exp(-x)$, one is reduced to calculating the residue of

$$\frac{(1-y)^{-k-1}\,dy}{y^{r+1}},$$

that is, the coefficient of y^r in $(1-y)^{-k-1}$. For $-r \leq k < 0$, it is zero, and for $k = 0$, it is 1.

This completes the proof of the Riemann–Roch theorem.

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Exposé IX. Some computations of $K_\bullet$-groups

by P. Berthelot

This exposé corresponds to no oral exposé and will not be used in the sequel of the seminar. It contains a certain number of computations of the groups $K_\bullet$ for various fibrations: vector bundles, projective bundles, torsors under a linear group, and so on. Throughout the exposé, the base scheme will be assumed noetherian, and the notation $K_\bullet(X)$ will denote the Grothendieck group of coherent sheaves on a scheme X. Recall that, if X is noetherian, the homomorphism from the K of the category of coherent $\mathcal{O}_X$-modules to the K of the category of pseudo-coherent complexes on X is an isomorphism (IV 2.4).

The results set out here are the analogues of those one obtains for the Chow ring, and which are to be found in Grothendieck's exposés in the Chevalley seminar of 1958. The question also arises of determining to what extent statements of the same type can be obtained for the graded group associated with $K_\bullet$ for the filtration defined by the dimension of the support (cf. X). It does not seem that the answer is known, even for Proposition 1.1, the homotopy exact sequence, which is one of the properties most used in the computations which follow.

1. Vector bundles

Proposition 1.1 (homotopy exact sequence). Let X be a noetherian scheme, let X' be a closed subscheme of X, let U be the open complement of X' in X, let i be the immersion of X' in X, and let j be the immersion of U in X. Then the sequence

$$K_\bullet(X') \xrightarrow{i_*} K_\bullet(X) \xrightarrow{j^*} K_\bullet(U) \longrightarrow 0$$

is exact.

If F' is a coherent $\mathcal{O}_{X'}$-module, $i_*(F')$ is an $\mathcal{O}_X$-module whose support is contained in X'; consequently its restriction to U is zero, and $j^*i_*(\mathrm{cl}(F')) = 0$. Since the elements of the form $\mathrm{cl}(F')$ generate $K_\bullet(X')$, $j^*i_* = 0$. We therefore obtain a homomorphism

$$\rho : K_\bullet(X)/\operatorname{Im} K_\bullet(X') \longrightarrow K_\bullet(U).$$

We shall define an inverse homomorphism to ρ. Let G be a coherent $\mathcal{O}_U$-module. Since X is noetherian, there exists a coherent $\mathcal{O}_X$-module F whose restriction to U is isomorphic to G. If F' is a second coherent $\mathcal{O}_X$-module whose restriction to U is isomorphic to G, then in $K_\bullet(X)$ one has

$$\mathrm{cl}(F) - \mathrm{cl}(F') \in \operatorname{Im} K_\bullet(X').$$

To see this, consider G as a subsheaf of $j^*(F \oplus F')$ by means of the diagonal map. Then G extends to a subsheaf F'' of $F \oplus F'$, and one is reduced to proving that

$$\mathrm{cl}(F) - \mathrm{cl}(F'') \in \operatorname{Im} K_\bullet(X')$$

and likewise for F'. Let H and K be the kernel and cokernel of the composite homomorphism

$$F'' \longrightarrow F \oplus F' \longrightarrow F.$$

This homomorphism restricts on U to the identity of G, and hence H and K have support contained in X'. If I is the ideal of X' in X, it follows that there exist integers m and n such that $I^m H = 0$ and $I^n K = 0$ (EGA I 5.3.4). Thus one may dévissser H and K into extensions of modules annihilated by I, hence of $\mathcal{O}_{X'}$-modules, and

$$\mathrm{cl}(H) \in \operatorname{Im} K_\bullet(X'), \qquad \mathrm{cl}(K) \in \operatorname{Im} K_\bullet(X'),$$

which proves the assertion. Thus, to every coherent $\mathcal{O}_U$-module, assigning the class of one of its extensions to X defines a function on the category of coherent $\mathcal{O}_U$-modules with values in $K_\bullet(X)/\operatorname{Im} K_\bullet(X')$.

This function is additive. Indeed, let

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$

be an exact sequence of coherent $\mathcal{O}_U$-modules. Let F be an extension of G to X. There exists a submodule F' of F extending G', and F/F' extends G''. Hence the function factors through $K_\bullet(U)$, defining the desired homomorphism $K_\bullet(U) \to K_\bullet(X)/\operatorname{Im} K_\bullet(X')$. One verifies at once that it is inverse to ρ.

Proposition 1.2. Let X and S be noetherian schemes, and let $f : X \to S$ be a flat morphism. Let $(\xi_i)_{i\in I}$ be a family of elements of $K^\circ(X)$, and consider the homomorphism

$$\varphi : K_\bullet(S)^{(I)} \longrightarrow K_\bullet(X)$$

defined by

$$\varphi((x_i)_{i\in I}) = \sum_{i\in I} \xi_i\, f^*(x_i),$$

where f^* is the inverse-image homomorphism from $K_\bullet(S)$ to $K_\bullet(X)$ defined in IV 2.12, and where $K_\bullet(X)$ is considered as a $K^\circ(X)$-module by IV 2.10. Suppose that, for every point $s \in S$, the images of the ξ_i in $K_\bullet(X_s)$ by the homomorphism

$$K^\circ(X) \longrightarrow K^\circ(X_s) = K_\bullet(X_s)$$

generate $K_\bullet(X_s)$ as an abelian group. Then φ is surjective.

Let S' be a closed subscheme of S, and let U be the open complement of S' in S. Write X_U for the restriction of X over U, and $X_{S'}$ for the restriction of X over S'. We again denote by ξ_i the images of the ξ_i in $K^\circ(X_{S'})$ and $K^\circ(X_U)$, and write

$$\varphi_{S'} : K_\bullet(S')^{(I)} \to K_\bullet(X_{S'}), \qquad \varphi_U : K_\bullet(U)^{(I)} \to K_\bullet(X_U)$$

for the homomorphisms defined in the same way as φ.

Consider the diagram

$$\begin{array}{ccccccc} K_\bullet(S')^{(I)} & \xrightarrow{i_*} & K_\bullet(S)^{(I)} & \xrightarrow{j^*} & K_\bullet(U)^{(I)} & \to & 0 \\ \downarrow{\scriptstyle \varphi_{S'}} & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi_U} & & \\ K_\bullet(X_{S'}) & \xrightarrow{u_*} & K_\bullet(X) & \xrightarrow{v^*} & K_\bullet(X_U) & \to & 0, \end{array} \tag{1.2.1}$$

where i and j denote the immersions of S' and U into S, and u and v the immersions of $X_{S'}$ and X_U into X. We show that, if $\varphi_{S'}$ and φ_U are surjective, then so is φ.

The two horizontal rows are exact by 1.1. To prove that φ is surjective, it is therefore enough to prove that the two squares commute. For $x_i \in K_\bullet(S')$, one has

$$\varphi(i_*(x_i)) = \sum_i \xi_i f^* i_*(x_i),$$

whereas

$$u_*\varphi_{S'}((x_i)) = u_*\left(\sum_i \xi_i u^* f^*(x_i)\right) = \sum_i \xi_i u_*(u^* f^*(x_i)),$$

the last equality following from the projection formula IV 2.11.1.2. The equality $\varphi i_* = u_*\varphi_{S'}$ therefore follows from $f^*i_* = u_*u^*f^*$, which is true by IV 3.1.1 since i is proper and f is flat. Similarly, for $(x_i) \in K_\bullet(S)^{(I)}$,

$$\varphi_U(j^*(x_i)) = \sum_i \xi_i f_U^* j^*(x_i),$$

and

$$v^*\varphi((x_i)) = v^*\left(\sum_i \xi_i f^*(x_i)\right) = \sum_i \xi_i v^* f^*(x_i) = \sum_i \xi_i f_U^* j^*(x_i),$$

the middle equality being IV 2.12.2.

This result permits one to prove 1.2 by noetherian induction on the set of closed subschemes of S. If one assumes that, for every closed subscheme S' of S distinct from S, $\varphi_{S'}$ is surjective, the preceding argument shows that one may reduce to proving 1.2 for an irreducible affine base scheme, after possibly replacing S by an irreducible affine open. Moreover, the diagram

$$\begin{array}{ccc} K_\bullet(S_{\mathrm{red}})^{(I)} & \longrightarrow & K_\bullet(S)^{(I)} \\ \downarrow{\scriptstyle \varphi_{S_{\mathrm{red}}}} & & \downarrow{\scriptstyle \varphi} \\ K_\bullet(X_{S_{\mathrm{red}}}) & \longrightarrow & K_\bullet(X) \end{array}$$

where $K_\bullet(X_{S_{\mathrm{red}}}) \to K_\bullet(X)$ is surjective by 1.1, shows that one may also assume S reduced.

We are thus on an integral affine base scheme. Put $S = \mathrm{Spec}(A)$, and let K denote the fraction field of A. If s is the generic point of S, then $\mathcal{O}_{S,s} = K$. Let $y \in K_\bullet(X)$. By hypothesis, $K_\bullet(X_s)$ is generated by the images of the ξ_i; if y_s is the image of y in $K_\bullet(X_s)$, then one can find integers n_i such that

$$y_s = \sum_i \xi_i n_i.$$

On the other hand $K = \varinjlim_f A_f$, where f runs through the elements of A. Hence $\mathrm{Spec}(K) = \varprojlim_f \mathrm{Spec}(A_f)$ and

$$X_s = X \times_S \mathrm{Spec}(K) = \varprojlim_f X \times_S \mathrm{Spec}(A_f)$$

by EGA IV 8.2.3 and 8.2.5. Since the schemes considered are noetherian and the transition morphisms affine and flat, one has, by IV 3.2.3,

$$K_\bullet(X_s) = \varinjlim_f K_\bullet(X \times_S \mathrm{Spec}(A_f)).$$

As inverse-image homomorphisms on the $K_\bullet$ are linear with respect to K° (IV 2.12.2), there exists a non-empty open $U = \mathrm{Spec}(A_f)$ of S such that in $K_\bullet(X_U)$ one has

$$y_U = \sum_i \xi_i n_i.$$

Let S' be a closed subscheme of S with support $S - U$. Then $\varphi_{S'}$ is surjective by the induction hypothesis, and consequently diagram (1.2.1) shows that

$$y - \sum_i \xi_i n_i \in \mathrm{Im}(\varphi).$$

Since $n_i = f^*(n_i)$, φ is surjective.

Corollary 1.3. Let $f : X \to S$ be a flat morphism of noetherian schemes. Suppose that, for every point s of S, the homomorphism

$$K^\circ(X) \longrightarrow K_\bullet(X_s)$$

is surjective. Then the homomorphism deduced from f^*,

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \longrightarrow K_\bullet(X),$$

is surjective.

Take for the family (ξ_i) the family of all elements of $K^\circ(X)$. The hypotheses of 1.2 are then satisfied, and hence

$$\varphi : K_\bullet(S)^{(K^\circ(X))} \longrightarrow K_\bullet(X)$$

is surjective. By definition of φ, this is equivalent to the surjectivity of the displayed tensor-product homomorphism.

Corollary 1.4. Let $f : X \to S$ be a flat morphism of noetherian schemes. Suppose that, for every point s of S, the canonical homomorphism $\mathbb{Z} \to K_\bullet(X_s)$ is an isomorphism. Then

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(X)$$

is surjective. If, moreover, f has a section g of finite Tor-dimension, then f^* is an isomorphism.

Take for the family (ξ_i) the family reduced to the single element $1 \in K^\circ(X)$. Then $\varphi = f^*$, and the conditions of 1.2 being fulfilled, f^* is surjective. If f admits a section g of finite Tor-dimension, one may define a homomorphism

$$g^* : K_\bullet(X) \longrightarrow K_\bullet(S)$$

(IV 2.12). Since $g^*f^* = (fg)^* = \mathrm{Id}$ by IV 2.12.1.1, f^* is injective, hence an isomorphism.

Remark 1.5. For g to be of finite Tor-dimension, it is necessary and sufficient that the fibres of X be regular at the points of $g(S)$, i.e. that f be regular at the points of $g(S)$, or smooth at these points if one assumes f of finite presentation.

These properties being local on X and S, one may suppose them affine: $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$. Then

$$\mathrm{tor.dim}_B(A) = \mathrm{dim.proj}_B(A) = \sup_{s \in S} \mathrm{dim.proj}_{B \otimes_A k(s)}(k(s)),$$

by EGA IV 12.3.2, where the finite-presentation hypothesis on B may be replaced by noetherian hypotheses. But $k(s)$ is isomorphic to the residue field of $\mathcal{O}_{X_s, g(s)}$, and by EGA 0_{IV} 17.2.5 one has

$$\mathrm{dim.proj}_{B \otimes_A k(s)}(k(s)) = \mathrm{dim.proj}_{\mathcal{O}_{X_s, g(s)}}(k(s)).$$

It follows from EGA 0_{IV} 17.2.11 that the latter is finite if and only if $\mathcal{O}_{X_s, g(s)}$ has finite cohomological dimension, i.e. is regular.

Proposition 1.6. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of finite type, let X be the vector bundle associated with E, and let f be the structural morphism of X. Then

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(X)$$

is an isomorphism.

Since f is smooth and admits a section, namely the zero section, it is enough, to prove 1.6, to show that for every $s \in S$ one has $K_\bullet(X_s) \simeq \mathbb{Z}$ (1.4). Thus one is reduced to the case where the

base scheme is a field. In this case E is a direct sum of rank-one submodules, which permits one to decompose f into a sequence of rank-one fibrations, and hence to reduce to the case where X has rank one, namely $X = \mathrm{Spec}(k[T])$. The proposition then follows from the next lemma.

Lemma 1.7. Let A be a principal ideal domain and let $X = \mathrm{Spec}(A)$. Then the canonical homomorphism

$$\mathbb{Z} \longrightarrow K_\bullet(X)$$

is an isomorphism.

Since X is regular, the homomorphism $K^\circ(X) \to K_\bullet(X)$ is an isomorphism. Hence $K_\bullet(X)$ is generated by the classes of locally free $\mathcal{O}_X$-modules of finite type, i.e. by the classes of projective A-modules of finite type. Since A is principal, every projective module of finite type is free, which gives surjectivity. Injectivity follows from the fact that $K^\circ(X)$ is $\mathbb{Z}$-augmented.

Proposition 1.8. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank r, let X be the vector bundle associated with E, let f be the structural morphism of X, and let g be the zero section of X. Then the composite homomorphism

$$K_\bullet(S) \xrightarrow{g_*} K_\bullet(X) \xrightarrow{g^*} K_\bullet(S)$$

where g^* is defined by IV 2.12.1 and 1.5, is multiplication by the element $\lambda_{-1}(E)$ of $K^\circ(S)$, writing E for the class of E in $K^\circ(S)$ and putting

$$\lambda_{-1}(E) = \sum_{i=0}^{r} (-1)^i \lambda^i(E).$$

This statement is the analogue, for $K_\bullet$, of VII 2.7. Let F be a coherent $\mathcal{O}_S$-module, of class F in $K_\bullet(S)$. Then $g_*(F)$ is the class in $K_\bullet(X)$ of the complex $Rg_*(F) = g_*(F)$, hence of F considered as an $\mathcal{O}_X$-module through g. To compute $g^*(g_*(F))$, one must take the class of

$$Lg^*(F) = F \otimes^L_{\mathcal{O}_X} \mathcal{O}_S.$$

For this, consider the Koszul complex

$$0 \longrightarrow \Lambda^r(E) \otimes_{\mathcal{O}_S} S(E) \longrightarrow \Lambda^{r-1}(E) \otimes_{\mathcal{O}_S} S(E) \longrightarrow \cdots \longrightarrow E \otimes_{\mathcal{O}_S} S(E) \longrightarrow S(E) \longrightarrow 0,$$

where $S(E)$ is the symmetric algebra of E. This is a resolution of $\mathcal{O}_S$ over $S(E)$ (VI 1.11). The associated complex on X, denoted $L_\bullet$, has k-th term $\Lambda^k(f^*E)$ and is therefore a resolution of $\mathcal{O}_S$ by locally free $\mathcal{O}_X$-modules of finite type. Thus

$$L_\bullet \otimes_{\mathcal{O}_X} F = \mathcal{O}_S \otimes_{\mathcal{O}_X} F,$$

and hence

$$\begin{aligned} g^*(g_*(F)) = \mathrm{cl}(L_\bullet \otimes_{\mathcal{O}_X} F) &= \sum_{i=0}^{r} (-1)^i \,\mathrm{cl}(L_i \otimes_{\mathcal{O}_X} F) \\ &= \sum_{i=0}^{r} (-1)^i \,\mathrm{cl}(\Lambda^i(E) \otimes_{\mathcal{O}_S} F) = F\lambda_{-1}(E). \end{aligned}$$

The proposition follows by additivity.

2. Principal bundles under split tori

2.1. Let S be a noetherian scheme, and let T be a torus over S. Suppose T is split; thus one has an isomorphism $T \simeq D_S(M)$, where M is a free abelian group of rank r, or equivalently

$T \simeq \mathbb{G}_{m,S}^r$. Let P be a torsor under T. Then P is affine over S, defined by a graded $\mathcal{O}_S$-algebra A of type M. If

$$A = \bigoplus_{m \in M} A_m,$$

then A_m is an invertible $\mathcal{O}_S$-module for every m, and, for $m, m' \in M$, the homomorphism

$$A_m \otimes_{\mathcal{O}_S} A_{m'} \longrightarrow A_{m+m'}$$

defined by the multiplication of A is an isomorphism (SGA 3 VIII 4.1). One therefore defines a multiplicative homomorphism

$$\eta : M \longrightarrow K^\circ(S)$$

by assigning to $m \in M$ the class of A_m in $K^\circ(S)$.

Proposition 2.2. With the notation and hypotheses of 2.1, let $f : P \to S$ be the structural morphism of P. Then the homomorphism

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(P)$$

is surjective, and its kernel is the subgroup of $K_\bullet(S)$ generated by the subgroups $(\eta(m)-1)K_\bullet(S)$ for $m \in M$, or, equivalently, for m running through a system of generators of M.

First suppose $r = 1$, i.e. $T = \mathbb{G}_{m,S}$. Then $M = \mathbb{Z}$, and for every $m \in M$ one has

$$A_m \simeq A_1^{\otimes m}.$$

If $A_1 = L$, then $A = \bigoplus_{n \in \mathbb{Z}} L^{\otimes n}$, which shows that P identifies with the complement of the zero section in the bundle $X = \mathbb{V}(L)$. Considering S as a closed subscheme of X by the zero section, one has, by 1.1, the exact sequence

$$K_\bullet(S) \xrightarrow{g_*} K_\bullet(X) \longrightarrow K_\bullet(P) \longrightarrow 0,$$

where $g : S \to X$ is the zero section of X. Moreover f^* factors as

$$K_\bullet(S) \xrightarrow{f'^*} K_\bullet(X) \longrightarrow K_\bullet(P),$$

where f' is the structural morphism of X. By 1.6, f'^* is an isomorphism, inverse to g^*. Thus we have the exact sequence

$$K_\bullet(S) \xrightarrow{g^* g_*} K_\bullet(S) \xrightarrow{f^*} K_\bullet(P) \longrightarrow 0,$$

and by 1.8 one obtains

$$K_\bullet(P) \simeq K_\bullet(S)/\lambda_{-1}(L)K_\bullet(S) = K_\bullet(S)/(1-L)K_\bullet(S),$$

where L is the class of L in $K^\circ(S)$. Since, for every $n \in \mathbb{Z}$, $L^n - 1$ is a multiple of $1 - L$, this proves 2.2 when P has rank one.

Continue the proof by induction on the rank r of P. Suppose the result proved for torsors under split tori of rank $< r$. Write $M = M_1 \oplus M_2$, with $M_1 \simeq \mathbb{Z}^{r-1}$ and $M_2 \simeq \mathbb{Z}$. Put $A' = \bigoplus_{m \in M_1} A_m$ and $A'' = \bigoplus_{m \in M_2} A_m$, so that $A = A' \otimes_{\mathcal{O}_S} A''$. Let $P_1 = \mathrm{Spec}(A')$. Then P_1 is a torsor under $D_S(M_1)$, and P is, over P_1, a torsor under the group $\mathbb{G}_{m,P_1}$ with algebra the inverse image of A'' on P_1. By the preceding rank-one case, $K_\bullet(P)$ is the quotient of $K_\bullet(P_1)$ by the subgroup generated by the $(\eta(m_2) - 1)K_\bullet(P_1)$ for $m_2 \in M_2$. By the induction hypothesis, $K_\bullet(P_1)$ is the quotient of $K_\bullet(S)$ by the subgroup generated by the $(\eta(m_1) - 1)K_\bullet(S)$ for $m_1 \in M_1$. Taking account of the relation

$$\eta(m_1 + m_2) - 1 = \eta(m_1)\eta(m_2) - 1 = \eta(m_1)(\eta(m_2) - 1) + \eta(m_1) - 1,$$

we conclude that $K_\bullet(P)$ is the quotient of $K_\bullet(S)$ by the subgroup generated by the $(\eta(m) - 1)K_\bullet(S)$ for $m \in M$, or equivalently for m running through a system of generators of M.

Corollary 2.3. With the notation and hypotheses of 2.1, suppose S regular. Then one has a canonical isomorphism

$$K^\circ(P) \simeq K^\circ(S)/((\eta(m) - 1)_{m\in M}),$$

where $((\eta(m) - 1)_{m\in M})$ is the ideal of $K^\circ(S)$ generated by the $\eta(m) - 1$, for m running through a system of generators of M.

If S is regular, the canonical homomorphism $K^\circ(S) \to K_\bullet(S)$ is an isomorphism (IV 2.5). Likewise, P is then regular, and $K^\circ(P) \to K_\bullet(P)$ is an isomorphism. The corollary therefore follows immediately from 2.2.

3. Projective bundles and flag bundles

Proposition 3.1. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, let $\pi = (p_1, \ldots, p_k)$ be a sequence of integers such that $\sum p_i = n$, and let $X = D_\pi(E)$ be the scheme of flags of type π in E. Then the canonical homomorphism

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \longrightarrow K_\bullet(X)$$

is an isomorphism.

We resume the notation of VI 4.5. Thus L_i denotes the canonical quotients of the inverse image of E on X, F_j denotes the kernel of $L_j \to L_{j-1}$, and $\lambda_i^{(j)}$ denotes the class in $K^\circ(X)$ of $\Lambda^i(F_j)$. Then the $\lambda_i^{(j)}$ generate $K^\circ(X)$ as a $K^\circ(S)$-algebra (VI 4.6). Likewise, the images of the $\lambda_i^{(j)}$ in $K^\circ(X_s)$ generate $K^\circ(X_s)$ as an algebra over $K^\circ(k(s)) = \mathbb{Z}$ for every point s of S. Since X_s is a flag scheme over a field, it is regular, and hence $K^\circ(X_s) \to K_\bullet(X_s)$ is an isomorphism. It follows that $K^\circ(X) \to K_\bullet(X_s)$ is surjective. By 1.3, this implies that

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \longrightarrow K_\bullet(X)$$

with $K^\circ(X)$ in the second factor is surjective.

To prove injectivity, one may first reduce to the case of the complete flag bundle $D(E)$. Indeed, suppose that

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(D(E)) \longrightarrow K_\bullet(D(E))$$

is injective. Since $K_\bullet(S) \to K_\bullet(D(E))$ factors through $K_\bullet(X)$, the corresponding homomorphism factors as

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \otimes_{K^\circ(X)} K^\circ(D(E)) \longrightarrow K_\bullet(X) \otimes_{K^\circ(X)} K^\circ(D(E)) \longrightarrow K_\bullet(D(E)).$$

The first homomorphism is injective. Since, by VI 4.3, 4.6 and 4.7, $K^\circ(D(E))$ is free over $K^\circ(X)$, this implies injectivity for

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \to K_\bullet(X).$$

Thus assume $X = D(E)$. One proceeds by induction on the rank of E. The case $n = 1$ is trivial. Suppose the property proved for $n - 1$. Introduce $P = \mathbb{P}(E)$, and let F be the kernel of the canonical homomorphism $p^*(E) \to \mathcal{O}_P(1)$, where p is the structural morphism of P. Then X identifies with the scheme of flags of F over P, and by the induction hypothesis it is enough to prove 3.1 for P. Taking VI 1.1 into account, the result to prove may therefore be stated as the following corollary.

Corollary 3.2. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, let $X = \mathbb{P}(E)$ be the associated projective bundle, let f be the structural morphism of X, and let ξ be the class in $K^\circ(X)$ of the sheaf $\mathcal{O}_X(1)$. Then the homomorphism

$$\varphi : (K_\bullet(S))^n \longrightarrow K_\bullet(X)$$

defined by

$$\varphi((x_i)) = \sum_{i=0}^{n-1} \xi^{-i} f^*(x_i)$$

is an isomorphism.

We have already seen that this homomorphism is surjective, since the elements ξ^i, $i = 0, \ldots, -n+1$, form a basis of $K^\circ(X)$ over $K^\circ(S)$. To see that it is injective, use the homomorphism f_*, defined because f is projective. By IV 2.12.4,

$$f_*\left(\sum_{i=0}^{n-1} \xi^{-i} f^*(x_i)\right) = \sum_{i=0}^{n-1} x_i f_*(\xi^{-i}) = x_0,$$

the last equality following from the fact that $f_*(\xi^i) = 0$ for $-n+1 \leq i < 0$ (VI 2.8). If $\varphi((x_i)) = 0$, what precedes shows that $x_0 = 0$. Applying the same reasoning to the elements $\xi^i \varphi((x_i))$, for $i = 1, \ldots, n-1$, one obtains by induction that all the x_i are zero. Thus φ is injective.

3.3. Let S still be a noetherian scheme, and let E be a locally free $\mathcal{O}_S$-module of rank n. Write $V = \mathbb{V}(E)$ for the vector bundle associated with E, contravariantly in E (SGA 3 I 4.6), considered as a group scheme over S. Let P be a torsor under V; we wish to compute $K_\bullet(P)$.

Recall how one embeds P in a projective bundle over S. Torsors under V correspond to extensions F of E by $\mathcal{O}_S$. Indeed, if one has an extension

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{s} F \xrightarrow{u} E \longrightarrow 0, \tag{3.3.1}$$

one obtains a torsor under V by considering the homomorphism $\mathbb{V}(F) \to \mathbb{V}(\mathcal{O}_S)$ defined by s, the section of $\mathbb{V}(\mathcal{O}_S) = \mathrm{Spec}(\mathcal{O}_S[T])$ defined by $T = 1$, and taking

$$P = \mathbb{V}(F) \times_{\mathbb{V}(\mathcal{O}_S)} S,$$

the operations of V on P being induced by those of V on $\mathbb{V}(F)$.

Conversely, let P be a torsor under V. Denote by $\underline{P}$ the sheaf of germs of homomorphisms from S to P. It is a principal homogeneous sheaf under the sheaf of germs of homomorphisms from S to V, which is identified with the dual $\check{E}$ of E. In the classical way this yields an extension of $\mathcal{O}_S$ by $\check{E}$, and by passing to duals one obtains (3.3.1).

Thus let P be a torsor under V, corresponding to an extension F of E by $\mathcal{O}_S$. The algebra $A(P)$ of P over $\mathcal{O}_S$ is

$$S(F) \otimes_{S(\mathcal{O}_S)} \mathcal{O}_S,$$

that is,

$$S(F)/(s-1)S(F),$$

which identifies with the ring of homogeneous fractions of degree 0, $S(F)_{(s)}$. Consequently P identifies with the open $D_+(s)$ of $\mathbb{P}(F)$. The complement of P in $\mathbb{P}(F)$ is then $\mathrm{Proj}(S(F)/sS(F))$, which is identified, by (3.3.1), with $\mathbb{P}(E)$. Thus one has immersions

$$\begin{array}{ccccc} \mathbb{P}(E) & \hookrightarrow & \mathbb{P}(F) & \hookleftarrow & P \\ & \searrow & & \swarrow & \\ & & S & & \end{array}$$

where $\mathbb{P}(E)$ is closed in $\mathbb{P}(F)$, and P is open and complementary to $\mathbb{P}(E)$.

Proposition 3.4. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, let $V = \mathbb{V}(E)$ be the associated vector bundle, let P be a torsor under V, and let f be the structural morphism of P. Then the canonical homomorphism

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(P)$$

is an isomorphism.

With the identifications of 3.3, one has the exact sequence

$$K_\bullet(\mathbb{P}(E)) \xrightarrow{i_*} K_\bullet(\mathbb{P}(F)) \longrightarrow K_\bullet(P) \longrightarrow 0, \tag{3.4.1}$$

where i denotes the closed immersion $\mathbb{P}(E) \to \mathbb{P}(F)$ defined by u. By 3.2,

$$K_\bullet(\mathbb{P}(E)) = K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(\mathbb{P}(E)), \qquad K_\bullet(\mathbb{P}(F)) = K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(\mathbb{P}(F)).$$

Since $\mathbb{P}(E)$ and $\mathbb{P}(F)$ are smooth, i is a regular immersion, and hence we have a homomorphism

$$i_* : K^\circ(\mathbb{P}(E)) \longrightarrow K^\circ(\mathbb{P}(F)).$$

The homomorphism between the $K_\bullet$ groups is then obtained by tensoring the homomorphism between the K° groups by $K_\bullet(S)$. It is enough, to see this, to show that for $a \in K_\bullet(S)$ and $b \in K^\circ(\mathbb{P}(E))$ one has

$$i_*(b\, h^*(a)) = i_*(b) g^*(a),$$

where g and h are the structural morphisms of $\mathbb{P}(F)$ and $\mathbb{P}(E)$. Since $h = gi$,

$$i_*(b\, h^*(a)) = i_*(b\, i^* g^*(a)) = g^*(a) i_*(b)$$

by IV 2.12.4.

From (3.4.1) one therefore gets

$$K_\bullet(P) \simeq K_\bullet(S) \otimes_{K^\circ(S)} \left(K^\circ(\mathbb{P}(F))/\operatorname{Im} K^\circ(\mathbb{P}(E))\right).$$

Let ξ_F and ξ_E be the classes of the canonical invertible sheaves on $\mathbb{P}(F)$ and $\mathbb{P}(E)$. By VI 1.1, $K^\circ(\mathbb{P}(F))$ (respectively $K^\circ(\mathbb{P}(E))$) has as basis over $K^\circ(S)$ the elements ξ_F^k for $k = 0, \ldots, -n$ (respectively ξ_E^k for $k = 0, \ldots, -n+1$). The projection formula gives

$$i_*(\xi_E^k) = i_*(i^*(\xi_F^k)) = \xi_F^k i_*(1).$$

If I is the ideal of $\mathbb{P}(E)$ in $\mathbb{P}(F)$, then I is invertible and $i_*(1) = 1 - I$. Moreover I is defined by the ideal J of $S(F)$, kernel of $S(F) \to S(E)$, which is generated by a submodule of F isomorphic to $\mathcal{O}_S$. Hence

$$I \simeq \mathcal{O}_{\mathbb{P}(F)}(-1).$$

The image of $K^\circ(\mathbb{P}(E))$ is thus the $K^\circ(S)$-submodule generated by the $\xi_F^k - \xi_F^{k-1}$, for $k = 0, \ldots, -n+1$. Therefore

$$K^\circ(S) \longrightarrow K^\circ(\mathbb{P}(F))/\operatorname{Im} K^\circ(\mathbb{P}(E))$$

is an isomorphism, and consequently so is $K_\bullet(S) \to K_\bullet(P)$.

4. Principal bundles under the groups $\mathrm{GL}(n)_S$

Proposition 4.1. Let S be a noetherian scheme, let $G = \mathrm{GL}(n)_S$, let P be a torsor under G, and let E be the locally free $\mathcal{O}_S$-module of rank n corresponding to P. Denote by E the class of E in $K^\circ(S)$, and let J be the λ-ideal generated by $E - n$. If f is the structural morphism of P, then f^*, defined because f is flat, is surjective and defines an isomorphism

$$K_\bullet(S)/JK_\bullet(S) \xrightarrow{\sim} K_\bullet(P).$$

Let $E_0 = (\mathcal{O}_S)^n$, with evident basis $e_1, \ldots, e_n$, and let $E_0^{(i)}$ be the submodule of E_0 generated by $e_1, \ldots, e_i$. The subfunctor of G which to every S-scheme S' assigns the subgroup of automorphisms of $E_{0,S'}$ leaving invariant the submodules $E_{0,S'}^{(i)}$ for all i is represented by a group subscheme B of G, a Borel subgroup of G. The quotient sheaf G/B for the fpqc topology is represented by the flag scheme of E_0, denoted X_0. Let $L_0^{(i)}$ be the canonical quotients of the inverse image of E_0 on X_0, let $G_0^{(i)}$ be the kernel of $E_{0,X_0} \to L_0^{(i)}$, and put

$$F_0^{(i)} = G_0^{(i-1)}/G_0^{(i)} \simeq \ker(L_0^{(i)} \to L_0^{(i-1)}).$$

Moreover, B is the semi-direct product of its unipotent radical U and of its maximal subtorus T leaving the e_i invariant. Let $Y_0 = G/U$. Then Y_0 is a torsor over X_0 under the group T_{X_0}. If S' is an S-scheme, the datum of an S'-point of Y_0 is equivalent to the datum, on some faithfully flat quasi-compact S'' over S', of an automorphism of $E_{0,S''}$ modulo the unipotents of $U_{S''}$; that is, of a flag of $E_{0,S''}$ with free submodules, and, for each i, of a class of free elements f_i^α on $\mathcal{O}_{S''}$ such that $f_i^\alpha - f_i^\beta \in G_{0,S''}^{(i+1)}$. Equivalently, it is the datum of a flag of $E_{0,S''}$ and, for every i, of a basis of $F_{0,S''}^{(i)}$, this datum being subject to the gluing condition on $S'' \times_{S'} S''$. By fpqc descent, this amounts to the same datum on S'. Thus Y_0 is the affine X_0-scheme defined by the algebra

$$A_0 = \bigoplus_{k_i \in \mathbb{Z}} (F_0^{(1)})^{\otimes k_1} \otimes \cdots \otimes (F_0^{(n)})^{\otimes k_n}.$$

The group G operates on E_0, on X_0 provided with the sheaves $L_0^{(i)}$, and on Y_0. Taking the product of these objects with P modulo G, one obtains the sheaf E associated with P, the scheme $X = D(E) = P/B$ of flags of E, and the scheme $Y = P/U$. We denote the canonical sheaves on X by the same letters as their analogues on X_0. Thus one has the diagram

$$\begin{array}{ccccc} & & Y = P/U & & \\ & \nearrow & & \searrow & \\ P & & & & X = P/B = D(E) \\ & \searrow & & \swarrow & \\ & & S & & \end{array}$$

We note that P is a torsor over Y with structural group U_Y, and that Y is a torsor over X with structural group T_X, defined by the $\mathcal{O}_X$-algebra

$$A = \bigoplus_{k_i \in \mathbb{Z}} (F^{(1)})^{\otimes k_1} \otimes \cdots \otimes (F^{(n)})^{\otimes k_n}.$$

The homomorphism $K_\bullet(S) \to K_\bullet(P)$ therefore factors as

$$K_\bullet(S) \longrightarrow K_\bullet(X) \longrightarrow K_\bullet(Y) \longrightarrow K_\bullet(P).$$

By 3.1, $K_\bullet(X) = K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X)$. Let ξ_i be the class of $F^{(i)}$ in $K^\circ(X)$. By 2.2, $K_\bullet(Y)$ is the quotient of $K_\bullet(X)$ by the submodule generated by the $(\xi_i - 1)K_\bullet(X)$. If I is the ideal of $K^\circ(X)$ generated by the $\xi_i - 1$, then

$$K_\bullet(Y) = K_\bullet(S) \otimes_{K^\circ(S)} (K^\circ(X)/I).$$

By VI 4.8, $K^\circ(X)$ is the $K^\circ(S)$-algebra generated by the $\xi_i - 1$, subject to the relations deduced from

$$\gamma_t(E-n) = \prod_{i=1}^{n} \gamma_t(\xi_i - 1).$$

Therefore $K^\circ(X)/I = K^\circ(S)/J$, where J is the ideal generated by the $\gamma^j(E-n)$ for $j = 1, \ldots, n$, hence the λ-ideal generated by $E - n$. Thus

$$K_\bullet(Y) \simeq K_\bullet(S)/JK_\bullet(S).$$

The demonstration is completed by the following lemma.

Lemma 4.2. Let S be a noetherian scheme, let B be a parabolic subgroup of a reductive S-group, and let P be a torsor under the unipotent radical U of B. Then one has an isomorphism

$$K_\bullet(S) \xrightarrow{\sim} K_\bullet(P).$$

Indeed, there exists a finite sequence of subgroup schemes of U,

$$U_0 = U \supset U_1 \supset U_2 \supset \cdots \supset U_p = e,$$

closed in B, finite because B is noetherian, and such that, for every i, U_i/U_{i+1} is isomorphic to an S-vector group (SGA 3 XXVI 2.1). Consequently $P \to S$ factors as a sequence of torsors under vector groups, and the result follows from 3.4.

Corollary 4.3. Let S be a noetherian scheme and let $G = \mathrm{GL}(n)_S$. Then the canonical homomorphism

$$K_\bullet(S) \longrightarrow K_\bullet(G)$$

is an isomorphism.

This follows immediately from 4.1, applied to $P = G$, since then $E = \mathcal{O}_S^n$.

Corollary 4.4. Under the hypotheses of 4.1, suppose S regular. Then one has an isomorphism

$$K^\circ(S)/J \xrightarrow{\sim} K^\circ(P).$$

If S is regular, G is also regular, and P as well. Thus $K^\circ(S) \simeq K_\bullet(S)$ and $K^\circ(P) \simeq K_\bullet(P)$, whence the result follows from 4.1.

Exposé X. Formalism of Intersections on Proper Algebraic Schemes

by O. Jussila
(with an appendix by A. Grothendieck)

1. Compatibility of the filtrations with the composition law

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X).$$

1.1

First let X be a noetherian scheme, and let $K_\bullet(X)$ be the group of classes of coherent $\mathcal{O}_X$-modules. For every coherent $\mathcal{O}_X$-module F, we denote its class in $K_\bullet(X)$ by $\mathrm{cl}_\bullet(F)$.

Definition 1.1.1. For every integer $n \in \mathbb{Z}$, let $\mathrm{Filt}_n(X)$ denote the subgroup of $K_\bullet(X)$ generated by the classes $\mathrm{cl}_\bullet(F)$ of coherent $\mathcal{O}_X$-modules F such that

$$\dim(\mathrm{Supp}(F)) \leq n.$$

This defines an increasing filtration $(\mathrm{Filt}_n(X))_{n\in\mathbb{Z}}$ of $K_\bullet(X)$ such that $\mathrm{Filt}_n(X) = 0$ for $n < 0$.

Proposition 1.1.2. Let X be a noetherian scheme, let F be a coherent $\mathcal{O}_X$-module such that $\dim(\mathrm{Supp}(F)) \leq n$, and, for $n \in \mathbb{Z}$, let $X(n)$ be the subset

$$\{x \in X\,;\, \dim(\overline{\{x\}}) = n\}$$

of X. For every $x \in X$, let $Z(x)$ denote the integral closed subscheme associated with $\overline{\{x\}}$. Then the length of the $\mathcal{O}_{X,x}$-module F_x is finite for every $x \in X(n)$ and is nonzero for only finitely many $x \in X(n)$, and one has

$$\mathrm{cl}_\bullet(F) = \sum_{x\in X(n)} \mathrm{length}(F_x)\,\mathrm{cl}_\bullet(\mathcal{O}_{Z(x)}) + w, \tag{1.1.3}$$

where $w \in \mathrm{Filt}_{n-1}(X)$.

By immediate induction on n, one obtains:

Corollary 1.1.4. For every noetherian scheme X and every integer n, $\mathrm{Filt}_n(X)$ is the subgroup of $K_\bullet(X)$ generated by the classes $\mathrm{cl}_\bullet(\mathcal{O}_Z)$, where Z ranges over the integral closed subschemes of X of dimension at most n.

Proof of 1.1.2. Let $X' = \mathrm{Supp}(F)$. Since X' is a noetherian space of dimension at most n, the elements of $X' \cap X(n)$ are maximal in X' and hence finite in number; moreover, $\mathrm{length}(F_x)$ is finite for every $x \in X' \cap X(n)$. This proves the first assertion.

Let $\mathcal{K} = \mathrm{Coh}(X)$ be the category of coherent $\mathcal{O}_X$-modules, and let $\mathcal{K}'$ be the subset of $\mathrm{Ob}(\mathcal{K})$ formed by the coherent $\mathcal{O}_X$-modules concentrated on X' that satisfy condition (1.1.3). Plainly $\mathcal{K}'$ contains the sheaves $\mathcal{O}_{Z(x)}$ for $x \in X'$. It is therefore enough to show that $\mathcal{K}'$ is an "exact" subset of $\mathrm{Ob}(\mathcal{K})$, for then the devissage lemma (EGA III 3.1.2) gives $F \in \mathcal{K}'$.

Thus let

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \tag{1.1.5}$$

be an exact sequence of $\mathcal{O}_X$-modules having two of its terms in $\mathcal{K}'$. The sheaves G', G, and G'' are then all concentrated on X'. Hence the lengths ℓ'_x, ℓ_x, and ℓ''_x of the $\mathcal{O}_{X,x}$-modules G'_x, G_x, and G''_x are finite for every $x \in X(n)$, and

$$\ell_x = \ell'_x + \ell''_x \qquad (x \in X(n)). \tag{1.1.6}$$

Assertion (1.1.3) for all the terms of (1.1.5) now follows immediately from the hypothesis and the equation

$$\mathrm{cl}_\bullet(G) = \mathrm{cl}_\bullet(G') + \mathrm{cl}_\bullet(G''). \tag{1.1.7}$$

It is known that $K_\bullet$ is a covariant functor on the category of locally noetherian schemes with respect to proper morphisms. The following proposition asserts that our filtration is functorial.

Proposition 1.1.8. Let $f : X' \to X$ be a proper morphism of locally noetherian schemes, and let

$$f_* : K_\bullet(X') \longrightarrow K_\bullet(X)$$

be the morphism induced by f. Then

$$f_*(\mathrm{Filt}_n(X')) \subset \mathrm{Filt}_n(X) \tag{1.1.9}$$

for every integer n.

Proof. If F is a coherent $\mathcal{O}_{X'}$-module, then by definition (IV 2.11),

$$f_*(\mathrm{cl}_\bullet(F)) = \sum_{i\in\mathbb{N}} (-1)^i \,\mathrm{cl}_\bullet(R^i f_*(F)).$$

The $\mathcal{O}_X$-modules $R^i f_*(F)$ are concentrated on $f(\mathrm{Supp}(F))$, and by EGA IV 5.4.1(ii),

$$\dim(\mathrm{Supp}(F)) \geq \dim(f(\mathrm{Supp}(F))),$$

which proves the assertion.

1.2

Let now X be an arbitrary scheme, and consider the group $K^\circ(X)$ of classes of locally free $\mathcal{O}_X$-modules (henceforth understood to be of finite type). It is an augmented pre-λ-algebra over the ring $H^0(X,\mathbb{Z})$ (V 3.9.1). The map $f \rightsquigarrow f \cdot 1 \in K^\circ(X)$ allows us henceforth to identify $H^0(X,\mathbb{Z})$ with a subring of $K^\circ(X)$. If E is a locally free $\mathcal{O}_X$-module, we denote its class in $K^\circ(X)$ by $\mathrm{cl}^\circ(E)$. The operators λ^k $(k \in \mathbb{N})$ are defined by

$$\lambda^k(\mathrm{cl}^\circ(E)) = \mathrm{cl}^\circ(\wedge^k_{\mathcal{O}} E) \tag{1.2.1}$$

and by

$$\lambda^k(x+y) = \sum_{\substack{i+j=k \\ i,j\in\mathbb{N}}} \lambda^i(x)\lambda^j(y). \tag{1.2.2}$$

(V 2.2). Recall that in the λ-filtration $(\mathrm{Filt}^i(X))_{i\in\mathbb{Z}}$ of $K^\circ(X)$, one has $\mathrm{Filt}^i(X) = K^\circ(X)$ for $i \leq 0$, while for $i \geq 1$, $\mathrm{Filt}^i(X)$ is the ideal generated by the products

$$P = \gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k) \tag{1.2.3}$$

with $1 \leq i_1, \ldots, i_k$; $i_1 + \cdots + i_k \geq i$; and $x_1, \ldots, x_k \in I = \mathrm{Ker}(\varepsilon)$, where ε is the augmentation homomorphism (cf. Exposé V 3.10). It was proved in V 3.10 that $\mathrm{Filt}^i(X)$ is even the $H^0(X,\mathbb{Z})$-submodule, hence, when X is connected, the abelian subgroup generated by the products (1.2.3), where one may moreover suppose that the elements x_p $(p = 1, \ldots, k)$ have the form

$$x_p = \mathrm{cl}^\circ(E_p) - n_p, \tag{1.2.4}$$

with E_p a locally free $\mathcal{O}_X$-module of rank $n_p \in H^0(X,\mathbb{Z})$.

1.3

Recall (IV 2.10) that the bifunctor

$$(E, F) \rightsquigarrow E \otimes_{\mathcal{O}_X} F,$$

where E is a locally free $\mathcal{O}_X$-module and F is a coherent $\mathcal{O}_X$-module, induces a composition law

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X) \tag{1.3.1}$$

that makes $K_\bullet(X)$ a unital $K^\circ(X)$-module. The key result of the present exposé is the following.

Theorem 1.3.2. Let X be a noetherian scheme. For all $i, j \in \mathbb{Z}$, the composition law (1.3.1) induces by restriction a composition law

$$\mathrm{Filt}^i(X) \times \mathrm{Filt}_j(X) \longrightarrow \mathrm{Filt}_{j-i}(X). \tag{1.3.3}$$

Corollary 1.3.3. If X is a noetherian scheme of dimension at most j, then the canonical homomorphism

$$\Theta_X : K^\circ(X) \longrightarrow K_\bullet(X),$$

defined by $\Theta_X(x) = x \cdot \mathrm{cl}_\bullet(\mathcal{O}_X)$ $(x \in K^\circ(X))$, induces for every $i \in \mathbb{Z}$ a homomorphism

$$\mathrm{Filt}^i(X) \longrightarrow \mathrm{Filt}_{j-i}(X).$$

Proof. First let us show that (1.3.3), for $i + j \leq k \in \mathbb{Z}$ and every integral closed subscheme of dimension at most j, implies Theorem 1.3.2 for $i + j \leq k$. By (1.1.4), it is enough to show that, for every $x \in \mathrm{Filt}^i(X)$ and every integral closed subscheme Z of dimension at most j, one has

$$x \,\mathrm{cl}_\bullet(\mathcal{O}_Z) \in \mathrm{Filt}_{j-i}(X).$$

To avoid confusion, let 1_Z denote the class of $\mathcal{O}_Z$ in $K_\bullet(Z)$. If $\alpha : Z \to X$ is the canonical immersion, the projection formula (IV ...) gives

$$x \,\mathrm{cl}_\bullet(\mathcal{O}_Z) = x\alpha_*(1_Z) = \alpha_*(\alpha^*(x)1_Z).$$

Since $\alpha^*(x) \in \mathrm{Filt}^i(Z)$, (1.1.8) and (1.3.3), applied with $X = Z$ and $i + j \leq k$, give

$$\alpha_*(\alpha^*(x)1_Z) \in \mathrm{Filt}_{j-i}(X),$$

which proves the assertion.

Theorem 1.3.2 is trivial for $i + j \leq 0$. Suppose by induction that it has been proved for $i + j < k$ and for every noetherian scheme X. By the preceding remark, it remains to prove that, for every integral noetherian scheme X of dimension j and every product (1.2.3) satisfying (1.2.4), one has

$$P \,\mathrm{cl}_\bullet(\mathcal{O}_X) \in \mathrm{Filt}_{j-i}(X). \tag{1.3.4}$$

If $k > 1$ in (1.2.3), the assertion follows from the induction hypothesis and the associativity of the law (1.3.1). We may therefore suppose, since X is connected, that

$$x = P = \gamma^m(\mathrm{cl}^\circ(E) - n), \tag{1.3.5}$$

where $m \geq i$ and E is a locally free $\mathcal{O}_X$-module of rank $n > 0$. We distinguish two cases.

I. Suppose that there exist invertible $\mathcal{O}_X$-modules $L_1, \ldots, L_n$ such that

$$\mathrm{cl}^\circ(E) = \mathrm{cl}^\circ(L_1) + \cdots + \mathrm{cl}^\circ(L_n) \in K^\circ(X).$$

It follows from (V) that x is the m-th elementary symmetric function in the $\mathrm{cl}^\circ(L_r) - 1$ $(r = 1, \ldots, n)$, and hence is a homogeneous polynomial of degree m with integral coefficients in the variables $(\mathrm{cl}^\circ(L_r) - 1)$ $(r = 1, \ldots, n)$. Applying again the induction hypothesis and the associativity of the law (1.3.1), we are reduced to proving that, for every invertible $\mathcal{O}_X$-module L, one has

$$(\mathrm{cl}^\circ(L) - 1)\,\mathrm{cl}_\bullet(\mathcal{O}_X) \in \mathrm{Filt}_{j-1}(X). \tag{1.3.6}$$

Since X is integral, L and $\mathcal{O}_X$ are isomorphic to submodules of the constant $\mathcal{O}_X$-module R_X of rational functions on X. The $\mathcal{O}_X$-module sum $F = \mathcal{O}_X + L$ is coherent, and one obtains

$$\begin{aligned}(\mathrm{cl}^\circ(L) - 1)\,\mathrm{cl}_\bullet(\mathcal{O}_X) &= \mathrm{cl}_\bullet(L) - \mathrm{cl}_\bullet(\mathcal{O}_X)\\ &= (\mathrm{cl}_\bullet(F) - \mathrm{cl}_\bullet(\mathcal{O}_X)) - (\mathrm{cl}_\bullet(F) - \mathrm{cl}_\bullet(L))\\ &= \mathrm{cl}_\bullet(F/\mathcal{O}_X) - \mathrm{cl}_\bullet(F/L).\end{aligned}$$

But the fibers at the generic point of the sheaves $F/\mathcal{O}_X$ and F/L vanish; their supports in X are therefore nowhere dense (since X is irreducible), and hence have dimension at most $j - 1$. Assertion (1.3.6) follows immediately.

II. It remains to prove the general case. Introduce the flag bundle

$$g : D \longrightarrow X$$

of E over X, on which there exist invertible $\mathcal{O}_D$-modules $L_1, \ldots, L_n$ such that

$$g^*(\mathrm{cl}^\circ(E)) = \mathrm{cl}^\circ(L_1) + \cdots + \mathrm{cl}^\circ(L_n) \in K^\circ(D). \tag{1.3.7}$$

Let e be the generic point of X, and let $k(e)$ be its residue field. There exists a rational point u of the generic fiber D_e over $k(e)$,

because these points are in one-to-one correspondence with flags of type $(1, \ldots, 1)$ (n terms) in the vector space E_e of dimension $n > 0$ over $k(e)$. These rational points also correspond to germs of sections of g in a neighborhood of e, so there exists such a section $s : U \to D$ over an open subset U of X. Let $X' = \overline{s(U)}$, endowed with its reduced induced structure, and let $f : X' \to X$ be the restriction of g to X'. The morphism f is projective and, since it is an isomorphism over $s(U)$, it is birational.

Thus $\dim(X') \leq \dim(X)$ (EGA IV 5.6.6.1). On the other hand, $f(X') = g(X')$ is closed and contains U, so f is surjective and $\dim(X') \geq \dim(X)$ (EGA IV 5.4.1(ii)). The hypotheses of case I are therefore satisfied for X' and $x' = f^*(x) \in \mathrm{Filt}^i(X')$, and one obtains

$$f^*(x)\,\mathrm{cl}_\bullet(\mathcal{O}_{X'}) \in \mathrm{Filt}_{j-i}(X'). \tag{1.3.7'}$$

Consequently, by the projection formula and (1.1.8),

$$x f_*(\mathrm{cl}_\bullet(\mathcal{O}_{X'})) \in \mathrm{Filt}_{j-i}(X). \tag{1.3.8}$$

On the other hand, the canonical homomorphism

$$\mathcal{O}_X \longrightarrow f_*(\mathcal{O}_{X'})$$

is an isomorphism over the dense open subset U, and the $\mathcal{O}_X$-modules $R^i f_*(\mathcal{O}_{X'})$ $(i \geq 1)$ are concentrated on the nowhere-dense closed subset $X - U$. It follows that

$$f_*(\mathrm{cl}_\bullet(\mathcal{O}_{X'})) = \mathrm{cl}_\bullet(\mathcal{O}_X) + w, \tag{1.3.9}$$

where $w \in \mathrm{Filt}_{j-1}(X)$. By the induction hypothesis, $x \cdot w \in \mathrm{Filt}_{j-i-1}(X)$; hence, by (1.3.8) and (1.3.9),

$$x\,\mathrm{cl}_\bullet(\mathcal{O}_X) \in \mathrm{Filt}_{j-i}(X).$$

This proves the assertion.

Remark 1.4. Let X be a locally noetherian scheme. Observe that, in this section, we have used only the following properties of the "dimension function"

$$D(x) = D_Y(x) = \dim(\overline{\{x\}})$$

on proper X-schemes Y:

(i) For every proper X-scheme Y and every $x \in Y$, one has $D_Y(x) \in \mathbb{Z}$.

(ii) If $y \neq x$ is a specialization of x, then $D(y) < D(x)$.

(iii) For every X-morphism $f : Y_1 \to Y_2$ of proper X-schemes and every $y \in Y_1$, one has

$$D_{Y_2}(f(y)) \leq D_{Y_1}(y).$$

(iv) Equality holds in (iii) if f is an immersion, or if Y_1 is irreducible, f is birational and surjective, and y is the generic point of Y_1.

Thus all the results of this section, in particular (1.3.2) and (1.3.3), remain valid unchanged if "dimension" is replaced by a function D satisfying conditions (i)–(iv) above and the filtration of $K_\bullet(X)$ is defined using such a function D.

Example 1.5. Let S be a locally noetherian universally catenary scheme (for example, a subscheme of a Cohen–Macaulay scheme). Put

$$D_S(x) = n - \operatorname{codim}(\overline{\{x\}}, S) = n - \dim \mathcal{O}_{S,x} \qquad (x \in S), \tag{1.5.1}$$

and, for every proper S-scheme $s_Y : Y \to S$ and every $y \in Y$, also put

$$D_Y(y) = D_S(s_Y(y)) + \deg. \operatorname{tr}_{k(s_Y(y))} k(y). \tag{1.5.2}$$

If $f : Y_1 \to Y_2$ is an S-morphism of proper S-schemes, the additivity of transcendence degrees and (1.5.2) immediately give

$$D_{Y_1}(y) = D_{Y_2}(f(y)) + \deg. \operatorname{tr}_{k(f(y))} k(y) \qquad (y \in Y_1). \tag{1.5.3}$$

We show that the functions D_Y so defined satisfy conditions (i)–(iv) of 1.4. By (1.5.3), condition (ii) is the only nontrivial one. In view of (iv), we may plainly suppose that $Y = \overline{\{x\}}$. Factor the structural morphism s_Y through its image $s_Y(Y) = Z \hookrightarrow S$, say $s_Y = g \circ f$:

$$Y \xrightarrow{\ f\ } Z \xhookrightarrow{\ g\ } S,$$

where Y and Z are irreducible, f is proper and surjective, and g is a closed immersion. Put $x' = s_Y(x)$ and $y' = s_Y(y)$. From (1.5.2) and (1.5.1) one obtains

$$\begin{aligned} D_Y(x) - D_Y(y) = {} & \dim \mathcal{O}_{S,y'} - \dim \mathcal{O}_{S,x'} \\ & + \deg. \operatorname{tr}_{k(x')} k(x) - \deg. \operatorname{tr}_{k(y')} k(y). \end{aligned}$$

Since S is catenary, one has

$$\dim \mathcal{O}_{S,y'} - \dim \mathcal{O}_{S,x'} = \dim \mathcal{O}_{Z,y'} - \dim \mathcal{O}_{Z,x'}$$

(EGA 0_{IV} 14.3.2.1), whence

$$\begin{aligned} D_Y(x) - D_Y(y) = &[\dim \mathcal{O}_{Z,y'} - \deg.\,\mathrm{tr}_{k(y')}\, k(y)] \\ &- [\dim \mathcal{O}_{Z,x'} - \deg.\,\mathrm{tr}_{k(x')}\, k(x)]. \end{aligned} \tag{1.5.4}$$

Since Z is universally catenary, EGA IV 5.6.5.1 applies and gives

$$C_1 = \dim \mathcal{O}_{Y,y} - e, \qquad C_2 = \dim \mathcal{O}_{Y,x} - e,$$

where C_1 and C_2 are the two terms on the right of (1.5.4). It follows immediately from (1.5.4) that

$$D_Y(x) - D_Y(y) = \dim \mathcal{O}_{Y,y} - \dim \mathcal{O}_{Y,x} > 0,$$

as required.

Thus (1.3.2) and (1.3.3) remain valid for a noetherian universally catenary scheme X of dimension n if the filtration of $K_\bullet(X)$ is defined by means of (1.5.1), according to the codimension of supports. This filtration coincides with that of (1.1.1) when, in addition, X is biequidimensional.

1.6

One may ask whether it is always possible, on a noetherian scheme X of finite dimension, to define a function D possessing property (i) of 1.4 and also the following property, stronger than (ii):

(ii bis) If $x \in X$ and $y \neq x$ is an immediate specialization of x, then $D(x) = D(y) + 1$.

For this it is plainly necessary that X be catenary, and sufficient that X be biequidimensional (EGA 0_{IV} 14.3.3). There nevertheless exist noetherian schemes of finite dimension that are catenary and equidimensional but admit no such function D. To see this, modify the example of EGA IV 5.6.11 as follows. Take two copies Y_1 and Y_2 of the spectrum of the ring E of loc. cit., and let y_i and z_i $(i = 1, 2)$ be the closed points of Y_i corresponding respectively to the maximal ideals $\mathfrak{m}$ and $\mathfrak{m}'$ of E, of heights 2 and 1. Recall that the residue fields of $\mathfrak{m}$ and $\mathfrak{m}'$ are isomorphic. One can therefore construct a scheme X by gluing Y_1 and Y_2, identifying y_1 with z_2 and y_2 with z_1. The schemes Y_1 and Y_2 may then be identified with the two irreducible components of X; inside Z, $Y_1 \cap Y_2 = \{y_1, y_2\}$ has dimension zero. Thus X is a noetherian, catenary, equidimensional scheme of dimension 2. If (ii bis) held on X, one would have

$$D(y_1) = D(z_1) - 1 = D(y_2) - 1,$$

and, on the other hand,

$$D(y_1) = D(z_2) = D(y_2) + 1,$$

which is a contradiction.

2. Snapper polynomials

2.1

We first recall the definition and a few elementary properties of polynomial maps.

Definition 2.1.1. Let f be a map from an abelian group A to an abelian group B. We say that f is a polynomial map if, for every sequence $x_1, \ldots, x_r$ $(r \geq 1)$ of elements of A, there is a family

$$\left(y(x_1, \ldots, x_r; k_1, \ldots, k_r)\right)_{k_1, \ldots, k_r \geq 0}$$

of elements of B, almost all zero, satisfying the following condition: for every sequence $n_1, \ldots, n_r$ of integers, one has

$$f(n_1x_1 + \cdots + n_rx_r) = \sum_{k_1,\ldots,k_r \geq 0} n_1^{k_1} \cdots n_r^{k_r} y(x_1, \ldots, x_r; k_1, \ldots, k_r).$$

We say that f has degree $\leq m$ if, for every integer $r \geq 1$ and all $x_1, \ldots, x_r \in A$, one has

$$y(x_1, \ldots, x_r; k_1, \ldots, k_r) = 0$$

whenever $k_1 + \cdots + k_r > m$.

Lemma 2.1.2. Every homomorphism of abelian groups is a polynomial map of degree ≤ 1. If $h = f \circ g$, where f is a polynomial map of degree p and g is a polynomial map of degree q, then h is a polynomial map of degree $\leq pq$.

The proof is trivial.

Lemma 2.1.3. Let

$$0 \longrightarrow G' \longrightarrow G \xrightarrow{j} G'' \longrightarrow 0$$

be an exact sequence of abelian groups such that G' is p-divisible for some prime p and G'' is a torsion group. Then every polynomial map $f : G \to \mathbb{Q}$ that factors through $\mathbb{Z}$ is constant.

Proof. It is enough to show that $f(x) = f(0)$ for every $x \in G$.

(a) Suppose first that x is a torsion element, say $mx = 0$ ($m \in \mathbb{Z}$, $m \neq 0$). Then the polynomial $P_x(n) = f(nx) - f(0)$ has infinitely many roots, namely all integers divisible by m. Thus P_x is identically zero, and $f(x) = f(0)$.

(b) Suppose now that x is an element of infinite order in G, divisible by every power p^k ($k \in \mathbb{Z}$). Consider the polynomials P_k defined by

$$P_k(n) = f(nx/p^k) = \sum_{i \in \mathbb{N}} n^i y(x/p^k; i) \in \mathbb{Z} \qquad (n \in \mathbb{Z};\ y(x/p^k; i) \in \mathbb{Q}).$$

For every $n \in \mathbb{Z}$, one has the identity

$$P_k(n) = P_{k+1}(pn),$$

that is,

$$\sum_{i \in \mathbb{N}} n^i y(x/p^k; i) = \sum_{i \in \mathbb{N}} p^i n^i y(x/p^{k+1}; i).$$

Comparing coefficients gives

$$y(x/p^k; i) = p^i y(x/p^{k+1}; i) \qquad (i \in \mathbb{N}).$$

Every polynomial $P : \mathbb{Q} \to \mathbb{Q}$ is continuous for the topology induced by $\mathbb{R}$. Thus, for k sufficiently large,

$$|P_x(n/p^k) - P_x(0)| < 1.$$

But, by definition, $P_x(n/p^k)$ is an integer for all $n, k \in \mathbb{Z}$. It follows that $P_x(n/p^k) - P_x(0) = 0$ for sufficiently large k. Hence P_x is constant and, in particular, $P_x(1) = P_x(0)$, i.e. $f(x) = f(0)$.

The two cases above show that the restriction of f to G' is constant. We now show that f is constant on each coset of G' in G. For every $b \in G$, let $g_b : G' \to \mathbb{Q}$ be defined by

$$g_b(y) = f(b + y) \qquad (y \in G').$$

The map g_b is a polynomial map from G' to $\mathbb{Q}$ that factors through $\mathbb{Z}$, and is therefore constant by the preceding argument. Thus f is constant on the cosets of G' in G. Consequently one may write $f = h \circ j$, where $h : G'' \to \mathbb{Q}$ is a polynomial map that factors through $\mathbb{Z}$. Since G'' is a torsion group, h is constant. Therefore f is constant.

Lemma 2.1.4. Let A be a unital commutative ring and I a nilpotent ideal of A. Let P be the multiplicative subgroup of A^* consisting of the invertible elements of A of the form $x = 1 + y$ with $y \in I$, and let $i : P \to A$ be the natural injection.

Then the composite map

$$\alpha : P \xrightarrow{i} A \xrightarrow{j} A \otimes_{\mathbb{Z}} \mathbb{Q},$$

where $j(x) = x \otimes 1$, is a polynomial map. If $I^{m+1} = 0$, then α has degree $\leq m$.

Proof. Let $x_1, \ldots, x_r$ be a sequence of elements of P, with $x_i = 1 + y_i$ $(i = 1, \ldots, r)$. The binomial formula gives, for every $n_1, \ldots, n_r \in \mathbb{Z}$,

$$x_1^{n_1} \cdots x_r^{n_r} = \sum_{k_1, \ldots, k_r \geq 0} \binom{n_1}{k_1} \cdots \binom{n_r}{k_r} y_1^{k_1} \cdots y_r^{k_r}, \tag{2.1.5}$$

where the sum on the right is finite because the elements $y_1, \ldots, y_r$ are nilpotent. The coefficients $\binom{n_1}{k_1} \cdots \binom{n_r}{k_r}$ are polynomials with rational coefficients in $n_1, \ldots, n_r$, of degree $k_1 + \cdots + k_r$; hence α is indeed a polynomial map. If $I^{m+1} = 0$, then $y_1^{k_1} \cdots y_r^{k_r} = 0$ whenever $k_1 + \cdots + k_r > m$. Thus α has degree $\leq m$.

2.2

Now let X be a noetherian scheme and let $y \in \mathrm{Filt}_j(X)$ (1.1.1). The assignment $x \mapsto xy$ defines a homomorphism of $K^\circ(X)$-modules

$$\Theta_y : K^\circ(X) \longrightarrow K_\bullet(X),$$

which, by (1.3.2), factors through the canonical epimorphism

$$j : K^\circ(X) \longrightarrow K^\circ(X)/\,\mathrm{Filt}^{j+1}(X) = A.$$

Let $\beta_y : A \to K_\bullet(X)$ be the homomorphism so defined. If I denotes the ideal

$$I = \mathrm{Filt}^1(X)/\,\mathrm{Filt}^{j+1}(X)$$

of A, then $I^{j+1} = 0$. Keeping the notation of Lemma 2.1.4, one obtains a homomorphism of abelian groups

$$\gamma : \mathrm{Pic}(X) \longrightarrow P$$

that sends the class $\mathrm{cl}_P(L)$ of an invertible $\mathcal{O}_X$-module L in $\mathrm{Pic}(X)$ to the element $j(\mathrm{cl}^\circ(L))$ of P.

Lemmas 2.1.2 and 2.1.4 immediately give:

Proposition 2.2.1. For every noetherian scheme X and every $y \in \mathrm{Filt}_j(X)$ $(j \in \mathbb{Z})$, the composite map

$$\mathrm{Pic}(X) \longrightarrow K_\bullet(X) \otimes_{\mathbb{Z}} \mathbb{Q},$$

defined by

$$\mathrm{cl}_P(L) \longmapsto (\mathrm{cl}^\circ(L)y) \otimes 1,$$

is a polynomial map of degree $\leq j$.

2.3

If X is a proper scheme over a field k and $g : X \to \mathrm{Spec}(k)$ is the structural morphism, then the homomorphism

$$g_* : K_\bullet(X) \longrightarrow K_\bullet(\mathrm{Spec}(k)) \simeq \mathbb{Z}$$

sends the class $\mathrm{cl}_\bullet(F)$ of every coherent $\mathcal{O}_X$-module F to the Euler–Poincaré characteristic $\chi(F) \in \mathbb{Z}$ of F on X. Proposition 2.2.1 immediately gives:

Proposition 2.3.1. For every scheme X proper over k and every $y \in \mathrm{Filt}_j(X)$, the map

$$S_y : \mathrm{Pic}(X) \longrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q},$$

defined by

$$\mathrm{cl}_P(L) \longmapsto g_*(\mathrm{cl}^\circ(L)y) = \chi(\mathrm{cl}^\circ(L)y),$$

is a polynomial map of degree $\leq j$.

When $L_1, \ldots, L_n$ are invertible $\mathcal{O}_X$-modules and F is a coherent $\mathcal{O}_X$-module such that $\dim(\mathrm{Supp}(F)) = j$, Proposition 2.3.1 implies in particular that the Euler–Poincaré characteristic

$$\chi(L_1^{\otimes m_1} \otimes \cdots \otimes L_n^{\otimes m_n} \otimes F) \tag{2.3.2}$$

is a numerical polynomial in $m_1, \ldots, m_n \in \mathbb{Z}$, of degree $\leq \dim(\mathrm{Supp}(F))$, sometimes called the Snapper polynomial of F and the sequence $L_1, \ldots, L_n$.

2.4

Recall [2] that, for the fpqc topology on $\mathrm{Sch}\,/k$, the Picard functor $\mathrm{Pic}_{X/k}$ is the sheaf of groups associated with the presheaf

$$T \longmapsto \mathrm{Pic}(X \times_k T) \qquad (T \in \mathrm{Ob}(\mathrm{Sch}\,/k)).$$

When k is algebraically closed, the canonical homomorphism

$$\mathrm{Pic}(X) \longrightarrow \mathrm{Pic}_{X/k}(\mathrm{Spec}(k))$$

is an isomorphism.

When X is proper over k, it is known [4] that the Picard functor is representable by a group scheme, locally of finite type over k, also denoted $\mathrm{Pic}_{X/k}$. The neutral component $\mathrm{Pic}^0_{X/k}$ of $\mathrm{Pic}_{X/k}$ is a connected commutative algebraic group over k.

Definition 2.4.1. Let X be a scheme proper over a field k, and let L be an invertible $\mathcal{O}_X$-module (respectively, let $x \in \mathrm{Pic}(X)$). We say that L (respectively, x) is τ-equivalent to zero if there is an integer $n \neq 0$ such that $L^{\otimes n}$ (respectively, nx) is algebraically equivalent to zero. The elements of $\mathrm{Pic}(X)$ algebraically equivalent (respectively, τ-equivalent) to zero form a subgroup $\mathrm{Pic}^0(X)$ (respectively, $\mathrm{Pic}^\tau(X)$) of $\mathrm{Pic}(X)$.

Lemma 2.4.3. For every proper scheme X over an algebraically closed field k, the group $\mathrm{Pic}^0(X)$ is divisible by every integer n prime to the characteristic of k.

Proof. Since k is algebraically closed, one has

$$\mathrm{Pic}^0(X) \simeq \mathrm{Pic}^0_{X/k}(k).$$

Since $\mathrm{Pic}^0_{X/k}$ is connected and of finite type, the n-th-power endomorphism is surjective when n is prime to the characteristic of k (SGA 3 XV 1.3 and VI 1.3.2). Since k is algebraically closed, the corresponding endomorphism of $\mathrm{Pic}^0_{X/k}(k)$ is surjective.

It follows from 2.4.3 that, in the exact sequence

$$0 \longrightarrow \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^\tau(X) \longrightarrow T \longrightarrow 0,$$

$\mathrm{Pic}^0(X)$ is p-divisible for some prime p and T is a torsion group. Thus Lemma 2.1.3 applies to the restriction of the polynomial map (2.3.1) to $\mathrm{Pic}^\tau(X)$. One obtains:

Proposition 2.4.4. The polynomial map S_y of (2.3.1) is constant on the cosets of $\mathrm{Pic}^\tau(X)$ in $\mathrm{Pic}(X)$.

Proof. If K is an algebraically closed extension of k, one has a commutative diagram

$$\begin{array}{ccccccc} \mathrm{Pic}(X) & \xrightarrow{S_y} & K_\bullet(\mathrm{Spec}(k)) & \longrightarrow & \mathbb{Z} & \hookrightarrow & \mathbb{Q} \\ \downarrow & & & \nearrow & & & \\ \mathrm{Pic}(X_K) & \xrightarrow[\overline{S}_y]{} & K_\bullet(\mathrm{Spec}(K)) & & & & \end{array}$$

where $\overline{S}_y$ is the polynomial map on $\mathrm{Pic}(X_K)$ corresponding to S_y. Hence one may suppose that k is algebraically closed. The assertion then follows immediately from Lemmas 2.1.3 and 2.4.3.

Remark 2.4.5. In Exposé XIII this result will be proved without using representability of the functor $\mathrm{Pic}_{X/k}$.

3. Projection formulas for the associated graded groups

3.1

Let X be a noetherian scheme. Denote by $\mathrm{Gr}^\circ(X)$ the graded group associated with the λ-filtration of $K^\circ(X)$, and by $\mathrm{Gr}_\bullet(X)$ the graded group associated with the filtration 1.1.1 of $K_\bullet(X)$. By Theorem 1.3.2, the composition law (1.3.1)

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X)$$

induces, on passing to the associated graded groups, composition laws

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_j(X) \longrightarrow \mathrm{Gr}_{j-i}(X) \qquad (i, j \in \mathbb{Z}). \tag{3.1.1}$$

Thus $\mathrm{Gr}_\bullet(X)$ becomes a module over the ring $\mathrm{Gr}^\circ(X)$, and even a graded module over the graded ring $\mathrm{Gr}^\circ(X)$, provided that the sign of the grading of $\mathrm{Gr}_\bullet(X)$ is changed by putting

$$\mathrm{Gr}^i_\bullet(X) = \mathrm{Gr}_{-i}(X) \qquad (i \in \mathbb{Z}).$$

Now let

$$f : X' \longrightarrow X$$

be a proper morphism of noetherian schemes. Then f induces a homomorphism

$$f^* : \mathrm{Gr}^\circ(X) \longrightarrow \mathrm{Gr}^\circ(X')$$

of graded rings (Exposé V), and, by Proposition 1.1.8, a homomorphism

$$f_* : \mathrm{Gr}_\bullet(X') \longrightarrow \mathrm{Gr}_\bullet(X)$$

of graded abelian groups. Passing the projection formula (IV 2.11) and (3.1.1) to the associated graded groups gives the graded projection formula

$$x \cdot f_*(y) = f_*(f^*(x) \cdot y) \qquad (x \in \mathrm{Gr}^\circ(X),\ y \in \mathrm{Gr}_\bullet(X')). \tag{3.1.2}$$

3.2

Let $f : X' \to X$ be a projective complete-intersection morphism (Exposé VIII), of virtual dimension r, between noetherian schemes, with X carrying an ample invertible $\mathcal{O}_X$-module. In Exposé VIII it was proved that f induces a homomorphism

$$f_* : \mathrm{Filt}^k(X')_{\mathbb{Q}} \longrightarrow \mathrm{Filt}^{k-r}(X)_{\mathbb{Q}} \qquad (k \in \mathbb{Z}), \tag{3.2.1}$$

which moreover satisfies the projection formula (Exposé IV):

$$f_*(f^*(x) \cdot y) = x \cdot f_*(y) \qquad (x \in K^\circ(X)_{\mathbb{Q}},\ y \in K^\circ(X')_{\mathbb{Q}}). \tag{3.2.2}$$

Passing to the associated graded groups in (3.2.1) and (3.2.2), one obtains a homomorphism of degree $-r$:

$$f_* : \mathrm{Gr}^\circ(X')_{\mathbb{Q}} \longrightarrow \mathrm{Gr}^\circ(X)_{\mathbb{Q}}, \tag{3.2.3}$$

and, by Theorem 1.3.2, a graded projection formula

$$f_*(f^*(x) \cdot y) = x \cdot f_*(y) \qquad (x \in \mathrm{Gr}^\circ(X)_{\mathbb{Q}},\ y \in \mathrm{Gr}^\circ(X')_{\mathbb{Q}}).$$

3.3

Finally, let $f : X' \to X$ be a morphism of finite Tor-dimension between noetherian schemes. It is known (Exposé IV) that f induces a homomorphism

$$f^* : K_\bullet(X) \longrightarrow K_\bullet(X')$$

such that, for every coherent $\mathcal{O}_X$-module F,

$$f^*(\mathrm{cl}_\bullet(F)) = \sum_i (-1)^i \,\mathrm{cl}_\bullet(\mathrm{Tor}_i^{\mathcal{O}_X}(F, \mathcal{O}_{X'})).$$

The $\mathcal{O}_{X'}$-modules $\mathrm{Tor}_i^{\mathcal{O}_X}(F, \mathcal{O}_{X'})$ are supported on $f^{-1}(\mathrm{Supp}(F))$. If the fibers of f are everywhere of dimension $\leq d$, then (EGA IV 5.6.7)

$$\dim\big(f^{-1}(\mathrm{Supp}(F))\big) \leq \dim(\mathrm{Supp}(F)) + d.$$

It follows that f induces, for every $k \in \mathbb{Z}$, a homomorphism

$$f^* : \mathrm{Filt}_k(X) \longrightarrow \mathrm{Filt}_{k+d}(X'),$$

hence a homomorphism of degree d of graded abelian groups.

Thus one has

$$f^* : \mathrm{Gr}_\bullet(X) \longrightarrow \mathrm{Gr}_\bullet(X'). \tag{3.3.1}$$

3.4

Suppose finally that the morphisms (3.2.3) and (3.3.1) are both defined for a morphism $f : X' \to X$ and that $r = d$. This is the case, for example, if X is quasi-compact with an ample invertible $\mathcal{O}_X$-module and if $X' = \mathbf{P}_X(E)$, where E is a locally free $\mathcal{O}_X$-module of rank $r+1$. The projection formula in the derived category then gives, for every $x \in \mathrm{Filt}^i(X')_{\mathbb{Q}}$ and $y \in \mathrm{Filt}_j(X)_{\mathbb{Q}}$,

$$f_*(x \cdot f^*(y)) = f_*(x) \cdot y \in \mathrm{Filt}_{j+r-i}(X)_{\mathbb{Q}}.$$

When X is noetherian, Theorem 1.3.2 yields the corresponding graded formula.

4. Intersection numbers

4.1

Let X be a proper scheme over a field k. Composing the product (3.1.1)

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathrm{Gr}_0(X) = \mathrm{Filt}_0(X) \qquad (i \in \mathbb{Z})$$

with the Euler–Poincaré characteristic

$$\chi : \mathrm{Filt}_0(X) \longrightarrow \mathbb{Z}$$

gives a pairing called the *intersection number*:

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathbb{Z}. \tag{4.1.1}$$

We denote this pairing by

$$(x, y) \longmapsto \langle x, y \rangle \in \mathbb{Z} \qquad (i \in \mathbb{Z},\ x \in \mathrm{Gr}^i(X),\ y \in \mathrm{Gr}_i(X)).$$

Example 4.2. Suppose that X has dimension n. Let $y \in \mathrm{Gr}_n(X)$, for example $y = \mathrm{cl}_\bullet(\mathcal{O}_Z)$, where Z is an integral closed subscheme of X of dimension n. Let P also be a homogeneous polynomial of weight n,

$$P \in \mathbb{Z}[T_1, \ldots, T_n],$$

where T_i is assigned weight i $(i = 1, \ldots, n)$; for example, P may contain a monomial $T_{i_1} \cdots T_{i_r}$ with $i_1 + \cdots + i_r = n$. Substituting for each variable T_i the Chern class

$$c_i(x) = \mathrm{cl}^i(\gamma^i(x - \varepsilon(x))) \in \mathrm{Gr}^i(X) \qquad (x \in K^\circ(X))$$

gives an element

$$P(c_1(x), \ldots, c_n(x)) \in \mathrm{Gr}^n(X).$$

The integer

$$P_y(x) = \langle P(c_1(x), \ldots, c_n(x)), y \rangle$$

is called the *Chern number* of x relative to y. When $y = \mathrm{cl}_\bullet(\mathcal{O}_X)$, the reference to y is omitted from this terminology.

4.3

Without any hypothesis on the dimension of X, consider invertible $\mathcal{O}_X$-modules $\mathcal{L}_1, \ldots, \mathcal{L}_d$ and a coherent $\mathcal{O}_X$-module $\mathcal{F}$ such that $\dim(\mathrm{Supp}(\mathcal{F})) \leq d$. Put

$$L_i = \mathrm{cl}^\circ(\mathcal{L}_i) \quad (i = 1, \ldots, d), \qquad F = \mathrm{cl}_\bullet(\mathcal{F}).$$

Proposition 4.3.1. With the preceding notation, for every sequence of nonnegative integers $\alpha_1, \ldots, \alpha_d$ such that $\sum_{i=1}^d \alpha_i = d$, the coefficient of

$$\binom{n_1}{\alpha_1} \cdots \binom{n_d}{\alpha_d}$$

in the Snapper polynomial

$$\chi(\mathcal{L}_1^{\otimes n_1} \otimes \cdots \otimes \mathcal{L}_d^{\otimes n_d} \otimes \mathcal{F})$$

is equal to

$$\langle c_1(L_1)^{\alpha_1}\cdots c_1(L_d)^{\alpha_d}, |F|\rangle, \tag{4.3.2}$$

where $|F|$ is the class of F in $\mathrm{Gr}_d(X)$. Thus, in the ordinary expansion of this polynomial in monomials in the n_i $(i=1,\dots,d)$, the coefficient of $n_1^{\alpha_1}\cdots n_d^{\alpha_d}$ is

$$\frac{1}{\alpha_1!\cdots\alpha_d!}\langle c_1(L_1)^{\alpha_1}\cdots c_1(L_d)^{\alpha_d}, |F|\rangle.$$

In particular, one obtains immediately:

Corollary 4.3.3. The coefficient of the monomial $n_1n_2\cdots n_d$ in the Snapper polynomial

$$\chi(\mathcal{L}_1^{\otimes n_1}\otimes\cdots\otimes\mathcal{L}_d^{\otimes n_d}\otimes\mathcal{F})$$

is equal to

$$\langle c_1(L_1)\cdots c_1(L_d), |F|\rangle = \big(c_1(L_1)\cdots c_1(L_d), |F|\big). \tag{4.3.4}$$

Proof. Put $L_i = 1 + M_i$ for $i = 1,\dots,d$. By the definitions, the integer (4.3.2) is equal to

$$\chi(M_1^{\alpha_1}\cdots M_d^{\alpha_d}F).$$

Substituting

$$A = K^\circ(X)/\operatorname{Filt}^{d+1}(X), \qquad I = \operatorname{Filt}^1(X)/\operatorname{Filt}^{d+1}(X)$$

in Lemma 2.1.4, formula (2.1.5) gives, since $M_i \in \operatorname{Filt}^1(X)$ for every $i = 1,\dots,d$, the congruence

$$L_1^{n_1}\cdots L_d^{n_d} \equiv \sum_{\substack{\alpha_1,\dots,\alpha_d\geq 0\\ \alpha_1+\cdots+\alpha_d\leq d}} \binom{n_1}{\alpha_1}\cdots\binom{n_d}{\alpha_d} M_1^{\alpha_1}\cdots M_d^{\alpha_d} \pmod{\operatorname{Filt}^{d+1}(X)}. \tag{4.3.5}$$

Hence the coefficient of $\binom{n_1}{\alpha_1}\cdots\binom{n_d}{\alpha_d}$ in $\chi(\mathcal{L}_1^{\otimes n_1}\otimes\cdots\otimes\mathcal{L}_d^{\otimes n_d}\otimes\mathcal{F})$ is $\chi(M_1^{\alpha_1}\cdots M_d^{\alpha_d}F)$, as required.

4.4

The integer (4.3.4) is sometimes called the *intersection number* of the invertible $\mathcal{O}_X$-modules $\mathcal{L}_1,\dots,\mathcal{L}_d$ with $\mathcal{F}$, and is denoted by

$$(\mathcal{L}_1,\dots,\mathcal{L}_d;\mathcal{F}) \qquad (\textit{cf.}\ [3]).$$

Expression (4.3.4) immediately gives the following properties.

1) The integer (4.3.4) is a symmetric d-linear function of the classes $L_1,\dots,L_d \in \mathrm{Pic}(X)$, is zero if $\dim(\mathrm{Supp}(\mathcal{F})) < d$, and is additive in $\mathcal{F}$, that is, in its class $\mathrm{cl}_\bullet(\mathcal{F}) = F$.

2) By Proposition 1.1.2, one obtains

$$(\mathcal{L}_1,\dots,\mathcal{L}_d;\mathcal{F}) = \sum_{x\in X(d)} \mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{F}_x)(\mathcal{L}_1,\dots,\mathcal{L}_d;\mathcal{O}_{Z(x)}). \tag{4.4.1}$$

3) Let Y be a closed subscheme of X, and let $\mathcal{G}$ be a coherent $\mathcal{O}_Y$-module such that $\mathcal{F} = f_*(\mathcal{G})$, where $f: Y \to X$ is the canonical immersion. Then

$$(f^*\mathcal{L}_1,\dots,f^*\mathcal{L}_d;\mathcal{G}) = (\mathcal{L}_1,\dots,\mathcal{L}_d;\mathcal{F}).$$

More generally, the projection formula gives

$$\chi(f^*(x)\cdot y) = \chi\big(f_*(f^*(x)\cdot y)\big) = \chi(x\cdot f_*(y)) \qquad (x\in K^\circ(X),\ y\in K_\bullet(Y)) \tag{4.4.2}$$

for every k-morphism $f: Y \to X$ of proper k-schemes.

4) Suppose that, in the morphism $f : Y \to X$ above, X and Y are irreducible of dimension $\leq d$. If f has degree n, then

$$(f^*\mathcal{L}_1, \ldots, f^*\mathcal{L}_d; \mathcal{O}_Y) = n(\mathcal{L}_1, \ldots, \mathcal{L}_d; \mathcal{O}_X). \tag{4.4.3}$$

Recall the definition of the degree n of f [3]: it is zero unless

$$\dim(X) = \dim(Y) = \dim(f(Y)).$$

In the latter case, one puts

$$n = \frac{\text{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{Y,y})}{\text{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x})}, \tag{4.4.4}$$

where x is the generic point of X and y is the generic point of Y.

In the proof of (4.4.3), the case $\dim(X) = \dim(Y)$ is the only nontrivial one. If $\dim(f(Y)) < d$, then

$$R^i f_*(\mathcal{O}_Y) = 0 \tag{4.4.5}$$

for $i > 0$ in a neighborhood of x. Condition (4.4.5) also holds when $\dim(f(Y)) = d$, since the generic fiber then has dimension zero (EGA III 4.2.2 and EGA IV 5.5.6). Proposition 1.1.2 therefore gives the congruences

$$f_*(\text{cl}_\bullet(\mathcal{O}_Y)) \equiv \text{cl}_\bullet(f_*\mathcal{O}_Y) \equiv \text{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{Y,y})\, \text{cl}_\bullet(\mathcal{O}_{X_{\text{red}}}) \pmod{\text{Filt}_{d-1}(X)}.$$

By (4.4.2), it follows that

$$(f^*\mathcal{L}_1, \ldots, f^*\mathcal{L}_d; \mathcal{O}_Y) = \text{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{Y,y})(\mathcal{L}_1, \ldots, \mathcal{L}_d; \mathcal{O}_{X_{\text{red}}}).$$

On the other hand, item 2 gives

$$(\mathcal{L}_1, \ldots, \mathcal{L}_d; \mathcal{O}_X) = \text{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x})(\mathcal{L}_1, \ldots, \mathcal{L}_d; \mathcal{O}_{X_{\text{red}}}),$$

which proves (4.4.3).

4.5

The intersection product $\langle\, , \,\rangle$ defines an equivalence relation called *numerical equivalence*.

Definition 4.5.1. Two elements x and y of $\text{Gr}^i(X)$ $(i \in \mathbb{Z})$, respectively of $\text{Gr}_i(X)$, are said to be *numerically equivalent* if they define the same homomorphism

$$\text{Gr}_i(X) \longrightarrow \mathbb{Z}, \qquad \text{respectively} \qquad \text{Gr}^i(X) \longrightarrow \mathbb{Z}.$$

Two invertible $\mathcal{O}_X$-modules are said to be numerically equivalent if their first Chern classes in $\text{Gr}^1(X)$ are numerically equivalent.

Proposition 4.5.2. An invertible $\mathcal{O}_X$-module $\mathcal{L}$ is numerically equivalent to zero if and only if

$$\chi(\mathcal{L}^{\otimes n} \otimes_{\mathcal{O}_X} \mathcal{O}_Z) = \chi(\mathcal{O}_Z) \tag{4.5.3}$$

for every $n \in \mathbb{Z}$ and every integral closed curve $Z \subset X$.

This follows from Propositions 1.1.2 and 4.3.1.

Let $\text{Pic}^N(X)$ denote the subgroup of $\text{Pic}(X)$ consisting of the classes of invertible $\mathcal{O}_X$-modules numerically equivalent to zero. Proposition 2.4.4 gives:

Corollary 4.5.3.

$$\text{Pic}^\tau(X) \subset \text{Pic}^N(X).$$

It will be proved in Exposé XIII that in fact

$$\text{Pic}^N(X) = \text{Pic}^\tau(X).$$

5. The isomorphism $\mathrm{Pic}(X) \simeq \mathrm{Gr}^1(X)$

5.1

Let X be an arbitrary ringed topos. The ring $K^\circ(X)$ and the filtration $(\mathrm{Filt}^i(X))_{i\in\mathbb{Z}}$ can be defined exactly as for schemes. Likewise, define $\mathrm{Pic}(X)$ to be the group of isomorphism classes of invertible $\mathcal{O}_X$-modules. Let

$$i : \mathrm{Pic}(X) \longrightarrow K^\circ(X) \tag{5.1.1}$$

be the map sending the class $\mathrm{cl}_P(\mathcal{L}) \in \mathrm{Pic}(X)$ of an invertible $\mathcal{O}_X$-module $\mathcal{L}$ to $\mathrm{cl}^\circ(\mathcal{L}) \in K^\circ(X)$. This map is plainly a homomorphism from $\mathrm{Pic}(X)$ to the multiplicative group $K^\circ(X)^*$ of units of $K^\circ(X)$. Henceforth the group law of $\mathrm{Pic}(X)$ is written multiplicatively.

The map i has a left inverse defined by the determinant functor. If $\mathcal{E}$ is a locally free $\mathcal{O}_X$-module of rank n, put

$$\det(\mathcal{E}) = \bigwedge^n \mathcal{E}.$$

If

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}'' \longrightarrow 0 \tag{5.1.2}$$

is an exact sequence of locally free $\mathcal{O}_X$-modules, there is a canonical isomorphism

$$\det(\mathcal{E}) \xrightarrow{\sim} \det(\mathcal{E}') \otimes_{\mathcal{O}_X} \det(\mathcal{E}''),$$

arising from (6.2.9) and the fact that (5.1.2) is locally split. One therefore obtains a homomorphism of abelian groups

$$\det : K^\circ(X) \longrightarrow \mathrm{Pic}(X) \tag{5.1.3}$$

which is the required left inverse of i. In particular, i is injective, and henceforth $\mathrm{Pic}(X)$ is identified with its image in $K^\circ(X)^*$.

5.2

Define another map

$$\delta : \mathrm{Pic}(X) \longrightarrow \mathrm{Filt}^1(X) \tag{5.2.1}$$

by putting, for every $L \in \mathrm{Pic}(X)$,

$$\delta(L) = 1 - L^{-1}.$$

Remarks 5.2.2.

a) Suppose that X is the locally ringed topos of Zariski sheaves on a scheme S. If D is an effective Cartier divisor on S and

$$L = \mathrm{cl}^\circ(\mathcal{O}_S(D)),$$

then

$$\Theta_S(\delta(L)) = \mathrm{cl}_\bullet(\mathcal{O}_D), \tag{5.2.3}$$

where $\Theta_S : K^\circ(X) \to K_\bullet(S)$ is the canonical homomorphism of 1.2.3. This follows immediately from the exact sequence

$$0 \longrightarrow \mathcal{O}_S(-D) \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

b) Under the hypotheses of 5.1 and 5.2, one has, for every $L \in \mathrm{Pic}(X)$,

$$\delta(L) \equiv L - 1 \pmod{\mathrm{Filt}^2(X)}, \tag{5.2.4}$$

which justifies the notation $c_1(L)$ introduced below. Indeed,

$$(L-1) - (1 - L^{-1}) = (L-1)(1-L^{-1}) \in \mathrm{Filt}^2(X).$$

5.3

Composing δ with the canonical epimorphism

$$\mathrm{Filt}^1(X) \longrightarrow \mathrm{Gr}^1(X)$$

gives the *first Chern class* on $\mathrm{Pic}(X)$:

$$c_1 : \mathrm{Pic}(X) \longrightarrow \mathrm{Gr}^1(X). \tag{5.3.1}$$

Theorem 5.3.2. The map c_1 in (5.3.1) is an injective homomorphism, and the homomorphism

$$\det : \mathrm{Filt}^1(X) \longrightarrow \mathrm{Pic}(X)$$

induces on passing to the quotient a left inverse

$$\alpha : \mathrm{Gr}^1(X) \longrightarrow \mathrm{Pic}(X)$$

of c_1. If every section n of the constant sheaf $\mathbb{Z}_X$ is bounded—for example, if X is quasi-compact, or if the final object e of X is a finite disjoint sum of connected objects—then c_1 is an isomorphism and α is its inverse.

Proof. The map c_1 is a homomorphism, since for all $L, L' \in \mathrm{Pic}(X)$ one has

$$(1 - (LL')^{-1}) - (1 - L^{-1}) - (1 - L'^{-1}) = -(1 - L^{-1})(1 - L'^{-1}) \in \mathrm{Filt}^2(X).$$

For the moment, admit that

$$\det(\mathrm{Filt}^2(X)) = \{1\} \subset \mathrm{Pic}(X),$$

so that α is well defined. Then

$$\alpha \circ c_1 = \mathrm{id}_{\mathrm{Pic}(X)},$$

because, by 5.2.2 b),

$$\det(1 - L^{-1}) = \det(L^{-1})^{-1} = L$$

for every $L \in \mathrm{Pic}(X)$. Next we show that

$$c_1 \circ \alpha = \mathrm{id}_{\mathrm{Gr}^1(X)}$$

when every section n of $\mathbb{Z}_X$ is bounded. Since $\mathrm{Filt}^1(X)$ is the abelian group generated by the elements $\mathrm{cl}^\circ(\mathcal{E}) - n$, where $\mathcal{E}$ is a locally free $\mathcal{O}_X$-module of rank n, it is enough, in view of 5.2.2 b), to show that

$$\det(\mathrm{cl}^\circ(\mathcal{E})) - \mathrm{cl}^\circ(\mathcal{E}) + n - 1 \in \mathrm{Filt}^2(X). \tag{5.3.3}$$

Since n is bounded, the final object e of X decomposes as a finite disjoint sum $\coprod_{i=1}^{p} e_i$ of subobjects such that n is constant on each subtopos X/e_i. Since the functor

$$Y \longmapsto K^\circ(X/Y) \qquad (Y \in \mathrm{Ob}(X))$$

takes finite disjoint sums to finite direct products, one may suppose that n is constant on X, say $n \in \mathbb{Z}$. Put $E = \mathrm{cl}^\circ(\mathcal{E})$. Then (cf. Exposé V):

$$\det(E) = \lambda^n(E) = \gamma^n((E-n)+1) = \sum_{0 \le i \le n} \gamma^i(E-n).$$

This proves (5.3.3), since its left-hand side is

$$\sum_{2 \le i \le n} \gamma^i(E-n).$$

It remains to prove the following lemma.

Lemma 5.3.4. For every ringed topos X and every $x \in \mathrm{Filt}^2(X)$,

$$\det(x) = 1.$$

Proof. Let $\varepsilon : K^\circ(X) \to \Gamma(\mathbb{Z}_X)$ be the augmentation homomorphism. Formula (6.2.8), together with 6.5.3 below, gives by linearity

$$\det(xy) = \det(x)^{\varepsilon(y)} \det(y)^{\varepsilon(x)} \qquad (x, y \in K^\circ(X)). \tag{5.3.5}$$

It is therefore enough to show that, for every $k > 1$,

$$\det(\gamma^k(E-n)) = 1,$$

where, as above, $E = \mathrm{cl}^\circ(\mathcal{E})$ and $n = \varepsilon(E)$. For every $x \in K^\circ(X)$,

$$\gamma^k(x) = \lambda^k(x+k-1),$$

and hence

$$\begin{aligned} \det(\gamma^k(E-n)) &= \det(\lambda^k(E-n+k-1)) \\ &= \det\left(\sum_{p+q=k} \binom{-n+k-1}{p} \lambda^q(E)\right). \end{aligned}$$

But (6.2.10) and 6.5.3 give

$$\det(\lambda^q(E)) = \det(E)^{\binom{n-1}{q-1}} \qquad (q > 0).$$

It follows that

$$\det(\gamma^k(E-n)) = \det(E)^r,$$

where

$$r = \sum_{\substack{p+q=k \\ q \ge 1}} \binom{-n+k-1}{p} \binom{n-1}{q-1}.$$

Now r is a section of $\mathbb{Z}_X$, everywhere equal to the coefficient of t^{k-1} in $(1+t)^{k-2}$, and is therefore zero. This proves the lemma.

6. Appendix: Calculation of Determinants of Locally Free Sheaves

6.1

First let $X = (\mathcal{C}, \mathcal{O})$ be an arbitrary ringed topos. For every positive section $n \in \Gamma(X, \mathbb{Z}_X)$ of the constant sheaf and every $\mathcal{O}$-module $M \in \mathrm{Mod}_{\mathcal{O}}(X)$, one can define the $\mathcal{O}$-modules

$$M^n, \qquad \bigotimes^n M, \qquad \bigwedge^n M, \quad \text{etc.} \tag{6.1.1}$$

The functors $(\ )^n$, $\bigotimes^n$, and $\bigwedge^n$ carry free modules, respectively locally free modules, to free modules, respectively locally free modules.

More generally, let $n_1, \ldots, n_r$ be sections of $\mathbb{Z}_X$ and let T be an r-variable multifunctor, covariant for definiteness,

$$T : \mathrm{Mod}_{\mathcal{O}}^{n_1}(X) \times \cdots \times \mathrm{Mod}_{\mathcal{O}}^{n_r}(X) \longrightarrow \mathrm{Mod}_{\mathcal{O}}(X),$$

where $\mathrm{Mod}_{\mathcal{O}}^{n}(X)$ denotes the category fibered over X of locally free $\mathcal{O}$-modules of rank n. Put

$$M = T(\mathcal{O}^{n_1}, \ldots, \mathcal{O}^{n_r})$$

and

$$G = \mathrm{GL}(n_1) \times \cdots \times \mathrm{GL}(n_r), \qquad \mathrm{GL}(n_i) = \mathrm{Aut}_{\mathcal{O}}(\mathcal{O}^{n_i}).$$

It follows immediately that T defines a representation

$$\alpha_T : G \longrightarrow \mathrm{Aut}_{\mathcal{O}}(M). \tag{6.1.2}$$

If M is itself locally free of rank m, then for every composite functor $T' \circ T$, where

$$T' : \mathrm{Mod}_{\mathcal{O}}^{m}(X) \longrightarrow \mathrm{Mod}_{\mathcal{O}}(X),$$

one has

$$\alpha_{T' \circ T} = \alpha_{T'} \circ \alpha_T. \tag{6.1.3}$$

Conversely, from a representation $\alpha = \alpha_T : G \to \mathrm{Aut}_{\mathcal{O}}(M)$ one recovers $T(E_1, \ldots, E_r)$

for every sequence $E_1, \ldots, E_r$ of locally free $\mathcal{O}$-modules of ranks $n_1, \ldots, n_r$, by gluing the free module $T(\mathcal{O}^{n_1}, \ldots, \mathcal{O}^{n_r})$ on local trivializing charts through α. More precisely, the sequence $E_1, \ldots, E_r$ determines a right G-torsor $P = P_{E_1, \ldots, E_r}$ and canonical isomorphisms

$$E_i \xrightarrow{\sim} P \wedge^G \mathcal{O}^{n_i} \qquad (i = 1, \ldots, r), \tag{6.1.3}$$

where $\wedge^G$ denotes the contracted product and G acts on $\mathcal{O}^{n_i}$ through the i-th projection $G \to \mathrm{GL}(n_i)$. Every representation

$$\alpha : G \longrightarrow \mathrm{Aut}_{\mathcal{O}}(M)$$

therefore defines a multifunctor $T = T_\alpha$ such that

$$T_\alpha(E_1, \ldots, E_r) = P_{E_1, \ldots, E_r} \wedge^G M. \tag{6.1.4}$$

Moreover, for every composite representation $\alpha = \alpha_1 \circ \alpha_2$, one has

$$T_\alpha = T_{\alpha_1} \circ T_{\alpha_2}. \tag{6.1.5}$$

6.2

For every locally free $\mathcal{O}$-module E of rank n, write

$$\det(E) = \bigwedge^n E.$$

With the notation of 6.1, the functor $\det(T(E_1, \dots, E_r))$ is determined, up to canonical isomorphism, by the corresponding representation

$$\det \circ \alpha_T : G \longrightarrow \mathbb{G}_{\mathrm{m}} = \mathrm{GL}(1). \tag{6.2.1}$$

Suppose now that X is the ringed topos of sheaves on the ringed site $(\mathrm{Sch}\,/S)$ endowed with the étale topology. Then $\det(T(E_1, \dots, E_r))$, or equivalently the representation $\det \circ \alpha_T$, can be calculated by means of the following proposition.

Proposition 6.2.2. For every positive section n of $\mathbb{Z}$, the étale sheaf of groups $\mathrm{Der}(\mathrm{GL}(n))$ of commutators of $\mathrm{GL}(n)$ is represented by

$$\mathrm{SL}(n) = \mathrm{Ker}\big(\det : \mathrm{GL}(n) \longrightarrow \mathbb{G}_{\mathrm{m}}\big).$$

Corollary 6.2.3. Every representation

$$\alpha : G = \mathrm{GL}(n_1) \times \cdots \times \mathrm{GL}(n_r) \longrightarrow \mathbb{G}_{\mathrm{m}} \tag{6.2.4}$$

is of the form

$$\alpha(Y)(s) = \det(s_1)^{m_1} \cdots \det(s_r)^{m_r}, \tag{6.2.5}$$

where $Y \in \mathrm{Ob}(\mathrm{Sch}\,/S)$, $s = (s_1, \dots, s_r) \in G(Y)$, and $m_1, \dots, m_r$ are sections of $\mathbb{Z}$ uniquely determined by α.

Corollary 6.2.6. With the notation of 6.1,

$$\det\big(T_\alpha(E_1, \dots, E_r)\big) = \det(E_1)^{\otimes m_1} \otimes \cdots \otimes \det(E_r)^{\otimes m_r}, \tag{6.2.6}$$

where $m_1, \dots, m_r$ are the sections of $\mathbb{Z}$ uniquely determined by α.

Remark 6.2.7. The exponents $m_1, \dots, m_r$ in (6.2.5) and (6.2.6) can be determined by successively substituting in (6.2.5)

$$s_i = \lambda \ \mathrm{id}_{\mathcal{O}_Y^{n_i}} \quad (\lambda \in \mathbb{G}_{\mathrm{m}}(Y)), \qquad s_j = \mathrm{id}_{\mathcal{O}_Y^{n_j}} \quad (j \neq i),$$

for $i = 1, \dots, r$. Then

$$\det(s_i) = \lambda^{n_i}, \qquad \alpha(Y)(s) = \lambda^{n_i m_i}.$$

For example, (6.2.6) gives

$$\det(E_1 \otimes E_2) = \det(E_1)^{\otimes n_2} \otimes \det(E_2)^{\otimes n_1}, \tag{6.2.8}$$

$$\det(E_1 \oplus E_2) = \det(E_1) \otimes \det(E_2), \tag{6.2.9}$$

$$\det\Big(\bigwedge^m E_1\Big) = \det(E_1)^{\otimes \binom{n_1 - 1}{m-1}}. \tag{6.2.10}$$

6.3

Let us first show how 6.2.3 follows from 6.2.2. It is plainly enough to prove (6.2.5) for

$$s_{(i)} = (1, \dots, s_i, \dots, 1) \qquad (i = 1, \dots, r).$$

Thus one may suppose that $r = 1$ and $G = \mathrm{GL}(n_1)$. By 6.2.2, the representation α factors through $\mathrm{GL}(n_1)/\mathrm{SL}(n_1)$; equivalently, there is a commutative diagram

$$\begin{array}{ccc} \mathrm{GL}(n_1) & \xrightarrow{\alpha} & \mathbb{G}_{\mathrm{m}} \\ {\scriptstyle j}\downarrow & \nearrow_{\beta} & \\ \mathrm{GL}(n_1)/\mathrm{SL}(n_1) & & \end{array}$$

in which j is the canonical epimorphism and β is a representation of $\mathrm{GL}(n_1)/\mathrm{SL}(n_1)$ in $\mathbb{G}_{\mathrm{m}}$. The homomorphism

$$\det : \mathrm{GL}(n_1) \longrightarrow \mathbb{G}_{\mathrm{m}}$$

has a section. Consequently, the monomorphism

$$\mathrm{GL}(n_1)/\mathrm{SL}(n_1) \longrightarrow \mathbb{G}_{\mathrm{m}}$$

induced by det is an isomorphism. On the other hand, it is known (SGA 3 VIII 1.3) that every representation of $\mathbb{G}_{\mathrm{m}}$ in itself is of the form

$$s \longmapsto s^m,$$

where m is a section of $\mathbb{Z}$. The assertion follows.

Corollary 6.2.6 is simply a reformulation of 6.2.3. Notice that 6.2.3 also follows immediately from SGA 3 XXII 6.2.1, but we have wished to give an elementary proof here by means of 6.2.2.

6.4. Proof of 6.2.2

It is enough to prove that, for every $Y \in \mathrm{Ob}(\mathrm{Sch}/S)$, every $y \in Y$, and every $s \in \mathrm{SL}(n)(Y)$, there is an étale neighborhood $Y' \to Y$ of y such that the restriction of s to Y' is a product of commutators in $\mathrm{GL}(n)(Y')$. Since the question is local, one may suppose that Y is affine with ring A and that the section n is constant on Y. The case $n = 1$ is trivial; hence assume $n \geq 2$. Let $\mathfrak{p}$ be the prime ideal corresponding to y, and write $s = (s_{ij})_{i,j=1,\ldots,n}$. For every $\lambda \in A$ and every $h, k \in \{1, \ldots, n\}$ with $h \neq k$, let $c_{h,k,\lambda}$ be the matrix $C = (c_{ij})$ defined by

$$c_{hk} = \lambda, \qquad c_{ij} = \delta_{ij} \quad \text{for } (i,j) \neq (h,k),$$

where δ_{ij} is the Kronecker symbol.

Thus $c_{h,k,\lambda} = I + \lambda E_{hk}$. For every matrix a, left multiplication by $c_{h,k,\lambda}$ adds λ times row k of a to row h, whereas right multiplication adds λ times column h of a to column k. In particular,

$$\det(c_{h,k,\lambda}) = 1,$$

and

$$c_{h,k,\lambda} c_{h,k,\mu} = c_{h,k,\lambda+\mu} \tag{6.3.1}$$

for all $\lambda, \mu \in A$. Hence one obtains a representation

$$c_{h,k} : \mathbb{G}_{\mathrm{a}} \longrightarrow \mathrm{SL}(n) \qquad (h, k = 1, \ldots, n,\ h \neq k).$$

No invertibility assumption is needed in the row-reduction step. We reduce to invertible parameters before the commutator step.

The proof has two steps. First, we show that s is a product of matrices $c_{h,k,\lambda}$ in a sufficiently small Zariski neighborhood of y. Then we show that these matrices are products of commutators in $\mathrm{GL}(n)$ in a sufficiently small étale neighborhood of y.

For the first step, one must transform s into the identity matrix $I = (\delta_{ij})$ in a Zariski neighborhood of y by multiplying on the left and right by suitable matrices $c_{h,k,\lambda}$. We first make $s_{11} = 1$. If $s_{11} \neq 1$ and $s_{21} \in \mathfrak{p}$, then, since $\det(s) = 1 \notin \mathfrak{p}$, there is an index i for which $s_{i1} \notin \mathfrak{p}$. Replacing s by $c_{2,i,1}s$ replaces s_{21} by $s_{21} + s_{i1}$, which is not in $\mathfrak{p}$. Thus, after localizing, one may suppose that s_{21} is invertible.

If $s_{11} - 1 \in \mathfrak{p}$, replacing s by $c_{1,2,1}s$ replaces $s_{11} - 1$ by

$$(s_{11} + s_{21}) - 1 \notin \mathfrak{p}.$$

After a further localization, one may therefore suppose that $s_{11} - 1$ is invertible. Then

$$\lambda = \frac{1 - s_{11}}{s_{21}}$$

is invertible, and left multiplication by $c_{1,2,\lambda}$ makes the new entry s_{11} equal to 1.

Assume now that $s_{11} = 1$. Left multiplication by $c_{i,1,-s_{i1}}$ for $i > 1$ clears the remaining entries in the first column, and right multiplication by $c_{1,i,-s_{1i}}$ clears the remaining entries in the first row. Repeating this procedure on the lower-right block transforms s, in a Zariski neighborhood U of y, into a matrix $s' = (s'_{ij})$ satisfying

$$s'_{ij} = \delta_{ij} \qquad \text{whenever } i < n \text{ or } j < n.$$

By construction, $s' = c_1 s c_2$ on U, where c_1 and c_2 are products of matrices $c_{h,k,\lambda}$. Hence

$$\det(s') = \det(s) = 1,$$

so $s'_{nn} = 1$ and therefore $s' = I$. Since $c_{h,k,\lambda}^{-1} = c_{h,k,-\lambda}$, this proves that s is itself a product of elementary matrices of this form.

After shrinking about y, a parameter $a \notin \mathfrak{p}$ is a unit. If $a \in \mathfrak{p}$, then $a = 1 + (a - 1)$, and both 1 and $a - 1$ are units near y. By (6.3.1),

$$c_{h,k,a} = c_{h,k,1} c_{h,k,a-1}$$

Thus it remains only to treat invertible parameters.

It remains to prove that these elementary matrices are products of commutators in an étale neighborhood of y. For invertible λ and μ, the two matrices

$c_{h,k,\lambda}$ and $c_{h,k,\mu}$ are conjugate. More precisely, there is an invertible matrix $a = (a_{ij})$ such that

$$a\, c_{h,k,\lambda} = c_{h,k,\mu}\, a.$$

Indeed, the diagonal matrix defined by

$$a_{hh} = \lambda^{-1}, \qquad a_{kk} = \mu^{-1}, \qquad a_{ii} = 1 \quad (i \neq h, k)$$

has this property; recall that λ and μ are invertible and that $h \neq k$. Thus, for every homomorphism

$$\beta : \mathrm{GL}(n)(U) \longrightarrow B$$

to an abelian group B, written additively, one has

$$\beta(c_{h,k,\lambda}) = \beta(c_{h,k,\mu}).$$

If $\lambda + \mu$ is invertible, this equality and (6.3.1) imply

$$\beta(c_{h,k,\lambda}) = 0.$$

It follows that $c_{h,k,\lambda}$ belongs to the commutator subgroup. If the residue field $k(y)$ is not the prime field $\mathbb{F}_2 = \{0,1\}$, then, for every invertible $\lambda \in A$, there are a Zariski neighborhood $U = \mathrm{Spec}(B)$ of y and an invertible $\mu \in B$ such that $\lambda + \mu$ is invertible in B. Hence $c_{h,k,\lambda}$ is a product of commutators on U.

If $k(y) = \mathbb{F}_2$, then

$$V = \mathrm{Spec}(A[T]/(T^2 + T + 1))$$

is an étale neighborhood of y whose residue field is different from $\mathbb{F}_2$. This completes the proof.

6.5. Remarks

6.5.1. If no residue field of S is isomorphic to $\mathbb{F}_2$, the preceding proof shows that 6.2.2 remains valid for the ringed topos of Zariski sheaves on S. By contrast, already for $S = \mathrm{Spec}(\mathbb{F}_2)$ one has

$$\mathrm{SL}(2, \mathbb{F}_2) = \mathrm{GL}(2, \mathbb{F}_2),$$

which is different from the commutator subgroup of $\mathrm{GL}(2, \mathbb{F}_2)$. The latter has order 3, whereas $\mathrm{SL}(2, \mathbb{F}_2)$ has order 6 (cf. [1]).

6.5.2. On an arbitrary locally ringed topos X, it is not true in general that every representation $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ is given by a

power. For example, on the topos (Ens) ringed by the field $\mathbb{Q}$, the representation $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ defined by

$$r \longmapsto |r| \qquad (r \in \mathbb{Q}^*)$$

is not a power. In particular, 6.2.3 does not remain valid for an arbitrary locally ringed topos X and an arbitrary representation $\alpha : G \to \mathbb{G}_\mathrm{m}$.

6.5.3. On the other hand, 6.2.3 remains valid, on an arbitrary ringed topos $X = (\mathcal{C}, \mathcal{O})$, for the representations associated with formulas (6.2.8), (6.2.9), and (6.2.10), because these representations are *algebraic* in the following sense.

Let e be the final object of X, put $S = \mathrm{Spec}(\mathcal{O}(e))$, and consider the functor

$$u : \mathcal{C} \longrightarrow (\mathrm{Sch}\,/S)$$

which sends $Y \in \mathrm{Ob}(\mathcal{C})$ to the affine S-scheme $\mathrm{Spec}(\mathcal{O}(Y))$. For every positive integer n, the presheaf $\mathrm{GL}(n)_X$ on $\mathcal{C}$ is obtained, by composition with u, from the corresponding presheaf $\mathrm{GL}(n)_S$ on $(\mathrm{Sch}\,/S)$. A representation on X

$$\alpha : \mathrm{GL}(n) \longrightarrow \mathbb{G}_\mathrm{m} \tag{6.5.4}$$

is said to be *algebraic* if it is induced via u by a representation

$$\alpha_S : \mathrm{GL}(n)_S \longrightarrow \mathbb{G}_{\mathrm{m},S} \tag{6.5.5}$$

of group presheaves on $(\mathrm{Sch}\,/S)$; that is, if

$$\alpha(s) = \alpha_S(s)$$

for every $Y \in \mathrm{Ob}(\mathcal{C})$ and every $s \in \mathrm{GL}(n)(Y) = \mathrm{GL}(n)_S(u(Y))$. The representation α in (6.5.4) is of course completely determined by α_S. It follows immediately that 6.2.3 remains valid in an arbitrary ringed topos whenever α is algebraic. In particular, formulas (6.2.8), (6.2.9), and (6.2.10) remain valid on every ringed topos.

7. Appendix: Specialization in Intersection Theory

by A. Grothendieck

Lemma 7.1. Let $i : Y \to X$ be a regular closed immersion (VII 1.4) of bounded codimension. Then i is a perfect morphism (VII 1.9), so the inverse-image homomorphism (IV 2.12)

$$i^* : K_\bullet(X) \longrightarrow K_\bullet(Y)$$

is defined. Let N be the conormal sheaf of i. It is a locally free sheaf of finite type on Y (VII 1.6), and hence one can form

$$\lambda_{-1}(N) \in K^\circ(Y).$$

Consider also the direct-image homomorphism (IV 2.11)

$$i_* : K_\bullet(Y) \longrightarrow K_\bullet(X).$$

Then

$$i^* i_*(x) = x\,\lambda_{-1}(N) \qquad (x \in K_\bullet(Y)),$$

where the product on the right is taken with respect to the $K^\circ(Y)$-module structure on $K_\bullet(Y)$ defined in IV 2.10.

The proof is, essentially without change, the same as that of the corresponding formula for $K^\circ(X)$ and $K^\circ(Y)$ established in VII 2.7.

Corollary 7.2. Let $f : X \to S$ be flat, where S is a trait, that is, the spectrum of a discrete valuation ring A. Let s be the closed point of S, let t be its generic point, and let $i : X_s \to X$ be the canonical inclusion. Then i has finite Tor-dimension, so (IV 2.12) the homomorphism

$$i^* : K_\bullet(X) \longrightarrow K_\bullet(X_s)$$

is defined.

On the other hand, (IV 2.11) gives a homomorphism

$$i_* : K_\bullet(X_s) \longrightarrow K_\bullet(X).$$

The composite

$$i^* i_* : K_\bullet(X_s) \longrightarrow K_\bullet(X) \longrightarrow K_\bullet(X_s)$$

is zero.

Indeed, $i : X_s \to X$ is a regular immersion of codimension 1. Its conormal sheaf N is free of rank 1, because X_s is defined by the global equation $f^*(g) = 0$, where g is a uniformizer of A. Thus $\lambda_{-1}(N) = 0$, and the assertion follows from 7.1.

For the rest of this appendix, suppose that X is noetherian and flat over S. Apply the exact homotopy sequence (IX 1.1) to X, with the open subscheme X_t and complementary closed subscheme X_s:

$$K_\bullet(X_s) \xrightarrow{i_*} K_\bullet(X) \xrightarrow{j^*} K_\bullet(X_t) \longrightarrow 0,$$

where $j : X_t \to X$ is the open immersion. Corollary 7.2 gives:

Proposition 7.3. With the notation of 7.2, and assuming X noetherian, there is a unique group homomorphism

$$\sigma_\bullet : K_\bullet(X_t) \longrightarrow K_\bullet(X_s), \tag{7.3.1}$$

called the *specialization homomorphism*, for which the diagram

$$\begin{array}{ccc} K_\bullet(X) & \xrightarrow{j^*} & K_\bullet(X_t) \\ {\scriptstyle i^*}\downarrow & \swarrow{\scriptstyle \sigma_\bullet} & \\ K_\bullet(X_s) & & \end{array} \tag{7.3.2}$$

commutes.

Regard the two sides of (7.3.1) as $K^\circ(X)$-modules via the ring homomorphisms

$$K^\circ(X) \xrightarrow{j^*} K^\circ(X_t), \qquad K^\circ(X) \xrightarrow{i^*} K^\circ(X_s) \qquad \text{(IV 2.7 b)}$$

and via the $K^\circ(X_t)$- and $K^\circ(X_s)$-module structures on $K_\bullet(X_t)$ and $K_\bullet(X_s)$, respectively (IV 2.10). Then $\sigma_\bullet$ is a homomorphism of $K^\circ(X)$-modules.

This last assertion follows because $i^* : K_\bullet(X) \to K_\bullet(X_s)$ is $K^\circ(X)$-linear, as is the epimorphism $j^* : K_\bullet(X) \to K_\bullet(X_t)$, by IV 2.12.2.

We now review other properties of the specialization homomorphism.

7.4. Compatibility with filtrations

Endow the groups $K_\bullet$ with the filtration $\mathrm{Filt}_\bullet K_\bullet$ by the dimension of supports (1.1.1). If f is of finite type, and more generally if X is catenary, then the specialization homomorphism is compatible with these filtrations and consequently induces a homomorphism, again called specialization,

$$\sigma_\bullet : \mathrm{Gr}_\bullet(X_t) \longrightarrow \mathrm{Gr}_\bullet(X_s). \tag{7.4.1}$$

To prove this, by 1.14 one is reduced to showing that, if Y_t is an integral closed subscheme of dimension d of X_t, then $\sigma_\bullet(\mathrm{cl}_\bullet(\mathcal{O}_{Y_t}))$ has filtration at most d. Let Y be the schematic closure of Y_t in X. Then Y is flat over S (EGA IV 2.8.1), and one readily obtains

$$i^*(\mathrm{cl}_\bullet(\mathcal{O}_Y)) = \mathrm{cl}_\bullet(\mathcal{O}_{Y_s}).$$

It follows that

$$\sigma_\bullet(\mathrm{cl}_\bullet(\mathcal{O}_{Y_t})) = \mathrm{cl}_\bullet(\mathcal{O}_{Y_s}).$$

Since X is catenary, and hence Y is catenary, one checks that

$$\dim Y_s \leq \dim Y_t.$$

When f is of finite type, this follows immediately, after reducing to the case in which X is integral, from EGA IV 7.1.13. This proves the assertion.

7.5. "Conservation of number"

Suppose now that f is proper. Then X_t and X_s are proper over $k(t)$ and $k(s)$, respectively, and one has the homomorphisms defined in 2.3,

$$\chi_t : K_\bullet(X_t) \longrightarrow \mathbb{Z}, \qquad \chi_s : K_\bullet(X_s) \longrightarrow \mathbb{Z}.$$

For every $x \in K_\bullet(X_t)$ one has

$$\chi_s(\sigma_\bullet(x)) = \chi_t(x). \tag{7.5.1}$$

This is the *conservation-of-number theorem*; equivalently, the diagram

$$\begin{array}{ccc} K_\bullet(X_t) & \xrightarrow{\sigma_\bullet} & K_\bullet(X_s) \\ & \searrow^{\chi_t} & \downarrow{\scriptstyle \chi_s} \\ & & \mathbb{Z} \end{array} \tag{7.5.2}$$

commutes.

To prove (7.5.1), one is again reduced to the case $x = \mathrm{cl}_\bullet(\mathcal{O}_{Y_t})$, where Y_t is an integral closed subscheme of X_t. With the notation of the proof of 7.4, it remains to prove

$$\chi(Y_s) = \chi(Y_t),$$

which is well known (EGA III 7.9.4, or Exposé III 5.7).

7.6

It is immediate that

$$\sigma_\bullet(\mathrm{cl}_\bullet(\mathcal{O}_{X_t})) = \mathrm{cl}_\bullet(\mathcal{O}_{X_s}). \tag{7.6.1}$$

Since $\sigma_\bullet$ is $K^\circ(X)$-linear, the following diagram

$$\begin{array}{ccccc} & & K^\circ(X) & & \\ & {\scriptstyle j^*}\swarrow & & \searrow{\scriptstyle i^*} & \\ K^\circ(X_t) & & & & K^\circ(X_s) \\ \downarrow & & & & \downarrow \\ K_\bullet(X_t) & & \xrightarrow[\sigma_\bullet]{} & & K_\bullet(X_s) \end{array} \tag{7.6.2}$$

commutes.

The vertical arrows are the canonical homomorphisms $K^\circ \to K_\bullet$ (IV 2.5.1).

7.7. Specialization for $K^\bullet$ and $\mathrm{Gr}^\bullet$

When X_t and X_s are regular (for example, when f is smooth), the two preceding vertical arrows are isomorphisms (IV 2.5). The specialization homomorphism can then be interpreted as a homomorphism

$$\sigma^\bullet : K^\bullet(X_t) \longrightarrow K^\bullet(X_s), \tag{7.7.1}$$

which is equivalently the unique homomorphism making the diagram

$$\begin{array}{ccc} K^\bullet(X) & \xrightarrow{j^*} & K^\bullet(X_t) \\ {\scriptstyle i^*}\downarrow & \swarrow{\scriptstyle \sigma^\bullet} & \\ K^\bullet(X_s) & & \end{array} \tag{7.7.2}$$

commute.

In this diagram i^* and j^* are ring homomorphisms (IV 2.7 b), and j^* is surjective. Hence (7.7.1) is a ring homomorphism. For the same reason, if X is separated, it is even a homomorphism

of λ-rings. Indeed, a regular separated noetherian scheme is divisorial (II 2.2.7.1), so its $K^\bullet$ is canonically identified with its "naive" $K^\bullet$ (IV 2.9(ii)), which carries a natural λ-ring structure (VI 3.2). Moreover, the regularity of X_t and X_s implies that X is regular (EGA 0_{IV} 17.1.8), and

$$i^* : K^\bullet(X) \longrightarrow K^\bullet(X_s), \qquad j^* : K^\bullet(X) \longrightarrow K^\bullet(X_t)$$

are λ-homomorphisms. Therefore the same is true of (7.7.1), which is obtained from i^* by passage to the quotient.

Since X is regular, hence normal, its irreducible components are its connected components, and they dominate S. Consequently the homomorphism

$$H^0(X, \mathbb{Z}) \longrightarrow H^0(X_t, \mathbb{Z})$$

induced by $X_t \to X$ is bijective. The homomorphism induced by $X_s \to X$ can therefore also be regarded as a specialization homomorphism

$$H^0(X_t, \mathbb{Z}) \longrightarrow H^0(X_s, \mathbb{Z}). \tag{7.7.3}$$

It follows immediately from the definitions that (7.7.1) is compatible, via (7.7.3), with the augmented-algebra structures of its source and target over $H^0(X_t, \mathbb{Z})$ and $H^0(X_s, \mathbb{Z})$, respectively, defined in IV 2.8.

Still assuming that X is separated, (7.7.1) induces a homomorphism of graded rings

$$\sigma^\bullet : \mathrm{Gr}^\bullet(X_t) \longrightarrow \mathrm{Gr}^\bullet(X_s). \tag{7.7.4}$$

Using the Chern-class construction of V 6, one obtains the commutative diagram

$$\begin{array}{ccc} K^\bullet(X_t) & \xrightarrow{c_{X_t}} & \mathrm{Ch}\ \mathrm{Gr}^\bullet(X_t) \\ \downarrow & & \downarrow \\ K^\bullet(X_s) & \xrightarrow[c_{X_s}]{} & \mathrm{Ch}\ \mathrm{Gr}^\bullet(X_s). \end{array}$$

7.8. Specialization for Pic

Retain the hypotheses of 7.7: X is separated, and X_t and X_s are regular. Using the canonical isomorphism (5.3.2)

$$\mathrm{Gr}^1 \simeq \mathrm{Pic},$$

the homomorphism (7.7.4) gives a canonical homomorphism

$$\mathrm{Pic}(X_t) \longrightarrow \mathrm{Pic}(X_s). \tag{7.8.1}$$

An equivalent description is as follows. Since X is regular, its local rings are factorial by the Auslander–Buchsbaum theorem (EGA IV 21.11.1). Thus every invertible sheaf L_t on X_t extends to an invertible sheaf L on X (EGA IV 21.6.11). The image of $\mathrm{cl}(L_t) \in \mathrm{Pic}(X_t)$ under (7.8.1) is $\mathrm{cl}(L_s) \in \mathrm{Pic}(X_s)$, where $L_s = L|_{X_s}$.

In fact, assume only that the local rings of X are factorial and that X_s is locally integral, without assuming regularity or separatedness. Then $\mathrm{cl}(L_s)$ is independent of the chosen extension L of L_t to an invertible sheaf on X, and this gives (7.8.1) directly. To prove the independence, it is enough to show that

$$L_t \simeq \mathcal{O}_{X_t} \quad \Longrightarrow \quad L_s \simeq \mathcal{O}_{X_s}.$$

The hypothesis means that $L \simeq \mathcal{O}_X(D)$ for a divisor D on X whose support is contained in X_s (EGA IV 21.2.11). One reduces at once to the case in which X_s is integral. Then $D = nX_s$, and $D = \mathrm{div}(g^n)$, where g is a uniformizer of A. Hence $\mathcal{O}_X(D) \simeq \mathcal{O}_X$, and a fortiori $L_s \simeq \mathcal{O}_{X_s}$.

7.9. Specialization and numerical equivalence

Suppose that f is proper and that X_s and X_t are regular. Consider the two specialization homomorphisms (7.4.1) and (7.7.4):

$$\begin{aligned} \sigma_\bullet &: \mathrm{Gr}_\bullet(X_t) \longrightarrow \mathrm{Gr}_\bullet(X_s), \\ \sigma^\bullet &: \mathrm{Gr}^\bullet(X_t) \longrightarrow \mathrm{Gr}^\bullet(X_s). \end{aligned} \tag{7.9.1}$$

They are induced by the homomorphism

$$\sigma^\bullet : K^\bullet(X_t) \longrightarrow K^\bullet(X_s). \tag{7.9.2}$$

Since (7.9.2) is compatible with the ring structures, formula (7.5.1) gives

$$\langle \sigma^\bullet(y_t), \sigma_\bullet(x_t) \rangle = \langle y_t, x_t \rangle\,, \qquad x_t \in \mathrm{Gr}_\bullet(X_t), \quad y_t \in \mathrm{Gr}^\bullet(X_t), \tag{7.9.3}$$

where the pairings between $\mathrm{Gr}^\bullet$ and $\mathrm{Gr}_\bullet$ are those of 4.1. It follows formally that, if $x \in \mathrm{Gr}_\bullet(X_t)$ (respectively, $x \in \mathrm{Gr}^\bullet(X_t)$) and $\sigma_\bullet(x)$ (respectively, $\sigma^\bullet(x)$) is numerically equivalent to zero (4.5.1), then x is numerically equivalent to zero. The converse holds if (7.9.2), or, equivalently, either of the homomorphisms in (7.9.1) obtained from it by passage to the associated graded groups, is surjective modulo torsion. This condition occurs only in exceptional cases.

7.9.4. It is plausible, at least when f is proper and smooth, that the converse just considered always holds: if x is numerically equivalent to zero, then so is $\sigma_\bullet(x)$ (respectively, $\sigma^\bullet(x)$). This would imply that the homomorphisms (7.9.1) induce monomorphisms on the corresponding groups modulo numerical equivalence,

$$\mathrm{Gr}_\bullet^{\mathrm{num}} \qquad \text{and} \qquad \mathrm{Gr}^\bullet_{\mathrm{num}}\,.$$

At least the preceding result shows that $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_t)$ is isomorphic to a quotient of a subgroup of $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_s)$; in particular, its rank is at most that of $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_s)$. The same holds for $\mathrm{Gr}^\bullet_{\mathrm{num}}$, and also for $K^\bullet_{\mathrm{num}}$.

7.10. Compatibility with direct and inverse images

Let

$$g : X \longrightarrow Y$$

be a morphism of noetherian schemes flat over S. Denote by $\sigma_{X,\bullet}$ and $\sigma_{Y,\bullet}$ the specialization homomorphisms for X and Y, and by i_X, j_X (respectively, i_Y, j_Y)

the inclusions of the special and generic fibers into the ambient schemes. Write

$$g_t : X_t \longrightarrow Y_t, \qquad g_s : X_s \longrightarrow Y_s$$

for the induced morphisms. They give the Cartesian squares

$$\begin{array}{ccc} X_t & \xrightarrow{g_t} & Y_t \\ {\scriptstyle j_X}\downarrow & & \downarrow{\scriptstyle j_Y} \\ X & \xrightarrow{g} & Y \end{array} \qquad\qquad \begin{array}{ccc} X_s & \xrightarrow{g_s} & Y_s \\ {\scriptstyle i_X}\downarrow & & \downarrow{\scriptstyle i_Y} \\ X & \xrightarrow{g} & Y. \end{array} \tag{7.10.0}$$

First suppose that g is proper. Then the diagram

$$\begin{array}{ccc} K_\bullet(X_t) & \xrightarrow{g_{t*}} & K_\bullet(Y_t) \\ {\scriptstyle \sigma_{X,\bullet}}\downarrow & & \downarrow{\scriptstyle \sigma_{Y,\bullet}} \\ K_\bullet(X_s) & \xrightarrow[g_{s*}]{} & K_\bullet(Y_s) \end{array} \tag{7.10.1}$$

commutes. This generalizes 7.5: in the present notation take $Y = S$ (equivalently, take the scheme denoted by X in (7.3.2) to be S); the specialization homomorphism for S is the identity on $\mathbb{Z}$. If X and Y satisfy the conditions of 7.7, that is, if their fibers are regular, then g_t and g_s automatically have finite Tor-dimension, and (7.10.1) may be read as a commutative square whose four terms are $K^\bullet$-groups. We leave this reformulation to the reader.

Using the definitions of $\sigma_{X,\bullet}$ and $\sigma_{Y,\bullet}$, the commutativity of (7.10.1) reduces to that of the two front faces of the prism

$$\begin{array}{ccccc} K_\bullet(X_t) & & \xrightarrow{\quad g_{t*} \quad} & & K_\bullet(Y_t) \\ & \nwarrow^{j_X^*} & & \nwarrow^{j_Y^*} & \\ & K_\bullet(X) & \xrightarrow{\quad g_* \quad} & K_\bullet(Y) & \\ & \swarrow_{i_X^*} & & \swarrow_{i_Y^*} & \\ K_\bullet(X_s) & & \xrightarrow[\quad g_{s*} \quad]{} & & K_\bullet(Y_s). \end{array} \tag{$*$}$$

These two faces are the base-change squares obtained from (7.10.0), so IV 3 applies. It remains only to verify that they correspond to the pairs (X, Y_t) and (X, Y_s) of schemes over Y, and that these pairs are Tor-independent over Y. This is clear for the first pair because $Y_t \to Y$ is flat. It is also immediate for the second, because $Y_s \to Y$ is a regular immersion of codimension 1 and remains so after base change by $X \to Y$.

Now drop the properness assumption and instead suppose that g has finite Tor-dimension. Then g_t and g_s also have finite Tor-dimension. This is immediate for g_t; for g_s it follows from the Tor-independence of X and Y_s over Y just noted. Hence one can form the square

$$\begin{array}{ccc} K_\bullet(Y_t) & \xrightarrow{g_t^*} & K_\bullet(X_t) \\ {\scriptstyle \sigma_{Y,\bullet}}\downarrow & & \downarrow{\scriptstyle \sigma_{X,\bullet}} \\ K_\bullet(Y_s) & \xrightarrow[g_s^*]{} & K_\bullet(X_s), \end{array} \tag{7.10.2}$$

whose horizontal arrows are those defined in IV 2.12. If X and Y satisfy the conditions of 7.7, that is, if their fibers are regular, then Y is regular and g automatically has finite Tor-dimension; the square may then be interpreted as a commutative square of $K^\bullet$-rings.

To prove the commutativity of (7.10.2), it is again enough to prove that of the two front faces of the prism analogous to $(*)$, with the direct-image homomorphisms g_{t*}, g_*, g_{s*} replaced by the inverse-image homomorphisms g_t^*, g^*, g_s^*. Their commutativity follows

immediately from that of the corresponding squares in (7.10.0), using the transitivity of inverse images (IV 2.12).

Remark 7.10.3. When g is proper, the commutativity of (7.10.1) immediately implies that of the following diagram, obtained by passage to the associated graded groups:

$$\begin{array}{ccc} \mathrm{Gr}_\bullet(X_t) & \xrightarrow{g_{t*}} & \mathrm{Gr}_\bullet(Y_t) \\ {\scriptstyle \sigma_{X,\bullet}}\downarrow & & \downarrow{\scriptstyle \sigma_{Y,\bullet}} \\ \mathrm{Gr}_\bullet(X_s) & \xrightarrow[g_{s*}]{} & \mathrm{Gr}_\bullet(Y_s). \end{array} \tag{7.10.1 bis}$$

Likewise, if X and Y are separated and have regular fibers, the commutativity of (7.10.2) implies that of

$$\begin{array}{ccc} \mathrm{Gr}^\bullet(Y_t) & \xrightarrow{g_t^*} & \mathrm{Gr}^\bullet(X_t) \\ {\scriptstyle \sigma_Y^\bullet}\downarrow & & \downarrow{\scriptstyle \sigma_X^\bullet} \\ \mathrm{Gr}^\bullet(Y_s) & \xrightarrow[g_s^*]{} & \mathrm{Gr}^\bullet(X_s). \end{array} \tag{7.10.2 bis}$$

If X and Y are moreover quasi-projective and smooth over S, then g_t^* and g_s^* are compatible with the filtrations by codimension of supports. Thus one obtains a commutative diagram of the same form as (7.10.2 bis), with $\mathrm{Gr}^\bullet$ replaced by $\mathrm{Gr}^\bullet_{\mathrm{top}}$ (Exposé 0, 2.1).

Finally, if X and Y are quasi-projective and separated over S and have regular fibers, then g_t and g_s are automatically complete-intersection morphisms (VIII 1.1). Using VIII 3.2, one obtains a commutative diagram analogous to (7.10.1 bis), with $\mathrm{Gr}_\bullet$ replaced by $\mathrm{Gr}^\bullet(-)_{\mathbb{Q}}$.

7.11. For a scheme T over an algebraically closed field k, one can define the subgroup of $K^\bullet(T)$ consisting of elements algebraically equivalent to zero as the subgroup of elements of the form

$$E(x_1) - E(x_0), \tag{7.11.1}$$

where x_0 and x_1 are two k-rational points of a proper, regular, connected algebraic curve C over k, where $E \in K^\bullet(T \times_k C)$, and where, for every k-rational point x of C, one sets

$$E(x) = i_x^*(E) \in K^\bullet(T), \qquad \text{where} \qquad i_x : T \longrightarrow T \times_k C$$

is the morphism $t \mapsto (t, x)$ defined by x. That the elements of the form (7.11.1) constitute a subgroup of $K^\bullet(T)$ follows by a standard argument, using a product $C \times_k C'$, the two points (x_1, x_1') and (x_0, x_0') of that product, and the fact that one can always find an irreducible algebraic curve through any two points of an irreducible quasi-projective algebraic scheme over k (here $C \times_k C'$). (A reader unwilling to admit this fact may define algebraic equivalence by the group generated by the elements of the form (7.11.1).)

The elements algebraically equivalent to zero form an ideal of $K^\bullet(T)$, as one sees at once. This ideal is moreover stable under the operations λ^i whenever they are defined, that is, when one works with the "naive" $K^\bullet$; it is also contained in the augmentation ideal of $K^\bullet(T)$, as is immediate. One can therefore define a quotient ring

$$K^\bullet_{\mathrm{alg}}(T), \tag{7.11.2}$$

which is even a λ-ring when one works with naive $K^\bullet$-groups.

In the same way one can define a subgroup of elements algebraically equivalent to zero in $K_\bullet(T)$, taking the element E above in $K_\bullet(T \times_k C)$ and observing that

$i_x : T \to T \times_k C$ is a perfect morphism: it is obtained, by flat base change along $T \times_k C \to C$, from the morphism $j_x : x \to C$, which is perfect because C is regular. One thereby obtains a quotient group

$$K_\bullet^{\mathrm{alg}}(T) \tag{7.11.3}$$

of $K_\bullet(T)$. This group is naturally a module over $K^\bullet_{\mathrm{alg}}(T)$, since the definitions show at once that the product of an element of $K^\bullet(T)$ with an element of $K_\bullet(T)$ is algebraically equivalent to zero if either factor is, using (IV 2.12.2) for i_x.

When one works with naive $K^\bullet$, define the ideal in $\mathrm{Gr}^\bullet(T)$ of elements algebraically equivalent to zero to be the ideal generated by the $c^i(x)$, where $i \geq 1$ and $x \in K^\bullet(T)$ is algebraically equivalent to zero. The quotient graded ring

$$\mathrm{Gr}^\bullet_{\mathrm{alg}}(T) \tag{7.11.4}$$

is then precisely the Chern ring associated with the λ-ring $K^{\bullet}_{\mathrm{alg}}(T)$.

Finally, when T is locally noetherian, define the graded subgroup of $\mathrm{Gr}_{\bullet}(T)$ consisting of elements algebraically equivalent to zero by requiring its component of dimension i to be the image in $\mathrm{Gr}_i(T)$ of the subgroup of $\mathrm{Fil}_i\, K_{\bullet}(T)$ formed by the elements algebraically equivalent to zero in $K_{\bullet}(T)$. It follows at once that the quotient graded group

$$\mathrm{Gr}^{\mathrm{alg}}_{\bullet}(T) \tag{7.11.5}$$

is a module over the ring $\mathrm{Gr}^{\bullet}_{\mathrm{alg}}(T)$, by passage to the quotient from the module structure of $\mathrm{Gr}_{\bullet}(T)$ over $\mathrm{Gr}^{\bullet}(T)$ (3.1).

7.11.6. The preceding definitions would still make sense without assuming k algebraically closed, provided that C above is taken geometrically connected over k. It is not clear, however, that these definitions would be very useful. In particular, it does not appear that, for the group $\mathrm{Pic}(T)$, one would

recover the classical "geometric" notion of algebraic equivalence considered in 2.4.1. We therefore prefer to say that an element of $K^{\bullet}(T)$ is algebraically equivalent to zero if its image in $K^{\bullet}(T \otimes_k \bar{k})$ is so, where $\bar{k}$ is an algebraic closure of k, and to proceed in the same way when defining algebraic equivalence in $K_{\bullet}(T)$, $\mathrm{Gr}^{\bullet}(T)$, and $\mathrm{Gr}_{\bullet}(T)$.

7.12. With these definitions in place, I claim that the *specialization homomorphism* (7.3.1) *carries elements algebraically equivalent to zero to elements algebraically equivalent to zero*, and consequently induces, by passage to the quotient, a homomorphism

$$K^{\mathrm{alg}}_{\bullet}(X_t) \longrightarrow K^{\mathrm{alg}}_{\bullet}(X_s). \tag{7.12.1}$$

To see this, one reduces immediately to the case in which A is complete and $k(s)$ is algebraically closed, using the straightforward compatibility made explicit in 7.15. It is enough to show that $\sigma_{\bullet}$ carries an element of the form (7.11.1) in $K_{\bullet}(X_t)$ to an element algebraically equivalent to zero in $K_{\bullet}(X_s)$. Here $T = X_t$, and accordingly we write C_t, E_t in place of C, E above. Since C_t is projective over $k(t)$, EGA IV 2.8.5 gives a projective flat scheme C over S whose generic fiber is C_t. Using Abhyankar's resolution of singularities for arithmetic surfaces [6], one sees that C can even be chosen regular. (N.B. Flatness is preserved under resolution, by EGA IV 14.3.8.) Since C is proper over S, the $k(t)$-rational points x_{0t} and x_{1t} of C_t extend to sections x_0, x_1 of C over S. Since C is regular, it is well known that C is smooth over S at the points of $x_0(S)$ and $x_1(S)$ (EGA IV 17.12.1(c)⇒(a), and 19.1.1). Consequently,

the immersions $x_i : S \to C$ $(i = 0, 1)$ are regular and remain regular immersions $x_i : s \to C_s$ after base change by $s \to S$. Note that, by the exact homotopy sequence (IX 1.1), the element

$$E_t \in K_{\bullet}(X_t \times_{k(t)} C_t)$$

comes from an element $E \in K_{\bullet}(X \times_S C)$. Thus the element $E_t(x_{1t}) - E_t(x_{0t})$ of $K_{\bullet}(X_t)$ comes from the element $E(x_1) - E(x_0)$ of $K_{\bullet}(X)$, and its image in $K_{\bullet}(X_s)$ under (7.3.1) is therefore

$$i^*\big(E(x_1) - E(x_0)\big). \tag{*}$$

For a section x of C over S, of course, we set

$$E(x) = i_x^*(E), \qquad \text{where} \qquad i_x : X \longrightarrow X \times_S C$$

is defined as above. It may be viewed as obtained from x by the flat base change $X \times_S C \to C$; it is therefore a regular immersion and, a fortiori, a perfect morphism, so that the expression

$E(x)$ is well defined. For the same reason $E_s(x_s)$ is defined. By transitivity of inverse images (IV 2.12) in the following commutative diagram of perfect morphisms,

$$\begin{array}{ccc} X & \xrightarrow{i_x} & X \times_S C \\ {\scriptstyle i}\uparrow & & \uparrow \\ X_s & \xrightarrow{i_{xs}} & X_s \times_{k(s)} C_s \end{array},$$

the expression $(*)$ is equal to

$$E_s(x_{1s}) - E_s(x_{0s}). \qquad (**)$$

Because $k(s)$ is algebraically closed, the reduced irreducible components of C_s are geometrically integral over $k(s)$, and their intersection points are rational over $k(s)$. It follows readily that $(**)$ is algebraically equivalent to zero in $K_\bullet(X_s)$, using the connectedness

of C_s, which follows from the geometric connectedness of C_t.

From (7.12.1), using 7.4, one obtains a homomorphism

$$\mathrm{Gr}^{\mathrm{alg}}_\bullet(X_t) \longrightarrow \mathrm{Gr}^{\mathrm{alg}}_\bullet(X_s). \qquad (7.12.2)$$

If X_s and X_t are moreover regular, then (7.12.1) can also be interpreted as a ring homomorphism

$$K^\bullet_{\mathrm{alg}}(X_t) \longrightarrow K^\bullet_{\mathrm{alg}}(X_s), \qquad (7.12.3)$$

which, if X is also separated, is a homomorphism of λ-rings and therefore induces a homomorphism

$$\mathrm{Gr}^\bullet_{\mathrm{alg}}(X_t) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{alg}}(X_s). \qquad (7.12.4)$$

7.12.5. Let T be a smooth proper algebraic scheme over a field k. It is easy to verify that an element $x \in K^\bullet(T)$ for which there is an integer $n \geq 1$ such that nx is algebraically equivalent to zero is numerically equivalent to zero. It is plausible that the converse is always true, at least when T is projective. In considering this question, one may plainly assume that k is algebraically closed and replace $K^\bullet(T)$ by $\mathrm{Gr}^\bullet(T)$ or $\mathrm{Gr}_\bullet(T)$ without changing the conjecture. One obtains an equivalent conjecture by working with the Chow ring $A(T)$ (cf. XIV 4.1), namely the more or less classical conjecture that, if Z is an algebraic cycle on T numerically equivalent to zero, then there is an integer $n \geq 1$ such that nZ is algebraically equivalent to zero. If this conjecture holds for X_t, the compatibility of (7.3.1) with algebraic equivalence would imply that this specialization homomorphism, when $f : X \to S$ is projective and smooth, is also compatible with numerical equivalence, as conjectured in 7.9.

7.13. Relations with ℓ-adic cohomology

Recall (SGA 5 VII) that, for every locally free module of finite type E on a scheme T, and every prime number ℓ distinct from the residue characteristics of T, one defines ℓ-adic Chern classes (for $i \geq 0$)

$$c^i(E) \in H^{2i}(T, \mathbb{Z}_\ell(i)),$$

satisfying the additivity formula

$$c(E) = c(E')c(E'')$$

when E is an extension of a locally free module E' by a locally free module E''. It follows from this additivity that the total Chern class $c(E) = \sum_i c^i(E)$ arises from a homomorphism

$$c : K^\bullet(T) \longrightarrow \widehat{G}(H^{2*}(T, \mathbb{Z}_\ell(*))) = 1 + \prod_{i \geq 1} H^{2i}(T, \mathbb{Z}_\ell(i)), \qquad (7.13.0)$$

where the target is a group under the product explained in V 1, and where $K^\bullet(T)$ is the "naive" $K^\bullet$. One obtains canonical maps

$$c^i : K^\bullet(T) \longrightarrow H^{2i}(T, \mathbb{Z}_\ell(i)). \tag{7.13.1}$$

Now let X be a proper flat scheme over S, with regular fibers, and suppose that S is henselian. Assume that ℓ is different from the residue characteristic of S. One can then form the diagram

$$\begin{array}{ccc} K^\bullet(X_t) & \xrightarrow{c^i} & H^{2i}(X_t, \mathbb{Z}_\ell(i)) \\ \Big\downarrow{\scriptstyle \sigma} & & \Big\uparrow{\scriptstyle \sigma^{-1}} \\ K^\bullet(X_s) & \xrightarrow[c^i]{} & H^{2i}(X_s, \mathbb{Z}_\ell(i)) \end{array} , \tag{7.13.2}$$

where the horizontal arrows are the Chern-class maps for X_t and X_s, respectively, and where σ^{-1} is the composite homomorphism

$$H(X_s) \xleftarrow[\sim]{i^*} H(X) \xrightarrow{j^*} H(X_t)\,;$$

the arrows displayed here are restriction homomorphisms, and one uses the fact that i^* is an isomorphism by the proper base change theorem (SGA 4 XII 5.5). I claim that the square (7.13.2) is commutative. The proof is essentially the same as the proof of the commutativity of (7.10.2): it is enough to use the compatibility of the formation of Chern classes with inverse images for the morphisms $j : X_t \to X$ and $i : X_s \to X$.

7.13.3. The square (7.13.2) is obtained by passage to the projective limit from the analogous squares in which cohomology with coefficients in the ℓ-adic sheaf $\mathbb{Z}_\ell(i)$ is replaced by cohomology with coefficients in the torsion sheaves $\mu_n^{\otimes i}$, where n runs through the powers of ℓ. Thus the commutativity of (7.13.2) merely expresses the commutativity of the squares involving Chern classes modulo n, for every power n of ℓ.

Choose an algebraic closure $\bar{K}$ of $K = k(t)$, and hence an algebraic closure $\bar{k}$ of $k = k(s)$, namely the residue field of the integral closure of A in $\bar{K}$. For every finite subextension K_α of $\bar{K}$, one can consider the analogous commutative diagram, for fixed n, in the situation obtained from $f : X \to S$ by the base change $S_\alpha \to S$, where S_α is the normalization of S in K_α. By Krull–Akizuki and the henselian hypothesis on S, the scheme S_α is indeed a trait. Here one must moreover assume that the fibers of X over S are geometrically regular, that is, that X is smooth over S, so that this hypothesis is preserved by base change. As α varies, one obtains an inductive system of commutative squares. Passing to the inductive limit and using SGA 4 VII 5.8 and IV 3.2, one obtains a commutative square (cf. 7.16):

$$\begin{array}{ccc} K^\bullet(X_{\bar{t}}) & \xrightarrow{c^i} & H^{2i}(X_{\bar{t}}, \mu_n^{\otimes i}) \\ \Big\downarrow{\scriptstyle \sigma} & & \Big\downarrow{\scriptstyle \sigma'} \\ K^\bullet(X_{\bar{s}}) & \xrightarrow[c^i]{} & H^{2i}(X_{\bar{s}}, \mu_n^{\otimes i}) \end{array} ,$$

where $X_{\bar{t}}$ and $X_{\bar{s}}$ are the geometric fibers of X. Moreover, the specialization homomorphism σ^{-1} for cohomology is now known to be an isomorphism (SGA 4 XVI 2.2). Hence the commutativity of the preceding square can also be expressed as the commutativity of the square obtained by reversing that arrow. Passing to the projective limit over n in the resulting commutative squares

gives a commutative square

$$\begin{array}{ccc} K^\bullet(X_{\bar{t}}) & \longrightarrow & H^{2i}(X_{\bar{t}}, \mathbb{Z}_\ell(i)) \\ {\scriptstyle\sigma}\downarrow & & \downarrow{\scriptstyle\sim} \\ K^\bullet(X_{\bar{s}}) & \longrightarrow & H^{2i}(X_{\bar{s}}, \mathbb{Z}_\ell(i)) \end{array} . \tag{7.13.4}$$

7.13.5. Another way to express this commutativity is to say that, if $x_t \in K^\bullet(X_t)$ and x_s is its image in $K^\bullet(X_s)$, then $c^i(x_{\bar{t}})$ and $c^i(x_{\bar{s}})$ are the values at the geometric points $\bar{t}$ and $\bar{s}$, respectively, of a section of $R^{2i}f_*(\mathbb{Z}_\ell(i))$. This section is easily interpreted: take an element $x \in K^\bullet(X)$ inducing x_t. Then $c^i(x) \in H^{2i}(X, \mathbb{Z}_\ell(i))$ defines the required section of $R^{2i}f_*(\mathbb{Z}_\ell(i))$.

7.13.6. Let T be a scheme. By passage to the associated graded groups, the homomorphism (7.13.0) induces homomorphisms

$$\psi_i : \mathrm{Gr}^i(T) \longrightarrow H^{2i}(T, \mathbb{Z}_\ell(i)),$$

the image under ψ_i of an element $x \in \mathrm{Filt}^i K^\bullet(T)$ being, by definition, $c^i(x)$. (The compatibility of (7.13.0) with the λ-filtration on $K^\bullet(T)$ and the degree filtration on the target follows readily from V 6.6.4 and from the fact that the c^i define a Chern theory on $K^\bullet(T)$ in the sense of V 6.13, when $K^\bullet(T)$ is regarded as a λ-algebra augmented over the binomial ring $H^0(T, \mathbb{Z})$.) One thereby obtains maps

$$\mathrm{cl} : \mathrm{Gr}^i(T) \longrightarrow H^{2i}(T, \mathbb{Q}_\ell(i)) \overset{\mathrm{dfn}}{=} H^{2i}(T, \mathbb{Z}_\ell(i)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell, \tag{7.13.7}$$

by the formula

$$\mathrm{cl}(x^{(i)}) = \frac{1}{(-1)^{i-1}(i-1)!}\, \psi_i(x^{(i)}), \tag{7.13.8}$$

compare XIV 5.1 and 5.2. The commutativity of (7.13.2) and (7.13.4) immediately implies that of the analogous diagrams

$$\begin{array}{ccc} \mathrm{Gr}^i(X_t)_{\mathbb{Q}} & \xrightarrow{\mathrm{cl}} & H^{2i}(X_t, \mathbb{Q}_\ell(i)) \\ {\scriptstyle\sigma_{\mathbb{Q}}}\downarrow & & \uparrow{\scriptstyle\sigma'} \\ \mathrm{Gr}^i(X_s)_{\mathbb{Q}} & \xrightarrow[\mathrm{cl}]{} & H^{2i}(X_s, \mathbb{Q}_\ell(i)) \end{array} , \tag{7.13.9}$$

and

$$\begin{array}{ccc} \mathrm{Gr}^i(X_{\bar{t}})_{\mathbb{Q}} & \xrightarrow{\mathrm{cl}} & H^{2i}(X_{\bar{t}}, \mathbb{Q}_\ell(i)) \\ \downarrow & & \downarrow{\scriptstyle\sim} \\ \mathrm{Gr}^i(X_{\bar{s}})_{\mathbb{Q}} & \xrightarrow[\mathrm{cl}]{} & H^{2i}(X_{\bar{s}}, \mathbb{Q}_\ell(i)) \end{array} . \tag{7.13.10}$$

Remark 7.13.11. The commutativity of the last two diagrams can be expressed by saying that specialization of algebraic cycles is compatible with formation of the associated ℓ-adic cohomology classes. This statement is not very satisfactory in the form just given, because we had to replace the coefficients $\mathbb{Z}_\ell(i)$ by $\mathbb{Q}_\ell(i)$, that is, to neglect torsion. To obtain a statement that does not neglect torsion, it seems that one will be forced to work with the Chow ring, since it is on that ring that the homomorphism cl is naturally defined with values in $H^{2*}(-, \mathbb{Z}_\ell(*))$; compare XIV 4.1. Unfortunately, no theory of specialization for the Chow ring has yet been developed (cf. 7.14) that would allow one to write commutative squares analogous to (7.13.9)

and (7.13.10) in terms of Chow rings and cohomology with coefficients in $\mathbb{Z}_\ell(*)$. All one can say when $f : X \to S$ is smooth and proper is that there is a commutative square

$$\begin{array}{ccc} Z^i(X_t) & \xrightarrow{\text{cl}} & H^{2i}(X_t, \mathbb{Z}_\ell(i)) \\ \Big\downarrow{\scriptstyle \sigma} & & \Big\uparrow \\ Z^i(X_s) & \xrightarrow[\text{cl}]{} & H^{2i}(X_s, \mathbb{Z}_\ell(i)) \end{array} , \tag{7.13.12}$$

analogous to (7.13.9), together with a corresponding square analogous to (7.13.10), which we leave to the reader to write down. Here Z^i denotes the group of algebraic cycles of codimension i, modulo no relation at all; the horizontal arrows are the associated-cohomology-class homomorphisms defined in SGA 5 IV; and σ is the specialization homomorphism for algebraic cycles, associating with a cycle Z_t the cycle Z_s, where Z is the schematic closure of Z_t in X (cf. 7.14). This commutativity follows from the following fact, proved in SGA 5 IV. If Z is a closed subscheme of X of codimension at least i on every fiber, one can associate with it a cohomology class

$$\text{cl}(Z) \in H^{2i}(X, \mathbb{Z}_\ell(i))$$

whose formation commutes with every base change and whose restriction to X_t (respectively, X_s) is therefore $\text{cl}(Z_t)$ (respectively, $\text{cl}(Z_s)$). Of course, once it is known that the homomorphism σ in (7.13.12) carries cycles rationally equivalent to zero to cycles rationally equivalent to zero, with X projective over S, one can replace the Z^i by the A^i in this diagram and obtain a diagram of the desired kind in terms of the Chow rings of X_t and X_s.

Remark 7.13.13. Let T be a smooth projective scheme over an algebraically closed field. It follows immediately from the formalism developed in SGA 5 IV that, if Z is a cycle of codimension i on T whose image in $H^{2i}(T, \mathbb{Q}_\ell(i))$ is zero, then Z is numerically equivalent to zero. One of the standard conjectures on algebraic cycles asserts that the converse is also true. By VII 4.11, one obtains an equivalent conjecture by replacing the group of algebraic cycles with $\text{Gr}^\bullet(T)_\mathbb{Q}$, using the definition (7.13.8), or with $K^\bullet(T)$, by stating that an element of $K^\bullet(T)$ is numerically equivalent to zero if and only if its augmentation and its Chern classes in the groups $H^{2i}(T, \mathbb{Q}_\ell(i))$ vanish.

It follows immediately from the commutativity of (7.13.10) that, if $X_{\bar{t}}$ satisfies the preceding conjecture, then the specialization homomorphism

$$\text{Gr}^\bullet(X_{\bar{t}}) \longrightarrow \text{Gr}^\bullet(X_{\bar{s}})$$

(and, a fortiori, $\text{Gr}^\bullet(X_t) \to \text{Gr}^\bullet(X_s)$) is compatible with numerical equivalence, as conjectured in 7.9. The conjecture used here (numerical equivalence implies $\mathbb{Q}_\ell$-cohomological equivalence) is weaker than the conjecture considered in 7.12.5.

7.14. Specialization homomorphism for groups of cycles and cycle classes

If T is a noetherian scheme, denote by $Z_i(T)$ (respectively, $Z^i(T)$) the group of algebraic cycles on T that are purely of dimension i (respectively, purely of codimension i). When $f : X \to S$ is of finite type, one defines a specialization homomorphism

$$\sigma_i : Z_i(X_t) \longrightarrow Z_i(X_s) \tag{7.14.1}$$

as follows. For every prime cycle Z_t of X_t, purely of dimension i and identified with an integral subscheme of X_t, define $\sigma_i(Z_t)$ to be the cycle associated with the closed subscheme Z_s of X_s, where Z is the schematic closure of Z_t in X. This Z is a closed subscheme of X flat over S

(EGA IV 2.8), with generic fiber equal to the original Z_t, and it satisfies $\dim Z_s = \dim Z_t$ (EGA IV 7.1.13). When X_s is locally equidimensional, that is, when all irreducible components passing through a given point have the same dimension, the specialization homomorphism just defined immediately induces homomorphisms

$$\sigma^i : Z^i(X_t) \longrightarrow Z^i(X_s). \tag{7.14.1 bis}$$

Now suppose that f is quasi-projective, so that rational equivalence of cycles is available on X_t and X_s (cf. [5], where the theory is developed over an algebraically closed field). It seems very plausible that (7.14.1) carries cycles rationally equivalent to zero to cycles rationally equivalent to zero and therefore induces a corresponding homomorphism on the "Chow" groups of cycle classes,

$$\sigma_i : A_i(X_t) \longrightarrow A_i(X_s), \tag{7.14.2}$$

or, when X_s is locally equidimensional,

$$\sigma^i : A^i(X_t) \longrightarrow A^i(X_s). \tag{7.14.2 bis}$$

One might try to prove this by a method paralleling that of 7.12, but here resolution of singularities and reduction to an algebraically closed residue field would be unnecessary, since $C = \mathbb{P}^1_S$ is available as a "parameter curve" along which to deform cycles. A difficulty nevertheless arises. With the notation of 7.12, E must now denote a cycle on $\mathbb{P}^1_S \times_S X$ for which $E_t(x_{1t}) - E_t(x_{0t})$ is "defined," but it does not follow that the cycle $E_s(x_{1s}) - E_s(x_{0s})$ is also defined. As was to be expected, one is thus reduced to formulating a statement of the kind of Chow's moving lemma in order to define (7.14.2). Observe, however, that the existence of (7.4.1) shows that (7.14.1) is compatible with the equivalence relation on cycles defined by the surjective homomorphism

$$Z_\bullet(T) \longrightarrow \mathrm{Gr}_\bullet(T),$$

which is only slightly coarser than linear equivalence. Moreover, if T is quasi-projective and smooth over a field, this relation differs from linear equivalence only by torsion (XIV 4.1, 4.2). In that case one therefore obtains a homomorphism analogous to (7.14.2), or better to (7.14.2 bis), with both sides tensored by $\mathbb{Q}$.

The most natural way to define (7.14.2) directly would seem to be to proceed as in 7.3. This would require a theory of Chow groups A_i (or of the Chow ring in the regular case) for schemes more general than schemes quasi-projective over a field, broad enough to include X, together with a contravariant law for these groups making it possible to define the restriction homomorphism to X_s,

$$A_i(X) \longrightarrow A_i(X_s).$$

Compare XIV 8. Once such a sufficiently general theory of Chow groups and rings is available, it should be possible to transcribe into Chow theory all the results stated above for the specialization homomorphism in K-theory.

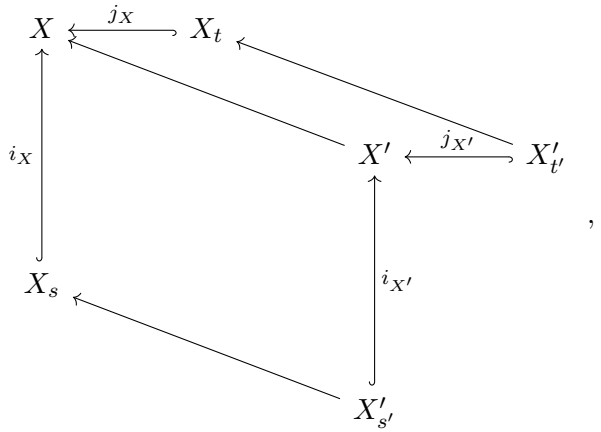
7.15. Varying S

Let S' be a second trait, and let

$$S' \longrightarrow S$$

be a morphism of traits, that is, a dominant local morphism, carrying the closed point to the closed point and the generic point to the generic point. Denote the closed point (respectively, the

generic point) of S' by s' (respectively, t'), and set $X' = X \times_S S'$. Suppose that X' is noetherian, as is the case, for example, when X is of finite type over S. There is a commutative diagram



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in which the oblique arrows are flat morphisms. One can therefore take the inverse-image homomorphisms on $K_\bullet$ for every arrow of the diagram (IV 2.12). Applying Definition 7.3, one finds that the diagram

$$\begin{array}{ccc} K_\bullet(X_t) & \xrightarrow{\sigma_X} & K_\bullet(X_s) \\ \downarrow & & \downarrow \\ K_\bullet(X'_{t'}) & \xrightarrow[\sigma_{X'}]{} & K_\bullet(X'_{s'}) \end{array} \tag{7.15.1}$$

is commutative, where the vertical arrows are the inverse-image homomorphisms. One immediately obtains an analogous commutative diagram with $\mathrm{Gr}_\bullet$ in place of $K_\bullet$. When X_s and X_t are regular, one may likewise write $K^\bullet$ and $\mathrm{Gr}^\bullet$ in place of $K_\bullet$ and $\mathrm{Gr}_\bullet$.

7.16. Passage to geometric fibers

Let $\bar{K}$ be an algebraic closure of $K = k(t)$, expressed as the inductive limit of its finite subextensions K_i, and let $\bar{A} = \varinjlim_i \bar{A}_i$ be the normalization of A in $\bar{K}$, the inductive limit of its normalizations $\bar{A}_i$ in the K_i. Choose a maximal ideal $\bar{\mathfrak{m}}$ of $\bar{A}$; it induces a maximal ideal $\mathfrak{m}_i$ on each $\bar{A}_i$. Set $B_i = (\bar{A}_i)_{\mathfrak{m}_i}$. By Krull–Akizuki, B_i is again a discrete valuation ring. Denote its spectrum by S_i and its closed and generic points by s_i and t_i, respectively. The residue field $\bar{k} = \bar{A}/\bar{\mathfrak{m}}$ is an algebraic closure of $k = k(s)$ and is the inductive limit of the residue fields $k_i = \bar{A}_i/\mathfrak{m}_i = k(s_i)$ of the B_i. Set $\bar{t} = \mathrm{Spec}(\bar{K})$ and $\bar{s} = \mathrm{Spec}(\bar{k})$.

Assume now that X is of finite type over S. Then $X_{\bar{t}}$ and $X_{\bar{s}}$ are of finite type over $\bar{t}$ and $\bar{s}$, respectively, and hence are noetherian. By IV 3.2.3,

$$K_\bullet(X_{\bar{t}}) \xleftarrow{\sim} \varinjlim_i K_\bullet(X_{t_i}), \qquad K_\bullet(X_{\bar{s}}) \xleftarrow{\sim} \varinjlim_i K_\bullet(X_{s_i}).$$

On the other hand, 7.15 gives a homomorphism from the inductive system of the $K_\bullet(X_{t_i})$ to that of the $K_\bullet(X_{s_i})$. Passing to the limit gives a canonical homomorphism

$$K_\bullet(X_{\bar{t}}) \longrightarrow K_\bullet(X_{\bar{s}}), \tag{7.16.1}$$

again, as expected, called the *specialization homomorphism*. By 7.4, it induces a homomorphism $\mathrm{Gr}_\bullet(X_{\bar{t}}) \to \mathrm{Gr}_\bullet(X_{\bar{s}})$, using the fact (whose verification is left to the reader) that the functor $\mathrm{Gr}_\bullet$, like $K_\bullet$, commutes with the limits occurring here. Similarly, when f is smooth, one may

interpret (7.16.1) as a homomorphism on $K^\bullet$; it is then a ring homomorphism, and even a λ-ring homomorphism if X is separated. In that case, it therefore induces a homomorphism on $\mathrm{Gr}^\bullet$.

7.16.2. We shall not review here, for geometric fibers, the extension of all the properties of specialization homomorphisms proved in the preceding sections for the "arithmetic" fibers; they all follow immediately from the latter by passage to the limit. Let us mention only the following interesting consequence of 7.9.4. When f is smooth and proper, $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_{\bar{t}})$ is isomorphic to a quotient of a subgroup of $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_{\bar{s}})$; in particular, its rank is bounded above by that of the latter group. This is therefore a semi-continuity result for the rank of the group of cycles modulo numerical equivalence.

7.17. Case of an arbitrary base

Let

$$f : X \longrightarrow S$$

be a flat and quasi-compact morphism of noetherian preschemes, and let s, t be two points of S such that s is a specialization of t. We wish to relate the groups $K_\bullet$, etc., for the two fibers X_t and X_s, as well as for the corresponding geometric fibers. For this, note (EGA II 7.1.9) that there exists a morphism

$$g : S' \longrightarrow S,$$

where S' is a trait and g carries the closed point s' to s and the generic point t' to t; one may even assume that $k(t) \xrightarrow{\sim} k(t')$. Suppose that $X' = X \times_S S'$ is noetherian. This is the case if f is of finite type or if S' is essentially of finite type over S (EGA IV 1.3.8), as one may assume when S itself is universally Japanese (EGA 0_{IV} 23.1.1). The specialization homomorphism for X' then defines a homomorphism

$$\sigma_\bullet : K_\bullet(X_t) \xrightarrow{\sim} K_\bullet(X_{t'}) \longrightarrow K_\bullet(X_{s'}). \tag{7.17.1}$$

Thus one obtains a specialization homomorphism on $K_\bullet(X_t)$, but with values not in $K_\bullet(X_s)$, but in $K_\bullet(X_{s'})$, where $X_{s'}$ is obtained from X_s by an extension $k(s) \to k(s')$ of the base fields (which may be assumed to be of finite type if S is universally Japanese). Passing to geometric fibers as explained in 7.16, one likewise obtains a homomorphism

$$\sigma_\bullet : K_\bullet(X_{\bar{t}}) \longrightarrow K_\bullet(X_{\overline{s'}}). \tag{7.17.2}$$

Here again, the "geometric" specialization homomorphism takes its values not in $K_\bullet(X_{\bar{s}})$, but in $K_\bullet(X_{\overline{s'}})$, where $X_{\overline{s'}}$ is obtained from $X_{\bar{s}}$ by the extension of algebraically closed fields $k(\bar{s}) \to k(\overline{s'})$.

7.17.3. In general, the homomorphism $K_\bullet(X_{\bar{s}}) \to K_\bullet(X_{\overline{s'}})$ is not an isomorphism, because of the possible presence of "continuous" components in $K_\bullet(X_{\bar{s}})$. It is, however, easy to verify that it is an isomorphism modulo algebraic equivalence (7.11), and also modulo numerical equivalence (4.5.1) if f is smooth and proper. Consequently, using 7.12, one finds that (7.17.2) induces a specialization homomorphism

$$K_\bullet^{\mathrm{alg}}(X_{\bar{t}}) \longrightarrow K_\bullet^{\mathrm{alg}}(X_{\bar{s}}); \tag{7.17.3.1}$$

and, if f is proper and smooth, provided that the specialization homomorphism (7.17.2) is compatible with numerical equivalence (cf. 7.9.4), one likewise obtains a homomorphism

$$K_\bullet^{\mathrm{num}}(X_{\bar{t}}) \longrightarrow K_\bullet^{\mathrm{num}}(X_{\bar{s}}). \tag{7.17.3.2}$$

We leave to the reader the task of developing, if needed, the corresponding constructions for $\mathrm{Gr}_\bullet$, $K^\bullet$, and $\mathrm{Gr}^\bullet$, and the properties of these specialization homomorphisms that follow immediately

from those developed in the preceding sections. Let us mention only that, by 7.16.2, $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_{\bar{t}})$ is isomorphic to a quotient of a subgroup of $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_{\bar{s}})$ when f is proper and smooth; in particular, its rank is again bounded above by the rank of the latter group.

7.17.4. When S is regular at s and t is a maximal point, one can construct S' by blowing up s in the localization of S at s, and then localizing at the generic point of the fiber of s in the blow-up. One then obtains a *pure extension* $k(s')$ of $k(s)$. Now it is not difficult to show that, if k' is a pure extension of a field k, then for every noetherian scheme T over k the canonical homomorphism

$$K_\bullet(T) \longrightarrow K_\bullet(T \otimes_k k')$$

is an isomorphism: represent k' as the inductive limit of the coordinate rings of nonempty affine open subsets of schemes of the form $\operatorname{Spec} k[T_1, \ldots, T_n]$, and use IX 1.6. It follows that (7.17.1) may be viewed as a canonical homomorphism

$$K_\bullet(X_t) \longrightarrow K_\bullet(X_s). \tag{7.17.4.1}$$

One should note, however, that this does not yield a homomorphism $K_\bullet(X_{\bar{t}}) \to K_\bullet(X_{\bar{s}})$: after passing to a finite extension K_i of $K = k(t)$ and normalizing S in K_i, the regularity hypothesis on S is lost.

Another way to define (7.17.4.1) is to reduce to the case in which S is local with closed point s (which is immediate), and to consider an increasing sequence of regular closed subschemes S_i of S, with generic points t_i, such that

$$\dim S_i = i, \qquad 0 \leq i \leq n = \dim S.$$

For every i, $0 \leq i \leq n-1$, the local ring $\mathcal{O}_{S_{i+1},t_i}$ is a discrete valuation ring with residue field $k(t_i)$ and fraction field $k(t_{i+1})$. There is therefore a specialization homomorphism

$$K_\bullet(X_{t_{i+1}}) \longrightarrow K_\bullet(X_{t_i}),$$

and the composite of these homomorphisms for $0 \leq i \leq n-1$ defines (7.17.4.1). By an argument analogous to that of 7.3, one can show that this is the same homomorphism as the one defined above by means of a blow-up. The advantage of the second method is that one even has homomorphisms of the form (7.16.1):

$$K_\bullet(X_{\bar{t}_{i+1}}) \longrightarrow K_\bullet(X_{\bar{t}_i}),$$

with suitable coherent choices of algebraic closures $k(\bar{t}_i)$ of the $k(t_i)$. Their composite defines a homomorphism

$$K_\bullet(X_{\bar{t}}) \longrightarrow K_\bullet(X_{\bar{s}}), \tag{7.17.4.2}$$

which probably depends essentially on the choice of the S_i, that is, of the t_i.

7.17.5. The construction of a specialization homomorphism (7.17.4.2) can be generalized to the case in which S is no longer assumed regular at s and t is no longer assumed to be a maximal point, provided that resolution of singularities is available for the integral closed subscheme T of S with generic point t, or merely for its localization at s. This hypothesis holds, in particular, if $\mathcal{O}_{S,s}$ is excellent of characteristic zero, by Hironaka. Indeed, one can then find a morphism

$$S' \longrightarrow \operatorname{Spec}(\mathcal{O}_{T,s})$$

with S' regular. Taking a closed point of the fiber S'_s, one can apply the preceding construction to $X' = X \times_S S'$ over S', its generic point t', and its closed point s'. This yields a homomorphism of the form (7.17.4.2), since $k(s')$ is a finite extension of $k(s)$ and therefore has the same algebraic closure.

7.17.6. In all the constructions considered in 7.17, one has reduced the construction of a specialization homomorphism to the case in which the base is a trait. It follows that the properties of the specialization homomorphism reviewed in the preceding sections for a trait as base extend automatically to the case of a more general base. Of course, the methods of definition considered apply equally well to $\mathrm{Gr}_\bullet$, to $K^\bullet$ when f is smooth, and to $\mathrm{Gr}^\bullet$ when f is also separated.

7.17.7. Finally, note that the definition of a homomorphism (7.17.3.1) or (7.17.3.2), or of a homomorphism (7.17.4.2), depends essentially on certain arbitrary choices: a suitable trait over S in 7.17.3, a sequence (S_i) in 7.17.4, and, in addition, the choice of a resolution of singularities in 7.17.5. The same is true of the variants for $\mathrm{Gr}_\bullet$, etc. Obvious examples can be constructed with s, say, an ordinary double point of an algebraic curve and X projective and smooth over S (for example, an abelian scheme of relative dimension 2), in which the specialization homomorphism on $\mathrm{Gr}^1 = \mathrm{Pic}$, and even on $\mathrm{Gr}^1_{\mathrm{alg}} = \mathrm{NS}$ (the Néron–Severi group), actually depends on the trait S' chosen over S, or, what amounts to the same thing here, on the choice of a point above s in the normalization of S.

7.18. Appendix to the Appendix: base-field extension for certain functors with values in commutative groups

We spell out here a technical point raised in 7.17.4.

7.18.1. Let k be a field, let C be the category of integral k-algebras and flat homomorphisms of such algebras, and let F be a functor on C with values in the category (Ens) of sets. Suppose that F commutes with filtered inductive limits. Given an extension K of k, we seek criteria for

$$F(k) \longrightarrow F(K)$$

to be injective (respectively, surjective).

7.18.2. Suppose that one of the following two conditions holds:

a) The field k is separably closed, and K/k is a separable extension.

b) The field k is infinite, and K is a pure extension of k.

Then the map $F(k) \to F(K)$ is injective.

Indeed, K is the filtered inductive limit of its finite-type k-subalgebras, with flat transition morphisms. Using this fact and the commutation of F with filtered inductive limits, one is reduced in case a) to proving the following: if A is an integral k-algebra of finite type whose fraction field K is separable over k, then $F(k) \to F(A)$ is injective. The hypotheses on k and K imply that there is a k-homomorphism $A \to k$; the composite

$$F(k) \longrightarrow F(A) \longrightarrow F(k)$$

is therefore the identity, and the assertion follows. In case b), passage to the limit similarly reduces the question to an integral finite-type k-algebra A whose fraction field is a pure extension of k. Since k is infinite, the hypothesis again implies the existence of a k-homomorphism $A \to k$, and one concludes as above.

7.18.3. Suppose now that F takes values in the category (Ab) of abelian groups, and suppose that for every finite extension k' of k a norm homomorphism

$$N_{k'/k} : F(k') \longrightarrow F(k)$$

has been defined, such that the composite

$$F(k) \longrightarrow F(k') \xrightarrow{N_{k'/k}} F(k)$$

is multiplication by $[k' : k]$. Under these conditions:

(i) for every extension K of k, the kernel of $F(k) \to F(K)$ is a torsion group;

(ii) for every pure extension K of k, the homomorphism $F(k) \to F(K)$ is injective. (Note that k is not assumed infinite.)

For (i), it is enough to prove that every element x in the kernel of $F(k) \to F(A)$, where A is an integral k-algebra of finite type, is a torsion element. Using a k-homomorphism $A \to k'$, where k' is a finite k-algebra, one reduces to the case in which $A = k'$ is a finite extension of k. The norm property above then gives $[k' : k]x = 0$, proving that x is torsion. One even has $x = 0$ when the greatest common divisor of the degrees of the finite extensions k' of k that are quotients of A is 1. This is the case when the fraction field K of A is a pure extension of k, as is easily checked (only the case in which k is finite requires a new verification). This proves (ii).

7.18.4. Suppose that, for every nonempty affine open subset U of the affine space $\operatorname{Spec} k[T_1, \ldots, T_n]$, if A is the ring of U, the homomorphism $F(k) \to F(A)$ is surjective (compare IX 1.6). Using the commutation of F with filtered inductive limits, one then finds that $F(k) \to F(K)$ is surjective for every pure extension K of k. If, in addition, k is infinite, or if norm homomorphisms can be defined as in 7.18.3, this homomorphism is therefore bijective.

7.18.5. As an example, let X be a scheme of finite type over k, and set

$$F(A) = K_\bullet(X \otimes_k A),$$

which is a functor on the category C by IV 2.12. By IV 3.2.3, it commutes with filtered inductive limits. Moreover, if k' is a finite extension of k, the projection

$$p : X \otimes_k k' \longrightarrow X$$

is finite, flat, and of finite presentation, and therefore defines a direct image homomorphism

$$p_* = N_{k'/k} : K_\bullet(X \otimes_k k') \longrightarrow K_\bullet(X).$$

The condition on the norm homomorphisms in 7.18.3 is immediately verified. Finally, if A is the affine algebra of an open subset U of the affine space $\operatorname{Spec} k[T_1, \ldots, T_n]$, then $F(k) \to F(A)$ identifies with the map

$$K_\bullet(X) \longrightarrow K_\bullet(X \times_k U),$$

which is surjective for X noetherian, by IX 1.6. The results proved above therefore give:

(i) If K/k is a pure extension of k, then $K_\bullet(X) \to K_\bullet(X_K)$ is injective, and is even bijective if X is noetherian.

(ii) If k is separably closed and K/k is a separable extension, then $K_\bullet(X) \to K_\bullet(X_K)$ is injective.

(iii) For every extension K of k, the kernel of $K_\bullet(X) \to K_\bullet(X_K)$ is a torsion group.

These results remain valid if one assumes only that X is noetherian, provided one restricts to extensions K of k of finite type (so that X_K is still noetherian). One obtains essentially the same results for $K^\bullet$ in place of the functor $K_\bullet$, with the sole difference that in (i) bijectivity

cannot be asserted in general, because there is no homotopy theorem for $K^\bullet$ in place of $K_\bullet$. To conclude bijectivity, one would have to assume that X is noetherian and regular; then the groups $K^\bullet$ and $K_\bullet$ agree for X and, respectively, for X_K.

One obtains essentially the same results for $\mathrm{Gr}^\bullet$ in place of $K_\bullet$ or $K^\bullet$, with the same reservation concerning the bijectivity assertion in (i). If one restricts to $\mathrm{Gr}^1 = \mathrm{Pic}$, that assertion is valid when X is noetherian and normal: in that case, for every open subset U of an affine space over k, one sees at once that

$$\mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(X \times_k U)$$

is surjective (EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.13).

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Exposé XII. A relative representability theorem for the Picard functor

by M. Raynaud (written up by S. Kleiman)

Warning

The purpose of Exposés XII and XIII is to prove the nine finiteness theorems for the Picard functor stated by A. Grothendieck in the comments (pp. C 07–C 11) to his *Fondements de la Géométrie Algébrique* (cited as FGA). In the first exposé, the first finiteness theorem is proved in the following form, which is more precise than the statement in FGA. Let X and Y be proper schemes over an integral noetherian base S, and let $f : X \to Y$ be a surjective S-morphism. Then there is a nonempty open subset V of S over which the induced morphism $f^* : \mathrm{Pic}_{Y/S} \to \mathrm{Pic}_{X/S}$ is representable by a quasi-affine morphism of finite presentation. In the second exposé, the remaining assertions are proved.

The two exposés differ notably in their proof techniques. In the first, Oort's dévissage, non-flat descent, and the representability of unramified functors are used in an essential way. In the second, one first reduces to the case in which the schemes X/S under consideration are projective and flat with geometrically integral fibers, or, when necessary, normal or smooth, always by applying the first finiteness theorem to a suitable morphism $X' \to X$. One then uses the tool of (b)-sheaves, forged by I. Castelnuovo and sharpened and used by T. Matsusaka, D. Mumford, Y. Nakai, and S. Kleiman. In particular, this tool makes it possible to dispense with Chow coordinates, the Frobenius morphism, and the Lefschetz theorems used in Grothendieck's original proofs.

The present Exposé XII was presented orally by M. Raynaud following Grothendieck; the editor wishes to thank him for communicating his personal notes.

1. Statement of the main theorem and applications

Theorem 1.1. Let S be a noetherian scheme, let X and Y be two proper S-schemes, and let $f : X \to Y$ be a morphism. Suppose that S is integral and that f is surjective. Then there is a nonempty open subset V of S such that the morphism

$$f^* : \mathrm{Pic}_{Y/S}\,|_V \longrightarrow \mathrm{Pic}_{X/S}\,|_V$$

between the Picard functors (localized for the fppf topology) is representable by a quasi-affine morphism of finite presentation.

Corollary 1.2. Let X be a proper scheme over a noetherian scheme S, and suppose that S is integral. Then there is a nonempty open subset V of S such that the Picard functor $\mathrm{Pic}_{X/S}\,|_V$ is representable by a V-group scheme that is a sum of quasi-projective open subschemes of finite type, and hence, *a fortiori*, is separated over V.

Indeed, recall the following fact (FGA no. 232). Let S be a scheme and let X be a projective S-scheme, of finite presentation and integral over S (that is, flat with geometrically integral fibers). Then the Picard functor is represented by an S-group scheme $\mathrm{Pic}_{X/S}$, an increasing union of quasi-projective open subschemes of finite presentation. Consideration of the "Hilbert polynomials" decomposes $\mathrm{Pic}_{X/S}$ here as a sum of open subschemes $\mathrm{Pic}^q_{X/S}$, $q \in \mathbb{Q}[t]$. In the following exposé it is proved, under the additional hypothesis that S is noetherian, that the

$\mathrm{Pic}^q_{X/S}$ are of finite type over S. Thus 1.2 is known when X is in addition projective and integral over S; the general case will be deduced using the following lemma.

Lemma 1.3. Let X be of finite type over an integral noetherian scheme S. Then there are a nonempty open subset V of S, an integral scheme S', and a finite flat morphism $S' \to V$ such that every irreducible component X'_i of $X_{S'}$, endowed with the induced reduced structure, is integral over S'.

Indeed, there is a finite extension L of the function field $k(S)$ such that every irreducible component of $(X_L)_{\mathrm{red}}$ is geometrically integral (EGA IV 4.6.8). Let S_1 be an integral S-scheme such that $k(S_1) = L$ and such that there are a nonempty open subset V_1 of S and a finite flat morphism $h : S_1 \to V_1$. There is a nonempty open subset S_2 of S_1 such that every irreducible component of X_{S_2}, endowed with the induced reduced structure, is integral over S_2 (EGA IV 9.7.7). Set

$$V = V_1 \setminus h(S_1 \setminus S_2), \qquad S' = h^{-1}(V).$$

We now prove 1.2. Let $h : S' \to V_1$ and the X'_i be as in 1.3. For each i, let $f_i : Z_i \to X'_i$ be a Chow presentation, with Z_i integral and f_i birational. After shrinking V and S', one may suppose that Z_i is integral over S' (EGA IV 9.7.7). Let

$$f : Z = \coprod_i Z_i \longrightarrow X' = X_{S'}$$

be the canonical morphism. Then $\mathrm{Pic}_{Z/S'} = \prod_i \mathrm{Pic}_{Z_i/S'}$ is representable by a group scheme that is a sum of quasi-projective open subschemes of finite type. Finally, 1.1 says that, after shrinking V and S', the morphism

$$f^* : \mathrm{Pic}_{X'/S'} \longrightarrow \mathrm{Pic}_{Z/S'}$$

is representable by a quasi-affine morphism. Consequently, $\mathrm{Pic}_{X'/S'}$ is representable by a group scheme that is a sum of quasi-projective open subschemes of finite type; by finite flat descent (SGA 1 VIII 7.6 and EGA II 6.6.1), the same is true of $\mathrm{Pic}_{X/S}\,|_V$.

Lemma 1.4. Let k be a field, let P and Q be k-group schemes locally of finite type, and let $f : P \to Q$ be a quasi-affine morphism. Then f is affine.

Indeed, let $R = \mathrm{Ker}(f)$. Then R is a quasi-affine k-group scheme of finite type; by SGA 3 $\mathrm{VI_B}$ 11.11, R is affine. Thus $P \to P/R$, which makes P an R-torsor over P/R, is affine. Moreover, by SGA 3 I 4.2, $P/R \to Q$ is a closed immersion, hence affine; therefore the composite $P \to P/R \to Q$ is affine.

Corollary 1.5. Let k be a field.

a) If X is a proper k-scheme, then $\mathrm{Pic}_{X/k}$ is representable (J. P. Murre).

b) If $f : X \to Y$ is a surjective morphism between two proper k-schemes, then the induced morphism

$$f^* : \mathrm{Pic}_{Y/k} \longrightarrow \mathrm{Pic}_{X/k}$$

is affine.

Corollary 1.6. Let S be an integral noetherian scheme, and let $f : X \to Y$ be a surjective S-morphism between two proper S-schemes.

a) If Y is normal over S (that is, flat with geometrically normal fibers), then, after shrinking S, $f^* : \mathrm{Pic}_{Y/S} \to \mathrm{Pic}_{X/S}$ is quasi-finite.

b) If $\mathrm{Pic}_{Y/S}$ is essentially proper over S (that is, if it satisfies the valuative criterion of properness; for example, this holds if Y is smooth over S), then, after shrinking S, $f^* : \mathrm{Pic}_{Y/S} \to \mathrm{Pic}_{X/S}$ is finite (and, in particular, affine).

Indeed, in (a), for every $s \in S$, $R_s = \mathrm{Ker}(f_s^*)$ is an affine algebraic group; its identity component is a proper algebraic group over $k(s)$ (FGA 236, Theorem 2.1 (iii)), so R_s is finite. In (b), f^* is both proper and quasi-affine, hence finite.

Remark 1.7. It is not known whether 1.1 remains valid when "quasi-affine" is replaced by "affine," as 1.5 and 1.6 might suggest.

2. First reductions

2.1. Apply 1.3 to X/S and Y/S. One obtains a nonempty open subset V of S, an integral scheme S', and a finite flat morphism $S' \to V$ such that every irreducible component X'_i of $X_{S'}$ and Y'_j of $Y_{S'}$, endowed with the induced reduced structure, is integral over S'. By fpqc descent (EGA IV 2.7.1 (xiv) and SGA 1 VIII 7.9), one may replace S by S'. It then follows from the commutative diagram

$$\begin{array}{ccc} X_{\mathrm{red}} & \xrightarrow{f_{\mathrm{red}}} & Y_{\mathrm{red}} \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ X & \xrightarrow[f]{} & Y \end{array}$$

which implies the commutative diagram

$$\begin{array}{ccc} \mathrm{Pic}_{X_{\mathrm{red}}/S} & \longleftarrow & \mathrm{Pic}_{Y_{\mathrm{red}}/S} \\ \uparrow & & \uparrow \\ \mathrm{Pic}_{X/S} & \longleftarrow & \mathrm{Pic}_{Y/S} \end{array}$$

and from elementary sorites concerning morphisms of functors representable by quasi-affine morphisms (EGA II 5.1.10 (ii), (v)) that it is enough to prove the following two assertions:

I. Theorem 1.1 when f is an immersion (necessarily nilpotent).

II. Theorem 1.1 when X and Y are reduced and every irreducible component X_i of X and Y_i of Y, endowed with the induced reduced structure, is integral over S.

2.2. Consider assertion II. For every i, let $j = \varphi(i)$ be an index such that $f(X_i) \subset Y_j$, and put $\Sigma_j = \coprod_{\varphi(i)=j} X_i$. Then one has the two diagrams

$$\begin{array}{ccc} Y & \longleftarrow & \coprod_j Y_j \\ \uparrow & & \uparrow \\ X & \longleftarrow & \coprod_j \Sigma_j \end{array}$$

and

$$\begin{array}{ccc} \mathrm{Pic}_{Y/S} & \xrightarrow{①} & \mathrm{Pic}_{\coprod_j Y_j/S} = \prod_j \mathrm{Pic}_{Y_j/S} \\ {\scriptstyle ②}\downarrow & & \downarrow{\scriptstyle ④} \\ \mathrm{Pic}_{X/S} & \xrightarrow[③]{} & \mathrm{Pic}_{\coprod_j \Sigma_j/S} = \prod_j \mathrm{Pic}_{\Sigma_j/S}\,. \end{array}$$

To establish II, it is enough to see that, after shrinking S, ①, ③, and ④ are quasi-affine.

Consider ④ first. Replacing Y by Y_j and X by Σ_j, one is reduced to the case in which Y is integral over S. Since f is surjective, one of the X_i, say X_0, dominates Y, and one has the two sequences

$$X_0 \hookrightarrow \coprod_i X_i \longrightarrow Y$$

and

$$\mathrm{Pic}_{X_0/S} \longleftarrow \mathrm{Pic}_{\coprod_i X_i/S} \longleftarrow \mathrm{Pic}_{Y/S}\,.$$

Thus, if the $\mathrm{Pic}_{X_i/S}$, and hence also $\mathrm{Pic}_{\coprod_i X_i/S}$, are known to be representable and the composite morphism is quasi-affine, then ④ is quasi-affine. In view of the proof of 1.2, proving ④ is therefore reduced to proving the following assertion:

II′. Theorem 1.1 when X and Y are integral over S.

For ① and ③, one must treat the case of a morphism $\coprod_i X_i \to X$, where X is reduced and the X_i are the irreducible components of X and are integral over S. One may suppose that X has connected fibers. Indeed, let η be a geometric generic point of S. Then the irreducible components of X_η are the $(X_i)_\eta$, and, after shrinking S, one may suppose that $X_i \cap X_j = \varnothing$ if and only if $(X_i)_\eta \cap (X_j)_\eta = \varnothing$. Let $Y_k = \bigcup_{i\in I(k)} X_i$ be the connected components of X. Then f decomposes as

$$f = \coprod_k f_k : \coprod_k \left(\coprod_{i\in I(k)} X_i \right) \longrightarrow \coprod_k Y_k,$$

and the assertion follows. Finally, an immediate induction reduces us to the case of a morphism $X_1 \amalg X_2 \to X$. After shrinking S, ① and ③ then follow from EGA III 7.8.6 and the following proposition.

Proposition 2.3. Let S be a scheme, let $g : Y \to S$ be a proper morphism, and let Y_1, Y_2 be two closed subschemes defined by the ideals I_1, I_2. Suppose that:

a) $Y, Y_1, Y_2, Y_1 \cap Y_2$ are faithfully flat and of finite presentation over S;

b) $g_*\mathcal{O}_Y = \mathcal{O}_S$ and $(g_i)_*\mathcal{O}_{Y_i} = \mathcal{O}_S$ universally, where $g_i = g|_{Y_i}$, $i = 1, 2$;

c) $I_1 \cap I_2 = 0$.

Then

$$\mathrm{Pic}_{Y/S} \longrightarrow \mathrm{Pic}_{(Y_1 \amalg Y_2)/S}$$

is representable by an affine morphism of finite presentation.

First note that the hypotheses are stable under every base change. For (c), this follows from the two exact sequences

$$0 \longrightarrow I_1 + I_2 \longrightarrow \mathcal{O}_Y \longrightarrow \mathcal{O}_{Y_1\cap Y_2} \longrightarrow 0$$

and

$$0 \longrightarrow I_1 \cap I_2 \longrightarrow I_1 \oplus I_2 \longrightarrow I_1 + I_2 \longrightarrow 0.$$

The first shows that $I_1 + I_2$ is flat over S, and the second then shows that $I_1 \cap I_2$ is stable under base change.

The assertion is local for the fppf topology. After the base change $Y_1 \cap Y_2 \to S$, one may therefore suppose that $Y_1 \cap Y_2$ has a section α, which defines sections α_i of Y_i and β of Y. After a suitable base change, one is reduced to showing that every point $\mu : S \to \mathrm{Pic}_{(Y_1 \amalg Y_2)/S}$ has as inverse

image a subfunctor of $\mathrm{Pic}_{Y/S}$ that is representable by an affine scheme over S. Now μ is defined by an invertible sheaf L_i on Y_i, furnished with a rigidification

$$u_i : \alpha_i^* L_i \xrightarrow{\sim} \mathcal{O}_S, \qquad i = 1, 2.$$

Likewise, a point $\lambda : S \to \mathrm{Pic}_{Y/S}$ corresponds to an invertible sheaf L on Y furnished with a rigidification $u : \beta^* L \xrightarrow{\sim} \mathcal{O}_S$. The image of λ in $\mathrm{Pic}_{(Y_1 \amalg Y_2)/S}$ is μ if and only if there are isomorphisms of rigidified sheaves $f_i^* L \simeq L_i$, where $f_i : Y_i \hookrightarrow Y$ is the inclusion, for $i = 1, 2$. These isomorphisms define an isomorphism

$$\varphi : L_1|_{Y_1 \cap Y_2} \xrightarrow{\sim} L_2|_{Y_1 \cap Y_2}$$

such that $u_1 = u_2 \circ \varphi$.

Conversely, if φ is such an isomorphism, there is a rigidified invertible sheaf L on Y obtained by gluing along φ. Explicitly, if $f_{12} : Y_1 \cap Y_2 \hookrightarrow Y$ denotes the common restriction of f_1 and f_2, then L is the kernel of the canonical homomorphism

$$(f_1)_* L_1 \oplus (f_2)_* L_2 \longrightarrow (f_{12})_* (L_2|_{Y_1 \cap Y_2})$$

defined by φ. This L then corresponds to a point λ of $\mathrm{Pic}_{Y/S}$ mapping to μ.

Finally, let I be the functor that assigns, to an S-scheme T, the set of isomorphisms

$$\varphi : (L_1)_T|_{(Y_1 \cap Y_2)_T} \xrightarrow{\sim} (L_2)_T|_{(Y_1 \cap Y_2)_T}$$

such that $(u_1)_T = (u_2)_T \circ \varphi$. We have just established an isomorphism between the inverse image of μ in $\mathrm{Pic}_{Y/S}$ and the functor I. It is therefore enough to apply the following lemma (with $X = Y_1 \cap Y_2$).

Lemma 2.4. Let X be a flat, proper scheme of finite presentation over S. Let L, M be two invertible sheaves on X, and let $\mathrm{Isom}(L, M)$ be the functor of isomorphisms $L_T \to M_T$, for T/S. Then:

i) $\mathrm{Isom}(L, M)$ is representable by an affine S-scheme of finite presentation.

ii) If, moreover, L and M are furnished with rigidifications (relative to a given section of X), the subfunctor of $\mathrm{Isom}(L, M)$ formed by the isomorphisms compatible with these rigidifications is representable by a closed subscheme of $\mathrm{Isom}(L, M)$.

Indeed, (ii) is clear. For (i), recall that the functors $T \mapsto \mathrm{Hom}(L_T, M_T)$ and $T \mapsto \mathrm{Hom}(M_T, L_T)$ are representable by vector bundles $\mathbb{V}(E)$ and $\mathbb{V}(F)$ (EGA III 7.7.8). Thus, if $u \in \mathrm{Hom}(L_T, M_T)$ and $v \in \mathrm{Hom}(M_T, L_T)$, the conditions $u \circ v = 1$ and $v \circ u = 1$ define a closed subscheme Z of $\mathbb{V}(E) \times_S \mathbb{V}(F)$, and it is clear that $Z \simeq \mathrm{Isom}(L, M)$.

2.5. Finally, let us reduce assertion II′ to the following assertion:

II*. Theorem 1.1 when X and Y are integral over S and the sequence

$$\mathcal{O}_Y \longrightarrow f_* \mathcal{O}_X \rightrightarrows g_* \mathcal{O}_{X \times_Y X} \qquad (*)$$

(where $g : X \times_Y X \to Y$ is the structural morphism) is exact.

For this, it is enough to prove that there is a factorization of f

$$X \xrightarrow{f_0} Y_0 \xrightarrow{f_1} Y_1 \longrightarrow \cdots \xrightarrow{f_n} Y_n = Y$$

such that every f_i satisfies (*) and, after possibly restricting S, every Y_i is integral over S.

For this purpose, put $Y_0 = \mathrm{Spec}(f_*\mathcal{O}_X)$, with f_0 arising from the Stein factorization of f (EGA III 4.3.3), so that $f_{0*}\mathcal{O}_X \xleftarrow{\sim} \mathcal{O}_{Y_0}$; hence f_0 does satisfy (*). To factor $Y_0 \to Y$ in the same way, one uses the following lemma.

Lemma 2.6. Let Y be a noetherian scheme, and let $f : X \to Y$ be a finite epimorphism (that is, $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is injective). Then f factors into a finite sequence of finite morphisms

$$X \xrightarrow{\sim} Y_0 \xrightarrow{f_1} Y_1 \longrightarrow \cdots \longrightarrow Y_n = Y$$

that are effective epimorphisms (that is, for each i the sequence

$$\mathcal{O}_{Y_i} \longrightarrow (f_i)_*\mathcal{O}_{Y_{i-1}} \rightrightarrows (g_i)_*(\mathcal{O}_{Y_{i-1}} \otimes_{\mathcal{O}_{Y_i}} \mathcal{O}_{Y_{i-1}})$$

is exact).

Put $A = \mathcal{O}_Y$, $B_0 = f_*\mathcal{O}_X$, and define recursively

$$B_i = \mathrm{Ker}\big(B_{i-1} \rightrightarrows B_{i-1} \otimes_A B_{i-1}\big).$$

One checks at once that

$$B_{i-1} \otimes_A B_{i-1} \xrightarrow{\sim} B_{i-1} \otimes_{B_i} B_{i-1}.$$

It remains to prove that $B_i = A$ for all sufficiently large i. Indeed, consider the decreasing sequence of B_0-modules $M_i = B_i/A$, and put $S_i = \mathrm{Supp}(M_i)$. The decreasing sequence of closed subsets S_i of Y is stationary; let S_∞ be its stationary value. We show that $S_\infty = \varnothing$. Otherwise, let s be a maximal point of S_∞. After replacing Y by $\mathrm{Spec}\,\mathcal{O}_{Y,s}$, we may suppose that Y is local with closed point s. By the choice of s, M_i is artinian for all sufficiently large i, and therefore the sequence (M_i) is stationary. Thus $B_i = B_{i+1}$ for some i. It follows that $\mathrm{Spec}(B_i) \to Y$ is a monomorphism, hence a closed immersion (Nakayama). Since $A \subset B_i$, one has $A = B_i$, contrary to $s \in S_\infty$.

3. Proof of I: Oort's dévissage

Proposition 3.1. (Oort, *Bull. soc. math. France*, vol. 90 (1962)). Let S be a locally noetherian scheme, let $f : X \to S$ be a proper morphism, and let $X' \subset X$ be a closed subscheme. Suppose that:

i) X' is defined by an ideal I contained in the nilradical N of X, and $NI = 0$.

ii) X' is flat over S and cohomologically flat over S in dimension 0 (EGA III 7.8.1).

iii) I is flat over S and cohomologically flat over S in dimensions 0 and 1, so that (by EGA III 7.4.6) there are a coherent $\mathcal{O}_S$-module Q and, functorially in the coherent $\mathcal{O}_S$-module M, an isomorphism

$$R^2 f_*(I \otimes f^*M) \simeq \mathrm{Hom}_{\mathcal{O}_S}(Q, M).$$

iv) The cokernel E of $f_*\mathcal{O}_{X'} \to R^1 f_*(I)$ is a locally free $\mathcal{O}_S$-module.

v) The homomorphism

$$(f_*\mathcal{O}_X)_{\mathrm{red}} \longrightarrow (f_*\mathcal{O}_{X'})_{\mathrm{red}}$$

is surjective.

Then the canonical morphism

$$\mathrm{Pic}_{X/S} \longrightarrow \mathrm{Pic}_{X'/S}$$

is representable by an affine morphism of finite presentation. More precisely, there is a canonical exact sequence of fppf sheaves

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \mathrm{Pic}_{X/S} \longrightarrow \mathrm{Pic}_{X'/S} \xrightarrow{u} \mathbb{V}(Q),$$

and, if $P = \mathrm{Ker}(u)$, the monomorphism $P \hookrightarrow \mathrm{Pic}_{X'/S}$ is representable by a closed immersion and $\mathrm{Pic}_{X/S}$ is a torsor over P under the group $\mathbb{V}(E^\vee)$ (that is, the sequence of fppf sheaves

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \mathrm{Pic}_{X/S} \longrightarrow P \longrightarrow 0$$

is exact).

For an S-scheme T, consider the following two diagrams with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_T & \longrightarrow & \mathcal{O}_{X_T} & \longrightarrow & \mathcal{O}_{X'_T} & \longrightarrow & 0 \\ & & \| & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & I_T & \xrightarrow{\alpha} & \mathcal{O}^*_{X_T} & \longrightarrow & \mathcal{O}^*_{X'_T} & \longrightarrow & 0, \end{array} \qquad \alpha(i) = 1 + i,$$

and

$$\begin{array}{ccccccccc} (f_T)_*\mathcal{O}_{X'_T} & \xrightarrow{\delta_T} & R^1(f_T)_*I_T & & & & & & \\ \uparrow & & \| & & & & & & \\ (f_T)_*\mathcal{O}^*_{X'_T} & \xrightarrow[\delta^*_T]{} & R^1(f_T)_*I_T & \to & R^1(f_T)_*\mathcal{O}^*_{X_T} & \to & R^1(f_T)_*\mathcal{O}^*_{X'_T} & \to & R^2(f_T)_*I_T. \end{array}$$

Note carefully that the left-hand square in this diagram is *not necessarily commutative*. Nevertheless,

3.2. $\qquad \mathrm{im}\,\delta^*_T = \mathrm{im}\,\delta_T.$

Indeed, we may plainly suppose that S and T are affine. Choose an affine covering $\{V_i\}$ of X_T. View $\mathcal{O}_{X'_T}$ as its direct image under the closed immersion $X'_T \hookrightarrow X_T$. Let $g \in \Gamma(X_T, \mathcal{O}^*_{X'_T})$, and let g_i be a lift of $g|_{V_i}$ in $\Gamma(V_i, \mathcal{O}^*_{X_T})$. Then $\delta^*_T(g)$ is the class of the 1-cocycle

$$\delta^*_{ij} = g_i g_j^{-1} - 1,$$

whereas $\delta_T(g)$ is the class of

$$\delta_{ij} = g_i - g_j = g_j\delta^*_{ij} = g\delta^*_{ij}.$$

Consequently,

$$\delta_T(g) = g\,\delta^*_T(g),$$

taking into account the action of $\Gamma(X_T, \mathcal{O}_{X'_T})$ on I_T, and hence on $H^1(X_T, I_T)$.

By (ii),

$$\Gamma(X_T, \mathcal{O}_{X'_T}) = \Gamma(X, \mathcal{O}_{X'}) \otimes_S \Gamma(T, \mathcal{O}_T),$$

and hence $g = \sum g_\beta \otimes t_\beta$, with $g_\beta \in \Gamma(X, \mathcal{O}_{X'})$ and $t_\beta \in \Gamma(T, \mathcal{O}_T)$. By (v), the map

$$(f_*\mathcal{O}_X)_{\mathrm{red}} \longrightarrow (f_*\mathcal{O}_{X'})_{\mathrm{red}}$$

is surjective. Thus $g_\beta = v(h_\beta) + n_\beta$, where $h_\beta \in \Gamma(X, \mathcal{O}_X)$, $v : \Gamma(X, \mathcal{O}_X) \to \Gamma(X, \mathcal{O}_{X'})$, and $n_\beta \in \mathrm{nilrad}(\Gamma(X, \mathcal{O}_{X'}))$.

By (i), $(n_\beta \otimes t_\beta) I_T = 0$. Therefore $\delta_T(g) = h\,\delta_T^*(g)$, where

$$h = \sum h_\beta \otimes t_\beta \in \Gamma(X_T, \mathcal{O}_{X_T}).$$

It follows that

$$\delta_T^*(g) = h^{-1}\delta_T(g) = \delta_T(h^{-1}g),$$

and hence $\operatorname{im}\delta_T^* \subset \operatorname{im}\delta_T$.

Finally, let $g = \sum g_\beta \otimes t_\beta \in \Gamma(X_T, \mathcal{O}_{X'_T})$. Write $g_\beta = v(h_\beta) + n_\beta$, with $n_\beta \in \operatorname{nilrad}\big(\Gamma(X, \mathcal{O}_{X'})\big)$. Then

$$\begin{aligned}\delta_T(g) &= \delta_T\Big(\sum n_\beta \otimes t_\beta\Big)\\ &= \delta_T\Big(1 + \sum n_\beta \otimes t_\beta\Big)\\ &= \delta_T^*\Big(1 + \sum n_\beta \otimes t_\beta\Big),\end{aligned}$$

whence $\operatorname{im}\delta_T \subset \operatorname{im}\delta_T^*$.

By 3.2, (iii), and (iv), there is an exact sequence, functorial in T,

$$0 \longrightarrow \mathbb{V}(E^\vee)(T) \longrightarrow R^1(f_T)_*\mathcal{O}_{X_T}^* \longrightarrow R^1(f_T)_*\mathcal{O}_{X'_T}^* \longrightarrow \mathbb{V}(Q)(T),$$

which, on passing to the associated fppf sheaves, gives an exact sequence

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \operatorname{Pic}_{X/S} \longrightarrow \operatorname{Pic}_{X'/S} \xrightarrow{u} \mathbb{V}(Q).$$

If $P = \operatorname{Ker}(u)$, the monomorphism $P \hookrightarrow \operatorname{Pic}_{X'/S}$ is obtained by base change from the unit section $S \hookrightarrow \mathbb{V}(Q)$. Since $\mathbb{V}(Q)$ is representable and separated, it follows that $P \hookrightarrow \operatorname{Pic}_{X'/S}$ is representable by a

closed immersion.

Finally,

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \operatorname{Pic}_{X/S} \xrightarrow{w} P \longrightarrow 0$$

is a torsor for the fppf topology. A fortiori, by fpqc descent from a "splitting" of the torsor, w is representable by a faithfully flat affine morphism of finite presentation (see Oort, *Commutative group schemes*, Proposition 17.4).

Corollary 3.3. Under the hypotheses of 3.1:

i) If

$$\operatorname{Pic}_{(\mathrm{zar})(X'/S)} \longrightarrow \operatorname{Pic}_{(\mathrm{fppf})(X'/S)}$$

is an isomorphism, then

$$\operatorname{Pic}_{(\mathrm{zar})(X/S)} \longrightarrow \operatorname{Pic}_{(\mathrm{fppf})(X/S)}$$

is an isomorphism.

ii) If the S-scheme T is affine and

$$\operatorname{Pic}(X'_T) \longrightarrow \operatorname{Pic}^{(T)}_{(\mathrm{zar})(X'/S)}$$

is surjective, then

$$\operatorname{Pic}(X_T) \longrightarrow \operatorname{Pic}^{(T)}_{(\mathrm{zar})(X/S)}$$

is surjective.

Indeed, (i) follows by applying the five lemma to the following diagram of sheaves for the Zariski topology:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{V}(E^\vee) & \longrightarrow & \mathrm{Pic}_{(\mathrm{zar})(X/S)} & \longrightarrow & \mathrm{Pic}_{(\mathrm{zar})(X'/S)} \longrightarrow \mathbb{V}(Q) \\ \downarrow & & \downarrow S & & \downarrow & & \downarrow S \quad \downarrow S \\ 0 & \longrightarrow & \mathbb{V}(E^\vee) & \longrightarrow & \mathrm{Pic}_{(\mathrm{fppf})(X/S)} & \longrightarrow & \mathrm{Pic}_{(\mathrm{fppf})(X'/S)} \longrightarrow \mathbb{V}(Q). \end{array}$$

Assertion (ii) follows from the following diagram with exact rows:

$$\begin{array}{ccccccc} H^1(X_T, I_T) & \longrightarrow & H^1(X_T, \mathcal{O}^*_{X_T}) & \longrightarrow & H^1(X_T, \mathcal{O}^*_{X'_T}) & \longrightarrow & H^2(X_T, I_T) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H^0(T, R^1(f_T)_* I_T) & \longrightarrow & H^0(T, R^1(f_T)_* \mathcal{O}^*_{X_T}) & \longrightarrow & H^0(T, R^1(f_T)_* \mathcal{O}^*_{X'_T}) & \longrightarrow & H^0(T, R^2(f_T)_* I_T), \end{array}$$

because the outside vertical arrows are isomorphisms when T is affine.

Corollary 3.4. Let k be a field, and let X be a proper k-scheme. Suppose that $\mathrm{Pic}_{X_{\mathrm{red}}/k}$ possesses a universal invertible sheaf. (For example, descent theory shows that this is the case if every connected component X'_i of X_{red} has a rational point; in particular, if k is algebraically closed, or more generally if the greatest common divisor of the degrees of the 0-cycles on every X'_i is 1.) Then, over every affine open subset V of $\mathrm{Pic}_{X/k}$, there is a universal invertible sheaf on $X \times_k V$.

Finally, we prove assertion I in the following more precise form.

Proposition 3.5. Let S be an integral noetherian scheme, and let $f : X \hookrightarrow Y$ be a nilpotent immersion of proper S-schemes. Then there is a nonempty open subset V of S such that the morphism

$$f^* : \mathrm{Pic}_{Y/S}\,|_V \longrightarrow \mathrm{Pic}_{X/S}\,|_V$$

is representable by an affine morphism of finite presentation. In fact, $N = \mathrm{Ker}(f^*)$ is an affine group scheme with unipotent fibers (in the sense of SGA 3 XVII).

Indeed, X is defined by a nilpotent ideal I. The ideal I has a filtration by ideals

$$I_0 = I \supset I_1 \supset \cdots \supset I_n = 0$$

such that $NI_{i-1} \subset I_i$ for $1 \leq i \leq n$, where N is the nilradical of $\mathcal{O}_Y$ (for example, $I_i = N^i I$). Let X_i be the closed subscheme of Y defined by I_i. By induction it is enough to study the immersions $X_{i-1} \hookrightarrow X_i$. We may therefore suppose that $NI = 0$; this is condition (i) of 3.1, and we are reduced finally to the case in which the other conditions (ii)–(v) are satisfied as well. By the generic flatness theorem (EGA IV 6.9.1), applied in the notation of 3.1 to $\mathcal{O}_{X'}$, the $R^i f_*(I)$, and E, and taking account of the Künneth relations of EGA III 6.9.9, after restricting S we may suppose that conditions (ii), (iii), and (iv) hold.

There is a finite extension K of $k(S)$ such that $(f_K)_* \mathcal{O}_{X'_K}$ has all its residue field extensions trivial. If S' is an integral scheme, finite and flat over an open subset V of S, with function field $k(S') = K$, there is an open subset V' of S' such that

$$\left[(f_{S'})_* \mathcal{O}_{X'_{S'}}\right]_{\mathrm{red}} \simeq \left[(f_{S'})_* \mathcal{O}_{X_{S'}}\right]_{\mathrm{red}}$$

over V'. After restricting S and S' further, we may suppose that $S = V$ and $S' = V'$. Since the assertion to be proved is local for the fpqc topology, we may replace S by S'. Finally, apply 3.1.

Remark 3.6. It follows in the same way, taking account of the exact sequences

$$\mathrm{Pic}_{X_i/S} \longrightarrow \mathrm{Pic}_{X_{i-1}/S} \longrightarrow \mathbb{V}(Q_i),$$

that, for every $s \in S$,

$$\operatorname{coker}(\operatorname{Pic}^0_{Y_s/s} \longrightarrow \operatorname{Pic}^0_{X_s/s})$$

is a unipotent group.

4. Proof of II*: the most delicate part of the proof

Recall assertion II*: Theorem 1.1 when X and Y are integral over S and the sequence

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \rightrightarrows g_*\mathcal{O}_{X\times_Y X}$$

is exact. Applying the generic flatness theorem, EGA IV 6.9.1, and EGA III 7.8.8, we are reduced to proving the following proposition.

Proposition 4.1. Let S be an integral noetherian scheme, let $\lambda : X \to S$ and $\mu : Y \to S$ be proper morphisms of finite presentation, let $f : X \to Y$ be a surjective S-morphism, and let $g : X \times_Y X \to Y$ be the canonical morphism. Suppose that:

1) X, Y, and $X \times_Y X$ are flat over S.

2) $\lambda_*\mathcal{O}_X = \mathcal{O}_S$ and $\mu_*\mathcal{O}_Y = \mathcal{O}_S$ universally.

3) For all $i, j \geq 0$, the modules $R^i f_*\mathcal{O}_X$ and $R^j g_*\mathcal{O}_{X\times_Y X}$ are flat over S.

4) The sequence

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \overset{\pi_1}{\underset{\pi_2}{\rightrightarrows}} g_*\mathcal{O}_{X\times_Y X} \qquad (*)$$

is exact, and $\operatorname{coker}(\pi_1 - \pi_2)$ is an $\mathcal{O}_Y$-module flat over S.

Then

$$f^* : \operatorname{Pic}_{Y/S} \longrightarrow \operatorname{Pic}_{X/S}$$

is representable by a quasi-affine morphism of finite presentation.

First note that (1)–(4) are stable under every base change $T \to S$. Only the stability of (3) and (4) is not evident. By the Künneth formula (EGA III 6.9.9), (1) and (3) imply that the formation of $R^i f_*\mathcal{O}_X$ and $R^i g_*\mathcal{O}_{X\times_Y X}$ commutes with base change $T \to S$; this immediately implies the stability of condition (3). Moreover, the formation of the cokernel of

$$\pi_1 - \pi_2 : f_*\mathcal{O}_X \longrightarrow g_*\mathcal{O}_{X\times_Y X}$$

also commutes with base change. Since it is flat over S by (4), $\operatorname{im}(\pi_1 - \pi_2)$ is flat over S and commutes with base change. Hence $\operatorname{Ker}(\pi_1 - \pi_2)$ commutes with base change $T \to S$ and is therefore universally equal to $\mathcal{O}_Y$.

The stability of the hypotheses implies two things:

a. Since the assertion to be proved is local for the fppf topology, by descent we may suppose that X has a section α; then Y has the section $\beta = f \circ \alpha$.

b. To prove the assertion, it is enough to consider an S-point of $\operatorname{Pic}_{X/S}$, that is, by (a) and (2), an invertible sheaf L on X endowed with a rigidification u relative to α, and to show that the fppf sheaf Z, the inverse image of this point in $\operatorname{Pic}_{Y/S}$, is representable by a quasi-affine S-scheme of finite presentation.

Lemma 4.2. Under the hypotheses of 4.1, f is a descent morphism for the category of locally free sheaves; that is, for every pair L, M of locally free sheaves on Y, the diagram

$$\mathrm{Hom}_Y(L,M) \to \mathrm{Hom}_X(f^*L, f^*M) \rightrightarrows \mathrm{Hom}_{X\times_Y X}(g^*L, g^*M)$$

is exact.

Indeed, if $N = \mathrm{Hom}_{\mathcal{O}_Y}(L,M)$, one must show that the sequence

$$\Gamma(Y,N) \longrightarrow \Gamma(X, f^*N) \rightrightarrows \Gamma(X\times_Y X, g^*N)$$

is exact, or equivalently (by EGA 0_{I} 5.4.10.1) that the sequence

$$\Gamma(Y,N) \longrightarrow \Gamma(Y, f_*\mathcal{O}_X \otimes_{\mathcal{O}_Y} N) \rightrightarrows \Gamma(Y, g_*\mathcal{O}_{X\times_Y X} \otimes_{\mathcal{O}_Y} N)$$

is exact. But the sequence (*)

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \rightrightarrows g_*\mathcal{O}_{X\times_Y X}$$

is exact by hypothesis. Applying the left exact functor $\Gamma(N\otimes -)$ gives the lemma, since N is flat over Y.

Let L be an invertible sheaf on X, endowed with a rigidification u relative to α. By 4.2 there is an equivalence between the category C of pairs (M,φ) consisting of an invertible sheaf M on Y and an isomorphism $\varphi : f^*M \xrightarrow{\sim} L$, and the category $\mathrm{Deff}(S)$ of effective descent data on L relative to f. If $f^*M \simeq L$, then, since L is rigidified relative to α, M is canonically endowed with a rigidification relative to β. Consequently, C is equivalent to the discrete category defined by the set of classes of β-rigidified invertible sheaves (M,v) on Y such that

$$f^*(M,v) \simeq (L,u).$$

But this is precisely the set $Z(S)$ of S-points of the inverse image in $\mathrm{Pic}_{Y/S}$ of the S-point of $\mathrm{Pic}_{X/S}$ defined by (L,u). Moreover, this description of $Z(S)$ is functorial in S. Hence Z identifies with Deff, and we may forget the rigidifications from now on.

There is a canonical monomorphism from Deff into the functor Rec of gluing data on L. By definition, Rec is the functor $\mathrm{Isom}(p_1^*L, p_2^*L)$, where the $p_i : X\times_Y X \to X$ are the projections; by 2.4 this functor is represented by an affine scheme I of finite presentation over S. It therefore remains to represent the monomorphism

$$i : \mathrm{Deff} \hookrightarrow \mathrm{Isom}(p_1^*L, p_2^*L)$$

by a morphism of finite presentation, which will necessarily be quasi-affine (by EGA IV 8.11.2).

Let us note in passing that, if desired, one may restrict $\mathrm{Isom}(p_1^*L, p_2^*L)$ by imposing the cocycle condition, but this is not needed below.

After the base change $I \to S$, we are thus reduced to the following problem: if L is an invertible sheaf on X, endowed with a gluing datum

$$\varphi : p_1^*L \xrightarrow{\sim} p_2^*L,$$

represent the subfunctor of S on which this datum becomes effective. We have the following lemma.

Lemma 4.3. Under the hypotheses of 4.1, let L be an invertible sheaf on X, endowed with a gluing datum

$$\varphi : p_1^*L \xrightarrow{\sim} p_2^*L.$$

This datum is effective if and only if the following five conditions hold.

1) For every $i \geq 0$, $R^i f_*(L)$ is flat over S.

2) For every $j \geq 0$, $R^j g_*(p_1^* L)$ is flat over S.

3) The cokernel of
$$\pi_1 - \pi_2 : f_*(L) \longrightarrow g_*(p_1^* L) \simeq g_*(p_2^* L)$$
is flat over S, where the latter isomorphism is $g_*(\varphi)$.

4) The kernel N of $\pi_1 - \pi_2 : f_*(L) \to g_*(p_1^* L)$ is invertible on Y.

5) The composite morphism
$$f^* N \longrightarrow f^* f_* L \longrightarrow L$$
is an isomorphism.

Indeed, these conditions are sufficient, since (4) and (5) alone imply effectivity. Conversely, let M be invertible on Y, and suppose that there is an isomorphism $f^*M \simeq L$ compatible with the gluing datum. We may cover Y by open subsets V over which M is isomorphic to $\mathcal{O}_Y$. The descent datum on L, restricted over $f^{-1}(V)$, is therefore isomorphic to the trivial descent datum on $\mathcal{O}_X|_{f^{-1}(V)}$. Finally, conditions (1)–(5) are local on Y, and the hypotheses of 4.1 show that they hold for $L = \mathcal{O}_X$; hence they hold for L.

It remains to represent conditions (1)–(5), one after another, by monomorphisms of finite presentation. For (1) and (2), we apply the following lemma.

Lemma 4.4. Let S be a scheme, let $\lambda : X \to S$ and $\mu : Y \to S$ be proper morphisms of finite presentation, and let $f : X \to Y$ be an S-morphism. Let F be a sheaf on X, of finite presentation and flat over S. For an integer n, let S_n be the subfunctor of S on which $R^i f_*(F)$ is flat over S for every $i \geq n$; more precisely, for a morphism $\phi : T \to S$,

$$S_n(T) = \begin{cases} \{\phi\}, & \text{if } R^i(f_T)_*(F_T) \text{ is flat for every } i \geq n, \\ \varnothing, & \text{otherwise.} \end{cases}$$

Then $S_n \hookrightarrow S$ is representable by a surjective monomorphism of finite presentation; moreover, for $i \geq n$, the formation of $R^i(f_{S_n})_*(F_{S_n})$ commutes with base changes $T \to S_n$. In addition, S_n coincides with the functor S'_n on which the functors

$$M \longmapsto R^i(f_V)_*(F_V \otimes \lambda_V^* M), \qquad i \geq n,$$

are exact, where M ranges over the quasi-coherent $\mathcal{O}_V$-modules, for every affine open subset V of S.

Indeed, the assertion is local on S, so we may suppose that S is quasi-compact. The relative dimension of X/Y is then bounded, say by r, and, for $n > r$,

$$R^n f_*(F_V \otimes \lambda_V^* M) = 0.$$

The assertion therefore holds for $n > r$.

Proceeding by descending induction, suppose it is known for $n+1$. After replacing S by $S_{n+1} = S'_{n+1}$, we may suppose that $R^i f_*(F)$ is flat over S for $i \geq n+1$. Since F is flat over S, EGA III 6.9.8 then shows that the formation of $R^n f_*(F)$ commutes with every base change. Consequently, the representability of S_n follows from the next lemma (see J. P. Murre, *Representation of unramified functors*, Séminaire Bourbaki, May 1965, p. 294-11, Theorem 2).

Lemma 4.5. Let $X \to S$ be a proper morphism of finite presentation, and let F be a sheaf of finite presentation on X. Then the subfunctor S_F of S on which F is flat over S is representable by a surjective monomorphism of finite presentation (which, of course, is not an immersion in general).

To represent condition (3) of 4.3, note that, by (1) and (2), the formation of $f_*(L)$ and $g_*(p_1^* L)$ commutes with the base changes $I \to S$; hence so does the formation of $\mathrm{Coker}(\pi_1 - \pi_2)$. We then apply 4.5.

To represent (4), note that, by (3), and because $f_*(L)$ and $g_*(p_1^* L)$ are flat and commute with base change,

$$N = \mathrm{Ker}(\pi_1 - \pi_2)$$

also commutes with base change. We then apply the following lemma.

Lemma 4.6. Let $\lambda : X \to S$ be a proper morphism of finite presentation, and let $u : F \to G$ be a homomorphism of $\mathcal{O}_X$-modules of finite presentation, with G flat over S. Then the subfunctor S_1 of S on which u is an isomorphism is representable by an open subscheme of S, of finite presentation.

As usual, we reduce to the case where S is noetherian, using EGA IV 8 and 11.2.6; this relieves us of the need to consider finite presentation. First let us represent the functor on which u is an epimorphism,

that is, on which $\operatorname{Coker} u$ is zero. Since the formation of $\operatorname{Coker} u$ commutes with base change, this functor is represented by the open subset

$$S - \lambda\big(\operatorname{Supp}(\operatorname{Coker} u)\big)$$

of S. We may therefore suppose that u is already an epimorphism. Using now the hypothesis that G is flat over S, we find that the formation of $\operatorname{Ker} u$ commutes with base change; consequently, the functor under consideration is represented by the open subset

$$S - \lambda\big(\operatorname{Supp}(\operatorname{Ker} u)\big)$$

of S.

Exposé XIII. Finiteness Theorems for the Picard Functor

by S. Kleiman

1. The (b)-sheaves

Let k be an algebraically closed field and X a projective k-scheme, endowed with an ample invertible sheaf $\mathcal{O}_X(1)$.

Definition 1.1 (Mumford). Let $m \in \mathbb{Z}$. A coherent sheaf F on X is said to be *m-regular* if, at every point of the support of F, $\mathcal{O}_X(1)$ is generated by its global sections and if, for every $q \geq 1$,

$$H^q(F(m-q)) = 0.$$

Lemma 1.2. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be an exact sequence of coherent sheaves on X.[24] If F is m-regular, then so is G.

Indeed, we have the corresponding exact sequence

$$H^q(F(m-q)) \longrightarrow H^q(G(m-q)) \longrightarrow H^{q+1}(F(m-q-1)).$$

Proposition 1.3. If a sheaf F on X is m-regular, then, for every $n \geq m$,

(i) F is n-regular.

(ii) The map

$$H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) \longrightarrow H^0(F(n+1))$$

is surjective.

(iii) $F(n)$ is generated by $H^0(F(n))$.

[24] Here, as below in analogous situations, it is understood that $F(-1) \to F$ is defined by tensoring with a given section of $\mathcal{O}_X(1)$.

The proof is by induction on the dimension s of the support of F; if $s = -1$, the assertion is trivial. Choose a section $\sigma \in H^0(\mathcal{O}_X(1))$ which vanishes at no point of the finite set $\mathrm{Ass}(F)$. Then σ defines an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0,$$

and G is m-regular by 1.2, with support of dimension $s - 1$.

Consider the corresponding exact sequence

$$H^q(F(n-q-1)) \longrightarrow H^q(F(n-q)) \longrightarrow H^q(G(n-q)).$$

By the induction hypothesis, $H^q(G(n-q)) = 0$ for $n \geq m$, so the vanishing of $H^q(F(n-q-1))$ implies that of $H^q(F(n-q))$. But $H^q(F(m-q)) = 0$, which proves (i).

To prove (ii), we chase through the following commutative diagram:

$$\begin{array}{ccccccc}
 & & H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) & \longrightarrow & H^0(G(n)) \otimes H^0(\mathcal{O}_X(1)) & \longrightarrow & 0 \\
 & \nearrow & \downarrow & & \downarrow & & \\
0 \longrightarrow H^0(F(n)) & \longrightarrow & H^0(F(n+1)) & \longrightarrow & H^0(G(n+1)) & & \\
 & & & & \downarrow & & \\
 & & & & 0 & &
\end{array}$$

Here the oblique arrow is defined by $\varphi \mapsto \varphi \otimes \sigma$; the horizontal rows are exact, the first by the n-regularity of F and by (i), already proved, and the second column is exact by the induction hypothesis.

Finally, (iii) follows from the commutative diagram

$$\begin{array}{ccc}
H^0(F(n))_X \otimes H^0(\mathcal{O}_X(1))_X & \xrightarrow{\alpha_n} & H^0(F(n)(1))_X \\
\downarrow & & \downarrow{\scriptstyle \beta_{n+1}} \\
[H^0(F(n))_X](1) & \xrightarrow{\beta_n \otimes \mathrm{id}} & F(n)(1).
\end{array}$$

Here, if M is a vector space over k, we write M_X for $M \otimes_k \mathcal{O}_X$. For $n \geq m$, α_n is surjective by (ii); therefore, if β_{n+1} is surjective, so is β_n. By Serre's theorem, β_{n+1} is surjective for $n \gg 0$; this proves (iii).

Proposition 1.4. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be an exact sequence on X. If G is m-regular, then:

(i) $H^q(F(n)) = 0$ for $q \geq 2$ and $n \geq m - q$.

(ii) $h^1(F(n-1)) \geq h^1(F(n))$ for $n \geq m - 1$.

(iii) $H^1(F(n)) = 0$ for $n \geq (m-1) + h^1(F(m-1))$.

In particular, F is $[m + h^1(F(m-1))]$-regular.

Indeed, in the exact sequence

$$H^{q-1}(G(n)) \longrightarrow H^q(F(n-1)) \longrightarrow H^q(F(n)) \longrightarrow H^q(G(n)),$$

the two outer groups are zero for $q \geq 2$ and $n \geq m - (q-1)$, by 1.3(i). Thus

$$H^q(F(m-q)) \xrightarrow{\sim} H^q(F(m-q+1)) \xrightarrow{\sim} H^q(F(m-q+2)) \xrightarrow{\sim} \cdots;$$

but, by Serre's theorem, $H^q(F(n)) = 0$ for $n \gg 0$, proving (i).

For $n \geq m - 1$, consider the exact sequence

$$0 \to H^0(F(n-1)) \to H^0(F(n)) \xrightarrow{\alpha_n} H^0(G(n)) \to H^1(F(n-1)) \to H^1(F(n)) \to 0.$$

It implies (ii). Now suppose that α_n is surjective, and consider the commutative diagram

$$\begin{array}{ccccc} H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) & \xrightarrow{\alpha_n \otimes \mathrm{id}} & H^0(G(n)) \otimes H^0(\mathcal{O}_X(1)) & \longrightarrow & 0 \\ \downarrow & & \downarrow \beta_n & & \\ H^0(F(n+1)) & \xrightarrow{\alpha_{n+1}} & H^0(G(n+1)). & & \end{array}$$

If $n \geq m$, then β_n is surjective by 1.3(ii), and hence α_{n+1} is surjective. Consequently,

$$H^1(F(n-1)) \xrightarrow{\sim} H^1(F(n)) \xrightarrow{\sim} \cdots \xrightarrow{\sim} 0.$$

On the other hand, if α_n is not surjective, then $h^1(F(n-1)) > h^1(F(n))$. Thus the function $n \longmapsto h^1(F(n))$ decreases *strictly*, starting from $h^1(F(m-1))$, until it becomes zero; this proves (iii).

Definition 1.5. Let F be a coherent sheaf on X, let $r \geq \dim \mathrm{Supp}(F)$, and let $(b) = (b_0, \ldots, b_r)$ be a sequence of $r+1$ integers. Then F is said to be a (b)-*sheaf* if it satisfies either of the following two equivalent sets of conditions.

(1.5.1)

(i) At every point of the support of F, $\mathcal{O}_X(1)$ is generated by $H^0(\mathcal{O}_X(1))$.

(ii) $h^0(F(-1)) \leq b_0$.

(iii) If $r \geq 1$, there is a section $\sigma \in H^0(\mathcal{O}_X(1))$ which defines an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

such that G is a $(b_1, \ldots, b_r)$-sheaf.

(1.5.2) There is an F-regular sequence of r sections $\sigma_1, \ldots, \sigma_r \in H^0(\mathcal{O}_X(1))$ such that, for $i = 0, \ldots, r$, if F_i denotes the restriction of F to the zero scheme of $\sigma_1, \ldots, \sigma_i$, then

$$h^0(F_i(-1)) \leq b_i.$$

Proposition 1.6. Let F be a (b)-sheaf on X. Then:

(i) For every sufficiently general sequence σ of r sections $\sigma_1, \ldots, \sigma_r \in H^0(\mathcal{O}_X(1))$, if $F_{\sigma,i}$ denotes the restriction of F to the zero scheme of $\sigma_1, \ldots, \sigma_i$, then $h^0(F_{\sigma,i}(-1)) \leq b_i$, for $0 \leq i \leq r$.

(ii) Every subsheaf G of F is a (b)-sheaf.

Indeed, let $S' = \mathbb{A}_k^N$ be the affine space whose k-points correspond to the sequences σ, and let T be the open subset of S' corresponding to the F-regular sequences. For fixed i, the sheaves $F_{\sigma,i}(-1)$ corresponding to the k-points of T are contained in a flat family over T, and T is nonempty by hypothesis. Thus (i) follows from upper semicontinuity of the function

$$\sigma \longmapsto h^0(F_{\sigma,i}(-1))$$

(EGA III 7.7.5). Finally, since every sufficiently general sequence is G-regular and is such that $G_{\sigma,i}$ is a subsheaf of $F_{\sigma,i}$, for every i, (ii) follows from (i).

Lemma 1.7. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be an exact sequence on X. If the Hilbert polynomial of F is written in the form

$$\chi(F(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i},$$

then

$$\chi(G(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

Indeed,

$$\begin{aligned}\chi(G(n)) &= \chi(F(n)) - \chi(F(n-1)) \\ &= \sum_{i=0}^{r} a_i \left[\binom{n+i}{i} - \binom{n-1+i}{i} \right] \\ &= \sum_{i=1}^{r} a_i \binom{n+i-1}{i-1},\end{aligned}$$

which proves the assertion.

Proposition 1.8. Let F be a (b)-sheaf on X, let $s = \dim \mathrm{Supp}(F)$, and let

$$\chi(F(n)) = \sum_{i=0}^{s} a_i \binom{n+i}{i}$$

be the Hilbert polynomial of F. Then:

(i) For $n \geq -1$,

$$h^0(F(n)) \leq \sum_{i=0}^{s} b_i \binom{n+i}{i}.$$

(ii) $a_s \leq b_s$, and F is also a $(b_0, \ldots, b_{s-1}, a_s)$-sheaf.

Indeed, we argue by induction on s. If $s = 0$, the assertion is clear, since $a_0 = h^0(F) = h^0(F(-1)) \leq b_0$. If $s \geq 1$, we have an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

with G a $(b_1, \dots, b_r)$-sheaf and $\dim \operatorname{Supp}(G) = s - 1$. Then

$$h^0(F(n)) - h^0(F(n-1)) \le h^0(G(n)),$$

whereas

$$h^0(G(n)) \le \sum_{i=0}^{s-1} b_{i+1} \binom{n+i}{i}$$

by induction on s, and $h^0(F(-1)) \le b_0$. Assertion (i) follows by induction on $n \ge -1$. Finally, by 1.7 and induction on s, $a_s \le b_s$, and G is a $(b_1, \dots, b_{s-1}, a_s)$-sheaf; this proves (ii).

Definition 1.9. The following polynomials with nonnegative rational coefficients, defined recursively, are called the (b)-*polynomials*:

$$\begin{cases} P_{-1} = 0, \\ P_r(X_0, \dots, X_r) = P_{r-1}(X_1, \dots, X_r) + \sum_{i=0}^{r} X_i \binom{P_{r-1}(X_1, \dots, X_r) - 1 + i}{i}. \end{cases}$$

Remark 1.10. For the (b)-polynomials, induction on $r \ge t$ immediately gives

$$P_r(X_0, \dots, X_t, 0, \dots, 0) = P_t(X_0, \dots, X_t).$$

Theorem 1.11. *(The main theorem on (b)-sheaves.)* Let F be a (b)-sheaf on X, and let

$$\chi(F(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$$

be the Hilbert polynomial of F. Let $(c) = (c_0, \dots, c_r)$ be a sequence of integers satisfying $(c) \ge (b) - (a)$, and let $m = P_r(c)$ be the integer given by the r-th (b)-polynomial (1.9). Then $m \ge 0$, and F is m-regular. In particular, if $s = \dim \operatorname{Supp}(F)$, then F is

$$P_{s-1}(c_0, \dots, c_{s-1})\text{-regular.}$$

Indeed, we argue by induction on r. If $r = 0$, the assertion is clear, since $m = 0$ and F is n-regular for every n. If $r \geq 1$, there is an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

in which G is a $(b_1, \ldots, b_r)$-sheaf. By 1.7 and the induction hypothesis, if $n = P_{r-1}(c_1, \ldots, c_r)$, then $n \geq 0$ and G is n-regular. Hence, by 1.4, F is $[n+h^1(F(n-1))]$-regular, and $h^q(F(n-1)) = 0$ for $q \geq 2$. Therefore

$$h^1(F(n-1)) = h^0(F(n-1)) - \chi(F(n-1)),$$

which, by 1.8(i), is at most

$$\sum_{i=0}^{r} (b_i - a_i) \binom{n-1+i}{i},$$

since $b_i \geq 0$. Finally, by 1.3(i), F is

$$\left[n + \sum_{i=0}^{r} c_i \binom{n-1+i}{i} \right] \text{-regular.}$$

This proves the first assertion; the second follows from 1.8(ii) and 1.10.

1.12. Let S be a noetherian scheme, and let X be an S-scheme of finite type. Let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibers of X/S. In other words, for every point $s \in S$ and every extension K of $k(s)$, coherent sheaves F_K are specified on the algebraic scheme X_K; and F_K and $F'_{K'}$ are declared to lie in the same class if there are $k(s)$-homomorphisms from K and K' into a common extension K'' of $k(s)$ such that

$$F_{K''} = F_K \otimes_K K'' \quad \text{and} \quad F'_{K''} = F'_{K'} \otimes_{K'} K''$$

are isomorphic on $X_{K''}$. The family $\mathcal{F}$ is said to be *bounded by the coherent sheaf F on $X_T = X \times_S T$*, where T is of finite type over S, if $\mathcal{F}$ is contained in the family of classes of the sheaves $F_{k(t)}$, with $t \in T$. The family $\mathcal{F}$ is said to be *bounded* if such T and F exist.

Suppose in addition that X/S is projective and endowed with a (relatively) ample invertible sheaf $\mathcal{O}_X(1)$. Then $\mathcal{F}$ is called a (b)-*family*, for a sequence of integers $(b) = (b_0, \ldots, b_r)$, if every class in $\mathcal{F}$ is represented by an F_K, with K algebraically closed, which is a (b)-sheaf.

Theorem 1.13. *Let S be a noetherian scheme, and let X be a projective S-scheme endowed with an ample sheaf $\mathcal{O}_X(1)$ such that the induced sheaves $\mathcal{O}_{X_s}(1)$, for $s \in S$, are generated by their $H^0(\mathcal{O}_{X_s}(1))$. Let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibers of X/S. Then the following conditions are equivalent:*

(i) *$\mathcal{F}$ is bounded. Moreover, if all the $F_K \in \mathcal{F}$ are locally free of rank p, then $\mathcal{F}$ may be assumed to be bounded by a sheaf F on some X_T, with F locally free of rank p.*

(ii) *The set of Hilbert polynomials $\chi(F_K(n))$, for $F_K \in \mathcal{F}$, is finite, and there exists a sequence of integers (b) such that $\mathcal{F}$ is a (b)-family.*

(iii) *The set of Hilbert polynomials $\chi(F_K(n))$, for $F_K \in \mathcal{F}$, is finite, and there exists an integer m such that all the $F_K \in \mathcal{F}$ are m-regular.*

(iv) *The set of Hilbert polynomials $\chi(F_K(n))$, for $F_K \in \mathcal{F}$, is finite, and $\mathcal{F}$ is contained in the family of classes of quotients of sheaves of the form E_K, where E is a coherent sheaf on some X_T. Moreover, one may assume that $T = S$ and that E has the form $\mathcal{O}_X(-m)^{\oplus M}$.*

(v) *$\mathcal{F}$ is contained in the family of classes of cokernels of homomorphisms of the form $E'_K \to E_K$, where E' and E are coherent sheaves on some X_T. Moreover, one may assume that $T = S$ and that E' and E have the forms $\mathcal{O}_X(-m')^{\oplus M'}$ and $\mathcal{O}_X(-m)^{\oplus M}$, respectively.*

Indeed, suppose that $\mathcal{F}$ is bounded by a sheaf F on some X_T. After partitioning T into pieces and applying the generic flatness theorem (EGA IV 6.9.1), one may assume that F is flat over T. Then, the number of polynomials $\chi(F(n))$ is at most the number of connected components of T (EGA III 7.9.5). It is also an elementary lemma that, if $t \in T$ and F_t is locally free of rank p, then F is locally free of rank p over an open neighborhood of t in T (see EGA IV 11.3.10, with $X = Y$ and $f = \mathrm{id}$). The hypotheses and EGA III 7.7.6 show, moreover, that after partitioning T still more finely one can find a sequence σ of sections

$$\sigma_1, \ldots, \sigma_r \in H^0(\mathcal{O}_X(1))$$

that is F-regular. Thus (ii) follows from the upper semicontinuity of the function

$$t \longmapsto h^0(F_{t,i}(-1)),$$

where, for $t \in T$ and $0 \leq i \leq r$, $F_{t,i}$ denotes the restriction of F_t to the zero scheme of $\sigma_1, \ldots, \sigma_i$ (EGA III 7.7.5).

The implication (ii) $\Rightarrow$ (iii) follows immediately from 1.11. The implication (iii) $\Rightarrow$ (iv) follows from 1.3 (iii), on taking

$$M = \max(\chi(F_K(m))), \qquad E = \mathcal{O}_X(-m)^{\oplus M}.$$

Suppose that $\mathcal{F}$ satisfies (iv). There are exact sequences

$$0 \longrightarrow F'_K \longrightarrow E_K \longrightarrow F_K \longrightarrow 0,$$

and the set of polynomials $\chi(F_K(n))$ is finite. By hypothesis, the family of classes of the E_K is bounded. Hence, by the already proved implication (i) $\Rightarrow$ (ii), the set of polynomials $\chi(E_K(n))$ is finite, and there exists a sequence of integers (b) such that all the E_K, with K algebraically closed, are (b)-sheaves. It follows that the set of polynomials $\chi(F'_K(n))$ is finite and, by 1.6, all the F'_K are (b)-sheaves. Applying the already proved implication (ii) $\Rightarrow$ (iv) to the family of classes of the F'_K, one obtains (v).

Finally, suppose that $\mathcal{F}$ satisfies (v), and let us prove that $\mathcal{F}$ is bounded. One may assume that E (respectively E') has the form $\mathcal{O}_X(-m)^{\oplus M}$. Indeed, by (i) $\Rightarrow$ (iv), take a surjection

$$L = \mathcal{O}_{X_T}(-m)^{\oplus M} \longrightarrow E.$$

After partitioning T into pieces so that E becomes flat over T, formation of the exact sequence

$$0 \longrightarrow I \xrightarrow{u} L \longrightarrow E \longrightarrow 0$$

commutes with passage to the fibers. By (i) $\Rightarrow$ (iii) and (i) $\Rightarrow$ (iv), choose m_1 sufficiently large that all the groups

$$\mathrm{Ext}^1(\mathcal{O}_{X_K}(-m_1), I_K) = H^1(I_K(m_1))$$

vanish and that there exist surjections

$$L_1 = \mathcal{O}_{X_T}(-m_1)^{\oplus M_1} \longrightarrow E', \qquad \alpha : L_2 = \mathcal{O}_{X_T}(-m_1)^{\oplus M_2} \longrightarrow I.$$

The maps

$$\mathrm{Hom}(L_{1,K}, L_K) \longrightarrow \mathrm{Hom}(L_{1,K}, E_K)$$

are then surjective. Let $\beta : E'_K \to E_K$ be a homomorphism, and let γ be the composite

$$L_{1,K} \longrightarrow E'_K \xrightarrow{\beta} E_K.$$

Then γ comes from a homomorphism $\delta : L_{1,K} \to L_K$, and

$$(\delta, u_K\alpha) : L_{1,K} \oplus L_{2,K} \longrightarrow L_K$$

and $\beta : E'_K \to E_K$ have the same cokernel. Consequently, $\mathcal{F}$ is contained in the family of classes of cokernels of homomorphisms of the form

$$\mathcal{O}_{X_K}(-m_1)^{\oplus(M_1+M_2)} \longrightarrow \mathcal{O}_{X_K}(-m)^{\oplus M}.$$

Still after partitioning T into pieces, one may assume in addition that X_T (and hence $\mathcal{H}om_{\mathcal{O}_{X_T}}(E', E)$) and all the

$$R^q(f_T)_*\mathcal{H}om_{\mathcal{O}_{X_T}}(E', E),$$

where $f_T : X_T \to T$ is the structural morphism, are flat over T, so that formation of

$$G = (f_T)_*\mathcal{H}om_{\mathcal{O}_{X_T}}(E', E)$$

commutes with every base change $T' \to T$ (EGA III 6.9.9.2). Let

$$R = \mathbb{V}(G^*).$$

Then X_R canonically carries a *universal* exact sequence

$$E'_R \longrightarrow E_R \longrightarrow F \longrightarrow 0,$$

and F bounds $\mathcal{F}$.

2. Several technical lemmas

2.1. Let k be a field and X a proper k-scheme. Let $L_1, \dots, L_s$ be invertible sheaves on X, let $c_1(L_i)$ be the Chern class of L_i, and let Y be a closed subscheme of X of dimension $\leq s$. Recall that the notation

$$\langle c_1(L_1) \cdots c_1(L_s) \cdot Y \rangle$$

denotes the intersection number, which may be calculated as the coefficient of the monomial $n_1 \cdots n_s$ in the Snapper polynomial

$$\chi\Big(L_1^{\otimes n_1} \otimes \cdots \otimes L_s^{\otimes n_s} \otimes \mathcal{O}_Y\Big)$$

(X 4.3.1).

2.2. From now on, suppose that k is algebraically closed, that X is projective and endowed with a very ample invertible sheaf $H = \mathcal{O}_X(1)$, and that X is integral of dimension r.

Let L be an invertible sheaf on X. The integer

$$\langle c_1(L) \cdot c_1(H)^{r-1} \rangle$$

is called the *degree* of L (relative to H) and is denoted $d(L)$.

In what follows we consider an invertible sheaf L and the Hilbert polynomials

$$\chi(L(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}, \qquad \chi(\mathcal{O}_X(n)) = \sum_{i=0}^{r} e_i \binom{n+i}{i}.$$

Lemma 2.3. *One has:*

(i) $a_r = e_r = d(H)$.

(ii) *If* $r \geq 1$, *then* $a_{r-1} - e_{r-1} = d(L)$.

Indeed, (i) and (ii) follow immediately by setting $m = 0, 1$ in the following polynomial:

$$\chi(L^{\otimes m} \otimes H^{\otimes n}) = \sum_{i+j \leq r} f_{ij} \binom{m+i}{i} \binom{n+j}{j}.$$

Lemma 2.4. *If* $d(L) \leq 0$ *and* $h^0(L) \neq 0$, *then* $L \simeq \mathcal{O}_X$; *hence* $d(L) = 0$ *and* $h^0(L) = 1$.

Indeed, since X is integral, there is a divisor $Y \geq 0$ such that $L \simeq \mathcal{O}_X(Y)$. If $Y > 0$, one would have

$$0 \geq \langle c_1(L) \cdot c_1(H)^{r-1} \rangle = \langle c_1(H)^{r-1} \cdot Y \rangle \geq 1.$$

Lemma 2.5. *If* $d(L) \leq 0$, *then* L *is a* $(0, \ldots, 0, a_r)$*-sheaf (with* $r - 1$ *zeros).*

Indeed, if $r = 0$, then L is plainly an a_0-sheaf. If $r \geq 1$, then by 2.4 one has $h^0(L(-1)) = 0$, since $d(L(-1)) < 0$. Let Y be a hyperplane section of X, and let

$$0 \longrightarrow L(-1) \longrightarrow L \longrightarrow L_Y \longrightarrow 0$$

be an exact sequence defined by Y. If $r = 1$, then by 1.7 one has $h^0(L_Y(-1)) = a_1$; hence L is a $(0, a_1)$-sheaf. If $r \geq 2$, one may take Y integral by Bertini's theorem and may thus assume inductively that L_Y is a $(0, \ldots, 0, a_r)$-sheaf; the assertion follows.

Lemma 2.7. *If* $r = 2$ *and* $d(L) \leq 0$, *then:*

(i) *For every* n,

$$h^1(L(n)) \leq a_1(a_1 + 1)a_2 - a_0.$$

(ii) $a_0 \leq a_1(a_1 + 1)$.

Let Y be an integral hyperplane section of X. By 1.7 and 2.6, L_Y is a $(0, a_2)$-sheaf with Hilbert polynomial

$$\chi(L_Y(n)) = a_1 + a_2 \binom{n+1}{1}.$$

Thus, by 1.11, $-a_1 \geq 0$, and L_Y is $[-a_1]$-regular.

Consider the following exact sequence:

$$H^0(L_Y(n)) \longrightarrow H^1(L(n-1)) \longrightarrow H^1(L(n)) \longrightarrow H^1(L_Y(n)).$$

For $n \geq -a_1 - 1$, one has $h^1(L(-a_1 - 2)) \geq h^1(L(n))$, since $h^1(L_Y(n)) = 0$ by 1.3 (i). For $n \leq -1$, one has $h^1(L(n-1)) \leq h^1(L(-1))$, since $h^0(L_Y(n)) = 0$ by 2.4. Finally, for every n, $h^0(L_Y(n-1)) \leq h^0(L_Y(n))$, since there is an injection $L_Y(n-1) \to L_Y(n)$, and

$$h^0(L_Y(-a_1 - 1)) = a_2 \binom{-a_1}{1} + a_1,$$

since $h^1(L_Y(-a_1 - 1)) = 0$. Hence, for $n \leq -a_1 - 2$,

$$h^1(L(n)) - h^1(L(n+1)) \leq -a_2 a_1 + a_1.$$

It follows that, for every n,

$$h^1(L(n)) \le h^1(L(-a_1-2)) + (-a_1-1)(-a_2a_1+a_1). \tag{2.7.1}$$

Since L_Y is $[-a_1]$-regular, one has $h^2(L(-a_1-2))=0$ by 1.4; hence

$$h^1(L(-a_1-2)) = h^0(L(-a_1-2)) - \chi(L(-a_1-2)).$$

But

$$h^0(L(-a_1-2)) \le a_2\binom{-a_1}{2}$$

by 1.8 (i), since L is a $(0,0,a_2)$-sheaf by 2.5. Therefore

$$0 \le h^1(L(-a_1-2)) \le -a_1(-a_1-1) - a_0. \tag{2.7.2}$$

Finally, 2.7.2 gives (ii), and 2.7.1 together with 2.7.2 gives (i).

Lemma 2.8. *Suppose that $r=2$ and $d(L)\le 0$. Put $\delta = d(L)$ and $\beta = \delta(e_2-1)+e_1$. Then:*

$$a_0 \ge \langle c_1(L)^2\rangle + 2e_0 + \sum_{i=1}^{2}(2e_i - a_i)\binom{\delta-1+i}{i} + \beta(\beta+1).$$

Indeed, write

$$\chi(L^{\otimes m}(\delta-1)) = \alpha_2 m^2 + \alpha_1 m + \alpha_0.$$

Comparison of polynomials gives

$$2\alpha_2 = \langle c_1(L)^2\rangle, \qquad \alpha_0 = \sum_{i=0}^{2} e_i\binom{\delta-1+i}{i},$$

and

$$\alpha_2+\alpha_1+\alpha_0 = \sum_{i=0}^{2} a_i\binom{\delta-1+i}{i}.$$

Write

$$\chi(L^{-1}(\delta)(n)) = \sum_{i=0}^{2}\beta_i\binom{n+i}{i}.$$

Then $\beta_0 = \alpha_2 - \alpha_1 + \alpha_0$, and by 2.3,

$$\beta_1 = d(L^{-1}(\delta)) + e_1 = \beta.$$

Thus

$$a_0 = 2\alpha_2 + 2\alpha_0 - \sum_{i=1}^{2} a_i\binom{\delta-1+i}{i} - \beta_0.$$

Since $d(L^{-1}(\delta)) \le 0$, one has $\beta_0 \le \beta(\beta+1)$ by 2.7 (ii); whence the assertion.

Lemma 2.9. *Suppose that $r\ge 2$, that $d(L)\le 0$, and that X is normal. Then:*

(i) *$h^1(L(-n)) = 0$ for $n \ge 2 + h_2(a_{r-2}, a_{r-1}, a_r)$;*

(ii) *$h^1(L(n)) \le h_r(a_{r-2}, a_{r-1}, a_r)$ for every n;*

(iii) *$|a_i| \le g_{r-i}(a_{r-2}, a_{r-1}, a_r)$ for $i = 0, \dots, r$, where $h_2(X_0,X_1,X_2), h_3(X_0,X_1,X_2), \dots$ and $g_0(X_0,X_1,X_2), g_1(X_0,X_1,X_2), \dots$ are universal polynomials with coefficients in $\mathbb{Q}$, independent of both k and X.*

Indeed, first consider the case $r = 2$. Put

$$g_i(X_0, X_1, X_2) = X_{2-i}^2 \quad (i = 0, 1, 2), \qquad h_2(X_0, X_1, X_2) = X_1(X_1 + 1)X_2 - X_0.$$

Then (iii) is clear, and (ii) is 2.7 (i).

Let $X \subset P = \mathbb{P}_k^N$, and put

$$\omega = \mathcal{E}xt_{\mathcal{O}_P}^{N-2}(\mathcal{O}_X, \Omega_{P/k}^N).$$

Then

$$h^1(L(-n)) = h^1(\omega \otimes L^{-1}(n)),$$

and ω is invertible outside the finite set Z of singular points of X (see FGA 149 § 6). Let Y be a normal hyperplane section of X not containing Z. Then $\omega_Y(1) \simeq \Omega_{Y/k}^1$, and there is an exact sequence

$$0 \longrightarrow \omega \otimes L^{-1}(-1) \longrightarrow \omega \otimes L^{-1} \longrightarrow \Omega_{Y/k}^1 \otimes L_Y^{-1}(-1) \longrightarrow 0.$$

Now

$$h^1(\Omega_{Y/k}^1 \otimes L_Y^{-1}(n)) = h^0(L_Y(-n))$$

for every n, and $h^0(L_Y(-n)) = 0$ for $n \geq 1$ by 2.5. Thus $\Omega_{Y/k}^1 \otimes L_Y^{-1}(-1)$ is 3-regular and, by 1.4, $\omega \otimes L^{-1}$ is

$$[3 + h^1(\omega \otimes L^{-1}(2))]\text{-regular.}$$

Consequently, $h^1(L(-n)) = 0$ for $n \geq 2 + h^1(L(-2))$; this proves (i).

Now suppose that $r \geq 3$, and argue by induction on r. Let Y be a normal hyperplane section of X, and consider the exact sequence

$$H^0(L_Y(-n)) \longrightarrow H^1(L(-n-1)) \longrightarrow H^1(L(-n)) \longrightarrow H^1(L_Y(-n)). \tag{2.9.1}$$

By 1.7,

$$\chi(L_Y(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

Thus, for $n \geq 2 + h_2(a_{r-2}, a_{r-1}, a_r)$, the induction hypothesis gives $h^1(L_Y(-n)) = 0$; and since $n \geq 2$ by (ii) for $r = 2$, 2.4 gives $h^0(L_Y(-n)) = 0$. Hence

$$h^1(L(-n-1)) = h^1(L(-n)).$$

But for $n \gg 0$, one has $h^1(L(n)) = 0$ (see FGA 149 § 6); hence (i).

As induction hypothesis, suppose that the polynomials g_i and h_i have been defined for $i \leq r - 1$. For every n, 2.9.1 gives

$$h^1(L(n)) \leq h^1(L(n-1)) + h_{r-1}(a_{r-2}, a_{r-1}, a_r).$$

Therefore, by (i) already proved, it follows that for every $m \geq -2$,

$$h^1(L(n)) \leq [2 + h_2(a_{r-2}, a_{r-1}, a_r) + m]h_{r-1}(a_{r-2}, a_{r-1}, a_r) \tag{2.9.2}$$

for every $n \leq m$.

By 2.5, L_Y is a $(0, \ldots, 0, a_r)$-sheaf. Put

$$c_i = g_{r-i}(a_{r-2}, a_{r-1}, a_r) \quad (i = 1, \ldots, r), \qquad m = P_{r-2}(c_1, \ldots, c_{r-1}).$$

By 1.11, $m \geq 0$, and L_Y is m-regular. Hence, by 1.4 (ii),

$$h^1(L(m-2)) \geq h^1(L(n)) \qquad \text{for every } n \geq m - 1.$$

Thus, if polynomials are defined by

$$M = P_{r-2}(g_{r-1}, \dots, g_1), \qquad h_r = (h_2 + M)h_{r-1},$$

then (ii) follows from 2.9.2.

Finally, since L_Y is m-regular, one has $h^q(L(m)) = 0$ for $q \geq 2$ by 1.4 (i). Hence

$$\sum_{i=0}^{r} a_i \binom{m+i}{i} = h^0(L(m)) - h^1(L(m)).$$

But, by 1.8 (i),

$$h^0(L(m)) \leq a_r \binom{m+r}{r}.$$

Therefore, if one defines

$$g_r = \sum_{i=1}^{r} g_{r-i} \binom{M+i}{i} + a_r \binom{M+r}{r} + h_r,$$

then (iii) follows.

Remark 2.10. For fixed m, consider an equality of polynomials in n:

$$\sum_{i=0}^{r} a_{m,i} \binom{n+i}{i} = \sum_{j=0}^{r} a_j \binom{m+n+j}{j}.$$

One readily finds the formula

$$a_{m,i} = \sum_{j=0}^{r-i} a_{j+i} \binom{m-1+i}{j}$$

by induction on r, reducing $a_{m,i}$ to $a_{m,0}$, and then setting $n = -1$.

Similarly, one finds a polynomial with rational coefficients

$$P_i^{(m,r)}(X_r, \dots, X_i),$$

increasing in X_i, such that in the equality

$$\sum_{i=0}^{r} a_i^{(m)} \binom{n+i}{i} = \sum_{j=0}^{r} a_j \binom{mn+j}{j},$$

one has

$$a_i^{(m)} = P_i^{(m,r)}(a_r, \dots, a_i).$$

Lemma 2.11. *Let S be a noetherian scheme and X a projective S-scheme endowed with a very ample sheaf $H = \mathcal{O}_X(1)$. Suppose that the fibers of X/S are geometrically integral and all have the same dimension r. Let $\mathcal{L}$ be a family of classes of invertible sheaves L_K on the fibers of X/S. For $\mathcal{L}$ to be bounded, it is necessary and sufficient that all the coefficients a_i in the Hilbert polynomials*

$$\chi(L_K(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$$

of the $L_K \in \mathcal{L}$ remain bounded. Moreover, if the fibers of X/S are also geometrically normal, then it is enough that the three coefficients a_r, a_{r-1}, a_{r-2} remain bounded.

Indeed, let

$$\delta = \sup\{d(L_K)\}.$$

If $\delta > 0$, one may replace $\mathcal{L}$ by the family of classes of the $L_K(-\delta)$, by 2.3 and 2.10; thus one may assume $\delta \leq 0$. Then $\mathcal{L}$ is a (b)-family by 2.5, and is therefore bounded if all the coefficients a_i remain bounded, by 1.13. If the fibers of X/S are geometrically normal, it is enough for this purpose that the three coefficients a_r, a_{r-1}, a_{r-2} remain bounded, by 2.9.

3. General finiteness theorems

Lemma 3.1. *Let S be a scheme and X a quasi-compact and quasi-separated S-scheme. Let $\underline{\mathrm{Pic}}_{X/S}$ be the Picard functor, localized for the fppf topology.*[25] *Then, for every filtered inverse system (Z_α) of quasi-compact and quasi-separated S-schemes with affine transition morphisms, the canonical map*

$$\varinjlim_\alpha \underline{\mathrm{Pic}}_{X/S}(Z_\alpha) \longrightarrow \underline{\mathrm{Pic}}_{X/S}\left(\varprojlim_\alpha Z_\alpha\right) \tag{3.1.1}$$

is bijective.

Consequently (EGA IV 8.14.2), *if the Picard functor is represented by an S-scheme* $\mathrm{Pic}_{X/S}$, *then* $\mathrm{Pic}_{X/S}$ *is locally of finite presentation.*

Indeed, an element λ_α of $\underline{\mathrm{Pic}}_{X/S}(Z_\alpha)$ is represented by an invertible sheaf L'_α on $X_{Z'_\alpha}$, where $Z'_\alpha \to Z_\alpha$ is an fppf morphism. Put

$$Z = \varprojlim_\alpha Z_\alpha.$$

The image λ of λ_α in $\underline{\mathrm{Pic}}_{X/S}(Z)$ is represented by the invertible sheaf L' on $X_{Z'}$ obtained by inverse image from L'_α, where

$$Z' = Z'_\alpha \times_{Z_\alpha} Z.$$

If λ is the neutral element of $\underline{\mathrm{Pic}}_{X/S}(Z)$, there exists an fppf morphism $Z'' \to Z'$ such that the inverse image of L' on $X_{Z''}$ is isomorphic to $\mathcal{O}_{X_{Z''}}$. By EGA IV 8.8.2, 8.10.5 (vi), 11.2.6 (ii), 8.5.2 (i), and 8.5.2.4, there exist $\beta \geq \alpha$ and an fppf morphism

$$Z''_\beta \longrightarrow Z'_\beta, \qquad Z'_\beta = Z'_\alpha \times_{Z_\alpha} Z_\beta,$$

such that the inverse image of L'_α on $X_{Z''_\beta}$ is isomorphic to $\mathcal{O}_{X_{Z''_\beta}}$. The image of λ_α is therefore already the neutral element in $\underline{\mathrm{Pic}}_{X/S}(Z_\beta)$. Thus 3.1.1 is injective.

The surjectivity of 3.1.1 is proved similarly.

Proposition 3.2. *Let S be a noetherian scheme and X a proper S-scheme. Suppose that the Picard functor is represented by an S-scheme* $\mathrm{Pic}_{X/S}$. *Then:*

(i) $\mathrm{Pic}_{X/S}$ *is locally of finite type over S.*

(ii) *Subsets Λ of* $\mathrm{Pic}_{X/S}$ *correspond bijectively to families $\mathscr{L}$ of classes of invertible sheaves on the fibers of X/S.*

(iii) *A subset Λ of* $\mathrm{Pic}_{X/S}$ *is quasi-compact if and only if the corresponding family $\mathscr{L}$ is bounded.*

[25] Cf. SGA 3 IV 6.3 and FGA no. 232 for the definition of $\underline{\mathrm{Pic}}_{X/S}$; here, as indicated, the fpqc topology should be replaced by the fppf topology.

Indeed, (i) follows from 3.1. Assertion (ii) follows from the following fact: if s is a point of S and K is an algebraically closed extension of $k(s)$, then the K-points of $\mathrm{Pic}_{X/S}$ correspond bijectively to the isomorphism classes of invertible sheaves on X_K.

For (iii), suppose that the subset Λ of $\mathrm{Pic}_{X/S}$ is quasi-compact. Then Λ is contained in an open subscheme U of finite type, and the inclusion $U \to \mathrm{Pic}_{X/S}$ corresponds to an invertible sheaf L on some X_T, where $T \to U$ is a surjective (flat) morphism of finite type. Finally $T \to S$ is of finite type, so L bounds $\mathscr{L}$.

Conversely, suppose that $\mathscr{L}$ is bounded by a coherent sheaf L on some X_T, with T of finite type over S. One may assume that L is invertible (1.13 (i)), so that L defines a morphism

$$f : T \longrightarrow \mathrm{Pic}_{X/S}\,.$$

The image of f contains Λ and is quasi-compact, hence noetherian by (i). It follows that Λ is quasi-compact, proving (iii).

By 3.2, the usual definition of morphisms of finite type between two Picard schemes generalizes as follows.

Definition 3.3. *Let X and Y be proper over a noetherian scheme S. A morphism between the Picard functors*

$$\underline{\mathrm{Pic}}_{Y/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is said to be of finite type if every bounded family of classes of invertible sheaves on the fibers of X/S has as inverse image a bounded family of classes of invertible sheaves on the fibers of Y/S.

Lemma 3.4. *Let X, Y, X', Y' be proper over a noetherian scheme S, and let $T \to S$ be surjective and of finite type. Then:*

(i) *A morphism $\Phi : \underline{\mathrm{Pic}}_{Y/S} \to \underline{\mathrm{Pic}}_{X/S}$ is of finite type if and only if $\Phi_T : \underline{\mathrm{Pic}}_{Y_T/T} \to \underline{\mathrm{Pic}}_{X_T/T}$ is of finite type.*

(ii) *In a commutative diagram*

$$\begin{array}{ccc} \underline{\mathrm{Pic}}_{Y'/S} & \xrightarrow{\Phi'} & \underline{\mathrm{Pic}}_{X'/S} \\ {\scriptstyle \Phi_Y}\uparrow & & \uparrow{\scriptstyle \Phi_X} \\ \underline{\mathrm{Pic}}_{Y/S} & \xrightarrow[\Phi]{} & \underline{\mathrm{Pic}}_{X/S} \end{array}$$

in which Φ' and Φ_Y are of finite type, Φ is of finite type.

Suppose that Φ_T is of finite type. Let $\mathscr{L}$ be a family on the fibers of X/S, bounded by an invertible sheaf L on some X_R. The family $\mathscr{L}_T = \mathscr{L} \times_S T$ on the fibers of X_T/T is bounded by the inverse image of L on $X_{R\times_S T}$. If $\Phi_T^{-1}(\mathscr{L}_T)$ is bounded by an invertible sheaf M on some Y_Q, then M also bounds $\Phi^{-1}(\mathscr{L})$. The converse in (i), and assertion (ii), are proved similarly.

Theorem 3.5. *Let S be a noetherian scheme, let X and Y be proper S-schemes, and let $f : X \to Y$ be a morphism. If f is surjective, then the induced morphism between the Picard functors*

$$f^* : \underline{\mathrm{Pic}}_{Y/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is of finite type.

By 3.4 (i), one may assume that S is reduced and irreducible. There is a nonempty open subset $U \subset S$ such that $\underline{\mathrm{Pic}}_{Y/S}$ and $\underline{\mathrm{Pic}}_{X/S}$ are representable and the restriction of f^* is quasi-affine

(XII 1.1), hence of finite type by 3.2 (i). Give $T = S - U$ its induced reduced structure. By noetherian induction, one may suppose that

$$\underline{\mathrm{Pic}}_{Y_T/T} \longrightarrow \underline{\mathrm{Pic}}_{X_T/T}$$

induced by f^* is of finite type. The assertion follows from 3.4 (i), applied to $U \amalg T \to S$.

Theorem 3.6. *Let S be a noetherian scheme, let X be a proper S-scheme, and let $\underline{\mathrm{Pic}}_{X/S}$ be its Picard functor. For every nonzero integer n, the n-th power homomorphism*

$$\Phi_n : \underline{\mathrm{Pic}}_{X/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is a morphism of finite type.

Let $f : X' \to X$ be a Chow presentation: X' is projective over S, and f is surjective. If the theorem is known for X', then 3.5 and 3.4 (ii) imply it for X. We may suppose X/S projective.

There are a nonempty open $U \subset S$, an integral S', and a finite surjective morphism $S' \to U$, such that every irreducible component X'_i of $X_{S'}$, with its induced reduced structure, is integral over S' (XII 1.3). Give $T = S - U$ its induced reduced structure. By noetherian induction, the theorem may be assumed for X_T/T. If it is assumed for all X_i/S', it follows for $\coprod_i X_i/S'$ by the following easy lemma, then for $X_{S'}/S'$ by 3.5 and 3.4 (ii), and finally for X/S by 3.4 (i).

Lemma 3.7. *Let $X = \coprod_i X_i$ be of finite type over a noetherian scheme S, and let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibers of X/S. Then $\mathcal{F}$ is bounded if and only if, for every i, the induced family $\mathcal{F}_i$ on the fibers of X_i/S is bounded.*

In the proof of 3.6 we reduced to X projective and integral over S; endow X/S with a very ample sheaf $\mathcal{O}_X(1)$. Let $\mathscr{L}$ be bounded on the fibers of X/S, and let L on some X_T, with T of finite type over S, bound it. After partitioning T, suppose L flat over T; after taking connected components, suppose T connected. Then

$$P(q, m) = \chi(L_t^{\otimes q}(m))$$

is independent of $t \in T$ (EGA III 7.9.4).

Put $\mathscr{L}_n = \Phi_n^{-1}(\mathscr{L})$. It consists of classes of invertible sheaves M_K whose n-th power class belongs to $\mathscr{L}$. Write

$$\chi(M_K^{\otimes p}(m)) = \sum_{i=0}^{r} a_i(p) \binom{m+i}{i},$$

where $a_i(p)$ has degree at most $r - i$ in p. Since $\chi(M_K^{\otimes qn}(m)) = P(q, m)$, every $a_i(qn)$, and hence every polynomial $a_i(p)$, is independent of M_K. Thus $\mathscr{L}_n$ has one Hilbert polynomial $\chi(M_K(m))$, and is bounded by 2.11.

Theorem 3.8. *Let S be noetherian, let Y be a projective S-scheme with a relatively ample invertible sheaf $\mathcal{O}_Y(1)$, let X be the zero scheme of a section φ of $\mathcal{O}_Y(1)$, and let $f : X \to Y$ be the inclusion. Assume*

that for every $s \in S$ and every irreducible component Y_s^i of Y_s, the irreducible components of $X_s \cap Y_s^i = V(\varphi|_{Y_s^i})$ have dimension at least 2 (as happens when the irreducible components of the fibers of Y have dimension at least 3). Then

$$f^* : \underline{\mathrm{Pic}}_{Y/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is of finite type. Moreover, if S is integral, there is a nonempty open $U \subset S$ such that the restricted Picard functors are representable and

$$\underline{\mathrm{Pic}}_{Y/S}|_U \longrightarrow \underline{\mathrm{Pic}}_{X/S}|_U$$

is quasi-affine (and of finite type).

For the first assertion, noetherian induction permits replacing S by any nonempty open. One reduces to S integral and Y integral over S, replacing X by $X \times_Y Y'$ when Y is replaced by Y'. One then reduces to Y normal over S, using the following lemma in place of XII 1.3.

Lemma 3.9. *Let S be integral noetherian and X an S-scheme integral over S. There are a nonempty open $U \subset S$, an integral scheme S' finite faithfully flat over U, an S'-scheme X' normal and integral over S', and a finite surjective S'-morphism $X' \to X_{S'}$.*

Let η be the generic point of S, and let K be an algebraic closure of $k(\eta)$. The scheme $\mathrm{Spec}(K)$ is the limit of the filtered inverse system (S_α) of integral affine schemes with finite faithfully flat morphisms $S_\alpha \to U_\alpha$, where U_α is a nonempty open of S. The normalization of X_K descends to an S_α-scheme X' for some α, and the finite surjective map $X'_K \to X_K$ descends to a finite surjective S_α-morphism $X' \to X_{S_\alpha}$ (EGA IV 8.8.2, 8.10.5). Since X'_K is normal and integral, there is an open $S' \subset S_\alpha$ such that $X'_{S'}$ is normal and integral over S' (EGA IV 9.9.5).

Remark 3.10. Retain 3.9 and suppose there is a resolving morphism $X'_K \to X_K$ (EGA IV 7.9.1), as happens when $\dim(X_K) \le 2$, by Abhyankar. A similar argument gives a finite faithfully flat $S' \to U$, a scheme X' smooth over S', and a proper surjective birational morphism $X' \to X_{S'}$.

In the proof of 3.8 we have reduced to S integral and Y normal and integral over S. If $\varphi = 0$, then $X = Y$. Otherwise, after restricting S, suppose that $\varphi_s \neq 0$ for every $s \in S$. Thus φ_s is $\mathcal{O}_{Y_s}$-regular, and the dimension r of Y_s is at least 3.

Let $\mathscr{L}$ be bounded on the fibers of X/S, and let M_K be invertible on a geometric fiber Y_K, with

$$\chi(M_K(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}.$$

The Hilbert polynomial of f^*M_K is

$$\sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

As the class of M_K varies through $(f^*)^{-1}(\mathscr{L})$, that is, as

the class of f^*M_K varies through $\mathscr{L}$, the coefficients a_r, a_{r-1}, a_{r-2} remain bounded by 1.13.

Let $H = \mathcal{O}_Y(m)$ be a very ample multiple of $\mathcal{O}_Y(1)$. In the Hilbert polynomials relative to H,

$$\chi(M_K(mn)) = \sum_{i=0}^{r} a_i^{(m)} \binom{n+i}{i},$$

the coefficients $a_r^{(m)}, a_{r-1}^{(m)}, a_{r-2}^{(m)}$ remain bounded by 2.10. Consequently $(f^*)^{-1}(\mathscr{L})$ is bounded by 2.11, proving the first part of 3.8.

For the second part, generic representability and separatedness of the Picard functors were established in XII 1.2. An argument like XII 2, using 3.9 and XII 1.1, reduces to Y normal over S. It is enough for $\underline{\mathrm{Pic}}_{Y/S} \to \underline{\mathrm{Pic}}_{X/S}$ to be separated and quasi-finite (EGA IV 8.12.8), which can be checked on geometric fibers and follows from:

Lemma 3.11. *Let k be algebraically closed, let Y be normal and projective with an ample sheaf $\mathcal{O}_Y(1)$, and let X be the zero scheme of a section φ. Suppose* $\dim(Y) \ge 3$ *(respectively,* $\dim(Y) \ge 2$*). Then*

$$\mathrm{Pic}_{Y/k} \longrightarrow \mathrm{Pic}_{X/k} \quad \text{(respectively, } \mathrm{Pic}^{\tau}_{Y/k} \longrightarrow \mathrm{Pic}^{\tau}_{X/k}\text{; see 4.1)}$$

is of finite type and has a finite kernel N, in general nonreduced.

Assume $\varphi \neq 0$; it is regular because Y is integral. The map is of finite type by 3.8 (respectively, 4.7 (iii)). Its kernel N is an algebraic group contained in $\mathrm{Pic}^{\tau}_{Y/k}$, hence is the same

in both cases. Since $\mathrm{Pic}^{\tau}_{Y/k}$ is proper because Y is normal (FGA no. 236, 2.13), and N is closed (SGA 3 VI$_B$ 1.9.2), it is enough to show that N contains no subgroup of type μ_ℓ ($= \mathbb{Z}/\ell\mathbb{Z}$), with $(\ell, \mathrm{char}(k)) = 1$.

Consider Kummer theory, arising from

$$0 \longrightarrow \mu_\ell \longrightarrow \mathbb{G}_m \xrightarrow{x \mapsto x^\ell} \mathbb{G}_m \longrightarrow 0$$

on Y (respectively, on X):

$$\begin{array}{ccccccc} & & & & H^1(Y,\mu_\ell) & \longrightarrow & H^1(Y,\mathbb{G}_m) \xrightarrow{\ell} H^1(Y,\mathbb{G}_m) \\ & & & & \downarrow & & \downarrow \qquad\qquad \downarrow \\ H^0(X,\mathbb{G}_m) & \xrightarrow{\ell} & H^0(X,\mathbb{G}_m) & \longrightarrow & H^1(X,\mu_\ell) & \longrightarrow & H^1(X,\mathbb{G}_m) \xrightarrow{\ell} H^1(X,\mathbb{G}_m). \end{array}$$

By Hilbert's Theorem 90, $H^1(Y,\mathbb{G}_m) = \mathrm{Pic}_{Y/k}(k)$ (respectively, $H^1(X,\mathbb{G}_m) = \mathrm{Pic}_{X/k}(k)$). Since X is proper and integral, $H^0(X,\mathbb{G}_m) = k^*$, so $H^1(X,\mu_\ell) \to H^1(X,\mathbb{G}_m)$ is injective.

It remains to show that

$$H^1(Y,\mathbb{Z}/\ell\mathbb{Z}) \longrightarrow H^1(X,\mathbb{Z}/\ell\mathbb{Z})$$

is injective. This follows from $\pi_1(X,\xi) \to \pi_1(Y,\xi)$ surjective; equivalently (SGA 1 V 6.9), every connected étale cover $\widetilde{Y}$ of Y with group $\mathbb{Z}/\ell\mathbb{Z}$ has connected restriction $\widetilde{X} = X \times_Y \widetilde{Y}$. This is a consequence of Bertini and Zariski connectedness (SGA 1 X 2.10).

Remark 3.12. Retain the hypotheses of 3.11.

(i) The kernel N is finite *unipotent* (SGA 3 XVII 1.1), since it contains no subgroup of multiplicative type μ_n, for any n (SGA 3 XVII 4.6.1 (vi)). This is proved similarly, using instead

of Kummer theory the following fact of Grothendieck (SGA 1 XI, last page): if X is proper over k, with $H^0(X,\mathcal{O}_X) = k$, then there is an isomorphism of k-group schemes

$$\underline{\mathrm{Hom}}_{\mathrm{grp}}(\mu_n, \underline{\mathrm{Pic}}_{X/k}) \simeq \underline{H^1(X,\mathbb{Z}/n\mathbb{Z})}.$$

Consequently, when k is algebraically closed, its k-points give

$$\mathrm{Hom}_{\mathrm{grp}}(\mu_n, \underline{\mathrm{Pic}}_{X/k}) \simeq H^1(X,\mathbb{Z}/n\mathbb{Z}).$$

(ii) If k has characteristic zero, then $N = 0$, since every finite unipotent group is zero (SGA 3 XVII 1.7). This approximately recovers Kodaira's vanishing theorem in Mumford's form [2].

Theorem 3.13. *Let S be noetherian and X a projective S-scheme with a relatively ample invertible sheaf $H = \mathcal{O}_X(1)$. Suppose the geometric fibers of X/S are integral of dimension r. For a family $\mathscr{L}$ of classes of invertible sheaves on the fibers, the following are equivalent:*

(i) *$\mathscr{L}$ is bounded.*

(ii) *In $\chi(L_K(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$, the coefficient a_{r-1} remains bounded and a_{r-2} remains bounded below.*

(iii) *The degree*

$$d(L_K) = \langle c_1(L_K)c_1(H_K)^{r-1}\rangle$$

remains bounded and $\langle c_1(L_K)^2c_1(H_K)^{r-2}\rangle$ *remains bounded below.*

(iv) *There is an integer* N *such that*

$$H^{-N} < \mathscr{L} < H^N;$$

that is, $H_K^{\otimes N} \otimes L_K$ *and* $H_K^{\otimes N} \otimes L_K^{-1}$ *are ample for every* $L_K \in \mathscr{L}$.

Suppose $\mathscr{L}$ is bounded by L on some X_T. Then (iv) holds by EGA II 4.6.12. Taking L flat over T, the polynomial $\chi(L_t^{\otimes m}(n))$ depends only on the connected component of T (EGA III 7.9.4). Hence (ii) holds, and 2.1 gives (iii).

The families of classes of H_K and $\mathcal{O}_{X_K}$ are bounded, so they satisfy (iii) and (ii), respectively. If $\mathscr{L}$ satisfies (iv), then

$$\langle [Nc_1(H_K) + c_1(L_K)]c_1(H_K)^{r-1}\rangle > 0,$$

so $-Nd(H_K) < d(L_K) < Nd(H_K)$. Moreover,

$$\langle [Nc_1(H_K) + c_1(L_K)]^2c_1(H_K)^{r-2}\rangle > 0,$$

whence

$$\langle c_1(L_K)^2c_1(H_K)^{r-2}\rangle > -2Nd(L_K) - N^2d(H_K).$$

Thus (iii) holds.

Finally, suppose (ii), respectively (iii), holds. By 2.10 and 2.3, replace H by a multiple and assume it very ample. Partition S so there is a linear section Y of X, integral over S, with two-dimensional fibers. By 3.8, replace $X, \mathscr{L}$ by $Y, \mathscr{L}|_Y$; 1.7, respectively the projection formula, preserves the relevant condition. Thus $r = 2$. Put

$$\delta = \sup\{d(L_K)\}.$$

If $\delta > 0$, replace $\mathscr{L}$ by the classes of $L_K(-\delta)$, using 2.3 and 2.10, so that $\delta \leq 0$.

If a_1 remains bounded and a_0 bounded below in $\sum_{i=0}^{2} a_i\binom{n+i}{i}$, all three coefficients remain bounded by 2.3 and 2.7. Thus $\mathscr{L}$ is bounded by 2.11, so (ii) implies

(i). On the other hand, (iii) implies (ii) by 2.3 and 2.8.

4. Finiteness theorems for $\mathrm{Pic}^{\tau}_{X/S}$

4.1. Let S be a scheme and G a commutative group functor over S. Denote by G^0, respectively G^τ, the subgroup functor whose T-points $f : T \to G$ satisfy: for every algebraically closed field K and K-point t of T, there are a connected K-scheme Z, two K-points z, z_0, and $g : Z \to G$, with $g(z) = f(t)$ and $g(z_0) = 0$ in $G(K)$; respectively, there is a nonzero n such that $nf(t) \in G^0(K)$.

Formation of G^0, respectively G^τ, commutes with base change, and $G \to G'$ induces $G^0 \to G'^0$, respectively $G^\tau \to G'^\tau$. If S is a field and G is represented by a group scheme locally of finite type, G^0 is represented by the connected component of the identity, an open and closed finite-type geometrically irreducible subscheme (SGA 3 VI$_A$ 2.4).

Lemma 4.2. *Let* $\Phi : G \to G'$ *be a morphism of commutative group functors over* S. *Suppose that for every algebraically closed* K *over* S, G_K, G'_K *are represented by group schemes locally of finite type and* Φ_K *by a finite-type (equivalently quasi-compact) morphism. Then*

$$\Phi^{-1}(G'^\tau) = G^\tau.$$

The inclusion $\Phi^{-1}(G'^{\tau}) \supset G^{\tau}$ has already been noted. Conversely, $g \in G(K)$ belongs to $G^{\tau}(K)$ if and only if the set of multiples ng, $n \in \mathbb{N}$, is quasi-compact. Since Φ_K is quasi-compact, it is enough that the set $\Phi(ng) = n\Phi(g)$ have this property. This proves the lemma.

Lemma 4.3. *Let X, Y be proper over S, and let $f : X \to Y$ be surjective (respectively, the inclusion of the zero scheme of a section φ of an ample $\mathcal{O}_Y(1)$). Then*

$$(f^*)^{-1}\underline{\mathrm{Pic}}^{\tau}_{X/S} = \underline{\mathrm{Pic}}^{\tau}_{Y/S}.$$

This follows formally from 4.2, XII 1.2, 3.1, and 3.5 (respectively, 3.8).

4.4. Let k be algebraically closed, X proper over k, and L invertible on X. The sheaf L is *algebraically equivalent* to zero if there are a connected finite-type k-scheme T, an invertible sheaf M on $X_T = X \times_k T$, and two k-points t, t_0, such that $L \simeq M_t$ and $\mathcal{O}_X \simeq M_{t_0}$. It is *$\tau$-equivalent* to zero if some nonzero tensor power is algebraically equivalent to zero. Equivalently, the point λ corresponding to L belongs to $\underline{\mathrm{Pic}}^0_{X/k}(k)$, respectively $\underline{\mathrm{Pic}}^{\tau}_{X/k}(k)$.

Recall that L is *numerically equivalent* to zero if

$$\langle c_1(L) \cdot Y \rangle = 0$$

for every closed integral curve $Y \subset X$ (see 2.1 or X 4.5).

Proposition 4.5 (Functorial behavior). *Let $f : X' \to X$ be a morphism between proper k-schemes, and let L be invertible on X. Then:*

(i) *If L is τ-equivalent to zero (respectively, numerically equivalent to zero), then so is f^*L.*

(ii) *Conversely, if f is surjective and f^*L is τ-equivalent to zero (respectively, numerically equivalent to zero), then so is L.*

The assertions for τ-equivalence follow from 4.1 and 4.3. Let Y' be a closed integral curve in X', and let $Y = f(Y')$, with its induced reduced structure. If $\dim(Y) = 1$, the projection formula (X 4.4.4) gives

$$\langle c_1(f^*L) \cdot Y' \rangle = [k(Y') : k(Y)] \, \langle c_1(L) \cdot Y \rangle. \tag{4.5.1}$$

If $\dim(Y) = 0$, then $(f^*L)_{Y'} \simeq \mathcal{O}_{Y'}$. This proves (i) for numerical equivalence.

Conversely, for a closed integral curve $Y \subset X$, choose a point $y' \in f^{-1}(y)$ algebraic over the function field $k(y)$ of the generic point y, and let Y' be its closure. Then $f(Y') = Y$, and 4.5.1 proves (ii).

Theorem 4.6 (The numerical characterization of $\mathrm{Pic}^{\tau}_{X/k}$). *Let k be algebraically closed, X proper over k, and L invertible on X. The following are equivalent:*

(i) *L is τ-equivalent to zero.*

(ii) *For every coherent F on X,*

$$\chi(F \otimes L) = \chi(F).$$

(iii) *L is numerically equivalent to zero.*

If X/k is projective with an ample $H = \mathcal{O}_X(1)$, these are also equivalent to:

(iv) *$L^{\otimes m}(1)$ is ample for every integer m.*

(v) *If X is irreducible, then*

$$\chi(L^{\otimes m}(n)) = \chi(\mathcal{O}_X(n))$$

for every m, n.

(vi) *If X is irreducible of dimension $r \geq 2$, then*

$$\langle c_1(L)c_1(H)^{r-1}\rangle = 0, \qquad \langle c_1(L)^2 c_1(H)^{r-2}\rangle = 0.$$

For (i) $\Rightarrow$ (ii), devissage (EGA III 3.1) and 4.5 reduce to X integral and $F = \mathcal{O}_X$. It remains to show that $\chi(L^{\otimes m})$ is constant. Replace L by a nonzero tensor power and suppose it algebraically equivalent to zero. If M on $X \times_k T$, with T connected of finite type, deforms L to $\mathcal{O}_X$, then $\chi(M_t^{\otimes m})$ is independent of $t \in T$ (EGA III 7.9.5).

For a closed integral curve Y, take $F = \mathcal{O}_Y$ in (ii). The Riemann formula

$$\chi(L_Y) = \langle c_1(L) \cdot Y\rangle + \chi(\mathcal{O}_Y)$$

shows that (ii) implies (iii).

Let $f : X' \to X$ be a Chow presentation. If (iii) implies (i) for X', then 4.5 gives it for X. Hence assume X projective with an ample H. Passing through the finite surjection $\coprod_i X_i \to X$, where the X_i are the reduced irreducible components, reduces (iii) $\Rightarrow$ (i),(iv) to X integral (4.5, EGA III 2.6.2).

For irreducible X, (ii) implies (v), (iii) implies (vi), and 2.1 gives (v) implies (vi); condition (vi) is unchanged on X_{red}. It remains to prove (vi) implies (i),(iv), and (iv) implies (i), for X integral. Under (vi), respectively (iv), the family of classes $L^{\otimes m}$, $m \in \mathbb{N}$, satisfies 3.13 (iii), respectively (iv), and is bounded. Hence some invertible $\mathcal{M}$ on X_T, with T of finite type over k, bounds it. Choose distinct p, q such that $L^{\otimes p}$ and $L^{\otimes q}$ correspond to the same connected component of T. Then $L^{\otimes(p-q)}$ is algebraically equivalent to zero, proving (i).

Condition (vi) also gives an N such that $L^{\otimes m} \otimes H^{\otimes N}$ is ample for every m. Therefore $(L^{\otimes m} \otimes H)^{\otimes N}$ is ample, so $L^{\otimes m} \otimes H$ is ample. Thus (vi) implies (iv).

Theorem 4.7. *Let S be noetherian and X proper over S. Then:*

(i) $\underline{\mathrm{Pic}}^{\tau}_{X/S} \to \underline{\mathrm{Pic}}_{X/S}$ *is representable by an open immersion.*

(ii) *If X is projective and integral over S, this map*

is also representable by a closed immersion.

(iii) $\underline{\mathrm{Pic}}^{\tau}_{X/S}$ *is of finite type over S; that is, the family of classes τ-equivalent to zero on the fibers is bounded.*

For (i), respectively (ii), take an arbitrary $T \to \underline{\mathrm{Pic}}_{X/S}$ and show that the inverse image of $\underline{\mathrm{Pic}}^{\tau}_{X/S}$ is an open, respectively open and closed, subscheme of T. Locally, $T = \mathrm{Spec}(B)$ over $\mathrm{Spec}(A) \subset S$, and T is the limit of $T_\alpha = \mathrm{Spec}(B_\alpha)$, for the finite-type A-subalgebras $B_\alpha \subset B$. By 3.1 the morphism descends to some T_α, so assume T of finite type over S.

The morphism corresponds to an fppf $R \to T$ and an invertible sheaf L on X_R. By fppf descent replace T by R, then base-change along $R \to S$. Compatibility with base change (4.1) reduces to a section $S \to \underline{\mathrm{Pic}}_{X/S}$ defined by L on X.

Let S_0 be the set of $s \in S$ such that L_s is τ-equivalent to zero. It must be open, and under (ii) also closed. Since S is noetherian, for $s \in S_0$ it suffices that

(a) every generalization of s lies in S_0;

(b) every specialization sufficiently near s lies in S_0;

(c) under (ii), every specialization of s lies in S_0.

Let $f : X' \to X$ be a Chow presentation. If (a),(b),(c), and (iii) hold for X', then 4.3 and 3.5 give them for X; assume X projective over S.

Let H be ample for X/S. If $(L^{\otimes m} \otimes H)_s$ is ample, then it remains ample at every generalization of s (EGA III 4.7.1). The equivalence 4.6 (i) $\iff$ (iv) gives (a).

For (b),(c), and (iii), reduce to S integral, with s generic in (b),(c); for (iii), apply 3.4 (i) to $\coprod_i S_i \to S$, where S_i are the reduced irreducible components.

There are an integral S' and an fppf $S' \to S$ such that every reduced irreducible component X'_i of $X_{S'}$ is integral over S' (XII 1.3). It is enough to prove (b),(iii) for $X_{S'}/S'$, by descent and by noetherian induction with 3.4 (i). The surjection $\coprod_i X'_i \to X_{S'}$, together with 4.3, 3.5, and 3.7, reduces (b),(iii) to (c),(iii) for each X'_i/S'.

Thus we are reduced to proving (c) and (iii), assuming that S is integral with generic point s, and that X is projective and integral over S. For a very ample $\mathcal{O}_X(1)$ for X/S, the values $\chi(L_t^{\otimes m}(n))$ and $\chi(\mathcal{O}_{X_t}(n))$, for every m, n, are independent of $t \in S$ (EGA III 7.9.4). Hence (c) follows from 4.6 (i) $\iff$ (v), which also says that the family of classes of invertible sheaves τ-equivalent to zero on the fibers of X/S has only one

Hilbert polynomial. Thus (iii) follows from 2.11.

Remark 4.8. Let X be proper and integral over a noetherian scheme S. Without the projectivity hypothesis on X/S, it is unknown whether $\underline{\mathrm{Pic}}^{\tau}_{X/S} \to \underline{\mathrm{Pic}}_{X/S}$ remains a *closed* immersion, except when X is normal over S and $\underline{\mathrm{Pic}}_{X/S}$ is *representable* (FGA 236, 2.3).

5. Finiteness theorems of "Néron–Séveri" type

Theorem 5.1. *Let X be proper over noetherian S. Then the Néron–Séveri groups $\underline{\mathrm{Pic}}_{X_K/K}(K)/\underline{\mathrm{Pic}}^0_{X_K/K}(K)$ of the geometric fibers X_K/K of X/S are finitely generated, and their ranks and the orders of their torsion subgroups $\underline{\mathrm{Pic}}^{\tau}_{X_K/K}(K)/\underline{\mathrm{Pic}}^0_{X_K/K}(K)$ remain bounded.*

By noetherian induction, replace S by an arbitrary nonempty open of S_{red}, and suppose $\underline{\mathrm{Pic}}^{\tau}_{X/S}$ represented by a finite-type S-scheme (XII 1.2, 4.7 (iii)). The order of $\underline{\mathrm{Pic}}^{\tau}_{X_K/K}(K)/\underline{\mathrm{Pic}}^0_{X_K/K}(K)$ is the number of connected components of $(\underline{\mathrm{Pic}}^{\tau}_{X/S})_K$, constant over a nonempty open of S (EGA IV 9.7.9).

It remains to bound the rank of $\underline{\mathrm{Pic}}_{X_K/K}(K)/\underline{\mathrm{Pic}}^{\tau}_{X_K/K}(K)$. Reduce as in 4.7 (iii) to S integral and X projective and integral. Let r be the common fiber dimension of X/S. For $r = 0$, respectively 1, the rank is 0, respectively 1. For $r \geq 2$, reduce to $r = 2$ and X smooth. After restricting S, take a linear section Y of X, still

integral over S, with two-dimensional fibers, and replace X by Y using 4.3. Then use 4.3 and 3.10, together with resolution of surface singularities, and replace S by a flat cover of a nonempty open, to make X smooth over S.

By 4.6 (i) $\iff$ (iii) and finiteness and specialization for étale cohomology (SGA 4 XVI 2.2), Theorem 5.1 follows from:

Proposition 5.2. *Let k be algebraically closed and X proper smooth over k. The group $C^p_{\mathrm{num}}(X)$ of codimension-p algebraic cycles modulo numerical equivalence is free of rank at most b^{2p}, the $2p$-th Betti number.*

Disregarding the action of roots of unity, étale cohomology $H^*(X,\mathbb{Q}_\ell) =_{\text{dfn}} \left[\varprojlim_\nu H^*(X,\mathbb{Z}/\ell^\nu\mathbb{Z})\right] \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, where $(\ell, \operatorname{char}(k)) = 1$, is a Weil cohomology: it satisfies Poincaré duality and the Künneth formula, and receives a functorial, nonzero homomorphism from the ring of algebraic cycles modulo rational equivalence $C^*_{\text{rat}}(X)$ (see X 7.13).

Choose $a_1, \dots, a_m \in C^{r-p}_{\text{rat}}(X)$, $r = \dim X$, whose images form a basis for the $\mathbb{Q}_\ell$-subspace of $H^{2(r-p)}(X,\mathbb{Q}_\ell)$ spanned by algebraic cycles. Define

$$\alpha : C^p_{\text{rat}}(X) \longrightarrow \mathbb{Z}^m.$$

It is defined by $\alpha(b) = (\langle b \cdot a_1\rangle, \dots, \langle b, a_m\rangle)$, where $\langle\ \rangle$ denotes the intersection number. Its image identifies with $C^p_{\text{num}}(X)$.

Remark 5.3. If X is a proper smooth surface over algebraically closed k, then its second Betti number is $b^2(X) = c_2(X) - 2 + 2b^1(X)$, where $b^1(X) = 2\dim \operatorname{Pic}^0_{X/k}$, and $c_2(X)$, the second Chern class, is the Euler-Poincaré characteristic, equivalently the self-intersection of the diagonal in $X \times X$. Thus $b^2(X)$ has a cohomology-independent interpretation. In characteristic > 0, the inequality $\operatorname{rank} C^1_{\text{num}}(X) \le b^2(X)$ in this form is due, before étale cohomology, to Igusa, who used deep results on the tame fundamental group of an algebraic curve minus some points.

Theorem 5.4. *Let k be algebraically closed and X proper over k. There are finitely many closed integral curves $Y_1, \dots, Y_p$ in X such that a subset Λ of $\operatorname{Pic}_{X/k}$ is quasi-compact if and only if the intersection numbers $\langle \lambda \cdot Y_i\rangle$, for $\lambda \in \Lambda$, remain bounded. Moreover, p may be taken to be the rank of the Néron–Séveri group $\operatorname{Pic}_{X/k}(k)/\operatorname{Pic}^0_{X/k}(k)$.*

Indeed, $\operatorname{Pic}^\tau_{X/k}$ is an open quasi-compact subgroup of $\operatorname{Pic}_{X/k}$ (4.7). Thus Λ is quasi-compact if and only if the corresponding set of points in $[\operatorname{Pic}_{X/k}(k)/\operatorname{Pic}^\tau_{X/k}(k)] \otimes_{\mathbb{Z}} \mathbb{Q}$ is finite. By 4.6 (i) $\iff$ (iii), τ-equivalence equals numerical equivalence. Choose curves whose images form a basis of the dual vector space; the assertion follows.

6. Appendix: Study of (b)-sheaves on $P = \mathbf{P}^N_k$.

6.1. Let k be an algebraically closed field and let $P = \mathbf{P}^N_k$. Let F be a coherent sheaf on P, with Hilbert polynomial

$$\chi(F(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i},$$

and let

$$(c) = (c_0, \dots, c_r)$$

be a sequence of integers with $(c) \ge (a)$. Put

$$a_{m,i} = \sum_{j=0}^{r-i} a_{j+i}\binom{m-1+j}{j} \quad \text{and} \quad c_{m,i} = \sum_{j=0}^{r-i} c_{j+i}\binom{m-1+j}{j}.$$

Finally, for every integer $q \ge 1$, denote by N_q the largest coherent subsheaf of F whose support has dimension at most $q-1$, and put $F_q = F/N_q$.

Lemma 6.2. *Under the conditions of 6.1, consider an exact sequence*

$$0 \longrightarrow I \longrightarrow \mathcal{O}_P^{\oplus M} \longrightarrow F(m) \longrightarrow 0$$

with $m \ge 0$. Then I is p-regular, where $p = P_r(c_{m,0}, \dots, c_{m,r})$, and P_r is the r-th (b)-polynomial of 1.9.

Indeed, $\mathcal{O}_P$ is plainly a $(0, \dots, 0, 1)$-sheaf ($N+1$ terms), so I is a $(0, \dots, 0, M)$-sheaf by 1.6. By 2.10, $\chi(I(n)) = M\binom{n+N}{N} - \sum_{i=0}^r a_{m,i}\binom{n+i}{i}$. The assertion follows from 1.11 and 1.10.

Proposition 6.3. *Under the conditions of 6.1, suppose that F is a (b)-sheaf. Then, by means of universal polynomials in the a_i, the b_i, and N, one can calculate integers m, M, m_1, M_1 for which there is an exact sequence*

$$\mathcal{O}_P(-m_1)^{\oplus M_1} \longrightarrow \mathcal{O}_P(-m)^{\oplus M} \longrightarrow F \longrightarrow 0.$$

Moreover, N enters only into the calculation of M_1.

Indeed, take $m = P_{r-1}(c_0, \ldots, c_{r-1})$, with $c_i = b_i - a_i$, and $M = \sum_i a_i \binom{m+i}{i}$, so that, by 1.11 and 1.3, there is an exact sequence

$$0 \longrightarrow I \longrightarrow \mathcal{O}_P^{\oplus M} \longrightarrow F(m) \longrightarrow 0.$$

Likewise, taking account of 6.2, take $m_1 = m + p$, where $p = P_r(a_{m,0}, \ldots, a_{m,r})$, and $M_1 = M\binom{p+N}{N} - \sum_i a_i \binom{m_1+i}{i} \quad (= \chi(I(p)))$.

Theorem 6.4. *Under the conditions of 6.1, suppose that F is a quotient of $\mathcal{O}_P(-m)^{\oplus M}$, with $m \geq 0$. Put $b_i = P_{r-i}(c_{m,i}, \ldots, c_{m,r})$ for $i = 0, \ldots, r$, where P_s is the s-th (b)-polynomial of 1.9; put $b = b_1 + m - 1$, and put $B = \sum_{i=1}^{r} a_i \binom{b+i}{i}$. Then:*

(i) *F is b-regular.*

(ii) $-B \leq a_0 = h^0(F(b)) - B$

(iii) *F is a $(b_0, \ldots, b_r)$-sheaf.*

Indeed, choose a section σ of $\mathcal{O}_P(1)$ for which one obtains the following commutative diagram with exact rows and columns:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & I(-1) & \longrightarrow & I & \longrightarrow & J & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathcal{O}_P(-1)^{\oplus M} & \longrightarrow & \mathcal{O}_P^{\oplus M} & \longrightarrow & \mathcal{O}_Y^{\oplus M} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(m-1) & \longrightarrow & F(m) & \longrightarrow & G(m) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

Since $\chi(G(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}$ by 1.7, J is b_1-regular by 6.2; hence, by 1.4(i), $H^q(I(p)) = 0$ for $q \geq 2$ and $p \geq b_1 - q$. The exact sequence $H^q(\mathcal{O}_P^{\oplus M}(p)) \longrightarrow H^q(F(m+p)) \longrightarrow H^{q+1}(I(p))$ then shows that $F(m)$ is $(b_1 - 1)$-regular; this proves (i).

Assertion (i) implies $0 \leq h^0(F(b)) = \chi(F(b)) = \sum_{i=0}^{r} a_i \binom{b+i}{i}$, which proves (ii). By 2.10 and 1.9,

$$\chi(F(b)) = \sum_{i=0}^{r} a_{m,i} \binom{b_1 - 1 + i}{i} \leq b_1 + \sum_{i=0}^{r} c_{m,i} \binom{b_1 - 1 + i}{i} = b_0.$$

Thus $h^0(F(-1)) \leq h^0(F(b)) \leq b_0$. By induction on r, one may suppose that G is a $(b_1, \ldots, b_r)$-sheaf, so (iii) follows from 1.6.2.

Lemma 6.5. *Under the conditions of 6.1, suppose that F has no associated prime cycle of dimension 0.*

(i) *If $h^0(F) \geq 1$, then $h^0(F(-1)) \leq h^0(F) - 1$.*

(ii) *$H^0(F(-n)) = 0$ for $n \geq h^0(F)$.*

(iii) *Let $0 \to F(-1) \to F \to G \to 0$ be exact, and let $n_0 \geq 0$ satisfy $H^0(G(-n_0)) = 0$. Then $H^0(F(-n_0)) = 0$ and $h^0(F) \leq n_0 \cdot h^0(G)$.*

For (i), let $0 \neq \sigma \in H^0(F)$. The section σ defines a nonzero subsheaf $F' = \mathcal{O}_X \cdot \sigma$ of F, so $s = \dim \mathrm{Supp}(F') \geq 1$. Consider the commutative diagram with exact rows and columns

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F'(-1) & \longrightarrow & F' & \longrightarrow & G' & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(-1) & \longrightarrow & F & \longrightarrow & G & \longrightarrow & 0.
\end{array}$$

Since $s \geq 1$, one has $G' \neq 0$; hence F' is not contained in $F(-1)$. In other words, the section σ is not in the image of the injection $H^0(F(-1)) \to H^0(F)$, which proves (i).

Assertion (ii) follows immediately from (i), bearing in mind that $h^0(F) = 0$ plainly implies $h^0(F(-1)) = 0$, since $h^0(F(-1)) \leq h^0(F)$. In (iii) one obtains $0 \leq h^0(F(n)) - h^0(F(n-1)) \leq h^0(G(n))$. Thus, for $n \geq n_0$, $h^0(F(-n_0)) = h^0(F(-n))$, since $h^0(G(-n_0)) = 0$; hence $h^0(F(-n_0)) = 0$ by (ii). Consequently, for $p \geq 1$, $h^0(F(p-n_0)) \leq h^0(G(1-n_0)) + \cdots + h^0(G(p-n_0)) \leq p \cdot h^0(G(p-n_0))$, which gives (iii).

Proposition 6.6. *Under the conditions of 6.1, suppose that F is a $(b_0, \ldots, b_r)$-sheaf. Then F_q is a $((b_{q-1})^q, \ldots, (b_{q-1})^2, b_{q-1}, b_q, \ldots, b_r)$-sheaf.*

Indeed, let σ be a "general" section of $\mathcal{O}_X(1)$. It defines a commutative diagram with exact rows and columns:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & N_q(-1) & \longrightarrow & N_q & \longrightarrow & G' & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(-1) & \longrightarrow & F & \longrightarrow & G & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F_q(-1) & \longrightarrow & F_q & \longrightarrow & G'' & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

where G is a $(b_1, \ldots, b_r)$-sheaf by 1.6(i), and where $G'' = G_{q-1}$ (a depth question).

Suppose first that $q = 1$. Then $\dim \mathrm{Supp}(N_q) = 0$, so $H^1(N_q(-1)) = 0$ and $h^0(F_q(-1)) \leq h^0(F(-1)) \leq b_0$. Moreover, $G' = 0$, and hence $G \simeq G''$. Thus F_1 is indeed a $(b_0, \ldots, b_r)$-sheaf.

In addition, 6.5(ii) shows that $H^0(F_1(-b_0)) = 0$. If $q \geq 2$, argue by induction on q. One may suppose that G_{q-1} is a $((b_{q-1})^{q-1}, \ldots, b_{q-1}, \ldots, b_r)$-sheaf and that $H^0(G_{q-1}(-b_{q-1})) = 0$. Lemma 6.5 then gives $H^0(F_q(-b_{q-1})) = 0$ and $h^0(F_q(-1)) \leq (b_{q-1})^q$, which proves the assertion.

Theorem 6.7. *There are two sequences* $\{A_i(X_0, \ldots, X_i; Y)\}$ *and* $\{A_i^{(q)}(X_0, \ldots, X_q; Y)\}$ *of polynomials having the following properties. Assume the conditions of 6.1 and suppose that* F *is a quotient of* $\mathcal{O}_P(-m)^{\oplus M}$*, with* $m \geq 0$*.*

(i) *If* F *is a* $(b_0, \ldots, b_r)$*-sheaf, then, for* $i = 0, \ldots, r$*,*

$$|a_i| \leq A_{r-i}(b_i, \ldots, b_r; m).$$

(ii) *If* $\chi(F_q(n)) = \sum_{i=0}^{r} a_i^{(q)} \binom{n+i}{i}$ *is the Hilbert polynomial of* F_q*, then, for* $i = 0, \ldots, q-1$*,*

$$|a_i^{(q)}| \leq A_{r-i}^{(r-q)}(c_{q-1}, \ldots, c_r; m).$$

(Notice that $a_{q-1}^{(q)} \leq a_{q-1}$ *and* $a_q^{(q)} = a_q, \ldots, a_r^{(q)} = a_r$*, since* $\dim \mathrm{Supp}(N_q) \leq q-1$*.)*

To prove (i), use induction on r and apply 6.4 to an exact sequence $0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$ in which G is a $(b_1, \ldots, b_r)$-sheaf. By induction on r, one may suppose that $A_0, \ldots, A_{r-1}$ have been defined, taking 1.7 into account. Then 6.4(ii), together with 1.8, displays the definition of A_r. Finally, (ii) follows formally from 6.4(iii), 6.6, and assertion (i), applied to F_q in place of F.

Corollary 6.8. (Grothendieck). *Let* X *be projective over a noetherian scheme* S*, let* $\mathcal{O}_X(1)$ *be very ample for* X/S*, and let* $\mathcal{J}$ *be a family of classes of coherent sheaves on the fibers of* X/S*. Suppose that:*

(a) *There is a coherent sheaf* E *on* X *such that* $\mathcal{J}$ *is contained in the family of classes of quotients of sheaves of the form* E_K*.*

(b_q) *In the Hilbert polynomials* $\chi(F_K(n))$*, for* $F_K \in \mathcal{J}$*, the coefficients in degrees at least* $q-1$ *remain bounded above.*

Then the $(F_K)_q$ *(as* F_K *ranges over* $\mathcal{J}$*) form a bounded family; moreover, the coefficients of* $\chi(F_K(n))$ *in degrees at least* $q-2$ *remain bounded below.*

Indeed, one may plainly suppose that $X = \mathbf{P}_S^N$ and $E = \mathcal{O}_X(-m)^{\oplus M}$. The first assertion follows formally from 6.7(ii) and 1.13. The second follows by induction, using an exact sequence $0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$ and 6.4(ii).

Definition 6.9. *Let* k *be a field. We call a* special positive k-cycle of dimension r *any projective* k*-scheme* X*, equipped with a very ample sheaf* $\mathcal{O}_X(1)$*, which is a union of closed subschemes* X_j *of dimension* r*, each* X_j *being obtained by base change* $\mathrm{Spec}(k) \to \mathrm{Spec}(k_j)$ *as the inverse image of an integral* k_j*-scheme* X_j' *(equipped with* $\mathcal{O}_{X_j'}(1)$*). We call the* degree *of* X *the coefficient* a_r *in the Hilbert polynomial* $\chi(\mathcal{O}_X(n)) = \sum_i a_i \binom{n+i}{i}$*.*

Lemma 6.10. *Let* k *be an algebraically closed field, and let* X *be a special positive* k*-cycle of dimension* r *and degree* d*. Then* $\mathcal{O}_X$ *is a* $(0, \ldots, 0, d)$*-sheaf.*

Indeed, view X as a subscheme of $\mathbf{P}_k^N$. After replacing k by the algebraic closure of a purely transcendental extension, which is permissible by 1.6(i), one may suppose that k contains $r(N+1)$ elements algebraically independent over all the k_j. The r successive "general" hyperplane sections of X defined by these elements are then always special positive k-cycles of degree d.

Arguing by induction on r, it remains only to observe that, when $r = 0$, one has $h^0(\mathcal{O}_X(-1)) = d$, while, when $r \geq 1$, one has $h^0(\mathcal{O}_X(-1)) = 0$. For the latter point, the injection $\mathcal{O}_X \longrightarrow \coprod_j \mathcal{O}_{X_j}$ allows one to suppose that X itself comes from an integral scheme X'. But then $h^0(\mathcal{O}_X(-1)) = h^0(\mathcal{O}_{X'}(-1)) = 0$, because $H^0(\mathcal{O}_{X'})$ is a field.

Corollary 6.11. (i) *Let r, d be two integers, put $c_i = A_{r-i}(0, \ldots, 0, d; 0)$ for $i = 0, \ldots, r$ (6.7(i)), and put $p = P_r(c_0, \ldots, c_r)$ (1.9). Let k be an algebraically closed field, and let X be a special positive k-cycle of dimension at most r and degree at most d, with Hilbert polynomial $\chi(\mathcal{O}_X(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$. Then, for $i = 0, \ldots, r$,*

$$|a_i| \leq c_i.$$

Moreover, X embeds as a subscheme of $\mathbf{P}_k^N$, with $N = d(r+1) - 1$, defined by (at most) $\binom{N+p}{N}$ equations of degree p.

(ii) (Chow) *Let S be a noetherian scheme. As K ranges over the algebraically closed extensions of the $k(s)$, for $s \in S$, the special positive K-cycles of bounded dimension and degree form a "bounded family" (either with the X regarded as subschemes of a fixed $\mathbf{P}_S^N$, or in an evident abstract sense).*

Indeed, in (i), the first assertion follows from 6.7(i); the second follows from 6.10 and 1.8(i), and from 6.2, applied to $0 \longrightarrow I \longrightarrow \mathcal{O}_P \longrightarrow \mathcal{O}_X \longrightarrow 0$, $P = \mathbf{P}_k^N$, together with 1.3. Finally, (ii) follows from (i) and 1.13.

Remark 6.12. Let k be an algebraically closed field and let X be a projective k-scheme, purely of dimension r, with no embedded associated prime cycle. If arbitrary nilpotent elements are allowed in $\mathcal{O}_X$, the coefficients in the Hilbert polynomial $\chi(\mathcal{O}_X(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$ can no longer be bounded in terms of a_r alone.

For example, let Z be a smooth curve of degree d in $\mathbf{P}_k^3$. For every $n \gg 0$, there is a smooth surface Y of degree n containing Z. Then Z is a Cartier divisor on Y; put $\mathcal{O}_{X_n} = \mathcal{O}_Y/\mathcal{O}_Y(-2Z)$. There is an exact sequence $0 \longrightarrow \mathcal{O}_{X_n}(-Z) \longrightarrow \mathcal{O}_{X_n} \longrightarrow \mathcal{O}_Z \longrightarrow 0$ and the formula $\langle Z^2 \rangle_Y = 2p_a(Z) - 2 - (n-4)d$. It follows that $a_0 = \chi(\mathcal{O}_{X_n}(-1)) \longrightarrow \infty$ as $n \longrightarrow \infty$, although $a_1 = 2d$.

Corollary 6.13. *Under the conditions of 6.1, consider an exact sequence*

$$\mathcal{O}_P(-m_1)^{\oplus M_1} \longrightarrow \mathcal{O}_P(-m)^{\oplus M} \xrightarrow{\alpha} F \longrightarrow 0$$

with $m_1 \geq m \geq 0$. Then, for $i = 0, \ldots, N$,

$$|a_i| \leq A_{N-i}(0, \ldots, 0, M; m_1) + M \binom{m}{N-i}.$$

Indeed, let $I = \mathrm{Ker}(\alpha)$. Plainly, I is a $(0, \ldots, 0, M)$-sheaf, and $\chi(I(n)) = \sum_{i=0}^{N} \big[M \binom{-m-1+N-i}{N-i} - a_i\big] \binom{n+i}{i}$. The assertion follows from 6.7(i) and the formula $\binom{-m-1+Q}{Q} = (-1)^Q \binom{m}{Q}$.

Corollary 6.14. (I. Hermann). *For every $N \geq 0$, there is a polynomial $R_N(m)$ such that, for every field k_0 and every ideal I of the polynomial ring $k_0[T_1, \ldots, T_N]$ generated by elements of degree at most m, the radical $\sqrt{I}$ is generated by elements of degree at most $R_N(m)$.*

Indeed, let k be the algebraic closure of k_0. After introducing an auxiliary indeterminate, let Y (respectively X) be the closed subscheme of $P = \mathbf{P}_k^N$ defined by the homogenized ideal of I (respectively of $\sqrt{I}$). In the Hilbert polynomial $\chi(\mathcal{O}_Y(n)) = \sum_{i=0}^{N} b_i \binom{n+i}{i}$, one has $|b_i| \leq c_i$, where $c_i = A_{N-i}(0, \ldots, 0, 1; m)$, by 6.13.

Let $X = \bigcup X^q$ be the decomposition by dimension. Then $\deg(X^q) \leq e_q$, where $e_q = P_{N-q}(c_q, \ldots, c_N)$. Indeed, after cutting X (respectively Y) by a "general" linear subspace of

codimension q, one may suppose that $q = 0$. Then $\deg(X^q) = h^0(\mathcal{O}_{X^q})$ and $h^0(\mathcal{O}_{X^q}) \leq h^0(\mathcal{O}_Y)$, and 6.4(iii) gives $h^0(\mathcal{O}_Y) \leq e_q$.

Consequently, $\mathcal{O}_{X^q}$ is a $(0, \ldots, 0, e_q)$-sheaf by 6.10. It follows from the injection $\mathcal{O}_X \longrightarrow \prod_q \mathcal{O}_{X^q}$ that $\mathcal{O}_X$ is an $(e_0, \ldots, e_N)$-sheaf. Hence, in the Hilbert polynomial $\chi(\mathcal{O}_X(n)) = \sum_{i=0}^N a_i \binom{n+i}{i}$, one has $|a_i| \leq f_i$, where $f_i = A_{N-i}(e_i, \ldots, e_N; 0)$, by 6.7. Finally, the ideal $\sqrt{I}$ of X is $R_N(m)$-regular, where $R_N(m) = P_N(f_0, \ldots, f_N)$, and the assertion follows from 1.3.

7. Appendix: The Hodge index theorem

Here we recover and complete some earlier results by a direct argument.

Theorem 7.1. *Let X be an irreducible surface, proper over an algebraically closed field. Suppose that there is an invertible sheaf H on X such that $\langle c_1(H)^2 \rangle > 0$ (for example, H ample). Then an invertible sheaf L on X satisfying*

$$\langle c_1(L) \cdot c_1(H) \rangle = 0, \qquad \langle c_1(L)^2 \rangle \geq 0,$$

is numerically equivalent to 0. Consequently, by 4.6, the intersection form on $A^1_{\mathrm{num}}(X)_{\mathbb{Q}} = [\mathrm{Pic}(X)/\mathrm{Pic}^\tau(X)] \otimes \mathbb{Q}$ is nondegenerate of type $(1, \rho - 1)$.

7.1.1. First suppose that X is reduced (hence integral), that H is ample, that $\langle c_1(L) \cdot c_1(H) \rangle = 0$, and that $\langle c_1(L)^2 \rangle > 0$. Arguing by contradiction, choose $m > 0$ such that $H_1 = L \otimes H^{\otimes m}$ is ample, and choose n such that, on putting $N = L^{\otimes n} \otimes H^{-1}$, one has $\langle c_1(N) \cdot c_1(H_1) \rangle = n\langle c_1(L)^2 \rangle - m\langle c_1(H)^2 \rangle > 0$. One also has $\langle c_1(N) \cdot c_1(H) \rangle = -\langle c_1(H)^2 \rangle < 0$ and $\langle c_1(N)^2 \rangle = n^2\langle c_1(L)^2 \rangle + \langle c_1(H)^2 \rangle > 0$, contradicting the following lemma.

Lemma 7.1.2. *Let X be an integral surface, projective over an algebraically closed field. For an invertible sheaf N on X such that $\langle c_1(N)^2 \rangle > 0$, the following conditions are equivalent:*

(i) for every ample sheaf H, one has $\langle c_1(N) \cdot c_1(H) \rangle > 0$;

(i′) there is an ample sheaf H such that $\langle c_1(N) \cdot c_1(H) \rangle > 0$;

(ii) there is an integer $n > 0$ such that $H^0(X, N^{\otimes n}) \neq 0$.

Indeed, (ii) implies that there is an effective divisor D such that $N^{\otimes n} \simeq \mathcal{O}_X(D)$, and $\langle c_1(N)^2 \rangle > 0$ implies $D > 0$; hence (i).

To see that (i′) implies (ii), one may plainly suppose that $H = \mathcal{O}_X(1)$ is very ample. There is then a closed integral locally principal curve Y such that $H = \mathcal{O}_X(Y)$. By the Riemann theorem, (i′) implies $H^1\big(N_Y^{\otimes n}(p)\big) = 0$ for $p \geq 0$ and $n \geq n_0$, where $n_0 = 2h^1(\mathcal{O}_Y) - 1$. Since $H^2(N^{\otimes n}(p)) = 0$ for $p \gg 0$, the exact sequence

$$H^1\big(N_Y^{\otimes n}(p)\big) \longrightarrow H^2\big(N^{\otimes n}(p-1)\big) \longrightarrow H^2\big(N^{\otimes n}(p)\big) \longrightarrow 0$$

gives $H^2(N^{\otimes n}) = 0$ for $n \geq n_0$. Thus, for $n \gg 0$,

$$h^0(N^{\otimes n}) = \chi(N^{\otimes n}) + h^1(N^{\otimes n}) \geq \chi(N^{\otimes n}) = \frac{1}{2}\langle c_1(N)^2 \rangle n^2 + \cdots > 0,$$

which completes the proof of the lemma.

7.1.3. Now suppose in 7.1 that there is an invertible sheaf M on X such that $\langle c_1(L) \cdot c_1(M) \rangle \neq 0$, and argue by contradiction. Replace X by the underlying reduced scheme of a Chow presentation of X, which is permissible by the projection formula X 4.4.4, and let G be ample on X. Choose $m > 0$ such that $H_1 = G^{\otimes m} \otimes M$ is ample and $\langle c_1(L) \cdot c_1(H_1) \rangle \neq 0$. Choose p, q, with $q \neq 0$, such

that, on putting $L_1 = L^{\otimes p} \otimes H^{\otimes q}$, one has $\langle c_1(L_1) \cdot c_1(H_1) \rangle = p\langle c_1(L) \cdot c_1(H_1) \rangle + q\langle c_1(H) \cdot c_1(H_1) \rangle = 0$. Then one also has $\langle c_1(L_1)^2 \rangle = p^2 \langle c_1(L)^2 \rangle + q^2 \langle c_1(H)^2 \rangle > 0$. This is the case 7.1.1, with $L = L_1$ and $H = H_1$.

7.1.4. Finally, consider the general case of 7.1. Still arguing by contradiction, suppose that L is not numerically equivalent to 0. Replace X by the normalization of X_{red}, so that X has

only finitely many singular points; this reduction is justified by the projection formula X 4.4.4 and by 4.5(ii). There is therefore a closed integral curve Y on X such that $\langle c_1(L) \cdot Y \rangle \neq 0$, and Y is locally principal outside a finite set Z. Let $f : X' \to X$ be the blow-up of the ideal $I \subset \mathcal{O}_X$ defining Y. The ideal $\mathcal{M} = I\mathcal{O}_{X'}$ is invertible and defines a subscheme of X' which is the union of two subschemes Y' and Y'' such that $f(Y') = Y$ and $f(Y'') \subset Z$. By the projection formula X 4.4.4, $\langle c_1(f^*L) \cdot c_1(\mathcal{M}) \rangle = -\langle c_1(f^*L) \cdot Y' \rangle - \langle c_1(f^*L) \cdot Y'' \rangle = -\langle c_1(L) \cdot Y \rangle - 0 \neq 0$. Thus one may replace X, L, H by X', f^*L, f^*H. This reduces the problem to the preceding case and completes the proof of 7.1.

Remark 7.2. It is not known whether an irreducible proper surface over an algebraically closed field which possesses an invertible sheaf H satisfying $\langle c_1(H)^2 \rangle > 0$ is necessarily projective. Nor is it known whether, for every irreducible proper surface, the intersection form on $A^1_{\mathrm{num}}(X)_{\mathbb{Q}}$ is nondegenerate. It follows from 7.1 that, when it is degenerate, it is necessarily negative (non-definite).

Corollary 7.3. *Under the conditions of 7.1, let M be an invertible sheaf on X which is not numerically equivalent to H, and put $d = \langle c_1(M) \cdot c_1(H) \rangle$ and $h = \langle c_1(H)^2 \rangle$. Then $\langle c_1(M)^2 \rangle < d^2/h$.*

Indeed, let $L = M^{\otimes h} \otimes H^{\otimes(-d)}$. Since $\langle c_1(L) \cdot c_1(H) \rangle = 0$, Theorem 7.1 gives $\langle c_1(L)^2 \rangle < 0$, which proves the assertion.

Corollary 7.4. *Let X be an irreducible scheme of dimension $r \geq 2$, projective over an algebraically closed field, and let H be an ample sheaf on X. Then:*

(i) *An invertible sheaf L on X is numerically equivalent to 0 if and only if*

$$\langle c_1(L) \cdot c_1(H)^{r-1} \rangle = 0, \qquad \langle c_1(L)^2 \cdot c_1(H)^{r-2} \rangle \geq 0;$$

(ii) *For every invertible sheaf M on X which is not numerically equivalent to H,*

$$\langle c_1(M)^2 \cdot c_1(H)^{r-2} \rangle < \frac{\langle c_1(M) \cdot c_1(H)^{r-1} \rangle^2}{\langle c_1(H)^r \rangle}.$$

Assertion (ii) follows immediately from (i), by the argument of 7.3. To prove (i), let Y be a closed integral curve in X; we wish to prove that $\langle c_1(L) \cdot Y \rangle = 0$. Embed X in $\mathbf{P}^N_k$ by means of $H^{\otimes n}$, with $n > 0$. Let $I = \bigoplus I_\nu$ be the homogeneous ideal of Y, and let ν_0 be an integer such that I_{ν_0} generates I_ν for $\nu \geq \nu_0$. Then the hypersurfaces Z of degree ν_0 containing Y have no common points outside Y. By induction on $r = \dim X$ and Bertini's theorem, one finds $r-2$ hypersurfaces $Z_1, \ldots, Z_{r-2}$ of degree ν_0 containing Y such that $X' = Z_1 \cap \cdots \cap Z_{r-2} \cap X$ is irreducible. Since $\langle c_1(L) \cdot Y \rangle = \langle c_1(L|_{X'}) \cdot Y \rangle$, it is enough to prove that $L|_{X'}$ is numerically equivalent to 0 on X'. By the projection formula, $\langle c_1(L|_{X'}) \cdot c_1(H|_{X'}) \rangle = 0$ and $\langle c_1(L|_{X'})^2 \rangle \geq 0$, so the assertion follows from 7.1.

Corollary 7.5. *Let k be an algebraically closed field, let Y be a projective k-scheme equipped with an ample sheaf H, and let X be the zero scheme of a section φ of H. Suppose that, for every irreducible component Y^i of Y, the irreducible components of $X \cap Y^i = V(\varphi|_{Y^i})$ have dimension at least 2. Then the morphism $A^1_{\mathrm{num}}(Y) \longrightarrow A^1_{\mathrm{num}}(X)$ is injective.*

Indeed, one may plainly suppose that Y is integral, and the assertion follows from 7.4(i) by the projection formula.

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Exposé XIV. Open problems in intersection theory

by A. Grothendieck

In the present exposé we review some of the problems connected with questions encountered in this seminar which deserve to be studied. We leave aside the problems connected with Serre's *local intersection theory* [14], which we have not had to use in the seminar, and refer in particular to *loc. cit.* for the two fundamental conjectures which remain open in that direction (*loc. cit.*, Chapter V, §4). We also pass over the problems, very important for arithmetic applications, connected with the *structure of groups of cycle classes*, especially on a smooth projective algebraic scheme. For discussion of these problems, we refer to S. Kleiman, "Algebraic cycles and the Weil conjectures," in *Dix exposés sur la cohomologie des schémas* (Masson–North Holland Publishing Co.). In no. 4 of the present exposé we also give some supplements to the seminar, concerning in particular the *Chow ring*, which had not found a place in the preceding exposés.

1. The operations Λ^i in the derived category $D(X)$[26]

Let X be a locally ringed topos, and let $K^\bullet(X)$ be the ring of classes of perfect complexes on X (Exposé IV). One would like to define on $K^\bullet(X)$ a λ-ring structure (Exposé V), functorial in X, such that the natural homomorphism

$$K^\circ_{\text{naive}}(X) \longrightarrow K^\bullet(X)$$

(where the first member is the ring of classes of locally free modules on X, equipped with the λ-structure defined in Exposé V) is a λ-homomorphism. For this purpose, one would have to define functors, in general nonadditive, of "exterior power"

$$\Lambda^i : D^-(X) \longrightarrow D^-(X),$$

[26]Cf. the note on p. 23.

transforming pseudo-coherent (respectively perfect) complexes into pseudo-coherent (respectively perfect) complexes and having the formal properties with respect to exact triangles needed to define a λ-structure on $K^\bullet(X)$ by the formula

$$\lambda^i(\mathrm{cl}(L^\bullet)) = \mathrm{cl}(\Lambda^i L^\bullet). \tag{1.1}$$

When X is of characteristic zero, that is, when its structure sheaf is a sheaf of $\mathbb{Q}$-algebras, the problem is solved simply by putting

$$\Lambda^i L^\bullet = \left(\otimes^i L^\bullet\right)^{\mathrm{alt}}, \tag{1.2}$$

where the i-th tensor power, in the sense of derived categories, of $L^\bullet$ is regarded as an object of $D^-(\mathcal{O}_i)$, with $\mathcal{O}_i = \mathcal{O}_X[\mathfrak{S}_i]$ the algebra of the symmetric group $\mathfrak{S}_i$ over $\mathcal{O}_X$, and where the superscript alt denotes passage to the "alternating part under the action of $\mathfrak{S}_i$," an operation which makes sense because of the characteristic-zero hypothesis. A formula of this type was already used in [8] (cited below as [RRR]) in the first proof of the Riemann–Roch theorem.

Without a characteristic hypothesis, when $L^\bullet$ is zero in degrees > 0, Dold–Puppe [6] gives a canonical homotopy from $L^\bullet$ to the complex associated with a semisimplicial module $L'^\bullet$ canonically attached to $L^\bullet$. For $L^\bullet$ flat, say, a fairly natural candidate for $\Lambda^i L^\bullet$ would be the complex associated with the semisimplicial module $\Lambda^i L'^\bullet$, whose degree-n component is $\Lambda^i L'_n$. But to extend the definition to complexes of $D^-(X)$ which are no longer assumed to be zero in degrees > 0, one would have to study the behavior of the preceding operation with respect to translation $L^\bullet \mapsto L^\bullet[1]$, or *suspension* in Dold–Puppe terminology [6]. Unfortunately, there does not seem to be an isomorphism $\Lambda^i(L^\bullet[1]) \simeq \Lambda^i(L^\bullet)[i]$ which would make the desired extension trivial.

2. The Riemann–Roch formula without projectivity hypotheses

2.1. Once one has a natural λ-structure on the $K^\bullet(X)$, it is possible to formulate the Riemann–Roch theorem, in the form contemplated in Exposé VIII, for any proper complete-intersection morphism $f : X \to Y$, and even, more generally, for a relative scheme X over a locally ringed topos Y, with X still subject to the corresponding relative conditions. The Todd class of f is defined by means of the class in $K^\bullet(X)$ of the relative cotangent complex T_f constructed in Exposé 0, 4.4; that complex is perfect here by the local hypothesis on f. It seems clear that a proof of the Riemann–Roch formula in this general case will require essentially new ideas, and it is possible that these same ideas could provide the key

to a proof of the analogous formula in the complex analytic (or rigid-analytic!) case. Recall that, in the complex analytic case, only the case of a locally free sheaf F on a compact nonsingular variety X, over a one-point Y, is currently known, by the Atiyah–Singer index formula; the case in which F is only assumed coherent still seems to escape presently known techniques.

2.2. As an essential ingredient of the Riemann–Roch formula, recall also the *finiteness problem* raised in Exposé III: if $f : X \to Y$ is a proper morphism and $L^\bullet$ is a complex on X which is pseudo-coherent relative to Y, is $Rf_*(L^\bullet)$ pseudo-coherent? A general proof of this conjecture will probably use algebraic ideas which will yield a proof of Grauert's "finiteness theorem" [7] under appreciably more general conditions than those of *loc. cit.*, working from the outset over topologically ringed spaces (or topoi) in $\mathbb{C}$-algebras Y more general than analytic spaces. This would include the cases in which Y is a locally compact space with the sheaf of continuous complex-valued functions, a C^m-manifold with the sheaf of complex-valued C^m-functions, a real analytic space, and so forth.

2.3. Finally, in these questions the properness hypothesis on f should be replaced by the hypothesis that the supports of the cohomology sheaves $H^i(L^\bullet)$ of the complexes under consideration on X are proper over Y.

3. A Riemann–Roch formula "without denominators" for an immersion

3.1. Let $f : Y \to X$ be a regular closed immersion (Exposé VII), with conormal sheaf $\underline{N}$. For simplicity, suppose that X admits an ample invertible sheaf. The Riemann–Roch formula for $y \in K^\bullet(Y)$,

$$\mathrm{ch}_X(f_*(y)) = f_*(\mathrm{ch}_Y(y)\,\mathrm{Todd}(-\check{\underline{N}})),$$

together with the general formula (Exposé VII)

$$f_*(y')f_*(y'') = f_*(y'y''c_d(\underline{N})), \qquad y', y'' \in \mathrm{Gr}^\bullet(Y)_\mathbb{Q},$$

allows one, assuming f has codimension d, to express the Chern classes of $f_*(y)$ in $\mathrm{Gr}^\bullet(X)_\mathbb{Q}$ in the form

$$c_i(f_*(y)) = f_*(P_i((c_\alpha(y))_\alpha, (c_\beta(\underline{N}))_\beta)), \tag{3.1}$$

where P_i is a universal polynomial in the $c_\alpha(y)$ and the $c_\beta(\underline{N})$, a polynomial which a priori has rational coefficients. In fact, one finds in [RRR] that these polynomials have *integer coefficients*. This raises the question of the validity of a Riemann–Roch formula "without denominators."

In fact, such a formula for $c_i(f_*(y))$ cannot in general be sought in $\mathrm{Gr}^\bullet(X)$ itself, since the homomorphism $f_* : K^\bullet(Y) \longrightarrow K^\bullet(X)$ is not itself compatible with the λ-filtrations, modulo a shift by d (4.6), but is compatible only up to torsion. Thus it is not possible to define a homomorphism $f_* : \mathrm{Gr}^\bullet(Y) \longrightarrow \mathrm{Gr}^\bullet(X)$ inducing the natural homomorphism $f_* : \mathrm{Gr}^\bullet(Y)_\mathbb{Q} \longrightarrow \mathrm{Gr}^\bullet(X)_\mathbb{Q}$ and making it possible to interpret (3.1) as a formula in $\mathrm{Gr}^\bullet(X)$ rather than only in $\mathrm{Gr}^\bullet(X)_\mathbb{Q}$.

When X is regular, however, so that $K^\bullet(X) \simeq K_\bullet(X)$ (IV 2.5), one can equip $K^\bullet(X)$ with the *topological filtration* by codimension of supports, inherited from the corresponding filtration of $K_\bullet(X)$. It is plausible that this filtration is compatible with the ring structure. This is proved at least in [RRR] when X is smooth over a field k, using the theory of the Chow ring [4]. When this is so, the associated graded

$$\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$$

is a *graded ring*; and, if Y is also regular, one has precisely what is needed to obtain a natural homomorphism of degree d of graded groups

$$f_* = f_*^{\mathrm{top}} : \mathrm{Gr}^\bullet_{\mathrm{top}}(Y) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X).$$

Since the λ-filtration of $K^\bullet(X)$ is finer than the topological filtration by codimension just considered (X 1.3.3), one obtains a canonical homomorphism of graded groups (and even of graded rings),

$$\mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X),$$

which is, moreover, an isomorphism modulo torsion (Exposé VIII). This makes it possible, as in [RRR], to consider a theory of Chern classes on X with values in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, and likewise on Y, and thus to interpret (3.1) as a formula in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$. In this form, the formula is proved in [RRR] when X is of *characteristic zero*, at least if X is smooth over a field (an assumption used only to ensure that the topological filtration of $K^\bullet(X)$ is compatible with the ring structure).

The characteristic-zero hypothesis is used essentially to have formulas of the type (1.1), (1.2), and essentially the same proof should apply without a characteristic hypothesis once the question raised in no. 1 has been answered affirmatively.

3.2. When X is smooth over a field, one also has the Chow ring

$$A(X),$$

and a theory of Chern classes on X with values in $A(X)$ [9], finer than the theory with values in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ just introduced. The latter Chern classes are the images of the Chow-style Chern classes under the canonical surjective homomorphism (cf. 4.1)

$$\varphi : A(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X).$$

This homomorphism is an isomorphism modulo torsion, but not an isomorphism in general (4.7). That said, if X and Y are smooth, the inclusion $f : Y \to X$ still defines a homomorphism of degree d of graded groups

$$f_* : A(Y) \longrightarrow A(X),$$

which makes it possible to interpret (3.1) as a formula in $A^i(X)$. This formula reduces to $0 = 0$ for $i < d$ and is therefore of interest only for $i > d$. Even for $i = d = 2$, this formula has not been proved at present (cf. 4.3), nor has the less precise formula deduced from it by replacing rational equivalence of algebraic cycles by algebraic equivalence. Unlike the analogous formula in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, there does not seem to be any convincing reason to suppose that the formula without denominators should be true in $A(X)$. A good way to test it, given our almost total ignorance of the nature of the groups $A^i(X)$[27] for $i > 2$, seems to be to test the formulas that can be deduced from it in groups $A^1(Z) = \mathrm{Pic}(Z)$, for example by applying to both sides of (3.1) $g_* : A(X) \longrightarrow A(Z)$, where $g : X \to Z$ is a proper morphism to a smooth quasi-projective Z of dimension $\dim X - i + 1$.

3.3. Finally, to give a meaning to (3.1), one may also use the cohomological theory of Chern classes (SGA 5 VII), whenever one has *Gysin homomorphisms*

$$f_* : H^p(Y, \mu_n^{\otimes i}) \longrightarrow H^{p+2d}(X, \mu_n^{\otimes (i+d)}).$$

This holds at least when Y and X are both smooth over a scheme S, by the theorem of relative cohomological purity (SGA 4 XVI 3.7); conjecturally, it also holds when Y and X are both regular, by the theorem of absolute cohomological purity, proved at least for excellent X of characteristic zero (SGA 4 XIX 3.2).

When X and Y are smooth over the field of complex numbers, one may also use the integral Chern class on the locally compact space $X(\mathbb{C})$ of complex points of X. Formula (3.1) was proved in this case by Atiyah and Hirzebruch [1], who more generally treat embeddings of complex analytic varieties (working with naive K'). It follows, by the Lefschetz principle and the general results of SGA 4, that the ℓ-adic formula (3.1) holds whenever Y and X are smooth over an algebraically closed field of characteristic zero. This makes the cohomological formula (3.1) extremely plausible. Nevertheless, it should be noted that, at present, it has not been proved even when X and Y are smooth over a field k of characteristic zero, if k is not algebraically closed.

3.4. One may also work with the cohomological theory of Chern classes, in the de Rham or Hodge style [13]. The question of the validity of (3.1) in this setting arises only when X is not of characteristic zero, since one knows in any case (no. 6) that the formula in question is true modulo torsion. For smooth quasi-projective schemes over a field k, the de Rham or Hodge

[27] The formula, as well as the variant mentioned in 3.3, has since been proved by P. Jouanolou (Nov. 1969). Cf.

version of (3.1) would moreover follow from the version using cycle classes modulo algebraic equivalence. Thus the de Rham or Hodge cohomological version provides another accessible way to test the validity of the cycle-class version of (3.1).

3.5. The proof [1] of the transcendental cohomological version of (3.1) is essentially transcendental in nature and uses tubular neighborhoods of Y in X. This suggests that a proof of a formula of type (3.1) may be tied to the systematic consideration of $K^\bullet$-groups and cohomology groups with support in Y, which might provide an adequate technical means of *localization around* Y.

4. Relations between $K^\bullet(X)$ and the Chow ring $A(X)$

4.1. Suppose for simplicity that X is a smooth quasi-projective scheme over a field k (see no. 8 for a more general case), so that the Chow ring $A(X)$ of cycle classes modulo rational equivalence is available [4]. Two variants of this ring, defined under appreciably more general conditions, are $\mathrm{Gr}^\bullet(X)$, the graded ring associated with the λ-filtration of $K^\bullet(X)$, and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, the graded ring associated with the filtration of $K^\bullet(X) \simeq K_\bullet(X)$ by codimension of supports. There are theories of Chern classes with values in each of these three rings (see [9] for the case of $A(X)$, and this seminar for $\mathrm{Gr}^\bullet(X)$, which maps into $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$). There is also an ℓ-adic cohomological theory (SGA 5 VII). Thus, when ℓ is prime to the characteristic of k,

$$c_i(E) \in H^{2i}(X, \mathbf{Z}_\ell(i)),$$

where E is a vector bundle on X, or more generally any element of $K^\bullet(X)$. These theories are related by the following canonical homomorphisms, which are compatible with the multiplicative structures:

$$\begin{array}{ccc} H^{2*}(X, \mathbf{Z}_\ell(*)) & & \\ \uparrow{\scriptstyle \gamma} & & \\ A(X) & \xrightarrow{\quad\varphi\quad} & \mathrm{Gr}^\bullet_{\mathrm{top}}(X) \\ & & \uparrow{\scriptstyle j} \\ & & \mathrm{Gr}^\bullet(X) \end{array} \tag{4.1}$$

where

$$H^{2*}(X, \mathbf{Z}_\ell(*)) = \bigoplus_i H^{2i}(X, \mathbf{Z}_\ell(i)).$$

The homomorphism j comes from the fact that the λ-filtration of $K^\bullet(X)$ is finer than the topological filtration by codimension of support (Expose X), and it is an isomorphism modulo torsion (Expose VII 4.11). The homomorphism φ, defined for example in [RRR], associates to the class of the codimension-i cycle carried by a closed subscheme $Y \subset X$ the element $\mathrm{cl}^i(\mathcal{O}_Y)$ of

$$\mathrm{Filt}^i_{\mathrm{top}} K^\bullet(X) / \mathrm{Filt}^{i+1}_{\mathrm{top}} K^\bullet(X).$$

It is surjective, and we shall see in 4.2 that it is an isomorphism modulo torsion. Hence, modulo torsion, the three rings $A(X)$, $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, and $\mathrm{Gr}^\bullet(X)$ are canonically isomorphic; consequently, the Riemann–Roch formulas expressed using any one of these rings after tensoring with $\mathbb{Q}$ are equivalent.

The graded ring $\mathrm{Gr}^\bullet(X)$, used in this seminar chiefly as a substitute for the Chow ring, has the considerable advantage of being defined for every locally ringed topos, and in particular for every scheme, whether regular or not. Its disadvantage (4.6) with respect to $A(X)$ and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ is

that it is not covariant as it stands for proper morphisms, even between smooth quasi-projective schemes over a field; it acquires that covariance only after tensoring with $\mathbb{Q}$.

Another advantage of $A(X)$ over its competitors $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ and $\mathrm{Gr}^\bullet(X)$ is the homomorphism

$$A(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*)),$$

which permits cohomological Chern classes to be interpreted as images of Chern classes with values in $A(X)$. In general this homomorphism does not factor through $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ (4.7), and it is at least doubtful that there is in general a homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*))$$

carrying Chern classes to Chern classes (see no. 5). These difficulties disappear after tensoring with $\mathbb{Q}$, and one obtains a homomorphism

$$\mathrm{Gr}^\bullet_{\mathrm{top}}(X) \longrightarrow H^{2*}(X, \mathbf{Q}_\ell(*)) = H^{2*}(X, \mathbf{Z}_\ell(*)) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell \tag{4.2}$$

that carries Chern classes to Chern classes.

4.2. To show that the surjective homomorphism φ in (4.1) is an isomorphism modulo torsion, one uses the Chern homomorphism obtained from the theory of Chern classes [9],

$$K^\bullet(X) \longrightarrow \mathrm{Ch}(A(X)),$$

where the target is the Chern ring of $A(X)$ as defined in Expose V. This homomorphism is compatible with the topological filtration of $K^\bullet(X)$ and the natural filtration of $\mathrm{Ch}(A(X))$, and therefore induces a homomorphism of graded groups

$$\psi : \mathrm{Gr}^\bullet_{\mathrm{top}}(X) \longrightarrow A(X). \tag{4.3}$$

One proves the formulas

$$\begin{cases} \psi\varphi(x) = (-1)^{i-1}(i-1)!\, x & (\text{mod torsion}), \quad x \in A^i(X), \\ \varphi\psi(x) = (-1)^{i-1}(i-1)!\, x & (\text{mod torsion}), \quad x \in \mathrm{Gr}^i_{\mathrm{top}}(X). \end{cases} \tag{4.4}$$

The first formula implies the second because φ is surjective. These formulas show that φ and ψ are isomorphisms modulo torsion and, up to the factor $(-1)^{i-1}(i-1)!$ in degree i, are inverse to one another.

To prove the first formula in (4.4), one may suppose that x is represented by an irreducible closed subscheme Y of X of codimension i. The formula is then equivalent to

$$c_i(\mathcal{O}_Y) = (-1)^{i-1}(i-1)!\ \mathrm{cl}(Y) \qquad (\text{mod torsion}). \tag{4.5}$$

On the left, $\mathcal{O}_Y$ is regarded as a coherent sheaf on X and hence defines an element of $K^\bullet(X)$; on the right, $\mathrm{cl}(Y)$ denotes the class of Y in $A^i(X)$. Since $A^i(X)$ is unchanged upon deleting from X a closed subset of codimension $> i$, one reduces to the case where Y is regular, and therefore smooth over the base field k if k is perfect. In that case, (4.5) is precisely the Riemann–Roch formula for the inclusion $Y \to X$ and the element $1 \in K^\bullet(Y)$, written using Chern classes with values in the Chow ring $A(X)$. Under the present hypotheses, this formula is proved in [2] by a method very close to the one used in this seminar for the analogous formula with values in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$.

Thus, at least when k is perfect, (4.4) follows, and so does the fact that φ is an isomorphism modulo torsion. This in turn implies, a posteriori, that Riemann–Roch with denominators for a proper morphism of smooth algebraic schemes over k is essentially the same whether it is

written using $A(X)$, as in [2], or using $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, as in this seminar. The case of an arbitrary k reduces easily to the algebraically closed case: for a finite extension k'/k of degree n, the kernel of the natural map

$$A(X) \xrightarrow{f_X^*} A(X \otimes_k k')$$

is annihilated by n. Indeed, the trace homomorphism

$$f_{X*} : A(X \otimes_k k') \longrightarrow A(X)$$

satisfies $f_{X*} f_X^* = n \cdot \mathrm{id}$.[28]

4.3. It seems quite plausible that formulas (4.4) hold exactly as stated, and not merely modulo torsion; equivalently, that formula (4.5) holds not merely modulo torsion. This question is a very special case of the question whether the denominator-free Riemann–Roch formula (3.1) is valid in $A(X)$ for an immersion $Y \to X$, namely the case $i = d$. This special case may reasonably be expected to hold. Indeed, we already noted in no. 3 that, when k has characteristic zero, the denominator-free Riemann–Roch formula is valid at least in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$. Consequently the two sides of (4.5) have the same image in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$; equivalently, the second formula in (4.4) holds. This result will extend automatically to arbitrary characteristic once

the question raised in no. 1 has been satisfactorily resolved. As evidence in support of formula (4.5) in $A(X)$ itself, let us note that it is clear a priori that the restriction of the left-hand side to $X - Y$ is zero. Thus, since Y is assumed irreducible, the left-hand side must a priori be an integral multiple of $\mathrm{cl}(Y)$:

$$c_i(Y) = n(Y)\,\mathrm{cl}(Y).$$

This already proves the exact formula (4.5) whenever $\mathrm{cl}(Y)$ is not a torsion element of $A^i(X)$. It is immediate that this is the case if X is projective (and not merely quasi-projective) over the field k: indeed, if $\xi \in A^1(X)$ is a polarization of X, then $\deg(Y \cdot \xi^i) > 0$. This proves the desired formula in that case. It follows that the formula is still valid whenever X is isomorphic to an open subscheme of a smooth projective scheme over k, as will be the case when an adequate solution of the problem of resolution of singularities is available. In particular, this holds when k has characteristic zero, by Hironaka [11].

The validity of the exact formula (4.5), and hence of (4.4), would imply that the kernel of

$$\varphi^i : A^i(X) \longrightarrow \mathrm{Gr}^i_{\mathrm{top}}(X)$$

is annihilated by $(i-1)!$, and in particular *that* $\varphi^2 : A^2(X) \to \mathrm{Gr}^2_{\mathrm{top}}(X)$ *is an isomorphism.* (That φ^1 is an isomorphism already follows immediately from X 5.3.2.) Unless I am mistaken, the assertion that φ^2 is an isomorphism has not yet been proved, even in the special case $\dim X = 3$. It was therefore in any event premature for the editor to assert the validity of (4.4) and (4.5) in [9, no. 4, 3°].

4.4. At present, the principal advantage of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ (or of $\mathrm{Gr}^\bullet(X)$) over the Chow ring $A(X)$ is that it is better suited to explicit calculations of certain intersection formulas between cycles in cases of excess intersection, that is, intersections of excessively large dimension. Indeed, the definition of the intersection class by means of Chow's lemma, which allows a cycle to be moved sufficiently within its class, is poorly suited to calculation. A striking example in this regard is the following fundamental formula [Exposé VII 2.7], for a closed immersion $f : Y \to X$ of smooth quasi-projective schemes over k:

$$f^* f_*(y) = y\, c_d(\underline{N}), \tag{4.6}$$

[28] Thus the formula is established, thanks to P. Jouanolou, for X smooth and quasi-projective over a field (see the note on p. 5).

a formula valid in $\mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$ for $y \in \mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$, where $\underline{N}$ is the conormal sheaf

of Y in X, assumed everywhere to have rank d. In the cited passage, this formula follows from a very simple computation of the sheaves $\underline{\mathrm{Tor}}$. By 4.2, the preceding formula implies the same formula in $A(Y)$, but only modulo torsion. It is quite remarkable that formula (4.6) in $A(Y)$ itself has not yet been proved,[29] nor even the special case obtained by taking $y = 1$, which gives the self-intersection of Y:

$$f^*(\mathrm{cl}(Y)) = c_d(\underline{N}), \tag{4.7}$$

even when k is the field of complex numbers.

An analogous problem arises when one attempts to compute the Chow ring of the variety X' obtained by blowing up X along Y, by imitating the method used to compute $\mathrm{Gr}^\bullet(X')_{\mathbb{Q}}$ [Exposé VII 4.8], which also works (without tensoring with $\mathbb{Q}$) for the computation of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X')$. With the notation of the cited passage, the key formula for this computation is

$$f^*(i_*(y)) = j_*(g^*(y)c_{d-1}(\underline{\check{F}})). \tag{4.8}$$

At present, this formula has been established in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X')$ (for $y \in \mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$) by a computation of $\underline{\mathrm{Tor}}$ (Exposé VII 3.4), while its validity in $A(X')$ (for $y \in A(Y)$) remains to be proved.

Let us note that even the ℓ-adic analogues of formulas (4.6), (4.7), and (4.8) have not yet been proved; when y is the cohomology class of an algebraic cycle, formulas (4.6) and (4.8) are in any event true modulo torsion.

4.5. In this section and the following ones, we shall give examples for which, in diagram (4.1), the homomorphism j, respectively the homomorphism φ, is not an isomorphism. For a given X, it is plainly equivalent to say that

$$j : \mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X)$$

is an isomorphism, a monomorphism, or an epimorphism. Moreover, the image of j is the subring of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ generated by the Chern classes of locally free sheaves on X. If $K^\bullet(X)$ is generated as a $\mathbb{Z}$-algebra by the classes of invertible sheaves on X, one sees at once that the subring in question is also the subring generated by $\mathrm{Gr}^1_{\mathrm{top}}(X)$, the degree-one component of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$.

Now take for X a flag variety

$$X = G/B,$$

where G is a simply connected semisimple algebraic group over the algebraically closed field k, and B is a Borel subgroup [5]. When $k = \mathbb{C}$, the Bruhat cell decomposition of X [5, Exposé 13] implies that the canonical homomorphism

$$K^\bullet(X) \longrightarrow K(X(\mathbb{C})) \tag{*}$$

is an isomorphism, as is the canonical homomorphism

$$A(X) \longrightarrow H^*(X(\mathbb{C}), \mathbb{Z}), \tag{**}$$

where $X(\mathbb{C})$ is the compact space of complex points of X, endowed with its usual topology induced by that of $\mathbb{C}$. On the other hand, L. Hodgkin [12] proved that $K(X(\mathbb{C}))$ is generated by the classes of invertible sheaves on X. The same is therefore true of $K^\bullet(X)$ itself; moreover, it is not difficult to deduce, by a specialization argument using semisimple groups split over $\mathbb{Z}$ (SGA 3), that the result remains valid over every algebraically closed field k. The cell decomposition of X (or, when $k = \mathbb{C}$, the bijectivity of $(**)$) also readily implies that $A(X)$ is torsion-free. Hence, by 4.2, the homomorphism

$$\varphi : A(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X)$$

[29] Cf. the note on p. 23.

is an isomorphism. It follows that j is an isomorphism if and only if $A(X)$ is generated by its elements of degree 1, or, when $k = \mathbb{C}$, if and only if the integral cohomology of $X(\mathbb{C})$ is generated by its elements of degree 2, i.e. if the classifying space $B_{G(\mathbb{C})}$ is without torsion. It is known [15, p. 217, Theorem B, (1) $\Leftrightarrow$ (2), and 2.5] that this holds if and only if G is without torsion, equivalently if G is isomorphic to a product of groups $\mathrm{SL}(n)$ and $\mathrm{Sp}(m)$. Thus it suffices to take for G a group $\mathrm{SO}(n)$ with $n \geq 7$, or a simple exceptional group, to obtain an example in which j is not an isomorphism.

4.6. In this example there can be no doubt that the good ring is the torsion-free ring

$$A(X) \simeq \mathrm{Gr}^\bullet_{\mathrm{top}}(X),$$

whose basis consists of the cells in the Bruhat decomposition and which, when $k = \mathbb{C}$, is also isomorphic to the integral cohomology of $X(\mathbb{C})$. By contrast, $\mathrm{Gr}^\bullet(X)$ appears as a ring having torsion of a rather pathological nature. This example therefore confirms the view that the invariant $\mathrm{Gr}^\bullet(X)$ is hardly interesting

except modulo torsion, i.e. after tensoring with $\mathbb{Q}$. At the same time, this example shows that $\mathrm{Gr}^\bullet(X)$ is covariant with respect to proper morphisms only after tensoring with $\mathbb{Q}$, even if one restricts to closed immersions of smooth quasi-projective schemes. Indeed, if X is such that j is not an isomorphism, there is an integral closed subscheme Y of X such that

$$\mathrm{cl}^\bullet(Y) \in K^\bullet(X)$$

does not lie in $\mathrm{Filt}^i(X)$, where $i = \mathrm{codim}(Y, X)$. We may choose Y of minimal dimension, so that

$$\mathrm{Filt}^{i+1}(X) = \mathrm{Filt}^{i+1}_{\mathrm{top}}(X).$$

If Z is the singular locus of Y, and $X' = X \setminus Z$, $Y' = Y \setminus Z$, it follows that

$$\mathrm{cl}^\bullet(Y') \in K^\bullet(X')$$

does not lie in $\mathrm{Filt}^i(X')$. Here one uses the fact that $K^\bullet(X) \to K^\bullet(X')$ is surjective and compatible with the augmentations, and that its kernel is

$$\mathrm{Im}\big(K(Z) \longrightarrow K(X)\big) \subset \mathrm{Filt}^{i+1}(X) = \mathrm{Filt}^{i+1}_{\mathrm{top}}(X).$$

Now consider the inclusion

$$f : Y' \longrightarrow X',$$

which is a closed immersion of codimension i between smooth quasi-projective schemes over k, *such that*

$$f_* : K^\bullet(Y') \longrightarrow K^\bullet(X')$$

is not compatible with the λ-filtrations, even up to a shift by i, since $f_*(1) \notin \mathrm{Filt}^i(X')$.

4.7. Retaining the notation of 4.5, note that Exposé IX shows that Hodgkin's theorem quoted above is equivalent to the isomorphism

$$K^\bullet(G) \simeq \mathbb{Z}, \qquad \text{hence} \qquad \mathrm{Gr}^\bullet_{\mathrm{top}}(G) \simeq \mathbb{Z}.$$

The homomorphism $K^\bullet(G/B) \to K^\bullet(G)$ is a priori surjective, and its kernel is the ideal generated by the elements $\mathrm{cl}^\bullet(L) - 1$, where L ranges over the invertible sheaves on $X = G/B$ associated with representations $B \to \mathbb{G}_{\mathrm{m}}$ (which in fact yield all invertible sheaves on X [5, Exposé 15]). An analogous argument shows that

$$A(G/B) \longrightarrow A(G)$$

is also surjective, and that its kernel is the ideal generated by the elements of $A^1(G/B)$ defined by representations $B \to \mathbb{G}_m$, hence by all of $A^1(G/B)$.[30] Thus $A(G)$ is reduced to $A^0(X) = \mathbb{Z}$ if and only if G is "without torsion", i.e. if and only if G is isomorphic to a product of groups $\mathrm{Sl}(n)$ and $\mathrm{Sp}(m)$. Thus, in this case and only in this case, the homomorphism

$$\varphi : A(G) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(G)$$

is an isomorphism. Consequently, if G has torsion, the underlying scheme provides an example of a non-bijective φ.

4.8. One likewise finds an example in which the homomorphism γ of (4.1)

$$\gamma : A(G) \longrightarrow H^{2*}(G, \mathbf{Z}_\ell(*)) \tag{4.9}$$

does not factor through $\mathrm{Gr}^\bullet_{\mathrm{top}}(G)$, or, what amounts to the same thing here, an example in which this homomorphism is nonzero in degrees > 0. Restricting (as one may, thanks to suitable specialization theorems) to the case $k = \mathbb{C}$, one is reduced to studying the homomorphism

$$A(G) \longrightarrow H^*(G(\mathbb{C}), \mathbb{Z}), \tag{4.10}$$

which is moreover obtained, by passage to the quotient, from the canonical homomorphism

$$A(G/B) \simeq H^*(G/B(\mathbb{C}), \mathbb{Z}) \longrightarrow H^*(G(\mathbb{C}), \mathbb{Z}).$$

It is tempting to conjecture that (4.10) is in fact injective, since it is also known that the torsion primes for $G(\mathbb{C})$ are exactly those dividing the order of the finite group

$$A^+(G) = \coprod_{i>0} A^i(G).$$

If this were so, whenever G had torsion, (4.10) would not factor through φ; hence, for a suitable ℓ (more precisely, for ℓ dividing the order of $A^+(G)$, i.e. for ℓ a torsion prime of G), the same would hold for (4.9). We can at least assert that, for the group $G = G_2$ over $\mathbb{C}$, (4.10) is nonzero in degree > 0: more precisely, there exists an element of $A^3(G)$ whose image in $H^6(G(\mathbb{C}), \mathbb{Z})$ has order 2. It follows that (4.9) does not factor through φ when $\ell = 2$. The fact used about G_2, which amounts to saying that there exists in $H^6(G/B(\mathbb{C}), \mathbb{Z})$ an element whose image in $H^6(G(\mathbb{C}), \mathbb{Z})$ is nonzero, is found in [3], via the familiar dictionary between complex semisimple algebraic groups and compact Lie groups; it was kindly communicated to me by J.-P. Serre.

5. Relations between $\mathrm{Gr}^\bullet(X)$ and $H^{2*}(X, \mathbf{Z}_\ell(*))$

5.1. The example given in 4.8 does not exclude the possibility that, for an arbitrary scheme X, one might find a ring homomorphism

$$g' : \mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*)),$$

such that, for every locally free module E on X, g^i sends the Chern class $c_i(E)$ of Exp. V to the cohomological Chern class of SGA 5 VII. Of course, ℓ here denotes a prime distinct from the residual characteristics of X. Since the ring $\mathrm{Gr}^\bullet(X)$ is generated by the $c^i(E)$, one sees in any case that g' is uniquely determined by the requirement that it be compatible with Chern classes.

5.2. Using the Chern homomorphism

$$K^\bullet(X) \longrightarrow \mathrm{Ch}(H^{2*}(X, \mathbf{Z}_\ell(*))),$$

[30] Cf. IX 3.4, 2.2.

one obtains, by passing to the associated gradeds respectively for the filtration on the first member and the filtration by degree on the second, a homomorphism of graded groups (but not of rings!)

$$\psi : \mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*)),$$

which satisfies

$$\psi(c_i(E)) = (-1)^{i-1}(i-1)!\, c_i^\ell(E),$$

where $c_i^\ell(E)$ denotes the ℓ-adic Chern class. Dividing the components ψ^i by the indicated factors therefore gives a ring homomorphism

$$g_{\mathbb{Q}} : \mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbf{Q}_\ell(*)) = H^{2*}(X, \mathbf{Z}_\ell(*)) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell,$$

sending Chern classes to Chern classes, and such that the image of the first member is contained in that of $H^2(X, \mathbf{Z}_\ell(*))$. Consequently, the question of the existence of g' has an affirmative answer provided $H^{2*}(X, \mathbf{Z}_\ell(*))$ is replaced by its quotient by its torsion subgroup. The first doubtful case for defining the components g^i of the hypothetical homomorphism g is that of g^3; a priori one succeeds only in defining $2g^3 = \psi^3$. The answer to the question of the existence of g^3 would be negative if, for example, one could find three invertible sheaves L_1, L_2, L_3 on X satisfying

$$(L_1 - 1)(L_2 - 1)(L_3 - 1) = 0 \quad \text{in } K^\bullet(X),$$

but such that

$$c_1(L_1)c_1(L_2)c_1(L_3) \neq 0 \quad \text{in } H^6(X, \mu_2^{\otimes 3}).$$

5.3. When X is smooth and quasi-projective over a field k, one may similarly ask whether there exists a ring homomorphism

$$g : \mathrm{Gr}^\bullet(X) \longrightarrow A(X)$$

taking values in the Chow ring $A(X)$ and sending Chern classes to Chern classes. The preceding observations may be repeated in this connection; they show in particular how to construct $(-1)^{i-1}(i-1)!g^i$ directly when g^i itself is unavailable.

5.4. Let us note, to conclude this section, that essentially the same question arises if, instead of a scheme X, one considers a compact topological space (say) X. One can still consider the associated graded $\mathrm{Gr}^\bullet(X)$ of $K^\bullet(X)$ endowed with its λ-filtration, and seek a ring homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbb{Z})$$

sending Chern classes to Chern classes.

6. Cohomological Riemann–Roch theorem and Gysin homomorphism

6.1. For simplicity, we shall confine ourselves here to schemes X admitting an ample invertible module, so that there is no need to distinguish between the naive $K^\bullet(X)$ and the other one. Let us begin with the Riemann–Roch formula "with denominators", for a proper complete-intersection morphism $f : X \longrightarrow Y$. For a prime ℓ prime to the residual characteristics of Y, one can define the ℓ-adic Chern classes of a locally free module E,

$$c_i(E) \in H^{2i}(X, \mathbf{Z}_\ell(i))$$

(SGA 5 VII), corresponding to a homomorphism

$$K^\bullet(X) \longrightarrow \mathrm{Ch}(H^{2*}(X, \mathbf{Z}_\ell(*))). \tag{6.1}$$

The question therefore arises of giving a cohomological version of the Riemann–Roch formula. The only ingredient missing in order to give meaning to the formula is a Gysin homomorphism

$$H^n(X, \mathbf{Q}_\ell(i)) \longrightarrow H^{n-2d}(Y, \mathbf{Q}_\ell(i-d)), \tag{6.2}$$

where d is the virtual relative dimension of f. Assuming this homomorphism to be defined, and taking into account the Riemann–Roch formula in the form of Exp. VIII, with values in $\mathrm{Gr}^\bullet(Y)_\mathbb{Q}$, one sees that the validity of the ℓ-adic Riemann–Roch formula for the morphism f is exactly equivalent to the commutativity of the square

$$\begin{array}{ccc} \mathrm{Gr}^\bullet(X)_\mathbb{Q} & \longrightarrow & H^{2*}(X, \mathbf{Q}_\ell(*)) \\ \downarrow & & \downarrow \\ \mathrm{Gr}^\bullet(Y)_\mathbb{Q} & \longrightarrow & H^{2*}(Y, \mathbf{Q}_\ell(*)), \end{array} \tag{6.3}$$

where the horizontal arrows are obtained from (6.1) by passage to the associated gradeds and division, so as to send Chern classes to Chern classes, while the vertical arrows are respectively the arrow of Exp. VIII and the Gysin homomorphism in cohomology (assumed given).

6.2. Note that the Gysin homomorphism is defined in all cases if Y is smooth over a field k (SGA 5 IV), using the theorem of relative cohomological purity over k. The same construction applies when Y is excellent, regular, and of characteristic zero, thanks to the theorem of absolute cohomological purity of SGA 4 XIX; and it will automatically apply to every excellent regular Y, without restriction on the characteristic, once the theorem of absolute cohomological purity has been proved in full generality. Finally, the same construction applies, thanks to the theorem of relative cohomological purity (SGA 4 XVI 3.7), when X and Y are both smooth over the same scheme S and f is an S-morphism. It is natural to conjecture that, in each of these cases in which the Gysin homomorphism is defined, diagram (6.3) is indeed commutative. The only case in which this assertion is currently verified is that in which X and Y are both smooth over a field k; it then follows from the commutativity of the analogous diagram

$$\begin{array}{ccc} A(X) & \longrightarrow & H^{2*}(X, \mathbf{Z}_\ell(*)) \\ \downarrow & & \downarrow \\ A(Y) & \longrightarrow & H^{2*}(Y, \mathbf{Z}_\ell(*)), \end{array} \tag{6.4}$$

involving the rings $A(X)$ and $A(Y)$ of cycle classes modulo rational equivalence, whose commutativity was established in SGA 5 IV.

6.3. Note that in each of the cases considered above in which one succeeds in defining a Gysin homomorphism, one in fact obtains a construction for cohomology with coefficients in $\mathbf{Z}_\ell(*)$, and not only in $\mathbf{Q}_\ell(*)$. This makes it possible, when f is an *immersion*, to pose the question of the validity of the Riemann–Roch formula "without denominators", as explained in no. 3. When X and Y are both smooth over a field k, the commutativity of (6.4) shows that it would suffice to prove the formula without denominators in the Chow ring $A(Y)$; cf. (3.3).

6.4. When no regularity hypothesis is made on Y, and even if X and Y are of finite type over $\mathbb{C}$, one no longer sees a plausible definition of a cohomological Gysin homomorphism (6.2), even for a regular immersion of codimension 1. Can one nevertheless find one with reasonable properties, in particular making square (6.3) commutative? The question is evidently connected with that of defining, for X a complete intersection of codimension i in Y, "its cohomology class"

$$\gamma(X) \in H^{2i}(Y, \mathbf{Z}_\ell(i)),$$

in such a way as to turn intersections (without excess components) into cup products. This latter question has an affirmative solution at least modulo torsion, i.e. in $H^{2*}(Y, \mathbf{Q}_\ell(*))$, by defining

$$\gamma(X) = \frac{1}{(-1)^{i-1}(i-1)!}\, c_i(\mathcal{O}_X).$$

But I do not know whether $c_i(\mathcal{O}_X)$ is divisible by $(i-1)!$ in $H^{2i}(Y, \mathbf{Z}_\ell(i))$. In particular, if Y is, say, a projective scheme of dimension 3 over the field of complex numbers and X is reduced to a (singular) point of Y, is $c_3(\mathcal{O}_X)$ divisible by 2 in $H^6(Y, \mathbb{Z}_2(3))$, and in particular is the Chern class zero in $H^6(Y, \mathbb{Z}/2\mathbb{Z})$? Let us also mention, along the same lines, the following question: let Y be a projective (singular) scheme of dimension 4, and X a closed subscheme of dimension 1 which is globally an intersection of three positive divisors D_1, D_2, D_3. Consider the product of their Chern classes modulo 2,

$$c_1(D_1)c_1(D_2)c_1(D_3) \in H^6(Y, \mathbb{Z}/2\mathbb{Z});$$

is this element of H^6 independent of the particular choice of the D_i whose intersection is X?

7. Chern classes of perfect complexes

7.1. When one poses the question of developing a cohomological Riemann–Roch formula, with or without denominators, outside the rather artificial hypothesis that an ample invertible module exists on the schemes under consideration, one is necessarily faced with the question of defining, for a perfect complex of modules $L^\bullet$ on a scheme X, the Chern classes

$$c_i(L^\bullet) \in H^{2i}(X, \mathbf{Z}_\ell(i)),$$

generalizing the Chern classes of locally free modules, SGA 5 VII. The same question arises, moreover, for general locally ringed topoi (compare [10]), as well as for perfect complexes on paracompact spaces ringed by continuous complex-valued functions, working then with ordinary integral cohomology. For a compact space, it is known that the $K^\bullet(X)$ constructed with perfect complexes is the same as the "naive" one constructed with locally free modules (Exp. IV), and in that case one is trivially reduced to the classical definition. But already for a locally compact space the problem posed seems nontrivial, since the integral cohomology of such a space is not always the inverse limit of the cohomologies of its compact subsets.

7.2. More precisely, when Y is a closed subset of X such that $L^\bullet|_{X-Y}$ is zero (in the sense of derived categories), i.e. such that the cohomology sheaves of $L^\bullet$ are supported in Y, one would like to define Chern classes *with support in* Y

$$c_i^Y(L^\bullet) \in H^{2i}_Y(X, \mathbf{Z}_\ell(i)) \quad \text{respectively} \quad H^{2i}_Y(X, \mathbb{Z}),$$

which would recover the "ordinary" (sic) Chern classes of 7.1 by means of the canonical homomorphism

$$H^*_Y(X, -) \longrightarrow H^*(X, -).$$

Even in the case where the latter reduce to the Chern classes of locally free modules (X a scheme with an ample invertible module, or a compact space), the construction of these Chern classes "with supports" has not been carried out and will probably only be possible together with that of the general Chern classes postulated in 7.1.[31]

[31] Added in October 1968. The problems raised here were solved in the framework of Hodge cohomology by L. Illusie; cf. [13].

It is probable that Chern classes with supports will have a role to play in the proof of the cohomological Riemann–Roch formula without denominators (cf. 3.5). On the other hand, they should make it possible to give an improved version of the Riemann–Roch formula (with or without denominators) by introducing families of supports into that formula.

8. Chow ring of regular schemes

The problem is to define rational equivalence of cycles and to study the resulting ring of classes (in particular its functorial properties) for a regular noetherian scheme X admitting an ample invertible module, thereby generalizing the known theory when X is smooth over a field [4]. The key technical result will again be a "moving lemma", ensuring that a cycle can be varied within its class so as to meet a given cycle in the proper dimension.

The problem under consideration is quite natural a priori, especially since we have seen (4.5 to 4.8) that the substitutes $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ for the Chow ring $A(X)$ are hardly satisfactory when one does not confine oneself to a theory "modulo torsion". When X is obtained from a smooth projective algebraic scheme Y over a field k by taking the local scheme at the vertex of a projective cone C and removing the origin, one finds (assuming that $A(X)$ is indeed the inductive limit of the Chow rings of $U - \{O\}$, where U ranges over the open neighborhoods in C of the vertex O) that $A(X)$ is canonically isomorphic to the quotient ring

$$A(Y)/\xi A(Y),$$

where $\xi \in A^1(Y)$ is induced by the chosen projective embedding of Y. This shows that knowing the Chow ring $A(X)$ (which is of a nonclassical type) is equivalent to knowing $A(Y)/\xi A(Y)$, which might be called the "primitive part" of the Chow ring $A(Y)$ (in a sense suggested by the Lefschetz–Hodge cohomological theory of nonsingular projective algebraic varieties). This example shows that the "local Chow rings" of local rings with isolated

singularities are geometric invariants that are by no means trivial and will undoubtedly have a role to play in the "arithmetic" study of these local rings.

Of course, once one has a reasonable notion of a Chow ring for a regular noetherian scheme admitting an ample invertible module, one should generalize to this case the relations established or conjectured in no. 4 between this ring and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, and formulate in terms of this ring the question of the "Riemann–Roch formula without denominators" stated in no. 3.

Bibliography of Exposé XIV

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[16](*) P. Deligne, Exposé XVIII in *Théorie des topos et cohomologie étale des schémas (SGA 6)*, by M. Artin, A. Grothendieck, and J. L. Verdier.

[17](*) U. Mausin, *Correspondances, motifs et transformations monoïdales* (in Russian). Mat. Sborik, t. 77 (119), no. 4, 1968.

(*) Added in February 1969.

Note for page 1

The theory of the operations Λ^i in $D^-(X)$, and of the operations λ^i in $K^\bullet(X)$, was written down by P. Deligne in 1968. He proves that $K^\bullet(X)$ thereby becomes a pre-λ-ring and, when $\mathcal{O}_X$ is a sheaf of algebras over a field, that $K^\bullet(X)$ is even a λ-ring. It is plausible that a refinement of Deligne's method would make it possible to dispense with the restrictive hypothesis on $\mathcal{O}_X$.

Notes for page 11

(*) (April 1969) D. Mumford has pointed out to us a proof of (4.6) in the *Chow ring* (without disregarding torsion), which he obtained in a seminar at Harvard in 1959! The reader will find the proof of this beautiful result in SGA 5 VII 9. This result obviously contains the corresponding cohomological result mentioned in note (**). The cohomological result, however, is proved by J.-P. Jouanolou under more general conditions (without a quasi-projective hyperplane, and in the relative smooth case over an arbitrary base).

(**) (January 1969) J.-P. Jouanolou has just proved these formulas for every y, without disregarding torsion; cf. SGA 5 VII § 4.9. Deligne and Manin [17] had previously observed independently, and by (easy) arguments of a rather different nature, that it was enough to establish the formulas for "algebraic" y, and even for $y = 1$ in [16], in order to conclude the same relation in the general case.

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of finite perfect amplitude (complex)	I	4.8
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of finite flat amplitude (complex)	I	5.2
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of finite perfect amplitude relative to f (complex)	III	4.1
of flat amplitude relative to f (or Y) contained in $[a, b]$ (complex)	III	3.1
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of finite presentation	I	2.8
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SGA 7

Exposé I

Summary of the First Lectures by A. Grothendieck

written up by P. Deligne

Contents

0. Preliminaries
1. Arithmetic proof of the monodromy theorem
2. Vanishing cycles
3. Geometric proof of the monodromy theorem
4. Vanishing criteria for the sheaves of vanishing cycles
5. Action of monodromy on the π_1
6. Appendix by P. Deligne: arithmetic proof of the stable reduction theorem

0. Preliminaries

0.0. Terminology. Recall the following definitions:

(0.0.1) trait: the spectrum of a discrete valuation ring. The generic and closed points of a trait are often denoted by η and s.

(0.0.2) strictly local: for a ring, local henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.0.3) geometric point of S (in this exposé): a morphism $\bar{x} : \mathrm{Spec}(\bar{k}) \longrightarrow S$ with image $x \in S$, where $k(\bar{x}) \underset{\text{dfn}}{=} \bar{k}$ is a separable closure of $k(x)$. For a henselian trait $S = \mathrm{Spec}(V)$, a generic geometric point of S (that is, one with image η) is often denoted by $\bar{\eta}$. The residue field of the valuation ring $\bar{V}$, the integral closure of V in $k(\bar{\eta})$, is a separably closed algebraic extension of $k(s)$ and defines a geometric point $\bar{s}$ of S centered at s.

(0.0.4) strict localization of X at the geometric point $\bar{x}$: SGA 4 VIII 4.4.

(0.0.5) essentially of finite type over S: for a local (respectively, henselian local; respectively, strictly local) S-scheme, the spectrum of a local ring (respectively, the henselization of a local ring; respectively, a strict localization) of a scheme of finite type over S.

(0.0.6) $F \longmapsto F(n)$: Tate twist.

(0.0.7) An endomorphism T of a finite-dimensional vector space is quasi-unipotent if its eigenvalues are roots of unity, that is, if there exist $N \geqslant 1$ and $M \geqslant 1$ such that $(T^N - 1)^M = 0$.

0.1. Review of ℓ-adic cohomology

Let X be a scheme of finite type over an algebraically closed field k of characteristic exponent p, and let ℓ be a prime different from p.

The ℓ-adic cohomology groups of X were defined in SGA 4 and SGA 5.

a) $H^i(X, \mathbb{Z}/\ell^n)$ is the i-th cohomology group of the ètale site X_{et} of X with coefficients in $\mathbb{Z}/\ell^n$;

b) by definition,

$$H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n);$$

c) by definition,

$$H^i(X, \mathbb{Q}_\ell) = H^i(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The ℓ-adic cohomology groups of X with proper supports are defined in the same way from the groups $H^i_c(X, \mathbb{Z}/\ell^n)$ of SGA 4 XVII.

In each of the following cases, one knows how to prove that these groups $H^i(X)$ have reasonable finiteness properties:

(0.1.1) for cohomology with proper supports;

(0.1.2) when X is proper over k;

(0.1.3) when X is smooth over k (by (0.1.1) and Poincaré duality, SGA 4 XVIII);

(0.1.4) when Hironaka's resolution of singularities is available for schemes of finite type over k of dimension at most $\dim(X)$;

(0.1.5) when $i = 0, 1$.

In all these cases, the groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, the groups $H^i(X, \mathbb{Z}_\ell)$ are finitely generated, the groups $H^i(X, \mathbb{Q}_\ell)$ are finite-dimensional, and (except perhaps in case (0.1.5)) there are split short exact sequences

$$0 \longrightarrow H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \longrightarrow H^i(X, \mathbb{Z}/\ell^n) \longrightarrow \mathrm{Tor}_1(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \longrightarrow 0.$$

Moreover, in each case the proof of the finiteness theorem also proves a generic constructibility theorem: if k is the field of fractions of an integral noetherian scheme S, and X is the generic fibre of a morphism $f_1 : X_1 \longrightarrow S$ of finite type, then there exists a nonempty open subset $U \subset S$ such that over U the sheaves $R^i f_* \mathbb{Z}/\ell^n$ are locally constant and their formation commutes with every base change.

0.2. We no longer suppose that k is algebraically closed. Let $\bar{k}$ be an algebraic closure of k (or a separable closure, which amounts to the same thing here). We consider the "geometric" cohomology groups $H^i(X_{\bar{k}})$, where $X_{\bar{k}}$ is obtained from X by extension of scalars from k to $\bar{k}$. By transport of structure, the Galois group $\mathrm{Gal}(\bar{k}/k)$ acts on these groups. In $\mathbb{Q}_\ell$-cohomology, in any of the cases (0.1.1)–(0.1.5), this gives continuous representations of $\mathrm{Gal}(\bar{k}/k)$ in a linear group $\mathrm{GL}(n, \mathbb{Q}_\ell)$.

0.3. Review of traits (0.0.1)

For the proofs, we refer to [4]. Let S be a henselian trait, and let η, s, V, $\bar{V}$, $\bar{\eta}$, and $\bar{s}$ be as in (0.0.1) and (0.0.3). Let p denote the characteristic exponent of $k(s)$, and let $\mathrm{Gal}(\bar{\eta}/\eta)$ (respectively, $\mathrm{Gal}(\bar{s}/s)$) denote the Galois group $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ (respectively, the corresponding group for $\bar{s}/s$). By transport of structure, $\mathrm{Gal}(\bar{\eta}/\eta)$ acts on $k(\bar{s})$; hence there is a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ onto $\mathrm{Gal}(\bar{s}/s)$ whose kernel is the inertia group I. The subfield of $k(\bar{\eta})$ fixed by I is a maximal unramified extension of $k(\eta)$.

The profinite group I has a largest pro-p subgroup P. The quotient I/P is the tame inertia group. Canonically,

$$I/P \simeq \varprojlim_n \mu_n(k(\bar{s})) \underset{\text{dfn}}{=} \widehat{\mathbb{Z}}(1)(k(\bar{s})).$$

If ϕ_n is the canonical map from I/P to $\mu_n(\bar{k})$ and u is an element of $\bar{V}$ of valuation $1/n$, then, for $\sigma \in I$,

$$\sigma(u) \equiv \phi_n(\sigma)\, u \qquad \text{(modulo elements of valuation greater than } 1/n).$$

In summary,

$$\begin{aligned} \mathrm{Gal}(\bar{\eta}/\eta) &\supset I \supset P, \\ \mathrm{Gal}(\bar{\eta}/\eta)/I &\simeq \mathrm{Gal}(\bar{s}/s), \\ I/P &\simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ P &: \text{ a pro-}p \text{ group} \quad (\{e\} \text{ if } p = 1), \end{aligned} \tag{0.3.1}$$

and the action (by inner automorphisms in $\mathrm{Gal}(\bar{\eta}/\eta)$) of $\mathrm{Gal}(\bar{\eta}/\eta)/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.4. More generally, let S be strictly local and regular, let D be a regular divisor in S, and let p' be the characteristic exponent of the generic point of D. Let $\bar{\eta}$ be a geometric generic point of S, and let $\bar{s}$ be the resulting geometric point centered at the closed point s. By Abhyankar's lemma, the fundamental group $\pi_1(S - D, \bar{\eta})$ has a filtration analogous to (0.3.1):

$$\begin{aligned} \pi_1(S - D, \bar{\eta}) &\supset I \supset P, \\ \pi_1(S - D, \bar{\eta})/I &\simeq \mathrm{Gal}(\bar{s}/s), \\ I/P &\simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ P &: \text{ with no quotient of order prime to } p'. \end{aligned} \tag{0.4.1}$$

The action by inner automorphisms of $\pi_1(S-D,\bar{\eta})/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.5. Néron desingularization

The method of Néron desingularization gives the following result.

Lemma 0.5.1. Let $S \longrightarrow S_o$ be a morphism of henselian traits, with $S = \mathrm{Spec}(V)$ and $S_o = \mathrm{Spec}(V_o)$, and let π be a uniformizer of V_o. Suppose that π is a uniformizer of V, and that the extensions $k(s)/k(s_o)$ and $k(\eta)/k(\eta_o)$ are separable. Then V is a filtered direct limit of smooth henselian V_o-algebras B_i essentially of finite type over V_o.

To construct the desired V_o-algebras, one takes $V_o \subset B \subset V$, with B of finite type, and desingularizes $\mathrm{Spec}(B)$ by Néron's method ([1], § 4).

Let $S = \mathrm{Spec}(V)$ be a henselian trait.

(0.5.2) If S is of equal characteristic, with uniformizer π, one may apply (0.5.1) with $S_o = \mathbf{F}_p[\pi]_{(\pi)}$. The extension s/s_o is separable because $\mathbf{F}_p$ is perfect; the extension η/η_o is separable because π, being a uniformizer, is not a p-th power (if $p \neq 1$), and hence belongs to a p-basis. It follows that S is a limit of spectra of smooth henselian $\mathbf{F}_p[\pi]_{(\pi)}$-algebras essentially of finite type.

(0.5.3) If S is complete and of mixed characteristic, with perfect residue field k, then V is an Eisenstein extension of the ring of Witt vectors (that is, the Cohen ring) $W(k)$:

$$V \simeq W(k)[\pi]/\left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i\right).$$

If the a_i are perturbed slightly, one obtains an isomorphic extension. We may therefore suppose that the a_i belong to $W(k')$, where k' is of finite type over $\mathbf{F}_p$. Let k_o be the perfect closure of k' in k, and set

$$V_o = W(k_o)[\pi]/\left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i\right).$$

Lemma 0.5.1 applies to V/V_o, and the residue field of V_o is a radicial extension of a field of finite type over $\mathbf{F}_p$.

1. Arithmetic proof of the monodromy theorem

The reader will find in [5], Appendix, a proof of the following result of A. Grothendieck.

Proposition 1.1. Let S be a henselian trait, let ℓ be a prime different from the characteristic exponent p of $k(s)$, and let ρ be a continuous representation of $\mathrm{Gal}(\bar{\eta}/\eta)$ in $\mathrm{GL}(n,\mathbb{Q}_\ell)$. Suppose that the following condition holds:

$(*_\ell)$ No finite extension of $k(s)$ contains all roots of unity of ℓ-power order.

Then there exists an open subgroup I_1 of the inertia group I such that $\rho(\sigma)$ is unipotent for every $\sigma \in I_1$.

Condition $(*_\ell)$ is automatically satisfied when $k(s)$ is of finite type over the prime field (or is radicial over such a field).

The following theorem follows immediately from 1.1.

Monodromy theorem 1.2. With the preceding notation, let X be a scheme of finite type over η, and let H be a cohomology space of $X_{\bar{\eta}}$ with coefficients in $\mathbb{Q}_\ell$, of one of the types (0.1.1)–(0.1.5). If condition $(*_\ell)$ is satisfied, the action on H of every element of I is quasi-unipotent.

The limit arguments of 0.5 allow us to dispense with hypothesis $(*_\ell)$.

Variant 1.3. Condition $(*_\ell)$ in 1.2 is superfluous.

Let H'_o be a $\mathbb{Z}_\ell$-cohomology space of $X_{\bar{\eta}}$ giving rise to H, and put $H_o = H'_o/\text{torsion}$. By hypothesis, $H = H_o \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and the action of $\mathrm{Gal}(\bar{\eta}/\eta)$ is induced by

$$\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{GL}(H_o).$$

Let $I_{\ell'}$ be the largest prime-to-ℓ subgroup of I. Its image in $\mathrm{GL}(H_o)$ is an ℓ-adic Lie group of order prime to ℓ, and is therefore finite.

Make an extension of traits $S' \longrightarrow S$. This does not change H_o (SGA 4 XVI 6.6), and $\rho(I')$ has finite index in $\rho(I)$, because I'/P' has finite index in I/P, by what precedes. It therefore suffices to prove the theorem over S'; we may suppose that

a) $\mathrm{Gal}(\bar{\eta}/\eta)$ acts trivially on $H_o/\ell H_o$;

b) if S is of mixed characteristic, then S is complete with perfect residue field (EGA O_{III} 10.3.1).

Put $S = \mathrm{Spec}(V)$, define V_o as in (0.5.2) and (0.5.3), and apply (0.5.1). We have

$$V = \varinjlim_i B_i.$$

Here $S_i = \mathrm{Spec}(B_i)$ is essentially of finite type and smooth over V_o. Let $D_i \subset S_i$ be defined by the equation $\pi = 0$. For sufficiently large i, H_o comes from a twisted constant $\mathbb{Z}_\ell$-sheaf $\mathcal{H}_i$ on $S_i - D_i$ (by "generic constructibility," (0.1)), and $\mathcal{H}_i/\ell\mathcal{H}_i$ is constant.

The filtrations (0.3.1) and (0.4.1) give rise to commutative diagrams:

(1.3.1)
$$\begin{array}{ccccc} \mathrm{Gal}(\bar{\eta}/\eta) & \supset & I & \supset & P \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(S_i - D_i, \bar{\eta}) & \supset & I_i & \supset & P_i \end{array} \qquad \text{and} \qquad \begin{array}{ccc} I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \\ \downarrow & & \| \\ I_i/P_i & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \end{array}$$

The representation on H_o of $\mathrm{Gal}(\bar{\eta}/\eta)$ is induced by an action of $\pi_1(S_i - D_i, \bar{\eta})$ on H_o that is trivial on $H_o/\ell H_o$. The group

$$\mathrm{Ker}\big(\mathrm{GL}(H_o) \longrightarrow \mathrm{GL}(H_o/\ell H_o)\big)$$

is a pro-ℓ group. Thus P_i acts trivially, and we apply the following variant of 1.1.

Variant 1.4. Retain the notation of 0.4, and let ρ be a continuous representation of $\mathrm{Gal}(\bar{\eta}/\eta)$ in $\mathrm{GL}(n, \mathbb{Q}_\ell)$. Suppose that $k(s)$ satisfies $(*_\ell)$ and that $\rho(P)$ is finite. Then the action of I is quasi-unipotent.

2. Vanishing cycles

2.1. Let S be a strictly local trait, and retain the notation of 0.3. Thus, by hypothesis, $s = \bar{s}$. Let $f : X \longrightarrow S$ be a scheme of finite type over S. The formalism of vanishing cycles is a tool for studying the relation between the cohomology of X_s and that of $X_{\bar{\eta}}$, as well as the action of the inertia group on the latter, using information about f that is local on X.

A key result of this kind is the specialization theorem (SGA 4 XVI 2.2), according to which X_s and $X_{\bar{\eta}}$ have the same cohomology when f is proper and smooth.

In the transcendental theory, if $f : X \longrightarrow D$ is a proper morphism from an analytic space X to the disk D, vanishing cycles arise as follows:

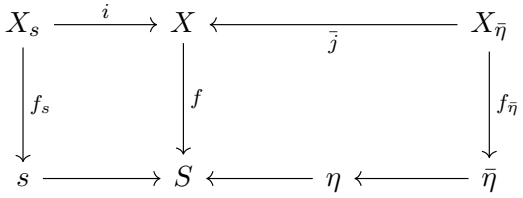
a) After shrinking D if necessary, one may suppose that X is a topological fibre bundle over the punctured disk D^* (Thom's isotopy theorem). In particular, the fibres

X_t $(t \in D^*)$ are all isomorphic.

b) It is not unreasonable to picture the special fibre X_s as obtained from "the" general fibre X_t by contracting polyhedra in X_t; one would then have a map $\pi : X_t \longrightarrow X_s$. The method of vanishing cycles consists in studying the difference between the cohomology of X_s and that of X_t by means of the Leray spectral sequence for π.

The geometric theory explained below is very close to this transcendental theory: the general point $t \in D$ must be replaced by $\bar{\eta}$, the geometric generic point of S. Conversely, the transcendental theory can be modeled on the geometric theory by replacing t or $\bar{\eta}$ with the universal cover $\widetilde{D}^*$ of D^*, and X_t or $X_{\bar{\eta}}$ with $X \times_D \widetilde{D}^*$. This eliminates many ε's and η's, but sometimes loses information.

2.2. Fix a torsion ring Λ in which p is invertible (for example, $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ prime to p). Consider the morphisms



The sheaves of vanishing cycles ψ^i on X_s are

$$\psi^i = i^* R^i \bar{j}_* \Lambda. \tag{2.2.1}$$

If f is proper, then (SGA 4 XII 5.5)

$$i^* : H^p(X, R^q \bar{j}_* \Lambda) \xrightarrow{\sim} H^p(X_s, i^* R^q \bar{j}_* \Lambda), \tag{2.2.2}$$

and the Leray spectral sequence for $\bar{j}$ becomes

$$H^p(X_s, \psi^q) \Longrightarrow H^{p+q}(X_{\bar{\eta}}, \Lambda). \tag{2.2.3}$$

When nothing happens at infinity, this spectral sequence can sometimes be obtained without a properness hypothesis. That some hypothesis at infinity is necessary is already clear from the trivial case $X_s = \varnothing$.

The definitions immediately give:

Proposition 2.3. Let $\bar{x}$ be a geometric point of X_s (for example, a closed point), let $X_{(\bar{x})}$ be the strict henselization of X at $\bar{x}$, and let $X_{(\bar{x})\bar{\eta}}$ be the geometric generic fibre of the projection $X_{(\bar{x})} \longrightarrow S$. Then

$$\psi^i_{\bar{x}} = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda).$$

Corollary 2.4. Wherever f is smooth, $\psi^i = 0$ for $i > 0$ and $\psi^0 = \Lambda$.

This is SGA 4 XV 2.1.

The group $\mathrm{Gal}(\bar{\eta}/\eta)$ (which, by hypothesis, equals the inertia group) acts on the ψ^i by transport of structure, and

Scholium 2.5. The spectral sequence (2.2.3) is $\mathrm{Gal}(\bar{\eta}/\eta)$-equivariant.

Variant 2.6. Put $\bar{S} = \mathrm{Spec}(\bar{V})$ and $\bar{X} = X \times_S \bar{S}$. The scheme $X_{\bar{\eta}}$ is the complement of $X_s = X_{\bar{s}}$ in $\bar{X}$. Put

$$\phi^i = \underline{H}^{i+1}_{X_s}(\bar{X}, \Lambda)$$

(the local-cohomology sheaves). There is an exact sequence of sheaves

$$0 \longrightarrow \phi^{-1} \longrightarrow \Lambda \longrightarrow \psi^0 \longrightarrow \phi^0 \longrightarrow 0,$$

and, for $i > 0$, isomorphisms $\psi^i \xrightarrow{\sim} \phi^i$. If f is smooth, the ϕ^i vanish.

Put $\phi^i_{\mathrm{gl}} = H^i_{X_s}(\bar{X}, \Lambda)$, the global analogue of ϕ^i. There is a spectral sequence

$$H^p(X_s, \phi^q) \Longrightarrow \phi^{p+q}_{\mathrm{gl}}, \tag{2.6.1}$$

and a long exact cohomology sequence which, when f is proper, becomes

$$\cdots \longrightarrow H^i(X_s, \Lambda) \longrightarrow H^i(X_{\bar{\eta}}, \Lambda) \longrightarrow \phi^i_{\mathrm{gl}} \longrightarrow \cdots. \tag{2.6.2}$$

Variant 2.7. Let K_t be the maximal tamely ramified extension of K in $\bar{K}$. Put $X_t = X \times_S \mathrm{Spec}(K_t)$, and let $\bar{j}_t$ be the projection from X_t to X. Define tame variants of the ψ^i by

$$\psi_t^i = R^i \bar{j}_{t*} \Lambda.$$

Since P is a pro-p group and p is invertible in Λ, one obtains

$$H^*(X_t, \Lambda) = H^*(X_{\bar{\eta}}, \Lambda)^P, \tag{2.7.1}$$

$$\psi_t^i = (\psi^i)^P. \tag{2.7.2}$$

When f is proper, the Leray spectral sequence for $\bar{j}_t$ is obtained from (2.2.3) by passing to P-invariants:

$$H^p(X_s, \psi_t^q) \Longrightarrow H^{p+q}(X_t, \Lambda) = H^{p+q}(X_{\bar{\eta}}, \Lambda)^P. \tag{2.7.3}$$

3. Geometric proof of the monodromy theorem

Let Λ be as in 2.2.

Conjecture 3.1 (purity). Let A be an excellent regular strictly local ring (0.0.2) and let D be a regular divisor in $\mathrm{Spec}(A)$ defined by a parameter x. Put $U = \mathrm{Spec}(A) - D = \mathrm{Spec}(A[1/x])$. Then

$$H^q(U, \Lambda) = \begin{cases} \Lambda & \text{if } q = 0, \\ \Lambda(-1) & \text{if } q = 1 \quad \text{(notation (0.0.6))}, \\ 0 & \text{if } q > 1. \end{cases} \tag{3.1.1}$$

Here are some cases in which this conjecture is known (SGA 4 XIX): for $q = 0$ or 1; when A has characteristic 0; when A has equal characteristic p and resolution of singularities is available in characteristic p in dimensions $\leqslant \dim(A)$; or when $(\mathrm{Spec}(A), D)$ is a smooth pair (the relative purity theorem, SGA 4 XVI 3.7).

When purity is available and $D = \sum_{i \in B} D_i$ is a normal-crossings divisor in $\mathrm{Spec}(A)$ defined by part of a regular system of parameters, the cohomology of $U = \mathrm{Spec}(A) - D$ is given by

$$\begin{aligned} &H^0(U, \Lambda) = \Lambda, \\ &H^1(U, \Lambda) = \Lambda(-1)^B, \\ &H^q(U, \Lambda) = \bigwedge^q H^1(U, \Lambda) \qquad \text{(via the cup product)}. \end{aligned} \tag{3.1.2}$$

These formulas are identical to those which, over $\mathbf{C}$, give the cohomology of $D^{*B} \times D^k$.

3.2. Let S be a strictly local trait, and retain the notation of (0.0.1) and (0.0.3). Let $f : X \longrightarrow S$ be a morphism of finite type. Suppose that X is regular, that X_η is smooth, and that X_s is a normal-crossings divisor in X. We shall suppose that this divisor is, globally and not merely locally for the étale topology, a sum of regular divisors:

$$(X_s)_{\mathrm{red}} = \sum_{i \in B} D_i$$

(the general case can be recovered by localization). Denote their multiplicities by m_i, so that $X_s = \sum_{i \in B} m_i D_i$.

Let x be a point of X_s, let $\bar{x}$ be a geometric point of X lying over x (for example, x may be closed and equal to $\bar{x}$), put $C = \{i \mid x \in D_i\}$, and let K be the complex concentrated in degrees 0 and -1

$$K : \mathbf{Z}^C \xrightarrow{d} \mathbf{Z}, \qquad d((n_j)) = \sum_j m_j n_j.$$

Theorem 3.3 (uses purity). The groups of tame vanishing cycles $\psi^i_{t\bar{x}}$ are as follows:
(i) There is a canonical isomorphism of graded Λ-algebras

$$\psi^*_{t\bar{x}} \simeq \bigwedge\nolimits^* \big(H_1(K) \otimes \Lambda(-1)\big) \otimes_\Lambda \psi^0_{t\bar{x}}.$$

(ii) $\psi^0_{t\bar{x}}$ is the Λ-algebra Λ^E, where E is a set on which the tame inertia group $I_t = I/P = \widehat{\mathbf{Z}}(1)(k(s))$ acts transitively, and whose cardinality is the largest prime-to-p divisor of $\#H_0(K)$.

Here is the method of proof.

a) Let $X_{(\bar{x})}$ be the strict localization of X at $\bar{x}$, let $U = X_{(\bar{x})} - X_{(\bar{x})s}$ (the complement of a normal-crossings divisor), let t_j be a local equation for D_j, put

$$X_n = X_{(\bar{x})}\big[(t_j^{1/n})\big] \qquad ((n,p) = 1),$$

which is a regular scheme, and let U_n be the inverse image of U in X_n. By (3.1.2),

$$H^q(U_n, \Lambda) = \bigwedge\nolimits^q \big(\Lambda(-1)^C\big).$$

For $m = nd$, the pullback map from $H^q(U_n, \Lambda)$ to $H^q(U_m, \Lambda)$ is multiplication by d^q. Putting $\widetilde{U} = \varprojlim_{(n,p)=1} U_n$, one therefore has

$$\text{(3.3.1)} \qquad \begin{cases} H^0(\widetilde{U}, \Lambda) = \Lambda, \\ H^q(\widetilde{U}, \Lambda) = \varinjlim_n H^q(U_n, \Lambda) = 0 \quad \text{if } q \neq 0. \end{cases}$$

b) $\widetilde{U}$ is a pro-covering of U with group $\widehat{\mathbf{Z}}(1)(k(s))^C$. By definition,

$$\psi^*_{t\bar{x}} = H^*(U_{\eta_t}, \Lambda).$$

$\widetilde{U}$ is a pro-covering, with group $H_1(K) \otimes \widehat{\mathbf{Z}}(1)(k(s))$, of each connected component of U_{η_t}, and these components are permuted transitively by the tame inertia group. Formula (3.3.1) and the Hochschild–Serre spectral sequence now give the formulas in 3.3.

Pictorially, one may say that $U_{\bar{\eta}}$ is the fibre of a morphism δ from the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s))^C, 1)$ represented by U to the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s)), 1)$ represented by η; the morphism δ induces

$$d : (n_j) \longmapsto \sum_j m_j n_j : \widehat{\mathbf{Z}}(1)(k(s))^C \longrightarrow \widehat{\mathbf{Z}}(1)(k(s)).$$

Corollary 3.4 (uses purity). Suppose that f is proper. Let N be the least common multiple of the multiplicities m_j ($j \in B$). Let $q \geqslant 0$, and let q' be the minimum of q and the greatest number of divisors D_j passing through one point. For T in the tame inertia group $I_t = I/P$, one has

$$(T^N - 1)^{q'+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \Lambda)^P.$$

This follows at once from 3.3, Scholium 2.5, and Variant 2.7.

The required purity results are available for H^1, in characteristic 0, or when S is essentially of finite type over a field.

Theorem 3.5. Let C be a smooth projective curve over the fraction field $k(\eta)$ of a discrete valuation ring V. The inertia group I acts on $H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell)$ by quasi-unipotent automorphisms of level 2: there exists N such that, for $T \in I$,

$$(T^N - 1)^2 = 0 \quad \text{on} \quad H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell).$$

We reduce to the case in which V is strictly local. The assertion is that there exists an open subgroup I_1 of I such that $(T-1)^2 = 0$ on H^1 for $T \in I_1$.

We may extend the scalars of $k(\eta)$ to a finite extension, replacing V by its normalization in that extension. Since the image of the pro-p group P in an ℓ-adic Lie group $\mathrm{GL}(n, \mathbf{Q}_\ell)$ is finite, we may in particular suppose that P acts trivially.

Let C_o be a minimal model of C over V (for the existence of C_o, which is based on Abhyankar's theorem, see the discussion in [3], § 2, p. 87; one could instead reduce to the case where V is excellent by passing to its completion and invoking SGA 4 XVI 1.6).

Let C_1 be a regular model of C, obtained from C_o by blowups, whose special fibre is a normal-crossings divisor. The arguments of 3.3 and 3.4 apply to C_1, ψ_t^0, ψ_t^1, and

$$H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n)^P = H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n),$$

with N independent of n; Theorem 3.5 follows.

Remark 3.6. Let A be an abelian variety over $k(\eta)$. Since A is a quotient of a Jacobian, it follows again that the action of I on $H^1(A_{\bar{\eta}}, \mathbb{Q}_\ell)$, or on $T_\ell(A)$, is quasi-unipotent of level 2.

Remark 3.7. The action of I on $H^1(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is in fact quasi-unipotent of level 2 for every scheme X of finite type over η.

Theorem 3.5 (or Remark 3.6) is the basis for the proof of the stable reduction theorem for abelian varieties that will be given in Exposé IX, § 3.

3.8. It is clear that whenever purity and resolution of singularities are available (for example, in characteristic 0), the preceding method proves the monodromy theorem and gives bounds for the nilpotence exponent. Thus, in equal characteristic 0, if X is proper and smooth over η and T belongs to an open subgroup I_1 of I, one has

$$(T-1)^{q+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \mathbb{Z}_\ell) \qquad (T \in I_1 \subset I).$$

4. Vanishing criteria for the sheaves of vanishing cycles

4.1. Let S be a strictly local trait and let $f : X \longrightarrow S$ be a scheme of finite type over S. Put $n = \dim(X_\eta)$.

If X' is the scheme-theoretic closure of X_η in X, then:

a) On $X_s - X'_s$, one has $\phi^{-1} = \Lambda$, and the other ϕ^i vanish;

b) on X'_s, the ϕ^i and ψ^i relative to X/S or to X'/S coincide, and $\phi^{-1} = 0$.

Since X' is flat over S, this often permits a reduction to the case in which X is flat over S.

Theorem 4.2.

(i) One has $\psi^i = 0$ for $i > n$.

(ii) More precisely, let x be a point of X_s whose closure has dimension k ($k = \mathrm{trdeg}_{k(s)}\, k(x)$), and let $\bar{x}$ be a geometric point lying over x. Then

$$\psi^i_{\bar{x}} = 0 \quad \text{for} \quad i > n - k, \qquad \text{i.e.} \qquad d(\psi^i) \leqslant n - i \quad (\text{notation of SGA 4 XIV 2.1}).$$

Indeed,

$$\psi^i_{\bar{x}} = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda),$$

and $X_{(\bar{x})\bar{\eta}}$ is an inverse limit of affine schemes of dimension at most n over $k(\bar{\eta})$. Assertion (i) therefore follows from the affine Lefschetz theorem (SGA 4 XIV 3.2) by passage to the limit.

To prove (ii), we may suppose that X is flat over S (4.1) and that there is a quasi-finite flat S-morphism

$$g : X \longrightarrow \mathbb{A}^k_S$$

such that $g(x)$ is the generic point of $\mathbb{A}^k_s$. Let S' be the strict localization of $\mathbb{A}^k_S$ at $\bar{x}$. Then:

a) $S'_{\bar{\eta}}$ is the spectrum of a field, and $\mathrm{Gal}(\bar{\eta}'/s'_{\bar{\eta}})$ is a p-group Q;

b) by (i), applied to g,

$$H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda) = 0 \qquad \text{for } i > n-k;$$

c) one has

$$H^i(X_{(\bar{x})\bar{\eta}}, \Lambda) = H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda)^Q,$$

which proves the theorem.

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension at most d, the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$$

is an isomorphism for $i > n+d+1$ and an epimorphism for $i = n+d+1$.

4.4. To prove, in certain cases, that the ϕ^i vanish for small i, we shall use local duality, either directly or through the local Lefschetz theorems of SGA 2 XIV. This tool is available only in characteristic 0, or in equal characteristic p when resolution of singularities à la Hironaka is available in the dimensions under consideration.

Theorem 4.5. Suppose that S has characteristic 0. Let $f : X \longrightarrow S$ be a morphism of finite type and let x be a closed point of X_s. Suppose that:

(a) x is an isolated non-smooth point in its fibre;

(b) X has relative geometric depth at least n at x: in a neighbourhood of x, the scheme X is identified with a subscheme defined by k equations in a smooth scheme of relative dimension N at x, with $n \leqslant N-k$.

Then $\phi^i_x = 0$ for $i < n$.

Let X_x be the strict localization of X at x, and put $V = X_x - \{x\}$. For every trait S' finite over S, put similarly

$$X'_x = X_x \times_S S', \qquad V' = V \times_S S'.$$

Let $\bar{S}$ be the normalization of S in $\bar{\eta}$, and put

$$\bar{X}_x = X_x \times_S \bar{S}, \qquad \bar{V} = V \times_S \bar{S}.$$

We have:

(1) If $\phi^i = 0$ on $(X_x)_s$ (i.e. at every generization of x in X_s) for $i \leqslant m$, then

$$H^{i-1}_{\{x\}}(\bar{X}_x) = \widetilde{H}^{\cdot,i}(\bar{V}) \xrightarrow{\sim} \widetilde{H}^{\cdot,i}(\bar{V}_{\bar{\eta}}) = \phi^i \qquad \text{for } i \leqslant m.$$

The ϕ^i are indeed the cohomology sheaves of $\bar{V}$ with support in V_s; under the hypotheses of 4.5, condition (a) makes (1) applicable with $m = \infty$ (SGA 4 XV 2.1).

(2) Hypothesis (b) of 4.5 is stable under base change. By SGA 2 XIV 5.6, it implies

$$\widetilde{H}^{\cdot,i}(V') = H^{i-1}_{\{x\}}(X'_x) = 0 \qquad \text{for } i < n.$$

We conclude by observing that $\widetilde{H}^{\cdot,i}(\bar{V})$ is the filtered colimit of the $\widetilde{H}^{\cdot,i}(V')$.

Combining 4.2 and 4.5 gives:

Corollary 4.6. Let S be a henselian trait of characteristic 0, let X/S be a relative complete intersection of relative dimension n, and let $x \in X_s$ be an isolated non-smooth point in its fibre. Then $\phi^i_x = 0$ for $i \neq n$.

Corollary 4.7. Under the hypotheses of 4.6, ϕ^n_x is a flat Λ-module.

Indeed, if Λ' is a finite Λ-algebra, a general nonsense (SGA 4 XVII 5.2.11) gives, for the $\phi^i_{\Lambda'}$,

$$\phi^{n-i}_{\Lambda'} = \mathrm{Tor}^{\Lambda}_i(\Lambda', \phi^n_{\Lambda}),$$

and we apply 4.6 to Λ'.

Variant 4.8. In 4.5, omit condition (a), and let Z be the non-smooth locus of f. One may still conclude that

$$\phi_x^i = 0 \qquad \text{for } i < n - \dim Z_s.$$

We proceed by descending induction on $\dim \overline{\{x\}}$. The induction hypothesis permits us to apply 4.5 (1) with $m = n - \dim Z_s$, and the proof is completed as before.

4.9. It is possible to obtain a vanishing criterion using only information about the geometric fibres of f. For complete intersections, however, the result is one degree weaker than 4.5.

Proposition 4.10. Under the general hypotheses of 4.5, suppose that:

(a) x is an isolated non-smooth point in its fibre;

(b) for every point y of $X_{\bar\eta}$ whose closure in $X_{\bar\eta}$ has dimension $d(y)$ and which is a generization of x, the étale depth satisfies

$$\operatorname{depth}_{\text{ét},y}(X_{\bar\eta}) \geqslant n - d(y), \quad \text{i.e.} \quad H^i_{\{y\}}(X_{\bar\eta}) = 0 \quad \text{for } i \leqslant n - d(y);$$

(c) $\operatorname{depth}_{\text{ét},x}(X_s) \geqslant n$.

Then $\phi_x^i = 0$ for $i < n - 1$.

Retain the notation of 4.5. We may apply 4.5 (1) with $m = \infty$. By the local Lefschetz theorem (SGA 2 XIV), hypothesis (b) gives

$$H^i(V') \xrightarrow{\sim} H^i(V_s) \qquad (\ i < n\text{-}1\ )$$

$$H^i(V') \hookrightarrow H^i(V_s) \qquad (\ i = n\text{-}1\ )$$

Condition (c) then gives

$$H^i(\bar V) = \varinjlim H^i(V') = 0 \qquad \text{for } i \leqslant n - 1,$$

which proves 4.10.

4.11. As in 4.8, one can give a variant of 4.10 in which (a) is not assumed.

5. Action of monodromy on π_1

In this section (Grothendieck's exposé of 19 November 1968), we shall use the semistable reduction theorem for curves. Artin and Winters [2] gave a direct proof of this theorem; it can also be deduced [3] from the analogous theorem for abelian varieties (IX § 3, using 3.6). We shall also use Schlessinger's theory (Rim's Exposé VI).

5.1. Let k be an algebraically closed field, let C be a projective curve over k, and let $x_1, \ldots, x_n$ be a finite set of closed points of C. The pointed curve

$$X = (C; x_1, \ldots, x_n)$$

is called stable if:

(a) C is reduced, and its only singularities are double points with distinct tangents;

(b) the x_i are distinct smooth points of C;

(c) if C_1 is an irreducible component of C, and E is the set of points of the normalization C_1' of C_1 lying above a singular point of C or above one of the x_i, then, if C_1' has genus 0 (respectively 1), the set E has at least 3 (respectively 1) elements.

Condition (c) is equivalent to each of the following:

(c') X has no nontrivial infinitesimal automorphisms.

(c") If ω is the invertible dualizing sheaf, the invertible sheaf

$$\omega' = \omega\Big(\sum_i x_i\Big)$$

is ample.

An S-scheme C equipped with sections $x_1, \ldots, x_n$ is a stable curve over S if its geometric fibres are stable curves. In [3], §§ 1–2, only the case $n = 0$ is considered. The extension of the results of loc. cit. to the general case is immediate:

(1) If X/k is stable, then $H^1(C, \omega'^{\otimes n}) = 0$ for $n \geqslant 2$, and $\omega'^{\otimes n}$ is very ample for $n \geqslant 3$ (cf. [3] 1.2).

(2) The formal moduli scheme M of X is smooth, and C remains singular over a normal-crossings divisor of M.

(3) If X and Y are stable over S, then $\mathrm{Isom}_S(X, Y)$ is finite and unramified over S (isomorphisms are required to respect the marked points).

(4) If X/k is stable, the map

$$\mathrm{Aut}_k(X) \longrightarrow \mathrm{Aut}_k \mathrm{Pic}^\circ(C)$$

is injective.

Finally, we have:

Proposition 5.1.1. Let $X_\eta = (C_\eta; x_1, \ldots, x_n)$ be stable over the generic point η of a trait S, with C_η smooth. There exist a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X' \otimes_{S'} \eta' = X_\eta \otimes_\eta \eta'.$$

This follows from the usual semistable reduction theorem, which permits one to choose η' so that the geometric special fibre of the minimal model C'' of $C_{\eta'}$ over S' satisfies (a). One then blows up, iteratively if necessary, smooth points of C'' so that (b) is satisfied. Next one contracts the chains of smooth rational components of self-intersection -2 which contain none of the $\bar{x}_i$ in the special fibre; the result is C' (cf. [3], proof of 2.3, p. 88).

As in loc. cit., one can check that C' exists as soon as the Jacobian of C_η has semistable reduction. The smoothness hypothesis on C_η is moreover unnecessary.

5.2. Let X_η be a geometrically connected scheme of finite type over the fraction field of a strictly local trait S. For a geometric point ξ of $X_{\bar{\eta}}$, there is an exact sequence

$$0 \longrightarrow \pi_1(X_{\bar{\eta}}, \xi) \longrightarrow \pi_1(X, \xi) \longrightarrow \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow 0. \tag{5.2.1}$$

It induces a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar{\eta}})$. Passing to the largest prime-to-p quotient $\pi_1^{(p)}(X_{\bar{\eta}})$ of $\pi_1(X_{\bar{\eta}})$, one obtains

$$\mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{Aut}_{\mathrm{ext}}\big(\pi_1^{(p)}(X_{\bar{\eta}})\big). \tag{5.2.2}$$

Suppose that X_η has a rational point x, and let $\bar{x}$ be the geometric point

$$\bar{\eta} \longrightarrow \eta \xrightarrow{x} X_\eta.$$

For $\xi = \bar{x}$, the exact sequence (5.2.1) splits canonically, giving

$$[x] : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{Aut}\big(\pi_1(X_{\bar{\eta}}, \bar{x})\big) \longrightarrow \mathrm{Aut}\big(\pi_1^{(p)}(X_{\bar{\eta}}, \bar{x})\big). \tag{5.2.3}$$

Theorem 5.3. The image of P under (5.2.2) (or under (5.2.3), which amounts to the same thing) is finite.

5.3.1. Reduction to the case in which X_η is geometrically normal and integral.

Let X'_η be the normalization of X_η, let $(X'_{i\eta})_i$ be the family of connected components of X'_η, put

$$X''_\eta = X'_\eta \times_{X_\eta} X'_\eta,$$

and let $(X''_{j\eta})_j$ be the family of connected components of X''_η. After a finite extension of $k(\eta)$, we may suppose that the $X'_{i\eta}$ are geometrically normal and integral and have rational points x_i, and that the $X''_{j\eta}$ are geometrically connected and have rational points y_j.

For $k = 1$ or 2, and whenever $p_k(y_j) \in X'_{i\eta}$, choose a class of paths ℓ_{kij} from $\mathrm{pr}_k(\bar{y}_j)$ to $\bar{x}_i$. It is an element of a torsor under $\pi_1(X'_{i\bar{\eta}}, \bar{x}_i)$. Let $\bar{\ell}_{kij}$ be the corresponding prime-to-p path (the same path modulo $\mathrm{Ker}(\pi \longrightarrow \pi^{(p)})$). We may choose $\bar{\ell}_{kij}$ to be invariant under P: the set of prime-to-p paths is an inverse limit of finite sets of cardinality prime to p on which the pro-p group P acts.

Consequently, if a finite-index subgroup P' of P acts trivially on the groups $\pi_1^{(p)}(X'_{i\bar{\eta}}, \bar{x}_i)$, the Van Kampen formulas—equivalently, the fact that the normalization map $X'_{\bar{\eta}} \longrightarrow X_{\bar{\eta}}$ is effective descent for étale coverings—show that P' acts trivially on $\pi_1^{(p)}(X_{\bar{\eta}}, \bar{x})$.

5.3.2. If $X_{\bar{\eta}}$ is normal and U is an open subset, then $\pi_1(U)$ maps onto $\pi_1(X_{\bar{\eta}})$; we may therefore suppose that $X_{\bar{\eta}}$ is smooth and affine. By cutting with generic hyperplane sections (which replaces S by its strict localization at the generic point of the special fibre of an $\mathbb{A}^n_S$) and applying Bertini, we reduce to the case in which X_η is smooth, geometrically connected, and of dimension one.

After a finite extension of $k(\eta)$, we may suppose that X_η is the complement in a smooth projective curve $\bar{X}_\eta$ of a finite set of rational points x_i, whose number we are free to increase. By the semistable reduction theorem (5.1.1), we reduce to the case in which there is a stable curve $X = (C; x_1, \ldots, x_n)$ over S such that

$$X_\eta = C_\eta - \{x_1, \ldots, x_n\}.$$

We then apply the following result.

Theorem 5.4. *Let S be a strictly local scheme, let $f : \bar{X} \longrightarrow S$ be a proper flat morphism whose special fibre is a curve with ordinary double points as its only non-smooth points, and let Y be a subscheme of $\bar{X}$ which is the union of a finite number of disjoint sections contained in the smooth locus. Let x be a rational point of $X = \bar{X} - Y$. Let U be the open subset of S over which X is smooth, and let ξ be a geometric point of U. Then the image of $\pi_1(U, \xi)$ in $\mathrm{Aut}(\pi_1^{(p)}(X_\xi, x\xi))$ is abelian and prime to p.*

Observe that $\pi_1(U, \xi)$ can act on $\pi_1^{(p)}(X_\xi, x\xi)$ only because the groups $\pi_1^{(p)}$ satisfy the specialization theorem.

We reduce first to the case in which S is complete, and then to the universal case in which S is a formal moduli scheme for the special fibre equipped with its sections. Then S is the spectrum of a formal power-series ring over a Cohen ring:

$$S = \mathrm{Spec}\big(W[[t_1, \ldots, t_n]]\big),$$

and U is the complement of a normal-crossings divisor with equation $t_1 \cdots t_m = 0$. In this universal case, Abhyankar's lemma shows that $\pi_1(U, \xi)$ is abelian and prime to p, which proves the theorem.

Remark 5.5. One can prove a result analogous to 5.4 in arbitrary relative dimension by taking a generic plane section of dimension one and applying Bertini.

6. Appendix: arithmetic proof of the stable reduction theorem

(by P. Deligne)

Let S be a trait and let A_η be an abelian variety over $k(\eta)$, of dimension $d \neq 0$. For a local morphism of traits $S' \longrightarrow S$, denote by $A_{\eta'}$ the abelian variety over $k(\eta')$ obtained from A_η by extension of scalars. Let $\mathcal{A}_{S'}$ be the Néron model of $A_{\eta'}$ over S', let $\mathcal{A}_{S',s'}$ be its special fibre, and let $\mathcal{A}^\circ_{S',s'}$ be the identity component of $\mathcal{A}_{S',s'}$. The stable reduction theorem is as follows.

Theorem 6.1. *For a suitable S', $A_{\eta'}$ has stable reduction, i.e. $\mathcal{A}^\circ_{S',s'}$ is an extension of an abelian variety by a torus.*

It follows fairly readily from 6.1 that one may choose S' such that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field (the same argument would work if $k(s)$ were radicial over such a field).

Let ℓ be a prime different from the residual characteristic exponent p, let $T_\ell(A)$ be the Tate module, let $V_\ell(A) = T_\ell(A) \otimes \mathbf{Q}_\ell$, and let

$$\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{GL}(T_\ell(A))$$

be the natural representation. After a preliminary extension of scalars, we are reduced to the case in which the inertia group I acts unipotently (1.2), and hence through its largest pro-ℓ quotient $\mathbf{Z}_\ell(1)$ (0.3). Let T be the image in $\mathrm{GL}(T_\ell(A))$ of a generator of $\mathbf{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that $(T-1)^r \neq 0$. We have:

Lemma 6.2. The map

$$(\sigma, v) \in I \times V_\ell(A) \longmapsto (\rho(\sigma) - 1)^r(v)$$

induces the morphism and the isomorphism in the commutative diagram

(6.2.1)
$$\begin{array}{ccc} V_\ell(A)_I \otimes \mathbf{Q}_\ell(r) & \xrightarrow{\quad M \quad} & V_\ell(A)^I \\ \downarrow & & \uparrow \\ \big(V_\ell(A)/\operatorname{Ker}(T-1)^r\big) \otimes \mathbf{Q}_\ell(r) & \xrightarrow{\quad \sim \quad} & \operatorname{Im}(T-1)^r \end{array}$$

$V_\ell(A)_I$ and $V_\ell(A)^I$ are $\mathrm{Gal}(\bar{s}/s)$-modules, and M is Galois-equivariant.

6.3. An ℓ-adic representation

$$\tau : \mathrm{Gal}(\bar{s}/s) \longrightarrow \mathrm{GL}(V)$$

is said to be of weight n ($n \in \mathbf{Z}$) if the following condition is satisfied:

(*) For a suitable model X of $k(s)$ (that is, an irreducible variety of finite type over the prime field such that $k(s) = k(X)$), τ factors through $\pi_1(X, \bar{s})$ and, for every closed point x of X, the eigenvalues of $\tau(F_x)$ are algebraic numbers all of whose complex conjugates have absolute value $|k(x)|^{n/2}$ (here F_x denotes the geometric Frobenius ϕ_x^{-1}).

If V (respectively W) has weight n (respectively m) and $n \neq m$, then $\mathrm{Hom}_{\mathrm{Gal}}(V, W) = 0$.

6.4. It follows from the universal property of the Néron model that $V_\ell(A)^I = V_\ell(\mathcal{A}^\circ_{S,s})$. If a is the dimension of the largest abelian quotient of $\mathcal{A}^\circ_{S,s}$ and m is the dimension of its multiplicative part, then, by Weil, $V_\ell(A)^I$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2.

6.5. Let A^* be the abelian variety dual to A. Recall that $V_\ell(A^*)$ and $V_\ell(A)$ (and hence $V_\ell(A^*)^I$ and $V_\ell(A)_I$) are in duality with values in $\mathbf{Q}_\ell(1)$. If a^* and m^* are defined for A^* as a and m are for A, it follows from 6.4 that $V_\ell(A)_I$ is an extension of a representation of dimension m^* and weight 0 by a representation of dimension $2a^*$ and weight -1.

6.6. The Galois module $\mathbf{Q}_\ell(r)$ has weight $-2r$. Since $\mathrm{Im}(T-1)^r \neq 0$, it follows from 6.4, 6.5, and the isomorphism in 6.2 that $r \leq 1$. If $r = 0$, then

$$\begin{cases} a + m \leq d, \\ 2a + m = 2d, \end{cases}$$

so $m = 0$, $a = d$, and A_η has good reduction. Suppose that $r = 1$. It then follows from 6.2 that $\mathrm{Im}(T-1)$ has weight -2 and that $V_\ell(A)/V_\ell(A)^I$ has weight 0; these spaces have the same dimension $m_1 \leq m$.

We then have

$$\dim V_\ell(A)^I = 2d - m_1 = 2a + m,$$

and

$$2 \dim \mathcal{A}_{S,s} \geq 2(a+m) \geq 2a + m + m_1 = 2d = 2 \dim \mathcal{A}_{S,s},$$

so A_η has stable reduction ($\dim \mathcal{A}_{S,s} = a + m$). At the same time, we have obtained $m_1 = m$; that is,

$$\mathrm{Im}(T-1) \subset V_\ell(A)^I \simeq V_\ell(\mathcal{A}^\circ_{S,s})$$

is the V_ℓ of the multiplicative part of $\mathcal{A}^\circ_{S,s}$.

6.7. To treat the general case, we shall use (0.5) (cf. 1.3). We reduce to the case in which:

(a) the conditions of (0.5.2) or (0.5.3) are satisfied;

(b) $\mathrm{Gal}(\bar\eta/\eta)$ acts trivially on A_ℓ;

(c) the action of I on $T_\ell(A)$ is unipotent.

Under these hypotheses, we shall prove that A_η has stable reduction. Hypothesis (a) allows us to apply (0.5.1) and write $S = \varprojlim S_i$ as in 1.3, whose notation we retain. Let s_i be the closed point of S_i.

For sufficiently large i, the identity component $\mathcal{A}^\circ$ of the Néron model $\mathcal{A}_S$ descends to a smooth group scheme $\mathcal{A}_i$ with connected fibres over S_i, and $(\mathcal{A}_i)_\ell$ is constant over $S_i - D_i$. The group $\pi_1(S_i - D_i, \bar\eta)$ acts on $T_\ell(A)$ and acts trivially on $T_\ell(A)/\ell T_\ell(A) \simeq A_\ell$. As in 1.3, one sees that P_i acts trivially. We then have, by (1.3.1),

$$T_\ell(A)_I = T_\ell(A)_{I_i}, \qquad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}^\circ_s) = T_\ell(A)^I = T_\ell(A)^{I_i},$$

and (6.2.1) is a diagram of $\mathrm{Gal}(\bar s_i/s_i)$-modules to which the arguments of 6.4–6.6 may be applied.

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Exposé II

Finiteness properties of the fundamental group

by Michèle Raynaud

Let k be a separably closed field of characteristic $p \geq 0$, let X be a connected scheme of finite type over k, and let $\Pi_1^{(p)}(X)$ be the largest "prime-to-p" quotient of $\Pi_1(X)$. In this exposé we prove that, when X is a surface, $\Pi_1^{(p)}(X)$ is topologically finitely presented, and that the same holds for arbitrary X if resolution of singularities is assumed.

For a surface, the proof method, due to J.-P. Murre, consists in reducing to a fibration over P^1 having properties (i), (ii), and (iii) of the following proposition.

Proposition 2.1. Let k be an algebraically closed field and let S be a projective, irreducible, smooth surface over k. Then there exists a blow-up

$$g : S' \longrightarrow S,$$

where S' is a regular surface equipped with a fibration

$$f : S' \longrightarrow P^1,$$

such that f has a section σ and the following conditions hold:

(i) f is smooth at the points of $\sigma(P^1)$.

(ii) The fibres of f are geometrically integral curves having only ordinary singularities.

(iii) There exists a nonempty open subset U of P^1 such that the fibres of f over the points of U are smooth.

The proposition follows from the existence of a projective embedding $S \longrightarrow P^r$ for which a Lefschetz pencil exists (XVII 2.5). Indeed, let $D = \{H_t\}_{t \in P^1}$ be such a pencil of hyperplanes. Recall that it has the following properties (loc. cit. 2.2):

(a) The axis Δ of D meets S transversely; the closed subscheme $\Delta \cap S$ is therefore reduced and consists of finitely many closed points $x_1, \ldots, x_n$ of S.

(b) There exists a nonempty open subset U of P^1 such that, for every $t \in U$, H_t meets S transversely.

(c) For every $t \in P^1 - U$, H_t meets S transversely except at one point, which is an ordinary quadratic singularity.

Through every point x of S not belonging to Δ there passes a unique hyperplane H_t. The map assigning $t \in P^1$ to x is a rational map $a : S \dashrightarrow P^1$ with domain of definition $S - (\Delta \cap S)$.

Consider the blow-up $g : S' \longrightarrow S$ of the closed subscheme $S \cap \Delta$. The scheme S' is regular, and the fibre S'_{x_i} over x_i is the projective bundle defined by the tangent space of S at x_i, hence is isomorphic to P^1 (EGA IV 19.4.2, 19.4.4). Moreover, the rational map $f = ag$ is everywhere defined (XVIII 3.1). Finally, to each x_i one can associate the section of f that sends $t \in P^1$ to the tangent vector at x_i of $H_t \cap S$.

Let us verify conditions (i), (ii), and (iii). For each $t \in P^1$, the curves S'_t and $S_t = H_t \cap S$ are isomorphic. Indeed, the morphism $S'_t \longrightarrow S_t$ is an isomorphism except possibly at the points x_i, where S_t is regular; moreover, since S'_t is a complete intersection in S', it is integral, and is therefore isomorphic to S_t.

The asserted properties of the fibres follow from (a)–(c). Finally, f is smooth at the points of σ, since f is flat and the fibres S'_t are geometrically regular at $\sigma(t)$.

Corollary 2.2. With the notation of 2.1, let Y be a normal-crossings divisor in S, and let $Y' = Y \times_S S'$. Then f, g, and U may be chosen so that $Y' \times_{P^1} U$ is a divisor with normal crossings relative to U.

It is known that, if a Lefschetz pencil exists for an embedding $S \longrightarrow P^r$, the Lefschetz pencils form an open subset of the variety of lines in P^r (XVII 3.2.1). The same is true of the set of

Lefschetz pencils whose axis does not meet Y and for which there is a nonempty open subset U_1 of P^1 such that $Y \cap H_t$ is smooth for $t \in U_1$. It is then enough to construct f and g from a pencil satisfying these two conditions and replace U by $U \cap U_1$.

2.3.0. Let G be a profinite group. Recall that G is said to be topologically finitely generated if there is an integer n such that G is isomorphic to a quotient of the free pro-group $F(n)$ on n generators. Put $K = \mathrm{Ker}(F(n) \longrightarrow G)$. The group G is said to be topologically finitely presented if, in addition, there are finitely many elements $k_1, \ldots, k_r$ of K such that the smallest closed normal subgroup of K containing $k_1, \ldots, k_r$ is K.

Theorem 2.3.1. Let k be a separably closed field of characteristic $p \geq 0$, and let X be a connected scheme of finite type over k. Suppose that schemes of finite type and dimension at most $\dim X$ over an algebraic closure of k admit strong resolution of singularities (SGA 5 I 3.1.5). This condition is satisfied when $\dim X = 2$, by [1], or when $\mathrm{char}\, k = 0$, by [2]. Then the fundamental group $\Pi_1^{(p)}(X)$ is topologically finitely presented.

By SGA 1 IX 4.10, we may suppose that k is algebraically closed.

1) Reduction to the case in which X is a regular surface.

If $f : X' \longrightarrow X$ is a morphism of finite type, put $X'' = X' \times_X X'$, and let $(X'_a)_{a \in A}$ and $(X''_b)_{b \in B}$ be the families of connected components of X' and X'', respectively. When f is an effective descent morphism for the category of finite étale coverings, the fundamental group $\Pi_1(X)$ can be computed explicitly

in terms of the fundamental groups $\Pi_1(X'_a)$ and $\Pi_1(X''_b)$. It follows (SGA 1 IX 5.2, 5.3) that if the $\Pi_1(X'_a)$ are topologically finitely generated, then so is $\Pi_1(X)$; and if the $\Pi_1(X'_a)$ are topologically finitely presented while the $\Pi_1(X''_b)$ are topologically finitely generated, then $\Pi_1(X)$ is topologically finitely presented.

Since X admits resolution of singularities, there is a regular scheme X' and a proper birational morphism $f : X' \longrightarrow X$. Since f is an effective descent morphism for finite étale coverings, it suffices to prove the theorem under the additional assumption that X is regular. One first deduces that $\Pi_1(X)$ is topologically finitely generated; knowing that $\Pi_1(X')$ is topologically finitely presented and that the $\Pi_1(X''_b)$ are topologically finitely generated then shows that $\Pi_1(X)$ is topologically finitely presented.

Now let $(U_i)_{i \in I}$ be a finite affine open covering of X, and let $f : \bigsqcup U_i \longrightarrow X$ be the canonical morphism. Since f is an effective descent morphism for finite étale coverings, it is enough to prove the theorem for the U_i. We are thus reduced to the case in which X is affine and smooth.

Assume henceforth that X is affine and smooth over k, with $\dim X > 2$. The Lefschetz theorem for quasi-projective schemes ([3] V 7.4) applies: if Y is a sufficiently general hyperplane section of X, then

$$\Pi_1(Y) \simeq \Pi_1(X).$$

It therefore suffices to prove the theorem for Y, reducing us to the case $\dim X = 2$.

2) The case in which X is a smooth surface over k.

By the strong resolution theorem for surfaces, there exist a smooth, integral, projective surface S and an open immersion $i : X \longrightarrow S$ such that $Y = S - X$ is a normal-crossings divisor. We apply Corollary 2.2 and retain its notation.

We show that it is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically finitely presented. Since X and X' are regular and

$$\mathrm{codim}\left(\bigcup_{1 \leq i \leq n} \{x_i\}, X\right) \geq 2,$$

every étale covering of X' induces on $X - \bigcup_{1 \leq i \leq n} \{x_i\}$ an étale covering which extends uniquely to an étale covering of X (SGA 2 1.11). This gives an isomorphism

$$\Pi_1(X') \simeq \Pi_1(X).$$

Consider the cartesian diagram

$$\begin{array}{ccc} X' & \longleftarrow & X'_U \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h_U} \\ P^1 & \longleftarrow & U \end{array}$$

where $h = f|_{X'}$. Let L be the set of primes different from p, and let $\bar{\eta}$ be a geometric point over the generic point η of P^1. We verify the hypotheses of SGA 1 XIII 4.8. The morphism h has a section σ, because f does. To prove condition (a) of SGA 1 XIII 4.7, note first that, since the fibres of h are geometrically integral, h is 0-acyclic and locally 0-acyclic (SGA 4 XV 4.1, 1.16). It remains to show that, for every locally constant sheaf of sets F on X', (F, h) is cohomologically proper in dimensions ≤ 0. If ξ is a closed point of P^1 and Z denotes the integral closure of $\operatorname{Spec} \mathcal{O}_{P^1,\xi}$ in $\bar{\eta}$, then $X'_Z = X' \times_{P^1} Z$ satisfies condition (S_2) at its closed points and is regular at every other point, hence is normal. Consequently, an étale covering of X'_Z whose restriction to $X'_{\bar{\eta}}$ is connected is necessarily connected. This proves the assertion, and hence condition (a).

Since X'_U is the complement in S'_U of a divisor with normal crossings relative to U, h_U is locally 1-aspherical for L (SGA 4 XV 2.1); and, for every finite constant sheaf of L-groups F on X'_U, (F, h_U) is cohomologically proper in dimensions ≤ 1 (SGA 1 XIII 2.4), which gives condition (b). Since P^1 is normal and $T = P^1 - U$ has codimension 1, we have

$$\operatorname{prof} \text{ét}_T(P^1) \geq 2,$$

which gives condition (c). The condition

$$\operatorname{prof} \operatorname{hop}_x(X') \geq 3$$

holds at every closed point of X' by the purity theorem, which gives condition (e). Finally, for every $t \in P^1$, h is smooth at $\sigma(t)$, and hence is locally 1-aspherical for L at $\sigma(t)$, which gives condition (d).

By SGA 1 XIII 4.7, if a is a geometric point of $X'_{\bar{\eta}}$, there is an exact sequence

$$1 \longrightarrow \Pi_1^{(p)}(X'_{\bar{\eta}}, a) \longrightarrow \Pi_1'(X'_U, a) \longrightarrow \Pi_1(U, a) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K be the image of $\Pi_1(U, a)$ in $\operatorname{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$. The group of coinvariants under K,

$$\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K,$$

is the quotient of $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ by the closed normal subgroup generated by the elements $x^{-1}q(x)$, where $x \in \Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ and $q \in K$. By loc. cit. (4.7.4), there is an isomorphism

$$\Pi_1^{(p)}(X', a) \simeq \Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K.$$

Write $P^1 - U = \coprod_{1 \leq i \leq n} \{t_i\}$. For each i, let $\bar{T}_i$ be the strict localization of P^1 at a geometric point $\bar{t}_i$ over t_i, and let s_i be the generic point of $\bar{T}_i$. An étale covering of U whose restriction to every s_i is trivial extends to an étale covering of P^1, and is therefore trivial. For each i, let $\overline{k(s_i)}$ be an algebraic closure of $k(s_i)$. The preceding assertion says that $\Pi_1(U, a)$ is the smallest closed normal subgroup containing the images of the Galois groups $\operatorname{Gal}(\overline{k(s_i)}/k(s_i))$ in $\Pi_1(U, a)$. By I 5.3, the image K_i of $\operatorname{Gal}(\overline{k(s_i)}/k(s_i))$ in $\operatorname{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$ is finite. Since K is the smallest closed normal subgroup containing all the K_i, and since $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ is topologically finitely presented (SGA 1 XIII 3.2), this proves that $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K$, and hence also the isomorphic group $\Pi_1^{(p)}(X)$, is topologically finitely presented.

Remark 2.3.2. Resolution of singularities is not needed to prove Theorem 2.3.1 when X is the open complement of a normal-crossings divisor in a projective scheme smooth over k.

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Exposé VI

FORMAL DEFORMATION THEORY

by D. S. Rim

1. Formal existence theorem

We recall the notion of a cofibered category (S.G.A. 1 IV). Let $\underline{C}$ be a fixed category. A cofibered category $\underline{A}$ over $\underline{C}$ is, by definition, a category $\underline{A}$ together with a functor $P : \underline{A} \longrightarrow \underline{C}$ such that:

Ax. 1. For every map f in $\underline{C}$ and every object a in $\underline{A}$ for which $P(a)$ is the source of f, there is a cocartesian f-morphism $\alpha : a \longrightarrow b$. Thus, for every f-morphism $\alpha' : a \longrightarrow b'$, there is a unique T-morphism $\tau : b \longrightarrow b'$ in $\underline{A}$ such that $\alpha' = \tau\alpha$, where T is the target of f.

Ax. 2. A composite of cocartesian morphisms in $\underline{A}$ is again cocartesian.

The following properties of a cofibered category $P : \underline{A} \longrightarrow \underline{C}$ follow immediately from the definitions.

(Cancellation) Let $a \overset{\alpha}{\underset{\beta}{\rightrightarrows}} a'$ be two cocartesian maps such that $P(\alpha) = P(\beta)$. If there is a cocartesian map $b \xrightarrow{\gamma} a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(Factorization) Given cocartesian maps $a \xrightarrow{\alpha} a'$ and $a \xrightarrow{\beta} a''$, there is a cocartesian map $a' \xrightarrow{\gamma} a''$ such that $\gamma\alpha = \beta$ if and only if there is a map $f : P(a') \longrightarrow P(a'')$ in $\underline{C}$ such that $fP(\alpha) = P(\beta)$. Moreover, γ is unique.

Let $\underline{A}$ be a cofibered category over $\underline{C}$. For each object S of $\underline{C}$, denote by $\underline{A}(S)$ the subcategory of $\underline{A}$ whose objects are the objects a for which $P(a) = S$, and in which

$$\mathrm{Hom}_{\underline{A}(S)}(a, a') = \{\alpha \in \mathrm{Hom}_{\underline{A}}(a, a') \mid P(\alpha) = \mathrm{id}_S\}.$$

A cofibered groupoid over $\underline{C}$ is a cofibered category $\underline{A}$ over $\underline{C}$ such that $\underline{A}(S)$

is a groupoid for every object S of $\underline{C}$. Recall that a groupoid is a category in which every morphism is an isomorphism. Thus a cofibered groupoid over $\underline{C}$ is a cofibered category $P : \underline{A} \longrightarrow \underline{C}$ in which every map of $\underline{A}$ is cocartesian. Furthermore, for every $\alpha \in \mathrm{Fl}(\underline{A})$, α is an isomorphism in $\underline{A}$ if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, fix a complete noetherian local ring Λ with residue field k, and let $\underline{C}_\Lambda$ be the category of artinian local Λ-algebras with residue field k, whose maps are local homomorphisms over Λ. From 1.2 onward, a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is always assumed to satisfy $\underline{A}(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of exactly one element for every two objects a, b of $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will simply be called a $\underline{C}_\Lambda$-map if $P(a') = P(a)$ and $P(\alpha) = 1_{P(a)}$. By abuse of language, a map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called surjective if $P(\alpha) : P(a') \longrightarrow P(a)$ is surjective in $\underline{C}_\Lambda$.

Ex. 1.1(a). Let $F : \underline{C} \longrightarrow$ (categories) be a functor. Define the category $\underline{F}$ as follows. An object of $\underline{F}$ is a pair (S, a) with $S \in \mathrm{ob}(\underline{C})$ and $a \in \mathrm{ob}F(S)$; a map $(S, a) \longrightarrow (S', a')$ is a pair (f, u'), where $f : S \longrightarrow S'$ is a map in $\underline{C}$ and $u' : F(f)a \longrightarrow a'$ is a map in $F(S')$. The composite of

$$(S, a) \xrightarrow{(f,u')} (S', a') \xrightarrow{(g,u'')} (S'', a'')$$

is

$$(gf,\ u'' \cdot F(g)u') : (S, a) \longrightarrow (S'', a'').$$

With respect to the functor $P : \underline{F} \longrightarrow \underline{C}$ given by $(S, a) \longmapsto S$, $\underline{F}$ becomes a cofibered category over $\underline{C}$. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(\underline{C})$, then $\underline{F}$ is a cofibered groupoid over $\underline{C}$. In particular, a functor $F : \underline{C} \longrightarrow$ (groups) yields a cofibered group, and a functor $F : \underline{C} \longrightarrow$ (Ens) yields a cofibered discrete groupoid. A natural transformation $\varphi : F \longrightarrow G$, where $F, G : \underline{C} \longrightarrow$ (categories), induces a cocartesian functor $\varphi : \underline{F} \longrightarrow \underline{G}$ over $\underline{C}$.

Ex. 1.1(b). Let X and Y be schemes over Λ together with a fixed

isomorphism

$$X \otimes_\Lambda k \xleftrightarrow{\tau} Y \otimes_\Lambda k.$$

Then the functor $\mathrm{Isom}_{X,Y} : \underline{C}_\Lambda \longrightarrow$ (groupoids) given by

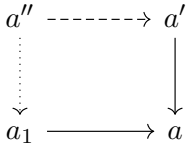
$$\mathrm{Isom}_{X,Y}(S) = \left\{ X \otimes_\Lambda S \xleftrightarrow{\sigma} Y \otimes_\Lambda S \;\middle|\; \sigma \otimes_\Lambda k = \tau \right\}$$

yields a cofibered groupoid $P : \underline{\mathrm{Isom}}_{X,Y} \longrightarrow \underline{C}_\Lambda$.

Ex. 1.1(c). Let X_0 be a fixed scheme over k, and let $\underline{M}_{X_0}$ be the category of deformations of X_0 over $\underline{C}_\Lambda$ (see §4). Thus an object of $\underline{M}_{X_0}$ is a pair (R, X), where $R \in \underline{C}_\Lambda$ and X is a deformation of X_0 to R. Then the functor $P : \underline{M}_{X_0} \longrightarrow \underline{C}_\Lambda$ given by $(R, X) \longmapsto R$ defines a cofibered groupoid over $\underline{C}_\Lambda$.

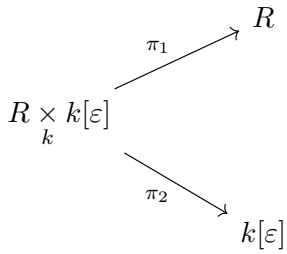
Ex. 1.1(d). For any groupoid X, denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects of X. If $P : \underline{F} \longrightarrow \underline{C}_\Lambda$ is a cofibered groupoid, we obtain a functor $[\underline{F}] : \underline{C}_\Lambda \longrightarrow$ (Ens) given by $[\underline{F}](S) = [\underline{F}(S)]$, and hence a cofibered discrete groupoid which, by abuse of notation, is again denoted by $[\underline{F}]$. Definition 1.2. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called semi-homogeneous if the following properties hold:

(1) Every diagram

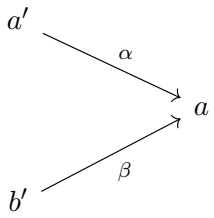


of solid arrows in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(2) Let



be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k, and let



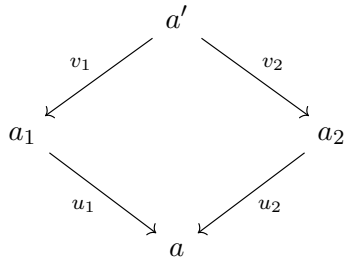
be π_1-maps in $\underline{A}$. Then there is a $\underline{C}_\Lambda$-map $\gamma : a' \longrightarrow b'$ such that $\alpha = \beta\gamma$ if and only if $(\pi_2)_*(a') \simeq (\pi_2)_*(b')$.

Remark 1.3(a). A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. In particular, 1.2(2) implies that the canonical map

$$[\underline{A}(R \times_k k[\varepsilon])] \longrightarrow [\underline{A}(R)] \times [\underline{A}(k[\varepsilon])]$$

is bijective.

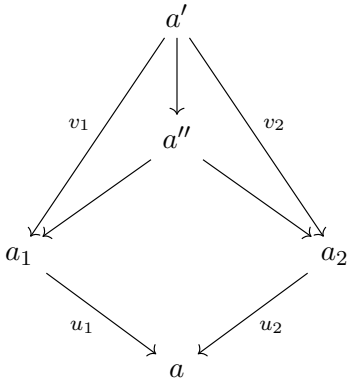
Remark 1.3(b). Consider a commutative diagram



in a cofibered category $\underline{A}$ over $\underline{C}_\Lambda$. Put $R = P(a)$, $R' = P(a')$, and $R_i = P(a_i)$ for $i = 1, 2$. If a'' is the direct image under the map

$$(P(v_1), P(v_2)) : R' \longrightarrow R_1 \times_R R_2,$$

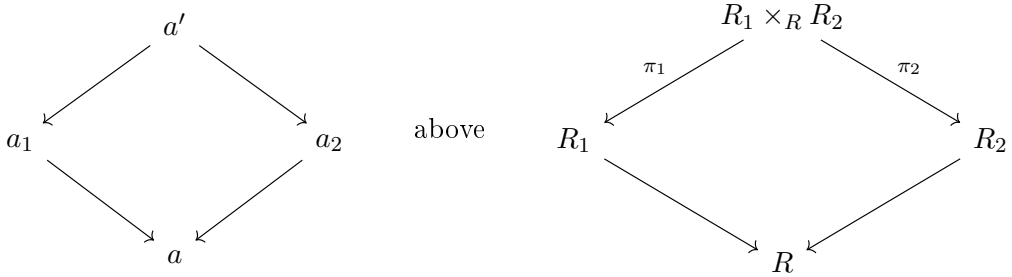
we obtain a commutative diagram



Thus 1.2(1) may be replaced by:

(1)′ Every diagram

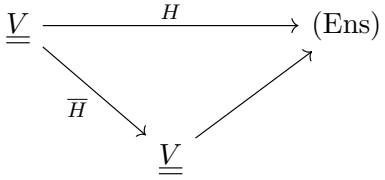
where $R_2 \longrightarrow R$ is surjective, can be completed to a commutative diagram



where π_i $(i = 1, 2)$ are the projections.

Remark 1.3(c). If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then $[\underline{A}(k[\varepsilon])]$ is canonically endowed with a vector space structure over k, where $k[\varepsilon]$ is the ring of dual numbers over k. Indeed, let $\underline{\underline{V}}$ be the category of

finite-dimensional vector spaces over k. Every product-preserving functor $H : \underline{\underline{V}} \longrightarrow$ (Ens) factors canonically as



where $\underline{\underline{V}} \longrightarrow (\mathrm{Ens})$ is the forgetful functor. On the other hand, the functor

$$G : \underline{\underline{V}} \longrightarrow \underline{C}_\Lambda, \qquad W \longmapsto k[W] = k \oplus W, \qquad W^2 = 0,$$

takes products in $\underline{\underline{V}}$ to fibre products over k in $\underline{C}_\Lambda$. If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then

$$[\underline{A}(k[V] \times_k k[W])] \longrightarrow [\underline{A}(k[V])] \times [\underline{A}(k[W])]$$

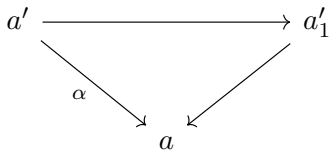
is bijective by 1.3(a). Hence the functor $V \longmapsto [\underline{A}(k[V])]$ commutes with products, proving the assertion.

Definition 1.4. (1) Let R be an object of $\underline{C}_\Lambda$. A small extension of R in $\underline{C}_\Lambda$ is a surjective map $R' \xrightarrow{f} R$ in $\underline{C}_\Lambda$ such that $(\ker f)^2 = 0$ and $\ker f \simeq k$. In other words, $R' \xrightarrow{f} R$ is a small extension of R if

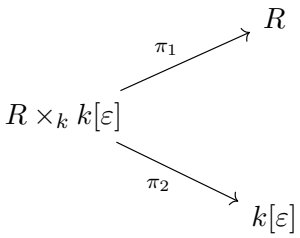
$$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$.

(2) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let a be an object of $\underline{A}$. A small prolongation of a is an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ such that $P(\alpha)$ is a small extension in $\underline{C}_\Lambda$. An arrow $a' \xrightarrow{\alpha} a$ will be called versal for small prolongations of a if, for every small prolongation $a'_1 \longrightarrow a$, there exists a commutative diagram

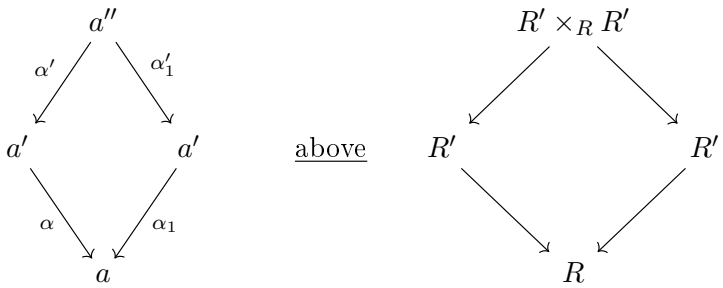


Proposition 1.5. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$.
(1) Let



be the projection maps in $\underline{C}_\Lambda$, and let $a' \xrightarrow{\alpha} a$ be a π_1-map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \longrightarrow (\pi_2)_* a'$.

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$, and consider a commutative diagram



Define the canonical map $\widetilde{f} : R' \times_R R' \longrightarrow k[\ker f]$ by

$$\widetilde{f}(r'_1, r'_2) = r'_1(0) + (r'_1 - r'_2).$$

If there exists a map $a' \longrightarrow (\widetilde{f})_*(a'')$, then there exists a map $\gamma : a' \longrightarrow a'$ such that $\alpha = \alpha_1\gamma$.

Proof. (1) The "only if" part is clear. Assume conversely that there exists a map $\varphi : a \longrightarrow (\pi_2)_* a'$. Set

$$h : R \longrightarrow R \times_k k[\varepsilon], \qquad h = 1 \times P(\varphi).$$

The composite $\pi_1 h$ is the identity. We therefore obtain maps

$$a \underset{\alpha_1}{\overset{\gamma}{\rightleftarrows}} h_* a, \qquad \alpha_1 \gamma = 1_a, \qquad P(\gamma) = h, \qquad P(\alpha_1) = \pi_1.$$

On the other hand, $\pi_2 h = P(\varphi)$, so there exists a π_2-map

$$h_* a \longrightarrow (\pi_2)_* a'.$$

Thus $(\pi_2)_*(h_* a) = (\pi_2)_* a'$, and 1.2(2) gives a $\underline{C}_\Lambda$-map $u : h_* a \longrightarrow a'$ such that $\alpha u = \alpha_1$. Define $\beta = u\gamma$. Then

$$\alpha\beta = \alpha u \gamma = \alpha_1 \gamma = 1_a.$$

(2) The map

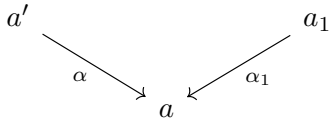
$$1 \times \widetilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\ker f]$$

is an isomorphism compatible with the projections onto the first factor, and $\pi_2(1 \times \widetilde{f}) = \widetilde{f}$, where $\pi_2 : R' \times_k k[\ker f] \longrightarrow k[\ker f]$ is the second projection. Since $R' \overset{f}{\to} R$ is a small extension, $k[\ker f] \simeq k[\varepsilon]$. Hence, if there exists a map

$$a' \longrightarrow (\widetilde{f})_* a'' = (\pi_2)_*\big((1 \times \widetilde{f})_* a''\big),$$

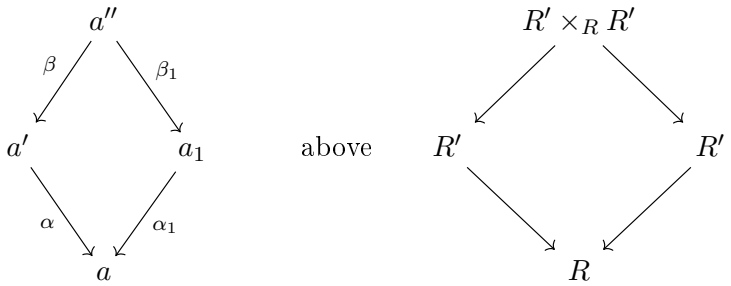
part (1) yields a map $\gamma' : a' \longrightarrow a''$ such that $\alpha'\gamma' = 1_{a'}$. Define $\gamma = \alpha_1' \gamma' : a' \longrightarrow a'$. Then $\alpha_1 \gamma = \alpha$.

Corollary 1.6. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram



in $\underline{A}$, where α_1 is a small prolongation. Assume that a is versal for small prolongations of e. Then there exists a map $\rho : a' \longrightarrow a_1$ such that $\alpha = \alpha_1 \rho$ if and only if there exists a map $f : P(a') \longrightarrow P(a_1)$ such that $P(\alpha) = P(\alpha_1) f$ in $\underline{C}_\Lambda$.

Proof. Assume that $P(\alpha) = P(\alpha_1) f$, where $f : P(a') \longrightarrow P(a_1)$. Replacing a' by its direct image under f, we may and shall assume that $P(\alpha) = P(\alpha_1)$. Put $R' = P(a_1)$ and $R = P(a)$. Since $\underline{A}$ is semi-homogeneous, we obtain a commutative diagram



Since a is versal for small prolongations of e, so is a'; the assertion therefore follows from 1.5(2).

Definition 1.7. For each object R of $\underline{C}_\Lambda$, put

$$\overline{\Omega}_R = M/(m_\Lambda R + M^2),$$

where m_Λ and M are the maximal ideals of Λ and R, respectively. This is a finite-dimensional vector space over k, and $R \longmapsto \overline{\Omega}_R$ defines a functor

$$\overline{\Omega} : \underline{C}_\Lambda \longrightarrow \underline{\underline{V}},$$

where $\underline{\underline{V}}$ is the category of finite-dimensional vector spaces over k. If $\underline{A}$ is a cofibered category over $\underline{C}_\Lambda$, the composite functor $\underline{A} \xrightarrow{P} \underline{C}_\Lambda \longrightarrow \underline{\underline{V}}$ is again denoted $\overline{\Omega}$, by abuse of notation.

Proposition 1.8. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$. Given an object a of $\underline{A}$ that is versal for small prolongations of e, there exists an arrow $a' \longrightarrow a$ such that:

(i) a' is versal for small prolongations of a;

(ii) $P(a') \longrightarrow P(a)$ is surjective, and its kernel is annihilated by the maximal ideal of $P(a)$;

(iii) $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

Proof. Put $R = P(a)$, and consider an exact sequence

$$0 \longrightarrow J \longrightarrow S \longrightarrow R \longrightarrow 0,$$

where S is a formal power-series ring over Λ in finitely many variables and $J \subset M^2$, with M the maximal ideal of S. Let Σ be the set of ideals J_1 of S such that

(i) $J \supset J_1 \supset MJ$, and

(ii) there exists a π_1-morphism $a_1 \longrightarrow a$ in $\underline{A}$, where $\pi_1 : S/J_1 \longrightarrow R$ is the canonical map.

Property 1.2(1) of $\underline{A}$, together with the fact that J/MJ is a finite-dimensional vector space over k, implies that Σ has a smallest ideal. Denote it by J', and choose a π'-morphism $a' \xrightarrow{g} a$, where $\pi' : S/J' \longrightarrow R$ is the canonical map. Since $J' \subset J \subset M^2$, the map $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

We claim that $a' \xrightarrow{g} a$ is versal for small prolongations of a. Indeed, let $a_1 \longrightarrow a$ be a small prolongation, and put $R_1 = P(a_1)$. We obtain an exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & J/MJ & \longrightarrow & S/MJ & \longrightarrow & R & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle h_1} & & \| & & \\ 0 & \longrightarrow & k & \longrightarrow & R_1 & \longrightarrow & R & \longrightarrow & 0. \end{array}$$

If $h = 0$, then $\pi_1 : R_1 \longrightarrow R$ admits a cross-section, so h_1 induces a map $f : S/J' \longrightarrow R_1$. If $h \neq 0$, then $h_1 : S/MJ \longrightarrow R_1$ is surjective and hence, by the definition of J', again induces $f : S/J' \longrightarrow R_1$. In either case we obtain a commutative diagram

$$\begin{array}{ccc} S/J' & \xrightarrow{\quad f \quad} & R_1 \\ & {\scriptstyle \pi'} \searrow \quad \swarrow {\scriptstyle \pi_1} & \\ & R & \end{array}$$

Since a is versal for small prolongations of e, Corollary 1.6 gives a commutative diagram

$$\begin{array}{ccc} a' & \xrightarrow{\quad \rho \quad} & a_1 \\ & \searrow \quad \swarrow & \\ & a & \end{array}$$

as required.

Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category in groupoids. It then gives a cofibered category in groupoids

$$\mathrm{pro}(P) : \mathrm{pro}(\underline{A}) \longrightarrow \mathrm{pro}(\underline{C}_\Lambda)$$

(for the definition of the category of pro-objects $\mathrm{pro}(\underline{A})$, see A. Grothendieck, Séminaire Bourbaki 195), fitting into the commutative diagram

$$\begin{array}{ccc} \underline{A} & \longrightarrow & \mathrm{pro}(\underline{A}) \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle \mathrm{pro}(P)} \\ \underline{C}_\Lambda & \longrightarrow & \mathrm{pro}(\underline{C}_\Lambda). \end{array}$$

Let $\widehat{\underline{C}}_\Lambda$ be the full subcategory of $\mathrm{pro}(\underline{C}_\Lambda)$ whose objects are systems $(R_i)_{i\in\mathbb{N}}$ with $R_i \in \underline{C}_\Lambda$ such that each map $\varphi_i : R_i \longrightarrow R_{i-1}$ is surjective and, for all sufficiently large i, induces an isomorphism

$$\frac{m_i}{m_\Lambda R_i + m_i^2} \xrightarrow{\sim} \frac{m_{i-1}}{m_\Lambda R_{i-1} + m_{i-1}^2},$$

where $\mathbb{N}$ denotes the ordered category of natural numbers and m_i is the maximal ideal of R_i.

Let $\widehat{\underline{A}}$ be the full subcategory of $\mathrm{pro}(\underline{A})$ defined by

$$\mathrm{ob}(\widehat{\underline{A}}) = \left\{ a \in \mathrm{ob}(\mathrm{pro}(\underline{A})) \;\middle|\; \mathrm{pro}(P)(a) \in \mathrm{ob}(\widehat{\underline{C}}_\Lambda) \right\}.$$

We obtain a cofibered category in groupoids $\widehat{P} : \widehat{\underline{A}} \longrightarrow \widehat{\underline{C}}_\Lambda$ and a commutative diagram

$$\begin{array}{ccc} \underline{A} & \hookrightarrow & \widehat{\underline{A}} \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle \widehat{P}} \\ \underline{C}_\Lambda & \hookrightarrow & \widehat{\underline{C}}_\Lambda. \end{array}$$

The category $\widehat{\underline{C}}_\Lambda$ is canonically equivalent to the category of complete local Λ-algebras of formally finite type over Λ with fixed residue field k; we henceforth make this identification.

Definition 1.9. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. An object ξ of $\widehat{\underline{A}}$ is called quasi-initial if, for every object a of $\widehat{\underline{A}}$, there exists a map $\xi \to a$ in $\widehat{\underline{A}}$. A quasi-initial object ξ is called minimal if the morphism $\xi : h_R \longrightarrow [\widehat{\underline{A}}]$ induces a bijection

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\sim} [\underline{A}(k[\varepsilon])], \qquad R = P(\xi).$$

An object ξ of $\widehat{\underline{A}}$ is called a versal formal object of $\underline{A}$ if every surjective map $\alpha : a' \to a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a') \longrightarrow \mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

A versal formal object ξ is called minimal if $\xi : h_R \longrightarrow [\widehat{\underline{A}}]$ induces the above bijection on $k[\varepsilon]$, where $R = P(\xi)$.

Proposition 1.10. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$.

(i) Every (minimally) versal formal object of $\underline{A}$ is a quasi-initial (minimal) object of $\widehat{\underline{A}}$.

(ii) Suppose that ξ is an object of $\widehat{\underline{A}}$ for which

$$\widetilde{\xi}(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow [\underline{A}(k[\varepsilon])]$$

is injective, where $P(\xi) = R$. Then every endomorphism of ξ in $\widehat{\underline{A}}$ is an automorphism. In particular, a quasi-initial minimal object of $\widehat{\underline{A}}$, if it exists, is unique up to noncanonical isomorphism.

Proof. (i) Let ξ be a versal formal object of $\underline{A}$. For an object a of $\widehat{\underline{A}}$, choose a presentation by surjective maps in $\underline{A}$,

$$\cdots \longrightarrow a_n \xrightarrow{\alpha_n} a_{n-1} \longrightarrow \cdots \longrightarrow a_1 \xrightarrow{\alpha_1} a_0, \qquad a = \varprojlim_n a_n.$$

Suppose maps $\varphi_i : \xi \longrightarrow a_i$ have been chosen with $\alpha_i\varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is versal and α_n is surjective, there is a map $\varphi_n : \xi \longrightarrow a_n$ with $\alpha_n\varphi_n = \varphi_{n-1}$. Thus $\varphi = \varprojlim_n \varphi_n$ defines a map $\xi \longrightarrow a$.

(ii) Let $\alpha : \xi \longrightarrow \xi$ be an endomorphism in $\widehat{\underline{A}}$. For every $\lambda \in \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon])$, we have

$$\lambda_*\xi \simeq (\lambda \circ P(\alpha))_*\xi.$$

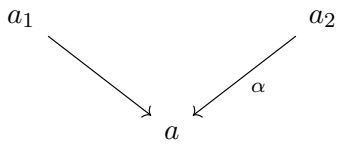
Injectivity therefore gives $\lambda = \lambda \circ P(\alpha)$ for every λ. It follows that $P(\alpha) : R \longrightarrow R$ is surjective. Every surjective endomorphism of a noetherian ring is an automorphism, so $P(\alpha)$, and hence α, is an automorphism.

Theorem 1.11 (Schlessinger). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then $\underline{A}$ admits a minimally versal formal object if and only if:

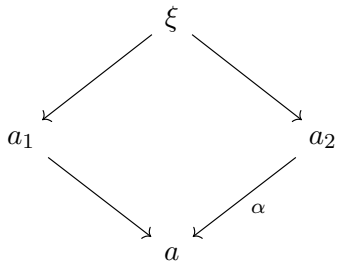
(i) $\underline{A}$ is semi-homogeneous;

(ii) $[\underline{A}(k[\varepsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

Proof. Necessity. Let $\xi \in \widehat{\underline{A}}$ be a minimally versal formal object of $\underline{A}$, and consider a diagram

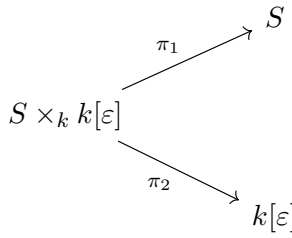


in $\underline{A}$, where α is surjective. Since ξ is quasi-initial, there is a map $\xi \longrightarrow a_1$. Since α is surjective, the composite $\xi \longrightarrow a_1 \longrightarrow a$ lifts to a_2; hence we obtain a commutative diagram

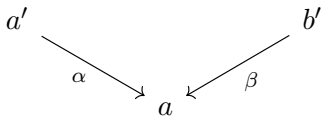


which proves condition 1.2(1).

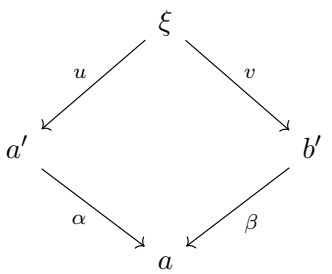
We now prove condition 1.2(2). Let



be the projections, and let



be π_1-maps in $\underline{A}$. By the definition of ξ, there is a commutative diagram



and therefore $\pi_1 P(u) = \pi_1 P(v)$. Suppose now that $(\pi_2)_* a' \simeq (\pi_2)_* b'$. Minimality of ξ gives $\pi_2 P(u) = \pi_2 P(v)$, whence

$$P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v).$$

Since $\underline{A}$ is a cofibered groupoid, there is a unique isomorphism $\sigma : a' \xrightarrow{\sim} b'$ in $\underline{A}(S \times_k k[\varepsilon])$ such that $\sigma u = v$. The equality $\beta\sigma u = \alpha u$ then implies $\beta\sigma = \alpha$.

Sufficiency. Let $c_1, \ldots, c_n$ be objects of $\underline{A}(k[\varepsilon])$ whose classes form a k-basis of $[\underline{A}(k[\varepsilon])]$. Choose

$$a_1 \in \underline{A}(k[\varepsilon] \times_k \cdots \times_k k[\varepsilon])$$

such that $(\pi_j)_* a_1 \simeq c_j$. Then

$$\widetilde{a}_1(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \longrightarrow [\underline{A}(k[\varepsilon])]$$

is bijective, where $R_1 = P(a_1)$. In particular, $a_1 \longrightarrow e$ is versal for small prolongations of e. Choose $a_2 \longrightarrow a_1$ versal for small prolongations of a_1 such that $\overline{\Omega}_{a_2} \longrightarrow \overline{\Omega}_{a_1}$ is bijective. Continue inductively, choosing $a_{n+1} \longrightarrow a_n$ versal for small prolongations and inducing a bijection on $\overline{\Omega}$, and set

$$\xi = \varprojlim_n a_n.$$

Then $\overline{\Omega}_\xi \longrightarrow \overline{\Omega}_{a_1}$ is bijective, so

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\sim} [\underline{A}(k[\varepsilon])]$$

is bijective.

It remains to prove versality. Consider a diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow{\scriptstyle \alpha} \\ \xi & \xrightarrow{h} & a \end{array}$$

where α is surjective. We must lift h to a'. By induction we may assume that a' is a small prolongation of a. Since $P(a)$ is artinian, h factors through some a_q,

$$\xi \longrightarrow a_q \longrightarrow a.$$

Semi-homogeneity gives a commutative diagram

$$\begin{array}{ccccc} & & a'' & \longrightarrow & a' \\ & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha} \\ \xi & \longrightarrow & a_q & \longrightarrow & a, \end{array}$$

where α' is also a small prolongation. Since $a_{q+1} \longrightarrow a_q$ is versal for small prolongations of a_q, there is a commutative diagram

$$\begin{array}{ccccc} & & & & a'' \\ & & & \nearrow & \downarrow{\scriptstyle \alpha'} \\ \xi & \longrightarrow & a_{q+1} & \longrightarrow & a_q, \end{array}$$

which supplies the required lift.

Remark 1.12. Let

$$\cdots \longrightarrow a_n \longrightarrow a_{n-1} \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow a_0 = e$$

be an infinite sequence of arrows in $\underline{A}$ such that

$$\begin{cases} a_i \text{ is versal for small prolongations of } a_{i-1}, \\ P(a_i) \longrightarrow P(a_{i-1}) \text{ is surjective,} \\ \overline{\Omega}_{a_i} \longrightarrow \overline{\Omega}_{a_{i-1}} \text{ is bijective for } i \gg 0. \end{cases}$$

The preceding proof shows that $\varprojlim_i a_i$ is a versal formal object of $\underline{A}$.

Remark 1.13. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. If $\underline{A}$ satisfies 1.11(i) and (ii), then so does $[\underline{A}]$; moreover, $\xi \in \mathrm{ob}(\widehat{\underline{A}})$ is a minimally versal formal object of $\underline{A}$ if and only if it is so for $[\underline{A}]$, by 1.10(ii).

Remark 1.14. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ. Any invariant of $R = P(\xi)$, a complete local Λ-algebra of formally finite type over Λ, is automatically an invariant of $\underline{A}$ over $\underline{C}_\Lambda$. Thus, for example, we say that $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ if R is (formally) smooth over Λ, that is, a formal power-series algebra over Λ; and we say that $\underline{A}$ is irreducible over $\underline{C}_\Lambda$ if $\mathrm{Spec}(R)$ is irreducible.

In the next section we introduce smoothness for a functor $\underline{A} \longrightarrow \underline{B}$ over $\underline{C}_\Lambda$, where $\underline{A}$ and $\underline{B}$ are cofibered categories in groupoids over $\underline{C}_\Lambda$. It follows from 2.4 that $\underline{A}$ being (formally) smooth over $\underline{C}_\Lambda$ is equivalent to the functor $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ being smooth, so no ambiguity arises. We define

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim(k \otimes_\Lambda R).$$

For this, let $\underline{C}_k$ be the category of artinian local k-algebras with residue field k. The functor $\underline{C}_k \longrightarrow \underline{C}_\Lambda$ is fully faithful, and its left adjoint is

$$\underline{C}_\Lambda \longrightarrow \underline{C}_k, \qquad R \longmapsto k \otimes_\Lambda R.$$

Let $\underline{A}|_{\underline{C}_k}$ be the pullback of $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ along $\underline{C}_k \longrightarrow \underline{C}_\Lambda$. If ξ is a minimally versal formal object for $\underline{A}$ over $\underline{C}_\Lambda$, then $k \otimes_\Lambda \xi$ is minimally versal for $\underline{A}|_{\underline{C}_k}$ over $\underline{C}_k$; the arrow $\xi \longrightarrow k \otimes_\Lambda \xi$ is cocartesian over $R \longrightarrow k \otimes_\Lambda R$. Consequently,

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim_{\underline{C}_k}(\underline{A}|_{\underline{C}_k}).$$

1.15. In the remainder of this paragraph, we briefly explain how to modify the preceding theory to allow residue-field extensions.

Let Λ and K be complete noetherian local rings, with K a Λ-algebra. Assume that:

(a) the natural map $\Lambda \longrightarrow K$ is a local homomorphism;

(b) the residue field K' of K is an extension of finite type of the residue field k of Λ.

The most interesting case is $K = K'$ with K a finite inseparable extension of k.

Denote by $\widehat{C}_{\Lambda,K}$ the category whose objects are complete noetherian local rings R equipped with a Λ-algebra structure and a Λ-epimorphism $s : R \longrightarrow K$. Let $C_{\Lambda,K}$ be its full subcategory consisting of pairs (R, s) for which $\ker(s)$ has finite length as an R-module. The category $\widehat{C}_{\Lambda,K}$ identifies with a full subcategory of $\mathrm{Pro}(C_{\Lambda,K})$. If K is artinian, then every object of $C_{\Lambda,K}$ is artinian.

For a finite-dimensional vector space V over K', write $D_V(K)$ for the algebra of dual numbers over K defined by V. Its elements have the form $x + v\varepsilon$, where $x \in K$, $v \in V$, and $\varepsilon^2 = 0$, with multiplication

$$(x + v\varepsilon)(x' + v'\varepsilon) = xx' + (xv' + x'v)\varepsilon.$$

The map

$$s : D_V(K) \longrightarrow K, \qquad s(x + v\varepsilon) = x,$$

makes $D_V(K)$ an object of $C_{\Lambda,K}$. The role previously played by $k[\varepsilon]$ is now played by $D_{K'}(K)$.

As in 1.8, every cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ extends naturally to a cofibered category $\widehat{\underline{A}}$ over $\widehat{C}_{\Lambda,K}$. A map in $\underline{A}$ or $\widehat{\underline{A}}$ is called surjective when the corresponding algebra homomorphism in $C_{\Lambda,K}$ or $\widehat{C}_{\Lambda,K}$ is surjective.

We assume that $\underline{A}(K) = \{e\}$.

Definition 1.16. A cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ is called semi-homogeneous if the following properties hold:

(I) Every diagram of solid arrows

$$\begin{array}{ccc} a'' & \dashrightarrow & a' \\ \vdots & & \downarrow \\ a_1 & \longrightarrow & a \end{array}$$

in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(II) For every R, the natural map

$$[\underline{A}(R \times_K D_{K'}(K))] \longrightarrow [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

is bijective.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$.

1.17. First restrict $\underline{A}$ to the full subcategory of $C_{\Lambda,K}$ whose objects are the $D_V(K)$. Put

$$\Omega = \Omega^1_{K/\Lambda} \otimes_K K', \qquad d : K \longrightarrow \Omega.$$

A map $f : D_V(K) \longrightarrow D_W(K)$ can be written

$$f(x + v\varepsilon) = x + \big(u(v) + D(dx)\big)\varepsilon,$$

with $u \in \mathrm{Hom}_{K'}(V, W)$ and $D \in \mathrm{Hom}_{K'}(\Omega, W)$. Identifying f with (u, D), composition is given by

$$(u, D) \circ (u', D') = (uu', D + uD').$$

Condition (II) implies:

(II′) the functor from K'-vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

Lemma 1.18. Assume that condition (II′) is satisfied by $\underline{A}$. Then:

(1) $[\underline{A}(D_{K'}(K))]$ carries a unique vector-space structure such that, functorially in V,

$$[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))].$$

(2) There is a linear map

$$\gamma : \mathrm{Hom}(\Omega, K') \longrightarrow [\underline{A}(D_{K'}(K))]$$

such that, for every $D \in \mathrm{Hom}(\Omega, K')$, the map

$$[\underline{A}((1, D))] : [\underline{A}(D_{K'}(K))] \longrightarrow [\underline{A}(D_{K'}(K))]$$

is $x \longmapsto x + \gamma(D)$.

(3) For every $(u, D) : D_V(K) \longrightarrow D_W(K)$, the induced map

$$(u, D) : [\underline{A}(D_V(K))] \longrightarrow [\underline{A}(D_W(K))],$$

that is,

$$(u, D) : V \otimes [\underline{A}(D_{K'}(K))] \longrightarrow W \otimes [\underline{A}(D_{K'}(K))],$$

is given by

$$x \longmapsto u(x) + \gamma(D).$$

Introduce the notation

$$\begin{aligned} F(V) &= [\underline{A}(D_V(K))], \\ A &= F(K'), \\ F(u, D) : F(V) &\longrightarrow F(W) \qquad \text{for the map induced by } (u, D) : D_V(K) \longrightarrow D_W(K), \\ F(D) : F(V) &\longrightarrow F(V), \qquad F(D) = F(1, D). \end{aligned}$$

By (II′), and because K' is a K'-vector-space object in the category of K'-vector spaces, $F(K')$ is a K'-vector-space object in the category of sets, hence a K'-vector space. This proves (1). We henceforth identify $F(V)$ with $V \otimes A$.

The identity

$$(u, D) = (1, D) \circ (u, 0)$$

gives

$$F(u, D) = F(D) \circ (u \otimes 1_A).$$

The functoriality of F amounts to the identities

$$\begin{aligned} F(0) &= \mathrm{Id}, \\ F(D) \circ F(D') &= F(D + D'), \\ (u \otimes 1) \circ F(D) &= F(uD) \circ (u \otimes 1_A). \end{aligned}$$

For any $D : \Omega \longrightarrow V$ and any $n \in \mathbb{N}$, let

$$i : V \hookrightarrow V \oplus K'^{,n}, \qquad p : V \oplus K'^{,n} \longrightarrow K'^{,n}$$

be the canonical inclusion and projection. Then $piD = 0$, and the following diagram is commutative:

$$\begin{array}{ccccc} V \otimes A & \longrightarrow & (V \oplus K'^{,n}) \otimes A & \longrightarrow & K'^{,n} \otimes A \\ \downarrow{\scriptstyle F(D)} & & \downarrow{\scriptstyle F(i,D)} & & \| \\ V \otimes A & \longrightarrow & (V \oplus K'^{,n}) \otimes A & \longrightarrow & K'^{,n} \otimes A. \end{array}$$

Let $x : K'^{,n} \longrightarrow V$ be a map, put $x_i = x(e_i)$, and let (a_i) be a family of elements of A. We have

$$(1_V + x) \circ i \circ D = D.$$

Consequently, if

$$F(i, D)\Big(\sum_i e_i a_i\Big) = \sum_i e_i a_i + a, \qquad a \in V \otimes A,$$

then

$$F(D)\Big(\sum_i x_i a_i\Big) = \sum_i x_i a_i + a,$$

and a is independent of the x_i. Hence

$$\begin{aligned} F(D)(y) &= y + G(D), \\ G(D + D') &= G(D) + G(D'), \\ uG(D) &= G(uD) \qquad \text{for every } u : V \longrightarrow W. \end{aligned}$$

Every such G is given, for some $\gamma \in \Omega \otimes A$, by

$$G(D) = (D \otimes 1_A)(\gamma).$$

This proves (2) and (3).

Definition 1.19. (1) An object $\xi \in \widehat{\underline{A}}$ is called a versal formal object if every surjective map $\alpha : a' \longrightarrow a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a') \longrightarrow \mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

(2) A versal formal object $\xi \in \widehat{\underline{A}}$, with image (R, s) in $\widehat{C}_{\Lambda,K}$, is called minimal if the following condition holds. For any two maps

$$\varphi_1, \varphi_2 : R \rightrightarrows D_{K'}(K),$$

if $\varphi_1(\xi)$ is isomorphic to $\varphi_2(\xi)$, then there is a derivation $D : K \longrightarrow K'$ such that

$$(1, D)(\varphi_1) = \varphi_2, \qquad \text{that is,} \qquad \varphi_2 - \varphi_1 = Ds.$$

Theorem 1.20. If $\underline{A}$ is semi-homogeneous and

$$\dim_{K'}[\underline{A}(D_{K'}(K))] < \infty,$$

then $\underline{A}$ admits a minimally versal formal object, and any two such objects are isomorphic.

With the notation of 1.18, let H^* be a subspace of $[\underline{A}(D_{K'}(K))]$ complementary to the image of γ, and let H be the dual of H^*. Set

$$R_0 = D_H(K).$$

Choose $\xi_0 \in \underline{A}(R_0)$ whose class, under the identifications

$$[\underline{A}(R_0)] \simeq H \otimes [\underline{A}(D_{K'}(K))] \simeq \mathrm{Hom}\big(H^*, [\underline{A}(D_{K'}(K))]\big),$$

is the inclusion $H^* \hookrightarrow [\underline{A}(D_{K'}(K))]$. Then:

(1) for every V, the maps

$$\mathrm{Hom}_{C_{\Lambda,K}}\big(D_H(K), D_V(K)\big) \longrightarrow [\underline{A}(D_V(K))]$$

are surjective;

(2) the minimality condition of 1.19(2) holds for (R_0, ξ_0).

Choose $S \in \widehat{C}_{\Lambda,K}$ formally smooth over Λ, together with an epimorphism $t : S \longrightarrow R_0$, such that

$$\Omega_{R_0/\Lambda} \otimes K' \xrightarrow{\sim} \Omega_{S/\Lambda} \otimes K'.$$

If M is the maximal ideal of S, define inductively, for $n \geq 1$,

$$R_n = S/I_n, \qquad \xi_n \in \mathrm{Ob}(\underline{A}(R_n)),$$

by the following conditions:

(a) I_{n+1} is the intersection of the ideals I such that

$$I_n M \subset I \subset I_n$$

and ξ_n extends to an object $\xi \in \underline{A}(S/I)$;

(b) ξ_{n+1} is an extension of ξ_n.

If

$$R = \varprojlim_n R_n, \qquad \xi = \varprojlim_n \xi_n,$$

then (R, ξ) is a minimally versal formal object, and it is unique up to isomorphism.

2. Prorepresentable cofibered groupoids

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ. In general, ξ alone does not determine $\underline{A}$ up to $\underline{C}_\Lambda$-equivalence. One must also know when two maps

$$R \underset{g}{\overset{f}{\rightrightarrows}} S$$

give rise to a $\underline{C}_\Lambda$-isomorphism $\xi(f) \longrightarrow \xi(g)$, where $R = P(\xi)$. As an application of 1.11 and 1.12, we now consider the problem of recovering $\underline{A}$, up to $\underline{C}_\Lambda$-equivalence. We obtain a satisfactory answer for a broad class of cofibered groupoids.

Many cofibered groupoids that arise in practice satisfy a property slightly stronger than semi-homogeneity; it will be defined explicitly in 2.5. To investigate them, it is convenient to recast the preceding discussion in terms of "smooth functors." All the notation and terminology of the preceding section remain in force throughout this section.

Definition 2.1. Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We say that φ is smooth if every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

(*)
$$\begin{array}{ccc} \varphi(a') & \longrightarrow & b' \\ & \searrow^{\varphi(\alpha)} & \downarrow{\scriptstyle \beta} \\ & & \varphi(a) \end{array}$$

for some $\alpha : a' \longrightarrow a$ in $\underline{A}$. We say that φ is minimally smooth if it is smooth and

$$\varphi(k[\varepsilon]) : [\underline{A}(k[\varepsilon])] \longrightarrow [\underline{B}(k[\varepsilon])]$$

is bijective.

Remark 2.2(a). Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. The following conditions are equivalent:

(i) φ is smooth;

(ii) every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a diagram $(*)$ in which $\varphi(a') \longrightarrow b'$ is a $\underline{C}_\Lambda$-isomorphism;

(iii) every small prolongation $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram $(*)$.

Remark 2.2(b). For every cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, the natural functor

$$\underline{A} \longrightarrow [\underline{A}]$$

is minimally smooth.

Remark 2.2(c). For each $R \in \widehat{\underline{C}}_\Lambda$, let

$$h_R : \underline{C}_\Lambda \longrightarrow (\mathrm{Ens})$$

be the functor defined by

$$h_R(S) = \mathrm{Hom}_{\Lambda\text{-alg}}(R, S) \qquad (S \in \underline{C}_\Lambda).$$

It determines a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$. A map $R \longrightarrow S$ in $\widehat{\underline{C}}_\Lambda$ induces a functor

$$\underline{h}_S \longrightarrow \underline{h}_R.$$

This functor is smooth if and only if S is a formal power-series ring over R.

Remark 2.2(d). Given $R \in \widehat{\underline{C}}_\Lambda$, denote by $(R, \underline{C}_\Lambda)$ the category of arrows in $\widehat{\underline{C}}_\Lambda$ with source R and target in $\underline{C}_\Lambda$. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Given $\xi \in \mathrm{Ob}\,\widehat{\underline{A}}$ with $P(\xi) = R$, let

$$\underline{A}' \longrightarrow (R, \underline{C}_\Lambda)$$

be the pullback of $\underline{A} \longrightarrow \underline{C}_\Lambda$ along the canonical functor $(R, \underline{C}_\Lambda) \longrightarrow \underline{C}_\Lambda$. Choose a section $\xi^\#$ such that $\xi^\#(1_R) = \xi$. Thus, for every map $f : R \longrightarrow S$, we choose an f-map $\xi \longrightarrow \xi(f)$.

We obtain a functor

$$\xi^\# : \underline{h}_R \longrightarrow \underline{A}$$

over $\underline{C}_\Lambda$ by sending $(R \xrightarrow{f} S)$ to $\xi(f)$. Up to natural equivalence, this functor does not depend on the choice of $\xi^\#$.

The assertion that $\xi^\# : \underline{h}_R \longrightarrow \underline{A}$ is (minimally) smooth means that every surjective map $a' \xrightarrow{\alpha} \xi(f)$ in $\underline{A}$ can be completed to a commutative diagram

$$\begin{array}{ccc} \xi(f') & \longrightarrow & a' \\ & {\scriptstyle \xi(\,)} \searrow & \downarrow {\scriptstyle \alpha} \\ & & \xi(f) \end{array}$$

(and, in the minimal case, that $\underline{h}_R(k[\varepsilon]) \longrightarrow \underline{A}(k[\varepsilon])$ is bijective). This means precisely that ξ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\Lambda$.

Thus the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor

$$\underline{h}_R \longrightarrow \underline{A}$$

over $\underline{C}_\Lambda$ for some $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$.

The following general properties of smoothness are immediate.

Proposition 2.3. Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$.

(a) If φ and ψ are both smooth, then so is $\psi \circ \varphi$.

(b) If $\psi \circ \varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth.

Remark 2.4. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a versal formal object ξ. By 2.2(d), this means that

$$\xi : \underline{h}_R \longrightarrow \underline{A}$$

is smooth, where $R = P(\xi)$. Therefore the following assertions are equivalent:

— $\underline{A}$ is formally smooth over $\underline{C}_\Lambda$ (cf. 1.13), that is, R is a formal power-series algebra over Λ;

— the composite functor

$$\underline{h}_R \xrightarrow{\xi} \underline{A} \xrightarrow{P} \underline{C}_\Lambda \ (= \underline{h}_\Lambda)$$

is smooth;

— $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ is smooth.

The last equivalence follows from Proposition 2.3(b).

We now consider a stronger property than semi-homogeneity.

Definition 2.5. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called homogeneous if every diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow \\ b' & \longrightarrow & a \end{array}$$

in which $a' \longrightarrow a$ is surjective admits a fibre product $a' \times_a b'$ in $\underline{A}$ such that

$$P(a' \times_a b') = P(a') \times_{P(a)} P(b').$$

Remark 2.6(a). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$(*) \qquad \begin{array}{ccccc} & & a' & & \\ & \swarrow & & \searrow & \\ a_1 & & & & a_2 \\ & \searrow & & \swarrow & \\ & & a & & \end{array} \qquad \text{in } \underline{A} \text{ above the diagram} \qquad \begin{array}{ccccc} & & R_1 \times_R R_2 & & \\ & \overset{\pi_1}{\swarrow} & & \overset{\pi_2}{\searrow} & \\ R_1 & & & & R_2 \\ & \searrow & & \swarrow & \\ & & R & & \end{array}$$

in $\underline{C}_\Lambda$, where the π_i are the projections. If a fibre product $a_1 \times_a a_2$ exists in $\underline{A}$, there is a unique map

$$\varphi : a' \longrightarrow a_1 \times_a a_2$$

compatible with the projections. If

$$P(a_1 \times_a a_2) = R_1 \times_R R_2,$$

then $P(\varphi)$ is the identity map of $R_1 \times_R R_2$; hence φ is a $\underline{C}_\Lambda$-isomorphism, that is, $a' = a_1 \times_a a_2$. Thus, if $\underline{A}$ is homogeneous, every diagram $(*)$ defines a fibre product $a_1 \times_a a_2$ whenever $R_1 \longrightarrow R$ is surjective. It follows at once

that every homogeneous cofibered groupoid is, a fortiori, semi-homogeneous.

Remark 2.6(b). Consider a diagram of categories and functors

$$\begin{array}{ccccc} X & & & & Y \\ & \searrow^{\varphi} & & \swarrow_{\psi} & \\ & & Z & & \end{array}$$

The ordinary fibre product $X \times_Z Y$ is the category whose objects are pairs (x, y) satisfying $\varphi(x) = \psi(y)$ and whose maps $(x, y) \longrightarrow (x', y')$ are pairs

$$\alpha : x \longrightarrow x', \qquad \beta : y \longrightarrow y'$$

such that $\varphi(\alpha) = \psi(\beta)$. The fibre product $X \times_Z^2 Y$ in the sense of 2-categories is the category whose objects are triples $(x, y; \gamma)$, where

$$\varphi(x) \xrightarrow{\gamma} \psi(y)$$

is an isomorphism. A map $(x, y; \gamma) \longrightarrow (x', y'; \gamma')$ is a pair $\alpha : x \longrightarrow x'$ and $\beta : y \longrightarrow y'$ for which the diagram

$$\begin{array}{ccc} \varphi(x) & \xrightarrow{\varphi(\alpha)} & \varphi(x') \\ {\scriptstyle\gamma}\downarrow & & \downarrow{\scriptstyle\gamma'} \\ \psi(y) & \xrightarrow[\psi(\beta)]{} & \psi(y') \end{array}$$

commutes.

The notions of a (semi-)homogeneous cofibered groupoid and of a smooth functor can be rephrased in terms of fibre products of categories in this 2-categorical sense. Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category, and consider $R_1 \times_R R_2$ in $\underline{C}_\Lambda$. Choose direct-image functors

$$(\pi_i)_* : \underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_i), \qquad i = 1, 2.$$

They induce a functor

$$\underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_1) \times^2_{\underline{A}(R)} \underline{A}(R_2).$$

Up to canonical natural equivalence, it is independent of the chosen direct-image functors. The groupoid $\underline{A}$ is homogeneous if and only if this functor is an equivalence whenever $R_1 \longrightarrow R$ is surjective; it is semi-homogeneous if and only if

$$[\underline{A}(R_1 \times_R R_2)] \longrightarrow [\underline{A}(R_1)] \times_{[\underline{A}(R)]} [\underline{A}(R_2)]$$

is surjective whenever $R_1 \longrightarrow R$ is surjective, and is bijective whenever $R = k$ and $R_1 = k[\varepsilon]$.

Now let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. For every map $R' \longrightarrow R$ in $\underline{C}_\Lambda$, φ induces, up to canonical natural equivalence, a functor

$$\underline{A}(R') \longrightarrow \underline{A}(R) \times^2_{\underline{B}(R)} \underline{B}(R').$$

The functor φ is smooth if and only if this functor is surjective on isomorphism classes of objects whenever $R' \longrightarrow R$ is surjective.

Remark 2.6(c). If $\underline{A}$ is homogeneous, then for any finite-dimensional k-vector spaces V and W the functor

$$\underline{A}(k[V \times W]) \longrightarrow \underline{A}(k[V]) \times \underline{A}(k[W])$$

is an equivalence of categories. Recall our convention $\underline{A}(k) = \{e\}$. In particular,

$$\mathrm{Aut}_{\underline{A}(k[V\times W])}(1_{V\times W}) \xrightarrow{\sim} \mathrm{Aut}_{\underline{A}(k[V])}(1_V) \times \mathrm{Aut}_{\underline{A}(k[W])}(1_W),$$

where 1_V is the direct image of e under the unique map $k \longrightarrow k[V]$ in $\underline{C}_\Lambda$. Consequently, if $\underline{A}$ is homogeneous, both $\mathrm{Aut}(1_\varepsilon)$ and $[\underline{A}(k[\varepsilon])]$ carry canonical k-vector-space structures.

Remark 2.6(d). For every $R \in \widehat{C}_\Lambda$, the cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. Homogeneity of $\underline{A}$, however, does not in general imply homogeneity of $[\underline{A}]$.

Proposition 2.7. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$. The following conditions are equivalent.

(i) $[\underline{A}]$ is homogeneous; that is,

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is bijective for every small extension $R' \longrightarrow R$.

(ii) For every small prolongation $a' \longrightarrow a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{\underline{A}(R')}(a') \longrightarrow \mathrm{Aut}_{\underline{A}(R)}(a)$$

is surjective, where $R' = P(a')$ and $R = P(a)$.

Proof. (i)⇒(ii). Let $f : R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $a' \longrightarrow a$ be an f-map in $\underline{A}$. Given $\tau \in \mathrm{Aut}_{\underline{A}(R)}(a)$, consider $a' \times_a a'$ and $a' \times_a a'_\tau$, where $a'_\tau \longrightarrow a$ is the composite

$$a' \longrightarrow a \xrightarrow{\tau} a.$$

Since

$$[\underline{A}(R' \times_R R')] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(R')]$$

is bijective, there is an isomorphism

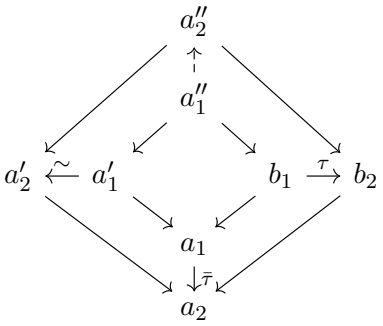
$$a' \times_a a' \xrightarrow{u} a' \times_a a'_\tau.$$

Projection onto the second factor then gives an isomorphism $\tau' : a' \longrightarrow a'$ lifting τ.

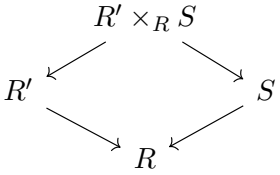
(ii)⇒(i). Let a''_i $(i = 1, 2)$ be objects of $\underline{A}(R' \times_R S)$ with the same image under

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)].$$

We obtain a diagram of solid arrows



above



where $\bar{\tau} : a_1 \longrightarrow a_2$ is the isomorphism induced by τ, so the lower-right square commutes. Under condition (ii), the given isomorphism $a_1' \xrightarrow{\sim} a_2'$ can be replaced so that the left square also commutes. Since

$$a_i'' = a_i' \times_{a_i} b_i \qquad (i = 1, 2)$$

by Remark 2.6(a), we obtain an isomorphism $a_1'' \xrightarrow{\sim} a_2''$. Hence

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is injective, and therefore bijective because $\underline{A}$ is homogeneous by hypothesis. □

Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let $f : R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$. There is a canonical map

$$\widetilde{f} : R' \times_R R' \longrightarrow k[\operatorname{Ker} f], \qquad (r_1', r_2') \longmapsto r_1'(0) + (r_1' - r_2'),$$

and

$$1 \times \widetilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\operatorname{Ker} f]$$

is an isomorphism. Given two f-maps $a_i \longrightarrow a$ $(i = 1, 2)$ in $\underline{A}$, the fibre product $a_1 \times_a a_2$ is an object over $R' \times_R R'$. We may therefore form

$$\tau(a_1, a_2) = (\widetilde{f})_*(a_1 \times_a a_2) \in \underline{A}(k[\operatorname{Ker} f]).$$

Since $1 \times \widetilde{f}$ is an isomorphism, there is an isomorphism

$$a_1 \times_a a_2 \simeq a_1 \times \tau(a_1, a_2)$$

compatible with projection onto the first factor. Statement 1.5(2) now takes the following more definite form.

Lemma 2.8. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let

$$a_i \xrightarrow{\alpha_i} a \qquad (i = 1, 2)$$

be f-maps in $\underline{A}$.

(1) There exists $\rho : a_1 \longrightarrow a_2$ in $\underline{A}$ with $\alpha_2 \circ \rho = \alpha_1$ if and only if there is an arrow $a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$.

(2) There exists such a ρ with $P(\rho) = \mathrm{id}_{R'}$ if and only if

$$\tau(a_1, a_2) = 0 \quad \text{in} \quad [\underline{A}(k[\operatorname{Ker} f])].$$

Proof. We prove both assertions simultaneously. The existence of $\rho : a_1 \longrightarrow a_2$ satisfying $\alpha_2 \circ \rho = \alpha_1$ (and $P(\rho) = \mathrm{id}_{R'}$) is equivalent successively to the existence of

$$h : a_1 \longrightarrow a_1 \times_a a_2, \qquad \mathrm{pr}_1 \circ h = \mathrm{id}_{a_1},$$

with $P(h)$ the diagonal map; to the existence of

$$g : a_1 \longrightarrow a_1 \times \tau(a_1, a_2), \qquad \mathrm{pr}_1 \circ g = \mathrm{id}_{a_1},$$

with

$$P(g) : R' \longrightarrow R' \times_k k[\mathrm{Ker}\, f], \qquad r' \longmapsto (r', r'(0));$$

and to the existence of an arrow $\gamma : a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$, with

$$P(\gamma) : R' \longrightarrow k[\mathrm{Ker}\, f], \qquad r' \longmapsto r'(0).$$

Such an arrow factors through $R' \longrightarrow k \longrightarrow k[\mathrm{Ker}\, f]$ if and only if $\tau(a_1, a_2) = 0$. □

Proposition 2.9. (a) Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$. Assume that

(i) $\underline{B}$ is semi-homogeneous and $\underline{C}$ is homogeneous;

(ii) the map

$$\psi(k[\varepsilon]) : [\underline{B}(k[\varepsilon])] \longrightarrow [\underline{C}(k[\varepsilon])]$$

is injective.

Then $\psi \circ \varphi$ is smooth if and only if φ and ψ are both smooth.

(b) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is semi-homogeneous and $\underline{B}$ is homogeneous, then

$$\widehat{\varphi} : \widehat{\underline{A}} \longrightarrow \widehat{\underline{B}}$$

is smooth. Explicitly, for every surjective map $b' \xrightarrow{\beta} \widehat{\varphi}(a)$ in $\widehat{\underline{B}}$, there are a map $\alpha : a' \longrightarrow a$ in $\widehat{\underline{A}}$ and a $\widehat{\underline{C}}_\Lambda$-isomorphism

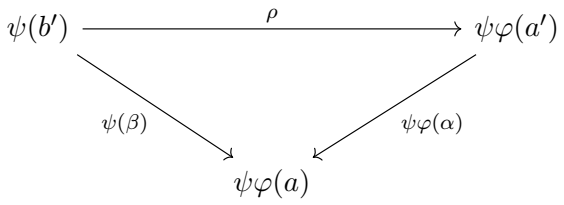
$$\rho : \widehat{\varphi}(a') \xrightarrow{\sim} b'$$

such that $\beta \circ \rho = \varphi(\alpha)$.

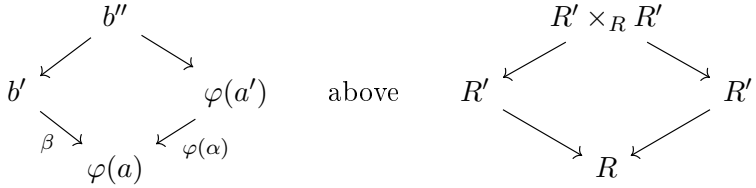
Proof. (a) Suppose $\psi \circ \varphi$ is smooth. If φ is smooth, then ψ is smooth by Proposition 2.3(b), so it remains to prove that φ is smooth. Let $b' \xrightarrow{\beta} \varphi(a)$ be a small prolongation in $\underline{B}$. Since $\psi \circ \varphi$ is smooth, there are an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism

$$\rho : \psi(b') \longrightarrow \psi\varphi(a')$$

for which



commutes. Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram



where $R = P(a)$ and $R' = P(b') = P(a')$. Homogeneity of $\underline{C}$ gives

$$\psi(b'') = \psi(b') \times_{\psi\varphi(a)} \psi\varphi(a')$$

by Remark 2.6(a). Lemma 2.8(2) then gives

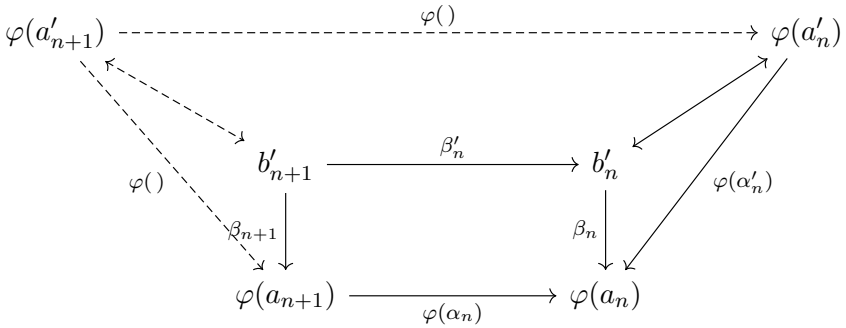
$$0 = \tau(\psi\varphi(a'), \psi(b')) = (\widetilde{f})_*\psi(b'').$$

By hypothesis (ii), it follows that $(\widetilde{f})_*(b'') = 0$. Statement 1.5(2) now provides a map $\gamma : b' \longrightarrow \varphi(a')$ satisfying $\beta = \varphi(\alpha) \circ \gamma$. Thus φ is smooth, and then so is ψ. The converse follows from Proposition 2.3(a).

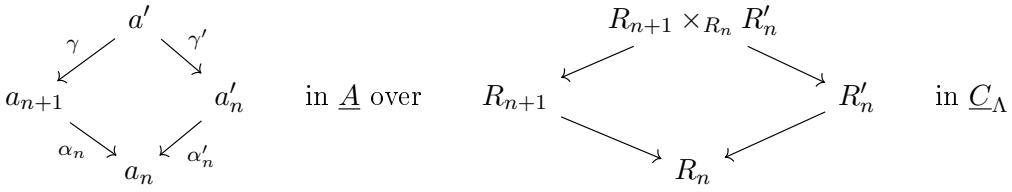
(b) Let a surjective map $\beta : b' \longrightarrow \widehat{\varphi}(a)$ in $\underline{\widehat{B}}$ be given. It is represented by a sequence of commutative diagrams

$$\begin{array}{ccccccc}
\dashrightarrow & b'_3 & \xrightarrow{\beta'_2} & b'_2 & \xrightarrow{\beta'_1} & b'_1 \\
 & \downarrow{\scriptstyle \beta_3} & & \downarrow{\scriptstyle \beta_2} & & \downarrow{\scriptstyle \beta_1} \\
\dashrightarrow & \varphi(a_3) & \xrightarrow[\varphi(\alpha_2)]{} & \varphi(a_2) & \xrightarrow[\varphi(\alpha_1)]{} & \varphi(a_1)
\end{array}$$

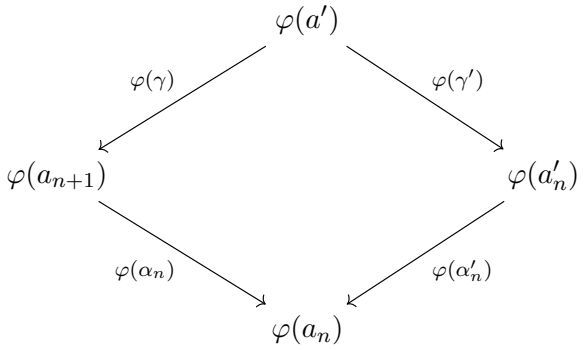
in $\underline{B}$, where every α_v, β_v, and β'_v is surjective and $\beta = \varprojlim_v \beta_v$. It is enough to show inductively that every diagram of solid arrows below, with α_n, β_n, β_{n+1}, and β'_n surjective, can be completed by the dotted arrows:



By hypothesis $\underline{A}$ is semi-homogeneous and $\alpha_n : a'_n \longrightarrow a_n$ is surjective. Hence there is a commutative diagram



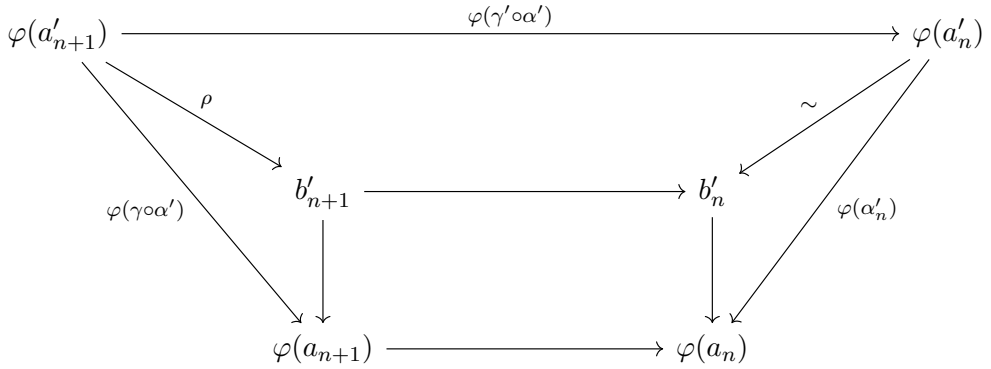
Homogeneity of $\underline{B}$ implies that



is a fibre-product diagram in $\underline{B}$. Thus there is a unique map $\delta : b'_{n+1} \longrightarrow \varphi(a')$ with the required commutativity. Since β_{n+1} and β'_n are surjective, so is δ. Smoothness of φ therefore gives a map $a'_{n+1} \xrightarrow{\alpha'} a'$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism

$$\rho : \varphi(a'_{n+1}) \xrightarrow{\sim} b'_{n+1}$$

such that $\delta \circ \rho = \varphi(\alpha')$. We obtain the commutative diagram



which completes the induction and proves (b). □

Corollary 2.10. (a) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. Let ξ be a versal formal object of $\underline{A}$, and let η be a minimally versal formal object of $\underline{B}$. If $\underline{B}$ is homogeneous, then for every map $\beta : \eta \longrightarrow \varphi(\xi)$ in $\widehat{\underline{B}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\beta) : P(\eta) \longrightarrow P(\varphi(\xi)) = P(\xi).$$

(b) Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let ξ and η be versal formal objects of $\underline{A}$. If η is minimal, then for every map $\alpha : \eta \longrightarrow \xi$ in $\widehat{\underline{A}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\alpha) : P(\eta) \longrightarrow P(\xi).$$

Proof. (a) A map $\beta : \eta \longrightarrow \varphi(\xi)$ induces a commutative diagram

$$\begin{array}{ccc} \underline{h}_{P(\xi)} & \xrightarrow{\widetilde{\beta}} & \underline{h}_{P(\eta)} \\ \downarrow{\scriptstyle \widetilde{\xi}} & & \downarrow{\scriptstyle \widetilde{\eta}} \\ \underline{A} & \xrightarrow[\varphi]{} & \underline{B}. \end{array}$$

Since $\widetilde{\xi}$ and φ are smooth, the composite $\widetilde{\eta} \circ \widetilde{\beta} = \varphi \circ \widetilde{\xi}$ is smooth. On the other hand,

$$\widetilde{\eta}(k[\varepsilon]) : \underline{h}_{P(\eta)}(k[\varepsilon]) \longrightarrow [\underline{B}(k[\varepsilon])]$$

is an isomorphism. Proposition 2.9(a) therefore shows that $\widetilde{\beta}$ is smooth, which means precisely that $P(\xi)$ is a formal power-series ring over $P(\eta)$.

(b) Apply (a) with $\underline{A} = \underline{B}$. □ We conclude by considering the "representability" of a cofibered groupoid. Let $\underline{C}$ be a fixed category. Every object R of $\underline{C}$ defines a functor

$$h_R : \underline{C} \longrightarrow (\mathrm{Sets}), \qquad h_R(S) = \mathrm{Hom}_{\underline{C}}(R, S),$$

and hence a fully faithful functor

$$h : \underline{C}^{\mathrm{op}} \longrightarrow \underline{\mathrm{Hom}}(\underline{C}, (\mathrm{Sets})).$$

A *cocategory in* $\underline{C}$ is a diagram

$$h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1} \underset{s_2}{\overset{s_1}{\underset{\longleftarrow \epsilon \longrightarrow}{\rightrightarrows}}} h_{R_0} \qquad \text{with } s_i \circ \epsilon = \mathrm{id} \quad (i = 1, 2),$$

where the first arrow is denoted by c and

$$h_{R_1} \times_{h_{R_0}} h_{R_1} = (h_{R_1})_{s_1} \times_{h_{R_0}} {}_{s_2}(h_{R_1}),$$

such that

(i) c is associative and compatible with s_i for $i = 1, 2$;

(ii) both composites

$$h_{R_1} \xrightarrow{\sim} h_{R_0} \times_{h_{R_0}} h_{R_1} \xrightarrow{\epsilon \times 1} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1},$$

$$h_{R_1} \xrightarrow{\sim} h_{R_1} \times_{h_{R_0}} h_{R_0} \xrightarrow{1 \times \epsilon} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

are identities.

If $(R_0, R_1, s_1, s_2, \epsilon, c)$ is a cocategory in $\underline{C}$, then for every $S \in \underline{C}$ we define a category $h_{(R_0,R_1,\ldots)}(S)$ by

$$\operatorname{ob} h_{(R_0,R_1,\ldots)}(S) = \operatorname{Hom}_{\underline{C}}(R_0, S), \qquad \operatorname{Fl} h_{(R_0,R_1,\ldots)}(S) = \operatorname{Hom}_{\underline{C}}(R_1, S),$$

with composition supplied by

$$c : h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1}.$$

The categories so obtained depend functorially on S and define a cofibered category $\underline{h}_{(R_0,R_1,\ldots)}$ over $\underline{C}$. For example, every object R of $\underline{C}$ defines a discrete cocategory

$$\underline{h}_R = \underline{h}_{(R,R,s_1,s_2,\epsilon,c)}$$

by taking s_1, s_2, ϵ, and c all to be identity maps. A homomorphism

$$(R_0, R_1, \ldots) \longrightarrow (R_0', R_1', \ldots)$$

of cocategories is a pair of maps of represented functors $h_{R_0} \longrightarrow h_{R_0'}$ and $h_{R_1} \longrightarrow h_{R_1'}$ whose induced map

$$h_{(R_0,R_1,\ldots)}(S) \longrightarrow h_{(R_0',R_1',\ldots)}(S)$$

is a functor for every $S \in \underline{C}$. Every cocategory $(R_0, R_1, \ldots)$ has, in particular, a homomorphism

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}.$$

A *cogroupoid in* $\underline{C}$ is a cocategory $(R_0, R_1, \ldots)$ for which $h_{(R_0,R_1,\ldots)}(S)$ is a groupoid for every $S \in \underline{C}$; equivalently, $\underline{h}_{(R_0,R_1,\ldots)}$ is a category cofibered in groupoids over $\underline{C}$.

In practice, cocategories and cogroupoids arise as follows. Let $P : \underline{F} \longrightarrow \underline{C}$ be a cofibered category, and fix an object ξ of $\underline{F}$. Put $R_0 = P(\xi)$, and let $(R_0, \underline{C})$ denote the category of arrows in $\underline{C}$ with source R_0. Let $P : \underline{F}' \longrightarrow (R_0, \underline{C})$ be the pullback of $\underline{F}$ along the canonical functor $(R_0, \underline{C}) \longrightarrow \underline{C}$. Choose a cocartesian section $\xi^{\#}$ of $\underline{F}'$ over $(R_0, \underline{C})$ such that $\xi^{\#}(\mathrm{id}_{R_0}) = \xi$. Thus, for every arrow $R_0 \xrightarrow{f} S$ in $\underline{C}$, we choose a cocartesian f-map $\xi \longrightarrow \xi^{\#}(f)$.

Define

$$\operatorname{Hom}_{\xi^{\#}} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\mathrm{Sets})$$

by

$$X = (f_1, f_2) \longmapsto \operatorname{Hom}_{\underline{F}(S)}(\xi^{\#}(f_1), \xi^{\#}(f_2)),$$

where S is the common target of f_1 and f_2. Up to canonical natural equivalence, this functor is independent of the choice of $\xi^{\#}$; henceforth we write simply ξ for $\xi^{\#}$.

Suppose the functor is represented by an object

$$X_0 = \left(R_0 \underset{s_2}{\overset{s_1}{\rightrightarrows}} R_1 \right)$$

of $(R_0 \amalg R_0, \underline{C})$. Thus there is an R_0-map $\varphi : \xi(s_1) \longrightarrow \xi(s_2)$ in $\underline{F}$ such that

$$\widetilde{\varphi} : \mathrm{Hom}_{(R_0 \amalg R_0, \underline{C})}(X_0, X) \longrightarrow \mathrm{Hom}_{\xi}(X)$$

is bijective for every X in $(R_0 \amalg R_0, \underline{C})$. Then X_0 naturally defines a cocategory $(R_0, R_1, \ldots)$ in $\underline{C}$. Indeed, a triple of arrows

$$R_0 \underset{w}{\overset{u}{\underset{v}{\rightrightarrows}}} S$$

in $\underline{C}$ defines a composition

$$\mathrm{Hom}_{\underline{F}(S)}(\xi(v), \xi(w)) \times \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(v)) \longrightarrow \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(w)),$$

and hence a composition rule

$$h_{R_1}(S) \times_{h_{R_0}(S)} h_{R_1}(S) \longrightarrow h_{R_1}(S),$$

functorial in S. Consequently, Hom_{ξ} defines the cocategory $(R_0, R_1, \ldots)$ in $\underline{C}$ and the cofibered category $\underline{h}_{(R_0, R_1, \ldots)}$ over $\underline{C}$. Explicitly,

$$\mathrm{ob}\, \underline{h}_{(R_0, R_1, \ldots)} = \coprod_{S \in \mathrm{ob}(\underline{C})} \mathrm{Hom}_{\underline{C}}(R_0, S).$$

An arrow

$$(R_0 \xrightarrow{f} S) \longrightarrow (R_0 \xrightarrow{f'} S')$$

is a pair of arrows $S \xrightarrow{h} S'$ and $R_1 \xrightarrow{u} S'$ in $\underline{C}$ such that

$$u \circ s_1 = h \circ f, \qquad u \circ s_2 = f'.$$

The structural functor $P : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{C}$ sends $(R_0 \xrightarrow{f} S)$ to S.

The pair (ξ, φ) now defines a functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{F} \quad \text{over } \underline{C}$$

by

$$(R_0 \xrightarrow{f} S) \longmapsto \xi(f)$$

on objects and by

$$\left[(R_0 \xrightarrow{f} S) \xrightarrow{(h,u)} (R_0 \xrightarrow{f'} S') \right] \longmapsto \left[\xi(f) \xrightarrow{h_*} \xi(hf) \xrightarrow{u(\varphi)} \xi(f') \right]$$

on arrows. For every $S \in \underline{C}$, the functor

$$(\xi, \varphi)(S) : \underline{h}_{(R_0, R_1, \ldots)}(S) \longrightarrow \underline{F}(S)$$

is fully faithful by the representability of Hom_{ξ}. It follows that

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{F}$$

is a $\underline{C}$-equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$, that is, if every object a of $\underline{F}$ admits a cocartesian arrow $\xi \longrightarrow a$.

One may instead consider the functor

$$\mathrm{Isom}_{\xi} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\mathrm{Sets}),$$

given by

$$X = (f_1, f_2) \longmapsto \mathrm{Isom}_{\underline{F}(S)}(\xi(f_1), \xi(f_2)),$$

where S is the common target of f_1 and f_2. If this functor is represented by an object $(R_0 \rightrightarrows R_1)$ of $(R_0 \amalg R_0, \underline{C})$, then it defines a cogroupoid $(R_0, R_1, \ldots)$ in $\underline{C}$, together with a functor

$$(\xi, \varphi) : \underline{h}_{(R_0,R_1,\ldots)} \longrightarrow \underline{F}.$$

When $\underline{F}$ is a cofibered groupoid, this functor is a $\underline{C}$-equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$. Definition and Proposition 2.11. A cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\underline{C}}_\Lambda$ is called smooth if either, and hence both, of the following equivalent conditions holds:

(i) R_1 is a formal power-series ring over R_0 via s_1 (and via s_2);

(ii) the functor

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$$

is smooth over $\underline{C}_\Lambda$.

A minimally smooth cogroupoid, also called a normalized smooth cogroupoid, in $\widehat{\underline{C}}_\Lambda$ is a smooth cogroupoid $(R_0, R_1, \ldots)$ for which $\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])$ is totally disconnected; equivalently,

$$s_1(k[\varepsilon]) = s_2(k[\varepsilon]).$$

Proof. Smoothness of $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,R_1,\ldots)}$ means that, for every surjective map

$$(S', f') \xrightarrow{(h,u)} (S, f)$$

in $\underline{h}_{(R_0,R_1,\ldots)}$, there are a lift f'' of f to S' and an isomorphism

$$(S', f'') \xrightarrow{(1,u')} (S', f').$$

Equivalently, given $u \in h_{R_1}(S)$ and a lift f' of $s_1(u)$ to S', there is a lift u' of u to S' such that $s_1(u') = f'$, for every surjection $S' \to S$ in $\underline{C}_\Lambda$. This is precisely the smoothness of

$$s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$$

over $\underline{C}_\Lambda$, and that in turn says that R_1 is a formal power-series ring over R_0 via s_1. □

Proposition 2.12. Let $(R_0, R_1, \ldots)$ be a cogroupoid in $\widehat{\underline{C}}_\Lambda$.

(a) If $(R_0, R_1, \ldots)$ is smooth, then $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous.

(b) If $s_1(k[\varepsilon]) = s_2(k[\varepsilon])$, then the following conditions are equivalent:

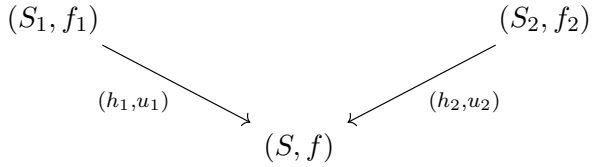
(i) $(R_0, R_1, \ldots)$ is smooth;

(ii) $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous;

(iii) $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous.

(c) If $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then it is equivalent to $\underline{h}_{(S_0,S_1,\ldots)}$ for some minimally smooth cogroupoid S_*.

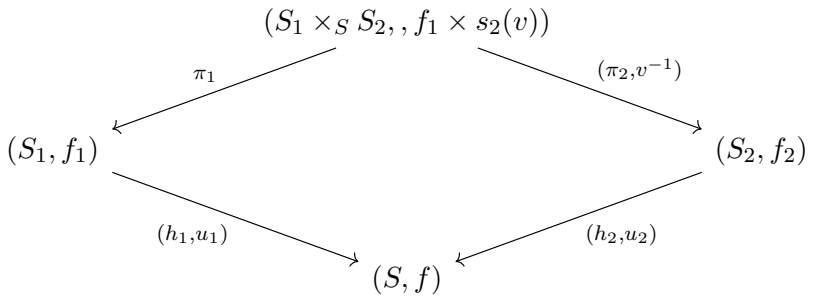
Proof. (a) Consider a diagram



in $\underline{h}_{(R_0,R_1,\ldots)}$, where $h_2 : S_2 \longrightarrow S$ is surjective in $\underline{C}_\Lambda$. Since $s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$ is smooth, $u_1^{-1}u_2$ lifts to an element $v \in h_{R_1}(S_2)$ with $s_1(v) = f_2$. We have

$$h_2 \circ s_2(v) = h_1 \circ f_1,$$

so $f_1 \times s_2(v)$ is an element of $h_{R_0}(S_1 \times_S S_2)$ and gives a commutative diagram



We claim that its top object is the fibre product

$$(S_1, f_1) \times_{(S,f)} (S_2, f_2)$$

in $\underline{h}_{(R_0,R_1,\ldots)}$. Indeed, for every object (T, g), a map to the top object is equivalently a tuple $(h_1' \times h_2', w_1, w_2)$ satisfying

$$\begin{aligned} h_1 w_1 &= h_2 w_2, & s_1(w_1) &= h_1' g, & s_2(w_1) &= f_1, \\ s_1(w_2) &= h_2' g, & s_2(w_2) &= s_2(v). \end{aligned}$$

Putting $w_2' = v^{-1} w_2$ gives a bijection with the tuples $(h_1' \times h_2', w_1, w_2')$ satisfying

$$\begin{aligned} u_2 h_2 w_2' &= u_1 h_1 w_1, & s_1(w_1) &= h_1' g, & s_2(w_1) &= f_1, \\ s_1(w_2') &= h_2' g, & s_2(w_2') &= f_2. \end{aligned}$$

These are exactly the pairs of arrows from (T, g) to (S_1, f_1) and (S_2, f_2) whose composites to (S, f) coincide. Thus

$$\begin{aligned} &\mathrm{Hom}((T,g), (S_1 \times_S S_2, f_1 \times s_2(v))) \\ &\qquad \simeq \mathrm{Hom}((T,g),(S_1,f_1)) \times_{\mathrm{Hom}((T,g),(S,f))} \mathrm{Hom}((T,g),(S_2,f_2)), \end{aligned}$$

which proves (a).

(b) It remains only to prove (iii)⇒(i). If $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then

$$\dim[\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])] = \dim h_{R_0}(k[\varepsilon]) < \infty,$$

so it admits a minimally versal formal object. On the other hand, $(R_0, 1_{R_0})$ is a quasi-initial minimal object of $\widehat{\underline{h}}_{(R_0,R_1,\ldots)}$, because

$$h_{R_0}(k[\varepsilon]) \longrightarrow [\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])]$$

is bijective. Proposition 1.10(ii) therefore makes $(R_0, 1_{R_0})$ a minimally versal formal object of $\underline{h}_{(R_0,R_1,\ldots)}$. Equivalently,

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$$

is minimally smooth. This proves (i). (c) Let (S_0, ξ_0) be a minimally versal formal object of $\underline{h}_{(R_0,R_1,\ldots)}$ (Theorem 1.11). For any object

$$\xi : R_0 \longrightarrow T \qquad \text{of } \underline{h}_{(R_0,R_1,\ldots)}(T),$$

consider the functor

$$\mathrm{Isom}(\xi, \xi) : \underline{C}_\Lambda / (T \widehat{\otimes}_\Lambda T) \longrightarrow (\mathrm{Sets}).$$

It assigns to R the set of isomorphisms between the two images of ξ on R obtained from

$$R_0 \xrightarrow{\xi} T \rightrightarrows T \widehat{\otimes}_\Lambda T \longrightarrow R.$$

This functor is represented by

$$R_1 \otimes_{R_0 \widehat{\otimes}_\Lambda R_0} (T \widehat{\otimes}_\Lambda T).$$

Passing to the inverse limit shows that $\mathrm{Isom}(\xi_0, \xi_0)$ is prorepresented by an object S_1, and that

$$\underline{h}_{(R_0,R_1,\ldots)} \simeq \underline{h}_{(S_0,S_1,\ldots)}.$$

By (b), the cogroupoid S_* is minimally smooth. □

Definition 2.13. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called (minimally) prorepresentable if it is $\underline{C}_\Lambda$-equivalent to $\underline{h}_{(R_0,R_1,\ldots)}$ for some (normalized) cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\underline{C}}_\Lambda$.

Proposition 2.14. If $(R_0, R_1, \ldots)$ is a normalized cogroupoid in $\widehat{\underline{C}}_\Lambda$, then every homomorphism

$$(\alpha_0, \alpha_1) : (R_0, R_1, \ldots) \longrightarrow (R_0, R_1, \ldots)$$

which induces a $\underline{C}_\Lambda$-equivalence

$$\underline{h}_{(\alpha_0,\alpha_1)} : \underline{h}_{(R_0,R_1,\ldots)} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$$

is an isomorphism. In

particular, a normalized prorepresentation of a cofibered groupoid over $\underline{C}_\Lambda$, if it exists, is unique up to isomorphism.

Proof. Consider the commutative diagram

$$\begin{array}{ccc} \underline{h}_{(R_0,R_1,\ldots)} & \xrightarrow{\underline{h}_{(\alpha_0,\alpha_1)}} & \underline{h}_{(R_0,R_1,\ldots)} \\ \uparrow{\scriptstyle \epsilon} & & \uparrow{\scriptstyle \epsilon} \\ \underline{h}_{R_0} & \xrightarrow[\underline{h}_{\alpha_0}]{} & \underline{h}_{R_0} \end{array}$$

in which $\underline{h}_{(\alpha_0,\alpha_1)}$ is a $\underline{C}_\Lambda$-equivalence. Since $\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])$ is totally disconnected, both

$$[\underline{h}_{(\alpha_0,\alpha_1)}(k[\varepsilon])] \quad \text{and} \quad [\epsilon(k[\varepsilon])]$$

are bijections, and hence so is $\underline{h}_{\alpha_0}(k[\varepsilon])$. Thus $\alpha_0 : R_0 \longrightarrow R_0$ is a surjective endomorphism and therefore an automorphism. It follows that α_1 is also an automorphism of R_1. □ Lemma 2.15. Let A and B be discrete groupoids over $\underline{C}_\Lambda$, and let $\varphi : A \longrightarrow B$ be a minimally smooth functor. If A and B are homogeneous, then φ is a $\underline{C}_\Lambda$-equivalence.

Proof. We must show that $\varphi(S) : A(S) \longrightarrow B(S)$ is bijective for every $S \in \mathrm{ob}(\underline{C}_\Lambda)$. It is surjective for every S and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S. Let $S' \xrightarrow{f} S$ be a small extension in $\underline{C}_\Lambda$. Since A and B are homogeneous discrete groupoids, the canonical isomorphism

$$1 \times \widetilde{f} : S' \times_S S' \xrightarrow{\sim} S' \times_k k[\mathrm{Ker}\, f]$$

induces a commutative diagram

$$\begin{array}{ccc} A(S') \times_{A(S)} A(S') & \xrightarrow{\sim} & A(S') \times A(k[\mathrm{Ker}\, f]) \\ \downarrow & & \downarrow \\ B(S') \times_{B(S)} B(S') & \xrightarrow[\sim]{} & B(S') \times B(k[\mathrm{Ker}\, f]). \end{array}$$

Thus $A(S')$ and $B(S')$ are principal homogeneous sets over $A(S)$ and $B(S)$, with structural groups identified by

$$A(k[\mathrm{Ker}\, f]) \xrightarrow{\sim} B(k[\mathrm{Ker}\, f]).$$

If $\varphi(S)$ is bijective, it follows that $\varphi(S')$ is injective, and hence bijective. This completes the induction. □

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_\Lambda$ such that

$$\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$$

is finite-dimensional. For every $\xi \in \mathrm{ob}(\widehat{\underline{A}})$, the functor

$$\mathrm{Isom}_\xi : \widehat{\underline{C}}_{R\widehat{\otimes}_\Lambda R} \longrightarrow (\mathrm{Sets})$$

is prorepresentable, where $R = P(\xi)$.

Proof. If Isom_ξ is homogeneous, Theorem 1.11 supplies a minimally smooth functor

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}}_\xi.$$

Lemma 2.15 makes Φ a $\widehat{\underline{C}}_{R\widehat{\otimes}_\Lambda R}$-equivalence, and hence Isom_ξ is prorepresentable. It remains to prove homogeneity. Consider a diagram

$$\begin{array}{ccc} T_1 & & T_2 \\ & \searrow \quad \swarrow & \\ & T & \end{array}$$

in $\widehat{\underline{C}}_{R\widehat{\otimes}_\Lambda R}$, with $T_2 \longrightarrow T$ surjective. Since $\underline{A}$ is homogeneous, the fibre products

$$\xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$$

exist in $\underline{A}$, and there are canonical isomorphisms

$$\xi_i(T_1 \times_T T_2) \xrightarrow{\sim} \xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2) \qquad (i = 1, 2)$$

(cf. 2.6(a)). Therefore

$$\mathrm{Isom}_\xi(T_1 \times_T T_2) \longrightarrow \mathrm{Isom}_\xi(T_1) \times_{\mathrm{Isom}_\xi(T)} \mathrm{Isom}_\xi(T_2)$$

is an isomorphism. □

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if

(1) $\underline{A}$ is homogeneous;

(2) $[\underline{A}(k[\varepsilon])]$ and $\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$ are finite-dimensional vector spaces over k.

Proof. Suppose first that $\underline{A}$ is prorepresented by a smooth cogroupoid. Thus $\underline{A}$ is $\underline{C}_\Lambda$-equivalent to

$$\underline{h} = \underline{h}_{(R_0, R_1, \ldots)}$$

for a smooth cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\underline{C}}_\Lambda$. Proposition 2.12(a) shows that $\underline{h}$ is homogeneous. Moreover,

$$[\underline{h}(k[\varepsilon])] = \mathrm{Coker}\Big(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \underset{s_2}{\overset{s_1}{\rightrightarrows}} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \Big),$$

$$\begin{aligned} \mathrm{Aut}_{\underline{h}}(1_\varepsilon) &= \mathrm{Ker}\Big(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \Big) \\ &= \mathrm{Ker}\Big(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \Big). \end{aligned}$$

Both are finite-dimensional vector spaces over k, so the same is true for $\underline{A}$.

Conversely, assume (1) and (2). Since $\underline{A}$ is homogeneous and $[\underline{A}(k[\varepsilon])]$ is finite-dimensional, it has a minimally versal formal object ξ. Put $P(\xi) = R_0$ and consider

$$\mathrm{Isom}_\xi : \widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0} \longrightarrow (\mathrm{Sets}),$$

given by

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T), \xi_2(T)),$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \longrightarrow T$ are the composite maps in

$$R_0 \underset{s_2}{\overset{s_1}{\rightrightarrows}} R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow T.$$

Corollary 2.16 prorepresents this functor by an object

$$R_1 \in \mathrm{ob}(\widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0})$$

and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ for which

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}}_\xi$$

is an isomorphism over $\widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0}$. The pair (R_0, R_1) therefore defines a cogroupoid, and

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{A}$$

is a $\underline{C}_\Lambda$-equivalence. Because ξ is minimally versal, the composite

$$\underline{h}_{(R_0, R_0, \ldots)} \xrightarrow{\epsilon} \underline{h}_{(R_0, R_1, \ldots)} \xrightarrow{(\xi, \varphi)} \underline{A}$$

is minimally smooth. Since (ξ, φ) is an equivalence, ϵ is minimally smooth as well. Thus $(R_0, R_1, \ldots)$ is a normalized smooth cogroupoid in $\widehat{\underline{C}}_\Lambda$. □

Corollary 2.18. A functor $F : \underline{C}_\Lambda \longrightarrow (\mathrm{Sets})$ is prorepresentable if and only if it is homogeneous and $\dim F(k[\varepsilon]) < \infty$.

Corollary 2.19. Every smooth cogroupoid in $\widehat{\underline{C}}_\Lambda$ can be normalized up to isomorphism.

Given a cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\underline{C}}_\Lambda$, set $\bar{R}_1 = R_1/J$, where J is the ideal of R_1 generated by

$$\{s_1(r) - s_2(r) \mid r \in R_0\}.$$

Then $(R_0, \bar{R}_1, \ldots)$ defines a formal group scheme over R_0 with the induced composition law.

Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \ldots)$ via (ξ, φ). Let

$$\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$$

be the embeddings corresponding to the surjections

$$R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow R_0 \longrightarrow k,$$

and let Aut_ξ and Aut^o be the restrictions of Isom_ξ to $\underline{C}_{R_0}$ and $\underline{C}_k$, respectively. Thus $\mathrm{Aut}_\xi(S)$ is the automorphism group of $\xi(S)$ for $S \in \mathrm{ob}(\underline{C}_{R_0})$, while $\mathrm{Aut}^o(S)$ is the automorphism group of 1_S for $S \in \mathrm{ob}(\underline{C}_k)$, where 1_S is the direct image of $1 \in \underline{A}(k)$ under $k \longrightarrow S$.

Since R_1 prorepresents Isom_ξ on $\underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$, it follows that $\bar{R}_1$ prorepresents Aut_ξ on $\underline{C}_{R_0}$, and that $k \otimes_{R_0} \bar{R}_1$ prorepresents Aut^o on $\underline{C}_k$. Although $\underline{A}$ is prorepresented by the smooth cogroupoid $(R_0, R_1, \ldots)$, neither $(R_0, \bar{R}_1, \ldots)$ nor $(k, k \otimes_{R_0} \bar{R}_1, \ldots)$ is smooth in general. More precisely:

Corollary 2.20. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \ldots)$ via (ξ, φ).

(a) $(R_0, \bar{R}_1, \ldots)$ and $(k, k \otimes_{R_0} \bar{R}_1, \ldots)$ prorepresent Aut_ξ and Aut^o, respectively.

(b) The following statements are equivalent:

(i) the formal group scheme $\underline{h}_{(R_0,\bar{R}_1,\ldots)}$ is formally smooth over R_0;

(ii) for every surjective arrow $a' \longrightarrow a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{P(a')}(a') \longrightarrow \mathrm{Aut}_{P(a)}(a)$$

is surjective;

(iii) R_0 prorepresents the functor $[\underline{A}]$ over $\underline{C}_\Lambda$.

Proof. We have already noted (a). For (b), the equivalence (i)⇔(ii) is clear, whereas (ii)⇔(iii) follows from 2.7 and 2.15, since

$$\widetilde{\xi} : \underline{h}_{R_0} \longrightarrow [\underline{A}]$$

is minimally smooth. □

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|\underline{C}_k$ be the pullback of $p : \underline{A} \longrightarrow \underline{C}_\Lambda$ along $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. If $\underline{A}$ is prorepresented by $(R_0, R_1, \ldots)$ via (ξ, φ), then $\underline{A}|\underline{C}_k$ is prorepresented by

$$(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \ldots)$$

via $(k \otimes_\Lambda \xi, k \otimes_\Lambda \varphi)$. Here $k \otimes_\Lambda \xi$ is the direct image of ξ under $R_0 \longrightarrow k \otimes_\Lambda R_0$, and $k \otimes_\Lambda \varphi$ is the image of φ under

$$\mathrm{Isom}_{\underline{A}(R_0)}\big(\xi(s_1), \xi(s_2)\big) \longrightarrow \mathrm{Isom}_{\underline{A}(k\otimes_\Lambda R_1)}\big(k \otimes_\Lambda \xi(s_1), k \otimes_\Lambda \xi(s_2)\big).$$

If $(R_0, R_1, \ldots)$ is normalized, then so is $(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \ldots)$; moreover, the relative dimension of R_1 over R_0 equals that of $k \otimes_\Lambda R_1$ over $k \otimes_\Lambda R_0$, since R_1 is smooth over R_0.

Remark 2.22. Suppose that $\underline{A}$ is prorepresented by smooth cogroupoids $(R_0, R_1, \ldots)$ and $(R'_0, R'_1, \ldots)$, with $(R_0, R_1, \ldots)$ normalized. We obtain a homomorphism

$$(i_0, i_1) : (R_0, R_1, \ldots) \longrightarrow (R'_0, R'_1, \ldots).$$

By 2.10(b), R'_0 is a formal power-series ring over R_0, and (i_0, i_1) induces an isomorphism

$$R'_0 \otimes_{R_0} (R_0, R_1, \ldots) \xrightarrow{\sim} (R'_0, R'_1, \ldots).$$

3. The coherent sheaves $\underline{D}^i_X$

One important functor which is usually not prorepresentable, but nevertheless admits a formal versal object, is the deformation functor. With this in view, we recall some basic facts about extensions of commutative rings and their obstruction theory. This section contains nothing new, except perhaps for its systematic use of Grothendieck cohomology on a site. Its purpose is to make this exposé as self-contained as possible; in fact, we prove somewhat more than will be needed later. The basic results are stated in Theorem 3.12. Throughout, all rings are commutative and have a unit.

Given a ring R, let $\underline{C}_R$ denote the category of R-algebras. For every $A \in \mathrm{Ob}(\underline{C}_R)$, put

$$\mathrm{Cov}(A) = \{\text{surjective arrows } \varphi : A' \longrightarrow A \text{ in } \underline{C}_R\}.$$

This defines a pretopology, and hence a site, on $\underline{C}_R$. For every $A \in \mathrm{Ob}(\underline{C}_R)$, let $\underline{C}_{A|R}$ be the category of arrows in $\underline{C}_R$ with target A. We endow $\underline{C}_{A|R}$ with the topology induced by the forgetful functor

$$j_A : \underline{C}_{A|R} \longrightarrow \underline{C}_R.$$

The category $\underline{C}_{A|R}$ has a final object $(1_A : A \longrightarrow A)$ and an initial object $(i_A : R \longrightarrow A)$, and it admits both fibre products and coproducts. When there is no risk of confusion, we write simply B for an object $(B \longrightarrow A)$ of $\underline{C}_{A|R}$.

Let $\mathscr{C}$ be an abelian category. A $\mathscr{C}$-valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor

$$F : \underline{C}^{\mathrm{op}}_{A|R} \longrightarrow \mathscr{C}$$

such that, for every surjective arrow $B' \longrightarrow B$ in $\underline{C}_{A|R}$, the sequence

$$0 \longrightarrow F(B) \longrightarrow F(B') \rightrightarrows F(B' \times_B B')$$

is exact. The additive presheaves are those which preserve coproducts; thus

$$F(B_1 \otimes_R B_2) = F(B_1) \oplus F(B_2).$$

For example, an A-module M defines the functor

$$\mathrm{Der}_R(-, M) : \underline{C}^{\mathrm{op}}_{A|R} \longrightarrow (A\text{-mod}),$$

which sends B to the R-linear derivations $\mathrm{Der}_R(B, M)$, with M regarded as a B-module via the structural map $B \longrightarrow A$. This is plainly an additive sheaf on $\underline{C}_{A|R}$. Here $(A\text{-mod})$ denotes the category of A-modules.

A presheaf F on $\underline{C}_{A|R}$ with values in $(A\text{-mod})$ is said to be of finite type if $F(B)$ is an A-module of finite type whenever B is an R-algebra of finite type. For example, if A is noetherian and M is an A-module of finite type, then $\mathrm{Der}_R(-, M)$ is an additive sheaf of finite type on $\underline{C}_{A|R}$.

We record a little more notation and a few basic facts from homological algebra.

3.1(a). Throughout the remainder of this section, $\widetilde{\underline{C}}_{A|R}$ and $\overset{\vee}{\underline{C}}_{A|R}$ denote, respectively, the categories of sheaves and presheaves with values in $(A\text{-mod})$. There is a left-exact inclusion functor

$$i_{A|R} : \widetilde{\underline{C}}_{A|R} \longrightarrow \overset{\vee}{\underline{C}}_{A|R}$$

and an exact associated-sheaf functor

$$G_{A|R} : \overset{\vee}{\underline{C}}_{A|R} \longrightarrow \widetilde{\underline{C}}_{A|R},$$

left adjoint to $i_{A|R}$.

An object $B \in \mathrm{Ob}(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \longrightarrow \underline{C}_{A|R}$ and hence a commutative diagram

(*)
$$\begin{array}{ccc} \widetilde{\underline{C}}_{A|R} & \xrightarrow{\widetilde{j}} & \widetilde{\underline{C}}_{B|R} \\ {\scriptstyle i_{A|R}}\downarrow & & \downarrow \\ \overset{\vee}{\underline{C}}_{A|R} & \xrightarrow[j^{\vee}]{} & \overset{\vee}{\underline{C}}_{B|R}. \end{array}$$

The restriction functor $\widetilde{j}$, as well as $j^{\vee}$, is exact, because the site $\underline{C}_{B|R}$ is simply the category of arrows in $\underline{C}_{A|R}$ with target B, endowed with the induced topology.

A ring homomorphism $S \longrightarrow R$ induces a continuous functor $\beta : \underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$ and hence a commutative diagram

(**)
$$\begin{array}{ccc} \widetilde{\underline{C}}_{A|S} & \xrightarrow{\widetilde{\beta}} & \widetilde{\underline{C}}_{A|R} \\ {\scriptstyle i_{A|S}}\downarrow & & \downarrow \\ \overset{\vee}{\underline{C}}_{A|S} & \xrightarrow[\beta^{\vee}]{} & \overset{\vee}{\underline{C}}_{A|R}. \end{array}$$

Notice that $\widetilde{\beta}$ need not be exact.

3.1(b). The right-derived functor of the inclusion

$$i_{A|R} : \widetilde{\underline{C}}_{A|R} \longrightarrow \overset{\vee}{\underline{C}}_{A|R}$$

will be denoted by $\underline{H}^{\bullet}_{A|R}$. For every sheaf F on $\underline{C}_{A|R}$, put

$$H^{\bullet}(A|R, F) = [\underline{H}^{\bullet}_{A|R}(F)](A).$$

For $B' \longrightarrow B$ in $\mathrm{Cov}(B)$, define

$$H^0(\{B' \longrightarrow B\}, F) = \ker\big(F(B') \rightrightarrows F(B' \times_B B')\big).$$

We denote the right-derived functors of $H^0(\{B' \longrightarrow B\}, -)$ by $H^{\bullet}(\{B' \longrightarrow B\}, -)$. These are the cohomology groups of the Čech semisimplicial module

$$F(B') \;\substack{\longrightarrow\\ \longrightarrow}\; F(B' \times_B B') \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; F(B' \times_B B' \times_B B') \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots .$$

The right-derived functor of

$$\check{\underline{H}}^0_{A|R} : \overset{\vee}{\underline{C}}_{A|R} \longrightarrow \overset{\vee}{\underline{C}}_{A|R}$$

will be denoted by $\check{\underline{H}}^{\bullet}_{A|R}$, where

$$[\check{\underline{H}}^0_{A|R}(F)](B) = \varinjlim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \longrightarrow B\}, F).$$

For every presheaf F on $\underline{C}_{A|R}$, put

$$\check{H}^{\bullet}(A|R, F) = [\check{\underline{H}}^{\bullet}_{A|R}(F)](A) = \varinjlim_{\{A' \to A\} \in \mathrm{Cov}(A)} H^{\bullet}(\{A' \longrightarrow A\}, F).$$

For every sheaf F on $\underline{C}_{A|R}$ there is the usual spectral sequence

$$\check{H}^p\big(A|R, \underline{H}^q_{A|R}(F)\big) \Longrightarrow H^{p+q}(A|R, F).$$

Here $\check{H}^0(A|R, \underline{H}^q_{A|R}(F)) = 0$ for every $q > 0$. Consequently,

$$\check{H}^1(A|R, F) = H^1(A|R, F),$$

and the sequence

$$0 \longrightarrow \check{H}^2(A|R, F) \longrightarrow H^2(A|R, F) \longrightarrow \check{H}^1\big(A|R, \underline{H}^1_{A|R}(F)\big) \longrightarrow \check{H}^3(A|R, F)$$

is exact. 3.1(c). If B is an object of $\underline{C}_{A|R}$, diagram $(*)$ in 3.1(a), together with the exactness of $\widetilde{j}$ and $j^{\vee}$, gives a canonical isomorphism

$$H^{\bullet}(B|R, \widetilde{j}F) = [\underline{H}^{\bullet}_{A|R}(F)](B).$$

On the other hand, a ring homomorphism $S \longrightarrow R$ induces diagram $(**)$ in 3.1(a). The functor $\widetilde{\beta}$ need not be exact; however, β preserves fibre products, and we obtain a spectral sequence

$$\underline{H}^p_{A|R}(R^q\widetilde{\beta}F) \Longrightarrow \beta^{\vee}\underline{H}^{p+q}_{A|S}(F),$$

or, on evaluating at A,

$$H^p(A|R, R^q\widetilde{\beta}F) \Longrightarrow H^{p+q}(A|S, F).$$

Since there is a canonical isomorphism

$$\widetilde{\beta} = G_{A|R} \circ \beta^{\vee} \circ i_{A|S}$$

and the functor $\beta^{\#} = G_{A|R} \circ \beta^{\vee}$ is exact, we have

$$R^{\bullet}\widetilde{\beta}F = \beta^{\#}\underline{H}^{\bullet}_{A|S}(F).$$

Thus the preceding spectral sequence may be written

$$H^p\big(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)\big) \Longrightarrow H^{p+q}(A|S, F).$$

Lemma 3.2. Let $B \in \mathrm{Ob}(\underline{C}_{A|R})$, and choose a surjective arrow $P \longrightarrow B$ in $\underline{C}_{B|R}$, where P is a polynomial algebra over R. Then, for every $F \in \mathrm{Ob}(\overset{\vee}{\underline{C}}_{A|R})$, the natural map

$$H^{\bullet}(\{P \longrightarrow B\}, F) \longrightarrow [\underline{\check{H}}^{\bullet}(F)](B)$$

is an isomorphism.

Proof. It suffices to prove that the natural map

$$H^0(\{P \longrightarrow B\}, F) \longrightarrow [\underline{\check{H}}^0(F)](B) = \varinjlim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \longrightarrow B\}, F)$$

is an isomorphism. This is clear because, for every $\{B' \longrightarrow B\} \in \mathrm{Cov}(B)$, there is a map of covers

$$\{P \longrightarrow B\} \longrightarrow \{B' \longrightarrow B\}.$$

□

Corollary 3.3.

(a) Suppose that A is noetherian. If $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A|R})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for every n.

(b) If P is a polynomial algebra over R, then

$$H^n(P|R, F) = 0$$

for every $n > 0$ and every $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{P|R})$.

(c) A sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

in $\widetilde{\underline{C}}_{A|R}$ is exact if and only if

$$0 \longrightarrow F'(P) \longrightarrow F(P) \longrightarrow F''(P) \longrightarrow 0$$

is exact for every polynomial R-algebra $P \in \mathrm{Ob}(\underline{C}_{A|R})$.

Proof.

(a) In view of the spectral sequence

$$\check{H}^p\big(B|R, \underline{H}^q(F)\big) \Longrightarrow H^{p+q}(B|R, F)$$

and the equality $\check{H}^0(\underline{H}^q(F)) = 0$ for $q > 0$, it is enough to show that $\underline{\check{H}}^n(G)$ is of finite type for every n and every presheaf G of finite type. By Lemma 3.2,

$$\check{H}^{\bullet}(B|R, G) = H^{\bullet}(\{P \longrightarrow B\}, G),$$

where P is a polynomial algebra over R and $\{P \longrightarrow B\} \in \mathrm{Cov}(B)$. The latter is the cohomology of the complex

$$\cdots G(P \times_B P \times_B P) \;\substack{\longleftarrow\\[-1ex]\longleftarrow\\[-1ex]\longleftarrow}\; G(P \times_B P) \;\substack{\longleftarrow\\[-1ex]\longleftarrow}\; G(P).$$

If B is an R-algebra of finite type, P may be chosen of finite type over R. All the iterated fibre products $P \times_B \cdots \times_B P$ are then R-algebras of finite type. If G is of finite type, their images under G are A-modules of finite type. Hence $H^\bullet(\{P \longrightarrow B\}, G)$ is an A-module of finite type because A is noetherian.

(b) Let P be a polynomial algebra over R. Every surjective R-algebra map $B \longrightarrow P$ admits a section, so

$$H^i(\{B \longrightarrow P\}, G) = 0 \qquad (i > 0).$$

It follows that $\check{H}^i(P|R, G) = 0$ for every $i > 0$ and every presheaf G on $\underline{C}_{P|R}$. Consequently the spectral sequence

$$\check{H}^p\big(P|R, \underline{H}^q_{P|R}(F)\big) \Longrightarrow H^{p+q}(P|R, F)$$

degenerates, and

$$\check{H}^0\big(P|R, \underline{H}^n(F)\big) \simeq H^n(P|R, F)$$

for every n. For a sheaf F, however, $\check{H}^0(P|R, \underline{H}^n(F)) = 0$ whenever $n > 0$. Thus $H^n(P|R, F) = 0$ for every $n > 0$.

(c) If $0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$ is exact in $\widetilde{\underline{C}}_{A|R}$, then the sequence

$$0 \longrightarrow F'(B) \longrightarrow F(B) \longrightarrow F''(B) \longrightarrow H^1(B|R, F') \longrightarrow H^1(B|R, F) \longrightarrow \cdots$$

is exact. If $P \in \mathrm{Ob}(\underline{C}_{A|R})$ is a polynomial algebra over R, then $H^1(P|R, F') = 0$ by (b), and hence

$$0 \longrightarrow F'(P) \longrightarrow F(P) \longrightarrow F''(P) \longrightarrow 0$$

is exact.

Conversely, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ with

$$\mathrm{Ob}(\underline{P}_{A|R}) = \{P \in \mathrm{Ob}(\underline{C}_{A|R}) \mid P \text{ is a polynomial algebra over } R\},$$

endowed with the induced topology, and let $p : \underline{P}_{A|R} \longrightarrow \underline{C}_{A|R}$ be the inclusion. Every object of $\underline{C}_{A|R}$ is covered by an object of $\underline{P}_{A|R}$; the comparison lemma therefore shows that

$$\widetilde{p} : \widetilde{\underline{C}}_{A|R} \longrightarrow \widetilde{\underline{P}}_{A|R}$$

is an equivalence of categories. If the sequence obtained by evaluating $F' \longrightarrow F \longrightarrow F''$ at every $P \in \mathrm{Ob}(\underline{P}_{A|R})$ is exact, then

$$0 \longrightarrow \widetilde{p}F' \longrightarrow \widetilde{p}F \longrightarrow \widetilde{p}F'' \longrightarrow 0$$

is exact in $\widetilde{\underline{P}}_{A|R}$, and therefore the original sequence is exact in $\widetilde{\underline{C}}_{A|R}$. □

Remark 3.4. Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows at once that every presheaf on $\underline{P}_{A|R}$ is a sheaf; that is,

$$\widetilde{\underline{P}}_{A|R} = \overset{\vee}{\underline{P}}_{A|R}.$$

Consequently, the restriction functor

$$\widetilde{\underline{C}}_{A|R} \longrightarrow \overset{\vee}{\underline{P}}_{A|R}$$

is an equivalence of categories. Lemma 3.5. Consider a diagram of R-algebras

$$\begin{array}{ccc} & & C \\ & & \big\downarrow \\ B' & \xrightarrow[\text{surjective}]{} & B \end{array}$$

and let S be an R-algebra.

(i) If $\mathrm{Tor}_1^R(B,S)=0$, then the canonical map

$$(B'\times_B C)\otimes_R S\longrightarrow (B'\otimes_R S)\times_{B\otimes_R S}(C\otimes_R S)$$

is an isomorphism.

(ii) Suppose that B' is flat over R. If

$$\mathrm{Tor}_i^R(C,S)=0\quad(0<i<n)\qquad\text{and}\qquad \mathrm{Tor}_i^R(B,S)=0\quad(0<i\leq n),$$

then

$$\mathrm{Tor}_i^R(B'\times_B C,S)=0\qquad(0<i<n).$$

Proof. Let $0\longrightarrow J\longrightarrow B'\longrightarrow B\longrightarrow 0$ be exact.
(i) If $\mathrm{Tor}_1^R(B,S)=0$, then

$$0\longrightarrow J\otimes_R S\longrightarrow B'\otimes_R S\longrightarrow B\otimes_R S\longrightarrow 0$$

is exact, and hence so is

$$0\longrightarrow J\otimes_R S\longrightarrow (B'\otimes_R S)\times_{B\otimes_R S}(C\otimes_R S)\longrightarrow C\otimes_R S\longrightarrow 0.$$

We therefore obtain a commutative diagram with exact rows:

$$\begin{array}{ccccccc}
J\otimes_R S & \longrightarrow & (B'\times_B C)\otimes_R S & \longrightarrow & C\otimes_R S & \longrightarrow & 0\\
\| & & \downarrow & & \| & & \\
0\longrightarrow J\otimes_R S & \longrightarrow & (B'\otimes_R S)\times_{B\otimes_R S}(C\otimes_R S) & \longrightarrow & C\otimes_R S & \longrightarrow & 0.
\end{array}$$

It follows that

$$(B'\times_B C)\otimes_R S\longrightarrow (B'\otimes_R S)\times_{B\otimes_R S}(C\otimes_R S)$$

is an isomorphism.

(ii) Since B' is flat over R and $\mathrm{Tor}_i^R(B,S)=0$ for $0<i\leq n$, the long exact Tor sequence gives

$$\mathrm{Tor}_i^R(J,S)=0\qquad(0<i<n).$$

Applying the same sequence to

$$0\longrightarrow J\longrightarrow B'\times_B C\longrightarrow C\longrightarrow 0$$

and using the hypothesis on C gives

$$\mathrm{Tor}_i^R(B'\times_B C,S)=0\qquad(0<i<n).$$

□

Corollary 3.6.

(a) Let A and S be R-algebras. If $\mathrm{Tor}_1^R(A,S)=0$, then for every $G\in \mathrm{Ob}(\underline{\overset{\vee}{C}}_{A\otimes_R S|S})$ the canonical map

$$\check{H}^\bullet(A\otimes_R S|S,G)\longrightarrow \check{H}^\bullet(A|R,\check{S}G)$$

is an isomorphism. Here

$$S:\underline{C}_{A|R}\longrightarrow \underline{C}_{A\otimes_R S|S},\qquad X\longmapsto X\otimes_R S.$$

In particular, if S is flat over R, then

$$\check{S}\,\underline{\check{H}}^\bullet(G)\longrightarrow \underline{\check{H}}^\bullet(\check{S}G)$$

is an isomorphism for every such G.

(b) Let $P \longrightarrow B$ be a surjective map of R-algebras, where P is a polynomial algebra over R. If

$$\mathrm{Tor}_i^R(B, S) = 0 \qquad (0 < i \leq n),$$

then, for every iterated fibre product of copies of P over B,

$$\mathrm{Tor}_i^R(P \times_B P \times_B \cdots \times_B P, S) = 0 \qquad (0 < i < n).$$

Proof.

(a) Given $B \in \mathrm{Ob}(\underline{C}_{A|R})$, choose a cover $\{P \longrightarrow B\} \in \mathrm{Cov}(B)$ with P a polynomial algebra over R. If $\mathrm{Tor}_1^R(B, S) = 0$, Lemma 3.5(i) shows that the natural map

$$\left(\overbrace{P \times_B \cdots \times_B P}^{n}\right) \otimes_R S \longrightarrow \overbrace{(P \otimes_R S) \times_{B \otimes_R S} \cdots \times_{B \otimes_R S} (P \otimes_R S)}^{n}$$

is an isomorphism for every n. Consequently,

$$\begin{aligned} \check{H}^\bullet(B|R, \check{S}G) &= H^\bullet(\{P \longrightarrow B\}, \check{S}G) \\ &= H^\bullet(\{P \otimes_R S \longrightarrow B \otimes_R S\}, G) \\ &= \check{H}^\bullet(B \otimes_R S|S, G). \end{aligned}$$

If S is flat over R, then $\mathrm{Tor}_1^R(X, S) = 0$ for every $X \in \mathrm{Ob}(\underline{C}_{A|R})$. The preceding isomorphism is therefore natural in X, and gives

$$\check{S}\,\underline{\check{H}}^\bullet(G) \xrightarrow{\sim} \underline{\check{H}}^\bullet(\check{S}G).$$

(b) Put

$$P^{(m)} = \overbrace{P \times_B \cdots \times_B P}^{m+1}.$$

For $m = 0$ there is nothing to prove, since P is flat over R. Suppose inductively that $\mathrm{Tor}_i^R(P^{(m-1)}, S) = 0$ for $0 < i < n$. Since $P \longrightarrow B$ is surjective and $P^{(m)} = P \times_B P^{(m-1)}$, Lemma 3.5(ii) gives

$$\mathrm{Tor}_i^R(P^{(m)}, S) = 0 \qquad (0 < i < n).$$

□

Proposition 3.7.

(a) Let A and S be R-algebras. If

$$\mathrm{Tor}_i^R(A, S) = 0 \qquad (0 < i \leq n),$$

then, for every sheaf F on $\underline{C}_{A \otimes_R S|S}$, the canonical map

$$H^i(A \otimes_R S|S, F) \longrightarrow H^i(A|R, \widetilde{S}F)$$

is an isomorphism for every $i \leq n$. Here

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \qquad X \longmapsto X \otimes_R S.$$

(b) Let P be a polynomial algebra over R. Then, for every additive sheaf F on $\underline{C}_{A \otimes_R P|R}$, the canonical map

$$H^n(A \otimes_R P|R, F) \longrightarrow H^n(A|R, F)$$

is an isomorphism for every $n > 0$.

(c) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every additive sheaf F on $\underline{C}_{A|S}$, there is an exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(A|R, F_R) &\longrightarrow H^0(A|S, F) \longrightarrow H^0(R|S, F) \longrightarrow H^1(A|R, F_R) \\ &\longrightarrow H^1(A|S, F) \longrightarrow H^1(R|S, F) \longrightarrow \cdots \longrightarrow H^n(A|R, F_R) \\ &\longrightarrow H^n(A|S, F) \longrightarrow H^n(R|S, F) \longrightarrow H^{n+1}(A|R, F_R) \longrightarrow \cdots, \end{aligned}$$

where F_R is the sheaf on $\underline{C}_{A|R}$ defined by

$$F_R(X) = \ker\big(F(X) \longrightarrow F(R)\big)$$

for every $X \in \mathrm{Ob}(\underline{C}_{A|R})$, via the forgetful functor $\underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$.

Proof.
(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A\otimes_R S|S}, \qquad S(X) = X \otimes_R S.$$

Since S sends polynomial algebras over R to polynomial algebras over S, the functor

$$\widetilde{S} : \underline{\widetilde{C}}_{A\otimes_R S|S} \longrightarrow \underline{\widetilde{C}}_{A|R}$$

is exact by 3.3(c). We therefore obtain the canonical map

$$\check{S}\, \underline{H}^\bullet_{A\otimes_R S|S}(F) \longrightarrow \underline{H}^\bullet_{A|R}(\widetilde{S}F)$$

and the following commutative diagram of spectral sequences:

$$\begin{array}{ccc} E_S^{p,q}(S(X)) = \check{H}^p\big(X \otimes_R S|S, \underline{H}^q_{A\otimes_R S|S}(F)\big) & \Longrightarrow & H^{p+q}(X \otimes_R S|S, F) \\ \downarrow & & \\ \check{H}^p\big(X|R, \check{S}\, \underline{H}^q_{A\otimes_R S|S}(F)\big) & & \downarrow \\ \downarrow & & \\ E_R^{p,q}(X) = \check{H}^p\big(X|R, \underline{H}^q_{A|R}(\widetilde{S}F)\big) & \Longrightarrow & H^{p+q}(X|R, \widetilde{S}F). \end{array}$$

By 3.6(a), the map

$$E_S^{p,q}(S(X)) \longrightarrow \check{H}^p\big(X|R, \check{S}\, \underline{H}^q_{A\otimes_R S|S}(F)\big)$$

is an isomorphism whenever $\mathrm{Tor}_1^R(X, S) = 0$. Suppose inductively that

$$[\check{S}\, \underline{H}^q_{A\otimes_R S|S}(F)](X) \longrightarrow [\underline{H}^q_{A|R}(\widetilde{S}F)](X)$$

is an isomorphism for every $q < n$ and every $X \in \mathrm{Ob}(\underline{C}_{A|R})$ satisfying

$$\mathrm{Tor}_i^R(X, S) = 0 \qquad (0 < i \leq n).$$

It follows from 3.6(b) that $E_S^{p,q}(S(X)) \longrightarrow E_R^{p,q}(X)$ is an isomorphism for every $q < n$ and every p, under the same Tor hypothesis. Moreover, $E_S^{0,n}(S(X)) \longrightarrow E_R^{0,n}(X)$ is an isomorphism, since both sides are zero. Hence

$$H^n(X \otimes_R S|S, F) \longrightarrow H^n(X|R, \widetilde{S}F)$$

is an isomorphism whenever $\mathrm{Tor}_i^R(X, S) = 0$ for $0 < i \leq n$. Equivalently,

$$[\check{S}\, \underline{H}^i_{A\otimes_R S|S}(F)](X) \longrightarrow [\underline{H}^i_{A|R}(\widetilde{S}F)](X)$$

is an isomorphism for every $i \leq n$ under that hypothesis.

(b) Let

$$f : \underline{C}_{A|R} \longrightarrow \underline{C}_{A\otimes_R P|R}, \qquad f(X) = X \otimes_R P.$$

Choose a surjective arrow $P' \longrightarrow X$ in $\underline{C}_{A|R}$, with P' polynomial over R. Then

$$P' \otimes_R P \longrightarrow X \otimes_R P$$

is a surjective arrow in $\underline{C}_{A\otimes_R P|R}$, and $P' \otimes_R P$ is again polynomial over R. Since P is flat over R, 3.5(i) gives

$$\begin{aligned} \check{H}^\bullet(X|R, \check{f}G) &= H^\bullet(\{P' \longrightarrow X\}, \check{f}G) \\ &= H^\bullet(\{P' \otimes_R P \longrightarrow X \otimes_R P\}, G) \\ &= \check{H}^\bullet(X \otimes_R P|R, G) \end{aligned}$$

for every $G \in \mathrm{Ob}(\overset{\vee}{\underline{C}}_{A\otimes_R P|R})$. Thus $\widetilde{f}$ sends acyclic sheaves to acyclic sheaves. On the other hand, f sends polynomial algebras over R to polynomial algebras over R, so

$$\widetilde{f} : \widetilde{\underline{C}}_{A\otimes_R P|R} \longrightarrow \widetilde{\underline{C}}_{A|R}$$

is exact by 3.3(c). It follows that

$$H^\bullet(A \otimes_R P|R, F) \longrightarrow H^\bullet(A|R, \widetilde{f}F)$$

is an isomorphism for every $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A\otimes_R P|R})$.

For each $X \in \mathrm{Ob}(\underline{C}_{A|R})$ there is a canonical map

$$(\widetilde{f}F)(X) = F(X \otimes_R P) \longrightarrow F(X) \oplus F(P),$$

and hence a canonical map $\widetilde{f}F \longrightarrow F \oplus F(P)$, where $F(P)$ denotes the constant sheaf and F on the right denotes the restriction of F to $\underline{C}_{A|R}$. If F is additive, this map is an isomorphism. Therefore, for every $n > 0$,

$$H^n(A|R, \widetilde{f}F) \xrightarrow{\sim} H^n(A|R, F \oplus F(P)) = H^n(A|R, F),$$

because the constant sheaf $F(P)$ is flasque. This proves (b).

(c) This argument was communicated by Rinehart. Consider the forgetful functor $\beta : \underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$. There is a spectral sequence

$$E^{p,q} = H^p\big(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)\big) \Longrightarrow H^{p+q}(A|S, F),$$

where

$$\beta^{\#} = G_{A|R} \circ \beta^{\vee}, \qquad G_{A|R} : \overset{\vee}{\underline{C}}_{A|R} \longrightarrow \widetilde{\underline{C}}_{A|R}$$

is the associated-sheaf functor.

Suppose now that F is additive. We claim that $\beta^{\#}\underline{H}^q_{A|S}(F)$ is a constant sheaf for every $q > 0$. Indeed, if P is a polynomial algebra over R, then $P = R \otimes_S P'$ for a polynomial algebra P' over S, and

$$[\beta^{\#}\underline{H}^q_{A|S}(F)](P) = H^q(P|S, F) = H^q(R \otimes_S P'|S, F) \longrightarrow H^q(R|S, F)$$

is an isomorphism for $q > 0$ by (b), and is surjective for $q = 0$. By 3.3(c), the canonical map

$$\beta^{\#}\underline{H}^q_{A|S}(F) \longrightarrow H^q(R|S, F)$$

to the corresponding constant sheaf is therefore an isomorphism for $q > 0$ and a surjection for $q = 0$. Hence $E^{p,q} = 0$ whenever $p, q > 0$, and the spectral sequence yields the exact sequence

$$(*) \quad 0 \longrightarrow E^{1,0} \longrightarrow H^1 \longrightarrow E^{0,1} \longrightarrow E^{2,0} \longrightarrow H^2 \longrightarrow E^{0,2} \longrightarrow E^{3,0} \longrightarrow H^3 \longrightarrow \cdots .$$

Here

$$E^{0,q} = [\beta^{\#}\underline{H}^q_{A|S}(F)](A) \xrightarrow{\sim} H^q(R|S,F) \quad (q>0), \qquad E^{p,0} = H^p(A|R,\widetilde{\beta}F).$$

On the other hand, the exact sequence

$$0 \longrightarrow F_R \longrightarrow \widetilde{\beta}F \longrightarrow F(R) \longrightarrow 0$$

gives the exact diagram

$$\begin{array}{ccccccccc} 0 \to H^0(A|R,F_R) & \to & H^0(A|R,\widetilde{\beta}F) & \to & H^0(A|R,F(R)) & \to & H^1(A|R,F_R) & \to & H^1(A|R,\widetilde{\beta}F) \to 0 \\ & & \| & & \| & & & & \\ & & H^0(A|S,F) & & H^0(R|S,F) & & & & \end{array}$$

labelled (∗∗) in the source. Combining (∗) and (∗∗) gives the desired exact sequence. □

Corollary 3.8.

(a) Let A and B be R-algebras such that

$$\mathrm{Tor}^R_i(A,B) = 0 \qquad (0 < i \leq n).$$

Then, for every additive sheaf $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$, the canonical map

$$H^i(A\otimes_R B|R,F) \longrightarrow H^i(A|R,F) \oplus H^i(B|R,F)$$

is an isomorphism for every $i \leq n$.

(b) Let A be an R-algebra such that $\mathrm{Tor}^R_i(A,A) = 0$ for every $i > 0$. If the map $A\otimes_R A \longrightarrow A$ is an isomorphism, then, for every additive sheaf $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A|R})$, one has

$$H^i(A|R,F) = 0$$

for every i.

Proof.

(a) Let $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$ be additive. Applied to $R \longrightarrow A \longrightarrow A\otimes_R B$, Proposition 3.7(c) gives the exact sequence

$$(*) \quad \cdots \longrightarrow H^i(A\otimes_R B|A,F_A) \longrightarrow H^i(A\otimes_R B|R,F) \longrightarrow H^i(A|R,F) \longrightarrow \cdots .$$

Consider the commutative diagram

$$\begin{array}{ccc} H^i(A\otimes_R B|A,F_A) & \longrightarrow & H^i(A\otimes_R B|R,F) \\ \downarrow & & \downarrow \\ H^i(B|R,\widetilde{A}F_A) & \xrightarrow{\approx} & H^i(B|R,F). \end{array}$$

Here

$$A : \underline{C}_{B|R} \longrightarrow \underline{C}_{A\otimes_R B|A}, \qquad X \longmapsto A\otimes_R X,$$

and the bottom map is induced by $\widetilde{A}F_A \longrightarrow F$, which is an isomorphism because F is additive. By functoriality, the commutativity of the diagram reduces to the case $i = 0$.

If $\mathrm{Tor}^R_i(A,B) = 0$ for $0 < i \leq n$, then Proposition 3.7(a) shows that the left vertical map is an isomorphism for $i \leq n$. Consequently, both composites

$$H^i(A\otimes_R B|A,F_A) \longrightarrow H^i(A\otimes_R B|R,F) \longrightarrow H^i(B|R,F)$$

and, by symmetry,

$$H^i(A \otimes_R B|B, F_B) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F)$$

are isomorphisms for $i \leq n$. It follows that $(*)$ breaks into split short exact sequences

$$0 \longrightarrow H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \longrightarrow 0$$

for $i \leq n$, and hence that

$$H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism in that range.

(b) By (a), the map

$$H^i(A \otimes_R A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F)$$

is an isomorphism for every i. On the other hand, the assumed isomorphism $A \otimes_R A \longrightarrow A$ induces an isomorphism

$$H^i(A|R, F) \longrightarrow H^i(A \otimes_R A|R, F).$$

Their composite is the diagonal map

$$H^i(A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F).$$

Since it is an isomorphism, $H^i(A|R, F) = 0$ for every i. □

Corollary 3.9. Let S be a multiplicative subset of an R-algebra A. Suppose that

$$F \in \mathrm{Ob}(\widetilde{\underline{C}}_{S^{-1}A|R})$$

is additive and that F_A is also additive. Then the canonical map

$$H^i(S^{-1}A|R, F) \longrightarrow H^i(A|R, F)$$

is an isomorphism for every i. Here $F_A \in \mathrm{Ob}(\widetilde{\underline{C}}_{S^{-1}A|A})$ is defined by

$$F_A(X) = \ker\big(F(X) \longrightarrow F(A)\big)$$

for every $X \in \mathrm{Ob}(\underline{C}_{S^{-1}A|A})$.

Proof. Applied to $R \longrightarrow A \longrightarrow S^{-1}A$, Proposition 3.7(c) gives the exact sequence

$$\cdots \longrightarrow H^n(S^{-1}A|A, F_A) \longrightarrow H^n(S^{-1}A|R, F) \longrightarrow H^n(A|R, F) \longrightarrow \cdots.$$

The map

$$S^{-1}A \otimes_A S^{-1}A \longrightarrow S^{-1}A$$

is an isomorphism, and F_A is additive by hypothesis. Corollary 3.8(b) therefore gives $H^n(S^{-1}A|A, F_A) = 0$ for every n. The middle map in the exact sequence is consequently an isomorphism for every n. □

Let A be an R-algebra. An A-module M defines an additive sheaf

$$\mathrm{Der}_R(-, M) \in \mathrm{Ob}(\widetilde{\underline{C}}_{A|R}), \qquad X \longmapsto \mathrm{Der}_R(X, M).$$

We make the following definition.

Definition 3.10. Let A be an R-algebra. For every A-module M, set

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)).$$

We now compute the groups of low degree, namely $D^i(A|R,M)$ for $i=1,2$. Choose a surjective arrow $P \longrightarrow A$ in $\underline{C}_{A|R}$ with P a polynomial algebra over R, and write

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0.$$

Lemma 3.10.

$$D^1(A|R,M) = \mathrm{Coker}\big(\mathrm{Der}_R(P,M) \longrightarrow \mathrm{Hom}_P(I,M)\big),$$
$$D^2(A|R,M) = H^1\big(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))\big).$$

Proof. Consider the simplicial algebra

$$(*) \qquad \cdots P_A^{(2)} \rightrightarrows P_A^{(1)} \rightrightarrows P_A^{(0)},$$

where

$$P_A^{(n)} = \overbrace{P \times_A \cdots \times_A P}^{n+1}.$$

For every n there is an exact sequence

$$0 \longrightarrow I^{(n)} \longrightarrow P_A^{(n)} \longrightarrow A \longrightarrow 0, \qquad I^{(n)} = \overbrace{I \times \cdots \times I}^{n+1}.$$

Put $\overline{I} = I/I^2$, and let Δ denote the diagonal map. It is straightforward to verify that, for every n, the sequence

$$\text{(a)} \qquad 0 \longrightarrow \mathrm{Hom}_A\big(\overline{I}^{(n)}/\Delta(\overline{I}), M\big) \longrightarrow \mathrm{Der}_R(P_A^{(n)}, M) \xrightarrow{\mathrm{Der}(\Delta,M)} \mathrm{Der}_R(P,M) \longrightarrow 0$$

is exact. We also have the exact sequence

$$\text{(b)} \qquad 0 \longrightarrow \mathrm{Hom}_A\big(\overline{I}^{(n)}/\Delta(\overline{I}), M\big) \longrightarrow \mathrm{Hom}_A(\overline{I}^{(n)}, M) \xrightarrow{\mathrm{Hom}(\Delta,M)} \mathrm{Hom}_A(\overline{I}, M) \longrightarrow 0.$$

The sequences (a) and (b) yield the corresponding exact sequences of simplicial A-modules. Since $\{I \longrightarrow 0\}$ and $\{I \longrightarrow I\}$ admit sections,

$$H^i(\{I \longrightarrow 0\}, \mathrm{Hom}_A(-,M)) = H^i(\{I \longrightarrow I\}, \mathrm{Hom}_A(-,M)) = 0 \qquad (i>0).$$

It follows that

$$\begin{cases} 0 \longrightarrow \mathrm{Der}_R(A,M) \longrightarrow \mathrm{Der}_R(P,M) \longrightarrow \mathrm{Hom}_A(\overline{I},M) \\ \qquad\qquad \longrightarrow H^1(\{P \longrightarrow A\}, \mathrm{Der}_R(-,M)) \longrightarrow 0 & \text{is exact,} \\ H^i(\{P \longrightarrow A\}, \mathrm{Der}_R(-,M)) = 0 & (i>1). \end{cases}$$

By 3.1(b), therefore,

$$\begin{aligned} D^1(A|R,M) &= \check{H}^1(A|R, \mathrm{Der}_R(-,M)) \\ &= H^1(\{P \longrightarrow A\}, \mathrm{Der}_R(-,M)) \\ &= \mathrm{Coker}\big(\mathrm{Der}_R(P,M) \longrightarrow \mathrm{Hom}_A(\overline{I},M)\big) \\ &= \mathrm{Coker}\big(\mathrm{Der}_R(P,M) \longrightarrow \mathrm{Hom}_P(I,M)\big), \\ D^2(A|R,M) &= H^1\big(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))\big). \end{aligned}$$

□

Choose a free P-module F and a P-linear map $\gamma : F \longrightarrow P$ such that $\gamma(F) = I$. Associated with γ is a Koszul complex, denoted by K_I by abuse of notation. Lemma 3.11. There is an isomorphism

$$D^2(A|R,M) \simeq \mathrm{Coker}\big(H^1(K_I,M) \longrightarrow \mathrm{Hom}_A(H_1(K_I),M)\big).$$

Proof. We have

$$\begin{aligned} D^2(A|R,M) &= H^1\big(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))\big) \\ &= \ker\Big(H^1(P^{(1)}_A|R,M) \rightrightarrows H^1(P^{(2)}_A|R,M)\Big), \end{aligned}$$

because

$$[\underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))](P^{(0)}_A) = [\underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))](P) = 0.$$

Let

$$0 \longrightarrow K \longrightarrow F \xrightarrow{\gamma} I \longrightarrow 0$$

be exact. Write $P[F]$ for the symmetric algebra over P generated by F. Then $P[F]$ and $P[F] \otimes_P P[F]$ are polynomial algebras over R, and there are surjective maps

$$\begin{aligned} P[F] &\longrightarrow P^{(1)}_A = P \times_A P, & X &\longmapsto (0, \gamma(X)), \\ P[F] \otimes_P P[F] &\longrightarrow P^{(2)}_A = P \times_A P \times_A P, & X \otimes 1 &\longmapsto (0, \gamma(X), 0), \\ & & 1 \otimes X &\longmapsto (0, 0, \gamma(X)), \end{aligned}$$

for $X \in F$. Put

$$J = \ker(P[F] \longrightarrow P^{(1)}_A), \qquad J' = \ker(P[F] \otimes_P P[F] \longrightarrow P^{(2)}_A).$$

If F_γ denotes the ideal generated by $\{X - \gamma(X) \mid X \in F\}$, then

$$\begin{aligned} J &= (F) \cap (F_\gamma) = (K) + (F \cdot F_\gamma), \\ J' &= (F \otimes 1 + 1 \otimes F) \cap (F_\gamma \otimes 1 + 1 \otimes F) \cap (F \otimes 1 + 1 \otimes F_\gamma) \\ &= (K \otimes 1 + 1 \otimes K) + (F \cdot F_\gamma \otimes 1 + 1 \otimes F \cdot F_\gamma) + (F \otimes F). \end{aligned}$$

The three maps $\pi_i : P^{(2)}_A \longrightarrow P^{(1)}_A$ admit lifts $\widehat{\pi}_i : P[F] \otimes_P P[F] \longrightarrow P[F]$ given, for $X \in F$, by

$$\widehat{\pi}_0 : \begin{cases} X \otimes 1 \longmapsto \gamma(X) - X, \\ 1 \otimes X \longmapsto X, \end{cases} \qquad \widehat{\pi}_1 : \begin{cases} X \otimes 1 \longmapsto 0, \\ 1 \otimes X \longmapsto X, \end{cases} \qquad \widehat{\pi}_2 : \begin{cases} X \otimes 1 \longmapsto X, \\ 1 \otimes X \longmapsto 0. \end{cases}$$

A direct computation shows that $\widehat{\pi}_0 - \widehat{\pi}_1 + \widehat{\pi}_2$ annihilates

$$K \otimes 1 + 1 \otimes K + 1 \otimes F \cdot F_\gamma$$

and maps $F \cdot F_\gamma \otimes 1 + F \otimes F$ onto $F \cdot F_\gamma$. For every

$$D \in \mathrm{Der}_R(P[F] \otimes_P P[F], M)$$

one has $D(F \cdot F_\gamma \otimes 1 + F \otimes F) = 0$, because $I \cdot M = 0$. Consequently,

$$\begin{aligned} D^2(A|R,M) &= \ker\Big(D^1(P^{(1)}_A|R,M) \rightrightarrows D^1(P^{(2)}_A|R,M)\Big) \\ &= \{[f] \in D^1(P^{(1)}_A|R,M) \mid f(F \cdot F_\gamma) = 0\} \\ &= \mathrm{Coker}\left(\mathrm{Der}_R(P[F],M) \longrightarrow \mathrm{Hom}_{P[F]}\left(\frac{(F) \cap (F_\gamma)}{F \cdot F_\gamma}, M\right)\right) \\ &= \mathrm{Coker}\left(\mathrm{Der}_P(P[F],M) \longrightarrow \mathrm{Hom}_{P[F]}\left(\frac{(F) \cap (F_\gamma)}{(F)(F_\gamma)}, M\right)\right). \end{aligned}$$

The sequences

$$0 \longrightarrow (F) \longrightarrow P[F] \xrightarrow{P[0]} P \longrightarrow 0, \qquad 0 \longrightarrow (F_\gamma) \longrightarrow P[F] \xrightarrow{P[\gamma]} P_\gamma \longrightarrow 0$$

are exact. Hence

$$\frac{(F)\cap(F_\gamma)}{(F)(F_\gamma)} = \operatorname{Tor}_1^{P[F]}(P, P_\gamma) = H_1(K_I),$$

where K_I is the Koszul complex associated with $F \xrightarrow{\gamma} I \subset P$. We therefore obtain

$$\begin{aligned} D^2(A|R,M) &= \operatorname{Coker}\big(\operatorname{Hom}_P(F,M) \longrightarrow \operatorname{Hom}_P(H_1(K_I),M)\big) \\ &= \operatorname{Coker}\big(H^1(K_I,M) \longrightarrow \operatorname{Hom}_A(H_1(K_I),M)\big). \end{aligned}$$

□

We can now state the basic facts needed below, summarizing the results already established.

Theorem 3.12.

(1) If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A-modules, then the sequence

$$\begin{aligned} 0 \longrightarrow D^0(A|R,M') \longrightarrow D^0(A|R,M) \longrightarrow D^0(A|R,M'') \longrightarrow D^1(A|R,M') \\ \longrightarrow D^1(A|R,M) \longrightarrow D^1(A|R,M'') \longrightarrow \cdots \longrightarrow D^n(A|R,M') \\ \longrightarrow D^n(A|R,M) \longrightarrow D^n(A|R,M'') \longrightarrow \cdots \end{aligned}$$

is exact.

(2) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every A-module M, the sequence

$$\begin{aligned} 0 \longrightarrow D^0(A|R,M) \longrightarrow D^0(A|S,M) \longrightarrow D^0(R|S,M) \longrightarrow D^1(A|R,M) \\ \longrightarrow D^1(A|S,M) \longrightarrow D^1(R|S,M) \longrightarrow \cdots \longrightarrow D^n(A|R,M) \\ \longrightarrow D^n(A|S,M) \longrightarrow D^n(R|S,M) \longrightarrow \cdots \end{aligned}$$

is exact.

(3) Let A and S be R-algebras such that $\operatorname{Tor}_i^R(A,S) = 0$ for every $i > 0$. Then, for every $A \otimes_R S$-module M, the canonical map

$$D^i(A \otimes_R S|S,M) \longrightarrow D^i(A|R,M)$$

is an isomorphism for every i. In particular, if A is flat over R, then

$$D^i(A \otimes_R S|S,M) \xrightarrow{\sim} D^i(A|R,M)$$

for every i, every R-algebra S, and every $A \otimes_R S$-module M.

(4) Let S be a multiplicative subset of A. For every $S^{-1}A$-module N, the canonical map

$$D^i(S^{-1}A|R,N) \longrightarrow D^i(A|R,N)$$

is an isomorphism for every i. Consequently, for every A-module M, there is a canonical isomorphism

$$D^i(S^{-1}A|R,S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R,M)$$

for every i.

(5) Let A and B be R-algebras such that $\operatorname{Tor}_i^R(A,B) = 0$ for $0 < i \leq n$. Then, for every $A \otimes_R B$-module M, the canonical map

$$D^i(A \otimes_R B|R,M) \longrightarrow D^i(A|R,M) \oplus D^i(B|R,M)$$

is an isomorphism for $i \leq n$. In particular, suppose that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that the diagonal

$$\Delta : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$$

is an open immersion—for example, suppose that A is *étale* over R. Then

$$D^i(A|R, M) = 0$$

for every $i > 0$ and every A-module M.

(6) Let R be noetherian, let A be an R-algebra essentially of finite type, and let E be a flat A-module. Then the canonical map

$$E \otimes_A D^i(A|R, M) \xrightarrow{\sim} D^i(A|R, E \otimes_A M)$$

is an isomorphism for every $i \geq 0$ and every A-module M.

(7) If R is noetherian, A is an R-algebra essentially of finite type, and M is an A-module of finite type, then $D^i(A|R, M)$ is an A-module of finite type for every i.

(8) Let P be a polynomial algebra over R, and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. Then

$$D^0(A|R, M) = \mathrm{Der}_R(A, M),$$

$$\begin{aligned} D^1(A|R, M) &= \mathrm{Coker}\big(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\big) \\ &= \big\{ \text{ isomorphism classes of commutative } R\text{-algebra extensions of } A \text{ by } M \ \big\}, \\ D^2(A|R, M) &= \mathrm{Coker}\big(H^1(K_I, M) \longrightarrow \mathrm{Hom}_A(H_1(K_I), M)\big). \end{aligned}$$

Proof.

(1) If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A-modules, then, for every polynomial algebra P over R, the sequence

$$0 \longrightarrow \mathrm{Der}_R(P, M') \longrightarrow \mathrm{Der}_R(P, M) \longrightarrow \mathrm{Der}_R(P, M'') \longrightarrow 0$$

is exact. Hence, by 3.3(c),

$$0 \longrightarrow \mathrm{Der}_R(-, M') \longrightarrow \mathrm{Der}_R(-, M) \longrightarrow \mathrm{Der}_R(-, M'') \longrightarrow 0$$

is an exact sequence of sheaves.

(2) The functor $\mathrm{Der}_S(-, M)$ is an additive sheaf on $\underline{C}_{A|S}$, and

$$\mathrm{Der}_R(B, M) = \ker\big(\mathrm{Der}_S(B, M) \longrightarrow \mathrm{Der}_S(R, M)\big)$$

for every $B \in \mathrm{Ob}(\underline{C}_{A|R})$. The assertion therefore follows from 3.7(c).

(3) Let

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \qquad X \longmapsto X \otimes_R S.$$

Then $\widetilde{S}\,\mathrm{Der}_S(-, M) = \mathrm{Der}_R(-, M)$, so the assertion follows from 3.7(a).

(4) This follows from 3.9. Indeed, $F = \mathrm{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and $F_A \in \mathrm{Ob}(\widetilde{\underline{C}}_{S^{-1}A|A})$ is also additive, since

$$\begin{aligned} F_A(X) &= \ker\big(F(X) \longrightarrow F(A)\big) \\ &= \ker\big(\mathrm{Der}_R(X, N) \longrightarrow \mathrm{Der}_R(A, N)\big) \\ &= \mathrm{Der}_A(X, N). \end{aligned}$$

Thus $F_A = \mathrm{Der}_A(-, N)$.

(5) The first assertion follows from 3.8(a). Now suppose that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that

$$\Delta : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$$

is an open immersion. Let $\varphi : A \otimes_R A \longrightarrow A$ be the multiplication map. For every maximal ideal $\mathfrak{m}$ of A, the induced map

$$(A \otimes_R A)_{\varphi^{-1}(\mathfrak{m})} \longrightarrow A_{\mathfrak{m}}$$

is an isomorphism. The first assertion, together with 3.12(4), therefore implies that the diagonal map

$$\Delta_{\mathfrak{m}} : D^i(A|R, M)_{\mathfrak{m}} \longrightarrow D^i(A|R, M)_{\mathfrak{m}} \oplus D^i(A|R, M)_{\mathfrak{m}}$$

is an isomorphism for every maximal ideal $\mathfrak{m}$ of A. Consequently,

$$\Delta : D^i(A|R, M) \longrightarrow D^i(A|R, M) \oplus D^i(A|R, M)$$

is an isomorphism, and hence $D^i(A|R, M) = 0$ for every $i > 0$.

(6) In view of 3.12(4), we may assume that A is an R-algebra of finite type. Consider the morphism of sheaves

$$E \otimes_A \mathrm{Der}_R(-, M) \longrightarrow \mathrm{Der}_R(-, E \otimes_A M)$$

on $\widetilde{\underline{C}}_{A|R}$. Since E is flat over A, the map

$$E \otimes_A \mathrm{Der}_R(B, M) \xrightarrow{\sim} \mathrm{Der}_R(B, E \otimes_A M)$$

is an isomorphism whenever $B \in \mathrm{Ob}(\underline{C}_{A|R})$ is of finite type over R. By 3.3(b), it follows that

$$D^i\big(A|R, E \otimes_A \mathrm{Der}_R(-, M)\big) \longrightarrow D^i\big(A|R, \mathrm{Der}_R(-, E \otimes_A M)\big)$$

is an isomorphism. On the other hand, $E \otimes_A -$ is exact, and therefore

$$E \otimes_A D^i(A|R, M) = D^i\big(A|R, E \otimes_A \mathrm{Der}_R(-, M)\big).$$

This proves the assertion.

(7) This follows from 3.3(a).

(8) In view of 3.10 and 3.11, it suffices to show that

$$\mathrm{Coker}\big(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\big) = \mathrm{Exalcomm}(A|R, M),$$

where $\mathrm{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R-algebra extensions of A by M. Let

$$0 \longrightarrow M \longrightarrow A' \longrightarrow A \longrightarrow 0$$

be such an extension. Since P is a polynomial algebra over R, we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0. \end{array}$$

The map $I \longrightarrow M$ is determined by the extension A', up to the action of $\mathrm{Der}_R(P, M)$. We thus obtain a map

$$\Phi : \mathrm{Exalcomm}(A|R, M) \longrightarrow \mathrm{Coker}\big(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\big).$$

Conversely, a P-linear map $I \longrightarrow M$ yields, by pushout, an extension fitting into a diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0. \end{array}$$

Two P-linear maps $I \rightrightarrows M$ that differ by an element of $\mathrm{Der}_R(P, M)$ yield isomorphic extensions. Hence there is a map

$$\Psi : \mathrm{Coker}\big(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\big) \longrightarrow \mathrm{Exalcomm}(A|R, M).$$

It is straightforward to verify that Φ and Ψ are inverse to each other.

□

Corollary 3.13.

(a) The R-algebra A is formally smooth over R if and only if $D^1(A|R, M) = 0$ for every A-module M.

(b) Let R be noetherian and let A be an R-algebra of finite type. Then A is a relative complete intersection over R if and only if $D^2(A|R, M) = 0$ for every A-module M.

(c) Let R be noetherian and let A be an R-algebra of finite type. For every multiplicative subset S of A, there is a canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M) \qquad (i \geq 0).$$

(d) Let R be noetherian and let A be a local R-algebra essentially of finite type. Write $\widehat{A}$ for the $\mathfrak{m}$-adic completion of A, where $\mathfrak{m}$ is the maximal ideal of A. Then, for every A-module E of finite type, there is a canonical isomorphism

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \xrightarrow{\sim} \widehat{A} \otimes_A D^1(A|R, E).$$

Proof.
Part (a) follows immediately from $D^1(A|R, M) = \mathrm{Exalcomm}(A|R, M)$.
For (b), let P be a polynomial algebra over R in finitely many variables, and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. After localization, we may suppose that I is generated by a P-regular sequence. Then $H_1(K_I) = 0$, and hence

$$D^2(A|R, M) = \mathrm{Coker}\big(H^1(K_I, M) \longrightarrow \mathrm{Hom}_A(H_1(K_I), M)\big) = 0.$$

Conversely, suppose that $D^2(A|R, M) = 0$ for every A-module M. Choose an exact sequence

$$0 \longrightarrow K \longrightarrow F \longrightarrow I \longrightarrow 0,$$

where F is a free P-module of finite type. If

$$K_0 = \mathrm{Im}\big(\textstyle\bigwedge^2 F \longrightarrow \bigwedge^1 F\big),$$

then $H_1(K_I) = K/K_0$. Since $D^2(A|R, H_1(K_I)) = 0$, the map

$$H^1(K_I, H_1(K_I)) \longrightarrow \mathrm{Hom}_A(H_1(K_I), H_1(K_I))$$

is surjective. Consequently, the exact sequence

$$0 \longrightarrow K/K_0 \longrightarrow F/IF \longrightarrow I/I^2 \longrightarrow 0$$

splits as a sequence of A-modules. In particular, I/I^2 is projective. After localizing A once more, we may suppose that I/I^2 is free and that $F/IF \longrightarrow I/I^2$ is an isomorphism. The split exact sequence then gives $H_1(K_I) = K/K_0 = 0$. Thus I is generated by a P-regular sequence, so A is a relative complete intersection over R.

Part (c) follows immediately from 3.12(4) and (6).

For (d), 3.12(6) first gives canonical isomorphisms

$$D^i(A|R, \widehat{A} \otimes_A E) \xrightarrow{\sim} \widehat{A} \otimes_A D^i(A|R, E)$$

for every i. It therefore remains to prove that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is an isomorphism. We use the description of D^1 in terms of algebra extensions.

Let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow A' \xrightarrow{\varphi} \widehat{A} \longrightarrow 0$$

be an R-algebra extension whose restriction to A splits. Since $\widehat{A} \otimes_A E$ is an $\widehat{A}$-module of finite type, A' is a local R-algebra complete for its $\mathfrak{m}'$-adic topology, where $\mathfrak{m}'$ is the maximal ideal of A'. If $\psi : A \longrightarrow A'$ is an R-algebra map lifting the canonical map $A \longrightarrow \widehat{A}$, then ψ extends uniquely to a map $\widehat{\psi} : \widehat{A} \longrightarrow A'$ satisfying $\varphi \circ \widehat{\psi} = \mathrm{id}_{\widehat{A}}$. This proves that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is injective.

Now let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow B \xrightarrow{\varphi} A \longrightarrow 0$$

be an R-algebra extension, and put $\widehat{E} = \widehat{A} \otimes_A E$. We first claim that the topology on $\widehat{E}$ induced by the $\mathfrak{m}_B$-adic topology on B coincides with its topology as an $\widehat{A}$-module. Indeed, since A is essentially of finite type over R, there is a subalgebra $B_0 \subset B$ such that B_0 is a local R-algebra essentially of finite type and $\varphi(B_0) = A$. Put $E_0 = \widehat{E} \cap B_0$, and denote the maximal ideals of B, B_0, and A by $\mathfrak{m}_B$, $\mathfrak{m}_0$, and $\mathfrak{m}$, respectively. Then

$$\mathfrak{m}_B = \mathfrak{m}_0 + \widehat{E}, \qquad \mathfrak{m}_B^{n+1} = \mathfrak{m}_0^{n+1} + \mathfrak{m}^n \widehat{E}$$

for every n. Since B_0 is noetherian, the Artin–Rees theorem gives an integer $r > 0$ such that, for every $n \geq r$,

$$\begin{aligned} \widehat{E} \cap \mathfrak{m}_B^{n+1} &= \mathfrak{m}^n \widehat{E} + \widehat{E} \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + E_0 \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + \mathfrak{m}^{n+1-r}(E_0 \cap \mathfrak{m}_0^r) \subset \mathfrak{m}^{n+1-r} \widehat{E}. \end{aligned}$$

This proves the assertion about the two topologies. In particular, $\widehat{E}$ is complete for the topology induced by the $\mathfrak{m}_B$-adic topology on B, and hence completion gives an exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \widehat{E} & \longrightarrow & B & \longrightarrow & A & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \widehat{E} & \longrightarrow & \widehat{B} & \longrightarrow & \widehat{A} & \longrightarrow & 0. \end{array}$$

Here $\widehat{B}$ denotes the $\mathfrak{m}_B$-adic completion of B. Thus

$$D^1(\widehat{A}|R,\widehat{E}) \longrightarrow D^1(A|R,\widehat{E})$$

is surjective, and the proof is complete. □

Corollary 3.14. Let R be noetherian, and let A be an R-algebra of finite type that is smooth everywhere except at one closed point $x \in \mathrm{Spec}(A)$. Then there are isomorphisms

$$D^1(A|R,A) \simeq D^1(A_x|R,A_x) \simeq D^1(\widehat{A}_x|R,\widehat{A}_x).$$

Proof. Since A is smooth over R away from x, $D^1(A|R,A)$ is an A-module of finite type supported on $\{x\}$. Hence

$$D^1(A|R,A) = D^1(A|R,A)_x = D^1(\widehat{A|R,A})_x.$$

The assertion now follows from 3.13. □

Remark 3.15. Let R be a topological ring and A a topological R-algebra. For every A-module E, define

$$D^i_{\mathrm{top}}(A|R,E) = \varinjlim_{\alpha} D^i(A_\alpha|R_\alpha, A_\alpha \otimes_A E),$$

where $A_\alpha = A/J_\alpha$, $R_\alpha = R/I_\alpha$, and J_α and I_α are open in A and R, respectively. Thus A is formally smooth as a topological R-algebra precisely when $D^1_{\mathrm{top}}(A|R,E) = 0$ for every A-module E.

Now let X be a scheme over a ring R, and let $\underline{E}$ be a quasi-coherent $\mathcal{O}_X$-module. We define sheaves

$$\underline{D}^i_{X|R}(\underline{E}) = \underline{D}^i(X|R,\underline{E})$$

on X from the presheaves on affine open subsets $U \subset X$ given by

$$\big[\underline{D}^i_{X|R}(\underline{E})\big](U) = D^i\big(\Gamma(U,\mathcal{O}_X)|R, \Gamma(U,\underline{E})\big).$$

If R is noetherian and X is of finite type over R, the localization property 3.13(c) shows that these presheaves canonically form quasi-coherent $\mathcal{O}_X$-modules.

When no confusion is possible, we abbreviate

$$\underline{D}^i_{X|R} = \underline{D}^i_{X|R}(\mathcal{O}_X).$$

All the assertions of 3.13 have corresponding formulations for the quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$. We leave those formulations to the reader and record below the properties most relevant to our purpose.

Corollary 3.15. Let R be noetherian and let X be an R-scheme of finite type.

(a) For every coherent $\mathcal{O}_X$-module $\underline{E}$, $\underline{D}^i_{X|R}(\underline{E})$ is again coherent for every i. Moreover,

$$\mathrm{Supp}\, \underline{D}^i_{X|R}(\underline{E}) \subset X^{\mathrm{Sing}} \qquad (i>0),$$

where

$$X^{\mathrm{Sing}} = \{x \in X \mid X \text{ is not smooth over } R \text{ at } x\}.$$

(b) If X is flat over R, then for every R-algebra S there is a canonical isomorphism

$$\underline{D}^i_{X\otimes_R S\,|\,S}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$$

for every i.

(c) Suppose that X is a relative complete intersection over R. For every surjective ring homomorphism $S \longrightarrow R$, there is an exact sequence

$$0 \longrightarrow \underline{D}^1_{X|R}(\underline{E}) \longrightarrow \underline{D}^1_{X|S}(\underline{E}) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_X}\big((I/I^2) \otimes_R \mathcal{O}_X, \underline{E}\big) \longrightarrow 0,$$

where $I = \ker(S \longrightarrow R)$. If, moreover, R is local with residue field k and X is flat over R, then

$$0 \longrightarrow \underline{D}^1_{X_0|k}(\mathcal{O}_{X_0}) \longrightarrow \underline{D}^1_{X|S}(\mathcal{O}_{X_0}) \longrightarrow D^1(R|S,k) \otimes_k \mathcal{O}_{X_0} \longrightarrow 0,$$

where $X_0 = X \otimes_R k$.

Proof. Part (a) follows from the finiteness and low-degree descriptions in 3.12, while (b) follows from 3.12(3).

For (c), because X is a relative complete intersection over R, we have $\underline{D}^i_{X|R}(\underline{E}) = 0$ for every $i \geq 2$. The transitivity sequence 3.12(2) therefore gives an exact sequence

$$0 \longrightarrow \underline{D}^0_{X|R}(\underline{E}) \longrightarrow \underline{D}^0_{X|S}(\underline{E}) \longrightarrow \underline{D}^0_{R|S}(\underline{E}) \longrightarrow \underline{D}^1_{X|R}(\underline{E}) \longrightarrow \underline{D}^1_{X|S}(\underline{E}) \longrightarrow \underline{D}^1_{R|S}(\underline{E}) \longrightarrow 0,$$

where the quasi-coherent sheaves $\underline{D}^i_{R|S}(\underline{E})$ on X are defined on every affine open subset $U \subset X$ by

$$\big[\underline{D}^i_{R|S}(\underline{E})\big](U) = D^i(R|S, \underline{E}(U)).$$

Since $S \longrightarrow R$ is surjective, $\underline{D}^0_{R|S}(\underline{E}) = 0$; and the low-degree description in 3.12(8) gives

$$\underline{D}^1_{R|S}(\underline{E}) = \underline{\mathrm{Hom}}_{\mathcal{O}_X}\big((I/I^2) \otimes_R \mathcal{O}_X, \underline{E}\big).$$

This proves the first exact sequence.

If X is flat over R, then, for every quasi-coherent $\mathcal{O}_{X_0}$-module $\underline{E}$, there are canonical isomorphisms

$$\underline{D}^i_{X_0|k}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$$

for every i. In particular,

$$\underline{D}^1_{X_0|k}(\mathcal{O}_{X_0}) \xrightarrow{\sim} \underline{D}^1_{X|R}(\mathcal{O}_{X_0}),$$

from which the second exact sequence follows. □

Remark 3.16. The quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$ defined above should occur as the terms $E_2^{0,i}$ of a spectral sequence relating the local theory to the global theory. L. Illusie has obtained such a theory in *Lecture Notes in Mathematics* 239. His method uses "homotopical algebra," generalizing the methods of M. André and D. Quillen, rather than "cohomological algebra." It seems that the approach taken here can be globalized by means of Grothendieck cohomology on the category of schemes, equipped with an appropriate topology generalizing the one on the affine category.

4. Formal moduli of deformations

Among the natural cofibered categories whose isomorphism classes of objects are generally not prorepresentable but nevertheless admit a formal versal object is the cofibered category of deformations. We return to the situation and notation of section 1. Thus, throughout this section, Λ is a fixed complete noetherian local ring with residue field k, and $\underline{C}_\Lambda$ is the category of artinian local Λ-algebras with residue field k.

Let X_0 be a fixed scheme over k. By an (infinitesimal) deformation of X_0 we mean a pair (R, X), where $R \in \mathrm{Ob}(\underline{C}_\Lambda)$ and X is a flat R-scheme, together with a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow{\scriptstyle \text{flat}} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that the induced morphism

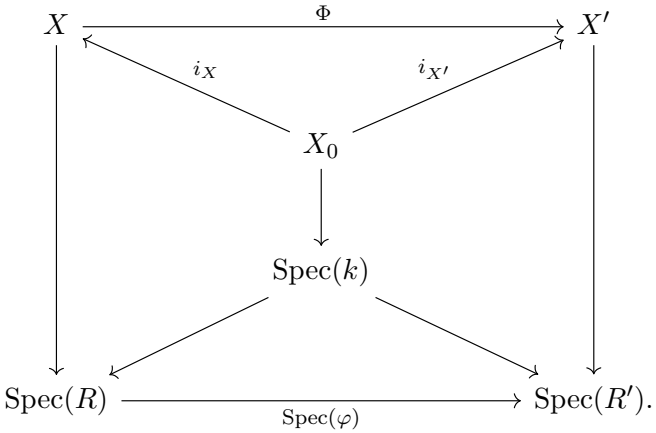
$$X_0 \longrightarrow X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$$

is an isomorphism. Since R is artinian local, $i_X : X_0 \longrightarrow X$ is necessarily a closed immersion and a homeomorphism on the underlying topological spaces.

If (R, X) and (R', X') are deformations of X_0 over R and R', respectively, an arrow

$$(R', X') \longrightarrow (R, X)$$

consists of a map $\varphi : R' \longrightarrow R$ in $\underline{C}_\Lambda$ and a morphism of schemes $\Phi : X \longrightarrow X'$ fitting into a commutative diagram



This defines the category $\underline{M}_{X_0}$ of infinitesimal deformations of X_0. It is equipped with the functor

$$p : \underline{M}_{X_0} \longrightarrow \underline{C}_\Lambda, \qquad (R, X) \longmapsto R.$$

Let $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$ and let $\varphi : R \longrightarrow S$ be a map in $\underline{C}_\Lambda$. There is a commutative diagram

$$\begin{array}{ccc} X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(S) & \xrightarrow{p_1} & X \\ \downarrow{\scriptstyle \text{flat}} & & \downarrow{\scriptstyle \text{flat}} \\ \mathrm{Spec}(S) & \xrightarrow[\mathrm{Spec}(\varphi)]{} & \mathrm{Spec}(R), \end{array}$$

and the φ-arrow

$$(\varphi, p_1) : (R, X) \longrightarrow \big(S, X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(S)\big)$$

is cocartesian in $\underline{M}_{X_0}$. Hence $\underline{M}_{X_0}$ is a cofibered category over $\underline{C}_\Lambda$. We record the following facts.

4.1(a). Let

$$(1_R, \Phi) : (R, X_1) \longrightarrow (R, X_2)$$

be an arrow in $\underline{M}_{X_0}$. Thus $\Phi : X_2 \longrightarrow X_1$ is an R-morphism whose reduction

$$X_2 \otimes_R k \longrightarrow X_1 \otimes_R k$$

is an isomorphism. Since R is artinian and the X_i are flat over R, Φ is an isomorphism. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b). Suppose that X_0 is affine, say $X_0 = \mathrm{Spec}(A_0)$. Then every $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$ has X affine as well. Consequently, for every $R \in \mathrm{Ob}(\underline{C}_\Lambda)$, taking global sections identifies $\underline{M}_{X_0}(R)$ with the following category. An object is a flat R-algebra A together with an R-algebra map $j_A : A \longrightarrow A_0$ inducing an isomorphism

$$A \otimes_R k \xrightarrow{\sim} A_0.$$

A morphism $\Phi : A_1 \longrightarrow A_2$ is an R-algebra map for which the diagram

$$\begin{array}{ccc} A_1 & \xrightarrow{\Phi} & A_2 \\ {\scriptstyle j_{A_1}}\downarrow & & \downarrow{\scriptstyle j_{A_2}} \\ A_0 & = & A_0 \end{array}$$

commutes.

4.1(c). Consider the cofibered category $\widehat{\underline{M}}_{X_0}$ in groupoids over $\widehat{\underline{C}}_\Lambda$. By definition, for every $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$, the category $\widehat{\underline{M}}_{X_0}(R)$ is canonically equivalent to the category of formal schemes $\mathfrak{X}$, adic over $\mathrm{Spf}(R)$, such that

$$\mathfrak{X}_n = \mathfrak{X} \otimes_R R/\mathfrak{m}^{n+1}$$

is flat over $R/\mathfrak{m}^{n+1}$ for every $n \geq 0$. If X_0 is noetherian, this is equivalent to requiring $\mathfrak{X}$ to be flat over $\mathrm{Spf}(R)$. Thus, in that case, $\widehat{\underline{M}}_{X_0}$ is canonically equivalent over $\widehat{\underline{C}}_\Lambda$ to the category of deformations of X_0 over the objects of $\widehat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d). Let $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$, so that there is a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow{\scriptstyle \text{flat}} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

whose induced map $X_0 \longrightarrow X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$ is an isomorphism.

Let $U_0 \subset X_0$ be an open subscheme. Since $i_X : X_0 \longrightarrow X$ is a closed immersion and a homeomorphism on underlying spaces, restriction gives a commutative diagram

$$\begin{array}{ccc} U_0 & \xrightarrow{i_X|_{U_0}} & X|_{U_0} \\ \downarrow & & \downarrow{\scriptstyle \text{flat}} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R). \end{array}$$

The induced map

$$U_0 \longrightarrow (X|_{U_0}) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k) = (X \otimes_R k)|_{U_0}$$

is an isomorphism, so $(R, X|_{U_0})$ is a deformation of U_0 over R. Moreover, every arrow $(\varphi, \Phi) : (R, X) \longrightarrow (R', X')$ in $\underline{M}_{X_0}$ restricts to an arrow

$$(\varphi, \Phi|_{U_0}) : (R, X|_{U_0}) \longrightarrow (R', X'|_{U_0})$$

in $\underline{M}_{U_0}$. We therefore have a restriction functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0}$$

over $\underline{C}_\Lambda$.

One may likewise localize and complete at a point. Let $x_0 \in X_0$. If $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$, then $\mathrm{Spec}(\mathcal{O}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\mathcal{O}_{X_0,x_0})$ over R, while $\mathrm{Spec}(\widehat{\mathcal{O}}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})$ over R; the hats denote completion for the topology defined by powers of the maximal ideal. Thus there are functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})}.$$

We first prove that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. Consider a diagram

$$\begin{array}{ccc} & & R' \\ & & \downarrow \\ R & \xrightarrow[\text{surjective}]{} & \bar{R} \end{array}$$

in $\underline{C}_\Lambda$, and let E and E' be flat modules over R and R', respectively. Then, for every R'-homomorphism

$$E' \longrightarrow \bar{E} = E \otimes_R \bar{R},$$

the fiber product $E \times_{\bar{E}} E'$ is flat as a module over $R \times_{\bar{R}} R'$.

Proof. Let

$$0 \longrightarrow I \longrightarrow R \longrightarrow \bar{R} \longrightarrow 0$$

be exact, so that

$$0 \longrightarrow I \longrightarrow R \times_{\bar{R}} R' \longrightarrow R' \longrightarrow 0$$

is exact. Since E is flat over R,

$$0 \longrightarrow I \otimes_R E \longrightarrow E \longrightarrow \bar{E} \longrightarrow 0$$

is exact, and hence we obtain the exact sequence

$$(*) \qquad 0 \longrightarrow I \otimes_R E \longrightarrow E \times_{\bar{E}} E' \longrightarrow E' \longrightarrow 0.$$

Set $S = R \times_{\bar{R}} R'$. Since I is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S-module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R'-module and

$$\mathrm{Tor}_1^S(E \times_{\bar{E}} E', R') = 0.$$

But

$$I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E'), \qquad E' = R' \otimes_S (E \times_{\bar{E}} E'),$$

and therefore the exact sequence $(*)$ may be rewritten as

$$0 \longrightarrow I \otimes_S (E \times_{\bar{E}} E') \longrightarrow E \times_{\bar{E}} E' \longrightarrow R' \otimes_S (E \times_{\bar{E}} E') \longrightarrow 0.$$

Thus

$$\mathrm{Tor}_1^S(R', E \times_{\bar{E}} E') = 0, \qquad R' \otimes_S (E \times_{\bar{E}} E') = E',$$

and E' is flat over R'. It follows that $E \times_{\bar{E}} E'$ is flat over S. □

Corollary 4.3. Given a diagram

$$\begin{array}{ccc} (R,X) & & (R',X') \\ & {\scriptstyle(\varphi,\Phi)}\searrow \quad \swarrow{\scriptstyle(\varphi',\Phi')} & \\ & (\bar{R},\bar{X}) & \end{array}$$

in $\underline{M}_{X_0}$, the amalgamated sum $X \amalg_{\bar{X}} X'$ (in the category of schemes) is a flat scheme over $R \times_{\bar{R}} R'$, provided that $\varphi : R \longrightarrow \bar{R}$ is surjective. Consequently, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Proof. For every affine open $U_0 \subset X_0$, by definition we have

$$(X \amalg_{\bar{X}} X')|_{U_0} \simeq \mathrm{Spec}\Big(\Gamma(U_0,X) \times_{\Gamma(U_0,\bar{X})} \Gamma(U_0,X')\Big).$$

Consequently, if $\varphi : R \longrightarrow \bar{R}$ is surjective, $X \amalg_{\bar{X}} X'$ is flat over $R \times_{\bar{R}} R'$ by Lemma 4.2. Moreover, the morphism

$$i_X \amalg_{i_{\bar{X}}} i_{X'} : X_0 \longrightarrow X \amalg_{\bar{X}} X'$$

induces an isomorphism

$$X_0 \longrightarrow (X \amalg_{\bar{X}} X') \otimes_S k, \qquad S = R \times_{\bar{R}} R'.$$

Therefore

$$\big(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X'\big) \in \mathrm{Ob}(\underline{M}_{X_0}),$$

and, in $\underline{M}_{X_0}$, we have

$$\big(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X'\big) = (R,X) \times_{(\bar{R},\bar{X})} (R',X').$$

Thus $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$. □

Lemma 4.4. For every scheme X_0 over k, there is a canonical exact sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}),$$

where the $\underline{D}^i_{X_0}$ are as defined in section 3.

Proof. First suppose that X_0 is affine, and put $\Gamma(X_0, \mathcal{O}_{X_0}) = A_0$. As noted in 4.1(b), the category $\underline{M}_{X_0}(k[\varepsilon])$ is canonically equivalent to a category $\underline{E}$ whose objects are flat $k[\varepsilon]$-algebras A equipped with a $k[\varepsilon]$-algebra map $j_A : A \longrightarrow A_0$ inducing an isomorphism

$$k \otimes_{k[\varepsilon]} A \xrightarrow{\sim} A_0.$$

Equivalently, an object of $\underline{E}$ is a k-algebra A equipped with an exact sequence

$$0 \longrightarrow A_0 \xrightarrow{\varepsilon} A \xrightarrow{j_A} A_0 \longrightarrow 0$$

in which $\varepsilon(A_0)^2 = 0$ as an ideal of A.

An arrow $A_1 \longrightarrow A_2$ in $\underline{E}$, in the first description, is a $k[\varepsilon]$-algebra map $\Phi : A_1 \longrightarrow A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$. In the second description it is a k-algebra map $\Phi : A_1 \longrightarrow A_2$ yielding a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_0 & \xrightarrow{\varepsilon} & A_1 & \xrightarrow{j_{A_1}} & A_0 & \longrightarrow & 0 \\ & & \| & & \downarrow{\scriptstyle\Phi} & & \| & & \\ 0 & \longrightarrow & A_0 & \longrightarrow & A_2 & \xrightarrow{j_{A_2}} & A_0 & \longrightarrow & 0. \end{array}$$

It follows that, via $\Gamma(X_0, -)$, $[\underline{M}_{X_0}(k[\varepsilon])]$ is canonically isomorphic to the set of isomorphism classes of k-algebra extensions of A_0 by A_0. In other words, there is a canonical bijection

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(X_0, \underline{D}^1_{X_0})$$

which plainly respects the vector-space structures on both sides.

In general, for every affine open $U_0 \subset X_0$, restriction and the affine case give the composite map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{U_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(U_0, \underline{D}^1_{X_0}).$$

These maps give a canonical map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}).$$

Its kernel consists of the isomorphism classes of locally trivial deformations of X_0 over $k[\varepsilon]$, that is, of deformations locally isomorphic to $X_0 \otimes_k k[\varepsilon]$. The sheaf of automorphisms of $X_0 \otimes_k k[\varepsilon]$ inducing the identity on X_0 is

$$\underline{D}^0_{X_0} = \underline{\mathrm{Der}}_k(\mathcal{O}_{X_0}, \mathcal{O}_{X_0}).$$

The isomorphism classes of such locally trivial deformations are therefore canonically identified with $H^1(X_0, \underline{D}^0_{X_0})$, which gives the asserted exact sequence. □

Theorem 4.5. Let X_0 be a scheme of finite type over k for which either

(i) X_0 is proper over k, or

(ii) X_0 is affine and

$$X_0^{\mathrm{sing}} = \{x \in X_0 \mid X_0 \text{ is not smooth over } k \text{ at } x\}$$

is finite.

(a) The category $\underline{M}_{X_0}$ admits a formal versal object over $\underline{C}_\Lambda$, called a formal versal deformation of X_0. If $(R, \mathcal{X})$ is a formal versal deformation of X_0, then for every $(S, \mathcal{Y}) \in \mathrm{Ob}(\widehat{\underline{M}}_{X_0})$ there is a map

$$(\varphi, \Phi) : (R, \mathcal{X}) \longrightarrow (S, \mathcal{Y})$$

in $\widehat{\underline{M}}_{X_0}$ such that the following diagram is cartesian:

$$\begin{array}{ccc} \mathcal{Y} & \xrightarrow{\Phi} & \mathcal{X} \\ \downarrow & & \downarrow \\ \mathrm{Spf}(S) & \xrightarrow[\mathrm{Spf}(\varphi)]{} & \mathrm{Spf}(R). \end{array}$$

Furthermore, a formal versal deformation $(R, \mathcal{X})$ of X_0 prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \longrightarrow R$ in $\underline{C}_\Lambda$ and every deformation (R', X') of X_0 over R', the homomorphism

$$\mathrm{Aut}_{R'}(X' \mid X_0) \longrightarrow \mathrm{Aut}_R(X' \otimes_{R'} R \mid X_0)$$

is surjective.

(b) If X_0 is proper over k, then $\underline{M}_{X_0}$ is smoothly pseudo-representable over $\underline{C}_\Lambda$.

Proof. **(a)** By Corollary 4.3, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$. By 1.11 and 2.9(b), it therefore suffices to prove that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k.

Consider the exact sequence

$$(*) \qquad 0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}).$$

If X_0 is proper over k, both $H^1(X_0, \underline{D}^0_{X_0})$ and $H^0(X_0, \underline{D}^1_{X_0})$ are finite-dimensional, since the $\underline{D}^i_{X_0}$ are coherent for all $i \geq 0$ by 2.13(a).

Now suppose that X_0 is affine and X_0^{sing} is finite. The sheaf $\underline{D}^1_{X_0}$ is coherent and

$$\operatorname{Supp} \underline{D}^1_{X_0} \subset X_0^{\text{sing}}$$

by 2.13(a). Hence $H^0(X_0, \underline{D}^1_{X_0})$ is finite-dimensional over k, whereas $H^1(X_0, \underline{D}^0_{X_0}) = 0$ because X_0 is affine. In either case, the exact sequence $(*)$ shows that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k. The second assertion of part (a) follows from 2.7.

(b) If X_0 is proper, then

$$\operatorname{Aut}(X_0 \otimes_k k[\varepsilon] \mid X_0) = H^0(X_0, \underline{D}^0_{X_0})$$

is finite-dimensional, and the assertion follows from 2.13. □

Remark 4.6(a). Let X_0 be affine. A formal versal deformation of X_0 exists if and only if the support of $\underline{D}^1_{X_0}$ is a finite set. This does not imply, however, that all nonsmooth points of X_0 are isolated: there are affine schemes X_0 over k having a nonisolated nonsmooth point for which $\operatorname{Supp} \underline{D}^1_{X_0}$ is finite.

Remark 4.6(b). Henceforth, a deformation of X_0 means an object of $\underline{M}_{X_0}$. Where confusion is possible, such an object will be called an infinitesimal deformation of X_0.

Remark 4.6(c). Let X_0 be a scheme over k. We have fixed a complete noetherian local ring Λ with residue field k and considered deformations of X_0 over $\underline{C}_\Lambda$.

In practice one usually takes $\Lambda = k$, or $\Lambda = W(k)$ when k is a perfect field of positive characteristic. The inclusion $\underline{C}_k \subset \underline{C}_\Lambda$ is fully faithful, and $\underline{M}_{X_0}|_{\underline{C}_k}$ is simply the category of deformations of X_0 over the objects of $\underline{C}_k$. Consequently, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_\Lambda$, then

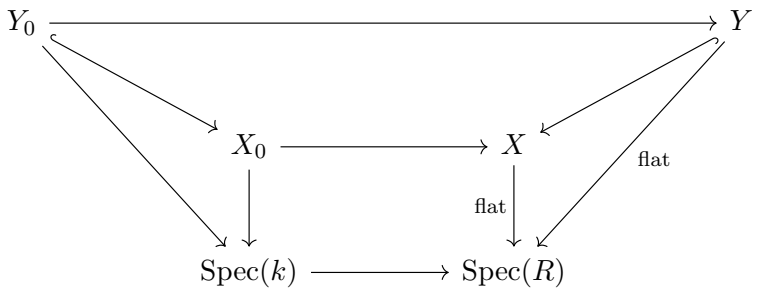
$$(k \otimes_\Lambda R,\ k \otimes_\Lambda \mathfrak{X})$$

is a formal versal deformation of X_0 over $\underline{C}_k$. It follows, for example, that

$$\dim_{\underline{C}_\Lambda} \underline{M}_{X_0} = \dim(k \otimes_\Lambda R)$$

does not depend on the choice of Λ, but only on X_0.

The preceding theorem admits many generalizations. For example, consider a pair $Y_0 \longrightarrow X_0$, where X_0 is a scheme of finite type over k and Y_0 is a fixed closed subscheme of X_0. For $R \in \operatorname{Ob}(\underline{C}_\Lambda)$, a deformation of (X_0, Y_0) over R is a commutative diagram



such that the induced map

$$(X_0, Y_0) \longrightarrow \big(X \times_{\operatorname{Spec}(R)} \operatorname{Spec}(k), Y \times_{\operatorname{Spec}(R)} \operatorname{Spec}(k)\big)$$

is an isomorphism. Defining morphisms

$$(R'; X', Y') \longrightarrow (R; X, Y)$$

of deformations of (X_0, Y_0) in the evident way gives a cofibered category $\underline{M}_{X_0, Y_0}$ in groupoids over $\underline{C}_\Lambda$. This construction is frequently used to rigidify automorphisms in deformation-moduli

problems. The argument of Corollary 4.3, based on Lemma 4.2, immediately shows that $\underline{M}_{X_0,Y_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Lemma 4.7. The sequence

$$0 \longrightarrow H^0\Big(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\Big) \longrightarrow [\underline{M}_{X_0,Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]$$

is exact, where $\underline{I}$ is the ideal sheaf defining Y_0 in X_0; thus

$$0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$$

is exact.

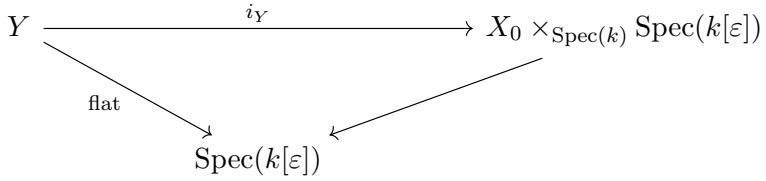
Proof. Let $\underline{B}$ be the full subcategory of $\underline{M}_{X_0,Y_0}(k[\varepsilon])$ whose objects have the form

$$\big(k[\varepsilon]; X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]), Y\big).$$

Then

$$[\underline{B}] = \ker\big([\underline{M}_{X_0,Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]\big)\,.$$

The category $\underline{B}$ is the groupoid of deformations of Y_0 over $k[\varepsilon]$ inside $X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon])$. Thus an object is a commutative diagram



whose special fiber is the fixed closed immersion i_{Y_0}. An arrow $(Y_1, i_{Y_1}) \longrightarrow (Y_2, i_{Y_2})$ is a $k[\varepsilon]$-morphism $\Phi : Y_1 \longrightarrow Y_2$ (necessarily an isomorphism) whose special fiber is id_{Y_0} and for which $i_{Y_2} \circ \Phi = i_{Y_1}$.

Equivalently, an object of $\underline{B}$ is a pair (Y, j_Y), where Y is a deformation of Y_0 over $k[\varepsilon]$ and $j_Y : Y \longrightarrow X_0$ is a k-morphism whose composite $Y_0 \longrightarrow Y \xrightarrow{j_Y} X_0$ is i_{Y_0}. An arrow $(Y_1, j_{Y_1}) \longrightarrow (Y_2, j_{Y_2})$ is a morphism $\Phi : Y_1 \longrightarrow Y_2$ whose special fiber is id_{Y_0} and for which $j_{Y_2} \circ \Phi = j_{Y_1}$.

The assignment sending (Y, j_Y) to the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \underline{I} & \longrightarrow & \mathcal{O}_{X_0} & \longrightarrow & \mathcal{O}_{Y_0} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow{\scriptstyle j_Y} & & \| & & \\
0 & \longrightarrow & \mathcal{O}_{Y_0} & \xrightarrow{\varepsilon} & \mathcal{O}_Y & \longrightarrow & \mathcal{O}_{Y_0} & \longrightarrow & 0
\end{array}$$

gives an isomorphism

$$\begin{aligned}
[\underline{B}] &\xrightarrow{j_Y|_{\underline{I}}} \mathrm{Hom}_{\mathcal{O}_{X_0}}(\underline{I}, \mathcal{O}_{Y_0}) \\
&= \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0}) \\
&= H^0\Big(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\Big)\,.
\end{aligned}$$

This proves exactness. □

Remark 4.8. For a more general formula implying Lemma 4.7, see SGA 3, III, 4.4. One may also note the following. For every S-scheme X and every quasi-coherent $\mathcal{O}_X$-module E, one can define quasi-coherent $\mathcal{O}_X$-modules $\underline{D}^i_{X|S}(E)$ for all $i \geq 0$, generalizing the case in which S is affine. As usual, we write

$$\underline{D}^i_{X|S}(\mathcal{O}_X) = \underline{D}^i_{X|S}.$$

For the closed immersion $Y_0 \hookrightarrow X_0$, we then have

$$H^0\Big(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\Big) = H^0(Y_0, \underline{D}^1_{Y_0|X_0}),$$

where $0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$ is exact.

Theorem 4.9. Let X_0 be a k-scheme of finite type and Y_0 a closed subscheme of X_0. Suppose either

(i) X_0 is proper over k, or
(ii) X_0 is affine and smooth over k outside finitely many points, and Y_0 is proper over k.

(a) The category $\underline{M}_{X_0,Y_0}$ admits a formal versal deformation over $\underline{C}_\Lambda$. It prorepresents the functor $[\underline{M}_{X_0,Y_0}]$ if and only if, for every surjective map $R' \longrightarrow R$ in $\underline{C}_\Lambda$ and every deformation $(R'; X', Y')$, the map

$$\mathrm{Aut}_{R'}((X', Y') \mid X_0) \longrightarrow \mathrm{Aut}_R\big((X' \otimes_{R'} R, Y' \otimes_{R'} R) \mid X_0\big)$$

is surjective.

(b) If, in addition,

$$H^0\Big(X_0, \underline{D}^0_{Y_0|k} \times_{\underline{D}^0_{X_0|k}(\mathcal{O}_{Y_0})} \underline{D}^0_{X_0|k}\Big)$$

is finite-dimensional, then $\underline{M}_{X_0,Y_0}$ is smoothly pseudo-representable.

Proof. This follows immediately from Lemmas 4.4 and 4.7 together with 1.11, 2.7, and 2.13. □

We now consider localization and completion in the deformation theory of schemes. We first fix notation. Let X_0 be a k-scheme of finite type and let $x_0 \in X_0$ be a fixed point. There are natural functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})},$$

and hence functors

$$\widehat{\underline{M}}_{X_0} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})}.$$

As observed in 4.1(c), $\widehat{\underline{M}}_{X_0}$ is canonically equivalent to the category whose objects are pairs $(R, \mathfrak{X})$, where $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$ and $\mathfrak{X}$ is a flat R-adic formal scheme equipped with a fixed isomorphism $\mathfrak{X}_0 \xrightarrow{\sim} X_0$.

When there is no danger of confusion, we denote the images of $(R, \mathfrak{X})$ under localization and completion by $(R, \mathfrak{X}_{(x_0)})$ and $(R, \mathfrak{X}_{(\widehat{x}_0)})$, respectively. Thus, if

$$\mathfrak{X} = \varinjlim_\nu \mathfrak{X}_\nu, \qquad \mathfrak{X}_\nu = \mathfrak{X} \times_{\mathrm{Spf}(R)} \mathrm{Spec}(R/\mathfrak{m}^{\nu+1}),$$

then

$$\mathfrak{X}_{(x_0)} = \varinjlim_\nu \mathrm{Spec}(\mathcal{O}_{\mathfrak{X}_\nu,x_0})$$

and

$$\mathfrak{X}_{(\widehat{x}_0)} = \varinjlim_\nu \mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X}_\nu,x_0}).$$

Theorem 4.10. Let X_0 be an affine scheme of finite type over k.

(a) If X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth over $\underline{C}_\Lambda$.

(b) Suppose that, for a fixed closed point $x \in X_0$, the open $X_0 - \{x\}$ is smooth over k. Then both functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}$$

are minimally smooth. In particular, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0, then $(R, \mathfrak{X}_{(x)})$ and $(R, \mathfrak{X}_{(\widehat{x})})$ are formal versal deformations of $\mathrm{Spec}(\mathcal{O}_{X_0,x})$ and $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})$, respectively.

Proof. We use the notation for the cohomology of commutative algebras from section 3, and write $X_0 = \mathrm{Spec}(A_0)$.

(a) It suffices to show that $P : [\underline{M}_{X_0}] \longrightarrow \underline{C}_\Lambda$ is smooth. Let $(R, \mathrm{Spec}(A))$ be a deformation of $\mathrm{Spec}(A_0)$ over $R \in \mathrm{Ob}(\underline{C}_\Lambda)$. Given a small extension $R' \longrightarrow R$ in $\underline{C}_\Lambda$, so that

$$0 \longrightarrow k \longrightarrow R' \longrightarrow R \longrightarrow 0$$

is exact, we must lift $(R, \mathrm{Spec}(A))$ to R'. The transitivity sequence for D^i with coefficients in A_0, applied to $R' \longrightarrow R \longrightarrow A$, gives

$$\cdots \longrightarrow D^1(A|R, A_0) \longrightarrow D^1(A|R', A_0) \longrightarrow D^1(R|R', k) \otimes_k A_0 \longrightarrow D^2(A|R, A_0) \longrightarrow \cdots .$$

Because $R' \longrightarrow R$ is a small extension, $D^1(R|R', k)$ is one-dimensional over k, generated by the class $[R']$ represented by $0 \longrightarrow k \longrightarrow R' \longrightarrow R \longrightarrow 0$. Since A is flat over R,

$$D^i(A|R, A_0) = D^i(A_0|k, A_0) \qquad (i > 0).$$

In particular, $D^2(A|R, A_0) = 0$ because A_0 is locally a complete intersection, by the low-degree description in 3.12(8). We therefore obtain an exact sequence

$$\longrightarrow D^1(A_0|k, A_0) \longrightarrow D^1(A|R', A_0) \longrightarrow D^1(R|R', k) \otimes_k A_0 \longrightarrow 0.$$

Hence there is an R'-algebra extension A' of A by A_0 whose class maps to $[R'] \otimes 1$. Equivalently, $(R, \mathrm{Spec}(A))$ lifts to the deformation $(R', \mathrm{Spec}(A'))$.

(b) By 3.14 there are isomorphisms

$$D^1(A_0|k, A_0) \xrightarrow{\sim} D^1((A_0)_x|k, (A_0)_x) \xrightarrow{\sim} D^1(\widehat{(A_0)_x}|k, \widehat{(A_0)_x}),$$

and hence

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}(k[\varepsilon])].$$

By 2.7(a), it remains to show that

$$[\underline{M}_{X_0}] \longrightarrow [\underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}] \longrightarrow [\underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}]$$

are smooth functors.

Let $(R, \mathrm{Spec}(A))$ be a deformation of $X_0 = \mathrm{Spec}(A_0)$, and let $0 \longrightarrow k \longrightarrow R' \longrightarrow R \longrightarrow 0$ be a small extension. As above, the homomorphisms $R' \longrightarrow R \longrightarrow A$ give an exact commutative diagram

$$\begin{array}{ccccccccc}
\longrightarrow & D^1(A_0|k, A_0) & \longrightarrow & D^1(A|R', A_0) & \xrightarrow{p} & D^1(R|R', k) \otimes_k A_0 & \longrightarrow & D^2(A_0|k, A_0) & \longrightarrow \cdots \\
 & \downarrow{\scriptstyle \sim} & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \sim} & \\
\longrightarrow & D^1(A_0|k, (A_0)_x) & \longrightarrow & D^1(A|R', (A_0)_x) & \to & D^1(R|R', k) \otimes_k (A_0)_x & \longrightarrow & D^2(A_0|k, (A_0)_x) & \longrightarrow \cdots \\
 & \uparrow & & \uparrow & & \uparrow & & \uparrow & \\
\to & D^1((A_0)_x|k, (A_0)_x) & \to & D^1(A_x|R', (A_0)_x) & \xrightarrow{p_x} & D^1(R|R', k) \otimes_k (A_0)_x & \to & D^2((A_0)_x|k, (A_0)_x) & \to \cdots
\end{array}$$

The indicated isomorphisms follow from the fact that A_0 is smooth over k away from the single closed point x, so that

$$D^i(A_0|k, (A_0)_x) = D^i(A_0|k, A_0)_x = D^i(A_0|k, A_0) \qquad (i > 0).$$

Suppose we are given an arrow

$$(R', \mathrm{Spec}(B)) \longrightarrow (R, \mathrm{Spec}(A_x))$$

in $[\underline{M}_{\mathrm{Spec}((A_0)_x)}]$. This amounts to a class $[B] \in D^1(A_x|R', (A_0)_x)$ satisfying $p_x([B]) = [R']$. A diagram chase gives a class $[A'] \in D^1(A|R', A_0)$ such that $p([A']) = [R']$ and $A'_x = B$. Thus $(R', \mathrm{Spec}(A'))$ is a deformation of $\mathrm{Spec}(A_0)$ over R' whose localization at x is $\mathrm{Spec}(B)$. This proves that

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}$$

is smooth.

The smoothness of

$$\underline{M}_{\mathrm{Spec}((A_0)_x)} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{(A_0)_x})}$$

follows from the analogous diagram and the identities

$$D^i((A_0)_x|k, \widehat{(A_0)_x}) = D^i(\widehat{(A_0)_x|k}, (A_0)_x) = D^i((A_0)_x|k, (A_0)_x) \qquad (i > 0),$$

since x is the only possible nonsmooth point. □

Theorem 4.11. Let X_0 be a separated scheme of finite type over k, smooth away from a finite closed subset

$$X_0^{\mathrm{Sing}} = \{x_1, \ldots, x_r\}.$$

Let $U_0^1, \ldots, U_0^r$ be affine open neighbourhoods of $x_1, \ldots, x_r$ such that $U_0^i \cap X_0^{\mathrm{Sing}} = \{x_i\}$. If $H^2(X_0, \underline{D}^0_{X_0}) = 0$, then the natural functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0^1} \times \cdots \times \underline{M}_{U_0^r}$$

is smooth.

Proof. Complete $U_0^1, \ldots, U_0^r$ to an affine open covering of X_0 by choosing $U_0^{r+1}, \ldots, U_0^N$ such that $U_0^j \cap X_0^{\mathrm{Sing}} = \varnothing$ for every $j > r$. Let $R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $(R', U'^1), \ldots, (R', U'^r)$ be deformations of $U_0^1, \ldots, U_0^r$ over R' such that

$$U'^i \otimes_{R'} R = X|_{U_0^i}, \qquad 1 \le i \le r,$$

for some deformation (R, X) of X_0 over R. We must find a deformation (R', X') of X_0 over R' such that

$$X' \otimes_{R'} R = X, \qquad X'|_{U_0^i} = U'^i \quad (1 \le i \le r).$$

For $j > r$, the scheme U_0^j is smooth over k; hence, by 4.9(a), $X|_{U_0^j}$ lifts to a deformation (R', U'^j) of U_0^j over R'. Thus

$$U'^j \otimes_{R'} R = X|_{U_0^j} \qquad (j > r).$$

For every pair $i \neq j$, the intersection $U_0^i \cap U_0^j$ is affine and smooth over k. We therefore obtain an isomorphism

$$\tau_{ij} : U'^i|_{U_0^i \cap U_0^j} \xrightarrow{\sim} U'^j|_{U_0^i \cap U_0^j}$$

such that $\tau_{ij} \otimes_{R'} R = \mathrm{id}_{X|_{U_0^i \cap U_0^j}}$. On triple intersections, the automorphisms

$$\tau_{ij}\tau_{jk}\tau_{ik}^{-1}$$

define a cohomology class in $H^2(X_0, \underline{D}^0_{X_0})$. This class is the obstruction to gluing the U'^i to a global deformation. It vanishes by hypothesis, so the U'^i glue to a deformation (R', X') with $X' \otimes_{R'} R = X$ and $X'|_{U_0^i} \simeq U'^i$. □

Remark 4.12. The preceding proof assumes that X_0 is separated, but this assumption is unnecessary. Indeed, fix a deformation (R, X) of X_0 and deformations (R', U'^i) of the members U_0^i of

an open covering. The deformations (R', V') of open subsets V of X over R' that are compatible with the already-given deformations (R', U'^i) form a "gerbe" (in Giraud's terminology) whose structure sheaf is $\underline{D}^0_{X_0}$. The obstruction to the existence of a section of this gerbe—that is, to a global deformation (R', X') of X compatible with the given data (R', U'^i)—lies in $H^2(X_0, \underline{D}^0_{X_0})$, and hence vanishes under the hypothesis of the theorem.

Corollary 4.13. Let X_0 be a scheme of finite type over k, smooth away from a finite subset

$$X_0^{\mathrm{Sing}} = \{x_1, \ldots, x_r\},$$

and suppose that $H^2(X_0, \underline{D}^0_{X_0}) = 0$. Then:

(a) The functor

$$\underline{M}_{X_0} \longrightarrow \prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}$$

is smooth. In particular,

$$\dim \underline{M}_{X_0} = \dim_k H^1(X_0, \underline{D}^0_{X_0}) + \sum_{i=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}.$$

(b) If, in addition, X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth.

Proof. **(a)** The first assertion follows from 4.10 and 4.9(b). Let $(R, \mathfrak{X})$ and $(R_1, \mathfrak{X}_1), \ldots, (R_r, \mathfrak{X}_r)$ be formal versal deformations of X_0 and of

$$\mathrm{Spec}(\mathcal{O}_{X_0,x_1}), \ldots, \mathrm{Spec}(\mathcal{O}_{X_0,x_r}),$$

respectively. We obtain a commutative diagram

$$\begin{array}{ccc} \underline{h}_R & \longrightarrow & \underline{h}_{R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r} \\ \downarrow & & \downarrow \\ \underline{M}_{X_0} & \longrightarrow & \displaystyle\prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}. \end{array}$$

The composite functor

$$\underline{h}_R \longrightarrow \underline{M}_{X_0} \longrightarrow \prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}$$

is smooth, whereas

$$\underline{h}_{R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r} \longrightarrow \prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}$$

is minimally smooth. It follows from 2.7(a) that

$$\underline{h}_R \longrightarrow \underline{h}_{R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r}$$

is smooth; equivalently, R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r$.

By 4.4, the sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow \prod_{i=1}^{r} [\underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}(k[\varepsilon])]$$

is exact. Equivalently,

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow \mathrm{Hom}_{\Lambda\text{-alg}}(R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r, k[\varepsilon])$$

is exact. Consequently, R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r$ in d variables, where

$$d = \dim_k H^1(X_0, \underline{D}^0_{X_0}).$$

Therefore

$$\begin{aligned}
\dim \underline{M}_{X_0} &= \dim(k \otimes_\Lambda R) \\
&= d + \dim\big(k \otimes_\Lambda (R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r)\big) \\
&= d + \dim\big((k \otimes_\Lambda R_1) \widehat{\otimes}_k \cdots \widehat{\otimes}_k (k \otimes_\Lambda R_r)\big) \\
&= d + \sum_{i=1}^{r} \dim(k \otimes_\Lambda R_i) \\
&= d + \sum_{i=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}.
\end{aligned}$$

(b) This follows from 4.10 and 4.9(a). □

4.14. We close this section by constructing explicitly a formal versal deformation in a simple case. When X_0 is affine and a complete intersection, such a deformation is immediate to write down. Let

$$A_0 = k[x_1, \ldots, x_n]/(f_1, \ldots, f_r) = S/J,$$

where $S = k[x_1, \ldots, x_n]$, $J = (f_1, \ldots, f_r)$, and $f_1, \ldots, f_r$ is an S-regular sequence. We have an exact sequence

$$\cdots \longrightarrow \mathrm{Der}_k(S, A_0) \longrightarrow \mathrm{Hom}_{A_0}(J/J^2, A_0) \longrightarrow D^1(A_0|k, A_0) \longrightarrow 0.$$

The A_0-module J/J^2 is free of rank r, generated by the classes of $f_1, \ldots, f_r$, because these elements form an S-regular sequence.

Assume now that $D^1(A_0|k, A_0)$ is finite-dimensional over k; for example, assume that A_0 has only isolated nonsmooth points. Put

$$d = \dim_k D^1(A_0|k, A_0),$$

and choose maps

$$p_i : J/J^2 \longrightarrow A_0 \qquad (1 \le i \le d)$$

whose classes form a k-basis of $D^1(A_0|k, A_0)$. For $1 \le i \le d$ and $1 \le j \le r$, choose $P_{ij} \in \Lambda[x_1, \ldots, x_n]$ such that

$$P_{ij} \equiv p_i(f_j) \pmod{\mathfrak{m}_\Lambda + J},$$

where $\mathfrak{m}_\Lambda$ is the maximal ideal of Λ. Set

$$R = \Lambda[[t_1, \ldots, t_d]], \qquad B = R[x_1, \ldots, x_n]/(G_1, \ldots, G_r),$$

where

$$G_j = F_j - \sum_{i=1}^{d} t_i P_{ij},$$

and $F_j \in \Lambda[x_1, \ldots, x_n]$ is obtained from f_j by lifting its coefficients from k to Λ.

The algebra $R[x_1, \ldots, x_n]$ is flat over R, and the images of $G_1, \ldots, G_r$ modulo $\mathfrak{m}_R$ form the regular sequence $f_1, \ldots, f_r$ in $k \otimes_R R[x_1, \ldots, x_n]$. Hence B is R-flat and $k \otimes_R B = A_0$. By construction, the map

$$\mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow D^1(A_0|k, A_0) = [\underline{M}_{\mathrm{Spec}(A_0)}(k[\varepsilon])]$$

is bijective. Thus, if

$$\mathrm{Spf}(B) = \varinjlim_{\nu} \mathrm{Spec}\big((R/\mathfrak{m}_R^{\nu+1}) \otimes_R B\big),$$

then $(R, \mathrm{Spf}(B))$ is an object of $\widehat{\underline{M}}_{\mathrm{Spec}(A_0)}$. The functor

$$\underline{h}_R \longrightarrow \underline{M}_{\mathrm{Spec}(A_0)}$$

that it induces is smooth by 1.6, since R is a formal power-series ring over Λ. In other words, $(R, \mathrm{Spf}(B))$ is a formal versal object of $\underline{M}_{\mathrm{Spec}(A_0)}$.

By 4.10(b), if instead we put

$$B = R[[x_1, \ldots, x_n]]/(G_1, \ldots, G_r),$$

then $(R, \mathrm{Spf}(B))$ is a formal versal deformation of $\mathrm{Spec}(A_0)$ for

$$A_0 = k[[x_1, \ldots, x_n]]/(f_1, \ldots, f_r).$$

Remark 4.15. Let X_0 be a scheme of finite type over k with only isolated nonsmooth points, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. The preceding construction shows that, when X_0 is affine and locally a complete intersection, $\mathfrak{X}$ is the formalization of an R-scheme of finite type. The same fact extends readily to the nonaffine case provided that $H^2(X_0, \underline{D}^0_{X_0}) = 0$.

Remark 4.16. The global theory of the relative cotangent complex—here reduced to the truncated complex of length one, which nevertheless suffices for many purposes in deformation theory—generalizes and simplifies results such as 4.10. Let S be a scheme, and let $S_0 \subset S' \subset S$ be subschemes defined by ideals $J \supset I$ with $JI = 0$. Let X' be flat over S', and write

$$L_\bullet = L^{X_0/S_0}$$

for the relative cotangent complex of X_0 over S_0, regarded as an object of $D(X_0, \mathcal{O}_{X_0})$. Let $\underline{F}$ be the category of flat S-schemes X extending X'/S'; thus an object is equipped with an S'-isomorphism

$$X \times_S S' \xrightarrow{\sim} X'.$$

Then:

(a) For an object X of $\underline{F}$, its group of automorphisms inducing the identity on X' is canonically isomorphic to

$$\mathrm{Ext}^0(X_0; L_\bullet, I_{X_0}),$$

where I_{X_0} is the inverse image on X_0 of the $\mathcal{O}_{S_0}$-module I.

(b) The set of isomorphism classes of objects of $\underline{F}$ is either empty or a torsor (a principal homogeneous space) under $\mathrm{Ext}^1(X_0; L_\bullet, I_{X_0})$.

(c) There is a canonical element

$$c \in \mathrm{Ext}^2(X_0; L_\bullet, I_{X_0})$$

whose vanishing is necessary and sufficient for $\underline{F}$ to be nonempty.

To study these groups, it is often convenient to write

$$\mathrm{Ext}^i(X_0; L_\bullet, I_{X_0}) = H^i\big(X_0, R\underline{\mathrm{Hom}}(L_\bullet, I_{X_0})\big).$$

The cohomology sheaves of $R\underline{\mathrm{Hom}}(L_\bullet, I_{X_0})$ are precisely the sheaves $\underline{D}^i_{X_0}(I_{X_0})$ introduced in section 3. For $i = 0, 1, 2$, they may be viewed respectively as the sheaf of automorphisms of an object X of $\underline{F}$ (automorphisms that induce the identity on X'), the sheaf measuring the indeterminacy of isomorphism classes of local solutions to the flat-deformation problem, and the sheaf of obstructions to the existence of local solutions.

The standard spectral sequence is

$$E_2^{p,q} = H^p\big(X_0, \underline{D}^q_{X_0}(I_{X_0})\big) \Longrightarrow \mathrm{Ext}^{p+q}(X_0; L_\bullet, I_{X_0}).$$

It gives the familiar isomorphisms

$$\begin{aligned}\mathrm{Aut}(X/X') &\simeq H^0\big(X_0, \underline{D}^0_{X_0}(I_{X_0})\big)\\ &\simeq H^0\big(X_0, \underline{\mathrm{Hom}}(\Omega^1_{X_0/S_0}, I_{X_0})\big)\\ &= \mathrm{Hom}(\Omega^1_{X_0/S_0}, I_{X_0}),\end{aligned}$$

and the low-degree exact sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}(I_{X_0})) \longrightarrow \mathrm{Ext}^1 \longrightarrow H^0(X_0, \underline{D}^1_{X_0}(I_{X_0})) \longrightarrow H^2(X_0, \underline{D}^0_{X_0}(I_{X_0})) \longrightarrow \mathrm{Ext}^2\,.$$

In particular, if $H^2(X_0, \underline{D}^0_{X_0}(I_{X_0})) = 0$ and local flat deformations of X'/S' exist, then compatible local deformations glue to a global flat deformation. More generally, for a prescribed section of the sheaf of isomorphism classes of local flat deformations, the argument of 4.10 places the obstruction to a compatible global deformation in $H^2(X_0, \underline{D}^0_{X_0}(I_{X_0}))$.

The same spectral sequence shows that Ext^2 vanishes whenever

$$(*) \qquad H^0(X_0, \underline{D}^2_{X_0}) = 0, \qquad H^1(X_0, \underline{D}^1_{X_0}) = 0, \qquad H^2(X_0, \underline{D}^0_{X_0}) = 0.$$

The first equality always holds when X_0 is a relative complete intersection over S_0, since then $\underline{D}^2_{X_0}(M) = 0$ for every $\mathcal{O}_{X_0}$-module M. The second holds when the support of $\underline{D}^1_{X_0}$ is discrete, as happens when X_0 is smooth over S_0 away from a discrete subset. If all three equalities in $(*)$ hold, then $\underline{F}$ is nonempty by (c), which generalizes 4.13(b).

There is also a geometric interpretation, requiring only the globalized functors $\underline{D}^i_{X_0}$ rather than the global theory of the relative cotangent complex. The three successive obstructions to extending X'/S' flatly over S lie in the three groups displayed in $(*)$. The local theory defines the first obstruction

$$c_1 \in H^0(X_0, \underline{D}^2_{X_0})$$

as the obstruction to the existence of local solutions. If $c_1 = 0$, the sheaf $\underline{F}$ of isomorphism classes of local solutions is canonically a torsor under $\underline{D}^1_{X_0}$. Its obstruction to having a section is

$$c_2 \in H^1(X_0, \underline{D}^1_{X_0});$$

such a section is a coherent system of isomorphism classes of local solutions. If $c_2 = 0$, choose a section of $\underline{F}$. The obstruction to a global solution realizing the chosen local classes is then

$$c_3 \in H^2(X_0, \underline{D}^0_{X_0}).$$

Changing the section of $\underline{F}$ by a section of $\underline{D}^1_{X_0}$ gives a map

$$d : H^0(X_0, \underline{D}^1_{X_0}) \longrightarrow H^2(X_0, \underline{D}^0_{X_0}).$$

A solution exists if and only if $c_1 = 0$, $c_2 = 0$, and the class of c_3 in $\mathrm{Coker}\, d$ is zero (each c_i being defined only after the preceding obstructions vanish).

The three obstruction groups

$$H^0(X_0, \underline{D}^2_{X_0}), \qquad H^1(X_0, \underline{D}^1_{X_0}), \qquad H^2(X_0, \underline{D}^0_{X_0})/\operatorname{Im} d$$

are the initial terms $E_2^{0,2}$, $E_2^{1,1}$, and $E_3^{2,0}$ of the preceding spectral sequence; up to sign, d is its differential

$$d_2 : E_2^{0,1} \longrightarrow E_2^{2,0}.$$

5. Formal Jacobian subschemes

Let X_0 be a scheme of finite type over a field k, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. The space $\mathfrak{X}$ is not an ordinary R-scheme, but an R-adic formal scheme. The purpose of this section is to clarify the meaning of the "nonsmooth locus of $\mathfrak{X}$ over R."

We first recall the Fitting ideals of a finite-type module. Let A be a commutative ring and let E be a finite-type A-module. Choose a presentation

$$K \longrightarrow F \longrightarrow E \longrightarrow 0,$$

where F is locally free of constant rank n. For $p = 0, 1, 2, \ldots$, define

$$I_A^p(E) = \operatorname{Im}\left(\bigwedge^{n-p} K \otimes_A \bigwedge^{n-p} F^* \longrightarrow A\right).$$

These ideals are independent of the chosen presentation. They are invariants of the A-module E, called its *Fitting ideals*. The following properties follow directly from the definition.

5.1. **(a)** For every ring homomorphism $A \longrightarrow B$, the canonical map

$$B \otimes_A A/I_A^p(E) \xrightarrow{\sim} B/I_B^p(B \otimes_A E)$$

is an isomorphism for every p. In particular,

$$I_{S^{-1}A}^p(S^{-1}E) = S^{-1}I_A^p(E)$$

for every multiplicative subset $S \subset A$.

(b) There are inclusions

$$I_A^0(E) \subset I_A^1(E) \subset I_A^2(E) \subset \cdots.$$

For every $x \in \operatorname{Spec}(A)$,

$$I_A^p(E)_x = A_x \qquad \text{for all } p \geq \dim_{\kappa(x)} E(x),$$

where $E(x) = \kappa(x) \otimes_A E$.

(c) A point $x \in \operatorname{Spec}(A) - \operatorname{Supp}(A/I_A^p(E))$ if and only if, for any (equivalently, for some) presentation

$$K \longrightarrow F \longrightarrow E \longrightarrow 0, \qquad F \simeq A^n,$$

the module K_x contains a copy of A_x^{n-p} that is a direct summand of F_x. In particular,

$$\operatorname{Supp}(A/I_A^0(E)) = \operatorname{Supp}(E).$$

(d) For any two finite-type A-modules,

$$I_A^r(E_1 \oplus E_2) = \sum_{p+q=r} I_A^p(E_1) I_A^q(E_2).$$

(e) If A is a noetherian adic ring, then

$$A/I_A^p(E) = \varprojlim_\nu A_\nu/I_{A_\nu}^p(E_\nu),$$

where $A = \varprojlim_\nu A_\nu$ and $E_\nu = E \otimes_A A_\nu$.

Property 5.1(a) globalizes the definition. If X is a scheme and $\underline{E}$ is a quasi-coherent $\mathcal{O}_X$-module of finite presentation, there are quasi-coherent ideal sheaves $\underline{I}_X^p(\underline{E})$, $p \geq 0$, characterized on every affine open $U \subset X$ by

$$\Gamma(U, \underline{I}_X^p(\underline{E})) \xrightarrow{\sim} I_{\Gamma(U,\mathcal{O}_X)}^p(\Gamma(U, \underline{E})).$$

For a morphism of schemes $X \longrightarrow Y$, the relative module of Kähler differentials $\Omega_{X|Y} = \Omega^1_{X|Y}$ is quasi-coherent. If the morphism is essentially of finite presentation, then $\Omega_{X|Y}$ is of finite presentation, so the ideal sheaves $\underline{I}^p_X(\Omega_{X|Y})$ are defined.

Definition 5.2. Let $X \longrightarrow Y$ be a morphism essentially of finite presentation. For every integer $p \geq 0$, define $J^p_X(X|Y)$ to be the closed subscheme of X
defined by the ideal sheaf

$$\underline{I}^p_X(X|Y) = \underline{I}^p_X(\Omega_{X|Y}).$$

It is called the pth *Jacobian subscheme* of X over Y. For every $x \in X$, put

$$X_x = \mathrm{Spec}(\mathcal{O}_{X,x}).$$

Theorem 5.3. Let $X \longrightarrow Y$ be a morphism essentially of finite presentation. Then:
(1) For every base change $Y' \longrightarrow Y$, the canonical map

$$J^p_{X'}(X'|Y') \xrightarrow{\sim} J^p_X(X|Y) \times_Y Y'$$

is an isomorphism for every p, where $X' = X \times_Y Y'$.
(2) For every $x \in X$, there is a canonical isomorphism

$$J^p_{X_x}(X_x|Y) \xrightarrow{\sim} J^p_X(X|Y)_x.$$

(3) There are inclusions

$$J^0_X(X|Y) \supset J^1_X(X|Y) \supset J^2_X(X|Y) \supset \cdots,$$

and

$$x \notin J^p_X(X|Y) \qquad \text{for every } p \geq \dim_{\kappa(x)} \Omega_{X|Y}(x).$$

(4) A point x does not belong to $J^p_X(X|Y)$ if and only if it has an open neighbourhood U that admits a closed immersion $i : U \hookrightarrow U'$ into a Y-scheme U' such that U' is smooth over Y at $x' = i(x)$ and

$$\dim_{x'} U'_{f(x)} = p.$$

(5) For the projections $\pi_i : X_1 \times_Y X_2 \longrightarrow X_i$,

$$\underline{I}^r_{X_1 \times_Y X_2}(X_1 \times_Y X_2|Y) = \sum_{p+q=r} \pi_1^* \underline{I}^p_{X_1}(X_1|Y)\, \pi_2^* \underline{I}^q_{X_2}(X_2|Y).$$

Proof. Assertion (3) follows from 5.1(b). Assertion (1) follows from 5.1(a) and the canonical isomorphism

$$P^* \Omega_{X|Y} \xrightarrow{\sim} \Omega_{X'|Y'}, \qquad P : X \times_Y Y' \longrightarrow X.$$

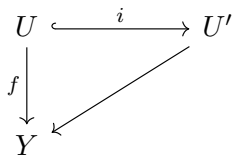
Likewise, (2) follows from 5.1(a) and

$$\mathcal{O}_{X,x} \otimes_{\mathcal{O}_X} \Omega_{X|Y} \xrightarrow{\sim} \Omega_{X_x|Y}.$$

Finally, (5) follows from 5.1(d) and

$$\pi_1^* \Omega_{X_1|Y} \oplus \pi_2^* \Omega_{X_2|Y} \xrightarrow{\sim} \Omega_{X_1 \times_Y X_2|Y}.$$

For (4), first suppose that there is a diagram



where U' is smooth over Y at $x' = i(x)$ and $\dim_{x'} U'_{f(x)} = p$. Then

$$\dim_{\kappa(x')} \Omega_{U'|Y}(x') = p,$$

and the sequence

$$i^*\Omega_{U'|Y} \longrightarrow \Omega_{U|Y} \longrightarrow 0$$

is exact. Hence $\dim_{\kappa(x)} \Omega_{U|Y}(x) \leq p$, so $x \notin J^p_U(U|Y)$ by (3). Since

$$J^p_U(U|Y) = J^p_X(X|Y)|_U,$$

we have $x \notin J^p_X(X|Y)$.

Conversely, suppose that $x \notin J^p_X(X|Y)$. The question is local, so write $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$. Choose an A-presentation

$$0 \longrightarrow J \longrightarrow P \longrightarrow B \longrightarrow 0, \qquad P = A[x_1, \ldots, x_n].$$

There is an exact sequence

$$J/J^2 \xrightarrow{d} B \otimes_P \Omega_{P|A} \longrightarrow \Omega_{B|A} \longrightarrow 0,$$

and $B \otimes_P \Omega_{P|A}$ is free of rank n over B. Thus

$$I^p_B(\Omega_{B|A}) = \mathrm{Im}\left(\bigwedge^{n-p} (J/J^2) \otimes_B (B \otimes_P \bigwedge^{n-p} \Omega_{P|A})^* \longrightarrow B \right).$$

Consequently, $x \notin J^p_X(X|Y)$ precisely when there are $f_1, \ldots, f_{n-p} \in J$ and a choice of $n-p$ among the variables $x_1, \ldots, x_n$ for which the corresponding Jacobian minor

$$\det\left(\frac{\partial f_i}{\partial x_j}\right)(x)$$

is nonzero. Equivalently,

$$\mathrm{Spec}\big(P/(f_1, \ldots, f_{n-p})\big) \longrightarrow \mathrm{Spec}(A)$$

is smooth at the image x' of x. Taking

$$U' = \mathrm{Spec}\big(P/(f_1, \ldots, f_{n-p})\big)$$

gives the required closed immersion locally. □

Theorem 5.4. Let $X \xrightarrow{f} Y$ be a morphism essentially of finite presentation. Suppose that there is a fixed integer $n \geq 0$ for which one of the following conditions holds:

(1) f is flat and $\dim_x X_{f(x)} = n$ for every $x \in X$;

(2) f is a relative complete intersection of virtual dimension n (SGA 6, VIII, 1.9).

Then $x \notin J^n_X(X|Y)$ if and only if f is smooth at x.

Proof. By 5.3(4), $x \notin J^n_X(X|Y)$ if and only if there is a diagram

$$\begin{array}{ccc} U & \xhookrightarrow{\ i\ } & U' \\ {\scriptstyle f}\downarrow & \swarrow_{f'} & \\ Y & & \end{array}$$

such that f' is smooth at x', with $\dim_{x'} U'_{f(x)} = n$, while i is a closed immersion and $x' = i(x)$. It therefore suffices to show that under either (1) or (2), i is an isomorphism at x.

Set

$$A = \mathcal{O}_{Y,f(x)}, \qquad B' = \mathcal{O}_{U',x'}, \qquad B = \mathcal{O}_{U,x},$$

and let

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be exact. We must show that $I = 0$.

Assume (1). Since B is flat over A, the sequence

$$0 \longrightarrow I/\mathfrak{m}I \longrightarrow B'/\mathfrak{m}B' \longrightarrow B/\mathfrak{m}B \longrightarrow 0$$

is exact, where $\mathfrak{m}$ is the maximal ideal of A. By hypothesis,

$$\dim_x U_{f(x)} = \dim_{x'} U'_{f(x)},$$

and hence $\dim(B'/\mathfrak{m}B') = \dim(B/\mathfrak{m}B)$. Since $B'/\mathfrak{m}B'$ is a regular local ring, it follows that $I/\mathfrak{m}I = 0$, and therefore $I = 0$.

Now assume (2), and consider the exact sequence

$$I/I^2 \xrightarrow{d} B \otimes_{B'} \Omega_{B'|A} \longrightarrow \Omega_{B|A} \longrightarrow 0.$$

Since X is a relative complete intersection over Y and $U' \longrightarrow Y$ is smooth at x', the map

$$d : I/I^2 \longrightarrow B \otimes_{B'} \Omega_{B'|A}$$

is injective at x, and both I/I^2 and $B \otimes_{B'} \Omega_{B'|A}$ are free at x. By the definition of the virtual dimension of a relative complete intersection, we must therefore have

$$n = \dim_{\kappa(x)} \Omega_{B'|A}(x') - \dim_{\kappa(x)}(I/I^2)(x) = n - \dim_{\kappa(x)}(I/I^2)(x).$$

Thus $I/I^2 = 0$, and hence $I = 0$. □ **Definition 5.5.** Let $X \xrightarrow{f} Y$ be a morphism essentially of finite presentation. If there is a fixed integer n satisfying (1) or (2) of 5.4, set

$$\underline{I}_X(X|Y) = \underline{I}^n_X(X|Y), \qquad J_X(X|Y) = J^n_X(X|Y).$$

This notion extends naturally to formal schemes. Let $\mathfrak{X}$ be a noetherian formal scheme and $\mathcal{E}$ a coherent $\mathfrak{X}$-module. For every affine open $U \subset \mathfrak{X}$, define

$$\underline{I}^p_{\mathfrak{X}}(\mathcal{E})(U) = I^p_{\mathfrak{X}(U)}\big(\mathcal{E}(U)\big).$$

If $\mathfrak{X} = \varinjlim_\nu X_\nu$, then $\mathcal{E} = \varprojlim_\nu \mathcal{E}_\nu$, where

$$\mathcal{E}_\nu = \mathcal{E} \otimes_{\mathcal{O}_{\mathfrak{X}}} \mathcal{O}_{X_\nu},$$

and noetherianness gives

$$\underline{I}^p_{\mathfrak{X}}(\mathcal{E}) = \varprojlim_\nu \underline{I}^p_{X_\nu}(\mathcal{E}_\nu).$$

Now let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme essentially of finite type. Let $\underline{I}_{\mathfrak{X}}$ be the ideal sheaf defining the closed diagonal immersion

$$\Delta : \mathfrak{X} \longrightarrow \mathfrak{X} \times_S \mathfrak{X}$$

(where the fibre product is taken in the category of formal schemes), so that

$$0 \longrightarrow \underline{I}_{\mathfrak{X}} \longrightarrow \mathcal{O}_{\mathfrak{X}\times_S\mathfrak{X}} \longrightarrow \mathcal{O}_{\mathfrak{X}} \longrightarrow 0$$

is exact. Define

$$\Omega_{\mathfrak{X}|S} = \underline{I}_{\mathfrak{X}}/\underline{I}_{\mathfrak{X}}^2.$$

Writing

$$\mathfrak{X} = \varinjlim_{\nu} X_\nu, \qquad X_\nu = \mathfrak{X} \times_S S_\nu,$$

we have

$$\Omega_{\mathfrak{X}|S} \simeq \varprojlim_{\nu} \Omega_{X_\nu|S_\nu},$$

so $\Omega_{\mathfrak{X}|S}$ is a coherent $\mathfrak{X}$-module. For every formal S-scheme S' of finite type there is a canonical isomorphism

$$\Omega_{\mathfrak{X}\times_S S'|S'} \simeq \mathcal{O}_{S'} \otimes_{\mathcal{O}_S} \Omega_{\mathfrak{X}|S}.$$

Definition 5.6. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme essentially of finite type. Define

$$\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S) = \underline{I}^p_{\mathfrak{X}}(\Omega_{\mathfrak{X}|S}),$$

and let $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ be the closed formal subscheme of $\mathfrak{X}$ defined by this ideal sheaf.

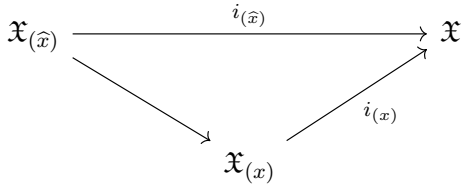
Since $\Omega_{\mathfrak{X}|S} = \varprojlim_{\nu} \Omega_{X_\nu|S_\nu}$, we have

$$\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S) = \varprojlim_{\nu} \underline{I}^p_{X_\nu}(X_\nu|S_\nu)$$

and

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \varinjlim_{\nu} J^p_{X_\nu}(X_\nu|S_\nu).$$

The formal scheme $\mathfrak{X}_{(x)}$ is again S-adic and essentially of finite type, and there is a commutative diagram



If x is an isolated point of $\mathfrak{X}$, then $i_{(x)}$ and $i_{(\widehat{x})}$ are both isomorphisms onto the connected component of $\mathfrak{X}$ containing x.

In particular, if

$$|\mathfrak{X}| = \{x_1, \ldots, x_m\}$$

is a finite discrete space, then there is a canonical identification

$$\mathfrak{X} = \coprod_{1\le i\le m} \mathfrak{X}_{(x_i)} \qquad \text{(disjoint union)}.$$

Corollary 5.7. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme essentially of finite type. Then:

(1) Formation of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ commutes with base change. More precisely, if

$$\mathfrak{X}' = \mathfrak{X} \times_S S',$$

then

$$J^p_{\mathfrak{X}'}(\mathfrak{X}'|S') = J^p_{\mathfrak{X}}(\mathfrak{X}|S) \times_S S'.$$

(2) The connected components of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ are in bijective correspondence with those of $J^p_{X_0}(X_0|S_0)$. For every

$$x \in \left|J^p_{X_0}(X_0|S_0)\right|$$

there is a canonical isomorphism

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S)_{(x)} \simeq J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S).$$

In particular, if

$$\big|J^p_{X_0}(X_0|S_0)\big| = \{x_1, \dots, x_m\}$$

is discrete, then

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \coprod_{1\le i\le m} J^p_{\mathfrak{X}_{(x_i)}}(\mathfrak{X}_{(x_i)}|S) \qquad \text{(disjoint union)}.$$

(3) Let S be an ordinary noetherian scheme and X an S-scheme essentially of finite type. Given a closed subscheme $S_0 \subset S$, let $\widehat{S}$ be the formal completion of S along S_0. If

$$\widehat{X} = X \times_S \widehat{S}$$

is the formal completion of X along $X_0 = X \times_S S_0$, then

$$J^p_{\widehat{X}}(\widehat{X}|\widehat{S}) = J^p_X(X|S) \times_S \widehat{S}.$$

Equivalently, it is the formal completion of $J^p_X(X|S)$ along $J^p_{X_0}(X_0|S_0)$.

(4) Suppose that a fixed integer $n \ge 0$ satisfies condition 5.4(1) or 5.4(2) for every $X_\nu \longrightarrow S_\nu$. Then

$$x \notin J^n_{\mathfrak{X}}(\mathfrak{X}|S)$$

if and only if $\mathfrak{X}$ is formally smooth over S at x.

Proof. Assertion (1) follows from

$$\Omega_{\mathfrak{X}'|S'} \simeq \mathcal{O}_{S'} \widehat{\otimes}_{\mathcal{O}_S} \Omega_{\mathfrak{X}|S},$$

whereas (2) follows from

$$\mathcal{O}_{\mathfrak{X}_{(x)}} \widehat{\otimes}_{\mathcal{O}_{\mathfrak{X}}} \Omega_{\mathfrak{X}|S} \simeq \Omega_{\mathfrak{X}_{(x)}|S}.$$

For (3),

$$\begin{aligned} J^p_{\widehat{X}}(\widehat{X}|\widehat{S}) &= \varinjlim_{\nu} J^p_{X\times_S S_\nu}(X \times_S S_\nu|S_\nu) \\ &= \varinjlim_{\nu} \big(J^p_X(X|S) \times_S S_\nu\big) = J^p_X(X|S) \times_S \widehat{S}. \end{aligned}$$

Finally, since $\mathfrak{X}$ is S-adic,

$$\begin{aligned} \mathfrak{X} \to S \text{ is formally smooth at } x &\iff X_\nu \to S_\nu \text{ is formally smooth at } x \text{ for every } \nu \\ &\iff x \notin J^n_{X_\nu}(X_\nu|S_\nu) \text{ for every } \nu \\ &\iff x \notin \varinjlim_{\nu} J^n_{X_\nu}(X_\nu|S_\nu) = J^n_{\mathfrak{X}}(\mathfrak{X}|S). \end{aligned}$$

□ **Remark 5.8.** Let $\mathfrak{X}$ be an S-adic formal scheme essentially of finite type. For a point $x \in \mathfrak{X}$, consider also the S-adic formal scheme

$$\mathfrak{X}_{(\widehat{x})} = \varinjlim_{\nu} \operatorname{Spec}\big(\widehat{\mathcal{O}_{X_\nu,x}}\big).$$

There are natural morphisms

$$\mathfrak{X}_{(\widehat{x})} \longrightarrow \mathfrak{X}_{(x)} \longrightarrow \mathfrak{X}.$$

In general, $\widehat{\mathcal{O}_{X_0,x}}$ is not essentially of finite type over $S_{(y)}$, where y is the image of x under $\mathfrak{X} \longrightarrow S$.

For simplicity, put

$$A_\nu = \mathcal{O}_{X_\nu,x}, \qquad R_\nu = \mathcal{O}_{S_\nu,y}.$$

The natural map

$$\Omega_{A_\nu|R_\nu} \longrightarrow \widehat{\Omega}_{A_\nu|R_\nu}$$

induces an isomorphism

$$\widehat{\Omega_{A_\nu|R_\nu}} \xrightarrow{\sim} \widehat{\widehat{\Omega}_{A_\nu|R_\nu}},$$

and therefore

$$\varprojlim_\nu \widehat{\Omega_{A_\nu|R_\nu}} \xrightarrow{\sim} \varprojlim_\nu \widehat{\widehat{\Omega}_{A_\nu|R_\nu}}.$$

Here the outer hat denotes the separated completion of a module for the topology induced by the ring. Consequently, if we define

$$\widehat{\Omega}_{\mathfrak{X}_{(\widehat{x})}|S} = \varprojlim_\nu \widehat{\Omega}_{\widehat{\mathcal{O}_{X_\nu,x}|\mathcal{O}_{S_\nu,y}}},$$

then $\widehat{\Omega}_{\mathfrak{X}_{(\widehat{x})}|S}$ is a coherent $\mathfrak{X}_{(\widehat{x})}$-module.

Define $J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S)$ to be the closed formal subscheme of $\mathfrak{X}_{(\widehat{x})}$ defined by

$$\underline{I}^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S) = \underline{I}^p_{\mathfrak{X}_{(\widehat{x})}}(\widehat{\Omega}_{\mathfrak{X}_{(\widehat{x})}|S}).$$

Then

$$J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)_{(\widehat{x})} \simeq J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S).$$

If x is an isolated point of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$, the natural morphism

$$J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S) \longrightarrow J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$$

is an isomorphism.

To form the "image" of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ under $f : \mathfrak{X} \longrightarrow S$, one could simply take the formal subscheme of S defined by the ideal sheaf

$$\ker\left(\mathcal{O}_S \longrightarrow \mathcal{O}_{\mathfrak{X}}/\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S)\right).$$

For our purposes, however, it is more convenient to use the Fitting ideal once again. **Definition 5.9.** Let S be a noetherian formal scheme and $f : \mathfrak{X} \longrightarrow S$ an S-adic formal scheme essentially of finite type. Suppose that $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ is proper over S, so that

$$f_*\mathcal{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)}$$

is a coherent $\mathcal{O}_S$-module. Define

$$\underline{I}^p_S(\mathfrak{X}|S) = \underline{I}^0_S\Big(f_*\mathcal{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)}\Big),$$

and let $J^p_S(\mathfrak{X}|S)$ be the closed formal subscheme of S defined by $\underline{I}^p_S(\mathfrak{X}|S)$.

If there is a fixed integer n satisfying condition 5.4(1) or 5.4(2) for every $X_\nu \longrightarrow S_\nu$, set

$$\underline{I}_{\mathfrak{X}}(\mathfrak{X}|S) = \underline{I}^n_{\mathfrak{X}}(\mathfrak{X}|S), \qquad J_{\mathfrak{X}}(\mathfrak{X}|S) = J^n_{\mathfrak{X}}(\mathfrak{X}|S),$$
$$\underline{I}_S(\mathfrak{X}|S) = \underline{I}^n_S(\mathfrak{X}|S), \qquad J_S(\mathfrak{X}|S) = J^n_S(\mathfrak{X}|S).$$

Corollary 5.10. Let S be a noetherian formal scheme and $f : \mathfrak{X} \longrightarrow S$ an S-adic formal scheme essentially of finite type such that $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ is proper over S. Then:

(1)

$$\big|J^p_S(\mathfrak{X}|S)\big| = f\big(\,|J^p_{\mathfrak{X}}(\mathfrak{X}|S)|\,\big).$$

(2) Formation of $J^p_S(\mathfrak{X}|S)$ commutes with base change: if $\mathfrak{X}' = \mathfrak{X} \times_S S'$, then

$$J^p_{S'}(\mathfrak{X}'|S') = J^p_S(\mathfrak{X}|S) \times_S S'.$$

(3) If

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \coprod_{1\le i\le m} \mathfrak{Y}^{(i)}$$

is a disjoint union, then

$$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1\le i\le m} \underline{I}^0_S\big(f_*\mathcal{O}_{\mathfrak{Y}^{(i)}}\big).$$

In particular, if

$$\big|J^p_{X_0}(X_0|S_0)\big| = \{x_1, \dots, x_m\}$$

is discrete, then

$$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1\le i\le m} \underline{I}^p_S(\mathfrak{X}_{(x_i)}|S) = \prod_{1\le i\le m} \underline{I}^p_S(\mathfrak{X}_{(\widehat{x}_i)}|S).$$

(4) Let S be an ordinary noetherian scheme, X an S-scheme essentially of finite type, and $S_0 \hookrightarrow S$ a closed subscheme. Let $\widehat{S}$ be the formal completion of S along S_0. If

$$\widehat{X} = X \times_S \widehat{S}$$

is the formal completion of X along $X_0 = X \times_S S_0$, then

$$J^p_{\widehat{S}}(\widehat{X}|\widehat{S}) = J^p_S(X|S) \times_S \widehat{S}.$$

Thus it is the formal completion of $J^p_S(X|S)$ along $J^p_{S_0}(X_0|S_0)$.

Proof. By 5.1(c),

$$\operatorname{Supp} f_*\mathcal{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)} = \big|J^p_S(\mathfrak{X}|S)\big|,$$

which proves (1). Assertions (2) and (4) follow from 5.1(a), while (3) follows from 5.1(d) and, for an isolated point $x \in J^p_{\mathfrak{X}}(\mathfrak{X}|S)$, the isomorphism

$$J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S) \xrightarrow{\sim} J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S).$$

□

Remark 5.11. Our definition of the Jacobian ideal, adopted from the classical definition over a base field, commutes with base change and, under the mild hypothesis of 5.4, gives a smoothness criterion. In a general setting, however, one can probably define in many ways a canonical subscheme whose support is the nonsmooth locus. We outline one such definition.

For an A-algebra B, put

$$I_{B|A} = \bigcap_E \operatorname{Ann} D^1(B|A, E),$$

where E ranges over all finite-type B-modules. By 3.13(c), the ideal $I_{B|A}$ commutes with localization. Thus a morphism $f : X \longrightarrow Y$ determines an ideal sheaf $\underline{I}_{X|Y}$. If f is essentially of finite type, then by 3.13(a), f is smooth at x if and only if

$$x \notin \operatorname{Supp}\big(\mathcal{O}_X/\underline{I}_{X|Y}\big).$$

This construction carries over naturally to adic formal schemes essentially of finite type, and is therefore equally suitable for deformation theory.

6. Non-degenerate quadratic singularities

The purpose of this section is to study deformations of a k-scheme whose only singularities are non-degenerate quadratic singularities. In view of 4.13, this becomes, in a suitable situation, a purely local problem.

We return to the notation of §4. Thus Λ is a fixed complete noetherian local ring with residue field k, and $\underline{C}_\Lambda$ denotes the category of local artinian Λ-algebras having residue field k.

Let X be a k-scheme of finite type, and let x be a closed point of X. Recall that x is called a *non-degenerate quadratic singularity rational over k* if

$$\widehat{\mathcal{O}}_{X,x} \simeq k[[x_1, \ldots, x_n]]/(f)$$

and

$$\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) = (x_1, \ldots, x_n).$$

One may always take f to be a quadratic form. Indeed:

Lemma 6.1. Let $f \in k[[x_1, \ldots, x_n]]$, and put $M = (x_1, \ldots, x_n)$. If

$$\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) = M,$$

then there are elements $x_1', \ldots, x_n'$ satisfying $x_i' \equiv x_i \pmod{M^2}$ such that

$$f = q(x_1', \ldots, x_n')$$

for a quadratic form q.

Proof. We use successive approximation. Write

$$f = f_2 + h_3,$$

where f_2 is quadratic and $h_3 \in M^3$. Since the partial derivatives of f generate M, there are $\varepsilon_1^{(1)}, \ldots, \varepsilon_n^{(1)} \in M^2$ such that

$$\sum_i \varepsilon_i^{(1)} \frac{\partial f}{\partial x_i} \equiv h_3 \pmod{M^4}.$$

Put $x_i^{(1)} = x_i - \varepsilon_i^{(1)}$. Then

$$\begin{aligned} f(x_1^{(1)}, \ldots, x_n^{(1)}) &= f(x_1, \ldots, x_n) - \sum_i \varepsilon_i^{(1)} \frac{\partial f}{\partial x_i} + \cdots \\ &\equiv f_2(x_1, \ldots, x_n) \pmod{M^4}. \end{aligned}$$

Thus

$$f(x_1^{(1)}, \ldots, x_n^{(1)}) = f_2(x_1, \ldots, x_n) + h_4, \qquad h_4 \in M^4.$$

Moreover, the derivatives with respect to the new variables still generate M. Hence one can choose $\varepsilon_1^{(2)}, \ldots, \varepsilon_n^{(2)} \in M^3$ such that

$$\sum_i \varepsilon_i^{(2)} \frac{\partial f}{\partial x_i^{(1)}} \equiv h_4 \pmod{M^5}.$$

It follows that

$$\begin{aligned} &f(x_1^{(1)} - \varepsilon_1^{(2)}, \ldots, x_n^{(1)} - \varepsilon_n^{(2)}) \\ &\quad = f(x_1^{(1)}, \ldots, x_n^{(1)}) - \sum_i \varepsilon_i^{(2)} \frac{\partial f}{\partial x_i^{(1)}} + \cdots \\ &\quad \equiv f_2(x_1, \ldots, x_n) \pmod{M^5}. \end{aligned}$$

Continuing in this way, one finds elements $\varepsilon_i^{(\nu)} \in M^{\nu+1}$ such that, on setting

$$x_i^{(\nu)} = x_i - \big(\varepsilon_i^{(1)} + \cdots + \varepsilon_i^{(\nu)}\big),$$

one has

$$f(x_1^{(\nu)}, \ldots, x_n^{(\nu)}) \equiv f_2(x_1, \ldots, x_n) \pmod{M^{\nu+3}}.$$

Let $\varepsilon_i = \sum_\nu \varepsilon_i^{(\nu)}$. Completeness then gives

$$f(x_1 - \varepsilon_1, \ldots, x_n - \varepsilon_n) = f_2(x_1, \ldots, x_n),$$

or, equivalently,

$$f(x_1, \ldots, x_n) = f_2(x_1 + \varepsilon_1, \ldots, x_n + \varepsilon_n).$$

Taking $x_i' = x_i + \varepsilon_i$ proves the assertion. □

Lemma 6.2. Let A_0 be a local k-algebra essentially of finite type such that

$$\widehat{A}_0 \simeq k[[x_1, \ldots, x_n]]/(f),$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \mathrm{Spec}(A_0)$. Then:

(1) The canonical maps

$$J_{\mathfrak{X}}(\mathfrak{X}|R) \longrightarrow J_R(\mathfrak{X}|R) \longrightarrow \mathrm{Spec}(\Lambda)$$

are both isomorphisms.

(2) The ideal $I_R(\mathfrak{X}|R)$ is principal and generated by a free formal generator of R over Λ; in other words,

$$I_R(\mathfrak{X}|R) = (t), \qquad R = \Lambda[[t]].$$

(3) The morphism $\mathfrak{X} \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth.

Proof. Write $\mathfrak{X} = \mathrm{Spf}(A)$. By 4.10(b), $(R, \mathrm{Spf}(\widehat{A}))$ is a formal versal deformation of $\widehat{X}_0 = \mathrm{Spec}(\widehat{A}_0)$, where

$$\widehat{A} = \varprojlim_\nu A \otimes_R R_\nu.$$

Now

$$\widehat{A}_0 \simeq k[[x_1, \ldots, x_n]]/(f), \qquad \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) = (x_1, \ldots, x_n).$$

It follows immediately that

$$\big[\underline{M}_{\widehat{X}_0}(k[\varepsilon])\big] \simeq k.$$

Consequently, by 4.14,

$$R \simeq \Lambda[[t]], \qquad \widehat{A} \simeq R[[x_1, \ldots, x_n]]/(F - t),$$

where F is the quadratic polynomial in $\Lambda[x_1, \ldots, x_n] \subset R[[x_1, \ldots, x_n]]$ obtained from f by lifting its coefficients to Λ.

The sequence

$$0 \longrightarrow \widehat{A} \xrightarrow{dF} \widehat{A}\,dx_1 \oplus \cdots \oplus \widehat{A}\,dx_n \longrightarrow \Omega_{\widehat{A}|R} \longrightarrow 0$$

is exact, where

$$1 \longmapsto \frac{\partial F}{\partial x_1}dx_1 + \cdots + \frac{\partial F}{\partial x_n}dx_n.$$

Therefore

$$I_{\widehat{A}}(\widehat{A}|R) = \left(\frac{\partial F}{\partial x_1}, \ldots, \frac{\partial F}{\partial x_n}\right) = (x_1, \ldots, x_n),$$

and hence

$$\begin{aligned}\widehat{A}/I_{\widehat{A}}(\widehat{A}|R) &= \widehat{A}/(x_1,\dots,x_n)\\ &= R[[x_1,\dots,x_n]]/(x_1,\dots,x_n,F-t)\simeq\Lambda.\end{aligned}$$

Thus the canonical map $\Lambda\longrightarrow\widehat{A}/I_{\widehat{A}}(\widehat{A}|R)$ is an isomorphism. The completion comparison identifies this quotient with $A/I_A(A|R)$, and consequently

$$\Lambda\xrightarrow{\sim}R/I_R(A|R).$$

This proves (1) and (2). Finally, (3) follows from the isomorphisms

$$\begin{aligned}\widehat{A}&\simeq R[[x_1,\dots,x_n]]/(F-t)\\ &\simeq\Lambda[[t,x_1,\dots,x_n]]/(F-t)\simeq\Lambda[[x_1,\dots,x_n]]\end{aligned}$$

of Λ-algebras. □

Corollary 6.3. Let X_0 be a k-scheme of finite type, and let $x\in X_0$ be a non-degenerate quadratic singularity rational over k.

(1) Let $(R,\mathfrak{X})$ and $(R,\mathfrak{Y})$ be two deformations of X_0, with $R\in\mathrm{ob}(\widehat{\underline{C}}_\Lambda)$. Assume that $k=k^2$. Then

$$\mathfrak{X}_{(\widehat{x})}\simeq\mathfrak{Y}_{(\widehat{x})}$$

as R-adic formal schemes if and only if

$$I_R(\mathfrak{X}_{(x)}|R)=I_R(\mathfrak{Y}_{(x)}|R).$$

(2) Let $R\in\mathrm{ob}(\widehat{\underline{C}}_\Lambda)$ be a regular local ring, and let $(R,\mathfrak{X})$ be a deformation of X_0. Then $\mathcal{O}_{\mathfrak{X},x}$ is regular if and only if

$$I_R(\mathfrak{X}_{(x)}|R)\not\subset\mathfrak{m}_R^2;$$

equivalently, $I_R(\mathfrak{X}_{(x)}|R)$ is a regular divisor in $\mathrm{Spf}(R)$.

Proof. Let (R_0,Z) be a formal versal deformation of $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})$. Since x is a non-degenerate quadratic singularity rational over k, Lemma 6.2 allows us to take

$$R_0\simeq\Lambda[[t]],\qquad Z\simeq\mathrm{Spf}\big(R_0[[X_1,\dots,X_n]]/(F-t)\big).$$

For (1), the deformations $(R,\mathfrak{X}_{(\widehat{x})})$ and $(R,\mathfrak{Y}_{(\widehat{x})})$ are induced from (R_0,Z) by continuous local Λ-algebra homomorphisms

$$\varphi,\psi:R_0\longrightarrow R.$$

Put $a=\varphi(t)$ and $b=\psi(t)$. Then

$$\begin{aligned}\mathfrak{X}_{(\widehat{x})}&\simeq\mathrm{Spf}\big(R[[X_1,\dots,X_n]]/(F-a)\big),\\ \mathfrak{Y}_{(\widehat{x})}&\simeq\mathrm{Spf}\big(R[[X_1,\dots,X_n]]/(F-b)\big).\end{aligned}$$

Suppose that $I_R(\mathfrak{X}_{(x)}|R)=I_R(\mathfrak{Y}_{(x)}|R)$; equivalently, $(a)=(b)$ in R. Then $a=ub$ for some unit $u\in R$. Since $k=k^2$ and R is complete, $u=v^2$ for some unit $v\in R$, and hence $a=v^2b$. Thus

$$(F-a)\longmapsto v^2(F-b)$$

under $X_i\mapsto vX_i$, and this substitution induces an R-algebra isomorphism

$$R[[X_1,\dots,X_n]]/(F-a)\xrightarrow{\sim}R[[X_1,\dots,X_n]]/(F-b),$$

because F is quadratic. The converse is immediate.

For (2), there is a continuous local Λ-algebra homomorphism

$$\varphi:R_0=\Lambda[[t]]\longrightarrow R$$

such that

$$\mathfrak{X}_{(\widehat{x})} \simeq \mathrm{Spf}\big(R[[X_1,\dots,X_n]]/(F-\varphi(t))\big).$$

Moreover,

$$I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{X}_{(\widehat{x})}|R) = (\varphi(t)),$$

and

$$R[[X_1,\dots,X_n]]/(F-\varphi(t)) = \widehat{\mathcal{O}}_{\mathfrak{X},x}$$

is regular if and only if $\mathcal{O}_{\mathfrak{X},x}$ is regular. Since R is regular, $R[[X_1,\dots,X_n]]$ is a regular local ring. If M denotes its maximal ideal, then its quotient by $F-\varphi(t)$ is regular if and only if

$$F-\varphi(t) \notin M^2,$$

which is equivalent to $\varphi(t) \notin \mathfrak{m}_R^2$, and hence to

$$I_R(\mathfrak{X}_{(x)}|R) \not\subset \mathfrak{m}_R^2.$$

□

Theorem 6.4. Let X_0 be a k-scheme of finite type whose nonsmooth points are isolated, and let $(R,\mathfrak{X})$ be a formal versal deformation of X_0. Denote the set of nonsmooth points of X_0 by $\{z_1,\dots,z_m\}$. Assume that

$$H^2(X_0, \underline{D}^0_{X_0}) = 0$$

and that every z_i is a non-degenerate quadratic singularity rational over k. Then:

(1) R is formally smooth over Λ; that is, R is a formal power-series ring over Λ.

(2) There is a disjoint-union decomposition

$$J_{\mathfrak{X}}(\mathfrak{X}|R) = \coprod_{1\le i\le m} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R),$$

and the canonical morphism

$$J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) \xrightarrow{\sim} J_R(\mathfrak{X}_{(z_i)}|R)$$

is an isomorphism.

(3) Each ideal $I_R(\mathfrak{X}_{(z_i)}|R)$ is principal, as is

$$I_R(\mathfrak{X}|R) = \prod_{1\le i\le m} I_R(\mathfrak{X}_{(z_i)}|R).$$

If $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i)$, then $t_1,\dots,t_m$ can be extended to a system of free formal generators of R over Λ.

(4) The morphism $\mathfrak{X} \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth; equivalently,

$$J_{\mathfrak{X}}(\mathfrak{X}|\Lambda) = \varnothing.$$

(5) If X_0 is a complete irreducible curve, then

$$\dim \underline{M}_{X_0} = 3g-3+a,$$

where

$$g = \dim_k H^1(X_0,\mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0,\underline{D}^0_{X_0}).$$

Proof. Assertion (1) follows from (2) and (3). The first assertion in (2) follows from 5.7(2). For each i, let $(R_i,\mathfrak{Y}_i)$ be a formal versal deformation of $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,z_i})$. Since $(R,\mathfrak{X}_{(\widehat{z}_i)})$ is such a deformation, there is a continuous local Λ-algebra homomorphism

$$\varphi_i : R_i \longrightarrow R$$

and a cartesian diagram

$$\begin{array}{ccc} \mathfrak{X}_{(\widehat{z}_i)} & \longrightarrow & \mathfrak{Y}_i \\ \downarrow & & \downarrow \\ \mathrm{Spf}(R) & \xrightarrow[\mathrm{Spf}(\varphi_i)]{} & \mathrm{Spf}(R_i). \end{array}$$

Consequently,

$$\begin{aligned} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) &= J_{\mathfrak{X}_{(\widehat{z}_i)}}(\mathfrak{X}_{(\widehat{z}_i)}|R) \\ &\simeq J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R), \end{aligned}$$

and

$$\begin{aligned} J_R(\mathfrak{X}_{(z_i)}|R) &= J_R(\mathfrak{X}_{(\widehat{z}_i)}|R) \\ &\simeq J_{R_i}(\mathfrak{Y}_i|R_i) \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R). \end{aligned}$$

By Lemma 6.2(1),

$$J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \longrightarrow J_{R_i}(\mathfrak{Y}_i|R_i)$$

is an isomorphism, proving the remaining assertion in (2).

By Lemma 6.2(2),

$$I_{R_i}(\mathfrak{Y}_i|R_i) = (u_i), \qquad R_i = \Lambda[[u_i]].$$

It follows that

$$I_R(\mathfrak{X}_{(z_i)}|R) = I_R(\mathfrak{X}_{(\widehat{z}_i)}|R) = (t_i), \qquad t_i = \varphi_i(u_i),$$

and therefore

$$I_R(\mathfrak{X}|R) = \prod_{1\le i\le m} I_R(\mathfrak{X}_{(z_i)}|R) = \left(\prod_{i=1}^{m} t_i\right).$$

By 4.13, the natural morphism

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{X_0,(\widehat{z}_1)} \times \cdots \times \underline{M}_{X_0,(\widehat{z}_m)}$$

is smooth. Hence

$$R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_m \longrightarrow R$$

is formally smooth. Thus $t_1, \ldots, t_m$ extend to a system of free formal generators of R over Λ, proving (1) and (3).

For (4), the projection

$$\mathfrak{X}_{(\widehat{z}_i)} \xrightarrow{\sim} \mathfrak{Y}_i \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R) \longrightarrow \mathfrak{Y}_i$$

is formally smooth because $\mathrm{Spf}(R) \longrightarrow \mathrm{Spf}(R_i)$ is formally smooth. Lemma 6.2(3) shows that $\mathfrak{Y}_i \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth. Hence every $\mathfrak{X}_{(\widehat{z}_i)} \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth. Since the z_i are precisely the nonsmooth points of X_0, it follows that $\mathfrak{X} \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth. For (5), R is formally smooth over Λ, and by 4.4 and 4.11 the sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}) \longrightarrow 0$$

is exact. Thus

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim_k H^1(X_0, \underline{D}^0_{X_0}) + \dim_k H^0(X_0, \underline{D}^1_{X_0}) \\ &= \dim_k H^1(X_0, \underline{D}^0_{X_0}) + m. \end{aligned}$$

It is therefore enough to show that

$$\dim_k H^1(X_0, \underline{D}^0_{X_0}) + m = 3g - 3 + \dim_k H^0(X_0, \underline{D}^0_{X_0}),$$

or equivalently

$$\chi(\underline{D}^0_{X_0}) = 3 - 3g + m,$$

where, for a coherent $\mathcal{O}_{X_0}$-module $\mathcal{E}$, we put

$$\chi(\mathcal{E}) = \dim_k H^0(X_0, \mathcal{E}) - \dim_k H^1(X_0, \mathcal{E}).$$

The Euler characteristic of $\underline{D}^0_{X_0}$ is unchanged by extension of the base field to its algebraic closure, so we may assume that k is algebraically closed. Let z be a nonsmooth point of X_0, put

$$A = \widehat{\mathcal{O}}_{X_0,z},$$

and let A' be the normalization of A. By hypothesis,

$$A \simeq k[[x,y]]/(f),$$

where f is a non-degenerate quadratic form. Over the algebraically closed field k we may take $f = xy$, so that

$$A \simeq k[[x,y]]/(xy).$$

It follows at once that

$$\ell(A'/A) = 1, \qquad \mathrm{Der}_k(A, \mathfrak{m}_A) = \mathrm{Der}_k(A, A).$$

Let C be the conductor of A in A'. Since A is a complete intersection,

$$\ell(A/C) = \ell(A'/A) = 1,$$

and hence $C = \mathfrak{m}_A$. Therefore

$$A' = \{a' \in K \mid a'\mathfrak{m}_A \subset \mathfrak{m}_A\},$$

where K is the total fraction ring of A.

We claim that

$$\mathrm{Der}_k(A, A) \subset \mathrm{Der}_k(A', A')$$

inside $\mathrm{Der}_k(K, K)$. Indeed, let $D \in \mathrm{Der}_k(A, A)$. For $a' \in A'$ and $x \in \mathfrak{m}_A$, one has $xa' \in \mathfrak{m}_A$, so $D(xa') \in \mathfrak{m}_A$ and $a'D(x) \in \mathfrak{m}_A$. The identity

$$D(xa') = a'D(x) + xD(a')$$

therefore gives $xD(a') \in \mathfrak{m}_A$ for every $x \in \mathfrak{m}_A$, whence $D(a') \in A'$.

Moreover, the exact sequence

$$0 \longrightarrow \mathrm{Der}_k(A, A) \longrightarrow \mathrm{Der}_k(A', A') \longrightarrow A'/\mathfrak{m}_A A' \longrightarrow 0$$

may be written

$$0 \longrightarrow \mathrm{Der}_k(A, A) \longrightarrow \mathrm{Der}_k(A', A') \longrightarrow A'/C \longrightarrow 0.$$

If X'_0 is the normalization of X_0, these local sequences give an exact sequence

$$0 \longrightarrow \underline{D}^0_{X_0} \longrightarrow \underline{D}^0_{X'_0} \longrightarrow \mathcal{O}_{X'_0}/C \longrightarrow 0.$$

Since X_0 is locally a complete intersection,

$$\ell(\mathcal{O}_{X'_0}/C) = 2\ell(\mathcal{O}_{X'_0}/\mathcal{O}_{X_0}) = 2m,$$

whereas the Riemann–Roch theorem gives

$$\chi(\underline{D}^0_{X'_0}) = 3 - 3(g - m).$$

Consequently,

$$\begin{aligned}\chi(\underline{D}^0_{X_0}) &= \chi(\underline{D}^0_{X_0'}) - \ell(\mathcal{O}_{X_0'}/C)\\ &= 3 - 3(g-m) - 2m = 3 - 3g + m.\end{aligned}$$

This completes the proof. □

Remark 6.5. In the formula of Theorem 6.4(5), the number $a = \dim_k H^0(X_0, \underline{D}^0_{X_0})$ is easily computed by the Riemann–Roch theorem:

$$a = \begin{cases} 0, & g \geq 2,\\ 1, & g = 1,\\ 3, & g = 0.\end{cases}$$

We finish with a remark on characteristic 2. If $\operatorname{char} k = 2$, there is no non-degenerate quadratic form in an odd number of variables. Equivalently, there is no non-degenerate quadratic singularity of even dimension in characteristic 2. In this case one may instead consider a quadratic form of maximal non-degeneracy.

Definition 6.6. Let X be a k-scheme of finite type. A closed point $z \in X$ is called an *ordinary quadratic singularity rational over* k if the following conditions hold.

(i) If either $\operatorname{char} k \neq 2$, or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is odd, then z is a non-degenerate quadratic singularity rational over k.

(ii) If $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is even, then

$$\overline{k} \otimes_k \widehat{\mathcal{O}}_{X,z} \simeq \overline{k}[[X_0, X_1, \ldots, X_{2n}]]/(f),$$

where $\overline{k}$ is an algebraic closure of k and

$$f = X_0^2 + X_1X_2 + X_3X_4 + \cdots + X_{2n-1}X_{2n}.$$

Lemma 6.7. Let k be a field of characteristic 2, and put

$$A_0 = k[[X_0, X_1, \ldots, X_{2n}]]/(f), \qquad f = X_0^2 + X_1X_2 + X_3X_4 + \cdots + X_{2n-1}X_{2n}.$$

Let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec}(A_0)$. Then

$$R = \Lambda[[t, t_1]],$$

where t, t_1 are free formal generators of R over Λ, and

$$I_R(\mathfrak{X}|R) = (t^2).$$

Moreover, $\mathfrak{X} \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth.

Proof. Write $\mathfrak{X} = \operatorname{Spf}(A)$. Since

$$\left(f, \frac{\partial f}{\partial X_0}, \ldots, \frac{\partial f}{\partial X_{2n}}\right) = (X_0^2, X_1, \ldots, X_{2n}),$$

we have

$$D^1(A_0|k, A_0) \simeq k[X_0]/(X_0^2).$$

Thus $[\underline{M}_{X_0}(k[\varepsilon])]$ is a two-dimensional k-vector space. By 4.14,

$$R \simeq \Lambda[[t, t_1]], \qquad A \simeq R[[X_0, X_1, \ldots, X_{2n}]]/(F),$$

where

$$F = f + t_1 + tX_0.$$

The exact sequence

$$0 \longrightarrow A \xrightarrow{dF} A\,dX_0 \oplus \cdots \oplus A\,dX_{2n} \longrightarrow \widehat{\Omega}_{A|R} \longrightarrow 0,$$

in which

$$dF:\ 1 \longmapsto \sum_{0\le i\le 2n} \frac{\partial f}{\partial X_i}\,dX_i + t\,dX_0,$$

gives

$$I_A(A|R) = (t, X_1, \ldots, X_{2n}) = (X_0^2 + t_1, t, X_1, \ldots, X_{2n}).$$

Hence

$$\begin{aligned} A/I_A(A|R) &\simeq A/(X_0^2 + t_1, t, X_1, \ldots, X_{2n}) \\ &\simeq \Lambda[[t_1, X_0]]/(X_0^2 + t_1). \end{aligned}$$

As an R-module this last module is isomorphic to $R/(t) \oplus R/(t)$. It follows that

$$I_R(A|R) = (t^2),$$

which proves the first assertion. On the other hand,

$$\begin{aligned} A &\simeq \Lambda[[t, t_1, X_0, X_1, \ldots, X_{2n}]]/(X_0^2 + X_1X_2 + \cdots + X_{2n-1}X_{2n} + tX_0 + t_1) \\ &\simeq \Lambda[[t, X_0, X_1, \ldots, X_{2n}]], \end{aligned}$$

so A is formally smooth over Λ. □

Lemma 6.7 gives the following analogue of Theorem 6.4; we leave its proof to the reader.

Corollary 6.8. Let X_0 be a k-scheme of finite type whose nonsmooth points are isolated, assume that

$$H^2(X_0, \underline{D}^0_{X_0}) = 0,$$

and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Denote the set of nonsmooth points by $\{z_1, \ldots, z_m\}$. Suppose that $\operatorname{char} k = 2$, that $\dim X_0$ is even, and that all the z_i are ordinary quadratic singularities. Then:

(1) R is formally smooth over Λ.

(2) Each ideal $I_R(\mathfrak{X}_{(z_i)}|R)$ is principal, as is

$$I_R(\mathfrak{X}|R) = \prod_{1\le i\le m} I_R(\mathfrak{X}_{(z_i)}|R).$$

More precisely,

$$I_R(\mathfrak{X}_{(z_i)}|R) = (t_i^2),$$

and $t_1, \ldots, t_m$ can be extended to a system of free formal generators of R over Λ.

(3) The morphism $\mathfrak{X} \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth.

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then

$$\dim \underline{M}_{X_0} = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, \underline{D}^0_{X_0}).$$

SGA 7

Exposé VII

BIEXTENSIONS OF SHEAVES OF GROUPS

by A. Grothendieck

0. Introduction

The notion of a biextension was introduced by D. Mumford [6], in the context of formal groups, as a convenient way of expressing the relations between the formal groups associated with an abelian scheme and the dual abelian scheme. In the present exposé we undertake a systematic study of this notion in the general context of sheaves of groups on a given site. In particular, we show that it fits remarkably simply into the classical cohomological formalism of $\overset{\mathrm{L}}{\otimes}$ and RHom (3.6.4 and 3.6.5). In the following exposé we shall make explicit the most interesting special cases, notably that of biextensions of group schemes by the multiplicative group $\underline{G}_m$ (working with such sites as the Zariski site, the fpqc site, or an intermediate site). We shall see, in particular, that if A and B are two abelian schemes over a scheme S, biextensions of (A, B) by $\underline{G}_m$ admit a remarkable interpretation as, essentially, the classical divisorial correspondences on A, B. This gives a particularly convenient explicit cohomological meaning to the notion of divisorial correspondence (already implicit in Tate duality for abelian varieties over p-adic fields). This interpretation of divisorial correspondences is also convenient for following the fate of such a correspondence when A and B (assumed to be defined over the fraction field of a discrete valuation ring) are made to degenerate by means of Néron theory.

The results of Exposé VIII will show, for example, that one obtains a biextension of the pair of identity components of the Néron models, hence a biextension of their connected special fibres, which plays the role of a specialization of the original divisorial correspondence. These results will be used in Exposé IX, in conjunction with the monodromy theorem (I 3), to prove certain results about the Néron models of abelian varieties, the most important being the "semistable reduction theorem" for the Néron model.

Throughout what follows, unless expressly stated otherwise, we work in a fixed $\underline{U}$-topos denoted by $\underline{T}$; that is (SGA 4, Exposé IV), $\underline{T}$ is a category equivalent to the category of sheaves on a suitable $\underline{U}$-site (which may, if desired, be taken to be $\underline{T}$ itself, endowed with its canonical topology). In our terminology we shall often identify the objects of $\underline{T}$ with the sheaves of sets on $\underline{T}$. All groups, extensions of groups, and so forth considered below are relative to this topos $\underline{T}$.

1. Complements on extensions of groups

1.1. Let P and G be two groups of $\underline{T}$, and let

$$1 \longrightarrow G \xrightarrow{\ i\ } E \xrightarrow{\ j\ } P \longrightarrow 1 \tag{1.1.1}$$

be an extension of P by G. Thus E is a group, $j : E \longrightarrow P$ is an epimorphism of groups, and i induces an isomorphism of G with $\ker j$. We shall reinterpret the datum of such an extension in different terms, which will be convenient for the definition and study of biextensions.

1.1.2. First, the homomorphism i makes G act on E in two ways, on the left and on the right, by setting

$$gxg' = i(g)xi(g')$$

for $g, g' \in G(S)$ and $x \in E(S)$, where S is any object of $\underline{T}$. If, by means of j, we regard E as an object "over P", that is, as an object of the induced topos $\underline{T}_{/P}$, this amounts to saying that the group $G_P = G \times P$ induced by G in $\underline{T}_{/P}$ acts on E on the left and on the right. Moreover, it is well known that each of these actions makes E a torsor (a left torsor and a right torsor, respectively) over P, with group G_P; more precisely still, it makes E a bitorsor under (G_P, G_P). Recall [2]

that this means that E is a right torsor E_d under the right action of G_P, and that the left action induces an isomorphism of G_P with the group of automorphisms of E_d (the sheaf of groups of automorphisms of E that commute with the right action of G_P on E). This condition is equivalent to the symmetric condition: the left action of G_P on E makes E a left torsor E_s, and the right action induces an isomorphism of the opposite group G_P^o with the group of automorphisms of E_s.

1.1.2.1. When $\underline{T}$ is the "point topos", that is, the category of sets (equivalent to the category of sheaves on the one-point topological space), so that P and G are ordinary groups, the datum of a bitorsor E under (G_P, G_P) is plainly equivalent to the datum, for each $p \in P$, of a bitorsor E_p under (G, G), namely the fibre of E at p. An analogous interpretation is available in the general case. Indeed, giving a bitorsor E under (G_P, G_P) is equivalent to giving, for every "point" $p \in P(S)$, where S is any object of $\underline{T}$, a bitorsor E_p under (G_S, G_S) over S, in such a way that the formation of E_p is "compatible with base change along every morphism $S' \longrightarrow S$". Thus, if $p' : S' \longrightarrow P$ is the composite morphism, one is also given an isomorphism of bitorsors from $E_{p'}$ to the inverse image of E_p under $S' \longrightarrow S$, and these isomorphisms satisfy the usual transitivity condition for a composite $S'' \longrightarrow S' \longrightarrow S$.

When stated in canonical language (SGA 1, Exposé VI), this reduces to the following tautology: giving a bitorsor under (G_P, G_P) over P is equivalent to giving a cartesian section, over the category $\underline{T}_{/P}$, of the fibred category of bitorsors under (G_S, G_S) over a variable base $S \in \mathrm{ob}\,\underline{T}_{/P}$. More precisely, the "value at P" functor induces an equivalence between the category of the cartesian sections in question and the category of bitorsors under (G_P, G_P). This follows trivially from the fact that P is a final object of the base category $\underline{T}_{/P}$ of our fibred category: for every category fibred over $\underline{T}_{/P}$, the "value at P" functor is an equivalence between the category of cartesian sections of that fibred category and its fibre category over P. Despite its tautological nature, however, this interpretation will be useful for expressing certain compatibilities in language close to the set-theoretic case, using points with values in an object S in place of ordinary points.

1.1.3. In terms of the extension structure (1.1.1), the bitorsor structure of 1.1.2 on E under (G_P, G_P) used only part of the multiplicative structure of E: namely, multiplication of pairs of "points" of which at least one comes from a point of G (via i). We shall now interpret the multiplicative structure of E as an additional structure on the bitorsor under consideration.

For this purpose, first consider two sections p, p' of P. With the notation of 1.1.2.1, these give two bitorsors $E_p, E_{p'}$ under (G, G), namely the inverse images in E of the sections $p, p' : e \longrightarrow P$ (where e denotes the final object of $\underline{T}$). It is then clear that multiplication in E induces a morphism of objects of $\underline{T}$

$$\varphi : E_p \times E_{p'} \longrightarrow E_{pp'},$$

which, in terms of the left and right structures of the three bitorsors involved (and, as usual, using points with values in an arbitrary object $S \in \mathrm{ob}\,\underline{T}$), satisfies

$$\text{(1.1.3.0)} \qquad \begin{cases} \varphi(gx, x') = g\varphi(x, x'), & \varphi(x, x'g') = \varphi(x, x')g', \\ \varphi(xg, x') = \varphi(x, gx'). \end{cases}$$

Here $g, g' \in G(S)$, $x \in E_p(S)$, and $x' \in E_{p'}(S)$. In terms of the contracted product [2]

$$E_p \overset{G}{\times} E_{p'},$$

the third relation says that φ factors as

$$E_p \times E_{p'} \longrightarrow E_p \overset{G}{\times} E_{p'} \xrightarrow{\varphi_{p,p'}} E_{pp'},$$

where the first arrow is the canonical quotient morphism. The first two relations, on the other hand, say that the morphism thus obtained,

$$\text{(1.1.3.1)} \qquad \varphi_{p,p'} : E_p \overset{G}{\times} E_{p'} \longrightarrow E_{pp'},$$

is a homomorphism of bitorsors, where the bitorsor structure on the source is the one for which the left (respectively, right) operations of G come from the left (respectively, right) operations of G on the first factor E_p (respectively, on the second factor $E_{p'}$).

Notice that, like every homomorphism of bitorsors under a fixed pair of groups, $\varphi_{p,p'}$ is necessarily an isomorphism.

The definition of the preceding isomorphism (1.1.3.1) immediately extends to the case in which p, p' are points of P with values in an arbitrary object S of $\underline{T}$, rather than sections of P: one simply applies the preceding discussion to the induced topos $\underline{T}_{/S}$ and to the extension of groups over S obtained from (1.1.1) by base change. The formation of the isomorphisms $\varphi_{p,p'}$ is then compatible with every base change $S' \longrightarrow S$. In other words, it corresponds to a homomorphism of cartesian sections of the fibred category over the category $\underline{T}_{/P\times P}$ of pairs of arrows $p, p' : S \rightrightarrows P$ with target P.

Giving a system of $\varphi_{p,p'}$ satisfying this condition is, of course, equivalent to giving $\varphi_{p,p'}$ in the "universal" case $S = P \times P$, where p, p' are the two projections pr_i $(i = 1, 2)$ from $P \times P$ to P. It is then an isomorphism of bitorsors under $(G_{P\times P}, G_{P\times P})$:

$$\varphi : \mathrm{pr}_1^*(E)\ \mathrm{pr}_2^*(E) \xrightarrow{\sim} \pi^*(E), \tag{1.1.3.2}$$

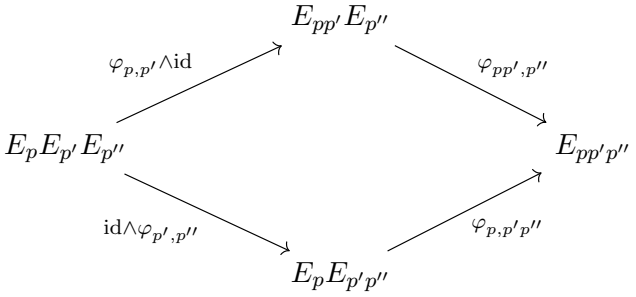
where

$$\pi : P \times P \longrightarrow P$$

is the morphism of objects of $\underline{T}$ defining the multiplicative structure of the group P. The compatibility properties below are most conveniently stated, however, in terms of the data (1.1.3.1), avoiding diagrams that are less close to intuition.

1.1.4. In writing the relations (1.1.3.0), which enabled us to express the given composition law on E in terms of isomorphisms of bitorsors (1.1.3.1), we used associativity of this composition law only for a product of three "points" of E of which at least one comes from a point of G. To express full associativity in terms of the $\varphi_{p,p'}$, one must require, for three points p, p', p'' of P with values in an object S of $\underline{T}$, that the following diagram be commutative:

(1.1.4.1)



In this diagram we have written the contracted product of bitorsors without the signs $\overset{G_S}{\times}$. It is understood that the contracted product $Q.R$ of two bitorsors is formed using, in passing to the quotient in $Q \times R$, the action of the structural group G on the right of Q and on the left of R. This contracted-product operation is associative up to a canonical isomorphism [2], and we make the identification

$$(E_pE_{p'})E_{p''} \simeq E_p(E_{p'}E_{p''}),$$

writing both sides simply as $E_pE_{p'}E_{p''}$. Obvious conventions of this kind will be used throughout the rest of this exposé without being expressly noted each time.

1.1.4.2. Just as the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) are determined by the "universal" case $S = P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$ considered in (1.1.3.2), so the associativity condition (1.1.4.1) may likewise be checked only in the universal case $S = P \times P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$, $p'' = \mathrm{pr}_3$. We leave it to the reader to write the diagram in whatever notation is preferred.

1.1.5. Conversely, suppose that the groups P and G of $\underline{T}$ are given, together with a bitorsor

$$j : E \longrightarrow P \tag{1.1.5.1}$$

under (G_P, G_P) (the two actions of G_P on E, or equivalently of G on E, being understood in the notation), and a system of isomorphisms of bitorsors over a variable base S,

$$\varphi_{p,p'} : E_p E_{p'} \xrightarrow{\sim} E_{pp'} \qquad (p, p' \in P(S)), \tag{1.1.5.2}$$

compatible with base change $S' \longrightarrow S$; in other words, an isomorphism of bitorsors (1.1.3.2). Suppose that the corresponding diagrams (1.1.4.1) commute. I claim that there is a unique extension structure on E of P by G such that the preceding data are obtained from it by the constructions of 1.1.2 and 1.1.3.

First observe that, on taking p, p' in (1.1.5.2) to be the identity section 1 of P, one obtains an isomorphism

$$\varphi_{1,1} : E_1 \xleftarrow{\sim} E_1 \overset{G}{\times} E_1. \tag{1.1.5.3}$$

Taking the contracted product, on the left say, of both sides with the inverse bitorsor E_1^{-1} of E_1 [2] therefore gives a canonical isomorphism

$$E_1 \simeq \text{the unit bitorsor } {}_sG_d \tag{1.1.5.4}$$

(that is, G acting on itself by left and right translations), in view of the canonical isomorphism

$$E_1^{-1}.E_1 \simeq \text{the unit bitorsor } {}_sG_d$$

of loc. cit., valid for every bitorsor. Equivalently, one has a section 1_E of E_1, hence of E,

$$1_E \in \Gamma(E_1) = \mathrm{Hom}(e, E_1) \subset \Gamma(E), \tag{1.1.5.5}$$

namely the image under (1.1.5.4) of the distinguished section of ${}_sG_d$, satisfying

$$g.1_E = 1_E.g \qquad (g \in G(S)).$$

By construction this section is characterized by

$$\varphi_{1,1}(1_E, 1_E) = 1_E. \tag{1.1.5.6}$$

On the other hand, giving the morphism of torsors φ of (1.1.3.2) is plainly equivalent to giving a morphism

$$\pi_E : E \times E \longrightarrow E, \qquad \text{also denoted } (x, y) \longmapsto xy, \tag{1.1.5.7}$$

satisfying

$$j(xy) = j(x)j(y) \qquad (x, y \in E(S)), \tag{1.1.5.8}$$

so that j is compatible with the multiplicative structures π_E on E and π on P, together with the three relations (1.1.3.0) (preferably beginning with the last of these). Finally define a morphism of objects of $\underline{T}$,

$$i : G \longrightarrow E,$$

by the formula

$$i(g) = g.1_E = 1_E.g. \tag{1.1.5.9}$$

The equality of the last two terms was already observed in constructing 1_E. I now claim that the multiplication law π_E on E makes E a group with identity section 1_E, and that $j : E \longrightarrow P$ and $i : G \longrightarrow E$ are homomorphisms of groups making E an extension of P by G.

Indeed, associativity of the composition law on E is precisely the commutativity of the diagrams (1.1.4.1). The construction of π_E by means of the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) also shows that, for every $x \in E(S)$, left and right translation by x in $E(S)$ are injective. Since $1_E 1_E = 1_E$ by (1.1.5.6), it follows at once that 1_E is a two-sided identity of E. We leave to the reader the verification that an $x \in E(S)$ lying over $p \in P(S)$ has an inverse; one uses the canonical isomorphism deduced from $\varphi_{p,p^{-1}}$,

$$E_{p^{-1}} \simeq (E_p)^{\circ}. \tag{1.1.5.10}$$

Thus E is indeed a group. We have already observed that j is compatible with the composition laws (1.1.5.8), and the same is true of i, since, for $g, g' \in G(S)$,

$$\begin{aligned} i(g)i(g') &= (g.1_E)(g'.1_E) \\ &= (g.1_E.g')1_E && \text{(the last relation of (1.1.3.0))} \\ &= g.1_E.g' && \text{(since } 1_E \text{ is an identity)} \\ &= g.(1_E.g') = g.(g'.1_E) \\ &= (gg').1_E && \text{by (1.1.5.9)} \\ &= i(gg'). \end{aligned}$$

Since j is the structural projection of a torsor, it is an epimorphism of objects of $\underline{T}$. It remains only to see that i induces an isomorphism of G with $\ker j = E_1$, which is clear from the definition (1.1.5.9) and the construction of 1_E.

1.1.6. We shall accept that the preceding arguments prove the following. For a fixed topos $\underline{T}$, the preceding constructions give an equivalence between the category of extensions of groups of $\underline{T}$ as in (1.1.1), where P, G, E are all variable, and the category of systems (P, G, E, φ) in which P and G are groups of $\underline{T}$, E is a bitorsor under (G_P, G_P) over P, and φ is an isomorphism of bitorsors (1.1.3.2) (or, equivalently, a system of base-change-compatible isomorphisms of bitorsors (1.1.3.1)), satisfying the associativity condition expressed by commutativity of diagram (1.1.4.2) (or, equivalently, of the diagrams (1.1.4.1)).

We leave it to the reader to spell out, using the preceding dictionary, the evident notion of inverse image (respectively, direct image) of an extension (1.1.1) under a homomorphism of groups

$$P' \longrightarrow P \qquad \big(\text{respectively, } G \longrightarrow G'\big),$$

or under both operations simultaneously. We likewise leave to the reader the compatibility of the preceding equivalence of categories with the inverse-image functor f^* associated with a morphism of topoi

$$f : \underline{T}' \longrightarrow \underline{T}.$$

For this one should use the datum φ in the form (1.1.3.2), in preference to the $\varphi_{p,p'}$ of (1.1.3.1); it then defines the corresponding datum $f^*(\varphi)$ on the topos $\underline{T}'$.

1.2. The commutative case. From 1.4 onward we shall be concerned only with commutative extensions of groups. We therefore assume that P and G are commutative, and spell out here, in terms of the bitorsor description (P, G, E, φ) of an extension, the condition that E be a commutative group.

An obviously necessary first condition—which in fact expresses that $i(G)$ be a central subgroup of E—is that the bitorsors E_p, for $p \in P(S)$, be ordinary torsors; that is, the left and right actions of G on E_p must coincide. Equivalently, E over P is an ordinary torsor under G_P. (Thus there will henceforth be no need to speak of bitorsors, and we shall consider only ordinary torsors.)

Assuming this preliminary condition, commutativity of E is expressed by commutativity of the following diagrams, for $p, p' \in P(S)$:

$$\begin{array}{ccc} E_p E_{p'} & \xrightarrow{\varphi_{p,p'}} & E_{pp'} \\ {\scriptstyle \mathrm{sym}}\downarrow & & \downarrow{\scriptstyle \mathrm{id}} \\ E_{p'} E_p & \xrightarrow{\varphi_{p',p}} & E_{p'p}. \end{array} \tag{1.2.1}$$

Here the first vertical arrow is the canonical commutativity isomorphism (defined because G is now commutative); moreover $pp' = p'p$ because P is commutative. This condition is plainly equivalent to the corresponding condition in the universal situation $S = P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$, which we need not write out again.

1.3. Relations with rigidified or birigidified torsors

1.3.1.0. Let P be an object of $\underline{T}$ equipped with a section e_P, let G be a group of $\underline{T}$, and let E be a torsor under G_P. A rigidification of E, relative to the given section e_P of P, is a trivialization of the torsor $E_1 = e_P^*(E)$ under G;

that is, a section of this torsor. This terminology is inspired by the case where G is commutative and

$$\mathrm{Hom}_{\mathrm{pt}}(P, G) \simeq (e), \tag{1.3.1.1}$$

where the left-hand side denotes the set of morphisms carrying e_P to the identity section e_G of G. This relation is equivalent to saying that the natural homomorphism

$$\Gamma(G) \xrightarrow{\sim} \mathrm{Hom}(P, G), \tag{1.3.1.2}$$

which associates with a section g of G the constant morphism $P \longrightarrow G$ with value g, is an isomorphism. It is immediate that these equivalent conditions are also equivalent to saying that every automorphism of the torsor E preserving the given rigidification is the identity automorphism (such automorphisms correspond precisely to the elements of the left-hand side of (1.3.1.1)). In particular, under condition (1.3.1.1), the category of torsors under G_P rigidified relative to e_P is "rigid".

1.3.2. Let, in addition, Q be an object of $\underline{T}$ equipped with a section e_Q, and now let E denote a torsor under $G_{P\times Q}$. Since e_Q defines a section $e_{P\times Q/P}$ of $P \times Q$ over P, one may consider rigidifications α of E relative to this section, working in the induced topos $\underline{T}_{/P}$; explicitly, these are sections of the torsor induced by E on $P \times e_Q$. Interchanging the roles of P and Q, one may likewise consider rigidifications β of E relative to the section $e_{P\times Q/Q}$ of $P \times Q$ over Q defined by e_P. Two such partial rigidifications are said to be compatible if the sections α and β agree on $e_P \times e_Q$, that is, if they define the same section of the torsor $E_{1,1}$ under G, the inverse image

of E under the section (e_P, e_Q) of $P \times Q$. A birigidification of the torsor E relative to the pair of sections (e_P, e_Q) is a pair of compatible partial rigidifications.

If G is commutative, then E admits a birigidification (α, β) if and only if it admits partial rigidifications α and β; moreover one may find a birigidification in which α (or β) is prescribed in advance. Indeed, it suffices to correct β by the morphism $Q \longrightarrow G$ obtained as the composite

$$Q \longrightarrow e \xrightarrow{\lambda} G,$$

where λ is defined by $\beta e_Q = (\alpha e_P)\lambda$.

Still in the commutative case, if E, E' are two birigidified torsors under $G_{P\times Q}$, saying that they are isomorphic merely as $G_{P\times Q}$-torsors is equivalent to saying that they are isomorphic as birigidified $G_{P\times Q}$-torsors. In other words, if there is an isomorphism $u : E \longrightarrow E'$ of $G_{P\times Q}$-torsors, then there is one $v : E \longrightarrow E'$ that preserves the birigidifications. Indeed, measuring the discrepancy of u from the given birigidifications gives morphisms $f : P \longrightarrow G$ and $g : Q \longrightarrow G$

such that $f(e_P) = g(e_Q) = a \in G(e)$. There is then a morphism $P \times Q \longrightarrow G$ whose restrictions along $e_{P\times Q/P}$ and $e_{P\times Q/Q}$ are respectively f and g, namely

$$h(p,q) = f(p)g(q)a^{-1},$$

and one may take $v = uh$.

1.3.3. The terminology of 1.3.1 extends immediately to the case of a bitorsor E under (G_P, G_P). A rigidification of the bitorsor E is then required to be an isomorphism from $e_P^*(E)$ to the trivial bitorsor ${}_sG_d$; equivalently, it is defined by a central section e_E of $e_P^*(E)$, one satisfying

$$g.e_E = e_E.g \qquad (g \in G(S)).$$

The terminology of birigidifications extends in the same way to bitorsors.

1.3.4. Return to the case in which two groups P and G are given, and choose the identity section e_P in order to define rigidifications of bitorsors over P and birigidifications of bitorsors over $P \times P$. There are natural functors

$$\textbf{EXT}(P;G) \xrightarrow{i} \mathrm{BITORSRIG}(P,G) \xrightarrow{\delta} \textbf{BITORSBIRIG}(P,P;G), \tag{1.3.4.1}$$

where the three expressions denote, respectively, the category of extensions of the group P by G, the category of bitorsors under (G_P, G_P) rigidified at e_P, and the category of bitorsors under $(G_{P\times P}, G_{P\times P})$ birigidified at (e_P, e_P). The functor i associates with an extension E of P by G the corresponding bitorsor over P (1.1.6), rigidified by the identity section of E. The functor δ is defined by

$$\delta(E) = \pi^*(E)\ \mathrm{pr}_1^*(E)^{-1}\ \mathrm{pr}_2^*(E)^{-1}, \tag{1.3.4.2}$$

where $\pi : P \times P \longrightarrow P$ is the composition law of P, and $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections. There is a canonical birigidification of $\delta(E)$. Indeed, the restriction of $\delta(E)$ to $P \times e_P$ is canonically isomorphic to

$$\mathrm{id}_P^*(E)\,\mathrm{id}_P^*(E)^{-1}\,e^*(E)^{-1},$$

where $e = \mathrm{pr}_2 \mathrm{inj}_1$ is the constant morphism $P \longrightarrow P$ with value e_P. The given rigidification of E provides an isomorphism from $e^*(E)$ to the trivial bitorsor, and hence an isomorphism of the restriction of $\delta(E)$ to $P \times e_P$ with the trivial bitorsor. Symmetrically one defines a partial rigidification along $e_P \times P$, and the two partial rigidifications are immediately seen to be compatible.

1.3.4.3. Observe that the composite δi of the functors in (1.3.4.1) is canonically isomorphic to the trivial functor, that is, to the constant functor whose value is the trivial birigidified bitorsor over $P \times P$. Indeed, for every extension E of P by G, the isomorphism φ of (1.1.3.2) plainly defines a trivialization of $\delta i(E)$ compatible with its canonical birigidification.

Proposition 1.3.5. Let P and G be two groups of T.

a) Suppose that every morphism of sheaves of sets $P \times P \longrightarrow G$ which is trivial on $P \times e_P$ and on $e_P \times P$ is trivial. Then the functor i of (1.3.4.1) is fully faithful. In particular, on a bitorsor E under (G_P, G_P) rigidified relative to e_P, that is, equipped with a central section e_E of $E_1 = e_P^*(E)$, there is at most one extension structure of P by G compatible with its bitorsor structure and having e_E as identity section.

b) Suppose in addition that every morphism of sheaves of sets $P \times P \times P \longrightarrow G$ which is trivial on the partial products $P \times P \times e_P$, $P \times e_P \times P$, and $e_P \times P \times P$ is trivial. Let E be a bitorsor under (G_P, G_P). In order that E admit an extension structure of P by G compatible with its bitorsor structure, it is necessary and sufficient that the bitorsor under $(G_{P\times P}, G_{P\times P})$ underlying

$$\delta(E) = \pi^*(E)\ \mathrm{pr}_1^*(E)^{-1}\ \mathrm{pr}_2^*(E)^{-1}$$

be trivial. When this is so, for every central section e_E of $E_1 = e_P^*(E)$, that is, every rigidification of E relative to e_P, there is a unique extension structure on E compatible with its bitorsor

structure and having e_E as identity section. In other words, E, regarded as a bitorsor rigidified relative to e_P, lies in the essential image of i. Moreover, the birigidified bitorsor $\delta(E)$ has one and only one trivialization.

Thus, in terms of diagram (1.3.4.1), the essential image of the fully faithful functor i is "the kernel" of δ. More picturesquely, diagram (1.3.4.1) is an "exact sequence of pointed categories," and even of categories endowed with pseudo-group structures (in the terminology introduced in 2.5.1 below).

Let us prove 1.3.5. For a), let E, E' be two extensions of P by G, and let $f : E \longrightarrow E'$ be a morphism of the underlying rigidified bitorsors. We show that f is a morphism of extensions, or, what plainly amounts to the same thing, a homomorphism of groups. We must prove $f(xy) = f(x)f(y)$. If $j' : E' \longrightarrow P$ denotes the structural morphism of the torsor E' over P, then $j'f(xy) = j'(f(x)f(y))$: the two sides are respectively $j(xy)$ and $j(x)j(y)$, which are equal because j is multiplicative. Hence

$$f(xy) = u(j(x), j(y))f(x)f(y),$$

where

$$u : P \times P \longrightarrow G$$

is a uniquely determined morphism. Setting first $x = 1$ and then $y = 1$ in the preceding relation, and using $f(1) = 1$, gives

$$u(p, 1) = u(1, p) = 1.$$

Thus u is trivial on $P \times e_P$ and on $e_P \times P$; by hypothesis it is trivial, and therefore $f(xy) = f(x)f(y)$, as required.

For b), necessity of the stated condition for E to admit an extension structure compatible with its bitorsor structure is immediate from 1.3.4.3. It remains to prove sufficiency for the first assertion. Pulling a trivialization φ of $\delta(E)$ back along the identity section of $P \times P$ gives a trivialization of $E_1 = e_P^*(E)$ (compare 1.5.1), that is, a rigidification of E relative to e_P. Fix this rigidification. A given trivialization φ of $\delta(E)$ need not be compatible with the corresponding birigidification. Relative to the natural partial rigidifications of $\delta(E)$, its restrictions to $P \times e_P$ and $e_P \times P$ are described by central morphisms

$$f : P \longrightarrow G, qquad g : P \longrightarrow G,$$

whose values at the identity section of P coincide: $f(1) = g(1) = a \in \Gamma G$. The central morphism

$$h : P \times P \longrightarrow G, qquad h(p, q) = f(p)g(q)a^{-1},$$

has restrictions f and g to $P \times e_P$ and $e_P \times P$. Correcting φ by h, that is, replacing it by $\varphi(p, q)h(p, q)^{-1}$, therefore gives a trivialization of $\delta(E)$ compatible with its birigidification. Such a trivialization is unique: two of them differ by a central morphism $P \times P \longrightarrow G$ trivial on $P \times e_P$ and $e_P \times P$, hence trivial by hypothesis. This proves the last assertion of 1.3.5 b).

It remains to prove that φ, viewed as an isomorphism of the form (1.1.3.2), satisfies the associativity condition (1.1.4.1); by our choice of φ, e_E is already a two-sided identity for the composition law defined by φ. The two composites in (1.1.4.1) are isomorphisms of bitorsors under $(G_{P\times P\times P}, G_{P\times P\times P})$, so they differ by a central morphism

$$u : P \times P \times P \longrightarrow G$$

through the formula

$$(xy)z = u(jx, jy, jz)\, x(yz).$$

Setting successively $z = 1$, $x = 1$, and $y = 1$, and using the fact that 1 is a two-sided identity for the composition law, shows that u is trivial on the three partial products $P \times P \times e_P$, $e_P \times P \times P$, and $P \times e_P \times P$. The hypothesis then implies that u is trivial, so $(xy)z = x(yz)$. This completes the proof.

Corollary 1.3.6. Suppose that every morphism of sheaves of sets $P \times G \longrightarrow G$ which is trivial on $P \times e_G$ and on $e_P \times G$ is trivial. Then, if G is commutative, G is central in every extension of P by G. If P is also commutative and the hypothesis of 1.3.5 a) holds, then every extension E of P by G is commutative.

For the first assertion, we must prove that the action of P on G defined by the extension E is trivial. This action is expressed by a morphism $u : P \times G \longrightarrow G$. The restrictions to $P \times e_G$ and $e_P \times G$ of the corresponding morphism

$$(p, g) \longmapsto u(p, g)g^{-1}$$

are plainly trivial. Hence this last morphism is trivial, so $u(p, g) = g$, as required. Now, if E is an extension of P by G, its opposite group E^o is an extension of P^o by G^o, and the bitorsor structure underlying $i(E^o)$ is obtained from $i(E)$ by interchanging the left and right actions. When E is a central extension, one therefore has $i(E) = i(E^o)$; if, in addition, P is commutative, so that $P = P^o$, the uniqueness assertion 1.3.5 a) gives $E = E^o$. Thus E is commutative, proving the corollary.

1.3.7. One interesting special case in which all the hypotheses of 1.3.5 and 1.3.6 hold is

$$\underline{\mathrm{Hom}}_{\mathrm{pt}}(P, G) = e \quad \text{(the final sheaf)}, \tag{1.3.7.1}$$

where the left-hand side denotes the sheaf of morphisms of pointed sheaves of sets from P to G. Independently of the marked sections, this condition may also be written in the form, analogous to (1.3.1.2),

$$G \simeq \underline{\mathrm{Hom}}_{\mathrm{sets}}(P, G), \tag{1.3.7.2}$$

where the second member denotes the sheaf of morphisms between the underlying sheaves of sets of P and G. These conditions also mean that, for every object Q of $\underline{T}$, the map $u \longmapsto u \circ \mathrm{pr}_2$

$$\mathrm{Hom}_{\mathrm{sets}}(Q, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{sets}}(P \times Q, G) \tag{1.3.7.3}$$

is bijective. This property of P (with G fixed) is stable under Cartesian products; in particular it implies the same property for $P \times P$, $P \times P \times P$, and so forth. It follows immediately that it implies the hypotheses made in 1.3.5 and 1.3.6. It also implies that all the categories occurring in (1.3.4.1) are rigid, that is, every automorphism of an object of any one of them is the identity.

Moreover, since condition (1.3.7.1) on P, G is stable under every base change $S \longrightarrow e$, descent together with the uniqueness result 1.3.5 a) shows that a rigidified bitorsor E over P comes from an extension of P by G as soon as it does so locally over e. By 1.3.5 b), this may also be expressed by the vanishing of the section

$$\underline{\delta}(E) \in \Gamma R^1 h_*(G_{P \times P})$$

defined by $\delta(E)$, where

$$h : P \times P \longrightarrow e$$

is the structural morphism.

1.3.8. Let $p : P \longrightarrow S$ be a morphism of schemes which is coherent (that is, quasi-compact and quasi-separated) and flat, and suppose that

$$p_*(\mathcal{O}_P) \xleftarrow{\sim} \mathcal{O}_S. \tag{1.3.8.1}$$

Since the formation of $p_*(\mathcal{O}_P)$ commutes with every flat base change (by coherence of p), the stated conditions on the relative scheme P/S are stable under Cartesian products and hence hold, in particular, for the Cartesian powers $P \times P$, $P \times P \times P$, and so forth. They are also stable under every flat base change.

Now let G be an affine scheme over S. Relation (1.3.7.2) follows at once from (1.3.8.1), since

$$\mathrm{Hom}_S(P, G) = \mathrm{Hom}_{\mathrm{Alg}}(\underline{A}, p_*(\mathcal{O}_P)) \simeq \mathrm{Hom}_{\mathrm{Alg}}(\underline{A}, \mathcal{O}_S),$$

where $\underline{A}$ is the quasi-coherent algebra over S such that $G = \mathrm{Spec}(\underline{A})$. By the preceding observation, the same conclusion applies to $P \times P$, $P \times P \times P$, and so on. Likewise, if G is flat over S, the same conclusion applies to the relative scheme P_G over G with respect to the relative scheme G_G over G.

When P and G are group schemes over S, and one works with a topology on $(\mathrm{Sch})_{/S}$ coarser than the canonical topology, so that P and G may be identified with sheaves on this site, every morphism from P, $P \times P$, $P \times P \times P$, and so on to G which is compatible with the marked sections is trivial. Similarly, if G is flat, every S-morphism $P \times G \longrightarrow G$ compatible with the marked sections is trivial, as one sees by interpreting it as a G-morphism from P_G to G_G. Thus the hypotheses of 1.3.5 hold, as do those of 1.3.6 when G is flat. Moreover, the three categories occurring in (1.3.4.1) are rigid.

1.3.8.2. The most interesting case in which (1.3.8.1) holds, and continues to hold after every extension of the base, is that in which P is an abelian scheme over S; then the conditions of 1.3.7 themselves hold. In this case the conclusions of 1.3.5 and 1.3.6 are therefore valid for every affine group G over S. This is the case originally considered by J. P. Serre, and the preceding developments were plainly inspired by it. An interesting case in which (1.3.8.1) holds without holding universally is that in which P is the Néron model of an abelian scheme over the fraction field of a discrete valuation ring with spectrum S.

1.4. Extensions of a free commutative group

Let I be an object of $\underline{T}$, and put

$$P = \mathbb{Z}[I], \tag{1.4.1}$$

the free $\mathbb{Z}$-module generated by I (SGA 4, IV, 11). With every extension E of P by a given $\mathbb{Z}$-module G (from now on only commutative group extensions will occur), associate the corresponding torsor under G_P, and then restrict it to I along the canonical morphism

$$I \longrightarrow P = \mathbb{Z}[I].$$

This gives a canonical functor

$$\mathrm{EXT}(P, G) \longrightarrow \mathrm{TORS}(I, G_I). \tag{1.4.2}$$

I claim that this functor is an equivalence of categories. Equivalently, for every G, the maps

$$\mathrm{Hom}_{\mathbb{Z}}(P, G) \longrightarrow H^0(I, G_I) = G(I) \tag{1.4.3}$$

and

$$\mathrm{Ext}^1_{\mathbb{Z}}(P, G) \longrightarrow H^1(I, G_I) \tag{1.4.4}$$

are isomorphisms. More generally, if A is a ring (not necessarily commutative) of $\underline{T}$ and $P = A[I]$ is the free A-module generated by I, there are canonical isomorphisms, functorial in the variable A-module G,

$$\mathrm{Ext}^i_A(A[I], G) \xrightarrow{\sim} H^i(I, G_I). \tag{1.4.5}$$

For $A = \mathbb{Z}$ and $i = 0, 1$, these coincide with (1.4.3) and (1.4.4). To define (1.4.5), recall that its left-hand side is the derived functor of

$$G \longmapsto \mathrm{Hom}_A(A[I], G),$$

which, by the definition of $A[I]$, is isomorphic to $G \longmapsto G(I) = H^0(I, G_I)$. Since $G \longmapsto G_I$ carries injective modules to injective modules of $\underline{T}_{/I}$ (SGA 4, V), the assertion follows. We leave to the reader the verification of the stated compatibility with (1.4.3) and (1.4.4).

Finally, the functor (1.4.2) is compatible with inverse images under a morphism of topoi

$$f : \underline{T}' \longrightarrow \underline{T}.$$

2. The notion of a biextension of abelian sheaves

2.0. Throughout this section, P, Q, G denote commutative groups of the topos $\underline{T}$. We shall frequently use the Cartesian diagram

$$\begin{array}{ccc} P & \longleftarrow & P \times Q \\ \downarrow & & \downarrow \\ e & \longleftarrow & Q \end{array} \tag{2.0.1}$$

and the groups G_P (respectively G_Q, $G_{P\times Q}$) over P (respectively Q, $P \times Q$), obtained by inverse image from the group G over e. We identify $G_{P\times Q}$ either with the inverse image $(G_P)_{P\times Q}$ under pr_1, or with the inverse image $(G_Q)_{P\times Q}$ under pr_2. In this notation the group structures of $P, Q, P \times Q$ play no role; indeed, we shall almost never use the product-group structure on $P \times Q$. We shall instead use on $P \times Q$ the structure of a group over P obtained by inverse image from Q over e,

$$P \times Q = Q_P, \tag{2.0.2}$$

and, similarly, the structure of a group over Q obtained by inverse image from P over e,

$$P \times Q = P_Q. \tag{2.0.3}$$

Thus over P we have the two groups G_P, Q_P, and over Q the two groups G_Q, P_Q.

If E is a torsor over $P \times Q$ under $G_{P\times Q}$, it may be viewed either as a torsor over Q_P under $G_{P\times Q} = (G_P)_{Q_P}$, or as a torsor over P_Q under $G_{P\times Q} = (G_Q)_{P_Q}$. It therefore makes sense to equip E with a commutative extension structure in either variable. Applied in the induced topoi $\underline{T}_{/P}$ and $\underline{T}_{/Q}$, 1.1.6 gives a convenient explicit description, using the canonical isomorphisms

$$\underline{T}_{/P\times Q} \simeq (\underline{T}_{/P})_{/P\times Q}, \qquad \underline{T}_{/P\times Q} \simeq (\underline{T}_{/Q})_{/P\times Q}.$$

An extension structure in the P-direction—that is, of P_Q by G_Q—is equivalent to giving, for every object S of $\underline{T}$ and elements $p, p' \in P(S)$, $q \in Q(S)$, an isomorphism of torsors under G_S

$$\varphi_{p,p';q} : E_{p,q}E_{p',q} \xrightarrow{\sim} E_{pp',q}. \tag{2.0.4}$$

As usual (cf. 1.1.2.1), for $(p, q) \in (P \times Q)(S)$, $E_{p,q}$ denotes the torsor under G_S obtained from E by inverse image along $(p, q) : S \longrightarrow P \times Q$. The data (2.0.4) must be functorial in S, associative, and commutative. The associativity condition is the commutativity of

$$\begin{array}{ccccc} & & E_{pp',q}E_{p'',q} & & \\ & \overset{\varphi_{p,p';q}\wedge\mathrm{id}}{\nearrow} & & \overset{\varphi_{pp',p'';q}}{\searrow} & \\ E_{p,q}E_{p',q}E_{p'',q} & & & & E_{pp'p'',q} \\ & \underset{\mathrm{id}\wedge\varphi_{p',p'';q}}{\searrow} & & \underset{\varphi_{p,p'p'';q}}{\nearrow} & \\ & & E_{p,q}E_{p'p'',q} & & \end{array} \tag{2.0.5}$$

for $p, p', p'' \in P(S)$ and $q \in Q(S)$. Commutativity is expressed by

$$\begin{array}{ccc} E_{p,q}E_{p',q} & \xrightarrow{\varphi_{p,p';q}} & E_{pp',q} \\ {\scriptstyle \mathrm{sym}}\downarrow & & \downarrow{\scriptstyle \mathrm{id}} \\ E_{p',q}E_{p,q} & \xrightarrow[\varphi_{p',p;q}]{} & E_{p'p,q}. \end{array} \tag{2.0.6}$$

Here $p, p' \in P(S)$ and $q \in Q(S)$.

Symmetrically, an extension structure in the Q-direction—that is, of Q_P by G_P—is described by isomorphisms of torsors under G_S

$$\psi_{p;q,q'} : E_{p,q}E_{p,q'} \xrightarrow{\sim} E_{p,qq'}, \tag{2.0.7}$$

for $p \in P(S)$ and $q, q' \in Q(S)$. These are functorial in S and satisfy the symmetric associativity and commutativity conditions

$$\begin{array}{ccccc} & & E_{p,qq'}E_{p,q''} & & \\ & {\scriptstyle \psi_{p;q,q'}\wedge\mathrm{id}}\nearrow & & \searrow{\scriptstyle \psi_{p;qq',q''}} & \\ E_{p,q}E_{p,q'}E_{p,q''} & & & & E_{p,qq'q''} \\ & {\scriptstyle \mathrm{id}\wedge\psi_{p;q',q''}}\searrow & & \nearrow{\scriptstyle \psi_{p;q,q'q''}} & \\ & & E_{p,q}E_{p,q'q''} & & \end{array} \tag{2.0.8}$$

for $p \in P(S)$ and $q, q', q'' \in Q(S)$, and

$$\begin{array}{ccc} E_{p,q}E_{p,q'} & \xrightarrow{\psi_{p;q,q'}} & E_{p,qq'} \\ {\scriptstyle \mathrm{sym}}\downarrow & & \downarrow{\scriptstyle \mathrm{id}} \\ E_{p,q'}E_{p,q} & \xrightarrow[\psi_{p;q',q}]{} & E_{p,q'q}. \end{array} \tag{2.0.9}$$

Here $p \in P(S)$ and $q, q' \in Q(S)$.

Definition 2.1. Let P, Q, G be abelian groups of $\underline{T}$, and let E be a torsor under $G_{P\times Q}$ equipped with an extension structure in the P-direction, defined by the isomorphisms (2.0.4), and an extension structure in the Q-direction, defined by the isomorphisms (2.0.7), each functorial in S and subject to the associativity and commutativity conditions above. The two structures are called compatible if, for every object S of $\underline{T}$ and $p, p' \in P(S)$, $q, q' \in Q(S)$, the following diagram commutes:

$$\begin{array}{ccccc} & & E_{p,qq'}E_{p',qq'} & & \\ & {\scriptstyle \psi_{p;q,q'}\wedge\psi_{p';q,q'}}\nearrow & & \searrow{\scriptstyle \varphi_{p,p';qq'}} & \\ E_{p,q}E_{p,q'}E_{p',q}E_{p',q'} & & & & E_{pp',qq'} \\ \downarrow & & & & \\ E_{p,q}E_{p',q}E_{p,q'}E_{p',q'} & & & \nearrow{\scriptstyle \psi_{pp';q,q'}} & \\ & {\scriptstyle \varphi_{p,p';q}\wedge\varphi_{p,p';q'}}\searrow & & & \\ & & E_{pp',q}E_{pp',q'} & & \end{array} \tag{2.1.1}$$

where the vertical arrow is induced by the symmetry isomorphism

$$E_{p,q'}E_{p',q} \xrightarrow{\sim} E_{p',q}E_{p,q'}.$$

A *biextension* of (P,Q) by G is a torsor E under $G_{P\times Q}$ equipped with compatible extension structures in the P- and Q-directions in the preceding sense.

2.1.1. Equivalently, a biextension structure on the $G_{P\times Q}$-torsor E is given by two partially defined composition laws $\varphi(x,y)$ and $\psi(x,y)$ on E. The first is defined when $\mathrm{pr}_2\, j(x) = \mathrm{pr}_2\, j(y)$, and the second when $\mathrm{pr}_1\, j(x) = \mathrm{pr}_1\, j(y)$, where $j : E \longrightarrow P \times Q$ is the structural morphism of the torsor. Besides associativity relations of the type

$$\varphi(gx,y) = \varphi(x,gy) = g\varphi(x,y) \qquad (x,y \in E(S),\ g \in G(S)) \tag{2.1.1.1}$$

and the analogous relations for ψ, the axioms say that these partial composition laws are associative and commutative and satisfy

$$\varphi(\psi(x,y),\psi(z,t)) = \psi(\varphi(x,z),\varphi(y,t)), \tag{2.1.1.2}$$

where x,y,z,t lie respectively in $E_{p,q}(S), E_{p,q'}(S), E_{p',q}(S), E_{p',q'}(S)$.

2.2. Identity sections. Consider first the identity section of E regarded as an extension of Q_P by G_P,

$$e_{E/P} \in \Gamma(E/P) = \mathrm{Hom}_P(P,E), \tag{2.2.1}$$

and, symmetrically, the identity section of E regarded as an extension of P_Q by G_Q,

$$e_{E/Q} \in \Gamma(E/Q) = \mathrm{Hom}_Q(Q,E). \tag{2.2.2}$$

In particular, the torsor E over $P\times Q$ under $G_{P\times Q}$ is canonically trivialized over $P\times 1_Q$ by $e_{E/P}$, and over $1_P\times Q$ by $e_{E/Q}$. These trivializations coincide over the identity section $1_P\times 1_Q$ of $P\times Q$, and hence define the same section

$$e_E \in \Gamma(E), \tag{2.2.3}$$

which, by abuse of language, will be called the identity section (or neutral section) of E.

To prove the asserted equality, let e,e' be the two sections of E deduced respectively from $e_{E/P}$ and $e_{E/Q}$. They are characterized as the identity sections in the inverse image $E_{1_P,1_Q}$ of E along the identity section of $P\times Q$, for the two composition laws φ and ψ, respectively. Applying (2.1.1.2) to the four sections e,e',e',e of $E_{1,1}$ gives e on the left and e' on the right; hence $e=e'$. It follows moreover that the two composition laws φ and ψ on $E_{1,1}$ are identical. Indeed, under transport of structure by the morphisms $g \mapsto g.e$ and $g \mapsto g.e'$, respectively, both correspond to the group law of G, and $e=e'$.

The extension of Q_P by G_P over the base P, restricted over the identity section 1_P of P, gives an extension of Q by G on the inverse image of $1_P\times Q$ in E. The homomorphism $e_{E/Q}$ is immediately seen to be multiplicative for this extension structure, so the extension is canonically split. Symmetrically, the extension of P by G obtained by restricting the extension of P_Q by G_Q over Q to the identity section 1_Q is canonically split by $e_{E/P}$.

2.3. Morphisms of biextensions. Let E be a biextension of (P,Q) by G, and let E' likewise be a biextension of (P',Q') by G', where P',Q',G' are again commutative groups of $\underline{T}$. A morphism of biextensions from E to E' is a system (u,v,w,f), where

$$u : P \longrightarrow P', \qquad v : Q \longrightarrow Q', \qquad w : G \longrightarrow G' \tag{2.3.1}$$

are morphisms of groups, and

$$f : E \longrightarrow E' \tag{2.3.2}$$

is a morphism of the sheaves of sets underlying E and E', such that f is simultaneously a homomorphism of group extensions over P and P', respectively, associated with the relative group homomorphisms

$$u \times v : Q_P \longrightarrow Q'_{P'}, \qquad u \times w : G_P \longrightarrow G'_{P'},$$

and a homomorphism of group extensions over Q and Q', respectively, associated with the relative group homomorphisms

$$u \times v : P_Q \longrightarrow P'_{Q'}, \qquad v \times w : G_Q \longrightarrow G'_{Q'}.$$

Notice moreover that u, v, w are determined by f alone. Indeed, passing to the quotient recovers $u \times v$ from f, and then recovers u and v by restriction to $P \times 1_Q$ and $1_P \times Q$, respectively; restricting f to $G.e_E$ recovers w. Thus no ambiguity arises from identifying the morphism (u, v, w, f) with f.

After taking the usual precaution of adjoining their sources and targets to morphisms (the canonical genuflection), one sees that the biextensions of the topos $\underline{T}$ are the objects of a category, whose objects and morphisms are those defined in 2.1 and 2.3 and whose composition of morphisms is defined in the evident way. Moreover, $(P, Q, G, E) \mapsto (P, Q, G)$ gives a canonical functor

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}), \tag{2.3.3}$$

where $\text{BIEXT}(\underline{T})$ denotes the category of biextensions over $\underline{T}$, and $\text{GRAB}(\underline{T})$ the category of abelian groups over $\underline{T}$. It is useful to decompose this canonical functor into the two functors $E \mapsto (P, Q)$,

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}), \tag{2.3.4}$$

and $E \mapsto G$,

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}). \tag{2.3.5}$$

2.4. The category $\text{BIEXT}(P, Q; G)$ for variable P, Q, G.

I claim that the canonical functor (2.3.4) is fibred, and that the canonical functor (2.3.5) is cofibred. The first assertion means, in particular, that if E is a biextension of (P, Q) by G, and if

$$u : P' \longrightarrow P, \qquad v : Q' \longrightarrow Q$$

are morphisms of abelian groups, then there is a biextension E' of (P', Q') by G and a morphism of biextensions

$$f : E' \longrightarrow E$$

compatible with u, v, id_G, having the usual universal property for morphisms of biextensions into E compatible with u, v, id_G.

The second assertion is expressed similarly. In particular, if E is a biextension of (P, Q) by G, and if

$$w : G \longrightarrow G'$$

is a morphism of abelian groups, then there is a biextension E' of (P, Q) by G' and a morphism of biextensions

$$g : E \longrightarrow E'$$

compatible with $\text{id}_P, \text{id}_Q, w$, having the usual universal property for morphisms of biextensions from E compatible with $\text{id}_P, \text{id}_Q, w$. We leave it to the reader to spell out the two constructions whose existence has just been asserted; they are obtained easily by paraphrasing the familiar analogous constructions for extensions of ordinary groups.

Thus, for three abelian groups P, Q, G of $\underline{T}$, write

$$\text{BIEXT}(P, Q; G) \tag{2.4.1}$$

for the category of biextensions of (P,Q) by G, defined as the fibre category of (2.3.3) over (P,Q,G) (so its morphisms are the morphisms of biextensions compatible with the identity morphisms of P,Q,G). Then this category depends covariantly on G and contravariantly on P,Q. Moreover, with these variances, the category $\mathrm{BIEXT}(P,Q;G)$ depends additively on each of the three arguments P,Q,G. For example, for the argument G, this means precisely:

a) $\mathrm{BIEXT}(P,Q;0)$ is equivalent to the one-point category;

b) for two abelian groups G,G' of $\underline{T}$, the functor

$$\mathrm{BIEXT}(P,Q;G\times G') \longrightarrow \mathrm{BIEXT}(P,Q;G)\times \mathrm{BIEXT}(P,Q;G') \tag{2.4.2}$$

deduced from the two projections $G\times G'\longrightarrow G$ and $G\times G'\longrightarrow G'$ is an equivalence of categories.

In the language of [4], one also says that, for fixed P,Q, the category of biextensions of (P,Q) by variable groups G is, via the functor $E\mapsto G$ induced by (2.3.5), an additive cofibred category over $\mathrm{GRAB}(\underline{T})$.

Remark 2.4.3. In the terminology of *loc. cit.*, the preceding cofibred category is not merely additive but even left exact [4, 2.2]. We shall not verify this property here, since it will reappear below in 3.6 as a consequence of more precise facts. We only point out that the general theory of *loc. cit.* would then describe the structure of this cofibred category, up to equivalence of fibred categories, in terms of a chain complex $L_\bullet$ in $\mathrm{GRAB}(\underline{T})$ depending only on P and Q. We shall also give below a direct construction of this complex as

$$P \overset{\mathbb{L}}{\otimes} Q,$$

the tensor product taken in a derived category.

2.5. Structure of the category $\mathrm{BIEXT}(P,Q;G)$.

The general formal considerations of [4, no. 1] give several useful consequences of the additive dependence of the categories $\mathrm{BIEXT}(P,Q;G)$ on G.

2.5.1. First, for fixed G, the category $\mathrm{BIEXT}(P,Q;G)$ has a composition law which assigns functorially to two biextensions E,E' of (P,Q) by G a product biextension, denoted $E.E'$. This construction is simply the analogue of the familiar product of two extensions of one commutative group by another. It is compatible with that construction whether E,E' are viewed as group extensions of Q_P by G_P, or as group extensions of P_Q by G_Q. In particular, as a torsor under $G_{P\times Q}$, EE' is the product of the torsors under $G_{P\times Q}$ defined by E and E'.

The formation of $E.E'$ is moreover associative and commutative up to canonical isomorphisms, and this composition law on $\mathrm{BIEXT}(P,Q;G)$ has a two-sided neutral object, called the trivial biextension of (P,Q) by G. It may be constructed, as in *loc. cit.*, from the unique biextension E_0 of (P,Q) by the zero group 0, up to isomorphism, by means of the homomorphism $0\longrightarrow G$. From this description one immediately obtains an alternative description of the trivial biextension as the trivial torsor

$$E = P\times Q\times G.$$

Its two group structures over P and over Q are obtained, respectively, from the product group structures on $Q\times G$ and $P\times G$ by the base changes $P\longrightarrow e$ and $Q\longrightarrow e$.

Finally, for every biextension E of (P,Q) by G, one can define the inverse biextension, denoted E^{-1}, together with an isomorphism from EE^{-1} to the trivial biextension. This determines E^{-1} up to unique isomorphism and makes its dependence on E functorial.

In short, the category $\mathrm{BIEXT}(P,Q;G)$ behaves like a group object in the category **Cat** of categories, except that associativity, commutativity, unit, and inverse relations hold only up to specified isomorphisms. One may call it a pseudo-group in the 2-category **Cat**.

2.5.2. For fixed P,Q,G, the preceding properties imply the following facts. For every biextension E of (P,Q) by G, the monoid of endomorphisms of E is canonically isomorphic to the

monoid of endomorphisms of the trivial biextension $\underline{1}$: an endomorphism u of $\underline{1}$ is sent to $u.\mathrm{id}_E$ on $\underline{1}.E \simeq E$. The endomorphism monoid of $\underline{1}$ is in fact a commutative group, denoted

$$\mathrm{Biext}^0(P,Q;G). \tag{2.5.2.1}$$

Its composition law is also given by $(u,v) \mapsto u.v \in \mathrm{End}(\underline{1}.\underline{1}) \simeq \mathrm{End}(\underline{1})$.

Likewise, the composition law $(E,E') \mapsto EE'$ equips the set

$$\mathrm{Biext}^1(P,Q;G) \tag{2.5.2.2}$$

of isomorphism classes of biextensions of (P,Q) by G with the structure of a commutative group; the inverse of the class of E is the class of E^{-1}.

2.5.3. Given a group homomorphism $G \longrightarrow G'$, the corresponding functor

$$\mathrm{BIEXT}(P,Q;G) \longrightarrow \mathrm{BIEXT}(P,Q;G') \tag{2.5.3.1}$$

is compatible, up to canonical isomorphisms, with the operations $E.E'$ and E^{-1} of 2.5.1. Consequently, the two evident maps

$$\mathrm{Biext}^i(P,Q;G) \longrightarrow \mathrm{Biext}^i(P,Q;G') \qquad (i=0,1) \tag{2.5.3.2}$$

are group homomorphisms. Moreover, if E is a biextension of (P,Q) by G and E' the corresponding biextension by G', the natural homomorphism

$$\mathrm{Hom}(E,E) = \mathrm{Aut}(E) \longrightarrow \mathrm{Hom}(E',E') = \mathrm{Aut}(E') \tag{2.5.3.3}$$

is precisely (2.5.3.2) for $i=0$.

2.5.4. The group $\mathrm{Biext}^0(P,Q;G)$ admits a simple explicit description. If E is a biextension of (P,Q) by G, an automorphism u of E is first of all an automorphism of the underlying torsor under $G_{P\times Q}$, and hence is given by a morphism

$$u_0 : P \times Q \longrightarrow G$$

through the formula

$$u(x) = u_0(j(x))x.$$

Compatibility of u with the first, respectively the second, extension structure on E is equivalent to additivity of u_0 in the second, respectively the first, argument. Thus u is an automorphism of the biextension structure of E if and only if u_0 is bilinear. This gives a canonical bijection

$$\mathrm{Biext}^0(P,Q;G) \simeq \mathrm{Hom}(P\otimes Q, G). \tag{2.5.4.1}$$

It is readily checked that this bijection is independent of E, respects the group structures, and is functorial in G for the functoriality on the left described in (2.5.3.2).

Below we shall determine $\mathrm{Biext}^1(P,Q;G)$ and find that, canonically and functorially in G, it is isomorphic to

$$\mathrm{Ext}^1\big(P \overset{\mathbb{L}}{\otimes} Q, G\big).$$

2.6. In 2.5 only G was varied, while P and Q were held fixed. We now say a few words about variation in P,Q. The essential fact, from which the others follow formally, is that the contravariant dependence of $\mathrm{BIEXT}(P,Q;G)$ on P,Q commutes with its covariant dependence on G. Thus the categories in question may be viewed as the fibre categories of a cofibred category

$$\mathrm{BIEXT}'(\underline{T}) \longrightarrow \mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T}), \tag{2.6.1}$$

where the superscript op denotes the opposite category and the structural functor is $E \mapsto (P,Q,G)$. This cofibred category is additive in each of its three arguments, or, as we shall say, tri-additive.

By paraphrasing standard arguments, which we shall not reproduce, it follows that the composition $E.E'$ in the fibre category $\mathrm{BIEXT}(P,Q;G)$ may be described using the additive structure of any one of the three arguments: the three resulting pseudo-group structures are canonically isomorphic. Consequently, the abelian groups

$$\mathrm{Biext}^i(P,Q;G), \qquad i=0,1, \tag{2.6.2}$$

may be regarded as two trilinear functors in P,Q,G, respectively in $\mathrm{GRAB}(\underline{T})^{\mathrm{op}}$, $\mathrm{GRAB}(\underline{T})^{\mathrm{op}}$, and $\mathrm{GRAB}(\underline{T})$. Equivalently, they are trilinear functors of variable P,Q,G in $\mathrm{GRAB}(\underline{T})$, contravariant in P,Q and covariant in G, with values in the category of abelian groups. For this variance, the isomorphism (2.5.4.1) is functorial in all of P,Q,G, and not merely in G. The analogous assertion will hold below for the description of Biext^1 by an Ext^1.

2.7. The symmetric biextension of a given biextension.

If E is a biextension of (P,Q) by G, define its symmetric biextension ${}^sE$, which is a biextension of (Q,P) by G. The underlying sheaf of sets of ${}^sE$ is that of E, and the actions of G on E and ${}^sE$ are the same. The structural morphism

$${}^sE \longrightarrow Q \times P$$

is the composite of $E \longrightarrow P \times Q$ with the symmetry $P \times Q \longrightarrow Q \times P$; finally, the left and right extension structures of ${}^sE$ are, respectively, the right and left extension structures of E. Evidently $E \mapsto {}^sE$ is an automorphism of the category $\mathrm{BIEXT}(\underline{T})$ of biextensions, and indeed an involutive automorphism:

$${}^s({}^sE) = E.$$

Every statement about bitorsors $E,\ldots$ therefore gives a symmetric statement upon being applied to ${}^sE,\ldots$. We may call it the statement obtained by *exchanging left and right*. In general we shall spell out only one of the two symmetric statements, choosing between left and right, for example, according to our political preferences.

2.8. Change of topos. Localization.

Let

$$f : \underline{T}' \longrightarrow \underline{T}$$

be a morphism of topoi. The inverse-image functor f^* transforms a biextension of (P,Q) by G into a biextension $f^*(E)$ of (f^*P, f^*Q) by f^*G, as always for a structure expressible in terms of finite projective limits and arbitrary inductive limits. To prove compatibility of the two extension structures on $f^*(E)$ induced by those on E, one expresses compatibility by commutativity of (2.1.1) in the universal case

$$S = P \times P \times Q \times Q,$$

where p,p',q,q' are the four projections. As expected, $E \mapsto f^*(E)$ gives a canonical functor

$$f^* : \mathrm{BIEXT}(\underline{T}) \longrightarrow \mathrm{BIEXT}(\underline{T}').$$

In what follows we shall use without further mention the fact that every construction made with biextensions is compatible with inverse images under morphisms of topoi. This applies in particular to the product EE' and inverse E^{-1} considered in 2.5.

One may take for $\underline{T}'$ an induced topos $\underline{T}_{/S'}$, and for f the canonical morphism, called the localization morphism (SGA 4 IV 5),

$$f = i_{S'} : \underline{T}_{/S'} \longrightarrow \underline{T}.$$

Then $f^*(E)$ is called the biextension induced by E on S', and is also denoted, as usual, by $E|_{S'}$ or $E_{S'}$.

Since restriction has the usual transitivity properties, the categories $\mathrm{BIEXT}(\underline{T}_{/S})$, for variable $S \in \underline{T}$, form the fibres of a fibred category over $\underline{T}$ (SGA 1 VI 8). In fact, this category is even

a stack over $\underline{T}$ in the sense of Giraud [2]: morphisms and objects in its fibres can be glued. For example, to give a biextension over S up to canonical isomorphism, it suffices to give a covering family (that is, an epimorphic family) $S_i \longrightarrow S$, a biextension E_i over each S_i (an object of $\underline{T}_{/S_i}$), and, for every pair (i,j), an isomorphism between $E_i|_{S_j}$ and $E_j|_{S_i}$, subject to the usual cocycle condition.

We call this the stack of biextensions over $\underline{T}$. Replacing $\underline{T}$ in (2.3.3), and hence also in (2.3.4) and (2.3.5), by $\underline{T}_{/S}$ for variable S gives morphisms of stacks over $\underline{T}$. The same applies to the operations $E.E'$ and E^{-1} of (2.5.1) in the categories $\mathrm{BIEXT}(P_S, Q_S; G_S)$ for variable S, where P, Q, G are fixed commutative groups over $\underline{T}$.

2.9. Relations with birigidified torsors.

In the terminology of 1.3.2, we saw in 2.2 that, for every biextension E of (P,Q) by G, the underlying torsor structure under $G_{P\times Q}$ is canonically birigidified relative to the pair of identity sections of P and Q. This gives a canonical functor

$$\mathrm{BIEXT}(P,Q;G) \xrightarrow{i} \mathrm{TORSBIRIG}(P,Q;G), \tag{2.9.1}$$

where the category on the right consists of birigidified torsors over $P\times Q$ under $G_{P\times Q}$.

Using the identity sections of P and Q, for every product of n objects each isomorphic to P or Q one obtains a canonical morphism inj_i from the i-th factor into that product. Thus, for a morphism from such a product to G, one may speak of its restriction to the i-th factor. Consider the products

$$\begin{cases} P\times Q, \\ P\times Q\times Q, \quad P\times Q\times Q\times Q, \\ P\times P\times Q, \quad P\times P\times P\times Q, \\ P\times P\times Q\times Q. \end{cases} \tag{2.9.2}$$

Proposition 2.9.3. Suppose that every morphism from one of the products (2.9.2) to G which is trivial on each factor of that product (embedded by the identity sections as just explained) is trivial. Then:

a) The canonical functor (2.9.1) is fully faithful, and both sides of (2.9.1) are rigid categories. In particular, if E is a torsor under $G_{P\times Q}$ birigidified relative to (e_P, e_Q), there is at most one biextension structure on E compatible with its birigidified torsor structure.

b) Such a biextension structure exists on E if and only if E admits an extension structure of Q_P by G_P compatible with its torsor structure under G_P, and an extension structure of P_Q by G_Q compatible with its torsor structure under G_Q (cf. 1.3.5(b)). In that case there is a unique extension structure of Q_P by G_P compatible with the torsor structure under G_P and the given rigidification over $P\times e_Q$, and a unique extension structure of P_Q by G_Q compatible with the torsor structure under G_Q and the given rigidification over $e_P\times Q$; these two extension structures are compatible in the sense of 2.1.

Proof. a) If E is a birigidified torsor over $P\times Q$, its automorphism group identifies with the group of morphisms $P\times Q\longrightarrow G$ trivial on the two partial factors. By the hypothesis applied to $P\times Q$, this group contains only the identity automorphism, proving rigidity. Now let E, E' be biextensions of (P,Q) by G. To see that every morphism $f: E\longrightarrow E'$ of birigidified torsors is a morphism of biextensions, apply 1.3.5(a) to the two extension structures defined by E, E' over P and over Q, respectively. This uses the hypothesis for $P\times Q\times Q$ and $P\times P\times Q$ in (2.9.2).
b) Necessity is clear. For sufficiency, 1.3.5(b) first shows that the stated condition gives unique extension structures of Q_P by G_P and of P_Q by G_Q, compatible with the torsor structure and the rigidifications. Here the hypothesis is used for $P\times Q\times Q\times Q$ and $P\times P\times P\times Q$. It remains to prove that the two extension structures are compatible, that is, that diagram (2.1.1) commutes. Its two composites are isomorphisms of torsors over $P\times P\times Q\times Q$, and hence differ by a morphism

$$u: P\times P\times Q\times Q\longrightarrow G.$$

Because the partial trivializations of E are identity sections for the laws φ and ψ, respectively, u is trivial on each of the four factors. The hypothesis for $P \times P \times Q \times Q$ therefore makes u trivial. This proves the proposition.

Corollary 2.9.4. Let P, Q, G be commutative groups satisfying

$$\underline{\mathrm{Hom}}_{\mathrm{pt}}(P, G) = e, \qquad \underline{\mathrm{Hom}}_{\mathrm{pt}}(Q, G) = e \tag{2.9.4.1}$$

(cf. 1.3.7). Then the hypothesis of Proposition 2.9.3 is satisfied, and conclusions (a) and (b) of that proposition hold. Moreover, if E is a torsor under $G_{P\times Q}$ birigidified relative to (e_P, e_Q), it admits a biextension structure of (P, Q) by G compatible with its given structures if and only if the two morphisms of pointed sheaves which it defines,

$$Q \longrightarrow R^1 p_*(G_P), \qquad P \longrightarrow R^1 q_*(G_Q), \tag{2.9.4.2}$$

where $p : P \longrightarrow e$ and $q : Q \longrightarrow e$ are the structural morphisms, are morphisms of groups.

The first assertion was proved in 1.3.7. For the second, necessity is clear. To prove sufficiency, apply Proposition 2.9.3(b) and 1.3.5(b). Multiplicativity of the first morphism in (2.9.4.2) says that the rigidified torsor E under G_P satisfies the condition of 1.3.5(b) locally over P. As observed in 1.3.7, that condition is then true globally over P, so E admits an extension structure of Q_P by G_P compatible with its rigidified torsor structure. Symmetrically, E admits an extension structure of P_Q by G_Q compatible with its rigidified torsor structure under G_Q. The conclusion follows from Proposition 2.9.3(b).

Example 2.9.5. Let S be a scheme, let P and Q be abelian schemes over S, and let $G = \underline{G}_{m,S}$ be the multiplicative group scheme over S. We noted in 1.3.8 that (2.9.4.1) then holds. A birigidified torsor E over $P \times Q$ is equivalently a birigidified invertible sheaf over $P \times Q$. Recall that a divisorial correspondence on P, Q is the isomorphism class of such a birigidified invertible sheaf (see, for example, [1]). In the fppf topology, the morphisms (2.9.4.2) are

$$Q \longrightarrow \underline{\mathrm{Pic}}_{P/S}, \qquad P \longrightarrow \underline{\mathrm{Pic}}_{Q/S}, \tag{2.9.5.1}$$

defined by the divisorial correspondence. The theorem of the square (*loc. cit.*) says that these morphisms necessarily respect the group structures. Thus 2.9.4 shows that the category of biextensions of (P, Q) by $G = \underline{G}_{m,S}$ is equivalent to the category of birigidified invertible sheaves on $P \times Q$; it is therefore a rigid category, and its set of isomorphism classes identifies with the set of divisorial correspondences on (P, Q). This identification plainly respects the natural group laws on the two sets, the first being that of (2.5.2.2).

Remark 2.9.6. Additivity of the first morphism in (2.9.4.2) is equivalent to factorization of the second through the subsheaf $\underline{\mathrm{Ext}}^1(Q, G)$; note that, by 1.3.5, the canonical morphism

$$\underline{\mathrm{Ext}}^1(Q, G) \longrightarrow R^1 q_*(G_Q)$$

is injective. Likewise, additivity of the second morphism in (2.9.4.2) means that the first factors through $\underline{\mathrm{Ext}}^1(P, G)$. This gives group isomorphisms

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}\big(Q, \underline{\mathrm{Ext}}^1(P, G)\big) \simeq \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q, G)\big). \tag{2.9.6.1}$$

In the situation of 2.9.5 these reduce to the familiar isomorphisms

$$\mathrm{Corr}(P, Q) \simeq \mathrm{Hom}(P, Q') \simeq \mathrm{Hom}(Q, P'), \tag{2.9.6.2}$$

where

$$P' = \underline{\mathrm{Ext}}^1(P, \underline{G}_m), \qquad Q' = \underline{\mathrm{Ext}}^1(Q, \underline{G}_m)$$

are the dual abelian schemes of P and Q, identified respectively with $\underline{\mathrm{Pic}}^0_{P/S}$ and $\underline{\mathrm{Pic}}^0_{Q/S}$. In both formulas, Hom denotes the group of group homomorphisms.

2.10. Various generalizations of the notion of biextension. We mention them for completeness, although we shall not use them.

2.10.1. One can define a biextension of (P,Q) by G without assuming P or Q commutative, only G being commutative. It is a torsor under $G_{P\times Q}$ equipped with central extension structures of Q_P by G_P and of P_Q by G_Q, each compatible with the torsor structure and compatible with one another, meaning that they make (2.1.1) commute. Most of the preceding developments remain valid, *mutatis mutandis*, for this more general notion. Even when P,Q are commutative, it is more general than the notion of 2.1, which is the case in which the two underlying relative group structures on E are commutative. In the noncommutative case, the category of biextensions of (P,Q) by G still depends additively on G, but not on P and Q.

2.10.2. Paraphrasing 2.1, for a fixed integer $n \geq 1$ one defines an n-extension of a sequence of n commutative groups $(P_1, P_2, \ldots, P_n)$ by a commutative group G. It consists of a torsor E under $G_{P_1\times\cdots\times P_n}$, equipped with n partial extension structures over the partial products of $n-1$ factors, with a compatibility relation of type (2.1.1) for every pair of distinct indices between 1 and n.

If $n \geq 3$, the P_i are abelian schemes over a scheme S, and $G = \underline{G}_{m,S}$, every n-extension of $(P_1, \ldots, P_n)$ by G is trivial.

2.10.3. The notion of biextension in 2.1 may be viewed as a variant of the notion of a commutative extension of commutative groups, that is, of an extension of $\mathbb{Z}$-modules. Let A be a ring of $\underline{T}$, assumed commutative for simplicity, and let P,Q,G be three A-modules. More generally one can define an A-module biextension of (P,Q) by G, a variant of an extension of A-modules.

Such a biextension is a biextension in the sense of 2.1 whose underlying partial extension structures are enriched to extension structures of A_P-modules and A_Q-modules, respectively, compatible with the relative module structures already present on Q_P, G_P and P_Q, G_Q. The two relative module structures on E may be interpreted as two actions of the underlying multiplicative monoid of A on the sheaf of sets underlying E, and these two actions are required to commute. This condition is added to the compatibility expressed by (2.1.1), and is itself expressed by an analogous diagram, which we leave to the reader. Finally, the first action of A on E is required to be distributive with respect not only to the first composition law on E but also to the second; the same is required of the second action.

The dictionary of 1.1.6 also has an A-linear variant. An extension of the A-module P by the A-module G amounts to a family of torsors E_p under G_p, parametrized by points $p \in P(S)$ for variable $S \in \mathrm{ob}\,\underline{T}$, with E_p varying A-linearly, rather than merely additively, in p. Besides the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1), this is expressed by isomorphisms

$$\mu_{a,p} : {}^aE_p \xrightarrow{\sim} E_{ap} \qquad (p \in P(S),\ a \in A(S)),$$

where ${}^aE_p$ denotes the image of the G_S-torsor E_p under extension of the structure group along

$$g \longmapsto ag : G_S \longrightarrow G_S.$$

We leave to the reader the conditions on φ and μ which make these data correspond to an extension of A-modules E of P by G (cf. 3.5.4(b)).

3. Supplements on homological algebra

The principal purpose of this section is to define the isomorphism

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$$

announced above. For this we shall need certain developments in homological algebra which are of independent interest and will occupy sections 3.1 through 3.4. The developments in 3.1 and 3.3 may be regarded as supplements to [4], especially to section 6.9 therein.

3.1. Construction of an additive cofibred category $\underline{\Psi}_{L_\bullet}$ in terms of a chain complex $L_\bullet$ in an abelian category $\underline{C}$

Let $\underline{C}$ be an abelian category and let $L_\bullet$ be a chain complex in $\underline{C}$,

$$L_2 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \tag{3.1.1}$$

written with lower-degree conventions, so that the differential has degree -1. Thus $L_i = 0$ for $i < 0$. We shall define a cofibred category

$$\underline{\Psi} = \underline{\Psi}_{L_\bullet}$$

over $\underline{C}$; up to canonical equivalence of cofibred categories it is the category of the same name in [4, 6.9]. The objects of $\underline{\Psi}_{L_\bullet}$ are pairs

$$(X, \alpha),$$

where X is an extension of L_0 by an object A of $\underline{C}$, depending on X,

$$0 \longrightarrow A \xrightarrow{i} X \xrightarrow{j} L_0 \longrightarrow 0, \tag{3.1.2}$$

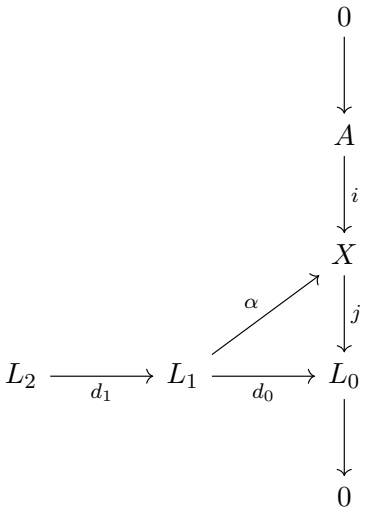
and α is a trivialization of the inverse image $d_0^*(X)$ of this extension by $d_0 : L_1 \longrightarrow L_0$. It is required that the corresponding trivialization of the inverse image $d_1^* d_0^*(X)$ by $d_1 : L_2 \longrightarrow L_1$ be the evident one, obtained from the transitivity isomorphism

$$d_1^* d_0^*(X) \simeq (d_0 d_1)^*(X)$$

and the relation $d_0 d_1 = 0$.

Equivalently, α is a lifting of $d_0 : L_1 \longrightarrow L_0$ to a morphism $\alpha : L_1 \longrightarrow X$, subject to $\alpha d_1 = 0$:

(3.1.3)



, $\alpha d_1 = 0$. $(*)$

Here is a simpler description, pointed out by Deligne: $\underline{\Psi}_{L_\bullet}(A)$ is the category of extensions of the complex $L_\bullet$ by A, regarded as a complex concentrated in degree zero.

In this form one sees that the description of the objects (X, α) depends only on the chain complex $[L_\bullet]$ truncated in degree 1, obtained from $L_\bullet$ by "killing $H_i(L_\bullet)$ for $i \geq 2$": its terms are zero in degrees at least 2, agree with those of $L_\bullet$ in degrees at most 0, and its degree-1 term is the cokernel of $d_1 : L_2 \longrightarrow L_1$.

A homomorphism from (X, α) to (X', α') is a homomorphism of extensions $u : X \longrightarrow X'$, compatible with the identity on L_0, such that $u\alpha = \alpha'$. Homomorphisms are composed by composing the corresponding homomorphisms of extensions. This gives the announced category $\underline{\Psi}_{L_\bullet}$. Up to canonical isomorphism, it depends only on the degree-1 truncation $[L_\bullet]$ of $L_\bullet$.

There is an evident functor $(X, \alpha) \longmapsto A$,

$$\pi : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{C}. \tag{3.1.4}$$

This functor is cofibred. Indeed, let $u : A \longrightarrow B$ be a morphism of $\underline{C}$, and let (X, α) be an object of the fibre $\underline{\Psi}_{L_\bullet}(A)$. Thus X is an extension of L_0 by A. Form the extension $u_*(X)$ of L_0 by B, the direct image of X under u, and equip it with the partial trivialization $\beta = f\alpha$, where $f : X \longrightarrow u_*(X)$ is the canonical morphism. It follows immediately that

$$f : (X, \alpha) \longrightarrow (u_*(X), \beta)$$

is cocartesian for (3.1.4): it has the usual universal property for u-morphisms in $\underline{\Psi}_{L_\bullet}$ with source (X, α). This is just the familiar and elementary corresponding universal property of $f : X \longrightarrow u_*(X)$.

Finally, the cofibred category $\underline{\Psi}_{L_\bullet}$ is additive; that is, it satisfies

$$\underline{\Psi}_{L_\bullet}(0) \quad \text{is equivalent to the one-point category,} \tag{3.1.5}$$

and, for arbitrary A, B in $\underline{C}$,

$$\underline{\Psi}_{L_\bullet}(A \times B) \longrightarrow \underline{\Psi}_{L_\bullet}(A) \times \underline{\Psi}_{L_\bullet}(B) \quad \text{is an equivalence.} \tag{3.1.6}$$

This follows immediately from the corresponding properties of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$. We shall freely use the consequences of the additivity of $\underline{\Psi}_{L_\bullet}$ established in [4, 1] and recalled above in 2.5.

3.1.7. Among these consequences, for every object A of $\underline{C}$ the category $\underline{\Psi}_{L_\bullet}(A)$ has a zero, or trivial, object (2.5.1). It is the trivial extension $X = L_0 + A$ of L_0 by A, equipped with the partial trivialization $\alpha : L_1 \longrightarrow X$ whose components are d_0 and 0.

Moreover, by 2.5.2, for every (X, α) in $\underline{\Psi}_{L_\bullet}(A)$ the automorphism group of this object is canonically isomorphic to the automorphism group of the trivial object. Denote the latter commutative group by $\Psi^0_{L_\bullet}(A)$. The explicit description above gives the canonical isomorphisms

$$\begin{aligned} \Psi^0_{L_\bullet}(A) = \mathrm{Ker}\big(\mathrm{Hom}(L_0, A) \longrightarrow \mathrm{Hom}(L_1, A)\big) &\simeq \mathrm{Hom}(H_0(L_\bullet), A) \\ &\simeq \mathrm{Ext}^0(L_\bullet, A), \end{aligned} \tag{3.1.8}$$

induced by the usual identification of the automorphism group of the extension X with $\mathrm{Hom}(L_0, A)$. The isomorphism (3.1.8) is plainly functorial in A.

3.1.9. Finally, recall from 2.5.2 that the natural multiplicative structure on the fibre category $\underline{\Psi}_{L_\bullet}(A)$ endows its set of isomorphism classes with a natural group structure. We denote this group by $\Psi^1_{L_\bullet}(A)$; it is an additive functor of A.

3.2. Determination of $\Psi^1_{L_\bullet}(A)$

Consider an object (X, α) (3.1.3) of $\underline{\Psi}_{L_\bullet}(A)$. Consider the cochain complex of length one

$$[X \longrightarrow L_0] : \quad 0 \longrightarrow X \longrightarrow L_0 \longrightarrow 0 \longrightarrow \cdots \tag{3.2.1}$$

defined by the morphism $j : X \longrightarrow L_0$, with X as its degree-zero component. It is a resolution of A, and hence may be identified with A as an object of the derived category $D(\underline{C})$ [5]. The diagram (3.1.3) then gives a degree-one morphism of complexes

$$L_\bullet \longrightarrow [X \longrightarrow L_0], \tag{3.2.2}$$

made explicit by the following diagram with anticommutative squares:

$$\begin{array}{ccccccccc} \cdots & & L_2 & \xrightarrow{d_1} & L_1 & \xrightarrow{d_0} & L_0 & \longrightarrow 0 & \cdots \\ & & & & \downarrow{\scriptstyle u^{-1}=-\alpha} & & \downarrow{\scriptstyle u^0=\mathrm{id}} & & \\ \cdots & & 0 & \longrightarrow & X & \xrightarrow{j} & L_0 & \longrightarrow 0 & \cdots \end{array} \tag{3.2.3}$$

(We use the usual convention which associates with the chain complex $L_\bullet$ a complex whose differential has degree 1 and whose degree-i component is L_{-i}.) We thus obtain a degree-one morphism in the derived category

$$c(X,\alpha) : L_\bullet \longrightarrow A \qquad \big(\text{a degree-one morphism in } D(\underline{C})\big), \tag{3.2.3}$$

which plainly depends only on the isomorphism class of (X,α). We have thereby defined a canonical map

$$\Psi^1_{L_\bullet}(A) \longrightarrow \mathrm{Ext}^1(L_\bullet, A) \overset{\mathrm{def}}{=} \mathrm{Hom}_{D(\underline{C})}(L_\bullet, A[1]). \tag{3.2.4}$$

Theorem 3.2.5. The map (3.2.4) is an isomorphism of groups, functorial in the object A of $\underline{C}$.

We leave to the reader the verification that (3.2.4) is additive and functorial in A, and shall confine ourselves to proving that it is bijective. For injectivity, let us first show that if the morphism of complexes u in (3.2.3) is zero in $D(\underline{C})$, then it is already zero in the homotopy category $K(\underline{C})$; that is, there exists a morphism

$$s : L_0 \longrightarrow X$$

such that

$$js = u^0 = \mathrm{id}_{L_0}, \qquad sd_0 = -u^{-1} = \alpha. \tag{3.2.5.1}$$

Indeed, the hypothesis says that the conclusion becomes true after the morphism of complexes under consideration is composed with a suitable quasi-isomorphism v

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & X & \xrightarrow{j} & L_0 & \longrightarrow 0 & & & & & \\ & & \downarrow & & \downarrow & & & & & & \\ 0 & \longrightarrow & C^0 & \xrightarrow{j'} & C^1 & \longrightarrow & C^2 & \longrightarrow & C^3 & \longrightarrow & \cdots \end{array} \tag{*}$$

(and, after replacing C^i by 0 for $i \geq 2$ and C^1 by $\mathrm{Im}(C^0 \longrightarrow C^1) = \mathrm{Ker}(C^1 \longrightarrow C^2)$ if necessary, we may suppose that $C^i = 0$ for $i \geq 2$). But the morphism (*) induces the identity on A, and therefore identifies X with the fibre product of L_0 and C^0 over C^1. A morphism $s' : L_0 \longrightarrow C^0$ defining a homotopy $vu \sim 0$, and hence satisfying $j's' = v^1u^0$, consequently factors through a morphism $s : L_0 \longrightarrow X$ satisfying $js = u^0 = \mathrm{id}_{L_0}$. Thus s splits the extension X of L_0 by A, so the complex $X \longrightarrow L_0$ is isomorphic in $K(\underline{C})$ to the complex $K^\bullet$ concentrated in A. For the latter complex it is clear that

$$\mathrm{Hom}_{K(\underline{C})}(L_\bullet, K^\bullet) \longrightarrow \mathrm{Hom}_{D(\underline{C})}(L_\bullet, K^\bullet)$$

is an isomorphism, proving the assertion.

Now the choice of a morphism $s : L_0 \longrightarrow X$ satisfying (3.2.5.1) is precisely equivalent to a trivialization of the extension X of L_0 by A compatible with the partial trivialization α. This proves the injectivity of (3.2.4).

For surjectivity, start with an element of the right-hand side. By definition it may be described by a degree-one morphism of complexes

$$\begin{array}{ccccccccc} \ldots & & L_2 & \xrightarrow{d_1} & L_1 & \xrightarrow{d_0} & L_0 & \longrightarrow & 0 \\ & & & & \downarrow{\scriptstyle u^{-1}} & & \downarrow{\scriptstyle u^0} & & \\ & & 0 & \longrightarrow & C^0 & \longrightarrow & C^1 & \longrightarrow & \ldots \end{array} \tag{3.2.5.2}$$

with anticommutative squares, where $C^\bullet$ is a resolution of A and, as above, we may suppose that $C^i = 0$ for $i \geq 2$. Thus C^0 is an extension of C^1 by A, and we may form the extension X obtained by pulling it back along $L_0 \longrightarrow C^1$. Anticommutativity in (3.2.5.2) shows that $-u^{-1}$ factors through a morphism $\alpha : L_1 \longrightarrow X$ which is a partial splitting, that is, $j\alpha = d_0$ (with j as in (3.1.2)). Moreover, the relation $u^{-1}d_1 = 0$ in (3.2.5.2) implies $\alpha d_1 = 0$. Hence (X, α) is indeed an object of $\underline{\Psi}_{L_\bullet}(A)$. By construction, the morphism (3.2.5.2) is the composite of the morphism (3.2.3) associated with (X, α) and the morphism

$$[X \longrightarrow L_0] \longrightarrow C^\bullet = [C^0 \longrightarrow C^1]$$

whose components are (f, u^0), where $f : X \longrightarrow C^0$ is the canonical morphism obtained from the definition of X as the pullback of the extension C^0.

The preceding morphism is a morphism of resolutions of A and induces the identity on A. It follows, by the definition $\mathrm{Ext}^1 = \mathrm{Hom}^1_{D(\underline{C})}$, that the given element of $\mathrm{Ext}^1(L_\bullet, A)$ described by (3.2.5.2) is precisely $c(X, \alpha)$. This proves the surjectivity of (3.2.4) and completes the proof of 3.2.5.

3.3. Properties of the cofibred category $\underline{\Psi}_{L_\bullet}$

In this section, which is not essential for understanding what follows, we freely use the notions of [4], our purpose being to compare the preceding results with those of that work.

Theorem 3.3.1. The additive cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{C}$ is left exact ([4, 2.2]), admits a quasi-maximal object ([4, 4.4]), and, for every object $E = (X, \alpha)$ of this category, there exists a typical complex $L^E_\bullet$ for E ([4, 3.1]). Moreover, the typical pro-complex of $\underline{\Psi}_{L_\bullet}$ ([4, 3.7 and 4.4]) is canonically isomorphic in $D(\underline{C})$ to the truncated complex $[L_\bullet]$ considered in 3.1.

We now spell out the meaning of the several assertions in 3.3.1.

3.3.2. To say that the cofibred category $\underline{\Psi}_{L_\bullet}$ is left exact means that it is additive and also satisfies the following condition:

3.3.2.1. For every exact sequence

$$0 \longrightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \longrightarrow 0$$

in $\underline{C}$, the natural functor (in the notation of [4, 1.8])

$$\underline{\Psi}_{L_\bullet}(A) \longrightarrow \underline{\Psi}_{L_\bullet}([A \longrightarrow A''])$$

is an equivalence of categories.

Recall that if $\underline{\Psi}$ is a cofibred category over $\underline{C}$ and $v : A \longrightarrow A''$ is a morphism in $\underline{C}$, then $\underline{\Psi}([A \longrightarrow A''])$ is the category of pairs (E, ρ) in which E is an object of $\underline{\Psi}(A)$ and ρ is a "trivialization" of the corresponding object $v_*(E)$ of $\underline{\Psi}(A'')$; that is, an isomorphism from it to the zero object (2.5.1) of $\underline{\Psi}(A'')$. The definition of the functor

$$\underline{\Psi}(A') \longrightarrow \underline{\Psi}([A \longrightarrow A''])$$

in 3.3.2.1 is evident. When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the objects of $\underline{\Psi}_{L_\bullet}([A \longrightarrow A''])$ are explicitly the triples (X, α, ρ), where (X, α) is as in (3.1.3) and $\rho : X \longrightarrow A''$ is a morphism in $\underline{C}$ satisfying

$$(*) \qquad \rho i = v, \qquad \rho\alpha = 0.$$

If $v : A \longrightarrow A''$ belongs to a short exact sequence as in 3.3.2.1, then, as is well known, giving a morphism satisfying the first relation (*) is equivalent, by the left exactness of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$, to giving an extension X' of L_0 by $A' \simeq \operatorname{Ker} v$ which is a subextension of X: one takes $X' = \operatorname{Ker} \rho$. The second relation (*) says that α factors through a morphism $\alpha' : L_1 \longrightarrow X'$. Up to the compatibilities whose verification is immediate, this proves the left exactness stated in 3.3.2.1.

3.3.3. Recall that an object E_0 of a cofibred category $\underline{\Psi}$ over $\underline{C}$ is called quasi-maximal if, for every object E of $\underline{\Psi}$ lying over an object A of $\underline{C}$, there is a morphism $E \longrightarrow E'$ in $\underline{\Psi}$ inducing a monomorphism $A \longrightarrow A'$ in $\underline{C}$, such that E is dominated by E_0, meaning that there is a morphism $E_0 \longrightarrow E'$. If $\underline{C}$ has enough injective objects, this is equivalent to saying that every object of $\underline{\Psi}$ over an injective object of $\underline{C}$ is dominated by E_0.

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ there is the following canonical quasi-maximal object. Put

$$L'_1 = \operatorname{Coker}(d_1 : L_2 \longrightarrow L_1)$$

and let $\bar{d}_0 : L'_1 \longrightarrow L_0$ be the morphism induced by $d_0 : L_1 \longrightarrow L_0$. Let $E_0 = (X_0, \alpha_0)$ be the object of $\underline{\Psi}_{L_\bullet}(L'_1)$ for which $X_0 = L_0 \oplus L'_1$ is the trivial extension of L_0 by L'_1 and α_0 is the composite

$$L_1 \twoheadrightarrow L'_1 \xrightarrow{(\bar{d}_0, \mathrm{id}_{L'_1})} L_0 \oplus L'_1.$$

To see that E_0 is quasi-maximal, observe that for every object (X, α) of $\underline{\Psi}_{L_\bullet}$ one can find a morphism from it to an object (X', α'), inducing a monomorphism $A \longrightarrow A'$ in $\underline{C}$, such that the extension X' of L_0 by A' is split. Indeed, it suffices to take $A' = X$ and $(X', \alpha') = i_*(X, \alpha)$, with the notation of (3.1.2). On the other hand, it is immediate that any object (X', α') of $\underline{\Psi}$ for which the extension X' is split is dominated by E_0.

3.3.4. Let E be an object of an additive cofibred category $\underline{\Psi}$ over $\underline{C}$, lying above an object A of $\underline{C}$, and consider the following functor of the variable object B of $\underline{C}$:

$$B \longmapsto \operatorname{Hom}(E, \theta_B), \tag{3.3.4.1}$$

where θ_B denotes the trivial object of $\underline{\Psi}(B)$. We say that E admits a typical complex if this functor is representable by an object Ω_E of $\underline{C}$. Since the functor in question maps canonically to the functor $B \longmapsto \operatorname{Hom}(A, B)$ represented by A, one obtains a canonical morphism

$$A \longrightarrow \Omega_E, \tag{3.3.4.2}$$

and the corresponding length-one chain complex $A \longrightarrow \Omega_E$ in $\underline{C}$ is called the typical complex of E.

When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ and $E = (X, \alpha)$, one checks immediately that the functor (3.3.4.1) is represented by $\operatorname{Coker} \alpha$, and that the canonical morphism (3.3.4.2) is simply the composite

$$A \xrightarrow{i} X \xrightarrow{\mathrm{can}} \operatorname{Coker} \alpha.$$

3.3.5. If an additive cofibred category $\underline{\Psi}$ over $\underline{C}$ admits a quasi-maximal object (3.3.3), and the typical complexes of its objects exist, then the typical complex $T_\bullet$ of a quasi-maximal object E_0, regarded as an object of the derived category $D(\underline{C})$, is independent, up to canonical isomorphism, of the choice of E_0 ([4, 4.4]). It is called the typical complex of the additive cofibred category $\underline{\Psi}$. If, in addition, $\underline{\Psi}$ is left exact (3.3.2), then $\underline{\Psi}$ can be reconstructed from the typical complex $T_\bullet$, up to an equivalence of fibred categories over $\underline{C}$ unique up to unique isomorphism ([4, 6.10]). There are also canonical isomorphisms, functorial in $A \in \underline{C}$ ([4, 5.4]),

$$\Psi^0(A) \xrightarrow{\sim} \operatorname{Ext}^0(T_\bullet, A), \qquad \Psi^1(A) \xrightarrow{\sim} \operatorname{Ext}^1(T_\bullet, A). \tag{3.3.5.1}$$

To complete the comparison with [4], we shall prove that when $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ its typical complex is the truncation $[L_\bullet]$. This gives another definition of the isomorphisms (3.1.8) and (3.2.4) in terms of (3.3.5.1), and one checks that it is compatible with the definitions in 3.1 and 3.2.

For this assertion we may plainly restrict to the case where $L_\bullet$ is already truncated in degree 1, since 3.1 noted that truncation does not change $\underline{\Psi}_{L_\bullet}$ up to an isomorphism of cofibred categories. The explicit construction in 3.3.4 then shows immediately that the typical complex of the quasi-maximal object described in 3.3.3 is canonically isomorphic to $L_\bullet$.

Remark 3.3.6. Let

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

be an exact sequence in $\underline{C}$. It gives an exact sequence of Ext^i groups

$$0 \longrightarrow \mathrm{Ext}^0(L_\bullet, A') \longrightarrow \mathrm{Ext}^0(L_\bullet, A) \longrightarrow \cdots \longrightarrow \mathrm{Ext}^1(L_\bullet, A) \longrightarrow \mathrm{Ext}^1(L_\bullet, A''),$$

which could of course be continued to the right by introducing Ext^i for $i \geq 2$. Using the isomorphisms (3.1.8) and (3.2.4), we obtain the exact sequence

$$\begin{aligned}(3.3.6.1)\quad 0 \longrightarrow \Psi^0_{L_\bullet}(A') \longrightarrow \Psi^0_{L_\bullet}(A) \longrightarrow \Psi^0_{L_\bullet}(A'') \xrightarrow{\partial} \Psi^1_{L_\bullet}(A') \\ \longrightarrow \Psi^1_{L_\bullet}(A) \longrightarrow \Psi^1_{L_\bullet}(A'').\end{aligned}$$

All arrows other than ∂ are defined simply by functoriality of $\Psi^0_{L_\bullet}$ and $\Psi^1_{L_\bullet}$. On the other hand, [4, 2.5.2] defines an exact sequence (3.3.6.1), for every left-exact cofibred category $\underline{\Psi}$ over $\underline{C}$, through a morphism

$$(3.3.6.2)\qquad \partial : \Psi^0(A'') \longrightarrow \Psi^1(A').$$

It associates with an element u of the left-hand side the class, in the right-hand side, of the object of $\underline{\Psi}(A')$, determined up to canonical isomorphism, whose image in $\underline{\Psi}([A \longrightarrow A''])$ (notation of 3.3.2) is the pair (X_0, u), where X_0 is the trivial object of $\underline{\Psi}(A)$.

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the morphism (3.3.6.2) defined by this "geometric" construction is the negative of the boundary morphism ∂ in (3.3.6.1). Thus, after the identifications above and a change of sign in ∂, the latter exact sequence is precisely that of [4, 2.5.2]. This follows from the final assertion of [4, 5.4], in view of the compatibilities noted in 3.3.5.

Remark 3.3.6.3. We take this opportunity to clarify a point in [4, no. 3.11] concerning the fundamental isomorphisms (3.11.3) and (3.11.4), which play an essential role in the proof of [4, 5.4]. The diagram on page 35 of [4] is commutative, not anticommutative, and therefore does not as it stands define a morphism

$$u : [L_1^X \longrightarrow L_0^X] \longrightarrow K^\bullet[1].$$

It must be made anticommutative by replacing u^1 by $-u^1$. One could instead adopt the opposite convention and replace u^0 by $-u^0$; however, the stated convention, in which u^1 becomes $-u^1$, is the one for which (3.11.3) generalizes the isomorphism (3.5.3) of [4], as claimed there. With this clarification, the claim made in [4, 5.4] must also be corrected: the two boundary morphisms considered there differ by a sign.

3.3.7. In [4, 6.13] the variance of the cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{C}$ as $L_\bullet$ varies was also made explicit, in particular by reconstructing the category of cocartesian functors from $\underline{\Psi}_{L_\bullet}$ to $\underline{\Psi}_{L'_\bullet}$ in terms of the chain complexes $L_\bullet$ and $L'_\bullet$. The set of isomorphism classes of such cocartesian functors is canonically in bijection with

$$\mathrm{Hom}_{D(\underline{C})}(L'_\bullet, L_\bullet).$$

We shall use only a small part of these results, which we develop in the next section using the simple description of $\underline{\Psi}_{L_\bullet}$ given in 3.1.

3.4. Variance of $\underline{\Psi}_{L_\bullet}$ with $L_\bullet$

Consider a morphism of complexes

$$f : L'_\bullet \longrightarrow L_\bullet \tag{3.4.1}$$

given by a commutative diagram

$$\begin{array}{ccccccccc} \longrightarrow & L'_2 & \longrightarrow & L'_1 & \longrightarrow & L'_0 & \longrightarrow & 0 \\ & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0} & & \\ \longrightarrow & L_2 & \longrightarrow & L_1 & \longrightarrow & L_0 & \longrightarrow & 0. \end{array} \tag{3.4.2}$$

We associate with it a canonical functor

$$f^* : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{\Psi}_{L'_\bullet}. \tag{3.4.3}$$

It is in fact a cocartesian functor of cofibred categories over $\underline{C}$: it is compatible with the structural functors to $\underline{C}$ and carries cocartesian morphisms to cocartesian morphisms; see SGA 1, Exposé VI.

If (X, α) is an object of $\underline{\Psi}_{L_\bullet}$, define $f^*(X, \alpha) = (X', \alpha')$ as follows. The extension X' of L'_0 by A is the pullback, along f_0, of the given extension X of L_0 by A, and α' is the partial trivialization deduced from α by transitivity in the second square of (3.4.2). The compatibility of α' with the relation $d'_0 d'_1 = 0$ follows immediately from the analogous compatibility for α. By construction, (3.4.3) commutes with the structural functors to $\underline{C}$. Its cocartesian character follows at once from the fact that, in the category of extensions of objects of $\underline{C}$, direct images of extensions commute with inverse images of extensions.

The functor (3.4.3) induces morphisms of functors

$$f^i : \Psi^i_{L_\bullet} \longrightarrow \Psi^i_{L'_\bullet}, \qquad i = 0, 1 \tag{3.4.4}$$

(compare 2.6). The morphism (3.4.1), on the other hand, induces morphisms

$$f^i : \mathrm{Ext}^i(L_\bullet, -) \longrightarrow \mathrm{Ext}^i(L'_\bullet, -), \qquad i \in \mathbb{Z}. \tag{3.4.5}$$

For $i = 0, 1$, the morphisms (3.4.4) and (3.4.5) are compatible with the isomorphisms (3.1.8) and (3.2.4). In other words, for $A \in \mathrm{Ob}\,\underline{C}$ and $i = 0, 1$, the following diagram is commutative:

$$\begin{array}{ccc} \Psi^i_{L_\bullet}(A) & \xrightarrow{\sim} & \mathrm{Ext}^i(L_\bullet, A) \\ \downarrow & & \downarrow \\ \Psi^i_{L'_\bullet}(A) & \xrightarrow{\sim} & \mathrm{Ext}^i(L'_\bullet, A). \end{array} \tag{3.4.6}$$

The verification of this compatibility is essentially immediate and is left to the reader.

Proposition 3.4.7. The functor (3.4.3) is faithful (respectively fully faithful, respectively an equivalence of categories) if and only if

$$H_0(f) : H_0(L'_\bullet) \longrightarrow H_0(L_\bullet)$$

is an epimorphism (respectively, if $H_0(f)$ is an isomorphism and $H_1(f)$ is an epimorphism; respectively, if $H_0(f)$ and $H_1(f)$ are isomorphisms).

Indeed, with f^i denoting the functorial morphism (3.4.4), the required condition is that f^0 be a monomorphism (respectively, that f^0 be an isomorphism and f^1 a monomorphism; respectively, that f^0 and f^1 be isomorphisms). By the commutativity of (3.4.6), these conditions may be

expressed by the analogous conditions on the functors Ext^i, $i = 0, 1$. The proposition then reduces to a straightforward statement of homological algebra, which we leave to the reader as an exercise.

3.4.8. Consider now an exact sequence of chain complexes

$$0 \longrightarrow L_\bullet \xrightarrow{f} L'_\bullet \xrightarrow{g} L''_\bullet \longrightarrow 0. \tag{3.4.8.1}$$

It gives functors of the form (3.4.3),

$$\underline{\Psi}_{L''_\bullet} \xrightarrow{g^*} \underline{\Psi}_{L'_\bullet} \xrightarrow{f^*} \underline{\Psi}_{L_\bullet}.$$

The transitivity property implicit in functors of the form (3.4.3) gives an isomorphism of functors

$$f^*g^* \simeq (gf)^* = (0_{L_\bullet \to L''_\bullet})^*.$$

The last member is evidently canonically isomorphic to the cocartesian zero functor from $\underline{\Psi}_{L''_\bullet}$ to $\underline{\Psi}_{L_\bullet}$. The preceding isomorphism therefore defines a canonical functor

$$\underline{\Psi}_{L''_\bullet} \longrightarrow \underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}. \tag{3.4.8.2}$$

Here $\underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}$ denotes the category of pairs (X', t) in which X' is an object of $\underline{\Psi}_{L'_\bullet}$ and t is a trivialization of its inverse image $f^*(X')$, that is, an isomorphism

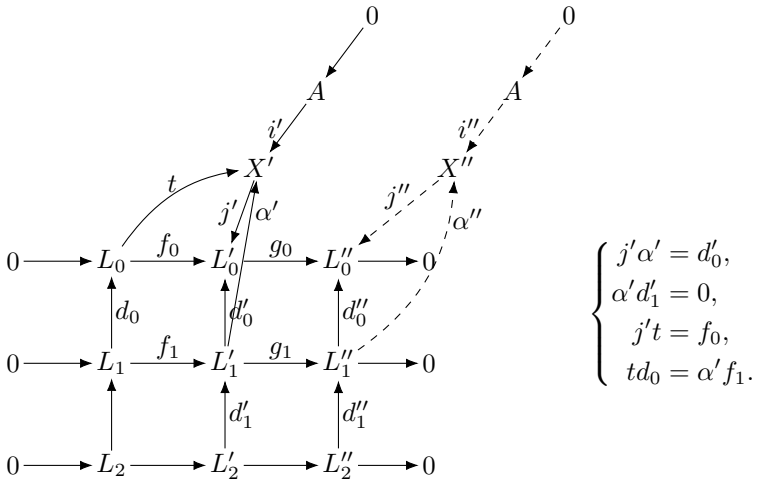
$$f^*(X') \simeq 0_{L_\bullet, A},$$

where $0_{L_\bullet, A}$ denotes the zero object of $\underline{\Psi}_{L_\bullet}(A)$ and A is the object of $\underline{C}$ underlying X'. We claim that the functor (3.4.8.2) is an equivalence of categories. Together with the left exactness in A established in 3.3.1, this completes the exactness property in $L_\bullet$ of the expression $\underline{\Psi}_{L_\bullet}(A)$.

Let us sketch the proof. To give an object of the target of (3.4.8.2) is to give an extension

$$0 \longrightarrow A \xrightarrow{i'} X' \xrightarrow{j'} L'_0 \longrightarrow 0$$

of L'_0 by an object A of $\underline{C}$, together with a morphism $\alpha' : L'_1 \longrightarrow X'$ satisfying $j'\alpha' = d'_0$ and $\alpha' d'_1 = 0$, and a morphism $t : L_0 \longrightarrow X'$ satisfying $j't = f_0$ and $td_0 = \alpha' f_1$. These data are displayed in the following diagram:



The familiar left exactness, in the first argument, of extensions of an object L by an object A, applied to the first row of the diagram, shows that a morphism t satisfying $j't = f_0$ allows one to interpret X' as the inverse image of an extension X'' of L''_0 by A. Under this interpretation, t corresponds to the natural trivialization of the inverse image of that extension along $g_0 f_0 = 0$.

The morphism α' satisfying $j'\alpha' = d'_0$ and $\alpha' f_1 = td_0$ is interpreted by essentially the same dictionary, now applied to the second row of the diagram, as arising from a morphism $\alpha'' : L''_1 \longrightarrow X''$ satisfying $j''\alpha'' = d''_0$. Here α' and α'' are regarded as trivializations of the inverse images of X' and X'' along d'_0 and d''_0, respectively. Finally, the relation $\alpha' d'_1 = 0$ is then equivalent to $\alpha'' d''_1 = 0$. Thus the data (X', α', t), an object of $\underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}$, are equivalent to the data (X'', α''), an object of $\underline{\Psi}_{L''_\bullet}$. This proves, in substance, that (3.4.8.2) is indeed an equivalence of categories.

3.4.9. Continue to consider the exact sequence (3.4.8.1), and the exact sequence of Ext^i groups that it defines:

$$(3.4.9.1)\quad 0 \longrightarrow \mathrm{Ext}^0(L''_\bullet, A) \xrightarrow{g^0} \mathrm{Ext}^0(L'_\bullet, A) \xrightarrow{f^0} \mathrm{Ext}^0(L_\bullet, A) \xrightarrow{\partial^0} \mathrm{Ext}^1(L''_\bullet, A) \xrightarrow{g^1} \mathrm{Ext}^1(L'_\bullet, A) \xrightarrow{f^1} \mathrm{Ext}^1(L_\bullet, A) \longrightarrow \cdots .$$

The canonical isomorphisms of the forms (3.1.8) and (3.2.4) give a geometric interpretation of the first six terms of this exact sequence in terms of the categories $\underline{\Psi}_{L_\bullet}$, $\underline{\Psi}_{L'_\bullet}$, and $\underline{\Psi}_{L''_\bullet}$. Compatibility (3.4.6) likewise gives a geometric interpretation of the arrows f^0, g^0, f^1, g^1. We shall now give such an interpretation of ∂^0, using the result just proved that (3.4.8.2) is an equivalence of categories.

Let

$$u \in \mathrm{Ext}^0(L_\bullet, A), \qquad \text{i.e.} \qquad u : L_0 \longrightarrow A, \quad ud_0 = 0.$$

Consider the object $(0, \overline{u})$ of $\underline{\Psi}_{[L_\bullet \to L'_\bullet]}$. Here 0 is the zero object of $\underline{\Psi}_{L'_\bullet}$, whose inverse image along f_0 is therefore identified with the zero object of $\underline{\Psi}_{L_\bullet}$, and $\overline{u}$ is the automorphism of the latter defined by u through the canonical isomorphism (3.1.8). Since (3.4.8.2) is an equivalence, $(0, \overline{u})$ comes from an object of $\underline{\Psi}_{L''_\bullet}(A)$, determined up to unique isomorphism. Its isomorphism class is an element of

$$\Psi^1_{L''_\bullet}(A) \simeq \mathrm{Ext}^1(L''_\bullet, A).$$

We claim that this element is precisely $\partial^0(u)$.

We prove this compatibility by reducing it to the analogous one in [4, Cor. 5.5], where the sign must be corrected as indicated in 3.3.6.3. The argument also gives another proof of the equivalence (3.4.8.2), by reducing it to [4, Prop. 2.7]. Consider the opposite category $\underline{C}^{\mathrm{o}}$ and interpret the construction of $\underline{\Psi}_{L_\bullet}(A)$, for fixed A, together with the constructions of 3.4.8 and the present section, in terms of $\underline{C}^{\mathrm{o}}$ and the category $\underline{E}$ cofibred over $\underline{C}^{\mathrm{o}}$ whose objects are extensions of the fixed object A of $\underline{C}^{\mathrm{o}}$ by a variable object L of $\underline{C}^{\mathrm{o}}$. This is an additive cofibred category over $\underline{C}^{\mathrm{o}}$, and is even left exact there (in the sense of [4, 2.2]).

A chain complex $L_\bullet$ of $\underline{C}$ may be interpreted as a cochain complex of $\underline{C}^{\mathrm{o}}$. Under this interpretation, the construction of $\underline{\Psi}_{L_\bullet}(A)$ coincides with the opposite of the category $\underline{\widehat{E}}(L_\bullet)$ in the sense of [4, 1.8]. The functor (3.4.8.2) is then identified with the opposite of the functor of the form [4, (2.1.2)], with E replaced by its extension $\underline{\widehat{E}}$ to the category of cochain complexes. Consequently, the assertion that this functor is an equivalence is the special case [4, 2.7, a)⇒d)].

On the other hand, the exact sequence (3.4.9.1) is identified with the exact sequence of the groups $\mathrm{Ext}^i(A, L''_\bullet)$, $\mathrm{Ext}^i(A, L'_\bullet)$, and $\mathrm{Ext}^i(A, L_\bullet)$, apart from a factor $(-1)^i$ on ∂^i (Verdier dixit). Similarly, the isomorphisms (3.1.8) and (3.2.4) are identified with the analogous isomorphisms (3.11.5) and (3.11.4). The asserted compatibility is therefore the special case of [4, Cor. 5.5] corrected as above.

Remark 3.4.9.2. The editor, being very afraid of signs, confesses that he does not have complete confidence in the preceding proof: he fears that signs may have been omitted in passing from C to C^{o}. He recommends that a courageous reader verify the asserted compatibility directly (or, respectively, its opposite). Fortunately, these sign questions will play no role in the following exposés.

3.5. A canonical partial flat resolution of an arbitrary $\mathbb{Z}$-module

We apply the preceding results when $\underline{C}$ is the category of commutative groups of the topos $\underline{T}$, that is, the category of $\mathbb{Z}$-modules of $\underline{T}$. Let P be an object of $\underline{C}$. We shall define, functorially in

P, a chain complex $L_\bullet(P)$ augmented to P:

$$0 \longrightarrow \overbrace{\mathbb{Z}[P\times P]\times\mathbb{Z}[P\times P\times P]}^{L_2} \xrightarrow{d_1} \overbrace{\mathbb{Z}[P\times P]}^{L_1} \xrightarrow{d_0} \overbrace{\mathbb{Z}[P]}^{L_0} \longrightarrow 0 \qquad (3.5.1)$$
$$\varepsilon \downarrow$$
$$P$$

with $L_i(P)=0$ for $i\geq 3$. The components are displayed in (3.5.1), using the notation of 1.4. To describe the differentials d_0, d_1 and the augmentation ε, write $[p]$ for the section of $\mathbb{Z}[P](S)$ defined by $p\in P(S)$, and similarly write $[p,q]$ and $[p,q,r]$ for the corresponding sections of $\mathbb{Z}[P\times P](S)$ and $\mathbb{Z}[P\times P\times P](S)$. Then set

$$\begin{cases} \varepsilon[p] = p, \\ d_0[p,q] = [p+q]-[p]-[q], \\ d_1[p,q] = [p,q]-[q,p], \\ d_1[p,q,r] = [p+q,r]-[p,q+r]+[p,q]-[q,r]. \end{cases} \qquad (3.5.2)$$

These homomorphisms of commutative groups do define a complex $L_\bullet(P)$ augmented to P:

$$\varepsilon d_0 = 0, \qquad d_0 d_1 = 0.$$

The first relation is immediate. The second splits into two relations according to the two factors of $L_2(P)$; they express, respectively, commutativity and associativity of the composition law on P.

The augmented complex depends functorially on P. A morphism $u : P \longrightarrow P'$ gives a commutative diagram

$$\begin{array}{ccc} L_\bullet(P) & \xrightarrow{L_\bullet(u)} & L_\bullet(P') \\ \downarrow & & \downarrow \\ P & \xrightarrow{u} & P'. \end{array}$$

Its components are flat, as is every free $\mathbb{Z}$-module.

Proposition 3.5.3. One has $H_1(L_\bullet(P))=0$, and the augmentation ε induces an isomorphism

$$H_0(L_\bullet(P)) \xrightarrow{\sim} P.$$

For the proof one can make the standard reduction to the case in which $\underline{T}$ is the category of sets and P is an ordinary $\mathbb{Z}$-module, where (3.5.1) is likely familiar. We instead give a direct proof that does not use this reduction, applying 3.4.7 to the augmentation

$$\varepsilon : L_\bullet(P) \longrightarrow P,$$

where P is regarded, as usual, as a chain complex concentrated in degree zero. It remains to prove that, for every object G of $\underline{C}$, the functor

$$(*) \qquad \varepsilon^* : \underline{\Psi}_P(G) \longrightarrow \underline{\Psi}_{L_\bullet(P)}(G)$$

is an equivalence of categories.

The source of (*) is plainly isomorphic to the category of extensions of the $\mathbb{Z}$-module P by G. To describe its target, use 1.4 to identify the categories of extensions of L_0, L_1, L_2 by G, respectively, with the category of torsors under G_P, the category of torsors under $G_{P\times P}$, and the category of pairs consisting of a torsor under $G_{P\times P}$ and a torsor under $G_{P\times P\times P}$.

Thus an object of the target of (*) is a pair (E, φ), where E is a torsor under G_P and φ is a trivialization of the torsor E' on $P \times P$ corresponding to the inverse image along d_0 of the extension of $\mathbb{Z}[P]$ by G defined by E. The trivialization is subject to the compatibility with $d_0 d_1 = 0$ described below. Formula (3.5.2) immediately gives

$$E' = \pi^* E \ \mathrm{pr}_1^*(E)^{-1} \ \mathrm{pr}_2^*(E)^{-1},$$

and a trivialization of this torsor is the same as an isomorphism, also denoted by φ,

$$\varphi : \mathrm{pr}_1^* E \ \mathrm{pr}_2^* E \xrightarrow{\sim} \pi^* E.$$

Expanding the compatibility of φ with $d_0 d_1 = 0$ gives the commutativity condition (1.2.1) and the associativity condition (1.1.4.1), corresponding to the two factors of L_2. The assertion that (*) is an equivalence is therefore precisely the dictionary of 1.1.6. This proves 3.5.3.

Remark 3.5.4. a) It would be interesting to strengthen 3.5.3 by constructing a resolution $L_\bullet(P)$ of P, functorial in P, whose components are functorially isomorphic to sums of free $\mathbb{Z}$-modules generated by cartesian powers of P. In view of isomorphisms of the form (1.4.5), this would give a method for partially computing $\mathrm{Ext}^i(P, A)$ as the abutment of a spectral sequence whose initial term is expressed through groups

$$H^j(P \times P \times \cdots \times P, A).$$

Here we obtain only very partial information about Ext^1, and no information about the higher Ext^i. There is, in any case, a natural way to continue the partial resolution $L_\bullet(P)$ by one degree. [1]

b) There is a variant of (3.5.1) and 3.5.3 in the category of A-modules, where A is a fixed ring of $\underline{T}$. The form of $L_\bullet(P)$ is more complicated because the description 1.1.6 must be enlarged (compare 2.10.3). One obtains

$$\text{(3.5.5)} \qquad \begin{cases} L_0 = A[P], \\ L_1 = A[P \times P] + A[A \times P], \\ L_2 = A[P \times P \times P] + A[P \times P] + A[A \times A \times P] + A[A \times P \times P]. \end{cases}$$

The augmented-complex structure is given by (3.5.2), supplemented by

$$\text{(3.5.6)} \qquad \begin{cases} d_0[\lambda, p] = [\lambda p] - \lambda[p], \\ d_1[\lambda, \mu, p] = [\lambda, \mu p] - [\lambda\mu, p] + \lambda[\mu, p], \\ d_1[\lambda, p, q] = [\lambda, p+q] - [\lambda, p] - [\lambda, q] + \lambda[p, q] - [\lambda p, \lambda q]. \end{cases}$$

The relation $d_0 d_1 = 0$ follows from the analogous relation on the components appearing in (3.5.2), which, as we saw, is equivalent to associativity and commutativity of addition on P, together with the relations

$$d_0 d_1[\lambda, \mu, p] = 0, \qquad d_0 d_1[\lambda, p, q] = 0.$$

The latter express, respectively, associativity of the given action of A on P,

$$\lambda(\mu p) = (\lambda\mu)p,$$

and its distributivity over addition,

$$\lambda(p+q) = \lambda p + \lambda q.$$

For a torsor E under G_P, a trivialization of $d_0^* E$ amounts to an internal composition law on E (satisfying the compatibility conditions already described for the internal law on E and its torsor structure), together with an external composition law of A on E, compatible with the action of A

1. L. Breen constructed such a resolution (editor's note).

on P and on G. The compatibility of this trivialization with the evident trivialization of $d_1^* d_0^* E$ is equivalent to the conditions already described above, which say that the internal law on E is a commutative-group law extending P by G, together with associativity of the external law and distributivity of the latter with respect to the internal law. These conditions say exactly that E, with its two composition laws and its torsor structure, is an A-module extension of the A-modules P and G.

It follows, by essentially the same method as in 3.5.3, that $L_\bullet$ is a partial resolution of the module P, that is, $H_1(L_\bullet) = 0$. It would again be interesting to extend this partial resolution to an infinite resolution of the same kind, with components of the form

$$A[A \times A \times \cdots \times P \times P \times \cdots].$$

Compare the analogous question in a) for commutative groups.

3.6. Application to the description of the cofibred category of biextensions of a given pair (P, Q) of commutative groups

Let P and Q be two $\mathbb{Z}$-modules of the topos $\underline{T}$. Choose a flat resolution $L'_\bullet(P)$ of P that agrees in degrees at most 2 with the augmented complex $L_\bullet(P)$ of 3.5. This is possible by 3.5.3 and because every $\mathbb{Z}$-module on $\underline{T}$ is a quotient of a free, hence flat, $\mathbb{Z}$-module. Similarly choose a flat resolution $L'_\bullet(Q)$ of Q agreeing with $L_\bullet(Q)$ in degrees at most 2. We have canonical homomorphisms

$$L_\bullet(P) \longrightarrow L'_\bullet(P), \qquad L_\bullet(Q) \longrightarrow L'_\bullet(Q), \tag{3.6.1}$$

and hence a canonical homomorphism

$$\begin{aligned} L_\bullet(P,Q) &\overset{\text{def}}{=} L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow L'_\bullet(P,Q) \\ &\overset{\text{def}}{=} L'_\bullet(P) \otimes L'_\bullet(Q) \simeq P \overset{\mathbb{L}}{\otimes} Q. \end{aligned} \tag{3.6.2}$$

It is an isomorphism on components in degrees at most 2. Consequently, by definition,

$$\underline{\Psi}_{L_\bullet(P,Q)} = \underline{\Psi}_{L'_\bullet(P,Q)}. \tag{3.6.3}$$

Theorem 3.6.4. The additive cofibred category over the category

$$\underline{C} = \mathrm{GRAB}(\underline{T})$$

of commutative groups of $\underline{T}$, consisting of biextensions of (P, Q) by a variable abelian group G (2.4), is canonically equivalent to the additive cofibred category

$$\underline{\Psi}_{L_\bullet(P,Q)} = \underline{\Psi}_{L'_\bullet(P,Q)}$$

defined from $L_\bullet(P,Q) = L_\bullet(P) \otimes L_\bullet(Q)$ as in 3.1.

This and 3.2.5 give the principal result of the present section, announced above:

Corollary 3.6.5. For fixed P and Q there is a canonical isomorphism, functorial in G,

$$\mathrm{Biext}^1(P, Q; G) \xrightarrow{\sim} \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G). \tag{3.6.5.1}$$

To prove 3.6.5, first make explicit the components of degrees $0, 1, 2$ of $L_\bullet(P,Q)$:

$$\begin{cases} L_0(P,Q) = L_0(P) \otimes L_0(Q) = \mathbb{Z}[P \times Q], \\ L_1(P,Q) = L_0(P) \otimes L_1(Q) + L_1(P) \otimes L_0(Q) \\ \qquad\quad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times P \times Q], \\ L_2(P,Q) = L_0(P) \otimes L_2(Q) + L_1(P) \otimes L_1(Q) + L_2(P) \otimes L_0(Q) \\ \qquad\quad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times Q \times Q \times Q] \\ \qquad\qquad + \mathbb{Z}[P \times P \times Q \times Q] \\ \qquad\qquad + \mathbb{Z}[P \times P \times Q] + \mathbb{Z}[P \times P \times P \times Q]. \end{cases} \tag{3.6.6}$$

To describe the cofibred category $\underline{\Psi}_{L_\bullet(P,Q)}$ through its fibres $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, use 1.4 to express extensions of $L_i(P,Q)$, for $i = 0, 1, 2$, by G in terms of torsors. An extension of $L_0(P,Q)$ by G is a torsor under $G_{P\times Q}$. An extension of $L_1(P,Q)$ by G is a pair consisting of a torsor under $G_{P\times Q\times Q}$ and one under $G_{P\times P\times Q}$. An extension of $L_2(P,Q)$ by G is similarly a system of five torsors, under the groups obtained from G by base change to

$$P\times Q\times Q,\quad P\times Q\times Q\times Q,\quad P\times P\times Q\times Q,$$
$$P\times P\times Q,\quad P\times P\times P\times Q.$$

Under this dictionary, a pair (X,α) consisting of an extension of $L_0(P,Q)$ by G and a trivialization of d_0^*X on $L_1(P,Q)$ becomes a torsor E on $P\times Q$ equipped with two trivializations (φ,ψ) of the torsors (E_1,E_2) on $P\times Q\times Q$ and $P\times P\times Q$, respectively, obtained from E by inverse image along

$$d_0 : L_1(P,Q) \longrightarrow L_0(P,Q).$$

Expanding d_0, as in the proof of 3.5.3, identifies φ and ψ with isomorphisms

$$\varphi : q_1^*E\, q_2^*E \xrightarrow{\sim} \pi_Q^*E, \qquad \psi : p_1^*E\, p_2^*E \xrightarrow{\sim} \pi_P^*E.$$

Here $\pi_Q, q_1, q_2 : P\times Q\times Q \longrightarrow P\times Q$ are induced by the composition law and the first and second projections $Q\times Q\longrightarrow Q$; the maps

$$\pi_P, p_1, p_2 : P\times P\times Q \longrightarrow P\times Q$$

are defined symmetrically. Thus φ and ψ are data of the forms (2.0.4) and (2.0.7) appearing in 2.1.

Finally, the condition that (X,α) be an object of $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, namely compatibility of the partial trivialization α with $d_0d_1 = 0$, becomes five compatibility relations for the structure (E,φ,ψ) on torsors over, respectively,

$$P\times Q\times Q,\quad P\times Q\times Q\times Q,\quad P\times P\times Q\times Q,$$
$$P\times P\times Q,\quad P\times P\times P\times Q.$$

They correspond to the five terms in the expression (3.6.6) for $L_2(P,Q)$. As in the proof of 3.5.3, the first two are precisely the commutativity relation (2.0.6) and the associativity relation (2.0.5); together they say that φ gives E the structure of a commutative extension of Q_P by G_P. Symmetrically, the last two say that ψ gives E the structure of a commutative extension of P_Q by G_Q. The middle relation, on torsors over $P\times P\times Q\times Q$, is exactly the commutativity of diagram (2.1.1). It says that φ and ψ are compatible, hence, in view of the other conditions, define on E a biextension structure of (P,Q) by G.

The verification of these interpretations is immediate and is left to the reader. The preceding discussion indeed defines an equivalence of categories cofibred over $\underline{C} = \mathrm{GRAB}(\underline{T})$, as asserted in 3.6.4.

Remark 3.6.7. Each of the three generalizations of biextensions mentioned in 2.10 has a corresponding variant of 3.6.4 and hence of 3.6.5. The generalization to any number of factors (2.10.2) is immediate, as is the generalization to biextensions of A-modules (2.10.3), once the variant of $L_\bullet(P)$ described in 3.5.4 b) has been constructed. From the viewpoint of homological algebra, the case in which P and Q are not assumed commutative (2.10.1) is more interesting.

Associate with a not necessarily commutative group P of $\underline{T}$ a complex $K_\bullet(P)$ truncated in degree 2. It is described as $L_\bullet(P)$ was in 3.5, except that

$$K_2(P) = \mathbb{Z}[P\times P\times P],$$

the first factor occurring in $L_2(P)$ being omitted; d_0 and d_1 are defined as in (3.5.2). Then

$$H_0(K_\bullet(P)) = P/[P,P] \simeq \underline{H}_1(P,\mathbb{Z}),$$

and similarly

$$H_1(K_\bullet(P)) \simeq \underline{H}_2(P, \mathbb{Z}).$$

Indeed, $K_\bullet(P)$ is precisely the truncation in degree 2 of the chain complex $C_\bullet(P, \mathbb{Z})$ of P with coefficients in $\mathbb{Z}$.

The argument proving 3.5.3 also shows that the cofibred category $\underline{\Psi}_{K_\bullet(P)}$ over $\underline{C} = \mathrm{GRAB}(\underline{T})$ is canonically equivalent to the category of central extensions of P by variable commutative groups G. Likewise, for two groups P, Q, the cofibred category of biextensions of (P, Q) in the sense of 2.10.1 is equivalent to $\underline{\Psi}_{K_\bullet(P,Q)}$, where

$$K_\bullet(P, Q) = K_\bullet(P) \overset{\mathbb{L}}{\otimes} K_\bullet(Q).$$

These considerations suggest that, when P is commutative, the canonical resolution sought in 3.5.4 a) might be obtained by a suitable modification of $C_\bullet(P, \mathbb{Z})$.

3.7. Various compatibilities

In this section we examine the behaviour, for variable P and Q, of the equivalence of cofibred categories in 3.6.4. The variance of the cofibred category of biextensions of (P, Q) by a variable G with respect to variable P, Q is defined in principle in 2.4; it may also be expressed by saying that one has a cofibred functor (2.6.1). Notice that the variance in P, Q is contravariant. The variance of the cofibred category $\underline{\Psi}_{L_\bullet(P,Q)}$ with respect to variable P, Q is also clear, since $L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q)$ depends functorially on the pair (P, Q), while $\underline{\Psi}_{L_\bullet}$ depends contravariantly on $L_\bullet$ (3.4). With this understood, a first compatibility is expressed, for every triple of morphisms

$$P' \longrightarrow P, \qquad Q' \longrightarrow Q, \qquad G \longrightarrow G', \tag{3.7.1}$$

by the fact that the diagram of functors

$$\begin{array}{ccc} \underline{\Psi}_{L_\bullet(P,Q)}(G) & \longrightarrow & \underline{\Psi}_{L_\bullet(P',Q')}(G') \\ \downarrow & & \downarrow \\ \mathrm{BIEXT}(P, Q; G) & \longrightarrow & \mathrm{BIEXT}(P', Q'; G') \end{array} \tag{3.7.2}$$

is commutative up to a canonical isomorphism. We shall admit this compatibility: making the commutativity isomorphism explicit could only be as obvious as it is tedious (as, indeed, could the complete explication it presupposes of the equivalence of cofibred categories in 3.6.4).

3.7.3. It follows in particular from the preceding compatibility that the isomorphism (3.6.5.1) is functorial not only in G, but also in P and Q. One may state, similarly and more trivially, the analogous functoriality of the isomorphism defined through (3.1.8), just as (3.6.5.1) was defined through (3.2.4):

$$\mathrm{Biext}^0(P, Q; G) \xrightarrow{\sim} \mathrm{Ext}^0(P \overset{\mathbb{L}}{\otimes} Q, G) \simeq \mathrm{Hom}(P \otimes Q, G). \tag{3.7.4}$$

But such an isomorphism was already obtained in (2.5.4.1), and was observed in 2.6 to be functorial in its three arguments. It is immediate to verify that the two isomorphisms (3.7.4) and (2.5.4.1) are equal.

3.7.5. The value of the isomorphism in 3.6.5 is that it expresses the remarkable group of geometric origin $\mathrm{Biext}^1(P, Q; G)$ in terms of the very simple homological-algebra invariant $\mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$. The latter belongs to the sequence of invariants $\mathrm{Ext}^i(P \overset{\mathbb{L}}{\otimes} Q, G)$, to which the usual formalism of homological algebra applies. In particular, the sequence of these Ext^i is a cohomological functor

in the sense of [3] (or a connected sequence of functors in the terminology of [5]) in each of the three arguments P, Q, G. Thus an exact sequence

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \tag{3.7.5.1}$$

(respectively,

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0, \qquad \text{or} \qquad 0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0) \tag{3.7.5.2}$$

gives rise to an infinite exact sequence of the Ext^i. Its first six terms, which involve only Ext^0 and Ext^1, may be interpreted in terms of biextensions by 3.6.5 and 3.7.4. We have already indicated in 3.3.6 a geometric interpretation of the exact sequence deduced from (3.7.5.1), and especially of the boundary map

$$\partial : \mathrm{Biext}^0(P, Q; G'') \longrightarrow \mathrm{Biext}^1(P, Q; G'), \tag{3.7.5.3}$$

within the framework of a left exact cofibred category. One may seek an analogous interpretation of the six-term exact sequence associated with either short exact sequence in (3.7.5.2), say the first; equivalently, in view of 3.7.3, one may seek a geometric interpretation of the boundary homomorphism

$$\partial : \mathrm{Biext}^0(P', Q; G) \longrightarrow \mathrm{Biext}^1(P'', Q; G). \tag{3.7.5.4}$$

Indeed, we have the following result.

Proposition 3.7.6. The cofibred category over

$$\mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})$$

of biextensions of (P, Q) by G, with P, Q, G variable (cf. (2.6.1)), is left exact not only in the argument G, but also in each of the arguments P and Q. Moreover, the boundary homomorphism (3.7.5.4) deduced from the exact sequence of the Ext^i associated with (3.7.5.2), by means of the isomorphisms 3.6.5 and 3.7.4, is the boundary homomorphism in the theory of left exact cofibred categories over $\underline{C} = \mathrm{GRAB}(\underline{T})^{\mathrm{op}}$ recalled in 3.3.6, applied to the cofibred category of biextensions of (P, Q) by G, where Q and G are fixed and P varies (subject to a possible sign error in 3.4.9; cf. 3.4.9.2).

We sketch the proof of 3.7.6. By 3.6.4 and compatibility (3.7.2), we interpret biextensions of (P, Q) by G as objects of $\underline{\Psi}_{L_\bullet(P,Q)}(G)$. Consider the homomorphism

$$L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow P \otimes L_\bullet(Q)$$

defined by the augmentation $L_\bullet(P) \longrightarrow P$. It follows from 3.5.3 and the flatness of $L_\bullet(Q)$ that this homomorphism induces isomorphisms on H_0 and H_1, and hence by 3.4.7 an equivalence

$$\underline{\Psi}_{P \otimes L_\bullet(Q)} \longrightarrow \underline{\Psi}_{L_\bullet(P,Q)}.$$

We may therefore interpret biextensions of (P, Q) by G as objects of $\underline{\Psi}_{P \otimes L_\bullet(Q)}(G)$. The exact sequence (3.7.5.2) now gives an exact sequence of chain complexes

$$0 \longrightarrow P' \otimes L_\bullet(Q) \longrightarrow P \otimes L_\bullet(Q) \longrightarrow P'' \otimes L_\bullet(Q) \longrightarrow 0. \tag{3.7.6.1}$$

The asserted left exactness follows from 3.4.8 applied to this exact sequence of complexes. (A direct verification would in any event be immediate.) Moreover, the homomorphism (3.7.5.4) is precisely the boundary homomorphism $\mathrm{Ext}^0 \longrightarrow \mathrm{Ext}^1$ deduced from this exact sequence by taking the Ext^i with values in G. The last assertion of 3.7.6 is consequently a special case of 3.4.9.

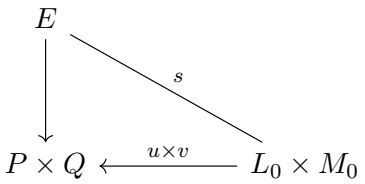
3.8. Biextensions of (P, Q) trivialized over extensions of P and Q

First consider two exact sequences of abelian groups

$$0 \longrightarrow L_1 \xrightarrow{i} L_0 \xrightarrow{u} P \longrightarrow 0, \qquad 0 \longrightarrow M_1 \xrightarrow{j} M_0 \xrightarrow{v} Q \longrightarrow 0. \tag{3.8.1}$$

We shall describe the category of biextensions E of (P, Q) by an abelian group G for which the inverse image $(u, v)^*(E)$ is trivial; that is, those for which there exists a homomorphism s making the following triangle commutative:

(3.8.2)



and satisfying the additivity condition required in each of its two arguments. The choice of such an s therefore identifies $(u, v)^*(E)$ with the trivial biextension $E_0 = L_0 \times M_0 \times G$ of (L_0, M_0) by G. The left exactness in P, Q of the cofibred category of biextensions of (P, Q) by fixed G (3.7.6) then shows that to give the biextension E of (P, Q), together with the trivialization s of its inverse image, is essentially equivalent to giving two partial trivializations of the trivial biextension E_0 of (L_0, M_0) by G, over (L_1, M_0) and (L_0, M_1) respectively, which agree over (L_1, M_1). Since the biextensions induced by E_0 over (L_1, M_0), (L_0, M_1), and (L_1, M_1) identify with the trivial biextensions, a partial trivialization over (L_1, M_0) (respectively over (L_0, M_1)) is equivalent to giving a homomorphism (or pairing)

$$f : L_1 \otimes M_0 \longrightarrow G \qquad \big(\text{respectively } g : L_0 \otimes M_1 \longrightarrow G\big). \tag{3.8.3}$$

The compatibility in question simply means that the two homomorphisms agree over $L_1 \otimes M_1$, namely

$$f(x_1 \otimes y_1) = g(x_1 \otimes y_1), \tag{3.8.4}$$

for sections x_1, y_1 of L_1, M_1 over an object of $\underline{T}$. In terms of the homomorphism s of (3.8.2), the forms f and g are characterized by

$$s(x_1, y_0) = f(x_1 \otimes y_0) e_{E/Q}(y_0), \qquad s(x_0, y_1) = g(x_0 \otimes y_1) e_{E/P}(x_0), \tag{3.8.5}$$

which again makes clear the necessity of condition (3.8.4).

Once the trivialization s has been chosen, the other trivializations are the morphisms $s' : L_0 \times M_0 \longrightarrow E$ of the form

$$s'(x_0, y_0) = s(x_0, y_0) + h(x_0 \otimes y_0), \tag{3.8.6}$$

where

$$h : L_0 \otimes M_0 \longrightarrow G$$

is an arbitrary form. It is then clear from (3.8.5) that the forms f, g defined by s are changed into

$$\begin{aligned} f'(x_1 \otimes y_0) &= f(x_1 \otimes y_0) + h(x_1 \otimes y_0), \\ g'(x_0 \otimes y_1) &= g(x_0 \otimes y_1) + h(x_0 \otimes y_1). \end{aligned} \tag{3.8.7}$$

Thus we obtain:

Proposition 3.8.8. The subgroup of $\mathrm{Biext}^1(P, Q; G)$ formed by the classes of biextensions whose inverse image under (u, v) is a trivial biextension of (L_0, M_0) (that is, the kernel of $\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(L_0, M_0; G)$) is canonically isomorphic, by the preceding description,

to the quotient of the group of pairs of forms (f,g) in (3.8.3) satisfying compatibility condition (3.8.4), by the pairs of forms which arise by restriction from a form $h : L_0 \otimes M_0 \longrightarrow G$.

We now interpret this isomorphism in terms of homological algebra, taking into account the canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$$

of 3.6.5. To this end, interpret the data (3.8.1) as two complexes resolving P and Q, respectively:

$$L_\bullet :\ 0 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \qquad M_\bullet :\ 0 \longrightarrow M_1 \longrightarrow M_0 \longrightarrow 0, \tag{3.8.9}$$

and consider the tensor-product complex $L_\bullet \otimes M_\bullet$

$$0 \longrightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \longrightarrow 0.$$

There is, moreover, a canonical homomorphism in the derived category

$$P \overset{\mathbb{L}}{\otimes} Q \longrightarrow L_\bullet \otimes M_\bullet,$$

and hence an equally canonical homomorphism

$$H^1(\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, G)) \longrightarrow \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G). \tag{3.8.10}$$

The group of 1-cocycles of $\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, G)$ identifies with the group of pairs (f,g) as in (3.8.3) which satisfy compatibility condition (3.8.4): to such a pair one associates the homomorphism $F : L_1 \otimes M_0 + L_0 \otimes M_1 \longrightarrow G$ with components (f,g),

$$F(x_1 \otimes y_0 + x_0 \otimes y_1) = f(x_1 \otimes y_0) + g(x_0 \otimes y_1). \tag{3.8.11}$$

Moreover, the coboundary of the 0-cochain $h : L_0 \otimes M_0 \longrightarrow G$ is precisely the pair of its restrictions to $L_1 \otimes M_0$ and $L_0 \otimes M_1$. Consequently, the group of pairs (f,g) modulo the pairs (h,h) described in 3.8.8 is canonically isomorphic to the first term of (3.8.10). With this understood, one has the following compatibility.

Proposition 3.8.12. If

$$\xi \in \mathrm{Ker}\big(\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Biext}^1(L_0,M_0;G)\big)$$

is described by the pair (f,g) as in 3.8.8, then the image of ξ in $\mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$ under the isomorphism (3.6.5.1) is the image under (3.8.10) of the element of its first term defined by the cocycle (3.8.11).

The verification of this compatibility is left to the reader.

3.8.13. We shall apply this especially in the case

$$L_0 = \mathbb{Z}[P], \qquad M_0 = \mathbb{Z}[Q],$$

where L_1 and M_1 are respectively the kernels of the canonical homomorphisms $\mathbb{Z}[P] \longrightarrow P$ and $\mathbb{Z}[Q] \longrightarrow Q$. Notice the canonical isomorphisms

$$L_0 \overset{\mathbb{L}}{\otimes} M_0 \simeq L_0 \otimes M_0 \simeq \mathbb{Z}[P \times Q].$$

Since L_0 and M_0 are flat, (1.4.5) gives canonical isomorphisms

$$\mathrm{Ext}^i(L_0 \overset{\mathbb{L}}{\otimes} M_0, G) \simeq H^i(P \times Q, G_{P\times Q}).$$

It follows from this and 3.6.5, or directly from the definitions and 1.4, that the functor which assigns to each biextension of (L_0,M_0) by G the $G_{P\times Q}$-torsor over $P \times Q$ that it induces is

an equivalence of categories. (This remains valid without using an additive structure on P, Q.) Consequently, if E is a biextension of (P, Q) by G, giving a trivialization of its inverse image over (L_0, M_0) is equivalent to giving a section of E considered only as an object over $P \times Q$, that is, a trivialization of E considered as a torsor under $G_{P\times Q}$.

Biextensions equipped with this additional structure are therefore described by the pairs (f, g) of (3.8.3) satisfying compatibility (3.8.4), and their classification up to isomorphism is the one given in 3.8.8.

With the notation of 3.5, one may also regard L_1 as the cokernel of $L_2(P) \longrightarrow L_1(P)$, and similarly for M_1. This makes it possible to express the data (f, g) in terms of a system of homomorphisms

$$\begin{cases} f' : L_1(P) \otimes L_0(Q) = \mathbb{Z}(P \times P \times Q) \longrightarrow G, \\ g' : L_0(P) \otimes L_1(Q) = \mathbb{Z}(P \times Q \times Q) \longrightarrow G, \end{cases}$$

that is, homomorphisms of sheaves of sets

$$P \times P \times Q \longrightarrow G, \qquad P \times Q \times Q \longrightarrow G,$$

satisfying five compatibility conditions. These express that f' and g' factor through f and g—which gives 2×2 equations, in view of the expression of $L_2(P)$ and $L_2(Q)$ as sums of two terms—together with condition (3.8.4). One then sees that f' and g' are precisely the partial multiplication data defining a biextension, and that the five conditions in question are exactly the five axioms for biextensions considered in 2.1 (cf. the proof of 3.6.4).

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Exposé VIII

COMPLEMENTS ON BIEXTENSIONS. GENERAL PROPERTIES OF BIEXTENSIONS OF GROUP SCHEMES

by A. Grothendieck

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0. Introduction

0.1. We continue here the general study of biextensions begun in the preceding exposé. First we give some complements in the case of an arbitrary base topos, developing in particular, with the detail it deserves, the formalism of the pairings associated with biextensions (Section 2), whose importance in the theory of abelian schemes is well known. We then begin the study of biextensions of general group schemes, the case of greatest interest for us being that of biextensions by the multiplicative group $\underline{G}_m$. We shall nevertheless be obliged, as a technical intermediate step, also to study biextensions by other types of group schemes (Sections 5 and 6).

The most interesting result for what follows is given in 7.1, which gives conditions under which, for P and Q smooth over a trait with generic point η, a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$ can be prolonged to a biextension of (P, Q) by $\underline{G}_m$, determined up to a unique isomorphism. This result will be very convenient in the following exposé, devoted to the theory of the Néron model of abelian varieties. It is likely that one could manage there at less expense, without the imposing paperweight of Exposés VII and VIII (which will no longer be used in the remainder of the Seminar); it nevertheless seems likely to us that the developments given in these exposés are destined to be useful elsewhere as well.

0.2. When we work with group schemes over a scheme S, it will be understood that we work with the fppf topos of S. This is the topos associated with the $\underline{U}$-site of schemes locally of finite presentation over S (and belonging to $\underline{U}$), endowed with the fppf topology (SGA 3 IV 6.3). The reader will readily verify that most of the statements we shall give would also be valid (and essentially equivalent to the fppf statements) for other topologies on S, whose underlying site contains at least the schemes locally of finite presentation over S, such as the Zariski, étale, and fpqc topologies. One reason is that, by descent theory, the category of extensions of a group scheme P locally of finite presentation over S by the group scheme $\underline{G}_m$ does not depend, up to equivalence, on the chosen topology; the same statement holds for the notion of biextension.

The technical reason for preferring the fppf topology (or a finer topology, such as fpqc) to coarser topologies such as the Zariski and étale topologies is that, for every integer $n > 0$, the homomorphism $n \cdot \mathrm{id}$ from $\underline{G}_m$ to itself is an fppf epimorphism (because it is faithfully flat and of finite presentation), but in general is not a Zariski or an étale epimorphism. Notice that it is an étale epimorphism if and only if n is prime to the residue characteristics of S, and a Zariski epimorphism if and only if S is empty.

0.3. Finally, most of the statements concerning extensions or biextensions by $\underline{G}_m$ would remain valid if that group were replaced by a torus G. Since such a G is locally isomorphic (for the fppf, and even for the étale topology) to a group $\underline{G}_m^r$, this is an essentially trivial generalization.

1. Various special cases over an arbitrary topos

1.1. In this section and the next, as in Exposé VII, T denotes a topos, and P, Q, G denote three abelian groups of T. Thus there is a canonical isomorphism (VII 3.6.5)

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right). \tag{1.1.1}$$

Recall the canonical isomorphisms (called the Cartan isomorphisms)

$$\begin{aligned}\mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) &\simeq \mathrm{Ext}^1(P, R\underline{\mathrm{Hom}}(Q,G)) \\ &\simeq \mathrm{Ext}^1(Q, R\underline{\mathrm{Hom}}(P,G)).\end{aligned} \tag{1.1.2}$$

All these isomorphisms are functorial in the three arguments P, Q, G.

Taking the standard truncation of the derived tensor product, whose homology objects are the $\mathrm{Tor}_i(P,Q)$, one obtains from (1.1.1) the following infinite exact sequence:

$$\begin{aligned}0 \longrightarrow \mathrm{Ext}^1(P\otimes Q, G) \longrightarrow \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}(\mathrm{Tor}_1(P,Q),G) \\ \longrightarrow \mathrm{Ext}^2(P\otimes Q, G) \longrightarrow \mathrm{Ext}^2\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) \longrightarrow \cdots.\end{aligned} \tag{1.1.3}$$

On the other hand, since the cohomology objects of $R\underline{\mathrm{Hom}}(Q,G)$ are the $\underline{\mathrm{Ext}}^i(Q,G)$, the first equality in (1.1.2) gives rise to the five-term exact sequence

$$\begin{aligned}0 \longrightarrow \mathrm{Ext}^1(P, \underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q,G)) \\ \longrightarrow \mathrm{Ext}^2(P, \underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{Ext}^2(P, R\underline{\mathrm{Hom}}(Q,G)),\end{aligned} \tag{1.1.4}$$

associated with the usual spectral sequence

$$\mathrm{Ext}^*(P, R\underline{\mathrm{Hom}}(Q,G)) \Longleftarrow \mathrm{Ext}^p(P, \underline{\mathrm{Ext}}^q(Q,G)).$$

We leave it to the reader, should the need arise, to make the first arrow of (1.1.3), respectively of (1.1.4), explicit by means of the natural fully faithful functor

$$\mathrm{EXT}(P\otimes Q, G) \longrightarrow \mathrm{BIEXT}(P,Q;G), \tag{1.1.5}$$

respectively

$$\mathrm{EXT}(P, \underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{BIEXT}(P,Q;G), \tag{1.1.6}$$

where EXT and BIEXT denote the evident categories of extensions and biextensions.

Of course, the isomorphism

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1(Q, R\underline{\mathrm{Hom}}(P,G))$$

gives rise to an exact sequence analogous to (1.1.3), with the roles of P and Q interchanged. Its first arrow can be made explicit by means of the fully faithful functor

$$\mathrm{EXT}(Q, \underline{\mathrm{Hom}}(P,G)) \longrightarrow \mathrm{BIEXT}(P,Q;G) \tag{1.1.7}$$

analogous to (1.1.6).

1.2. Suppose that

$$\mathrm{Tor}_1(P,Q)=0. \tag{1.2.1}$$

Then (1.1.3) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G)\simeq \mathrm{Ext}^1(P\otimes Q,G). \tag{1.2.2}$$

This holds, for example, if P or Q is a flat $\mathbb{Z}$-module. More precisely, under condition (1.2.1), the functor (1.1.5) is an equivalence of categories.

1.3. Suppose that

$$P\otimes Q=0. \tag{1.3.1}$$

Then (1.1.3) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G)\simeq \mathrm{Hom}(\mathrm{Tor}_1(P,Q),G). \tag{1.3.2}$$

The case of greatest interest to us in which (1.3.1) holds is that in which there is an integer $n>0$ such that

$$nP=P,\qquad Q=\varinjlim_i {}_{n^i}Q; \tag{1.3.3}$$

that is, P is divisible by n, and Q is the limit of subgroups annihilated by powers of n. Put, as usual,

$$T_n(P)=({}_{n^i}P)_{i\geq 0}\quad\text{(a projective system)},\qquad {}_{n^\infty}P=\varinjlim_i {}_{n^i}P.$$

From (1.3.3) one obtains the following formulas, more precise than (1.3.1):

$$\begin{aligned} &P\otimes Q=0,\\ \mathrm{Tor}_1(P,Q)&\simeq \varinjlim_i\big({}_{n^i}P\otimes {}_{n^i}Q\big)\simeq T_n(P)\otimes Q\\ &\simeq {}_{n^\infty}P\otimes T_n(Q). \end{aligned} \tag{1.3.4}$$

Thus the isomorphism (1.3.2) here assumes the more concrete form

$$\begin{aligned} \mathrm{Biext}^1(P,Q;G)&\simeq \mathrm{Hom}(T_n(P),D(Q))\\ &\simeq \mathrm{Hom}(Q,D(T_n(P)))\\ &\simeq \mathrm{Hom}(T_n(P)\otimes T_n(Q),T_n(G)). \end{aligned} \tag{1.3.5}$$

Let us specify the meaning of these formulas.

a) In the inductive system occurring in (1.3.4), the transition homomorphism

$${}_{n^i}P\otimes {}_{n^i}Q\longrightarrow {}_{n^j}P\otimes {}_{n^j}Q\qquad (i\leq j)$$

is described by identifying the first member with ${}_{n^j}P\otimes {}_{n^i}Q$ by means of $n^{j-i}\otimes\mathrm{id}$, and then using the homomorphism

$$\mathrm{id}\otimes\mathrm{inj}:{}_{n^j}P\otimes {}_{n^i}Q\longrightarrow {}_{n^j}P\otimes {}_{n^j}Q,$$

where $\mathrm{inj}:{}_{n^i}Q\longrightarrow {}_{n^j}Q$ denotes the inclusion.

b) By definition, put

$$T_n(P)\otimes Q\overset{\mathrm{def}}{=}\varinjlim_i\big(T_n(P)\otimes {}_{n^i}Q\big),$$

where

$$T_n(P)\otimes M\overset{\mathrm{def}}{=}{}_{n^i}P\otimes M\qquad \text{if } n^iM=0.$$

The meaning of the third term written in (1.3.4) is specified in the same way when $nQ=Q$ is also assumed, so that $T_n(Q)$, like $T_n(P)$, is a strict projective system. The last term of (1.3.5) is a Hom between projective systems of groups.

c) The letter D denotes $\underline{\mathrm{Hom}}(-,G)$. The object $D(Q)$ is defined as the projective system of groups

$$D(Q) = \big(D({}_{n^i}Q)\big)_{i\geq 0} = \big(\underline{\mathrm{Hom}}({}_{n^i}Q, G)\big)_{i\geq 0},$$

the second member of (1.3.5) being the group of homomorphisms of projective systems of groups. The following definition gives a simpler interpretation of the next member:

$$D(T_n(P)) = \varinjlim_i D({}_{n^i}P),$$

where the transition morphisms of this inductive system are obtained by transposition from the transition morphisms

$$n^{j-i} : {}_{n^j}P \longrightarrow {}_{n^i}P$$

of $T_n(P)$.

We leave it to the reader to make explicit the calculation of $\mathrm{Tor}_1(P,Q)$ indicated in (1.3.4), and to derive (1.3.5) from it. We note only that one uses the relations, deduced from the second hypothesis in (1.3.3),

$$\mathrm{Tor}_p(P,Q) \simeq \varinjlim_i \mathrm{Tor}_p(P, {}_{n^i}Q).$$

For $p=0$ this immediately gives $P\otimes Q = 0$, and it reduces the calculation of $\mathrm{Tor}_1(P,Q)$ to that of the $\mathrm{Tor}_1(P,{}_{n^i}Q)$ and their transition morphisms. Thus, when Q is annihilated by an integer $m = n^i$ such that $mP = P$, it remains to make explicit an isomorphism

$$\mathrm{Tor}_1(P,Q) \xleftarrow{\sim} {}_mP \otimes {}_mQ \qquad (mP = P,\ mQ = 0). \tag{1.3.6}$$

There are several ways to do this. One consists in using the exact Tor sequence associated with

$$0 \longrightarrow {}_mP \longrightarrow P \xrightarrow{m} P \longrightarrow 0.$$

One may also use the homomorphism (2.1.1) below; the fact that the transition morphisms are those specified in part (a) is then precisely 2.1.11. In any case, the choice of (1.3.6) amounts to the choice of a sign; see 2.1.10.

1.4. Suppose that

$$\underline{\mathrm{Hom}}(Q,G) = 0. \tag{1.4.1}$$

Then (1.1.4) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G) \xrightarrow{\sim} \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q,G)). \tag{1.4.2}$$

This case occurs in particular when Q is an abelian scheme and $G = \underline{G}_m$; see VII 1.3.8. Hence the functor in P,

$$P \longmapsto \mathrm{Biext}^1(P,Q;G),$$

is <u>representable</u> by $Q^* = \underline{\mathrm{Ext}}^1(Q,G)$. In particular, there is a canonical biextension E^Q of (Q^*,Q) by G, determined up to a unique isomorphism, since its automorphism group is

$$\mathrm{Hom}(Q^*, \underline{\mathrm{Hom}}(Q,G)) = 0.$$

This biextension, or its symmetric biextension ${}^sE^Q$ of (Q,Q^*) by G, may be called the <u>Weil biextension</u> of (Q^*,Q), respectively (Q,Q^*), by G.

1.5. Suppose that

$$\mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q,G)) = 0. \tag{1.5.1}$$

This holds in particular if $\underline{\mathrm{Ext}}^1(Q,G)=0$; for example, if Q is a finite locally free group scheme or a group of multiplicative type (SGA 3 IX), and $G=\underline{G}_m$; see 3.4 below. Then (1.1.4) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G) \xleftarrow{\sim} \mathrm{Ext}^1(P,\underline{\mathrm{Hom}}(Q,G)), \tag{1.5.2}$$

where $\underline{\mathrm{Hom}}(Q,G)$ plays the role of a "dual group" of Q (relative to G). More precisely, it follows from (1.5.2) that the fully faithful functor (1.1.6) is an equivalence of categories.

1.6. Suppose that the two relations

$$\underline{\mathrm{Hom}}(P,G)=0, \qquad \mathrm{Hom}(P,\underline{\mathrm{Ext}}^1(Q,G))=0 \tag{1.6.1}$$

hold simultaneously. By 1.4 and 1.5, respectively, these two relations give isomorphisms

$$\mathrm{Biext}^1(P,Q;G) \xrightarrow{\sim} \mathrm{Hom}(Q,\underline{\mathrm{Ext}}^1(P,G)),$$

$$\mathrm{Biext}^1(P,Q;G) \xleftarrow{\sim} \mathrm{Ext}^1(P,\underline{\mathrm{Hom}}(Q,G)).$$

Their composite is the canonical isomorphism

$$\mathrm{Ext}^1(P,\underline{\mathrm{Hom}}(Q,G)) \xrightarrow{\sim} \mathrm{Hom}(Q,\underline{\mathrm{Ext}}^1(P,G)) \quad \big(\simeq \mathrm{Biext}^1(P,Q;G)\big), \tag{1.6.2}$$

which plays the role of a duality formula.

Notice also that the first relation in (1.6.1) plainly implies $\mathrm{Hom}(P\otimes Q,G)=0$. Hence the category $\mathrm{EXT}(P,\underline{\mathrm{Hom}}(Q,G))$ of commutative extensions of P by $\underline{\mathrm{Hom}}(Q,G)$ is rigid (every automorphism of an object is the identity), as is $\mathrm{BIEXT}(P,Q;G)$. The structure of these categories is therefore known once the group of isomorphism classes of objects is known; by (1.6.2), this group is canonically isomorphic to

$$\mathrm{Hom}(Q,\underline{\mathrm{Ext}}^1(P,G)).$$

The map (1.6.2) is in fact defined in all cases, without assuming (1.6.1), and can also be described directly, without passing through $\mathrm{Biext}^1(P,Q;G)$, in terms of the natural pairing

$$Q\times Q' \longrightarrow G, \qquad Q'=\underline{\mathrm{Hom}}(Q,G).$$

Interpreting it as a homomorphism $Q\longrightarrow \underline{\mathrm{Hom}}(Q',G)$ gives a composite

$$\begin{aligned}\mathrm{Ext}^1(P,Q')\times Q &\longrightarrow \mathrm{Ext}^1(P,Q')\times \mathrm{Hom}(Q',G)\\ &\longrightarrow \mathrm{Ext}^1(P,G),\end{aligned} \tag{1.6.3}$$

which recovers (1.6.2).

For a useful case in which the considerations of 1.6 apply, see 3.7 below.

2. Pairings defined by a biextension

2.1. Let P and Q be two abelian groups of T, and let $n>0$ be an integer. Recall that ${}_nP$ denotes the kernel of multiplication by $n\cdot \mathrm{id}_P$. We shall define a canonical homomorphism

$$\alpha=\alpha^n_{P,Q}: {}_nP\otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P,Q), \tag{2.1.1}$$

which will be the composite

$${}_nP\otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1({}_nP,{}_nQ) \longrightarrow \underline{\mathrm{Tor}}_1(P,Q), \tag{2.1.2}$$

where the first arrow is $\alpha^n_{{}_nP,{}_nQ}$ and the second is induced by the inclusions ${}_nP\longrightarrow P$ and ${}_nQ\longrightarrow Q$. We are thus reduced to defining (2.1.1) when $nP=0$ and $nQ=0$, that is, when P and Q are $\mathbb{Z}/n\mathbb{Z}$-modules.

Suppose only that $nQ = 0$. Taking a flat resolution $L_\bullet$ of P (as a $\mathbb{Z}$-module), one obtains canonical isomorphisms

$$P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} Q \simeq L_\bullet \otimes_{\mathbb{Z}} Q \simeq (L_\bullet \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} Q,$$

which may be written as an isomorphism in the derived category

$$P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} Q \simeq \left(P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}\right) \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}/n\mathbb{Z}} Q. \tag{2.1.3}$$

This gives rise to the homological Künneth-type spectral sequence (cf. SGA 1 IV 5.2)

$$\underline{\mathrm{Tor}}^{\mathbb{Z}}_*(P, Q) \Longleftarrow E^2_{p,q} = \underline{\mathrm{Tor}}^{\mathbb{Z}/n\mathbb{Z}}_p\big(\underline{\mathrm{Tor}}^{\mathbb{Z}}_q(P, \mathbb{Z}/n\mathbb{Z}), Q\big). \tag{2.1.4}$$

Recall also the canonical isomorphisms

$$\underline{\mathrm{Tor}}^{\mathbb{Z}}_0(P, \mathbb{Z}/n\mathbb{Z}) = P_n := P/nP, \qquad \underline{\mathrm{Tor}}^{\mathbb{Z}}_1(P, \mathbb{Z}/n\mathbb{Z}) \simeq {}_nP,$$

and

$$\underline{\mathrm{Tor}}^{\mathbb{Z}}_i(P, \mathbb{Z}/n\mathbb{Z}) = 0 \qquad \text{for } i \geq 2.$$

They follow by computing these objects from the canonical flat resolution of $(\mathbb{Z}/n\mathbb{Z})_T$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z}_T & \xrightarrow{n} & \mathbb{Z}_T & \longrightarrow & 0 \\ & & \| & & \| & & \\ & & L_1 & & L_0 & & \end{array}$$

whose augmentation $L_0 \longrightarrow (\mathbb{Z}/n\mathbb{Z})_T$ sends 1 to 1. Consequently, the spectral sequence (2.1.4) reduces essentially to the exact sequence

$$\begin{aligned} 0 \longrightarrow \underline{\mathrm{Tor}}^{\mathbb{Z}/n\mathbb{Z}}_2(P_n, Q) \longrightarrow {}_nP \otimes Q \xrightarrow{\alpha} \underline{\mathrm{Tor}}^{\mathbb{Z}}_1(P, Q) \\ \longrightarrow \underline{\mathrm{Tor}}^{\mathbb{Z}/n\mathbb{Z}}_1(P_n, Q) \longrightarrow 0, \end{aligned} \tag{2.1.5}$$

where $P_n = P \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = P/nP$. The desired homomorphism α is the edge homomorphism $E^2_{0,1} \longrightarrow \underline{\mathrm{Tor}}^{\mathbb{Z}}_1$ in this exact sequence. Since the sequence is plainly functorial in P and Q, functoriality in P shows that the homomorphism in question is indeed the composite

$${}_nP \otimes Q \longrightarrow \underline{\mathrm{Tor}}_1({}_nP, Q) \longrightarrow \underline{\mathrm{Tor}}_1(P, Q),$$

where the first homomorphism is the α just defined, with (P, Q) replaced by $({}_nP, Q)$, and the second is induced by the inclusion ${}_nP \longrightarrow P$.

2.1.6. This therefore defines, for arbitrary P and Q, a homomorphism (2.1.1) satisfying the factorization condition (2.1.2). Moreover, when $nQ = 0$, the exact sequence (2.1.5) shows that α is an isomorphism whenever P_n is flat over $\mathbb{Z}/n\mathbb{Z}$. This holds in particular if $P_n = 0$, that is, if $nP = P$ and P is n-divisible, or if P itself is a flat $\mathbb{Z}/n\mathbb{Z}$-module. It also holds if Q is a flat $\mathbb{Z}/n\mathbb{Z}$-module.

2.1.7. Let us make the calculation of the homomorphism α in (2.1.5) explicit. To determine it, it suffices, for every section q of Q (over an object S of T; take the final object for simplicity), to determine the corresponding homomorphism

$${}_nP \longrightarrow \underline{\mathrm{Tor}}^{\mathbb{Z}}_1(P, Q).$$

The section q may be regarded as the image of the section 1 of $(\mathbb{Z}/n\mathbb{Z})_T$ under the uniquely determined homomorphism

$$u_q : (\mathbb{Z}/n\mathbb{Z})_T \longrightarrow Q.$$

Functoriality of α in Q therefore reduces us to the case $Q = (\mathbb{Z}/n\mathbb{Z})_T$, with q its unit section. In this case, identify

$$\underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q) = \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z})$$

with ${}_nP$ as explained after (2.1.4). The homomorphism in question is then the identity. Equivalently, the edge homomorphism

$$E^2_{0,q} = \underline{\mathrm{Tor}}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \longrightarrow \underline{\mathrm{Tor}}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z})$$

is the identity. This follows immediately from the definitions. The analogous assertion remains valid when $\mathbb{Z}$ and its quotient $\mathbb{Z}/n\mathbb{Z}$ are replaced by any ring A on T and any quotient ring A/I.

2.1.8. The homomorphism α of (2.1.1) may also be determined by describing, for every pair of sections p of ${}_nP$ and q of ${}_nQ$ (over an object S of T, again taken to be final to simplify the notation), the section of $\underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q)$ determined by (p, q). By functoriality in P and Q, as above, we are reduced to the case

$$P = Q = (\mathbb{Z}/n\mathbb{Z})_T,$$

with p and q the unit section. Thus we need only determine

$$\alpha : (\mathbb{Z}/n\mathbb{Z})_T \otimes (\mathbb{Z}/n\mathbb{Z})_T \longrightarrow \underline{\mathrm{Tor}}_1^{\mathbb{Z}}\big((\mathbb{Z}/n\mathbb{Z})_T, (\mathbb{Z}/n\mathbb{Z})_T\big).$$

The description in 2.1.7 shows that this is the isomorphism

$$(\mathbb{Z}/n\mathbb{Z})_T \xrightarrow{\sim} \underline{\mathrm{Tor}}_1^{\mathbb{Z}}\big((\mathbb{Z}/n\mathbb{Z})_T, (\mathbb{Z}/n\mathbb{Z})_T\big) \tag{2.1.8.1}$$

deduced from the canonical flat resolution

$$0 \longrightarrow \mathbb{Z}_T \xrightarrow{n} \mathbb{Z}_T \longrightarrow 0 \tag{2.1.8.2}$$

of the first factor $P = (\mathbb{Z}/n\mathbb{Z})_T$ occurring in $\underline{\mathrm{Tor}}_1$.

2.1.9. The definition given in (2.1.1) is not symmetric. By interchanging the roles of P and Q in the preceding construction, one may define an analogous homomorphism

$$\beta^n_{P,Q} : {}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P, Q), \tag{2.1.9.1}$$

first reducing to the case $nP = 0$ and then using a flat resolution of Q. Equivalently, β may be defined in terms of α by requiring the following diagram to commute:

$$\begin{array}{ccc} {}_nP \otimes {}_nQ & \xrightarrow{\sim} & {}_nQ \otimes {}_nP \\ \downarrow{\scriptstyle \beta^n_{P,Q}} & & \downarrow{\scriptstyle \alpha^n_{Q,P}} \\ \underline{\mathrm{Tor}}_1(P, Q) & \xrightarrow{\sim} & \underline{\mathrm{Tor}}_1(Q, P). \end{array} \tag{2.1.9.2}$$

Here the horizontal arrows are the symmetry isomorphisms. With this convention, the relation between α and β is

$$\beta^n_{P,Q} = -\alpha^n_{P,Q}. \tag{2.1.9.3}$$

In other words:

Lemma 2.1.10. Let P and Q be two commutative groups and let $n > 0$ be an integer. Then the following diagram is anticommutative, where α is defined in (2.1.1):

$$\begin{array}{ccc} {}_nP \otimes {}_nQ & \xrightarrow{\sim} & {}_nQ \otimes {}_nP \\ \downarrow{\scriptstyle \alpha^n_{P,Q}} & - & \downarrow{\scriptstyle \alpha^n_{Q,P}} \\ \underline{\mathrm{Tor}}_1(P, Q) & \xrightarrow{\sim} & \underline{\mathrm{Tor}}_1(Q, P). \end{array} \tag{2.1.10.1}$$

The horizontal arrows are the symmetry isomorphisms.

Proof. The procedure of 2.1.8 immediately reduces us to the case $P = Q = (\mathbb{Z}/n\mathbb{Z})_T$. We must then prove that the two isomorphisms (2.1.8.1), obtained by resolving respectively the first and second arguments of $\underline{\mathrm{Tor}}_1$ by (2.1.8.2), differ by a sign (equivalently, that the symmetry isomorphism on $\underline{\mathrm{Tor}}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ is $-\mathrm{id}$). We may plainly suppose that T is the punctual topos, so that we work with ordinary commutative groups. Let $L_\bullet$ be the complex

$$L_\bullet = \big(0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0\big),$$

regarded as a flat resolution of $P = Q = \mathbb{Z}/n\mathbb{Z}$. Use $L_\bullet \otimes L_\bullet$ to compute

$$\mathrm{Tor}_1(P, Q) = \mathrm{Tor}_1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}).$$

The isomorphisms with $H_1(L_\bullet \otimes \mathbb{Z}/n\mathbb{Z})$, which defines α, and with $H_1(\mathbb{Z}/n\mathbb{Z} \otimes L_\bullet)$, which defines β, are induced respectively by the augmentations $L_\bullet \longrightarrow \mathbb{Z}/n\mathbb{Z}$ of the second and first factors of $L_\bullet \otimes L_\bullet$. It follows at once that $\alpha(1,1)$ is represented by the cycle

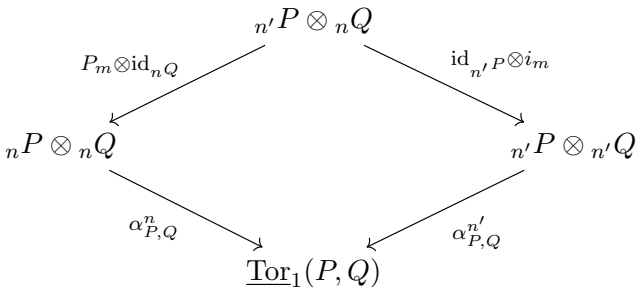
$$-1_0 \otimes 1_1 + 1_1 \otimes 1_0 \in (L_\bullet \otimes L_\bullet)_1 = L_0 \otimes L_1 + L_1 \otimes L_0,$$

where 1_0 and 1_1 denote the element 1 of L_0 and L_1 respectively, whereas $\beta(1,1)$ is represented by $1_0 \otimes 1_1 - 1_1 \otimes 1_0$. This proves 2.1.10.

To finish these generalities on the pairings α, we must still specify the relations among the homomorphisms $\alpha^n_{P,Q}$ as n varies. The main result is the following.

Lemma 2.1.11. Let P, Q be two commutative groups of T, and let n, n' be positive integers such that n' is a multiple of n, say $n' = mn$. Then the following diagram is commutative:

(2.1.11.1)



Here $P_m : {}_{n'}P \longrightarrow {}_nP$ is the homomorphism induced by $m \cdot \mathrm{id}_P$, and $i_m : {}_nQ \longrightarrow {}_{n'}Q$ is the inclusion. Likewise, the diagram (2.1.11.2) symmetric to (2.1.11.1) is commutative; it has the same vertices except that the upper vertex ${}_{n'}P \otimes {}_nQ$ is replaced by ${}_nP \otimes {}_{n'}Q$.

It suffices to prove the first assertion; the second follows by symmetry in view of 2.1.10. By the procedure of 2.1.8, we are reduced to the punctual topos, where P and Q are ordinary commutative groups and

$$P = \mathbb{Z}/n'\mathbb{Z}, \qquad Q = \mathbb{Z}/n'\mathbb{Z}.$$

In fact, for P we use only the hypothesis $n'P = 0$. Using the resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n'} \mathbb{Z} \longrightarrow 0$$

of Q, identify $\mathrm{Tor}_1(P, Q)$ with ${}_{n'}P = P$. Likewise, using

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0$$

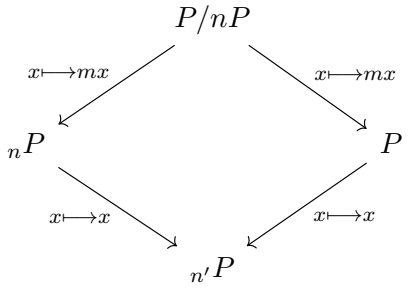
to resolve ${}_nQ \simeq \mathbb{Z}/n\mathbb{Z}$, identify $\mathrm{Tor}_1({}_nP, {}_nQ)$ with ${}_nP$. The augmentation in the latter resolution sends 1_0 to $m \cdot 1_Q$, where 1_Q denotes the element 1 of $Q = \mathbb{Z}/n'\mathbb{Z}$. Under these identifications the

homomorphism

$$\begin{array}{ccc} \mathrm{Tor}_1({}_nP, {}_nQ) & \longrightarrow & \mathrm{Tor}_1(P,Q) \\ \wr\big| & & \wr\big| \\ {}_nP & & {}_{n'}P = P \end{array}$$

induced by the inclusions ${}_nP \hookrightarrow P$ and ${}_nQ \hookrightarrow Q$ is simply the natural inclusion ${}_nP \hookrightarrow {}_{n'}P$.

Consequently, identify the upper vertex of (2.1.11.1) with P/nP using the generator $m \cdot 1_Q$ of ${}_nQ$, which identifies ${}_nQ$ with $\mathbb{Z}/n\mathbb{Z}$. The diagram becomes



which is plainly commutative.

Corollary 2.1.12. With the notation of 2.1.11, the following diagram is commutative:

$$\begin{array}{ccc} {}_{n'}P \otimes {}_{n'}Q & \xrightarrow{m\otimes m} & {}_nP \otimes {}_nQ \\ \big\downarrow{\scriptstyle \alpha^{n'}_{P,Q}} & & \big\downarrow{\scriptstyle \alpha^{n}_{P,Q}} \\ {}_{n'}\underline{\mathrm{Tor}}_1(P,Q) & \xrightarrow{m} & {}_n\underline{\mathrm{Tor}}_1(P,Q). \end{array} \tag{2.1.12.1}$$

In the description of the horizontal arrows, m denotes the homomorphisms ${}_{n'}E \longrightarrow {}_nE$ induced by $m \cdot \mathrm{id}_E$, for $E = P, Q$, and $\underline{\mathrm{Tor}}_1(P,Q)$ respectively.

Indeed, by 2.1.11, for $x \in {}_{n'}P(S)$ and $y \in {}_{n'}Q(S)$ one has

$$\alpha^n(mx \otimes my) = \alpha^{n'}(x \otimes my).$$

The right-hand side equals $m\alpha^{n'}(x \otimes y)$, since $x \otimes my = m(x \otimes y)$. This proves the commutativity of (2.1.12.1).

2.1.13. Let ℓ be a prime number. For every commutative group E, denote by $T_\ell(E)$ the inverse system

$$T_\ell(E) = (\ell^\nu E)_{\nu \geq 0}, \tag{2.1.13.1}$$

which plainly depends functorially on E. The compatibilities of 2.1.12, for n, n' of the form $\ell^\nu, \ell^{\nu'}$ with $\nu' \geq \nu$, show that the homomorphisms $\alpha^{\ell^\nu}_{P,Q}$, as ν varies, define a homomorphism of inverse systems, functorial in P, Q:

$$\alpha^{(\ell)}_{P,Q} : T_\ell(P) \otimes T_\ell(Q) \longrightarrow T_\ell\big(\underline{\mathrm{Tor}}_1(P,Q)\big). \tag{2.1.13.2}$$

Knowledge of these homomorphisms for every prime ℓ is plainly equivalent to knowledge of the $\alpha^n_{P,Q}$ for every integer n. The antisymmetry relations of 2.1.10 then become the analogous ℓ-adic antisymmetry relations, for every ℓ, expressed by the anticommutativity of the diagrams

$$\begin{array}{ccc} T_\ell(P) \otimes T_\ell(Q) & \longrightarrow & T_\ell(Q) \otimes T_\ell(P) \\ \big\downarrow{\scriptstyle \alpha^{(\ell)}_{P,Q}} & - & \big\downarrow{\scriptstyle \alpha^{(\ell)}_{Q,P}} \\ T_\ell\big(\underline{\mathrm{Tor}}_1(P,Q)\big) & \longrightarrow & T_\ell\big(\underline{\mathrm{Tor}}_1(Q,P)\big). \end{array} \tag{2.1.13.3}$$

Here the horizontal arrows are the symmetry arrows.

Remark 2.1.14. The construction of the homomorphism (2.1.1), and the results 2.1.10 and 2.1.11 concerning it, may be generalized to the situation in which one is given a commutative ring $\underline{A}$ on T and two ideals $\underline{a}$ and $\underline{b}$ of $\underline{A}$. In the case treated above, both ideals are the ideal $n\mathbb{Z}_T$ of the constant ring $\mathbb{Z}_T$ on T. One then defines a homomorphism, functorial in the $\underline{A}$-modules P, Q,

$$ {}_{\underline{a}}P \otimes {}_{\underline{b}}Q \xrightarrow{\alpha} \underline{\mathrm{Hom}}_{\underline{A}}\left(\frac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1^{\underline{A}}(P,Q)\right), \tag{2.1.14.1} $$

as the composite of the homomorphisms

$$ \begin{aligned} {}_{\underline{a}}P \otimes {}_{\underline{b}}Q &\longrightarrow \underline{\mathrm{Hom}}(\underline{A}/\underline{a}, P) \otimes \underline{\mathrm{Hom}}(\underline{A}/\underline{b}, Q) \\ &\longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{b}), \underline{\mathrm{Tor}}_1(P,Q)), \end{aligned} $$

using the well-known canonical isomorphism

$$ \underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{b}) \simeq \frac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}. \tag{2.1.14.2} $$

In fact, the sign in 2.1.10 is "hidden" in the choice of this isomorphism, according as one works with the resolution $0 \longrightarrow \underline{a} \longrightarrow \underline{A} \longrightarrow 0$ of $\underline{A}/\underline{a}$, or with the resolution $0 \longrightarrow \underline{b} \longrightarrow \underline{A} \longrightarrow 0$ of $\underline{A}/\underline{b}$. Here it becomes the anticommutativity of the diagram

$$ \begin{array}{ccc} {}_{\underline{a}}P \otimes {}_{\underline{b}}Q & \longrightarrow & {}_{\underline{b}}Q \otimes {}_{\underline{a}}P \\ \downarrow & -1 & \downarrow \\ \underline{\mathrm{Hom}}\left(\dfrac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1(P,Q)\right) & \longrightarrow & \underline{\mathrm{Hom}}\left(\dfrac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1(Q,P)\right). \end{array} \tag{2.1.14.3} $$

The horizontal arrows are the symmetry arrows. When $\underline{a} = \underline{b}$, this anticommutativity may also be expressed by saying that the symmetry automorphism on $\underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{a})$ is $-\mathrm{id}$.

In the special case where $\underline{a} = \underline{b} = n\underline{A}$, with n a section of $\underline{A}$, multiplication by n induces an epimorphism

$$ \underline{A}/\underline{a} \longrightarrow \frac{\underline{a}\cap\underline{a}}{\underline{a}\cdot\underline{a}} = \underline{a}/\underline{a}^2, $$

and hence a functorial injective homomorphism, in the $\underline{A}$-module T,

$$ \underline{\mathrm{Hom}}(\underline{a}/\underline{a}^2, T) \longrightarrow {}_{\underline{a}}T. $$

Thus in this case the homomorphism (2.1.14.1) may be interpreted as a homomorphism

$$ {}_nP \otimes {}_nQ \longrightarrow {}_n\underline{\mathrm{Tor}}_1(P,Q), \tag{2.1.14.4} $$

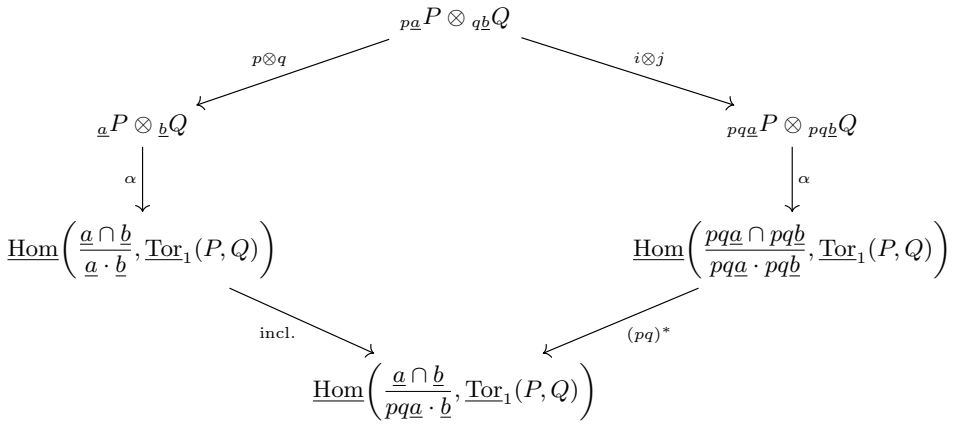
which is closer in form to (2.1.1).

Continue to suppose that $\underline{a} = \underline{b} = n\underline{A}$, and that n is a regular section of $\underline{A}$. The spectral sequence (2.1.4) still makes sense in the present setting and yields the following exact sequence of low-degree terms, generalizing (2.1.5), where Q is now an $\underline{A}$-module annihilated by $\underline{a}$:

$$ \begin{aligned} \underline{\mathrm{Tor}}_2^{\underline{A}}(P,Q) \longrightarrow \underline{\mathrm{Tor}}_2^{\underline{A}/n\underline{A}}(P_n, Q) \longrightarrow {}_nP \otimes Q \xrightarrow{\alpha} \underline{\mathrm{Tor}}_1^{\underline{A}}(P,Q) \\ \longrightarrow \underline{\mathrm{Tor}}_1^{\underline{A}/n\underline{A}}(P_n, Q) \longrightarrow 0. \end{aligned} \tag{2.1.14.5} $$

Return now to arbitrary $\underline{a}$ and $\underline{b}$, and let p and q be two sections of $\underline{A}$. We then obtain the commutative diagram below; (2.1.11.1) is essentially the special case obtained by taking $q = 1$:

(2.1.14.6)



Here

$$i : {}_{p\underline{a}}P \longrightarrow {}_{pq\underline{a}}P, \qquad j : {}_{q\underline{b}}Q \longrightarrow {}_{pq\underline{b}}Q$$

are the canonical inclusions, and $(pq)^*$ denotes the injective homomorphism obtained, by passing to $\underline{\mathrm{Hom}}$, from the surjective homomorphism

$$\frac{\underline{a}\cap\underline{b}}{pq\underline{a}\cdot\underline{b}} \longrightarrow \frac{pq\underline{a}\cap pq\underline{b}}{pq\underline{a}\cdot pq\underline{b}}$$

induced by multiplication by pq. When $\underline{a} = \underline{b} = n\underline{A}$, this diagram may be rewritten in a simpler form, given in (2.1.11.1) when $q = 1$.

2.2. Consider now, in addition to P and Q, a commutative group G. In (1.1.3) we defined a canonical homomorphism, functorial in P, Q, and G,

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(\mathrm{Tor}_1(P,Q), G\big). \tag{2.2.1}$$

On the other hand, if n is a positive integer (respectively, if ℓ is a prime number), the homomorphism $\alpha^n_{P,Q}$ of (2.1.1) (respectively, the homomorphism $\alpha^{(\ell)}_{P,Q}$ of (2.1.13.2)) gives a homomorphism

$$\mathrm{Hom}\big(\mathrm{Tor}_1(P,Q), G\big) \longrightarrow \mathrm{Hom}({}_nP \otimes {}_nQ, G)$$

(respectively, a homomorphism

$$\mathrm{Hom}\big(\mathrm{Tor}_1(P,Q), G\big) \longrightarrow \mathrm{Hom}\big(T_\ell(P) \otimes T_\ell(Q), T_\ell(G)\big)).$$

These last homomorphisms are also functorial in P, Q, G. Composing them with (2.2.1), we obtain canonical homomorphisms, functorial in P, Q, G,

$$\varphi^n = \varphi^n_{P,Q} : \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}({}_nP \otimes {}_nQ, G), \tag{2.2.2}$$

$$\varphi^{(\ell)} = \varphi^{(\ell)}_{P,Q} : \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(T_\ell(P) \otimes T_\ell(Q), T_\ell(G)\big). \tag{2.2.3}$$

For

$$\xi \in \mathrm{Biext}^1(P,Q;G), \tag{2.2.4}$$

denote its image under (2.2.2) (respectively, under (2.2.3)) by φ^n_ξ (respectively, by $\varphi^{(\ell)}_\xi$):

$$\varphi^n_\xi : {}_nP \otimes {}_nQ \longrightarrow G \qquad \text{(or, equivalently, } {}_nG\text{)}, \tag{2.2.5}$$

(2.2.6) $$\varphi_\xi^{(\ell)} : T_\ell(P) \otimes T_\ell(Q) \longrightarrow T_\ell(G).$$

2.2.7. Plainly, knowledge of $\varphi_\xi^{(\ell)}$ is equivalent to knowledge of the φ_ξ^n for all $n = \ell^\nu$, with $\nu \geq 0$. For fixed n, knowledge of φ_ξ^n is equivalent to knowledge, on the image of

$$\alpha_{P,Q}^n : {}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P,Q),$$

of the homomorphism

(2.2.8) $$\underline{\mathrm{Tor}}_1(P,Q) \longrightarrow G$$

which is the image of ξ under (2.2.1).

Observe that if $P = nP$ and $Q = nQ$, then the preceding image is ${}_n\underline{\mathrm{Tor}}_1(P,Q)$. This follows readily by considering the exact sequence (1.3.5) and the analogous exact sequence for Q. Furthermore, if P and Q are divisible, that is, if $nP = P$ and $nQ = Q$ for every positive integer n, one readily checks that $\underline{\mathrm{Tor}}_1(P,Q)$ is torsion, that is,

$$\underline{\mathrm{Tor}}_1(P,Q) = \varinjlim_n {}_n\underline{\mathrm{Tor}}_1(P,Q).$$

Thus, in this case, knowledge of the φ_ξ^n for all n is equivalent to knowledge of (2.2.8), whereas knowledge of $\varphi_\xi^{(\ell)}$ for a given prime ℓ is equivalent to knowledge of the restriction of (2.2.8) to the ℓ-primary component of $\underline{\mathrm{Tor}}_1(P,Q)$.

We now make explicit the symmetry properties of the pairings (2.2.5) and (2.2.6) associated with biextensions. First observe that there is a commutative diagram

(2.2.9)
$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow & & \downarrow \\ \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(P,Q),G\big) & \xrightarrow{\sim} & \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(Q,P),G\big), \end{array}$$

where the horizontal arrows are the symmetry arrows (the first of which was defined in VII 2.7), and the vertical arrows are (2.2.1). The commutativity follows from that of the two constituent squares in the fuller diagram

(2.2.10)
$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow & & \downarrow \\ \mathrm{Ext}^1\Big(P \overset{\mathbb{L}}{\otimes} Q, G\Big) & \xrightarrow{\sim} & \mathrm{Ext}^1\Big(Q \overset{\mathbb{L}}{\otimes} P, G\Big) \\ \downarrow & & \downarrow \\ \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(P,Q),G\big) & \xrightarrow{\sim} & \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(Q,P),G\big). \end{array}$$

The first vertical arrow in each column is the canonical isomorphism of VII 3.6.5, and the second is the evident homomorphism. The second square in (2.2.10) is trivially commutative: by definition, the symmetry isomorphism

$$P \overset{\mathbb{L}}{\otimes} Q \xrightarrow{\sim} Q \overset{\mathbb{L}}{\otimes} P$$

induces the symmetry isomorphism $\underline{\mathrm{Tor}}_1(P,Q) \simeq \underline{\mathrm{Tor}}_1(Q,P)$, and, for a variable complex $L_\bullet$, the canonical isomorphism

$$\mathrm{Ext}^1(L_\bullet, G) \simeq \mathrm{Hom}\big(H_1(L_\bullet), G\big)$$

is functorial in $L_\bullet$.

As for the commutativity of the first square of (2.2.10), it follows by making explicit simultaneously the symmetry isomorphisms and the vertical isomorphisms defined in VII 3.6.5. We leave the details of the verification to the reader. One uses the fact that the symmetry isomorphism

$$L_\bullet(P) \otimes L_\bullet(Q) \simeq L_\bullet(Q) \otimes L_\bullet(P)$$

induces in degree 1 the isomorphism

$$L_0(P) \otimes L_1(Q) + L_1(P) \otimes L_0(Q) \simeq L_0(Q) \otimes L_1(P) + L_1(Q) \otimes L_0(P),$$

which, on corresponding summands, is given by the ordinary symmetry isomorphism of the underlying ungraded modules (and hence introduces no sign).

In view of the commutativity of (2.2.9), the anticommutativity of (2.1.10.1), in terms of the φ^n or $\varphi^{(\ell)}$, becomes the following relation.

Proposition 2.2.11. The following diagrams are anticommutative:

(2.2.11.1)
$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow{\scriptstyle \varphi^n_{P,Q}} & - & \downarrow{\scriptstyle \varphi^n_{Q,P}} \\ \mathrm{Hom}({}_nP \otimes {}_nQ, G) & \xrightarrow{\sim} & \mathrm{Hom}({}_nQ \otimes {}_nP, G), \end{array}$$

(2.2.11.2)
$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow{\scriptstyle \varphi^{(\ell)}_{P,Q}} & - & \downarrow{\scriptstyle \varphi^{(\ell)}_{Q,P}} \\ \mathrm{Hom}\big(T_\ell(P) \otimes T_\ell(Q), T_\ell(G)\big) & \xrightarrow{\sim} & \mathrm{Hom}\big(T_\ell(Q) \otimes T_\ell(P), T_\ell(G)\big), \end{array}$$

where the horizontal arrows are the symmetry arrows.

We record the important corollary.

Corollary 2.2.12. Suppose that $P = Q$, and suppose that ξ is the class of a symmetric (respectively, antisymmetric) biextension of (P,P) by G; that is, suppose ${}^s\xi = \xi$ (respectively, ${}^s\xi = -\xi$), where ${}^s\xi$ denotes the image of ξ under symmetry. Then the bilinear maps φ^n_ξ, on ${}_nP \times {}_nP$, and $\varphi^{(\ell)}_\xi$, on $T_\ell(P) \times T_\ell(P)$, are antisymmetric (respectively, symmetric).

Remark 2.2.13. Suppose that ξ is symmetric. If n (respectively, ℓ) is odd, then 2.2.12 shows that φ^n_ξ (respectively, $\varphi^{(\ell)}_\xi$) is in fact an alternating form. This conclusion is not valid in general without the restriction that n (respectively, ℓ) be odd, as is already seen when P is the ordinary group $\mathbb{Z}/2\mathbb{Z}$ and $G = \mathbb{Z}/2\mathbb{Z}$.

Suppose, however, that $2P = P$. This is the case in particular when P is an abelian scheme, working as usual in the fppf or fpqc topology. The preceding forms are then still alternating. It is enough to prove that $\varphi^{2^\nu}_\xi(x,x) = 0$. Since $2P = P$, a given element $x \in {}_{2^\nu}P(S)$ may be written as $x = 2y$, with $y \in {}_{2^{\nu+1}}P(S)$. Hence

$$\varphi^{2^\nu}_\xi(x,x) = 2\varphi^{2^{\nu+1}}_\xi(y,y) = 0,$$

because $\varphi_\xi^{2\nu+1}$ is antisymmetric.

2.3. Relation with "Cartier duality". The description given in 2.2 of the pairings defined by a biextension, through the isomorphism theorem VII 3.6.5 and the homomorphism (2.2.1), is plainly not the definition formerly used for abelian schemes, at a time when the very notion of biextension did not yet exist. That definition proceeds through Cartier duality, and we now give its relation to the present one.

Consider again the homomorphism

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q,G)\big) \tag{2.3.1}$$

arising from the exact sequence (1.4). It is in fact an isomorphism when

$$\underline{\mathrm{Hom}}(Q,G) = 0. \tag{2.3.2}$$

This condition holds in particular when Q is an abelian scheme over a scheme S and G is the multiplicative group $\underline{G}_m$ (VII 1.3.8).

There is also an evident "restriction to nQ" homomorphism

$$\underline{\mathrm{Ext}}^1(Q,G) \longrightarrow \underline{\mathrm{Ext}}^1(nQ,G),$$

and hence a composite homomorphism

$$\begin{aligned}\mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q,G)\big) &\longrightarrow \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(nQ,G)\big)\\ &\longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ,G)\big),\end{aligned} \tag{2.3.4}$$

where the second arrow is restriction to ${}_nP$. Composing with (2.3.1), we obtain a homomorphism

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ,G)\big). \tag{2.3.5}$$

On the other hand, the exact sequence

$$0 \longrightarrow {}_nQ \longrightarrow Q \xrightarrow{n} nQ \longrightarrow 0 \tag{2.3.6}$$

gives rise to a long exact sequence of the $\underline{\mathrm{Ext}}^i(-,G)$, and hence to the short exact sequence

$$0 \longrightarrow \underline{\mathrm{Hom}}(Q,G)_n \longrightarrow \underline{\mathrm{Hom}}({}_nQ,G) \xrightarrow{\partial} {}_n\underline{\mathrm{Ext}}^1(nQ,G) \longrightarrow 0, \tag{2.3.7}$$

where, as usual, E_n denotes E/nE. We retain from this sequence the canonical surjective homomorphism

$$\underline{\mathrm{Hom}}({}_nQ,G) \xrightarrow{\qquad\partial\qquad} {}_n\underline{\mathrm{Ext}}^1(nQ,G), \tag{2.3.8}$$

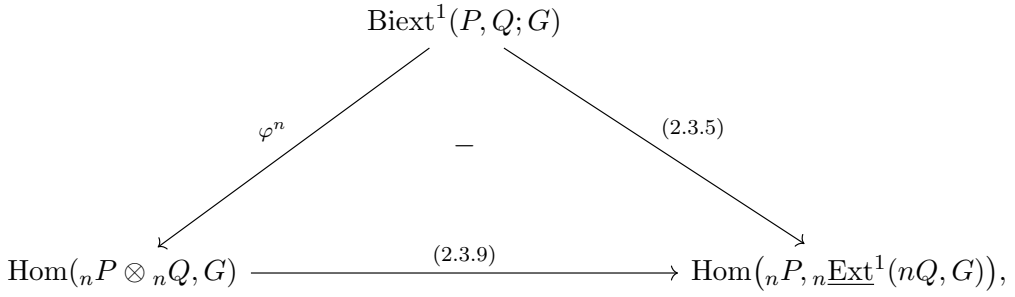
which is an isomorphism if $\underline{\mathrm{Hom}}(Q,G)_n = 0$, and a fortiori if (2.3.2) holds. From (2.3.8) we deduce a homomorphism

$$\begin{aligned}\mathrm{Hom}\big({}_nP, \underline{\mathrm{Hom}}({}_nQ,G)\big) &\simeq \mathrm{Hom}({}_nP \otimes {}_nQ, G)\\ &\longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ,G)\big),\end{aligned} \tag{2.3.9}$$

which is an isomorphism when (2.3.2) holds.

All the homomorphisms under consideration are functorial in P, Q, G. The homomorphisms (2.3.5), (2.3.9), and (2.2.2) form the diagram

(2.3.10)



where the horizontal arrow (2.3.9) is an isomorphism if (2.3.2) holds.

Proposition 2.3.11. The diagram (2.3.10) is anticommutative.

When $\underline{\mathrm{Hom}}(Q,G)=0$, this statement therefore recovers a characterization of the homomorphism φ^n of (2.2.2) in terms of the homomorphisms (2.3.5) and (2.3.9). This is the characterization formerly used in the theory of abelian schemes, where moreover $nQ=Q$.

Proof of 2.3.11.[2] Let ξ be an element of $\mathrm{Biext}^1(P,Q;G)$, let

$$\varphi_\xi : {}_nP \otimes {}_nQ \longrightarrow G$$

be the associated pairing, and let

$$u_\xi : {}_nP \longrightarrow {}_n\underline{\mathrm{Ext}}^1(nQ,G)$$

be the image of ξ under (2.3.5). We must prove that $-u_\xi$ is obtained from φ_ξ by the arrow (2.3.9), that is, that it is the composite

$$(*) \qquad {}_nP \xrightarrow{\varphi_\xi} \underline{\mathrm{Hom}}({}_nQ,G) \xrightarrow{(2.3.8)} {}_n\underline{\mathrm{Ext}}^1(nQ,G).$$

To prove this, we may take a section x of ${}_nP$ and prove that its image under $-u_\xi$ equals its image under the composite (*). If desired, this would allow us to reduce to the case $P=(\mathbb{Z}/n\mathbb{Z})_T$, but we shall not need that reduction.

a) Choose flat resolutions $L_\bullet$ and $M_\bullet$,

$$0 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \qquad 0 \longrightarrow M_1 \longrightarrow M_0 \longrightarrow 0,$$

of P and Q, respectively. A flat resolution of P, for example, is obtained by writing P as a quotient of a flat $\mathbb{Z}_T$-module L_0 and taking L_1 to be the kernel of $L_0 \longrightarrow P$; this kernel is flat because it is torsion-free. We obtain the complex $L_\bullet \otimes M_\bullet$

$$0 \longrightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \longrightarrow 0,$$

whose sheaf $\underline{H}_1$ will serve as $\underline{\mathrm{Tor}}_1(P,Q)$.

Let L_0' (respectively, M_0') be the subsheaf of L_0 (respectively, M_0) which is the inverse image of ${}_nP$ (respectively, ${}_nQ$). Thus

$$0 \longrightarrow L_1 \longrightarrow L_0' \longrightarrow 0 \qquad \big(\text{respectively, } 0 \longrightarrow M_1 \longrightarrow M_0' \longrightarrow 0\big)$$

is a flat resolution of ${}_nP$ (respectively, of ${}_nQ$). The homomorphism

$${}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P,Q) = \underline{H}_1(L_\bullet \otimes M_\bullet)$$

may then be described, by passage to the quotient, from the homomorphism

$$L_0' \otimes M_0' \longrightarrow \underline{H}_1(L_\bullet \otimes M_\bullet)$$

given by

$$x_0 \otimes y_0 \longmapsto \text{the class of the 1-cycle } [nx_0] \otimes y_0 - x_0 \otimes [ny_0].$$

Here the brackets indicate that the element in question is regarded as a section of the subsheaf L_1 of L_0 (respectively, M_1 of M_0). This description follows from the calculation made in the proof of 2.1.10.

b) Take an injective resolution

$$C^\bullet = \big(0 \longrightarrow C^0 \xrightarrow{d^0} C^1 \longrightarrow \cdots\big)$$

2. For a more elegant proof, see 2.3.19 below.

of G, and identify

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1\Big(P \overset{\mathbb{L}}{\otimes} Q, G\Big) = H^1(\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, C^{\cdot}))\,.$$

The element ξ may therefore be represented by a pair of homomorphisms f_1, f_0, giving the anti-commutative diagram

$$\begin{array}{ccc} L_1 \otimes M_0 + L_0 \otimes M_1 & \xrightarrow{d_1} & L_0 \otimes M_0 \\ \downarrow{\scriptstyle f_1} & - & \downarrow{\scriptstyle f_0} \\ C^0 & \xrightarrow{d^0} & C^1. \end{array} \tag{2.3.12}$$

By definition, the pairing

$$\varphi_\xi : {}_nP \otimes {}_nQ \longrightarrow G$$

is obtained, by passage to the quotient, from the homomorphism

$$\varphi : L'_0 \otimes M'_0 \longrightarrow G$$

given by

$$\varphi(x_0 \otimes y_0) = f_1\big([nx_0] \otimes y_0 - x_0 \otimes [ny_0]\big) \in G = \ker(C^0 \longrightarrow C^1). \tag{2.3.13}$$

Since assertion 2.3.11 is local on T, we may suppose that the given section x of ${}_nP$ lifts to a section x_0 of L'_0. Regarding y_0 as variable, the preceding formula then defines a homomorphism

$$\varphi' : M'_0 \longrightarrow G, \qquad y_0 \longmapsto f_1\big([nx_0] \otimes y_0 - x_0 \otimes [ny_0]\big), \tag{2.3.14}$$

which passes to the quotient to give

$${}_nQ \longrightarrow G, \qquad y \longmapsto \varphi_\xi(x,y),$$

the homomorphism associated with φ_ξ and x.

c) We must take the image of this homomorphism under the connecting operator ∂ of (2.3.7), associated with the exact sequence (2.3.6). Now $nQ \simeq Q/{}_nQ$ is plainly also isomorphic to M_0/M'_0, so that we have a homomorphism of exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M'_0 & \longrightarrow & M_0 & \longrightarrow & M_0/M'_0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \wr\downarrow & & \\ 0 & \longrightarrow & {}_nQ & \longrightarrow & Q & \longrightarrow & nQ & \longrightarrow & 0 \end{array} \quad ,$$

where the last vertical arrow is an isomorphism. This shows that the desired section of $\underline{\mathrm{Ext}}^1(nQ, G)$ may also be computed as the image of φ' in (2.3.14) under the connecting operator

$$\underline{\mathrm{Hom}}(M'_0, G) \xrightarrow{\partial} \underline{\mathrm{Ext}}^1(M_0/M'_0, G),$$

or, equivalently, as the section of the second term defined by the element of global Ext^1 which is the image of φ' under the connecting operator

$$\mathrm{Hom}(M'_0, G) \longrightarrow \mathrm{Ext}^1(M_0/M'_0, G).$$

The usual construction of the connecting operator gives the following recipe. Extend the composite (2.3.13)

$$M_0' \xrightarrow{\varphi'} G \longrightarrow C^0$$

to a homomorphism

$$\psi : M_0 \longrightarrow C^0,$$

which is possible since C^0 is injective, and consider the unique homomorphism

$$\lambda : M_0/M_0' \longrightarrow C^1 \tag{2.3.15}$$

which makes the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_0' & \longrightarrow & M_0 & \longrightarrow & M_0/M_0' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \varphi'} & & \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \lambda} & & \\ 0 & \longrightarrow & G & \longrightarrow & C^0 & \longrightarrow & C^1 & & \end{array} \quad .$$

commutative. It is clear that (2.3.15) takes values in the 1-cocycles—that is, it is a 1-cocycle—and that this cocycle represents the desired element of

$$\mathrm{Ext}^1(nQ, G) = \mathrm{Ext}^1(M_0/M_0', G).$$

d) It remains to describe explicitly the section of $\underline{\mathrm{Ext}}^1(nQ, G)$ obtained from ξ and x by means of arrow (2.3.5) in diagram (2.3.10). For this purpose we may disregard the fact that the section x of P is annihilated by n. In terms of the chosen lift x_0 of x to a section of L_0, the description is as follows. This lift defines an augmentation homomorphism

$$u : M_\bullet \longrightarrow L_\bullet \otimes M_\bullet, \qquad y_0 \longmapsto x_0 \otimes y_0, \quad y_1 \longmapsto x_0 \otimes y_1.$$

The section of $\underline{\mathrm{Ext}}^1(Q, G)$ obtained from (2.3.1) is therefore the one obtained from the element ξ of $\mathrm{Ext}^1(L_\bullet \otimes M_\bullet, G)$ by composition with the preceding homomorphism. Thus it is represented by the degree-one homomorphism

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_1 & \xrightarrow{d_1} & M_0 & \longrightarrow & 0 & & \\ & & \downarrow{\scriptstyle y_1 \mapsto f_1(x_0 \otimes y_1)} & - & \downarrow{\scriptstyle y_0 \mapsto f_0(x_0 \otimes y_0)} & & & & \\ 0 & \longrightarrow & C^0 & \xrightarrow{d^0} & C^1 & & \cdots & & \end{array} \quad .$$

To restrict this class to $nQ \hookrightarrow Q$, use the isomorphism $nQ \simeq M_0/M_0'$, which gives a flat resolution

$$0 \longrightarrow M_0' \longrightarrow M_0 \longrightarrow 0$$

of nQ. The inclusion $nQ \hookrightarrow Q$ lifts to the following homomorphism of resolutions:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_0' & \longrightarrow & M_0 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle y_0' \mapsto [n y_0']} & + & \downarrow{\scriptstyle y_0 \mapsto n y_0} & & \\ 0 & \longrightarrow & M_1 & \longrightarrow & M_0 & \longrightarrow & 0 \end{array} \quad .$$

By composition, the desired section of $\underline{\mathrm{Ext}}^1(nQ, G)$ is therefore associated with the element of $\mathrm{Ext}^1(nQ, G)$ represented by the degree-one homomorphism of complexes

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_0' & \xrightarrow{d_1} & M_0 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle g_1} & & \downarrow{\scriptstyle g_0} & & \\ 0 & \longrightarrow & C^0 & \longrightarrow & C^1 & \longrightarrow & \dots \end{array} \quad ,$$

where

$$g_1(y_0') = f_1\big(x_0 \otimes [ny_0']\big), \qquad g_0(y_0) = nf_0(x_0 \otimes y_0).$$

e) Using now the fact that x is a section of ${}_nP$, we show that the preceding element of $\mathrm{Ext}^1(nQ, G)$ is the negative of the element defined by (2.3.15). For any homomorphism

$$h : M_0 \longrightarrow C^0,$$

the element of Ext^1 considered in d) is unchanged if g_1, g_0 are replaced, respectively, by

$$g_1' = g_1 + hd_1, \qquad g_0' = g_0 - d^0h.$$

We have at our disposal the homomorphism $\psi : M_0 \longrightarrow C^0$ constructed in c), which by definition satisfies

$$\psi(y_0') = f_1\big([nx_0] \otimes y_0' - x_0 \otimes [ny_0']\big)$$

for every section y_0' of M_0'. Equivalently,

$$f_1\big(x_0 \otimes [ny_0']\big)\big(= g_1(y_0')\big) = f_1\big([nx_0] \otimes y_0'\big) - \psi(y_0').$$

Define $h : M_0 \longrightarrow C^0$ by

$$h(y_0) = -f_1\big([nx_0] \otimes y_0\big) + \psi(y_0).$$

By construction, then,

$$g_1' = 0, \qquad g_0'(y_0) = nf_0(x_0 \otimes y_0) + d^0f_1\big([nx_0] \otimes y_0\big) - d^0\psi(y_0).$$

Since $d^0f_1 = -f_0d_1$, we have

$$d^0f_1\big([nx_0] \otimes y_0\big) = -f_0(nx_0 \otimes y_0),$$

and hence

$$g_1' = 0, \qquad g_0' = -d^0\psi.$$

Thus the element in question is the one defined by the homomorphism

$$M_0/M_0' \simeq nQ \longrightarrow C^1$$

induced by $-d^0\psi$ upon passage to the quotient. We therefore obtain the negative of the element defined by (2.3.15).

Remarks 2.3.16. The preceding proof actually gives a slightly more precise compatibility than the one stated in 2.3.11, obtained by choosing a section x of ${}_nP$. This section then defines a homomorphism

$${}_nQ \longrightarrow G, \qquad y \longmapsto \varphi_\xi(x, y),$$

and hence, by the connecting homomorphism associated with (2.3.6), an element of $\mathrm{Ext}^1(nQ, G)$. On the other hand, if we forget that x is annihilated by n, we consider the homomorphism $\mathbb{Z}_T \longrightarrow P$ defined by x. Its inverse image is a biextension of $(\mathbb{Z}_T, Q)$ by G, whose class may be

interpreted as an element of $\mathrm{Ext}^1(Q,G)$ and therefore induces an element of $\mathrm{Ext}^1(nQ,G)$. With these conventions, the two elements obtained in $\mathrm{Ext}^1(nQ,G)$ are opposite.

2.3.17. Let us use the explicit cocycle description of the pairing

$$\varphi_\xi^n : {}_nP \otimes {}_nQ \longrightarrow G$$

given in (2.3.12) to obtain another expression for it. First suppose that the biextension E of class ξ admits a section s over $P \times Q$. As in VII 3.8.13, take for the flat resolution of P the one with $L_0 = \mathbb{Z}[P]$ and L_1 the kernel of the canonical homomorphism $\mathbb{Z}[P] \longrightarrow P$, and define $M_\bullet$ similarly. There is then a unique biextension trivialization of the inverse image of E on (L_0, M_0) whose restriction to $P \times Q$ is given by s; denote this trivialization again by s (cf. VII 3.8.2). By VII 3.8.12, the class of ξ in

$$\mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right)$$

is represented by the cocycle with components (f,g) characterized by

$$s(x_1,y_0) = f(x_1 \otimes y_0)e_{E/Q}(y_0), \qquad s(x_0,y_1) = g(x_0 \otimes y_1)e_{E/P}(x_0).$$

If x and y are sections of ${}_nP$ and ${}_nQ$, lift them to the corresponding sections $x_0 = [x]$ and $y_0 = [y]$ of L_0 and M_0. Applying formula (2.3.13), and putting $z = s(x,y)$, gives

$$\varphi_\xi^n(x,y) = {}^nz/e_{E/Q}(y) - z^n/e_{E/P}(x). \tag{2.3.18}$$

Here the exponent n written on the left, respectively on the right, of z denotes the nth power for the left, respectively right, multiplicative structure on E. It therefore gives a section of E over $(0,y)$, respectively $(x,0)$; the meaning of the "quotient" of such an element by $e_{E/Q}(y)$, respectively $e_{E/P}(x)$, is clear.

The preceding formula (2.3.18) still makes sense whenever z is a global section of E lifting the sections x, y of ${}_nP$, ${}_nQ$. I claim that (2.3.18) is valid for every such section, without assuming that a section of E over $P \times Q$ exists. The section (x,y) of $P \times Q$ always lifts locally to a section of E, since $E \longrightarrow P \times Q$ is an epimorphism; thus the formula computes $\varphi_\xi^n(x,y)$ in every case. Moreover, if z is replaced by another section z' of E lifting (x,y), then $z' = z + g$ for a section g of G. The two terms on the right-hand side of (2.3.18) corresponding to z' are the terms corresponding to z, each increased by ng. Their difference is therefore independent of z, showing that (2.3.18) defines a well-determined homomorphism

$${}_nP \otimes {}_nQ \longrightarrow G.$$

It remains to show that this homomorphism is indeed φ_ξ^n. For this we reduce at once to the case

$$P = Q = (\mathbb{Z}/n\mathbb{Z})_T.$$

Then $P \times Q$ is of the form I_T, where $I = \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ is a finite set. In this case the torsor E under $G_{P\times Q}$—which identifies with a family of torsors E_i under G, for $i \in I$—plainly becomes trivial locally on T. We are thus reduced to the already treated case in which E has a global section over $P \times Q$.

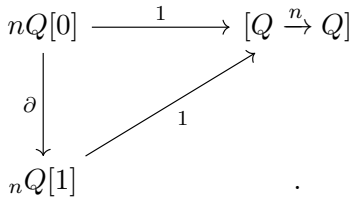
2.3.19. Another proof of 2.3.11 (by Deligne).

a) Reduction to $P = \mathbb{Z}/n\mathbb{Z}$ and to $T = (\mathrm{Sets})$.

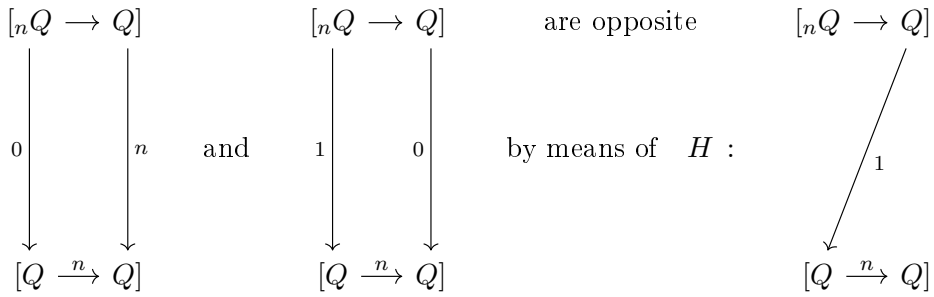
b) Let ∂ be the following arrow from nQ to ${}_nQ[1]$:

$$nQ \underset{\text{“}n\text{”}}{\overset{\sim}{\leftarrow}} [\,{}_nQ \longrightarrow Q\,] \longrightarrow {}_nQ[1].$$

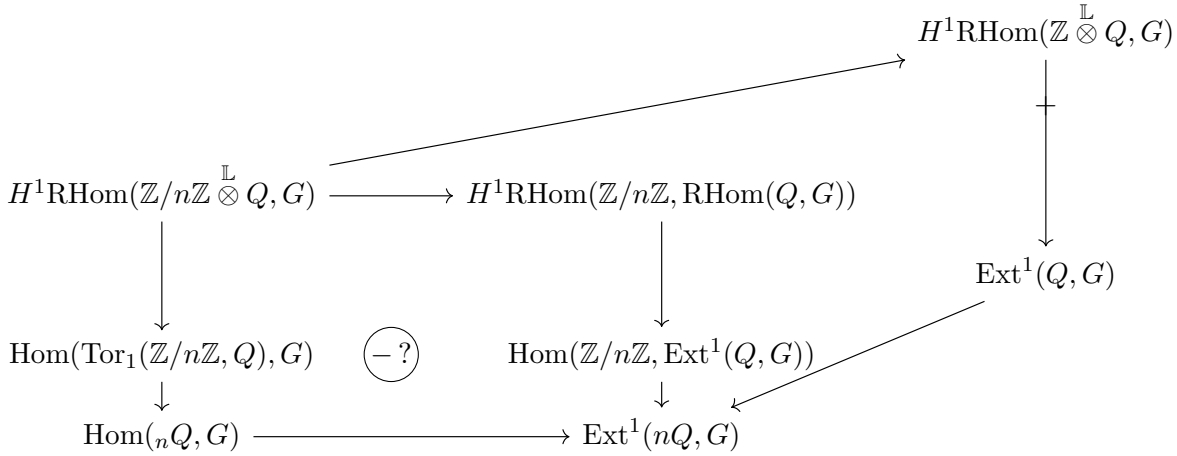
The following diagram is anticommutative:



Indeed, the arrows



c) Diagram 2.3.10 is



and:

c1) We have an isomorphism

$$\mathbb{Z}/n\mathbb{Z} \overset{\mathbb{L}}{\otimes} Q \simeq [Q \xrightarrow{n} Q],$$

and, under this isomorphism, the first composite vertical arrow is induced by

$${}_nQ[1] \longrightarrow [Q \xrightarrow{n} Q].$$

c2) The second horizontal arrow is the transpose of the arrow ∂ displayed above in b).

c3) If the diagram is traversed clockwise, it is clear that one obtains the arrow induced by

$$nQ[0] \longrightarrow [Q \xrightarrow{n} Q].$$

This proves 2.3.11.

2.4. Let

$$f : \underline{T}' \longrightarrow \underline{T}$$

be a morphism of topoi. There is then a canonical homomorphism (VII 2.8)

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Biext}^1(P',Q';G'),$$

where primes denote inverse-image groups. Since, for every commutative group E of $\underline{T}$, there is plainly a canonical isomorphism

$$({}_nE)' \simeq {}_n(E'),$$

the homomorphisms (2.2.2) over $\underline{T}$ and $\underline{T}'$ give a canonical diagram

$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \longrightarrow & \mathrm{Biext}^1(P',Q';G') \\ \downarrow{\scriptstyle \varphi^n_{P,Q}} & & \downarrow{\scriptstyle \varphi^n_{P',Q'}} \\ \mathrm{Hom}({}_nP \otimes {}_nQ, G) & \longrightarrow & \mathrm{Hom}({}_nP' \otimes {}_nQ', G'). \end{array} \tag{2.4.1}$$

Similarly, the homomorphisms (2.2.3) give an analogous diagram in terms of the T_ℓ. I claim that these diagrams are commutative. The verification is essentially immediate from the definitions and is left to the reader. More generally, notice that all the canonical homomorphisms considered in this section and the preceding one are compatible with change of topos, in the evident sense. We shall henceforth use compatibilities of this kind without further mention.

3. Biextensions of group schemes (P,Q) by $\underline{G}_m$: generalities

Throughout this section we fix a base scheme S and, as announced in Section 0, work in the fppf topos of S.

3.1. Let us first recall the most important example of a group $\mathrm{Biext}^1(P,Q;\underline{G}_m)$: the case where P and Q are two abelian schemes over S, and where Biext^1 is interpreted as the group of divisorial correspondences on (P,Q) (VII 2.9.4). The ℓ-adic pairings of the preceding section are then the familiar pairings [1] in the theory of abelian schemes.

3.2. More generally, suppose only that Q is an abelian scheme. It is well known (VII 1.3.8) that

$$\underline{\mathrm{Hom}}(Q,\underline{G}_m) = 0, \tag{3.2.1}$$

and hence (1.4) there is a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;\underline{G}_m) \simeq \mathrm{Hom}(P,Q^*), \tag{3.2.2}$$

where

$$Q^* = \underline{\mathrm{Ext}}^1(Q,\underline{G}_m) \tag{3.2.3}$$

is the dual abelian scheme of Q. (For the representability of Q^*, due to M. Raynaud in the general case, see [1].)

When P is also an abelian scheme, symmetrically to (3.2.2), there is a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;\underline{G}_m) \simeq \mathrm{Hom}(Q,P^*). \tag{3.2.4}$$

Thus the datum of a biextension ξ of (P,Q) by $\underline{G}_m$ may be interpreted equivalently as the datum of a homomorphism

$$u : P \longrightarrow Q^*$$

or of a homomorphism

$$v : Q \longrightarrow P^*.$$

Recall that a homomorphism of abelian schemes $w : A \longrightarrow B$ is an isomorphism if and only if, for every integer $n > 0$, the induced homomorphism ${}_nw : {}_nA \longrightarrow {}_nB$ is an isomorphism. In the case of u and v above, ${}_nu : {}_nP \longrightarrow {}_nQ^*$ is identified with the homomorphism ${}_nP \longrightarrow D({}_nQ)$ defined by the pairing

$$\varphi_\xi^n : {}_nP \times {}_nQ \longrightarrow \underline{G}_m$$

(see 2.3.11), whereas ${}_nv : {}_nQ \longrightarrow {}_nP^*$ is identified with the homomorphism ${}_nQ \longrightarrow D({}_nP)$ defined by the pairing

$$\psi_\xi^n : {}_nP \times {}_nQ \longrightarrow \underline{G}_m.$$

Now $\varphi_\xi^n = -\psi_\xi^n$ (2.1.10), so, up to sign, ${}_nu$ and ${}_nv$ are transposes of one another. Since ${}_nP$ and ${}_nQ$ are finite and locally free over S, and therefore satisfy Cartier duality, it follows that ${}_nu$ is an isomorphism if and only if ${}_nv$ is an isomorphism. Consequently u is an isomorphism if and only if v is an isomorphism.

An equivalent, though slightly less symmetric, formulation is the following. Consider the canonical biextension (the Weil biextension) of (P, P^*) by $\underline{G}_m$, corresponding to the identity homomorphism $v : P^* \longrightarrow P^*$. Then the corresponding canonical homomorphism

$$u : P \longrightarrow P^{**}$$

is an isomorphism. This is the biduality theorem of Barsotti and Cartier.

3.3. We now turn to the case where Q is finite locally free over S, or is a group scheme of multiplicative type over S. The key lemma in this case is the following well-known result.

Proposition 3.3.1. Let Q be a commutative group scheme over S, and suppose that Q is finite locally free over S, or that Q is of finite type and of multiplicative type over S (SGA 3 IX). Then

$$\underline{\mathrm{Ext}}^1(Q, \underline{G}_m) = 0.$$

Proof. We distinguish the two cases.

a) Suppose that Q is finite locally free over S. We must prove that every extension E of Q by $\underline{G}_m$ splits locally for the fppf topology. We may plainly suppose S affine; there is then an integer $n > 0$ such that ${}_nQ = 0$ (SGA 3 IX 6.1). Moreover, $n \cdot \mathrm{id}$ is an epimorphism of $\underline{G}_m$ for the fppf topology, and hence the Kummer exact sequence

$$0 \longrightarrow \mu_n \longrightarrow \underline{G}_m \xrightarrow{n} \underline{G}_m \longrightarrow 0$$

gives an exact sequence of the $\mathrm{Ext}^i(Q, -)$ and, in particular, an epimorphism

$$\mathrm{Ext}^1(Q, \mu_n) \longrightarrow {}_n\,\mathrm{Ext}^1(Q, \underline{G}_m) = \mathrm{Ext}^1(Q, \underline{G}_m).$$

The last equality follows because multiplication by n, being zero on Q, is also zero on $\mathrm{Ext}^1(Q, \underline{G}_m)$. Thus the extension E comes from an extension F of Q by μ_n.

The group F is represented by a finite locally free scheme over S (fpqc descent theory, SGA 1 VII). By the known properties of Cartier duality for finite locally free groups (SGA 3 VII 3),

$$D(F) = \underline{\mathrm{Hom}}(F, \underline{G}_m)$$

is an extension of $D(\mu_n)$ by $D(Q)$. Here one uses the following fact: if $A \longrightarrow B$ is a monomorphism of commutative finite locally free groups over S, then $D(B) \longrightarrow D(A)$ is faithfully flat. Since $D(B)$ and $D(A)$ are themselves finite locally free over S, the fibrewise criterion for faithful flatness reduces us to the case where S is the spectrum of a field k. In that case the assertion says that the homomorphism of affine coordinate rings

$$k(D(A)) \longrightarrow k(D(B))$$

is injective (SGA 3 VI$_B$ 11.14). This homomorphism is the transpose of $k(B) \longrightarrow k(A)$, which is surjective because $A \longrightarrow B$ is a closed immersion (SGA 3 VI$_B$ 1.4.2). Hence $k(D(A)) \longrightarrow k(D(B))$ is indeed injective.

The inclusion $\mu_n \longrightarrow \underline{G}_m$ identifies with a section of

$$D(\mu_n) = \underline{\mathrm{Hom}}(\mu_n, \underline{G}_m),$$

and it remains only to prove that, locally for the fppf topology on S, this inclusion extends to a homomorphism $F \longrightarrow \underline{G}_m$; equivalently, that the section of $D(\mu_n)$ just considered lifts fppf-locally to a section of $D(F)$. This is clear, since $D(F) \longrightarrow D(\mu_n)$ is faithfully flat and of finite presentation.

b) By SGA 3 X 4.5, after replacing S by an ètale cover, we may suppose that Q is diagonalizable, say $Q = D(M)$, where M is a finitely generated $\mathbb{Z}$-module. Let T be the torsion submodule of M, and put $L = M/T$, so that $L \simeq \mathbb{Z}^r$. The exact sequence

$$0 \longrightarrow T \longrightarrow M \longrightarrow L \longrightarrow 0$$

then gives an exact sequence (SGA 3 IX 3.1)

$$0 \longrightarrow D(L) \longrightarrow D(M) \longrightarrow D(T) \longrightarrow 0.$$

It is therefore enough to prove separately that the $\underline{\mathrm{Ext}}^1$ of $D(L)$ and of $D(T)$ by $\underline{G}_m$ vanish. The case of $D(T)$ follows from a), and the case of $D(L)$ reduces to $L = \mathbb{Z}$. Thus it remains to check that

$$\underline{\mathrm{Ext}}^1(\underline{G}_m, \underline{G}_m) = 0,$$

that is, that every extension E of $\underline{G}_m$ by $\underline{G}_m$ is fppf-locally trivial.

Such an extension is represented by an affine flat group scheme over S (fpqc descent theory, SGA 1 VII) whose fibres are tori; it is therefore a torus (SGA 3 X 4.9). By SGA 3 X 4.5 we may suppose that E is a split torus

$$E = D(R) \simeq \underline{G}_m^2.$$

After a Zariski localization, we may also suppose that the monomorphism $\underline{G}_m \hookrightarrow E = D(R)$ comes from a constant morphism $R \longrightarrow \mathbb{Z}_S$ (SGA 3 VIII 1.4), which is necessarily an epimorphism (SGA 3 VIII 3.2 b)). Choosing $r \in R$ above $1 \in \mathbb{Z}$ gives a homomorphism $E \longrightarrow \underline{G}_m$ splitting the extension, and proves the assertion.

Corollary 3.3.2. Let Q be as in 3.3.1. Then, for every P, there is a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; \underline{G}_m) \xleftarrow{\sim} \mathrm{Ext}^1(P, D(Q)), \tag{3.3.2.1}$$

where $D(Q)$ denotes the Cartier dual $\underline{\mathrm{Hom}}(Q, \underline{G}_m)$ of Q. More precisely, the canonical functor (1.1.6) is an equivalence from the category of extensions of P by $D(Q)$ to the category of biextensions of (P, Q) by $\underline{G}_m$.

This is the special case of 1.5 obtained by applying 3.3.1.

Remarks 3.3.3. Under the hypotheses of 3.3.1, $D(Q)$ is represented by a separated group scheme, locally of finite presentation and locally quasi-finite over S. More precisely, if Q is finite locally free over S, then so is $D(Q)$, as is well known (SGA 3 VII 3). If Q is of multiplicative type, then $D(Q)$ is a finitely generated twisted constant group (SGA 3 X 5.1), as was seen in SGA 3 IX 4.5 and X 5.7. If P is a group scheme, every extension of P by $D(Q)$ is therefore representable, by a familiar lemma from fpqc descent theory (SGA 3 X 5.4).

3.3.4. It is false that $\underline{\mathrm{Ext}}^1(Q, \underline{G}_m) = 0$ for every affine flat group scheme Q of finite presentation over S. This already fails for

$$Q = \underline{G}_{a_S}, \qquad S = \mathrm{Spec}\, k[t]/(t^2),$$

where k is a field of characteristic $p > 0$.

Proposition 3.4. Let P be a smooth group scheme of finite presentation over S, with connected fibres, and let Q be a torus over S. Then the category of biextensions of (P, Q) by $\underline{G}_m$ is equivalent to the one-point category; that is,

$$\text{Biext}^0(P, Q; \underline{G}_m) = \text{Biext}^1(P, Q; \underline{G}_m) = 0. \tag{3.4.1}$$

Proof. Put $M = D(Q)$. This is a twisted constant group over S whose fibres are isomorphic to groups $\mathbb{Z}^r$ (3.3.3). By 3.3.2, the category in question is equivalent to the category of extensions of P by M. This category is rigid: the automorphism group of any object is the trivial group. This is the first equality in (3.4.1), which may also be written

$$\text{Hom}(P, M) = 0.$$

It follows at once from the fact that M is etale over S, whereas P has connected fibres. It remains to prove

$$\text{Ext}^1(P, M) = 0,$$

that is, that every extension E of P by M is trivial.

By the uniqueness of a splitting when one exists, it is enough to prove that E is locally trivial. We may therefore reduce to the case $M = \mathbb{Z}_S^r$, and then to $M = \mathbb{Z}_S$. When S is the spectrum of a field, the conclusion follows from 5.5 (i) below. (Notice that we shall not use 3.4 before IX 6, so in particular there is no vicious circle.) We now reduce the general case to this special case.

It is enough to find a section f of E over P such that $f(0) = 0$, for such a section is necessarily multiplicative. Indeed,

$$f(x + y) - f(x) - f(y) = 0 :$$

the left-hand side defines a morphism $P \times_S P \longrightarrow \mathbb{Z}_S$ which vanishes on the zero section of $P \times_S P$. It therefore vanishes identically, since $P \times_S P$ has connected fibres (the geometric fibres of P being connected). The required conclusion now follows from the following lemma.

Lemma 3.4.2. Let S be a scheme, P a smooth S-scheme of finite presentation with geometrically connected fibres, Z a group, and E a Z_P-torsor (for the fppf, etale, or fpqc topology; these notions are equivalent here, cf. SGA 3 X 5.4). Let e be a section of P over S, and let e' be a section of E over S above e (that is, a section of $e^{-1}(E)$ over S).

The torsor E is trivial if and only if, for every $s \in S$, the induced torsor E_s over P_s is trivial. In that case there is a unique section f of E over P such that $f \circ e = e'$.

Proof. Uniqueness is clear. Two such sections f and g "differ" by a morphism

$$h : P \longrightarrow Z_S, \qquad h \circ e = 1,$$

and such a morphism is constantly equal to 1, since P has connected fibres. Moreover, if a section f of E over P exists, it can always be corrected by a morphism $P \longrightarrow Z_S$ coming from a morphism $z : S \longrightarrow Z_S$, so as to obtain a section f' satisfying $f' \circ e = e'$. One need only choose z so that

$$f(e(t)) = e'(t)z(t).$$

Applying this observation fibrewise, for every $s \in S$ we obtain a well-determined section f_s of E_s over P_s taking e_s to e'_s. It remains to prove that the f_s arise from a section f of E over P. Equivalently (EGA IV 17.9.3), we must prove that

$$U = \bigcup_{s \in S} f_s(P_s) \subset E$$

is open in E.

Suppose first that S is noetherian. Then U is constructible. To see this, for each $s \in S$ let T be the closure of s, endowed with its reduced induced structure. There is an open neighbourhood T' of s in T and a section of $E_{T'}$ over $P_{T'}$ extending f_s and compatible with the marked sections e and e' (as follows directly from EGA IV 8.8.2). By the criterion of EGA IV 1.10.1, it remains to show that U is stable under generization. The usual argument using EGA II 7.1.9 reduces this to the case in which S is the spectrum of a discrete valuation ring.

Since P is smooth over S, it is regular and hence normal; and since it has connected fibres, it is connected. The given section of E over the generic fibre of P over S therefore extends uniquely to a section f of E over P (EGA IV 18.10.19 and 18.10.20), necessarily compatible with e and e'. Thus in this case $U = f(P)$ is open. This proves 3.4 when S is noetherian. The general case reduces readily to this one by the usual limit arguments of EGA IV, 8 and 9; we leave the details to the reader.

Remark 3.4.3. Lemma 3.4.2 can be "crushed" under learned references, namely SGA 4 XV 1.15 (the case $n = 1$) and 4.1.

Corollary 3.5. Let P be a smooth group scheme of finite presentation over S, with connected fibres, and let

$$0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0$$

be an exact sequence of commutative groups over S, with Q' a torus. Then the inverse-image functor on biextensions

$$\mathrm{BIEXT}(P, Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{BIEXT}(P, Q; \underline{G}_m) \tag{3.5.1}$$

is an equivalence of categories. In particular, the following natural homomorphisms are isomorphisms:

$$\mathrm{Biext}^i(P, Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^i(P, Q; \underline{G}_m), \qquad i = 0, 1. \tag{3.5.2}$$

Indeed, up to equivalence, $\mathrm{BIEXT}(P, Q''; \underline{G}_m)$ identifies with the category of biextensions E of (P, Q) by $\underline{G}_m$ equipped with a trivialization of the induced biextension E' of (P, Q') by $\underline{G}_m$ (VII 3.7.6). By 3.4 there is exactly one such trivialization of E' for each given biextension E. One can of course also prove directly that (3.5.2) is an isomorphism for $i = 0, 1$ - which is equivalent to the conclusion of 3.5 - by using the exact sequence of the Biext^i in VII 3.7.5 associated with

$$0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0.$$

Remark 3.6. The symmetric statement obtained from 3.5 by interchanging left and right is of course also valid; it also follows formally from 3.5 by passing to symmetric biextensions (VII 2.7). Suppose, in addition to the data and hypotheses of 3.5, that the symmetric data and hypotheses are given: namely, that Q has connected fibres (equivalently, that Q'' has connected fibres, since Q' has connected fibres), and that there is an exact sequence

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0$$

with P' a torus. Successive application of 3.5 and of its symmetric form, applied to this last sequence and to Q'', shows that the functor

$$\mathrm{BIEXT}(P'', Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{BIEXT}(P, Q; \underline{G}_m) \tag{3.6.1}$$

is an equivalence of categories. In particular, the following homomorphism is an isomorphism:

$$\mathrm{Biext}^i(P'', Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^i(P, Q; \underline{G}_m), \qquad i = 0, 1. \tag{3.6.2}$$

Proposition 3.7. Let A be an abelian scheme over S, let T be a group scheme of multiplicative type (SGA 3 IX) and of finite type over S, and let

$$\underline{M} = \underline{\mathrm{Hom}}(T, \underline{G}_m)$$

be the Cartier dual of T (a finitely generated twisted constant group over S, by 3.3.3). Then the categories $\mathrm{EXT}(A,T)$ and $\mathrm{BIEXT}(A, \underline{M}; \underline{G}_m)$, formed respectively by commutative extensions of A by T and by biextensions of $(A, \underline{M})$ by $\underline{G}_m$, are rigid and canonically equivalent. Their groups of isomorphism classes are canonically isomorphic to $\mathrm{Hom}(\underline{M}, A^*)$, where

$$A^* = \underline{\mathrm{Ext}}^1(A, \underline{G}_m)$$

is the dual abelian scheme of A (3.2). In particular, there is a canonical isomorphism

$$\mathrm{Ext}^1(A,T) \xrightarrow{\sim} \mathrm{Hom}(\underline{M}, A^*). \tag{3.7.1}$$

This is the special case of 1.6 obtained by taking $P = A$ and $Q = \underline{M}$. By the biduality theorem (SGA 3 X 5.7), we do indeed have

$$T \simeq \underline{\mathrm{Hom}}(\underline{M}, \underline{G}_m),$$

and it remains only to verify the conditions (1.6.1), which here read

$$\underline{\mathrm{Hom}}(A, \underline{G}_m) = 0, \qquad \mathrm{Hom}\big(A, \underline{\mathrm{Ext}}^1(\underline{M}, \underline{G}_m)\big) = 0.$$

The first equality was recalled in (3.2.1); the second may be strengthened to

$$\underline{\mathrm{Ext}}^1(\underline{M}, \underline{G}_m) = 0.$$

This assertion is local on S, so we may suppose that $\underline{M}$ is constant, $\underline{M} = M_S$. Decomposing M as a direct sum of cyclic $\mathbb{Z}$-modules reduces us to $M = \mathbb{Z}$, which is trivial, and to $M = \mathbb{Z}/n\mathbb{Z}$, which follows from the n-divisibility of $\underline{G}_m$ (or, more learnedly, from 3.3.1).

When $T = \underline{G}_m$, and hence $\underline{M} = \mathbb{Z}_S$, (3.7.1) reduces to the identity isomorphism (3.2.3). Of course, a direct proof of (3.7.1) would be essentially trivial, without introducing the group

$$\mathrm{Ext}^1\big(A \overset{\mathbb{L}}{\otimes} \underline{M}, \underline{G}_m\big) = \mathrm{Biext}^1(A, \underline{M}; \underline{G}_m).$$

4. Extensions and biextensions of smooth connected group schemes over a field

We recall the following lemma due to Rosenlicht [3 bis, Proposition 3].

Lemma 4.1. Let k be a field, and let X and Y be two separable, geometrically connected k-schemes of finite type, equipped respectively with k-rational points e_X and e_Y. Then every morphism

$$X \times Y \longrightarrow \underline{G}_m$$

which is trivial on the two subschemes $X \times e_Y$ and $e_X \times Y$ is trivial.

We shall use this result only when X and Y are group schemes over k; in that case the hypotheses imply that X and Y are in fact smooth over k. By induction, 4.1 also implies the analogous result for any number of factors. We may therefore apply the results of VII 1.3 and VII 2.9, and obtain the following two propositions.

Proposition 4.2. Let P be a smooth connected group over the field k. The natural functor from the category of group extensions of P by $\underline{G}_m$ to the category of line bundles on P rigidified at the identity element e_P is fully faithful. Its essential image consists of the rigidified line bundles $\mathbb{L}$ on P for which

$$\pi^*\mathbb{L} \simeq \mathrm{pr}_1^*\,\mathbb{L} \otimes \mathrm{pr}_2^*\,\mathbb{L},$$

where $\pi : P \times P \longrightarrow P$ is the group law of P, and $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections.

Proposition 4.3. Let P and Q be two smooth connected algebraic groups over the field k. The natural functor from the category of biextensions of (P, Q) by $\underline{G}_m$ to the category of line bundles on $P \times Q$ birigidified with respect to the identity sections e_P and e_Q of P and Q is fully faithful; moreover, both categories are rigid. A birigidified line bundle $\mathbb{L}$ on $P \times Q$ comes from a biextension if and only if its pullbacks to $P \times P \times Q$ and $P \times Q \times Q$ by $\pi_P \times \mathrm{id}_Q$ and $\mathrm{id}_P \times \pi_Q$ satisfy the "theorem of the cube" ([1] or [2]).

This last condition is simply the translation of the conditions in VII 2.9.3 b), in the explicit form given in VII 3.5 b).

4.4. I do not know whether the cube condition in 4.3 is automatically satisfied. At least it is known [2, IV 2.6] that, for a given $\mathbb{L}$, there is always an integer $n \geq 1$, a power of the characteristic exponent of k, such that $\mathbb{L}^{\otimes n}$ satisfies that condition and hence comes from a biextension of (P, Q) by $\underline{G}_m$. It is also always possible to find a purely inseparable extension k' of k such that the required condition becomes true after the base extension $k \longrightarrow k'$; after this extension, therefore, $\mathbb{L}$ is a birigidified line bundle arising from a biextension of (P, Q) by $\underline{G}_m$.

In particular, when k is perfect, the condition in 4.3 is always satisfied. This last fact, which formally implies the preceding statements by a trace argument, can also be seen by applying 4.7 below; cf. 4.9.

We recall, in the form of a lemma, the following result from EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.13.

Lemma 4.5. Let $f : X \longrightarrow Y$ be a smooth surjective morphism with integral fibres, where Y is normal and integral with generic point y. Let $\mathbb{L}$ be a line bundle on X whose restriction to X_y is trivial. Then there is a line bundle $\mathbb{M}$ on Y such that

$$\mathbb{L} \simeq f^*\mathbb{M}.$$

Proposition 4.6. Let P, Q be two smooth connected algebraic groups over the field k, with P affine. Suppose in addition either that P admits a composition series each of whose factors is a torus or is isomorphic to the additive group $\underline{G}_a$ (a condition satisfied in particular when k is perfect [SGA 3, XVII 7.2.1, 4.1.1 (i)$\Longleftrightarrow$ (ii), and 4.1.5]), or that Q is an extension of an abelian variety B by a smooth connected affine algebraic group satisfying the preceding condition. Under either hypothesis, every birigidified line bundle $\mathbb{L}$ on $P \times Q$ is trivial; a fortiori, by 4.3, every biextension of (P, Q) by $\underline{G}_m$ is trivial.

Proof. First observe that every automorphism of a birigidified line bundle is the identity (4.3). Descent theory therefore shows that it is enough to prove that $\mathbb{L}$ becomes trivial after a finite separable extension of k. Passing to the limit immediately reduces us to the case where k is separably closed.

Every torus over such a field is split. Hence, if P admits a composition series as in 4.6, it also admits one whose quotients are isomorphic to $\underline{G}_a$ or $\underline{G}_m$. Repeated application of 4.5, together with

$$\mathrm{Pic}(\underline{G}_{a_K}) = \mathrm{Pic}(\underline{G}_{m_K}) = 0$$

for every field K, gives

$$\mathrm{Pic}(P) = 0.$$

This equality remains true after every extension of the base field k. Applying 4.5 to

$$\mathrm{pr}_2 : P \times Q \longrightarrow Q$$

and to $\mathbb{L}$, we find a line bundle $\mathbb{M}$ on Q such that $\mathbb{L} \simeq \mathrm{pr}_2^*\mathbb{M}$. Since the restriction of $\mathbb{L}$ to $e_P \times Q$ is trivial, $\mathbb{M}$ is trivial, and so is $\mathbb{L}$. This proves the first case.

In the second case, let Q be an extension of an abelian scheme B by a group M admitting a composition series with quotients isomorphic to $\underline{G}_m$ or $\underline{G}_a$. Apply 4.5 to

$$f : X = P \times Q \longrightarrow Y = P \times B.$$

The fibre over the generic point y of $P\times B$ is an M_y-torsor. It is necessarily trivial by the particular structure of M and the classical equalities

$$H^1(K,\underline{G}_a)=0,\qquad H^1(K,\underline{G}_m)=0,$$

valid for every field K, here taken to be $k(y)$. Since $\mathrm{Pic}(M)=0$, we have $\mathrm{Pic}(X_y)=0$; thus 4.5 applies and shows that $\mathbb{L}$ is the pullback of a line bundle $\mathbb{N}$ on $Y=P\times B$.

Let $\mathbb{N}_1$ and $\mathbb{N}_2$ be the restrictions of $\mathbb{N}$ to $P\times e_B$ and $e_P\times B$, respectively. Replacing $\mathbb{N}$ by

$$\mathbb{N}\otimes \mathrm{pr}_1^*(\mathbb{N}_1)^{-1}\otimes \mathrm{pr}_2^*(\mathbb{N}_2)^{-1},$$

we may suppose that $\mathbb{N}$ is birigidified and that $\mathbb{L}$ is its birigidified pullback. We are thereby reduced to the case in which Q itself is an abelian scheme.

Then $\mathbb{L}$ defines a canonical morphism

$$(*)\qquad P\longrightarrow \underline{\mathrm{Pic}}_{Q/k}$$

taking the identity to the identity. Since P is connected, this morphism factors through the dual abelian scheme

$$Q'=\underline{\mathrm{Pic}}^0_{Q/k}.$$

It is well known that every morphism of schemes from a smooth connected affine algebraic group to an abelian scheme is constant. This is a geometric assertion, so we may suppose k algebraically closed; a devissage then reduces it immediately to $P=\underline{G}_a$ or $P=\underline{G}_m$, and every rational map from $\mathbb{P}^1_k$ to an abelian scheme over k is constant.

The constancy, hence vanishing, of (*) means that $\mathbb{L}$ is the pullback of a line bundle $\mathbb{M}$ on P. Since $\mathbb{L}$ is trivial on $P\times e_Q$, $\mathbb{M}$ is trivial, and therefore so is $\mathbb{L}$. This completes the proof.

Proposition 4.7. Let P and Q be two smooth connected algebraic groups over the field k. Suppose that P is an extension of an abelian scheme A by a smooth connected affine algebraic group L admitting a composition series each of whose factors is isomorphic to a torus or to $\underline{G}_a$ (a condition always satisfied when k is perfect). Then the functor considered in 1.3 is an equivalence of categories; that is, every birigidified line bundle $\mathbb{L}$ on $P\times Q$ comes from a biextension of (P,Q). Moreover, the functor between the rigid categories

$$\mathrm{BIEXT}(A,Q;\underline{G}_m)\longrightarrow \mathrm{BIEXT}(P,Q;\underline{G}_m)$$

is an equivalence of categories; in other words, it induces an isomorphism of groups

$$(4.7.1)\qquad \mathrm{Biext}^1(A,Q;\underline{G}_m)\xrightarrow{\sim}\mathrm{Biext}^1(P,Q;\underline{G}_m).$$

For the first assertion, descent and 4.3 again reduce us to the case where k is separably closed. Proceeding as in the proof of 4.6, we deduce from 4.5 and the hypothesis on L that $\mathbb{L}$ is isomorphic to the pullback of a birigidified line bundle $\mathbb{M}$ on $A\times Q$. For the first assertion of 4.7 we are thus reduced to the case $A=P$. The line bundle $\mathbb{L}$ then defines a morphism of pointed schemes

$$f:Q\longrightarrow \underline{\mathrm{Pic}}_{P/k},$$

which necessarily factors through the dual abelian scheme P' of P. Let $\mathbb{L}'$ be the pullback of the Weil sheaf on $P\times P'$ along

$$\mathrm{id}_P\times f: P\times Q\longrightarrow P\times P'.$$

This is a birigidified line bundle which, by the definition of f, differs from $\mathbb{L}$ only by the pullback of a line bundle $\mathbb{M}$ on P. Since both $\mathbb{L}'$ and $\mathbb{L}$ have trivial restriction to $P\times e_Q$, it follows that $\mathbb{M}$ is trivial and hence that $\mathbb{L}\simeq\mathbb{L}'$. We noted in VII 2.9.4 that divisorial correspondences on a product of two abelian schemes arise from biextensions. Consequently $\mathbb{L}'$, and therefore $\mathbb{L}$, comes from a biextension.

Returning to the general case, where P is an extension of A by L, it remains to prove that (4.7.1) is an isomorphism. This follows from the exact sequence of the Biext^i, for $i = 0, 1$, associated with

$$0 \longrightarrow L \longrightarrow P \longrightarrow A \longrightarrow 0,$$

and from the relations

$$\mathrm{Biext}^0(L, Q; \underline{G}_m) = 0, \qquad \mathrm{Biext}^1(L, Q; \underline{G}_m) = 0.$$

The first is a special case of the rigidity assertion in 4.3, and the second is precisely the first case of 4.6.

Corollary 4.8. Under the hypotheses of 4.7, suppose that Q is an extension of an algebraic group B (necessarily smooth and connected) by a smooth connected affine algebraic group M. Then the functor between the rigid categories

$$\mathrm{BIEXT}(A, B; \underline{G}_m) \longrightarrow \mathrm{BIEXT}(P, Q; \underline{G}_m)$$

is an equivalence of categories; in other words, it induces an isomorphism of groups

$$\mathrm{Biext}^1(A, B; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(P, Q; \underline{G}_m). \tag{4.8.1}$$

In view of (4.7.1), it remains to prove that

$$\mathrm{Biext}^1(A, B; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(A, Q; \underline{G}_m)$$

is an isomorphism. As in the proof of (4.7.1), this follows from the exact sequence of the Biext^i associated with

$$0 \longrightarrow M \longrightarrow Q \longrightarrow B \longrightarrow 0,$$

together with the relations

$$\mathrm{Biext}^0(P, M; \underline{G}_m) = 0, \qquad \mathrm{Biext}^1(P, M; \underline{G}_m) = 0.$$

Indeed, (4.7.1), applied with M in place of Q, identifies these groups with the corresponding groups having A in the first variable, as required by the preceding exact sequence. The second vanishing displayed above is a special case of the second case of 4.6, after interchanging the two factors.

Remark 4.9. Let P and Q be two smooth connected algebraic groups over a perfect field k. It is then known [3] that each is an extension of an abelian scheme by a smooth connected affine group, the latter being the product of a torus with an iterated extension of copies of $\underline{G}_a$ (cf. the reference given in 4.6):

$$0 \longrightarrow L \longrightarrow P \longrightarrow A \longrightarrow 0, \qquad 0 \longrightarrow M \longrightarrow Q \longrightarrow B \longrightarrow 0.$$

Thus 4.8 applies and shows that the theory of biextensions of (P, Q) by $\underline{G}_m$ reduces to the analogous (and much more classical!) theory for the pair (A, B) of abelian varieties.

Corollary 4.10. Let P and Q be two smooth connected algebraic groups over k, let L be a smooth connected affine algebraic subgroup of P, and let E be a biextension of (P, Q) by $\underline{G}_m$. By 2.2, E induces a pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\underline{G}_m).$$

Then the induced pairing

$$T_\ell(L) \times T_\ell(Q) \longrightarrow T_\ell(\underline{G}_m)$$

is zero.

Indeed, we may plainly suppose that k is algebraically closed. We are then under the hypotheses of 4.6, which imply that the restriction of E to (L, Q) is the trivial biextension. The asserted conclusion follows, since the induced pairing in question is the pairing defined by this restriction.

4.11. Let P and Q be algebraic groups smooth over k, let E be a biextension of (P, Q) by $\underline{G}_m$, and let T be the largest subtorus of P. We shall determine the left annihilator of the pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\underline{G}_{m,k}), \tag{4.11.1}$$

where ℓ is again a fixed prime number. By 4.10 this annihilator always contains the subgroup $T_\ell(T)$ of $T_\ell(P)$.

Choose an extension k' of k over which P and Q can each be written as an extension of an abelian scheme by an affine group scheme that is an iterated extension of tori and copies of $\underline{G}_a$. If P and Q have unipotent rank zero, we may take $k' = k$; in general, we may take for k' the perfect closure of k (or a finite radicial extension). Let A' and B' be the abelian parts of $P_{k'}$ and $Q_{k'}$, respectively. By 4.7, the biextension $E_{k'}$ of $(P_{k'}, Q_{k'})$ comes from a biextension F' of (A', B'), uniquely determined up to unique isomorphism. Consequently the pairing

$$T_\ell(P_{k'}) \times T_\ell(Q_{k'}) \longrightarrow T_\ell(\underline{G}_{m,k'}), \tag{4.11.2}$$

obtained from (4.11.1) by the base-field extension $k \to k'$, and also defined by the biextension $E_{k'}$, is obtained from the pairing

$$T_\ell(A') \times T_\ell(B') \longrightarrow T_\ell(\underline{G}_{m,k'}) \tag{4.11.3}$$

defined by F', by composing with the canonical homomorphisms (which are in fact epimorphisms if P and Q are ℓ-divisible)

$$T_\ell(P_{k'}) \longrightarrow T_\ell(A'), \qquad T_\ell(Q_{k'}) \longrightarrow T_\ell(B'). \tag{4.11.4}$$

It follows that the left annihilator of (4.11.2), which is obtained from the left annihilator of (4.11.1) by base change, is the inverse image under the first homomorphism in (4.11.4) of the left annihilator of (4.11.3). Let C' be the largest abelian subscheme of A' such that the restriction of F' to (C', B') is the trivial biextension. Equivalently, C' is the reduced identity component of the kernel of the homomorphism

$$A' \longrightarrow (B')^* \tag{4.11.5}$$

defined by F', where $(B')^*$ is the dual abelian scheme of B'. The left annihilator of (4.11.3) identifies with $T_\ell(C')$; therefore the left annihilator of (4.11.1) is precisely the inverse image of

$$T_\ell(C') \hookrightarrow T_\ell(A')$$

under the first morphism in (4.11.4).

In particular, to say that the left annihilator of (4.11.1) is exactly $T_\ell(T)$ – equivalently, that the left annihilator of (4.11.2) is the kernel $T_\ell(T')$ of the first morphism in (4.11.4) – is to say that (4.11.3) is nondegenerate on the left, that is, that $C' = 0$, or equivalently that (4.11.5) has finite kernel. In this last form the condition is visibly independent of ℓ, and in any of its forms it is plainly independent of the choice of k'. When it holds, we shall simply say that the given biextension E of (P, Q) by $\underline{G}_m$ is nondegenerate on the left. By duality, this also means that the homomorphism

$$B' \longrightarrow (A')^* \tag{4.11.6}$$

defined by F' is an epimorphism, that is, is surjective.

The notion of a biextension nondegenerate on the right is defined symmetrically: (4.11.6) must have finite kernel, or, equivalently, (4.11.5) must be an epimorphism. The biextension E of (P, Q) by $\underline{G}_{m,k}$ is called nondegenerate if it is nondegenerate on both the left and the right. This is equivalent to saying that (4.11.5), or equivalently (4.11.6), is an isogeny.

It is immediate from the definition that these notions are invariant under every extension of the base field k.

4.12. Let P be a smooth connected group scheme, let $\mathbb{L}$ be a line bundle on P, and put

$$\delta(\mathbb{L}) = \pi^*\mathbb{L} \otimes \mathrm{pr}_1^*(\mathbb{L})^{-1} \otimes \mathrm{pr}_2^*(\mathbb{L})^{-1}, \tag{4.12.1}$$

with the notation of 4.2. As already observed in VII 1.3.4, this is a birigidified line bundle on $P \times P$. By 4.3, if $\delta(\mathbb{L})$ comes from a biextension, that biextension is determined up to unique isomorphism. In particular, by 4.7, it is defined when P is an extension of an abelian scheme A by an iterated extension of tori and copies of $\underline{G}_a$. Whenever it is defined, this biextension is plainly symmetric.

In the preceding favourable case, and assuming that the maximal torus of P is split, the construction of $\delta(\mathbb{L})$ can be reduced to the case where P is an abelian scheme. Indeed, the construction is functorial in P, while the homomorphism

$$\mathrm{Pic}(A) \longrightarrow \mathrm{Pic}(P) \tag{4.12.2}$$

induced by the canonical quotient homomorphism $f : P \longrightarrow A$ is surjective by 4.5 (cf. the beginning of the proof of 4.6). We can therefore find an isomorphism

$$\mathbb{L} \simeq f^*\mathbb{M},$$

where $\mathbb{M}$ is a line bundle on A. Then $\delta(\mathbb{L})$ is the pullback of the biextension $\delta(\mathbb{M})$ of (A, A) by $\underline{G}_{m,k}$.

Proposition 4.13. Under the hypotheses of 4.12, suppose that $\mathbb{L}$ is an ample line bundle on P. If the biextension $\delta(\mathbb{L})$ of (P, P) by $\underline{G}_{m,k}$ is defined, then it is nondegenerate in the sense of 4.11. If P is an extension of an abelian scheme A by a smooth connected affine group L satisfying the condition of 4.7, then the biextension of (A, A) from which $\delta(\mathbb{L})$ comes by 4.8 is of the form $\delta(\mathbb{M})$, where $\mathbb{M}$ is an ample line bundle on A.

We reduce to the case where k is algebraically closed. By the definition, it is then enough, with the notation of 4.12, to prove that the biextension $\delta(\mathbb{M})$ of (A, A) is nondegenerate. M. Raynaud proves in his thesis [2, XI 1.11] that if $\mathbb{L} = f^*\mathbb{M}$ is ample, then $\mathbb{M}$ is ample as well. The theory of A. Weil then gives the familiar conclusion that $\delta(\mathbb{M})$ is nondegenerate [1, 2], that is, that the homomorphism from A to its dual abelian scheme is an isogeny.

Remark 4.14. Suppose we are in the favourable case where P is an extension of an abelian scheme A by a group L which is an iterated extension of tori and copies of $\underline{G}_a$, and consider the canonical composite

$$\mathrm{NS}(A) \xrightarrow{\delta} \mathrm{Biext}^1(A, A; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(P, P; \underline{G}_m). \tag{4.14.1}$$

By 4.8 the second arrow is bijective. On the other hand, the theory of abelian varieties shows that the first arrow identifies the Neron–Severi group $\mathrm{NS}(A)$ of A with the subgroup of

$$\mathrm{Biext}^1(A, A; \underline{G}_m) = \mathrm{Corr}(A, A)$$

consisting of the classes of symmetric biextensions, or equivalently symmetric divisorial correspondences. Thus (4.14.1) identifies $\mathrm{NS}(A)$ with the group of symmetric biextensions of (P, P) by $\underline{G}_m$. By 4.7 this group can equally be described as the group of classes of symmetric divisorial correspondences on (P, P).

It is natural to call a divisorial correspondence on (P,P) ample, and likewise to call the corresponding biextension of (P,P) by $\underline{G}_m$ ample, when it comes from an ample element of $\mathrm{NS}(A)$. This condition is plainly invariant under extension of the base field; the notion therefore still makes sense without the preceding restrictive hypothesis on P, assuming only that P is smooth and connected.

Suppose further that L is "split solvable over k", in the sense that it admits a devissage by copies of $\underline{G}_a$ and $\underline{G}_m$. The preceding observation and the surjectivity of (4.12.2) show that the image of

$$\mathrm{Pic}(P) \xrightarrow{\delta} \mathrm{Biext}^1(P,P;\underline{G}_m) \tag{4.14.2}$$

consists of the symmetric biextensions, and that its kernel is the subgroup $\mathrm{Pic}^0(P)$ of $\mathrm{Pic}(P)$ which is the image of $\mathrm{Pic}^0(A)$. Consequently, if we put

$$\mathrm{NS}(P) = \mathrm{Pic}(P)/\mathrm{Pic}^0(P) \simeq \mathrm{NS}(A),$$

we may also interpret (4.14.1) as a canonical isomorphism

$$\mathrm{NS}(P) \xrightarrow{\sim} \mathrm{Biext}^1(P,P;\underline{G}_m)^{\mathrm{sym}}. \tag{4.14.3}$$

Finally, the theorem of Raynaud cited in 4.13 says that an element ξ of $\mathrm{Pic}(P)$ is ample if and only if its image $\delta(\xi)$ under (4.14.2) is the class of an ample biextension.

5. Extensions and biextensions by torsion-free truncated constant group schemes.

In this section and the next we give some auxiliary results that will be used in section 7 in the case $Z = \mathbb{Z}$.

Proposition 5.1. Let S be an irreducible geometrically unibranch scheme with generic point η, and let Z be a torsion-free group, hence giving a constant group scheme Z_S over S. Then every torsor E under Z_S is trivial, and the functor

$$E \longmapsto E(\eta)$$

from the category of torsors under Z_S to the category of torsors under $Z = Z_S(\eta)$ is an equivalence of categories.

A familiar lemma of descent theory (SGA 3 X 5.4) shows that every torsor E under Z_S is representable. In fact, that result applies to every group scheme over S which is separated, locally of finite presentation, and locally quasi-finite over S. Since Z_S is etale over S, fpqc descent shows that the same is true of every torsor E under Z_S; such a torsor is therefore locally trivial for the etale topology (and not merely for the fppf topology). To prove that E is trivial, it is consequently enough to prove

$$H^1(S_{\text{ét}}, Z_S) = e.$$

Likewise, the full-faithfulness assertion of 5.1 reduces to the relation

$$Z \xrightarrow{\sim} H^0(S_{\text{ét}}, Z_S).$$

The latter is immediate. To prove the former, consider the inclusion

$$i : \eta \longrightarrow S.$$

Because S is geometrically unibranch, we have

$$Z_S \xrightarrow{\sim} i_*(Z_\eta)$$

(SGA 4 IX 2.14.1). The noncommutative low-degree exact sequence that replaces the Leray spectral sequence for $i : \eta \to S$ (SGA 4 XII 3.2) therefore begins

$$e \longrightarrow H^1(S, i_*(Z_\eta)) \longrightarrow H^1(\eta, Z_\eta) \longrightarrow \cdots,$$

and shows that

$$H^1(S, Z_S) \longrightarrow H^1(\eta, Z_\eta)$$

is injective. We are thus reduced to the case in which S is the spectrum of a field k. If π is the Galois group of a separable closure of k, then

$$H^1(k, Z) = \mathrm{Hom}_{\mathrm{cont}}(\pi, Z)/Z = e,$$

because Z is torsion-free. This completes the proof.

Corollary 5.2. With the notation of 5.1, let P be a smooth scheme over S. Then every torsor E under Z_P is trivial, and the restriction functor

$$E \longmapsto E_\eta$$

from the category of torsors on P under Z_P to the category of torsors on P_η under Z_{P_η} is an equivalence of categories.

Indeed, P is geometrically unibranch (EGA IV 17.5.7). Moreover, every irreducible component of P dominates S, because P is flat over S, and the set of irreducible components of P is locally finite. Thus P is the disjoint sum of irreducible geometrically unibranch schemes P_i. The assertion now follows immediately from 5.1 applied to the various P_i and their generic fibres $P_{i\eta}$, noting that the generic point p_i of P_i is also the generic point of $P_{i\eta}$.

5.2.1. The preceding corollary immediately gives the analogous statement for bitorsors on P under (Z_P, Z_P): restriction to the generic fibre induces an equivalence from their category to the category of bitorsors on P_η under (Z_{P_η}, Z_{P_η}). Indeed, it is enough to express the bitorsor structure on E in terms of its structure as a right torsor under Z_P, together with an isomorphism from Z_P to the commutant of Z_P for its action on E. Since E is necessarily trivial, that commutant is itself isomorphic to Z_P. We are thus reduced, on each irreducible component X of P, to the following immediate observation: if X is a nonempty connected scheme, the functor $Z \mapsto Z_X$ from sets to X-schemes is fully faithful.

Proposition 5.3. Let S and Z be as in 5.1, and let P be a smooth group scheme over S. Consider extensions E of group schemes of P by Z_S. Then restriction to the generic fibre

$$E \longmapsto E_\eta$$

is an equivalence from the category of these extensions to the category of extensions of P_η by Z_η. If P and Z are commutative, it also induces an equivalence between the corresponding categories of commutative extensions.

First observe that every extension of sheaves of groups of P by Z_S (and similarly of P_η by Z_η) gives, by 5.2, a trivial torsor, hence in particular a representable one. The categories in 5.3 may therefore equally well be interpreted as categories of extensions of sheaves of groups. The description in VII 1.1.6 and 1.2 expresses these in terms of bitorsors on P, $P \times P$, and $P \times P \times P$ (respectively on P_η, $P_\eta \times P_\eta$, and $P_\eta \times P_\eta \times P_\eta$), under the pairs of groups obtained from (Z, Z) by base change. Applying 5.2 in the commutative case and 5.2.1 in the general case to the smooth S-schemes P, $P \times P$, and $P \times P \times P$ gives the asserted conclusion. The same argument gives the following result.

Corollary 5.4. Let S, Z be as in 5.1, and let P, Q be two smooth group schemes over S. Suppose that P, Q, and Z are commutative, and consider biextensions E of (P, Q) by Z_S. Then restriction to the generic fibre

$$E \longmapsto E_\eta$$

is an equivalence from the category of these biextensions to the category of biextensions of (P_η, Q_η) by Z_η.

The preceding results reduce the determination of categories of extensions or biextensions by a torsion-free constant group to the case in which the base scheme is the spectrum of a field. In that case we obtain the following.

Proposition 5.5. Let k be a field, Z a torsion-free group, P a k-group scheme locally of finite type, P° its identity component, and $\Phi = P/P^\circ$ its group of connected components (SGA 3 VI$_A$ 4). Then:

(i) The natural functor from the category of group schemes which are extensions (respectively commutative extensions) of Φ by Z_k to the category of group schemes which are extensions (respectively commutative extensions) of P by Z_k is an equivalence. In particular, if P is connected, every extension of P by Z_k is trivial, and the category of such extensions is rigid.

(ii) Suppose that Φ is a torsion group and that P and Z are commutative. Then the category of extensions of P by Z_k is rigid: every automorphism of an object of this category is trivial. If, in addition, there is an integer $n > 0$ such that $n\mathrm{id}_\Phi = 0$—for example, if Φ is finite, as happens when P is of finite type—then the group of classes of commutative extensions of P by Z_k is given by the canonical isomorphism

$$\mathrm{Ext}^1(P, Z_k) \xleftarrow{\sim} \mathrm{Hom}\big(\Phi, (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z\big), \tag{5.5.1}$$

defined by the boundary operator in the exact sequence of Ext groups associated with the exact sequence

$$0 \longrightarrow Z \longrightarrow Z \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z \longrightarrow 0. \tag{5.5.2}$$

The canonical morphism

$$P \longrightarrow \Phi$$

plainly induces an isomorphism on the sets of irreducible components, since it is flat with irreducible fibres. By 5.1, the induced functor from torsors under Z_Φ to torsors under Z_P is therefore an equivalence. (Recall that a group scheme locally of finite type over k is geometrically unibranch.) The same conclusion applies to bitorsors and to the morphisms

$$P \times P \longrightarrow \Phi \times \Phi, \qquad P \times P \times P \longrightarrow \Phi \times \Phi \times \Phi.$$

The description in VII 1.1.6 of extensions of P by Z_k and of Φ by Z_k (and VII 1.3 in the commutative case) now gives the assertions in (i).

We restrict attention to commutative extensions. The automorphism group of an extension of P by Z_k is

$$\mathrm{Hom}(P, Z_k) \simeq \mathrm{Hom}(\Phi, Z_k),$$

which is plainly zero when Φ is a torsion group. Likewise, if $V = Z \otimes_{\mathbb{Z}} \mathbb{Q}$ and Φ is annihilated by the integer $n > 0$, then

$$\mathrm{Ext}^i(\Phi, V_k) = 0 \qquad \text{for every } i \in \mathbb{Z},$$

because the groups in question are $\mathbb{Q}$-vector spaces annihilated by n. The isomorphism (5.5.1) follows at once.

The same method proves, essentially, the following.

Corollary 5.6. Let k be a field, Z a torsion-free commutative group, P and Q two commutative group schemes locally of finite type over k, and let $\Phi = P/P^\circ$ and $\Psi = Q/Q^\circ$ be their groups of connected components. Then:

(i) The natural functor from the category of biextensions of (Φ, Ψ) by Z_k to the category of biextensions of (P, Q) by Z_k is an equivalence. In particular, if P or Q is connected, the category of biextensions of (P, Q) by Z_k is equivalent to the terminal category.

(ii) Suppose that Φ or Ψ is a torsion group. Then the category of biextensions of (P,Q) by Z_k is rigid. If, in addition, there is an integer $n>0$ such that $n\mathrm{id}_\Phi=0$ or $n\mathrm{id}_\Psi=0$, then the group of classes of biextensions of (P,Q) by Z_k is given by the canonical isomorphism

$$\mathrm{Biext}^1(P,Q;Z_k)\xleftarrow{\sim}\mathrm{Hom}\big(\Phi\otimes\Psi,(Z\otimes_{\mathbb{Z}}\mathbb{Q})/Z\big), \tag{5.6.1}$$

defined by the boundary operator in the exact sequence of Biext^i groups $(i=0,1)$ associated with (5.5.2).

5.7. Let S be a scheme, T a closed subscheme, and Z a set. The constant scheme over S, truncated along T, with value Z, denoted $Z_{S,T}$, is the scheme obtained by taking the sum of copies S_z of S, one for each $z\in Z$, and gluing all these copies along the open subscheme $U=S\setminus T$. Thus

$$Z_{S,T}\times_S U\simeq U,\qquad Z_{S,T}\times_S T\simeq Z_T, \tag{5.7.1}$$

and $Z_{S,T}$ is etale over S. For a variable S-scheme S' there is an isomorphism, functorial in S',

$$\mathrm{Hom}_S(S',Z_{S,T})\simeq\mathrm{Hom}_T(S'_T,Z_T). \tag{5.7.2}$$

The map is restriction to S'_T, using the second isomorphism in (5.7.1); we leave it to the reader to check that the map is bijective. If

$$i:T\longrightarrow S$$

denotes the inclusion, the functorial isomorphism (5.7.2) may also be written as an isomorphism of sheaves

$$Z_{S,T}\simeq i_*(Z_T). \tag{5.7.3}$$

This formula, or the definition directly, shows that as Z varies, $Z_{S,T}$ depends functorially on Z, and that this functor commutes with finite inverse limits. In particular, it takes groups to group schemes and commutative groups to commutative group schemes, as is also clear from the form (5.7.2) of the functor represented by $Z_{S,T}$.

It is also immediate from the definition, or from (5.7.3), that the formation of $Z_{S,T}$ commutes with every base change $f:S'\to S$. This gives an isomorphism, functorial in Z,

$$Z_{S,T}\times_S S'\simeq Z_{S',T'},\qquad T'=T\times_S S'=T_{S'}. \tag{5.7.4}$$

5.8. We shall study torsors E under a group scheme of the form $Z_{S,T}$, where Z is a group, assumed commutative here for simplicity. There is a natural functor

$$E\longmapsto E_T \tag{5.8.0}$$

from the category of torsors under $Z_{S,T}$ to the category of torsors under Z_T. We claim that this functor is an equivalence of categories. Its full faithfulness means that for two torsors E,F under $Z_{S,T}$, the map

$$\mathrm{Hom}(E,F)\longrightarrow\mathrm{Hom}(E_T,F_T)$$

is bijective. By fpqc descent this reduces immediately to the case in which E and F are trivial, hence to $E=F=Z_{S,T}$, where the assertion is precisely the bijectivity of (5.7.2) for $S'=S$.

It remains to prove that (5.8.0) is essentially surjective. Let E_T be a torsor under Z_T. We already observed in the proof of 5.1 that E_T is representable and locally trivial for the etale topology. It therefore suffices to prove that

$$H^1(S_{\text{ét}},Z_{S,T})\longrightarrow H^1(T_{\text{ét}},Z_T)$$

is bijective. This follows from (5.7.3) and the relation

$$R^1i_{\text{ét},*}(Z_T)=0,$$

which is the special case SGA 4 VIII 5.6.

5.8.1. Let now P be a commutative group scheme over S. We wish to determine the category of extensions, as sheaves of commutative groups, of P by $Z_{S,T}$. For this we use the description in VII 1.1.6 and 1.3 in terms of torsors on P, $P \times P$, and $P \times P \times P$, under the inverse images of $Z_{S,T}$. By (5.7.4), these inverse images are again of the same form as $Z_{S,T}$. Applying the preceding result to those torsors shows that they correspond to torsors on P_T, $P_T \times_T P_T$, and $P_T \times_T P_T \times_T P_T$ under the inverse images of Z_T. We therefore obtain the following.

Proposition 5.9. Let S be a scheme, T a closed subscheme, Z a commutative group, $Z_{S,T}$ the corresponding truncated constant group scheme over S defined in 5.7, and P a commutative group scheme over S. Then restriction to T induces an equivalence between the category of sheaves of commutative groups which are extensions of P by $Z_{S,T}$ and the category of sheaves of commutative groups which are extensions of P_T by Z_T.

An essentially identical argument gives the following.

Corollary 5.10. With the notation of 5.9, let Q be another commutative group scheme over S. Then restriction to T induces an equivalence between the category of biextensions of (P, Q) by $Z_{S,T}$ and the category of biextensions of (P_T, Q_T) by Z_T.

6. Extensions and biextensions by the Neron model of G_m.

6.1. Let S be a scheme, T a closed subscheme, and Z a set, giving a truncated constant scheme $Z_{S,T}$ over S (5.7). There is a canonical homomorphism, functorial in Z, of etale schemes over S,

$$Z_S \longrightarrow Z_{S,T}, \tag{6.1.1}$$

which is a surjective morphism, as follows at once from (5.7.1). Its functoriality in Z shows that when Z is a group, (6.1.1) is a homomorphism of etale group schemes over S. Its kernel is therefore an etale group scheme over S, denoted $Z_{S,U}$, and there is a canonical exact sequence, functorial in Z,

$$0 \longrightarrow Z_{S,U} \longrightarrow Z_S \longrightarrow Z_{S,T} \longrightarrow 0. \tag{6.1.2}$$

It follows immediately from (5.7.1) that $Z_{S,U}$ is the sum of a family of S-schemes S_n, $n \in Z$, where $S_n = S$ for $n = 0$ and $S_n = U$ for $n \neq 0$. This description of $Z_{S,U}$ readily shows that for every group F over S, restriction to U induces a bijection

$$\mathrm{Hom}_{\mathrm{gr}}(Z_{S,U}, F) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(Z_U, F|U). \tag{6.1.3}$$

6.1.4. Henceforth we suppose that Z is commutative, and we wish to examine extensions of commutative groups of $Z_{S,T}$ by a given commutative group G over S. By the exact sequence (6.1.2), the category of these extensions is equivalent to the category of extensions F of Z_S by G, furnished with a trivialization of the inverse image of the extension along $Z_{S,U} \to Z_S$. By (6.1.3), such a trivialization amounts to a trivialization of the extension $F|U$ of Z_U by $G|U$.

For what follows we take Z to be the group $\mathbb{Z}$ of integers. Thus $\mathbb{Z}_S$ is the free group on one generator, and VII 1.4 gives an equivalence between the category of extensions of $\mathbb{Z}_S$ by G and the category of torsors under G. Applying the same result to $\mathbb{Z}_U$ and $G|U$ gives the following.

Lemma 6.2. Let S be a scheme, T a closed subscheme, $U = S \setminus T$, and G a commutative group over S. The category of commutative groups over S which are extensions of $\mathbb{Z}_{S,T}$ (defined in (5.7)) by G is equivalent to the category of pairs (F, s) consisting of a torsor F under G and a section s of $F|U$. If $\alpha : S \to \mathbb{Z}_{S,T}$ is the section of $\mathbb{Z}_{S,T}$ which is the image of the section 1 of $\mathbb{Z}_S$ over S, then the torsor corresponding to an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by G is $F = \alpha^*(\overline{G})$ under G, furnished with the section of $F|U$ given by restricting the unit section of $\overline{G}$ to U.

6.2.1. If $\overline{G}$ corresponds to the pair (F, s), then the inverse image $f_n^*(\overline{G})$ in $\overline{G}$ of the section f_n of $\mathbb{Z}_{S,T}$ defined by the constant section of $\mathbb{Z}_S$ with value $n \in \mathbb{Z}$ is canonically identified, as a torsor under G, with the n-th power $F^{(n)}$ of the G-torsor F. The canonical isomorphism of G_U-torsors

$$f_n^*(\overline{G})|U \simeq G_U,$$

defined by the first relation in (5.7.1), is determined by the section $s^{(n)}$ of $F_U^{(n)}$, the n-th power of the given section s of F_U. This makes the following description of the sections of $\overline{G}$ over S explicit. A section $\overline{g}$ of $\overline{G}$ over S is determined by:

a) a section $\underline{n}$ of $\mathbb{Z}_{S,T}$ over S, which may be interpreted as a section of $\mathbb{Z}_T$, that is, as a locally constant function on T with values in $\mathbb{Z}$, or equivalently as a partition

$$T = \coprod_{i \in \mathbb{Z}} T_i$$

of T into a disjoint sum of open subsets indexed by $i \in \mathbb{Z}$;

b) a family of sections

$$\overline{g}_i \in \Gamma(F^{(i)}/S'_i), \qquad i \in \mathbb{Z},$$
$$S'_i = S \setminus T'_i, \qquad T'_i = \bigcup_{\substack{j \in \mathbb{Z} \\ j \neq i}} T_j,$$

subject to the following compatibility condition. For $i, j \in \mathbb{Z}$, $\overline{g}_i$ and $\overline{g}_j$ agree on $S'_i \cap S'_j = U$ after $F^{(i)}|U$ and $F^{(j)}|U$ have been identified with G_U by means of the sections $s^{(i)}$ and $s^{(j)}$, respectively. In other words, there must exist a necessarily unique section $h_{ij} \in \Gamma(G/U)$ such that

$$\overline{g}_i = h_{ij} s^{(i)}, \qquad \overline{g}_j = h_{ij} s^{(j)}.$$

Since the description in 6.2 of an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by G is plainly compatible with every base change $S' \to S$, the preceding gives a complete description of the sheaf $\overline{G}$ as a functor of the argument $S' \in \mathrm{Ob}(\mathrm{Sch}_{/S})$.

6.2.2. If T is a scheme which is a sum of nonempty connected schemes T_i, $i \in I$, the description in 6.2.1 of the sections of $\overline{G}$ over S may be simplified. Such a section is determined by a family of integers $(n_i)_{i \in I}$ indexed by I, and a family of sections

$$\overline{g}_i \in \Gamma(F^{(n_i)}/S'_i), \qquad i \in I,$$

where

$$S'_i = S \setminus T'_i, \qquad T'_i = \coprod_{\substack{j \in I \\ j \neq i}} T_j.$$

The $\overline{g}_i$ satisfy the following compatibility condition: for every pair $i, j \in I$, the restrictions of $\overline{g}_i$ and $\overline{g}_j$ to $S'_i \cap S'_j = U$ agree under the identifications of $F^{(n_i)}|U$ and $F^{(n_j)}|U$ deduced from the respective sections $s^{(n_i)}$ and $s^{(n_j)}$ of these torsors.

6.2.3. Finally, if T is assumed nonempty and connected, giving a section $\overline{g}$ of $\overline{G}$ over S amounts to giving an integer n and a section $\overline{g}$ of $F^{(n)}$ over S. There is no longer any compatibility condition to impose, so the section s of $F|U$ does not occur in the description of the group of sections of $\overline{G}$ over S. (Of course, s occurs again as soon as one seeks to describe the points of $\overline{G}$ with values in a general S-scheme S'.)

6.2.4. Since $\overline{G}$ is the union of the "open subschemes" $f_n^*(\overline{G})$, $n \in \mathbb{Z}$, described in 6.2.1, it follows that $\overline{G}$ is representable if and only if the torsors $F^{(n)}$ are representable. For example, it suffices that G be representable and affine over S, by the descent theory of SGA 1 VIII 2.1.

6.3. In what follows we suppose that $G = \underline{G}_{mS}$. Every extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by $G = \underline{G}_{mS}$ is then representable (6.2.4), and is of course smooth over S, being an extension of the etale group scheme $\mathbb{Z}_{S,T}$ by the smooth group scheme $\underline{G}_{mS}$. Moreover, giving a torsor F under $\underline{G}_m$ is equivalent to giving an invertible Module $\mathbb{L}$ over S (by associating to the latter the torsor of invertible sections of $\mathbb{L}$). Thus $\overline{G}$ is described by a pair $(\mathbb{L}, s)$, where $\mathbb{L}$ is an invertible Module over S and s is a trivialization of $\mathbb{L}$ over $U = S \setminus T$, that is, an invertible section of $\mathbb{L}|U$.

When U is schematically dense in S, and S is locally noetherian, or is reduced with a locally finite set of irreducible components, giving such a pair is equivalent to giving a divisor D on S such that

$$\operatorname{supp} D \subset T, \tag{6.3.1}$$

by associating to such a divisor the pair

$$\big(\underline{O}_S(-D), s_{-D}|U\big) \tag{6.3.2}$$

formed by the invertible Module $\underline{O}_S(-D)$ associated with $-D$ (EGA IV 21.2.8.1) and the restriction to U of its canonical rational section s_{-D} (EGA IV 21.2.11 and 20.2.11(ii)). In any event, without any hypothesis on S, a divisor D on S satisfying (6.3.1) defines by (6.3.2) a pair $(\mathbb{L}, s)$ and hence an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by $\underline{G}_m$. We call it the Neron–Raynaud model of $\underline{G}_m$ associated with the divisor D and the closed subscheme T of S satisfying (6.3.1). If $T = \operatorname{supp} D$, we omit T from the terminology.

It is immediate that the formation of this extension $\overline{G}$ commutes with every base change $f : S' \to S$ for which the inverse-image divisor $f^*(D)$ is defined (EGA IV 21.4.2), and in particular with every flat base change.

6.3.3. The general description in 6.2.1 of the sections of $\overline{G}$ immediately shows that they may be identified with the sections f of the sheaf $\mathcal{M}_S^*$ of regular meromorphic sections on S (EGA IV 20.1.8) whose divisor $\operatorname{div}(f)$ is locally, for the Zariski topology, an integral multiple of D. If V is an open subset of S on which the restriction of $\operatorname{div}(f)$ equals that of nD, where $n \in \mathbb{Z}$, then the section of $\mathbb{Z}_{S,T}$ defined by f is simply the image in (6.1.2) of the constant section of $\mathbb{Z}_S$ with value n. The preceding description of the group of sections of $\overline{G}$ over S remains valid over every open subset of S, and more generally for the group of points of $\overline{G}$ with values in any flat S-scheme S'. Since $\overline{G}$ itself is flat (indeed smooth) over S, this description again gives a complete characterization of the Neron–Raynaud model of $\underline{G}_m$ defined by D and $T \supset \operatorname{supp} D$.

6.4. Suppose now that S is normal, that its set of irreducible components is locally finite (so that S is a sum of normal integral schemes), and that D is a positive divisor with all multiplicities equal to 1 (EGA IV 21.6), that $T = \operatorname{supp} D$, and that T is locally irreducible, that is, its irreducible components are disjoint. Then for every rational function f on S which is regular on $U = S \setminus T$, the divisor $\operatorname{div}(f)$ is locally a multiple of D. Thus in this case there is a canonical isomorphism of groups

$$\Gamma(S, \overline{G}) \xrightarrow{\sim} \Gamma(U, \underline{G}_{mU}). \tag{6.4.1}$$

Suppose further that T is geometrically unibranch and locally integral, a hypothesis stable under smooth base change (EGA IV 17.5.7). Since all the hypotheses are then stable under base change, (6.4.1) implies that for every S' smooth over S there is a canonical group isomorphism, functorial in S',

$$\operatorname{Hom}_S(S', \overline{G}) \xrightarrow{\sim} \operatorname{Hom}_U(S'_U, \underline{G}_{mU}) \simeq \Gamma(S'_U, \underline{O}^*_{S'_U}). \tag{6.4.2}$$

Since $\overline{G}$ is itself smooth over S, this isomorphism characterizes the group scheme $\overline{G}$ over S up to unique isomorphism. It also shows that $\overline{G}$ plays the role of a "Neron model" for the group scheme

$\underline{G}_{mU}$ over the open subset U of S. Together with the fact that it was first introduced and used by M. Raynaud, this justifies the terminology adopted in 6.3 for $\overline{G}$.

Theorem 6.5. Let S be a normal locally integral scheme and let D be a positive divisor on S, without multiple components, whose support is geometrically unibranch and locally integral. Let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$ over S relative to D (6.3), and consider the category of torsors E over S under the group scheme $\overline{G}$. The restriction functor

$$E \longmapsto E_U = E|U$$

from this category to the category of torsors over U under the group scheme $\underline{G}_{mU}$ is fully faithful. If S is regular (more generally, if its local rings are factorial), this functor is an equivalence of categories.

Let us prove that the functor in question is fully faithful, that is, that for two torsors E, F under $\overline{G}$, the restriction map

$$(*) \qquad \mathrm{Hom}(E, F) \longrightarrow \mathrm{Hom}(E_U, F_U)$$

is bijective, where Hom denotes the set of homomorphisms of torsors. First observe that this question is plainly local on S for the Zariski topology. Indeed, suppose that S is the union of open subsets S_i such that for every open subset S' of S contained in some S_i (and hence in particular for $S' = S_i$ or $S_i \cap S_j$), the map

$$\mathrm{Hom}(E_{S'}, F_{S'}) \longrightarrow \mathrm{Hom}(E_{S' \cap U}, F_{S' \cap U})$$

is bijective. Then (*) is bijective. On the other hand, it follows at once from 5.8 applied to $S, T, \mathbb{Z}$, and from 5.1 applied to $T, \mathbb{Z}$, that every torsor under $\overline{G}$ is locally trivial for the Zariski topology. These two observations reduce us to proving the bijectivity of (*) when $E = F = \overline{G}$. In that case the map is precisely (6.4.1), whose bijectivity has already been established.

We now prove that the functor $E \mapsto E_U$ is essentially surjective when the local rings of S are factorial. Since $\underline{G}_{mS}$ is identified with an open subgroup scheme of $\overline{G}$, it is enough to show that every torsor under $\underline{G}_{mU}$ is isomorphic to the restriction to U of a torsor under $\underline{G}_{mS}$. Equivalently, every invertible Module over U must be isomorphic to the restriction of an invertible Module over S. This follows from the hypothesis on S (EGA IV 21.6.11, where only the locally noetherian case is treated; that is the only case needed below). This proves 6.5.

6.5.1. Suppose now, in addition, that S is locally noetherian and regular. Since S is then locally factorial, giving D is equivalent to giving a reduced, purely 1-codimensional closed subscheme, again denoted D, which is geometrically unibranch. Notice also that the hypotheses on S and D are stable under every smooth base change $S' \to S$, so that the conclusion of 6.5 applies to S' with the subscheme $D' = D \times_S S'$. The description in VII 1.1.6 of extension groups now immediately gives the following consequence.

Corollary 6.6. Let S be a locally noetherian regular scheme, let D be a reduced, geometrically unibranch, purely 1-codimensional closed subscheme (thus defining a divisor on S), let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$ relative to D, and let P be a smooth commutative group scheme over S. Consider the category of commutative extensions, as sheaves of groups, of P by $\overline{G}$. Restriction to U, $E \mapsto E|U$, induces an equivalence between this category and the category of extensions of P_U by $\underline{G}_{mU}$.

6.6.1. The same method gives the analogous statement for extensions which are not necessarily commutative (with P no longer assumed commutative). We leave the details of the proof to the reader; they are slightly more involved because bitorsors must be considered in place of torsors.

By essentially the same proof as for 6.6, one likewise obtains:

Corollary 6.7. Let $S, D, \overline{G}$ be as in 6.6, and let P, Q be two smooth commutative group schemes over S. Consider the category of biextensions of (P, Q) by $\overline{G}$. Then the restriction functor to U,

$E \mapsto E|U$, from this category to the category of biextensions of (P_U, Q_U) by $\underline{G}_{mU}$ is an equivalence of categories.

Remarks 6.8. a) We shall apply 6.6 and 6.7 when S is a trait and D is the divisor defined by the inclusion of the closed point s of S into S. In this case U is the spectrum of the fraction field K of S, and $\overline{G}$ is the usual Neron model of the multiplicative group over K, introduced by Raynaud. Notice that even in this special case, the proof of 6.6 and 6.7 uses 6.5 over bases more general than traits (such as P and its Cartesian powers). Nevertheless, in this case it is enough to know the existence of the Neron model $\overline{G}$ in the special situation under consideration: 6.5 for base schemes smooth over S, and hence 6.6 and 6.7, then follow formally. For us, the introduction of $\overline{G}$ will chiefly serve as a convenient intermediate step in studying, in the next section, the problem of extending extensions and biextensions by the multiplicative group.

b) A Neron–Raynaud model $\overline{G}$ of $\underline{G}_{mU}$, satisfying the fundamental condition (6.4.2) for every S' smooth over S, can be defined under much more general conditions than those considered above, with U merely a schematically dense open subset of a normal locally noetherian scheme S. In this case $\overline{G}$ is an fppf sheaf, not necessarily representable. Let $\underline{D}$ denote the sheaf, on the small etale site of S, of divisors on S whose support is contained in $T = S \setminus U$, and, by abuse of language, also denote by the same letter its canonical extension to a sheaf on $(\mathrm{Sch})_{/S}$. Its sections over a relative scheme $f : S' \to S$ are the sections of $f^*_{\text{ét}}(\underline{D})$ over S'. The desired model is an extension of groups

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow \overline{G} \longrightarrow \underline{D} \longrightarrow 0. \tag{6.8.1}$$

The construction of a canonical extension (6.8.1), without any normality hypothesis on S, presents no difficulty when one uses the general description VII 1.1.6; at the same time one obtains (6.4.2) for S' etale over S. For the same relation to hold for every S' smooth over S, it is equivalent to verify that the formation of the sheaf $\underline{D}$ commutes with every smooth base change $f : S' \to S$. For normal S, this follows from EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.9.

6.9. "Self-criticism" (observations of P. Deligne).[3]

Sections 5 and 6 are really rather uninspiring. One periodically has the impression that the formalism of i_*, i^* is being redone "by hand", without seeing what is geometric and what is "general nonsense".

I propose extracting the "geometric" statements a), b), c):

a) (cf. 6.8 b)) Let $j : U \hookrightarrow S$ be a schematically dense open subset of a normal noetherian scheme S. The etale sheaf $\underline{G}_{mS}$ is then identified with a subsheaf of the etale sheaf $j_*\underline{G}_{mU}$. Let $\underline{D}$ be the quotient etale sheaf in the exact sequence

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow j_*\underline{G}_{mU} \longrightarrow \underline{D} \longrightarrow 0.$$

Then the formation of $\underline{D}$ commutes with every smooth base change (EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.9); moreover, $R^1 j_*\underline{G}_m = 0$ (Theorem 90) if the strictly local rings of S are factorial.

b) If $U = S \setminus Y$, where Y is a geometrically unibranch divisor, and if i is the inclusion of Y into S, then

$$\underline{D} = i_*\mathbb{Z}_Y.$$

c) If Y is a closed subscheme of a scheme X, and E is a set, then the etale sheaf $i_*\underline{E}_Y$ is represented by an etale scheme over X, whose formation commutes with every base change.

Corollary. Let S be a normal noetherian scheme, let Y be a geometrically unibranch divisor, and let $U = S \setminus Y$:

3. The author of the present Expose having admitted defeat, the reader is invited, if necessary, to rewrite sections 5 and 6 according to the accompanying directions.

$$U \xhookrightarrow[j]{} S \xleftarrow{i} Y \qquad .$$

There exists a unique smooth group scheme $\mathcal{G}_m$ over S such that after every smooth base change one has $\mathcal{G}_m|S_{\text{ét}} = j_*(\underline{G}_m)_U$.

Indeed, the preceding requirements determine $\mathcal{G}_m$ as a sheaf, for the etale topology, on the category of schemes smooth over S. By a) and b), the sequence

$$0 \longrightarrow \underline{G}_m \longrightarrow \mathcal{G}_m \xrightarrow{q} i_*\mathbb{Z}_Y \longrightarrow 0$$

is exact. By c), $i_*\mathbb{Z}_Y$ is representable. Finally, the morphism q is representable.

Use of the preceding "small smooth site" should make it possible to deduce the statements of section 6 formally from a).

7. Canonical extensions of extensions and biextensions by $\underline{G}_m$.

7.0. We now reap the fruit of our preparations in sections 5 and 6. First let the base scheme S be a trait, that is, the spectrum of a discrete valuation ring; let η be its generic point, s its closed point, and $k = k(s)$. Let P be a smooth commutative group scheme over S, and write $P_0 = P_s$ for its special fibre. In the present section we consider only commutative groups; in particular, all group extensions under consideration are commutative.

Theorem 7.1. The notation is that recalled in 7.0.

a) Suppose that the group of connected components of the special fibre of P,

$$\Phi = P_0/P_0^\circ,$$

is a torsion group—as is the case, in particular, if P is of finite type over S. Then the functor $E \mapsto E_\eta$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_η by $\underline{G}_{m\eta}$ is fully faithful. Suppose that there is an integer $n > 0$ such that $n\Phi = 0$. Then every extension E_η of P_η by $\underline{G}_{m\eta}$ has an associated canonical homomorphism

$$d(E_\eta) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{7.1.1}$$

whose vanishing is necessary and sufficient for E_η to be isomorphic to the generic fibre of an extension of P by $\underline{G}_{mS}$. In particular, if P_0 is connected, the preceding functor $E \mapsto E_\eta$ is an equivalence of categories.

b) Let P and Q be two smooth group schemes over S, put

$$\Phi = P_0/P_0^\circ, \qquad \Psi = Q_0/Q_0^\circ,$$

and suppose that either Φ or Ψ is a torsion group (for example, that either P or Q is of finite type over S). Then the functor $E \mapsto E_\eta$ from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of biextensions of (P_η, Q_η) by $\underline{G}_{m\eta}$ is fully faithful. Suppose that there is an integer $n > 0$ such that $n\Phi = 0$ or $n\Psi = 0$. Then every biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$ has an associated canonical pairing

$$d(E_\eta) : \Phi \otimes_{\mathbb{Z}} \Psi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{7.1.2}$$

whose vanishing is necessary and sufficient for E_η to be isomorphic to the generic fibre of a biextension E of (P, Q) by $\underline{G}_{mS}$. In particular, if either P_0 or Q_0 is connected, the preceding functor $E \mapsto E_\eta$ is an equivalence of categories.

Proof. a) Let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$. The functor under consideration is the composite of the functor "extension of the kernel $\underline{G}_{mS} \to \overline{G}$" with the functor "restriction to η" defined on the category of extensions by $\overline{G}$. The latter functor is an equivalence of categories by 6.6, so it remains to study the first. For this purpose we use the exact sequence of group schemes

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow \overline{G} \longrightarrow \mathbb{Z}_{S,s} \longrightarrow 0, \tag{7.1.3}$$

where the notation $\mathbb{Z}_{S,s}$ is that of 5.7. The exact sequence of Ext^i, $i = 0, 1$, associated with (7.1.3) shows that the functor in question is fully faithful if and only if

$$\mathrm{Hom}(P, \mathbb{Z}_{S,s}) = 0, \tag{7.1.4}$$

and that the obstruction for an extension $\overline{E}$ of P by $\overline{G}$ to come from an extension E of P by $\underline{G}_{mS}$ lies in

$$\mathrm{Ext}^1(P, \mathbb{Z}_{S,s}). \tag{7.1.5}$$

By (5.7.2), the first group in (7.1.4) is isomorphic to $\mathrm{Hom}(P_0, \mathbb{Z}_k)$, which in turn is isomorphic to $\mathrm{Hom}(\Phi, \mathbb{Z}_k)$. Thus (7.1.4) is equivalent to

$$\mathrm{Hom}(\Phi, \mathbb{Z}_k) = 0. \tag{7.1.4 bis}$$

This relation certainly holds if Φ is a torsion group. Moreover, if $n\Phi = 0$, then 5.9 and 5.5 give canonical isomorphisms

$$\mathrm{Ext}^1(P, \mathbb{Z}_{S,s}) \simeq \mathrm{Ext}^1(P_0, \mathbb{Z}_k) \simeq \mathrm{Hom}\big(\Phi, (\mathbb{Q}/\mathbb{Z})_k\big), \tag{7.1.6}$$

which completes the proof of 7.1(a).

b) The proof is essentially the same. By 6.7, we reduce to studying the functor "extension of the kernel $\underline{G}_{mS} \to \overline{G}$" from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of biextensions of (P, Q) by $\overline{G}$. We use the exact sequence (7.1.3) and the associated exact sequence of Biext^i, $i = 0, 1$ (VII 3.7.5). Full faithfulness of the functor is equivalent to

$$\mathrm{Hom}(P \otimes Q, \mathbb{Z}_{S,s}) = 0. \tag{7.1.7}$$

By (5.7.2), the left side is isomorphic to $\mathrm{Hom}(P_0 \otimes Q_0, \mathbb{Z}_k)$ and hence to $\mathrm{Hom}(\Phi \otimes \Psi, \mathbb{Z}_k)$; this group is indeed zero if either Φ or Ψ is a torsion group. On the other hand, for a given biextension $\overline{E}$ of (P, Q) by $\overline{G}$, the obstruction to its coming from a biextension E of (P, Q) by $\underline{G}_{mS}$ lies in

$$\mathrm{Biext}^1(P, Q; \mathbb{Z}_{S,s}).$$

By 5.10 this is isomorphic to $\mathrm{Biext}^1(P_0, Q_0; \mathbb{Z}_k)$, which, when $n\Phi = 0$ or $n\Psi = 0$, is itself isomorphic by 5.6 to

$$\mathrm{Hom}\big(\Phi \otimes \Psi, (\mathbb{Q}/\mathbb{Z})_k\big).$$

This completes the proof of 7.1.

Remark 7.2. Given a smooth group scheme P over a connected regular locally noetherian scheme S with generic point η, one may ask to study the functor $E \mapsto E_\eta$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_η by $\underline{G}_{m\eta}$. This problem is readily reduced to the one treated in 7.1(a). For simplicity, suppose that P is of finite type over S. It follows easily from 7.1 that this functor is fully faithful.[4] One readily deduces that if E_η is an extension of P_η by $\underline{G}_{m\eta}$, then it is isomorphic to the generic fibre of an extension E of P by $\underline{G}_{mS}$ if and only if, for every $s \in S$ such that

$$\dim \mathrm{Spec}(\underline{O}_{S,s}) = 1,$$

the analogous extension problem obtained by base change $\mathrm{Spec}(\underline{O}_{S,s}) \to S$ has a solution. By 7.1(a), this is expressed by the vanishing of a certain homomorphism

$$d(E_\eta, s) : P_s/P_s^\circ \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k(s)}. \tag{7.2.1}$$

The preceding assertion plainly reduces to the following one. Let U be an open subset of S whose complement T satisfies

$$\mathrm{codim}(T, S) \geq 2.$$

4. For this it is enough that S be normal rather than regular.

Then the functor $E \mapsto E|U$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_U by $\underline{G}_{mU}$ is an equivalence of categories. In terms of the description VII 1.1.6 this assertion is essentially trivial: for every smooth S-scheme X (such as P, $P \times P$, $P \times P \times P$, and so on), X is locally factorial and $X_T = X \setminus X_U$ has codimension at least 2 in X. Consequently, the functor $F \mapsto F|X_U$ from the category of torsors under $\underline{G}_{mX}$ to the category of torsors under $\underline{G}_{mX_U}$ is an equivalence of categories.

There are entirely analogous results for the problem of extending biextensions; we leave their formulation to the reader.

7.3. We shall make explicit some functorial properties of the obstructions defined in (7.1.1) and (7.1.2), with P, Q, S variable, and some relations among them. In what follows, we shall suppose that the conditions allowing these obstructions to be defined are satisfied. Since the verifications present no difficulty, we leave them to the reader.

7.3.1. Let

$$u : P' \longrightarrow P$$

be a morphism of smooth group schemes over S, and let E_η be an extension of P_η by $\underline{G}_{m\eta}$. Pullback by $u_\eta : P'_\eta \to P_\eta$ gives an extension E'_η of P'_η by $\underline{G}_{m\eta}$. With this notation, the following diagram is commutative:

(7.3.1.1)
$$\begin{array}{ccc} P'_0/P'^\circ_0 & & \\ \downarrow{\scriptstyle u_0} & \searrow^{d(E'_\eta)} & \\ & & (\mathbb{Q}/\mathbb{Z})_k \\ & \nearrow_{d(E_\eta)} & \\ P_0/P^\circ_0 & & \end{array} \qquad .$$

Similarly, if we have morphisms of smooth group schemes over S

$$u : P' \longrightarrow P, \qquad v : Q' \longrightarrow Q,$$

and a biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$, whose pullback by (u_η, v_η) is a biextension E'_η of (P'_η, Q'_η) by $\underline{G}_{m\eta}$, then the following diagram is commutative:

(7.3.1.2)
$$\begin{array}{ccc} P'_0/P'^\circ_0 \otimes Q'_0/Q'^\circ_0 & & \\ \downarrow{\scriptstyle u_0 \otimes v_0} & \searrow^{d(E'_\eta)} & \\ & & (\mathbb{Q}/\mathbb{Z})_k \\ & \nearrow_{d(E_\eta)} & \\ P_0/P^\circ_0 \otimes Q_0/Q^\circ_0 & & \end{array}$$

The essentially mechanical verification of these compatibilities is left to the reader.

7.3.2. It follows from the compatibility (7.3.1.1) that, if E_η is an extension of P_η by $\underline{G}_{m\eta}$, there is a largest group subscheme V of P such that the restriction of E_η to V_η extends to an extension of V by $\underline{G}_{mS}$. It is the open subgroup V of P containing P_η and defined by the condition

(7.3.2.1)
$$V_0/V_0^\circ = \ker d(E_\eta).$$

If $u : P' \to P$ is then a morphism of smooth group schemes over S, the extension E'_η, obtained by pulling E_η back along u_η, extends to an extension of P' by $\underline{G}_{mS}$ if and only if

(7.3.2.2)
$$u(P') \subset V.$$

In the case of a biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$, there is likewise a largest group subscheme V of P such that the restriction of E_η to (V_η, Q_η) extends to a biextension of (V, Q) by $\underline{G}_{mS}$. It is the open subgroup of P containing the generic fibre P_η and characterized by the fact that V_0/V_0° is the left annihilator in Φ of the pairing (7.1.2), that is, the kernel of the corresponding homomorphism from Φ to $\underline{\mathrm{Hom}}(\Psi, (\mathbb{Q}/\mathbb{Z})_k)$:

$$V_0/V_0^\circ = \ker\left(\Phi \xrightarrow{d(E_\eta)} \underline{\mathrm{Hom}}(\Psi, (\mathbb{Q}/\mathbb{Z})_k)\right). \tag{7.3.2.3}$$

If $u : P' \to P$ is a morphism of smooth group schemes over S, the biextension of (P'_η, Q_η) by $\underline{G}_{m\eta}$ obtained by pulling E_η back along u_η extends to a biextension of (P', Q) by $\underline{G}_{mS}$ if and only if (7.3.2.2) holds. Symmetric results hold upon interchanging the roles of P and Q.

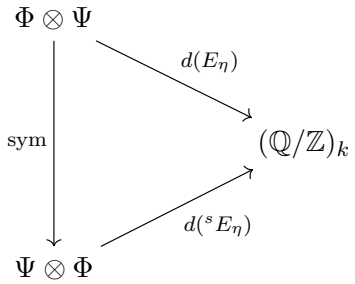
Of course, in general there is no largest pair of subgroups $P' \to P$, $Q' \to Q$ for which the restriction E'_η of E_η to a biextension of (P'_η, Q'_η) by $\underline{G}_{m\eta}$ extends to a biextension of (P', Q') by $\underline{G}_{mS}$. One can only say that, for a given pair (P', Q'), or more generally for a given pair of morphisms $P' \to P$, $Q' \to Q$ of smooth group schemes, the desired extension exists if and only if the restriction of the form $d(E_\eta)$ to

$$P'_0/P'^\circ_0 \otimes Q'_0/Q'^\circ_0$$

is zero.

7.3.3. Symmetry. Let E_η be a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$, and consider the transposed biextension ${}^sE_\eta$ (VII 2.7), which is a biextension of (Q_η, P_η) by $\underline{G}_{m\eta}$. The following diagram is commutative:

(7.3.3.1)



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where the vertical arrow is the canonical symmetry. (Compare this statement with the antisymmetry statement of 2.2.10.) It follows in particular that, if $P = Q$ and E_η is a symmetric biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$, that is, if E_η is isomorphic to ${}^sE_\eta$, then the bilinear form $d(E_\eta)$ on $\Phi \times \Phi$ with values in $(\mathbb{Q}/\mathbb{Z})_k$ is symmetric (and not alternating, as in 2.2.12!). If it is zero, that is, if E_η extends to a biextension E of (P, P) by $\underline{G}_{mS}$, the latter is necessarily symmetric as well: by 7.1(a), every isomorphism ${}^sE_\eta \to E_\eta$ extends uniquely to an isomorphism ${}^sE \to E$.

7.3.4. Reduction of obstructions of type (7.1.2) to those of type (7.1.1).

Suppose, for definiteness, that there is an integer $n > 0$ such that $n\Phi = 0$, and let

$$q \in Q(S)$$

be a section of Q. By an abuse of notation, write

$$q_0 \in \Psi(k)$$

for the image in $\Psi(k)$ of the value of q at s. Regarding E_η as an extension of $P_\eta \times_\eta Q_\eta$ by $\underline{G}_{mQ_\eta}$ over Q_η, pullback gives an extension

$$q_\eta^*(E_\eta), \qquad \text{an extension of } P_\eta \text{ by } \underline{G}_{m\eta},$$

and hence an obstruction

$$d_q \overset{\text{dfn}}{=} d\big(q_\eta^*(E_\eta)\big) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k. \tag{7.3.4.1}$$

With $d = d(E_\eta)$, one has the relation

$$d_q(x) = d(x, q_0), \tag{7.3.4.2}$$

valid for every point x of Φ with values in a k-scheme T.

7.3.4.3. For every $q \in Q(S)$, this formula determines d_q in terms of the pairing d; conversely, it plainly recovers d from the invariants d_q, at least when every point of Ψ comes from a section of Q. By Hensel's lemma, this condition holds automatically when S is strictly local. The pairing d is known once it is known over the separable closure of k, and its formation commutes with etale localization on S by 7.3.5.3 below. Thus, after replacing S by its strict localization if necessary, the determination of d reduces to that of the invariants d_q.

In particular, when every point of Ψ comes from a section q of Q, the pairing d is zero—that is, E_η extends to a biextension E of (P, Q) by $\underline{G}_m$—if and only if d_q is zero for every section q of Q, that is, if and only if $q_\eta^*(E_\eta)$ extends to an extension of P by $\underline{G}_m$. With no hypothesis on Ψ, the analogous criterion remains valid provided that the q are allowed to be points with values in local etale S-schemes S'.

7.3.4.4. Conversely, starting from the situation of 7.1(a), the obstruction (7.1.1) may be interpreted as an obstruction of type (7.1.2): take $Q = \mathbb{Z}_S$ and the biextension of $(P_\eta, \mathbb{Z}_\eta)$ deduced from E_η. Indeed, it suffices to apply the preceding construction with q equal to the section 1 of $\mathbb{Z}_S$.

7.3.5. Effect of a base change $S' \to S$. Suppose given a dominant local morphism of traits

$$f : S' \longrightarrow S,$$

and denote by η' the generic point of S', by s' its closed point, and put $k' = k(s')$. If $S = \mathrm{Spec}(V)$ and $S' = \mathrm{Spec}(V')$, then V' is a discrete valuation ring dominating the discrete valuation ring V. Put

$$e(S'/S) = e(V'/V) = \mathrm{length}_{V'}(V'/\mathfrak{m}V'), \tag{7.3.5.1}$$

where $\mathfrak{m}$ is the maximal ideal of V.

Let now P be as in 7.1(a), let E_η be an extension of P_η by $\underline{G}_{m\eta}$, put $P' = P \times_S S'$, and let $E'_{\eta'}$ be the extension of $P'_{\eta'}$ by $\underline{G}_{m\eta'}$ obtained by pulling back E_η. With the notation of 4.1(a), there is a canonical isomorphism

$$\Phi' \overset{\text{dfn}}{=} P'_0/P'^\circ_0 \simeq \Phi \otimes_k k' \simeq \Phi_{k'},$$

and hence one may consider

$$d(E_\eta)_{k'} : \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k'}.$$

Then

$$d(E'_{\eta'}) = e(S'/S)\, d(E_\eta)_{k'}. \tag{7.3.5.2}$$

Likewise, under the hypotheses of 7.1(b) and with the evident notation, (7.3.5.2) holds with $d(E_\eta)$ now denoting the pairing (7.1.2), $d(E_\eta)_{k'}$ the pairing

$$\Phi' \otimes \Psi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k'}$$

obtained from it by base change $k \to k'$, and $E'_{\eta'}$ the biextension of $(P'_{\eta'}, Q'_{\eta'})$ by $\underline{G}_{m\eta'}$ obtained from the biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$ through the base change $\eta' \to \eta$.

7.3.5.3. An interesting case is $e(S'/S) = 1$, that is,

$$\mathfrak{m}' = \mathfrak{m}V',$$

where $\mathfrak{m}'$ denotes the maximal ideal of V'. This holds in particular if V' is etale over V, or if V' is the henselization, strict henselization, or completion of V. In this case (7.3.5.2) simply says that the formation of obstruction (7.1.1), respectively (7.1.2), commutes with the base change under consideration $S' \to S$.

7.3.5.4. On the other hand, (7.3.5.2) in the general case shows that, for every extension E_η as in 7.1(a) (respectively every biextension E_η as in 7.1(b)), there is a finite morphism of traits $S' \to S$ such that $k(\eta')$ is a separable extension of $k(\eta)$ and such that the extension (respectively biextension) $E'_{\eta'}$ obtained from E_η by base change $\eta' \to \eta$ extends to an extension (respectively biextension) E' over all of S'. Indeed, it is enough to choose S'/S so that $e(S'/S)$ is a multiple of the integer $n > 0$ occurring in 7.1, and such an extension can always be found with $k(\eta')$ separable over $k(\eta)$.

Proposition 7.4. Let S be a reduced scheme.

a) Let P be a smooth group scheme over S whose fibres at the maximal points of S are connected. Then the natural functor

$$\text{EXT}(P, \underline{G}_{mS}) \longrightarrow \text{TORSRIG}(P, e_P; \underline{G}_{mS}) \tag{7.4.1}$$

from the category of extensions of P by $\underline{G}_{mS}$ to the category of torsors under $\underline{G}_{mP}$ rigidified along the identity section of P is fully faithful. If S is normal and P has connected fibres, an object E of the target of (7.4.1) belongs to the essential image of this functor if and only if, for every maximal point η of S, the rigidified torsor E_η on P_η comes from an extension of P_η by $\underline{G}_{m\eta}$.

b) Let P, Q be two smooth group schemes over S whose fibres at the maximal points of S are connected. Then the natural functor

$$\text{BIEXT}(P, Q; \underline{G}_{mS}) \longrightarrow \text{TORSBIRIG}(P, Q; \underline{G}_{mS}) \tag{7.4.2}$$

from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of torsors under $\underline{G}_{m,P\times_S Q}$ birigidified along the pair of identity sections of P and Q is fully faithful. If S is normal and either P or Q has connected fibres, an object E of the target of (7.4.2) belongs to the essential image of the functor if and only if, for every maximal point η of S, E_η comes from a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$.

We prove (a); the essentially identical proof of (b) is left to the reader. For the full-faithfulness assertion, note that 4.1 remains valid if X and Y are flat schemes over a reduced scheme S whose fibres at the maximal points satisfy the conditions stated in 4.1. Indeed, the family of maximal fibres of $X \times_S Y$ (which is flat over the reduced scheme S) is schematically dense in $X \times_S Y$. Hence a morphism from $X \times_S Y$ to the separated S-scheme $\underline{G}_{mS}$ is trivial as soon as it is trivial on the maximal fibres. Thus the full faithfulness of (7.4.1) is proved as in 4.2, and the characterization of its essential image given there remains valid.

It remains to prove that, if the rigidified torsor E on P is such that the E_η come from extensions, then so does E, assuming that S is normal and P has connected fibres. The latter condition implies that P is of finite presentation over S (SGA 3 VI$_B$ 5.5). Full faithfulness also shows that the question is local on S, so that we may suppose S affine and local. The usual limit argument, using Nagata's theorem on the finiteness of the integral closure of a reduced finitely generated $\mathbb{Z}$-algebra (EGA IV 7.8), reduces us further to the case in which S is noetherian.

Then 7.1(a) shows that there is an open subset $U \subset S$ whose complement has codimension at least 2 and such that E_U comes from an extension of P_U by $\underline{G}_{mU}$. In other words, the desired isomorphism

$$\pi^*(E) \simeq \text{pr}_1^*(E)\,\text{pr}_2^*(E)$$

can be constructed over $(P \times_S P)_U$. This is an open subset of the normal scheme $P \times_S P$ whose complement has codimension at least 2. The isomorphism on $(P \times_S P)_U$ therefore extends over all of $P \times_S P$, completing the proof.

Corollary 7.5. Let S be a normal scheme and let P and Q be two smooth group schemes over S, with connected fibres at the maximal points, either P or Q having connected fibres. Suppose in addition that, for every maximal point η of S, either P_η or Q_η is an extension of an abelian scheme by a torus (or, more generally, admits a composition series whose factors are abelian schemes, tori, or groups G_a). Then the functor (7.4.2) is an equivalence of categories.

This is an immediate consequence of 7.4(b) and 4.7.

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Exposé IX

NÉRON MODELS AND MONODROMY

by A. Grothendieck
(with an appendix by M. Raynaud)

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5. The orthogonality theorem for $\ell = p$ (semistable case). Duality of the abelian schemes B_o and B'_o. Characterization of the fixed part. Criteria for good reduction, essentially good reduction, and semistable reduction
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7. The Raynaud extension $G^\natural$ over S attached to the abelian scheme A_K with semistable reduction. Duality of the abelian schemes B, B' over S
8. The Raynaud extension $A_K^{\natural o}$ in the semistable case, and the ind-group $A_K^\natural$
9. Definition of the monodromy pairing $\underline{M}_\ell \otimes \underline{M}'_\ell \longrightarrow \mathbb{Z}_{\ell S}$
10. Properties of the monodromy pairing. Integrality and positivity theorem
11. Connected components of the Néron model: duality relation, asymptotic behavior
12. Néron model of a Jacobian and the Picard–Lefschetz formula
13. Relations with transcendental theory: the complex-analytic case
    - 13.1. Review of analytic groups over S
    - 13.2. Formal completion along a fiber
    - 13.3. The analytic Raynaud extension
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    - 13.5. The relatively algebraic case
14. Relations with transcendental theory: the rigid-analytic case. The canonical homomorphism $\underline{M}_K \longrightarrow A_K^{\natural o}$

0. Introduction

0.1. In the present exposé we apply the results of the preceding exposés to the study of the Néron model (1.1) A of an abelian scheme A_K defined over the function field K of the Henselian trait S. We shall see that the most important properties of the Néron model can be interpreted in terms of the action of the local monodromy group I on the Tate module $T_\ell(A_K(\overline{K}))$ (where ℓ is a prime number distinct from p). Since the dual of the latter can be interpreted as the ℓ-adic cohomology group $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$, and since, for every smooth projective scheme X over K, $H^1(X_{\overline{K}}, \mathbb{Z}_\ell)$ is identified with $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$, where $A_{\overline{K}}$ is the Albanese variety of $X_{\overline{K}}$, the present exposé may also be viewed as a detailed study of local monodromy phenomena for ℓ-adic H^1 (or, better still, for "motivic" H^1) of smooth projective varieties over K. From this point of view, it

seems clear that the principal results of the present exposé are destined to be subsumed within a "Néron theory" for motives of arbitrary weight, that is, for H^i (ℓ-adic, de Rham, Hodge, and so forth) with arbitrary i, of which at present we are only beginning to catch a glimpse. (See [9] in this connection, and more particularly Deligne's Conjectures 9.8 through 9.13 in the cited report.)

0.2. The most important result of the present exposé is the "semistable reduction theorem" 3.6, which asserts that, after a finite separable extension K' of K, the Néron model of $A_{K'}$ has a special fiber whose identity component is an extension of an abelian scheme by a torus. This result was conjectured by J.-P. Serre in 1964, and the principle of the proof given here [5] goes back to the same year. It appears in 3.5 that this property of "semistable reduction" is equivalent to the action of the monodromy group I on $T_\ell(A_K(\overline{K}))$ being "essentially unipotent of step 2" (where ℓ is a fixed prime distinct from the residual characteristic p of S). In this form it was proved, in terms of ℓ-adic H^1, in Exposé III as a special case of the "monodromy theorem", using resolution of singularities for excellent schemes of dimension 2 (due to S. Abhyankar). To prove the equivalence of the two properties, we shall also need an "orthogonality theorem" 2.4, due to Igusa [11] in the special case in which A_K is the Jacobian of a smooth curve X_K over K, the generic fiber of a regular connected projective scheme X over S whose geometric special fiber has no singular points other than ordinary quadratic singularities. This is indeed a semistable-reduction case, but it would not suffice for our purposes; it is chiefly in order to establish the orthogonality theorem in its general form that we use the convenient tool supplied by the theory of biextensions developed in the two preceding exposés. Let us also point out that D. Mumford has found a very different proof of the semistable reduction theorem, one which does not use resolution of singularities and instead uses his theory of theta functions (unpublished); unfortunately, he must restrict himself to residual characteristic different from 2.

0.3. The semistable reduction theorem is the principal tool used in the subsequent sections to study Néron models, which we undertake here from the "algebraic" point of view. It also seems indispensable for studying abelian varieties over K from the "rigid-analytic" point of view, or from the point of view of "theta functions" (these terms apparently being synonymous when abelian varieties over complete valued fields are in question), as developed recently by D. Mumford and M. Raynaud following work of Tate and Morikawa. (In Sections 13 and 14 we shall say a few words about the relations between the algebraic and transcendental points of view.) Finally, the same theorem is a key ingredient in the proof of the "semistable reduction theorem for algebraic curves" of Artin–Deligne–Mumford [3], [6] which we have already had occasion to use in the study of the action of monodromy on the fundamental group (the "essentially tame action theorem" of Exposé V), and which is used again in the proof of the integrality and positivity theorem 10.4 in the present exposé. [7]

0.4. The present exposé differs from the other exposés of the Seminar in that, in studying the Tate module $T_\ell(A_K)$, we have systematically taken into account also the case in which ℓ equals the residual characteristic p. It turns out that the language of Barsotti–Tate groups [33] [8] makes it possible to extend to this case all the most important phenomena established for $\ell \neq p$, subject on occasion to having available Tate's fundamental theorem [33, Theorem 4], which at present has been established only when K has characteristic zero. Of course, in the latter case the pro-p Barsotti–Tate group $T_p(A_K)$ may still be interpreted in terms of the action of the monodromy group I on the p-adic module $T_p(A_K(\overline{K}))$, but one should bear in mind that most of the statements suggested by this superficial analogy with the case $\ell \neq p$ are grossly false. (Thus it is no longer true that the monodromy action is essentially unipotent when $\ell = p$.) By using more judicious translations, however (such as 5.13 below), it is already clear that the study of pro-p Barsotti–

5. It is developed in a letter from the lecturer to J.-P. Serre dated 30 October 1964.

6. P. Deligne gave the proof of this theorem in the oral Seminar, but it was not considered useful to reproduce it in the Seminar texts.

7. There is another proof of the latter theorem, by way of rigid-analytic theory, which does not use the semistable reduction theorem.

8. Or "p-divisible groups" in Tate's terminology.

Tate groups defined geometrically over the function field of a trait of residual characteristic $p > 0$, whether or not the trait has mixed characteristic, must be regarded as part of the theory of "local monodromy", in which these groups play the role of p-adic local systems on $\mathrm{Spec}(K)$. In fact, these are essentially the "local systems" that enter into the study, from the "p-adic" point of view, of H^1 of smooth projective varieties over K. For the still-to-be-developed study of higher H^i from the p-adic point of view, it appears that the category of pro-p Barsotti–Tate groups over K, respectively over S, will have to be suitably enlarged, taking inspiration from the viewpoints provided by "crystalline cohomology".

0.5. The exposé given here follows the oral lectures very closely up to Section 4, but is markedly more detailed and complete from the following sections onward, which had been only partially sketched. It will not be used in the subsequent exposés. Some results of the present exposé (10.4 and 12.5), on the other hand, use the Picard–Lefschetz formula, which will be established only later.

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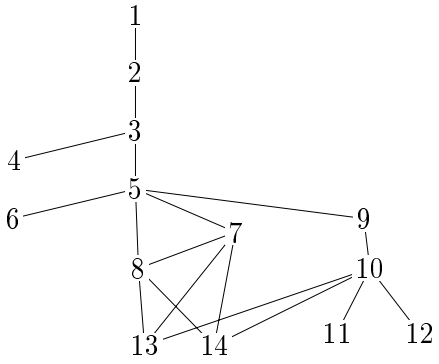


1. Neron models of abelian schemes: notation; the canonical pairing

$$\Phi_o \times \Phi'_o \longrightarrow (\mathbb{Q}/\mathbb{Z})_k,$$

and the canonical biextension W^o of (A^o, A'^o).

1.0. Throughout this exposé we fix a trait S, for which we use the usual notation s, η, k, K, and so forth, of I 1. Let A_K be an abelian scheme over K, and denote its dual abelian scheme [5] by A'_K. The duality between A_K and A'_K is given by a divisorial correspondence W_K on (A_K, A'_K), that is, an invertible sheaf on $A_K \times A'_K$, birigidified with respect to the identity sections of A_K and A'_K. It is determined up to unique isomorphism by its class (cf. VIII 3.1)

$$w_K \in \mathrm{Biext}^1(A_K, A'_K; \mathbb{G}_{mK}). \tag{1.0.1}$$

An important role will be played by the corresponding pairing (VIII 2.2)

$$\varphi : T_\ell(A_K) \times T_\ell(A'_K) \longrightarrow \mathbb{Z}_\ell(1)_K \overset{\mathrm{def}}{=} T_\ell(\mathbb{G}_{mK}), \tag{1.0.2}$$

where ℓ is an arbitrary prime number. When $\ell \neq p = \mathrm{char}\, k$, giving φ is equivalent to giving a pairing of finite free $\mathbb{Z}_\ell$-modules

$$\varphi : T_\ell(A_K(\overline{K})) \times T_\ell(A'_K(\overline{K})) \longrightarrow T_\ell(\overline{K}^*), \tag{1.0.3}$$

compatible with the action of

$$\pi = \mathrm{Gal}(\overline{K}/K)$$

on the three modules in question, where $\overline{K}$ is a separable closure of K. It is known (VIII 3.2) that this pairing is a perfect duality: it identifies each of the modules $T_\ell(A_K(\overline{K}))$ and $T_\ell(A'_K(\overline{K}))$ with the dual of the other, with values in the invertible module $\mathbb{Z}_\ell(1) = T_\ell(\overline{K}^*)$.

At times we shall suppose that a polarization of A_K has been chosen; thus we are given an element

$$\xi \in \mathrm{NS}(A_K/K) = \underline{\mathrm{NS}}_{A_K/K}(K), \tag{1.0.4}$$

where

$$\underline{\mathrm{NS}}_{A_K/K} \overset{\mathrm{def}}{=} \underline{\mathrm{Pic}}_{A_K/K}/\big(\underline{\mathrm{Pic}}^{o}_{A_K/K} = A'_K\big).$$

In the usual way it defines a homomorphism

$$\psi_\xi : A_K \longrightarrow A'_K, \tag{1.0.5}$$

hence a homomorphism

$$T_\ell(\psi_\xi), \quad \text{or simply} \quad \psi_\xi : T_\ell(A_K) \longrightarrow T_\ell(A'_K), \tag{1.0.6}$$

and a bilinear form φ_ξ

$$\begin{aligned} \varphi_\xi(x,y) &= \varphi(x,\psi_\xi(y)), \\ \varphi_\xi : T_\ell(A_K) \times T_\ell(A_K) &\longrightarrow \mathbb{Z}_\ell(1)_K, \end{aligned} \tag{1.0.7}$$

which, for $\ell \neq p$, may also be interpreted as a pairing of finite free $\mathbb{Z}_\ell$-modules with operators:

$$\varphi_\xi : T_\ell(A_K(\overline{K})) \times T_\ell(A_K(\overline{K})) \longrightarrow T_\ell(\overline{K}^*). \tag{1.0.8}$$

These constructions make sense for every element of the Neron–Severi group of A_K, and yield an alternating form φ_ξ, in either interpretation (1.0.7) or (1.0.8) (VIII 2.2.11). The fact that ξ is a polarization implies that ψ_ξ in (1.0.5) is an isogeny; equivalently, since $\dim A_K = \dim A'_K$, it is an epimorphism of groups. This is also equivalent to the injectivity of $T_\ell(\psi_\xi)$ in (1.0.6) (or to its having finite cokernel), and finally to the form φ_ξ in (1.0.7) being separating.

1.1. Recall that there exists a smooth scheme A over S, called the Neron model of A_K, which represents the functor

$$S' \longmapsto \mathrm{Hom}(S'_\eta, A_K) \tag{1.1.1}$$

on the category of smooth S-schemes. Thus, by definition, there is an isomorphism, functorial in the smooth relative S-scheme S',

$$\mathrm{Hom}_S(S', A) \simeq \mathrm{Hom}_K(S'_\eta, A_K). \tag{1.1.2}$$

Neron [19] constructed A when the residue field k of S is perfect. Raynaud extended the construction to the general case in an unwritten seminar held at IHES in 1966–67 (see [21] for the statement of Raynaud's principal results, which complete those of Neron in loc. cit.). We shall admit this existence result here, together with the fact that A is a quasi-projective scheme over S. Applying (1.1.2) when S' is smooth over η, one also finds that the following canonical morphism is an isomorphism:

$$A \times_S \mathrm{Spec}(K) \xrightarrow{\sim} A_K. \tag{1.1.3}$$

This justifies the paired notation A, A_K; when convenient we shall also write A_η in place of A_K. Using (1.1.3) as an identifying isomorphism, the canonical bijection (1.1.2) is simply the map given by restriction to the generic fibers.

Since the functor (1.1.1) plainly has the structure of a commutative group, the Neron model A representing it itself has the structure of a smooth commutative group scheme over S. The remainder of this exposé is devoted to the study of this group scheme.

An important invariant arising from the Neron model is, of course, the smooth commutative group scheme of finite type over k

$$A_o = A \times_S s = A_k. \tag{1.1.4}$$

Following the general notation, we denote its identity component by A_o^o and also consider the group scheme

$$\Phi_o = A_o/A_o^o \tag{1.1.5}$$

of connected components of A_o, which is finite and etale over k. In terms of a separable closure $\overline{k}$ of k, it is therefore determined by the ordinary finite group

$$\Phi_o(\overline{k}), \tag{1.1.6}$$

together with the natural action of

$$\pi_o = \mathrm{Gal}(\overline{k}/k)$$

on this group.

Along with the Neron model A of A_K, we shall study

$$A' = \text{the Neron model of } A'_K, \tag{1.1.7}$$

which is likewise a smooth commutative group scheme of finite type over S. We shall denote by

$$\Phi'_o = A'_o/A'^o_o \tag{1.1.8}$$

the group of connected components of its special fiber A'_o.

1.2. Apply VIII 7.1(b) to the pair (A, A') and to the biextension W_K of (A_K, A'_K) by $\mathbb{G}_{mK}$ in (1.0.1). We thereby obtain a canonical pairing

$$\Phi_o \times \Phi'_o \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{1.2.1}$$

that is, a pairing of π_o-modules

$$\Phi_o(\overline{k}) \times \Phi'_o(\overline{k}) \longrightarrow \mathbb{Q}/\mathbb{Z}. \tag{1.2.2}$$

Here the pairing appears as the obstruction to extending W_K to a biextension W of (A, A') by $\mathbb{G}_{mS}$; the meaning of this point of view is explained in VIII 7.3.2. We record at once the following conjecture concerning this pairing.

Conjecture 1.3. The pairing (1.2.1) is a perfect duality: it identifies each of the component groups Φ_o and Φ'_o with the dual of the other, with values in $(\mathbb{Q}/\mathbb{Z})_k$.

1.3.1. The formation of (1.2.1) commutes with formally etale base change (VIII 7.3.5.3), and the formation of Neron models has the analogous property. Consequently, to prove Conjecture 1.3 it would suffice to treat the case in which S is complete and has separably closed residue field. Artin and Mazur proved the conjecture when A_K is the Jacobian of a proper smooth curve over K, using their general, unpublished theory of the self-duality of the relative Jacobian (in the sense of derived categories) of a proper curve scheme over S. They also proved Conjecture 1.3 when the residue field is finite, using Tate duality [32]. We shall likewise see in 11.4 and 11.5 (subject to compatibility checks) that 1.3 holds on the ℓ-primary components for $\ell \neq p$, and that it holds in the case of "semistable reduction". These results make 1.3 extremely plausible. Moreover, 1.3 may also be regarded as a by-product of a general duality theory for group schemes over fraction fields of complete discrete valuation rings, formally analogous to SGA 5 I. Such a theory remains heuristic at present; it would also contain local class field theory in the form given to it by Serre [27].

1.3.2. For a given abelian scheme A_K, the validity of 1.3 means that the pairing (1.2.1) is separating on both the left and the right. By VIII 7.3.2, separation on the left says precisely that the only open subgroup scheme V of A for which W_K can be extended to a biextension of (V, A') by $\mathbb{G}_{mS}$ is the identity component A^o of A. There is a symmetric interpretation of separation on the right.

1.4. More generally, suppose that

$$U \subset A \qquad \text{and} \qquad U' \subset A' \tag{1.4.1}$$

are open subgroup schemes such that W_K extends to a biextension of (U, U') by $\mathbb{G}_{mS}$. By VIII 7.3.2, this says explicitly that the subgroups

$$F_o = U_o/U_o^o \subset \Phi_o, \qquad F'_o = U'_o/U'^o_o \subset \Phi'_o \tag{1.4.2}$$

defined by U and U' are orthogonal under (1.2.1).[9] We denote this extension (unique up to unique isomorphism) of W_K by W. It is determined up to unique isomorphism by its class

$$w \in \mathrm{Biext}^1(U, U'; \mathbb{G}_{mS}), \tag{1.4.3}$$

which is itself characterized by the condition that its image in $\mathrm{Biext}^1(A_K, A'_K; \mathbb{G}_{mK})$ be the class w_K of (1.0.1).

For example, one may take for U and U' the identity components A^o and A'^o of A and A'. One thereby obtains a canonical biextension W^o, or W, of (A^o, A'^o) by $\mathbb{G}_{mS}$, giving rise to a class

$$w = w^o \in \mathrm{Biext}^1(A^o, A'^o; \mathbb{G}_{mS}), \tag{1.4.4}$$

with which we shall work chiefly. Evidently, the restrictions to (A^o, A'^o) of the classes w in (1.4.3), for the various pairs (U, U'), are all equal to w^o in (1.4.4).

Proposition 1.5. Consider the special fiber W_s of the canonical biextension W of (A^o, A'^o) by $\mathbb{G}_{mS}$ in (1.4.4), so that W_s is a biextension of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$. This biextension is separating (VIII 4.11).

Indeed, choose an ample invertible sheaf $\mathbb{L}$ on A^o. Then $\mathbb{L}_\eta$ defines a homomorphism (cf. 1.0.5)

$$A_K \longrightarrow A'_K,$$

which extends to a homomorphism $A \longrightarrow A'$, and hence gives

$$\psi : A^o \longrightarrow A'^o.$$

We may consider the biextension $W_{\mathbb{L}}$ of (A^o, A^o) by $\mathbb{G}_{mS}$ obtained by pulling back W along $(\mathrm{id}_{A^o}, \psi)$. Its restriction to the generic fibers is precisely the biextension $\delta(\mathbb{L}_\eta)$ of VIII 4.12. Hence, by VIII 7.4, $W_{\mathbb{L}}$ is associated with the birigidified invertible sheaf $\delta(\mathbb{L})$, and consequently $(W_{\mathbb{L}})_s$ is defined by $\delta(\mathbb{L}_s)$. On the other hand, it may also be described as the pullback of W_s by the morphisms

$$\mathrm{id}_{A^o_o} : A^o_o \longrightarrow A^o_o, \qquad \psi_o : A^o_o \longrightarrow A'^o_o.$$

Since $\mathbb{L}$, and hence $\mathbb{L}_s$, is ample, VIII 4.13 shows that $W_{\mathbb{L}_s}$ is separating, in particular separating on the left. A fortiori W_s is therefore separating on the left. By symmetry it is also separating on the right, which proves 1.5.

We shall substantially sharpen 1.5 when A^o_o has unipotent rank zero (5.4).

2. Fixed and toric parts of $T_\ell(A_K)$. Criteria for good reduction. Orthogonality theorem for $\ell \neq p$.

9. The printed formula, confirmed in a direct high-resolution check, reuses Φ'_o for the first subgroup and writes $\Psi'_o \subset \Psi_o$ for the second, colliding with the established notation for the component group of A'. The neutral local names F_o, F'_o above make the intended two subgroups unambiguous without changing the assertion.

2.1. Like every commutative algebraic group over k, A_o has a largest subtorus, whose formation commutes with every extension of the base field [SGA 3 XVII 7.2.1, XII 1.12]:

$$T_o \subset A_o. \tag{2.1.1}$$

When k is perfect, T_o is characterized by the condition that A_o^o/T_o be an extension of an abelian scheme B_o over k by a smooth connected unipotent group [25]. The latter necessarily has a composition series all of whose quotients are isomorphic to $\mathbb{G}_{ak}$ (SGA 3 XVII 4.1.3), and one has an exact sequence

$$0 \longrightarrow U_o \longrightarrow A_o^o/T_o \longrightarrow B_o \longrightarrow 0. \tag{2.1.2}$$

In any case, whether or not k is perfect, if A_o^o is an extension of an abelian scheme B_o by a smooth connected affine scheme L_o,

$$0 \longrightarrow L_o \longrightarrow A_o^o \longrightarrow B_o \longrightarrow 0, \tag{2.1.3}$$

this extension structure is known to be unique [25]. Moreover, $T_o \subset L_o$, and T_o is the largest subtorus of L_o. Thus one has the exact sequence (2.1.2), with

$$U_o = L_o/T_o. \tag{2.1.4}$$

2.1.5. Let us note that A_o^o has unipotent rank zero (that is, after passage to an algebraic closure of k, it has no subgroup isomorphic to $\mathbb{G}_a$) if and only if A_o^o/T_o is an abelian scheme B_o. In that case the preceding hypotheses apply with $U_o = 0$.

Let ℓ be a prime different from the characteristic of k, so that A_o^o is ℓ-divisible. The inclusion (2.1.1) then gives an exact sequence of pro-group schemes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & T_\ell(T_o) & \longrightarrow & T_\ell(A_o^o) & \longrightarrow & T_\ell(A_o^o/T_o) & \longrightarrow & 0 \\ & & & & \downarrow\wr & & \downarrow\wr & & \\ & & & & T_\ell(A_o) & & T_\ell(A_o/T_o) & & \end{array} , \tag{2.1.6}$$

whose terms may also be interpreted as twisted constant free ℓ-adic sheaves. When an exact sequence (2.1.2) is available—for example, when k is perfect or when A_o^o has unipotent rank zero—one has

$$T_\ell(A_o^o/T_o) \xrightarrow{\sim} T_\ell(B_o), \tag{2.1.7}$$

so that the exact sequence (2.1.6) becomes

$$0 \longrightarrow T_\ell(T_o) \longrightarrow T_\ell(A_o^o) \longrightarrow T_\ell(B_o) \longrightarrow 0. \tag{2.1.8}$$

Put

$$n = \dim A_K = \dim A_o, \qquad \mu = \dim T_o, \qquad \lambda = \dim U_o, \qquad \alpha = \dim B_o. \tag{2.1.9}$$

Then

$$n = \mu + \lambda + \alpha, \tag{2.1.10}$$

and

$$\begin{cases} \operatorname{rank} T_\ell(T_o) = \mu, & \operatorname{rank} T_\ell(A_o^o/T_o) = \operatorname{rank} T_\ell(B_o) = 2\alpha, \\ \operatorname{rank} T_\ell(A_o^o) = \mu + 2\alpha. \end{cases} \tag{2.1.11}$$

The second relation is a well-known theorem of Weil [5, 15].

2.1.12. The integers n, μ, λ, α are plainly defined, after passing to an algebraic closure of k, even without assuming the existence of an exact sequence (2.1.2). They are respectively called the dimension of A_o, the reductive rank of A_o (equal to the dimension of T_o), the unipotent rank of A_o, and the abelian rank of A_o. They are still related by (2.1.10) and (2.1.11), with the term $\operatorname{rank} T_\ell(B_o)$ omitted from the latter.

2.1.13. Suppose that k has characteristic $p > 0$ and that A_o^o is p-divisible, or equivalently that it has unipotent rank zero (cf. 2.1.5). One should then also consider the pro-group $T_p(A_o^o)$, which is a Barsotti–Tate group (or a p-divisible group, in the terminology of [33]). The exact sequence (2.1.8) remains valid for $\ell = p$ and is then an exact sequence of Barsotti–Tate groups. Here $\lambda = 0$, and hence

$$n = \mu + \alpha. \tag{2.1.13.1}$$

The formulas (2.1.11) also remain valid, provided that rank is interpreted as the rank (or "height") of the Barsotti–Tate groups in question.

2.1.14. For ${A'}_o^o$ we likewise introduce the invariants

$$T'_o \subset {A'}_o^o, \qquad U'_o, \qquad B'_o,$$

where T'_o is the toric part of ${A'}_o^o$, B'_o its abelian part (defined in particular when k is perfect or when ${A'}_o^o$ has unipotent rank zero), and U'_o its unipotent part (defined, for example, when k is perfect, or trivially when ${A'}_o^o$ has unipotent rank zero). There is a corresponding exact sequence of the form (2.1.6), or (2.1.8), which we shall not write again. One could introduce integers μ', λ', α' for ${A'}_o^o$ as in (2.1.9), but we shall see below (2.2.7) that they are the same as those for A_o^o; this spares us from introducing new notation for them.

2.2. We propose to make more precise the relations between $T_\ell(A_K)$ and $T_\ell(A_o) = T_\ell(A_o^o)$ by bringing $T_\ell(A^o)$ into play. First recall the following.

Lemma 2.2.1. Let $m > 0$ be an integer. Then the following conditions are equivalent:
(i) $m\,\mathrm{id}_{A^o}$ is flat.
(ii) $m\,\mathrm{id}_{A^o}$ is surjective.
(iii) $m\,\mathrm{id}_{A^o}$ is quasi-finite.
(iv) m is prime to the characteristic of k, or A_o^o has unipotent rank zero (cf. 2.1.5).
Moreover, if these conditions hold for m, they hold for m^ν for every $\nu > 0$; the groups

$$ {}_{m^\nu}A^o \underset{\text{dfn}}{=} \ker(m^\nu\,\mathrm{id}_{A^o})$$

are quasi-finite, separated, and flat over S, and, as ν varies, they form an inverse system of group schemes whose transition morphisms

$$x \longmapsto m^{\nu'-\nu}x : {}_{m^{\nu'}}A^o \longrightarrow {}_{m^\nu}A^o \qquad (\nu' \geq \nu)$$

are faithfully flat.

Indeed, the fiberwise criterion for flatness (EGA IV 11.3.5) shows that (i), like (ii) and (iii), is a condition on the fibers; these three conditions are equivalent by SGA 3 VI$_A$ 5.6, since the fibers in question are smooth and connected. In the form (ii), a classical theorem of Weil [5, 15] shows that the conditions hold on the generic fiber, which is an abelian scheme; it is therefore enough to state them on the special fiber. Their equivalence with (iv) is then also well known from the structure theory. In any of these forms, stability of the condition under $m \mapsto m^\nu$ is immediate. It follows that the ${}_{m^\nu}A^o$ are quasi-finite flat group schemes over S, and they are plainly separated over S because A^o is. Finally, faithful flatness of the transition morphisms—equivalently, the fact

that they are epimorphisms for the fppf topology—follows formally from the fact that $m\,\mathrm{id}_{A^o}$ is faithfully flat, hence an epimorphism for that topology.

2.2.2. Consequently, we shall take m to be a prime ℓ satisfying the conditions of 2.2.1. We then denote by

$$T_\ell(A^o) = ({}_{\ell^\nu}A^o)_{\nu>0} \tag{2.2.2.1}$$

the inverse system described in 2.2.1. Under the two base changes $\eta \longrightarrow S$ and $s \longrightarrow S$, it recovers the inverse systems $T_\ell(A_K)$ and $T_\ell(A^o_o) = T_\ell(A_o)$ considered in 1.0 and 2.1 respectively.

2.2.3. Suppose now that S is henselian. Every quasi-finite separated scheme X over S then has a canonical decomposition as a disjoint union

$$X = X^f \coprod X', \tag{2.2.3.1}$$

where X^f is finite over S and $X'_o = \varnothing$, that is, X' is confined to its generic fiber X'_η; thus X^f has the same special fiber as X. This decomposition is plainly functorial in X. In particular, if X is a group scheme over S, then X^f is a subgroup scheme. If X is flat, respectively étale, over S, then so is X^f.

We apply this to the components ${}_{\ell^\nu}A^o$ of $T_\ell(A^o)$. This gives a projective subsystem of groups

$$\left(({}_{\ell^\nu}A^o)^f\right)_{\nu>0} \underset{\text{dfn}}{=} T_\ell(A^o)^f \subset T_\ell(A^o), \tag{2.2.3.2}$$

called the fixed part of the Tate module $T_\ell(A^o)$. By construction, there are canonical isomorphisms, functorial in A,

$$T_\ell(A^o)^f \times_S s \xrightarrow{\sim} T_\ell(A^o) \times_S s \xrightarrow{\sim} T_\ell(A^o_o). \tag{2.2.3.3}$$

On the other hand, the generic fiber of $T_\ell(A^o)^f$ defines a canonical sub-pro-group of $T_\ell(A_K)$, again called the "fixed part" of this Tate module:

$$T_\ell(A_K)^f \subset T_\ell(A_K). \tag{2.2.3.4}$$

2.2.4. When $\ell \neq p = \mathrm{char}\, k$, the fixed part $T_\ell(A^o)^f$ may be interpreted as a twisted constant ℓ-adic sheaf on S (the $({}_{\ell^\nu}A^o)^f$ being finite étale over S). Since the functor

$$X \longmapsto X_o = X \times_S s$$

from finite étale schemes over S to finite étale schemes over k is an equivalence of categories when S is henselian, there is an analogous equivalence for the categories of twisted constant ℓ-adic sheaves over S and over k. Hence (2.2.3.3), which gives the special fiber of $T_\ell(A^o)^f$, characterizes this fixed part up to unique isomorphism. On the other hand, as an ℓ-adic subsheaf of $T_\ell(A^o)$ it is plainly characterized by its generic fiber (2.2.3.4), or equivalently by the corresponding $\mathbb{Z}_\ell$-submodule, stable under $\pi = \mathrm{Gal}(\overline{K}/K)$:

$$T_\ell(A_K)^f(\overline{K}) \underset{\text{dfn}}{=} T_\ell(A(\overline{K}))^f \subset T_\ell(A(\overline{K})). \tag{2.2.4.1}$$

In fact, one has the following result.

Proposition 2.2.5. Suppose, as above, that $\ell \neq p$ and that S is henselian. Then the submodule (2.2.4.1) of $T_\ell(A(\overline{K})) = T_\ell(A_K)(\overline{K})$ corresponding to the fixed part $T_\ell(A_K)^f$ in (2.2.3.4) is the submodule $T_\ell(A(\overline{K}))^I$ of invariants under the inertia group $I \subset \pi$.

This follows, by passage to the inverse limit, from the analogous relation valid for every integer $m > 0$ prime to $p = \mathrm{char}\, k$:

$$({}_mA)^f(\overline{K}) = ({}_mA(\overline{K}))^I. \tag{2.2.5.1}$$

To prove this relation, introduce the strict localization $\widetilde{S}$ of S, whose fraction field $\widetilde{K}$ is the maximal unramified subextension of $\overline{K}/K$. Since ${}_mA)^f$ is finite étale over S, its points with values in $\overline{K}$ already have values in $\widetilde{K}$. Thus the left-hand side of (2.2.5.1) is $\mathrm{Hom}_S(\widetilde{S}, ({}_mA)^f)$, whereas the right-hand side is, by definition of $\widetilde{K}$, equal to ${}_mA(\widetilde{K})$. By the definition of $({}_mA)^f$, and since $\widetilde{S}$ is an inverse limit of finite étale S-schemes S_i, the former is also equal to

$$\mathrm{Hom}_S(\widetilde{S}, {}_mA) = {}_m\,\mathrm{Hom}_S(\widetilde{S}, A).$$

By the defining property of the Néron model, $\mathrm{Hom}_S(\widetilde{S}, A) = A(\widetilde{K})$, which proves (2.2.5.1).

To deduce 2.2.5, observe that (2.2.5.1) gives

$$(*) \qquad T_\ell(A(\overline{K}))^I = \varprojlim_\nu (\ell^\nu A)^f(\overline{K}) \supset \varprojlim_\nu (\ell^\nu A^o)^f(\overline{K}).$$

If (x_ν) belongs to the rightmost term, then for every $\nu \geq 0$, x_ν is a section of A that is infinitely divisible by ℓ. Its value at s is therefore an element of $A_o(k)$ infinitely divisible by ℓ, and consequently lies in $A_o^o(k)$. This proves equality of the last two terms of (*) and completes the proof of 2.2.5.

Proposition 2.2.6. Let C_K be a second abelian scheme over K, with Néron model C, and consider a homomorphism $u_K : C_K \longrightarrow A_K$, which therefore extends to a homomorphism $u : C \longrightarrow A$. Suppose that u_K is an isogeny. Then, for every prime ℓ, $T_\ell(u_K) : T_\ell(C_K) \longrightarrow T_\ell(A_K)$ induces an isogeny of fixed parts

$$(2.2.6.1) \qquad T_\ell(C_K)^f \longrightarrow T_\ell(A_K)^f,$$

and likewise the map induced by $u_o : C_o \longrightarrow A_o$

$$(2.2.6.2) \qquad T_\ell(u_o^o) : T_\ell(C_o^o) \longrightarrow T_\ell(A_o^o)$$

is an isogeny. Finally, $u_o : C_o \longrightarrow A_o$ induces an isogeny from the largest subtorus of C_o to the largest subtorus T_o of A_o; and over an extension k'/k on which the "abelian parts" of C_o and A_o are defined (for example, over the perfect closure of k), u_o induces an isogeny between the abelian parts of $C_{ok'}$ and $A_{ok'}$.

This statement follows formally from the fact that $T_\ell(A_K)^f$, $T_\ell(A_o^o)$, the largest subtorus of A_o, and the "abelian part" of $A_{ok'}$ are additive functors of the varying A_K, and from the fact that u_K is an isogeny precisely in the sense that there is a morphism $v_K : A_K \longrightarrow C_K$ such that

$$u_K v_K = n\,\mathrm{id}, \qquad v_K u_K = n\,\mathrm{id}$$

for some integer $n > 0$.

Corollary 2.2.7. If A_K and C_K are isogenous abelian schemes over K, then the corresponding invariants μ, λ, α in (2.1.11) are the same for A_K and C_K. In particular, they are the same for A_K and its dual A'_K.

Remarks 2.2.8. In the two preceding statements, 2.2.6 and 2.2.7, we assumed that S is henselian. It is immediate, however, that 2.2.7, as well as the assertions of 2.2.6 concerning the properties of the induced homomorphism $u_o : C_o \longrightarrow A_o$, holds without this hypothesis. Indeed, one reduces to the case already treated by the base change $S^h \longrightarrow S$, where S^h is the henselization of S, using the fact—immediate from the definitions—that formation of Néron models commutes with this base change.

Corollary 2.2.9. (Good reduction.) Let S be a trait, let A_K be an abelian scheme over its fraction field K, let A be the Néron model of A_K, and let ℓ be a prime different from $\mathrm{char}\, k$. The following conditions are equivalent.

(i) There exists an abelian scheme over S whose generic fiber is isomorphic to A_K.

(ii) A is an abelian scheme.
(ii^o) A^o is an abelian scheme.
(iii) A_o is an abelian scheme.
(iii^o) A^o_o is an abelian scheme, that is, with the notation of (2.1.11), $\lambda = \mu = 0$.
(iv) A is proper over S.
(iv^o) A^o is proper over S.
(v) $T_\ell(A_K)$ is "unramified over S", that is, it extends to a twisted constant ℓ-adic sheaf over S, or equivalently the inertia group I acts trivially on $T_\ell(A_K(\overline{K}))$.
(v bis) For every integer $\nu \geq 0$, ${}_{\ell^\nu}A_K$ is "unramified over S", that is, it extends to an étale covering of S, or equivalently the inertia group I acts trivially on ${}_{\ell^\nu}A_K(\overline{K})$.
Finally, an abelian scheme over S as in (i) is a Néron model of A_K, and in particular is determined up to unique isomorphism.

2.2.9.1. We say that A_K has good reduction over S if it satisfies condition (i) above. Criterion (v), or (v bis), is known as the Néron–Ogg–Shafarevich criterion for good reduction.

Let us prove 2.2.9. It is well known (cf., for example, [5]) that an abelian scheme over S satisfies the universal property of a Néron model; this reduces to the fact that a rational section of an abelian scheme over a regular base is defined everywhere. This proves the equivalence $(i) \iff (ii) \iff (ii^o)$, as well as the last assertion of 2.2.9. These conditions plainly imply (iii), (iv), and (iv^o), each of which implies (iii^o). Let us prove that (iii^o)$\Longrightarrow$(ii^o), which will establish the equivalence of (i) through (iv^o). This follows from the fact that a smooth group scheme over S whose fibers are abelian schemes is proper over S (and hence is an abelian scheme), by EGA IV 15.7.10.

On the other hand, in view of 2.2.5 and the fact that

$$T_\ell(A_K(\overline{K}))/T_\ell(A_K(\overline{K}))^I$$

is plainly a torsion-free $\mathbb{Z}_\ell$-module, condition (v) says that $T_\ell(A_K(\overline{K}))$ and its fixed part have the same rank. By (2.1.11) and (2.2.3.3), this is the condition

$$\mu + 2\alpha = 2(\lambda + \mu + \alpha),$$

that is, $2\lambda + \mu = 0$, or $\lambda = \mu = 0$, which is precisely (iii^o). Finally, the equivalence of (v) and (v bis) is clear from the definitions. This completes the proof of 2.2.9.

2.3. Suppose once again that S is henselian and that $\ell \neq p = \operatorname{char} k$. We now use the inclusion

$$T_\ell(T_o) \hookrightarrow T_\ell(A^o_o) \tag{2.3.1}$$

of (2.1.6), together with the equivalence of categories recalled in 2.2.4. We obtain a canonical twisted constant ℓ-adic subsheaf of $T_\ell(A^o)^f$, called the toric part of $T_\ell(A^o)$,

$$T_\ell(A^o)^t \hookrightarrow T_\ell(A^o)^f \hookrightarrow T_\ell(A^o), \tag{2.3.2}$$

characterized by the condition that its special fiber is the inclusion (2.3.1):

$$T_\ell(A^o)^t \times_S s = T_\ell(T_o) \hookrightarrow T_\ell(A^o_o). \tag{2.3.3}$$

Later we shall give an interpretation of the toric part of $T_\ell(A^o)$ as the Tate module $T_\ell(T)$ of a torus T over S lifting the torus T_o over k. That interpretation will allow the definition of the toric part to be extended to the case $\ell = p$. For the moment, and to fix ideas, we confine ourselves to the case $\ell \neq p$. In this case the toric part of $T_\ell(A^o)$ may be described by the corresponding $\mathbb{Z}_\ell$-submodule of $T_\ell(A(\overline{K}))$:

$$T_\ell(A(\overline{K}))^t \hookrightarrow T_\ell(A(\overline{K}))^f \hookrightarrow T_\ell(A(\overline{K})). \tag{2.3.4}$$

It is stable under the action of π; its first term is defined as $T_\ell(A^o)^t(\overline{K})$ and is, as one would expect, again called the toric submodule of the Tate module $T_\ell(A(\overline{K}))$. Thus this Tate module carries the canonical three-step filtration (2.3.4). Applying the same considerations to the dual abelian scheme A'_K, we obtain the analogous three-step filtration

$$T_\ell(A'(\overline{K}))^t \hookrightarrow T_\ell(A'(\overline{K}))^f \hookrightarrow T_\ell(A'(\overline{K})). \tag{2.3.5}$$

The middle terms in (2.3.4) and (2.3.5), the "fixed parts," are described in terms of the π-module structures of the Tate modules of $A(\overline{K})$ and $A'(\overline{K})$ as their submodules of invariants under the inertia group I (2.2.5). The toric parts are then completely described in terms of the fixed parts and the pairing φ of (1.0.2), by the following theorem (a special case of which already appears in [11]).

Theorem 2.4. (Orthogonality theorem.) Let S be a henselian trait, let A_K be an abelian scheme over its fraction field K, let A'_K be the dual abelian scheme, and let ℓ be a prime different from the residual characteristic of S. The Tate modules $T_\ell(A(\overline{K}))$ and $T_\ell(A'(\overline{K}))$ then carry the canonical filtrations (2.3.4) and (2.3.5), respectively, by the "fixed part" and the "toric part." The toric part $T_\ell(A(\overline{K}))^t$ is the intersection of the fixed part $T_\ell(A(\overline{K}))^f$ with the orthogonal of the fixed part $T_\ell(A'(\overline{K}))^f = T_\ell(A'(\overline{K}))^I$ with respect to the canonical pairing φ of (1.0.3). Symmetrically, the toric part $T_\ell(A'(\overline{K}))^t$ is the intersection of $T_\ell(A'(\overline{K}))^f$ with the orthogonal of the fixed part $T_\ell(A(\overline{K}))^f = T_\ell(A(\overline{K}))^I$. (Here the superscript I denotes the submodule of invariants under the action of the inertia group I.)

Proof. a) Let us prove that $T_\ell(A(\overline{K}))^t$ is orthogonal to $T_\ell(A'(\overline{K}))^f$. To do so, we interpret the restriction of φ to the fixed parts of the Tate modules using the canonical biextension $W^o = W$ of (1.4.4). This is the canonical extension of the biextension W_K of (A_K, A'_K) by $\mathbb{G}_{mK}$ to a biextension of (A^o, A'^o) by $\mathbb{G}_{mS}$. By VIII 2.2, W gives rise to a pairing

$$\varphi_W : T_\ell(A^o) \times T_\ell(A'^o) \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.1}$$

from which the pairing φ of (1.0.3) is obtained by passing to $\overline{K}$-valued points. It follows that the restriction of φ to the fixed parts considered in 2.4 is obtained, by passing to $\overline{K}$-valued points, from the pairing

$$T_\ell(A^o)^f \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.2}$$

the restriction of (2.4.1). Since the two factors on the left of (2.4.2) are now twisted constant ℓ-adic sheaves, this pairing may be studied on the special fibers. The fibers of the two factors are precisely the Tate modules $T_\ell(A^o_o)$ and $T_\ell(A'^o_o)$ (2.2.3.3), and the pairing induced on special fibers by (2.4.2) is

$$T_\ell(A^o_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{2.4.3}$$

obtained from the biextension W_o of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$, the restriction of W to the special fibers.

Recall from 2.3 that the toric part of $T_\ell(A(\overline{K}))$ is obtained by passing to $\overline{K}$-valued points in the twisted constant ℓ-adic subsheaf $T_\ell(A^o)^t$ of $T_\ell(A^o)^f$. Thus the orthogonality relation to be proved says that the pairing induced by (2.4.2),

$$T_\ell(A^o)^t \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.4}$$

is identically zero. It is enough to verify this for the corresponding pairing on the special fibers, which by the definition of the toric part is

$$T_\ell(T_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{2.4.5}$$

induced by (2.4.3). By VIII 4.10 this pairing is necessarily zero, which proves the asserted orthogonality.

b) Let us prove that the inclusion

$$T_\ell(A(\overline{K}))^t \subset T_\ell(A(\overline{K}))^f \cap \big(T_\ell(A'(\overline{K}))^f\big)^\perp \tag{2.4.6}$$

is an equality, where $\perp$ denotes the orthogonal. Equivalently, in the pairing (2.4.2), the annihilator of the second factor—which, as we have just seen, contains the toric part $T_\ell(A^o)^t$—is equal to that toric part. Since these are again relations between twisted constant ℓ-adic sheaves, it is enough to check the assertion on the special fibers: namely, that the inclusion

$$T_\ell(T_o) \subset T_\ell(A'^o_o)^\perp \tag{2.4.7}$$

is an equality. In other words, we must verify that the biextension W_o of (A^o_o, A'^o_o) is nondegenerate on the left (VIII 4.11). But this is precisely what we already observed in 1.5, as a consequence of a result from Raynaud's thesis.

2.5. The orthogonality theorem has an equivalent formulation using a polarization ξ of A_K and the alternating bilinear form (1.0.8), compatible with the operations of π,

$$\varphi_\xi : T_\ell(A(\overline{K})) \times T_\ell(A(\overline{K})) \longrightarrow T_\ell(\overline{K}^*), \tag{2.5.1}$$

obtained from the form φ of (1.0.3) by transporting structure in the second argument along the isogeny of $\mathbb{Z}_\ell$-modules

$$T_\ell(A(\overline{K})) \longrightarrow T_\ell(A'(\overline{K}))$$

induced by the isogeny $\psi_\xi : A_K \longrightarrow A'_K$ defined by ξ. The equality in (2.4.6) then takes the simple form

$$W = V \cap V^\perp, \tag{2.5.2}$$

where we have set

$$\begin{cases} U = T_\ell(A(\overline{K})), \qquad V = U^I = \text{the "fixed part" of } T_\ell(A(\overline{K})), \\ \qquad\qquad W = \text{the toric part of } T_\ell(A(\overline{K})), \end{cases} \tag{2.5.3}$$

and $\perp$ denotes the orthogonal with respect to the polarization form (2.5.1). These modules give the following diagram of inclusions:

$$\begin{array}{ccccccc} & & & & V & & & & \\ & & & \overset{2\lambda}{\swarrow} & & \overset{2\alpha}{\nwarrow} & & & \\ U & \overset{\mu}{\hookleftarrow} & W^\perp = V + V^\perp & & & & W = V \cap V^\perp & \overset{\mu}{\hookleftarrow} & 0 \\ & & & \underset{2\alpha}{\nwarrow} & & \underset{2\lambda}{\swarrow} & & & \\ & & & & V^\perp & & & & \end{array} \tag{2.5.4}$$

where the integers 2α, 2λ, and μ are the ranks of the quotients of consecutive modules in the diagram, with the notation of (2.1.9). (That W has rank μ is the definition of μ; that V/W has rank 2α follows from the second relation in (2.1.11) together with the exact sequence (2.1.6). Hence, by orthogonality, $\operatorname{rank}(U/W^\perp) = \mu$ and $\operatorname{rank}(W^\perp/V^\perp) = 2\alpha$. Finally, the formulas

$$\operatorname{rank}\big(V/(V \cap V^\perp)\big) = \operatorname{rank}(V/W) = 2\alpha \tag{2.5.5}$$

and the dual formula follow from the preceding ones, in view of (2.1.10) and $\operatorname{rank} U = n$.)

Remarks 2.6. a) Here we obtained the orthogonality theorem 2.4 as an easy consequence of the theory of biextensions developed in the preceding sections. The inclusion (2.4.6) can be proved

without using the theory of biextensions, by an arithmetic argument in the style of the proof of the "monodromy theorem" given in Exposé I. One reduces to a situation in which S has finite residue field, where the orthogonality result follows from a simple consideration of weights for the Frobenius operation (using A. Weil's result on the eigenvalues of that Frobenius). There is no need to invoke resolution of singularities here: one may state the desired result for group schemes A over schemes S under sufficiently general conditions. Unfortunately, it does not appear possible to obtain by this method equality in the preceding inclusion. That equality will apparently be used in an essential way in the proof of the "semistable reduction theorem" in the next section.

b) The orthogonality theorem 2.4 allows the most important properties of the Néron model of an abelian variety A_K to be expressed in terms of the representation of the inertia group I on the Tate module $T_\ell(A(\overline{K}))$, where ℓ is a fixed prime different from the residual characteristic. (The case $\ell = p$ will be considered separately later, in section 5.) As a first statement of this kind, though one that does not use the orthogonality theorem, recall the good-reduction criterion 2.2.9. We shall give many further examples below, beginning in the next section (3.5).

3. The case of semistable reduction. The semistable reduction theorem.

We first give an auxiliary result in the style of SGA 3.

Proposition 3.1. Let S be a scheme, let G be a commutative S-group scheme that is flat, separated, and of finite presentation over S, let H be a quasi-separated group scheme locally of finite type over S, let $u : G \longrightarrow H$ be a homomorphism of S-group schemes, and put $K = \ker u$. For every $s \in S$, let $\bar{s}$ denote the spectrum of an algebraic closure of $k(s)$, and denote by $\mu(s)$ (respectively, $\alpha(s)$) the reductive rank (respectively, the abelian rank) of K_s, that is, the dimension of the largest subtorus (respectively, of the largest abelian-variety quotient) of $(K_{\bar{s}})^o_{\mathrm{red}}$. Let $s \in S$, let $t \in S$ be a generalization of s, and let p be the characteristic exponent of $k(s)$.

a) For every integer $n > 0$ prime to p, ${}_nK$ is etale, of finite type, and separated over S in a neighborhood of s, and one has the divisibility relation

$$\mathrm{rank}({}_nK_s) \mid \mathrm{rank}({}_nK_t). \tag{3.1.1}$$

b) One has

$$\mu(s) + 2\alpha(s) \leq \mu(t) + 2\alpha(t). \tag{3.1.2}$$

c) If $(K_{\bar{t}})^o_{\mathrm{red}}$ is unipotent, then so is $(K_{\bar{s}})^o_{\mathrm{red}}$.

d) Suppose that G_s has unipotent rank zero, that H is smooth over S, and that u_t is an isogeny. Then u is flat and quasi-finite over an open neighborhood U of s; a fortiori, $K|_U$ is flat and quasi-finite.

e) Under the hypotheses of d), suppose in addition that u_t is a monomorphism (and hence an isomorphism from G_t onto an open subgroup of H_t). Then u_s is an isomorphism from G_s onto an open subgroup of H_s. If, moreover, S is irreducible with generic point t, then there is an open neighborhood U of s such that u induces an isomorphism from $G|_U$ onto an open subgroup of $H|_U$.

Proof.

a) We may suppose that n is invertible on S. Then $n\,\mathrm{id}_G$ is etale on every fibre (SGA 3, XV, 1.3); since G is flat and of finite presentation over S, the fibrewise criterion for etaleness (EGA IV, 17.8.2) shows that $n\,\mathrm{id}_G$ is etale. Its kernel ${}_nG$ is therefore etale; it is also separated and of finite presentation over S, since G is. Similarly, ${}_nH$ is locally of finite type, quasi-separated, and formally unramified over S,[10] so its unit section is an open immersion (EGA IV, 17.4.2 b')). Consequently, the kernel of

$${}_nu : {}_nG \longrightarrow {}_nH$$

10. At least when H is commutative. In the general case, ${}_nH$ is still unramified over S along its unit section, which is all that is needed here.

is an open subscheme of ${}_nG$, hence is etale over S; it is quasi-compact in ${}_nG$ and therefore of finite presentation over S, and it is separated over S because G is. This kernel is plainly ${}_nK$.

Formula (3.1.1) follows from these properties of ${}_nK$. Indeed, after a base change we may reduce to the case where S is henselian local with closed point s, and then use the decomposition of ${}_nK$ into ${}_nK^f$ and a scheme with empty special fibre (cf. 2.2.3). More precisely, the same argument gives a monomorphism on geometric fibres

$$ {}_nK(\bar{s}) \hookrightarrow {}_nK(\bar{t}). $$

b) Choose an integer $n > 0$ prime to the characteristic exponent of $k(s)$ and to the ranks of the quotients of the geometric fibres of K at s and t by their reduced identity components. The two sides of (3.1.1) are then obtained by raising n, respectively, to the powers whose exponents are the two sides of (3.1.2), as follows from the known structure of commutative algebraic groups. Thus b) follows from a).

c) The hypothesis that $(K_{\bar{t}})^o_{\mathrm{red}}$ is unipotent is equivalent to the equalities $\mu(t) = \alpha(t) = 0$, and likewise for $K_{\bar{s}}$. Assertion c) therefore follows formally from b).

d) Since u_t is an isogeny, K_t is finite. We are therefore under the hypotheses of c), and $(K_{\bar{s}})^o_{\mathrm{red}}$ is unipotent. As G_s has unipotent rank zero, it follows that K_s is finite. Moreover, the fact that u_t is an isogeny gives $\dim G_t = \dim H_t$. Since G and H are flat and locally of finite presentation, one has

$$ \dim G_s = \dim G_t, \qquad \dim H_s = \dim H_t \quad (\text{SGA 3, VI}_B\text{, 4.3}), $$

and hence $\dim G_s = \dim H_s$. Since $\dim \ker u_s = 0$, it follows that $\dim H_s = \dim u_s(G_s)$ (SGA 3, VI$_B$, 1.2), which, because H_s is smooth, is equivalent to u_s being flat (SGA 3, VI$_A$, 5.6). There is therefore an open neighborhood U of s such that u is flat and quasi-finite at the points of the unit section over U (SGA 3, VI$_B$, 2.2(i) and 2.5(ii)). As for u_s, it follows that $u_{s'}$ is flat for every $s' \in U$; hence $u|_{G_U}$ is flat and quasi-finite, since it has these properties on every fibre.

e) For the first assertion, a base change immediately reduces us to the case where S is integral with generic point t, and hence to proving the second assertion of e). That assertion follows at once from d) and the following lemma.

Lemma 3.1.3. Let S be a scheme, let K be a quasi-finite S-scheme of finite presentation that is separated and flat over S, let $s \in S$, and let $t \in S$ be a generalization of s. Then

$$ \operatorname{rank} K_s \leq \operatorname{rank} K_t. $$

If $\operatorname{rank} K_\eta \leq 1$ for every maximal point η of S, then $K \longrightarrow S$ is an open immersion (and hence an isomorphism if it admits a section).

For the first assertion, strict localization reduces us to the case where S is strictly local. Decomposing K as the disjoint union of a finite scheme K^f and a scheme with empty special fibre, we obtain

$$ \operatorname{rank} K_s = \operatorname{rank} K^f_s = \operatorname{rank} K^f_t \leq \operatorname{rank} K_t. $$

The second equality follows because K^f is flat, finite, and of finite presentation over S, and therefore locally free over S.

For the second assertion, the preceding argument already shows that $\operatorname{rank} K_s \leq 1$ for every $s \in S$. Thus K_s is empty or isomorphic to $\operatorname{Spec} k(s)$. Consequently, $K \longrightarrow S$ is a monomorphism (SGA 3, VI$_B$, 2.11), and, being flat and of finite presentation, it is an open immersion (EGA IV, 17.9.1). This proves the lemma.

Remark 3.1.4. Part 3.1 a), and the remainder of Proposition 3.1, give no information about ${}_nK_s$ when n is a power of p (for $p > 1$). At least when G is smooth (and S is locally noetherian or H is of finite presentation over S), essentially the same method as in the proof of a) gives such

information. Indeed, reducing to the case where S is complete local noetherian and using SGA 3, XV, 1.6 c), X, 3.2, IX, 3.6, and IX, 5.2, one obtains

(3.1.4.1) $$\mathrm{rank}_{\mathrm{mult}}({}_nK_s) \mid \mathrm{rank}_{\mathrm{mult}}({}_nK_t),$$

where $\mathrm{rank}_{\mathrm{mult}}$ denotes the rank of the largest subgroup of multiplicative type of the commutative algebraic group under consideration (SGA 3, XVII, 7.2.1). Notice that a torsion algebraic group of multiplicative type is finite. Formula (3.1.4.1) is valid for every integer $n \geq 1$. It follows from SGA 3, XVII, 4.6.1(vi) that, in c), if K_t is unipotent (SGA 3, VII, 1.1), then K_s is unipotent as well.

Proposition 3.2. Let S be a regular integral locally noetherian scheme of dimension 1, let K be its field of rational functions, let A_K be an abelian scheme over K, and let A be its Néron model over S, which is a smooth separated group scheme over S (and is quasi-projective over S if S is noetherian [23, VIII, 2, 3°]). The following conditions are equivalent:

(i) For every $s \in S$, A_s^o has unipotent rank zero; that is, it is an extension of an abelian scheme by a torus (2.1.5).

(ii) There exists a smooth separated group scheme G of finite type over S, extending A_K, whose fibres have unipotent rank zero.

Moreover, if G is as in (ii), then the unique morphism $G \longrightarrow A$ inducing the given isomorphism on the generic fibres is an isomorphism from G onto an open subgroup of A; it therefore induces an isomorphism of identity components

$$G^o \xrightarrow{\sim} A^o.$$

3.2.1. Let us note at once that the definition of the Néron model A of A_K is the same as in the local case recalled in 1.1. The existence of this Néron model, and the fact that it is separated and of finite type over S, are well known and in fact reduce immediately to the local case. Of course, for every closed point $s \in S$,

$$S' = \mathrm{Spec}(\mathcal{O}_{S,s})$$

is a trait, and $A \times_S S'$ is precisely the Néron model of A_K over S' in the sense considered in 1.1.

Let us prove 3.2. It is trivial that (i) implies (ii), since it suffices to take $G = A$. It remains to prove that if G is as in (ii), then $G \longrightarrow A$ is an open immersion. This is indeed a special case of 3.1 e).

Corollary 3.3. With the notation of 3.2, suppose that the identity components of the fibres of the Néron model have unipotent rank zero. Let S' be a scheme satisfying the same hypotheses as S, let K' be its field of rational functions, let $f : S' \longrightarrow S$ be a dominant morphism, let

$$A_{K'} \simeq A_K \otimes_K K' \simeq \text{the generic fibre of } A \times_S S',$$

let A' be the Néron model of $A_{K'}$, and consider the unique morphism

$$u : A \times_S S' \longrightarrow A'$$

inducing the identity on the generic fibres. Then u is an open immersion and therefore induces an isomorphism on identity components

(3.3.1) $$A^o \times_S S' \xrightarrow{\sim} A'^o.$$

Indeed, it suffices to apply 3.2 over S', taking $G \simeq A \times_S S'$.

3.3.2. Thus the equivalent properties of 3.2 imply a partial "stability" property for Néron models. One should bear in mind that when A_K does not have good reduction, that is, when A is not

an abelian scheme, it is not true that, for every S' as in 3.3 (even for S' finite over S and K'/K separable), the morphism

$$A \times_S S' \longrightarrow A'$$

itself is an isomorphism, or that it induces isomorphisms on the groups of connected components of the fibres (see 11.10 below for the behavior of these groups of connected components under extension of the base). These observations, reinforced by 3.9 below, justify the following terminology, introduced by Mumford (and which also appears to agree with the notions of the same name studied in his book [17]).

Definition 3.4. With the notation of 3.2, the abelian scheme A_K over K is said to have semistable reduction over S if the equivalent conditions (i) and (ii) of 3.2 are satisfied.

3.4.0. Notice that in the form (ii), this condition does not invoke the notion of a Neron model. Moreover, by localization it reduces immediately to the case where S is a trait, as follows for example from 3.2.1. Furthermore, since formation of the Neron model commutes with base change from S to its henselization or its strict henselization, the notion even reduces to the case where S is a strictly local trait.

Proposition 3.5. (Galois criterion for semistable reduction). Suppose that S is a trait. The following conditions on the abelian scheme A_K are equivalent (the notation being that of (2.5.3), applied to the situation deduced from (A_K, S) by passing to the henselization of S):

(i) A_K has semistable reduction over S (3.4).

(ii) With the notation of (2.5.3) and (2.5.4), one has

$$V^\perp \subset V.$$

(iii) For the toric part W of the Tate module one has

$$W = V^\perp.$$

(iv) The action of the inertia group I on $U = T_\ell(A(\overline{K}))$ is unipotent of level 2 (that is, there is an I-stable submodule U' of U such that I acts trivially on U' and on U/U').

We may suppose that S is henselian (3.4.0). With the notation of 2.1.9, (i) means that

$$\lambda = 0.$$

By the rank formulas following (2.5.4), this is equivalent to $V^\perp = V \cap V^\perp$, that is, to (ii). Since $W = V \cap V^\perp$ by the orthogonality theorem 2.4, conditions (ii) and (iii) are equivalent. Moreover, (iv) also says that I acts trivially on $U/U^I = U/V$. By duality this means that it acts trivially on $V^\perp$, that is, that $V^\perp \subset U^I = V$, which is (ii). This completes the proof.

Remarks 3.5.1. Conditions (ii) and (iv) plainly depend only on the action of I on the $\mathbb{Q}_\ell$-vector space $U \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and could have been formulated in terms of that action. Let us also note that these conditions are manifestly stable under every base change by a morphism of traits $S' \longrightarrow S$. Finally, if (iv) holds for A_K, it plainly holds for every abelian subscheme and every quotient abelian scheme of A_K, and in particular for every abelian scheme isogenous to A_K (for example, for A'_K).

Corollary 3.5.2. Let g be a topological generator of $\mathbb{Z}_\ell(1) = T_\ell(\overline{k}^*)$. The equivalent conditions of 3.5 are also equivalent to the following condition:

(v) The action of the inertia group I on $U = T_\ell(A(\overline{K}))$ factors through the maximal pro-ℓ quotient $I(\ell)$ of I, which is canonically isomorphic to $\mathbb{Z}_\ell(1)$, and the action g_U of the generator g on U satisfies

$$(1 - g_U)^2 = 0. \tag{3.5.3}$$

Indeed, for every continuous unipotent action of a profinite group I on an ℓ-adic vector space, one sees at once that the image of I is a pro-ℓ group. It follows that (v) is equivalent to (iv).

We can now prove the central result of the present expose:

Theorem 3.6. (Semistable reduction theorem). Let S be a connected regular noetherian scheme of dimension 1, let K be the field of rational functions of S, and let A_K be an abelian scheme over K. Then there is a finite Galois extension K' of K such that the abelian scheme $A_{K'} = A_K \otimes_K K'$ has semistable reduction (3.4) over the normalization S' of S in K'.

Notice that S' is connected, regular, noetherian, and of dimension 1, so that the stated conclusion makes sense. To prove 3.6, choose a separable closure $\overline{K}$ of K and write $\overline{K}$ as the filtered union of its finite Galois subextensions K_i. Let S_i be the normalization of S in K_i, and let U be the largest open subset of S over which A_K has good reduction. Since S is connected, noetherian, and one-dimensional, and $U \neq \varnothing$, the complement $T = S - U$ is a finite set of closed points of S, and $\mathcal{O}_{S,t}$ is a discrete valuation ring for every $t \in T$. It is enough to find, for every $t \in T$, an index $i = i(t)$ such that A_{K_i} has semistable reduction at the points of S_i above t: one then takes for K' the compositum of the extensions $K_{i(t)}$, for $t \in T$. Thus we are reduced to the case where S is a trait.

Moreover, since the Galois group permutes the points of the normalization lying above the closed point s of S, it is enough to find K' such that $A_{K'}$ has semistable reduction at *one* point of the normalization S'. Here it is unnecessary at first to require that K'/K be Galois: it is enough to find a finite etale extension K'/K satisfying the preceding condition, since its Galois closure satisfies the same condition. Using criterion 3.5(iv), we see that Theorem 3.6 reduces to the following assertion.

Corollary 3.7. Suppose again that S is a trait. Then there is an open subgroup π' of π such that the action of $I' = I \cap \pi'$ on $U = T_\ell(A(\overline{K}))$ is unipotent of level 2; equivalently, there is an open subgroup I' of I whose action on U is unipotent of level 2.

There is a canonical isomorphism, compatible with the actions of π,

$$(*) \qquad H^1(A_{\overline{K}}, \mathbb{Z}_\ell) \simeq \mathrm{Hom}\big(T_\ell(A(\overline{K})), \mathbb{Z}_\ell\big),$$

which shows that assertion 3.7 is equivalent to the same assertion for the actions of I on the left-hand side of (*). But that assertion was proved in Expose III (using resolution of singularities for excellent schemes of dimension 2).

Corollary 3.8. The equivalent conditions (i)–(iv) of 3.5 are also equivalent to the following condition:

(vi) The action of I on $U = T_\ell(A(\overline{K}))$ is unipotent.

Indeed, to prove that (vi) implies (iv), observe that if a group I acting unipotently on a finite-dimensional vector space E over a field of characteristic zero (here $\mathbb{Q}_\ell$) has a subgroup I' of finite index acting with level m (here $m = 2$), then I itself acts with level m. In fact, since I' has finite index in I, the algebraic group $G' \subset \mathrm{GL}(E)$ that it defines has finite index in the group G defined by I; but G is unipotent and hence connected (in characteristic zero), so $G = G'$. Thus I and I' have the same "level of unipotence."

We also obtain a converse to 3.3:

Corollary 3.9. The notation is that of 3.2. The conditions (i) and (ii) of 3.2 are also equivalent to the following condition:

(iii) For every finite etale extension K' of K, let S' denote the normalization of S in K', and let A' denote the Neron model of $A_{K'} = A \otimes_S K' = A_K \otimes_K K'$. Then the canonical morphism

$$A \times_S S' \longrightarrow A'$$

induces an isomorphism on identity components.

The necessity of the condition was proved, in a stronger form, in 3.3. For sufficiency, choose K' as in 3.6. Since the fibres of A'^o have unipotent rank zero, the same is true of the fibres of $(A \times_S S')^o = A^o \times_S S'$, and hence also of the fibres of A^o, from which the former are obtained by extension of the base field.

4. Application to a conjecture of Serre–Tate and to the conductor

4.1. We suppose that S is henselian, and retain the notation of (2.5.3), with $\ell \neq p$.[11] To study the action of $\pi = \mathrm{Gal}(\overline{K}/K)$ and of its inertia subgroup I on $U = T_\ell(A(\overline{K}))$, we shall use the fact that there exists an open subgroup I' of I, which we may moreover take to be normal in π, such that the action of I' on U is unipotent of level two; that is,

$$U^{I'} \supset (U^{I'})^{\perp}.$$

It is then clear that $U^{I'}$ does not depend on the particular choice of I'. We might call it the essentially fixed submodule of U, and shall denote it by U^{ef}. Its orthogonal will be called the essentially toric submodule of U, and will be denoted by U^{et}. We thus have a three-step filtration

$$U \supset U^{\mathrm{ef}} \supset U^{\mathrm{et}} \supset 0, \tag{4.1.1}$$

which is plainly invariant under the action of π.

Let us also note that there is a largest subgroup of I acting unipotently (or, equivalently here, unipotently of level two) on U: it is the subgroup I' consisting of the elements acting unipotently. It is also the kernel of the representation of I on the graded module associated with the filtration (4.1.1), or, equivalently in the present case, the kernel of the induced representation of I on U^{ef}. By the Galois criterion for semistable reduction (3.5), this subgroup I' corresponds to the smallest Galois extension $\widetilde{K}'$ of $\widetilde{K}$ (the maximal unramified extension of K) such that $A_{\widetilde{K}'}$ has semistable reduction. This description also shows that the subgroup I' of I is independent of the choice of the prime $\ell \neq p$.

Notice that, once the representation of π on U^{ef}, and hence on U^{et}, is known, the representation on U/U^{ef} is known up to isogeny, by means of the isogeny induced by the polarization pairing (2.5.1):

$$U/U^{\mathrm{ef}} \longrightarrow (U^{\mathrm{et}})^\vee(1) \qquad \text{an isogeny,} \tag{4.1.2}$$

where (1) denotes the Tate twist by $\mathbb{Z}_\ell(1) = T_\ell(\overline{K}^*)$; this homomorphism is plainly compatible with the operations of π. More precisely, there is a canonical isomorphism

$$U/U^{\mathrm{ef}} \simeq ((U')^{\mathrm{et}})^\vee(1), \qquad U' = T_\ell(A'(\overline{K})).$$

It therefore remains to study more closely the action of π on U^{ef}.

By construction this action is trivial on the normal subgroup I' of I, and hence factors through a representation of the quotient group

$$\pi' = \pi/I', \tag{4.1.3}$$

an action of which we shall now give a description "independent of ℓ."

4.2. Let K' be a Galois extension of K corresponding to an open normal subgroup V of π such that $V \cap I = I'$. The Galois group G of K'/K then occurs in an exact sequence

$$1 \longrightarrow \Gamma \longrightarrow G \longrightarrow G_o \longrightarrow 1, \tag{4.2.1}$$

where

$$\Gamma = I/I' \tag{4.2.2}$$

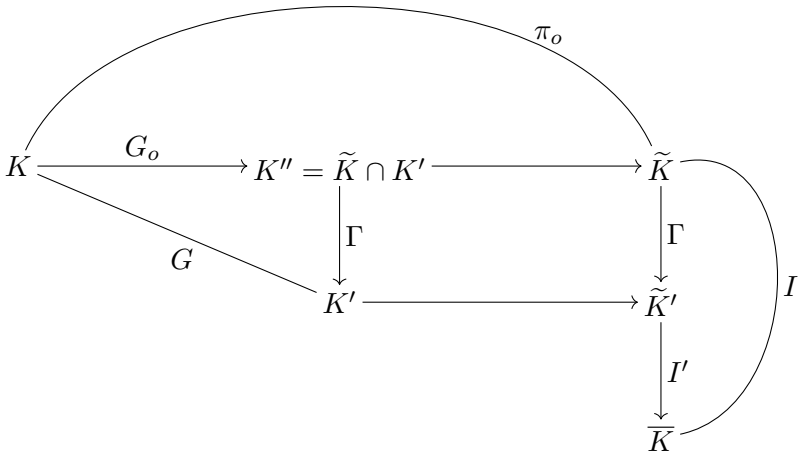
11. The case $\ell = p$ will be studied in §§5 and 6 below.

is the inertia subgroup, and G_o is the Galois group of the largest unramified subextension K'' of K'/K. Thus (4.2.1) is the image of the analogous exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \pi' \longrightarrow \pi_o \longrightarrow 1, \tag{4.2.3}$$

where π' is the group (4.1.3) acting on U^{ef}. We therefore have the following diagram of field inclusions:

(4.2.4)



Let S' be the normalization of S in K', and let A^* be the Neron model of $A_{K'}$. Then, by the Galois criterion for semistable reduction (3.5), $A_{K'}$ has semistable reduction over S'; that is, A^{*o}_o is an extension of an abelian scheme by a torus. On the other hand, by transport of structure, G acts on A^*, and hence on A^{*o}, by automorphisms in a manner compatible with the given action of G on the base scheme S' of A^*. It is then clear that the action of π' on

$$U^{\mathrm{ef}} = T_\ell(A^{*o})^f(\overline{K})$$

is obtained by transport of structure from the action of π' on $(A^{*o}/K', \widetilde{K}')$, where π' acts on A^* through G. Taking into account the canonical isomorphism

$$U^{\mathrm{ef}} \simeq T_\ell(A^{*o})^f(\overline{K}) \simeq T_\ell(A^{*o}_o)(\overline{k}) \tag{4.2.6}$$

(valid for every twisted constant ℓ-adic sheaf over S'; cf. Exposé I [reference omitted in the source]), we obtain the following description of the action of π' on U^{ef}.

Consider the smooth connected group scheme A^{*o}_o over the residue field $k' \hookrightarrow \overline{k}$ of S', which is a quasi-Galois extension of k with Galois group G_o. The quotient group G of π' acts on A^{*o}_o compatibly with its action on k' through G_o. Moreover, π' acts on $\overline{k}$ through its quotient π_o, and hence acts on the situation $(A^{*o}_o/k', \overline{k})$. Thus, by transport of structure, π' acts on

$$U^{\mathrm{ef}} = T_\ell(A^{*o}_o)(\overline{k}),$$

and this is the desired action of π'. In particular, the subgroup Γ of G, which acts trivially on k', acts on A^{*o}_o by k'-automorphisms, and the corresponding representation of $\Gamma = I/I'$ on $T_\ell(A^{*o}_o)(\overline{k})$ is the representation deduced from the natural representation of the inertia group I on U^{ef}.

On the other hand, the essentially toric part U^{et} of U corresponds simply to the submodule

$$U^{\mathrm{et}} \simeq T_\ell(T^*_o)(\overline{k}) \subset U^{\mathrm{ef}} \simeq T_\ell(A^{*o}_o)(\overline{k}), \tag{4.2.7}$$

where T^*_o is the largest subtorus of A^{*o}_o. Likewise, put

$$B^*_o = A^{*o}_o/T^*_o \tag{4.2.8}$$

for the abelian part of A_o^{*o}. The action of G on A_o^{*o} induces separately actions of G on T_o^* and on B_o^*, and the action of π' on U^{et} and on $U^{\mathrm{ef}}/U^{\mathrm{et}}$ is obtained from these actions by means of the isomorphism (4.2.7) and the corresponding isomorphism

$$U^{\mathrm{ef}}/U^{\mathrm{et}} \simeq T_\ell(B_o^*)(\overline{k}). \tag{4.2.9}$$

It follows from this and the isogeny (4.1.2) that the character $g \longmapsto \operatorname{Tr} g_U$ of the action of π' (or of π) on U is given by

$$\operatorname{Tr} g_{U_\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + \chi_\ell(g) \operatorname{Tr} g_{T_o^*,\ell}^{-1}. \tag{4.2.10}$$

Here U_ℓ has been written for U on the left to recall the chosen prime $\ell \neq p$; $g_{B_o^*,\ell}$ and $g_{T_o^*,\ell}$ denote, respectively, the actions of g on the Tate modules $T_\ell(B_o^*)(\overline{k})$ and $T_\ell(T_o^*)(\overline{k})$ induced by its actions on B_o^* and T_o^* just described; and

$$\chi_\ell : \pi' \longrightarrow \pi_o \longrightarrow \mathbb{Z}_\ell^* \tag{4.2.11}$$

is the character of π' (or of π_o) induced by the action on $\mathbb{Z}_\ell(1) = T_\ell(\overline{k}^*)$—a character that is trivial on Γ.

We are now in a position to prove a special case of a conjecture of Serre–Tate [29, Appendix]. [12]

Theorem 4.3. Let S be a henselian trait with residue field k and function field K, and let A_K be an abelian scheme over K. Consider the representation of $\pi = \mathrm{Gal}(\overline{K}/K)$ on the Tate module $U_\ell = T_\ell(A(\overline{K}))$, where ℓ is a prime distinct from the residual characteristic.

a) There exists an open subgroup I' of the inertia group that acts unipotently on U_ℓ, so that the character of the representation of π on U_ℓ is in fact a function on π/I'. Moreover, on $\Gamma = I/I'$ this function takes integral values and is independent of ℓ. More generally, for every $g \in I$, the coefficients of the characteristic polynomial of g_{U_ℓ} are rational integers independent of ℓ.

b) Suppose that k is a finite field with q elements, so that

$$\pi_o = \pi/I \simeq \mathrm{Gal}(\overline{k}/k)$$

is identified with the profinite completion $\widehat{\mathbb{Z}}$ of $\mathbb{Z}$, the generator 1 of $\mathbb{Z}$ corresponding to the Frobenius automorphism

$$f_q : x \longmapsto x^q$$

of $\overline{k}/k$. Then, for every $g \in \pi$ whose image in π_o lies in $\mathbb{Z}$, the trace (and, more generally, every coefficient of the characteristic polynomial) of g_{U_ℓ} is a rational number independent of ℓ. It is even an integer if the image of g in π_o is of the form f_q^n, with $n \geq 0$.

Let us prove, for example, the assertions concerning traces. The assertions concerning the coefficients of the characteristic polynomial could be proved in the same way. One may also note that they follow formally from the assertions about traces, by expressing these coefficients in terms of the traces of the iterates (although those formulas introduce denominators and therefore do not give the asserted integrality), together with a lemma of Dwork–Fatou, for which a pleasant proof is given by Kleiman in [14, 2.8].

a) We use formula (4.2.10), which here becomes

$$\operatorname{Tr} g_{U_\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell}^{-1}. \tag{4.3.0}$$

Here $g_{B_o^*}$ and $g_{T_o^*}$ are now k'-automorphisms of B_o^* and T_o^*, respectively. It therefore suffices to show that the two corresponding traces are rational integers independent of ℓ, which is a general "geometric" assertion about algebraic groups. For $g_{B_o^*}$ this is a well-known result of Weil in the theory of abelian varieties [5] [15]. For $g_{T_o^*}$ it is practically immediate: one describes the

12. This is essentially the case $i = 1$ of those conjectures for the H^i.

representation of Γ on $T = T_o^*$ by means of the corresponding representation on the dual group $M' \simeq \mathbb{Z}^\mu$. The required trace is that of the contragredient of the latter representation, by the canonical isomorphism

$$T_\ell(T) \simeq \check{M}_\ell(1) \overset{\text{def}}{=} \check{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1).$$

b) If the image of g in π_o is f_q^n, formula (4.2.10) now becomes

$$\operatorname{Tr} g_{U,\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + q^n \operatorname{Tr} g_{T_o^*,\ell}^{-1}. \tag{4.3.1}$$

We are thus reduced to proving the following lemma, in which k is to be replaced by an algebraic closure of k', and A by $A_o^{*o} \otimes_{k'} \overline{k}$.

Lemma 4.3.2. Let k be the algebraic closure of a finite field, let A be a commutative group scheme of finite type over k, and let g be an automorphism of this group scheme compatible with the automorphism f of $\operatorname{Spec}(k)$ obtained by transport of structure from an iterated Frobenius automorphism

$$x \longmapsto x^q, \qquad q = p^r, \quad r \geq 0,$$

where q is a power of the characteristic. Thus the action of g on functions φ satisfies

$$g^*(\lambda\varphi) = f^*(\lambda)g^*(\varphi) = \lambda^{q^{-1}} g^*(\varphi) = \lambda^{p^{-r}} g^*(\varphi) \qquad (\lambda \in k).$$

Let g_ℓ be the automorphism induced by g on $T_\ell(A(k))$ by "transport of structure," where ℓ is a prime different from p. Then the trace of g_ℓ (and, more generally, every coefficient of its characteristic polynomial) is a rational integer independent of ℓ, and its eigenvalues are algebraic integers of absolute value $q^{1/2}$ or q. If A is an abelian scheme, their absolute values are $q^{1/2}$; if A is affine, the eigenvalues are of the form $q\varepsilon_i$, where the ε_i are roots of unity.

Indeed, consider the "q-th power" endomorphism f_q^A of the underlying absolute scheme of A, defined as the identity on points and as the q-th power map on the structure sheaf. It is an endomorphism of the group scheme A compatible with the endomorphism $f_q^{\operatorname{Spec}(k)} = f^{-1}$ of the base $\operatorname{Spec}(k)$. Consequently the composite

$$F = f_q^A g \tag{4.3.3}$$

is an endomorphism of the group scheme A compatible with the identity on $\operatorname{Spec}(k)$; in other words, it is a k-endomorphism of the group scheme A. I claim that the automorphism Γ_g of $A(k)$ defined by g through transport of structure, namely

$$\Gamma_g(u) = g \circ u \circ f^{-1} \qquad \text{for } u \in A(k), \tag{4.3.4}$$

may also be described by

$$\Gamma_g(u) = F(u) \qquad \text{for } u \in A(k). \tag{4.3.5}$$

To verify this, it is enough to observe that the underlying set maps of the morphisms $\operatorname{Spec}(k) \longrightarrow A$ represented by the two sides are the same. This reduces us to the equality obtained from (4.3.5) by composing its left-hand member on the left with f_q^A and its right-hand member on the right with f^{-1}, since f_q^A and f^{-1} induce the identity on the underlying sets of the source schemes. The resulting formula follows immediately from (4.3.3) and (4.3.4).

Applying (4.3.5) to $u \in {}_{\ell^\nu}A(k)$ and passing to the inverse limit over ν, we find that g_ℓ may also be interpreted as the endomorphism $T_\ell(F)$ of $T_\ell(A(k))$ associated with the endomorphism F of A. It is known, on the other hand, that $T_\ell(F)$ has a characteristic polynomial with integral coefficients independent of ℓ. By devissage, this reduces to the case in which A is either an abelian scheme or a torus, where the required result was already used in part a) of the proof. The same argument also proves the assertions about the eigenvalues, in view, in the abelian-scheme case, of the classical results of Weil [15].

This proves 4.3.2 and completes the proof of 4.3. At the same time it proves the following result.

Corollary 4.4. Suppose that the hypotheses of 4.3 b) hold. Then, for every $g \in \pi'$ whose image in π_o is of the form f_q^n with $n \geq 0$, the eigenvalues of the induced automorphisms of

$$U^{\mathrm{et}}, \qquad U^{\mathrm{ef}}/U^{\mathrm{et}}, \qquad U/U^{\mathrm{ef}}$$

are algebraic integers whose absolute values are, respectively, q^n, $q^{n/2}$, and 1. For the two extreme pieces one may say more precisely that the corresponding eigenvalues are, respectively,

$$(q^n\varepsilon_i)_{1\leq i\leq \mu} \qquad \text{and} \qquad (\varepsilon_i)_{1\leq i\leq \mu},$$

where the ε_i $(1 \leq i \leq \mu)$ are roots of unity.

Indeed, formula (4.1.2) gives for the last piece the eigenvalues

$$\frac{q^n}{q^n\varepsilon_i} = \varepsilon_i^{-1}.$$

But since ε_i is a root of unity, one has $\varepsilon_i^{-1} = \overline{\varepsilon_i}$ (complex conjugate), and since the family of the ε_i is stable under complex conjugation (being the family of roots of an equation with integral coefficients), the family of the ε_i^{-1} is also the family of the ε_i. This proves the last assertion of 4.4.

4.5. We shall give an application of 4.3 a) to the theory of the local conductor of abelian schemes. Suppose that k is algebraically closed. [13] Recall (SGA 5 X 6.2, or [20]) that, for every commutative etale ℓ-group scheme F_K over K $(\ell \neq p)$, one defines an integer

$$\alpha_s(F_K) = \alpha_s(i_*F_K), \tag{4.5.1}$$

where $i : \eta \longrightarrow S$ denotes the inclusion, as follows. One chooses a finite Galois extension K' of K, with group Γ, that splits F_K, so that F_K is represented by a finite ℓ-group M on which Γ acts. On the other hand, one associates with the extension K'/K a certain projective $\mathbb{Z}_\ell[\Gamma]$-module, defined up to isomorphism,

$$\mathrm{Sw} = \mathrm{Sw}(K'/K, \mathbb{Z}_\ell), \tag{4.5.2}$$

and sets

$$\alpha_s(F_K) = \mathrm{length}\,\mathrm{Hom}_{\mathbb{Z}_\ell(\Gamma)}(\mathrm{Sw}, M). \tag{4.5.3}$$

One checks (loc. cit.) that this expression is independent of the choice of the extension K'/K. Moreover, the projectivity of Sw as a $\mathbb{Z}_\ell[\Gamma]$-module implies that $\alpha_s(F_K)$ is additive in exact sequences.

We apply this definition to the sheaf ${}_\ell A_K$, where A_K is our abelian scheme over K. In this case one therefore obtains an integer

$$\alpha_s(A_K;\ell) = \mathrm{length}\,\mathrm{Hom}_{\mathbb{Z}_\ell(\Gamma)}\big(\mathrm{Sw}, U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell\big), \qquad U_\ell = T_\ell(A(\overline{K})), \tag{4.5.4}$$

where Γ is chosen as above (a choice that a priori depends on ℓ). The integer (4.5.4) is called the *wild conductor exponent* of the abelian scheme A_K relative to the discrete valuation ring $V \subset K$. [14]

13. It would in fact suffice for k to be perfect, but, after passing to the strict henselization of S, one is immediately reduced to the case considered here.

14. One should also define the *tame conductor exponent* of A_K relative to V as the difference between the lengths over $\mathbb{Z}/\ell\mathbb{Z}$ of the geometric generic and special fibres of ${}_\ell A^o$. It is independent of ℓ and equals $2\lambda+\mu$ in the notation of (1.1.11). The (total) conductor exponent of A_K is the sum of the tame and wild exponents; it is the local term occurring in the Euler–Poincare formula for ${}_\ell A^o$ in SGA 5 X 6.2 and [20].

After passage to the henselization it is of course defined without assuming that V is henselian, and one is immediately reduced to the case in which S is strictly local. We intend to prove the following result.

Corollary 4.6. The wild conductor exponent (4.5.4) is independent of the choice of the prime $\ell \neq p$.

We may suppose that S is strictly local. We compute (4.5.4), replacing Sw by any other projective $\mathbb{Z}_\ell(\Gamma)$-module L. Observe that the filtration (4.1.1) of U_ℓ consists of $\mathbb{Z}_\ell$-submodules that are manifestly direct summands. It therefore induces a corresponding filtration of

$$U(\ell) = U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell,$$

whose successive factors are obtained, by tensoring with $\mathbb{F}_\ell$, from the factors V_i $(i = 1, 2, 3)$

$$U^{\mathrm{et}}, \qquad U^{\mathrm{ef}}/U^{\mathrm{et}}, \qquad U/U^{\mathrm{ef}}$$

appearing in (4.1.1). This filtration is, of course, stable under Γ. Additivity thus decomposes

$$\operatorname{length} \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell))$$

as a sum of three terms, the lengths of the groups

$$(*) \qquad \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell) \simeq \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i) \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell.$$

The isomorphism in (*) again follows from the assumed projectivity of L; it is only necessary to choose K' large enough for Γ to act on the V_i. Now $\operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i)$ is plainly a torsion-free $\mathbb{Z}_\ell$-module, so its rank is the length of the right-hand side of (*). If φ_i is the character of the representation of Γ on V_i, and ψ the character of the representation of Γ on $\check{L}$, we obtain

$$(4.6.1) \qquad \operatorname{length} \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell)) = \sum_{1 \leq i \leq 3} N^{-1} \sum_{g \in \Gamma} \psi(g)\varphi_i(g)$$

$$= N^{-1} \sum_{g \in \Gamma} \psi(g)\varphi(g), \qquad N = \operatorname{card} \Gamma,$$

where $\varphi(g) = \sum_i \varphi_i(g)$ is also the function on Γ obtained, by passage to the quotient, from the character of the monodromy representation of the inertia group I on U. By 4.3 a), this character is an integral-valued rational function independent of ℓ. Taking $L = \mathrm{Sw}$, the same is true of the character $\psi(g)$ of L, which is

$$a(g) - N\delta(g) + 1,$$

where δ is the Dirac function on Γ and a is the Artin conductor of K'/K [26]. This proves 4.6 and at the same time gives the following expression, independent of ℓ, for the wild conductor exponent:

$$(4.6.2) \qquad \alpha_s(A_K) = N^{-1} \sum_{g \in \Gamma} \big(a(g) - N\delta(g) + 1\big)\varphi(g).$$

Here Γ is the Galois group of a finite Galois extension K' of K that splits U^{ef} (for example the smallest such extension; cf. 4.1), $a(g)$ is the Artin conductor on Γ, and $\varphi(g)$ is the trace function on Γ described in (4.3.0). In the semistable-reduction case one may take $K' = K$, hence $\Gamma = \{e\}$, and obtains

$$(4.6.3) \qquad \alpha_s(A_K) = 0 \qquad \text{(semistable-reduction case)}.$$

To conclude, let us state explicitly a criterion for semistable reduction that follows from the general considerations of 4.2.

Proposition 4.7. (M. Raynaud's criterion for semistable reduction.) Let S be a trait, let A_K be an abelian scheme over its function field K, and let $n > 1$ be an integer prime to the residual characteristic p. Suppose that ${}_nA_K$ is unramified relative to S, that is, that the inertia group acts trivially on ${}_nA(\overline{K})$. Then either A_K has semistable reduction over S, or $n = 2$; in the latter case, if S is strictly local, A_K acquires semistable reduction over a finite Galois extension K' of K whose Galois group is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$. [15]

We may plainly suppose that S is strictly local. With the notation of 4.1 and 4.2, take K' as small as possible; in other words, suppose that Γ acts faithfully on A_o^{*o}. By hypothesis Γ acts trivially on ${}_nA(\overline{K})$, and hence a fortiori on

$$ {}_nA(\overline{K})^{\mathrm{ef}} \simeq {}_nA_o^*(k) \supset {}_nA_o^{*o}(k). $$

The conclusion follows from the next two lemmas.

Lemma 4.7.1. (J.-P. Serre.) Let G be a commutative group scheme over a field k, which is an extension of an abelian scheme by a torus, and let $n \geq 2$ be an integer. Every finite-order automorphism of G that induces the identity on ${}_nG$ is the identity if $n \neq 2$, and is either the identity or has order two if $n = 2$.

Lemma 4.7.2. A finite group in which every element distinct from 1 has order 2 is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$ for a suitable $r \geq 0$.

4.7.3. Let us recall the well-known proof of 4.7.1. Since $n\,\mathrm{id}_G$ is an epimorphism, the hypothesis that u induces the identity on ${}_nG$, or equivalently that $1 - u$ induces zero on ${}_nG$, is expressed by the condition

$$ 1 - u = nv $$

in the endomorphism ring of G. To prove the asserted conclusion, we reduce, by decomposing the finite cyclic group generated by u into its p-primary components for the various primes p dividing the order of u, to the case in which u has order p^N.

Let E be the subring of $\mathrm{End}(G)$ generated by v. Since $u = 1 - nv$ and $u^{p^N} = 1$, it is a quotient of

$$ \mathbb{Z}[T]/\big((1 - nT)^{p^N} - 1\big); $$

it is therefore of finite type over $\mathbb{Z}$. It is moreover torsion-free (a formal consequence of the divisibility of G), and thus embeds in

$$ E_{\mathbb{Q}} = E \otimes_{\mathbb{Z}} \mathbb{Q}. $$

The latter is a quotient, after the change of variable $u = 1 - nT$, of $\mathbb{Q}[u]/(u^{p^N} - 1)$, which is a direct product of finite field extensions of $\mathbb{Q}$; the same is therefore true of $E_{\mathbb{Q}}$. Considering the images of E, u, and v in the component fields K_i of $E_{\mathbb{Q}}$, it remains to prove the following assertion. Let $K = \mathbb{Q}(u)$ be generated by a primitive p^m-th root of unity u, and let $n \geq 2$ be an integer such that

$$ u - 1 \equiv 0 \pmod{n} $$

in the ring of integers of K. Then $u = 1$, or $u = -1$ and $n = 2$.

This follows at once from the familiar fact that, if $u \neq 1$, that is, $m \geq 1$, the ideal $(u - 1)$ is prime and (p) is a power of it, with exponent greater than 1 except when $p = 2$ and $m = 1$, that is, when $u = -1$. Since $(u - 1)$ is prime and divides p, an integer $n \geq 2$ can divide $u - 1$ only if $n = p$; and p can divide $u - 1$ only if $(u - 1) = (p)$, that is, if $u = -1$. Hence $n = 2$.

4.7.4. Let us prove 4.7.2. It is enough to prove that Γ is commutative. If $x, y \in \Gamma$, then $(xy)(xy) = 1$; since $x^2 = 1$ and $y^2 = 1$, this may also be written

$$ xy = (xy)^{-1} = y^{-1}x^{-1} = yx, $$

15. As follows from an example communicated to me by J. Tate, 4.7 is no longer valid when n equals the residual characteristic p, even if K has characteristic zero and A_K is an elliptic curve with complex multiplication.

as required.

Remark 4.7.5. The proof given in 4.7.3, together with 4.7.2, shows that if G is a divisible commutative group in a topos, $n \geq 2$ is an integer, and Γ is a finite group acting faithfully on G while inducing the identity on ${}_nG$, then Γ is the trivial group if $n \geq 3$, and is isomorphic to a group $(\mathbb{Z}/2\mathbb{Z})^r$ if $n = 2$.

5. The orthogonality theorem in the case $\ell = p$ (semistable case). Duality of the abelian schemes B_o and B'_o. Characterization of the fixed part. Criteria for good reduction, essentially good reduction, and semistable reduction.

5.1. Let us place ourselves in the situation of 2.2.3, where S is henselian and A^o is ℓ-divisible, so that the fixed part

$$T_\ell(A^o)^f \hookrightarrow T_\ell(A)$$

is defined as the pro-ℓ Barsotti–Tate group formed by the $({}_{\ell^\nu}A^o)^f$. We shall now give a definition of the toric part (2.3.2) of $T_\ell(A^o)$ which is independent of the hypothesis $\ell \neq p$ made in 2.3, but which instead assumes that S is complete.[16] For this purpose, for every integer $n \geq 0$ introduce the local Artinian scheme

$$S_n = \mathrm{Spec}(V/\mathfrak{m}^{n+1}), \tag{5.1.1}$$

the n-th infinitesimal neighbourhood of s in S, and, for every S-scheme X, put $X_n = X \times_S S_n$. Since $(A^o)_n = (A_n)^o$ is smooth over S_n, it is known (SGA 3 IX 3.6 bis) that T_o is the special fibre of a unique subtorus

$$T_n \subset A^o_n. \tag{5.1.2}$$

By the uniqueness of the T_n, they are obtained from one another by the base changes $S_{n'} \longrightarrow S_n$. In other words, they define a formal subtorus $\widehat{T}$ of the group $\widehat{A}^o$ (in the category of formal schemes over S), the formal completion of A^o along the special fibre:

$$\widehat{T} \subset \widehat{A}^o. \tag{5.1.3}$$

Since the functor $X \longmapsto \widehat{X}$ (formal completion along the special fibre) induces an equivalence between the category of finite schemes over S and that of finite formal schemes over S (EGA III 4.8), we shall simply identify a finite scheme over S with the corresponding formal scheme. With this identification, the definition of the *fixed parts* immediately gives, for every $\nu \geq 0$,

$$({}_{\ell^\nu}A^o)^f = {}_{\ell^\nu}(\widehat{A}^o), \tag{5.1.4}$$

which may also be written as an identity of projective systems:

$$T_\ell(A^o)^f = T_\ell(\widehat{A}^o). \tag{5.1.5}$$

We therefore obtain a Barsotti–Tate pro-subgroup of the fixed part of $T_\ell(A^o)$ by taking $T_\ell(\widehat{T}) \subset T_\ell(\widehat{A}^o)$. We accordingly define the toric part of $T_\ell(A^o)$ by

$$T_\ell(A^o)^t \overset{\text{def}}{=} T_\ell(\widehat{T}) \subset T_\ell(\widehat{A}^o) = T_\ell(A^o)^f. \tag{5.1.6}$$

Thus, by definition,

$$T_\ell(A^o)^t = ({}_{\ell^\nu}\widehat{T})_{\nu \geq 0}, \tag{5.1.7}$$

the finite flat S-group ${}_{\ell^\nu}\widehat{T}$ itself being described as the inductive limit of the finite flat S_n-groups

$${}_{\ell^\nu}(T_n), \qquad n \text{ variable}; \tag{5.1.8}$$

16. This hypothesis is in fact not essential, as is shown in 6.1 below.

that is, its affine algebra is described as the projective limit of the affine algebras of the groups (5.1.8). Of course,

$$\big(T_\ell(A^o)^t\big)_n = T_\ell(T_n). \tag{5.1.9}$$

For $n = 0$ this is precisely formula (2.3.3), by which we characterized the toric part (as a profinite etale group) in the case $\ell \neq p$. This shows that definition (5.1.6) is compatible with that given in 2.3 when $\ell \neq p$.

Passing to generic fibres, the fixed and toric parts of $T_\ell(A^o)$ define fixed and toric parts in $T_\ell(A_K)$:

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K). \tag{5.1.10}$$

These are inclusions of pro-ℓ Barsotti–Tate groups. Consider the analogous filtration corresponding to the dual abelian scheme A'_K:

$$T_\ell(A'_K)^t \subset T_\ell(A'_K)^f \subset T_\ell(A'_K). \tag{5.1.11}$$

We then have the following result, which generalizes 2.4 (at least in the case where S is complete, to which one could easily have reduced in 2.4).

Theorem 5.2. Suppose that S is a complete trait, and let A_K be an abelian scheme over K, A'_K the dual abelian scheme, and ℓ a prime such that the connected Neron model A^o is ℓ-divisible (cf. 2.2.1), so that one then has the filtrations (5.1.10) and (5.1.11) of the pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$. Then the toric parts are recovered from the fixed parts by the formula

$$T_\ell(A_K)^t = T_\ell(A_K)^f \cap \big(T_\ell(A'_K)^f\big)^\perp, \tag{5.2.1}$$

and by the analogous formula obtained by interchanging the roles of A_K and A'_K, where $\perp$ denotes the orthogonal complement with respect to the canonical pairing (1.0.2).

Finally, when A_K has semistable reduction over S (which follows from the other hypotheses in the case $\ell = p$), one even has the formula

$$T_\ell(A_K)^t = \big(T_\ell(A'_K)^f\big)^\perp. \tag{5.2.2}$$

5.2.3. The meaning of formula (5.2.1) is quite clear when ℓ is prime to the characteristic of K: as in 2.4, it can then be interpreted by passing to the ordinary $\mathbb{Z}_\ell$-modules, with an action of $\pi = \mathrm{Gal}(\overline{K}/K)$, formed by the $\overline{K}$-points of the pro-groups involved. Let us spell it out in the general case, using the notion of the dual

$$D(G) = \mathrm{Hom}\big(G, T_\ell(\mathbb{G}_m)\big) \tag{5.2.3.1}$$

of a pro-ℓ Barsotti–Tate group $G = (G(\nu))$. By definition, this is the projective system of Cartier duals $D(G(\nu))$ (SGA 3 VII 3.3.1 and VIII 1), whose transition morphisms are defined in the evident way (for example, by transposition from the inclusion morphisms among the $G(\nu)$). The duality theory of abelian varieties tells us that the pairing mentioned in 5.2 actually defines an isomorphism

$$T_\ell(A'_K) \simeq D\big(T_\ell(A_K)\big), \tag{5.2.3.2}$$

and hence, by composition, a morphism

$$T_\ell(A_K)^f \longrightarrow D\big(T_\ell(A'_K)^f\big). \tag{5.2.3.3}$$

Formula (5.2.1) says that the kernel of the preceding homomorphism (which a priori is a pro-subgroup of its first term) is precisely the toric part $T_\ell(A_K)^t$. Notice that this is an isomorphism

of pro-objects, and not of projective systems: we do not assert (and it would in general be false) that, for every integer ν, $T_\ell(A_K)^t(\nu)$ equals the kernel of

$$T_\ell(A_K)^f(\nu) \longrightarrow D\big(T_\ell(A'_K)^f\big)(\nu);$$

the latter can a priori be larger. We shall see below, however, that it is not larger if A_K has semistable reduction; by the hypotheses made in 5.2, this is in particular the case when $\ell = p$.

5.2.4. The proof of (5.2.1) merely repeats, with practically no change, the proof of 2.4. Observe that the induced pairing

$$T_\ell(A_K)^f \times T_\ell(A'_K)^f \longrightarrow T_\ell(\mathbb{G}_{mK}) \tag{5.2.4.1}$$

which gives rise to (5.2.3.3) is induced by a pairing

$$T_\ell(A^o)^f \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{5.2.4.2}$$

of pro-Barsotti–Tate groups over S, itself induced by the pairing of pro-groups

$$T_\ell(A^o) \times T_\ell(A'^o) \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{5.2.4.3}$$

deduced from the biextension W of (A^o, A'^o) by $\mathbb{G}_{mS}$. Consequently, the pairing

$$T_\ell(A^o_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{5.2.4.4}$$

obtained from (5.2.4.2) by passing to the special fibres, is precisely the pairing deduced from the biextension W_o of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$. By VIII 4.11 and 1.5, the left annihilator of the pairing (5.2.4.4) is therefore $T_\ell(T_o)$, which is precisely the special fibre of the toric part $T_\ell(A^o)^t$. Returning to the definition of left annihilators given in 5.2.3, we conclude by means of the following lemma.

Lemma 5.2.5. *Let $u : G \longrightarrow H$ be a homomorphism of pro-ℓ Barsotti–Tate groups over S. If $u_o : G_o \longrightarrow H_o$ is zero, then u is zero. If $u_o : G_o \longrightarrow H_o$ is an isogeny (respectively an isomorphism), then u is an isogeny (respectively an isomorphism).*

Indeed, apply the first assertion of 5.2.5 to the composite morphism

$$T_\ell(A^o)^t \longrightarrow T_\ell(A^o)^f \longrightarrow D\big(T_\ell(A'^o)^f\big),$$

where the second morphism is deduced from the pairing (5.2.4.2). The composite is zero, so the pairing under consideration vanishes on the toric part on the left. By symmetry it also vanishes on the toric part on the right, and hence descends to a pairing on the quotients, or *abelian pieces*,

$$T_\ell(A^o)^{ab} \times T_\ell(A'^o)^{ab} \longrightarrow T_\ell(\mathbb{G}_m).$$

These quotients are again pro-Barsotti–Tate groups, as one checks readily. It remains to prove that the morphism corresponding to this pairing,

$$T_\ell(A^o)^{ab} \longrightarrow D\big(T_\ell(A'^o)^{ab}\big),$$

is an isogeny, which again follows from 5.2.5.

We leave to the reader the easy proof of 5.2.5, a result valid over any connected Noetherian base. Moreover, below in 7.4.7, by a somewhat more precise analysis of the biextensions W_n of (A^o_n, A'^o_n), we shall find a quite evident direct proof of the two special cases in which we had to use 5.2.5.

5.2.6. Suppose now that A_K has semistable reduction over S, so that ℓ may be any prime whatsoever (equal or unequal to p). Moreover, by the Galois criterion for semistable reduction 3.5, for $\ell \neq p$ one has

$$\big(T_\ell(A'_K)^f\big)^\perp \ \big(= T_\ell(A_K)^t\big) \subset T_\ell(A_K)^f. \tag{5.2.7}$$

To finish the proof of 5.2, it remains to show that this inclusion is valid without restriction on ℓ.

Indeed, the left-hand term is canonically isomorphic to

$$D\big(T_\ell(A'_K)/T_\ell(A'_K)^f\big)$$

and therefore has *rank* (or *height*) $2n-2\alpha-\mu=\mu$, in the notation of (2.1.9), since by hypothesis $\lambda=0$. This is also the rank of $T_\ell(A_K)^t$. Since the inclusion

$$(*) \qquad T_\ell(A_K)^t \subset T_\ell(A_K)$$

is strict, in the sense that it already induces a monomorphism on every term of the projective systems under consideration, it follows readily that the left-hand term of (*) is maximal among the pro-Barsotti–Tate subgroups of $T_\ell(A_K)$ having the same rank. Thus equality holds in (5.2.7); by the orthogonality theorem this is equivalent to the inclusion between the two extreme terms of (5.2.7).

5.3. Using the orthogonality correspondence between strict pro-Barsotti–Tate subgroups (in the sense specified above) of the dual pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$, one formally deduces from (5.2.2) that the pairing (1.0.2) between these pro-groups induces a pairing

$$(5.3.2) \qquad T_\ell(A_K)^{ab} \times T_\ell(A'_K)^{ab} \longrightarrow T_\ell(\mathbb{G}_{mK})$$

which is a perfect duality, that is, it identifies each of the two factors on the left with the dual of the other (in the sense of 5.2.3). [17] It is also clear that (5.3.2) is induced by the pairing

$$(5.3.3) \qquad T_\ell(A^o)^{ab} \times T_\ell(A'^o)^{ab} \longrightarrow T_\ell(\mathbb{G}_{mS})$$

obtained from the pairing (5.2.4.2) by passing to the quotients. Since this is a perfect duality on the generic fibres, that is, since the corresponding homomorphism

$$(5.3.4) \qquad T_\ell(A^o)^{ab} \longrightarrow D\big(T_\ell(A'^o)^{ab}\big)$$

is an isomorphism on the generic fibres, one naturally wishes to deduce that it is so over the whole of S, or equivalently on the special fibres. This deduction is trivially valid if $\ell \neq p$. It is also valid, at least in unequal characteristics when $\ell = p$, by a deep theorem of Tate [33, Th. 4]. We shall nevertheless establish it directly below in 7.4.6, without using Tate's theorem.

To make precise the geometric meaning of the result we wish to prove, observe that the pairing obtained from (5.3.3) by passage to the special fibres is precisely the pairing

$$(5.3.5) \qquad T_\ell(B_o) \times T_\ell(B'_o) \longrightarrow T_\ell(\mathbb{G}_{mk})$$

deduced from the biextension W_o of (A^o_o, A'^o_o) by passage to the quotient, where B_o and B'_o denote respectively the abelian parts of A^o_o and A'^o_o. By VIII 4.8, the biextension W_o comes from a biextension W^{ab}_o of (B_o, B'_o) by $\mathbb{G}_{mk}$, unique up to a unique isomorphism. Thus the pairing (5.3.5) may also be interpreted as the one deduced from the divisorial correspondence W^{ab}_o, and the corresponding homomorphism

$$(5.3.6) \qquad T_\ell(B_o) \longrightarrow D\big(T_\ell(B'_o)\big) \simeq T_\ell\big(D(B'_o)\big)$$

(where, in the last term, D denotes the dual abelian scheme) is precisely the morphism obtained by applying T_ℓ functorially to the homomorphism

$$(5.3.7) \qquad B_o \longrightarrow D(B'_o)$$

defined by W^{ab}_o. Consequently, by the standard dictionary (cf. [5]), saying that (5.3.5) is a perfect duality is also saying that the kernel of (5.3.7), which is a finite group over k by 1.5, has zero

17. The superscripts *ab* denote the *abelian parts*, the quotient of the fixed part by the toric part.

ℓ-primary component. Saying that this holds for every ℓ therefore means that the isogeny (5.3.7) is an isomorphism, that is, that the divisorial correspondence W_o^{ab} is a perfect duality between B_o and B'_o. The preceding considerations show only that the kernel of (5.3.7) is a p-group. We therefore state the more complete result.

Theorem 5.4. Let S be a trait, and let A_K be an abelian scheme over K having semistable reduction over S. Let B_o and B'_o denote respectively the abelian parts of the Neron models of A_K and of the dual abelian scheme A'_K. Then the canonical divisorial correspondence W_o^{ab} on (B_o, B'_o), deduced by VIII 4.8 from the canonical biextension W_o of $(A_o^o, A_o'^o)$ (the special fibre of the biextension W of 1.4), is a perfect duality between B_o and B'_o; that is, it defines an isomorphism of each of these abelian schemes with the dual of the other.

The proof of 5.4 in the general case will be given in section 7, as a consequence of a very natural and important construction due to Raynaud. Let us merely observe here that, in proving 5.4, one may assume that S is complete, by 3.3.

5.5. To express conveniently the extreme factors in the filtrations (5.1.10) and (5.1.11), in the case of semistable reduction, let us introduce the twisted constant group schemes over S

$$\underline{M'}_o = D(T_o), \qquad \underline{M}_o = D(T'_o), \tag{5.5.1}$$

dual to the maximal tori T_o, T'_o of the special fibres of A, A' (SGA 3, X 5.7). Since S is complete local, these group schemes may be regarded as the special fibres of twisted constant schemes

$$\underline{M'}, \qquad \underline{M} \quad \text{over } S. \tag{5.5.2}$$

Thus, if the torus T_o is split (for example, if k is separably closed), that is, if $\underline{M'}_o$ is the constant group defined by the ordinary group

$$M' = \mathrm{Hom}_{k\text{-gr}}(T_o, \mathbb{G}_{mk}), \tag{5.5.3}$$

then $\underline{M'}$ is the constant group defined by M':

$$\underline{M'} = M'_S.$$

One may moreover observe that, since T_o and T'_o are isogenous (2.2.6), T'_o is split whenever T_o is. In this case one should therefore also introduce

$$M = \mathrm{Hom}_{k\text{-gr}}(T'_o, \mathbb{G}_{mk}), \tag{5.5.4}$$

and then

$$\underline{M} = M_S.$$

With this notation, for every integer n there are evidently isomorphisms

$$\underline{M'}_n \simeq D(T_n), \qquad \underline{M}_n \simeq D(T'_n), \tag{5.5.5}$$

where T'_n is defined in terms of A' as T_n is in terms of A (5.1.2). Hence there follow canonical isomorphisms (where $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{mS})$)

$$T_\ell(T_n) \simeq \check{\underline{M}}'_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1), \qquad T_\ell(T'_n) \simeq \check{\underline{M}}_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1), \tag{5.5.6}$$

and, on passing to the limit over n, canonical isomorphisms

$$\begin{cases} T_\ell(A^o)^t = T_\ell(\widehat{T}) \simeq \check{\underline{M}}' \otimes \mathbb{Z}_\ell(1), \\ T_\ell(A'^o)^t = T_\ell(\widehat{T}') \simeq \check{\underline{M}} \otimes \mathbb{Z}_\ell(1). \end{cases} \tag{5.5.7}$$

In the case of semistable reduction, duality, together with the orthogonality theorem 5.2 and relation (5.2.2), then gives canonical isomorphisms

$$\begin{cases} T_\ell(A_K)/T_\ell(A_K)^f \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell, \\ T_\ell(A'_K)/T_\ell(A'_K)^f \simeq \underline{M}'_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell. \end{cases} \tag{5.5.8}$$

Likewise, the same formula (5.2.2) gives canonical isomorphisms

$$\begin{cases} T_\ell(A_K)/T_\ell(A_K)^t \simeq D\big(T_\ell(A'_K)^f\big), \\ T_\ell(A'_K)/T_\ell(A'_K)^t \simeq D\big(T_\ell(A_K)^f\big), \end{cases} \tag{5.5.9}$$

where one should recall that the groups $T_\ell(A_K)^f$ and $T_\ell(A'_K)^f$ occurring on the right are the restrictions to η of the pro-ℓ Barsotti–Tate groups $T_\ell(A^o)^f$ and $T_\ell(A'^o)^f$, respectively, defined over S. Here D denotes duality in the sense of pro-Barsotti–Tate groups, namely $\underline{\mathrm{Hom}}(-, \mathbb{Z}_\ell(1))$. Thus we have:

Proposition 5.6. Suppose that S is complete and that A_K has semistable reduction over S. Then, for every prime number ℓ (which may equal the residue characteristic of S), the pro-ℓ Barsotti–Tate group $T_\ell(A_K)^t$ is the generic fibre of a pro-Barsotti–Tate group over S "of toric type", given by (5.5.7), whereas $T_\ell(A_K)/T_\ell(A_K)^f$ is the generic fibre of a pro-Barsotti–Tate group over S of pro-étale type, given by (5.5.8). Finally, the groups $T_\ell(A_K)^f$ and $T_\ell(A_K)/T_\ell(A_K)^t$ are the generic fibres of pro-Barsotti–Tate groups defined over S, equal respectively to $T_\ell(A^o)^f$ and to $D(T_\ell(A'^o)^f) = \underline{\mathrm{Hom}}(T_\ell(A'^o)^f, \mathbb{Z}_\ell(1))$.

5.6.1. Observe that, for an étale covering S' of S, splitting $T_\ell(A_K)^t$ is equivalent to splitting $T_\ell(A_K)/T_\ell(A_K)^f$: these conditions say respectively that S' splits the twisted constant groups $\underline{M}$ and $\underline{M}'$, and these are isogenous because T_o and T'_o are. At the same time one sees that there is a smallest étale extension S' of S which does the job.

We also record the following easy consequence of the definitions in 2.2.3.

Proposition 5.7. Suppose that A^o is ℓ-divisible and that S is henselian. Let F_K be a constructible ℓ-adic sheaf over K which is "unramified", that is, which extends to a twisted constant ℓ-adic sheaf F over S, and let $u : F_K \longrightarrow T_\ell(A_K)$ be a homomorphism of pro-groups. Then u factors through a homomorphism from F to $T_\ell(A_K)^f$.

Indeed, write

$$u = (u(\nu))_{\nu \geqslant 0}, \qquad u(\nu) : F_K(\nu) \longrightarrow T_\ell(A_K)(\nu) = {}_{\ell^\nu}A_K. \tag{5.7.1}$$

By the universal property of A, since $F(\nu)$ is étale and hence smooth over S, the morphism $u(\nu)$ comes from a homomorphism

$$v(\nu) : F(\nu) \longrightarrow {}_{\ell^\nu}A.$$

By definition of the fixed part, this homomorphism factors as

$$v(\nu) : F(\nu) \longrightarrow ({}_{\ell^\nu}A)^f.$$

Since the image of the induced morphism on special fibres lies in the infinitely ℓ-divisible part of A_o, it follows that $v(\nu)$ in fact factors through $({}_{\ell^\nu}A^o)^f$. Passing to the limit over ν gives a homomorphism

$$v : F \longrightarrow T_\ell(A^o)^f,$$

and the induced morphism on generic fibres plainly gives the required factorization.

Remarks 5.7.2. When $\ell \neq p$, property 5.7 uniquely characterizes the sub-pro-group $T_\ell(A_K)^f$ of $T_\ell(A_K)$ as its largest unramified ℓ-adic subsheaf. When $\ell = p$, the statement of 5.7 ceases to be very satisfactory. We shall give a considerably stronger property, which can serve as an intrinsic

characterization of the fixed part of $T_p(A_K)$ (in terms of the pro-Barsotti–Tate group structure of $T_p(A_K)$ alone), at least in unequal characteristics.

Theorem 5.8 (M. Raynaud). Let A_K be an abelian scheme with semistable reduction over the function field K of a henselian trait S. Let ℓ be a prime number (the interesting case being $\ell = p$), and consider the pro-ℓ Barsotti–Tate group $X_\eta = T_\ell(A_K)$ and its "fixed part" X_η^f, which is the generic fibre of the pro-ℓ Barsotti–Tate group $X^f = T_\ell(A^o)^f$. If $\ell = p$, suppose that $\operatorname{char} K = 0$ (so that Tate's theorem [33, Th. 4] is available over S). Let F be a pro-ℓ Barsotti–Tate group over S. Then every homomorphism $F_\eta \longrightarrow X_\eta$ factors through X_η^f.

Passing to the ind-ℓ groups associated with the pro-ℓ Barsotti–Tate groups under consideration, the preceding statement becomes equivalent, by the cited theorem of Tate, to the following.

Corollary 5.9. If F is an ind-ℓ Barsotti–Tate group, then every homomorphism $u_\eta : F_\eta \longrightarrow A_\eta$ extends uniquely to a homomorphism $u : F \longrightarrow A$.

In other words, the characteristic Neron property (1.1.2) of the Neron model remains valid for homomorphisms from ind-group schemes of Barsotti–Tate type, with no smoothness hypothesis on them. (Only the case of ind-étale groups would follow directly from the definition.)

Remark 5.9.1. In fact, M. Raynaud proves 5.9 without assuming that A_K has semistable reduction, and even for every Neron group scheme A over S (whose generic fibre need not be an abelian scheme). Of course, the assumption that S is henselian is also unnecessary in the formulation of 5.9. Raynaud's proof of this more general case reduces it to the special case considered in 5.8, using an unpublished property in the construction of Neron models. Here we shall give only Raynaud's proof of 5.9.

Proof of 5.9. It is known [33] that F is an extension of an ind-étale Barsotti–Tate group F' by a Barsotti–Tate group F'' with connected special fibre, so that for every integer n there is an exact sequence

$$0 \longrightarrow F''(n) \longrightarrow F(n) \longrightarrow F'(n) \longrightarrow 0,$$

with $F'(n)$ étale and $F''(n)_o$ purely infinitesimal. The given homomorphism u_η is defined by a sequence of homomorphisms

$$u_\eta(n) : F(n)_\eta \longrightarrow A_\eta,$$

and it is enough to extend these separately to homomorphisms $u(n) : F(n) \longrightarrow A$. Lemma 5.9.2 below then reduces us to proving that the homomorphism $u''_\eta : F''_\eta \longrightarrow A_\eta$ induced by u_η extends to $u'' : F'' \longrightarrow A$. Consequently, we may suppose that F has connected special fibre.

We may regard u_η as a homomorphism from F_η to the ind-ℓ Barsotti–Tate group Y_η associated with $X_\eta = T_\ell(A_K)$, and then consider the composite

$$(*) \qquad v_\eta : F_\eta \longrightarrow Y_\eta \longrightarrow Y_\eta / Y_\eta^f.$$

By 5.6, the latter group extends to an ind-étale Barsotti–Tate group Z over S, namely

$$\underline{M} \otimes_{\mathbb{Z}} (\mathbb{Q}_\ell / \mathbb{Z}_\ell).$$

By Tate's theorem cited in 5.8, the homomorphism (*) then extends to a homomorphism

$$v : F \longrightarrow Z.$$

Since F has connected special fibre and Z is ind-étale, this morphism is zero. Hence v_η is zero as well, and u_η therefore factors through Y_η^f, as required.

It remains to prove the following lemma.

Lemma 5.9.2. Let S be a trait, let A be a Neron group scheme over S (1.1.2), let

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1$$

be an extension of group schemes over S, with G'' smooth, let $u : G' \longrightarrow A$ be a group homomorphism, and let $v_\eta : G_\eta \longrightarrow A_\eta$ be a group homomorphism extending u_η. Suppose that the extension of G'' by A obtained from the extension G by pushout along u is representable [18]—as is the case, for example, if $G \longrightarrow G''$ admits a section locally for the "finite locally free" topology (cf. SGA 3, IV 6.3) and if A is quasi-projective over S. Then there exists a unique group homomorphism $v : G \longrightarrow A$ extending both v_η and u.

Indeed, the last hypothesis reduces us to the case in which $G' = A$, u is the identity, and $v_\eta : G_\eta \longrightarrow A_\eta$ is a projection which we seek to extend to a projection $v : G \longrightarrow A$. But G is smooth, being an extension of two smooth group schemes, and the conclusion follows from the definition of the Neron property of A.

Corollary 5.10. Let A_K be an abelian scheme over the function field K of a trait S, and let ℓ be a prime number. When $\ell = p$, suppose that $\operatorname{char} K = 0$, so that Tate's theorem already invoked in 5.8 is available. Then A_K has good reduction over S if and only if the pro-ℓ Barsotti–Tate group $T_\ell(A_K)$ "has good reduction over S", that is, extends to a Barsotti–Tate group over S.

(This is the result conjectured in [8, 4.9].)

Necessity being clear, it remains to prove sufficiency; observe that the case $\ell \neq p$ has already been treated in 2.2.9. We may suppose that S is henselian (2.2.9). First suppose that A_K has semistable reduction over S. Applying 5.8 with $F = X$ gives

$$T_\ell(A_K) = T_\ell(A_K)^f.$$

This also says that the maximal torus of A_o is zero, that is, that A_o^o is an abelian scheme; the conclusion then follows from 2.2.9 (iii^o).

We now treat the general case. By 2.2.9 we may plainly assume that S is strictly local. Let K'/K be a finite Galois extension such that $A_{K'}$ has semistable reduction over the normalization S' of S in K' (3.6). If A^* denotes the corresponding Neron model, $G = \operatorname{Gal}(K'/K)$ acts on the situation (A^*, S'), and hence on A_o^{*o}. By 4.1 it is enough to prove that G acts trivially on the latter; this indeed implies that A_K has semistable reduction, reducing us to the case already treated.

Let X be a Barsotti–Tate group over S extending $X_\eta = T_\ell(A_K)$. By Tate's uniqueness theorem already invoked, there is a unique isomorphism

$$X \times_S S' \simeq T_\ell(A^{*o})$$

which induces the identity on generic fibres. By transport of structure, this isomorphism is compatible with the actions of G. Now G plainly acts trivially on the special fibre of $X \times_S S'$, and hence on $T_\ell(A_o^{*o})$. It follows at once that G acts trivially on A_o^{*o}, completing the proof.

Remark 5.11. Once Tate's theorem has been proved without any restrictive hypothesis on $\operatorname{char} K$, the preceding results 5.8, 5.9, and 5.10, as well as 5.13 below, will plainly hold without such a restriction.

We record a definition which will also be useful elsewhere.

Definition 5.12. Let S be a henselian trait, let ℓ be a prime number, and let X_η be a pro-ℓ Barsotti–Tate group over the generic point η of S. We say that X_η has *good reduction over* S if it extends to a Barsotti–Tate group over S (cf. 5.10). We say that it has *virtual good reduction over* S if there exists a filtration

$$\left(\operatorname{Fil}^i(X_\eta)\right)_{i \geq 0}$$

by strict Barsotti–Tate subgroups [19] such that the Barsotti–Tate groups

$$\operatorname{Gr}^i(X_\eta) = \operatorname{Fil}^i(X_\eta)/\operatorname{Fil}^{i+1}(X_\eta)$$

18. Unpublished results of M. Raynaud imply, moreover, that this condition is always satisfied.

19. That is, subobjects as projective systems, or equivalently subobjects term by term.

have good reduction over S. If the filtration can be chosen to have length at most n (that is, if at most n of the Gr^i are nonzero), we say that X_η has *virtual good reduction of level n over S*.

Finally, we say that X_η has *essential good reduction over S* (respectively, *essential virtual good reduction*, respectively, *essential virtual good reduction of level n*) if there exists a finite extension K'/K such that $X_{K'}$ has good reduction (respectively, virtual good reduction, respectively, virtual good reduction of level n) over the normalization S' of S in K'.

5.12.1. When $\ell \neq p$, the categories of pro-ℓ Barsotti–Tate groups over η and over S may be interpreted in terms of ℓ-adic representations of

$$\pi = \pi_1(\eta), \qquad \pi_o = \pi_1(S),$$

respectively. The six notions just introduced can then be expressed in terms of the natural representation of the inertia group I on $X_\eta(\overline{K})$: the representation is, respectively, trivial, unipotent, unipotent of level n, essentially trivial, essentially unipotent, or essentially unipotent of level n.

5.12.2. Take $X_\eta = T_\ell(A_K)$, where A_K is an abelian scheme over K. We saw in 5.6 that, if A_K has semistable reduction over S, then X_η has virtual good reduction of level 2 (and we shall see the converse in a moment). Consequently, by the semistable reduction theorem 3.6, X_η has essential virtual good reduction of level 2. This statement should be regarded as the "right" formulation of the geometric monodromy theorem in Exposé III [20] for H^1 of a proper smooth scheme over K, without excluding the case $\ell = p$ (which was not considered there). The geometric monodromy theorem—or, for the present, conjecture—for the higher H^i given in Exposé III should likewise be reformulated so as to include "p-adic" coefficients, as was observed in 0.4.

We now summarize the relations between the reduction properties of abelian schemes over K and those of the associated Barsotti–Tate groups.

Proposition 5.13. Let S be a henselian trait, let A_K be an abelian scheme over its fraction field, and let ℓ be a prime number. When $\ell = p$, assume that K has characteristic zero (cf. 5.11).

a) The abelian scheme A_K has good reduction over S (2.2.9) if and only if $T_\ell(A_K)$ has good reduction over S (5.12).

b) The abelian scheme A_K has essential good reduction over S—that is, there exists a finite extension K'/K such that $A_{K'}$ has good reduction over the normalization S' of S in K'—if and only if $T_\ell(A_K)$ has essential good reduction over S (5.12). In that case K' may be chosen to be a finite separable extension of K.

c) The following conditions are equivalent:

1) A_K has semistable reduction over S (3.4);

2) $T_\ell(A_K)$ has virtual good reduction of level 2 over S (5.12);

3) $T_\ell(A_K)$ has virtual good reduction over S.

Proof. Part a) is just 5.10 and has been included for reference.

For b), the first assertion follows immediately from a) and the definitions. To prove the final assertion, we may suppose that $\ell \neq p$. In view of a), it then suffices to take for K' the finite separable extension of K corresponding to an open subgroup π' of

$$\pi = \mathrm{Gal}(\overline{K}/K)$$

whose intersection with the inertia group I is contained in the kernel of the representation of I on $T_\ell(A_K(\overline{K}))$; this kernel is open by hypothesis (cf. 5.12.1).

For c), we have already noted that 1) implies 2) (5.12.2), and it is immediate that 2) implies 3). It remains to prove that 3) implies 1). For $\ell \neq p$, this is precisely the Galois criterion for semistable reduction 3.5. For $\ell = p$, argue as follows. Choose a prime $\ell' \neq p$ and apply

20. The source leaves the precise internal reference blank.

the considerations of 4.1 to it. We may suppose that S is strictly local (3.4.0), and choose a finite Galois extension K'/K, with group G, such that $A_{K'}$ has semistable reduction. We must prove that the representation of G on each of the three quotients occurring in (4.1.1) is trivial. Equivalently, their direct-sum representation is trivial, or again its trace is the "trivial" trace, that is, is constant. We saw in 4.3(a) that this trace function is independent of the choice of $\ell' \neq p$, and in 6.4 below we indicate a method of calculating it in terms of $T_p(A_K)$ as well. That calculation plainly gives a constant function when $T_p(A_K)$ has virtual good reduction, and this completes the proof.

Remark 5.13.1. It seems plausible that, if X_η is a pro-ℓ Barsotti–Tate group over η having virtual good reduction over S and essential virtual good reduction of level n, then it has virtual good reduction of level n. (This is true when $\ell \neq p$.)

The following result, an easy consequence of 5.7 and of the semistable reduction theorem 3.6, was pointed out to me by J.-P. Serre.

Proposition 5.14 (Criterion for essential good reduction). Let S be a trait with function field K, let A_K be an abelian scheme over K, and let ℓ be a prime number distinct from $\operatorname{char} K$.

a) Suppose that $\ell \neq p = \operatorname{char} k$. Then the following conditions are equivalent:

(i) A_K has essential good reduction over S; that is, there exists a finite separable extension K'/K such that $A_{K'}$ has good reduction over the normalization S' of S in K';

(ii) the action of the inertia group I on $U_\ell = T_\ell(A(\overline{K}))$ is essentially trivial; that is, it is trivial on an open subgroup of I, or equivalently factors through a finite quotient Γ of I;

(iii) the action of I on $U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ is semisimple.

b) Suppose that $\ell = p$. If $n = \dim A_K \neq 0$, then the image of I in the automorphism group of U_ℓ is not finite; more precisely, its image in the group $\mathbb{Z}_\ell^\times$ of automorphisms of $\bigwedge^{2n} U_\ell$ has finite index. A sufficient (but by no means necessary) condition for A_K to have essential good reduction is that the action of I on $U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ be semisimple.

Proof. When $\ell \neq p$, the equivalence of (i) and (ii) follows immediately from the good-reduction criterion 2.2.9, and (ii) implies (iii) because $\mathbb{Q}_\ell$ has characteristic zero. The homomorphism

$$\pi = \operatorname{Gal}(\overline{K}/K) \longrightarrow \mathbb{Z}_\ell^\times \simeq \operatorname{Aut}\left(\bigwedge^{2n} U_\ell\right)$$

is χ^n, where χ is induced by the character

$$\mathbb{Z}_\ell^\times = \operatorname{Aut}(\mathbb{Z}_\ell(1))$$

arising from the action on the ℓ^ν-th roots of unity. It is known that, when $\ell = p$, the image of I under this character has finite index [33]. The fact that the action of I on $U_p \otimes \mathbb{Q}_p$ need not be semisimple even when A_K is an elliptic curve with good reduction is exhibited in [28, IV A 2.4].

It remains to prove that (iii) implies (i), for every ℓ. We use the semistable reduction theorem 3.6, which immediately reduces us to the case where A_K has semistable reduction over S: hypothesis (iii) is preserved upon replacing K by a finite separable extension K', or equivalently I by an open subgroup I'. When $\ell \neq p$, the action of I on $U_\ell \otimes \mathbb{Q}_\ell$ is unipotent (3.5(v)) and semisimple, hence trivial, and the good-reduction criterion 2.2.9 gives the result. We now give an argument valid with no restriction on ℓ, which yields a slightly sharper statement.

Corollary 5.15. With the notation of 5.14, suppose that A_K has semistable reduction over S, and that the fixed part

$$(U_{\ell\mathbb{Q}_\ell})^I \overset{\text{def}}{=} (U_\ell^I)_{\mathbb{Q}_\ell}$$

admits an I-stable complement in $U_{\ell\mathbb{Q}_\ell}$. Then A_K has good reduction.

It is enough to prove under these hypotheses that $U_\ell = U_\ell^I$. By 5.6 this says precisely that the toric part T_o of A_o^o is zero, so that A_o^o is an abelian scheme; hence A is an abelian scheme by 2.2.9.

We may plainly suppose that k is separably closed. The intersection of the asserted complement in 5.15 with U_ℓ then defines a pro-ℓ subgroup F_K of $T_\ell(A_K)$ which is isogenous to the cotoric part

$$T_\ell(A_K)/T_\ell(A_K)^f.$$

The latter is pro-étale and unramified by 5.6, and the same is therefore true of F_K. It follows from 5.7 that F_K is contained in $T_\ell(A_K)^f$; thus $F_K = 0$, proving that U_ℓ is equal to its fixed part.

6. Remarks on the construction of the toric part and the fixed part of $T_p(A)$ in the case of non-semistable reduction.

6.1. The construction of the toric part of $T_\ell(A_K)$ given in 5.1 still makes sense as soon as S is complete, without assuming that A^o is ℓ-divisible; that is (when $p = \ell$), without assuming semistable reduction. That hypothesis had been used only in order to formulate the orthogonality theorem 5.2. Moreover, one can construct $T_p(A_K)^t$ while assuming only that S is henselian: one shows that the toric part $T_p(A_{\widehat{K}})^t$ over the completion, constructed by the method of 5.1, comes from a Barsotti–Tate pro-subgroup of $T_p(A_K)$, necessarily determined uniquely.

More precisely, one can construct a pro-p Barsotti–Tate group $T_p(A)^t$ over S and a homomorphism

$$T_p(A)^t \longrightarrow T_p(A)$$

of projective systems which is a monomorphism term by term and whose pullback to $\widehat{S}$ gives the analogous objects constructed by the method of 5.1. This follows by paraphrasing the construction of 5.1, except that, instead of lifting the whole maximal torus T_o of A_o, for each integer $\nu \geq 0$ one merely lifts the subgroup ${}_{p^\nu}T_o$ to a multiplicative-type subgroup $({}_{p^\nu}A^o)^t$ of ${}_{p^\nu}A^o$.

For the existence and uniqueness of such a lifting, note that the functor

$$(\mathrm{Sch})^o_{/S} \longrightarrow \mathrm{Sets}, \qquad S' \longmapsto \{\text{finite multiplicative-type subgroups of } A^o_{S'}\}$$

is represented by an S-scheme X of finite type (cf. the proof of SGA 3 XI 3.12(a)), which is étale over S by SGA 3 IX 3.6 bis. Consequently, there is exactly one section of X over S extending a given section of X_s over s, which proves the assertion.

The same argument shows more generally that, for every finite multiplicative-type subgroup H_o of A_o, there is a unique finite multiplicative-type subgroup H of A extending it. Consequently, for every toric Barsotti–Tate p-subgroup H_o of A_o (that is, one whose components are of multiplicative type), there is a unique Barsotti–Tate p-subgroup H of A extending it. This in turn gives a Barsotti–Tate p-subgroup of ${}_{p^\infty}A_K$, or, equivalently, a pro-p Barsotti–Tate subgroup of $T_p(A_K)$.

In particular, this procedure applies when H_o is the largest toric Barsotti–Tate p-subgroup of A_o. It is obtained by taking the largest p-divisible subgroup $\overline{A}_o$ of A^o_o ($\overline{A}_o = p^N A^o_o$ for N large) and the "toric part" of ${}_{p^\infty}\overline{A}_o$ (defined, for example, by SGA 3 XVII 7.2.1). We thus obtain a Barsotti–Tate subgroup of ${}_{p^\infty}A_K$ which might be called the toric part of the latter.

6.2. Again let $\overline{A}_o$ be the largest p-divisible subgroup scheme of A^o_o, and consider the Barsotti–Tate p-group

$$F_o = {}_{p^\infty}(\overline{A}_o).$$

It is immediate that F_o may be characterized as the largest Barsotti–Tate p-subgroup of A_o (or of A^o_o). It is natural to ask whether this Barsotti–Tate group can be extended to a Barsotti–Tate subgroup F of A, which would then play the role of the fixed part $T_\ell(A)^f$ of (2.2.3.2) in the case in which A^o is ℓ-divisible.

In the semistable case, where $\overline{A}_o = A^o_o$, the method of 2.2 answers the question affirmatively. In the general case, whenever the problem has a solution, it is readily seen to be unique. The corresponding pro-subgroups of $T_p(A_K)$ and $T_p(A'_K)$ (which again deserve to be called the fixed parts of the pro-groups in question) still satisfy the orthogonality theorem 5.2, and the fixed part $T_p(A_K)^f$ may still be characterized as in 5.8.

6.2.1. We shall nevertheless show that in general the question has no affirmative answer; that is, the fixed part $T_p(A)^f$ cannot in general be defined in the case of non-semistable reduction. Indeed, by a lifting theorem of Serre–Tate [30], when S is complete, the existence of the fixed part is equivalent to the existence of infinitesimal liftings $\overline{A}_n$ of $\overline{A}_o$ in the A_n, or again to the existence of a smooth group scheme $\overline{A}$ over S, an extension of an abelian scheme by a torus T, having special fibre $\overline{A}_o$, together with a formal homomorphism

$$\widehat{\overline{A}} \longrightarrow \widehat{A}$$

extending the inclusion on the special fibres.

Thus, when $\overline{A}_o$ is an abelian scheme, that is, when $T_o = 0$, this is equivalent to the existence of an abelian scheme $\overline{A}$ extending $\overline{A}_o$ and a homomorphism $\overline{A} \longrightarrow A$ extending the inclusion of the special fibres. It will therefore suffice to give an example in which $T_o = 0$, A_o^o is neither unipotent nor proper, and A_K is simple over K.

6.2.2. Here is an example, due to M. Raynaud. It is based on the observation that, if S is a trait, S' is the normalization of S in a finite separable extension K' of K, and A' is an abelian scheme over S', then the group scheme

$$A = \prod_{S'/S} A'/S'$$

over S is a Néron model: it identifies with the Néron model of its generic fibre, which is indeed an abelian scheme. This follows immediately from the definitions and from the fact that A' itself is a Néron model over S'. Then

$$A_o = \prod_{S'_s/s} (A' \times_{S'} S'_s)/S'_s$$

is an extension of an abelian scheme, nonzero if $A' \neq 0$, by a connected unipotent scheme, nonzero if $A' \neq 0$ and S' is ramified over S.

It therefore suffices to give an example in which $A' \neq 0$, S' is ramified over S (a complete local trait), and

$$A_K = \prod_{K'/K} A'_{K'}/K'$$

is simple over K. Take K'/K to be a ramified quadratic extension (such an extension always exists), and take A' to be an elliptic curve over S'. Then A has relative dimension two, and saying that A_K is simple over K means that there is no nonzero homomorphism

$$B_K \longrightarrow A_K,$$

where B_K is an abelian scheme of dimension one over K. Giving such a homomorphism is equivalent to giving a nonzero homomorphism

$$B_{K'} \longrightarrow A'_{K'},$$

which would necessarily be an isogeny. Thus A_K is simple over K precisely when $A'_{K'}$ is not isogenous to an elliptic curve coming from K.

If the contrary held, the j-invariant of A' would be algebraically dependent (over the prime field) on its conjugate $\overline{j}$ under the nontrivial K-automorphism of K'. If j is chosen so that

$$a = j + \overline{j} = \mathrm{Tr}_{K'/K}\, j$$

is transcendental over the prime field, this is equivalent to saying that j is algebraically dependent on a. By cardinality considerations, however, there exists an element $a \in V$ transcendental over the prime subfield P of K, and then an element $b \in V'$ (the normalization of V in K') which is transcendental over $P(a)$ and satisfies

$$\mathrm{Tr}_{V'/V}\, b = 0.$$

It is enough to take $j = a + b \in V'$. We may moreover suppose that $b \in \mathfrak{r}(V')$, and that the image of a in $k = V/\mathfrak{m}$ is a prescribed element of k, chosen distinct from the exceptional values $j_o = 0$ and $j_o = 12^3$. This is possible even when k has only two elements, because then these two contemplated values of j_o coincide. Thus the image j_o of j in $k(s')$ is different from 0 and 12^3, and it is known that there then exists an elliptic curve over S' with j-invariant j.

6.3. Return to the case in which S is merely assumed henselian. In analogy with the definitions of 4.1 for $\ell \neq p$, one can then define a filtration

$$T_p(A_K) \supset T_p(A_K)^{\mathrm{ef}} \supset T_p(A_K)^{\mathrm{et}} \supset 0 \tag{6.3.1}$$

by strict Barsotti–Tate pro-subgroups, called respectively the essentially fixed part and the essentially toric part of $T_p(A_K)$.

To do so, choose a finite Galois extension K' of K such that $A_{K'}$ has semistable reduction over the normalization S' of S in K'. The fixed part $T_p(A_{K'})^f$ of $T_p(A_{K'}) = (T_p(A_K))_{K'}$ is then defined (2.2.3), as is the toric part (6.1). The latter is moreover the orthogonal complement of the fixed part $T_p(A'_{K'})^f$ in $T_p(A'_{K'})$ (5.2). Plainly

$$T_p(A_{K'})^t \subset T_p(A_{K'})^f$$

(this follows from 5.3, or directly from the construction of the toric part in 6.1), and the Galois group of K'/K preserves these two strict pro-subgroups of $(T_p(A_K))_{K'}$. Descent therefore gives the desired filtration (6.3.1). It is independent of the choice of K', by the semistable-reduction property 3.3. Descent also plainly gives the orthogonality property

$$\begin{aligned} T_p(A_K)^{\mathrm{et}} &= \big(T_p(A'_{K'})^{\mathrm{ef}}\big)^{\perp}, \\ T_p(A_K)^{\mathrm{ef}} &= \big(T_p(A'_{K'})^{\mathrm{et}}\big)^{\perp}. \end{aligned} \tag{6.3.2}$$

One could give an intrinsic characterization of the filtration (6.3.1) in terms of the Barsotti–Tate pro-group $X = T_p(A_K)$ over the fraction field K of S, in the spirit of 5.8—at least, for the moment, in unequal characteristics. The essentially fixed part is the largest Barsotti–Tate pro-subgroup having "essentially good reduction over S": after a finite separable extension K' of K, it gives a Barsotti–Tate pro-group $X_{K'}$ having good reduction over the normalization S' of S in K', that is, extending to a Barsotti–Tate pro-group over S'. Such an extension is necessarily unique up to a unique isomorphism, by the already-cited theorem of Tate [33, Th. 4]. The essentially toric part is then defined in terms of the essentially fixed part by orthogonality (6.3.2).

Let us also record the following immediate consequence of 5.8 and the definitions; for $\ell = p$ it reduces to 5.12(a).

Proposition 6.3.3. Let ℓ be a prime number. If $\ell = p$, assume that $\operatorname{char} K = 0$.[21] Then A_K has essential good reduction over S (5.12(a)) if and only if its essentially fixed part $T_\ell(A_K)^{\mathrm{ef}}$ (4.1.1 and 6.3.1) is all of $T_\ell(A_K)$.

6.4. Recall that, when $\ell \neq p$, in 4.3(a) we exhibited a remarkable integer-valued function on the inertia group I of S, independent of the choice of ℓ. When the residue field is perfect, knowledge of this function suffices to reconstruct the conductor exponent of A (4.5): one takes, on a sufficiently large quotient $\Gamma = I/I'$ of I, the inner product of this function with the Artin conductor on Γ (cf. (4.6.2)). It is useful to observe that this integer-valued function on I can also be recovered from the knowledge of $T_\ell(A_K)$ for $\ell = p$—at least, for the time being, in unequal characteristics.

When the residue field k is finite, an analogous observation applies to the $\mathbb{Q}$-valued function considered in 4.3(b), defined on the subgroup π' of π consisting of the elements whose image in π_o has the form f_q^n, $n \in \mathbb{Z}$. We indicate the principle of the verification. Using the filtration (6.3.1), one is reduced to attaching a suitable function (on I, respectively on π') to a pro-p Barsotti–Tate

21. This hypothesis is probably unnecessary; cf. 5.11.

group Y over K having "essentially good reduction over S" in the sense of 5.12. Here Y will be one of the three quotients arising from the filtration (6.3.1).

In the case of 4.3(a), a suitable residue-field extension within S (EGA 0_{III} 10.3.1) reduces us to the case of a perfect residue field, or even an algebraically closed one if desired. Choose a finite Galois extension K' of K such that $Y_{K'}$ has good reduction over S'. We then have a Barsotti–Tate group $Y_{S'}$ over S' and an action of G on $Y_{S'}$ compatible with its action on S'. If k' is the residue field of S', then G acts, not necessarily faithfully, on k' (whose invariant field is k), and on $Y_{k'}$ compatibly with its action on k'.

By Dieudonné theory [16 bis], the Barsotti–Tate pro-p group $Y_{k'}$ has a canonically associated finite free module M (of rank equal to the rank of $Y_{k'}$) over the ring $W(k')$ of Witt vectors of k'. The group G plainly acts on M compatibly with its actions on $W(k')$, through its actions on k'. Hence the inertia subgroup Γ of G acts $W(k')$-linearly on M; since Γ is a quotient of the inertia group I, this gives a trace function on Γ, and hence on I, which is the desired function.

A priori this function takes values in $W(k')$,[22] but it agrees with the analogous function defined in terms of the T_ℓ for $\ell \neq p$ whenever the action of G on $Y_{S'}$ is obtained from an action of G on a smooth p-divisible group scheme C' over S' by taking $Y = T_p(C')$ (cf. [25 bis, Th. 7, p. 737]). It is therefore integer-valued in this case. The case of 4.3(b) is handled similarly: to define a trace function on π', one uses the method of the proof of 4.3.2 to "linearize" on M the action defined by an element $g \in \pi$ lying above an f_q^n, with $n \geq 0$.

7. The Raynaud extension $G^\natural$ over S associated with the abelian scheme A_K having semistable reduction. Duality of the abelian schemes B, B' over S.

7.1. Let S be a Noetherian scheme which is the spectrum of a ring V, separated and complete for the topology defined by an ideal $\underline{m}$ of V. Let A be a smooth group scheme over S, let A^o be its identity component, and suppose that A_o/A_o^o is finite (hence finite etale) over S_o, and that A_o^o is an extension of an abelian scheme by a torus. As usual, we have put $A_n = A \times_S S_n$, where $S_n = \operatorname{Spec}(V/\underline{m}^{n+1})$. We shall be interested chiefly in the case where S is a trait, the generic fiber A_K is an abelian scheme having semistable reduction over S, and A is the Neron model of A_K.

Let T_o be the maximal torus of A_o, so that $B_o = A_o^o/T_o$ is an abelian scheme. The construction of 5.1 then applies and gives a formal subtorus $\widehat{T}$ of the formal completion $\widehat{A}$. For every n, the quotient

$$B_n = A_n^o/T_n \tag{7.1.1}$$

is representable (SGA 3 VI_A 3.2); more precisely, it is a flat scheme of finite type over S_n whose pullback to S_o is the abelian scheme B_o. Consequently B_n is an abelian scheme over S_n. As n varies, the B_n are obtained from one another by base change and therefore define a formal abelian scheme

$$\widehat{B} = (B_n)_{n \geqslant 0}\,. \tag{7.1.2}$$

We shall give conditions implying that this formal abelian scheme is algebraizable (EGA III 5.4.1), more precisely, that it is the formal completion of a projective abelian scheme over S, which we shall then denote by B. By loc. cit., B is determined up to a unique isomorphism (as an S-scheme; the group structure of $\widehat{B}$ then comes uniquely from a group structure on B, again by loc. cit.). This abelian scheme B will then be called the abelian part of the scheme A, and will sometimes be denoted by A^{ab}.

7.1.3. By EGA III 5.4.5, projectivity of $\widehat{B}$ means that one can find an invertible sheaf $\widehat{\mathbb{M}}$ on $\widehat{B}$, that is, a sequence of invertible sheaves $\mathbb{M}_n$ on the B_n which glue as n varies, and which is relatively ample over S, that is, such that $\mathbb{M}_o$ is ample. The natural idea would be to start with an ample invertible sheaf $\mathbb{L}$ on A and show that, for every n, its restriction $\mathbb{L}_n$ to A_n is the pullback

22. In fact it is immediate that this function always takes values in $W(k)$, since it is invariant under $\operatorname{Gal}(k'/k)$.

of an invertible sheaf $\mathbb{M}_n$ on B_n.[23] I do not know whether such $\mathbb{M}_n$ necessarily exist (and strongly doubt it). We shall follow a slightly different method, constructing a polarization of $\widehat{B}$, that is, a projective system of polarizations ρ_n of the B_n which glue. It is known [5] that the doubles of these polarizations canonically define (symmetric) invertible sheaves $\mathbb{M}_n$ on the B_n. By their canonical nature these sheaves necessarily glue and furnish the desired $\widehat{\mathbb{M}}$.

We shall interpret an element of $\mathrm{NS}(B_n/S_n)$ as a symmetric divisorial correspondence on (B_n, B_n) [5], that is, as a biextension of (B_n, B_n) by $\mathbb{G}_{mS_n}$. By VIII 3.5, these biextensions correspond bijectively to biextensions of (A_n, A_n) by $\mathbb{G}_{mS_n}$. To obtain a coherent system of such biextensions W_n, it suffices to start with

(7.1.4) $\qquad W$, a symmetric biextension of (A, A) by $\mathbb{G}_{mS}$,

and take the W_n obtained from W by the base changes $S_n \longrightarrow S$. We must in addition suppose that the element of $\mathrm{NS}(B_o)$ defined by W_o is a polarization of B_o; we thereby obtain a polarization of $\widehat{B}$.

Proposition 7.1.5. With the preceding notation for $S, A, \widehat{B}$, suppose that S is normal. Then $\widehat{B}$ is algebraizable, and the abelian scheme B over S from which it comes is projective over S.

We may plainly suppose that $A = A^o$ and that S is integral. By Raynaud [23, XI 1.13], the hypothesis that S is normal implies that A is quasi-projective over S. Choose a relatively ample invertible sheaf $\mathbb{L}$ on A, and form the birigidified invertible sheaf

$$\delta(\mathbb{L}) = \pi^*(\mathbb{L})\,\mathrm{pr}_1^*(\mathbb{L})^{-1}\,\mathrm{pr}_2^*(\mathbb{L})^{-1}$$

on $A \times_S A$, as in VIII 4.12. Since the fibers of A_o have unipotent rank zero, the same is true of the fibers of A (this follows easily from 3.1). Therefore, by VIII 7.5, $\delta(\mathbb{L})$ defines a biextension W of (A, A), plainly symmetric, which by VIII 4.13 (that is, essentially by Raynaud [23, XI 1.11]) satisfies the required ampleness condition at the points of S_o.

Remark 7.1.6. Without assuming that S is normal, one can prove that $\widehat{B}$ is always algebraizable by reducing to the normal case as in 13.5.3(a).

7.2. We now record the following lemma.

Lemma 7.2.1. Let S be the spectrum of a Noetherian ring V, separated and complete for the topology defined by an ideal $\underline{m}$ of V. Let C be the category of group schemes Λ over S, smooth and of finite type over S, such that Λ/Λ^o is a finite etale group scheme over S and Λ^o is an extension of an abelian scheme B by an isotrivial torus T (SGA 3 IX 1.1). Let C' be the category of group objects in the category of formal schemes of finite type (EGA I 10.3.3) over $\widehat{S} = \mathrm{Spf}(V)$, and consider the functor

$$A \longmapsto \widehat{A}$$

from C to C' which takes the formal completion along $A_o = A \times_S S_o$, where $S_o = \mathrm{Spec}(V/\underline{m})$.[24] Then:

(a) The preceding functor is fully faithful.

(b) Let $\mathcal{A}$ be an object of C', smooth over $\widehat{S}$; that is, suppose that the ordinary schemes

$$A_n = \mathcal{A} \times_{\widehat{S}} S_n, \qquad S_n = \mathrm{Spec}(V/\underline{m}^{n+1}),$$

are smooth over S_n. Suppose that A_o/A_o^o is a finite etale group scheme over S_o, and that A_o^o is an extension of an abelian scheme B_o by an isotrivial torus T_o. Put

$$\mathcal{A}^o = (A_n^o)_{n\geq 0}, \qquad \mathcal{A}/\mathcal{A}^o = (A_n/A_n^o)_{n\geq 0}.$$

23. The printed text changes incoherently here from an invertible sheaf $\mathbb{M}$ on G to the restriction $\mathbb{L}_n$ of $\mathbb{L}$ to G_n. The notation above is the type-correct reading fixed by the surrounding construction and by Proposition 7.1.5.

24. At both occurrences in this statement the print writes $\mathrm{Specf}(A)$, and it also writes $\mathrm{Spec}(A/\underline{m}^{n+1})$ below. Since V is the adic base ring and A denotes the varying group scheme, the type-correct base is $\mathrm{Spf}(V)$ and $S_n = \mathrm{Spec}(V/\underline{m}^{n+1})$.

Then $\mathcal{A}/\mathcal{A}^o$ is finite etale over $\widehat{S}$, and $\mathcal{A}^o$ is, in an essentially unique manner and functorially in $\mathcal{A}$, an extension of a formal abelian scheme $\mathcal{B}$ over $\widehat{S}$ by a formal torus $\mathcal{T}$ over $\widehat{S}$, reducing respectively to B_o and T_o. The formal group scheme $\mathcal{A}$ belongs to the essential image of $C \longrightarrow C'$—or, as we shall also say, is "algebraizable"—if and only if $\mathcal{B}$ is algebraizable.

7.2.1.1. First observe, over an arbitrary scheme S, that if A is a smooth group scheme such that A^o is an extension of an abelian scheme by a torus, this extension structure is essentially unique and is functorial in A in the evident sense. Indeed, the assertion reduces at once to the fact that every morphism from a torus T to an abelian scheme B is zero. This is well known on geometric fibers [15], and SGA 3 IX 5.3 then gives the conclusion. Applying the result to the compatible systems of smooth group schemes A_n over the S_n occurring in 7.2.1 gives the analogous result for formal group schemes over S. Moreover, we already observed in 7.1 that, if A_o^o is an extension of an abelian scheme by a torus, the same is true of A_n^o. This proves the first assertion of (b).

7.2.1.2. Suppose next that the formal group scheme $\mathcal{A}$ over S is of the form $\widehat{A}$, with $A \in \mathrm{ob}\, C$. If A^o is an extension of an abelian scheme B by a torus T, then plainly $\mathcal{B} \simeq \widehat{B}$. Thus algebraizability of $\mathcal{A}$ implies algebraizability of $\mathcal{B}$.

Conversely, suppose that $\mathcal{B}$ is algebraizable, that is, that it comes from an abelian scheme B, and let us prove that $\mathcal{A}$ is algebraizable. We first prove that $\mathcal{A}^o = (A_n^o)$ is algebraizable. By VIII 3.7, the extensions A_n^o of B_n by T_n may be interpreted as homomorphisms

$$(*) \qquad \underline{M}_n \longrightarrow B'_n,$$

where $\underline{M}_n$ is the twisted constant group over S_n dual to T_n (SGA 3 X 5.7), and B'_n is the abelian scheme dual to B_n. Restriction to S_o, or to any S_n, induces an equivalence from the category of isotrivial tori over S to the corresponding category over S_o (SGA 3 X 3.2). Hence the T_n come from a torus T over S, determined up to a unique isomorphism. Denote its dual group by $\underline{M}$; its restriction to S_n is canonically isomorphic to $\underline{M}_n$.

The isotriviality of T means that $\underline{M}$ is a disjoint union of S-schemes $\underline{M}^i$ which are finite over S (SGA 3 X 5.11). It follows at once that the homomorphisms $(*)$ come from a unique homomorphism

$$(**) \qquad \underline{M} \longrightarrow B',$$

where B' is the abelian scheme dual to B. In turn, this homomorphism is interpreted as an extension of B by T, say A^o, which plainly supplies an algebraization of $\mathcal{A}^o$.

7.2.1.3. On the other hand, by hypothesis A_o/A_o^o is a finite group scheme Φ_o over S_o, and the A_n/A_n^o are likewise finite group schemes Φ_n over S_n; they are necessarily etale over S_n.[25] There is therefore an etale group scheme Φ over S, determined up to a unique isomorphism, whose formal completion is the formal group scheme $\mathcal{A}/\mathcal{A}^o$ defined by the Φ_n.

Since $G = A^o$ is divisible, the category of extensions, as fppf sheaves, of a finite locally constant group over S such as Φ by the group scheme A^o is canonically equivalent to the category of torsors under the sheaf

$$H = \underline{\mathrm{Hom}}_{\mathrm{gr}}(\Phi, G),$$

and this equivalence is compatible with every base change. Applying the same description to extensions of Φ_n by G_n in terms of torsors under

$$H_n \simeq \underline{\mathrm{Hom}}_{\mathrm{gr}}(\Phi_n, G_n),$$

the A_n define a coherent system of torsors P_n under the H_n. Because G is an extension of an abelian scheme by a torus, it follows readily that H is a group scheme finite and locally free over S. Consequently, the functor

$$P \longmapsto \widehat{P} = (P_n)$$

25. The print writes S rather than the fiber bases S_o and S_n in these sentences, and leaves blank the parenthetical citation from which representability of Φ_n is said to follow. The bases are restored from the statement of the lemma; the missing citation is not guessed.

is an equivalence between the category of torsors over S under H and the category of "formal" torsors $\widehat{P}$ under $\widehat{H}$, that is, coherent systems of torsors P_n under the H_n.

Thus the P_n defined by the extensions A_n come from a single torsor P over S under H. We obtain an extension A of Φ by $G = A^o$ which algebraizes $\mathcal{A}$, except that its representability remains to be checked. Regard A as a torsor over Φ under A^o_Φ. This torsor becomes trivial after base change by the finite, surjective, locally free morphism

$$P \times_S \Phi \longrightarrow \Phi,$$

since the base change $P \longrightarrow S$ splits the extension A of Φ by A^o. One may therefore conclude by one of Raynaud's many criteria for effectiveness of descent [23, XI 3.1(6)].

7.2.1.4. This completes the proof of (b), and only the full faithfulness assertion (a) remains. We leave it to the reader to verify that this assertion is essentially contained in the method of proof of (b), or, if preferred, to find an independent direct proof.

7.2.1.5. Notice that the appeal to Raynaud's delicate descent result becomes unnecessary if A^o is assumed quasi-projective over S, or, equivalently, if B is projective over S. This condition holds in particular when one starts from the formal group scheme $\widehat{A}$ over S, with A as in 7.1.5. This is the only case in which we shall use 7.2.1.

7.2.2. We apply 7.2.1 to the formal group scheme $\widehat{A}$ considered in 7.1, in the favorable case treated there in which the formal abelian scheme $\widehat{B}$ is algebraizable (cf. 7.1.5), and we suppose in addition that the torus T_o over S_o is isotrivial. We thereby obtain a group scheme $A^\natural$ over S, called the Raynaud group of A over S, characterized up to isomorphism by a canonical isomorphism of formal group schemes over $\mathrm{Spf}(V) = \widehat{S}$:

$$(A^\natural)^\wedge \simeq \widehat{A}. \tag{7.2.3}$$

It follows from 7.2.1 that

$$\Phi = A^\natural / A^{\natural o} \tag{7.2.4}$$

is a finite etale group over S, and that the identity component

$$A^{\natural o} = A^{o\natural}$$

is endowed with the structure of an extension

$$0 \longrightarrow T \longrightarrow A^{\natural o} \longrightarrow B \longrightarrow 0, \tag{7.2.5}$$

where B is the abelian scheme over S whose formal completion is the formal abelian scheme $\widehat{B}$ constructed in 7.1, and T is characterized, up to a unique isomorphism, as the isotrivial torus over S lifting T_o. On restriction to the S_n, the extension structure (7.2.5) gives the extension structures on $A^{\natural o}_n = A^o_n$ defined by (7.1.1).

7.2.6. It also follows immediately from 7.2.1 that the Raynaud group $A^\natural$, as well as the exact sequence (7.2.5), depends functorially on A (satisfying the conditions under consideration), and that its formation commutes with every base change $S' \longrightarrow S$ associated with a homomorphism of adic topological rings $V \longrightarrow V'$.

7.3. Let us now consider, for a fixed prime number ℓ, the pro-ℓ group $T_\ell(A^o)$. Identifying a finite scheme over S with the corresponding formal scheme, and defining the "fixed part" $T_\ell(A^o)^f$ as usual, we obtain an isomorphism as in (5.1.5):

$$T_\ell(A^o)^f \simeq T_\ell(\widehat{A}^o) \quad , \tag{7.3.1}$$

and, defining the "toric part" as in (5.1.6),

$$T_\ell(A^o)^t = T_\ell(\widehat{T}) \simeq T_\ell(T) \subset T_\ell(A^o)^f \simeq T_\ell(\widehat{A}^o) \quad , \tag{7.3.2}$$

we immediately obtain, for the "abelian part," an isomorphism

$$T_\ell(A^o)^{ab} \overset{\text{def}}{=} T_\ell(A^o)^f / T_\ell(A^o)^t \simeq T_\ell(\widehat{B}) \quad . \tag{7.3.3}$$

We have not yet used here the hypothesis that $\widehat{B}$ is algebraizable, the hypothesis that allowed us to construct the abelian scheme B and the Raynaud extension of B by T. Since there is, of course, a canonical isomorphism (the special case of (7.3.1) applied to the abelian scheme B!) $T_\ell(\widehat{B}) \simeq T_\ell(B)$, the isomorphism (7.3.3) also gives us

$$T_\ell(A^o)^{ab} \simeq T_\ell(B) \quad . \tag{7.3.4}$$

Let us also note in passing that (7.2.3) and (7.3.1) imply a canonical isomorphism

$$T_\ell(A^o)^f \simeq T_\ell(A^{\natural o}) \quad , \tag{7.3.5}$$

since the right-hand side is already a profinite group, that is, identical to its fixed part. The canonical exact sequence

$$0 \longrightarrow T_\ell(A^o)^t \longrightarrow T_\ell(A^o)^f \longrightarrow T_\ell(B) \longrightarrow 0 \quad , \tag{7.3.6}$$

which can in any case (without the algebraizability hypothesis on $\widehat{B}$) be interpreted as the exact sequence obtained by applying the functor T_ℓ to the exact sequence

$$0 \longrightarrow \widehat{T} \longrightarrow \widehat{A}^o \longrightarrow \widehat{B} \longrightarrow 0 \quad ,$$

can also be regarded as obtained by applying the functor T_ℓ to the canonical exact sequence of ordinary group schemes

$$0 \longrightarrow T \longrightarrow A^{\natural o} \longrightarrow B \longrightarrow 0 \qquad .$$

7.4. Suppose now that a second group scheme A' over S is given, satisfying the same hypotheses as A, and suppose that A and A' have connected fibres. Let W be a biextension of (A, A') by $\mathbb{G}_{mS}$. We thereby obtain a pairing (VIII 2.2)

$$T_\ell(A) \times T_\ell(A') \longrightarrow T_\ell(\mathbb{G}_{mS}) = \mathbb{Z}_\ell(1) \tag{7.4.1}$$

and, by restriction to the fixed parts, a pairing

$$T_\ell(A)^f \times T_\ell(A')^f = T_\ell(\widehat{A}) \times T_\ell(\widehat{A'}) \longrightarrow \mathbb{Z}_\ell(1) \qquad . \tag{7.4.2}$$

This can also be interpreted as the pairing defined by the biextension $\widehat{W}$ of $(\widehat{A}, \widehat{A'})$ by $\widehat{\mathbb{G}}_{mS}$ obtained from W by completion (here, for simplicity, defining a biextension of formal group schemes as a coherent family of biextensions W_n of (A_n, A'_n) by $\mathbb{G}_{mS_n}$). Now we know (VIII 3.5) that W_n comes from a biextension W_n^{ab} of (B_n, B'_n), where B'_n is the abelian part of A'_n, so that (7.4.2) factors through

(7.4.3) $$T_\ell(A)^{ab} \times T_\ell(A')^{ab} = T_\ell(\widehat{B}) \times T_\ell(\widehat{B'}) \longrightarrow \mathbb{Z}_\ell(1) \qquad .$$

This is also the pairing defined by the biextension

$$\widehat{W}^{ab} = (W_n^{ab})$$

of $(\widehat{B}, \widehat{B'})$ by $\widehat{\mathbb{G}}_{mS}$. When $\widehat{B}$ and $\widehat{B'}$ are algebraizable, so that abelian schemes B, B' over S are available, EGA IV 5.1.4 shows that $\widehat{W}^{ab}$ comes from a biextension, defined up to a unique isomorphism,

$$W^{ab} \quad \text{of} \quad (B, B') \quad \text{by} \quad \mathbb{G}_{mS},$$

and the corresponding pairing

(7.4.4) $$T_\ell(B) \times T_\ell(B') \longrightarrow T_\ell(\mathbb{G}_{mS})$$

is canonically isomorphic to (7.4.3).

Writing $D(B')$ for the abelian scheme dual to B', W^{ab} defines a homomorphism

(7.4.5) $$u : B \longrightarrow D(B') \qquad ,$$

whose properties reflect those of the pairings (7.4.4), that is, (7.4.3). For example, the pairing (7.4.3) is a perfect duality at a point $t \in S$ if and only if $u_t : B_t \longrightarrow D(B')_t$ is an isogeny whose kernel has rank prime to ℓ. This implies that u itself is an isogeny (since S is local, hence connected), and $\ker u$ is flat and finite over S, so that $\ker u$ has the same rank at every point. Thus the condition just considered is independent of the chosen point t.

7.4.6. This completes in particular the proof of 5.4. Indeed, under the conditions of that theorem (where S may be assumed complete), applying the preceding discussion to A^o and A'^o shows that, for every ℓ, the pairing (7.4.3) is a perfect duality at the generic point η, and hence also at the point s.

7.4.7. The preceding considerations likewise make the orthogonality theorem 5.2 clear, without recourse to Lemma 5.2.4. Indeed, along the way in 7.4 we established that the pairing (5.2.3.1) = (7.4.2) vanishes on the toric parts on both sides, and therefore factors through the pairing (7.4.3). It remains only to establish that this pairing is nondegenerate on the generic fibres of the pro-groups under consideration, knowing that it is so on the special fibres. This follows from the fact that if (7.4.5) induces an isogeny on the special fibres, then it is an isogeny.

7.4.8. Return to the general conditions of 7.4, and suppose that an open subset $U \subset S - S_o$ is given such that A_U and A'_U are abelian schemes, dual to one another through the biextension W_U. Restricting (7.3.6), and the analogous exact sequence for A', to U, we obtain canonical filtrations on the Barsotti–Tate pro-ℓ groups $T_\ell(A_U)$ and $T_\ell(A'_U)$:

(7.4.9) $$\begin{cases} T_\ell(A_U) \supset T_\ell(A_U)^f \supset T_\ell(A_U)^t \supset 0, \\ \\ T_\ell(A'_U) \supset T_\ell(A'_U)^f \supset T_\ell(A'_U)^t \supset 0 \qquad . \end{cases}$$

I claim that these two filtrations are dual to one another with respect to the pairing (7.4.1), which here is a perfect duality (VIII 3.2); that is, relation (5.2.2) and its symmetric counterpart still hold, with A_K, A'_K replaced by A_U, A'_U. We saw above that this orthogonality assertion is also

equivalent to saying that the biextension W^{ab} of (B, B') by $\mathbb{G}_{mS}$ is a perfect duality over U (or, equivalently when U is dense in S, over S). It is enough to verify this property at every point η of U; the usual argument then reduces us to the already known case in which S is a trait with generic point η.

The orthogonality relation also allows us to extend formulas (5.5.7), (5.5.8), and (5.5.9) to the present case, replacing restriction to the generic point η by restriction to the open subset U. Consequently, 5.6 likewise extends to the present case. We refrain from rewriting the preceding formulas and statement here in this context.

Let us summarize, from the point of view of Néron models, the results obtained concerning the abelian parts and their pairings.

Proposition 7.5. Let S be a complete trait with function field K. To every abelian scheme A_K over K having semistable reduction over S, the procedure of 7.1, applied to the identity component A^o of the Néron model A of A_K, associates its abelian part $B = A^{ab}$, which is an abelian scheme over S depending functorially on A_K. For every other abelian scheme A'_K with semistable reduction, one can associate to every divisorial correspondence W_K on (A_K, A'_K) a unique divisorial correspondence W^{ab} on (A^{ab}, A'^{ab}) such that, for every integer $n \geq 0$, the pullback of W_n^{ab} along $A_n^o \longrightarrow B_n = A_n^{oab}$ and $A'^o_n \longrightarrow B'_n = A'^{oab}_n$ is the biextension W_n of (A_n^o, A'^o_n) obtained from W by reduction, where W is the canonical extension (1.4) of W_K to (A^o, A'^o). The formation of W^{ab} is compatible with the formation of pullbacks of correspondences by morphisms of abelian schemes $\overline{A}_K \longrightarrow A_K$ and $\overline{A'}_K \longrightarrow A'_K$. If W_K is a perfect duality, then so is W^{ab}. If W_K is the divisorial correspondence on (A_K, A_K) associated with a polarization of A_K, then W^{ab} is associated with a polarization of A^{ab}.

The compatibility of the formation of the W^{ab} with pullbacks is immediate from the indicated characterization of the W^{ab}. The additional assertion concerning polarizations was established in 7.1.5 at the same time as the algebraizability of $\widehat{B}$.

7.6. Continue to suppose that S is a complete trait, with A the Néron model of an abelian scheme A_K over K. Set

$$A_K^\natural = (A^\natural)_K \quad , \tag{7.6.1}$$

where $A^\natural$ is the Raynaud group scheme (7.2.2): it is an extension of the finite etale group Φ by $A^{\natural o}$, while $A^{\natural o}$ is an extension of the abelian scheme B by the torus T. Thus $A_K^\natural$ is itself an extension of Φ_K by $(A_K^\natural)^o = (A^{\natural o})_K = (A^{o\natural})_K$, which is an extension of the abelian scheme B_K by the torus T_K. For every prime number ℓ, (7.3.5) and (7.3.2) give

$$T_\ell(A_K^\natural) \simeq T_\ell(A_K^{\natural o}) \simeq T_\ell(A_K)^f \ , \qquad T_\ell(T_K) \simeq T_\ell(A_K)^t \quad . \tag{7.6.2}$$

7.6.3. By 3.3, the formation of $A_K^{\natural o}$ commutes with every base change associated with a morphism of complete traits $S' \longrightarrow S$.

8. The Raynaud extension $A_K^{\natural o}$ in the non-semistable case, and the ind-group $A_K^\flat$

8.1. Continue to suppose that the trait S is complete, but no longer suppose necessarily that A_K has semistable reduction. As usual, let K' be a finite Galois extension of K such that $A_{K'}$ has semistable reduction relative to the normalization S' of S in K'. Consider the Raynaud extension $A_{K'}^\natural$ associated with $A_{K'}$ (7.6.1). It is clear that the Galois group of K'/K acts by transport of structure on this group scheme; hence, by Galois descent, it gives a group scheme over K, denoted $A_{K,K'}^\natural$, which depends on the choice of the extension K'. Its identity component, however, is independent of this choice up to canonical isomorphism, by 7.6.3, and will simply be denoted $A_K^{\natural o}$. Descent shows that $A_K^{\natural o}$ is an extension

(8.1.1) $$0 \longrightarrow T_K \longrightarrow A_K^{\natural o} \longrightarrow B_K \longrightarrow 0 \quad ,$$

where T_K is a torus and B_K an abelian scheme over K. Likewise, by the definition of the essentially fixed and essentially toric parts of $T_\ell(A_K)$ (4.1 and 6.5), the isomorphisms (7.6.2) give canonical isomorphisms

(8.1.2) $$T_\ell(A_K^\natural) \simeq T_\ell(A_K)^{ef} \quad , \qquad T_\ell(T_K) \simeq T_\ell(A_K)^{et} \quad ,$$

and hence, on passing to the quotient,

(8.1.3) $$T_\ell(B_K) \simeq T_\ell(A_K)^{ef}/T_\ell(A_K)^{et} \quad .$$

This therefore determines two of the quotients occurring in the filtration

(8.1.4) $$T_\ell(A_K) \supset T_\ell(A_K)^{ef} \supset T_\ell(A_K)^{et} \supset 0$$

Thus the extension of group schemes (8.1.1) describes two quotients in this filtration. For the third quotient $T_\ell(A_K)/T_\ell(A_K)^{ef}$, (6.5.2) gives a canonical isomorphism

(8.1.5) $$T_\ell(A_K)/T_\ell(A_K)^{ef} \simeq D(T_\ell(A'_K)^{et}) \quad ,$$

where $D = \underline{\mathrm{Hom}}(-, T_\ell(\mathbb{G}_m))$ denotes Cartier duality (in the sense of Barsotti–Tate pro-groups), and A'_K denotes the abelian scheme dual to A_K. Consider then the exact sequence of type (8.1.1)

(8.1.6) $$0 \longrightarrow T'_K \longrightarrow {A'_K}^{\natural o} \longrightarrow B'_K \longrightarrow 0$$

associated with the abelian scheme A'_K over K, together with the corresponding isomorphisms of type (8.1.2). Formula (8.1.5) then gives

(8.1.7) $$T_\ell(A_K)/T_\ell(A_K)^{ef} \simeq D(T_\ell(T'_K)) \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad ,$$

where, generalizing the notation introduced in (5.5.2), we have set

(8.1.8) $$\underline{M}_K = D(T'_K) \quad , \qquad \underline{M}'_K = D(T_K) \quad ,$$

and where D now denotes Cartier duality $\underline{\mathrm{Hom}}(-, \mathbb{G}_{mK})$ in the sense of ordinary group schemes. Thus $\underline{M}_K$ and $\underline{M}'_K$ are twisted constant groups over K which become constant over K' and isomorphic to groups of the form $\mathbb{Z}^r$, where $r = \dim T = \dim T'$.

8.2. The hypotheses are those of 8.1. It is clear that the extension (8.1.1) depends functorially on the abelian scheme A_K, and that its formation commutes with every morphism of traits $\widetilde{S} \longrightarrow S$. Choose a polarization of A_K, hence an isogeny $A_K \longrightarrow A'_K$. It induces a homomorphism from (8.1.1) to (8.1.6), which is immediately seen, by looking at the T_ℓ, to be an isogeny term by term. Likewise, using 7.5 and descent, one sees that every biextension by $\mathbb{G}_{mK}$ of abelian schemes A_K and $\overline{A}_K$ over K defines a biextension of $(B_K, \overline{B}_K)$ by $\mathbb{G}_{mK}$, where B_K and $\overline{B}_K$ denote the abelian quotients of the Raynaud extensions corresponding to A_K and $\overline{A}_K$, respectively. When W is a perfect duality, so is the corresponding biextension of $(B_K, \overline{B}_K)$. In particular, with the notation of 7.6, the abelian schemes B_K and B'_K are canonically dual to one another.

The correspondence between biextensions on $(A_K, \overline{A}_K)$ and biextensions on $(B_K, \overline{B}_K)$ has evident functoriality properties as A_K and $\overline{A}_K$ vary. These follow by descent from the analogous properties stated in 7.5, and we leave their formulation to the reader. We also leave to the reader the compatibility of the preceding construction with every change of traits $\widetilde{S} \longrightarrow S$.

8.3. Return to the group $A^{\natural}_{K,K'}$ and study its variation with K'. It is clear that two isomorphic extensions K' and K'_1 of K give canonically isomorphic groups, so it is enough to consider the subextensions K' of the separable closure $\overline{K}$ of K. If $K' \hookrightarrow K''$ are two such finite Galois subextensions of $\overline{K}$, it follows immediately from 3.3 that there is a canonical open immersion

$$A^{\natural}_{K,K'} \hookrightarrow A^{\natural}_{K,K''} \qquad , \tag{8.3.1}$$

which identifies the first group with an open subgroup of the second. As K' varies, the $A^{\natural}_{K,K'}$ thus form an inductive system of smooth groups of finite type over K, whose transition morphisms are open immersions. If we set

$$\Phi_{K,K'} = A^{\natural}_{K,K'}/{A^{\natural}_{K,K'}}^{o} \qquad , \tag{8.3.2}$$

the $\Phi_{K,K'}$ themselves form an inductive system of finite etale groups over K, which will be studied in greater detail below.

We shall set

$$A^{\flat}_K = \varinjlim_{K'} A^{\natural}_{K,K'} \qquad , \tag{8.3.3}$$

where the limit is taken in the category of fppf sheaves over K. Since the transition morphisms in this inductive system are open immersions, it is clear moreover that this limit is represented by a group scheme over K (in general non-quasi-compact), which we shall naturally denote by the same symbol $A^{\flat}_K$. For this group scheme one has

$${A^{\flat}_K}^{o} = A^{\natural o}_K \quad , \qquad A^{\flat}_K/{A^{\flat}_K}^{o} \simeq \varinjlim_{K'} \Phi_{K,K'} \qquad . \tag{8.3.4}$$

The latter group will be determined in 11.10: it is canonically isomorphic to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$, with the notation of (8.1.8).

9. Definition of the monodromy pairing $\underline{M}_\ell \otimes \underline{M}'_\ell \longrightarrow \mathbb{Z}_{\ell S}$.

9.1. In this section, suppose that S is henselian and that A_K has semistable reduction over S, or, equivalently, that A'_K has semistable reduction over S (3.5.1). We then have over S two twisted constant $\mathbb{Z}$-modules $\underline{M}$ and $\underline{M}'$ (5.5.2). Over a suitable finite etale covering of S (and hence over S itself if S is strictly local, that is, if k is separably closed), they split into constant groups with values M and M', respectively, noncanonically isomorphic to $\mathbb{Z}^r$, where r is the common dimension of the maximal tori of A_o and A'_o. For every prime number ℓ, we shall consider the corresponding ℓ-adic sheaves

$$\underline{M}_\ell = \underline{M} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad , \qquad \underline{M}'_\ell = \underline{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \qquad , \tag{9.1.1}$$

whose geometric meaning is made explicit by formulas (5.5.7) and (5.5.8). In this section we shall define a canonical pairing, called the monodromy pairing,

$$u_\ell : \underline{M}_\ell \otimes \underline{M}'_\ell \longrightarrow \mathbb{Z}_\ell \qquad , \tag{9.1.2}$$

which will be defined in terms of the Barsotti–Tate pro-ℓ-groups $T_\ell(A^o)$ and $T_\ell(A'^o)$, and even in terms of the generic fiber $T_\ell(A_K)$ of the first of these groups (subject, when $\ell = p$, to the availability of Tate's theorem [33, Th. 4], that is, for the time being, provided we are then in unequal characteristic). When $\ell \neq p$, this is equivalent to saying that the pairing (9.1.2) is described in terms of the Galois π-module

$$U = T_\ell(A_K)(\overline{K}) = T_\ell(A_{\overline{K}}) \tag{9.1.3}$$

considered in (2.5.3). More precisely, we shall see that knowing the pairing is essentially equivalent to knowing the action of the monodromy group I on U.

9.2. The case $\ell \neq p$. Begin with the simpler case $\ell \neq p$, using the notation introduced in (2.5.3). Note that, up to torsion, the relation $V = U^I$ also means that $V^\perp$ is the module generated by the elements $gx - x$, where $x \in U$ and $g \in I$, and in any case one has

$$gx - x \in V^\perp \quad \text{for } x \in U,\ g \in I \qquad . \tag{9.2.1}$$

Thus, for $g \in I$,

$$gx = x + u(g)\,.\,x \qquad , \tag{9.2.2}$$

where $g \longmapsto u(g)$ is a homomorphism from U to $V^\perp$ which vanishes on $V = U^I$, or equivalently

$$u \in \mathrm{Hom}(U/V, V^\perp) \simeq (U/V)^\vee \otimes V^\perp \qquad . \tag{9.2.3}$$

In the case of semistable reduction under consideration, that is, when $V^\perp \subset V$, formula (9.2.2) says that the action of g on U is a transvection parallel to $V^\perp = W$, and composition of these transvections is simply addition of the corresponding homomorphisms u in (9.2.3). Consequently, the action of the inertia group I on U is completely described by a group homomorphism

$$I \longrightarrow (U/V)^\vee \otimes V^\perp \qquad . \tag{9.2.4}$$

The orthogonality theorem, applied both to A_K and to A'_K, gives

$$V^\perp = W \quad , \qquad (U/V)^\vee \simeq W'(-1) \qquad ,$$

where W' is the toric part of $U' = T_\ell(A'(\overline{K}))$, and (-1) denotes the Tate twist. Moreover, the homomorphism (9.2.4) factors through $I(\ell) \simeq \mathbb{Z}_\ell(1)$, so that (9.2.4) may be interpreted as a canonical homomorphism

$$u : \mathbb{Z}_\ell(1) \longrightarrow W' \otimes W(-1) \qquad , \tag{9.2.5}$$

or, equivalently, as an element

$$u \in W' \otimes W(-2) \qquad . \tag{9.2.6}$$

Recall that W and W' are the groups of $\overline{K}$-valued points of the two twisted constant ℓ-adic sheaves $T_\ell(A^o)^t$ and $T_\ell(A'^o)^t$, whose special fibers are $T_\ell(T_o)$ and $T_\ell(T'_o)$, respectively. We obtain the canonical isomorphisms

(9.2.7)
$$\begin{cases} W \simeq T_\ell(T_o)(\overline{k}) \simeq \check{M} \otimes \mathbb{Z}_\ell(1), \\ W' \simeq T_\ell(T'_o)(\overline{k}) \simeq \check{M}' \otimes \mathbb{Z}_\ell(1), \end{cases}$$

where M and M' denote the character groups over $\overline{k}$ of T_o and T'_o, respectively:

(9.2.8)
$$T_o = D(M) \quad , \qquad T'_o = D(M') \qquad .$$

It follows that the element u of (9.2.6), which describes the action of the inertia group I on the Tate module U of A_K, is interpreted as a canonical element of $(\check{M}' \otimes_{\mathbb{Z}} \check{M}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. It is moreover invariant under $\pi_o = \pi/I = \mathrm{Gal}(\overline{k}/k)$, since by definition it is invariant under transport of structure:

(9.2.9)
$$u \in \left((\check{M}' \otimes_{\mathbb{Z}} \check{M}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell\right)^I \simeq \mathrm{Hom}_{\mathbb{Z}}(\underline{M} \otimes \underline{M}', \mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(\underline{M}_\ell \otimes \underline{M}'_\ell, \mathbb{Z}_\ell) \quad .$$

Up to writing u in place of u_ℓ, this is the element (9.1.2) that we wished to define.

To define (9.1.2) in the general case, we shall need a few preliminaries, which will be developed in the next two sections.

9.3. Mixed extensions.

Let $\underline{C}$ be an abelian category, and let

(9.3.1)
$$P,\ R,\ Q$$

be three objects of $\underline{C}$. We are interested in classifying the objects X of $\underline{C}$ endowed with a three-step filtration

(*)
$$X^0 = X \supset X^1 \supset X^2 \supset X^3 = 0$$

and with given isomorphisms [26]

$$X^0/X^1 \xrightarrow{\sim} Q, \qquad X^1/X^2 \xrightarrow{\sim} R, \qquad X^2/X^3 = X^2 \xrightarrow{\sim} P.$$

Putting

$$E = X^0/X^2, \qquad F = X^1/X^3 = X^1,$$

we see that X gives rise to two ordinary extensions

(9.3.2)
$$0 \longrightarrow P \longrightarrow F \longrightarrow R \longrightarrow 0, \qquad 0 \longrightarrow R \longrightarrow E \longrightarrow Q \longrightarrow 0,$$

of R by P and of Q by R, respectively. Our point of view here will be to start, in addition to the objects (9.3.1), with the extension structures (9.3.2), which are likewise assumed to be given, and to consider the corresponding three-step filtered structures X.

More precisely, a mixed extension of the extension E by the extension F will mean an object X of $\underline{C}$, endowed with a three-step filtration (*) and with given isomorphisms

26. The printed source assigns P, R, Q, in that order, to these three graded pieces. The immediately following exact sequences and every subsequent construction require Q, R, P; the evident interchange of P and Q has therefore been corrected here.

(9.3.3) $$E \simeq X^0/X^2, \qquad F \simeq X^1/X^3 = X^1,$$

compatible with the filtrations: they carry X^1/X^2 to $\mathrm{Im}(R \longrightarrow E)$ and X^2 to $\mathrm{Im}(P \longrightarrow F)$, respectively, and the isomorphism $R \simeq F/P$ (and hence $R \simeq X^1/X^2$) deduced from (9.3.3) is required to equal the one deduced from the first exact sequence in (9.3.2).

For fixed P, Q, R and fixed extensions E and F, the mixed extensions of E by F form a category in the evident way; it is in fact a groupoid (every arrow is invertible). The automorphism group of a mixed extension X of E by F is canonically isomorphic to $\mathrm{Hom}(Q, P)$ through

(9.3.4) $$\mathrm{Hom}(Q, P) \xrightarrow{\sim} \mathrm{Aut}_{\mathrm{extpan}}(X), \qquad u \longmapsto \mathrm{id}_X + \bar{u},$$

where $\bar{u}$ is the endomorphism of the object of $\underline{C}$ underlying X obtained by composing $X \longrightarrow Q \xrightarrow{u} P \longrightarrow X$; the outer arrows are respectively the canonical epimorphism and the canonical monomorphism coming from the given mixed-extension structure on X.

One could make the structure of the category $\mathrm{EXTPAN}(E, F)$ of mixed extensions of E by F precise by regarding it as a "torsor" under the "group-category" (VII 2.5) $\mathrm{EXT}(Q, P)$ of extensions of Q by P, using a natural functor analogous to the Brauer composition of ordinary extensions:

(9.3.5) $$\begin{aligned} \omega : \mathrm{EXT}(Q, P) \times \mathrm{EXTPAN}(E, F) &\longrightarrow \mathrm{EXTPAN}(E, F), \\ \omega(G, X) &= G \wedge X. \end{aligned}$$

For every object X of $\mathrm{EXTPAN}(E, F)$, the partial functor $G \longmapsto G \wedge X$ is an equivalence from $\mathrm{EXT}(Q, P)$ to $\mathrm{EXTPAN}(E, F)$.

We shall give only a brief description of ω. Regard a mixed extension X of E by F as defining an ordinary extension, also denoted X, of Q by F (viewed as an object of $\underline{C}$). For every extension G of Q by P, let $\bar{G}$ denote the extension of Q by F induced by the inclusion $P \longrightarrow F$. Set

(9.3.6) $$G \wedge X = \omega(G, X) \overset{\mathrm{dfn}}{=} \bar{G} \wedge X,$$

where $\wedge$ denotes Brauer composition in the category of extensions of Q by F. The three-step filtration of the right-hand side X' of (9.3.6) is clear, as is the second isomorphism of type (9.3.3) attached to X'.

To define the first isomorphism of type (9.3.3) for X', observe that the functor

$$Y \longmapsto \widetilde{Y}, \qquad \mathrm{EXT}(Q, F) \longrightarrow \mathrm{EXT}(Q, R),$$

induced by the given morphism $F \longrightarrow R$ in (9.3.2) is compatible with Brauer composition. It therefore sends X' to an extension $\widetilde{X'}$ canonically isomorphic to $\widetilde{\bar{G}} \wedge \widetilde{X}$. The first isomorphism (9.3.3) gives an isomorphism $\widetilde{X} \simeq E$, while $\widetilde{\bar{G}}$—the extension induced from G by the composite $P \longrightarrow F \longrightarrow R$—is canonically isomorphic to the trivial extension, since that composite $P \longrightarrow R$ is zero. Consequently there is a canonical isomorphism $\widetilde{X'} \simeq E$, which specifies on the right-hand side of (9.3.6) its mixed-extension structure of E by F. [27]

We leave to the reader the definition of ω on the arrows of $\mathrm{EXT}(Q, P) \times \mathrm{EXTPAN}(E, F)$, the verification that (9.3.5) has the asserted property—namely, for every fixed mixed extension X, the functor $G \longmapsto G \wedge X$ is an equivalence from $\mathrm{EXT}(Q, P)$ to $\mathrm{EXTPAN}(E, F)$—and the proof of the associativity formula

27. At both occurrences in this paragraph the printed source has F where the objects involved are extensions of Q by R and hence must be identified with E.

(9.3.7) $$G_1 \wedge (G_2 \wedge X) \simeq (G_1 \wedge G_2) \wedge X,$$

where $G_1 \wedge G_2$ denotes Brauer composition in $\mathrm{EXT}(Q, P)$.

The preceding discussion immediately gives parts (a) and (b) of the following result.

Proposition 9.3.8. Given two extensions E and F as in (9.3.2) in the abelian category $\underline{C}$, consider the category $\mathrm{EXTPAN}(E, F)$ of mixed extensions of E by F. Then:

a) $\mathrm{EXTPAN}(E, F)$ is a groupoid. For every one of its objects the automorphism group is canonically isomorphic to $\mathrm{Ext}^0(Q, P) = \mathrm{Hom}(Q, P)$.

b) The set $\mathrm{Extpan}(E, F)$ of isomorphism classes of objects of $\mathrm{EXTPAN}(E, F)$ is empty, or is naturally a torsor under the group $\mathrm{Ext}^1(Q, P)$.

c) One can canonically define a class

(9.3.9) $$c(E, F) \in \mathrm{Ext}^2(Q, P),$$

whose vanishing is necessary and sufficient for $\mathrm{Extpan}(E, F) \neq \varnothing$, that is, for the existence of a mixed extension of E by F.

It remains to prove (c). [28] A mixed extension of E by F may also be described as an extension X of E by P together with an isomorphism of extensions of R by P

$$X^1 \simeq F,$$

where X^1 denotes the image of the extension X under the morphism $R \longrightarrow E$ in (9.3.2). Such a mixed extension therefore exists if and only if there is an element of $\mathrm{Ext}^1(E, P)$ whose image in $\mathrm{Ext}^1(R, P)$ is the class ξ of F. Consider the exact sequence of the $\mathrm{Ext}^i(-, P)$ associated with the second exact sequence in (9.3.2) [29]:

$$\cdots \longrightarrow \mathrm{Ext}^1(E, P) \longrightarrow \mathrm{Ext}^1(R, P) \xrightarrow{\partial} \mathrm{Ext}^2(Q, P) \longrightarrow \cdots .$$

It is enough to define $c(E, F)$ by

(9.3.10) $$c(E, F) = \partial\xi, \qquad \xi = \mathrm{cl}(F) \in \mathrm{Ext}^1(R, P).$$

Now suppose that we are given an exact functor

(9.3.11) $$r : \underline{C} \longrightarrow \underline{C}',$$

where $\underline{C}'$ is another abelian category. Write P', Q', R', E', F' for the objects of $\underline{C}'$ obtained from P, Q, R, E, F by applying r, so that E' is an extension of Q' by R' and F' is an extension of R' by P'. Up to canonical isomorphism, the following diagram of functors is commutative:

(9.3.12)
$$\begin{array}{ccc} \mathrm{EXT}(Q, P) \times \mathrm{EXTPAN}(E, F) & \xrightarrow{\omega} & \mathrm{EXTPAN}(E, F) \\ \downarrow & & \downarrow \\ \mathrm{EXT}(Q', P') \times \mathrm{EXTPAN}(E', F') & \xrightarrow[\omega']{} & \mathrm{EXTPAN}(E', F'). \end{array}$$

28. The printed source says "prove (d)", although Proposition 9.3.8 has only parts (a), (b), and (c).

29. The printed final term is $\mathrm{Ext}^2(R, Q)$; applying $\mathrm{Ext}(-, P)$ to $0 \to R \to E \to Q \to 0$ gives the displayed $\mathrm{Ext}^2(Q, P)$, which is also the group specified in (9.3.9).

The vertical arrows are the evident functors induced by the exact functor r, and ω' is the arrow analogous to ω in (9.3.5), with $\underline{C}$ replaced by $\underline{C}'$. The lower arrow in the printed diagram is labelled ω without a prime; the explanatory sentence identifies it as ω', which is the notation used here. We obtain analogous compatibilities, with the functor r, of the structures considered in 9.3.8; their explicit formulation is left to the reader.

We shall use the following consequence of this commutativity. Suppose that $\mathrm{EXTPAN}(E,F)$ is nonempty and choose an object X in it. Then, by means of $G' \longmapsto G' \wedge X'$, where $X' = r(X)$, the category $\mathrm{EXTPAN}(E',F')$ is equivalent to $\mathrm{EXT}(Q',P')$. Thus an object Y' of $\mathrm{EXTPAN}(E',F')$ is known up to isomorphism when the isomorphism class of the corresponding object of $\mathrm{EXT}(Q',P')$ is known, that is, when the corresponding element

$$c_X(Y') \in \mathrm{Ext}^1(Q',P')$$

is known.

If X is replaced by another object $X_1 = H \wedge X$, where H is an extension of Q by P, then one immediately obtains [30]

$$c_{X_1}(Y') = c_X(Y') - \mathrm{cl}(H')$$

from the associativity formula (9.3.7). Hence, for fixed Y' and varying X, the elements $c_X(Y')$ run through exactly one coset

$$c(Y') \in \mathrm{Ext}^1(Q',P')/\operatorname{Im}\mathrm{Ext}^1(Q,P), \tag{9.3.13}$$

which is therefore canonically associated with the mixed extension Y' of E' by F'. By construction, its vanishing is necessary and sufficient for Y' to be isomorphic to the transform under r of a mixed extension X of E by F.

9.4. Let S be a henselian trait with generic point η, let n be an integer > 0, put $\Lambda = \mathbb{Z}/n\mathbb{Z}$, and let $\underline{C}$ be the category of Λ-modules on S (for the fppf site on S; cf. VIII 0.2) and $\underline{C}'$ the category of Λ-modules on η. Let P,R,Q be objects of $\underline{C}$, and let E,F be extensions of Q by R and of R by P, respectively (9.3.2). Suppose that Q is a locally free Λ_S-module of finite type and that P is represented by a finite group of multiplicative type over S (SGA 3 IX), so that it is of the form

$$P = D(Q^*),$$

where $D = \underline{\mathrm{Hom}}(-,\mathbb{G}_{mS})$ denotes Cartier duality. Let $P_\eta, R_\eta, Q_\eta, E_\eta, F_\eta$ denote the restrictions of P,R,Q,E,F to η, and suppose that a mixed extension X_η of E_η by F_η is given. Under these conditions we shall canonically define a pairing

$$c(X_\eta) : Q \otimes_\Lambda Q^* \longrightarrow \Lambda_S \qquad (\Lambda = \mathbb{Z}/n\mathbb{Z}). \tag{9.4.1}$$

For this purpose, first suppose that S is strictly local. Then Q and Q^* are constant, say $Q = Q_{oS}$ and $Q^* = Q^*_{oS}$, and giving (9.4.1) amounts to giving an element

$$c(X_\eta) \in \check{Q}_o \otimes \check{Q}^*_o, \tag{9.4.2}$$

where the check denotes the ordinary dual $\mathrm{Hom}(-,\Lambda)$. We apply the considerations of 9.3 with the restriction functor $r : \underline{C} \longrightarrow \underline{C}'$. First observe that a mixed extension of E by F exists. Indeed, the obstruction to its existence lies in $\mathrm{Ext}^2(Q,P)$ (9.3.8(c)), and it is enough to prove

$$\mathrm{Ext}^2(Q,P) = 0.$$

30. The printed right-hand side of the following formula has $c_X(Y)$; the fixed object in this construction is Y'.

This is a special case of the following lemma.

Lemma 9.4.3. Under the preceding hypotheses, with S strictly local, one has:

a) canonical isomorphisms

$$\mathrm{Ext}^i_\Lambda(Q,P) \simeq H^i(S, \check{Q}\otimes P), \qquad \mathrm{Ext}^i_\Lambda(Q_\eta,P_\eta) \simeq H^i(\eta, \check{Q}\otimes P);$$

b) a canonical isomorphism

$$\check{Q}\otimes P \simeq D(Q\otimes Q^*); \tag{b}$$

c) for every finite constant group $H = H_{oS}$ over S,

$$H^i(S, D(H)) \simeq H^i(\eta, D(H)_\eta) = 0 \qquad \text{for } i \geq 2; \tag{c}$$

d) canonical isomorphisms

$$\begin{aligned} H^1(S, D(H)) &\simeq \mathbb{G}_m(S)\otimes \check{H}_o = V^*\otimes \check{H}_o, \\ H^1(\eta, D(H)_\eta) &\simeq \mathbb{G}_m(\eta)\otimes \check{H}_o = K^*\otimes \check{H}_o, \end{aligned} \tag{d}$$

where the check in (d) denotes $\mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$.

The formulas in (a) and (b) follow immediately from the fact that Q is a locally free Λ_S-module of finite type. To prove (c), we plainly reduce to the case $H_o = \mathbb{Z}/m\mathbb{Z}$, that is, $D(H) = \mu_m$. The cohomology sequence associated with the Kummer exact sequence

$$0 \longrightarrow \mu_m \longrightarrow \mathbb{G}_m \xrightarrow{m\cdot \mathrm{id}} \mathbb{G}_m \longrightarrow 0$$

then reduces us to proving

$$H^i(S, \mathbb{G}_m) = H^i(\eta, \mathbb{G}_m) = 0 \qquad \text{for } i \geq 1.$$

These groups are the groups $H^i_{\text{ét}}$ computed on the étale site [6, Theorem 11.7], and therefore vanish for S, since S is strictly local. Their vanishing for η is likewise well known [6, Theorem 1.1].

It remains to describe the isomorphisms in (d). Over an arbitrary base U there is a canonical homomorphism

$$\mathbb{G}_m(U)\otimes \check{H}_o \longrightarrow H^1(U, D(H_{oU})). \tag{9.4.3.1}$$

If H_o is annihilated by an integer $m > 0$, this homomorphism is described using the boundary map $\mathbb{G}_m(U) \xrightarrow{\partial} H^1(U, \mu_m)$ associated with the Kummer sequence: it is the composite

$$\begin{aligned} \mathbb{G}_m(U)\otimes \check{H}_o &\longrightarrow H^1(U,\mu_m)\otimes \check{H}_o \\ &\longrightarrow H^1(U, \mu_m\otimes \check{H}_o) \longrightarrow H^1(U, D(H_{oU})). \end{aligned}$$

Here the second arrow is deduced from the evident homomorphism $\check{H}_o \longrightarrow \mathrm{Hom}(\mu_m, \mu_m\otimes \check{H}_o)$, and the third is a special case of (b), with m in place of n. We leave it to the reader to check that (9.4.3.1) is independent of the choice of the integer m such that $mH_o = 0$. We also claim that (9.4.3.1) is always injective and that its cokernel vanishes if $\mathrm{Pic}(U) = 0$. (In fact, the cokernel is always isomorphic to $\mathrm{Hom}(H_o, \mathrm{Pic}(U))$, but this is immaterial here.) Indeed, we reduce to the case $H_o = \mathbb{Z}/n\mathbb{Z}$, and the assertion then follows immediately from the cohomology sequence of the Kummer sequence above. This proves (d), and immediately yields:

Corollary 9.4.4. With the notation of 9.4.3, there is a canonical isomorphism
e)

$$H^1(\eta, D(H)_\eta)/\operatorname{Im} H^1(S, D(H)) \simeq \check{H}_o,$$

and hence, on combining (a) and (b), a canonical isomorphism

$$\operatorname{Ext}^1(Q_\eta, P_\eta)/\operatorname{Im}\operatorname{Ext}^1(Q, P) \simeq \check{Q}_o \otimes \check{Q}_o^*. \tag{9.4.5}$$

Returning to the definition of (9.4.2) when S is strictly local, the relation $\operatorname{Ext}^2(Q, P) = 0$ makes it possible to define the canonical element (9.3.13), taking for Y' the given mixed extension X_η of E_η by F_η. Formula (9.4.5) identifies this element $c(X_\eta)$ with an element of $\check{Q}_o \otimes \check{Q}_o^*$, thereby completing the definition of (9.4.2). The general considerations of 9.3 also show that this element is the obstruction to extending X_η to a mixed extension X of E by F.

9.4.6. The definition of (9.4.1) when S is henselian now reduces immediately to the strictly local case by Galois descent from the strict henselization of S.

Proposition 9.4.7. Let P, Q, R, E, F be as above, except that the trait S need not be henselian. Consider the functor

$$X \longmapsto X_\eta \tag{9.4.7.1}$$

from the category of mixed extensions of E by F to the category of mixed extensions of E_η by F_η. For every mixed extension X_η of E_η by F_η, consider the pairing

$$c_o(X_\eta) : Q_o \otimes Q_o^* \longrightarrow \Lambda_s \tag{9.4.7.2}$$

induced by the pairing of type (9.4.1) defined over the henselization S^h of S. Then the functor (9.4.7.1) is fully faithful, and a mixed extension X_η of E_η by F_η belongs to its essential image if and only if the pairing (9.4.7.2) is zero.

The printed target of (9.4.7.2) is Λ_s; the French transcription's $\wedge_s$ is a character-recognition error.

9.4.7.3. This statement still makes sense and remains valid if one assumes only that S is a connected regular noetherian scheme of dimension 1. Instead of the single pairing (9.4.7.2), one must then consider a pairing $Q_s \otimes Q_s^* \longrightarrow \Lambda_s$ for each closed point s of S. It will be most convenient to prove 9.4.7 in this more general form.

a) For every mixed extension X, the map

$$\operatorname{End}(X) \longrightarrow \operatorname{End}(X_\eta)$$

is bijective. The two sides are respectively the sections over S and η of $\underline{\operatorname{Hom}}(Q, P) = D(Q \otimes Q^*)$; hence the assertion follows from the normality of S and the finiteness of $D(Q \otimes Q^*)$ over S.[31] It follows at once that if X and Y are two mixed extensions already known to be isomorphic, then

$$\operatorname{Hom}(X, Y) \longrightarrow \operatorname{Hom}(X_\eta, Y_\eta)$$

is bijective.

b) If X and Y are two mixed extensions, every isomorphism $X_\eta \xrightarrow{\sim} Y_\eta$ comes from an isomorphism $X \longrightarrow Y$. By (a), the question is local for the étale topology, so we may reduce to the case in which S is strictly local. In that case 9.4.3 shows that the map

31. The printed proof has $\underline{\operatorname{Hom}}(P, Q)$ here. Proposition 9.3.8(a), formula (9.3.4), and the displayed identification with $D(Q \otimes Q^*)$ all require $\underline{\operatorname{Hom}}(Q, P)$.

$$\mathrm{Ext}^1_\Lambda(Q,P) \longrightarrow \mathrm{Ext}^1_\Lambda(Q_\eta,P_\eta)$$

identifies with

$$V^* \otimes \check{H}_o \longrightarrow K^* \otimes \check{H}_o, \qquad H_o = Q_o \otimes Q_o^*.$$

The latter is injective, since the exact sequence $0 \longrightarrow V^* \longrightarrow K^* \longrightarrow \mathbb{Z} \longrightarrow 0$ is split. By the discussion of 9.3 this implies that if X_η and Y_η are isomorphic, then so are X and Y; the conclusion now follows from (a). [32]

Thus (a) and (b) show that $X \longmapsto X_\eta$ is fully faithful.

c) Let X_η be a mixed extension over η. We prove that it extends to S if and only if the pairings (9.4.7.2) vanish. Here again, the uniqueness already proved shows that the question is local for the étale topology, reducing us once more to the strictly local case. The conclusion then holds by construction, as observed immediately before 9.4.6. [33]

This completes the proof of 9.4.7.

9.5. Fix a prime number ℓ, pro-ℓ Barsotti–Tate groups over S

$$P = (P(n))_{n\geq 0}, \qquad R = (R(n))_{n\geq 0}, \qquad Q = (Q(n))_{n\geq 0}, \tag{9.5.1}$$

and extensions of pro-ℓ Barsotti–Tate groups

$$0 \longrightarrow P \longrightarrow F \longrightarrow R \longrightarrow 0, \qquad 0 \longrightarrow R \longrightarrow E \longrightarrow Q \longrightarrow 0. \tag{9.5.2}$$

Thus, for every integer $n \geq 0$, these define extensions of Λ_n-modules, where $\Lambda_n = \mathbb{Z}/\ell^{n+1}\mathbb{Z}$:

$$\begin{aligned} &0 \longrightarrow P(n) \longrightarrow F(n) \longrightarrow R(n) \longrightarrow 0, \\ &0 \longrightarrow R(n) \longrightarrow E(n) \longrightarrow Q(n) \longrightarrow 0. \end{aligned} \tag{9.5.3}$$

We assume that Q is of étale type and P of multiplicative type. Finally, suppose that a pro-ℓ Barsotti–Tate group X_η is given over η which is a mixed extension of E_η by F_η. By this we mean that it is endowed, as a pro-ℓ Barsotti–Tate group, with a three-step filtration whose partial extensions are supplied with isomorphisms to E_η and F_η, respectively. It follows that, for every n, $X_\eta(n)$ is endowed with the structure of a mixed extension of $E(n)_\eta$ by $F(n)_\eta$. We may therefore associate with it a pairing of type (9.4.1):

$$c(X(n)_\eta) : Q(n) \otimes Q^*(n) \longrightarrow \Lambda_n = \mathbb{Z}/\ell^{n+1}\mathbb{Z}, \tag{9.5.4}$$

where the $Q^*(n)$ are the finite étale groups over S defined by

$$P(n) \simeq D(Q^*(n)), \qquad \text{i.e.} \qquad Q^*(n) = D(P(n)),$$

and D again denotes Cartier duality. The $Q^*(n)$ form a pro-ℓ group in the evident way, denoted by Q^*. We leave it to the reader to verify that, as n varies, the homomorphisms (9.5.4) define a homomorphism of projective systems, that is, a homomorphism of twisted constant ℓ-adic sheaves over S:

$$c(X_\eta) : Q \otimes Q^* \longrightarrow \mathbb{Z}_{\ell S}. \tag{9.5.5}$$

32. The printed Ext groups in this paragraph have the arguments (P,Q). The action of $\mathrm{Ext}^1(Q,P)$ on mixed extensions in Proposition 9.3.8(b), and Lemma 9.4.3 itself, require (Q,P).

33. The printed proof cites the nonexistent pairings "(9.4.6.2)"; these are the pairings (9.4.7.2).

We mention in passing the following result, which will not be used below.

Proposition 9.5.6. Let P, Q, R, E, F be as above, except that the trait S need not be henselian. Consider the functor

$$X \longmapsto X_\eta \tag{9.5.6.1}$$

from the category of Barsotti–Tate mixed extensions of E by F to the category of mixed extensions of E_η by F_η. For every Barsotti–Tate group X_η which is a mixed extension of E_η by F_η, consider the corresponding pairing

$$Q_o \otimes Q_o^* \longrightarrow \mathbb{Z}_{\ell s} \tag{9.5.6.2}$$

induced by the pairing of type (9.5.5) defined over the henselization of S. Then the functor (9.5.6.1) is fully faithful, and its essential image consists of those Barsotti–Tate groups X_η which are mixed extensions of E_η by F_η and for which the corresponding pairing (9.5.6.2) is zero. Moreover, when $\operatorname{char} K = 0$ (so that Tate's theorem [33, Theorem 4] applies), X_η belongs to the essential image if and only if it extends to a Barsotti–Tate group over S.

9.5.6.3. We leave it to the reader to extend this statement to the case in which S is assumed only to be a connected regular noetherian scheme of dimension 1.

The first assertion of 9.5.6 follows immediately from 9.4.7, applied to the components of the Barsotti–Tate groups under consideration.[34] It remains to prove that if a Barsotti–Tate group X_η, which is a mixed extension of E_η by F_η, extends to a Barsotti–Tate group X over S, then it also extends as a mixed extension.[35] By Tate's theorem [33], the homomorphism $X_\eta \longrightarrow Q_\eta$ extends uniquely to a homomorphism

$$X \longrightarrow Q.$$

For every integer n, the corresponding homomorphism $X(n) \longrightarrow Q(n)$ is surjective on the generic fibres and hence is surjective. Since $Q(n)$ is étale over S, the map $X(n) \longrightarrow Q(n)$ is flat, and consequently it is an epimorphism. Thus $X \longrightarrow Q$ is an epimorphism component by component.[36] Let $\overline{F}$ be the kernel of $X \longrightarrow Q$. Then $\overline{F}$ is a Barsotti–Tate group and X is an extension of Q by $\overline{F}$. The isomorphism $F_\eta \xrightarrow{\sim} \overline{F}_\eta$ extends, by Tate's theorem, to an isomorphism $F \simeq \overline{F}$. Hence X is an extension of Q by F. Consider now the quotient $\overline{E} = X/P$. A further application of Tate's theorem gives an isomorphism $E \simeq \overline{E}$. It follows that X is a mixed extension of E by F extending the mixed extension X_η, as required.

Remarks 9.5.6.4.

a) Of course, the last assertion of 9.5.6 should hold without restriction on K, since the same should be true of the invoked theorem of Tate.

b) One can give another proof of the good-reduction criterion 5.10. As in 5.10, reduce to the case where A_K has semistable reduction, and then apply 9.5.6 to $T_\ell(A_K)$ regarded as a mixed extension (cf. 9.6). This implies $u_\ell = 0$; using 11.6.2(a) below then gives $T_\ell(A_K)^t = 0$.

9.5.8. When $\ell \neq p$, the Barsotti–Tate groups under consideration over S and over K, respectively, may simply be interpreted in terms of twisted constant ℓ-adic sheaves over S, or equivalently in terms of representations of $\pi_o = \operatorname{Gal}(\overline{k}/k)$ and $\pi = \operatorname{Gal}(\overline{K}/K)$, respectively, on free $\mathbb{Z}_\ell$-modules

34. The printed proof cites 9.4.6, which only performs the henselian descent construction; the full-faithfulness and essential-image statement being applied componentwise is Proposition 9.4.7.

35. At this point the printed source gives the extension over S the name X_η as well. We reserve X_η for the generic fibre and write X for its extension over S.

36. The printed sentence says that $Q(n)$ is étale over η. The group $Q(n)$ was defined over S and is assumed of étale type there; the base required for this flatness argument is S.

of finite type. One then sees that the homomorphism (9.5.5) is the canonical homomorphism expressing the action of the inertia subgroup $I \subset \pi$ on $X_\eta(\overline{K})$, as defined by the method of 9.2.[37] To verify this, one reduces immediately to the situation of 9.4, taking n there to be a power of ℓ and R to be a locally constant Λ-module (as are Q and P); the same is then true of E, F, and X_η. Verification that the two definitions agree is left to the reader.

9.6. We are now in a position to define the announced canonical pairing (9.1.2), without any restriction on ℓ, in such a way that, for $\ell \neq p$, this definition agrees with the one given in 9.2. For this purpose, we apply the considerations of 9.5 to the following situation:

$$(9.6.1)\qquad \begin{cases} P = T_\ell(A^o)^t = \underline{\check{M}}'_\ell(1), \quad Q = \underline{M}_\ell, & R = T_\ell(A^o)^f/T_\ell(A^o)^t, \\ F = T_\ell(A^o)^f, & E = D(F') = D\big(T_\ell(A'^o)^f\big). \end{cases}$$

In the last formula, D denotes Cartier duality in the sense of pro-ℓ Barsotti–Tate groups; that is, $D(Y)$ is the projective system of the $D(Y(n))$, with the usual transition morphisms. The structure of F as an extension of R by P is clear. Likewise,

$$F' = T_\ell(A'^o)^f$$

has the structure of an extension of R' by $P' = T_\ell(A'^o)^t = \underline{\check{M}}_\ell(1)$. Applying Cartier duality, we conclude that $E = D(F')$ is endowed with the structure of an extension of $D(P') = \underline{M}_\ell$ by $D(R')$. Now $D(R')$ is canonically isomorphic to R by 5.5 (proved in 7.4.6); this specifies the structure of E as an extension of Q by R.

As the mixed extension X_η of E_η by F_η, we take

$$(9.6.2)\qquad X_\eta = T_\ell(A_K) = X_\eta^0 \supset X_\eta^1 = T_\ell(A_K)^f \supset X_\eta^2 = T_\ell(A_K)^t \supset 0.$$

The isomorphism of extensions of R_η by P_η

$$X_\eta^1 \simeq F_\eta$$

is clear. The isomorphism of extensions of Q_η by R_η

$$X_\eta/X_\eta^2 \simeq E_\eta = D(F'_\eta)$$

is clear from the orthogonality theorem 5.2.[38]

We therefore have the data required to define a pairing (9.5.5), which here, since $Q^* = \underline{M}'_\ell$, takes the form (9.1.2).[39] Its compatibility with the definition of 9.2 follows from 9.5.8. The considerations of 5.8 also show, subject to the validity of Tate's theorem [33] (hence at least when $\operatorname{char} K = 0$), that the pairing in question can be described in terms of the single pro-ℓ Barsotti–Tate group $X_\eta = T_\ell(A_K)$ over η: its three-step filtration, and the extensions F and E extending X_η^1 and X_η/X_η^2, can be defined canonically in terms of X_η alone.[40]

Remark 9.7. We no longer necessarily assume that A_K has semistable reduction over the henselian trait S. Let K' be a finite Galois extension of K such that $A_{K'}$ has semistable reduction over

37. The printed source refers here to "the homomorphism (9.5.6)"; the homomorphism constructed in this section is (9.5.5), whereas 9.5.6 is the proposition.

38. The printed formula has $F_\eta = D(F'_\eta)$. Since $E = D(F')$ and the displayed quotient is the extension of Q_η by R_η, the required term is E_η.

39. The printed source calls this a pairing "(9.5.6)"; the pairing is the homomorphism (9.5.5), whereas 9.5.6 is the subsequent proposition.

40. The printed sentence says that X_η is "over S"; its notation and definition make it the generic-fibre group over η. It also lacks its closing parenthesis.

the normalization S' of S in K' (3.6). With the notation of (8.1.8), Galois descent from the monodromy pairing for $A_{K'}$ then defines a pairing

$$(*) \qquad v_\ell^{K'} : \underline{M}_K \otimes \underline{M}'_K \longrightarrow \mathbb{Z}_{\ell K}.$$

A priori this pairing depends on the choice of K', but it follows from (10.4.3) below that the pairing

$$(9.7.1) \qquad u_\ell : \underline{M}_K \otimes \underline{M}'_K \longrightarrow \mathbb{Q}_{\ell K}$$

given by

$$u_\ell = \frac{1}{e(S'/S)}\, v_\ell^{K'}$$

is independent of it, where

$$e(S'/S) = \text{length}(V'/\mathfrak{m}V').$$

We may again call (9.7.1) the monodromy pairing for the abelian scheme A_K over K and the prime number ℓ under consideration: when A_K itself has semistable reduction, it plainly coincides with the restriction to η of the monodromy pairing (9.1.2). It follows from 10.4 below that this pairing in fact always has values in $\mathbb{Q}_K$ and is independent of ℓ.

One may ask whether it has values in $\mathbb{Z}_K$, which is equivalent to saying that (9.7.1) has values in $\mathbb{Z}_{\ell K}$ for every prime number ℓ. This is so at least when A_K is an elliptic curve. The only case requiring verification is then the one in which $\underline{M}_K \simeq \underline{M}'_K$ has rank 1, so that $\underline{M}_K \otimes \underline{M}'_K$ is canonically isomorphic to $\mathbb{Z}_K$ and the monodromy pairing (9.7.1) can simply be interpreted as an ordinary rational number. It follows from the results of Néron [19] that this number is $v(j)$, where j is the absolute invariant of A_K and v is the normalized valuation of K.

10. Properties of the monodromy pairing. Integrality and positivity theorem

In this section we shall first give certain essentially trivial formal properties of the monodromy pairings, leaving the (somewhat tedious) details of the verifications to the reader. We shall also state a deeper theorem (10.4) concerning these pairings and indicate how its proof may be reduced to a special case of Jacobians, to be treated (by transcendental methods) in section 12, which is in fact logically independent of section 11.

10.1. Transport of structure. It is clear from the definition of the monodromy pairing (9.1.2) that this pairing is compatible with transport of structure in the situation (S, A_K, A'_K).

10.2. Symmetry. Starting with the dual scheme A'_K of A_K and applying the construction of section 9 to it, we must introduce the dual scheme $(A'_K)'$ of A'_K, which is canonically isomorphic to A_K by the biduality theorem ([5] or VIII 3.2). Denote by $u_\ell^{A_K}$ the monodromy pairing (9.1.2) relative to A_K. In the same way we obtain a pairing

$$(10.2.1) \qquad u_\ell^{A'_K} : \underline{M}'_\ell \otimes \underline{M}_\ell \longrightarrow \mathbb{Z}_\ell.$$

I claim that the latter is obtained from $u_\ell^{A_K}$ by the symmetry

$$s : \underline{M}'_\ell \otimes \underline{M}_\ell \xrightarrow{\sim} \underline{M}_\ell \otimes \underline{M}'_\ell;$$

that is,

$$(10.2.2) \qquad u_\ell^{A'_K} = u_\ell^{A_K} \circ s.$$

Taking account of the known anticommutativity property of the pairings φ_ξ ([5] or VIII 2.2.11), the symmetry formula (10.2.2) may be reformulated as follows in terms of a fixed polarization ξ of A_K. It defines a homomorphism

$$\widetilde{\xi} : A_K \longrightarrow A'_K$$

and hence a corresponding homomorphism (t standing for "toric")

$$\xi_t : \underline{M} \longrightarrow \underline{M}'. \tag{10.2.3}$$

We may therefore form the pairing

$$\begin{aligned} \varphi^t_{\xi,\ell} : \underline{M}_\ell \otimes \underline{M}_\ell &\longrightarrow \mathbb{Z}_\ell, \\ \varphi^t_{\xi,\ell}(x,y) &= u_\ell^{A_K}\big(x, \xi_t(y)\big). \end{aligned} \tag{10.2.4}$$

The preceding form $\varphi^t_{\xi,\ell}$ is symmetric:

$$\varphi^t_{\xi,\ell}(x,y) = \varphi^t_{\xi,\ell}(y,x). \tag{10.2.5}$$

In fact, this assertion is valid for every element ξ of the Néron–Severi group of A_K, not necessarily a polarization. But only when ξ is nondegenerate—that is, defines an isogeny $A_K \longrightarrow A'_K$, and hence also an isogeny (10.2.3)—is the symmetry relation (10.2.5) equivalent to (10.2.2). When ξ is a polarization, we shall moreover sharpen (10.2.5) considerably in 10.4(b) below.

Let us finally mention a third interpretation of the symmetry relation. Consider a biextension (that is, a divisorial correspondence) ξ on a pair $(A_K, \overline{A}_K)$ of abelian schemes with semistable reduction over K. It defines a homomorphism

$$A_K \longrightarrow \overline{A}'_K.$$

Passing to the Néron models, this defines a homomorphism on the maximal tori of the special fibres, and therefore, in the reverse direction, a homomorphism on their character groups:

$$\xi_t : \overline{\underline{M}} \longrightarrow \underline{M}', \tag{10.2.6}$$

where $\overline{\underline{M}}$ is defined from $\overline{A}_K$ as $\underline{M}$ is defined from A_K. We can therefore form the pairing

$$\begin{aligned} \varphi^t_{\xi} : \underline{M}_\ell \otimes \overline{\underline{M}}_\ell &\longrightarrow \mathbb{Z}_\ell, \\ \varphi^t_{\xi}(x,y) &= u_\ell^{A_K}\big(x, \xi_t(y)\big). \end{aligned} \tag{10.2.7}$$

[41] Notice in passing that, for a variable abelian scheme A_K with semistable reduction, the twisted constant group $\underline{M}$ is a covariant functor of A_K. Formation of the pairings (10.2.7) associated with biextensions of abelian schemes with semistable reduction over (K, S) is then compatible, in the evident sense, with taking inverse images of biextensions.

When ξ is the canonical ("Weil") biextension of (A_K, A'_K), the pairing (10.2.7) is of course precisely the monodromy pairing (9.1.2). [42] Now consider the transpose ${}^s\xi$ of ξ, a biextension of $(\overline{A}_K, A_K)$, and hence the pairing

41. In the printed right-hand side the underlying pairing is denoted only by φ^t, without a definition. It is the monodromy pairing $u_\ell^{A_K}$ used in the identical construction (10.2.4).

42. The printed source refers here to (9.2.1), which is the inertia orthogonality formula $gx - x \in V^\perp$, not the monodromy pairing. The pairing is numbered (9.1.2).

$$\varphi^t_{s\xi} : \overline{M}_\ell \otimes \underline{M}_\ell \longrightarrow \mathbb{Z}_\ell.$$

The symmetry relation is

$$\varphi^t_{s\xi} = \varphi^t_\xi \circ s, \qquad s : \overline{M}_\ell \otimes \underline{M}_\ell \xrightarrow{\sim} \underline{M}_\ell \otimes \overline{M}_\ell, \tag{10.2.8}$$

where s is again the symmetry isomorphism. Compare this relation with the antisymmetry relation of VIII 2.2.11.

10.3. Change of trait. Let

$$g : S' \longrightarrow S \tag{10.3.1}$$

be a dominant morphism of henselian traits, associated with an injective local homomorphism of discrete valuation rings

$$V \longrightarrow V'.$$

As usual, put

$$e(S'/S) = \mathrm{length}_V\big(V'/\underline{m}V'\big), \tag{10.3.2}$$

where $\underline{m}$ is the maximal ideal of V. It is then known that the homomorphism induced by (10.3.1) on the "tame" parts of the inertia groups,

$$g_* : I'_t \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell(1)(k') \longrightarrow I_t = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)(k), \tag{10.3.3}$$

is given by

$$g_*(x) = e(S'/S)x, \tag{10.3.4}$$

after the two sides of (10.3.3) have been identified by means of the evident isomorphisms $\mathbb{Z}_\ell(1)(k) \xrightarrow{\sim} \mathbb{Z}_\ell(1)(k')$. [43]

On the other hand, if A_K is an abelian scheme over K with semistable reduction over S, formation of the twisted lattices $\underline{M}$ and $\underline{M}'$ commutes with every base change $S' \longrightarrow S$ as above, as follows immediately from the definitions (especially 3.4). We may therefore regard the pairing $u_\ell^{A_{K'}}$ defined by $A_{K'}$, where K' is the fraction field of V', as a pairing

$$u_\ell^{A_{K'}} : \underline{M}_{\ell S'} \otimes \underline{M}'_{\ell S'} \longrightarrow \mathbb{Z}_{\ell S'},$$

and compare it with the pairing $u_{\ell S'}^{A_K}$ obtained from $u_\ell^{A_K}$ by base change. When $\ell \neq p$, the construction of the pairings u_ℓ given in 9.2, together with formula (10.3.4), shows at once that [44]

$$u_\ell^{A_{K'}} = e(S'/S) u_\ell^{A_K}. \tag{10.3.5}$$

In fact this formula remains valid for $\ell = p$. To see this, return to the general construction of 9.3–9.6 and reduce to the analogous compatibility assertion for the isomorphism 9.4.4(e), under a change of strictly local traits $S' \longrightarrow S$. Notice that the isomorphism in question is induced by the normalized valuation $K^*/V^* \simeq \mathbb{Z}$, and therefore gives the commutative diagram [45]

43. The printed source leaves the internal reference following "Exposé I" blank.

44. The printed source cites (10.3.3), the displayed map g_*; the multiplication formula it uses is (10.3.4).

45. The printed source says that $K^*/V^* \simeq \mathbb{Z}$ is deduced from the "norm." The displayed map is the normalized discrete valuation.

$$\begin{array}{ccc} K^*/V^* & \xrightarrow{\sim} & \mathbb{Z} \\ \downarrow & & \downarrow{\scriptstyle e(S'/S)} \\ K'^*/V'^* & \xrightarrow{\sim} & \mathbb{Z}. \end{array}$$

Here is a deeper result concerning the monodromy pairings. We shall prove it by transcendental methods in section 12, relying on results that will be proved later in the Seminar. (We shall not, in fact, use 10.4 in the remainder of the Seminar, any more than we shall use the other results of the present exposé.)

Theorem 10.4. Let S be a henselian trait, and let A_K be an abelian scheme over K with semistable reduction over S. Consider the lattices $\underline{M}$ and $\underline{M}'$ associated as in 9.1 (cf. 5.5.2). Then the following assertions hold:

a) There exists a unique pairing

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S \tag{10.4.1}$$

such that, for every prime number ℓ, $u \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ is the monodromy pairing u_ℓ of (9.1.2). This pairing is separating.

b) Let ξ be a polarization of A_K, thus defining a homomorphism (10.2.3)

$$\xi_t : \underline{M} \longrightarrow \underline{M}',$$

and consider the form

$$\begin{aligned} \varphi_\xi^t &: \underline{M} \otimes \underline{M} \longrightarrow \mathbb{Z}_S, \\ &\varphi_\xi^t(x, y) = u\big(x, \xi_t(y)\big). \end{aligned} \tag{10.4.2}$$

This form is symmetric, positive, and nondegenerate.

Of course, in terms of the geometric fibre M_o of $\underline{M}$ at a given geometric point of S, the preceding assertion says that the corresponding form

$$M_o \otimes M_o \longrightarrow \mathbb{Z}$$

is symmetric, positive, and nondegenerate. The pairing (10.4.1) will again be called the monodromy pairing.

10.5. Reduction of 10.4 to a particular case

10.5.1. The uniqueness of a u satisfying the stated condition (even for a single ℓ) is clear. On the other hand, existence simply means that every u_ℓ in fact maps $\underline{M} \otimes \underline{M}'$ into $\mathbb{Z}_S$, and that the homomorphism thereby obtained

$$\underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S$$

is independent of ℓ. In fact it even suffices to prove the analogous assertion for the $u_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ and for a homomorphism

$$u_{\mathbb{Q}} : \underline{M}_{\mathbb{Q}} \otimes \underline{M}'_{\mathbb{Q}} \longrightarrow \mathbb{Q}_S.$$

Suppose, then, that existence has been established after a base change $S' \longrightarrow S$ as in 9.3. It follows immediately from (10.3.5) that there exists a homomorphism

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \frac{1}{e(S'/S)}\mathbb{Z}_S \subset \mathbb{Q}_S$$

compatible with the $u_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$; this proves a). As for b), it is then plain that, if it holds for $A_{K'}$, it also holds for A_K.

Consequently, in proving 10.4 we may make any dominant change of traits $S' \longrightarrow S$. In particular, this allows us to suppose that S is complete with algebraically closed residue field. In that case $\underline{M}$ and $\underline{M}'$ are constant group schemes, with values M and M', finitely generated free $\mathbb{Z}$-modules, and a) and b) may be reformulated more conveniently in terms of the latter.

10.5.2. Suppose now that A_K is isomorphic to a direct factor of an abelian scheme $\overline{A}_K$ over K with semistable reduction, and that 10.4 has been proved for the latter. It follows at once that 10.4 also holds for A_K, in view of the evident functorial behaviour of the lattices $\underline{M}, \underline{M}'$ and of the monodromy pairings with respect to direct sums of abelian schemes with semistable reduction. [46]

It is equally immediate from 10.5.1 that the validity of 10.4 depends only on the isogeny class of A_K. Together with the preceding remark and the "Poincaré reducibility theorem" [15], this shows that, if A_K is a quotient of an abelian scheme $\overline{A}_K$ for which 10.4 holds, then 10.4 also holds for A_K. [47]

10.5.3. We now use the well-known fact [31, Th. 11 (p. 20)] that, after making a finite extension of K if necessary (which is permissible by 10.5.1), every abelian scheme A_K is a quotient of the Jacobian

$$\underline{\mathrm{Pic}}^0_{X_K/K}$$

of a proper, smooth, geometrically connected curve X_K over K. Thus, by 10.5.2, we are reduced to proving 10.4 for such a Jacobian. [48]

10.5.4. By a theorem of Artin–Mumford [3] (already cited and used in Exposé V), after making a further finite extension K' of K if necessary, we may suppose that X_K extends to a projective and flat scheme over S which is regular and whose geometric special fibre $X_{\bar{s}}$ has no singularities other than ordinary quadratic singularities (that is, here, normal crossings of two branches). [49] In this case we shall prove 10.4 a) in 12.5 below, by describing the monodromy pairing (10.4.1) completely in terms of the geometric data of the situation; at the same time we shall verify 10.4 b) for the canonical polarization of the Jacobian.

10.5.5. It remains to see that, if for a given A_K with semistable reduction assertion 10.4 a) holds and assertion 10.4 b) holds for one given polarization ξ_o of A_K, then the same conclusion holds for every polarization ξ of A_K. Indeed, let

$$\widetilde{\xi}_o, \widetilde{\xi} : A_K \longrightarrow A'_K$$

be the homomorphisms defined by ξ_o and ξ; these are isogenies. Then

$$\widetilde{\xi} = \widetilde{\xi}_o \alpha = {}^t(\alpha^*)\widetilde{\xi}_o, \qquad \alpha \in \mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q},$$

where $\alpha \longmapsto \alpha^*$ denotes the canonical involution of the algebra $\mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q}$ defined by the polarization ξ_o, and the superscript t in ${}^t(\alpha^*)$ denotes transpose (associating with an endomorphism of A_K an endomorphism of A'_K). On the other hand, it is known [15, Chap. V, Th. 3] that

$$\alpha = \sum_i \beta_i^* \beta_i, \qquad \beta_i \in \mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q},$$

and hence

$$\widetilde{\xi} = \sum_i {}^t\beta_i \, \widetilde{\xi}_o \, \beta_i.$$

46. The printed source says that $\overline{A}_K$ is "over k"; the notation and the direct-factor argument require "over K."

47. The printed source leaves the pinpoint following citation [15] blank.

48. The printed source says that the curve X_K is "over X"; the displayed relative Picard scheme and the argument require "over K."

49. The printed source leaves the internal reference following "Exposé V" blank.

Equivalently, in terms of the biextensions (that is, divisorial correspondences) associated with the polarizations under consideration (VIII 4.14),

$$\delta(\xi) = \sum_i \delta(\xi_o) \circ (\beta_i, \beta_i). \tag{10.5.5.1}$$

By what was said after (10.2.7), this formula implies (with the notation of (10.4.2))

$$\varphi_\xi^t = \sum_i \varphi_{\xi_o}^t(\beta_i, \beta_i) \in \mathrm{Hom}\big(M_{\mathbb{Q}} \otimes_{\mathbb{Q}} M_{\mathbb{Q}}, \mathbb{Q}\big). \tag{10.5.5.2}$$

It indeed follows that, if $\varphi_{\xi_o}^t$ is symmetric and positive, then so is φ_ξ^t.

Finally, the nondegeneracy of $\varphi_{\xi_o}^t$ is plainly equivalent to the assertion that the pairing (10.4.1) is separating (since $\xi_{ot} : M \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow M' \otimes_{\mathbb{Z}} \mathbb{Q}$ is an isomorphism, $\widetilde{\xi}_o$ being an isogeny), and is therefore likewise independent of the chosen polarization. One may also observe that, in order to verify that (10.4.1) is separating, it suffices to verify the same assertion for the monodromy pairing u_ℓ of (9.1.2) for one prime number ℓ. Taking $\ell \neq p$, one may use the description of u_ℓ in 9.2; the fact that u_ℓ is separating on the left (and hence on both sides, for reasons of rank) is then immediate.

11. Connected components of the Néron model: duality relation and asymptotic behaviour

11.0. The purpose of the present section is to study the finite étale k-group

$$\Phi_o = A_o/A_o^o \tag{11.0.1}$$

already considered in section 1. For every prime number ℓ, denote by

$$\Phi_o(\ell)$$

its ℓ-primary component. Thus knowledge of Φ_o is equivalent to knowledge of the $\Phi_o(\ell)$, by the formula

$$\Phi_o = \coprod_\ell \Phi_o(\ell). \tag{11.0.2}$$

Notice that the groups Φ_o and $\Phi_o(\ell)$ are determined, respectively, by the ordinary finite groups $\Phi_o(\bar{k})$ and $\Phi_o(\ell)(\bar{k})$ together with the natural action of

$$\pi_o = \mathrm{Gal}(\bar{k}/k)$$

on them. On the other hand, it is clear that $\Phi_o(\bar{k})$ is also the group of connected components of the Néron model of $A_{\widetilde{K}}$, where $\widetilde{K}$ is the fraction field of the strict henselization of S relative to the chosen separable closure $\bar{k}$ of k. Thus, in studying Φ_o, we may reduce to the case where S is strictly local; the actions of

$$\pi_o \simeq \mathrm{Gal}(\widetilde{K}/K)$$

on $\Phi_o(\bar{k})$ are then recovered automatically by transport of structure. We could likewise reduce further to the case where S is complete, using the base change $\widehat{S} \longrightarrow S$ and the fact that formation of the Néron model commutes with it [21, Th. 3.4 b)].

11.1. Suppose, then, that S is strictly local, so that Φ_o may be identified with the ordinary finite group $\Phi_o(k)$. We obtain

(11.1.1) $$\Phi_o(k) \simeq A_o(k)/A_o^o(k) \simeq A(S)/A^o(S) \simeq A(K)/A(K)^o,$$

where the first equality follows from the fact that A_o is smooth over k, the second from Hensel's lemma since A is smooth over S and S is henselian, and the third from the relation $A(S) \xrightarrow{\sim} A(K)$ and from the convention

(11.1.2) $$A(K)^o := A^o(S).$$

Thus $A(K)^o$ appears as a natural subgroup of finite index in $A(K)$ (and is not to be confused with $A^o(K) = A(K)$). The isomorphisms (11.1.1) are plainly functorial in A and compatible with transport of structure in (S, A_K), and in particular with the action of any Galois groups.

Now let ℓ be a prime number different from p. Then A^o is ℓ-divisible; more precisely, $\ell\,\mathrm{id}_{A^o}$ is an étale surjective morphism. It consequently induces a surjective endomorphism of $A^o(S) = A(K)^o$, which is therefore an ℓ-divisible group. It follows at once that, for every integer $n \geq 0$ prime to $p = \mathrm{char}(k)$,

(11.1.3) $$ {}_n\Phi_o(k) \simeq {}_nA(K)/{}_nA(K)^o, \qquad (n,p) = 1.$$

By the definition (2.2.3) of the fixed part of ${}_nA_K$, we have

(11.1.4) $$ {}_nA(K)^o = ({}_nA_K)^f(\bar{K}) \simeq T_n(A_K)^f(\bar{K}) \otimes_{\mathbb{Z}_n} \mathbb{Z}/n\mathbb{Z}.$$

Taking n to be a power of ℓ, and putting as in (2.5.3)

(11.1.5) $$U = T_\ell\big(A(\bar{K})\big), \qquad T_\ell\big(A(\bar{K})\big)^f = U^I,$$

where U^I denotes the submodule of U consisting of invariants under the inertia group I, we may write (11.1.3) in the form

(11.1.6) $$ {}_n\Phi_o(k) \simeq (U_n)^I/(U^I)_n, \qquad n = \ell^{r+1},$$

where the subscript n denotes reduction modulo n, that is, tensor product over $\mathbb{Z}_\ell$ with $\mathbb{Z}_\ell/n\mathbb{Z}_\ell$. The transition morphisms between the groups occurring on the two sides of (11.1.6) being evident, passage to the inductive limit over the powers of ℓ gives the following result.

Proposition 11.2. Suppose that S is strictly local, and let ℓ be a prime number different from p. Then there is a canonical isomorphism

(11.2.1) $$\Phi_o(\ell) \simeq \frac{(U \otimes D_\ell)^I}{U^I \otimes D_\ell},$$

where U is defined in (11.1.5) and $D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell$.

Thus the group $\Phi_o(\ell)$ can be described solely in terms of the Tate module $U = T_\ell(A(\bar{K}))$, regarded as a Galois module under the action of the inertia group I.

11.3. We shall use the explicit expression (11.2.1) to sketch the definition of a canonical pairing

(11.3.1) $$\Phi_o(\ell) \otimes \Phi'_o(\ell) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

which is a perfect duality; here Φ'_o denotes, as in section 1, the group of connected components associated with the Néron model of the dual A'_K of A_K. There is little doubt (although the author has dispensed with the verification) that this pairing is none other than, possibly up to sign, the one defined in (1.2.1), and the reader is invited, as an exercise, to establish the required compatibility (with the correct sign!).

Put

$$U' = T_\ell\big(A'(\bar{K})\big), \qquad T = T_\ell(1)(\bar{K}) = T_\ell(\bar{K}^*).$$

There is a perfect pairing of finitely generated free $\mathbb{Z}_\ell$-modules, compatible with the actions of I,

$$\varphi : U \otimes U' \longrightarrow T, \qquad (T \simeq \mathbb{Z}_\ell \text{ noncanonically}), \tag{11.3.2}$$

and, by the known structure of I, an exact sequence [50]

$$1 \longrightarrow R \longrightarrow I \longrightarrow T \longrightarrow 1, \tag{11.3.3}$$

where R is a profinite group prime to ℓ. Starting now from the data (11.3.2) and (11.3.3), introduce the groups

$$\Phi = \frac{(U \otimes D_\ell)^I}{U^I \otimes D_\ell}, \qquad \Phi' = \frac{(U' \otimes D_\ell)^I}{U'^I \otimes D_\ell}, \qquad D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.3.4}$$

We shall define a perfect duality

$$\Phi \otimes \Phi' \longrightarrow \mathbb{Q}/\mathbb{Z}, \tag{11.3.5}$$

which will in particular give the announced pairing (11.3.1). Consider the exact sequence of I-modules

$$0 \longrightarrow U \xrightarrow{i} U \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \xrightarrow{j} U \otimes D_\ell \longrightarrow 0 \tag{11.3.6}$$

and the associated exact cohomology sequence, where the cohomology of I is defined by continuous cochains with coefficients in the topological groups under consideration:

(11.3.7)

$$\begin{array}{ccccccccc}
0 \longrightarrow & H^o(I,U) & \xrightarrow{i^o} & H^o(I, U\otimes\mathbb{Q}_\ell) & \xrightarrow{j^o} & H^o(I, U\otimes D_\ell) & \xrightarrow{\partial} & H^1(I,U) & \xrightarrow{i^1} H^1(I, U\otimes\mathbb{Q}_\ell) \\
& \big\downarrow\wr & & \big\downarrow\wr & & \big\downarrow\wr & & & \big\downarrow\wr \\
& U^I & & (U\otimes\mathbb{Q}_\ell)^I \simeq U^I\otimes_{\mathbb{Z}}\mathbb{Q} & & (U\otimes D_\ell)^I & & & H^1(I,U)\otimes_{\mathbb{Z}}\mathbb{Q}.
\end{array}$$

It is immediate that the group Φ defined by (11.3.4) is precisely $\operatorname{coker} j^o$, and is therefore canonically isomorphic to $\ker i^1$. Thus we obtain a canonical exact sequence

$$0 \longrightarrow \Phi \longrightarrow H^1(I,U) \longrightarrow H^1(I,U)\otimes_{\mathbb{Z}}\mathbb{Q}, \tag{11.3.8}$$

which shows that Φ is canonically isomorphic to the torsion subgroup of $H^1(I,U)$.

50. The printed source leaves the internal reference following the first occurrence of I blank.

On the other hand, from the exact sequence (11.3.3) we deduce, for every $n = \ell^r$, an isomorphism

$$H^1(I, T_n) \simeq \mathrm{Hom}(I, T_n) \xleftarrow{\sim} \mathrm{Hom}(T, T_n) \xleftarrow{\sim} \mathbb{Z}/n\mathbb{Z}. \tag{11.3.9}$$

(Notice that I acts trivially on T, and hence on $T_n = T/nT$.) It follows that, for every finite I-$\mathbb{Z}_\ell$-module M, if

$$M' = \mathrm{Hom}(M, T \otimes D_\ell) \tag{11.3.10}$$

denotes its "Cartier dual," there is a canonical pairing

$$H^i(I, M) \otimes H^{1-i}(I, M') \longrightarrow H^1(I, T \otimes D_\ell) \simeq D_\ell, \tag{11.3.11}$$

where the last isomorphism is deduced from the isomorphisms (11.3.9) by passage to the inductive limit over the powers n of ℓ. It is well known that these pairings are perfect dualities (see, for example, SGA 5 I 5). Taking M in these pairings to be of the form U_n, so that $M' = U'_n$, and passing to the limit, we obtain a perfect pairing

$$H^o(I, U \otimes D_\ell) \times H^1(I, U') \longrightarrow D_\ell, \tag{11.3.12}$$

where the second factor on the left is a finitely generated $\mathbb{Z}_\ell$-module. A further passage to the limit in the pairings (11.3.12), for transition morphisms of the form $\ell^r\,\mathrm{id}$, gives a second perfect pairing

$$H^o(I, U \otimes \mathbb{Q}_\ell) \times H^1(I, U' \otimes \mathbb{Q}_\ell) \longrightarrow D_\ell \tag{11.3.13}$$

of finite-dimensional vector spaces over $\mathbb{Q}_\ell$. Let i' and j' denote the morphisms defined as in (11.3.6), with U replaced by U'. It is then easy to see that the following homomorphisms

$$H^o(I, U \otimes \mathbb{Q}_\ell) \xrightarrow{\quad j^o \quad} H^o(I, U \otimes D_\ell)$$

$$H^1(I, U' \otimes \mathbb{Q}_\ell) \xleftarrow{\quad i'^1 \quad} H^1(I, U')$$

are transposes of one another under the pairings (11.3.12) and (11.3.13). It follows immediately that $\mathrm{coker}\, j^o$ and $\ker i'^1$ are dual to one another by a canonical pairing with values in D_ℓ; this is the announced pairing (11.3.5).

11.4. The preceding description of a pairing (11.3.1), by means of the expression (11.2.1) for the ℓ-primary component of the group of connected components, is valid only when $\ell \neq p$. Suppose now that A_K has semistable reduction over S. We shall again define pairings that are perfect dualities

$$\Phi_o(\ell) \otimes \Phi'_o(\ell) \longrightarrow \mathbb{Q}/\mathbb{Z} \tag{11.4.1}$$

without any restriction on ℓ. As in 11.3,[51] one should establish that this recovers (up to a sign, perhaps, which ought to be specified) the pairing defined in (1.2.1). This would therefore establish Conjecture 1.3 in the case of semistable reduction, just as it does when one restricts to the ℓ-primary components for $\ell \neq p$.

To define (11.4.1), we shall give, essentially, an expression for $\Phi_o(\ell)$ in terms of the Barsotti–Tate group $T_\ell(A_K)$ over K; in the present case this will play the same role as 11.2 did in the case treated above.

Theorem 11.5. Suppose that S is henselian and that A_K has semistable reduction over S. Let ℓ be a prime and consider the monodromy pairing (9.1.2), and hence the canonical homomorphism

$$\widetilde{u}_\ell : \underline{M}_\ell \longrightarrow \check{\underline{M}}'_\ell.$$

Put $\Phi(\ell) = \operatorname{coker} \widetilde{u}_\ell$. Then there is a canonical isomorphism

$$\Phi_o(\ell) \xrightarrow{\sim} \Phi(\ell) \times_S s. \tag{11.5.1}$$

This statement immediately gives the definition of a perfect pairing (11.4.1). Indeed, applying 11.5 to A'_K in place of A_K and using (10.2.2), we also obtain a canonical isomorphism

$$\Phi'_o(\ell) \xrightarrow{\sim} \Phi'(\ell) \times_S s, \qquad \Phi'(\ell) = \operatorname{coker}\big({}^t\widetilde{u}_\ell : \underline{M}'_\ell \longrightarrow \check{\underline{M}}_\ell\big).$$

Remarks 11.5.2. a) Theorem 11.5 implies in particular that $\widetilde{u}_\ell$ is an isogeny, that is, that u_ℓ is separating, including in the case $\ell = p$, where this assertion is not immediate from the definition in 9.6. Notice that we do not use here the (considerably deeper) Theorem 10.4, which will be proved in the next section.

b) Using 10.4, which supplies us with a monodromy pairing

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S,$$

and denoting by Φ the cokernel of the corresponding homomorphism

$$\widetilde{u} : \underline{M} \longrightarrow \check{\underline{M}}',$$

we obtain an equivalent formulation of 11.5, independent of the choice of ℓ, in the canonical isomorphism

$$\Phi_o \xrightarrow{\sim} \Phi \times_S s. \tag{11.5.3}$$

11.6. To define the isomorphism (11.5.1), we may restrict ourselves to the case in which S is strictly local (11.0), so that $\underline{M}$ and $\underline{M}'$ are constant, with values

$$M = D(T'_o), \qquad M' = D(T_o), \tag{11.6.1}$$

where T_o and T'_o are respectively the maximal tori of A_o and A'_o. Likewise identifying Φ_o with an ordinary finite group, we must define a canonical isomorphism between its component $\Phi_o(\ell)$ and

$$\Phi(\ell) = \operatorname{coker}\big(\widetilde{u}_\ell : M_\ell \longrightarrow \check{M}'_\ell\big).$$

In fact, we shall directly define a canonical isomorphism

51. The printed source reads "as in 11.4" here, within section 11.4 itself. The reference is corrected to the immediately preceding construction in 11.3.

(11.6.2) $$\Phi_o(\ell) \xrightarrow{\sim} \ker\bigl(\widetilde{u}_\ell \otimes D_\ell : M_\ell \otimes D_\ell \longrightarrow \check{M}'_\ell \otimes D_\ell\bigr).$$

Since $\Phi_o(\ell)$ is finite and M_ℓ and $\check{M}'_\ell$ have the same rank, it will first follow that $\widetilde{u}_\ell : M_\ell \to \check{M}'_\ell$ is an isogeny. It then follows that the right-hand side of (11.6.2) is canonically isomorphic to $\Phi(\ell)$:

(11.6.3) $$\Phi(\ell) = \operatorname{coker} \widetilde{u}_\ell \xrightarrow{\sim} \ker(\widetilde{u}_\ell \otimes D_\ell).$$

Indeed, identifying M_ℓ by means of $\widetilde{u}_\ell$ with a finite-index submodule of $\check{M}'_\ell$, we may write the source and target of $\widetilde{u}_\ell \otimes D_\ell$ as [52]

$$M_\ell \otimes D_\ell = V/M_\ell, \qquad \check{M}'_\ell \otimes D_\ell = V/\check{M}'_\ell,$$

where

$$V = M_\ell \otimes \mathbb{Q}_\ell \xrightarrow{\sim} \check{M}'_\ell \otimes \mathbb{Q}_\ell.$$

Composing (11.6.2) with (11.6.3) now immediately gives the isomorphism (11.5.1) to be defined, so everything reduces to defining the canonical isomorphism (11.6.2).

Writing $M_\ell \otimes D_\ell$ (respectively, $\check{M}'_\ell \otimes D_\ell$) as the inductive limit of the M_n (respectively, the $\check{M}'_n$), where n ranges over the powers of ℓ, it is equivalent to define canonical isomorphisms

(11.6.4) $$ {}_n\Phi_o \xrightarrow{\sim} \ker\bigl(\widetilde{u}_n : M_n \longrightarrow \check{M}'_n\bigr), \qquad n = \ell^{r+1},$$

compatible with the evident transition morphisms as n varies. Here, by an abuse of notation, $\widetilde{u}_n$ now denotes the homomorphism obtained by reducing $\widetilde{u}_\ell : M_\ell \to \check{M}'_\ell$ modulo n.

For this purpose, consider the group scheme ${}_nA$ over S. By the semistable-reduction hypothesis it is flat and quasi-finite over S (2.2.1). There is a canonical filtration

(11.6.5) $$ {}_nA \supset ({}_nA^o)^f \supset ({}_nA^o)^t \supset 0,$$

where the superscript f (respectively, t) denotes the "fixed part" (2.2.3) (respectively, the "toric part" (5.1)) of the group scheme to which it is applied. In the definition of the toric part in 5.1, we had assumed S complete; this is permissible here (11.0). (We saw moreover in 6.1 that the construction generalizes to the henselian case, if one insists.) Put

(11.6.6) $$ {}_n\Psi = {}_nA/({}_nA^o)^f.$$

This is a flat, separated, quasi-finite group scheme over S (SGA 3 VI$_A$ 3.2 and VI$_B$ 9.2), with generic fibre

$$ {}_n\Psi_K = ({}_nA_K)/({}_nA_K)^f,$$

where, by definition, $({}_nA_K)^f = ({}_nA^o)^f_K$ denotes the r-th constituent ($n = \ell^{r+1}$) of the Barsotti–Tate pro-group $T_\ell(A_K)^f$. The first of relations (5.5.8) therefore gives a canonical isomorphism

(11.6.7) $$ {}_n\Psi_K \xrightarrow{\sim} \underline{M}_{nK}, \qquad \underline{M}_n := \underline{M} \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}).$$

52. At this point the printed source labels V/M_ℓ as $\Phi(\ell)$ and $V/\check{M}'_\ell$ as $\Phi'(\ell)$. Those quotients are instead the source and target of the map whose kernel is the finite cokernel $\check{M}'_\ell/M_\ell$; the contradictory labels are not reproduced.

In fact, we have:

Lemma 11.6.8. The group scheme ${}_n\Psi$ over S defined in (11.6.6) is *étale* over S, and the isomorphism (11.6.7) extends uniquely to a morphism

$$ {}_n\Psi \longrightarrow \underline{M}_n \tag{11.6.8.1} $$

which is an open immersion.

To see that ${}_n\Psi$ is étale, in view of (11.6.7) it remains only to verify this at the points of its special fibre, or equivalently to verify that this fibre is étale. This is immediate, since the fibre in question is

$$ {}_nA_o/{}_nA_o^o \subset A_o/A_o^o = \Phi_o. $$

It follows that ${}_n\Psi$ is normal; and since $\underline{M}_n$ is finite over S, the existence of the extension (11.6.8.1) follows at once. This is a morphism of étale group schemes over S, so its kernel is étale over S. Since ${}_n\Psi$, and hence this kernel, is separated, while its generic fibre is the unit group by (11.6.7), the kernel is the unit group (3.1.3). Thus (11.6.8.1) is an étale monomorphism, hence an open immersion (EGA IV 17.9.1), proving the lemma.

Now put

$$ {}_n\Phi = ({}_nA)^f/({}_nA^o)^f \subset {}_n\Psi. \tag{11.6.9} $$

Thus ${}_n\Phi$ is a finite open subscheme of ${}_n\Psi$, since $({}_nA)^f$ is a finite open subgroup scheme of ${}_nA$. More precisely, it is immediate that

$$ {}_n\Phi \xrightarrow{\sim} ({}_n\Psi)^f. \tag{11.6.10} $$

Composing (11.6.9) with (11.6.8.1), we obtain an open immersion

$$ {}_n\Phi \longrightarrow \underline{M}_n. \tag{11.6.11} $$

On the other hand,

$$ {}_n\Phi \times_S s = {}_n\Psi \times_S s = {}_nA_o/{}_nA_o^o, $$

and, since A_o^o is n-divisible (2.2.1), the last member is immediately identified with ${}_n\Phi_o$. We therefore obtain an isomorphism

$$ {}_n\Phi \times_S s \xrightarrow{\sim} {}_n\Phi_o, \tag{11.6.12} $$

which shows that ${}_n\Phi$ is characterized, up to unique isomorphism, as the finite étale scheme over S whose special fibre is ${}_n\Phi_o$. Thus, to define the desired isomorphism (11.6.4) by means of the open immersion (11.6.11), it is enough to prove the following.

Lemma 11.6.13. The image of the canonical homomorphism (11.6.11) is equal to the kernel of

$$ \widetilde{u}_n : \underline{M}_n \longrightarrow \check{\underline{M}}'_n. $$

The required compatibility of the homomorphisms (11.6.11) with the transition homomorphisms as n varies is immediate from the definitions. Consequently Lemma 11.6.13 will complete the proof of Theorem 11.5.

To prove the lemma, note that

$$K_n = \ker \widetilde{u}_n$$

is also the left annihilator of the canonical pairing

$$u_n : \underline{M}_n \times \underline{M}'_n \longrightarrow (\mathbb{Z}/n\mathbb{Z})_S,$$

whose explicit definition is given in 9.4 and 9.6 in terms of the extensions

(11.6.14)

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \check{\underline{M}}'_n(1) & \longrightarrow & F_n & \longrightarrow & R_n & \longrightarrow & 0 \\ & & & & \| & & & & \\ & & & & ({}_nA^o)^f & & & & \end{array} \quad , \quad \begin{array}{ccccccccc} 0 & \longrightarrow & \check{\underline{M}}_n(1) & \longrightarrow & F'_n & \longrightarrow & R'_n & \longrightarrow & 0 \\ & & & & \| & & & & \\ & & & & ({}_nA'^o)^f & & & & \end{array}$$

(where R_n and R'_n are Cartier dual to one another), and in terms of the mixed extension $X_{n\eta}$ of $E_{n\eta}$ by $F_{n\eta}$, which is the generic fibre of (11.6.5),[53] where we put

(11.6.15)

$$E_n = D(F_n), \qquad \begin{array}{ccccccccc} 0 & \longrightarrow & R_n & \longrightarrow & E_n & \longrightarrow & \underline{M}_n & \longrightarrow & 0 \\ & & \wr\downarrow & & & & & & \\ & & D(R'_n) & & & & & & \end{array} \quad .$$

It follows immediately from Proposition 9.4.7[54] that K_n is the largest among the finite étale subgroups K'_n of $\underline{M}_n$ for which the inverse image $\overline{X}_{n\eta}$ of K'_n in the mixed extension $X_{n\eta}$, regarded as a mixed extension of $\overline{E}_{n\eta}$ by $F_{n\eta}$, extends to a mixed extension of $\overline{E}_n$ by F_n. Here $\overline{E}_n$ is the evident subgroup extension of K'_n by R_n.[55]

This condition is plainly satisfied for the subgroup ${}_n\Phi$ of $\underline{M}_n$ in (11.6.11), since the corresponding mixed extension $({}_nA)^f$ over S is available. Hence ${}_n\Phi \subset K_n$. To prove the opposite inclusion, consider the mixed extension $\overline{X}_n$ of $\overline{E}_n$ by F_n extending $\overline{X}_{n\eta}$, defined as above by taking $K'_n = K_n$. Thus it is an extension

$$0 \longrightarrow F_n \longrightarrow \overline{X}_n \longrightarrow K_n \longrightarrow 0,$$

and there is a canonical homomorphism

$$(*) \qquad F_n = ({}_nA^o)^f \hookrightarrow A,$$

which on the generic fibre is induced by the canonical homomorphism

$$(**) \qquad \overline{X}_{n\eta} \hookrightarrow X_{n\eta} = {}_nA_K \hookrightarrow A_K.$$

It follows immediately from (5.9.2) that the homomorphism $(**)$ extends to a unique homomorphism

53. The printed source refers here to nonexistent formula (11.7.5); the filtration whose generic fibre is being used is (11.6.5).

54. The printed source cites 9.4.6, but the required extension criterion is Proposition 9.4.7.

55. The printed source says "of K_n" in this general criterion, although the subgroup being varied at this point is K'_n; the specialization $K'_n = K_n$ is made only below.

(***) $$\overline{X}_n \longrightarrow A$$

inducing (∗). Since this last homomorphism maps the generic fibre into ${}_nA$, which is a closed subscheme of A, it factors through $\overline{X}_n \to {}_nA$; and since $\overline{X}_n$ is finite, it factors through $({}_nA)^f$. Looking at the induced homomorphism on generic fibres, we conclude that

$$\overline{X}_{n\eta} \subset ({}_nA)^f_\eta, \quad \text{i.e.} \quad K_{n\eta} \subset {}_n\Phi_\eta, \quad \text{i.e.} \quad K_n \subset {}_n\Phi.$$

This proves the lemma.

Remarks 11.7. The construction made in 11.6 [56] shows that, from the extensions (11.6.14) over S and the mixed extension $X_{n\eta} = {}_nA_K$ of E_n by F_n, one can reconstruct, up to unique isomorphism, the mixed extension (11.6.5), and hence also the subgroup

$$({}_nA)^f \supset ({}_nA^o)^f$$

of ${}_nA$. Indeed, by 9.4 these data determine the pairing

$$u_n : \underline{M}_n \times \underline{M}'_n \longrightarrow (\mathbb{Z}/n\mathbb{Z})_S,$$

and the preceding construction gives $({}_nA)^f$ as the mixed extension $\overline{X}_n$ of $\overline{E}_n$ by F_n, extending the mixed extension $\overline{X}_{n\eta}$ induced by $X_{n\eta}$. Here $\overline{E}_n$ is the inverse image of $\ker \widetilde{u}_n$ under $E_n \to \underline{M}_n$. By (9.4.8), this mixed extension $\overline{X}_n$ is thus determined up to unique isomorphism. We then recover ${}_nA$ by gluing $\overline{X}_n$ and $X_{n\eta}$ along their common open subscheme $\overline{X}_{n\eta}$.

Thus, if the structural data (9.6.1) and the mixed extension

$$X_\eta = T_\ell(A_K)$$

of E_η by F_η are known—equivalently, if the extensions (11.6.14) and $X_{n\eta}$ are known for every $n = \ell^{r+1}$, $r \geq 0$—then one can reconstruct the inductive (or, if preferred, projective) system of mixed extensions (11.6.5).

11.8. As in 8.3, regard the separable closure $\overline{K}$ of K as the inductive limit of its finite subextensions K'. For every such K', let $A[K']$ denote the Néron model of

$$A_{K'} = A_K \otimes_K K'.$$

It gives a finite étale group scheme

(11.8.1) $$\Phi_o[K'] = A[K']_o/A[K']^o_o$$

of connected components, defined over the residue field of the normalization $S[K']$ of S in K'. For simplicity, suppose that the residue field of S is separably closed, so that the same is true of the residue fields of the $S[K']$. The groups (11.8.1) may then be identified with ordinary finite groups $\Phi[K']$, and it is immediate that, as K' varies, these form an inductive system, which we now propose to study. We shall be interested in particular in the group

(11.8.2) $$\Psi = \varinjlim_{K'} \Phi[K'].$$

We know from 3.6 that, for K' sufficiently large, say $K' \supset K'_o$, $A[K']$ has semistable reduction. Hence for $K' \supset K'_o$ the transition morphisms are injective (3.3). Moreover, with the notation of 5.5, retained in the present section, there is for $K' \supset K'_o$ a canonical isomorphism

56. The printed source says "the construction made in 11.7" at the opening of 11.7; the reference is to the construction just completed in 11.6.

(11.8.3) $$\Phi[K'](\ell) \simeq \ker\big(\widetilde{u}[K']_\ell \otimes D_\ell : M_\ell \otimes D_\ell \longrightarrow \check{M}'_\ell \otimes D_\ell\big),$$

coming from (11.6.2) applied to $A[K']$. One also verifies immediately that the corresponding immersions

$$\Phi[K'](\ell) \longrightarrow M_\ell \otimes D_\ell$$

are compatible with the transition morphisms as K' varies. Passing to the limit therefore gives a canonical monomorphism

(11.8.4) $$\Psi(\ell) \overset{\text{def}}{=} \varinjlim_{K'} \Phi[K'](\ell) \longrightarrow M_\ell \otimes D_\ell$$

from the ℓ-primary component of Ψ; summing over all ℓ gives

(11.8.5) $$\Psi \overset{\text{def}}{=} \varinjlim_{K'} \Phi[K'] \longrightarrow M \otimes \mathbb{Q}/\mathbb{Z}.$$

Theorem 11.9. The canonical homomorphism (11.8.5) is an isomorphism.

We already know that it is a monomorphism. To prove that it is an isomorphism, we must use the characterization (11.8.3) of the image of $\Phi[K'](\ell)$ and the behaviour of $\widetilde{u}[K']$ as K' varies. That behaviour is given by (10.3.4).

We may reduce to the case in which S is strictly local. To prove 11.9 we are then reduced to the following familiar assertion, applied to $S[K'_o]$: if S is a strictly local trait with fraction field K, then for every integer $n \geq 1$ there is a separable extension K' of K whose degree is divisible by n.[57]

Corollary 11.10. Under the hypotheses (S complete) and with the notation of 8.3, the ind-finite group scheme over K of connected components of the group scheme $A^\flat_K$, namely $A^\flat_K/A^{\flat\,o}_K$, is canonically isomorphic to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$, where $\underline{M}_K$ is the twisted constant group over K defined in (8.1.8).

This follows immediately from (8.2.4) and 11.9, in view of the definition of $\underline{M}_K$.

Remarks 11.11. a) If one restricts to the ℓ-primary components for $\ell \neq p$, Theorem 11.9 is immediate in terms of 11.2 and does not require the more delicate considerations of 11.6 (via 9.4). Likewise, if K is assumed to be of characteristic zero, 11.9 can be proved without using (11.8.3): it is immediate that the image of ${}_n\Phi[K'](\ell)$ in

$$M_n = {}_n(M_\ell \otimes D_\ell)$$

contains the image of the group ${}_nA(K')$ of K'-valued points of ${}_nA$. For K' sufficiently large this group is equal to ${}_nA(\overline{K})$, and therefore maps onto M_n. If $\operatorname{char} K = p$, on the other hand, it appears that there is no "trivial" proof—that is, no proof avoiding 9.4—of 11.10 for the p-primary component.

b) The existence of an isomorphism

$$\Psi \simeq (\mathbb{Q}/\mathbb{Z})^r,$$

together with the correct value of r, was conjectured in 1964 by J.-P. Serre. 12. Néron model of a Jacobian and the Picard–Lefschetz formula

57. The printed source appends "cf. I" followed by a blank internal reference. No target is supplied, so none is invented here.

Let us first recall the following general results of M. Raynaud [22, Ths. 5 and 7; 22 bis].

Theorem 12.1 (M. Raynaud). Let S be a strictly local trait, and let $f : X \to S$ be a scheme proper and flat over S, with one-dimensional fibres, such that

(12.1.1) $$f_*\underline{\mathcal{O}}_X \simeq \underline{\mathcal{O}}_S,$$

and such that the local rings of X are factorial. For every maximal point x_i of X_o, put

(12.1.2) $$d_i = \text{length}\,\underline{\mathcal{O}}_{X_o,x_i}, \qquad d_i^t = \mu_i d_i,$$

where μ_i is the radicial multiplicity of $k(x_i)$ over k (EGA IV, 4.7.4) (thus a power of the characteristic exponent of k). Let d (respectively d^t) be the gcd of the integers d_i (respectively d_i^t); hence $d = d^t$ if k is perfect. Put

(12.1.3) $$P = \underline{\text{Pic}}_{X/S} : (\text{Sch})^{o}_{/S} \longrightarrow (\text{Ab}).$$

a) Consider the following conditions:

1) $d^t = 1$.

2) X has a zero-cycle of degree 1.

3) There exists an invertible sheaf L on $X \times P^o$ such that the corresponding homomorphism $P^o_\eta \to \underline{\text{Pic}}_{X_\eta/\eta} = P_\eta$ is the canonical inclusion. (It is known, by J.-P. Murre, that P_η is representable by a group scheme locally of finite type over η; see, for example, SGA 6, XII, 1.5.)

4) X is cohomologically flat over S (EGA III, 7.8.1) (equivalently, cohomologically flat in dimension zero, or equivalently $k \xrightarrow{\sim} H^o(X_o, \underline{\mathcal{O}}_{X_o})$), and $d = 1$.

5) P is representable (equivalently, representable by a group scheme smooth over S).

6) P^o is representable (equivalently, representable by a group scheme smooth over S). (For the definition of P^o, see SGA 3, VI_B, 3.1.)

7) P^o is separated over S, that is, it satisfies the valuative criterion of separatedness of EGA II, 7.2.3 (equivalently, every section s of P^o over S for which $s(\eta) = 0$ is zero).

8) $d = 1$.

Then one has the implications

(12.1.4) $$1) \iff 2) \implies 3) \implies 4) \iff 5) \iff 6) \iff 7) \implies 8).$$

In particular, if $d = d^t$, for example if k is perfect, the conditions 1) through 8) are equivalent.

b) Suppose that $d^t = 1$, so that P is representable by a group scheme smooth over S. Then the schematic closure of the identity section of P is a group subscheme E which is ètale over S. If I is the set of reduced irreducible components X_o^i of X_o, one obtains an isomorphism

(12.1.5) $$\mathbb{Z}^I \simeq \Gamma(S, E)$$

by associating to every $i \in I$ the section S_i of P over S (zero over η) defined by the invertible sheaf $\underline{\mathcal{O}}_X(X_o^i)$ on X [58]. (Because the local rings of X are factorial by hypothesis, the cycles X_o^i are indeed divisors; thus $\underline{\mathcal{O}}_X(X_o^i)$ has its stated meaning as the invertible sheaf associated with that divisor; see EGA IV, 21.2.8.1.) One has

58. The printed source says "on S" here and later writes X_i^o for the same divisor. The sheaf is the invertible sheaf on X associated with the component X_o^i; the English text keeps that notation consistently.

(12.1.6) $$P^o \cap E = \text{the zero section of } P.$$

c) Under the conditions of b), the quotient

(12.1.7) $$Q = P/E$$

is representable by a smooth separated group scheme over S, and this group scheme is Néron, that is, it satisfies the universal property (1.1.2) in terms of its special fibre.

d) Under the conditions of c), suppose in addition that the geometric generic fibre of X is integral. Then there is a canonical exact sequence

(12.1.8) $$0 \longrightarrow A \longrightarrow Q \xrightarrow{\deg} \mathbb{Z}_S \longrightarrow 0,$$

where deg is induced by the "degree" homomorphism $P \to \mathbb{Z}_S$, and where A is a smooth separated group scheme of finite type over S, which is likewise Néron, that is, a Néron model of

(12.1.9) $$A_\eta = P^o_\eta = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}.$$

Moreover, the composite homomorphism $P^o \to P \to Q$ induces an isomorphism

(12.1.10) $$P^o \xrightarrow{\sim} A^o.$$

12.1.11. The case of interest to us is that in which X_η is smooth over η, so that, in view of (12.1.1), the geometric generic fibre of X is smooth and connected (hence integral), and in which X_η has a zero-cycle of degree one. All the conditions considered in 12.1 (and in particular conditions 1) through 8) of a)) then hold. Moreover, $A_\eta = P^o_\eta$ is then an abelian scheme, and d) gives us a very explicit construction of its Néron model A. In particular, (12.1.10) gives an isomorphism

(12.1.12) $$A^o_o \simeq \underline{\mathrm{Pic}}^o_{X_o/k},$$

which shows in particular that A has semistable reduction over S if and only if $\underline{\mathrm{Pic}}^o_{X_o/k}$ has unipotent rank zero. When X is regular and X_o is separable over k, this is known to hold if and only if the geometric special fibre $\overline{X}_o$ has no singular points other than ordinary double points (that is, points whose completed local ring is the same as that of the union of the two coordinate axes of affine 2-space at the origin). Verification of this fact, together with the "well-known" description of the Picard scheme of a proper separable curve over a field k, using EGA IV, 21.8.5, is left to the reader as an exercise. Notice also that the implication "$\overline{X}_o$ has ordinary double points" $\Longrightarrow$ "$A_\eta = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}$ has semistable reduction" follows immediately from the "geometric monodromy theorem" III[59], in view of the Galois criterion for semistable reduction in 3.5. This argument, however, does not give the important isomorphism (12.1.10) and its consequence (12.1.12).

12.2. For the proof of 12.1, we refer to [22 bis]; in fact we shall need only the case where X_η is smooth over η. Let us also note that [22 bis] proves relations between Picard schemes and Néron models under conditions substantially more general than those of 12.1 c) and d). As their statements are more technical, we have chosen not to include them in this review. We mention only that b), c), and d) are easy consequences of a), and that in a) the only delicate points are the

59. The printed source leaves the internal locator following "III" blank; no target is invented here.

implications 3) $\Longrightarrow$ 4) $\Longrightarrow$ 5) and 7) $\Longrightarrow$ 4), the first being the most difficult. This point is trivial, however, if one assumes that X_o is separable over k (which implies that X has a section over S, since S is strictly local, and hence that condition 1), $d^t = 1$, holds). We advise the reader to give the proof in this particularly simple case, assuming in addition, if necessary, that X_η is smooth, so that P^o_η is an abelian scheme. One may then use its Néron model A, and A^o represents $\underline{\mathrm{Pic}}^o_{X/S}$. We shall use 12.1 only in this particular case, and under the further assumption that $\overline{X}_o$ has no singularities other than ordinary double points, that is, by (12.1.11), that A_η has semistable reduction. This is the case considered in 12.5.4, to which we restrict ourselves from now on. 12.3.

Let C be a proper separable curve over a field k, which for simplicity we suppose separably closed (or even algebraically closed, if the reader prefers). If $\widetilde{C}$ is the normalization of C, it is known that $\underline{\mathrm{Pic}}^o_{\widetilde{C}/k}$ has "reductive rank" zero (that is, its maximal torus is zero). Consequently the maximal torus of $\underline{\mathrm{Pic}}^o_{C/k}$ is contained in

$$N = \ker\big(\underline{\mathrm{Pic}}_{C/k} \longrightarrow \underline{\mathrm{Pic}}_{\widetilde{C}/k}\big),$$

and is therefore canonically isomorphic to the maximal torus of this last group. The structure of N is easily made explicit by EGA IV, 21.8.5: N is canonically an extension of a torus T, which we shall describe, by a connected unipotent group U (which itself has, over k, a composition series whose quotients are isomorphic to the additive group $\mathbb{G}_a$). Moreover, if T' is the maximal torus of N, then $T' \cap U$ is both unipotent and of multiplicative type, and is therefore zero (SGA 3, XVII, 2.4), while the image of T' in T is all of T (SGA 3, XII, 7.1 e)). Thus $T' \to T$ is an isomorphism. Hence T is canonically isomorphic to the maximal torus of N, and therefore to that of $\underline{\mathrm{Pic}}^o_{C/k}$.[60]

To make the structure of N explicit, it is convenient to suppose that k is separably closed. The irreducible components C_i of C are then geometrically irreducible, and the nonnormal points of C are radicial over k. Let J be the set of irreducible components of $\widetilde{C}$, and for every $x \in C$ let $J(x)$ be the set of "branches" of C through x (that is, the points of $\widetilde{C}$ above x). Finally put

$$R(x) = \mathbb{Z}^{J(x)} / \operatorname{Im} \mathbb{Z}, \tag{12.3.1}$$

where $\mathbb{Z} \to \mathbb{Z}^{J(x)}$ is the diagonal map, and

$$R = \left(\bigoplus_x R(x) \right) / \operatorname{Im} \mathbb{Z}^J. \tag{12.3.2}$$

Clearly it is enough to take the sum over the finite set of points of C which belong to at least two branches of C. Thus R is a finitely generated $\mathbb{Z}$-module. There is then a canonical isomorphism

$$T \ \big(\simeq \text{ the maximal torus of } \underline{\mathrm{Pic}}^o_{C/k}\big) \simeq R \otimes \mathbb{G}_{m,k}. \tag{12.3.3}$$

In other words, the character group M of T is

$$M \overset{\text{def}}{=} D(T) = \check{R} \simeq \ker\left(\bigoplus_x \check{R}(x) \longrightarrow \mathbb{Z}^J \right), \tag{12.3.4}$$

where $\check{R}(x)$ may also be interpreted as the set of integer-valued functions on $J(x)$ whose values have sum zero.[61]

60. The printed source has $\underline{\mathrm{Pic}}^o_{X/k}$ at the end of this argument. The curve throughout 12.3 is C, and the preceding and following occurrences both read $\underline{\mathrm{Pic}}^o_{C/k}$; the English text restores C.

61. The printed source numbers these two displays (12.2.3) and (12.2.4). They belong to section 12.3 and are cited below as (12.3.3) and (12.3.4); those coherent numbers are used here.

12.3.5. We shall need only the case in which at most two branches of C pass through any point x (as happens, in particular, for ordinary double points). If $\operatorname{card} J(x) = 2$, then $R(x)$ is isomorphic to $\mathbb{Z}$, but the isomorphism depends on an ordering of the two branches (C', C'') through x, or equivalently on the choice of one branch C': to C' one associates the element of $R(x)$ which is the canonical image of $0_{C'} + 1_{C''}$. One then obtains an isomorphism

$$R = \mathbb{Z}^I / \operatorname{Im} \mathbb{Z}^J, \qquad T \simeq \mathbb{G}_m^I / \operatorname{Im} \mathbb{G}_m^J, \qquad M = \ker\bigl(\mathbb{Z}^I \longrightarrow \mathbb{Z}^J\bigr), \tag{12.3.6}$$

where I is the set of points $x \in C$ which belong to two branches of C, after a choice of signs as just explained.

Remarks 12.3.7. A sometimes convenient way to describe the character group M of the maximal torus T of $\underline{\mathrm{Pic}}^o_{C/k}$ is the following. Consider the "unoriented one-dimensional graph" $\Gamma(C)$ (N. Bourbaki, *Groupes et algèbres de Lie*, Chapter IV, Appendix), whose vertices are the irreducible components of $\widetilde{C}$. Two vertices $\widetilde{C}_i$ and $\widetilde{C}_j$ are joined by a set of edges in bijective correspondence with the pairs of distinct branches of C, contained respectively in $\widetilde{C}_i$ and $\widetilde{C}_j$, which pass through the same point of C. This set is $\widetilde{C}_i \times_C \widetilde{C}_j$ when $i \neq j$; when $i = j$, it is

$$\bigl(\widetilde{C}_i \times_C \widetilde{C}_i - \Delta(\widetilde{C}_i)\bigr)/(\text{symmetry}),$$

where $\Delta : \widetilde{C} \to \widetilde{C} \times_C \widetilde{C}$ is the diagonal. (It is therefore empty if C_i is geometrically unibranch.) One then has canonical isomorphisms

$$M = D(T) \xleftarrow{\sim} H_1(\Gamma(C), \mathbb{Z}), \qquad \check{M} = R \simeq H^1(\Gamma(C), \mathbb{Z}). \tag{12.3.7}$$

We shall not need this combinatorial interpretation of M, and shall confine ourselves to defining the canonical homomorphism

$$H_1(\Gamma(C), \mathbb{Z}) \longrightarrow M, \tag{12.3.8}$$

leaving it to the reader to prove that this is an isomorphism. Let I be the set of points of C which belong to at least two branches of C, let J, as above, be the set of irreducible components of $\widetilde{C}$ (or of C, which amounts to the same thing), and let K be the set of "branches" of C at the points of I, that is, the inverse image of I in $\widetilde{C}$. There is a canonical map

$$h : K \longrightarrow I \times J, \qquad h = (f, g), \tag{12.3.9}$$

and the description (12.3.4) of M is expressed in terms of h as

$$M \simeq \ker\left(\mathbb{Z}^{(h)} : \mathbb{Z}^{(K)} \longrightarrow \mathbb{Z}^{(I \times J)}\right). \tag{12.3.10}$$

Indeed, the expression (12.3.3) for $R = \check{M}$ may plainly be interpreted as the cokernel of the homomorphism $\mathbb{Z}^{I \times J} \to \mathbb{Z}^K$ induced by h, which is the transpose of $\mathbb{Z}^{(h)}$. The graph $\Gamma(C)$ is likewise described in terms of h by

$$\Gamma_0 = J, \qquad \Gamma_1 = (K \times_I K - \Delta(K))/(\text{symmetry}).$$

Choose an orientation on each edge of $\Gamma(C)$, and denote by Γ_1 the set of these oriented edges (x, C', C''). Then

(12.3.11) $$H_1(\Gamma(C),\mathbb{Z}) = \ker\left(d : \mathbb{Z}^{(\Gamma_1)} \longrightarrow \mathbb{Z}^{(J)}\right),$$

where the differential is

(12.3.12) $$d(x, C', C'') = g(C'') - g(C').$$

The homomorphism (12.3.8) is induced by the following homomorphism v of chain complexes:[62]

(12.3.13) $$\begin{aligned} v = (v_0, v_1) : C_\bullet(\Gamma(C)) = &\left[\mathbb{Z}^{(\Gamma_1)} \longrightarrow \mathbb{Z}^{(J)}\right] \\ &\longrightarrow \left[\mathbb{Z}^{(K)} \longrightarrow \mathbb{Z}^{(I\times J)}\right], \end{aligned}$$

given by

(12.3.14) $$v_1(x, C', C'') = (x, C'') - (x, C'), \qquad v_0(\widetilde{C}_j) = (0, \widetilde{C}_j).$$

12.4. We now place ourselves in the case considered at the end of 12.2. By (12.1.12) and (12.3.6), applied with $C = X_o$, the character group M of the maximal torus T_o of the special fibre A_o of the Néron model A of $A = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}$ is given by

(12.4.1) $$M = \ker\left(\mathbb{Z}^I \xrightarrow{d} \mathbb{Z}^J\right),$$

where I is the set of singular points of X_o, and J is the set of its irreducible components, the homomorphism $\mathbb{Z}^I \to \mathbb{Z}^J$ being given by

(12.4.2) $$d(x_i) = C_i' - C_i''.$$

Here C_i' and C_i'' are the two irreducible components of $\widetilde{C}$ corresponding to the branches through the double point x_i. The operator d depends on the choice of an order (C_i', C_i'') on these two branches, and the isomorphism (12.4.1) plainly depends on this system of choices. Nevertheless, if, for $i \in I$, we denote by

(12.4.3) $$\varphi_{x_i} : M \longrightarrow \mathbb{Z}$$

the composite

$$M \longrightarrow \mathbb{Z}^I \xrightarrow{\mathrm{pr}_i} \mathbb{Z},$$

where the first homomorphism comes from (12.4.1), then, on returning to the expression (12.3.4), one sees at once that φ_{x_i} is well-defined up to sign. More precisely, it depends only on the order chosen between the two branches C_i', C_i'' through x_i, and changes sign when that order is reversed. It follows in particular that the bilinear form

62. The printed display defining (12.3.8) carries no number even though the following text cites it, and (12.3.13) writes $\mathbb{Z}^{(\Gamma)}$ where the edge chain group is $\mathbb{Z}^{(\Gamma_1)}$. The English text supplies the cited label and the required subscript. The working transcription's $\widetilde{C}_i \times_C \widetilde{C}_1$ in the loop-edge description is also restored from the scan as $\widetilde{C}_i \times_C \widetilde{C}_i$.

(12.4.4) $$\varphi_{x_i} \otimes \varphi_{x_i} \in \check{M} \otimes_{\mathbb{Z}} \check{M} \simeq \mathrm{Hom}(M \otimes M, \mathbb{Z})$$

is determined canonically by the singular point x_i, without any sign ambiguity. We may therefore put

(12.4.5) $$u = \sum_i \varphi_{x_i} \otimes \varphi_{x_i} : M \otimes_{\mathbb{Z}} M \longrightarrow \mathbb{Z}.$$

This gives a well-defined pairing, independent of every choice of signs. It is plainly positive definite, since it is the restriction to $M \hookrightarrow \mathbb{Z}^I$ of the standard quadratic form $\sum_i X_i^2$ on $\mathbb{Z}^I$.

The notation M for the character group of T_o is not, a priori, consistent with that of the preceding sections, where this group was written M', while M denoted the analogous group associated with the dual abelian scheme A'_K of A_K. But since A_K is a Jacobian, it has a canonical polarization ξ [5] [63], which induces a canonical isomorphism

(12.4.6) $$\psi_\xi : A_K \xrightarrow{\sim} A'_K.$$

By means of this isomorphism we shall identify A_K and A'_K, so that there is no longer any need to distinguish M from M'. In particular, for every prime number ℓ, the monodromy pairing (9.1.2) may be regarded as a pairing

(12.4.7) $$u_\ell : M_\ell \otimes_{\mathbb{Z}_\ell} M_\ell \longrightarrow \mathbb{Z}_\ell.$$

With this understood, we can state the principal result of the present section:

Theorem 12.5 (Picard–Lefschetz formula). Let S be a strictly local trait, and let X be a regular scheme, projective and flat over S, with one-dimensional fibres, whose geometric generic fibre is smooth and connected and whose geometric special fibre has no singularities other than ordinary quadratic singularities. With the notation introduced in 12.4, for every prime number ℓ, the monodromy pairing (12.4.7) is equal to the pairing obtained from (12.4.5) by extension of scalars from $\mathbb{Z}$ to $\mathbb{Z}_\ell$. 12.6. Notice that this statement proves 10.4 in the particular case under consideration when, in 10.4 b), one takes for ξ the canonical polarization considered there. As we showed in 10.5, this suffices to imply 10.4 in the general case.

12.7. Proof of 12.5. The case $\ell \neq p$. It relies essentially on the cohomological Picard–Lefschetz formula in the situation considered in 12.5, a formula giving the action of the monodromy group I on the following group: [64]

(12.7.1) $$H^1(X_{\bar\eta}, \mathbb{Z}_\ell) \simeq H^1(A_{\bar\eta}, \mathbb{Z}_\ell) \simeq \mathrm{Hom}\big(T_\ell(A_{\bar\eta}), \mathbb{Z}_\ell\big)$$

when $\ell \neq p$. This formula, which will be proved in a later exposé (Exposé XV), and indeed in a substantially more general case, is essentially identical to assertion 12.5 for the prime ℓ in question, in view of the definition in 9.2 of u_ℓ for $\ell \neq p$. It therefore remains to treat the case $\ell = p$. We shall do this by a method which reduces it to characteristic zero (and, moreover, to the case where X_o has only one singular point), which we now explain.

63. The printed citation following the canonical polarization is "[5]" followed by a blank pinpoint. The English text preserves the reference without inventing a missing locator.

64. The working transcription adds a dual superscript to $T_\ell(A_{\bar\eta})$ in (12.7.1). A direct 5,000-dpi inspection of the printed formula shows no such superscript, in agreement with the standard identification $H^1(A_{\bar\eta}, \mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell(A_{\bar\eta}), \mathbb{Z}_\ell)$.

Remarks 12.7.2. Notice also that, once the announced reduction has been made, the usual arguments would allow us to reduce to the case in which the base field is the field $\mathbb{C}$ of complex numbers, and then to invoke the Picard–Lefschetz formula in its classical form—except that there does not seem to be a satisfactory reference for that formula. The proof of the Picard–Lefschetz formula given in our Seminar will proceed similarly, by reduction to the transcendental case, which is amenable to properly transcendental methods. One may therefore say that, in any event, the proof of 12.5 (and hence of 10.4) proposed here is transcendental in nature. Let us also point out that the only other known proof of 10.4 comes from rigid-analytic theory [24], and must for that reason likewise be regarded as essentially "transcendental." It would be interesting to find purely algebraic proofs of these results (although it is by no means evident that this is possible!).

12.8. Proof of 12.5: reduction to the case in which S has characteristic zero. We shall use an argument based on the same principle as the proof of the "theorem on the essentially tame action of monodromy on π_1" in Exposé V.[65] Notice that we may assume S complete, by 10.3. Let W be a Cohen p-ring with residue field k, and choose a homomorphism $W \longrightarrow V$ inducing the identity on residue fields (EGA 0_{IV}, 19.8.6). Let $(x_\lambda)_{\lambda\in L}$ be the set of singular points of X_o, and consider the formal moduli scheme $\underline{S}$ of X_o over W, together with the "universal curve"

$$\underline{f} : \underline{X} \longrightarrow \underline{S}$$

whose special fibre is X_o (Exposé VI).[66] We saw in the cited place that $\underline{S}$ is isomorphic to the formal spectrum of a power-series ring $W[[T_1, \ldots, T_N]]$, and that in $\underline{S}$ one can find a normal-crossings divisor

$$D = \sum_{\lambda\in L} D_\lambda, \tag{12.8.1}$$

whose irreducible components D_λ are indexed by L, each D_λ being described by an equation of the form $T_{i_\lambda} = 0$. (If desired, one may suppose $i_\lambda = \lambda$ after indexing L by a set of integers $[1, r]$.) The component D_λ is the image under $\underline{f} : \underline{X} \longrightarrow \underline{S}$ of the component D'_λ through the singular point x_λ of X_o in the non-smoothness locus

$$D' = \bigcup_{\lambda\in L} D'_\lambda$$

of $\underline{f}$. By the construction itself, $f : X \longrightarrow S$ is isomorphic to the pullback of $\underline{f} : \underline{X} \longrightarrow \underline{S}$ by a suitable W-morphism $g : S \longrightarrow \underline{S}$:

$$\begin{array}{ccc} \underline{X} & \longleftarrow & X \\ {\scriptstyle \underline{f}}\downarrow & & \downarrow{\scriptstyle f} \\ \underline{S} & \xleftarrow{\ g\ } & S \end{array} \qquad . \tag{12.8.2}$$

Set

$$P = \underline{\mathrm{Pic}}^o_{\underline{X}/\underline{S}}. \tag{12.8.3}$$

Since the special fibre of $\underline{f}$ is geometrically reduced of dimension 1, it follows that $\underline{f}$ is cohomologically flat in dimension zero,[67] by EGA III, 7.8.1, and that P is formally smooth over $\underline{S}$ (the

65. The printed source leaves the internal pinpoint following "V" blank; no section target is supplied.

66. The printed source likewise leaves the internal pinpoint following "VI" blank.

67. The working transcription reads $\underline{F}$ here. Direct inspection of the printed page shows $\underline{f}$, as also required by the sentence's reference to the special fibre of the universal curve.

latter fact follows from $H^2(X_o, \underline{\mathcal{O}}_{X_o}) = 0$, since X_o has dimension 1). For simplicity, we shall assume that P is representable by a group scheme over $\underline{S}$; it will necessarily be a smooth group scheme with connected fibres. This is a special case of an unpublished result of Mumford cited in [22] (and considerably generalized by Raynaud in the note cited there, when the base is a trait). Since no proof of either Mumford's or Raynaud's result has been published, the reader may prefer to use the general results of M. Artin [1]. They give the representability of P—under the sole assumption that $\underline{f}$ is proper, flat, and cohomologically flat in dimension zero—by an "algebraic space" in Artin's sense, which would suffice for our purpose. Put

$$U = \underline{S} - \operatorname{supp} D. \tag{12.8.4}$$

Thus U is the largest open subset of $\underline{S}$ over which $\underline{X}$ is smooth, and hence

$$P|_U = \underline{\mathrm{Pic}}^o_{\underline{X}_U/U}$$

is an abelian scheme over U. Its pullback to η by $g : \eta \longrightarrow U$ is the abelian scheme A_η over η that we wish to study. More precisely, there is a canonical isomorphism over S (12.1.10):

$$P \times_{\underline{S}} S \simeq A^o. \tag{12.8.5}$$

Since the special fibre $P_o = A^o_o$ is ℓ-divisible for every ℓ, the same holds for P itself; all its fibres are therefore extensions of an abelian variety by a torus. Consider the pro-ℓ group $T_\ell(P)$ over $\underline{S}$, whose pullback by g is precisely $T_\ell(A^o)$. Applying the constructions of 7.3, one defines a filtration of this pro-group by a "fixed part" and a "toric part":

$$T_\ell(P) \supset T_\ell(P)^f = T_\ell(\widehat{P}) \supset T_\ell(P)^t = T_\ell(T) \supset 0. \tag{12.8.6}$$

Its pullback by g is the analogous filtration of $T_\ell(A^o)$ studied in sections 2 and 5.

Consider the restriction of this filtration over the open subset U of $\underline{S}$. There we have the canonical self-duality of P_U, as the Jacobian of a smooth curve scheme over U, and hence a corresponding pairing

$$\varphi : T_\ell(P_U) \otimes T_\ell(P_U) \longrightarrow T_\ell(\mathbb{G}_{mU}). \tag{12.8.7}$$

This is a self-duality, in the sense of Barsotti–Tate groups, of $T_\ell(P_U)$.[68] We saw in 7.4.8 that, for this pairing, $T_\ell(P_U)^f$ and $T_\ell(P_U)^t$ are each other's orthogonal complements. It follows that

$$E_U = T_\ell(P_U)/T_\ell(P_U)^t$$

is canonically isomorphic to the Barsotti–Tate pro-group dual to $T_\ell(P_U)^f = F_U$. It therefore extends canonically to a Barsotti–Tate group E over $\underline{S}$, namely

$$E = D(F), \qquad \text{where} \qquad F = T_\ell(P)^f. \tag{12.8.8}$$

We are now in a typical situation of mixed extensions (9.3), and can repeat the constructions of 9.4–9.6 (for whose having been written in the stale and piss-poor setting of a base scheme

68. The printed source reads $T_\ell(B_U)$ here, although no B_U has been introduced and the pairing and every adjacent term involve P_U. The English text restores the internally forced P_U.

reduced to a trait we apologize!): $T_\ell(P_U)$ is a mixed extension of E_U by F_U, where E and F are Barsotti–Tate groups over $\underline{S}$ which are extensions, dual to one another:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & P & \longrightarrow & F & \longrightarrow & R & \longrightarrow & 0 \\ & & \wr\downarrow & & \wr\downarrow & & & & \\ & & T_\ell(P)^t \simeq \underline{\mathrm{Hom}}\big(M, T_\ell(\mathbb{G}_{m\underline{S}})\big) & & T_\ell(P)^f & & & & \end{array}$$

(12.8.9)

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \longrightarrow & E & \longrightarrow & Q & \longrightarrow & 0 \\ & & \wr\downarrow & & & & \wr\downarrow & & \\ & & D(R) & & & & M \otimes \mathbb{Z}_{\ell\,\underline{S}} & & \end{array} \quad .$$

The printed upper identification in (12.8.9) omits the outer closing parenthesis after $\underline{\mathrm{Hom}}(M, T_\ell(\mathbb{G}_{m\underline{S}}))$; the balanced expression has been supplied here. Notice that R is canonically self-dual, as follows from the self-duality (12.8.7) and the orthogonality statement noted above. The only difference from the cited construction is that the pairing obtained on $M_\ell \times M_\ell$ takes values in the group

$$\big(\mathbb{G}_m(U)/\mathbb{G}_m(\underline{S})\big) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

which is no longer canonically isomorphic to $\mathbb{Z}_\ell$, but rather to $\mathbb{Z}_\ell^L$, by means of the isomorphism

$$\mathbb{Z}^L \simeq \mathbb{G}_m(U)/\mathbb{G}_m(\underline{S}). \tag{12.8.10}$$

It maps the element indexed by λ in the canonical basis of $\mathbb{Z}^L$ to the class of T_{i_λ} in the right-hand side, where T_{i_λ} is an equation for the divisor D_λ. We thus obtain a canonical pairing

$$\underline{u}_\ell : M_\ell \otimes M_\ell \longrightarrow \mathbb{Z}_\ell^L \quad . \tag{12.8.11}$$

It is essentially immediate that the monodromy pairing u_ℓ (12.4.7) over S which we propose to study is the composite of $\underline{u}_\ell$ with the homomorphism $\mathbb{Z}_\ell^L \longrightarrow \mathbb{Z}_\ell$ obtained, by tensoring with $\mathbb{Z}_\ell$ over $\mathbb{Z}$, from the homomorphism

$$\begin{array}{ccc} \mathbb{G}_m(U)/\mathbb{G}_m(\underline{S}) & \longrightarrow & \mathbb{G}_m(\eta)/\mathbb{G}_m(S) \\ \wr\big\downarrow & & \wr\big\downarrow \\ \mathbb{Z}^L & & \mathbb{Z} \end{array}$$

induced by $g : S \longrightarrow \underline{S}$ (which sends η into U). This homomorphism is given by an element

$$(d_\lambda)_{\lambda\in L} \in \mathbb{Z}^L, \qquad d_\lambda \geqslant 1 \quad \text{for every } \lambda \in L, \tag{12.8.12}$$

where, for every $\lambda \in L$, d_λ is the multiplicity of the inverse image in S of the divisor D_λ on $\underline{S}$. It in fact depends only on the completion $\widehat{\mathcal{O}}_{X,x_\lambda}$ as a V-algebra (Exposé VI).[69] Moreover, it is known (Exposé VI) that, for a given λ, one has $d_\lambda = 1$ if and only if X is regular at x_λ.[70] Since in 12.5 we assumed X regular, that is, all the d_λ equal to 1, assertion 12.5 will therefore follow from the following lemma, in which X, S, g no longer occur.

Lemma 12.8.13. If $\lambda \in L$, the composite of (12.8.11) with pr_λ is equal to the form $\varphi_{x_\lambda} \otimes \varphi_{x_\lambda}$ of (12.4.4).

69. Both printed citations to Exposé VI in this sentence leave their internal section pinpoints blank.

70. The printed source has the lowercase letter x as the subject of "is regular." The following sentence and the mathematical statement require the scheme X.

To prove this last assertion, consider the completed strict localization $\underline{S}^\lambda$ of $\underline{S}$ at the generic point of D_λ,[71] and put

$$\underline{X}^\lambda = \underline{X} \times_{\underline{S}} \underline{S}^\lambda.$$

The special fibre $\underline{X}^\lambda_o$ then has only one singular point y_λ, which is again an ordinary quadratic singularity. In the corresponding monodromy pairing

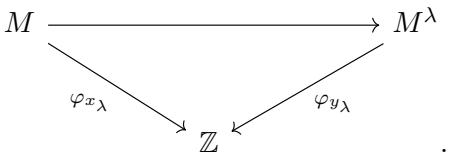
$$u^\lambda_\ell : M^\lambda_\ell \otimes M^\lambda_\ell \longrightarrow \mathbb{Z}_\ell \qquad , \tag{12.8.14}$$

M^λ consequently has rank 1 or 0. We leave to the reader the verification of the following easy facts, which make explicit, in the special case in which we need them, the behaviour under localization of the preceding constructions.

a) There is a unique subtorus T_λ of $P \times_{\underline{S}} D_\lambda$ whose restriction to the generic point s_λ of D_λ is the maximal torus of P_{s_λ} (which has dimension 0 or 1).

b) Consider the transpose homomorphism $M \longrightarrow M^\lambda$ on character groups of the inclusion $T_{\lambda o} \hookrightarrow T_o$, where T_o is the maximal torus of P_o, and the form $\varphi_{y_\lambda} : M^\lambda \longrightarrow \mathbb{Z}$, defined up to sign as in 12.4.3 (with X replaced by $\underline{X}^\lambda$). Then, up to sign, the following diagram is commutative:

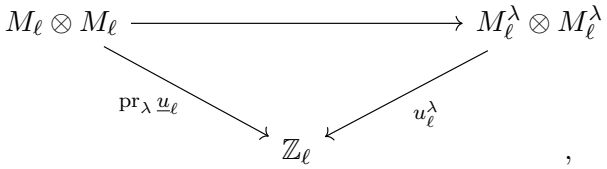
(12.8.15)



.

The working transcription misreads the right diagonal label in (12.8.15) as $\varphi_{\lambda_\lambda}$. The printed diagram, the immediately preceding definition, and the formula below all give φ_{y_λ}. (In fact, choosing an ordering of the two branches of X_o through x_λ determines an analogous choice at the point y_λ of $\underline{X}^\lambda_o$, and for choices corresponding in this way the preceding diagram is commutative.)

c) The following diagram is commutative:



,

where the notation is that of 12.8.13 and (12.8.14). These relations show at once that, to prove 12.8.13, it is enough to prove the analogous relation for $\underline{X}^\lambda$:

$$u^\lambda_\ell = (\varphi_{y_\lambda} \otimes \varphi_{y_\lambda}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

which is nothing other than 12.5 for $\underline{X}^\lambda$ over $\underline{S}^\lambda$. Now $\underline{S}^\lambda$ is a trait of characteristic zero.[72] This completes the announced reduction.

Remarks 12.9. Notice that this reduction simultaneously reduces us to the case in which X_o has only one singular point, the case classically considered in the Picard–Lefschetz formula. Notice also that, in this case, X_o has one or two irreducible components, corresponding respectively to

71. In this reduction the printed source drops the subscript λ from D_λ and prints an un-underlined X as the right-hand factor of the following fibre product. Both marks are restored: the localization is at the component D_λ, and the fibre product is formed from the universal curve $\underline{X}$ over $\underline{S}$.

72. At this final step the printed source writes $\underline{S}$ rather than the localized trait $\underline{S}^\lambda$ used throughout the reduction. The English text restores the mathematically required superscript.

rank $M = 1$ and rank $M = 0$ (the latter being the case in which the Jacobian of the generic fibre has good reduction).

12.10. The proof given in 12.8 yields a result more general than 12.5, valid without assuming that X is regular, by introducing the integers $d_\lambda \geqslant 1$ of (12.8.12): all the monodromy pairings are then deduced from the pairing

$$u = \sum_{\lambda \in L} d_\lambda \, \varphi_{x_\lambda} \otimes \varphi_{x_\lambda} : M \otimes M \longrightarrow \mathbb{Z} \qquad . \tag{12.10.1}$$

13. Links with transcendental theory: the complex analytic case.

For the general notions concerning analytic spaces, we refer to [2].

13.1. Review of analytic groups over S. Let S be an analytic space, and let A be an analytic space over S, endowed with the structure of a commutative group over S (that is, A is a commutative group in the category of analytic spaces over S). We assume that A is smooth over S, and denote by

$$\underline{E} = \underline{\mathrm{Lie}}(A/S) \tag{13.1.1}$$

the locally free module on S obtained by pulling the relative tangent sheaf of A over S back along the zero section. We shall admit here the definition of the exponential homomorphism

$$\exp : E \longrightarrow A \qquad , \tag{13.1.2}$$

where E is the analytic vector bundle on S whose module of holomorphic sections is $\underline{E}$. In a neighbourhood of a point $s \in S$, this homomorphism is first defined near the zero section of E by the familiar methods of the theory of formal groups in characteristic zero, and is then extended in the evident way, by multiplicativity, to all of E. It is a homomorphism of relative analytic groups over S; it is étale, and its image is the open subspace A^o of A which is the union of the identity components of the fibres of A over S. If R is the analytic subspace which is the kernel of exp, we therefore have an exact sequence of relative analytic groups

$$0 \longrightarrow R \longrightarrow E \longrightarrow A^o \longrightarrow 0 \qquad . \tag{13.1.3}$$

Here the morphism $E \longrightarrow A^o$ is surjective and étale, so R is a relative analytic group étale over S. It is an analytic subspace of E, and is closed in E if and only if A^o is separated over S, that is, if and only if the zero section of A^o is closed in A^o.

13.1.4. Of course, the exact sequence (13.1.3) is functorial in A, and commutes with every base change $S' \longrightarrow S$. In this way one obtains an equivalence of categories, compatible with base changes, between the category of relative analytic groups over S which are smooth and have connected fibres, and the category of pairs (E, R) consisting of a locally free vector bundle E over S and an analytic subgroup R of E which is étale over S.

13.1.5. When A is an "abeloid" group over S, by which we mean that it is smooth, proper, and separated over S, with connected fibres—so that in particular $A = A^o$—then R is a locally constant group over S, whose rank at each point s (as a $\mathbb{Z}$-module) is equal to $2\,\mathrm{rank}_s(E)$, and conversely. In this way one obtains the familiar interpretation of an abeloid space over S in terms of a "local system" R over S and a complex analytic vector-bundle structure on the real analytic bundle $R \otimes_{\mathbb{Z}} \mathbb{R}$ defined by R.

13.2. Formal completion along a fibre. We now fix a point $s \in S$, and shall work locally in a neighbourhood of s. For every integer $n \geqslant 0$, let S_n denote the n-th infinitesimal neighbourhood of s in S, and, for every analytic space X over S, put likewise $X_n = X \times_S S_n$. As n varies, the system of the X_n then forms an inductive system of analytic spaces over S, which we denote by $\widehat{X}$ and call the formal completion of X along X_s. It clearly depends functorially on X, and its formation commutes with finite projective limits; hence $X \longmapsto \widehat{X}$ takes groups to groups. Notice that $\widehat{X}$ in fact depends only on the restriction of X to an arbitrary open neighbourhood U of s in S.

Lemma 13.2.1. The data are those of 13.1, and suppose that the following conditions hold: a) $A = A^o$; b) R is constant in a neighbourhood of s; and c) R_s spans the complex vector space E_s. Let A' be a second smooth analytic group over S, and consider the map

$$u \longmapsto \widehat{u} : \mathrm{Hom}_{S\text{-gr}}(A, A') \longrightarrow \mathrm{Hom}_{\widehat{S}\text{-gr}}(\widehat{A}, \widehat{A}') \quad ,$$

where the notation is that of 13.2, and the analogous map obtained by restricting over a variable open neighbourhood U of s and passing to the inductive limit:

$$u \longmapsto \widehat{u} : \mathrm{Hom}_{(S,s)\text{-gr}}(A, A') \longrightarrow \mathrm{Hom}_{\widehat{S}\text{-gr}}(\widehat{A}, \widehat{A}') \quad . \tag{13.2.1.1}$$

(Here (S, s) denotes the germ of the analytic space S at s.) Then the latter map is bijective.

Indeed, consider the exact sequence

$$0 \longrightarrow R' \longrightarrow E' \longrightarrow A'^o \longrightarrow C$$

analogous to (13.1.3) and defined by A'. Since the homomorphisms $A \longrightarrow A'$ necessarily factor through A'^o, and the same holds after the base changes $U \longrightarrow S$ and $S_n \longrightarrow S$, we may assume that $A' = A'^o$. By the dictionary of 13.1.4, it remains to prove that giving a germ of a vector bundle homomorphism $E \longrightarrow E'$ which takes R into R' is equivalent to giving a coherent system of homomorphisms $E_n \longrightarrow E'_n$ which take R_n into R'_n.

If $\underline{\mathcal{O}}_s$ denotes the local ring of S at s, giving a coherent system of homomorphisms $E_n \longrightarrow E'_n$ is equivalent to giving a homomorphism of $\underline{\widehat{\mathcal{O}}}_s$-modules

$$\widehat{F} : \underline{\widehat{E}}_s \longrightarrow \underline{\widehat{E}}'_s,$$

where $\underline{E}_s$ and $\underline{E}'_s$ are the free $\underline{\mathcal{O}}_s$-modules obtained at s from the $\underline{\mathcal{O}}_s$-modules $\underline{E}$ and $\underline{E}'$, corresponding to E and E'. On the other hand, a germ of a vector bundle homomorphism $E \longrightarrow E'$ is equivalent to giving a homomorphism of $\underline{\mathcal{O}}_s$-modules

$$F : \underline{E}_s \longrightarrow \underline{E}'_s \quad .$$

This already shows that (13.2.1.1) is injective. For surjectivity, it is enough to prove that if $\widehat{F}$ is a homomorphism as above such that, for every n, the induced homomorphism $F_n : E_n \longrightarrow E'_n$ takes R_n into R'_n, then $\widehat{F}$ comes from an F—that is, it takes $\underline{E}_s$ into $\underline{E}'_s$—and the corresponding germ of homomorphisms $E \longrightarrow E'$ takes R into R'.

By hypothesis c), $R_s = R_o$ spans E_s; it follows from Nakayama's lemma that the group R_o (identified either with the group of sections of $\widehat{R}$ over $\widehat{S}$, or with the group of germs of sections of R over S) generates $\widehat{E}_s$. Thus, to prove that $\widehat{F}(\underline{E}_s) \subset \underline{E}'_s$, it is enough to prove that $\widehat{F}(R_o) \subset \underline{E}'_s$. But $\widehat{F}(R_o) \subset R'_o$, where R'_o is itself interpreted as the group of germs of sections of $R' \subset E'$ in a neighbourhood of s. This gives the asserted inclusion, and hence the existence of F. Moreover, we see that $F(R_o) \subset R'_o$, where R_o and R'_o are regarded as groups of germs of sections of E and E', respectively. By b), R_o generates R in a neighbourhood of s; it follows that $F(R) \subset R'$ in a neighbourhood of s, which completes the proof.

Corollary 13.2.2. If A satisfies conditions a), b), and c) of 13.2.1, it is determined up to unique isomorphism in a neighbourhood of s by the knowledge of the $\widehat{S}$-group $\widehat{A}$.[73] 13.3. The analytic Raynaud extension.

We are now in a position to paraphrase, in analytic geometry, the construction of the Raynaud extension given in 7. Suppose that A is as in 13.1, with $A = A^o$,[74] and consider $R_o = R_s$, which is a free $\mathbb{Z}$-module of finite type. After restricting S to a sufficiently small open neighbourhood of s, and interpreting R_o as the group of germs of sections of R near s, one obtains a homomorphism

$$R_{oS} \longrightarrow R \subset E, \tag{13.3.1}$$

where R_{oS} denotes the constant analytic group over S with value R_o. On choosing the neighbourhood of s small enough that every section of $\underline{\mathcal{O}}_S$ whose germ at s is zero is itself zero, one sees that the preceding homomorphism is injective. Its image is therefore a constant analytic subgroup of R; it is plainly the largest constant subgroup of R, and is called the fixed part of R and denoted by R^f:

$$R_{oS} \simeq R^f \hookrightarrow R, \qquad \text{inducing an isomorphism } R^f_s \simeq R_s. \tag{13.3.2}$$

We may then consider the quotient group

$$A^\natural = E/R^f, \tag{13.3.3}$$

and, in view of (13.1.3), obtain a canonical exact sequence

$$0 \longrightarrow R/R^f \xrightarrow{\ i\ } A^\natural \longrightarrow A \longrightarrow 0. \tag{13.3.4}$$

Notice that, after restricting S to a neighbourhood of s, R^f is always closed in E, and hence $A^\natural$ is separated over S. Notice also that the homomorphism $A^\natural \longrightarrow A$ is étale, and therefore induces an isomorphism

$$\underline{\mathrm{Lie}}(A^\natural) \xrightarrow{\sim} \underline{\mathrm{Lie}}(A) = \underline{E}. \tag{13.3.5}$$

Moreover, for every $n \geqslant 0$, the canonical homomorphism $A^\natural \longrightarrow A$ induces an isomorphism $A^\natural_n \simeq A_n$ (since $R^f_n \xrightarrow{\sim} R_n$), and consequently an isomorphism

$$\widehat{A^\natural} \xrightarrow{\sim} \widehat{A}. \tag{13.3.6}$$

This last fact shows that $A^\natural$ does indeed play the role of the group, likewise denoted by $A^\natural$, that we considered in 7. Here it has been constructed without any hypothesis on the fibre $A_s = A_o$ of the form "an extension of an abeloid group by a torus". Such a hypothesis will nevertheless be necessary if we wish to recover on $A^\natural$ the important extension structure (7.2.3). By contrast, the homomorphism $A^\natural \longrightarrow A$ in (13.3.4) is essentially transcendental in nature and has no counterpart in the purely algebraic theory of § 7. In § 14 we shall return to the question of a purely algebraic interpretation of the homomorphism $i : R/R^f \longrightarrow A^\natural$ in (13.3.4).

Suppose first that we are given a complex subtorus

73. The printed source refers here to the conditions of 13.2.2 itself; the three conditions are introduced in Lemma 13.2.1.

74. The printed source has $A = A_o$ here. The required hypothesis, and the one used below, is $A = A^o$; compare 13.1 and Lemma 13.2.1.

(13.3.7) $$T_o \subset A_o,$$

where by a "complex torus" we mean an analytic group isomorphic to a Cartesian power of $\mathbb{C}^*$. Let

(13.3.8) $$T = T_o \times S$$

be the constant torus over S with value T_o. Proceeding as in 5.1, one obtains a unique homomorphism (indeed, an immersion) of inductive systems of analytic groups

$$\widehat{T} \longrightarrow \widehat{A^\natural},$$

which over S_o reduces to (13.3.7). By 13.2.2, this homomorphism comes from a unique germ of homomorphisms

(13.3.9) $$T \longrightarrow A^\natural,$$

and, after taking S small enough around s, we may suppose that it is defined over all of S. The corresponding homomorphism

$$\underline{\mathrm{Lie}}(T) \longrightarrow \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{E}$$

is injective over each S_n, hence over $\widehat{S}$, and therefore over $\underline{\mathcal{O}}_s$ by flat descent. Thus there is a neighbourhood of s in S, which we may take to be all of S, on which $\underline{\mathrm{Lie}}(T)$ identifies with a locally direct summand $\underline{E}^t$ of $\underline{E}$; denote the corresponding vector subbundle of E by E^t:

(13.3.10) $$\underline{\mathrm{Lie}}(T) \xrightarrow{\sim} \underline{E}^t \subset \underline{E} \simeq \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{\mathrm{Lie}}(A), \qquad E^t \subset E.$$

Likewise, let

(13.3.11) $$R_o^t \subset R_o, \qquad R^t = R_{oS}^t \subset R_{oS} \simeq R^f$$

denote respectively the sublattice of R_o defined by the subtorus T_o of A_o, and the corresponding constant analytic group over S. Then the homomorphism (13.3.9) corresponds simply to the homomorphism

$$E^t/R_{oS}^t = E^t/R^t \longrightarrow E/R_{oS} = E/R^f$$

induced by the inclusions

$$E^t \hookrightarrow E, \qquad R^t \hookrightarrow R^f.$$

Using the fact that (13.3.9) induces an immersion on the fibre over s, we conclude that the homomorphism

$$R_o/R_o^t \longrightarrow E_o/E_o^t = (E/E^t)_o$$

is injective. It follows at once that

$$R_o/R_o^t \longrightarrow (E/E^t)_{s'}$$

is still injective for s' near s. Thus, after shrinking S around s, we may suppose that (13.3.9) is a closed immersion.

We are therefore led to introduce

$$B = A^\natural/T \simeq (E/E^t)/(R^f/R^t), \tag{13.3.12}$$

so that $A^\natural$ acquires an extension structure

$$0 \longrightarrow T \longrightarrow A^\natural \longrightarrow B \longrightarrow 0, \tag{13.3.13}$$

where B is a relative analytic group over S, smooth and separated over S, with connected fibres. By construction we therefore have

$$\underline{\mathrm{Lie}}(B) \simeq \underline{E}/\underline{E}^t, \tag{13.3.14}$$

and the exponential exact sequence for B is

$$0 \longrightarrow R^f/R^t \longrightarrow E/E^t \longrightarrow B \longrightarrow 0. \tag{13.3.15}$$

Notice that

$$R^f/R^t = (R_o/R_o^t)_S \tag{13.3.16}$$

is a constant analytic group over S. It follows from 13.1.5 that B is an abeloid group over S in a neighbourhood of s if and only if its fibre at s,

$$B_o = A_o/T_o,$$

is an abeloid analytic group; in that case A_o appears as an extension of such a group B_o by a torus T_o. Every way of writing A_o as an extension of an abeloid group by a complex torus $(*)$ therefore gives rise on A, in a neighbourhood of s, to a unique extension structure (13.3.13) of an abeloid group B by a torus T, inducing the given extension structure at s. 13.4. Interpretation in terms of Deligne's "mixed Hodge structures".

Let us first return to the dictionary of 13.1.4, which we shall reformulate in a slightly different way, better suited to introducing the "Hodge structures" relevant to the present situation. For this purpose, let $\underline{R}$ be the sheaf of sections of R over S, and introduce the $\underline{\mathcal{O}}_S$-module

$$\mathcal{E}(A) = \mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \quad . \tag{13.4.1}$$

It is locally free of finite type when R is locally constant. (Note that $\mathcal{E}$ need not be coherent when R is not locally constant.) The homomorphism (13.1.3) $\underline{R} \longrightarrow \underline{E}$ then defines a canonical homomorphism

$$\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \longrightarrow \underline{E} \quad . \tag{13.4.2}$$

We denote its kernel by $\underline{F}(A)$, or simply by $\underline{F}$; thus there is an exact sequence

$$0 \longrightarrow \underline{F} \longrightarrow \mathcal{E} \longrightarrow \underline{E} \quad . \tag{13.4.3}$$

The most interesting case is that in which (13.4.2) is an epimorphism, that is, in which R_s spans the complex vector space E_s for every $s \in S$. We then have an exact sequence

$$0 \longrightarrow \underline{F} \longrightarrow \mathcal{E} \longrightarrow \underline{E} \longrightarrow 0 \qquad . \tag{13.4.4}$$

If R is locally constant, this is a short exact sequence of locally free modules over S. Knowledge of the pair (E, R) is then equivalent to knowledge of the pair $(R, \underline{F})$, where R is a locally constant analytic group over S whose fibres are free $\mathbb{Z}$-modules of finite type, and $\underline{F}$ is a locally direct-summand submodule of $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S}$, subject in addition to

$$\underline{F} \cap \underline{R} = 0 \qquad . \tag{13.4.5}$$

From the standpoint of Hodge structures, $\mathcal{E}$ should be regarded as filtered by the "Hodge filtration"

$$\mathrm{Fil}^1(\mathcal{E}) = \mathcal{E}, \qquad \mathrm{Fil}^0(\mathcal{E}) = \underline{F}, \qquad \mathrm{Fil}^{-1}(\mathcal{E}) = 0 \qquad . \tag{13.4.6}$$

Notice that the exact sequence (13.4.3) still depends functorially on the analytic group A over S, and commutes with arbitrary base change.

Returning now to the conditions of 13.3, put

$$\mathcal{E}^f = \underline{R}^f \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S}, \qquad \mathcal{E}^t = \underline{R}^t \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S} \qquad . \tag{13.4.7}$$

These modules are defined on a suitable neighbourhood of s, which we may suppose to be all of S, and define a filtration of $\mathcal{E}$, called the weight filtration,

$$\mathcal{E} \supset \mathcal{E}^f \supset \mathcal{E}^t \supset 0 \qquad . \tag{13.4.8}$$

Here $\mathcal{E}^t$, $\mathcal{E}^f$, and $\mathcal{E}$ are to be regarded as having weights -2, -1, and 0, respectively; thus this is an increasing filtration. Like R^t, the module $\mathcal{E}^t$ depends on the choice of a subtorus T_o of $A_o = A_s$. Since R^f and R^t are locally constant, $\mathcal{E}^f$ and $\mathcal{E}^t$ are locally free; they may also be interpreted as

$$\mathcal{E}^f = \mathcal{E}(A^\natural), \qquad \mathcal{E}^t = \mathcal{E}(T) \qquad . \tag{13.4.9}$$

Applying the functoriality of (13.4.3), as A varies, to the homomorphisms

$$T \longrightarrow A^\natural \longrightarrow A$$

shows at once that $\underline{F}(A^\natural)$ and $\underline{F}(T)$ identify respectively with

$$\underline{F}(A^\natural) \simeq \underline{F}^f \overset{\text{def}}{=} \underline{F} \cap \mathcal{E}^f, \qquad \underline{F}(T) = \underline{F}^t = \underline{F} \cap \mathcal{E}^t = 0 \qquad . \tag{13.4.10}$$

The last equality follows immediately from the fact that $\underline{F}(T) = 0$ for a torus T: in this case the exact sequence of type (13.4.3) is $0 \longrightarrow 0 \longrightarrow \mathcal{E}(T) \longrightarrow \underline{\mathrm{Lie}}(T) \longrightarrow 0$. At the same time we see that the composite homomorphism

$$\mathcal{E}^t \longrightarrow \mathcal{E} \longrightarrow \underline{E}$$

induces an isomorphism

(13.4.11) $$\mathcal{E}^t \xrightarrow{\sim} \underline{E}^t \simeq \underline{\mathrm{Lie}}(T) \quad .$$

Suppose now that R_s spans the complex vector space E_s, as happens in particular in the favourable case 13.3.16, where A_s is an extension of an abeloid group by a complex torus T_o. After restricting S to a suitable neighbourhood of s, the canonical homomorphism of type (13.4.2), with A replaced by $A^\natural$,

$$\mathcal{E}^f = \mathcal{E}(A^\natural) \longrightarrow \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{E}$$

is an epimorphism; a fortiori, so is (13.4.2). We therefore have the exact sequences (13.4.4) and

(13.4.12) $$0 \longrightarrow \underline{F}^f \longrightarrow \mathcal{E}^f \longrightarrow \underline{E} \longrightarrow 0 \quad .$$

In particular,

(13.4.13) $$\underline{F} + \mathcal{E}^f = \mathcal{E} \quad ,$$

since $\mathcal{E}/\underline{F} \simeq \underline{E}$. Moreover, since $B = A^\natural/T$ is a quotient of $A^\natural$,[75] it satisfies the analogous surjectivity condition, and hence gives an exact sequence of type (13.4.4):

(13.4.14) $$0 \longrightarrow \underline{F}(B) \longrightarrow \mathcal{E}(B) \longrightarrow \underline{\mathrm{Lie}}(B) \longrightarrow 0 \quad .$$

Its interpretation in terms of the modules already introduced is immediate. Indeed, by (13.3.12),

(13.4.15) $$\begin{cases} \mathcal{E}(B) \simeq (\underline{R}^f/\underline{R}^t) \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \simeq \mathcal{E}^f/\mathcal{E}^t, \\ \underline{\mathrm{Lie}}(B) \simeq \underline{E}/\underline{E}^t. \end{cases}$$

It follows from (13.4.11) and the relation $\underline{F} \cap \mathcal{E}^t = 0$ in (13.4.10) that $\underline{F}(B) = \ker(\mathcal{E}^f/\mathcal{E}^t \longrightarrow \underline{E}/\underline{E}^t)$ is isomorphic to $\underline{F}^f$ through the homomorphism

(13.4.16) $$\underline{F}^f \simeq \underline{F}(A^\natural) \xrightarrow{\sim} \underline{F}(B) \quad .$$

Let us summarize the properties just obtained for the weight filtration (13.4.8), and its relation with the Hodge filtration (13.4.6) of $\mathcal{E}$ defined by the submodule $\underline{F}$, in the favourable case 13.3.16 where A_s is an extension of an abeloid analytic group B_o by a torus T_o. We use this torus to define R^t, so that B is an abeloid analytic group over S.

Proposition 13.4.17. Under the preceding conditions, the Hodge and weight filtrations on $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S$,

(13.4.17.1) $$\mathcal{E} \supset \underline{F}, \qquad \mathcal{E} \supset \mathcal{E}^f \supset \mathcal{E}^t \supset 0,$$

satisfy the following conditions:

a) The weight filtration is obtained, by tensoring with $\underline{\mathcal{O}}_S$ over $\mathbb{Z}$, from the analogous filtration on $\underline{R}$

75. The printed source says here that B is a quotient of A; the displayed definition $B = A^\natural/T$ gives the intended group.

$$\underline{R} \supset \underline{R}^f \supset \underline{R}^t \supset 0,$$

where $\underline{R}^f$ and $\underline{R}^t$ are locally constant analytic groups over S, with values free $\mathbb{Z}$-modules of finite type, namely $R^f_o = R_s = \pi_1(A_s)$ and $R^t_o = \pi_1(T_o)$.

b) One has the relations

$$(*) \qquad \underline{F} \cap \mathcal{E}^t = 0, \qquad \underline{F} + \mathcal{E}^f = \mathcal{E} \qquad .$$

In other words, the filtration on $\mathcal{E}^t$ induced by the Hodge filtration, with the degree conventions of (13.4.6), is concentrated in degree -1, whereas the filtration on $\mathcal{E}/\mathcal{E}^f$ induced by the Hodge filtration is concentrated in degree 0.

c) Consider on

$$\mathcal{E}^{\mathrm{ab}} \overset{\mathrm{def}}{=} \mathcal{E}^f/\mathcal{E}^t \simeq \underline{R}^{\mathrm{ab}} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S, \qquad \underline{R}^{\mathrm{ab}} = \underline{R}^f/\underline{R}^t \quad \text{(cf. a))},$$

the filtration induced by the Hodge filtration on $\mathcal{E}$. It is therefore defined by the submodule $\underline{F}^{\mathrm{ab}}$ of $\mathcal{E}^{\mathrm{ab}}$ which is the image of

$$\underline{F}^f \overset{\mathrm{def}}{=} \underline{F} \cap \mathcal{E}^f \longrightarrow \mathcal{E}^{\mathrm{ab}}.$$

The arrow is a monomorphism by the first relation $(*)$. Then the pair

$$\left(\underline{R}^{\mathrm{ab}}, \underline{F}^{\mathrm{ab}} \subset \underline{R}^{\mathrm{ab}} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S = \mathcal{E}^{\mathrm{ab}}\right)$$

is isomorphic to the pair $(\underline{R}(B), \underline{F}(B))$ associated with an abeloid group over S, as explained after (13.4.4). Thus

$$B \simeq \mathcal{E}^{\mathrm{ab}}/(F^{\mathrm{ab}} + R^{\mathrm{ab}}),$$

where the symbols without underlining denote the corresponding analytic groups over S.

Remark 13.4.18. In Deligne's terminology [4], the conclusion of the preceding proposition, after restriction to an open subset U of S on which R is locally constant, says that R, endowed with the two filtrations (13.4.17.1) on $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S$, is an "analytic family of mixed Hodge structures over U". The situation just described served as the model for Deligne's conjectures on Néron theory in higher dimensions, stated in [9, 9.8–9.13].

There is, however, an important feature of the algebraic theory studied in § 7 which is missing from the description in 13.4.17. Suppose that A is an abeloid group over an open subset U complementary to a nowhere-dense analytic subset Z of S—say $Z = \{s\}$ when $\dim S = 1$—so that, by (13.1.5), $R|U$ is a local system over U. Then the local system R/R^t over U ought to be constant; more precisely, it ought to be dual to the local system R'^f, where R' is the analytic group, ètale over S, which would correspond (as R corresponds to A) to a suitable analytic group A' over S such that $A'|U$ is the abeloid group dual to $A|U$. In the notation of § 5, the algebraic analogue is the duality between $T_\ell(A_K)/T_\ell(A_K)^t$ and $T_\ell(A'_K)^f$ in (5.5.9), which follows from the duality theorem 5.2.

I do not know whether such a statement holds under only the hypotheses of 13.4.17, or even whether the action of $\pi_1(U,t)$ on R_t is unipotent for $t \in U$. Nor do the naive methods used in the present paragraph seem capable of proving such a result, even under the additional hypothesis that $A|U$ is endowed with a "polarization", that is, with an invertible sheaf ample on every fibre. This is, however, something that the "semi-simple group people" apparently know how to prove, at least when S is nonsingular of dimension 1; see the report [9]. In the next subsection we shall show that everything works under relative algebraicity hypotheses on A, and at the same time relate the result to the construction of § 7. 13.5. The case of a relatively algebraic situation. We

now suppose that A comes from a relative group scheme over S which is smooth and separated, and whose fibres are extensions of abelian schemes by tori. We denote this relative scheme by the same letter A. (We allow ourselves this abuse of notation because the functor $G \longmapsto G^{\mathrm{an}}$, from the category of relative group schemes of the type under consideration to the category of analytic groups over S, is fully faithful, as would be easy to prove using results of relative GAGA type [10].) For everything concerning the notion of a relative scheme and the relations between relative schemes over S and analytic spaces over S, we refer to [10] (see also [2, Exp. 15]).

If $\underline{O}$ denotes the local ring of S at s, then the relative scheme A is determined near s by the ordinary scheme $A_{\underline{O}}$ over $\mathrm{Spec}(\underline{O})$ obtained from A by localization at s. It is then clear that the formal completion of $A_{\underline{O}}$ along its special fibre (isomorphic to $A_s = A_o$) is isomorphic to the formal completion of the analytic group A along A_s (13.2), where, by the abuse of notation introduced above, the $A_n = A \times_S S_n$ are identified with algebraic schemes over S. In the favourable case in which $\hat{A}$, viewed as a formal group scheme over $\underline{\hat{O}}$, is algebraizable (7.2.1), as is the case in particular when S is normal at s and A is quasi-projective over S (7.1.5), we therefore obtain a group scheme $A^{\natural}_{\underline{\hat{O}}}$ over $\mathrm{Spec}(\underline{\hat{O}})$, which is an extension of an abelian scheme $B_{\underline{\hat{O}}}$ by a torus. For simplicity, we suppose that $B_{\underline{\hat{O}}}$ is projective over $\underline{\hat{O}}$ (as is indeed the case, we know, if $\underline{O}$, and hence $\underline{\hat{O}}$, is normal). We shall then prove:

Proposition 13.5.1. Under the preceding conditions, the analytic group $A^{\natural}$ of (13.3.3) comes, in a neighbourhood of s, from a relative group scheme determined up to a unique isomorphism, which is an extension of a projective relative abelian scheme over S by a torus. If $A^{\natural}_{\underline{O}}$ denotes the group scheme it defines over $\underline{O} = \underline{O}_{S,s}$, then the group scheme

$$A^{\natural}_{\underline{O}} \otimes_{\underline{O}} \underline{\hat{O}}$$

is canonically isomorphic, with its extension structure, to the Raynaud extension $A^{\natural}_{\underline{\hat{O}}}$ introduced above.

The latter assertion follows immediately from the first, in view of 7.2.1, since, by (13.3.6), the formal completion of $A^{\natural}_{\underline{O}} \otimes_{\underline{O}} \underline{\hat{O}}$ along its special fibre is $\hat{A}_{\underline{O}}$. To prove the algebraicity of $A^{\natural}$ near s, proceeding as in 7.2.1, we are immediately reduced to proving that of the abelian quotient B of $A^{\natural}$, which follows from the next lemma.

Lemma 13.5.2. Let B be an abeloid analytic group over the analytic space S, and let s be a point of S such that the completion $\hat{B}$ along the fibre B_s is algebraizable; that is, the $B_n = B \times_S S_n$ are algebraizable and $\hat{B}$, now viewed as a formal scheme over $\underline{\hat{O}}_{S,s}$, is algebraizable, or equivalently comes from an abelian scheme $B_{\underline{\hat{O}}}$ over $\underline{\hat{O}}$. Suppose moreover that $B_{\underline{\hat{O}}}$ is projective over $\underline{\hat{O}}$. Then B is projective over S in a neighbourhood of s.

The hypothesis means that there is a polarization of the formal analytic space $\hat{B}$, that is, a sequence of polarizations p_n of the B_n which "glue together". The conclusion means that there is a polarization p of B over a neighbourhood of s. Let P be the analytic space over S of relative polarizations of B over S, which represents the functor

$$S' \longmapsto \{\text{polarizations of } B_{S'} \text{ over } S'\}$$

on the category of analytic spaces S' over S. This is an open subfunctor of the Néron–Severi functor $\underline{\mathrm{NS}}_{B/S}$ over S. The simplest standard arguments, which we omit here, show that $\underline{\mathrm{NS}}_{B/S}$, and hence also P, is represented by an unramified analytic space over S. The hypothesis then says that there is a formal section p of P over S at s. Since P is unramified and therefore locally quasi-finite over S, it follows from [2, Exp. 18, no. 1] that every formal section of P over S at s comes from an analytic section near s, that is, from a polarization of B over a neighbourhood of s. This proves the assertion.

Remarks 13.5.3.

a) If in 13.5.2 we omit the hypothesis that $B_{\hat{\underline{O}}}$ is projective over $\hat{\underline{O}}$,[76] we can still conclude that B is algebraizable near s. For this one uses an unpublished result (due independently to M. Artin and P. Deligne, using the theory of the sheaves $\underline{T}^i$ of VI 2) which asserts that, if X is proper over S and X_{red} is algebraizable, then X is algebraizable. This reduces us to the case where S is reduced. On the other hand, when S is normal, so that $\hat{\underline{O}}$ is normal, every abelian scheme over $\hat{\underline{O}}$ is known to be projective over $\hat{\underline{O}}$ (see, for example, [23]), and 13.5.2 applies. To obtain the case of an arbitrary reduced S, one uses non-flat descent from the normalization S' of S, by means of the following (delicate) result, which we shall not prove here: a proper morphism of noetherian schemes $f : S' \longrightarrow S$ which is an epimorphism (that is, such that $\underline{O}_S \longrightarrow f_*(\underline{O}_{S'})$ is injective) is an effective descent morphism for the fibred category of relative abelian schemes.

b) It appears that the proof of 13.5.2 should also prove the analogous result for every analytic space B proper over S. Probably more delicate is the question whether an analytic space X proper over S whose formal completion $\hat{X}$ along X_s is algebraizable is itself algebraizable over S near s. We merely point out that here the notion of relative algebraizability of X over S should be taken in a broader sense than in [10]: near s, X should be required to come from an algebraic space over $\underline{O}_{S,s}$ in the sense of M. Artin [1].

13.5.4. Let us place ourselves under the conditions of 13.5.1. For every integer $n > 0$, choose also the standard isomorphism of groups over $\mathbb{C}$

$$\mu_{n\mathbb{C}} \simeq (\mathbb{Z}/n\mathbb{Z})_{\mathbb{C}}, \qquad m \longmapsto \exp(2\pi i m/n) \qquad .$$

By passing to the inverse limit over the powers of a fixed prime ℓ, this gives a canonical isomorphism of ℓ-adic sheaves over $\mathbb{C}$,

$$\mathbb{Z}_\ell(1)_{\mathbb{C}} \simeq \mathbb{Z}_{\ell\mathbb{C}} \qquad ,$$

which we shall use to identify the two sides.

Applying the exact sequence of $\underline{\mathrm{Ext}}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, -)$ to the exact sequence (13.1.3), we obtain a canonical isomorphism

$$(13.5.5) \qquad {}_nA^{\mathrm{o}} \simeq \underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \qquad ,$$

which, as n runs through the powers of a given prime ℓ, gives an isomorphism of inverse systems of groups over S,

$$(13.5.6) \qquad T_\ell(A^{\mathrm{o}}) \simeq \underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \qquad ,$$

where $\mathbb{Z}_\ell$ is regarded as the inverse system of the $\mathbb{Z}/\ell^{n+1}\mathbb{Z}$. In the situation considered here, in which A comes from a separated group scheme over S, it is then immediately clear that the $\underline{R} \otimes \mathbb{Z}/n\mathbb{Z}$ may be regarded as relative schemes. This gives a meaning to the symbol $(\underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}}$ (whereas $R_{\underline{O}}$ would in general have none!), and yields a canonical isomorphism

$$(13.5.7) \qquad T_\ell(A_{\underline{O}}) \simeq (\underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}} \qquad .$$

It follows immediately from the constructions of paragraph 2 and 13.3 that the preceding isomorphism carries the fixed part to the fixed part and the toric part to the toric part:

$$(13.5.8) \qquad T_\ell(A_{\underline{O}})^f \simeq (\underline{R}^f \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}}, \qquad T_\ell(A_{\underline{O}})^t \simeq (\underline{R}^t \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}} \qquad .$$

76. The printed source says "projective over S" here; the hypothesis of Lemma 13.5.2 and the ensuing normality argument require "projective over $\hat{\underline{O}}$".

If A' is a relative analytic group over S satisfying the same hypotheses as A, and if $W_{\underline{O}}$ is a biextension of $(A_{\underline{O}}, A'_{\underline{O}})$ by $\mathbb{G}_{m\underline{O}}$, hence defines over a neighbourhood of s a biextension of (A, A') by $\mathbb{G}_{mS}$ (now using the notion in the complex-analytic sense, obtained by working in the site of analytic spaces over S for a suitable topology), then, by VIII 2.2, W defines pairings

$$(\underline{R} \otimes \mathbb{Z}_\ell) \times (\underline{R}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell) \simeq T_\ell(A) \times T_\ell(A') \longrightarrow \mathbb{Z}_\ell(1)_S \simeq \mathbb{Z}_{\ell S} \qquad , \tag{13.5.9}$$

and it is again clear that the corresponding pairing

$$T_\ell(A_{\underline{O}}) \times T_\ell(A'_{\underline{O}}) \longrightarrow \mathbb{Z}_\ell(1)_S \tag{13.5.10}$$

is precisely the one associated, by loc. cit., with the biextension $W_{\underline{O}}$ of $(A_{\underline{O}}, A'_{\underline{O}})$. It is also not difficult to see that, after restricting to a neighbourhood of s, the pairings (13.5.9) arise from a unique pairing (no longer depending on ℓ)

$$\underline{R} \otimes \underline{R}' \longrightarrow \mathbb{Z} \qquad . \tag{13.5.11}$$

With these preliminaries, the algebraic results concerning the $T_\ell(A_{\underline{O}})$ can be used to draw conclusions about the sheaf $\underline{R}$. Thus the pairing (13.5.11) is zero on $\underline{R}^t \times \underline{R}'^f$ and on $\underline{R}^f \times \underline{R}'^t$ (7.4.8). Moreover, if S is nonsingular of dimension 1 and, over $U = S - \{s\}$, A and A' are relative abelian schemes dual to one another through W, then $\underline{R}^t_U$ and $\underline{R}'^f_U$, as well as $\underline{R}^f_U$ and $\underline{R}'^t_U$, are exact annihilators of one another over a sufficiently small neighbourhood of s (7.4.8). Since (13.5.11) is a perfect duality over U, it follows that there is a duality over U between $\underline{R}/\underline{R}^t$ and $\underline{R}'^f$. In particular, $\underline{R}/\underline{R}^t$ is constant, and a fortiori the action of $\pi_1(U,t)$ on $\underline{R}_t$ is unipotent of level 2. Similarly, over U there is a perfect duality between $\underline{R}/\underline{R}^f$ and $\underline{R}'^t$. Since the latter is also isomorphic to the dual torus of the lattice M_S over S, where M is the group defined in (5.5.4), we obtain a canonical isomorphism

$$\underline{R}_U/\underline{R}^f_U \simeq M_U \qquad , \tag{13.5.12}$$

and the homomorphism i of (13.3.4) may be written as a homomorphism

$$i : M_U \longrightarrow A^\natural \qquad , \tag{13.5.13}$$

which we shall encounter again in a slightly different context in the next paragraph. 14. Links with transcendental theory: the rigid-analytic case. The canonical homomorphism $\underline{M}_K \longrightarrow A^{\natural o}_K$.

14.1. We assume here the foundations of the theory of rigid-analytic spaces, for which one may consult [12], [13], and [34].

Let S be a complete trait, and let A_K be an abelian scheme over its field of fractions. In 8.1, by descent from the Raynaud extension in the semistable case, we constructed an extension $G_K = A^{\natural o}_K$ (8.1.1)

$$0 \longrightarrow T_K \longrightarrow G_K \longrightarrow B_K \longrightarrow 0 \tag{14.1.1}$$

of an abelian scheme B_K by a torus T_K. Considering the dual abelian scheme A'_K, we likewise have an extension $G'_K = A'^{\natural o}_K$

$$0 \longrightarrow T'_K \longrightarrow G'_K \longrightarrow B'_K \longrightarrow 0 \qquad , \tag{14.1.2}$$

where B'_K is moreover identified with the abelian scheme dual to B_K (8.2). As in (8.1.8), let $\underline{M}_K$ and $\underline{M}'_K$ denote the lattices over K dual to T'_K and T_K, respectively:

$$\underline{M}_K = D(T'_K), \qquad \underline{M}'_K = D(T_K) \qquad . \tag{14.1.3}$$

Thus the extension structures (14.1.2) and (14.1.1) may respectively be interpreted (VIII 3.7) as equivalent to the data of homomorphisms

$$\underline{M}_K \xrightarrow{p_K} B_K, \qquad \underline{M}'_K \xrightarrow{p'_K} B'_K \qquad . \tag{14.1.4}$$

Let $W_K^\natural$ be the biextension of (B_K, B'_K) by $\mathbb{G}_{mK}$ which defines the canonical duality between these abelian schemes (8.2), and let E_K be the biextension of $(\underline{M}_K, \underline{M}'_K)$ by $\mathbb{G}_{mK}$ obtained by pulling $W_K^\natural$ back along the homomorphisms (14.1.4). One checks immediately that trivializations of E_K correspond canonically and bijectively to homomorphisms

$$i : \underline{M}_K \longrightarrow G_K \tag{14.1.5}$$

lifting $p_K : \underline{M}_K \longrightarrow B_K$, or, symmetrically, to homomorphisms

$$i' : \underline{M}'_K \longrightarrow G'_K \tag{14.1.6}$$

lifting $p'_K : \underline{M}'_K \longrightarrow B'_K$.

With this understood, write a superscript $^{\mathrm{an}}$ for the rigid-analytic groups over K associated with the algebraic groups under consideration [13]. In [24] one defines a canonical exact sequence

$$0 \longrightarrow \underline{M}_K^{\mathrm{an}} \xrightarrow{i^{\mathrm{an}}} G_K^{\mathrm{an}} \xrightarrow{e} A_K^{\mathrm{an}} \longrightarrow 0 \qquad . \tag{14.1.7}$$

Here i^{an}, which may also be interpreted as a homomorphism $i : \underline{M}_K \longrightarrow G_K$ of ordinary group schemes, lifts the homomorphism p in (14.1.4), that is, it is of the form considered in (14.1.5). This exact sequence is the rigid-analytic analogue of the exact sequence (13.3.4). It depends functorially on A_K, and its formation is compatible with every change of traits $S' \longrightarrow S$.

The latter fact also shows that, in order to define and study (14.1.7), finite descent reduces us to the case where A_K has semistable reduction. In this case the group schemes $T_K, G_K, B_K, \underline{M}_K, \ldots$ are the generic fibres of group schemes $T, G, B, M, \ldots$ over S, with G an extension of the abelian scheme B by the torus T (paragraph 7). We may then consider the formal completion $\hat{G}$ of G along its special fibre, and by construction there is a canonical isomorphism (7.2.6)

$$\hat{G} \xrightarrow{\sim} \widehat{A^{\mathrm{o}}} \qquad . \tag{14.1.8}$$

Passing to the "generic fibres", which are rigid-analytic groups over K, gives a canonical isomorphism

$$\hat{G}^{\mathrm{an}} \xrightarrow{\sim} \widehat{A^{\mathrm{o}}}^{\,\mathrm{an}} \qquad . \tag{14.1.9}$$

On the other hand, there are canonical homomorphisms (rigid-analytic open immersions, but not in general isomorphisms)

$$\hat{G}^{\mathrm{an}} \hookrightarrow G_K^{\mathrm{an}}, \qquad \widehat{A^{\mathrm{o}}}^{\,\mathrm{an}} \hookrightarrow A_K^{\mathrm{an}} \qquad . \tag{14.1.10}$$

The homomorphism e in (14.1.7) is uniquely characterized by the commutativity of the diagram

$$\begin{array}{ccc} \hat{G}^{\mathrm{an}} & \longrightarrow & \widehat{A^{\mathrm{o}}}^{\mathrm{an}} \\ \downarrow & & \downarrow \\ G_K^{\mathrm{an}} & \longrightarrow & A_K^{\mathrm{an}} \end{array} \tag{14.1.11}$$

where the unnamed arrows are those of (14.1.9) and (14.1.10).

Remarks 14.1.12.

a) Notice that the uniqueness of e making the preceding diagram commute is immediate, since the rigid-analytic space G_K^{an} is connected. The existence of e, on the other hand, is by no means trivial, and on this point we merely refer to the cited note of M. Raynaud. Moreover, the definition of i^{an} given in loc. cit. is likewise transcendental in nature, although it may be interpreted as an object of an essentially algebraic nature (14.1.5), once the construction of G itself, and hence of G_K, has been obtained through the constructions of formal geometry in paragraph 7. I know of no purely algebro-geometric description of it, but below (14.4) I shall propose an "arithmetic" characterization, obtained by reducing to a situation of finite type over the ring $\mathbb{Z}$. Thus, from the standpoint of algebraic geometry, the homomorphism (14.1.5) seems to be a more hidden piece of structure than those encountered in the preceding paragraphs.

b) As explained above, the homomorphism i is canonically associated with a well-determined trivialization of the biextension E_K of $(\underline{M}_K, \underline{M}'_K)$, or equivalently with a well-determined homomorphism (14.1.6) lifting p'_K. One then proves that the latter is precisely the homomorphism occurring in the exact sequence

$$0 \longrightarrow \underline{M}'^{\mathrm{an}}_K \xrightarrow{i'^{\mathrm{an}}} G'^{\mathrm{an}}_K \longrightarrow A'^{\mathrm{an}}_K \longrightarrow 0 \tag{14.1.13}$$

analogous to (14.1.7), associated with the abelian scheme A'_K dual to A_K (unless it is the opposite???).

14.2. Relations of the homomorphism $i : \underline{M}_K \longrightarrow G_K$ with monodromy.

Let n be an integer > 0. From the exact sequence (14.1.7), by means of the exact sequence of the $\underline{\mathrm{Ext}}^i(\mathbb{Z}/n\mathbb{Z}, -)$ on the site of rigid-analytic spaces over K, one then obtains a canonical exact sequence (taking account of the fact that $\underline{M}_K^{\mathrm{an}}$ is torsion-free and G_K^{an} is n-divisible):

$$0 \longrightarrow {}_nG_K^{\mathrm{an}} \longrightarrow {}_nA_K^{\mathrm{an}} \longrightarrow \underline{M}_{K,n}^{\mathrm{an}} \longrightarrow 0 \qquad ,$$

where $\underline{M}_{K,n} \overset{\mathrm{dfn}}{=} \underline{M}_K/n\underline{M}_K$. Since all the terms occurring in this exact sequence come from finite group schemes over K, it may be rewritten as arising from an exact sequence

$$0 \longrightarrow {}_nG_K \longrightarrow {}_nA_K \longrightarrow \underline{M}_{K,n} \longrightarrow 0 \qquad . \tag{14.2.1}$$

As n varies, these exact sequences behave in the expected way, and, on passing to the inverse limit over the powers of a prime number ℓ, one obtains the analogous exact sequence

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longrightarrow 0 \qquad . \tag{14.2.2}$$

But one already has such an exact sequence by purely algebraic means, thanks to the first formula (8.1.2) and to (8.1.7), which give

$$T_\ell(G_K) \xrightarrow{\sim} T_\ell(A_K)^{\mathrm{ef}} \subset T_\ell(A_K), \qquad T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}} \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \qquad . \tag{14.2.3}$$

With this understood, I claim that the homomorphisms (14.2.3) arising from the algebraic theory are precisely those deduced from the exact sequence (14.2.2) arising from the transcendental theory. For the first homomorphism, which corresponds to the first homomorphism of (14.2.2), our assertion follows at once, after reduction to the semistable situation, from the commutativity of (14.1.11), since the vertical arrows there induce isomorphisms on the T_ℓ, while applying T_ℓ to the horizontal arrows gives, respectively, the arrow under consideration in (14.2.3) and that in (14.2.2). We shall admit the asserted compatibility for the two isomorphisms obtained between $T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}}$ and $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. Notice, moreover, that this compatibility (for a given ℓ) already suffices to characterize uniquely, by transcendental means, the homomorphism (14.1.5) occurring in (14.1.7), interpreted as an isomorphism from $\underline{M}_K^{\mathrm{an}}$ onto $\operatorname{Ker} e$.

One may also say that the homomorphism e of (14.1.7) identifies the $T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}}$ with the ℓ-adic completions $\underline{R}_K \otimes \mathbb{Z}_\ell$ of a single lattice $\underline{R}_K = \operatorname{Ker} e$ over K, and that this "integral structure" on the preceding system of pro-ℓ groups is compatible, in the evident sense, with the integral structure arising from formulas (8.1.7) (orthogonality theorem 5.2). This then defines canonically an isomorphism

$$i : \underline{M}_K \xrightarrow{\sim} \operatorname{Ker} e \qquad .$$

Since the homomorphism i (14.1.5) enables one to reconstruct the usual extension structure on $T_\ell(A_K)$, namely (14.2.2), and hence also its structure as a blended extension, described in 9.6, by means of the exact sequence deduced from (14.1.1),

$$0 \longrightarrow \qquad T_\ell(T_K) \qquad \longrightarrow \quad T_\ell(G_K) \quad \longrightarrow \quad T_\ell(B_K) \quad \longrightarrow \quad 0$$

$$\underline{M}'_K \otimes \mathbb{Z}_\ell(1)$$

it also enables one, in the semistable case, to reconstruct the monodromy pairings (9.1.2)

$$u_\ell : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_\ell \qquad , \tag{14.2.4}$$

these having been defined precisely in terms of blended extensions in 9.6. Let us merely indicate the result of translating the construction of 9.6. For this purpose, observe that in the semistable case p_K and p'_K arise from homomorphisms

$$p : \underline{M} \longrightarrow B, \qquad p' : \underline{M}' \longrightarrow B' \qquad ,$$

where B, B' are abelian schemes over S, dual to one another. Pulling back by (p, p') therefore gives a biextension E of $(\underline{M}, \underline{M}')$ by $\mathbb{G}_{mS}$ whose generic fibre is the biextension E_K already considered in 4.1.

With this said, suppose for simplicity that $\underline{M}$ and $\underline{M}'$ are constant, with values M and M'. For every pair $(m, m') \in M \times M'$, $E_{m,m'}$ is a torsor under $\mathbb{G}_{mS}$, and hence corresponds to an invertible module $L_{m,m'}$ over the ring V of S. It therefore admits a basis u_0, determined up to multiplication by a unit of V. Consequently one obtains a canonical "valuation" map

$$\underline{v} : E_{m,m'}(K) \longrightarrow \mathbb{Z}$$

which associates with every basis u of $L_{m,m'} \otimes_V K$, written $u = \lambda u_0$ ($\lambda \in K^*$), the element

$$\underline{v}(u) = \underline{v}(\lambda) \in \mathbb{Z} \qquad ,$$

where $\underline{v}$ on the right denotes the valuation associated with V (normalized in the usual way, so that its value group is $\mathbb{Z}$); the resulting integer is plainly independent of the choice of u_0. If φ is now a trivialization of E, one may consider the corresponding map

(14.2.5) $$(m, m') \longmapsto \underline{v}\big(\varphi(m, m')\big) : M \times M' \longrightarrow \mathbb{Z} \quad .$$

With this understood, one finds that the monodromy homomorphism (14.2.4) takes its values in $\mathbb{Z}$ and is precisely the homomorphism (14.2.5), in which φ is the splitting of E_K associated with the homomorphism i (14.1.5).

Remarks 14.2.6. In particular, one recovers (by means of the transcendental rigid-analytic theory) the fact that the monodromy pairing is integral and independent of ℓ (10.4 a)), proved by a different (and likewise transcendental) method in 12.5. According to M. Raynaud, rigid-analytic theory also recovers the positivity assertion 10.4 b) (which is likewise contained in 12.5).

Remarks 14.2.7. Regard $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Q}$ as an extension of $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$ by $\underline{M}_K$. Using the homomorphism (14.1.5) $i : \underline{M}_K \longrightarrow G_K$, one deduces an extension A_K^b

(14.2.8) $$0 \longrightarrow G_K \longrightarrow A_K^b \longrightarrow \underline{M}_K \otimes \mathbb{Q}/\mathbb{Z} \longrightarrow 0 \quad .$$

Observe moreover that, since G_K is a divisible group, the datum of such an extension is equivalent to that of a family of extensions of the form (14.2.2), namely those deduced from (14.2.8) by applying the T_ℓ. Since i is a monomorphism, the extensions in question which define (14.2.8) are precisely the extensions (14.2.2) arising from the exact sequence (14.2.1). As these latter extensions are defined by purely algebraic means, the same is true of the extension (14.2.8). In fact, the characterization just given shows at once that this extension is canonically isomorphic to the group, likewise denoted A_K^b, introduced in (8.2.3), in view of the isomorphism 11.11 from the group of connected components of the latter to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$.

14.3. Application to an arithmetic characterization of $i : \underline{M}_K \longrightarrow G_K$.

The characterization we have in mind rests on the observation that, for every prime number ℓ, the extension (14.2.2) of $\underline{M}_K \otimes \mathbb{Z}_\ell$ by $T_\ell(G_K)$ deduced from i is known by purely algebraic means. In particular, the image of i under the canonical homomorphism

(14.3.1) $$\mathrm{Hom}(\underline{M}_K, G_K) \longrightarrow \prod_\ell \mathrm{Ext}^1\big(\underline{M}_K \otimes \mathbb{Z}_\ell, T_\ell(G_K)\big) \quad .$$

is known. In the case (to which one may always reduce by descent) in which $\underline{M}_K$ is constant, with value M, the preceding homomorphism may be interpreted as

(14.3.2) $$\mathrm{Hom}\big(M, G_K(K)\big) \longrightarrow \prod_\ell \mathrm{Hom}\big(M, H^1(K, T_\ell(G_K))\big),$$

deduced from the canonical homomorphism

(14.3.3) $$G_K(K) \longrightarrow \prod_\ell H^1(K, T_\ell(G_K)) \quad ,$$

which comes from the connecting homomorphisms in the Kummer exact sequences

$$0 \longrightarrow {}_nG_K \longrightarrow G_K \xrightarrow{n} G_K \longrightarrow 0 \quad .$$

(The homomorphism (14.3.1) could in fact always be interpreted as the connecting homomorphism analogous to (14.3.3) for the twisted group $\check{M}_K \otimes G_K = \underline{\mathrm{Hom}}(\underline{M}_K, G_K)$.) The kernel of (14.3.3) plainly consists of the elements of $G_K(K)$ which are divisible by n for every integer $n \geq 1$, or, as we shall say, which are infinitely divisible. Thus this already determines i modulo

an element of $\mathrm{Hom}(M, g)$, where g is the subgroup of $G_K(K)$ formed by its infinitely divisible elements:

$$g = \bigcap_{n \geq 1} nG_K(K) \qquad .$$

We now have the following result.

Lemma 14.3.4. Let V be a discrete valuation ring whose residue field k has characteristic $p > 0$ and is of finite type over its prime field, and let G_K be a group scheme of finite type over K, of unipotent rank zero if $\mathrm{char}\, K = 0$. Then the group g of infinitely divisible elements of $G_K(K)$ is zero; that is, (if G_K is divisible) the homomorphism (14.3.3) is injective.

After making a finite extension K'/K and replacing V by its normalization in K', we may suppose that G_K admits a composition series whose successive quotients are of one of the following types: a constant finite group, an abelian variety, $\mathbb{G}_{mK}$, or $\mathbb{G}_{aK}$. We are therefore immediately reduced to the case where G_K itself is of one of these types. The case of a constant finite group, or of $\mathbb{G}_{aK}$, is trivial. In the abelian case, G_K may be regarded as the generic fibre of its Néron model G, so we are reduced to showing that, if G is a scheme of finite type over S, the subgroup of $G(S)$ formed by the infinitely divisible points is zero.

This is true of the group $G(k)$, as follows at once from the usual dévissage of G_o and from the Mordell–Weil theorem [16] (which says that $G_o(k)$ is a finitely generated $\mathbb{Z}$-module if G_o is an abelian scheme over the field of finite type k). Consequently g is contained in the kernel $G(S)^{(1)}$ of $G(S) \longrightarrow G(k)$. The kernel of $G(S)^{(1)} \longrightarrow G(V/\underline{m}^{n+1})$ has a filtration whose quotients are isomorphic to vector groups over k (after supposing, as we may, that S is complete), and these are killed by p. It follows that g is contained in the kernel of the preceding homomorphism for every n, and hence that it is zero. The case $G_K = \mathbb{G}_{mK}$ reduces to the preceding one, since $G_K(K) = K^*$ is an extension of $\mathbb{Z}$ (which has no nonzero infinitely divisible elements) by $V^* = \mathbb{G}_m(S)$, and the latter satisfies the same condition by what we have just proved.

Remarks 14.3.4.1. Notice that in 14.3.4 it is essential to require divisibility by all integers ≥ 1, and not only by the powers of a fixed prime ℓ; equivalently, one must consider the kernel of (14.3.3), rather than restricting the right-hand side to a single factor $H^1(K, T_\ell(G_K))$. Indeed, if G_K comes from a smooth group scheme G over S and S is henselian, every element of the kernel of $G(S) \longrightarrow G(k)$ is infinitely ℓ-divisible for every $\ell \neq p$! The argument proving 14.3.4 shows that the subgroup of $G_K(K)$ formed by the infinitely p-divisible elements (that is, the kernel of $G_K(K) \longrightarrow H^1(K, T_p(G_K))$ when G_K is p-divisible) is a finite group.

14.3.5. The preceding considerations show that, when the residue field k has characteristic $p > 0$ and is of finite type in the absolute sense, the homomorphism i (14.1.5) is completely determined by its image under (14.3.1), that is, by the extensions (14.2.2). To deduce from this a principle characterizing i in the general case, with no restriction on k, observe that in paragraph 7 we were able to construct the scheme G over S under conditions considerably more general than those in which S is a complete trait and one has an abelian variety with semistable reduction over its field of fractions.

More generally, let S denote the spectrum of a normal noetherian ring V, separated and complete for an $\underline{m}$-adic topology, and consider a smooth quasi-projective group scheme A over S, with connected fibres. Suppose that $A_o = A \times_S S_o$, where $S_o = \mathrm{Spec}(V/\underline{m})$, is an extension of an abelian scheme B_o by an isotrivial torus T_o, and that A_U is an abelian scheme, where $U \subset S - S_o$ is a nonempty open subset. In 7.22 we then constructed a scheme G over S, an extension of an abelian scheme B by a torus T, characterized up to unique isomorphism by the isomorphism (14.1.8)

$$\hat{G} \xrightarrow{\sim} \hat{A} \tag{14.3.5.1}$$

on the formal completions along G_o and A_o. For every prime number ℓ one deduces an isomorphism (7.3.5)

$$T_\ell(G) \xrightarrow{\sim} T_\ell(A)^f \qquad , \tag{14.3.5.2}$$

and hence, on restricting to U, the structure of an extension of pro-ℓ Barsotti–Tate groups

$$\begin{array}{ccccccccc} 0 & \longrightarrow & T_\ell(G_U) & \longrightarrow & T_\ell(A_U) & \longrightarrow & T_\ell(A_U)/T_\ell(A_U)^f & \longrightarrow & 0 \\ & & \| & & & & & & \\ & & T_\ell(A_U)^f & & & & & & \end{array} \quad . \tag{14.3.5.3}$$

Suppose now that a second group scheme A' over S is given, satisfying the same hypotheses as A, together with a biextension W of (A, A') by $\mathbb{G}_m$ which over U induces a duality between the abelian schemes A_U and A'_U. (In fact, by VIII 7.1, giving W is equivalent to giving such a duality between A_U and A'_U.) Then W defines pairings

$$T_\ell(A_U) \times T_\ell(A'_U) \longrightarrow \mathbb{Z}_\ell(1) \qquad , \tag{14.3.5.4}$$

which give rise to the orthogonality theorem (7.4.8): $T_\ell(A_U)^f$ is the orthogonal of $T_\ell(A'_U)^t = T_\ell(T'_U)$, where T' is the toric part of the Raynaud extension $G' = A'^\natural$ associated with A'. Thus one again obtains an isomorphism

$$T_\ell(A_U)/T_\ell(A_U)^f \simeq \underline{M}_U \otimes \mathbb{Z}_\ell \qquad , \tag{14.3.5.5}$$

where $\underline{M} = D(T')$ is the lattice over S dual to the torus T'. Consequently (14.3.5.3) and (14.3.5.5) give an extension

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow \underline{M}_U \otimes \mathbb{Z}_\ell \longrightarrow 0 \qquad . \tag{14.3.5.6}$$

With this understood, we may state the following conjecture.

Conjecture 14.4. The hypotheses being those just set out in 14.3.5, there exists a homomorphism

$$i : \underline{M}_U \longrightarrow G_U \tag{14.4.1}$$

such that, for every prime number ℓ, the extension (14.3.5.6) is isomorphic to the extension deduced from i as explained for (14.2.2). More precisely, there is one and only one way of associating with every situation of the type considered in 14.3.5 a homomorphism (14.4.1) satisfying the preceding condition, in such a way that the formation of i is compatible with every base change $S' \longrightarrow S$ associated with a continuous homomorphism $V \longrightarrow V'$ of noetherian adic rings.

14.4.2. It seems that the existence of i, compatibly with base change, follows from M. Raynaud's rigid-analytic theory, in view of the fact that he is developing it in a framework more general than that stated in 14.1, working, as here, over general noetherian adic rings. Moreover, to prove the existence and uniqueness asserted in 14.4, one is immediately reduced to considering only those S for which S_o is a scheme of finite type over $\mathbb{Z}$. Indeed, V is the filtered inductive limit of subrings V_i of finite type over $\mathbb{Z}$; these may be supposed normal by Nagata's theorem (EGA IV 7.8), and, for i sufficiently large, the situation (A, A', W) comes from an analogous situation over

V_i, or alternatively over the separated completion of V_i for the topology defined by $m_i = \underline{m}V_i$. This gives the asserted reduction.

14.4.3. Observe now that $S - S_o$ is a Jacobson scheme (EGA IV 10.5.8). Thus, to see that two solutions i, i' of the problem, for a given situation (A, A', W), are equal, it is enough to prove that these homomorphisms are equal at every closed point η of U. Let S' be the normalization of the closure of η in S. Then S' is a complete trait, and the uniqueness result of 4.3 may be applied to the pullback to S' of the situation over S, since the residue field of S' is finite. Once uniqueness is thus established, it implies compatibility of the formation of i with base change, subject to existence.

Remarks 14.5. It would remain to formulate with the desired generality, and to establish, a compatibility between the homomorphism i introduced in 14.4 by a rigid-analytic method and the analogous homomorphism (13.5.13) of complex-analytic theory—or, better still, to find a common generalization of them!

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INTERSECTIONS ON REGULAR SURFACES

by P. Deligne

In the first part of this exposé, after some preliminaries from intersection theory (contained in [5] ∪ [7]), we give Artin's proof of a special case of a theorem of Raynaud [9] (1.13), and an a priori proof of a theorem which Néron [8] had verified case by case (1.15). The theorems of [9] used in [2], and hence in IX, follow fairly easily from (1.13).

In the second part, continuing in the same vein, I explain how the "Italians" [1], [3] calculated intersection numbers on surfaces. The main results, 2.9 and 2.11, may also be found in Northcott [12] and [13]. To conclude, in 2.27, 2.30, and 2.34 I give a conjectural relation between the dimension of the versal formal moduli scheme of a singularity of a reduced curve, the invariant δ of the singularity, and an invariant measuring its symmetry.

This exposé will not be used in the remainder of the seminar.

1. Intersection numbers on arithmetic surfaces

1.0. Throughout this section, let X be a regular scheme of dimension 2, let Y be a reduced divisor on X, and let k be a subfield of $H^0(Y, \underline{O}_Y)$. We assume that Y is a curve (EGA II 7.4.1) proper over k. Here are three examples of this situation.

Example 1.1. Let (S, η, s) be a trait and let $f : X \longrightarrow S$ be a regular, proper, and flat S-scheme whose fibres are purely one-dimensional. We take $Y = (f^{-1}(s))_{\mathrm{red}}$ and $k = k(s)$. From 1.10 onward we shall restrict ourselves to this case.

Example 1.2. Let X_o be the spectrum of a normal local ring of dimension 2, let s be the closed point of X_o, and let $f : X \longrightarrow X_o$ be a regular connected X_o-scheme, proper over X_o, distinct from X_o but agreeing with X_o over $X_o \setminus \{s\}$. We take $Y = (f^{-1}(s))_{\mathrm{red}}$ and $k = k(s)$.

Example 1.3. We take X to be a complete regular surface over a field k, and Y to be a reduced curve on X.

Recall the following lemma (SGA 6 IV, Ex. 1.9).

Lemma 1.4. Let Z_o be a closed subscheme of a Noetherian scheme Z. Then the inclusion of Z_o in Z induces an isomorphism from the Grothendieck group of the category of coherent $\underline{O}_{Z_o}$-modules to the Grothendieck group of the category of coherent sheaves on Z with support in Z_o.

This lemma allows us to define the Euler–Poincaré characteristic $\chi(\mathcal{F})$ of a coherent sheaf $\mathcal{F}$ on X supported on Y so that:

(i) If $\mathcal{F}$ is a coherent sheaf of $\underline{O}_Y$-modules, then

$$\chi(\mathcal{F}) = \dim_k H^0(\mathcal{F}) - \dim_k H^1(\mathcal{F}). \tag{1.4.1}$$

(ii) χ is additive with respect to short exact sequences.

If the cohomology sheaves of a complex K of $\underline{O}_X$-modules are coherent sheaves supported on Y, we set

$$\chi(K) = \sum_i (-1)^i \chi(\underline{H}^i(K)). \tag{1.4.2}$$

If $\mathcal{F}$ is supported at a point y of Y, then

$$\chi(\mathcal{F}) = \mathrm{length}_{\underline{O}_y}(\mathcal{F}_y)[k(y) : k]. \tag{1.4.3}$$

Under the hypotheses of (1.1), (1.2), or (1.3), one has

$$\chi(\mathcal{F}) = \mathrm{length}\, f_*\mathcal{F} - \mathrm{length}\, R^1 f_*\mathcal{F}. \tag{1.4.4}$$

Let D be a divisor ≥ 0 supported on Y, and let $\mathcal{L}$ be an invertible sheaf on D. There is then a unique way to define the degree $\deg_D(\mathcal{L})$ such that

$$\begin{cases} \deg_D(\mathcal{L}' \otimes \mathcal{L}'') = \deg_D(\mathcal{L}') + \deg_D(\mathcal{L}''), \\ \text{for } \mathcal{L} \subset \underline{O}, \text{ one has } \deg_D(\mathcal{L}) = -\chi(\underline{O}/\mathcal{L}). \end{cases} \tag{1.4.5}$$

The Riemann–Roch formula on D is then

$$\chi(\mathcal{L}) = \chi(\underline{O}_D) + \deg_D(\mathcal{L}). \tag{1.4.6}$$

If $D = \sum n_i D_i$ is the decomposition of D into reduced irreducible divisors, then

$$\deg_D(\mathcal{L}) = \sum n_i \deg_{D_i}(\mathcal{L}). \tag{1.4.7}$$

Definition 1.5. Let D and E be two divisors ≥ 0 on X, and suppose that D is supported on Y. The intersection number of D and E (relative to the field k (1.0)) is defined by

$$(D, E) = \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_E) = \chi(\underline{O}_D \otimes \underline{O}_E) - \chi(\mathrm{Tor}^1(\underline{O}_D, \underline{O}_E)).$$

Indeed, under the hypotheses made here, $\mathrm{Tor}^2(\underline{O}_D, \underline{O}_E) = 0$.

Proposition 1.6. Under the hypotheses of 1.5, one has:

(i) The intersection number (D, E) is additive in the divisors D and E; this allows us, by linearity, to define the intersection number of two divisors which are not necessarily positive.

(ii) $(D, E) = \deg_D(\underline{O}(E))$.

(iii) If D and E meet at only finitely many points, then $(D, E) = \chi(\underline{O}_D \otimes \underline{O}_E) \geq 0$, and equality can occur only if D and E are disjoint.

(iv) If D and E are supported on Y, then $(D, E) = (E, D)$.

(v) If N is the conormal sheaf of D, then

$$D^2 = (D, D) = -\deg_D(N).$$

(vi) If E, not necessarily positive, is the divisor of a meromorphic function on X, then $(D, E) = 0$.

Proof. The exact sequence

$$0 \longrightarrow \underline{O}(-E) \longrightarrow \underline{O}_X \longrightarrow \underline{O}_E \longrightarrow 0$$

gives, by (1.4.6),

$$\begin{aligned}(D, E) &\overset{\text{def}}{=} \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_E) \\ &= \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_X) - \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}(-E)) \\ &= \chi(\underline{O}_D) - \chi(\underline{O}(-E)|D) = -\deg_D(\underline{O}(-E)) = \deg_D(\underline{O}(E)).\end{aligned}$$

This proves (ii), (vi), and biadditivity, by (1.4.5) and (1.4.7). Under the hypotheses of (iii), one has $\mathrm{Tor}^1(\underline{O}_D, \underline{O}_E) = 0$, which gives the conclusion. Under the hypotheses of (v), one has

$$\underline{O}_D \otimes \underline{O}_D = \underline{O}_D \quad \text{and} \quad \mathrm{Tor}^1(\underline{O}_D, \underline{O}_D) \simeq N \quad \text{(SGA 6 VII 2.5)},$$

hence

$$(D, D) = \chi(\underline{O}_D) - \chi(N) = -\deg_D(N).$$

Finally, (iv) follows from the symmetry of the tensor product.

Notice that (ii), (iv), and (vi) remain true without assuming the divisors positive.

Proposition 1.7. Let $(Y_i)_{1 \leq i \leq k}$ be the irreducible components of Y, f a meromorphic function on X, and

$$\mathrm{div}(f) = \sum_i n_i Y_i + Z$$

its divisor, with $|Z| \cap Y$ finite. Then, putting $D_i = n_i Y_i$, one has, for $(a_i) \in \mathbb{Q}^k$:

$$\left(\sum_i a_i D_i\right)^2 = -\sum_{i<j} (a_i - a_j)^2 (D_i, D_j) - \sum_i a_i^2 (D_i, Z).$$

Proof (after [7]). By 1.6 (v), (vi),

$$\begin{aligned}\left(\sum_i a_i D_i\right)^2 &= \sum_i a_i \left(D_i, \sum_j a_j D_j\right) = \sum_i a_i \left(D_i, \sum_j a_j D_j - a_i \operatorname{div} f\right)\\ &= \sum_i a_i \left(D_i, \sum_{j\neq i} (a_j - a_i) D_j - a_i Z\right)\\ &= \sum_{i\neq j} a_i (a_j - a_i)(D_i, D_j) - \sum_i a_i^2 (D_i, Z)\\ &= -\sum_{i<j} (a_i - a_j)^2 (D_i, D_j) - \sum_i a_i^2 (D_i, Z).\end{aligned}$$

Corollary 1.8. Under the hypotheses of Example 1.1, take for f a uniformizer π of S. Here $Z = 0$ and

$$\left(\sum_i k_i Y_i\right)^2 = -\sum_{i<j} \frac{1}{n_i n_j} (k_i n_j - k_j n_i)^2 (Y_i, Y_j);$$

thus the intersection form is negative, and if Y is connected, its kernel consists precisely of the multiples of $\operatorname{div}(\pi)$.

Corollary 1.9 (Mumford [7]). Under the hypotheses of Example 1.2, take for f a nonzero element of the maximal ideal of the local ring $\underline{O}_s$. Here $Z > 0$, so the intersection form is negative definite.

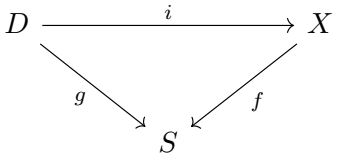
1.10. For the remainder of this section we restrict ourselves to the case considered in Example 1.1.

Lichtenbaum [5] proved that, in this case, X is automatically projective over S, so that there is no difficulty in defining the complex $Rf^!\underline{O}_S$ ([4] III 8.7, p. 190). Since f has relative dimension 1 and X and S are regular (Gorenstein would suffice), there exists ([4] V 8.3 and V 9.1) an invertible sheaf $\omega_{X/S}$ on X, unique up to unique isomorphism, such that

$$Rf^!\underline{O}_S \simeq \omega_{X/S}[1] \quad . \tag{1.10.1}$$

We denote by K any divisor (effective or not) such that $\omega_{X/S} \simeq \underline{O}(K)$.

Let D be a divisor ≥ 0 supported on Y, and let N be its conormal sheaf:



One has

$$Rg^!\underline{O}_S = Ri^!Rf^!\underline{O}_S = Ri^!\omega_{X/S}[1] = \omega_{X/S}\big|_D \otimes N^\vee[0]. \tag{1.10.2}$$

By the duality theorem ([4] V 11.1, p. 210),

$$Rg_*Rg^!\underline{O}_S \simeq \mathrm{RHom}(Rg_*\underline{O}_D, \underline{O}_S) \quad . \tag{1.10.3}$$

Moreover, for every $\underline{O}_S$-module M supported at the closed point,

(1.10.4) $$\chi\big(\mathrm{RHom}(M, \underline{O}_S)\big) = -\chi(M),$$

so that here, using (1.4.4) and (1.10.3), we obtain

$$\chi(Rg^!\underline{O}_S) = -\chi(\underline{O}_D) \quad .$$

Since, by (1.4.6),

$$\chi(Rg^!\underline{O}_S) = \chi(\underline{O}_D) + \deg_D(Rg^!\underline{O}_S),$$

we conclude from (1.10.2) that

$$-2\chi(\underline{O}_D) = \deg_D(Rg^!\underline{O}_S) = \deg_D(\omega_{X/S}|_D \otimes N^\vee) = \deg_D(\omega_{X/S}) - \deg_D(N),$$

which, by 1.6(ii) and (v), may be rewritten as follows.

Proposition 1.11. Under the hypotheses of (1.1), for every divisor $D \geq 0$ supported on Y, one has

$$\chi(\underline{O}_D) = -\frac{1}{2}(D, D+K) \quad .$$

1.12. We denote by d the greatest common divisor of the geometric multiplicities of the irreducible components of $X_s = f^{-1}(s)$; that is, the greatest common divisor of the lengths of the local rings at the generic points of the geometric special fibre of f. If X_η has a rational point, or equivalently if $f : X \longrightarrow S$ has a section x, then $d = 1$; more precisely, f is smooth at $x(s)$. Indeed, x is a closed immersion of regular schemes, hence a regular immersion, and EGA IV 17.12.1 applies.

Here is Artin's proof of a special case of a theorem of Raynaud [9], which only assumes that d is prime to the characteristic p of k.

Theorem 1.13. With the preceding notation, if Y is geometrically connected and $d = 1$, then X is cohomologically flat over S in dimension 0 (EGA III 7.8.1); more precisely (EGA III 7.6.9(ii)), one has

$$k \xrightarrow{\sim} H^0(X_s, \underline{O}_{X_s}).$$

It is easy to reduce to the case in which S is strictly local. Since Y is geometrically connected, $H^0(Y, \underline{O}_Y)$ is a purely inseparable extension of k; and since $(d, p) = 1$, at least one irreducible component of Y is

geometrically reduced. It follows readily that $H^0(Y, \underline{O}_Y) = k$.

Let D be a divisor on X such that

$$Y \leq D < X_s \quad .$$

Since $d = 1$, D is not a rational multiple of X_s; hence, by 1.8,

$$(D, X_s - D) = -(D, D) > 0 \quad .$$

The divisor Y therefore has an irreducible component E such that

$$0 < E \leq X_s - D \quad \text{and} \quad (D, E) > 0 \quad .$$

Consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \underline{O}(-D-E) & \longrightarrow & \underline{O} & \longrightarrow & \underline{O}_{D+E} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle\varphi} & & \| & & \downarrow{\scriptstyle\psi} & & \\ 0 & \longrightarrow & \underline{O}(-D) & \longrightarrow & \underline{O} & \longrightarrow & \underline{O}_D & \longrightarrow & 0 \end{array}$$

The snake lemma gives an isomorphism between $\mathrm{Ker}(\psi)$ and $\mathrm{Coker}(\varphi)$. This cokernel is precisely the restriction of $\underline{O}(-D)$ to E, whence an exact sequence

$$0 \longrightarrow \underline{O}(-D)|_E \longrightarrow \underline{O}_{D+E} \longrightarrow \underline{O}_D \longrightarrow 0 \quad .$$

By construction, $\deg_E(\underline{O}(-D)|_E) = -(D,E) < 0$, so that $H^0(\underline{O}(-D)|_E) = 0$ and the restriction map

$$0 \longrightarrow H^0(\underline{O}_{D+E}) \longrightarrow H^0(\underline{O}_D)$$

is injective. This allows us, by induction, to construct a sequence of divisors

$$Y = D_0 \leq D_1 \leq \cdots \leq D_n = X_s$$

such that $H^0(\underline{O}_{D_i})$ injects into $H^0(\underline{O}_{D_{i-1}})$. Thus $H^0(\underline{O}_{X_s}) = H^0(\underline{O}_Y) = k$, which completes the proof of 1.13.

The following fact is proved in [5] or [10].

Theorem 1.14 (Castelnuovo, Enriques, ...). Under the hypotheses of (1.1), let D be a reduced irreducible divisor contained in Y. Suppose that $H^1(D, \underline{O}_D) = 0$ and that, on putting $k' = H^0(D, \underline{O}_D)$, one has

$$(D,D) = -[k' : k] \quad .$$

Then D is a projective line over k', and there exist a regular S-scheme X_1, a closed point x_1 of X_1 with residue field k', and an isomorphism from X to the S-scheme obtained from X_1 by blowing up x_1, under which D is identified with the fibre over x_1.

A divisor D having these properties is called an exceptional curve of the first kind.

I cannot resist the pleasure of proving the following proposition of Néron [8], Proposition 1′, p. 90.

Proposition 1.15. Suppose that X_η is an elliptic curve (in particular, that it has a section), and that X contains no exceptional curve of the first kind. Then, with the notation of (1.10.1), the canonical map

$$f^* f_* \omega_{X/S} \xrightarrow{\sim} \omega_{X/S}$$

is universally an isomorphism.

Since $d = 1$ (1.12), by (1.13) it is enough to prove that $\omega_{X/S}$ is isomorphic to $\underline{O}_X$. Since $\omega_{X/S}$ is trivial on the elliptic curve X_η, we know a priori that

$$\omega_{X/S} \simeq \underline{O}\Big(\sum_i n_i Y_i\Big),$$

where the Y_i are the irreducible components of Y; and since $\mathrm{div}(\pi) \sim 0$, we may assume the n_i positive. If $\sum_i n_i Y_i$ is proportional to $\mathrm{div}(\pi)$, it is an integral multiple of $\mathrm{div}(\pi)$ and we are done. Otherwise,

$$\Big(\sum_i n_i Y_i\Big)^2 < 0,$$

there is an i such that

$$\Big(Y_i, \sum_j n_j Y_j\Big) < 0,$$

and the following lemma applies.

Lemma 1.16. Under the hypotheses of 1.1, suppose that Y is connected and that either Y is reducible or $\chi(\underline{O}_Y) \leq 0$. Then an irreducible component Y_i of Y is an exceptional curve of the first kind if and only if $(Y_i, K) < 0$.

Let $k' = H^0(Y_i, \underline{O}_{Y_i})$. The Riemann–Roch formula 1.11 gives

$$\chi(\underline{O}_{Y_i}) = [k' : k]\big(1 - \dim_{k'} H^1(Y_i, \underline{O}_{Y_i})\big) = -\frac{1}{2}\big((Y_i, Y_i) + (Y_i, K)\big) \quad .$$

If Y_i is exceptional of the first kind, it follows that $(Y_i, K) = -[k' : k] < 0$. Conversely, if $(Y_i, K) < 0$, then $\chi(\underline{O}_{Y_i}) > 0$, which implies $\chi(\underline{O}_{Y_i}) = [k' : k]$ and that Y is reducible. Hence, by 1.8, $(Y_i, Y_i) < 0$. The negative integers (Y_i, Y_i) and (Y_i, K) are divisible by $[k' : k]$, and therefore necessarily

$$(Y_i, Y_i) = (Y_i, K) = -[k' : k] \quad .$$

Exercise. (a) Under the hypotheses of (1.1), suppose that X_η is a principal homogeneous space under an elliptic curve and that X contains no exceptional curve of the first kind. Show that on X the Riemann–Roch formula (1.11) takes the form

$$\chi(\underline{O}_D) = -\frac{1}{2}(D, D).$$

(b) Suppose in addition that the residue field $k(s)$ is algebraically closed. Show that the diagram of irreducible components of X_s is a multiple of one of the diagrams considered by Néron [8], pp. 124–125, fourth column (cases a through c8). The diagram determines the genera of the reduced irreducible components, their multiplicities in X_s, and the nature and disposition of the singularities of $(X_s)_{\text{red}}$.

First treat the case in which X_s is irreducible. Once this case has been excluded, prove that every reduced irreducible component is a projective line of self-intersection -2. If $X_s = \sum_i n_i D_i$ is the decomposition of X_s into reduced irreducible divisors, rewrite this condition as

$$(*) \qquad n_i = \frac{1}{2}\sum_{j \neq i} n_j (D_i, D_j) \quad .$$

On the other hand, X_s is connected, so that:

(**) For any i and j, there exists a chain $i = i_1, i_2, \ldots, i_n = j$ with $(D_{i_k}, D_{i_{k+1}}) \neq 0$.

Now find all solutions in integers of $(*) + (**)$. First show that, except in Néron's cases b2 and c2, the intersection numbers (D_i, D_j) are either zero or one.

2. Italian intersections

Let S' and S be two smooth projective surfaces over a field k, and let $f : S' \longrightarrow S$ be a birational morphism. If D and E are two divisors on S, then

$$(D, E) = (f^*D, f^*E),$$

as one sees, for example, by moving D and E.

It is therefore natural to seek a "birational" expression for the intersection number. I propose to explain how the Italians proceeded. My principal source of information is [1], which contains a long bibliography. See also Manin [6] and Northcott [12], [13].

(2.0) Let S be the spectrum of a regular local ring of dimension 2, let s be the closed point of S, and let k be a subfield of $k(s)$ over which $k(s)$ is finite. Intersection numbers and Euler–Poincaré characteristics will be computed over k.

In this section, an S-scheme $p : S' \longrightarrow S$ will be called a birational transform of S if it is regular and connected, and if p is proper and induces an isomorphism $S' \setminus p^{-1}(s) \longrightarrow S \setminus \{s\}$. Everything said about S will also apply to the localizations of its birational transforms at a closed point.

(2.1) Let $f : (S'', p'') \longrightarrow (S', p')$ be a morphism of S-schemes between birational transforms of S. It is known [5] that f has a unique factorization

$$f = f_n \cdots f_1, \qquad f_i : S_{i+1} \longrightarrow S_i,$$

such that S_{i+1} is obtained from S_i by blowing up a finite set X_i of closed points, and $f_i(X_{i+1}) \subset X_i$. The set X_i is the set of points of S_i at which the projection of S'' onto S_i is not an isomorphism.

In particular, f induces an isomorphism over the complement of a finite subset of S' containing the generic points of the irreducible components of $p'^{-1}(s)$. Hence f^{-1} defines an injection

from the set $\mathrm{Irr}(p'^{-1}(s))$ of irreducible components of $p'^{-1}(s)$ into $\mathrm{Irr}(p''^{-1}(s))$.

The birational transforms of S have no S-automorphisms. We write I for the set of S-isomorphism classes of birational transforms of S, ordered by domination. The set I is filtered to the right and has least element S.

Definition 2.2. The set S^* of points of S infinitely near to s is the direct-limit set

$$S^* = \varinjlim_I \mathrm{Irr}(p^{-1}(s)) \quad .$$

If an element $t \in S^*$ is represented on a birational transform S' of S by a complete reduced irreducible curve T, the field $H^0(T, \underline{O}_T)$ depends only on t, not on the choice of S'. It is called the residue field $k(t)$ of t.

Let S_1 be the blow-up of S along s. The projective line s^*, the inverse image of $\{s\}$ in S_1, is called the curvetta of s and defines an element s^* of S^* with residue field $k(s)$.

Let S' be a birational transform of S, let t be a closed point of S', and let S'_t be the spectrum of the local ring of S' at t. There is an evident injection $(S'_t)^* \hookrightarrow S^*$; in particular, t defines an element t^* of S^* with the same residue field. We write $(S')^*$ for the union in S^* of the $(S'_t)^*$ as t ranges over the closed points of S'.

The following proposition follows from the description 2.1 of the birational transforms of S.

Proposition 2.3. For every $t \in S^*$, there is a unique sequence of birational transforms $(S_i)_{0 \le i \le n+1}$ of S and closed points $t_i \in S_i$ $(0 \le i \le n)$ such that

(i) $S_0 = S$, and S_{i+1} is the blow-up of S_i along $\{t_i\}$;

(ii) $t_{i+1} \in t_i^*$ $(0 \le i < n)$, and $t = t_n^*$.

Moreover, for every birational transform (S', p') of S, the element t belongs to the image of $\mathrm{Irr}(p'^{-1}(s))$ in S^* if and only if S' dominates S_{n+1}.

The integer n is called the order of t. We write $S(t)$ (respectively $S_+(t)$) for the birational transform S_n (respectively S_{n+1}) of S. By abuse of notation, we also write t for the closed point t_n of S_n, and t^* for its inverse image t_n^* in S_{n+1}.

We equip S^* with the following order:

$$t_1 \prec t_2 \quad \Longleftrightarrow \quad S_+(t_2) \text{ dominates } S_+(t_1) \quad .$$

If $t_1 \prec t_2$, then $k(t_2)$ is an extension of $k(t_1)$. The ordered set S^* has least element s^*, and every bounded subset of S^* is finite and totally ordered (S^* is a "tree rooted at s^*"). Moreover, the order n of $t \in S^*$ is the distance in the tree S^* from s^* to t.

The following proposition makes the description 2.1 of the birational transforms of S explicit.

Proposition 2.4. The function assigning to a birational transform $p : S' \to S$ the subset $\mathrm{Irr}(p^{-1}(s))$ of S^* induces a bijection between I (2.1) and the finite subsets of S^* which, together with each element,

contain all its predecessors.

Let D be a divisor ≥ 0 on S, and let $t \in S^*$. We say that t belongs to D if the closed point t of $S(t)$ (2.3) belongs to the strict transform (the schematic closure of $D - \{s\}$) of D on $S(t)$.

Definition 2.5. Let $t, t' \in S^*$. We say that t' is proximate to t if $t' \in S_+(t)^*$ (2.2) and belongs to the curvetta t^* of t (2.3).

If t' is an immediate successor of t, then t' is proximate to t. If in a sequence $(t_i)_{0 \leq i \leq n}$ every t_{i+1} is an immediate successor of t_i, and if t_n is proximate to t_0, then every t_i with $0 < i \leq n$ is proximate to t_0, while t_n is not proximate to t_i for $0 < i \leq n-1$.

(2.6) Let A be a Noetherian local ring of dimension n, let T be its spectrum, let m be its maximal ideal, let $k = A/m$ be its residue field, and let $P = \sum_{0 \leq i < n} a_i X^i$ be its Hilbert–Samuel polynomial:

$$P(r) = \dim_k(m^r/m^{r+1}) \qquad \text{for sufficiently large } r.$$

Recall that the multiplicity μ_s of the closed point s in T is defined by

$$a_{n-1} = \frac{\mu_s}{(n-1)!} \quad .$$

When $n = 1$, this simply says

$$\mu_s = \dim_k(m^r/m^{r+1}) \qquad \text{for sufficiently large } r.$$

Geometrically, μ_s may be defined as the degree, in the projective space $\mathbf{P}_k(m/m^2)$, of the special fibre of the blow-up of T along the closed subscheme $\{s\}$.

Recall that if Y and Z are two closed subschemes of a scheme X, then the strict transform of Y in the blow-up $\widetilde{X}$ of X along Z—the schematic closure of $Y \setminus Z$ in $\widetilde{X}$—is precisely the blow-up of Y along $Y \cap Z$.

Suppose that T is a closed subscheme of a regular scheme U. Let $p : U_1 \longrightarrow U$ be the blow-up of U along $\{s\}$, let s^* be the inverse image of s in U_1, and let T' be the strict transform of T in U_1. The preceding discussion shows that the multiplicity of T at s is the degree, in the projective space s^*, of the subscheme $s^* \cap T'$.

Suppose further that T is a divisor on U. Let T^* be its total transform in U_1 (the inverse-image divisor), and define ν by $T^* = \nu s^* + T'$. If X is a line in s^*, and Euler–Poincaré characteristics are calculated as alternating sums of dimensions over k, then

$$(X, s^*) = \chi(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{s^*}) = \deg_X \underline{O}(s^*) = -1, \tag{2.6.1}$$

and we have seen that

$$(X, T') = \chi(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{T'}) = \deg_{s^*}(s^* \cap T') = \mu_s \quad . \tag{2.6.2}$$

The projection formula

$$Rp_*(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{T^*}) = Rp_*(\underline{O}_X \overset{\mathbb{L}}{\otimes} Lp^* \underline{O}_T) = Rp_* \underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_T$$

and its corollary

$$(X, T^*) = \nu(X, s^*) + (X, T') = 0$$

give, by (2.6.1) and (2.6.2), the identity

$$\mu_s = \nu \quad . \tag{2.6.3}$$

Proposition 2.7. Let T be a divisor ≥ 0 on a regular scheme U, s a closed point of U, U_1 the blow-up of U along $\{s\}$, s^* the inverse image of s in U_1, T' the strict transform of T on U_1, and T^* its total transform.

The following three quantities are equal:

(i) the multiplicity, in Samuel's sense, of the point s on T;

(ii) the degree, in the projective space s^*, of $s^* \cap T'$;

(iii) the multiplicity of s^* in the divisor T^*.

2.8. Return to the hypotheses of 2.0. Let D be a divisor ≥ 0 on S and let $t \in S^*$. The multiplicity $\mu_t(D)$ of D at t is, by definition, the multiplicity at t of the strict transform of D on $S(t)$ (2.3). The support of the function $t \mapsto \mu_t(D)$ is the set of points infinitely near to s that belong to D (2.4). When D is regular, the numbers $\mu_t(D)[k(t) : k(s)]$ are zero or one: indeed, the strict transforms D' of D are isomorphic to D, and one applies the definition (2.6) of multiplicity.

Intersection numbers can be calculated in terms of multiplicities.

Theorem 2.9. If the divisors D and E meet only at s, then

$$(D, E) = \sum_{t \in S^*} \mu_t(D)\mu_t(E)[k(t) : k],$$

where the sum is over the points infinitely near to s which belong to both D and E. Proof. Let $p : S' \longrightarrow S$ be a birational transform of S. We denote the total transforms of D and E by D^* and E^*, and their strict transforms by D' and E'.

Lemma 2.9.1. (i) The total transform D^* is characterized by:

(a) $D^* - D'$ is supported on $p^{-1}(s)$;

(b) if X is supported on $p^{-1}(s)$, then $(D^*, X) = 0$.

(ii) $(D, E) = (D^*, E^*) = (D^*, E') = (D', E^*)$.

Assertion (i)(a) is immediate, and (i)(b) follows from 1.6(vi). That (a) and (b) determine D^* follows from 1.9. By construction, the cokernel of the map $\underline{O}_E \longrightarrow p_*\underline{O}_{E'}$ is finite; hence

$$\begin{aligned}(D^*, E') &= \chi(Lp^*\underline{O}_D \otimes \underline{O}_{E'}) \\ &= \chi(\underline{O}_D \otimes Rp_*\underline{O}_{E'}) = \chi(\underline{O}_D \otimes \underline{O}_E) = (D, E),\end{aligned}$$

which, by (i), implies (ii).

Take for S' the blow-up of S at s. By (2.9.1), (2.6.3), and 2.7,

$$\begin{aligned}(D, E) &= (D^*, E') = (D' + \mu_s(D)s^*, E') \\ &= (D', E') + \mu_s(D)(s^*, E') \\ &= (D', E') + \mu_s(D)\mu_s(E)[k(s) : k].\end{aligned} \tag{2.9.2}$$

Theorem 2.9 now follows by induction on (D, E), through repeated application of (2.9.2).

(2.10) Let A be a reduced Noetherian local ring of dimension one, let s be the closed point of $\mathrm{Spec}(A)$, let $\widetilde{A}$ be the normalization of A, and let k be a field over which $k(s)$ is finite. Suppose that $\widetilde{A}$ is finite over A (equivalently, that the completion $\widehat{A}$ is reduced). The δ-invariant of the singularity of $\mathrm{Spec}(A)$ at s is the length over k of the A-module $\widetilde{A}/A$.

Theorem 2.11. Let D be a reduced curve on S. Then:

(i) for every $t \in S^*$, with the notation of 2.8, one has

$$\mu_t(D) = \sum_{\substack{t' \text{ proximate} \\ \text{to } t}} \mu_{t'}(D)[k(t') : k(t)] \quad .$$

(ii) Suppose that the normalization $\widetilde{D}$ of D is finite over D, and let $\delta = \chi(\underline{O}_{\widetilde{D}}/\underline{O}_D)$ be the invariant of the singularity of D at s. For all but finitely many $t \in S^*$, $\mu_t(D)$ is zero or one, and

$$\delta = \sum_{t \in S^*} \frac{1}{2} \mu_t(D)\big(\mu_t(D) - 1\big)[k(t) : k] \quad .$$

Let $p : S_1 \longrightarrow S$ be the blow-up of S along $\{s\}$, let D^* be the total transform of D, and let D' be its strict transform. By (2.7),

$$D^* = D' + \mu_s(D)s^* \tag{2.11.1}$$

and

$$\mu_s(D) = (s^*, D')\,[k(s) : k]^{-1} \quad . \tag{2.11.2}$$

The curvetta s^* is nonsingular. Thus, by 2.8 and Definition 2.5, formula 2.9, applied to the divisors s^* and D' on S_1, becomes

$$(s^*, D') = \sum_{\substack{t' \text{ proximate} \\ \text{to } s}} \mu_{t'}(D)[k(t') : k] \quad . \tag{2.11.3}$$

Assertion (i) for $s = t$ follows from (2.11.2) and (2.11.3); the general case is obtained by applying this result to the localization of $S(t)$ at t (2.3).

Let m be the maximal ideal defining $\{s\}$, and let $n \geq 0$. It is known (SGA VII 3.5) that

$$p_*\underline{O}_{S_1}(-ns^*) = m^n \quad \text{and} \quad R^i p_*\underline{O}_{S_1}(-ns^*) = 0 \quad (i > 0). \tag{2.11.4}$$

It follows from this and from the exact sequence

$$0 \longrightarrow \underline{O}_{S_1}(-ns^*) \longrightarrow \underline{O}_{S_1} \longrightarrow \underline{O}_{ns^*} \longrightarrow 0$$

that

$$p_*\underline{O}_{ns^*} = \underline{O}_S/m^n \quad \text{and} \quad R^i p_*\underline{O}_{ns^*} = 0 \quad (i > 0), \tag{2.11.5}$$

and that

$$\chi(\underline{O}_{ns^*}) = \chi(\underline{O}_S/m^n) = \frac{n(n+1)}{2}[k(s) : k] \quad . \tag{2.11.6}$$

Put $n = 0$ in (2.11.4). For every complex K of coherent sheaves on S, the projection formula gives

$$K \xrightarrow{\sim} Rp_*Lp^*K \qquad \big(\simeq K \overset{\mathbb{L}}{\otimes} Rp_*\underline{O}_{S_1}\big).$$

In particular, since $Lp^*\underline{O}_D = p^*\underline{O}_D = \underline{O}_{D^*}$, one has

$$p_*\underline{O}_{D^*} = \underline{O}_D \quad \text{and} \quad R^i p_*\underline{O}_{D^*} = 0 \quad (i > 0). \tag{2.11.7}$$

Put $E = \mu_s(D)s^*$. By (2.11.1), (2.9.1)(i)(b), and (2.6.1),

$$\begin{aligned} \chi(\underline{O}_{D' \cap E}) &= (D', E) = (D^* - E, E) = -(E, E) \\ &= -\mu_s(D)^2(s^*, s^*) = \mu_s(D)^2[k(s) : k]. \end{aligned} \tag{2.11.8}$$

The exact sequence

$$0 \longrightarrow \underline{O}_{D^*} \longrightarrow \underline{O}_{D'} \oplus \underline{O}_E \longrightarrow \underline{O}_{D' \cap E} \longrightarrow 0$$

gives, by direct image and (2.11.7), the exact sequence

$$0 \longrightarrow \underline{O}_D \longrightarrow p_* \underline{O}_{D'} \oplus p_* \underline{O}_E \longrightarrow p_* \underline{O}_{D' \cap E} \longrightarrow 0,$$

and hence

$$0 \longrightarrow p_* \underline{O}_E \longrightarrow p_* \underline{O}_{D' \cap E} \longrightarrow p_* \underline{O}_{D'} / \underline{O}_D \longrightarrow 0 \quad .$$

Consequently, by (2.11.6) and (2.11.8),

$$\begin{aligned} \chi(p_* \underline{O}_{D'} / \underline{O}_D) &= \mu_s(D)^2 [k(s) : k] - \frac{1}{2} \mu_s(D) \big(\mu_s(D) + 1\big) [k(s) : k] \\ &= \frac{1}{2} \mu_s(D) \big(\mu_s(D) - 1\big) [k(s) : k] \quad . \end{aligned} \tag{2.11.9}$$

The schemes D and D' are reduced, differ only over s, and D' is finite over D. They therefore have the same normalization, and the δ-invariant of the singularity of D at s is the sum of $\chi(p_* \underline{O}_{D'} / \underline{O}_D)$ and the invariants δ of the singularities of D' at the points of $D' \cap s^*$. We prove 2.11(ii) by induction on δ. If $\mu_s(D) > 1$, the induction hypothesis applies to D', and 2.11(ii) follows from (2.11.9). If $\mu_s(D) = 1$, then $(D', s^*) = 1$ and $D' \cap s^*$ is a regular divisor on D'; by EGA IV 19.1.1, D' is regular. Formula (2.11.9) then shows that D is itself regular, and 2.11(ii) follows from 2.8.

2.12. Let $t \in S^*$, and let S' be a birational transform of S which dominates $S_+(t)$ (2.3). The total transform of t on S', again denoted by t^*, is the total transform on S' of the divisor t^* on $S_+(t)$.

Let $\nu : S^* \longrightarrow \mathbf{N}$ be a function of finite support, and suppose that, for every $t \in S^*$,

$$\nu(t) \geq \sum_{\substack{t' \text{ proximate} \\ \text{to } t}} \nu(t') [k(t') : k(t)] \tag{2.12.1}$$

(the proximate-point inequality). Let $f : S' \longrightarrow S$ be a birational transform of S which dominates $S_+(t)$ for each t in the support of ν, and let

$$\nu^* = \sum_{t \in S^*} \nu(t) t^*$$

be the corresponding divisor on S'.

We write $I(\nu)$ for the ideal of $\underline{O}_S$ obtained by direct image under f of the ideal of $\underline{O}_{S'}$ defining ν^*:

$$I(\nu) = f_* \underline{O}_{S'}(-\nu^*) \quad .$$

By (2.11.4), this ideal does not depend on the choice of S'. The ideals of S obtained in this way are those which Zariski [11] calls "complete."

Theorem 2.13. Under the preceding hypotheses, one has

$$\chi(\underline{O}_S / I(\nu)) = \sum_{t \in S^*} \frac{\nu(t)(\nu(t) + 1)}{2} [k(t) : k],$$

and

$$R^i f_* \underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0.$$

In old French, this is expressed briefly by saying: "requiring a plane curve to pass n times through a point t, whether proper or infinitely near, imposes $n(n+1)/2$ conditions, generally independent, on the coefficients of its equation."

We first prove the following lemma.

Lemma 2.14. Let $p : S_1 \longrightarrow S$ be the blow-up of S along $\{s\}$; let T be a closed subscheme of S_1 with ideal q; let n be an integer; let $J = \underline{O}_{S_1}(-s^*)$ be the ideal defining the divisor s^*; put $Y = \mathrm{Spec}(\underline{O}_{S_1}/J^n q)$; and let D be a regular divisor on S, with strict transform D' on S_1. Suppose that $T \cap s^*$ and $T \cap D'$ are finite.

(i) If $\mathrm{length}_{k(s)}(T \cap s^*) \leq n+1$, then

$$R^i p_*(qJ^n) = 0 \quad \text{for } i > 0,$$

and, if T is Artinian,

$$\mathrm{length}_{k(s)}\big(\underline{O}_S/p_*(qJ^n)\big) = \frac{n(n+1)}{2} + \mathrm{length}_{k(s)}(T).$$

(ii) If $\mathrm{length}_{k(s)}(T \cap s^*) \leq n$, then

$$\mathrm{length}_{k(s)}\Big(D \cap \mathrm{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)\Big) = n + \mathrm{length}_{k(s)}(D' \cap T) \quad .$$

Consider the exact sequence on S_1

$$0 \longrightarrow J^n q \longrightarrow J^n \xrightarrow{u} J^n \otimes \underline{O}_T \longrightarrow 0 \quad . \tag{2.14.1}$$

To prove that $R^i p_*(J^n q) = 0$ for $i > 0$, it is enough, by (2.11.4), to show that $p_*(u)$ is surjective. By the theorem on formal functions, it suffices to prove that the maps

$$u_k : p_*(J^n \otimes \underline{O}/J^k) \longrightarrow p_*(J^n \otimes \underline{O}_T \otimes \underline{O}/J^k)$$

are surjective.

Write Gr_J for the associated graded object of the J-adic filtration. The module J^m/J^{m+1} is isomorphic to $\underline{O}_{s^*}(m)$; hence $R^1 p_*(J^m/J^{m+1}) = 0$, and the map

$$p_*(J^n/J^{n+k+1}) \longrightarrow p_*(J^n/J^{n+k})$$

is surjective. Therefore, to prove by induction on k that the maps u_k are surjective, it suffices to prove the surjectivity of

$$v_k : p_*\big(J^n \otimes \mathrm{Gr}_J^k(\underline{O})\big) \longrightarrow p_*\big(J^n \otimes \mathrm{Gr}_J^k(\underline{O}_T)\big) \qquad (k \geq 0).$$

The module $\mathrm{Gr}_J^k(\underline{O}_T)$ is a quotient of $J^k/J^{k+1} \otimes \underline{O}_T$, and thus has length at most $n+1$ over $k(s)$. The kernel N of the surjective map

$$v'_k : J^n \otimes \mathrm{Gr}_J^k(\underline{O}) \simeq \underline{O}_{s^*}(n+k) \longrightarrow J^n \otimes \mathrm{Gr}_J^k(\underline{O}_T)$$

is therefore an invertible sheaf on the projective line s^* of degree ≥ -1. Hence $R^1 p_* N = 0$, and $v_k = p_*(v'_k)$ is surjective.

If T is Artinian, (2.14.1) now gives

$$\mathrm{length}_{k(s)}\big(p_* J^n/p_*(J^n q)\big) = \mathrm{length}_{k(s)}(\underline{O}_T),$$

and 2.14(i) follows from (2.11.4) and (2.11.6).

We now prove (ii). Since D is regular, its strict transform D' is isomorphic to D and meets s^* with length one. The assumption $\mathrm{length}_{k(s)}(T \cap s^*) \leq n$ therefore implies

$$\mathrm{length}_{k(s)}\big((D' \cup T) \cap s^*\big) \leq n+1,$$

so that $D' \cup T$ satisfies the hypothesis of (i).

The projection p induces an isomorphism between D and D'. Under this isomorphism, the ideal of $\underline{O}_D$ defining the subscheme

$$D \cap \mathrm{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)$$

corresponds to the ideal of D' generated by the global sections of qJ^n on S_1. This ideal is contained in the ideal of D' defining $D' \cap Y$. It is in fact equal to it, because every section of $\underline{O}_{D' \cup Y}$ extends to a section of $\underline{O}_{S_1}$: if $I_{D' \cup Y}$ is the ideal defining $D' \cup Y$, the obstruction to extension lies in $R^1p_*(I_{D' \cup Y})$. One has

$$\begin{aligned} I_{D' \cup Y} &= \underline{O}(-D') \cap qJ^n \\ &= (\underline{O}(-D') \cap J^n) \cap qJ^n \\ &= \underline{O}(-D')J^n \cap qJ^n \\ &= (\underline{O}(-D') \cap q)J^n, \end{aligned}$$

and by (i), applied to $D' \cup T$, there is no obstruction.

Thus

$$\mathrm{length}_{k(s)}\Big(D \cap \mathrm{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)\Big) = \mathrm{length}_{k(s)}(D' \cap Y) \quad . \tag{2.14.2}$$

The isomorphism

$$J^n \otimes \big(\underline{O}_{D'}/q\underline{O}_{D'}\big) \xrightarrow{\sim} \underline{O}_{D'}J^n/\underline{O}_{D'}qJ^n$$

gives the exact sequence

$$0 \longrightarrow J^n \otimes \underline{O}_{D' \cap T} \longrightarrow \underline{O}_{D' \cap Y} \longrightarrow \underline{O}_{D' \cap ns^*} \longrightarrow 0,$$

and hence

$$\mathrm{length}_{k(s)}(D' \cap Y) = n(D', s^*) + \mathrm{length}_{k(s)}(D' \cap T). \tag{2.14.3}$$

Since $(D', s^*) = 1$, assertion 2.14(ii) follows from (2.14.2) and (2.14.3).

We now prove, by induction on the number of elements in the support of ν, the assertions of 2.13 and the following assertion:

(2.14.4) if D is a regular divisor on S, then $$\chi\Big(D \cap \mathrm{Spec}(\underline{O}_S/I(\nu))\Big) = \sum_{\substack{t \in S^* \\ t \text{ on } D}} \nu(t)[k(t):k].$$

If $\nu = 0$, then $\nu^* = 0$, $I(\nu) = \underline{O}_S$, and 2.13 and (2.14.4) are immediate (the second assertion of 2.13 follows by repeated application of (2.11.4) with $n = 0$). Otherwise $\nu(s) > 0$, and there is an S-morphism $g : S' \longrightarrow S_1$.

For each closed point a of S_1, let S_{1a} be the localization of S_1 at a, let ν_a be the restriction of ν to S_{1a}^*, let $S'_a = S' \times_{S_1} S_{1a}$, let $I(\nu_a) = g_*\underline{O}_{S'_a}(-\nu_a^*)$, and put

$$T_a = \mathrm{Spec}(\underline{O}_{S_1}/I(\nu_a)), \qquad T = \coprod_a T_a,$$

where the union is finite and disjoint. Let q be the ideal of T. With the notation of 2.14, one has

$$g_*\underline{O}_{S'}(-\nu^*) = qJ^{\nu(s)}. \tag{2.14.5}$$

The induction hypothesis applies to (S_{1a}, S'_a, ν_a). Hence

$$R^i g_*\underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0, \tag{2.14.6}$$

$$\chi(T) = \sum_{t \in S^* \setminus \{s\}} \frac{\nu(t)(\nu(t)+1)}{2}[k(t):k], \tag{2.14.7}$$

and, by (2.14.4) applied to the divisor s^* on S_1 and by (2.12.1),

$$\operatorname{length}_{k(s)}(T \cap s^*) = \sum_{\substack{t \text{ proximate} \\ \text{to } s}} \nu(t)[k(t):k(s)] \leq \nu(s).$$

This inequality allows us to apply 2.14. From (2.14.5) we obtain

$$R^i p_* g_*\underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0$$

and

$$\operatorname{length}_{k(s)}\big(\underline{O}_S / p_* g_*\underline{O}_{S'}(-\nu^*)\big) = \frac{\nu(s)(\nu(s)+1)}{2} + \operatorname{length}_{k(s)}(T).$$

The assertions of 2.13 therefore follow from (2.14.6) and (2.14.7). Finally, 2.14(ii) gives

$$\operatorname{length}_{k(s)}\Big(D \cap \operatorname{Spec}(\underline{O}_S / I(\nu))\Big) = \nu(s) + \operatorname{length}_{k(s)}(D' \cap T).$$

Applying (2.14.4) to D' gives

$$\begin{aligned}\chi\Big(D \cap \operatorname{Spec}(\underline{O}_S / I(\nu))\Big) &= \nu(s)[k(s):k] + \sum_{\substack{t \text{ on } D \\ t \neq s}} \nu(t)[k(t):k] \\ &= \sum_{\substack{t \in S^* \\ t \text{ on } D}} \nu(t)[k(t):k].\end{aligned}$$

This completes the proof of 2.13. (2.15) Fragments of the preceding discussion generalize to higher dimension. Let S be the spectrum of a regular local ring of dimension n, with closed point s, where $k(s)$ is a finite extension of k. We may again consider the set I of schemes $f : S' \longrightarrow S$ obtained from S by successive blow-ups of closed points, and put, as in 2.1,

$$S^* = \varinjlim_{S' \in I} \operatorname{Irr}\big(f^{-1}(s)\big).$$

The set S^* will again be called the set of points of S infinitely near to s.

If X is a closed subscheme of S and $t \in S^*$, the multiplicity of X at t is defined as in 2.8.

Let $f : S' \longrightarrow S$ be a composite of blow-up morphisms at closed points, let D be a divisor on S, and let $D^* = f^*D$ be its total transform on S'.

Let E be a curve on S, let E' be its strict transform on S', and suppose that $D \cap E$ is supported at s. The projection formula again gives

$$(D, E) = (D^*, E'). \tag{2.16}$$

Proceeding as in 2.9, one proves with the aid of (2.16) that

$$(D,E)=\sum_{t\in S^*}\mu_t(D)\mu_t(E)[k(t):k] \quad . \tag{2.17}$$

Define proximate points as in 2.5. Proceeding as in 2.11, one verifies that

$$\mu_t(E)=\sum_{\substack{t' \text{ proximate}\\ \text{to } t}}\mu_{t'}(E)[k(t'):k] \quad . \tag{2.18}$$

On the other hand, formula 2.11(ii) is false in general (see the examples in 2.25).

2.19. Let S be a regular projective surface over a field k. Write S^* for the disjoint sum, taken over the closed points of S,

$$S^*=\coprod_{s\in S}\mathrm{Spec}(\underline{O}_{s,S})^*,$$

and call S^* the set of points infinitely near to the points of S.

Let I be the set of surfaces obtained from S by successive blow-ups of closed points. The set I is cofinal in the set of regular models of the function field of S, ordered by domination. For $i\in I$, let S_i denote the corresponding surface, equipped with $p_i : S_i \longrightarrow S$. We write $\mathrm{Div}^{\pm}(S_i)$ for the group of divisors, effective or not, on S_i.

Definition 2.20. (i) The group of "curves" on S is the inductive limit

$$\varinjlim_{i\in I}\mathrm{Div}^{\pm}(S_i).$$

(ii) The category of "invertible sheaves" on S is the inductive limit (SGA 4 VI 6) of the categories of invertible sheaves on the $(S_i)_{i\in I}$.

These objects are birational invariants of S.

Example 2.21. (i) Every "curve" C on S defines an "invertible sheaf" $\underline{O}(C)$ on S.

(ii) Every divisor C on S defines a "curve" on S, denoted by C^*.

(iii) If $t\in S^*$, define the surface $S_+(t)$ and the divisor t^* on $S_+(t)$ as in 2.3; the "curve" defined by this divisor will again be denoted by t^*.

(2.22) One verifies easily that a "curve" C (respectively an "invertible sheaf" $\mathcal{L}$) can be written in one and only one way in the form

$$C=C_0^*-\sum_{t\in S^*}i_t t^* \qquad \text{respectively} \qquad \mathcal{L}=\mathcal{L}_0\otimes\underline{O}\left(-\sum_{t\in S^*}i_t t^*\right),$$

where C_0 (respectively $\mathcal{L}_0$) is a divisor (respectively an invertible sheaf) on S, and where the integers i_t are almost all zero. We say that C (respectively $\mathcal{L}$) is the curve C_0 (respectively the invertible sheaf $\mathcal{L}_0$) endowed with the virtual multiplicities i_t.

The completion $\mathrm{Div}^{\pm}(S)\times\mathbb{Z}^{S^*}$ of the group $\mathrm{Div}^{\pm}(S)\times\mathbb{Z}^{(S^*)}$ of "curves" on S is again a birational invariant of S; it is called the group of "generalized curves" on S. On S, a "generalized curve" C is represented by a divisor C_0 endowed with a possibly infinite collection of virtual multiplicities.

"Generalized invertible sheaves" on S are defined in the same way.

Example 2.23. Suppose that k is perfect. Then the invertible sheaf $\Omega^2_{S/k}$, endowed with virtual multiplicity -1 at every point $t\in S^*$, is a birational invariant of S; denote it by ${\Omega^2_{S/k}}^*$.

Example 2.24. Let $f : S_1 \longrightarrow S_2$ be a dominant morphism of regular projective surfaces over k. The inverse image of a "curve" or an "invertible sheaf" on S_2 is defined without difficulty by passing to a model of S_2 on which this "curve" or "invertible sheaf" is an ordinary divisor or invertible

sheaf. By passage to the limit (I have not checked whether this passage to the limit is possible), one similarly defines the inverse image of a "generalized curve" or a "generalized invertible sheaf." Suppose that k is perfect and that f is generically étale. The different $\mathfrak{d}(f)$ of f is then the effective divisor on S_1 characterized by

$$f^*\Omega^2_{S_2/k}\big(\mathfrak{d}(f)\big) \xrightarrow{\sim} \Omega^2_{S_1/k} \quad .$$

This different is not a birational invariant of f; to obtain a birational invariant one must consider the generalized coincidence curve

$$\mathfrak{d}^*(f) = \mathfrak{d}(f) + \sum_{t\in S_1^*} t^* - f^*\left(\sum_{t\in S_2^*} t^*\right),$$

or, briefly,

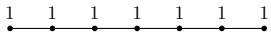
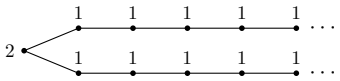
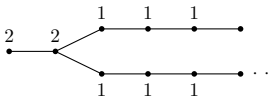
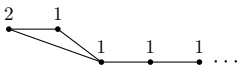
$$\mathfrak{d}^*(f) = {\Omega^2_{S_1/k}}^* - f^*{\Omega^2_{S_2/k}}^* \quad .$$

This "generalized curve" is supported on the closed subset of S_1 where f is not étale.

(2.25) Example.

Let k be an algebraically closed field and let E_n be affine n-space over k. Let C be a reduced curve in E_n having a singularity at the origin. With this singularity we associate its Enriques diagram (cf. [3]): its vertices are the points of C infinitely near the origin (2.4 and 2.15), and each vertex is joined by a line to the vertices proximate to it (2.5 and 2.15). This diagram determines the multiplicities of these points on C (2.11 and 2.18) and, when $n = 2$, the invariant δ (2.10 and 2.11) of the singularity of C at the origin.

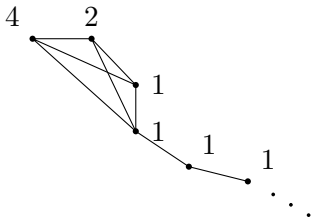
For the simplest plane singularities one obtains the following table. A number next to a vertex is its multiplicity.

Sketch	*Equation*	δ	*Enriques diagram*
	$y = 0$	0	
	$xy = 0$	1	
	$y(y - x^2) = 0$	2	
	$y^2 = x^3$	1	

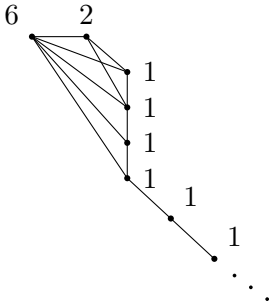
If $n > 2$, the invariant δ is no longer determined by this diagram. For example, the union of three concurrent coplanar lines (respectively, noncoplanar lines) has invariant $\delta = 3$ (respectively, $\delta = 2$), although the Enriques diagram is the same. I do not know whether this phenomenon also occurs for unibranch curves.

Here are two examples of diagrams in the case $n = 3$ which cannot occur for plane curves (cf. 2.5). We give, in order, the equations of the curve, the equations of the pure transform after blowing up the origin, parametric equations of the curve, the invariant δ, and the Enriques diagram.

$$\left\{\begin{array}{l} \begin{cases} y^2 = xz^4, \\ z^3 = x^2, \end{cases} \qquad \begin{cases} (y/z)^2 = (x/z)^3, \\ z = (x/z)^2, \end{cases} \\ \qquad\qquad \begin{cases} x = u^6, \\ y = u^7, \qquad \delta = 5 \\ z = u^4, \end{cases} \end{array}\right. \tag{2.25.1}$$



(2.25.2)
$$\left\{\begin{array}{l}\left\{\begin{array}{l}y^2z=x^3,\\ z^3=y^2,\end{array}\right. \qquad \left\{\begin{array}{l}(y/z)^2=(x/z)^3,\\ z=(y/z)^2,\end{array}\right.\\ \left\{\begin{array}{l}x=u^8,\\ y=u^9,\\ z=u^6,\end{array}\right. \qquad \delta=10\end{array}\right.$$



2.26. Theorem 2.27 below was stated as a false conjecture in the first version of this exposé (circulated by the IHES). It owes much to conversations with D. S. Rim and includes some of his results (D. S. Rim, "Torsion Differentials and Deformations," Theorem 2.7(i)).

Let C be a reduced projective curve over an algebraically closed field k, and let $s \in C(k)$. We shall write

A_s — for the local ring of C at s (or its completion, which would make no difference below);

$\widetilde{A}_s$ — for the normalization of A_s;

$\delta(s)$ — for $\dim_k(\widetilde{A}_s/A_s)$;

$m_0(s)$ — for the dimension of the formal versal scheme T of deformations of $\mathrm{Spec}(A_s)$.

For an irreducible component E of T, put likewise $m_0^E(s) = \dim(E)$.

The module $\widetilde{\tau}$ of derivations of $\widetilde{A}_s$ and the module τ of derivations of A_s are torsion-free. If A_s is integral, they embed in the module of derivations of the fraction field of A_s; in general, they embed in the module of derivations of the total quotient ring of A_s (a product of fields). Moreover, through these embeddings, τ and $\widetilde{\tau}$ are commensurable: both $\tau/(\tau \cap \widetilde{\tau})$ and $\widetilde{\tau}/(\tau \cap \widetilde{\tau})$ are finite-dimensional over k. Set

$$m_1(s) = [\widetilde{\tau} : \tau] \underset{\text{def}}{=} \dim_k \frac{\widetilde{\tau}}{\tau \cap \widetilde{\tau}} - \dim_k \frac{\tau}{\tau \cap \widetilde{\tau}}.$$

Theorem 2.27. Let E be an irreducible component of the formal versal scheme T of deformations of A_s. If the deformation of A_s over the generic point e of E is smooth—that is, if the image of the formal non-smoothness subscheme (defined by a Jacobian ideal: VI 5), which is finite over T, does not contain e—then

$$3\delta(s) = m_0^E(s) + m_1(s).$$

The validity of 2.27 depends only on the completion of A_s. Replacing (C, s) by a pair (C', s) formally isomorphic to it at s, we may suppose that the following conditions hold:

(a) s is the only singular point of C;

(b) C has no infinitesimal automorphisms.

Condition (b) ensures that the deformation functor of C is pro-representable. It is even effectively pro-representable, since there is no obstruction to lifting a prescribed ample invertible sheaf on C. Let $f : X_0 \longrightarrow T_0$ be the universal deformation of C, and let t_0 be the closed point of T_0, so that $C \xrightarrow{\sim} X_{0t_0}$.

Lemma 2.28. There exist a scheme T_1 of finite type over k, a point $t_1 \in T_1(k)$, and a proper flat morphism $f : X_1 \longrightarrow T_1$, whose fiber at t_1 is C, such that for every $u \in T_1(k)$ the completion of T_1 at u is a formal moduli scheme for the fiber X_{1u}.

Here is a direct proof of this corollary of general results of Artin. Let $\mathcal{L}$ be a very ample invertible sheaf on C such that $H^1(C, \mathcal{L}) = 0$, and put $\mathbb{P} = \mathbb{P}(H^0(C, \mathcal{L}))$. We have an embedding $v : C \hookrightarrow \mathbb{P}$. Let T_2 be the Hilbert scheme parametrizing the subvarieties of $\mathbb{P}$ having the same Hilbert polynomial as $v(C) \subset \mathbb{P}$, and let $t_2 \in T_2$ correspond to $v(C)$. The completion $\widehat{T}_2$ of T_2 at t_2 is smooth over T_0. The vertical tangent bundle of $\widehat{T}_2/T_0$ is induced by the subbundle L of the tangent bundle of T_2 which is the image of $\mathrm{Lie}(\underline{\mathrm{Aut}}(\mathbb{P}))$ under the action of $\underline{\mathrm{Aut}}(\mathbb{P})$ on T_2.

In a neighborhood of t_2, let $f_1, \ldots, f_n$, with $n = \dim \underline{\mathrm{Aut}}(\mathbb{P})$, be equations defining a subscheme T_1' of T_2 which passes through t_2 and whose completion at t_2 is étale over T_0. Take for T_1 a neighborhood of t_2 in T_1' on which the tangent bundle is transverse to L.

2.29. The morphism of functors

$$(\text{deformations of } C) \longrightarrow (\text{deformations of } A_s)$$

corresponds to a morphism $p : T_0 \longrightarrow T$. By VI 2.10, p is smooth. Let t be the closed point of T. The Zariski tangent space of $p^{-1}(t)$ at t_0 classifies the locally trivial deformations of C over $k[\varepsilon]$, where $\varepsilon^2 = 0$. Putting $E_0 = p^{-1}(E)$, we obtain

$$\dim(E_0) = \dim(E) + \dim H^1(C, T^1_{C/k}). \tag{2.29.1}$$

Since X_0/T_0 is flat, all its fibers have the same arithmetic genus p, where $1 - p = \chi(\underline{O})$. By hypothesis, the fiber at the generic point of E_0 is smooth. The formal moduli scheme of a smooth proper curve being smooth of dimension $3p - 3$, Lemma 2.28 gives

$$\dim(E_0) = 3p - 3. \tag{2.29.2}$$

Let $\widetilde{C}$ be the normalization of C, of arithmetic genus $\widetilde{p} = p - \delta(s)$. We have

$$\begin{aligned} \dim H^1(C, T^1_{C/k}) &= -\chi(C, T^1_{C/k}) \\ &= -\chi(\widetilde{C}, T^1_{\widetilde{C}/k}) + m_1(s) \\ &= -[3(p - \delta(s)) - 3] + m_1(s). \end{aligned} \tag{2.29.3}$$

Formulas (2.29.1), (2.29.2), and (2.29.3) prove 2.27.

Conjecture 2.30. Every reduced proper curve has a smooth deformation.

This conjecture implies, by 2.27 and 2.28, that the formal versal scheme of deformations of A_s is always equidimensional, of dimension $3\delta(s) - m_1(s)$.

2.31. Let C be a reduced proper curve over k, of arithmetic genus p. We shall write

M_0 for the dimension of the formal versal scheme T of deformations of C;

M_1 for the dimension of the Lie algebra of $\underline{\mathrm{Aut}}(C)$.

For an irreducible component E of T, write M_0^E for the dimension of E. The global analogue of 2.27 is the following.

Proposition 2.32. If the deformation of C over the generic point of E is smooth, then

$$3p - 3 = M_0^E - M_1.$$

Let $X \subset C(k)$ be a finite set of smooth points of C such that every vector field on C which vanishes at every point $x \in X$ is zero. The deformation functor of (C, X) is pro-representable and satisfies a variant of 2.28. The formal moduli scheme of (C, X) is smooth over T, of relative dimension $\#X - M_1$, and the analogue of 2.28 gives

$$M_0^E + (\#X - M_1) = 3p - 3 + \#X,$$

which proves 2.32.

Suppose now that k has characteristic zero. Then $M_1 = \dim \underline{\mathrm{Aut}}(C)$. Here is an analogous interpretation of $m_1(s)$.

Lemma 2.33. If k has characteristic zero, every derivation of A_s extends to a derivation of $\widetilde{A}_s$.

If D is a derivation of A_s, then

$$\exp(Dt) = \sum_{n \geq 0} \frac{D^n t^n}{n!}$$

is an automorphism of the completion at $(s, 0)$ of $A_s \otimes_k k[t]$. By transport of structure, this automorphism acts on the normalization $(\widetilde{A}_s \otimes_k k[t])^\wedge$ of that ring. Reducing modulo t^2, we obtain an automorphism $1 + Dt$ of $\widetilde{A}_s^\wedge \otimes_k k[t]/(t^2)$, and D thereby extends to a derivation of $\widetilde{A}_s^\wedge$. Since it also extends to the fraction field of A_s (to the total quotient ring when A_s is not integral), it extends to $\widetilde{A}_s$.

Suppose that s is singular. A similar argument then shows that the vector field on $\mathrm{Spec}(\widetilde{A}_s)$ defined by D vanishes at every point above s.

Let c be an ideal of $\widetilde{A}_s$ contained in the conductor of $\widetilde{A}_s/A_s$. Put

$$\widetilde{G}_c = \underline{\mathrm{Aut}}(\widetilde{A}_s/c),$$

and let G_c be the subgroup of $\widetilde{G}_c$ which stabilizes the subalgebra A_s/c. If $d \subset c$, then $\widetilde{G}_d$ stabilizes c/d and maps onto $\widetilde{G}_c$. Notice that $g \in \widetilde{G}_c$ is determined by $g(t)$ for a uniformizer t, and that $g(t)$ may be any uniformizer. Consequently the pointed homogeneous space $\widetilde{G}_c/G_c$ is independent of c; denote it by $\widetilde{G}/G$.

Let $\widetilde{\tau}_0$ be the $\widetilde{A}_s$-module of vector fields on $\mathrm{Spec}(\widetilde{A}_s)$ which vanish at the points above s. When s is singular, one has

$$\dim_k(\widetilde{\tau}/\widetilde{\tau}_0) = \text{the number of branches},$$

and

$$\widetilde{\tau}_0/\tau \xrightarrow{\sim} \text{the tangent space at the origin of } \widetilde{G}/G.$$

Proposition 2.34. In characteristic zero, when s is singular, the integer $m_1(s)$ is the sum of the number of branches and the dimension of $\widetilde{G}/G$.

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SGA 7

EXPOSÉ XI

COHOMOLOGY OF COMPLETE INTERSECTIONS

by P. DELIGNE

1. Cohomology of projective space and complete intersections

Let X be a nonsingular complex projective variety, purely of dimension n, and let $\underline{O}(1)$ be an ample invertible sheaf on X. By Akizuki and Nakano [1], Kodaira's vanishing theorem [4] implies that

$$(*) \qquad H^i(X, \Omega^j_X(-1)) = 0 \qquad \text{for } i+j<n.$$

Again by [1], this result implies the "weak Lefschetz theorem" in complex cohomology. We propose to prove, by brute force, that if X is a projective space over a field of arbitrary characteristic, then (*) remains true. This will allow us, following [1], to compare the Hodge and de Rham cohomology of complete intersections over a field of arbitrary characteristic with that of projective space.

We shall then spell out some consequences of the results obtained. In §2 we give qualitative results deduced from Hirzebruch's numerical formula (2.2). I wish to thank N. Katz for his help in the preparation of that section.

1.0. Let $\underline{E}$ be a locally free module of rank $r+1$ on a scheme S. We assume that $r>0$. Recall that $\mathbb{P}(\underline{E})$ denotes the relative projective space over S of hyperplanes of $\underline{E}$. Let $p:\mathbb{P}(\underline{E})\to S$ be the projection, and let

$$\eta \in H^0\big(S, R^1p_*\Omega^1_{\mathbb{P}(\underline{E})/S}\big)$$

be the first Chern class, in the Hodge sense, of the invertible sheaf $\underline{O}(1)$.

Recall that η is the image of the class of $\underline{O}(1)$ in $H^0(S, R^1p_*\underline{O}^*)$ under the logarithmic differential

$$df/f : \underline{O}^* \longrightarrow \Omega^1_{\mathbb{P}(\underline{E})/S}.$$

For every sheaf $\underline{F}$ of $\underline{O}$-modules on $\mathbb{P}(\underline{E})$, put

$$\underline{F}(n) = \underline{F}\otimes \underline{O}(1)^{\otimes n} \qquad (n\in\mathbb{Z}).$$

Theorem 1.1. (i) The sheaves $R^ip_*(\Omega^j_{\mathbb{P}(\underline{E})/S}(n))$ are locally free, and their formation is compatible with arbitrary base change.

(ii) For $0\le i\le r$, the section η^i of $R^ip_*(\Omega^i_{\mathbb{P}(\underline{E})/S})$ defines an isomorphism from $\underline{O}_S$ onto this sheaf. Moreover,

$$R^ip_*\Omega^j_{\mathbb{P}(\underline{E})/S}=0 \qquad \text{if } i\neq j \text{ or } i>r.$$

(iii) If $n\neq 0$, then, outside the following two cases, $R^ip_*(\Omega^j_{\mathbb{P}(\underline{E})/S}(n))$ vanishes:

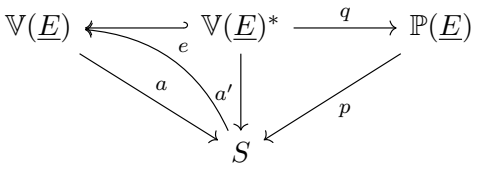
$$(\text{iii}_a) \qquad i=0 \quad \text{and} \quad n>j,$$

$$(\text{iii}_b) \qquad i=r \quad \text{and} \quad n<j-r.$$

Let

$$\mathbb{V}(\underline{E}) = \mathrm{Spec}\big(\mathrm{Sym}^*\,\underline{E}\big)$$

be the affine space defined by $\underline{E}$, let e be its zero section, and put $\mathbb{V}(\underline{E})^* = \mathbb{V}(\underline{E}) \setminus e(S)$. Let a and a' be the projections of $\mathbb{V}(\underline{E})$ and $\mathbb{V}(\underline{E})^*$ onto S, and let q be the projection of $\mathbb{V}(\underline{E})^*$ onto $\mathbb{P}(\underline{E})$:



The punctured affine space $\mathbb{V}(\underline{O}(1))^*$ over $\mathbb{P}(\underline{E})$ is canonically isomorphic to $\mathbb{V}(\underline{E})^*$. Consequently, for every quasi-coherent sheaf $\underline{F}$ of $\underline{O}$-modules on $\mathbb{P}(\underline{E})$, there is a canonical isomorphism

$$q_*q^*\underline{F} \simeq \bigoplus_{n\in\mathbb{Z}} \underline{F}(n). \tag{1.1.1}$$

This grading of $q_*q^*\underline{F}$ can also be obtained from the action on this sheaf of the group $\mathbb{G}_m$ of homotheties of $\mathbb{V}(\underline{E})$.

Relative vector fields on $\mathbb{V}(\underline{E})$ are identified with S-morphisms from $\mathbb{V}(\underline{E})$ to itself, the target being identified with the tangent space at the origin. The vector field X defined by the identity map of $\mathbb{V}(\underline{E})$ is invariant under homotheties. It is tangent to the fibers of q and defines a trivialization

$$X : \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \longrightarrow \underline{O}_{\mathbb{V}(\underline{E})^*}. \tag{1.1.2}$$

If the x_i form a basis of $\underline{E}$, then

$$X = \sum_i x_i\, dx^i.$$

Since $\Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})}$ has rank one, the exact sequence

$$0 \longrightarrow q^*\Omega^1_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \longrightarrow 0$$

induces exact sequences

$$0 \longrightarrow q^*\Omega^i_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^i_{\mathbb{V}(\underline{E})^*/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \otimes q^*\Omega^{i-1}_{\mathbb{P}(\underline{E})/S} \longrightarrow 0.$$

Using (1.1.2), these become

$$0 \longrightarrow q^*\Omega^i_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^i_{\mathbb{V}(\underline{E})^*/S} \xrightarrow{\iota_X} q^*\Omega^{i-1}_{\mathbb{P}(\underline{E})/S} \longrightarrow 0. \tag{1.1.3}$$

Canonically,

$$\Omega^i_{\mathbb{V}(\underline{E})/S} = a^* \bigwedge^i \underline{E}. \tag{1.1.4}$$

The short exact sequences (1.1.3) are therefore obtained by splitting the following long exact sequence on $\mathbb{V}(\underline{E})^*$:

$$(1.1.5)\qquad \cdots \longrightarrow a'^{*}\bigwedge^{i}\underline{E} \xrightarrow{\iota_X} a'^{*}\bigwedge^{i-1}\underline{E} \xrightarrow{\iota_X} a'^{*}\bigwedge^{i-2}\underline{E} \longrightarrow \cdots .$$

This long exact sequence is simply the restriction to $\mathbb{V}(\underline{E})^*$ of the Koszul resolution of $\underline{O}_{e(S)}$ on $\mathbb{V}(\underline{E})$.

Push the long exact sequence (1.1.5) forward to $\mathbb{P}(\underline{E})$ by the affine morphism q. By (1.1.1) and the formula $q_*a'^{*} = q_*q^*p^*$, one obtains a long exact sequence

$$(1.1.6)\qquad \cdots \longrightarrow \bigoplus_{n\in\mathbb{Z}} p^*\bigwedge^{i}\underline{E}(n) \xrightarrow{\iota_X} \bigoplus_{n\in\mathbb{Z}} p^*\bigwedge^{i-1}\underline{E}(n) \longrightarrow \cdots .$$

It splits into short exact sequences, the images of (1.1.3),

$$(1.1.7)\qquad 0 \longrightarrow \bigoplus_{n\in\mathbb{Z}} \Omega^{i}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow \bigoplus_{n\in\mathbb{Z}} p^*\bigwedge^{i}\underline{E}(n) \xrightarrow{\iota_X} \bigoplus_{n\in\mathbb{Z}} \Omega^{i-1}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow 0.$$

The exact sequences (1.1.7) are direct sums of the exact sequences

$$(1.1.8)\qquad 0 \longrightarrow \Omega^{i}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow p^*\bigwedge^{i}\underline{E}(n-i) \xrightarrow{\iota_X} \Omega^{i-1}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow 0.$$

Recall (EGA III, 2.1.15, or FAC) that the sheaves $R^ip_*\underline{O}(k)$ are locally free and their formation is compatible with arbitrary base change, and that

$$(1.1.9)\qquad R^ip_*\underline{O}(k) = 0 \quad \text{if } i \neq 0, r, \quad \text{or if } i = 0, k < 0, \quad \text{or if } i = r, k > -r-1,$$

and hence that $R^ip_*p^*\bigwedge^j \underline{E}(k)$ vanishes under the same conditions.

The exact sequences (1.1.8) give a resolution

$$(1.1.10)\qquad 0 \longrightarrow \Omega^{i}_{\mathbb{P}(\underline{E})/S}(n) \xrightarrow{E} p^*\bigwedge^{i}\underline{E}(n-i) \xrightarrow{\iota_X} \cdots \longrightarrow p^*\bigwedge^{0}\underline{E}(n) \longrightarrow 0$$

of $\Omega^{i}_{\mathbb{P}(\underline{E})/S}(n)$. The spectral sequence defined by this resolution degenerates ($E_1 = E_\infty$) for degree reasons by (1.1.9); for $0 < j < r$, it gives

$$(1.1.11)\qquad R^jp_*\Omega^{i}_{\mathbb{P}(\underline{E})/S}(n) = \mathcal{H}\left(\cdots \xrightarrow{\iota_X} p_*\bigwedge^{i-j}\underline{E}(n-i+j) \xrightarrow{\iota_X} \cdots\right).$$

Since $r \geq 1$, a depth argument shows that

$$(1.1.12)\qquad \bigoplus_{n\in\mathbb{Z}} p_*p^*\bigwedge^{i}\underline{E}(n) = p_*q_*q^*p^*\bigwedge^{i}\underline{E} = a'_*a'^{*}\bigwedge^{i}\underline{E} = a_*a^*\bigwedge^{i}\underline{E}.$$

Thus $p_*p^*\bigwedge^i \underline{E}(n)$ is the homogeneous part of weight $n+i$ in $a_*a^*\bigwedge^i \underline{E}$, namely $\mathrm{Sym}^n(\underline{E}) \otimes \bigwedge^i \underline{E}$. Remembering that (1.1.5) comes from the Koszul resolution, (1.1.11) shows that, for $0 <$

$j < r$, the sheaf $R^j p_* \Omega^i_{\mathbb{P}(\underline{E})/S}(n)$ can be nonzero only when $i - j = 0$ and $n - i + j = 0$, that is, only when $n = 0$ and $i = j$. In that case, moreover, $R^i p_* \Omega^i_{\mathbb{P}(\underline{E})/S}$ is isomorphic to $\underline{O}_S$.

For $j = 0$, (1.1.10) also gives

$$(1.1.13) \qquad p_* \Omega^i_{\mathbb{P}(\underline{E})/S}(n) = \ker\left(\iota_X : p_* \bigwedge^i \underline{E}(n-i) \longrightarrow p_* \bigwedge^{i-1} \underline{E}(n-i+1)\right),$$

and the same argument shows that ι_X is injective when $n - i \leq 0$.

By (1.1.8), with $n = 0$ and $i = r + 1$, one has

$$\Omega^r_{\mathbb{P}(\underline{E})/S} \simeq p^* \bigwedge^{r+1} \underline{E}(-r-1).$$

When S is the spectrum of a field, the vanishing assertions in 1.1(ii) and (iii) now follow from the preceding argument for $j \neq r$, and follow from it by duality ([2], V, 11.2, p. 211) for $j = r$. In particular, for any j and n there is at most one i for which $R^i p_* \Omega^j_{\mathbb{P}(\underline{E})/S}(n) \neq 0$. This proves (i), since $\Omega^j_{\mathbb{P}(\underline{E})/S}$ is flat over S (the hardest part is locating the reference in EGA III; we suggest 6.10.5, 7.4.1, and 7.5.5, followed by a passage to the limit).

Knowing that $R^i p_* \Omega^i_{\mathbb{P}(\underline{E})/S}$ is an invertible sheaf for $0 \leq i \leq r$, it remains only to prove that, when S is the spectrum of a field, $\eta^i \neq 0$. Indeed, η^r is the cohomology class of a rational point (the intersection of r hyperplanes), hence a basis of $H^r(\mathbb{P}^r, \Omega^r)$ (FGA, Exposé 149, no. 4).

1.2. The Euler–Poincaré characteristics of the $\Omega^i(n)$ may be calculated directly from (1.1.10) or (1.1.13). The rather unappetizing result is

$$\chi\big(\mathbb{P}^r_k, \Omega^j_{\mathbb{P}^r_k}(n)\big) = \sum_{0 \leq i \leq j} (-1)^{i-j} \binom{r+1}{i} \binom{r+n-i}{r}.$$

Proposition 1.3. Let $f : X \to S$ be a proper smooth morphism, let $\underline{W}$ be a locally free sheaf of rank c on X, let $\underline{F}$ be a coherent sheaf on X, let $d > 0$ be an integer, and let $s : \underline{W} \to \underline{O}_X$ be a section of $\underline{W}^\vee$. Assume that S is noetherian and that:

(a) The closed subscheme H of X defined by the vanishing of s is smooth over S.

(b) Locally on X, if the s_i are the coordinates of s in a basis of $\underline{W}^\vee$, the s_i form both an $\underline{O}_X$-regular sequence and an $\underline{F}$-regular sequence.

(c) For every sequence of integers k_ℓ, not all zero, one has

$$R^i f_* \left(\bigotimes_\ell \bigwedge^{k_\ell} \underline{W} \otimes \Omega^j_{X/S} \otimes \underline{F} \right) = 0 \qquad \text{for } i + j < d.$$

Then:

(i) $$R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F} \right) \xrightarrow{\sim} R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F} \right) \qquad \text{for } i + j < d - c,$$

(ii) $$R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F} \right) \hookrightarrow R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F} \right) \qquad \text{for } i + j = d - c.$$

The conormal sheaf $N_{H/X}$ of H in X is isomorphic to the restriction of $\underline{W}$ to H, so the exact sequence

$$(1.3.1) \qquad 0 \longrightarrow N_{H/X} \longrightarrow \Omega^1_{X/S} \otimes \underline{O}_H \longrightarrow \Omega^1_{H/S} \longrightarrow 0$$

defines a decreasing filtration on $\Omega^j_{X/S} \otimes \underline{O}_H$ whose successive quotients are

$$\mathrm{Gr}^k\big(\Omega^j_{X/S} \otimes \underline{O}_H\big) \simeq \bigwedge^k \underline{W} \otimes \Omega^{j-k}_{H/S}. \tag{1.3.2}$$

It follows from (b) that one also has

$$\mathrm{Gr}^k\big(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\big) = \bigwedge^k \underline{W} \otimes \Omega^{j-k}_{H/S} \otimes \underline{F}. \tag{1.3.3}$$

We prove (i) and (ii) by induction on j, simultaneously for all $\underline{F}$ satisfying (b) and (c). The assertions are empty for $j < 0$.

Suppose, then, that (i) and (ii) have been proved for $j' < j$. The induction hypothesis applies to the sheaves $\bigwedge^k \underline{W} \otimes \underline{F}$ and, by (c), gives

$$R^i f_*\left(\bigwedge^k \underline{W} \otimes \Omega^{j-k}_{H/S} \otimes \underline{F}\right) = 0 \quad \text{for } k > 0 \text{ and } i + j < d - c + k. \tag{1.3.4}$$

The filtration of $\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}$ considered above defines a spectral sequence

$$E_1^{p,i-p} = R^i f_*\left(\bigwedge^p \underline{W} \otimes \Omega^{j-p}_{H/S} \otimes \underline{F}\right) \Longrightarrow R^i f_*\left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right).$$

By (1.3.4), $E_1^{p,i-p} = 0$ when $p > 0$ and $i + j \leq d - c$; hence

$$\begin{cases} R^i f_*\left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right) \xrightarrow{\sim} R^i f_*\left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j < d - c, \\ R^i f_*\left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right) \hookrightarrow R^i f_*\left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j = d - c. \end{cases} \tag{1.3.5}$$

By hypothesis (b), with a suitable differential, the sheaves $\bigwedge^k \underline{W} \otimes \Omega^j_{X/S} \otimes \underline{F}$ form a left Koszul resolution of $\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}$. This filtration defines a spectral sequence

$$E_1^{-p,q} = R^q f_*\left(\bigwedge^p \underline{W} \otimes \Omega^j_{X/S} \otimes \underline{F}\right) \Longrightarrow R^{q-p} f_*\left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right). \tag{1.3.6}$$

By hypothesis (c), $E_1^{-p,q} = 0$ for $p > 0$ and $q + j < d$, and trivially for $p \notin [0, c]$. Consequently,

$$\begin{cases} R^i f_*\left(\Omega^j_{X/S} \otimes \underline{F}\right) \xrightarrow{\sim} R^i f_*\left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j < d - c, \\ R^i f_*\left(\Omega^j_{X/S} \otimes \underline{F}\right) \hookrightarrow R^i f_*\left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j = d - c. \end{cases} \tag{1.3.7}$$

Assertions (i) and (ii) now follow from (1.3.5) and (1.3.7).

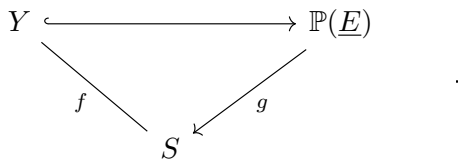
1.4. Let $\underline{E}$ be a locally free sheaf of rank $r + 1$ on a scheme S, let $\mathbb{P}(\underline{E})$ be the corresponding projective bundle, let $d \leq r$ be an integer, and let $\underline{a} = (a_1, \ldots, a_d)$ be a sequence of integers. A relative complete intersection of multidegree $\underline{a}$ in $\mathbb{P}(\underline{E})$ is a closed subscheme which has codimension d in every fiber and which, locally on S for the étale topology, is the intersection of d hypersurfaces of respective degrees $a_1, \ldots, a_d$.

By 1.1, hypotheses (b) and (c) of 1.3 hold for $X/S = \mathbb{P}(\underline{E})/S$, $\underline{F} = \underline{O}_X$,

$$\underline{W} = \bigoplus_i \underline{O}(-a_i),$$

and a section s of $\underline{W}^\vee$ defining a relative complete intersection $Y \subset \mathbb{P}(\underline{E})$ of multidegree $\underline{a}$.

Theorem 1.5. Under the hypotheses of 1.4, let $n = r - d$ be the relative dimension of Y, and let $\eta \in H^0(S, R^1 f_* \Omega^1_{Y/S})$ be the first Chern class of $\underline{O}(1)$ in the diagram



If Y is smooth over S, then:

(i) The sheaves $R^i f_* \Omega^j_{Y/S}$ are locally free, and their formation is compatible with arbitrary base change.

(ii) The spectral sequence

$$E_1^{p,q} = R^q f_* \Omega^p_{Y/S} \Longrightarrow R^{p+q} f_* (\Omega^*_{Y/S}),$$

relating Hodge and de Rham cohomology degenerates ($E_1 = E_\infty$); its abutment is locally free, and its formation is compatible with arbitrary base change.

(iii)$_a$ One has

$$R^i f_* \Omega^j_{Y/S} = 0 \qquad \text{if } i \neq j \text{ and } i + j \neq n.$$

(iii)$_b$ η^i is a basis of $R^i f_* \Omega^i_{Y/S}$ if $2i < n$.

(iii)$_c$ If $n < 2i \leq 2n$, and if α_i is the basis of $R^i f_* \Omega^i_{Y/S}$ dual to the basis η^{n-i} of $R^{n-i} f_* \Omega^{n-i}_{Y/S}$, then

$$\eta^i = \left(\prod_{h=1}^{d} a_h \right) \alpha_i.$$

(iii)$_d$ If $n = 2m$, then $\eta^m \neq 0$ (provided $S \neq \varnothing$), and the quotient $R^m f_* \Omega^m_{Y/S} / \underline{O}_S \eta^m$ is locally free.

It is enough to prove 1.5 in a "universal" case, which permits us to assume that S is smooth over $\mathrm{Spec}(\mathbb{Z})$. In this case (ii) follows from (i), since it holds over $\mathbb{C}$, and therefore over every field of characteristic zero, by Hodge theory. Moreover, since S is reduced, (i) follows from (iii), assumed proved when S is the spectrum of a field. Indeed, by (iii), for each j the functions

$$s \longmapsto \dim_{k(s)} H^i(Y_s, \Omega^j_{Y_s})$$

are constant on S, except perhaps when $i = j$; hence they are all constant, because their alternating sum is constant (EGA III, 7.9.2). The conclusion then follows from EGA III, 6.10.5, 7.6.9, and 7.7.5.

By 1.4 and 1.3, one has

$$\begin{cases} R^i g_* \Omega^j_{\mathbb{P}(\underline{E})/S} \xrightarrow{\sim} R^i f_* \Omega^j_{Y/S} & \text{for } i + j < n, \\ R^i g_* \Omega^j_{\mathbb{P}(\underline{E})/S} \hookrightarrow R^i f_* \Omega^j_{Y/S} & \text{for } i + j = n. \end{cases} \tag{1.5.1}$$

When S is the spectrum of a field, assertions (iii)$_a$, (iii)$_b$, and (iii)$_d$ follow from (1.5.1) and Serre duality ([2], V, 11.2, p. 211). The cohomology class

η^n is the class of a zero-cycle of degree $\prod_{h=1}^d a_h$ (the degree of the complete intersection), so

$$\eta^n = \left(\prod_{h=1}^{d} a_h\right)\alpha_n.$$

By definition, $\eta^{n-i}\alpha_i = \alpha_n$, and $(\text{iii})_c$ follows.

Thus (i) and (ii) hold; in the universal case, the assertions in (iii) follow from (i) and from (iii) in the case already proved, where S is the spectrum of a field.

Theorem 1.6. Keep the hypotheses and notation of 1.4 and 1.5, assume that Y is smooth, let ℓ be a prime invertible in all residue fields of S, and let $\eta_\ell \in H^0(S, R^2 f_* \mathbb{Z}_\ell(1))$ denote the ℓ-adic first Chern class of $\underline{O}(1)$ (SGA 5, VII). Then:

(i) $R^i f_* \mathbb{Z}_\ell = 0$ if i is odd and $i \neq n$.

(ii) If $2i < n$, then $R^{2i} f_* \mathbb{Z}_\ell(i)$ is isomorphic to $\mathbb{Z}_\ell$ and has η_ℓ^i as a basis.

(iii) If $n < 2i \leq 2n$, then $R^{2i} f_* \mathbb{Z}_\ell(i)$ is isomorphic to $\mathbb{Z}_\ell$ and has

$$\eta_\ell^i \Big/ \prod_{h=1}^{d} a_h$$

as a basis.

(iv) The $\mathbb{Z}_\ell$-sheaf $R^n f_* \mathbb{Z}_\ell$ is constructible, twisted constant, and torsion-free. If n is even, then $\eta_\ell^{n/2} \neq 0$ (provided $S \neq \varnothing$), and $R^n f_* \mathbb{Z}_\ell(n/2)/\mathbb{Z}_\ell \eta_\ell^{n/2}$ is torsion-free.

The $\mathbb{Z}_\ell$-sheaves $R^i f_* \mathbb{Z}_\ell$ are constructible and twisted constant, and their formation is compatible with arbitrary base change, by SGA 4, XVI, 2.2. It therefore suffices to prove 1.6 when S is the spectrum of an algebraically closed field. Repeated application of SGA 4, XIV, 3.3 then shows, for every m, that the canonical map

$$H_Y^q\big(\mathbb{P}(\underline{E}), \mathbb{Z}/\ell^m\big) \longrightarrow H^q(Y, \mathbb{Z}/\ell^m)$$

is bijective for $q > r + d$ and surjective for $q = r + d$. From this and Poincaré duality (SGA 4, XVIII), one deduces (SGA 5, VII) that the restriction map

$$H^q\big(\mathbb{P}(\underline{E}), \mathbb{Z}_\ell\big) \longrightarrow H^q(Y, \mathbb{Z}_\ell)$$

is bijective for $q < n$, and injective with torsion-free cokernel for $q = n$. This proves (i) for $i < n$, as well as (ii) and (iv). As in 1.5, one checks that η_ℓ^n is $\prod_{h=1}^{d} a_h$ times the canonical class; (i) and (iii) then follow by Poincaré duality (SGA 4, XVIII).

1.7. If Y is a complete intersection of dimension at least three in a projective space $\mathbb{P}^r(k)$ over a field k, then $\mathrm{Pic}(Y)$ is known to be isomorphic to $\mathbb{Z}$ and to be generated by $\underline{O}(1)$ (SGA 2, XII, 3.7). In dimension two, when Y is smooth, one can still prove the following.

Theorem 1.8. Let Y be a smooth complete-intersection surface in the projective space $\mathbb{P}^r(k)$ over an algebraically closed field k. Then $\underline{\mathrm{Pic}}^\tau(Y) = 0$, and the class of $\underline{O}(1)$ is not divisible in the Néron–Severi group $\mathrm{NS}(Y)$.

Since $H^1(Y, \underline{O}) = 0$ by $1.5(\text{iii})_a$, one knows that $\underline{\mathrm{Pic}}^0(Y) = 0$. The vanishing of $\underline{\mathrm{Pic}}^\tau(Y)$ then follows from the familiar lemma below, the fact that $H^2(Y, \mathbb{Z}_\ell(1))$ is torsion-free by 1.6(iv), and the fact that $H^0(Y, \Omega_Y^1) = 0$ by $1.5(\text{iii})_a$.

Lemma 1.9. Let Y be a complete algebraic variety over an algebraically closed field k.

(i) If ℓ is prime to the characteristic of k, then the ℓ-torsion of $\mathrm{NS}(Y)$ identifies with the ℓ-torsion of $H^2(Y, \mathbb{Z}_\ell(1))$.

(ii) If k has nonzero characteristic p and Y is smooth, then the kernel of multiplication by p in $\mathrm{Pic}(Y)$ identifies with the subgroup of $H^0(Y, \Omega^1_Y)$ consisting of locally logarithmic differentials.

The Kummer exact sequences

$$0 \longrightarrow \mu_{\ell^n} \longrightarrow \mathbb{G}_m \xrightarrow{x \mapsto x^{\ell^n}} \mathbb{G}_m \longrightarrow 0$$

give exact sequences

$$0 \longrightarrow \mathrm{Pic}(Y)/\ell^n \,\mathrm{Pic}(Y) \longrightarrow H^2(Y, \mu_{\ell^n}) \longrightarrow H^2(Y, \mathbb{G}_m)_{\ell^n} \longrightarrow 0,$$

where

$$\mathrm{Pic}(Y)/\ell^n \,\mathrm{Pic}(Y) = \mathrm{NS}(Y) \otimes \mathbb{Z}/\ell^n.$$

Passing to the inverse limit gives

$$0 \longrightarrow \mathrm{NS}(Y) \otimes \mathbb{Z}_\ell \longrightarrow H^2(Y, \mathbb{Z}_\ell(1)) \longrightarrow \mathrm{Hom}\big(\mathbb{Q}_\ell/\mathbb{Z}_\ell, H^2(Y, \mathbb{G}_m)\big) \longrightarrow 0.$$

The cokernel is torsion-free because $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is divisible. This proves (i).

To prove (ii), it is enough to observe that the exact sequence

$$0 \longrightarrow \underline{O}^*_Y \xrightarrow{x \mapsto x^p} \underline{O}^*_Y \xrightarrow{df/f} \Omega^1_Y$$

defines a short exact sequence

$$0 \longrightarrow \underline{O}^*_Y \longrightarrow \underline{O}^*_Y \longrightarrow \Omega^1_{Y,\mathrm{loc\ log}} \longrightarrow 0,$$

and then to take the corresponding long exact cohomology sequence. This argument is due to Cartier, who proves moreover that, if C denotes the Cartier operator on $H^0(Y, \Omega^1_Y)$, then the locally logarithmic differentials form the kernel of $C - \mathrm{Id}$.

Suppose, for a contradiction, that the class of $\underline{O}(1)$ in $\mathrm{NS}(Y)$ has the form nu, with $n > 1$. If ℓ divides n and is prime to the characteristic, then in ℓ-adic cohomology one again has

$$\eta_\ell = n\, c_1(u),$$

contradicting 1.6(iv).

Finally, if the characteristic p of k is nonzero and p divides n, then in Hodge cohomology one again has

$$\eta = n\, c_1(u) = 0,$$

contradicting $1.5(\mathrm{iii})_d$ and completing the proof.

2. Numerical results

2.1. Let $\underline{a} = (a_i)_{1 \le i \le d}$ be a sequence of integers ≥ 1, and let $V_n(\underline{a})$ denote a smooth complete intersection of dimension n and multidegree $\underline{a}$ in a projective space $\mathbb{P}^{n+d}$ over a field k. Put

$$h^{p,q}(\underline{a}) = \dim_k H^q\big(V_{p+q}(\underline{a}), \Omega^p\big)$$

and

$$h^{p,q}_0(\underline{a}) = h^{p,q}(\underline{a}) - \delta_{p,q}.$$

Thus the $h^{p,q}_0(\underline{a})$ are the Hodge numbers of the middle-dimensional primitive cohomology of a complete intersection of multidegree $\underline{a}$. They depend only on $\underline{a}$, p, and q, by 1.5. Set

(2.1.1) $$H(\underline{a}) = \sum_{p,q\geq 0} h_0^{p,q}(\underline{a}) y^p z^q \in \mathbb{Z}[[y,z]].$$

For $d = 1$ and $\underline{a} = (a)$, put $h^{p,q}(a) = h^{p,q}(\underline{a})$ and $H(a) = H(\underline{a})$. For $d = 0$, put $\sup(a_i) = 1$. The following formula is the special case $k = 0$ of formula (2), p. 160, in Hirzebruch [3]:

(2.2) $$\sum_{n=0}^{\infty} \sum_{p} \chi\big(V_n(\underline{a}), \Omega^p\big) y^p z^{n+d} = \frac{1}{(1+yz)(1-z)} \prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + y(1-z)^{a_i}}.$$

We shall transform this into the following formula.
Theorem 2.3 (Hirzebruch). With the notation of 2.1, one has

$$H(\underline{a}) = \frac{1}{(1+y)(1+z)} \left[\prod_{i=1}^{d} \frac{(1+y)^{a_i} - (1+z)^{a_i}}{-(1+y)^{a_i} z + (1+z)^{a_i} y} - 1 \right].$$

By 1.5,

$$\chi\big(V_n(\underline{a}), \Omega^p\big) = (-1)^p + (-1)^{n-p} h_0^{p,n-p}(\underline{a}).$$

Consequently, the left-hand side I of (2.2) can be rewritten as

$$\begin{aligned} I &= \sum_{n=0}^{\infty} \sum_{p+q=n} \left((-1)^p + (-1)^q h_0^{p,q}(\underline{a}) \right) y^p z^{p+q+d} \\ &= \sum_{p,q\geq 0} (-1)^p (yz)^p z^q z^d + \sum_{p,q\geq 0} (-1)^q h_0^{p,q}(\underline{a}) (yz)^p z^q z^d \\ &= z^d \left[\frac{1}{(1+yz)(1-z)} + H(\underline{a})(yz, -z) \right]. \end{aligned}$$

The right-hand side II of (2.2) can be rewritten as

$$II = \frac{z^d}{(1+yz)(1-z)} \prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + yz(1-z)^{a_i}}.$$

After cancelling z^d in $I = II$, one obtains

$$H(\underline{a})(yz, -z) = \frac{1}{(1+yz)(1-z)} \left[\prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + (1-z)^{a_i} yz} - 1 \right],$$

and 2.3 follows by a change of variables.
Corollary 2.4. With the notation of 2.1:

(i) For $d = 1$, one has

$$\begin{aligned} H(a) &= -\frac{(1+y)^{a-1} - (1+z)^{a-1}}{(1+y)^a z - (1+z)^a y} \\ &= \frac{\displaystyle\sum_{i,j\geq 0} \binom{a-1}{i+j+1} y^i z^j}{\displaystyle 1 - \sum_{i,j\geq 1} \binom{a}{i+j} y^i z^j}. \end{aligned}$$

(ii) Writing $|P|$ for the number of elements of a subset P of $[1,d]$, one has

$$H(\underline{a}) = \sum_{\substack{P\subset[1,d]\\ P\neq\varnothing}} (1+y)^{|P|-1}(1+z)^{|P|-1}\prod_{i\in P} H(a_i).$$

By 2.3,

$$\begin{aligned} H(a) &= -\frac{1}{(1+y)(1+z)}\left[\frac{(1+y)^a-(1+z)^a}{(1+y)^a z-(1+z)^a y}+1\right] \\ &= -\frac{1}{(1+y)(1+z)}\left[\frac{(1+y)^a(1+z)-(1+z)^a(1+y)}{(1+y)^a z-(1+z)^a y}\right] \\ &= -\frac{(1+y)^{a-1}-(1+z)^{a-1}}{(1+y)^a z-(1+z)^a y}. \end{aligned}$$

Let N and D be the numerator and denominator in this last expression, so that $H(a) = -N/D$. One has

$$N = \sum_k \binom{a-1}{k}(y^k - z^k) = (y-z)\sum_{i,j\geq 0}\binom{a-1}{i+j+1}y^i z^j$$

and

$$\begin{aligned} D &= \sum_k \binom{a}{k}(y^k z - yz^k) = (z-y) + yz\sum_{k\geq 1}\binom{a}{k}(y^{k-1}-z^{k-1}) \\ &= (y-z)\left(-1+yz\sum_{i,j\geq 0}\binom{a}{i+j+2}y^i z^j\right) \\ &= (y-z)\left(-1+\sum_{i,j\geq 1}\binom{a}{i+j}y^i z^j\right). \end{aligned}$$

This proves (i).
Now invert the formula in 2.3 when $d=1$. One finds

$$-\frac{(1+y)^a-(1+z)^a}{(1+y)^a z-(1+z)^a y} = 1+(1+y)(1+z)H(a).$$

Substituting this identity into 2.3 gives

$$\begin{aligned} H(\underline{a}) &= \frac{1}{(1+y)(1+z)}\left[\prod_{i=1}^{d}(1+(1+y)(1+z)H(a_i))-1\right] \\ &= \frac{1}{(1+y)(1+z)}\left[\sum_{\substack{P\subset[1,d]\\ P\neq\varnothing}}(1+y)^{|P|}(1+z)^{|P|}\prod_{i\in P}H(a_i)\right], \end{aligned}$$

which proves (ii). Theorem 2.5. With the notation of 2.1:

(i) If $p+q = p'+q'$ and $p\leq p'\leq q'\leq q$, then

$$h_0^{p,q}(\underline{a}) \leq h_0^{p',q'}(\underline{a}).$$

(ii) If $p \leq q$, then $h_0^{p,q}(\underline{a}) \neq 0$ if and only if

$$p > \left\lfloor \frac{p+q+d-\sum_i a_i}{\sup_i(a_i)} \right\rfloor .$$

Here $\lfloor\ \rfloor$ denotes the integer part of the quotient.

Let (*) be the following property of a formal power series $P = \sum a_{p,q} y^p z^q$:

(*) the $a_{p,q}$ are nonnegative, $a_{p,q} = a_{q,p}$, and whenever $p+q = p'+q'$ and $p \leq p' \leq q' \leq q$, one has $a_{p,q} \leq a_{p',q'}$.

Assertion (i) follows at once from the formulas in 2.4, the identity $(1-x)^{-1} = \sum_{n \geq 0} x^n$, and the following lemma, whose proof is left to the reader.

Lemma 2.5.1. The set of formal power series satisfying (*) is stable under addition and multiplication.

We first prove (ii) in the special case of hypersurfaces ($d = 1$). For $p \leq q$, the desired formula can then be rewritten as

$$h_0^{p,q}(a) \neq 0 \quad \Longleftrightarrow \quad p \geq \left\lfloor \frac{p+q+1}{a} \right\rfloor \tag{2.5.2}$$

or, equivalently, $(p+1)a \geq p+q+2$. From the second formula in 2.4(i), one sees that $h_0^{p,q}(a)$ is nonzero if and only if there exist pairs (e,f) and (c_i,d_i) such that

$$\begin{cases} (e,f) \geq (0,0), & e+f \leq a-2, \\ (c_i,d_i) \geq (1,1), & c_i+d_i \leq a, \\ (p,q) = (e,f) + \sum_i (c_i,d_i). & \end{cases} \tag{2.5.3}$$

For fixed $p+q$, the smallest possible p is obtained by taking $e = 0$ and every $c_i = 1$; it is the number of indices i. Thus it is the lower bound of those p for which

$$p \geq \frac{(p+q)-(a-2)}{a}, \qquad \text{i.e.} \qquad (p+1)a \geq p+q+2.$$

We now pass to the general case. If $\underline{a}'$ is obtained from $\underline{a}$ by deleting the a_i equal to one, then $H(\underline{a}) = H(\underline{a}')$. In view of the form of 2.5(ii), we may therefore assume that every $a_i > 1$. If $d = 0$, then $h_0^{p,q} = 0$ and the assertion is correct; we lose no generality by assuming $d > 0$ and $a_d = \sup_i(a_i)$.

By 2.4(ii) and (2.5.2), the condition $h_0^{p,q}(\underline{a}) \neq 0$ is equivalent to the existence of a nonempty subset P of $[1,d]$, a pair (e,f), and pairs $(c_i,d_i)_{i \in P}$ such that

$$\begin{cases} (|P|-1, |P|-1) \geq (e,f) \geq (0,0), & \\ c_i, d_i \geq \left\lfloor \frac{c_i+d_i+1}{a_i} \right\rfloor, & (c_i,d_i) \geq (0,0), \\ (p,q) = (e,f) + \sum_{i \in P} (c_i,d_i). & \end{cases} \tag{2.5.4}$$

For fixed $n = p+q$, seek the smallest possible value of p. Put $n_i = c_i + d_i$. For such a p, conditions (2.5.4) become

$$
\begin{cases}
a_i = 2 \Longrightarrow n_i \text{ is even} & \text{(otherwise (2.5.4) is impossible)},\\
c_i \geq \left\lfloor \dfrac{n_i+1}{a_i} \right\rfloor, & c_i \leq n_i,\\
(|P|-1, |P|-1) \geq (e,f) \geq (0,0),\\
n = e + f + \displaystyle\sum_{i\in P} n_i,\\
p = e + \displaystyle\sum_{i\in P} c_i.
\end{cases} \tag{2.5.5}
$$

This description first shows that the smallest p is obtained for $P = [1,d]$, to which case we henceforth restrict. If

$$
\sum_i (a_i - 2) + (d-1) \geq n, \tag{2.5.6}
$$

then a solution exists with $p = c_i = e = 0$. Otherwise, if $a_d \neq 2$, the best choice is $n_i = a_i - 2$ for $i \neq d$, so that $c_i = 0$ for $i \neq d$, $e = 0$, $f = d-1$, and

$$
n_d = n - (d-1) - \sum_{i\neq d} (a_i - 2).
$$

Then

$$
p = \left\lfloor \frac{n - (d-1) - \sum_{i\neq d}(a_i-2) + 1}{a_d} \right\rfloor, \tag{2.5.7}
$$

and hence, for $p \leq q$, one has $h_0^{p,q}(\underline{a}) \neq 0$ if and only if

$$
p \geq \left\lfloor \frac{n - \sum_{i\neq d}(a_i - 1) + 1}{a_d} \right\rfloor. \tag{2.5.8}
$$

Formula (2.5.8) reduces to (2.5.2) for $d = 1$ and is trivially true when (2.5.6) holds. If $d \geq 2$, (2.5.6) fails, and all the a_i equal two, the best choice is instead

$n_i = 0$ for $i \neq d$, $e = 0$, and $f = d-1$ or $d-2$, chosen congruent to n modulo two, with $n_d = n - f$. If $n \equiv d-1 \pmod 2$, this proves (2.5.7) as above. Otherwise, observe that

$$
\left\lfloor \frac{n-(d-2)+1}{2} \right\rfloor = \left\lfloor \frac{n-(d-1)+1}{2} \right\rfloor,
$$

so (2.5.7), and hence (2.5.8), remains valid when $a_d = 2$.

It remains to verify that (2.5.8) is equivalent to 2.5(ii). Indeed,

$$
\left\lfloor \frac{n - \sum_{i\neq d}(a_i-1)+1}{a_d} \right\rfloor = \left\lfloor \frac{n+d-\sum_{i\neq d} a_i}{a_d} \right\rfloor = \left\lfloor \frac{p+q+d-\sum_i a_i}{a_d} \right\rfloor + 1.
$$

2.6. If X is a complete intersection of dimension n in $\mathbb{P}^r$, it follows at once from (2.5.8) that the primitive part of $H^n(X)$ vanishes only when X is an odd-dimensional quadric.

Definition 2.7. If X is a proper smooth algebraic variety over $\mathbb{C}$, the Hodge level of $H^n(X)$ is the supremum of the numbers $q-p$ for which $p+q = n$ and $h^{p,q}(X) \neq 0$.

The Hodge level ℓ is therefore congruent to n modulo two, and is at most

$$
\min\bigl(n, 2\dim(X) - n\bigr).
$$

It is nonnegative unless $H^n(X) = 0$, in which case it is $-\infty$. [1]

Corollary 2.8. Retain the notation of 2.1, put $k = \mathbb{C}$, and suppose that $a_d = \sup_i(a_i)$. Let ℓ be an integer with $0 \leq \ell < n$, congruent to n modulo two. The Hodge level of $H^n(V_n(\underline{a}))$ is at most ℓ if and only if

$$\sum_{i \neq d}(a_i - 1) \leq (\ell + 1) - (a_d - 2)\frac{n - \ell}{2}.$$

By (2.5.8), if the Hodge level is nonnegative, it is the maximum of zero and

$$n - 2\left\lfloor \frac{n - \sum_{i \neq d}(a_i - 1) + 1}{a_d} \right\rfloor.$$

The inequality

$$n - 2\left\lfloor \frac{n - \sum_{i \neq d}(a_i - 1) + 1}{a_d} \right\rfloor \leq \ell$$

may be rewritten as

$$\frac{n - \ell}{2} \leq \left\lfloor \frac{n - \sum_{i \neq d}(a_i - 1) + 1}{a_d} \right\rfloor,$$

or equivalently as

$$\sum_{i \neq d}(a_i - 1) \leq 1 + n - a_d\frac{n - \ell}{2} = \ell + 1 - (a_d - 2)\frac{n - \ell}{2}.$$

2.9. Using formula 2.8, it is easy, for small ℓ, to enumerate the multidegrees of complete intersections in $\mathbb{P}^r$ of dimension $n > 1$ whose H^n has level ℓ. The table given in the first version of this exposé was, however, incorrect; the table below is extracted from the more complete one given by M. Rapoport in his article "Complément à l'article de P. Deligne *La conjecture de Weil pour les surfaces K3*", *Inventiones Math.* 15, 227–236 (1972). For $\ell \leq 1$, the complete intersections $V_n(\underline{a})$ of dimension n and multidegree $\underline{a} = (a_1, \ldots, a_d)$, with $n > \ell$, $d \geq 1$, and $a_i \geq 2$, having Hodge level ℓ, are the following:

$$\begin{array}{ll}
\ell = -\infty & V_{2k+1}(2) \qquad (k \geq 0), \\
\ell = 0 & V_{2k}(2)\ (k \geq 1), \quad V_{2k}(2,2)\ (k \geq 1), \quad V_2(3), \\
\ell = 1 & V_{2k+1}(2,2)\ (k \geq 1), \quad V_{2k+1}(2,2,2)\ (k \geq 1), \\
 & V_3(3), \quad V_5(3), \quad V_3(3,2), \quad V_3(4).
\end{array}$$

2.10. For small $\underline{a}$, the generating functions of the $h_0^{p,q}(\underline{a})$ are, by 2.4,

$$\begin{aligned}
H(2) &= \frac{1}{1 - yz}, \\
H(3) &= \frac{2 + y + z}{1 - 3yz - y^2z - yz^2}, \\
H(2,2) &= \frac{y + z}{(1 - yz)^2} + \frac{2}{(1 - yz)^2} + \frac{1}{1 - yz},
\end{aligned}$$

so that

[1] The printed source reads $\min(n, \dim(X) - n)$. The factor 2 is required by the Hodge ranges $0 \leq p, q \leq \dim(X)$; the English text corrects this evident source typo.

$$
\begin{aligned}
h_0^{p,p}(2) &= 1 \quad , \\
h_0^{1,1}(3) &= 6 \quad , \\
h_0^{p,p}(2,2) &= 2p+3 \quad .
\end{aligned}
$$

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S G A 7

E X P O S É XII

QUADRICS

by P. DELIGNE

0.0. The present exposé collects several well-known results about quadrics which we shall need in order to prove the Picard–Lefschetz formula.

0.1. Let m be a section of an $\mathcal{O}_S$-module M on a scheme S. We shall say that m is nowhere zero on S if, for every $s \in S$, the image of m in the $k(s)$-vector space M_s obtained from M by the base change $\{s\} \longrightarrow S$ is nonzero. We use the same terminology when $S = \mathrm{Spec}(A)$ and m is an element of an A-module M.

1. Quadratic forms

1.1. Let $S = \mathrm{Spec}(A)$ be an affine scheme, let V be a locally free A-module of rank n, and let Q be a quadratic form on V (Bourbaki, *Algèbre*, Ch. 9, §3, no. 4). Recall that Q is called nondegenerate if the associated bilinear form

$$\Phi(x, y) = Q(x + y) - Q(x) - Q(y)$$

is nondegenerate, that is, if it induces an isomorphism from V to its dual V^*. If 2 is not invertible in A, this can occur only when n is even (loc. cit.).

The quadratic forms on V form a projective A-module; a quadratic form Q is "nowhere zero" on S (0.1) if and only if the elements $Q(v)$ generate the unit ideal of A.

The notation $\mathbb{P}(V)$ is used in the sense of EGA; recall that if A is a field, then $\mathbb{P}(V)$ is the projective space whose set of rational points is the quotient $(V^* \setminus \{0\})/A^*$.

The quadratic form Q identifies with a section of the invertible sheaf $\mathcal{O}(2)$ on the projective space $\mathbb{P}(V^*)$ over S. When Q is nowhere zero, the ("homogeneous") equation $Q = 0$ defines a subscheme of $\mathbb{P}(V^*)$, flat and purely of relative dimension $n - 2$ over S. This scheme is called the quadric defined by Q. The form Q is called ordinary if it is nowhere zero on S and the quadric that it defines is smooth over S.

The form Q is ordinary if and only if every form obtained from it by extending the scalars from A to a field is ordinary. If A is a field, then:

a) if n is even or $\mathrm{char}(A) \neq 2$, then Q is ordinary if and only if Q is nondegenerate;

b) if n is odd and $\mathrm{char}(A) = 2$, then Q is ordinary if and only if the kernel N of the associated (alternating) bilinear form Φ has dimension one and Q does not vanish on N.

Proposition 1.2. If Q is ordinary and $n = 2m$ (respectively, $n = 2m + 1$), then, locally for the étale topology on S, V has a basis $e_1, \ldots, e_n$ such that

$$Q\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{m} x_i x_{i+m}$$

(respectively,

$$Q\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{m} x_i x_{i+m} + \lambda x_{2m+1}^2, \qquad (\lambda \text{ invertible}).$$

[2]

We prove 1.2 by induction on m. If $m = 0$, the assertion is clear. If $m > 0$, the quadric with equation $Q = 0$ is smooth over S and has nonempty geometric fibers; it therefore has sections locally for the étale topology. Locally, such a section is defined by an element $e \in V$, nowhere zero on $\mathrm{Spec}(A)$, with $Q(e) = 0$. At no point of S does e belong to the kernel of Φ; hence, locally, one can find f' such that $\Phi(e, f') = 1$. If

$$f = -Q(f')e + f',$$

then

$$Q(e) = Q(f) = 0, \qquad \Phi(e, f) = 1.$$

[3]

If V_2 is the orthogonal complement of the submodule V_1 generated by e and f, then $V = V_1 \oplus V_2$, and the conclusion follows by applying the induction hypothesis to V_2.

1.3. For the definition and principal properties of the Clifford algebra of Q, we refer to Bourbaki, *Algèbre*, Ch. 9, §9. We denote by ρ the canonical map from V to $C(Q)$.

Lemma 1.3.1. Under the hypotheses of 1.1, for every $\lambda \in A$ there exists a unique homomorphism

$$[\lambda] \colon C^+(\lambda Q) \longrightarrow C^+(Q)$$

such that, for $x, y \in V$,

$$[\lambda]\big(\rho(x)\rho(y)\big) = \lambda\rho(x)\rho(y).$$

This homomorphism is induced on passing to the quotient by the endomorphism $[\lambda]$ of the even part $T^+(V)$ of the tensor algebra of V, which acts as λ^r on $T^{2r}(V)$. Indeed,

$$[\lambda]\big(x \otimes x - \lambda Q(x)\big) = \lambda\big(x \otimes x - Q(x)\big).$$

Suppose that Q is nowhere zero on S. Lemma 1.3.1 then defines a transitive system of isomorphisms between the algebras $C^+(\lambda Q)$ for $\lambda \in A^*$.

Consequently, the algebra $C^+(Q)$ depends (up to unique isomorphism) only on V and on Q up to multiplication by a factor; in other words, $C^+(Q)$ depends only on V and on the subscheme of $\mathbb{P}(V^*)$ with equation $Q = 0$. Every homothety with invertible ratio is an automorphism of the pair $(V, X \subset \mathbb{P}(V^*))$; one checks that these automorphisms act trivially on $C^+(V, X)$. It follows that the algebra $C^+(V, X)$ depends only on the pair of schemes $X \subset \mathbb{P}(V^*)$ over A. We shall denote it by $C^+(X, \mathbb{P}(V^*))$. 1.4. Let V be a locally free A-module of rank $2m > 0$, and let Q be a nondegenerate quadratic form on V. Suppose given a decomposition $V = W_1 \oplus W_2$ into two totally singular subspaces. The form $Q(w_1 + w_2)$ then places W_1 and W_2 in perfect duality. Recall that, for $f \in W_2$ (identified with W_1^*), contraction by f is the endomorphism of degree -1 of $\bigwedge W_1$ defined by

$$f \lrcorner (e_1 \wedge \cdots \wedge e_p) = \sum_{i=1}^{p} (-1)^{i-1} \langle e_i, f \rangle e_1 \wedge \cdots \wedge \widehat{e_i} \wedge \cdots \wedge e_p.$$

[4]

[2] In both displayed sums the printed source has upper limit $m - 1$. That leaves a two-dimensional radical and contradicts the statement; the standard ordinary normal form, and the induction that follows, require the upper limit m.

[3] The final term in the printed definition of f is f; it is the evident source typo f', as is confirmed by the displayed conclusion.

[4] The printed formula has $(-1)^i$ and begins the remaining wedge with an undefined e_0. The English restores the standard contraction formula $(-1)^{i-1} e_1 \wedge \cdots \widehat{e_i} \cdots \wedge e_p$, required by the Clifford action displayed immediately below. The reading was checked in a targeted high-detail enlargement of folio 65.

Endow the A-module $\bigwedge W_1$ with the $\mathbb{Z}/2$-grading for which

$$(\bigwedge W_1)^+ = \bigoplus_i \bigwedge^{2i} W_1, \qquad (\bigwedge W_1)^- = \bigoplus_i \bigwedge^{2i+1} W_1.$$

The algebra $\mathrm{End}(\bigwedge W_1)$ is then $\mathbb{Z}/2$-graded: its even (respectively, odd) endomorphisms are those that preserve the grading (respectively, have degree one modulo two).

We refer to Bourbaki, *Algèbre*, Ch. 9, §9, no. 4, for the proof of the existence of a unique isomorphism s of $\mathbb{Z}/2$-graded algebras from $C(Q)$ to $\mathrm{End}(\bigwedge W_1)$ such that

$$s\big(\rho((e,0))\big) = e \wedge (-), \qquad s\big(\rho((0,f))\big) = f \lrcorner (-).$$

Temporarily denote by $e(W_1, W_2)$ the element of $C^+(Q)$ for which $s(e(W_1,W_2))$ is the identity on $(\bigwedge W_1)^+$ and zero on $(\bigwedge W_1)^-$. The center $Z(Q)$ of $C^+(Q)$ is isomorphic to $A \oplus A$, this decomposition being defined by the idempotents $e(W_1,W_2)$ and $1-e(W_2,W_1)$. Moreover, $C^+(Q)$ is a matrix algebra over $Z(Q)$.

From this and 1.2 one deduces, as in loc. cit.:

Proposition 1.5. Under the hypotheses of 1.1, if $n = 2m > 0$ and Q is nondegenerate, then the center $Z(Q)$ of $C^+(Q)$ is an étale algebra, locally free of rank two over A, and $C^+(Q)$ is an Azumaya algebra over $Z(Q)$.

1.6. Let V and Q be as in 1.4, and let W be a totally singular direct summand of V, locally free of rank m. Let W_1' be a complement of W, so that $V = W \oplus W_1'$. Every complement W_2' of W identifies with the graph of a map $u : W_1' \to W$. The form Φ places W and W_1' in duality, so u is uniquely determined by the bilinear form on W_1'

$$B(x,y) = \Phi(x, u(y)),$$

and W_2' is totally singular if and only if

$$B(x,x) = -Q(x)$$

on W_1'. Since such forms B always exist, W has totally singular complements. Having chosen one of them, say W', the set of totally singular complements W_0' of W identifies with the set of alternating forms on W', or equivalently with the set of sections of the affine space $\mathbb{E}(\bigwedge^2 W')$ over S.

For each section x of $\mathbb{E}(\bigwedge^2 W')$, let $W_0'(x)$ be the corresponding complement of W and let $e(x) = e(W, W_0'(x))$. The function $x \mapsto e(x)$ is compatible with every extension of scalars and hence defines a morphism of S-schemes

$$e : \mathbb{E}(\bigwedge^2 W') \longrightarrow \mathrm{Spec}(Z(Q)).$$

The S-scheme $\mathbb{E}(\bigwedge^2 W')$ is flat with geometrically connected fibers, whereas $\mathrm{Spec}(Z(Q))$ is étale over S. The morphism e therefore factors through S, hence through a section of $\mathrm{Spec}(Z(Q))$ over S; equivalently, $e(x)$ is independent of x. [5] This justifies the following notation.

Notation 1.7. For W as above and any totally singular complement W' of W, denote by $e(W)$ the idempotent $e(W,W') \in Z(Q)$.

1.8. We saw in 1.3 that, for invertible λ, $C^+(Q)$ identifies with $C^+(\lambda Q)$. A continuity argument analogous to the preceding one shows that this identification carries

$$e(W) \in Z(Q) \qquad \text{to} \qquad e(W) \in Z(\lambda Q).$$

[5] The printed sentence says that e factors through a section of the affine space $\mathbb{E}(\bigwedge^2 W')$. The connected-to-étale argument instead gives the factorization through S and a section of $\mathrm{Spec}(Z(Q))$, exactly as the following conclusion requires.

6

1.9. Recall that if B and C are two $\mathbb{Z}/2$-graded algebras, then multiplication in their graded tensor product $B \otimes' C$ is defined by the Koszul rule

$$(b_1 \otimes c_1)(b_2 \otimes c_2) = (-1)^{\deg(c_1)\deg(b_2)} b_1 b_2 \otimes c_1 c_2$$

when the b_i and c_i are homogeneous.

If (V_1, Q_1) and (V_2, Q_2) are two quadratic modules, then

$$C(Q_1 \oplus Q_2) = C(Q_1) \otimes' C(Q_2)$$

(Bourbaki, loc. cit.).

If P and Q are two finite-type, locally free, $\mathbb{Z}/2$-graded A-modules, we identify the $\mathbb{Z}/2$-graded algebras $\mathrm{End}(P) \otimes' \mathrm{End}(Q)$ and $\mathrm{End}(P \otimes Q)$ by setting, for homogeneous u, v, p, q,

$$(u \otimes v)(p \otimes q) = (-1)^{\deg(v)\deg(p)} u(p) \otimes v(q).$$

Lemma 1.10. Let (V, Q) be the quadratic module which is the sum of two quadratic modules (V_1, Q_1) and (V_2, Q_2), where V_i is locally free of rank $2m_i$ and Q_i is nondegenerate. Let $V_i = W_i \oplus W_i'$ be a decomposition of V_i into two totally singular subspaces $(i = 1, 2)$. Then the diagram

$$\begin{array}{ccccc} C(Q) & & \xrightarrow{\qquad\sim\qquad} & & C(Q_1) \otimes' C(Q_2) \\ \Big| \wr & & & & \Big\| \\ \mathrm{End}(\bigwedge(W_1 \oplus W_2)) & \simeq & \mathrm{End}(\bigwedge W_1 \otimes \bigwedge W_2) & \simeq & \mathrm{End}(\bigwedge W_1) \otimes' \mathrm{End}(\bigwedge W_2) \end{array}$$

is commutative.

The verification is left to the reader. One deduces at once from this lemma that

$$e(W_1 \oplus W_2) = e(W_1)e(W_2) + (1 - e(W_1))(1 - e(W_2)). \tag{1.10.1}$$

1.11. The following construction is borrowed from Micali and Villamayor. If $p : Z \longrightarrow S$ is a double étale covering of S, then Z may be regarded as a $\mathbb{Z}/2$-torsor over S. Denote the addition of torsors by $*$; for a section s of Z, let $e(s)$ be the idempotent of $p_*\mathcal{O}_Z$ that equals one on s and zero elsewhere. With the notation of 1.10, there is a unique isomorphism

$$\mathrm{Spec}(Z(Q_1)) * \mathrm{Spec}(Z(Q_2)) \longrightarrow \mathrm{Spec}(Z(Q)), \tag{1.11.1}$$

whose underlying addition morphism is

$$+ : \mathrm{Spec}(Z(Q_1)) \times_S \mathrm{Spec}(Z(Q_2)) \longrightarrow \mathrm{Spec}(Z(Q)),$$

and which, after every base change, satisfies

$$e(s + t) = e(s)e(t) + (1 - e(s))(1 - e(t)).$$

If $s(W)$ is the section of $\mathrm{Spec}(Z(Q))$ such that $e(s(W)) = e(W)$, then formula (1.10.1) becomes

$$s(W_1 \oplus W_2) = s(W_1) + s(W_2).$$

[6] The printed cross-reference is 1.6; the scalar-change identification is the construction of 1.3.1.

Proposition 1.12. Let A be a field, let V be a vector space of dimension $2m > 0$ over A, let Q be a nondegenerate quadratic form on V, and let W_1, W_2 be two totally isotropic subspaces of V of dimension m. Then $e(W_1) = e(W_2)$ if and only if $\dim(W_1/(W_1 \cap W_2))$ is even.
[7]

(Cf. Bourbaki, *Algèbre*, Ch. 9, §6, Exercise 28.) Let $e_1, \ldots, e_m, f_1, \ldots, f_m$ be a basis of V such that

a) $Q(\sum_i x_i e_i + \sum_i y_i f_i) = \sum_i x_i y_i$;

b) W_1 is generated by the e_i;

c) W_2 is generated by $(e_i)_{1 \leq i \leq s}$ and $(f_i)_{s < i \leq m}$.

Then V is the sum of the $V_i = Ae_i + Af_i$, and

$$W_\alpha = \bigoplus_i (W_\alpha \cap V_i), \qquad \alpha = 1, 2.$$

The addition formula 1.11 therefore reduces the proof to the case $\dim V = 2$. If $W_1 = W_2$, the assertion is clear. Otherwise, if $W_1 = Ae$ and $W_2 = Af$ with $\Phi(e, f) = 1$, then $e(W_1) = fe$ and $e(W_2) = ef = 1 - fe$.

2. Quadrics

2.1. Let k be an algebraically closed field and V a vector space of dimension $n + 2$ over k. A smooth quadric of dimension n over k is a k-scheme X isomorphic to the subscheme of $\mathbb{P}(V^*)$ defined by the equation $Q = 0$, for some ordinary quadratic form Q over k.

Remark 2.2. For $n = 0, 1, 2$, X is a smooth quadric of dimension n over k if and only if

$$\begin{aligned} n = 0 &\quad : \quad X \simeq \mathrm{Spec}(k) \amalg \mathrm{Spec}(k), \\ n = 1 &\quad : \quad X \simeq \mathbb{P}^1_k, \\ n = 2 &\quad : \quad X \simeq \mathbb{P}^1_k \times \mathbb{P}^1_k. \end{aligned}$$

Lemma 2.3. With the notation of 2.1, suppose that X is defined by $Q = 0$.

(i) $\Omega^n_X \simeq \mathcal{O}_X(-n)$.

(ii)
$$\mathrm{Pic}(X) \simeq \begin{cases} 0, & n = 0, \\ \mathbb{Z}, & n = 1;\ \mathcal{O}_X(1) \text{ is the square of a generator}, \\ \mathbb{Z} \oplus \mathbb{Z}, & n = 2, \\ \mathbb{Z}, & n \geq 3;\ \mathcal{O}_X(1) \text{ is a generator}. \end{cases}$$

(iii) $H^i(X, \mathcal{O}_X) = 0$ for $i > 0$, and $H^1(X, \mathcal{O}_X(1)) = 0$.

(iv) The natural map
$$H^0\big(\mathbb{P}(V^*), \mathcal{O}_{\mathbb{P}(V^*)}(1)\big) = V^* \xrightarrow{\sim} H^0(X, \mathcal{O}_X(1))$$
is an isomorphism; and, for $n > 0$, $H^0(X, \mathcal{O}_X) = k$.

Assertion (i) follows from the fact that the conormal sheaf of X in $\mathbb{P}(V^*)$ is $\mathcal{O}_X(-2)$ and that the canonical sheaf of $\mathbb{P}^r$ is $\mathcal{O}(-r-1)$. Assertion (ii) follows from 2.2 when $n \leq 2$, and from SGA 2, XII, 3.7 when $n \geq 3$. Finally, (iii) and (iv) follow from FAC, no. 78.

Definition 2.4. A smooth quadric of dimension n over a scheme S is an S-scheme $f : X \longrightarrow S$, proper and smooth over S, whose geometric fibres are smooth quadrics.

[7] The printed proposition says that V has dimension $2m$ "over V"; the base field introduced in the same sentence is A.

For $n = 0$, a smooth quadric of dimension zero over S is simply a double étale covering of S. For $n = 1$, it is simply a Severi–Brauer variety of dimension one over S.

2.5. Let $\rho : X \longrightarrow S$ be a smooth quadric of dimension n over S. By 2.3(i), the inverse of the invertible sheaf $\Omega^n_{X/S}$ is ample.

By 2.3(ii), the Picard scheme $\mathrm{Pic}_{X/S}$ is smooth over S (because $H^2(X, \mathcal{O}_X) = 0$), unramified over S (because $H^1(X, \mathcal{O}_X) = 0$), and essentially proper over S (because X/S is smooth). Thus, locally on S for the étale topology, it is a constant discrete group having the structure described in 2.3(ii).[8]

In particular, if X has a section, there is an invertible sheaf L on X such that $L^{\otimes(-n)}$ and $\Omega^n_{X/S}$ have the same image in $\mathrm{Pic}_{X/S}$; any two such sheaves are locally isomorphic on S. By 2.3(iv) and [1], 0.5, p. 19, $\rho_* L$ is locally free and its formation commutes with every base change. Hence $\rho_* L$ is locally free of rank $n + 2$, the map $\rho^* \rho_* L \longrightarrow L$ is surjective, and the canonical map

$$X \longrightarrow \mathbb{P}(\rho_* L)$$

is a closed immersion. It identifies X with a relative divisor of degree two in $\mathbb{P}(\rho_* L)$; locally on S, X is the quadric in $\mathbb{P}(\rho_* L)$ defined by an ordinary quadratic form on $(\rho_* L)^*$, unique up to an invertible factor.

[9]

2.6. Suppose that $n > 0$. If L_1 and L_2 are two invertible sheaves of the type considered above, then locally on S there is an isomorphism $u : L_1 \xrightarrow{\sim} L_2$. It is unique up to an invertible factor, since $\rho_* \mathcal{O}_X = \mathcal{O}_S$ by 2.3(iv). Consequently all the isomorphisms $u : L_1 \xrightarrow{\sim} L_2$ induce the same isomorphism

$$\mathbb{P}(u) : \mathbb{P}(\rho_* L_1) \xrightarrow{\sim} \mathbb{P}(\rho_* L_2).$$

Thus $\mathbb{P}(\rho_* L_1)$ is canonically isomorphic to $\mathbb{P}(\rho_* L_2)$ and depends only on X/S, not on the choice of L. We denote it by $P(X)$.

The projective space $P(X)$ is defined locally on S for the étale topology. By descent it defines a Severi–Brauer variety over S, again denoted $P(X)$, in which X is a relative divisor of degree two.

Locally on S, the pair $X \hookrightarrow P(X)$ defines a Clifford algebra $C^+(X, P(X))$ (1.3). Descent yields an algebra over S, which we denote simply by $C^+(X)$.

2.7. Suppose that $n = 2m > 0$. The centre Z of $C^+(X)$ is then an $\mathcal{O}_S$-algebra defining a double étale covering $Z(X)$ of S.

A generator of X is a subscheme D of X which is a linear subspace of dimension m in $P(X)$.

Locally on S, when $P(X) \simeq \mathbb{P}(V^*)$ and X is defined by $Q = 0$, a generator D of X is, by definition, identified with a locally direct-summand submodule $W(D)$ of V, totally singular and of rank $m+1$. By 1.8, the idempotent $e(W(D))$ in the centre of $C^+(X)$ depends only on $D \subset X$. Let $e(D)$ denote the corresponding section of $Z(X)$, namely the one on which $e(W(D))$ has value one.

The generators of X/S are the S-points of a scheme $\mathrm{Gen}(X)$ projective over S, and e defines a morphism of S-schemes

$$e : \mathrm{Gen}(X) \longrightarrow Z(X). \tag{2.7.1}$$

When $n = 0$, we set $Z(X) = X$, call a section of X a generator of X, and define e to be the identity.

Proposition 2.8. Under the hypotheses of 2.7, $p : \mathrm{Gen}(X) \longrightarrow S$ is projective and smooth, and (2.7.1) is its Stein factorization.

[8] At both occurrences in this sentence the print cites 2.3(iii); the Picard-group computation is 2.3(ii).

[9] After introducing the projection as ρ, the print changes its symbol to p in the direct-image formulas and cites 2.3(iii) where the required base-change statement is 2.3(iv). The notation and cross-reference have been regularized here.

The question is local for the étale topology on S. Locally, if D is a generator, then by 1.6 the generators D' disjoint from D are identified with the S-points of a suitable affine space over S. In this way $\mathrm{Gen}(X)$ is covered by open subschemes smooth over S.

To prove that the geometric fibres of e are geometrically connected, it is enough to consider the case in which S is the spectrum of an algebraically closed field k and X is defined by $Q = 0$ in $\mathbb{P}(V^*)$.

Let W_1 and W_2 be two totally isotropic subspaces of rank $m+1$ in V such that $e(W_1) = e(W_2)$, that is, such that $W_1/(W_1 \cap W_2)$ has even dimension (1.12). Proceeding as in 1.12, one obtains an orthogonal decomposition

$$V = V_0 \oplus \bigoplus_i V_i$$

such that

a) $W_\alpha = (W_\alpha \cap V_0) \oplus \bigoplus_i (W_\alpha \cap V_i)$ for $\alpha = 1, 2$;

b) $W_1 \cap V_0 = W_2 \cap V_0$;

c) $\dim V_i = 4$ and $V_i = (W_1 \cap V_i) \oplus (W_2 \cap V_i)$.

It remains to prove that $W_1 \cap V_i$ and $W_2 \cap V_i$ belong to the same connected family of maximal totally singular subspaces of V_i. In other words, the assertion reduces to the case of quadrics of dimension two. Such a quadric is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, and its generators are the subschemes $\{t\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{t\}$; the proposition is then evident.

3. Cohomology of quadrics

3.1. Let $p : X \longrightarrow S$ be a smooth quadric of dimension n over S, and let ℓ be a prime number invertible on S. We write

$$\eta \in H^0\big(S, R^2 p_* \mathbb{Z}_\ell(1)\big)$$

for the "cohomology class of a hyperplane section." Locally on S for the étale topology, $P(X) \simeq \mathbb{P}(V^*)$; by definition, η is then the image in $H^0(S, R^2 p_* \mathbb{Z}_\ell(1))$ of the first Chern class

$$c_1(\mathcal{O}(1)) \in H^2(X, \mathbb{Z}_\ell(1)).$$

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3.2. Suppose that $n = 2m$. A generator D of X then defines a cohomology class $c\ell'(D) \in H^n(X, \mathbb{Z}_\ell(m))$, whose image in $H^0(S, R^n p_* \mathbb{Z}_\ell(m))$ is denoted $c\ell(D)$. By 2.8, the latter depends only on the section $e(D)$ of $Z(X)$. We therefore obtain a morphism of étale sheaves on S

$$c\ell : Z(X) \longrightarrow R^n p_* \mathbb{Z}_\ell(m). \tag{3.2.1}$$

Theorem 3.3. Let $p : X \longrightarrow S$ be a smooth quadric of dimension n over S, and let ℓ be a prime number invertible on S.

(i) $R^{2i+1} p_* \mathbb{Z}_\ell = 0$.

(ii) If $0 \leq 2i < n$ (respectively, if $n < 2i \leq 2n$), then $R^{2i} p_* \mathbb{Z}_\ell(i)$ is canonically isomorphic to the constant sheaf $\mathbb{Z}_\ell$, and is generated by η^i (respectively, by $\eta^i/2$).

(iii) If $n = 2m$, the map deduced from (3.2.1),

$$\mathbb{Z}_\ell^{Z(X)} \longrightarrow R^n p_* \mathbb{Z}_\ell(m), \tag{3.3.1}$$

[10] The displayed definition in the print has $R^1 p_*$, whereas the following sentence and the degree of $c_1(\mathcal{O}(1))$ both give $R^2 p_*$.

is an isomorphism. If α and β are two disjoint sections of $Z(X)$ (assumed isomorphic to $S \amalg S$, as is locally the case), then

(a) $\eta^m = c\ell(\alpha) + c\ell(\beta)$,

(b) $\begin{cases} \mathrm{Tr}(c\ell(\alpha)^2) = \mathrm{Tr}(c\ell(\beta)^2) = 1, & c\ell(\alpha)c\ell(\beta) = 0, \quad m \text{ even}, \\ c\ell(\alpha)^2 = c\ell(\beta)^2 = 0, & \mathrm{Tr}(c\ell(\alpha)c\ell(\beta)) = 1, \quad m \text{ odd}, \end{cases}$

(c) $\eta\big(c\ell(\alpha) - c\ell(\beta)\big) = 0.$

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Assertions (i) and (ii) are recalled only for reference (XI, 1.6, supplemented by XI, 2.6 when $2i + 1 = n$ in (i)).

It is enough to verify (iii) when S is the spectrum of an algebraically closed field k (one could even take $S = \mathrm{Spec}(\mathbb{C})$).

Formula (a). We may suppose that X is the quadric in $\mathbb{P}^{2m+1}(k)$ with equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} = 0.$$

The class η^m is the cohomology class of the plane section defined by

$$x_i = 0 \qquad (0 \leq i < m),$$

and this cycle is the sum of the generators D_1, D_2 defined by

$$\begin{aligned} D_1 : \quad & x_i = 0 && (0 \leq i \leq m), \\ D_2 : \quad & x_i = 0 && (0 \leq i < m), \qquad x_{2m+1} = 0. \end{aligned}$$

By 1.12, $e(D_1) \neq e(D_2)$, and hence

$$\eta^m = c\ell(D_1) + c\ell(D_2) = c\ell(\alpha) + c\ell(\beta).$$

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Formula (b). If generators D_1, D_2 are disjoint (respectively, meet transversally in one point), then $c\ell(D_1)c\ell(D_2) = 0$ (respectively, $\mathrm{Tr}(c\ell(D_1)c\ell(D_2)) = 1$). By 1.12, when m is even we have $e(D_1) \neq e(D_2)$ (respectively, $e(D_1) = e(D_2)$); when m is odd we have $e(D_1) = e(D_2)$ (respectively, $e(D_1) \neq e(D_2)$). Formula (b) follows.

Formula (c). This is clear when $n = 0$. Otherwise, by (ii), it is enough to prove

$$\eta^m\big(c\ell(\alpha) - c\ell(\beta)\big) = 0.$$

Indeed,

$$\mathrm{Tr}(\eta^m(c\ell(\alpha) - c\ell(\beta))) = \mathrm{Tr}\big(c\ell(\alpha)^2 - c\ell(\beta)^2\big) = 0.$$

Verification. When m is even,

$$2 = \mathrm{Tr}(\eta^{2m}) = \mathrm{Tr}\big(c\ell(\alpha)^2 + 2c\ell(\alpha)c\ell(\beta) + c\ell(\beta)^2\big) = 1 + 1;$$

when m is odd,

$$2 = \mathrm{Tr}(\eta^{2m}) = \mathrm{Tr}\big(c\ell(\alpha)^2 + 2c\ell(\alpha)c\ell(\beta) + c\ell(\beta)^2\big) = 2.$$

[11] The print repeats the tag (3.2.1) on the map in part (iii), but the proof itself refers to this map as (3.3.1).

[12] The printed ambient projective space is $\mathbb{P}^{m+1}$, although the displayed nondegenerate equation has $2m + 2$ coordinates. The required ambient space is $\mathbb{P}^{2m+1}$.

[13]

That (3.3.1) is an isomorphism now follows from the fact that the two bilinear forms in two variables

$$B(x,y) = x_1y_2 + x_2y_1, \qquad B(x,y) = x_1y_1 + x_2y_2$$

have discriminant ± 1.

Verification 3.4. Let X be a smooth quadric of dimension n over a finite field $\mathbb{F}_q$. In view of 3.3, the Lefschetz trace formula gives the classical formula

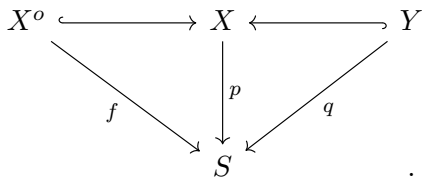
$$\#X(\mathbb{F}_q) = \begin{cases} \sum_{i=0}^{n} q^i, & n \text{ odd}, \\ \sum_{i=0}^{n} q^i + \varepsilon q^m, & n = 2m, \quad \varepsilon = \pm 1. \end{cases}$$

When n is even, $\varepsilon = 1$ if X is split, that is, if it has a rational generator, and $\varepsilon = -1$ otherwise.

3.5. Let $f : X \longrightarrow S$ be a smooth quadric of dimension $n = 2m$. The primitive part of $R^n f_* \mathbb{Z}_\ell(m)$ is the orthogonal complement of η^m; with the notation of 3.3, it is generated by $c\ell(\alpha) - c\ell(\beta)$. The primitive quotient of $R^n f_* \mathbb{Z}_\ell(m)$ is the quotient by $\mathbb{Z}_\ell \eta^m$. The primitive part and primitive quotient of $R^n f_* \mathbb{Z}_\ell$ are obtained by tensoring with $\mathbb{Z}_\ell(-m)$.

3.6. Let V be a locally free vector bundle of rank $n+2$ on S, and let X be the smooth quadric over S defined by an ordinary quadratic form Q on V. Let H moreover be a hyperplane in $\mathbb{P}(V^*)$ meeting X transversally, and put $X^o = X - H$. The schemes X and $Y = X \cap H$ are smooth quadrics over S, of dimensions n and $n-1$. We may therefore use 3.3 to compute the cohomology of $X^o = X - Y$.

(3.6.1)



.

If $n = 0$, then $Y = \varnothing$ and $X = X^o$. Suppose that $n > 0$. We have two long exact sequences dual to one another:

$$\cdots \xrightarrow{\partial} R^i f_! \mathbb{Z}_\ell \longrightarrow R^i p_* \mathbb{Z}_\ell \xrightarrow{r_i} R^i q_* \mathbb{Z}_\ell \longrightarrow \cdots \tag{3.6.2}$$

and

$$\cdots \xrightarrow{\partial} R^{i-2} q_* \mathbb{Z}_\ell(-1) \longrightarrow R^i p_* \mathbb{Z}_\ell \longrightarrow R^i f_* \mathbb{Z}_\ell \longrightarrow \cdots . \tag{3.6.3}$$

For n even, the restriction homomorphism r_i is an isomorphism for $i \neq n, 2n$; for n odd, it is an isomorphism for $i \neq n-1$. If $n = 2m$ and α is the cohomology class of a generator, then

$$r_n(\alpha) = \tfrac{1}{2}\eta^m,$$

so r_n is surjective and its kernel is the primitive part of $R^n p_* \mathbb{Z}_\ell$. If $n = 2m+1$, then $r_{2m}(\eta^m) = \eta^m$, so r_{2m} is injective and its cokernel is the primitive quotient of $R^{2m} q_* \mathbb{Z}_\ell$. It follows that $R^i f_! \mathbb{Z}_\ell = 0$ for $i \neq n, 2n$, and that $R^i f_! \mathbb{Z}_\ell$ is a twisted constant free $\mathbb{Z}_\ell$-sheaf of rank one for $i = n, 2n$. By duality, or from (3.6.3), one likewise shows that $R^i f_* \mathbb{Z}_\ell = 0$ for $i \neq 0, n$, that $f_* \mathbb{Z}_\ell = \mathbb{Z}_\ell$, and that $R^n f_* \mathbb{Z}_\ell$ has rank one.

[14]

Suppose that $n = 2m$ and retain the notation of 3.3.

[13] In these two top-degree intersection computations the print has η^2 in place of η^{2m}, and once prints $c\ell(\alpha^2)$ in place of $c\ell(\alpha)^2$.

[14] This paragraph prints $r = 2m+1$ instead of $n = 2m+1$ and cites 3.5.3 instead of the displayed sequence (3.6.3).

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R^n f_! \mathbb{Z}_\ell(m) & \longrightarrow & R^n p_* \mathbb{Z}_\ell(m) & \longrightarrow & R^n q_* \mathbb{Z}_\ell(m) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \varphi} & & \| & & & & \\ 0 & \longleftarrow & R^n f_* \mathbb{Z}_\ell(m) & \longleftarrow & R^n p_* \mathbb{Z}_\ell(m) & \longleftarrow & R^{n-2} q_* \mathbb{Z}_\ell(m-1) & \longleftarrow & 0. \end{array}$$

The sheaf $R^n f_! \mathbb{Z}_\ell(m)$ then has two natural opposite generators $\pm\delta$, whose images in $R^n p_* \mathbb{Z}_\ell(m)$ are $\pm(c\ell(\alpha) - c\ell(\beta))$. By 3.3(iii)(b),

$$\mathrm{Tr}(\delta^2) = (c\ell(\alpha) - c\ell(\beta))^2 = (-1)^m \cdot 2,$$

and $\pm\varphi(\delta)$ is twice the natural dual generators $\pm\delta'$ of $R^n f_* \mathbb{Z}_\ell(m)$, namely the opposite images of $c\ell(\alpha)$ and $c\ell(\beta)$.

Suppose now that $n = 2m+1$ and apply the notation of 3.3 to Y.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R^{2m} p_* \mathbb{Z}_\ell(m) & \longrightarrow & R^{2m} q_* \mathbb{Z}_\ell(m) & \xrightarrow{\partial} & R^n f_! \mathbb{Z}_\ell(m) & \longrightarrow & 0 \\ & & & & & & \downarrow{\scriptstyle \varphi} & & \\ & & R^{2m+2} p_* \mathbb{Z}_\ell(m+1) & \longleftarrow & R^{2m} q_* \mathbb{Z}_\ell(m) & \longleftarrow & R^n f_* \mathbb{Z}_\ell(m+1) & \longleftarrow & 0. \end{array}$$

[15]

The sheaf $R^n f_! \mathbb{Z}_\ell(m)$ then has two natural opposite generators $\pm\delta$, namely $\partial c\ell(\alpha)$ and $\partial c\ell(\beta)$. We have

$$\delta\,\varphi(\delta) = \delta^2 = (\partial c\ell(\alpha))^2 = \partial\big(c\ell(\alpha)\,\partial c\ell(\alpha)\big) = 0,$$

and therefore $\varphi(\delta) = 0$. On the other hand, $R^n f_* \mathbb{Z}_\ell(m+1)$ has two natural opposite generators $\pm\delta'$, whose images in $R^{2m} q_* \mathbb{Z}_\ell(m)$ are $\pm(c\ell(\alpha) - c\ell(\beta))$, and $\mathrm{Tr}(\delta\delta') = \pm 1$.

These results are assembled in the following table.

Table 3.7. Cohomology of affine quadrics of dimension n, $X^o = X - Y$.

A. The cohomology is torsion-free. The nonzero Betti numbers are

$$\begin{aligned} \text{ordinary cohomology:} \quad & b_0 = b_n = 1, \\ \text{cohomology with compact supports:} \quad & b_{2n} = b_n = 1, \end{aligned}$$

except that $b_0 = 2$ when $n = 0$.

B. If $n = 2m > 0$, then

$$\begin{aligned} R^n f_! \mathbb{Z}_\ell(m) &= \text{the primitive part of } R^{2m} p_* \mathbb{Z}_\ell(m), \\ R^n f_* \mathbb{Z}_\ell(m) &= \text{the primitive quotient of } R^{2m} p_* \mathbb{Z}_\ell(m). \end{aligned}$$

If $n = 2m+1$, then

$$\begin{aligned} R^n f_! \mathbb{Z}_\ell(m) &= \text{the primitive quotient of } R^{2m} q_* \mathbb{Z}_\ell(m), \\ R^n f_* \mathbb{Z}_\ell(m+1) &= \text{the primitive part of } R^{2m} q_* \mathbb{Z}_\ell(m). \end{aligned}$$

Locally, these groups have natural generators defined up to sign; δ denotes those for cohomology with compact supports, and δ' those for ordinary cohomology.

C. The map

$$\varphi : R^n f_! \mathbb{Z}_\ell \longrightarrow R^n f_* \mathbb{Z}_\ell$$

[15] The leftmost term in the printed lower row has twist (m); duality and the other three terms require the displayed twist $(m+1)$.

is zero when n is odd, and sends $\pm\delta$ to $\pm 2\delta'$ when $n > 0$ is even. Moreover,

$$\mathrm{Tr}(\delta\delta') = \pm 1, \qquad \mathrm{Tr}(\delta^2) = \begin{cases} 0, & n = 2m+1, \\ (-1)^m \cdot 2, & n = 2m. \end{cases}$$

Verification 3.8. The complex affine quadric S in $\mathbb{C}^n$ with equation $\sum_i z_i^2 = 1$ is diffeomorphic to the tangent bundle of the real sphere $S \cap \mathbb{R}^n$. This agrees with A and C.

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SGA 7 — EXPOSÉ XIII

The Formalism of Vanishing Cycles

by P. Deligne

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Introduction

In this exposé we take up again the formalism introduced in no. 2 of Grothendieck's Exposé I. The only new point is the introduction of the algebraic analogue of variation in the classical theory.

The topological picture I have in mind is as follows. Let X be an analytic space, let $D \subset \mathbb{C}$ be a disc centred at 0, put $D^* = D - \{0\}$, and let $f : X \longrightarrow D$ be a proper morphism (the nonproper case will be considered later). Replace D by a sufficiently small disc centred at 0, replace X by the inverse image of this small disc, and continue to denote the result by $f : X \longrightarrow D$. According to the complex-analytic variant of Thom's isotopy theorem, $f^{-1}(D^*)$ is a topological fibre bundle over D^*. I do not know whether one can always construct a pair (Γ, r) of the kind described below, but it seems plausible that one can.

(a) Γ is a locally coherent system of local trivializations of the bundle $f^{-1}(D^*)$ over D^*. In other words, Γ defines a $\pi_1(D^*)$-equivariant trivialization of the inverse image of X on a universal covering $\widetilde{D}^*$ of D^*.

(b) r is a retraction of X onto the special fibre $X_0 = f^{-1}(0)$, compatible with Γ: if, on a connected open subset U of D^*, Γ identifies $X_U = f^{-1}(U)$ with $X_1 \times U$, then $r = r(x_1, u)$ is independent of u.

Let (X, f, Γ, r) be given. Let $t \in D$ and $X_t = f^{-1}(t)$ be the "general fibre." The bundle $\varphi^* X$, obtained from X/D by pullback along

$$\varphi : [0,1] \longrightarrow D^*, \qquad x \longmapsto \exp(2\pi i x)\, t,$$

is trivialized by Γ. The monodromy $T : X_t \longrightarrow X_t$ is the homeomorphism

$$X_t = (\varphi^* X)_0 \xrightarrow[\Gamma]{\sim} (\varphi^* X)_1 = X_t.$$

Let $r_t : X_t \longrightarrow X_0$ be the restriction of r to X_t. Then $r_t T = r_t$. One reconstructs (X, f) from (X_t, X_0, T, r_t) as follows:

(a) identify the ends $X_t \times \{0\}$ and $X_t \times \{1\}$ of $X_t \times [0,1]$ using T, obtaining $f' : X' \longrightarrow S^1$;

(b) r_t defines $r' : X' \longrightarrow X_0$, and (X, f) is identified with the map of mapping cones

$$\mathrm{cone}(f') : \mathrm{cone}(r' : X' \longrightarrow X_0) \longrightarrow D = \mathrm{cone}(S^1 \longrightarrow \mathrm{pt}).$$

Put $\Psi = (r, f) : X \longrightarrow X_0 \times D$. The higher Leray direct images $R^i\Psi_*\mathbb{Z}$ satisfy the following condition:

> (*) The space D^* is covered by open subsets U such that the restriction of the sheaf F to $X_0 \times U$ is pulled back from a sheaf on X_0.

Construction 0.1. The sheaves F on $X_0 \times D$ satisfying (*) are identified with systems (F_0, F_t, T, α) in which

a) F_0 is a sheaf on X_0;

b) F_t is a sheaf on X_0, endowed with an action of the generator T of $\pi_1(D^*, t)$;

c) α is a morphism from F_0 to the sheaf of T-invariants of F_t.

Let G be a sheaf on $X_0 \times D^*$ satisfying (*), and let $\widetilde{D}^*$ be the universal covering of D^*, with base point t. The pullback of G to $X_0 \times \widetilde{D}^*$ is, in one and only one way, pulled back from a sheaf G_t on X_0. The sheaf G_t is endowed with an action of $\pi_1(D^*, t)$, and G is obtained from $\mathrm{pr}_1^* G_t$ on $X_0 \times \widetilde{D}^*$ by taking the quotient by $\pi_1(D^*, t)$.

A sheaf F on $X_0 \times D$ is determined by

1) its restriction F_0 to X_0;

2) its restriction G to $X_0 \times D^*$;

3) a morphism $\alpha : F_0 \longrightarrow i^* j_* G$, where $i : X_0 \hookrightarrow X_0 \times D$ and $j : X_0 \times D^* \hookrightarrow X_0 \times D$.

If F satisfies (*), then G is described by (G_t, T) on X_0, and the asserted description of F follows.

For the sheaves $R^i\Psi_*\mathbb{Z}$ we have

$$(R^i\Psi_*\mathbb{Z})_0 = \begin{cases} \mathbb{Z}, & i = 0, \\ 0, & i > 0, \end{cases}$$

whereas, for $i > 0$, the $(R^i\Psi_*\mathbb{Z})_t$ deserve the name sheaves of vanishing cycles (cf. the discussion at the beginning of paragraph 2 of Exposé I).

This discussion is useful only after it is refined by passing to the derived category: the $R^i\Psi_*\mathbb{Z}$ are obtained from an object of $D^+(X_0 \times D)$, and indeed from an object $R\Psi_*\mathbb{Z}$ of the derived category of the category of abelian sheaves on $X_0 \times D$ satisfying (*). This object determines not only the sheaves of vanishing cycles and their monodromy, but also the classical "variation" morphisms (cf. 1.4 and XIV, 3.1).

We shall define its analogue in algebraic geometry. If S is a strictly henselian trait and $f : X \longrightarrow S$ is a morphism of schemes, we shall define an object $R\Psi(\mathbb{Z}/\ell^n)$ of the derived category of the category of abelian sheaves on the product topos $(X_0)_{\mathrm{et}} \times S_{\mathrm{et}}$. A sheaf on this topos may be interpreted as a system (F_0, F_t, T, α) in which

a) F_0 is a sheaf on X_0;

b) F_t is a sheaf on X_0, endowed with a continuous action T of the inertia group I;

c) α is a morphism from F_0 to the sheaf of I-invariants of F_t.

We shall have to work in a slightly more general framework than that described above:

a) The construction of $R\Psi(\mathbb{Z}/\ell^n)$ does not require f to be proper. For a topological interpretation, one would repeat the preceding construction after replacing X by a neighbourhood of the special fibre X_0.

b) To construct $R\Psi(\mathbb{Z}/\ell^n)$ and prove its properties, one must consider not only the constant sheaf $\mathbb{Z}/\ell^n$, but also general sheaves and even complexes of sheaves.

c) When the residue field k of the henselian trait S is not assumed separably closed, the product topos $(X_0)_{\text{et}} \times S_{\text{et}}$ must be replaced by the fibred product topos

$$(X_0)_{\text{et}} \times_{\text{Spec}(k)_{\text{et}}} S_{\text{et}}.$$

The reader who dislikes 2-fibred products of topoi may take the Galois description of this topos in 1.2.4 as its definition.

0.2. Notation and terminology.

(0.2.1) trait: the spectrum of a discrete valuation ring.

(0.2.2) henselian: for a scheme, the spectrum of a henselian local ring.

(0.2.3) strictly henselian: for a ring, henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.2.4) geometric point of S: a morphism $\bar{x} : \text{Spec}(\bar{k}) \longrightarrow S$ with image $x \in S$, where $k(\bar{x}) =_{\text{def}} \bar{k}$ is a separable closure of $\kappa(x)$. We say that $\bar{x}$ is localized at x. By abuse of language, we shall sometimes also call a morphism $\bar{x} : \text{Spec}(\bar{k}) \longrightarrow S$ a geometric point when $\bar{k}$ is a separably closed extension of $\kappa(x)$; such a morphism factors uniquely through a geometric point of S in the proper sense.

(0.2.5) $s, \eta, \bar{s}, \bar{\eta}, \bar{S}$: If $S = \text{Spec}(V)$ is a henselian trait, we shall normally denote by s the closed point of S, by η its generic point, by $\bar{\eta}$ a generic geometric point of S (localized at η), and by $\bar{s}$ the corresponding geometric point localized at s (I, 0.0.3). Let $\bar{S}$ denote the spectrum of the normalization of V in $k(\bar{\eta})$; its residue field is an inseparable extension of $k(\bar{s})$. Write $\text{Gal}(\bar{\eta}/\eta)$ for the Galois group of $k(\bar{\eta})$ over $k(\eta)$, and similarly for s. There is an exact sequence

$$0 \longrightarrow I \longrightarrow \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Gal}(\bar{s}/s) \longrightarrow 0,$$

where I is the inertia group.

(0.2.6) sheaf: on a scheme X, always means a sheaf on the étale site X_{et} of X.

(0.2.7) prime to: a torsion sheaf F on a scheme S (or a torsion abelian group F) is prime to the residue characteristics of S if, for every $s \in S$, multiplication by the characteristic exponent of $k(s)$ is an automorphism of F.

1. Sheaves of sets

The sheaves of vanishing cycles will be defined as values of derived functors of the left exact functor Ψ defined in no. 3 for every sheaf of sets.

1.1. Galois sheaves.

1.1.1. Let Y be a scheme over a field k with separable closure $\bar{k}$, and put $\bar{Y} = Y \otimes_k \bar{k}$. Let G be a profinite group and let

$$u : G \longrightarrow \text{Gal}(\bar{k}/k)$$

be a continuous homomorphism from G to the Galois group of $\bar{k}$ over k. The Galois group $\text{Gal}(\bar{k}/k)$ acts on the left on $\bar{Y}$ by transport of structure. The group G acts on $\bar{Y}$ through u.

Let $\mathcal{F}$ be a sheaf of sets on (the étale site of) $\bar{Y}$. An action of G on $\mathcal{F}$ compatible with the action of G on $\bar{Y}$ is an action of G by automorphisms of the pair $(\bar{Y}, \mathcal{F})$ which induces the given action on $\bar{Y}$. In other words, it is a system of isomorphisms

$$\sigma(g) : u(g)_*\mathcal{F} \longrightarrow \mathcal{F}$$

satisfying $\sigma(gh) = \sigma(g)\,\sigma(h)$. If G acts on $\mathcal{F}$, then for every U étale over Y, with $\bar{U} = U \otimes_k \bar{k}$ étale over $\bar{Y}$, the group G acts on $\mathcal{F}(\bar{U})$.

Definition 1.1.2. We say that G acts continuously on $\mathcal{F}$ if, for every quasi-compact U étale over Y, the action of G on the discrete set $\mathcal{F}(\bar{U})$ is continuous.

It would amount to the same thing to consider only affine U.

If $\mathcal{F}$ is a sheaf of sets on Y, with inverse image $\bar{\mathcal{F}}$ on $\bar{Y}$, then $\mathrm{Gal}(\bar{k}/k)$ acts on $\bar{\mathcal{F}}$ by transport of structure, compatibly with its action on $\bar{Y}$.

Reminder 1.1.3.

(i) The action of $\mathrm{Gal}(\bar{k}/k)$ on $\bar{\mathcal{F}}$ is continuous.

(ii) The functor

$$\mathcal{F} \longmapsto \bar{\mathcal{F}}$$

where $\bar{\mathcal{F}}$ is endowed with the action of $\mathrm{Gal}(\bar{k}/k)$, is an equivalence from the category of sheaves of sets on Y to the category of sheaves of sets on $\bar{Y}$ endowed with a continuous action of $\mathrm{Gal}(\bar{k}/k)$ compatible with the action of $\mathrm{Gal}(\bar{k}/k)$ on $\bar{Y}$.

Write $\bar{k}$ as the direct limit of finite Galois extensions k_i. Let

$$\varphi_i : Y_i = Y \otimes_k k_i \longrightarrow Y, \qquad \bar{\varphi} : \bar{Y} \longrightarrow Y$$

be the projections. For U étale over Y, assumed affine, put $U_i = U \otimes_k k_i$. Then

$$\bar{\mathcal{F}}(\bar{U}) = \varinjlim_i (\varphi_i^*\mathcal{F})(U_i).$$

On $(\varphi_i^*\mathcal{F})(U_i)$, the group $\mathrm{Gal}(\bar{k}/k)$ acts through $\mathrm{Gal}(k_i/k)$, which proves (i).

The functor in (ii) has as right adjoint the functor

$$\mathcal{G} \longmapsto (\bar{\varphi}_*\mathcal{G})^{\mathrm{Gal}(\bar{k}/k)}.$$

For every $\mathcal{F}$ on Y, one has

$$\bar{\varphi}_*\bar{\varphi}^*\mathcal{F} = \varinjlim_i \varphi_{i*}\varphi_i^*\mathcal{F}.$$

Thus, to prove that

$$\mathcal{F} \xrightarrow{\sim} (\bar{\varphi}_*\bar{\varphi}^*\mathcal{F})^{\mathrm{Gal}(\bar{k}/k)},$$

it is enough to prove, for every i, that

$$\mathcal{F} \xrightarrow{\sim} (\varphi_{i*}\varphi_i^*\mathcal{F})^{\mathrm{Gal}(k_i/k)}.$$

It is even enough to prove that this arrow becomes an isomorphism after extension of scalars from k to k_i; then $\mathrm{Spec}(k_i)$ becomes a sum of copies of $\mathrm{Spec}(k)$ permuted simply transitively by $\mathrm{Gal}(k_i/k)$, and the assertion becomes trivial.

The exact functor in (ii) is therefore fully faithful. To conclude, it remains to verify that the arrow

$$\bar{\varphi}^*(\bar{\varphi}_*\mathcal{G})^{\mathrm{Gal}(\bar{k}/k)} \longrightarrow \mathcal{G}$$

is an epimorphism. Let $\bar{\varphi}_i : \bar{Y} \longrightarrow Y_i$. The hypothesis implies that

$$\varinjlim_i \bar{\varphi}_i^*(\bar{\varphi}_{i*}\mathcal{G})^{\mathrm{Gal}(\bar{k}/k_i)} \longrightarrow \mathcal{G}$$

is an epimorphism.

We have

$$\bar{\varphi}^*(\bar{\varphi}_*\mathcal{G})^{\mathrm{Gal}(\bar{k}/k)} = \bar{\varphi}_i^*\left(\varphi_i^*\varphi_{i*}\left((\bar{\varphi}_{i*}\mathcal{G})^{\mathrm{Gal}(\bar{k}/k_i)}\right)^{\mathrm{Gal}(k_i/k)}\right),$$

and the conclusion follows by observing, by the same Galois-descent argument as above, that for $\mathcal{G}'$ on Y_i, endowed with an action of $\mathrm{Gal}(k_i/k)$ compatible with its action on Y_i, one has

$$\varphi_i^*(\varphi_{i*}\mathcal{G}')^{\mathrm{Gal}(k_i/k)} \xrightarrow{\sim} \mathcal{G}'.$$

1.2. The topos $Y \times_s S$

1.2.1.

Let S be a henselian local scheme with closed point $i : s \hookrightarrow S$. The functor

$$S' \longmapsto S' \times_S s = i^*S'$$

is an equivalence from the category of finite étale S-schemes S' to the category of finite étale s-schemes s' (spectra of étale $k(s)$-algebras). The inverse functor sp^*, from the étale site of s to the étale site of S, is a morphism of sites

$$\mathrm{sp} : S \longrightarrow s,$$

the specialization morphism (cf. SGA 4, VIII, 7). If s' is étale over s, one says that $\mathrm{sp}^*(s')$ is obtained from S by extension of the residue field. For a sheaf F on S, there is a functorial isomorphism

$$\mathrm{sp}_*F = i^*F. \tag{1.2.1.1}$$

For every open subscheme of S, $j : U \hookrightarrow S$, or for every point of S, $j : \eta \hookrightarrow S$, we shall again write sp for the composite morphism $\mathrm{sp} \circ j$. For a sheaf F on U (respectively on η), one has

$$\mathrm{sp}_*F = i^*j_*F. \tag{1.2.1.2}$$

1.2.2.

Let S be a henselian trait, and let $s, \eta, \bar{s}, \bar{\eta}, I$ be as in 0.2.5.

$$s \xhookrightarrow{i} S \xhookleftarrow{j} \eta \quad .$$

Here is the Galois-theoretic interpretation of sheaves on s, η, and S, and of the specialization morphism.

(a) Sheaves F on s (respectively on η) are identified, by $F \mapsto F_{\bar{s}}$ (respectively $F \mapsto F_{\bar{\eta}}$), with sets endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ (respectively $\mathrm{Gal}(\bar{\eta}/\eta)$) (SGA 4, VIII, 2.1, or 1.1.3).

(b) A sheaf F on S defines sheaves $F_s = i^*F$ and $F_\eta = j^*F$ on s and η, and an adjunction arrow $F \longrightarrow j_*j^*F$ inducing

$$\varphi : F_s = i^*F \longrightarrow i^*j_*j^*F = \mathrm{sp}_*F_\eta.$$

It is known (SGA 4, IV, 9.5.4) that the functor

$$F \longmapsto (F_s, F_\eta, \varphi)$$

is an equivalence from the category of sheaves on S to the category of triples (F_s, F_η, φ), where F_s is a sheaf on s, F_η is a sheaf on η, and $\varphi : F_s \longrightarrow i^*j_*F_\eta$.

Under the equivalence in (a), the functor i^*j_* is identified with the functor of "invariants under I". Thus sheaves on S are identified with triples consisting of a set $F_{\bar{s}}$ endowed with an action of $\mathrm{Gal}(\bar{s}/s)$, a set $F_{\bar{\eta}}$ endowed with an action of $\mathrm{Gal}(\bar{\eta}/\eta)$, and an equivariant morphism $\varphi : F_{\bar{s}} \longrightarrow F_{\bar{\eta}}$.

(c) Under (a) and (b), the morphisms sp, j, i, and $\mathrm{sp} = \mathrm{sp} \circ j$ are expressed as follows:

$$\begin{array}{lll}
\mathrm{sp} : S \longrightarrow s : & \mathrm{sp}^*(F_{\bar{s}}) & = (F_{\bar{s}}, F_{\bar{s}}, \mathrm{Id}), \\
& \mathrm{sp}_*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{s}}, \\
j : \eta \longrightarrow S : & j^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{\eta}}, \\
& j_*(F_{\bar{\eta}}) & = (F_{\bar{\eta}}^I, F_{\bar{\eta}}, \mathrm{inclusion}), \\
i : s \longrightarrow S : & i^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{s}}, \\
& i_*(F_{\bar{s}}) & = (F_{\bar{s}}, \{0\}, \mathrm{projection}), \\
\mathrm{sp} = \mathrm{sp} \circ j : \eta \longrightarrow s : & \mathrm{sp}^*(F_{\bar{s}}) & = F_{\bar{s}}, \\
& \mathrm{sp}_*(F_{\bar{\eta}}) & = F_{\bar{\eta}}^I.
\end{array}$$

1.2.3.

Let S be a henselian trait as in 1.2.2, and let Y be a scheme over s. We shall have to work with the 2-fiber product $Y \times_s S$ of the étale topoi of Y and S over the étale topos of s. By Giraud [1], this 2-fiber product exists, and it is readily seen to admit the Galois-theoretic description given below. Those who dislike 2-fiber products may take this description as the definition.

Construction 1.2.4.

Sheaves F on $Y \times_s S$ are identified with triples (F_s, F_η, φ) in which:

(a) F_s is a sheaf on Y, or equivalently a sheaf $F_{\bar{s}}$ on $\bar{Y} = Y \otimes_s \bar{s}$, endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ compatible with the action of $\mathrm{Gal}(\bar{s}/s)$ on $\bar{Y}$;

(b) F_η is a sheaf $F_{\bar{\eta}}$ on $\bar{Y}$, endowed with a continuous action of $\mathrm{Gal}(\bar{\eta}/\eta)$ compatible with the action (via 1.1.1) of $\mathrm{Gal}(\bar{\eta}/\eta)$ on $\bar{Y}$;

(c) φ is an equivariant morphism from $F_{\bar{s}}$ to $F_{\bar{\eta}}$.

Similarly, a sheaf F_η on $Y \times_s \eta$ is an object as in 1.2.4(b). There is a diagram of topoi

(1.2.4.1)
$$\begin{array}{ccc} Y & & \\ \downarrow & \xleftarrow{\mathrm{sp}} & \\ s & & \end{array} \qquad\qquad \begin{array}{ccccc} Y & \xhookrightarrow{i} & Y \times_s S & \xhookleftarrow{j} & Y \times_s \eta \\ \downarrow & & \downarrow & & \downarrow \\ s & \xhookrightarrow{i} & S & \xhookleftarrow{j} & \eta \end{array} .$$

The topos $Y \times_s S$ is the union of the open subtopos $Y \times_s \eta$ and its closed complement Y. The formulas in 1.2.2(c) remain valid for $\mathrm{sp} : Y \times_s S \longrightarrow Y$ (also denoted pr_1), $j : Y \times_s \eta \hookrightarrow Y \times_s S$, $i : Y \hookrightarrow Y \times_s S$, and $\mathrm{sp} : Y \times_s \eta \longrightarrow Y$ (also denoted pr_1).

1.2.4.2. If 1.2.4 is taken as the definition of $Y \times_s S$, one must show by a direct construction that, up to unique isomorphism, $Y \times_s S$ does not depend on the choice of a separable closure $k(\bar{\eta})$ of $k(\eta)$. This is done by interpreting the F_η of 1.2.4(b) as a functor which, to each separable closure $k(\bar{\eta})$ of $k(\eta)$ defining a separable closure $k(\bar{s})$ of $k(s)$, associates a sheaf on $Y \otimes_{k(s)} k(\bar{s})$; the F_η of 1.2.4(b) is the restriction of this functor to $\{\bar{\eta}\}$.

1.2.5.

Every geometric point $\bar{x}$ of $Y_{\bar{s}}$ defines a geometric point $(\bar{x}, \bar{\eta})$ of $Y \times_s \eta$, whose corresponding fiber functor is

$$F_\eta \longmapsto (F_{\bar{\eta}})_{\bar{x}}.$$

It defines geometric points $(\bar{x}, \bar{\eta})$ and $(\bar{x}, \bar{s})$ of $Y \times_s S$ by

$$F_{(\bar{x},\bar{\eta})} = (F_{\bar{\eta}})_{\bar{x}} \qquad \text{and} \qquad F_{(\bar{x},\bar{s})} = (F_{\bar{s}})_{\bar{x}}.$$

One verifies that all the points of the topoi $Y \times_s \eta$ and $Y \times_s S$ are obtained in this way. We shall not use this. One also verifies at once that these sets of points are conservative.

***Remark 1.2.6.**

The construction in 1.2.5 defines an equivalence from the category $\mathrm{Point}(Y \times_s \eta)$ (respectively $\mathrm{Point}(Y \times_s S)$) of points of $Y \times_s \eta$ (respectively of $Y \times_s S$) to the 2-fiber product

$$\mathrm{Point}(Y) \times_{\mathrm{Point}(s)} \mathrm{Point}(\eta).$$

This equivalence also follows from the fact that $Y \times_s \eta$ (respectively $Y \times_s S$) is a 2-fiber product of topoi.*

1.2.7.

By the functoriality of 2-fiber products, diagram (1.2.4.1) is functorial in Y and S. In particular:

a) Every morphism $f : Y \longrightarrow Y'$ of schemes over s induces a morphism of topoi from $Y \times_s S$ to $Y' \times_s S$. One has

$$f^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^*F_{\bar{s}}, f^*F_{\bar{\eta}}, f^*(\varphi)), \tag{1.2.7.1}$$

and, when f is quasi-compact,

$$f_*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f_*F_{\bar{s}}, f_*F_{\bar{\eta}}, f_*(\varphi)). \tag{1.2.7.2}$$

Analogous formulas hold for $f : Y \times_s \eta \longrightarrow Y' \times_s \eta$.

b) Every surjective morphism of henselian traits $f : S' \longrightarrow S$ induces a morphism of topoi from $Y_{s'} \times_{s'} S'$ to $Y \times_s S$. Again writing f for the natural morphism $Y_{s'} = Y \otimes_{k(s)} k(s') \longrightarrow Y$, one has

$$f^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^*F_{\bar{s}}, f^*F_{\bar{\eta}}, f^*(\varphi)). \tag{1.2.7.3}$$

When f is finite, f_* is expressed as an induced representation.

1.2.8.

Let $f : Y \longrightarrow Y'$ be a separated morphism, locally of finite type, between s-schemes. Define a "direct image with proper support" functor

$$f_! : (\text{sheaves of pointed sets on } Y \times_s S) \longrightarrow (\text{sheaves of pointed sets on } Y' \times_s S)$$

by the formula

$$f_!(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f_!F_{\bar{s}}, f_!F_{\bar{\eta}}, f_!(\varphi)). \tag{1.2.8.1}$$

When f is a locally closed immersion, $f_!$ is called the extension-by-zero functor.

An analogous functor is defined for $Y \times_s \eta$.

1.2.9.

Let $f : Y \hookrightarrow Y'$ be a locally closed immersion of s-schemes. The functor $f_!$ has a right adjoint $f^!$, "sections with support in Y", satisfying

$$f^!(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^! F_{\bar{s}}, f^! F_{\bar{\eta}}, f^!(\varphi)). \tag{1.2.9.1}$$

Suppose only that f is quasi-finite. The functor $f_!$ from abelian sheaves on $Y \times_s S$ to abelian sheaves on $Y' \times_s S$ still has a right adjoint $f^!$, given by (1.2.9.1) (cf. SGA 4, XVIII, 3.1.8). If f is étale, then $f^! = f^*$.

Analogous functors are defined for $Y \times_s \eta$.

1.3. The functor Ψ

1.3.1.

Let S be a henselian trait as in 0.2.5, and let $p : X \longrightarrow S$. We propose to define a left exact functor Ψ from sheaves on X to sheaves on the topos $X_s \times_s S$ constructed in 1.2. In general, this functor is not the direct-image functor of a morphism of topoi.

We shall also define

$$\Psi_\eta : (\text{sheaves on } X_\eta) \longrightarrow (\text{sheaves on } X_s \times_s \eta).$$

1.3.2.

Put $\bar{X} = X \times_S \bar{S}$. There is a Cartesian diagram

$$\begin{array}{ccccc} X_{\bar{s}} & \xhookrightarrow{\bar{i}} & \bar{X} & \xhookleftarrow{\bar{j}} & X_{\bar{\eta}} \\ \downarrow & & \downarrow & & \downarrow \\ X_s & \hookrightarrow & X & \hookleftarrow & X_\eta \end{array} \tag{1.3.2.1}$$

over the diagram

$$\begin{array}{ccccc} \bar{s} & \hookrightarrow & \bar{S} & \hookleftarrow & \bar{\eta} \\ \downarrow & & \downarrow & & \downarrow \\ s & \hookrightarrow & S & \hookleftarrow & \eta \end{array} .$$

Let F be a sheaf on X_η, with inverse image $F_{\bar{\eta}}$ on $X_{\bar{\eta}}$. Put

$$\Psi_\eta(F) = \bar{i}^* \bar{j}_* F_{\bar{\eta}}. \tag{1.3.2.2}$$

By transport of structure, this sheaf is endowed with a continuous action of $\mathrm{Gal}(\bar{\eta}/\eta)$, compatible with the action of $\mathrm{Gal}(\bar{\eta}/\eta)$ on $X_{\bar{s}}$. The functor

$$\Psi_\eta : (X_{\eta,\mathrm{et}})^\sim \longrightarrow (X_s \times_s \eta)^\sim \tag{1.3.2.3}$$

is left exact.

1.3.3.

Let F be a sheaf on X, and let F_η and F_s be its restrictions to X_η and X_s. Put

$$(\Psi(F))_\eta = \Psi_\eta(F_\eta) \tag{1.3.3.1}$$

and

$$(\Psi(F))_s = F_s. \tag{1.3.3.2}$$

Let $\bar{F}$ be the inverse image of F on $\bar{X}$, and let φ' be the adjunction morphism

$$\varphi' : \bar{F} \longrightarrow \bar{j}_*\bar{j}^*\bar{F}.$$

This morphism induces

$$\varphi : F_{\bar{s}} = \bar{i}^*\bar{F} \longrightarrow \bar{i}^*\bar{j}_*\bar{j}^*\bar{F} = (\Psi_\eta(F_\eta))_{\bar{\eta}}.$$

The triple

$$\Psi(F) = ((\Psi(F))_s, (\Psi(F))_\eta, \varphi)$$

is a sheaf on $X_s \times_s S$. The functor

$$\Psi : (X_{\mathrm{et}})^\sim \longrightarrow (X_s \times_s S)^\sim \tag{1.3.3.3}$$

is left exact.

1.3.4.

When there is no risk of confusion, we shall write

$$\Psi_\eta(F) = (\Psi(F))_\eta, \qquad \Psi_{\bar{\eta}}(F) = (\Psi(F))_{\bar{\eta}}, \qquad \text{and} \qquad \Psi_{\bar{\eta}}(F) = (\Psi_\eta(F))_{\bar{\eta}}.$$

1.3.5.

For the remainder of this section, we list functoriality properties of Ψ (cf. 1.2.7–1.2.9). They will be useful chiefly in their derived form (2.1).

Let $f : X \longrightarrow X'$ be a morphism of S-schemes, giving a diagram

$$\begin{array}{ccccc} X_{\bar{s}} & \xrightarrow{\bar{i}} & \bar{X} & \xleftarrow{\bar{j}} & X_{\bar{\eta}} \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} \\ X'_{\bar{s}} & \xrightarrow{\bar{i}} & \bar{X}' & \xleftarrow{\bar{j}} & X'_{\bar{\eta}} \end{array} \tag{1.3.5.1}$$

(the printed diagram inadvertently omits the bar from its top-middle $\bar{X}$). The ambiguity in the notation for the arrows simplifies the formulas and causes no confusion: only one interpretation makes sense of the composites below. Likewise, we write Ψ for the functor (1.3.3.3) for both X and X'.

The morphisms of functors described below for Ψ have variants for Ψ_η, which we generally omit to state explicitly.

1.3.6.

The base-change morphism $\bar{i}^*f_* \longrightarrow f_*\bar{i}^*$ induces a morphism of functors

$$\Psi f_* \longrightarrow f_*\Psi, \tag{1.3.6.1}$$

which is an isomorphism when f is proper (SGA 4, XII, 5.1(i)).

The essential case is $X' = S$. One obtains a morphism

$$f_* \longrightarrow f_*\Psi \tag{1.3.6.2}$$

whose essential component is

$$\Gamma(X_{\bar{\eta}}, F) \longrightarrow \Gamma(X_{\bar{s}}, \Psi_{\bar{\eta}}(F)); \tag{1.3.6.3}$$

this morphism is an isomorphism when f is proper.

1.3.7.

The base-change morphism $f^*\bar{j}_* \longrightarrow \bar{j}_* f^*$ induces a morphism of functors

$$f^*\Psi \longrightarrow \Psi f^*. \tag{1.3.7.1}$$

1.3.8.

Suppose that f is quasi-finite. The functors $f_!$ commute with base change; the morphism $f_!\bar{j}_* \longrightarrow \bar{j}_* f_!$ induces a morphism of functors

$$f_!\Psi \longrightarrow \Psi f_!. \tag{1.3.8.1}$$

When f is finite, this is the inverse of (1.3.6.1).

1.3.9.

The situation is dual for $f^!$ (we suppose that f is quasi-finite when working with abelian sheaves, or that it is an immersion when working with sheaves of pointed sets). This functor commutes with direct images, and the morphism $\bar{i}^* f^! \longrightarrow f^! \bar{i}^*$ induces a morphism of functors

$$\Psi f^! \longrightarrow f^! \Psi. \tag{1.3.9.1}$$

When f is étale, this is the inverse of (1.3.7.1).

1.3.10.

Finally, a base change $S' \xrightarrow{f} S$ gives a base-change morphism

$$f^*\Psi \longrightarrow \Psi f^*. \tag{1.3.10.1}$$

1.4. The variation

1.4.1.

Let S be a henselian trait as in 0.2.5, and let Y be a scheme over s. Let Λ be a ring. More generally, one could take for Λ a sheaf of rings on Y. For every object K of the derived category $D(Y \times_s S, \Lambda)$ of the category of sheaves of Λ-modules on $Y \times_s S$, we propose to define a variation morphism.

1.4.2.

A complex of sheaves of Λ-modules K on $Y \times_s S$ is identified with a triple (K_s, K_η, φ) in which

a) K_s (respectively K_η) is a complex of sheaves $K_{\bar{s}}$ (respectively $K_{\bar{\eta}}$) on $\bar{Y}$, endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ (respectively $\mathrm{Gal}(\bar{\eta}/\eta)$);

b) φ is an equivariant morphism $\varphi : K_{\bar{s}} \longrightarrow K_{\bar{\eta}}$.

Here and below we retain the scheme Y fixed in 1.4.1; the printed opening sentence of 1.4.2 inadvertently writes $X \times_s S$.

Such a complex K is always homotopic to a complex $K' = (K'_s, K'_\eta, \varphi')$ such that φ' is injective and the exact sequence

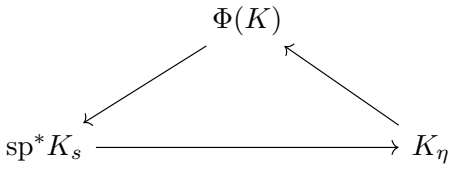
$$0 \longrightarrow K'_{\bar{s}} \longrightarrow K'_{\bar{\eta}} \longrightarrow \mathrm{coker}(\varphi') \longrightarrow 0 \tag{1.4.2.1}$$

is split degree by degree. It is enough to take $K'_s = K_s$ and to take for K'_η the direct sum of K_η and the cone of $\mathrm{sp}^* K_s$. Put

$$\Phi(K) = \mathrm{coker}(\varphi').$$

The exact sequence (1.4.2.1) defines a distinguished triangle

(1.4.2.2)



in the category $K(Y \times_s \eta, \Lambda)$ of complexes of sheaves of Λ-modules on $Y \times_s \eta$, up to homotopy. This triangle depends only on $K \in \mathrm{Ob}\, K(Y \times_s S, \Lambda)$.

The construction passes to the derived category and associates with every object K of $D(Y \times_s S, \Lambda)$ a distinguished triangle of $D(Y \times_s \eta, \Lambda)$.

Triangle (1.4.2.2) is useful chiefly through the long exact hypercohomology sequence that it induces:

$$\cdots \longrightarrow H^i(\bar{Y}, K_{\bar{s}}) \longrightarrow H^i(\bar{Y}, K_{\bar{\eta}}) \longrightarrow H^i(\bar{Y}, \Phi(K)_{\bar{\eta}}) \longrightarrow \cdots .$$

1.4.3.

The inertia group I acts trivially on $K'_{\bar{s}}$. Hence, for every $\sigma \in I$, the endomorphism $\sigma - 1$ of $K'_{\bar{\eta}}$ (notation (1.4.2.1)) factors through a morphism

$$\mathrm{Var}(\sigma) : \mathrm{coker}(\varphi') \longrightarrow K'_{\bar{\eta}}.$$

This construction passes to the derived category: for $K \in \mathrm{Ob}\, D(Y \times_s S, \Lambda)$ and $\sigma \in I$, it defines the variation

$$\mathrm{Var}(\sigma) : \Phi(K)_{\bar{\eta}} \longrightarrow K_{\bar{\eta}}. \tag{1.4.3.1}$$

Writing q for the arrow from K_η to $\Phi(K)$, one has

$$\sigma = 1 + \mathrm{Var}(\sigma) q \quad \text{on } K_{\bar{\eta}}, \tag{1.4.3.2}$$

$$\sigma = 1 + q\,\mathrm{Var}(\sigma) \quad \text{on } \Phi(K)_{\bar{\eta}}, \tag{1.4.3.3}$$

and

$$\mathrm{Var}(\sigma\tau) = \mathrm{Var}(\sigma) q\,\mathrm{Var}(\tau) + \mathrm{Var}(\sigma) + \mathrm{Var}(\tau). \tag{1.4.3.4}$$

2. Vanishing cycles

2.1. The functor $R\Psi$

2.1.1.

Let S be a henselian trait as in 0.2.5, and let Λ be a torsion ring whose torsion is prime to the residual characteristic p of S (0.2.7). More generally, one could take for Λ a torsion sheaf whose torsion is prime to the residual characteristics; the most important case is $\Lambda = \mathbb{Z}/\ell^n$ with ℓ an invertible prime in $\mathcal{O}_S$.

Let X be a scheme over S. The functor Ψ (respectively Ψ_η) carries sheaves of Λ-modules on X (respectively on X_η) to sheaves of Λ-modules on $X_s \times_s S$ (respectively on $X_s \times_s \eta$). Let $R\Psi$ and $R\Psi_\eta$ be the derived functors

$$R\Psi : D^+(X, \Lambda) \longrightarrow D^+(X_s \times_s S, \Lambda),$$

$$R\Psi_\eta : D^+(X_\eta, \Lambda) \longrightarrow D^+(X_s \times_s \eta, \Lambda),$$

and let $R\Phi$ be the composite functor $\Phi \circ R\Psi$:

$$R\Phi : D^+(X, \Lambda) \longrightarrow D^+(X_s \times_s \eta, \Lambda).$$

2.1.2.

Consider the commutative diagram (1.3.2.1)

(1.3.2.1)
$$\begin{array}{ccccc} X_{\bar{s}} & \xhookrightarrow{\bar{i}} & \bar{X} & \xhookleftarrow{\bar{j}} & X_{\bar{\eta}} \\ \downarrow & & \downarrow & & \downarrow \\ X_s & \xhookrightarrow{i} & X & \xhookleftarrow{j} & X_\eta \end{array}$$

If F is an injective sheaf of modules on X, its restriction to the open subscheme X_η is still injective. If F_η is injective on X_η, its inverse image $F_{\bar{\eta}}$ is acyclic. Thus the derived forms of (1.3.3.1) and (1.3.3.2) are

$$R\Psi(K)_s = i^*K \qquad (K \in \mathrm{Ob}\, D^+(X,\Lambda)), \tag{2.1.2.1}$$

$$R\Psi(K)_\eta = R\Psi_\eta(K_\eta) \qquad (K \in \mathrm{Ob}\, D^+(X,\Lambda)), \tag{2.1.2.2}$$

and

$$R\Psi_\eta(K)_{\bar{\eta}} = \bar{i}^* R\bar{j}_* K_{\bar{\eta}} \qquad (K \in \mathrm{Ob}\, D^+(X_\eta,\Lambda)). \tag{2.1.2.3}$$

This allows us to write, without conflict,

$$R\Psi_\eta(K) = (R\Psi(K))_\eta, \qquad R\Psi_{\bar{\eta}}(K) = (R\Psi(K))_{\bar{\eta}}, \qquad \text{and} \qquad R\Psi_{\bar{\eta}}(K) = (R\Psi_\eta(K))_{\bar{\eta}}.$$

With this notation, the distinguished triangle (1.4.2.2) becomes

(2.1.2.4)
$$\begin{array}{ccc} & R\Phi(K) & \\ \swarrow & & \nwarrow \\ \mathrm{sp}^* i^* K & \longrightarrow & R\Psi_\eta(K_\eta) \end{array} .$$

The printed bottom-right term has K_s; the defining identity (2.1.2.2) requires K_η, as written here.

For $\sigma \in I$, there is a variation morphism

$$\mathrm{Var}(\sigma) : R\Phi(K)_{\bar{\eta}} \longrightarrow R\Psi_{\bar{\eta}}(K). \tag{2.1.2.5}$$

The sheaves $R^i\Phi(\Lambda)$, or more generally $R^i\Phi(K)$, are the sheaves of vanishing cycles.

2.1.3.

Let $(\bar{x},\bar{\eta})$ be the geometric point of $X_s \times_s \eta$ defined by a geometric point $\bar{x}$ of $X_{\bar{s}}$ (1.2.5). We compute the fiber of $R\Psi_\eta(K_\eta)$ at $(\bar{x},\bar{\eta})$ using (2.1.2.3) (cf. I, 2.3). The strict henselization $X_{(\bar{x})}$ of X at $\bar{x}$ is a scheme over the strict henselization S_{nr} of S, with geometric generic point η_{nr}.

Proposition 2.1.4.

One has

$$(R\Psi(K_\eta))_{(\bar{x},\bar{\eta})} = R\Gamma\big(X_{(\bar{x})} \times_{\eta_{\mathrm{nr}}} \bar{\eta}, K\big)\,;$$

in particular,

$$R^i\Psi(K_\eta)_{(\bar{x},\bar{\eta})} = \mathbb{H}^i\big(X_{(\bar{x})} \times_{\eta_{\mathrm{nr}}} \bar{\eta}, K\big)\,.$$

Apply SGA 4, XV, 2.1 (the local-acyclicity theorem for smooth morphisms). This gives the following restatement.

Restatement 2.1.5.

If f is smooth and F is a locally constant sheaf, then

$$R\Phi(F) = 0.$$

More generally, if $K \in \mathrm{Ob}\, D^+(X, \Lambda)$, then $R\Phi(K) = 0$ wherever f is smooth and the sheaves $\mathcal{H}^i(K)$ are locally constant. See also 2.1.11.

2.1.6.

Let $f : Y \longrightarrow Y'$ be a morphism of schemes over s. The functors f^*, f_*, $f^!$, and $f_!$ (1.2.7–1.2.9) have natural analogues in the derived category.

a) The functor f^* is exact and therefore derives trivially.

b) The functor f_* has a right derived functor Rf_*. When f is quasi-compact, one has

(2.1.6.1) $$(Rf_*K)_{\bar{s}} = Rf_{\bar{s}*}(K_{\bar{s}})$$

and

(2.1.6.2) $$(Rf_*K)_{\bar{\eta}} = Rf_{\bar{s}*}(K_{\bar{\eta}}).$$

c) When f is quasi-finite, the functor $f_!$ is exact and therefore derives trivially. In general, the derived functor of $f_!$ is pathological, and one proceeds as in SGA 4, XVII. Suppose that f is separated and of finite type and that Y' is noetherian. There is then [2] a factorization $f = \widetilde{f}\, j$, where j is an open immersion and $\widetilde{f}$ is proper, and one defines

$$Rf_! = R\widetilde{f}_* \circ j_!.$$

The arguments of SGA 4, XVII show that $Rf_!$ is independent of the choice of the factorization $\widetilde{f}\, j$. The analogues of (2.1.6.1) and (2.1.6.2) hold.

d) We shall use $Rf^!$ only for a quasi-finite morphism, in which case it is the derived functor of $f^!$. Again, the analogues of (2.1.6.1) and (2.1.6.2) hold.

2.1.7.

We now review the functoriality properties of $R\Psi$, that is, we derive 1.3.5–1.3.10. Let $f : X \longrightarrow X'$ be a morphism of S-schemes, and retain the notation of 1.3.5.

The morphism (1.3.6.1) defines

(2.1.7.1) $$R\Psi\, Rf_* \longrightarrow Rf_*\, R\Psi.$$

By the proper base-change theorem (SGA 4, XII, 5.1), this morphism is an isomorphism if f is proper.

The morphism (1.3.7.1) defines

(2.1.7.2) $$f^*R\Psi \longrightarrow R\Psi f^*.$$

By the smooth base-change theorem (SGA 4, XVI, 1.2), this morphism is an isomorphism if f is smooth.

The isomorphism (2.1.7.1) for f proper and the morphism (1.3.8.1) define a morphism

(2.1.7.3) $$Rf_!R\Psi \longrightarrow R\Psi Rf_!,$$

which, when f is proper, is the inverse of (2.1.7.1). The print mistakenly refers here to the later special case (2.1.8.1).

When f is quasi-finite, (1.3.9.1) derives to

$$R\Psi Rf^! \longrightarrow Rf^! R\Psi, \tag{2.1.7.4}$$

which, when f is étale, is the inverse of (2.1.7.2). The printed reference to (2.1.7.1) is incompatible with the two functors involved.

Let $f : S' \longrightarrow S$ be a change of trait, and put $X' = X \times_S S'$:

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ S' & \xrightarrow{f} & S \end{array}$$

(for the ambiguity in the notation, see 1.3.5). Suppose that f is surjective. The base-change arrow (1.3.10.1) derives to

$$f^* R\Psi \longrightarrow R\Psi f^*. \tag{2.1.7.5}$$

The same statements hold for $R\Psi_\eta$.

2.1.8.

The essential case of (2.1.7.1) and (2.1.7.3) is $X' = S$. For $K \in \mathrm{Ob}\, D^+(X_{\bar\eta}, \Lambda)$ one then obtains morphisms

$$Rf_* \longrightarrow Rf_* R\Psi \tag{2.1.8.1}$$

and

$$Rf_! R\Psi \longrightarrow Rf_!, \tag{2.1.8.2}$$

hence

$$\mathbb{H}^i(X_{\bar\eta}, K) \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \tag{2.1.8.3}$$

and

$$\mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \longrightarrow \mathbb{H}^i_c(X_{\bar\eta}, K), \tag{2.1.8.4}$$

and the commutativity of the diagrams

$$\begin{array}{ccc} (R^i f_* K)_{\bar s} & \xrightarrow{\mathrm{sp}} & \mathbb{H}^i(X_{\bar\eta}, K) \\ \downarrow & & \downarrow \\ \mathbb{H}^i(X_{\bar s}, K) & \longrightarrow & \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \end{array} \tag{2.1.8.5}$$

and

$$\begin{array}{ccc} \mathbb{H}^i_c(X_{\bar s}, K) & \xrightarrow{\qquad\mathrm{sp}\qquad} & \mathbb{H}^i_c(X_{\bar\eta}, K) \\ & \searrow \quad \nearrow & \\ & \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) & \end{array} \tag{2.1.8.6}$$

.

For $K \in \mathrm{Ob}\, D^+(X_\eta, \Lambda)$ and $\sigma \in I$, the following diagrams are commutative:

$$\begin{array}{ccc} \mathbb{H}^i(X_{\bar\eta}, K) & \xrightarrow{\qquad\sigma-1\qquad} & \mathbb{H}^i(X_{\bar\eta}, K) \\ \downarrow & & \downarrow \\ \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) & \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Phi_{\bar\eta}(K)) \xrightarrow{\mathrm{Var}(\sigma)} & \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \end{array} \tag{2.1.8.7}$$

and

$$\begin{array}{ccc} \mathbb{H}^i_c(X_{\bar\eta}, K) & \xrightarrow{\qquad\sigma-1\qquad} & \mathbb{H}^i_c(X_{\bar\eta}, K) \\ \uparrow & & \uparrow \\ \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) & \longrightarrow \mathbb{H}^i_c(X_{\bar s}, R\Phi_{\bar\eta}(K)) \xrightarrow{\mathrm{Var}(\sigma)} & \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \end{array} \tag{2.1.8.8}$$

.

In (2.1.8.8) the English restores the bar omitted from the printed bottom-left $X_{\bar s}$ and the missing closing parenthesis in the bottom-middle term. In (2.1.8.7), however, the upper connector really is printed as a rule without an arrowhead, whereas (2.1.8.8) has an arrow; this source-visible distinction is retained.

When f is proper, (2.1.8.1) and (2.1.8.2), as well as (2.1.8.3) and (2.1.8.4), are mutually inverse isomorphisms. The distinguished triangle (2.1.2.4) gives a long exact sequence

$$\cdots \longrightarrow \mathbb{H}^i(X_{\bar s}, K) \longrightarrow \mathbb{H}^i(X_{\bar\eta}, K) \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Phi_{\bar\eta}(K)) \longrightarrow \cdots . \tag{2.1.8.9}$$

Thus the vanishing complex $R\Phi(K)$ expresses the difference between the cohomologies of the geometric fibers $X_{\bar s}$ and $X_{\bar\eta}$, with coefficients in the inverse image of K. By (2.1.8.7), the variation $\mathrm{Var}(\sigma)$ determines the action of the inertia group I on $\mathbb{H}^i(X_{\bar\eta}, K)$. These facts underlie all applications of the method of vanishing cycles.

The properness hypothesis imposed above on f may be replaced by regularity-at-infinity conditions on f and K. For example, one has the following result (for $K \in \mathrm{Ob}\, D(X_{\bar\eta}, \Lambda)$).

Proposition 2.1.9.

Suppose that f admits a factorization $f = f_1 k$

$$X \xhookrightarrow{\quad k \quad} X_1 \xrightarrow{\quad f_1 \quad} S,$$

where f_1 is proper and k identifies X with the complement in X_1 of a relative normal-crossings divisor D (thus, by hypothesis, f_1 is smooth on a neighborhood of D). If, on a neighborhood U of D, the sheaves $\mathcal{H}^i(K)$ are locally constant on $U - (D \cup X_0)$ and tamely ramified along the normal-crossings divisor $D \cup X_0$, then (2.1.8.1) and (2.1.8.2) are isomorphisms.

The hypercohomology spectral sequence for K shows that it is enough to treat the case in which K consists of a sheaf F_η in degree zero. After replacing S by its normalization in a finite extension (which is permissible, as one sees directly from the definitions) and applying Abhyankar's lemma, we may suppose that F_η is the restriction to X_η of a sheaf F on X, locally constant on $U \cap X$ and tamely ramified along D. With the notation of 2.1.2, let $\overline{F}$ be the inverse image of F on $\overline{X}$, and write $\bar f : \overline{X} \longrightarrow \overline{S}$.

To prove that (2.1.8.1) is an isomorphism, it is enough to prove that

$$R\bar f_*\big(R\bar j_* \bar j^* \overline{F}\big)$$

commutes with base change by $\bar s \longrightarrow \overline{S}$. In the distinguished triangle

$$\longrightarrow \overline{F} \longrightarrow R\bar j_* \bar j^* \overline{F} \longrightarrow \Delta \longrightarrow,$$

the sheaves $\mathcal{H}^i(\Delta)$ are supported on $X_{\bar s}$; for Δ, the base-change statement in question is tautological. It therefore remains to prove that $R\bar f_*(\overline{F})$ commutes with base change by $\bar s \longrightarrow \overline{S}$, or, in

view of the base-change theorem for the proper morphism f_1, that $Rk_*(\overline{F})$ commutes with base change $X_{1\bar{s}} \longrightarrow X_1$. This is Lemma 2.1.10 below.

To prove that (2.1.8.2) is an isomorphism, it is enough to prove, for the sheaf F, that extension by zero $k_!$ commutes with the functor $R\bar{j}_*$; that is, on a neighborhood of D,

$$R\Phi(k_!F) = 0 \qquad \text{on } X_{1\bar{s}}.$$

This is Lemma 2.1.11 below.

Lemma 2.1.10.

Let $f : X \longrightarrow S$ be smooth, let $D \subset X$ be a relative normal-crossings divisor, put $U = X - D \hookrightarrow^j X$, and let F be a sheaf of Λ-modules on U which is locally constant and tamely ramified along D. Formation of Rj_*F commutes with arbitrary base change $T \longrightarrow S$.

The question is local on X for the étale topology. We may therefore suppose that there is a morphism $g : X' \longrightarrow X$, where X' is the normalization of X in a finite étale surjective covering $g' : U' \longrightarrow U$, tamely ramified along D, such that g'^*F is constant on U'. By Abhyankar's lemma, we may further suppose that X' is smooth over S and that U' is the complement in X' of a relative normal-crossings divisor D'.

The sheaf F injects into $g'_*g'^*F$, and the cokernel again satisfies the hypotheses of the lemma. Iterating this construction (the same g continues to work) gives a resolution of F by sheaves of the form g'_*G, where G is constant and g' is as above. It therefore suffices to prove the assertion for such a sheaf, or, equivalently, for $(X'/S, D', G)$ with G constant on U'. This is a special case of SGA 5, II, 4.4.

Lemma 2.1.11.

Let S be a henselian trait, let $f : X \longrightarrow S$ be smooth, let $D \subset X$ be a relative normal-crossings divisor, put $U = X - D \hookrightarrow^j X$, and let F be a sheaf of Λ-modules on U which is locally constant and tamely ramified along D. Then

$$R\Phi(j_!F) = 0.$$

The dévissage just used reduces us to the case in which F is constant, with value G. We may suppose that D is the union of smooth divisors $(D_i)_{i\in I}$. For $J \subset I$, let D_J be the intersection of the D_i for $i \in J$. The sheaf F has a resolution $F^\bullet$ whose terms are sums of sheaves G_J, constant on D_J and extended by zero to X. It is therefore enough to prove the lemma for one such sheaf, and this follows from (2.1.5), applied to D_J/S.

I conjecture that (2.1.7.5) is an isomorphism when S and S' are excellent. Here is a partial result (see also 2.4.2).

Proposition 2.1.12.

The morphism (2.1.7.5) is an isomorphism in the following cases:

(a) S is excellent and S' is its completion;
(b) S and S' have equal characteristic zero.

In case (b), we reduce to the situation in which S and S' are strictly local and a uniformizer of S is still a uniformizer of S' (extend the scalars without changing $\bar{S}' \longrightarrow \bar{S}$). In both cases (a) and (b), writing an overbar for normalization in a separable closure of the fraction field, one then has

$$\bar{S}' \simeq S' \times_S \bar{S}.$$

Returning to the definitions, it is enough to prove that $S' \longrightarrow S$ is universally locally acyclic (apply SGA 4, XVI, 1.1). This follows from I, 0.5.1 (based on Artin's version of Néron desingularization) and SGA 4, XV, 2.1 (the local-acyclicity theorem for smooth morphisms).

2.1.13.

Suppose that $f : X \longrightarrow S$ is of finite type. The cohomological dimension (SGA 4, XVII, 1.2.9) of $R\Psi$ is then at most $\dim(X_\eta)$, as follows from 2.1.4 and SGA 4, XIV, 3.2 (cf. I, 4.2(i)). Thus $R\Psi$ extends to the entire derived category,

$$R\Psi : D(X, \Lambda) \longrightarrow D(X_s \times_s S, \Lambda),$$

and it carries D^- into D^-. Let $L \in \mathrm{Ob}\, D^-(\Lambda)$ be a bounded-above complex of right Λ-modules, and let $K \in \mathrm{Ob}\, D^-(X, \Lambda)$. There is then an isomorphism

$$L \overset{\mathbf{L}}{\otimes}_\Lambda R\Psi(K) \xrightarrow{\sim} R\Psi\big(L \overset{\mathbf{L}}{\otimes}_\Lambda K\big). \tag{2.1.13.1}$$

To verify (2.1.13.1), first replace L by a complex of free Λ-modules. Since $R\Psi$ is triangulated and has finite cohomological dimension, it is enough to verify the assertion after replacing L by one of its terms. Passage to the filtered colimit reduces us to the case in which L is a finite free module, and then to the tautological case $L = \Lambda$.

It follows from (2.1.13.1), by the usual argument (cf. SGA 4, XVII, 5.2.11), that if $K \in \mathrm{Ob}\, D(X_\eta, \Lambda)$ has finite Tor-dimension at most d, where $d \in \mathbb{Z}$ (SGA 4, XVII, 4.1.9), then $R\Psi_\eta(K)$ likewise has Tor-dimension at most d.

2.2. A compatibility (trace morphisms)

2.2.1.

Let S be a henselian trait, and let $u : X_s \longrightarrow Y_s$ be a separated, quasi-finite, flat morphism of finite presentation between s-schemes. For every abelian sheaf F on Y_s, there is a trace morphism

$$\mathrm{Tr}_u : u_! u^* F \longrightarrow F$$

(SGA 4, XVII, 6.2.3). If F is an abelian sheaf on $Y_s \times_s S$ or on $Y_s \times_s \eta$, this gives an analogous morphism $\mathrm{Tr}_u : u_! u^* F \longrightarrow F$.

Now let $u : X \longrightarrow Y$ be a separated, quasi-finite, flat morphism of finite presentation between S-schemes. We again write u for the induced morphism

$$X_s \times_s S \longrightarrow Y_s \times_s S,$$

and we use Ψ for the functors of 1.3.3, both for X and for Y.

Proposition 2.2.2.

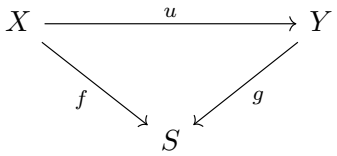
For every abelian sheaf F on Y, the following diagram is commutative:

$$\begin{array}{ccccc} u_! u^* \Psi F & \xrightarrow{(1.3.7.1)} & u_! \Psi u^* F & \xrightarrow{(1.3.8.1)} & \Psi u_! u^* F \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & & & & \downarrow{\scriptstyle \mathrm{Tr}_u} \\ \Psi F & = & = & = & \Psi F. \end{array}$$

The verification is left to the reader. There is a variant for Ψ_η and for F on Y_η.

2.2.3.

Let Λ be as in 2.1.1, and let $u : X \longrightarrow Y$ be a quasi-finite flat morphism between separated, finite-type, flat S-schemes:



The print refers here to 2.2.1, but the coefficient ring Λ is fixed in 2.1.1. We again write f and g for the morphisms

$$X_s \times_s S \longrightarrow S, \qquad Y_s \times_s S \longrightarrow S,$$

and suppose that $\dim Y_\eta \leq n$. For every $K \in \mathrm{Ob}\, D(S, \Lambda)$, there is a trace morphism (SGA 4, XVIII, 2.9)

$$\mathrm{Tr}_f : Rf_!f^*K(n)[2n] \longrightarrow K.$$

Corollary 2.2.4.

The following diagram is commutative:

$$\begin{array}{ccccc} Rg_!u_!u^*R\Psi\, g^*K(n)[2n] & \xrightarrow{(1.3.7.1)} & Rf_!R\Psi\, f^*K(n)[2n] & \xrightarrow{(2.1.10.1)} & Rf_!f^*K(n)[2n] \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & & & & \downarrow{\scriptstyle \mathrm{Tr}_f} \\ Rg_!R\Psi\, g^*K(n)[2n] & \xrightarrow{(2.1.10.1)} & Rg_!g^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_g} & K \end{array}$$

If u is étale, (1.3.7.1) is an isomorphism.

With the arguments suppressed, the diagram (2.2.4) expands as follows:

$$\begin{array}{ccccccc} Rg_!u_!u^*R\Psi\, g^* & \longrightarrow & Rg_!R\Psi\, u_!u^*g^* & \longrightarrow & Rg_!u_!u^*g^* & \xrightarrow{\sim} & Rf_!f^* \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & (1) & \downarrow{\scriptstyle \mathrm{Tr}_u} & (2) & \downarrow{\scriptstyle \mathrm{Tr}_u} & (3) & \downarrow{\scriptstyle \mathrm{Tr}_f} \\ Rg_!R\Psi\, g^* & = & Rg_!R\Psi\, g^* & \longrightarrow & Rg_!g^* & \xrightarrow{\mathrm{Tr}_g} & K. \end{array}$$

The commutativity of (1) follows from 2.2.2; that of (2) is the functoriality of (2.1.10.2); and that of (3) is the transitivity of the trace (SGA 4, XVIII, 2.9, Variant 3).

2.2.4.1.

There is an analogous result for Ψ_η and $K \in \mathrm{Ob}\, D(\eta, \Lambda)$.

2.2.5.

Let Φ be a subset of Y_s which is proper over s, and write Rg_Φ for the "direct image with support in Φ" functor. Let Tr_g denote the composite

$$\begin{aligned} \mathrm{Tr}_g : Rg_\Phi R\Psi g^*K(n)[2n] &\longrightarrow Rg_!R\Psi g^*K(n)[2n] \\ &\longrightarrow Rg_!g^*K(n)[2n] \longrightarrow K, \end{aligned} \tag{2.2.5.1}$$

and similarly for Ψ_η and $K \in D(\eta, \Lambda)$. The print has Rf_Φ as the first functor in (2.2.5.1); the immediately preceding definition and the remainder of the composite require Rg_Φ, which is used here.

If Φ is a finite set of closed points, then

$$Rg_\Phi \circ u_! = Rf_{u^{-1}\Phi},$$

and hence there is a trace morphism

$$\mathrm{Tr}_u : Rf_{u^{-1}\Phi}u^* = Rg_\Phi u_! u^* \longrightarrow Rg_\Phi.$$

By 2.2.4, the following diagram is commutative:

$$\begin{array}{ccccc} Rf_{u^{-1}\Phi}u^*R\Psi\, g^*K(n)[2n] & \longrightarrow & Rf_{u^{-1}\Phi}R\Psi\, f^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_f} & K \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & & & & \| \\ Rg_\Phi R\Psi\, g^*K(n)[2n] & & \xrightarrow[\mathrm{Tr}_g]{} & & K. \end{array}$$

Suppose that u is étale, let $x \in X_s$, and suppose that x and $y = u(x)$ have the same residue field. Then

$$Rf_{\{x\}} \xrightarrow{\sim} Rg_{\{y\}}.$$

The print has $\{x\}$ also on the right, but the support for g lies in Y_s and is $\{y\}$, as the following diagram confirms. This isomorphism, given by Tr_u, expresses the étale locality of cohomology with support at x. The diagram

$$\begin{array}{ccc} Rf_{\{x\}}R\Psi\, f^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_f} & K \\ \| & & \| \\ Rg_{\{y\}}R\Psi\, g^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_g} & K \end{array} \qquad (2.2.5.2)$$

is commutative.

Scholium 2.2.6.

The compatibility (2.2.5.2) says that the trace morphism (2.2.5.1), for $\Phi = \{y\}$, is a local invariant: it depends only on the henselization of the S-scheme X at x. The same holds for $R\Psi_\eta$. The print's references to (2.3.5.2) and (2.3.5.1) are the immediately preceding (2.2.5.2) and (2.2.5.1); likewise, $\Phi \subset Y_s$ makes $\{y\}$, rather than the printed $\{x\}$, the relevant support.

2.3. Finiteness theorem (in equal characteristic zero)

Theorem 2.3.1.

Let S be a henselian trait as in 0.2.5, let Λ be a noetherian torsion ring, and let $f : X \longrightarrow S$ be a morphism of finite type. Suppose that S has equal characteristic zero. Then, for every constructible sheaf of Λ-modules F on X_η, the sheaves $R^i\Psi_{\bar\eta}(F)$ on $X_{\bar s}$ are constructible.

We begin by repeating the opening of the proof of SGA 4, XIV, 1.10. Apply SGA 4, IX, 2.14(ii). There is a resolution $F^\bullet$ of F

$$\begin{array}{ccccccccc} & & & & F & & & & \\ & & & & \downarrow & & & & \\ 0 & \longrightarrow & F^0 & \longrightarrow & F^1 & \longrightarrow & \cdots & & \end{array}$$

by sheaves of the form $\bigoplus_i p'_{i*}G_i$, where $p'_i : Y_{i\eta} \longrightarrow X_\eta$ is finite and G_i is a constructible constant sheaf. We may suppose that $Y_{i\eta}$ is reduced; let $p_i : Y_i \longrightarrow X$ be the normalization of X in $Y_{i\eta}$. Using the spectral sequence of the resolution $F^\bullet$ and applying (2.1.7.1) in the elementary case of the finite morphism p_i, we see that it suffices to prove the theorem for each pair (Y_i, G_i).

We next prove the theorem when F is the constant sheaf defined by a Λ-module M. Recall that Λ is a $\mathbb{Z}/n$-algebra for a suitable n. The module M may therefore be dévissé into modules M_i over $\Lambda/\ell_i\Lambda$, where each $\ell_i \mid n$ is a suitable prime. This reduces us to the case in which Λ is a $\mathbb{Z}/\ell$-algebra, with ℓ prime. Formula (2.1.13.1) then gives

$$M \otimes_{\mathbb{Z}/\ell} R^i\Psi_{\bar\eta}(\mathbb{Z}/\ell) \xrightarrow{\sim} R^i\Psi_{\bar\eta}(M).$$

It remains to treat the constant sheaf $F = \mathbb{Z}/\ell$.

Choose a descending sequence of closed subsets

$$X_\eta = X'_0 \supset \cdots \supset X'_n$$

such that $(X'_i - X'_{i+1})_{\mathrm{red}}$ is smooth over η, and let $X_i = \overline{X'_i}$ be the closure in X. The constant sheaf $\mathbb{Z}/\ell$ is dévissé into the constant sheaves $\mathbb{Z}/\ell$ on the strata $X'_i - X'_{i+1}$, extended by zero. Thus:

Reduction 1. It is enough to prove Theorem 2.3.1 for a sheaf $j_!\mathbb{Z}/\ell$, where $j : U \hookrightarrow X_\eta$ is the inclusion of a smooth open subset.

By resolution of singularities there is a morphism $g : X_1 \longrightarrow X$ such that

(a) g is proper and induces an isomorphism $g^{-1}(U) \xrightarrow{\sim} U$;
(b) X_1 is regular and $X_1 - g^{-1}(U)$ is a normal-crossings divisor in X_1, which is a union of regular divisors D_α.

Let $j_1 : g^{-1}(U) \hookrightarrow X_1$. Then

$$Rg_* j_{1!}\mathbb{Z}/\ell = j_!\mathbb{Z}/\ell.$$

Applying (2.1.7.1) and the finiteness theorem SGA 4, XIV, 1.1, we see that it suffices to prove Theorem 2.3.1 for $(X_1, j_{1!}\mathbb{Z}/\ell)$.

There is a resolution

$$0 \longrightarrow j_{1!}\mathbb{Z}/\ell \longrightarrow \mathbb{Z}/\ell \longrightarrow \bigoplus_\alpha (\mathbb{Z}/\ell)_{D_\alpha} \longrightarrow \bigoplus_{\alpha,\beta} (\mathbb{Z}/\ell)_{D_\alpha \cap D_\beta} \longrightarrow \cdots,$$

and hence:

Reduction 2. It is enough to prove Theorem 2.3.1 for the constant sheaf $\mathbb{Z}/\ell$ on a regular S-scheme X such that X_s is a normal-crossings divisor in X.

In this final case one uses the explicit computation I, 3.3, which in turn uses the purity theorem SGA 4, XIX, 3.2 and the characteristic-zero hypothesis.

2.4. Isolated singularities

2.4.1.

Throughout this section, for simplicity, we suppose that S is a strictly henselian trait. Let Λ be a noetherian torsion ring whose torsion is prime to the residual characteristic of S, let $f : X \longrightarrow S$ be a morphism of finite type, and let F be a constructible sheaf of Λ-modules on X. A point $x \in X$ is called a smooth point of (X, F) if f is smooth at x and F is locally constant in a neighborhood of x. Let Σ denote the set of points where (X, F) is not smooth, and let Σ_s be its special fiber. We know that $R\Phi(F) = 0$ outside Σ_s.

Theorem 2.4.2.

At every isolated point x of Σ_s, the modules $R^i\Phi_{\bar{\eta}}(F)_x$ are Λ-modules of finite type and are invariant under change of trait (via 2.1.7.1).

In equal characteristic zero, these assertions were proved in 2.1.12 and 2.3.1 without assuming that x is isolated.

Let x be an isolated point of Σ_s. The theorem is local in a neighborhood of x, so we may suppose that f is proper. We then have the long exact sequence (2.1.8.9)

$$\cdots \longrightarrow H^i(X_s, F) \longrightarrow H^i(X_{\bar{\eta}}, F) \longrightarrow \mathbb{H}^i\big(X_s, R\Phi_{\bar{\eta}}(F)\big) \longrightarrow \cdots .$$

In this sequence, $H^i(X_s, F)$ and $H^i(X_{\bar{\eta}}, F)$ are invariant under change of trait and are Λ-modules of finite type (SGA 4, XIV, 1.1). The modules $\mathbb{H}^i(X_s, R\Phi_{\bar{\eta}}(F))$ therefore have the same properties. Recall the following lemma.

Lemma 2.4.3.

Let (T, Λ) be a ringed topos, let Z be a closed subtopos of T, and let $i : Z \hookrightarrow T$ be the inclusion. Then the functor

$$i_* : D(Z, \Lambda) \longrightarrow D(T, \Lambda)$$

is an equivalence from $D(Z, \Lambda)$ onto the full subcategory of $D(T, \Lambda)$ consisting of the complexes K for which $\mathcal{H}^i(K)$ is supported on Z.

Indeed, for such a complex K, the morphism $K \longrightarrow i_*i^*K$ is a quasi-isomorphism.

We may consequently regard $R\Phi_{\bar{\eta}}(F)$ as an object of

$$D^b(\Sigma_s, \Lambda) = D^b(\Sigma_s - \{x\}, \Lambda) \oplus D^b(\{x\}, \Lambda).$$

Passing to a direct summand, we find that

$$\mathbb{H}^i\big(\{x\}, R\Phi_{\bar{\eta}}(F)\big) = R^i\Phi_{\bar{\eta}}(F)_x$$

has the properties stated in 2.4.2. This completes the proof.

Remark 2.4.4.

The same proof shows that, if K is constructible in $D^+(X, \Lambda)$ and x is isolated in the support of $R\Phi_{\bar{\eta}}(K)$, then the cohomology modules of the stalk $R\Phi_{\bar{\eta}}(K)_x$ are of finite type over Λ. If, after a change of trait, x remains an isolated point of the support, these modules are invariant under that change of trait.

2.4.5.

Let $x \in X_s$ be an isolated point where (X, F) is not smooth. More generally, let $K \in \mathrm{Ob}\, D^b(X, \Lambda)$, and let x be an isolated point of the support of $R\Phi(K)$. Let $i : \{x\} \hookrightarrow X_s$ be the inclusion. The complex $i_*i^*R\Phi(K)$, which is isomorphic to $R\Phi(K)$ in a neighborhood of x, is a direct summand of $R\Phi(K)$ by 2.4.3. Thus, for $\sigma \in I$, the variation morphism induces

$$\mathrm{Var}(\sigma) : i_*i^*R\Phi_{\bar{\eta}}(K) \longrightarrow R\Psi_{\bar{\eta}}(K),$$

that is,

$$\mathrm{Var}(\sigma) : \big(R\Phi_{\bar{\eta}}(K)\big)_x \longrightarrow R\Gamma_{\{x\}}\big(R\Psi_{\bar{\eta}}(K)\big). \tag{2.4.5.1}$$

The printed $R\Phi(F)$ in the direct-summand sentence is $R\Phi(K)$ in this general formulation.

2.4.6.

Suppose that the support Σ of $R\Phi(K)$ consists of finitely many isolated points. For every $\sigma \in I$, the endomorphism $\sigma - 1$ of $R\Psi_{\bar{\eta}}(K)$ is then the composite displayed in the following diagram:
(2.4.6.1)

$$\begin{array}{ccc} R\Psi_{\bar{\eta}}(K) & \longrightarrow & R\Phi_{\bar{\eta}}(K) = \bigoplus_{x\in\Sigma} i_{x*}\big(R\Phi_{\bar{\eta}}(K)\big)_x \\ \Big\downarrow{\scriptstyle \sigma-1} & & \Big\downarrow{\scriptstyle \bigoplus i_{x*}\,\mathrm{Var}(\sigma)} \\ R\Psi_{\bar{\eta}}(K) & \longleftarrow & \bigoplus_{x\in\Sigma} i_{x*}R\Gamma_{\{x\}}R\Psi_{\bar{\eta}}(K) \end{array}$$

If f is proper, applying H^i to (2.4.6.1) shows that the endomorphism $\sigma - 1$ of $\mathbb{H}^i(X_{\bar{\eta}}, K)$ is the composite
(2.4.6.2)

$$\begin{array}{ccccc} \mathbb{H}^i(X_{\bar{\eta}}, K) & \xrightarrow{\sim} & \mathbb{H}^i\big(X_s, R\Psi_{\bar{\eta}}(K)\big) & \longrightarrow & \mathbb{H}^i\big(X_s, R\Phi_{\bar{\eta}}(K)\big) = \bigoplus_{x\in\Sigma} R^i\Phi_{\bar{\eta}}(K)_x \\ \Big\downarrow{\scriptstyle \sigma-1} & & & & \Big\downarrow{\scriptstyle \bigoplus \mathrm{Var}(\sigma)} \\ \mathbb{H}^i(X_{\bar{\eta}}, K) & \xleftarrow{\sim} & \mathbb{H}^i\big(X_s, R\Psi_{\bar{\eta}}(K)\big) & \longleftarrow & \bigoplus_{x\in\Sigma} \mathbb{H}^i_{\{x\}}\big(X_s, R\Psi_{\bar{\eta}}(K)\big) \end{array}$$

This describes the action of I on $\mathbb{H}^i(X_{\bar{\eta}}, K)$; its essential ingredients are the variation morphisms (2.4.5.1), which are local on X. We also have the long exact sequence

$$\longrightarrow \mathbb{H}^i(X_s, K) \longrightarrow \mathbb{H}^i(X_{\bar{\eta}}, K) \longrightarrow \bigoplus_{x\in\Sigma} R^i\Phi_{\bar{\eta}}(K)_x \longrightarrow \cdots . \tag{2.4.6.3}$$

The same argument applies when f is not necessarily proper, provided that (X, K) satisfies 2.1.9. The diagram (2.4.6.5) below defines a morphism

$$\mathrm{Var}(\sigma) : R\Gamma(X_{\bar{\eta}}, K) \longrightarrow R\Gamma_c(X_{\bar{\eta}}, K) \tag{2.4.6.4}$$

such that the endomorphism $\sigma - 1$ of $R\Gamma(X_{\bar{\eta}}, K)$ (respectively, $R\Gamma_c(X_{\bar{\eta}}, K)$) is the composite $\alpha\,\mathrm{Var}(\sigma)$ (respectively, $\mathrm{Var}(\sigma)\alpha$), where

$$\alpha : R\Gamma_c(X_{\bar{\eta}}, K) \longrightarrow R\Gamma(X_{\bar{\eta}}, K).$$

(2.4.6.5)

$$\begin{array}{ccccc}
R\Gamma(X_{\bar\eta},K) & \xrightarrow{\sim} & R\Gamma\big(X_s,R\Psi_{\bar\eta}(K)\big) & \longrightarrow & \bigoplus_{x\in\Sigma}\big(R\Phi_{\bar\eta}(K)\big)_x \\
\Big\downarrow{\scriptstyle \mathrm{Var}(\sigma)} & & & & \Big\downarrow{\scriptstyle \oplus} \\
R\Gamma_c(X_{\bar\eta},K) & \xrightarrow[\sim]{} & R\Gamma_c\big(X_s,R\Psi_{\bar\eta}(K)\big) & \longleftarrow & \bigoplus_{x\in\Sigma} R\Gamma_{\{x\}}\big(X_s,R\Psi_{\bar\eta}(K)\big)
\end{array} .$$

In this case we also have the long exact sequences
(2.4.6.6)

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & \mathbb{H}^i_c(X_s,K) & \longrightarrow & \mathbb{H}^i_c(X_{\bar\eta},K) & \longrightarrow & \bigoplus_{x\in\Sigma}\big(R^i\Phi_{\bar\eta}(K)\big)_x & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \| & & \\
\cdots & \longrightarrow & \mathbb{H}^i(X_s,K) & \longrightarrow & \mathbb{H}^i(X_{\bar\eta},K) & \longrightarrow & \bigoplus_{x\in\Sigma} R^i\Phi_{\bar\eta}(K)_x & \longrightarrow & \cdots
\end{array}$$

The printed reference (2.4.4.1) in the discussion following (2.4.6.2) is the variation morphism (2.4.5.1).

2.4.6. (second occurrence in the source)

Let $X_{s(x)}$ be the henselization of X_s at x, and put

$$X^*_{s(x)} = X_{s(x)} - \{x\}.$$

The long hypercohomology exact sequences of $(X_{s(x)},\{x\})$, with coefficients in K_s, $R\Psi_{\bar\eta}(K)$, and $R\Phi_{\bar\eta}(K)$, give the following commutative diagram of long exact sequences:
(2.4.6.1)

$$\begin{array}{ccccccccccc}
 & & 0 & \longrightarrow & R^{n-1}\Phi_{\bar\eta}(K)_x & \longrightarrow & R^{n-1}\Phi_{\bar\eta}(K)_x & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
\longrightarrow & \mathbb{H}^{n-1}(X^*_{s(x)},K) & \xrightarrow{j} & \mathbb{H}^n_{\{x\}}(X_s,K) & \longrightarrow & \underline{H}^n(K)_x & \longrightarrow & \mathbb{H}^n(X^*_{s(x)},K) & \longrightarrow & & \\
 & \| & & \downarrow & & \downarrow & & \| & & & \\
\longrightarrow & \mathbb{H}^{n-1}(X^*_{s(x)},K) & \xrightarrow[①]{j} & \mathbb{H}^n_{\{x\}}\big(X_s,R\Psi_{\bar\eta}(K)\big) & \longrightarrow & R^n\Psi_{\bar\eta}(K)_x & \longrightarrow & \mathbb{H}^n(X_{s(x)},K) & \longrightarrow & & \\
 & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & \\
 & 0 & \longrightarrow & R^n\Phi_{\bar\eta}(K)_x & = & R^n\Phi_{\bar\eta}(K)_x & \longrightarrow & 0 & & &
\end{array}$$

The image of the arrow ① is contained in the inertia invariants.

The source prints both this paragraph and the preceding paragraph as 2.4.6, and prints both principal diagrams as (2.4.6.1). The repeated numbers are retained here; this second occurrence is the one cited in Exposé XV, 2.2.6.

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SGA 7 — EXPOSÉ XIV

Comparison with Transcendental Theory

by P. Deligne

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Introduction

To compare the theory of vanishing cycles in Exposé XIII with its transcendental analogue over $\mathbb{C}$, we proceed in two stages.

(a) We give a transcendental formalism modeled on that of Exposé XIII and prove a comparison theorem by the usual methods (already used to prove the finiteness theorem in XIII, 2.3).

(b) In the case of isolated singularities, we explain in more concrete terms the meaning of the transcendental formalism introduced here.

Besides the comparison theorem, this exposé contains a proof of the Picard–Lefschetz formula in the transcendental case.

In §4 we give a result preliminary to the Hodge theory of vanishing cycles. That section will not be used later in the seminar.

1. Transcendental formalism of vanishing cycles

1.1. The fundamental group of the punctured disc: sign conventions

We shall have to consider punctured discs D^* and their fundamental group $\mathbb{Z}$. Since this group is abelian, sign ambiguities may arise. We eliminate them by forgetting that it is abelian.

1.1.1.

Let X be a nice connected topological space and let $x \in X$. The elements of the fundamental group are homotopy classes of loops based at x. If $\alpha, \beta \in \pi_1(X, x)$ are represented by loops a, b, then $\alpha\beta$ is represented by the loop obtained by traversing b and then a.

1.1.2.

Let F be a locally constant sheaf on X. For every path $a : [0,1] \longrightarrow X$, the inverse image a^*F on $[0,1]$ is locally constant, hence constant, and therefore gives a bijection

$$a : F_{a(0)} \xleftarrow{\sim} H^0([0,1], a^*F) \xrightarrow{\sim} F_{a(1)}.$$

This bijection depends only on the homotopy class of a. Through this construction, $\pi_1(X,x)$ acts on the left on F_x. The functor $F \longmapsto F_x$ is an equivalence from the category of locally constant sheaves on X to the category of sets equipped with an action of $\pi_1(X,x)$.

1.1.3.

As x varies, the groups $\pi_1(X,x)$ are the fibers of a locally constant sheaf of groups $\Pi_1(X)$, which acts on every locally constant sheaf F. If $\pi_1(X,x)$ is abelian, then $\Pi_1(X)$ is constant; its value is denoted by $\pi_1(X)$.

1.1.4.

The group $\pi_1(X,x)$ acts on the universal covering

$$p : (\widetilde{X}, \widetilde{x}) \longrightarrow (X,x).$$

Regarded as an étalé space over X, and hence as a sheaf (the sheaf of its local sections), this covering corresponds under 1.1.2 to the set $\pi_1(X,x)$ with distinguished element e, on which $\pi_1(X,x)$ acts by left translation. The group also acts on this π_1-set by automorphisms of π_1-sets: an element $\sigma \in \pi_1(X,x)$ acts by

$$g \longmapsto g\sigma^{-1}.$$

This construction identifies

$$\pi_1(X,x) = \mathrm{Aut}_X(\widetilde{X}).$$

If F is locally constant on X, the bijection

$$H^0(\widetilde{X}, p^*F) \longrightarrow (p^*F)_{\widetilde{x}} = F_x \tag{1.1.4.1}$$

commutes with the action of $\pi_1(X,x)$: on the left the action is transport of structure through its action on $\widetilde{X}$, while on the right it is the action of 1.1.2. The equivalence of 1.1.2 may therefore be reformulated by saying that

$$F \longmapsto \big(\widetilde{X} \longmapsto H^0(\widetilde{X}, p^*F)\big)$$

is an equivalence from the category of locally constant sheaves on X to the category of functors

$$(\text{universal coverings of } X) \longrightarrow (\text{Sets}).$$

1.1.5.

If $\pi_1(X,x)$ is abelian and hence independent of x, the action of 1.1.4 of $\pi_1(X)$ on $\widetilde{X}$ is independent of the base point: $\pi_1(X)$ acts on every universal covering $\widetilde{X}$ of X.

One must bear in mind that this action, which is the only one we shall use, corresponds to the inverse of the action in 1.1.3.

1.1.6.

Let D be a disc, that is, a topological space homeomorphic to the standard disc

$$\{z \in \mathbb{C} : |z| < 1\}.$$

Let $0 \in D$ and put $D^* = D - \{0\}$. Suppose that D is equipped with the structure of an analytic manifold, and hence is oriented. The fundamental group $\pi_1(D^*)$ is then canonically isomorphic to $\mathbb{Z}$. For the standard disc, with center 0 and $t \in D^*$, the positive generator of $\pi_1(D^*,t)$ is represented by the loop

$$x \longmapsto \exp(2\pi i x)t.$$

1.1.7.

Let $z : D \longrightarrow \mathbb{C}$ be a uniformizing parameter, so that $(dz)_0 \neq 0$, and let $\widetilde{D}^*$ be a universal covering of D^*. Write $\log z$ for any function on $\widetilde{D}^*$ such that

$$\exp(\log z) = z.$$

The pair $(\widetilde{D}^*, \log z)$ is unique up to unique isomorphism.

For $\alpha \in \mathbb{C}$, put

$$z^\alpha = \exp(\alpha \log z).$$

Let T be the positive generator of $\pi_1(D^*)$, acting on $\widetilde{D}^*$ as in 1.1.5 and on functions on $\widetilde{D}^*$ by transport of structure. Then

$$T \log z = \log z + 2\pi i, \tag{1.1.7.1}$$

and

$$Tz^\alpha = \exp(2\pi i\alpha) z^\alpha. \tag{1.1.7.2}$$

1.1.8.

Let D be a disc, let $0 \in D$, and put $D^* = D - \{0\}$. If $D' \subset D$ is a smaller disc containing 0, restriction to D'^* is an equivalence from the category of locally constant sheaves on D^* to the category of locally constant sheaves on D'^*. Indeed, for $t \in D'^*$,

$$\pi_1(D'^*, t) \xrightarrow{\sim} \pi_1(D^*, t).$$

Suppose now that D is equipped with a differentiable-manifold structure. Let T be the tangent space to D at 0, and put $T^* = T - \{0\}$. For $D' \subset D$ as above and an embedding $f : D' \hookrightarrow T$ satisfying $f(0) = 0$ and $(df)_0 = \mathrm{id}$, the composite

$$(\text{restriction to } D')^{-1} \circ f^*$$

is an equivalence $[f]$ from the category of locally constant sheaves on T^* to the category of locally constant sheaves on D^*. Up to canonical isomorphism it depends only on the germ of f at 0, subject to $(df)_0 = \mathrm{id}$. A path from one such germ to another defines an isomorphism of equivalences, and homotopic paths define the same isomorphism. Since the space of such germs is contractible—by the retraction to a fixed germ f_0 given by $(1-\lambda)f + \lambda f_0$—we obtain an equivalence, well defined up to unique isomorphism, from locally constant sheaves on T^* to locally constant sheaves on D^*.

1.1.9.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, and let $p : \widetilde{D}^* \longrightarrow D^*$ be a universal covering. We write

$$p : \widetilde{D} \longrightarrow D$$

for the topological space over D obtained from $\widetilde{D}^*$ by adjoining a point 0, in such a way that:

(a) $\widetilde{D}^*$ is open in $\widetilde{D}$;

(b) $p : \widetilde{D} \longrightarrow D$ extends $p : \widetilde{D}^* \longrightarrow D^*$ and $p(0) = 0$;

(c) the sets $p^{-1}(U)$, as U ranges over the neighborhoods of 0 in D, form a neighborhood basis of 0 in $\widetilde{D}$.

1.1.10.

Let X be a nonsingular complex algebraic curve, let $0 \in X$, and let D be a disc neighborhood of 0 in X. The ring of germs at 0 of holomorphic functions on D which are algebraic over the field $K(X)$ of rational functions on X is the henselization $\mathcal{O}_0^h$ of the local ring $\mathcal{O}_0$ of X at 0. Let $\widetilde{D}^*$ be a universal covering of $D^* = D - \{0\}$. The field of germs at $0 \in \widetilde{D}$ of meromorphic functions on $\widetilde{D}^*$ which are algebraic over $K(X)$ is an algebraic closure $\overline{K}$ of $K(X)$, and hence also of the fraction field K_0^h of $\mathcal{O}_0^h$.

The group $\pi_1(D^*)$ of D-automorphisms of $\widetilde{D}^*$ (1.1.5) acts by transport of structure on $\overline{K}$, giving

$$\begin{array}{ccc} \pi_1(D^*) & \longrightarrow & \mathrm{Gal}(\overline{K}/K_0^h) \\ \| & & \| \\ \mathbb{Z} & & \widehat{\mathbb{Z}}(1) \end{array} \qquad (1.1.10.1)$$

.

It follows from (1.1.7.2) that this diagram is compatible with the natural inclusion, over $\mathbb{C}$, of $\mathbb{Z}$ into $\widehat{\mathbb{Z}}(1)$. This inclusion is the composite of $\mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}}$ and the isomorphism

$$\widehat{\mathbb{Z}} \xrightarrow{\sim} \widehat{\mathbb{Z}}(1),$$

which is the inverse limit of the isomorphisms

$$\mathbb{Z}/n \longrightarrow \mu_n, \qquad k \longmapsto \exp\left(\frac{2\pi i}{n}k\right).$$

1.1.11.

Every sheaf F on $\mathrm{Spec}(K_0^h)$ defines a locally constant sheaf on D^*. We use the equivalence of 1.1.4 and, to each universal covering $\widetilde{D}^*$ of D^*, associate $F_{\overline{K}}$ for the field $\overline{K}$ constructed in 1.1.10.

1.2. Sheaves of sets

1.2.1.

Let D be a disc (1.1.6), let $0 \in D$, and put $D^* = D - \{0\}$. We denote by $\mathcal{D}^*$ the topos of locally constant sheaves of sets on D^*. By 1.1.8 this topos does not "change" if D is replaced by a smaller disc D' containing 0, nor, when D is endowed with the structure of a differentiable manifold, if D is replaced by its tangent space at 0.

For $t \in D^*$, the functor $F \longmapsto F_t$ identifies this topos with the topos of sets equipped with an action of $\pi_1(D^*)$ (1.1.2). By 1.1.4, the functor

$$F \longmapsto \left(\widetilde{D}^* \longmapsto H^0(\widetilde{D}^*, p^*F)\right)$$

identifies it with the topos of functors

$$(\text{universal coverings of } D^*) \longrightarrow (\mathrm{Sets}),$$

and these identifications are related by (1.1.4.1).

1.2.2.

Let F be a sheaf on D whose restriction to D^* is locally constant. There is a unique morphism of sheaves

$$\underline{F_0} \xrightarrow{\alpha} F$$

whose map on the stalk at 0 is the identity. As explained in XIII, 0.1 (with $X_0 = (P^t)$), the functor

$$F \longmapsto (F_0, F|_{D^*}, \alpha)$$

identifies these sheaves with triples consisting of a set F_0, a locally constant sheaf F_η on D^*, and a morphism

$$\alpha : (\text{constant sheaf with value } F_0 \text{ on } D^*) \longrightarrow F_\eta.$$

1.2.3.

We denote by $\mathcal{D}$ the topos of sheaves of sets on D whose restriction to D^* is locally constant. The inclusion of $\mathcal{D}$ into the topos of sheaves on D is the inverse-image functor of a morphism of topoi $D \longrightarrow \mathcal{D}$, which fits into the commutative diagram

$$\begin{array}{ccccc} \{0\} & \hookrightarrow & D & \hookleftarrow & D^* \\ \| & & \downarrow\beta & & \downarrow\beta \\ \{0\} & \hookrightarrow & \mathcal{D} & \hookleftarrow & \mathcal{D}^* \end{array} \quad .$$

As in 1.1.1, one checks that this topos does not change when D is replaced by a smaller disc or by the tangent space to D at 0.

1.2.4.

Let Y be a topological space. The product topos $Y \times \mathcal{D}$ has the following three descriptions (for their equivalence, see XIII, 0.1).

(a) $Y \times \mathcal{D}$ is the topos of sheaves F on $Y \times D$ satisfying

$(*)$ D^* is covered by open subsets U such that $F|_{Y\times U}$ is the inverse image of a sheaf on Y.

(b) Let $t \in D^*$. The topos $Y \times \mathcal{D}$ is the topos of triples (F_0, F_t, α) in which

$\alpha)$ F_0 is a sheaf on Y;
$\beta)$ F_t is a sheaf on Y equipped with an action of $\pi_1(D^*)$;
$\gamma)$ $\alpha : F_0 \longrightarrow F_t$ is a morphism whose image is contained in the invariants.

(c) The topos $Y \times \mathcal{D}$ is the topos of triples (F_0, F_η, α) in which

$\alpha')$ F_0 is a sheaf on Y;
$\beta')$ F_η is a functor, with values denoted by $F_{\bar\eta}$,

$$(\text{universal coverings of } D^*) \longrightarrow (\text{sheaves on } Y);$$

$\gamma')$ α is a morphism from F_0 (regarded as the constant functor with value F_0 from the universal coverings of D^* to the sheaves on Y) to F_η.

It is immaterial here whether this topos is literally the product topos of Y and $\mathcal{D}$. Likewise, we define $Y \times \mathcal{D}^*$, at our choice, as

(a^*) the topos of sheaves on $Y \times D^*$ satisfying $(*)$;
(b^*) for $t \in D^*$, the topos of sheaves F_t on Y equipped with an action of $\pi_1(D^*)$;
(c^*) the topos of functors

$$(\text{universal coverings of } D^*) \longrightarrow (\text{sheaves on } Y).$$

1.2.5.

The topos $Y \times \mathcal{D}$ has the same functoriality properties in Y as the topoi $Y \times_S S$ of Exposé XIII, and we shall not list them. To fix notation, we merely record the diagram

$$\begin{array}{ccccccc} Y & & Y & \xhookrightarrow{i} & Y \times \mathcal{D} & \xhookleftarrow{j} & Y \times \mathcal{D}^* \\ \downarrow & \xleftarrow{\text{sp}} & \downarrow & & \downarrow & & \downarrow \\ (P^t) & & \{0\} & \xhookrightarrow{i} & \mathcal{D} & \xhookleftarrow{j} & \mathcal{D}^* \end{array}$$

1.2.6.

Let K be a complex of sheaves of abelian groups on $Y \times \mathcal{D}$. As in XIII, 1.4.2, one obtains from K a distinguished triangle in $K(Y \times \mathcal{D}^*)$:

$$\begin{array}{ccc} & \Phi(K) & \\ \swarrow & & \nwarrow \\ \text{sp}^* K_0 & \longrightarrow & K_\eta \end{array}$$

This construction passes to the derived category and defines a functor

$$D(Y \times \mathcal{D}) \longrightarrow (\text{distinguished triangles of } D(Y \times \mathcal{D}^*)).$$

Suppose that D is oriented (for example, complex analytic), and let T be the positive generator of $\pi_1(D^*)$ (1.1.6). For $K \in \text{Ob}\, D(Y \times \mathcal{D})$, we define, as in XIII, 1.4.3, the variation $V = \text{Var}(T)$,

$$V : \Phi(K) \longrightarrow K_\eta.$$

The endomorphism $T - \text{Id}$ of K_η is the composite

$$T - \text{Id} : K_\eta \longrightarrow \Phi(K) \longrightarrow K_\eta.$$

1.2.7.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, let $f : X \longrightarrow D$ be a continuous map, and put $X_0 = f^{-1}(0)$. We shall define a functor

$$\Psi : (\text{sheaves on } X) \longrightarrow (\text{sheaves on } X_0 \times \mathcal{D}).$$

Let $\widetilde{D}^*$ be a universal covering of D^*, let $\widetilde{D}$ be as in 1.1.9, and put

$$\overline{X} = X \times_D \widetilde{D}, \qquad X^* = X - X_0 = X \times_D D^*, \qquad \overline{X}^* = \overline{X} - X_0 = X \times_D \widetilde{D}^*.$$

There is a commutative diagram

$$\begin{array}{ccccc} X_0 & \xhookrightarrow{\bar{i}} & \overline{X} & \xhookleftarrow{\bar{j}} & \overline{X}^* \\ \| & & \downarrow p & & \downarrow p' \\ X_0 & \xhookrightarrow{i} & X & \xhookleftarrow{j} & X^* \end{array} \tag{1.2.7.1}$$

cartesian over the diagram

$$\begin{array}{ccccc} \{0\} & \xhookrightarrow{\bar{i}} & \widetilde{D} & \xhookleftarrow{\bar{j}} & \widetilde{D}^* \\ \| & & \downarrow p & & \downarrow p' \\ \{0\} & \xhookrightarrow{i} & D & \xhookleftarrow{j} & D^* \end{array} \tag{1.2.7.2}$$

Editorial note. In (1.2.7.1) the printed top-right node lacks the overline. The preceding definition and the map p' force that node to be $\overline{X}^*$, which is the notation used here.

For a sheaf F on X, we set (with the notation of 1.2.4(c))

$$\Psi(F) = \big(i^*F,\ \overline{i}^*\overline{j}_*((jp')^*F),\ \alpha\big),$$

where

$$\begin{aligned}\alpha : i^*F = \overline{i}^*p^*F &\longrightarrow \overline{i}^*\overline{j}_*((jp')^*F)\\ &= \overline{i}^*\overline{j}_*\overline{j}^*p^*F\end{aligned}$$

is obtained, after applying $\overline{i}^*$, from the adjunction map $\mathrm{Id} \longrightarrow \overline{j}_*\overline{j}^*$.

We also define a functor

$$\Psi_\eta : (\text{sheaves on } X^*) \longrightarrow (\text{sheaves on } X_0 \times \mathcal{D}^*)$$

by

$$\Psi_\eta(F) = \overline{i}^*\overline{j}_*(p'^*F).$$

1.2.8.

The functoriality properties of Ψ are the same as those of its analogue studied in XIII, 1.3.5–1.3.9.

1.3. The functor $R\Psi$

1.3.1.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, let $f : X \longrightarrow D$ be a continuous map, and put $X_0 = f^{-1}(0)$. The functors $R\Psi$ and $R\Psi_\eta$ are the derived functors of Ψ and Ψ_η:

$$R\Psi : D^+(X) \longrightarrow D^+(X_0 \times \mathcal{D}),$$

$$R\Psi_\eta : D^+(X^*) \longrightarrow D^+(X_0 \times \mathcal{D}^*).$$

We denote by $R\Phi$ the composite functor $\Phi \circ R\Psi$:

$$R\Phi : D^+(X) \longrightarrow D^+(X_0 \times \mathcal{D}^*).$$

The properties of these functors are exactly analogous to those of the functors with the same names in XIII, 2.1. Their proofs are much more elementary, since in the topological setting the base-change theorems for a proper morphism or for a smooth morphism (that is, one locally isomorphic to $\mathbb{R}^n \times S \longrightarrow S$) are easy to prove. The minor changes required are as follows.

1.3.2.

The functor $Rf_!$ is defined only for a separated morphism between locally compact spaces, and is the derived functor of the direct-image functor with proper supports $f_!$.

1.3.3.

For a proper (and separated) morphism $f : X \longrightarrow D$, the fundamental isomorphism (XIII, (2.1.8.3)) must be rewritten as follows. Let $p : \widetilde{D}^* \longrightarrow D^*$ be a universal covering of D^*. For every neighborhood U of 0 in D, put

$$\widetilde{U}^* = p^{-1}(U), \qquad X_{\widetilde{U}^*} = X \times_D \widetilde{U}^*.$$

For $K \in \mathrm{Ob}\, D^+(f^{-1}(D^*))$, one has

$$\varinjlim_{U} \mathbb{H}^i(X_{\widetilde{U}^*}, K) \xrightarrow{\sim} \mathbb{H}^i(X_0, R\Psi_{\bar{\eta}}(K)). \tag{1.3.3.1}$$

This isomorphism is especially useful when $X^* = f^{-1}(D^*)$ is a topological fiber bundle over D^*. As a disc $U \subset D$ shrinks to 0, the Leray spectral sequence for $X_{\widetilde{U}^*} \longrightarrow \widetilde{U}^*$ then shows that, for every locally constant sheaf Λ on X^* and every $t \in \widetilde{U}^*$,

$$H^i(X_{\widetilde{U}^*}, \Lambda) \xrightarrow{\sim} H^i(X_t, \Lambda).$$

Thus, for $t \in D^*$,

$$H^i(X_t, \Lambda) \xrightarrow{\sim} H^i(X_0, R\Psi_t(\Lambda)), \tag{1.3.3.2}$$

provided that X/D is proper, X^* is a fiber bundle, and Λ is locally constant.

In practice, to verify that X^* is a fiber bundle over D^* one uses Thom's isotopy theorem, one form of which is stated below. For the proof of this theorem, none of whose consequences will be used in the subsequent exposés, we refer to Whitney and Mather.

1.3.4. Preliminary.

Let $f : X \longrightarrow Y$ be a smooth morphism (by definition, one with no critical point) of differentiable manifolds. A connection ∇ on X/Y is a vector subbundle of the tangent bundle of X complementary to $\ker(df)$. For a vector field u on Y, there is a unique vector field $\nabla(u)$ on X which lifts u and lies in this subbundle. The connection is called integrable if

$$\nabla([u, v]) = [\nabla(u), \nabla(v)].$$

When f is proper, an integrable connection gives local trivializations over Y of X/Y,

$$\alpha : f^{-1}(U) \xrightarrow{\sim} X_0 \times U,$$

compatible with projection to U and satisfying $d\alpha(\nabla(u)) = (0, u)$. When f is not proper, the same result holds under suitable conditions at infinity; for example, it is enough that $\nabla(u)g$ vanish for a suitable exhaustion function g.

1.3.5. Thom's isotopy theorem.

Let

$$D = \{z \in \mathbb{C} : |z| < 1\}, \qquad D^* = D - \{0\},$$

and let $f : X \longrightarrow D$ be a proper morphism of analytic spaces. After replacing D by a smaller disc and X by its inverse image, there exists a stratification Σ of $X^* = f^{-1}(D^*)$ having the following properties:

(a) *Σ is a filtration*

$$X^* = F_0 \supset F_1 \supset \cdots \supset F_n \supset F_{n+1} = \varnothing$$

of X^* by closed subsets such that the closure $\overline{F}_i$ of F_i in X is an analytic subspace of X.

(b) *The open strata $F_i - F_{i+1}$ are smooth over D^*.*

(c) *Σ satisfies Whitney's conditions (a) and (b).*

For such a stratification Σ, there are integrable connections ∇_i on the $F_i - F_{i+1}$ such that

(i) *locally on D^*, ∇_i defines a trivialization of $F_i - F_{i+1}$;*

(ii) *these trivializations glue to give local trivializations of X^*/D^*.*

Finally, if (Y_α) is a finite family of closed analytic subspaces of X, then, after shrinking D further, one can choose Σ so that every Y_α is a union of connected components of open strata $(F_i - F_{i+1})$. A connection ∇ as above then locally trivializes $(X, (Y_\alpha))$.

1.3.6.

The isomorphism (1.3.3.2) remains valid for hypercohomology with values in a complex of sheaves K when X is proper over D and the following condition holds:

$(*)$ For every $i \in \mathbb{Z}$, the space D^* is covered by open subsets U for which there exist an isomorphism compatible with projection to U,

$$\varphi : f^{-1}(U) \xrightarrow{\sim} Y \times U,$$

and a sheaf G on Y such that, on $f^{-1}(U)$, $\underline{H}^i(K)$ is isomorphic to $\varphi^* \operatorname{pr}_1^* G$.

Under this hypothesis, the triangle of 1.2.6 gives a long exact sequence

$$\cdots \longrightarrow \mathbb{H}^i(X_0, K) \longrightarrow \mathbb{H}^i(X_t, K) \longrightarrow \mathbb{H}^i(X_0, R\Phi_t(K)) \longrightarrow \cdots . \tag{1.3.6.1}$$

Editorial note. The print cites the triangle "1.2.1" here; the distinguished triangle used is the one constructed in 1.2.6.

By Thom's theorem, condition $(*)$ holds (after shrinking D) whenever X admits a complex analytic stratification such that all the $\underline{H}^i(K)$ are locally constant on every open stratum.

1.4. A special case

This section will not be used later in the seminar. It formalizes a construction sketched in the Introduction to Exposé XIII.

1.4.1.

Let D be an oriented disc, let $0 \in D$, and put $D^* = D - \{0\}$. A topological space $\mathcal{X}^*$ over $\mathcal{D}^*$ is a functor

$$(\text{universal coverings of } D^*) \longrightarrow (\text{topological spaces}).$$

After a universal covering $\widetilde{D}^*$ has been chosen, this is a topological space X_η equipped with an action of $\pi_1(D^*)$; when $\widetilde{D}^*$ is the universal covering of (D^*, t), we denote it by $\mathcal{X}_t^*$.

A sheaf on $\mathcal{X}^*$ is a functor

$$(\text{universal coverings of } D^*) \longrightarrow (\text{topological spaces equipped with a sheaf}),$$

or, equivalently, a $\pi_1(D^*)$-equivariant sheaf on X_η.

To $\mathcal{X}^*$ we associate the topological space

$$\beta^*\mathcal{X}^* = X_\eta \times \widetilde{D}^*/\pi_1(D^*)$$

over D^*. Every sheaf F on $\mathcal{X}^*$ descends to a sheaf β^*F on $\beta^*\mathcal{X}^*$, giving a commutative diagram of topoi

$$\begin{array}{ccc} \beta^*\mathcal{X}^* & \longrightarrow & \mathcal{X}^* \\ \downarrow & & \downarrow \\ D^* & \xrightarrow{\beta} & \mathcal{D}^*. \end{array}$$

1.4.2.

A topological space $\mathcal{X}$ over $\mathcal{D}$ consists of

(a) a topological space $\mathcal{X}^*$ over $\mathcal{D}^*$;

(b) a topological space X_0;

(c) a $\pi_1(D^*)$-invariant morphism

$$\mathrm{sp} : X_\eta \longrightarrow X_0,$$

where $\pi_1(D^*)$ acts trivially on X_0.

A sheaf on $\mathcal{X}$ consists of a sheaf F_0 on X_0, a sheaf on $\mathcal{X}^*$ given by F_η, and an equivariant morphism

$$\mathrm{sp} : \mathrm{sp}^* F_0 \longrightarrow F_\eta \qquad (\text{or } F_0 \longrightarrow \mathrm{sp}_* F_\eta).$$

There is a commutative diagram

$$\begin{array}{ccccc} X_0 & \hookrightarrow & \mathcal{X} & \hookleftarrow & \mathcal{X}^* \\ \downarrow & & \downarrow & & \downarrow \\ \{0\} & \hookrightarrow & \mathcal{D} & \hookleftarrow & \mathcal{D}^* \end{array}$$

and, in addition, a morphism of topoi

$$\Psi : \mathcal{X} \longrightarrow X_0 \times \mathcal{D}.$$

It is obtained from the functoriality of the preceding construction, applied to

$$(X_\eta, X_0, \mathrm{sp}) \longrightarrow (X_0, X_0, \mathrm{Id}),$$

and satisfies

$$\Psi(F_0, F_\eta, \mathrm{sp}) = (F_0, \mathrm{sp}_* F_\eta, \mathrm{sp}).$$

This morphism induces

$$\Psi_\eta : \mathcal{X}^* \longrightarrow X_0 \times \mathcal{D}^*.$$

1.4.3.

To a topological space $\mathcal{X}$ over $\mathcal{D}$ we associate a topological space $X = \beta^* \mathcal{X}$ over D as follows. To $X^* = \beta^* \mathcal{X}^*$ we adjoin the fiber X_0 over 0. A fundamental system of neighborhoods of $x_0 \in X_0$ is formed by the sets

$$V \cup \big(\mathrm{sp}^{-1}(V) \times \widetilde{U}^* / \pi_1(D^*)\big),$$

where V is a neighborhood of x_0 in X_0, U is a neighborhood of 0 in D, and $\widetilde{U}^* = U \times_D \widetilde{D}^*$. Let j (respectively i) denote the inclusion of X^* (respectively X_0) into X. For a sheaf F on $\mathcal{X}^*$, one has

$$i^* j_* \beta^* F = (\mathrm{sp}_* F_\eta)^{\pi_1(D^*)}.$$

It follows that there is a commutative diagram of topoi

$$\begin{array}{ccc} X & \xrightarrow{\beta} & \mathcal{X} \\ \downarrow & & \downarrow \\ D & \xrightarrow{\beta} & \mathcal{D} \end{array}$$

such that, for every sheaf F on $\mathcal{X}$, there is an isomorphism

$$\Psi(F) \xrightarrow{\sim} \Psi(\beta^* F). \tag{1.4.3.1}$$

Proposition 1.4.4.

The map (1.4.3.1) derives to an isomorphism

$$R\Psi \xrightarrow{\sim} R\Psi \circ \beta^*$$

between functors from $D^+(\mathcal{X})$ to $D^+(X_0 \times \mathcal{D})$.

Indeed, let $\overline{j}$ be the inclusion of

$$\overline{X}^* = X \times_D \widetilde{D}^* \simeq X_\eta \times \widetilde{D}^*$$

into $\overline{X} = X \times_D \widetilde{D}$. The proposition reduces to checking that, for every flabby sheaf of abelian groups F on X_η, the sheaf $\mathrm{pr}_1^* F$ is acyclic for $\overline{j}_*$. This in turn reduces (with $B = \widetilde{D}^*$) to the following familiar assertion, proved by a homotopy argument. If F is a sheaf of abelian groups on a space Y and B is a disc, then

$$H^i(Y, F) \xrightarrow{\sim} H^i(Y \times B, \mathrm{pr}_1^* F).$$

1.4.5.

Proposition 1.4.4 gives a method for computing $R\Psi$ for sheaves on X of the form $\beta^* F$, reducing the computation of $R\Psi$ essentially to that of $R\,\mathrm{sp}_*$.

1.4.6.

Later we shall consider the case in which X_η is locally compact and X_0 is obtained from X_η by contracting a compact subset $V \subset X_\eta$ to a point 0, with sp the contraction map and the action of $\pi_1(D^*)$ trivial outside V. Proper base change for sp then gives, for $K \in \mathrm{Ob}\, D^+(\mathcal{X}^*)$,

$$R\Psi_\eta(\beta^* K)_0 = R\Gamma(V, K) \tag{1.4.6.1}$$

and

$$R\Gamma_{\{0\}} R\Psi_\eta(\beta^* K) = R\Gamma_V(X_\eta, K), \tag{1.4.6.2}$$

the latter being cohomology with supports.

For K on $\mathcal{X}$, suppose that

$$\mathrm{sp} : \mathrm{sp}^* K_0 \longrightarrow K_\eta$$

is a quasi-isomorphism outside V. Then $R\Phi(K) = 0$ outside 0. In particular, for a constant sheaf Λ on $\mathcal{X}$,

$$R^i\Phi(\Lambda) = \widetilde{H}^i(V, \Lambda), \qquad \text{supported at } 0,$$

where the right-hand side is reduced cohomology.

1.4.7.

The case of interest to us is that in which X_η is a differentiable manifold and $V \subset X_\eta$ is a submanifold with boundary and interior V°. Then $X - \{0\}$ is a differentiable manifold obtained by gluing $\beta^* X_\eta$ to the constant bundle over D with fiber $X_\eta - V$. The map

$$H_c^i(V^\circ, \Lambda) \xrightarrow{\sim} H_V^i(X_\eta, \Lambda) \tag{1.4.7.1}$$

is an isomorphism. These groups are also denoted by $H^i(V \bmod \partial V, \Lambda)$. The isomorphism (1.4.7.1) defines an isomorphism

$$R\Gamma_c(V^\circ, \Lambda) \xrightarrow{\sim} R\Gamma_{\{0\}} R\Psi_\eta(\Lambda), \tag{1.4.7.2}$$

and the variation is defined, by the procedure of XIII, 2.4.5 and through (1.4.7.2), as a morphism whose source is the reduced cohomology of V:

$$\text{Var} : R\widetilde{\Gamma}(V, \Lambda) \longrightarrow R\Gamma_c(V^\circ, \Lambda). \tag{1.4.7.3}$$

The automorphism induced by T on the pair $(V, \partial V)$ is again denoted by T, and its restriction ∂T to ∂V is the identity. The variation (1.4.7.3) should coincide with the classical variation morphism (3.1.9) attached to such an automorphism T. A statement of this kind will be proved in 3.1.10.

2. The comparison theorem

2.1.

For every scheme X of finite type over $\mathbb{C}$, we denote by X_{cl} the space $X(\mathbb{C})$ endowed with its usual topology.

Let S be a smooth curve over $\mathbb{C}$, that is, a smooth separated scheme of finite type over $\mathbb{C}$, pure of dimension 1. Let $0 \in S(\mathbb{C})$, let $f : X \longrightarrow S$ be a separated S-scheme of finite type, and let Λ be a noetherian torsion ring (for example $\mathbb{Z}/n\mathbb{Z}$). Put $X_0 = f^{-1}(0)$.

2.2.

We denote by S^h the henselized henselian trait of S at 0, and by η the generic point of S^h. There is a functor

$$\begin{aligned}\Psi : (\text{sheaves on } X) &\xrightarrow{\text{restriction}} (\text{sheaves on } X \times_S S^h) \\ &\xrightarrow{\Psi} (\text{sheaves on } X_0 \times S^h).\end{aligned}$$

Its derived functor is the composite

$$R\Psi : D(X, \Lambda) \longrightarrow D(X \times_S S^h, \Lambda) \xrightarrow{R\Psi} D(X_0 \times S^h, \Lambda).$$

2.3.

Let $D \subset S_{\text{cl}}$ be a disc containing 0. The composite functor Ψ_{cl} (or simply Ψ)

$$\begin{aligned}\Psi_{\text{cl}} : (\text{sheaves on } X_{\text{cl}}) &\xrightarrow{\text{restriction}} (\text{sheaves on } X_{\text{cl}} \times_{S_{\text{cl}}} D) \\ &\xrightarrow{\Psi} (\text{sheaves on } (X_0)_{\text{cl}} \times \mathcal{D})\end{aligned}$$

is independent, up to unique isomorphism, of the choice of D; recall that $\mathcal{D}$ does not change when D is replaced by a smaller disc. Its derived functor is the composite

$$R\Psi_{\text{cl}} : D(X_{\text{cl}}, \Lambda) \longrightarrow D(f_{\text{cl}}^{-1}(D), \Lambda) \xrightarrow{R\Psi} D((X_0)_{\text{cl}} \times \mathcal{D}, \Lambda).$$

2.4.

Likewise, we define

$$\begin{aligned}R\Psi_\eta &: D(X - X_0, \Lambda) \longrightarrow D(X_0 \times \eta, \Lambda), \\ R\Psi_{\eta\,\text{cl}} &: D((X - X_0)_{\text{cl}}, \Lambda) \longrightarrow D((X_0)_{\text{cl}} \times \mathcal{D}^*, \Lambda).\end{aligned}$$

2.5.

The construction of 1.1.11 defines a morphism of topoi from $\mathcal{D}^*$ to η, which extends to

$$\varepsilon : \mathcal{D} \longrightarrow S^h.$$

Together with $\varepsilon : (X_0)_{\mathrm{cl}} \longrightarrow X_0$ (SGA 4, XI, 4), this morphism defines

$$\varepsilon : (X_0)_{\mathrm{cl}} \times \mathcal{D} \longrightarrow X_0 \times S^h.$$

If $\widetilde{D}^*$ is a universal covering of D^* and $k(\bar{\eta})$ is the corresponding algebraic closure (1.1.10) of $k(\eta)$, then

$$\varepsilon^*(F_0, F_\eta, \alpha) = \left(\varepsilon^* F_0, \varepsilon^* F_{\bar{\eta}}, \varepsilon^* \alpha\right).$$

2.6.

Let $\widetilde{D}^*$ be a universal covering of D^*, let $k(\bar{\eta})$ be the corresponding algebraic closure of $k(\eta)$, and let $\overline{S}$ be the normalization of S^h in $k(\bar{\eta})$. Write $\overline{S}$ as an inverse limit of affine S-schemes S_i. For every i, after replacing D by a smaller disc D_i, there is a natural morphism of S_{cl}-spaces

$$a_i : \widetilde{D}_i \longrightarrow (S_i)_{\mathrm{cl}}.$$

Let $j_i : U_i \hookrightarrow S_i$ be the open complement in S_i of the image 0 of the closed point of $\overline{S}$, and let $i : 0 \hookrightarrow S_i$.

Let F be a sheaf on X, and put $F_{\mathrm{cl}} = \varepsilon^* F$. We again denote by j_i the inclusion of $X \times_S U_i$ into $X \times_S S_i$, by i the inclusion of X_0 into $X \times_S S_i$, and by F the inverse image of F on $X \times_S S_i$. Then

$$\Psi(F) = \left(i^* F,\ \varinjlim_i i^* j_{i*} j_i^* F,\ \varinjlim_i \varphi_i \right).$$

The functors ε^* and inverse image commute. The morphism

$$\varepsilon^* j_{i*} \longrightarrow (j_{i,\mathrm{cl}})_* \varepsilon^*$$

therefore gives

$$c_i : (i^* j_{i*} j_i^* F)_{\mathrm{cl}} \longrightarrow i^* j_{i*} j_i^* (F_{\mathrm{cl}}).$$

Let $\overline{j}$ (respectively $\overline{j}_i$) be the inclusion of $\widetilde{D}^*$ (respectively $\widetilde{D}_i^*$) into $\widetilde{D}$ (respectively $\widetilde{D}_i$), as well as the induced inclusions

$$X \times_S \widetilde{D}^* \hookrightarrow X \times_S \widetilde{D}, \qquad X \times_S \widetilde{D}_i^* \hookrightarrow X \times_S \widetilde{D}_i.$$

The adjunction map $\mathrm{Id} \longrightarrow a_{i*} a_i^*$ defines

$$i^* j_{i*} j_i^* F_{\mathrm{cl}} \longrightarrow i^* \overline{j}_{i*} \overline{j}_i^* \overline{F}_{\mathrm{cl}} = i^* \overline{j}_* \overline{j}^* \overline{F}_{\mathrm{cl}}.$$

Passing to the limit gives

$$c_\eta : \Psi_{\bar{\eta}}(F)_{\mathrm{cl}} \longrightarrow \Psi_{\mathrm{cl}}(F_{\mathrm{cl}}).$$

This morphism defines the comparison morphism

$$c : \varepsilon^* \Psi(F) \longrightarrow \Psi_{\mathrm{cl}}(\varepsilon^* F). \tag{2.6.1}$$

Similarly, one defines

$$c_\eta : \varepsilon^* \Psi_\eta(F) \longrightarrow \Psi_\eta(\varepsilon^* F). \tag{2.6.2}$$

2.7.

The morphisms (2.6.1) and (2.6.2) derive to

$$c : \varepsilon^* R\Psi \longrightarrow R\Psi_{\mathrm{cl}} \circ \varepsilon^*$$

and

$$c_\eta : \varepsilon^* R\Psi_\eta \longrightarrow R\Psi_{\eta\,\mathrm{cl}} \circ \varepsilon^*.$$

Theorem 2.8.

Let $K \in \mathrm{Ob}\, D(X, \Lambda)$ (respectively, $K \in \mathrm{Ob}\, D(X - X_0, \Lambda)$) be a complex of sheaves of Λ-modules whose cohomology sheaves $\underline{H}^i(K)$ are constructible. Then the comparison morphism

$$c : \varepsilon^* R\Psi(K) \longrightarrow R\Psi_{\mathrm{cl}}(\varepsilon^* K)$$

(respectively,

$$c : \varepsilon^* R\Psi_\eta(K) \longrightarrow R\Psi_{\eta\,\mathrm{cl}}(\varepsilon^* K))$$

is an isomorphism.

It suffices to prove the assertion for $R\Psi_\eta$. Moreover, both sides of 2.7 are triangulated functors of finite cohomological dimension, so it is enough to treat the case in which K is a single sheaf F placed in degree 0. The proof is exactly analogous to that of the finiteness theorem XIII, 2.3.1. Since $R\Psi$ and $R\Psi_{\mathrm{cl}}$ have the same variance properties, the dévissages used there (with permission to shrink S) reduce the problem to the case in which X is regular, $f^{-1}(0)$ is a normal crossings divisor in X, and $F = \mathbb{Z}/\ell\mathbb{Z}$. One then compares Exposé I, 3.3 with its transcendental analogue.

3. Isolated singularities

3.1. The manifold of vanishing cycles

3.1.1.

Let

$$D = \{z \in \mathbb{C} \mid |z| < 1\},$$

let X be an analytic space with $0 \in X$, and let $f : X \to D$ be a morphism such that $f(0) = 0$. We suppose that f is smooth away from 0 (in particular, X is smooth away from 0). We intend to study the vanishing cycles near 0; hence there is no loss of generality in supposing that X is embedded in $\mathbb{C}^n \times D \subset \mathbb{C}^{n+1}$ compatibly with the projection to D, and in such a way that 0 maps to 0.

3.1.2.

Let $\delta(x)$ be a real-analytic function on X (the restriction to X of a real-analytic function on $\mathbb{C}^{n+1}$) such that, for $x \in X_0 = f^{-1}(0)$,

(a) $\delta(x) \geq 0$;

(b) $\delta(x) = 0$ if and only if $x = 0$.

For example, we may take δ to be the square of the Euclidean distance from x to 0 in $\mathbb{C}^{n+1}$. The critical locus of the restriction of δ to X_0 is real analytic and therefore, in a suitable neighborhood of 0, has only finitely many connected components. Replacing X by an open neighborhood of 0 and applying Sard's theorem, we may consequently obtain the following situation:

(c) for a suitable $r > 0$, the space $X_0 \cap \delta^{-1}([0, r])$ is compact;

(d) on X_0, δ has no critical value in $(0, r]$.

After shrinking X further, we then have:

(e) there exists a C^∞ function $\tau(s)$, $0 \leq s \leq r$, with $\tau(0) = 0$ and $\tau'(s) > 0$ for $s > 0$, such that, for $0 < s \leq r$ and $0 \leq |t|^2 \leq \tau(s)$, the fiber $X_t = f^{-1}(t)$ is transverse to the real hypersurface $\delta(x) = s$ in X, and the box

$$B_{t,d} = \{x \in X \mid \delta(x) \leq d \text{ and } |f(x)|^2 < t\}$$

is proper over the open disc D_t of radius $t^{1/2}$.

Let $L = X_0 \cap \delta^{-1}(r)$. By (c) and (d), there is a homeomorphism from $X_0 \cap \delta^{-1}([0, r])$ to the cone

$$L \times [0, r] \big/ \big(L \times \{0\} \text{ contracted to a point}\big)$$

which transforms δ into the second coordinate and is C^∞ away from the origin.

3.1.3.

Let $d \leq r$ and $t \leq \tau(d)$. Away from 0, $B_{t,d}$ is a manifold with corners, and the projection $f : B_{t,d} \to D_t$ is a fibration. In particular:

(a) over $D_t^* = D_t - \{0\}$, $B_{t,d}$ induces a C^∞ bundle of compact manifolds with boundary;

(b) the boundary $\delta(x) = d$ of this bundle extends to a C^∞ bundle $\partial B_{t,d}$ of compact manifolds over D_t.

Let $y \in D_t^*$. We call the manifold with boundary $V = f^{-1}(y)$ the manifold of vanishing cycles. Since the disc is contractible, the bundle $\partial B_{t,d}$ over D_t is isomorphic to the constant bundle with fiber ∂V, and the space of bundle isomorphisms

$$\alpha : \partial B_{t,d} \xrightarrow{\sim} \partial V \times D_t$$

that restrict to the identity on the fiber over y is contractible. Choose one, denoted by α. Lift the vector field $\partial/\partial\theta$ on $S_{|y|}$, the circle of radius $|y|$, to a vector field u on $f^{-1}(S_{|y|})$ which is compatible with α on the boundary, so that

$$d\alpha(u) = (0, \partial/\partial\theta).$$

The space of such lifts is contractible. Integrating u once around the circle in the positive direction defines a diffeomorphism T of V onto itself which is the identity on ∂V; this is the monodromy transformation. It is well defined up to isotopy relative to ∂V.

In a sense whose precise formulation is left to the reader, (V, T) is independent of the choices of t and d. I do not know whether it is independent of the choice of δ. This is nevertheless the case if we restrict δ to be the restriction to X of a real-analytic function g on $\mathbb{C}^{n+1}$ such that, on the Zariski tangent space of X_0 at 0, one has $dg = 0$ and $d^2g > 0$ (positive definite). Indeed, for δ and δ' of this kind, r can be chosen as in 3.1.2(c),(d), uniformly for every $\lambda\delta + (1 - \lambda)\delta'$, $0 \leq \lambda \leq 1$.

3.1.4.

Let (V_1, T_1) be obtained from (V, T) by adjoining a collar:

$$V_1 - V^\circ \simeq \partial V \times [0,1], \qquad T_1|_{V_1 - V^\circ} = \mathrm{id}\,.$$

Define the following space over D_t (cf. 1.4.3):

$$f' : W \longrightarrow D_t.$$

(a) Let $\widetilde{D}_t^*$ be a universal covering of D_t^* with base point y, and let W^* be the bundle over D_t^* obtained as the quotient of $V_1 \times \widetilde{D}_t^*$ by the equivalence relation generated by

$$(v, y) \sim (T_1 v, e^{2\pi i} y).$$

(b) Since $T_1|_{V_1 - V^\circ}$ is trivial, W^* can be glued to the trivial bundle over D_t with fiber $V_1 - V$, giving a space W_1 over D_t.

(c) Put $W = W_1 \cup \{0\}$ and $f'(0) = 0$. Give W the unique topology inducing that of W_1, for which f' is continuous and, for $t' < t$, $f'^{-1}(\overline{D}_{t'})$ is the one-point compactification of $W_1 \cap f'^{-1}(\overline{D}_{t'})$.

Equip $W - \{0\}$ with the integrable connection of 1.3.4 for which the trivialization, supplied by (a), of the inverse image of W on $\widetilde{D}^*$ is horizontal.

Proposition 3.1.5.

There exists a homeomorphism

$$\Psi : B_{t,d} \xrightarrow{\sim} W$$

compatible with the projections to D_t and carrying 0 to 0; this homeomorphism may be chosen to be a diffeomorphism away from 0.

This result will not be used in the remainder of the seminar.

Let $d_0 < d_1 \leq r$, let $t_0 \leq \tau(d_0)$, and let $y \in D_{t_0}^*$. We have

$$B_{t_0,d_0} \subset B_{t_0,d_1},$$

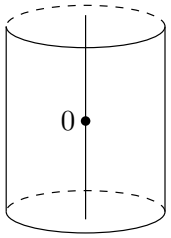
and $B_{t_0,d_1} - B^\circ_{t_0,d_0}$ is a bundle over D_{t_0}. Trivialize it and, over the circle $S_{|y|}$, lift $\partial/\partial\theta$ compatibly with this trivialization. Let T_1 be the corresponding endomorphism of

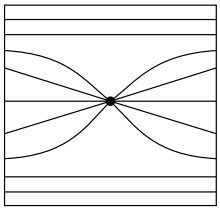
$$V_1 = f^{-1}(y) \cap B_{t_0,d_1}, \qquad V_0 = f^{-1}(y) \cap B_{t_0,d_0}.$$

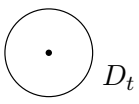
Then $(V_0 \subset V_1, T_1)$ is isomorphic to $(V \subset V_1, T_1)$ of 3.1.4. Let W be constructed from $(V_0 \subset V_1, T_1)$. We have $f'^{-1}(y) = V_1$ canonically. The proof will show that Ψ can be chosen to induce the identity

$$f^{-1}(y) \xrightarrow{\sim} f'^{-1}(y).$$

The following drawings of f and of its restriction to $f^{-1}(\mathbb{R})$ summarize the proof.







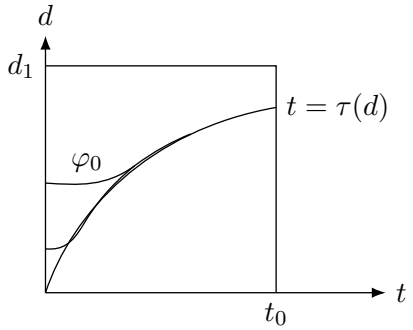
Let $B^+ \subset B_{t_0,d_1}$ be the set of points x such that

$$|f(x)|^2 \leq \tau(\delta(x)).$$

Let φ_n be a sequence of C^∞ functions from $[0,t_0]$ to $(0,d_1]$ such that, for a sequence $t_0 > r_1 > r_2 > \cdots$ tending to 0,

(a) for $|t|^2 \geq r_i$, φ_i is the inverse function of τ;

(b) for $|t|^2 < r_i$ and $j > i$, $\varphi_j(t) < \varphi_i(t)$.



The space $B^+ - \{0\}$ is the union of the bundles B_i^+ over D_t defined by

$$\delta(x) \geq \varphi_i(|f(x)|^2).$$

A standard argument, using the fact that δ has no critical points on the fibers of $B^+ - \{0\} \to D_t$, shows that there is a diffeomorphism $\alpha_i : B_i^+ \to B_{i+1}^+$, compatible with the projection to D_t and equal to the identity outside a small neighborhood of the subspace

$$\delta(x) = \varphi_i(|f(x)|^2), \qquad |f(x)|^2 \leq r_i,$$

of B_i^+. Composing the α_i and passing to the limit gives a diffeomorphism

$$\alpha : B_0^+ - \{x \mid f(x) = 0,\ \delta(x) = \varphi_0(0)\} \xrightarrow{\sim} B^+ - \{0\}.$$

Transport a trivialization of the bundle B_0^+ over D_t by α to obtain an integrable connection ∇' on $B^+ - \{0\}$. We may then:

(a) extend the restriction of this connection to $f^{-1}(S_{|y|}) \cap B^+$ to a connection on $f^{-1}(S_{|y|})/S_{|y|}$; this extension gives T_1;

(b) extend to B_{t_0,d_1} the lift, supplied by ∇' on $B^+ - \{0\}$, of the radial vector field on D_t.

There is then a unique integrable connection ∇ on B_{t_0,d_1} extending ∇', (a), and (b), and a unique horizontal diffeomorphism from $B_{t_0,d_1} - \{0\}$ to $W - \{0\}$ that is the identity on $f^{-1}(y)$. This is the required diffeomorphism.

3.1.6. A generalization.

The preceding discussion generalizes to the case in which $f : X \to D$ is a morphism such that $0 \in X$ is an isolated non-smooth point in X_0. After shrinking X and D, the non-smooth locus of f is then finite over D. The manifold of vanishing cycles V has a finite set Σ of critical points; $V - \Sigma$ is a differentiable manifold with boundary (its boundary is compact, although the manifold is not). The monodromy transformation T is the identity on the boundary and stabilizes Σ. Finally, Proposition 3.1.5 remains valid: the fibered homeomorphism Ψ is a diffeomorphism away from the critical locus, which it carries to the part of W arising from Σ.

3.1.7.

Proposition 3.1.5 allows us to use the calculations of 1.4. We obtain:

(A) The vanishing-cycle groups $(R^i\Phi(\Lambda))_0$ are the reduced cohomology groups, with coefficients in Λ, of the manifold of vanishing cycles V.

(B) The groups $H^i_{\{0\}}(X_0, R\Psi(\Lambda))$ are the cohomology groups with proper support of V°, with coefficients in Λ.

(C) The variation

$$\mathrm{Var} : \widetilde{H}^i(V,\Lambda) \longrightarrow H^i_c(V^\circ,\Lambda) = H^i(V \bmod \partial V, \Lambda)$$

is the classical variation defined by the automorphism T of V, which is the identity on ∂V.

3.1.8.

Assertions (A), (B), and (C) are in fact more elementary than Proposition 3.1.5. Here is a direct proof and construction which spares us from checking that the isomorphisms in (A) and (B) are independent of the choices made in the construction of 3.1.5.

Let $d \le r$ and $t \le \tau(d)$, and let B be the open box

$$B = \{x \mid \delta(x) < d \text{ and } |f(x)|^2 < t\}.$$

Let $\widetilde{D}^*$ be a universal covering of D^*, and consider the diagram

(3.1.8.1)
$$\begin{array}{ccccc} \{0\} & \xhookrightarrow{\bar\imath} & \widetilde{D}_t & \xhookleftarrow{\bar\jmath} & \widetilde{D}_t^* \\ \| & & \downarrow p & & \downarrow p' \\ \{0\} & \xhookrightarrow{i} & D_t & \xhookleftarrow{j} & D_t^* \end{array}$$

obtained from (1.2.7.2) by restriction to D_t. Let

(3.1.8.2)
$$\begin{array}{ccccc} B_0 & \xhookrightarrow{\bar\imath} & \overline{B} & \xhookleftarrow{\bar\jmath} & \overline{B}^* \\ \| & & \downarrow p & & \downarrow p' \\ B_0 & \xhookrightarrow{i} & B & \xhookleftarrow{j} & B^* \end{array}$$

be the inverse image of $f : B \to D$ over (3.1.8.1). We again write f for the projection from (3.1.8.2) to (3.1.8.1).

(A) By definition, $R^i\Psi_\eta(\Lambda)_0$ is the direct limit, as U runs through the neighborhoods of 0 in X, of the cohomology groups

$$H^i(U \times_D \widetilde{D}^*, \Lambda).$$

Let U run through the fundamental system of neighborhoods of 0 formed by boxes B as above. The fiber product $U \times_D \widetilde{D}^* = \overline{B}^*$ is a bundle with fiber V° over the contractible space $\widetilde{D}_t^*$. Hence the direct system of the groups $H^i(\overline{B}^*, \Lambda)$ is constant, with value

$$H^i(V^\circ, \Lambda) = H^i(V, \Lambda).$$

The resulting isomorphisms

(3.1.8.3)
$$R^i\Psi_\eta(\Lambda)_0 \simeq H^i(V,\Lambda)$$

may be refined to the following composite isomorphism (cf. (B) below):

$$\begin{array}{ccccccc} R\Psi_\eta(\Lambda)_0 & \xleftarrow{\sim} & R\Gamma(B_0, R\Psi_\eta(\Lambda)) & \xleftarrow{\sim} & (Rf_*R\bar{\jmath}_*\Lambda)_0 & \xleftarrow{\sim} & R\Gamma(\widetilde{D}_t, Rf_*R\bar{\jmath}_*\Lambda) \\ & & & & & & \| \\ & & R\Gamma(V,\Lambda) & \xrightarrow{\sim} & R\Gamma(V^\circ,\Lambda) & \xleftarrow{\sim} & R\Gamma(\overline{B}^*,\Lambda) \end{array} \tag{3.1.8.4}$$

The complex $R\Phi(\Lambda)_0$ is defined by the distinguished triangle

$$\Lambda \longrightarrow (R\Psi_\eta\Lambda)_0 \longrightarrow R\Phi(\Lambda)_0 \longrightarrow .$$

Consequently, the groups $R^i\Phi(\Lambda)_0$ are the reduced cohomology groups of V.

(B) Let $\mathfrak{F}$ be the family of closed subsets of $\overline{B}^*$ whose closure in $\overline{B}$ is proper over $\widetilde{D}$. The arrows in the following diagram are isomorphisms:

$$\begin{array}{ccccc} R\Gamma_{\{0\}}(R\Psi_\eta(\Lambda)) & \underset{①}{\xrightarrow{\sim}} & R\Gamma_c(B_0, R\Psi_\eta(\Lambda)) & \underset{②}{\xleftarrow{\sim}} & (Rf_!R\bar{\jmath}_*\Lambda)_0 \\ & & & & \uparrow\wr\ ③ \\ R\Gamma_c(V^\circ,\Lambda) & \underset{⑤}{\xleftarrow{\sim}} & R\Gamma_{\mathfrak{F}}(\overline{B}^*,\Lambda) & \underset{④}{=\!=} & R\Gamma(\widetilde{D}_t, Rf_!R\bar{\jmath}_*\Lambda) \end{array} \tag{3.1.8.5}$$

Indeed, ① is an isomorphism because B_0 is a cone (3.1.2) and $R^i\Psi_\eta(\Lambda)$ is constant away from 0; ② by the proper base-change theorem; ③ because $Rf_!R\bar{\jmath}_*\Lambda$ is constant away from 0; ④ is the derived-category version of the spectral sequence of a composite functor; and ⑤ because $\widetilde{D}^*$ is contractible.

The isomorphism (3.1.8.5) supplies the desired isomorphisms

$$R^i\Gamma_{\{0\}}(R\Psi_\eta(\Lambda)) \simeq H^i_c(V^\circ,\Lambda). \tag{3.1.8.6}$$

We shall verify (C) in 3.1.10.

3.1.9.

Cf. [3]. Let X be a topological space, let $i : Y \hookrightarrow X$ be a closed subset, let T be an automorphism of X inducing the identity on Y, and let Λ be an abelian group. To define the variation Var, choose a T-equivariant resolution Λ^*_X of Λ on X, a resolution Λ^*_Y of the constant sheaf Λ on Y on which, by convention, T acts trivially, and an equivariant morphism

$$\varphi : \Lambda^*_X \longrightarrow i_*\Lambda^*_Y$$

resolving the evident morphism $\varphi_0 : \Lambda_X \to i_*\Lambda_Y$. Suppose that the sheaves Λ^i_X and Λ^i_Y are acyclic for the functors to be considered. Passing to global sections gives

$$\Gamma(\varphi) : \Gamma(\Lambda^*_X) \longrightarrow \Gamma(\Lambda^*_Y).$$

For simplicity, suppose that every $\Gamma(\varphi)^i$ is surjective; otherwise, make it surjective by adding the cone of $\Gamma(\Lambda^*_Y)$ to $\Gamma(\Lambda^*_X)$. Put

$$R\Gamma(X \bmod Y, \Lambda) = \mathrm{Ker}(\Gamma(\varphi)).$$

Under reasonable hypotheses, if $\mathfrak{F}$ is the family of closed subsets of X disjoint from Y, there is a quasi-isomorphism

$$\Gamma_{\mathfrak{F}}(X - Y, \Lambda^*_X) \xrightarrow{\sim} R\Gamma(X \bmod Y, \Lambda).$$

The endomorphism $T-1$ of $\Gamma(\Lambda_X^*)$, where T acts by transport of structure, is zero on Λ (the constants) and factors through $\mathrm{Ker}(\Gamma(\varphi))$. Hence, in the derived category and under the same hypotheses, it gives a variation morphism

$$\mathrm{Var} : R\widetilde{\Gamma}(X,\Lambda) \longrightarrow R\Gamma_{\mathfrak{F}}(X-Y,\Lambda).$$

Without these hypotheses, one obtains instead

$$\mathrm{Var} : R\widetilde{\Gamma}(X,\Lambda) \longrightarrow R\Gamma(X \bmod Y,\Lambda).$$

This description simplifies when T is the identity on a neighborhood of Y and acts as the identity on Λ_X^* in that neighborhood. The endomorphism $T-1$ of $\Gamma(\Lambda_X^*)$ then factors through $\Gamma_{\mathfrak{F}}(\Lambda_X^*)$ and supplies the variation.

Proposition 3.1.10.

Under the isomorphisms of 3.1.8(A),(B), the variation

$$(R\Phi_\eta(\Lambda))_0 \longrightarrow R\Gamma_{\{0\}}(R\Psi_\eta(\Lambda))$$

is equal to the variation of 3.1.9 defined by the automorphism T of V, which is the identity on ∂V.

The reader is invited to draw a small picture rather than read the proof. Let ${}_fB$ be the closed box of 3.1.2 and let ${}_fC \subset {}_fB$ be a collar along the boundary; thus ${}_fC \simeq \partial_f B \times [0,1]$ as a bundle over D_t, through an isomorphism carrying $\partial_f B \times \{1\}$ identically onto $\partial_f B$. Let

$${}_f\alpha : {}_fC \xrightarrow{\sim} (\partial V \times [0,1]) \times D_t$$

be a trivialization of the bundle ${}_fC$ over D_t whose fiber at $y \in D_t$ induces the identity

$$\partial V = \partial(f^{-1}(y)) \xrightarrow{\sim} \partial V \times \{1\}.$$

Put $C = {}_fC \cap B$ and let

$$\alpha : C \xrightarrow{\sim} (\partial V \times [0,1)) \times D_t$$

be induced by ${}_f\alpha$. We retain the notation of (3.1.8.2).

Let R be the equivalence relation on $\overline{B}$ defined by xRy if either $x=y$, or $p(x), p(y) \in C$ and $\alpha(p(x))$, $\alpha(p(y))$ have the same image in $\partial V \times [0,1)$. Put

$${}_1\overline{B} = \overline{B}/R, \qquad {}_1\overline{B}^* = \overline{B}^*/R, \qquad {}_1\overline{C}^* = p^{-1}(C)/R.$$

There is a commutative diagram

$$\begin{array}{ccccc} B_0 & \xhookrightarrow{\bar{\imath}} & \overline{B} & \xhookleftarrow{\bar{\jmath}} & \overline{B}^* \\ \| & & \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle q'} \\ B_0 & \xhookrightarrow{\bar{\imath}} & {}_1\overline{B} & \xhookleftarrow{{}_1\bar{\jmath}} & {}_1\overline{B}^* \end{array}$$

Let Λ^* be a $\pi_1(D^*)$-equivariant resolution of Λ by acyclic sheaves on $\overline{B}^*$. By definition, the variation is computed from the morphism of complexes

$$\Lambda[0] \longrightarrow \bar{\imath}^*\bar{\jmath}_*(\Lambda^*).$$

Since $\Lambda \xrightarrow{\sim} Rq_*\Lambda$ and $\Lambda \xrightarrow{\sim} Rq'_*\Lambda$, it may equally be computed using an equivariant resolution of Λ on ${}_1\overline{B}^*$ and the morphism

$$\Lambda[0] \longrightarrow \bar{\imath}^*{}_1\bar{\jmath}_*(\Lambda).$$

Under (A) and (B), the variation is thereby identified with the composite morphism

$$\begin{array}{ccccc} R\Gamma(V,\Lambda) & \xrightarrow{\sim} & R\Gamma(V^\circ,\Lambda) & \xleftarrow[①]{\sim} & R\Gamma({}_1\overline{B}^*,\Lambda) \\ & & & & \Big\downarrow \mathrm{Var} \\ & & R\Gamma_c(V^\circ,\Lambda) & \xleftarrow[②]{\sim} & R\Gamma({}_1\overline{B}^* \bmod {}_1\overline{C}^*,\Lambda) \end{array} \tag{3.1.10.1}$$

where ① and ② are induced by the inclusion of V° in ${}_1\overline{B}^*$, and where Var is the variation of 3.1.9 for ${}_1\overline{B}^*$ and the action of the positive generator of $\pi_1(D^*)$. This group acts on ${}_1\overline{B}^*$ through the action described in 1.1.5 and acts trivially on ${}_1\overline{C}^*$.

Choose a connection ∇ on B^*/D^* compatible with the trivialization of C. This connection trivializes $\overline{B}^*$ and, on passing to the quotient, defines a map $u : {}_1\overline{B}^* \to V$. For T defined as in 3.1.3, the diagram

$$\begin{array}{ccc} {}_1\overline{B}^* & \xrightarrow{u} & V \\ {\scriptstyle T}\downarrow & & \downarrow{\scriptstyle T} \\ {}_1\overline{B}^* & \xrightarrow{u} & V \end{array}$$

commutes. The map u induces isomorphisms in cohomology, so Proposition 3.1.10 follows from (3.1.10.1) and the functoriality of variation.

Proposition 3.1.10 is meant to justify an explicit calculation to be made later in which, as here, T is the identity on a neighborhood of ∂V. We shall take a cohomology class c represented by a current C and use the fact that $\mathrm{Var}(c)$ is represented by the current $TC - C$, compactly supported in V°.

3.1.11.

In practice, T is often calculated in two steps.

(a) Fix t and d with $t \leq \tau(d)$, choose $y \in D_t^*$, and choose a connection ∇ on $f^{-1}(S_{|y|}) \cap B_{t,d}$. This connection defines a diffeomorphism T_0 of

$$V_0 = f^{-1}(y) \cap B_{t,d}$$

onto itself, which need not be the identity on the boundary.

(b) Choose a trivialization of $\partial B_{t,d}$ over D_t. Integrating ∇ gives a diffeomorphism $\varphi(\theta)$ from ∂V to

$$f^{-1}(e^{2\pi i\theta}y) \cap \partial B_{t,d},$$

which is identified with ∂V by the trivialization. For $0 \leq \theta \leq 1$, $\varphi(\theta)$ is a path in $\mathrm{Diff}(\partial V)$ joining the identity to ∂T_0.

The resulting pair (T_0, φ) depends on the arbitrary choices made. Since the space of these choices is contractible, it is independent of them up to homotopy. For a transformation T of the kind considered in 3.1.3, $(T, \text{constant path})$ is a pair of the kind considered here.

3.1.12.

The map

$$\mathrm{Diff}(V) \longrightarrow \mathrm{Diff}(\partial V)$$

is a fibration. Consequently, the assignment $T \mapsto (T, \text{constant path})$ induces a bijection between:

a) diffeomorphisms $T : V \to V$ satisfying $\partial T = \mathrm{id}$, up to isotopy; and

b) pairs (T_0, φ), where $T_0 : V \xrightarrow{\sim} V$ and φ is a homotopy class of paths from id to ∂T_0, up to isotopy.

A pair (T_0, φ) calculated by 3.1.11(a),(b) therefore determines T. Concretely, calculate T from (T_0, φ) as follows.

(α) Choose a collar embedding

$$[0,1] \times \partial V \xhookrightarrow{\alpha} V, \qquad \alpha(1, x) = x,$$

and a representative $\varphi : [0,1] \to \mathrm{Diff}(\partial V)$ such that its extension by id for negative parameters is C^∞. On the collar put

$$\beta(x) = \mathrm{pr}_1\, \alpha^{-1}(x).$$

Thus β has no critical point, takes values in $[0,1]$, and equals 1 on ∂V.

(β) Take $T = T_0$ away from the collar and, on the collar, set

$$T(x) = T_0\big(\varphi(\beta(x))^{-1}(x)\big).$$

3.1.13.

Here is a variant of 3.1.11(b) for calculating the homotopy class of the path φ. On every ray $\arg(z) = \theta$, choose a trivialization of the bundle $\partial B_{t,d}$ which varies continuously with θ. Use the trivialization over the ray through y to identify

$$(\partial V)_0 = f^{-1}(0) \cap \partial B_{t,d}$$

with ∂V. Let

$$\Psi_\theta : V \longrightarrow f^{-1}(e^{i\theta} y) \cap B_{t,d}$$

be obtained by integrating ∇. Transporting $\partial \Psi_\theta$ along the rays gives a diffeomorphism of $(\partial V)_0$, hence of ∂V, onto itself. This is $\varphi(\theta)$.

3.2. The Picard–Lefschetz formula (transcendental case)

3.2.1.

We shall make the results of 3.1 explicit for the map

$$f = \sum_{i=1}^{n+1} z_i^2 : \mathbb{C}^{n+1} \longrightarrow \mathbb{C}.$$

For δ take the function $\sum |z_i|^2$, and take the box $B = B_{a,b}$ defined by

$$\Big|\sum z_i^2\Big| \le a^2, \qquad \sum |z_i|^2 \le b^2,$$

where $0 < a < b$. It is only to simplify the notation that we use a closed box rather than a box as in 3.1.2.

3.2.2.

The manifold of vanishing cycles V has equations

$$\begin{aligned} \sum z_i^2 &= a^2, \\ \sum |z_i|^2 &\leq b^2. \end{aligned} \tag{3.2.2.1}$$

Write $z = (z_i)$ in terms of its real and imaginary parts, $z = x + iy$. The preceding equations become

$$\begin{aligned} x^2 - y^2 &= a^2, \\ x \cdot y &= 0 \qquad (\text{i.e. } x \perp y \text{ in } \mathbb{R}^{n+1}), \\ x^2 + y^2 &\leq b^2. \end{aligned} \tag{3.2.2.2}$$

[16] Introduce on V the new coordinate system

$$X = \frac{x}{|x|}, \qquad Y = y\left(\frac{b^2 - a^2}{2}\right)^{-1/2}.$$

Then $x = X(a^2 + y^2)^{1/2}$, and (3.2.2.2) becomes

$$\begin{aligned} X^2 &= 1, \\ X \cdot Y &= 0, \\ Y^2 &\leq 1. \end{aligned} \tag{3.2.2.3}$$

This identifies V with the tangent ball bundle of the sphere S^n.

3.2.3.

The map

$$\theta \longmapsto \exp(2\pi i\theta)a$$

identifies $\mathbb{R}$ with the universal covering of the circle S_a in $\mathbb{C}$ having radius a and center 0. The pullback to $\mathbb{R}$ of the bundle B is isomorphic to $\mathbb{R} \times V$. We trivialize it by means of the isomorphism

$$v \longmapsto \exp(\pi i\theta)v$$

from V to the fiber of B over $\exp(2\pi i\theta)a$. Let T_0 be the map furnished by this trivialization as in 3.1.11. In the coordinates (3.2.2.3),

$$T_0(X, Y) = (-X, -Y).$$

3.2.4.

The bundle over the disc D_a of radius a which forms the boundary of the "bundle" B has equations

$$\sum |z_i|^2 = b^2, \qquad f(z) \in D_a,$$

the projection to the disc being f. We shall describe a trivialization of the inverse image of this bundle over each ray

$$e^{2\pi i\theta}\mathbb{R}^+ \cap D_a.$$

Over the ray $\mathbb{R}^+ \cap D_a$, take the isomorphism with $(\mathbb{R}^+ \cap D_a) \times \partial V$ which, in the coordinates (3.2.2.3) on ∂V, is given by

$$z = x + iy \longmapsto (f(z), X, Y) = \left(f(z), \frac{x}{|x|}, \frac{y}{|y|}\right).$$

[16] The printed source has $\mathbb{R}^n$ in (3.2.2.2), while z has $n + 1$ coordinates. The English text corrects this evident dimension typo to $\mathbb{R}^{n+1}$.

Over $(e^{2\pi i\theta}\mathbb{R}^+) \cap D_a$, use the image of this trivialization under multiplication by $e^{\pi i\theta}$. Apply 3.1.13 and use on ∂V the coordinates (X, Y) satisfying

$$X^2 = Y^2 = 1, \qquad X \cdot Y = 0. \tag{3.2.4.1}$$

If $\varphi(\theta)$ takes (X, Y) to (X_θ, Y_θ), one finds

$$X_\theta + iY_\theta = e^{\pi i\theta}(X + iY) \qquad (0 \leq \theta \leq 1). \tag{3.2.4.2}$$

3.2.5.

In summary, the manifold of vanishing cycles V is the tangent ball bundle of S^n. The monodromy transformation is represented by (T_0, φ), where T_0 is the differential of the antipodal map and φ is the following path from Id to ∂T_0: under $\varphi(\theta)$, each unit tangent vector is parallel-transported along the geodesic which it determines through a distance $+\pi\theta$ (the plus sign means "in the direction in which the vector points"). At $\theta = 1$ one reaches the antipodal vector.

By 3.1.13 this gives the following description of T. Let $\varphi : [0, 1] \to [0, 1]$ be a C^∞ map which is 0 near 0 and 1 near 1. The map T moves u, along the geodesic which u determines, through the distance

$$-\pi\big(1 - \varphi(|u|)\big).$$

Near $|u| = 1$ this is the identity; near $|u| = 0$ it is the antipodal map.

3.2.7.

Let W be a differentiable manifold of dimension d, and let

$$p : T(W) \longrightarrow W$$

be its tangent bundle, regarded as a manifold. If $(x_1, \ldots, x_d)$ is a local coordinate system on W, then

$$(x_1, \ldots, x_d, dx_1, \ldots, dx_d)$$

is a local coordinate system on $T(W)$. Let ε' be the orientation of $T(W)$ defined by this coordinate system. It is independent of the choice of the x_i, and is therefore globally defined.

Suppose that W is orientable. The submanifold $W \subset T(W)$ (the zero section), endowed with an orientation (respectively, with an orientation of its normal bundle), defines an element w of $H_d(T(W))$ (respectively, of $H^d_c(T(W))$). The self-intersection of $W \subset T(W)$ is the Euler–Poincare characteristic:

$$\int w \wedge w = \chi(W)$$

for the orientation ε'.

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For the manifold of vanishing cycles V, its complex structure defines an orientation ε. It is related to the orientation ε' of the tangent bundle of S^n by

$$\varepsilon = (-1)^{\frac{n(n-1)}{2}}\varepsilon'.$$

3.2.8.

The vanishing cycle δ is well defined up to sign. For $n \neq 0$, it is the element of $H^n_c(V^\circ)$ defined by the submanifold $S^n \subset V^\circ$. For $n = 0$, the space V consists of two points, and δ is their difference. With the complex orientation of V, 3.2.7 gives

$$(\delta, \delta) = (-1)^{\frac{n(n-1)}{2}}\big(1 + (-1)^n\big). \tag{3.2.8.1}$$

[17] The printed sentence mixes three dimension symbols: it calls the dimension x, indexes the coordinates by n, and places the classes in degree d. The English text consistently uses d in the general statement; the specialization below uses $d = n$.

3.2.9.

Put

$$\widetilde{H}^i_c(V^\circ) = \mathrm{Ker}\Big(H^i_c(V^\circ) \xrightarrow{\mathrm{Tr}} \mathbb{Z}\Big)\,;$$

this is $H^i_c(V^\circ)$ for $i \neq 2n$. Then

$$\begin{aligned} \widetilde{H}^i_c(V^\circ) &= 0 \quad \text{for } i \neq n, \\ \widetilde{H}^n_c(V^\circ) &= \mathbb{Z}, \quad \text{generated by } \delta. \end{aligned}$$

Dually,

$$\begin{aligned} \widetilde{H}^i(V) &= 0 \qquad\qquad \text{for } i \neq n, \\ \widetilde{H}^n(V) &\xrightarrow[\sim]{(\,\cdot\,,\delta)} \mathbb{Z}. \end{aligned}$$

For $n \neq 0$, the image of $\delta \in H^n_c(V^\circ)$ in $H^n(V)$ is zero when n is odd, and is divisible by two (and no further) when n is even.

3.2.10.

Let us calculate the variation

$$\mathrm{Var} : \widetilde{H}^n(V) \longrightarrow H^n_c(V^\circ).$$

We first use the orientation ε' and suppose that $n \neq 0$. Let α be an orientation of the tangent bundle of S^n, and let $\delta \in H^n_c(V^\circ)$ be defined by $S^n \subset V^\circ$ together with the orientation α of its normal bundle. The cocycle $\delta' \in H^n(V)$ such that $(\delta', \delta) = 1$ is represented by a fiber of the fibration of V° over S^n. The normal bundle of this fiber is the constant bundle with value the tangent space at one point of S^n, and it is endowed with the orientation α.

This assertion is independent of the sign convention chosen to define the map

$$(\text{submanifold} + \text{orientation of its normal bundle}) \longrightarrow \text{cohomology}.$$

Indeed, changing the convention changes δ to $-\delta$ and δ' to $-\delta'$. The assertion is valid when

$$\mathrm{Tr} : H^{2n}_c(V, \mathbb{R}) \longrightarrow \mathbb{R}$$

is defined as follows:

(α) since Ω^*_V is a soft resolution of $\underline{\mathbb{R}}$, $\Gamma_c(\Omega^*_V)$ represents $R\Gamma_c(V, \mathbb{R})$, whence an isomorphism

$$H^i\big(\Gamma_c(V, \Omega^*_V)\big) \simeq H^i_c(V, \mathbb{R})$$

(the iterated coboundary is not used);

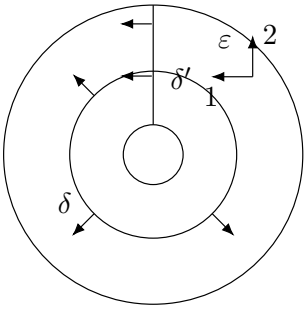
(β) Tr is integration.

The class $T(\delta')$ coincides with δ' in a neighborhood of ∂V, so that $\mathrm{Var}(\delta') = T(\delta') - \delta'$ has compact support in V°. We have

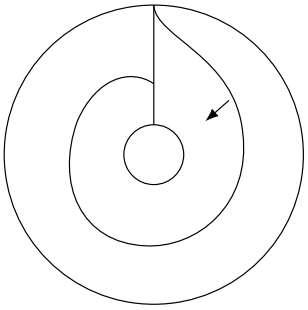
$$T(\delta') - \delta' \sim -\delta, \qquad \text{and hence}$$

$$\mathrm{Var}(\delta') = -(\delta', \delta)\,\delta.$$

Here is the picture for $n = 1$. We draw the tangent bundle of S^1 so that vectors pointing in the chosen counterclockwise orientation of S^1 point outward.



δ, δ' and ε



$\mathrm{Var}(\delta')$

For the orientation ε, we therefore have

$$\mathrm{Var}(x) = -\frac{\varepsilon}{\varepsilon'}(x,\delta)\,\delta.$$

Theorem 3.2.11.

The group $\widetilde{H}^n_c(V^\circ)$ is isomorphic to $\mathbb{Z}$, generated by the vanishing cycle δ. The group $\widetilde{H}^n(V)$ is also isomorphic to $\mathbb{Z}$ and is dual to the preceding group. All the other groups $\widetilde{H}^i_c(V^\circ)$ and $\widetilde{H}^i(V)$ vanish.

The variation is given by

$$\mathrm{Var}(x) = -(-1)^{\frac{n(n-1)}{2}}(x,\delta)\,\delta, \qquad \text{and}$$

$$(\delta,\delta) = (-1)^{\frac{n(n-1)}{2}}\big(1+(-1)^n\big).$$

According to the congruence class of n modulo 4, we have the following table:

$n \bmod 4$	0	1	2	3
(δ,δ)	2	0	-2	0
$\mathrm{Var}(x) = \pm(x,\delta)\delta$	$-$	$-$	$+$	$+$
$T(\delta) = (-1)^{n+1}\delta$	$-\delta$	δ	$-\delta$	δ

As a check, we indeed have

$$T(\delta) = \delta + \mathrm{Var}(\delta).$$

4. De Rham cohomology

4.1.

Let D be a complex analytic disc, let $0 \in D$, put $D^* = D - \{0\}$, and let $\widetilde{D}^*$ be a universal covering of D^*. Let X be an analytic space, let $f : X \longrightarrow D$ be a morphism, and consider the diagram of (1.2.7.1):

$$\begin{array}{ccccc} X_0 & \xrightarrow{\bar{i}} & \bar{X} & \xleftarrow{\bar{j}} & \bar{X}^* \\ \| & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p'} \\ X_0 & \xrightarrow{i} & X & \xleftarrow{j} & X^* \end{array} \tag{4.1.1}$$

Suppose that $X^* = X - X_0$ is nonsingular (that is, smooth). Its covering $\bar{X}^*$ is therefore smooth. Denote by $\Omega^*_{\bar{X}^*}$ the De Rham complex of sheaves of holomorphic differential forms on $\bar{X}^*$. It is a resolution of the constant sheaf $\mathbb{C}$ (the holomorphic Poincare lemma), and hence gives a morphism

$$\bar{j}_*\Omega^*_{\bar{X}^*} \longrightarrow R\bar{j}_*\mathbb{C} \tag{4.1.2}$$

in $D^+(\bar{X},\mathbb{C})$.

Proposition 4.2.

The morphism (4.1.2) is an isomorphism.

It is clear that (4.1.2) is a quasi-isomorphism at every point of $\bar{X}^* \subset \bar{X}$.[18] It is therefore enough to show that every point $x \in X_0$ has a fundamental system of neighborhoods U in $\bar{X}$ such that

$$H^q\big(\bar{j}^{-1}(U), \Omega^p\big) = 0 \qquad \text{for } q > 0.$$

As V runs through a fundamental system of neighborhoods of x in X, the sets $p^{-1}(V)$ form a fundamental system of neighborhoods of x in $\bar{X}$. In particular, one may take the $p^{-1}(V)$ for V a Stein neighborhood of x in X. If V is Stein, then $V - X_0$ is still Stein (since $X_0 \cap V$ is defined by one equation in V), as is its covering

$$p'^{-1}(V - X_0) = \bar{j}^{-1}p^{-1}(V).$$

Consequently, for every coherent sheaf $\mathcal{F}$ on $\bar{X}^*$,

$$H^q\big(\bar{j}^{-1}p^{-1}(V), \mathcal{F}\big) = 0 \qquad \text{for } q > 0,$$

which proves the proposition.

Corollary 4.3.

The isomorphism (4.1.2) induces an isomorphism

$$\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*} \xrightarrow{\sim} R\Psi_\eta(\mathbb{C}). \tag{4.3.1}$$

Variant 4.4.

Let V be a locally constant sheaf of finite-dimensional complex vector spaces on X^* (a complex local system). The quasi-isomorphism

$$V \longrightarrow \Omega^*_{X^*}(V)$$

of the Poincare lemma induces a quasi-isomorphism

$$\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*}(p'^*V) \xrightarrow{\sim} R\Psi_\eta(V). \tag{4.4.1}$$

We shall show that (4.3.1) remains true when $\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*}$ is replaced by a subcomplex defined by growth and monodromy conditions.

4.5.

Let $f : X^* \longrightarrow D^*$ be a topological space over D^*, let F be a sheaf of $f^\circ\mathcal{O}_{D^*}$-modules on X^*, and let $\bar{F}$ be its inverse image on

$$\bar{X}^* = X^* \times_{D^*} \widetilde{D}^*.$$

Let $z : D \hookrightarrow \mathbb{C}$ be a holomorphic embedding with $z(0) = 0$. We shall always be free to shrink D below; for a sufficiently small disc the existence of z is trivial. Choose on $\widetilde{D}^*$ a function $\log z$ such that $\exp(\log z) = z$, and put

$$z^\alpha = \exp(\alpha \log z) \qquad (\alpha \in \mathbb{C}).$$

The group $\pi_1(D^*)$ acts, by transport of structure, on $H^0(\bar{X}^*, \bar{F})$. An element $f \in H^0(\bar{X}^*, \bar{F})$ is of finite determination if its transforms under $\pi_1(D^*)$ span a finite-dimensional complex vector

[18] The printed proof says (4.1.1), which is the preceding diagram rather than the morphism under discussion; the English restores the evident reference to (4.1.2).

space $V(f)$. Let T be the positive generator of $\pi_1(D^*)$. The section f is of finite determination if and only if there is a nonzero polynomial $P \in \mathbb{C}[X]$ such that $P(T)(f) = 0$. The polynomial of least degree with this property is then both the minimal polynomial and the characteristic polynomial of T acting on $V(f)$.

We say that f is of unipotent finite determination if there is an A such that

$$(T-1)^A f = 0.$$

On the space of these sections define

$$N = \frac{\log T}{2\pi i}$$

by the finite sum

$$N(f) = -\frac{1}{2\pi i} \sum_{n>0} \frac{(1-T)^n}{n} f.$$

If f (respectively, g) is a section of unipotent finite determination in $H^0(D^*, \mathcal{O})$ (respectively, in $H^0(\bar{X}^*, \bar{F})$), then

$$N(fg) = N(f)g + fN(g),$$

since the logarithm of an automorphism is a derivation.

We say that f is of quasi-unipotent finite determination if there are A and $B > 0$ such that

$$(T^B - 1)^A f = 0.$$

Examples 4.6.

For $X = D^*$ and $F = \mathcal{O}$, one has

$$\big(T - \exp(2\pi i\alpha)\big)^{k+1} \big(z^\alpha (\log z)^k\big) = 0,$$

$$N(\log z) = 1,$$

and

$$N\left(\frac{(\log z)^k}{k!}\right) = \frac{(\log z)^{k-1}}{(k-1)!}, \qquad (0 \text{ for } k = 0).$$

Proposition 4.7.

A section $f \in H^0(\bar{X}^*, \bar{F})$ is of finite determination if and only if it can be written

$$f = \sum_{\alpha,k} f_{\alpha,k}\, z^\alpha \frac{(\log z)^k}{k!}, \tag{4.7.1}$$

where $0 \le \mathrm{Re}(\alpha) < 1$, $k \in \mathbb{N}$, and the $f_{\alpha,k} \in H^0(X^*, F)$ are almost all zero. For f of finite determination, the decomposition (4.7.1) is unique.

By 4.6, every section of the form (4.7.1) is of finite determination. Conversely, let f be of finite determination, with $P(T)(f) = 0$. Write

$$P(X) = \prod_{\beta \in \mathbb{C}} (X-\beta)^{n'(\beta)} = \prod_{0 \le \mathrm{Re}(\alpha) < 1} \big(X - \exp(2\pi i\alpha)\big)^{n(\alpha)},$$

where $n(\alpha) = n'(\exp(2\pi i\alpha))$ is almost always zero. Decomposing the $\mathbb{C}[X]$-module $V(f)$ into its isotypic components, one finds a unique decomposition

$$f = \sum_\alpha f'_\alpha$$

such that

$$\big(T - \exp(2\pi i\alpha)\big)^{n(\alpha)} f'_\alpha = 0.$$

Choose a polynomial $Q_\alpha \in \mathbb{C}[X]$ congruent to 1 modulo $\big(X - \exp(2\pi i\alpha)\big)^{n(\alpha)}$ and to 0 modulo $\big(X - \exp(2\pi i\alpha')\big)^{n(\alpha')}$ for every $\alpha' \neq \alpha$. Then $f'_\alpha = Q_\alpha(T)(f)$. Put

$$f_\alpha = z^{-\alpha} f'_\alpha = z^{-\alpha} Q_\alpha(T)(f). \tag{4.7.2}$$

Since

$$T(z^{-\alpha} f'_\alpha) = \exp(-2\pi i\alpha) z^{-\alpha} T(f'_\alpha),$$

we have

$$(T-1)^{n(\alpha)} f_\alpha = 0.$$

If

$$f_\alpha = \sum_k f_{\alpha,k} \frac{(\log z)^k}{k!}, \qquad f_{\alpha,k} \in H^0(X^*, F),$$

then 4.6 and the identity $\sum_j (-1)^j \binom{k}{j} = \delta_{0,k}$ give

$$f_{\alpha,k} = \sum_{j\geq 0} (-1)^j \frac{(\log z)^j}{j!} N^{j+k} f_\alpha, \tag{4.7.3}$$

which proves uniqueness.

Conversely, if f_α is a section of unipotent finite determination, the $f_{\alpha,k}$ defined by (4.7.3) vanish for large k (because $N^k f_\alpha = 0$ for large k) and are invariant under T; hence they identify with sections of F on X^*. Equivalently, we must show that $N f_{\alpha,k} = 0$. Apply to $N^k f_\alpha$ the identity

$$\begin{aligned} N \sum_{j\geq 0} (-1)^j \frac{(\log z)^j}{j!} N^j &= \sum_{j>0} (-1)^j \frac{(\log z)^{j-1}}{(j-1)!} N^j \\ &\quad + \sum_{j\geq 0} (-1)^j \frac{(\log z)^j}{j!} N^{j+1} = 0. \end{aligned}$$

Finally,

$$\begin{aligned} \sum_{k\geq 0} f_{\alpha,k} \frac{(\log z)^k}{k!} &= \sum_{j,k\geq 0} (-1)^j \frac{(\log z)^{j+k}}{j!k!} N^{j+k} f_\alpha \\ &= \sum_{n\geq 0} \left(\sum_j (-1)^j \binom{n}{j} \right) \frac{(\log z)^n}{n!} N^n f_\alpha = f_\alpha, \end{aligned}$$

which proves 4.7.

Remark 4.8.1.

The condition $\mathrm{Re}(\alpha) \in [0,1)$ in 4.7 may be replaced by $\alpha \in C$, where C is any fundamental domain for $\mathbb{C} \longrightarrow \mathbb{C}/\mathbb{Z}$.

Remark 4.8.2.

Let f be of finite determination, with decomposition (4.7.1). For $0 \leq \mathrm{Re}(\alpha) < 1$ and for α such that some $f_{\alpha,k} \neq 0$, put

$$m(\alpha) = 1 + \sup\{k \mid f_{\alpha,k} \neq 0\}.$$

If all the $f_{\alpha,k}$ vanish, put $m(\alpha) = 0$. The characteristic polynomial of T acting on $V(f)$ is

$$\mathrm{Pc}(T, V(f)) = \prod_{\alpha} \big(X - \exp(2\pi i\alpha)\big)^{m(\alpha)}.$$

In particular, f is of quasi-unipotent finite determination if and only if the α in (4.7.1) may all be taken in $\mathbb{Q}$.

4.9.

Return to the more restrictive hypotheses of 4.1, and let F be a coherent sheaf on X whose restriction to X^* is locally free.

Proposition 4.10.

Let f be a section of finite determination of the inverse image $\bar{F}$ of F on $\bar{X}^*$. Then f has moderate growth along X_0 if and only if the $f_{\alpha,k}$ in the decomposition (4.7.1) are meromorphic along X_0.

For the definition of "moderate growth," we refer to [1], 2.10. It is clear from (4.7.1), (4.7.2), and (4.7.3) that f has moderate growth if and only if the $f_{\alpha,k}$ do. For sections of F on X^*, moderate growth along X_0 is known to be equivalent to being meromorphic along X_0.

For $X = D$ and $F = \mathcal{O}$, 4.10 has the following elementary corollary.

Corollary 4.11.

Let K be the field of germs at 0 of meromorphic functions on D, and let K^m (respectively, K^{mu} and K^{mqu}) be the K-algebra of germs at 0 of holomorphic functions on $\widetilde{D}^*$ which have moderate growth at 0 and finite determination (respectively, unipotent finite determination and quasi-unipotent finite determination). Then the following are bases over K:

$$\begin{aligned}
K^m &: \quad z^\alpha(\log z)^k, \quad 0 \le \mathrm{Re}(\alpha) < 1,\ k \in \mathbb{N},\\
K^{mu} &: \quad (\log z)^k, \quad k \in \mathbb{N},\\
K^{mqu} &: \quad z^\alpha(\log z)^k, \quad \alpha \in \mathbb{Q},\ 0 \le \alpha < 1,\ k \in \mathbb{N}.
\end{aligned}$$

Under the hypotheses of 4.9, denote by $\Psi_\eta^m(F)$ (respectively, $\Psi_\eta^{mu}(F)$ and $\Psi_\eta^{mqu}(F)$) the subsheaf of $\bar{i}^*\bar{j}_*\bar{F}$ consisting of sections of $\bar{F}$, near points of X_0, which have moderate growth along X_0 and finite determination (respectively, unipotent finite determination and quasi-unipotent finite determination). Let

$$F^m \subset j_*j^*F$$

be the sheaf on X of sections of j_*j^*F which are meromorphic along X_0. Proposition 4.10 has the following corollary, independent of the arbitrary choices of z and $\log z$.

Corollary 4.12.

The natural morphisms

$$\begin{aligned}
&\text{(i)} \qquad K^m \otimes_K i^*F^m \longrightarrow \Psi_\eta^m(F),\\
&\text{(ii)} \qquad K^{mu} \otimes_K i^*F^m \longrightarrow \Psi_\eta^{mu}(F),\\
&\text{(iii)} \qquad K^{mqu} \otimes_K i^*F^m \longrightarrow \Psi_\eta^{mqu}(F)
\end{aligned}$$

are isomorphisms.

The main result of this section is the following.

Theorem 4.13.

The natural morphism induced by (4.3.1),

$$\Psi_\eta^{mqu}(\Omega_X^*) \longrightarrow R\Psi_\eta(\mathbb{C}), \tag{4.13.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

We shall in fact prove the more general results 4.15 and 4.17. Their statements require a few preliminaries which a reader interested only in 4.13 may omit.

4.14.

Let V be a locally constant sheaf of finite-dimensional complex vector spaces on X^* (a local system). Let

$$\mathcal{V} = V \otimes_{\mathbb{C}} \mathcal{O}$$

be the corresponding locally free $\mathcal{O}$-module, and let ∇ be the integrable connection on $\mathcal{V}$ for which the local sections of V are horizontal. It is known that $\mathcal{V}$ admits an extension $\widetilde{\mathcal{V}}$ to a coherent sheaf on X such that ∇ is regular along X_0 relative to this extension. It is also known that any two such extensions are meromorphically equivalent ([1], II, 5.7). Thus the sheaf $\Psi_\eta^m(\widetilde{\mathcal{V}})$, and its variants Ψ^{mu} and Ψ^{mqu}, does not depend on the chosen extension; we denote it by $\Psi_\eta^m(\mathcal{V})$. Likewise define

$$\Psi_\eta^m\big(\Omega_{X^*}^p(V)\big) = \Psi_\eta^m(\Omega_X^p \otimes \mathcal{V}),$$

and its variants Ψ^{mu} and Ψ^{mqu}.

Proposition 4.15.

The natural morphism induced by (4.4.1),

$$\Psi_\eta^m\big(\Omega_{X^*}^*(V)\big) \longrightarrow R\Psi_\eta(V), \tag{4.15.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

4.16.

A local system V as in 4.14 is said to be unipotent (respectively, quasi-unipotent) along X_0 if, for every morphism u from the standard disc D_0 to X such that $u^{-1}(X_0) = \{0\}$, the monodromy transformation of the local system u^*V on D_0^* is unipotent (respectively, quasi-unipotent).

Proposition 4.17.

If V is quasi-unipotent, the natural morphism induced by (4.4.1),

$$\Psi_\eta^{mqu}\big(\Omega_{X^*}^*(V)\big) \longrightarrow R\Psi_\eta(V), \tag{4.17.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

Theorem 4.13 is the special case of 4.17 in which $V = \underline{\mathbb{C}}$.

4.18.

We shall first treat the case in which X is smooth and $(X_0)_{\mathrm{red}}$ is a normal-crossings divisor in X. Since the theorems are local, we may suppose that f is the map

$$f(\underline{z}) = \prod_{i=1}^{d} z_i^{n_i}, \qquad 0 < d \leq e, \quad n_i > 0,$$

from $X = D^e$ to D, where this time D denotes the standard disc $D = \{z \in \mathbb{C} \mid |z| < 1\}$. It is enough to prove that the fiber at $0 \in D^e$ of (4.15.1) (respectively, (4.17.1)) is a quasi-isomorphism.

Let U be a neighborhood of 0 in D^e of the form $|z_i| < a$, where $0 < a \leq 1$. Define U_0, $\bar{U}$, U^*, and $\bar{U}^*$ by taking the inverse image of U in (4.1.1). The cohomology of U^* or $\bar{U}^*$ is independent of a, and hence

$$H^*(\bar{U}^*, V) \xrightarrow{\sim} \big(R^*\Psi_\eta(V)\big)_0.$$

The fundamental group $\mathbb{Z}^d$ of $X^* = (D^*)^d \times D^{e-d}$ (which is also that of U^*) is abelian. Therefore V has a finite filtration whose successive quotients are rank-one local systems. By devissage, we may thus suppose that V has rank one.

For $\widetilde{\mathcal{V}}$ as in 4.14, write $j_*^m\Omega^*_{X^*}(V)$ for the subcomplex of $j_*\Omega^*_{X^*}(V)$ whose degree-p term is $(\Omega^p_X \otimes \widetilde{\mathcal{V}})^m$. By [1], II 6.4.1 and 6.8, the natural map induced by the quasi-isomorphism $V \xrightarrow{\simeq} \Omega^*_{X^*}(V)$ (the holomorphic Poincare lemma),

$$j_*^m\Omega^*_{X^*}(V) \longrightarrow Rj_*V, \tag{4.18.1}$$

is a quasi-isomorphism. Thus

$$H^*\big((j_*^m\Omega^*_{X^*}(V))_0\big) \xrightarrow{\sim} (R^*j_*V)_0 \xleftarrow{\sim} H^*(U^*, V). \tag{4.18.2}$$

[19] This will be the only analytic ingredient in this part 4.18 of the proof.

Lemma 4.18.3.

For a rank-one V, one has

(i) if V is nontrivial: $H^*(U^*, V) = 0$,

(ii) if $V = \mathbb{C}$: $H^*(U^*, V) = \bigwedge^* \mathbb{C}^d$.

Indeed, U^* has the homotopy type of a torus.

Let η be the generator of $H^1(D^*, \mathbb{Z})$, and also its inverse image in $H^1(U^*, \mathbb{Z})$.

Lemma 4.18.4.

For a rank-one V, the natural map

$$H^*(U^*, V) \xrightarrow[(4.18.1)]{\sim} H^*\big((j_*^m\Omega^*_{X^*}(V))_0\big) \longrightarrow H^*\big(\Psi_\eta^{mu}(\Omega^*_{X^*}(V))_0\big)$$

identifies $H^*(\Psi_\eta^{mu}(\Omega^*_{X^*}(V))_0)$ with $H^*(U^*, V)/\eta H^*(U^*, V)$. [20]

Use 4.12(ii) and the identity $d\log f = \sum_i n_i\, dz_i/z_i$. The complex $\Psi_\eta^{mu}(\Omega^*_{X^*}(V))_0$ identifies with the simple complex $\mathrm{s}K$ associated (using direct sums) with the double complex K whose terms are

$$K^{-p,n+p} = \frac{(\log f)^p}{p!} \cdot (j_*^m\Omega^n_{X^*}(V))_0.$$

This complex lies in a bad quadrant ($p \leq 0, q \geq 0, p+q \geq 0$). Nevertheless, $\mathrm{s}K$ is the inductive limit of the subcomplexes $\mathrm{s}\,\sigma_{p\geq p_0}(K)$ whose terms are

$$(\mathrm{s}\,\sigma_{p\geq p_0}K)^n = \bigoplus_{\substack{p+q=n \\ p\geq p_0}} K^{pq},$$

and this suffices to justify the argument below.

[19] The printed right-hand term in (4.18.2) is $H^*(U, V)$, although V is defined on U^*; the English restores the evident U^*.

[20] The printed left-hand term is $H^*(U^*, \mathbb{C})$, while both the hypothesis and the conclusion concern the general rank-one system V; the English restores V.

The spectral sequence for the filtration by the first degree is

$$E_1^{-p,n+p} = H^n(U^*, V) \qquad (p \geq 0),$$

with

$$d_1 = 2\pi i\, \eta\wedge : H^n(U^*, V) \longrightarrow H^{n+1}(U^*, V).$$

If V is nontrivial, then $E_1 = 0$ by 4.18.3, and 4.18.4 reduces to $0 = 0$. Suppose that $V = \underline{\mathbb{C}}$. The complexes (E_1, d_1) are then Koszul complexes, possibly truncated, so that, by acyclicity of the Koszul complex,

$$E_2^{pq} = 0 \qquad \text{for } p \neq 0,$$

and

$$E_2^{0n} = H^n(U^*, V)/\eta H^{n-1}(U^*, V),$$

which proves 4.18.4.

For $\beta \in \mathbb{C}^*$, let V_β denote the rank-one local system on D^* whose inverse image on $\widetilde{D}^*$ is trivialized and whose monodromy is β. We use the same notation for its inverse image on U^*. On $\bar{U}^*$, the inverse image of $V \otimes V_\beta$ is canonically isomorphic to that of V.

Lemma 4.18.5.

For a rank-one V, the map

$$\bigoplus_{\beta\in\mathbb{C}^*} H^*(U^*, V \otimes V_\beta)/\eta H^*(U^*, V \otimes V_\beta) \longrightarrow H^*(\bar{U}^*, V)$$

is bijective.

Let k be the greatest common divisor of the n_i. The space $\bar{U}^*$ has k connected components, indexed by the kth roots of unity. On the component $\bar{U}^*_\xi$ indexed by ξ, the inverse image of the function $z^{1/k}$ on $\widetilde{D}^*$ coincides with the inverse image of the function

$$\xi \prod_{i=1}^{d} z_i^{n_i/k}$$

on U^*. The component $\bar{U}^*_\xi$ has the homotopy type of a torus, and the sequence

$$0 \longrightarrow \pi_1(\bar{U}^*_\xi) \longrightarrow \pi_1(U^*) \longrightarrow \pi_1(D^*)$$

is exact. Write $\varphi : \mathbb{Z}^d \to \mathbb{Z}$ for the map $\pi_1(U^*) \to \pi_1(D^*)$. If $\ker(\varphi)$ acts nontrivially on V, both sides of 4.18.5 vanish.[21] Otherwise, after replacing V by a suitable $V \otimes V_\beta$, we may suppose that V is trivial, that is, $V = \mathbb{C}$. In complex cohomology we then have

$$H^0(\bar{U}^*) \otimes H^*(U^*)/\eta H^*(U^*) \xrightarrow{\sim} H^*(\bar{U}^*).$$

The result follows on observing that V_β is trivial on U^* when $\beta^k = 1$, that

$$\bigoplus_{\beta^k=1} H^0(U^*, V_\beta) = \bigoplus_{\beta^k=1} \mathbb{C} \xrightarrow{\sim} H^0(\bar{U}^*, \mathbb{C}),$$

and that $H^*(U^*, V_\beta) = 0$ when $\beta^k \neq 1$.

[21] The print refers here to 4.18.4; the two sides under discussion are those of 4.18.5.

Remark 4.18.6.

If V is quasi-unipotent, the sum over $\mathbb{C}^*$ in 4.18.5 may be replaced by a sum over the roots of unity; the proof is the same.

Applying 4.12 gives a commutative diagram

$$\begin{array}{ccccc} H^*\Psi_\eta^m(\Omega^*_{X^*}(V))_0 & \xleftarrow[4.12]{\sim} & \bigoplus_\beta H^*\Psi_\eta^{mu}(\Omega^*_{X^*}(V\otimes V_\beta)) & \xleftarrow[4.18.4]{\sim} & \bigoplus_\beta H^*(U^*, V\otimes V_\beta)/\eta H^*(U^*, V\otimes V_\beta) \\ \downarrow & & & & \wr \downarrow 4.18.5 \\ H^*R\Psi(V)_0 & \xleftarrow{\qquad\qquad\sim\qquad\qquad} & & & H^*(\bar{U}^*, V) \end{array}$$

from which 4.15 follows. Proposition 4.17 is obtained by replacing Ψ^m with Ψ^{mqu}, the sum over β with a sum over the roots of unity, and 4.18.5 with 4.18.6.

4.19.

Let us deduce 4.15 and 4.17 from the special case 4.18. Let $u : Y \longrightarrow X$ be a resolution of (X, X_0), that is, a proper morphism inducing an isomorphism $Y^* \xrightarrow{\sim} X^*$, such that Y is smooth and $(Y_0)_{\mathrm{red}}$ is a normal-crossings divisor in Y. If V is quasi-unipotent along X_0, its inverse image u^*V is quasi-unipotent along Y_0.

For every coherent sheaf F on X, denote by F^m, or by $j_*^m j^*F$, the subsheaf of j_*j^*F formed by the meromorphic sections. If f is an equation for X_0, then $j_*^m j^*F$ is the inductive limit of the system

$$F \xrightarrow{f} F \xrightarrow{f} F \longrightarrow \cdots . \tag{4.19.1}$$

The following lemma is proved in [2].

Lemma 4.19.2.

For every coherent sheaf F on X, one has

$$F^m \xrightarrow{\sim} u_*(u^*F)^m$$

and $R^iu_*(u^*F)^m = 0$ for $i > 0$.

By Grauert's theorem, the $R^iu_*u^*F$ are coherent sheaves. The kernels and cokernels of $F \longrightarrow u_*u^*F$, as well as the $R^iu_*u^*F$ for $i > 0$, vanish outside X_0 and are therefore locally annihilated by a power of f. Since u is proper, R^iu_* commutes with filtered inductive limits, in particular with the inductive limit (4.19.1), and (4.19.2) follows.

From 4.19.2 and 4.12 one deduces that, under the hypotheses just stated and still denoting by u the map $Y_0 \to X_0$ induced by u, one has the following.

Lemma 4.19.3.

The morphisms

$$\Psi_\eta^m(F) \longrightarrow u_*\Psi_\eta^m(u^*F) \longrightarrow Ru_*\Psi_\eta^m(u^*F)$$

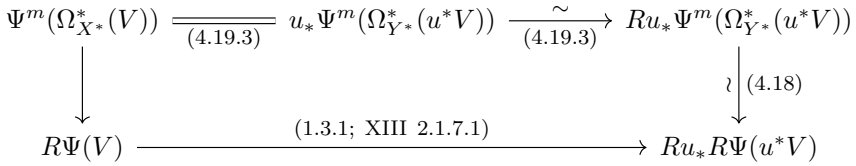
are isomorphisms.

That the second arrow is an isomorphism means that

$$R^iu_*\Psi_\eta^m(u^*F) = 0 \qquad (i > 0).$$

This follows from 4.12 and from $R^iu_*(i^*u^*F^m) = 0$ for $i > 0$, which in turn follows from $R^iu_*(u^*F^m) = 0$ for $i > 0$ and the proper base-change theorem for u.

To prove 4.15, it remains to consider the commutative diagram



The proof of 4.17 is the same, with Ψ^m replaced everywhere by Ψ^{mqu}.

Remark 4.20.

Theorem 4.13 gives an incarnation of the monodromy theorem at the level of complexes.

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SGA 7 — EXPOSÉ XV

The Picard–Lefschetz Formula

by P. Deligne

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1. Ordinary quadratic singularities (normal forms)

In this section we take up the results of VI 6 again, adopting a "henselian" rather than a "formal" point of view. We give normal forms for ordinary quadratic singularities and their deformations, into which they may be put locally for the etale topology.

These results follow easily from VI 6, the implicit-function theorem (in Tougeron's form), and R. Elkik's theorem on versal henselian moduli of isolated singularities. At the time of the oral seminar, the latter was not yet available; a weaker theorem, covering only the case of complete intersections (which was enough for us), had been proved and will not be published.

1.1. Reminders

1.1.1.

Let $f : X \longrightarrow S$ be a morphism, with X locally describable by p equations in relative affine space $\mathbb{A}_S^{n+p}$ of dimension $n+p$ over S. The Jacobian ideal $J^n(X/S)$ was defined in VI 5. If X is the closed subscheme of $\mathbb{A}_S^{n+p}$ defined by the equations $f_i = 0$ $(1 \leq i \leq p)$, it is the ideal sheaf generated by the $p \times p$ minors of the Jacobian matrix $(\partial f_i/\partial x_j)$. The following theorem is well known. A special case of its formal analogue is proved in Bourbaki, *Algèbre commutative*, III, §4, no. 6, Corollary 3 to Theorem 2; the C^∞ analogue is proved in Tougeron, *Idéaux de fonctions différentiables* (thesis, Rennes, May 1967); the case considered here is treated in Artin, *Algebraic approximation of structures over complete local rings*, Publ. Math. IHES 36 (Zariski's volume), Lemma 5.10.

Theorem 1.1.2 (Tougeron–Artin).

Let S be the spectrum of a henselian local ring A, let $f : X \to S$ be a morphism as in 1.1.1, let a and I be two proper ideals of A, and let δ' be an ideal sheaf of X contained in $J^n(X/S)$. Suppose that I is of finite type. Let s be a section of f over S/I, that is, $s \in X(A/I)$. Denote by $s^*\delta'^2$ the ideal of A inverse to the ideal $s^*\delta'^2$ of A/I.

If $(s^*\delta'^2)\, a \supset I$, then there is a section $s_1 \in X(A)$ of f congruent to s modulo $(s^*\delta')\, a$.

1.1.3.

Let S be the spectrum of a henselian local ring, with closed point s. Let X_0' be a scheme of finite type over $k(s)$, smooth away from a closed point x_0, and let X_0 be the henselization of X_0' at x_0. We are interested in systems

$$\begin{array}{ccccc} X_0 & \hookrightarrow & X & & \\ \downarrow & & \downarrow & & \\ s & \hookrightarrow & T & \longrightarrow & S \end{array}$$

of the following kind:

(a) T is henselian local and essentially of finite presentation over S (I 0.0.5); the morphism $T \to S$ is local and induces an isomorphism of residue fields;

(b) X is henselian local, flat, and essentially of finite presentation over T;

(c) the special fiber of X/T is equipped with a $k(s)$-isomorphism to X_0.

These are the henselian S-deformations of X_0. A morphism from (X,T) to (X',T') is a commutative diagram of S-schemes

$$\begin{array}{ccc} X & \longrightarrow & X' \\ \downarrow & & \downarrow \\ T & \longrightarrow & T' \end{array}$$

which induces the identity of X_0 after passage to the special fibers. A deformation X/T is called versal (minimally versal) if:

(a) for every morphism $X''/T'' \longrightarrow X'/T'$ for which $T'' \to T'$ is a closed immersion, the map

$$\mathrm{Hom}(X'/T', X/T) \longrightarrow \mathrm{Hom}(X''/T'', X/T)$$

is surjective. In particular, on taking $X''/T'' = X_0/s$, one has $\mathrm{Hom}(X'/T', X/T) \neq \varnothing$;

(b) for $T' = \mathrm{Spec}(k(s)[\varepsilon]/(\varepsilon^2))$, the "inverse-image" map

$$\mathrm{Hom}(T', T) \longrightarrow \{\text{isomorphism classes of deformations of } X_0 \text{ over } T'\}$$

is bijective.

Theorem 1.1.4 (R. Elkik).

There exists a versal henselian S-deformation of X_0 (and it is unique up to a nonunique isomorphism). If S is noetherian, a deformation X/T is such a deformation if and only if X induces over the completion $\widehat{T}$ of T a versal formal S-deformation of X_0 (a minimally formal versal deformation; VI 1.9 and VI 4).

For the proof of this delicate theorem, we refer to R. Elkik (ENS seminar, 1971/72).

1.2. Ordinary quadratic singularities

Let y be a closed point of a scheme Y of finite type over a field k of characteristic p, and let n be the dimension of Y at y.

Definition 1.2.1 (= VI 6.6).

(i) If k is algebraically closed, y is called an ordinary quadratic point of Y if the completion $\widehat{\mathcal{O}}_{Y,y}$ of the local ring of Y at y is isomorphic to the quotient of $k[[x_1, \ldots, x_{n+1}]]$ by the ideal generated by a single formal power series f which can be written

$$f(\underline{X}) = Q(\underline{X}) + \text{terms of order } > 2, \tag{1.2.1.1}$$

where $Q(\underline{X})$ is an ordinary quadratic form. [22]

(ii) If $\bar{k}$ is an algebraic closure of k, y is called an ordinary quadratic point of Y if the points of $\bar{Y} = Y \otimes_k \bar{k}$ lying over y are ordinary quadratic points of $\bar{Y}$.

1.2.2.

Replacing "ordinary quadratic form" by "nondegenerate quadratic form" in this definition gives the notion of a nondegenerate quadratic point (VI 6.0). A point y is a nondegenerate quadratic point of Y if and only if it is an ordinary quadratic point and either $p \neq 2$ or n is odd.

Example 1.2.3.

Let $Q(\underline{X})$ be a nondegenerate quadratic form in $n+1$ variables over k. Let Y_0 be the quadratic cone in $\mathbb{A}_k^{n+1}$ with equation $Q = 0$. Then the origin, denoted by y_0, is a nondegenerate quadratic point of Y_0.

Example 1.2.4.

Suppose that $p = 2$ and $n = 2m$. Let $a \in k$, and let Q be a polynomial

$$Q(\underline{X}) = (X_0^2 - a) + \sum_{0<i\leq j\leq 2m} a_{ij} x_i x_j.$$

Suppose that the quadratic form in $2m$ variables $\sum_{0<i\leq j} a_{ij} x_i x_j$ is nondegenerate, and denote by y_0 the closed point of $\mathbb{A}_k^{n+1}$ with coordinates $(\sqrt{a}, 0, \ldots, 0)$. Then y_0 is an ordinary quadratic point of the closed subscheme $Y_0 \subset \mathbb{A}_k^{n+1}$ with equation $Q = 0$.

Example 1.2.5.

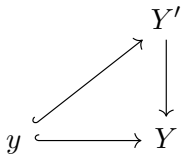
Let k' be a finite separable extension of k, and let Y_0 be a k'-scheme of the kind considered in 1.2.3 (respectively, 1.2.4). If Y_0 is regarded as a k-scheme, then y_0 is a nondegenerate (respectively, ordinary) quadratic point of Y_0. The extension k'/k identifies with the largest separable subextension of $k(y_0)/k$.

Theorem 1.2.6.

Suppose that y is an ordinary quadratic point of the k-scheme Y, and let k' be the largest separable subextension of $k(y)/k$. There then exist a k'-scheme $Y_0 \subset \mathbb{A}_{k'}^{n+1}$ of the kind in 1.2.3 or 1.2.4 and a k-isomorphism φ between the henselization $Y_{(y)}$ of Y at y and the henselization $Y_{0(y_0)}$ of Y_0 at y_0.

First observe that there is an etale neighborhood Y' of y in Y

[22] The printed display is numbered (4.2.1) and has an extra equals sign after $Q(\underline{X})$; the English restores the local numbering and the asserted expansion $f = Q +$ higher-order terms.



which is a k'-scheme. Indeed, $Y_{(y)}$ is canonically a k'-scheme. Replacing (k, Y, y) by (k', Y', y) reduces us to the case $k = k'$, that is, to the case in which $k(y)/k$ is radicial. The $k(y)$-vector space $(\Omega^1_{Y/k})_y$ has dimension $n+1$. It is enough to check this after extending scalars from $k(y)$ to its algebraic closure and completing at y, where it is immediate. A neighborhood of y in Y is therefore a closed subscheme of a smooth k-scheme Z of dimension $n+1$. Moreover, after replacing Y and Z by suitable neighborhoods of y, the ideal in $\mathcal{O}_Z$ defining Y is principal (the same argument as above), say with equation f.

Lemma 1.2.7.

Suppose that y is a nondegenerate quadratic point of Y and that $k(y)/k$ is radicial. Then, in a neighborhood of y, $J^n_{Y/k}$ (1.1.1) is generated by one element, supported at y, and of rank one over k. In particular, $k(y) = k$.

It is enough to prove the first assertion after passing to an algebraic closure of k and completing at y. It is then immediate by direct calculation: with the notation of 1.2.1, Nakayama's lemma shows that the $\partial f/\partial x_i$ (and hence, a fortiori, f and the $\partial f/\partial x_i$) generate the maximal ideal.

Let us prove 1.2.6 when y is nondegenerate. We may and shall suppose that $k(y) = k$, that Y is defined by an equation f in a smooth scheme Z, and that there is an etale morphism

$$x : Z \longrightarrow \mathbb{A}^{n+1}_k$$

(a "system of local coordinates x_i") taking y to 0. There is then a nondegenerate quadratic form Q in $n+1$ variables such that, at y,

$$f = Q(x_i) + \text{terms of order } > 2.$$

Let Y_0 be defined by Q as in 1.2.3.[23]

On Y, the system (x_i) is an "approximate solution" of the equation $Q(x_i) = 0$: if $\mathfrak{m}$ is the ideal defining y, then

$$Q(x_i) \in \mathfrak{m}^3 \qquad \text{on } Y.$$

Since Q is nondegenerate, the ideal of $\mathbb{A}^{n+1}_k$ generated by the $\partial Q/\partial x_i$ is the maximal ideal at 0, and its inverse image under $x = (x_i) : Y \to \mathbb{A}^{n+1}_k$ is $\mathfrak{m}$. By the implicit-function theorem 1.1.2 (applied with $a = \mathfrak{m}$ and $\delta' = J$), there are functions x'_i on the henselization $Y_{(y)}$, congruent to the x_i modulo $\mathfrak{m}^2$, which satisfy $Q(x'_i) = 0$. They give the required isomorphism between $Y_{(y)}$ and $Y_{0(y_0)}$.

Lemma 1.2.8.

Suppose that y is a degenerate ordinary quadratic point ($p = 2$ and n even), and that $k(y)/k$ is radicial. Then, in a neighborhood of y, $T^1_{Y/k}$ is generated by one element, supported at y, and of rank two over k. In particular, either $k(y) = k$, or $k(y)$ is isomorphic to $k(\sqrt{a})$ for some $a \in k$, $a \notin k^2$.

Proceed as in 1.2.7. With the notation of 1.2.1, it is enough to check that

$$\dim_k \left(k[[x_1, \ldots, x_{n+1}]]/(f, \partial f/\partial x_i)\right) = 2.$$

[23] The print refers to 4.3 here; the normal form just invoked is Example 1.2.3.

In suitable coordinates $x_0, x_1, \ldots, x_{2m}$, the formal power series f can be written

$$f = x_0^2 + \sum_{i=1}^{m} x_i x_{i+m} + R, \qquad \mathrm{ord}(R) \geq 3.$$

Let $\mathfrak{n}'$ be the ideal generated by f and the $\partial f/\partial x_i$, let $\mathfrak{n}$ be the ideal generated by x_0^2 and the x_i for $i \neq 0$, and let $\mathfrak{q}$ be the maximal ideal. We prove that $\mathfrak{n} = \mathfrak{n}'$. Plainly $\mathfrak{n}' \subset \mathfrak{n}$, and by Nakayama it is enough to check that

$$\mathfrak{n}' + \mathfrak{n}\mathfrak{q} = \mathfrak{n}.$$

Modulo $\mathfrak{n}\mathfrak{q}$, one has $f \equiv x_0^2$; modulo $\mathfrak{n}\mathfrak{q} + (x_0^2)$, one has

$$\frac{\partial f}{\partial x_i} \equiv x_{i+m}, \qquad \frac{\partial f}{\partial x_{i+m}} \equiv x_i \qquad (1 \leq i \leq m),$$

which proves the assertion.

Under the hypotheses of 1.2.8, if $k = k(y)$, we shall let a denote any square in k.

Lemma 1.2.9.

Let V be a vector bundle on a scheme S, and let T be a closed subscheme of $\mathbb{A}_S(V)$ locally free of rank two over S. Then T is contained in one and only one line in $\mathbb{A}_S(V)$.

The question is local for the fppf topology on S, so we may suppose that T/S has a section. By translation, we may suppose that T contains the zero section. Let $\mathfrak{n}$ be the ideal of T defining 0; it is locally free of rank one over S. Every homogeneous linear form $v \in V$ on $\mathbb{A}_S(V)$ defines an element of $\mathfrak{n}$, and hence an epimorphism $V \twoheadrightarrow \mathfrak{n}$. Its kernel defines the required line.

1.2.10.

In particular, if A is a radicial rank-two closed subscheme of affine space $\mathbb{A}_k^{n+1}$ over a field k of characteristic 2, there is an $a \in k$ such that, in suitable affine coordinates, A has equations

$$x_0^2 - a = 0, \qquad x_i = 0 \quad (1 \leq i \leq n).$$

It is enough to take the x_i which vanish on the line defined by A.

1.2.11.

Let us prove 1.2.6 for a degenerate y ($p = 2$ and $n = 2m$). We may and shall suppose that $k(y) = k(\sqrt{a})$, that Y is defined by an equation f in a smooth scheme Z, and that there is a system of local coordinates at y

$$x = (x_i)_{0 \leq i \leq 2m} : Z \longrightarrow \mathbb{A}_k^{2m+1}$$

taking y to the closed point $(\sqrt{a}, 0, \ldots, 0)$, such that, in a neighborhood of y,

$$J^n_{Y/k} \simeq \mathcal{O}_Y/(x_0^2 - a, x_1, \ldots, x_{2m}).$$

After replacing f by a suitable multiple, one has

$$f = (x_0^2 - a) + \sum_{1 \leq i \leq j \leq 2m} a_{ij} x_i x_j + R,$$

where

(a) the quadratic form in $2m$ variables $\sum a_{ij} x_i x_j$ is nondegenerate;

(b) after extending scalars to $\bar{k}$, R has order at least 3 in $x_0 - \sqrt{a}$ and the x_i.

Let Y_0 be defined, as in 1.2.4, by

$$Q = x_0^2 - a + \sum a_{ij} x_i x_j.$$

On Y, the system (x_i) may again be regarded as an approximate solution of $Q(X_i) = 0$. This time, the ideal δ, the inverse image under x of the ideal generated by the $\partial Q/\partial X_i$, satisfies $\mathcal{O}_Y/\delta \simeq J^n_{Y/k}$; at y it is

$$\delta = (x_0^2 - a, (x_i)_{1 \le i}).$$

Let $\mathfrak{q}$ be the maximal ideal defining y. To conclude as in the nondegenerate case by applying the implicit-function theorem, it remains to verify that

$$Q(x_i) \equiv 0 \pmod{\delta^2 \mathfrak{q}} \qquad \text{on } Y. \tag{1.2.11.1}$$

Let $\bar{\mathfrak{q}}$ be the ideal defining y after scalar extension to an algebraic closure of k. One checks that

$$\delta^2 \bar{\mathfrak{q}} \cap \mathcal{O}_{Y,y} = \delta^2 \mathfrak{q}.$$

It is therefore enough to verify (1.2.11.1) after extending scalars to $\bar{k}$; we may suppose that $a = 0$.[24] At y one then has

$$\delta^2 \mathfrak{q} = (x_0^5, (x_0^3 x_i)_{1 \le i}, (x_0 x_i x_j)_{1 \le i,j}, (x_i x_j x_k)_{1 \le i,j,k}).$$

It follows that $\delta^2 \mathfrak{q} \supset \mathfrak{q}^5$ and that

$$x_0^2 f \equiv x_0^4 \pmod{\delta^2 \mathfrak{q}}, \qquad \delta^2 \mathfrak{q} + (x_0^2 f) \supset \mathfrak{q}^4,$$

$$x_0 f \equiv x_0^3 \pmod{\delta^2 \mathfrak{q} + \mathfrak{q}^4}, \qquad \delta^2 \mathfrak{q} + (x_0 f) \supset \mathfrak{q}^3.$$

Finally,

$$Q(x_i) \equiv 0 \pmod{\delta^2 \mathfrak{q} + (f)} \qquad \text{on } Z,$$

which is (1.2.11.1).

Remark 1.2.12.

Let Q be a not necessarily homogeneous, nonzero quadratic form in $n+1$ variables, and suppose that the closed subscheme $Y \subset \mathbb{A}_k^{n+1}$ with equation Q has an ordinary quadratic singularity at a point y. A nonhomogeneous linear change of variables then puts Y into the form 1.2.3 or 1.2.4. In the nondegenerate case $k(y) = k$, and it is enough to put y at the origin. In the degenerate case it is enough that the support of $T^1_{Y/k}$ lie on the line $x_i = 0$ $(1 \le i \le 2m)$ (cf. 1.2.9).

1.3. Moduli of quadratic singularities

The following proposition follows from R. Elkik's theorem 1.1.4 and from explicit calculations left to the reader (and essentially contained in VI 6).

Proposition 1.3.1.

Let $S = \mathrm{Spec}(A)$ be a henselian local scheme with closed point s, and let $Y_0 \subset \mathbb{A}^{n+1}_{k(s)}$ be a scheme of the kind described in 1.2.3 or 1.2.4.

[24] The print refers at this point to (4.11.1); the displayed congruence under proof is (1.2.11.1).

(i) Suppose that Y_0 is of the kind described in 1.2.3, defined by the equation

$$Q(\underline{X}) = \sum_{i \le j} \bar{a}_{ij} x_i x_j = 0 \qquad (\bar{a}_{ij} \in k(s)),$$

and lift the $\bar{a}_{ij}$ to elements a_{ij} of A. Then the versal henselian S-scheme of deformations of the singularity of Y_0 at y_0 is

$$T = \mathrm{Spec}(A\{b\}),$$

the versal deformation being the henselization at y_0 of the T-scheme $Y \subset \mathbb{A}^{n+1}_T$ with equation

$$\sum a_{ij} x_i x_j - b = 0.$$

(ii) Suppose that Y_0 is of the kind described in 1.2.4, defined by the equation

$$Q(\underline{x}) = (x_0^2 - \bar{a}) + \sum \bar{a}_{ij} x_i x_j = 0 \qquad (\bar{a}, \bar{a}_{ij} \in k(s)),$$

and lift the $\bar{a}_{ij}$ and $\bar{a}$ to elements a_{ij} and a of A. Then the versal henselian S-scheme of deformations of the singularity of Y_0 at y_0 is

$$T = \mathrm{Spec}(A\{b, c\}),$$

the versal deformation being the henselization at y_0 of the T-scheme $Y \subset \mathbb{A}^{n+1}_T$ with equation

$$(x_0^2 - a) + \sum a_{ij} x_i x_j + b x_0 + c = 0.$$

The following corollary follows at once from 1.2.6 and 1.3.1.

Corollary 1.3.2.

Let $S = \mathrm{Spec}(A)$ be a henselian local scheme with closed point s, and let $f : X \to S$ be a flat morphism of finite presentation. Suppose that, at the closed point x of X_s, the scheme X_s has an ordinary quadratic singularity, that $k(x)$ is radicial over $k(s)$, and that X_s has dimension n.

(i) If x is nondegenerate, there exist a nondegenerate quadratic form Q in $n+1$ variables with coefficients in A, and an element b in the maximal ideal of A, such that the henselized S-scheme of X at x is isomorphic to the henselization at the origin of the closed subscheme of $\mathbb{A}^{n+1}_S$ with equation $Q - b = 0$, as in 1.3.1(i).

(ii) If x is degenerate ($n = 2m$, $\mathrm{char}(k(s)) = 2$), there exists a nonhomogeneous quadratic form in $n+1$ variables with coefficients in A

$$Q(\underline{X}) = x_0^2 + b x_0 + c + \sum_{1 \le i \le j \le 2m} a_{ij} x_i x_j, \tag{1.3.2.1}$$

such that b lies in the maximal ideal, the quadratic form $\sum a_{ij} x_i x_j$ in $2m$ variables is nondegenerate, and the henselized S-scheme of X at x is isomorphic to the henselization at $(\sqrt{c}, 0, \ldots, 0)$ of the closed subscheme of $\mathbb{A}^{n+1}_S$ with equation $Q = 0$. [25]

[25] The print labels this display (4.14.2) and writes x_0 in its first term. The surrounding normal forms and Remark 1.3.3 require the local tag (1.3.2.1) and the quadratic term x_0^2; the English restores both.

Remark 1.3.3.

In case (i), suppose that the henselization of X_s at x is isomorphic, by an isomorphism φ, to the henselization at O of the closed subscheme $X_0 \subset \mathbb{A}_k^{n+1}$ with equation

$$\sum \bar{a}_{ij} x_i x_j = 0,$$

and suppose that the a_{ij} lift the $\bar{a}_{ij}$. One may then take

$$Q = \sum a_{ij} x_i x_j.$$

In case (ii), suppose that the henselization of X_s at x is isomorphic, by an isomorphism φ, to the henselization at $(\sqrt{a}, 0, \ldots, 0)$ of the closed subscheme $X_0 \subset \mathbb{A}_k^{n+1}$ with equation

$$(x_0^2 + \bar{a}) + \sum \bar{a}_{ij} x_i x_j = 0,$$

and suppose that the a_{ij} lift the $\bar{a}_{ij}$. One may then take

$$Q = x_0^2 + b x_0 + c + \sum a_{ij} x_i x_j, \qquad c \equiv \bar{a} \pmod{\mathfrak{m}_A}.$$

In both cases, the isomorphism promised in 1.3.2 may be chosen to extend φ.

Corollary 1.3.4.

Let $f : X \to S$ be as in 1.3.2. There is a neighborhood U of x in X such that the nonsmooth points of f contained in U are ordinary quadratic points of their respective fibers.

2. Computation of vanishing cycles in a standard ordinary quadratic case

2.1. Preliminary: the cohomology of a cone

2.1.1.

Let $Y \subset \mathbb{P}^r$ be a projective variety over an algebraically closed field k. We use the following notation.

The affine cone over Y is denoted by $X \subset \mathbb{A}^{r+1}$, and its vertex by O;
$X_{(O)}$ is the henselization of X at O;
$X^* = X - \{O\}$ and $X^*_{(O)} = X_{(O)} - \{O\}$;
$X_1 \subset \mathbb{P}^{r+1}$ is the projective cone over Y, so $X = X_1 - Y$;
$\widetilde{X}$, $\widetilde{X}_{(O)}$, and $\widetilde{X}_1$ are the blowups of X, $X_{(O)}$, and X_1 at O;
Y_0 is the inverse image of O in $\widetilde{X}$, $\widetilde{X}_{(O)}$, or $\widetilde{X}_1$;
$h : \widetilde{X}_1 \to Y$ is the natural projection;
$y_0, y_\infty : Y \to \widetilde{X}_1$ are the sections of h whose images are Y_0 and Y;
F is an abelian torsion group of order prime to the characteristic of k. All cohomology groups considered below have coefficients in F, which will not be displayed.

Our purpose in this subsection is to relate the cohomology of $X^*_{(O)}$ to that of Y.

Proposition 2.1.2.

There are isomorphisms

$$\text{(i)} \qquad H^i(X) \xrightarrow{\sim} H^i(\{O\}) = \begin{cases} F, & i = 0, \\ 0, & i > 0; \end{cases}$$

$$\text{(ii)} \qquad H^i_{\{O\}}(X) \xrightarrow{\sim} H^i_c(X).$$

This proposition is proved by a homotopy argument, which we now formalize.

Let k be an algebraically closed field, let U and V be two k-schemes, let Λ be a torsion ring of order prime to the characteristic of k, and let

$$K \in \mathrm{Ob}\, D^+(U, \Lambda), \qquad L \in \mathrm{Ob}\, D^+(V, \Lambda).$$

By a "morphism" f from (U, K) to (V, L) we mean a pair consisting of a morphism $f : U \to V$ and a morphism $\varphi : f^*L \to K$. Such a morphism induces

$$f^* : H^*(V, L) \longrightarrow H^*(U, K).$$

We say that two "morphisms" f_0 and f_1 are homotopic if there exist a connected k-scheme T of finite type, two points 0 and 1 of T, and a "morphism" H from $(U \times T, \mathrm{pr}_1^* K)$ to (V, L) whose fibers at 0 and 1 induce f_0 and f_1.

Lemma 2.1.3.

If f_0 is homotopic to f_1, then $f_0^* = f_1^*$.

There is a sequence of points $0 = x_0, x_1, \ldots, x_n = 1$ of T, smooth connected curves Γ_i, and morphisms $a_i : \Gamma_i \to T$ such that x_i and x_{i+1} belong to the image of a_i. To construct the Γ_i, normalize a connected one-dimensional closed subscheme of T containing 0 and 1. This reduces us to the case in which T is a smooth curve. Consider the commutative square

$$\begin{array}{ccc} U \times T & \xrightarrow{\mathrm{pr}_1} & U \\ {\scriptstyle \mathrm{pr}_2}\downarrow & & \downarrow{\scriptstyle f} \\ T & \xrightarrow[t]{} & k. \end{array}$$

By smooth base change, applied to t, one has

$$t^* R f_* K \xrightarrow{\sim} R\,\mathrm{pr}_{2*}(\mathrm{pr}_1^* K).$$

The map f_i^* $(i = 0, 1)$ therefore factors through

$$\begin{array}{ccccc} H^n(V, L) & \xrightarrow{H^*} & H^n(U \times T, \mathrm{pr}_1^* K) & \longrightarrow & H^0\big(T, R^n\,\mathrm{pr}_{2*}(\mathrm{pr}_1^* K)\big) \\ & & & \nearrow \wr & \downarrow \text{fiber at } i \\ & & H^0\big(T, t^* H^n(U, K)\big) & \swarrow \wr & \\ & & & & H^n(U, K) \end{array}$$

and is consequently independent of i.

2.1.3.

Let us prove 2.1.2. Scalar multiplication gives a homotopy between the identity map of X and the constant map from X to X with value O; this proves (i).

Using scalar multiplications whose factors tend to infinity, one sees that Y is a deformation retract of X_1^*. Assertion (ii) then follows from the five lemma and the diagram

$$\begin{array}{ccccccccc} \xrightarrow{\partial} & H^i_{\{O\}}(X) & \longrightarrow & H^i(X_1) & \longrightarrow & H^i(X_1^*) & \xrightarrow{\partial} & & \\ & \downarrow & & \| & & \wr\downarrow & & & \\ \xrightarrow{\partial} & H^i_c(X) & \longrightarrow & H^i(X_1) & \longrightarrow & H^i(Y) & \xrightarrow{\partial} & . & \end{array}$$

Corollary 2.1.4.

$$H^i(X^*) \xrightarrow{\sim} H^i(X^*_{(O)}).$$

Indeed, apply the five lemma to

$$\begin{array}{ccccccccc}
\longrightarrow & H^i_{\{O\}}(X) & \longrightarrow & H^i(X) & \longrightarrow & H^i(X^*) & \longrightarrow \\
 & \downarrow \wr & & \downarrow \wr \ (2.1.2(i)) & & \downarrow & \\
\longrightarrow & H^i_{\{O\}}(X_{(O)}) & \longrightarrow & H^i(X_{(O)}) & \longrightarrow & H^i(X^*_{(O)}) & \longrightarrow .
\end{array}$$

2.1.5.

The Leray spectral sequences for the restrictions of h to $\widetilde{X}_1 - Y_0$ and $\widetilde{X}_1 - Y$ show that the restriction maps

$$H^i(\widetilde{X}_1 - Y_0) \xrightarrow{\sim} H^i(Y), \qquad H^i(\widetilde{X}) \xrightarrow{\sim} H^i(Y_0) = H^i(Y)$$

are isomorphisms. Under these isomorphisms, the long exact cohomology sequences of the pairs $(\widetilde{X}_1 - Y_0, Y)$ and $(\widetilde{X}_1 - Y, Y_0)$ form the rows of the diagram

$$\begin{array}{ccccccccc}
 & H^{i-1}(X^*) & \longrightarrow & H^{i-2}(Y)(-1) & \xrightarrow{\eta} & H^i(Y) & \longrightarrow & H^i(X^*) & \longrightarrow \\
(2.1.5.1) & \| & & \| & & \big|{-1} & & \| & \\
 & H^{i-1}(X^*) & \longrightarrow & H^{i-2}(Y)(-1) & \xrightarrow{-\eta} & H^i(Y) & \longrightarrow & H^i(X^*) & \longrightarrow .
\end{array}$$

Lemma 2.1.6.

The arrows denoted by η and $-\eta$ in (2.1.5.1) are cup products with the cohomology class of a hyperplane section. Diagram (2.1.5.1) is commutative.

The first assertion follows because η (respectively $-\eta$) is the restriction to Y (respectively to $Y_0 \simeq Y$) of the cohomology class of the invertible sheaf $\mathcal{O}(Y)$ (respectively $\mathcal{O}(Y_0)$) on $\widetilde{X}_1 - Y_0$ (respectively $\widetilde{X}_1 - Y$).

Use the Leray spectral sequence for h. The two rows of (2.1.5.1) are obtained by applying the functor $H(Y, -)$ to the distinguished triangles

$$y_{\infty *} R y^!_\infty F \longrightarrow R(h|\widetilde{X}_1 - Y_0)_* F \xrightarrow{①} R(h|X^*)_* F \xrightarrow{②}$$

and

$$y_{0*} R y^!_0 F \longrightarrow R(h|\widetilde{X}_1 - Y)_* F \longrightarrow R(h|X^*)_* F \longrightarrow .$$

The nonzero cohomology sheaves of the complexes at the vertices of these triangles occur in degrees (2), (0), and (0 and 1). The arrows ① and ② are therefore determined by their effect on cohomology sheaves. This reduces us to the case in which Y is a point, which is left to the reader.

2.1.7.

The long exact sequence of the pair $(\widetilde{X}_{(O)}, Y_0)$ is a local analogue of the second row of (2.1.5.1). One has

$$H^i(\widetilde{X}_{(O)}) \xrightarrow{\sim} H^i(Y_0)$$

by proper base change for the morphism $\widetilde{X}_{(O)} \to X_{(O)}$. The long exact sequence in question may therefore be written

$$\begin{array}{ccccccccc} H^{i-1}(X^*_{(O)}) & \longrightarrow & H^{i-2}(Y_0)(-1) & \xrightarrow{-\eta} & H^i(Y_0) & \longrightarrow & H^i(X^*_{(O)}) \\ \wr\uparrow & & \| & & \| & & \wr\downarrow \\ H^{i-1}(X^*) & \longrightarrow & H^{i-2}(Y)(-1) & \xrightarrow{-\eta} & H^i(Y) & \longrightarrow & H^i(X^*) \end{array} \quad , \tag{2.1.7.1}$$

the second row of this commutative diagram being the second row of (2.1.5.1). [26]

The following technical lemma will be used in 2.2.7.

Lemma 2.1.8.

$$\begin{array}{ccccccccc} H^{n-1}(Y_0) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}) & \longrightarrow & H^{n-1}(X^*) & \longrightarrow & H^n_{\{x\}}(X) & \longrightarrow & H^n_c(X_s) \\ \| & & & & & & & & \| \\ H^{n-1}(Y) & & & & \xrightarrow{\qquad\qquad\partial\qquad\qquad} & & & & H^n_c(X_s) \end{array}$$

is anticommutative. [27]

By 2.1.7, this is equivalent to proving the commutativity of

$$\begin{array}{ccc} H^{n-1}(X_1 - \{O\}) & \xrightarrow{\partial} & H^n_{\{O\}}(X_1) = H^n_{\{O\}}(X) \\ \downarrow\wr & & \downarrow \\ H^{n-1}(Y) & \xrightarrow{\partial} & H^n_c(X). \end{array}$$

This is general nonsense.

2.2. Vanishing cycles

2.2.1.

Let S be a henselian trait, and let $s, \eta, \bar{s}, \bar{\eta}$ have the meaning of XIII, 0.2.5. Put $\Lambda = \mathbb{Z}/k$, where k is prime to the residual characteristic of S. Let $\mathbb{A}^{n+1}_S$ be standard affine $(n+1)$-space over S, and let X be the closed subscheme of $\mathbb{A}^{n+1}_S$ defined by a quadratic equation $Q = 0$ which is nonzero modulo a uniformizer:

$$Q(\underline{x}) = \sum_{i \le j} a_{ij} x_i x_j + \sum_i b_i x_i + c.$$

Regard $\mathbb{A}^{n+1}_S$ as the complement in $\mathbb{P}^{n+1}_S$ of the hyperplane at infinity H. Then X is obtained from the quadric $X_1 \subset \mathbb{P}^{n+1}_S$ with homogeneous equation

$$\sum_{i \le j} a_{ij} x_i x_j + \sum_i b_i x_i z + c z^2 = 0 \tag{2.2.1.1}$$

by removing $Y = X_1 \cap H$.

We make the following assumptions:

(a) Y is a smooth quadric over S, that is, the quadratic form $\sum_{i \le j} a_{ij} x_i x_j$ is ordinary;

(b) the quadric X_s is singular, that is, $X_{\bar{s}}$ is a quadratic cone.

[28]

Let x_0 denote the singular point of X_s, and let $f : X \to S$ be the projection.

[26] The printed tag omits the dot between 1 and 7; the English uses the local number (2.1.7.1).

[27] The print numbers this lemma 2.7.8. Its position and the forward reference in 2.2.7 require 2.1.8.

[28] The print has X'_s in (b). The equivalent clause and the immediately following definition of the singular point show that X_s is intended.

Remark 2.2.2.

If x_0 is a rational point, a translation puts it at the origin of the coordinates. If $x_0 = 0$, the b_i and c are divisible by a uniformizer.

The subscheme A of X_s defined by the equations $\partial Q/\partial x_i$ is concentrated at x_0. Except in the exceptional case where $k(s)$ has characteristic 2 and $n+1$ is even, this scheme has degree one over $k(s)$, and hence x_0 is rational. In the exceptional case it is ramified over $k(s)$ and has rank 2, so that $k(x_0)$ is either $k(s)$ or a purely inseparable quadratic extension of $k(s)$. The assertions about A are checked most easily after passing to the algebraic closure $k(\bar{s})$ of $k(s)$.

Proposition 2.2.3.

Under the assumptions of 2.2.1, the morphisms

$$H^i(X_{\bar{\eta}}, \Lambda) \xrightarrow{\sim} \mathbb{H}^i\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big) \xrightarrow{\sim} R^i\psi_{\bar{\eta}}(\Lambda)_{x_0}$$

$$H^i_c(X_{\bar{\eta}}, \Lambda) \xleftarrow{\sim} \mathbb{H}^i_c\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big) \xleftarrow{\sim} \mathbb{H}^i_{\{x_0\}}\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big)$$

are isomorphisms.

That the left-hand arrows are isomorphisms follows from 2.1.8.6 and 2.1.10.5; the assertion for the right-hand arrows is a corollary of 2.1.2. Indeed, there is a distinguished triangle

$$(\Lambda \text{ on } X_{\bar{s}})[0] \longrightarrow R\psi_{\bar{\eta}}(\Lambda) \longrightarrow R\Phi_{\bar{\eta}}(\Lambda),$$

and 2.1.2 applies to $(\Lambda$ on $X_{\bar{s}})$, whereas $R\Phi_{\bar{\eta}}(\Lambda)$ is supported at x_0.

Corollary 2.2.4.

If the geometric generic fiber $X_{\bar{\eta}}$ is singular, that is, if it is still a quadratic cone, then all the sheaves $R^i\Phi(\Lambda)$ vanish.

By 2.1.2,

$$\Lambda = H^0(X_{\bar{s}}, \Lambda) \xrightarrow{\sim} H^0(X_{\bar{\eta}}, \Lambda), \qquad 0 = H^i(X_{\bar{s}}, \Lambda) = H^i(X_{\bar{\eta}}, \Lambda) \quad (i > 0).$$

We then apply the long exact sequence XIII, 2.1.8.9 (valid by XIII, 2.1.9). [29]

2.2.5.

Suppose now that

$$(*) \qquad \text{the generic fiber } X_\eta \text{ is smooth.}$$

For simplicity we shall also assume that S is strictly henselian. The sheaves of vanishing cycles are easily computed with the aid of 2.2.3 and XII, 3.7. The cohomology with coefficients in Λ considered here follows from the ℓ-adic cohomology of the cited passage by the universal-coefficient formula (cf. 2.1.13). The results are as follows.

A. Cohomology without supports.

One has $R^i\psi_{\bar{\eta}}(\Lambda) = 0$ for $i \neq 0, n$.

1) If $n \neq 0$: $\psi_{\bar{\eta}}(\Lambda) = \Lambda$. Moreover, $R^n\psi_{\bar{\eta}}(\Lambda)$ is (noncanonically) isomorphic to the sheaf Λ on x_0, extended by zero to X_s.

2) For arbitrary n: even when $n = 0$, one checks that $R^i\Phi_{\bar{\eta}}(\Lambda) = 0$ for $i \neq n$, and that $R^n\Phi_{\bar{\eta}}(\Lambda)$ is (noncanonically) isomorphic to the sheaf Λ on x_0, extended by zero to X_s.

[29] The print uses the letter o in the two occurrences of H^o; the cohomological degree is 0.

B. Cohomology with support at x_0.

One has

$$H^i_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda)\big) = 0 \qquad (i \neq n, 2n).$$

There is, moreover, a trace morphism

$$\mathrm{tr} : H^{2n}_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big) \longrightarrow \Lambda. \tag{2.2.5.1}$$

If $n \neq 0$: the morphism (2.2.5.1) is an isomorphism. Moreover,

$$H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big)$$

is (noncanonically) isomorphic to Λ.

C. Duality. For

$$a \in R^n\psi_{\bar\eta}(\Lambda)_{x_0}, \qquad b \in H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big),$$

put $(a, b) = \mathrm{Tr}(a \wedge b)$, where Tr is the morphism XIII, 2.2.6

$$\mathrm{Tr} : H^{2n}_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big) \longrightarrow \Lambda.$$

It follows from Poincaré duality on $X_{\bar\eta}$ and the isomorphisms 2.2.3 that (a, b) gives a perfect duality between the free Λ-modules

$$R^n\psi_{\bar\eta}(\Lambda)_{x_0} \quad \text{and} \quad H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big).$$

[30] D. $n = 2m > 0$.

By XII, 3.7, the groups

$$H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(m))\big) \quad \text{and} \quad R^n\psi_{\bar\eta}(\Lambda(m))_{x_0}$$

have natural generators, defined up to sign; denote them by δ and δ', respectively. Let ϕ be the natural map from the first of these groups to the second. The signs of δ and δ' may be normalized so that

$$(\delta', \delta) = 1; \tag{2.2.5.2}$$

then

$$\phi(\delta) = (-1)^m\, 2\, \delta'. \tag{2.2.5.3}$$

Let Z be the separable quadratic extension of $k(\eta)$ which is the center of the even part of the Clifford algebra of the quadratic form (2.2.1.1), and let ϵ be the corresponding character, of order 1 or 2, of the inertia group I. For $\sigma \in I$, one has

$$\sigma\delta = \epsilon(\sigma)\delta \tag{2.2.5.4}$$

and

$$\sigma\delta' = \epsilon(\sigma)\delta'. \tag{2.2.5.5}$$

Formulas (2.2.5.2), (2.2.5.3), and (2.2.5.4), in cohomology with coefficients in $\mathbb{Z}/2k$, imply that the variation is

$$\mathrm{Var}(\sigma)(a) = \frac{\epsilon(\sigma) - 1}{2}\,(-1)^m\,(a, \delta)\,\delta. \tag{2.2.5.6}$$

[30] The print drops the bar from $\bar\eta$ in the displayed trace map; the surrounding formulas determine the indicated term.

E. $n = 2m+1$—Vanishing cycles.

By XII, 3.7, the groups

$$H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(m))\big) \quad \text{and} \quad R^n\psi_{\bar\eta}(\Lambda(m+1))_{x_0}$$

have natural generators, defined up to sign; denote them by δ and δ', respectively. With the notation of D, their signs may be normalized so that

$$(\delta', \delta) = 1; \tag{2.2.5.7}$$

then

$$\phi(\delta) = 0. \tag{2.2.5.8}$$

F. $n = 2m+1$—The variation.

The inertia group I acts trivially on the cohomology of X_η. This follows from XII, 3.7 and from the fact that I acts trivially on the cohomology of Y_η, since Y is proper and smooth over S.

For σ in the inertia group one necessarily has

$$D(\sigma)(\delta') = \lambda(\sigma)\delta, \qquad \lambda(\sigma) \in \Lambda(1). \tag{2.2.5.9}$$

The additivity formula (XIII, 1.4.3.4) becomes

$$\lambda(\sigma\tau) = \lambda(\sigma) + \lambda(\tau).$$

Let

$$\epsilon : I \longrightarrow \Lambda(1) = \mu_k$$

be the homomorphism such that

$$\sigma\big(\sqrt[k]{t}\big) = \epsilon(\sigma)\sqrt[k]{t}$$

for a uniformizer t. In view of the structure of the tame inertia group, λ is necessarily a multiple of ϵ. Formula (2.2.5.9) may therefore be rewritten

$$\mathrm{Var}(\sigma)(a) = \lambda_X \epsilon(\sigma)(a, \delta)\delta \tag{2.2.5.10}$$

for a suitable $\lambda_X \in \Lambda$ (depending on X/S). We shall determine λ_X later. [31]

2.2.6.

We shall give the cycles δ and δ' of D and E a description which is local for the ètale topology.

The case of even n is easy. If $2 \nmid k$ ($\Lambda = \mathbb{Z}/k$), then $\pm\delta$ is characterized by (2.2.5.3) and (2.2.5.4), that is, by

$$(\delta, \delta) = (-1)^m\, 2. \tag{2.2.6.1}$$

In general, $\pm\delta$ is the image of any cycle $\widetilde{\delta}$, in cohomology with coefficients in $\mathbb{Z}/2^a k$ for a sufficiently large, which satisfies (2.2.6.1).

Suppose that n is odd. Then x_0 is a rational point (2.2.2). Let $X_{s(0)}$ be the henselization of X_s at x_0, let $\widetilde{X}_{s(0)}$ be its blowup at x_0, let Y_0 be the inverse image of x_0 in $\widetilde{X}_{s(0)}$ (a quadric), and put

$$X^*_{s(0)} = X_{s(0)} - \{x_0\} = \widetilde{X}_{s(0)} - Y_0.$$

Consider the composite map (cf. XIII, 2.4.6.1)

(2.2.6.2)

$$\begin{array}{ccccc}
H^{n-1}(Y_0, \Lambda(m)) & \xleftarrow{\ \sim\ } & H^{n-1}(\widetilde{X}_{s(0)}, \Lambda(m)) & \longrightarrow & H^{n-1}(X^*_{s(0)}, \Lambda(m)) \\
 & & & & \big\downarrow \\
H^n_{\{x_0\}}\big(X_s, R\Psi_{\bar\eta}(\Lambda(m))\big) & \longleftarrow & H^n_{\{x_0\}}(X_s, \Lambda(m)) & &
\end{array}$$

[31] In (2.2.5.6) and (2.2.5.10), the print suppresses the comma in (a, δ); this is the pairing defined in C.

Lemma 2.2.7.

The map (2.2.6.2) identifies $H^n_{\{x_0\}}(X_s, R\Psi_{\bar\eta}(\Lambda(m)))$ with the primitive quotient of the middle-dimensional cohomology of the $2m$-dimensional quadric Y_0; the two classes $\pm\delta$ are the images of the natural generators of this primitive quotient.

The cycles $\pm\delta$ are characterized by their images in $H^n_c(X_{\bar\eta}, \Lambda(m))$. Using (2.1.7.1), it is therefore enough to prove that the composite map

$$\begin{array}{ccccc} H^{n-1}(Y_0,\Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_s,\Lambda(m)) & \longrightarrow & H^{n-1}(X_s^*,\Lambda(m)) \\ & & & & \downarrow \\ H^n_c(X_{\bar\eta},\Lambda(m)) & \xleftarrow{\text{sp}} & H^n_c(X_s,\Lambda(m)) & \longleftarrow & H^n_{\{x_0\}}(X_s,\Lambda(m)) \end{array}$$

sends the distinguished generators of the primitive quotient of the first group to the distinguished generators of the last group.

Apply 2.1.8. Up to sign, the map may be rewritten

$$\begin{array}{ccc} H^{n-1}(Y_s,\Lambda(m)) & \xrightarrow{\partial} & H^n_c(X_s,\Lambda(m)) \\ \| & & \downarrow \\ H^{n-1}(Y_{\bar\eta},\Lambda(m)) & \longrightarrow & H^n_c(X_{\bar\eta},\Lambda(m)), \end{array}$$

and 2.2.7 follows. [32]

3. The Picard–Lefschetz formula

3.1. Preliminary results

3.1.1.

Let S be a henselian trait and let $f : X \to S$ be a flat morphism of finite type, purely of relative dimension n. Suppose that the special fiber X_s is smooth except at a finite set of points Σ', at which X_s has an ordinary quadratic singularity.

Let Σ be the set of points $x \in \Sigma'$ such that the generic fiber X_η is smooth in a neighborhood of x. Finally, let Λ be a torsion ring whose order is prime to the residual characteristic of S.

Proposition 3.1.2.

(i) The sheaves $R^i\Phi(\Lambda)$ vanish for $i \neq n$.

(ii) The sheaf $R^n\Phi(\Lambda)$ is zero outside Σ. Its restriction to Σ is a sheaf of Λ-modules of rank one.

(iii) For every geometric point $\bar x$ of Σ, the pairing (a, b) of 2.2.5(C) puts the Λ-modules

$$\big(R^n\psi(\Lambda)\big)_{\bar x} \quad \text{and} \quad H^n_{\{\bar x\}}\big(X_{\bar s}, R\psi_{\bar\eta}(\Lambda)\big)$$

in duality. They are free of rank one if $n \neq 0$, and of rank two if $n = 0$.

[32] In this proof the print has H^i_c for the group in which the degree-n cycle δ is characterized; the displayed maps and all surrounding terms require H^n_c. The print also abbreviates the support point in (2.2.6.2) as x; the English uses the fixed singular point x_0.

[33]

The assertions of 3.1.2 are local for the ètale topology. We may therefore treat only the case where S is strictly henselian and X is the hypersurface in relative affine space $\mathbb{A}_S^{n+1}$ defined by a quadratic equation satisfying 2.2.1(a) and (b) (apply 1.3.2).

The morphism f is smooth at every point of $X_s - \Sigma'$, hence $R\Phi = 0$ outside Σ'. By 2.2.4, $R\Phi$ even vanishes outside Σ. We may now assume 2.2.5(*); assertions (ii) and (iii) then follow from 2.2.5(A), (B), and (C).

3.2. Even relative dimension

We retain the notation of 3.1 and take $n = 2m$ to be even. For simplicity we suppose that S is strictly henselian (the general case would follow by descent).

Proposition 3.2.1.

(i) For $x \in \Sigma$, the group $H^n_{\{x\}}(X_s, R\psi_{\bar\eta}(\Lambda(m)))$ has a natural generator δ, well defined up to sign, characterized by being natural in Λ and by satisfying

$$(\delta, \delta) = (-1)^m\, 2.$$

For $n = 0$, add the condition $\mathrm{Tr}(\delta) = 0$.

(ii) For a suitable character, independent of Λ,

$$\varepsilon_x : I \longrightarrow \{\pm 1\}$$

of the inertia group I, the variation is

$$\mathrm{Var}(\sigma)(a) = (-1)^m \left(\frac{\varepsilon_x(\sigma) - 1}{2}\right)(a, \delta)\delta$$

for $a \in R^n\Phi_{\bar\eta}(\Lambda(m))$. Consequently,

$$\sigma(\delta) = \varepsilon_x(\sigma)\delta.$$

[34]

Passing to the inverse limit, it is enough to treat the universal case $\Lambda = \mathbb{Z}_\ell$. Up to sign, only one δ can then satisfy the conditions in (i). As in 3.1.2, we reduce to the case 2.2.5(*) and apply 2.2.5(D).

Complement 3.2.2.

Suppose that, at $x \in \Sigma$, the henselization $X_{(x)}$ of X at x is isomorphic to the henselization at x_0 of the projective scheme Q with equation

$$\sum_{i \le j} a_{ij} x_i x_j = 0.$$

Then the character ε_x is defined by the separable quadratic extension $Z(C^+(Q))$ of the fraction field. This follows from 2.2.5(D).

[33] The print says that $R^n\Phi(\Lambda)$ is "alone" outside Σ (*seul*); the proof and the asserted support on Σ require "zero" (*nul*). In (iii) it prints $H^n_{\{\bar x\}}(R^n\psi_{\bar\eta}(\Lambda))$; the pairing cited from 2.2.5(C) has the second term $H^n_{\{\bar x\}}(X_{\bar s}, R\psi_{\bar\eta}(\Lambda))$.

[34] The print has $H^n_{\{x\}}(R^n\psi_{\bar\eta}(\Lambda(m)))$ in (i). The proof explicitly invokes 2.2.5(D), where the natural generator belongs to $H^n_{\{x\}}(X_s, R\psi_{\bar\eta}(\Lambda(m)))$.

3.2.3.

In residual characteristic different from 2, Complement 3.2.2 can be made more explicit. In that case I has a unique nontrivial character ε of order 2. For a uniformizer t, one has

$$\sigma(\sqrt{t}) = \varepsilon(\sigma)\sqrt{t}.$$

On the other hand, $X_{(x)}$ is then isomorphic to the henselization at 0 of an affine scheme

$$\sum a_{ij}x_ix_j = b,$$

where b lies in the maximal ideal and $Q(x) = \sum a_{ij}x_ix_j$ is nondegenerate. The ideal (b) defines the image of the subscheme of $X_{(x)}$ which is the support of $J^n_{X_{(x)}/S}$. The center of the Clifford algebra is isomorphic to

$$k(\eta)\left(\left((-1)^{m+1}2b\det(a_{ij})\right)^{1/2}\right)$$

(Bourbaki, *Algèbre*, Ch. 9, §9, no. 4, final remark), whence

$$\varepsilon_x = \varepsilon^{v(b)}.$$

The variation is zero if b has even valuation; otherwise $\varepsilon_x = \varepsilon$.

3.3. Odd relative dimension

To prove the Picard–Lefschetz formula in odd relative dimension, we shall have to reduce to the transcendental case treated in XIV, 3.2.1.1. The specialization argument required for this forces us to work over a base which is no longer one-dimensional. We shall use ad hoc arguments adapted to the precise case we need: I do not have a usable theory over a base of dimension greater than one, and I have preferred not to cover my ignorance with generalities.

3.3.1.

Let $S = \mathrm{Spec}(A)$ be a strictly local noetherian scheme, and let $f : X \to S$ be a flat morphism of finite type, purely of relative dimension n, whose special fiber X_s has no nonsmooth points other than nondegenerate quadratic points. Let s be the closed point of S, let X_s be the special fiber, and let $x_0 \in X_s$ be a singular point.

By 1.3.2(i), in a neighborhood of x_0 the scheme X/S is locally isomorphic to an S-scheme defined in $\mathbb{A}^{n+1}_S$ by an equation $Q - b = 0$, where b belongs to the maximal ideal and Q is a nondegenerate quadratic form. The ideal (b) of A depends only on $(X/S, x_0)$: in a neighborhood of x_0 there is a section

$$s \in X(A/(b))$$

of X over $\mathrm{Spec}(A/(b))$ such that

$$J^n_{X/S} \simeq \mathcal{O}_{s(\mathrm{Spec}(A/(b)))}.$$

We denote this section by s_x, and denote by $b(x)$ any generator of (b).

3.3.2.

Suppose that S is normal, and let $b \in A$. If n is invertible in A and $b \neq 0$, write $\varepsilon^{(n)}_b$ for the following homomorphism from the Galois group of the fraction field of A to $\mathbb{Z}/(n)(1)$:

$$\varepsilon^{(n)}_b(\sigma) = (\sigma\sqrt[n]{b})/\sqrt[n]{b}.$$

One has

$$\left(\varepsilon^{(nm)}_b\right)^m = \varepsilon^{(n)}_b.$$

If Λ is a $\mathbb{Z}/n$-algebra, we again denote by ε_b the composite

$$\mathrm{Gal} \longrightarrow \mathbb{Z}/(n)(1) \longrightarrow \Lambda(1).$$

It is independent of n.

The character ε_b factors through

$$\pi_1\big(S - (\text{the subscheme } b = 0)\big).$$

3.3.3.

Suppose that S is a strictly henselian trait and that n is odd, $n = 2m+1$. Let x be a singular point of X_s in a neighborhood of which X_η is smooth, and let X_1/S be a subscheme of $\mathbb{A}_S^{n+1}$ with equation $Q - b = 0$, as in 1.3.2(i). The assumption on x means that b, although zero modulo the maximal ideal, is not zero.

Let $X_{s(x)}$ be the henselization of X_s at x, let $\widetilde{X}_{s(x)}$ be the blowup of X_s at x, let Y be the quadric which is the inverse image of x, and put

$$X^*_{s(x)} = X_{s(x)} - \{x\} = \widetilde{X}_{s(x)} - Y.$$

The composite morphism

$$\begin{array}{ccccc} H^{n-1}(Y,\Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_{s(x)},\Lambda(m)) & \longrightarrow & H^{n-1}(X^*_{s(x)},\Lambda(m)) \\ & & & & \downarrow \\ & & H^n_{\{x\}}\big(X_s, R\Psi_{\bar\eta}(\Lambda(m))\big) & \longleftarrow & H^n_{\{x\}}(X_s,\Lambda(m)) \end{array}$$

identifies the latter group with the primitive quotient (XII, 3.5) of the cohomology of the quadric Y. Indeed, this follows from 2.2.7 in a locally isomorphic situation. The images of the two natural generators of this quotient (the cohomology classes of the rulings of Y) are the two basis vectors

$$\pm\delta \in H^n_{\{x\}}\big(X_s, R\Psi_{\bar\eta}(\Lambda(m))\big).$$

35

3.3.4.

Return to the more general assumptions of 3.3.1, with $n = 2m+1$. For every singular point x of X_s, we again denote by $\pm\delta_x$ the sections of $R^n f_!\Lambda(m)$ over S which are the images of the generators of the primitive quotient of $H^{n-1}(Y,\Lambda(m))$ (with the same notation as above) under the composite morphism

$$\begin{array}{ccccc} H^{n-1}(Y,\Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_{s(x)},\Lambda(m)) & \longrightarrow & H^{n-1}(X^*_{s(x)},\Lambda(m)) \\ & & & & \downarrow \\ H^0\big(S, R^n f_!\Lambda(m)\big) & \xleftarrow{\sim} & H^n_c(X_s,\Lambda(m)) & \longleftarrow & H^n_{\{x\}}(X_s,\Lambda(m)) \end{array}$$

[35] At the definition of the punctured henselization, the print places a tilde on the first $X^*_{s(x)}$. The equality with $X_{s(x)} - \{x\}$ and the identical construction in 2.2.6 determine the untilded term used here.

The class $\pm\delta_x$ is the vanishing cycle at x.

When S is a trait, the class

$$\pm\delta \in H^n_c(X_{\bar\eta}, \Lambda(m))$$

is the image under XIII, 2.1.8.4 of the class of 3.3.3

$$\pm\delta \in H^n_{\{x\}}\big(X_s, R\Psi_{\bar\eta}(\Lambda(m))\big).$$

We shall prove the following two results simultaneously.

Proposition 3.3.5.

Let S be a henselian trait and let X/S be as in 3.3.3. The variation is given by

$$\mathrm{Var}(\sigma)(a) = (-1)^{m+1}\varepsilon_{b(x)}(\sigma)(a,\delta)\delta.$$

Proposition 3.3.6.

Let S be strictly local and regular, and let $f : X \to S$ be a proper flat morphism of relative dimension $n = 2m+1$ such that X_s has exactly one nonsmooth point x, assumed to be a nondegenerate quadratic point. Suppose also that $b(x)$ is a parameter. Then Galois acts on the cohomology of the geometric generic fiber by

$$\sigma(a) = a + (-1)^{m+1}\varepsilon_{b(x)}(\sigma)(a,\delta_x)\delta_x.$$

(A) Let k be an algebraically closed field of characteristic 0, and let S be the henselization of $\mathrm{Spec}(k[T])$ at 0. We prove 3.3.5 when $b(x)$ is a uniformizer.

By the Lefschetz principle, we reduce to the case $k = \mathbb{C}$. Since the question is local, we may suppose that $f : X \to S$ is obtained by base change from the map

$$\sum_i z_i^2 : \mathbb{C}^{n+1} \longrightarrow \mathbb{C}.$$

The construction of $\pm\delta$ given above has a transcendental analogue in integral cohomology. Via XIV, 2.1, the class $\pm\delta$ therefore corresponds to a transcendental vanishing cycle, the image of

$$\pm\delta_{\mathrm{cl}} \in H^n_{\{0\}}\big(X_0, R\Psi_{\mathrm{cl}\eta}(\mathbb{Z})\big).$$

By XIV, 2.1 and the corresponding property of the algebraic cycles $\pm\delta$, the image of $\pm\delta_{\mathrm{cl}}$ in

$$H^n_{\{0\}}\big(X_0, R\Psi_{\mathrm{cl}\eta}(\mathbb{Z})\big) \otimes \mathbb{Z}_\ell$$

is a generator of this group. Thus $\pm\delta_{\mathrm{cl}}$ are the two generators of $H^n_{\{0\}}(X_0, R\Psi_{\mathrm{cl}\eta}(\mathbb{Z}))$, and coincide with the vanishing cycles of XIV, 3.2. Formula 3.3.5 consequently follows from XIV, 3.2.11. Strictly speaking, one should have checked compatibility between the isomorphism of XIV, 2.1 and cup products and traces.

(B) We prove 3.3.6 for X/S as in (A), with X/S proper in addition. This follows from XIII, 2.4.6.2.

(C) Let p be a prime number, and let S_0 be the strict henselization of $\mathrm{Spec}(\mathbb{Z}[T])$ at (p,T). We prove 3.3.6 for X/S_0 such that $b(x) = T$.

The restriction of the sheaf $R^i f_*\Lambda$ to the strict henselization of $\mathrm{Spec}(\mathbb{Z}[T])$ at (T) is governed by (B). This sheaf is locally constant on the complement of the smooth divisor $T = 0$, which has mixed characteristic. Assertion (C) therefore follows from (B) and Abhyankar's lemma.

(D) In at least one case to which (C) applies, δ_x is not a torsion element in ℓ-adic cohomology.

Let S_1 be the strict henselization of $\mathrm{Spec}(\mathbb{Z})$ at (p), and let $V \subset \mathbb{P}^r/S_1$ be a smooth projective variety over S_1, purely of relative dimension $n+1$. Suppose that there is a Lefschetz pencil of hyperplane sections for the special fiber V_p of V/S_1 (XVII, 2.5). This pencil is defined by a line d' in the special fiber of the projective space $\check{\mathbb{P}}^{,r}/S_1$ dual to $\mathbb{P}^r/S_1$. Lift d' to a line d in $\check{\mathbb{P}}^{,r}/S_1$, and let $X \to d$ be the pencil of hyperplane sections of V parametrized by d (XVIII, 3.1). The exceptional locus Δ of d is ètale over S_1 (XVII, 6). After ètale localization at a point of $\Delta_p \subset d$, we therefore obtain a situation of type (C).

To check that δ_x is not torsion, it is enough to do so after extending scalars from S_1 to $\mathbb{C}$. Take $V = \mathbb{P}^{n+1}$ with the projective embedding defined by $\mathcal{O}(N)$. For $N \geq 2$, XVII, 2.5 applies (it is valid for every N in this particular case). We then use the following lemma.

Lemma 3.3.7.

For a Lefschetz pencil of hypersurfaces of sufficiently large degree N in $\mathbb{P}^{n+1}(\mathbb{C})$ ($N \geq 3$ suffices), a vanishing cycle in $\mathbb{Q}_\ell$-cohomology is nonzero.

This lemma follows from the theory of Lefschetz pencils, for example from XVIII, 6.6.1. There is no circularity in using here this result of XVIII, based on the "Picard–Lefschetz formula," since (A) is already available. One may instead use the more direct, purely transcendental variant XIX, no. 4, which depends only on XIV, 3.2.

(E) We prove 3.3.5 in residual characteristic p for $(X/S, x \in X_s)$.

Let X_0/S_0 be as in (D). By 1.3.2(i), there is a morphism $g : S \to S_0$ such that the henselization at x_0 of

$$g^*X_0 = X_0 \times_{S_0} S$$

is S-isomorphic to the henselization of X at x. It is therefore enough to prove 3.3.5 for g^*X_0. By base change from (C) for X_0/S_0, one obtains

$$\sigma(a) = a - (-1)^m \varepsilon_{b(x)}(\sigma)(a, \delta_x)\delta_x$$

in $H^n(X_{0\bar{\eta}})$, for σ in the inertia group. Let π be a uniformizer and compare this formula with (2.2.5.9), where $\varepsilon = \varepsilon_\pi$. We find

$$\lambda_X \varepsilon_\pi(\sigma)(a, \delta_x)\delta_x = -(-1)^m \varepsilon_{b(x)}(\sigma)(a, \delta_x)\delta_x$$

in $H^n(X_{0\bar{\eta}})$. Since δ_x is not torsion, we may cancel $(a, \delta_x)\delta_x$ from this identity. Formula (2.2.5.9), with the resulting value of λ_X, is precisely 3.3.5.

(F) Proof of 3.3.6 in the general case.

The general case of 3.3.6 reduces to the general case of 3.3.5 just as (B) and (C) reduce to (A).

3.4. Summary

Propositions 3.1.2, 3.2.1, 3.2.2, 3.2.3, and 3.3.5 constitute the "Picard–Lefschetz formula." This formula is the algebraic analogue of XIV, 3.2.11, but takes very different forms according to the parity of the relative dimension.

In the most interesting case, where X/S is proper, the results may be summarized as follows.

Theorem 3.4 (Picard–Lefschetz formula).

Let S be a strictly henselian trait and let $f : X \to S$ be a proper flat morphism, purely of relative dimension $n = 2m$ or $2m+1$. Suppose that X_η is smooth and that X_s has only ordinary quadratic singularities. Let Σ be the singular locus of X_s.

(i) To every $x \in \Sigma$ is attached a vanishing cycle

$$\pm\delta_x \in H_n(X_{\bar{\eta}})(m).$$

The δ_x are pairwise orthogonal. When n is even,

$$(\delta_x, \delta_x) = (-1)^m, 2.$$

(ii) There are isomorphisms

$$H^i(X_s) \xrightarrow{\sim} H^i(X_{\bar{\eta}})$$

for $i \neq n, n+1$, and an exact sequence

$$0 \longrightarrow H^n(X_s) \longrightarrow H^n(X_{\bar{\eta}}) \xrightarrow{((\,\cdot\,,\delta_x))_{x\in\Sigma}} \bigoplus_{x\in\Sigma} \Lambda(n-m) \longrightarrow H^{n+1}(X_s) \longrightarrow H^{n+1}(X_{\bar{\eta}}) \longrightarrow 0.$$

(iii) There are defined local characters of the inertia group I:

$$\begin{aligned} n \text{ even}: &\quad \varepsilon_x : I \longrightarrow \{\pm 1\}, \\ n \text{ odd}: &\quad \varepsilon_x : I \longrightarrow \Lambda(1). \end{aligned}$$

When X is regular at x, these characters are surjective; when x is nondegenerate, they are respectively the unique character of order 2 and the canonical character.

The action of the inertia group I is given by

$$n = 2m \qquad \sigma(a) = a + (-1)^m \sum_{x\in\Sigma} \frac{\varepsilon_x(\sigma) - 1}{2} (a, \delta_x)\delta_x,$$

$$n = 2m+1 \qquad \sigma(a) = a - (-1)^m \sum_{x\in\Sigma} \varepsilon_x(\sigma)(a, \delta_x)\delta_x.$$

In particular,

$$\begin{aligned} n \text{ even}: &\quad \sigma(\delta_x) = \varepsilon_x(\sigma)\delta_x, \\ n \text{ odd}: &\quad \sigma(\delta_x) = \delta_x. \end{aligned}$$

SGA 7 — EXPOSÉ XVI

The Milnor Formula

by P. Deligne

In this exposé, Milnor's formula [3] (see also [2]) is transposed to characteristic p. In certain cases this formula makes it possible to compute the dimensions of spaces of vanishing cycles. The proof is global in nature and treats only the "geometric" case in which the base is a henselian trait of equal characteristic with perfect residue field.

1. Statement of the problem

1.1.

Let S be a regular scheme, purely of dimension one, let s be a closed point of S with perfect residue field, let $f : X \to S$ be a flat morphism of finite type, let x_0 be a closed point of the fiber X_s, and let $\bar{x}_0$ be a geometric point of X localized at x_0. Suppose that

(a) x_0 is an isolated point at which f is not smooth: there are an integer n and a neighborhood U of x_0, purely of relative dimension n, such that $U - \{x_0\}$ is smooth over S;

(b) X is regular at x_0.

Under these assumptions, X is a relative complete intersection over S at x_0. We refer to VI, 3.10 for the definition of $D^1_{X/S}$. On the neighborhood U one has

$$D^1_{X/S} = \mathcal{H}^1\big(R\mathcal{H}om(L\Omega_{X/S}, \mathcal{O}_X)\big) = \mathcal{E}xt^1(\Omega^1_{X/S}, \mathcal{O}_X);$$

on this neighborhood, $D^1_{X/S}$ has finite length and support contained in $\{x_0\}$.

Definition 1.2.

The Milnor number $\mu = \mu(X/S, \bar{x}_0)$ of X/S at $\bar{x}_0$ is the length of the $\mathcal{O}_{X,x_0}$-module defined by $D^1_{X/S}$.

Example 1.3.

Take S to be a smooth curve over an algebraically closed field k. After replacing X by a neighborhood of x_0, we may suppose that X is smooth over k. The cotangent complex then reduces to the following resolution of $\Omega^1_{X/S}$:

$$L\Omega_{X/S} = \big[f^*\Omega^1_{S/k} \longrightarrow \Omega^1_{X/k}\big].$$

Choose a local coordinate $t : S \to \mathbb{E}^1_k$ on S at s (with t étale at s), again denote by f the composite $t \circ f$, and let $(x_i)_{1 \le i \le n+1}$ be a system of local coordinates on X at x_0. At x_0, the module $f^*\Omega^1_{S/k}$ then has basis df, while $\Omega^1_{X/k}$ has basis the dx_i. Thus $D^1_{X/S}$ is identified with the quotient of $\mathcal{O}_X$ by the ideal generated by the $\partial f/\partial x_i$, and

$$\mu = \dim_k\left(\mathcal{O}_{X,x_0}/\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_{n+1}}\right)\right). \tag{1.3.1}$$

This is the definition given by Milnor.

Example 1.4.

Suppose that X is the closed subscheme of $\mathbb{E}_S^{n+1}$ defined by an equation g, and let $i : X \hookrightarrow \mathbb{E}_S^{n+1}$ be the inclusion. The cotangent complex then reduces to

$$\big[(g)/(g)^2 \longrightarrow i^*\Omega^1_{\mathbb{E}^{n+1}/S}\big],$$

and $D^1_{X/S}$ is isomorphic to the quotient of $\mathcal{O}_X$ by the ideal generated by $(\partial g/\partial x_i)_{1\leq i\leq n+1}$. Equivalently, it is the quotient of $\mathcal{O}_{\mathbb{E}^{n+1}}$ by the ideal generated by g and the $\partial g/\partial x_i$.

Example 1.5.

Let S be the spectrum of a discrete valuation ring V with perfect residue field and fraction field K, and let X be the normalization of S in a separable extension K' of K. By 1.4 and [5], III, 7.14, p. 68, $D^1_{X/S}$ is then isomorphic to the quotient of $\mathcal{O}_X$ by the different $\mathfrak{d}_{X/S}$. If $\bar{x}_0$ is a geometric point of X_s, localized at $x_0 \in X_s$, its Milnor number is therefore

$$\mu = \mathrm{length}\big(\mathcal{O}_{X,x_0}/\mathfrak{d}_{X/S}\big). \tag{1.5.1}$$

Let $\bar{s}$ be a geometric point of S localized at the closed point s, and let $d_{K'/K} \in V$ generate the discriminant ideal, with valuation $v(d_{K'/K})$. It follows from (1.5.1) that

$$v(d_{K'/K}) = \sum_{x\in X(\bar{s})} \mu(X/S, x). \tag{1.5.2}$$

1.6.

Retain the assumptions of 1.1. Let $\bar{s}$ be the geometric point of S which is the image of $\bar{x}_0$, let $S_{(\bar{s})}$ be the spectrum of the strict henselization of $\mathcal{O}_{S,s}$ at $\bar{s}$, and let $X_{(\bar{s})}$ be the inverse image of X over $S_{(\bar{s})}$. Let $\bar{\eta}$ be the spectrum of an algebraic closure of the fraction field $k(\eta)$ of $S_{(\bar{s})}$, and put

$$I = \mathrm{Gal}(k(\bar{\eta})/k(\eta)).$$

Let ℓ be a prime number different from the residual characteristic of S. In a neighborhood of $\bar{x}_0$ in $X_{\bar{s}}$, the sheaves of vanishing cycles

$$\Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell) = R^i\Phi_{\bar{\eta}}(\mathbb{Z}/\ell)$$

relative to the constant sheaf $\mathbb{Z}/\ell$ on $X_{(\bar{s})}$ are concentrated at $\bar{x}_0$. Moreover, the inertia group I acts on the $\mathbb{Z}/\ell$-vector spaces

$$\Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0}.$$

These spaces are finite-dimensional (XIII, 2.4.2).

1.7.

Let I_1 be a finite quotient of the inertia group. Recall that the Swan character Sw of I_1 is the difference $a - u$ between the Artin character a and the character u of the augmentation representation. Recall also that, up to isomorphism, there is a unique projective $\mathbb{Z}_\ell[I_1]$-module $S(I_1)$ with character Sw (see [6]).

Let V be a finite-dimensional $\mathbb{Z}/\ell$-vector space with an action $\rho : I \to \mathrm{GL}(V)$, and suppose that ρ factors through I_1. The wild dimension of V is then

$$\dim \mathrm{Sw}(V) = \dim_{\mathbb{Z}/\ell} \mathrm{Hom}_{I_1}(S(I_1) \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, , V)\,. \tag{1.7.1}$$

This integer is independent of the choice of I_1, since for a normal subgroup H of I_1 one has $S(I_1/H) \sim S(I_1)_H$.

We further put

$$\dim \mathrm{tot}(V) = \dim \mathrm{Sw}(V) + \dim_{\mathbb{Z}/\ell}(V)$$

and call this number the total dimension of V.

Definition 1.8.

With the terminology of 1.6 and 1.7, the number of vanishing cycles of X/S at $\bar{x}_0$ (respectively the wild number of vanishing cycles, respectively the total number of vanishing cycles) is the integer

$$(-1)^n \dim \Phi_{\bar{x}_0} \underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \dim \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0},$$

$$(-1)^n \dim \mathrm{Sw}\, \Phi_{\bar{x}_0} \underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \dim \mathrm{Sw}\, \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0},$$

$$\begin{aligned}(-1)^n \dim \mathrm{tot}\, \Phi_{\bar{x}_0} \underset{\text{def}}{=} (-1)^n \sum_i (-1)^i & \\ \cdot \dim \mathrm{tot}\, \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0} & \\ = (-1)^n \big(\dim \Phi_{\bar{x}_0} + \dim \mathrm{Sw}\, \Phi_{\bar{x}_0}\big). &\end{aligned}$$

Conjecture 1.9.

Under the assumptions of 1.1, the Milnor number μ of X/S at $\bar{x}_0$ is equal to the total number of vanishing cycles of X/S at $\bar{x}_0$.

Remark 1.10.

Let S be a henselian trait with imperfect residue field and let $f : X \to S$ be a morphism which is smooth except at $x_0 \in X_s$. Suppose that X is regular at x_0. Definition 1.2 of the Milnor number μ still makes sense under these assumptions, but I no longer have a geometric interpretation of μ to propose.

1.11.

For the remainder of this section, suppose that $S = \mathrm{Spec}(V)$ is a henselian trait with algebraically closed residue field and closed point s, that $f : X \to S$ is a flat morphism of finite type, purely of relative dimension n, that X is regular, that x_0 is a closed point of X_s, and that $X - \{x_0\}$ is smooth over S.

This entails no loss of generality.

Proposition 1.12.

If $n = 0$ (that is, if X/S is finite), Conjecture 1.9 holds for $(X/S, x_0)$.

We may suppose that X is the normalization of S in a finite extension K' of the fraction field K. By 1.5, $\mu = v(d_{K'/K})$.

The extension K' is defined by an open subgroup H of the inertia group I. Let $H' \subset H$ be an open normal subgroup of I, and put $I' = I/H'$. Let $s_{I/H}$ be the character of the permutation representation of I' on I/H, let $a_{I'}$ be the Artin character, let $\mathrm{Sw}_{I'}$ be the Swan character, and let $u_{I'}$ be the character of the augmentation representation. By Artin's "Führerdiskriminantenproduktformel," the discriminant ideal is the Artin conductor of $s_{I/H}$ ([5], VI, 3, Corollary 1, p. 111):

$$\mu = (a_{I'}, s_{I/H}) = |I'|^{-1} \sum_{i \in I'} a_{I'}(i) s_{I/H}(i).$$

One has

$$a_{I'} = u_{I'} + \mathrm{Sw}_{I'}, \qquad (u_{I'}, s_{I/H}) = \#(I/H) - 1,$$

and

$$(\mathrm{Sw}_{I'}, s_{I/H}) = \dim_{\mathbb{Z}/\ell} \mathrm{Hom}_I\Big(S(I') \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, (\mathbb{Z}/\ell)^{I/H}\Big)$$
$$= \dim_{\mathbb{Z}/\ell} \mathrm{Hom}_I\Big(S(I') \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, \mathrm{Coker}\big(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H}\big)\Big).$$

Consequently,

$$\mu = \#(I/H) - 1 + \dim \mathrm{Sw}\,\mathrm{Coker}\big(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H}\big).$$

One checks immediately that $\Phi^i_{x_0} = 0$ for $i \neq 0$ and that

$$\Phi^0_{x_0} = \mathrm{Coker}\big(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H}\big);$$

the formula follows.

Proposition 1.13.

If X_s has an ordinary quadratic singularity at x_0, Conjecture 1.9 holds for $(X/S, x_0)$.

Case 1: $n = \dim(X_s) = 2m - 1$.

The problem is local for the étale topology on X. We may therefore suppose that X is the closed subscheme of $\mathbb{E}^{n+1}_S$ defined by an equation

$$f = \sum_{i=1}^{m} x_i x_{i+m} + a = 0,$$

where a lies in the maximal ideal of V and x_0 is the origin of the coordinates. Moreover, X is regular at x_0 if and only if a is a uniformizer. By 1.4,

$$\mu = \mathrm{length}\left(\mathcal{O}_{\mathbb{E}^{2m}_S, x_0} / \left(f, \frac{\partial f}{\partial x_i}\right)\right) = 1.$$

By the ordinary-quadratic vanishing-cycle calculation of Exposé XV, $\Phi^i_{x_0} = 0$ for $i \neq n$, while $\Phi^n_{x_0}$ has dimension one and inertia acts trivially on it.[36] The wild number of vanishing cycles is therefore zero and the total number is 1, which verifies 1.9.

Case 2: $n = \dim(X_s) = 2m$ and $\mathrm{char}(k(s)) \neq 2$.

Under these assumptions we may take $X \subset \mathbb{E}^{n+1}_S$ to be defined by

$$f = \sum_{i=1}^{m} x_i x_{i+m} + x^2_{2m+1} + a = 0,$$

where a is a uniformizer. Again $\mu = 1$, $\Phi^i_{x_0} = 0$ for $i \neq n$, and $\Phi^n_{x_0}$ has dimension one. This time the inertia group acts by ± 1 on $\Phi^n_{x_0}$, hence tamely. The wild number of vanishing cycles is again zero, and 1.9 is verified.

Case 3: $n = \dim(X_s) = 2m$ and $\mathrm{char}(k(s)) = 2$.

This time we may take $X \subset \mathbb{E}^{n+1}_S$ to be defined by

$$f = \sum_{i=1}^{m} x_i x_{i+m} + (x^2_{2m+1} + a x_{2m+1} + b) = 0,$$

where a and b lie in the maximal ideal. The scheme X is regular if and only if b is a uniformizer, and its generic fiber is smooth if and only if $a^2 - 4b \neq 0$.

[36] After the printed word "By," the source leaves the expected reference blank. The stated concentration and inertia action are the ordinary-quadratic vanishing-cycle calculation established in Exposé XV.

The Milnor number of X/S is the same as that of $X_0 \subset \mathbb{E}^1_S$ defined by

$$x^2 + ax + b = 0. \tag{1}$$

Likewise, $\Phi^i_{x_0} = 0$ for $i \neq n$, $\Phi^n_{x_0}$ has dimension one, and I acts on it by ± 1; the index-two subgroup of I acting trivially corresponds to the extension of K defined by (1). The total number of vanishing cycles of X/S at x_0 therefore coincides with the one for X_0/S, and 1.9 in this case follows from 1.12. The Milnor number is given by the discriminant formula of 1.5:

$$\mu = v(a^2 - 4b). \tag{1.13.1}$$

2. The "geometric" case

We shall prove that Conjecture 1.9 holds in equal characteristic. The key geometric result is the following. Its possible use was suggested to me by conversations with B. Saint-Donat.

Proposition 2.1.

Let S be a smooth connected projective curve over an algebraically closed field k, let X be a smooth proper variety over k, purely of dimension $n+1$, and let $f : X \to S$ be a morphism which is smooth away from a finite set F of points of X. Then the Euler–Poincaré characteristics in étale cohomology of S, X, and the geometric generic fiber $X_{\bar\eta}$ of f satisfy

$$\chi(X) = \chi(S)\chi(X_{\bar\eta}) - (-1)^n \sum_{x \in F} \mu(X/S, x). \tag{2.1.1}$$

The Euler–Poincaré characteristic of X is the trace (or integral over X) of the top Chern class of the tangent bundle:

$$\chi(X) = \int_X c_{n+1}(\Theta_X) = (-1)^{n+1} \int_X c_{n+1}(\Omega^1_X).$$

Write

$$\Omega^1_X = \left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \otimes f^*\Omega^1_S.$$

The total Chern class of the invertible sheaf $f^*\Omega^1_S$ is the inverse image of the total Chern class of Ω^1_S on S:

$$c^*(f^*\Omega^1_S) = 1 + f^*c^1(S, \Omega^1_S),$$

and the cup powers of c^1 vanish:

$$\left(f^*c^1(S,\Omega^1_S)\right)^n = f^*\left(c^1(S,\Omega^1_S)^n\right) = 0 \qquad (n > 1).$$

It follows that

$$\begin{aligned} c^{n+1}(\Omega^1_X) = c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*c_1(S,\Omega^1_S) \\ + c^{n+1}\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right). \end{aligned}$$

If S has genus g, then $c_1(S,\Omega^1_S)$ is $(2g-2)$ times the cohomology class of any closed point s of S. Consequently, when X_s is nonsingular,

$$\begin{aligned} \mathrm{I} &= \int_X c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*c_1(S,\Omega^1_S) \\ &= (2g-2) \int_X c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*\,\mathrm{cl}(s) \\ &= (2g-2) \int_{X_s} c^n\left(X_s, \Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1} \otimes \mathcal{O}_{X_s}\right). \end{aligned}$$

Moreover, on X_s the sheaf $f^*\Omega^1_S$ is isomorphic to $\mathcal{O}_{X_s}$, and $\Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{O}_{X_s}$ occurs in the exact sequence

$$0 \longrightarrow f^*\Omega^1_S \otimes \mathcal{O}_{X_s} \longrightarrow \Omega^1_X \otimes \mathcal{O}_{X_s} \longrightarrow \Omega^1_{X_s} \longrightarrow 0.$$

Hence

$$c^n\big(X_s, \Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1} \otimes \mathcal{O}_{X_s}\big) = c^n(X_s, \Omega^1_{X_s}),$$

and

$$\mathrm{I} = (2g-2)\int_{X_s} c^n(X_s, \Omega^1_{X_s}) = (-1)^{n+1}\chi(S)\chi(X_s).$$

The canonical map

$$f^* : f^*\Omega^1_S \longrightarrow \Omega^1_X$$

is identified with a section a of $\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}$. After choosing systems of local coordinates on S and X, thereby identifying this bundle with $\mathcal{O}_X^{n+1}$, the coordinates of a are the partial derivatives $\partial f/\partial x_i$. Thus a vanishes only on F, and by 1.3 the multiplicity of its zero at $x \in F$ is the Milnor number $\mu(X/S, x)$.

Recall that if a section a of a vector bundle G of rank d on a smooth proper k-scheme X, purely of dimension d, vanishes only at isolated points, then $\int_X c^d(G)$ is the number of zeros of a, counted with multiplicity. Here

$$\mathrm{II} = \int_X c^{n+1}\big(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\big) = \sum_{x \in F} \mu(X/S, x).$$

Collecting terms gives

$$\chi(X) = (-1)^{n+1}(\mathrm{I} + \mathrm{II}) = \chi(S)\chi(X_s) - (-1)^n \sum_{x \in F} \mu(X/S, x),$$

which proves 2.1, since $\chi(X_s) = \chi(X_{\bar\eta})$.

Corollary 2.2.

Under the preceding assumptions, if $n(X/S, x)$ is the total number of vanishing cycles of X/S at x (1.8), then

$$\sum_{x \in F} \mu(X/S, x) = \sum_{x \in F} n(X/S, x).$$

The Leray spectral sequence for f gives

$$\begin{aligned}\chi(X) &= \sum_{p,q} (-1)^{p+q} \dim_{\mathbb{Z}/\ell} H^p(S, R^q f_* \mathbb{Z}/\ell) \\ &= \sum_q (-1)^q \chi(S, R^q f_* \mathbb{Z}/\ell).\end{aligned}$$

For every constructible sheaf G of $\mathbb{Z}/\ell$-modules on S, one has ([4] or SGA 5, X, 7.2)

$$\chi(S, G) = \chi(S)\dim_{\mathbb{Z}/\ell}(G_{\bar\eta}) - \sum_{s \in S} \varepsilon_s(G),$$

where, for each $s \in S$, after identifying $\bar\eta$ with the algebraic closure of the fraction field of the henselization of S at s,

$$\varepsilon_s(G) = \dim(G_{\bar\eta}) - \dim(G_s) + \dim \mathrm{Sw}(G_{\bar\eta}).$$

[37] The print has a minus sign before the Swan term. The plus sign is forced both by the standard Artin-conductor formula cited in the preceding sentence and by the printed identity (c), which identifies the alternating sum with the *total* dimension dim + dim Sw.

For $s \in S$, the exact sequences XIII, 2.4.6.3,

$$\cdots \longrightarrow (R^q f_* \mathbb{Z}/\ell)_s \longrightarrow (R^q f_* \mathbb{Z}/\ell)_{\bar{\eta}} \longrightarrow \bigoplus_{x \in F(s)} \Phi^q_{\bar{\eta}}(\mathbb{Z}/\ell)_x \longrightarrow \cdots$$

give

$$\sum_q (-1)^q \left(\dim(R^q f_* \mathbb{Z}/\ell)_{\bar{\eta}} - \dim(R^q f_* \mathbb{Z}/\ell)_s\right) = \sum_{x \in F(s)} \sum_q (-1)^q \dim \Phi^q_{\bar{\eta}}(\mathbb{Z}/\ell)_x,$$

and

$$\sum_q (-1)^q \dim \mathrm{Sw}(R^q f_* \mathbb{Z}/\ell)_{\bar{\eta}} = \sum_{x \in F(s)} \sum_q (-1)^q \dim \mathrm{Sw}\, \Phi^q_{\bar{\eta}}(\mathbb{Z}/\ell)_x.$$

Consequently,

$$\text{(c)} \qquad \sum_q (-1)^q \varepsilon_s(R^q f_* \mathbb{Z}/\ell) = \sum_{x \in F(s)} \dim \mathrm{tot}\, \Phi_x.$$

The Euler–Poincaré formula therefore becomes

$$\begin{aligned}
\chi(X) &= \sum_q (-1)^q \chi(S, R^q f_* \mathbb{Z}/\ell) \\
&= \sum_q (-1)^q \left(\chi(S) \dim(R^q f_* \mathbb{Z}/\ell)_{\bar{\eta}} - \sum_s \varepsilon_s(R^q f_* \mathbb{Z}/\ell) \right) \\
&= \chi(S) \sum_q (-1)^q \dim(R^q f_* \mathbb{Z}/\ell)_{\bar{\eta}} - \sum_s \sum_q (-1)^q \varepsilon_s(R^q f_* \mathbb{Z}/\ell) \\
&= \chi(S)\chi(X_{\bar{\eta}}) - \sum_s \sum_{x \in F(s)} \dim \mathrm{tot}\, \Phi_x \\
&= \chi(S)\chi(X_{\bar{\eta}}) - (-1)^n \sum_{x \in F} n(X/S, x).
\end{aligned}$$

Comparison with (2.1.1) proves 2.2.

Corollary 2.3.

Under the preceding assumptions and with the preceding notation, suppose that $x_0 \in F$ and that, for every $x \in F$ distinct from x_0, the fiber of f through x has an ordinary quadratic singularity at x. Then

$$\mu(X/S, x_0) = n(X/S, x_0).$$

For $x \neq x_0$, Proposition 1.13 gives

$$\mu(X/S, x) = n(X/S, x),$$

and it remains to apply 2.2.

Theorem 2.4.

Let S be a henselian trait of equal characteristic with perfect residue field, let $f : X \to S$ be a flat morphism of finite type, and let $\bar{x}$ be a geometric point of X whose image in X is a closed point x of X_s at which X/S is not smooth, and which is isolated as such. Suppose that X is regular at

x. Then the total number (1.8) of vanishing cycles of X/S at $\bar{x}$ coincides with the Milnor number $\mu(X/S, x)$.

Let S' be the henselian trait obtained from S by extending the residue field $k(s)$ to its separable closure, and put $X' = X \times_S S'$. The assertion of 2.4 for X/S is equivalent to the assertion for X'/S'. Since $k(s)$ is already assumed perfect, we may therefore suppose that the residue field of S is algebraically closed.

For such an S, let S' be its completion and put $X' = X \times_S S'$. The Milnor number at x is the same for X/S and X'/S', and by XIII, 2.4.2 the same is true of the total number of vanishing cycles. We may thus suppose in addition that S is complete, hence isomorphic to $k[[t]]$, with k algebraically closed. If X_s has dimension n at x, we then have:

Proposition 2.5.

There exist a smooth projective variety Y over k, purely of dimension $n+1$, a morphism $f : Y \to \mathbb{P}^1_k$ which is smooth outside a finite set F of points of Y, and a point $y_0 \in F$ such that:

(a) if $y \in F$ and $y \neq y_0$, then y is an ordinary quadratic singular point of the fiber of f through y;

(b) $f(y_0) = 0$, and if $a : S \to \mathbb{P}^1_k$ is the natural identification of S with the completion of $\mathbb{P}^1_k$ at 0, there is an S-isomorphism between the henselization of X at x and the henselization of a^*Y at y_0.

Let us show that 2.5 implies 2.4. By 2.3 and 2.5(a), the Milnor number of $Y/\mathbb{P}^1_k$ at y_0 coincides with the total number of vanishing cycles.[38] As above, XIII, 2.4.2 gives the same result for a^*Y/S, hence also for $(X/S, x)$, which by 2.5(b) is locally isomorphic to $(a^*Y/S, y_0)$ for the étale topology. We prove 2.5. As a $k[[t]]$-algebra, the completion of the local ring $\mathcal{O}_{X,x}$ is isomorphic to a quotient

$$\widehat{\mathcal{O}}_{X,x} = k[[t, x_1, \ldots, x_{n+1}]]/(f).$$

The implicit-function theorem XV, 1.1.2 (that is, [1], Lemma 5.10) then shows that there is an integer N such that condition 2.5(b) holds whenever

$$\widehat{\mathcal{O}}_{Y,y_0} \simeq k[[t, x_1, \ldots, x_{n+1}]]/(f_1)$$

as $k[[t]]$-algebras, with $f_1 - f$ in the Nth power $\mathfrak{m}^N$ of the maximal ideal of $k[[t, x_1, \ldots, x_{n+1}]]$.

Let (t_0, t_1) and $(x_0, \ldots, x_{n+1})$ be the usual homogeneous coordinate systems on $\mathbb{P}^1$ and $\mathbb{P}^{n+1}$. We take $Y/\mathbb{P}^1$ to be the subvariety of $\mathbb{P}^1 \times \mathbb{P}^{n+1}$ defined by an equation

$$g(t_0, t_1; x_0, \ldots, x_{n+1}) = 0, \tag{2.5.1}$$

homogeneous of degree d' in (t_0, t_1) and of degree d'' in the x_i. More intrinsically, g is a section of

$$\mathcal{O}(d', d'') = \mathrm{pr}_1^*\, \mathcal{O}(d') \otimes \mathrm{pr}_2^*\, \mathcal{O}(d'').$$

Condition (b) holds if

$$g(1, t; 1, x_1, \ldots, x_{n+1}) \equiv f \pmod{\mathfrak{m}^N}. \tag{2.5.2}$$

Fix d' and d''. The sections g of type (2.5.1) are the k-rational points of an affine space A. For $u \in \mathbb{P}^1 \times \mathbb{P}^{n+1}$, let $B(u, r)$ denote the affine space whose points are the r-jets at u of sections of $\mathcal{O}(d', d'')$. Put $u_0 = (1, 0; 1, 0, \ldots, 0)$. If $u \neq u_0$, then for sufficiently large d' and d'' the always-linear restriction map

$$A \longrightarrow B(u_0, N) \times B(u, 2) \tag{2.5.3}$$

is surjective.

Let $C(u) \subset B(u, 2)$ be the set of 2-jets which are the 2-jets either of the zero germ or of a germ $\widetilde{h}$ such that

[38] The print says Y/T here, but no base T is introduced; Proposition 2.5 defines the morphism $Y \to \mathbb{P}^1_k$.

(a) u lies on the variety V defined by $\widetilde{h} = 0$;

(b) u is neither a smooth point nor an ordinary quadratic point of the fiber through u of $V/\mathbb{P}^1$.

It is readily checked that $C(u)$ is closed of codimension $n+3$ in $B(u,2)$.

Let D be the affine subspace of A consisting of the g which satisfy (2.5.2). For $u \neq u_0$, let $D(u)$ be the subspace of D consisting of the zero g and of those g such that $g(u) = 0$ and u is neither a smooth point nor an ordinary quadratic point of the fiber through u of the variety V defined by $g = 0$. If d' and d'' are sufficiently large for (2.5.3) to be surjective, then $D(u)$ has codimension $n+3$ in D. The union of the $D(u)$ for $u \neq u_0$ therefore has codimension at least

$$1 = (n+3) - (n+2),$$

and a general element g of D lies in none of the $D(u)$. It solves the problem.

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SGA 7 — EXPOSÉ XVII

Lefschetz Pencils: Existence Theorem

by N. M. Katz

0. Introduction

In the present exposé (which, in the oral seminar, came before Exposés XIV and XV devoted to the Picard–Lefschetz formula), we introduce the "Lefschetz pencils" of hyperplane sections of a smooth projective scheme and give conditions for their existence. In the following exposé we make an elementary cohomological study of them, based on the known structure of the cohomology of blown-up varieties. The fact that certain hypotheses on Lefschetz pencils which we shall introduce are often satisfied will follow from the Picard–Lefschetz formula of Exposé XV, which is less elementary in nature than the results of the present exposé and the next.

Some developments of the present exposé and the next will be given elsewhere in a more general setting (general base schemes, quasi-projectivity hypotheses in place of projectivity, and so forth): in particular, the precohomological study of pencils of hyperplane sections, and especially of Lefschetz pencils, in EGA V, and the cohomological study of blown-up schemes in Jouanolou's Exposé VII of SGA 5. The reader will also find in EGA V proofs in which intrinsic diagram chasing replaces the use of coordinates favored by the writer of the present exposé.

I wish to thank P. Deligne and P. Griffiths for the help they gave me in conceiving this exposé. Throughout it, k denotes an algebraically closed field unless expressly stated otherwise.

1. A reminder on quadratic singularities (cf. VI, 6)

1.1.

Let Y be a k-scheme pure of dimension $n-1$. [39] A closed point $y \in Y$ is called an ordinary quadratic singularity of Y if:

(a) the completion $\widehat{\mathcal{O}}_{Y,y}$ of the local ring of Y at y is isomorphic to

$$k[[T_1, \ldots, T_n]]/(Q(T)),$$

the quotient of the formal power-series ring in $n = 1 + \dim Y$ variables by the ideal generated by an element $Q(T)$;

(b) Q begins in degree two, and the subvariety of $\mathbb{P}_k^{n-1}$ defined by the vanishing of the homogeneous degree-two part of $Q(T)$ is smooth over k when $n \geq 2$, while the degree-two part of $Q(T)$ is nonzero when $n = 1$.

If the characteristic of k is different from two, or if n is even, condition (b) is equivalent to:

(b′) Q begins in degree two and its Hessian matrix

$$\left(\frac{\partial^2 Q}{\partial T_i \partial T_j}(0)\right)_{1 \leq i,j \leq n}$$

is invertible.

[39] The print first says "of dimension n" and then, in the hypersurface presentation below, imposes $n = 1 + \dim Y$. The latter is the convention used by the displayed Hessian and normal form.

This follows from the Jacobian criterion for smoothness.

The standard form for the homogeneous degree-two part Q_2 of a series Q satisfying (b), obtainable after a linear change of variables, is

$$(1.1.1)\qquad \begin{cases} Q_2(T) = \sum_{i=1}^{n/2} T_{2i-1}T_{2i} & \text{if } n \text{ is even,} \\ Q_2(T) = T_n^2 + \sum_{i=1}^{(n-1)/2} T_{2i-1}T_{2i} & \text{if } n \text{ is odd.} \end{cases}$$

This follows from the theory of quadratic forms (Bourbaki, *Algèbre*, Ch. X).

1.2.

A closed point $y \in Y$ satisfying (a) and (b) is called a nondegenerate quadratic singularity. Since the determinant of the Hessian matrix is identically zero in characteristic two when n is odd, nondegenerate quadratic singularities presented in n variables exist only when n is even or the characteristic of k is different from two; in those cases they are precisely the ordinary quadratic singularities.

2. Lefschetz pencils: statement of the results

2.1.

Let $\mathbb{P}^r$ denote projective r-space over k, $\check{\mathbb{P}}^r$ the dual projective space of hyperplanes in $\mathbb{P}^r$, and $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$ the variety of lines in $\check{\mathbb{P}}^r$. A point $D \in \mathrm{Gr}(1, \check{\mathbb{P}}^r)$ is called a pencil of hyperplanes in $\mathbb{P}^r$. Regarding D as a line in $\check{\mathbb{P}}^r$, we write, by abuse of notation, $D = \{H_t\}_{t\in D}$, where the H_t are the hyperplanes of $\mathbb{P}^r$ which are the points of the line D.

The axis of D is the codimension-two linear subvariety of $\mathbb{P}^r$ which is the intersection $H_0 \cap H_\infty$ of any two distinct members of the pencil. Regarding this axis as a point of $\mathrm{Gr}(r-2, \mathbb{P}^r)$, the variety of $(r-2)$-dimensional linear subvarieties of $\mathbb{P}^r$, the map sending D to its axis is the familiar orthogonality isomorphism

$$\mathrm{Gr}(1, \check{\mathbb{P}}^r) \xrightarrow{\sim} \mathrm{Gr}(r-2, \mathbb{P}^r).$$

2.2.

Let X be a proper, smooth, irreducible k-scheme of dimension $n \geq 1$, equipped with an embedding $X \hookrightarrow \mathbb{P}^r$. A pencil $D = \{H_t\}_{t\in D}$ of hyperplanes in $\mathbb{P}^r$ is called a Lefschetz pencil (relative to this embedding) if:

(a) the axis of D meets X transversely (EGA IV, 17.13.7);

(b) there is a dense open subset $U \subset D$ such that, for every $t \in U$, the hyperplane H_t meets X transversely;

(c) for $t_0 \in D - U$, the hyperplane H_{t_0} meets X transversely except at one point, which is an ordinary quadratic singular point of $X \cap H_{t_0}$.

We shall see below in 3.2.1 that the Lefschetz pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$.

2.3.

The embedding $X \hookrightarrow \mathbb{P}^r$ is called a Lefschetz embedding if the Lefschetz pencils form a dense open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$. [40]

[40] More precisely, they form a dense open subset in the set of closed points of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$. We shall permit ourselves such abuses of language below.

2.4.

Recall the notion of a multiple of a given embedding $X \hookrightarrow \mathbb{P}^r$. It is the composite of that embedding with the d-th Veronese embedding [41]

$$S_d : \mathbb{P}^r \hookrightarrow \mathbb{P}^{\binom{r+d}{d}-1}, \qquad (X_0, \ldots, X_r) \longmapsto (\ldots, X^W, \ldots),$$

where

$$X^W = X_0^{W_0} \cdots X_r^{W_r} \quad \text{and} \quad W_0 + \cdots + W_r = d.$$

Thus a "section of X by a hypersurface of degree d" for the given embedding—in English, a hypersurface section of degree d—is simply a hyperplane section for the d-th multiple of that embedding.

Theorem 2.5.

Let X be irreducible, proper, and smooth over k, of dimension $n \geq 1$, and equipped with an embedding $i : X \hookrightarrow \mathbb{P}^r$. Then:

2.5.1. every d-th multiple of i, with $d \geq 2$, is a Lefschetz embedding;

2.5.2. if k has characteristic zero, the given embedding i is a Lefschetz embedding.

The proof is divided into several pieces, which follow.

3. The dual variety

3.1.

Let I be the ideal in $\mathbb{P}^r$ defining X. Write $N = I/I^2$ for the conormal sheaf, locally free of rank $r - n$ on X, and let $\mathbb{P}(N)$ be the projective bundle over X formed by the lines in $\check{N}$. (When $X = \mathbb{P}^r$, the bundle $\mathbb{P}(N)$ is empty.) The bundle $\mathbb{P}(N)$ identifies with the subvariety of $X \times \check{\mathbb{P}}^r$ formed by the pairs (x, H) for which H is tangent to X at x, by the map

$$\nu : \mathbb{P}(N) \longrightarrow X \times \check{\mathbb{P}}^r.$$

More explicitly,

$$\text{(3.1.1)} \qquad \mathbb{P}(I/I^2)_x \ni F \longmapsto \left(x, \text{ the hyperplane with equation } \sum_i \frac{\partial F}{\partial X_i}(x) T_i = 0 \right),$$

where F denotes a local section of I/I^2. Define

$$\text{(3.1.2)} \qquad \varphi : \mathbb{P}(N) \longrightarrow \check{\mathbb{P}}^r$$

to be the composite

$$\mathbb{P}(N) \xrightarrow{\nu} X \times \check{\mathbb{P}}^r \xrightarrow{\mathrm{pr}_2} \check{\mathbb{P}}^r.$$

3.1.3.

Let $\check{X}$ denote the image of φ. It is the set of hyperplanes which do not meet X transversely, and is called the dual variety of X for the embedding $i : X \hookrightarrow \mathbb{P}^r$.

[41] The print calls this the d-th Segre embedding; the displayed monomial map is the Veronese embedding.

Proposition 3.1.4.

The dual variety $\check{X}$ is irreducible and proper, of dimension at most $r-1$.
Proof. It is the image under φ of $\mathbb{P}(N)$, which is itself irreducible of dimension $r-1$ if $X \neq \mathbb{P}^r$, and empty if $X = \mathbb{P}^r$.

Remark 3.1.5.

The interpretation of $\mathbb{P}(N)$ can be made more precise as follows. Let $\mathcal{Y} \subset X \times \check{\mathbb{P}}^r$ be the incidence subscheme whose fiber over every point $H \in \check{\mathbb{P}}^r(k)$ is $X \cap H$. The scheme $\mathcal{Y}$ is plainly smooth of dimension $n+r-1$, being a projective bundle of relative dimension $r-1$ over X. Thus $\mathcal{Y} \to \check{\mathbb{P}}^r$ makes $\mathcal{Y}$ a relative complete intersection of virtual relative dimension $(n+r-1)-r = n-1$.

The Jacobian subscheme

$$J_{n-1}(\mathcal{Y}/\check{\mathbb{P}}^r)$$

(VI, 5.2) is then precisely $\mathbb{P}(N)$. Formation of the Jacobian subscheme of a relative complete intersection is compatible with arbitrary base change. Consequently, for every line $D \subset \check{\mathbb{P}}^r$, the Jacobian subscheme $J_{n-1}(\widetilde{X}/D)$, where

$$\widetilde{X} = \mathcal{Y} \times_{\check{\mathbb{P}}^r} D,$$

is precisely $\mathbb{P}(N) \times_{\check{\mathbb{P}}^r} D$.

Proposition 3.2.

The subset $B \subset \check{X}$ formed by the hyperplanes H such that $X \cap H$ has exactly one singular point, and that point is an ordinary quadratic singularity, is open.

This is a special case of XV, 1.3.2, based on R. Elkik's theory of "henselian modules" (XV, 1.1.4). The case in which one restricts to nondegenerate quadratic points—that is, where $\mathrm{char}(k) \neq 2$ or $n = \dim X$ is even—is also immediate from 3.3 below, whose proof does not use 3.2.

Corollary 3.2.1.

Under the hypotheses of 3.2, let $F = \check{X} - B$, a closed subset of $\check{X}$. Then the Lefschetz pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$, and the following conditions are equivalent:

3.2.2. the given embedding $i : X \hookrightarrow \mathbb{P}^r$ is a Lefschetz embedding;

3.2.3. there exists a Lefschetz pencil of hyperplanes for the given embedding;

3.2.4. the codimension of F in $\check{\mathbb{P}}^r$ is at least two.

Proof. A line $D \subset \check{\mathbb{P}}^r$ is a Lefschetz pencil if and only if:

(3.2.5) the axis of D meets X transversely;

(3.2.6) D is not contained in $\check{X}$;

(3.2.7) D does not meet F.

Conditions (3.2.5) and (3.2.6) define open subsets of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$; since F is closed, so does (3.2.7). Hence the Lefschetz pencils form an open subset.

Because $\dim X \leq r-1$, the open subset in (3.2.6) is nonempty and therefore dense. The open subset in (3.2.5) is likewise nonempty and dense. Indeed, choose a hyperplane H meeting X transversely; such an H exists because $\dim X \leq r-1$. Since $\dim(X \cap H) \leq r-1$, there is a hyperplane G meeting $X \cap H$ transversely, and the line D through H and G then has an axis meeting X transversely. Thus the embedding is Lefschetz if and only if the open subset in (3.2.7) is nonempty, hence dense. Since F is closed, this occurs if and only if F has codimension at least two in $\check{\mathbb{P}}^r$ (cf. EGA V, or Mumford, *Introduction to Algebraic Geometry*, p. 88, Cor. 3).

Corollary 3.2.8.

Under the hypotheses of 3.2, suppose that $i : X \hookrightarrow \mathbb{P}^r$ is a Lefschetz embedding. Let H_0 be a hyperplane meeting X transversely, and let $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$ be the subvariety of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$ formed by the lines through H_0. Then the Lefschetz pencils through H_0 form a dense open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$.

Proof. By 3.2.1 these pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$. The argument just given to prove that the open subsets (3.2.5) and (3.2.6) are nonempty also shows that their intersections with this subvariety are nonempty. It remains to prove that some lines through H_0 do not meet F.

Choose a hyperplane $T \subset \check{\mathbb{P}}^r$ not containing H_0. There is an isomorphism

$$\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0} \xleftarrow{\ \sim\ } T$$

$$\begin{matrix}\text{the line through} \\ H_0 \text{ and } \tau\end{matrix} \longleftarrow\!\shortmid\ \tau$$

so $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$ "is" a projective space of dimension $r-1$. The lines in it which pass through a point of F form a closed subset of dimension at most $r-2$, being the image of the morphism

$$F \longrightarrow \mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$$

$$f \longmapsto \begin{matrix}\text{the line through } H_0 \\ \text{and through } f\end{matrix}.$$

Proposition 3.3.

The morphism $\varphi : \mathbb{P}(N) \to \check{\mathbb{P}}^r$ is unramified at a closed point (x_0, H) if and only if x_0 is a nondegenerate quadratic singularity (1.2) of $X \cap H$. In particular, the subset of $\mathbb{P}(N)$ formed by the points (x, H) for which x is a nondegenerate quadratic singularity of $X \cap H$ is open.

Proof of Proposition 3.3. We may restrict to the case $X \neq \mathbb{P}^r$. The two properties asserted to be equivalent are independent of the particular choice of coordinates, so we may describe them using coordinates chosen in a very special way. Since X is a smooth subvariety of $\mathbb{P}^r$ of dimension $n \geq 1$, choose homogeneous coordinates $X_0, \ldots, X_r$ so that:

3.3.1. the point $x_0 \in \mathbb{P}^r$ is $(1, 0, \ldots, 0)$;

3.3.2. putting $x_i = X_i/X_0$ for $i = 1, \ldots, r$, so that $\widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}$ identifies with $k[[x_1, \ldots, x_r]]$, the ideal defining the trace of X in $\operatorname{Spec} \widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}$ is generated by elements g_i $(i = 1, \ldots, r-n)$ which, in the completion, have the form

$$g_i = x_{n+i} - h_i(x_1, \ldots, x_n), \qquad i = 1, \ldots, r-n,$$

where the h_i are power series in $x_1, \ldots, x_n$ alone and lie in the square of the maximal ideal of $k[[x_1, \ldots, x_n]]$;

3.3.3. under (3.1.1), the hyperplane H comes from g_{r-n}.

Put $V = \operatorname{Spec}(\widehat{\mathcal{O}}_{X, x_0})$. It follows from 3.3.2 that there is a trivialization

$$(I/I^2)|_V \simeq \mathcal{O}_V^{r-n}$$

given by

$$(\Lambda_1, \ldots, \Lambda_{r-n}) \longmapsto \sum_{i=1}^{r-n} \Lambda_i g_i. \tag{3.3.4}$$

Together with 3.3.3 this gives a neighborhood W of (x_0, H) in $\mathbb{P}(N)|_V$ and an isomorphism $V \times \mathbb{A}^{r-n-1} \simeq W$, given by

$$\begin{aligned} V \times \mathbb{A}^{r-n-1} &\longrightarrow \mathbb{P}(I/I^2)|_V, \\ (v, (\lambda_1, \ldots, \lambda_{r-n-1})) &\longmapsto \left(v, \left[g_{r-n} + \sum_{i=1}^{r-n-1} \lambda_i g_i \right] \right), \end{aligned} \tag{3.3.5}$$

where $\lambda_i = \Lambda_i / \Lambda_{r-n}$.

In particular, the completion of the local ring of $\mathbb{P}(N)$ at (x_0, H) identifies with

$$k[[x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_{r-n-1}]].$$

We now express φ in these coordinates at the level of completions:

$$\varphi(x, \lambda) = \text{the hyperplane with equation } \sum_{i=0}^{r} \varphi_i(x, \lambda) T_i = 0. \tag{3.3.6}$$

Since this image hyperplane contains the point x, whose homogeneous coordinates are

$$x = (1, x_1, \ldots, x_n, h_1(x), \ldots, h_{r-n}(x)), \tag{3.3.7}$$

we have

$$\varphi_0(x, \lambda) = -\sum_{i=1}^{n} x_i \varphi_i(x, \lambda) - \sum_{j=1}^{r-n} h_j(x) \varphi_{n+j}(x, \lambda) \tag{3.3.8}$$

in $\widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}$. Moreover, by (3.1.1), if

$$D_i = \frac{\partial}{\partial x_i} \in \mathrm{Der}_k(\widehat{\mathcal{O}}_{\mathbb{P}^r,x_0}, \widehat{\mathcal{O}}_{\mathbb{P}^r,x_0}),$$

then for $1 \leq i \leq r$,

$$\varphi_i(x, \lambda) = D_i \left(g_{r-n} + \sum_{j=1}^{r-n-1} \lambda_j g_j \right)(x) \quad \text{in } \widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}. \tag{3.3.9}$$

Recalling that $g_i = x_{n+i} - h_i(x_1, \ldots, x_n)$, one obtains

$$\varphi_i(x, \lambda) = -\frac{\partial}{\partial x_i} \left(h_{r-n} + \sum_{j=1}^{r-n-1} \lambda_j h_j \right), \qquad 1 \leq i \leq n, \tag{3.3.10}$$

$$\varphi_{n+j}(x, \lambda) = \lambda_j, \qquad 1 \leq j \leq r-n-1, \tag{3.3.11}$$

and

$$\varphi_r(x, \lambda) = 1. \tag{3.3.12}$$

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Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}$. Since $h_i \equiv 0 \pmod{\mathfrak{m}^2}$ for $i = 1, \ldots, r-n$, we find

$$\varphi_0(x, \lambda) \equiv 0 \pmod{\mathfrak{m}^2} \tag{3.3.13}$$

[42] The print gives incompatible ranges in (3.3.11). The coordinates in (3.3.5), the coefficient formula (3.3.9), and the identity block of the printed Jacobian matrix force the indices shown here.

and

$$\varphi_i(x,\lambda) \equiv -\frac{\partial h_{r-n}}{\partial x_i} \pmod{\mathfrak{m}^2}, \qquad i = 1,\dots,n. \tag{3.3.14}$$

Putting (3.3.11)–(3.3.14) together, the Jacobian matrix of φ at (x_0, H) has the block form

$$J_{(x_0,H)}(\varphi) = \left(\begin{array}{c|c} 0_{1\times n} & 0_{1\times(r-n-1)} \\ \hline -\operatorname{Hess}(h_{r-n})(0) & 0_{n\times(r-n-1)} \\ \hline 0_{(r-n-1)\times n} & I_{r-n-1} \\ \hline 0_{1\times n} & 0_{1\times(r-n-1)} \end{array}\right). \tag{3.3.15}$$

Thus φ is unramified at (x_0, H) if and only if the Hessian matrix

$$\left(\frac{\partial^2 h_{r-n}}{\partial x_i \partial x_j}(0)\right)_{1\le i,j\le n}$$

is invertible. On the other hand, 3.3.2 and 3.3.3 show that the completion of the local ring of $X \cap H$ at x_0 is

$$k[[x_1,\dots,x_n]]/\big(h_{r-n}(x_1,\dots,x_n)\big), \tag{3.3.16}$$

and the result follows from Definition 1.2.

Example 3.4.

Suppose that X is a hypersurface in $\mathbb{P}^r$, defined by a homogeneous equation $F(X_0,\dots,X_r) = 0$. Then $\mathbb{P}(N) = X$, and in homogeneous coordinates $\varphi : X \to \check{\mathbb{P}}^r$ is given by

$$\varphi(X_0,\dots,X_r) = \left(\frac{\partial F}{\partial X_0},\dots,\frac{\partial F}{\partial X_r}\right). \tag{3.4.1}$$

This is the Gauss morphism.

Here is an example in which φ is everywhere ramified. Let p be the characteristic of k, let $r \ge 1$, and take the hypersurface X with equation

$$\sum_{i=0}^{r} X_i^{p^r+1} = 0. \tag{3.4.2}$$

Then

$$\varphi(X_0,\dots,X_r) = (X_0^{p^r},\dots,X_r^{p^r}). \tag{3.4.3}$$

Thus $\check{X}$ is reduced to a point if $p = 0$, while if $p > 0$ one has $\check{X} \simeq X$ and φ is the Frobenius morphism. It follows from 3.3 (or from 3.5.0 below) that if $p > 2$, this embedding is not a Lefschetz embedding.

We now return to the general case.

Proposition 3.5.

Suppose that the morphism $\varphi : \mathbb{P}(N) \to \check{\mathbb{P}}^r$ is generically unramified (that is, is not everywhere ramified). Then φ induces a birational morphism from $\mathbb{P}(N)$ onto $\check{X}$. More precisely, the following three open subsets of $\check{X}$ are identical:

(a) the largest open subset on which $\check{X}$ is smooth;

(b) the largest open subset $V \subset \check{X}$ such that φ induces an isomorphism $\varphi^{-1}(V) \xrightarrow{\sim} V$;

(c) the open subset (cf. 3.2) formed by those hyperplanes H for which $X \cap H$ has exactly one singular point, that point being a nondegenerate quadratic singularity.

Corollary 3.5.0.

Suppose that $\mathrm{char}(k) \neq 2$ or that $\dim X$ is even. The given embedding is a Lefschetz embedding if and only if one of the following two conditions holds:

(a) φ is generically unramified;

(b) $\dim \check{X} \leq r-2$.

Indeed, sufficiency follows from criterion (3.2.4). In case (a), Proposition 3.5 identifies F with the singular locus of $\check{X}$, so

$$\dim F < \dim \check{X} = r-1;$$

in case (b), the criterion is immediate. Conversely, suppose that D is a Lefschetz pencil and $\dim \check{X} = r-1$. There is a point $H \in D \cap \check{X}$, and 3.3 together with Definition 2.2(c) shows that φ is unramified at the points of $\varphi^{-1}(H)$. Hence it is generically unramified.

Proof of Proposition 3.5. The hypothesis implies

$$\dim \check{X} = \dim \mathbb{P}(N) = r-1,$$

and therefore 3.2 shows that $\check{X}$ is an irreducible hypersurface in $\check{\mathbb{P}}^r$.

Let R be the ramification locus of φ. By hypothesis it is a closed subset of $\mathbb{P}(N)$ of dimension at most $r-2$; hence $\varphi(R)$ is a closed subset of $\check{X}$ of dimension at most $r-2$. Let V' be the dense open subset

$$V' = (\check{X} - \varphi(R)) \cap (\check{X} - \mathrm{Sing}(\check{X})) \tag{3.5.1}$$

of $\check{X}$, and denote its complement by F:

$$F = \varphi(R) \cup \mathrm{Sing}(\check{X}). \tag{3.5.1.1}$$

We first prove that φ induces an isomorphism $\varphi^{-1}(V') \simeq V'$ by constructing an inverse $\psi : V' \to \varphi^{-1}(V')$. Regard $\varphi^{-1}(V')$ as embedded in $X \times V'$ by (3.1.1), so that φ is the second projection. Constructing ψ is then equivalent to constructing a morphism $\psi_1 : V' \to \mathbb{P}^r$ which makes the diagram

(3.5.2)
$$\begin{array}{ccccc} \mathbb{P}(N) & & \supset \varphi^{-1}(V') & \xrightarrow{\varphi} & V' \\ \downarrow & & & & \downarrow \\ & & & & \psi_1 \\ \downarrow & & & & \downarrow \\ X & \hookrightarrow & & & \mathbb{P}^r \end{array}$$

commute; one then puts $\psi(v) = (\psi_1(v), v) \in \mathbb{P}^r \times V'$.

For any hypersurface in projective space, the Gauss morphism (3.4.1) is defined at every smooth point. Apply this to the hypersurface $\check{X} \subset \check{\mathbb{P}}^r$. Since V' lies in the smooth locus V of $\check{X}$, the Gauss morphism gives

$$\psi_1 : V' \longrightarrow \check{\check{\mathbb{P}}}^r \simeq \mathbb{P}^r. \tag{3.5.3}$$

It remains to check that (3.5.2) commutes. Let (x_0, H_0) be a closed point of $\varphi^{-1}(V')$. We must show that x_0, viewed as a hyperplane in $\check{\mathbb{P}}^r$, is tangent to $\check{X}$ at H_0: in other words, tangency of H_0 to X at x_0 implies tangency of x_0 to $\check{X}$ at H_0. Since φ is unramified at (x_0, H_0) and $\check{X}$ is smooth at $H_0 = \varphi(x_0, H_0)$, its differential is an isomorphism

$$d\varphi : T_{\mathbb{P}(N),(x_0,H_0)} \xrightarrow{\sim} T_{\check{X},H_0}. \tag{3.5.4}$$

The first part of the proof is therefore completed by the following lemma.

Lemma 3.6 (Euler).

Let (x_0, H_0) be a closed point of $\mathbb{P}(N)$, and let T_L denote the tangent bundle of the hyperplane $L \subset \check{\mathbb{P}}^r$ defined by x_0. Then, inside $T_{\check{\mathbb{P}}^r, H_0}$,

$$d\varphi\big(T_{\mathbb{P}(N),(x_0,H_0)}\big) \subset T_{L,H_0}.$$

Proof. Choose homogeneous coordinates $X_0, \dots, X_r$ on $\mathbb{P}^r$ in which x_0 is the point $\underline{A} = (A_0, \dots, A_r)$. A point of T_{X,x_0} is a point of $\mathbb{P}^r$ with values in $k[\epsilon]/(\epsilon^2)$ of the form

$$\underline{A} + \epsilon \underline{B}, \tag{3.6.1}$$

such that, for every local section H of I,

$$\sum_{i=0}^{r} B_i \frac{\partial H}{\partial X_i}(\underline{A}) = 0. \tag{3.6.2}$$

Choose also a local section H of I which gives rise to H_0 by (3.1.1). Since $\mathbb{P}(N)$ is locally trivial over X, a point of $T_{\mathbb{P}(N),(x_0,H_0)}$ can be represented as

$$(\underline{A} + \epsilon \underline{B},\ H + \epsilon F), \tag{3.6.3}$$

where F is a local section of I and $\underline{A}, \underline{B}$ are as above. Its image under φ is the hyperplane over $k[\epsilon]/(\epsilon^2)$ with equation

$$\sum_{i=0}^{r} \frac{\partial (H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon \underline{B}) T_i = 0. \tag{3.6.4}$$

Thus the assertion is equivalent to

$$\sum_{i=0}^{r} A_i \frac{\partial (H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon \underline{B}) = 0. \tag{3.6.5}$$

Because $\epsilon^2 = 0$, Taylor's formula gives

$$\frac{\partial (H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon \underline{B}) = \frac{\partial H}{\partial X_i}(\underline{A}) + \epsilon \frac{\partial F}{\partial X_i}(\underline{A}) + \epsilon \sum_{j=0}^{r} B_j \frac{\partial^2 H}{\partial X_j \partial X_i}(\underline{A}). \tag{3.6.6}$$

It is therefore enough to verify

$$\sum_{i=0}^{r} A_i \frac{\partial H}{\partial X_i}(\underline{A}) = 0, \tag{3.6.7}$$

$$\sum_{i=0}^{r} A_i \frac{\partial F}{\partial X_i}(\underline{A}) = 0, \tag{3.6.8}$$

and

$$\sum_{i,j=0}^{r} A_i B_j \frac{\partial^2 H}{\partial X_i \partial X_j}(\underline{A}) = 0. \tag{3.6.9}$$

Here H and F are represented by homogeneous functions of degree zero, so Euler's formula gives

$$\sum_i X_i \frac{\partial H}{\partial X_i} = 0, \qquad \sum_i X_i \frac{\partial F}{\partial X_i} = 0. \tag{3.6.10}$$

Since $\partial H/\partial X_j$ is homogeneous of degree -1, the same formula gives

$$\sum_i X_i \frac{\partial^2 H}{\partial X_i \partial X_j} = -\frac{\partial H}{\partial X_j}.$$

Consequently, (3.6.9) becomes

$$-\sum_j B_j \frac{\partial H}{\partial X_j}(\underline{A}) = 0, \tag{3.6.11}$$

which is precisely (3.6.2).

This proves that φ induces an isomorphism

$$\varphi^{-1}(V') \xrightarrow{\sim} V', \tag{3.6.12}$$

whose inverse is ψ. Now observe that the morphism

$$\begin{cases} \psi : V' \longrightarrow \mathbb{P}^r \times V' \subset \mathbb{P}^r \times \check{X}, \\ \psi(v) = (\psi_1(v), v) \end{cases} \tag{3.6.13}$$

extends to a morphism

$$\begin{cases} \widetilde{\psi} : V \longrightarrow \mathbb{P}^r \times V, \\ \widetilde{\psi}(v) = (\widetilde{\psi}_1(v), v), \end{cases} \tag{3.6.14}$$

where

$$\widetilde{\psi}_1 : V \longrightarrow \mathbb{P}^r \tag{3.6.15}$$

is the Gauss morphism, now regarded as defined on all of V.

We have just proved

$$\psi(V') \subset \mathbb{P}(N). \tag{3.6.16}$$

Since V' is dense in V and $\mathbb{P}(N)$ is closed in $\mathbb{P}^r \times \check{X}$, this implies

$$\widetilde{\psi}(V) \subset \mathbb{P}(N), \tag{3.6.17}$$

and in fact plainly

$$\widetilde{\psi}(V) \subset \varphi^{-1}(V). \tag{3.6.18}$$

Again by continuity, the identities

$$\psi \circ \varphi = \mathrm{id}_{\varphi^{-1}(V')}, \qquad \varphi \circ \psi = \mathrm{id}_{V'}$$

extend to

$$\widetilde{\psi} \circ \varphi = \mathrm{id}_{\varphi^{-1}(V)}, \qquad \varphi \circ \widetilde{\psi} = \mathrm{id}_V .$$

Thus the open subsets in 3.5(a) and 3.5(b) are identical.

We next show that the open subsets in 3.5(a) and 3.5(c) are identical. By 3.3, the latter is the subset $U \subset \check{X}$ defined by

$$U = \left\{ H \in \check{X} \;\middle|\; \begin{array}{l} \varphi^{-1}(H) \text{ consists of exactly one point, and} \\ \varphi \text{ is unramified at that point} \end{array} \right\}. \tag{3.6.19}$$

The scheme $\varphi^{-1}(U)$ is normal, since it is smooth, and the induced morphism

$$\varphi : \varphi^{-1}(U) \longrightarrow U$$

is birational, as already proved, radicial by the definition of U, and proper. It is therefore the normalization of U. In particular, U is geometrically unibranch. Since this morphism is unramified, EGA IV, 18.10.1 shows that it is étale; EGA IV, 18.10.18 then shows that it is an open immersion. It is consequently an isomorphism, and U is smooth. This proves $3.5(c) \subset 3.5(a)$. Formula (3.6.19) makes the inclusion $3.5(b) \subset 3.5(c)$ clear, and the equality $3.5(a) = 3.5(b)$ completes the proof.

3.7.

Theorem 2.5 is true when $\mathrm{char}(k) \neq 2$, and also when X has even dimension.
Proof. In view of 3.5.0 and 3.3, it is enough to verify the following assertion:

(3.7.1) For every closed point $x_0 \in X$ and every integer $d \geq 2$, there is a hypersurface H of degree d, tangent to X at x_0, such that x_0 is an ordinary quadratic singularity of $X \cap H$.

Choose homogeneous coordinates $X_0, \ldots, X_r$ so that $x_0 = (1, 0, \ldots, 0)$ and the functions X_i/X_0, $i = 1, \ldots, n$, are local coordinates on X near x_0. One may take H to be the hypersurface with equation

$$H = X_0^{d-2} \sum_{i=1}^{n} X_i^2 \qquad \text{if } \mathrm{char}(k) \neq 2, \tag{3.7.1.1}$$

or

$$H = X_0^{d-2} \sum_{i=1}^{n/2} X_i X_{n/2+i} \qquad \text{if } n \text{ is even.} \tag{3.7.1.2}$$

When n is odd (under the first hypothesis), the equivalent standard choice recorded in the source is

$$H = X_0^{d-2} \left(X_n^2 + \sum_{i=1}^{\lfloor n/2 \rfloor} X_i X_{\lfloor n/2 \rfloor + i} \right). \tag{3.7.1.3}$$

For assertion 2.5.2, note that either φ is generically unramified, in which case 3.5.0 applies, or it is everywhere ramified. In characteristic zero the latter case implies $\dim \check{X} \leq r - 2$, so criterion (3.2.4) is then automatically satisfied.[43]

4. The general case

We shall sketch another method of proving 2.5, valid in every characteristic, assuming the following statement, which is close to 3.1 and is a special case of XV, 1.1.4.

(4.1) The subset of $\mathbb{P}(N)$ formed by the points (x, H) such that x is an ordinary quadratic singularity of $X \cap H$ is open.

Observe that 4.1 follows from 3.3, except when $\mathrm{char}(k) = 2$ and $\dim X$ is odd.

4.1.1.

Let R_1 be the closed complement in $\mathbb{P}(N)$ of the open set 4.1, and let

$$F_1 = \varphi(R_1) \subset \check{X} \subset \check{\mathbb{P}}^r$$

be its image.

Lemma 4.1.2.

If the open set 4.1 is nonempty, then $\mathrm{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2$.
Proof. By hypothesis,

$$\dim R_1 < \dim \mathbb{P}(N) = r - 1,$$

and $F_1 = \varphi(R_1)$.

[43] The print has $\dim X \leq r - 2$ here. The alternative in Corollary 3.5.0 and the criterion being invoked require $\dim \check{X} \leq r - 2$.

4.1.3.

Let F_2 be the subset of $\check{X}$ formed by the hyperplanes which are tangent to X at at least two points. Notice that F_2 is not closed, but that

$$\overline{F_2} \subset F_1 \cup F_2. \tag{4.1.4}$$

This follows from the fact that $F_1 \cup F_2 = F$ (cf. 3.2) is closed. (Inclusion (4.1.4) "says" that a confluence of two singularities is not an ordinary quadratic singularity.) In any case, the construction in 4.2.4 will show that F_2 is constructible.

To prove 2.5.1 we shall apply criterion 3.2.4. For this we need to know that

$$\operatorname{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2 \quad \text{and} \quad \operatorname{codim}_{\check{\mathbb{P}}^r}(F_2) \geq 2.$$

On the one hand, 3.7.1 and 4.1.2 show that, after replacing the given embedding by its d-th multiple, $d \geq 2$, one has $\operatorname{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2$. [44] Thus 2.5.1 will follow once we know the following proposition.

Proposition 4.2.

Let X be a smooth irreducible subvariety of $\mathbb{P}^r$, of dimension $n \geq 1$. For every integer $d \geq 2$, the subset F_2 of

$$\check{\mathbb{P}}^N = \check{\mathbb{P}}^{\binom{r+d}{d}-1}$$

formed by the hypersurfaces of degree d in $\mathbb{P}^r$ which are tangent to X at at least two points has codimension at least 2 in $\check{\mathbb{P}}^N$. [45]

4.2.1.

Begin with the special case $d = 2$ and X a linear subvariety of $\mathbb{P}^r$. It is enough to find a line in $\check{\mathbb{P}}^N$ which does not meet F (cf. 3.2.4). Choose homogeneous coordinates $X_0, \ldots, X_r$ on $\mathbb{P}^r$ so that X is defined by

$$X_{n+1} = \cdots = X_r = 0,$$

and homogeneous coordinates (λ, μ) on $\mathbb{P}^1$.

For n even, $n = 2k$, take the pencil of quadrics

$$H_{(\lambda,\mu)} = \lambda X_0^2 + (\lambda X_k + \mu X_0)X_{2k} + \sum_{i=1}^{k-1} (\lambda X_i + \mu X_{i+1})X_{k+i}. \tag{4.2.2}$$

For n odd, $n = 2k + 1$, take the pencil

$$H_{(\lambda,\mu)} = (\lambda X_1 + \mu X_0)X_0 + \sum_{i=1}^{k} (\lambda X_{2i} + \mu X_i)X_{2i+1}. \tag{4.2.3}$$

In both cases one checks that $X \cap H_{(\lambda,\mu)}$ is smooth for $(\lambda, \mu) \neq (0, 1)$, while $X \cap H_{(0,1)}$ has just one singular point, which is moreover an ordinary quadratic singularity. Thus the pencil does not meet F.

[44] The print reads "3.8.1" and F_2 in this sentence. The ordinary-quadratic construction is (3.7.1), and Lemma 4.1.2 applies to F_1; Proposition 4.2 supplies the required assertion for F_2.

[45] The print has $\binom{n+d}{d} - 1$ here and $\binom{r+d}{d} - d$ in Proposition 4.3. The degree-d Veronese embedding defined in 2.4 gives $N = \binom{r+d}{d} - 1$ in both places.

4.2.4.

Let $\mathrm{Btg}^d(X)$ be the subvariety of

$$\check{\mathbb{P}}^N \times \big((X \times X) - \Delta(X \times X)\big)$$

formed by triples (H, x_1, x_2) such that H is a hypersurface of degree d in $\mathbb{P}^r$, tangent to X at x_1 and at x_2, with $x_1 \neq x_2$. The image of $\mathrm{Btg}^d(X)$ under pr_1 is the set F_2 of Proposition 4.2, so it is enough to prove

$$\dim \mathrm{Btg}^d(X) \leq N - 2 \quad \text{if } d \geq 3, \text{ or if } d = 2 \text{ and } X \text{ is not linear.} \tag{4.2.5}$$

For this purpose, let $\mathrm{Lin}(X \times X)$ be the subvariety of $(X \times X) - \Delta(X \times X)$ given by

$$\begin{aligned} \mathrm{Lin}(X \times X) &= \big\{(x_1, x_2) \in X \times X \mid x_1 \neq x_2, \\ &\qquad\qquad \text{the line spanned by } x_1, x_2 \text{ is tangent to } X \text{ at } x_1\big\} \\ &= \big\{(x_1, x_2) \in X \times X \mid x_1 \neq x_2, \\ &\qquad\qquad x_2 \in X \cap \mathrm{Tg}_X(x_1)\big\}. \end{aligned} \tag{4.2.6}$$

Clearly:

(4.2.7) If X is not a linear subvariety of $\mathbb{P}^r$, then

$$\dim \mathrm{Lin}(X \times X) < 2 \dim X.$$

Consequently, to prove (4.2.5) it is enough to show that the fibers of the morphism

$$\mathrm{Btg}^d(X) \xrightarrow{\mathrm{pr}_{2,3}} (X \times X) - \Delta(X \times X), \tag{4.2.8}$$

regarded as subvarieties of $\check{\mathbb{P}}^N$, have

$$\begin{cases} \text{codimension at least } 2 \dim X + 1 & \text{over } \mathrm{Lin}(X \times X), \\ \text{codimension at least } 2 \dim X + 2 & \text{over } (X \times X) - \Delta(X \times X) - \mathrm{Lin}(X \times X). \end{cases} \tag{4.2.9}$$

Statement (4.2.9) follows from the next proposition by taking L_i to be the tangent space to X at x_i, for $i = 1, 2$.

Proposition 4.3.

Let x_1 and x_2 be two distinct closed points of $\mathbb{P}^r$, and let L_1 and L_2 be two linear subspaces of $\mathbb{P}^r$ such that $x_1 \in L_1$ and $x_2 \in L_2$. In the space

$$\check{\mathbb{P}}^N = \check{\mathbb{P}}^{\binom{r+d}{d}-1}$$

of hypersurfaces of degree d, the linear subspace formed by the hypersurfaces tangent to L_1 at x_1 and to L_2 at x_2 has codimension

$$\begin{cases} \dim L_1 + \dim L_2 + 2, & \text{if } d \geq 3, \text{ or if } d = 2 \text{ and } L_1 \cap L_2 \text{ does not} \\ & \text{contain the line } (x_1, x_2) \text{ through } x_1, x_2, \\ \dim L_1 + \dim L_2 + 1, & \text{if } d = 2 \text{ and } L_1 \cap L_2 \text{ contains the line } (x_1, x_2). \end{cases}$$

Proof. For $d \geq 3$, one must prove that $\dim L_1 + \dim L_2 + 2$ linear conditions are independent. It is therefore enough to treat the case $L_1 = L_2 = \mathbb{P}^r$, that is, to determine how many independent conditions make a hypersurface singular at both x_1 and x_2. Choose homogeneous coordinates so that

$$x_1 = (1, 0, \ldots, 0), \qquad x_2 = (0, 1, 0, \ldots, 0).$$

The hypersurface with equation $\sum_W A_W X^W = 0$ is singular at $(1,0,\dots,0)$ if and only if all the coefficients of the monomials $X_0^{d-1}X_i$, $i = 0,\dots,r$, vanish. It is singular at $(0,1,0,\dots,0)$ if and only if all the coefficients of the monomials $X_1^{d-1}X_i$, $i = 0,\dots,r$, vanish. For $d \geq 3$ all these monomials are distinct, which proves the assertion in this case.

For $d = 2$, the same calculation gives

$$\operatorname{codim} \geq \dim L_1 + \dim L_2 + 1,$$

because the monomial X_0X_1 is counted twice.

Suppose first that $L_1 \cap L_2$ does not contain the line (x_1, x_2). We reduce to the case $L_1 = \mathbb{P}^r$ and L_2 is a hyperplane in $\mathbb{P}^r$ which does not contain (x_1, x_2). Choose homogeneous coordinates $X_0,\dots,X_r$ so that

$$x_1 = (1,0,\dots,0), \qquad x_2 = (0,1,0,\dots,0), \qquad L_2 = (X_0 = 0).$$

A quadric $\sum_{i\leq j} A_{ij}X_iX_j$ is singular at $(1,0,\dots,0)$ if and only if $A_{0j} = 0$ for $j = 0,\dots,r$, and is tangent to $X_0 = 0$ at $(0,1,\dots,0)$ if and only if $A_{1j} = 0$ for $j = 1,\dots,r$. Together these are

$$2r + 1 = \dim L_1 + \dim L_2 + 2$$

independent conditions.

Finally, suppose that $L_1 \cap L_2$ contains the line (x_1, x_2). We reduce to the case $L_1 = L_2 = (x_1, x_2)$. In homogeneous coordinates with

$$x_1 = (1,0,\dots,0), \qquad x_2 = (0,1,0,\dots,0),$$

the quadric $\sum_{i\leq j} A_{ij}X_iX_j$ passes through $(1,0,\dots,0)$ and is tangent there to L_1 if and only if $A_{00} = A_{01} = 0$. It passes through $(0,1,0,\dots,0)$ and is tangent there to L_2 if and only if $A_{11} = A_{01} = 0$. Thus there are altogether only

$$3 = \dim L_1 + \dim L_2 + 1$$

conditions.

5. The degree of the dual variety by an "elementary" method

(Cf. XVIII, 3.2 for another method when $\dim X$ is even or $\operatorname{char}(k) \neq 2$.)

We continue to use the notation of 2.2: thus $X \hookrightarrow \mathbb{P}^r$ is a smooth irreducible subvariety of dimension n.

5.1.

We first consider the morphism (3.1.1)

$$\nu : \mathbb{P}(N) \longrightarrow X \times \check{\mathbb{P}}^r. \tag{5.1.0}$$

By definition, the conormal sheaf $\check{N}$ occurs in the exact sequence of coherent sheaves on X

$$0 \longrightarrow \check{N} \longrightarrow \Omega^1_{\mathbb{P}^r/k}|_X \longrightarrow \Omega^1_X \longrightarrow 0. \tag{5.1.1}$$

On the other hand, writing differential forms on $\mathbb{P}^r$ in homogeneous coordinates gives an exact sequence of sheaves on $\mathbb{P}^r$

$$0 \longrightarrow \Omega^1_{\mathbb{P}^r/k} \longrightarrow \mathcal{O}_{\mathbb{P}^r}(-1)^{r+1} \longrightarrow \mathcal{O}_{\mathbb{P}^r} \longrightarrow 0. \tag{5.1.2}$$

Its restriction to X is therefore the exact sequence

$$0 \longrightarrow \Omega^1_{\mathbb{P}^r/k}|_X \longrightarrow \mathcal{O}_X(-1)^{r+1} \longrightarrow \mathcal{O}_X \longrightarrow 0. \tag{5.1.3}$$

[46] Combining (5.1.1) and (5.1.3) gives an inclusion

$$\check{N} \hookrightarrow \mathcal{O}_X(-1)^{r+1}, \tag{5.1.4}$$

which, by transposition, gives a surjection

$$\mathcal{O}_X(1)^{r+1} \longrightarrow N. \tag{5.1.5}$$

Hence there is a closed immersion

$$\alpha : \mathbb{P}(N) \longrightarrow \mathbb{P}\big(\mathcal{O}_X(1)^{r+1}\big) \simeq X \times \check{\mathbb{P}}^r, \tag{5.1.6}$$

which is precisely (5.1.0).

5.2. The idea of the calculation.

From now on suppose that $\check{X} = \varphi(\mathbb{P}(N))$ has dimension $r-1$, so that it is a hypersurface in $\check{\mathbb{P}}^r$. Its degree is naturally calculated by means of Chern classes [1].

For every smooth projective variety T, let $\mathrm{CH}(T)$ denote its Chow ring, graded by codimension. For $z \in \mathrm{CH}(T)$, write

$$\int_T z$$

for the degree of its zero-dimensional component. The class in $\mathrm{CH}^1(T)$ of an invertible sheaf $\mathcal{L}$ on T will also be denoted by $\mathcal{L}$ whenever no confusion can arise (for example, under the integral sign).

Let L be the class of a line in $\mathrm{CH}(\check{\mathbb{P}}^r)$. Then

$$\deg(\check{X}) = \int_{\check{\mathbb{P}}^r} L \cdot \check{X}. \tag{5.2.1}$$

Set-theoretically, $\check{X}$ is the image of $\mathbb{P}(N)$ under φ (3.1.2). From the viewpoint of the proper pushforward

$$\varphi_* : \mathrm{CH}(\mathbb{P}(N)) \longrightarrow \mathrm{CH}(\check{\mathbb{P}}^r),$$

however, one has

$$\varphi_*[\mathbb{P}(N)] = \deg(\varphi)[\check{X}].$$

Consequently, (5.2.1) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\check{\mathbb{P}}^r} L \cdot \varphi_*[\mathbb{P}(N)]. \tag{5.2.2}$$

Here, by abuse of notation, φ in $\deg(\varphi)$ denotes the morphism $\mathbb{P}(N) \to \check{X}$ induced by (3.1.2); its degree is a nonnegative integer, taken to be zero precisely when $\dim \check{X} < r-1 = \dim \mathbb{P}(N)$. [47] The projection formula rewrites (5.2.2) as

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \varphi^*(L). \tag{5.2.3}$$

[46] The print leaves the last term as $\mathcal{O}_{\mathbb{P}^r}$ after restricting the sequence to X; the restricted term is $\mathcal{O}_X$.

[47] The print has $\dim X < r-1$. The condition for the induced map to the dual variety to be generically finite is $\dim \check{X} = r-1$.

Let $\eta \in \mathrm{CH}(\check{\mathbb{P}}^r)$ be the class of a hyperplane. Since $L = \eta^{r-1}$, equation (5.2.3) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \big(\varphi^*(\eta)\big)^{r-1}. \tag{5.2.4}$$

We now calculate $\varphi^*(\eta)$ concretely.

Let ξ (respectively τ) denote the canonical invertible sheaf $\mathcal{O}_{\mathbb{P}(N)}(1)$ on $\mathbb{P}(N)$ (respectively $\mathcal{O}_{\mathbb{P}(\mathcal{O}_X(1)^{r+1})}(1)$ on $\mathbb{P}(\mathcal{O}_X(1)^{r+1})$). Functoriality gives, by (5.1.6),

$$\alpha^*(\tau) = \xi. \tag{5.2.5}$$

Under the identification

$$\mathbb{P}\big(\mathcal{O}_X(1)^{r+1}\big) \simeq X \times \check{\mathbb{P}}^r,$$

and writing $H = \mathcal{O}_X(1)$, one plainly has

$$\tau = \mathrm{pr}_1^*(H) \otimes \mathrm{pr}_2^*(\eta). \tag{5.2.6}$$

Let $\pi : \mathbb{P}(N) \to X$ be the structural morphism. Then

$$\pi = \mathrm{pr}_1 \circ \alpha, \qquad \varphi = \mathrm{pr}_2 \circ \alpha.$$

Applying α^* to (5.2.6) and using (5.2.5), we obtain

$$\xi = \alpha^*(\tau) = \pi^*(H) \otimes \varphi^*(\eta). \tag{5.2.6$'$}$$

[48] Thus, in $\mathrm{CH}(\mathbb{P}(N))$,

$$\varphi^*(\eta) = \xi - \pi^*(H), \tag{5.2.7}$$

and (5.2.3) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \big(\xi - \pi^*(H)\big)^{r-1}. \tag{5.2.8}$$

The right-hand side of (5.2.8) is calculated by means of the following lemma (cf. 5.3.3).

Lemma 5.3.

Let X be a smooth connected quasi-projective variety of dimension n, and let M be a locally free sheaf of finite rank a on X. Let ξ denote the invertible sheaf $\mathcal{O}_{\mathbb{P}(M)}(1)$ on $\mathbb{P}(M)$, and let $\pi : \mathbb{P}(M) \to X$ be the structural morphism. Recall that $\mathrm{CH}(\mathbb{P}(M))$ is a free $\mathrm{CH}(X)$-module (via π^*), with basis $1, \xi, \dots, \xi^{a-1}$, where ξ satisfies

$$\xi^a - C_1\xi^{a-1} + C_2\xi^{a-2} - C_3\xi^{a-3} + \cdots = 0, \tag{5.3.1}$$

the C_i being the Chern classes of M in $\mathrm{CH}(X)$. Put $C(M) = 1 + C_1 + C_2 + \cdots$, the total Chern class of M; it is an invertible element of $\mathrm{CH}(X)$. Then:

(i) one has

$$\pi_*\left(\frac{\xi^{a-1}}{1+\xi}\right) = \frac{1}{C(M)}; \tag{5.3.2}$$

[48] The print numbers both this display and the preceding one (5.2.6); the prime distinguishes the second occurrence.

(ii) let $F(T) \in \mathrm{CH}(X)[T]$ be a polynomial such that the coefficient of T^i is homogeneous of degree $n+a-1-i$. Thus $F(\xi)$ has maximal degree $n+a-1 = \dim \mathbb{P}(M)$ in $\mathrm{CH}(\mathbb{P}(M))$. Then

$$\int_{\mathbb{P}(M)} F(\xi) = (-1)^{a-1} \int_X \frac{F(-1)}{C(M)}. \tag{5.3.3}$$

Proof. We first deduce (5.3.3) from (5.3.2). Write

$$\frac{1}{C(M)} = \sum_{j \geq 0} b_j, \qquad b_0 = 1, \qquad b_j \in \mathrm{CH}^j(X), \qquad b_j = 0 \text{ for } j < 0. \tag{5.3.4}$$

Then (5.3.2) is equivalent to

$$\pi_*(\xi^{a-1+j}) = (-1)^j b_j \qquad \text{whenever } a-1+j \geq 0, \tag{5.3.5}$$

since $\pi_*(\xi^q) = 0$ for $q < a-1$ for dimensional reasons.

By linearity it suffices to prove (5.3.3) when $F(T) = uT^{a-1+j}$, with $u \in \mathrm{CH}^{n-j}(X)$ homogeneous. By (5.3.5) and the projection formula,

$$\begin{aligned} \int_{\mathbb{P}(M)} F(\xi) &= \int_{\mathbb{P}(M)} \pi^*(u) \xi^{a-1+j} \\ &= \int_X u\, \pi_*(\xi^{a-1+j}) = (-1)^j \int_X u\, b_j. \end{aligned} \tag{5.3.6}$$

On the other hand,

$$\begin{aligned} (-1)^{a-1} \int_X \frac{F(-1)}{C(M)} &= (-1)^{a-1} \int_X \frac{(-1)^{a-1+j} u}{C(M)} \\ &= (-1)^j \int_X \sum_i u\, b_i. \end{aligned} \tag{5.3.7}$$

Because $\deg u = n-j$ and $\deg b_i = i$, one has

$$\int_X u\, b_i = 0 \qquad \text{for } i \neq j.$$

Thus (5.3.7) reduces to

$$(-1)^{a-1} \int_X \frac{F(-1)}{C(M)} = (-1)^j \int_X u\, b_j, \tag{5.3.8}$$

which proves the asserted implication.

It remains to prove (5.3.2), or equivalently (5.3.5).

For dimensional reasons,

$$\pi_*(\xi^j) = 0 \qquad \text{if } j < a-1,$$

whereas

$$\pi_*(\xi^{a-1}) = 1_X$$

for the geometric reason that ξ^{a-1} meets each fiber in one point. The projection formula therefore gives

$$\pi_*\left(\sum_{i=0}^{a-1} \pi^*(U_i) \xi^i \right) = U_{a-1}. \tag{5.3.9}$$

[49] By (5.3.9), assertion (5.3.5) follows from the next lemma, in which ξ is replaced by T and C_i by S_i, in view of (5.3.1).

[49] The printed display mixes the indices i, j and gives U_{j-i} on the right. The two immediately preceding push-forward identities force the coefficient U_{a-1}.

Lemma 5.4.

Let $a > 0$ be an integer and let $A = \mathbb{Z}[[X_1, \dots, X_a]]$ be the ring of formal power series in a indeterminates. Let B be the A-algebra

$$B = A[[T]] \Big/ \left(\prod_{i=1}^{a} (T - X_i) \right).$$

Thus B is a free A-module with basis $1, T, \dots, T^{a-1}$:

$$B = AT^{a-1} \oplus AT^{a-2} \oplus \cdots \oplus A. \tag{5.4.1}$$

Let $S_i \in A$ be the i-th elementary symmetric function

$$S_i = \sum_{1 \le \alpha_1 < \cdots < \alpha_i \le a} X_{\alpha_1} \cdots X_{\alpha_i},$$

so that

$$\sum_{i=0}^{a} (-1)^i S_i T^{a-i} = 0 \qquad \text{in } B. \tag{5.4.2}$$

The element $1 + S_1 + \cdots + S_a$ is invertible in A; write its inverse as

$$\frac{1}{\sum_{i=0}^{a} S_i} = \sum_{j \ge 0} b_j, \qquad b_j \text{ homogeneous of degree } j \text{ in } A, \qquad b_0 = 1. \tag{5.4.3}$$

For every integer $j \ge 0$, define a homogeneous element $a_j \in A$ of degree j by writing, using (5.4.1),

$$T^{a-1+j} = a_j T^{a-1} + ?\, T^{a-2} + ?\, T^{a-3} + \cdots + ?, \qquad ? \in A. \tag{5.4.4}$$

Then, for every $j \ge 0$,

$$a_j = (-1)^j b_j. \tag{5.4.5}$$

Proof. In view of the definition (5.4.3) of the b_j, it is enough to prove

$$\begin{cases} a_0 = 1, \\ \displaystyle\sum_{i=0}^{a} (-1)^i S_i \, a_{j-i} = 0 \quad \text{if } j > 0. \end{cases} \tag{5.4.6}$$

Clearly $a_0 = 1$. For a fixed $j > 0$, the second relation is obtained by taking the coefficient of T^{a-1}, using (5.4.1), in the relation

$$\sum_{i=0}^{a} (-1)^i S_i T^{a-1+j-i} = 0 \qquad \text{in } B. \tag{5.4.7}$$

[50] This relation is obtained by multiplying (5.4.2) by T^{j-1}, which proves the lemma.

[50] The print omits the upper summation bound a in (5.4.7); the relation is (5.4.2) multiplied by T^{j-1}.

Corollary 5.5.

One has

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \int_X \frac{(1+H)^{r-1}}{C(N)}. \tag{5.5.1}$$

Proof. Apply (5.3.3), with $M = N$ the normal sheaf of $X \hookrightarrow \mathbb{P}^r$, to (5.2.8).

For a smooth quasi-projective variety S, write

$$C(S) \stackrel{\text{def}}{=} C(T_S) \tag{5.5.2}$$

for its total Chern class, where T_S denotes the tangent bundle of S.

Dualizing the exact sequence (5.1.2) gives

$$C(\mathbb{P}^r) = (1+H)^{r+1} \qquad \text{in } \mathrm{CH}(\mathbb{P}^r). \tag{5.5.3}$$

Dualizing the exact sequence (5.1.1) gives

$$C(N)\, C(X) = C(\mathbb{P}^r)|_X \qquad \text{in } \mathrm{CH}(X). \tag{5.5.4}$$

Combining (5.5.3) and (5.5.4), formula (5.5.1) becomes:

Corollary 5.6.

One has

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \int_X \frac{C(X)}{(1+H)^2}. \tag{5.6.1}$$

5.7.

For a smooth projective variety S, define its Euler–Poincaré characteristic by

$$\chi(S) = \int_S C(S). \tag{5.7.1}$$

Proposition 5.7.2.

Let Y be a smooth hyperplane section of X, and let Δ be a smooth section of X by a codimension-two linear subspace of $\mathbb{P}^r$. Then

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \big[\chi(X) + \chi(\Delta) - 2\chi(Y)\big]. \tag{5.7.3}$$

Proof. Let $i : Y \hookrightarrow X$ be the inclusion. One has

$$N_{Y/X} = H|_Y = i^*H.$$

The exact sequence

$$0 \longrightarrow T_Y \longrightarrow i^*T_X \longrightarrow i^*H \longrightarrow 0$$

[51] The print has $(-1)^r$. Substitution of (5.5.3) and (5.5.4) into (5.5.1) does not alter its sign $(-1)^n$; formula (5.7.3) immediately below confirms the same exponent.

therefore gives

$$C(Y) = i^*\left(\frac{C(X)}{1+H}\right). \tag{5.7.4}$$

Hence

$$\chi(Y) = \int_Y i^*\left(\frac{C(X)}{1+H}\right),$$

and the projection formula gives

$$\chi(Y) = \int_X i_* i^*\left(\frac{C(X)}{1+H}\right) = \int_X \frac{H\,C(X)}{1+H}. \tag{5.7.5}$$

Iterating this calculation for Δ, one obtains

$$\begin{aligned}\chi(\Delta) &= \int_Y \frac{H\,C(Y)}{1+H} \\ &= \int_Y i^*\left(\frac{H}{1+H}\cdot\frac{C(X)}{1+H}\right) = \int_X \frac{H^2 C(X)}{(1+H)^2}.\end{aligned} \tag{5.7.6}$$

It follows that

$$\begin{aligned}\chi(X) - 2\chi(Y) + \chi(\Delta) &= \int_X C(X)\left[1 - \frac{2H}{1+H} + \frac{H^2}{(1+H)^2}\right] \\ &= \int_X \frac{C(X)}{(1+H)^2}.\end{aligned}$$

Together with (5.6.1), this proves the proposition.

6. The case of a general base

6.1.

Let S be a scheme and let $\bar{s}$ be a geometric point of S, defined by an algebraic closure of the residue field $k(s)$ for a point $s \in S$. Let $\mathbb{P}^r_S$ be projective r-space over S, let $\check{\mathbb{P}}^r_S$ be its dual projective space, and let

$$I \subset \mathbb{P}^r_S \times_S \check{\mathbb{P}}^r_S$$

be the incidence variety. Let $D \subset \check{\mathbb{P}}^r_S$ be a line, defined by an S-valued point of the appropriate Grassmannian, and let Δ be the axis of the corresponding pencil of hyperplanes (cf. 2.1).

Let X be a proper smooth S-scheme, pure of dimension $n \geq 1$, equipped with a projective embedding $X \hookrightarrow \mathbb{P}^r_S$. Put

$$\widetilde{X} = (X \times_S D) \cap I.$$

Thus $\widetilde{X}$ is the scheme over D whose fibers are the hyperplane sections of X parametrized by D; let $u : \widetilde{X} \to D$ be the projection.

We use a subscript $\bar{s}$ to denote base change by $\bar{s} \to S$. Suppose that $\Delta_{\bar{s}}$ is transverse to $X_{\bar{s}}$. Then, over a neighborhood U of s in S, Δ is transverse to X. In what follows we replace S by this neighborhood and write $U = S$. The scheme $\widetilde{X}$ is then smooth over S, and a neighborhood in $\widetilde{X}$ of the inverse image of Δ is smooth over D.

Suppose moreover that one of the hyperplane sections of $X_{\bar{s}}$ belonging to the pencil $D_{\bar{s}}$ is smooth. After again replacing S by a suitable neighborhood of s, the closed subscheme $F \subset \widetilde{X}$ defined by the Jacobian ideal $J^{n-1}(\widetilde{X}/D)$ is finite and flat over S.

6.2.

The exceptional locus of the pencil D is the relative Cartier divisor E of D/S defined by the finite flat $\mathcal{O}_D$-module $u_*\mathcal{O}_F$. Locally on D, if

$$\mathcal{O}_D^m \xrightarrow{v} \mathcal{O}_D^m \longrightarrow u_*\mathcal{O}_F \longrightarrow 0$$

is a presentation of $u_*\mathcal{O}_F$, then E has equation $\det(v) = 0$. Formation of E commutes with arbitrary base change.

The divisor $E_{\bar{s}}$ on $D_{\bar{s}}$ can be written

$$E_{\bar{s}} = \sum_P n(P)[P],$$

where $n(P)$ is the sum of the colengths of the Jacobian ideal $J^{n-1}(\widetilde{X}/D)$ at the critical points of u mapping to P.

Proposition 6.3.

Suppose that $D_{\bar{s}}$ is a Lefschetz pencil of hyperplane sections of $X_{\bar{s}}$ and that $\operatorname{char} k(s) \neq 2$ or n is even. There is a neighborhood U of s in S such that:

(a) for every geometric point $\bar{t}$ of U, $D_{\bar{t}}$ is a Lefschetz pencil of hyperplane sections of $X_{\bar{t}}$;

(b) over U, the exceptional locus E is étale over S.

To prove (a) and (b), it is enough to observe that, under these hypotheses, $D_{\bar{s}}$ satisfies condition 2.2(c) if and only if the divisor $E_{\bar{s}}$ is reduced. Invoking 3.3, one could use only the easier implication from left to right.

6.3 (exceptional case).

[52] The proposition is false in the exceptional case just mentioned. Nevertheless, if S has characteristic two and n is odd, there is a neighborhood U of s over which (a) holds, and one can define a divisor E' étale over U such that $E = 2E'$. We shall not use this result.

References

[1] A. Grothendieck, *La théorie des classes de Chern*, Bull. Soc. Math. France **86** (1958), 137–154.

[52] The print repeats the number 6.3 after Proposition 6.3.

SGA 7 — EXPOSÉ XVIII

Cohomological Study of Lefschetz Pencils

by N. Katz[53]

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Introduction

In the present exposé we give the elementary cohomological structure of Lefschetz pencils. Sections 1–4 give the "well-known" facts concerning the cohomological structure of projective bundles and blow-ups. In Section 5 we carry out the study named in the title, assuming that X satisfies the "Lefschetz vache theorem" (5.2) and that the pencil D in question satisfies a certain condition (A), local in nature on D (5.3). The principal results are 5.6, 5.7, and 5.8.7; the last plays an essential role in the proof of Griffiths's theorem in Exposé XX. In the final section, which is distinctly less elementary, we use Mme Raynaud's Exposé XIII of SGA 1 on the tame fundamental group, and the Picard–Lefschetz formula (XVI), to prove:

(1) condition (A) is often satisfied (6.3, 6.4);

(2) except when $\dim(X)$ is odd and the characteristic is 2, the monodromy group of a Lefschetz pencil acts irreducibly on the "vanishing cohomology," and transitively (up to sign) on the set of "local vanishing cycles" associated, via the Picard–Lefschetz formula, with the various "critical points" of the pencil (6.6 and 6.7).

1. The cohomology of a projective bundle

Theorem 1.1 (Deligne).

Let $f : X \longrightarrow S$ be a proper morphism, let ℓ be a prime number invertible on S, let $u \in H^2(X, \mathbb{Z}_\ell(1))$ (respectively, $u \in H^2(X, \mathbb{Q}_\ell(1))$) be a cohomology class on X, and let $\mathfrak{F}$ be a $\mathbb{Z}_\ell$-sheaf (respectively, a $\mathbb{Q}_\ell$-sheaf) on X. Suppose that there is an integer n such that, for every integer j and every integer $i > 0$, the map

$$R^{n-i}f_*\big(\mathfrak{F}(j)\big) \xrightarrow{\wedge u^i} R^{n+i}f_*\big(\mathfrak{F}(j+i)\big) \tag{1.1.1}$$

[53] Based on succinct notes of Grothendieck.

is an isomorphism of $\mathbb{Z}_\ell$-sheaves (respectively, $\mathbb{Q}_\ell$-sheaves) on S. Then, for every morphism $g : S \longrightarrow T$, the Leray spectral sequence

$$E_2^{p,q} = R^p g_* R^q f_*(\mathfrak{F}) \Longrightarrow R^{p+q}(gf)_*(\mathfrak{F})$$

degenerates.

The proof is found in [1, Prop. 2.4]. Notice that, by the proper base-change theorem (SGA 4 XII, 5.2), the hypothesis can be checked fibre by fibre.

Theorem 1.2 (cf. SGA 5 VII, 2.2.1).

Let $\mathcal{E} \longrightarrow X$ be a vector bundle, locally free of rank $1+r$. Let $f : \mathbb{P}(\mathcal{E}) \longrightarrow X$ denote the projective bundle defined by $\mathcal{E}$, and let $u \in H^2(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell(1))$ be the first Chern class of the canonical invertible sheaf $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$. For ℓ a prime number invertible on X, one has:

a) the spectral sequence

(1.2.1) $$E_2^{p,q} = H^p\big(X, R^q f_* \mathbb{Z}_\ell\big) \Longrightarrow H^{p+q}\big(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell\big)$$

degenerates;

b) the map

(1.2.2) $$\bigoplus_{0 \le j \le r} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big) \longrightarrow H^q\big(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell\big), \qquad (\tau_{q-2j})_j \longmapsto \sum_j f^*(\tau_{q-2j}) \wedge u^j$$

is an isomorphism.

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Proof. The fibres of f are projective spaces of dimension r, and one knows that

(1.2.3) $$R^i f_* \mathbb{Z}_\ell \xleftarrow[\sim]{\wedge^{i/2} u} \begin{cases} \mathbb{Z}_\ell\left(-\dfrac{i}{2}\right), & \text{if } i \text{ is even and } 0 \le i \le 2r, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 1.1 applies to $(f, \ell, u, \mathbb{Z}_\ell, r)$, which proves (a). By (1.2.3),

(1.2.4) $$E_2^{a,b} = \begin{cases} H^a\left(X, \mathbb{Z}_\ell\left(-\dfrac{b}{2}\right)\right), & \text{if } b \text{ is even and } 0 \le b \le 2r, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the graded object associated with $H^q(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell)$ for the filtration coming from the spectral sequence is

$$\bigoplus_{0 \le j \le r} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big).$$

If this direct sum is equipped with the filtration

$$F^i = \bigoplus_{\substack{0 \le j \le r \\ q-2j \le i}} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big),$$

the map (1.2.2) is compatible with the filtrations and induces the identity on the associated graded objects. This proves (b).

[54] In (1.2.2) the print has the upper bound n, the degree $q-2-j$, and one fewer closing parenthesis. The rank $1+r$, the defining map, and the proof immediately below require $0 \le j \le r$ and $q-2j$; the corrected formula is printed here.

2. The cohomology of blown-up varieties (cf. SGA 5 VII, 8)

2.1.

Let X be a smooth scheme over a field k, and let $\Delta \subset X$ be an everywhere smooth subscheme of codimension $1+r$. It is known (EGA IV, 19.4.9) that there is a smooth scheme $\widetilde{X}$, equipped with a projective morphism

$$f : \widetilde{X} \longrightarrow X, \tag{2.1.1}$$

such that, on setting

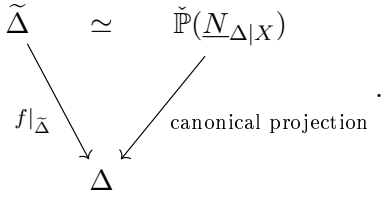
$$\widetilde{\Delta} = f^{-1}(\Delta) = \Delta \times_X \widetilde{X}, \tag{2.1.2}$$

f induces an isomorphism

$$f|_{\widetilde{X}-\widetilde{\Delta}} : \widetilde{X} - \widetilde{\Delta} \xrightarrow{\sim} X - \Delta, \tag{2.1.3}$$

and there is a Δ-isomorphism between $\widetilde{\Delta}$ and $\check{\mathbb{P}}(\underline{N}_{\Delta/X})$, the projective bundle of lines in the normal bundle $\underline{N}_{\Delta/X}$ of Δ in X. Thus one has a commutative diagram

(2.1.4)



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2.1.5.

In the special case

$$X = \mathrm{Spec}(A), \qquad \Delta = \mathrm{Spec}(A/I), \qquad I = (g_0, \ldots, g_r),$$

$\widetilde{X}$ is obtained by the Hopf construction (cf. EGA IV, 19.4.11). One defines a morphism from $X - \Delta$ to $\mathbb{P}^r$ by

$$x \in X - \Delta \longmapsto \big(g_0(x), \ldots, g_r(x)\big) \in \mathbb{P}^r$$

and regards its graph Γ as embedded in $X \times \mathbb{P}^r$. Then $\widetilde{X}$ is defined to be the schematic closure of Γ in $X \times \mathbb{P}^r$, and the morphism $f : \widetilde{X} \longrightarrow X$ is the composite

$$\widetilde{X} \hookrightarrow X \times \mathbb{P}^r \xrightarrow{\mathrm{pr}_1} X.$$

2.1.6.

Similarly (cf. again EGA IV, 19.4.11), if X is projective and Δ is the intersection of X with $1+r$ hypersurfaces, say with equations $G_i = 0$, $i = 0, \ldots, r$, one again considers the morphism

$$X - \Delta \longrightarrow \mathbb{P}^r, \qquad x \longmapsto \big(G_0(x), \ldots, G_r(x)\big),$$

and obtains $\widetilde{X}$ as the closure of its graph in $X \times \mathbb{P}^r$. Concretely, in terms of homogeneous coordinates $\lambda_0, \ldots, \lambda_r$ on $\mathbb{P}^r$, $\widetilde{X}$ is the subscheme of $X \times \mathbb{P}^r$ defined by the equations

$$\lambda_i G_j(X) - \lambda_j G_i(X), \qquad 0 \leq i < j \leq r. \tag{2.1.6.1}$$

These equations show that the point (x, λ) lies in $\widetilde{\Delta}$ if and only if $x \in \Delta$; in other words, $\check{\mathbb{P}}(\underline{N}_{\Delta/X})$ is the trivial projective bundle[55]

(2.1.6.2)
$$\begin{array}{ccc} \check{\mathbb{P}}(\underline{N}_{\Delta/X}) & \simeq & \Delta \times \mathbb{P}^r \\ & \searrow^{\text{canonical projection}} \quad \swarrow_{\mathrm{pr}_1} & \\ & \Delta & \end{array} .$$

One has the following commutative diagram:

(2.1.6.3)
$$\begin{array}{ccccccc} \Delta \times \mathbb{P}^r & \xrightarrow{\sim} & \widetilde{\Delta} & \hookrightarrow & \widetilde{X} & \hookrightarrow & X \times \mathbb{P}^r \\ & \searrow^{\mathrm{pr}_1} & \downarrow{\scriptstyle f|_{\widetilde{\Delta}}} & & \downarrow{\scriptstyle f} & \swarrow_{\mathrm{pr}_1} & \\ & & \Delta & \hookrightarrow & X & & \end{array} ,$$

and the composite inclusion $\Delta \times \mathbb{P}^r \hookrightarrow X \times \mathbb{P}^r$ is obtained from $\Delta \hookrightarrow X$ by base change.

Theorem 2.2.

Let X be smooth over a field k, let Δ be an everywhere smooth subscheme of codimension $1 + r$, and let $f : \widetilde{X} \longrightarrow X$ be the blow-up of X along Δ. Then:

(2.2.1) The Leray spectral sequence

$$E_2^{p,q} = H^p\big(X, R^q f_* \mathbb{Z}_\ell\big) \Longrightarrow H^{p+q}\big(\widetilde{X}, \mathbb{Z}_\ell\big)$$

degenerates.

(2.2.2) There is an additive isomorphism

$$\begin{aligned} H^q(\widetilde{X}, \mathbb{Z}_\ell) &\xleftarrow{\sim} H^q(X, \mathbb{Z}_\ell) \oplus \frac{H^q(\widetilde{\Delta}, \mathbb{Z}_\ell)}{\operatorname{Im} H^q(\Delta, \mathbb{Z}_\ell)} \\ &\simeq \left(\bigoplus_{j=1}^{r} H^{q-2j}(\Delta, \mathbb{Z}_\ell(-j)) \right) \oplus H^q(X, \mathbb{Z}_\ell). \end{aligned}$$

Proof. Let

(2.2.3)
$$g : \widetilde{\Delta} \longrightarrow \Delta$$

be the restriction of f to $\widetilde{\Delta}$, and let

(2.2.4)
$$\begin{cases} j : \Delta \hookrightarrow X, \\ i : \widetilde{\Delta} \hookrightarrow \widetilde{X} \end{cases}$$

be the inclusions, so that one has the commutative diagram

(2.2.5)
$$\begin{array}{ccc} \widetilde{X} & \xleftarrow{\ i\ } & \widetilde{\Delta} \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} . \\ X & \xleftarrow{\ j\ } & \Delta \end{array}$$

[55] The print labels the right-hand projection in (2.1.6.2) pr_i. Its target is Δ, and the same projection is labelled pr_1 in (2.1.6.3); the corrected label is used here.

It induces a homomorphism between the Leray spectral sequences for the morphisms f and g, with the sheaf $\mathbb{Z}_\ell$ on $\widetilde{X}$ and $\widetilde{\Delta}$:

$$E_s^{p,q}(f) \xrightarrow{(i^*,j^*)} E_s^{p,q}(g). \tag{2.2.6}$$

On the other hand, since f induces an isomorphism from $\widetilde{X}-\widetilde{\Delta}$ to $X-\Delta$, the proper base-change theorem gives

$$R^q f_*\mathbb{Z}_\ell \simeq \begin{cases} \mathbb{Z}_\ell, & q=0, \\ j_*R^q g_*\mathbb{Z}_\ell, & q>0. \end{cases} \tag{2.2.7}$$

Hence

$$E_2^{p,q}(f) \xrightarrow[\sim]{(i^*,j^*)} E_2^{p,q}(g) \qquad (q\neq 0). \tag{2.2.8}$$

We shall prove by induction on s that

$$E_s^{p,q}(f) \xrightarrow[\sim]{(i^*,j^*)} E_s^{p,q}(g) \qquad (q\neq 0,\ s\geq 2) \tag{2.2.9}$$

and

$$d_s^{p,q}(f)=0 \qquad (s\geq 2). \tag{2.2.10}$$

For the moment, admit the formula

$$d_s^{p,s-1}(f)=0 \qquad (s\geq 2,\ \text{for every } p). \tag{2.2.11}$$

Begin with $s=2$. We have a commutative diagram

$$\begin{array}{ccc} E_2^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_2^{p,q}(g) \\ \downarrow{\scriptstyle d_2^{p,q}(f)} & & \downarrow{\scriptstyle 0=d_2^{p,q}(g)} \\ E_2^{p+2,q-1}(f) & \xrightarrow{(i^*,j^*)} & E_2^{p+2,q-1}(g) \end{array} . \tag{2.2.12}$$

For $q\leq 0$, one has $E_2^{p+2,q-1}(f)=0$, hence $d_2^{p,q}(f)=0$. For $q=1$, formula (2.2.11) gives $d_2^{p,1}(f)=0$. For $q>1$, the horizontal arrows in (2.2.12) are isomorphisms, while $d_2^{p,q}(g)=0$ by 1.2 and 2.1.4. This proves (2.2.10) for $s=2$, and (2.2.9) for $s=2$ is exactly (2.2.8).

Suppose (2.2.9) and (2.2.10) are known for some $s\geq 2$. Hypothesis (2.2.10) implies $E_s^{p,q}(f)\simeq E_{s+1}^{p,q}(f)$, and by 1.2 one has $E_s^{p,q}(g)\simeq E_{s+1}^{p,q}(g)$, so there is a commutative diagram

$$\begin{array}{ccc} E_s^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_s^{p,q}(g) \\ \wr\downarrow & & \downarrow\wr \\ E_{s+1}^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_{s+1}^{p,q}(g) \end{array} , \tag{2.2.13}$$

which proves (2.2.9) for $s+1$, in view of (2.2.9) for s. To deduce (2.2.10) for $s+1$, consider the commutative diagram

$$\begin{array}{ccc} E_{s+1}^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_{s+1}^{p,q}(g) \\ \downarrow{\scriptstyle d_{s+1}^{p,q}(f)} & & \downarrow{\scriptstyle 0=d_{s+1}^{p,q}(g)} \\ E_{s+1}^{p+s+1,q-s}(f) & \xrightarrow{(i^*,j^*)} & E_{s+1}^{p+s+1,q-s}(g) \end{array} . \tag{2.2.14}$$

For $q \leq s-1$, one has $E_{s+1}^{p+s+1,q-s}(f) = 0$, hence $d_{s+1}^{p,q}(f) = 0$. For $q = s$, one has (2.2.11), and for $q > s$ the horizontal arrows in (2.2.14) are isomorphisms and $d_{s+1}^{p,q}(g) = 0$ by 1.2. This gives the desired conclusion. [56]

It remains to verify (2.2.11), which says that the composite edge homomorphism

$$E_2^{p,0}(f) \longrightarrow E_3^{p,0}(f) \longrightarrow E_4^{p,0}(f) \longrightarrow \cdots \longrightarrow E_\infty^{p,0}(f) \tag{2.2.15}$$

is an isomorphism; that is, that

$$f^* : H^p(X, \mathbb{Z}_\ell) \longrightarrow H^p(\widetilde{X}, \mathbb{Z}_\ell) \tag{2.2.16}$$

is injective. But it has a left inverse, namely the Gysin homomorphism (SGA 5 IV)

$$f_* : H^p(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^p(X, \mathbb{Z}_\ell), \tag{2.2.17}$$

which indeed satisfies

$$f_* f^* = \mathrm{id}. \tag{2.2.18}$$

This follows from the projection formula and the fact that $f_*(1) = 1$, since f is birational and therefore has degree one. This proves (2.2.1).

As for (2.2.2), passing to cohomology in diagram (2.2.5) gives a commutative diagram

$$\begin{array}{ccc} H^p(\widetilde{X}, \mathbb{Z}_\ell) & \xrightarrow{i^*} & H^p(\widetilde{\Delta}, \mathbb{Z}_\ell) \\ \uparrow{\scriptstyle f^*} & & \uparrow{\scriptstyle g^*} \\ H^p(X, \mathbb{Z}_\ell) & \xrightarrow{j^*} & H^p(\Delta, \mathbb{Z}_\ell) \end{array} , \tag{2.2.19}$$

hence a morphism

$$\frac{H^p(\widetilde{X}, \mathbb{Z}_\ell)}{f^* H^p(X, \mathbb{Z}_\ell)} \xrightarrow{(i^*, j^*)} \frac{H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}{g^* H^p(\Delta, \mathbb{Z}_\ell)}. \tag{2.2.20}$$

Let F^i denote the filtrations on $H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$ and $H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell)$ coming from the Leray spectral sequences. Then (2.2.20) can be rewritten as[57]

$$\frac{H^p(\widetilde{X}, \mathbb{Z}_\ell)}{F^p H^p(\widetilde{X}, \mathbb{Z}_\ell)} \xrightarrow{(i^*, j^*)} \frac{H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}{F^p H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}. \tag{2.2.21}$$

By (2.2.9), the map induced by (2.2.21) on associated graded objects is an isomorphism, so (2.2.20) is an isomorphism. On the other hand, (2.2.18) gives a direct-sum decomposition

$$H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \underset{f_* \oplus (1 - f^* f_*)}{\overset{f^* + \text{inclusion}}{\rightleftarrows}} H^\bullet(X, \mathbb{Z}_\ell) \oplus \operatorname{Ker} f_* \ , \tag{2.2.22}$$

and (1.2.2) gives the direct-sum decomposition

$$\frac{H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell)}{g^* H^\bullet(\Delta, \mathbb{Z}_\ell)} \xleftarrow{\sim} \bigoplus_{1 \leq j \leq r} H^{\bullet - 2j}(\Delta, \mathbb{Z}_\ell(-j)). \tag{2.2.23}$$

Combining the isomorphism (2.2.20) and decompositions (2.2.22) and (2.2.23) gives (2.2.2), as required.

[56] At this point the print writes $d_s^{p,q}(f) = 0$, although the inductive step and diagram (2.2.14) concern $d_{s+1}^{p,q}(f)$.

[57] In the right-hand denominator of (2.2.21), the print reverses the notation to $H^p F^p$. The filtration used in the preceding sentence and on the left is $F^p H^p$; that notation is restored here.

Remark 2.2.24.

Let us sketch a second, distinctly simpler proof of (2.2.2), valid when X is proper over k. More generally, without any hypothesis on X, one obtains the corresponding result (2.2.26) in cohomology with proper supports. The exact sequences of cohomology with proper supports, denoted $H_c^\bullet$, for the pairs $(\widetilde{X}, \widetilde{\Delta})$ and (X, Δ) form the commutative diagram

$$
\begin{array}{lccccccc}
(2.2.25) & \longrightarrow H_c^i(\widetilde{X}-\widetilde{\Delta},\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\widetilde{X},\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\widetilde{\Delta},\mathbb{Z}_\ell) & \longrightarrow & H_c^{i+1}(\widetilde{X}-\widetilde{\Delta},\mathbb{Z}_\ell) \\
& \wr\uparrow & & \uparrow f^* & & \uparrow g^* & & \wr\uparrow \\
(2.2.26) & \longrightarrow H_c^i(X-\Delta,\mathbb{Z}_\ell) & \longrightarrow & H_c^i(X,\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\Delta,\mathbb{Z}_\ell) & \longrightarrow & H_c^{i+1}(X-\Delta,\mathbb{Z}_\ell)
\end{array}
$$

The quotient of the acyclic complex (2.2.25) by the acyclic subcomplex (2.2.26) gives the acyclic complex

$$
(2.2.27) \qquad \longrightarrow 0 \longrightarrow \frac{H_c^i(\widetilde{X},\mathbb{Z}_\ell)}{f^*H_c^i(X,\mathbb{Z}_\ell)} \longrightarrow \frac{H_c^i(\widetilde{\Delta},\mathbb{Z}_\ell)}{g^*H_c^i(\Delta,\mathbb{Z}_\ell)} \longrightarrow 0 \longrightarrow .
$$

Thus, without considering spectral sequences, one finds that (2.2.20) is an isomorphism. The arguments just given then apply and yield (2.2.2).

Corollary 2.3.

Set

$$
b_q(X) = \dim_{\mathbb{Q}_\ell} H^q(X,\mathbb{Q}_\ell), \qquad \chi(X) = \sum_q (-1)^q b_q(X).
$$

Then[58]

$$
(2.3.1) \qquad b_q(\widetilde{X}) = b_q(X) + \sum_{j=1}^{r} b_{q-2j}(\Delta),
$$

$$
(2.3.2) \qquad \chi(\widetilde{X}) = \chi(X) + r\,\chi(\Delta).
$$

3. The blow-up associated with a pencil

3.1.

Let X be projective, smooth, and irreducible over an algebraically closed field k, of dimension n, and let $D = \{H_\tau\}_{\tau\in\mathbb{P}^1}$ be a pencil of hypersurfaces whose axis meets X transversely, in Δ. We consider the blow-up of X along Δ, which we shall now describe explicitly.[59]

In terms of equations, let (λ,μ) be homogeneous coordinates on $\mathbb{P}^1$, and let F and G be two linear forms such that $F = G = 0$ defines the axis of the pencil. The pencil is written

$$
(3.1.1) \qquad \lambda F - \mu G.
$$

By definition, the subscheme Δ is the intersection of X with the linear subvariety $G = F = 0$. By 2.1.6,

$$
(3.1.2) \qquad \widetilde{X} = \{(x,(\lambda,\mu)) \in X\times\mathbb{P}^1 \mid \lambda F - \mu G = 0\};
$$

[58] The print has $b_{q-j}(\Delta)$ in (2.3.1). The decomposition (2.2.2) shifts cohomological degree by $2j$, so the correct summand is $b_{q-2j}(\Delta)$.

[59] The print writes u for the dimension in this sentence. Section 3.2 immediately denotes the same dimension by n, which also governs the signs in (3.2.3)–(3.2.4); that notation is used here.

it is the scheme-theoretic closure, in $X \times \mathbb{P}^1$, of the graph of the map

$$X - \Delta \longrightarrow \mathbb{P}^1$$

(3.1.3)

$$x \longrightarrow (G(x), F(x)).$$

The great advantage of $\widetilde{X}$ is that it is projective (being embedded in $X \times \mathbb{P}^1$) and smooth (because Δ is smooth) over k, and admits a morphism ρ to $\mathbb{P}^1$, namely

$$\rho : \widetilde{X} \longrightarrow X \times \mathbb{P}^1 \xrightarrow{\mathrm{pr}_2} \mathbb{P}^1, \tag{3.1.4}$$

whose fibers are the hyperplane sections of X by the members of our pencil:

$$\rho^{-1}(t) = X_t \overset{\text{def}}{=} X \cdot H_t.$$

3.2. The degree of the dual variety: topological method

Suppose in addition that the pencil of 3.1 is such that

(3.2.1) X_t is smooth for every t in a nonempty open subset of $\mathbb{P}^1$,

and

(3.2.2)
if X_t is singular, its only singularities are nondegenerate quadratic singularities (XVII 1.1).

These conditions hold in particular for a Lefschetz pencil (XVII 2.21) if $\operatorname{char} k \neq 2$ or if X has even dimension.

Let η be the generic point of $\mathbb{P}^1$, let $\bar{\eta}$ be a geometric point above it, let $n = \dim X$, and let χ be the Euler–Poincaré characteristic with coefficients in $\mathbb{Q}_\ell$. The Euler–Poincaré formula XVI 2.1 applied to ρ gives

$$\chi(\widetilde{X}) = \chi(\mathbb{P}^1)\,\chi(X_{\bar{\eta}}) + (-1)^n \sum_{\substack{\text{singular points} \\ \text{of the fibers}}} 1, \tag{3.2.3}$$

which, in view of (2.2.2), can be rewritten

$$(-1)^n \left[\begin{array}{c} \text{total number of} \\ \text{singular points} \end{array} \right] = \chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}}). \tag{3.2.4}$$

Notice that the quantity $\chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}})$ is independent of the particular choice of pencil. It depends only on the embedding of X in some $\mathbb{P}^r$, and indeed only on the Chern class of $\mathcal{O}_X(1)$; see XVII 5.7.5 and 5.7.6.

3.2.5.

Let us make this explicit in a concrete case. Take $X = \mathbb{P}^2$ and let $\{H_\tau\}$ be a pencil of plane curves of degree d satisfying 3.1, (3.2.1), and (3.2.2). Then

$$\begin{aligned} &\chi(X) = 3 && \text{(by 1.2)}, \\ &\chi(X_{\bar{\eta}}) = -d(d-3) && \text{(see SGA 5 VII 7.3)}, \\ &\chi(\Delta) = d^2, \end{aligned} \tag{3.2.6}$$

whence

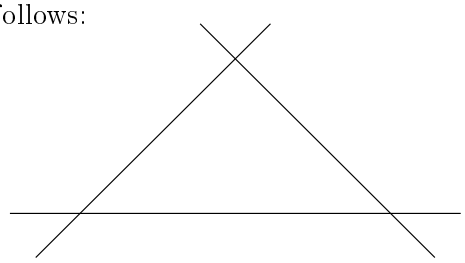
(3.2.7) the total number of singular points is $3(d-1)^2$.

Take the special case $d = 3$ and consider the family of elliptic curves (in characteristic different from 3)

(3.2.8) $$X^3 + Y^3 + Z^3 + 3\mu XYZ.$$

The four singular fibers lie over $\mu^3 = -1$ and $\mu = \infty$, respectively,[60] and each singular fiber is the union of three lines as follows:

(3.2.9)



,

It has exactly three singular points, all nondegenerate quadratic singularities. Altogether this gives twelve singular points and verifies (3.2.7) in this example.

Proposition 3.2.10.

Let $X \hookrightarrow \mathbb{P}^r$ be a Lefschetz embedding (XVII 2.3), and suppose that $n = \dim X$ is even, or that $\operatorname{char} k \neq 2$. Then every Lefschetz pencil, regarded as a line in $\check{\mathbb{P}}^r$, meets $\check{X}$ in

(3.2.11) $$(-1)^n\big[\chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}})\big]$$

points, and meets it transversely (EGA IV 17.13.7).

Proof. By hypothesis, every singularity of a fiber of the Lefschetz pencil is a nondegenerate quadratic singularity. Formula (3.2.4) therefore shows that the number of singular fibers in the pencil D—each containing exactly one singular point—is the number (3.2.11). Now the fiber X_τ is singular if and only if $\tau \in \check{X} \cap D$.

To see that the intersection is transverse, observe that (3.2.11) is independent of the particular Lefschetz pencil. Since the generic pencil is a Lefschetz pencil, (3.2.11) is the degree of $\check{X}$; that is, its degree as a hypersurface in $\check{\mathbb{P}}^r$, taken by convention to be 0 if $\dim \check{X} \leq r-2$. If a line D meets a hypersurface of degree $d > 0$ in d distinct points x_i $(1 \leq i \leq d)$, the intersection must be transverse. Indeed, the local intersection multiplicity

$$\mu_i = \operatorname{length}_{x_i}(D \cap \check{X})$$

is at least 1, while $\sum_{i=1}^{d} \mu_i = d$. Hence $\mu_i = 1$ for every i; equivalently, $D \cap \check{X}$ is smooth, which is precisely transversality.[61]

Remark 3.2.12.

For a Lefschetz pencil D already assumed to meet $\check{X}$ transversely (a condition satisfied in any case by sufficiently general D), the result 3.2.10 may also be regarded as the conjunction of XVII 5.7.2 and the fact that $\deg \varphi = 1$ contained in XVII 3.5. Of course, the fact proved here that every Lefschetz pencil D meets $\check{X}$ transversely—by means of the Euler–Poincaré formula XVI 2.1—may also be verified directly and elementarily, using the fact that the axis of D meets X transversely.

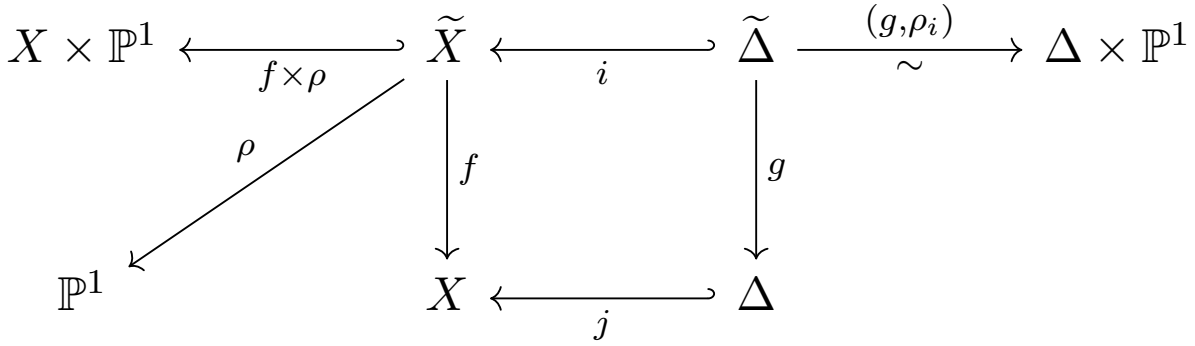
[60] The print says that the finite parameters are the cube roots of unity. With the printed plus sign in (3.2.8), the Hesse factorization gives $\mu^3 = -1$; cube roots of $+1$ correspond to the polynomial with coefficient -3μ.

[61] The print denotes the local length by rg; the standard local-length notation and the stated intersection-multiplicity argument require lg (or, equivalently, length).

4. Cohomology of the blow-up associated with a pencil

4.1.

We return to the situation of 3.1, summarized by the commutative diagram (see 2.1.6.3):
(4.1.1)



We begin by making (2.2.2) more explicit in this case.

Proposition 4.2.

Under the hypotheses of 3.1:
(4.2.1) The homomorphisms of $\mathbb{Z}_\ell$-modules

$$H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \underset{f^* + \mathrm{incl}}{\overset{f_* \oplus (1 - f^* f_*)}{\rightleftarrows}} H^\bullet(X, \mathbb{Z}_\ell) \oplus \mathrm{Ker}\Big(f_* : H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \to H^\bullet(X, \mathbb{Z}_\ell)\Big)$$

are mutually inverse isomorphisms.
(4.2.2) The homomorphisms of $\mathbb{Z}_\ell$-modules

$$H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \underset{f^* + i_* g^*}{\overset{f_* \oplus (-g_* i^*)}{\rightleftarrows}} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet - 2}(\Delta, \mathbb{Z}_\ell(-1))$$

are mutually inverse isomorphisms.[62]
(4.2.3) The algebra structure on

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet - 2}(\Delta, \mathbb{Z}_\ell(-1)),$$

transported by (4.2.2) from the cup product on $H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$, is expressed, for

$$a, b \in H^\bullet(X, \mathbb{Z}_\ell), \qquad x, y \in H^{\bullet - 2}(\Delta, \mathbb{Z}_\ell(-1)),$$

by the formulas

(4.2.3.1) $$(0 \oplus x) \wedge (0 \oplus y) = -j_*(xy) \oplus 2xy\, c_1(\mathcal{O}_\Delta(1)),$$

(4.2.3.2) $$(a \oplus 0) \wedge (b \oplus 0) = ab \oplus 0,$$

(4.2.3.3) $$(a \oplus 0) \wedge (0 \oplus y) = 0 \oplus j^*(a) y,$$

(4.2.3.4) $$(0 \oplus x) \wedge (b \oplus 0) = 0 \oplus x j^*(b).$$

[62] The print omits the minus sign before $g_* i^*$ in (4.2.2), (4.2.6), (4.2.9), and (4.2.10). But Lemma 4.3 gives $g_* i^* i_* g^* = -\mathrm{id}$, and the printed product calculation itself uses $-g_* i^*$. The sign is therefore restored consistently.

In particular, if

$$a \oplus x \in H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1))$$

and

$$b \oplus y \in H^{2n-\bullet}(X, \mathbb{Z}_\ell) \oplus H^{2n-2-\bullet}(\Delta, \mathbb{Z}_\ell(-1)),$$

then

(4.2.3.5) $$(a \oplus x) \wedge (b \oplus y) = ab - j_*(xy).$$

In other words, the total intersection matrix of $H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$ is

$$\left(\begin{array}{c|c} \begin{array}{c}\text{total intersection}\\ \text{matrix on } X\end{array} & 0 \\ \hline 0 & -\begin{array}{c}\text{total intersection}\\ \text{matrix on } \Delta\end{array} \end{array}\right).$$

Proof. We have already proved (4.2.1); see (2.2.22). For (4.2.2), first observe that, since $g : \widetilde{\Delta} \to \Delta$ is a projective-line bundle, the sequence

(4.2.4) $$0 \longrightarrow H^\bullet(\Delta, \mathbb{Z}_\ell) \xrightarrow{g^*} H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell) \xrightarrow{g_*} H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \longrightarrow 0$$

is exact, by (1.2.2). Together with (2.2.26), this gives the commutative diagram of exact sequences
(4.2.5)

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^\bullet(X, \mathbb{Z}_\ell) & \xrightarrow{f^*} & H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) & \xrightarrow{g_* i^*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle j^*} & & \downarrow{\scriptstyle i^*} & & \| & & \\ 0 & \longrightarrow & H^\bullet(\Delta, \mathbb{Z}_\ell) & \xrightarrow{g^*} & H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell) & \xrightarrow{g_*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longrightarrow & 0. \end{array}$$

In view of the splitting (4.2.1), this yields the isomorphism

(4.2.6) $$f_* \oplus (-g_* i^*) : H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{\sim} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)).$$

The transpose of (4.2.5), after the degree reversal $\bullet \mapsto 2n - \bullet$, gives the commutative diagram of exact sequences
(4.2.7)

$$\begin{array}{ccccccccc} 0 & \longleftarrow & H^\bullet(X, \mathbb{Z}_\ell) & \xleftarrow{f_*} & H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) & \xleftarrow{i_* g^*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longleftarrow & 0 \\ & & \uparrow{\scriptstyle j_*} & & \uparrow{\scriptstyle i_*} & & \| & & \\ 0 & \longleftarrow & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-2)) & \xleftarrow{g_*} & H^{\bullet-2}(\widetilde{\Delta}, \mathbb{Z}_\ell(-1)) & \xleftarrow{g_*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longleftarrow & 0 \end{array}$$

and hence the isomorphism

(4.2.8) $$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow[\sim]{f^* + i_* g^*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell).$$

It remains to prove that the composite

(4.2.9) $$\begin{aligned} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) &\xrightarrow{f^* + i_* g^*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \\ &\xrightarrow{f_* \oplus (-g_* i^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \end{aligned}$$

is the identity. Now

$$f_* f^* = \mathrm{id} \quad \text{on } H^\bullet(X, \mathbb{Z}_\ell) \quad \text{by (2.2.18)},$$

$$g_* i^* f^* = 0 \quad \text{by (4.2.5)}, \qquad f_* i_* g^* = 0 \quad \text{by (4.2.7)}.$$

It therefore suffices to prove

$$-g_* i^* i_* g^*(x) = x \quad \text{for } x \in H^\bullet(\Delta, \mathbb{Z}_\ell(-1)). \tag{4.2.10}$$

To verify (4.2.10), first observe that $i : \widetilde{\Delta} \hookrightarrow \widetilde{X}$ is the inclusion of a smooth subvariety of codimension one. Thus (SGA 5 VII)

$$i^* i_*(x) = c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\, x, \tag{4.2.11}$$

and

$$i_*(x) i_*(y) = i_*\Big(xy\, c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big). \tag{4.2.12}$$

Write $\xi_X = \underline{\mathcal{O}}_X(1)$, $\xi_\Delta = \underline{\mathcal{O}}_\Delta(1)$, and $\xi_{\mathbb{P}^1} = \underline{\mathcal{O}}_{\mathbb{P}^1}(1)$. By EGA II 8.1.7, if $\widetilde{\mathfrak{J}}$ is the ideal of $\widetilde{\Delta}$ in $\widetilde{X}$, then

$$\begin{aligned} \check{\underline{N}}_{\widetilde{\Delta}/\widetilde{X}} &\simeq \widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}^2 \simeq \widetilde{\mathfrak{J}} \otimes_{\underline{\mathcal{O}}_{\widetilde{X}}} \underline{\mathcal{O}}_{\widetilde{\Delta}} \\ &\simeq \underline{\mathcal{O}}_{\widetilde{X}}(1) \otimes_{\underline{\mathcal{O}}_{\widetilde{X}}} \underline{\mathcal{O}}_{\widetilde{\Delta}} = \underline{\mathcal{O}}_{\widetilde{\Delta}}(1). \end{aligned} \tag{4.2.13}$$

Here $\widetilde{\Delta}$ is viewed as $\mathbb{P}(\mathfrak{J}/\mathfrak{J}^2)$, where $\mathfrak{J}$ is the ideal of Δ in X. Since Δ is the intersection of X with two given hyperplanes, there is a canonical isomorphism

$$\mathfrak{J}/\mathfrak{J}^2 \simeq \underline{\mathcal{O}}_\Delta^2(-1),$$

which induces

$$\widetilde{\Delta} = \mathbb{P}(\mathfrak{J}/\mathfrak{J}^2) \simeq \mathbb{P}(\underline{\mathcal{O}}_\Delta^2(-1)) \xrightarrow[\sim]{\alpha} \mathbb{P}(\underline{\mathcal{O}}_\Delta^2) \simeq \Delta \times \mathbb{P}^1.$$

The canonical isomorphism α of EGA II 4.1.4 carries the tautological line bundle to

$$(\rho i)^* \underline{\mathcal{O}}_{\mathbb{P}^1}(1) \otimes g^* \underline{\mathcal{O}}_\Delta(-1).$$

Consequently, (4.2.13) becomes

$$\check{\underline{N}}_{\widetilde{\Delta}/\widetilde{X}} \simeq (\rho i)^* \underline{\mathcal{O}}_{\mathbb{P}^1}(1) \otimes g^* \underline{\mathcal{O}}_\Delta(-1). \tag{4.2.14}$$

Lemma 4.3.

One has

$$c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}}) = g^* c_1(\xi_\Delta) - (\rho i)^* c_1(\xi_{\mathbb{P}^1}).$$

Return now to the proof of (4.2.10). By (4.2.11) and Lemma 4.3, it is enough to prove

$$g_*(g^*(x)\,[(\rho i)^* c_1(\xi_{\mathbb{P}^1}) - g^* c_1(\xi_\Delta)]) = x.$$

By (4.2.4), $g_* g^* = 0$, while

$$\widetilde{\Delta} \xrightarrow[\sim]{g \times \rho_i} \Delta \times \mathbb{P}^1,$$

so

$$g_*(g^*(x)(\rho i)^* c_1(\xi_{\mathbb{P}^1})) = x, \tag{4.3.1}$$

and hence

$$g_*\Big(g^*(x) c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) = -x. \tag{4.3.2}$$

We now compute the algebra structure on

$$H^\bullet(X,\mathbb{Z}_\ell)\oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1))$$

transported from the cup product by (4.2.2). Let a,b belong to $H^\bullet(X,\mathbb{Z}_\ell)$ and x,y to $H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1))$. Begin with (4.2.3.1), and write

$$(0\oplus x)\wedge(0\oplus y)=A\oplus B.$$

By (4.2.12) and (4.3.2),

$$\begin{aligned}A&=f_*((i_*g^*x)(i_*g^*y))\\&=f_*i_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=j_*g_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)=-j_*(xy),\end{aligned}$$

and

$$\begin{aligned}B&=-g_*i^*((i_*g^*x)(i_*g^*y))\\&=-g_*i^*i_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=-g_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})^2\Big).\end{aligned}$$

Lemma 4.3 gives

$$c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})^2=g^*\big(c_1(\xi_\Delta)^2\big)-2g^*c_1(\xi_\Delta)(\rho i)^*c_1(\xi_{\mathbb{P}^1}),$$

because $c_1(\xi_{\mathbb{P}^1})^2=0$. Therefore, by (4.2.4) and (4.3.1),

$$B=0+2xy\,c_1(\xi_\Delta),$$

which proves (4.2.3.1). Formula (4.2.3.2) expresses the fact that f^* is an algebra homomorphism.

To prove (4.2.3.3), the projection formula shows that $\operatorname{Ker} f_*$ is stable under multiplication by $\operatorname{im} f^*$. Hence

$$(a\oplus 0)\wedge(0\oplus y)=0\oplus C,$$

where

$$C=-g_*i^*(f^*(a)i_*g^*(y))\,. \tag{4.3.3}$$

Using (4.3.2),

$$\begin{aligned}C&=-g_*\Big(i^*f^*(a)g^*(y)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=-g_*\Big(g^*j^*(a)g^*(y)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=j^*(a)y.\end{aligned}$$

This proves (4.2.3.3). Finally, (4.2.3.4) follows from (4.2.3.3) by graded commutativity of the cup product.

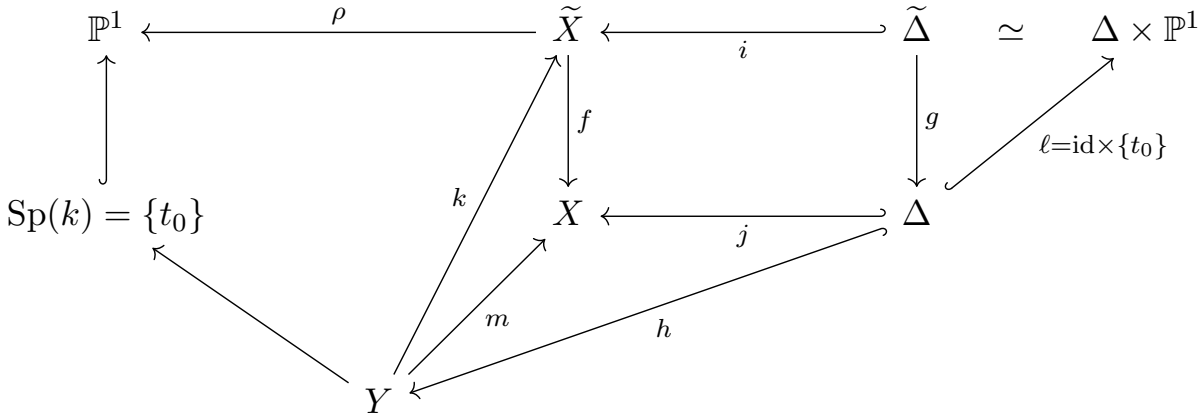
5. Cohomological study of the morphism $\rho:\widetilde{X}\longrightarrow\mathbb{P}^1$ associated with a pencil

5.1. The restriction homomorphism $H^*(\widetilde{X})\longrightarrow H^*(Y)$

Let Y denote one of the smooth fibers of $\rho:\widetilde{X}\to\mathbb{P}^1$, say

$$Y=\rho^{-1}(t_0)=X\cdot H_{t_0}.$$

We have the commutative diagram



Proposition 5.1.1.

Under the hypotheses of (3.1), with Y denoting a smooth fiber:

(5.1.2) The restriction homomorphism

$$k^* : H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^\bullet(Y, \mathbb{Z}_\ell)$$

is expressed (via (4.2.2)) by

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{m^*+h_*} H^\bullet(Y, \mathbb{Z}_\ell).$$

(5.1.3) The Gysin map

$$k_* : H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \longrightarrow H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$$

is expressed by

$$H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \xrightarrow{m_*\oplus(-h^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)).$$

Proof. The smooth subschemes $\widetilde{\Delta}$ and Y of $\widetilde{X}$ intersect transversely in Δ; in other words, the square of smooth schemes

(5.1.4)
$$\begin{array}{ccc} \Delta & \xrightarrow{\ell} & \widetilde{\Delta} \\ \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle i} \\ Y & \xrightarrow[k]{} & \widetilde{X} \end{array}$$

is Cartesian. It therefore implies the formula [SGA 5, IV]

(5.1.5) $$h_*\ell^* = k^*i_*, \qquad \ell_*h^* = i^*k_*.$$

The composite

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{f^*+i_*g^*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{k^*} H^\bullet(Y, \mathbb{Z}_\ell)$$

is the sum of $k^*f^* = m^*$ and $k^*i_*g^* = h_*\ell^*g^*$ by (5.1.5). Since $g\ell = \mathrm{id}_\Delta$, this proves (5.1.2).

The composite

$$H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \xrightarrow{k_*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{f_*\oplus(-g_*i^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1))$$

is the sum of

$$f_*k_* = m_* \qquad \text{and} \qquad -g_*i^*k_* = -g_*\ell_*h^* = -h^*,$$

which proves (5.1.3).

Corollary 5.1.6.

Under the hypotheses of 5.1.1, for $i \leq n-1$, the images of

$$m^* : H^i(X, \mathbb{Z}_\ell) \longrightarrow H^i(Y, \mathbb{Z}_\ell)$$

and of

$$k^* : H^i(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^i(Y, \mathbb{Z}_\ell)$$

are equal.

Proof. By the weak Lefschetz theorem for hyperplane sections (SGA 5, VII 7.1), for $i \leq n-1$ the map

$$H^{i-2}(X, \mathbb{Z}_\ell(-1)) \xrightarrow{j^*} H^{i-2}(\Delta, \mathbb{Z}_\ell(-1)) \tag{5.1.7}$$

is an isomorphism. Thus, by (5.1.2), the image of k^* is the sum of the image of m^* and the image of the composite

$$H^{i-2}(X, \mathbb{Z}_\ell(-1)) \xrightarrow{j^*} H^{i-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{h_*} H^i(Y, \mathbb{Z}_\ell).$$

But the projection formula gives

$$\begin{aligned} h_* j^*(x) = h_*\big(h^* m^*(x)\big) &= m^*(x) h_*(1) = \xi_Y \, m^*(x) \\ &= m^*(x \xi_X), \qquad \text{since } m^*(\xi_X) = \xi_Y. \end{aligned}$$

Consequently $\operatorname{Im} k^* \subset \operatorname{Im} m^*$, and the reverse inclusion follows directly from (5.1.2). This proves the corollary.

5.2. The hard Lefschetz theorem (LV)

5.2.1.

Let S be projective of dimension n over the algebraically closed field k, and let $S \hookrightarrow \mathbb{P}$ be a projective embedding. Denote by

$$u \in H^2(S, \mathbb{Z}_\ell(1))$$

the cohomology class of a hyperplane section. Thus u is the pullback along $S \hookrightarrow \mathbb{P}$ of the first Chern class of $\mathcal{O}_{\mathbb{P}}(1)$ in $H^2(\mathbb{P}, \mathbb{Z}_\ell(1))$. Write

$$L_S : H^\bullet(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{\bullet+2}(S, \mathbb{Q}_\ell(j+1))$$

for multiplication by u.

5.2.2.

We say that S (equipped with u) satisfies (LV), the hard Lefschetz theorem, if, for every (or for one) integer j and every integer $i > 0$, the maps

$$L_S^i : H^{n-i}(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{n+i}(S, \mathbb{Q}_\ell(i+j))$$

are isomorphisms.

5.2.2.1.

It is conjectured that this condition holds when S is smooth. It is true when k has characteristic zero [6]. The specialization theorem [SGA 4, XVI 2.2] therefore gives the same conclusion when S, together with its projective embedding, lifts to characteristic zero. In particular, it holds for a smooth complete intersection. It is also known for a smooth S of dimension at most two [2], and for an abelian variety, either directly (Lieberman [4] and [2]) or by Mumford's theorem [5] that a polarized abelian variety lifts, together with its polarization, to characteristic zero.

Recall that, for a projective variety S and $i \leq n$, the primitive part of cohomology is defined by

$$\text{(5.2.2.2)} \qquad \mathrm{Prim}^i(S, \mathbb{Q}_\ell(j)) = \mathrm{Ker}\big(L^{n-i+1} : H^i(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{2n-i+2}(S, \mathbb{Q}_\ell(j+n-i+1))\big)\,.$$

If S satisfies (LV), then for $i \leq n$ one has the decomposition

$$\text{(5.2.2.3)} \qquad H^i(S, \mathbb{Q}_\ell(j)) \xleftarrow[\sim]{1+L} \mathrm{Prim}^i(S, \mathbb{Q}_\ell(j)) \oplus H^{i-2}(S, \mathbb{Q}_\ell(j-1)).$$

Proposition 5.2.3.

Let X be smooth and projective and suppose that it satisfies (LV). Let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Then

$$\text{(5.2.3.1)} \qquad \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) = \mathrm{Ker}(\tau^* : H^n(X, \mathbb{Q}_\ell(j)) \longrightarrow H^n(Y, \mathbb{Q}_\ell(j)))\,.$$

Proof. The projection formula for $\tau : Y \to X$ gives the commutative diagram

$$\text{(5.2.3.2)} \qquad \begin{array}{ccc} H^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\qquad L \qquad} & H^{n+2}(X, \mathbb{Q}_\ell(j+1)) \\ & {\scriptstyle \tau^*} \searrow \quad \swarrow {\scriptstyle \tau_*} & \\ & H^n(Y, \mathbb{Q}_\ell(j)) & \end{array}$$

By duality and weak Lefschetz,

$$\tau_* : H^n(Y, \mathbb{Q}_\ell(j)) \longrightarrow H^{n+2}(X, \mathbb{Q}_\ell(j+1))$$

is the transpose of an isomorphism and hence is itself an isomorphism. The assertion follows.

Lemma 5.2.4.

Let X be smooth and projective, and let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Then X satisfies (LV) if and only if the following two conditions hold:

5.2.4.1. Y satisfies (LV).

5.2.4.2. The restriction of the cup product on $H^{n-1}(Y, \mathbb{Q}_\ell)$ to the subspace $\mathrm{Im}\, H^{n-1}(X, \mathbb{Q}_\ell)$ is nondegenerate.

Proof. We have

$$\text{(5.2.4.3)} \qquad \begin{cases} L_X = \tau_* \tau^*, \\ L_Y = \tau^* \tau_* \qquad \text{(by 4.2.11)}, \end{cases}$$

and consequently

$$\text{(5.2.4.4)} \qquad L_X^{i+1} = \tau_* L_Y^i \tau^*.$$

In the commutative diagram

$$\begin{array}{ccc} H^{n-i-1}(X,\mathbb{Q}_\ell(j)) & \xrightarrow{L_X^{i+1}} & H^{n+i+1}(X,\mathbb{Q}_\ell(j+i+1)) \\ \downarrow{\scriptstyle \tau^*} & & \uparrow{\scriptstyle \tau_*} \\ H^{n-1-i}(Y,\mathbb{Q}_\ell(j)) & \xrightarrow{L_Y^{i}} & H^{n-1+i}(Y,\mathbb{Q}_\ell(j+i)) \end{array} \tag{5.2.4.5}$$

τ^* is an isomorphism for $i \geq 1$ by weak Lefschetz, and τ_* is an isomorphism by duality. Hence X satisfies (LV) if and only if Y satisfies (LV) and

$$L_X : H^{n-1}(X,\mathbb{Q}_\ell(j)) \longrightarrow H^{n+1}(X,\mathbb{Q}_\ell(j+1)) \tag{5.2.4.6}$$

is an isomorphism.

It remains to identify (5.2.4.6) with 5.2.4.2. Let $a, b \in H^{n-1}(X,\mathbb{Q}_\ell(j))$. By Poincaré duality on X, (5.2.4.6) is equivalent to nondegeneracy of the bilinear form

$$\begin{aligned} H^{n-1}(X,\mathbb{Q}_\ell(j)) \times H^{n-1}(X,\mathbb{Q}_\ell(j)) &\longrightarrow H^{2n}(X,\mathbb{Q}_\ell(2j+1)), \\ (a,b) &\longmapsto a\, L_X(b). \end{aligned} \tag{5.2.4.7}$$

But by (5.2.4.3) and the projection formula,

$$aL_X(b) = a\tau_*\tau^*(b) = \tau_*\big(\tau^*(a)\tau^*(b)\big).$$

This proves the equivalence of (5.2.4.7) and 5.2.4.2.

Corollary 5.2.5.

Let X be smooth and projective of dimension n, and let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Suppose that X satisfies (LV). Then, for $q \geq n$, the map

$$\tau^* : H^q(X,\mathbb{Q}_\ell(j)) \longrightarrow H^q(Y,\mathbb{Q}_\ell(j))$$

is surjective.

Proof. By (5.2.4.3), for $i \geq 0$ one has

$$L_Y^{i+1} = \tau^* L_X^i \tau_*,$$

and hence the commutative diagram

$$\begin{array}{ccc} H^{n-i}(X,\mathbb{Q}_\ell(j+1)) & \xrightarrow{L_X^{i}} & H^{n+i}(X,\mathbb{Q}_\ell(j+i+1)) \\ \uparrow{\scriptstyle \tau_*} & & \downarrow{\scriptstyle \tau^*} \\ H^{n-i-2}(Y,\mathbb{Q}_\ell(j)) & \xrightarrow{L_Y^{i+1}} & H^{n+i}(Y,\mathbb{Q}_\ell(j+i+1)). \end{array}$$

This gives the required result.

Lemma 5.2.6.

Let $f : F \to B$ be a smooth proper morphism with B connected. Let $\mathcal{O}_F(1)$ be a relatively ample invertible sheaf on F, and equip each geometric fiber $F_{\bar{b}}$ with the ample invertible sheaf induced by $\mathcal{O}_F(1)$. If one geometric fiber satisfies (LV), then so do all the others.

Proof. Let $\xi_F \in H^2(F, \mathbb{Z}_\ell(1))$ be the first Chern class of $\mathcal{O}_F(1)$. This class and its powers $\xi_F^i \in H^{2i}(F, \mathbb{Z}_\ell(i))$ induce, by cup product, the homomorphisms

$$\xi_F^i : R^{n-i} f_* \mathbb{Q}_\ell(j) \longrightarrow R^{n+i} f_* \mathbb{Q}_\ell(j+i), \tag{5.2.6.1}$$

where n is the relative dimension of the fibers. By proper base change [SGA 4, XII 5.2], all geometric fibers satisfy (LV) if and only if all the maps (5.2.6.1) are isomorphisms. By the specialization theorem [SGA 4, XVI 2.2], the sheaves $R^i f_* \mathbb{Q}_\ell(j)$ are twisted constant sheaves. Thus the maps (5.2.6.1) are isomorphisms if and only if they are so over one given geometric point.

Corollary 5.2.7.

Let X be smooth and projective. If one smooth hyperplane section satisfies (LV), then every smooth hyperplane section does.

Proof. The universal family of smooth hyperplane sections of X satisfies the hypotheses of Lemma 5.2.6.

5.3. Condition (A)

5.3.1.

Let

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

be the "fibration" associated with a pencil whose axis meets X transversely and whose generic fiber is assumed to be smooth.

Let U be a nonempty open subset of $\mathbb{P}^1$ over which ρ is smooth, and let

$$\nu : U \hookrightarrow \mathbb{P}^1 \tag{5.3.1.1}$$

be the inclusion. Since the sheaves $R^i \rho_* \mathbb{Q}_\ell(j)$ are twisted constant over U, the sheaves

$$\nu_* \nu^* R^i \rho_* \mathbb{Q}_\ell(j)$$

are seen to be essentially independent of the choice of U; the canonical "adjunction" homomorphisms

$$R^i \rho_* \mathbb{Q}_\ell(j) \longrightarrow \nu_* \nu^* R^i \rho_* \mathbb{Q}_\ell(j) \tag{5.3.2}$$

are likewise independent of the choice of U.

5.3.3.

Observe that, for an arbitrary pencil of hyperplane sections, the weak Lefschetz theorem implies that, for $i \leq n-2$, the immersion

$$\widetilde{X} \overset{(f \times \rho)}{\hookrightarrow} X \times \mathbb{P}^1$$

gives an isomorphism of sheaves on $\mathbb{P}^1$

$$R^i \rho_* \mathbb{Q}_\ell(j) \overset{\sim}{\longleftarrow} H^i(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1}, \qquad i \leq n-2.$$

In other words, $R^i \rho_* \mathbb{Q}_\ell(j)$ is constant for $i \leq n-2$ and, a fortiori, the maps (5.3.2) are isomorphisms for $i \leq n-2$.

5.3.4.

The local theory of vanishing cycles, and in particular the Picard–Lefschetz formula [SGA 7, XV 3.4], shows that, for a Lefschetz pencil, the homomorphisms (5.3.2) are isomorphisms for $i \neq n$, and are surjective for $i = n$.

5.3.5.

We say that ρ, or the pencil from which it arises, satisfies condition (A) if all the homomorphisms (5.3.2) are isomorphisms (equivalently, if this holds for every i with $j = 0$).

We shall see below (6.3, 6.4) that condition (A) has a strong tendency to hold.

Lemma 5.4.

Suppose that one smooth hyperplane section of X (and hence every such section, by Corollary 5.2.7) satisfies (LV)—for example, suppose that X satisfies (LV), cf. Lemma 5.2.4—and suppose that ρ satisfies condition (A). Then every fiber (including every singular fiber) satisfies (LV), and the sheaves $R^i\rho_*\mathbb{Q}_\ell(j)$ are constant for $i \neq n-1$.

Proof. Let

$$L : R^i\rho_*\mathbb{Q}_\ell(j) \longrightarrow R^{i+2}\rho_*\mathbb{Q}_\ell(j+1)$$

denote multiplication by $f^*c_1(\mathcal{O}_X(1)) \in H^2(\widetilde{X}, \mathbb{Z}_\ell(1))$; fiber by fiber, this is the operation considered in (5.2.1). By hypothesis, over U we have an isomorphism

$$\nu^*(L^i) : \nu^*R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} \nu^*R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i) \tag{5.4.1}$$

for every $i > 0$ and every j. Applying the functor ν_* gives the isomorphisms

$$(\nu_*\nu^*)L^i : \nu_*\nu^*R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} \nu_*\nu^*R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i). \tag{5.4.2}$$

In view of (A), these identify with the morphisms

$$L^i : R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i). \tag{5.4.3}$$

The conclusion follows from 5.3.3.

5.4.4.

Let Y be a smooth fiber of $\rho : \widetilde{X} \to \mathbb{P}^1$. By weak Lefschetz, the restriction map

$$H^{n-1}(X, \mathbb{Q}_\ell(j)) \overset{m^*}{\hookrightarrow} H^{n-1}(Y, \mathbb{Q}_\ell(j))$$

is injective. We saw in 5.2.4 that, if X satisfies (LV), then the cup product on $H^{n-1}(Y, \mathbb{Q}_\ell(j))$ remains nondegenerate when restricted to $m^*H^{n-1}(X, \mathbb{Q}_\ell(j))$. The orthogonal complement of $H^{n-1}(X, \mathbb{Q}_\ell(j))$ in $H^{n-1}(Y, \mathbb{Q}_\ell(j))$ is therefore a complementary subspace to $H^{n-1}(X, \mathbb{Q}_\ell(j))$ in $H^{n-1}(Y, \mathbb{Q}_\ell(j))$; following the classical terminology, it is called the vanishing-cohomology space of Y.

Let $\nu : U \hookrightarrow \mathbb{P}^1$ be any nonempty open subset over which ρ is smooth. The corresponding subsheaf of $\nu_*\nu^*R^{n-1}\rho_*\mathbb{Q}_\ell(j)$, the "vanishing-cohomology sheaf" $E^{n-1}\rho_*\mathbb{Q}_\ell(j)$ on $\mathbb{P}^1$, is defined by

$$E^{n-1}\rho_*\mathbb{Q}_\ell(j) = \nu_*\left\{\begin{array}{l}\text{the orthogonal complement in } \nu^*R^{n-1}\rho_*\mathbb{Q}_\ell(j) \text{ of the} \\ \text{constant subsheaf } H^{n-1}(X, \mathbb{Q}_\ell(j))_U\end{array}\right\}. \tag{5.4.4.1}$$

It is clear that this sheaf is independent of the choice of U.

Lemma 5.5.

Suppose that X satisfies (LV) and that the maps (5.3.2) are isomorphisms for $i = n-1$—for example, suppose that ρ arises from a Lefschetz pencil, cf. 5.3.4. Then there is a direct-sum decomposition

$$R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1} \oplus E^{n-1}\rho_*\mathbb{Q}_\ell(j). \tag{5.5.1}$$

Proof. By (LV), over U there is a decomposition

$$\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_U \oplus \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j), \tag{5.5.2}$$

to which we apply ν_*. This proves the assertion.

Theorem 5.6.

Let X be a smooth, irreducible, projective variety satisfying (LV) (5.2.2). Let $\{H_\tau\}$ be a pencil of hyperplanes whose generic member and axis meet X transversely, in X_η and Δ respectively. Suppose that the projection $\rho : \widetilde{X} \to \mathbb{P}^1$ satisfies condition (A). Then: [63]

(5.6.1) the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \Longrightarrow H^{p+q}(\widetilde{X}, \mathbb{Q}_\ell)$$

[64] degenerates.

(Note.) This remains valid if, instead of assuming that X satisfies (LV), one assumes only that a smooth hyperplane section Y does.

5.6.2.

Let η denote the generic point of $\mathbb{P}^1$, $\bar{\eta}$ the geometric generic point, $X_{\bar{\eta}}$ the geometric generic fiber of ρ, and π the Galois group of $k(\bar{\eta})/k(\eta)$.

For $q \neq n-1$ there are canonical isomorphisms

$$E_2^{0,q} = H^0(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \simeq H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell), \tag{5.6.2.1}$$

$$E_2^{2,q} = H^2(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \simeq H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1)), \tag{5.6.2.2}$$

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) = 0, \qquad p \neq 0, 2. \tag{5.6.2.3}$$

For $q = n-1$ one has

$$E_2^{0,n-1} = H^0(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)^\pi \simeq H^{n-1}(X, \mathbb{Q}_\ell), \tag{5.6.2.4}$$

$$E_2^{2,n-1} = H^2(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1))^\pi \simeq H^{n-1}(X, \mathbb{Q}_\ell(-1)). \tag{5.6.2.5}$$

(The missing value $E_2^{1,n-1}$ will be found in 5.7 below.)

Proof. By (5.4.3), Theorem 1.1 applies and gives (5.6.1). The first three formulas (5.6.2.1)–(5.6.2.3) follow from the constancy of $R^q\rho_*\mathbb{Q}_\ell$ for $q \neq n-1$ (Lemma 5.4); for (5.6.2.3) one also uses, of course, the simple connectedness of $\mathbb{P}^1$.

The first isomorphism in (5.6.2.4) follows from

$$R^{n-1}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} \nu_*\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell.$$

Thus

$$H^0(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^0(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell),$$

[63] The print says "condition (5.3.2)", but (5.3.2) is the displayed family of adjunction maps; condition (A) is defined from those maps in 5.3.5.

[64] The printed abutment has H^{m+q}; the bidegree and the parallel statement in Theorem 5.6.8 require H^{p+q}.

and $\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell$ is a "twisted constant" sheaf on U, hence corresponds to the π-module $H^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell)$. The second isomorphism in (5.6.2.4) follows from degeneration together with Corollary 5.1.6.

We turn to (5.6.2.5). Let $F^\bullet$ denote the filtration on $H^\bullet(\widetilde{X},\mathbb{Q}_\ell)$ defined by the spectral sequence. By degeneration, $F^2H^{n+1}(\widetilde{X},\mathbb{Q}_\ell) = E_2^{2,n-1}$ and $\mathrm{gr}^1 H^{n+1}(\widetilde{X},\mathbb{Q}_\ell) = E_2^{1,n} = 0$ by (5.6.2.3). Consequently (cf. 5.1.0),

$$\begin{aligned} E_2^{2,n-1} &= F^1H^{n+1}(\widetilde{X},\mathbb{Q}_\ell) \\ &= \mathrm{Ker}\Big(H^{n+1}(\widetilde{X},\mathbb{Q}_\ell) \longrightarrow H^0(\mathbb{P}^1, R^{n+1}\rho_*\mathbb{Q}_\ell)\Big) \\ &\overset{(A)}{=} \mathrm{Ker}\Big(k^* : H^{n+1}(\widetilde{X},\mathbb{Q}_\ell) \longrightarrow H^{n+1}(X_{\bar\eta},\mathbb{Q}_\ell)\Big)\,. \end{aligned} \tag{5.6.3}$$

By (5.1.2), k^* may be written as

$$H^{n+1}(X,\mathbb{Q}_\ell) \oplus H^{n-1}(\Delta,\mathbb{Q}_\ell(-1)) \xrightarrow{m^*+h_*} H^{n+1}(X_{\bar\eta},\mathbb{Q}_\ell). \tag{5.6.4}$$

Using (5.2.2.3), this becomes

$$\begin{aligned} \mathrm{Prim}^{n-1}(X,\mathbb{Q}_\ell(-1)) \oplus H^{n-3}(X,\mathbb{Q}_\ell(-2)) \oplus H^{n-1}(\Delta,\mathbb{Q}_\ell(-1)) \\ \xrightarrow{m^*L+m^*L^2+h_*} H^{n+1}(X_{\bar\eta},\mathbb{Q}_\ell). \end{aligned} \tag{5.6.5}$$

[65] Now the map

$$H^{n-3}(X,\mathbb{Q}_\ell(-2)) \xrightarrow{m^*L^2} H^{n+1}(X_{\bar\eta},\mathbb{Q}_\ell)$$

is an isomorphism by (LV), since $m^*L_X^2 = L_{X_{\bar\eta}}^2 m^*$ (cf. 5.2.4 and 5.2.5). Hence the kernel of (5.6.5) identifies, by $\mathrm{pr}_1 \oplus \mathrm{pr}_3$, with

$$\mathrm{Prim}^{n-1}(X,\mathbb{Q}_\ell(-1)) \oplus H^{n-1}(\Delta,\mathbb{Q}_\ell(-1)).$$

By (LV) and weak Lefschetz,

$$H^{n-1}(\Delta,\mathbb{Q}_\ell(-1)) \underset{\sim}{\xleftarrow{L}} H^{n-3}(\Delta,\mathbb{Q}_\ell(-2)) \underset{\sim}{\longleftarrow} H^{n-3}(X,\mathbb{Q}_\ell(-2)).$$

Together with (5.2.2.3), this gives

$$E_2^{2,n-1} \simeq H^{n-1}(X,\mathbb{Q}_\ell(-1)). \tag{5.6.6}$$

The other isomorphism in (5.6.2.5) follows by twisting (5.6.2.4).

Remark 5.6.7.

The assertion (5.6.1) can be proved under substantially weaker hypotheses, as Deligne explained to me:

Theorem 5.6.8.

Let X be smooth of dimension n over an algebraically closed field k, and let

$$\rho : X \longrightarrow \mathbb{P}^1_k$$

be a projective morphism whose generic fiber is smooth. Suppose that:

[65] The printed (5.6.5) omits several closing parentheses and the plus sign before the third-summand map h_*. The three displayed source summands and (5.6.4) force the corrected three-term map shown here.

(5.6.8.1) the geometric generic fiber, equipped with the restriction of the relatively ample invertible sheaf $\mathcal{O}_X(1)$, satisfies (LV);

(5.6.8.2) the maps (5.3.2) are isomorphisms for $i \neq n$ and are surjective for $i = n$ (a condition satisfied in particular when ρ arises from a Lefschetz pencil, cf. 5.3.4). Then the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \Longrightarrow H^{p+q}(X, \mathbb{Q}_\ell)$$

degenerates.

Remark. The same theorem, with the same proof, holds with $\mathbb{P}^1_k$ replaced by any smooth, irreducible, projective curve.

Proof. For the proof we identify the sheaf $\mathbb{Q}_\ell(1)$ with $\mathbb{Q}_\ell$; this is possible because k is algebraically closed. We may plainly suppose that X is connected and that ρ is surjective, hence flat with fibers of dimension $n-1$. Let

$$\xi_X \in H^2(X, \mathbb{Q}_\ell)$$

be the first Chern class of $\mathcal{O}_X(1)$, and let

$$\nu : U \hookrightarrow \mathbb{P}^1$$

be the inclusion of an open subset over which ρ is smooth. By (5.6.8.1) and Lemma 5.2.6, the cup-product maps

$$\xi_X^i : \nu^* R^{n-1-i}\rho_*\mathbb{Q}_\ell \longrightarrow \nu^* R^{n-1+i}\rho_*\mathbb{Q}_\ell$$

are isomorphisms for $i \geq 1$. Applying ν_* and using (5.6.8.2) gives, for $i \geq 2$, the isomorphisms

$$\xi_X^i : R^{n-1-i}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} R^{n-1+i}\rho_*\mathbb{Q}_\ell, \tag{5.6.8.3}$$

and, for $i = 1$, the isomorphism

$$\xi_X : R^{n-2}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell. \tag{5.6.8.4}$$

The latter factors as

$$R^{n-2}\rho_*\mathbb{Q}_\ell \xrightarrow{\xi_X} R^n\rho_*\mathbb{Q}_\ell \xrightarrow{\text{adjunction}} \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell. \tag{5.6.8.5}$$

Consequently there is a canonical decomposition

$$R^n\rho_*\mathbb{Q}_\ell \xleftarrow[\sim]{\xi_X + \text{incl}} R^{n-2}\rho_*\mathbb{Q}_\ell \oplus \text{Local}^n\,\mathbb{Q}_\ell, \tag{5.6.8.6}$$

where

$$\text{Local}^n\,\mathbb{Q}_\ell = \text{Ker}(R^n\rho_*\mathbb{Q}_\ell \longrightarrow \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell)\,. \tag{5.6.8.7}$$

The sheaf $\text{Local}^n\,\mathbb{Q}_\ell$ is concentrated at the points of $\mathbb{P}^1$ where the fiber of ρ is not smooth. Combining this fact with (5.6.8.3), we obtain

$$\xi_X^{i+1} : H^j(\mathbb{P}^1, R^{n-2-i}\rho_*\mathbb{Q}_\ell) \xrightarrow{\sim} H^j(\mathbb{P}^1, R^{n+i}\rho_*\mathbb{Q}_\ell), \qquad i \geq 0, \quad j \geq 1. \tag{5.6.8.8}$$

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We now prove degeneration. Since $E_2^{p,q} = 0$ unless $p = 0, 1, 2$, it is enough to prove that all the differentials

$$d_2 : H^0(\mathbb{P}^1, R^i\rho_*\mathbb{Q}_\ell) \longrightarrow H^2(\mathbb{P}^1, R^{i-1}\rho_*\mathbb{Q}_\ell)$$

[66] The printed (5.6.8.8) has an odd degree shift, labels the map only ξ_X, and starts at $i = 1$. The corrected form is forced by (5.6.8.3), (5.6.8.6), the lower arrow of the next commutative square, and the later specialization $E_2^{1,n-2} \simeq E_2^{1,n}$.

vanish. For $q \leq n-1$, set

$$\mathrm{Prim}(E_2^{p,q}) = \mathrm{Ker}\Big(\xi_X^{n-q} : E_2^{p,q} \longrightarrow E_2^{p,2n-q}\Big). \tag{5.6.8.9}$$

Equations (5.6.8.3) and (5.6.8.6), in particular the injectivity of

$$\xi_X : H^p(\mathbb{P}^1, R^{n-2}\rho_*\mathbb{Q}_\ell) \longrightarrow H^p(\mathbb{P}^1, R^n\rho_*\mathbb{Q}_\ell),$$

give, for $q \leq n-1$, the decomposition

$$E_2^{p,q} \xleftarrow[\sim]{1+\xi_X+\xi_X^2+\cdots} \mathrm{Prim}(E_2^{p,q}) \oplus \mathrm{Prim}(E_2^{p,q-2}) \oplus \mathrm{Prim}(E_2^{p,q-4}) \oplus \cdots . \tag{5.6.8.10}$$

[67] To prove that

$$d_2^{p,q} = 0 \qquad (q \leq n-1),$$

consider the commutative diagram

$$\begin{array}{ccc} \mathrm{Prim}(E_2^{p,q}) & \xrightarrow{\xi_X^{n-q}=0} & E_2^{p,2n-q} \\ \downarrow{\scriptstyle d_2^{p,q}} & & \downarrow{\scriptstyle d_2^{p,2n-q}} \\ E_2^{p+2,q-1} & \xrightarrow[\sim]{\xi_X^{n-1-(q-1)}} & E_2^{p+2,2n-q-1}. \end{array}$$

The lower arrow is an isomorphism by (5.6.8.8). The diagram therefore shows that $d_2^{p,q}$ kills $\mathrm{Prim}(E_2^{p,q})$ for $q \leq n-1$, and (5.6.8.10) then shows that d_2 kills $E_2^{p,q}$ for $q \leq n-1$. By (5.6.8.3), it also kills $E_2^{p,q}$ for $q \geq n+1$. It remains only to prove that

$$d_2^{0,n} : E_2^{0,n} \longrightarrow E_2^{2,n-1}$$

is zero, or equivalently that

$$\dim E_3^{2,n-1} = \dim E_2^{2,n-1}. \tag{5.6.8.12}$$

We have already proved

$$\dim E_2^{p,q} = \dim E_3^{p,q} \qquad \text{for } (p,q) \neq (0,n), (2,n-1), \tag{5.6.8.13}$$

and for degree reasons

$$E_3^{p,q} = E_\infty^{p,q} \qquad \text{for every pair } (p,q). \tag{5.6.8.14}$$

It follows that

$$b_i(X) = \dim H^i(X, \mathbb{Q}_\ell) = \sum_{p+q=i} \dim E_3^{p,q}.$$

In view of (5.6.8.13), we therefore have

$$b_{n-1}(X) = \dim E_2^{0,n-1} + \dim E_2^{1,n-2} + \dim E_2^{2,n-3} \tag{5.6.8.15}$$

and

$$b_{n+1}(X) = \dim E_2^{0,n+1} + \dim E_2^{1,n} + \dim E_3^{2,n-1}. \tag{5.6.8.16}$$

[67] The print displays the first three primitive summands while its arrow label already contains an ellipsis; the final $\oplus\cdots$ makes the same finite decomposition explicit.

Poincaré duality gives

$$b_{n-1}(X) = b_{n+1}(X). \tag{5.6.8.17}$$

First, by (5.6.8.8),

$$\dim E_2^{1,n} = \dim E_2^{1,n-2}.$$

Thus (5.6.8.17) shows that the proof of (5.6.8.12) is complete once we know

$$\dim E_2^{2,n-3} = \dim E_2^{0,n+1}, \tag{5.6.8.18}$$

$$\dim E_2^{0,n-1} = \dim E_2^{2,n-1}. \tag{5.6.8.19}$$

These two formulas follow from (5.6.8.2), Poincaré duality on the smooth fibers of ρ, and the following lemma applied to $\mathbb{P}^1$ and the sheaves $R^{n-3}\rho_*\mathbb{Q}_\ell$ and $R^{n-1}\rho_*\mathbb{Q}_\ell$.

Lemma 5.6.9.

Let D be a smooth, connected, proper curve over an algebraically closed field k. Let $i : \eta \hookrightarrow D$ be the inclusion of the generic point, and let F be a constructible $\mathbb{Q}_\ell$-sheaf on D such that $F \xrightarrow{\sim} i_*i^*F$. Then

$$H^2(D, F) \simeq H^0(D, \check{F}(-1))^\vee,$$

where $\check{F} \simeq i_*((i^*F)^\vee)$.

Corollary 5.6.10.

(Uses Picard–Lefschetz.) Let $\rho : \widetilde{X} \to \mathbb{P}^1$ be the "fibration" arising from a Lefschetz pencil, and suppose that the smooth fibers satisfy (LV). Then

$$H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)^\pi = \mathrm{Im}\big(H^i(X, \mathbb{Q}_\ell) \longrightarrow H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)\big)\,. \tag{5.6.10.1}$$

Proof. By 5.3.4, which uses the Picard–Lefschetz formula, Theorem 5.6.8 applies and gives

$$H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)^\pi = \mathrm{Im}\Big(H^i(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)\Big)\,.$$

It remains to check that $H^i(X, \mathbb{Q}_\ell)$ and $H^i(\widetilde{X}, \mathbb{Q}_\ell)$ have the same image in $H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)$. This holds for $i \leq n-1$ by Corollary 5.1.6, and for $i \geq n$ by Corollary 5.2.5.

We now turn to the most important computation:

Theorem 5.7.

Under the hypotheses of 5.6, set

$$E^{n-2}(\Delta, \mathbb{Q}_\ell(j)) = \begin{array}{c}\text{the orthogonal complement in } H^{n-2}(\Delta, \mathbb{Q}_\ell(j)) \\ \text{of the image of } H^{n-2}(X, \mathbb{Q}_\ell(j)).\end{array}$$

(This is the "vanishing" part of the cohomology of Δ; cf. 5.4.4.) Then there is a direct-sum decomposition (cf. 4.2)

$$E_2^{1,n-1} = H^1(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \xleftarrow[f^*+i_*g^*]{\sim} \mathrm{Prim}^n(X, \mathbb{Q}_\ell) \oplus E^{n-2}(\Delta, \mathbb{Q}_\ell(-1)). \tag{5.7.1}$$

Proof. By degeneration,

$$E_2^{1,n-1} \simeq F^1H^n(\widetilde{X}, \mathbb{Q}_\ell)/F^2H^n(\widetilde{X}, \mathbb{Q}_\ell).$$

By (A),

$$F^1H^n(\widetilde{X},\mathbb{Q}_\ell)=\ker\Big(k^*:H^n(\widetilde{X},\mathbb{Q}_\ell)\longrightarrow H^n(X_{\bar\eta},\mathbb{Q}_\ell)\Big), \tag{5.7.2}$$

which, by (5.1.2), identifies with

$$\ker\Big(H^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1))\xrightarrow{m^*+h_*}H^n(X_{\bar\eta},\mathbb{Q}_\ell)\Big).$$

Under (LV), k^* can also be written as

$$\mathrm{Prim}^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(X,\mathbb{Q}_\ell(-1))\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1))\xrightarrow{m^*+m^*L+h_*}H^n(X_{\bar\eta},\mathbb{Q}_\ell), \tag{5.7.3}$$

and the middle arrow

$$H^{n-2}(X,\mathbb{Q}_\ell(-1))\xrightarrow{m^*L}H^n(X_{\bar\eta},\mathbb{Q}_\ell) \tag{5.7.4}$$

is an isomorphism (cf. 5.2.3 and (5.2.2.3)). Thus, denoting by

$$\wp:H^n(X,\mathbb{Q}_\ell)\longrightarrow\mathrm{Prim}^n(X,\mathbb{Q}_\ell) \tag{5.7.5}$$

the projection onto the primitive part, we obtain [68]

$$F^1H^n(\widetilde{X},\mathbb{Q}_\ell)\xrightarrow[\wp f_*\oplus(-g_*i^*)]{\sim}\mathrm{Prim}^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1)). \tag{5.7.6}$$

By (5.6.2.2) and degeneration,

$$F^2H^n(\widetilde{X},\mathbb{Q}_\ell)\simeq E_2^{2,n-2}\simeq H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1)). \tag{5.7.7}$$

On the other hand, the inclusion of $F^2H^n(\widetilde{X},\mathbb{Q}_\ell)=H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))$ in $H^n(\widetilde{X},\mathbb{Q}_\ell)$ is the Gysin map k_*. [69] Strictly speaking, the Gysin map is defined only as a map

$$H^{i-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))\longrightarrow H^i(X\times_k k(\bar\eta),\mathbb{Q}_\ell),$$

but, in view of the canonical isomorphism (SGA 4 XII 5.4)

$$H^i(X,\mathbb{Q}_\ell)\xrightarrow{\sim}H^i(X\otimes_k K,\mathbb{Q}_\ell)$$

for every algebraically closed field K/k, we hope the reader will allow us such abuses. By (5.1.3), the Gysin map is expressed as

$$H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))\xrightarrow{m_*\oplus(-h^*)}H^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1)). \tag{5.7.8}$$

By weak Lefschetz,

$$m_*H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))=m_*m^*H^{n-2}(X,\mathbb{Q}_\ell(-1))=L_XH^{n-2}(X,\mathbb{Q}_\ell(-1))$$

is the orthogonal complement of the primitive part. Hence we have the commutative diagram

$$\begin{array}{ccc} F^2H^n(\widetilde{X},\mathbb{Q}_\ell) & \hookrightarrow & F^1H^n(\widetilde{X},\mathbb{Q}_\ell) \\ \wr\uparrow k_* & & \wr\downarrow \wp f_*\oplus(-g_*i^*) \\ H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1)) & \xrightarrow{0\oplus h^*} & \mathrm{Prim}^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1)). \end{array} \tag{5.7.9}$$

This proves (5.7.1), as required.

[68] The printed label in (5.7.6) has 2^* after g_*. The inverse blow-up decomposition (4.2.2), used here verbatim, requires i^*.

[69] The print has F^2H^N in the first sentence and omits the Tate twist (-1) from the displayed source of the Gysin map. The surrounding identifications and the degree-two Gysin shift force the readings shown here.

Remark 5.7.10.

In 5.7, hypothesis (A) may be replaced by the weaker assumption that, for $q = n$, the maps

$$H^q(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^0(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \tag{5.7.11}$$

and

$$H^q(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell) \tag{5.7.12}$$

have the same kernel, as our proof shows. Without this condition, the conclusion can fail; cf. 5.9.

5.8. The Griffiths homomorphism

5.8.1.

Let $f : T \to S$ be a morphism of schemes. The Leray spectral sequence defines a filtration (F^i) on the groups $H^\bullet(T, \mathbb{Q}_\ell(j))$. Define

$$\mathrm{Prim}^\bullet(T/S, \mathbb{Q}_\ell(j)) = F^1H^\bullet(T, \mathbb{Q}_\ell(j)). \tag{5.8.1.1}$$

Because the Leray spectral sequence lies in the first quadrant, we have

$$E_\infty^{0,\bullet} \hookrightarrow E_2^{0,\bullet}, \tag{5.8.1.2}$$

and hence

$$\mathrm{Prim}^\bullet(T/S, \mathbb{Q}_\ell(j)) = \ker\big(H^\bullet(T, \mathbb{Q}_\ell(j)) \longrightarrow H^0(S, R^\bullet f_*\mathbb{Q}_\ell(j))\big)\,. \tag{5.8.1.3}$$

For the same reason,

$$E_\infty^{1,\bullet} \hookrightarrow E_2^{1,\bullet}. \tag{5.8.1.4}$$

The composite

$$\begin{array}{ccc} \mathrm{Prim}^i(T/S, \mathbb{Q}_\ell(j)) = F^1H^i(T, \mathbb{Q}_\ell(j)) & \longrightarrow & \mathrm{gr}^1\, H^i(T, \mathbb{Q}_\ell(j)) = E_\infty^{1,i-1} \\ & & \big\downarrow \\ & & H^1(S, R^{i-1}f_*\mathbb{Q}_\ell(j)) = E_2^{1,i-1} \end{array} \tag{5.8.1.5}$$

will be denoted [70]

$$U^i : \mathrm{Prim}^i(T/S, \mathbb{Q}_\ell(j)) \longrightarrow H^1(S, R^{i-1}f_*\mathbb{Q}_\ell(j)). \tag{5.8.1.6}$$

5.8.2.

Now let X be projective, smooth, and irreducible, and let $D = \{H_\tau\}$ be a pencil of hyperplanes whose generic member and axis meet X transversely. Let $U \xrightarrow{\nu} \mathbb{P}^1$ be a nonempty open subset over which the projection $\rho : \widetilde{X} \to \mathbb{P}^1$ is smooth, and put $\widetilde{X}|_U = \widetilde{X} \times_{\mathbb{P}^1} U$. We have the commutative diagram

$$\begin{array}{ccccc} & & \widetilde{X} & \xleftarrow{\alpha} & \widetilde{X}|_U \\ & \swarrow^{f} & \big\downarrow{\scriptstyle \rho} & & \big\downarrow{\scriptstyle \rho|_U} \\ X & & \mathbb{P}^1 & \xleftarrow{\nu} & U. \end{array} \tag{5.8.2.1}$$

[70] The print has a baseline dot in place of the degree index in (5.8.1.6). The source and target of (5.8.1.5) force $R^{i-1}f_*$.

Since $\rho|_U : \widetilde{X}|_U \to U$ is proper and smooth, the specialization theorem for cohomology (SGA 4 XVI 2.2) gives

$$\mathrm{Prim}^\bullet(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)) = \ker\Big(H^\bullet(\widetilde{X}|_U, \mathbb{Q}_\ell(j)) \longrightarrow H^\bullet(X_{\bar{\eta}}, \mathbb{Q}_\ell(j))\Big). \tag{5.8.2.2}$$

Consequently, by (5.2.3.1), we have:

Proposition 5.8.2.3.

Under the hypotheses of 5.8.2, for $x \in H^n(X, \mathbb{Q}_\ell(j))$, the following conditions are equivalent:

$$x \in \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \iff \alpha^* f^*(x) \in \mathrm{Prim}^n(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)). \tag{5.8.2.4}$$

If, in addition, ρ satisfies hypothesis (A) of 5.3, then

$$x \in \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \iff f^*(x) \in \mathrm{Prim}^n(\widetilde{X}/\mathbb{P}^1, \mathbb{Q}_\ell(j)). \tag{5.8.2.5}$$

Proof. Formula (5.8.2.4) is proved. For (5.8.2.5), hypothesis (A) implies that

$$H^0(\mathbb{P}^1, R^n\rho_*\mathbb{Q}_\ell(j)) \xrightarrow[\nu^*]{\sim} H^0(U, \nu^* R^n\rho_*\mathbb{Q}_\ell(j)).$$

Thus $f^*(x)$ is primitive relative to $\mathbb{P}^1$ if and only if $\alpha^* f^*(x)$ is primitive relative to U. [71]

Definition 5.8.3.

In the situation of 5.8.2, the composite homomorphism [72]

(5.8.3.1)
$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\alpha^* f^*} & \mathrm{Prim}^n(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)) \\ & \searrow^{\mathrm{Grif}} & \downarrow{\scriptstyle U^n} \\ & & H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \end{array}$$

is called the Griffiths homomorphism (for X, relative to the given pencil D).

Adjoin to diagram (5.8.2.1) the constant family over U with fiber X:

(5.8.3.2)
$$\begin{array}{ccccc} \widetilde{X} & \xleftarrow{\alpha} & \widetilde{X}|_U & \xrightarrow{f\times\rho|_U} & X \times U \\ & {\scriptstyle f}\swarrow & \downarrow{\scriptstyle \rho|_U} & \swarrow{\scriptstyle \mathrm{pr}_2} & \\ X & & & & \\ & & \mathbb{P}^1 & \xleftarrow{\nu} & U \end{array} .$$

[71] The last printed implication has a stray α after $\mathbb{P}^1$ and substitutes α for the class x in $\alpha^* f^*(x)$. The displayed equivalence and the preceding restriction isomorphism determine the corrected sentence.

[72] The ink beside the vertical arrow is not independently legible in the scan. The map is nevertheless determined uniquely: (5.8.1.6) has just defined U^i with exactly this source and target, so the degree-n label is U^n.

Theorem 5.8.4.

Suppose that X satisfies (LV) (5.2.2), and that ρ satisfies hypothesis (A) (5.3). Then, for every nonempty smoothness open $U \xrightarrow{\nu} \mathbb{P}^1$ of ρ, the Griffiths homomorphism "modulo the fixed part", which is by definition the composite

(5.8.4.1)

$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\qquad\qquad\text{Grif}\qquad\qquad} & H^1(U, \nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(U, \nu^* E^{n-1}\rho_* \mathbb{Q}_\ell(j)) \simeq \dfrac{H^1(U, \nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j))}{(f \times \rho|_U)^* H^1(U, H^{n-1}(X, \mathbb{Q}_\ell(j))_U)} \end{array}$$

is injective. [73]

Proof. By (5.5.1), over $\mathbb{P}^1$ we have

$$R^{n-1}\rho_* \mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1} \oplus E^{n-1}\rho_* \mathbb{Q}_\ell(j), \tag{5.8.4.2}$$

and over U we have

$$\nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_U \oplus \nu^* E^{n-1}\rho_* \mathbb{Q}_\ell(j). \tag{5.8.4.3}$$

This permits us to rewrite (5.8.4.1) as indicated:

$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\text{Grif}} & H^1(U, \nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(U, \nu^* E^{n-1}\rho_* \mathbb{Q}_\ell(j)). \end{array} \tag{5.8.4.4}$$

By (5.7.1), the map (defined only by virtue of hypothesis (A))

$$u : \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \longrightarrow H^1(\mathbb{P}^1, R^{n-1}\rho_* \mathbb{Q}_\ell(j)) \tag{5.8.4.5}$$

is injective. Since $\mathbb{P}^1$ is simply connected, decomposition (5.8.4.2) gives [74]

$$H^1(\mathbb{P}^1, R^{n-1}\rho_* \mathbb{Q}_\ell(j)) \simeq H^1(\mathbb{P}^1, E^{n-1}\rho_* \mathbb{Q}_\ell(j)). \tag{5.8.4.6}$$

We can therefore complete (5.8.4.4) to a commutative square:

$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\qquad\text{Grif}\qquad} & H^1(U, \nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j)) \\ \downarrow{\scriptstyle u} & & \downarrow \\ H^1(\mathbb{P}^1, E^{n-1}\rho_* \mathbb{Q}_\ell(j)) & \xrightarrow{\qquad\nu^*\qquad} & H^1(U, \nu^* E^{n-1}\rho_* \mathbb{Q}_\ell(j)). \end{array} \tag{5.8.4.7}$$

It remains only to see that the lower arrow is injective. This follows from

$$E^{n-1}\rho_* \mathbb{Q}_\ell(j) \simeq \nu_* \nu^* E^{n-1}\rho_* \mathbb{Q}_\ell(j) \tag{5.8.4.8}$$

and the following well-known lemma.

[73] The print omits ν^* from the lower-left sheaf in (5.8.4.1) and leaves its coefficient parenthesis unclosed. Formula (5.8.4.3) and the same lower node in (5.8.4.4) give the corrected reading.

[74] The print omits the twist (j) from the targets in (5.8.4.5) and from the left-hand side of (5.8.4.6), although the domain and the decomposition (5.8.4.2) retain it.

Lemma 5.8.5.

Let $\nu : A \to B$ be a morphism of topoi, and let F be a sheaf on A. Then the canonical map

$$H^1(B, \nu_* F) \longrightarrow H^1(A, F) \tag{5.8.5.1}$$

is injective.

Proof. In a first-quadrant spectral sequence,

$$E_2^{1,0} \simeq E_\infty^{1,0}. \tag{5.8.5.2}$$

Thus, for the Leray spectral sequence

$$E_2^{p,q} = H^p(B, R^q \nu_* F) \Longrightarrow H^{p+q}(A, F),$$

we have

$$E_2^{1,0} = H^1(B, \nu_* F) \simeq E_\infty^{1,0} = F^1 H^1(A, F). \tag{5.8.5.3}$$

Taking $F = \nu^* E$, where E is a sheaf on B, gives

$$H^1(B, \nu_* \nu^* E) \hookrightarrow H^1(A, \nu^* E). \tag{5.8.5.4}$$

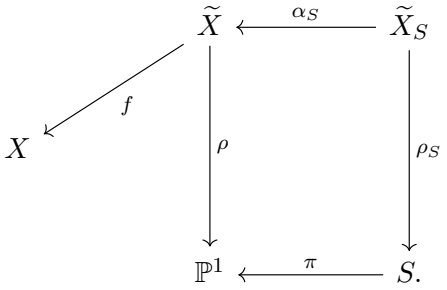
5.8.6.

Let $L/k(\eta)$ be a finite extension of the rational function field of $\mathbb{P}^1$, and put

$$S = \text{the normalization of } \mathbb{P}^1 \text{ in } L. \tag{5.8.6.1}$$

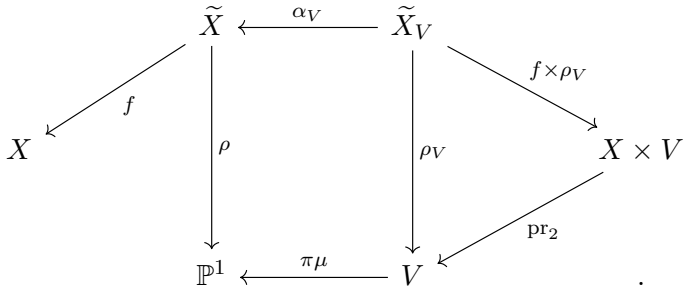
Make the base change $S \to \mathbb{P}^1$ in situation (5.8.3.2):

(5.8.6.1 bis)



For every open immersion $V \overset{\mu}{\hookrightarrow} S$ over which ρ_S is smooth, base change by $V \xrightarrow{\pi\mu} \mathbb{P}^1$ in (5.8.3.2) gives

(5.8.6.2)



We still have (cf. (5.8.2.3))

$$\alpha_V^* f^* \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \subset \operatorname{Prim}^n(\widetilde{X}_V / V, \mathbb{Q}_\ell(j)), \tag{5.8.6.3}$$

so that we may again speak of the Griffiths homomorphism

$$\operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \xrightarrow{\text{Grif}} H^1(V, R^{n-1} \rho_{V*} \mathbb{Q}_\ell(j)). \tag{5.8.6.4}$$

Corollary 5.8.7.

With the preceding notation and under the hypotheses of 5.8.4, the Griffiths homomorphism "modulo the fixed part," that is, the composite

(5.8.7.1)

$$\begin{array}{ccc} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) & \xrightarrow{\quad\mathrm{Grif}\quad} & H^1(V,R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(V,(\pi\mu)^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \simeq \dfrac{H^1(V,R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j))}{(f\times\rho_V)^*H^1(V,H^{n-1}(X,\mathbb{Q}_\ell(j))_V)} \end{array}$$

is injective.

Proof. By the specialization theorem for cohomology,

$$R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j) = (\pi\mu)^*R^{n-1}\rho_*\mathbb{Q}_\ell(j).$$

Thus decomposition (5.8.4.2) rewrites (5.8.7.1) as indicated:

$$\begin{array}{ccc} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) & \xrightarrow{\mathrm{Grif}} & H^1(V,(\pi\mu)^*R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(V,(\pi\mu)^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{array}$$

It is therefore enough to prove the injectivity of

$$\mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) \longrightarrow H^1(V,\mu^*\pi^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \tag{5.8.7.3}$$

for arbitrarily small open immersions $V \overset{\mu}{\hookrightarrow} S$. We are reduced to the case

$$\begin{cases} U \overset{\nu}{\hookrightarrow} \mathbb{P}^1 & \text{is an open subset over which } \rho \text{ is smooth,} \\ V \xrightarrow{\pi_U} U & \text{is the normalization of } U \text{ in } L. \end{cases}$$

From the cartesian square

$$\begin{array}{ccc} V & \xhookrightarrow{\quad\mu\quad} & S \\ {\scriptstyle\pi_U}\downarrow & & \downarrow{\scriptstyle\pi} \\ U & \xhookrightarrow{\quad\nu\quad} & \mathbb{P}^1 \end{array}$$

we obtain

$$\mu^*\pi^*E^{n-1}\rho_*\mathbb{Q}_\ell(j) = \pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j).$$

Consequently, (5.8.7.3) factors as

$$\begin{array}{ccc} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) & \longrightarrow & H^1(U,\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \\ & & \downarrow{\scriptstyle\pi_U^*} \\ & & H^1(V,\pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{array}$$

By (5.8.4), it is enough to verify that

$$\pi_U^* : H^1(U,\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \longrightarrow H^1(V,\pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j))$$

is injective. Since $\pi_U : V \to U$ is finite and flat, trace theory [SGA 4 XVII 6.2] supplies, for every $\mathbb{Q}_\ell$-sheaf F on U (for example $F = \nu^* E^{n-1} \rho_* \mathbb{Q}_\ell(j)$), a map

$$\mathrm{tr}_{\pi_U} : H^1(V, \pi_U^* F) \longrightarrow H^1(U, F)$$

such that

$$\mathrm{tr}_{\pi_U} \circ \pi_U^* = \deg(\pi_U)\ \mathrm{id}\,.$$

It follows that $H^*(U, F) \to H^*(V, \pi_U^* F)$ is injective, completing the proof. [75]

5.9. An example in which hypothesis (A) is not satisfied.

Let X be a smooth quadric surface in $\mathbb{P}^3$. Then X satisfies (LV) (5.2.4). Take a pencil (a Lefschetz pencil, if desired) of hyperplanes whose generic member and axis meet X transversely. Then (as Deligne pointed out to me) the projection

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

does not satisfy (A). Indeed, $R^2 \rho_* \mathbb{Q}_\ell$ is not constant: the singular fibers are singular conics, hence reducible, and in a Lefschetz pencil each is the union of two lines meeting at one point (this is the definition of a Lefschetz pencil in this simple case). Thus

$$\dim_{\mathbb{Q}_\ell} H^2(\text{singular fiber}, \mathbb{Q}_\ell) = 2,$$

whereas

$$\dim_{\mathbb{Q}_\ell} H^2(\text{smooth fiber}, \mathbb{Q}_\ell) = 1.$$

Moreover, the conclusion of Theorem 5.8.4 is false here. Let $U \overset{\nu}{\hookrightarrow} \mathbb{P}^1$ be a nonempty open subset over which ρ is smooth. Since the fibers of ρ over U are smooth curves of genus zero, we have

$$\nu^* R^1 \rho_* \mathbb{Q}_\ell = 0$$

and hence

$$H^1(U, \nu^* R^1 \rho_* \mathbb{Q}_\ell) = 0.$$

Consequently, the Griffiths homomorphism

$$\mathrm{Prim}^2(X, \mathbb{Q}_\ell) \xrightarrow{\mathrm{Grif}} H^1(U, \nu^* R^1 \rho_* \mathbb{Q}_\ell)$$

is zero. But on a quadric surface,

$$\dim_{\mathbb{Q}_\ell} \mathrm{Prim}^2(X, \mathbb{Q}_\ell) = 1.$$

Likewise, the conclusion of 5.7 fails here.

6. Application of the local theory of vanishing cycles (Picard–Lefschetz)

Proposition 6.1.

Let Z be a hypersurface in projective space $\mathbb{P}^r_k$, $r \geq 2$, where k is an algebraically closed field. Let

$$i_o : D_o \hookrightarrow \mathbb{P}^r$$

[75] The print drops the twist (j) from one base-change display and the frozen transcription corrupts both the source and target of the trace map. The corrected formulas are the standard finite-flat pullback–trace identities used by the concluding argument.

be the inclusion of a line which meets Z only in its smooth locus and meets it transversely. Choose a geometric point $\bar{a}$ of $D_o - D_o \cap Z$, and denote by

$$\pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a})$$

the quotient of $\pi_1(D_o - D_o \cap Z, \bar{a})$ classifying the étale coverings of $D_o - D_o \cap Z$ which are tamely ramified at every point of $D_o \cap Z$. Then:

6.1.1.

The canonical homomorphism

$$i_{o*} : \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a}) \tag{6.1.1.1}$$

is surjective.

6.1.2.

Every choice of a geometric point $\bar{x}_o \in D_o \cap Z$ and of a "path" in D_o from $\bar{x}_o$ to $\bar{a}$ determines a homomorphism

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \tag{6.1.2.1}$$

whose image is "the" inertia group at $\bar{x}_o$. Composing this morphism with i_{o*} gives a homomorphism

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a}). \tag{6.1.2.2}$$

If Z is irreducible, all these homomorphisms are conjugate to one another.

Proof. Let T be the subscheme of the Grassmannian formed by the lines in $\mathbb{P}^r$ which

(i) pass through $\bar{a}$;

(ii) meet Z only in its smooth locus, and meet it there transversely.

Let η be its generic point, and let

$$i_\eta : D_\eta \longrightarrow \mathbb{P}^r \otimes_k k(\eta)$$

be the corresponding line (thus a line "joining $\bar{a}$ to the generic point of $\mathbb{P}^r_k$"). Choose also a geometric point $\bar{\eta}$ over η, and write $D_{\bar{\eta}}$ for the corresponding "geometric" line. Let $o \in T$ be the point corresponding to the given line D_o. By [SGA 1, XIII 2.8], there is a specialization homomorphism (depending on a path in T from $\bar{\eta}$ to o, and hence defined up to conjugacy)

$$\pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \xrightarrow{\bar{\eta} \longrightarrow o} \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \tag{6.1.3}$$

such that the following assertions hold.

6.1.4.

The local-monodromy homomorphisms (6.1.2.1)

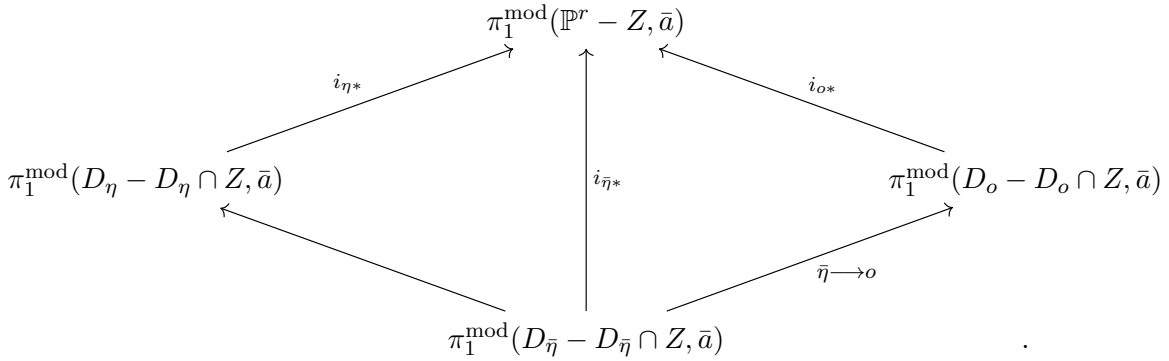
$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a})$$

are obtained by composing the specialization arrow (6.1.3) with the local monodromy homomorphisms for $D_{\bar{\eta}}$:

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \xrightarrow{\bar{\eta} \longrightarrow o} \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}). \tag{6.1.4.1}$$

6.1.5.

The following diagram is commutative:
(6.1.5)



.

Assume this for the moment, and first prove 6.1.1. By the commutativity of 6.1.5, it is enough to prove that

$$i_{\bar{\eta}*} : \pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a})$$

is surjective. For this, it is enough to prove that the analogous homomorphism of the ordinary fundamental groups

$$i_{\bar{\eta}*} : \pi_1(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \longrightarrow \pi_1(\mathbb{P}^r - Z, \bar{a}) \tag{6.1.6}$$

is surjective. In other words, for every étale covering $E \to \mathbb{P}^r - Z$ with E connected (and hence irreducible, since $\mathbb{P}^r - Z$ is smooth and therefore unibranch [EGA IV, 18.10.3]), its inverse image

$$E_{\bar{\eta}} = E \times_{\mathbb{P}^r - Z} (D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z)$$

over $D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z$ must remain geometrically irreducible. It is enough to use the following variant of Bertini's theorem ([EGA V, citation left blank in the source]) for the composite $E \to \mathbb{P}^r - Z \hookrightarrow \mathbb{P}^r$ and $s = 1$.

6.1.6.1. Bertini's theorem.

Let k be a field, let $\mathbb{P}^r = \mathbb{P}^r_k$, and let $f : X \to \mathbb{P}^r$ be a nonconstant morphism, where X is a geometrically irreducible algebraic scheme. For every integer $s < \dim \overline{f(X)}$, let L_t denote the generic linear subspace of codimension s in $\mathbb{P}^r$. Then $X_t = f^{-1}(L_t)$ is geometrically irreducible.
We now prove 6.1.2. By 6.1.4 and the commutativity of (6.1.5), it is enough to prove that all the local-monodromy homomorphisms (6.1.3) for $D_{\bar{\eta}}$,

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}), \tag{6.1.7.1}$$

become conjugate after composition with the homomorphism

$$\pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \longrightarrow \pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a}). \tag{6.1.7.2}$$

To prove this, observe that for every finite Galois extension K of $k(\eta)$, with

$$k(\eta) \subset K \subset k(\bar{\eta}),$$

the scheme

$$D_K \overset{\mathrm{def}}{=} D_\eta \times_{k(\eta)} K$$

is an étale covering of D_η. Thus $D_K - D_K \cap Z$ is a tame covering of $D_\eta - D_\eta \cap Z$, and hence $\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a})$ acts on the scheme $D_K \cap Z$. We take K sufficiently large that every point of $D_K \cap Z$ is K-rational; this is possible because, by hypothesis, $D_\eta \cap Z$ is étale over $k(\eta)$.

It remains to prove that $\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a})$ acts transitively on the points of $D_K \cap Z$. This will prove that the composite local-inertia homomorphisms

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a})$$

attached to the various points of $D_K \cap Z$ are conjugate. Now the action on $D_K \cap Z$ factors through the canonical quotient [76]

$$\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a}) \longrightarrow \pi_1(\eta, \bar{\eta}) = \mathrm{Gal}\big(k(\bar{\eta})/k(\eta)\big).$$

The action of $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ on $D_K \cap Z$ is the one compatible with the inclusion

$$D_K \cap Z \hookrightarrow \mathbb{P}^r_k \times_k K,$$

where the Galois group acts on $\mathbb{P}^r_k \times_k K$ only through the second factor. It follows that the desired assertion is equivalent to saying that $D_\eta \cap Z$ has only one point, that is, that it is the spectrum of a field. In elementary terms, this is precisely the following Bertini-type fact (take an affine equation $F = 0$ for Z and suppose that $\bar{a}$ is the origin).

Fact 6.1.8.

Let k be an algebraically closed field, let $r \geq 2$, and let $F(X_1, \ldots, X_r) \in k[X_1, \ldots, X_r]$ be irreducible. Then

$$G(T) = F(X_1 T, \ldots, X_r T) \in k(X_1, \ldots, X_r)[T]$$

is irreducible as a polynomial in T with coefficients in $k(X_1, \ldots, X_r)$.

This completes the proof of 6.1.2.

Remark 6.1.9.

Another method of proving 6.1.2, valid under considerably more general conditions, would be simply to establish a general compatibility for "monodromy homomorphisms": the composite homomorphisms in 6.1.2 are conjugate to the monodromy homomorphism at the generic point of Z, and therefore are conjugate to one another.

Corollary 6.1.10.

Let X be a smooth connected projective variety of dimension n over an algebraically closed field k. Suppose that $n = \dim X$ is even, or that $\mathrm{char}(k) \neq 2$.[77] Let D be a Lefschetz pencil for X, let η be its generic point, and let $\bar{\eta}$ be a geometric point over η. Then the natural representation of $\pi_1(D - D \cap \check{X}, \bar{\eta})$ on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ factors through $\pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$. If this representation is nontrivial, then $D \cap \check{X} \neq \varnothing$, and the inertia group at every point of $D \cap \check{X}$ acts nontrivially.

[76] The numerator of the printed Galois group is blank. The field tower immediately preceding the display determines it as $k(\bar{\eta})$.

[77] The print has $\mathrm{card}(k) \neq 2$; the parallel hypothesis in (6.3.2) and the tame-monodromy argument require the characteristic.

Proof.

If $\dim(\check{X}) \leq r-2$, then $D \cap \check{X}$ is empty and there is nothing to prove, since $\mathbb{P}^1$ is simply connected. Otherwise $\check{X}$ is a hypersurface in $\mathbb{P}^r$ [XVII, 3.1.4], and 6.1.1 applies by 3.2.10 and XVII 3.5, with $Z = \check{X}$ and $D_o = D$. The fact that the action of π_1 on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is tame was established in the local theory by the Picard–Lefschetz formula [XV, 3.4], recalled below. Finally, the nontrivial fact that $\pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$ is generated by the inertia groups at the various points of $D \cap \check{X}$ [SGA 1, XIII 2.10] gives the required conclusion.

We shall now use the preceding result to study condition (A) of (5.3.5). First recall the following consequence of the Picard–Lefschetz formula [XV, 3.4].

Proposition 6.2.

Let X be a smooth connected projective variety of dimension n over the algebraically closed field k. Let D be a Lefschetz pencil, let η be its generic point, and let $\bar{\eta}$ be a geometric point over η. Then the pencil D satisfies condition (A) if and only if the inertia group at every point of $D \cap \check{X}$ acts nontrivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

Theorem 6.3.

Under the hypotheses of 6.2, the following assertions hold.

(6.3.1) If $n = \dim X$ is odd, condition (A) is satisfied by every Lefschetz pencil for X.

(6.3.2) Suppose that $n = \dim X$ is even, or that $\mathrm{char}(k) \neq 2$. Then the following conditions are equivalent:

(i) D does not satisfy condition (A) of (5.3.5);

(ii) $\pi_1(D - D \cap \check{X})$ acts trivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$, and $D \cap \check{X} \neq \varnothing$;

(iii) there is a point of $D \cap \check{X}$ whose inertia group acts trivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$;

(iv) the sheaf $R^{n-1}\rho_* \mathbb{Q}_\ell$ is constant on D, and $D \cap \check{X}$ is nonempty;

(v) the restriction $R^{n-1}\rho_* \mathbb{Q}_\ell|_U$ is constant on U, where U is a nonempty open subset of D over which ρ is smooth, and ρ is not smooth.

(6.3.3) If, in addition, X satisfies condition (LV) of (5.2.2), conditions (i)–(v) are also equivalent to:

(vi) the restriction map

$$H^{n-1}(X, \mathbb{Q}_\ell) \longrightarrow H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$$

is surjective (and hence is an isomorphism by the weak Lefschetz theorem).

Proof.

Assertion 6.3.1 follows from XV 3.4. For the equivalences in 6.3.2, (i) is equivalent to (iii) by 6.2, (ii) to (iii) by 6.1.6, (ii) to (v) tautologically, and (iv) to (v) by 5.3.4. Finally, if X satisfies condition (LV), 5.6.10 gives the equivalence of (ii) and (vi).

Remark 6.3.4.

Condition (A) does not depend on the choice of the Lefschetz pencil D, but only on the chosen (Lefschetz) immersion $i : X \longrightarrow \mathbb{P}^r$. This is clear from (6.3.1) when $\dim X$ is odd. When $\dim X$ is even, condition (A) also says that the local system on $\mathbb{P}^r - \check{X}$ formed by the groups $H^{n-1}(X \cap H_s, \mathbb{Q}_\ell)$ is constant. This follows from criterion (ii), from the fact that the local system in question is tamely ramified at the generic point of $\check{X}$, and from the surjectivity in (6.1.2).

Corollary 6.4.

Under the hypotheses of 6.2, suppose that $n = \dim X$ is even or $\mathrm{char}(k) \neq 2$, and that X satisfies condition (LV).[78] Then there is an integer d_o such that, for $d \geq d_o$, every Lefschetz pencil of sections of X by hypersurfaces of degree d satisfies condition (A).

Proof.

By the equivalence (i)$\Longleftrightarrow$(vi) of (6.3.2), it is enough to prove that, if $Y(d)$ denotes a smooth hypersurface section of X of degree d, then

$$\dim H^{n-1}(Y(d), \mathbb{Q}_\ell) \longrightarrow \infty \qquad \text{as } d \longrightarrow \infty. \tag{6.4.1}$$

Indeed, we have the following precise result.

Lemma 6.4.2.

Let X be a smooth connected projective variety of dimension $n \geq 1$ over an algebraically closed field k, and let $Y(d)$ denote a hypersurface section of degree d. Then there is a polynomial $f(T) \in \mathbb{Z}[T]$ such that

$$f(T) = \deg(X)T^n + \text{a polynomial of degree at most } n-1, \tag{6.4.2.1}$$

$$f(d) = \dim H^{n-1}(Y(d), \mathbb{Q}_\ell) \qquad (d \geq 1). \tag{6.4.2.2}$$

Proof.

By XVII 5.7.5, if $C(X)$ denotes the total Chern class of X and H the first Chern class of a hyperplane, the Euler–Poincaré characteristic of $Y(d)$ is

$$\chi(Y(d)) = \int_X \frac{dH\, C(X)}{1+dH} = \sum_{i=1}^{n} (-1)^{i-1} d^i \int_X H^i C_{n-i}. \tag{6.4.2.3}$$

On the other hand, Poincaré duality and weak Lefschetz give

$$\chi(Y(d)) = (-1)^{n-1} \dim H^{n-1}(Y(d), \mathbb{Q}_\ell) + 2\sum_{i\geq 1} (-1)^{n-1-i} \dim H^{n-1-i}(X, \mathbb{Q}_\ell). \tag{6.4.2.4}$$

Combining (6.4.2.3) and (6.4.2.4) gives the result.

6.5. A reminder on the Picard–Lefschetz formula (XV 3.4)

Let S be the spectrum of a strictly henselian valuation ring, with generic point η, geometric generic point $\bar{\eta}$, and closed point s. Let

$$\rho : X \longrightarrow S$$

be a proper morphism, where X is a connected regular scheme of dimension n. Suppose that ρ is smooth away from a single point x in the special fiber and that x is an ordinary quadratic singularity of that fiber.

Let I denote the inertia group of S at s. By XV 3.4, I acts trivially on $H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ for $i \neq n-1$, and its action on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is described as follows.

[78] In the print, the placement of the conjunction attaches condition (LV) only to the second alternative. The proof invokes the equivalence (i)$\Longleftrightarrow$(vi) of (6.3.2), so condition (LV) is required in either case.

6.5.1.

If $n = 2m+1$, then I acts through a quotient isomorphic to $\mathbb{Z}/2\mathbb{Z}$ (if $\mathrm{char}(k) \neq 2$, there is only one such quotient). There is an element

$$\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)),$$

determined uniquely up to sign and called the vanishing cycle at s, such that

$$\delta \cdot \delta = (-1)^m 2 \quad \text{in} \quad H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(n-1)) \simeq \mathbb{Q}_\ell. \tag{6.5.1.1}$$

The nontrivial element σ of $\mathbb{Z}/2\mathbb{Z}$ acts on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ by the reflection

$$\sigma x = x - (-1)^m (x \cdot \delta)\delta. \tag{6.5.1.2}$$

[79] Here $x \cdot \delta$ belongs to $H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \simeq \mathbb{Q}_\ell(-m)$, so $(x \cdot \delta)\delta$ defines an element of $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

6.5.2.

If $n = 2m+2$, then I acts through its canonical quotient $\mathbb{Z}_\ell(1)$. There is an element

$$\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)),$$

canonically determined up to sign and called the vanishing cycle at s, such that $\sigma \in \mathbb{Z}_\ell(1)$ acts on $x \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ by the transvection

$$\sigma(x) = x - (-1)^m \sigma\,(x \cdot \delta)\delta. \tag{6.5.2.1}$$

Indeed, $x \cdot \delta$ belongs to $H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \simeq \mathbb{Q}_\ell(-m-1)$. Multiplication by $\sigma \in \mathbb{Z}_\ell(1)$ therefore gives an element $\sigma(x \cdot \delta) \in \mathbb{Q}_\ell(-m)$, whose product with $\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$ lies in $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

We return to the global situation.

Theorem 6.6.

Let X be a smooth connected projective variety of dimension n over an algebraically closed field k. Suppose that n is even or that $\mathrm{char}(k) \neq 2$. Let D be a Lefschetz pencil for X, let η be its generic point, let $\bar{\eta}$ be a geometric point over η, and let $s_1, \ldots, s_N$ be the points of $D \cap \check{X}$.

For each i, choose an isomorphism ℓ_i between $\bar{\eta}$ and the geometric generic point of S_i, the strict henselization of D at s_i. Via ℓ_i, let I_i be the inertia group at s_i acting on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(j))$, and let δ_i be the vanishing cycle at s_i defined in 6.5.[80] Thus

$$\delta_i \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)), \qquad m = \begin{cases} (n-1)/2, & n \text{ odd}, \\ n/2 - 1, & n \text{ even}. \end{cases}$$

[81] Choose the ℓ_i so that the subgroups

$$I_i \subset \pi = \pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$$

topologically generate π, as is possible by SGA 1, XIII 2.10. Then:

[79] The print numbers the reflection formula (6.5.2.2), although it is the second formula of 6.5.1 and is followed by (6.5.2.1).

[80] The print calls I_i the inertia group "of δ_i" before δ_i is defined. The standard construction is the inertia group at s_i, embedded by the chosen path ℓ_i.

[81] The print omits the generic-fiber subscript in this display; the preceding action and the definition in 6.5 place every δ_i in the cohomology of $X_{\bar{\eta}}$.

6.6.1. If X satisfies condition (LV), the vanishing cohomology space

$$E^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m)) \overset{\text{def}}{=} H^{n-1}(X,\mathbb{Q}_\ell(m))^\perp \subset H^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m))$$

(cf. 5.4.4) is generated by the vanishing cycles δ_i, $i=1,\dots,N$.

6.6.2. The vanishing cycles δ_i are conjugate, up to sign, under the global monodromy group $\pi_1(D-D\cap\check X)$.

Proof.

For 6.6.1, the Picard–Lefschetz formula 6.5 and the fact that the I_i generate $\pi=\pi_1(D-D\cap\check X)$ give

$$H^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m))^\pi = \big\langle \delta_1,\dots,\delta_N\big\rangle^\perp \quad \text{in } H^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m)). \tag{6.6.3}$$

By 5.6.10,

$$H^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m))^\pi = \mathrm{Im}\big(H^{n-1}(X,\mathbb{Q}_\ell(m)) \longrightarrow H^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m))\big). \tag{6.6.4}$$

Taking orthogonals in (6.6.3) proves 6.6.1.

For 6.6.2, note simply that the vanishing cycle δ_i is determined up to sign by the representation of the inertia group at s_i. Apply 6.1.2, which gives conjugacy in $\pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r-\check X,\bar\eta)$ of the composite inertia homomorphisms

$$\begin{array}{ccc} \mathbb{Z}_\ell(1) & \longrightarrow & \pi_1^{\mathrm{mod}}(D-D\cap\check X,\bar\eta) \\ & & \big\downarrow \\ & & \pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r-\check X,\bar\eta) \end{array} \quad .$$

Consequently, the vanishing cycles determined by these homomorphisms are conjugate up to sign by elements of $\pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r-\check X,\bar\eta)$. The proof is completed by 6.1.1, which gives the surjection

$$\pi_1^{\mathrm{mod}}(D-D\cap\check X) \longrightarrow \pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r-\check X).$$

Corollary 6.7.

Under the hypotheses of 6.6, suppose that X satisfies condition (LV). Then:

6.7.1. The representation of $\pi_1(D-D\cap\check X)$ on $E^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell)$ is absolutely irreducible.

6.7.2. If n is even, every finite-index subgroup of $\pi_1(D-D\cap\check X)$ acts absolutely irreducibly on $E^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell)$.

Proof.

We may replace the module in question by its twist

$$E = E^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m)).$$

Let $\bar E = E\otimes_{\mathbb{Q}_\ell}\bar{\mathbb{Q}}_\ell$, and choose $0\neq\chi\in\bar E$. To prove 6.7.1, we must show that the $\bar{\mathbb{Q}}_\ell$-linear subspace of $\bar E$ generated by χ and its conjugates under π is all of $\bar E$. By 6.6.1 and 6.6.2, it is enough to prove that this subspace contains one of the vanishing cycles δ_i.

Since X satisfies (LV), the restriction of the cup-product form to

$$H^{n-1}(X,\mathbb{Q}_\ell(m)) \hookrightarrow H^{n-1}(X_{\bar\eta},\mathbb{Q}_\ell(m))$$

is nondegenerate; hence so is its restriction to the orthogonal $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(m))$. By 6.6.1, there is therefore at least one i such that

$$\chi \cdot \delta_i \neq 0.$$

For this i, 6.5 supplies an element σ_i of the inertia group at s_i such that

$$\sigma_i(\chi) - \chi = \text{a nonzero multiple of } \delta_i. \tag{6.7.3}$$

This proves 6.7.1.

For 6.7.2, $n-1$ is odd, so $\delta_i \cdot \delta_i = 0$. Iterating 6.5.2 gives, for every k,

$$\sigma_i^k(\chi) = \chi + k\,\sigma_i(\chi \cdot \delta_i)\delta_i. \tag{6.7.4}$$

Thus a $\mathbb{Q}_\ell$-subspace M of E is stable under the action of π if and only if it is stable under a finite-index subgroup: both conditions are equivalent to

$$\chi \in M \quad \Longrightarrow \quad \delta_i \in M \quad \text{for every } i \text{ with } \chi \cdot \delta_i \neq 0.$$

The assertion therefore reduces to 6.7.1.

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[82] The print gives the journal volume as 15; the article appeared in volume 90.

SGA 7 — EXPOSÉ XIX

The Noether Theorem

by P. Deligne

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1. Statement of the theorems

The present exposé updates Lefschetz's proof [3, p. 108] of the Noether theorem (1.2) stated below.

1.1.

Let k be an algebraically closed field of characteristic p. Let $d \geq 1$ and let

$$\underline{a} = (a_i)_{1 \leq i \leq d}$$

be a sequence of integers at least 1. A complete intersection V of dimension n and multidegree $\underline{a}$ in the projective space $\mathbb{P}^{n+d}$ over k will be called generic if, after an extension of k, it is projectively isomorphic to a complete intersection defined by equations whose coefficients are algebraically independent over the prime field. In what follows, unless expressly stated otherwise, we suppose that $d \geq 1$ and $a_i \geq 2$.

Theorem 1.2.

Let $V \subset \mathbb{P}^{2+d}$ be a generic complete-intersection surface of multidegree $\underline{a}$. Suppose that

$$\underline{a} \neq (2), (3), (2,2).$$

Then every (reduced) curve on V is the scheme-theoretic intersection of V with a hypersurface of $\mathbb{P}^{2+d}$.

Let us show that the theorem follows from its corollary

$$\mathrm{Pic}(\mathbb{P}^{2+d}) = \mathbb{Z} \xrightarrow{\sim} \mathrm{Pic}(V). \tag{1.2.1}$$

If C is a curve on V, (1.2.1) implies that $\mathcal{O}(C) \simeq \mathcal{O}(n)$ for a suitable n. If H is a hyperplane section, then

$$n = \left(\textstyle\prod_i a_i\right)^{-1}(C, H) > 0.$$

It remains to prove that the linear system induced on V by $|nH|$ is complete. Write V as the intersection of hypersurfaces H_i of degree a_i, and put $V_i = H_1 \cap \cdots \cap H_i$. Induction on i shows that the linear system induced by $|nH|$ on V_i is complete: in the exact sequence

$$H^0(V_i, \mathcal{O}(n)) \longrightarrow H^0(V_{i+1}, \mathcal{O}(n)) \longrightarrow H^1(V_i, \mathcal{O}(n - a_{i+1})),$$

the last group vanishes (FAC).

Theorem 1.3.

Let $V \subset \mathbb{P}^{2n+d}$ be a generic complete intersection of nonzero even dimension $2n$ and multidegree $\underline{a}$. Suppose that one of the following conditions holds:

(a) k is not of characteristic 2, and $(2n; \underline{a})$ is distinct from $(2n; 2)$, $(2n; 2, 2)$, and $(2; 3)$;

(b) the Hodge number $h^{2n,0}$ is nonzero, i.e. $\sum_i (a_i - 1) > 2n$.

Then every algebraic cohomology class in $H^{2n}(V, \mathbb{Q}_\ell(n))$ is a rational multiple of the class η^n of a linear section.

Let us deduce (1.2.1) from 1.3. The exceptional cases in 1.2 are exactly those for which $\sum_i (a_i - 1) \leq 2$; hence 1.3(b) applies to the surface V of 1.2.[83] We know that $\mathrm{Pic}^\tau(V) = 0$ (XI 1.8), so $\mathrm{Pic}(V)$ injects into $H^2(V, \mathbb{Q}_\ell(1))$. By 1.3 every element of $\mathrm{Pic}(V)$ is a rational multiple of the class η of $\mathcal{O}(1)$, and (XI 1.8) gives the conclusion.

It is clear that the validity of 1.2 and 1.3 does not depend on the chosen generic complete intersection, but only on p, n, and $\underline{a}$.

Theorem 1.4.

Let $X \subset \mathbb{P}$ be a smooth projective variety, pure of dimension $2n+1$ over k. Let $\ell \subset \check{\mathbb{P}}$ be a Lefschetz pencil of hyperplane sections of X. If k has characteristic 2, suppose in addition that ℓ is a generic Lefschetz pencil.

Suppose that X satisfies condition (LV) (XVIII 5.2.2), and let V be the generic hyperplane section in the pencil ℓ. One of the following two conditions holds:

(a) no nonzero cohomology class in the vanishing part $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ of $H^{2n}(V, \mathbb{Q}_\ell(n))$ is algebraic;

(b) the algebraic cohomology classes generate $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$.

Let us make precise the meaning of "generic" in 1.4.

(a) Let G be the Grassmannian of lines in the projective space $\check{\mathbb{P}}$ dual to $\mathbb{P}$. By a "geometric point" of G we mean a morphism from the spectrum of an algebraically closed field to G. Every geometric point $\bar{s} \to G$ defines a pencil ℓ of hyperplane sections of the variety obtained from X by extending scalars from k to $k(\bar{s})$. We call ℓ generic if $\bar{s}$ lies over the generic point of G.

(b) Let ℓ be a pencil of hyperplane sections (a line in $\check{\mathbb{P}}$). Every geometric point $\bar{s} \to \ell$ defines a hyperplane section of the variety obtained from X by extending scalars from k to $k(\bar{s})$. We call this section generic if $\bar{s}$ lies over the generic point of ℓ.

Let $\bar{s}$ be as above, with image the generic point s of ℓ, and let $S \subset \ell$ be the finite set of points giving singular hyperplane sections. The monodromy group $\pi_1(\ell - S, \bar{s})$, a quotient of $\mathrm{Gal}(k(\bar{s})/k(s))$, acts on $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$. If k is not of characteristic 2, this action is irreducible (XVIII 6.7); in Section 2 below we shall show that the assertion remains true without a characteristic hypothesis for a generic pencil. The subspace of $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ generated by algebraic cohomology classes is stable under monodromy (that is, under Galois). It is therefore either zero or the whole space, which proves 1.4.

[83] The print writes d_i in this sum, although the multidegree has been denoted throughout by (a_i).

Corollary 1.5.

Let $V \subset \mathbb{P}^{2n+d}$ be a generic complete intersection of even dimension $2n$ and multidegree $\underline{a}$. Then either

(a) every algebraic cohomology class in $H^{2n}(V, \mathbb{Q}_\ell(n))$ is a rational multiple of η^n; or

(b) $H^{2n}(V, \mathbb{Q}_\ell(n))$ is generated by algebraic cohomology classes.

If (b) holds, then for every complete intersection of dimension $2n$ and multidegree $\underline{a}$ over an algebraically closed field of characteristic p, the group $H^{2n}(V, \mathbb{Q}_\ell(n))$ is generated by algebraic classes.

Apply 1.4 with X a generic complete intersection of dimension $2n+1$ and multidegree $(a_1, \ldots, a_{d-1})$, and with ℓ a generic pencil of sections of X by hypersurfaces of degree a_d (cf. XVII 2.4). One could equally take X of multidegree $(a_1, \ldots, a_d)$ and ℓ a generic pencil of hyperplane sections. The final assertion follows because an algebraic cohomology class remains algebraic under specialization.

2. A complement to XVIII

Proposition 2.1.

Let $X \subset \mathbb{P}$ be a smooth projective variety, pure of odd dimension $2n+1$ over an algebraically closed field k, and let $\ell \subset \check{\mathbb{P}}$ be a generic Lefschetz pencil of hyperplane sections of X. Let s be a geometric point of ℓ, let H_s be the corresponding hyperplane section, let $S \subset \ell$ be the set of exceptional values of the pencil, and let $\pi_1(\ell - S, s)$ be the monodromy group. Suppose that X satisfies condition (LV). Then the representation of $\pi_1(\ell - S, s)$ on the vanishing cohomology

$$\mathrm{Ev}(H_s, \mathbb{Q}_\ell) \subset H^{2n}(H_s, \mathbb{Q}_\ell)$$

is absolutely irreducible.[84]

Recall that $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ is the orthogonal complement of the image of $H^{2n}(X, \mathbb{Q}_\ell)$ in $H^{2n}(H_s, \mathbb{Q}_\ell)$, and that this image equals the subgroup of monodromy invariants (XVIII 5.6.10).

Proof.

The projective line $\mathbb{P}^1$ is simply connected. Hence $\pi_1(\ell - S, s)$ is topologically generated by the inertia subgroups at the points of S and their conjugates. Its image in $\mathrm{GL}(H^{2n}(H_s, \mathbb{Q}_\ell))$ is therefore topologically generated by the transformations

$$x \longmapsto x - (-1)^n (x \cdot \delta)\delta,$$

where δ is conjugate to a vanishing cycle.[85] Consequently, the image of $H^{2n}(X, \mathbb{Q}_\ell)$ in $H^{2n}(H_s, \mathbb{Q}_\ell)$ is the orthogonal complement of the subspace generated by the vanishing cycles and their conjugates; thus $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ is generated by those cycles and their conjugates.

For a generic Lefschetz pencil, the arguments of XVIII §6 show that the vanishing cycles attached to the different points of S are conjugate to one another.

Let $W \subset \mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ be a nonzero invariant subspace. There exist a conjugate δ of a vanishing cycle and an element $v \in W$ such that $(v, \delta) \neq 0$; this is where condition (LV) is used. Since

$$v - (-1)^n (v \cdot \delta)\delta \in W,$$

[84] The print says that X has even dimension $2n$. The displayed degree $H^{2n}(H_s)$, the reflection formula, Remark 2.2, and the application to Theorem 1.4 all force odd dimension $2n+1$.

[85] The print gives $H^n(H_s)$ at this point, although every surrounding occurrence and the statement of the proposition use the middle degree $H^{2n}(H_s)$.

we have $\delta \in W$.[86] The subspace W, containing one vanishing cycle and all its conjugates, is therefore $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$. The argument remains valid after extending the coefficient field $\mathbb{Q}_\ell$, proving absolute irreducibility.

Remark 2.2.

If condition (LV) is omitted, the preceding arguments still show that every invariant subspace of $H^{2n}(H_s, \mathbb{Q}_\ell)$ either contains $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ or is contained in the image of $H^{2n}(X, \mathbb{Q}_\ell)$. The same assertion holds for X of even dimension (and for every Lefschetz pencil).

3. Proof of 1.3

3.1.

Under the hypotheses of the theorem, it remains to prove that 1.5(b) does not hold. The case $h^{0,2n} \neq 0$ will be treated by Katz in XXI 4.1 (the condition $h^{0,2n} \neq 0$ is equivalent to

$$\sum_i a_i - (2n + r + 1) \geq 0,$$

this number being the integer m such that $\Omega_V^{2n} \simeq \mathcal{O}(m)$). [87] The results of the present exposé will not be used again in the remainder of the seminar, so there is no vicious circle.

3.2.

When $k = \mathbb{C}$, one proves 1.3 by observing that if 1.5(b) holds, the cohomology of V is purely of type (n, n), since algebraic cohomology is of type (n, n). This occurs only in the exceptional cases excluded in 1.3 (XI 2.9). By the Lefschetz principle, 1.3 is therefore true in characteristic zero.

Proposition 3.3.

Assume the hypotheses of 1.4, and suppose that (b) holds. Let $\mathrm{Ev}(V, \mathbb{Q})$ be the $\mathbb{Q}$-vector subspace of $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ generated by the algebraic cohomology classes.

(a) There is an isomorphism

$$\mathrm{Ev}(V, \mathbb{Q}) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} \mathrm{Ev}(V, \mathbb{Q}_\ell(n)).$$

On $\mathrm{Ev}(V, \mathbb{Q})$ the intersection form is rational and definite, of sign $(-1)^n$.

(b) The vanishing cycles and their conjugates belong to $\mathrm{Ev}(V, \mathbb{Q})$; there they form an irreducible root system of type A, D, or E.

(c) The monodromy group (the image of $\pi_1(\ell - S, \bar{s})$ in $\mathrm{GL}(\mathrm{Ev}(V, \mathbb{Q}))$) is finite; it is the Weyl group of the root system in (b).[88]

Apart from definiteness of the intersection form, (a) follows because the intersection form is nondegenerate on $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ (by (LV)) and rational on $\mathrm{Ev}(V, \mathbb{Q})$.

Condition 1.4(b) says equivalently that there are algebraic cycles on V, modulo numerical equivalence, which are not invariant under monodromy; this condition is therefore independent of

[86] The print uses the undefined exponent m in this second occurrence; the immediately preceding reflection and the Picard–Lefschetz formula give n.

[87] The print writes d_i in this sum, although the multidegree is denoted by (a_i).

[88] The print repeatedly says "groupe de Weil"; the group attached to a root system is its Weyl group.

ℓ. Let x be such an algebraic cycle, and let δ be the vanishing cycle associated with $s_0 \in S$ and with a path c from $\bar{s}$ (which defines V) to s_0. For a suitable c, $(x,\delta) \neq 0$. The Galois transform

$$\sigma(x) = x - (-1)^n (x,\delta)\delta$$

is again algebraic; hence $(x,\delta)\delta$ is algebraic, the image of a cycle y independent of ℓ. The rational number

$$\frac{(-1)^n}{2}(y,y) = (x,\delta)^2$$

is consequently independent of ℓ. It is a square in every $\mathbb{Q}_\ell$, hence a square in $\mathbb{Q}$; thus, up to sign, (x,δ) belongs to $\mathbb{Q}$ and is independent of ℓ. This shows not only that $\delta \in \mathrm{Ev}(V,\mathbb{Q})$, but also that the algebraic cycle modulo numerical equivalence

$$\pm\delta = \frac{\sigma(x) - x}{(x,\delta)}$$

is independent of ℓ.

An algebraic cycle on a variety V is always defined over a finite extension of any field of definition of V. It follows that $\pi_1(\ell - S, s)$ acts on $\mathrm{Ev}(V,\mathbb{Q})$ through a finite group. This group is generated by the reflections associated with the vanishing cycles δ and their conjugates (XVIII 6.6.1 and Section 2). The cycles δ and their conjugates therefore form a root system whose Weyl group is the monodromy group. Moreover, all roots of this root system are conjugate under the Weyl group; it is of type A, D, or E.

Up to a scalar, the intersection form is the unique invariant form under the Weyl group. It is therefore definite. Its sign is determined by computing

$$(\delta,\delta) = 2(-1)^n.$$

Proposition 3.4.

If $k = \mathbb{C}$, then under the hypotheses of 1.4 the following conditions are equivalent:

(i) the monodromy group is finite;

(ii) the vanishing cohomology $\mathrm{Ev}(V)$ is purely of type (n,n).

Let $\widetilde{X} \to \ell$ be the family whose fibers are the hyperplane sections in the pencil ℓ, and let V_s be the fiber at $s \in \ell(\mathbb{C})$. The $\mathrm{Ev}(V_s)$ form an algebraic variation of Hodge structures on ℓ. By the fixed-part theorem [2, 4.1.2], applied to a finite covering of ℓ (cf. [2, 4.1.3.3]), (i) implies that the Hodge structure on $\mathrm{Ev}(V_s)$ is locally constant. The monodromy group therefore acts by automorphisms of the Hodge structure. For a Picard–Lefschetz transformation T, $\mathrm{Im}(T-1)$ is a rank-one Hodge substructure, hence is of type (n,n) and is generated by a vanishing cycle δ. Since the vanishing cycles are of type (n,n), all of $\mathrm{Ev}(V_s)$ is of that type.

Conversely, if $\mathrm{Ev}(V)$ is purely of type (n,n), the intersection form on $\mathrm{Ev}(V)$ is definite and the monodromy group is finite (cf. [2, 4.1.3.3]).

Remark 3.5.

If the equivalent conditions of 3.3 hold, the argument of 3.3 shows that the vanishing cycles and their conjugates form a root system of type A, D, or E, whose Weyl group is the monodromy group.

3.6.

Let us prove 1.3 when k has finite characteristic different from 2. Let $S = \mathrm{Spec}(\mathcal{V})$ be the spectrum of a discrete valuation ring of mixed characteristic with residue field k, and let $X \subset \mathbb{P}_S^{2n+d}$ be a complete intersection of relative dimension $2n+1$ and multidegree $(a_1, \ldots, a_{d-1})$ over S. Let ℓ be a Lefschetz pencil of sections of X by hypersurfaces of degree a_d (in the special fiber, ℓ_s is a Lefschetz pencil for X_s). Let

$$f : \widetilde{X} \longrightarrow \ell \simeq \mathbb{P}_S^1$$

be the family whose fibers are the sections in the pencil. The exceptional locus $T \subset \ell$ is ètale over S (XVII 6; here one uses $p \neq 2$). The sheaf $R^{2n}f_*\mathbb{Q}_\ell$ is locally constant on $\ell - T$; it is therefore tamely ramified, and its monodromy is the same in characteristic p as in characteristic zero. If 1.5(b) holds, it follows from 3.3(c) and 3.4 that the cohomology group $H^{2n}(V)$ of complex complete intersections of dimension $2n$ and multidegree $\underline{a}$ is purely of type (n,n). This occurs only in the cases excluded by 1.3 (XI 2.9).

4. A second complement to XVIII

In this section we work over $\mathbb{C}$ and with integral, classical cohomology. Using methods closer to Lefschetz's, we revisit some of the results of XVIII. There they were proved only after tensoring with $\mathbb{Q}$ (or rather with $\mathbb{Q}_\ell$, which amounts to the same thing).

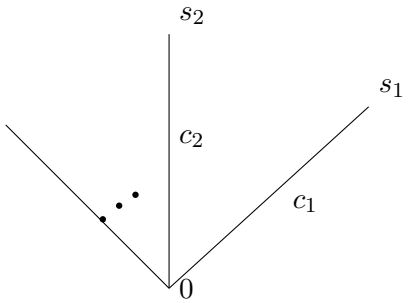
4.1.

Let $X \subset \mathbb{P}$ be a smooth complex projective variety, pure of dimension $n+1$, and let ℓ be a Lefschetz pencil of hyperplane sections of X, with axis A. We also write ℓ for the parameter space of the pencil, which is isomorphic to $\mathbb{P}^1$. Let $S \subset \ell$ be the set of exceptional values. For $s \in \ell$, write H_s for the corresponding hyperplane section of X (of dimension n).

Let $T(A)$ be a closed tubular neighborhood of $A \cap X$ in X, whose boundary is transverse to every H_s.

Choose two points $0, \infty \in \ell - S$, and let $\alpha : \ell \to \mathbb{P}^1(\mathbb{C})$ be an orientation-preserving diffeomorphism that sends 0 to 0, ∞ to ∞, and S to the circle of radius one centered at 0. [89] The composite αf of α with the projection to ℓ makes $X - T(A)^\circ$ a singular fibration in manifolds with boundary over $\mathbb{P}^1(\mathbb{C})$ (one singularity above each point of $\alpha(S)$). Write c_s for the straight segment joining 0 to $\alpha(s)$.

Suppressing α from the notation, the picture is



Near $s \in S$ and the singular point x_s of $H_s \subset X$, suitable local coordinates put us in the situation studied in XIV 3.2. If α is chosen holomorphic near s, one can even choose local coordinates on X and ℓ in which f is written $\sum z_i^2$ and $\alpha^{-1}(c_s)$ is given by "the z_i are real and positive." Above the intersection of a neighborhood of s with $\alpha^{-1}(c_s) - \{s\}$, consider the bundle of real spheres $\Delta'_s \subset X$ defined by "the z_i are real." As $t \to s$, the fiber Δ'_{st} of Δ'_s at t tends to x_s.

[89] The print gives $\ell - S$ as the domain of α, although the same sentence evaluates α on S; the required domain is ℓ.

Let Δ_s denote the union of $\{x_s\}$ and a sphere bundle over $\alpha^{-1}(c_s)-\{s\}$, contained in $X-T(A)$ and extending Δ'_s. Then Δ_s is an embedded disk in X whose boundary $H_0 \cap \Delta_s$ is a sphere representing the vanishing cycle δ_s at s, transported to H_0 along the path $\alpha^{-1}(c_s)$.

Apply Morse theory to the function $|\alpha f|^2$ to study how the homotopy type of

$$\{x \in X \mid x \in T(A) \text{ or } |\alpha f(x)|^2 \leq r\}$$

changes as r increases. Put

$$X^- = \{x \in X \mid x \in T(A) \text{ or } \alpha f(x) \neq \infty\}.$$

One finds that the inclusions

$$H_0 \cup \bigcup_{s\in S} \Delta_s \hookrightarrow T(A) \cup H_0 \cup \bigcup_{s\in S} \Delta_s \hookrightarrow X^-$$

are homotopy equivalences. Morse theory also shows that $X - X^-$ can be deformation-retracted onto a subcomplex of real dimension at most n. Altogether this gives the following result.

Theorem 4.2.

There exists a compact subcomplex Σ of real dimension at most n in $H_\infty^{\text{aff}} = H_\infty - A$ such that the inclusion

$$H_0 \cup \bigcup_{s\in S} \Delta_s \longrightarrow X - \Sigma$$

is a homotopy equivalence.

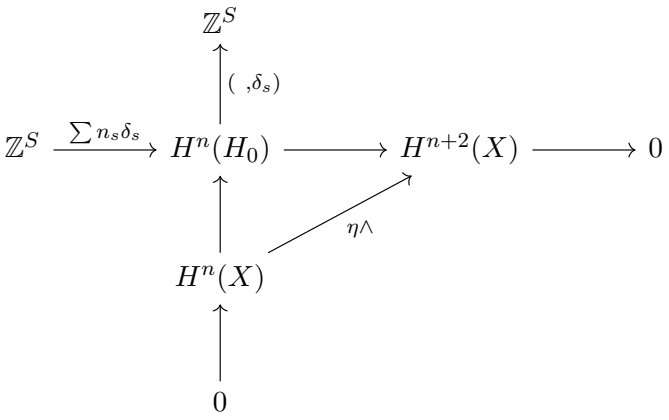
4.3.

Removing Σ has no effect on the cohomology, homology, or homotopy groups of X in degrees $i \leq n$. Since $H_0 \cup \bigcup_{s\in S} \Delta_s$ is obtained from H_0 by attaching $(n+1)$-disks along spheres representing the vanishing cycles, homology gives[90]

$$\mathbb{Z}^S \longrightarrow H_n(H_0) \longrightarrow H_n(X) \longrightarrow 0.$$

The analogous exact sequence in cohomology, together with the sequence deduced from it by Poincaré duality, forms the commutative diagram

(4.3.1)



5. The exceptional cases of the Noether theorem

In this section we continue to work over $\mathbb{C}$, with integral cohomology.

[90] The print writes V_0 in this sequence and in the diagram below, but Section 4 defines the fiber as H_0.

5.1.

Let $2n$, d, and $\underline{a} = (a_1, \ldots, a_d)$, with $d \geq 1$ and $a_i \geq 2$, be such that the middle cohomology of smooth complete intersections V of dimension $2n$ and multidegree $\underline{a}$ is purely of type (n, n). In other words, either $n = 0$, or $(2n; \underline{a})$ is one of $(2n; 2)$, $(2n; 2, 2)$, and $(2; 3)$ (XI 2.9).

Suppose that V is obtained from a complete intersection $X \subset \mathbb{P}^{2n+d}$ (respectively, $X \subset \mathbb{P}^{2n+d+1}$) of dimension $2n+1$ and multidegree $(a_1, \ldots, a_{d-1})$ (respectively, $(a_1, \ldots, a_d)$) by taking a section by a hypersurface of degree a_d (respectively, a hyperplane section) belonging to a Lefschetz pencil ℓ.

By (4.3.1), the vanishing part $\mathrm{Ev}(V)$ of $H^{2n}(V)$, which equals its primitive part, is generated over $\mathbb{Z}$ by the vanishing cycles. It is the orthogonal complement of the power η^n of the cohomology class η of a hyperplane section.

Since the intersection form on $H^{2n}(V)$ is unimodular and η^{n}[91] is not by, with $b > 1$, for another cohomology class y (XI 1.6(iv)), the discriminant of the intersection form on $\mathrm{Ev}(V)$ is

$$(\eta^n, \eta^n) = \prod_i a_i.$$

Let R be the root system in $H^{2n}(V)$ formed by the vanishing cycles and their conjugates. We have the following information about R:

(a) R is empty or is of type A, D, or E;

(b) the rank of R is given in XI 1.10;

(c) its discriminant is $\prod_i a_i$. In the tables, this discriminant is the order of the center of the corresponding simply connected complex simple group. The argument by which we computed the discriminant shows that this center, dual to the quotient of the weight lattice by the root lattice, is cyclic.

In every case this information determines R completely if $n \neq 0$. One finds:

Proposition 5.2.

(i) If V is an even-dimensional quadric, R is of type A_1.

(ii) If V is a cubic surface, R is of type E_6.

(iii) If V has dimension $2n$ and is the intersection of two quadrics, R is of type D_{2n+3}.

5.3.

Suppose that $n \neq 0$. The set R^1 of vectors of length ± 2 in $\mathrm{Ev}(V)$ is also a root system and has the same properties (a), (b), and (c) as R. Hence $R = R^1$. Let π_1 be the fundamental group of the parameter space of all complete intersections $V \subset \mathbb{P}^{2n+d}$ of the type under consideration. The image of π_1 in $\mathrm{GL}(\mathrm{Ev}(V))$ is the Weyl group of R. In every case it is the automorphism group of the triple

$$\big(H^{2n}(V), \eta^n, \text{intersection form}\big).$$

[91] The print has η^{2n} in the primitivity assertion, but that class lies in degree $4n$; both the surrounding discussion and the next display require $\eta^n \in H^{2n}(V)$.

5.4.

With R replaced by R^1 from 5.3, assertion 5.2(ii) is well known [1], and for small values of n, 5.2(iii) was known, at least to Manin ($n = 0$ is trivial; a surface that is the intersection of two quadrics is a del Pezzo surface).[92] A detailed study of the intersection of two quadrics was made by M. Reid [4]. He deduces 5.2(iii) from a study of the geometry of the n-dimensional linear subspaces on an intersection of two quadrics $V \subset \mathbb{P}^{2n+2}$.

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[92] The print refers here to 4.2(ii) and 4.2(iii), but the assertions cited are the immediately preceding cases of Proposition 5.2.

SGA 7 — EXPOSÉ XX

The Griffiths Theorem

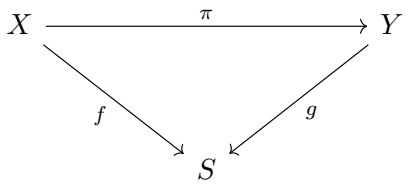
by N. Katz

0. Introduction

We prove Griffiths's theorem [1], which gives a systematic (though nonconstructive) way of "finding" algebraic cycles that are $\mathbb{Q}_\ell$-cohomologous to zero, but no multiple of which is algebraically equivalent to zero. The proof given here (due to Grothendieck) is the translation into purely algebraic terms of Griffiths's original, more or less transcendental, proof. In both proofs, the irreducibility theorem XVIII 6.7 is a key point (cf. 3.1.6 and 3.2).

1. The formalism of primitive classes

Throughout what follows, a prime number ℓ is fixed. All schemes under consideration are assumed to have characteristic different from ℓ. Consider the situation

(1.0.1)


in which we assume

(1.0.3) $\qquad f,\ g \quad$ smooth of relative dimensions $\quad d_1$ and d_2,

and

(1.0.4) $\qquad \pi \quad$ proper.

Since X and Y are smooth over S, the properness of π gives a Gysin morphism (SGA 5, Exposé IV; the further citation is left blank in the print)

$$\pi_* : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q-2(d_1-d_2)}(Y, \mathbb{Q}_\ell(i - d_1 + d_2)). \tag{1.0.5}$$

Localizing on S in the ètale topology, we likewise obtain a morphism of sheaves on S[93]

$$\pi_* : R^q f_* \mathbb{Q}_\ell(i) \longrightarrow R^{q-2(d_1-d_2)} g_* \mathbb{Q}_\ell(i - d_1 + d_2), \tag{1.0.6}$$

and, on applying the functor $H^p(S, -)$, a morphism

$$\pi_* : H^p(S, R^q f_* \mathbb{Q}_\ell(i)) \longrightarrow H^p(S, R^{q-2(d_1-d_2)} g_* \mathbb{Q}_\ell(i - d_1 + d_2)). \tag{1.0.7}$$

Definition 1.0.8.

Let $(E_r^{p,q}, d_r^{p,q})$ and $('E_r^{p,q}, 'd_r^{p,q})$ be two spectral sequences in a given abelian category, and let $(a, b) \in \mathbb{Z} \times \mathbb{Z}$. The shift of $('E_r^{p,q}, 'd_r^{p,q})$ by (a, b) is the spectral sequence $(''E_r^{p,q}, ''d_r^{p,q})$ given by[94]

$$\begin{cases} ''E_r^{p,q} = 'E_r^{p+a,q+b}, \\ ''d_r^{p,q} = 'd_r^{p+a,q+b}. \end{cases} \tag{1.0.9}$$

[93] In (1.0.6) the print has an apostrophe in place of the closing parenthesis in the exponent and has $i - d_1 - d_2$ in the target twist. Formulae (1.0.5) and (1.0.7), as well as the Gysin degree, give the reading displayed here.

[94] The print omits the prime on the right-hand differential in (1.0.9). The definition of the shift of the primed spectral sequence requires the primed differential.

1.0.10.

A morphism of spectral sequences from $(E_r^{p,q}, d_r^{p,q})$ to $({}''E_r^{p,q}, {}''d_r^{p,q})$ is called a morphism of bidegree (a,b) from $(E_r^{p,q}, d_r^{p,q})$ to $({}'E_r^{p,q}, {}'d_r^{p,q})$.

The following lemma is an immediate consequence of the theory of the Gysin morphism (SGA 5, Exposé IV).

Lemma 1.1.

There is a canonical morphism π_* of bidegree $(0, -2(d_1 - d_2))$[95] between the Leray spectral sequences

$$R^p\alpha_* R^q f_* \mathbb{Q}_\ell(i) \Longrightarrow R^{p+q}(\alpha f)_* \mathbb{Q}_\ell(i) \tag{1.1.0}$$

and

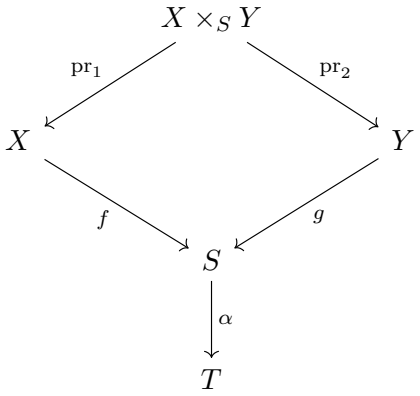
$$R^p\alpha_* R^q g_* \mathbb{Q}_\ell(i - d_1 + d_2) \Longrightarrow R^{p+q}(\alpha g)_* \mathbb{Q}_\ell(i - d_1 + d_2), \tag{1.1.1}$$

which is given on the E_2 term by (1.0.7), and on the abutment by (1.0.5).

1.2.

Now consider the situation

(1.2.1)



in which S, X, f, Y, g are as in (1.0.1)–(1.0.4), but now f is also assumed proper.[96] Let

$$z \in H^a(X \times_S Y, \mathbb{Q}_\ell(b)). \tag{1.2.2}$$

Then z induces a map, again denoted by z,

$$z : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1}(Y, \mathbb{Q}_\ell(i + b - d_1)) \tag{1.2.3}$$

by the formula

$$z(x) = \mathrm{pr}_{2*}\big(z \wedge \mathrm{pr}_1^*(x)\big), \tag{1.2.4}$$

where pr_{2*} is the Gysin homomorphism. By ètale localization on S, the class z likewise defines

$$z : R^q f_* \mathbb{Q}_\ell(i) \longrightarrow R^{q+a-2d_1} g_* \mathbb{Q}_\ell(i + b - d_1), \tag{1.2.5}$$

and, after applying $H^p(S, -)$, maps

$$H^p(S, R^q f_* \mathbb{Q}_\ell(i)) \longrightarrow H^p(S, R^{q+a-2d_1} g_* \mathbb{Q}_\ell(i + b - d_1)). \tag{1.2.6}$$

[95] The print has $(a, -2(d_1 - d_2))$. The map (1.0.7) leaves the first Leray index p unchanged, so the first component of the bidegree is 0.

[96] The print begins this range at the nonexistent (1.0.2); the displayed situation is (1.0.1).

Lemma 1.3.

The class z defines a morphism of bidegree $(0, a-2d_1)$[97] between the Leray spectral sequences

$$H^p(S, R^q f_*\mathbb{Q}_\ell(i)) \Longrightarrow H^{p+q}(X, \mathbb{Q}_\ell(i)) \tag{1.2.7}$$

and

$$H^p(S, R^q g_*\mathbb{Q}_\ell(i+b-d_1)) \Longrightarrow H^{p+q}(Y, \mathbb{Q}_\ell(i+b-d_1)), \tag{1.2.8}$$

which is given on the E_2 terms by (1.2.6), and on the abutments by (1.2.3).

Proof. In view of (1.2.4), and since "multiplication by z" defines a morphism of bidegree $(0, a)$ between the Leray spectral sequences

$$H^p\big(S, R^q(f\circ \mathrm{pr}_1)_*\mathbb{Q}_\ell(i)\big) \Longrightarrow H^{p+q}(X \times_S Y, \mathbb{Q}_\ell(i)) \tag{1.2.9}$$

$$\Big\downarrow z$$

$$H^p\big(S, R^{q+a}(f\circ \mathrm{pr}_1)_*\mathbb{Q}_\ell(i+b)\big) \Longrightarrow H^{p+q+a}(X \times_S Y, \mathbb{Q}_\ell(i+b)), \tag{1.2.10}$$

it suffices to apply Lemma 1.1 to the situation[98]

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{\quad\mathrm{pr}_2\quad} & Y \\ & {\scriptstyle f\circ \mathrm{pr}_1}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array} \qquad . \tag{1.2.11}$$

Corollary 1.4.

In the situation (1.2.1)–(1.2.2), the morphism (1.2.3)

$$z : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1}(Y, \mathbb{Q}_\ell(i+b-d_1)) \tag{1.4.1}$$

takes primitive classes (XVIII 5.8.1) to primitive classes, and the following diagram is commutative:

$$\begin{array}{ccc} \mathrm{Prim}^q(X/S, \mathbb{Q}_\ell(i)) & \xrightarrow{\ z\ } & \mathrm{Prim}^{q+a-2d_1}(Y/S, \mathbb{Q}_\ell(i+b-d_1)) \\ \Big\downarrow{\scriptstyle u_f} & & \Big\downarrow{\scriptstyle u_g} \\ H^1(S, R^{q-1}f_*\mathbb{Q}_\ell(i)) & \xrightarrow{\ z\ } & H^1(S, R^{q+a-2d_1-1}g_*\mathbb{Q}_\ell(i+b-d_1)). \end{array} \tag{1.4.2}$$

Here u_f and u_g are the canonical homomorphisms of XVIII 5.8.1.

1.5.

Let $\bar\eta$ be a geometric point of S (lying over a point η of S), and let

$$z_{\bar\eta} \in H^a\big(X_{\bar\eta} \times_{k(\bar\eta)} Y_{\bar\eta}, \mathbb{Q}_\ell(b)\big) \tag{1.5.1}$$

be the class induced by z on the corresponding geometric fiber of

$$X \times_S Y \longrightarrow S. \tag{1.5.2}$$

[97] The print has $(0, b-2d_1)$. The maps (1.2.3), (1.2.5), and (1.2.6) change cohomological degree by $a-2d_1$; b controls the Tate twist.

[98] In (1.2.9) and (1.2.10) the print writes $(\mathrm{pr}_1 f)_*$, whereas the map to S is $f\circ\mathrm{pr}_1$, as the printed diagram (1.2.11) itself shows.

Proposition 1.5.3.

Suppose that f and g are proper, that S is connected,[99] and let $q \in \mathbb{Z}$. If the class $z_{\bar{\eta}}$ induces the zero homomorphism

$$0 = z_{\bar{\eta}} : H^{q-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1-1}(Y_{\bar{\eta}}, \mathbb{Q}_\ell(i+b-d_1)), \tag{1.5.3}$$

then, for every $x \in \mathrm{Prim}^q(X/S, \mathbb{Q}_\ell(i))$, one has

$$u_g(z(x)) = 0 \quad \text{in} \quad H^1\big(S, R^{q+a-2d_1-1}g_*\mathbb{Q}_\ell(i+b-d_1)\big). \tag{1.5.4}$$

Proof. By the commutativity of (1.4.2), it is enough to see that z induces the zero homomorphism

$$z : R^{q-1}f_*\mathbb{Q}_\ell(i) \xrightarrow{0} R^{q+a-2d_1-1}g_*\mathbb{Q}_\ell(i+b-d_1). \tag{1.5.5}$$

Indeed, since f and g are proper and smooth, the sheaves $R^\bullet f_*\mathbb{Q}_\ell(\cdot)$ and $R^\bullet g_*\mathbb{Q}_\ell(\cdot)$ are twisted constant sheaves (SGA 4, XVI 2.2), hence are equivalent to their values at $\bar{\eta}$, regarded as modules under $\pi_1(S, \bar{\eta})$.

1.6. A construction of Manin

Suppose that two sections of $f : X \longrightarrow S$ are given:

$$S \underset{t_2}{\overset{t_1}{\leftarrow}} X \quad , \tag{1.6.0}$$

and associate with each its cohomology class

$$\mathrm{cl}(t_i) = t_{i*}(1) \in H^{2d_1}(X, \mathbb{Q}_\ell(d_1)). \tag{1.6.1}$$

If the fibers of $f : X \longrightarrow S$ are geometrically connected (more generally, if, for $s \in S$, $t_1(s)$ and $t_2(s)$ belong to the same connected component of X_s), it is clear that[100]

$$\mathrm{cl}(t_1) - \mathrm{cl}(t_2) \in \mathrm{Prim}^{2d_1}(X/S, \mathbb{Q}_\ell(d_1)). \tag{1.6.2}$$

We denote by

$$u_f(t_1 - t_2) \in H^1(S, R^{2d_1-1}f_*\mathbb{Q}_\ell(d_1)) \tag{1.6.3}$$

its image under

$$u_f : \mathrm{Prim}^{2d_1}(X/S, \mathbb{Q}_\ell(d_1)) \longrightarrow H^1(S, R^{2d_1-1}f_*\mathbb{Q}_\ell(d_1)). \tag{1.6.4}$$

(It is an analogue $\mu(t_1 - t_2)$ of this element that Manin studies in [5].)

[99] The print has the nonsensical adjective "annexe" in place of "connexe", and uses the capital X for the element in the conclusion. The proof requires connectedness of S, while (1.5.4) itself uses the element x.

[100] The print says $f(s)$ and $g(s)$ here, although the two maps under discussion are the sections t_1 and t_2.

1.6.5.

Now let z be an algebraic cycle on $X \times_S Y$, of codimension at least b on each fiber of $X \times_S Y$,[101] which determines a cohomology class (SGA 5, Exposé IV; the further citation is blank in the print)

$$\mathrm{cl}(z) \in H^{2b}(X \times_S Y, \mathbb{Q}_\ell(b)). \tag{1.6.6}$$

This class defines a map (cf. (1.2.3))

$$\mathrm{cl}(z) \cdot H^{2d_1}(X, \mathbb{Q}_\ell(d_1)) \longrightarrow H^{2b}(Y, \mathbb{Q}_\ell(b)), \tag{1.6.7}$$

and, if the cycles $z(t_1)$ and $z(t_2)$ are defined, the compatibility of the cup product with intersections of cycles (SGA 5, Exposé IV; the further citation is blank in the print) gives

$$\mathrm{cl}(z)\big(\mathrm{cl}(t_1) - \mathrm{cl}(t_2)\big) = \mathrm{cl}\big(z(t_1) - z(t_2)\big). \tag{1.6.8}$$

The commutativity of (1.4.2) gives us

$$\mathrm{cl}(z)\big(u_f(t_1 - t_2)\big) = u_g\big(\mathrm{cl}(z(t_1) - z(t_2))\big). \tag{1.6.9}$$

As a corollary of Proposition 1.5.3, we therefore have:

Proposition 1.7.

Let t_1 and t_2 be two sections of $f : X \longrightarrow S$, with f assumed to have connected fibers, and let z be an algebraic cycle on $X \times_S Y$ of codimension b, such that the algebraic cycle

$$z(t_1) - z(t_2) \tag{1.7.1}$$

is defined. Then its cohomology class is primitive (XVIII 5.8.1). If the algebraic cycle $z_{\bar\eta}$ on $X_{\bar\eta} \times Y_{\bar\eta}$ induces the zero homomorphism

$$z_{\bar\eta} : H^{2d_1-1}(X_{\bar\eta}, \mathbb{Q}_\ell(d_1)) \xrightarrow{0} H^{2b-1}(Y_{\bar\eta}, \mathbb{Q}_\ell(b)), \tag{1.7.2}$$

then

$$u_g\big(z(t_1) - z(t_2)\big) = 0 \quad \text{in} \quad H^1(S, R^{2b-1}g_*\mathbb{Q}_\ell(b)), \tag{1.7.3}$$

where u_g is the homomorphism of XVIII 5.8.1 occurring in 1.4. [102]

2. A reminder on level

(cf. [2], 9.7 and 10.1).

2.0.1.

Let X be proper and smooth over an algebraically closed field k. We saw in (1.2.3) that, for every

$$T \quad \text{proper, smooth, and connected over } k, \text{ of dimension } d, \tag{2.0.2}$$

and every

$$\text{algebraic cycle } z \text{ of codimension } b \text{ on } T \times X, \tag{2.0.3}$$

[101] If S is smooth over a scheme T (for example $T = \mathrm{Spec}\, k$, with k a field), it would be enough to suppose that z has codimension at least b on each fiber over T.

[102] The sentence in the print uses an upper-case $Z_{\bar\eta}$, whereas (1.7.2) and the cycle used throughout the section are denoted by $z_{\bar\eta}$.

the cohomology class

$$\mathrm{cl}(z) \in H^{2b}(T \times X, \mathbb{Q}_\ell(b)) \tag{2.0.4}$$

gives rise to maps

$$\mathrm{cl}(z) : H^q(T, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+2b-2d}(X, \mathbb{Q}_\ell(i+b-d)). \tag{2.0.5}$$

The coniveau filtration (N^j) on $H^q(X, \mathbb{Q}_\ell(i))$ is defined by

$$N^j H^q(X, \mathbb{Q}_\ell(i)) = \sum_{(T,z)} \mathrm{Im}\big(\mathrm{cl}(z) : H^{q-2j}(T, \mathbb{Q}_\ell(i-j)) \longrightarrow H^q(X, \mathbb{Q}_\ell(i))\big)\,, \tag{2.0.6}$$

where the sum extends over the pairs

$$(T, z), \qquad \begin{array}{l} T/k \text{ proper, smooth, and connected, of dimension } d, \\ z \text{ an algebraic cycle on } T \times X \text{ of codimension } j+d. \end{array} \tag{2.0.7}$$

It should be noted that one sometimes speaks of the increasing filtration F^j "by level", defined by

$$F^j H^q(X, \mathbb{Q}_\ell(i)) = N^{\left[\frac{q-j}{2}\right]} H^q(X, \mathbb{Q}_\ell(i)). \tag{2.0.8}$$

Replacing (T, z) by $(\mathbb{P}^1 \times T, \{\mathrm{pt}\} \times z)$, one sees that

$$N^j H^q(X, \mathbb{Q}_\ell(i)) \subset N^{j-1} H^q(X, \mathbb{Q}_\ell(i)). \tag{2.0.9}$$

Obviously, one has:

Proposition 2.1.

Let X be proper, smooth, and geometrically connected over a field k, let $\bar{k}$ be an algebraic closure of k, and put $X_{\bar{k}} = X \otimes_k \bar{k}$. Then $\mathrm{Gal}(\bar{k}/k)$, acting on $H^q(X_{\bar{k}}, \mathbb{Q}_\ell(i))$, preserves the coniveau filtration.

Standard arguments (cf. [2]) prove the following statements.

Proposition 2.2.

The level filtration is invariant under an extension k'/k of algebraically closed fields.

Proposition 2.3.

Let S be a reduced and irreducible scheme, with generic point η, and let

$$\pi : X \longrightarrow S$$

be a proper and smooth morphism with geometrically connected fibers. If, for given integers j and q, one has on the geometric generic fiber $X_{\bar{\eta}}$

$$H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell) = N^j H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell), \tag{2.3.1}$$

then there exists a nonempty open subset $U \subset S$ such that, at every point $u \in U$, one has on the geometric fiber $X_{\bar{u}}$

$$H^q(X_{\bar{u}}, \mathbb{Q}_\ell) = N^j H^q(X_{\bar{u}}, \mathbb{Q}_\ell). \tag{2.3.2}$$

3. Application to Lefschetz pencils

3.1.

From now on, we fix

(3.1.0) $$k, \quad \text{an algebraically closed field,}$$

and

(3.1.1) $$X \subset \mathbb{P}^r, \quad \text{a projective, smooth, connected } k\text{-scheme of even dimension } n, \text{ satisfying (LV) (XVIII 5.2.2).}$$

Take

(3.1.2) $$D = \{H_t\}_{t\in\mathbb{P}^1}, \quad \text{a Lefschetz pencil of hypersurfaces of degree } d,$$

with generic fiber

(3.1.3) $$X_\eta, \qquad \eta = \text{the generic point of } \mathbb{P}^1.$$

Put (cf. XVIII 5.4.4)

(3.1.4) $$m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta), \quad \text{the inclusion,}$$

and

(3.1.5) $$E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = \big(m^* H^{n-1}(X, \mathbb{Q}_\ell(i))\big)^\perp \subset H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)).$$

We saw in XVIII 6.7.2 that the Picard–Lefschetz formula implies[103]

(3.1.6) $$\text{If } n \text{ is even, every finite-index subgroup of } \mathrm{Gal}(k(\bar\eta)/k(\eta)) \text{ acts irreducibly on } E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)).$$

Proposition 3.2.

Let $D = \{H_t\}$ be a Lefschetz pencil, and suppose that $n = 2m$ is even. Then either

(3.2.1) $$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \subset m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + N^{m-1} H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)),$$

or else:

(3.2.2)

For every curve C proper and smooth over $k(\bar\eta)$ and every algebraic cycle z on $C \times X_{\bar\eta}$ of codimension m, the induced homomorphism (1.2.3)

(3.2.3) $$\mathrm{cl}(z) : H^1(C, \mathbb{Q}_\ell(1 + i - m)) \longrightarrow H^{2m-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i))$$

has its image in $m^* H^{2m-1}(X, \mathbb{Q}_\ell(i))$.

Proof. Suppose that the image of such a homomorphism (3.2.3) is not contained in $m^* H^{n-1}(X, \mathbb{Q}_\ell(i))$, or, equivalently, that its image in $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i))$ is nonzero. This image is evidently stable under the action of a finite-index subgroup of $\mathrm{Gal}(k(\bar\eta)/k(\eta))$. The irreducibility (3.1.6) of $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i))$ therefore says that

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + \mathrm{Im}(3.2.3).$$

[103] The print has E^{2n-1} in (3.1.6). The space defined in (3.1.5), and used throughout the proof of Proposition 3.2, is E^{n-1}.

By definition,

$$\mathrm{Im}(3.2.3) = \mathrm{cl}(z)\big(H^1(C, \mathbb{Q}_\ell(1+i-m))\big) \subset N^{m-1}H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)).$$

Consequently,

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = m^*H^{n-1}(X, \mathbb{Q}_\ell(i)) + N^{m-1}H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)),$$

which is precisely the first alternative.

3.3. Condition (B)

3.3.1. We shall say that the inclusion

$$m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta)$$

satisfies condition (B) if there exists

(3.3.2) $\qquad V_{\bar\eta}$, an algebraic cycle of codimension $n-1$ on $X_{\bar\eta} \times \big(X \times_k k(\bar\eta)\big)$,

such that the map it defines

(3.3.3) $$V_{\bar\eta} : H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{n-1}(X, \mathbb{Q}_\ell(i))$$

is a left inverse of

(3.3.4) $$m^* : H^{n-1}(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)),$$

that is, such that

(3.3.5) $$V_{\bar\eta}\, m^* = \mathrm{id}.$$

3.4.

Condition (B) is satisfied if the operation $\wedge$ of [4, 1.4.2.1] is algebraic (cf. [4, 2.12(ii)]). It is conjectured that $\wedge$ is always algebraic, and hence that condition (B) always holds; this is known in many cases [4]. It holds trivially when X is a complete intersection of even dimension, since in that case we know (IX 1.6) that

(3.4.1) $$H^{n-1}(X, \mathbb{Q}_\ell(i)) = 0.$$

4. Griffiths's theorem

Theorem 4.1.

Let X be projective, smooth, and connected over k, of even dimension $n = 2m \geq 4$. Let D be a Lefschetz pencil of sections by hypersurfaces of degree d. Assume that:

(4.1.0) X satisfies (LV) (XVIII 5.2.2);
(4.1.1) the pencil D satisfies hypothesis (A) (XVIII 5.3);
(4.1.2) the pencil D satisfies alternative (3.2.2), that is, in the setting of Proposition 3.2, the vanishing cohomology $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell)$ is not of level ≤ 1;
(4.1.3) the inclusion $m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta)$ satisfies hypothesis (B) (3.3).[104]

[104] The print writes m^* before this geometric inclusion. Condition (B) in 3.3 is imposed on m; m^* denotes its induced map in cohomology.

Then, if an algebraic cycle z of codimension m on X has a restriction to $X_{\bar\eta}$ which, as an algebraic cycle, satisfies

(4.1.4) $\qquad z_{\bar\eta} = z|_{X_{\bar\eta}}$ is algebraically equivalent to zero,

then

(4.1.5) $\qquad \mathrm{cl}(z) = 0 \quad$ in $H^{2m}(X, \mathbb{Q}_\ell(m))$;

in other words, z is $\mathbb{Q}_\ell$-homologous to zero.

4.2. A remark on the case where X is a complete intersection

As already noted (cf. XVIII 5.2.5 and (3.4)), hypotheses (4.1.0) and (4.1.3) then hold automatically. We saw in XVIII 6.3.4 and 6.4 that hypothesis (4.1.1) holds for a pencil of hypersurfaces of degree $d > 2n$. Moreover, in the next exposé we shall prove condition (4.1.2) for a "general" pencil of hypersurfaces of degree $d > 2n$ (XXI 5.2). Consequently, a "general" pencil of hypersurfaces of degree $d > 2n$ satisfies all the hypotheses of Theorem 4.1.

4.3.

Returning to the case of arbitrary X, it seems very plausible that, for d sufficiently large, all the conditions of 4.1 hold for D, at least when D is sufficiently general. We have already seen this for (4.1.1) (XVIII 6.4), and we noted the conjecture that (4.1.0) and (4.1.3) always hold (with (4.1.0) known in every case when $\mathrm{char}(k) = 0$). As for condition (4.1.2), Hodge theory proves it for large d when $\mathrm{char}(k) = 0$,[105] while heuristic arguments, given in the appendix to the next exposé, make the same conclusion plausible when $\mathrm{char}(k) > 0$.

More generally, these arguments—Hodge theory and, respectively, crystalline cohomology—prove, and respectively make plausible, that for a smooth projective variety X of dimension n, and for every Lefschetz pencil D of hypersurfaces of sufficiently large degree d (respectively, every sufficiently general pencil D of hypersurfaces of degree d), the vanishing cohomology $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell)$ has level exactly $n-1$; that is, it is not of level $\leq n-3$, or equivalently is not of coniveau 1.

4.4. Proof of Griffiths's theorem

Let z be an algebraic cycle of codimension m on X such that

(4.4.1) $\qquad z_{\bar\eta} = z|_{X_{\bar\eta}}$ is algebraically equivalent to zero.

By definition, there exist

(4.4.2) $\qquad$ a proper, smooth, connected curve $C_{\bar\eta}$ over $k(\bar\eta)$,

(4.4.3) $\qquad t_i(\bar\eta) : \mathrm{Spec}\, k(\bar\eta) \longrightarrow C_{\bar\eta}, \qquad i = 1, 2,$

and

(4.4.4) $\qquad$ an algebraic cycle $P_{\bar\eta}$ of codimension m on $C_{\bar\eta} \times X_{\bar\eta}$,

such that

(4.4.5) $\qquad z_{\bar\eta} = P_{\bar\eta}\big(t_1(\bar\eta)\big) - P_{\bar\eta}\big(t_2(\bar\eta)\big) \quad$ on $X_{\bar\eta}$.

[105] The print has $\mathrm{char}(k) \neq 0$ in this Hodge-theoretic clause. The following clause treats positive characteristic, so the required reading here is $\mathrm{char}(k) = 0$.

The print writes target C rather than $C_{\bar\eta}$ in (4.4.3), and both above and in (4.4.4) it writes codimension n rather than m; the defined curve, Theorem 4.1, and the degree shift below force the readings used here.

Regard $P_{\bar\eta}$ as an algebraic correspondence from $C_{\bar\eta}$ to $X_{\bar\eta}$. It defines a cohomological correspondence, again denoted by $P_{\bar\eta}$, or equivalently a homomorphism

$$P_{\bar\eta} : H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)). \tag{4.4.6}$$

The print omits the final closing parenthesis in the target of (4.4.6).

Likewise, consider the algebraic correspondence $V_{\bar\eta}$ of (3.3.2), from $X_{\bar\eta}$ to $X \otimes_k k(\bar\eta)$, supplied by hypothesis (B) in (4.1.3). It defines

$$V_{\bar\eta} : H^*(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)) \longrightarrow H^*(X \otimes_k k(\bar\eta), \mathbb{Q}_\ell(i+m-1)). \tag{4.4.7}$$

The composite algebraic correspondence $Q_{\bar\eta} = V_{\bar\eta} \circ P_{\bar\eta}$ from $C_{\bar\eta}$ to $X \otimes_k k(\bar\eta)$ therefore defines

$$Q_{\bar\eta} = V_{\bar\eta} \circ P_{\bar\eta} : H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}(X \otimes_k k(\bar\eta), \mathbb{Q}_\ell(i+m-1)). \tag{4.4.8}$$

Consider the inclusion

$$m : X_{\bar\eta} \longrightarrow X \otimes_k k(\bar\eta).$$

It gives the algebraic correspondence m^* from $X \otimes_k k(\bar\eta)$ to $X_{\bar\eta}$, hence the composite correspondence $m^*Q_{\bar\eta}$ from $C_{\bar\eta}$ to $X_{\bar\eta}$, defining

$$m^*Q_{\bar\eta} = m^*V_{\bar\eta}P_{\bar\eta} : H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)). \tag{4.4.9}$$

The print repeats an extraneous $\bar\eta$ after $m^*Q_{\bar\eta}$ in the preceding sentence.

Now consider the algebraic correspondence

$$\begin{aligned} R_{\bar\eta} = P_{\bar\eta} - m^*Q_{\bar\eta} = (\mathrm{id} - m^*V_{\bar\eta})P_{\bar\eta} :& \\ H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}&(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)). \end{aligned} \tag{4.4.10}$$

Since $V_{\bar\eta}$ is a left inverse of m^*, $\mathrm{id} - m^*V_{\bar\eta}$ is a projector on $H^*(X_{\bar\eta}, -)$ whose image is a complement of

$$\mathrm{Im}\big(m^* : H^*(X \otimes_k k(\bar\eta), -) \longrightarrow H^*(X_{\bar\eta}, -)\big).$$

By hypothesis (4.1.2), the image of

$$R_{\bar\eta} : H^1(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{2m-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)) \tag{4.4.11}$$

lies in $\mathrm{Im}\, m^*$. Consequently,

(4.4.12) the homomorphism (4.4.11) is zero.

Choose a finite extension

$$L/k(\eta)$$

over which the objects

$$C_{\bar\eta}, \quad t_1(\bar\eta), \quad t_2(\bar\eta), \quad Q_{\bar\eta}, \quad R_{\bar\eta}$$

are already defined, and let U be a nonempty open subset of the normalization of $\mathbb{P}^1$ in L. After shrinking U, there is a proper smooth relative curve

$$C \longrightarrow U$$

with geometrically connected fibers and geometric generic fiber $C_{\bar\eta}$, together with two sections

$$t_i : U \longrightarrow C, \qquad i = 1, 2,$$

whose values at $\bar\eta$ are $t_i(\bar\eta)$.

Let

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

be the projection of XVIII 3.1.4 whose fibers are the members of the pencil, put

$$\widetilde{X}_U = \widetilde{X} \times_{\mathbb{P}^1} U,$$

and let

$$M : \widetilde{X}_U \hookrightarrow X \times U$$

be the canonical immersion into the constant family over U with fiber X. Set

$$z_U = M^* \operatorname{pr}_1^*(z),$$

the algebraic cycle on $\widetilde{X}_U$ induced by z.

After shrinking U once more, there are an algebraic cycle R on $C \times_U \widetilde{X}_U$ whose value on $C_{\bar\eta} \times X_{\bar\eta}$ is $R_{\bar\eta}$, and an algebraic cycle Q on $C \times X \times U$ whose value on $C_{\bar\eta} \times (X \otimes_k k(\bar\eta))$ is $Q_{\bar\eta}$, such that

$$z_U = R(t_1) - R(t_2) + (\mathrm{id} \times M)^*\big[Q(t_1) - Q(t_2)\big]. \tag{4.4.13}$$

To prove the theorem it suffices, by the injectivity of the Griffiths homomorphism "modulo the fixed part" (XVIII 5.8.7),

$$\operatorname{Prim}^{2m}(X, \mathbb{Q}_\ell(m)) \xrightarrow{\text{grif}} \frac{H^1(U, R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m))}{M^*H^1(U, H^{2m-1}(X, \mathbb{Q}_\ell(m)))},$$

to show that the image of z_U under

$$u_0 : \operatorname{Prim}^{2m}(\widetilde{X}_U/U, \mathbb{Q}_\ell(m)) \longrightarrow H^1(U, R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m))$$

lies in $M^*H^1(U, H^{2m-1}(X, \mathbb{Q}_\ell(m)))$. The print gives the twist n in the domain of the Griffiths map; the codimension and all adjacent terms require m.

In view of (4.4.13) and the commutativity of (1.4.2), we have the commutative diagram

$$\begin{array}{ccccc}
\operatorname{Prim}^2(C/U, \mathbb{Q}_\ell(1)) & \xrightarrow{Q} & \operatorname{Prim}^{2m}(X \times U/U, \mathbb{Q}_\ell(m)) & \xrightarrow{M^*} & \operatorname{Prim}^{2m}(\widetilde{X}_U/U, \mathbb{Q}_\ell(m)) \\
 & & \downarrow{\scriptstyle u_{\mathrm{pr}_2}} & & \downarrow{\scriptstyle u_0} \\
 & & H^1(U, H^{2m-1}(X, \mathbb{Q}_\ell(m))) & \xrightarrow{M^*} & H^1(U, R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m)),
\end{array}$$

so it is enough to prove

$$u_0\big(R(t_1) - R(t_2)\big) = 0 \quad \text{in} \quad H^1(U, R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m)).$$

This follows from (4.4.12), by 1.7. The print has $2n-1$ and twist n in the last group and cites the nonexistent (4.3.12); the coherent readings are $2m-1$, twist m, and (4.4.12).

5. The Griffiths group

5.0.

Let S be proper and smooth over an algebraically closed field k. Denote by

(5.0.1) $Z^i(S)$ the group of algebraic cycles of codimension i on S;

(5.0.2) $Z^i_{\mathrm{alg}}(S)$ the subgroup of $Z^i(S)$ consisting of cycles algebraically equivalent to zero; and, for every prime number $\ell \neq \mathrm{char}(k)$,
(5.0.3) $Z^i_{\ell\text{-coh}}(S)$ the subgroup of $Z^i(S)$ consisting of cycles whose cohomology class in $H^{2i}(S, \mathbb{Q}_\ell(i))$ is zero.

It is conjectured in [4] that this group is independent of ℓ; this is known at least when $\mathrm{char}(k) = 0$. Set

$$\mathrm{Grif}^i_\ell(S) = Z^i_{\ell\text{-coh}}(S)/Z^i_{\mathrm{alg}}(S). \tag{5.0.4}$$

If, moreover, S is projective and is equipped with a projective embedding, set

(5.0.5) $Z^{i,\mathrm{prim}}(S)$ equal to the subgroup of $Z^i(S)$ consisting of primitive cycles (that is, cycles whose cohomology class is primitive), and
(5.0.6) $\mathrm{Prim}^{2i}_{\mathrm{alg}}(S, \mathbb{Q}_\ell(i))$ equal to the $\mathbb{Q}$-vector subspace of $H^{2i}(S, \mathbb{Q}_\ell(i))$ spanned by the cohomology classes of primitive algebraic cycles.

Corollary 5.1.

Under the hypotheses of Theorem 4.1, one has

$$\dim_{\mathbb{Q}}(\mathrm{Grif}^n_\ell(X_{\bar\eta}) \otimes_{\mathbb{Z}} \mathbb{Q}) \geq \dim_{\mathbb{Q}}\big(\mathrm{Prim}^{2n}_{\mathrm{alg}}(X, \mathbb{Q}_\ell(n))\big)\,. \tag{5.1.1}$$

Indeed, by (4.1.5), the kernel of the composite map

$$\mathbb{Q} \otimes_{\mathbb{Z}} Z^{n,\mathrm{prim}}(X) \xrightarrow{\text{res. to } X_{\bar\eta}} \mathbb{Q} \otimes_{\mathbb{Z}} Z^n_{\ell\text{-coh}}(X_{\bar\eta}) \longrightarrow \mathrm{Grif}^n_\ell(X_{\bar\eta})$$

is contained in $Z^n_{\ell\text{-coh}}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$. Consequently, $\mathrm{Prim}^{2n}_{\mathrm{alg}}(X, \mathbb{Q}_\ell(n))$ is a quotient of a subspace of $\mathrm{Grif}^n_\ell(X_{\bar\eta}) \otimes_{\mathbb{Z}} \mathbb{Q}$, which proves the assertion.

Remark 5.2.

For X a quadric hypersurface in $\mathbb{P}^{2m+1}$, one has

$$\dim_{\mathbb{Q}_\ell}\big(\mathrm{Prim}^{2m}_{\mathrm{alg}}(X, \mathbb{Q}_\ell(m))\big) = 1.$$

The print has twist n here and omits the outer closing parenthesis; the middle degree $2m$ forces the twist m.

In particular, when $m = 2$, for X a quadric in $\mathbb{P}^5$, the section of X by a general hypersurface of degree > 5 contains a curve which is $\mathbb{Q}_\ell$-homologous to zero but of which no nonzero multiple is algebraically equivalent to zero.[106]

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[1] P. Griffiths, "On the Periods of Certain Rational Integrals III," forthcoming in *Annals of Mathematics*.

[2] A. Grothendieck, "Le Groupe de Brauer III" (especially §§9–10), in *Dix Exposés sur la Cohomologie des Schémas*, North-Holland, 1968.

[3] J.-P. Jouanolou, *Cohomologie ℓ-adique*, thesis, Paris, 1969.

[4] S. Kleiman, "Algebraic Cycles and the Weil Conjectures," in *Dix Exposés sur la Cohomologie des Schémas*, North-Holland, 1968.

[5] J. Manin, "Rational Points of Algebraic Curves over Function Fields," *Translations of the American Mathematical Society*.

[106] The print inserts the literal parenthetical "(4.2)" at this point. Direct inspection of the native scan at 9000-dpi-equivalent enlargement confirms the glyphs; the numerical specialization from $\mathbb{P}^{2m+1}$ to $\mathbb{P}^5$ is $m = 2$.

EXPOSÉ XXI

THE LEVEL OF THE COHOMOLOGY OF COMPLETE INTERSECTIONS

by N. Katz

with an Appendix by P. Deligne (§5)

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0. Introduction

Let X be a smooth projective variety of dimension $n \geq 3$, let $Y(d)$ be a section by a hypersurface of degree d, and let $E^{n-1}(Y(d), \mathbb{Q}_\ell)$ be the "vanishing" cohomology of $Y(d)$ (XVIII 5.4.4). The aim of this exposé is to prove that, for d sufficiently large and $Y(d)$ sufficiently general, it is not true that the whole of $E^{n-1}(Y(d), \mathbb{Q}_\ell)$ "comes" from the groups H^i, $i < n-1$, of smooth proper algebraic varieties by means of algebraic correspondences. In characteristic zero this follows trivially from the Hodge bigrading on complex cohomology and from the fact that, in coherent cohomology,

$$\dim H^{n-1}(Y(d), \mathcal{O}_{Y(d)}) \longrightarrow \infty \qquad \text{as } d \longrightarrow \infty.$$

The print has the evident typographical forms *et de démontrer* and *suffisament*; the readings used above are *est de démontrer* and *suffisamment*.

In characteristic p, the method again relates the coherent group $H^{n-1}(Y(d), \mathcal{O}_{Y(d)})$ to the étale group $H^{n-1}_{\text{et}}(Y(d), \mathbb{Q}_\ell)$. To do so, one works over a finite base field. The two are then linked by the zeta function of $Y(d)$: this function is expressed in terms of ℓ-adic cohomology, while its "reduction modulo p" is expressed in terms of coherent cohomology (cf. XXII 3.9). For complete intersections this method works without difficulty, owing to the detailed knowledge of both kinds of cohomology (étale and coherent) in that case. It yields the main result, Theorem 4.2, as a consequence of a result on Hasse–Witt operations, Theorem 1.4 (see 1.0).

There are heuristic arguments due to A. Grothendieck, based on the yoga of crystalline cohomology, which make the general statement plausible for every smooth projective X, by essentially the same method.

Even in the special case of complete intersections, the proof makes essential use in 2.2 of Theorem 2.1 on the integrality of the eigenvalues of Frobenius, recently proved by P. Deligne and proved here in the Appendix (§5). The print calls this "Appendix I," while the title and the document structure identify Deligne's Appendix as §5.

1. The Hasse–Witt matrix of a complete intersection

Definition 1.0.

A proper scheme over a field k of characteristic exponent p is called *special* if the additive p-linear operation

$$\mathfrak{F}_p : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad g \longmapsto g^p, \tag{1.0.1}$$

induces a nilpotent operation

$$\mathfrak{F}_p : H^{\dim X}(X, \mathcal{O}_X) \longrightarrow H^{\dim X}(X, \mathcal{O}_X). \tag{1.0.2}$$

Equivalently, if $M = \dim_k H^{\dim X}(X, \mathcal{O}_X)$, the p^M-linear operation

$$\mathfrak{F}_p^M = \mathfrak{F}_{p^M} : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad g \longmapsto g^{p^M}, \tag{1.0.3}$$

annihilates $H^{\dim X}(X, \mathcal{O}_X)$.

Recall from [2] that, for every $i \in \mathbb{Z}$, the homomorphism

$$H^i(X, \mathcal{O}_X) \xrightarrow{\mathfrak{F}_p} H^i(X, \mathcal{O}_X)$$

induced by (1.0.1) is classically called the *Hasse–Witt homomorphism* on $H^i(X, \mathcal{O}_X)$. Notice that, when $p = 1$, saying that X is non-special means simply that $H^{\dim X}(X, \mathcal{O}_X) \neq 0$.

Theorem 1.1.

Fix a characteristic exponent p, a dimension $n \geq 1$, an integer $r \geq 1$, and a multidegree $(d_1, \ldots, d_r)$. If

$$\sum_{i=1}^{r} d_i \geq (n+r)+1, \tag{1.1.0}$$

then, among all complete intersections of dimension n in $\mathbb{P}^{n+r}$ having multidegree $(d_1, \ldots, d_r)$ in characteristic p, a "sufficiently general" member is non-special in the sense of 1.0.

Proof. Let

$$f : X_U \longrightarrow U \tag{1.1.1}$$

denote the "universal family" of complete intersections in $\mathbb{P}^{n+r}$ of multidegree $(d_1, \ldots, d_r)$ in characteristic p, together with their defining equations. Thus U is an open subset of a product of r projective spaces over $\mathbb{F}_p$, of impressive dimension.

The $\mathcal{O}_U$-module

$$R^n f_* \mathcal{O}_{X_U} \tag{1.1.2}$$

is locally free, and its formation commutes with every base change (XI 1.5(i)). By (1.1.0), its rank M is nonzero (XI 2.5). When $p = 1$, this proves the theorem for every complete intersection satisfying (1.1.0).

Suppose now that $p \geq 2$. The iterated Hasse–Witt operation

$$\mathfrak{F}_p^M = \mathfrak{F}_{p^M} : R^n f_* \mathcal{O}_{X_U} \longrightarrow R^n f_* \mathcal{O}_{X_U} \tag{1.1.3}$$

is p^M-linear over $\mathcal{O}_U$. Since $R^n f_* \mathcal{O}_{X_U}$ is locally free, (1.1.3) is represented locally on U by an (M, M)-matrix H with entries in $\mathcal{O}_U$. Consequently,

$$\{u \in U \mid X_u \text{ is non-special}\} \quad \text{is open in } U. \tag{1.1.4}$$

Indeed, this is the set of u for which $H(u) \neq 0$. Since U is irreducible, it is therefore enough to find one value of u for which X_u is non-special; we shall do so over a finite field.

Proposition 1.1.5.

Let X be a complete intersection over $\mathbb{F}_q$. If X has no $\mathbb{F}_q$-rational points, then X is non-special.

Proof. It is plainly enough to prove, for a complete intersection X, the implication

$$X \text{ special} \quad \Longrightarrow \quad \operatorname{card}\big(X(\mathbb{F}_q)\big) \equiv 1 \pmod{p}. \tag{1.1.5.1}$$

By the trace formula, proved in the following Exposé XXII, 3.2,

$$\operatorname{card}\big(X(\mathbb{F}_q)\big) \equiv \sum_{i \geq 0} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\big) \pmod{p}.$$

Now $H^i(X, \mathcal{O}_X) = 0$ unless $i = 0$ or $i = n = \dim X$, while $H^0(X, \mathcal{O}_X) \simeq \mathbb{F}_q$ (XI 1.5(iii)); hence this becomes

$$\operatorname{card}\big(X(\mathbb{F}_q)\big) \equiv 1 + (-1)^n \operatorname{tr}\big(\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)\big) \pmod{p}.$$

If X is special, the operation of $\mathfrak{F}_q$ on $H^n(X, \mathcal{O}_X)$ is nilpotent by definition and therefore has trace zero. This proves (1.1.5.1), and hence the proposition.

Source note. The diplomatic French retains the printed forms *parmi toute les*, *théroème*, *intersection complète vérifiante*, and *n'a pas des points*; the editorial French and the translation use the grammatically coherent readings recorded in the source-defect register.

Proposition 1.2.

For every power q of p, if

$$\sum_{i=1}^{r} d_i \geq n + r + 1,$$

there exists a complete intersection in $\mathbb{P}^{n+r}_{\mathbb{F}_q}$ of multidegree $(d_1, \ldots, d_r)$ having no $\mathbb{F}_q$-rational point.

Proof. Begin with the case $r = 1$. For every $d \geq n + 2$, we must find a homogeneous form of degree d,

$$H(X_1, \ldots, X_{n+2}) \in \mathbb{F}_q[X_1, \ldots, X_{n+2}], \tag{1.2.1}$$

such that

$$(\alpha_1, \ldots, \alpha_{n+2}) \in (\mathbb{F}_q)^{n+2}, \qquad H(\alpha_1, \ldots, \alpha_{n+2}) = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_{n+2} = 0. \tag{1.2.2}$$

Set $\mathbb{E} = \mathbb{F}_{q^d}$, and choose elements $\beta_1, \ldots, \beta_{n+2}$ of $\mathbb{E}$ that are linearly independent over $\mathbb{F}_q$. Take

$$H(X_1, \ldots, X_{n+2}) = N_{\mathbb{E}/\mathbb{F}_q}\left(\sum_{i=1}^{n+2} \beta_i X_i\right). \tag{1.2.3}$$

In the general case, since

$$\sum_{i=1}^{r} d_i \geq n + r + 1, \tag{1.2.4}$$

we can set $a_0 = 0$ and choose r integers

$$1 \leq a_1 < a_2 < \cdots < a_r = n + r + 1 \tag{1.2.5}$$

such that

$$d_i \geq a_i - a_{i-1} \qquad (1 \leq i \leq r). \tag{1.2.6}$$

For each $i = 1, \ldots, r$, choose a homogeneous form of degree d_i,

$$H_i(X_{a_{i-1}+1}, \ldots, X_{a_i}) \in \mathbb{F}_q[X_{a_{i-1}+1}, \ldots, X_{a_i}],$$

in the $a_i - a_{i-1}$ displayed variables and having property (1.2.2). The required complete intersection is defined by the homogeneous forms

$$H_1(X_1, \ldots, X_{a_1}), \quad \ldots, \quad H_r(X_{a_{r-1}+1}, \ldots, X_{a_r}).$$

Source note. The diplomatic French retains the printed misplaced parenthesis in (1.2.1), the literal bound $d_1 \geq a_o$ in (1.2.6), and the missing $+1$ in the terminal H_r block. Direct native-pixel inspection confirms these as print features; the construction forces the coherent forms displayed above.

Theorem 1.3.

Let X be an irreducible complete intersection of dimension $n \geq 1$ over a field k of characteristic exponent p. For every integer $d \geq n+1$, a sufficiently general section of X by a hypersurface of degree d is non-special in the sense of 1.0.

Proof. The sections $X.H_u$ of X by hypersurfaces of degree d are parametrized by the lines in $H^0(X, \mathcal{O}_X(d))$, that is, by

$$U = \mathbb{P}\big(H^0(X, \mathcal{O}_X(d))\big). \tag{1.3.1}$$

Over U we have the "universal section of X by a hypersurface of degree d,"

$$f : X.H_u \longrightarrow U. \tag{1.3.2}$$

Let U' be the open subset of those $u \in U$ for which $\dim(X_u) = n-1$. By XI 1.5(i) and 2.5, the restriction

$$R^{n-1} f_* \mathcal{O}_{X.H_u}\big|_{U'} \tag{1.3.3}$$

is locally free, its formation commutes with every base change, and its rank M is at least one. If $p = 1$, this proves the theorem for every section by a hypersurface of degree at least $n+1$.

Suppose therefore that $p > 0$. The iterated Hasse–Witt operation $\mathfrak{F}_{p^M}$ is again represented locally by an $M \times M$ matrix with entries in $\mathcal{O}_U$. Consequently,

$$\{u \in U' \mid X.H_u \text{ is non-special}\} \quad \text{is open in } U'. \tag{1.3.4}$$

It remains only to prove that the set in (1.3.4) is nonempty.

First consider the case

$$k = \mathbb{F}_q. \tag{1.3.5}$$

After replacing $\mathbb{F}_q$ by a finite extension, we may suppose that

$$\operatorname{card}\big(X(\mathbb{F}_q)\big) > 0 \tag{1.3.6}$$

and that

$$\begin{aligned} &\text{there are homogeneous coordinates} \\ &\quad X_1, \ldots, X_{n+r+1} \text{ in the ambient space } \mathbb{P}_k^{n+r} \\ &\text{such that } X \text{ does not meet the linear subvariety} \\ &\quad X_1 = \cdots = X_{n+1} = 0. \end{aligned} \tag{1.3.7}$$

Choose a homogeneous form of degree d,

$$H(X_1, \ldots, X_{n+1}) \in \mathbb{F}_q[X_1, \ldots, X_{n+1}], \tag{1.3.8}$$

having property (1.2.2), which is possible because $d \geq n+1$. Let H also denote the hypersurface in $\mathbb{P}^{n+r}$ defined by $H(X_1, \ldots, X_{n+1}) = 0$. By construction, $X.H$ has no $\mathbb{F}_q$-rational point. In view of (1.3.6), this also shows that $X \not\subset H$; since X is irreducible, $X.H$ has dimension $n-1$, so $H \in U'$. Proposition 1.1.5 now shows that $X.H$ is non-special.

Source note. The diplomatic French retains the printed exponent p^n, the base sheaf $\mathcal{O}_u$, and the shortened citation "(1.5)." The rank-M nilpotence test, the base U, and Proposition 1.1.5 force the coherent readings used above. The closing parenthesis in (1.3.1), absent only from the inherited TeX, has been restored directly from the scan.

We now pass to the general case. Looking at the equations defining X, we may suppose that k is finitely generated over $\mathbb{F}_p$. There is then a subring $B \subset k$, finitely generated as an algebra over $\mathbb{F}_p$ and having fraction field k, such that

$$X \subset \mathbb{P}^{n+r}_k \tag{1.3.9}$$

is the generic fiber of a relative complete intersection $\widetilde{X} \subset \mathbb{P}^{n+r}_B$ which, fiber by fiber, has the same multidegree as X and is geometrically irreducible.

Write

$$\rho : \widetilde{X} \longrightarrow S = \mathrm{Spec}(B) \tag{1.3.10}$$

for the projection,

$$E = \rho_* \mathcal{O}_{\widetilde{X}}(d) \tag{1.3.11}$$

for the locally free $\mathcal{O}_S$-module, and

$$\pi : T = \check{\mathbb{P}}(E) \longrightarrow S \tag{1.3.12}$$

for the associated bundle of lines in E. The generic fiber of (1.3.12) is precisely

$$U = \mathbb{P}\big(H^0(X, \mathcal{O}_X(d))\big).$$

Over T we have the universal section of $\widetilde{X} \subset \mathbb{P}^{n+r}_B$ by hypersurfaces of degree d, denoted

$$\alpha : \widetilde{X}.H_T \longrightarrow T. \tag{1.3.13}$$

These constructions fit into the following diagram of cartesian squares:

$$\begin{array}{ccc}
\widetilde{X}.H_T & \longleftarrow & X.H_u = (\widetilde{X}.H_T) \times_T U \\
\big\downarrow{\scriptstyle \alpha} & & \big\downarrow \\
T & \longleftarrow & U = T \times_S \mathrm{Sp}(k) \\
\big\downarrow{\scriptstyle \pi} & & \big\downarrow \\
S & \longleftarrow & \mathrm{Sp}(k)
\end{array} \tag{1.3.14}$$

The $\mathcal{O}_T$-module

$$R^{n-1}\alpha_* \mathcal{O}_{\widetilde{X}.H_T} \tag{1.3.15}$$

is locally free over the open subset $T' \subset T$ consisting of the t for which $\dim \alpha^{-1}(t) = n-1$, and its formation there commutes with every base change. Since T and U have the same generic point, it is enough, in order to show that (1.3.4) is nonempty, to prove that the open subset of T'

$$\big\{ t \in T' \mid \widetilde{X}_{\pi(t)}.H_t \text{ is non-special} \big\} \tag{1.3.16}$$

is nonempty. But the already established case (1.3.5) shows that, for every closed point $s \in S$, the fiber $\pi^{-1}(s)$ has nonempty intersection with the open set (1.3.16). This proves the theorem.

Source note. The diplomatic French retains the printed top-right object $(\widetilde{X}.H_T) \times_U T$ and the phrase *se formation*. The cartesian square and the definitions of T and its generic fiber U force $(\widetilde{X}.H_T) \times_T U$ and *sa formation*, as used above. The plain E, the exponent 0 in H^0, and the outer parenthesis defining U are transcription repairs made directly from the scan rather than editorial emendations.

Theorem 1.4.

Let $X \subset \mathbb{P}^N$ be a projective Cohen–Macaulay scheme, equidimensional of dimension $n \geq 1$, over a field k. There is an integer d_0 such that, for every $d \geq d_0$, if $i : X.H \longrightarrow X$ denotes the inclusion of a section by a general hypersurface of degree d, then the action of

$$\mathfrak{F}_p \quad \text{on} \quad \frac{H^{n-1}(X.H, \mathcal{O}_{X.H})}{i^* H^{n-1}(X, \mathcal{O}_X)} \tag{1.4.0}$$

is non-nilpotent.

Proof. As above, we reduce to the case $k = \mathbb{F}_q$ and to the problem of finding just one H—for any fixed sufficiently large degree for which $\dim H^{n-1}(X.H, \mathcal{O}_{X.H})$ is independent of H; see (1.4.4)—for which (1.4.0) holds. After enlarging q, we may suppose that the irreducible components X_i of X are geometrically irreducible, that

$$\operatorname{card}\big(X_i(\mathbb{F}_q)\big) \geq 2, \tag{1.4.1}$$

and that

$$\begin{aligned} &\text{there are projective coordinates } X_0, \dots, X_N \text{ such that} \\ &\quad X \text{ does not meet the linear subvariety } X_0 = \cdots = X_n = 0. \end{aligned} \tag{1.4.2}$$

By a vanishing theorem of Serre ([1], or SGA 2 XII 1.4), there is an integer d_1 such that, for every $d \geq d_1$,

$$H^i(X, \mathcal{O}_X(-d)) = 0 \qquad \text{if } i \neq n = \dim X. \tag{1.4.3}$$

Thus, if H is a hypersurface of degree $d \geq d_1$, then (1.4.3) gives

$$\begin{aligned} H^i(X, \mathcal{O}_X) &\xrightarrow{\sim} H^i(X.H, \mathcal{O}_{X.H}) \qquad \text{if } i \leq n-2, \\ H^{n-1}(X, \mathcal{O}_X) &\hookrightarrow H^{n-1}(X.H, \mathcal{O}_{X.H}) \quad \text{injectively.} \end{aligned} \tag{1.4.4}$$

We prove the theorem with $d_0 = \max(N+1, d_1)$. For every $d \geq d_0$, we must find a hypersurface H of degree d, defined over $\mathbb{F}_q$, which contains no X_i and for which

$$\operatorname{tr}\left(\mathfrak{F}_q \;\middle|\; \frac{H^{n-1}(X.H, \mathcal{O}_{X.H})}{H^{n-1}(X, \mathcal{O}_X)}\right) \neq 0. \tag{1.4.5}$$

By the trace formula (XX 3.2.1), modulo p we have

$$\begin{aligned} \operatorname{card}\big((X.H)(\mathbb{F}_q)\big) &\equiv \sum_{i=0}^{n-1} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X.H, \mathcal{O}_{X.H})\big) \\ &\overset{(1.4.4)}{\equiv} \sum_{i=0}^{n-1} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\big) \\ &\quad + (-1)^{n-1} \operatorname{tr}\left(\mathfrak{F}_q \;\middle|\; \frac{H^{n-1}(X.H, \mathcal{O}_{X.H})}{H^{n-1}(X, \mathcal{O}_X)}\right). \end{aligned} \tag{1.4.6}$$

Let $a \in \mathbb{Z}/p\mathbb{Z}$ denote the quantity

$$a = \sum_{i=0}^{n-1} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\big).$$

It is therefore enough, for every $d \geq d_0$, to find a hypersurface H of degree d, defined over $\mathbb{F}_q$, which does not contain X and such that

$$\operatorname{card}\big((X.H)(\mathbb{F}_q)\big) \not\equiv a \pmod{p}. \tag{1.4.7}$$

There are two cases. If $a \neq 0$, it is enough to take an H over $\mathbb{F}_q$ for which $(X.H)(\mathbb{F}_q)$ is empty, just as in (1.3.8). Such an H satisfies (1.4.7) and contains no X_i, because $\mathrm{card}(X_i(\mathbb{F}_q)) \geq 2$ by hypothesis.

If $a = 0$, it is enough to find an H such that $\mathrm{card}((X.H)(\mathbb{F}_q)) = 1$. Then (1.4.7) holds and $X_i \not\subset H$, since $\mathrm{card}(X_i(\mathbb{F}_q)) \geq 2$. Choose projective coordinates $X_0, \ldots, X_N$ such that $(1, 0, \ldots, 0)$ is a point of X. Take H to have equation $H(X_1, \ldots, X_N) = 0$, where H is a homogeneous form satisfying (cf. (1.2.2))

$$\alpha_1, \ldots, \alpha_N \in \mathbb{F}_q, \quad H(\alpha_1, \ldots, \alpha_N) = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_N = 0. \tag{1.4.8}$$

The hypersurface defined by H in $\mathbb{P}^N$ then has only the point $(1, 0, \ldots, 0)$ over $\mathbb{F}_q$; a fortiori, $X.H$ has only this point over $\mathbb{F}_q$. This proves the theorem.

Source note. The diplomatic French restores the scan's *coordonnées*, the two printed equality signs in (1.4.6), and the three-dot ellipsis in (1.4.8), all of which differ from the inherited transcription. The print uses the letter-like subscript *o* for the zero indices in d_o and X_o, says *une H défini*, and uses irregular point-count parentheses. The corrected French and English regularize these as d_0, X_0, masculine agreement, and ordinary evaluation notation without changing the argument.

2. Coniveau over a finite field of definition

2.0.

Let X be a separated scheme of finite type over $\mathbb{F}_q$. For every prime number $\ell \neq \mathrm{char}(\mathbb{F}_q)$, write

$$H_c^i(\overline{X}, \mathbb{Q}_\ell) \tag{2.0.1}$$

for the compactly supported cohomology groups of $\overline{X} = X \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q$. These are finite-dimensional $\mathbb{Q}_\ell$-vector spaces on which the Galois group $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ acts. Let

$$\varphi_q \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \tag{2.0.2}$$

denote the Frobenius automorphism. It is known (SGA 5 XIII 3.2.1) that the zeta function of X (XXII 3.0.1) is given by

$$Z(t, X/\mathbb{F}_q) = \prod_{i \geq 0} \det\bigl(1 - t\varphi_q^{-1} \mid H_c^i(\overline{X}, \mathbb{Q}_\ell)\bigr)^{(-1)^{i+1}}. \tag{2.0.3}$$

Deligne has recently proved, generalizing a result of Washnitzer for smooth projective X, the following theorem, which is proved in the appendix below (5.5.3).

Theorem 2.1.

The eigenvalues of φ_q^{-1} on $H_c^i(\overline{X}, \mathbb{Q}_\ell)$ are algebraic integers; if $i > \dim X$, they are divisible, as algebraic integers, by $q^{i - \dim X}$.

In particular, tautologically, we obtain:

(2.1 bis). The eigenvalues of φ_q^{-1} on $H_c^i(\overline{X}, \mathbb{Q}_\ell(-j))$ are algebraic integers divisible by q^j.

Corollary 2.2.

Let X be proper, smooth, and geometrically connected over $\mathbb{F}_q$. Then the eigenvalues of φ_q^{-1} on $N^j H^q(\overline{X}, \mathbb{Q}_\ell)$ (notation of XX 2.0.6) are algebraic integers divisible by q^j.

Proof. The assertion is invariant under a finite extension $\mathbb{F}_{q^M}/\mathbb{F}_q$, which amounts to replacing the eigenvalues under consideration by their M-th powers. After making such a base change,

the definition gives finitely many projective, smooth, geometrically connected schemes T_i over $\mathbb{F}_q$, of dimensions d_i, and algebraic cycles Z_i on $T_i \times_{\mathbb{F}_q} X$ of codimension $j + d_i$, such that the homomorphism of $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$-modules

$$\bigoplus_i H^{q-2j}(\overline{T}_i, \mathbb{Q}_\ell(-j)) \xrightarrow{\oplus_i \mathrm{Cl}(Z_i)} N^j H^q(\overline{X}, \mathbb{Q}_\ell)$$

is surjective. The conclusion follows from (2.1 bis).

Source note. The diplomatic French preserves the printed terms $\mathbb{Q}_\ell$-*vectoriels*, *fonction zêta ... donné*, the overstruck phrase following the extension $\mathbb{F}_{q^M}/\mathbb{F}_q$, the Galois group with a blank numerator, and the displayed domain without an explicit direct-sum sign. The corrected French and English use standard vector-space terminology and agreement, state the M-th-power operation plainly, restore $\overline{\mathbb{F}}_q$ in the Galois group, and display the direct sum required by $\oplus_i \mathrm{Cl}(Z_i)$.

3. Application to complete intersections

3.0.

Let X be a smooth complete intersection of dimension $n \geq 1$ in $\mathbb{P}^N$ over a field k. For $\ell \neq \mathrm{char}(k)$, weak Lefschetz and duality give (XI 1.6)

$$H^i(\overline{X}, \mathbb{Q}_\ell) = \begin{cases} 0 & \text{if } i \text{ is odd and } i \neq n, \\ \mathbb{Q}_\ell\Big(-\dfrac{i}{2}\Big) & \text{if } i \text{ is even, } i \leq 2n, \text{ and } i \neq n, \\ H^n(\overline{X}, \mathbb{Q}_\ell) & \text{if } i = n. \end{cases} \tag{3.0.1}$$

For $k = \mathbb{F}_q$ this gives the following formula for the zeta function:

$$Z(t, X/\mathbb{F}_q) \prod_{i=0}^{n} (1 - q^i t) = \det\big(1 - t\varphi_q^{-1} \mid \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell)\big)^{(-1)^{n+1}} . \tag{3.0.2}$$

Here we have used the primitive decomposition and (3.0.1).

Corollary 3.0.3.

Under the hypotheses of 3.0, with $k = \mathbb{F}_q$, there is a congruence modulo p

$$\det\big(1 - t\varphi_q^{-1} \mid H^n(\overline{X}, \mathbb{Q}_\ell)\big) \equiv \det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)) .$$

Proof. By the congruence formula (XXII 3.2.1), and since

$$H^i(X, \mathcal{O}_X) \simeq \begin{cases} \mathbb{F}_q & i = 0, \\ 0 & 1 \leq i \leq n-1, \\ 0 & i \geq n+1, \end{cases} \tag{3.0.4}$$

we have

$$Z(t, X/\mathbb{F}_q) \equiv (1-t)^{-1} \det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X))^{(-1)^{n+1}} \pmod{p}. \tag{3.0.5}$$

Consequently, (3.0.5) and (3.0.2) give

$$\det\big(1 - t\varphi_q^{-1} \mid \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell)\big) \equiv \det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)) \pmod{p}. \tag{3.0.6}$$

The desired conclusion follows because

$$H^n(\overline{X}, \mathbb{Q}_\ell) \simeq \begin{cases} \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell) & \text{if } n \text{ is odd}, \\ \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell) \oplus \mathbb{Q}_\ell\Big(-\dfrac{n}{2}\Big) & \text{if } n \text{ is even}. \end{cases} \tag{3.0.7}$$

Corollary 3.1.

Let X be a smooth complete intersection of dimension $n \geq 1$ over $\mathbb{F}_q$. If

$$H^n(\overline{X}, \mathbb{Q}_\ell) = N^1 H^n(\overline{X}, \mathbb{Q}_\ell), \tag{3.1.0}$$

then

$$\mathfrak{F}_p \quad \text{is nilpotent on} \quad H^n(X, \mathcal{O}_X). \tag{3.1.1}$$

Proof. By Corollary 2.2, (3.1.0) implies

$$\det\bigl(1 - t\varphi_q^{-1} \mid H^n(\overline{X}, \mathbb{Q}_\ell)\bigr) \equiv 1 \pmod{p} \qquad \text{(indeed, even modulo } q\text{)}. \tag{3.1.2}$$

Together with Corollary 3.0.3, this gives

$$\det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)) = 1. \tag{3.1.3}$$

Thus $\mathfrak{F}_q$ is nilpotent on $H^n(X, \mathcal{O}_X)$, and hence so is $\mathfrak{F}_p$. This proves the corollary.

Source note. The diplomatic French preserves the printed word order *intersection complète dans* $\mathbb{P}^N$ *lisse*, the typo *da décomposition primitive*, and the omitted index in $1 \leq \ \leq n-1$ in (3.0.4). The corrected French and English use normal word order, *la décomposition primitive*, and restore the forced index i in the cohomology range.

Proposition 3.2.

Let X be a smooth complete intersection of dimension $n \geq 1$ over a field k. If

$$H^n(\overline{X}, \mathbb{Q}_\ell) = N^1 H^n(\overline{X}, \mathbb{Q}_\ell), \tag{3.2.1}$$

then X is special; that is,

$$\mathfrak{F}_p \quad \text{is nilpotent on} \quad H^n(X, \mathcal{O}_X). \tag{3.2.2}$$

Proof. We may suppose that k is finitely generated over $\mathbb{F}_p$ (the case $p = 0$ being trivial), and that there is a finitely generated $\mathbb{F}_p$-subalgebra B of k such that X/k is the generic fiber of a smooth relative complete intersection $\widetilde{X}$:

$$\begin{array}{ccc} \widetilde{X} & \longleftarrow & X = \widetilde{X}_\eta \\ \Big\downarrow{\scriptstyle \pi} & & \Big\downarrow \\ \mathrm{Spec}(B) = S & \longleftarrow & \eta = \mathrm{Sp}(k)\ . \end{array} \tag{3.2.3}$$

The B-module $R^n\pi_*(\mathcal{O}_{\widetilde{X}})$ is locally free, and its formation commutes with every base change. Consequently the set of points

$$\{\, s \in S \mid X_s \text{ is special} \,\} \tag{3.2.4}$$

is closed in S.

By (XX 2.3), after shrinking S we may suppose that

$$\forall s \in S, \qquad H^n(X_{\bar{s}}, \mathbb{Q}_\ell) = N^1 H^n(X_{\bar{s}}, \mathbb{Q}_\ell). \tag{3.2.5}$$

By Corollary 3.1, X_s is special for every closed point s of S. It follows that the closed subset (3.2.4) is all of S and hence, in particular, that $X = \widetilde{X}_\eta$ is special.

Source note. In diagram (3.2.3) the diplomatic French preserves the printed $X = \widetilde{X}/_\eta$ notation. The corrected French and English use the standard generic-fiber notation $X = \widetilde{X}_\eta$, agreeing with the concluding sentence of the proof.

4. Conclusions

Combining Theorem 1.1 with Proposition 3.2 gives the following result.

Theorem 4.1.

Fix a characteristic p, a dimension n, and a multidegree $(d_1, \dots, d_r)$. Let

$$X_V \longrightarrow V \tag{4.1.0}$$

denote the universal smooth complete intersection of dimension n and multidegree $(d_1, \dots, d_r)$ in characteristic p. If

$$\sum_{i=1}^{r} d_i \geq n + r + 1, \tag{4.1.1}$$

then there is a dense open subset $U \subset V$ such that

$$\text{for every } u \in U, \qquad H^n(X_{\bar{u}}, \mathbb{Q}_\ell) \neq N^1 H^n(X_{\bar{u}}, \mathbb{Q}_\ell). \tag{4.1.2}$$

Combining Theorem 1.3 with Proposition 3.2 likewise gives the following.

Theorem 4.2.

Let X be a smooth complete intersection of dimension $n \geq 1$ over a field k. Fix an integer

$$d \geq n + 1. \tag{4.2.1}$$

Let

$$X \cdot H_u \longrightarrow U \tag{4.2.2}$$

denote the universal family of sections of X by hypersurfaces of degree d that are smooth of dimension $n - 1$. There is a dense open subset $T \subset U$ such that

$$\text{for every } t \in T, \qquad H^{n-1}(X \cdot H_{\bar{t}}, \mathbb{Q}_\ell) \neq N^1 H^{n-1}(X \cdot H_{\bar{t}}, \mathbb{Q}_\ell). \tag{4.2.3}$$

Bibliography

[1] J.-P. Serre, *Faisceaux algébriques cohérents*, Annals of Mathematics **61** (1955), 197–278.

[2] J.-P. Serre, *Sur la topologie des variétés algébriques en caractéristique p*, in *Symposium Internacional de Topología Algebraica*, Mexico, 1958.

Source note. The diplomatic French preserves the printed lower limit 1 in (4.1.1) and the unaccented phrase *de meme*. The corrected French and English make the summation index explicit as $i = 1$ and use the ordinary accented form.

5. Appendix: Integrality Theorems

By P. Deligne

5.0.

Let p be a prime number, q a power of p, $\mathbb{F}_q$ a field with q elements, E the completion of a number field at a place dividing a prime number $\ell \neq p$, and T a set of prime numbers (for example, $T = \varnothing$).

An element α of an algebraic closure of E is called *T-integral* if it is algebraic over $\mathbb{Q}$ and integral over $\mathbb{Z}[(1/r)_{r\in T}]$. An endomorphism F of a finite-dimensional E-vector space is called T-integral if its eigenvalues are T-integral.

Let k be a finite field with q' elements, let $\bar{k}$ be an algebraic closure of k, and let

$$\mathfrak{G} = \mathrm{Gal}(\bar{k}/k)$$

be the Galois group of k. We denote by $f \in \mathfrak{G}$ the Frobenius substitution

$$f(x) = x^{q'},$$

and by F the "geometric" Frobenius, the inverse of f in $\mathfrak{G}$. A representation ρ of $\mathfrak{G}$ on a finite-dimensional E-vector space is called T-integral if $\rho(F)$ is T-integral.

If X is a scheme of finite type over $\mathbb{F}_q$ and x is a closed point of X, we denote by F_x the geometric Frobenius element of the Galois group of $k(x)$. If $\bar{x}$ is a geometric point of X lying over x and $\underline{G}$ is an E-sheaf on X, then F_x acts, by transport of structure, on the fiber $G_{\bar{x}}$.

Definition 5.1.

A constructible E-sheaf $\underline{G}$ on a scheme X of finite type over $\mathbb{F}_q$ is called T-integral if, for every geometric point $\bar{x}$ of X lying over a closed point x of X, F_x is a T-integral automorphism of $G_{\bar{x}}$.

If $a : X \longrightarrow \mathrm{Spec}(\mathbb{F}_q)$ is a scheme of finite type over $\mathbb{F}_q$ and $\underline{G}$ is a constructible E-sheaf on X, we write $\underline{H}^*(X, \underline{G})$ for the E-sheaves $R^*a_*\underline{G}$ on $\mathrm{Spec}(\mathbb{F}_q)_{\acute{e}t}$. If $\overline{\mathbb{F}}_q$ is an algebraic closure of $\mathbb{F}_q$ and $\overline{X} = X \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$, these sheaves identify with the cohomology groups $H^*(\overline{X}, \underline{G})$ regarded as Galois modules, that is, endowed with an action of the geometric Frobenius F. We use the same convention for cohomology with proper support.

Lemma 5.2.1.

Let $\underline{G}$ be a constructible E-sheaf on a scheme X of finite type over $\mathbb{F}_q$, of dimension at most n.

(i) There is a closed subset Y of X, of dimension 0, such that the evident map

$$\underline{H}^0(X, \underline{G}) \longrightarrow \underline{H}^0(Y, \underline{G})$$

is injective.

(ii) If X is separated and U is an open subset of X whose complement Z has dimension less than n, there is a closed subset Y of U, of dimension 0, and a surjective map

$$\underline{H}^0(Y, \underline{G})(-n) \longrightarrow \underline{H}^{2n}_c(X, \underline{G}).$$

Proof. Assertion (i) is evident. We prove (ii). After replacing X by X_{red} and shrinking U, we may suppose that U is smooth and that $\underline{G}|_U$ is twisted constant. Since $\dim Z < n$, we have

$$0 = \underline{H}^{2n-1}_c(Z, \underline{G}) \longrightarrow \underline{H}^{2n}_c(U, \underline{G}) \xrightarrow{\sim} \underline{H}^{2n}_c(X, \underline{G}) \longrightarrow \underline{H}^{2n}_c(Z, \underline{G}) = 0,$$

and Poincaré duality (SGA 4, Exposé XVIII, and [2]) gives

$$\underline{H}^{2n}_c(U, \underline{G}) \simeq \underline{H}^0(U, \underline{G}^\vee(n))^\vee.$$

The conclusion follows from (i).

Theorem 5.2.2.

Let X be a separated scheme of finite type over $\mathbb{F}_q$, of dimension at most n, and let $\underline{G}$ be a constructible T-integral E-sheaf. Then the endomorphisms F and $q^{n-i}F$ of $\underline{H}^i_c(X,\underline{G})$ are T-integral.

Proof. The case $\dim X \leq 0$ is trivial, and the cases $i = 0$ and $i = 2n$ follow from Lemma 5.2.1. Let $Z(X,\underline{G},t)$ be the formal power series

$$Z(X,\underline{G},t) = \prod_{x\in X} \det\Big(1 - F_x t^{\deg(x)}\,;\, \underline{G}_x\Big)^{-1} \in E[[t]].$$

By SGA 5, Exposé XV, 3.2, one has

$$Z(X,\underline{G},t) = \prod_i \det\big(1 - Ft\,;\, \underline{H}^i_c(X,\underline{G})\big)^{(-1)^{i+1}}. \tag{5.2.3}$$

We first prove the theorem for $n = 1$. Then $\underline{H}^i_c(X,\underline{G}) = 0$ for $i > 2$ (SGA 4, Exposé X, 4.3, or Exposé XVII, 5.2.8.1). By hypothesis the coefficients of $Z(X,\underline{G},t)$ are T-integral. By the cases $i = 0$ and $i = 2n = 2$ already established, the product of $Z(X,\underline{G},t)$ with

$$\det\big(1 - Ft\,;\, \underline{H}^0_c(X,\underline{G})\big)\det\big(1 - Ft\,;\, \underline{H}^2_c(X,\underline{G})\big)$$

still has T-integral coefficients. This product is precisely $\det(1 - Ft\,;\, \underline{H}^1_c(X,\underline{G}))$, proving the theorem for $n = 1$.

We now proceed by induction on $n = \dim X$. Let U be a Zariski-open subset of X whose complement Z has dimension less than n, and suppose there is a morphism $f : U \longrightarrow Y$ to a curve, with fibers of dimension at most $n-1$:

$$\begin{array}{ccccc} U & \overset{j}{\hookrightarrow} & X & \overset{i}{\hookleftarrow} & Z \\ {\scriptstyle f}\big\downarrow & & & & \\ Y & & & & \end{array}.$$

The exact sequence

$$0 \longrightarrow j_!j^*\underline{G} \longrightarrow \underline{G} \longrightarrow i_!i^*\underline{G} \longrightarrow 0$$

induces exact sequences

$$\underline{H}^i_c(U,\underline{G}) \longrightarrow \underline{H}^i_c(X,\underline{G}) \longrightarrow \underline{H}^i_c(Z,\underline{G}).$$

By induction it is therefore enough to verify the theorem for $\underline{G}$ on U. Applying the induction hypothesis to the fibers of f, together with proper base change, shows that the sheaves $R^if_!\underline{G}$ and $R^if_!\underline{G}(i-(n-1))$ on Y are T-integral. The Leray spectral sequence for f is

$$E_2^{pq} = \underline{H}^p_c(Y, R^qf_!\underline{G}) \Longrightarrow \underline{H}^{p+q}_c(U,\underline{G}).$$

By what precedes and by the theorem already proved for Y, both E_2^{pq} and $E_2^{pq}(p+q-n)$ are T-integral. The theorem follows.

Corollary 5.3.

Let X be a scheme of finite type over $\mathbb{F}_q$, of dimension at most 1, let U be an open subset of X, let $j : U \hookrightarrow X$ be the inclusion, and let $\underline{G}$ be a constructible E-sheaf on U.

(a) If $\underline{G}$ is T-integral, then $j_*\underline{G}$ is T-integral.

(b) If $G_x^\vee$ is T-integral for every $x \in U$, then $(j_*\underline{G})_x^\vee$ is T-integral for every $x \in X$.

Proof. We reduce first to the case where X is affine, and then to the case where X is projective. Define $\underline{H}$ by the exact sequence

$$0 \longrightarrow j_!\underline{G} \longrightarrow j_*\underline{G} \longrightarrow \underline{H} \longrightarrow 0.$$

If Y is the finite support of $\underline{H}$, the corresponding cohomology sequence gives an exact sequence

$$\underline{H}^0(U, \underline{G}) \longrightarrow \underline{H}^0(Y, \underline{H}) \longrightarrow \underline{H}^1_c(U, \underline{G}).$$

Lemma 5.2.1 and Theorem 5.2.2 show that the two outer terms are T-integral. Hence $\underline{H}^0(Y, \underline{H})$ is T-integral, which proves (a).

Let X' be the normalization of the pure one-dimensional part of X_{red}, and choose an open subset U' of U, dense in the smooth one-dimensional locus of U_{red}, such that $\underline{G}|_{U'}$ is twisted constant:

$$\begin{array}{ccccc} U' & \overset{j'}{\hookrightarrow} & X' & & \\ \downarrow & & \downarrow p & & \\ U & \overset{j}{\hookrightarrow} & X & \hookleftarrow & Y \end{array} \quad ,$$

Then

$$(p_* j'_* \underline{G})|_Y \simeq (j_*\underline{G})|_Y.$$

It is therefore enough to treat the case in which X is smooth of dimension one and $\underline{G}|_U$ is twisted constant. The sheaf $\underline{G}^\vee$ is then T-integral, so we already know that $(j_*\underline{G}^\vee)_x$ is T-integral for every $x \in X$.

Let $\mathfrak{G}_x$ denote the Galois group of the fraction field of $\widehat{\mathcal{O}}_{X,x}$, and let $\mathfrak{I}_x$ be the inertia group. The sheaf $\underline{G}$ defines a representation of $\mathfrak{G}_x$, again denoted by $\underline{G}$. The $\mathfrak{G}_x/\mathfrak{I}_x = \mathrm{Gal}(\overline{k(x)}/k(x))$-module $(j_*\underline{G}^\vee)_x$ (respectively, $(j_*\underline{G})^\vee_x$) identifies with the module of invariants (respectively, coinvariants) in $\underline{G}^\vee$. It remains to prove:

Source note. The diplomatic French preserves the printed coefficient field $F[[t]]$ in the zeta series and the printed restriction of $\underline{G}$ to U in the normalization reduction. The corrected French and English use the coefficient field E fixed in 5.0 and the newly chosen open U'. Direct scan comparison also repairs the inherited joined words, Fraktur Galois symbols, and doubled consonant in *Désignons* to the printed forms.

Lemma 5.3.1.

Let K be a local field with residue field $\mathbb{F}_q$, let G be its Galois group, let I be the inertia group, and let ρ be a continuous representation of G on an n-dimensional E-vector space V (cf. 5.0). Let V^I and V_I denote respectively the space of invariants and the space of coinvariants under I, regarded as G/I-modules. Put $r = \dim_E V^I$, and let $\alpha_1, \ldots, \alpha_r$ be the eigenvalues of F on V^I. Then there are integers $m_i \geq 0$ such that the eigenvalues of F on V_I are

$$\alpha_i q^{m_i} \qquad (1 \leq i \leq r).$$

Proof. Let I' be the largest subgroup of I of order prime to ℓ. The group $\rho(I')$ is finite, since it is a compact subgroup of $\mathrm{GL}(V)$ of order prime to ℓ. Hence

$$V_I \simeq \left(V^{\rho(I')}\right)_I.$$

After replacing V by $V^{I'}$, we may therefore suppose that I' acts trivially on V. As G/I-modules one has

$$I/I' \simeq \mathbb{Z}_\ell(1).$$

Let T be a topological generator of I/I'. For each α, let $V^{(\alpha)}$ be the generalized eigenspace of T on V with eigenvalue α. Then

$$V = \bigoplus_{\alpha} V^{(\alpha)},$$

and the decomposition

$$V = V^{(1)} \oplus \bigoplus_{\alpha \neq 1} V^{(\alpha)}$$

is stable under G and compatible with duality. We may thus replace V by $V^{(1)}$ and suppose that I acts unipotently.

Lemma 5.3.2.

Let V be a finite-dimensional vector space over a field k of characteristic 0, let I be a one-parameter additive group, let $\rho : I \longrightarrow \mathrm{GL}(V)$ be a representation, and let N be an infinitesimal generator of I. Set

$$V^{I,k} = V^I \cap \mathrm{Im}(N^k)$$

and

$$V_{I,k} = \mathrm{Im}\Big(\mathrm{Ker}(N^{k+1}) \longrightarrow V_I\Big).$$

Then N^k induces an isomorphism

$$N^k : V_{I,k}/V_{I,k-1} \xrightarrow{\sim} V^{I,k}/V^{I,k+1},$$

and defines canonical isomorphisms

$$V_{I,k}/V_{I,k-1} \otimes I^{\otimes k} \simeq V^{I,k}/V^{I,k+1}.$$

Proof. Via N, representations V of I identify with $k[T]$-modules killed by a power of T. It is enough to verify Lemma 5.3.2 when V corresponds to an indecomposable $k[T]$-module $k[T]/T^k$, where the assertion is immediate.

We now finish the proof of Lemma 5.3.1. Lemma 5.3.2 supplies increasing and decreasing filtrations $V_{I,k}$ and $V^{I,k}$ of V_I and V^I, together with isomorphisms of G/I-modules

$$V^{I,k}/V^{I,k+1} \simeq \big(V_{I,k}/V_{I,k-1}\big)(k).$$

The assertion follows.

Source note. The diplomatic French enumerates the eigenvalues on V^I by the dimension n of V, cites the lemma as "3.2", and calls $k[T]/T^k$ irreducible. The corrected French and English use $r = \dim_E V^I$, the exact locator 5.3.2, and the mathematically correct term *indecomposable*.

Theorem 5.4.

Let X be a separated scheme of finite type over $\mathbb{F}_q$, of dimension n, and let $\underline{G}$ be a constructible E-sheaf on X. Suppose that, for every $x \in X$, the endomorphism F_x of $G_x^\vee$ is T-integral. Then the endomorphisms $q^i F^\vee$ and $q^n F^\vee$ of $\underline{H}^i_c(X, \underline{G})^\vee$ are T-integral.

Proof. As in Theorem 5.2.2, we reduce to the case in which X is a smooth curve and $\underline{G}$ is twisted constant on X. Let j be the inclusion of X in a smooth projective curve X' compactifying it.

By Poincaré duality, $\underline{H}^i(X', j_*\underline{G})$ is dual to $\underline{H}^{2-i}(X', j_*\underline{G}^\vee(1))$. By Corollary 5.3(a), Theorem 5.2.2 applies to $j_*\underline{G}^\vee$:

$$\begin{aligned}
\underline{H}^0(X', j_*\underline{G}^\vee) &\simeq \underline{H}^2(X', j_*\underline{G})^\vee(-1) && \text{is } T\text{-integral,}\\
\underline{H}^1(X', j_*\underline{G}^\vee) &\simeq \underline{H}^1(X', j_*\underline{G})^\vee(-1) && \text{is } T\text{-integral,}\\
\underline{H}^2(X', j_*\underline{G}^\vee)(1) &\simeq \underline{H}^0(X', j_*\underline{G})^\vee && \text{is } T\text{-integral.}
\end{aligned}$$

This proves the theorem for $j_*\underline{G}$.

Define $\underline{H}$ by the exact sequence

$$0 \longrightarrow j_!\underline{G} \longrightarrow j_*\underline{G} \longrightarrow \underline{H} \longrightarrow 0.$$

If Y is the support of $\underline{H}$, the long exact cohomology sequence provides

$$\underline{H}^0_c(X,\underline{G}) \hookrightarrow \underline{H}^0(X', j_*\underline{G}) \quad ,$$

$$\underline{H}^0(Y,\underline{H}) \longrightarrow \underline{H}^1_c(X,\underline{G}) \longrightarrow \underline{H}^1(X', j_*\underline{G}) \quad ,$$

$$\underline{H}^2_c(X,\underline{G}) \xrightarrow{\sim} \underline{H}^2(X', j_*\underline{G}) \quad ,$$

Thus the theorem for $(X,\underline{G})$ follows from the theorem for $(X', j_*\underline{G})$.

5.5.

The preceding method of fibration into curves also gives Lang–Weil estimates [3] for the complex absolute values of the eigenvalues of Frobenius.

Let X be a scheme of finite type over $\mathbb{F}_q$, let $\underline{G}$ be a constructible E-sheaf on X, and let $\lambda \in \mathbb{R}$. Suppose that, for every point $x \in X$, with $\#k(x) = q_x$, the eigenvalues of F_x acting on G_x are algebraic numbers α all of whose complex absolute values satisfy

$$|\alpha| \leq q_x^{\lambda}.$$

We shall then write

$$\log|\underline{G}| \leq \lambda.$$

Lemma 5.5.1.

Let X be a curve over $\mathbb{F}_q$, by which we mean here a separated scheme X of finite type over $\mathbb{F}_q$ and of dimension at most 1. Let $\underline{G}$ be a constructible E-sheaf on X such that $\log|\underline{G}| \leq \lambda$, with $\lambda \in \mathbb{R}$. Then the complex absolute values of the eigenvalues α of F acting on $\underline{H}^1_c(X,G)$ satisfy

$$|\alpha| \leq q^{\lambda+1}.$$

Proof. Let σ be a complex embedding of E, or simply a complex embedding of the algebraic closure of $\mathbb{Q}$ in E. It is enough to verify that, for every σ, the zeros β of the rational function $\sigma Z(X,\underline{G},t)$ satisfy

$$|\beta^{-1}| \leq q^{\lambda+1}.$$

Indeed, Lemma 5.2.1 already shows that, if α is an eigenvalue of Frobenius acting on $\underline{H}^0_c(X,\underline{G})$ or on $\underline{H}^2_c(X,\underline{G})$, then $|\sigma\alpha| \leq q^{\lambda+1}$.

We verify that the series

$$\log \sigma Z(X,\underline{G},t) = \sum_k \frac{t^k}{k} \sum_{x \in X(\mathbb{F}_{q^k})} \sigma \operatorname{Tr}(F, x^*\underline{G})$$

converges for $|t| < q^{-\lambda-1}$. Representing X as a ramified covering of $\mathbb{P}^1$, one obtains the elementary estimate

$$\#X(\mathbb{F}_{q^k}) \leq Aq^k.$$

On the other hand, the hypothesis gives

$$|\sigma \operatorname{Tr}(F, x^*\underline{G})| \leq B(q^k)^\lambda.$$

Thus the coefficient c_k of t^k in $\log \sigma Z(X, \underline{G}, t)$ satisfies

$$|c_k| \leq \frac{1}{k} C(q^{1+\lambda})^k,$$

and the assertion follows.

Theorem 5.5.2.

Let X be a separated scheme of finite type over $\mathbb{F}_q$, of dimension at most n, and let $\underline{G}$ be a constructible E-sheaf on X such that $\log |\underline{G}| \leq \lambda$, with $\lambda \in \mathbb{R}$. Then the complex absolute values of the eigenvalues α of Frobenius acting on $\underline{H}^i_c(X, \underline{G})$ satisfy

$$|\alpha| \leq q^{\lambda+i} \qquad \text{and} \qquad |\alpha| \leq q^{\lambda+n}.$$

Proof. If $n \leq 0$, the theorem is trivial. If $n = 1$, its cases $i = 0$ and $i = 2$ follow from Lemma 5.2.1, while the case $i = 1$ follows from Lemma 5.5.1. We then proceed by induction on n, as in the proof of Theorem 5.2.2.

Corollary 5.5.3.

Let X be a separated scheme of finite type and dimension n over $\mathbb{F}_q$. Then the eigenvalues α of the geometric Frobenius endomorphism F of $\underline{H}^i_c(X, \mathbb{Q}_\ell)$ are algebraic integers. Moreover:

(i) for $\ell' \neq p$, α is an ℓ'-adic unit;

(ii) for $i > n$, α is divisible by q^{i-n};

(iii) for $i \leq n$ (respectively, $i \geq n$), $q^i\alpha^{-1}$ (respectively, $q^n\alpha^{-1}$) is an algebraic integer;

(iv) for $0 < i$, the complex absolute values of α satisfy

$$|\alpha| \leq q^{i-\frac{1}{2}} \qquad \text{and} \qquad |\alpha| \leq q^n.$$

Proof. By Theorem 5.2.2, α and $q^{n-i}\alpha$ are algebraic integers. Assertions (i)–(iii) then follow from Theorem 5.4. For $i > n$, assertion (iv) follows from Theorem 5.5.2. To obtain (iv) when $i \leq n$, we must combine Theorem 5.5.2 with the Riemann hypothesis for curves.

We argue by induction on n, the case $n = 0$ being trivial. If U is an open subset of X whose complement Z has dimension less than n, the exact sequence

$$\underline{H}^i_c(U, \mathbb{Q}_\ell) \longrightarrow \underline{H}^i_c(X, \mathbb{Q}_\ell) \longrightarrow \underline{H}^i_c(Z, \mathbb{Q}_\ell)$$

shows that it is enough to prove the desired assertion for U. After first replacing X by X_{red}, this reduces us to the case in which X is smooth and affine and there is a smooth morphism $f : X \longrightarrow S$ satisfying:

(a) $\dim S \leq n - 1$;

(b) the fibers of f are purely one-dimensional.

Since the fibers of f are affine curves, one has

$$R^0 f_! \mathbb{Q}_\ell = 0.$$

By Weil [4],

$$\log|R^1 f_! \mathbb{Q}_\ell| \leq \frac{1}{2}, \tag{5.5.3.1}$$

and Lemma 5.2.1 applied to the fibers of f gives

$$\log|R^2 f_! \mathbb{Q}_\ell| \leq 1. \tag{5.5.3.2}$$

The Leray spectral sequence for f reduces here to a long exact sequence; the relevant part is

$$\underline{H}_c^{i-2}(S, R^2 f_! \mathbb{Q}_\ell) \longrightarrow \underline{H}_c^i(X, \mathbb{Q}_\ell) \longrightarrow \underline{H}_c^{i-1}(S, R^1 f_! \mathbb{Q}_\ell).$$

By (5.5.3.1) (respectively, (5.5.3.2)) and Theorem 5.5.2, the complex absolute values of the eigenvalues α of Frobenius on $\underline{H}_c^{i-1}(S, R^1 f_! \mathbb{Q}_\ell)$ (respectively, on $\underline{H}_c^{i-2}(S, R^2 f_! \mathbb{Q}_\ell)$) satisfy

$$|\alpha| \leq q^{i-1+\frac{1}{2}} = q^{i-\frac{1}{2}}$$

(respectively,

$$|\alpha| \leq q^{i-2+1} = q^{i-1} \leq q^{i-\frac{1}{2}}).$$

The corollary follows.

Source note. The diplomatic French preserves three printed slips: the coefficient c_k is said to belong to σZ although the displayed series and the factor $1/k$ make it a coefficient of $\log \sigma Z$; the proof of Theorem 5.5.2 refers to itself rather than to the parallel induction in Theorem 5.2.2; and the proof of Corollary 5.5.3 says to prove Theorem 5.2.2 for U although the argument is proving assertion (iv) of the corollary. The corrected French and English give the coherent readings. Direct scan comparison also restores the notation underlines omitted by the inherited French TeX in the statement of Theorem 5.5.2.

5.6.

To obtain results analogous to the preceding ones for cohomology without supports, it seems necessary to have resolution of singularities available. In dimension at most 1, the problem is solved by Corollary 5.3.

Theorem 5.6.

Let N be an integer, and suppose that strong resolution of singularities in the sense of Hironaka is available for schemes of finite type over $\mathbb{F}_q$ of dimension at most N (at present one may therefore take $N = 2$ [1]). Let $f : X \longrightarrow Y$ be a morphism of schemes of finite type over $\mathbb{F}_q$, and let $\underline{G}$ be a constructible T-integral E-sheaf on X. If $\dim X \leq N$, then the E-sheaves $R^i f_* \underline{G}$ on Y are T-integral.

Proof. We argue by induction on the dimension $n \leq N$ of X. Let $j : U \hookrightarrow X$ be an open subset of X, smooth and affine, such that $\underline{G}|_U$ is twisted constant and the dimension of its complement Z is less than n. By resolution of singularities, the sheaves $R^i j_*(j^* \underline{G})$ are constructible (SGA 5, Exposé I; [2]); for $i > 0$ they are supported on Z, and for $i = 0$, $j_*(j^* \underline{G})$ differs from $\underline{G}$ only on Z. The induction hypothesis therefore shows that it is enough to prove the theorem for the sheaf $j^* \underline{G}$ and the morphisms j and $f \circ j$. Thus we may further suppose that X is smooth and affine and that $\underline{G}$ is twisted constant, that is, comes from a continuous representation of the fundamental group of X on a finite-dimensional E-vector space.

Let R be the ring of integers of E, let λ be a uniformizer, let e be the absolute ramification index, and choose an integer $b > e/(\ell-1)$. Since X is smooth, hence normal, and its fundamental group is profinite, there is a twisted-constant R-sheaf $\underline{G}^\circ$ such that

$$\underline{G} \simeq \underline{G}^\circ \otimes_R E.$$

Let $p : X' \longrightarrow X$ be the étale covering that trivializes the locally constant sheaf $\underline{G}^\circ \otimes_R R/\lambda^b$. The sheaf $\underline{G}$ is a direct summand of $p_* p^* \underline{G}$, so it is enough to prove the theorem for $p^*\underline{G}$ and $f \circ p$. It remains to treat the case in which the following conditions hold:

(∗) X is smooth and affine, and there is a twisted-constant R-sheaf $\underline{G}^\circ$ such that $\underline{G} \simeq \underline{G}^\circ \otimes_R E$ and $\underline{G}^\circ \otimes_R R/\lambda^b$ is constant.

Factor f as a proper morphism followed by an open immersion, $f = \bar{f} \circ j$:

$$\begin{array}{ccc} X & \xrightarrow[j]{} & X' \\ {\scriptstyle f}\downarrow & \swarrow_{\bar{f}} & \\ Y & & \end{array} \quad .$$

The theorem has already been proved for $\bar{f}$ (5.2), so it is enough to prove it for j. Applying resolution of singularities to X', we may suppose that X' is smooth and that X is the complement in X' of a normal-crossings divisor which is a union of smooth divisors. Hypothesis (∗) implies that $\underline{G}$ is tamely ramified along this divisor. We then have:

Lemma 5.6.1.

Let X be a smooth scheme of finite type over $\mathbb{F}_q$, let D be a normal-crossings divisor in X which is a union of smooth divisors D_i, let $j : U \hookrightarrow X$ be the complement of D in X, and let $\underline{G}$ be a twisted-constant E-sheaf on U, tamely ramified along D. If $\underline{G}$ is T-integral, then the sheaves $R^i j_* \underline{G}$ are T-integral.

We shall admit the following result (cf. SGA 1, Exposé XIII).

Lemma 5.6.2.

(i) Under the hypotheses of Lemma 5.6.1, let D' be a component of D, let D'' be the union of the other components, and put $U' = X - D''$ and $D'^\circ = D' \cap U'$:

$$\begin{array}{ccccc} & & D'^\circ & \xrightarrow[j'_D]{} & D' \\ & & \downarrow{\scriptstyle i^\circ} & & \downarrow{\scriptstyle i} \\ U & \xrightarrow[j_1]{} & U' & \xrightarrow[j']{} & X \end{array} \quad .$$

Then the base-change map

$$i^* R j'_* (R j_{1*} \underline{G}) \longrightarrow R j'_{D*}\, i^{\circ *} (R j_{1*} \underline{G})$$

is an isomorphism, and the sheaves $i^{\circ *} R j_{1*} \underline{G}$ are twisted constant and tamely ramified along $D' - D'^\circ$.

(ii) If C is a smooth curve on X, transverse to D at a smooth point x of D,

$$\begin{array}{ccc} C \cap U & \xrightarrow[j_c]{} & C \\ \downarrow{\scriptstyle k'} & & \downarrow{\scriptstyle k} \\ U & \xrightarrow[j]{} & X \end{array} \quad ,$$

then the base-change map

$$k^* R j_* \underline{G} \longrightarrow R j_{c*}\, k'^* \underline{G}$$

is an isomorphism at x.

Let us deduce Lemma 5.6.1 from Lemma 5.6.2 by induction on the dimension, temporarily admitting Lemma 5.6.1 for curves. Lemma 5.6.2(ii) and this hypothesis show that the sheaves $R^i j_* \underline{G}$ are T-integral at every smooth point of D. In particular, in the notation of Lemma 5.6.2(i), the sheaves $R^i j_{1*} \underline{G}$ are T-integral. Lemma 5.6.2(i), together with the induction hypothesis applied to $i^{\circ *} R j_{1*} \underline{G}$ and to j'_D, therefore shows that the sheaves $i^* R^i j_* \underline{G}$ are T-integral, whichever component D' of D is chosen. This proves Lemma 5.6.1.

It remains to handle curves. Under the hypotheses of Lemma 5.6.1, let x be a point of $X - U$, let $\mathfrak{G}$ be the Galois group of the fraction field of $\widehat{\mathcal{O}}_{x,X}$, and let $\mathfrak{J}$ be the inertia group. Denote again by $\underline{G}$ the representation of $\mathfrak{G}$ defined by the sheaf, and let $\bar{x}$ be a geometric point of X localized at x. There are isomorphisms of Galois $\mathfrak{G}/\mathfrak{J}$-modules

$$(j_* \underline{G})_{\bar{x}} \simeq \underline{G}^{\mathfrak{J}},$$

$$(R^1 j_* \underline{G})_{\bar{x}} \simeq \underline{G}_{\mathfrak{J}}(-1),$$

and

$$(R^i j_* \underline{G})_{\bar{x}} = 0 \qquad \text{for } i \geq 2.$$

The assertion follows from Corollary 5.3(a) and Lemma 5.3.1.

Source note. Direct authority comparison restores the factorization diagram's j label to the lower side of its arrow and restores the printed Fraktur $\mathfrak{G}$ in the curve case. The diplomatic French preserves the printed singular *morphisme*, the punctuation collision before hypothesis $(*)$, and the shortened reference 6.2(i). Corrected French and English use plural *morphismes*, ordinary sentence punctuation, and the exact reference Lemma 5.6.2(i).

EXPOSÉ XXII

A CONGRUENCE FORMULA FOR THE ZETA FUNCTION

by N. Katz

0. Introduction

Let X be a proper scheme over $\mathbb{F}_q$. Its zeta function, regarded as an element of $\mathbb{Z}[[T]]$, may be reduced modulo p; we wish to compute it modulo p. We obtain an expression of the kind one might have expected from fixed-point formulas of "Woods Hole" type: an alternating product of characteristic polynomials of a suitable "Frobenius" acting on any one of the following three cohomology spaces:

1) coherent cohomology $H^i(X, \mathcal{O}_X)$ (3.1.1);

2) de Rham cohomology $H^i_{\mathrm{DR}}(X) = \mathbb{H}^i(X, \Omega^\bullet_X)$ (3.1.1);

3) étale cohomology $H^i_{\mathrm{et}}(X, \mathbb{Z}/p\mathbb{Z})$ (3.1.2).

This expression should be viewed as an improvement of a special case of the Woods Hole trace formula, namely that of the structure sheaf and the Frobenius endomorphism. That case is indeed a corollary of the congruence formula (cf. 3.2.1).

The method of this exposé is completely naïve. It consists in reducing to the case of a hypersurface and then "reducing modulo p" Dwork's computation of the actual zeta function of a hypersurface. By "reducing" carefully, one obtains the improvement 5.0.6, due to Ax [1] (and also to Dwork in the smooth case [4, §7]), of the well-known Warning–Chevalley theorem [8].

Another method, entirely cohomological and in the spirit of SGA 5, which also applies to "twisted coefficients," has just been found by Deligne and ought to appear in the new edition of SGA 5. It rests on the trace formula of Woods Hole type and on Deligne's "symmetric Künneth formula" (SGA 4, XVII 5.5.21).

1. A reminder on "p-linear" operations

1.0.

Let V be a finite-dimensional vector space over a field k of characteristic $p > 0$, and let q be a power of p. A map

$$\varphi : V \longrightarrow V \tag{1.0.1}$$

is called *q-linear* if it is additive and satisfies

$$\varphi(\lambda v) = \lambda^q \varphi(v) \qquad \text{for every } \lambda \in k,\ v \in V. \tag{1.0.2}$$

1.0.3.

Notice that the n-th iterate φ^n of a q-linear endomorphism is q^n-linear and that, if k is not perfect, the image $\varphi(V)$ of a q-linear endomorphism need not be a k-vector subspace.

1.0.4.

For every extension K/k, a q-linear endomorphism φ of V gives rise to a unique q-linear endomorphism φ_K of $V_K = V \otimes_k K$ satisfying

$$\varphi_K(\mu \otimes v) = \mu^q \varphi(v) \qquad \text{for } \mu \in K,\ v \in V. \tag{1.0.5}$$

1.0.6.

A q-linear endomorphism φ of V is called *semisimple* if its image spans V over k, and *nilpotent* if some iterate is zero. Define the following φ-stable k-subspaces:

$$V_{\mathrm{ss}} = \bigcap_{n \geq 1} \langle \varphi^n(V) \rangle_k, \tag{1.0.7}$$

$$V_{\mathrm{nilp}} = \bigcup_{n \geq 1} \mathrm{Ker}(\varphi^n : V \longrightarrow V). \tag{1.0.8}$$

Clearly, the restriction of φ to V_{ss} is semisimple, while its restriction to V_{nilp} is nilpotent. If k is perfect, then

$$V = V_{\mathrm{ss}} \oplus V_{\mathrm{nilp}}. \tag{1.0.9}$$

If k is not perfect, however, (1.0.9) is false, since there exist injective q-linear endomorphisms which are not semisimple. For an arbitrary extension K/k, one has

$$(V_K)_{\mathrm{ss}} = V_{\mathrm{ss}} \otimes_k K. \tag{1.0.10}$$

Proposition 1.1.

Let φ be a q-linear endomorphism of a finite-dimensional vector space V over a separably closed field k. Suppose that φ is semisimple, and put

$$V^{1-\varphi} = \text{the } \mathbb{F}_q\text{-vector space of fixed points of } \varphi = \mathrm{Ker}(1 - \varphi).$$

Then

$$V \xleftarrow{\sim} V^{1-\varphi} \otimes_{\mathbb{F}_q} k. \tag{1.1.1}$$

Corollary 1.1.2.

There is an isomorphism

$$V^{1-\varphi^n} \xleftarrow{\sim} V^{1-\varphi} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^n}.$$

Proof of Proposition 1.1. (Lang, cf. [6], p. 119.) Let e be a k-basis of V, and write

$$\varphi(e) = Ae, \qquad A \in \mathrm{GL}(n,k). \tag{1.1.3}$$

Let

$$F : \mathrm{GL}(n,k) \longrightarrow \mathrm{GL}(n,k) \tag{1.1.4}$$

denote the map which raises every coordinate to the q-th power. For $g \in \mathrm{GL}(n,k)$, the basis ge of V consists of fixed points of φ if and only if

$$F(g)Ag^{-1} = 1. \tag{1.1.5}$$

Let $\bar{k}$ be an algebraic closure of k. The group $\mathrm{GL}(n,\bar{k})$ acts on the variety $\mathrm{GL}(n,\bar{k})$ by

$$\begin{cases} \mathrm{GL}(n,\bar{k}) \times \mathrm{GL}(n,\bar{k}) \longrightarrow \mathrm{GL}(n,\bar{k}), \\ \qquad\qquad (g,A) \longmapsto F(g)Ag^{-1}. \end{cases} \tag{1.1.6}$$

For fixed A, the map

$$g \longmapsto F(g)Ag^{-1} \tag{1.1.7}$$

is étale, as one sees by computing its tangent map. Its orbits are therefore open. Since $\mathrm{GL}(n,\bar{k})$ is irreducible, there is only one orbit; in other words, the map

$$\begin{aligned} \mathrm{GL}(n,\bar{k}) &\longrightarrow \mathrm{GL}(n,\bar{k}), \\ g &\longmapsto F(g^{-1})\, g \end{aligned} \tag{1.1.8}$$

is surjective. Since it is also étale and k is separably closed, the map

$$\begin{aligned} \mathrm{GL}(n,k) &\longrightarrow \mathrm{GL}(n,k), \\ g &\longmapsto F(g^{-1})\, g \end{aligned} \tag{1.1.9}$$

is surjective. Thus V has a k-basis consisting of fixed points of φ. Since φ is q-linear, every fixed point of φ lies in the $\mathbb{F}_q$-vector space spanned by such a k-basis. This proves Proposition 1.1.

Corollary 1.1.10.

Let φ be a q-linear endomorphism of a finite-dimensional vector space V over a separably closed field k. There is a canonical isomorphism

$$V_{\mathrm{ss}} \xleftarrow{\sim} V^{1-\varphi} \otimes_{\mathbb{F}_q} k. \tag{1.1.11}$$

Indeed, it is enough to apply Proposition 1.1 to V_{ss}.

Proposition 1.2.

Let φ be a q-linear endomorphism of a finite-dimensional vector space over a separably closed field k. Then the additive map

$$1-\varphi : V \longrightarrow V$$

is surjective.

Proof. There is an exact sequence of abelian groups

$$0 \longrightarrow V_{\mathrm{ss}} \longrightarrow V \longrightarrow V/V_{\mathrm{ss}} \longrightarrow 0.$$

on which $1-\varphi$ acts. By the snake lemma, it is enough to treat V_{ss}, which is immediate from Proposition 1.1 by using a basis consisting of fixed points, and V/V_{ss}, on which φ is nilpotent and hence $1-\varphi$ is invertible.

Source note. The diplomatic French preserves the print's hyphen in *peut-être*, the singular phrase *coefficient tordus*, *Una application*, the omitted verb in *si elle additive*, and two spacing/punctuation collisions. The corrected French and English regularize those surface defects without changing the mathematical content. Formula (1.0.7) was read directly from a high-detail authority crop: its large operator is an intersection.

2. Coherent cohomology and étale cohomology

Proposition 2.0.

Let $\bar{X}$ be a proper scheme over a separably closed field k, and let

$$\mathfrak{F}_p : H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \tag{2.0.1}$$

denote the p-linear map induced by

$$\mathfrak{F}_p : \mathcal{O}_{\bar{X}} \longrightarrow \mathcal{O}_{\bar{X}}, \qquad \mathfrak{F}_p(f) = f^p. \tag{2.0.2}$$

Denote by

$$H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})$$

the étale cohomology groups with coefficients in the constant sheaf $\mathbb{Z}/p\mathbb{Z}$. Then there are canonical isomorphisms

$$H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \simeq H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\mathfrak{F}_p}. \tag{2.0.3}$$

Proof. There is an exact sequence of sheaves on $\bar{X}_{\mathrm{et}}$:

$$0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathcal{O}_{\bar{X}} \xrightarrow{1-\mathfrak{F}_p} \mathcal{O}_{\bar{X}} \longrightarrow 0, \tag{2.0.4}$$

which gives an exact cohomology sequence

$$\cdots \longrightarrow H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \longrightarrow H^i_{\mathrm{et}}(\bar{X}, \mathcal{O}_{\bar{X}}) \xrightarrow{1-\mathfrak{F}_p} H^i_{\mathrm{et}}(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow \cdots . \tag{2.0.5}$$

One has $H^i_{\text{et}}(\bar{X}, \mathcal{O}_{\bar{X}}) \simeq H^i(\bar{X}, \mathcal{O}_{\bar{X}})$; this is a finite-dimensional k-vector space because $\bar{X}$ is proper. Applying Proposition 1.2 gives exact sequences

$$0 \longrightarrow H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \longrightarrow H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \xrightarrow{1-\mathfrak{F}_p} H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow 0, \tag{2.0.6}$$

which proves the assertion. Proposition 1.1.10 now gives the following corollary of Proposition 2.0.

Corollary 2.1.

For $\bar{X}$ proper over a separably closed field k, there is a canonical isomorphism

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}})_{\text{ss}} \xleftarrow{\sim} H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \otimes_{\mathbb{F}_p} k. \tag{2.1.1}$$

2.2.

Let now $f : X \longrightarrow \text{Spec}(\mathbb{F}_q)$ be a proper scheme over $\mathbb{F}_q$. Let k be an algebraic closure of $\mathbb{F}_q$, and put

$$\bar{X} = X \times_{\mathbb{F}_q} k. \tag{2.2.1}$$

The étale sheaf $R^i f_*(\mathbb{Z}/p\mathbb{Z})$ on $\text{Spec}(\mathbb{F}_q)$ identifies with $H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})$, viewed as a $\text{Gal}(k/\mathbb{F}_q)$-module. Let

$$\mathfrak{F}_q : H^i(X, \mathcal{O}_X) \longrightarrow H^i(X, \mathcal{O}_X) \tag{2.2.2}$$

denote the linear operation (it is q-linear, but the ground field is $\mathbb{F}_q$) induced by

$$\mathfrak{F}_q : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad \mathfrak{F}_q(f) = f^q. \tag{2.2.3}$$

Proposition 2.2.4.

Let $\varphi_q \in \text{Gal}(k/\mathbb{F}_q)$ be the "Frobenius" element $\varphi_q(\lambda) = \lambda^q$, and let $i \in \mathbb{Z}$. There is an identity of characteristic polynomials

$$\det_{\mathbb{F}_p}\big(1 - t\varphi_q^{-1} \mid H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})\big) = \det_{\mathbb{F}_q}\big(1 - t\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\big)\,. \tag{2.2.4.1}$$

Here the left-hand side is the characteristic polynomial of the element φ_q^{-1} of $\pi_1(\text{Spec}(\mathbb{F}_q))$ acting on the Galois module $H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})$.

Proposition 2.2.4 is a corollary of the following result.

Proposition 2.2.5.

Let X be proper over $\text{Spec}(\mathbb{F}_q)$. There is a canonical isomorphism

$$H^i(X, \mathcal{O}_X)_{\text{ss}} \otimes_{\mathbb{F}_q} k \simeq H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \otimes_{\mathbb{F}_p} k \tag{2.2.6}$$

which carries the automorphism induced by $\mathfrak{F}_q$ on the left-hand side to the automorphism induced by φ_q^{-1} on the right-hand side. The latter may also be interpreted as $(\text{Fr}_q^X)^* \otimes_{\mathbb{F}_p} k$, where $\text{Fr}_q^X : \bar{X} \longrightarrow \bar{X}$ is the Frobenius k-endomorphism (cf. SGA 5, XVIII 2.3.4).

Proof. Since

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \simeq H^i(X, \mathcal{O}_X) \otimes_{\mathbb{F}_q} k, \tag{2.2.7}$$

Proposition 1.0.10 gives

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}})_{\mathrm{ss}} \simeq H^i(X, \mathcal{O}_X)_{\mathrm{ss}} \otimes_{\mathbb{F}_q} k. \tag{2.2.8}$$

Thus the isomorphism (2.2.6) is obtained by composing (2.2.8) with (2.1.1).
In view of (2.2.7), the q-linear operation

$$\mathfrak{F}_q : H^i(X, \mathcal{O}_X) \longrightarrow H^i(X, \mathcal{O}_X) \tag{2.2.9}$$

has a q-linear extension (cf. 1.0.5)

$$\overline{\mathfrak{F}}_q : H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow H^i(\bar{X}, \mathcal{O}_{\bar{X}}), \tag{2.2.10}$$

and also a k-linear extension, again denoted by $\overline{\mathfrak{F}}_q$,

$$\overline{\mathfrak{F}}_q : H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow H^i(\bar{X}, \mathcal{O}_{\bar{X}}). \tag{2.2.11}$$

Let

$$\varphi_q : H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \tag{2.2.12}$$

denote the q-linear operation whose fixed points are $H^i(X, \mathcal{O}_X)$. Under (2.2.7), the operators $\mathfrak{F}_q$, φ_q, and $\overline{\mathfrak{F}}_q$ may also be written respectively as $\mathfrak{F}_q \otimes \mathrm{id}_k$, $\mathrm{id} \otimes \varphi_q$, and $\mathfrak{F}_q \otimes \varphi_q$. They therefore commute with one another, and

$$\overline{\mathfrak{F}}_q = \varphi_q \mathfrak{F}_q = \mathfrak{F}_q \circ \varphi_q. \tag{2.2.13}$$

They all commute with $\mathfrak{F}_p$, so $\mathfrak{F}_q$ and φ_q act on both $H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q}$ and $H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\mathfrak{F}_p}$. By (2.2.13), $\mathfrak{F}_q$ and φ_q are inverse to one another on $H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q}$. Hence the action of $\mathfrak{F}_q$ on

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}})_{\mathrm{ss}} \simeq H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q} \otimes_{\mathbb{F}_q} k \qquad \text{(cf. 1.1.11)}$$

is inverse to the action of $\varphi_q \otimes \mathrm{id}_k$ on

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q} \otimes_{\mathbb{F}_q} k.$$

Writing

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q} \simeq H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\mathfrak{F}_p} \otimes_{\mathbb{F}_p} \mathbb{F}_q, \tag{2.2.14}$$

the automorphism φ_q of the left-hand side of (2.2.14) becomes, via (2.2.14), the automorphism induced by φ_q on $H^i(\bar{X}, \mathcal{O}_{\bar{X}})^{1-\mathfrak{F}_p} \otimes_{\mathbb{F}_p} \mathbb{F}_q$. Consequently the automorphism induced by $\mathfrak{F}_q$ on $H^i(\bar{X}, \mathcal{O}_{\bar{X}})_{\mathrm{ss}}$ becomes, via (2.1.1), the automorphism induced by φ_q^{-1} on $H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \otimes_{\mathbb{F}_p} k$. This proves Proposition 2.2.5.

Source note. The final printed sentence of the proof says φ_q where the proposition and the immediately preceding inverse relation require φ_q^{-1}. The English follows the mathematically coherent reading and the corrected French workpass; the diplomatic French retains the printed wording.

3. The congruence formula: statements and elementary equivalences

I wish to thank Grothendieck and Deligne for explaining to me the variants (3.1.2), (3.2.1)–(3.2.2), and (3.3.1) of (3.1.1).

3.0.

Let X be a scheme of finite type over $\mathbb{F}_q$. The degree of a closed point $y \in X$ is defined to be the degree of its residue field $\mathbb{F}_q(y)/\mathbb{F}_q$. The zeta function of $X/\mathbb{F}_q$, as an element of $\mathbb{Z}[[t]]$, is defined by the formula (cf. also SGA 5, XIII 3.1)

$$Z(t, X/\mathbb{F}_q) = \prod_y (1 - t^{\deg(y)})^{-1}. \tag{3.0.1}$$

By the "zeta function modulo p" we shall always mean the image of $Z(t, X/\mathbb{F}_q)$ in $\mathbb{F}_q[[t]]$ under the "reduction modulo p" map

$$\mathbb{Z}[[t]] \longrightarrow \mathbb{F}_q[[t]].$$

Theorem 3.1.

Let X be a proper scheme over $\mathbb{F}_q$. Then

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i \geq 0} \det\bigl(1 - t\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\bigr)^{(-1)^{i+1}} \pmod{p}, \tag{3.1.1}$$

or, equivalently by Proposition 2.2.4,

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i \geq 0} \det\bigl(1 - t\varphi_q^{-1} \mid H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})\bigr)^{(-1)^{i+1}} \pmod{p}. \tag{3.1.2}$$

Corollary 3.2.

Let X be a proper scheme over $\mathbb{F}_q$. Then

$$\sum_{i \geq 0} (-1)^i \operatorname{Tr}\bigl(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\bigr) \equiv \#X(\mathbb{F}_q) \pmod{p}, \tag{3.2.1}$$

or, equivalently by Proposition 2.2.4,

$$\sum_{i \geq 0} (-1)^i \operatorname{Tr}\bigl(\varphi_q^{-1} \mid H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})\bigr) \equiv \#X(\mathbb{F}_q) \pmod{p}. \tag{3.2.2}$$

Indeed, take the coefficients of t in the congruences (3.1.1) and (3.1.2).

Theorem 3.3.

Let X be a proper scheme over $\mathbb{F}_q$, and denote by $H^i_{\mathrm{DR}}(X/\mathbb{F}_q)$ the $\mathbb{F}_q$-vector spaces $H^i(X, \Omega^\bullet_{X/\mathbb{F}_q})$, on which $\mathfrak{F}_q$ acts. Then

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i\geq 0} \det\bigl(1 - t\mathfrak{F}_q \mid H^i_{\mathrm{DR}}(X/\mathbb{F}_q)\bigr)^{(-1)^{i+1}} \pmod p. \tag{3.3.1}$$

Indeed, consider the spectral sequence

$$E_1^{p,q} = H^q(X, \Omega^p_{X/\mathbb{F}_q}) \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/\mathbb{F}_q). \tag{3.3.2}$$

The operator $\mathfrak{F}_q$ acts on it through the endomorphism $\mathfrak{F}_q = (\mathrm{Fr}_q^X)^*$ induced by $\mathrm{Fr}_q^X : X \longrightarrow X$, and kills $E_1^{p,q}$ for $p \neq 0$. It follows that (3.3.1) is equivalent to (3.1.1).

Let us add a small complement. Since formation of the "semisimple part" (cf. 1.0.7) is an exact functor, (3.3.2) gives a spectral sequence

$$(E_1^{p,q})_{\mathrm{ss}} \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/\mathbb{F}_q)_{\mathrm{ss}}. \tag{3.3.3}$$

Since $(E_1^{p,q})_{\mathrm{ss}} = 0$ for $p \neq 0$, we obtain for every i an isomorphism

$$H^i_{\mathrm{DR}}(X/\mathbb{F}_q)_{\mathrm{ss}} \xrightarrow{\sim} H^i(X, \mathcal{O}_X)_{\mathrm{ss}} \tag{3.3.4}$$

that commutes with $\mathfrak{F}_q$. Hence

$$\det\bigl(1 - t\mathfrak{F}_q \mid H^i_{\mathrm{DR}}(X/\mathbb{F}_q)\bigr) = \det\bigl(1 - t\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\bigr)\,. \tag{3.3.5}$$

Theorem 3.4.

Let X be a separated scheme of finite type over $\mathbb{F}_q$. Denoting by H^i_c étale cohomology with proper supports (SGA 4, XVII 5.1.9(iii)), one has

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i\geq 0} \det\bigl(1 - t\varphi_q^{-1} \mid H^i_c(\bar{X}, \mathbb{Z}/p\mathbb{Z})\bigr)^{(-1)^{i+1}} \pmod p. \tag{3.4.1}$$

We begin by proving the equivalence of (3.4.1) and (3.1.1). For the moment, write

$$Z(X) \quad \text{in place of } Z(t, X/\mathbb{F}_q) \tag{3.4.2}$$

and

$$Z'(X) \quad \text{in place of} \quad \prod_i \det\bigl(1 - t\varphi_q^{-1} \mid H^i_c(\bar{X}, \mathbb{Z}/p\mathbb{Z})\bigr)^{(-1)^{i+1}}\,. \tag{3.4.3}$$

Lemma 3.5.

If $\{U_i\}$ is a finite covering of X by open subschemes, then

$$Z'(X) = \prod_{j\geq 0} \prod_{1\leq i_0<\cdots<i_j\leq n} Z'(U_{i_0}\cap\cdots\cap U_{i_j})^{(-1)^j}. \tag{3.5.1}$$

Proof. This follows immediately from the localization spectral sequence (SGA 4, XVII 6.2.10)

$$E_1^{-a,b} = \bigoplus_{i_0<\cdots<i_b} H_c^a(\bar{U}_{i_0}\cap\cdots\cap\bar{U}_{i_b},\mathbb{Z}/p\mathbb{Z}) \Longrightarrow H_c^{-a+b}(\bar{X},\mathbb{Z}/p\mathbb{Z}).$$

It may also be obtained by induction on n, using Lemma 3.6 below.

Lemma 3.6.

If U is an open subscheme of X and Y is the closed subscheme $X-U$, then

$$Z'(X) = Z'(U)Z'(Y). \tag{3.6.1}$$

Proof. This follows from the exact cohomology sequence

$$\longrightarrow H_c^i(\bar{U},\mathbb{Z}/p\mathbb{Z}) \longrightarrow H_c^i(\bar{X},\mathbb{Z}/p\mathbb{Z}) \longrightarrow H_c^i(\bar{Y},\mathbb{Z}/p\mathbb{Z}) \longrightarrow . \tag{3.6.2}$$

Lemma 3.7 (Möbius).

Let B be a finite set and let $\{A_i\}$ be a finite covering by subsets. Then

$$\mathrm{card}(B) = \sum_{j\geq 0}(-1)^j \sum_{i_0<\cdots<i_j} \mathrm{card}(A_{i_0}\cap\cdots\cap A_{i_j}). \tag{3.7.1}$$

Proof. There is a Leray spectral sequence for the rational cohomology of the discrete space B and the open covering $\{A_i\}$,

$$E_1^{p,q} = \bigoplus_{i_0<\cdots<i_q} H^p(A_{i_0}\cap\cdots\cap A_{i_q},\mathbb{Q}) \Longrightarrow H^{p+q}(B,\mathbb{Q}),$$

and one computes the Euler–Poincaré characteristic of B.

Lemma 3.8.

If $\{U_i\}$ is a finite covering of X by subschemes, then

$$Z(X) = \prod_{j\geq 0} \prod_{i_0<\cdots<i_j} Z(U_{i_0}\cap\cdots\cap U_{i_j})^{(-1)^j}. \tag{3.8.1}$$

Proof. One has

$$\mathrm{card}(X(\mathbb{F}_{q^n})) = \sum_{r|n} r\ \mathrm{card}\{y\in X \mid \deg(y) = r\}, \tag{3.8.2}$$

which gives

$$Z(X) = \exp\left(\sum_{n\geq 1} \operatorname{card}(X(\mathbb{F}_{q^n}))\frac{t^n}{n}\right). \tag{3.8.3}$$

The conclusion follows from (3.7.1).

Lemma 3.9.

Suppose that formula (3.4.1), or equivalently (3.1.2) and hence (3.1.1), holds when X is a projective hypersurface. Then it holds for every separated scheme X of finite type.

Proof. By Lemmas 3.8 and 3.6, (3.4.1) holds for the complement $\mathbb{P}^n - H$ of a hypersurface. Since

$$\bigcap_i (\mathbb{P}^n - H_i) = \mathbb{P}^n - \bigcup_i H_i$$

is again the complement of a hypersurface, the open covering $\{\mathbb{P}^n - H_i\}$ of $\mathbb{P}^n - \bigcap_i H_i$ allows us to apply Lemmas 3.8 and 3.5 and deduce (3.4.1) for $\mathbb{P}^n - \bigcap_i H_i$. By Lemmas 3.8 and 3.6, (3.4.1) then holds for $\bigcap_i H_i$, and hence for every projective X. Another application of Lemmas 3.8 and 3.5 gives (3.4.1) for every quasi-projective X, in particular for every affine X. Finally, if X is separated and of finite type, choose a finite covering by affine open subschemes and apply Lemmas 3.8 and 3.5.

Source note. The printed opening references three nonexistent or malformed variants of (3.1.1); the corrected French and English cite the actual formulas (3.1.2), (3.2.1)–(3.2.2), and (3.3.1). In the proof of Theorem 3.3 the print writes fr_p^{X*} although the map displayed beside it is fr_q^X, and twice says that Frobenius kills spectral-sequence terms for $q \neq 0$ rather than for positive differential-form degree $p \neq 0$. Finally, the proof of Lemma 3.9 twice cites nonexistent (3.1.3) where its own argument is proving (3.4.1). The diplomatic French retains all these printed forms; corrected French and English use the coherent readings just stated.

4. The calculation of the zeta function of a hypersurface by Dwork's method

4.0.

Let

$$f(X) = f(X_1, \ldots, X_{n+2}) \in \mathbb{F}_q[X] \tag{4.0.1}$$

be homogeneous of degree d, and let its zero locus define the hypersurface $V \subset \mathbb{P}^{n+1}$. Denote by V^* the open subset of V on which $\prod_{i=1}^{n+2} X_i$ is invertible, and by V^*_{aff} the affine cone over V^*.

Choose a nontrivial additive character

$$\chi : \mathbb{F}_{q^\nu} \longrightarrow \mu_p(C), \tag{4.0.2}$$

where C is a fixed algebraically closed field of characteristic zero (for example $C = \mathbb{C}$). Thus

$$\text{for } x \in \mathbb{F}_{q^\nu}, \qquad \sum_{y \in \mathbb{F}_{q^\nu}} \chi(xy) = \begin{cases} q^\nu, & x = 0, \\ 0, & x \neq 0. \end{cases} \tag{4.0.3}$$

For $x = (x_1, \ldots, x_{n+2}) \in (\mathbb{F}_{q^\nu}^*)^{n+2}$ we therefore have

$$1 + \sum_{x_0 \in \mathbb{F}_{q^\nu}^*} \chi(x_0 f(x)) = \sum_{x_0 \in \mathbb{F}_{q^\nu}} \chi(x_0 f(x)) = \begin{cases} q^\nu, & f(x) = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{4.0.4}$$

Summing (4.0.4) over $x \in (\mathbb{F}_{q^\nu}^*)^{n+2}$ gives

$$\operatorname{card} V_{\mathrm{aff}}^*(\mathbb{F}_{q^\nu}) = \frac{1}{q^\nu} \sum_{x \in (\mathbb{F}_{q^\nu}^*)^{n+2}} \left\{ 1 + \sum_{x_0 \in \mathbb{F}_{q^\nu}^*} \chi(x_0 f(x)) \right\}. \tag{4.0.5}$$

Since

$$\operatorname{card} V_{\mathrm{aff}}^*(\mathbb{F}_{q^\nu}) = (q^\nu - 1) \operatorname{card} V^*(\mathbb{F}_{q^\nu}), \tag{4.0.6}$$

formula (4.0.5) becomes

$$(q^\nu - 1) \operatorname{card} V^*(\mathbb{F}_{q^\nu}) = \frac{(q^\nu - 1)^{n+2}}{q^\nu} + \frac{1}{q^\nu} \sum_{(x_0, x) \in (\mathbb{F}_{q^\nu}^*)^{n+3}} \chi(x_0 f(x)). \tag{4.0.7}$$

Write $f(X)$ explicitly as a sum of monomials,

$$f(X) = \sum_w a_w X^w, \tag{4.0.8}$$

where $w = (w_1, \ldots, w_{n+2})$, $\sum_i w_i = d$, and $X^w := X_1^{w_1} \cdots X_{n+2}^{w_{n+2}}$. Then (4.0.7) may be rewritten as

$$(q^\nu - 1) \operatorname{card} V^*(\mathbb{F}_{q^\nu}) = \frac{(q^\nu - 1)^{n+2}}{q^\nu} + \frac{1}{q^\nu} \sum_{(x_0, x) \in (\mathbb{F}_{q^\nu}^*)^{n+3}} \prod_w \chi(a_w x_0 X^w). \tag{4.0.9}$$

4.1.

We must now make explicit our "additive character with values in characteristic zero." One would like simply to take

$$\chi : x \longmapsto \exp\Big(2\pi i \operatorname{tr}_{\mathbb{F}_{q^\nu}/\mathbb{F}_p}(x)\Big) = \xi_p^{\operatorname{tr}_{\mathbb{F}_{q^\nu}/\mathbb{F}_p}(x)}. \tag{4.1.0}$$

Let

$$\Omega = \text{an algebraic closure of } \mathbb{Q}_p, \tag{4.1.1}$$

$$\operatorname{ord} : \Omega^* \longrightarrow \mathbb{Q} \qquad \text{be the valuation normalized by } \operatorname{ord}(p) = 1, \tag{4.1.2}$$

$$\xi_p \in \Omega \qquad \text{be a primitive } p\text{-th root of unity.} \tag{4.1.3}$$

One checks (cf. [2]) that there exists

$$\pi \in \mathbb{Q}_p(\xi_p), \qquad \operatorname{ord}(\pi) = \frac{1}{p-1}, \tag{4.1.4}$$

such that

$$\sum_{n\geq 0} \frac{\pi^{p^n}}{p^n} = 0. \tag{4.1.5}$$

(Notice also that the left-hand side converges in $\mathbb{Q}_p(\xi_p)$.)
Define the Artin–Hasse exponential

$$E(X) = \exp\left(\sum_{n\geq 0} \frac{X^{p^n}}{p^n}\right) \in \mathbb{Q}_p[[X]], \tag{4.1.6}$$

which, miraculously, lies in $\mathbb{Z}_p[[X]]$. Define

$$\Theta(X) = E(\pi X). \tag{4.1.7}$$

Its fundamental properties are summarized in the following lemma.

Lemma 4.2.

$$\Theta(X) = \sum_n B_n X^n, \qquad B_0 = 1, \quad B_1 = \pi, \quad \operatorname{ord}(B_n) \geq \frac{n}{p-1}. \tag{4.2.1}$$

$$\Theta(1) = \xi_p, \qquad \text{a primitive } p\text{-th root of unity.} \tag{4.2.2}$$

Let $t \in \Omega$, and suppose that, for a given integer $a > 0$,

$$t = t^{p^a}; \tag{4.2.3}$$

that is, t is a Teichmüller representative in $W(\mathbb{F}_{p^a}) \subset \Omega$. Then $\sum_{j=0}^{a-1} t^{p^j} \in \mathbb{Z}_p$, so the formal series

$$(\Theta(X))^{\sum_{j=0}^{a-1} t^{p^j}}$$

is defined, and one has the identity of formal power series

$$\prod_{j=0}^{a-1} \Theta(t^{p^j} X) = (\Theta(X))^{\sum_{j=0}^{a-1} t^{p^j}}. \tag{4.2.3.1}$$

Proof. Since $E(X) \in \mathbb{Z}_p[[X]]$, (4.2.1) follows from the definition. For (4.2.2), we have

$$\log(\Theta(1)) = 0 \qquad \text{by the choice of } \pi \text{ in (4.1.5)}, \tag{4.2.4}$$

and

(4.2.5) $$\operatorname{ord}(\Theta(1)-1)=\frac{1}{p-1}\qquad\text{by (4.2.1).}$$

By (4.2.4), there is an integer $n\geq 0$ such that $\Theta(1)^{p^n}=1$ (this is the characterization of the kernel of the logarithm), and (4.2.5) implies that $n=1$.

As for (4.2.3.1), after taking logarithms it follows by an immediate calculation, once one knows that $\sum_{j=0}^{a-1} t^{p^j}\in\mathbb{Z}_p$. But the t^{p^j}, $0\leq j\leq a-1$, are precisely the conjugates over $\mathbb{Z}_p$ of $t\in W(\mathbb{F}_{p^a})$; hence

$$\sum_{j=0}^{a-1} t^{p^j}=\operatorname{Tr}_{W(\mathbb{F}_{p^a})/\mathbb{Z}_p}(t)\in\mathbb{Z}_p.$$

Evaluating the two sides of (4.2.3.1) at $X=1$ gives the following result.

Corollary 4.3.

Let $t\in\Omega$ satisfy $t^{p^a}=t$, and denote its residue class by $\bar{t}\in\mathbb{F}_{p^a}$. Then

(4.3.1) $$\prod_{j=0}^{a-1}\Theta(t^{p^j})=\xi_p^{\operatorname{tr}_{\mathbb{F}_{p^a}/\mathbb{F}_p}(\bar{t})}.$$

Source note. The print has three mutually exposed notation slips in Lemma 4.2: it writes $A_1=\pi$ in the coefficient expansion in (4.2.1), places the Teichmüller representative in $W(\mathbb{F}_{q^a})$ although both $t^{p^a}=t$ and Corollary 4.3 use $\mathbb{F}_{p^a}$, and prints $\sum t^j$ in the exponent of (4.2.3.1) although the product and the following proof use $\sum t^{p^j}$. The diplomatic French preserves those forms; corrected French and English use B_1, $\mathbb{F}_{p^a}$, and t^{p^j}.

4.4.

Recall that the residue field of the ring of integers of $\mathbb{Q}_p(\xi_p)$ is $\mathbb{F}_p$. Let

(4.4.0) $$K=\text{the unramified extension of } \mathbb{Q}_p(\xi_p) \text{ with residue field } \mathbb{F}_q,$$

and let

(4.4.1) $$\tau\in\operatorname{Gal}(K/\mathbb{Q}_p(\xi_p))$$

be the "Frobenius", that is, the automorphism reducing to the Frobenius automorphism of $\mathbb{F}_q/\mathbb{F}_p$, $x\longmapsto x^p$.

Let

(4.4.2) $$F(X)=\sum_w A_w X^w\in K[X_1,\dots,X_{n+2}]$$

be the lifting of $f(X)$ whose coefficients satisfy $A_w^q=A_w$. Putting (4.3.1) and (4.0.9) together, and taking into account the definition (4.1.0) of χ, one obtains

$$(q^\nu-1)\operatorname{card} V^*(\mathbb{F}_{q^\nu})-\frac{(q^\nu-1)^{n+2}}{q^\nu}$$

$$=\frac{1}{q^\nu}\sum_{(X_0,X)\in(\mu_{q^\nu-1}(\Omega))^{n+3}}\prod_w\prod_{j=0}^{\operatorname{ord}(q^\nu)-1}\Theta\Big(A_w^{p^j}X_0^{p^j}X^{p^jw}\Big)\,. \tag{4.4.3}$$

Define

$$H(X)=H(X_0,\dots,X_{n+2})\in K[[X_0,\dots,X_{n+2}]]$$

by

$$H(X)=\prod_w\Theta\big(A_wX_0X_1^{w_1}\cdots X_{n+2}^{w_{n+2}}\big)\,. \tag{4.4.4}$$

We let $\mathrm{Gal}(K/\mathbb{Q}_p(\xi_p))$ act on $K[[X_0,\dots,X_{n+2}]]$ through its action on the coefficients; the componentwise action of $\sigma\in\mathrm{Gal}(K/\mathbb{Q}_p(\xi_p))$ will be denoted

$$G(X)\longmapsto G^\sigma(X). \tag{4.4.5}$$

In particular,

$$H^{\tau^j}(X)=\prod_w\Theta\Big(A_w^{\tau^j}X_0X_1^{w_1}\cdots X_{n+2}^{w_{n+2}}\Big)\,.$$

Since $A_w^{\tau^j}=A_w^{p^j}$, formula (4.4.3) may therefore be rewritten as

$$(q^\nu-1)\operatorname{card} V^*(\mathbb{F}_{q^\nu})=\frac{(q^\nu-1)^{n+2}}{q^\nu}+\frac{1}{q^\nu}\sum_{(X_0,X)\in(\mu_{q^\nu-1}(\Omega))^{n+3}}\prod_{j=0}^{\operatorname{ord}(q^\nu)-1}H^{\tau^j}(X^{p^j}). \tag{4.4.6}$$

4.5.

We shall now express the sum in (4.4.6) as a trace.

For each $b\in\mathbb{Q}$, define a p-adic Banach K-algebra

$$L(b)=\text{the subring of } K[[X_0,\dots,X_{n+2}]] \text{ consisting of the series} \tag{4.5.1}$$

which can be written

$$\sum_{dU_0=\sum_{i\geq1}U_i}B_UX_0^{U_0}X_1^{U_1}\cdots X_{n+2}^{U_{n+2}},\qquad U=(U_0,\dots,U_{n+2}), \tag{4.5.2}$$

and which satisfy

$$\inf_U\big(\operatorname{ord}(B_U)-bU_0\big)>-\infty. \tag{4.5.3}$$

It is a p-adic Banach algebra for the order function (the negative of the logarithm of the norm)

$$\inf_U \big(\mathrm{ord}(B_U) - bU_0\big) = \mathrm{ord}\left(\sum_U B_U X^U\right). \tag{4.5.4}$$

It follows from (4.4.4) and (4.2.1) that

$$H(X) \in L\left(\frac{1}{p-1}\right).$$

Let us introduce some operators. For every integer $k > 0$, define a continuous operator between Banach spaces over K by

$$\begin{cases} \psi_k : L(b) \longrightarrow L(kb), \\ \psi_k\left(\sum_U B_U X^U\right) = \sum_U B_{kU} X^U. \end{cases} \tag{4.5.5}$$

For every pair (b, b') with $b' < b$, there is a "restriction" operator, which is completely continuous (that is, a uniform limit of finite-rank operators):

$$\begin{cases} L(b) \xrightarrow{\text{restriction}} L(b'), \\ \sum_U B_U X^U \longmapsto \sum_U B_U X^U. \end{cases} \tag{4.5.6}$$

Each element G of the algebra $L(b)$ gives the continuous endomorphism of multiplication by G,

$$\begin{cases} G : L(b) \longrightarrow L(b), \\ \eta \longmapsto \eta G. \end{cases} \tag{4.5.7}$$

These operators are related by the following lemma, whose proof is found in [2] (see also [4]).

Lemma 4.5.8 (Dwork trace lemma).

Let $b \in \mathbb{Q}$, $b > 0$, let $k \in \mathbb{Z}$, $k \geq 2$, and let $G \in L(b/k)$. Then the composite

$$L(b) \xrightarrow{\text{restriction}} L(b/k) \xrightarrow{\times G} L(b/k) \xrightarrow{\psi_k} L(b),$$

denoted by $\psi_k \circ G$, is completely continuous, and its trace (cf. [4]) is given by

$$\mathrm{tr}(\psi_k \circ G) = \frac{1}{(k-1)^{n+2}} \sum_{X \in (\mu_{k-1}(\Omega))^{n+3}} G(X). \tag{4.5.9}$$

Apply this lemma with

$$\begin{cases} k = q^\nu, \\ G(X) = \prod_{j=0}^{\mathrm{ord}(q^\nu)-1} H^{\tau^j}(X^{p^j}) \in L\left(\frac{p}{(p-1)q^\nu}\right), \\ b = \frac{1}{p-1}. \end{cases}$$

Formula (4.4.6) becomes

$$
\begin{aligned}
(q^\nu - 1)\operatorname{card} V^*(\mathbb{F}_{q^\nu}) &= \frac{(q^\nu-1)^{n+2}}{q^\nu} \\
&\quad + \frac{(q^\nu-1)^{n+3}}{q^\nu}\operatorname{tr}\left(\psi_{q^\nu}\circ \prod_{j=0}^{\operatorname{ord}(q^\nu)-1} H^{\tau^j}(X^{p^j})\,\middle|\, L\left(\frac{1}{p-1}\right)\right).
\end{aligned}
\tag{4.5.10}
$$

Let $L^0(b)$ be the ideal of the elements of $L(b)$ which "vanish at zero", that is, which have no constant term. It is stable under $\psi_k \circ G$, and evidently

$$
\operatorname{tr}(\psi_k\circ G) = G(0) + \operatorname{tr}\big(\psi_k\circ G|\, L^0(b)\big)\,. \tag{4.5.11}
$$

Since $H(0)=1$, formula (4.5.10) may be rewritten as

$$
\begin{aligned}
\operatorname{card} V^*(\mathbb{F}_{q^\nu}) &= (q^\nu-1)^{n+1} \\
&\quad + \frac{(q^\nu-1)^{n+2}}{q^\nu}\operatorname{tr}\left(\psi_{q^\nu}\circ \prod_{j=0}^{\operatorname{ord}(q^\nu)-1} H^{\tau^j}(X^{p^j})\,\middle|\, L^0\left(\frac{1}{p-1}\right)\right).
\end{aligned}
\tag{4.5.12}
$$

Consider now the τ^{-1}-linear operator

$$
\tau^{-1}\circ\psi_p\circ H(X) : L(b) \longrightarrow L(b). \tag{4.5.13}
$$

Its iterates are

$$
(\tau^{-1}\circ\psi_p\circ H(X))^a = \tau^{-a}\circ\psi_{p^a}\circ H(X)H^\tau(X^p)\cdots H^{\tau^{a-1}}(X^{p^{a-1}}), \tag{4.5.14}
$$

and, if $\operatorname{ord}(q)$ divides a, then $\tau^{-a} = \mathrm{id}$. Define the τ^{-1}-linear operation

$$
\begin{cases}
\alpha : L^0\left(\dfrac{1}{p-1}\right) \longrightarrow L^0\left(\dfrac{1}{p-1}\right), \\
\qquad\qquad \alpha = \dfrac{1}{p}\,\tau^{-1}\circ\psi_p\circ H(X).
\end{cases}
\tag{4.5.15}
$$

Then (4.5.12) takes the form

$$
\operatorname{card} V^*(\mathbb{F}_{q^\nu}) = (q^\nu-1)^{n+1} + (q^\nu-1)^{n+2}\operatorname{tr}\left(\alpha^{\operatorname{ord}(q^\nu)}\,\middle|\, L^0\left(\frac{1}{p-1}\right)\right). \tag{4.5.16}
$$

Put

$$
S = \{1,\dots,n+2\}. \tag{4.5.17}
$$

For every nonempty subset $A\subset S$, define the subspace

$$
\begin{aligned}
L^A(b) = \{\,&\text{series } \sum_U B_U X^U \in L(b) \text{ such that, for every monomial } X^U \text{ with } B_U\neq 0, \\
&\text{the indices } i \text{ for which } U_i>0 \text{ are exactly 0 and the } j\in A\,\}.
\end{aligned}
\tag{4.5.18}
$$

Then

$$L^0(b) = \bigoplus_{\substack{A \subset S \\ A \neq \varnothing}} L^A(b). \tag{4.5.19}$$

Source note. The print drops ν from the factor $q^\nu - 1$ in (4.5.10) and from the denominator q^ν in (4.5.12), although both formulas are direct rewrites of (4.4.6). It also transposes the formula number (4.5.14) as (5.4.14). The diplomatic French preserves all three printed forms; corrected French and English use the algebraically forced readings displayed above.

And, for every nonempty subset $B \subset S$, one has

$$\bigoplus_{A \supset B} L^A(b) = L(b) \cap \left(\prod_{i \in B} X_i \right) K[[X]]. \tag{4.5.20}$$

Clearly every subspace $\bigoplus_{A \supset B} L^A(b)$ is stable under α. It follows that the same is true of $\bigoplus_{A \not\supset B} L^A(b)$, which is a sum of subspaces of the preceding type. Define

(4.5.21)

$$\alpha_B = \text{the } \tau^{-1}\text{-linear endomorphism induced by } \alpha \text{ on } \frac{\displaystyle\bigoplus_{A \supset B} L^A\left(\frac{1}{p-1}\right)}{\displaystyle\bigoplus_{\substack{A \supset B \\ A \neq B}} L^A\left(\frac{1}{p-1}\right)} \simeq L^B\left(\frac{1}{p-1}\right).$$

Explicitly, α_B is given by

$$\alpha_B = \frac{\tau^{-1}}{p} \psi_p H_B(X), \tag{4.5.22}$$

an operation under which L^B is stable, where $H_B(X)$ is obtained from $H(X)$ by substituting $X_i = 0$ for $i \notin \{0\} \cup B$.

Rewriting (4.5.16) once more gives

$$\operatorname{card} V^*(\mathbb{F}_{q^\nu}) = (q^\nu - 1)^{n+1} + (q^\nu - 1)^{n+2} \sum_{\substack{A \subset S \\ A \neq \varnothing}} \operatorname{tr}\Big((\alpha_A)^{\operatorname{ord}(q^\nu)} \Big). \tag{4.5.23}$$

For $A \subset S$, $A \neq \varnothing$, define the hypersurface

$$V_A \subset \mathbb{P}^{\operatorname{card}(A)-1} \simeq \mathbb{P}^{n+1} \cap \{X_i = 0 \text{ for every } i \notin A\} \tag{4.5.24}$$

by $f_A(X)$, the polynomial obtained from $f(X)$ by substituting $X_i = 0$ for $i \notin A$.

By compatibility of the formation of the series H from $f(X)$ with substitution of 0 for some of the variables, (4.5.23) gives, for every $A \subset S$, $A \neq \varnothing$,

$$\operatorname{card} V_A^*(\mathbb{F}_{q^\nu}) = (q^\nu - 1)^{\operatorname{card}(A)-1} + (q^\nu - 1)^{\operatorname{card}(A)} \sum_{B \subset A} \operatorname{tr}\Big((\alpha_B)^{\operatorname{ord}(q^\nu)} \Big). \tag{4.5.25}$$

On the other hand,

$$\operatorname{card} V(\mathbb{F}_{q^\nu}) = \sum_{\substack{A\subset S\\ A\neq\varnothing}} \operatorname{card} V_A^*(\mathbb{F}_{q^\nu}). \tag{4.5.26}$$

After an easy summation, (4.5.25) therefore gives

$$\begin{aligned}\operatorname{card} V(\mathbb{F}_{q^\nu}) &= 1 + q^\nu + q^{2\nu} + \cdots + q^{\nu(\operatorname{card}(S)-1)} \\ &\quad + \sum_{\substack{B\subset S\\ B\neq\varnothing}} (q^\nu - 1)^{\operatorname{card}(B)} q^{\nu\operatorname{card}(S-B)} \operatorname{tr}\Big((\alpha_B)^{\operatorname{ord}(q^\nu)}\Big).\end{aligned} \tag{4.5.27}$$

Since α, and hence the α_B, are τ^{-1}-linear, the operators $\alpha_B^{\operatorname{ord}(q)}$ are K-linear. Define the characteristic series

$$\Delta_B(t) \in 1 + tK[[t]]$$

by

$$\Delta_B(t) = \det\left(1 - t\alpha_B^{\operatorname{ord}(q)} \;\middle|\; L^B\left(\frac{1}{p-1}\right)\right). \tag{4.5.28}$$

Introduce the following endomorphisms of $1 + tK[[t]]$:

$$\varphi : g(t) \longmapsto g^\varphi(t) = g(qt) \tag{4.5.29}$$

and

$$\delta = 1 - \varphi : g(t) \longmapsto g^\delta(t) = \frac{g(t)}{g(qt)}. \tag{4.5.30}$$

Taking account of the identities

$$Z(t, V/\mathbb{F}_q) = \exp\left(\sum_{\nu\geq 1} \operatorname{card} V(\mathbb{F}_{q^\nu}) \frac{t^\nu}{\nu}\right) \tag{4.5.31}$$

and

$$\det\Big(1 - t\alpha_B^{\operatorname{ord}(q)}\Big) = \exp\left(-\sum_{\nu\geq 1} \operatorname{tr}\Big((\alpha_B)^{\nu\operatorname{ord}(q)}\Big) \frac{t^\nu}{\nu}\right), \tag{4.5.32}$$

formula (4.5.27) can be rewritten as

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1} (1 - q^i t) = \prod_{j=1}^{n+2} \left(\prod_{\substack{B\subset S\\ \operatorname{card}(B)=j}} \Delta_B(t)\right)^{-(-\delta)^j \varphi^{n+2-j}}. \tag{4.5.33}$$

Source note. The print numbers (4.5.23) as (4.5.2.3), abbreviates $\operatorname{card}(A)$ as $\operatorname{car}(A)$ in the first exponent of (4.5.25), and reuses the number (4.5.23) for the final zeta identity. The corrected French and English layers use the sequential numbers and the full cardinality notation displayed above; the diplomatic French preserves the printed forms.

4.6.

We now study the integrality of the series Δ_B. Put

(4.6.1) $$\mathcal{O} = \text{the ring of integers of } K,$$

and define a filtration on $1 + t\mathcal{O}[[t]]$ by

(4.6.2) $$F^i(1 + t\mathcal{O}[[t]]) = 1 + q^i t\mathcal{O}[[q^i t]] = \varphi^i(1 + t\mathcal{O}[[t]]).$$

One has

(4.6.3) $$\varphi(F^i) \subset F^{i+1},$$

(4.6.4) $$\delta(F^i) \subset F^i,$$

(4.6.5) $$\delta \text{ induces the identity on } F^i/F^{i+1}.$$

Lemma 4.6.6.

One has $\Delta_B(t) \in 1 + t\mathcal{O}[[t]]$; moreover, if an integer j satisfies

(4.6.7) $$jd < \mathrm{card}(B),$$

then

(4.6.8) $$\Delta_B(t) \in F^j(1 + t\mathcal{O}[[t]]).$$

Proof. The Banach spaces $L^B\left(\frac{1}{p-1}\right)$ are all of type $B(I)$ in the sense of Serre [7]; that is, they admit orthonormal bases, namely the monomials

(4.6.9) $$\pi^{U_0} X_0^{U_0} X_1^{U_1} \cdots X_{n+2}^{U_{n+2}}$$

which belong to them. To say that this is an orthonormal basis means, by definition, that

(4.6.10) $$\mathrm{ord}\left(\sum_U C_U \pi^{U_0} X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}\right) = \min_U \mathrm{ord}(C_U).$$

Let

(4.6.11) $$R_B = \text{the } \mathcal{O}\text{-submodule of } L^B\left(\frac{1}{p-1}\right) \text{ formed by the elements of order} \geq 0.$$

To prove the lemma, it is enough to show

(4.6.12) $$jd < \mathrm{card}(B) \quad \Longrightarrow \quad (\alpha_B)^{\mathrm{ord}(q)} R_B \subset q^j R_B,$$

or, a fortiori, to show

(4.6.13) $$jd < \mathrm{card}(B) \quad \Longrightarrow \quad \alpha_B(R_B) \subset p^j R_B.$$

By (4.5.22),

(4.6.14) $$\alpha_B = \frac{1}{p}\,\tau^{-1} \circ \psi_p \circ H_B(X).$$

Since

(4.6.15) $$H(X) \in L\left(\frac{1}{p-1}\right), \qquad \mathrm{ord}(H(X)) \geq 0,$$

we have

(4.6.16) $$H_B(X)\, R_B \subset R_B$$

and, plainly,

(4.6.17) $$\tau^{-1}(R_B) = R_B.$$

Consequently, (4.6.13) follows from

(4.6.18) $$jd < \mathrm{card}(B) \quad \Longrightarrow \quad \psi_p(R_B) \subset p^{j+1} R_B.$$

We verify (4.6.18) monomial by monomial. We have

(4.6.19) $$\begin{cases} \psi_p\Big(\pi^{U_0} X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}\Big) = 0, & \text{unless } p \mid U_i \text{ for every } i, \\ \psi_p\Big(\pi^{pU_0} X_0^{pU_0} \cdots X_{n+2}^{pU_{n+2}}\Big) = \pi^{(p-1)U_0}\, \underbrace{\pi^{U_0} X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}}_{\in R_B}. \end{cases}$$

The conditions for $\pi^{U_0} X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}$ to belong to $L^B\left(\frac{1}{p-1}\right)$ are

(4.6.20) $$\begin{cases} dU_0 = \displaystyle\sum_{i \geq 1} U_i, \\ U_i > 0 \Longleftrightarrow i \in B, \qquad i \geq 1. \end{cases}$$

In particular,

(4.6.21) $$dU_0 \geq \mathrm{card}(B).$$

Hence

(4.6.22) $$\mathrm{ord}\Big(\pi^{(p-1)U_0}\Big) = U_0 \geq \frac{\mathrm{card}(B)}{d} > j,$$

and, since $U_0 \in \mathbb{Z}$,

(4.6.23) $$\mathrm{ord}\Big(\pi^{(p-1)U_0}\Big) = U_0 \geq j + 1.$$

Together with (4.6.19), this proves (4.6.18).

5. A theorem of Ax [1] (see also [4], Section 7)

Theorem 5.0.

Let V be a hypersurface in $\mathbb{P}^{n+1}$ of degree $d > 0$, defined over $\mathbb{F}_q$, and let j be an integer. If

$$jd < n + 2, \tag{5.0.1}$$

then

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1} (1 - q^i t) \in 1 + q^j t\mathbb{Z}[[q^j t]], \tag{5.0.2}$$

and the reciprocal of every zero or pole of

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1} (1 - q^i t) \tag{5.0.3}$$

is of the form q^j times an algebraic integer.

Theorem 5.0 bis.

Let V_{aff} be a hypersurface in $\mathbb{A}^{n+2}$ defined by a homogeneous form of degree d, over $\mathbb{F}_q$. If

$$jd < n + 2, \tag{5.0.4}$$

then

$$Z(t, V_{\text{aff}}/\mathbb{F}_q) \in 1 + q^j t\mathbb{Z}[[q^j t]]. \tag{5.0.5}$$

In particular,

$$\operatorname{card}\big(V_{\text{aff}}(\mathbb{F}_q)\big) \equiv 0 \pmod{q^j}, \tag{5.0.6}$$

and consequently

$$\operatorname{card}\big(V(\mathbb{F}_q)\big) \geq \frac{q^j - 1}{q - 1} > 0, \tag{5.0.6 bis}$$

which improves the Chevalley–Warning theorem [8].

Source note. The print numbers the hypothesis in Theorem 5.0 bis as (5.0.7), duplicating the number subsequently used for the point-count identity. The corrected French and English layers use the sequential number (5.0.4); the diplomatic French preserves the printed number.

Proof. The implication (5.0.5) $\Rightarrow$ (5.0.6) is clear on taking the coefficient of t in the zeta function. Assertions (5.0.4) and (5.0.2) are equivalent, since

$$\operatorname{card} V_{\text{aff}}(\mathbb{F}_{q^\nu}) = 1 + (q^\nu - 1) \operatorname{card} V(\mathbb{F}_{q^\nu}), \tag{5.0.7}$$

whence

$$Z(t, V_{\mathrm{aff}}/\mathbb{F}_q) = \frac{Z(t, V/\mathbb{F}_q)}{Z(qt, V/\mathbb{F}_q)(1-t)} \tag{5.0.8}$$

and

$$Z(t, V/\mathbb{F}_q) = \prod_{i\geq 0} \left[(1-q^i t) Z(q^i t, V_{\mathrm{aff}}/\mathbb{F}_q)\right]. \tag{5.0.9}$$

Assertions (5.0.2) and (5.0.3) are likewise equivalent, by the theorem of Fatou [3] and the rationality of the zeta function. Since

$$Z(t, V/\mathbb{F}_q) \in \mathbb{Z}[[t]], \tag{5.0.10}$$

it is enough to prove

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1} (1-q^i t) \in F^j(1 + t\mathcal{O}[[t]]). \tag{5.0.11}$$

In view of (4.5.33) and (4.6.4)–(4.6.5), it is enough to show, for $\varnothing \neq B \subset \{1, \ldots, n+2\}$, that

$$\Delta_B(t) \in F^{j+\mathrm{card}(B)-(n+2)}. \tag{5.0.12}$$

We may finish the proof by applying Lemma 4.6.6 once we have proved

$$jd < n+2 \quad \Longrightarrow \quad \big(j + \mathrm{card}(B) - (n+2)\big)d < \mathrm{card}(B). \tag{5.0.13}$$

This may be rewritten as

$$jd < n+2 \quad \Longrightarrow \quad (j - (n+2))d < \mathrm{card}(B)(1-d). \tag{5.0.14}$$

But

$$\inf_{\varnothing \neq B \subset \{1,\ldots,n+2\}} \mathrm{card}(B)(1-d) = (n+2)(1-d),$$

so it is enough to prove

$$jd < n+2 \quad \Longrightarrow \quad (j - (n+2))d < (n+2)(1-d), \tag{5.0.15}$$

which is true.

Source note. In (5.0.9) the print has the impossible factor $(1-q^i f)$ rather than $(1-q^i t)$, and the last paragraph cites Lemma 4.4.6 rather than the integrality lemma 4.6.6 used there. Both are corrected above; the diplomatic French retains the printed forms.

Remark 5.1.

Condition

$$jd < n+2 \tag{5.1.1}$$

admits a cohomological interpretation. Let V be a smooth hypersurface of degree d and dimension n. As in XI, 2.1, put, for $a+b=n$,

$$h_0^{a,b} = \dim H^b(V, \Omega_V^a) - \delta_{a,b} = \dim E^{a,b}(V), \tag{5.1.2}$$

where $E^{a,b}(V)$ is the bidegree-(a,b) component of the vanishing Hodge cohomology (compare XVII, 5.4.4)

$$E^{a,b}(V) = \mathrm{Coker}\Big(H^b(\mathbb{P}^{n+1}, \Omega^a_{\mathbb{P}^{n+1}}) \longrightarrow H^b(V, \Omega_V^a)\Big). \tag{5.1.3}$$

Then, by XI, 2.8,

$$jd < n+2 \iff h_0^{a,n-a} = 0 \quad \text{for } a < j \text{ and } a > n-j. \tag{5.1.4}$$

In other words, the condition says that the Hodge level (XI, 2.7) of the vanishing de Rham cohomology of V,

$$E^n_{\mathrm{DR}}(V) = \mathrm{Coker}\big(H^n_{\mathrm{Hdg}}(\mathbb{P}^{n+1}) \longrightarrow H^n_{\mathrm{Hdg}}(V)\big),$$

is at most $n-2j$, or equivalently that its coniveau is at least j. On the other hand, the conclusion also says that the eigenvalues of the Frobenius $\mathfrak{F}_q$ acting on the vanishing ℓ-adic cohomology ($\ell \neq p$) are algebraic integers divisible by q^j (XXI, A.2). Thus Ax's result 5.0 supports the following conjecture, suggested by the philosophy of the level of motives and relating the classical Hodge and Tate conjectures [5].

Conjecture 5.2.

Let $f : X \to S$ be a proper smooth morphism, with S integral, of finite type over $\mathrm{Spec}(\mathbb{Z})$, and dominant over $\mathrm{Spec}(\mathbb{Z})$; let i, j be integers. Suppose that the Hodge coniveau of H^i of the generic fiber of f is at least j. Then, for every point $s \in S$, the coniveau (XX, 2.0.6) of $H^i(X_s, \mathbb{Q}_\ell)$ is at least j. Hence (XX, 2.2), if s is a closed point of S, the eigenvalues of Frobenius acting on $H^i(X_s, \mathbb{Q}_\ell)$ are algebraic integers divisible by q^j, where $q = \mathrm{card}(k(s))$.

6. End of the proof of the congruence formula

Taking account of (4.6.6) and (4.6.3)–(4.6.4), formula (4.5.33) gives

$$Z(t, V/\mathbb{F}_q)(1-t) \equiv \Delta_S(t)^{(-1)^{n+1}} \pmod{1 + qt\mathcal{O}[[t]]}. \tag{6.0.1}$$

We now calculate $\Delta_S(t)$ modulo $\pi\mathcal{O}[[t]]$. Define, as in (4.6.11),

$$W_S = R_S \otimes_{\mathcal{O}} \mathbb{F}_q. \tag{6.0.2}$$

Source note. The print cites (4.5.24), but the congruence follows from the terminal zeta identity (4.5.33). It also defines W_S using R_B, although the entire section and every following formula concern R_S. The corrected French and English layers use (4.5.33) and R_S; the diplomatic French retains the printed forms.

$$\bar{\alpha}_S : W_S \longrightarrow W_S \tag{6.0.3}$$

denotes the τ^{-1}-linear operation induced by α_S;

$$R_S(i) = \text{the submodule of } R_S \text{ formed by the elements of degree } i \text{ in } X_0, \tag{6.0.4}$$

and

$$W_S(i) = R_S(i) \otimes_{\mathcal{O}} \mathbb{F}_q. \tag{6.0.5}$$

Proposition 6.0.6.

One has

$$\alpha_S(R_S) \subset R_S(1) + pR_S.$$

Proof. We have the factorization (4.6.14)

$$\alpha_S = \frac{1}{p}\,\tau^{-1} \circ \psi_p \circ H(X). \tag{6.0.7}$$

We know (see (4.6.15)) that

$$H(X)\,R_S \subset R_S, \tag{6.0.8}$$

and plainly

$$\tau^{-1}\big(R_S(1) + pR_S\big) = R_S(1) + pR_S. \tag{6.0.9}$$

It is therefore enough to verify

$$\frac{1}{p}\,\psi_p(R_S) \subset R_S(1) + pR_S, \tag{6.0.10}$$

which is clear from (4.6.19).

Corollary 6.0.11.

One has

$$\bar{\alpha}_S(W_S) \subset W_S(1).$$

Corollary 6.0.12.

One has the congruence

$$\Delta_S(t) \equiv \det\Big(1 - t(\bar{\alpha}_S)^{\mathrm{ord}(q)} \;\Big|\; W_S(1)\Big) \pmod{\pi\mathcal{O}[[t]]}.$$

Proposition 6.1.

The τ^{-1}-linear operation $\bar{\alpha}_S : W_S(1) \to W_S(1)$ is equal to

$$\frac{1}{p}\,\tau^{-1} \circ \psi_p \circ (X_0 f(X))^{p-1}. \tag{6.1.1}$$

Proof. Write

$$H(X) = 1 + \sum_{i \geq 1} H_i(X), \tag{6.1.2}$$

where

$$H_i(X) \text{ has degree } i \text{ in } X_0. \tag{6.1.3}$$

Again by (4.6.19),

$$\frac{1}{p}\,\psi_p R_S(i) \subset pR_S \qquad \text{if } i \neq p. \tag{6.1.4}$$

Since

$$H_i(X)\, R_S(j) \subset R_S(i+j), \tag{6.1.5}$$

we have

$$\left(\alpha_S - \frac{1}{p}\,\tau^{-1} \circ \psi_p \circ H_{p-1}(X)\right)(R_S(1)) \subset pR_S. \tag{6.1.6}$$

Consequently,

$$\left[(\alpha_S)^{\mathrm{ord}(q)} - \left(\frac{1}{p}\,\tau^{-1} \circ \psi_p \circ H_{p-1}(X)\right)^{\mathrm{ord}(q)}\right] R_S(1) \subset pR_S. \tag{6.1.7}$$

It remains to calculate $H_{p-1}(X)$. From (4.4.4),

$$\begin{aligned} H(X) &= \prod_W \Theta(\pi A_W X_0 X^W) \\ &= \prod_W \exp\left(\sum_{r \geq 0} \frac{(\pi A_W X_0 X^W)^{p^r}}{p^r}\right) \\ &= \prod_W \prod_{r \geq 0} \exp\left(\frac{(\pi A_W X_0 X^W)^{p^r}}{p^r}\right) \\ &\equiv \prod_W \exp(\pi A_W X_0 X^W) \pmod{\text{terms of degree } \geq p} \\ &= \exp\left(\pi \sum_W A_W X_0 X^W\right) \pmod{\text{terms of degree } \geq p} \\ &= \exp(\pi X_0 f(X)) \pmod{\text{terms of degree } \geq p}. \end{aligned} \tag{6.1.8}$$

It follows that

$$H_{p-1}(X) = \frac{\pi^{p-1} X_0^{p-1}}{(p-1)!}\, f(X)^{p-1}. \tag{6.1.9}$$

One easily checks that

$$\frac{\pi^{p-1}}{p} \equiv -1 \pmod{p}, \tag{6.1.10}$$

and hence

$$\frac{1}{p}\, H_{p-1}(X) \equiv (X_0 f(X))^{p-1} \pmod{p}. \tag{6.1.11}$$

This completes the proof of Proposition 6.1.

Corollary 6.1.12.

The linear operator

$$(\bar{\alpha}_S)^{\mathrm{ord}(q)} : W_S(1) \longrightarrow W_S(1)$$

is precisely

$$\psi_q \circ (X_0 f(X))^{q-1}.$$

Proof. By (6.1.1),

$$\begin{aligned}
(\bar{\alpha}_S)^{\mathrm{ord}(q)} &= \left(\frac{1}{p}\,\tau^{-1} \circ \psi_p \circ (X_0 f(X))^{p-1}\right)^{\mathrm{ord}(q)} \\
&= \psi_q \circ \prod_{j=0}^{\mathrm{ord}(q)-1} \left(X_0^{p^j} f^{\tau^j}(X^{p^j})\right)^{p-1} \\
&= \psi_q \circ \prod_{j=0}^{\mathrm{ord}(q)-1} (X_0 f(X))^{p^j(p-1)} \\
&= \psi_q \circ (X_0 f(X))^{q-1}.
\end{aligned}$$

Source note. The first product in the printed proof has the undefined upper bound $\mathrm{ord}(a) - 1$; the next line and the definition of q require $\mathrm{ord}(q) - 1$. The corrected French and English layers use the latter; the diplomatic French preserves the print.

Corollary 6.1.13.

Modulo p, one has the congruence

$$Z(t, V/\mathbb{F}_q) \equiv \frac{\det\Big(1 - t\,\psi_q \circ (X_0 f(X))^{q-1} \;\Big|\; W_S(1)\Big)^{(-1)^{n+1}}}{1-t}.$$

6.2.

It remains to calculate the cohomology groups of $\mathcal{O}_V$. Put

$$\mathfrak{F}_q = \text{the operation of raising to the } q\text{-th power in } \mathcal{O}_V. \tag{6.2.1}$$

There is an exact sequence of sheaves on $\mathbb{P}^{n+1} = \mathbb{P}$,

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}}(-d) \xrightarrow{f} \mathcal{O}_{\mathbb{P}} \longrightarrow \mathcal{O}_V \longrightarrow 0, \tag{6.2.2}$$

and the following diagram is commutative:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{O}_{\mathbb{P}}(-d) & \xrightarrow{f} & \mathcal{O}_{\mathbb{P}} & \longrightarrow & \mathcal{O}_V & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f^{q-1}\mathfrak{F}_q} & & \downarrow{\scriptstyle \mathfrak{F}_q} & & \downarrow{\scriptstyle \mathfrak{F}_q} & & \\ 0 & \longrightarrow & \mathcal{O}_{\mathbb{P}}(-d) & \xrightarrow{f} & \mathcal{O}_{\mathbb{P}} & \longrightarrow & \mathcal{O}_V & \longrightarrow & 0. \end{array} \tag{6.2.3}$$

Taking account of

$$\begin{cases} H^i(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}) = 0, & i \neq 0, \\ H^i(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) = 0, & i \neq 0, n+1, \end{cases} \tag{6.2.4}$$

the long exact cohomology sequence associated with (6.2.2) gives

$$\begin{cases} H^0(V, \mathcal{O}_V) \simeq \mathbb{F}_q, & \\ H^i(V, \mathcal{O}_V) = 0, & i \neq 0, n, \\ H^n(V, \mathcal{O}_V) \simeq H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)). & \end{cases} \tag{6.2.5}$$

The commutative diagram (6.2.3) therefore gives

$$\det\Big(1 - t\mathfrak{F}_q \;\Big|\; H^0(V, \mathcal{O}_V)\Big) = 1 - t \tag{6.2.6}$$

and

$$\det\Big(1 - t\mathfrak{F}_q \;\Big|\; H^n(V, \mathcal{O}_V)\Big) = \det\Big(1 - tf^{q-1}\mathfrak{F}_q \;\Big|\; H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d))\Big). \tag{6.2.7}$$

It remains only to prove

$$\begin{aligned} &\det\Big(1 - tf^{q-1}\mathfrak{F}_q \;\Big|\; H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d))\Big) \\ &\quad = \det\Big(1 - t\,\psi_q \circ (X_0 f(X))^{q-1} \;\Big|\; W_S(1)\Big), \end{aligned} \tag{6.2.8}$$

which is a consequence of the following proposition.

Proposition 6.3.

There is an isomorphism

$$W_S(1) \simeq \mathrm{Hom}_{\mathbb{F}_q}\big(H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)), \mathbb{F}_q\big) \tag{6.3.1}$$

with respect to which the operator

$$\psi_q \circ (X_0 f(X))^{q-1} : W_S(1) \longrightarrow W_S(1) \tag{6.3.2}$$

is the transpose of

$$f(X)^{q-1}\mathfrak{F}_q : H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) \longrightarrow H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)). \tag{6.3.3}$$

Proof. Computing $H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d))$ by means of the standard covering of $\mathbb{P}^{n+1}$ gives

$$\begin{aligned} H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) &= \text{the } \mathbb{F}_q\text{-vector space with basis the monomials} \\ &\qquad X_1^{W_1}\cdots X_{n+2}^{W_{n+2}}, \qquad W_i \in \mathbb{Z}, \quad \sum_i W_i = -d, \\ &= \text{the } \mathbb{F}_q\text{-subspace with basis those monomials} \\ &\qquad X_1^{W_1}\cdots X_{n+2}^{W_{n+2}} \text{ for which } \exists i \text{ with } W_i \geq 0. \end{aligned} \tag{6.3.4}$$

On the other hand, $W_S(1) = R_S(1) \otimes_{\mathcal{O}} \mathbb{F}_q$, and

$$\begin{aligned} R_S(1) &= \text{the free } \mathcal{O}\text{-module with basis the monomials} \\ &\qquad (\pi X_0) X_1^{U_1}\cdots X_{n+2}^{U_{n+2}}, \qquad \sum_i U_i = d, \quad U_i > 0 \ \forall i. \end{aligned} \tag{6.3.5}$$

The pairing

$$(\ ,\ ) : W_S(1) \times H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) \longrightarrow \mathbb{F}_q \tag{6.3.6}$$

defined by

$$\Big(\mathrm{class}\Big(\pi X_0 X_1^{U_1}\cdots X_{n+2}^{U_{n+2}}\Big), X_1^{W_1}\cdots X_{n+2}^{W_{n+2}}\Big) = \begin{cases} 1, & U_i = -W_i \text{ for every } i, \\ 0, & \text{otherwise,} \end{cases} \tag{6.3.7}$$

is plainly a perfect duality of $\mathbb{F}_q$-vector spaces. One checks immediately that $\psi_q \circ (X_0 f(X))^{q-1}$ is the transpose of $f(X)^{q-1}\mathfrak{F}_q$, which proves Proposition 6.3.

Source note. In (6.3.4) the print has H^{i+1}, while Proposition 6.3, (6.3.3), and the sentence introducing the computation all require H^{n+1}. The corrected French and English layers use $n+1$; the diplomatic French preserves the printed exponent.

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